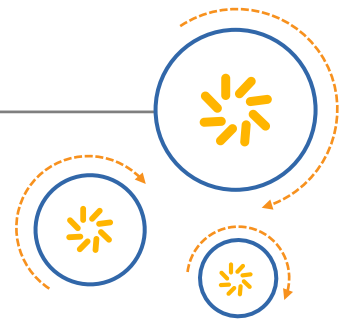




Qualcomm Technologies, Inc.



Qualcomm® Snapdragon™ 410 Processor APQ8016

Hardware Register Description

January 2016

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LM80-P0436-13 Rev. C

Revision history

Revision	Date	Description
A	June 30, 2015	Initial release
B	July 23, 2015	Updated Overview section.
C	January 22, 2016	Removed chapter: Multimedia Subsystem Oxili

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1 Introduction

1.1 Overview

This hardware register description document lists the programmable registers for the APQ8016 processor chip. It is intended for embedded OEMs or developers who require information about how to program the APQ8016 hardware.

Complementary ICs within this APQ chipset include:

- Audio codec
- Power manager
- WLAN and BT connectivity

1.2 System memory map

A[31:0]= 0x FFFF FFFF	DDR (2048)
A[31:0]= 0x 8000 0000	RESERVED
A[31:0]= 0x 6000 0000	RESERVED
A[31:0]= 0x 4000 0000	RESERVED
A[31:0]= 0x 1000 0000	RESERVED
A[31:0]= 0x 0C00 0000	RESERVED
A[31:0]= 0x 0B80 0000	APSS AHB (8MB)
A[31:0]= 0x 0B00 0000	WCSS AHB (16MB)
A[31:0]= 0x 0A00 0000	QDSS STM (16MB)
A[31:0]= 0x 0900 0000	RESERVED
A[31:0]= 0x 0880 2000	APSS AHB Timeout (4KB)
A[31:0]= 0x 0880 1000	WCSS Bus Timeout (4KB)
A[31:0]= 0x 0880 0000	RESERVED
A[31:0]= 0x 0862 0000	RESERVED
A[31:0]= 0x 0860 4000	IMEM (16KB)
A[31:0]= 0x 0860 0000	RESERVED
A[31:0]= 0x 0844 0000	RESERVED
A[31:0]= 0x 0842 0000	GMEM (128KB)
A[31:0]= 0x 0840 0000	RESERVED
A[31:0]= 0x 0820 0000	APSS L2/TCM (2 MB)
A[31:0]= 0x 0800 0000	PCNOC (128MB)
A[31:0]= 0x 0000 0000	

Figure 1-1 System memory map

1.3 PCNOC memory map

A[31:0]= 0x 07FF FFFF	RESERVED
A[31:0]= 0x 0790 0000	USB2.0 (256KB)
A[31:0]= 0x 078C 0000	BLSP1 (256KB)
A[31:0]= 0x 0788 0000	SDC2 (256KB)
A[31:0]= 0x 0784 0000	SDC1 (256KB)
A[31:0]= 0x 0780 0000	RESERVED
A[31:0]= 0x 0772 0000	AUDIO & DIGITAL CODEC (128KB)
A[31:0]= 0x 0770 0000	RESERVED
A[31:0]= 0x 0600 0000	RESERVED
A[31:0]= 0x 0480 0000	MSS (8MB)
A[31:0]= 0x 0400 0000	PMIC ARB (32MB)
A[31:0]= 0x 0200 0000	TCU CFG (2MB)
A[31:0]= 0x 01E0 0000	VIDEO CORE (1MB)
A[31:0]= 0x 01D0 0000	GPU (1MB)
A[31:0]= 0x 01C0 0000	CAMERA SS (1MB)
A[31:0]= 0x 01B0 0000	DISP_SS MDP (1MB)
A[31:0]= 0x 01A0 0000	TCSR (1MB)
A[31:0]= 0x 0190 0000	CLK CTRL (1MB)
A[31:0]= 0x 0180 0000	RESERVED
A[31:0]= 0x 0140 0000	TLMM (4MB)
A[31:0]= 0x 0100 0000	RESERVED
A[31:0]= 0x 0090 0000	QDSS CFG (1MB)
A[31:0]= 0x 0080 0000	RESERVED
A[31:0]= 0x 0074 0000	Crypto CFG (256KB)
A[31:0]= 0x 0070 0000	RESERVED
A[31:0]= 0x 0060 0000	RESERVED
A[31:0]= 0x 0058 0000	RESERVED
A[31:0]= 0x 0050 0000	RESERVED
A[31:0]= 0x 004C 0000	BIMC DEHR (64KB)
A[31:0]= 0x 004B 0000	MPM (64KB)
A[31:0]= 0x 004A 0000	EBI1PHY_CFG (128KB)
A[31:0]= 0x 0048 0000	BIMC_CFG (512KB)
A[31:0]= 0x 0040 0000	RESERVED
A[31:0]= 0x 0030 0000	RPM (1MB)
A[31:0]= 0x 0020 0000	BOOT ROM (1MB)
A[31:0]= 0x 0010 0000	RESERVED
A[31:0]= 0x 0006 C000	PDM (16KB)
A[31:0]= 0x 0842 0000	RESERVED
A[31:0]= 0x 0006 0000	RESERVED
A[31:0]= 0x 0005 8000	IMEM CFG(32KB)
A[31:0]= 0x 0005 0000	RESERVED
A[31:0]= 0x 0000 0000	RESERVED

Figure 1-2 PCNOC memory map

2 Bus Integrated Memory Controller

2.1 BIMC_MISC

BIMC Misc Global

2.1.1 BIMC_BIMC_MISC

BIMC Misc Global

0x00400020 BIMC_CORE_CLK_SEL

Type: RW

Clock: Config Clock

Reset State: 0x00000000

Control bit in this register enables legacy mode clocking scheme. In this scheme, BIMC block gets a 2x clock input from the clock controller. BIMC locally divides the clock extract 1x clock.

BIMC_CORE_CLK_SEL

Bits	Name	Description
1	CLK_2X_MODE	0: bmc_core_clk input is 1x when CLK_2X_MODE_CTRL_EN is 1 1: bmc_core_clk input is 2x clock when CLK_2X_MODE_CTRL_EN is 1
0	CLK_2X_MODE_CTRL_EN	0: CLK_2X_MODE select control used from BIMC interface 1: CLK_2X_MODE select control used from this register bit 1 (programmed)

0x00400024 BIMC_CORE_CLK_PERIOD

Type: RW

Clock: Config Clock

Reset State: 0x00000000

Core Clock Period Register

BIMC_CORE_CLK_PERIOD

Bits	Name	Description
28	CLK_PERIOD_RESOLUTION	1: DDR clock period defined in period field of this register is in 1ps unit. 0: DDR clock period is in 100ps unit.
16:0	PERIOD	value of core clock period defined in unit described by encoding in bit 28. When CLK_PERIOD_RESOLUTION is 100ps, maximum value that can be programmed for the PERIOD is 255

**0x00400100+ BIMC_SCMOn_LOCAL_CGC_THRESHOLD, n=[0..1]
0x4*n****Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000

used to compare with core clk period info from time based feature design for disabling specific SCMO internal core clk CGC(s).

BIMC_SCMOn_LOCAL_CGC_THRESHOLD

Bits	Name	Description
28	THRESHOLD_ENA	1: enable frequency based clock gating. 0: disable frequency based clock gating.
7:0	THRESHOLD_VAL	value of core clk period in 100ps unit. If the period info from time based module is smaller than the value from these bits, then SCMO internal core clk CGCs will be disabled and clk is not gated. The following are some examples: 0x00: the minimum value of the setting and the comparator output is always low. This means clk gating will be enabled if THRESHOLD_ENA of this register is also set to 0 0x01: 100ps=10GHz 0x0c: 1200ps=833MHz 0xFF: 6553.6ns=15.25kHz (max. setting)

**0x00400140+ BIMC_DPEn_LOCAL_CGC_THRESHOLD, n=[0..1]
0x4*n****Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000

used to compare with core clk period info from time based feature design for disabling DPE internal core clk CGC(s).

BIMC_DPEn_LOCAL_CGC_THRESHOLD

Bits	Name	Description
28	THRESHOLD_ENA	1: enable frequency based clock gating. 0: disable frequency based clock gating.
7:0	THRESHOLD_VAL	value of core clk period in 100ps unit. If the period info from time based module is smaller than the value from these bits, then DPE internal core clk CGCs will be disabled and clk is not gated. The following are some examples: 0x00: the minimum value of the setting and the comparator output is always low. This means clk gating will be enabled if THRESHOLD_ENA of this register is also set to 0 0x01: 100ps=10GHz 0x0c: 1200ps=833MHz 0xFF: 6553.6ns=15.25kHz (max. setting)

0x00400180 BIMC_MPORT_CORE_DOMAIN_LOCAL_CGC_THRESHOLD**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000

Frequency threshold value to disable internal clock gating of core clk domain logic of masterport sub-module.

BIMC_MPORT_CORE_DOMAIN_LOCAL_CGC_THRESHOLD

Bits	Name	Description
28	THRESHOLD_ENA	1: enable frequency based clock gating. 0: disable frequency based clock gating.
7:0	THRESHOLD_VAL	value of core clk period in 100ps unit. If the period info from time based module is smaller than the value from these bits, then DPE internal core clk CGCs will be disabled and clk is not gated. The following are some examples: 0x00: the minimum value of the setting and the comparator output is always low. This means clk gating will be enabled if THRESHOLD_ENA of this register is also set to 0 0x01: 100ps=10GHz 0x0c: 1200ps=833MHz 0xFF: 6553.6ns=15.25kHz (max. setting)

**0x004001C0+ BIMC_MPORTn_CLK_LOCAL_CGC_THRESHOLD, n=[0..6]
0x4*n**
Type: RW**Clock:** Config Clock**Reset State:** 0x00000000

used to compare with master clk period info from time based feature design for disabling internal clk CGC(s) for master clk domain logic in masterport submodule.

BIMC_MPORTn_CLK_LOCAL_CGC_THRESHOLD

Bits	Name	Description
28	THRESHOLD_ENA	1: enable frequency based clock gating. 0: disable frequency based clock gating.
7:0	THRESHOLD_VAL	value of master clk period in 100ps unit. If the period info from time based module is smaller than the value from these bits, then masterport internal master clk CGCs will be disabled and clk is not gated. The following are some examples: 0x00: the minimum value of the setting and the comparator output is always low. This means clk gating will be enabled if THRESHOLD_ENA of this register is also set to 0 0x01: 100ps=10GHz 0x0c: 1200ps=833MHz 0xFF: 6553.6ns=15.25kHz (max. setting)

0x00400200 BIMC_SWAY_LOCAL_CGC_THRESHOLD
Type: RW**Clock:** Config Clock**Reset State:** 0x00000000

Frequency threshold value to disable internal clock gating of masterport sub-module.

BIMC_SWAY_LOCAL_CGC_THRESHOLD

Bits	Name	Description
28	THRESHOLD_ENA	1: enable frequency based clock gating. 0: disable frequency based clock gating.

BIMC_SWAY_LOCAL_CGC_THRESHOLD (cont.)

Bits	Name	Description
7:0	THRESHOLD_VAL	value of core clk period in 100ps unit. If the period info from time based module is smaller than the value from these bits, then DPE internal core clk CGCs will be disabled and clk is not gated. The following are some examples: 0x00: the minimum value of the setting and the comparator output is always low. This means clk gating will be enabled if THRESHOLD_ENA of this register is also set to 0 0x01: 100ps=10GHz 0x0c: 1200ps=833MHz 0xFF: 6553.6ns=15.25kHz (max. setting)

0x00400240+ BIMC_SCMOn_CGC_CFG, n=[0..1]
0x4*n
Type: RW**Clock:** Config Clock**Reset State:** 0x00000000

Usage: used to compare with core clk period info from time based feature design for disabling BIMC level CGC for SCMO instance. (note: this is not for SCMO internal CGCs.)

BIMC_SCMOn_CGC_CFG

Bits	Name	Description
28	CLK_DIS	A 1 in this bit will disable SCMO core clock When THRESHOLD_ENA is 0. This is for SW to turn on/off clock.
24	THRESHOLD_ENA	1: enable frequency based clock gating. 0: disable frequency based clock gating.
7:0	THRESHOLD_VAL	value of core clk period in 100ps unit. If the period info from time based module is smaller than the value from these bits, then DPE internal core clk CGCs will be disabled and clk is not gated. The following are some examples: 0x00: the minimum value of the setting and the comparator output is always low. This means clk gating will be enabled if THRESHOLD_ENA of this register is also set to 0 0x01: 100ps=10GHz 0x0c: 1200ps=833MHz 0xFF: 6553.6ns=15.25kHz (max. setting)

**0x00400280+ BIMC_DPEn_CGC_CFG, n=[0..1]
0x4*n****Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000

Usage: used to compare with core clk period info from time based feature design for disabling BIMC level CGC for DPE instance. (note: this is not for DPE internal CGCs.)

BIMC_DPEn_CGC_CFG

Bits	Name	Description
28	CLK_DIS	A 1 in this bit will disable DPE core clock when THRESHOLD_ENA is 0. This is for SW to turn on/off clock.
24	THRESHOLD_ENA	1: enable frequency based clock gating. 0: disable frequency based clock gating.
7:0	THRESHOLD_VAL	value of core clk period in 100ps unit. If the period info from time based module is smaller than the value from these bits, then DPE internal core clk CGCs will be disabled and clk is not gated. The following are some examples: 0x00: the minimum value of the setting and the comparator output is always low. This means clk gating will be enabled if THRESHOLD_ENA of this register is also set to 0 0x01: 100ps=10GHz 0x0c: 1200ps=833MHz 0xFF: 6553.6ns=15.25kHz (max. setting)

0x00400400 BIMC_CALIBRATION_CFG**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000

Calibration configuration

BIMC_CALIBRATION_CFG

Bits	Name	Description
28	RAPID_CAL_ENA	Enable/disable rapid calibration. When rapid calibration is enabled, the calibrate pulses are generated every alternate core clock cycle.
23:0	COUNT_VAL	Count value used to generate calibrate pulses periodically. The counter is based of XO clock cycles. For example when the COUNT_VAL is 1, the calibrate pulses are generated every alternate XO clock cycle. Note: When the COUNT_VAL is 0, the calibrate pulses aren't generated.

**0x00400600+ BIMC_DFIn_LP_CFG, n=[0..1]
0x4*n****Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000

Used to signal DDR PHY to low power

BIMC_DFIn_LP_CFG

Bits	Name	Type	Description
15:12	RFU	RW	Spare field - reserved for future use.
8	LP_ACK	R	Indicates the acknowledge status for the low power request.
4	LP_REQ	RW	Request to PHY to go to low power mode.
3:0	LP_WAKEUP	RW	Specifies the clock cycles within which PHY has to ensure that it comes out of low power.

0x00400800 BIMC_HFSC_CFG**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000

Used for SW driven DDR clock frequency switch

BIMC_HFSC_CFG

Bits	Name	Description
0	DISABLE	Disable/Enable the hardware driven clock frequency switch control sequence. When disabled, the SW can control the clock frequency switch handshake sequence.

0x0040080C BIMC_HFSC_STATUS**Type:** R**Clock:** Config Clock**Reset State:** 0x00000000

Used for frequency switch FSM debug

BIMC_HFSC_STATUS

Bits	Name	Description
0	BUSY	Status indicating that frequency switch FSM is in non-idle state.

0x00400840 BIMC_HFSC_STAGE1_CTRL**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000

Used for SW driven DDR clock frequency switch

BIMC_HFSC_STAGE1_CTRL

Bits	Name	Type	Description
4	ACK	R	Status indicating that STAGE1 preparation is done for clock frequency switch.
0	REQ	RW	Sends a request to the STAGE1 components to get them prepared for clock frequency switch.

0x00400860 BIMC_HFSC_STAGE2_CTRL**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000

Used for SW driven DDR clock frequency switch

BIMC_HFSC_STAGE2_CTRL

Bits	Name	Type	Description
4	ACK	R	Status indicating that STAGE2 preparation is done for clock frequency switch.
0	REQ	RW	Sends a request to the STAGE2 components to get them prepared for clock frequency switch.

2.2 GLOBAL1

BRIC Global #1

2.2.1 BIMC_BRIC_GLOBAL1

BRIC Global #1

0x00401000 BIMC_BRIC_GLOBAL1_COMPONENT_INFO

Type: R

Clock: Config Clock

Reset State: 0x00000101

Component information registers provide discoverability information on the type of component and instance of each component. These registers are always the very first register in each component's configuration address space and allow for a software routine to walk the configuration address space in steps of 4KB to identify the location of each configuration register space and to which component each 4KB configures.

BIMC_BRIC_GLOBAL1_COMPONENT_INFO

Bits	Name	Description
15:8	SUB_TYPE	Identifies this configuration space as belonging to global1 configuration registers 0x1: GLOBAL1
7:0	TYPE	Identifies this configuration space as belonging to a global/common component, that does not specifically connect to a bus master or slave. 0x1: GLOBAL

0x00401020 BIMC_BRIC_CONFIGURATION_INFO_0

Type: R

Clock: Config Clock

Reset State: 0x00000808

Configuration information registers provide information on the capability and configuration of the interconnect. See register field descriptions for more details.

BIMC_BRIC_CONFIGURATION_INFO_0

Bits	Name	Description
31:24	ADDR_WIDTH	Indicates the width of addresses generated by masters connected to the interconnect.
23:16	BUSID	Indicates the BUSID added to all commands originating from masters that are directly connected to this interconnect. Generally assigned to be a unique value in the SoC.

BIMC_BRIC_CONFIGURATION_INFO_0 (cont.)

Bits	Name	Description
15:8	NUM_SWAYS	Indicates the number of slaveways in the interconnect. Also includes any internal slaveways that perform controller functionality to on-chip and off-chip memory, including DDR-SDRAM. Each slaveway connected to a single bus slave or another level of interconnect.
7:0	NUM_MPORTS	Indicates the number of masterports in the interconnect. Each masterport connects to a single bus master or another level of interconnect.

0x00401030 BIMC_BRIC_CONFIGURATION_INFO_1**Type:** R**Clock:** Config Clock**Reset State:** 0x00000000

Configuration information registers provide information on the capability and configuration of the interconnect. See register field descriptions for more details.

BIMC_BRIC_CONFIGURATION_INFO_1

Bits	Name	Description
31:16	DATA_WIDTH	Indicates the width of read and write data used in the core of the interconnect.
15:8	TID_WIDTH	Indicates the width of transfer ID in the core of the interconnect. TIDs are made as wide as necessary to accommodate TIDs from all masters and to accommodate any addition transaction tagging required by the interconnect.
7:0	MID_WIDTH	Indicates the width of master ID in the core of the interconnect, including any bus and port IDs as inserted by different levels of interconnects.

0x00401040 BIMC_BRIC_CONFIGURATION_INFO_2A**Type:** R**Clock:** Config Clock**Reset State:** 0x00000000

Configuration information registers provide information on the capability and configuration of the interconnect. See register field descriptions for more details.

BIMC_BRIC_CONFIGURATION_INFO_2A

Bits	Name	Description
31:0	CFG_ADDR_BASE_GLOBAL	In configuration space, indicates the base address of first global CSR region. Upto 8 global CSR slaves occupy 8 4KB regions.

0x00401044 BIMC_BRIC_CONFIGURATION_INFO_2B**Type:** R**Clock:** Config Clock**Reset State:** 0x00000000

Configuration information registers provide information on the capability and configuration of the interconnect. See register field descriptions for more details.

BIMC_BRIC_CONFIGURATION_INFO_2B

Bits	Name	Description
31:0	CFG_ADDR_BASE_MPORTS	In configuration space, indicates the base address of first masterport CSR region. Each masterport occupies 4 4KB regions.

0x00401048 BIMC_BRIC_CONFIGURATION_INFO_2C**Type:** R**Clock:** Config Clock**Reset State:** 0x00000000

Configuration information registers provide information on the capability and configuration of the interconnect. See register field descriptions for more details.

BIMC_BRIC_CONFIGURATION_INFO_2C

Bits	Name	Description
31:0	CFG_ADDR_BASE_SWAYS	In configuration space, indicates the base address of first slaveway CSR region. Each large slaveway occupies 12 4KB regions and each regular slaveway occupies 8 4KB regions.

0x0040104C BIMC_BRIC_CONFIGURATION_INFO_2D**Type:** R**Clock:** Config Clock**Reset State:** 0x00000808

Configuration information registers provide information on the capability and configuration of the interconnect. See register field descriptions for more details.

BIMC_BRIC_CONFIGURATION_INFO_2D

Bits	Name	Description
23:16	CFG_NUM_SWAYS_LARGE	Indicates the number of slaweways (large) that are allocated 12 4KB regions. Remaining slaweways (regular) are allocated only 8 4KB regions.
15:8	CFG_NUM_SWAYS	Indicates the number of slaweways (large and regular) that are allocated with CSR regions. Usually always equals the number of slaweways indicated in BRIC_CONFIGURATION_INFO_0.
7:0	CFG_NUM_MPORTS	Indicates the number of masterports that are allocated with CSR regions. Usually always equals the number of slaweways indicated in BRIC_CONFIGURATION_INFO_0.

0x00401100 BIMC_BRIC_INTERRUPT_STATUS_0

Type: R

Clock: Config Clock

Reset State: 0x00000000

Interrupt status registers collect interrupts from each component and are part of a central set of interrupt status and control registers to indicate and control interrupts from which components are propagated to the interrupt controller.

BIMC_BRIC_INTERRUPT_STATUS_0

Bits	Name	Description
7:0	MPORT	Indicates interrupts from each masterport. 1-bit per masterport. When set (1), indicates that the corresponding masterport has generated an interrupt that requires servicing.

0x00401108 BIMC_BRIC_INTERRUPT_CLEAR_0

Type: W

Clock: Config Clock

Reset State: 0x00000000

Interrupt clear registers provide the capability to clear an interrupt status, once an interrupt has been acknowledged or serviced. It is part of a central set of interrupt status and control registers to indicate and control interrupts from which components are propagated to the interrupt controller.

BIMC_BRIC_INTERRUPT_CLEAR_0

Bits	Name	Description
7:0	MPORT	Write '1' to specific bits to clear interrupt status. 1-bit per masterport. Only interrupt status that are set are cleared. Writes of '1' to interrupt status that are not set are ignored. Writes of '0' are ignored.

0x0040110C BIMC_BRIC_INTERRUPT_ENABLE_0**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000

Interrupt enable registers provide control over which interrupts are propagated to the interrupt controller. It is part of a central set of interrupt status and control registers to indicate and control interrupts from which components are propagated to the interrupt controller.

BIMC_BRIC_INTERRUPT_ENABLE_0

Bits	Name	Description
7:0	MPORT	When set (1), configures the corresponding interrupt to be propagated to the interrupt controller. 1-bit per masterport.

0x00401110 BIMC_BRIC_INTERRUPT_STATUS_1**Type:** R**Clock:** Config Clock**Reset State:** 0x00000000

Interrupt status registers collect interrupts from each component and are part of a unified set of interrupt status and control registers to indicate and control interrupts from which components are propagated out of the interconnect.

BIMC_BRIC_INTERRUPT_STATUS_1

Bits	Name	Description
7:0	SWAY_0	Indicates interrupt #0 from each slaveway. 1-bit per slaveway. When set (1), indicates that the corresponding slaveway has generated an interrupt that requires servicing.

0x00401118 BIMC_BRIC_INTERRUPT_CLEAR_1**Type:** W**Clock:** Config Clock**Reset State:** 0x00000000

Interrupt clear registers provide the capability to clear an interrupt status, once an interrupt has been acknowledged or serviced. It is part of a central set of interrupt status and control registers to indicate and control interrupts from which components are propagated to the interrupt controller

BIMC_BRIC_INTERRUPT_CLEAR_1

Bits	Name	Description
7:0	SWAY_0	Write '1' to specific bits to clear interrupt status. 1-bit per slaway. Only interrupt status that are set are cleared. Writes of '1' to interrupt status that are not set are ignored. Writes of '0' are ignored.

0x0040111C BIMC_BRIC_INTERRUPT_ENABLE_1**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000

Interrupt enable registers provide control over which interrupts are propagated to the interrupt controller. It is part of a central set of interrupt status and control registers to indicate and control interrupts from which components are propagated to the interrupt controller.

BIMC_BRIC_INTERRUPT_ENABLE_1

Bits	Name	Description
7:0	SWAY_0	When set (1), configures the corresponding interrupt to be propagated to the interrupt controller. 1-bit per slaway.

0x00401120 BIMC_BRIC_INTERRUPT_STATUS_2**Type:** R**Clock:** Config Clock**Reset State:** 0x00000000

Interrupt status registers collect interrupts from each component and are part of a unified set of interrupt status and control registers to indicate and control interrupts from which components are propagated out of the interconnect.

BIMC_BRIC_INTERRUPT_STATUS_2

Bits	Name	Description
7:0	SWAY_1	Indicates interrupt #1 from each slaveway. 1-bit per slaveway. When set (1), indicates that the corresponding slaveway has generated an interrupt that requires servicing.

0x00401128 BIMC_BRIC_INTERRUPT_CLEAR_2**Type:** W**Clock:** Config Clock**Reset State:** 0x00000000

Interrupt clear registers provide the capability to clear an interrupt status, once an interrupt has been acknowledged or serviced. It is part of a central set of interrupt status and control registers to indicate and control interrupts from which components are propagated to the interrupt controller

BIMC_BRIC_INTERRUPT_CLEAR_2

Bits	Name	Description
7:0	SWAY_1	Write '1' to specific bits to clear interrupt status. 1-bit per slaveway. Only interrupt status that are set are cleared. Writes of '1' to interrupt status that are not set are ignored. Writes of '0' are ignored.

0x0040112C BIMC_BRIC_INTERRUPT_ENABLE_2**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000

Interrupt enable registers provide control over which interrupts are propagated to the interrupt controller. It is part of a central set of interrupt status and control registers to indicate and control interrupts from which components are propagated to the interrupt controller.

BIMC_BRIC_INTERRUPT_ENABLE_2

Bits	Name	Description
7:0	SWAY_1	When set (1), configures the corresponding interrupt to be propagated to the interrupt controller. 1-bit per slaveway.

0x00401130 BIMC_BRIC_INTERRUPT_STATUS_3**Type:** R**Clock:** Config Clock**Reset State:** 0x00000000

Interrupt status registers collect interrupts from each component and are part of a unified set of interrupt status and control registers to indicate and control interrupts from which components are propagated out of the interconnect.

BIMC_BRIC_INTERRUPT_STATUS_3

Bits	Name	Description
7:0	SWAY_2	Indicates interrupt #2 from each slaveway. 1-bit per slaveway. When set (1), indicates that the corresponding slaveway has generated an interrupt that requires servicing.

0x00401138 BIMC_BRIC_INTERRUPT_CLEAR_3**Type:** W**Clock:** Config Clock**Reset State:** 0x00000000

Interrupt clear registers provide the capability to clear an interrupt status, once an interrupt has been acknowledged or serviced. It is part of a central set of interrupt status and control registers to indicate and control interrupts from which components are propagated to the interrupt controller.

BIMC_BRIC_INTERRUPT_CLEAR_3

Bits	Name	Description
7:0	SWAY_2	Write '1' to specific bits to clear interrupt status. 1-bit per slaveway. Only interrupt status that are set are cleared. Writes of '1' to interrupt status that are not set are ignored. Writes of '0' are ignored.

0x0040113C BIMC_BRIC_INTERRUPT_ENABLE_3**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000

Interrupt enable registers provide control over which interrupts are propagated to the interrupt controller. It is part of a central set of interrupt status and control registers to indicate and control interrupts from which components are propagated to the interrupt controller.

BIMC_BRIC_INTERRUPT_ENABLE_3

Bits	Name	Description
7:0	SWAY_2	When set (1), configures the corresponding interrupt to be propagated to the interrupt controller. 1-bit per slaveway.

0x00401200 BIMC_BRIC_REF_TIMER_INTERVAL**Type:** RW**Clock:** Config & Reference Clocks**Reset State:** 0x00000000**BIMC_BRIC_REF_TIMER_INTERVAL**

Bits	Name	Description
15:0	INTERVAL	Configures the number of clock cycles of implementation specified reference clock to generate a reference timer tick on for use in arbitration and other functions in the interconnect. The reference timer tick defines the time units that arbitration intervals are specified in. A single interval definition allows for this reference time unit to be adjusted on any change in the frequency of the reference clock. When configured with either '0' or '1', indicates an interval of 1-cycle. When configured with values greater than '1', a timer tick is generated once every configured number of reference clock cycles.

2.3 GLOBAL2

BRIC Global #2

2.3.1 BIMC_BRIC_GLOBAL2

BRIC Global #2

0x00402000 BIMC_BRIC_GLOBAL2_COMPONENT_INFO**Type:** R**Clock:** Config Clock**Reset State:** 0x00000201

Component information registers provide discoverability information on the type of component and instance of each component. These registers are always the very first register in each component's configuration address space and allow for a software routine to walk the configuration address space in steps of 4KB to identify the location of each configuration register space and to which component each 4KB configures.

BIMC_BRIC_GLOBAL2_COMPONENT_INFO

Bits	Name	Description
15:8	SUB_TYPE	Identifies this configuration space as belonging to global2 configuration registers 0x2: GLOBAL2
7:0	TYPE	Identifies this configuration space as belonging to a global/common component, that does not specifically connect to a bus master or slave. 0x1: GLOBAL

0x004021F0 BIMC_BRIC_DEFAULT_SEGMENT**Type:** RW**Clock:** Config Clock**Reset State:** 0x80060007

Default segment configuration register defines the slaveways to direct all commands that do not decode to a specific bus slave in the system and commands that request specific routing.

BIMC_BRIC_DEFAULT_SEGMENT

Bits	Name	Description
31	REDIRECT_EN	When set (1), enables redirection of commands that request cacheable/coherent redirection (AREDIRECT=1). When cleared (0), command redirection is ignored and address of command determines the slave to which the command is routed to.
19:16	REDIRECT	Configures the slaveway to send all commands that request cacheable/coherent redirection. Defaults to the slaveway interfacing to the bus slave capable of handling cachable/coherent commands.
3:0	DEFAULT	Configures the slaveway to send all commands that do not decode to one of the specified address ranges. Defaults to an internal default slave capable of generating decode error, but can be configured to be any other slaveway.

**0x00402200+ BIMC_BRIC_SEGMENTn_ADDR_BASE_A_LOWER, n=[0..7]
0x80*n****Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000

Segment configuration registers define the address range in the memory map for each bus slave. Each bus slave has three sets of address ranges (A, B, C), with each address range defined with a pair of BASE and MASK registers. Each BASE/MASK pair of registers defines a power-of-2 address range that is at a minimum 1MB in size. For e.g., a BASE of 0x80000000 and MASK of

0xF0000000 defines the address range 0x80000000-0x8FFFFFFF. Each BASE and MASK register is broken into UPPER and LOWER to accommodate upto 64-bit address.

BIMC_BRIC_SEGMENTn_ADDR_BASE_A_LOWER

Bits	Name	Description
31:20	BASE_31_20	Configures ADDR_BASE[31:20] to match for in address of a command. Used in conjunction with ADDR_MASK[31:20], which controls which bits of ADDR_BASE[31:20] are matched for in address of a command.
10	BASE_10	Configures ADDR_BASE[10] to match for in address of a command. Used in conjunction with ADDR_MASK[10], which controls whether ADDR_BASE[10] is matched for in address of a command.
3:2	RFU	Reserved for future use
1	TYPE	When cleared (0), identifies BASE_A/MASK_A pair of address range registers as adding to the address space defined for bus slave 'n'. When set (1), identifies BASE_A/MASK_A pair of address range registers as subtracting from the address space defined for bus slave 'n'. For e.g., allows for the address space to be defined as 128MB - 16MB for a total of 114MB.
0	ENABLE	When set (1), enables use of BASE_A/MASK_A pair of address range registers for determining whether a command decodes to bus slave 'n' or not. When cleared (0), disables use of BASE_A/MASK_A pair of address range registers.

0x00402204+ BIMC_BRIC_SEGMENTn_ADDR_BASE_A_UPPER, n=[0..7] 0x80*n

Type: R

Clock: Config Clock

Reset State: 0x00000000

Segment configuration registers define the address range in the memory map for each bus slave. Each bus slave has three sets of address ranges (A, B, C), with each address range defined with a pair of BASE and MASK registers. Each BASE/MASK pair of registers defines a power-of-2 address range that is at a minimum 1MB in size. For e.g., a BASE of 0x80000000 and MASK of 0xF0000000 defines the address range 0x80000000-0x8FFFFFFF. Each BASE and MASK register is broken into UPPER and LOWER to accommodate upto 64-bit address.

BIMC_BRIC_SEGMENTn_ADDR_BASE_A_UPPER

Bits	Name	Description
3:0	UNUSED	

**0x00402208+ BIMC_BRIC_SEGMENTn_ADDR_MASK_A_LOWER, n=[0..7]
0x80*n****Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000

Segment configuration registers define the address range in the memory map for each bus slave. Each bus slave has three sets of address ranges (A, B, C), with each address range defined with a pair of BASE and MASK registers. Each BASE/MASK pair of registers defines a power-of-2 address range that is at a minimum 1MB in size. For e.g., a BASE of 0x80000000 and MASK of 0xF0000000 defines the address range 0x80000000-0x8FFFFFFF. Each BASE and MASK register is broken into UPPER and LOWER to accommodate upto 64-bit address.

BIMC_BRIC_SEGMENTn_ADDR_MASK_A_LOWER

Bits	Name	Description
31:20	MASK_31_20	Configures ADDR_MASK[31:20]. When a bit position is set (1), enables matching of that bit position in address of a command with the same bit position in ADDR_BASE. When a bit position is cleared (0), that bit position in address of a command is ignored and is not used for address decoding.
10	MASK_10	Configures ADDR_MASK[10]. When set (1), enables matching of address[10] of a command with ADDR_BASE[10]. When cleared (0), address[10] of a command is ignored and is not used for address decoding.

**0x0040220C+ BIMC_BRIC_SEGMENTn_ADDR_MASK_A_UPPER, n=[0..7]
0x80*n****Type:** R**Clock:** Config Clock**Reset State:** 0x00000000

Segment configuration registers define the address range in the memory map for each bus slave. Each bus slave has three sets of address ranges (A, B, C), with each address range defined with a pair of BASE and MASK registers. Each BASE/MASK pair of registers defines a power-of-2 address range that is at a minimum 1MB in size. For e.g., a BASE of 0x80000000 and MASK of 0xF0000000 defines the address range 0x80000000-0x8FFFFFFF. Each BASE and MASK register is broken into UPPER and LOWER to accommodate upto 64-bit address.

BIMC_BRIC_SEGMENTn_ADDR_MASK_A_UPPER

Bits	Name	Description
3:0	UNUSED	

**0x00402210+ BIMC_BRIC_SEGMENTn_ADDR_BASE_B_LOWER, n=[0..7]
0x80*n****Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000

Segment configuration registers define the address range in the memory map for each bus slave. Each bus slave has three sets of address ranges (A, B, C), with each address range defined with a pair of BASE and MASK registers. Each BASE/MASK pair of registers defines a power-of-2 address range that is at a minimum 1MB in size. For e.g., a BASE of 0x80000000 and MASK of 0xF0000000 defines the address range 0x80000000-0x8FFFFFFF. Each BASE and MASK register is broken into UPPER and LOWER to accommodate upto 64-bit address.

BIMC_BRIC_SEGMENTn_ADDR_BASE_B_LOWER

Bits	Name	Description
31:20	BASE_31_20	Configures ADDR_BASE[31:20] to match for in address of a command. Used in conjunction with ADDR_MASK[31:20], which controls which bits of ADDR_BASE[31:20] are matched for in address of a command.
10	BASE_10	Configures ADDR_BASE[10] to match for in address of a command. Used in conjunction with ADDR_MASK[10], which controls whether ADDR_BASE[10] is matched for in address of a command.
3:2	RFU	Reserved for future use
1	TYPE	When cleared (0), identifies BASE_B/MASK_B pair of address range registers as adding to the address space defined for bus slave 'n'. When set (1), identifies BASE_B/MASK_B pair of address range registers as subtracting from the address space defined for bus slave 'n'. For e.g., allows for the address space to be defined as 128MB - 16MB for a total of 114MB.
0	ENABLE	When set (1), enables use of BASE_B/MASK_B pair of address range registers for determining whether a command decodes to bus slave 'n' or not. When cleared (0), disables use of BASE_B/MASK_B pair of address range registers.

**0x00402214+ BIMC_BRIC_SEGMENTn_ADDR_BASE_B_UPPER, n=[0..7]
0x80*n****Type:** R**Clock:** Config Clock**Reset State:** 0x00000000

Segment configuration registers define the address range in the memory map for each bus slave. Each bus slave has three sets of address ranges (A, B, C), with each address range defined with a pair of BASE and MASK registers. Each BASE/MASK pair of registers defines a power-of-2 address range that is at a minimum 1MB in size. For e.g., a BASE of 0x80000000 and MASK of 0xF0000000 defines the address range 0x80000000-0x8FFFFFFF. Each BASE and MASK register is broken into UPPER and LOWER to accommodate upto 64-bit address.

BIMC_BRIC_SEGMENTn_ADDR_BASE_B_UPPER

Bits	Name	Description
3:0	UNUSED	

**0x00402218+ BIMC_BRIC_SEGMENTn_ADDR_MASK_B_LOWER, n=[0..7]
0x80*n****Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000

Segment configuration registers define the address range in the memory map for each bus slave. Each bus slave has three sets of address ranges (A, B, C), with each address range defined with a pair of BASE and MASK registers. Each BASE/MASK pair of registers defines a power-of-2 address range that is at a minimum 1MB in size. For e.g., a BASE of 0x80000000 and MASK of 0xF0000000 defines the address range 0x80000000-0x8FFFFFFF. Each BASE and MASK register is broken into UPPER and LOWER to accommodate upto 64-bit address.

BIMC_BRIC_SEGMENTn_ADDR_MASK_B_LOWER

Bits	Name	Description
31:20	MASK_31_20	Configures ADDR_MASK[31:20]. When a bit position is set (1), enables matching of that bit position in address of a command with the same bit position in ADDR_BASE. When a bit position is cleared (0), that bit position in address of a command is ignored and is not used for address decoding.
10	MASK_10	ADDR_MASK[10]

**0x0040221C+ BIMC_BRIC_SEGMENTn_ADDR_MASK_B_UPPER, n=[0..7]
0x80*n****Type:** R**Clock:** Config Clock**Reset State:** 0x00000000

Segment configuration registers define the address range in the memory map for each bus slave. Each bus slave has three sets of address ranges (A, B, C), with each address range defined with a pair of BASE and MASK registers. Each BASE/MASK pair of registers defines a power-of-2 address range that is at a minimum 1MB in size. For e.g., a BASE of 0x80000000 and MASK of 0xF0000000 defines the address range 0x80000000-0x8FFFFFFF. Each BASE and MASK register is broken into UPPER and LOWER to accommodate upto 64-bit address.

BIMC_BRIC_SEGMENTn_ADDR_MASK_B_UPPER

Bits	Name	Description
3:0	UNUSED	

**0x00402220+ BIMC_BRIC_SEGMENTn_ADDR_BASE_C_LOWER, n=[0..7]
0x80*n****Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000

Segment configuration registers define the address range in the memory map for each bus slave. Each bus slave has three sets of address ranges (A, B, C), with each address range defined with a pair of BASE and MASK registers. Each BASE/MASK pair of registers defines a power-of-2 address range that is at a minimum 1MB in size. For e.g., a BASE of 0x80000000 and MASK of 0xF0000000 defines the address range 0x80000000-0x8FFFFFFF. Each BASE and MASK register is broken into UPPER and LOWER to accommodate upto 64-bit address.

BIMC_BRIC_SEGMENTn_ADDR_BASE_C_LOWER

Bits	Name	Description
31:20	BASE_31_20	Configures ADDR_BASE[31:20] to match for in address of a command. Used in conjunction with ADDR_MASK[31:20], which controls which bits of ADDR_BASE[31:20] are matched for in address of a command.
10	BASE_10	Configures ADDR_BASE[10] to match for in address of a command. Used in conjunction with ADDR_MASK[10], which controls whether ADDR_BASE[10] is matched for in address of a command.
3:2	RFU	Reserved for future use
1	TYPE	When cleared (0), identifies BASE_C/MASK_C pair of address range registers as adding to the address space defined for bus slave 'n'. When set (1), identifies BASE_C/MASK_C pair of address range registers as subtracting from the address space defined for bus slave 'n'. For e.g., allows for the address space to be defined as 128MB - 16MB for a total of 114MB.
0	ENABLE	When set (1), enables use of BASE_C/MASK_C pair of address range registers for determining whether a command decodes to bus slave 'n' or not. When cleared (0), disables use of BASE_C/MASK_C pair of address range registers.

**0x00402224+ BIMC_BRIC_SEGMENTn_ADDR_BASE_C_UPPER, n=[0..7]
0x80*n****Type:** R**Clock:** Config Clock**Reset State:** 0x00000000

Segment configuration registers define the address range in the memory map for each bus slave. Each bus slave has three sets of address ranges (A, B, C), with each address range defined with a pair of BASE and MASK registers. Each BASE/MASK pair of registers defines a power-of-2 address range that is at a minimum 1MB in size. For e.g., a BASE of 0x80000000 and MASK of 0xF0000000 defines the address range 0x80000000-0x8FFFFFFF. Each BASE and MASK register is broken into UPPER and LOWER to accommodate upto 64-bit address.

BIMC_BRIC_SEGMENTn_ADDR_BASE_C_UPPER

Bits	Name	Description
3:0	UNUSED	

**0x00402228+ BIMC_BRIC_SEGMENTn_ADDR_MASK_C_LOWER, n=[0..7]
0x80*n****Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000

Segment configuration registers define the address range in the memory map for each bus slave. Each bus slave has three sets of address ranges (A, B, C), with each address range defined with a pair of BASE and MASK registers. Each BASE/MASK pair of registers defines a power-of-2 address range that is at a minimum 1MB in size. For e.g., a BASE of 0x80000000 and MASK of 0xF0000000 defines the address range 0x80000000-0x8FFFFFFF. Each BASE and MASK register is broken into UPPER and LOWER to accommodate upto 64-bit address.

BIMC_BRIC_SEGMENTn_ADDR_MASK_C_LOWER

Bits	Name	Description
31:20	MASK_31_20	Configures ADDR_MASK[31:20]. When a bit position is set (1), enables matching of that bit position in address of a command with the same bit position in ADDR_BASE. When a bit position is cleared (0), that bit position in address of a command is ignored and is not used for address decoding.
10	MASK_10	Configures ADDR_MASK[10]. When set (1), enables matching of address[10] of a command with ADDR_BASE[10]. When cleared (0), address[10] of a command is ignored and is not used for address decoding.

**0x0040222C+ BIMC_BRIC_SEGMENTn_ADDR_MASK_C_UPPER, n=[0..7]
0x80*n****Type:** R**Clock:** Config Clock**Reset State:** 0x00000000

Segment configuration registers define the address range in the memory map for each bus slave. Each bus slave has three sets of address ranges (A, B, C), with each address range defined with a pair of BASE and MASK registers. Each BASE/MASK pair of registers defines a power-of-2 address range that is at a minimum 1MB in size. For e.g., a BASE of 0x80000000 and MASK of 0xF0000000 defines the address range 0x80000000-0x8FFFFFFF. Each BASE and MASK register is broken into UPPER and LOWER to accommodate upto 64-bit address.

BIMC_BRIC_SEGMENTn_ADDR_MASK_C_UPPER

Bits	Name	Description
3:0	UNUSED	

**0x00402A00+ BIMC_BRIC_REMAPn_ADDR_BASE_LOWER, n=[0..3]
0x40*n****Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000

Remap configuration registers define allow for one address range to be remapped to a different address range to facilitate easier decoding in bus slaves and/or allow for multiple disjoint address ranges to be collected together to make them appear to be a single contiguous space for the slave. It is particularly useful in external DRAM access, where a single device could be allowed to be accessed with interleaved addresses for a portion and linear addresses for the remainder. Refer HPG for example configurations.

BIMC_BRIC_REMAPn_ADDR_BASE_LOWER

Bits	Name	Description
31:28	BASE_31_28	Configures ADDR_BASE[31:28] to match for in address of a command. Used in conjunction with ADDR_MASK[31:28], which controls which bits of ADDR_BASE[31:28] are matched for in address of a command.
10	BASE_10	Configures ADDR_BASE[10] to match for in address of a command. Used in conjunction with ADDR_MASK[10], which controls whether ADDR_BASE[10] is matched for in address of a command.
3:1	RFU	Reserved for future use
0	ENABLE	When set (1), enables use of BASE/MASK pair of address range registers for determining whether a command decodes to a specific address range and enables remapping of matched addresses.

**0x00402A04+ BIMC_BRIC_REMAPn_ADDR_BASE_UPPER, n=[0..3]
0x40*n****Type:** R**Clock:** Config Clock**Reset State:** 0x00000000

Remap configuration registers define allow for one address range to be remapped to a different address range to facilitate easier decoding in bus slaves and/or allow for multiple disjoint address ranges to be collected together to make them appear to be a single contiguous space for the slave. It is particularly useful in external DRAM access, where a single device could be allowed to be

accessed with interleaved addresses for a portion and linear addresses for the remainder. Refer HPG for example configurations.

BIMC_BRIC_REMAPn_ADDR_BASE_UPPER

Bits	Name	Description
3:0	UNUSED	

0x00402A08+ BIMC_BRIC_REMAPn_ADDR_MASK_LOWER, n=[0..3] 0x40*n

Type: RW

Clock: Config Clock

Reset State: 0x00000000

Remap configuration registers define allow for one address range to be remapped to a different address range to facilitate easier decoding in bus slaves and/or allow for multiple disjoint address ranges to be collected together to make them appear to be a single contiguous space for the slave. It is particularly useful in external DRAM access, where a single device could be allowed to be accessed with interleaved addresses for a portion and linear addresses for the remainder. Refer HPG for example configurations.

BIMC_BRIC_REMAPn_ADDR_MASK_LOWER

Bits	Name	Description
31:28	MASK_31_28	Configures ADDR_MASK[31:28]. When a bit position is set (1), enables matching of that bit position in address of a command with the same bit position in ADDR_BASE. When a bit position is cleared (0), that bit position in address of a command is ignored and is not used for address decoding.
10	MASK_10	Configures ADDR_MASK[10]. When set (1), enables matching of address[10] of a command with ADDR_BASE[10]. When cleared (0), address[10] of a command is ignored and is not used for address decoding.

0x00402A0C+ BIMC_BRIC_REMAPn_ADDR_MASK_UPPER, n=[0..3] 0x40*n

Type: R

Clock: Config Clock

Reset State: 0x00000000

Remap configuration registers define allow for one address range to be remapped to a different address range to facilitate easier decoding in bus slaves and/or allow for multiple disjoint address ranges to be collected together to make them appear to be a single contiguous space for the slave. It is particularly useful in external DRAM access, where a single device could be allowed to be accessed with interleaved addresses for a portion and linear addresses for the remainder. Refer HPG for example configurations.

BIMC_BRIC_REMAPn_ADDR_MASK_UPPER

Bits	Name	Description
3:0	UNUSED	

**0x00402A20+ BIMC_BRIC_REMAPn_OP0, n=[0..3]
0x40*n****Type:** RW**Clock:** Config Clock**Reset State:** 0x000A0000

Remap operation configuration registers define the operation to perform on a matched address. Two operations are currently supported, de-interleaving and offset change. Each operation can be independently enabled. When an operation is not enabled, no change is performed on the matched address. Incoming command address is input for remap operation #0. The result of remap operation #0 is fed as input to remap operation #1. The result of remap operation #1 is forwarded to internal slaves.

BIMC_BRIC_REMAPn_OP0

Bits	Name	Type	Description
23:16	OPERAND	R	De-interleaving granularity. For eg., when de-interleaving on 1KByte granularity, addr[10] is dropped and bits higher than addr[10] are shifted right. A '0' is shifted into the MSB. 0xA: GRANULARITY_1KB
4	OPCODE	R	De-interleave matched addresses on granularity selected in operand. 0x0: DE_INTERLEAVE
0	OPCODE_ENABLE	RW	When set(1), enables conversion of matched addresses based on opcode selected

**0x00402A24+ BIMC_BRIC_REMAPn_OP1, n=[0..3]
0x40*n****Type:** RW**Clock:** Config Clock**Reset State:** 0x00000010

Remap operation configuration registers define the operation to perform on a matched address. Two operations are currently supported, de-interleaving and offset change. Each operation can be independently enabled. When an operation is not enabled, no change is performed on the matched address. Incoming command address is input for remap operation #0. The result of remap operation #0 is fed as input to remap operation #1. The result of remap operation #1 is forwarded to internal slaves.

BIMC_BRIC_REMAPn_OP1

Bits	Name	Type	Description
23:16	OPERAND	RW	Offset to add or subtract from matched addresses in 256MB granularity. For eg., +1, adds 256MB to matched address, -1 subtracts 256MB from matched address.
4	OPCODE	R	Change offset of matched addresses, in 256MB granularity. 0x1: OFFSET
0	OPCODE_ENABLE	RW	When set(1), enables conversion of matched addresses based on opcode selected

2.4 BIMC_M_APP_MPORT

BRIC Masterport

2.4.1 BIMC_M_APP_MPORT_BRIC_MASTERPORT

BRIC Masterport

0x00408000 BIMC_M_APP_MPORT_COMPONENT_INFO**Type:** R**Clock:** Config Clock**Reset State:** 0x00000102

Component information registers provide discoverability information on the type of component and instance of each component. These registers are always the very first register in each component's configuration address space and allow for a software routine to walk the configuration address space in steps of 4KB to identify the location of each configuration register space and to which component each 4KB configures.

BIMC_M_APP_MPORT_COMPONENT_INFO

Bits	Name	Description
23:16	INSTANCE	Indicates the instance number of this masterport.
15:8	SUB_TYPE	Identifies this configuration space as belonging to a masterport 0x1: MASTERPORT
7:0	TYPE	Identifies this configuration space as belonging to a component that interfaces to a bus master. 0x2: MASTERPORT

0x00408020 BIMC_M_APP_MPORT_CONFIGURATION_INFO_0**Type:** R**Clock:** Config Clock**Reset State:** 0x01000000

Configuration information registers provide information on the capability and configuration of each masterport instance. See register field descriptions for more details.

BIMC_M_APP_MPORT_CONFIGURATION_INFO_0

Bits	Name	Description
31:24	SYNC_MODE	Indicates the clock relationship between the clock used in the master side and the clock used in the core side of the masterport. When ASYNC, identifies the master clock and core clock as unrelated and asynchronous to each other. When SYNC, identifies the master clock and core clock as synchronous to each other. When ISOSYNC, identifies the master clock and core clock as related, but of different frequencies, which are of integer ratios, i.e., they are iso-synchronous to each other. 0x0: ASYNC 0x1: SYNC 0x2: ISOSYNC
15:8	CONNECTION_TYPE	Indicates the type of master connecting to this masterport. When DIRECT, indicates that a bus master is directly connected to this masterport. When CASCADE, indicates that bus masters connect to this masterport through another level of interconnect. 0x0: DIRECT 0x1: CASCADE
7:0	FUNCTIONALITY	Indicates the functionality and type of bus traffic handled by this masterport. When INORDER, identifies this masterport as optimized for in-order bus traffic. When OOO_ONLY, identifies this masterport as optimized for out-of-order bus traffic, where a single master is capable of handling responses return in any order. When REORDER, identifies this masterport as having a reorder buffer for re-ordering responses to command order, while allowing bus slaves to return responses in any order they desire. When STUB, identifies this masterport has a stub. 0x0: STUB 0x1: OOO_ONLY 0x2: INORDER 0x3: REORDER

0x00408030 BIMC_M_APP_MPORT_CONFIGURATION_INFO_1**Type:** R**Clock:** Config Clock**Reset State:** 0x00000000

Configuration information registers provide information on the capability and configuration of each masterport instance. See register field descriptions for more details.

BIMC_M_APP_MPORT_CONFIGURATION_INFO_1

Bits	Name	Description
31:0	SWAY_CONNECTIVITY	Indicates slaveways that this masterport can route commands to. 1-bit per slaveway. When a specific bit is set (1), indicates that commands can be routed to the slaveway corresponding to this bit. When a specific bit is clear (0), indicates that commands are not routed to the slaveway corresponding to this bit.

0x00408040 BIMC_M_APP_MPORT_CONFIGURATION_INFO_2A

Type: R

Clock: Config Clock

Reset State: 0x00000000

Configuration information registers provide information on the capability and configuration of each masterport instance. See register field descriptions for more details.

BIMC_M_APP_MPORT_CONFIGURATION_INFO_2A

Bits	Name	Description
31:16	M_DATA_WIDTH	Indicates the width of read and write data from the bus master.
15:8	M_TID_WIDTH	Indicates the width of transfer IDs from the bus master.
7:0	M_MID_WIDTH	Indicates the width of master ID from the bus master (includes any bus and port IDs if bus master is an interconnect.)

0x00408044 BIMC_M_APP_MPORT_CONFIGURATION_INFO_2B

Type: R

Clock: Config Clock

Reset State: 0x00000000

Configuration information registers provide information on the capability and configuration of each masterport instance. See register field descriptions for more details.

BIMC_M_APP_MPORT_CONFIGURATION_INFO_2B

Bits	Name	Description
15:0	C_DATA_WIDTH	Indicates the width of read and write data used in core of this masterport.

0x00408050 BIMC_M_APP_MPORT_CONFIGURATION_INFO_3**Type:** R**Clock:** Config Clock**Reset State:** 0x00000000

Configuration information registers provide information on the capability and configuration of each masterport instance. See register field descriptions for more details.

BIMC_M_APP_MPORT_CONFIGURATION_INFO_3

Bits	Name	Description
31:24	RCH_DEPTH	Indicates the depth of read channel buffers in this masterport. A beat of read data occupies one entry in the read channel buffer, until accepted by the bus master.
23:16	BCH_DEPTH	Indicates the depth of write response channel buffers in this masterport. Each write response occupies one entry in the write response channel buffer, until accepted by the bus master.
15:8	WCH_DEPTH	Indicates the depth of write channel buffers in this masterport. A beat of write data occupies one entry in the write channel buffer, until accepted by the arbiter.
7:0	ACH_DEPTH	Indicates the depth of address channel buffers in this masterport. Each command occupies one entry in the address channel buffer, until accepted by the arbiter.

0x00408060 BIMC_M_APP_MPORT_CONFIGURATION_INFO_4**Type:** R**Clock:** Config Clock**Reset State:** 0x00000000

Configuration information registers provide information on the capability and configuration of each masterport instance. See register field descriptions for more details.

BIMC_M_APP_MPORT_CONFIGURATION_INFO_4

Bits	Name	Description
15:8	REORDER_BUFFER_DEPTH	Indicates the number of read data beats that can be re-ordered in this masterport (if the masterport provides response re-ordering functionality.)
7:0	REORDER_TABLE_DEPTH	Indicates the number of read and write commands whose responses can be re-ordered in this masterport (if the masterport provides responses re-ordering functionality.)

0x00408070 BIMC_M_APP_MPORT_CONFIGURATION_INFO_5A**Type:** R**Clock:** Config Clock**Reset State:** 0x00000000

Configuration information registers provide information on the capability and configuration of each masterport instance. See register field descriptions for more details.

BIMC_M_APP_MPORT_CONFIGURATION_INFO_5A

Bits	Name	Description
31:0	ACH_PIPELINE_STAGES	Indicates pipeline stages enabled on the address channel. Each digit identifies a specific pipeline stage in the masterport, where '0' indicates no pipeline stage, '1' indicates 1-deep pipeline stage where forward path is registered, return path is fed through, '2' indicates 2-deep pipeline stage where both forward and return path are pipelined. Digit 3 identifies pipelining on the master to masterport interface, digit 2 identifies pipelining after address decode, digit 1 identifies pipelining in the address channel buffer and digit 0 identifies pipelining on the masterport to arbiter interface.

0x00408074 BIMC_M_APP_MPORT_CONFIGURATION_INFO_5B**Type:** R**Clock:** Config Clock**Reset State:** 0x00000000

Configuration information registers provide information on the capability and configuration of each masterport instance. See register field descriptions for more details.

BIMC_M_APP_MPORT_CONFIGURATION_INFO_5B

Bits	Name	Description
31:0	WCH_PIPELINE_STAGES	Indicates pipeline stages enabled on the write channel. Each digit identifies a specific pipeline stage in the masterport, where '0' indicates no pipeline stage, '1' indicates 1-deep pipeline stage where forward path is registered, return path is fed through, '2' indicates 2-deep pipeline stage where both forward and return path are pipelined. Digit 3 identifies pipelining on the master to masterport interface, digit 2 identifies pipelining after address decode, digit 1 identifies pipelining in the write channel buffer and digit 0 identifies pipelining on the masterport to arbiter interface.

0x00408078 BIMC_M_APP_MPORT_CONFIGURATION_INFO_5C**Type:** R**Clock:** Config Clock**Reset State:** 0x00000000

Configuration information registers provide information on the capability and configuration of each masterport instance. See register field descriptions for more details.

BIMC_M_APP_MPORT_CONFIGURATION_INFO_5C

Bits	Name	Description
31:0	BCH_PIPELINE_STAGES	Indicates pipeline stages enabled on the write response channel. Each digit identifies a specific pipeline stage in the masterport, where '0' indicates no pipeline stage, '1' indicates 1-deep pipeline stage where forward path is registered, return path is fed through, '2' indicates 2-deep pipeline stage where both forward and return path are pipelined. Digit 2 identifies pipelining on the slaveway to masterport interface after arbitration of write response channels from multiple slaveways, digit 1 identifies pipelining in the write response channel buffer and digit 0 identifies pipelining on the masterport to master interface.

0x0040807C BIMC_M_APP_MPORT_CONFIGURATION_INFO_5D**Type:** R**Clock:** Config Clock**Reset State:** 0x00000000

Configuration information registers provide information on the capability and configuration of each masterport instance. See register field descriptions for more details.

BIMC_M_APP_MPORT_CONFIGURATION_INFO_5D

Bits	Name	Description
31:0	RCH_PIPELINE_STAGES	Indicates pipeline stages enabled on the read channel. Each digit identifies a specific pipeline stage in the masterport, where '0' indicates no pipeline stage, '1' indicates 1-deep pipeline stage where forward path is registered, return path is fed through, '2' indicates 2-deep pipeline stage where both forward and return path are pipelined. Digit 2 identifies pipelining on the slaveway to masterport interface after arbitration of read channels from multiple slaveways, digit 1 identifies pipelining in the read channel buffer and digit 0 identifies pipelining on the masterport to master interface.

0x00408100 BIMC_M_APP_MPORT_INTERRUPT_STATUS**Type:** R**Clock:** Config Clock**Reset State:** 0x00000000

Interrupt status registers provide indication of which interrupts from this masterport require servicing.

BIMC_M_APP_MPORT_INTERRUPT_STATUS

Bits	Name	Description
1:0	RFU	Reserved for future use

0x00408108 BIMC_M_APP_MPORT_INTERRUPT_CLEAR**Type:** W**Clock:** Config Clock**Reset State:** 0x00000000

Interrupt clear registers provide the mechanism to turn off interrupt sources once they have been acknowledged or serviced. A write of '1' to the bits corresponding to the interrupts indicated in interrupt status is required to turn off interrupts.

BIMC_M_APP_MPORT_INTERRUPT_CLEAR

Bits	Name	Description
1:0	RFU	Reserved for future use

0x0040810C BIMC_M_APP_MPORT_INTERRUPT_ENABLE**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000

Interrupt enable registers provide control over which interrupts are allowed to be propagated to an interrupt controller for servicing.

BIMC_M_APP_MPORT_INTERRUPT_ENABLE

Bits	Name	Description
1:0	RFU	Reserved for future use

0x00408200 BIMC_M_APP_MPORT_CLOCK_CTRL**Type:** RW**Clock:** Config, Core & Master Clocks**Reset State:** 0x00000003**BIMC_M_APP_MPORT_CLOCK_CTRL**

Bits	Name	Description
1	MASTER_CLOCK_GATING_EN	When set (1), enables activity based clock gating of all registers clocked by master clock. Activity based clock gating uses architectural clock gates to reduce dynamic power consumption by gating clocks during periods of in-activity. When cleared (0), disables all activity based clock gating by forcing all architectural clock gates to not gate clocks even during periods of in-activity.
0	CORE_CLOCK_GATING_EN	When set (1), enables activity based clock gating of all registers clocked by core clock. Activity based clock gating uses architectural clock gates to reduce dynamic power consumption by gating clocks during periods of in-activity. When cleared (0), disables all activity based clock gating by forcing all architectural clock gates to not gate clocks even during periods of in-activity.

0x00408208 BIMC_M_APP_MPORT_CDC_CTRL**Type:** RW**Clock:** Config, Core & Master Clocks**Reset State:** 0x00000000

Clock domain crossing (CDC) control register provides configuration for clock domain crossing logic.

BIMC_M_APP_MPORT_CDC_CTRL

Bits	Name	Description
1	MASTER_CLOCK_DEMETA_SEL	When cleared (0), selects DEMETA synchronizer that requires atleast 3 clock edges (two rising and one falling). When set (1), selects DEMETA synchronizer that requires atleast 2 clock edges (one falling and one rising) to synchronize to master clock in all buffers.
0	CORE_CLOCK_DEMETA_SEL	When cleared (0), selects DEMETA synchronizer that requires atleast 3 clock edges (two rising and one falling). When set (1), selects DEMETA synchronizer that requires atleast 2 clock edges (one falling and one rising) to synchronize to core clock in all buffers.

0x00408210 BIMC_M_APP_MPORT_MODE**Type:** RW**Clock:** Config, Core & Master Clocks**Reset State:** 0x00000012**BIMC_M_APP_MPORT_MODE**

Bits	Name	Description
31:28	WR_GATHER_BEATS	Configures the number of write data beats to gather/buffer in write channel buffer before allowing the corresponding write command to be routed to arbiters. When configured to 0, write commands are held until the first beat of write data has been buffered. When configured to 1, write commands are held until two beats of write data have been buffered and so on. When configured with values larger than the number of beats in a write command, the command is held until the last beat has been buffered. When configured with values larger than the depth of write channel buffers, the command is held until all entries in the write channel buffer have been occupied.
19:16	RD_GATHER_BEATS	Configures the number of read data beats to gather/buffer in read channel buffer before allowing the read data to proceed to the master. When configured to 0, read data is allowed to proceed to the master as soon as the first beat is buffered in the read channel buffer. When configured to 1, read data is allowed to proceed to the master as soon as two beats of read data have been buffered and so on. When configured with values larger than the number of beats in the read burst, read data is held until the last beat has been buffered. When configured with values larger than the depth of read channel buffers, read data is held until all entries in the read channel buffer have been occupied.
15	RCSSH	Redirect Cacheable System SHareable When set (1), enables redirection of all cacheable transactions that are system shareable to the redirect slaweway. Note: Only masters connecting using AMBA 4 ACE protocol are able to indicate system shareability for transactions.
14	RCOSH	Redirect Cacheable Outer SHareable When set (1), enables redirection of all cacheable transactions that are outer shareable to the redirect slaweway.
13	RCISH	Redirect Cacheable Inner SHareable When set (1), enables redirection of all cacheable transactions that are inner shareable to the redirect slaweway.
12	RCNSH	Redirect Cacheable Non-SHareable When set (1), enables redirection of all cacheable transactions that are non-shareable to the redirect slaweway.
11	RNCSSH	Redirect Non-Cacheable System SHareable When set (1), enables redirection of all non-cacheable transactions that are system shareable to the redirect slaweway. Note: Only masters connecting using AMBA 4 ACE protocol are able to indicate system shareability for transactions.

BIMC_M_APP_MPORT_MODE (cont.)

Bits	Name	Description
10	RNCOSH	Redirect Non-Cacheable Outer SHareable When set (1), enables redirection of all non-cacheable transactions that are outer shareable to the redirect slaway.
9	RNCISH	Redirect Non-Cacheable Inner SHareable When set (1), enables redirection of all non-cacheable transactions that are inner shareable to the redirect slaway.
8	RNCNSH	Redirect Non-Cacheable Non-SHareable When set (1), enables redirection of all non-cacheable transactions that are non-shareable to the redirect slaway.
4	NARROW_WRITES	When set (1), enables converting of 1-beat bus-width sized writes down to 32-bit ASIZE based on write strobes. It is recommended to be enabled for directly connected CPU masters. When cleared (0), disables conversion and all writes are forwarded downstream as generated by the master.
2	FIODV	Force In-Order Device When set (1), forces commands to device memory to be executed one at a time. In other words, commands to device memory are held until all previous commands have completed (responses have been received at the masterport) and all commands after the device command are held until the device completes (response has been received at the masterport.)
1	FIOSO	Force In-Order Strongly-Ordered When set (1), forces commands to strongly ordered memory to be executed one at a time. In other words, commands to strongly ordered memory are held until all previous commands have completed (responses have been received at the masterport) and all commands after the strongly ordered command are held until the strongly ordered completes (response has been received at the masterport.)
0	ORDERING_MODEL	Configures how commands to memories of type Strongly ordered and Device (SoDv) are handled in the masterport. When configured as STRICT, all commands to SoDv memory are held until all previous commands have completed (responses have been received at the masterport) and all commands after the SoDv command are held until the SoDv completes (i.e. response has been received at the masterport.) When configured as RELAXED, all commands to SoDv memory are treated as accessing Normal memory. 0x0: RELAXED 0x1: STRICT

0x00408220 BIMC_M_APP_MPORT_WR_WAY_CROSSING**Type:** RW**Clock:** Config & Master Clocks**Reset State:** 0x00000000

Configures the level of strictness or relaxedness of way crossing checks for writes.

BIMC_M_APP_MPORT_WR_WAY_CROSSING

Bits	Name	Description
7:0	REG	1-bit per slaveway. When 2 or more bits are set (1), allows the corresponding slaveways to be treated as one slaveway when write commands are considered as way crossing, i.e.. write commands to these 2 (or more) slaveways are not treated as way crossing when one write command targets one slaveway and the next write command targets the second slaveway. Not treating write commands as way crossing allows the write commands to go to the slaveway ahead of the write data for potentially better overall system performance. When all bits are set (1), disables all way crossing checks. When all bits are cleared (0), way crossing is strict with each way crossing write command held until write data for all previous writes have been routed to their corresponding slaveways.

0x00408230 BIMC_M_APP_MPORT_PRIORITYLVL_OVERRIDE

Type: RW

Clock: Config & Master Clocks

Reset State: 0x00000000

Override registers provide the capability to override one or more bus signals to specific values.

BIMC_M_APP_MPORT_PRIORITYLVL_OVERRIDE

Bits	Name	Description
9:8	PRIORITYLVL	Configures the values to override APRIORITYLVL from bus master with when OVERRIDE_PRIORITYLVL is set (1).
0	OVERRIDE_PRIORITYLVL	When set (1), overrides APRIORITYLVL from the bus master with values configured in PRIORITYLVL register field. When cleared (0), allows APRIORITYLVL from the bus master to be propagated downstream.

0x00408240 BIMC_M_APP_MPORT_READ_COMMAND_OVERRIDE

Type: RW

Clock: Config & Master Clocks

Reset State: 0x00000000

Override registers provide the capability to override one or more bus signals to specific values.

BIMC_M_APP_MPORT_READ_COMMAND_OVERRIDE

Bits	Name	Description
25:24	AREQPRIORITY	Configures the values to override AREQPRIORITY from bus master with for read commands when OVERRIDE_AREQPRIORITY is set (1).
18:16	AMEMTYPE	Configures the values to override AMEMTYPE from bus master with for read commands when OVERRIDE_AMEMTYPE is set (1).
12	ATTRANSIENT	Configures the values to override ATTRANSIENT from bus master with for read commands when OVERRIDE_ATTRANSIENT is set (1).
11	ASHARED	Configures the values to override ASHARED from bus master with for read commands when OVERRIDE_ASHARED is set (1).
10	AREDIRECT	Configures the values to override AREDIRECT from bus master with for read commands when OVERRIDE_AREDIRECT is set (1).
9	AOOO	Configures the values to override AOOO from bus master with for read commands when OVERRIDE_AOOO is set (1).
8	AINNERSHARED	Configures the values to override AINNERSHARED from bus master with for read commands when OVERRIDE_AINNERSHARED is set (1).
6	OVERRIDE_AREQPRIORITY	When set (1), overrides AREQPRIORITY from the bus master for read commands with values configured in AREQPRIORITY register field. When cleared (0), allows AREQPRIORITY from the bus master to be propagated downstream.
5	OVERRIDE_AMEMTYPE	When set (1), overrides AMEMTYPE from the bus master for read commands with values configured in AMEMTYPE register field. When cleared (0), allows AMEMTYPE from the bus master to be propagated downstream.
4	OVERRIDE_ATTRANSIENT	When set (1), overrides ATTRANSIENT from the bus master for read commands with values configured in ATTRANSIENT register field. When cleared (0), allows ATTRANSIENT from the bus master to be propagated downstream.
3	OVERRIDE_ASHARED	When set (1), overrides ASHARED from the bus master for read commands with values configured in ASHARED register field. When cleared (0), allows ASHARED from the bus master to be propagated downstream.
2	OVERRIDE_AREDIRECT	When set (1), overrides AREDIRECT from the bus master for read commands with values configured in AREDIRECT register field. When cleared (0), allows AREDIRECT from the bus master to be propagated downstream.
1	OVERRIDE_AOOO	When set (1), overrides AOOO from the bus master for read commands with values configured in AOOO register field. When cleared (0), allows AOOO from the bus master to be propagated downstream.

BIMC_M_APP_MPORT_READ_COMMAND_OVERRIDE (cont.)

Bits	Name	Description
0	VERRIDE_AINNERSHARE D	When set (1), overrides AINNERSHARED from the bus master for read commands with values configured in AINNERSHARED register field. When cleared (0), allows AINNERSHARED from the bus master to be propagated downstream.

0x00408250 BIMC_M_APP_MPORT_WRITE_COMMAND_OVERRIDE**Type:** RW**Clock:** Config & Master Clocks**Reset State:** 0x00000000

Override registers provide the capability to override one or more bus signals to specific values.

BIMC_M_APP_MPORT_WRITE_COMMAND_OVERRIDE

Bits	Name	Description
25:24	AREQPRIORITY	Configures the values to override AREQPRIORITY from bus master with for write commands when OVERRIDE_AREQPRIORITY is set (1).
18:16	AMEMTYPE	Configures the values to override AMEMTYPE from bus master with for write commands when OVERRIDE_AMEMTYPE is set (1).
12	ATRANSIENT	Configures the values to override ATRANSIENT from bus master with for write commands when OVERRIDE_ATRANSIENT is set (1).
11	ASHARED	Configures the values to override ASHARED from bus master with for write commands when OVERRIDE_ASHARED is set (1).
10	AREDIRECT	Configures the values to override AREDIRECT from bus master with for write commands when OVERRIDE_AREDIRECT is set (1).
9	AOOO	Configures the values to override AOOO from bus master with for write commands when OVERRIDE_AOOO is set (1).
8	AINNERSHARED	Configures the values to override AINNERSHARED from bus master with for write commands when OVERRIDE_AINNERSHARED is set (1).
6	VERRIDE_AREQPRIORITY	When set (1), overrides AREQPRIORITY from the bus master for write commands with values configured in AREQPRIORITY register field. When cleared (0), allows AREQPRIORITY from the bus master to be propagated downstream.
5	VERRIDE_AMEMTYPE	When set (1), overrides AMEMTYPE from the bus master for write commands with values configured in AMEMTYPE register field. When cleared (0), allows AMEMTYPE from the bus master to be propagated downstream.

BIMC_M_APP_MPORT_WRITE_COMMAND_OVERRIDE (cont.)

Bits	Name	Description
4	OVERRIDE_ATRANSIENT	When set (1), overrides ATRANSIENT from the bus master for write commands with values configured in ATRANSIENT register field. When cleared (0), allows ATRANSIENT from the bus master to be propagated downstream.
3	OVERRIDE_ASHARED	When set (1), overrides ASHARED from the bus master for write commands with values configured in ASHARED register field. When cleared (0), allows ASHARED from the bus master to be propagated downstream.
2	OVERRIDE_AREDIRECT	When set (1), overrides AREDIRECT from the bus master for write commands with values configured in AREDIRECT register field. When cleared (0), allows AREDIRECT from the bus master to be propagated downstream.
1	OVERRIDE_AOOO	When set (1), overrides AOOO from the bus master for write commands with values configured in AOOO register field. When cleared (0), allows AOOO from the bus master to be propagated downstream.
0	OVERRIDE_AINNERSHARE D	When set (1), overrides AINNERSHARED from the bus master for write commands with values configured in AINNERSHARED register field. When cleared (0), allows AINNERSHARED from the bus master to be propagated downstream.

0x00408300 BIMC_M_APP_MPORT_BKE_ENABLE**Type:** RW**Clock:** Config & Master Clocks**Reset State:** 0x00000000**BIMC_M_APP_MPORT_BKE_ENABLE**

Bits	Name	Description
0	ENABLE	When set (1), enables the book keeping element (BKE) to determine AREQPRIORITY and APRIORITYLVL of commands from the bus master by comparing the amount of bus traffic from the bus master to a target, and determining a health level for the bus master. The health level determines the relative priority of commands from the bus master to allow the bus master to catch-up if it is not getting serviced, or slow down to allow other bus masters to get their share of system resources. When cleared (0), disables the book keeping element and AREQPRIORITY and APRIORITYLVL from the bus master are propagated downstream.

0x00408304 BIMC_M_APP_MPORT_BKE_GRANT_PERIOD**Type:** RW**Clock:** Config & Reference Clocks**Reset State:** 0x00000000**BIMC_M_APP_MPORT_BKE_GRANT_PERIOD**

Bits	Name	Description
9:0	GRANT_PERIOD	Defines the period over which the book keeping element accumulates grants to the bus master for determining it's health level. Grant period is defined in number of reference timer ticks. Refer configuration of reference timer tick for exact time unit.

0x00408308 BIMC_M_APP_MPORT_BKE_GRANT_COUNT**Type:** RW**Clock:** Config & Master Clocks**Reset State:** 0x00000000**BIMC_M_APP_MPORT_BKE_GRANT_COUNT**

Bits	Name	Description
15:0	GRANT_COUNT	Defines the target grant count for the bus master. The target grant count is the number of beats of read and write data the bus master is expected to write and read in the grant period and is used as a reference to determine if the bus master is behind or ahead of where it ought to be at any given time.

0x00408320 BIMC_M_APP_MPORT_BKE_THRESHOLD_HIGH**Type:** RW**Clock:** Config & Master Clocks**Reset State:** 0x00000000**BIMC_M_APP_MPORT_BKE_THRESHOLD_HIGH**

Bits	Name	Description
15:0	THRESHOLD	Defines a high threshold to determine how much behind or ahead a bus master is with reference to where it ought to be at any given time.

0x00408324 BIMC_M_APP_MPORT_BKE_THRESHOLD_MEDIUM**Type:** RW**Clock:** Config & Master Clocks**Reset State:** 0x00000000**BIMC_M_APP_MPORT_BKE_THRESHOLD_MEDIUM**

Bits	Name	Description
15:0	THRESHOLD	Defines a medium threshold to determine how much behind or ahead a bus master is with reference to where it ought to be at any given time.

0x00408328 BIMC_M_APP_MPORT_BKE_THRESHOLD_LOW**Type:** RW**Clock:** Config & Master Clocks**Reset State:** 0x00000000**BIMC_M_APP_MPORT_BKE_THRESHOLD_LOW**

Bits	Name	Description
15:0	THRESHOLD	Defines a low threshold to determine how much behind or ahead a bus master is with reference to where it ought to be at any given time.

0x00408340 BIMC_M_APP_MPORT_BKE_HEALTH_0**Type:** RW**Clock:** Config & Master Clocks**Reset State:** 0x00000000

BKE health registers configure the request priority and realtime priority level to use for commands from the bus master when it has been determined to be of a certain health.

BIMC_M_APP_MPORT_BKE_HEALTH_0

Bits	Name	Description
31	LIMIT_COMMANDS	When set (1), defines that no new commands may be accepted from the bus master as long as it is determined to be at health 0.
9:8	AREQPRIORITY	Defines the request priority of commands from the bus master when it has been determined to be at health 0.
1:0	PRIORITYLVL	Defines the realtime priority level to use for commands from the bus master when it has been determined to be at health 0.

0x00408344 BIMC_M_APP_MPORT_BKE_HEALTH_1**Type:** RW**Clock:** Config & Master Clocks**Reset State:** 0x00000000

BKE health registers configure the request priority and realtime priority level to use for commands from the bus master when it has been determined to be of a certain health.

BIMC_M_APP_MPORT_BKE_HEALTH_1

Bits	Name	Description
31	LIMIT_COMMANDS	When set (1), defines that no new commands may be accepted from the bus master as long as it is determined to be at health 1.
9:8	AREQPRIORITY	Defines the request priority of commands from the bus master when it has been determined to be at health 1.
1:0	PRIORITYLVL	Defines the realtime priority level to use for commands from the bus master when it has been determined to be at health 1.

0x00408348 BIMC_M_APP_MPORT_BKE_HEALTH_2**Type:** RW**Clock:** Config & Master Clocks**Reset State:** 0x00000000

BKE health registers configure the request priority and realtime priority level to use for commands from the bus master when it has been determined to be of a certain health.

BIMC_M_APP_MPORT_BKE_HEALTH_2

Bits	Name	Description
31	LIMIT_COMMANDS	When set (1), defines that no new commands may be accepted from the bus master as long as it is determined to be at health 2.
9:8	AREQPRIORITY	Defines the request priority of commands from the bus master when it has been determined to be at health 2.
1:0	PRIORITYLVL	Defines the realtime priority level to use for commands from the bus master when it has been determined to be at health 2.

0x0040834C BIMC_M_APP_MPORT_BKE_HEALTH_3**Type:** RW**Clock:** Config & Master Clocks**Reset State:** 0x00000000

BKE health registers configure the request priority and realtime priority level to use for commands from the bus master when it has been determined to be of a certain health. Health 3 is the default health for the bus master.

BIMC_M_APP_MPORT_BKE_HEALTH_3

Bits	Name	Description
9:8	AREQPRIORITY	Defines the request priority of commands from the bus master when it has been determined to be at health 3.
1:0	PRIORITYLVL	Defines the realtime priority level to use for commands from the bus master when it has been determined to be at health 3.

0x00408420 BIMC_M_APP_MPORT_STATUS_2A

Type: R

Clock: Config, Master & Pipe0 Clocks

Reset State: 0x00000000

Status registers indicate the state of several elements in the masterport, including the number of transactions outstanding/pending waiting for a response, emptiness, fullness of several buffers, etc. The state information is available only as a debug indication and could be delayed by a few clock cycles from when the state is reached to being reflected in the status registers. One or more of the state registers are read live and could capture intermediate values as they change.

BIMC_M_APP_MPORT_STATUS_2A

Bits	Name	Description
23:16	WR_PENDING_PRE_BUF	Indicates the number of write transactions that are outstanding/pending waiting for a response from the slave. This transaction count is from before address channel buffers, which interfaces to frontend of the masterport.
7:0	RD_PENDING_PRE_BUF	Indicates the number of read transactions that are outstanding/pending waiting for a response from the slave. This transaction count is from before address channel buffers, which interfaces to frontend of the masterport.

0x00408424 BIMC_M_APP_MPORT_STATUS_2B

Type: R

Clock: Config, Master & Pipe0 Clocks

Reset State: 0x00000000

Status registers indicate the state of several elements in the masterport, including the number of transactions outstanding/pending waiting for a response, emptiness, fullness of several buffers, etc. The state information is available only as a debug indication and could be delayed by a few clock

cycles from when the state is reached to being reflected in the status registers. One or more of the state registers are read live and could capture intermediate values as they change.

BIMC_M_APP_MPORT_STATUS_2B

Bits	Name	Description
23:16	WR_PENDING_POST_BUF	Indicates the number of write transactions that are outstanding/pending waiting for a response from the slave. This transaction count is from after address channel buffers, which interfaces to core of the interconnect, i.e., arbiter and slaveway.
7:0	RD_PENDING_POST_BUF	Indicates the number of read transactions that are outstanding/pending waiting for a response from the slave. This transaction count is from after address channel buffers, which interfaces to core of the interconnect, i.e., arbiter and slaveway.

0x00408430 BIMC_M_APP_MPORT_STATUS_3A

Type: R

Clock: Config, Master & Pipe0 Clocks

Reset State: 0x55111111

Status registers indicate the state of several elements in the masterport, including the number of transactions outstanding/pending waiting for a response, emptiness, fullness of several buffers, etc. The state information is available only as a debug indication and could be delayed by a few clock cycles from when the state is reached to being reflected in the status registers. One or more of the state registers are read live and could capture intermediate values as they change.

BIMC_M_APP_MPORT_STATUS_3A

Bits	Name	Description
31	RCB_DATA_RD_FULL	Indicates that the data part of read channel buffer is full when seen from it's read side, which interfaces to frontend of the masterport.
30	RCB_DATA_RD_EMPTY	Indicates that the data part of read channel buffer is empty when seen from it's read side, which interfaces to frontend of the masterport.
29	RCB_CTRL_RD_FULL	Indicates that the control part of read channel buffer is full when seen from it's read side, which interfaces to frontend of the masterport.
28	RCB_CTRL_RD_EMPTY	Indicates that the control part of read channel buffer is empty when seen from it's write side, which interfaces to frontend of the masterport.
27	RCB_DATA_WR_FULL	Indicates that the data part of read channel buffer is full when seen from it's write side, which interfaces to core of the interconnect, i.e., arbiter and slaveway.
26	RCB_DATA_WR_EMPTY	Indicates that the data part of read channel buffer is empty when seen from it's write side, which interfaces to core of the interconnect, i.e., arbiter and slaveway.

BIMC_M_APP_MPORT_STATUS_3A (cont.)

Bits	Name	Description
25	RCB_CTRL_WR_FULL	Indicates that the control part of read channel buffer is full when seen from it's write side, which interfaces to core of the interconnect, i.e., arbiter and slaveway.
24	RCB_CTRL_WR_EMPTY	Indicates that the control part of read channel buffer is empty when seen from it's write side, which interfaces to core of the interconnect, i.e., arbiter and slaveway.
21	BCB_RD_FULL	Indicates that the write response channel buffer is full when seen from it's read side, which interfaces to frontend of the masterport.
20	BCB_RD_EMPTY	Indicates that the write response channel buffer is empty when seen from it's read side, which interfaces to frontend of the masterport.
17	BCB_WR_FULL	Indicates that the write response channel buffer is full when seen from it's write side, which interfaces to core of the interconnect, i.e., arbiter and slaveway.
16	BCB_WR_EMPTY	Indicates that the write response channel buffer is empty when seen from it's write side, which interfaces to core of the interconnect, i.e., arbiter and slaveway.
13	WCB_RD_FULL	Indicates that the write channel buffer is full when seen from it's read side, which interfaces to core of the interconnect, i.e., arbiter and slaveway.
12	WCB_RD_EMPTY	Indicates that the write channel buffer is empty when seen from it's read side, which interfaces to core of the interconnect, i.e., arbiter and slaveway.
9	WCB_WR_FULL	Indicates that the write channel buffer is full when seen from it's write side, which interfaces to frontend of the masterport.
8	WCB_WR_EMPTY	Indicates that the write channel buffer is empty when seen from it's write side, which interfaces to frontend of the masterport.
5	ACB_RD_FULL	Indicates that the address channel buffer is full when seen from it's read side, which interfaces to core of the interconnect, i.e., arbiter and slaveway.
4	ACB_RD_EMPTY	Indicates that the address channel buffer is empty when seen from it's read side, which interfaces to core of the interconnect, i.e., arbiter and slaveway.
1	ACB_WR_FULL	Indicates that the address channel buffer is full when seen from it's write side, which interfaces to frontend of the masterport.
0	ACB_WR_EMPTY	Indicates that the address channel buffer is empty when seen from it's write side, which interfaces to frontend of the masterport.

0x00408434 BIMC_M_APP_MPORT_STATUS_3B**Type:** R**Clock:** Config, Master & Pipe0 Clocks**Reset State:** 0x00111111

Buffer status register indicate the state of all buffers in the masterport, whether they are empty or full on either side of the buffer. Some buffers carry information from master to core and others carry information from core to master. Both write side and read side status are indicated separately, since they would be not be similar when master clock and core clock are not synchronous to each other.

BIMC_M_APP_MPORT_STATUS_3B

Bits	Name	Description
21	RGB_RD_FULL	Indicates that the read gathering buffer is full when seen from it's read side, which interfaces to frontend of the masterport.
20	RGB_RD_EMPTY	Indicates that the read gathering buffer is empty when seen from it's read side, which interfaces to frontend of the masterport.
17	RGB_WR_FULL	Indicates that the read gathering buffer is full when seen from it's write side, which interfaces to core of the interconnect, i.e., arbiter and slaweway.
16	RGB_WR_EMPTY	Indicates that the read gathering buffer is empty when seen from it's write side, which interfaces to core of the interconnect, i.e., arbiter and slaweway.
13	W2AB_RD_FULL	Indicates that the write channel to address channel user signals buffer is full when seen from it's read side, which interfaces to core of the interconnect, i.e., arbiter and slaweway.
12	W2AB_RD_EMPTY	Indicates that the write channel to address channel user signals buffer is empty when seen from it's read side, which interfaces to core of the interconnect, i.e., arbiter and slaweway.
9	W2AB_WR_FULL	Indicates that the write channel to address channel user signals buffer is full when seen from it's write side, which interfaces to frontend of the masterport.
8	W2AB_WR_EMPTY	Indicates that the write channel to address channel user signals buffer is empty when seen from it's write side, which interfaces to frontend of the masterport.
5	A2WB_RD_FULL	Indicates that the address channel to write channel user signals buffer is full when seen from it's read side, which interfaces to core of the interconnect, i.e., arbiter and slaweway.
4	A2WB_RD_EMPTY	Indicates that the address channel to write channel user signalsbuffer is empty when seen from it's read side, which interfaces to core of the interconnect, i.e., arbiter and slaweway.
1	A2WB_WR_FULL	Indicates that the address channel to write channel user signals buffer is full when seen from it's write side, which interfaces to frontend of the masterport.

BIMC_M_APP_MPORT_STATUS_3B (cont.)

Bits	Name	Description
0	A2WB_WR_EMPTY	Indicates that the address channel to write channel user signals buffer is empty when seen from it's write side, which interfaces to frontend of the masterport.

2.5 BIMC_M_APP_PROF

BRIC Traffic Profiling

2.5.1 BIMC_M_APP_PROF_BRIC_PROFILING

BRIC Traffic Profiling

0x00409000 BIMC_M_APP_PROF_COMPONENT_INFO**Type:** R**Clock:** Config Clock**Reset State:** 0x00000202

Component information registers provide discoverability information on the type of component and instance of each component. These registers are always the very first register in each component's configuration address space and allow for a software routine to walk the configuration address space in steps of 4KB to identify the location of each configuration register space and to which component each 4KB configures.

BIMC_M_APP_PROF_COMPONENT_INFO

Bits	Name	Description
23:16	INSTANCE	Indicates the instance number of this traffic profiling.
15:8	SUB_TYPE	Identifies this configuration space as belonging to traffic profiling 0x2: PROFILING
7:0	TYPE	Identifies this configuration space as belonging to a component that interfaces to a bus master. 0x2: MASTERPORT

0x00409020 BIMC_M_APP_PROF_CONFIGURATION_INFO_0**Type:** R**Clock:** Config Clock**Reset State:** 0x00000000

Configuration information registers provide information on the capability and configuration of each traffic profiling instance. See register field descriptions for more details.

BIMC_M_APP_PROF_CONFIGURATION_INFO_0

Bits	Name	Description
31:24	TENURE_WIDTH	Indicates the width of tenures (latencies) measured in traffic profiling.
23:16	TRACKING_TABLE_DEPTH	Indicates the number of entries in the table that tracks outstanding commands for determining how many commands have larger turnaround latencies that a programmed threshold.
7:0	SCALING_FACTOR	Indicates how many extra scaling is performed on events generated by traffic profiling if such events have to be synchronized from the tenure clock to the event clock. Any value that is non-power-of-2 is converted to power-of-2 by hardware. When the value read from this register field is used in software, similar conversion of non-power-of-2 values to next higher power-of-2 value is necessary.

0x00409030 BIMC_M_APP_PROF_CONFIGURATION_INFO_1

Type: R

Clock: Config Clock

Reset State: 0x00000000

Configuration information registers provide information on the capability and configuration of each traffic profiling instance. See register field descriptions for more details.

BIMC_M_APP_PROF_CONFIGURATION_INFO_1

Bits	Name	Description
23:16	PIPE2_ACH_DEPTH	Indicates the depth of address channel buffer in pipe2 of masterport in which this traffic profiling instance is paired with. Applies only to traffic profiling instances in XR type masterports, which contains three distinct paths to slaves, in pipe0, pipe1, pipe2. In non-XR type masterports, value contained in field does not refer to any actual address channel buffer.
15:8	PIPE1_ACH_DEPTH	Indicates the depth of address channel buffer in pipe1 of masterport in which this traffic profiling instance is paired with. Applies only to traffic profiling instances in XR type masterports, which contains three distinct paths to slaves, in pipe0, pipe1, pipe2. In non-XR type masterports, value contained in field does not refer to any actual address channel buffer.
7:0	PIPE0_ACH_DEPTH	Indicates the depth of address channel buffer in pipe0 of masterport in which this traffic profiling instance is paired with. Applies only to traffic profiling instances in XR type masterports, which contains three distinct paths to slaves, in pipe0, pipe1, pipe2. In non-XR type masterports, pipe0 refers to the common address channel buffer used for all slave accesses.

0x00409200 BIMC_M_APP_PROF_CLOCK_CTRL**Type:** RW**Clock:** Config, Event & Tenure Clocks**Reset State:** 0x00000003**BIMC_M_APP_PROF_CLOCK_CTRL**

Bits	Name	Description
1	TENURE_CLOCK_GATING_EN	When set (1), enables activity based clock gating of all registers clocked by tenure clock. Activity based clock gating uses architectural clock gates to reduce dynamic power consumption by gating clocks during periods of in-activity. When cleared (0), disables all activity based clock gating by forcing all architectural clock gates to not gate clocks even during periods of in-activity.
0	EVENT_CLOCK_GATING_EN	When set (1), enables activity based clock gating of all registers clocked by event clock. Activity based clock gating uses architectural clock gates to reduce dynamic power consumption by gating clocks during periods of in-activity. When cleared (0), disables all activity based clock gating by forcing all architectural clock gates to not gate clocks even during periods of in-activity.

0x00409210 BIMC_M_APP_PROF_MODE**Type:** RW**Clock:** Config, Event & Tenure Clocks**Reset State:** 0x00000000

BIMC_M_APP_PROF_MODE

Bits	Name	Description
25:24	WR_TENURE_MODE	Configures the start and end events for monitoring write latencies/tenures. When configured to WR_CMD_TO_BRESP, starting event is first cycle of write command from bus master as seen by traffic profiling and ending event is first cycle of write response to bus master as seen by traffic profiling. When configured to WR_CMD_TO_BRESP_HANDSHAKE, starting event is first cycle of write command from bus master as seen by traffic profiling and ending event is transfer handshake of write response to bus master as seen by traffic profiling. When configured to WR_CMD_HANDSHAKE, starting event is first cycle of write command from bus master as seen by traffic profiling and ending event is transfer handshake of the same write command with bus master as seen by traffic profiling. When configured to WR_RESP_HANDSHAKE, starting event is first cycle of write response to bus master as seen by traffic profiling and ending event is transfer handshake of write response to bus master as seen by traffic profiling. 0x0: WR_CMD_TO_BRESP 0x1: WR_CMD_TO_BRESP_HANDSHAKE 0x2: WR_CMD_HANDSHAKE 0x3: WR_RESP_HANDSHAKE
17:16	RD_TENURE_MODE	Configures the start and end events for monitoring read latencies/tenures. When configured to RD_CMD_TO_FIRST_RDATA, starting event is first cycle of read command from bus master as seen by traffic profiling and ending event is first cycle of first beat of read data to bus master as seen by traffic profiling. When configured to RD_CMD_TO_LAST_RDATA, starting event is first cycle of read command from bus master as seen by traffic profiling and ending event is transfer handshake of last beat of read data to bus master as seen by traffic profiling. When configured to RD_CMD_HANDSHAKE, starting event is first cycle of read command from bus master as seen by traffic profiling and ending event is transfer handshake of the same read command with bus master as seen by traffic profiling. When configured to RD_DATA_HANDSHAKE, starting event is first cycle of first beat of read data to bus master as seen by traffic profiling and ending event is transfer handshake of last beat of read data to bus master as seen by traffic profiling. 0x0: RD_CMD_TO_FIRST_RDATA 0x1: RD_CMD_TO_LAST_RDATA 0x2: RD_CMD_HANDSHAKE 0x3: RD_DATA_HANDSHAKE
0	PROFILING_EN	When set (1), enables traffic profiling to monitor bus transactions from bus master and generate events for counting in performance monitors. When cleared (0), disables traffic profiling.

0x00409214 BIMC_M_APP_PROF_RD_TENURE_THRESHOLD**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_APP_PROF_RD_TENURE_THRESHOLD**

Bits	Name	Description
15:0	THRESHOLD	Configures the threshold over which read latencies/tenures need to be before an event is generated to performance monitors for counting max latencies.

0x00409218 BIMC_M_APP_PROF_WR_TENURE_THRESHOLD**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_APP_PROF_WR_TENURE_THRESHOLD**

Bits	Name	Description
15:0	THRESHOLD	Configures the threshold over which write latencies/tenures need to be before an event is generated to performance monitors for counting max latencies.

0x00409220 BIMC_M_APP_PROF_SWAY_SELECT**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_APP_PROF_SWAY_SELECT**

Bits	Name	Description
7:0	SWAY	Selects the slaveways to which commands are routed to profile. 1-bit per slaveway.

0x00409230 BIMC_M_APP_PROF_MID_MASK**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000

BIMC_M_APP_PROF_MID_MASK

Bits	Name	Description
15:0	MID_MASK	When a bit position is set (1), enables matching of that bit position in MID of command with same bit position in MID_MATCH. Commands with MIDs that do not match are not considered for profiling.

0x00409234 BIMC_M_APP_PROF_MID_MATCH**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_APP_PROF_MID_MATCH**

Bits	Name	Description
15:0	MID_MATCH	Configures the value to match for in AMID in commands. Used in conjunction with MID_MASK, which controls which bits of MID are compared.

0x00409238 BIMC_M_APP_PROF_MID_INV_MATCH**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_APP_PROF_MID_INV_MATCH**

Bits	Name	Description
0	MID_INV_MATCH	When set (1), configures MID matching to result in an inverse match, i.e., only commands whose MIDs that do not match MID_MATCH are considered for profiling.

0x00409240 BIMC_M_APP_PROF_ADDR_MASK_LOWER**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000

BIMC_M_APP_PROF_ADDR_MASK_LOWER

Bits	Name	Description
31:10	ADDR_MASK_LOWER	Configures ADDR_MASK[31:0]. When a bit position is set (1), enables matching of that bit position of address of command with same bit position in ADDR_MATCH[31:0].

0x00409244 BIMC_M_APP_PROF_ADDR_MASK_UPPER**Type:** R**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_APP_PROF_ADDR_MASK_UPPER**

Bits	Name	Description
3:0	UNUSED	

0x00409248 BIMC_M_APP_PROF_ADDR_MATCH_LOWER**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_APP_PROF_ADDR_MATCH_LOWER**

Bits	Name	Description
31:10	ADDR_MATCH_LOWER	Configures the value to match for in address of commands. Used in conjunction with ADDR_MASK[31:0], which controls which bits of address are compared.

0x0040924C BIMC_M_APP_PROF_ADDR_MATCH_UPPER**Type:** R**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_APP_PROF_ADDR_MATCH_UPPER**

Bits	Name	Description
3:0	UNUSED	

0x00409250 BIMC_M_APP_PROF_ADDR_INV_MATCH**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_APP_PROF_ADDR_INV_MATCH**

Bits	Name	Description
0	ADDR_INV_MATCH	When set (1), configures address matching to result in an inverse match, i.e., only commands whose addresses do not match ADDR_MATCH are considered for profiling.

0x00409260 BIMC_M_APP_PROF_ACH_MASK_0**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_APP_PROF_ACH_MASK_0**

Bits	Name	Description
24	AINNERSHARED_MASK	When set (1), enables matching of AINNERSHARED in commands with AINNERSHARED_MATCH.
16	AFULL_MASK	When set (1), enables matching of AFULL in commands with AFULL_MATCH.
8	AEXCLUSIVE_MASK	When set (1), enables matching of AEXCLUSIVE in commands with AEXCLUSIVE_MATCH.
0	ABURST_MASK	When set (1), enables matching of ABURST in commands with ABURST_MATCH.

0x00409264 BIMC_M_APP_PROF_ACH_MATCH_0**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_APP_PROF_ACH_MATCH_0**

Bits	Name	Description
24	AINNERSHARED_MATCH	Configures the value to match for in AINNERSHARED in commands. Used in conjunction with AINNERSHARED_MASK, which controls whether AINNERSHARED in commands is compared

BIMC_M_APP_PROF_ACH_MATCH_0 (cont.)

Bits	Name	Description
16	AFULL_MATCH	Configures the value to match for in AFULL in commands. Used in conjunction with AFULL_MASK, which controls whether AFULL in commands is compared.
8	AEXCLUSIVE_MATCH	Configures the value to match for in AEXCLUSIVE in commands. Used in conjunction with AEXCLUSIVE_MASK, which controls whether AFULL in commands is compared.
0	ABURST_MATCH	Configures the value to match for in ABURST in commands. Used in conjunction with ABURST_MASK, which controls whether ABURST in commands is compared.

0x00409270 BIMC_M_APP_PROF_ACH_MASK_1**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_APP_PROF_ACH_MASK_1**

Bits	Name	Description
24	AMSSSELFAUTH_MASK	When set (1), enables matching of AMSSSELFAUTH in commands with AMSSSELFAUTH_MATCH.
18:16	AMEMTYPE_MASK	When a bit position is set (1), enables matching of corresponding bit position of AMEMTYPE with AMEMTYPE_MATCH.
11:8	ALEN_MASK	When a bit position is set (1), enables matching of corresponding bit position of ALEN with ALEN_MATCH.
0	AINST_MASK	When set (1), enables matching of AINST in commands with AINST_MASK.

0x00409274 BIMC_M_APP_PROF_ACH_MATCH_1**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_APP_PROF_ACH_MATCH_1**

Bits	Name	Description
24	AMSSSELFAUTH_MATCH	Configures the value to match for in AMSSSELFAUTH in commands. Used in conjunction with AMSSSELFAUTH_MASK, which controls whether AMSSSELFAUTH in commands is compared.

BIMC_M_APP_PROF_ACH_MATCH_1 (cont.)

Bits	Name	Description
18:16	AMEMTYPE_MATCH	Configures the value to match for in AMEMTYPE in commands. Used in conjunction with AMEMTYPE_MASK, which controls which bits of AMEMTYPE in commands are compared.
11:8	ALEN_MATCH	Configures the value to match for in ALEN in commands. Used in conjunction with ALEN_MASK, which controls which bits of ALEN in commands are compared.
0	AINST_MATCH	Configures the value to match for in AINST in commands. Used in conjunction with AINST_MASK, which controls whether AINST in commands is compared.

0x00409278 BIMC_M_APP_PROF_ACH_INV_MATCH_1**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_APP_PROF_ACH_INV_MATCH_1**

Bits	Name	Description
16	AMEMTYPE_INV_MATCH	When set (1), configures AMEMTYPE matching to result in an inverse match, i.e., only commands whose AMEMTYPE does not match AMEMTYPE_MATCH are considered for profiling.
8	ALEN_INV_MATCH	When set (1), configures ALEN matching to result in an inverse match, i.e., only commands when ALEN does not match ALEN_MATCH are considered for profiling.

0x00409280 BIMC_M_APP_PROF_ACH_MASK_2**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_APP_PROF_ACH_MASK_2**

Bits	Name	Description
24	APROTNS_MASK	When set (1), enables matching of APROTNS in commands with APROTNS_MATCH.
16	APRIV_MASK	When set (1), enables matching of APRIV in commands with APRIV_MATCH.
8	AOOO_MASK	When set (1), enables matching of AOOO in commands with AOOO_MATCH.
0	ANOALLOCATE_MASK	When set (1), enables matching of ANOALLOCATE in commands with ANOALLOCATE_MATCH.

0x00409284 BIMC_M_APP_PROF_ACH_MATCH_2**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_APP_PROF_ACH_MATCH_2**

Bits	Name	Description
24	APROTNS_MATCH	Configures the value to match for in APROTNS in commands. Used in conjunction with APROTNS_MASK, which controls whether APROTNS in commands is compared
16	APRIV_MATCH	Configures the value to match for in APRIV in commands. Used in conjunction with APRIV_MASK, which controls whether APRIV in commands is compared
8	AOOO_MATCH	Configures the value to match for in AOOO in commands. Used in conjunction with AOOO_MASK, which controls whether AOOO in commands is compared
0	ANOALLOCATE_MATCH	Configures the value to match for in ANOALLOCATE in commands. Used in conjunction with ANOALLOCATE_MASK, which controls whether ANOALLOCATE in commands is compared

0x00409290 BIMC_M_APP_PROF_ACH_MASK_3**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_APP_PROF_ACH_MASK_3**

Bits	Name	Description
24	ASHARED_MASK	When set (1), enables matching of ASHARED in commands with ASHARED_MATCH.
17:16	AREQPRIORITY_MASK	When a bit position is set (1), enables matching of corresponding bit position of AREQPRIORITY with AREQPRIORITY_MATCH.
8	AREDIRECT_MASK	When set (1), enables matching of AREDIRECT in commands with AREDIRECT_MATCH.
0	ARDBEATNDXEN_MASK	When set (1), enables matching of ARDBEATNDXEN in commands with ARDBEATNDXEN_MATCH.

0x00409294 BIMC_M_APP_PROF_ACH_MATCH_3**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_APP_PROF_ACH_MATCH_3**

Bits	Name	Description
24	ASHARED_MATCH	Configures the value to match for in ASHARED in commands. Used in conjunction with ASHARED_MASK, which controls whether ASHARED in commands is compared
17:16	AREQPRIORITY_MATCH	Configures the value to match for in AREQPRIORITY in commands. Used in conjunction with AREQPRIORITY_MASK, which controls which bits of AREQPRIORITY in commands are compared.
8	AREDIRECT_MATCH	Configures the value to match for in AREDIRECT in commands. Used in conjunction with AREDIRECT_MASK, which controls whether AREDIRECT in commands is compared
0	ARDBEATNDXEN_MATCH	Configures the value to match for in ARDBEATNDXEN in commands. Used in conjunction with ARDBEATNDXEN_MASK, which controls whether ARDBEATNDXEN in commands is compared

0x00409298 BIMC_M_APP_PROF_ACH_INV_MATCH_3**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_APP_PROF_ACH_INV_MATCH_3**

Bits	Name	Description
16	AREQPRIORITY_INV_MATCH	When set (1), configures AREQPRIORITY matching to result in an inverse match, i.e., only commands whose AREQPRIORITY does not match AREQPRIORITY_MATCH are considered for profiling

0x004092A0 BIMC_M_APP_PROF_ACH_MASK_4**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000

BIMC_M_APP_PROF_ACH_MASK_4

Bits	Name	Description
24	AWRITE_MASK	When set (1), enables matching of AWRITE in commands with AWRITE_MATCH.
20:16	AVMID_MASK	When a bit position is set (1), enables matching of corresponding bit position in AVMID with AVMID_MATCH
8	ATRANSIENT_MASK	When set (1), enables matching of ATRANSIENT in commands with ATRANSIENT_MATCH.
2:0	ASIZE_MASK	When a bit position is set (1), enables matching of corresponding bit position in ASIZE with ASIZE_MATCH.

0x004092A4 BIMC_M_APP_PROF_ACH_MATCH_4**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_APP_PROF_ACH_MATCH_4**

Bits	Name	Description
24	AWRITE_MATCH	Configures the value to match for in AWRITE in commands. Used in conjunction with AWRITE_MASK, which controls whether AWRITE in commands is compared.
20:16	AVMID_MATCH	Configures the value to match for in AVMID in commands. Used in conjunction with AVMID_MASK, which controls which bits of AVMID in commands are compared.
8	ATRANSIENT_MATCH	Configures the value to match for in ATRANSIENT in commands. Used in conjunction with ATRANSIENT_MASK, which controls whether ATRANSIENT in commands is compared.
2:0	ASIZE_MATCH	Configures the value to match for in ASIZE in commands. Used in conjunction with ASIZE_MASK, which controls which bits of ASIZE in commands are compared.

0x004092A8 BIMC_M_APP_PROF_ACH_INV_MATCH_4**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000

BIMC_M_APP_PROF_ACH_INV_MATCH_4

Bits	Name	Description
16	AVMID_INV_MATCH	When set (1), configures AVMID matching to result in an inverse match, i.e., only commands whose AVMID does not match AVMID_MATCH are considered for profiling.
0	ASIZE_INV_MATCH	When set (1), configures ASIZE matching to result in an inverse match, i.e., only commands whose AVMID does not match ASIZE_MATCH are considered for profiling.

0x00409300 BIMC_M_APP_PROF_BCH_MASK**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_APP_PROF_BCH_MASK**

Bits	Name	Description
1:0	BRESP_MASK	When a bit position is set (1), enables matching of corresponding bit position in BRESP with BRESP_MATCH.

0x00409304 BIMC_M_APP_PROF_BCH_MATCH**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_APP_PROF_BCH_MATCH**

Bits	Name	Description
1:0	BRESP_MATCH	Configures the value to match for in BRESP in write responses. Used in conjunction with BRESP_MASK, which controls which bits of BRESP in writes responses are compared.

0x00409308 BIMC_M_APP_PROF_BCH_INV_MATCH**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000

BIMC_M_APP_PROF_BCH_INV_MATCH

Bits	Name	Description
0	BRESP_INV_MATCH	When set (1), configures BRESP matching to result in an inverse match, i.e., only write responses whose BRESP does not match BRESP_MATCH are considered for profiling.

0x00409340 BIMC_M_APP_PROF_RCH_MASK**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_APP_PROF_RCH_MASK**

Bits	Name	Description
8	RLAST_MASK	When set (1), enables matching of RLAST in commands with corresponding setting in RCH_MATCH register.
1:0	RRESP_MASK	When a bit position is set (1), enables matching of corresponding bit position in RRESP with RRESP_MATCH.

0x00409344 BIMC_M_APP_PROF_RCH_MATCH**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_APP_PROF_RCH_MATCH**

Bits	Name	Description
8	RLAST_MATCH	Configures the value to match for in RLAST in read responses. Used in conjunction with RLAST_MASK, which controls whether RLAST in read responses is compared.
1:0	RRESP_MATCH	Configures the value to match for in RRESP in read responses. Used in conjunction with RRESP_MASK, which controls which bits of RRESP in read responses are compared.

0x00409348 BIMC_M_APP_PROF_RCH_INV_MATCH**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000

BIMC_M_APP_PROF_RCH_INV_MATCH

Bits	Name	Description
0	RRESP_INV_MATCH	When set (1), configures RRESP matching to result in an inverse match, i.e., only read responses whose RRESP does not match RRESP_MATCH are considered for profiling.

**0x00409400+ BIMC_M_APP_PROF_RD_STATUS_n, n=[0..0]
0x4*n**
Type: R**Clock:** Config & Tenure Clocks**Reset State:** 0x00000000**BIMC_M_APP_PROF_RD_STATUS_n**

Bits	Name	Description
31	RD_OUTSTANDING_OVERFLOW	Indicates that the number of outstanding reads crossed the maximum supported (255) during when traffic profiling was last enabled. It is recommended that this register be read only after traffic profiling has been disabled.
30	RD_OUTSTANDING_UNDERFLOW	Indicates that the number of outstanding reads went below 0 during when traffic profiling was last enabled. It indicates that more command completions were seen compared to commands starts. The most likely scenario is an overflow of number of outstanding reads occurred leading to some command starts not being tracked. It is recommended that this register be read only after traffic profiling has been disabled.
7:0	RD_OUTSTANDING_MAX	Indicates the maximum number of outstanding reads that were seen when traffic profiling was last enabled. Only reads that match the MID filter and slaweway select filter are accounted for. It is recommended that this register be read only after traffic profiling has been disabled.

**0x00409410+ BIMC_M_APP_PROF_WR_STATUS_n, n=[0..0]
0x4*n**
Type: R**Clock:** Config & Tenure Clocks**Reset State:** 0x00000000**BIMC_M_APP_PROF_WR_STATUS_n**

Bits	Name	Description
31	WR_OUTSTANDING_OVERFLOW	Indicates that the number of outstanding writes crossed the maximum supported (255) during when traffic profiling was last enabled. It is recommended that this register be read only after traffic profiling has been disabled.

BIMC_M_APP_PROF_WR_STATUS_n (cont.)

Bits	Name	Description
30	WR_OUTSTANDING_UNDE RFLOW	Indicates that the number of outstanding writes went below 0 during when traffic profiling was last enabled. It indicates that more command completions were seen compared to commands starts. The most likely scenario is an overflow of number of outstanding writes occurred leading to some command starts not being tracked. It is recommended that this register be read only after traffic profiling has been disabled.
7:0	WR_OUTSTANDING_MAX	Indicates the maximum number of outstanding writes that were seen when traffic profiling was last enabled. Only writes that match the MID filter and slaweway select filter are accounted for. It is recommended that this register be read only after traffic profiling has been disabled.

2.6 BIMC_M_DSP_MPORT

BRIC Masterport

2.6.1 BIMC_M_DSP_MPORT_BRIC_MASTERPORT

BRIC Masterport

0x0040C000 BIMC_M_DSP_MPORT_COMPONENT_INFO**Type:** R**Clock:** Config Clock**Reset State:** 0x00000102

Component information registers provide discoverability information on the type of component and instance of each component. These registers are always the very first register in each component's configuration address space and allow for a software routine to walk the configuration address space in steps of 4KB to identify the location of each configuration register space and to which component each 4KB configures.

BIMC_M_DSP_MPORT_COMPONENT_INFO

Bits	Name	Description
23:16	INSTANCE	Indicates the instance number of this masterport.
15:8	SUB_TYPE	Identifies this configuration space as belonging to a masterport 0x1: MASTERPORT
7:0	TYPE	Identifies this configuration space as belonging to a component that interfaces to a bus master. 0x2: MASTERPORT

0x0040C020 BIMC_M_DSP_MPORT_CONFIGURATION_INFO_0**Type:** R**Clock:** Config Clock**Reset State:** 0x01000000

Configuration information registers provide information on the capability and configuration of each masterport instance. See register field descriptions for more details.

BIMC_M_DSP_MPORT_CONFIGURATION_INFO_0

Bits	Name	Description
31:24	SYNC_MODE	Indicates the clock relationship between the clock used in the master side and the clock used in the core side of the masterport. When ASYNC, identifies the master clock and core clock as unrelated and asynchronous to each other. When SYNC, identifies the master clock and core clock as synchronous to each other. When ISOSYNC, identifies the master clock and core clock as related, but of different frequencies, which are of integer ratios, i.e., they are iso-synchronous to each other. 0x0: ASYNC 0x1: SYNC 0x2: ISOSYNC
15:8	CONNECTION_TYPE	Indicates the type of master connecting to this masterport. When DIRECT, indicates that a bus master is directly connected to this masterport. When CASCADE, indicates that bus masters connect to this masterport through another level of interconnect. 0x0: DIRECT 0x1: CASCADE
7:0	FUNCTIONALITY	Indicates the functionality and type of bus traffic handled by this masterport. When INORDER, identifies this masterport as optimized for in-order bus traffic. When OOO_ONLY, identifies this masterport as optimized for out-of-order bus traffic, where a single master is capable of handling responses return in any order. When REORDER, identifies this masterport as having a reorder buffer for re-ordering responses to command order, while allowing bus slaves to return responses in any order they desire. When STUB, identifies this masterport has a stub. 0x0: STUB 0x1: OOO_ONLY 0x2: INORDER 0x3: REORDER

0x0040C030 BIMC_M_DSP_MPORT_CONFIGURATION_INFO_1**Type:** R**Clock:** Config Clock**Reset State:** 0x00000000

Configuration information registers provide information on the capability and configuration of each masterport instance. See register field descriptions for more details.

BIMC_M_DSP_MPORT_CONFIGURATION_INFO_1

Bits	Name	Description
31:0	SWAY_CONNECTIVITY	Indicates slaveways that this masterport can route commands to. 1-bit per slaveway. When a specific bit is set (1), indicates that commands can be routed to the slaveway corresponding to this bit. When a specific bit is clear (0), indicates that commands are not routed to the slaveway corresponding to this bit.

0x0040C040 BIMC_M_DSP_MPORT_CONFIGURATION_INFO_2A

Type: R

Clock: Config Clock

Reset State: 0x00000000

Configuration information registers provide information on the capability and configuration of each masterport instance. See register field descriptions for more details.

BIMC_M_DSP_MPORT_CONFIGURATION_INFO_2A

Bits	Name	Description
31:16	M_DATA_WIDTH	Indicates the width of read and write data from the bus master.
15:8	M_TID_WIDTH	Indicates the width of transfer IDs from the bus master.
7:0	M_MID_WIDTH	Indicates the width of master ID from the bus master (includes any bus and port IDs if bus master is an interconnect.)

0x0040C044 BIMC_M_DSP_MPORT_CONFIGURATION_INFO_2B

Type: R

Clock: Config Clock

Reset State: 0x00000000

Configuration information registers provide information on the capability and configuration of each masterport instance. See register field descriptions for more details.

BIMC_M_DSP_MPORT_CONFIGURATION_INFO_2B

Bits	Name	Description
15:0	C_DATA_WIDTH	Indicates the width of read and write data used in core of this masterport.

0x0040C050 BIMC_M_DSP_MPORT_CONFIGURATION_INFO_3**Type:** R**Clock:** Config Clock**Reset State:** 0x00000000

Configuration information registers provide information on the capability and configuration of each masterport instance. See register field descriptions for more details.

BIMC_M_DSP_MPORT_CONFIGURATION_INFO_3

Bits	Name	Description
31:24	RCH_DEPTH	Indicates the depth of read channel buffers in this masterport. A beat of read data occupies one entry in the read channel buffer, until accepted by the bus master.
23:16	BCH_DEPTH	Indicates the depth of write response channel buffers in this masterport. Each write response occupies one entry in the write response channel buffer, until accepted by the bus master.
15:8	WCH_DEPTH	Indicates the depth of write channel buffers in this masterport. A beat of write data occupies one entry in the write channel buffer, until accepted by the arbiter.
7:0	ACH_DEPTH	Indicates the depth of address channel buffers in this masterport. Each command occupies one entry in the address channel buffer, until accepted by the arbiter.

0x0040C060 BIMC_M_DSP_MPORT_CONFIGURATION_INFO_4**Type:** R**Clock:** Config Clock**Reset State:** 0x00000000

Configuration information registers provide information on the capability and configuration of each masterport instance. See register field descriptions for more details.

BIMC_M_DSP_MPORT_CONFIGURATION_INFO_4

Bits	Name	Description
15:8	REORDER_BUFFER_DEPTH	Indicates the number of read data beats that can be re-ordered in this masterport (if the masterport provides response re-ordering functionality.)
7:0	REORDER_TABLE_DEPTH	Indicates the number of read and write commands whose responses can be re-ordered in this masterport (if the masterport provides responses re-ordering functionality.)

0x0040C070 BIMC_M_DSP_MPORT_CONFIGURATION_INFO_5A**Type:** R**Clock:** Config Clock**Reset State:** 0x00000000

Configuration information registers provide information on the capability and configuration of each masterport instance. See register field descriptions for more details.

BIMC_M_DSP_MPORT_CONFIGURATION_INFO_5A

Bits	Name	Description
31:0	ACH_PIPELINE_STAGES	Indicates pipeline stages enabled on the address channel. Each digit identifies a specific pipeline stage in the masterport, where '0' indicates no pipeline stage, '1' indicates 1-deep pipeline stage where forward path is registered, return path is fed through, '2' indicates 2-deep pipeline stage where both forward and return path are pipelined. Digit 3 identifies pipelining on the master to masterport interface, digit 2 identifies pipelining after address decode, digit 1 identifies pipelining in the address channel buffer and digit 0 identifies pipelining on the masterport to arbiter interface.

0x0040C074 BIMC_M_DSP_MPORT_CONFIGURATION_INFO_5B**Type:** R**Clock:** Config Clock**Reset State:** 0x00000000

Configuration information registers provide information on the capability and configuration of each masterport instance. See register field descriptions for more details.

BIMC_M_DSP_MPORT_CONFIGURATION_INFO_5B

Bits	Name	Description
31:0	WCH_PIPELINE_STAGES	Indicates pipeline stages enabled on the write channel. Each digit identifies a specific pipeline stage in the masterport, where '0' indicates no pipeline stage, '1' indicates 1-deep pipeline stage where forward path is registered, return path is fed through, '2' indicates 2-deep pipeline stage where both forward and return path are pipelined. Digit 3 identifies pipelining on the master to masterport interface, digit 2 identifies pipelining after address decode, digit 1 identifies pipelining in the write channel buffer and digit 0 identifies pipelining on the masterport to arbiter interface.

0x0040C078 BIMC_M_DSP_MPORT_CONFIGURATION_INFO_5C

Type: R
Clock: Config Clock
Reset State: 0x00000000

Configuration information registers provide information on the capability and configuration of each masterport instance. See register field descriptions for more details.

BIMC_M_DSP_MPORT_CONFIGURATION_INFO_5C

Bits	Name	Description
31:0	BCH_PIPELINE_STAGES	Indicates pipeline stages enabled on the write response channel. Each digit identifies a specific pipeline stage in the masterport, where '0' indicates no pipeline stage, '1' indicates 1-deep pipeline stage where forward path is registered, return path is fed through, '2' indicates 2-deep pipeline stage where both forward and return path are pipelined. Digit 2 identifies pipelining on the slaveway to masterport interface after arbitration of write response channels from multiple slaveways, digit 1 identifies pipelining in the write response channel buffer and digit 0 identifies pipelining on the masterport to master interface.

0x0040C07C BIMC_M_DSP_MPORT_CONFIGURATION_INFO_5D

Type: R
Clock: Config Clock
Reset State: 0x00000000

Configuration information registers provide information on the capability and configuration of each masterport instance. See register field descriptions for more details.

BIMC_M_DSP_MPORT_CONFIGURATION_INFO_5D

Bits	Name	Description
31:0	RCH_PIPELINE_STAGES	Indicates pipeline stages enabled on the read channel. Each digit identifies a specific pipeline stage in the masterport, where '0' indicates no pipeline stage, '1' indicates 1-deep pipeline stage where forward path is registered, return path is fed through, '2' indicates 2-deep pipeline stage where both forward and return path are pipelined. Digit 2 identifies pipelining on the slaveway to masterport interface after arbitration of read channels from multiple slaveways, digit 1 identifies pipelining in the read channel buffer and digit 0 identifies pipelining on the masterport to master interface.

0x0040C100 BIMC_M_DSP_MPORT_INTERRUPT_STATUS**Type:** R**Clock:** Config Clock**Reset State:** 0x00000000

Interrupt status registers provide indication of which interrupts from this masterport require servicing.

BIMC_M_DSP_MPORT_INTERRUPT_STATUS

Bits	Name	Description
1:0	RFU	Reserved for future use

0x0040C108 BIMC_M_DSP_MPORT_INTERRUPT_CLEAR**Type:** W**Clock:** Config Clock**Reset State:** 0x00000000

Interrupt clear registers provide the mechanism to turn off interrupt sources once they have been acknowledged or serviced. A write of '1' to the bits corresponding to the interrupts indicated in interrupt status is required to turn off interrupts.

BIMC_M_DSP_MPORT_INTERRUPT_CLEAR

Bits	Name	Description
1:0	RFU	Reserved for future use

0x0040C10C BIMC_M_DSP_MPORT_INTERRUPT_ENABLE**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000

Interrupt enable registers provide control over which interrupts are allowed to be propagated to an interrupt controller for servicing.

BIMC_M_DSP_MPORT_INTERRUPT_ENABLE

Bits	Name	Description
1:0	RFU	Reserved for future use

0x0040C200 BIMC_M_DSP_MPORT_CLOCK_CTRL**Type:** RW**Clock:** Config, Core & Master Clocks**Reset State:** 0x00000003**BIMC_M_DSP_MPORT_CLOCK_CTRL**

Bits	Name	Description
1	MASTER_CLOCK_GATING_EN	When set (1), enables activity based clock gating of all registers clocked by master clock. Activity based clock gating uses architectural clock gates to reduce dynamic power consumption by gating clocks during periods of in-activity. When cleared (0), disables all activity based clock gating by forcing all architectural clock gates to not gate clocks even during periods of in-activity.
0	CORE_CLOCK_GATING_EN	When set (1), enables activity based clock gating of all registers clocked by core clock. Activity based clock gating uses architectural clock gates to reduce dynamic power consumption by gating clocks during periods of in-activity. When cleared (0), disables all activity based clock gating by forcing all architectural clock gates to not gate clocks even during periods of in-activity.

0x0040C208 BIMC_M_DSP_MPORT_CDC_CTRL**Type:** RW**Clock:** Config, Core & Master Clocks**Reset State:** 0x00000000

Clock domain crossing (CDC) control register provides configuration for clock domain crossing logic.

BIMC_M_DSP_MPORT_CDC_CTRL

Bits	Name	Description
1	MASTER_CLOCK_DEMETA_SEL	When cleared (0), selects DEMETA synchronizer that requires atleast 3 clock edges (two rising and one falling). When set (1), selects DEMETA synchronizer that requires atleast 2 clock edges (one falling and one rising) to synchronize to master clock in all buffers.
0	CORE_CLOCK_DEMETA_SEL	When cleared (0), selects DEMETA synchronizer that requires atleast 3 clock edges (two rising and one falling). When set (1), selects DEMETA synchronizer that requires atleast 2 clock edges (one falling and one rising) to synchronize to core clock in all buffers.

0x0040C210 BIMC_M_DSP_MPORT_MODE**Type:** RW**Clock:** Config, Core & Master Clocks**Reset State:** 0x00000012**BIMC_M_DSP_MPORT_MODE**

Bits	Name	Description
31:28	WR_GATHER_BEATS	Configures the number of write data beats to gather/buffer in write channel buffer before allowing the corresponding write command to be routed to arbiters. When configured to 0, write commands are held until the first beat of write data has been buffered. When configured to 1, write commands are held until two beats of write data have been buffered and so on. When configured with values larger than the number of beats in a write command, the command is held until the last beat has been buffered. When configured with values larger than the depth of write channel buffers, the command is held until all entries in the write channel buffer have been occupied.
19:16	RD_GATHER_BEATS	Configures the number of read data beats to gather/buffer in read channel buffer before allowing the read data to proceed to the master. When configured to 0, read data is allowed to proceed to the master as soon as the first beat is buffered in the read channel buffer. When configured to 1, read data is allowed to proceed to the master as soon as two beats of read data have been buffered and so on. When configured with values larger than the number of beats in the read burst, read data is held until the last beat has been buffered. When configured with values larger than the depth of read channel buffers, read data is held until all entries in the read channel buffer have been occupied.
15	RCSSH	Redirect Cacheable System SHareable When set (1), enables redirection of all cacheable transactions that are system shareable to the redirect slaweway. Note: Only masters connecting using AMBA 4 ACE protocol are able to indicate system shareability for transactions.
14	RCOSH	Redirect Cacheable Outer SHareable When set (1), enables redirection of all cacheable transactions that are outer shareable to the redirect slaweway.
13	RCISH	Redirect Cacheable Inner SHareable When set (1), enables redirection of all cacheable transactions that are inner shareable to the redirect slaweway.
12	RCNSH	Redirect Cacheable Non-SHareable When set (1), enables redirection of all cacheable transactions that are non-shareable to the redirect slaweway.
11	RNCSSH	Redirect Non-Cacheable System SHareable When set (1), enables redirection of all non-cacheable transactions that are system shareable to the redirect slaweway. Note: Only masters connecting using AMBA 4 ACE protocol are able to indicate system shareability for transactions.

BIMC_M_DSP_MPORT_MODE (cont.)

Bits	Name	Description
10	RNCOSH	Redirect Non-Cacheable Outer SHareable When set (1), enables redirection of all non-cacheable transactions that are outer shareable to the redirect slaway.
9	RNCISH	Redirect Non-Cacheable Inner SHareable When set (1), enables redirection of all non-cacheable transactions that are inner shareable to the redirect slaway.
8	RNCNSH	Redirect Non-Cacheable Non-SHareable When set (1), enables redirection of all non-cacheable transactions that are non-shareable to the redirect slaway.
4	NARROW_WRITES	When set (1), enables converting of 1-beat bus-width sized writes down to 32-bit ASIZE based on write strobes. It is recommended to be enabled for directly connected CPU masters. When cleared (0), disables conversion and all writes are forwarded downstream as generated by the master.
2	FIODV	Force In-Order Device When set (1), forces commands to device memory to be executed one at a time. In other words, commands to device memory are held until all previous commands have completed (responses have been received at the masterport) and all commands after the device command are held until the device completes (response has been received at the masterport.)
1	FIOSO	Force In-Order Strongly-Ordered When set (1), forces commands to strongly ordered memory to be executed one at a time. In other words, commands to strongly ordered memory are held until all previous commands have completed (responses have been received at the masterport) and all commands after the strongly ordered command are held until the strongly ordered completes (response has been received at the masterport.)
0	ORDERING_MODEL	Configures how commands to memories of type Strongly ordered and Device (SoDv) are handled in the masterport. When configured as STRICT, all commands to SoDv memory are held until all previous commands have completed (responses have been received at the masterport) and all commands after the SoDv command are held until the SoDv completes (i.e. response has been received at the masterport.) When configured as RELAXED, all commands to SoDv memory are treated as accessing Normal memory. 0x0: RELAXED 0x1: STRICT

0x0040C220 BIMC_M_DSP_MPORT_WR_WAY_CROSSING**Type:** RW**Clock:** Config & Master Clocks**Reset State:** 0x00000000

Configures the level of strictness or relaxedness of way crossing checks for writes.

BIMC_M_DSP_MPORT_WR_WAY_CROSSING

Bits	Name	Description
7:0	REG	1-bit per slaveway. When 2 or more bits are set (1), allows the corresponding slaveways to be treated as one slaveway when write commands are considered as way crossing, ie. write commands to these 2 (or more) slaveways are not treated as way crossing when one write command targets one slaveway and the next write command targets the second slaveway. Not treating write commands as way crossing allows the write commands to go to the slaveway ahead of the write data for potentially better overall system performance. When all bits are set (1), disables all way crossing checks. When all bits are cleared (0), way crossing is strict with each way crossing write command held until write data for all previous writes have been routed to their corresponding slaveways.

0x0040C230 BIMC_M_DSP_MPORT_PRIORITYLVL_OVERRIDE

Type: RW

Clock: Config & Master Clocks

Reset State: 0x00000000

Override registers provide the capability to override one or more bus signals to specific values.

BIMC_M_DSP_MPORT_PRIORITYLVL_OVERRIDE

Bits	Name	Description
9:8	PRIORITYLVL	Configures the values to override APRIORITYLVL from bus master with when OVERRIDE_PRIORITYLVL is set (1).
0	OVERRIDE_PRIORITYLVL	When set (1), overrides APRIORITYLVL from the bus master with values configured in PRIORITYLVL register field. When cleared (0), allows APRIORITYLVL from the bus master to be propagated downstream.

0x0040C240 BIMC_M_DSP_MPORT_READ_COMMAND_OVERRIDE

Type: RW

Clock: Config & Master Clocks

Reset State: 0x00000000

Override registers provide the capability to override one or more bus signals to specific values.

BIMC_M_DSP_MPORT_READ_COMMAND_OVERRIDE

Bits	Name	Description
25:24	AREQPRIORITY	Configures the values to override AREQPRIORITY from bus master with for read commands when OVERRIDE_AREQPRIORITY is set (1).
18:16	AMEMTYPE	Configures the values to override AMEMTYPE from bus master with for read commands when OVERRIDE_AMEMTYPE is set (1).
12	ATRANSIENT	Configures the values to override ATRANSIENT from bus master with for read commands when OVERRIDE_ATRANSIENT is set (1).
11	ASHARED	Configures the values to override ASHARED from bus master with for read commands when OVERRIDE_ASHARED is set (1).
10	AREDIRECT	Configures the values to override AREDIRECT from bus master with for read commands when OVERRIDE_AREDIRECT is set (1).
9	AOOO	Configures the values to override AOOO from bus master with for read commands when OVERRIDE_AOOO is set (1).
8	AINNERSHARED	Configures the values to override AINNERSHARED from bus master with for read commands when OVERRIDE_AINNERSHARED is set (1).
6	OVERRIDE_AREQPRIORITY	When set (1), overrides AREQPRIORITY from the bus master for read commands with values configured in AREQPRIORITY register field. When cleared (0), allows AREQPRIORITY from the bus master to be propagated downstream.
5	OVERRIDE_AMEMTYPE	When set (1), overrides AMEMTYPE from the bus master for read commands with values configured in AMEMTYPE register field. When cleared (0), allows AMEMTYPE from the bus master to be propagated downstream.
4	OVERRIDE_ATRANSIENT	When set (1), overrides ATRANSIENT from the bus master for read commands with values configured in ATRANSIENT register field. When cleared (0), allows ATRANSIENT from the bus master to be propagated downstream.
3	OVERRIDE_ASHARED	When set (1), overrides ASHARED from the bus master for read commands with values configured in ASHARED register field. When cleared (0), allows ASHARED from the bus master to be propagated downstream.
2	OVERRIDE_AREDIRECT	When set (1), overrides AREDIRECT from the bus master for read commands with values configured in AREDIRECT register field. When cleared (0), allows AREDIRECT from the bus master to be propagated downstream.
1	OVERRIDE_AOOO	When set (1), overrides AOOO from the bus master for read commands with values configured in AOOO register field. When cleared (0), allows AOOO from the bus master to be propagated downstream.

BIMC_M_DSP_MPORT_READ_COMMAND_OVERRIDE (cont.)

Bits	Name	Description
0	VERRIDE_AINNERSHARE D	When set (1), overrides AINNERSHARED from the bus master for read commands with values configured in AINNERSHARED register field. When cleared (0), allows AINNERSHARED from the bus master to be propagated downstream.

0x0040C250 BIMC_M_DSP_MPORT_WRITE_COMMAND_OVERRIDE**Type:** RW**Clock:** Config & Master Clocks**Reset State:** 0x00000000

Override registers provide the capability to override one or more bus signals to specific values.

BIMC_M_DSP_MPORT_WRITE_COMMAND_OVERRIDE

Bits	Name	Description
25:24	AREQPRIORITY	Configures the values to override AREQPRIORITY from bus master with for write commands when OVERRIDE_AREQPRIORITY is set (1).
18:16	AMEMTYPE	Configures the values to override AMEMTYPE from bus master with for write commands when OVERRIDE_AMEMTYPE is set (1).
12	ATRANSIENT	Configures the values to override ATRANSIENT from bus master with for write commands when OVERRIDE_ATRANSIENT is set (1).
11	ASHARED	Configures the values to override ASHARED from bus master with for write commands when OVERRIDE_ASHARED is set (1).
10	AREDIRECT	Configures the values to override AREDIRECT from bus master with for write commands when OVERRIDE_AREDIRECT is set (1).
9	AOOO	Configures the values to override AOOO from bus master with for write commands when OVERRIDE_AOOO is set (1).
8	AINNERSHARED	Configures the values to override AINNERSHARED from bus master with for write commands when OVERRIDE_AINNERSHARED is set (1).
6	VERRIDE_AREQPRIORIT Y	When set (1), overrides AREQPRIORITY from the bus master for write commands with values configured in AREQPRIORITY register field. When cleared (0), allows AREQPRIORITY from the bus master to be propagated downstream.
5	VERRIDE_AMEMTYPE	When set (1), overrides AMEMTYPE from the bus master for write commands with values configured in AMEMTYPE register field. When cleared (0), allows AMEMTYPE from the bus master to be propagated downstream.

BIMC_M_DSP_MPORT_WRITE_COMMAND_OVERRIDE (cont.)

Bits	Name	Description
4	OVERRIDE_ATRANSIENT	When set (1), overrides ATRANSIENT from the bus master for write commands with values configured in ATRANSIENT register field. When cleared (0), allows ATRANSIENT from the bus master to be propagated downstream.
3	OVERRIDE_ASHARED	When set (1), overrides ASHARED from the bus master for write commands with values configured in ASHARED register field. When cleared (0), allows ASHARED from the bus master to be propagated downstream.
2	OVERRIDE_AREDIRECT	When set (1), overrides AREDIRECT from the bus master for write commands with values configured in AREDIRECT register field. When cleared (0), allows AREDIRECT from the bus master to be propagated downstream.
1	OVERRIDE_AOOO	When set (1), overrides AOOO from the bus master for write commands with values configured in AOOO register field. When cleared (0), allows AOOO from the bus master to be propagated downstream.
0	OVERRIDE_AINNERSHARE D	When set (1), overrides AINNERSHARED from the bus master for write commands with values configured in AINNERSHARED register field. When cleared (0), allows AINNERSHARED from the bus master to be propagated downstream.

0x0040C300 BIMC_M_DSP_MPORT_BKE_ENABLE**Type:** RW**Clock:** Config & Master Clocks**Reset State:** 0x00000000**BIMC_M_DSP_MPORT_BKE_ENABLE**

Bits	Name	Description
0	ENABLE	When set (1), enables the book keeping element (BKE) to determine AREQPRIORITY and APRIORITYLVL of commands from the bus master by comparing the amount of bus traffic from the bus master to a target, and determining a health level for the bus master. The health level determines the relative priority of commands from the bus master to allow the bus master to catch-up if it is not getting serviced, or slow down to allow other bus masters to get their share of system resources. When cleared (0), disables the book keeping element and AREQPRIORITY and APRIORITYLVL from the bus master are propagated downstream.

0x0040C304 BIMC_M_DSP_MPORT_BKE_GRANT_PERIOD**Type:** RW**Clock:** Config & Reference Clocks**Reset State:** 0x00000000**BIMC_M_DSP_MPORT_BKE_GRANT_PERIOD**

Bits	Name	Description
9:0	GRANT_PERIOD	Defines the period over which the book keeping element accumulates grants to the bus master for determining it's health level. Grant period is defined in number of reference timer ticks. Refer configuration of reference timer tick for exact time unit.

0x0040C308 BIMC_M_DSP_MPORT_BKE_GRANT_COUNT**Type:** RW**Clock:** Config & Master Clocks**Reset State:** 0x00000000**BIMC_M_DSP_MPORT_BKE_GRANT_COUNT**

Bits	Name	Description
15:0	GRANT_COUNT	Defines the target grant count for the bus master. The target grant count is the number of beats of read and write data the bus master is expected to write and read in the grant period and is used as a reference to determine if the bus master is behind or ahead of where it ought to be at any given time.

0x0040C320 BIMC_M_DSP_MPORT_BKE_THRESHOLD_HIGH**Type:** RW**Clock:** Config & Master Clocks**Reset State:** 0x00000000**BIMC_M_DSP_MPORT_BKE_THRESHOLD_HIGH**

Bits	Name	Description
15:0	THRESHOLD	Defines a high threshold to determine how much behind or ahead a bus master is with reference to where it ought to be at any given time.

0x0040C324 BIMC_M_DSP_MPORT_BKE_THRESHOLD_MEDIUM**Type:** RW**Clock:** Config & Master Clocks**Reset State:** 0x00000000**BIMC_M_DSP_MPORT_BKE_THRESHOLD_MEDIUM**

Bits	Name	Description
15:0	THRESHOLD	Defines a medium threshold to determine how much behind or ahead a bus master is with reference to where it ought to be at any given time.

0x0040C328 BIMC_M_DSP_MPORT_BKE_THRESHOLD_LOW**Type:** RW**Clock:** Config & Master Clocks**Reset State:** 0x00000000**BIMC_M_DSP_MPORT_BKE_THRESHOLD_LOW**

Bits	Name	Description
15:0	THRESHOLD	Defines a low threshold to determine how much behind or ahead a bus master is with reference to where it ought to be at any given time.

0x0040C340 BIMC_M_DSP_MPORT_BKE_HEALTH_0**Type:** RW**Clock:** Config & Master Clocks**Reset State:** 0x00000000

BKE health registers configure the request priority and realtime priority level to use for commands from the bus master when it has been determined to be of a certain health.

BIMC_M_DSP_MPORT_BKE_HEALTH_0

Bits	Name	Description
31	LIMIT_COMMANDS	When set (1), defines that no new commands may be accepted from the bus master as long as it is determined to be at health 0.
9:8	AREQPRIORITY	Defines the request priority of commands from the bus master when it has been determined to be at health 0.
1:0	PRIORITYLVL	Defines the realtime priority level to use for commands from the bus master when it has been determined to be at health 0.

0x0040C344 BIMC_M_DSP_MPORT_BKE_HEALTH_1**Type:** RW**Clock:** Config & Master Clocks**Reset State:** 0x00000000

BKE health registers configure the request priority and realtime priority level to use for commands from the bus master when it has been determined to be of a certain health.

BIMC_M_DSP_MPORT_BKE_HEALTH_1

Bits	Name	Description
31	LIMIT_COMMANDS	When set (1), defines that no new commands may be accepted from the bus master as long as it is determined to be at health 1.
9:8	AREQPRIORITY	Defines the request priority of commands from the bus master when it has been determined to be at health 1.
1:0	PRIORITYLVL	Defines the realtime priority level to use for commands from the bus master when it has been determined to be at health 1.

0x0040C348 BIMC_M_DSP_MPORT_BKE_HEALTH_2**Type:** RW**Clock:** Config & Master Clocks**Reset State:** 0x00000000

BKE health registers configure the request priority and realtime priority level to use for commands from the bus master when it has been determined to be of a certain health.

BIMC_M_DSP_MPORT_BKE_HEALTH_2

Bits	Name	Description
31	LIMIT_COMMANDS	When set (1), defines that no new commands may be accepted from the bus master as long as it is determined to be at health 2.
9:8	AREQPRIORITY	Defines the request priority of commands from the bus master when it has been determined to be at health 2.
1:0	PRIORITYLVL	Defines the realtime priority level to use for commands from the bus master when it has been determined to be at health 2.

0x0040C34C BIMC_M_DSP_MPORT_BKE_HEALTH_3**Type:** RW**Clock:** Config & Master Clocks**Reset State:** 0x00000000

BKE health registers configure the request priority and realtime priority level to use for commands from the bus master when it has been determined to be of a certain health. Health 3 is the default health for the bus master.

BIMC_M_DSP_MPORT_BKE_HEALTH_3

Bits	Name	Description
9:8	AREQPRIORITY	Defines the request priority of commands from the bus master when it has been determined to be at health 3.
1:0	PRIORITYLVL	Defines the realtime priority level to use for commands from the bus master when it has been determined to be at health 3.

0x0040C420 BIMC_M_DSP_MPORT_STATUS_2A

Type: R

Clock: Config, Master & Pipe0 Clocks

Reset State: 0x00000000

Status registers indicate the state of several elements in the masterport, including the number of transactions outstanding/pending waiting for a response, emptiness, fullness of several buffers, etc. The state information is available only as a debug indication and could be delayed by a few clock cycles from when the state is reached to being reflected in the status registers. One or more of the state registers are read live and could capture intermediate values as they change.

BIMC_M_DSP_MPORT_STATUS_2A

Bits	Name	Description
23:16	WR_PENDING_PRE_BUF	Indicates the number of write transactions that are outstanding/pending waiting for a response from the slave. This transaction count is from before address channel buffers, which interfaces to frontend of the masterport.
7:0	RD_PENDING_PRE_BUF	Indicates the number of read transactions that are outstanding/pending waiting for a response from the slave. This transaction count is from before address channel buffers, which interfaces to frontend of the masterport.

0x0040C424 BIMC_M_DSP_MPORT_STATUS_2B

Type: R

Clock: Config, Master & Pipe0 Clocks

Reset State: 0x00000000

Status registers indicate the state of several elements in the masterport, including the number of transactions outstanding/pending waiting for a response, emptiness, fullness of several buffers, etc. The state information is available only as a debug indication and could be delayed by a few clock

cycles from when the state is reached to being reflected in the status registers. One or more of the state registers are read live and could capture intermediate values as they change.

BIMC_M_DSP_MPORT_STATUS_2B

Bits	Name	Description
23:16	WR_PENDING_POST_BUF	Indicates the number of write transactions that are outstanding/pending waiting for a response from the slave. This transaction count is from after address channel buffers, which interfaces to core of the interconnect, i.e., arbiter and slaveway.
7:0	RD_PENDING_POST_BUF	Indicates the number of read transactions that are outstanding/pending waiting for a response from the slave. This transaction count is from after address channel buffers, which interfaces to core of the interconnect, i.e., arbiter and slaveway.

0x0040C430 BIMC_M_DSP_MPORT_STATUS_3A

Type: R

Clock: Config, Master & Pipe0 Clocks

Reset State: 0x55111111

Status registers indicate the state of several elements in the masterport, including the number of transactions outstanding/pending waiting for a response, emptiness, fullness of several buffers, etc. The state information is available only as a debug indication and could be delayed by a few clock cycles from when the state is reached to being reflected in the status registers. One or more of the state registers are read live and could capture intermediate values as they change.

BIMC_M_DSP_MPORT_STATUS_3A

Bits	Name	Description
31	RCB_DATA_RD_FULL	Indicates that the data part of read channel buffer is full when seen from it's read side, which interfaces to frontend of the masterport.
30	RCB_DATA_RD_EMPTY	Indicates that the data part of read channel buffer is empty when seen from it's read side, which interfaces to frontend of the masterport.
29	RCB_CTRL_RD_FULL	Indicates that the control part of read channel buffer is full when seen from it's read side, which interfaces to frontend of the masterport.
28	RCB_CTRL_RD_EMPTY	Indicates that the control part of read channel buffer is empty when seen from it's write side, which interfaces to frontend of the masterport.
27	RCB_DATA_WR_FULL	Indicates that the data part of read channel buffer is full when seen from it's write side, which interfaces to core of the interconnect, i.e., arbiter and slaveway.
26	RCB_DATA_WR_EMPTY	Indicates that the data part of read channel buffer is empty when seen from it's write side, which interfaces to core of the interconnect, i.e., arbiter and slaveway.

BIMC_M_DSP_MPORT_STATUS_3A (cont.)

Bits	Name	Description
25	RCB_CTRL_WR_FULL	Indicates that the control part of read channel buffer is full when seen from it's write side, which interfaces to core of the interconnect, i.e., arbiter and slaveway.
24	RCB_CTRL_WR_EMPTY	Indicates that the control part of read channel buffer is empty when seen from it's write side, which interfaces to core of the interconnect, i.e., arbiter and slaveway.
21	BCB_RD_FULL	Indicates that the write response channel buffer is full when seen from it's read side, which interfaces to frontend of the masterport.
20	BCB_RD_EMPTY	Indicates that the write response channel buffer is empty when seen from it's read side, which interfaces to frontend of the masterport.
17	BCB_WR_FULL	Indicates that the write response channel buffer is full when seen from it's write side, which interfaces to core of the interconnect, i.e., arbiter and slaveway.
16	BCB_WR_EMPTY	Indicates that the write response channel buffer is empty when seen from it's write side, which interfaces to core of the interconnect, i.e., arbiter and slaveway.
13	WCB_RD_FULL	Indicates that the write channel buffer is full when seen from it's read side, which interfaces to core of the interconnect, i.e., arbiter and slaveway.
12	WCB_RD_EMPTY	Indicates that the write channel buffer is empty when seen from it's read side, which interfaces to core of the interconnect, i.e., arbiter and slaveway.
9	WCB_WR_FULL	Indicates that the write channel buffer is full when seen from it's write side, which interfaces to frontend of the masterport.
8	WCB_WR_EMPTY	Indicates that the write channel buffer is empty when seen from it's write side, which interfaces to frontend of the masterport.
5	ACB_RD_FULL	Indicates that the address channel buffer is full when seen from it's read side, which interfaces to core of the interconnect, i.e., arbiter and slaveway.
4	ACB_RD_EMPTY	Indicates that the address channel buffer is empty when seen from it's read side, which interfaces to core of the interconnect, i.e., arbiter and slaveway.
1	ACB_WR_FULL	Indicates that the address channel buffer is full when seen from it's write side, which interfaces to frontend of the masterport.
0	ACB_WR_EMPTY	Indicates that the address channel buffer is empty when seen from it's write side, which interfaces to frontend of the masterport.

0x0040C434 BIMC_M_DSP_MPORT_STATUS_3B**Type:** R**Clock:** Config, Master & Pipe0 Clocks**Reset State:** 0x00111111

Buffer status register indicate the state of all buffers in the masterport, whether they are empty or full on either side of the buffer. Some buffers carry information from master to core and others carry information from core to master. Both write side and read side status are indicated separately, since they would be not be similar when master clock and core clock are not synchronous to each other.

BIMC_M_DSP_MPORT_STATUS_3B

Bits	Name	Description
21	RGB_RD_FULL	Indicates that the read gathering buffer is full when seen from it's read side, which interfaces to frontend of the masterport.
20	RGB_RD_EMPTY	Indicates that the read gathering buffer is empty when seen from it's read side, which interfaces to frontend of the masterport.
17	RGB_WR_FULL	Indicates that the read gathering buffer is full when seen from it's write side, which interfaces to core of the interconnect, i.e., arbiter and slaweway.
16	RGB_WR_EMPTY	Indicates that the read gathering buffer is empty when seen from it's write side, which interfaces to core of the interconnect, i.e., arbiter and slaweway.
13	W2AB_RD_FULL	Indicates that the write channel to address channel user signals buffer is full when seen from it's read side, which interfaces to core of the interconnect, i.e., arbiter and slaweway.
12	W2AB_RD_EMPTY	Indicates that the write channel to address channel user signals buffer is empty when seen from it's read side, which interfaces to core of the interconnect, i.e., arbiter and slaweway.
9	W2AB_WR_FULL	Indicates that the write channel to address channel user signals buffer is full when seen from it's write side, which interfaces to frontend of the masterport.
8	W2AB_WR_EMPTY	Indicates that the write channel to address channel user signals buffer is empty when seen from it's write side, which interfaces to frontend of the masterport.
5	A2WB_RD_FULL	Indicates that the address channel to write channel user signals buffer is full when seen from it's read side, which interfaces to core of the interconnect, i.e., arbiter and slaweway.
4	A2WB_RD_EMPTY	Indicates that the address channel to write channel user signalsbuffer is empty when seen from it's read side, which interfaces to core of the interconnect, i.e., arbiter and slaweway.
1	A2WB_WR_FULL	Indicates that the address channel to write channel user signals buffer is full when seen from it's write side, which interfaces to frontend of the masterport.

BIMC_M_DSP_MPORT_STATUS_3B (cont.)

Bits	Name	Description
0	A2WB_WR_EMPTY	Indicates that the address channel to write channel user signals buffer is empty when seen from it's write side, which interfaces to frontend of the masterport.

2.7 BIMC_M_DSP_PROF

BRIC Traffic Profiling

2.7.1 BIMC_M_DSP_PROF_BRIC_PROFILING

BRIC Traffic Profiling

0x0040D000 BIMC_M_DSP_PROF_COMPONENT_INFO**Type:** R**Clock:** Config Clock**Reset State:** 0x00000202

Component information registers provide discoverability information on the type of component and instance of each component. These registers are always the very first register in each component's configuration address space and allow for a software routine to walk the configuration address space in steps of 4KB to identify the location of each configuration register space and to which component each 4KB configures.

BIMC_M_DSP_PROF_COMPONENT_INFO

Bits	Name	Description
23:16	INSTANCE	Indicates the instance number of this traffic profiling.
15:8	SUB_TYPE	Identifies this configuration space as belonging to traffic profiling 0x2: PROFILING
7:0	TYPE	Identifies this configuration space as belonging to a component that interfaces to a bus master. 0x2: MASTERPORT

0x0040D020 BIMC_M_DSP_PROF_CONFIGURATION_INFO_0**Type:** R**Clock:** Config Clock**Reset State:** 0x00000000

Configuration information registers provide information on the capability and configuration of each traffic profiling instance. See register field descriptions for more details.

BIMC_M_DSP_PROF_CONFIGURATION_INFO_0

Bits	Name	Description
31:24	TENURE_WIDTH	Indicates the width of tenures (latencies) measured in traffic profiling.
23:16	TRACKING_TABLE_DEPTH	Indicates the number of entries in the table that tracks outstanding commands for determining how many commands have larger turnaround latencies than a programmed threshold.
7:0	SCALING_FACTOR	Indicates how many extra scaling is performed on events generated by traffic profiling if such events have to be synchronized from the tenure clock to the event clock. Any value that is non-power-of-2 is converted to power-of-2 by hardware. When the value read from this register field is used in software, similar conversion of non-power-of-2 values to next higher power-of-2 value is necessary.

0x0040D030 BIMC_M_DSP_PROF_CONFIGURATION_INFO_1

Type: R

Clock: Config Clock

Reset State: 0x00000000

Configuration information registers provide information on the capability and configuration of each traffic profiling instance. See register field descriptions for more details.

BIMC_M_DSP_PROF_CONFIGURATION_INFO_1

Bits	Name	Description
23:16	PIPE2_ACH_DEPTH	Indicates the depth of address channel buffer in pipe2 of masterport in which this traffic profiling instance is paired with. Applies only to traffic profiling instances in XR type masterports, which contains three distinct paths to slaves, in pipe0, pipe1, pipe2. In non-XR type masterports, value contained in field does not refer to any actual address channel buffer.
15:8	PIPE1_ACH_DEPTH	Indicates the depth of address channel buffer in pipe1 of masterport in which this traffic profiling instance is paired with. Applies only to traffic profiling instances in XR type masterports, which contains three distinct paths to slaves, in pipe0, pipe1, pipe2. In non-XR type masterports, value contained in field does not refer to any actual address channel buffer.
7:0	PIPE0_ACH_DEPTH	Indicates the depth of address channel buffer in pipe0 of masterport in which this traffic profiling instance is paired with. Applies only to traffic profiling instances in XR type masterports, which contains three distinct paths to slaves, in pipe0, pipe1, pipe2. In non-XR type masterports, pipe0 refers to the common address channel buffer used for all slave accesses.

0x0040D200 BIMC_M_DSP_PROF_CLOCK_CTRL**Type:** RW**Clock:** Config, Event & Tenure Clocks**Reset State:** 0x00000003**BIMC_M_DSP_PROF_CLOCK_CTRL**

Bits	Name	Description
1	TENURE_CLOCK_GATING_EN	When set (1), enables activity based clock gating of all registers clocked by tenure clock. Activity based clock gating uses architectural clock gates to reduce dynamic power consumption by gating clocks during periods of in-activity. When cleared (0), disables all activity based clock gating by forcing all architectural clock gates to not gate clocks even during periods of in-activity.
0	EVENT_CLOCK_GATING_EN	When set (1), enables activity based clock gating of all registers clocked by event clock. Activity based clock gating uses architectural clock gates to reduce dynamic power consumption by gating clocks during periods of in-activity. When cleared (0), disables all activity based clock gating by forcing all architectural clock gates to not gate clocks even during periods of in-activity.

0x0040D210 BIMC_M_DSP_PROF_MODE**Type:** RW**Clock:** Config, Event & Tenure Clocks**Reset State:** 0x00000000

BIMC_M_DSP_PROF_MODE

Bits	Name	Description
25:24	WR_TENURE_MODE	Configures the start and end events for monitoring write latencies/tenures. When configured to WR_CMD_TO_BRESP, starting event is first cycle of write command from bus master as seen by traffic profiling and ending event is first cycle of write response to bus master as seen by traffic profiling. When configured to WR_CMD_TO_BRESP_HANDSHAKE, starting event is first cycle of write command from bus master as seen by traffic profiling and ending event is transfer handshake of write response to bus master as seen by traffic profiling. When configured to WR_CMD_HANDSHAKE, starting event is first cycle of write command from bus master as seen by traffic profiling and ending event is transfer handshake of the same write command with bus master as seen by traffic profiling. When configured to WR_RESP_HANDSHAKE, starting event is first cycle of write response to bus master as seen by traffic profiling and ending event is transfer handshake of write response to bus master as seen by traffic profiling. 0x0: WR_CMD_TO_BRESP 0x1: WR_CMD_TO_BRESP_HANDSHAKE 0x2: WR_CMD_HANDSHAKE 0x3: WR_RESP_HANDSHAKE
17:16	RD_TENURE_MODE	Configures the start and end events for monitoring read latencies/tenures. When configured to RD_CMD_TO_FIRST_RDATA, starting event is first cycle of read command from bus master as seen by traffic profiling and ending event is first cycle of first beat of read data to bus master as seen by traffic profiling. When configured to RD_CMD_TO_LAST_RDATA, starting event is first cycle of read command from bus master as seen by traffic profiling and ending event is transfer handshake of last beat of read data to bus master as seen by traffic profiling. When configured to RD_CMD_HANDSHAKE, starting event is first cycle of read command from bus master as seen by traffic profiling and ending event is transfer handshake of the same read command with bus master as seen by traffic profiling. When configured to RD_DATA_HANDSHAKE, starting event is first cycle of first beat of read data to bus master as seen by traffic profiling and ending event is transfer handshake of last beat of read data to bus master as seen by traffic profiling. 0x0: RD_CMD_TO_FIRST_RDATA 0x1: RD_CMD_TO_LAST_RDATA 0x2: RD_CMD_HANDSHAKE 0x3: RD_DATA_HANDSHAKE
0	PROFILING_EN	When set (1), enables traffic profiling to monitor bus transactions from bus master and generate events for counting in performance monitors. When cleared (0), disables traffic profiling.

0x0040D214 BIMC_M_DSP_PROF_RD_TENURE_THRESHOLD**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_DSP_PROF_RD_TENURE_THRESHOLD**

Bits	Name	Description
15:0	THRESHOLD	Configures the threshold over which read latencies/tenures need to be before an event is generated to performance monitors for counting max latencies.

0x0040D218 BIMC_M_DSP_PROF_WR_TENURE_THRESHOLD**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_DSP_PROF_WR_TENURE_THRESHOLD**

Bits	Name	Description
15:0	THRESHOLD	Configures the threshold over which write latencies/tenures need to be before an event is generated to performance monitors for counting max latencies.

0x0040D220 BIMC_M_DSP_PROF_SWAY_SELECT**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_DSP_PROF_SWAY_SELECT**

Bits	Name	Description
7:0	SWAY	Selects the slaveways to which commands are routed to profile. 1-bit per slaveway.

0x0040D230 BIMC_M_DSP_PROF_MID_MASK**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000

BIMC_M_DSP_PROF_MID_MASK

Bits	Name	Description
15:0	MID_MASK	When a bit position is set (1), enables matching of that bit position in MID of command with same bit position in MID_MATCH. Commands with MIDs that do not match are not considered for profiling.

0x0040D234 BIMC_M_DSP_PROF_MID_MATCH**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_DSP_PROF_MID_MATCH**

Bits	Name	Description
15:0	MID_MATCH	Configures the value to match for in AMID in commands. Used in conjunction with MID_MASK, which controls which bits of MID are compared.

0x0040D238 BIMC_M_DSP_PROF_MID_INV_MATCH**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_DSP_PROF_MID_INV_MATCH**

Bits	Name	Description
0	MID_INV_MATCH	When set (1), configures MID matching to result in an inverse match, i.e., only commands whose MIDs that do not match MID_MATCH are considered for profiling.

0x0040D240 BIMC_M_DSP_PROF_ADDR_MASK_LOWER**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000

BIMC_M_DSP_PROF_ADDR_MASK_LOWER

Bits	Name	Description
31:10	ADDR_MASK_LOWER	Configures ADDR_MASK[31:0]. When a bit position is set (1), enables matching of that bit position of address of command with same bit position in ADDR_MATCH[31:0].

0x0040D244 BIMC_M_DSP_PROF_ADDR_MASK_UPPER**Type:** R**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_DSP_PROF_ADDR_MASK_UPPER**

Bits	Name	Description
3:0	UNUSED	

0x0040D248 BIMC_M_DSP_PROF_ADDR_MATCH_LOWER**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_DSP_PROF_ADDR_MATCH_LOWER**

Bits	Name	Description
31:10	ADDR_MATCH_LOWER	Configures the value to match for in address of commands. Used in conjunction with ADDR_MASK[31:0], which controls which bits of address are compared.

0x0040D24C BIMC_M_DSP_PROF_ADDR_MATCH_UPPER**Type:** R**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_DSP_PROF_ADDR_MATCH_UPPER**

Bits	Name	Description
3:0	UNUSED	

0x0040D250 BIMC_M_DSP_PROF_ADDR_INV_MATCH**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_DSP_PROF_ADDR_INV_MATCH**

Bits	Name	Description
0	ADDR_INV_MATCH	When set (1), configures address matching to result in an inverse match, i.e., only commands whose addresses do not match ADDR_MATCH are considered for profiling.

0x0040D260 BIMC_M_DSP_PROF_ACH_MASK_0**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_DSP_PROF_ACH_MASK_0**

Bits	Name	Description
24	AINNERSHARED_MASK	When set (1), enables matching of AINNERSHARED in commands with AINNERSHARED_MATCH.
16	AFULL_MASK	When set (1), enables matching of AFULL in commands with AFULL_MATCH.
8	AEXCLUSIVE_MASK	When set (1), enables matching of AEXCLUSIVE in commands with AEXCLUSIVE_MATCH.
0	ABURST_MASK	When set (1), enables matching of ABURST in commands with ABURST_MATCH.

0x0040D264 BIMC_M_DSP_PROF_ACH_MATCH_0**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_DSP_PROF_ACH_MATCH_0**

Bits	Name	Description
24	AINNERSHARED_MATCH	Configures the value to match for in AINNERSHARED in commands. Used in conjunction with AINNERSHARED_MASK, which controls whether AINNERSHARED in commands is compared

BIMC_M_DSP_PROF_ACH_MATCH_0 (cont.)

Bits	Name	Description
16	AFULL_MATCH	Configures the value to match for in AFULL in commands. Used in conjunction with AFULL_MASK, which controls whether AFULL in commands is compared.
8	AEXCLUSIVE_MATCH	Configures the value to match for in AEXCLUSIVE in commands. Used in conjunction with AEXCLUSIVE_MASK, which controls whether AFULL in commands is compared.
0	ABURST_MATCH	Configures the value to match for in ABURST in commands. Used in conjunction with ABURST_MASK, which controls whether ABURST in commands is compared.

0x0040D270 BIMC_M_DSP_PROF_ACH_MASK_1**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_DSP_PROF_ACH_MASK_1**

Bits	Name	Description
24	AMSSSELFAUTH_MASK	When set (1), enables matching of AMSSSELFAUTH in commands with AMSSSELFAUTH_MATCH.
18:16	AMEMTYPE_MASK	When a bit position is set (1), enables matching of corresponding bit position of AMEMTYPE with AMEMTYPE_MATCH.
11:8	ALEN_MASK	When a bit position is set (1), enables matching of corresponding bit position of ALEN with ALEN_MATCH.
0	AINST_MASK	When set (1), enables matching of AINST in commands with AINST_MASK.

0x0040D274 BIMC_M_DSP_PROF_ACH_MATCH_1**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_DSP_PROF_ACH_MATCH_1**

Bits	Name	Description
24	AMSSSELFAUTH_MATCH	Configures the value to match for in AMSSSELFAUTH in commands. Used in conjunction with AMSSSELFAUTH_MASK, which controls whether AMSSSELFAUTH in commands is compared.

BIMC_M_DSP_PROF_ACH_MATCH_1 (cont.)

Bits	Name	Description
18:16	AMEMTYPE_MATCH	Configures the value to match for in AMEMTYPE in commands. Used in conjunction with AMEMTYPE_MASK, which controls which bits of AMEMTYPE in commands are compared.
11:8	ALEN_MATCH	Configures the value to match for in ALEN in commands. Used in conjunction with ALEN_MASK, which controls which bits of ALEN in commands are compared.
0	AINST_MATCH	Configures the value to match for in AINST in commands. Used in conjunction with AINST_MASK, which controls whether AINST in commands is compared.

0x0040D278 BIMC_M_DSP_PROF_ACH_INV_MATCH_1**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_DSP_PROF_ACH_INV_MATCH_1**

Bits	Name	Description
16	AMEMTYPE_INV_MATCH	When set (1), configures AMEMTYPE matching to result in an inverse match, i.e., only commands whose AMEMTYPE does not match AMEMTYPE_MATCH are considered for profiling.
8	ALEN_INV_MATCH	When set (1), configures ALEN matching to result in an inverse match, i.e., only commands when ALEN does not match ALEN_MATCH are considered for profiling.

0x0040D280 BIMC_M_DSP_PROF_ACH_MASK_2**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_DSP_PROF_ACH_MASK_2**

Bits	Name	Description
24	APROTNS_MASK	When set (1), enables matching of APROTNS in commands with APROTNS_MATCH.
16	APRIV_MASK	When set (1), enables matching of APRIV in commands with APRIV_MATCH.
8	AOOO_MASK	When set (1), enables matching of AOOO in commands with AOOO_MATCH.
0	ANOALLOCATE_MASK	When set (1), enables matching of ANOALLOCATE in commands with ANOALLOCATE_MATCH.

0x0040D284 BIMC_M_DSP_PROF_ACH_MATCH_2**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_DSP_PROF_ACH_MATCH_2**

Bits	Name	Description
24	APROTNS_MATCH	Configures the value to match for in APROTNS in commands. Used in conjunction with APROTNS_MASK, which controls whether APROTNS in commands is compared
16	APRIV_MATCH	Configures the value to match for in APRIV in commands. Used in conjunction with APRIV_MASK, which controls whether APRIV in commands is compared
8	AOOO_MATCH	Configures the value to match for in AOOO in commands. Used in conjunction with AOOO_MASK, which controls whether AOOO in commands is compared
0	ANOALLOCATE_MATCH	Configures the value to match for in ANOALLOCATE in commands. Used in conjunction with ANOALLOCATE_MASK, which controls whether ANOALLOCATE in commands is compared

0x0040D290 BIMC_M_DSP_PROF_ACH_MASK_3**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_DSP_PROF_ACH_MASK_3**

Bits	Name	Description
24	ASHARED_MASK	When set (1), enables matching of ASHARED in commands with ASHARED_MATCH.
17:16	AREQPRIORITY_MASK	When a bit position is set (1), enables matching of corresponding bit position of AREQPRIORITY with AREQPRIORITY_MATCH.
8	AREDIRECT_MASK	When set (1), enables matching of AREDIRECT in commands with AREDIRECT_MATCH.
0	ARDBEATNDXEN_MASK	When set (1), enables matching of ARDBEATNDXEN in commands with ARDBEATNDXEN_MATCH.

0x0040D294 BIMC_M_DSP_PROF_ACH_MATCH_3**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_DSP_PROF_ACH_MATCH_3**

Bits	Name	Description
24	ASHARED_MATCH	Configures the value to match for in ASHARED in commands. Used in conjunction with ASHARED_MASK, which controls whether ASHARED in commands is compared
17:16	AREQPRIORITY_MATCH	Configures the value to match for in AREQPRIORITY in commands. Used in conjunction with AREQPRIORITY_MASK, which controls which bits of AREQPRIORITY in commands are compared.
8	AREDIRECT_MATCH	Configures the value to match for in AREDIRECT in commands. Used in conjunction with AREDIRECT_MASK, which controls whether AREDIRECT in commands is compared
0	ARDBEATNDXEN_MATCH	Configures the value to match for in ARDBEATNDXEN in commands. Used in conjunction with ARDBEATNDXEN_MASK, which controls whether ARDBEATNDXEN in commands is compared

0x0040D298 BIMC_M_DSP_PROF_ACH_INV_MATCH_3**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_DSP_PROF_ACH_INV_MATCH_3**

Bits	Name	Description
16	AREQPRIORITY_INV_MATCH	When set (1), configures AREQPRIORITY matching to result in an inverse match, i.e., only commands whose AREQPRIORITY does not match AREQPRIORITY_MATCH are considered for profiling

0x0040D2A0 BIMC_M_DSP_PROF_ACH_MASK_4**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000

BIMC_M_DSP_PROF_ACH_MASK_4

Bits	Name	Description
24	AWRITE_MASK	When set (1), enables matching of AWRITE in commands with AWRITE_MATCH.
20:16	AVMID_MASK	When a bit position is set (1), enables matching of corresponding bit position in AVMID with AVMID_MATCH
8	ATRANSIENT_MASK	When set (1), enables matching of ATRANSIENT in commands with ATRANSIENT_MATCH.
2:0	ASIZE_MASK	When a bit position is set (1), enables matching of corresponding bit position in ASIZE with ASIZE_MATCH.

0x0040D2A4 BIMC_M_DSP_PROF_ACH_MATCH_4**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_DSP_PROF_ACH_MATCH_4**

Bits	Name	Description
24	AWRITE_MATCH	Configures the value to match for in AWRITE in commands. Used in conjunction with AWRITE_MASK, which controls whether AWRITE in commands is compared.
20:16	AVMID_MATCH	Configures the value to match for in AVMID in commands. Used in conjunction with AVMID_MASK, which controls which bits of AVMID in commands are compared.
8	ATRANSIENT_MATCH	Configures the value to match for in ATRANSIENT in commands. Used in conjunction with ATRANSIENT_MASK, which controls whether ATRANSIENT in commands is compared.
2:0	ASIZE_MATCH	Configures the value to match for in ASIZE in commands. Used in conjunction with ASIZE_MASK, which controls which bits of ASIZE in commands are compared.

0x0040D2A8 BIMC_M_DSP_PROF_ACH_INV_MATCH_4**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000

BIMC_M_DSP_PROF_ACH_INV_MATCH_4

Bits	Name	Description
16	AVMID_INV_MATCH	When set (1), configures AVMID matching to result in an inverse match, i.e., only commands whose AVMID does not match AVMID_MATCH are considered for profiling.
0	ASIZE_INV_MATCH	When set (1), configures ASIZE matching to result in an inverse match, i.e., only commands whose AVMID does not match ASIZE_MATCH are considered for profiling.

0x0040D300 BIMC_M_DSP_PROF_BCH_MASK**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_DSP_PROF_BCH_MASK**

Bits	Name	Description
1:0	BRESP_MASK	When a bit position is set (1), enables matching of corresponding bit position in BRESP with BRESP_MATCH.

0x0040D304 BIMC_M_DSP_PROF_BCH_MATCH**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_DSP_PROF_BCH_MATCH**

Bits	Name	Description
1:0	BRESP_MATCH	Configures the value to match for in BRESP in write responses. Used in conjunction with BRESP_MASK, which controls which bits of BRESP in writes responses are compared.

0x0040D308 BIMC_M_DSP_PROF_BCH_INV_MATCH**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000

BIMC_M_DSP_PROF_BCH_INV_MATCH

Bits	Name	Description
0	BRESP_INV_MATCH	When set (1), configures BRESP matching to result in an inverse match, i.e., only write responses whose BRESP does not match BRESP_MATCH are considered for profiling.

0x0040D340 BIMC_M_DSP_PROF_RCH_MASK**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_DSP_PROF_RCH_MASK**

Bits	Name	Description
8	RLAST_MASK	When set (1), enables matching of RLAST in commands with corresponding setting in RCH_MATCH register.
1:0	RRESP_MASK	When a bit position is set (1), enables matching of corresponding bit position in RRESP with RRESP_MATCH.

0x0040D344 BIMC_M_DSP_PROF_RCH_MATCH**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_DSP_PROF_RCH_MATCH**

Bits	Name	Description
8	RLAST_MATCH	Configures the value to match for in RLAST in read responses. Used in conjunction with RLAST_MASK, which controls whether RLAST in read responses is compared.
1:0	RRESP_MATCH	Configures the value to match for in RRESP in read responses. Used in conjunction with RRESP_MASK, which controls which bits of RRESP in read responses are compared.

0x0040D348 BIMC_M_DSP_PROF_RCH_INV_MATCH**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000

BIMC_M_DSP_PROF_RCH_INV_MATCH

Bits	Name	Description
0	RRESP_INV_MATCH	When set (1), configures RRESP matching to result in an inverse match, i.e., only read responses whose RRESP does not match RRESP_MATCH are considered for profiling.

**0x0040D400+ BIMC_M_DSP_PROF_RD_STATUS_n, n=[0..0]
0x4*n**
Type: R**Clock:** Config & Tenure Clocks**Reset State:** 0x00000000**BIMC_M_DSP_PROF_RD_STATUS_n**

Bits	Name	Description
31	RD_OUTSTANDING_OVERFLOW	Indicates that the number of outstanding reads crossed the maximum supported (255) during when traffic profiling was last enabled. It is recommended that this register be read only after traffic profiling has been disabled.
30	RD_OUTSTANDING_UNDERFLOW	Indicates that the number of outstanding reads went below 0 during when traffic profiling was last enabled. It indicates that more command completions were seen compared to commands starts. The most likely scenario is an overflow of number of outstanding reads occurred leading to some command starts not being tracked. It is recommended that this register be read only after traffic profiling has been disabled.
7:0	RD_OUTSTANDING_MAX	Indicates the maximum number of outstanding reads that were seen when traffic profiling was last enabled. Only reads that match the MID filter and slaweway select filter are accounted for. It is recommended that this register be read only after traffic profiling has been disabled.

**0x0040D410+ BIMC_M_DSP_PROF_WR_STATUS_n, n=[0..0]
0x4*n**
Type: R**Clock:** Config & Tenure Clocks**Reset State:** 0x00000000**BIMC_M_DSP_PROF_WR_STATUS_n**

Bits	Name	Description
31	WR_OUTSTANDING_OVERFLOW	Indicates that the number of outstanding writes crossed the maximum supported (255) during when traffic profiling was last enabled. It is recommended that this register be read only after traffic profiling has been disabled.

BIMC_M_DSP_PROF_WR_STATUS_n (cont.)

Bits	Name	Description
30	WR_OUTSTANDING_UNDE RFLOW	Indicates that the number of outstanding writes went below 0 during when traffic profiling was last enabled. It indicates that more command completions were seen compared to commands starts. The most likely scenario is an overflow of number of outstanding writes occurred leading to some command starts not being tracked. It is recommended that this register be read only after traffic profiling has been disabled.
7:0	WR_OUTSTANDING_MAX	Indicates the maximum number of outstanding writes that were seen when traffic profiling was last enabled. Only writes that match the MID filter and slaweway select filter are accounted for. It is recommended that this register be read only after traffic profiling has been disabled.

2.8 BIMC_M_DSP_PROF

BRIC Traffic Profiling

2.8.1 BIMC_M_DSP_PROF_BRIC_PROFILING

BRIC Traffic Profiling

0x0040D000 BIMC_M_DSP_PROF_COMPONENT_INFO**Type:** R**Clock:** Config Clock**Reset State:** 0x00000202

Component information registers provide discoverability information on the type of component and instance of each component. These registers are always the very first register in each component's configuration address space and allow for a software routine to walk the configuration address space in steps of 4KB to identify the location of each configuration register space and to which component each 4KB configures.

BIMC_M_DSP_PROF_COMPONENT_INFO

Bits	Name	Description
23:16	INSTANCE	Indicates the instance number of this traffic profiling.
15:8	SUB_TYPE	Identifies this configuration space as belonging to traffic profiling 0x2: PROFILING
7:0	TYPE	Identifies this configuration space as belonging to a component that interfaces to a bus master. 0x2: MASTERPORT

0x0040D020 BIMC_M_DSP_PROF_CONFIGURATION_INFO_0**Type:** R**Clock:** Config Clock**Reset State:** 0x00000000

Configuration information registers provide information on the capability and configuration of each traffic profiling instance. See register field descriptions for more details.

BIMC_M_DSP_PROF_CONFIGURATION_INFO_0

Bits	Name	Description
31:24	TENURE_WIDTH	Indicates the width of tenures (latencies) measured in traffic profiling.
23:16	TRACKING_TABLE_DEPTH	Indicates the number of entries in the table that tracks outstanding commands for determining how many commands have larger turnaround latencies that a programmed threshold.
7:0	SCALING_FACTOR	Indicates how many extra scaling is performed on events generated by traffic profiling if such events have to be synchronized from the tenure clock to the event clock. Any value that is non-power-of-2 is converted to power-of-2 by hardware. When the value read from this register field is used in software, similar conversion of non-power-of-2 values to next higher power-of-2 value is necessary.

0x0040D030 BIMC_M_DSP_PROF_CONFIGURATION_INFO_1**Type:** R**Clock:** Config Clock**Reset State:** 0x00000000

Configuration information registers provide information on the capability and configuration of each traffic profiling instance. See register field descriptions for more details.

BIMC_M_DSP_PROF_CONFIGURATION_INFO_1

Bits	Name	Description
23:16	PIPE2_ACH_DEPTH	Indicates the depth of address channel buffer in pipe2 of masterport in which this traffic profiling instance is paired with. Applies only to traffic profiling instances in XR type masterports, which contains three distinct paths to slaves, in pipe0, pipe1, pipe2. In non-XR type masterports, value contained in field does not refer to any actual address channel buffer.

BIMC_M_DSP_PROF_CONFIGURATION_INFO_1 (cont.)

Bits	Name	Description
15:8	PIPE1_ACH_DEPTH	Indicates the depth of address channel buffer in pipe1 of masterport in which this traffic profiling instance is paired with. Applies only to traffic profiling instances in XR type masterports, which contains three distinct paths to slaves, in pipe0, pipe1, pipe2. In non-XR type masterports, value contained in field does not refer to any actual address channel buffer.
7:0	PIPE0_ACH_DEPTH	Indicates the depth of address channel buffer in pipe0 of masterport in which this traffic profiling instance is paired with. Applies only to traffic profiling instances in XR type masterports, which contains three distinct paths to slaves, in pipe0, pipe1, pipe2. In non-XR type masterports, pipe0 refers to the common address channel buffer used for all slave accesses.

0x0040D200 BIMC_M_DSP_PROF_CLOCK_CTRL**Type:** RW**Clock:** Config, Event & Tenure Clocks**Reset State:** 0x00000003**BIMC_M_DSP_PROF_CLOCK_CTRL**

Bits	Name	Description
1	TENURE_CLOCK_GATING_EN	When set (1), enables activity based clock gating of all registers clocked by tenure clock. Activity based clock gating uses architectural clock gates to reduce dynamic power consumption by gating clocks during periods of in-activity. When cleared (0), disables all activity based clock gating by forcing all architectural clock gates to not gate clocks even during periods of in-activity.
0	EVENT_CLOCK_GATING_EN	When set (1), enables activity based clock gating of all registers clocked by event clock. Activity based clock gating uses architectural clock gates to reduce dynamic power consumption by gating clocks during periods of in-activity. When cleared (0), disables all activity based clock gating by forcing all architectural clock gates to not gate clocks even during periods of in-activity.

0x0040D210 BIMC_M_DSP_PROF_MODE**Type:** RW**Clock:** Config, Event & Tenure Clocks**Reset State:** 0x00000000

BIMC_M_DSP_PROF_MODE

Bits	Name	Description
25:24	WR_TENURE_MODE	Configures the start and end events for monitoring write latencies/tenures. When configured to WR_CMD_TO_BRESP, starting event is first cycle of write command from bus master as seen by traffic profiling and ending event is first cycle of write response to bus master as seen by traffic profiling. When configured to WR_CMD_TO_BRESP_HANDSHAKE, starting event is first cycle of write command from bus master as seen by traffic profiling and ending event is transfer handshake of write response to bus master as seen by traffic profiling. When configured to WR_CMD_HANDSHAKE, starting event is first cycle of write command from bus master as seen by traffic profiling and ending event is transfer handshake of the same write command with bus master as seen by traffic profiling. When configured to WR_RESP_HANDSHAKE, starting event is first cycle of write response to bus master as seen by traffic profiling and ending event is transfer handshake of write response to bus master as seen by traffic profiling. 0x0: WR_CMD_TO_BRESP 0x1: WR_CMD_TO_BRESP_HANDSHAKE 0x2: WR_CMD_HANDSHAKE 0x3: WR_RESP_HANDSHAKE
17:16	RD_TENURE_MODE	Configures the start and end events for monitoring read latencies/tenures. When configured to RD_CMD_TO_FIRST_RDATA, starting event is first cycle of read command from bus master as seen by traffic profiling and ending event is first cycle of first beat of read data to bus master as seen by traffic profiling. When configured to RD_CMD_TO_LAST_RDATA, starting event is first cycle of read command from bus master as seen by traffic profiling and ending event is transfer handshake of last beat of read data to bus master as seen by traffic profiling. When configured to RD_CMD_HANDSHAKE, starting event is first cycle of read command from bus master as seen by traffic profiling and ending event is transfer handshake of the same read command with bus master as seen by traffic profiling. When configured to RD_DATA_HANDSHAKE, starting event is first cycle of first beat of read data to bus master as seen by traffic profiling and ending event is transfer handshake of last beat of read data to bus master as seen by traffic profiling. 0x0: RD_CMD_TO_FIRST_RDATA 0x1: RD_CMD_TO_LAST_RDATA 0x2: RD_CMD_HANDSHAKE 0x3: RD_DATA_HANDSHAKE
0	PROFILING_EN	When set (1), enables traffic profiling to monitor bus transactions from bus master and generate events for counting in performance monitors. When cleared (0), disables traffic profiling.

0x0040D214 BIMC_M_DSP_PROF_RD_TENURE_THRESHOLD**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_DSP_PROF_RD_TENURE_THRESHOLD**

Bits	Name	Description
15:0	THRESHOLD	Configures the threshold over which read latencies/tenures need to be before an event is generated to performance monitors for counting max latencies.

0x0040D218 BIMC_M_DSP_PROF_WR_TENURE_THRESHOLD**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_DSP_PROF_WR_TENURE_THRESHOLD**

Bits	Name	Description
15:0	THRESHOLD	Configures the threshold over which write latencies/tenures need to be before an event is generated to performance monitors for counting max latencies.

0x0040D220 BIMC_M_DSP_PROF_SWAY_SELECT**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_DSP_PROF_SWAY_SELECT**

Bits	Name	Description
7:0	SWAY	Selects the slaveways to which commands are routed to profile. 1-bit per slaveway.

0x0040D230 BIMC_M_DSP_PROF_MID_MASK**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000

BIMC_M_DSP_PROF_MID_MASK

Bits	Name	Description
15:0	MID_MASK	When a bit position is set (1), enables matching of that bit position in MID of command with same bit position in MID_MATCH. Commands with MIDs that do not match are not considered for profiling.

0x0040D234 BIMC_M_DSP_PROF_MID_MATCH**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_DSP_PROF_MID_MATCH**

Bits	Name	Description
15:0	MID_MATCH	Configures the value to match for in AMID in commands. Used in conjunction with MID_MASK, which controls which bits of MID are compared.

0x0040D238 BIMC_M_DSP_PROF_MID_INV_MATCH**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_DSP_PROF_MID_INV_MATCH**

Bits	Name	Description
0	MID_INV_MATCH	When set (1), configures MID matching to result in an inverse match, i.e., only commands whose MIDs that do not match MID_MATCH are considered for profiling.

0x0040D240 BIMC_M_DSP_PROF_ADDR_MASK_LOWER**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000

BIMC_M_DSP_PROF_ADDR_MASK_LOWER

Bits	Name	Description
31:10	ADDR_MASK_LOWER	Configures ADDR_MASK[31:0]. When a bit position is set (1), enables matching of that bit position of address of command with same bit position in ADDR_MATCH[31:0].

0x0040D244 BIMC_M_DSP_PROF_ADDR_MASK_UPPER**Type:** R**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_DSP_PROF_ADDR_MASK_UPPER**

Bits	Name	Description
3:0	UNUSED	

0x0040D248 BIMC_M_DSP_PROF_ADDR_MATCH_LOWER**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_DSP_PROF_ADDR_MATCH_LOWER**

Bits	Name	Description
31:10	ADDR_MATCH_LOWER	Configures the value to match for in address of commands. Used in conjunction with ADDR_MASK[31:0], which controls which bits of address are compared.

0x0040D24C BIMC_M_DSP_PROF_ADDR_MATCH_UPPER**Type:** R**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_DSP_PROF_ADDR_MATCH_UPPER**

Bits	Name	Description
3:0	UNUSED	

0x0040D250 BIMC_M_DSP_PROF_ADDR_INV_MATCH**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_DSP_PROF_ADDR_INV_MATCH**

Bits	Name	Description
0	ADDR_INV_MATCH	When set (1), configures address matching to result in an inverse match, i.e., only commands whose addresses do not match ADDR_MATCH are considered for profiling.

0x0040D260 BIMC_M_DSP_PROF_ACH_MASK_0**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_DSP_PROF_ACH_MASK_0**

Bits	Name	Description
24	AINNERSHARED_MASK	When set (1), enables matching of AINNERSHARED in commands with AINNERSHARED_MATCH.
16	AFULL_MASK	When set (1), enables matching of AFULL in commands with AFULL_MATCH.
8	AEXCLUSIVE_MASK	When set (1), enables matching of AEXCLUSIVE in commands with AEXCLUSIVE_MATCH.
0	ABURST_MASK	When set (1), enables matching of ABURST in commands with ABURST_MATCH.

0x0040D264 BIMC_M_DSP_PROF_ACH_MATCH_0**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_DSP_PROF_ACH_MATCH_0**

Bits	Name	Description
24	AINNERSHARED_MATCH	Configures the value to match for in AINNERSHARED in commands. Used in conjunction with AINNERSHARED_MASK, which controls whether AINNERSHARED in commands is compared

BIMC_M_DSP_PROF_ACH_MATCH_0 (cont.)

Bits	Name	Description
16	AFULL_MATCH	Configures the value to match for in AFULL in commands. Used in conjunction with AFULL_MASK, which controls whether AFULL in commands is compared.
8	AEXCLUSIVE_MATCH	Configures the value to match for in AEXCLUSIVE in commands. Used in conjunction with AEXCLUSIVE_MASK, which controls whether AFULL in commands is compared.
0	ABURST_MATCH	Configures the value to match for in ABURST in commands. Used in conjunction with ABURST_MASK, which controls whether ABURST in commands is compared.

0x0040D270 BIMC_M_DSP_PROF_ACH_MASK_1**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_DSP_PROF_ACH_MASK_1**

Bits	Name	Description
24	AMSSSELFAUTH_MASK	When set (1), enables matching of AMSSSELFAUTH in commands with AMSSSELFAUTH_MATCH.
18:16	AMEMTYPE_MASK	When a bit position is set (1), enables matching of corresponding bit position of AMEMTYPE with AMEMTYPE_MATCH.
11:8	ALEN_MASK	When a bit position is set (1), enables matching of corresponding bit position of ALEN with ALEN_MATCH.
0	AINST_MASK	When set (1), enables matching of AINST in commands with AINST_MASK.

0x0040D274 BIMC_M_DSP_PROF_ACH_MATCH_1**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_DSP_PROF_ACH_MATCH_1**

Bits	Name	Description
24	AMSSSELFAUTH_MATCH	Configures the value to match for in AMSSSELFAUTH in commands. Used in conjunction with AMSSSELFAUTH_MASK, which controls whether AMSSSELFAUTH in commands is compared.

BIMC_M_DSP_PROF_ACH_MATCH_1 (cont.)

Bits	Name	Description
18:16	AMEMTYPE_MATCH	Configures the value to match for in AMEMTYPE in commands. Used in conjunction with AMEMTYPE_MASK, which controls which bits of AMEMTYPE in commands are compared.
11:8	ALEN_MATCH	Configures the value to match for in ALEN in commands. Used in conjunction with ALEN_MASK, which controls which bits of ALEN in commands are compared.
0	AINST_MATCH	Configures the value to match for in AINST in commands. Used in conjunction with AINST_MASK, which controls whether AINST in commands is compared.

0x0040D278 BIMC_M_DSP_PROF_ACH_INV_MATCH_1**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_DSP_PROF_ACH_INV_MATCH_1**

Bits	Name	Description
16	AMEMTYPE_INV_MATCH	When set (1), configures AMEMTYPE matching to result in an inverse match, i.e., only commands whose AMEMTYPE does not match AMEMTYPE_MATCH are considered for profiling.
8	ALEN_INV_MATCH	When set (1), configures ALEN matching to result in an inverse match, i.e., only commands when ALEN does not match ALEN_MATCH are considered for profiling.

0x0040D280 BIMC_M_DSP_PROF_ACH_MASK_2**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_DSP_PROF_ACH_MASK_2**

Bits	Name	Description
24	APROTNS_MASK	When set (1), enables matching of APROTNS in commands with APROTNS_MATCH.
16	APRIV_MASK	When set (1), enables matching of APRIV in commands with APRIV_MATCH.
8	AOOO_MASK	When set (1), enables matching of AOOO in commands with AOOO_MATCH.
0	ANOALLOCATE_MASK	When set (1), enables matching of ANOALLOCATE in commands with ANOALLOCATE_MATCH.

0x0040D284 BIMC_M_DSP_PROF_ACH_MATCH_2**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_DSP_PROF_ACH_MATCH_2**

Bits	Name	Description
24	APROTNS_MATCH	Configures the value to match for in APROTNS in commands. Used in conjunction with APROTNS_MASK, which controls whether APROTNS in commands is compared
16	APRIV_MATCH	Configures the value to match for in APRIV in commands. Used in conjunction with APRIV_MASK, which controls whether APRIV in commands is compared
8	AOOO_MATCH	Configures the value to match for in AOOO in commands. Used in conjunction with AOOO_MASK, which controls whether AOOO in commands is compared
0	ANOALLOCATE_MATCH	Configures the value to match for in ANOALLOCATE in commands. Used in conjunction with ANOALLOCATE_MASK, which controls whether ANOALLOCATE in commands is compared

0x0040D290 BIMC_M_DSP_PROF_ACH_MASK_3**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_DSP_PROF_ACH_MASK_3**

Bits	Name	Description
24	ASHARED_MASK	When set (1), enables matching of ASHARED in commands with ASHARED_MATCH.
17:16	AREQPRIORITY_MASK	When a bit position is set (1), enables matching of corresponding bit position of AREQPRIORITY with AREQPRIORITY_MATCH.
8	AREDIRECT_MASK	When set (1), enables matching of AREDIRECT in commands with AREDIRECT_MATCH.
0	ARDBEATNDXEN_MASK	When set (1), enables matching of ARDBEATNDXEN in commands with ARDBEATNDXEN_MATCH.

0x0040D294 BIMC_M_DSP_PROF_ACH_MATCH_3**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_DSP_PROF_ACH_MATCH_3**

Bits	Name	Description
24	ASHARED_MATCH	Configures the value to match for in ASHARED in commands. Used in conjunction with ASHARED_MASK, which controls whether ASHARED in commands is compared
17:16	AREQPRIORITY_MATCH	Configures the value to match for in AREQPRIORITY in commands. Used in conjunction with AREQPRIORITY_MASK, which controls which bits of AREQPRIORITY in commands are compared.
8	AREDIRECT_MATCH	Configures the value to match for in AREDIRECT in commands. Used in conjunction with AREDIRECT_MASK, which controls whether AREDIRECT in commands is compared
0	ARDBEATNDXEN_MATCH	Configures the value to match for in ARDBEATNDXEN in commands. Used in conjunction with ARDBEATNDXEN_MASK, which controls whether ARDBEATNDXEN in commands is compared

0x0040D298 BIMC_M_DSP_PROF_ACH_INV_MATCH_3**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_DSP_PROF_ACH_INV_MATCH_3**

Bits	Name	Description
16	AREQPRIORITY_INV_MATCH	When set (1), configures AREQPRIORITY matching to result in an inverse match, i.e., only commands whose AREQPRIORITY does not match AREQPRIORITY_MATCH are considered for profiling

0x0040D2A0 BIMC_M_DSP_PROF_ACH_MASK_4**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000

BIMC_M_DSP_PROF_ACH_MASK_4

Bits	Name	Description
24	AWRITE_MASK	When set (1), enables matching of AWRITE in commands with AWRITE_MATCH.
20:16	AVMID_MASK	When a bit position is set (1), enables matching of corresponding bit position in AVMID with AVMID_MATCH
8	ATRANSIENT_MASK	When set (1), enables matching of ATRANSIENT in commands with ATRANSIENT_MATCH.
2:0	ASIZE_MASK	When a bit position is set (1), enables matching of corresponding bit position in ASIZE with ASIZE_MATCH.

0x0040D2A4 BIMC_M_DSP_PROF_ACH_MATCH_4**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_DSP_PROF_ACH_MATCH_4**

Bits	Name	Description
24	AWRITE_MATCH	Configures the value to match for in AWRITE in commands. Used in conjunction with AWRITE_MASK, which controls whether AWRITE in commands is compared.
20:16	AVMID_MATCH	Configures the value to match for in AVMID in commands. Used in conjunction with AVMID_MASK, which controls which bits of AVMID in commands are compared.
8	ATRANSIENT_MATCH	Configures the value to match for in ATRANSIENT in commands. Used in conjunction with ATRANSIENT_MASK, which controls whether ATRANSIENT in commands is compared.
2:0	ASIZE_MATCH	Configures the value to match for in ASIZE in commands. Used in conjunction with ASIZE_MASK, which controls which bits of ASIZE in commands are compared.

0x0040D2A8 BIMC_M_DSP_PROF_ACH_INV_MATCH_4**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000

BIMC_M_DSP_PROF_ACH_INV_MATCH_4

Bits	Name	Description
16	AVMID_INV_MATCH	When set (1), configures AVMID matching to result in an inverse match, i.e., only commands whose AVMID does not match AVMID_MATCH are considered for profiling.
0	ASIZE_INV_MATCH	When set (1), configures ASIZE matching to result in an inverse match, i.e., only commands whose AVMID does not match ASIZE_MATCH are considered for profiling.

0x0040D300 BIMC_M_DSP_PROF_BCH_MASK**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_DSP_PROF_BCH_MASK**

Bits	Name	Description
1:0	BRESP_MASK	When a bit position is set (1), enables matching of corresponding bit position in BRESP with BRESP_MATCH.

0x0040D304 BIMC_M_DSP_PROF_BCH_MATCH**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_DSP_PROF_BCH_MATCH**

Bits	Name	Description
1:0	BRESP_MATCH	Configures the value to match for in BRESP in write responses. Used in conjunction with BRESP_MASK, which controls which bits of BRESP in writes responses are compared.

0x0040D308 BIMC_M_DSP_PROF_BCH_INV_MATCH**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000

BIMC_M_DSP_PROF_BCH_INV_MATCH

Bits	Name	Description
0	BRESP_INV_MATCH	When set (1), configures BRESP matching to result in an inverse match, i.e., only write responses whose BRESP does not match BRESP_MATCH are considered for profiling.

0x0040D340 BIMC_M_DSP_PROF_RCH_MASK**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_DSP_PROF_RCH_MASK**

Bits	Name	Description
8	RLAST_MASK	When set (1), enables matching of RLAST in commands with corresponding setting in RCH_MATCH register.
1:0	RRESP_MASK	When a bit position is set (1), enables matching of corresponding bit position in RRESP with RRESP_MATCH.

0x0040D344 BIMC_M_DSP_PROF_RCH_MATCH**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_DSP_PROF_RCH_MATCH**

Bits	Name	Description
8	RLAST_MATCH	Configures the value to match for in RLAST in read responses. Used in conjunction with RLAST_MASK, which controls whether RLAST in read responses is compared.
1:0	RRESP_MATCH	Configures the value to match for in RRESP in read responses. Used in conjunction with RRESP_MASK, which controls which bits of RRESP in read responses are compared.

0x0040D348 BIMC_M_DSP_PROF_RCH_INV_MATCH**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000

BIMC_M_DSP_PROF_RCH_INV_MATCH

Bits	Name	Description
0	RRESP_INV_MATCH	When set (1), configures RRESP matching to result in an inverse match, i.e., only read responses whose RRESP does not match RRESP_MATCH are considered for profiling.

**0x0040D400+ BIMC_M_DSP_PROF_RD_STATUS_n, n=[0..0]
0x4*n**
Type: R**Clock:** Config & Tenure Clocks**Reset State:** 0x00000000**BIMC_M_DSP_PROF_RD_STATUS_n**

Bits	Name	Description
31	RD_OUTSTANDING_OVERFLOW	Indicates that the number of outstanding reads crossed the maximum supported (255) during when traffic profiling was last enabled. It is recommended that this register be read only after traffic profiling has been disabled.
30	RD_OUTSTANDING_UNDERFLOW	Indicates that the number of outstanding reads went below 0 during when traffic profiling was last enabled. It indicates that more command completions were seen compared to commands starts. The most likely scenario is an overflow of number of outstanding reads occurred leading to some command starts not being tracked. It is recommended that this register be read only after traffic profiling has been disabled.
7:0	RD_OUTSTANDING_MAX	Indicates the maximum number of outstanding reads that were seen when traffic profiling was last enabled. Only reads that match the MID filter and slaweway select filter are accounted for. It is recommended that this register be read only after traffic profiling has been disabled.

**0x0040D410+ BIMC_M_DSP_PROF_WR_STATUS_n, n=[0..0]
0x4*n**
Type: R**Clock:** Config & Tenure Clocks**Reset State:** 0x00000000**BIMC_M_DSP_PROF_WR_STATUS_n**

Bits	Name	Description
31	WR_OUTSTANDING_OVERFLOW	Indicates that the number of outstanding writes crossed the maximum supported (255) during when traffic profiling was last enabled. It is recommended that this register be read only after traffic profiling has been disabled.

BIMC_M_DSP_PROF_WR_STATUS_n (cont.)

Bits	Name	Description
30	WR_OUTSTANDING_UNDE RFLOW	Indicates that the number of outstanding writes went below 0 during when traffic profiling was last enabled. It indicates that more command completions were seen compared to commands starts. The most likely scenario is an overflow of number of outstanding writes occurred leading to some command starts not being tracked. It is recommended that this register be read only after traffic profiling has been disabled.
7:0	WR_OUTSTANDING_MAX	Indicates the maximum number of outstanding writes that were seen when traffic profiling was last enabled. Only writes that match the MID filter and slaweway select filter are accounted for. It is recommended that this register be read only after traffic profiling has been disabled.

2.9 BIMC_M_GPU_PROF

BRIC Traffic Profiling

2.9.1 BIMC_M_GPU_PROF_BRIC_PROFILING

BRIC Traffic Profiling

0x00411000 BIMC_M_GPU_PROF_COMPONENT_INFO

Type: R**Clock:** Config Clock**Reset State:** 0x00000202

Component information registers provide discoverability information on the type of component and instance of each component. These registers are always the very first register in each component's configuration address space and allow for a software routine to walk the configuration address space in steps of 4KB to identify the location of each configuration register space and to which component each 4KB configures.

BIMC_M_GPU_PROF_COMPONENT_INFO

Bits	Name	Description
23:16	INSTANCE	Indicates the instance number of this traffic profiling.
15:8	SUB_TYPE	Identifies this configuration space as belonging to traffic profiling 0x2: PROFILING
7:0	TYPE	Identifies this configuration space as belonging to a component that interfaces to a bus master. 0x2: MASTERPORT

0x00411020 BIMC_M_GPU_PROF_CONFIGURATION_INFO_0**Type:** R**Clock:** Config Clock**Reset State:** 0x00000000

Configuration information registers provide information on the capability and configuration of each traffic profiling instance. See register field descriptions for more details.

BIMC_M_GPU_PROF_CONFIGURATION_INFO_0

Bits	Name	Description
31:24	TENURE_WIDTH	Indicates the width of tenures (latencies) measured in traffic profiling.
23:16	TRACKING_TABLE_DEPTH	Indicates the number of entries in the table that tracks outstanding commands for determining how many commands have larger turnaround latencies that a programmed threshold.
7:0	SCALING_FACTOR	Indicates how many extra scaling is performed on events generated by traffic profiling if such events have to be synchronized from the tenure clock to the event clock. Any value that is non-power-of-2 is converted to power-of-2 by hardware. When the value read from this register field is used in software, similar conversion of non-power-of-2 values to next higher power-of-2 value is necessary.

0x00411030 BIMC_M_GPU_PROF_CONFIGURATION_INFO_1**Type:** R**Clock:** Config Clock**Reset State:** 0x00000000

Configuration information registers provide information on the capability and configuration of each traffic profiling instance. See register field descriptions for more details.

BIMC_M_GPU_PROF_CONFIGURATION_INFO_1

Bits	Name	Description
23:16	PIPE2_ACH_DEPTH	Indicates the depth of address channel buffer in pipe2 of masterport in which this traffic profiling instance is paired with. Applies only to traffic profiling instances in XR type masterports, which contains three distinct paths to slaves, in pipe0, pipe1, pipe2. In non-XR type masterports, value contained in field does not refer to any actual address channel buffer.

BIMC_M_GPU_PROF_CONFIGURATION_INFO_1 (cont.)

Bits	Name	Description
15:8	PIPE1_ACH_DEPTH	Indicates the depth of address channel buffer in pipe1 of masterport in which this traffic profiling instance is paired with. Applies only to traffic profiling instances in XR type masterports, which contains three distinct paths to slaves, in pipe0, pipe1, pipe2. In non-XR type masterports, value contained in field does not refer to any actual address channel buffer.
7:0	PIPE0_ACH_DEPTH	Indicates the depth of address channel buffer in pipe0 of masterport in which this traffic profiling instance is paired with. Applies only to traffic profiling instances in XR type masterports, which contains three distinct paths to slaves, in pipe0, pipe1, pipe2. In non-XR type masterports, pipe0 refers to the common address channel buffer used for all slave accesses.

0x00411200 BIMC_M_GPU_PROF_CLOCK_CTRL**Type:** RW**Clock:** Config, Event & Tenure Clocks**Reset State:** 0x00000003**BIMC_M_GPU_PROF_CLOCK_CTRL**

Bits	Name	Description
1	TENURE_CLOCK_GATING_EN	When set (1), enables activity based clock gating of all registers clocked by tenure clock. Activity based clock gating uses architectural clock gates to reduce dynamic power consumption by gating clocks during periods of in-activity. When cleared (0), disables all activity based clock gating by forcing all architectural clock gates to not gate clocks even during periods of in-activity.
0	EVENT_CLOCK_GATING_EN	When set (1), enables activity based clock gating of all registers clocked by event clock. Activity based clock gating uses architectural clock gates to reduce dynamic power consumption by gating clocks during periods of in-activity. When cleared (0), disables all activity based clock gating by forcing all architectural clock gates to not gate clocks even during periods of in-activity.

0x00411210 BIMC_M_GPU_PROF_MODE**Type:** RW**Clock:** Config, Event & Tenure Clocks**Reset State:** 0x00000000

BIMC_M_GPU_PROF_MODE

Bits	Name	Description
25:24	WR_TENURE_MODE	Configures the start and end events for monitoring write latencies/tenures. When configured to WR_CMD_TO_BRESP, starting event is first cycle of write command from bus master as seen by traffic profiling and ending event is first cycle of write response to bus master as seen by traffic profiling. When configured to WR_CMD_TO_BRESP_HANDSHAKE, starting event is first cycle of write command from bus master as seen by traffic profiling and ending event is transfer handshake of write response to bus master as seen by traffic profiling. When configured to WR_CMD_HANDSHAKE, starting event is first cycle of write command from bus master as seen by traffic profiling and ending event is transfer handshake of the same write command with bus master as seen by traffic profiling. When configured to WR_RESP_HANDSHAKE, starting event is first cycle of write response to bus master as seen by traffic profiling and ending event is transfer handshake of write response to bus master as seen by traffic profiling. 0x0: WR_CMD_TO_BRESP 0x1: WR_CMD_TO_BRESP_HANDSHAKE 0x2: WR_CMD_HANDSHAKE 0x3: WR_RESP_HANDSHAKE
17:16	RD_TENURE_MODE	Configures the start and end events for monitoring read latencies/tenures. When configured to RD_CMD_TO_FIRST_RDATA, starting event is first cycle of read command from bus master as seen by traffic profiling and ending event is first cycle of first beat of read data to bus master as seen by traffic profiling. When configured to RD_CMD_TO_LAST_RDATA, starting event is first cycle of read command from bus master as seen by traffic profiling and ending event is transfer handshake of last beat of read data to bus master as seen by traffic profiling. When configured to RD_CMD_HANDSHAKE, starting event is first cycle of read command from bus master as seen by traffic profiling and ending event is transfer handshake of the same read command with bus master as seen by traffic profiling. When configured to RD_DATA_HANDSHAKE, starting event is first cycle of first beat of read data to bus master as seen by traffic profiling and ending event is transfer handshake of last beat of read data to bus master as seen by traffic profiling. 0x0: RD_CMD_TO_FIRST_RDATA 0x1: RD_CMD_TO_LAST_RDATA 0x2: RD_CMD_HANDSHAKE 0x3: RD_DATA_HANDSHAKE
0	PROFILING_EN	When set (1), enables traffic profiling to monitor bus transactions from bus master and generate events for counting in performance monitors. When cleared (0), disables traffic profiling.

0x00411214 BIMC_M_GPU_PROF_RD_TENURE_THRESHOLD**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_GPU_PROF_RD_TENURE_THRESHOLD**

Bits	Name	Description
15:0	THRESHOLD	Configures the threshold over which read latencies/tenures need to be before an event is generated to performance monitors for counting max latencies.

0x00411218 BIMC_M_GPU_PROF_WR_TENURE_THRESHOLD**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_GPU_PROF_WR_TENURE_THRESHOLD**

Bits	Name	Description
15:0	THRESHOLD	Configures the threshold over which write latencies/tenures need to be before an event is generated to performance monitors for counting max latencies.

0x00411220 BIMC_M_GPU_PROF_SWAY_SELECT**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_GPU_PROF_SWAY_SELECT**

Bits	Name	Description
7:0	SWAY	Selects the slaveways to which commands are routed to profile. 1-bit per slaveway.

0x00411230 BIMC_M_GPU_PROF_MID_MASK**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000

BIMC_M_GPU_PROF_MID_MASK

Bits	Name	Description
15:0	MID_MASK	When a bit position is set (1), enables matching of that bit position in MID of command with same bit position in MID_MATCH. Commands with MIDs that do not match are not considered for profiling.

0x00411234 BIMC_M_GPU_PROF_MID_MATCH**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_GPU_PROF_MID_MATCH**

Bits	Name	Description
15:0	MID_MATCH	Configures the value to match for in AMID in commands. Used in conjunction with MID_MASK, which controls which bits of MID are compared.

0x00411238 BIMC_M_GPU_PROF_MID_INV_MATCH**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_GPU_PROF_MID_INV_MATCH**

Bits	Name	Description
0	MID_INV_MATCH	When set (1), configures MID matching to result in an inverse match, i.e., only commands whose MIDs that do not match MID_MATCH are considered for profiling.

0x00411240 BIMC_M_GPU_PROF_ADDR_MASK_LOWER**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000

BIMC_M_GPU_PROF_ADDR_MASK_LOWER

Bits	Name	Description
31:10	ADDR_MASK_LOWER	Configures ADDR_MASK[31:0]. When a bit position is set (1), enables matching of that bit position of address of command with same bit position in ADDR_MATCH[31:0].

0x00411244 BIMC_M_GPU_PROF_ADDR_MASK_UPPER**Type:** R**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_GPU_PROF_ADDR_MASK_UPPER**

Bits	Name	Description
3:0	UNUSED	

0x00411248 BIMC_M_GPU_PROF_ADDR_MATCH_LOWER**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_GPU_PROF_ADDR_MATCH_LOWER**

Bits	Name	Description
31:10	ADDR_MATCH_LOWER	Configures the value to match for in address of commands. Used in conjunction with ADDR_MASK[31:0], which controls which bits of address are compared.

0x0041124C BIMC_M_GPU_PROF_ADDR_MATCH_UPPER**Type:** R**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_GPU_PROF_ADDR_MATCH_UPPER**

Bits	Name	Description
3:0	UNUSED	

0x00411250 BIMC_M_GPU_PROF_ADDR_INV_MATCH**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_GPU_PROF_ADDR_INV_MATCH**

Bits	Name	Description
0	ADDR_INV_MATCH	When set (1), configures address matching to result in an inverse match, i.e., only commands whose addresses do not match ADDR_MATCH are considered for profiling.

0x00411260 BIMC_M_GPU_PROF_ACH_MASK_0**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_GPU_PROF_ACH_MASK_0**

Bits	Name	Description
24	AINNERSHARED_MASK	When set (1), enables matching of AINNERSHARED in commands with AINNERSHARED_MATCH.
16	AFULL_MASK	When set (1), enables matching of AFULL in commands with AFULL_MATCH.
8	AEXCLUSIVE_MASK	When set (1), enables matching of AEXCLUSIVE in commands with AEXCLUSIVE_MATCH.
0	ABURST_MASK	When set (1), enables matching of ABURST in commands with ABURST_MATCH.

0x00411264 BIMC_M_GPU_PROF_ACH_MATCH_0**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_GPU_PROF_ACH_MATCH_0**

Bits	Name	Description
24	AINNERSHARED_MATCH	Configures the value to match for in AINNERSHARED in commands. Used in conjunction with AINNERSHARED_MASK, which controls whether AINNERSHARED in commands is compared

BIMC_M_GPU_PROF_ACH_MATCH_0 (cont.)

Bits	Name	Description
16	AFULL_MATCH	Configures the value to match for in AFULL in commands. Used in conjunction with AFULL_MASK, which controls whether AFULL in commands is compared.
8	AEXCLUSIVE_MATCH	Configures the value to match for in AEXCLUSIVE in commands. Used in conjunction with AEXCLUSIVE_MASK, which controls whether AFULL in commands is compared.
0	ABURST_MATCH	Configures the value to match for in ABURST in commands. Used in conjunction with ABURST_MASK, which controls whether ABURST in commands is compared.

0x00411270 BIMC_M_GPU_PROF_ACH_MASK_1**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_GPU_PROF_ACH_MASK_1**

Bits	Name	Description
24	AMSSSELFAUTH_MASK	When set (1), enables matching of AMSSSELFAUTH in commands with AMSSSELFAUTH_MATCH.
18:16	AMEMTYPE_MASK	When a bit position is set (1), enables matching of corresponding bit position of AMEMTYPE with AMEMTYPE_MATCH.
11:8	ALEN_MASK	When a bit position is set (1), enables matching of corresponding bit position of ALEN with ALEN_MATCH.
0	AINST_MASK	When set (1), enables matching of AINST in commands with AINST_MASK.

0x00411274 BIMC_M_GPU_PROF_ACH_MATCH_1**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_GPU_PROF_ACH_MATCH_1**

Bits	Name	Description
24	AMSSSELFAUTH_MATCH	Configures the value to match for in AMSSSELFAUTH in commands. Used in conjunction with AMSSSELFAUTH_MASK, which controls whether AMSSSELFAUTH in commands is compared.

BIMC_M_GPU_PROF_ACH_MATCH_1 (cont.)

Bits	Name	Description
18:16	AMEMTYPE_MATCH	Configures the value to match for in AMEMTYPE in commands. Used in conjunction with AMEMTYPE_MASK, which controls which bits of AMEMTYPE in commands are compared.
11:8	ALEN_MATCH	Configures the value to match for in ALEN in commands. Used in conjunction with ALEN_MASK, which controls which bits of ALEN in commands are compared.
0	AINST_MATCH	Configures the value to match for in AINST in commands. Used in conjunction with AINST_MASK, which controls whether AINST in commands is compared.

0x00411278 BIMC_M_GPU_PROF_ACH_INV_MATCH_1**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_GPU_PROF_ACH_INV_MATCH_1**

Bits	Name	Description
16	AMEMTYPE_INV_MATCH	When set (1), configures AMEMTYPE matching to result in an inverse match, i.e., only commands whose AMEMTYPE does not match AMEMTYPE_MATCH are considered for profiling.
8	ALEN_INV_MATCH	When set (1), configures ALEN matching to result in an inverse match, i.e., only commands when ALEN does not match ALEN_MATCH are considered for profiling.

0x00411280 BIMC_M_GPU_PROF_ACH_MASK_2**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_GPU_PROF_ACH_MASK_2**

Bits	Name	Description
24	APROTNS_MASK	When set (1), enables matching of APROTNS in commands with APROTNS_MATCH.
16	APRIV_MASK	When set (1), enables matching of APRIV in commands with APRIV_MATCH.
8	AOOO_MASK	When set (1), enables matching of AOOO in commands with AOOO_MATCH.
0	ANOALLOCATE_MASK	When set (1), enables matching of ANOALLOCATE in commands with ANOALLOCATE_MATCH.

0x00411284 BIMC_M_GPU_PROF_ACH_MATCH_2**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_GPU_PROF_ACH_MATCH_2**

Bits	Name	Description
24	APROTNS_MATCH	Configures the value to match for in APROTNS in commands. Used in conjunction with APROTNS_MASK, which controls whether APROTNS in commands is compared
16	APRIV_MATCH	Configures the value to match for in APRIV in commands. Used in conjunction with APRIV_MASK, which controls whether APRIV in commands is compared
8	AOOO_MATCH	Configures the value to match for in AOOO in commands. Used in conjunction with AOOO_MASK, which controls whether AOOO in commands is compared
0	ANOALLOCATE_MATCH	Configures the value to match for in ANOALLOCATE in commands. Used in conjunction with ANOALLOCATE_MASK, which controls whether ANOALLOCATE in commands is compared

0x00411290 BIMC_M_GPU_PROF_ACH_MASK_3**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_GPU_PROF_ACH_MASK_3**

Bits	Name	Description
24	ASHARED_MASK	When set (1), enables matching of ASHARED in commands with ASHARED_MATCH.
17:16	AREQPRIORITY_MASK	When a bit position is set (1), enables matching of corresponding bit position of AREQPRIORITY with AREQPRIORITY_MATCH.
8	AREDIRECT_MASK	When set (1), enables matching of AREDIRECT in commands with AREDIRECT_MATCH.
0	ARDBEATNDXEN_MASK	When set (1), enables matching of ARDBEATNDXEN in commands with ARDBEATNDXEN_MATCH.

0x00411294 BIMC_M_GPU_PROF_ACH_MATCH_3**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_GPU_PROF_ACH_MATCH_3**

Bits	Name	Description
24	ASHARED_MATCH	Configures the value to match for in ASHARED in commands. Used in conjunction with ASHARED_MASK, which controls whether ASHARED in commands is compared
17:16	AREQPRIORITY_MATCH	Configures the value to match for in AREQPRIORITY in commands. Used in conjunction with AREQPRIORITY_MASK, which controls which bits of AREQPRIORITY in commands are compared.
8	AREDIRECT_MATCH	Configures the value to match for in AREDIRECT in commands. Used in conjunction with AREDIRECT_MASK, which controls whether AREDIRECT in commands is compared
0	ARDBEATNDXEN_MATCH	Configures the value to match for in ARDBEATNDXEN in commands. Used in conjunction with ARDBEATNDXEN_MASK, which controls whether ARDBEATNDXEN in commands is compared

0x00411298 BIMC_M_GPU_PROF_ACH_INV_MATCH_3**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_GPU_PROF_ACH_INV_MATCH_3**

Bits	Name	Description
16	AREQPRIORITY_INV_MATCH	When set (1), configures AREQPRIORITY matching to result in an inverse match, i.e., only commands whose AREQPRIORITY does not match AREQPRIORITY_MATCH are considered for profiling

0x004112A0 BIMC_M_GPU_PROF_ACH_MASK_4**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000

BIMC_M_GPU_PROF_ACH_MASK_4

Bits	Name	Description
24	AWRITE_MASK	When set (1), enables matching of AWRITE in commands with AWRITE_MATCH.
20:16	AVMID_MASK	When a bit position is set (1), enables matching of corresponding bit position in AVMID with AVMID_MATCH
8	ATRANSIENT_MASK	When set (1), enables matching of ATRANSIENT in commands with ATRANSIENT_MATCH.
2:0	ASIZE_MASK	When a bit position is set (1), enables matching of corresponding bit position in ASIZE with ASIZE_MATCH.

0x004112A4 BIMC_M_GPU_PROF_ACH_MATCH_4**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_GPU_PROF_ACH_MATCH_4**

Bits	Name	Description
24	AWRITE_MATCH	Configures the value to match for in AWRITE in commands. Used in conjunction with AWRITE_MASK, which controls whether AWRITE in commands is compared.
20:16	AVMID_MATCH	Configures the value to match for in AVMID in commands. Used in conjunction with AVMID_MASK, which controls which bits of AVMID in commands are compared.
8	ATRANSIENT_MATCH	Configures the value to match for in ATRANSIENT in commands. Used in conjunction with ATRANSIENT_MASK, which controls whether ATRANSIENT in commands is compared.
2:0	ASIZE_MATCH	Configures the value to match for in ASIZE in commands. Used in conjunction with ASIZE_MASK, which controls which bits of ASIZE in commands are compared.

0x004112A8 BIMC_M_GPU_PROF_ACH_INV_MATCH_4**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000

BIMC_M_GPU_PROF_ACH_INV_MATCH_4

Bits	Name	Description
16	AVMID_INV_MATCH	When set (1), configures AVMID matching to result in an inverse match, i.e., only commands whose AVMID does not match AVMID_MATCH are considered for profiling.
0	ASIZE_INV_MATCH	When set (1), configures ASIZE matching to result in an inverse match, i.e., only commands whose AVMID does not match ASIZE_MATCH are considered for profiling.

0x00411300 BIMC_M_GPU_PROF_BCH_MASK**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_GPU_PROF_BCH_MASK**

Bits	Name	Description
1:0	BRESP_MASK	When a bit position is set (1), enables matching of corresponding bit position in BRESP with BRESP_MATCH.

0x00411304 BIMC_M_GPU_PROF_BCH_MATCH**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_GPU_PROF_BCH_MATCH**

Bits	Name	Description
1:0	BRESP_MATCH	Configures the value to match for in BRESP in write responses. Used in conjunction with BRESP_MASK, which controls which bits of BRESP in writes responses are compared.

0x00411308 BIMC_M_GPU_PROF_BCH_INV_MATCH**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000

BIMC_M_GPU_PROF_BCH_INV_MATCH

Bits	Name	Description
0	BRESP_INV_MATCH	When set (1), configures BRESP matching to result in an inverse match, i.e., only write responses whose BRESP does not match BRESP_MATCH are considered for profiling.

0x00411340 BIMC_M_GPU_PROF_RCH_MASK**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_GPU_PROF_RCH_MASK**

Bits	Name	Description
8	RLAST_MASK	When set (1), enables matching of RLAST in commands with corresponding setting in RCH_MATCH register.
1:0	RRESP_MASK	When a bit position is set (1), enables matching of corresponding bit position in RRESP with RRESP_MATCH.

0x00411344 BIMC_M_GPU_PROF_RCH_MATCH**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_GPU_PROF_RCH_MATCH**

Bits	Name	Description
8	RLAST_MATCH	Configures the value to match for in RLAST in read responses. Used in conjunction with RLAST_MASK, which controls whether RLAST in read responses is compared.
1:0	RRESP_MATCH	Configures the value to match for in RRESP in read responses. Used in conjunction with RRESP_MASK, which controls which bits of RRESP in read responses are compared.

0x00411348 BIMC_M_GPU_PROF_RCH_INV_MATCH**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000

BIMC_M_GPU_PROF_RCH_INV_MATCH

Bits	Name	Description
0	RRESP_INV_MATCH	When set (1), configures RRESP matching to result in an inverse match, i.e., only read responses whose RRESP does not match RRESP_MATCH are considered for profiling.

0x00411400+ BIMC_M_GPU_PROF_RD_STATUS_n, n=[0..0]
0x4*n
Type: R**Clock:** Config & Tenure Clocks**Reset State:** 0x00000000**BIMC_M_GPU_PROF_RD_STATUS_n**

Bits	Name	Description
31	RD_OUTSTANDING_OVERFLOW	Indicates that the number of outstanding reads crossed the maximum supported (255) during when traffic profiling was last enabled. It is recommended that this register be read only after traffic profiling has been disabled.
30	RD_OUTSTANDING_UNDERFLOW	Indicates that the number of outstanding reads went below 0 during when traffic profiling was last enabled. It indicates that more command completions were seen compared to commands starts. The most likely scenario is an overflow of number of outstanding reads occurred leading to some command starts not being tracked. It is recommended that this register be read only after traffic profiling has been disabled.
7:0	RD_OUTSTANDING_MAX	Indicates the maximum number of outstanding reads that were seen when traffic profiling was last enabled. Only reads that match the MID filter and slaweway select filter are accounted for. It is recommended that this register be read only after traffic profiling has been disabled.

0x00411410+ BIMC_M_GPU_PROF_WR_STATUS_n, n=[0..0]
0x4*n
Type: R**Clock:** Config & Tenure Clocks**Reset State:** 0x00000000**BIMC_M_GPU_PROF_WR_STATUS_n**

Bits	Name	Description
31	WR_OUTSTANDING_OVERFLOW	Indicates that the number of outstanding writes crossed the maximum supported (255) during when traffic profiling was last enabled. It is recommended that this register be read only after traffic profiling has been disabled.

BIMC_M_GPU_PROF_WR_STATUS_n (cont.)

Bits	Name	Description
30	WR_OUTSTANDING_UNDE RFLOW	Indicates that the number of outstanding writes went below 0 during when traffic profiling was last enabled. It indicates that more command completions were seen compared to commands starts. The most likely scenario is an overflow of number of outstanding writes occurred leading to some command starts not being tracked. It is recommended that this register be read only after traffic profiling has been disabled.
7:0	WR_OUTSTANDING_MAX	Indicates the maximum number of outstanding writes that were seen when traffic profiling was last enabled. Only writes that match the MID filter and slaweway select filter are accounted for. It is recommended that this register be read only after traffic profiling has been disabled.

2.10 BIMC_M_GPU_MPORT

BRIC Masterport

2.10.1 BIMC_M_GPU_MPORT_BRIC_MASTERPORT

BRIC Masterport

0x00410000 BIMC_M_GPU_MPORT_COMPONENT_INFO**Type:** R**Clock:** Config Clock**Reset State:** 0x00000102

Component information registers provide discoverability information on the type of component and instance of each component. These registers are always the very first register in each component's configuration address space and allow for a software routine to walk the configuration address space in steps of 4KB to identify the location of each configuration register space and to which component each 4KB configures.

BIMC_M_GPU_MPORT_COMPONENT_INFO

Bits	Name	Description
23:16	INSTANCE	Indicates the instance number of this masterport.
15:8	SUB_TYPE	Identifies this configuration space as belonging to a masterport 0x1: MASTERPORT
7:0	TYPE	Identifies this configuration space as belonging to a component that interfaces to a bus master. 0x2: MASTERPORT

0x00410020 BIMC_M_GPU_MPORT_CONFIGURATION_INFO_0**Type:** R**Clock:** Config Clock**Reset State:** 0x01000000

Configuration information registers provide information on the capability and configuration of each masterport instance. See register field descriptions for more details.

BIMC_M_GPU_MPORT_CONFIGURATION_INFO_0

Bits	Name	Description
31:24	SYNC_MODE	Indicates the clock relationship between the clock used in the master side and the clock used in the core side of the masterport. When ASYNC, identifies the master clock and core clock as unrelated and asynchronous to each other. When SYNC, identifies the master clock and core clock as synchronous to each other. When ISOSYNC, identifies the master clock and core clock as related, but of different frequencies, which are of integer ratios, i.e., they are iso-synchronous to each other. 0x0: ASYNC 0x1: SYNC 0x2: ISOSYNC
15:8	CONNECTION_TYPE	Indicates the type of master connecting to this masterport. When DIRECT, indicates that a bus master is directly connected to this masterport. When CASCADE, indicates that bus masters connect to this masterport through another level of interconnect. 0x0: DIRECT 0x1: CASCADE
7:0	FUNCTIONALITY	Indicates the functionality and type of bus traffic handled by this masterport. When INORDER, identifies this masterport as optimized for in-order bus traffic. When OOO_ONLY, identifies this masterport as optimized for out-of-order bus traffic, where a single master is capable of handling responses return in any order. When REORDER, identifies this masterport as having a reorder buffer for re-ordering responses to command order, while allowing bus slaves to return responses in any order they desire. When STUB, identifies this masterport has a stub. 0x0: STUB 0x1: OOO_ONLY 0x2: INORDER 0x3: REORDER

0x00410030 BIMC_M_GPU_MPORT_CONFIGURATION_INFO_1**Type:** R**Clock:** Config Clock**Reset State:** 0x00000000

Configuration information registers provide information on the capability and configuration of each masterport instance. See register field descriptions for more details.

BIMC_M_GPU_MPORT_CONFIGURATION_INFO_1

Bits	Name	Description
31:0	SWAY_CONNECTIVITY	Indicates slaweways that this masterport can route commands to. 1-bit per slaweway. When a specific bit is set (1), indicates that commands can be routed to the slaweway corresponding to this bit. WHen a specific bit is clear (0), indicates that commands are not routed to the slaweway corresponding to this bit.

0x00410040 BIMC_M_GPU_MPORT_CONFIGURATION_INFO_2A

Type: R

Clock: Config Clock

Reset State: 0x00000000

Configuration information registers provide information on the capability and configuration of each masterport instance. See register field descriptions for more details.

BIMC_M_GPU_MPORT_CONFIGURATION_INFO_2A

Bits	Name	Description
31:16	M_DATA_WIDTH	Indicates the width of read and write data from the bus master.
15:8	M_TID_WIDTH	Indicates the width of transfer IDs from the bus master.
7:0	M_MID_WIDTH	Indicates the width of master ID from the bus master (includes any bus and port IDs if bus master is an interconnect.)

0x00410044 BIMC_M_GPU_MPORT_CONFIGURATION_INFO_2B

Type: R

Clock: Config Clock

Reset State: 0x00000000

Configuration information registers provide information on the capability and configuration of each masterport instance. See register field descriptions for more details.

BIMC_M_GPU_MPORT_CONFIGURATION_INFO_2B

Bits	Name	Description
15:0	C_DATA_WIDTH	Indicates the width of read and write data used in core of this masterport.

0x00410050 BIMC_M_GPU_MPORT_CONFIGURATION_INFO_3**Type:** R**Clock:** Config Clock**Reset State:** 0x00000000

Configuration information registers provide information on the capability and configuration of each masterport instance. See register field descriptions for more details.

BIMC_M_GPU_MPORT_CONFIGURATION_INFO_3

Bits	Name	Description
31:24	RCH_DEPTH	Indicates the depth of read channel buffers in this masterport. A beat of read data occupies one entry in the read channel buffer, until accepted by the bus master.
23:16	BCH_DEPTH	Indicates the depth of write response channel buffers in this masterport. Each write response occupies one entry in the write response channel buffer, until accepted by the bus master.
15:8	WCH_DEPTH	Indicates the depth of write channel buffers in this masterport. A beat of write data occupies one entry in the write channel buffer, until accepted by the arbiter.
7:0	ACH_DEPTH	Indicates the depth of address channel buffers in this masterport. Each command occupies one entry in the address channel buffer, until accepted by the arbiter.

0x00410060 BIMC_M_GPU_MPORT_CONFIGURATION_INFO_4**Type:** R**Clock:** Config Clock**Reset State:** 0x00000000

Configuration information registers provide information on the capability and configuration of each masterport instance. See register field descriptions for more details.

BIMC_M_GPU_MPORT_CONFIGURATION_INFO_4

Bits	Name	Description
15:8	REORDER_BUFFER_DEPTH	Indicates the number of read data beats that can be re-ordered in this masterport (if the masterport provides response re-ordering functionality.)
7:0	REORDER_TABLE_DEPTH	Indicates the number of read and write commands whose responses can be re-ordered in this masterport (if the masterport provides responses re-ordering functionality.)

0x00410070 BIMC_M_GPU_MPORT_CONFIGURATION_INFO_5A**Type:** R**Clock:** Config Clock**Reset State:** 0x00000000

Configuration information registers provide information on the capability and configuration of each masterport instance. See register field descriptions for more details.

BIMC_M_GPU_MPORT_CONFIGURATION_INFO_5A

Bits	Name	Description
31:0	ACH_PIPELINE_STAGES	Indicates pipeline stages enabled on the address channel. Each digit identifies a specific pipeline stage in the masterport, where '0' indicates no pipeline stage, '1' indicates 1-deep pipeline stage where forward path is registered, return path is fed through, '2' indicates 2-deep pipeline stage where both forward and return path are pipelined. Digit 3 identifies pipelining on the master to masterport interface, digit 2 identifies pipelining after address decode, digit 1 identifies pipelining in the address channel buffer and digit 0 identifies pipelining on the masterport to arbiter interface.

0x00410074 BIMC_M_GPU_MPORT_CONFIGURATION_INFO_5B**Type:** R**Clock:** Config Clock**Reset State:** 0x00000000

Configuration information registers provide information on the capability and configuration of each masterport instance. See register field descriptions for more details.

BIMC_M_GPU_MPORT_CONFIGURATION_INFO_5B

Bits	Name	Description
31:0	WCH_PIPELINE_STAGES	Indicates pipeline stages enabled on the write channel. Each digit identifies a specific pipeline stage in the masterport, where '0' indicates no pipeline stage, '1' indicates 1-deep pipeline stage where forward path is registered, return path is fed through, '2' indicates 2-deep pipeline stage where both forward and return path are pipelined. Digit 3 identifies pipelining on the master to masterport interface, digit 2 identifies pipelining after address decode, digit 1 identifies pipelining in the write channel buffer and digit 0 identifies pipelining on the masterport to arbiter interface.

0x00410078 BIMC_M_GPU_MPORT_CONFIGURATION_INFO_5C**Type:** R**Clock:** Config Clock**Reset State:** 0x00000000

Configuration information registers provide information on the capability and configuration of each masterport instance. See register field descriptions for more details.

BIMC_M_GPU_MPORT_CONFIGURATION_INFO_5C

Bits	Name	Description
31:0	BCH_PIPELINE_STAGES	Indicates pipeline stages enabled on the write response channel. Each digit identifies a specific pipeline stage in the masterport, where '0' indicates no pipeline stage, '1' indicates 1-deep pipeline stage where forward path is registered, return path is fed through, '2' indicates 2-deep pipeline stage where both forward and return path are pipelined. Digit 2 identifies pipelining on the slaveway to masterport interface after arbitration of write response channels from multiple slaveways, digit 1 identifies pipelining in the write response channel buffer and digit 0 identifies pipelining on the masterport to master interface.

0x0041007C BIMC_M_GPU_MPORT_CONFIGURATION_INFO_5D**Type:** R**Clock:** Config Clock**Reset State:** 0x00000000

Configuration information registers provide information on the capability and configuration of each masterport instance. See register field descriptions for more details.

BIMC_M_GPU_MPORT_CONFIGURATION_INFO_5D

Bits	Name	Description
31:0	RCH_PIPELINE_STAGES	Indicates pipeline stages enabled on the read channel. Each digit identifies a specific pipeline stage in the masterport, where '0' indicates no pipeline stage, '1' indicates 1-deep pipeline stage where forward path is registered, return path is fed through, '2' indicates 2-deep pipeline stage where both forward and return path are pipelined. Digit 2 identifies pipelining on the slaveway to masterport interface after arbitration of read channels from multiple slaveways, digit 1 identifies pipelining in the read channel buffer and digit 0 identifies pipelining on the masterport to master interface.

0x00410100 BIMC_M_GPU_MPORT_INTERRUPT_STATUS**Type:** R**Clock:** Config Clock**Reset State:** 0x00000000

Interrupt status registers provide indication of which interrupts from this masterport require servicing.

BIMC_M_GPU_MPORT_INTERRUPT_STATUS

Bits	Name	Description
1:0	RFU	Reserved for future use

0x00410108 BIMC_M_GPU_MPORT_INTERRUPT_CLEAR**Type:** W**Clock:** Config Clock**Reset State:** 0x00000000

Interrupt clear registers provide the mechanism to turn off interrupt sources once they have been acknowledged or serviced. A write of '1' to the bits corresponding to the interrupts indicated in interrupt status is required to turn off interrupts.

BIMC_M_GPU_MPORT_INTERRUPT_CLEAR

Bits	Name	Description
1:0	RFU	Reserved for future use

0x0041010C BIMC_M_GPU_MPORT_INTERRUPT_ENABLE**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000

Interrupt enable registers provide control over which interrupts are allowed to be propagated to an interrupt controller for servicing.

BIMC_M_GPU_MPORT_INTERRUPT_ENABLE

Bits	Name	Description
1:0	RFU	Reserved for future use

0x00410200 BIMC_M_GPU_MPORT_CLOCK_CTRL**Type:** RW**Clock:** Config, Core & Master Clocks**Reset State:** 0x00000003**BIMC_M_GPU_MPORT_CLOCK_CTRL**

Bits	Name	Description
1	MASTER_CLOCK_GATING_EN	When set (1), enables activity based clock gating of all registers clocked by master clock. Activity based clock gating uses architectural clock gates to reduce dynamic power consumption by gating clocks during periods of in-activity. When cleared (0), disables all activity based clock gating by forcing all architectural clock gates to not gate clocks even during periods of in-activity.
0	CORE_CLOCK_GATING_EN	When set (1), enables activity based clock gating of all registers clocked by core clock. Activity based clock gating uses architectural clock gates to reduce dynamic power consumption by gating clocks during periods of in-activity. When cleared (0), disables all activity based clock gating by forcing all architectural clock gates to not gate clocks even during periods of in-activity.

0x00410208 BIMC_M_GPU_MPORT_CDC_CTRL**Type:** RW**Clock:** Config, Core & Master Clocks**Reset State:** 0x00000000

Clock domain crossing (CDC) control register provides configuration for clock domain crossing logic.

BIMC_M_GPU_MPORT_CDC_CTRL

Bits	Name	Description
1	MASTER_CLOCK_DEMETA_SEL	When cleared (0), selects DEMETA synchronizer that requires atleast 3 clock edges (two rising and one falling). When set (1), selects DEMETA synchronizer that requires atleast 2 clock edges (one falling and one rising) to synchronize to master clock in all buffers.
0	CORE_CLOCK_DEMETA_SEL	When cleared (0), selects DEMETA synchronizer that requires atleast 3 clock edges (two rising and one falling). When set (1), selects DEMETA synchronizer that requires atleast 2 clock edges (one falling and one rising) to synchronize to core clock in all buffers.

0x00410210 BIMC_M_GPU_MPORT_MODE**Type:** RW**Clock:** Config, Core & Master Clocks**Reset State:** 0x00000012**BIMC_M_GPU_MPORT_MODE**

Bits	Name	Description
31:28	WR_GATHER_BEATS	Configures the number of write data beats to gather/buffer in write channel buffer before allowing the corresponding write command to be routed to arbiters. When configured to 0, write commands are held until the first beat of write data has been buffered. When configured to 1, write commands are held until two beats of write data have been buffered and so on. When configured with values larger than the number of beats in a write command, the command is held until the last beat has been buffered. When configured with values larger than the depth of write channel buffers, the command is held until all entries in the write channel buffer have been occupied.
19:16	RD_GATHER_BEATS	Configures the number of read data beats to gather/buffer in read channel buffer before allowing the read data to proceed to the master. When configured to 0, read data is allowed to proceed to the master as soon as the first beat is buffered in the read channel buffer. When configured to 1, read data is allowed to proceed to the master as soon as two beats of read data have been buffered and so on. When configured with values larger than the number of beats in the read burst, read data is held until the last beat has been buffered. When configured with values larger than the depth of read channel buffers, read data is held until all entries in the read channel buffer have been occupied.
15	RCSSH	Redirect Cacheable System SHareable When set (1), enables redirection of all cacheable transactions that are system shareable to the redirect slaweway. Note: Only masters connecting using AMBA 4 ACE protocol are able to indicate system shareability for transactions.
14	RCOSH	Redirect Cacheable Outer SHareable When set (1), enables redirection of all cacheable transactions that are outer shareable to the redirect slaweway.
13	RCISH	Redirect Cacheable Inner SHareable When set (1), enables redirection of all cacheable transactions that are inner shareable to the redirect slaweway.
12	RCNSH	Redirect Cacheable Non-SHareable When set (1), enables redirection of all cacheable transactions that are non-shareable to the redirect slaweway.
11	RNCSSH	Redirect Non-Cacheable System SHareable When set (1), enables redirection of all non-cacheable transactions that are system shareable to the redirect slaweway. Note: Only masters connecting using AMBA 4 ACE protocol are able to indicate system shareability for transactions.

BIMC_M_GPU_MPORT_MODE (cont.)

Bits	Name	Description
10	RNCOSH	Redirect Non-Cacheable Outer SHareable When set (1), enables redirection of all non-cacheable transactions that are outer shareable to the redirect slaway.
9	RNCISH	Redirect Non-Cacheable Inner SHareable When set (1), enables redirection of all non-cacheable transactions that are inner shareable to the redirect slaway.
8	RNCNSH	Redirect Non-Cacheable Non-SHareable When set (1), enables redirection of all non-cacheable transactions that are non-shareable to the redirect slaway.
4	NARROW_WRITES	When set (1), enables converting of 1-beat bus-width sized writes down to 32-bit ASIZE based on write strobes. It is recommended to be enabled for directly connected CPU masters. When cleared (0), disables conversion and all writes are forwarded downstream as generated by the master.
2	FIODV	Force In-Order Device When set (1), forces commands to device memory to be executed one at a time. In other words, commands to device memory are held until all previous commands have completed (responses have been received at the masterport) and all commands after the device command are held until the device completes (response has been received at the masterport.)
1	FIOSO	Force In-Order Strongly-Ordered When set (1), forces commands to strongly ordered memory to be executed one at a time. In other words, commands to strongly ordered memory are held until all previous commands have completed (responses have been received at the masterport) and all commands after the strongly ordered command are held until the strongly ordered completes (response has been received at the masterport.)
0	ORDERING_MODEL	Configures how commands to memories of type Strongly ordered and Device (SoDv) are handled in the masterport. When configured as STRICT, all commands to SoDv memory are held until all previous commands have completed (responses have been received at the masterport) and all commands after the SoDv command are held until the SoDv completes (i.e. response has been received at the masterport.) When configured as RELAXED, all commands to SoDv memory are treated as accessing Normal memory. 0x0: RELAXED 0x1: STRICT

0x00410220 BIMC_M_GPU_MPORT_WR_WAY_CROSSING**Type:** RW**Clock:** Config & Master Clocks**Reset State:** 0x00000000

Configures the level of strictness or relaxedness of way crossing checks for writes.

BIMC_M_GPU_MPORT_WR_WAY_CROSSING

Bits	Name	Description
7:0	REG	1-bit per slaveway. When 2 or more bits are set (1), allows the corresponding slaveways to be treated as one slaveway when write commands are considered as way crossing, ie. write commands to these 2 (or more) slaveways are not treated as way crossing when one write command targets one slaveway and the next write command targets the second slaveway. Not treating write commands as way crossing allows the write commands to go to the slaveway ahead of the write data for potentially better overall system performance. When all bits are set (1), disables all way crossing checks. When all bits are cleared (0), way crossing is strict with each way crossing write command held until write data for all previous writes have been routed to their corresponding slaveways.

0x00410230 BIMC_M_GPU_MPORT_PRIORITYLVL_OVERRIDE

Type: RW

Clock: Config & Master Clocks

Reset State: 0x00000000

Override registers provide the capability to override one or more bus signals to specific values.

BIMC_M_GPU_MPORT_PRIORITYLVL_OVERRIDE

Bits	Name	Description
9:8	PRIORITYLVL	Configures the values to override APRIORITYLVL from bus master with when OVERRIDE_PRIORITYLVL is set (1).
0	OVERRIDE_PRIORITYLVL	When set (1), overrides APRIORITYLVL from the bus master with values configured in PRIORITYLVL register field. When cleared (0), allows APRIORITYLVL from the bus master to be propagated downstream.

0x00410240 BIMC_M_GPU_MPORT_READ_COMMAND_OVERRIDE

Type: RW

Clock: Config & Master Clocks

Reset State: 0x00000000

Override registers provide the capability to override one or more bus signals to specific values.

BIMC_M_GPU_MPORT_READ_COMMAND_OVERRIDE

Bits	Name	Description
25:24	AREQPRIORITY	Configures the values to override AREQPRIORITY from bus master with for read commands when OVERRIDE_AREQPRIORITY is set (1).
18:16	AMEMTYPE	Configures the values to override AMEMTYPE from bus master with for read commands when OVERRIDE_AMEMTYPE is set (1).
12	ATTRANSIENT	Configures the values to override ATTRANSIENT from bus master with for read commands when OVERRIDE_ATTRANSIENT is set (1).
11	ASHARED	Configures the values to override ASHARED from bus master with for read commands when OVERRIDE_ASHARED is set (1).
10	AREDIRECT	Configures the values to override AREDIRECT from bus master with for read commands when OVERRIDE_AREDIRECT is set (1).
9	AOOO	Configures the values to override AOOO from bus master with for read commands when OVERRIDE_AOOO is set (1).
8	AINNERSHARED	Configures the values to override AINNERSHARED from bus master with for read commands when OVERRIDE_AINNERSHARED is set (1).
6	OVERRIDE_AREQPRIORITY	When set (1), overrides AREQPRIORITY from the bus master for read commands with values configured in AREQPRIORITY register field. When cleared (0), allows AREQPRIORITY from the bus master to be propagated downstream.
5	OVERRIDE_AMEMTYPE	When set (1), overrides AMEMTYPE from the bus master for read commands with values configured in AMEMTYPE register field. When cleared (0), allows AMEMTYPE from the bus master to be propagated downstream.
4	OVERRIDE_ATTRANSIENT	When set (1), overrides ATTRANSIENT from the bus master for read commands with values configured in ATTRANSIENT register field. When cleared (0), allows ATTRANSIENT from the bus master to be propagated downstream.
3	OVERRIDE_ASHARED	When set (1), overrides ASHARED from the bus master for read commands with values configured in ASHARED register field. When cleared (0), allows ASHARED from the bus master to be propagated downstream.
2	OVERRIDE_AREDIRECT	When set (1), overrides AREDIRECT from the bus master for read commands with values configured in AREDIRECT register field. When cleared (0), allows AREDIRECT from the bus master to be propagated downstream.
1	OVERRIDE_AOOO	When set (1), overrides AOOO from the bus master for read commands with values configured in AOOO register field. When cleared (0), allows AOOO from the bus master to be propagated downstream.

BIMC_M_GPU_MPORT_READ_COMMAND_OVERRIDE (cont.)

Bits	Name	Description
0	OVERRIDE_AINNERSHARE D	When set (1), overrides AINNERSHARED from the bus master for read commands with values configured in AINNERSHARED register field. When cleared (0), allows AINNERSHARED from the bus master to be propagated downstream.

0x00410250 BIMC_M_GPU_MPORT_WRITE_COMMAND_OVERRIDE**Type:** RW**Clock:** Config & Master Clocks**Reset State:** 0x00000000

Override registers provide the capability to override one or more bus signals to specific values.

BIMC_M_GPU_MPORT_WRITE_COMMAND_OVERRIDE

Bits	Name	Description
25:24	AREQPRIORITY	Configures the values to override AREQPRIORITY from bus master with for write commands when OVERRIDE_AREQPRIORITY is set (1).
18:16	AMEMTYPE	Configures the values to override AMEMTYPE from bus master with for write commands when OVERRIDE_AMEMTYPE is set (1).
12	ATRANSIENT	Configures the values to override ATRANSIENT from bus master with for write commands when OVERRIDE_ATRANSIENT is set (1).
11	ASHARED	Configures the values to override ASHARED from bus master with for write commands when OVERRIDE_ASHARED is set (1).
10	AREDIRECT	Configures the values to override AREDIRECT from bus master with for write commands when OVERRIDE_AREDIRECT is set (1).
9	AOOO	Configures the values to override AOOO from bus master with for write commands when OVERRIDE_AOOO is set (1).
8	AINNERSHARED	Configures the values to override AINNERSHARED from bus master with for write commands when OVERRIDE_AINNERSHARED is set (1).
6	OVERRIDE_AREQPRIORITY	When set (1), overrides AREQPRIORITY from the bus master for write commands with values configured in AREQPRIORITY register field. When cleared (0), allows AREQPRIORITY from the bus master to be propagated downstream.
5	OVERRIDE_AMEMTYPE	When set (1), overrides AMEMTYPE from the bus master for write commands with values configured in AMEMTYPE register field. When cleared (0), allows AMEMTYPE from the bus master to be propagated downstream.

BIMC_M_GPU_MPORT_WRITE_COMMAND_OVERRIDE (cont.)

Bits	Name	Description
4	OVERRIDE_ATRANSIENT	When set (1), overrides ATRANSIENT from the bus master for write commands with values configured in ATRANSIENT register field. When cleared (0), allows ATRANSIENT from the bus master to be propagated downstream.
3	OVERRIDE_ASHARED	When set (1), overrides ASHARED from the bus master for write commands with values configured in ASHARED register field. When cleared (0), allows ASHARED from the bus master to be propagated downstream.
2	OVERRIDE_AREDIRECT	When set (1), overrides AREDIRECT from the bus master for write commands with values configured in AREDIRECT register field. When cleared (0), allows AREDIRECT from the bus master to be propagated downstream.
1	OVERRIDE_AOOO	When set (1), overrides AOOO from the bus master for write commands with values configured in AOOO register field. When cleared (0), allows AOOO from the bus master to be propagated downstream.
0	OVERRIDE_AINNERSHARE D	When set (1), overrides AINNERSHARED from the bus master for write commands with values configured in AINNERSHARED register field. When cleared (0), allows AINNERSHARED from the bus master to be propagated downstream.

0x00410300 BIMC_M_GPU_MPORT_BKE_ENABLE**Type:** RW**Clock:** Config & Master Clocks**Reset State:** 0x00000000**BIMC_M_GPU_MPORT_BKE_ENABLE**

Bits	Name	Description
0	ENABLE	When set (1), enables the book keeping element (BKE) to determine AREQPRIORITY and APRIORITYLVL of commands from the bus master by comparing the amount of bus traffic from the bus master to a target, and determining a health level for the bus master. The health level determines the relative priority of commands from the bus master to allow the bus master to catch-up if it is not getting serviced, or slow down to allow other bus masters to get their share of system resources. When cleared (0), disables the book keeping element and AREQPRIORITY and APRIORITYLVL from the bus master are propagated downstream.

0x00410304 BIMC_M_GPU_MPORT_BKE_GRANT_PERIOD**Type:** RW**Clock:** Config & Reference Clocks**Reset State:** 0x00000000**BIMC_M_GPU_MPORT_BKE_GRANT_PERIOD**

Bits	Name	Description
9:0	GRANT_PERIOD	Defines the period over which the book keeping element accumulates grants to the bus master for determining it's health level. Grant period is defined in number of reference timer ticks. Refer configuration of reference timer tick for exact time unit.

0x00410308 BIMC_M_GPU_MPORT_BKE_GRANT_COUNT**Type:** RW**Clock:** Config & Master Clocks**Reset State:** 0x00000000**BIMC_M_GPU_MPORT_BKE_GRANT_COUNT**

Bits	Name	Description
15:0	GRANT_COUNT	Defines the target grant count for the bus master. The target grant count is the number of beats of read and write data the bus master is expected to write and read in the grant period and is used as a reference to determine if the bus master is behind or ahead of where it ought to be at any given time.

0x00410320 BIMC_M_GPU_MPORT_BKE_THRESHOLD_HIGH**Type:** RW**Clock:** Config & Master Clocks**Reset State:** 0x00000000**BIMC_M_GPU_MPORT_BKE_THRESHOLD_HIGH**

Bits	Name	Description
15:0	THRESHOLD	Defines a high threshold to determine how much behind or ahead a bus master is with reference to where it ought to be at any given time.

0x00410324 BIMC_M_GPU_MPORT_BKE_THRESHOLD_MEDIUM**Type:** RW**Clock:** Config & Master Clocks**Reset State:** 0x00000000**BIMC_M_GPU_MPORT_BKE_THRESHOLD_MEDIUM**

Bits	Name	Description
15:0	THRESHOLD	Defines a medium threshold to determine how much behind or ahead a bus master is with reference to where it ought to be at any given time.

0x00410328 BIMC_M_GPU_MPORT_BKE_THRESHOLD_LOW**Type:** RW**Clock:** Config & Master Clocks**Reset State:** 0x00000000**BIMC_M_GPU_MPORT_BKE_THRESHOLD_LOW**

Bits	Name	Description
15:0	THRESHOLD	Defines a low threshold to determine how much behind or ahead a bus master is with reference to where it ought to be at any given time.

0x00410340 BIMC_M_GPU_MPORT_BKE_HEALTH_0**Type:** RW**Clock:** Config & Master Clocks**Reset State:** 0x00000000

BKE health registers configure the request priority and realtime priority level to use for commands from the bus master when it has been determined to be of a certain health.

BIMC_M_GPU_MPORT_BKE_HEALTH_0

Bits	Name	Description
31	LIMIT_COMMANDS	When set (1), defines that no new commands may be accepted from the bus master as long as it is determined to be at health 0.
9:8	AREQPRIORITY	Defines the request priority of commands from the bus master when it has been determined to be at health 0.
1:0	PRIORITYLVL	Defines the realtime priority level to use for commands from the bus master when it has been determined to be at health 0.

0x00410344 BIMC_M_GPU_MPORT_BKE_HEALTH_1**Type:** RW**Clock:** Config & Master Clocks**Reset State:** 0x00000000

BKE health registers configure the request priority and realtime priority level to use for commands from the bus master when it has been determined to be of a certain health.

BIMC_M_GPU_MPORT_BKE_HEALTH_1

Bits	Name	Description
31	LIMIT_COMMANDS	When set (1), defines that no new commands may be accepted from the bus master as long as it is determined to be at health 1.
9:8	AREQPRIORITY	Defines the request priority of commands from the bus master when it has been determined to be at health 1.
1:0	PRIORITYLVL	Defines the realtime priority level to use for commands from the bus master when it has been determined to be at health 1.

0x00410348 BIMC_M_GPU_MPORT_BKE_HEALTH_2**Type:** RW**Clock:** Config & Master Clocks**Reset State:** 0x00000000

BKE health registers configure the request priority and realtime priority level to use for commands from the bus master when it has been determined to be of a certain health.

BIMC_M_GPU_MPORT_BKE_HEALTH_2

Bits	Name	Description
31	LIMIT_COMMANDS	When set (1), defines that no new commands may be accepted from the bus master as long as it is determined to be at health 2.
9:8	AREQPRIORITY	Defines the request priority of commands from the bus master when it has been determined to be at health 2.
1:0	PRIORITYLVL	Defines the realtime priority level to use for commands from the bus master when it has been determined to be at health 2.

0x0041034C BIMC_M_GPU_MPORT_BKE_HEALTH_3**Type:** RW**Clock:** Config & Master Clocks**Reset State:** 0x00000000

BKE health registers configure the request priority and realtime priority level to use for commands from the bus master when it has been determined to be of a certain health. Health 3 is the default health for the bus master.

BIMC_M_GPU_MPORT_BKE_HEALTH_3

Bits	Name	Description
9:8	AREQPRIORITY	Defines the request priority of commands from the bus master when it has been determined to be at health 3.
1:0	PRIORITYLVL	Defines the realtime priority level to use for commands from the bus master when it has been determined to be at health 3.

0x00410420 BIMC_M_GPU_MPORT_STATUS_2A

Type: R

Clock: Config, Master & Pipe0 Clocks

Reset State: 0x00000000

Status registers indicate the state of several elements in the masterport, including the number of transactions outstanding/pending waiting for a response, emptiness, fullness of several buffers, etc. The state information is available only as a debug indication and could be delayed by a few clock cycles from when the state is reached to being reflected in the status registers. One or more of the state registers are read live and could capture intermediate values as they change.

BIMC_M_GPU_MPORT_STATUS_2A

Bits	Name	Description
23:16	WR_PENDING_PRE_BUF	Indicates the number of write transactions that are outstanding/pending waiting for a response from the slave. This transaction count is from before address channel buffers, which interfaces to frontend of the masterport.
7:0	RD_PENDING_PRE_BUF	Indicates the number of read transactions that are outstanding/pending waiting for a response from the slave. This transaction count is from before address channel buffers, which interfaces to frontend of the masterport.

0x00410424 BIMC_M_GPU_MPORT_STATUS_2B

Type: R

Clock: Config, Master & Pipe0 Clocks

Reset State: 0x00000000

Status registers indicate the state of several elements in the masterport, including the number of transactions outstanding/pending waiting for a response, emptiness, fullness of several buffers, etc. The state information is available only as a debug indication and could be delayed by a few clock

cycles from when the state is reached to being reflected in the status registers. One or more of the state registers are read live and could capture intermediate values as they change.

BIMC_M_GPU_MPORT_STATUS_2B

Bits	Name	Description
23:16	WR_PENDING_POST_BUF	Indicates the number of write transactions that are outstanding/pending waiting for a response from the slave. This transaction count is from after address channel buffers, which interfaces to core of the interconnect, i.e., arbiter and slaveway.
7:0	RD_PENDING_POST_BUF	Indicates the number of read transactions that are outstanding/pending waiting for a response from the slave. This transaction count is from after address channel buffers, which interfaces to core of the interconnect, i.e., arbiter and slaveway.

0x00410430 BIMC_M_GPU_MPORT_STATUS_3A

Type: R

Clock: Config, Master & Pipe0 Clocks

Reset State: 0x55111111

Status registers indicate the state of several elements in the masterport, including the number of transactions outstanding/pending waiting for a response, emptiness, fullness of several buffers, etc. The state information is available only as a debug indication and could be delayed by a few clock cycles from when the state is reached to being reflected in the status registers. One or more of the state registers are read live and could capture intermediate values as they change.

BIMC_M_GPU_MPORT_STATUS_3A

Bits	Name	Description
31	RCB_DATA_RD_FULL	Indicates that the data part of read channel buffer is full when seen from it's read side, which interfaces to frontend of the masterport.
30	RCB_DATA_RD_EMPTY	Indicates that the data part of read channel buffer is empty when seen from it's read side, which interfaces to frontend of the masterport.
29	RCB_CTRL_RD_FULL	Indicates that the control part of read channel buffer is full when seen from it's read side, which interfaces to frontend of the masterport.
28	RCB_CTRL_RD_EMPTY	Indicates that the control part of read channel buffer is empty when seen from it's write side, which interfaces to frontend of the masterport.
27	RCB_DATA_WR_FULL	Indicates that the data part of read channel buffer is full when seen from it's write side, which interfaces to core of the interconnect, i.e., arbiter and slaveway.
26	RCB_DATA_WR_EMPTY	Indicates that the data part of read channel buffer is empty when seen from it's write side, which interfaces to core of the interconnect, i.e., arbiter and slaveway.

BIMC_M_GPU_MPORT_STATUS_3A (cont.)

Bits	Name	Description
25	RCB_CTRL_WR_FULL	Indicates that the control part of read channel buffer is full when seen from it's write side, which interfaces to core of the interconnect, i.e., arbiter and slaveway.
24	RCB_CTRL_WR_EMPTY	Indicates that the control part of read channel buffer is empty when seen from it's write side, which interfaces to core of the interconnect, i.e., arbiter and slaveway.
21	BCB_RD_FULL	Indicates that the write response channel buffer is full when seen from it's read side, which interfaces to frontend of the masterport.
20	BCB_RD_EMPTY	Indicates that the write response channel buffer is empty when seen from it's read side, which interfaces to frontend of the masterport.
17	BCB_WR_FULL	Indicates that the write response channel buffer is full when seen from it's write side, which interfaces to core of the interconnect, i.e., arbiter and slaveway.
16	BCB_WR_EMPTY	Indicates that the write response channel buffer is empty when seen from it's write side, which interfaces to core of the interconnect, i.e., arbiter and slaveway.
13	WCB_RD_FULL	Indicates that the write channel buffer is full when seen from it's read side, which interfaces to core of the interconnect, i.e., arbiter and slaveway.
12	WCB_RD_EMPTY	Indicates that the write channel buffer is empty when seen from it's read side, which interfaces to core of the interconnect, i.e., arbiter and slaveway.
9	WCB_WR_FULL	Indicates that the write channel buffer is full when seen from it's write side, which interfaces to frontend of the masterport.
8	WCB_WR_EMPTY	Indicates that the write channel buffer is empty when seen from it's write side, which interfaces to frontend of the masterport.
5	ACB_RD_FULL	Indicates that the address channel buffer is full when seen from it's read side, which interfaces to core of the interconnect, i.e., arbiter and slaveway.
4	ACB_RD_EMPTY	Indicates that the address channel buffer is empty when seen from it's read side, which interfaces to core of the interconnect, i.e., arbiter and slaveway.
1	ACB_WR_FULL	Indicates that the address channel buffer is full when seen from it's write side, which interfaces to frontend of the masterport.
0	ACB_WR_EMPTY	Indicates that the address channel buffer is empty when seen from it's write side, which interfaces to frontend of the masterport.

0x00410434 BIMC_M_GPU_MPORT_STATUS_3B**Type:** R**Clock:** Config, Master & Pipe0 Clocks**Reset State:** 0x00111111

Buffer status register indicate the state of all buffers in the masterport, whether they are empty or full on either side of the buffer. Some buffers carry information from master to core and others carry information from core to master. Both write side and read side status are indicated separately, since they would be not be similar when master clock and core clock are not synchronous to each other.

BIMC_M_GPU_MPORT_STATUS_3B

Bits	Name	Description
21	RGB_RD_FULL	Indicates that the read gathering buffer is full when seen from it's read side, which interfaces to frontend of the masterport.
20	RGB_RD_EMPTY	Indicates that the read gathering buffer is empty when seen from it's read side, which interfaces to frontend of the masterport.
17	RGB_WR_FULL	Indicates that the read gathering buffer is full when seen from it's write side, which interfaces to core of the interconnect, i.e., arbiter and slaweway.
16	RGB_WR_EMPTY	Indicates that the read gathering buffer is empty when seen from it's write side, which interfaces to core of the interconnect, i.e., arbiter and slaweway.
13	W2AB_RD_FULL	Indicates that the write channel to address channel user signals buffer is full when seen from it's read side, which interfaces to core of the interconnect, i.e., arbiter and slaweway.
12	W2AB_RD_EMPTY	Indicates that the write channel to address channel user signals buffer is empty when seen from it's read side, which interfaces to core of the interconnect, i.e., arbiter and slaweway.
9	W2AB_WR_FULL	Indicates that the write channel to address channel user signals buffer is full when seen from it's write side, which interfaces to frontend of the masterport.
8	W2AB_WR_EMPTY	Indicates that the write channel to address channel user signals buffer is empty when seen from it's write side, which interfaces to frontend of the masterport.
5	A2WB_RD_FULL	Indicates that the address channel to write channel user signals buffer is full when seen from it's read side, which interfaces to core of the interconnect, i.e., arbiter and slaweway.
4	A2WB_RD_EMPTY	Indicates that the address channel to write channel user signalsbuffer is empty when seen from it's read side, which interfaces to core of the interconnect, i.e., arbiter and slaweway.
1	A2WB_WR_FULL	Indicates that the address channel to write channel user signals buffer is full when seen from it's write side, which interfaces to frontend of the masterport.

BIMC_M_GPU_MPORT_STATUS_3B (cont.)

Bits	Name	Description
0	A2WB_WR_EMPTY	Indicates that the address channel to write channel user signals buffer is empty when seen from it's write side, which interfaces to frontend of the masterport.

2.11 BIMC_M_SYS0_PROF

BRIC Traffic Profiling

2.11.1 BIMC_M_SYS0_PROF_BRIC_PROFILING

BRIC Traffic Profiling

**0x00415400+ BIMC_M_SYS0_PROF_RD_STATUS_n, n=[0..0]
0x4*n**

Type: R**Clock:** Config & Tenure Clocks**Reset State:** 0x00000000**BIMC_M_SYS0_PROF_RD_STATUS_n**

Bits	Name	Description
31	RD_OUTSTANDING_OVERFLOW	Indicates that the number of outstanding reads crossed the maximum supported (255) during when traffic profiling was last enabled. It is recommended that this register be read only after traffic profiling has been disabled.
30	RD_OUTSTANDING_UNDERFLOW	Indicates that the number of outstanding reads went below 0 during when traffic profiling was last enabled. It indicates that more command completions were seen compared to commands starts. The most likely scenario is an overflow of number of outstanding reads occurred leading to some command starts not being tracked. It is recommended that this register be read only after traffic profiling has been disabled.
7:0	RD_OUTSTANDING_MAX	Indicates the maximum number of outstanding reads that were seen when traffic profiling was last enabled. Only reads that match the MID filter and slaweway select filter are accounted for. It is recommended that this register be read only after traffic profiling has been disabled.

0x00415410+ BIMC_M_SYS0_PROF_WR_STATUS_n, n=[0..0] 0x4*n

Type: R

Clock: Config & Tenure Clocks

Reset State: 0x00000000

BIMC_M_SYS0_PROF_WR_STATUS_n

Bits	Name	Description
31	WR_OUTSTANDING_OVERFLOW	Indicates that the number of outstanding writes crossed the maximum supported (255) during when traffic profiling was last enabled. It is recommended that this register be read only after traffic profiling has been disabled.
30	WR_OUTSTANDING_UNDERFLOW	Indicates that the number of outstanding writes went below 0 during when traffic profiling was last enabled. It indicates that more command completions were seen compared to commands starts. The most likely scenario is an overflow of number of outstanding writes occurred leading to some command starts not being tracked. It is recommended that this register be read only after traffic profiling has been disabled.
7:0	WR_OUTSTANDING_MAX	Indicates the maximum number of outstanding writes that were seen when traffic profiling was last enabled. Only writes that match the MID filter and slaveway select filter are accounted for. It is recommended that this register be read only after traffic profiling has been disabled.

2.12 BIMC_M_SYS0_MPORT

BRIC Masterport

2.12.1 BIMC_M_SYS0_MPORT_BRIC_MASTERPORT

BRIC Masterport

0x00414000 BIMC_M_SYS0_MPORT_COMPONENT_INFO

Type: R

Clock: Config Clock

Reset State: 0x00000102

Component information registers provide discoverability information on the type of component and instance of each component. These registers are always the very first register in each component's configuration address space and allow for a software routine to walk the configuration address space in steps of 4KB to identify the location of each configuration register space and to which component each 4KB configures.

BIMC_M_SYS0_MPORT_COMPONENT_INFO

Bits	Name	Description
23:16	INSTANCE	Indicates the instance number of this masterport.
15:8	SUB_TYPE	Identifies this configuration space as belonging to a masterport 0x1: MASTERPORT
7:0	TYPE	Identifies this configuration space as belonging to a component that interfaces to a bus master. 0x2: MASTERPORT

0x00414020 BIMC_M_SYS0_MPORT_CONFIGURATION_INFO_0**Type:** R**Clock:** Config Clock**Reset State:** 0x01000000

Configuration information registers provide information on the capability and configuration of each masterport instance. See register field descriptions for more details.

BIMC_M_SYS0_MPORT_CONFIGURATION_INFO_0

Bits	Name	Description
31:24	SYNC_MODE	Indicates the clock relationship between the clock used in the master side and the clock used in the core side of the masterport. When ASYNC, identifies the master clock and core clock as unrelated and asynchronous to each other. When SYNC, identifies the master clock and core clock as synchronous to each other. When ISOSYNC, identifies the master clock and core clock as related, but of different frequencies, which are of integer ratios, i.e., they are iso-synchronous to each other. 0x0: ASYNC 0x1: SYNC 0x2: ISOSYNC
15:8	CONNECTION_TYPE	Indicates the type of master connecting to this masterport. When DIRECT, indicates that a bus master is directly connected to this masterport. When CASCADE, indicates that bus masters connect to this masterport through another level of interconnect. 0x0: DIRECT 0x1: CASCADE

BIMC_M_SYS0_MPORT_CONFIGURATION_INFO_0 (cont.)

Bits	Name	Description
7:0	FUNCTIONALITY	Indicates the functionality and type of bus traffic handled by this masterport. When INORDER, identifies this masterport as optimized for in-order bus traffic. When OOO_ONLY, identifies this masterport as optimized for out-of-order bus traffic, where a single master is capable of handling responses return in any order. When REORDER, identifies this masterport as having a reorder buffer for re-ordering responses to command order, while allowing bus slaves to return responses in any order they desire. When STUB, identifies this masterport has a stub. 0x0: STUB 0x1: OOO_ONLY 0x2: INORDER 0x3: REORDER

0x00414030 BIMC_M_SYS0_MPORT_CONFIGURATION_INFO_1**Type:** R**Clock:** Config Clock**Reset State:** 0x00000000

Configuration information registers provide information on the capability and configuration of each masterport instance. See register field descriptions for more details.

BIMC_M_SYS0_MPORT_CONFIGURATION_INFO_1

Bits	Name	Description
31:0	SWAY_CONNECTIVITY	Indicates slaveways that this masterport can route commands to. 1-bit per slaveway. When a specific bit is set (1), indicates that commands can be routed to the slaveway corresponding to this bit. When a specific bit is clear (0), indicates that commands are not routed to the slaveway corresponding to this bit.

0x00414040 BIMC_M_SYS0_MPORT_CONFIGURATION_INFO_2A**Type:** R**Clock:** Config Clock**Reset State:** 0x00000000

Configuration information registers provide information on the capability and configuration of each masterport instance. See register field descriptions for more details.

BIMC_M_SYS0_MPORT_CONFIGURATION_INFO_2A

Bits	Name	Description
31:16	M_DATA_WIDTH	Indicates the width of read and write data from the bus master.

BIMC_M_SYS0_MPORT_CONFIGURATION_INFO_2A (cont.)

Bits	Name	Description
15:8	M_TID_WIDTH	Indicates the width of transfer IDs from the bus master.
7:0	M_MID_WIDTH	Indicates the width of master ID from the bus master (includes any bus and port IDs if bus master is an interconnect.)

0x00414044 BIMC_M_SYS0_MPORT_CONFIGURATION_INFO_2B**Type:** R**Clock:** Config Clock**Reset State:** 0x00000000

Configuration information registers provide information on the capability and configuration of each masterport instance. See register field descriptions for more details.

BIMC_M_SYS0_MPORT_CONFIGURATION_INFO_2B

Bits	Name	Description
15:0	C_DATA_WIDTH	Indicates the width of read and write data used in core of this masterport.

0x00414050 BIMC_M_SYS0_MPORT_CONFIGURATION_INFO_3**Type:** R**Clock:** Config Clock**Reset State:** 0x00000000

Configuration information registers provide information on the capability and configuration of each masterport instance. See register field descriptions for more details.

BIMC_M_SYS0_MPORT_CONFIGURATION_INFO_3

Bits	Name	Description
31:24	RCH_DEPTH	Indicates the depth of read channel buffers in this masterport. A beat of read data occupies one entry in the read channel buffer, until accepted by the bus master.
23:16	BCH_DEPTH	Indicates the depth of write response channel buffers in this masterport. Each write response occupies one entry in the write response channel buffer, until accepted by the bus master.
15:8	WCH_DEPTH	Indicates the depth of write channel buffers in this masterport. A beat of write data occupies one entry in the write channel buffer, until accepted by the arbiter.

BIMC_M_SYS0_MPORT_CONFIGURATION_INFO_3 (cont.)

Bits	Name	Description
7:0	ACH_DEPTH	Indicates the depth of address channel buffers in this masterport. Each command occupies one entry in the address channel buffer, until accepted by the arbiter.

0x00414060 BIMC_M_SYS0_MPORT_CONFIGURATION_INFO_4**Type:** R**Clock:** Config Clock**Reset State:** 0x00000000

Configuration information registers provide information on the capability and configuration of each masterport instance. See register field descriptions for more details.

BIMC_M_SYS0_MPORT_CONFIGURATION_INFO_4

Bits	Name	Description
15:8	REORDER_BUFFER_DEPT H	Indicates the number of read data beats that can be re-ordered in this masterport (if the masterport provides response re-ordering functionality.)
7:0	REORDER_TABLE_DEPTH	Indicates the number of read and write commands whose responses can be re-ordered in this masterport (if the masterport provides responses re-ordering functionality.)

0x00414070 BIMC_M_SYS0_MPORT_CONFIGURATION_INFO_5A**Type:** R**Clock:** Config Clock**Reset State:** 0x00000000

Configuration information registers provide information on the capability and configuration of each masterport instance. See register field descriptions for more details.

BIMC_M_SYS0_MPORT_CONFIGURATION_INFO_5A

Bits	Name	Description
31:0	ACH_PIPELINE_STAGES	Indicates pipeline stages enabled on the address channel. Each digit identifies a specific pipeline stage in the masterport, where '0' indicates no pipeline stage, '1' indicates 1-deep pipeline stage where forward path is registered, return path is fed through, '2' indicates 2-deep pipeline stage where both forward and return path are pipelined. Digit 3 identifies pipelining on the master to masterport interface, digit 2 identifies pipelining after address decode, digit 1 identifies pipelining in the address channel buffer and digit 0 identifies pipelining on the masterport to arbiter interface.

0x00414074 BIMC_M_SYS0_MPORT_CONFIGURATION_INFO_5B**Type:** R**Clock:** Config Clock**Reset State:** 0x00000000

Configuration information registers provide information on the capability and configuration of each masterport instance. See register field descriptions for more details.

BIMC_M_SYS0_MPORT_CONFIGURATION_INFO_5B

Bits	Name	Description
31:0	WCH_PIPELINE_STAGES	Indicates pipeline stages enabled on the write channel. Each digit identifies a specific pipeline stage in the masterport, where '0' indicates no pipeline stage, '1' indicates 1-deep pipeline stage where forward path is registered, return path is fed through, '2' indicates 2-deep pipeline stage where both forward and return path are pipelined. Digit 3 identifies pipelining on the master to masterport interface, digit 2 identifies pipelining after address decode, digit 1 identifies pipelining in the write channel buffer and digit 0 identifies pipelining on the masterport to arbiter interface.

0x00414078 BIMC_M_SYS0_MPORT_CONFIGURATION_INFO_5C**Type:** R**Clock:** Config Clock**Reset State:** 0x00000000

Configuration information registers provide information on the capability and configuration of each masterport instance. See register field descriptions for more details.

BIMC_M_SYS0_MPORT_CONFIGURATION_INFO_5C

Bits	Name	Description
31:0	BCH_PIPELINE_STAGES	Indicates pipeline stages enabled on the write response channel. Each digit identifies a specific pipeline stage in the masterport, where '0' indicates no pipeline stage, '1' indicates 1-deep pipeline stage where forward path is registered, return path is fed through, '2' indicates 2-deep pipeline stage where both forward and return path are pipelined. Digit 2 identifies pipelining on the slaveway to masterport interface after arbitration of write response channels from multiple slaveways, digit 1 identifies pipelining in the write response channel buffer and digit 0 identifies pipelining on the masterport to master interface.

0x0041407C BIMC_M_SYS0_MPORT_CONFIGURATION_INFO_5D

Type: R
Clock: Config Clock
Reset State: 0x00000000

Configuration information registers provide information on the capability and configuration of each masterport instance. See register field descriptions for more details.

BIMC_M_SYS0_MPORT_CONFIGURATION_INFO_5D

Bits	Name	Description
31:0	RCH_PIPELINE_STAGES	Indicates pipeline stages enabled on the read channel. Each digit identifies a specific pipeline stage in the masterport, where '0' indicates no pipeline stage, '1' indicates 1-deep pipeline stage where forward path is registered, return path is fed through, '2' indicates 2-deep pipeline stage where both forward and return path are pipelined. Digit 2 identifies pipelining on the slaveway to masterport interface after arbitration of read channels from multiple slaveways, digit 1 identifies pipelining in the read channel buffer and digit 0 identifies pipelining on the masterport to master interface.

0x00414100 BIMC_M_SYS0_MPORT_INTERRUPT_STATUS

Type: R
Clock: Config Clock
Reset State: 0x00000000

Interrupt status registers provide indication of which interrupts from this masterport require servicing.

BIMC_M_SYS0_MPORT_INTERRUPT_STATUS

Bits	Name	Description
1:0	RFU	Reserved for future use

0x00414108 BIMC_M_SYS0_MPORT_INTERRUPT_CLEAR

Type: W
Clock: Config Clock
Reset State: 0x00000000

Interrupt clear registers provide the mechanism to turn off interrupt sources once they have been acknowledged or serviced. A write of '1' to the bits corresponding to the interrupts indicated in interrupt status is required to turn off interrupts.

BIMC_M_SYS0_MPORT_INTERRUPT_CLEAR

Bits	Name	Description
1:0	RFU	Reserved for future use

0x0041410C BIMC_M_SYS0_MPORT_INTERRUPT_ENABLE**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000

Interrupt enable registers provide control over which interrupts are allowed to be propagated to an interrupt controller for servicing.

BIMC_M_SYS0_MPORT_INTERRUPT_ENABLE

Bits	Name	Description
1:0	RFU	Reserved for future use

0x00414200 BIMC_M_SYS0_MPORT_CLOCK_CTRL**Type:** RW**Clock:** Config, Core & Master Clocks**Reset State:** 0x00000003**BIMC_M_SYS0_MPORT_CLOCK_CTRL**

Bits	Name	Description
1	MASTER_CLOCK_GATING_EN	When set (1), enables activity based clock gating of all registers clocked by master clock. Activity based clock gating uses architectural clock gates to reduce dynamic power consumption by gating clocks during periods of in-activity. When cleared (0), disables all activity based clock gating by forcing all architectural clock gates to not gate clocks even during periods of in-activity.
0	CORE_CLOCK_GATING_EN	When set (1), enables activity based clock gating of all registers clocked by core clock. Activity based clock gating uses architectural clock gates to reduce dynamic power consumption by gating clocks during periods of in-activity. When cleared (0), disables all activity based clock gating by forcing all architectural clock gates to not gate clocks even during periods of in-activity.

0x00414208 BIMC_M_SYS0_MPORT_CDC_CTRL**Type:** RW**Clock:** Config, Core & Master Clocks**Reset State:** 0x00000000

Clock domain crossing (CDC) control register provides configuration for clock domain crossing logic.

BIMC_M_SYS0_MPORT_CDC_CTRL

Bits	Name	Description
1	MASTER_CLOCK_DEMETA_SEL	When cleared (0), selects DEMETA synchronizer that requires atleast 3 clock edges (two rising and one falling). When set (1), selects DEMETA synchronizer that requires atleast 2 clock edges (one falling and one rising) to synchronize to master clock in all buffers.
0	CORE_CLOCK_DEMETA_SEL	When cleared (0), selects DEMETA synchronizer that requires atleast 3 clock edges (two rising and one falling). When set (1), selects DEMETA synchronizer that requires atleast 2 clock edges (one falling and one rising) to synchronize to core clock in all buffers.

0x00414210 BIMC_M_SYS0_MPORT_MODE**Type:** RW**Clock:** Config, Core & Master Clocks**Reset State:** 0x00000012**BIMC_M_SYS0_MPORT_MODE**

Bits	Name	Description
31:28	WR_GATHER_BEATS	Configures the number of write data beats to gather/buffer in write channel buffer before allowing the corresponding write command to be routed to arbiters. When configured to 0, write commands are held until the first beat of write data has been buffered. When configured to 1, write commands are held until two beats of write data have been buffered and so on. When configured with values larger than the number of beats in a write command, the command is held until the last beat has been buffered. When configured with values larger than the depth of write channel buffers, the command is held until all entries in the write channel buffer have been occupied.

BIMC_M_SYS0_MPORT_MODE (cont.)

Bits	Name	Description
19:16	RD_GATHER_BEATS	Configures the number of read data beats to gather/buffer in read channel buffer before allowing the read data to proceed to the master. When configured to 0, read data is allowed to proceed to the master as soon as the first beat is buffered in the read channel buffer. When configured to 1, read data is allowed to proceed to the master as soon as two beats of read data have been buffered and so on. When configured with values larger than the number of beats in the read burst, read data is held until the last beat has been buffered. When configured with values larger than the depth of read channel buffers, read data is held until all entries in the read channel buffer have been occupied.
15	RCSSH	Redirect Cacheable System SHareable When set (1), enables redirection of all cacheable transactions that are system shareable to the redirect slaveway. Note: Only masters connecting using AMBA 4 ACE protocol are able to indicate system shareability for transactions.
14	RCOSH	Redirect Cacheable Outer SHareable When set (1), enables redirection of all cacheable transactions that are outer shareable to the redirect slaveway.
13	RCISH	Redirect Cacheable Inner SHareable When set (1), enables redirection of all cacheable transactions that are inner shareable to the redirect slaveway.
12	RCNSH	Redirect Cacheable Non-SHareable When set (1), enables redirection of all cacheable transactions that are non-shareable to the redirect slaveway.
11	RNCSSH	Redirect Non-Cacheable System SHareable When set (1), enables redirection of all non-cacheable transactions that are system shareable to the redirect slaveway. Note: Only masters connecting using AMBA 4 ACE protocol are able to indicate system shareability for transactions.
10	RNCOSH	Redirect Non-Cacheable Outer SHareable When set (1), enables redirection of all non-cacheable transactions that are outer shareable to the redirect slaveway.
9	RNCISH	Redirect Non-Cacheable Inner SHareable When set (1), enables redirection of all non-cacheable transactions that are inner shareable to the redirect slaveway.
8	RNCNSH	Redirect Non-Cacheable Non-SHareable When set (1), enables redirection of all non-cacheable transactions that are non-shareable to the redirect slaveway.
4	NARROW_WRITES	When set (1), enables converting of 1-beat bus-width sized writes down to 32-bit ASIZE based on write strobes. It is recommended to be enabled for directly connected CPU masters. When cleared (0), disables conversion and all writes are forwarded downstream as generated by the master.

BIMC_M_SYS0_MPORT_MODE (cont.)

Bits	Name	Description
2	FIO DV	Force In-Order Device When set (1), forces commands to device memory to be executed one at a time. In other words, commands to device memory are held until all previous commands have completed (responses have been received at the masterport) and all commands after the device command are held until the device completes (response has been received at the masterport.)
1	FIO SO	Force In-Order Strongly-Ordered When set (1), forces commands to strongly ordered memory to be executed one at a time. In other words, commands to strongly ordered memory are held until all previous commands have completed (responses have been received at the masterport) and all commands after the strongly ordered command are held until the strongly ordered completes (response has been received at the masterport.)
0	ORDERING_MODEL	Configures how commands to memories of type Strongly ordered and Device (SoDv) are handled in the masterport. When configured as STRICT, all commands to SoDv memory are held until all previous commands have completed (responses have been received at the masterport) and all commands after the SoDv command are held until the SoDv completes (i.e. response has been received at the masterport.) When configured as RELAXED, all commands to SoDv memory are treated as accessing Normal memory. 0x0: RELAXED 0x1: STRICT

0x00414220 BIMC_M_SYS0_MPORT_WR_WAY_CROSSING**Type:** RW**Clock:** Config & Master Clocks**Reset State:** 0x00000000

Configures the level of strictness or relaxedness of way crossing checks for writes.

BIMC_M_SYS0_MPORT_WR_WAY_CROSSING

Bits	Name	Description
7:0	REG	1-bit per slaveway. When 2 or more bits are set (1), allows the corresponding slaveways to be treated as one slaveway when write commands are considered as way crossing, ie. write commands to these 2 (or more) slaveways are not treated as way crossing when one write command targets one slaveway and the next write command targets the second slaveway. Not treating write commands as way crossing allows the write commands to go to the slaveway ahead of the write data for potentially better overall system performance. When all bits are set (1), disables all way crossing checks. When all bits are cleared (0), way crossing is strict with each way crossing write command held until write data for all previous writes have been routed to their corresponding slaveways.

0x00414230 BIMC_M_SYS0_MPORT_PRIORITYLVL_OVERRIDE**Type:** RW**Clock:** Config & Master Clocks**Reset State:** 0x00000000

Override registers provide the capability to override one or more bus signals to specific values.

BIMC_M_SYS0_MPORT_PRIORITYLVL_OVERRIDE

Bits	Name	Description
9:8	PRIORITYLVL	Configures the values to override APRIORITYLVL from bus master with when OVERRIDE_PRIORITYLVL is set (1).
0	OVERRIDE_PRIORITYLVL	When set (1), overrides APRIORITYLVL from the bus master with values configured in PRIORITYLVL register field. When cleared (0), allows APRIORITYLVL from the bus master to be propagated downstream.

0x00414240 BIMC_M_SYS0_MPORT_READ_COMMAND_OVERRIDE**Type:** RW**Clock:** Config & Master Clocks**Reset State:** 0x00000000

Override registers provide the capability to override one or more bus signals to specific values.

BIMC_M_SYS0_MPORT_READ_COMMAND_OVERRIDE

Bits	Name	Description
25:24	AREQPRIORITY	Configures the values to override AREQPRIORITY from bus master with for read commands when OVERRIDE_AREQPRIORITY is set (1).
18:16	AMEMTYPE	Configures the values to override AMEMTYPE from bus master with for read commands when OVERRIDE_AMEMTYPE is set (1).
12	ATTRANSIENT	Configures the values to override ATTRANSIENT from bus master with for read commands when OVERRIDE_ATTRANSIENT is set (1).
11	ASHARED	Configures the values to override ASHARED from bus master with for read commands when OVERRIDE_ASHARED is set (1).
10	AREDIRECT	Configures the values to override AREDIRECT from bus master with for read commands when OVERRIDE_AREDIRECT is set (1).
9	AOOO	Configures the values to override AOOO from bus master with for read commands when OVERRIDE_AOOO is set (1).
8	AINNERSHARED	Configures the values to override AINNERSHARED from bus master with for read commands when OVERRIDE_AINNERSHARED is set (1).
6	OVERRIDE_AREQPRIORITY	When set (1), overrides AREQPRIORITY from the bus master for read commands with values configured in AREQPRIORITY register field. When cleared (0), allows AREQPRIORITY from the bus master to be propagated downstream.
5	OVERRIDE_AMEMTYPE	When set (1), overrides AMEMTYPE from the bus master for read commands with values configured in AMEMTYPE register field. When cleared (0), allows AMEMTYPE from the bus master to be propagated downstream.
4	OVERRIDE_ATTRANSIENT	When set (1), overrides ATTRANSIENT from the bus master for read commands with values configured in ATTRANSIENT register field. When cleared (0), allows ATTRANSIENT from the bus master to be propagated downstream.
3	OVERRIDE_ASHARED	When set (1), overrides ASHARED from the bus master for read commands with values configured in ASHARED register field. When cleared (0), allows ASHARED from the bus master to be propagated downstream.
2	OVERRIDE_AREDIRECT	When set (1), overrides AREDIRECT from the bus master for read commands with values configured in AREDIRECT register field. When cleared (0), allows AREDIRECT from the bus master to be propagated downstream.
1	OVERRIDE_AOOO	When set (1), overrides AOOO from the bus master for read commands with values configured in AOOO register field. When cleared (0), allows AOOO from the bus master to be propagated downstream.

BIMC_M_SYS0_MPORT_READ_COMMAND_OVERRIDE (cont.)

Bits	Name	Description
0	OVERWRITE_AINNERSHARE D	When set (1), overrides AINNERSHARED from the bus master for read commands with values configured in AINNERSHARED register field. When cleared (0), allows AINNERSHARED from the bus master to be propagated downstream.

0x00414250 BIMC_M_SYS0_MPORT_WRITE_COMMAND_OVERRIDE**Type:** RW**Clock:** Config & Master Clocks**Reset State:** 0x00000000

Override registers provide the capability to override one or more bus signals to specific values.

BIMC_M_SYS0_MPORT_WRITE_COMMAND_OVERRIDE

Bits	Name	Description
25:24	AREQPRIORITY	Configures the values to override AREQPRIORITY from bus master with for write commands when OVERRIDE_AREQPRIORITY is set (1).
18:16	AMEMTYPE	Configures the values to override AMEMTYPE from bus master with for write commands when OVERRIDE_AMEMTYPE is set (1).
12	ATRANSIENT	Configures the values to override ATRANSIENT from bus master with for write commands when OVERRIDE_ATRANSIENT is set (1).
11	ASHARED	Configures the values to override ASHARED from bus master with for write commands when OVERRIDE_ASHARED is set (1).
10	AREDIRECT	Configures the values to override AREDIRECT from bus master with for write commands when OVERRIDE_AREDIRECT is set (1).
9	A000	Configures the values to override A000 from bus master with for write commands when OVERRIDE_A000 is set (1).
8	AINNERSHARED	Configures the values to override AINNERSHARED from bus master with for write commands when OVERRIDE_AINNERSHARED is set (1).
6	OVERWRITE_AREQPRIORITY	When set (1), overrides AREQPRIORITY from the bus master for write commands with values configured in AREQPRIORITY register field. When cleared (0), allows AREQPRIORITY from the bus master to be propagated downstream.
5	OVERWRITE_AMEMTYPE	When set (1), overrides AMEMTYPE from the bus master for write commands with values configured in AMEMTYPE register field. When cleared (0), allows AMEMTYPE from the bus master to be propagated downstream.

BIMC_M_SYS0_MPORT_WRITE_COMMAND_OVERRIDE (cont.)

Bits	Name	Description
4	OVERRIDE_ATRANSIENT	When set (1), overrides ATRANSIENT from the bus master for write commands with values configured in ATRANSIENT register field. When cleared (0), allows ATRANSIENT from the bus master to be propagated downstream.
3	OVERRIDE_ASHARED	When set (1), overrides ASHARED from the bus master for write commands with values configured in ASHARED register field. When cleared (0), allows ASHARED from the bus master to be propagated downstream.
2	OVERRIDE_AREDIRECT	When set (1), overrides AREDIRECT from the bus master for write commands with values configured in AREDIRECT register field. When cleared (0), allows AREDIRECT from the bus master to be propagated downstream.
1	OVERRIDE_AOOO	When set (1), overrides AOOO from the bus master for write commands with values configured in AOOO register field. When cleared (0), allows AOOO from the bus master to be propagated downstream.
0	OVERRIDE_AINNERSHARE D	When set (1), overrides AINNERSHARED from the bus master for write commands with values configured in AINNERSHARED register field. When cleared (0), allows AINNERSHARED from the bus master to be propagated downstream.

0x00414300 BIMC_M_SYS0_MPORT_BKE_ENABLE**Type:** RW**Clock:** Config & Master Clocks**Reset State:** 0x00000000**BIMC_M_SYS0_MPORT_BKE_ENABLE**

Bits	Name	Description
0	ENABLE	When set (1), enables the book keeping element (BKE) to determine AREQPRIORITY and APRIORITYLVL of commands from the bus master by comparing the amount of bus traffic from the bus master to a target, and determining a health level for the bus master. The health level determines the relative priority of commands from the bus master to allow the bus master to catch-up if it is not getting serviced, or slow down to allow other bus masters to get their share of system resources. When cleared (0), disables the book keeping element and AREQPRIORITY and APRIORITYLVL from the bus master are propagated downstream.

0x00414304 BIMC_M_SYS0_MPORT_BKE_GRANT_PERIOD**Type:** RW**Clock:** Config & Reference Clocks**Reset State:** 0x00000000**BIMC_M_SYS0_MPORT_BKE_GRANT_PERIOD**

Bits	Name	Description
9:0	GRANT_PERIOD	Defines the period over which the book keeping element accumulates grants to the bus master for determining it's health level. Grant period is defined in number of reference timer ticks. Refer configuration of reference timer tick for exact time unit.

0x00414308 BIMC_M_SYS0_MPORT_BKE_GRANT_COUNT**Type:** RW**Clock:** Config & Master Clocks**Reset State:** 0x00000000**BIMC_M_SYS0_MPORT_BKE_GRANT_COUNT**

Bits	Name	Description
15:0	GRANT_COUNT	Defines the target grant count for the bus master. The target grant count is the number of beats of read and write data the bus master is expected to write and read in the grant period and is used as a reference to determine if the bus master is behind or ahead of where it ought to be at any given time.

0x00414320 BIMC_M_SYS0_MPORT_BKE_THRESHOLD_HIGH**Type:** RW**Clock:** Config & Master Clocks**Reset State:** 0x00000000**BIMC_M_SYS0_MPORT_BKE_THRESHOLD_HIGH**

Bits	Name	Description
15:0	THRESHOLD	Defines a high threshold to determine how much behind or ahead a bus master is with reference to where it ought to be at any given time.

0x00414324 BIMC_M_SYS0_MPORT_BKE_THRESHOLD_MEDIUM**Type:** RW**Clock:** Config & Master Clocks**Reset State:** 0x00000000**BIMC_M_SYS0_MPORT_BKE_THRESHOLD_MEDIUM**

Bits	Name	Description
15:0	THRESHOLD	Defines a medium threshold to determine how much behind or ahead a bus master is with reference to where it ought to be at any given time.

0x00414328 BIMC_M_SYS0_MPORT_BKE_THRESHOLD_LOW**Type:** RW**Clock:** Config & Master Clocks**Reset State:** 0x00000000**BIMC_M_SYS0_MPORT_BKE_THRESHOLD_LOW**

Bits	Name	Description
15:0	THRESHOLD	Defines a low threshold to determine how much behind or ahead a bus master is with reference to where it ought to be at any given time.

0x00414340 BIMC_M_SYS0_MPORT_BKE_HEALTH_0**Type:** RW**Clock:** Config & Master Clocks**Reset State:** 0x00000000

BKE health registers configure the request priority and realtime priority level to use for commands from the bus master when it has been determined to be of a certain health.

BIMC_M_SYS0_MPORT_BKE_HEALTH_0

Bits	Name	Description
31	LIMIT_COMMANDS	When set (1), defines that no new commands may be accepted from the bus master as long as it is determined to be at health 0.
9:8	AREQPRIORITY	Defines the request priority of commands from the bus master when it has been determined to be at health 0.
1:0	PRIORITYLVL	Defines the realtime priority level to use for commands from the bus master when it has been determined to be at health 0.

0x00414344 BIMC_M_SYS0_MPORT_BKE_HEALTH_1**Type:** RW**Clock:** Config & Master Clocks**Reset State:** 0x00000000

BKE health registers configure the request priority and realtime priority level to use for commands from the bus master when it has been determined to be of a certain health.

BIMC_M_SYS0_MPORT_BKE_HEALTH_1

Bits	Name	Description
31	LIMIT_COMMANDS	When set (1), defines that no new commands may be accepted from the bus master as long as it is determined to be at health 1.
9:8	AREQPRIORITY	Defines the request priority of commands from the bus master when it has been determined to be at health 1.
1:0	PRIORITYLVL	Defines the realtime priority level to use for commands from the bus master when it has been determined to be at health 1.

0x00414348 BIMC_M_SYS0_MPORT_BKE_HEALTH_2**Type:** RW**Clock:** Config & Master Clocks**Reset State:** 0x00000000

BKE health registers configure the request priority and realtime priority level to use for commands from the bus master when it has been determined to be of a certain health.

BIMC_M_SYS0_MPORT_BKE_HEALTH_2

Bits	Name	Description
31	LIMIT_COMMANDS	When set (1), defines that no new commands may be accepted from the bus master as long as it is determined to be at health 2.
9:8	AREQPRIORITY	Defines the request priority of commands from the bus master when it has been determined to be at health 2.
1:0	PRIORITYLVL	Defines the realtime priority level to use for commands from the bus master when it has been determined to be at health 2.

0x0041434C BIMC_M_SYS0_MPORT_BKE_HEALTH_3**Type:** RW**Clock:** Config & Master Clocks**Reset State:** 0x00000000

BKE health registers configure the request priority and realtime priority level to use for commands from the bus master when it has been determined to be of a certain health. Health 3 is the default health for the bus master.

BIMC_M_SYS0_MPORT_BKE_HEALTH_3

Bits	Name	Description
9:8	AREQPRIORITY	Defines the request priority of commands from the bus master when it has been determined to be at health 3.
1:0	PRIORITYLVL	Defines the realtime priority level to use for commands from the bus master when it has been determined to be at health 3.

0x00414420 BIMC_M_SYS0_MPORT_STATUS_2A

Type: R

Clock: Config, Master & Pipe0 Clocks

Reset State: 0x00000000

Status registers indicate the state of several elements in the masterport, including the number of transactions outstanding/pending waiting for a response, emptiness, fullness of several buffers, etc. The state information is available only as a debug indication and could be delayed by a few clock cycles from when the state is reached to being reflected in the status registers. One or more of the state registers are read live and could capture intermediate values as they change.

BIMC_M_SYS0_MPORT_STATUS_2A

Bits	Name	Description
23:16	WR_PENDING_PRE_BUF	Indicates the number of write transactions that are outstanding/pending waiting for a response from the slave. This transaction count is from before address channel buffers, which interfaces to frontend of the masterport.
7:0	RD_PENDING_PRE_BUF	Indicates the number of read transactions that are outstanding/pending waiting for a response from the slave. This transaction count is from before address channel buffers, which interfaces to frontend of the masterport.

0x00414424 BIMC_M_SYS0_MPORT_STATUS_2B

Type: R

Clock: Config, Master & Pipe0 Clocks

Reset State: 0x00000000

Status registers indicate the state of several elements in the masterport, including the number of transactions outstanding/pending waiting for a response, emptiness, fullness of several buffers, etc. The state information is available only as a debug indication and could be delayed by a few clock

cycles from when the state is reached to being reflected in the status registers. One or more of the state registers are read live and could capture intermediate values as they change.

BIMC_M_SYS0_MPORT_STATUS_2B

Bits	Name	Description
23:16	WR_PENDING_POST_BUF	Indicates the number of write transactions that are outstanding/pending waiting for a response from the slave. This transaction count is from after address channel buffers, which interfaces to core of the interconnect, i.e., arbiter and slaveway.
7:0	RD_PENDING_POST_BUF	Indicates the number of read transactions that are outstanding/pending waiting for a response from the slave. This transaction count is from after address channel buffers, which interfaces to core of the interconnect, i.e., arbiter and slaveway.

0x00414430 BIMC_M_SYS0_MPORT_STATUS_3A

Type: R

Clock: Config, Master & Pipe0 Clocks

Reset State: 0x55111111

Status registers indicate the state of several elements in the masterport, including the number of transactions outstanding/pending waiting for a response, emptiness, fullness of several buffers, etc. The state information is available only as a debug indication and could be delayed by a few clock cycles from when the state is reached to being reflected in the status registers. One or more of the state registers are read live and could capture intermediate values as they change.

BIMC_M_SYS0_MPORT_STATUS_3A

Bits	Name	Description
31	RCB_DATA_RD_FULL	Indicates that the data part of read channel buffer is full when seen from it's read side, which interfaces to frontend of the masterport.
30	RCB_DATA_RD_EMPTY	Indicates that the data part of read channel buffer is empty when seen from it's read side, which interfaces to frontend of the masterport.
29	RCB_CTRL_RD_FULL	Indicates that the control part of read channel buffer is full when seen from it's read side, which interfaces to frontend of the masterport.
28	RCB_CTRL_RD_EMPTY	Indicates that the control part of read channel buffer is empty when seen from it's write side, which interfaces to frontend of the masterport.
27	RCB_DATA_WR_FULL	Indicates that the data part of read channel buffer is full when seen from it's write side, which interfaces to core of the interconnect, i.e., arbiter and slaveway.
26	RCB_DATA_WR_EMPTY	Indicates that the data part of read channel buffer is empty when seen from it's write side, which interfaces to core of the interconnect, i.e., arbiter and slaveway.

BIMC_M_SYS0_MPORT_STATUS_3A (cont.)

Bits	Name	Description
25	RCB_CTRL_WR_FULL	Indicates that the control part of read channel buffer is full when seen from it's write side, which interfaces to core of the interconnect, i.e., arbiter and slaveway.
24	RCB_CTRL_WR_EMPTY	Indicates that the control part of read channel buffer is empty when seen from it's write side, which interfaces to core of the interconnect, i.e., arbiter and slaveway.
21	BCB_RD_FULL	Indicates that the write response channel buffer is full when seen from it's read side, which interfaces to frontend of the masterport.
20	BCB_RD_EMPTY	Indicates that the write response channel buffer is empty when seen from it's read side, which interfaces to frontend of the masterport.
17	BCB_WR_FULL	Indicates that the write response channel buffer is full when seen from it's write side, which interfaces to core of the interconnect, i.e., arbiter and slaveway.
16	BCB_WR_EMPTY	Indicates that the write response channel buffer is empty when seen from it's write side, which interfaces to core of the interconnect, i.e., arbiter and slaveway.
13	WCB_RD_FULL	Indicates that the write channel buffer is full when seen from it's read side, which interfaces to core of the interconnect, i.e., arbiter and slaveway.
12	WCB_RD_EMPTY	Indicates that the write channel buffer is empty when seen from it's read side, which interfaces to core of the interconnect, i.e., arbiter and slaveway.
9	WCB_WR_FULL	Indicates that the write channel buffer is full when seen from it's write side, which interfaces to frontend of the masterport.
8	WCB_WR_EMPTY	Indicates that the write channel buffer is empty when seen from it's write side, which interfaces to frontend of the masterport.
5	ACB_RD_FULL	Indicates that the address channel buffer is full when seen from it's read side, which interfaces to core of the interconnect, i.e., arbiter and slaveway.
4	ACB_RD_EMPTY	Indicates that the address channel buffer is empty when seen from it's read side, which interfaces to core of the interconnect, i.e., arbiter and slaveway.
1	ACB_WR_FULL	Indicates that the address channel buffer is full when seen from it's write side, which interfaces to frontend of the masterport.
0	ACB_WR_EMPTY	Indicates that the address channel buffer is empty when seen from it's write side, which interfaces to frontend of the masterport.

0x00414434 BIMC_M_SYS0_MPORT_STATUS_3B**Type:** R**Clock:** Config, Master & Pipe0 Clocks**Reset State:** 0x00111111

Buffer status register indicate the state of all buffers in the masterport, whether they are empty or full on either side of the buffer. Some buffers carry information from master to core and others carry information from core to master. Both write side and read side status are indicated separately, since they would be not be similar when master clock and core clock are not synchronous to each other.

BIMC_M_SYS0_MPORT_STATUS_3B

Bits	Name	Description
21	RGB_RD_FULL	Indicates that the read gathering buffer is full when seen from it's read side, which interfaces to frontend of the masterport.
20	RGB_RD_EMPTY	Indicates that the read gathering buffer is empty when seen from it's read side, which interfaces to frontend of the masterport.
17	RGB_WR_FULL	Indicates that the read gathering buffer is full when seen from it's write side, which interfaces to core of the interconnect, i.e., arbiter and slaweway.
16	RGB_WR_EMPTY	Indicates that the read gathering buffer is empty when seen from it's write side, which interfaces to core of the interconnect, i.e., arbiter and slaweway.
13	W2AB_RD_FULL	Indicates that the write channel to address channel user signals buffer is full when seen from it's read side, which interfaces to core of the interconnect, i.e., arbiter and slaweway.
12	W2AB_RD_EMPTY	Indicates that the write channel to address channel user signals buffer is empty when seen from it's read side, which interfaces to core of the interconnect, i.e., arbiter and slaweway.
9	W2AB_WR_FULL	Indicates that the write channel to address channel user signals buffer is full when seen from it's write side, which interfaces to frontend of the masterport.
8	W2AB_WR_EMPTY	Indicates that the write channel to address channel user signals buffer is empty when seen from it's write side, which interfaces to frontend of the masterport.
5	A2WB_RD_FULL	Indicates that the address channel to write channel user signals buffer is full when seen from it's read side, which interfaces to core of the interconnect, i.e., arbiter and slaweway.
4	A2WB_RD_EMPTY	Indicates that the address channel to write channel user signalsbuffer is empty when seen from it's read side, which interfaces to core of the interconnect, i.e., arbiter and slaweway.
1	A2WB_WR_FULL	Indicates that the address channel to write channel user signals buffer is full when seen from it's write side, which interfaces to frontend of the masterport.

BIMC_M_SYS0_MPORT_STATUS_3B (cont.)

Bits	Name	Description
0	A2WB_WR_EMPTY	Indicates that the address channel to write channel user signals buffer is empty when seen from it's write side, which interfaces to frontend of the masterport.

2.13 BIMC_M_TCU0_MPORT

BRIC Masterport

2.13.1 BIMC_M_TCU0_MPORT_BRIC_MASTERPORT

BRIC Masterport

0x0041C000 BIMC_M_TCU0_MPORT_COMPONENT_INFO**Type:** R**Clock:** Config Clock**Reset State:** 0x00000102

Component information registers provide discoverability information on the type of component and instance of each component. These registers are always the very first register in each component's configuration address space and allow for a software routine to walk the configuration address space in steps of 4KB to identify the location of each configuration register space and to which component each 4KB configures.

BIMC_M_TCU0_MPORT_COMPONENT_INFO

Bits	Name	Description
23:16	INSTANCE	Indicates the instance number of this masterport.
15:8	SUB_TYPE	Identifies this configuration space as belonging to a masterport 0x1: MASTERPORT
7:0	TYPE	Identifies this configuration space as belonging to a component that interfaces to a bus master. 0x2: MASTERPORT

0x0041C020 BIMC_M_TCU0_MPORT_CONFIGURATION_INFO_0**Type:** R**Clock:** Config Clock**Reset State:** 0x01000000

Configuration information registers provide information on the capability and configuration of each masterport instance. See register field descriptions for more details.

BIMC_M_TCU0_MPORT_CONFIGURATION_INFO_0

Bits	Name	Description
31:24	SYNC_MODE	Indicates the clock relationship between the clock used in the master side and the clock used in the core side of the masterport. When ASYNC, identifies the master clock and core clock as unrelated and asynchronous to each other. When SYNC, identifies the master clock and core clock as synchronous to each other. When ISOSYNC, identifies the master clock and core clock as related, but of different frequencies, which are of integer ratios, i.e., they are iso-synchronous to each other. 0x0: ASYNC 0x1: SYNC 0x2: ISOSYNC
15:8	CONNECTION_TYPE	Indicates the type of master connecting to this masterport. When DIRECT, indicates that a bus master is directly connected to this masterport. When CASCADE, indicates that bus masters connect to this masterport through another level of interconnect. 0x0: DIRECT 0x1: CASCADE
7:0	FUNCTIONALITY	Indicates the functionality and type of bus traffic handled by this masterport. When INORDER, identifies this masterport as optimized for in-order bus traffic. When OOO_ONLY, identifies this masterport as optimized for out-of-order bus traffic, where a single master is capable of handling responses return in any order. When REORDER, identifies this masterport as having a reorder buffer for re-ordering responses to command order, while allowing bus slaves to return responses in any order they desire. When STUB, identifies this masterport has a stub. 0x0: STUB 0x1: OOO_ONLY 0x2: INORDER 0x3: REORDER

0x0041C030 BIMC_M_TCU0_MPORT_CONFIGURATION_INFO_1

Type: R

Clock: Config Clock

Reset State: 0x00000000

Configuration information registers provide information on the capability and configuration of each masterport instance. See register field descriptions for more details.

BIMC_M_TCU0_MPORT_CONFIGURATION_INFO_1

Bits	Name	Description
31:0	SWAY_CONNECTIVITY	Indicates slaweways that this masterport can route commands to. 1-bit per slaweway. When a specific bit is set (1), indicates that commands can be routed to the slaweway corresponding to this bit. WHEN a specific bit is clear (0), indicates that commands are not routed to the slaweway corresponding to this bit.

0x0041C040 BIMC_M_TCU0_MPORT_CONFIGURATION_INFO_2A**Type:** R**Clock:** Config Clock**Reset State:** 0x00000000

Configuration information registers provide information on the capability and configuration of each masterport instance. See register field descriptions for more details.

BIMC_M_TCU0_MPORT_CONFIGURATION_INFO_2A

Bits	Name	Description
31:16	M_DATA_WIDTH	Indicates the width of read and write data from the bus master.
15:8	M_TID_WIDTH	Indicates the width of transfer IDs from the bus master.
7:0	M_MID_WIDTH	Indicates the width of master ID from the bus master (includes any bus and port IDs if bus master is an interconnect.)

0x0041C044 BIMC_M_TCU0_MPORT_CONFIGURATION_INFO_2B**Type:** R**Clock:** Config Clock**Reset State:** 0x00000000

Configuration information registers provide information on the capability and configuration of each masterport instance. See register field descriptions for more details.

BIMC_M_TCU0_MPORT_CONFIGURATION_INFO_2B

Bits	Name	Description
15:0	C_DATA_WIDTH	Indicates the width of read and write data used in core of this masterport.

0x0041C050 BIMC_M_TCU0_MPORT_CONFIGURATION_INFO_3**Type:** R**Clock:** Config Clock**Reset State:** 0x00000000

Configuration information registers provide information on the capability and configuration of each masterport instance. See register field descriptions for more details.

BIMC_M_TCU0_MPORT_CONFIGURATION_INFO_3

Bits	Name	Description
31:24	RCH_DEPTH	Indicates the depth of read channel buffers in this masterport. A beat of read data occupies one entry in the read channel buffer, until accepted by the bus master.
23:16	BCH_DEPTH	Indicates the depth of write response channel buffers in this masterport. Each write response occupies one entry in the write response channel buffer, until accepted by the bus master.
15:8	WCH_DEPTH	Indicates the depth of write channel buffers in this masterport. A beat of write data occupies one entry in the write channel buffer, until accepted by the arbiter.
7:0	ACH_DEPTH	Indicates the depth of address channel buffers in this masterport. Each command occupies one entry in the address channel buffer, until accepted by the arbiter.

0x0041C060 BIMC_M_TCU0_MPORT_CONFIGURATION_INFO_4**Type:** R**Clock:** Config Clock**Reset State:** 0x00000000

Configuration information registers provide information on the capability and configuration of each masterport instance. See register field descriptions for more details.

BIMC_M_TCU0_MPORT_CONFIGURATION_INFO_4

Bits	Name	Description
15:8	REORDER_BUFFER_DEPTH	Indicates the number of read data beats that can be re-ordered in this masterport (if the masterport provides response re-ordering functionality.)
7:0	REORDER_TABLE_DEPTH	Indicates the number of read and write commands whose responses can be re-ordered in this masterport (if the masterport provides responses re-ordering functionality.)

0x0041C070 BIMC_M_TCU0_MPORT_CONFIGURATION_INFO_5A**Type:** R**Clock:** Config Clock**Reset State:** 0x00000000

Configuration information registers provide information on the capability and configuration of each masterport instance. See register field descriptions for more details.

BIMC_M_TCU0_MPORT_CONFIGURATION_INFO_5A

Bits	Name	Description
31:0	ACH_PIPELINE_STAGES	Indicates pipeline stages enabled on the address channel. Each digit identifies a specific pipeline stage in the masterport, where '0' indicates no pipeline stage, '1' indicates 1-deep pipeline stage where forward path is registered, return path is fed through, '2' indicates 2-deep pipeline stage where both forward and return path are pipelined. Digit 3 identifies pipelining on the master to masterport interface, digit 2 identifies pipelining after address decode, digit 1 identifies pipelining in the address channel buffer and digit 0 identifies pipelining on the masterport to arbiter interface.

0x0041C074 BIMC_M_TCU0_MPORT_CONFIGURATION_INFO_5B**Type:** R**Clock:** Config Clock**Reset State:** 0x00000000

Configuration information registers provide information on the capability and configuration of each masterport instance. See register field descriptions for more details.

BIMC_M_TCU0_MPORT_CONFIGURATION_INFO_5B

Bits	Name	Description
31:0	WCH_PIPELINE_STAGES	Indicates pipeline stages enabled on the write channel. Each digit identifies a specific pipeline stage in the masterport, where '0' indicates no pipeline stage, '1' indicates 1-deep pipeline stage where forward path is registered, return path is fed through, '2' indicates 2-deep pipeline stage where both forward and return path are pipelined. Digit 3 identifies pipelining on the master to masterport interface, digit 2 identifies pipelining after address decode, digit 1 identifies pipelining in the write channel buffer and digit 0 identifies pipelining on the masterport to arbiter interface.

0x0041C078 BIMC_M_TCU0_MPORT_CONFIGURATION_INFO_5C**Type:** R**Clock:** Config Clock**Reset State:** 0x00000000

Configuration information registers provide information on the capability and configuration of each masterport instance. See register field descriptions for more details.

BIMC_M_TCU0_MPORT_CONFIGURATION_INFO_5C

Bits	Name	Description
31:0	BCH_PIPELINE_STAGES	Indicates pipeline stages enabled on the write response channel. Each digit identifies a specific pipeline stage in the masterport, where '0' indicates no pipeline stage, '1' indicates 1-deep pipeline stage where forward path is registered, return path is fed through, '2' indicates 2-deep pipeline stage where both forward and return path are pipelined. Digit 2 identifies pipelining on the slaveway to masterport interface after arbitration of write response channels from multiple slaveways, digit 1 identifies pipelining in the write response channel buffer and digit 0 identifies pipelining on the masterport to master interface.

0x0041C07C BIMC_M_TCU0_MPORT_CONFIGURATION_INFO_5D**Type:** R**Clock:** Config Clock**Reset State:** 0x00000000

Configuration information registers provide information on the capability and configuration of each masterport instance. See register field descriptions for more details.

BIMC_M_TCU0_MPORT_CONFIGURATION_INFO_5D

Bits	Name	Description
31:0	RCH_PIPELINE_STAGES	Indicates pipeline stages enabled on the read channel. Each digit identifies a specific pipeline stage in the masterport, where '0' indicates no pipeline stage, '1' indicates 1-deep pipeline stage where forward path is registered, return path is fed through, '2' indicates 2-deep pipeline stage where both forward and return path are pipelined. Digit 2 identifies pipelining on the slaveway to masterport interface after arbitration of read channels from multiple slaveways, digit 1 identifies pipelining in the read channel buffer and digit 0 identifies pipelining on the masterport to master interface.

0x0041C100 BIMC_M_TCU0_MPORT_INTERRUPT_STATUS

Type: R
Clock: Config Clock
Reset State: 0x00000000

Interrupt status registers provide indication of which interrupts from this masterport require servicing.

BIMC_M_TCU0_MPORT_INTERRUPT_STATUS

Bits	Name	Description
1:0	RFU	Reserved for future use

0x0041C108 BIMC_M_TCU0_MPORT_INTERRUPT_CLEAR

Type: W
Clock: Config Clock
Reset State: 0x00000000

Interrupt clear registers provide the mechanism to turn off interrupt sources once they have been acknowledged or serviced. A write of '1' to the bits corresponding to the interrupts indicated in interrupt status is required to turn off interrupts.

BIMC_M_TCU0_MPORT_INTERRUPT_CLEAR

Bits	Name	Description
1:0	RFU	Reserved for future use

0x0041C10C BIMC_M_TCU0_MPORT_INTERRUPT_ENABLE

Type: RW
Clock: Config Clock
Reset State: 0x00000000

Interrupt enable registers provide control over which interrupts are allowed to be propagated to an interrupt controller for servicing.

BIMC_M_TCU0_MPORT_INTERRUPT_ENABLE

Bits	Name	Description
1:0	RFU	Reserved for future use

0x0041C200 BIMC_M_TCU0_MPORT_CLOCK_CTRL**Type:** RW**Clock:** Config, Core & Master Clocks**Reset State:** 0x00000003**BIMC_M_TCU0_MPORT_CLOCK_CTRL**

Bits	Name	Description
1	MASTER_CLOCK_GATING_EN	When set (1), enables activity based clock gating of all registers clocked by master clock. Activity based clock gating uses architectural clock gates to reduce dynamic power consumption by gating clocks during periods of in-activity. When cleared (0), disables all activity based clock gating by forcing all architectural clock gates to not gate clocks even during periods of in-activity.
0	CORE_CLOCK_GATING_EN	When set (1), enables activity based clock gating of all registers clocked by core clock. Activity based clock gating uses architectural clock gates to reduce dynamic power consumption by gating clocks during periods of in-activity. When cleared (0), disables all activity based clock gating by forcing all architectural clock gates to not gate clocks even during periods of in-activity.

0x0041C208 BIMC_M_TCU0_MPORT_CDC_CTRL**Type:** RW**Clock:** Config, Core & Master Clocks**Reset State:** 0x00000000

Clock domain crossing (CDC) control register provides configuration for clock domain crossing logic.

BIMC_M_TCU0_MPORT_CDC_CTRL

Bits	Name	Description
1	MASTER_CLOCK_DEMETA_SEL	When cleared (0), selects DEMETA synchronizer that requires atleast 3 clock edges (two rising and one falling). When set (1), selects DEMETA synchronizer that requires atleast 2 clock edges (one falling and one rising) to synchronize to master clock in all buffers.
0	CORE_CLOCK_DEMETA_SEL	When cleared (0), selects DEMETA synchronizer that requires atleast 3 clock edges (two rising and one falling). When set (1), selects DEMETA synchronizer that requires atleast 2 clock edges (one falling and one rising) to synchronize to core clock in all buffers.

0x0041C210 BIMC_M_TCU0_MPORT_MODE**Type:** RW**Clock:** Config, Core & Master Clocks**Reset State:** 0x00000012**BIMC_M_TCU0_MPORT_MODE**

Bits	Name	Description
31:28	WR_GATHER_BEATS	Configures the number of write data beats to gather/buffer in write channel buffer before allowing the corresponding write command to be routed to arbiters. When configured to 0, write commands are held until the first beat of write data has been buffered. When configured to 1, write commands are held until two beats of write data have been buffered and so on. When configured with values larger than the number of beats in a write command, the command is held until the last beat has been buffered. When configured with values larger than the depth of write channel buffers, the command is held until all entries in the write channel buffer have been occupied.
19:16	RD_GATHER_BEATS	Configures the number of read data beats to gather/buffer in read channel buffer before allowing the read data to proceed to the master. When configured to 0, read data is allowed to proceed to the master as soon as the first beat is buffered in the read channel buffer. When configured to 1, read data is allowed to proceed to the master as soon as two beats of read data have been buffered and so on. When configured with values larger than the number of beats in the read burst, read data is held until the last beat has been buffered. When configured with values larger than the depth of read channel buffers, read data is held until all entries in the read channel buffer have been occupied.
15	RCSSH	Redirect Cacheable System SHareable When set (1), enables redirection of all cacheable transactions that are system shareable to the redirect slaweway. Note: Only masters connecting using AMBA 4 ACE protocol are able to indicate system shareability for transactions.
14	RCOSH	Redirect Cacheable Outer SHareable When set (1), enables redirection of all cacheable transactions that are outer shareable to the redirect slaweway.
13	RCISH	Redirect Cacheable Inner SHareable When set (1), enables redirection of all cacheable transactions that are inner shareable to the redirect slaweway.
12	RCNSH	Redirect Cacheable Non-SHareable When set (1), enables redirection of all cacheable transactions that are non-shareable to the redirect slaweway.
11	RNCSSH	Redirect Non-Cacheable System SHareable When set (1), enables redirection of all non-cacheable transactions that are system shareable to the redirect slaweway. Note: Only masters connecting using AMBA 4 ACE protocol are able to indicate system shareability for transactions.

BIMC_M_TCU0_MPORT_MODE (cont.)

Bits	Name	Description
10	RNCOSH	Redirect Non-Cacheable Outer SHareable When set (1), enables redirection of all non-cacheable transactions that are outer shareable to the redirect slaway.
9	RNCISH	Redirect Non-Cacheable Inner SHareable When set (1), enables redirection of all non-cacheable transactions that are inner shareable to the redirect slaway.
8	RNCNSH	Redirect Non-Cacheable Non-SHareable When set (1), enables redirection of all non-cacheable transactions that are non-shareable to the redirect slaway.
4	NARROW_WRITES	When set (1), enables converting of 1-beat bus-width sized writes down to 32-bit ASIZE based on write strobes. It is recommended to be enabled for directly connected CPU masters. When cleared (0), disables conversion and all writes are forwarded downstream as generated by the master.
2	FIODV	Force In-Order Device When set (1), forces commands to device memory to be executed one at a time. In other words, commands to device memory are held until all previous commands have completed (responses have been received at the masterport) and all commands after the device command are held until the device completes (response has been received at the masterport.)
1	FIOSO	Force In-Order Strongly-Ordered When set (1), forces commands to strongly ordered memory to be executed one at a time. In other words, commands to strongly ordered memory are held until all previous commands have completed (responses have been received at the masterport) and all commands after the strongly ordered command are held until the strongly ordered completes (response has been received at the masterport.)
0	ORDERING_MODEL	Configures how commands to memories of type Strongly ordered and Device (SoDv) are handled in the masterport. When configured as STRICT, all commands to SoDv memory are held until all previous commands have completed (responses have been received at the masterport) and all commands after the SoDv command are held until the SoDv completes (i.e. response has been received at the masterport.) When configured as RELAXED, all commands to SoDv memory are treated as accessing Normal memory. 0x0: RELAXED 0x1: STRICT

0x0041C220 BIMC_M_TCU0_MPORT_WR_WAY_CROSSING**Type:** RW**Clock:** Config & Master Clocks**Reset State:** 0x00000000

Configures the level of strictness or relaxedness of way crossing checks for writes.

BIMC_M_TCU0_MPORT_WR_WAY_CROSSING

Bits	Name	Description
7:0	REG	1-bit per slaveway. When 2 or more bits are set (1), allows the corresponding slaveways to be treated as one slaveway when write commands are considered as way crossing, ie. write commands to these 2 (or more) slaveways are not treated as way crossing when one write command targets one slaveway and the next write command targets the second slaveway. Not treating write commands as way crossing allows the write commands to go to the slaveway ahead of the write data for potentially better overall system performance. When all bits are set (1), disables all way crossing checks. When all bits are cleared (0), way crossing is strict with each way crossing write command held until write data for all previous writes have been routed to their corresponding slaveways.

0x0041C230 BIMC_M_TCU0_MPORT_PRIORITYLVL_OVERRIDE

Type: RW

Clock: Config & Master Clocks

Reset State: 0x00000000

Override registers provide the capability to override one or more bus signals to specific values.

BIMC_M_TCU0_MPORT_PRIORITYLVL_OVERRIDE

Bits	Name	Description
9:8	PRIORITYLVL	Configures the values to override APRIORITYLVL from bus master with when OVERRIDE_PRIORITYLVL is set (1).
0	OVERRIDE_PRIORITYLVL	When set (1), overrides APRIORITYLVL from the bus master with values configured in PRIORITYLVL register field. When cleared (0), allows APRIORITYLVL from the bus master to be propagated downstream.

0x0041C240 BIMC_M_TCU0_MPORT_READ_COMMAND_OVERRIDE

Type: RW

Clock: Config & Master Clocks

Reset State: 0x00000000

Override registers provide the capability to override one or more bus signals to specific values.

BIMC_M_TCU0_MPORT_READ_COMMAND_OVERRIDE

Bits	Name	Description
25:24	AREQPRIORITY	Configures the values to override AREQPRIORITY from bus master with for read commands when OVERRIDE_AREQPRIORITY is set (1).
18:16	AMEMTYPE	Configures the values to override AMEMTYPE from bus master with for read commands when OVERRIDE_AMEMTYPE is set (1).
12	ATransient	Configures the values to override ATRANSIENT from bus master with for read commands when OVERRIDE_ATransient is set (1).
11	AShared	Configures the values to override ASHARED from bus master with for read commands when OVERRIDE_ASHARED is set (1).
10	AREDIRECT	Configures the values to override AREDIRECT from bus master with for read commands when OVERRIDE_AREDIRECT is set (1).
9	A000	Configures the values to override A000 from bus master with for read commands when OVERRIDE_A000 is set (1).
8	AINNERSHARED	Configures the values to override AINNERSHARED from bus master with for read commands when OVERRIDE_AINNERSHARED is set (1).
6	OVERRIDE_AREQPRIORITY	When set (1), overrides AREQPRIORITY from the bus master for read commands with values configured in AREQPRIORITY register field. When cleared (0), allows AREQPRIORITY from the bus master to be propagated downstream.
5	OVERRIDE_AMEMTYPE	When set (1), overrides AMEMTYPE from the bus master for read commands with values configured in AMEMTYPE register field. When cleared (0), allows AMEMTYPE from the bus master to be propagated downstream.
4	OVERRIDE_ATransient	When set (1), overrides ATRANSIENT from the bus master for read commands with values configured in ATRANSIENT register field. When cleared (0), allows ATRANSIENT from the bus master to be propagated downstream.
3	OVERRIDE_AShared	When set (1), overrides ASHARED from the bus master for read commands with values configured in ASHARED register field. When cleared (0), allows ASHARED from the bus master to be propagated downstream.
2	OVERRIDE_AREDIRECT	When set (1), overrides AREDIRECT from the bus master for read commands with values configured in AREDIRECT register field. When cleared (0), allows AREDIRECT from the bus master to be propagated downstream.
1	OVERRIDE_A000	When set (1), overrides A000 from the bus master for read commands with values configured in A000 register field. When cleared (0), allows A000 from the bus master to be propagated downstream.

BIMC_M_TCU0_MPORT_READ_COMMAND_OVERRIDE (cont.)

Bits	Name	Description
0	OVERRIDE_AINNERSHARE D	When set (1), overrides AINNERSHARED from the bus master for read commands with values configured in AINNERSHARED register field. When cleared (0), allows AINNERSHARED from the bus master to be propagated downstream.

0x0041C250 BIMC_M_TCU0_MPORT_WRITE_COMMAND_OVERRIDE**Type:** RW**Clock:** Config & Master Clocks**Reset State:** 0x00000000

Override registers provide the capability to override one or more bus signals to specific values.

BIMC_M_TCU0_MPORT_WRITE_COMMAND_OVERRIDE

Bits	Name	Description
25:24	AREQPRIORITY	Configures the values to override AREQPRIORITY from bus master with for write commands when OVERRIDE_AREQPRIORITY is set (1).
18:16	AMEMTYPE	Configures the values to override AMEMTYPE from bus master with for write commands when OVERRIDE_AMEMTYPE is set (1).
12	ATRANSIENT	Configures the values to override ATRANSIENT from bus master with for write commands when OVERRIDE_ATRANSIENT is set (1).
11	ASHARED	Configures the values to override ASHARED from bus master with for write commands when OVERRIDE_ASHARED is set (1).
10	AREDIRECT	Configures the values to override AREDIRECT from bus master with for write commands when OVERRIDE_AREDIRECT is set (1).
9	AOOO	Configures the values to override AOOO from bus master with for write commands when OVERRIDE_AOOO is set (1).
8	AINNERSHARED	Configures the values to override AINNERSHARED from bus master with for write commands when OVERRIDE_AINNERSHARED is set (1).
6	OVERRIDE_AREQPRIORITY	When set (1), overrides AREQPRIORITY from the bus master for write commands with values configured in AREQPRIORITY register field. When cleared (0), allows AREQPRIORITY from the bus master to be propagated downstream.
5	OVERRIDE_AMEMTYPE	When set (1), overrides AMEMTYPE from the bus master for write commands with values configured in AMEMTYPE register field. When cleared (0), allows AMEMTYPE from the bus master to be propagated downstream.

BIMC_M_TCU0_MPORT_WRITE_COMMAND_OVERRIDE (cont.)

Bits	Name	Description
4	OVERWRITE_ATRANSIENT	When set (1), overrides ATRANSIENT from the bus master for write commands with values configured in ATRANSIENT register field. When cleared (0), allows ATRANSIENT from the bus master to be propagated downstream.
3	OVERWRITE_ASHARED	When set (1), overrides ASHARED from the bus master for write commands with values configured in ASHARED register field. When cleared (0), allows ASHARED from the bus master to be propagated downstream.
2	OVERWRITE_AREDIRECT	When set (1), overrides AREDIRECT from the bus master for write commands with values configured in AREDIRECT register field. When cleared (0), allows AREDIRECT from the bus master to be propagated downstream.
1	OVERWRITE_AOOO	When set (1), overrides AOOO from the bus master for write commands with values configured in AOOO register field. When cleared (0), allows AOOO from the bus master to be propagated downstream.
0	OVERWRITE_AINNERSHARE D	When set (1), overrides AINNERSHARED from the bus master for write commands with values configured in AINNERSHARED register field. When cleared (0), allows AINNERSHARED from the bus master to be propagated downstream.

0x0041C300 BIMC_M_TCU0_MPORT_BKE_ENABLE**Type:** RW**Clock:** Config & Master Clocks**Reset State:** 0x00000000**BIMC_M_TCU0_MPORT_BKE_ENABLE**

Bits	Name	Description
0	ENABLE	When set (1), enables the book keeping element (BKE) to determine AREQPRIORITY and APRIORITYLVL of commands from the bus master by comparing the amount of bus traffic from the bus master to a target, and determining a health level for the bus master. The health level determines the relative priority of commands from the bus master to allow the bus master to catch-up if it is not getting serviced, or slow down to allow other bus masters to get their share of system resources. When cleared (0), disables the book keeping element and AREQPRIORITY and APRIORITYLVL from the bus master are propagated downstream.

0x0041C304 BIMC_M_TCU0_MPORT_BKE_GRANT_PERIOD**Type:** RW**Clock:** Config & Reference Clocks**Reset State:** 0x00000000**BIMC_M_TCU0_MPORT_BKE_GRANT_PERIOD**

Bits	Name	Description
9:0	GRANT_PERIOD	Defines the period over which the book keeping element accumulates grants to the bus master for determining it's health level. Grant period is defined in number of reference timer ticks. Refer configuration of reference timer tick for exact time unit.

0x0041C308 BIMC_M_TCU0_MPORT_BKE_GRANT_COUNT**Type:** RW**Clock:** Config & Master Clocks**Reset State:** 0x00000000**BIMC_M_TCU0_MPORT_BKE_GRANT_COUNT**

Bits	Name	Description
15:0	GRANT_COUNT	Defines the target grant count for the bus master. The target grant count is the number of beats of read and write data the bus master is expected to write and read in the grant period and is used as a reference to determine if the bus master is behind or ahead of where it ought to be at any given time.

0x0041C320 BIMC_M_TCU0_MPORT_BKE_THRESHOLD_HIGH**Type:** RW**Clock:** Config & Master Clocks**Reset State:** 0x00000000**BIMC_M_TCU0_MPORT_BKE_THRESHOLD_HIGH**

Bits	Name	Description
15:0	THRESHOLD	Defines a high threshold to determine how much behind or ahead a bus master is with reference to where it ought to be at any given time.

0x0041C324 BIMC_M_TCU0_MPORT_BKE_THRESHOLD_MEDIUM**Type:** RW**Clock:** Config & Master Clocks**Reset State:** 0x00000000**BIMC_M_TCU0_MPORT_BKE_THRESHOLD_MEDIUM**

Bits	Name	Description
15:0	THRESHOLD	Defines a medium threshold to determine how much behind or ahead a bus master is with reference to where it ought to be at any given time.

0x0041C328 BIMC_M_TCU0_MPORT_BKE_THRESHOLD_LOW**Type:** RW**Clock:** Config & Master Clocks**Reset State:** 0x00000000**BIMC_M_TCU0_MPORT_BKE_THRESHOLD_LOW**

Bits	Name	Description
15:0	THRESHOLD	Defines a low threshold to determine how much behind or ahead a bus master is with reference to where it ought to be at any given time.

0x0041C340 BIMC_M_TCU0_MPORT_BKE_HEALTH_0**Type:** RW**Clock:** Config & Master Clocks**Reset State:** 0x00000000

BKE health registers configure the request priority and realtime priority level to use for commands from the bus master when it has been determined to be of a certain health.

BIMC_M_TCU0_MPORT_BKE_HEALTH_0

Bits	Name	Description
31	LIMIT_COMMANDS	When set (1), defines that no new commands may be accepted from the bus master as long as it is determined to be at health 0.
9:8	AREQPRIORITY	Defines the request priority of commands from the bus master when it has been determined to be at health 0.
1:0	PRIORITYLVL	Defines the realtime priority level to use for commands from the bus master when it has been determined to be at health 0.

0x0041C344 BIMC_M_TCU0_MPORT_BKE_HEALTH_1**Type:** RW**Clock:** Config & Master Clocks**Reset State:** 0x00000000

BKE health registers configure the request priority and realtime priority level to use for commands from the bus master when it has been determined to be of a certain health.

BIMC_M_TCU0_MPORT_BKE_HEALTH_1

Bits	Name	Description
31	LIMIT_COMMANDS	When set (1), defines that no new commands may be accepted from the bus master as long as it is determined to be at health 1.
9:8	AREQPRIORITY	Defines the request priority of commands from the bus master when it has been determined to be at health 1.
1:0	PRIORITYLVL	Defines the realtime priority level to use for commands from the bus master when it has been determined to be at health 1.

0x0041C348 BIMC_M_TCU0_MPORT_BKE_HEALTH_2**Type:** RW**Clock:** Config & Master Clocks**Reset State:** 0x00000000

BKE health registers configure the request priority and realtime priority level to use for commands from the bus master when it has been determined to be of a certain health.

BIMC_M_TCU0_MPORT_BKE_HEALTH_2

Bits	Name	Description
31	LIMIT_COMMANDS	When set (1), defines that no new commands may be accepted from the bus master as long as it is determined to be at health 2.
9:8	AREQPRIORITY	Defines the request priority of commands from the bus master when it has been determined to be at health 2.
1:0	PRIORITYLVL	Defines the realtime priority level to use for commands from the bus master when it has been determined to be at health 2.

0x0041C34C BIMC_M_TCU0_MPORT_BKE_HEALTH_3**Type:** RW**Clock:** Config & Master Clocks**Reset State:** 0x00000000

BKE health registers configure the request priority and realtime priority level to use for commands from the bus master when it has been determined to be of a certain health. Health 3 is the default health for the bus master.

BIMC_M_TCU0_MPORT_BKE_HEALTH_3

Bits	Name	Description
9:8	AREQPRIORITY	Defines the request priority of commands from the bus master when it has been determined to be at health 3.
1:0	PRIORITYLVL	Defines the realtime priority level to use for commands from the bus master when it has been determined to be at health 3.

0x0041C420 BIMC_M_TCU0_MPORT_STATUS_2A

Type: R

Clock: Config, Master & Pipe0 Clocks

Reset State: 0x00000000

Status registers indicate the state of several elements in the masterport, including the number of transactions outstanding/pending waiting for a response, emptiness, fullness of several buffers, etc. The state information is available only as a debug indication and could be delayed by a few clock cycles from when the state is reached to being reflected in the status registers. One or more of the state registers are read live and could capture intermediate values as they change.

BIMC_M_TCU0_MPORT_STATUS_2A

Bits	Name	Description
23:16	WR_PENDING_PRE_BUF	Indicates the number of write transactions that are outstanding/pending waiting for a response from the slave. This transaction count is from before address channel buffers, which interfaces to frontend of the masterport.
7:0	RD_PENDING_PRE_BUF	Indicates the number of read transactions that are outstanding/pending waiting for a response from the slave. This transaction count is from before address channel buffers, which interfaces to frontend of the masterport.

0x0041C424 BIMC_M_TCU0_MPORT_STATUS_2B

Type: R

Clock: Config, Master & Pipe0 Clocks

Reset State: 0x00000000

Status registers indicate the state of several elements in the masterport, including the number of transactions outstanding/pending waiting for a response, emptiness, fullness of several buffers, etc. The state information is available only as a debug indication and could be delayed by a few clock

cycles from when the state is reached to being reflected in the status registers. One or more of the state registers are read live and could capture intermediate values as they change.

BIMC_M_TCU0_MPORT_STATUS_2B

Bits	Name	Description
23:16	WR_PENDING_POST_BUF	Indicates the number of write transactions that are outstanding/pending waiting for a response from the slave. This transaction count is from after address channel buffers, which interfaces to core of the interconnect, i.e., arbiter and slaveway.
7:0	RD_PENDING_POST_BUF	Indicates the number of read transactions that are outstanding/pending waiting for a response from the slave. This transaction count is from after address channel buffers, which interfaces to core of the interconnect, i.e., arbiter and slaveway.

0x0041C430 BIMC_M_TCU0_MPORT_STATUS_3A

Type: R

Clock: Config, Master & Pipe0 Clocks

Reset State: 0x55111111

Status registers indicate the state of several elements in the masterport, including the number of transactions outstanding/pending waiting for a response, emptiness, fullness of several buffers, etc. The state information is available only as a debug indication and could be delayed by a few clock cycles from when the state is reached to being reflected in the status registers. One or more of the state registers are read live and could capture intermediate values as they change.

BIMC_M_TCU0_MPORT_STATUS_3A

Bits	Name	Description
31	RCB_DATA_RD_FULL	Indicates that the data part of read channel buffer is full when seen from it's read side, which interfaces to frontend of the masterport.
30	RCB_DATA_RD_EMPTY	Indicates that the data part of read channel buffer is empty when seen from it's read side, which interfaces to frontend of the masterport.
29	RCB_CTRL_RD_FULL	Indicates that the control part of read channel buffer is full when seen from it's read side, which interfaces to frontend of the masterport.
28	RCB_CTRL_RD_EMPTY	Indicates that the control part of read channel buffer is empty when seen from it's write side, which interfaces to frontend of the masterport.
27	RCB_DATA_WR_FULL	Indicates that the data part of read channel buffer is full when seen from it's write side, which interfaces to core of the interconnect, i.e., arbiter and slaveway.
26	RCB_DATA_WR_EMPTY	Indicates that the data part of read channel buffer is empty when seen from it's write side, which interfaces to core of the interconnect, i.e., arbiter and slaveway.

BIMC_M_TCU0_MPORT_STATUS_3A (cont.)

Bits	Name	Description
25	RCB_CTRL_WR_FULL	Indicates that the control part of read channel buffer is full when seen from it's write side, which interfaces to core of the interconnect, i.e., arbiter and slaveway.
24	RCB_CTRL_WR_EMPTY	Indicates that the control part of read channel buffer is empty when seen from it's write side, which interfaces to core of the interconnect, i.e., arbiter and slaveway.
21	BCB_RD_FULL	Indicates that the write response channel buffer is full when seen from it's read side, which interfaces to frontend of the masterport.
20	BCB_RD_EMPTY	Indicates that the write response channel buffer is empty when seen from it's read side, which interfaces to frontend of the masterport.
17	BCB_WR_FULL	Indicates that the write response channel buffer is full when seen from it's write side, which interfaces to core of the interconnect, i.e., arbiter and slaveway.
16	BCB_WR_EMPTY	Indicates that the write response channel buffer is empty when seen from it's write side, which interfaces to core of the interconnect, i.e., arbiter and slaveway.
13	WCB_RD_FULL	Indicates that the write channel buffer is full when seen from it's read side, which interfaces to core of the interconnect, i.e., arbiter and slaveway.
12	WCB_RD_EMPTY	Indicates that the write channel buffer is empty when seen from it's read side, which interfaces to core of the interconnect, i.e., arbiter and slaveway.
9	WCB_WR_FULL	Indicates that the write channel buffer is full when seen from it's write side, which interfaces to frontend of the masterport.
8	WCB_WR_EMPTY	Indicates that the write channel buffer is empty when seen from it's write side, which interfaces to frontend of the masterport.
5	ACB_RD_FULL	Indicates that the address channel buffer is full when seen from it's read side, which interfaces to core of the interconnect, i.e., arbiter and slaveway.
4	ACB_RD_EMPTY	Indicates that the address channel buffer is empty when seen from it's read side, which interfaces to core of the interconnect, i.e., arbiter and slaveway.
1	ACB_WR_FULL	Indicates that the address channel buffer is full when seen from it's write side, which interfaces to frontend of the masterport.
0	ACB_WR_EMPTY	Indicates that the address channel buffer is empty when seen from it's write side, which interfaces to frontend of the masterport.

0x0041C434 BIMC_M_TCU0_MPORT_STATUS_3B**Type:** R**Clock:** Config, Master & Pipe0 Clocks**Reset State:** 0x00111111

Buffer status register indicate the state of all buffers in the masterport, whether they are empty or full on either side of the buffer. Some buffers carry information from master to core and others carry information from core to master. Both write side and read side status are indicated separately, since they would be not be similar when master clock and core clock are not synchronous to each other.

BIMC_M_TCU0_MPORT_STATUS_3B

Bits	Name	Description
21	RGB_RD_FULL	Indicates that the read gathering buffer is full when seen from it's read side, which interfaces to frontend of the masterport.
20	RGB_RD_EMPTY	Indicates that the read gathering buffer is empty when seen from it's read side, which interfaces to frontend of the masterport.
17	RGB_WR_FULL	Indicates that the read gathering buffer is full when seen from it's write side, which interfaces to core of the interconnect, i.e., arbiter and slaweway.
16	RGB_WR_EMPTY	Indicates that the read gathering buffer is empty when seen from it's write side, which interfaces to core of the interconnect, i.e., arbiter and slaweway.
13	W2AB_RD_FULL	Indicates that the write channel to address channel user signals buffer is full when seen from it's read side, which interfaces to core of the interconnect, i.e., arbiter and slaweway.
12	W2AB_RD_EMPTY	Indicates that the write channel to address channel user signals buffer is empty when seen from it's read side, which interfaces to core of the interconnect, i.e., arbiter and slaweway.
9	W2AB_WR_FULL	Indicates that the write channel to address channel user signals buffer is full when seen from it's write side, which interfaces to frontend of the masterport.
8	W2AB_WR_EMPTY	Indicates that the write channel to address channel user signals buffer is empty when seen from it's write side, which interfaces to frontend of the masterport.
5	A2WB_RD_FULL	Indicates that the address channel to write channel user signals buffer is full when seen from it's read side, which interfaces to core of the interconnect, i.e., arbiter and slaweway.
4	A2WB_RD_EMPTY	Indicates that the address channel to write channel user signalsbuffer is empty when seen from it's read side, which interfaces to core of the interconnect, i.e., arbiter and slaweway.
1	A2WB_WR_FULL	Indicates that the address channel to write channel user signals buffer is full when seen from it's write side, which interfaces to frontend of the masterport.

BIMC_M_TCU0_MPORT_STATUS_3B (cont.)

Bits	Name	Description
0	A2WB_WR_EMPTY	Indicates that the address channel to write channel user signals buffer is empty when seen from it's write side, which interfaces to frontend of the masterport.

2.14 BIMC_M_SYS1_PROF

BRIC Traffic Profiling

2.14.1 BIMC_M_SYS1_PROF_BRIC_PROFILING

BRIC Traffic Profiling

0x00419000 BIMC_M_SYS1_PROF_COMPONENT_INFO**Type:** R**Clock:** Config Clock**Reset State:** 0x00000202

Component information registers provide discoverability information on the type of component and instance of each component. These registers are always the very first register in each component's configuration address space and allow for a software routine to walk the configuration address space in steps of 4KB to identify the location of each configuration register space and to which component each 4KB configures.

BIMC_M_SYS1_PROF_COMPONENT_INFO

Bits	Name	Description
23:16	INSTANCE	Indicates the instance number of this traffic profiling.
15:8	SUB_TYPE	Identifies this configuration space as belonging to traffic profiling 0x2: PROFILING
7:0	TYPE	Identifies this configuration space as belonging to a component that interfaces to a bus master. 0x2: MASTERPORT

0x00419020 BIMC_M_SYS1_PROF_CONFIGURATION_INFO_0**Type:** R**Clock:** Config Clock**Reset State:** 0x00000000

Configuration information registers provide information on the capability and configuration of each traffic profiling instance. See register field descriptions for more details.

BIMC_M_SYS1_PROF_CONFIGURATION_INFO_0

Bits	Name	Description
31:24	TENURE_WIDTH	Indicates the width of tenures (latencies) measured in traffic profiling.
23:16	TRACKING_TABLE_DEPTH	Indicates the number of entries in the table that tracks outstanding commands for determining how many commands have larger turnaround latencies that a programmed threshold.
7:0	SCALING_FACTOR	Indicates how many extra scaling is performed on events generated by traffic profiling if such events have to be synchronized from the tenure clock to the event clock. Any value that is non-power-of-2 is converted to power-of-2 by hardware. When the value read from this register field is used in software, similar conversion of non-power-of-2 values to next higher power-of-2 value is necessary.

0x00419030 BIMC_M_SYS1_PROF_CONFIGURATION_INFO_1**Type:** R**Clock:** Config Clock**Reset State:** 0x00000000

Configuration information registers provide information on the capability and configuration of each traffic profiling instance. See register field descriptions for more details.

BIMC_M_SYS1_PROF_CONFIGURATION_INFO_1

Bits	Name	Description
23:16	PIPE2_ACH_DEPTH	Indicates the depth of address channel buffer in pipe2 of masterport in which this traffic profiling instance is paired with. Applies only to traffic profiling instances in XR type masterports, which contains three distinct paths to slaves, in pipe0, pipe1, pipe2. In non-XR type masterports, value contained in field does not refer to any actual address channel buffer.
15:8	PIPE1_ACH_DEPTH	Indicates the depth of address channel buffer in pipe1 of masterport in which this traffic profiling instance is paired with. Applies only to traffic profiling instances in XR type masterports, which contains three distinct paths to slaves, in pipe0, pipe1, pipe2. In non-XR type masterports, value contained in field does not refer to any actual address channel buffer.
7:0	PIPE0_ACH_DEPTH	Indicates the depth of address channel buffer in pipe0 of masterport in which this traffic profiling instance is paired with. Applies only to traffic profiling instances in XR type masterports, which contains three distinct paths to slaves, in pipe0, pipe1, pipe2. In non-XR type masterports, pipe0 refers to the common address channel buffer used for all slave accesses.

0x00419200 BIMC_M_SYS1_PROF_CLOCK_CTRL**Type:** RW**Clock:** Config, Event & Tenure Clocks**Reset State:** 0x00000003**BIMC_M_SYS1_PROF_CLOCK_CTRL**

Bits	Name	Description
1	TENURE_CLOCK_GATING_EN	When set (1), enables activity based clock gating of all registers clocked by tenure clock. Activity based clock gating uses architectural clock gates to reduce dynamic power consumption by gating clocks during periods of in-activity. When cleared (0), disables all activity based clock gating by forcing all architectural clock gates to not gate clocks even during periods of in-activity.
0	EVENT_CLOCK_GATING_EN	When set (1), enables activity based clock gating of all registers clocked by event clock. Activity based clock gating uses architectural clock gates to reduce dynamic power consumption by gating clocks during periods of in-activity. When cleared (0), disables all activity based clock gating by forcing all architectural clock gates to not gate clocks even during periods of in-activity.

0x00419210 BIMC_M_SYS1_PROF_MODE**Type:** RW**Clock:** Config, Event & Tenure Clocks**Reset State:** 0x00000000

BIMC_M_SYS1_PROF_MODE

Bits	Name	Description
25:24	WR_TENURE_MODE	Configures the start and end events for monitoring write latencies/tenures. When configured to WR_CMD_TO_BRESP, starting event is first cycle of write command from bus master as seen by traffic profiling and ending event is first cycle of write response to bus master as seen by traffic profiling. When configured to WR_CMD_TO_BRESP_HANDSHAKE, starting event is first cycle of write command from bus master as seen by traffic profiling and ending event is transfer handshake of write response to bus master as seen by traffic profiling. When configured to WR_CMD_HANDSHAKE, starting event is first cycle of write command from bus master as seen by traffic profiling and ending event is transfer handshake of the same write command with bus master as seen by traffic profiling. When configured to WR_RESP_HANDSHAKE, starting event is first cycle of write response to bus master as seen by traffic profiling and ending event is transfer handshake of write response to bus master as seen by traffic profiling. 0x0: WR_CMD_TO_BRESP 0x1: WR_CMD_TO_BRESP_HANDSHAKE 0x2: WR_CMD_HANDSHAKE 0x3: WR_RESP_HANDSHAKE
17:16	RD_TENURE_MODE	Configures the start and end events for monitoring read latencies/tenures. When configured to RD_CMD_TO_FIRST_RDATA, starting event is first cycle of read command from bus master as seen by traffic profiling and ending event is first cycle of first beat of read data to bus master as seen by traffic profiling. When configured to RD_CMD_TO_LAST_RDATA, starting event is first cycle of read command from bus master as seen by traffic profiling and ending event is transfer handshake of last beat of read data to bus master as seen by traffic profiling. When configured to RD_CMD_HANDSHAKE, starting event is first cycle of read command from bus master as seen by traffic profiling and ending event is transfer handshake of the same read command with bus master as seen by traffic profiling. When configured to RD_DATA_HANDSHAKE, starting event is first cycle of first beat of read data to bus master as seen by traffic profiling and ending event is transfer handshake of last beat of read data to bus master as seen by traffic profiling. 0x0: RD_CMD_TO_FIRST_RDATA 0x1: RD_CMD_TO_LAST_RDATA 0x2: RD_CMD_HANDSHAKE 0x3: RD_DATA_HANDSHAKE
0	PROFILING_EN	When set (1), enables traffic profiling to monitor bus transactions from bus master and generate events for counting in performance monitors. When cleared (0), disables traffic profiling.

0x00419214 BIMC_M_SYS1_PROF_RD_TENURE_THRESHOLD**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_SYS1_PROF_RD_TENURE_THRESHOLD**

Bits	Name	Description
15:0	THRESHOLD	Configures the threshold over which read latencies/tenures need to be before an event is generated to performance monitors for counting max latencies.

0x00419218 BIMC_M_SYS1_PROF_WR_TENURE_THRESHOLD**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_SYS1_PROF_WR_TENURE_THRESHOLD**

Bits	Name	Description
15:0	THRESHOLD	Configures the threshold over which write latencies/tenures need to be before an event is generated to performance monitors for counting max latencies.

0x00419220 BIMC_M_SYS1_PROF_SWAY_SELECT**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_SYS1_PROF_SWAY_SELECT**

Bits	Name	Description
7:0	SWAY	Selects the slaveways to which commands are routed to profile. 1-bit per slaveway.

0x00419230 BIMC_M_SYS1_PROF_MID_MASK**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000

BIMC_M_SYS1_PROF_MID_MASK

Bits	Name	Description
15:0	MID_MASK	When a bit position is set (1), enables matching of that bit position in MID of command with same bit position in MID_MATCH. Commands with MIDs that do not match are not considered for profiling.

0x00419234 BIMC_M_SYS1_PROF_MID_MATCH**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_SYS1_PROF_MID_MATCH**

Bits	Name	Description
15:0	MID_MATCH	Configures the value to match for in AMID in commands. Used in conjunction with MID_MASK, which controls which bits of MID are compared.

0x00419238 BIMC_M_SYS1_PROF_MID_INV_MATCH**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_SYS1_PROF_MID_INV_MATCH**

Bits	Name	Description
0	MID_INV_MATCH	When set (1), configures MID matching to result in an inverse match, i.e., only commands whose MIDs that do not match MID_MATCH are considered for profiling.

0x00419240 BIMC_M_SYS1_PROF_ADDR_MASK_LOWER**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000

BIMC_M_SYS1_PROF_ADDR_MASK_LOWER

Bits	Name	Description
31:10	ADDR_MASK_LOWER	Configures ADDR_MASK[31:0]. When a bit position is set (1), enables matching of that bit position of address of command with same bit position in ADDR_MATCH[31:0].

0x00419244 BIMC_M_SYS1_PROF_ADDR_MASK_UPPER**Type:** R**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_SYS1_PROF_ADDR_MASK_UPPER**

Bits	Name	Description
3:0	UNUSED	

0x00419248 BIMC_M_SYS1_PROF_ADDR_MATCH_LOWER**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_SYS1_PROF_ADDR_MATCH_LOWER**

Bits	Name	Description
31:10	ADDR_MATCH_LOWER	Configures the value to match for in address of commands. Used in conjunction with ADDR_MASK[31:0], which controls which bits of address are compared.

0x0041924C BIMC_M_SYS1_PROF_ADDR_MATCH_UPPER**Type:** R**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_SYS1_PROF_ADDR_MATCH_UPPER**

Bits	Name	Description
3:0	UNUSED	

0x00419250 BIMC_M_SYS1_PROF_ADDR_INV_MATCH**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_SYS1_PROF_ADDR_INV_MATCH**

Bits	Name	Description
0	ADDR_INV_MATCH	When set (1), configures address matching to result in an inverse match, i.e., only commands whose addresses do not match ADDR_MATCH are considered for profiling.

0x00419260 BIMC_M_SYS1_PROF_ACH_MASK_0**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_SYS1_PROF_ACH_MASK_0**

Bits	Name	Description
24	AINNERSHARED_MASK	When set (1), enables matching of AINNERSHARED in commands with AINNERSHARED_MATCH.
16	AFULL_MASK	When set (1), enables matching of AFULL in commands with AFULL_MATCH.
8	AEXCLUSIVE_MASK	When set (1), enables matching of AEXCLUSIVE in commands with AEXCLUSIVE_MATCH.
0	ABURST_MASK	When set (1), enables matching of ABURST in commands with ABURST_MATCH.

0x00419264 BIMC_M_SYS1_PROF_ACH_MATCH_0**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_SYS1_PROF_ACH_MATCH_0**

Bits	Name	Description
24	AINNERSHARED_MATCH	Configures the value to match for in AINNERSHARED in commands. Used in conjunction with AINNERSHARED_MASK, which controls whether AINNERSHARED in commands is compared

BIMC_M_SYS1_PROF_ACH_MATCH_0 (cont.)

Bits	Name	Description
16	AFULL_MATCH	Configures the value to match for in AFULL in commands. Used in conjunction with AFULL_MASK, which controls whether AFULL in commands is compared.
8	AEXCLUSIVE_MATCH	Configures the value to match for in AEXCLUSIVE in commands. Used in conjunction with AEXCLUSIVE_MASK, which controls whether AFULL in commands is compared.
0	ABURST_MATCH	Configures the value to match for in ABURST in commands. Used in conjunction with ABURST_MASK, which controls whether ABURST in commands is compared.

0x00419270 BIMC_M_SYS1_PROF_ACH_MASK_1**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_SYS1_PROF_ACH_MASK_1**

Bits	Name	Description
24	AMSSSELFAUTH_MASK	When set (1), enables matching of AMSSSELFAUTH in commands with AMSSSELFAUTH_MATCH.
18:16	AMEMTYPE_MASK	When a bit position is set (1), enables matching of corresponding bit position of AMEMTYPE with AMEMTYPE_MATCH.
11:8	ALEN_MASK	When a bit position is set (1), enables matching of corresponding bit position of ALEN with ALEN_MATCH.
0	AINST_MASK	When set (1), enables matching of AINST in commands with AINST_MASK.

0x00419274 BIMC_M_SYS1_PROF_ACH_MATCH_1**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_SYS1_PROF_ACH_MATCH_1**

Bits	Name	Description
24	AMSSSELFAUTH_MATCH	Configures the value to match for in AMSSSELFAUTH in commands. Used in conjunction with AMSSSELFAUTH_MASK, which controls whether AMSSSELFAUTH in commands is compared.

BIMC_M_SYS1_PROF_ACH_MATCH_1 (cont.)

Bits	Name	Description
18:16	AMEMTYPE_MATCH	Configures the value to match for in AMEMTYPE in commands. Used in conjunction with AMEMTYPE_MASK, which controls which bits of AMEMTYPE in commands are compared.
11:8	ALEN_MATCH	Configures the value to match for in ALEN in commands. Used in conjunction with ALEN_MASK, which controls which bits of ALEN in commands are compared.
0	AINST_MATCH	Configures the value to match for in AINST in commands. Used in conjunction with AINST_MASK, which controls whether AINST in commands is compared.

0x00419278 BIMC_M_SYS1_PROF_ACH_INV_MATCH_1**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_SYS1_PROF_ACH_INV_MATCH_1**

Bits	Name	Description
16	AMEMTYPE_INV_MATCH	When set (1), configures AMEMTYPE matching to result in an inverse match, i.e., only commands whose AMEMTYPE does not match AMEMTYPE_MATCH are considered for profiling.
8	ALEN_INV_MATCH	When set (1), configures ALEN matching to result in an inverse match, i.e., only commands when ALEN does not match ALEN_MATCH are considered for profiling.

0x00419280 BIMC_M_SYS1_PROF_ACH_MASK_2**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_SYS1_PROF_ACH_MASK_2**

Bits	Name	Description
24	APROTNS_MASK	When set (1), enables matching of APROTNS in commands with APROTNS_MATCH.
16	APRIV_MASK	When set (1), enables matching of APRIV in commands with APRIV_MATCH.
8	AOOO_MASK	When set (1), enables matching of AOOO in commands with AOOO_MATCH.
0	ANOALLOCATE_MASK	When set (1), enables matching of ANOALLOCATE in commands with ANOALLOCATE_MATCH.

0x00419284 BIMC_M_SYS1_PROF_ACH_MATCH_2**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_SYS1_PROF_ACH_MATCH_2**

Bits	Name	Description
24	APROTNS_MATCH	Configures the value to match for in APROTNS in commands. Used in conjunction with APROTNS_MASK, which controls whether APROTNS in commands is compared
16	APRIV_MATCH	Configures the value to match for in APRIV in commands. Used in conjunction with APRIV_MASK, which controls whether APRIV in commands is compared
8	AOOO_MATCH	Configures the value to match for in AOOO in commands. Used in conjunction with AOOO_MASK, which controls whether AOOO in commands is compared
0	ANOALLOCATE_MATCH	Configures the value to match for in ANOALLOCATE in commands. Used in conjunction with ANOALLOCATE_MASK, which controls whether ANOALLOCATE in commands is compared

0x00419290 BIMC_M_SYS1_PROF_ACH_MASK_3**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_SYS1_PROF_ACH_MASK_3**

Bits	Name	Description
24	ASHARED_MASK	When set (1), enables matching of ASHARED in commands with ASHARED_MATCH.
17:16	AREQPRIORITY_MASK	When a bit position is set (1), enables matching of corresponding bit position of AREQPRIORITY with AREQPRIORITY_MATCH.
8	AREDIRECT_MASK	When set (1), enables matching of AREDIRECT in commands with AREDIRECT_MATCH.
0	ARDBEATNDXEN_MASK	When set (1), enables matching of ARDBEATNDXEN in commands with ARDBEATNDXEN_MATCH.

0x00419294 BIMC_M_SYS1_PROF_ACH_MATCH_3**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_SYS1_PROF_ACH_MATCH_3**

Bits	Name	Description
24	ASHARED_MATCH	Configures the value to match for in ASHARED in commands. Used in conjunction with ASHARED_MASK, which controls whether ASHARED in commands is compared
17:16	AREQPRIORITY_MATCH	Configures the value to match for in AREQPRIORITY in commands. Used in conjunction with AREQPRIORITY_MASK, which controls which bits of AREQPRIORITY in commands are compared.
8	AREDIRECT_MATCH	Configures the value to match for in AREDIRECT in commands. Used in conjunction with AREDIRECT_MASK, which controls whether AREDIRECT in commands is compared
0	ARDBEATNDXEN_MATCH	Configures the value to match for in ARDBEATNDXEN in commands. Used in conjunction with ARDBEATNDXEN_MASK, which controls whether ARDBEATNDXEN in commands is compared

0x00419298 BIMC_M_SYS1_PROF_ACH_INV_MATCH_3**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_SYS1_PROF_ACH_INV_MATCH_3**

Bits	Name	Description
16	AREQPRIORITY_INV_MATCH	When set (1), configures AREQPRIORITY matching to result in an inverse match, i.e., only commands whose AREQPRIORITY does not match AREQPRIORITY_MATCH are considered for profiling

0x004192A0 BIMC_M_SYS1_PROF_ACH_MASK_4**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000

BIMC_M_SYS1_PROF_ACH_MASK_4

Bits	Name	Description
24	AWRITE_MASK	When set (1), enables matching of AWRITE in commands with AWRITE_MATCH.
20:16	AVMID_MASK	When a bit position is set (1), enables matching of corresponding bit position in AVMID with AVMID_MATCH
8	ATRANSIENT_MASK	When set (1), enables matching of ATRANSIENT in commands with ATRANSIENT_MATCH.
2:0	ASIZE_MASK	When a bit position is set (1), enables matching of corresponding bit position in ASIZE with ASIZE_MATCH.

0x004192A4 BIMC_M_SYS1_PROF_ACH_MATCH_4**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_SYS1_PROF_ACH_MATCH_4**

Bits	Name	Description
24	AWRITE_MATCH	Configures the value to match for in AWRITE in commands. Used in conjunction with AWRITE_MASK, which controls whether AWRITE in commands is compared.
20:16	AVMID_MATCH	Configures the value to match for in AVMID in commands. Used in conjunction with AVMID_MASK, which controls which bits of AVMID in commands are compared.
8	ATRANSIENT_MATCH	Configures the value to match for in ATRANSIENT in commands. Used in conjunction with ATRANSIENT_MASK, which controls whether ATRANSIENT in commands is compared.
2:0	ASIZE_MATCH	Configures the value to match for in ASIZE in commands. Used in conjunction with ASIZE_MASK, which controls which bits of ASIZE in commands are compared.

0x004192A8 BIMC_M_SYS1_PROF_ACH_INV_MATCH_4**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000

BIMC_M_SYS1_PROF_ACH_INV_MATCH_4

Bits	Name	Description
16	AVMID_INV_MATCH	When set (1), configures AVMID matching to result in an inverse match, i.e., only commands whose AVMID does not match AVMID_MATCH are considered for profiling.
0	ASIZE_INV_MATCH	When set (1), configures ASIZE matching to result in an inverse match, i.e., only commands whose AVMID does not match ASIZE_MATCH are considered for profiling.

0x00419300 BIMC_M_SYS1_PROF_BCH_MASK**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_SYS1_PROF_BCH_MASK**

Bits	Name	Description
1:0	BRESP_MASK	When a bit position is set (1), enables matching of corresponding bit position in BRESP with BRESP_MATCH.

0x00419304 BIMC_M_SYS1_PROF_BCH_MATCH**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_SYS1_PROF_BCH_MATCH**

Bits	Name	Description
1:0	BRESP_MATCH	Configures the value to match for in BRESP in write responses. Used in conjunction with BRESP_MASK, which controls which bits of BRESP in writes responses are compared.

0x00419308 BIMC_M_SYS1_PROF_BCH_INV_MATCH**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000

BIMC_M_SYS1_PROF_BCH_INV_MATCH

Bits	Name	Description
0	BRESP_INV_MATCH	When set (1), configures BRESP matching to result in an inverse match, i.e., only write responses whose BRESP does not match BRESP_MATCH are considered for profiling.

0x00419340 BIMC_M_SYS1_PROF_RCH_MASK**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_SYS1_PROF_RCH_MASK**

Bits	Name	Description
8	RLAST_MASK	When set (1), enables matching of RLAST in commands with corresponding setting in RCH_MATCH register.
1:0	RRESP_MASK	When a bit position is set (1), enables matching of corresponding bit position in RRESP with RRESP_MATCH.

0x00419344 BIMC_M_SYS1_PROF_RCH_MATCH**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_SYS1_PROF_RCH_MATCH**

Bits	Name	Description
8	RLAST_MATCH	Configures the value to match for in RLAST in read responses. Used in conjunction with RLAST_MASK, which controls whether RLAST in read responses is compared.
1:0	RRESP_MATCH	Configures the value to match for in RRESP in read responses. Used in conjunction with RRESP_MASK, which controls which bits of RRESP in read responses are compared.

0x00419348 BIMC_M_SYS1_PROF_RCH_INV_MATCH**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000

BIMC_M_SYS1_PROF_RCH_INV_MATCH

Bits	Name	Description
0	RRESP_INV_MATCH	When set (1), configures RRESP matching to result in an inverse match, i.e., only read responses whose RRESP does not match RRESP_MATCH are considered for profiling.

0x00419400+ BIMC_M_SYS1_PROF_RD_STATUS_n, n=[0..0]
0x4*n
Type: R**Clock:** Config & Tenure Clocks**Reset State:** 0x00000000**BIMC_M_SYS1_PROF_RD_STATUS_n**

Bits	Name	Description
31	RD_OUTSTANDING_OVERFLOW	Indicates that the number of outstanding reads crossed the maximum supported (255) during when traffic profiling was last enabled. It is recommended that this register be read only after traffic profiling has been disabled.
30	RD_OUTSTANDING_UNDEFLOW	Indicates that the number of outstanding reads went below 0 during when traffic profiling was last enabled. It indicates that more command completions were seen compared to commands starts. The most likely scenario is an overflow of number of outstanding reads occurred leading to some command starts not being tracked. It is recommended that this register be read only after traffic profiling has been disabled.
7:0	RD_OUTSTANDING_MAX	Indicates the maximum number of outstanding reads that were seen when traffic profiling was last enabled. Only reads that match the MID filter and slaweway select filter are accounted for. It is recommended that this register be read only after traffic profiling has been disabled.

0x00419410+ BIMC_M_SYS1_PROF_WR_STATUS_n, n=[0..0]
0x4*n
Type: R**Clock:** Config & Tenure Clocks**Reset State:** 0x00000000**BIMC_M_SYS1_PROF_WR_STATUS_n**

Bits	Name	Description
31	WR_OUTSTANDING_OVERFLOW	Indicates that the number of outstanding writes crossed the maximum supported (255) during when traffic profiling was last enabled. It is recommended that this register be read only after traffic profiling has been disabled.

BIMC_M_SYS1_PROF_WR_STATUS_n (cont.)

Bits	Name	Description
30	WR_OUTSTANDING_UNDE RFLOW	Indicates that the number of outstanding writes went below 0 during when traffic profiling was last enabled. It indicates that more command completions were seen compared to commands starts. The most likely scenario is an overflow of number of outstanding writes occurred leading to some command starts not being tracked. It is recommended that this register be read only after traffic profiling has been disabled.
7:0	WR_OUTSTANDING_MAX	Indicates the maximum number of outstanding writes that were seen when traffic profiling was last enabled. Only writes that match the MID filter and slaweway select filter are accounted for. It is recommended that this register be read only after traffic profiling has been disabled.

2.15 BIMC_M_TCU0_MPORT

BRIC Masterport

2.15.1 BIMC_M_TCU0_MPORT_BRIC_MASTERPORT

BRIC Masterport

0x0041C000 BIMC_M_TCU0_MPORT_COMPONENT_INFO**Type:** R**Clock:** Config Clock**Reset State:** 0x00000102

Component information registers provide discoverability information on the type of component and instance of each component. These registers are always the very first register in each component's configuration address space and allow for a software routine to walk the configuration address space in steps of 4KB to identify the location of each configuration register space and to which component each 4KB configures.

BIMC_M_TCU0_MPORT_COMPONENT_INFO

Bits	Name	Description
23:16	INSTANCE	Indicates the instance number of this masterport.
15:8	SUB_TYPE	Identifies this configuration space as belonging to a masterport 0x1: MASTERPORT
7:0	TYPE	Identifies this configuration space as belonging to a component that interfaces to a bus master. 0x2: MASTERPORT

0x0041C020 BIMC_M_TCU0_MPORT_CONFIGURATION_INFO_0**Type:** R**Clock:** Config Clock**Reset State:** 0x01000000

Configuration information registers provide information on the capability and configuration of each masterport instance. See register field descriptions for more details.

BIMC_M_TCU0_MPORT_CONFIGURATION_INFO_0

Bits	Name	Description
31:24	SYNC_MODE	Indicates the clock relationship between the clock used in the master side and the clock used in the core side of the masterport. When ASYNC, identifies the master clock and core clock as unrelated and asynchronous to each other. When SYNC, identifies the master clock and core clock as synchronous to each other. When ISOSYNC, identifies the master clock and core clock as related, but of different frequencies, which are of integer ratios, i.e., they are iso-synchronous to each other. 0x0: ASYNC 0x1: SYNC 0x2: ISOSYNC
15:8	CONNECTION_TYPE	Indicates the type of master connecting to this masterport. When DIRECT, indicates that a bus master is directly connected to this masterport. When CASCADE, indicates that bus masters connect to this masterport through another level of interconnect. 0x0: DIRECT 0x1: CASCADE
7:0	FUNCTIONALITY	Indicates the functionality and type of bus traffic handled by this masterport. When INORDER, identifies this masterport as optimized for in-order bus traffic. When OOO_ONLY, identifies this masterport as optimized for out-of-order bus traffic, where a single master is capable of handling responses return in any order. When REORDER, identifies this masterport as having a reorder buffer for re-ordering responses to command order, while allowing bus slaves to return responses in any order they desire. When STUB, identifies this masterport has a stub. 0x0: STUB 0x1: OOO_ONLY 0x2: INORDER 0x3: REORDER

0x0041C030 BIMC_M_TCU0_MPORT_CONFIGURATION_INFO_1**Type:** R**Clock:** Config Clock**Reset State:** 0x00000000

Configuration information registers provide information on the capability and configuration of each masterport instance. See register field descriptions for more details.

BIMC_M_TCU0_MPORT_CONFIGURATION_INFO_1

Bits	Name	Description
31:0	SWAY_CONNECTIVITY	Indicates slaveways that this masterport can route commands to. 1-bit per slaveway. When a specific bit is set (1), indicates that commands can be routed to the slaveway corresponding to this bit. When a specific bit is clear (0), indicates that commands are not routed to the slaveway corresponding to this bit.

0x0041C040 BIMC_M_TCU0_MPORT_CONFIGURATION_INFO_2A

Type: R

Clock: Config Clock

Reset State: 0x00000000

Configuration information registers provide information on the capability and configuration of each masterport instance. See register field descriptions for more details.

BIMC_M_TCU0_MPORT_CONFIGURATION_INFO_2A

Bits	Name	Description
31:16	M_DATA_WIDTH	Indicates the width of read and write data from the bus master.
15:8	M_TID_WIDTH	Indicates the width of transfer IDs from the bus master.
7:0	M_MID_WIDTH	Indicates the width of master ID from the bus master (includes any bus and port IDs if bus master is an interconnect.)

0x0041C044 BIMC_M_TCU0_MPORT_CONFIGURATION_INFO_2B

Type: R

Clock: Config Clock

Reset State: 0x00000000

Configuration information registers provide information on the capability and configuration of each masterport instance. See register field descriptions for more details.

BIMC_M_TCU0_MPORT_CONFIGURATION_INFO_2B

Bits	Name	Description
15:0	C_DATA_WIDTH	Indicates the width of read and write data used in core of this masterport.

0x0041C050 BIMC_M_TCU0_MPORT_CONFIGURATION_INFO_3**Type:** R**Clock:** Config Clock**Reset State:** 0x00000000

Configuration information registers provide information on the capability and configuration of each masterport instance. See register field descriptions for more details.

BIMC_M_TCU0_MPORT_CONFIGURATION_INFO_3

Bits	Name	Description
31:24	RCH_DEPTH	Indicates the depth of read channel buffers in this masterport. A beat of read data occupies one entry in the read channel buffer, until accepted by the bus master.
23:16	BCH_DEPTH	Indicates the depth of write response channel buffers in this masterport. Each write response occupies one entry in the write response channel buffer, until accepted by the bus master.
15:8	WCH_DEPTH	Indicates the depth of write channel buffers in this masterport. A beat of write data occupies one entry in the write channel buffer, until accepted by the arbiter.
7:0	ACH_DEPTH	Indicates the depth of address channel buffers in this masterport. Each command occupies one entry in the address channel buffer, until accepted by the arbiter.

0x0041C060 BIMC_M_TCU0_MPORT_CONFIGURATION_INFO_4**Type:** R**Clock:** Config Clock**Reset State:** 0x00000000

Configuration information registers provide information on the capability and configuration of each masterport instance. See register field descriptions for more details.

BIMC_M_TCU0_MPORT_CONFIGURATION_INFO_4

Bits	Name	Description
15:8	REORDER_BUFFER_DEPTH	Indicates the number of read data beats that can be re-ordered in this masterport (if the masterport provides response re-ordering functionality.)
7:0	REORDER_TABLE_DEPTH	Indicates the number of read and write commands whose responses can be re-ordered in this masterport (if the masterport provides responses re-ordering functionality.)

0x0041C070 BIMC_M_TCU0_MPORT_CONFIGURATION_INFO_5A**Type:** R**Clock:** Config Clock**Reset State:** 0x00000000

Configuration information registers provide information on the capability and configuration of each masterport instance. See register field descriptions for more details.

BIMC_M_TCU0_MPORT_CONFIGURATION_INFO_5A

Bits	Name	Description
31:0	ACH_PIPELINE_STAGES	Indicates pipeline stages enabled on the address channel. Each digit identifies a specific pipeline stage in the masterport, where '0' indicates no pipeline stage, '1' indicates 1-deep pipeline stage where forward path is registered, return path is fed through, '2' indicates 2-deep pipeline stage where both forward and return path are pipelined. Digit 3 identifies pipelining on the master to masterport interface, digit 2 identifies pipelining after address decode, digit 1 identifies pipelining in the address channel buffer and digit 0 identifies pipelining on the masterport to arbiter interface.

0x0041C074 BIMC_M_TCU0_MPORT_CONFIGURATION_INFO_5B**Type:** R**Clock:** Config Clock**Reset State:** 0x00000000

Configuration information registers provide information on the capability and configuration of each masterport instance. See register field descriptions for more details.

BIMC_M_TCU0_MPORT_CONFIGURATION_INFO_5B

Bits	Name	Description
31:0	WCH_PIPELINE_STAGES	Indicates pipeline stages enabled on the write channel. Each digit identifies a specific pipeline stage in the masterport, where '0' indicates no pipeline stage, '1' indicates 1-deep pipeline stage where forward path is registered, return path is fed through, '2' indicates 2-deep pipeline stage where both forward and return path are pipelined. Digit 3 identifies pipelining on the master to masterport interface, digit 2 identifies pipelining after address decode, digit 1 identifies pipelining in the write channel buffer and digit 0 identifies pipelining on the masterport to arbiter interface.

0x0041C078 BIMC_M_TCU0_MPORT_CONFIGURATION_INFO_5C**Type:** R**Clock:** Config Clock**Reset State:** 0x00000000

Configuration information registers provide information on the capability and configuration of each masterport instance. See register field descriptions for more details.

BIMC_M_TCU0_MPORT_CONFIGURATION_INFO_5C

Bits	Name	Description
31:0	BCH_PIPELINE_STAGES	Indicates pipeline stages enabled on the write response channel. Each digit identifies a specific pipeline stage in the masterport, where '0' indicates no pipeline stage, '1' indicates 1-deep pipeline stage where forward path is registered, return path is fed through, '2' indicates 2-deep pipeline stage where both forward and return path are pipelined. Digit 2 identifies pipelining on the slaveway to masterport interface after arbitration of write response channels from multiple slaveways, digit 1 identifies pipelining in the write response channel buffer and digit 0 identifies pipelining on the masterport to master interface.

0x0041C07C BIMC_M_TCU0_MPORT_CONFIGURATION_INFO_5D**Type:** R**Clock:** Config Clock**Reset State:** 0x00000000

Configuration information registers provide information on the capability and configuration of each masterport instance. See register field descriptions for more details.

BIMC_M_TCU0_MPORT_CONFIGURATION_INFO_5D

Bits	Name	Description
31:0	RCH_PIPELINE_STAGES	Indicates pipeline stages enabled on the read channel. Each digit identifies a specific pipeline stage in the masterport, where '0' indicates no pipeline stage, '1' indicates 1-deep pipeline stage where forward path is registered, return path is fed through, '2' indicates 2-deep pipeline stage where both forward and return path are pipelined. Digit 2 identifies pipelining on the slaveway to masterport interface after arbitration of read channels from multiple slaveways, digit 1 identifies pipelining in the read channel buffer and digit 0 identifies pipelining on the masterport to master interface.

0x0041C100 BIMC_M_TCU0_MPORT_INTERRUPT_STATUS**Type:** R**Clock:** Config Clock**Reset State:** 0x00000000

Interrupt status registers provide indication of which interrupts from this masterport require servicing.

BIMC_M_TCU0_MPORT_INTERRUPT_STATUS

Bits	Name	Description
1:0	RFU	Reserved for future use

0x0041C108 BIMC_M_TCU0_MPORT_INTERRUPT_CLEAR**Type:** W**Clock:** Config Clock**Reset State:** 0x00000000

Interrupt clear registers provide the mechanism to turn off interrupt sources once they have been acknowledged or serviced. A write of '1' to the bits corresponding to the interrupts indicated in interrupt status is required to turn off interrupts.

BIMC_M_TCU0_MPORT_INTERRUPT_CLEAR

Bits	Name	Description
1:0	RFU	Reserved for future use

0x0041C10C BIMC_M_TCU0_MPORT_INTERRUPT_ENABLE**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000

Interrupt enable registers provide control over which interrupts are allowed to be propagated to an interrupt controller for servicing.

BIMC_M_TCU0_MPORT_INTERRUPT_ENABLE

Bits	Name	Description
1:0	RFU	Reserved for future use

0x0041C200 BIMC_M_TCU0_MPORT_CLOCK_CTRL**Type:** RW**Clock:** Config, Core & Master Clocks**Reset State:** 0x00000003**BIMC_M_TCU0_MPORT_CLOCK_CTRL**

Bits	Name	Description
1	MASTER_CLOCK_GATING_EN	When set (1), enables activity based clock gating of all registers clocked by master clock. Activity based clock gating uses architectural clock gates to reduce dynamic power consumption by gating clocks during periods of in-activity. When cleared (0), disables all activity based clock gating by forcing all architectural clock gates to not gate clocks even during periods of in-activity.
0	CORE_CLOCK_GATING_EN	When set (1), enables activity based clock gating of all registers clocked by core clock. Activity based clock gating uses architectural clock gates to reduce dynamic power consumption by gating clocks during periods of in-activity. When cleared (0), disables all activity based clock gating by forcing all architectural clock gates to not gate clocks even during periods of in-activity.

0x0041C208 BIMC_M_TCU0_MPORT_CDC_CTRL**Type:** RW**Clock:** Config, Core & Master Clocks**Reset State:** 0x00000000

Clock domain crossing (CDC) control register provides configuration for clock domain crossing logic.

BIMC_M_TCU0_MPORT_CDC_CTRL

Bits	Name	Description
1	MASTER_CLOCK_DEMETA_SEL	When cleared (0), selects DEMETA synchronizer that requires atleast 3 clock edges (two rising and one falling). When set (1), selects DEMETA synchronizer that requires atleast 2 clock edges (one falling and one rising) to synchronize to master clock in all buffers.
0	CORE_CLOCK_DEMETA_SEL	When cleared (0), selects DEMETA synchronizer that requires atleast 3 clock edges (two rising and one falling). When set (1), selects DEMETA synchronizer that requires atleast 2 clock edges (one falling and one rising) to synchronize to core clock in all buffers.

0x0041C210 BIMC_M_TCU0_MPORT_MODE**Type:** RW**Clock:** Config, Core & Master Clocks**Reset State:** 0x00000012**BIMC_M_TCU0_MPORT_MODE**

Bits	Name	Description
31:28	WR_GATHER_BEATS	Configures the number of write data beats to gather/buffer in write channel buffer before allowing the corresponding write command to be routed to arbiters. When configured to 0, write commands are held until the first beat of write data has been buffered. When configured to 1, write commands are held until two beats of write data have been buffered and so on. When configured with values larger than the number of beats in a write command, the command is held until the last beat has been buffered. When configured with values larger than the depth of write channel buffers, the command is held until all entries in the write channel buffer have been occupied.
19:16	RD_GATHER_BEATS	Configures the number of read data beats to gather/buffer in read channel buffer before allowing the read data to proceed to the master. When configured to 0, read data is allowed to proceed to the master as soon as the first beat is buffered in the read channel buffer. When configured to 1, read data is allowed to proceed to the master as soon as two beats of read data have been buffered and so on. When configured with values larger than the number of beats in the read burst, read data is held until the last beat has been buffered. When configured with values larger than the depth of read channel buffers, read data is held until all entries in the read channel buffer have been occupied.
15	RCSSH	Redirect Cacheable System SHareable When set (1), enables redirection of all cacheable transactions that are system shareable to the redirect slaweway. Note: Only masters connecting using AMBA 4 ACE protocol are able to indicate system shareability for transactions.
14	RCOSH	Redirect Cacheable Outer SHareable When set (1), enables redirection of all cacheable transactions that are outer shareable to the redirect slaweway.
13	RCISH	Redirect Cacheable Inner SHareable When set (1), enables redirection of all cacheable transactions that are inner shareable to the redirect slaweway.
12	RCNSH	Redirect Cacheable Non-SHareable When set (1), enables redirection of all cacheable transactions that are non-shareable to the redirect slaweway.
11	RNCSSH	Redirect Non-Cacheable System SHareable When set (1), enables redirection of all non-cacheable transactions that are system shareable to the redirect slaweway. Note: Only masters connecting using AMBA 4 ACE protocol are able to indicate system shareability for transactions.

BIMC_M_TCU0_MPORT_MODE (cont.)

Bits	Name	Description
10	RNCOSH	Redirect Non-Cacheable Outer SHareable When set (1), enables redirection of all non-cacheable transactions that are outer shareable to the redirect slaway.
9	RNCISH	Redirect Non-Cacheable Inner SHareable When set (1), enables redirection of all non-cacheable transactions that are inner shareable to the redirect slaway.
8	RNCNSH	Redirect Non-Cacheable Non-SHareable When set (1), enables redirection of all non-cacheable transactions that are non-shareable to the redirect slaway.
4	NARROW_WRITES	When set (1), enables converting of 1-beat bus-width sized writes down to 32-bit ASIZE based on write strobes. It is recommended to be enabled for directly connected CPU masters. When cleared (0), disables conversion and all writes are forwarded downstream as generated by the master.
2	FIODV	Force In-Order Device When set (1), forces commands to device memory to be executed one at a time. In other words, commands to device memory are held until all previous commands have completed (responses have been received at the masterport) and all commands after the device command are held until the device completes (response has been received at the masterport.)
1	FIOSO	Force In-Order Strongly-Ordered When set (1), forces commands to strongly ordered memory to be executed one at a time. In other words, commands to strongly ordered memory are held until all previous commands have completed (responses have been received at the masterport) and all commands after the strongly ordered command are held until the strongly ordered completes (response has been received at the masterport.)
0	ORDERING_MODEL	Configures how commands to memories of type Strongly ordered and Device (SoDv) are handled in the masterport. When configured as STRICT, all commands to SoDv memory are held until all previous commands have completed (responses have been received at the masterport) and all commands after the SoDv command are held until the SoDv completes (i.e. response has been received at the masterport.) When configured as RELAXED, all commands to SoDv memory are treated as accessing Normal memory. 0x0: RELAXED 0x1: STRICT

0x0041C220 BIMC_M_TCU0_MPORT_WR_WAY_CROSSING**Type:** RW**Clock:** Config & Master Clocks**Reset State:** 0x00000000

Configures the level of strictness or relaxedness of way crossing checks for writes.

BIMC_M_TCU0_MPORT_WR_WAY_CROSSING

Bits	Name	Description
7:0	REG	1-bit per slaveway. When 2 or more bits are set (1), allows the corresponding slaveways to be treated as one slaveway when write commands are considered as way crossing, ie. write commands to these 2 (or more) slaveways are not treated as way crossing when one write command targets one slaveway and the next write command targets the second slaveway. Not treating write commands as way crossing allows the write commands to go to the slaveway ahead of the write data for potentially better overall system performance. When all bits are set (1), disables all way crossing checks. When all bits are cleared (0), way crossing is strict with each way crossing write command held until write data for all previous writes have been routed to their corresponding slaveways.

0x0041C230 BIMC_M_TCU0_MPORT_PRIORITYLVL_OVERRIDE

Type: RW

Clock: Config & Master Clocks

Reset State: 0x00000000

Override registers provide the capability to override one or more bus signals to specific values.

BIMC_M_TCU0_MPORT_PRIORITYLVL_OVERRIDE

Bits	Name	Description
9:8	PRIORITYLVL	Configures the values to override APRIORITYLVL from bus master with when OVERRIDE_PRIORITYLVL is set (1).
0	OVERRIDE_PRIORITYLVL	When set (1), overrides APRIORITYLVL from the bus master with values configured in PRIORITYLVL register field. When cleared (0), allows APRIORITYLVL from the bus master to be propagated downstream.

0x0041C240 BIMC_M_TCU0_MPORT_READ_COMMAND_OVERRIDE

Type: RW

Clock: Config & Master Clocks

Reset State: 0x00000000

Override registers provide the capability to override one or more bus signals to specific values.

BIMC_M_TCU0_MPORT_READ_COMMAND_OVERRIDE

Bits	Name	Description
25:24	AREQPRIORITY	Configures the values to override AREQPRIORITY from bus master with for read commands when OVERRIDE_AREQPRIORITY is set (1).
18:16	AMEMTYPE	Configures the values to override AMEMTYPE from bus master with for read commands when OVERRIDE_AMEMTYPE is set (1).
12	ATTRANSIENT	Configures the values to override ATTRANSIENT from bus master with for read commands when OVERRIDE_ATTRANSIENT is set (1).
11	ASHARED	Configures the values to override ASHARED from bus master with for read commands when OVERRIDE_ASHARED is set (1).
10	AREDIRECT	Configures the values to override AREDIRECT from bus master with for read commands when OVERRIDE_AREDIRECT is set (1).
9	AOOO	Configures the values to override AOOO from bus master with for read commands when OVERRIDE_AOOO is set (1).
8	AINNERSHARED	Configures the values to override AINNERSHARED from bus master with for read commands when OVERRIDE_AINNERSHARED is set (1).
6	OVERRIDE_AREQPRIORITY	When set (1), overrides AREQPRIORITY from the bus master for read commands with values configured in AREQPRIORITY register field. When cleared (0), allows AREQPRIORITY from the bus master to be propagated downstream.
5	OVERRIDE_AMEMTYPE	When set (1), overrides AMEMTYPE from the bus master for read commands with values configured in AMEMTYPE register field. When cleared (0), allows AMEMTYPE from the bus master to be propagated downstream.
4	OVERRIDE_ATTRANSIENT	When set (1), overrides ATTRANSIENT from the bus master for read commands with values configured in ATTRANSIENT register field. When cleared (0), allows ATTRANSIENT from the bus master to be propagated downstream.
3	OVERRIDE_ASHARED	When set (1), overrides ASHARED from the bus master for read commands with values configured in ASHARED register field. When cleared (0), allows ASHARED from the bus master to be propagated downstream.
2	OVERRIDE_AREDIRECT	When set (1), overrides AREDIRECT from the bus master for read commands with values configured in AREDIRECT register field. When cleared (0), allows AREDIRECT from the bus master to be propagated downstream.
1	OVERRIDE_AOOO	When set (1), overrides AOOO from the bus master for read commands with values configured in AOOO register field. When cleared (0), allows AOOO from the bus master to be propagated downstream.

BIMC_M_TCU0_MPORT_READ_COMMAND_OVERRIDE (cont.)

Bits	Name	Description
0	VERRIDE_AINNERSHARE D	When set (1), overrides AINNERSHARED from the bus master for read commands with values configured in AINNERSHARED register field. When cleared (0), allows AINNERSHARED from the bus master to be propagated downstream.

0x0041C250 BIMC_M_TCU0_MPORT_WRITE_COMMAND_OVERRIDE**Type:** RW**Clock:** Config & Master Clocks**Reset State:** 0x00000000

Override registers provide the capability to override one or more bus signals to specific values.

BIMC_M_TCU0_MPORT_WRITE_COMMAND_OVERRIDE

Bits	Name	Description
25:24	AREQPRIORITY	Configures the values to override AREQPRIORITY from bus master with for write commands when OVERRIDE_AREQPRIORITY is set (1).
18:16	AMEMTYPE	Configures the values to override AMEMTYPE from bus master with for write commands when OVERRIDE_AMEMTYPE is set (1).
12	ATRANSIENT	Configures the values to override ATRANSIENT from bus master with for write commands when OVERRIDE_ATRANSIENT is set (1).
11	ASHARED	Configures the values to override ASHARED from bus master with for write commands when OVERRIDE_ASHARED is set (1).
10	AREDIRECT	Configures the values to override AREDIRECT from bus master with for write commands when OVERRIDE_AREDIRECT is set (1).
9	AOOO	Configures the values to override AOOO from bus master with for write commands when OVERRIDE_AOOO is set (1).
8	AINNERSHARED	Configures the values to override AINNERSHARED from bus master with for write commands when OVERRIDE_AINNERSHARED is set (1).
6	VERRIDE_AREQPRIORIT Y	When set (1), overrides AREQPRIORITY from the bus master for write commands with values configured in AREQPRIORITY register field. When cleared (0), allows AREQPRIORITY from the bus master to be propagated downstream.
5	VERRIDE_AMEMTYPE	When set (1), overrides AMEMTYPE from the bus master for write commands with values configured in AMEMTYPE register field. When cleared (0), allows AMEMTYPE from the bus master to be propagated downstream.

BIMC_M_TCU0_MPORT_WRITE_COMMAND_OVERRIDE (cont.)

Bits	Name	Description
4	OVERRIDE_ATRANSIENT	When set (1), overrides ATRANSIENT from the bus master for write commands with values configured in ATRANSIENT register field. When cleared (0), allows ATRANSIENT from the bus master to be propagated downstream.
3	OVERRIDE_ASHARED	When set (1), overrides ASHARED from the bus master for write commands with values configured in ASHARED register field. When cleared (0), allows ASHARED from the bus master to be propagated downstream.
2	OVERRIDE_AREDIRECT	When set (1), overrides AREDIRECT from the bus master for write commands with values configured in AREDIRECT register field. When cleared (0), allows AREDIRECT from the bus master to be propagated downstream.
1	OVERRIDE_AOOO	When set (1), overrides AOOO from the bus master for write commands with values configured in AOOO register field. When cleared (0), allows AOOO from the bus master to be propagated downstream.
0	OVERRIDE_AINNERSHARE D	When set (1), overrides AINNERSHARED from the bus master for write commands with values configured in AINNERSHARED register field. When cleared (0), allows AINNERSHARED from the bus master to be propagated downstream.

0x0041C300 BIMC_M_TCU0_MPORT_BKE_ENABLE**Type:** RW**Clock:** Config & Master Clocks**Reset State:** 0x00000000**BIMC_M_TCU0_MPORT_BKE_ENABLE**

Bits	Name	Description
0	ENABLE	When set (1), enables the book keeping element (BKE) to determine AREQPRIORITY and APRIORITYLVL of commands from the bus master by comparing the amount of bus traffic from the bus master to a target, and determining a health level for the bus master. The health level determines the relative priority of commands from the bus master to allow the bus master to catch-up if it is not getting serviced, or slow down to allow other bus masters to get their share of system resources. When cleared (0), disables the book keeping element and AREQPRIORITY and APRIORITYLVL from the bus master are propagated downstream.

0x0041C304 BIMC_M_TCU0_MPORT_BKE_GRANT_PERIOD**Type:** RW**Clock:** Config & Reference Clocks**Reset State:** 0x00000000**BIMC_M_TCU0_MPORT_BKE_GRANT_PERIOD**

Bits	Name	Description
9:0	GRANT_PERIOD	Defines the period over which the book keeping element accumulates grants to the bus master for determining it's health level. Grant period is defined in number of reference timer ticks. Refer configuration of reference timer tick for exact time unit.

0x0041C308 BIMC_M_TCU0_MPORT_BKE_GRANT_COUNT**Type:** RW**Clock:** Config & Master Clocks**Reset State:** 0x00000000**BIMC_M_TCU0_MPORT_BKE_GRANT_COUNT**

Bits	Name	Description
15:0	GRANT_COUNT	Defines the target grant count for the bus master. The target grant count is the number of beats of read and write data the bus master is expected to write and read in the grant period and is used as a reference to determine if the bus master is behind or ahead of where it ought to be at any given time.

0x0041C320 BIMC_M_TCU0_MPORT_BKE_THRESHOLD_HIGH**Type:** RW**Clock:** Config & Master Clocks**Reset State:** 0x00000000**BIMC_M_TCU0_MPORT_BKE_THRESHOLD_HIGH**

Bits	Name	Description
15:0	THRESHOLD	Defines a high threshold to determine how much behind or ahead a bus master is with reference to where it ought to be at any given time.

0x0041C324 BIMC_M_TCU0_MPORT_BKE_THRESHOLD_MEDIUM**Type:** RW**Clock:** Config & Master Clocks**Reset State:** 0x00000000**BIMC_M_TCU0_MPORT_BKE_THRESHOLD_MEDIUM**

Bits	Name	Description
15:0	THRESHOLD	Defines a medium threshold to determine how much behind or ahead a bus master is with reference to where it ought to be at any given time.

0x0041C328 BIMC_M_TCU0_MPORT_BKE_THRESHOLD_LOW**Type:** RW**Clock:** Config & Master Clocks**Reset State:** 0x00000000**BIMC_M_TCU0_MPORT_BKE_THRESHOLD_LOW**

Bits	Name	Description
15:0	THRESHOLD	Defines a low threshold to determine how much behind or ahead a bus master is with reference to where it ought to be at any given time.

0x0041C340 BIMC_M_TCU0_MPORT_BKE_HEALTH_0**Type:** RW**Clock:** Config & Master Clocks**Reset State:** 0x00000000

BKE health registers configure the request priority and realtime priority level to use for commands from the bus master when it has been determined to be of a certain health.

BIMC_M_TCU0_MPORT_BKE_HEALTH_0

Bits	Name	Description
31	LIMIT_COMMANDS	When set (1), defines that no new commands may be accepted from the bus master as long as it is determined to be at health 0.
9:8	AREQPRIORITY	Defines the request priority of commands from the bus master when it has been determined to be at health 0.
1:0	PRIORITYLVL	Defines the realtime priority level to use for commands from the bus master when it has been determined to be at health 0.

0x0041C344 BIMC_M_TCU0_MPORT_BKE_HEALTH_1**Type:** RW**Clock:** Config & Master Clocks**Reset State:** 0x00000000

BKE health registers configure the request priority and realtime priority level to use for commands from the bus master when it has been determined to be of a certain health.

BIMC_M_TCU0_MPORT_BKE_HEALTH_1

Bits	Name	Description
31	LIMIT_COMMANDS	When set (1), defines that no new commands may be accepted from the bus master as long as it is determined to be at health 1.
9:8	AREQPRIORITY	Defines the request priority of commands from the bus master when it has been determined to be at health 1.
1:0	PRIORITYLVL	Defines the realtime priority level to use for commands from the bus master when it has been determined to be at health 1.

0x0041C348 BIMC_M_TCU0_MPORT_BKE_HEALTH_2**Type:** RW**Clock:** Config & Master Clocks**Reset State:** 0x00000000

BKE health registers configure the request priority and realtime priority level to use for commands from the bus master when it has been determined to be of a certain health.

BIMC_M_TCU0_MPORT_BKE_HEALTH_2

Bits	Name	Description
31	LIMIT_COMMANDS	When set (1), defines that no new commands may be accepted from the bus master as long as it is determined to be at health 2.
9:8	AREQPRIORITY	Defines the request priority of commands from the bus master when it has been determined to be at health 2.
1:0	PRIORITYLVL	Defines the realtime priority level to use for commands from the bus master when it has been determined to be at health 2.

0x0041C34C BIMC_M_TCU0_MPORT_BKE_HEALTH_3**Type:** RW**Clock:** Config & Master Clocks**Reset State:** 0x00000000

BKE health registers configure the request priority and realtime priority level to use for commands from the bus master when it has been determined to be of a certain health. Health 3 is the default health for the bus master.

BIMC_M_TCU0_MPORT_BKE_HEALTH_3

Bits	Name	Description
9:8	AREQPRIORITY	Defines the request priority of commands from the bus master when it has been determined to be at health 3.
1:0	PRIORITYLVL	Defines the realtime priority level to use for commands from the bus master when it has been determined to be at health 3.

0x0041C420 BIMC_M_TCU0_MPORT_STATUS_2A

Type: R

Clock: Config, Master & Pipe0 Clocks

Reset State: 0x00000000

Status registers indicate the state of several elements in the masterport, including the number of transactions outstanding/pending waiting for a response, emptiness, fullness of several buffers, etc. The state information is available only as a debug indication and could be delayed by a few clock cycles from when the state is reached to being reflected in the status registers. One or more of the state registers are read live and could capture intermediate values as they change.

BIMC_M_TCU0_MPORT_STATUS_2A

Bits	Name	Description
23:16	WR_PENDING_PRE_BUF	Indicates the number of write transactions that are outstanding/pending waiting for a response from the slave. This transaction count is from before address channel buffers, which interfaces to frontend of the masterport.
7:0	RD_PENDING_PRE_BUF	Indicates the number of read transactions that are outstanding/pending waiting for a response from the slave. This transaction count is from before address channel buffers, which interfaces to frontend of the masterport.

0x0041C424 BIMC_M_TCU0_MPORT_STATUS_2B

Type: R

Clock: Config, Master & Pipe0 Clocks

Reset State: 0x00000000

Status registers indicate the state of several elements in the masterport, including the number of transactions outstanding/pending waiting for a response, emptiness, fullness of several buffers, etc. The state information is available only as a debug indication and could be delayed by a few clock

cycles from when the state is reached to being reflected in the status registers. One or more of the state registers are read live and could capture intermediate values as they change.

BIMC_M_TCU0_MPORT_STATUS_2B

Bits	Name	Description
23:16	WR_PENDING_POST_BUF	Indicates the number of write transactions that are outstanding/pending waiting for a response from the slave. This transaction count is from after address channel buffers, which interfaces to core of the interconnect, i.e., arbiter and slaveway.
7:0	RD_PENDING_POST_BUF	Indicates the number of read transactions that are outstanding/pending waiting for a response from the slave. This transaction count is from after address channel buffers, which interfaces to core of the interconnect, i.e., arbiter and slaveway.

0x0041C430 BIMC_M_TCU0_MPORT_STATUS_3A

Type: R

Clock: Config, Master & Pipe0 Clocks

Reset State: 0x55111111

Status registers indicate the state of several elements in the masterport, including the number of transactions outstanding/pending waiting for a response, emptiness, fullness of several buffers, etc. The state information is available only as a debug indication and could be delayed by a few clock cycles from when the state is reached to being reflected in the status registers. One or more of the state registers are read live and could capture intermediate values as they change.

BIMC_M_TCU0_MPORT_STATUS_3A

Bits	Name	Description
31	RCB_DATA_RD_FULL	Indicates that the data part of read channel buffer is full when seen from it's read side, which interfaces to frontend of the masterport.
30	RCB_DATA_RD_EMPTY	Indicates that the data part of read channel buffer is empty when seen from it's read side, which interfaces to frontend of the masterport.
29	RCB_CTRL_RD_FULL	Indicates that the control part of read channel buffer is full when seen from it's read side, which interfaces to frontend of the masterport.
28	RCB_CTRL_RD_EMPTY	Indicates that the control part of read channel buffer is empty when seen from it's write side, which interfaces to frontend of the masterport.
27	RCB_DATA_WR_FULL	Indicates that the data part of read channel buffer is full when seen from it's write side, which interfaces to core of the interconnect, i.e., arbiter and slaveway.
26	RCB_DATA_WR_EMPTY	Indicates that the data part of read channel buffer is empty when seen from it's write side, which interfaces to core of the interconnect, i.e., arbiter and slaveway.

BIMC_M_TCU0_MPORT_STATUS_3A (cont.)

Bits	Name	Description
25	RCB_CTRL_WR_FULL	Indicates that the control part of read channel buffer is full when seen from it's write side, which interfaces to core of the interconnect, i.e., arbiter and slaveway.
24	RCB_CTRL_WR_EMPTY	Indicates that the control part of read channel buffer is empty when seen from it's write side, which interfaces to core of the interconnect, i.e., arbiter and slaveway.
21	BCB_RD_FULL	Indicates that the write response channel buffer is full when seen from it's read side, which interfaces to frontend of the masterport.
20	BCB_RD_EMPTY	Indicates that the write response channel buffer is empty when seen from it's read side, which interfaces to frontend of the masterport.
17	BCB_WR_FULL	Indicates that the write response channel buffer is full when seen from it's write side, which interfaces to core of the interconnect, i.e., arbiter and slaveway.
16	BCB_WR_EMPTY	Indicates that the write response channel buffer is empty when seen from it's write side, which interfaces to core of the interconnect, i.e., arbiter and slaveway.
13	WCB_RD_FULL	Indicates that the write channel buffer is full when seen from it's read side, which interfaces to core of the interconnect, i.e., arbiter and slaveway.
12	WCB_RD_EMPTY	Indicates that the write channel buffer is empty when seen from it's read side, which interfaces to core of the interconnect, i.e., arbiter and slaveway.
9	WCB_WR_FULL	Indicates that the write channel buffer is full when seen from it's write side, which interfaces to frontend of the masterport.
8	WCB_WR_EMPTY	Indicates that the write channel buffer is empty when seen from it's write side, which interfaces to frontend of the masterport.
5	ACB_RD_FULL	Indicates that the address channel buffer is full when seen from it's read side, which interfaces to core of the interconnect, i.e., arbiter and slaveway.
4	ACB_RD_EMPTY	Indicates that the address channel buffer is empty when seen from it's read side, which interfaces to core of the interconnect, i.e., arbiter and slaveway.
1	ACB_WR_FULL	Indicates that the address channel buffer is full when seen from it's write side, which interfaces to frontend of the masterport.
0	ACB_WR_EMPTY	Indicates that the address channel buffer is empty when seen from it's write side, which interfaces to frontend of the masterport.

0x0041C434 BIMC_M_TCU0_MPORT_STATUS_3B**Type:** R**Clock:** Config, Master & Pipe0 Clocks**Reset State:** 0x00111111

Buffer status register indicate the state of all buffers in the masterport, whether they are empty or full on either side of the buffer. Some buffers carry information from master to core and others carry information from core to master. Both write side and read side status are indicated separately, since they would be not be similar when master clock and core clock are not synchronous to each other.

BIMC_M_TCU0_MPORT_STATUS_3B

Bits	Name	Description
21	RGB_RD_FULL	Indicates that the read gathering buffer is full when seen from it's read side, which interfaces to frontend of the masterport.
20	RGB_RD_EMPTY	Indicates that the read gathering buffer is empty when seen from it's read side, which interfaces to frontend of the masterport.
17	RGB_WR_FULL	Indicates that the read gathering buffer is full when seen from it's write side, which interfaces to core of the interconnect, i.e., arbiter and slaweway.
16	RGB_WR_EMPTY	Indicates that the read gathering buffer is empty when seen from it's write side, which interfaces to core of the interconnect, i.e., arbiter and slaweway.
13	W2AB_RD_FULL	Indicates that the write channel to address channel user signals buffer is full when seen from it's read side, which interfaces to core of the interconnect, i.e., arbiter and slaweway.
12	W2AB_RD_EMPTY	Indicates that the write channel to address channel user signals buffer is empty when seen from it's read side, which interfaces to core of the interconnect, i.e., arbiter and slaweway.
9	W2AB_WR_FULL	Indicates that the write channel to address channel user signals buffer is full when seen from it's write side, which interfaces to frontend of the masterport.
8	W2AB_WR_EMPTY	Indicates that the write channel to address channel user signals buffer is empty when seen from it's write side, which interfaces to frontend of the masterport.
5	A2WB_RD_FULL	Indicates that the address channel to write channel user signals buffer is full when seen from it's read side, which interfaces to core of the interconnect, i.e., arbiter and slaweway.
4	A2WB_RD_EMPTY	Indicates that the address channel to write channel user signalsbuffer is empty when seen from it's read side, which interfaces to core of the interconnect, i.e., arbiter and slaweway.
1	A2WB_WR_FULL	Indicates that the address channel to write channel user signals buffer is full when seen from it's write side, which interfaces to frontend of the masterport.

BIMC_M_TCU0_MPORT_STATUS_3B (cont.)

Bits	Name	Description
0	A2WB_WR_EMPTY	Indicates that the address channel to write channel user signals buffer is empty when seen from it's write side, which interfaces to frontend of the masterport.

2.16 BIMC_M_TCU0_PROF

BRIC Traffic Profiling

2.16.1 BIMC_M_TCU0_PROF_BRIC_PROFILING

BRIC Traffic Profiling

0x0041D000 BIMC_M_TCU0_PROF_COMPONENT_INFO**Type:** R**Clock:** Config Clock**Reset State:** 0x00000202

Component information registers provide discoverability information on the type of component and instance of each component. These registers are always the very first register in each component's configuration address space and allow for a software routine to walk the configuration address space in steps of 4KB to identify the location of each configuration register space and to which component each 4KB configures.

BIMC_M_TCU0_PROF_COMPONENT_INFO

Bits	Name	Description
23:16	INSTANCE	Indicates the instance number of this traffic profiling.
15:8	SUB_TYPE	Identifies this configuration space as belonging to traffic profiling 0x2: PROFILING
7:0	TYPE	Identifies this configuration space as belonging to a component that interfaces to a bus master. 0x2: MASTERPORT

0x0041D020 BIMC_M_TCU0_PROF_CONFIGURATION_INFO_0**Type:** R**Clock:** Config Clock**Reset State:** 0x00000000

Configuration information registers provide information on the capability and configuration of each traffic profiling instance. See register field descriptions for more details.

BIMC_M_TCU0_PROF_CONFIGURATION_INFO_0

Bits	Name	Description
31:24	TENURE_WIDTH	Indicates the width of tenures (latencies) measured in traffic profiling.
23:16	TRACKING_TABLE_DEPTH	Indicates the number of entries in the table that tracks outstanding commands for determining how many commands have larger turnaround latencies than a programmed threshold.
7:0	SCALING_FACTOR	Indicates how many extra scaling is performed on events generated by traffic profiling if such events have to be synchronized from the tenure clock to the event clock. Any value that is non-power-of-2 is converted to power-of-2 by hardware. When the value read from this register field is used in software, similar conversion of non-power-of-2 values to next higher power-of-2 value is necessary.

0x0041D030 BIMC_M_TCU0_PROF_CONFIGURATION_INFO_1

Type: R

Clock: Config Clock

Reset State: 0x00000000

Configuration information registers provide information on the capability and configuration of each traffic profiling instance. See register field descriptions for more details.

BIMC_M_TCU0_PROF_CONFIGURATION_INFO_1

Bits	Name	Description
23:16	PIPE2_ACH_DEPTH	Indicates the depth of address channel buffer in pipe2 of masterport in which this traffic profiling instance is paired with. Applies only to traffic profiling instances in XR type masterports, which contains three distinct paths to slaves, in pipe0, pipe1, pipe2. In non-XR type masterports, value contained in field does not refer to any actual address channel buffer.
15:8	PIPE1_ACH_DEPTH	Indicates the depth of address channel buffer in pipe1 of masterport in which this traffic profiling instance is paired with. Applies only to traffic profiling instances in XR type masterports, which contains three distinct paths to slaves, in pipe0, pipe1, pipe2. In non-XR type masterports, value contained in field does not refer to any actual address channel buffer.
7:0	PIPE0_ACH_DEPTH	Indicates the depth of address channel buffer in pipe0 of masterport in which this traffic profiling instance is paired with. Applies only to traffic profiling instances in XR type masterports, which contains three distinct paths to slaves, in pipe0, pipe1, pipe2. In non-XR type masterports, pipe0 refers to the common address channel buffer used for all slave accesses.

0x0041D200 BIMC_M_TCU0_PROF_CLOCK_CTRL**Type:** RW**Clock:** Config, Event & Tenure Clocks**Reset State:** 0x00000003**BIMC_M_TCU0_PROF_CLOCK_CTRL**

Bits	Name	Description
1	TENURE_CLOCK_GATING_EN	When set (1), enables activity based clock gating of all registers clocked by tenure clock. Activity based clock gating uses architectural clock gates to reduce dynamic power consumption by gating clocks during periods of in-activity. When cleared (0), disables all activity based clock gating by forcing all architectural clock gates to not gate clocks even during periods of in-activity.
0	EVENT_CLOCK_GATING_EN	When set (1), enables activity based clock gating of all registers clocked by event clock. Activity based clock gating uses architectural clock gates to reduce dynamic power consumption by gating clocks during periods of in-activity. When cleared (0), disables all activity based clock gating by forcing all architectural clock gates to not gate clocks even during periods of in-activity.

0x0041D210 BIMC_M_TCU0_PROF_MODE**Type:** RW**Clock:** Config, Event & Tenure Clocks**Reset State:** 0x00000000

BIMC_M_TCU0_PROF_MODE

Bits	Name	Description
25:24	WR_TENURE_MODE	Configures the start and end events for monitoring write latencies/tenures. When configured to WR_CMD_TO_BRESP, starting event is first cycle of write command from bus master as seen by traffic profiling and ending event is first cycle of write response to bus master as seen by traffic profiling. When configured to WR_CMD_TO_BRESP_HANDSHAKE, starting event is first cycle of write command from bus master as seen by traffic profiling and ending event is transfer handshake of write response to bus master as seen by traffic profiling. When configured to WR_CMD_HANDSHAKE, starting event is first cycle of write command from bus master as seen by traffic profiling and ending event is transfer handshake of the same write command with bus master as seen by traffic profiling. When configured to WR_RESP_HANDSHAKE, starting event is first cycle of write response to bus master as seen by traffic profiling and ending event is transfer handshake of write response to bus master as seen by traffic profiling. 0x0: WR_CMD_TO_BRESP 0x1: WR_CMD_TO_BRESP_HANDSHAKE 0x2: WR_CMD_HANDSHAKE 0x3: WR_RESP_HANDSHAKE
17:16	RD_TENURE_MODE	Configures the start and end events for monitoring read latencies/tenures. When configured to RD_CMD_TO_FIRST_RDATA, starting event is first cycle of read command from bus master as seen by traffic profiling and ending event is first cycle of first beat of read data to bus master as seen by traffic profiling. When configured to RD_CMD_TO_LAST_RDATA, starting event is first cycle of read command from bus master as seen by traffic profiling and ending event is transfer handshake of last beat of read data to bus master as seen by traffic profiling. When configured to RD_CMD_HANDSHAKE, starting event is first cycle of read command from bus master as seen by traffic profiling and ending event is transfer handshake of the same read command with bus master as seen by traffic profiling. When configured to RD_DATA_HANDSHAKE, starting event is first cycle of first beat of read data to bus master as seen by traffic profiling and ending event is transfer handshake of last beat of read data to bus master as seen by traffic profiling. 0x0: RD_CMD_TO_FIRST_RDATA 0x1: RD_CMD_TO_LAST_RDATA 0x2: RD_CMD_HANDSHAKE 0x3: RD_DATA_HANDSHAKE
0	PROFILING_EN	When set (1), enables traffic profiling to monitor bus transactions from bus master and generate events for counting in performance monitors. When cleared (0), disables traffic profiling.

0x0041D214 BIMC_M_TCU0_PROF_RD_TENURE_THRESHOLD**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_TCU0_PROF_RD_TENURE_THRESHOLD**

Bits	Name	Description
15:0	THRESHOLD	Configures the threshold over which read latencies/tenures need to be before an event is generated to performance monitors for counting max latencies.

0x0041D218 BIMC_M_TCU0_PROF_WR_TENURE_THRESHOLD**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_TCU0_PROF_WR_TENURE_THRESHOLD**

Bits	Name	Description
15:0	THRESHOLD	Configures the threshold over which write latencies/tenures need to be before an event is generated to performance monitors for counting max latencies.

0x0041D220 BIMC_M_TCU0_PROF_SWAY_SELECT**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_TCU0_PROF_SWAY_SELECT**

Bits	Name	Description
7:0	SWAY	Selects the slaveways to which commands are routed to profile. 1-bit per slaveway.

0x0041D230 BIMC_M_TCU0_PROF_MID_MASK**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000

BIMC_M_TCU0_PROF_MID_MASK

Bits	Name	Description
15:0	MID_MASK	When a bit position is set (1), enables matching of that bit position in MID of command with same bit position in MID_MATCH. Commands with MIDs that do not match are not considered for profiling.

0x0041D234 BIMC_M_TCU0_PROF_MID_MATCH**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_TCU0_PROF_MID_MATCH**

Bits	Name	Description
15:0	MID_MATCH	Configures the value to match for in AMID in commands. Used in conjunction with MID_MASK, which controls which bits of MID are compared.

0x0041D238 BIMC_M_TCU0_PROF_MID_INV_MATCH**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_TCU0_PROF_MID_INV_MATCH**

Bits	Name	Description
0	MID_INV_MATCH	When set (1), configures MID matching to result in an inverse match, i.e., only commands whose MIDs that do not match MID_MATCH are considered for profiling.

0x0041D240 BIMC_M_TCU0_PROF_ADDR_MASK_LOWER**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000

BIMC_M_TCU0_PROF_ADDR_MASK_LOWER

Bits	Name	Description
31:10	ADDR_MASK_LOWER	Configures ADDR_MASK[31:0]. When a bit position is set (1), enables matching of that bit position of address of command with same bit position in ADDR_MATCH[31:0].

0x0041D244 BIMC_M_TCU0_PROF_ADDR_MASK_UPPER**Type:** R**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_TCU0_PROF_ADDR_MASK_UPPER**

Bits	Name	Description
3:0	UNUSED	

0x0041D248 BIMC_M_TCU0_PROF_ADDR_MATCH_LOWER**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_TCU0_PROF_ADDR_MATCH_LOWER**

Bits	Name	Description
31:10	ADDR_MATCH_LOWER	Configures the value to match for in address of commands. Used in conjunction with ADDR_MASK[31:0], which controls which bits of address are compared.

0x0041D24C BIMC_M_TCU0_PROF_ADDR_MATCH_UPPER**Type:** R**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_TCU0_PROF_ADDR_MATCH_UPPER**

Bits	Name	Description
3:0	UNUSED	

0x0041D250 BIMC_M_TCU0_PROF_ADDR_INV_MATCH**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_TCU0_PROF_ADDR_INV_MATCH**

Bits	Name	Description
0	ADDR_INV_MATCH	When set (1), configures address matching to result in an inverse match, i.e., only commands whose addresses do not match ADDR_MATCH are considered for profiling.

0x0041D260 BIMC_M_TCU0_PROF_ACH_MASK_0**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_TCU0_PROF_ACH_MASK_0**

Bits	Name	Description
24	AINNERSHARED_MASK	When set (1), enables matching of AINNERSHARED in commands with AINNERSHARED_MATCH.
16	AFULL_MASK	When set (1), enables matching of AFULL in commands with AFULL_MATCH.
8	AEXCLUSIVE_MASK	When set (1), enables matching of AEXCLUSIVE in commands with AEXCLUSIVE_MATCH.
0	ABURST_MASK	When set (1), enables matching of ABURST in commands with ABURST_MATCH.

0x0041D264 BIMC_M_TCU0_PROF_ACH_MATCH_0**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_TCU0_PROF_ACH_MATCH_0**

Bits	Name	Description
24	AINNERSHARED_MATCH	Configures the value to match for in AINNERSHARED in commands. Used in conjunction with AINNERSHARED_MASK, which controls whether AINNERSHARED in commands is compared

BIMC_M_TCU0_PROF_ACH_MATCH_0 (cont.)

Bits	Name	Description
16	AFULL_MATCH	Configures the value to match for in AFULL in commands. Used in conjunction with AFULL_MASK, which controls whether AFULL in commands is compared.
8	AEXCLUSIVE_MATCH	Configures the value to match for in AEXCLUSIVE in commands. Used in conjunction with AEXCLUSIVE_MASK, which controls whether AFULL in commands is compared.
0	ABURST_MATCH	Configures the value to match for in ABURST in commands. Used in conjunction with ABURST_MASK, which controls whether ABURST in commands is compared.

0x0041D270 BIMC_M_TCU0_PROF_ACH_MASK_1**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_TCU0_PROF_ACH_MASK_1**

Bits	Name	Description
24	AMSSSELFAUTH_MASK	When set (1), enables matching of AMSSSELFAUTH in commands with AMSSSELFAUTH_MATCH.
18:16	AMEMTYPE_MASK	When a bit position is set (1), enables matching of corresponding bit position of AMEMTYPE with AMEMTYPE_MATCH.
11:8	ALEN_MASK	When a bit position is set (1), enables matching of corresponding bit position of ALEN with ALEN_MATCH.
0	AINST_MASK	When set (1), enables matching of AINST in commands with AINST_MASK.

0x0041D274 BIMC_M_TCU0_PROF_ACH_MATCH_1**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_TCU0_PROF_ACH_MATCH_1**

Bits	Name	Description
24	AMSSSELFAUTH_MATCH	Configures the value to match for in AMSSSELFAUTH in commands. Used in conjunction with AMSSSELFAUTH_MASK, which controls whether AMSSSELFAUTH in commands is compared.

BIMC_M_TCU0_PROF_ACH_MATCH_1 (cont.)

Bits	Name	Description
18:16	AMEMTYPE_MATCH	Configures the value to match for in AMEMTYPE in commands. Used in conjunction with AMEMTYPE_MASK, which controls which bits of AMEMTYPE in commands are compared.
11:8	ALEN_MATCH	Configures the value to match for in ALEN in commands. Used in conjunction with ALEN_MASK, which controls which bits of ALEN in commands are compared.
0	AINST_MATCH	Configures the value to match for in AINST in commands. Used in conjunction with AINST_MASK, which controls whether AINST in commands is compared.

0x0041D278 BIMC_M_TCU0_PROF_ACH_INV_MATCH_1**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_TCU0_PROF_ACH_INV_MATCH_1**

Bits	Name	Description
16	AMEMTYPE_INV_MATCH	When set (1), configures AMEMTYPE matching to result in an inverse match, i.e., only commands whose AMEMTYPE does not match AMEMTYPE_MATCH are considered for profiling.
8	ALEN_INV_MATCH	When set (1), configures ALEN matching to result in an inverse match, i.e., only commands when ALEN does not match ALEN_MATCH are considered for profiling.

0x0041D280 BIMC_M_TCU0_PROF_ACH_MASK_2**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_TCU0_PROF_ACH_MASK_2**

Bits	Name	Description
24	APROTNS_MASK	When set (1), enables matching of APROTNS in commands with APROTNS_MATCH.
16	APRIV_MASK	When set (1), enables matching of APRIV in commands with APRIV_MATCH.
8	AOOO_MASK	When set (1), enables matching of AOOO in commands with AOOO_MATCH.
0	ANOALLOCATE_MASK	When set (1), enables matching of ANOALLOCATE in commands with ANOALLOCATE_MATCH.

0x0041D284 BIMC_M_TCU0_PROF_ACH_MATCH_2**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_TCU0_PROF_ACH_MATCH_2**

Bits	Name	Description
24	APROTNS_MATCH	Configures the value to match for in APROTNS in commands. Used in conjunction with APROTNS_MASK, which controls whether APROTNS in commands is compared
16	APRIV_MATCH	Configures the value to match for in APRIV in commands. Used in conjunction with APRIV_MASK, which controls whether APRIV in commands is compared
8	AOOO_MATCH	Configures the value to match for in AOOO in commands. Used in conjunction with AOOO_MASK, which controls whether AOOO in commands is compared
0	ANOALLOCATE_MATCH	Configures the value to match for in ANOALLOCATE in commands. Used in conjunction with ANOALLOCATE_MASK, which controls whether ANOALLOCATE in commands is compared

0x0041D290 BIMC_M_TCU0_PROF_ACH_MASK_3**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_TCU0_PROF_ACH_MASK_3**

Bits	Name	Description
24	ASHARED_MASK	When set (1), enables matching of ASHARED in commands with ASHARED_MATCH.
17:16	AREQPRIORITY_MASK	When a bit position is set (1), enables matching of corresponding bit position of AREQPRIORITY with AREQPRIORITY_MATCH.
8	AREDIRECT_MASK	When set (1), enables matching of AREDIRECT in commands with AREDIRECT_MATCH.
0	ARDBEATNDXEN_MASK	When set (1), enables matching of ARDBEATNDXEN in commands with ARDBEATNDXEN_MATCH.

0x0041D294 BIMC_M_TCU0_PROF_ACH_MATCH_3**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_TCU0_PROF_ACH_MATCH_3**

Bits	Name	Description
24	ASHARED_MATCH	Configures the value to match for in ASHARED in commands. Used in conjunction with ASHARED_MASK, which controls whether ASHARED in commands is compared
17:16	AREQPRIORITY_MATCH	Configures the value to match for in AREQPRIORITY in commands. Used in conjunction with AREQPRIORITY_MASK, which controls which bits of AREQPRIORITY in commands are compared.
8	AREDIRECT_MATCH	Configures the value to match for in AREDIRECT in commands. Used in conjunction with AREDIRECT_MASK, which controls whether AREDIRECT in commands is compared
0	ARDBEATNDXEN_MATCH	Configures the value to match for in ARDBEATNDXEN in commands. Used in conjunction with ARDBEATNDXEN_MASK, which controls whether ARDBEATNDXEN in commands is compared

0x0041D298 BIMC_M_TCU0_PROF_ACH_INV_MATCH_3**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_TCU0_PROF_ACH_INV_MATCH_3**

Bits	Name	Description
16	AREQPRIORITY_INV_MATCH	When set (1), configures AREQPRIORITY matching to result in an inverse match, i.e., only commands whose AREQPRIORITY does not match AREQPRIORITY_MATCH are considered for profiling

0x0041D2A0 BIMC_M_TCU0_PROF_ACH_MASK_4**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000

BIMC_M_TCU0_PROF_ACH_MASK_4

Bits	Name	Description
24	AWRITE_MASK	When set (1), enables matching of AWRITE in commands with AWRITE_MATCH.
20:16	AVMID_MASK	When a bit position is set (1), enables matching of corresponding bit position in AVMID with AVMID_MATCH
8	ATRANSIENT_MASK	When set (1), enables matching of ATRANSIENT in commands with ATRANSIENT_MATCH.
2:0	ASIZE_MASK	When a bit position is set (1), enables matching of corresponding bit position in ASIZE with ASIZE_MATCH.

0x0041D2A4 BIMC_M_TCU0_PROF_ACH_MATCH_4**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_TCU0_PROF_ACH_MATCH_4**

Bits	Name	Description
24	AWRITE_MATCH	Configures the value to match for in AWRITE in commands. Used in conjunction with AWRITE_MASK, which controls whether AWRITE in commands is compared.
20:16	AVMID_MATCH	Configures the value to match for in AVMID in commands. Used in conjunction with AVMID_MASK, which controls which bits of AVMID in commands are compared.
8	ATRANSIENT_MATCH	Configures the value to match for in ATRANSIENT in commands. Used in conjunction with ATRANSIENT_MASK, which controls whether ATRANSIENT in commands is compared.
2:0	ASIZE_MATCH	Configures the value to match for in ASIZE in commands. Used in conjunction with ASIZE_MASK, which controls which bits of ASIZE in commands are compared.

0x0041D2A8 BIMC_M_TCU0_PROF_ACH_INV_MATCH_4**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000

BIMC_M_TCU0_PROF_ACH_INV_MATCH_4

Bits	Name	Description
16	AVMID_INV_MATCH	When set (1), configures AVMID matching to result in an inverse match, i.e., only commands whose AVMID does not match AVMID_MATCH are considered for profiling.
0	ASIZE_INV_MATCH	When set (1), configures ASIZE matching to result in an inverse match, i.e., only commands whose AVMID does not match ASIZE_MATCH are considered for profiling.

0x0041D300 BIMC_M_TCU0_PROF_BCH_MASK**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_TCU0_PROF_BCH_MASK**

Bits	Name	Description
1:0	BRESP_MASK	When a bit position is set (1), enables matching of corresponding bit position in BRESP with BRESP_MATCH.

0x0041D304 BIMC_M_TCU0_PROF_BCH_MATCH**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_TCU0_PROF_BCH_MATCH**

Bits	Name	Description
1:0	BRESP_MATCH	Configures the value to match for in BRESP in write responses. Used in conjunction with BRESP_MASK, which controls which bits of BRESP in writes responses are compared.

0x0041D308 BIMC_M_TCU0_PROF_BCH_INV_MATCH**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000

BIMC_M_TCU0_PROF_BCH_INV_MATCH

Bits	Name	Description
0	BRESP_INV_MATCH	When set (1), configures BRESP matching to result in an inverse match, i.e., only write responses whose BRESP does not match BRESP_MATCH are considered for profiling.

0x0041D340 BIMC_M_TCU0_PROF_RCH_MASK**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_TCU0_PROF_RCH_MASK**

Bits	Name	Description
8	RLAST_MASK	When set (1), enables matching of RLAST in commands with corresponding setting in RCH_MATCH register.
1:0	RRESP_MASK	When a bit position is set (1), enables matching of corresponding bit position in RRESP with RRESP_MATCH.

0x0041D344 BIMC_M_TCU0_PROF_RCH_MATCH**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_TCU0_PROF_RCH_MATCH**

Bits	Name	Description
8	RLAST_MATCH	Configures the value to match for in RLAST in read responses. Used in conjunction with RLAST_MASK, which controls whether RLAST in read responses is compared.
1:0	RRESP_MATCH	Configures the value to match for in RRESP in read responses. Used in conjunction with RRESP_MASK, which controls which bits of RRESP in read responses are compared.

0x0041D348 BIMC_M_TCU0_PROF_RCH_INV_MATCH**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000

BIMC_M_TCU0_PROF_RCH_INV_MATCH

Bits	Name	Description
0	RRESP_INV_MATCH	When set (1), configures RRESP matching to result in an inverse match, i.e., only read responses whose RRESP does not match RRESP_MATCH are considered for profiling.

**0x0041D400+ BIMC_M_TCU0_PROF_RD_STATUS_n, n=[0..0]
0x4*n**
Type: R**Clock:** Config & Tenure Clocks**Reset State:** 0x00000000**BIMC_M_TCU0_PROF_RD_STATUS_n**

Bits	Name	Description
31	RD_OUTSTANDING_OVERFLOW	Indicates that the number of outstanding reads crossed the maximum supported (255) during when traffic profiling was last enabled. It is recommended that this register be read only after traffic profiling has been disabled.
30	RD_OUTSTANDING_UNDERFLOW	Indicates that the number of outstanding reads went below 0 during when traffic profiling was last enabled. It indicates that more command completions were seen compared to commands starts. The most likely scenario is an overflow of number of outstanding reads occurred leading to some command starts not being tracked. It is recommended that this register be read only after traffic profiling has been disabled.
7:0	RD_OUTSTANDING_MAX	Indicates the maximum number of outstanding reads that were seen when traffic profiling was last enabled. Only reads that match the MID filter and slaweway select filter are accounted for. It is recommended that this register be read only after traffic profiling has been disabled.

**0x0041D410+ BIMC_M_TCU0_PROF_WR_STATUS_n, n=[0..0]
0x4*n**
Type: R**Clock:** Config & Tenure Clocks**Reset State:** 0x00000000**BIMC_M_TCU0_PROF_WR_STATUS_n**

Bits	Name	Description
31	WR_OUTSTANDING_OVERFLOW	Indicates that the number of outstanding writes crossed the maximum supported (255) during when traffic profiling was last enabled. It is recommended that this register be read only after traffic profiling has been disabled.

BIMC_M_TCU0_PROF_WR_STATUS_n (cont.)

Bits	Name	Description
30	WR_OUTSTANDING_UNDE RFLOW	Indicates that the number of outstanding writes went below 0 during when traffic profiling was last enabled. It indicates that more command completions were seen compared to commands starts. The most likely scenario is an overflow of number of outstanding writes occurred leading to some command starts not being tracked. It is recommended that this register be read only after traffic profiling has been disabled.
7:0	WR_OUTSTANDING_MAX	Indicates the maximum number of outstanding writes that were seen when traffic profiling was last enabled. Only writes that match the MID filter and slaweway select filter are accounted for. It is recommended that this register be read only after traffic profiling has been disabled.

2.17 BIMC_M_TCU1_MPORT

BRIC Masterport

2.17.1 BIMC_M_TCU1_MPORT_BRIC_MASTERPORT

BRIC Masterport

0x00420000 BIMC_M_TCU1_MPORT_COMPONENT_INFO**Type:** R**Clock:** Config Clock**Reset State:** 0x00000102

Component information registers provide discoverability information on the type of component and instance of each component. These registers are always the very first register in each component's configuration address space and allow for a software routine to walk the configuration address space in steps of 4KB to identify the location of each configuration register space and to which component each 4KB configures.

BIMC_M_TCU1_MPORT_COMPONENT_INFO

Bits	Name	Description
23:16	INSTANCE	Indicates the instance number of this masterport.
15:8	SUB_TYPE	Identifies this configuration space as belonging to a masterport 0x1: MASTERPORT
7:0	TYPE	Identifies this configuration space as belonging to a component that interfaces to a bus master. 0x2: MASTERPORT

0x00420020 BIMC_M_TCU1_MPORT_CONFIGURATION_INFO_0**Type:** R**Clock:** Config Clock**Reset State:** 0x01000000

Configuration information registers provide information on the capability and configuration of each masterport instance. See register field descriptions for more details.

BIMC_M_TCU1_MPORT_CONFIGURATION_INFO_0

Bits	Name	Description
31:24	SYNC_MODE	Indicates the clock relationship between the clock used in the master side and the clock used in the core side of the masterport. When ASYNC, identifies the master clock and core clock as unrelated and asynchronous to each other. When SYNC, identifies the master clock and core clock as synchronous to each other. When ISOSYNC, identifies the master clock and core clock as related, but of different frequencies, which are of integer ratios, i.e., they are iso-synchronous to each other. 0x0: ASYNC 0x1: SYNC 0x2: ISOSYNC
15:8	CONNECTION_TYPE	Indicates the type of master connecting to this masterport. When DIRECT, indicates that a bus master is directly connected to this masterport. When CASCADE, indicates that bus masters connect to this masterport through another level of interconnect. 0x0: DIRECT 0x1: CASCADE
7:0	FUNCTIONALITY	Indicates the functionality and type of bus traffic handled by this masterport. When INORDER, identifies this masterport as optimized for in-order bus traffic. When OOO_ONLY, identifies this masterport as optimized for out-of-order bus traffic, where a single master is capable of handling responses return in any order. When REORDER, identifies this masterport as having a reorder buffer for re-ordering responses to command order, while allowing bus slaves to return responses in any order they desire. When STUB, identifies this masterport has a stub. 0x0: STUB 0x1: OOO_ONLY 0x2: INORDER 0x3: REORDER

0x00420030 BIMC_M_TCU1_MPORT_CONFIGURATION_INFO_1**Type:** R**Clock:** Config Clock**Reset State:** 0x00000000

Configuration information registers provide information on the capability and configuration of each masterport instance. See register field descriptions for more details.

BIMC_M_TCU1_MPORT_CONFIGURATION_INFO_1

Bits	Name	Description
31:0	SWAY_CONNECTIVITY	Indicates slaveways that this masterport can route commands to. 1-bit per slaveway. When a specific bit is set (1), indicates that commands can be routed to the slaveway corresponding to this bit. When a specific bit is clear (0), indicates that commands are not routed to the slaveway corresponding to this bit.

0x00420040 BIMC_M_TCU1_MPORT_CONFIGURATION_INFO_2A

Type: R

Clock: Config Clock

Reset State: 0x00000000

Configuration information registers provide information on the capability and configuration of each masterport instance. See register field descriptions for more details.

BIMC_M_TCU1_MPORT_CONFIGURATION_INFO_2A

Bits	Name	Description
31:16	M_DATA_WIDTH	Indicates the width of read and write data from the bus master.
15:8	M_TID_WIDTH	Indicates the width of transfer IDs from the bus master.
7:0	M_MID_WIDTH	Indicates the width of master ID from the bus master (includes any bus and port IDs if bus master is an interconnect.)

0x00420044 BIMC_M_TCU1_MPORT_CONFIGURATION_INFO_2B

Type: R

Clock: Config Clock

Reset State: 0x00000000

Configuration information registers provide information on the capability and configuration of each masterport instance. See register field descriptions for more details.

BIMC_M_TCU1_MPORT_CONFIGURATION_INFO_2B

Bits	Name	Description
15:0	C_DATA_WIDTH	Indicates the width of read and write data used in core of this masterport.

0x00420050 BIMC_M_TCU1_MPORT_CONFIGURATION_INFO_3**Type:** R**Clock:** Config Clock**Reset State:** 0x00000000

Configuration information registers provide information on the capability and configuration of each masterport instance. See register field descriptions for more details.

BIMC_M_TCU1_MPORT_CONFIGURATION_INFO_3

Bits	Name	Description
31:24	RCH_DEPTH	Indicates the depth of read channel buffers in this masterport. A beat of read data occupies one entry in the read channel buffer, until accepted by the bus master.
23:16	BCH_DEPTH	Indicates the depth of write response channel buffers in this masterport. Each write response occupies one entry in the write response channel buffer, until accepted by the bus master.
15:8	WCH_DEPTH	Indicates the depth of write channel buffers in this masterport. A beat of write data occupies one entry in the write channel buffer, until accepted by the arbiter.
7:0	ACH_DEPTH	Indicates the depth of address channel buffers in this masterport. Each command occupies one entry in the address channel buffer, until accepted by the arbiter.

0x00420060 BIMC_M_TCU1_MPORT_CONFIGURATION_INFO_4**Type:** R**Clock:** Config Clock**Reset State:** 0x00000000

Configuration information registers provide information on the capability and configuration of each masterport instance. See register field descriptions for more details.

BIMC_M_TCU1_MPORT_CONFIGURATION_INFO_4

Bits	Name	Description
15:8	REORDER_BUFFER_DEPTH	Indicates the number of read data beats that can be re-ordered in this masterport (if the masterport provides response re-ordering functionality.)
7:0	REORDER_TABLE_DEPTH	Indicates the number of read and write commands whose responses can be re-ordered in this masterport (if the masterport provides responses re-ordering functionality.)

0x00420070 BIMC_M_TCU1_MPORT_CONFIGURATION_INFO_5A**Type:** R**Clock:** Config Clock**Reset State:** 0x00000000

Configuration information registers provide information on the capability and configuration of each masterport instance. See register field descriptions for more details.

BIMC_M_TCU1_MPORT_CONFIGURATION_INFO_5A

Bits	Name	Description
31:0	ACH_PIPELINE_STAGES	Indicates pipeline stages enabled on the address channel. Each digit identifies a specific pipeline stage in the masterport, where '0' indicates no pipeline stage, '1' indicates 1-deep pipeline stage where forward path is registered, return path is fed through, '2' indicates 2-deep pipeline stage where both forward and return path are pipelined. Digit 3 identifies pipelining on the master to masterport interface, digit 2 identifies pipelining after address decode, digit 1 identifies pipelining in the address channel buffer and digit 0 identifies pipelining on the masterport to arbiter interface.

0x00420074 BIMC_M_TCU1_MPORT_CONFIGURATION_INFO_5B**Type:** R**Clock:** Config Clock**Reset State:** 0x00000000

Configuration information registers provide information on the capability and configuration of each masterport instance. See register field descriptions for more details.

BIMC_M_TCU1_MPORT_CONFIGURATION_INFO_5B

Bits	Name	Description
31:0	WCH_PIPELINE_STAGES	Indicates pipeline stages enabled on the write channel. Each digit identifies a specific pipeline stage in the masterport, where '0' indicates no pipeline stage, '1' indicates 1-deep pipeline stage where forward path is registered, return path is fed through, '2' indicates 2-deep pipeline stage where both forward and return path are pipelined. Digit 3 identifies pipelining on the master to masterport interface, digit 2 identifies pipelining after address decode, digit 1 identifies pipelining in the write channel buffer and digit 0 identifies pipelining on the masterport to arbiter interface.

0x00420078 BIMC_M_TCU1_MPORT_CONFIGURATION_INFO_5C**Type:** R**Clock:** Config Clock**Reset State:** 0x00000000

Configuration information registers provide information on the capability and configuration of each masterport instance. See register field descriptions for more details.

BIMC_M_TCU1_MPORT_CONFIGURATION_INFO_5C

Bits	Name	Description
31:0	BCH_PIPELINE_STAGES	Indicates pipeline stages enabled on the write response channel. Each digit identifies a specific pipeline stage in the masterport, where '0' indicates no pipeline stage, '1' indicates 1-deep pipeline stage where forward path is registered, return path is fed through, '2' indicates 2-deep pipeline stage where both forward and return path are pipelined. Digit 2 identifies pipelining on the slaveway to masterport interface after arbitration of write response channels from multiple slaveways, digit 1 identifies pipelining in the write response channel buffer and digit 0 identifies pipelining on the masterport to master interface.

0x0042007C BIMC_M_TCU1_MPORT_CONFIGURATION_INFO_5D**Type:** R**Clock:** Config Clock**Reset State:** 0x00000000

Configuration information registers provide information on the capability and configuration of each masterport instance. See register field descriptions for more details.

BIMC_M_TCU1_MPORT_CONFIGURATION_INFO_5D

Bits	Name	Description
31:0	RCH_PIPELINE_STAGES	Indicates pipeline stages enabled on the read channel. Each digit identifies a specific pipeline stage in the masterport, where '0' indicates no pipeline stage, '1' indicates 1-deep pipeline stage where forward path is registered, return path is fed through, '2' indicates 2-deep pipeline stage where both forward and return path are pipelined. Digit 2 identifies pipelining on the slaveway to masterport interface after arbitration of read channels from multiple slaveways, digit 1 identifies pipelining in the read channel buffer and digit 0 identifies pipelining on the masterport to master interface.

0x00420100 BIMC_M_TCU1_MPORT_INTERRUPT_STATUS**Type:** R**Clock:** Config Clock**Reset State:** 0x00000000

Interrupt status registers provide indication of which interrupts from this masterport require servicing.

BIMC_M_TCU1_MPORT_INTERRUPT_STATUS

Bits	Name	Description
1:0	RFU	Reserved for future use

0x00420108 BIMC_M_TCU1_MPORT_INTERRUPT_CLEAR**Type:** W**Clock:** Config Clock**Reset State:** 0x00000000

Interrupt clear registers provide the mechanism to turn off interrupt sources once they have been acknowledged or serviced. A write of '1' to the bits corresponding to the interrupts indicated in interrupt status is required to turn off interrupts.

BIMC_M_TCU1_MPORT_INTERRUPT_CLEAR

Bits	Name	Description
1:0	RFU	Reserved for future use

0x0042010C BIMC_M_TCU1_MPORT_INTERRUPT_ENABLE**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000

Interrupt enable registers provide control over which interrupts are allowed to be propagated to an interrupt controller for servicing.

BIMC_M_TCU1_MPORT_INTERRUPT_ENABLE

Bits	Name	Description
1:0	RFU	Reserved for future use

0x00420200 BIMC_M_TCU1_MPORT_CLOCK_CTRL**Type:** RW**Clock:** Config, Core & Master Clocks**Reset State:** 0x00000003**BIMC_M_TCU1_MPORT_CLOCK_CTRL**

Bits	Name	Description
1	MASTER_CLOCK_GATING_EN	When set (1), enables activity based clock gating of all registers clocked by master clock. Activity based clock gating uses architectural clock gates to reduce dynamic power consumption by gating clocks during periods of in-activity. When cleared (0), disables all activity based clock gating by forcing all architectural clock gates to not gate clocks even during periods of in-activity.
0	CORE_CLOCK_GATING_EN	When set (1), enables activity based clock gating of all registers clocked by core clock. Activity based clock gating uses architectural clock gates to reduce dynamic power consumption by gating clocks during periods of in-activity. When cleared (0), disables all activity based clock gating by forcing all architectural clock gates to not gate clocks even during periods of in-activity.

0x00420208 BIMC_M_TCU1_MPORT_CDC_CTRL**Type:** RW**Clock:** Config, Core & Master Clocks**Reset State:** 0x00000000

Clock domain crossing (CDC) control register provides configuration for clock domain crossing logic.

BIMC_M_TCU1_MPORT_CDC_CTRL

Bits	Name	Description
1	MASTER_CLOCK_DEMETA_SEL	When cleared (0), selects DEMETA synchronizer that requires atleast 3 clock edges (two rising and one falling). When set (1), selects DEMETA synchronizer that requires atleast 2 clock edges (one falling and one rising) to synchronize to master clock in all buffers.
0	CORE_CLOCK_DEMETA_SEL	When cleared (0), selects DEMETA synchronizer that requires atleast 3 clock edges (two rising and one falling). When set (1), selects DEMETA synchronizer that requires atleast 2 clock edges (one falling and one rising) to synchronize to core clock in all buffers.

0x00420210 BIMC_M_TCU1_MPORT_MODE**Type:** RW**Clock:** Config, Core & Master Clocks**Reset State:** 0x00000012**BIMC_M_TCU1_MPORT_MODE**

Bits	Name	Description
31:28	WR_GATHER_BEATS	Configures the number of write data beats to gather/buffer in write channel buffer before allowing the corresponding write command to be routed to arbiters. When configured to 0, write commands are held until the first beat of write data has been buffered. When configured to 1, write commands are held until two beats of write data have been buffered and so on. When configured with values larger than the number of beats in a write command, the command is held until the last beat has been buffered. When configured with values larger than the depth of write channel buffers, the command is held until all entries in the write channel buffer have been occupied.
19:16	RD_GATHER_BEATS	Configures the number of read data beats to gather/buffer in read channel buffer before allowing the read data to proceed to the master. When configured to 0, read data is allowed to proceed to the master as soon as the first beat is buffered in the read channel buffer. When configured to 1, read data is allowed to proceed to the master as soon as two beats of read data have been buffered and so on. When configured with values larger than the number of beats in the read burst, read data is held until the last beat has been buffered. When configured with values larger than the depth of read channel buffers, read data is held until all entries in the read channel buffer have been occupied.
15	RCSSH	Redirect Cacheable System SHareable When set (1), enables redirection of all cacheable transactions that are system shareable to the redirect slaweway. Note: Only masters connecting using AMBA 4 ACE protocol are able to indicate system shareability for transactions.
14	RCOSH	Redirect Cacheable Outer SHareable When set (1), enables redirection of all cacheable transactions that are outer shareable to the redirect slaweway.
13	RCISH	Redirect Cacheable Inner SHareable When set (1), enables redirection of all cacheable transactions that are inner shareable to the redirect slaweway.
12	RCNSH	Redirect Cacheable Non-SHareable When set (1), enables redirection of all cacheable transactions that are non-shareable to the redirect slaweway.
11	RNCSSH	Redirect Non-Cacheable System SHareable When set (1), enables redirection of all non-cacheable transactions that are system shareable to the redirect slaweway. Note: Only masters connecting using AMBA 4 ACE protocol are able to indicate system shareability for transactions.

BIMC_M_TCU1_MPORT_MODE (cont.)

Bits	Name	Description
10	RNCOSH	Redirect Non-Cacheable Outer SHareable When set (1), enables redirection of all non-cacheable transactions that are outer shareable to the redirect slaway.
9	RNCISH	Redirect Non-Cacheable Inner SHareable When set (1), enables redirection of all non-cacheable transactions that are inner shareable to the redirect slaway.
8	RNCNSH	Redirect Non-Cacheable Non-SHareable When set (1), enables redirection of all non-cacheable transactions that are non-shareable to the redirect slaway.
4	NARROW_WRITES	When set (1), enables converting of 1-beat bus-width sized writes down to 32-bit ASIZE based on write strobes. It is recommended to be enabled for directly connected CPU masters. When cleared (0), disables conversion and all writes are forwarded downstream as generated by the master.
2	FIODV	Force In-Order Device When set (1), forces commands to device memory to be executed one at a time. In other words, commands to device memory are held until all previous commands have completed (responses have been received at the masterport) and all commands after the device command are held until the device completes (response has been received at the masterport.)
1	FIOSO	Force In-Order Strongly-Ordered When set (1), forces commands to strongly ordered memory to be executed one at a time. In other words, commands to strongly ordered memory are held until all previous commands have completed (responses have been received at the masterport) and all commands after the strongly ordered command are held until the strongly ordered completes (response has been received at the masterport.)
0	ORDERING_MODEL	Configures how commands to memories of type Strongly ordered and Device (SoDv) are handled in the masterport. When configured as STRICT, all commands to SoDv memory are held until all previous commands have completed (responses have been received at the masterport) and all commands after the SoDv command are held until the SoDv completes (i.e. response has been received at the masterport.) When configured as RELAXED, all commands to SoDv memory are treated as accessing Normal memory. 0x0: RELAXED 0x1: STRICT

0x00420220 BIMC_M_TCU1_MPORT_WR_WAY_CROSSING**Type:** RW**Clock:** Config & Master Clocks**Reset State:** 0x00000000

Configures the level of strictness or relaxedness of way crossing checks for writes.

BIMC_M_TCU1_MPORT_WR_WAY_CROSSING

Bits	Name	Description
7:0	REG	1-bit per slaveway. When 2 or more bits are set (1), allows the corresponding slaveways to be treated as one slaveway when write commands are considered as way crossing, i.e.. write commands to these 2 (or more) slaveways are not treated as way crossing when one write command targets one slaveway and the next write command targets the second slaveway. Not treating write commands as way crossing allows the write commands to go to the slaveway ahead of the write data for potentially better overall system performance. When all bits are set (1), disables all way crossing checks. When all bits are cleared (0), way crossing is strict with each way crossing write command held until write data for all previous writes have been routed to their corresponding slaveways.

0x00420230 BIMC_M_TCU1_MPORT_PRIORITYLVL_OVERRIDE

Type: RW

Clock: Config & Master Clocks

Reset State: 0x00000000

Override registers provide the capability to override one or more bus signals to specific values.

BIMC_M_TCU1_MPORT_PRIORITYLVL_OVERRIDE

Bits	Name	Description
9:8	PRIORITYLVL	Configures the values to override APRIORITYLVL from bus master with when OVERRIDE_PRIORITYLVL is set (1).
0	OVERRIDE_PRIORITYLVL	When set (1), overrides APRIORITYLVL from the bus master with values configured in PRIORITYLVL register field. When cleared (0), allows APRIORITYLVL from the bus master to be propagated downstream.

0x00420240 BIMC_M_TCU1_MPORT_READ_COMMAND_OVERRIDE

Type: RW

Clock: Config & Master Clocks

Reset State: 0x00000000

Override registers provide the capability to override one or more bus signals to specific values.

BIMC_M_TCU1_MPORT_READ_COMMAND_OVERRIDE

Bits	Name	Description
25:24	AREQPRIORITY	Configures the values to override AREQPRIORITY from bus master with for read commands when OVERRIDE_AREQPRIORITY is set (1).
18:16	AMEMTYPE	Configures the values to override AMEMTYPE from bus master with for read commands when OVERRIDE_AMEMTYPE is set (1).
12	ATRANSIENT	Configures the values to override ATRANSIENT from bus master with for read commands when OVERRIDE_ATRANSIENT is set (1).
11	ASHARED	Configures the values to override ASHARED from bus master with for read commands when OVERRIDE_ASHARED is set (1).
10	AREDIRECT	Configures the values to override AREDIRECT from bus master with for read commands when OVERRIDE_AREDIRECT is set (1).
9	AOOO	Configures the values to override AOOO from bus master with for read commands when OVERRIDE_AOOO is set (1).
8	AINNERSHARED	Configures the values to override AINNERSHARED from bus master with for read commands when OVERRIDE_AINNERSHARED is set (1).
6	OVERRIDE_AREQPRIORITY	When set (1), overrides AREQPRIORITY from the bus master for read commands with values configured in AREQPRIORITY register field. When cleared (0), allows AREQPRIORITY from the bus master to be propagated downstream.
5	OVERRIDE_AMEMTYPE	When set (1), overrides AMEMTYPE from the bus master for read commands with values configured in AMEMTYPE register field. When cleared (0), allows AMEMTYPE from the bus master to be propagated downstream.
4	OVERRIDE_ATRANSIENT	When set (1), overrides ATRANSIENT from the bus master for read commands with values configured in ATRANSIENT register field. When cleared (0), allows ATRANSIENT from the bus master to be propagated downstream.
3	OVERRIDE_ASHARED	When set (1), overrides ASHARED from the bus master for read commands with values configured in ASHARED register field. When cleared (0), allows ASHARED from the bus master to be propagated downstream.
2	OVERRIDE_AREDIRECT	When set (1), overrides AREDIRECT from the bus master for read commands with values configured in AREDIRECT register field. When cleared (0), allows AREDIRECT from the bus master to be propagated downstream.
1	OVERRIDE_AOOO	When set (1), overrides AOOO from the bus master for read commands with values configured in AOOO register field. When cleared (0), allows AOOO from the bus master to be propagated downstream.

BIMC_M_TCU1_MPORT_READ_COMMAND_OVERRIDE (cont.)

Bits	Name	Description
0	OVERWRITE_AINNERSHARE D	When set (1), overrides AINNERSHARED from the bus master for read commands with values configured in AINNERSHARED register field. When cleared (0), allows AINNERSHARED from the bus master to be propagated downstream.

0x00420250 BIMC_M_TCU1_MPORT_WRITE_COMMAND_OVERRIDE**Type:** RW**Clock:** Config & Master Clocks**Reset State:** 0x00000000

Override registers provide the capability to override one or more bus signals to specific values.

BIMC_M_TCU1_MPORT_WRITE_COMMAND_OVERRIDE

Bits	Name	Description
25:24	AREQPRIORITY	Configures the values to override AREQPRIORITY from bus master with for write commands when OVERRIDE_AREQPRIORITY is set (1).
18:16	AMEMTYPE	Configures the values to override AMEMTYPE from bus master with for write commands when OVERRIDE_AMEMTYPE is set (1).
12	ATRANSIENT	Configures the values to override ATRANSIENT from bus master with for write commands when OVERRIDE_ATRANSIENT is set (1).
11	ASHARED	Configures the values to override ASHARED from bus master with for write commands when OVERRIDE_ASHARED is set (1).
10	AREDIRECT	Configures the values to override AREDIRECT from bus master with for write commands when OVERRIDE_AREDIRECT is set (1).
9	A000	Configures the values to override A000 from bus master with for write commands when OVERRIDE_A000 is set (1).
8	AINNERSHARED	Configures the values to override AINNERSHARED from bus master with for write commands when OVERRIDE_AINNERSHARED is set (1).
6	OVERWRITE_AREQPRIORITY	When set (1), overrides AREQPRIORITY from the bus master for write commands with values configured in AREQPRIORITY register field. When cleared (0), allows AREQPRIORITY from the bus master to be propagated downstream.
5	OVERWRITE_AMEMTYPE	When set (1), overrides AMEMTYPE from the bus master for write commands with values configured in AMEMTYPE register field. When cleared (0), allows AMEMTYPE from the bus master to be propagated downstream.

BIMC_M_TCU1_MPORT_WRITE_COMMAND_OVERRIDE (cont.)

Bits	Name	Description
4	VERRIDE_ATRANSIENT	When set (1), overrides ATRANSIENT from the bus master for write commands with values configured in ATRANSIENT register field. When cleared (0), allows ATRANSIENT from the bus master to be propagated downstream.
3	VERRIDE_ASHARED	When set (1), overrides ASHARED from the bus master for write commands with values configured in ASHARED register field. When cleared (0), allows ASHARED from the bus master to be propagated downstream.
2	VERRIDE_AREDIRECT	When set (1), overrides AREDIRECT from the bus master for write commands with values configured in AREDIRECT register field. When cleared (0), allows AREDIRECT from the bus master to be propagated downstream.
1	VERRIDE_AOOO	When set (1), overrides AOOO from the bus master for write commands with values configured in AOOO register field. When cleared (0), allows AOOO from the bus master to be propagated downstream.
0	VERRIDE_AINNERSHARE D	When set (1), overrides AINNERSHARED from the bus master for write commands with values configured in AINNERSHARED register field. When cleared (0), allows AINNERSHARED from the bus master to be propagated downstream.

0x00420300 BIMC_M_TCU1_MPORT_BKE_ENABLE**Type:** RW**Clock:** Config & Master Clocks**Reset State:** 0x00000000**BIMC_M_TCU1_MPORT_BKE_ENABLE**

Bits	Name	Description
0	ENABLE	When set (1), enables the book keeping element (BKE) to determine AREQPRIORITY and APRIORITYLVL of commands from the bus master by comparing the amount of bus traffic from the bus master to a target, and determining a health level for the bus master. The health level determines the relative priority of commands from the bus master to allow the bus master to catch-up if it is not getting serviced, or slow down to allow other bus masters to get their share of system resources. When cleared (0), disables the book keeping element and AREQPRIORITY and APRIORITYLVL from the bus master are propagated downstream.

0x00420304 BIMC_M_TCU1_MPORT_BKE_GRANT_PERIOD**Type:** RW**Clock:** Config & Reference Clocks**Reset State:** 0x00000000**BIMC_M_TCU1_MPORT_BKE_GRANT_PERIOD**

Bits	Name	Description
9:0	GRANT_PERIOD	Defines the period over which the book keeping element accumulates grants to the bus master for determining it's health level. Grant period is defined in number of reference timer ticks. Refer configuration of reference timer tick for exact time unit.

0x00420308 BIMC_M_TCU1_MPORT_BKE_GRANT_COUNT**Type:** RW**Clock:** Config & Master Clocks**Reset State:** 0x00000000**BIMC_M_TCU1_MPORT_BKE_GRANT_COUNT**

Bits	Name	Description
15:0	GRANT_COUNT	Defines the target grant count for the bus master. The target grant count is the number of beats of read and write data the bus master is expected to write and read in the grant period and is used as a reference to determine if the bus master is behind or ahead of where it ought to be at any given time.

0x00420320 BIMC_M_TCU1_MPORT_BKE_THRESHOLD_HIGH**Type:** RW**Clock:** Config & Master Clocks**Reset State:** 0x00000000**BIMC_M_TCU1_MPORT_BKE_THRESHOLD_HIGH**

Bits	Name	Description
15:0	THRESHOLD	Defines a high threshold to determine how much behind or ahead a bus master is with reference to where it ought to be at any given time.

0x00420324 BIMC_M_TCU1_MPORT_BKE_THRESHOLD_MEDIUM**Type:** RW**Clock:** Config & Master Clocks**Reset State:** 0x00000000**BIMC_M_TCU1_MPORT_BKE_THRESHOLD_MEDIUM**

Bits	Name	Description
15:0	THRESHOLD	Defines a medium threshold to determine how much behind or ahead a bus master is with reference to where it ought to be at any given time.

0x00420328 BIMC_M_TCU1_MPORT_BKE_THRESHOLD_LOW**Type:** RW**Clock:** Config & Master Clocks**Reset State:** 0x00000000**BIMC_M_TCU1_MPORT_BKE_THRESHOLD_LOW**

Bits	Name	Description
15:0	THRESHOLD	Defines a low threshold to determine how much behind or ahead a bus master is with reference to where it ought to be at any given time.

0x00420340 BIMC_M_TCU1_MPORT_BKE_HEALTH_0**Type:** RW**Clock:** Config & Master Clocks**Reset State:** 0x00000000

BKE health registers configure the request priority and realtime priority level to use for commands from the bus master when it has been determined to be of a certain health.

BIMC_M_TCU1_MPORT_BKE_HEALTH_0

Bits	Name	Description
31	LIMIT_COMMANDS	When set (1), defines that no new commands may be accepted from the bus master as long as it is determined to be at health 0.
9:8	AREQPRIORITY	Defines the request priority of commands from the bus master when it has been determined to be at health 0.
1:0	PRIORITYLVL	Defines the realtime priority level to use for commands from the bus master when it has been determined to be at health 0.

0x00420344 BIMC_M_TCU1_MPORT_BKE_HEALTH_1**Type:** RW**Clock:** Config & Master Clocks**Reset State:** 0x00000000

BKE health registers configure the request priority and realtime priority level to use for commands from the bus master when it has been determined to be of a certain health.

BIMC_M_TCU1_MPORT_BKE_HEALTH_1

Bits	Name	Description
31	LIMIT_COMMANDS	When set (1), defines that no new commands may be accepted from the bus master as long as it is determined to be at health 1.
9:8	AREQPRIORITY	Defines the request priority of commands from the bus master when it has been determined to be at health 1.
1:0	PRIORITYLVL	Defines the realtime priority level to use for commands from the bus master when it has been determined to be at health 1.

0x00420348 BIMC_M_TCU1_MPORT_BKE_HEALTH_2**Type:** RW**Clock:** Config & Master Clocks**Reset State:** 0x00000000

BKE health registers configure the request priority and realtime priority level to use for commands from the bus master when it has been determined to be of a certain health.

BIMC_M_TCU1_MPORT_BKE_HEALTH_2

Bits	Name	Description
31	LIMIT_COMMANDS	When set (1), defines that no new commands may be accepted from the bus master as long as it is determined to be at health 2.
9:8	AREQPRIORITY	Defines the request priority of commands from the bus master when it has been determined to be at health 2.
1:0	PRIORITYLVL	Defines the realtime priority level to use for commands from the bus master when it has been determined to be at health 2.

0x0042034C BIMC_M_TCU1_MPORT_BKE_HEALTH_3**Type:** RW**Clock:** Config & Master Clocks**Reset State:** 0x00000000

BKE health registers configure the request priority and realtime priority level to use for commands from the bus master when it has been determined to be of a certain health. Health 3 is the default health for the bus master.

BIMC_M_TCU1_MPORT_BKE_HEALTH_3

Bits	Name	Description
9:8	AREQPRIORITY	Defines the request priority of commands from the bus master when it has been determined to be at health 3.
1:0	PRIORITYLVL	Defines the realtime priority level to use for commands from the bus master when it has been determined to be at health 3.

0x00420420 BIMC_M_TCU1_MPORT_STATUS_2A

Type: R

Clock: Config, Master & Pipe0 Clocks

Reset State: 0x00000000

Status registers indicate the state of several elements in the masterport, including the number of transactions outstanding/pending waiting for a response, emptiness, fullness of several buffers, etc. The state information is available only as a debug indication and could be delayed by a few clock cycles from when the state is reached to being reflected in the status registers. One or more of the state registers are read live and could capture intermediate values as they change.

BIMC_M_TCU1_MPORT_STATUS_2A

Bits	Name	Description
23:16	WR_PENDING_PRE_BUF	Indicates the number of write transactions that are outstanding/pending waiting for a response from the slave. This transaction count is from before address channel buffers, which interfaces to frontend of the masterport.
7:0	RD_PENDING_PRE_BUF	Indicates the number of read transactions that are outstanding/pending waiting for a response from the slave. This transaction count is from before address channel buffers, which interfaces to frontend of the masterport.

0x00420424 BIMC_M_TCU1_MPORT_STATUS_2B

Type: R

Clock: Config, Master & Pipe0 Clocks

Reset State: 0x00000000

Status registers indicate the state of several elements in the masterport, including the number of transactions outstanding/pending waiting for a response, emptiness, fullness of several buffers, etc. The state information is available only as a debug indication and could be delayed by a few clock

cycles from when the state is reached to being reflected in the status registers. One or more of the state registers are read live and could capture intermediate values as they change.

BIMC_M_TCU1_MPORT_STATUS_2B

Bits	Name	Description
23:16	WR_PENDING_POST_BUF	Indicates the number of write transactions that are outstanding/pending waiting for a response from the slave. This transaction count is from after address channel buffers, which interfaces to core of the interconnect, i.e., arbiter and slaveway.
7:0	RD_PENDING_POST_BUF	Indicates the number of read transactions that are outstanding/pending waiting for a response from the slave. This transaction count is from after address channel buffers, which interfaces to core of the interconnect, i.e., arbiter and slaveway.

0x00420430 BIMC_M_TCU1_MPORT_STATUS_3A

Type: R

Clock: Config, Master & Pipe0 Clocks

Reset State: 0x55111111

Status registers indicate the state of several elements in the masterport, including the number of transactions outstanding/pending waiting for a response, emptiness, fullness of several buffers, etc. The state information is available only as a debug indication and could be delayed by a few clock cycles from when the state is reached to being reflected in the status registers. One or more of the state registers are read live and could capture intermediate values as they change.

BIMC_M_TCU1_MPORT_STATUS_3A

Bits	Name	Description
31	RCB_DATA_RD_FULL	Indicates that the data part of read channel buffer is full when seen from it's read side, which interfaces to frontend of the masterport.
30	RCB_DATA_RD_EMPTY	Indicates that the data part of read channel buffer is empty when seen from it's read side, which interfaces to frontend of the masterport.
29	RCB_CTRL_RD_FULL	Indicates that the control part of read channel buffer is full when seen from it's read side, which interfaces to frontend of the masterport.
28	RCB_CTRL_RD_EMPTY	Indicates that the control part of read channel buffer is empty when seen from it's write side, which interfaces to frontend of the masterport.
27	RCB_DATA_WR_FULL	Indicates that the data part of read channel buffer is full when seen from it's write side, which interfaces to core of the interconnect, i.e., arbiter and slaveway.
26	RCB_DATA_WR_EMPTY	Indicates that the data part of read channel buffer is empty when seen from it's write side, which interfaces to core of the interconnect, i.e., arbiter and slaveway.

BIMC_M_TCU1_MPORT_STATUS_3A (cont.)

Bits	Name	Description
25	RCB_CTRL_WR_FULL	Indicates that the control part of read channel buffer is full when seen from it's write side, which interfaces to core of the interconnect, i.e., arbiter and slaveway.
24	RCB_CTRL_WR_EMPTY	Indicates that the control part of read channel buffer is empty when seen from it's write side, which interfaces to core of the interconnect, i.e., arbiter and slaveway.
21	BCB_RD_FULL	Indicates that the write response channel buffer is full when seen from it's read side, which interfaces to frontend of the masterport.
20	BCB_RD_EMPTY	Indicates that the write response channel buffer is empty when seen from it's read side, which interfaces to frontend of the masterport.
17	BCB_WR_FULL	Indicates that the write response channel buffer is full when seen from it's write side, which interfaces to core of the interconnect, i.e., arbiter and slaveway.
16	BCB_WR_EMPTY	Indicates that the write response channel buffer is empty when seen from it's write side, which interfaces to core of the interconnect, i.e., arbiter and slaveway.
13	WCB_RD_FULL	Indicates that the write channel buffer is full when seen from it's read side, which interfaces to core of the interconnect, i.e., arbiter and slaveway.
12	WCB_RD_EMPTY	Indicates that the write channel buffer is empty when seen from it's read side, which interfaces to core of the interconnect, i.e., arbiter and slaveway.
9	WCB_WR_FULL	Indicates that the write channel buffer is full when seen from it's write side, which interfaces to frontend of the masterport.
8	WCB_WR_EMPTY	Indicates that the write channel buffer is empty when seen from it's write side, which interfaces to frontend of the masterport.
5	ACB_RD_FULL	Indicates that the address channel buffer is full when seen from it's read side, which interfaces to core of the interconnect, i.e., arbiter and slaveway.
4	ACB_RD_EMPTY	Indicates that the address channel buffer is empty when seen from it's read side, which interfaces to core of the interconnect, i.e., arbiter and slaveway.
1	ACB_WR_FULL	Indicates that the address channel buffer is full when seen from it's write side, which interfaces to frontend of the masterport.
0	ACB_WR_EMPTY	Indicates that the address channel buffer is empty when seen from it's write side, which interfaces to frontend of the masterport.

0x00420434 BIMC_M_TCU1_MPORT_STATUS_3B**Type:** R**Clock:** Config, Master & Pipe0 Clocks**Reset State:** 0x00111111

Buffer status register indicate the state of all buffers in the masterport, whether they are empty or full on either side of the buffer. Some buffers carry information from master to core and others carry information from core to master. Both write side and read side status are indicated separately, since they would be not be similar when master clock and core clock are not synchronous to each other.

BIMC_M_TCU1_MPORT_STATUS_3B

Bits	Name	Description
21	RGB_RD_FULL	Indicates that the read gathering buffer is full when seen from it's read side, which interfaces to frontend of the masterport.
20	RGB_RD_EMPTY	Indicates that the read gathering buffer is empty when seen from it's read side, which interfaces to frontend of the masterport.
17	RGB_WR_FULL	Indicates that the read gathering buffer is full when seen from it's write side, which interfaces to core of the interconnect, i.e., arbiter and slaweway.
16	RGB_WR_EMPTY	Indicates that the read gathering buffer is empty when seen from it's write side, which interfaces to core of the interconnect, i.e., arbiter and slaweway.
13	W2AB_RD_FULL	Indicates that the write channel to address channel user signals buffer is full when seen from it's read side, which interfaces to core of the interconnect, i.e., arbiter and slaweway.
12	W2AB_RD_EMPTY	Indicates that the write channel to address channel user signals buffer is empty when seen from it's read side, which interfaces to core of the interconnect, i.e., arbiter and slaweway.
9	W2AB_WR_FULL	Indicates that the write channel to address channel user signals buffer is full when seen from it's write side, which interfaces to frontend of the masterport.
8	W2AB_WR_EMPTY	Indicates that the write channel to address channel user signals buffer is empty when seen from it's write side, which interfaces to frontend of the masterport.
5	A2WB_RD_FULL	Indicates that the address channel to write channel user signals buffer is full when seen from it's read side, which interfaces to core of the interconnect, i.e., arbiter and slaweway.
4	A2WB_RD_EMPTY	Indicates that the address channel to write channel user signalsbuffer is empty when seen from it's read side, which interfaces to core of the interconnect, i.e., arbiter and slaweway.
1	A2WB_WR_FULL	Indicates that the address channel to write channel user signals buffer is full when seen from it's write side, which interfaces to frontend of the masterport.

BIMC_M_TCU1_MPORT_STATUS_3B (cont.)

Bits	Name	Description
0	A2WB_WR_EMPTY	Indicates that the address channel to write channel user signals buffer is empty when seen from it's write side, which interfaces to frontend of the masterport.

2.18 BIMC_M_TCU1_PROF

BRIC Traffic Profiling

2.18.1 BIMC_M_TCU1_PROF_BRIC_PROFILING

BRIC Traffic Profiling

0x00421000 BIMC_M_TCU1_PROF_COMPONENT_INFO**Type:** R**Clock:** Config Clock**Reset State:** 0x00000202

Component information registers provide discoverability information on the type of component and instance of each component. These registers are always the very first register in each component's configuration address space and allow for a software routine to walk the configuration address space in steps of 4KB to identify the location of each configuration register space and to which component each 4KB configures.

BIMC_M_TCU1_PROF_COMPONENT_INFO

Bits	Name	Description
23:16	INSTANCE	Indicates the instance number of this traffic profiling.
15:8	SUB_TYPE	Identifies this configuration space as belonging to traffic profiling 0x2: PROFILING
7:0	TYPE	Identifies this configuration space as belonging to a component that interfaces to a bus master. 0x2: MASTERPORT

0x00421020 BIMC_M_TCU1_PROF_CONFIGURATION_INFO_0**Type:** R**Clock:** Config Clock**Reset State:** 0x00000000

Configuration information registers provide information on the capability and configuration of each traffic profiling instance. See register field descriptions for more details.

BIMC_M_TCU1_PROF_CONFIGURATION_INFO_0

Bits	Name	Description
31:24	TENURE_WIDTH	Indicates the width of tenures (latencies) measured in traffic profiling.
23:16	TRACKING_TABLE_DEPTH	Indicates the number of entries in the table that tracks outstanding commands for determining how many commands have larger turnaround latencies that a programmed threshold.
7:0	SCALING_FACTOR	Indicates how many extra scaling is performed on events generated by traffic profiling if such events have to be synchronized from the tenure clock to the event clock. Any value that is non-power-of-2 is converted to power-of-2 by hardware. When the value read from this register field is used in software, similar conversion of non-power-of-2 values to next higher power-of-2 value is necessary.

0x00421030 BIMC_M_TCU1_PROF_CONFIGURATION_INFO_1

Type: R

Clock: Config Clock

Reset State: 0x00000000

Configuration information registers provide information on the capability and configuration of each traffic profiling instance. See register field descriptions for more details.

BIMC_M_TCU1_PROF_CONFIGURATION_INFO_1

Bits	Name	Description
23:16	PIPE2_ACH_DEPTH	Indicates the depth of address channel buffer in pipe2 of masterport in which this traffic profiling instance is paired with. Applies only to traffic profiling instances in XR type masterports, which contains three distinct paths to slaves, in pipe0, pipe1, pipe2. In non-XR type masterports, value contained in field does not refer to any actual address channel buffer.
15:8	PIPE1_ACH_DEPTH	Indicates the depth of address channel buffer in pipe1 of masterport in which this traffic profiling instance is paired with. Applies only to traffic profiling instances in XR type masterports, which contains three distinct paths to slaves, in pipe0, pipe1, pipe2. In non-XR type masterports, value contained in field does not refer to any actual address channel buffer.
7:0	PIPE0_ACH_DEPTH	Indicates the depth of address channel buffer in pipe0 of masterport in which this traffic profiling instance is paired with. Applies only to traffic profiling instances in XR type masterports, which contains three distinct paths to slaves, in pipe0, pipe1, pipe2. In non-XR type masterports, pipe0 refers to the common address channel buffer used for all slave accesses.

0x00421200 BIMC_M_TCU1_PROF_CLOCK_CTRL**Type:** RW**Clock:** Config, Event & Tenure Clocks**Reset State:** 0x00000003**BIMC_M_TCU1_PROF_CLOCK_CTRL**

Bits	Name	Description
1	TENURE_CLOCK_GATING_EN	When set (1), enables activity based clock gating of all registers clocked by tenure clock. Activity based clock gating uses architectural clock gates to reduce dynamic power consumption by gating clocks during periods of in-activity. When cleared (0), disables all activity based clock gating by forcing all architectural clock gates to not gate clocks even during periods of in-activity.
0	EVENT_CLOCK_GATING_EN	When set (1), enables activity based clock gating of all registers clocked by event clock. Activity based clock gating uses architectural clock gates to reduce dynamic power consumption by gating clocks during periods of in-activity. When cleared (0), disables all activity based clock gating by forcing all architectural clock gates to not gate clocks even during periods of in-activity.

0x00421210 BIMC_M_TCU1_PROF_MODE**Type:** RW**Clock:** Config, Event & Tenure Clocks**Reset State:** 0x00000000

BIMC_M_TCU1_PROF_MODE

Bits	Name	Description
25:24	WR_TENURE_MODE	Configures the start and end events for monitoring write latencies/tenures. When configured to WR_CMD_TO_BRESP, starting event is first cycle of write command from bus master as seen by traffic profiling and ending event is first cycle of write response to bus master as seen by traffic profiling. When configured to WR_CMD_TO_BRESP_HANDSHAKE, starting event is first cycle of write command from bus master as seen by traffic profiling and ending event is transfer handshake of write response to bus master as seen by traffic profiling. When configured to WR_CMD_HANDSHAKE, starting event is first cycle of write command from bus master as seen by traffic profiling and ending event is transfer handshake of the same write command with bus master as seen by traffic profiling. When configured to WR_RESP_HANDSHAKE, starting event is first cycle of write response to bus master as seen by traffic profiling and ending event is transfer handshake of write response to bus master as seen by traffic profiling. 0x0: WR_CMD_TO_BRESP 0x1: WR_CMD_TO_BRESP_HANDSHAKE 0x2: WR_CMD_HANDSHAKE 0x3: WR_RESP_HANDSHAKE
17:16	RD_TENURE_MODE	Configures the start and end events for monitoring read latencies/tenures. When configured to RD_CMD_TO_FIRST_RDATA, starting event is first cycle of read command from bus master as seen by traffic profiling and ending event is first cycle of first beat of read data to bus master as seen by traffic profiling. When configured to RD_CMD_TO_LAST_RDATA, starting event is first cycle of read command from bus master as seen by traffic profiling and ending event is transfer handshake of last beat of read data to bus master as seen by traffic profiling. When configured to RD_CMD_HANDSHAKE, starting event is first cycle of read command from bus master as seen by traffic profiling and ending event is transfer handshake of the same read command with bus master as seen by traffic profiling. When configured to RD_DATA_HANDSHAKE, starting event is first cycle of first beat of read data to bus master as seen by traffic profiling and ending event is transfer handshake of last beat of read data to bus master as seen by traffic profiling. 0x0: RD_CMD_TO_FIRST_RDATA 0x1: RD_CMD_TO_LAST_RDATA 0x2: RD_CMD_HANDSHAKE 0x3: RD_DATA_HANDSHAKE
0	PROFILING_EN	When set (1), enables traffic profiling to monitor bus transactions from bus master and generate events for counting in performance monitors. When cleared (0), disables traffic profiling.

0x00421214 BIMC_M_TCU1_PROF_RD_TENURE_THRESHOLD**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_TCU1_PROF_RD_TENURE_THRESHOLD**

Bits	Name	Description
15:0	THRESHOLD	Configures the threshold over which read latencies/tenures need to be before an event is generated to performance monitors for counting max latencies.

0x00421218 BIMC_M_TCU1_PROF_WR_TENURE_THRESHOLD**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_TCU1_PROF_WR_TENURE_THRESHOLD**

Bits	Name	Description
15:0	THRESHOLD	Configures the threshold over which write latencies/tenures need to be before an event is generated to performance monitors for counting max latencies.

0x00421220 BIMC_M_TCU1_PROF_SWAY_SELECT**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_TCU1_PROF_SWAY_SELECT**

Bits	Name	Description
7:0	SWAY	Selects the slaveways to which commands are routed to profile. 1-bit per slaveway.

0x00421230 BIMC_M_TCU1_PROF_MID_MASK**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000

BIMC_M_TCU1_PROF_MID_MASK

Bits	Name	Description
15:0	MID_MASK	When a bit position is set (1), enables matching of that bit position in MID of command with same bit position in MID_MATCH. Commands with MIDs that do not match are not considered for profiling.

0x00421234 BIMC_M_TCU1_PROF_MID_MATCH**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_TCU1_PROF_MID_MATCH**

Bits	Name	Description
15:0	MID_MATCH	Configures the value to match for in AMID in commands. Used in conjunction with MID_MASK, which controls which bits of MID are compared.

0x00421238 BIMC_M_TCU1_PROF_MID_INV_MATCH**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_TCU1_PROF_MID_INV_MATCH**

Bits	Name	Description
0	MID_INV_MATCH	When set (1), configures MID matching to result in an inverse match, i.e., only commands whose MIDs that do not match MID_MATCH are considered for profiling.

0x00421240 BIMC_M_TCU1_PROF_ADDR_MASK_LOWER**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000

BIMC_M_TCU1_PROF_ADDR_MASK_LOWER

Bits	Name	Description
31:10	ADDR_MASK_LOWER	Configures ADDR_MASK[31:0]. When a bit position is set (1), enables matching of that bit position of address of command with same bit position in ADDR_MATCH[31:0].

0x00421244 BIMC_M_TCU1_PROF_ADDR_MASK_UPPER**Type:** R**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_TCU1_PROF_ADDR_MASK_UPPER**

Bits	Name	Description
3:0	UNUSED	

0x00421248 BIMC_M_TCU1_PROF_ADDR_MATCH_LOWER**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_TCU1_PROF_ADDR_MATCH_LOWER**

Bits	Name	Description
31:10	ADDR_MATCH_LOWER	Configures the value to match for in address of commands. Used in conjunction with ADDR_MASK[31:0], which controls which bits of address are compared.

0x0042124C BIMC_M_TCU1_PROF_ADDR_MATCH_UPPER**Type:** R**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_TCU1_PROF_ADDR_MATCH_UPPER**

Bits	Name	Description
3:0	UNUSED	

0x00421250 BIMC_M_TCU1_PROF_ADDR_INV_MATCH**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_TCU1_PROF_ADDR_INV_MATCH**

Bits	Name	Description
0	ADDR_INV_MATCH	When set (1), configures address matching to result in an inverse match, i.e., only commands whose addresses do not match ADDR_MATCH are considered for profiling.

0x00421260 BIMC_M_TCU1_PROF_ACH_MASK_0**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_TCU1_PROF_ACH_MASK_0**

Bits	Name	Description
24	AINNERSHARED_MASK	When set (1), enables matching of AINNERSHARED in commands with AINNERSHARED_MATCH.
16	AFULL_MASK	When set (1), enables matching of AFULL in commands with AFULL_MATCH.
8	AEXCLUSIVE_MASK	When set (1), enables matching of AEXCLUSIVE in commands with AEXCLUSIVE_MATCH.
0	ABURST_MASK	When set (1), enables matching of ABURST in commands with ABURST_MATCH.

0x00421264 BIMC_M_TCU1_PROF_ACH_MATCH_0**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_TCU1_PROF_ACH_MATCH_0**

Bits	Name	Description
24	AINNERSHARED_MATCH	Configures the value to match for in AINNERSHARED in commands. Used in conjunction with AINNERSHARED_MASK, which controls whether AINNERSHARED in commands is compared

BIMC_M_TCU1_PROF_ACH_MATCH_0 (cont.)

Bits	Name	Description
16	AFULL_MATCH	Configures the value to match for in AFULL in commands. Used in conjunction with AFULL_MASK, which controls whether AFULL in commands is compared.
8	AEXCLUSIVE_MATCH	Configures the value to match for in AEXCLUSIVE in commands. Used in conjunction with AEXCLUSIVE_MASK, which controls whether AFULL in commands is compared.
0	ABURST_MATCH	Configures the value to match for in ABURST in commands. Used in conjunction with ABURST_MASK, which controls whether ABURST in commands is compared.

0x00421270 BIMC_M_TCU1_PROF_ACH_MASK_1**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_TCU1_PROF_ACH_MASK_1**

Bits	Name	Description
24	AMSSSELFAUTH_MASK	When set (1), enables matching of AMSSSELFAUTH in commands with AMSSSELFAUTH_MATCH.
18:16	AMEMTYPE_MASK	When a bit position is set (1), enables matching of corresponding bit position of AMEMTYPE with AMEMTYPE_MATCH.
11:8	ALEN_MASK	When a bit position is set (1), enables matching of corresponding bit position of ALEN with ALEN_MATCH.
0	AINST_MASK	When set (1), enables matching of AINST in commands with AINST_MASK.

0x00421274 BIMC_M_TCU1_PROF_ACH_MATCH_1**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_TCU1_PROF_ACH_MATCH_1**

Bits	Name	Description
24	AMSSSELFAUTH_MATCH	Configures the value to match for in AMSSSELFAUTH in commands. Used in conjunction with AMSSSELFAUTH_MASK, which controls whether AMSSSELFAUTH in commands is compared.

BIMC_M_TCU1_PROF_ACH_MATCH_1 (cont.)

Bits	Name	Description
18:16	AMEMTYPE_MATCH	Configures the value to match for in AMEMTYPE in commands. Used in conjunction with AMEMTYPE_MASK, which controls which bits of AMEMTYPE in commands are compared.
11:8	ALEN_MATCH	Configures the value to match for in ALEN in commands. Used in conjunction with ALEN_MASK, which controls which bits of ALEN in commands are compared.
0	AINST_MATCH	Configures the value to match for in AINST in commands. Used in conjunction with AINST_MASK, which controls whether AINST in commands is compared.

0x00421278 BIMC_M_TCU1_PROF_ACH_INV_MATCH_1**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_TCU1_PROF_ACH_INV_MATCH_1**

Bits	Name	Description
16	AMEMTYPE_INV_MATCH	When set (1), configures AMEMTYPE matching to result in an inverse match, i.e., only commands whose AMEMTYPE does not match AMEMTYPE_MATCH are considered for profiling.
8	ALEN_INV_MATCH	When set (1), configures ALEN matching to result in an inverse match, i.e., only commands when ALEN does not match ALEN_MATCH are considered for profiling.

0x00421280 BIMC_M_TCU1_PROF_ACH_MASK_2**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_TCU1_PROF_ACH_MASK_2**

Bits	Name	Description
24	APROTNS_MASK	When set (1), enables matching of APROTNS in commands with APROTNS_MATCH.
16	APRIV_MASK	When set (1), enables matching of APRIV in commands with APRIV_MATCH.
8	AOOO_MASK	When set (1), enables matching of AOOO in commands with AOOO_MATCH.
0	ANOALLOCATE_MASK	When set (1), enables matching of ANOALLOCATE in commands with ANOALLOCATE_MATCH.

0x00421284 BIMC_M_TCU1_PROF_ACH_MATCH_2**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_TCU1_PROF_ACH_MATCH_2**

Bits	Name	Description
24	APROTNS_MATCH	Configures the value to match for in APROTNS in commands. Used in conjunction with APROTNS_MASK, which controls whether APROTNS in commands is compared
16	APRIV_MATCH	Configures the value to match for in APRIV in commands. Used in conjunction with APRIV_MASK, which controls whether APRIV in commands is compared
8	AOOO_MATCH	Configures the value to match for in AOOO in commands. Used in conjunction with AOOO_MASK, which controls whether AOOO in commands is compared
0	ANOALLOCATE_MATCH	Configures the value to match for in ANOALLOCATE in commands. Used in conjunction with ANOALLOCATE_MASK, which controls whether ANOALLOCATE in commands is compared

0x00421290 BIMC_M_TCU1_PROF_ACH_MASK_3**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_TCU1_PROF_ACH_MASK_3**

Bits	Name	Description
24	ASHARED_MASK	When set (1), enables matching of ASHARED in commands with ASHARED_MATCH.
17:16	AREQPRIORITY_MASK	When a bit position is set (1), enables matching of corresponding bit position of AREQPRIORITY with AREQPRIORITY_MATCH.
8	AREDIRECT_MASK	When set (1), enables matching of AREDIRECT in commands with AREDIRECT_MATCH.
0	ARDBEATNDXEN_MASK	When set (1), enables matching of ARDBEATNDXEN in commands with ARDBEATNDXEN_MATCH.

0x00421294 BIMC_M_TCU1_PROF_ACH_MATCH_3**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_TCU1_PROF_ACH_MATCH_3**

Bits	Name	Description
24	ASHARED_MATCH	Configures the value to match for in ASHARED in commands. Used in conjunction with ASHARED_MASK, which controls whether ASHARED in commands is compared
17:16	AREQPRIORITY_MATCH	Configures the value to match for in AREQPRIORITY in commands. Used in conjunction with AREQPRIORITY_MASK, which controls which bits of AREQPRIORITY in commands are compared.
8	AREDIRECT_MATCH	Configures the value to match for in AREDIRECT in commands. Used in conjunction with AREDIRECT_MASK, which controls whether AREDIRECT in commands is compared
0	ARDBEATNDXEN_MATCH	Configures the value to match for in ARDBEATNDXEN in commands. Used in conjunction with ARDBEATNDXEN_MASK, which controls whether ARDBEATNDXEN in commands is compared

0x00421298 BIMC_M_TCU1_PROF_ACH_INV_MATCH_3**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_TCU1_PROF_ACH_INV_MATCH_3**

Bits	Name	Description
16	AREQPRIORITY_INV_MATCH	When set (1), configures AREQPRIORITY matching to result in an inverse match, i.e., only commands whose AREQPRIORITY does not match AREQPRIORITY_MATCH are considered for profiling

0x004212A0 BIMC_M_TCU1_PROF_ACH_MASK_4**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000

BIMC_M_TCU1_PROF_ACH_MASK_4

Bits	Name	Description
24	AWRITE_MASK	When set (1), enables matching of AWRITE in commands with AWRITE_MATCH.
20:16	AVMID_MASK	When a bit position is set (1), enables matching of corresponding bit position in AVMID with AVMID_MATCH
8	ATRANSIENT_MASK	When set (1), enables matching of ATRANSIENT in commands with ATRANSIENT_MATCH.
2:0	ASIZE_MASK	When a bit position is set (1), enables matching of corresponding bit position in ASIZE with ASIZE_MATCH.

0x004212A4 BIMC_M_TCU1_PROF_ACH_MATCH_4**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_TCU1_PROF_ACH_MATCH_4**

Bits	Name	Description
24	AWRITE_MATCH	Configures the value to match for in AWRITE in commands. Used in conjunction with AWRITE_MASK, which controls whether AWRITE in commands is compared.
20:16	AVMID_MATCH	Configures the value to match for in AVMID in commands. Used in conjunction with AVMID_MASK, which controls which bits of AVMID in commands are compared.
8	ATRANSIENT_MATCH	Configures the value to match for in ATRANSIENT in commands. Used in conjunction with ATRANSIENT_MASK, which controls whether ATRANSIENT in commands is compared.
2:0	ASIZE_MATCH	Configures the value to match for in ASIZE in commands. Used in conjunction with ASIZE_MASK, which controls which bits of ASIZE in commands are compared.

0x004212A8 BIMC_M_TCU1_PROF_ACH_INV_MATCH_4**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000

BIMC_M_TCU1_PROF_ACH_INV_MATCH_4

Bits	Name	Description
16	AVMID_INV_MATCH	When set (1), configures AVMID matching to result in an inverse match, i.e., only commands whose AVMID does not match AVMID_MATCH are considered for profiling.
0	ASIZE_INV_MATCH	When set (1), configures ASIZE matching to result in an inverse match, i.e., only commands whose AVMID does not match ASIZE_MATCH are considered for profiling.

0x00421300 BIMC_M_TCU1_PROF_BCH_MASK**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_TCU1_PROF_BCH_MASK**

Bits	Name	Description
1:0	BRESP_MASK	When a bit position is set (1), enables matching of corresponding bit position in BRESP with BRESP_MATCH.

0x00421304 BIMC_M_TCU1_PROF_BCH_MATCH**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_TCU1_PROF_BCH_MATCH**

Bits	Name	Description
1:0	BRESP_MATCH	Configures the value to match for in BRESP in write responses. Used in conjunction with BRESP_MASK, which controls which bits of BRESP in writes responses are compared.

0x00421308 BIMC_M_TCU1_PROF_BCH_INV_MATCH**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000

BIMC_M_TCU1_PROF_BCH_INV_MATCH

Bits	Name	Description
0	BRESP_INV_MATCH	When set (1), configures BRESP matching to result in an inverse match, i.e., only write responses whose BRESP does not match BRESP_MATCH are considered for profiling.

0x00421340 BIMC_M_TCU1_PROF_RCH_MASK**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_TCU1_PROF_RCH_MASK**

Bits	Name	Description
8	RLAST_MASK	When set (1), enables matching of RLAST in commands with corresponding setting in RCH_MATCH register.
1:0	RRESP_MASK	When a bit position is set (1), enables matching of corresponding bit position in RRESP with RRESP_MATCH.

0x00421344 BIMC_M_TCU1_PROF_RCH_MATCH**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_M_TCU1_PROF_RCH_MATCH**

Bits	Name	Description
8	RLAST_MATCH	Configures the value to match for in RLAST in read responses. Used in conjunction with RLAST_MASK, which controls whether RLAST in read responses is compared.
1:0	RRESP_MATCH	Configures the value to match for in RRESP in read responses. Used in conjunction with RRESP_MASK, which controls which bits of RRESP in read responses are compared.

0x00421348 BIMC_M_TCU1_PROF_RCH_INV_MATCH**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000

BIMC_M_TCU1_PROF_RCH_INV_MATCH

Bits	Name	Description
0	RRESP_INV_MATCH	When set (1), configures RRESP matching to result in an inverse match, i.e., only read responses whose RRESP does not match RRESP_MATCH are considered for profiling.

0x00421400+ BIMC_M_TCU1_PROF_RD_STATUS_n, n=[0..0]
0x4*n
Type: R**Clock:** Config & Tenure Clocks**Reset State:** 0x00000000**BIMC_M_TCU1_PROF_RD_STATUS_n**

Bits	Name	Description
31	RD_OUTSTANDING_OVERFLOW	Indicates that the number of outstanding reads crossed the maximum supported (255) during when traffic profiling was last enabled. It is recommended that this register be read only after traffic profiling has been disabled.
30	RD_OUTSTANDING_UNDERFLOW	Indicates that the number of outstanding reads went below 0 during when traffic profiling was last enabled. It indicates that more command completions were seen compared to commands starts. The most likely scenario is an overflow of number of outstanding reads occurred leading to some command starts not being tracked. It is recommended that this register be read only after traffic profiling has been disabled.
7:0	RD_OUTSTANDING_MAX	Indicates the maximum number of outstanding reads that were seen when traffic profiling was last enabled. Only reads that match the MID filter and slaweway select filter are accounted for. It is recommended that this register be read only after traffic profiling has been disabled.

0x00421410+ BIMC_M_TCU1_PROF_WR_STATUS_n, n=[0..0]
0x4*n
Type: R**Clock:** Config & Tenure Clocks**Reset State:** 0x00000000**BIMC_M_TCU1_PROF_WR_STATUS_n**

Bits	Name	Description
31	WR_OUTSTANDING_OVERFLOW	Indicates that the number of outstanding writes crossed the maximum supported (255) during when traffic profiling was last enabled. It is recommended that this register be read only after traffic profiling has been disabled.

BIMC_M_TCU1_PROF_WR_STATUS_n (cont.)

Bits	Name	Description
30	WR_OUTSTANDING_UNDE RFLOW	Indicates that the number of outstanding writes went below 0 during when traffic profiling was last enabled. It indicates that more command completions were seen compared to commands starts. The most likely scenario is an overflow of number of outstanding writes occurred leading to some command starts not being tracked. It is recommended that this register be read only after traffic profiling has been disabled.
7:0	WR_OUTSTANDING_MAX	Indicates the maximum number of outstanding writes that were seen when traffic profiling was last enabled. Only writes that match the MID filter and slaveway select filter are accounted for. It is recommended that this register be read only after traffic profiling has been disabled.

2.19 BIMC_S_APP_SWAY

BRIC Slaveway

2.19.1 BIMC_S_APP_SWAY_BRIC_SLAVEWAY

BRIC Slaveway

0x00450000 BIMC_S_APP_SWAY_COMPONENT_INFO**Type:** R**Clock:** Config Clock**Reset State:** 0x00000103

Component information registers provide discoverability information on the type of component and instance of each component. These registers are always the very first register in each component's configuration address space and allow for a software routine to walk the configuration address space in steps of 4KB to identify the location of each configuration register space and to which component each 4KB configures.

BIMC_S_APP_SWAY_COMPONENT_INFO

Bits	Name	Description
23:16	INSTANCE	Indicates the instance number of this slaveway.
15:8	SUB_TYPE	Identifies this configuration space as belonging to a slaveway 0x1: SLAVEWAY
7:0	TYPE	Identifies this configuration space as belonging to a component that interfaces to a bus slave. 0x3: SLAVEWAY

0x00450020 BIMC_S_APP_SWAY_CONFIGURATION_INFO_0**Type:** R**Clock:** Config Clock**Reset State:** 0x01000001

Configuration information registers provide information on the capability and configuration of each slaveway instance. See register field descriptions for more details.

BIMC_S_APP_SWAY_CONFIGURATION_INFO_0

Bits	Name	Description
31:24	SYNC_MODE	Indicates the clock relationship between the clock used in the core side and the clock used in the slave side of the slaveway. When ASYNC, identifies the master clock and core clock as un-related and asynchronous to each other. When SYNC, identifies the core clock and slave clock as synchronous to each other. When ISOSYNC, identifies the core clock and slave clock as related, but of different frequencies, which are of integer ratios, i.e., they are iso-synchronous to each other. 0x0: ASYNC 0x1: SYNC 0x2: ISOSYNC
7:0	FUNCTIONALITY	Indicates the functionality and type of bus traffic handled by this slaveway. When DEFAULT_SLAVE, identifies this slaveway as a default slave which responds to all commands with decode error. When SLAVEWAY, identifies this slaveway as interfacing to a bus slave. When STUB, identifies this slaveway has a stub. 0x0: STUB 0x1: SLAVEWAY

0x00450030 BIMC_S_APP_SWAY_CONFIGURATION_INFO_1**Type:** R**Clock:** Config Clock**Reset State:** 0x00000000

Configuration information registers provide information on the capability and configuration of each slaveway instance. See register field descriptions for more details.

BIMC_S_APP_SWAY_CONFIGURATION_INFO_1

Bits	Name	Description
31:0	MPORT_CONNECTIVITY	Indicates which masterports can route commands to this slaveway instance. 1-bit per masterport. When a specific bit is set (1), indicates that the masterport corresponding to this bit can route commands to this slaveway instance. When a specific bit is cleared (0), indicates that the masterport corresponding to this bit does not route commands to this slaveway instance.

0x00450040 BIMC_S_APP_SWAY_CONFIGURATION_INFO_2**Type:** R**Clock:** Config Clock**Reset State:** 0x00000000

Configuration information registers provide information on the capability and configuration of each slaveway instance. See register field descriptions for more details.

BIMC_S_APP_SWAY_CONFIGURATION_INFO_2

Bits	Name	Description
31:16	S_DATA_WIDTH	Indicates the width of read and write data to the bus slave.
15:8	S_TID_WIDTH	Indicates the width of transfer IDs to the bus slave.
7:0	S_MID_WIDTH	Indicates the width of master ID to the bus slave (includes any bus and port IDs.)

0x00450050 BIMC_S_APP_SWAY_CONFIGURATION_INFO_3**Type:** R**Clock:** Config Clock**Reset State:** 0x00000000

Configuration information registers provide information on the capability and configuration of each slaveway instance. See register field descriptions for more details.

BIMC_S_APP_SWAY_CONFIGURATION_INFO_3

Bits	Name	Description
31:24	RCH0_DEPTH	Indicates the depth of read channel buffers #0 in this slaveway. A beat of read data occupies one entry in the read channel buffer, until accepted by the masterport.
23:16	BCH_DEPTH	Indicates the depth of write response channel buffers in this slaveway. Each write response occupies one entry in the write response channel buffer, until accepted by the masterport.
15:8	WCH_DEPTH	Indicates the depth of write channel buffers in this slaveway. A beat of write data occupies one entry in the write channel buffer, until accepted by the bus slave.
7:0	ACH_DEPTH	Indicates the depth of address channel buffers in this slaveway. Each command occupies one entry in the address channel buffer, until accepted by the bus slave.

0x00450060 BIMC_S_APP_SWAY_CONFIGURATION_INFO_4**Type:** R**Clock:** Config Clock**Reset State:** 0x00000000

Configuration information registers provide information on the capability and configuration of each slaveway instance. See register field descriptions for more details.

BIMC_S_APP_SWAY_CONFIGURATION_INFO_4

Bits	Name	Description
31	DUAL_RCH_EN	Indicates that this slaveway has dual read channel buffers.
7:0	RCH1_DEPTH	Indicates the depth of read channel buffers #1 in this slaveway. A beat of read data occupies one entry in the read channel buffer, until accepted by the masterport.

0x00450070 BIMC_S_APP_SWAY_CONFIGURATION_INFO_5**Type:** R**Clock:** Config Clock**Reset State:** 0x00000000

Configuration information registers provide information on the capability and configuration of each slaveway instance. See register field descriptions for more details.

BIMC_S_APP_SWAY_CONFIGURATION_INFO_5

Bits	Name	Description
31	QOS_EN	Indicates that this slaveway uses command request priorities to control the number of high and low priority commands forwarded to the bus slave.

0x00450080 BIMC_S_APP_SWAY_CONFIGURATION_INFO_6A**Type:** R**Clock:** Config Clock**Reset State:** 0x00000000

Configuration information registers provide information on the capability and configuration of each slaveway instance. See register field descriptions for more details.

BIMC_S_APP_SWAY_CONFIGURATION_INFO_6A

Bits	Name	Description
0	ACH_PIPELINE_STAGES	Indicates pipeline stages enabled on the address channel. Each digit identifies a specific pipeline stage in the slaveway, where '0' indicates no pipeline stage, '1' indicates 1-deep pipeline stage where forward path is registered, return path is fed through, '2' indicates 2-deep pipeline stage where both forward and return path are pipelined. Digit 1 identifies pipelining in the address channel buffer, digit 0 identifies pipelining on the slaveway to slave interface.

0x00450084 BIMC_S_APP_SWAY_CONFIGURATION_INFO_6B**Type:** R**Clock:** Config Clock**Reset State:** 0x00000000

Configuration information registers provide information on the capability and configuration of each slaveway instance. See register field descriptions for more details.

BIMC_S_APP_SWAY_CONFIGURATION_INFO_6B

Bits	Name	Description
0	WCH_PIPELINE_STAGES	Indicates pipeline stages enabled on the write channel. Each digit identifies a specific pipeline stage in the slaveway, where '0' indicates no pipeline stage, '1' indicates 1-deep pipeline stage where forward path is registered, return path is fed through, '2' indicates 2-deep pipeline stage where both forward and return path are pipelined. Digit 1 identifies pipelining in the write channel buffer, digit 0 identifies pipelining on the slaveway to slave interface.

0x00450088 BIMC_S_APP_SWAY_CONFIGURATION_INFO_6C**Type:** R**Clock:** Config Clock**Reset State:** 0x00000000

Configuration information registers provide information on the capability and configuration of each slaveway instance. See register field descriptions for more details.

BIMC_S_APP_SWAY_CONFIGURATION_INFO_6C

Bits	Name	Description
0	BCH_PIPELINE_STAGES	Indicates pipeline stages enabled on the write response channel. Each digit identifies a specific pipeline stage in the slaveway, where '0' indicates no pipeline stage, '1' indicates 1-deep pipeline stage where forward path is registered, return path is fed through, '2' indicates 2-deep pipeline stage where both forward and return path are pipelined. Digit 2 identifies pipelining on the slave to slaveway interface, digit 1 identifies in the write response channel buffer, digit 0 identifies pipelining on the slaveway to masterport interface.

0x004500BC BIMC_S_APP_SWAY_CONFIGURATION_INFO_6D**Type:** R**Clock:** Config Clock**Reset State:** 0x00000000

Configuration information registers provide information on the capability and configuration of each slaveway instance. See register field descriptions for more details.

BIMC_S_APP_SWAY_CONFIGURATION_INFO_6D

Bits	Name	Description
0	RCH_PIPELINE_STAGES	Indicates pipeline stages enabled on the read channel. Each digit identifies a specific pipeline stage in the slaveway, where '0' indicates no pipeline stage, '1' indicates 1-deep pipeline stage where forward path is registered, return path is fed through, '2' indicates 2-deep pipeline stage where both forward and return path are pipelined. Digit 2 identifies pipelining on the slave to slaveway interface, digit 1 identifies in the read channel buffer, digit 0 identifies pipelining on the slaveway to masterport interface.

0x00450100 BIMC_S_APP_SWAY_INTERRUPT_STATUS**Type:** R**Clock:** Config Clock**Reset State:** 0x00000000

Interrupt status registers provide indication of which interrupts from this masterport require servicing.

BIMC_S_APP_SWAY_INTERRUPT_STATUS

Bits	Name	Description
1:0	RFU	Reserved for future use

0x00450108 BIMC_S_APP_SWAY_INTERRUPT_CLEAR**Type:** W**Clock:** Config Clock**Reset State:** 0x00000000

Interrupt clear registers provide the mechanism to turn off interrupt sources once they have been acknowledged or serviced. A write of '1' to the bits corresponding to the interrupts indicated in interrupt status is required to turn off interrupts.

BIMC_S_APP_SWAY_INTERRUPT_CLEAR

Bits	Name	Description
1:0	RFU	Reserved for future use

0x0045010C BIMC_S_APP_SWAY_INTERRUPT_ENABLE**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000

Interrupt enable registers provide control over which interrupts are allowed to be propagated to an interrupt controller for servicing.

BIMC_S_APP_SWAY_INTERRUPT_ENABLE

Bits	Name	Description
1:0	RFU	Reserved for future use

0x00450200 BIMC_S_APP_SWAY_CLOCK_CTRL**Type:** RW**Clock:** Config, Core & Slave Clocks**Reset State:** 0x00000003**BIMC_S_APP_SWAY_CLOCK_CTRL**

Bits	Name	Description
1	SLAVE_CLOCK_GATING_EN	When set (1), enables activity based clock gating of all registers clocked by slave clock. Activity based clock gating uses architectural clock gates to reduce dynamic power consumption by gating clocks during periods of in-activity. When cleared (0), disables all activity based clock gating by forcing all architectural clock gates to not gate clocks even during periods of in-activity.

BIMC_S_APP_SWAY_CLOCK_CTRL (cont.)

Bits	Name	Description
0	CORE_CLOCK_GATING_EN	When set (1), enables activity based clock gating of all registers clocked by core clock. Activity based clock gating uses architectural clock gates to reduce dynamic power consumption by gating clocks during periods of in-activity. When cleared (0), disables all activity based clock gating by forcing all architectural clock gates to not gate clocks even during periods of in-activity.

0x00450208 BIMC_S_APP_SWAY_CDC_CTRL**Type:** RW**Clock:** Config, Core & Slave Clocks**Reset State:** 0x00000000

Clock domain crossing (CDC) control register provides configuration for clock domain crossing logic.

BIMC_S_APP_SWAY_CDC_CTRL

Bits	Name	Description
1	SLAVE_CLOCK_DEMETA_SEL	When cleared (0), selects DEMETA synchronizer that requires atleast 3 clock edges (two rising and one falling). When set (1), selects DEMETA synchronizer that requires atleast 2 clock edges (one falling and one rising) to synchronize to slave clock in all buffers.
0	CORE_CLOCK_DEMETA_SEL	When cleared (0), selects DEMETA synchronizer that requires atleast 3 clock edges (two rising and one falling). When set (1), selects DEMETA synchronizer that requires atleast 2 clock edges (one falling and one rising) to synchronize to core clock in all buffers.

0x00450210 BIMC_S_APP_SWAY_RCH_SELECT**Type:** R**Clock:** Config & Slave Clocks**Reset State:** 0x00000000

Read channel select register controls which read channel buffer to use for buffering read data for read data destined for each masterport in the interconnect.

BIMC_S_APP_SWAY_RCH_SELECT

Bits	Name	Description
7:0	UNUSED	

0x00450220 BIMC_S_APP_SWAY_MAX_OUTSTANDING_REQUESTS**Type:** RW**Clock:** Config & Core Clocks**Reset State:** 0x0000FFFF**BIMC_S_APP_SWAY_MAX_OUTSTANDING_REQUESTS**

Bits	Name	Description
15:8	WRITE	Configures the maximum number of write commands that can be outstanding at the slaveway at any time.
7:0	READ	Configures the maximum number of read commands that can be outstanding at the slaveway at any time.

0x00450230 BIMC_S_APP_SWAY_GATHERING**Type:** RW**Clock:** Config & Slave Clocks**Reset State:** 0x00000000**BIMC_S_APP_SWAY_GATHERING**

Bits	Name	Description
3:0	RD_GATHER_BEATS	Configures the number of read data beats to gather/buffer in read channel buffer before allowing the read data to be routed to masterports. When configured with 0, read data is allowed to proceed as soon as the first beat has been buffered. When configured with 1, read data is allowed to proceed as soon as the second beat has been buffered. When configured with values larger than the number of beats in the read burst, the read data is allowed to proceed as soon as the last beat has been buffered. When configured with values larger than the depth of read channel buffers, the read data is held until all entries in the read channel buffer have been occupied.

0x00450400 BIMC_S_APP_SWAY_STATUS_0A**Type:** R**Clock:** Config & Core Clocks**Reset State:** 0x00000000

Status registers indicate the state of several elements in the slaveway, including the number of transactions outstanding/pending waiting for a response, emptiness, fullness of several buffers, etc. The state information is available only as a debug indication and could be delayed by a few clock cycles from when the state is reached to being reflected in the status registers. One or more of the state registers are read live and could capture intermediate values as they change.

BIMC_S_APP_SWAY_STATUS_0A

Bits	Name	Description
23:16	WR_PENDING_PRE_BUF	Indicates the number of write transactions that are outstanding/pending waiting for a response from the slave. This transaction count is before address channel buffers, which interfaces to core of the interconnect.
7:0	RD_PENDING_PRE_BUF	Indicates the number of read transactions that are outstanding/pending waiting for a response from the slave. This transaction count is from PIPE0 of the masterport before address channel buffers, which interfaces to core of the interconnect.

0x00450404 BIMC_S_APP_SWAY_STATUS_0B**Type:** R**Clock:** Config & Slave Clocks**Reset State:** 0x00000000

Status registers indicate the state of several elements in the slaveway, including the number of transactions outstanding/pending waiting for a response, emptiness, fullness of several buffers, etc. The state information is available only as a debug indication and could be delayed by a few clock cycles from when the state is reached to being reflected in the status registers. One or more of the state registers are read live and could capture intermediate values as they change.

BIMC_S_APP_SWAY_STATUS_0B

Bits	Name	Description
23:16	WR_PENDING_POST_BUF	Indicates the number of write transactions that are outstanding/pending waiting for a response from the slave. This transaction count is from after address channel buffers, which interfaces to the bus slave.
7:0	RD_PENDING_POST_BUF	Indicates the number of read transactions that are outstanding/pending waiting for a response from the slave. This transaction count is from PIPE0 of the masterport after address channel buffers, which interfaces to the bus slave.

0x00450408 BIMC_S_APP_SWAY_STATUS_0C**Type:** R**Clock:** Config & Core Clocks**Reset State:** 0x00000000

Status registers indicate the state of several elements in the slaveway, including the number of transactions outstanding/pending waiting for a response, emptiness, fullness of several buffers, etc. The state information is available only as a debug indication and could be delayed by a few clock

cycles from when the state is reached to being reflected in the status registers. One or more of the state registers are read live and could capture intermediate values as they change.

BIMC_S_APP_SWAY_STATUS_0C

Bits	Name	Description
23:16	RDQ_FULL	Indicates the state of QoS feedback signalling to the arbiter. When set (1), indicates that reads can no longer be accepted from the arbiter. 1-bit per masterport, bit position corresponds to masterport enumeration.
8	WRQ_FULL	Indicates the state of QoS feedback signalling to the arbiter. When set (1), indicates that writes can no longer be accepted from the arbiter.
3:0	CMDQ_FULL	Indicates the state of QoS feedback signalling to the arbiter. When set (1), indicates that commands (either reads or writes) of a certain priority can no longer be accepted from the arbiter. 1-bit per priority, bit position corresponds to request priority enumeration.

0x0045040C BIMC_S_APP_SWAY_STATUS_0D

Type: R

Clock: Config & Core Clocks

Reset State: 0x00000000

Status registers indicate the state of several elements in the slaweway, including the number of transactions outstanding/pending waiting for a response, emptiness, fullness of several buffers, etc. The state information is available only as a debug indication and could be delayed by a few clock cycles from when the state is reached to being reflected in the status registers. One or more of the state registers are read live and could capture intermediate values as they change.

BIMC_S_APP_SWAY_STATUS_0D

Bits	Name	Description
23:16	RDQ_ALMOST_FULL	Indicates the state of QoS feedback signalling to the arbiter. When set (1), indicates that reads can no longer be accepted from the arbiter, after one more read. 1-bit per masterport, bit position corresponds to masterport enumeration.
8	WRQ_ALMOST_FULL	Indicates the state of QoS feedback signalling to the arbiter. When set (1), indicates that writes can no longer be accepted from the arbiter, after one more write.
3:0	CMDQ_ALMOST_FULL	Indicates the state of QoS feedback signalling to the arbiter. When set (1), indicates that commands (either reads or writes) of a certain priority can no longer be accepted from the arbiter, after one more command. 1-bit per priority, bit position corresponds to request priority enumeration.

0x00450410 BIMC_S_APP_SWAY_STATUS_1A**Type:** R**Clock:** Config, Core & Slave Clocks**Reset State:** 0x55111111

Status registers indicate the state of several elements in the slaveway, including the number of transactions outstanding/pending waiting for a response, emptiness, fullness of several buffers, etc. The state information is available only as a debug indication and could be delayed by a few clock cycles from when the state is reached to being reflected in the status registers. One or more of the state registers are read live and could capture intermediate values as they change.

BIMC_S_APP_SWAY_STATUS_1A

Bits	Name	Description
31	RCB0_DATA_RD_FULL	Indicates that the data part of read channel buffer #0 is full when seen from it's read side, which interfaces to core of the interconnect.
30	RCB0_DATA_RD_EMPTY	Indicates that the data part of read channel buffer #0 is empty when seen from it's read side, which interfaces to core of the interconnect.
29	RCB0_CTRL_RD_FULL	Indicates that the control part of read channel buffer #0 is full when seen from it's read side, which interfaces to core of the interconnect.
28	RCB0_CTRL_RD_EMPTY	Indicates that the control part of read channel buffer #0 is empty when seen from it's write side, which interfaces to core of the interconnect.
27	RCB0_DATA_WR_FULL	Indicates that the data part of read channel buffer #0 is full when seen from it's write side, which interfaces to the bus slave.
26	RCB0_DATA_WR_EMPTY	Indicates that the data part of read channel buffer #0 is empty when seen from it's write side, which interfaces to the bus slave.
25	RCB0_CTRL_WR_FULL	Indicates that the control part of read channel buffer #0 is full when seen from it's write side, which interfaces to the bus slave.
24	RCB0_CTRL_WR_EMPTY	Indicates that the control part of read channel buffer #0 is empty when seen from it's write side, which interfaces to the bus slave.
21	BCB_RD_FULL	Indicates that the write response channel buffer is full when seen from it's read side, which interfaces to core of the interconnect.
20	BCB_RD_EMPTY	Indicates that the write response channel buffer is empty when seen from it's read side, which interfaces to core of the interconnect.
17	BCB_WR_FULL	Indicates that the write response channel buffer is full when seen from it's write side, which interfaces to the bus slave.
16	BCB_WR_EMPTY	Indicates that the write response channel buffer is empty when seen from it's write side, which interfaces to the bus slave.
13	WCB_RD_FULL	Indicates that the write channel buffer is full when seen from it's read side, which interfaces to the bus slave.

BIMC_S_APP_SWAY_STATUS_1A (cont.)

Bits	Name	Description
12	WCB_RD_EMPTY	Indicates that the write channel buffer is empty when seen from it's read side, which interfaces to the bus slave.
9	WCB_WR_FULL	Indicates that the write channel buffer is full when seen from it's write side, which interfaces to core of the interconnect.
8	WCB_WR_EMPTY	Indicates that the write channel buffer is empty when seen from it's write side, which interfaces to core of the interconnect.
5	ACB_RD_FULL	Indicates that the address channel buffer is full when seen from it's read side, which interfaces to the bus slave.
4	ACB_RD_EMPTY	Indicates that the address channel buffer is empty when seen from it's read side, which interfaces to the bus slave.
1	ACB_WR_FULL	Indicates that the address channel buffer is full when seen from it's write side, which interfaces to core of the interconnect.
0	ACB_WR_EMPTY	Indicates that the address channel buffer is empty when seen from it's write side, which interfaces to core of the interconnect.

0x00450414 BIMC_S_APP_SWAY_STATUS_1B**Type:** R**Clock:** Config, Core & Slave Clocks**Reset State:** 0x11111111

Status registers indicate the state of several elements in the slaveway, including the number of transactions outstanding/pending waiting for a response, emptiness, fullness of several buffers, etc. The state information is available only as a debug indication and could be delayed by a few clock cycles from when the state is reached to being reflected in the status registers. One or more of the state registers are read live and could capture intermediate values as they change.

BIMC_S_APP_SWAY_STATUS_1B

Bits	Name	Description
29	WIB_RD_FULL	Indicates that the write channel IDs buffer is full when seen from it's read side, which interfaces to the bus slave.
28	WIB_RD_EMPTY	Indicates that the write channel IDs buffer is empty when seen from it's read side, which interfaces to the bus slave.
25	WIB_WR_FULL	Indicates that the write channel IDs buffer is full when seen from it's write side, which interfaces to core of the interconnect.
24	WIB_WR_EMPTY	Indicates that the write channel IDs buffer is empty when seen from it's write side, which interfaces to core of the interconnect.
21	RGB0_RD_FULL	Indicates that the read gathering buffer #0 is full when seen from it's read side, which interfaces to frontend of the masterport.

BIMC_S_APP_SWAY_STATUS_1B (cont.)

Bits	Name	Description
20	RGB0_RD_EMPTY	Indicates that the read gathering buffer #0 is empty when seen from it's read side, which interfaces to core of the interconnect.
17	RGB0_WR_FULL	Indicates that the read gathering buffer #0 is full when seen from it's write side, which interfaces to core of the interconnect.
16	RGB0_WR_EMPTY	Indicates that the read gathering buffer #0 is empty when seen from it's write side, which interfaces to the bus slave.
13	W2AB_RD_FULL	Indicates that the write channel to address channel user signals buffer is full when seen from it's read side, which interfaces to the bus slave.
12	W2AB_RD_EMPTY	Indicates that the write channel to address channel user signals buffer is empty when seen from it's read side, which interfaces to the bus slave.
9	W2AB_WR_FULL	Indicates that the write channel to address channel user signals buffer is full when seen from it's write side, which interfaces to core of the interconnect.
8	W2AB_WR_EMPTY	Indicates that the write channel to address channel user signals buffer is empty when seen from it's write side, which interfaces to core of the interconnect.
5	A2WB_RD_FULL	Indicates that the address channel to write channel user signals buffer is full when seen from it's read side, which interfaces to the bus slave.
4	A2WB_RD_EMPTY	Indicates that the address channel to write channel user signalsbuffer is empty when seen from it's read side, which interfaces to the bus slave.
1	A2WB_WR_FULL	Indicates that the address channel to write channel user signals buffer is full when seen from it's write side, which interfaces to core of the interconnect.
0	A2WB_WR_EMPTY	Indicates that the address channel to write channel user signals buffer is empty when seen from it's write side, which interfaces to core of the interconnect.

0x00450418 BIMC_S_APP_SWAY_STATUS_1C**Type:** R**Clock:** Config, Core & Slave Clocks**Reset State:** 0x00001155

Status registers indicate the state of several elements in the slaweway, including the number of transactions outstanding/pending waiting for a response, emptiness, fullness of several buffers, etc. The state information is available only as a debug indication and could be delayed by a few clock cycles from when the state is reached to being reflected in the status registers. One or more of the state registers are read live and could capture intermediate values as they change.

BIMC_S_APP_SWAY_STATUS_1C

Bits	Name	Description
13	RGB1_RD_FULL	Indicates that the read gathering buffer #1 is full when seen from it's read side, which interfaces to frontend of the masterport.
12	RGB1_RD_EMPTY	Indicates that the read gathering buffer #1 is empty when seen from it's read side, which interfaces to core of the interconnect.
9	RGB1_WR_FULL	Indicates that the read gathering buffer #1 is full when seen from it's write side, which interfaces to core of the interconnect.
8	RGB1_WR_EMPTY	Indicates that the read gathering buffer #1 is empty when seen from it's write side, which interfaces to the bus slave.
7	RCB1_DATA_RD_FULL	Indicates that the data part of read channel buffer #1 is full when seen from it's read side, which interfaces to core of the interconnect.
6	RCB1_DATA_RD_EMPTY	Indicates that the data part of read channel buffer #1 is empty when seen from it's read side, which interfaces to core of the interconnect.
5	RCB1_CTRL_RD_FULL	Indicates that the control part of read channel buffer #1 is full when seen from it's read side, which interfaces to core of the interconnect.
4	RCB1_CTRL_RD_EMPTY	Indicates that the control part of read channel buffer #1 is empty when seen from it's write side, which interfaces to core of the interconnect.
3	RCB1_DATA_WR_FULL	Indicates that the data part of read channel buffer #1 is full when seen from it's write side, which interfaces to the bus slave.
2	RCB1_DATA_WR_EMPTY	Indicates that the data part of read channel buffer #1 is empty when seen from it's write side, which interfaces to the bus slave.
1	RCB1_CTRL_WR_FULL	Indicates that the control part of read channel buffer #1 is full when seen from it's write side, which interfaces to the bus slave.
0	RCB1_CTRL_WR_EMPTY	Indicates that the control part of read channel buffer #1 is empty when seen from it's write side, which interfaces to the bus slave.

2.20 BIMC_S_SYS_SWAY

BRIC Slaveway

2.20.1 BIMC_S_SYS_SWAY_BRIC_SLAVEWAY

BRIC Slaveway

0x00458000 BIMC_S_SYS_SWAY_COMPONENT_INFO**Type:** R**Clock:** Config Clock**Reset State:** 0x00000103

Component information registers provide discoverability information on the type of component and instance of each component. These registers are always the very first register in each component's configuration address space and allow for a software routine to walk the configuration address space in steps of 4KB to identify the location of each configuration register space and to which component each 4KB configures.

BIMC_S_SYS_SWAY_COMPONENT_INFO

Bits	Name	Description
23:16	INSTANCE	Indicates the instance number of this slaveway.
15:8	SUB_TYPE	Identifies this configuration space as belonging to a slaveway 0x1: SLAVEWAY
7:0	TYPE	Identifies this configuration space as belonging to a component that interfaces to a bus slave. 0x3: SLAVEWAY

0x00458020 BIMC_S_SYS_SWAY_CONFIGURATION_INFO_0**Type:** R**Clock:** Config Clock**Reset State:** 0x01000001

Configuration information registers provide information on the capability and configuration of each slaveway instance. See register field descriptions for more details.

BIMC_S_SYS_SWAY_CONFIGURATION_INFO_0

Bits	Name	Description
31:24	SYNC_MODE	Indicates the clock relationship between the clock used in the core side and the clock used in the slave side of the slaveway. When ASYNC, identifies the master clock and core clock as un-related and asynchronous to each other. When SYNC, identifies the core clock and slave clock as synchronous to each other. When ISOSYNC, identifies the core clock and slave clock as related, but of different frequencies, which are of integer ratios, i.e., they are iso-synchronous to each other. 0x0: ASYNC 0x1: SYNC 0x2: ISOSYNC

BIMC_S_SYS_SWAY_CONFIGURATION_INFO_0 (cont.)

Bits	Name	Description
7:0	FUNCTIONALITY	Indicates the functionality and type of bus traffic handled by this slaveway. When DEFAULT_SLAVE, identifies this slaveway as a default slave which responds to all commands with decode error. When SLAVEWAY, identifies this slaveway as interfacing to a bus slave. When STUB, identifies this slaveway has a stub. 0x0: STUB 0x1: SLAVEWAY

0x00458030 BIMC_S_SYS_SWAY_CONFIGURATION_INFO_1**Type:** R**Clock:** Config Clock**Reset State:** 0x00000000

Configuration information registers provide information on the capability and configuration of each slaveway instance. See register field descriptions for more details.

BIMC_S_SYS_SWAY_CONFIGURATION_INFO_1

Bits	Name	Description
31:0	MPORT_CONNECTIVITY	Indicates which masterports can route commands to this slaveway instance. 1-bit per masterport. When a specific bit is set (1), indicates that the masterport corresponding to this bit can route commands to this slaveway instance. When a specific bit is cleared (0), indicates that the masterport corresponding to this bit does not route commands to this slaveway instance.

0x00458040 BIMC_S_SYS_SWAY_CONFIGURATION_INFO_2**Type:** R**Clock:** Config Clock**Reset State:** 0x00000000

Configuration information registers provide information on the capability and configuration of each slaveway instance. See register field descriptions for more details.

BIMC_S_SYS_SWAY_CONFIGURATION_INFO_2

Bits	Name	Description
31:16	S_DATA_WIDTH	Indicates the width of read and write data to the bus slave.
15:8	S_TID_WIDTH	Indicates the width of transfer IDs to the bus slave.
7:0	S_MID_WIDTH	Indicates the width of master ID to the bus slave (includes any bus and port IDs.)

0x00458050 BIMC_S_SYS_SWAY_CONFIGURATION_INFO_3**Type:** R**Clock:** Config Clock**Reset State:** 0x00000000

Configuration information registers provide information on the capability and configuration of each slaveway instance. See register field descriptions for more details.

BIMC_S_SYS_SWAY_CONFIGURATION_INFO_3

Bits	Name	Description
31:24	RCH0_DEPTH	Indicates the depth of read channel buffers #0 in this slaveway. A beat of read data occupies one entry in the read channel buffer, until accepted by the masterport.
23:16	BCH_DEPTH	Indicates the depth of write response channel buffers in this slaveway. Each write response occupies one entry in the write response channel buffer, until accepted by the masterport.
15:8	WCH_DEPTH	Indicates the depth of write channel buffers in this slaveway. A beat of write data occupies one entry in the write channel buffer, until accepted by the bus slave.
7:0	ACH_DEPTH	Indicates the depth of address channel buffers in this slaveway. Each command occupies one entry in the address channel buffer, until accepted by the bus slave.

0x00458060 BIMC_S_SYS_SWAY_CONFIGURATION_INFO_4**Type:** R**Clock:** Config Clock**Reset State:** 0x00000000

Configuration information registers provide information on the capability and configuration of each slaveway instance. See register field descriptions for more details.

BIMC_S_SYS_SWAY_CONFIGURATION_INFO_4

Bits	Name	Description
31	DUAL_RCH_EN	Indicates that this slaveway has dual read channel buffers.
7:0	RCH1_DEPTH	Indicates the depth of read channel buffers #1 in this slaveway. A beat of read data occupies one entry in the read channel buffer, until accepted by the masterport.

0x00458070 BIMC_S_SYS_SWAY_CONFIGURATION_INFO_5**Type:** R**Clock:** Config Clock**Reset State:** 0x00000000

Configuration information registers provide information on the capability and configuration of each slaveway instance. See register field descriptions for more details.

BIMC_S_SYS_SWAY_CONFIGURATION_INFO_5

Bits	Name	Description
31	QOS_EN	Indicates that this slaveway uses command request priorities to control the number of high and low priority commands forwarded to the bus slave.

0x00458080 BIMC_S_SYS_SWAY_CONFIGURATION_INFO_6A**Type:** R**Clock:** Config Clock**Reset State:** 0x00000000

Configuration information registers provide information on the capability and configuration of each slaveway instance. See register field descriptions for more details.

BIMC_S_SYS_SWAY_CONFIGURATION_INFO_6A

Bits	Name	Description
0	ACH_PIPELINE_STAGES	Indicates pipeline stages enabled on the address channel. Each digit identifies a specific pipeline stage in the slaveway, where '0' indicates no pipeline stage, '1' indicates 1-deep pipeline stage where forward path is registered, return path is fed through, '2' indicates 2-deep pipeline stage where both forward and return path are pipelined. Digit 1 identifies pipelining in the address channel buffer, digit 0 identifies pipelining on the slaveway to slave interface.

0x00458084 BIMC_S_SYS_SWAY_CONFIGURATION_INFO_6B**Type:** R**Clock:** Config Clock**Reset State:** 0x00000000

Configuration information registers provide information on the capability and configuration of each slaveway instance. See register field descriptions for more details.

BIMC_S_SYS_SWAY_CONFIGURATION_INFO_6B

Bits	Name	Description
0	WCH_PIPELINE_STAGES	Indicates pipeline stages enabled on the write channel. Each digit identifies a specific pipeline stage in the slaveway, where '0' indicates no pipeline stage, '1' indicates 1-deep pipeline stage where forward path is registered, return path is fed through, '2' indicates 2-deep pipeline stage where both forward and return path are pipelined. Digit 1 identifies pipelining in the write channel buffer, digit 0 identifies pipelining on the slaveway to slave interface.

0x00458088 BIMC_S_SYS_SWAY_CONFIGURATION_INFO_6C**Type:** R**Clock:** Config Clock**Reset State:** 0x00000000

Configuration information registers provide information on the capability and configuration of each slaveway instance. See register field descriptions for more details.

BIMC_S_SYS_SWAY_CONFIGURATION_INFO_6C

Bits	Name	Description
0	BCH_PIPELINE_STAGES	Indicates pipeline stages enabled on the write response channel. Each digit identifies a specific pipeline stage in the slaveway, where '0' indicates no pipeline stage, '1' indicates 1-deep pipeline stage where forward path is registered, return path is fed through, '2' indicates 2-deep pipeline stage where both forward and return path are pipelined. Digit 2 identifies pipelining on the slave to slaveway interface, digit 1 identifies in the write response channel buffer, digit 0 identifies pipelining on the slaveway to masterport interface.

0x004580BC BIMC_S_SYS_SWAY_CONFIGURATION_INFO_6D**Type:** R**Clock:** Config Clock**Reset State:** 0x00000000

Configuration information registers provide information on the capability and configuration of each slaveway instance. See register field descriptions for more details.

BIMC_S_SYS_SWAY_CONFIGURATION_INFO_6D

Bits	Name	Description
0	RCH_PIPELINE_STAGES	Indicates pipeline stages enabled on the read channel. Each digit identifies a specific pipeline stage in the slaveway, where '0' indicates no pipeline stage, '1' indicates 1-deep pipeline stage where forward path is registered, return path is fed through, '2' indicates 2-deep pipeline stage where both forward and return path are pipelined. Digit 2 identifies pipelining on the slave to slaveway interface, digit 1 identifies in the read channel buffer, digit 0 identifies pipelining on the slaveway to masterport interface.

0x00458100 BIMC_S_SYS_SWAY_INTERRUPT_STATUS**Type:** R**Clock:** Config Clock**Reset State:** 0x00000000

Interrupt status registers provide indication of which interrupts from this masterport require servicing.

BIMC_S_SYS_SWAY_INTERRUPT_STATUS

Bits	Name	Description
1:0	RFU	Reserved for future use

0x00458108 BIMC_S_SYS_SWAY_INTERRUPT_CLEAR**Type:** W**Clock:** Config Clock**Reset State:** 0x00000000

Interrupt clear registers provide the mechanism to turn off interrupt sources once they have been acknowledged or serviced. A write of '1' to the bits corresponding to the interrupts indicated in interrupt status is required to turn off interrupts.

BIMC_S_SYS_SWAY_INTERRUPT_CLEAR

Bits	Name	Description
1:0	RFU	Reserved for future use

0x0045810C BIMC_S_SYS_SWAY_INTERRUPT_ENABLE**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000

Interrupt enable registers provide control over which interrupts are allowed to be propagated to an interrupt controller for servicing.

BIMC_S_SYS_SWAY_INTERRUPT_ENABLE

Bits	Name	Description
1:0	RFU	Reserved for future use

0x00458200 BIMC_S_SYS_SWAY_CLOCK_CTRL**Type:** RW**Clock:** Config, Core & Slave Clocks**Reset State:** 0x00000003**BIMC_S_SYS_SWAY_CLOCK_CTRL**

Bits	Name	Description
1	SLAVE_CLOCK_GATING_EN	When set (1), enables activity based clock gating of all registers clocked by slave clock. Activity based clock gating uses architectural clock gates to reduce dynamic power consumption by gating clocks during periods of in-activity. When cleared (0), disables all activity based clock gating by forcing all architectural clock gates to not gate clocks even during periods of in-activity.
0	CORE_CLOCK_GATING_EN	When set (1), enables activity based clock gating of all registers clocked by core clock. Activity based clock gating uses architectural clock gates to reduce dynamic power consumption by gating clocks during periods of in-activity. When cleared (0), disables all activity based clock gating by forcing all architectural clock gates to not gate clocks even during periods of in-activity.

0x00458208 BIMC_S_SYS_SWAY_CDC_CTRL**Type:** RW**Clock:** Config, Core & Slave Clocks**Reset State:** 0x00000000

Clock domain crossing (CDC) control register provides configuration for clock domain crossing logic.

BIMC_S_SYS_SWAY_CDC_CTRL

Bits	Name	Description
1	SLAVE_CLOCK_DEMETA_SEL	When cleared (0), selects DEMETA synchronizer that requires atleast 3 clock edges (two rising and one falling). When set (1), selects DEMETA synchronizer that requires atleast 2 clock edges (one falling and one rising) to synchronize to slave clock in all buffers.
0	CORE_CLOCK_DEMETA_SEL	When cleared (0), selects DEMETA synchronizer that requires atleast 3 clock edges (two rising and one falling). When set (1), selects DEMETA synchronizer that requires atleast 2 clock edges (one falling and one rising) to synchronize to core clock in all buffers.

0x00458210 BIMC_S_SYS_SWAY_RCH_SELECT**Type:** R**Clock:** Config & Slave Clocks**Reset State:** 0x00000000

Read channel select register controls which read channel buffer to use for buffering read data for read data destined for each masterport in the interconnect.

BIMC_S_SYS_SWAY_RCH_SELECT

Bits	Name	Description
7:0	UNUSED	

0x00458230 BIMC_S_SYS_SWAY_GATHERING**Type:** RW**Clock:** Config & Slave Clocks**Reset State:** 0x00000000

BIMC_S_SYS_SWAY_GATHERING

Bits	Name	Description
3:0	RD_GATHER_BEATS	Configures the number of read data beats to gather/buffer in read channel buffer before allowing the read data to be routed to masterports. When configured with 0, read data is allowed to proceed as soon as the first beat has been buffered. When configured with 1, read data is allowed to proceed as soon as the second beat has been buffered. When configured with values larger than the number of beats in the read burst, the read data is allowed to proceed as soon as the last beat has been buffered. When configured with values larger than the depth of read channel buffers, the read data is held until all entries in the read channel buffer have been occupied.

0x00458400 BIMC_S_SYS_SWAY_STATUS_0A**Type:** R**Clock:** Config & Core Clocks**Reset State:** 0x00000000

Status registers indicate the state of several elements in the slaweway, including the number of transactions outstanding/pending waiting for a response, emptiness, fullness of several buffers, etc. The state information is available only as a debug indication and could be delayed by a few clock cycles from when the state is reached to being reflected in the status registers. One or more of the state registers are read live and could capture intermediate values as they change.

BIMC_S_SYS_SWAY_STATUS_0A

Bits	Name	Description
23:16	WR_PENDING_PRE_BUF	Indicates the number of write transactions that are outstanding/pending waiting for a response from the slave. This transaction count is before address channel buffers, which interfaces to core of the interconnect.
7:0	RD_PENDING_PRE_BUF	Indicates the number of read transactions that are outstanding/pending waiting for a response from the slave. This transaction count is from PIPE0 of the masterport before address channel buffers, which interfaces to core of the interconnect.

0x00458404 BIMC_S_SYS_SWAY_STATUS_0B**Type:** R**Clock:** Config & Slave Clocks**Reset State:** 0x00000000

Status registers indicate the state of several elements in the slaweway, including the number of transactions outstanding/pending waiting for a response, emptiness, fullness of several buffers, etc. The state information is available only as a debug indication and could be delayed by a few clock

cycles from when the state is reached to being reflected in the status registers. One or more of the state registers are read live and could capture intermediate values as they change.

BIMC_S_SYS_SWAY_STATUS_0B

Bits	Name	Description
23:16	WR_PENDING_POST_BUF	Indicates the number of write transactions that are outstanding/pending waiting for a response from the slave. This transaction count is from after address channel buffers, which interfaces to the bus slave.
7:0	RD_PENDING_POST_BUF	Indicates the number of read transactions that are outstanding/pending waiting for a response from the slave. This transaction count is from PIPE0 of the masterport after address channel buffers, which interfaces to the bus slave.

0x00458408 BIMC_S_SYS_SWAY_STATUS_0C

Type: R

Clock: Config & Core Clocks

Reset State: 0x00000000

Status registers indicate the state of several elements in the slaveway, including the number of transactions outstanding/pending waiting for a response, emptiness, fullness of several buffers, etc. The state information is available only as a debug indication and could be delayed by a few clock cycles from when the state is reached to being reflected in the status registers. One or more of the state registers are read live and could capture intermediate values as they change.

BIMC_S_SYS_SWAY_STATUS_0C

Bits	Name	Description
23:16	RDQ_FULL	Indicates the state of QoS feedback signalling to the arbiter. When set (1), indicates that reads can no longer be accepted from the arbiter. 1-bit per masterport, bit position corresponds to masterport enumeration.
8	WRQ_FULL	Indicates the state of QoS feedback signalling to the arbiter. When set (1), indicates that writes can no longer be accepted from the arbiter.
3:0	CMDQ_FULL	Indicates the state of QoS feedback signalling to the arbiter. When set (1), indicates that commands (either reads or writes) of a certain priority can no longer be accepted from the arbiter. 1-bit per priority, bit position corresponds to request priority enumeration.

0x0045840C BIMC_S_SYS_SWAY_STATUS_0D**Type:** R**Clock:** Config & Core Clocks**Reset State:** 0x00000000

Status registers indicate the state of several elements in the slaweway, including the number of transactions outstanding/pending waiting for a response, emptiness, fullness of several buffers, etc. The state information is available only as a debug indication and could be delayed by a few clock cycles from when the state is reached to being reflected in the status registers. One or more of the state registers are read live and could capture intermediate values as they change.

BIMC_S_SYS_SWAY_STATUS_0D

Bits	Name	Description
23:16	RDQ_ALMOST_FULL	Indicates the state of QoS feedback signalling to the arbiter. When set (1), indicates that reads can no longer be accepted from the arbiter, after one more read. 1-bit per masterport, bit position corresponds to masterport enumeration.
8	WRQ_ALMOST_FULL	Indicates the state of QoS feedback signalling to the arbiter. When set (1), indicates that writes can no longer be accepted from the arbiter, after one more write.
3:0	CMDQ_ALMOST_FULL	Indicates the state of QoS feedback signalling to the arbiter. When set (1), indicates that commands (either reads or writes) of a certain priority can no longer be accepted from the arbiter, after one more command. 1-bit per priority, bit position corresponds to request priority enumeration.

0x00458410 BIMC_S_SYS_SWAY_STATUS_1A**Type:** R**Clock:** Config, Core & Slave Clocks**Reset State:** 0x55111111

Status registers indicate the state of several elements in the slaweway, including the number of transactions outstanding/pending waiting for a response, emptiness, fullness of several buffers, etc. The state information is available only as a debug indication and could be delayed by a few clock cycles from when the state is reached to being reflected in the status registers. One or more of the state registers are read live and could capture intermediate values as they change.

BIMC_S_SYS_SWAY_STATUS_1A

Bits	Name	Description
31	RCB0_DATA_RD_FULL	Indicates that the data part of read channel buffer #0 is full when seen from it's read side, which interfaces to core of the interconnect.

BIMC_S_SYS_SWAY_STATUS_1A (cont.)

Bits	Name	Description
30	RCB0_DATA_RD_EMPTY	Indicates that the data part of read channel buffer #0 is empty when seen from it's read side, which interfaces to core of the interconnect.
29	RCB0_CTRL_RD_FULL	Indicates that the control part of read channel buffer #0 is full when seen from it's read side, which interfaces to core of the interconnect.
28	RCB0_CTRL_RD_EMPTY	Indicates that the control part of read channel buffer #0 is empty when seen from it's write side, which interfaces to core of the interconnect.
27	RCB0_DATA_WR_FULL	Indicates that the data part of read channel buffer #0 is full when seen from it's write side, which interfaces to the bus slave.
26	RCB0_DATA_WR_EMPTY	Indicates that the data part of read channel buffer #0 is empty when seen from it's write side, which interfaces to the bus slave.
25	RCB0_CTRL_WR_FULL	Indicates that the control part of read channel buffer #0 is full when seen from it's write side, which interfaces to the bus slave.
24	RCB0_CTRL_WR_EMPTY	Indicates that the control part of read channel buffer #0 is empty when seen from it's write side, which interfaces to the bus slave.
21	BCB_RD_FULL	Indicates that the write response channel buffer is full when seen from it's read side, which interfaces to core of the interconnect.
20	BCB_RD_EMPTY	Indicates that the write response channel buffer is empty when seen from it's read side, which interfaces to core of the interconnect.
17	BCB_WR_FULL	Indicates that the write response channel buffer is full when seen from it's write side, which interfaces to the bus slave.
16	BCB_WR_EMPTY	Indicates that the write response channel buffer is empty when seen from it's write side, which interfaces to the bus slave.
13	WCB_RD_FULL	Indicates that the write channel buffer is full when seen from it's read side, which interfaces to the bus slave.
12	WCB_RD_EMPTY	Indicates that the write channel buffer is empty when seen from it's read side, which interfaces to the bus slave.
9	WCB_WR_FULL	Indicates that the write channel buffer is full when seen from it's write side, which interfaces to core of the interconnect.
8	WCB_WR_EMPTY	Indicates that the write channel buffer is empty when seen from it's write side, which interfaces to core of the interconnect.
5	ACB_RD_FULL	Indicates that the address channel buffer is full when seen from it's read side, which interfaces to the bus slave.
4	ACB_RD_EMPTY	Indicates that the address channel buffer is empty when seen from it's read side, which interfaces to the bus slave.
1	ACB_WR_FULL	Indicates that the address channel buffer is full when seen from it's write side, which interfaces to core of the interconnect.
0	ACB_WR_EMPTY	Indicates that the address channel buffer is empty when seen from it's write side, which interfaces to core of the interconnect.

0x00458414 BIMC_S_SYS_SWAY_STATUS_1B**Type:** R**Clock:** Config, Core & Slave Clocks**Reset State:** 0x11111111

Status registers indicate the state of several elements in the slaveway, including the number of transactions outstanding/pending waiting for a response, emptiness, fullness of several buffers, etc. The state information is available only as a debug indication and could be delayed by a few clock cycles from when the state is reached to being reflected in the status registers. One or more of the state registers are read live and could capture intermediate values as they change.

BIMC_S_SYS_SWAY_STATUS_1B

Bits	Name	Description
29	WIB_RD_FULL	Indicates that the write channel IDs buffer is full when seen from it's read side, which interfaces to the bus slave.
28	WIB_RD_EMPTY	Indicates that the write channel IDs buffer is empty when seen from it's read side, which interfaces to the bus slave.
25	WIB_WR_FULL	Indicates that the write channel IDs buffer is full when seen from it's write side, which interfaces to core of the interconnect.
24	WIB_WR_EMPTY	Indicates that the write channel IDs buffer is empty when seen from it's write side, which interfaces to core of the interconnect.
21	RGB0_RD_FULL	Indicates that the read gathering buffer #0 is full when seen from it's read side, which interfaces to frontend of the masterport.
20	RGB0_RD_EMPTY	Indicates that the read gathering buffer #0 is empty when seen from it's read side, which interfaces to core of the interconnect.
17	RGB0_WR_FULL	Indicates that the read gathering buffer #0 is full when seen from it's write side, which interfaces to core of the interconnect.
16	RGB0_WR_EMPTY	Indicates that the read gathering buffer #0 is empty when seen from it's write side, which interfaces to the bus slave.
13	W2AB_RD_FULL	Indicates that the write channel to address channel user signals buffer is full when seen from it's read side, which interfaces to the bus slave.
12	W2AB_RD_EMPTY	Indicates that the write channel to address channel user signals buffer is empty when seen from it's read side, which interfaces to the bus slave.
9	W2AB_WR_FULL	Indicates that the write channel to address channel user signals buffer is full when seen from it's write side, which interfaces to core of the interconnect.
8	W2AB_WR_EMPTY	Indicates that the write channel to address channel user signals buffer is empty when seen from it's write side, which interfaces to core of the interconnect.
5	A2WB_RD_FULL	Indicates that the address channel to write channel user signals buffer is full when seen from it's read side, which interfaces to the bus slave.

BIMC_S_SYS_SWAY_STATUS_1B (cont.)

Bits	Name	Description
4	A2WB_RD_EMPTY	Indicates that the address channel to write channel user signalsbuffer is empty when seen from it's read side, which interfaces to the bus slave.
1	A2WB_WR_FULL	Indicates that the address channel to write channel user signals buffer is full when seen from it's write side, which interfaces to core of the interconnect.
0	A2WB_WR_EMPTY	Indicates that the address channel to write channel user signals buffer is empty when seen from it's write side, which interfaces to core of the interconnect.

0x00458418 BIMC_S_SYS_SWAY_STATUS_1C**Type:** R**Clock:** Config, Core & Slave Clocks**Reset State:** 0x00001155

Status registers indicate the state of several elements in the slaveway, including the number of transactions outstanding/pending waiting for a response, emptiness, fullness of several buffers, etc. The state information is available only as a debug indication and could be delayed by a few clock cycles from when the state is reached to being reflected in the status registers. One or more of the state registers are read live and could capture intermediate values as they change.

BIMC_S_SYS_SWAY_STATUS_1C

Bits	Name	Description
13	RGB1_RD_FULL	Indicates that the read gathering buffer #1 is full when seen from it's read side, which interfaces to frontend of the masterport.
12	RGB1_RD_EMPTY	Indicates that the read gathering buffer #1 is empty when seen from it's read side, which interfaces to core of the interconnect.
9	RGB1_WR_FULL	Indicates that the read gathering buffer #1 is full when seen from it's write side, which interfaces to core of the interconnect.
8	RGB1_WR_EMPTY	Indicates that the read gathering buffer #1 is empty when seen from it's write side, which interfaces to the bus slave.
7	RCB1_DATA_RD_FULL	Indicates that the data part of read channel buffer #1 is full when seen from it's read side, which interfaces to core of the interconnect.
6	RCB1_DATA_RD_EMPTY	Indicates that the data part of read channel buffer #1 is empty when seen from it's read side, which interfaces to core of the interconnect.
5	RCB1_CTRL_RD_FULL	Indicates that the control part of read channel buffer #1 is full when seen from it's read side, which interfaces to core of the interconnect.

BIMC_S_SYS_SWAY_STATUS_1C (cont.)

Bits	Name	Description
4	RCB1_CTRL_RD_EMPTY	Indicates that the control part of read channel buffer #1 is empty when seen from it's write side, which interfaces to core of the interconnect.
3	RCB1_DATA_WR_FULL	Indicates that the data part of read channel buffer #1 is full when seen from it's write side, which interfaces to the bus slave.
2	RCB1_DATA_WR_EMPTY	Indicates that the data part of read channel buffer #1 is empty when seen from it's write side, which interfaces to the bus slave.
1	RCB1_CTRL_WR_FULL	Indicates that the control part of read channel buffer #1 is full when seen from it's write side, which interfaces to the bus slave.
0	RCB1_CTRL_WR_EMPTY	Indicates that the control part of read channel buffer #1 is empty when seen from it's write side, which interfaces to the bus slave.

2.21 BIMC_S_DEFAULT_SWAY

BRIC Default Slave

2.21.1 BIMC_S_DEFAULT_SWAY_BRIC_SLAVEWAY_DEFAULT

BRIC Default Slave

0x00460000 BIMC_S_DEFAULT_SWAY_COMPONENT_INFO**Type:** R**Clock:** Config Clock**Reset State:** 0x00000103

Component information registers provide discoverability information on the type of component and instance of each component. These registers are always the very first register in each component's configuration address space and allow for a software routine to walk the configuration address space in steps of 4KB to identify the location of each configuration register space and to which component each 4KB configures.

BIMC_S_DEFAULT_SWAY_COMPONENT_INFO

Bits	Name	Description
23:16	INSTANCE	Indicates the instance number of this slaway.
15:8	SUB_TYPE	Identifies this configuration space as belonging to a slaway 0x1: SLAVEWAY
7:0	TYPE	Identifies this configuration space as belonging to a component that interfaces to a bus slave. 0x3: SLAVEWAY

0x00460020 BIMC_S_DEFAULT_SWAY_CONFIGURATION_INFO_0**Type:** R**Clock:** Config Clock**Reset State:** 0x00000002

Configuration information registers provide information on the capability and configuration of each slaway instance. See register field descriptions for more details.

BIMC_S_DEFAULT_SWAY_CONFIGURATION_INFO_0

Bits	Name	Description
7:0	FUNCTIONALITY	Indicates the functionality and type of bus traffic handled by this slaway. When DEFAULT_SLAVE, identifies this slaway as a default slave which responds to all commands with decode error. When SLAWEWAY, identifies this slaway as interfacing to a bus slave. When STUB, identifies this slaway has a stub. 0x2: DEFAULT_SLAVE

0x00460030 BIMC_S_DEFAULT_SWAY_CONFIGURATION_INFO_1**Type:** R**Clock:** Config Clock**Reset State:** 0x00000000

Configuration information registers provide information on the capability and configuration of each slaway instance. See register field descriptions for more details.

BIMC_S_DEFAULT_SWAY_CONFIGURATION_INFO_1

Bits	Name	Description
31:0	MPORT_CONNECTIVITY	Indicates which masterports can route commands to this slaway instance. 1-bit per masterport. When a specific bit is set (1), indicates that the masterport corresponding to this bit can route commands to this slaway instance. When a specific bit is cleared (0), indicates that the masterport corresponding to this bit does not route commands to this slaway instance.

0x00460100 BIMC_S_DEFAULT_SWAY_INTERRUPT_STATUS**Type:** R**Clock:** Config Clock**Reset State:** 0x00000000

Interrupt status registers provide indication of which interrupts from this masterport require servicing.

BIMC_S_DEFAULT_SWAY_INTERRUPT_STATUS

Bits	Name	Description
1	RFU	Reserved for future use
0	DECODE_ERROR	When set (1), indicates that a read or write command was responded with decode error.

0x00460108 BIMC_S_DEFAULT_SWAY_INTERRUPT_CLEAR**Type:** W**Clock:** Config Clock**Reset State:** 0x00000000

Interrupt clear registers provide the mechanism to turn off interrupt sources once they have been acknowledged or serviced. A write of '1' to the bits corresponding to the interrupts indicated in interrupt status is required to turn off interrupts.

BIMC_S_DEFAULT_SWAY_INTERRUPT_CLEAR

Bits	Name	Description
1	RFU	Reserved for future use
0	DECODE_ERROR	When written with '1', clears the decode error interrupt status if it is also set. If decode error interrupt status is not set, the write is ignored. All writes of '0' are ignored.

0x0046010C BIMC_S_DEFAULT_SWAY_INTERRUPT_ENABLE**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000

Interrupt enable registers provide control over which interrupts are allowed to be propagated to an interrupt controller for servicing.

BIMC_S_DEFAULT_SWAY_INTERRUPT_ENABLE

Bits	Name	Description
1	RFU	Reserved for future use
0	DECODE_ERROR	When set (1), indicates that any decode errors interrupts are allowed to be propagated to a system wide interrupt controller. When cleared (0), any decode error interrupt is not propagated to the system wide interrupt controller.

0x00460400 BIMC_S_DEFAULT_SWAY_DEFAULT_SLAVE_STATUS_0

Type: R
Clock: Config & Core Clocks
Reset State: 0x00000000

Default slave status registers recorded the last read or write command that was responded to with decode error.

BIMC_S_DEFAULT_SWAY_DEFAULT_SLAVE_STATUS_0

Bits	Name	Description
31:0	ADDR_LOWER	Indicates ADDR[31:0] of the last read or write command that was responded to with decode error.

0x00460410 BIMC_S_DEFAULT_SWAY_DEFAULT_SLAVE_STATUS_1

Type: R
Clock: Config & Core Clocks
Reset State: 0x00000000

Default slave status registers recorded the last read or write command that was responded to with decode error.

BIMC_S_DEFAULT_SWAY_DEFAULT_SLAVE_STATUS_1

Bits	Name	Description
31	WRITE	When set (1), indicates that the last command responded to with decode error was a write. When cleared (0), indicates that the last command responded to with decode error was a read.
3:0	ADDR_UPPER	Indicates ADDR[35:32] of the last read or write command that was responded to with decode error.

0x00460420 BIMC_S_DEFAULT_SWAY_DEFAULT_SLAVE_STATUS_2

Type: R
Clock: Config & Core Clocks
Reset State: 0x00000000

Default slave status registers recorded the last read or write command that was responded to with decode error.

BIMC_S_DEFAULT_SWAY_DEFAULT_SLAVE_STATUS_2

Bits	Name	Description
31:16	TID	Indicates the transfer ID of the last command that was responded to with decode error.
15:0	MID	Indicates the master ID of the last command that was responded to with decode error.

2.22 BIMC_S_APP_ARB

BRIC Arbiter

2.22.1 BIMC_S_APP_ARB_BRIC_ARBITER

BRIC Arbiter

0x00451000 BIMC_S_APP_ARB_COMPONENT_INFO**Type:** R**Clock:** Config Clock**Reset State:** 0x00000203

Component information registers provide discoverability information on the type of component and instance of each component. These registers are always the very first register in each component's configuration address space and allow for a software routine to walk the configuration address space in steps of 4KB to identify the location of each configuration register space and to which component each 4KB configures.

BIMC_S_APP_ARB_COMPONENT_INFO

Bits	Name	Description
23:16	INSTANCE	Indicates the instance number of this arbiter.
15:8	SUB_TYPE	Identifies this configuration space as belonging to an arbiter. 0x2: ARBITER
7:0	TYPE	Identifies this configuration space as belonging to a component that interfaces to a bus slave. 0x3: SLAVEWAY

0x00451020 BIMC_S_APP_ARB_CONFIGURATION_INFO_0**Type:** R**Clock:** Config Clock**Reset State:** 0x00000000

Configuration information registers provide information on the capability and configuration of each arbiter instance. See register field descriptions for more details.

BIMC_S_APP_ARB_CONFIGURATION_INFO_0

Bits	Name	Description
31	ARB2SW_PIPELINE_EN	When set (1), indicates that a pipeline stage is present between the arbiter and the slaveway.
15:8	WGB_DEPTH	Indicates the depth of write grants buffer, which allows for multiple write commands to be granted ahead of the corresponding write data has flowed through to the slaveway. The depth provides an indication on the range of values of WR_GRANTS_AHEAD register field that will have an effect on the performance of the arbiter.
7:0	FUNCTIONALITY	Indicates that this arbiter instance is capable of using plain or fair round-robin for arbitration of commands. 0x0: ROUND_ROBIN 0x1: FAIR_ROUND_ROBIN 0x2: PRIORITY_RR 0x3: PRIORITY_FRR 0x4: TIERED_RR

0x00451030 BIMC_S_APP_ARB_CONFIGURATION_INFO_1**Type:** R**Clock:** Config Clock**Reset State:** 0x00000000

Configuration information registers provide information on the capability and configuration of each arbiter instance. See register field descriptions for more details.

BIMC_S_APP_ARB_CONFIGURATION_INFO_1

Bits	Name	Description
31:0	MPORT_CONNECTIVITY	Indicates which masterports can route commands to this arbiter instance. 1-bit per masterport. When a specific bit is set (1), indicates that the masterport corresponding to this bit can route commands to this arbiter instance. When a specific bit is cleared (0), indicates that the masterport corresponding to this bit does not route commands to this arbiter instance.

0x00451200 BIMC_S_APP_ARB_CLOCK_CTRL**Type:** RW**Clock:** Config & Core Clocks**Reset State:** 0x00000001**BIMC_S_APP_ARB_CLOCK_CTRL**

Bits	Name	Description
0	CORE_CLOCK_GATING_EN	When set (1), enables activity based clock gating of all registers in this arbiter instance. Activity based clock gating uses architectural clock gates to reduce dynamic power consumption by gating clocks during periods of in-activity. When cleared (0), disables all activity based clock gating by forcing all architectural clock gates to not gate clocks even during periods of in-activity.

0x00451210 BIMC_S_APP_ARB_MODE**Type:** RW**Clock:** Config & Core Clocks**Reset State:** 0x00000000**BIMC_S_APP_ARB_MODE**

Bits	Name	Description
31:28	WR_GRANTS_AHEAD	Number of write commands that can be granted ahead of the corresponding write data bursts. For each write command granted ahead of the corresponding write data, the source route is stored in 'Write Grants Buffer' for routing write data bursts. Range of values is supported is dependent on the depth of 'Write Grants Buffers', as indicated in WGB_DEPTH in CONFIGURATION_INFO registers. Any setting larger than WGB_DEPTH is treated similar to a setting equal to WGB_DEPTH.

2.23 BIMC_S_DDR0_ARB

BRIC Arbiter

2.23.1 BIMC_S_DDR0_ARB_BRIC_ARBITER

BRIC Arbiter

0x00449000 BIMC_S_DDR0_ARB_COMPONENT_INFO**Type:** R**Clock:** Config Clock**Reset State:** 0x00000203

Component information registers provide discoverability information on the type of component and instance of each component. These registers are always the very first register in each component's configuration address space and allow for a software routine to walk the configuration address space in steps of 4KB to identify the location of each configuration register space and to which component each 4KB configures.

BIMC_S_DDR0_ARB_COMPONENT_INFO

Bits	Name	Description
23:16	INSTANCE	Indicates the instance number of this arbiter.
15:8	SUB_TYPE	Identifies this configuration space as belonging to an arbiter. 0x2: ARBITER
7:0	TYPE	Identifies this configuration space as belonging to a component that interfaces to a bus slave. 0x3: SLAVEWAY

0x00449020 BIMC_S_DDR0_ARB_CONFIGURATION_INFO_0**Type:** R**Clock:** Config Clock**Reset State:** 0x00000000

Configuration information registers provide information on the capability and configuration of each arbiter instance. See register field descriptions for more details.

BIMC_S_DDR0_ARB_CONFIGURATION_INFO_0

Bits	Name	Description
31	ARB2SW_PIPELINE_EN	When set (1), indicates that a pipeline stage is present between the arbiter and the slaveway.
15:8	WGB_DEPTH	Indicates the depth of write grants buffer, which allows for multiple write commands to be granted ahead of the corresponding write data has flowed through to the slaveway. The depth provides an indication on the range of values of WR_GRANTS_AHEAD register field that will have an effect on the performance of the arbiter.

BIMC_S_DDR0_ARB_CONFIGURATION_INFO_0 (cont.)

Bits	Name	Description
7:0	FUNCTIONALITY	Indicates that this arbiter instance is capable of using plain or fair round-robin for arbitration of commands. 0x0: ROUND_ROBIN 0x1: FAIR_ROUND_ROBIN 0x2: PRIORITY_RR 0x3: PRIORITY_FRR 0x4: TIERED_RR

0x00449030 BIMC_S_DDR0_ARB_CONFIGURATION_INFO_1**Type:** R**Clock:** Config Clock**Reset State:** 0x00000000

Configuration information registers provide information on the capability and configuration of each arbiter instance. See register field descriptions for more details.

BIMC_S_DDR0_ARB_CONFIGURATION_INFO_1

Bits	Name	Description
31:0	MPORT_CONNECTIVITY	Indicates which masterports can route commands to this arbiter instance. 1-bit per masterport. When a specific bit is set (1), indicates that the masterport corresponding to this bit can route commands to this arbiter instance. When a specific bit is cleared (0), indicates that the masterport corresponding to this bit does not route commands to this arbiter instance.

0x00449200 BIMC_S_DDR0_ARB_CLOCK_CTRL**Type:** RW**Clock:** Config & Core Clocks**Reset State:** 0x00000001**BIMC_S_DDR0_ARB_CLOCK_CTRL**

Bits	Name	Description
0	CORE_CLOCK_GATING_EN	When set (1), enables activity based clock gating of all registers in this arbiter instance. Activity based clock gating uses architectural clock gates to reduce dynamic power consumption by gating clocks during periods of in-activity. When cleared (0), disables all activity based clock gating by forcing all architectural clock gates to not gate clocks even during periods of in-activity.

0x00449210 BIMC_S_DDR0_ARB_MODE**Type:** RW**Clock:** Config & Core Clocks**Reset State:** 0x00000000**BIMC_S_DDR0_ARB_MODE**

Bits	Name	Description
31:28	WR_GRANTS_AHEAD	Number of write commands that can be granted ahead of the corresponding write data bursts. For each write command granted ahead of the corresponding write data, the source route is stored in 'Write Grants Buffer' for routing write data bursts. Range of values is supported is dependent on the depth of 'Write Grants Buffers', as indicated in WGB_DEPTH in CONFIGURATION_INFO registers. Any setting larger than WGB_DEPTH is treated similar to a setting equal to WGB_DEPTH.

2.24 BIMC_S_SYS_ARB

BRIC Arbiter

2.24.1 BIMC_S_SYS_ARB_BRIC_ARBITER

BRIC Arbiter

0x00459000 BIMC_S_SYS_ARB_COMPONENT_INFO**Type:** R**Clock:** Config Clock**Reset State:** 0x00000203

Component information registers provide discoverability information on the type of component and instance of each component. These registers are always the very first register in each component's configuration address space and allow for a software routine to walk the configuration address space in steps of 4KB to identify the location of each configuration register space and to which component each 4KB configures.

BIMC_S_SYS_ARB_COMPONENT_INFO

Bits	Name	Description
23:16	INSTANCE	Indicates the instance number of this arbiter.
15:8	SUB_TYPE	Identifies this configuration space as belonging to an arbiter. 0x2: ARBITER

BIMC_S_SYS_ARB_COMPONENT_INFO (cont.)

Bits	Name	Description
7:0	TYPE	Identifies this configuration space as belonging to a component that interfaces to a bus slave. 0x3: SLAVEWAY

0x00459020 BIMC_S_SYS_ARB_CONFIGURATION_INFO_0**Type:** R**Clock:** Config Clock**Reset State:** 0x00000000

Configuration information registers provide information on the capability and configuration of each arbiter instance. See register field descriptions for more details.

BIMC_S_SYS_ARB_CONFIGURATION_INFO_0

Bits	Name	Description
31	ARB2SW_PIPELINE_EN	When set (1), indicates that a pipeline stage is present between the arbiter and the slaveway.
15:8	WGB_DEPTH	Indicates the depth of write grants buffer, which allows for multiple write commands to be granted ahead of the corresponding write data has flowed through to the slaveway. The depth provides an indication on the range of values of WR_GRANTS_AHEAD register field that will have an effect on the performance of the arbiter.
7:0	FUNCTIONALITY	Indicates that this arbiter instance is capable of using plain or fair round-robin for arbitration of commands. 0x0: ROUND_ROBIN 0x1: FAIR_ROUND_ROBIN 0x2: PRIORITY_RR 0x3: PRIORITY_FRR 0x4: TIERED_RR

0x00459030 BIMC_S_SYS_ARB_CONFIGURATION_INFO_1**Type:** R**Clock:** Config Clock**Reset State:** 0x00000000

Configuration information registers provide information on the capability and configuration of each arbiter instance. See register field descriptions for more details.

BIMC_S_SYS_ARB_CONFIGURATION_INFO_1

Bits	Name	Description
31:0	MPORT_CONNECTIVITY	Indicates which masterports can route commands to this arbiter instance. 1-bit per masterport. When a specific bit is set (1), indicates that the masterport corresponding to this bit can route commands to this arbiter instance. When a specific bit is cleared (0), indicates that the masterport corresponding to this bit does not route commands to this arbiter instance.

0x00459200 BIMC_S_SYS_ARB_CLOCK_CTRL**Type:** RW**Clock:** Config & Core Clocks**Reset State:** 0x00000001**BIMC_S_SYS_ARB_CLOCK_CTRL**

Bits	Name	Description
0	CORE_CLOCK_GATING_EN	When set (1), enables activity based clock gating of all registers in this arbiter instance. Activity based clock gating uses architectural clock gates to reduce dynamic power consumption by gating clocks during periods of in-activity. When cleared (0), disables all activity based clock gating by forcing all architectural clock gates to not gate clocks even during periods of in-activity.

0x00459210 BIMC_S_SYS_ARB_MODE**Type:** RW**Clock:** Config & Core Clocks**Reset State:** 0x00000000**BIMC_S_SYS_ARB_MODE**

Bits	Name	Description
31:28	WR_GRANTS_AHEAD	Number of write commands that can be granted ahead of the corresponding write data bursts. For each write command granted ahead of the corresponding write data, the source route is stored in 'Write Grants Buffer' for routing write data bursts. Range of values is supported is dependent on the depth of 'Write Grants Buffers', as indicated in WGB_DEPTH in CONFIGURATION_INFO registers. Any setting larger than WGB_DEPTH is treated similar to a setting equal to WGB_DEPTH.

2.25 BIMC_S_DEFAULT_ARB

BRIC Arbiter

2.25.1 BIMC_S_DEFAULT_ARB_BRIC_ARBITER

BRIC Arbiter

0x00461000 BIMC_S_DEFAULT_ARB_COMPONENT_INFO

Type: R

Clock: Config Clock

Reset State: 0x00000203

Component information registers provide discoverability information on the type of component and instance of each component. These registers are always the very first register in each component's configuration address space and allow for a software routine to walk the configuration address space in steps of 4KB to identify the location of each configuration register space and to which component each 4KB configures.

BIMC_S_DEFAULT_ARB_COMPONENT_INFO

Bits	Name	Description
23:16	INSTANCE	Indicates the instance number of this arbiter.
15:8	SUB_TYPE	Identifies this configuration space as belonging to an arbiter. 0x2: ARBITER
7:0	TYPE	Identifies this configuration space as belonging to a component that interfaces to a bus slave. 0x3: SLAVEWAY

0x00461020 BIMC_S_DEFAULT_ARB_CONFIGURATION_INFO_0

Type: R

Clock: Config Clock

Reset State: 0x00000000

Configuration information registers provide information on the capability and configuration of each arbiter instance. See register field descriptions for more details.

BIMC_S_DEFAULT_ARB_CONFIGURATION_INFO_0

Bits	Name	Description
31	ARB2SW_PIPELINE_EN	When set (1), indicates that a pipeline stage is present between the arbiter and the slaveway.

BIMC_S_DEFAULT_ARB_CONFIGURATION_INFO_0 (cont.)

Bits	Name	Description
15:8	WGB_DEPTH	Indicates the depth of write grants buffer, which allows for multiple write commands to be granted ahead of the corresponding write data has flowed through to the slaved way. The depth provides an indication on the range of values of WR_GRANTS_AHEAD register field that will have an effect on the performance of the arbiter.
7:0	FUNCTIONALITY	Indicates that this arbiter instance is capable of using plain or fair round-robin for arbitration of commands. 0x0: ROUND_ROBIN 0x1: FAIR_ROUND_ROBIN 0x2: PRIORITY_RR 0x3: PRIORITY_FRR 0x4: TIERED_RR

0x00461030 BIMC_S_DEFAULT_ARB_CONFIGURATION_INFO_1**Type:** R**Clock:** Config Clock**Reset State:** 0x00000000

Configuration information registers provide information on the capability and configuration of each arbiter instance. See register field descriptions for more details.

BIMC_S_DEFAULT_ARB_CONFIGURATION_INFO_1

Bits	Name	Description
31:0	MPORT_CONNECTIVITY	Indicates which masterports can route commands to this arbiter instance. 1-bit per masterport. When a specific bit is set (1), indicates that the masterport corresponding to this bit can route commands to this arbiter instance. When a specific bit is cleared (0), indicates that the masterport corresponding to this bit does not route commands to this arbiter instance.

0x00461200 BIMC_S_DEFAULT_ARB_CLOCK_CTRL**Type:** RW**Clock:** Config & Core Clocks**Reset State:** 0x00000001

BIMC_S_DEFAULT_ARB_CLOCK_CTRL

Bits	Name	Description
0	CORE_CLOCK_GATING_EN	When set (1), enables activity based clock gating of all registers in this arbiter instance. Activity based clock gating uses architectural clock gates to reduce dynamic power consumption by gating clocks during periods of in-activity. When cleared (0), disables all activity based clock gating by forcing all architectural clock gates to not gate clocks even during periods of in-activity.

0x00461210 BIMC_S_DEFAULT_ARB_MODE**Type:** RW**Clock:** Config & Core Clocks**Reset State:** 0x00000000**BIMC_S_DEFAULT_ARB_MODE**

Bits	Name	Description
31:28	WR_GRANTS_AHEAD	Number of write commands that can be granted ahead of the corresponding write data bursts. For each write command granted ahead of the corresponding write data, the source route is stored in 'Write Grants Buffer' for routing write data bursts. Range of values is supported is dependent on the depth of 'Write Grants Buffers', as indicated in WGB_DEPTH in CONFIGURATION_INFO registers. Any setting larger than WGB_DEPTH is treated similar to a setting equal to WGB_DEPTH.

2.26 BIMC_S_DDR0_SCMO

SCMO configuration

2.26.1 BIMC_S_DDR0_SCMO_SCMO_CFGREG

SCMO configuration

0x00448020 BIMC_S_DDR0_SCMO_CONFIGURATION_INFO_0**Type:** R**Clock:** Config Clock**Reset State:** 0x00000000

BIMC_S_DDR0_SCMO_CONFIGURATION_INFO_0

Bits	Name	Description
31:16	DATA_WIDTH	Data bus width
15:8	TID_WIDTH	
7:0	MID_WIDTH	

0x00448030 BIMC_S_DDR0_SCMO_CONFIGURATION_INFO_1**Type:** R**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_S_DDR0_SCMO_CONFIGURATION_INFO_1**

Bits	Name	Description
31:0	MPORT_CONNECTIVITY	Master port connectivity bit-vector

0x00448040 BIMC_S_DDR0_SCMO_CONFIGURATION_INFO_2**Type:** R**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_S_DDR0_SCMO_CONFIGURATION_INFO_2**

Bits	Name	Description
23:16	NUM_GLOBAL_MONS	Number of Global monitor instances
7:0	VMID_WIDTH	

0x00448050 BIMC_S_DDR0_SCMO_CONFIGURATION_INFO_3**Type:** R**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_S_DDR0_SCMO_CONFIGURATION_INFO_3**

Bits	Name	Description
31:24	RCH0_CTRL_DEPTH	
23:16	RCH0_DEPTH	
15:8	BCH_DEPTH	

BIMC_S_DDR0_SCMO_CONFIGURATION_INFO_3 (cont.)

Bits	Name	Description
7:0	WCH_DEPTH	

0x00448060 BIMC_S_DDR0_SCMO_CONFIGURATION_INFO_4**Type:** R**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_S_DDR0_SCMO_CONFIGURATION_INFO_4**

Bits	Name	Description
15:8	RCH1_CTRL_DEPTH	
7:0	RCH1_DEPTH	

0x00448070 BIMC_S_DDR0_SCMO_CONFIGURATION_INFO_5**Type:** R**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_S_DDR0_SCMO_CONFIGURATION_INFO_5**

Bits	Name	Description
15:8	DPE_CQ_DEPTH	
7:0	DDR_BUS_WIDTH	

0x00448080 BIMC_S_DDR0_SCMO_CONFIGURATION_INFO_6**Type:** R**Clock:** Config Clock**Reset State:** 0x00000000**BIMC_S_DDR0_SCMO_CONFIGURATION_INFO_6**

Bits	Name	Description
12	WBUFC_PIPE	Write buffer coalescing logic pipe-line stage
8	RDOPT_PIPE	Read optimizer pipe-line stage
4	ACHAN_INTF_PIPE	Enabled when arbiter output is combinatorial
0	ADDR_DECODE_HT	

0x00448100 BIMC_S_DDR0_SCMO_INTERRUPT_STATUS**Type:** R**Clock:** Config Clock**Reset State:** 0x00000000

Interrupt status register indicating AXI bus error occurred

BIMC_S_DDR0_SCMO_INTERRUPT_STATUS

Bits	Name	Description
0	ERR_OCCURRED	Error Status flag to indicate that an error has occurred.

0x00448108 BIMC_S_DDR0_SCMO_INTERRUPT_CLEAR**Type:** W**Clock:** Config Clock**Reset State:** 0x00000000

Interrupt clear register

BIMC_S_DDR0_SCMO_INTERRUPT_CLEAR

Bits	Name	Description
0	IRQ_CLR	Writing 1 to this register will clear the interrupt signal.

0x0044810C BIMC_S_DDR0_SCMO_INTERRUPT_ENABLE**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000

Interrupt enable register

BIMC_S_DDR0_SCMO_INTERRUPT_ENABLE

Bits	Name	Description
0	IRQ_EN	When set, interrupt is generated to processor

0x00448120 BIMC_S_DDR0_SCMO_ESYN_ADDR

Type: R
Clock: Config Clock
Reset State: 0x00000000

Error Address Register.

This register contains the address of the request that caused the AXI bus error. The error address is locked with the first error address.

BIMC_S_DDR0_SCMO_ESYN_ADDR

Bits	Name	Description
31:0	ERR_ADDR_L	The address[31:0] of the request that caused the bus error

0x00448128 BIMC_S_DDR0_SCMO_ESYN_APACKET_0

Type: R
Clock: Config Clock
Reset State: 0x00000000

Error Command Attributes Register 0.

This register provides request ID that caused the AXI bus error. It is sticky register and holds information related to the first error.

BIMC_S_DDR0_SCMO_ESYN_APACKET_0

Bits	Name	Description
31:16	ERR_ATID	Transfer ID of the request that caused the bus error.
15:0	ERR_AMID	Master ID of the request that caused the bus error.

0x0044812C BIMC_S_DDR0_SCMO_ESYN_APACKET_1

Type: R
Clock: Config Clock
Reset State: 0x00000000

Error Command Attributes Register 1.

This register provides various information about the request that caused the AXI bus error. It is sticky register and holds information related to the first error.

BIMC_S_DDR0_SCMO_ESYN_APACKET_1

Bits	Name	Description
24	ERR_CODE	Indicates address decode error. Writes to this field are ignored. 0x0: NO_ERROR 0x1: ADDRESS_DECODE_ERROR
19:16	ERR_ALEN	The burst length of the request that caused bus error
15:13	ERR_ASIZE	The ASIZE of the request that caused the bus error 0x0: BYTE 0x1: HALF_WORD 0x2: WORD 0x3: DOUBLE_WORD
12	ERR_ABURST	The ABURST of the request that caused the bus error
8	ERR_AEXCLUSIVE	The AEXCLUSIVE of the request that caused the bus error
4	ERR_APROTNS	The APROTNS of the request that caused the bus error.
2	ERR_AOOORD	The AOOORD of the request that caused the bus error
1	ERR_AOOOWR	The AOOOWR of the request that caused the bus error
0	ERR_AWRITE	The awrite of the request that caused bus error

0x00448130 BIMC_S_DDR0_SCMO_ESYN_APACKET_2**Type:** R**Clock:** Config Clock**Reset State:** 0x00000000

Error Command Attributes Register 2.

This register provides request ID that caused the AXI bus error. It is sticky register and holds information related to the first error.

BIMC_S_DDR0_SCMO_ESYN_APACKET_2

Bits	Name	Description
4:0	ERR_AVMID	AVMID of the request that caused the bus error.

0x00448200 BIMC_S_DDR0_SCMO_CLOCK_CTRL**Type:** RW**Clock:** Config Clock**Reset State:** 0x00111111

This register contains Clock-gating enable signals.

BIMC_S_DDR0_SCMO_CLOCK_CTRL

Bits	Name	Description
31:21	RFU_31_21	Reserved for future use
20	CFG_WR_CORE_CG_EN	Dynamic clock gating enable for core clock to synchronize config settings whenever Config registers get updated.
16	PEN_CMD_CG_EN	Dynamic clock gating enable for pending command flow.
12	RCH_CG_EN	Dynamic clock gating enable for read data and response flow.
8	FLUSH_CG_EN	Dynamic clock gating enable for write buffer flush flow.
4	WCH_CG_EN	Dynamic clock gating enable for Write data and response flow.
0	ACH_CG_EN	Dynamic clock gating enable for Command flow.

0x00448400 BIMC_S_DDR0_SCMO_SLV_INTERLEAVE_CFG**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000

Interleaved Slave Configuration Register.

BIMC_S_DDR0_SCMO_SLV_INTERLEAVE_CFG

Bits	Name	Description
9:8	RANK_INTERLEAVE	Defines whether a channel is rank interleaved or not; Rank interleaving can be selected between 1KB, 2KB, 4KB boundary. 0x0: DISABLE 0x1: RANK_INTLV_1K 0x2: RANK_INTLV_2K 0x3: RANK_INTLV_4K
7:5	RFU_7_5	Reserved for future use
4	INTERLEAVE_CS1	Defines whether Rank1 is channel interleaved or not. It defines the way that bus system slave reconstructs the system address for MPU, error capture. 0x0: DISABLED 0x1: INT_1K
3:1	RFU_3_1	Reserved for future use
0	INTERLEAVE_CS0	Defines whether Rank0 is channel interleaved or not. It defines the way that bus system slave reconstructs the system address for MPU, error capture. 0x0: DISABLED 0x1: INT_1K

**0x00448410+ BIMC_S_DDR0_SCMO_ADDR_BASE_CS_n, n=[0..1]
0x4*n****Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000

Base Address Register for Rank n

This register defines the base address for the memory that associates to chip select n.

BIMC_S_DDR0_SCMO_ADDR_BASE_CS_n

Bits	Name	Description
15:10	RFU_15_10	Reserved for future use
9:2	ADDR_BASE	Defines the Base Address decode, in conjunction with the ADDR_MAP_CS _n , n = [0,1], to determine the address space of the targeted chip select. The Base Address is compared to the upper AXI address. If the portions of the two addresses match, then the particular CS is accessed.
1:0	RFU_1_0	Reserved for future use

**0x00448420+ BIMC_S_DDR0_SCMO_ADDR_MAP_CS_n, n=[0..1]
0x4*n****Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000

Address Mapping Register for Rank n.

This register defines the way to map AXI system address into rank/bank/row/column address to memory devices.

Once the number of bank, row size and column size are defined, the memory size of the associated rank is defined.

BIMC_S_DDR0_SCMO_ADDR_MAP_CS_n

Bits	Name	Description
15	RANK_EN	Rank has memory devices attached 0x0: RANK_NOT_PRESENT 0x1: RANK_IS_PRESENT
14:13	RFU_14_13	Reserved for future use

BIMC_S_DDR0_SCMO_ADDR_MAP_CS_n (cont.)

Bits	Name	Description
12	ADDR_MODE	Addressing Mode Configuration. Selects the Addressing Mode of the Rank and determines the mapping of the System Address to the Memory Address. Linear addressing mode is configuring as 'Rank,Bank,Row,Column'. Bank interleaved addressing mode is configuring as 'Rank,Row,Bank,Column' mode. RK = Rank, R = Row, B = Bank, C=Column. For high tier products linear addressing mode of RKBRC is not supported 0x0: RKRBC 0x1: RKBRC
11:9	RFU_11_9	Reserved for future use
8	BANK_SIZE	Specifies the number of banks in the memory device. NOTE: For high tier products only 8 banks are supported. 0x0: BANKS_4 0x1: BANKS_8
7:6	RFU_7_6	Reserved for future use
5:4	ROW_SIZE	Specifies the number of bits used for the row address of the external memory 0x0: ROWS_13 0x1: ROWS_14 0x2: ROWS_15 0x3: ROWS_16
3:2	RFU_3_2	Reserved for future use
1:0	COL_SIZE	Specifies the number of bits used for the column address of the external memory 0x0: COLS_8 0x1: COLS_9 0x2: COLS_10 0x3: COLS_11

0x00448430+ BIMC_S_DDR0_SCMO_ADDR_MASK_CS_n, n=[0..1]
0x4*n

Type: RW

Clock: Config Clock

Reset State: 0x00000000

Address Size Mask Register for Rank n

This register defines the memory capacity of memory that associates to chip select n.

BIMC_S_DDR0_SCMO_ADDR_MASK_CS_n

Bits	Name	Description
15:10	RFU_15_10	Reserved for future use
9:2	ADDR_MASK	Define the address size mask bits. Note: For high tier products 64MB and 128MB are reserved. All other values not specified are reserved. 0x0: RESERVED 0x80: DENSITY_8GB 0xC0: DENSITY_4GB 0xE0: DENSITY_2GB 0xF0: DENSITY_1GB 0xF8: DENSITY_512MB 0xFC: DENSITY_256MB 0xFE: DENSITY_128MB 0xFF: DENSITY_64MB
1:0	RFU_1_0	Reserved for future use

0x00448450 BIMC_S_DDR0_SCMO_SLV_STATUS**Type:** R**Clock:** Config Clock**Reset State:** 0x00000031

SCMO Status Register.

BIMC_S_DDR0_SCMO_SLV_STATUS

Bits	Name	Description
15:8	GLOBAL_MONS_IN_USE	Global monitors that have been reserved for use. This is defined in one-hot encoded value.
5:4	RANK_IDLE	This indicates that there are no read and write commands existing to the corresponding rank. But there could be read and write responses pending. Bit 0 corresponds to Rank 0 and bit 1 corresponds to Rank 1
0	SLAVE_IDLE	This indicates that there are no commands existing and all read and write responses have been sent.

0x00448460 BIMC_S_DDR0_SCMO_GLOBAL_MON_CFG**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000010

SCMO GLobal Monitor Configuration Register.

BIMC_S_DDR0_SCMO_GLOBAL_MON_CFG

Bits	Name	Description
4	EXCMD_ADDR_GRAN	Defines the exclusive access's address granularity to either 128-bit or 64-bit 0x1: GRANULARITY_128_BIT 0x0: GRANULARITY_64_BIT
3:2	RFU_3_2	Reserved for future use
1	CLR_EXMON_EN	Enable/disable request generation whenever any exclusive monitor gets cleared 0x0: DISABLE 0x1: ENABLE
0	EXRD_RESP	Defines exclusive read response to be sent when there are no global monitors available. When set to 1, EXRD response shall be EXOKAY irrespective of the global monitor availability. When set to 0, EXRD response shall be OKAY if there are no global monitors available. 0x0: EXOKAY_OR_OKAY 0x1: EXOKAY_ONLY

0x00448500 BIMC_S_DDR0_SCMO_CMD_BUF_CFG

Type: RW

Clock: Config Clock

Reset State: 0x00000001

SCMO Command buffers Configuration Register.

BIMC_S_DDR0_SCMO_CMD_BUF_CFG

Bits	Name	Description
11:10	RFU_11_10	Reserved for future use
9:8	CMD_ORDERING	Defines command reordering strategy 0x0: REORDERING 0x1: IN_ORDER_PER_MASTER 0x2: IN_ORDER_GLOBALLY 0x3: RESERVED
4	HP_CMD_AREQPRIO_MAP	Defines which commands are considered as high priority based on AXI Areqpriority. These commands are allocated to command buffers reserved for high priority requests. 0x0: PRIORITY3 0x1: PRIORITY3_AND_2

BIMC_S_DDR0_SCMO_CMD_BUF_CFG (cont.)

Bits	Name	Description
3	RFU_3	Reserved for future use
2:0	HP_CMD_Q_DEPTH	Defines number of command buffer entries reserved for high priority requests. NOTE: Configuration values 3,4,5 are mainly used for profiling/debug purposes 0x0: BUFFERS_0 0x1: BUFFERS_2 0x2: BUFFERS_3 0x3: BUFFERS_4 0x4: BUFFERS_5 0x5: BUFFERS_6

0x00448520 BIMC_S_DDR0_SCMO_CMD_BUF_STATUS**Type:** R**Clock:** Config Clock**Reset State:** 0x00000000

Command buffers status register.

BIMC_S_DDR0_SCMO_CMD_BUF_STATUS

Bits	Name	Description
7:4	HP_CMD_BUF_ENTRIES_IN_USE	The number of HP command buffer entries that have been reserved for use.
3:0	LP_CMD_BUF_ENTRIES_IN_USE	The number of LP command buffer entries that have been reserved for use.

0x00448540 BIMC_S_DDR0_SCMO_RCH_SELECT**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000

AXI read channel to Master port mapping configuration register

BIMC_S_DDR0_SCMO_RCH_SELECT

Bits	Name	Description
31:16	DYNAMIC_RCH_SELECT	Enables dynamic read channel selection based on fifo occupancy per each master port. Each bit index corresponds to port ID 0x0: DISABLE 0x1: ENABLE
15:0	RCH_PORTS	Defines each master port assignment to one of the dual read Channels. Each bit index corresponds to port ID 0x0: RCH0 0x1: RCH1

0x00448544 BIMC_S_DDR0_SCMO_RCH_BKPR_CFG**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000

Read Channel data fifos backpressure threshold configuration register.

NOTE: RCH0/1_FIFO_BKPR_HI_TH = RCH0/1_DEPTH - Max read transfer burst length from all masters to SCMO.

Example: RCH0/1_FIFO_BKPR_HI_TH = 32 - 16 = 16.

BIMC_S_DDR0_SCMO_RCH_BKPR_CFG

Bits	Name	Description
31:24	RCH1_FIFO_BKPR_HI_TH	Back pressure assertion threshold for the read channel1 data FIFO. Back pressure is asserted if the data FIFO occupancy is greater than or equal to this value. When back pressure is asserted, no more read commands can be executed
23:16	RCH1_FIFO_BKPR_LO_TH	Back pressure de-assertion threshold for the read channel1 data FIFO. Back pressure is de-asserted if the data FIFO occupancy is lower than or equal to this value. When back pressure is de-asserted, read commands are allowed to execute. NOTE: It must be less than or equal to high threshold. Recommended setting: Same value as RCH1_FIFO_BKPR_HI_TH
15:8	RCH0_FIFO_BKPR_HI_TH	Back pressure assertion threshold for the read channel0 data FIFO. Back pressure is asserted if the data FIFO occupancy is greater than or equal to this value. When back pressure is asserted, no more read commands can be executed

BIMC_S_DDR0_SCMO_RCH_BKPR_CFG (cont.)

Bits	Name	Description
7:0	RCH0_FIFO_BKPR_LO_TH	Back pressure de-assertion threshold for the read channel0 data FIFO. Back pressure is de-asserted if the data FIFO occupancy is lower than or equal to this value. When back pressure is de-asserted, read commands are allowed to execute. NOTE: It must be less than or equal to high threshold. Recommended setting: Same value as RCH0_FIFO_BKPR_HI_TH

0x00448560 BIMC_S_DDR0_SCMO_RCH_STATUS**Type:** R**Clock:** Config Clock**Reset State:** 0x00011111

Read Channel status register.

BIMC_S_DDR0_SCMO_RCH_STATUS

Bits	Name	Description
17	PRQ_FIFO_FULL	Full status of the pending read command FIFO.
16	PRQ_FIFO_EMPTY	Empty status of the pending read command FIFO.
13	RCH1_QUAL_FIFO_FULL	Full status of the CH1 read resp FIFO
12	RCH1_QUAL_FIFO_EMPTY	Empty status of the CH1 read resp FIFO.
9	RCH1_DATA_FIFO_FULL	Full status of the CH1 read data FIFO
8	RCH1_DATA_FIFO_EMPTY	Empty status of the CH1 read data FIFO.
5	RCH0_QUAL_FIFO_FULL	Full status of the CH0 read resp FIFO
4	RCH0_QUAL_FIFO_EMPTY	Empty status of the CH0 read resp FIFO.
1	RCH0_DATA_FIFO_FULL	Full status of the CH0 read data FIFO
0	RCH0_DATA_FIFO_EMPTY	Empty status of the CH0 read data FIFO.

0x00448580 BIMC_S_DDR0_SCMO_WCH_BUF_CFG**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000

Write Buffer Configuration Register

BIMC_S_DDR0_SCMO_WCH_BUF_CFG

Bits	Name	Description
7:5	RFU_7_5	Reserved for future use
4	WRITE_BLOCK_READ	When set, no read command is allowed to execute while there is any write data in the buffer. NOTE: When WRITE_BLOCK_READ = 1, CMD_ORDERING must be 2. This must be HIGH mainly for profiling/ debug purposes only.
3:1	RFU_3_1	Reserved for future use
0	COALESCE_EN	Enable write coalescing where in data from separate AXI bursts will be coalesced. This must be LOW mainly for profiling/ debug purposes only.

0x004485A0 BIMC_S_DDR0_SCMO_WCH_STATUS**Type:** R**Clock:** Config Clock**Reset State:** 0x00000111

Write data, response and Coalescing Write Buffer Status Registers

BIMC_S_DDR0_SCMO_WCH_STATUS

Bits	Name	Description
9	BRESP_FIFO_FULL	Full status of the Write response FIFO
8	BRESP_FIFO_EMPTY	Empty status of the Write response FIFO.
5	WDATA_FIFO_FULL	Full status of the Write data FIFO
4	WDATA_FIFO_EMPTY	Empty status of the Write data FIFO.
1	WBUF_FULL	Set when all of the Write Coalescing Buffers contains valid data.
0	WBUF_EMPTY	Set when none of the Write Coalescing Buffers contains any valid data.

0x004485C0 BIMC_S_DDR0_SCMO_FLUSH_CFG**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000

Write Buffer Flush Configuration Register.

BIMC_S_DDR0_SCMO_FLUSH_CFG

Bits	Name	Description
31	BL16_ALIGN_FLUSH_EN	For LPDDR4 only. When 0x1, Enables the Burst Length BL16 aligned FLUSH.
30:29	RFU_30_29	Reserved for future use
28	FLUSH_IN_ORDER	Write Coalescing Buffer lines are flushed in the order in which they were allocated. NOTE: When FLUSH_IN_ORDER = 1, COALESCE_EN must be 0. This must be HIGH mainly for profiling/debug purposes only
27:26	RFU_27_26	Reserved for future use
25:16	FLUSH_IDLE_DELAY	After the controller is idle for FLUSH_IDLE_DELAY AXI cycles, the Write Buffer will be flushed. Recommended setting: 0x7F. Min setting: 0x4
15:12	RFU_15_12	Reserved for future use
11:8	FLUSH_UPPER_LIMIT	The flush high water mark. Flushing will be given higher priority when the number of valid lines in the Write Coalescing Buffer is greater than FLUSH_UPPER_LIMIT. Recommended setting: 0xD
7:4	RFU_7_4	Reserved for future use
3:0	FLUSH_LOWER_LIMIT	The flush low water mark. Buffers are considered for flush when the number of valid lines is greater than FLUSH_LOWER_LIMIT. Once flushing has begun, lines may get flushed until the number of valid lines is less than or equal to FLUSH_LOWER_LIMIT. Recommended setting: 0x8

0x004485C4 BIMC_S_DDR0_SCMO_FLUSH_CMD**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000

SW FLush All Buffer Command Register

BIMC_S_DDR0_SCMO_FLUSH_CMD

Bits	Name	Description
3:2	RFU_3_2	Reserved for future use

BIMC_S_DDR0_SCMO_FLUSH_CMD (cont.)

Bits	Name	Description
1:0	FLUSH_ALL_BUF	SW: Set, HW: Clear SW trigger to flush the Write Coalescing Buffers of selected rank when corresponding bit set. Hardware will clear this field when flush of all the buffers to corresponding rank completed. SW must flush all buffers before entering sleep/ power collapse.

0x00448700 BIMC_S_DDR0_SCMO_CMD_OPT_CFG0**Type:** RW**Clock:** Config Clock**Reset State:** 0x00010000

DDR Command Optimization Configuration register 0.

This register is used to define the page management policy, RW optimizer preferences, and output command override controls.

BIMC_S_DDR0_SCMO_CMD_OPT_CFG0

Bits	Name	Description
23:21	RFU_23_21	Reserved for future use
20	IGNORE_BANK_UNAVL	When set, command execution ignores rank/bank unavailable inputs. When cleared, commands are allowed to be executed only if the corresponding rank/bank is available
19:18	RFU_19_18	Reserved for future use
17:16	MASK_CMDOUT_PRI	When 0x1, Mask priority of commands issued to DPE completely. When 0x0, AREQ=3 as well as timeout command priority is propagated to DPE. When 0x2, Mask priority of timeout commands issued to DPE (only AREQ=3 is passed on to DPE as high priority). NOTE: When DPE_CMD_REORDERING=0, MASK_CMDOUT_PRI must be 1 0x0: NO_MASK 0x1: AREQ_AND_TIMEOUT_MASK 0x2: TIMEOUT_MASK_ONLY 0x3: RESERVED
15:13	RFU_15_13	Reserved for future use
12	DPE_CMD_REORDERING	When cleared, forces output command order. When set, output commands are allowed to be reordered by protocol engine based corresponding dependency vector.
11:9	RFU_11_9	Reserved for future use

BIMC_S_DDR0_SCMO_CMD_OPT_CFG0 (cont.)

Bits	Name	Description
8	WR_OPT_EN	Indicates whether DDR page hit optimizations enabled in the write optimizer.
7:5	RFU_7_5	Reserved for future use
4	RD_OPT_EN	Indicates whether DDR page hit optimizations enabled in the read optimizer.
3:1	RFU_3_1	Reserved for future use
0	PAGE_MGMT_POLICY	Controls how HW manages DRAM pages. 0x0: KEEP_PAGE_OPEN 0x1: CLOSE_AFTER_EACH_ACCESS

0x00448704 BIMC_S_DDR0_SCMO_CMD_OPT_CFG1**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000

DDR Command Optimization Configuration register 1.

This register is used to define the read command time-out settings per priority level.

NOTE: Programmed values must be HSTP_CMD_TIMEOUT < HP_CMD_TIMEOUT < MP_CMD_TIMEOUT < LP_CMD_TIMEOUT

BIMC_S_DDR0_SCMO_CMD_OPT_CFG1

Bits	Name	Description
31:29	RFU_31_29	Reserved for future use
28:24	HSTP_CMD_TIMEOUT	Specifies the number of times highest-priority read command can be skipped for any other read command or write buffer flush. 0x0: RESERVED
23:21	RFU_23_21	Reserved for future use
20:16	HP_CMD_TIMEOUT	Specifies the number of times high-priority read command can be skipped for any other read command or write buffer flush. 0x0: RESERVED
15:13	RFU_15_13	Reserved for future use
12:8	MP_CMD_TIMEOUT	Specifies the number of times mid-priority read command can be skipped for any other read command or write buffer flush. 0x0: RESERVED
7:5	RFU_7_5	Reserved for future use

BIMC_S_DDR0_SCMO_CMD_OPT_CFG1 (cont.)

Bits	Name	Description
4:0	LP_CMD_TIMEOUT	Specifies the number of times low-priority read command can be skipped for any other read command or write buffer flush. 0x0: RESERVED

0x00448708 BIMC_S_DDR0_SCMO_CMD_OPT_CFG2**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000

DDR Command Optimization Configuration register 2.

This register is used to define the read/write command time-out settings for R/W Optimizer.

BIMC_S_DDR0_SCMO_CMD_OPT_CFG2

Bits	Name	Description
15:12	RFU_15_12	Reserved for future use
11:8	RWOPT_WR_CMD_TIMEOUT	Defines number of times high threshold write buffer flush command can be skipped. 0x0: RESERVED
7:4	RFU_7_4	Reserved for future use
3:0	RWOPT_RD_CMD_TIMEOUT	Defines number of times high priority/timed-out read command can be skipped. 0x0: RESERVED

0x0044870C BIMC_S_DDR0_SCMO_CMD_OPT_CFG3**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000

DDR Command Optimization Configuration register 3.

This register is used to define the flush command time-out settings for WR Optimizer.

BIMC_S_DDR0_SCMO_CMD_OPT_CFG3

Bits	Name	Description
31:21	RFU_31_21	Reserved for future use

BIMC_S_DDR0_SCMO_CMD_OPT_CFG3 (cont.)

Bits	Name	Description
20:16	WROPT_CMD_TIMEOUT	Defines number of times high threshold buffer flush not treated as high priority. 0x0: RESERVED
15:5	RFU_15_5	Reserved for future use
4:0	FLUSH_CMD_TIMEOUT	Defines number of times SW flush or flush on idle command can be skipped. 0x0: RESERVED

0x00448710 BIMC_S_DDR0_SCMO_CMD_OPT_CFG4**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000

DDR Command Optimization Configuration register 4.

This register is used to define the mask control settings for write buffer and read queues settings for Command Optimizer.

BIMC_S_DDR0_SCMO_CMD_OPT_CFG4

Bits	Name	Description
7:5	RFU_7_5	Reserved for future use
4	MASK_WRQ_FULL	When 0x1, Masks the write data buffer full, almost full or stall status (wrq_full) to the Arbiter completely. When 0x0, Allows the write data buffer status to propagate to the Arbiter. This helps to back pressure the Arbiter from not granting anymore writes.
3:1	RFU_3_1	Reserved for future use
0	MASK_RDQ_FULL	When 0x1, Masks the read channel FIFO status (rdq_full, rdq_almost_full) to the Arbiter completely. When 0x0, Allows the read channel status to propagate to the Arbiter. This helps to back pressure the Arbiter from not granting read request from those masters.

0x00448780 BIMC_S_DDR0_SCMO_FSPM_CMD_OPT_CFG0, m=[1..7]**Type:** RW**Clock:** Config Clock**Reset State:** 0x00010000

*m-1

BIMC_S_DDR0_SCMO_FSPM_CMD_OPT_CFG0

Bits	Name	Description
23:21	RFU_23_21	
20	IGNORE_BANK_UNAVL	
19:18	RFU_19_18	
17:16	MASK_CMDOUT_PRI	0x0: NO_MASK 0x1: AREQ_AND_TIMEOUT_MASK 0x2: TIMEOUT_MASK_ONLY 0x3: RESERVED
15:13	RFU_15_13	
12	DPE_CMD_REORDERING	
11:9	RFU_11_9	
8	WR_OPT_EN	
7:5	RFU_7_5	
4	RD_OPT_EN	
3:1	RFU_3_1	
0	PAGE_MGMT_POLICY	0x0: KEEP_PAGE_OPEN 0x1: CLOSE_AFTER_EACH_ACCESS

0x00448784 BIMC_S_DDR0_SCMO_FSPM_CMD_OPT_CFG1, m=[1..7]**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000

*m-1

BIMC_S_DDR0_SCMO_FSPM_CMD_OPT_CFG1

Bits	Name	Description
31:29	RFU_31_29	
28:24	HSTP_CMD_TIMEOUT	0x0: RESERVED
23:21	RFU_23_21	
20:16	HP_CMD_TIMEOUT	0x0: RESERVED
15:13	RFU_15_13	
12:8	MP_CMD_TIMEOUT	0x0: RESERVED
7:5	RFU_7_5	

BIMC_S_DDR0_SCMO_FSPM_CMD_OPT_CFG1 (cont.)

Bits	Name	Description
4:0	LP_CMD_TIMEOUT	0x0: RESERVED

0x00448788 BIMC_S_DDR0_SCMO_FSPM_CMD_OPT_CFG2, m=[1..7]**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000

*m-1

BIMC_S_DDR0_SCMO_FSPM_CMD_OPT_CFG2

Bits	Name	Description
15:12	RFU_15_12	
11:8	RWOPT_WR_CMD_TIMEOUT	0x0: RESERVED
7:4	RFU_7_4	
3:0	RWOPT_RD_CMD_TIMEOUT	0x0: RESERVED

0x0044878C BIMC_S_DDR0_SCMO_FSPM_CMD_OPT_CFG3, m=[1..7]**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000

*m-1

BIMC_S_DDR0_SCMO_FSPM_CMD_OPT_CFG3

Bits	Name	Description
31:21	RFU_31_21	
20:16	WROPT_CMD_TIMEOUT	0x0: RESERVED
15:5	RFU_15_5	
4:0	FLUSH_CMD_TIMEOUT	0x0: RESERVED

0x00448790 BIMC_S_DDR0_SCMO_FSPM_CMD_OPT_CFG4, m=[1..7]**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000

*m-1

BIMC_S_DDR0_SCMO_FSPM_CMD_OPT_CFG4

Bits	Name	Description
7:5	RFU_7_5	
4	MASK_WRQ_FULL	
3:1	RFU_3_1	
0	MASK_RDQ_FULL	

0x004487B0 BIMC_S_DDR0_SCMO_FSPM_RCH_SELECT, m=[1..7]**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000

*m-1

BIMC_S_DDR0_SCMO_FSPM_RCH_SELECT

Bits	Name	Description
31:16	DYNAMIC_RCH_SELECT	0x0: DISABLE 0x1: ENABLE
15:0	RCH_PORTS	0x0: RCH0 0x1: RCH1

0x00448A04 BIMC_S_DDR0_SCMO_FSPM_THRESHOLD_CFG, m=[1..7]**Type:** RW**Clock:** Config Clock**Reset State:** 0x00000000

*m-1

BIMC_S_DDR0_SCMO_FSPM_THRESHOLD_CFG

Bits	Name	Description
8	FSP_EN	

BIMC_S_DDR0_SCMO_FSPM_THRESHOLD_CFG (cont.)

Bits	Name	Description
7:0	FREQ_TH	

0x00448A40 BIMC_S_DDR0_SCMO_FSP_STATUS**Type:** R**Clock:** Config Clock**Reset State:** 0x00000000

Frequency Set Point configuration status register that is used to know the status of the chosen shadow register's index during clock frequency switching operation

BIMC_S_DDR0_SCMO_FSP_STATUS

Bits	Name	Description
2:0	FREQ_SET_POINT_REG_I N_USE	Frequency Set Point shadow register's index. Register's value is between 0 to 7, 0 being the original registers'

2.27 BIMC_S_DDR0_DPE_ABHN_DPE**0x0044C010 BIMC_S_DDR0_DPE_CONFIG_0****Type:** RW**Clock:** ACLK**Reset State:** 0x00000000

DDR Memory Configuration Register 0.

This register is used to setup the DRAM configuration in the DDR controller.

BIMC_S_DDR0_DPE_CONFIG_0

Bits	Name	Description
29:28	CA_BUS_OE_CTRL	When enabled, the design can indicate to the PHY to tri-state the command address bus. This is required as a shunt resistance is typically used in the command address path for PCDDR3 memory boards and hence will always burn power if the command address bus is always driven. The two scenarios wherein the bus can be tri-stated are (1) when all ranks are in self refresh and the memory clocks are off or (2) when all ranks are in powerdown. 0x0: DISABLED (DISABLED (default) (dpe_phy_pad_oe_ca is always driven to 1'b1. The CA bus is always in output mode.)) 0x1: SR_ONLY (SR_ONLY (dpe_phy_pad_oe_ca is only driven to 1'b0 when both ranks are in self refresh and the clock to the memory device has been turned off.)) 0x2: SR_OR_PWR_DN (SR_OR_PWR_DN (dpe_phy_pad_oe_ca is driven to 1'b0 when both ranks are in self refresh and the clock to the memory device has been turned off or when both ranks are in powerdown)) 0x3: RESERVED (RESERVED)
24	RD_BEAT_NDX_OOO	Setting this bit will enable the DPE to return read data beats to the front end out of order. dpe_cmd_rd_reseq will indicate that a transaction will have beats out of order, and the change in order will be reflected on the dpe_rdata_ndx port. This can be enabled for LPDDR2/3. This CSR has no effect on PCDDR3, as the read beat index will return out of order based on the column address by default for that memory type. 0x0: IN_ORDER (IN_ORDER (beats for a read are always returned in incrementing order) (Example: 0, 1, 2, 3, 4, 5, 6, 7)) 0x1: OOO (OOO (beats for a read can be returned in non-incrementing order) (Example: 0, 1, 6, 7, 2, 3, 4, 5))
20	DDR_WR_MASK_POLARITY	This bit is used to set the polarity for the write byte mask. LPDDR2/3 and PCDDR3 should use the default value '0'. 0x0: ACTIVE_HIGH (ACTIVE_HIGH (a high level will mask the corresponding byte from being written - default for LPDDR2)) 0x1: ACTIVE_LOW (ACTIVE_LOW (a low level will mask the corresponding byte from being written))
17:16	DEVICE_TYPE	Type of memory device 0x0: LPDDR2 (LPDDR2 (default)) 0x1: PCDDR3 (PCDDR3) 0x2: LPDDR3 (LPDDR3) 0x3: PCDDR4 (PCDDR4)

BIMC_S_DDR0_DPE_CONFIG_0 (cont.)

Bits	Name	Description
11:10	DEVICE_CFG_RANK1	Device Configuration for Rank 1. This is required for SRR in order to update the refresh rate correctly if multiple devices are involved. If all x8 individual memory devices are used, then select '00'. If all x16 individual memory devices are used, then select '01'. If all x32 individual memory devices are used, then select '10'. If all x64 individual memory devices are used, then select '11'. 0x0: X8_X8_X8_X8 (X8_X8_X8_X8 (default)) 0x1: X16_X16 (X16_X16) 0x2: X32 (X32) 0x3: X64 (X64)
9:8	DEVICE_CFG_RANK0	Device Configuration for Rank 0. This is required for SRR in order to update the refresh rate correctly if multiple devices are involved. If all x8 individual memory devices are used, then select '00'. If all x16 individual memory devices are used, then select '01'. If all x32 individual memory devices are used, then select '10'. If all x64 individual memory devices are used, then select '11'. 0x0: X8_X8_X8_X8 (X8_X8_X8_X8 (default)) 0x1: X16_X16 (X16_X16) 0x2: X32 (X32) 0x3: X64 (X64)
4	SHARED_CLK	Rank Clock Configuration. 0x0: UNSHARED_CLK (UNSHARED_CLK (default)) 0x1: SHARED_CLK (SHARED_CLK (Rank 0 and Rank 1 have the same clock))
0	DDR_CMD	Specify if Device command bus is Double Data Rate (in addition to the data bus). This needs to be enabled (1) for LPDDR2/3 SDRAM devices and disabled for PCDDR3 SDRAM devices. 0x0: SDR_ON_CMD (SDR_ON_CMD (default)) 0x1: DDR_ON_CMD (DDR_ON_CMD)

0x0044C014 BIMC_S_DDR0_DPE_CONFIG_1**Type:** RW**Clock:** ACLK**Reset State:** 0x00000000

DDR Memory Configuration Register 1.

This register is used to setup the DRAM configuration in the DDR controller.

BIMC_S_DDR0_DPE_CONFIG_1

Bits	Name	Description
31:24	PERIOD_BUS_THRESHOLD_POWER_MODE	Indicates the core clock frequency (in 100ps granularity) threshold for the controller to enable low power mode to the PHY. If the current core clock frequency is less than this threshold, then low power mode is enabled for the PHY and the corresponding pads. This value will be considered valid if CONFIG_1[PAD_POWER_MODE] is enabled and CONFIG_1[PAD_POWER_MODE_CTRL] is set to 0.
20	PAD_POWER_MODE_CTRL	Indicates if HW automatically controls the pad power mode to the PHY based on the current frequency of operation. If set to 1, then SW needs to program the values in DRAM_TIMING_16[PAD_MODE] depending on the current frequency of operation. SW would also be required to setup the new value before a frequency switch event. If set to 0, then HW automatically changes the power mode based on CONFIG_1[PERIOD_BUS_THRESHOLD_POWER_MODE] and the current frequency of operation and also would handle changes in the power mode based on frequency switch events. The value of this CSR is only valid if CONFIG_1[PAD_POWER_MODE] is enabled. If operating in cycle-based mode, the only allowed value is '1'. 0x0: HW_CTRL (HW_CTRL(default)) 0x1: SW_CTRL (SW_CTRL)
16	PAD_POWER_MODE	A value of '0' is required for chips which use Gen2 PHY whereas a value of '1' is recommended for chips which use Gen1 PHY. 0x0: DISABLED (DISABLED (default, Drive to high power mode)) 0x1: ENABLED (ENABLED (Power mode is controlled by the memory controller))
12:8	NUM_BANKS_RANK1	Number of banks for Rank 1
4:0	NUM_BANKS_RANK0	Number of banks for Rank 0

0x0044C018 BIMC_S_DDR0_DPE_CONFIG_2**Type:** RW**Clock:** ACLK**Reset State:** 0x00000000

DDR Memory Configuration Register 2.

This register is used to setup the DRAM configuration in the DDR controller.

BIMC_S_DDR0_DPE_CONFIG_2

Bits	Name	Description
29:28	TIMING_MODE	Command Interface timing relative to the chip select. This is necessary to provide an option to guarantee an additional cycle for the command interface signals to propagate to the DRAM device and settle (to meet input setup requirements) before being clocked in. This must be set to 00 for LPDDR2/3 (TM_1T). 1T, 2T and 3T is supported for PCDDR3. 0x0: TM_1T (TM_1T (Default: Address and command can be driven in the same cycle as the chip select)) 0x1: TM_2T (TM_2T (Address and command must be driven at least 1 cycle before chip select is asserted)) 0x2: TM_3T (TM_3T) 0x3: RESERVED (RESERVED)
24:20	DDR_BURST_LEN	Specifies the Memory Burst Length. For LPDDR2, only 4 and 8 are valid. LPDDR3 only supports burst length of 8. PCDDR3 supports burst length of 8 only. BL of 16 is not supported. 0x4: BL_OF_4 (BL_OF_4) 0x8: BL_OF_8 (BL_OF_8) 0x10: BL_OF_16 (BL_OF_16)
16	WR_DQS_EARLY	Indicates if the DQS needs to toggle one cycle before the write data is driven. This needs to be set based on the specific SDRAM device present. This is to be enabled for LPDDR3 and PCDDR3 devices, and disabled for LPDDR2 devices. 0x0: DISABLED (DISABLED (default)) 0x1: ENABLED (ENABLED)
12	DDR_CK_ON_DBG	Debug mode in which the clock to the memory device will continuously run regardless of the state of the memory (CSPD, Self Refresh). 0x0: DISABLED (DISABLED (default)) 0x1: ENABLED (ENABLED (Enable Debug Mode))
11:8	MRR_BURST_LEN	The Memory burst length of a Mode Register Read. This is used for LPDDR2/3 memory only. For LPDDR2, this is 4 beats, while for LPDDR3 this is 8 beats. 0x4: BL4 (BL4) 0x8: BL8 (BL8)
5:4	BURST_RD_START	Specifies the starting column address restrictions for a read operation. This field needs to be set depending on the specific SDRAM device. For LPDDR2/3 use '00'. For PCDDR3, use '01'. This is currently not supported. 0x0: ADDRS_0_2_4 (ADDRS_0_2_4 (default- addresses of 0,2,4,6 etc)) 0x1: ADDRS_0_4_8 (ADDRS_0_4_8 (addresses of 0,4, etc)) 0x2: ADDRS_0_ONLY (ADDRS_0_ONLY (addresses of 0 only) 11: (invalid))

BIMC_S_DDR0_DPE_CONFIG_2 (cont.)

Bits	Name	Description
1:0	BURST_WR_START	Specifies the starting column address restrictions for a write operation. This field needs to be set depending on the specific SDRAM device. For LPDDR2/3 use '00'. For PCDDR3, use '01'. This is currently not supported. 0x0: ADDRS_0_2_4 (ADDRS_0_2_4 (default- addresses of 0,2,4,6 etc)) 0x1: ADDRS_0_4_8 (ADDRS_0_4_8 (addresses of 0,4, etc)) 0x2: ADDRS_0_ONLY (ADDRS_0_ONLY (addresses of 0 only) 11: (invalid))

0x0044C01C BIMC_S_DDR0_DPE_CONFIG_3**Type:** RW**Clock:** ACLK**Reset State:** 0x00000000

DDR Memory Configuration Register 3.

This register is used to setup the DRAM configuration in the DDR controller.

BIMC_S_DDR0_DPE_CONFIG_3

Bits	Name	Description
28	DIST_LOAD_CTRL	This will enable duplication of cs_n, cke and odt signals in order to ease timing for multi-t by distributing the load of these signals. This is for single rank systems only. When enabled, cs_n[1] = cs_n[0], cke[1] = cke[0] and odt[1] = odt[0]. 0x0: DISABLED (DISABLED (default)) 0x1: ENABLED (ENABLED)
26:24	REF_DEFICIT_WTRMRK	Number of refreshes to postpone. This field allows the design to postpone up to 7 refreshes to increase performance. When set to 0, all refreshes are 'urgent' and will be issued immediately. When set to a non-zero value, the controller will postpone this number of refreshes in the presence of RD/WR traffic. Once the controller has postponed REF_DEFICIT_WTRMRK refreshes, each additional outstanding refresh will be treated as 'urgent' and will be issued to the memory device immediately. The design will catch up on the postponed refreshes only when there is no RD/WR traffic in the command queue for the rank and when all the pages to the device are closed (for example due the expiration of the PG_OPEN_TIMER).

BIMC_S_DDR0_DPE_CONFIG_3 (cont.)

Bits	Name	Description
20:16	TREFBW_XO_DIV_6_CYCLES	This field is used to define the amount of time in the burst refresh window for LPDDR2 and LPDDR3. See MAX_REFRESHES_TREFBW to disable tREFBW for PCDDR devices. This value will not vary with the frequency of the DPE core clock. The value programmed for TREFBW_XO_DIV_6_CYCLES is if largest_tRFCab < 210 ns { TREFBW_XO_DIV_6_CYCLES = ROUND_UP(32x tRFCab / (Txo*6)) + 1; } else { TREFBW_XO_DIV_6_CYCLES = ROUND_UP(32 x tRFCab / (Txo*6)); } where Txo = the period of the tcxo clock. largest_tRFCab = largest of the tRFCab values on either rank as specified in ns in the specification or data sheet. e.g., largest_tRFCab = 210 ns, tcxo=19.2 MHz gives RU(6720 ns/ 312.5 ns) = 22 decimal largest_tRFCab = 130 ns, tcxo = 19.2 MHz gives RU(4160/312.5) + 1 = 15 decimal.
14:12	MAX_REFRESHES_TREFBW	This indicates the maximum number of refreshes allowed in the TREFBW rolling window. This only applies to LPDDR2 and LPDDR3 as a means to limit the maximum current consumption. Typically, for LPDDR, write 3'b001 for up to 8 all-bank refreshes to be permitted in the tREFBW rolling window. For non-LPDDR devices, this field should be disabled by writing all zeros. Supported values are: 0x0: DISABLED (Disabled (default)) 0x1: EIGHT (EIGHT) 0x2: SEVEN (SEVEN) 0x3: SIX (SIX) 0x4: FIVE (FIVE)
8	DRAIN_CQ_ON_SHKE_CMD	This enables draining of commands in the command queue when a housekeeping command is received. The commands drained are only for the destination rank/bank of the housekeeping command. This is currently not supported. 0x0: DISABLE (DISABLE (When a housekeeping command is received, read/write commands to the DDR memory are halted and the house keeping command is driven)) 0x1: ENABLE (ENABLE (Read/write commands are drained only for the destination rank/bank of the housekeeping command))

BIMC_S_DDR0_DPE_CONFIG_3 (cont.)

Bits	Name	Description
4	LOAD_TIMING_ON_FREQ_CHNG	Indicates the active DDR timing registers will be loaded upon a frequency change request. 0x0: DISABLE (DISABLE (Frequency change request will not effect active DDR timing registers) (default)) 0x1: ENABLE (ENABLE (Frequency change request will load active DDR timing registers with values stored in read/writable timing registers.))
1	HW_FREQ_SW_SR_ONLY	This bit overrides the 'HW_FREQ_SWITCH' bit and forces all frequency switches to be done with self refresh regardless of the state of memory ranks. This must be enabled for PCDDR3, as CSPD is not supported. This can be enabled for LPDDR2/3. 0x0: DISABLE (DISABLE) 0x1: ENABLE (ENABLE)
0	HW_FREQ_SWITCH	This selects between the method for frequency switch, Clockstop Powerdown or Self Refresh. However, design will optimize frequency switch as follows. If any rank is already in self refresh, switch will be done with self refresh. If no rank is already in self refresh but any rank is already in clock stop power down, the switch will be done using clock stop power down. If no rank is already in self refresh or clock stop powerdown, switch will follow the setting of this bit. Note, if HW_FREQ_SW_SR_ONLY is set, the frequency switch will be done with self refresh regardless of this bit. 0x0: SELF_REFRESH (SELF REFRESH) 0x1: CSPD (CSPD)

0x0044C020 BIMC_S_DDR0_DPE_CONFIG_4**Type:** W**Clock:** ACLK**Reset State:** 0x00000000

DDR Memory Configuration Register 4.

This register is used to setup the DRAM configuration in the DDR controller.

BIMC_S_DDR0_DPE_CONFIG_4

Bits	Name	Description
4	RECALC_PS_PARAMS	Writing a 1 to this bit will cause all parameters which are programmed in 100ps units as specified by their corresponding *_IN_PS bit in CONFIG_5/6 to be recalculated by HW. They will not be loaded into the active registers by writing this bit. After writing this bit, the CYC_CALC_VALID bit in MEMC_STATUS_1 should be polled before writing a '1' to LOAD_ALL_CONFIG. Therefore, RECALC should not be written to this bit in the same write as '1' ('LOAD') is written to LOAD_ALL_CONFIG. 0x0: NOP (NOP) 0x1: RECALC (RECALC (1 will kick off calculation of fields specified in pico seconds))
0	LOAD_ALL_CONFIG	Setting this will load the DDR configuration registers for use by the controller. DPE maintains 2 sets of configuration registers: one read/write which can be set by software based on frequency requirements or for other reasons. These can be written at anytime and have no affect on current operation. Another shadow set of registers exists which is actively used by the DPE. Software must load the read/write set, then set this bit to propagate the new settings to the controller. After writing this bit, the SHADOW_REGS_STALE bit in MEMC_STATUS_2 will go to 'UP_TO_DATE' until another write is performed to the control registers. See MEMC_STATUS_2 for more details. 0x0: NOP (NOP (No change in active shadow register set) (default)) 0x1: LOAD (LOAD (1 will load the read only active shadow register set with the values in the R/W set))

0x0044C024 BIMC_S_DDR0_DPE_CONFIG_5**Type:** RW**Clock:** ACLK**Reset State:** 0x00000000

DDR Memory Configuration Register 5.

If the timing parameter value is only specified in clock cycles as per the JEDEC specification, then it is recommended to set that specific timing parameter in cycle based mode in the CSR below.

If the timing parameter value is only specified in ns as per the JEDEC specification, then it is recommended to set that specific timing parameter in time based mode in the CSR below.

If the timing parameter value is specified such that it has a minimum cycle time requirement as well as a timing parameter value in ns as per the JEDEC specification, then it is recommended to set that specific timing parameter in time based mode in the CSR below.

Each bit in this register indicates if the value stored in the DRAM_TIMING register is in 100ps intervals, or in DDR clock cycles.

BIMC_S_DDR0_DPE_CONFIG_5

Bits	Name	Description
31	ODTOFF_MIN_IN_PS	0x0: CLK_CYC (CLK_CYC (ODTOFF_MIN register value is in DDR clock cycles)) 0x1: PS (PS (ODTON_MAX register value is in 100ps intervals))
30	ODTOFF_MAX_IN_PS	0x0: CLK_CYC (CLK_CYC (ODTOFF_MAX register value is in DDR clock cycles)) 0x1: PS (PS (ODTON_MAX register value is in 100ps intervals))
29	ODTON_MIN_IN_PS	0x0: CLK_CYC (CLK_CYC (ODTON_MIN register value is in DDR clock cycles)) 0x1: PS (PS (ODTON_MAX register value is in 100ps intervals))
28	ODTON_MAX_IN_PS	0x0: CLK_CYC (CLK_CYC (ODTON_MAX register value is in DDR clock cycles)) 0x1: PS (PS (ODTON_MAX register value is in 100ps intervals))
27	TRASMIN_IN_PS	0x0: CLK_CYC (CLK_CYC (TRASMIN register value is in DDR clock cycles)) 0x1: PS (PS (TRASMIN register value is in 100ps intervals))
26	TWR_IN_PS	0x0: CLK_CYC (CLK_CYC (TWR register value is in DDR clock cycles)) 0x1: PS (PS (TWR register value is in 100ps intervals))
25	TRCD_IN_PS	0x0: CLK_CYC (CLK_CYC (TRCD register value is in DDR clock cycles)) 0x1: PS (PS (TRCD register value is in 100ps intervals))
24	TWTR_IN_PS	0x0: CLK_CYC (CLK_CYC (TWTR register value is in DDR clock cycles)) 0x1: PS (PS (TWTR register value is in 100ps intervals))
23	TRRD_IN_PS	0x0: CLK_CYC (CLK_CYC (TRRD register value is in DDR clock cycles)) 0x1: PS (PS (TRRD register value is in 100ps intervals))
22	TRFCAB_IN_PS	0x0: CLK_CYC (CLK_CYC (TRFCAB register value is in DDR clock cycles)) 0x1: PS (PS (TRFCAB register value is in 100ps intervals))
21	TRFCPB_IN_PS	0x0: CLK_CYC (CLK_CYC (TRFCPB register value is in DDR clock cycles)) 0x1: PS (PS (TRFCPB register value is in 100ps intervals))
20	TRTP_IN_PS	0x0: CLK_CYC (CLK_CYC (TRTP register value is in DDR clock cycles)) 0x1: PS (PS (TRTP register value is in 100ps intervals))
19	TRPAB_IN_PS	0x0: CLK_CYC (CLK_CYC (TRPAB register value is in DDR clock cycles)) 0x1: PS (PS (TRPAB register value is in 100ps intervals))
18	TRPPB_IN_PS	0x0: CLK_CYC (CLK_CYC (TRPPB register value is in DDR clock cycles)) 0x1: PS (PS (TRPPB register value is in 100ps intervals))

BIMC_S_DDR0_DPE_CONFIG_5 (cont.)

Bits	Name	Description
17	TFAW_IN_PS	0x0: CLK_CYC (CLK_CYC (TFAW register value is in DDR clock cycles)) 0x1: PS (PS (TFAW register value is in 100ps intervals))
16	TCKE_IN_PS	0x0: CLK_CYC (CLK_CYC (TCKE register value is in DDR clock cycles)) 0x1: PS (PS (TCKE register value is in 100ps intervals))
15	TZQCS_IN_PS	0x0: CLK_CYC (CLK_CYC (TZQCS register value is in DDR clock cycles)) 0x1: PS (PS (TZQCS register value is in 100ps intervals))
14	TZQCL_IN_PS	0x0: CLK_CYC (CLK_CYC (TZQCL register value is in DDR clock cycles)) 0x1: PS (PS (TZQCL register value is in 100ps intervals))
13	TMOD_IN_PS	0x0: CLK_CYC (CLK_CYC (TMOD register value is in DDR clock cycles)) 0x1: PS (PS (TMOD register value is in 100ps intervals))
12	MIN_SR_DURATION_IN_PS	0x0: CLK_CYC (CLK_CYC (MIN_SR_DURATION register value is in DDR clock cycles)) 0x1: PS (PS (MIN_SR_DURATION register value is in 100ps intervals))
11	TXSRD_IN_PS	0x0: CLK_CYC (CLK_CYC (TXSRD register value is in DDR clock cycles)) 0x1: PS (PS (TXSRD register value is in 100ps intervals))
10	TXSNR_IN_PS	0x0: CLK_CYC (CLK_CYC (TXSNR register value is in DDR clock cycles)) 0x1: PS (PS (TXSNR register value is in 100ps intervals))
9	TXPNR_ACT_PWR_DN_IN_PS	0x0: CLK_CYC (CLK_CYC (TXPNR_ACT_PWR_DN register value is in DDR clock cycles)) 0x1: PS (PS (TXPNR_ACT_PWR_DN register value is in 100ps intervals))
8	TXPR_ACT_PWR_DN_IN_PS	0x0: CLK_CYC (CLK_CYC (TXPR_ACT_PWR_DN register value is in DDR clock cycles)) 0x1: PS (PS (TXPR_ACT_PWR_DN register value is in 100ps intervals))
7	TXPNR_PCHG_PWR_DN_IN_PS	0x0: CLK_CYC (CLK_CYC (TXPNR_PCHG_PWR_DN register value is in DDR clock cycles)) 0x1: PS (PS (TXPNR_PCHG_PWR_DN register value is in 100ps intervals))
6	TXPR_PCHG_PWR_DN_IN_PS	0x0: CLK_CYC (CLK_CYC (TXPR_PCHG_PWR_DN register value is in DDR clock cycles)) 0x1: PS (PS (TXPR_PCHG_PWR_DN register value is in 100ps intervals))

BIMC_S_DDR0_DPE_CONFIG_5 (cont.)

Bits	Name	Description
5	CLK_AFTER_SR_ENTRY_IN_PS	0x0: CLK_CYC (CLK_CYC (CLK_AFTER_SR_ENTRY register value is in DDR clock cycles)) 0x1: PS (PS (CLK_AFTER_SR_ENTRY register value is in 100ps intervals))
4	CLK_BEFORE_SR_EXIT_IN_PS	0x0: CLK_CYC (CLK_CYC (CLK_BEFORE_SR_EXIT register value is in DDR clock cycles)) 0x1: PS (PS (CLK_BEFORE_SR_EXIT register value is in 100ps intervals))
3	IE_WINDOW_START_IN_PS	0x0: CLK_CYC (CLK_CYC (IE_WINDOW_START register value is in DDR clock cycles)) 0x1: PS (PS (IE_WINDOW_START register value is in 100ps intervals))
2	IE_WINDOW_END_IN_PS	0x0: CLK_CYC (CLK_CYC (IE_WINDOW_END register value is in DDR clock cycles)) 0x1: PS (PS (IE_WINDOW_END register value is in 100ps intervals))
1	RD_CAPTURE_START_IN_PS	0x0: CLK_CYC (CLK_CYC (RD_CAPTURE_START register value is in DDR clock cycles)) 0x1: PS (PS (RD_CAPTURE_START register value is in 100ps intervals))
0	RD_CAPTURE_END_IN_PS	0x0: CLK_CYC (CLK_CYC (RD_CAPTURE_END register value is in DDR clock cycles)) 0x1: PS (PS (RD_CAPTURE_END register value is in 100ps intervals))

0x0044C028 BIMC_S_DDR0_DPE_CONFIG_6**Type:** RW**Clock:** ACLK**Reset State:** 0x00000000

DDR Memory Configuration Register 6.

This register is used to setup the DRAM configuration in the DDR controller. These registers are used to enable/disable debug modes for IO stage and Calibrated Delay Circuit interfaces as well as to block activates if necessary for manual SW frequency switch.

BIMC_S_DDR0_DPE_CONFIG_6

Bits	Name	Description
12	DRAM_RD_BURST_ORDER	Some memory devices will return read data in differ net orders, based on the column address. This register is used to identify a specific order of read data. For example, LPDDR2 column read with column_address[2:1]=01 will return data in column order 2 3, 4 5, 6 7, 0 1. However, a PCDDR3 column read with column_address[2:1]=01 will return data in column order 2 3, 0 1, 6 7, 4 5. For PCDDR3, set this to '1'. For LPDDR2 and LPDDR3, set this to '0'. 0x0: INCR_ORDER (INCR_ORDER) 0x1: DECR_ORDER (DECR_ORDER)
9	RANK1_BLK_ACT	Writing a 1 to this bit will block new activates on Rank 1. Clearing allows activates to resume. 'LOAD_ALL_CONFIG' is required before changes to this bit take effect. Usage: Intended for manual SW Freq switch sequence. Set this bit after Rank 1 is in self refresh. Load_all_config. Switch frequency. Exit self refresh. Perform MRWs to update latencies. Clear this bit to allow traffic to resume. Load_all_config. 0x0: NOP (NOP) 0x1: BLOCK (BLOCK)
8	RANK0_BLK_ACT	Writing a 1 to this bit will block new activates on Rank 0. Clearing allows activates to resume. 'LOAD_ALL_CONFIG' is required before changes to this bit take effect. Usage: Intended for manual SW Freq switch sequence. Set this bit after Rank 1 is in self refresh. Load_all_config. Switch frequency. Exit self refresh. Perform MRWs to update latencies. Clear this bit to allow traffic to resume. Load_all_config. 0x0: NOP (NOP) 0x1: BLOCK (BLOCK)

0x0044C02C BIMC_S_DDR0_DPE_CONFIG_7**Type:** RW**Clock:** ACLK**Reset State:** 0x00000000

DDR Memory Configuration Register 7.

This register is used to setup the DRAM configuration in the DDR controller. These registers are used to indicate how the DQ is mapped to the data bytes. All fields in this register are currently unsupported.

BIMC_S_DDR0_DPE_CONFIG_7

Bits	Name	Description
7:6	MREG_DQ3_BYTE_MAP	0x0: BYTE_0_DQ3 (BYTE_0_DQ3 (Byte 0 is mapped to DQ3)) 0x1: BYTE_1_DQ3 (BYTE_1_DQ3 (Byte 1 is mapped to DQ3)) 0x2: BYTE_2_DQ3 (BYTE_2_DQ3 (Byte 2 is mapped to DQ3)) 0x3: BYTE_3_DQ3 (BYTE_3_DQ3 (Byte 3 is mapped to DQ3)) Currently not supported)
5:4	MREG_DQ2_BYTE_MAP	0x0: BYTE_0_DQ2 (BYTE_0_DQ2 (Byte 0 is mapped to DQ2)) 0x1: BYTE_1_DQ2 (BYTE_1_DQ2 (Byte 1 is mapped to DQ2)) 0x2: BYTE_2_DQ2 (BYTE_2_DQ2 (Byte 2 is mapped to DQ2)) 0x3: BYTE_3_DQ2 (BYTE_3_DQ2 (Byte 3 is mapped to DQ2)) Currently not supported)
3:2	MREG_DQ1_BYTE_MAP	0x0: BYTE_0_DQ1 (BYTE_0_DQ1 (Byte 0 is mapped to DQ1)) 0x1: BYTE_1_DQ1 (BYTE_1_DQ1 (Byte 1 is mapped to DQ1)) 0x2: BYTE_2_DQ1 (BYTE_2_DQ1 (Byte 2 is mapped to DQ1)) 0x3: BYTE_3_DQ1 (BYTE_3_DQ1 (Byte 3 is mapped to DQ1)) Currently not supported)
1:0	MREG_DQ0_BYTE_MAP	0x0: BYTE_0_DQ0 (BYTE_0_DQ0 (Byte 0 is mapped to DQ0)) 0x1: BYTE_1_DQ0 (BYTE_1_DQ0 (Byte 1 is mapped to DQ0)) 0x2: BYTE_2_DQ0 (BYTE_2_DQ0 (Byte 2 is mapped to DQ0)) 0x3: BYTE_3_DQ0 (BYTE_3_DQ0 (Byte 3 is mapped to DQ0)) Currently not supported)

0x0044C038 BIMC_S_DDR0_DPE_CONFIG_DQ_MAP**Type:** RW**Clock:** ACLK**Reset State:** 0x00000000

DQ Map configuration register.

This register issued to indicate how the DQ bytes on the controller are mapped to the DRAM data bytes. This feature is implemented only for LPDDR3 parts. Software needs to program this CSR based on the DQ byte mapping between the **MSM** and memory.

BIMC_S_DDR0_DPE_CONFIG_DQ_MAP

Bits	Name	Description
30:28	DQ7_BYTE_MAP	For 32 bit Controller 000: (Not used) 001: (Not used) 010: (Not used) 011: (Not used) 100: (Not used) 101: (Not used) 110: (Not used) 111: (Not used) For 64 bit Controller 000: (Controller DQ7 is mapped to DRAM Byte 7) 001: (Controller DQ7 is mapped to DRAM BYte 0) 010: (Controller DQ7 is mapped to DRAM Byte 1) 011: (Controller DQ7 is mapped to DRAM Byte 2) 100: (Controller DQ7 is mapped to DRAM Byte 3) 101: (Controller DQ7 is mapped to DRAM Byte 4) 110: (Controller DQ7 is mapped to DRAM Byte 5) 111: (Controller DQ7 is mapped to DRAM Byte 6)
26:24	DQ6_BYTE_MAP	For 32 bit Controller 000: (Not used) 001: (Not used) 010: (Not used) 011: (Not used) 100: (Not used) 101: (Not used) 110: (Not used) 111: (Not used) For 64 bit Controller 000: (Controller DQ6 is mapped to DRAM Byte 6) 001: (Controller DQ6 is mapped to DRAM Byte 7) 010: (Controller DQ6 is mapped to DRAM Byte 0) 011: (Controller DQ6 is mapped to DRAM Byte 1) 100: (Controller DQ6 is mapped to DRAM Byte 2) 101: (Controller DQ6 is mapped to DRAM Byte 3) 110: (Controller DQ6 is mapped to DRAM Byte 4) 111: (Controller DQ6 is mapped to DRAM Byte 5)

BIMC_S_DDR0_DPE_CONFIG_DQ_MAP (cont.)

Bits	Name	Description
22:20	DQ5_BYTE_MAP	For 32 bit Controller 000: (Not used) 001: (Not used) 010: (Not used) 011: (Not used) 100: (Not used) 101: (Not used) 110: (Not used) 111: (Not used) For 64 bit Controller 000: (Controller DQ5 is mapped to DRAM Byte 5) 001: (Controller DQ5 is mapped to DRAM Byte 6) 010: (Controller DQ5 is mapped to DRAM Byte 7) 011: (Controller DQ5 is mapped to DRAM Byte 0) 100: (Controller DQ5 is mapped to DRAM Byte 1) 101: (Controller DQ5 is mapped to DRAM Byte 2) 110: (Controller DQ5 is mapped to DRAM Byte 3) 111: (Controller DQ5 is mapped to DRAM Byte 4)
18:16	DQ4_BYTE_MAP	For 32 bit Controller 000: (Not used) 001: (Not used) 010: (Not used) 011: (Not used) 100: (Not used) 101: (Not used) 110: (Not used) 111: (Not used) For 64 bit Controller 000: (Controller DQ4 is mapped to DRAM Byte 4) 001: (Controller DQ4 is mapped to DRAM Byte 5) 010: (Controller DQ4 is mapped to DRAM Byte 6) 011: (Controller DQ4 is mapped to DRAM Byte 7) 100: (Controller DQ4 is mapped to DRAM Byte 0) 101: (Controller DQ4 is mapped to DRAM Byte 1) 110: (Controller DQ4 is mapped to DRAM Byte 2) 111: (Controller DQ4 is mapped to DRAM Byte 3)

BIMC_S_DDR0_DPE_CONFIG_DQ_MAP (cont.)

Bits	Name	Description
14:12	DQ3_BYTE_MAP	For 32 bit Controller 000: (Controller DQ3 is mapped to DRAM Byte 3) 001: (Controller DQ3 is mapped to DRAM Byte 0) 010: (Controller DQ3 is mapped to DRAM Byte 1) 011: (Controller DQ3 is mapped to DRAM Byte 2) 100: (Not used) 101: (Not used) 110: (Not used) 111: (Not used) For 64 bit Controller 000: (Controller DQ3 is mapped to DRAM Byte 3) 001: (Controller DQ3 is mapped to DRAM Byte 4) 010: (Controller DQ3 is mapped to DRAM Byte 5) 011: (Controller DQ3 is mapped to DRAM Byte 6) 100: (Controller DQ3 is mapped to DRAM Byte 7) 101: (Controller DQ3 is mapped to DRAM Byte 0) 110: (Controller DQ3 is mapped to DRAM Byte 1) 111: (Controller DQ3 is mapped to DRAM Byte 2)
10:8	DQ2_BYTE_MAP	For 32 bit Controller 000: (Controller DQ2 is mapped to DRAM Byte 2) 001: (Controller DQ2 is mapped to DRAM Byte 3) 010: (Controller DQ2 is mapped to DRAM Byte 0) 011: (Controller DQ2 is mapped to DRAM Byte 1) 100: (Not used) 101: (Not used) 110: (Not used) 111: (Not used) For 64 bit Controller 000: (Controller DQ2 is mapped to DRAM Byte 2) 001: (Controller DQ2 is mapped to DRAM Byte 3) 010: (Controller DQ2 is mapped to DRAM Byte 4) 011: (Controller DQ2 is mapped to DRAM Byte 5) 100: (Controller DQ2 is mapped to DRAM Byte 6) 101: (Controller DQ2 is mapped to DRAM Byte 7) 110: (Controller DQ2 is mapped to DRAM Byte 0) 111: (Controller DQ2 is mapped to DRAM Byte 1)

BIMC_S_DDR0_DPE_CONFIG_DQ_MAP (cont.)

Bits	Name	Description
6:4	DQ1_BYTE_MAP	For 32 bit Controller 000: (Controller DQ1 is mapped to DRAM Byte 1) 001: (Controller DQ1 is mapped to DRAM Byte 2) 010: (Controller DQ1 is mapped to DRAM Byte 3) 011: (Controller DQ1 is mapped to DRAM Byte 0) 100: (Not used) 101: (Not used) 110: (Not used) 111: (Not used) For 64 bit Controller 000: (Controller DQ1 is mapped to DRAM Byte 1) 001: (Controller DQ1 is mapped to DRAM Byte 2) 010: (Controller DQ1 is mapped to DRAM Byte 3) 011: (Controller DQ1 is mapped to DRAM Byte 4) 100: (Controller DQ1 is mapped to DRAM Byte 5) 101: (Controller DQ1 is mapped to DRAM Byte 6) 110: (Controller DQ1 is mapped to DRAM Byte 7) 111: (Controller DQ1 is mapped to DRAM Byte 0)
2:0	DQ0_BYTE_MAP	For 32 bit Controller 000: (Controller DQ0 is mapped to DRAM Byte 0) 001: (Controller DQ0 is mapped to DRAM Byte 1) 010: (Controller DQ0 is mapped to DRAM Byte 2) 011: (Controller DQ0 is mapped to DRAM Byte 3) 100: (Not used) 101: (Not used) 110: (Not used) 111: (Not used) For 64 bit Controller 000: (Controller DQ0 is mapped to DRAM Byte 0) 001: (Controller DQ0 is mapped to DRAM Byte 1) 010: (Controller DQ0 is mapped to DRAM Byte 2) 011: (Controller DQ0 is mapped to DRAM Byte 3) 100: (Controller DQ0 is mapped to DRAM Byte 4) 101: (Controller DQ0 is mapped to DRAM Byte 5) 110: (Controller DQ0 is mapped to DRAM Byte 6) 111: (Controller DQ0 is mapped to DRAM Byte 7)

0x0044C040 BIMC_S_DDR0_DPE_ODT_CONFIG_0**Type:** RW**Clock:** ACLK**Reset State:** 0x00000000

ODT Configuration Register 0.

This register is used to setup the ODT controls for PCDDR3/LPDDR3 memory devices.

BIMC_S_DDR0_DPE_ODT_CONFIG_0

Bits	Name	Description
9	RANK1_FORCE_ODT	Force Rank1 ODT to be driven high and continuously. Do not set except as part of write leveling sequence as described in the HPG. 0x0: DISABLED (DISABLED (default - normal operation)) 0x1: ENABLED (ENABLED (ODT pin at DRAM device force high continuously))
8	RANK0_FORCE_ODT	Force Rank0 ODT to be driven high and continuously. Do not set except as part of write leveling sequence as described in the HPG. 0x0: DISABLED (DISABLED (default- normal operation)) 0x1: ENABLED (ENABLED (ODT pin at DRAM device force high continuously))
6	RANK1_ODT_ON_WR_RANK1	Enable Rank1 ODT to be driven when write occurs on Rank1. 0x0: DISABLED (DISABLED (default)) 0x1: ENABLED (ENABLED)
5	RANK1_ODT_ON_WR_RANK0	Enable Rank1 ODT to be driven when write occurs on Rank0. 0x0: DISABLED (DISABLED (default)) 0x1: ENABLED (ENABLED)
4	RANK1_ODT_ON_RD_RANK0	Enable Rank1 ODT to be driven when read occurs from Rank0. 0x0: DISABLED (DISABLED (default)) 0x1: ENABLED (ENABLED)
2	RANK0_ODT_ON_WR_RANK1	Enable Rank0 ODT to be driven when write occurs on Rank1. 0x0: DISABLED (DISABLED (default)) 0x1: ENABLED (ENABLED)
1	RANK0_ODT_ON_WR_RANK0	Enable Rank0 ODT to be driven when write occurs on Rank0. 0x0: DISABLED (DISABLED (default)) 0x1: ENABLED (ENABLED)
0	RANK0_ODT_ON_RD_RANK1	Enable Rank0 ODT to be driven when read occurs from Rank1. 0x0: DISABLED (DISABLED (default)) 0x1: ENABLED (ENABLED)

0x0044C044 BIMC_S_DDR0_DPE_ODT_CONFIG_1

Type: RW

Clock: ACLK

Reset State: 0x00000000

ODT Configuration Register 1.

This register is used to setup the ODT controls for PCDDR3/LPDDR3 memory devices.

BIMC_S_DDR0_DPE_ODT_CONFIG_1

Bits	Name	Description
30:28	ODTH8	Specifies the timing parameter ODTH8 (for PCDDR3) in core clock cycles.
24	SYNC_ODT_DLY_MODE	Specifies whether the synchronous ODT delays specified in this register below are in actual cycles or in terms of the write latency offset (negative offset). For LPDDR3, this needs to be set in cycles whereas for PCDDR3, this needs to be set as write latency offset. 0x0: IN_CYCLES (IN_CYCLES(default)) 0x1: WL_OFFSET (WL_OFFSET)
22:20	SYNC_ODT_ON_DLY	Specifies the synchronous ODT on delay in either clock cycles or write latency offset (negative offset) depending on the value specified in field - SYNC_ODT_DLY_MODE. For LPDDR3, the ODT on delay is specified in ns, hence the value programmed here should be 0. For PCDDR3, this should be set to 2 cycles.
18:16	SYNC_ODT_OFF_DLY	Specifies the synchronous ODT off delay in either clock cycles or write latency offset (negative offset) depending on the value specified in field - SYNC_ODT_DLY_MODE. For LPDDR3, the ODT off delay is specified in ns, hence the value programmed here should be 0. For PCDDR3, this should be set to 2 cycles.
9:8	ODT_OFF_BEFORE_RD	The number of cycles ODT must be disabled before the DRAM returns read data for a read command to that rank. This CSR is specified in read latency offset. This should be set to '1' for LPDDR3, and '2' for PCDDR3.
4	ODT_ON_MIN_POS	Indicates whether the ODT on min delay specified in DRAM_TIMING_23 is positive or negative. For LPDDR3, this needs to be set to POSITIVE whereas for PCDDR3, this needs to be set to NEGATIVE. 0x0: POSITIVE (POSITIVE (default)) 0x1: NEGATIVE (NEGATIVE)
0	ODT_OFF_MIN_POS	Indicates whether the ODT off min delay specified in DRAM_TIMING_23 is positive or negative. For LPDDR3 and PCDDR3, this needs to be set to POSITIVE. 0x0: POSITIVE (POSITIVE (default)) 0x1: NEGATIVE (NEGATIVE)

0x0044C050 BIMC_S_DDR0_DPE_CA_TRAIN_PRE_CS**Type:** RW**Clock:** ACLK**Reset State:** 0x00000000

DPE CA Training sequence register 0.

This register is define the first 2 of 6 values driven onto the CA bus during the CA Training session. These two values are driven onto the CA bus before CS goes low (PRE CS). LPDDR3 only.

BIMC_S_DDR0_DPE_CA_TRAIN_PRE_CS

Bits	Name	Description
25:16	CA_BUS_2	This is the 2nd value driven onto the CA bus during the CA training session. LPDDR3 only.
9:0	CA_BUS_1	This is the 1st value driven onto the CA bus during the CA training session. LPDDR3 only.

0x0044C054 BIMC_S_DDR0_DPE_CA_TRAIN_CS**Type:** RW**Clock:** ACLK**Reset State:** 0x00000000

DPE CA Training sequence register 1.

This register is define the 3rd and 4th of 6 values driven onto the CA bus during the CA Training session. These two values are driven onto the CA bus while CS is low (During CS). These values should be captured into the DQ bus. LPDDR3 only.

BIMC_S_DDR0_DPE_CA_TRAIN_CS

Bits	Name	Description
25:16	CA_BUS_4	This is the 4th value driven onto the CA bus during the CA training session. LPDDR3 only.
9:0	CA_BUS_3	This is the 3rd value driven onto the CA bus during the CA training session. LPDDR3 only.

0x0044C058 BIMC_S_DDR0_DPE_CA_TRAIN_POST_CS**Type:** RW**Clock:** ACLK**Reset State:** 0x00000000

DPE CA Training sequence register 2.

This register is define the 5th and 6th of 6 values driven onto the CA bus during the CA Training session. These two values are driven onto the CA bus after CS went low (POST CS). LPDDR3 only.

BIMC_S_DDR0_DPE_CA_TRAIN_POST_CS

Bits	Name	Description
25:16	CA_BUS_6	This is the 6th value driven onto the CA bus during the CA training session. LPDDR3 only.

BIMC_S_DDR0_DPE_CA_TRAIN_POST_CS (cont.)

Bits	Name	Description
9:0	CA_BUS_5	This is the 5th value driven onto the CA bus during the CA training session. LPDDR3 only.

0x0044C060 BIMC_S_DDR0_DPE_RCW_CTRL**Type:** RW**Clock:** ACLK**Reset State:** 0x00000000

DPE Read Capture Window Control CSR.

This register controls the async reset for the read enable circuitry in the PHY for DQS qualification. It also includes a global disable control which results in the external read enable signal being always driven to 1'b1.

Note that the byte numbers below may not correspond to the actual DRAM byte. This is required so as to keep the byte number reference correct between the controller and the PHY (which has no information on the byte swap feature)

BIMC_S_DDR0_DPE_RCW_CTRL

Bits	Name	Description
31	RCW_ASYNC_RESET_BYT E7	This field is only valid for x64 DRAM devices. SW control for the async reset for the flops in the RCW generation circuit for PHYdata byte7. SW writes this field to 1'b1 if the flops in the PHY are required to be reset. SW is then required to write this field to 1'b0 to deassert the reset. 0x0: RESET_OFF (RESET_OFF) 0x1: RESET_ON (RESET_ON)
30	RCW_ASYNC_RESET_BYT E6	This field is only valid for x64 DRAM devices. SW control for the async reset for the flops in the RCW generation circuit for PHYdata byte6. SW writes this field to 1'b1 if the flops in the PHY are required to be reset. SW is then required to write this field to 1'b0 to deassert the reset. 0x0: RESET_OFF (RESET_OFF) 0x1: RESET_ON (RESET_ON)
29	RCW_ASYNC_RESET_BYT E5	This field is only valid for x64 DRAM devices. SW control for the async reset for the flops in the RCW generation circuit for PHYdata byte5. SW writes this field to 1'b1 if the flops in the PHY are required to be reset. SW is then required to write this field to 1'b0 to deassert the reset. 0x0: RESET_OFF (RESET_OFF) 0x1: RESET_ON (RESET_ON)

BIMC_S_DDR0_DPE_RCW_CTRL (cont.)

Bits	Name	Description
28	RCW_ASYNC_RESET_BYTE4	This field is only valid for x64 DRAM devices. SW control for the async reset for the flops in the RCW generation circuit for PHYdata byte4. SW writes this field to 1'b1 if the flops in the PHY are required to be reset. SW is then required to write this field to 1'b0 to deassert the reset. 0x0: RESET_OFF (RESET_OFF) 0x1: RESET_ON (RESET_ON)
27	RCW_ASYNC_RESET_BYTE3	SW control for the async reset for the flops in the RCW generation circuit for PHYdata byte3. SW writes this field to 1'b1 if the flops in the PHY are required to be reset. SW is then required to write this field to 1'b0 to deassert the reset. 0x0: RESET_OFF (RESET_OFF) 0x1: RESET_ON (RESET_ON)
26	RCW_ASYNC_RESET_BYTE2	SW control for the async reset for the flops in the RCW generation circuit for PHYdata byte2. SW writes this field to 1'b1 if the flops in the PHY are required to be reset. SW is then required to write this field to 1'b0 to deassert the reset. 0x0: RESET_OFF (RESET_OFF) 0x1: RESET_ON (RESET_ON)
25	RCW_ASYNC_RESET_BYTE1	SW control for the async reset for the flops in the RCW generation circuit for PHYdata byte1. SW writes this field to 1'b1 if the flops in the PHY are required to be reset. SW is then required to write this field to 1'b0 to deassert the reset. 0x0: RESET_OFF (RESET_OFF) 0x1: RESET_ON (RESET_ON)
24	RCW_ASYNC_RESET_BYTE0	SW control for the async reset for the flops in the RCW generation circuit for PHYdata byte0. SW writes this field to 1'b1 if the flops in the PHY are required to be reset. SW is then required to write this field to 1'b0 to deassert the reset. 0x0: RESET_OFF (RESET_OFF) 0x1: RESET_ON (RESET_ON)
20	RCW_ENABLE	When set to disable, the external read enable for precise DQS qualification is always driven to 1'b1 for all data bytes. When enabled (required if ODT is being used for LPDDR3 or PCDDR3), the design will send out a read enable pulse equal to the memory BL/2 - 1 cycles to the PHY data bytes. 0x0: DISABLE (DISABLE) 0x1: ENABLE (ENABLE)
7:0	EXP_PATTERN	This is the read data pattern from the memory device for LPDDR3/PCDDR3. Some of the example patterns could be '10101010', '00110011' (for LPDDR3) or '01010101' (for PCDDR3). These represent memory burst length of 8.

0x0044C068 BIMC_S_DDR0_DPE_PWR_CTRL_0**Type:** RW**Clock:** ACLK**Reset State:** 0x00000000

DPE Power Control Register 0.

This register is used to setup and activate the power controls of the DPE and memory.

BIMC_S_DDR0_DPE_PWR_CTRL_0

Bits	Name	Description
28	PWR_DN_MODE	Indicates that each rank will not power down until all ranks are capable of powerdown. If this bit is not set, each rank powers down on an individual basis. Concurrent Mode is currently not supported. 0x0: INDIVIDUAL (INDIVIDUAL (Each rank can power down individually when ready, default)) 0x1: CONCURRENT (CONCURRENT (All ranks power down together only when all ranks are ready for power down.) (This is currently not supported))
24	PWR_DN_EN	Enable power down mode for the memory devices. When this mode is enabled, the controller will power down the memory device when the proper conditions are met. 0x0: DISABLED (DISABLED (default)) 0x1: ENABLED (ENABLED (Enable power down mode))
20	CLK_STOP_PWR_DN_EN	Enable clock stop power down mode for the memory devices. When this mode is enabled, the controller will put the memory device into CSPD when conditions are met. CSPD is supported for LPDDR2/3 only. 0x0: DISABLED (DISABLED (default)) 0x1: ENABLED (ENABLED (Enable clock stop powerdown mode))
16	CLK_ON_EN	Enable clock on functionality. The controller can request that the clock remain on through the use of a clock on port. When this feature is enabled, the controller can indicate that it no longer needs a clock by de-asserting the clock on port. Certain controller idle conditions must be met. 0x0: DISABLED (DISABLED (Controller clock on port will remain high) (default)) 0x1: ENABLED (ENABLED (Controller clock on port will go low when certain idle conditions are met))
15:4	CLK_ON_IDLE_CNT	Number of clock cycles the controller must be in an idle state before the clock on port can be de-asserted. (0x0 = 0 cycles, 0x1 = cycle etc.) The minimum supported value for this register is 6 cycles.
0	PCHG_PD_DLL	Precharge Powerdown DLL setting. This is currently not supported.

0x0044C078 BIMC_S_DDR0_DPE_OPT_CTRL_0**Type:** RW**Clock:** ACLK**Reset State:** 0x00000000

DPE Optimization Control Register 0.

This register is used to setup and activate performance optimization features of the DPE.

BIMC_S_DDR0_DPE_OPT_CTRL_0

Bits	Name	Description
20	BC_OTF_EN	Enables the DPE to issue Burst Chop (On The Fly) commands to shorten the burst length of reads and writes where possible. This is for PCDDR3 memory devices only. Burst chop, on the fly, must be enabled through Mode Register 0 before enabling this feature. 0x0: DISABLED (DISABLED (default)) 0x1: ENABLED (ENABLED)
16	RD_INT_BST	Enables the DPE controller logic to drive a burst stop to reduce the length of the DDR Memory burst for a read command. This can be enabled for LPDDR2, but must be disabled for LPDDR3 and PCDDR3. 0x0: DISABLED (DISABLED (default)) 0x1: ENABLED (ENABLED)
12	WR_INT_BST	Enables the DPE controller logic to drive a burst stop to reduce the length of the DDR Memory burst for a write command. This can be enabled for LPDDR2, but must be disabled for LPDDR3 and PCDDR3. 0x0: DISABLED (DISABLED (default)) 0x1: ENABLED (ENABLED)
8	RD_INT_RD	Enables the DPE controller logic to interrupt a DDR Read command with another read command to the DDR memory when possible. This will improve the performance of the controller by removing un-needed DDR memory data beats. This can be enabled for LPDDR2, but must be disabled for LPDDR3 and PCDDR3. 0x0: DISABLED (DISABLED (default)) 0x1: ENABLED (ENABLED)
4	WR_INT_WR	Enables the DPE controller logic to interrupt a DDR Write command with another Write command to the DDR memory when possible. This will improve the performance of the controller by removing un-needed DDR memory data beats. This can be enabled for LPDDR2, but must be disabled for LPDDR3 and PCDDR3. 0x0: DISABLED (DISABLED (default)) 0x1: ENABLED (ENABLED)

BIMC_S_DDR0_DPE_OPT_CTRL_0 (cont.)

Bits	Name	Description
0	SPLIT_LONG_WR	Setting this to '1' will split long write transactions to allow read commands to the memory device. Long writes require multiple Write CAS commands. In some cases, long writes can dominate the usage of the memory. This control register allows the controller to stop sending write commands to allow read commands to reach the memory. This feature is currently not supported. 0x0: DISABLED (DISABLED (default)) 0x1: ENABLED (ENABLED)

0x0044C07C BIMC_S_DDR0_DPE_OPT_CTRL_1**Type:** RW**Clock:** ACLK**Reset State:** 0x00000000

DPE Optimization Control Register 1.

This register is used to setup and activate performance optimization features of the DPE.

BIMC_S_DDR0_DPE_OPT_CTRL_1

Bits	Name	Description
25:16	DDR_EFF_OPT_TIMER	DDR efficiency optimization timer. Specifies the number of cycles a higher priority non-optimally serviceable transaction may be delayed to continue with more optimal transactions. Supported values are 0-1023, with 0 disabling this feature.
12	PG_MGMT	Indicates if PAGE_OPEN_TIMER is used for managing open pages. 0x0: KEEP_OPEN (KEEP_OPEN (Keep page open unless forced to close it. e.g. forced to closed because of a refresh, different page to open in the same bank.) (default)) 0x1: TIMER (TIMER (Pages will be managed with all listed above plus PAGE_OPEN_TIMER))
11:0	PG_OPEN_TIMER	Maximum time interval in DDR cycles a page can remain open without any page hits. (0 = 0 cycles, 1 = 1 cycle etc.)

0x0044C08C BIMC_S_DDR0_DPE_DRAM_STATUS_0**Type:** R**Clock:** ACLK**Reset State:** 0x00000000

DDR Status Register 0.

This register contains information on the state of the SDRAM device.

BIMC_S_DDR0_DPE_DRAM_STATUS_0

Bits	Name	Description
25	RANK_1_SR	READ ONLY Indicates Rank 1 is in Self Refresh 0x0: NOT_IN_SR (NOT_IN_SR (default)) 0x1: IN_SR (IN_SR)
24	RANK_0_SR	READ ONLY Indicates Rank 0 is in Self Refresh 0x0: NOT_IN_SR (NOT_IN_SR (default)) 0x1: IN_SR (IN_SR)
21	RANK_1_DPD	READ ONLY Indicates Rank 1 is in Deep Power Down. This is supported by LPDDR2/3 devices only, this register will hold a value of '0' for PCDDR3. 0x0: NOT_IN_DPD (NOT_IN_DPD (default)) 0x1: IN_DPD (IN_DPD)
20	RANK_0_DPD	READ ONLY Indicates Rank 0 is in Deep Power Down. This is supported by LPDDR2/3 devices only, this register will hold a value of '0' for PCDDR3. 0x0: NOT_IN_DPD (NOT_IN_DPD (default)) 0x1: IN_DPD (IN_DPD)
17	RANK_1_DERATE	READ ONLY Indicates if Rank 1 has temperature derating applied to the timing values. This is supported by LPDDR2/3 devices only, this register will hold a value of '0' for PCDDR3. 0x0: NOT_IN_DERATE (NOT_IN_DERATE (Derate offset has not been added to timer values, default)) 0x1: IN_DERATE (IN_DERATE (Derate offset has been added to timer values))
16	RANK_0_DERATE	READ ONLY Indicates if Rank 0 has temperature derating applied to the timing values. This is supported by LPDDR2/3 devices only, this register will hold a value of '0' for PCDDR3. 0x0: NOT_IN_DERATE (NOT_IN_DERATE (Derate offset has not been added to timer values, default)) 0x1: IN_DERATE (IN_DERATE (Derate offset has been added to timer values))
13	RANK_1_TEMP_OOR	READ ONLY Indicates if RANK 1 has reached a temperature out of the range of the temperature sensor. This is supported by LPDDR2/3 devices only, this register will hold a value of '0' for PCDDR3. 0x0: NOT_OOR (NOT_OOR (default)) 0x1: OOR (OOR)

BIMC_S_DDR0_DPE_DRAM_STATUS_0 (cont.)

Bits	Name	Description
12	RANK_0_TEMP_OOR	READ ONLY Indicates if RANK 1 has reached a temperature out of the range of the temperature sensor. This is supported by LPDDR2/3 devices only, this register will hold a value of '0' for PCDDR3. 0x0: NOT_OOR (NOT_OOR (default)) 0x1: OOR (OOR)
9	RANK_1_CSPD	READ ONLY Indicates Rank 1 is in Clock Stop Power Down. This is supported by LPDDR2/3 devices only, this register will hold a value of '0' for PCDDR3. 0x0: NOT_IN_CSPD (NOT_IN_CSPD (default)) 0x1: IN_CSPD (IN_CSPD)
8	RANK_0_CSPD	READ ONLY Indicates Rank 0 is in Clock Stop Power Down. This is supported by LPDDR2/3 devices only, this register will hold a value of '0' for PCDDR3. 0x0: NOT_IN_CSPD (NOT_IN_CSPD (default)) 0x1: IN_CSPD (IN_CSPD)
5	RANK_1_PWR_DN	READ ONLY Indicates Rank 1 is in Power Down. 0x0: NOT_IN_PWR_DN (NOT_IN_PWR_DN (default)) 0x1: IN_PWR_DN (IN_PWR_DN)
4	RANK_0_PWR_DN	READ ONLY Indicates Rank 0 is in Power Down. 0x0: NOT_IN_PWR_DN (NOT_IN_PWR_DN (default)) 0x1: IN_PWR_DN (IN_PWR_DN)
1	RANK_1_BUSY	READ ONLY Indicates Rank 1 has pending commands in the command queue. 0x0: IN_IDLE (IN_IDLE (default)) 0x1: BUSY (BUSY)
0	RANK_0_BUSY	READ ONLY Indicates Rank 0 has pending commands in the command queue 0x0: IN_IDLE (IN_IDLE (default)) 0x1: BUSY (BUSY)

0x0044C09C BIMC_S_DDR0_DPE_MEMC_STATUS_0**Type:** R**Clock:** ACLK**Reset State:** 0x00000000

DPE Status Register 0.

This register contains information on the state of the memory controller.

BIMC_S_DDR0_DPE_MEMC_STATUS_0

Bits	Name	Description
15:0	RD_FIFO_BYTE_EMPTY	READ ONLY Empty Status of each byte lane in the Read Data FIFO

0x0044C0A0 BIMC_S_DDR0_DPE_MEMC_STATUS_1**Type:** R**Clock:** ACLK**Reset State:** 0x00000000

DPE Status Register 1.

This register contains information on the state of the memory controller. When the timing parameters in DRAM_TIMING registers are stored in multiples of 100ps, time is required for the calculation of DDR clock cycles. This register is the status of that calculation. The calculation is kicked off by either a req/ack interface in the DPE I/Os or writing RECALC ('1') to CONFIG_4[RECALC_PS_PARAMS]. The status bit will go to NO ('1') when the calculation has been kicked off and will go to YES ('0') when all cycle calculations for all DRAM_TIMING registers are complete.

BIMC_S_DDR0_DPE_MEMC_STATUS_1

Bits	Name	Description
0	CYC_CALC_VALID	0x0: YES (YES (The DPE logic is NOT currently calculating DRAM_TIMING values from 100ps intervals into DDR clock cycles)) 0x1: NO (NO (The DPE logic is currently calculating DRAM_TIMING values from 100ps intervals into DDR clock cycles))

0x0044C0A4 BIMC_S_DDR0_DPE_MEMC_STATUS_2**Type:** R**Clock:** ACLK**Reset State:** 0x00000000

DPE Status Register 2.

This register contains information on the state of the memory controller. This register indicates if the active shadow set of registers matches the set of Read/Write registers. Any time the Read/Write registers are written, this bit will go to '0' (STALE) to indicate the active shadow registers are stale with respect to the Read/Write registers. The bit will go back to '1' (UP_TO_DATE) once a '1' is written to the LOAD_ALL_CONFIG register in CONFIG_4 or the new values are loaded into the active shadow set due to a frequency switch. It will remain '1' (UP_TO_DATE) until the next time a Read/Write register is written by software.

BIMC_S_DDR0_DPE_MEMC_STATUS_2

Bits	Name	Description
0	SHADOW_REGS_STALE	0x0: STALE (STALE (Active shadow registers are stale)) 0x1: UP_TO_DATE (UP_TO_DATE (Active shadow registers match Read/Write registers))

0x0044C0B4 BIMC_S_DDR0_DPE_INTERRUPT_ENABLE**Type:** RW**Clock:** ACLK**Reset State:** 0x00000000

This register controls the conditions for the interrupt bit to assert. These registers are used for LPDDR2/3 only, and should be set to DISABLED for PCDDR3 devices.

BIMC_S_DDR0_DPE_INTERRUPT_ENABLE

Bits	Name	Description
6:4	LOW_TEMP_THRESHOLD	The external interrupt will be generated based on the following refresh rate requirements based on low temperature: 0x0: DISABLED (DISABLED (default, interrupt for low temp disabled)) 0x1: TREFI_2X_OR_4X (TREFI_2X_OR_4X (Interrupt is enabled if temp reaches 2X tREFI, 4X tREFI or LOW_LIMIT level)) 0x2: TREFI_4X (TREFI_4X (Interrupt is enabled if temp reaches 4X tREFI or LOW_LIMIT level)) 0x3: TREFI_2X (TREFI_2X (Interrupt is enabled if temp reaches 2X tREFI level)) 0x4: TREFI_1X (TREFI_1X (Interrupt is enabled if temp reaches 1X tREFI level)) 0x5: TREFI_0_5X (TREFI_0.5X (LPDDR3 only, Interrupt is enabled if temp reaches 0.5X tREFI level, 1X tREFI level, 2X tREFI level, 4X tREFI level, or LOW_LIMIT))

BIMC_S_DDR0_DPE_INTERRUPT_ENABLE (cont.)

Bits	Name	Description
2:0	HIGH_TEMP_THRESHOLD	The external interrupt will be generated based on the following refresh rate requirements based on high temperature: 0x0: DISABLED (DISABLED (default, interrupt for high temp disabled)) 0x1: TREFI_0_25X_DERATE (TREFI_0.25X_DERATE (Interrupt is enabled if temp reaches 0.25X tREFI, DERATE, or HIGH_LIMIT level)) 0x2: DERATE (DERATE (Interrupt is enabled if temp reaches derate or HIGH_LIMIT level)) 0x3: TREFI_0_25X (TREFI_0.25X (Interrupt is enabled if temp reaches 0.25X tREFI level)) 0x4: TREFI_1X (TREFI_1X (Interrupt is enabled if temp reaches 1X tREFI level)) 0x5: TREFI_0_5X (TREFI_0.5X (LPDDR3 only, Interrupt is enabled if temp reaches 0.5X tREFI level, 0.25X tREFI level, DERATE level or HIGH_LIMIT level))

0x0044C0B8 BIMC_S_DDR0_DPE_INTERRUPT_STATUS**Type:** R**Clock:** ACLK**Reset State:** 0x00000000

Interrupts are enabled in the INTERRUPT_ENABLE register. This register is read to determine which enabled, interrupt events have occurred and are causing an interrupt assertion. Interrupt will remain asserted until cleared via the INTERRUPT_CLR register. These registers are used for LPDDR2/3 only, and are required to remain at '0' (default value) for PCDDR3 devices.

BIMC_S_DDR0_DPE_INTERRUPT_STATUS

Bits	Name	Description
22:20	RANK1_TEMP_INFO	0x0: LOW_LIMIT (LOW_LIMIT (SDRAM Low temperature operating limit exceeded)) 0x1: TREFI_4X (TREFI_4X (4x tREFI, 4x tREFIpb, 4x tREFW)) 0x2: TREFI_2X (TREFI_2X (2x tREFI, 2x tREFIpb, 2x tREFW)) 0x3: TREFI_1X (TREFI_1X (1x tREFI, 1x tREFIpb, 1x tREFW (<=85C))) 0x4: TREFI_0_5X (TREFI_0.5X (0.5x tREFI, 0.5x tREFIpb, 0.5x tREFW, ONLY for LPDDR3)) 0x5: TREFI_0_25X (TREFI_0.25X (0.25x tREFI, 0.25x tREFIpb, 0.25x tREFW, do not de-rate SDRAM AC timing)) 0x6: TREFI_0_25X_DERATE (TREFI_0.25X_DERATE (0.25x tREFI, 0.25x tREFIpb, 0.25x tREFW, de-rate SDRAM AC timing)) 0x7: HIGH_LIMIT (HIGH_LIMIT (SDRAM HIGH temperature operating limit exceeded))

BIMC_S_DDR0_DPE_INTERRUPT_STATUS (cont.)

Bits	Name	Description
18:16	RANK0_TEMP_INFO	0x0: LOW_LIMIT (LOW_LIMIT (SDRAM Low temperature operating limit exceeded)) 0x1: TREFI_4X (TREFI_4X (4x tREFI, 4x tREFIpb, 4x tREFW)) 0x2: TREFI_2X (TREFI_2X (2x tREFI, 2x tREFIpb, 2x tREFW)) 0x3: TREFI_1X (TREFI_1X (1x tREFI, 1x tREFIpb, 1x tREFW (<=85C))) 0x4: TREFI_0_5X (TREFI_0.5X (0.5x tREFI, 0.5x tREFIpb, 0.5x tREFW, ONLY for LPDDR3)) 0x5: TREFI_0_25X (TREFI_0.25X (0.25x tREFI, 0.25x tREFIpb, 0.25x tREFW, do not de-rate SDRAM AC timing)) 0x6: TREFI_0_25X_DERATE (TREFI_0.25X_DERATE (0.25x tREFI, 0.25x tREFIpb, 0.25x tREFW, de-rate SDRAM AC timing)) 0x7: HIGH_LIMIT (HIGH_LIMIT (SDRAM HIGH temperature operating limit exceeded))
3	RANK1_LOW_TEMP	A '1' indicates that Rank 1's low temperature threshold was enabled and crossed. 0x0: NONE (NONE) 0x1: THRESHLD_CROSSED (THRESHLD_CROSSED (programmed threshold was crossed))
2	RANK1_HIGH_TEMP	A '1' indicates that Rank 1's high temperature threshold was enabled and crossed. 0x0: NONE (NONE) 0x1: THRESHLD_CROSSED (THRESHLD_CROSSED (programmed threshold was crossed))
1	RANK0_LOW_TEMP	A '1' indicates that Rank 0's low temperature threshold was enabled and crossed. 0x0: NONE (NONE) 0x1: THRESHLD_CROSSED (THRESHLD_CROSSED (programmed threshold was crossed))
0	RANK0_HIGH_TEMP	A '1' indicates that Rank 0's high temperature threshold was enabled and crossed. 0x0: NONE (NONE) 0x1: THRESHLD_CROSSED (THRESHLD_CROSSED (programmed threshold was crossed))

0x0044C0BC BIMC_S_DDR0_DPE_INTERRUPT_CLR**Type:** W**Clock:** ACLK**Reset State:** 0x00000000

This register allows control for SW to clear the interrupt. HW will set the interrupt based on the scenarios enabled in INTERRUPT_ENABLE register. Additional information about the interrupt is available via INTERRUPT_STATUS CSRs. The interrupt I/O will remain set (asserted) until SW writes a '1' to the corresponding field in this register. These registers are used for LPDDR2/3 only, and should not be used for PCDDR3 devices.

BIMC_S_DDR0_DPE_INTERRUPT_CLR

Bits	Name	Description
3	RANK1_LOW_TEMP	0x0: NOP (NOP) 0x1: CLR_INT (CLR_INT)
2	RANK1_HIGH_TEMP	0x0: NOP (NOP) 0x1: CLR_INT (CLR_INT)
1	RANK0_LOW_TEMP	0x0: NOP (NOP) 0x1: CLR_INT (CLR_INT)
0	RANK0_HIGH_TEMP	0x0: NOP (NOP) 0x1: CLR_INT (CLR_INT)

0x0044C0C0 BIMC_S_DDR0_DPE_INTERRUPT_SET**Type:** W**Clock:** ACLK**Reset State:** 0x00000000

This register gives SW the capability to force the interrupt to be driven. Writing a '1' to a field in this register will force the interrupt to be driven until cleared by writing the corresponding field in INTERRUPT_CTRL CSR. These registers are used for LPDDR2/3 only, and should be used for PCDDR3 devices.

BIMC_S_DDR0_DPE_INTERRUPT_SET

Bits	Name	Description
3	RANK1_LOW_TEMP	0x0: NOP (NOP) 0x1: SET_INT (SET_INT)
2	RANK1_HIGH_TEMP	0x0: NOP (NOP) 0x1: SET_INT (SET_INT)
1	RANK0_LOW_TEMP	0x0: NOP (NOP) 0x1: SET_INT (SET_INT)
0	RANK0_HIGH_TEMP	0x0: NOP (NOP) 0x1: SET_INT (SET_INT)

0x0044C0D0 BIMC_S_DDR0_DPE_ELEVATE_PRI_RD**Type:** RW**Clock:** ACLK**Reset State:** 0x00000000

This register indicates the number of DDR clock cycles for a read command to be elevated to priority level 1/2/3/4.

BIMC_S_DDR0_DPE_ELEVATE_PRI_RD

Bits	Name	Description
31:24	CYCLES_PRI0	DDR cycles divided by 8 after which a read with current priority level of 0 gets promoted to priority level of 1. A value of 0 indicates that the read command at current priority level 0 will never get promoted to priority level 1.
23:16	CYCLES_PRI1	DDR cycles divided by 8 after which a read with current priority level of 1 gets promoted to priority level of 2. A value of 0 indicates that the read command at current priority level 1 will never get promoted to priority level 2.
15:8	CYCLES_PRI2	DDR cycles divided by 8 after which a read with current priority level of 2 gets promoted to priority level of 3. A value of 0 indicates that the read command at current priority level 2 will never get promoted to priority level 3.
7:0	CYCLES_PRI3	DDR cycles divided by 8 after which a read with current priority level of 3 gets promoted to priority level of 4. A value of 0 indicates that the read command at current priority level 3 will never get promoted to priority level 4.

0x0044C0D4 BIMC_S_DDR0_DPE_ELEVATE_PRI_WR**Type:** RW**Clock:** ACLK**Reset State:** 0x00000000

This register indicates the number of DDR clock cycles for a write command to be elevated to priority level 1/2/3/4.

BIMC_S_DDR0_DPE_ELEVATE_PRI_WR

Bits	Name	Description
31:24	CYCLES_PRI0	DDR cycles divided by 8 after which a write with current priority level of 0 gets promoted to priority level of 1. A value of 0 indicates that the write command at current priority level 0 will never get promoted to priority level 1.
23:16	CYCLES_PRI1	DDR cycles divided by 8 after which a write with current priority level of 1 gets promoted to priority level of 2. A value of 0 indicates that the write command at current priority level 1 will never get promoted to priority level 2.
15:8	CYCLES_PRI2	DDR cycles divided by 8 after which a write with current priority level of 2 gets promoted to priority level of 3. A value of 0 indicates that the read command at current priority level 2 will never get promoted to priority level 3.

BIMC_S_DDR0_DPE_ELEVATE_PRI_WR (cont.)

Bits	Name	Description
7:0	CYCLES_PRI3	DDR cycles divided by 8 after which a write with current priority level of 3 gets promoted to priority level of 4. A value of 0 indicates that the write command at current priority level 3 will never get promoted to priority level 4.

0x0044C0E4 BIMC_S_DDR0_DPE_TIMER_0**Type:** RW**Clock:** ACLK**Reset State:** 0x00000000

DPE Clock cycle timer register 0.

This register contains timer values used for frequency independent timing parameters. Values are in clock cycles.

BIMC_S_DDR0_DPE_TIMER_0

Bits	Name	Description
31:28	WTR_DIFF_RANK	Additional sync time between a write to one Rank and a Read to a different Rank.
27:24	RTW_DIFF_RANK	Additional delay to add to the time interval between a Read to one Rank and a Write to a different Rank. This needs to be set to a minimum of 0010.
23:20	RTR_DIFF_RANK	Additional delay beyond DDR memory burst length/2 between a Read to one Rank and a Read to a different Rank. For LPDDR2/3 and PCDDR3 devices, this can be set to MAX(1, RU((tHZ(DQ)max - tLZ(DQ)min/tck) for optimal setting. tck here represents the time period for the fastest DDR1x clock frequency which will be used.
19:16	WTW_DIFF_RANK	Additional delay beyond DDR memory burst length/2 between a Write to one Rank and a Write to a different Rank.
15:12	RTW_SAME_RANK_ADD_DLY	Additional delay to add to the time interval of a Read then a Write to the same Rank. This needs to be set to a minimum of 0010.
11:8	WTR_SAME_RANK_ADD_DLY	Additional delay beyond DDR memory burst length/2 between a Write to one Rank and a Read to the same Rank.
7:4	RTR_SAME_RANK_ADD_DLY	Additional delay beyond DDR memory burst length/2 between a Read to one Rank and a Read to the same Rank.
3:0	WTW_SAME_RANK_ADD_DLY	Additional delay beyond DDR memory burst length/2 between a Write to one Rank and a Write to the same Rank.

0x0044C0E8 BIMC_S_DDR0_DPE_TIMER_1**Type:** RW**Clock:** ACLK**Reset State:** 0x00000000

DPE Clock cycle timer register 1.

This register contains timer values used for frequency independent timing parameters. These are to be programmed in DDR clock cycles.

BIMC_S_DDR0_DPE_TIMER_1

Bits	Name	Description
13:12	DERATE_OFFSET	Temperature Derating value for DDR memory timing parameters. TRCD, TRAS, TRP and TRRD will be derated to add DERATE_OFFSET extra clock cycles to the normal value. Not used for PCDDR, must program to 0x0. For LPDDR2/3 this must be programmed based on upcoming frequency prior to frequency switch: set to 0x2 for frequencies > 50 MHz set to 0x0 for frequencies <= 50 MHz (Below 50MHz, the min cycle fields will guarantee the derate requirements are met.)
8	TDQSS	DDR Timing parameter tDQSS. This is for LPDDR2/3 only. 0x0: ZERO (ZERO (0 cycles, default, PCDDR3)) 0x1: ONE (ONE (1 cycle, LPDDR2/3))
7:0	IDLE_PWR_DN_CNT	Minimum number of cycles a Rank must be idle before going into Power Down. If this is set to '0', the device will not wait any idle cycles before going into power down. Setting this to a value greater than zero will put the memory device into power down once a count down from the programmed value expires. (0x0 = 0 cycles, 0x1 = 1 cycle etc)

0x0044C0EC BIMC_S_DDR0_DPE_TIMER_2**Type:** RW**Clock:** ACLK**Reset State:** 0x00000000

DPE Clock cycle timer register 2.

This register contains timer values used for frequency independent timing parameters. It is to be programmed in DDR clock cycles.

BIMC_S_DDR0_DPE_TIMER_2

Bits	Name	Description
16:8	TRASMAX_XO_DIV_6_CYCLES	This field is used to define the amount of time to allow for tRASMAX for LPDDR2 and LPDDR3. tRASMAX is the maximum row active time. Generally, tRASMAX is specified as 70 usec for both LPDDR2 and LPDDR3 but refer to the device datasheet for actual requirement. The setting for this value will not vary with the frequency of the DPE core clock as the implementation actually uses the XO clock divided by 6. Set to 0x0 to disable tRASMAX. tRASMAX is enforced automatically via refreshes for PCDDR3 so this should be set to 0x0 for PCDDR3. The value programmed for TRASMAX_XO_DIV_6_CYCLES is $TRASMAX_XO_DIV_6_CYCLES = ROUND_DOWN(tRASMAX / (Txo * 6)) - 6$; where Txo = the period of the tcxo clock. $tRASMAX$ = tRAS(max) value specified in uSec in the specification or data sheet. e.g., $tRASMAX = 70 \text{ uSec}$, $tcxo = 19.2 \text{ MHz}$ gives $RD(70us / 312.5 \text{ ns}) - 6 = 218$ decimal
7:4	TMRW	Mode register write command period. For LPDDR2 devices, this should be set to tMRW. For LPDDR3 devices, this must be set to 'h15. For PCDDR3, this should be set to tMRD.
2:0	TMRR	DDR timing parameter tMRR. Mode register read command Period. This is for LPDDR2/3 only.

0x0044C0F0 BIMC_S_DDR0_DPE_TIMER_3**Type:** RW**Clock:** ACLK**Reset State:** 0x00000000

DPE Clock cycle timer register 3.

This register contains timer values used for frequency independent timing parameters. It is to be programmed in DDR clock cycles.

BIMC_S_DDR0_DPE_TIMER_3

Bits	Name	Description
11:8	BANK_AVAIL_OFFSET	Number of DDR cycles before tRFC expires that DPE will indicate bank(s) being refreshed are available by de asserting corresponding bits in bank_unavail vector.

BIMC_S_DDR0_DPE_TIMER_3 (cont.)

Bits	Name	Description
5:4	CLK_CYC_BEFORE_PD_EXIT	Number of DDR clock cycles the memory clock must toggle before exiting power down. This only applies if the clock was stopped during power down. For LPDDR2/3, set this to 2 cycles. This is for LPDDR2/3 only.
2:0	CLK_CYC_AFTER_PD_ENTRY	Number of DDR clock cycles for memory clock to toggle after power down. For LPDDR2 set this to a minimum of 1 cycle. For LPDDR3 set this to a minimum of 2 cycles. This is for LPDDR2/3 only.

0x0044C114 BIMC_S_DDR0_DPE_MRW_BEFORE_FREQ_SWITCH_0**Type:** RW**Clock:** ACLK**Reset State:** 0x00000000

DPE Mode Register Write before Frequency Switch 0.

This register contains the Mode Register Write address and data information. The MRW will be scheduled to occur before frequency switch, if the operation is valid (indicated by MRW_FREQ_SWITCH). This operation is currently not supported.

BIMC_S_DDR0_DPE_MRW_BEFORE_FREQ_SWITCH_0

Bits	Name	Description
27:20	W_ADDR	Mode Register address.
15:0	W_DATA	Mode Register Write Data.

0x0044C118 BIMC_S_DDR0_DPE_MRW_BEFORE_FREQ_SWITCH_1**Type:** RW**Clock:** ACLK**Reset State:** 0x00000000

DPE Mode Register Write before Frequency Switch 1.

This register contains the Mode Register Write address and data information. The MRW will be scheduled to occur before frequency switch, if the operation is valid (indicated by MRW_FREQ_SWITCH). This operation is currently not supported.

BIMC_S_DDR0_DPE_MRW_BEFORE_FREQ_SWITCH_1

Bits	Name	Description
27:20	W_ADDR	Mode Register address.

BIMC_S_DDR0_DPE_MRW_BEFORE_FREQ_SWITCH_1 (cont.)

Bits	Name	Description
15:0	W_DATA	Mode Register Write Data.

0x0044C11C BIMC_S_DDR0_DPE_MRW_BEFORE_FREQ_SWITCH_2**Type:** RW**Clock:** ACLK**Reset State:** 0x00000000

DPE Mode Register Write before Frequency Switch 2.

This register contains the Mode Register Write address and data information. The MRW will be scheduled to occur before frequency switch, if the operation is valid (indicated by MRW_FREQ_SWITCH). This operation is currently not supported.

BIMC_S_DDR0_DPE_MRW_BEFORE_FREQ_SWITCH_2

Bits	Name	Description
27:20	W_ADDR	Mode Register address.
15:0	W_DATA	Mode Register Write Data.

0x0044C120 BIMC_S_DDR0_DPE_MRW_BEFORE_FREQ_SWITCH_3**Type:** RW**Clock:** ACLK**Reset State:** 0x00000000

DPE Mode Register Write before Frequency Switch 3.

This register contains the Mode Register Write address and data information. The MRW will be scheduled to occur before frequency switch, if the operation is valid (indicated by MRW_FREQ_SWITCH). This operation is currently not supported.

BIMC_S_DDR0_DPE_MRW_BEFORE_FREQ_SWITCH_3

Bits	Name	Description
27:20	W_ADDR	Mode Register address.
15:0	W_DATA	Mode Register Write Data.

0x0044C124 BIMC_S_DDR0_DPE_MRW_AFTER_FREQ_SWITCH_0**Type:** RW**Clock:** ACLK**Reset State:** 0x00000000

DPE Mode Register Write after Frequency Switch 0.

This register contains the Mode Register Write address and data information. The MRW will be scheduled to occur after frequency switch, if the operation is valid (indicated by MRW_FREQ_SWITCH). This is typically used for changing Read/Write latencies on frequency switch.

BIMC_S_DDR0_DPE_MRW_AFTER_FREQ_SWITCH_0

Bits	Name	Description
27:20	W_ADDR	Mode Register address.
15:0	W_DATA	Mode Register Write Data.

0x0044C128 BIMC_S_DDR0_DPE_MRW_AFTER_FREQ_SWITCH_1**Type:** RW**Clock:** ACLK**Reset State:** 0x00000000

DPE Mode Register Write after Frequency Switch 1.

This register contains the Mode Register Write address and data information. The MRW will be scheduled to occur after frequency switch, if the operation is valid (indicated by MRW_FREQ_SWITCH). This is typically used for changing Read/Write latencies on frequency switch.

BIMC_S_DDR0_DPE_MRW_AFTER_FREQ_SWITCH_1

Bits	Name	Description
27:20	W_ADDR	Mode Register address.
15:0	W_DATA	Mode Register Write Data.

0x0044C12C BIMC_S_DDR0_DPE_MRW_AFTER_FREQ_SWITCH_2**Type:** RW**Clock:** ACLK**Reset State:** 0x00000000

DPE Mode Register Write after Frequency Switch 2.

This register contains the Mode Register Write address and data information. The MRW will be scheduled to occur after frequency switch, if the operation is valid (indicated by MRW_FREQ_SWITCH). This is typically used for changing Read/Write latencies on frequency switch.

BIMC_S_DDR0_DPE_MRW_AFTER_FREQ_SWITCH_2

Bits	Name	Description
27:20	W_ADDR	Mode Register address.
15:0	W_DATA	Mode Register Write Data.

0x0044C130 BIMC_S_DDR0_DPE_MRW_AFTER_FREQ_SWITCH_3

Type: RW

Clock: ACLK

Reset State: 0x00000000

DPE Mode Register Write after Frequency Switch 3.

This register contains the Mode Register Write address and data information. The MRW will be scheduled to occur after frequency switch, if the operation is valid (indicated by MRW_FREQ_SWITCH). This is typically used for changing Read/Write latencies on frequency switch.

BIMC_S_DDR0_DPE_MRW_AFTER_FREQ_SWITCH_3

Bits	Name	Description
27:20	W_ADDR	Mode Register address.
15:0	W_DATA	Mode Register Write Data.

0x0044C134 BIMC_S_DDR0_DPE_MRW_FREQ_SWITCH

Type: RW

Clock: ACLK

Reset State: 0x00000000

DPE Mode Register Write for Frequency Switch

This register indicates which Mode Register Writes are valid to execute upon a Frequency Switch event.

BIMC_S_DDR0_DPE_MRW_FREQ_SWITCH

Bits	Name	Description
19	BEFORE_3	Setting this to '1' indicates a Mode register write stored in MRW_BEFORE_FREQ_SWITCH_3 is valid and will be carried out upon a frequency switch event. This is Currently not supported. 0x0: NOP (NOP (default)) 0x1: EXEC (EXEC (perform the write))
18	BEFORE_2	Setting this to '1' indicates a Mode register write stored in MRW_BEFORE_FREQ_SWITCH_2 is valid and will be carried out upon a frequency switch event. This is Currently not supported. 0x0: NOP (NOP (default)) 0x1: EXEC (EXEC (perform the write))
17	BEFORE_1	Setting this to '1' indicates a Mode register write stored in MRW_BEFORE_FREQ_SWITCH_1 is valid and will be carried out upon a frequency switch event. This is Currently not supported. 0x0: NOP (NOP (default)) 0x1: EXEC (EXEC (perform the write))
16	BEFORE_0	Setting this to '1' indicates a Mode register write stored in MRW_BEFORE_FREQ_SWITCH_0 is valid and will be carried out upon a frequency switch event. This is Currently not supported. 0x0: NOP (NOP (default)) 0x1: EXEC (EXEC (perform the write))
3	AFTER_3	Setting this to '1' indicates a Mode register write stored in MRW_AFTER_FREQ_SWITCH_3 is valid and will be carried out upon a frequency switch event. 0x0: NOP (NOP (default)) 0x1: EXEC (EXEC (perform the write))
2	AFTER_2	Setting this to '1' indicates a Mode register write stored in MRW_AFTER_FREQ_SWITCH_2 is valid and will be carried out upon a frequency switch event. 0x0: NOP (NOP (default)) 0x1: EXEC (EXEC (perform the write))
1	AFTER_1	Setting this to '1' indicates a Mode register write stored in MRW_AFTER_FREQ_SWITCH_1 is valid and will be carried out upon a frequency switch event. 0x0: NOP (NOP (default)) 0x1: EXEC (EXEC (perform the write))
0	AFTER_0	Setting this to '1' indicates a Mode register write stored in MRW_AFTER_FREQ_SWITCH_0 is valid and will be carried out upon a frequency switch event. 0x0: NOP (NOP (default)) 0x1: EXEC (EXEC (perform the write))

0x0044C144 BIMC_S_DDR0_DPE_DRAM_TIMING_0**Type:** RW**Clock:** ACLK**Reset State:** 0x00000000

This register contains the parameters which are specified by the memory vendor in ns. These values are not used by the design, until they are loaded into the active register DRAM_TIMING_0_ACT by setting CONFIG_4[LOAD_ALL_CONFIG] to '1' or when CONFIG_3[LOAD_TIMING_ON_FREQ_CHNG] is set to '1' and a frequency change occurs.

The value of the parameter contained in the registers can be in DDR clock cycles or in multiples of 100ps, and may vary from one SDRAM vendor to another.

NOTE The following applies to all DRAM_TIMING_* CSRs

If the timing parameter value is only specified in clock cycles as per the JEDEC specification, then it is recommended to set that specific timing parameter in cycle based mode in CONIFG_5 CSR and program the timing parameter field as the cycle value, while the MIN_CYC value for that parameter would be 0 (SW could also program the MIN_CYC value if the parameter has one specified in the JEDEC specification).

If the timing parameter value is only specified in ns as per the JEDEC specification, then it is recommended to set that specific timing parameter in time based mode in CONIFG_5 CSR and to program the timing parameter field in multiples of 100ps and leave the MIN_CYC field (for that timing parameter) as 0.

If the timing parameter value is specified such that it has a minimum cycle time requirement as well as a timing parameter value in ns as per the JEDEC specification, then it is recommended to set that specific timing parameter in time based mode in CONIFG_5 CSR and to program the timing parameter field in multiples of 100ps and program the MIN_CYC field to reflect the minimum cycle time requirement for that timing parameter.

BIMC_S_DDR0_DPE_DRAM_TIMING_0

Bits	Name	Description
15:12	TRASMIN_MIN_CYC	Minimum TRASMIN value in DDR clock cycles.
9:0	TRASMIN	DDR timing parameter tRASmin.

0x0044C148 BIMC_S_DDR0_DPE_DRAM_TIMING_1**Type:** RW**Clock:** ACLK**Reset State:** 0x00000000

This register contains the parameters which are specified by the memory vendor in ns. These values are not used by the design, until they are loaded into the active register DRAM_TIMING_1_ACT by setting CONFIG_4[LOAD_ALL_CONFIG] to '1' or when CONFIG_3[LOAD_TIMING_ON_FREQ_CHNG] is set to '1' and a frequency change occurs.

The value of the parameter contained in the registers can be in DDR clock cycles or in multiples of 100ps, and may vary from one SDRAM vendor to another.

Refer to the note in DRAM_TIMING_0 CSR.

BIMC_S_DDR0_DPE_DRAM_TIMING_1

Bits	Name	Description
31:28	TWR_MIN_CYC	Minimum TWR value in DDR clock cycles.
23:16	TWR	DDR timing parameter tWR, write recovery time.
15:12	TRCD_MIN_CYC	Minimum TRCD value in DDR clock cycles.
8:0	TRCD	DDR timing parameter tRCD.

0x0044C14C BIMC_S_DDR0_DPE_DRAM_TIMING_2

Type: RW

Clock: ACLK

Reset State: 0x00000000

This register contains the parameters which are specified by the memory vendor in ns. These values are not used by the design, until they are loaded into the active register DRAM_TIMING_2_ACT by setting CONFIG_4[LOAD_ALL_CONFIG] to '1' or when CONFIG_3[LOAD_TIMING_ON_FREQ_CHNG] is set to '1' and a frequency change occurs.

The value of the parameter contained in the registers can be in DDR clock cycles or in multiples of 100ps, and may vary from one SDRAM vendor to another.

Refer to the note in DRAM_TIMING_0 CSR.

BIMC_S_DDR0_DPE_DRAM_TIMING_2

Bits	Name	Description
31:28	TWTR_MIN_CYC	Minimum TWTR value in DDR clock cycles.
22:16	TWTR	DDR timing parameter tWTR.
15:12	TRRD_MIN_CYC	Minimum TRRD value in DDR clock cycles.
7:0	TRRD	DDR timing parameter tRRD. Number of DDR clock cycles between Activate to Activate (different banks on the same chip select).

0x0044C150 BIMC_S_DDR0_DPE_DRAM_TIMING_3

Type: RW

Clock: ACLK

Reset State: 0x00000000

This register contains the parameters which are specified by the memory vendor in ns. These values are not used by the design, until they are loaded into the active register DRAM_TIMING_3_ACT by setting CONFIG_4[LOAD_ALL_CONFIG] to '1' or when CONFIG_3[LOAD_TIMING_ON_FREQ_CHNG] is set to '1' and a frequency change occurs.

The value of the parameter contained in the registers can be in DDR clock cycles or in multiples of 100ps, and may vary from one SDRAM vendor to another.

Refer to the note in DRAM_TIMING_0 CSR.

BIMC_S_DDR0_DPE_DRAM_TIMING_3

Bits	Name	Description
31:28	TRFCAB_MIN_CYC	Minimum TRFCAB value in DDR clock cycles.
27:16	TRFCAB	DDR timing parameter tRFCab. Number of DDR clock cycles between auto-refresh command and the next activate command.
15:12	TRFCPB_MIN_CYC	Minimum TRFCPB value in DDR clock cycles.
11:0	TRFCPB	DDR timing parameter tRFCpb. Number of DDR clock cycles between a per-bank refresh command and the next activate command. This is currently not supported.

0x0044C154 BIMC_S_DDR0_DPE_DRAM_TIMING_4

Type: RW

Clock: ACLK

Reset State: 0x00000000

This register contains the parameters which are specified by the memory vendor in ns. These values are not used by the design, until they are loaded into the active register DRAM_TIMING_4_ACT by setting CONFIG_4[LOAD_ALL_CONFIG] to '1' or when CONFIG_3[LOAD_TIMING_ON_FREQ_CHNG] is set to '1' and a frequency change occurs.

The value of the parameter contained in the registers can be in DDR clock cycles or in multiples of 100ps, and may vary from one SDRAM vendor to another.

Refer to the note in DRAM_TIMING_0 CSR.

BIMC_S_DDR0_DPE_DRAM_TIMING_4

Bits	Name	Description
15:12	TRTP_MIN_CYC	Minimum TRTP value in DDR clock cycles. *note* Current design restriction requires the minimum value for this parameter to be 4 (0100).
6:0	TRTP	DDR timing parameter tRTP. Number of additional DDR clock cycles between a read and a precharge command to the same page with DDR BL/2 as the initial base delay.

0x0044C158 BIMC_S_DDR0_DPE_DRAM_TIMING_5**Type:** RW**Clock:** ACLK**Reset State:** 0x00000000

This register contains the parameters which are specified by the memory vendor in ns. These values are not used by the design, until they are loaded into the active register DRAM_TIMING_5_ACT by setting CONFIG_4[LOAD_ALL_CONFIG] to '1' or when CONFIG_3[LOAD_TIMING_ON_FREQ_CHNG] is set to '1' and a frequency change occurs.

The value of the parameter contained in the registers can be in DDR clock cycles or in multiples of 100ps, and may vary from one SDRAM vendor to another.

Refer to the note in DRAM_TIMING_0 CSR.

BIMC_S_DDR0_DPE_DRAM_TIMING_5

Bits	Name	Description
31:28	TRPAB_MIN_CYC	Minimum TRPAB value in DDR cycles.
24:16	TRPAB	DDR timing parameter tRPab. Duration of time from all bank Precharge to Activate command to the same bank. PCDDR3 devices do not differentiate between per-bank and all-bank precharge timing, in that case program this to TRP. For LPDDR2/3, set this timing parameter to tRPab.
15:12	TRPPB_MIN_CYC	Minimum TRPPB value in DDR cycles.
8:0	TRPPB	DDR timing parameter tRPpb. Duration of time from per bank Precharge to Activate command to the same bank. PCDDR3 devices do not differentiate between per-bank and all-bank precharge timing, in that case program this to TRP. For LPDDR2/3, set this timing parameter to tRPpb.

0x0044C15C BIMC_S_DDR0_DPE_DRAM_TIMING_6**Type:** RW**Clock:** ACLK**Reset State:** 0x00000000

This register contains the parameters which are specified by the memory vendor in ns. These values are not used by the design, until they are loaded into the active register DRAM_TIMING_6_ACT by setting CONFIG_4[LOAD_ALL_CONFIG] to '1' or when CONFIG_3[LOAD_TIMING_ON_FREQ_CHNG] is set to '1' and a frequency change occurs.

The value of the parameter contained in the registers can be in DDR clock cycles or in multiples of 100ps, and may vary from one SDRAM vendor to another.

Refer to the note in DRAM_TIMING_0 CSR.

BIMC_S_DDR0_DPE_DRAM_TIMING_6

Bits	Name	Description
31:28	TFAW_MIN_CYC	Minimum TFAW value in DDR clock cycles.
25:16	TFAW	DDR timing parameter tFAW. Programmable rolling window in time during which no more than 4 activates are allowed.
15:12	TCKE_MIN_CYC	Minimum TCKE value in DDR clock cycles.
7:0	TCKE	DDR timing parameter tCKE. Minimum CKE pulse width (High or Low).

0x0044C160 BIMC_S_DDR0_DPE_DRAM_TIMING_7**Type:** RW**Clock:** ACLK**Reset State:** 0x00000000

This register contains the parameters which are specified by the memory vendor in ns. These values are not used by the design, until they are loaded into the active register DRAM_TIMING_7_ACT by setting CONFIG_4[LOAD_ALL_CONFIG] to '1' or when CONFIG_3[LOAD_TIMING_ON_FREQ_CHNG] is set to '1' and a frequency change occurs.

The value of the parameter contained in the registers can be in DDR clock cycles or in multiples of 100ps, and may vary from one SDRAM vendor to another.

Refer to the note in DRAM_TIMING_0 CSR.

BIMC_S_DDR0_DPE_DRAM_TIMING_7

Bits	Name	Description
22:16	TZQCS_MIN_CYC	Minimum TZQCS value in DDR clock cycles.
10:0	TZQCS	Number of DDR clock cycles required for ZQ short calibration. For LPDDR2/3, this is the timing parameter tZQCS. For PCDDR3, this is the timing Parameter tZQCS.

0x0044C164 BIMC_S_DDR0_DPE_DRAM_TIMING_8**Type:** RW**Clock:** ACLK**Reset State:** 0x00000000

This register contains the parameters which are specified by the memory vendor in ns. These values are not used by the design, until they are loaded into the active register DRAM_TIMING_8_ACT by setting CONFIG_4[LOAD_ALL_CONFIG] to '1' or when CONFIG_3[LOAD_TIMING_ON_FREQ_CHNG] is set to '1' and a frequency change occurs.

The value of the parameter contained in the registers can be in DDR clock cycles or in multiples of 100ps, and may vary from one SDRAM vendor to another.

Refer to the note in DRAM_TIMING_0 CSR.

BIMC_S_DDR0_DPE_DRAM_TIMING_8

Bits	Name	Description
15:12	TMOD_MIN_CYC	Minimum TMOD value in DDR clock cycles.
10:0	TMOD	Specifies the timing parameter tMOD in DDR clock cycles for PCDDR3 devices

0x0044C168 BIMC_S_DDR0_DPE_DRAM_TIMING_9

Type: RW

Clock: ACLK

Reset State: 0x00000000

This register contains the parameters which are specified by the memory vendor in ns. These values are not used by the design, until they are loaded into the active register DRAM_TIMING_9_ACT by setting CONFIG_4[LOAD_ALL_CONFIG] to '1' or when CONFIG_3[LOAD_TIMING_ON_FREQ_CHNG] is set to '1' and a frequency change occurs.

The value of the parameter contained in the registers can be in DDR clock cycles or in multiples of 100ps, and may vary from one SDRAM vendor to another.

Refer to the note in DRAM_TIMING_0 CSR.

BIMC_S_DDR0_DPE_DRAM_TIMING_9

Bits	Name	Description
15:12	MIN_SR_DURATION_MIN_CYC	Minimum MIN_SR_DURATION value in DDR clock cycles.
7:0	MIN_SR_DURATION	Specifies the minimum time the SDRAM device must remain in self refresh mode once a self refresh command is received. For LPDDR2/3, this is specified as tCKESR. For PCDDR3, this is specified as tCKE.

0x0044C16C BIMC_S_DDR0_DPE_DRAM_TIMING_10

Type: RW

Clock: ACLK

Reset State: 0x00000000

This register contains the parameters which are specified by the memory vendor in ns. These values are not used by the design, until they are loaded into the active register

DRAM_TIMING_10_ACT by setting CONFIG_4[LOAD_ALL_CONFIG] to '1' or when CONFIG_3[LOAD_TIMING_ON_FREQ_CHNG] is set to '1' and a frequency change occurs.

The value of the parameter contained in the registers can be in DDR clock cycles or in multiples of 100ps, and may vary from one SDRAM vendor to another.

Refer to the note in DRAM_TIMING_0 CSR.

BIMC_S_DDR0_DPE_DRAM_TIMING_10

Bits	Name	Description
31:28	TXSRD_MIN_CYC	Minimum TXSRD value in DDR clock cycles.
27:16	TXSRD	Time between self refresh exit and next valid Read command. This needs to be set based on the specific SDRAM device and the PHY generation as follows: LPDDR2/3, Gen1 PHY (with pad power mode support required): max(175ns, tXSR) LPDDR2/3, Gen2 PHY or Gen1 PHY (with no support for pad power mode): tXSR PCDDR3: tXSDLL (which is in cycles only)
15:12	TXSNR_MIN_CYC	Minimum TXSNR value in DDR clock cycles.
11:0	TXSNR	Number of DDR clock cycles between self refresh exit and next valid Non-Read command. This needs to be set based on the specific SDRAM device. For LPDDR2/3 devices, the value programmed in TXSNR should be tXSR. For PCDDR3, this needs to be programmed with tXS timing parameter.

0x0044C170 BIMC_S_DDR0_DPE_DRAM_TIMING_11

Type: RW

Clock: ACLK

Reset State: 0x00000000

This register contains the parameters which are specified by the memory vendor in ns. These values are not used by the design, until they are loaded into the active register DRAM_TIMING_11_ACT by setting CONFIG_4[LOAD_ALL_CONFIG] to '1' or when CONFIG_3[LOAD_TIMING_ON_FREQ_CHNG] is set to '1' and a frequency change occurs.

The value of the parameter contained in the registers can be in DDR clock cycles or in multiples of 100ps, and may vary from one SDRAM vendor to another.

Refer to the note in DRAM_TIMING_0 CSR.

BIMC_S_DDR0_DPE_DRAM_TIMING_11

Bits	Name	Description
31:28	TXPNR_ACT_PWR_DN_MIN_CYC	Minimum TXPNR_ACT_PWR_DN value in DDR clock cycles.
23:16	TXPNR_ACT_PWR_DN	Number of cycles between power down exit and next valid Non-Read command. This is used only if the Power down state was active power down. This distinction is required for PCDDR devices where in a slow/fast exit option the Mode Register could potentially mean different timers for read commands (which require a DLL) vs. a non-read command. For LPDDR2/3, program this to 'txPNR' if available, else set to 'txP'. For PCDDR3, set this to 'txP'.
15:12	TXPR_ACT_PWR_DN_MIN_CYC	Minimum TXPR_ACT_PWR_DN value in DDR clock cycles.
8:0	TXPR_ACT_PWR_DN	Number of DDR clock cycles between power down exit and next read command. This value gets used only if the Power down state was active power down. For LPDDR2/3, program this to 'txPR' if available, else set to 'txP'. For PCDDR3, set this to 'txP'.

0x0044C174 BIMC_S_DDR0_DPE_DRAM_TIMING_12**Type:** RW**Clock:** ACLK**Reset State:** 0x00000000

This register contains the parameters which are specified by the memory vendor in ns. These values are not used by the design, until they are loaded into the active register DRAM_TIMING_12_ACT by setting CONFIG_4[LOAD_ALL_CONFIG] to '1' or when CONFIG_3[LOAD_TIMING_ON_FREQ_CHNG] is set to '1' and a frequency change occurs.

The value of the parameter contained in the registers can be in DDR clock cycles or in multiples of 100ps, and may vary from one SDRAM vendor to another.

Refer to the note in DRAM_TIMING_0 CSR.

BIMC_S_DDR0_DPE_DRAM_TIMING_12

Bits	Name	Description
31:28	TXPNR_PCHG_PWR_DN_MIN_CYC	Minimum TXPNR_PCHG_PWR_DN value in DDR clock cycles.
23:16	TXPNR_PCHG_PWR_DN	Number of cycles between power down exit and next valid Non-Read command. This is used only if the Power down state was precharge power down. For LPDDR2/3, program this to 'txPNR' if available, else set to 'txP'. For PCDDR3, set this to 'txP'.
15:12	TXPR_PCHG_PWR_DN_MIN_CYC	Minimum TXPR_PCHG_PWR_DN value in DDR clock cycles.

BIMC_S_DDR0_DPE_DRAM_TIMING_12 (cont.)

Bits	Name	Description
8:0	TXPR_PCHG_PWR_DN	Number of DDR clock cycles between power down exit and next read command. This value gets used only if the Power down state was precharge power down. For LPDDR2/3, program this to 'TXPR' if available, else set to 'tXP'. For PCDDR3, if Mode register for 'DLL control for Precharge PD' is programmed to 'fast exit', set this memory timing parameter to 'tXP', else program this to 'tXPDLL'.

0x0044C178 BIMC_S_DDR0_DPE_DRAM_TIMING_13**Type:** RW**Clock:** ACLK**Reset State:** 0x00000000

This register contains the parameters which are specified by the memory vendor in ns. These values are not used by the design until they are loaded in the active register

DRAM_TIMING_13_ACT by setting CONFIG_4[LOAD_ALL_CONFIG] to '1' or when CONFIG_3[LOAD_TIMING_ON_FREQ_CHNG] is set to '1' and a frequency change occurs

The value of the parameter contained in the registers can be in DDR clock cycles or in multiples of 100ps, and may vary from one SDRAM vendor to another.

Refer to the note in DRAM_TIMING_0 CSR.

BIMC_S_DDR0_DPE_DRAM_TIMING_13

Bits	Name	Description
31:28	CLK_AFTER_SR_ENTRY_MIN_CYC	Minimum number of DDR clock cycles for memory clock to toggle after entering self refresh. For LPDDR2 devices, this should be set to '1', and for LPDDR3 devices, this should be set to '2'. For PCDDR3, this should be set to 5 DDR clock cycles.
22:16	CLK_AFTER_SR_ENTRY	Duration of time the DDR clock must toggle after entering self refresh. For LPDDR2 devices, this should be set to '1', and for LPDDR3 devices, this should be set to '2'. For PCDDR3, this should be set to ddr timing parameter tCKSRE.
15:12	CLK_BEFORE_SR_EXIT_MIN_CYC	Minimum number of DDR clock cycles the DDR clock must toggle before exiting self refresh. For LPDDR2/3, this needs to be set to at least 2 DDR clock cycles. For PCDDR3, this needs to be set to 5 DDR clock cycles.

BIMC_S_DDR0_DPE_DRAM_TIMING_13 (cont.)

Bits	Name	Description
10:0	CLK_BEFORE_SR_EXIT	Required duration of time DDR clock must toggle before exiting self refresh. For LPDDR2/3, this needs to be set to at least 2 DDR clock cycles. For PCDDR3, this needs to be set to DDR timing parameter tCKSRX. In addition, an extra delay of 10-20ns would need to be applied to this CSR, if using the pad low power mode ie if CONFIG_1[PAD_POWER_MODE] is 1. Furthermore, if the current mode in the pads is enabled and the system is configured to dynamically turn off the current mode during self refresh, then the timer needs to be programmed such that it provides at least 100ns of pad settling time.

0x0044C17C BIMC_S_DDR0_DPE_DRAM_TIMING_14**Type:** RW**Clock:** ACLK**Reset State:** 0x00000000

This register contains timing values for the pad input enable windows for capturing read data from the DDR memory. These values are not used by the design until they are loaded in the active register DRAM_TIMING_14_ACT by setting CONFIG_4[LOAD_ALL_CONFIG] to '1' or when CONFIG_3[LOAD_TIMING_ON_FREQ_CHNG] is set to '1' and a frequency change occurs.

The value of the parameter may vary from one SDRAM vendor to another.

Refer to the note in DRAM_TIMING_0 CSR.

BIMC_S_DDR0_DPE_DRAM_TIMING_14

Bits	Name	Description
31:28	IE_WINDOW_START_ADD_CYCLES	Additional clock cycles to delay the start of the pad input enable window.
25:24	IE_WINDOW_START_SUB_CYCLES	Number of clock cycles to subtract from the delay of the start of the pad input enable window. Allows the IE window to start up to 3 cycles sooner than RL cycles beyond the CAS command. For PCDDR3 devices, this is required to be set to 1 as the first DQS edge could arrive slightly before RL expires.
23:16	IE_WINDOW_START	Amount of time to delay the DQ/DQS pad input enable window beyond CAS cycle + Read Latency. The maximum value for this register is tDQSkmin + any routing delays to get to the pad + delays in the read data path from memory to the DQ pad (board/package delays) - pad IE turn on delay.
15:12	IE_WINDOW_END_ADD_CYCLES	Additional clock cycles to extend the pad input enable window.
7:0	IE_WINDOW_END	Amount of time to extend the DQ/DQS pad input enable window beyond CAS Cycle + Read Latency + burst length/2. This can be set to a minimum of tDQSKmax + delays in the read data path from memory to the DQ pad (board/package delays) + 500ps.

0x0044C180 BIMC_S_DDR0_DPE_DRAM_TIMING_15**Type:** RW**Clock:** ACLK**Reset State:** 0x00000000

This register contains timing values for the read fifo capture windows for capturing read data from the DDR memory. These values are not used by the design until they are loaded in the active register DRAM_TIMING_15_ACT by setting CONFIG_4[LOAD_ALL_CONFIG] to '1' or when CONFIG_3[LOAD_TIMING_ON_FREQ_CHNG] is set to '1' and a frequency change occurs

The value of the parameter may vary from one SDRAM vendor to another.

Refer to the note in DRAM_TIMING_0 CSR.

BIMC_S_DDR0_DPE_DRAM_TIMING_15

Bits	Name	Description
31:28	RD_CAPTURE_START_AD D_CYCLES	Additional clock cycles to delay the start of the read capture window. Typically, this needs to be set to the number of IOSTAGE pipeline registers placed between the memory controller and the PHY.
23:16	RD_CAPTURE_START	Amount of time to delay the start of the read capture window beyond the CAS cycle + Read Latency of the memory device. This can be set to a maximum of tDQSKmin + any round trip delays.
15:12	RD_CAPTURE_END_ADD_ CYCLES	Additional clock cycles to extend the read capture window. Typically, this needs to be set to the number of IOSTAGE pipeline registers placed between the memory controller and the PHY.
7:0	RD_CAPTURE_END	Amount of time to extend the read capture window beyond the CAS cycle + Read Latency + burst length/2 + delays in the read data path from the memory to the controller (includes board/package delays + delays in the PHY). This can be set to a minimum of tDQSKmax + any round trip delays + 500ps.

0x0044C184 BIMC_S_DDR0_DPE_DRAM_TIMING_16**Type:** RW**Clock:** ACLK**Reset State:** 0x00000000

This register contains timing values required by the memory vendor in DDR clock cycles. These values are not used by the design until they are loaded in the active register DRAM_TIMING_16_ACT by setting CONFIG_4[LOAD_ALL_CONFIG] to '1' or when CONFIG_3[LOAD_TIMING_ON_FREQ_CHNG] is set to '1' and a frequency change occurs

The value of the parameter may vary from one SDRAM vendor to another.

BIMC_S_DDR0_DPE_DRAM_TIMING_16

Bits	Name	Description
29:28	ODTON_MAX_ADD_CYCLE S	Additional clock cycles to add to the DRAM_TIMING_23[ODTON_MAX] timing parameter. ODT is supported for PCDDR3 and LPDDR3 devices only. The only supported values for this field are '0' and '1'.
25:24	ODTOFF_MIN_SUB_CYCLE S	Additional clock cycles to add to the DRAM_TIMING_23[ODTOFF_MIN] timing parameter. ODT is supported for PCDDR3 and LPDDR3 devices only. The only supported values for this field are '0' and '1'.
20	PAD_MODE	Indicates the power mode to be used for the PHY based on the frequency of operation. This value in the CSR is only valid if CONFIG_1[PAD_POWER_MODE] is enabled and CONFIG_1[PAD_POWER_MODE_CTRL] is set to SW_CTRL. 0x0: LOW_POWER_MODE (Low Power Mode (default)) 0x1: HIGH_POWER_MODE (High Power Mode)
19:16	TCCD	DDR timing parameter tCCD min, CAS to CAS delay. This is fixed at 2 for LPDDR2, 4 for LPDDR3 and PCDDR3.
13:8	RD_LATENCY	Read latency of the memory device.
5:0	WR_LATENCY	Write latency of the memory device.

0x0044C188 BIMC_S_DDR0_DPE_DRAM_TIMING_17**Type:** RW**Clock:** ACLK**Reset State:** 0x00000000

This register contains timing values required by the memory vendor in DDR clock cycles. These values are not used by the design until they are loaded in the active register DRAM_TIMING_17_ACT by setting CONFIG_4[LOAD_ALL_CONFIG] to '1' or when CONFIG_3[LOAD_TIMING_ON_FREQ_CHNG] is set to '1' and a frequency change occurs

The value of the parameter may vary from one SDRAM vendor to another.

Refer to the note in DRAM_TIMING_0 CSR.

BIMC_S_DDR0_DPE_DRAM_TIMING_17

Bits	Name	Description
25:16	TZQCL_MIN_CYC	Minimum TZQCL value in DDR clock cycles.
11:0	TZQCL	Number of DDR clock cycles required for the ZQ long calibration sequence to execute. For LPDDR2/3, this is the timing parameter tZQCL. For PCDDR3, this is the timing parameter TZQoper.

0x0044C18C BIMC_S_DDR0_DPE_DRAM_TIMING_18**Type:** RW**Clock:** ACLK**Reset State:** 0x00000000

This register contains values for read enable control in DDR clock cycles. These values are not used by the design until they are loaded in the active register DRAM_TIMING_18_ACT by setting CONFIG_4[LOAD_ALL_CONFIG] to '1' or when CONFIG_3[LOAD_TIMING_ON_FREQ_CHNG] is set to '1' and a frequency change occurs. The contents of this CSR is only valid for x64 DRAM devices.

Note that the byte numbers below may not correspond to the actual DRAM byte. This is required so as to keep the byte number reference correct between the controller and the PHY (which has no information on the byte swap feature)

BIMC_S_DDR0_DPE_DRAM_TIMING_18

Bits	Name	Description
28:24	RCW_START_DLY_BYTE7	Enables the external read enable n cycles from the CAS cycle for the read command. The minimum value supported is 1 cycle.
20:16	RCW_START_DLY_BYTE6	Enables the external read enable n cycles from the CAS cycle for the read command. The minimum value supported is 1 cycle.
12:8	RCW_START_DLY_BYTE5	Enables the external read enable n cycles from the CAS cycle for the read command. The minimum value supported is 1 cycle.
4:0	RCW_START_DLY_BYTE4	Enables the external read enable n cycles from the CAS cycle for the read command. The minimum value supported is 1 cycle.

0x0044C190 BIMC_S_DDR0_DPE_DRAM_TIMING_19**Type:** RW**Clock:** ACLK**Reset State:** 0x00000000

This register contains values for read enable control in DDR clock cycles. These values are not used by the design until they are loaded in the active register DRAM_TIMING_19_ACT by setting CONFIG_4[LOAD_ALL_CONFIG] to '1' or when CONFIG_3[LOAD_TIMING_ON_FREQ_CHNG] is set to '1' and a frequency change occurs.

Note that the byte numbers below may not correspond to the actual DRAM byte. This is required so as to keep the byte number reference correct between the controller and the PHY (which has no information on the byte swap feature)

BIMC_S_DDR0_DPE_DRAM_TIMING_19

Bits	Name	Description
28:24	RCW_START_DLY_BYTE3	Enables the external read enable n cycles from the CAS cycle for the read command. The minimum value supported is 1 cycle.
20:16	RCW_START_DLY_BYTE2	Enables the external read enable n cycles from the CAS cycle for the read command. The minimum value supported is 1 cycle.
12:8	RCW_START_DLY_BYTE1	Enables the external read enable n cycles from the CAS cycle for the read command. The minimum value supported is 1 cycle.
4:0	RCW_START_DLY_BYTE0	Enables the external read enable n cycles from the CAS cycle for the read command. The minimum value supported is 1 cycle.

0x0044C194 BIMC_S_DDR0_DPE_DRAM_TIMING_20**Type:** RW**Clock:** ACLK**Reset State:** 0x00000000

This register contains fine grain delay values for read enable control required by PHY. These values are not used by the design until they are loaded in the active register DRAM_TIMING_20_ACT by setting CONFIG_4[LOAD_ALL_CONFIG] to '1' or when CONFIG_3[LOAD_TIMING_ON_FREQ_CHNG] is set to '1' and a frequency change occurs. The contents of this CSR is only valid for x64 DRAM devices.

Note that the byte numbers below may not correspond to the actual DRAM byte. This is required so as to keep the byte number reference correct between the controller and the PHY (which has no information on the byte swap feature)

BIMC_S_DDR0_DPE_DRAM_TIMING_20

Bits	Name	Description
31:24	RCW_FINE_DLY_BYTE7	This value controls the fine-grain delay which will be applied by the PHY for the external read enable signal for precise DQS capture from the PCDDR3/LPDDR3 DRAM device.
23:16	RCW_FINE_DLY_BYTE6	This value controls the fine-grain delay which will be applied by the PHY for the external read enable signal for precise DQS capture from the PCDDR3/LPDDR3 DRAM device.
15:8	RCW_FINE_DLY_BYTE5	This value controls the fine-grain delay which will be applied by the PHY for the external read enable signal for precise DQS capture from the PCDDR3/LPDDR3 DRAM device.
7:0	RCW_FINE_DLY_BYTE4	This value controls the fine-grain delay which will be applied by the PHY for the external read enable signal for precise DQS capture from the PCDDR3/LPDDR3 DRAM device.

0x0044C198 BIMC_S_DDR0_DPE_DRAM_TIMING_21**Type:** RW**Clock:** ACLK**Reset State:** 0x00000000

This register contains fine grain delay values for read enable control required by PHY. These values are not used by the design until they are loaded in the active register DRAM_TIMING_21_ACT by setting CONFIG_4[LOAD_ALL_CONFIG] to '1' or when CONFIG_3[LOAD_TIMING_ON_FREQ_CHNG] is set to '1' and a frequency change occurs.

Note that the byte numbers below may not correspond to the actual DRAM byte. This is required so as to keep the byte number reference correct between the controller and the PHY (which has no information on the byte swap feature)

BIMC_S_DDR0_DPE_DRAM_TIMING_21

Bits	Name	Description
31:24	RCW_FINE_DLY_BYTE3	This value controls the fine-grain delay which will be applied by the PHY for the external read enable signal for precise DQS capture from the PCDDR3/LPDDR3 DRAM device.
23:16	RCW_FINE_DLY_BYTE2	This value controls the fine-grain delay which will be applied by the PHY for the external read enable signal for precise DQS capture from the PCDDR3/LPDDR3 DRAM device.
15:8	RCW_FINE_DLY_BYTE1	This value controls the fine-grain delay which will be applied by the PHY for the external read enable signal for precise DQS capture from the PCDDR3/LPDDR3 DRAM device.
7:0	RCW_FINE_DLY_BYTE0	This value controls the fine-grain delay which will be applied by the PHY for the external read enable signal for precise DQS capture from the PCDDR3/LPDDR3 DRAM device.

0x0044C19C BIMC_S_DDR0_DPE_DRAM_TIMING_22**Type:** RW**Clock:** ACLK**Reset State:** 0x00000000

This register contains coarse grain delay values for read enable control required by PHY. These values are not used by the design until they are loaded in the active register DRAM_TIMING_22_ACT by setting CONFIG_4[LOAD_ALL_CONFIG] to '1' or when CONFIG_3[LOAD_TIMING_ON_FREQ_CHNG] is set to '1' and a frequency change occurs.

Note that the byte numbers below may not correspond to the actual DRAM byte. This is required so as to keep the byte number reference correct between the controller and the PHY (which has no information on the byte swap feature)

BIMC_S_DDR0_DPE_DRAM_TIMING_22

Bits	Name	Description
29:28	RCW_COARSE_DLY_BYTE 7	This value controls the coarse delay which will be applied by the PHY for the external read enable signal for precise DQS capture from the PCDDR3/LPDDR3 DRAM device. 0x0: NO_DELAY (No delay (default)) 0x1: MINOR_DLY (MINOR_DLY) 0x2: ENUM_1_2_CYCLE (1/2_CYCLE) 0x3: ENUM_1_2_CYCLE_PLUS_MINOR_DLY (1/2_CYCLE_PLUS_MINOR_DLY)
25:24	RCW_COARSE_DLY_BYTE 6	This value controls the coarse delay which will be applied by the PHY for the external read enable signal for precise DQS capture from the PCDDR3/LPDDR3 DRAM device. 0x0: NO_DELAY (No delay (default)) 0x1: MINOR_DLY (MINOR_DLY) 0x2: ENUM_1_2_CYCLE (1/2_CYCLE) 0x3: ENUM_1_2_CYCLE_PLUS_MINOR_DLY (1/2_CYCLE_PLUS_MINOR_DLY)
21:20	RCW_COARSE_DLY_BYTE 5	This value controls the coarse delay which will be applied by the PHY for the external read enable signal for precise DQS capture from the PCDDR3/LPDDR3 DRAM device. 0x0: NO_DELAY (No delay (default)) 0x1: MINOR_DLY (MINOR_DLY) 0x2: ENUM_1_2_CYCLE (1/2_CYCLE) 0x3: ENUM_1_2_CYCLE_PLUS_MINOR_DLY (1/2_CYCLE_PLUS_MINOR_DLY)
17:16	RCW_COARSE_DLY_BYTE 4	This value controls the coarse delay which will be applied by the PHY for the external read enable signal for precise DQS capture from the PCDDR3/LPDDR3 DRAM device. 0x0: NO_DELAY (No delay (default)) 0x1: MINOR_DLY (MINOR_DLY) 0x2: ENUM_1_2_CYCLE (1/2_CYCLE) 0x3: ENUM_1_2_CYCLE_PLUS_MINOR_DLY (1/2_CYCLE_PLUS_MINOR_DLY)
13:12	RCW_COARSE_DLY_BYTE 3	This value controls the coarse delay which will be applied by the PHY for the external read enable signal for precise DQS capture from the PCDDR3/LPDDR3 DRAM device. 0x0: NO_DELAY (No delay (default)) 0x1: MINOR_DLY (MINOR_DLY) 0x2: ENUM_1_2_CYCLE (1/2_CYCLE) 0x3: ENUM_1_2_CYCLE_PLUS_MINOR_DLY (1/2_CYCLE_PLUS_MINOR_DLY)

BIMC_S_DDR0_DPE_DRAM_TIMING_22 (cont.)

Bits	Name	Description
9:8	RCW_COARSE_DLY_BYTE 2	This value controls the coarse delay which will be applied by the PHY for the external read enable signal for precise DQS capture from the PCDDR3/LPDDR3 DRAM device. 0x0: NO_DELAY (No delay (default)) 0x1: MINOR_DLY (MINOR_DLY) 0x2: ENUM_1_2_CYCLE (1/2_CYCLE) 0x3: ENUM_1_2_CYCLE_PLUS_MINOR_DLY (1/2_CYCLE_PLUS_MINOR_DLY)
5:4	RCW_COARSE_DLY_BYTE 1	This value controls the coarse delay which will be applied by the PHY for the external read enable signal for precise DQS capture from the PCDDR3/LPDDR3 DRAM device. 0x0: NO_DELAY (No delay (default)) 0x1: MINOR_DLY (MINOR_DLY) 0x2: ENUM_1_2_CYCLE (1/2_CYCLE) 0x3: ENUM_1_2_CYCLE_PLUS_MINOR_DLY (1/2_CYCLE_PLUS_MINOR_DLY)
1:0	RCW_COARSE_DLY_BYTE 0	This value controls the coarse delay which will be applied by the PHY for the external read enable signal for precise DQS capture from the PCDDR3/LPDDR3 DRAM device. 0x0: NO_DELAY (No delay (default)) 0x1: MINOR_DLY (MINOR_DLY) 0x2: ENUM_1_2_CYCLE (1/2_CYCLE) 0x3: ENUM_1_2_CYCLE_PLUS_MINOR_DLY (1/2_CYCLE_PLUS_MINOR_DLY)

0x0044C1A0 BIMC_S_DDR0_DPE_DRAM_TIMING_23**Type:** RW**Clock:** ACLK**Reset State:** 0x00000000

This register contains timing values required by the memory vendor in ns. These values are not used by the design until they are loaded in the active register DRAM_TIMING_23_ACT by setting CONFIG_4[LOAD_ALL_CONFIG] to '1' or when CONFIG_3[LOAD_TIMING_ON_FREQ_CHNG] is set to '1' and a frequency change occurs

The value of the parameter contained in the registers can be in DDR clock cycles or in multiples of 100ps and may vary from one SDRAM vendor to another.

Refer to the note in DRAM_TIMING_0 CSR.

BIMC_S_DDR0_DPE_DRAM_TIMING_23

Bits	Name	Description
29:24	ODTON_MAX	Specifies the ODT on max delay. For PCDDR3 devices, program this to tAONmax. For LPDDR3 devices, this should be programmed to tODTonmax - tDQSCKmin. If tDQSCKmin should exceed the tODTonmax, then program this to '0'.
20:16	ODTON_MIN	Specifies the ODT on min delay for LPDDR3/PCDDR3.
13:8	ODTOFF_MAX	Specifies the ODT off max delay for LPDDR3/PCDDR3.
4:0	ODTOFF_MIN	Specifies the ODT off min delay for LPDDR3/PCDDR3.

0x0044C1A4 BIMC_S_DDR0_DPE_DRAM_TIMING_0_ACT**Type:** R**Clock:** ACLK**Reset State:** 0x00000000

This register contains the current value of the timing parameter in DDR clock cycles and is a result of comparison to min value. However this does not include the derate offset if timing is being derated (where applicable based on memory device specification).

BIMC_S_DDR0_DPE_DRAM_TIMING_0_ACT

Bits	Name	Description
6:0	TRASMIN	READ ONLY

0x0044C1A8 BIMC_S_DDR0_DPE_DRAM_TIMING_1_ACT**Type:** R**Clock:** ACLK**Reset State:** 0x00000000

This register contains the current value of the timing parameter in DDR clock cycles and is a result of comparison to min value. However this does not include the derate offset if timing is being derated (where applicable based on memory device specification).

BIMC_S_DDR0_DPE_DRAM_TIMING_1_ACT

Bits	Name	Description
20:16	TWR	READ ONLY
5:0	TRCD	READ ONLY

0x0044C1AC BIMC_S_DDR0_DPE_DRAM_TIMING_2_ACT**Type:** R**Clock:** ACLK**Reset State:** 0x00000000

This register contains the current value of the timing parameter in DDR clock cycles and is a result of comparison to min value. However this does not include the derate offset if timing is being derated (where applicable based on memory device specification).

BIMC_S_DDR0_DPE_DRAM_TIMING_2_ACT

Bits	Name	Description
19:16	TWTR	READ ONLY
4:0	TRRD	READ ONLY

0x0044C1B0 BIMC_S_DDR0_DPE_DRAM_TIMING_3_ACT**Type:** R**Clock:** ACLK**Reset State:** 0x00000000

This register contains the current value of the timing parameter in DDR clock cycles and is a result of comparison to min value. However this does not include the derate offset if timing is being derated (where applicable based on memory device specification).

BIMC_S_DDR0_DPE_DRAM_TIMING_3_ACT

Bits	Name	Description
24:16	TRFCAB	READ ONLY
8:0	TRFCPB	READ ONLY

0x0044C1B4 BIMC_S_DDR0_DPE_DRAM_TIMING_4_ACT**Type:** R**Clock:** ACLK**Reset State:** 0x00000000

This register contains the current value of the timing parameter in DDR clock cycles and is a result of comparison to min value. However this does not include the derate offset if timing is being derated (where applicable based on memory device specification).

BIMC_S_DDR0_DPE_DRAM_TIMING_4_ACT

Bits	Name	Description
3:0	TRTP	READ ONLY

0x0044C1B8 BIMC_S_DDR0_DPE_DRAM_TIMING_5_ACT**Type:** R**Clock:** ACLK**Reset State:** 0x00000000

This register contains the current value of the timing parameter in DDR clock cycles and is a result of comparison to min value. However this does not include the derate offset if timing is being derated (where applicable based on memory device specification).

BIMC_S_DDR0_DPE_DRAM_TIMING_5_ACT

Bits	Name	Description
21:16	TRPAB	READ ONLY
5:0	TRPPB	READ ONLY

0x0044C1BC BIMC_S_DDR0_DPE_DRAM_TIMING_6_ACT**Type:** R**Clock:** ACLK**Reset State:** 0x00000000

This register contains the current value of the timing parameter in DDR clock cycles and is a result of comparison to min value. However this does not include the derate offset if timing is being derated (where applicable based on memory device specification).

BIMC_S_DDR0_DPE_DRAM_TIMING_6_ACT

Bits	Name	Description
22:16	TFAW	READ ONLY
4:0	TCKE	READ ONLY

0x0044C1C0 BIMC_S_DDR0_DPE_DRAM_TIMING_7_ACT**Type:** R**Clock:** ACLK**Reset State:** 0x00000000

This register contains the current value of the timing parameter in DDR clock cycles and is a result of comparison to min value. However this does not include the derate offset if timing is being derated (where applicable based on memory device specification).

BIMC_S_DDR0_DPE_DRAM_TIMING_7_ACT

Bits	Name	Description
7:0	TZQCS	READ ONLY

0x0044C1C4 BIMC_S_DDR0_DPE_DRAM_TIMING_8_ACT

Type: R

Clock: ACLK

Reset State: 0x00000000

This register contains the current value of the timing parameter in DDR clock cycles and is a result of comparison to min value. However this does not include the derate offset if timing is being derated (where applicable based on memory device specification).

BIMC_S_DDR0_DPE_DRAM_TIMING_8_ACT

Bits	Name	Description
7:0	TMOD	READ ONLY

0x0044C1C8 BIMC_S_DDR0_DPE_DRAM_TIMING_9_ACT

Type: R

Clock: ACLK

Reset State: 0x00000000

This register contains the current value of the timing parameter in DDR clock cycles and is a result of comparison to min value. However this does not include the derate offset if timing is being derated (where applicable based on memory device specification).

BIMC_S_DDR0_DPE_DRAM_TIMING_9_ACT

Bits	Name	Description
4:0	MIN_SR_DURATION	READ ONLY

0x0044C1CC BIMC_S_DDR0_DPE_DRAM_TIMING_10_ACT**Type:** R**Clock:** ACLK**Reset State:** 0x00000000

This register contains the current value of the timing parameter in DDR clock cycles and is a result of comparison to min value. However this does not include the derate offset if timing is being derated (where applicable based on memory device specification).

BIMC_S_DDR0_DPE_DRAM_TIMING_10_ACT

Bits	Name	Description
25:16	TXSRD	READ ONLY
8:0	TXSNR	READ ONLY

0x0044C1D0 BIMC_S_DDR0_DPE_DRAM_TIMING_11_ACT**Type:** R**Clock:** ACLK**Reset State:** 0x00000000

This register contains the current value of the timing parameter in DDR clock cycles and is a result of comparison to min value. However this does not include the derate offset if timing is being derated (where applicable based on memory device specification).

BIMC_S_DDR0_DPE_DRAM_TIMING_11_ACT

Bits	Name	Description
20:16	TXPNR_ACT_PWR_DN	READ ONLY
5:0	TXPR_ACT_PWR_DN	READ ONLY

0x0044C1D4 BIMC_S_DDR0_DPE_DRAM_TIMING_12_ACT**Type:** R**Clock:** ACLK**Reset State:** 0x00000000

This register contains the current value of the timing parameter in DDR clock cycles and is a result of comparison to min value. However this does not include the derate offset if timing is being derated (where applicable based on memory device specification).

BIMC_S_DDR0_DPE_DRAM_TIMING_12_ACT

Bits	Name	Description
20:16	TXPNR_PCHG_PWR_DN	READ ONLY
5:0	TXPR_PCHG_PWR_DN	READ ONLY

0x0044C1D8 BIMC_S_DDR0_DPE_DRAM_TIMING_13_ACT**Type:** R**Clock:** ACLK**Reset State:** 0x00000000

This register contains the current value of the timing parameter in DDR clock cycles and is a result of comparison to min value. However this does not include the derate offset if timing is being derated (where applicable based on memory device specification).

BIMC_S_DDR0_DPE_DRAM_TIMING_13_ACT

Bits	Name	Description
20:16	CLK_AFTER_SR_ENTRY	READ ONLY
7:0	CLK_BEFORE_SR_EXIT	READ ONLY

0x0044C1DC BIMC_S_DDR0_DPE_DRAM_TIMING_14_ACT**Type:** R**Clock:** ACLK**Reset State:** 0x00000000

This register contains the current value of the timing parameter in DDR clock cycles and is a result of comparison to min value. It also reflects the value in the ADD_CYCLES field in the DRAM_TIMING_14 CSR.

BIMC_S_DDR0_DPE_DRAM_TIMING_14_ACT

Bits	Name	Description
21:16	IE_WINDOW_START	READ ONLY
5:0	IE_WINDOW_END	READ ONLY

0x0044C1E0 BIMC_S_DDR0_DPE_DRAM_TIMING_15_ACT**Type:** R**Clock:** ACLK**Reset State:** 0x00000000

This register contains the current value of the timing parameter in DDR clock cycles and is a result of comparison to min value. It also reflects the value in the ADD_CYCLES field in the DRAM_TIMING_15 CSR.

BIMC_S_DDR0_DPE_DRAM_TIMING_15_ACT

Bits	Name	Description
21:16	RD_CAPTURE_START	READ ONLY
5:0	RD_CAPTURE_END	READ ONLY

0x0044C1E4 BIMC_S_DDR0_DPE_DRAM_TIMING_16_ACT**Type:** R**Clock:** ACLK**Reset State:** 0x00000000

This register contains the current value of the timing parameter in DDR clock cycles.

BIMC_S_DDR0_DPE_DRAM_TIMING_16_ACT

Bits	Name	Description
20	PAD_MODE	READ ONLY
19:16	TCCD	READ ONLY
13:8	RD_LATENCY	READ ONLY
5:0	WR_LATENCY	READ ONLY

0x0044C1E8 BIMC_S_DDR0_DPE_DRAM_TIMING_17_ACT**Type:** R**Clock:** ACLK**Reset State:** 0x00000000

This register contains the current value of the timing parameter in DDR clock cycles and is a result of comparison to min value. However this does not include the derate offset if timing is being derated (where applicable based on memory device specification).

BIMC_S_DDR0_DPE_DRAM_TIMING_17_ACT

Bits	Name	Description
9:0	TZQCL	READ ONLY

0x0044C1EC BIMC_S_DDR0_DPE_DRAM_TIMING_18_ACT**Type:** R**Clock:** ACLK**Reset State:** 0x00000000

This register contains the current value of the following read enable control in DDR clock cycles. The contents of this register are only valid for x64 DRAM devices.

Note that the byte numbers below may not correspond to the actual DRAM byte. This is required so as to keep the byte number reference correct between the controller and the PHY (which has no information on the byte swap feature)

BIMC_S_DDR0_DPE_DRAM_TIMING_18_ACT

Bits	Name	Description
28:24	RCW_START_DLY_BYTE7	READ ONLY Enables the external read enable n cycles from the CAS cycle for the read command.
20:16	RCW_START_DLY_BYTE6	READ ONLY Enables the external read enable n cycles from the CAS cycle for the read command.
12:8	RCW_START_DLY_BYTE5	READ ONLY Enables the external read enable n cycles from the CAS cycle for the read command.
4:0	RCW_START_DLY_BYTE4	READ ONLY Enables the external read enable n cycles from the CAS cycle for the read command.

0x0044C1F0 BIMC_S_DDR0_DPE_DRAM_TIMING_19_ACT**Type:** R**Clock:** ACLK**Reset State:** 0x00000000

This register contains the current value of the following read enable control in DDR clock cycles.

Note that the byte numbers below may not correspond to the actual DRAM byte. This is required so as to keep the byte number reference correct between the controller and the PHY (which has no information on the byte swap feature).

BIMC_S_DDR0_DPE_DRAM_TIMING_19_ACT

Bits	Name	Description
28:24	RCW_START_DLY_BYTE3	READ ONLY Enables the external read enable n cycles from the CAS cycle for the read command.
20:16	RCW_START_DLY_BYTE2	READ ONLY Enables the external read enable n cycles from the CAS cycle for the read command.
12:8	RCW_START_DLY_BYTE1	READ ONLY Enables the external read enable n cycles from the CAS cycle for the read command.
4:0	RCW_START_DLY_BYTE0	READ ONLY Enables the external read enable n cycles from the CAS cycle for the read command.

0x0044C1F4 BIMC_S_DDR0_DPE_DRAM_TIMING_20_ACT**Type:** R**Clock:** ACLK**Reset State:** 0x00000000

This register contains the current value of the fine grain delay values for read enable control required by PHY. The contents of this CSR is only valid for x64 DRAM devices.

Note that the byte numbers below may not correspond to the actual DRAM byte. This is required so as to keep the byte number reference correct between the controller and the PHY (which has no information on the byte swap feature)

BIMC_S_DDR0_DPE_DRAM_TIMING_20_ACT

Bits	Name	Description
31:24	RCW_FINE_DLY_BYTE7	READ ONLY This value controls the fine-grain delay which will be applied by the PHY for the external read enable signal for precise DQS capture from the PCDDR3/LPDDR3 DRAM device.
23:16	RCW_FINE_DLY_BYTE6	READ ONLY This value controls the fine-grain delay which will be applied by the PHY for the external read enable signal for precise DQS capture from the PCDDR3/LPDDR3 DRAM device.
15:8	RCW_FINE_DLY_BYTE5	READ ONLY This value controls the fine-grain delay which will be applied by the PHY for the external read enable signal for precise DQS capture from the PCDDR3/LPDDR3 DRAM device.

BIMC_S_DDR0_DPE_DRAM_TIMING_20_ACT (cont.)

Bits	Name	Description
7:0	RCW_FINE_DLY_BYTE4	READ ONLY This value controls the fine-grain delay which will be applied by the PHY for the external read enable signal for precise DQS capture from the PCDDR3/LPDDR3 DRAM device.

0x0044C1F8 BIMC_S_DDR0_DPE_DRAM_TIMING_21_ACT**Type:** R**Clock:** ACLK**Reset State:** 0x00000000

This register contains the current value of the fine grain delay values for read enable control required by PH.

Note that the byte numbers below may not correspond to the actual DRAM byte. This is required so as to keep the byte number reference correct between the controller and the PHY (which has no information on the byte swap feature).

BIMC_S_DDR0_DPE_DRAM_TIMING_21_ACT

Bits	Name	Description
31:24	RCW_FINE_DLY_BYTE3	READ ONLY This value controls the fine-grain delay which will be applied by the PHY for the external read enable signal for precise DQS capture from the PCDDR3/LPDDR3 DRAM device.
23:16	RCW_FINE_DLY_BYTE2	READ ONLY This value controls the fine-grain delay which will be applied by the PHY for the external read enable signal for precise DQS capture from the PCDDR3/LPDDR3 DRAM device.
15:8	RCW_FINE_DLY_BYTE1	READ ONLY This value controls the fine-grain delay which will be applied by the PHY for the external read enable signal for precise DQS capture from the PCDDR3/LPDDR3 DRAM device.
7:0	RCW_FINE_DLY_BYTE0	READ ONLY This value controls the fine-grain delay which will be applied by the PHY for the external read enable signal for precise DQS capture from the PCDDR3/LPDDR3 DRAM device.

0x0044C1FC BIMC_S_DDR0_DPE_DRAM_TIMING_22_ACT**Type:** R**Clock:** ACLK**Reset State:** 0x00000000

This register contains the current value of the coarse grain delay values for read enable control required by PHY.

Note that the byte numbers below may not correspond to the actual DRAM byte. This is required so as to keep the byte number reference correct between the controller and the PHY (which has no information on the byte swap feature)

BIMC_S_DDR0_DPE_DRAM_TIMING_22_ACT

Bits	Name	Description
29:28	RCW_COARSE_DLY_BYTE 7	READ ONLY This value controls the coarse delay which will be applied by the PHY for the external read enable signal for precise DQS capture from the PCDDR3/LPDDR3 DRAM device. This field is only valid for x64 DRAM devices. 0x0: NO_DELAY (No delay (default)) 0x1: MINOR_DLY (MINOR_DLY) 0x2: ENUM_1_2_CYCLE (1/2_CYCLE) 0x3: ENUM_1_2_CYCLE_PLUS_MINOR_DLY (1/2_CYCLE_PLUS_MINOR_DLY)
25:24	RCW_COARSE_DLY_BYTE 6	READ ONLY This value controls the coarse delay which will be applied by the PHY for the external read enable signal for precise DQS capture from the PCDDR3/LPDDR3 DRAM device. This field is only valid for x64 DRAM devices. 0x0: NO_DELAY (No delay (default)) 0x1: MINOR_DLY (MINOR_DLY) 0x2: ENUM_1_2_CYCLE (1/2_CYCLE) 0x3: ENUM_1_2_CYCLE_PLUS_MINOR_DLY (1/2_CYCLE_PLUS_MINOR_DLY)
21:20	RCW_COARSE_DLY_BYTE 5	READ ONLY This value controls the coarse delay which will be applied by the PHY for the external read enable signal for precise DQS capture from the PCDDR3/LPDDR3 DRAM device. This field is only valid for x64 DRAM devices. 0x0: NO_DELAY (No delay (default)) 0x1: MINOR_DLY (MINOR_DLY) 0x2: ENUM_1_2_CYCLE (1/2_CYCLE) 0x3: ENUM_1_2_CYCLE_PLUS_MINOR_DLY (1/2_CYCLE_PLUS_MINOR_DLY)
17:16	RCW_COARSE_DLY_BYTE 4	READ ONLY This value controls the coarse delay which will be applied by the PHY for the external read enable signal for precise DQS capture from the PCDDR3/LPDDR3 DRAM device. This field is only valid for x64 DRAM devices. 0x0: NO_DELAY (No delay (default)) 0x1: MINOR_DLY (MINOR_DLY) 0x2: ENUM_1_2_CYCLE (1/2_CYCLE) 0x3: ENUM_1_2_CYCLE_PLUS_MINOR_DLY (1/2_CYCLE_PLUS_MINOR_DLY)

BIMC_S_DDR0_DPE_DRAM_TIMING_22_ACT (cont.)

Bits	Name	Description
13:12	RCW_COARSE_DLY_BYTE 3	READ ONLY This value controls the coarse delay which will be applied by the PHY for the external read enable signal for precise DQS capture from the PCDDR3/LPDDR3 DRAM device. This field is only valid for x64 DRAM devices. 0x0: NO_DELAY (No delay (default)) 0x1: MINOR_DLY (MINOR_DLY) 0x2: ENUM_1_2_CYCLE (1/2_CYCLE) 0x3: ENUM_1_2_CYCLE_PLUS_MINOR_DLY (1/2_CYCLE_PLUS_MINOR_DLY)
9:8	RCW_COARSE_DLY_BYTE 2	READ ONLY This value controls the coarse delay which will be applied by the PHY for the external read enable signal for precise DQS capture from the PCDDR3/LPDDR3 DRAM device. This field is only valid for x64 DRAM devices. 0x0: NO_DELAY (No delay (default)) 0x1: MINOR_DLY (MINOR_DLY) 0x2: ENUM_1_2_CYCLE (1/2_CYCLE) 0x3: ENUM_1_2_CYCLE_PLUS_MINOR_DLY (1/2_CYCLE_PLUS_MINOR_DLY)
5:4	RCW_COARSE_DLY_BYTE 1	READ ONLY This value controls the coarse delay which will be applied by the PHY for the external read enable signal for precise DQS capture from the PCDDR3/LPDDR3 DRAM device. This field is only valid for x64 DRAM devices. 0x0: NO_DELAY (No delay (default)) 0x1: MINOR_DLY (MINOR_DLY) 0x2: ENUM_1_2_CYCLE (1/2_CYCLE) 0x3: ENUM_1_2_CYCLE_PLUS_MINOR_DLY (1/2_CYCLE_PLUS_MINOR_DLY)
1:0	RCW_COARSE_DLY_BYTE 0	READ ONLY This value controls the coarse delay which will be applied by the PHY for the external read enable signal for precise DQS capture from the PCDDR3/LPDDR3 DRAM device. This field is only valid for x64 DRAM devices. 0x0: NO_DELAY (No delay (default)) 0x1: MINOR_DLY (MINOR_DLY) 0x2: ENUM_1_2_CYCLE (1/2_CYCLE) 0x3: ENUM_1_2_CYCLE_PLUS_MINOR_DLY (1/2_CYCLE_PLUS_MINOR_DLY)

0x0044C200 BIMC_S_DDR0_DPE_DRAM_TIMING_23_ACT**Type:** R**Clock:** ACLK**Reset State:** 0x00000000

This register contains the current value of the timing parameter in DDR clock cycles. It also reflects the value in the ADD_CYCLES field in DRAM_TIMING_16 CSR.

BIMC_S_DDR0_DPE_DRAM_TIMING_23_ACT

Bits	Name	Description
31:28	ODTON_MAX	READ ONLY Specifies the ODT on max delay for LPDDR3/PCDDR3. Not Supported.
25:24	ODTON_MIN	READ ONLY Specifies the ODT on min delay for LPDDR3/PCDDR3. Not Supported.
22:20	ODTOFF_MAX	READ ONLY Specifies the ODT off max delay for LPDDR3/PCDDR3. Not Supported.
17:16	ODTOFF_MIN	READ ONLY Specifies the ODT off min delay for LPDDR3/PCDDR3. Not Supported.

0x0044C240 BIMC_S_DDR0_DPE_DQCAL_RDATA_RANK1_U

Type: R

Clock: ACLK

Reset State: 0x00000000

DQCAL Read Data Register for Rank1 Upper.

This register contains the upper 32 bits of a x64 DRAM mode register contents for a DQ calibration command triggered via SHKE to Rank1. The contents of this register are only valid for x64 DRAM devices. This is used to capture the read data from the DRAM device's mode register for a read DQ calibration command for LPDDR3 or an MPR read command for PCDDR3. The DQS data bits 0, 8, 16, 24, 32, 40, 48 and 56 are only captured on each DQS edge. The field below will capture the data for the last 2 cycles of DQS (last 4 DQS edges). Eg. DQCAL_RDATA[7:0] will have DQS data bits 0, 8, 16, 24, 32, 40, 48 and 56 for the fifth DQS edge. Hence 32 bits for the first 4 DQS edges.

BIMC_S_DDR0_DPE_DQCAL_RDATA_RANK1_U

Bits	Name	Description
31:0	DQCAL_RDATA	READ ONLY Upper 32 bits of Data for DQ calibration read for Rank1. The data captures the lowest bit for each byte across the memory burst. For eg. each cycle of memory burst is 16 bytes of data for x64 DRAM- 1 bit for each byte is captured which is 16 bits for every cycle of DQS.

0x0044C244 BIMC_S_DDR0_DPE_DQCAL_RDATA_RANK1_L**Type:** R**Clock:** ACLK**Reset State:** 0x00000000

DQCAL Read Data Register for Rank1 Lower.

This register contains the lower 32 bits of a x64 DRAM or 32 bits of a x32 DRAM mode register contents for a DQ calibration command triggered via SHKE to Rank1. This is used to capture the read data from the DRAM device's mode register for a read DQ calibration command for LPDDR3 or an MPR read command for PCDDR3. The DQS data bits 0, 8, 16 and 24 are only captured on each DQS edge for x32 DRAM device. For x32 LPDDR3 or PCDDR3 DRAM, DQCAL_RDATA[3:0] will have DQS data bits 0, 8, 16 and 24 for the first DQS edge. Hence 32 bits for the 8 DQS edges. For x64 LPDDR3 or PCDDR3 DRAM, the DQS data bits 0, 8, 16, 24, 32, 40, 48 and 56 are only captured on each DQS edge for x64 DRAM device. For x64 LPDDR3 or PCDDR3 DRAM, DQCAL_RDATA[7:0] will have DQS data bits 0, 8, 16, 24, 32, 40, 48 and 56 for every DQS edge (in this case DQS edges 1 - 4) resulting in 32 bits of captured data.

BIMC_S_DDR0_DPE_DQCAL_RDATA_RANK1_L

Bits	Name	Description
31:0	DQCAL_RDATA	READ ONLY Lower 32 bits of Data for DQ calibration read for Rank1. The data captures the lowest bit for each byte across the memory burst. For eg. each cycle of memory burst is 8 bytes of data for x32 DRAM (or 16 bytes of data for x64 DRAM) - 1 bit for each byte is captured which is 8 bits for every cycle of DQS.

0x0044C248 BIMC_S_DDR0_DPE_DQCAL_RDATA_RANK0_U**Type:** R**Clock:** ACLK**Reset State:** 0x00000000

DQCAL Read Data Register for Rank0 Upper.

This register contains the upper 32 bits of a x64 DRAM mode register contents for a DQ calibration command triggered via SHKE to Rank0. The contents of this register are only valid for x64 DRAM devices. This is used to capture the read data from the DRAM device's mode register for a read DQ calibration command for LPDDR3 or an MPR read command for PCDDR3. The DQS data bits 0, 8, 16, 24, 32, 40, 48 and 56 are only captured on each DQS edge. The field below will capture the data for the last 2 cycles of DQS (last 4 DQS edges). Eg. DQCAL_RDATA[7:0] will have DQS data bits 0, 8, 16, 24, 32, 40, 48 and 56 for the fifth DQS edge. Hence 32 bits for the first 4 DQS edges.

BIMC_S_DDR0_DPE_DQCAL_RDATA_RANK0_U

Bits	Name	Description
31:0	DQCAL_RDATA	READ ONLY Upper 32 bits of Data for DQ calibration read for Rank0. The data captures the lowest bit for each byte across the memory burst. For eg. each cycle of memory burst is 16 bytes of data for x64 DRAM- 1 bit for each byte is captured which is 16 bits for every cycle of DQS.

0x0044C24C BIMC_S_DDR0_DPE_DQCAL_RDATA_RANK0_L**Type:** R**Clock:** ACLK**Reset State:** 0x00000000

DQCAL Read Data Register for Rank0 Lower.

This register contains the lower 32 bits of a x64 DRAM or 32 bits of a x32 DRAM mode register contents for a DQ calibration command triggered via SHKE to Rank1. This is used to capture the read data from the DRAM device's mode register for a read DQ calibration command for LPDDR3 or an MPR read command for PCDDR3. The DQS data bits 0, 8, 16 and 24 are only captured on each DQS edge for x32 DRAM device. For x32 LPDDR3 or PCDDR3 DRAM, DQCAL_RDATA[3:0] will have DQS data bits 0, 8, 16 and 24 for the first DQS edge. Hence 32 bits for the 8 DQS edges. For x64 LPDDR3 or PCDDR3 DRAM, the DQS data bits 0, 8, 16, 24, 32, 40, 48 and 56 are only captured on each DQS edge for x64 DRAM device. For x64 LPDDR3 or PCDDR3 DRAM, DQCAL_RDATA[7:0] will have DQS data bits 0, 8, 16, 24, 32, 40, 48 and 56 for every DQS edge (in this case DQS edges 1 - 4) resulting in 32 bits of captured data.

BIMC_S_DDR0_DPE_DQCAL_RDATA_RANK0_L

Bits	Name	Description
31:0	DQCAL_RDATA	READ ONLY Lower 32 bits of Data for DQ calibration read for Rank0. The data captures the lowest bit for each byte across the memory burst. For eg. each cycle of memory burst is 8 bytes of data for x32 DRAM (or 16 bytes of data for x64 DRAM) - 1 bit for each byte is captured which is 8 bits for every cycle of DQS.

0x0044C260 BIMC_S_DDR0_DPE_DQCAL_STATUS0_RANK1**Type:** R**Clock:** ACLK**Reset State:** 0x00000000

DQCAL Status0 Register for Rank1.

Note that the byte numbers below may not correspond to the actual DRAM byte. This is required so as to keep the byte number reference correct between the controller and the PHY (which has no information on the byte swap feature)

BIMC_S_DDR0_DPE_DQCAL_STATUS0_RANK1

Bits	Name	Description
7	DQCAL_COMPARE_BYTE7	READ ONLY This field is only valid for x64 DRAM devices. Indicates whether the DQ calibration command for LPDDR3 or PCDDR3 triggered via SHKE_DRAM_MANUAL_0[RD_DQCAL] resulted in an expected pattern comparison pass/fail. The value in this CSR field is valid when HW deasserts SHKE_DRAM_MANUAL_0[RD_DQCAL] indicating that the command completed. SW also needs to ensure that the number of DQS edges received in DQCAL_STATUS1_RANK1[DQS_PE_BYTE7] and DQCAL_STATUS1_RANK1[DQS_NE_BYTE7] are correct. A valid pass is only when number of DQS edges received are correct and the PASS comparison bit is set. 0x0: FAIL (FAIL) 0x1: PASS (PASS)
6	DQCAL_COMPARE_BYTE6	READ ONLY This field is only valid for x64 DRAM devices. Indicates whether the DQ calibration command for LPDDR2/3 or PCDDR3 triggered via SHKE_DRAM_MANUAL_0[RD_DQCAL] resulted in an expected pattern comparison pass/fail. The value in this CSR field is valid when HW deasserts SHKE_DRAM_MANUAL_0[RD_DQCAL] indicating that the command completed. SW also needs to ensure that the number of DQS edges received in DQCAL_STATUS1_RANK1[DQS_PE_BYTE6] and DQCAL_STATUS1_RANK1[DQS_NE_BYTE6] are correct. A valid pass is only when number of DQS edges received are correct and the PASS comparison bit is set. 0x0: FAIL (FAIL) 0x1: PASS (PASS)
5	DQCAL_COMPARE_BYTE5	READ ONLY This field is only valid for x64 DRAM devices. Indicates whether the DQ calibration command for LPDDR2/3 or PCDDR3 triggered via SHKE_DRAM_MANUAL_0[RD_DQCAL] resulted in an expected pattern comparison pass/fail. The value in this CSR field is valid when HW deasserts SHKE_DRAM_MANUAL_0[RD_DQCAL] indicating that the command completed. SW also needs to ensure that the number of DQS edges received in DQCAL_STATUS1_RANK1[DQS_PE_BYTE5] and DQCAL_STATUS1_RANK1[DQS_NE_BYTE5] are correct. A valid pass is only when number of DQS edges received are correct and the PASS comparison bit is set. 0x0: FAIL (FAIL) 0x1: PASS (PASS)

BIMC_S_DDR0_DPE_DQCAL_STATUS0_RANK1 (cont.)

Bits	Name	Description
4	DQCAL_COMPARE_BYTE4	READ ONLY This field is only valid for x64 DRAM devices. Indicates whether the DQ calibration command for LPDDR2/3 or PCDDR3 triggered via SHKE_DRAM_MANUAL_0[RD_DQCAL] resulted in an expected pattern comparison pass/fail. The value in this CSR field is valid when HW deasserts SHKE_DRAM_MANUAL_0[RD_DQCAL] indicating that the command completed. SW also needs to ensure that the number of DQS edges received in DQCAL_STATUS1_RANK1[DQS_PE_BYTE4] and DQCAL_STATUS1_RANK1[DQS_NE_BYTE4] are correct. A valid pass is only when number of DQS edges received are correct and the PASS comparison bit is set. 0x0: FAIL (FAIL) 0x1: PASS (PASS)
3	DQCAL_COMPARE_BYTE3	READ ONLY Indicates whether the DQ calibration command for LPDDR2/3 or PCDDR3 triggered via SHKE_DRAM_MANUAL_0[RD_DQCAL] resulted in an expected pattern comparison pass/fail. The value in this CSR field is valid when HW deasserts SHKE_DRAM_MANUAL_0[RD_DQCAL] indicating that the command completed. SW also needs to ensure that the number of DQS edges received in DQCAL_STATUS2_RANK1[DQS_PE_BYTE3] and DQCAL_STATUS2_RANK1[DQS_NE_BYTE3] are correct. A valid pass is only when number of DQS edges received are correct and the PASS comparison bit is set. 0x0: FAIL (FAIL) 0x1: PASS (PASS)
2	DQCAL_COMPARE_BYTE2	READ ONLY Indicates whether the DQ calibration command for LPDDR2/3 or PCDDR3 triggered via SHKE_DRAM_MANUAL_0[RD_DQCAL] resulted in an expected pattern comparison pass/fail. The value in this CSR field is valid when HW deasserts SHKE_DRAM_MANUAL_0[RD_DQCAL] indicating that the command completed. SW also needs to ensure that the number of DQS edges received in DQCAL_STATUS2_RANK1[DQS_PE_BYTE2] and DQCAL_STATUS2_RANK1[DQS_NE_BYTE2] are correct. A valid pass is only when number of DQS edges received are correct and the PASS comparison bit is set. 0x0: FAIL (FAIL) 0x1: PASS (PASS)

BIMC_S_DDR0_DPE_DQCAL_STATUS0_RANK1 (cont.)

Bits	Name	Description
1	DQCAL_COMPARE_BYTE1	READ ONLY Indicates whether the DQ calibration command for LPDDR2/3 or PCDDR3 triggered via SHKE_DRAM_MANUAL_0[RD_DQCAL] resulted in an expected pattern comparison pass/fail. The value in this CSR field is valid when HW deasserts SHKE_DRAM_MANUAL_0[RD_DQCAL] indicating that the command completed. SW also needs to ensure that the number of DQS edges received in DQCAL_STATUS2_RANK1[DQS_PE_BYTE1] and DQCAL_STATUS2_RANK1[DQS_NE_BYTE1] are correct. A valid pass is only when number of DQS edges received are correct and the PASS comparison bit is set. 0x0: FAIL (FAIL) 0x1: PASS (PASS)
0	DQCAL_COMPARE_BYTE0	READ ONLY Indicates whether the DQ calibration command for LPDDR2/3 or PCDDR3 triggered via SHKE_DRAM_MANUAL_0[RD_DQCAL] resulted in an expected pattern comparison pass/fail. The value in this CSR field is valid when HW deasserts SHKE_DRAM_MANUAL_0[RD_DQCAL] indicating that the command completed. SW also needs to ensure that the number of DQS edges received in DQCAL_STATUS2_RANK1[DQS_PE_BYTE0] and DQCAL_STATUS2_RANK1[DQS_NE_BYTE0] are correct. A valid pass is only when number of DQS edges received are correct and the PASS comparison bit is set. 0x0: FAIL (FAIL) 0x1: PASS (PASS)

0x0044C264 BIMC_S_DDR0_DPE_DQCAL_STATUS1_RANK1**Type:** R**Clock:** ACLK**Reset State:** 0x00000000

DQCAL Status1 Register for Rank0. This CSR is only valid for x64 DRAM devices.

Note that the byte numbers below may not correspond to the actual DRAM byte. This is required so as to keep the byte number reference correct between the controller and the PHY (which has no information on the byte swap feature)

BIMC_S_DDR0_DPE_DQCAL_STATUS1_RANK1

Bits	Name	Description
30:28	DQS_PE_BYTE7	READ ONLY Indicates the number of DQS PE edges received for the DQ calibration command. The value in this CSR field is valid when HW deasserts SHKE_DRAM_MANUAL_0[RD_DQCAL] indicating that the command completed. For LPDDR3/PCDDR3, the calibration command has a memory burst length of 8. 0x0: NONE (None) 0x1: ENUM_1_POS_EDGE (1_POS_EDGE) 0x2: ENUM_2_POS_EDGES (2_POS_EDGES) 0x3: ENUM_3_POS_EDGES (3_POS_EDGES) 0x4: ENUM_4_POS_EDGES (4_POS_EDGES) 0x5: ENUM_5_POS_EDGES (5_POS_EDGES) 0x6: ENUM_6_POS_EDGES (6_POS_EDGES) 0x7: ENUM_7_OR_MORE_POS_EDGES (7_OR_MORE_POS_EDGES)
26:24	DQS_NE_BYTE7	READ ONLY Indicates the number of DQS NE edges received for the DQ calibration command. The value in this CSR field is valid when HW deasserts SHKE_DRAM_MANUAL_0[RD_DQCAL] indicating that the command completed. For LPDDR3/PCDDR3, the calibration command has a memory burst length of 8. 0x0: NONE (None) 0x1: ENUM_1_POS_EDGE (1_POS_EDGE) 0x2: ENUM_2_POS_EDGES (2_POS_EDGES) 0x3: ENUM_3_POS_EDGES (3_POS_EDGES) 0x4: ENUM_4_POS_EDGES (4_POS_EDGES) 0x5: ENUM_5_POS_EDGES (5_POS_EDGES) 0x6: ENUM_6_POS_EDGES (6_POS_EDGES) 0x7: ENUM_7_OR_MORE_POS_EDGES (7_OR_MORE_POS_EDGES)
22:20	DQS_PE_BYTE6	READ ONLY Indicates the number of DQS PE edges received for the DQ calibration command. The value in this CSR field is valid when HW deasserts SHKE_DRAM_MANUAL_0[RD_DQCAL] indicating that the command completed. For LPDDR3/PCDDR3, the calibration command has a memory burst length of 8. 0x0: NONE (None) 0x1: ENUM_1_POS_EDGE (1_POS_EDGE) 0x2: ENUM_2_POS_EDGES (2_POS_EDGES) 0x3: ENUM_3_POS_EDGES (3_POS_EDGES) 0x4: ENUM_4_POS_EDGES (4_POS_EDGES) 0x5: ENUM_5_POS_EDGES (5_POS_EDGES) 0x6: ENUM_6_POS_EDGES (6_POS_EDGES) 0x7: ENUM_7_OR_MORE_POS_EDGES (7_OR_MORE_POS_EDGES)

BIMC_S_DDR0_DPE_DQCAL_STATUS1_RANK1 (cont.)

Bits	Name	Description
18:16	DQS_NE_BYTE6	READ ONLY Indicates the number of DQS NE edges received for the DQ calibration command. The value in this CSR field is valid when HW deasserts SHKE_DRAM_MANUAL_0[RD_DQCAL] indicating that the command completed. For LPDDR3/PCDDR3, the calibration command has a memory burst length of 8. 0x0: NONE (None) 0x1: ENUM_1_POS_EDGE (1_POS_EDGE) 0x2: ENUM_2_POS_EDGES (2_POS_EDGES) 0x3: ENUM_3_POS_EDGES (3_POS_EDGES) 0x4: ENUM_4_POS_EDGES (4_POS_EDGES) 0x5: ENUM_5_POS_EDGES (5_POS_EDGES) 0x6: ENUM_6_POS_EDGES (6_POS_EDGES) 0x7: ENUM_7_OR_MORE_POS_EDGES (7_OR_MORE_POS_EDGES)
14:12	DQS_PE_BYTE5	READ ONLY Indicates the number of DQS PE edges received for the DQ calibration command. The value in this CSR field is valid when HW deasserts SHKE_DRAM_MANUAL_0[RD_DQCAL] indicating that the command completed. For LPDDR3/PCDDR3, the calibration command has a memory burst length of 8. 0x0: NONE (None) 0x1: ENUM_1_POS_EDGE (1_POS_EDGE) 0x2: ENUM_2_POS_EDGES (2_POS_EDGES) 0x3: ENUM_3_POS_EDGES (3_POS_EDGES) 0x4: ENUM_4_POS_EDGES (4_POS_EDGES) 0x5: ENUM_5_POS_EDGES (5_POS_EDGES) 0x6: ENUM_6_POS_EDGES (6_POS_EDGES) 0x7: ENUM_7_OR_MORE_POS_EDGES (7_OR_MORE_POS_EDGES)
10:8	DQS_NE_BYTE5	READ ONLY Indicates the number of DQS NE edges received for the DQ calibration command. The value in this CSR field is valid when HW deasserts SHKE_DRAM_MANUAL_0[RD_DQCAL] indicating that the command completed. For LPDDR3/PCDDR3, the calibration command has a memory burst length of 8. 0x0: NONE (None) 0x1: ENUM_1_POS_EDGE (1_POS_EDGE) 0x2: ENUM_2_POS_EDGES (2_POS_EDGES) 0x3: ENUM_3_POS_EDGES (3_POS_EDGES) 0x4: ENUM_4_POS_EDGES (4_POS_EDGES) 0x5: ENUM_5_POS_EDGES (5_POS_EDGES) 0x6: ENUM_6_POS_EDGES (6_POS_EDGES) 0x7: ENUM_7_OR_MORE_POS_EDGES (7_OR_MORE_POS_EDGES)

BIMC_S_DDR0_DPE_DQCAL_STATUS1_RANK1 (cont.)

Bits	Name	Description
6:4	DQS_PE_BYTE4	READ ONLY Indicates the number of DQS PE edges received for the DQ calibration command. The value in this CSR field is valid when HW deasserts SHKE_DRAM_MANUAL_0[RD_DQCAL] indicating that the command completed. For LPDDR3/PCDDR3, the calibration command has a memory burst length of 8. 0x0: NONE (None) 0x1: ENUM_1_POS_EDGE (1_POS_EDGE) 0x2: ENUM_2_POS_EDGES (2_POS_EDGES) 0x3: ENUM_3_POS_EDGES (3_POS_EDGES) 0x4: ENUM_4_POS_EDGES (4_POS_EDGES) 0x5: ENUM_5_POS_EDGES (5_POS_EDGES) 0x6: ENUM_6_POS_EDGES (6_POS_EDGES) 0x7: ENUM_7_OR_MORE_POS_EDGES (7_OR_MORE_POS_EDGES)
2:0	DQS_NE_BYTE4	READ ONLY Indicates the number of DQS NE edges received for the DQ calibration command. The value in this CSR field is valid when HW deasserts SHKE_DRAM_MANUAL_0[RD_DQCAL] indicating that the command completed. For LPDDR3/PCDDR3, the calibration command has a memory burst length of 8. 0x0: NONE (None) 0x1: ENUM_1_POS_EDGE (1_POS_EDGE) 0x2: ENUM_2_POS_EDGES (2_POS_EDGES) 0x3: ENUM_3_POS_EDGES (3_POS_EDGES) 0x4: ENUM_4_POS_EDGES (4_POS_EDGES) 0x5: ENUM_5_POS_EDGES (5_POS_EDGES) 0x6: ENUM_6_POS_EDGES (6_POS_EDGES) 0x7: ENUM_7_OR_MORE_POS_EDGES (7_OR_MORE_POS_EDGES)

0x0044C268 BIMC_S_DDR0_DPE_DQCAL_STATUS2_RANK1**Type:** R**Clock:** ACLK**Reset State:** 0x00000000

DQCAL Status2 Register for Rank1.

Note that the byte numbers below may not correspond to the actual DRAM byte. This is required so as to keep the byte number reference correct between the controller and the PHY (which has no information on the byte swap feature)

BIMC_S_DDR0_DPE_DQCAL_STATUS2_RANK1

Bits	Name	Description
30:28	DQS_PE_BYTE3	READ ONLY Indicates the number of DQS PE edges received for the DQ calibration command. The value in this CSR field is valid when HW deasserts SHKE_DRAM_MANUAL_0[RD_DQCAL] indicating that the command completed. For LPDDR3/PCDDR3, the calibration command has a memory burst length of 8. 0x0: NONE (None) 0x1: ENUM_1_POS_EDGE (1_POS_EDGE) 0x2: ENUM_2_POS_EDGES (2_POS_EDGES) 0x3: ENUM_3_POS_EDGES (3_POS_EDGES) 0x4: ENUM_4_POS_EDGES (4_POS_EDGES) 0x5: ENUM_5_POS_EDGES (5_POS_EDGES) 0x6: ENUM_6_POS_EDGES (6_POS_EDGES) 0x7: ENUM_7_OR_MORE_POS_EDGES (7_OR_MORE_POS_EDGES)
26:24	DQS_NE_BYTE3	READ ONLY Indicates the number of DQS NE edges received for the DQ calibration command. The value in this CSR field is valid when HW deasserts SHKE_DRAM_MANUAL_0[RD_DQCAL] indicating that the command completed. For LPDDR3/PCDDR3, the calibration command has a memory burst length of 8. 0x0: NONE (None) 0x1: ENUM_1_POS_EDGE (1_POS_EDGE) 0x2: ENUM_2_POS_EDGES (2_POS_EDGES) 0x3: ENUM_3_POS_EDGES (3_POS_EDGES) 0x4: ENUM_4_POS_EDGES (4_POS_EDGES) 0x5: ENUM_5_POS_EDGES (5_POS_EDGES) 0x6: ENUM_6_POS_EDGES (6_POS_EDGES) 0x7: ENUM_7_OR_MORE_POS_EDGES (7_OR_MORE_POS_EDGES)
22:20	DQS_PE_BYTE2	READ ONLY Indicates the number of DQS PE edges received for the DQ calibration command. The value in this CSR field is valid when HW deasserts SHKE_DRAM_MANUAL_0[RD_DQCAL] indicating that the command completed. For LPDDR3/PCDDR3, the calibration command has a memory burst length of 8. 0x0: NONE (None) 0x1: ENUM_1_POS_EDGE (1_POS_EDGE) 0x2: ENUM_2_POS_EDGES (2_POS_EDGES) 0x3: ENUM_3_POS_EDGES (3_POS_EDGES) 0x4: ENUM_4_POS_EDGES (4_POS_EDGES) 0x5: ENUM_5_POS_EDGES (5_POS_EDGES) 0x6: ENUM_6_POS_EDGES (6_POS_EDGES) 0x7: ENUM_7_OR_MORE_POS_EDGES (7_OR_MORE_POS_EDGES)

BIMC_S_DDR0_DPE_DQCAL_STATUS2_RANK1 (cont.)

Bits	Name	Description
18:16	DQS_NE_BYTE2	READ ONLY Indicates the number of DQS NE edges received for the DQ calibration command. The value in this CSR field is valid when HW deasserts SHKE_DRAM_MANUAL_0[RD_DQCAL] indicating that the command completed. For LPDDR3/PCDDR3, the calibration command has a memory burst length of 8. 0x0: NONE (None) 0x1: ENUM_1_POS_EDGE (1_POS_EDGE) 0x2: ENUM_2_POS_EDGES (2_POS_EDGES) 0x3: ENUM_3_POS_EDGES (3_POS_EDGES) 0x4: ENUM_4_POS_EDGES (4_POS_EDGES) 0x5: ENUM_5_POS_EDGES (5_POS_EDGES) 0x6: ENUM_6_POS_EDGES (6_POS_EDGES) 0x7: ENUM_7_OR_MORE_POS_EDGES (7_OR_MORE_POS_EDGES)
14:12	DQS_PE_BYTE1	READ ONLY Indicates the number of DQS PE edges received for the DQ calibration command. The value in this CSR field is valid when HW deasserts SHKE_DRAM_MANUAL_0[RD_DQCAL] indicating that the command completed. For LPDDR3/PCDDR3, the calibration command has a memory burst length of 8. 0x0: NONE (None) 0x1: ENUM_1_POS_EDGE (1_POS_EDGE) 0x2: ENUM_2_POS_EDGES (2_POS_EDGES) 0x3: ENUM_3_POS_EDGES (3_POS_EDGES) 0x4: ENUM_4_POS_EDGES (4_POS_EDGES) 0x5: ENUM_5_POS_EDGES (5_POS_EDGES) 0x6: ENUM_6_POS_EDGES (6_POS_EDGES) 0x7: ENUM_7_OR_MORE_POS_EDGES (7_OR_MORE_POS_EDGES)
10:8	DQS_NE_BYTE1	READ ONLY Indicates the number of DQS NE edges received for the DQ calibration command. The value in this CSR field is valid when HW deasserts SHKE_DRAM_MANUAL_0[RD_DQCAL] indicating that the command completed. For LPDDR3/PCDDR3, the calibration command has a memory burst length of 8. 0x0: NONE (None) 0x1: ENUM_1_POS_EDGE (1_POS_EDGE) 0x2: ENUM_2_POS_EDGES (2_POS_EDGES) 0x3: ENUM_3_POS_EDGES (3_POS_EDGES) 0x4: ENUM_4_POS_EDGES (4_POS_EDGES) 0x5: ENUM_5_POS_EDGES (5_POS_EDGES) 0x6: ENUM_6_POS_EDGES (6_POS_EDGES) 0x7: ENUM_7_OR_MORE_POS_EDGES (7_OR_MORE_POS_EDGES)

BIMC_S_DDR0_DPE_DQCAL_STATUS2_RANK1 (cont.)

Bits	Name	Description
6:4	DQS_PE_BYTE0	READ ONLY Indicates the number of DQS PE edges received for the DQ calibration command. The value in this CSR field is valid when HW deasserts SHKE_DRAM_MANUAL_0[RD_DQCAL] indicating that the command completed. For LPDDR3/PCDDR3, the calibration command has a memory burst length of 8. 0x0: NONE (None) 0x1: ENUM_1_POS_EDGE (1_POS_EDGE) 0x2: ENUM_2_POS_EDGES (2_POS_EDGES) 0x3: ENUM_3_POS_EDGES (3_POS_EDGES) 0x4: ENUM_4_POS_EDGES (4_POS_EDGES) 0x5: ENUM_5_POS_EDGES (5_POS_EDGES) 0x6: ENUM_6_POS_EDGES (6_POS_EDGES) 0x7: ENUM_7_OR_MORE_POS_EDGES (7_OR_MORE_POS_EDGES)
2:0	DQS_NE_BYTE0	READ ONLY Indicates the number of DQS NE edges received for the DQ calibration command. The value in this CSR field is valid when HW deasserts SHKE_DRAM_MANUAL_0[RD_DQCAL] indicating that the command completed. For LPDDR3/PCDDR3, the calibration command has a memory burst length of 8. 0x0: NONE (None) 0x1: ENUM_1_POS_EDGE (1_POS_EDGE) 0x2: ENUM_2_POS_EDGES (2_POS_EDGES) 0x3: ENUM_3_POS_EDGES (3_POS_EDGES) 0x4: ENUM_4_POS_EDGES (4_POS_EDGES) 0x5: ENUM_5_POS_EDGES (5_POS_EDGES) 0x6: ENUM_6_POS_EDGES (6_POS_EDGES) 0x7: ENUM_7_OR_MORE_POS_EDGES (7_OR_MORE_POS_EDGES)

0x0044C280 BIMC_S_DDR0_DPE_DQCAL_STATUS0_RANK0**Type:** R**Clock:** ACLK**Reset State:** 0x00000000

DQCAL Status0 Register for Rank0.

Note that the byte numbers below may not correspond to the actual DRAM byte. This is required so as to keep the byte number reference correct between the controller and the PHY (which has no information on the byte swap feature)

BIMC_S_DDR0_DPE_DQCAL_STATUS0_RANK0

Bits	Name	Description
7	DQCAL_COMPARE_BYTE7	READ ONLY This field is only valid for x64 DRAM devices. Indicates whether the DQ calibration command for LPDDR3 or PCDDR3 triggered via SHKE_DRAM_MANUAL_0[RD_DQCAL] resulted in an expected pattern comparison pass/fail. The value in this CSR field is valid when HW deasserts SHKE_DRAM_MANUAL_0[RD_DQCAL] indicating that the command completed. SW also needs to ensure that the number of DQS edges received in DQCAL_STATUS1_RANK1[DQS_PE_BYTE7] and DQCAL_STATUS1_RANK1[DQS_NE_BYTE7] are correct. A valid pass is only when number of DQS edges received are correct and the PASS comparison bit is set. 0x0: FAIL (FAIL) 0x1: PASS (PASS)
6	DQCAL_COMPARE_BYTE6	READ ONLY This field is only valid for x64 DRAM devices. Indicates whether the DQ calibration command for LPDDR3 or PCDDR3 triggered via SHKE_DRAM_MANUAL_0[RD_DQCAL] resulted in an expected pattern comparison pass/fail. The value in this CSR field is valid when HW deasserts SHKE_DRAM_MANUAL_0[RD_DQCAL] indicating that the command completed. SW also needs to ensure that the number of DQS edges received in DQCAL_STATUS1_RANK1[DQS_PE_BYTE6] and DQCAL_STATUS1_RANK1[DQS_NE_BYTE6] are correct. A valid pass is only when number of DQS edges received are correct and the PASS comparison bit is set. 0x0: FAIL (FAIL) 0x1: PASS (PASS)
5	DQCAL_COMPARE_BYTE5	READ ONLY This field is only valid for x64 DRAM devices. Indicates whether the DQ calibration command for LPDDR3 or PCDDR3 triggered via SHKE_DRAM_MANUAL_0[RD_DQCAL] resulted in an expected pattern comparison pass/fail. The value in this CSR field is valid when HW deasserts SHKE_DRAM_MANUAL_0[RD_DQCAL] indicating that the command completed. SW also needs to ensure that the number of DQS edges received in DQCAL_STATUS1_RANK1[DQS_PE_BYTE5] and DQCAL_STATUS1_RANK1[DQS_NE_BYTE5] are correct. A valid pass is only when number of DQS edges received are correct and the PASS comparison bit is set. 0x0: FAIL (FAIL) 0x1: PASS (PASS)

BIMC_S_DDR0_DPE_DQCAL_STATUS0_RANK0 (cont.)

Bits	Name	Description
4	DQCAL_COMPARE_BYTE4	READ ONLY This field is only valid for x64 DRAM devices. Indicates whether the DQ calibration command for LPDDR3 or PCDDR3 triggered via SHKE_DRAM_MANUAL_0[RD_DQCAL] resulted in an expected pattern comparison pass/fail. The value in this CSR field is valid when HW deasserts SHKE_DRAM_MANUAL_0[RD_DQCAL] indicating that the command completed. SW also needs to ensure that the number of DQS edges received in DQCAL_STATUS1_RANK1[DQS_PE_BYTE4] and DQCAL_STATUS1_RANK1[DQS_NE_BYTE4] are correct. A valid pass is only when number of DQS edges received are correct and the PASS comparison bit is set. 0x0: FAIL (FAIL) 0x1: PASS (PASS)
3	DQCAL_COMPARE_BYTE3	READ ONLY Indicates whether the DQ calibration command for LPDDR3 or PCDDR3 triggered via SHKE_DRAM_MANUAL_0[RD_DQCAL] resulted in an expected pattern comparison pass/fail. The value in this CSR field is valid when HW deasserts SHKE_DRAM_MANUAL_0[RD_DQCAL] indicating that the command completed. SW also needs to ensure that the number of DQS edges received in DQCAL_STATUS2_RANK1[DQS_PE_BYTE3] and DQCAL_STATUS2_RANK1[DQS_NE_BYTE3] are correct. A valid pass is only when number of DQS edges received are correct and the PASS comparison bit is set. 0x0: FAIL (FAIL) 0x1: PASS (PASS)
2	DQCAL_COMPARE_BYTE2	READ ONLY Indicates whether the DQ calibration command for LPDDR3 or PCDDR3 triggered via SHKE_DRAM_MANUAL_0[RD_DQCAL] resulted in an expected pattern comparison pass/fail. The value in this CSR field is valid when HW deasserts SHKE_DRAM_MANUAL_0[RD_DQCAL] indicating that the command completed. SW also needs to ensure that the number of DQS edges received in DQCAL_STATUS2_RANK1[DQS_PE_BYTE2] and DQCAL_STATUS2_RANK1[DQS_NE_BYTE2] are correct. A valid pass is only when number of DQS edges received are correct and the PASS comparison bit is set. 0x0: FAIL (FAIL) 0x1: PASS (PASS)

BIMC_S_DDR0_DPE_DQCAL_STATUS0_RANK0 (cont.)

Bits	Name	Description
1	DQCAL_COMPARE_BYTE1	READ ONLY Indicates whether the DQ calibration command for LPDDR3 or PCDDR3 triggered via SHKE_DRAM_MANUAL_0[RD_DQCAL] resulted in an expected pattern comparison pass/fail. The value in this CSR field is valid when HW deasserts SHKE_DRAM_MANUAL_0[RD_DQCAL] indicating that the command completed. SW also needs to ensure that the number of DQS edges received in DQCAL_STATUS2_RANK1[DQS_PE_BYTE1] and DQCAL_STATUS2_RANK1[DQS_NE_BYTE1] are correct. A valid pass is only when number of DQS edges received are correct and the PASS comparison bit is set. 0x0: FAIL (FAIL) 0x1: PASS (PASS)
0	DQCAL_COMPARE_BYTE0	READ ONLY Indicates whether the DQ calibration command for LPDDR3 or PCDDR3 triggered via SHKE_DRAM_MANUAL_0[RD_DQCAL] resulted in an expected pattern comparison pass/fail. The value in this CSR field is valid when HW deasserts SHKE_DRAM_MANUAL_0[RD_DQCAL] indicating that the command completed. SW also needs to ensure that the number of DQS edges received in DQCAL_STATUS2_RANK1[DQS_PE_BYTE0] and DQCAL_STATUS2_RANK1[DQS_NE_BYTE0] are correct. A valid pass is only when number of DQS edges received are correct and the PASS comparison bit is set. 0x0: FAIL (FAIL) 0x1: PASS (PASS)

0x0044C284 BIMC_S_DDR0_DPE_DQCAL_STATUS1_RANK0**Type:** R**Clock:** ACLK**Reset State:** 0x00000000

DQCAL Status1 Register for Rank0. The contents of this CSR are only valid for x64 DRAM devices.

Note that the byte numbers below may not correspond to the actual DRAM byte. This is required so as to keep the byte number reference correct between the controller and the PHY (which has no information on the byte swap feature)

BIMC_S_DDR0_DPE_DQCAL_STATUS1_RANK0

Bits	Name	Description
30:28	DQS_PE_BYTE7	READ ONLY Indicates the number of DQS PE edges received for the DQ calibration command. The value in this CSR field is valid when HW deasserts SHKE_DRAM_MANUAL_0[RD_DQCAL] indicating that the command completed. For LPDDR3/PCDDR3, the calibration command has a memory burst length of 8. 0x0: NONE (None) 0x1: ENUM_1_POS_EDGE (1_POS_EDGE) 0x2: ENUM_2_POS_EDGES (2_POS_EDGES) 0x3: ENUM_3_POS_EDGES (3_POS_EDGES) 0x4: ENUM_4_POS_EDGES (4_POS_EDGES) 0x5: ENUM_5_POS_EDGES (5_POS_EDGES) 0x6: ENUM_6_POS_EDGES (6_POS_EDGES) 0x7: ENUM_7_OR_MORE_POS_EDGES (7_OR_MORE_POS_EDGES)
26:24	DQS_NE_BYTE7	READ ONLY Indicates the number of DQS NE edges received for the DQ calibration command. The value in this CSR field is valid when HW deasserts SHKE_DRAM_MANUAL_0[RD_DQCAL] indicating that the command completed. For LPDDR3/PCDDR3, the calibration command has a memory burst length of 8. 0x0: NONE (None) 0x1: ENUM_1_POS_EDGE (1_POS_EDGE) 0x2: ENUM_2_POS_EDGES (2_POS_EDGES) 0x3: ENUM_3_POS_EDGES (3_POS_EDGES) 0x4: ENUM_4_POS_EDGES (4_POS_EDGES) 0x5: ENUM_5_POS_EDGES (5_POS_EDGES) 0x6: ENUM_6_POS_EDGES (6_POS_EDGES) 0x7: ENUM_7_OR_MORE_POS_EDGES (7_OR_MORE_POS_EDGES)
22:20	DQS_PE_BYTE6	READ ONLY Indicates the number of DQS PE edges received for the DQ calibration command. The value in this CSR field is valid when HW deasserts SHKE_DRAM_MANUAL_0[RD_DQCAL] indicating that the command completed. For LPDDR3/PCDDR3, the calibration command has a memory burst length of 8. 0x0: NONE (None) 0x1: ENUM_1_POS_EDGE (1_POS_EDGE) 0x2: ENUM_2_POS_EDGES (2_POS_EDGES) 0x3: ENUM_3_POS_EDGES (3_POS_EDGES) 0x4: ENUM_4_POS_EDGES (4_POS_EDGES) 0x5: ENUM_5_POS_EDGES (5_POS_EDGES) 0x6: ENUM_6_POS_EDGES (6_POS_EDGES) 0x7: ENUM_7_OR_MORE_POS_EDGES (7_OR_MORE_POS_EDGES)

BIMC_S_DDR0_DPE_DQCAL_STATUS1_RANK0 (cont.)

Bits	Name	Description
18:16	DQS_NE_BYTE6	READ ONLY Indicates the number of DQS NE edges received for the DQ calibration command. The value in this CSR field is valid when HW deasserts SHKE_DRAM_MANUAL_0[RD_DQCAL] indicating that the command completed. For LPDDR3/PCDDR3, the calibration command has a memory burst length of 8. 0x0: NONE (None) 0x1: ENUM_1_POS_EDGE (1_POS_EDGE) 0x2: ENUM_2_POS_EDGES (2_POS_EDGES) 0x3: ENUM_3_POS_EDGES (3_POS_EDGES) 0x4: ENUM_4_POS_EDGES (4_POS_EDGES) 0x5: ENUM_5_POS_EDGES (5_POS_EDGES) 0x6: ENUM_6_POS_EDGES (6_POS_EDGES) 0x7: ENUM_7_OR_MORE_POS_EDGES (7_OR_MORE_POS_EDGES)
14:12	DQS_PE_BYTE5	READ ONLY Indicates the number of DQS PE edges received for the DQ calibration command. The value in this CSR field is valid when HW deasserts SHKE_DRAM_MANUAL_0[RD_DQCAL] indicating that the command completed. For LPDDR3/PCDDR3, the calibration command has a memory burst length of 8. 0x0: NONE (None) 0x1: ENUM_1_POS_EDGE (1_POS_EDGE) 0x2: ENUM_2_POS_EDGES (2_POS_EDGES) 0x3: ENUM_3_POS_EDGES (3_POS_EDGES) 0x4: ENUM_4_POS_EDGES (4_POS_EDGES) 0x5: ENUM_5_POS_EDGES (5_POS_EDGES) 0x6: ENUM_6_POS_EDGES (6_POS_EDGES) 0x7: ENUM_7_OR_MORE_POS_EDGES (7_OR_MORE_POS_EDGES)
10:8	DQS_NE_BYTE5	READ ONLY Indicates the number of DQS NE edges received for the DQ calibration command. The value in this CSR field is valid when HW deasserts SHKE_DRAM_MANUAL_0[RD_DQCAL] indicating that the command completed. For LPDDR3/PCDDR3, the calibration command has a memory burst length of 8. 0x0: NONE (None) 0x1: ENUM_1_POS_EDGE (1_POS_EDGE) 0x2: ENUM_2_POS_EDGES (2_POS_EDGES) 0x3: ENUM_3_POS_EDGES (3_POS_EDGES) 0x4: ENUM_4_POS_EDGES (4_POS_EDGES) 0x5: ENUM_5_POS_EDGES (5_POS_EDGES) 0x6: ENUM_6_POS_EDGES (6_POS_EDGES) 0x7: ENUM_7_OR_MORE_POS_EDGES (7_OR_MORE_POS_EDGES)

BIMC_S_DDR0_DPE_DQCAL_STATUS1_RANK0 (cont.)

Bits	Name	Description
6:4	DQS_PE_BYTE4	READ ONLY Indicates the number of DQS PE edges received for the DQ calibration command. The value in this CSR field is valid when HW deasserts SHKE_DRAM_MANUAL_0[RD_DQCAL] indicating that the command completed. For LPDDR3/PCDDR3, the calibration command has a memory burst length of 8. 0x0: NONE (None) 0x1: ENUM_1_POS_EDGE (1_POS_EDGE) 0x2: ENUM_2_POS_EDGES (2_POS_EDGES) 0x3: ENUM_3_POS_EDGES (3_POS_EDGES) 0x4: ENUM_4_POS_EDGES (4_POS_EDGES) 0x5: ENUM_5_POS_EDGES (5_POS_EDGES) 0x6: ENUM_6_POS_EDGES (6_POS_EDGES) 0x7: ENUM_7_OR_MORE_POS_EDGES (7_OR_MORE_POS_EDGES)
2:0	DQS_NE_BYTE4	READ ONLY Indicates the number of DQS NE edges received for the DQ calibration command. The value in this CSR field is valid when HW deasserts SHKE_DRAM_MANUAL_0[RD_DQCAL] indicating that the command completed. For LPDDR3/PCDDR3, the calibration command has a memory burst length of 8. 0x0: NONE (None) 0x1: ENUM_1_POS_EDGE (1_POS_EDGE) 0x2: ENUM_2_POS_EDGES (2_POS_EDGES) 0x3: ENUM_3_POS_EDGES (3_POS_EDGES) 0x4: ENUM_4_POS_EDGES (4_POS_EDGES) 0x5: ENUM_5_POS_EDGES (5_POS_EDGES) 0x6: ENUM_6_POS_EDGES (6_POS_EDGES) 0x7: ENUM_7_OR_MORE_POS_EDGES (7_OR_MORE_POS_EDGES)

0x0044C288 BIMC_S_DDR0_DPE_DQCAL_STATUS2_RANK0**Type:** R**Clock:** ACLK**Reset State:** 0x00000000

DQCAL Status2 Register for Rank0.

Note that the byte numbers below may not correspond to the actual DRAM byte. This is required so as to keep the byte number reference correct between the controller and the PHY (which has no information on the byte swap feature)

BIMC_S_DDR0_DPE_DQCAL_STATUS2_RANK0

Bits	Name	Description
30:28	DQS_PE_BYTE3	READ ONLY Indicates the number of DQS PE edges received for the DQ calibration command. The value in this CSR field is valid when HW deasserts SHKE_DRAM_MANUAL_0[RD_DQCAL] indicating that the command completed. For LPDDR3/PCDDR3, the calibration command has a memory burst length of 8. 0x0: NONE (None) 0x1: ENUM_1_POS_EDGE (1_POS_EDGE) 0x2: ENUM_2_POS_EDGES (2_POS_EDGES) 0x3: ENUM_3_POS_EDGES (3_POS_EDGES) 0x4: ENUM_4_POS_EDGES (4_POS_EDGES) 0x5: ENUM_5_POS_EDGES (5_POS_EDGES) 0x6: ENUM_6_POS_EDGES (6_POS_EDGES) 0x7: ENUM_7_OR_MORE_POS_EDGES (7_OR_MORE_POS_EDGES)
26:24	DQS_NE_BYTE3	READ ONLY Indicates the number of DQS NE edges received for the DQ calibration command. The value in this CSR field is valid when HW deasserts SHKE_DRAM_MANUAL_0[RD_DQCAL] indicating that the command completed. For LPDDR3/PCDDR3, the calibration command has a memory burst length of 8. 0x0: NONE (None) 0x1: ENUM_1_POS_EDGE (1_POS_EDGE) 0x2: ENUM_2_POS_EDGES (2_POS_EDGES) 0x3: ENUM_3_POS_EDGES (3_POS_EDGES) 0x4: ENUM_4_POS_EDGES (4_POS_EDGES) 0x5: ENUM_5_POS_EDGES (5_POS_EDGES) 0x6: ENUM_6_POS_EDGES (6_POS_EDGES) 0x7: ENUM_7_OR_MORE_POS_EDGES (7_OR_MORE_POS_EDGES)
22:20	DQS_PE_BYTE2	READ ONLY Indicates the number of DQS PE edges received for the DQ calibration command. The value in this CSR field is valid when HW deasserts SHKE_DRAM_MANUAL_0[RD_DQCAL] indicating that the command completed. For LPDDR3/PCDDR3, the calibration command has a memory burst length of 8. 0x0: NONE (None) 0x1: ENUM_1_POS_EDGE (1_POS_EDGE) 0x2: ENUM_2_POS_EDGES (2_POS_EDGES) 0x3: ENUM_3_POS_EDGES (3_POS_EDGES) 0x4: ENUM_4_POS_EDGES (4_POS_EDGES) 0x5: ENUM_5_POS_EDGES (5_POS_EDGES) 0x6: ENUM_6_POS_EDGES (6_POS_EDGES) 0x7: ENUM_7_OR_MORE_POS_EDGES (7_OR_MORE_POS_EDGES)

BIMC_S_DDR0_DPE_DQCAL_STATUS2_RANK0 (cont.)

Bits	Name	Description
18:16	DQS_NE_BYTE2	READ ONLY Indicates the number of DQS NE edges received for the DQ calibration command. The value in this CSR field is valid when HW deasserts SHKE_DRAM_MANUAL_0[RD_DQCAL] indicating that the command completed. For LPDDR3/PCDDR3, the calibration command has a memory burst length of 8. 0x0: NONE (None) 0x1: ENUM_1_POS_EDGE (1_POS_EDGE) 0x2: ENUM_2_POS_EDGES (2_POS_EDGES) 0x3: ENUM_3_POS_EDGES (3_POS_EDGES) 0x4: ENUM_4_POS_EDGES (4_POS_EDGES) 0x5: ENUM_5_POS_EDGES (5_POS_EDGES) 0x6: ENUM_6_POS_EDGES (6_POS_EDGES) 0x7: ENUM_7_OR_MORE_POS_EDGES (7_OR_MORE_POS_EDGES)
14:12	DQS_PE_BYTE1	READ ONLY Indicates the number of DQS PE edges received for the DQ calibration command. The value in this CSR field is valid when HW deasserts SHKE_DRAM_MANUAL_0[RD_DQCAL] indicating that the command completed. For LPDDR3/PCDDR3, the calibration command has a memory burst length of 8. 0x0: NONE (None) 0x1: ENUM_1_POS_EDGE (1_POS_EDGE) 0x2: ENUM_2_POS_EDGES (2_POS_EDGES) 0x3: ENUM_3_POS_EDGES (3_POS_EDGES) 0x4: ENUM_4_POS_EDGES (4_POS_EDGES) 0x5: ENUM_5_POS_EDGES (5_POS_EDGES) 0x6: ENUM_6_POS_EDGES (6_POS_EDGES) 0x7: ENUM_7_OR_MORE_POS_EDGES (7_OR_MORE_POS_EDGES)
10:8	DQS_NE_BYTE1	READ ONLY Indicates the number of DQS NE edges received for the DQ calibration command. The value in this CSR field is valid when HW deasserts SHKE_DRAM_MANUAL_0[RD_DQCAL] indicating that the command completed. For LPDDR3/PCDDR3, the calibration command has a memory burst length of 8. 0x0: NONE (None) 0x1: ENUM_1_POS_EDGE (1_POS_EDGE) 0x2: ENUM_2_POS_EDGES (2_POS_EDGES) 0x3: ENUM_3_POS_EDGES (3_POS_EDGES) 0x4: ENUM_4_POS_EDGES (4_POS_EDGES) 0x5: ENUM_5_POS_EDGES (5_POS_EDGES) 0x6: ENUM_6_POS_EDGES (6_POS_EDGES) 0x7: ENUM_7_OR_MORE_POS_EDGES (7_OR_MORE_POS_EDGES)

BIMC_S_DDR0_DPE_DQCAL_STATUS2_RANK0 (cont.)

Bits	Name	Description
6:4	DQS_PE_BYTE0	READ ONLY Indicates the number of DQS PE edges received for the DQ calibration command. The value in this CSR field is valid when HW deasserts SHKE_DRAM_MANUAL_0[RD_DQCAL] indicating that the command completed. For LPDDR3/PCDDR3, the calibration command has a memory burst length of 8. 0x0: NONE (None) 0x1: ENUM_1_POS_EDGE (1_POS_EDGE) 0x2: ENUM_2_POS_EDGES (2_POS_EDGES) 0x3: ENUM_3_POS_EDGES (3_POS_EDGES) 0x4: ENUM_4_POS_EDGES (4_POS_EDGES) 0x5: ENUM_5_POS_EDGES (5_POS_EDGES) 0x6: ENUM_6_POS_EDGES (6_POS_EDGES) 0x7: ENUM_7_OR_MORE_POS_EDGES (7_OR_MORE_POS_EDGES)
2:0	DQS_NE_BYTE0	READ ONLY Indicates the number of DQS NE edges received for the DQ calibration command. The value in this CSR field is valid when HW deasserts SHKE_DRAM_MANUAL_0[RD_DQCAL] indicating that the command completed. For LPDDR3/PCDDR3, the calibration command has a memory burst length of 8. 0x0: NONE (None) 0x1: ENUM_1_POS_EDGE (1_POS_EDGE) 0x2: ENUM_2_POS_EDGES (2_POS_EDGES) 0x3: ENUM_3_POS_EDGES (3_POS_EDGES) 0x4: ENUM_4_POS_EDGES (4_POS_EDGES) 0x5: ENUM_5_POS_EDGES (5_POS_EDGES) 0x6: ENUM_6_POS_EDGES (6_POS_EDGES) 0x7: ENUM_7_OR_MORE_POS_EDGES (7_OR_MORE_POS_EDGES)

0x0044C2FC BIMC_S_DDR0_DPE_CGC_CTRL**Type:** RW**Clock:** ACLK**Reset State:** 0x00000000

DPE clock gating control register. This register is used to enable automatic, dynamic gating of clock trees within the DPE. Values written to the CSR_ACCESS and RECALC_TCYC fields take effect immediately, while changes to all other fields become active upon writing a value of '1' to CONFIG_4[LOAD_ALL_CONFIG] or when CONFIG_3[LOAD_TIMING_ON_FREQ_CHNG] is set to '1' and a frequency change occurs.

BIMC_S_DDR0_DPE_CGC_CTRL

Bits	Name	Description
16	BANKTIMERS	Bank timer clock tree dynamic gating. 0x0: DISABLE (Disable) 0x1: ENABLE (Enable)
15	COL_TIMERS	Column timer clock tree dynamic gating. 0x0: DISABLE (Disable) 0x1: ENABLE (Enable)
14	CQ	Command queue clock tree dynamic gating. 0x0: DISABLE (Disable) 0x1: ENABLE (Enable)
13	CSR_ACCESS	CSR access clock tree dynamic gating. 0x0: DISABLE (Disable) 0x1: ENABLE (Enable)
12	LOAD_TIMING	Load timing clock tree dynamic gating. 0x0: DISABLE (Disable) 0x1: ENABLE (Enable)
11	MEMCAS_CMD	Column logic clock tree dynamic gating. 0x0: DISABLE (Disable) 0x1: ENABLE (Enable)
10	NDX_FIFO	Read index FIFO clock tree dynamic gating. 0x0: DISABLE (Disable) 0x1: ENABLE (Enable)
9	OPT	Open page table clock tree dynamic gating. 0x0: DISABLE (Disable) 0x1: ENABLE (Enable)
8	PAGE_IDLE_TIMER	Page idle timer clock tree dynamic gating. 0x0: DISABLE (Disable) 0x1: ENABLE (Enable)
7	PWR_MGMT	Power management clock tree dynamic gating. 0x0: DISABLE (Disable) 0x1: ENABLE (Enable)
6	RANKTIMERS	Rank timer clock tree dynamic gating. 0x0: DISABLE (Disable) 0x1: ENABLE (Enable)
5	RDATA	Read data FIFO clock tree dynamic gating. 0x0: DISABLE (Disable) 0x1: ENABLE (Enable)
4	RECALC_TCYC	Timing-based parameter logic clock tree dynamic gating. 0x0: DISABLE (Disable) 0x1: ENABLE (Enable)

BIMC_S_DDR0_DPE_CGC_CTRL (cont.)

Bits	Name	Description
3	SHKECS	Housekeep command sequencer clock tree dynamic gating. 0x0: DISABLE (Disable) 0x1: ENABLE (Enable)
2	WR_BUFFER	Write data buffer clock tree dynamic gating. 0x0: DISABLE (Disable) 0x1: ENABLE (Enable)
1	WR_CTRL	Write control logic clock tree dynamic gating. 0x0: DISABLE (Disable) 0x1: ENABLE (Enable)
0	WRDATA_MGMT	Write data management clock tree dynamic clock gating. 0x0: DISABLE (Disable) 0x1: ENABLE (Enable)

2.28 BIMC_S_DDR0_SHKE_ABHN_SHKE**0x0044D004 BIMC_S_DDR0_SHKE_CONFIG****Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

Config register for SHKE.

BIMC_S_DDR0_SHKE_CONFIG

Bits	Name	Description
29	RANK1_EN	SW: RW, HW: R Rank1 Enable 0x0: DISABLE (Disable (default)) 0x1: ENABLE (Enable)
28	RANK0_EN	SW: RW, HW: R Rank0 Enable 0x0: DISABLE (Disable (default)) 0x1: ENABLE (Enable)
25	RANK1_INITCOMPLETE	SW: RW, HW: R After initialization of rank1 is complete, SW writes this field to 1. 0x0: INIT_INCOMPLETE (Init Incomplete (default)) 0x1: INIT_COMPLETE (Init Complete)

BIMC_S_DDR0_SHKE_CONFIG (cont.)

Bits	Name	Description
24	RANK0_INITCOMPLETE	SW: RW, HW: R After initialization of rank0 is complete, SW writes this field to 1. 0x0: INIT_INCOMPLETE (Init Incomplete (default)) 0x1: INIT_COMPLETE (Init Complete)
21:20	NUM_BANKS	SW: RW, HW: R Number of Banks in the memory device. 0x0: ENUM_4_BANKS (4 banks (default)) 0x1: ENUM_8_BANKS (8 banks)
19:16	IOCAL_UPDATE_PULSE_WIDTH	SW: RW, HW: R Number of DDR clock cycles for the pulse for IOCAL updates. 0x0: INVALID (Invalid (default))
15	IOCAL_UPDATE_PULSE_WIDTH_MULT	SW: RW, HW: R This field provides a multiplier (1x/2x) for the pulse width above. Valid values are 0 (multiplier of 1) and 1 (multiplier of 2) 0x0: MULT_1X (MULT_1x(default)) 0x1: MULT_2X (MULT_2x (default))
0	CLK_ON_EN	SW: RW, HW: R Enable clock on functionality. The controller can request that the clock remain on through the use of a clock on port. When this feature is enabled, the controller can indicate that it no longer needs the DPE clock by de-asserting the clock on port. 0x0: DISABLE (Disable (default)) 0x1: ENABLE (Enable)

0x0044D010 BIMC_S_DDR0_SHKE_CGC_CNTL**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

Clock Gating control register for SHKE.

BIMC_S_DDR0_SHKE_CGC_CNTL

Bits	Name	Description
0	CSR_ACCESS	SW: RW, HW: R CSR access clock tree dynamic gating 0x0: DISABLE (Disable (default)) 0x1: ENABLE (Enable)

0x0044D020 BIMC_S_DDR0_SHKE_DRAM_MANUAL_0**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

Manual Command Register 0.

This register enables the software to manually issue commands to the DDR SDRAM interface. A value of '1' should be written in the bit field of the command that the software wants to force the controller to perform. When the controller has finished executing the command, it will clear the bit. Only one bit should be active at any given time in the register below, except for the RANK_SEL field. This limits the SW to have only 1 outstanding SW command for execution by the memory controller. This CSR is typically used for initialization the memory rank along with the DRAM_MANUAL_1 register. During SDRAM device initialization, SW should also program the DRAM_MANUAL_1 register (optional) so HW can maintain timing between the memory commands else, this is SW's responsibility. Once the rank(s) are initialized, it is not typical for SW to trigger commands (except those which are required as part of sequences - eg. Deep Power down). However after rank initialization, the HW will maintain protocol legal timings for SW triggered commands.

BIMC_S_DDR0_SHKE_DRAM_MANUAL_0

Bits	Name	Description
29:28	RANK_SEL	SW: RW, HW: R Select which chip select should be targeted for the chosen manual command to be executed. 0x0: INVALID (Invalid (default)) 0x1: CS0_ONLY (CS0 only) 0x2: CS1_ONLY (CS1 only) 0x3: CS0_AND_CS1 (CS0 and CS1)
27	CK_ON	SW: RW, HW: R Turns on external clock CK, CKN. This bit is self-cleared once the command is executed. Until the rank is initialized, HW would control the clocks based on this setting. 0x0: NOP (Nop (default)) 0x1: EXECUTE (Execute)
26	CKE_ON	SW: RW, HW: R Turns on clock enable CKE. This bit is self-cleared once the command is executed. Until the rank is initialized, HW would control the CKE based on this setting. 0x0: NOP (Nop (default)) 0x1: EXECUTE (Execute)

BIMC_S_DDR0_SHKE_DRAM_MANUAL_0 (cont.)

Bits	Name	Description
25	RESETN_ON	SW: RW, HW: R Turns on reset pin to PCDDR3/4 devices only. This bit is self-cleared once the command is executed. Until the rank is initialized, HW would control the SDRAM reset based on this setting. 0x0: NOP (Nop (default)) 0x1: EXECUTE (Execute)
24	CK_OFF	SW: RW, HW: R Turns off external clock CK, CKN. This bit is self-cleared once the command is executed. Until the rank is initialized, HW would control the clocks based on this setting. 0x0: NOP (Nop (default)) 0x1: EXECUTE (Execute)
23	CKE_OFF	SW: RW, HW: R Turns off clock enable CKE. This bit is self-cleared once the command is executed. Until the rank is initialized, HW would control the CKE based on this setting. 0x0: NOP (Nop (default)) 0x1: EXECUTE (Execute)
22	RESETN_OFF	SW: RW, HW: R Turns off reset pin to PCDDR3/4 devices only. This bit is self-cleared once the command is executed. Until the rank is initialized, HW would control the SDRAM reset based on this setting. 0x0: NOP (Nop (default)) 0x1: EXECUTE (Execute)
21	ENTER_DEEP_PD	SW: RW, HW: RW Issue Deep Power Down. This bit is self-cleared once the command is executed. Deep Power down feature is only available on LPDDR2/3 SDRAM devices. HW will check to ensure all commands outstanding to the rank have completed before issuing the DPD command. The RANK_SEL field for this command cannot be set to '11'. 0x0: NOP (Nop (default)) 0x1: EXECUTE (Execute)

BIMC_S_DDR0_SHKE_DRAM_MANUAL_0 (cont.)

Bits	Name	Description
20	EXIT_DEEP_PD	SW: RW, HW: RW Issue Exit Deep Power Down. This bit is self-cleared once the command is executed. Deep Power down feature is only available on LPDDR2/3 SDRAM devices. The RANK_SEL field for this command cannot be set to '11'. When exiting Deep Power Down for a given Rank, it is required to re-initialize the LPDDR2/3 SDRAM devices by repeating the initialization sequence for the associated rank before that memory can be utilized or accessed. It is SW's responsibility to ensure that the minimum duration of DPD is met. Additionally SW needs to ensure that the clocks are stable before exiting DPD. Refer to the HW programming guide for Deep Power Down sequence 0x0: NOP (Nop (default)) 0x1: EXECUTE (Execute)
19	ZQCAL_SHORT	SW: RW, HW: RW Issue Short ZQ-Cal Command. This bit is self-cleared once the command is executed. The RANK_SEL field for this command cannot be set to '11'. 0x0: NOP (Nop (default)) 0x1: EXECUTE (Execute)
18	ZQCAL_LONG	SW: RW, HW: RW Issue Long ZQ-Cal Command. This bit is self-cleared once the command is executed. The RANK_SEL field for this command cannot be set to '11'. 0x0: NOP (Nop (default)) 0x1: EXECUTE (Execute)
17	AUTO_REFRESH	SW: RW, HW: RW NOT SUPPORTED. Auto-Refresh Command. This bit is self-cleared once the command is executed. 0x0: NOP (Nop (default)) 0x1: EXECUTE (Execute)
16	PRECHARGE_ALL	SW: RW, HW: RW NOT SUPPORTED. Precharge All Command. This bit is self-cleared once the command is executed. 0x0: NOP (Nop (default)) 0x1: EXECUTE (Execute)

BIMC_S_DDR0_SHKE_DRAM_MANUAL_0 (cont.)

Bits	Name	Description
15	ENTER_SELF_REFRESH_IDLE	SW: RW, HW: R Enter Self-Refresh when idle. This SW trigger for self refresh is only supported for all active ranks. SW needs to account for the same when setting the RANK_SEL field for this command (eg. if the other rank is uninitialized or in deep power down). This bit is self-cleared once the command is executed. 0x0: NOP (Nop (default)) 0x1: EXECUTE (Execute)
14	ENTER_SELF_REFRESH_ASAP	SW: RW, HW: R Enter Self-Refresh immediately. This SW trigger for self refresh is only supported for all active ranks. SW needs to account for the same when setting the RANK_SEL field for this command (eg. if the other rank is uninitialized or in deep power down). This bit is self-cleared once the command is executed. 0x0: NOP (Nop (default)) 0x1: EXECUTE (Execute)
13	EXIT_SELF_REFRESH	SW: RW, HW: R Exit Self-Refresh immediately. This SW trigger for self refresh is only supported for all active ranks. SW needs to account for the same when setting the RANK_SEL field for this command (eg. if the other rank is uninitialized or in deep power down). Also, this command is only supported if SW had triggered a self refresh entry event or a watchdog event occurred. The corresponding status can be observed via SHKE_DRAM_STATUS CSR. This bit is self-cleared once the command is executed. 0x0: NOP (Nop (default)) 0x1: EXECUTE (Execute)
12	MODE_REGISTER_READ	SW: RW, HW: RW Mode Register Read Command. This only applies to LPDDR2/3 MRR commands. The Mode Register Address information for the command is used from SHKE_MREG_ADDR_WDATA_CNTL[MREG_ADDR]. This bit is self-cleared once the command is executed. Additionally, the read data from the memory device is available in SHKE_MREG_RDATA_RANK1/0 CSR. The read status is available in SHKE_MREG_RDATA_STATUS CSR. The RANK_SEL field for this command cannot be set to '11'. 0x0: NOP (Nop (default)) 0x1: EXECUTE (Execute)

BIMC_S_DDR0_SHKE_DRAM_MANUAL_0 (cont.)

Bits	Name	Description
11	MODE_REGISTER_WRITE	SW: RW, HW: RW Mode Register Write Command. The Mode Register Address and data information for the command is used from SHKE_MREG_ADDR_WDATA_CNTL[MREG_ADDR] and SHKE_MREG_ADDR_WDATA_CNTL[MREG_WDATA] CSR fields. This bit is self-cleared once the command is executed. 0x0: NOP (Nop (default)) 0x1: EXECUTE (Execute)
8	STALL_CMDS	SW: RW, HW: R This allows SW to stop transactions on the DDR bus. Once the transactions are stalled, SW will be required to unstall by executing the RESUME_CMDS command. This feature has been provided for debug and the stall status is available via SHKE_DRAM_STATUS CSR. This SW trigger for stall is only supported for all active ranks. SW needs to account for the same when setting the RANK_SEL field for this command (e.g. if the other rank is uninitialized or in deep power down). This bit is self-cleared once the command is executed. 0x0: NOP (Nop (default)) 0x1: EXECUTE (Execute)
7	RESUME_CMDS	SW: RW, HW: R This allows SW to resume transactions on the DDR bus. This feature has been provided for debug. This SW trigger for resume is only supported for all active ranks. SW needs to account for the same when setting the RANK_SEL field for this command (eg. if the other rank is uninitialized or in deep power down). Also, this command is only supported if SW had triggered a stall. The corresponding status can be observed via SHKE_DRAM_STATUS CSR This bit is self-cleared once the command is executed. 0x0: NOP (Nop (default)) 0x1: EXECUTE (Execute)
6	EXTND_MODE_REGISTER_WRITE	SW: RW, HW: RW Extended Mode Register Write Command. The Mode Register Address and data information for the command is used from SHKE_MREG_ADDR_WDATA_CNTL[MREG_ADDR] and SHKE_MREG_ADDR_WDATA_CNTL[MREG_WDATA] CSR fields. This bit is self-cleared once the command is executed. This bit is only to be used for Mode register writes related to the LPDDR3 CA training sequence as this allows the CA bus to be stable before and after the command to allow for sufficient setup time for the DRAM to sample the command correctly. The RANK_SEL field for this command cannot be set to '11'. 0x0: NOP (Nop (default)) 0x1: EXECUTE (Execute)

BIMC_S_DDR0_SHKE_DRAM_MANUAL_0 (cont.)

Bits	Name	Description
5	CA_TRAIN_PATTERN_CMD	SW: RW, HW: RW LPDDR3 CA Pattern Command for CA Training. The specific address pattern to be driven in needs to be programmed in DPE_CA_TRAIN_PRE_CS, DPE_CA_TRAIN_CS and DPE_CA_TRAIN_POST_CS CSRs before triggering this command. This bit is self-cleared once the command is executed. The RANK_SEL field for this command cannot be set to '11'. 0x0: NOP (Nop (default)) 0x1: EXECUTE (Execute)
4	RD_DQCAL	SW: RW, HW: RW DQ Calibration Read Command. This relates to LPDDR3 MRR command & PCDDR3 MPR command. This is used for read training and/or rank auto detect. A pre-defined read pattern is sent out by the DRAM and captured by the controller. In case of LPDDR3, this is an MRR commands with SHKE_MREG_ADDR_WDATA_CNTL[MREG_ADDR] specified correctly (valid values are 0x32 & 0x40). In case of PCDDR3 devices, this is an MPR read command and hence SHKE_MREG_ADDR_WDATA_CNTL[MREG_ADDR] is a don't care. This bit is self-cleared once the command is executed. The read data status for the calibration read command is captured in the DPE status registers. The RANK_SEL field for this command cannot be set to '11'. 0x0: NOP (Nop (default)) 0x1: EXECUTE (Execute)

0x0044D024 BIMC_S_DDR0_SHKE_DRAM_MANUAL_1**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

Manual Command Register 1.

This register can only be used during memory rank initialization. As part of the memory device initialization, certain delays are required between commands. This can be enforced by SW or by setting up the following register before triggering a command to be executed via DRAM_MANUAL_0 CSR. When WAIT_TIMER_BEFORE_HW_CLEAR is set to a non-zero value any SW command issued during memory device initialization will wait the programmable number of timer/xo clock cycles before indicating the command completion response.

BIMC_S_DDR0_SHKE_DRAM_MANUAL_1

Bits	Name	Description
16	WAIT_TIMER_DOMAIN	SW: RW, HW: R Indicates which clock should be used in HW for the down counter. 0x0: TIMER_CLOCK (Timer Clock (default)) 0x1: XO_CLOCK (XO Clock)
15:0	WAIT_TIMER_BEFORE_HW_CLEAR	SW: RW, HW: R This indicates the amount of time in timer/xo clock cycles required to wait after sending the SW triggered command in SHKE_DRAM_MANUAL_0 CSR before indicating the command completion response.

0x0044D030 BIMC_S_DDR0_SHKE_MREG_ADDR_WDATA_CNTL**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

Mode Register Address and Write Data Control.

A SW triggered mode register read/write command via DRAM_MANUAL_0[MODE_REGISTER_READ/WRITE] CSR uses the mode register address and write data (if Mode Register Write) from the register below.

BIMC_S_DDR0_SHKE_MREG_ADDR_WDATA_CNTL

Bits	Name	Description
27:20	MREG_ADDR	SW: RW, HW: R Mode Register Address Information for the Mode Register Read/Write triggered via SW using SHKE_DRAM_MANUAL_0[MODE_REGISTER_READ/WRITE], SHKE_DRAM_MANUAL_0[EXTND_MODE_REGISTER_WRITE] or SHKE_DRAM_MANUAL_0[RD_DQCAL]
15:0	MREG_WDATA	SW: RW, HW: R Mode Register Write data information corresponding to the Mode Register Write command triggered via SW using SHKE_DRAM_MANUAL_0[MODE_REGISTER_WRITE/EXTND_MODE_REGISTER_WRITE] CSR.

0x0044D050 BIMC_S_DDR0_SHKE_MREG_RDATA_RANK1_U**Type:** R**Clock:** HCLK**Reset State:** 0x00000000

SHKE Mode Register Read (Upper 32 bits) from Memory Rank1. The data in this register is valid only for 64-bit memory devices. (does not apply for LPDDR2/3 devices)

BIMC_S_DDR0_SHKE_MREG_RDATA_RANK1_U

Bits	Name	Description
31:0	MREG_RDATA	This is the read data corresponding to either the Mode Register Read triggered via SW using DRAM_MANUAL_0[MODE_REGISTER_READ] CSR targeted to rank1. In case of an mode register read command, the read data represents 32 bits of the first beat of read data from the SDRAM device.

0x0044D054 BIMC_S_DDR0_SHKE_MREG_RDATA_RANK1_L

Type: R

Clock: HCLK

Reset State: 0x00000000

SHKE Mode Register Read (Lower 32 bits) from Memory Rank1. The data in this register is valid only for both 64-bit & 32-bit memory devices.

BIMC_S_DDR0_SHKE_MREG_RDATA_RANK1_L

Bits	Name	Description
31:0	MREG_RDATA	This is the read data corresponding to either the Mode Register Read triggered via SW using DRAM_MANUAL_0[MODE_REGISTER_READ] CSR targeted to rank1. In case of an mode register read command, the read data represents 32 bits of the first beat of read data from the SDRAM device.

0x0044D060 BIMC_S_DDR0_SHKE_MREG_RDATA_RANK0_U

Type: R

Clock: HCLK

Reset State: 0x00000000

SHKE Mode Register Read (Upper 32 bits) from Memory Rank0. The data in this register is valid only for 64-bit memory devices. (does not apply for LPDDR2/3 devices)

BIMC_S_DDR0_SHKE_MREG_RDATA_RANK0_U

Bits	Name	Description
31:0	MREG_RDATA	This is the read data corresponding to either the Mode Register Read triggered via SW using DRAM_MANUAL_0[MODE_REGISTER_READ] CSR targeted to rank1. In case of an mode register read command, the read data represents 32 bits of the first beat of read data from the SDRAM device.

0x0044D064 BIMC_S_DDR0_SHKE_MREG_RDATA_RANK0_L**Type:** R**Clock:** HCLK**Reset State:** 0x00000000

SHKE Mode Register Read (Lower 32 bits) from Memory Rank0. The data in this register is valid only for both 64-bit & 32-bit memory devices.

BIMC_S_DDR0_SHKE_MREG_RDATA_RANK0_L

Bits	Name	Description
31:0	MREG_RDATA	This is the read data corresponding to either the Mode Register Read triggered via SW using DRAM_MANUAL_0[MODE_REGISTER_READ] CSR targeted to rank0. In case of an mode register read command, the read data represents 32 bits of the first beat of read data from the SDRAM device.

0x0044D070 BIMC_S_DDR0_SHKE_MREG_RDATA_STATUS**Type:** R**Clock:** HCLK**Reset State:** 0x00000000

SHKE Mode Register Read status.

BIMC_S_DDR0_SHKE_MREG_RDATA_STATUS

Bits	Name	Description
0	MREG_TIMEOUT	This is the MRR status corresponding to the Mode Register Read triggered via SW using DRAM_MANUAL_0[MODE_REGISTER_READ] CSR. If the memory device does not respond with DQS edges (eg. if the memory device isn't present) then this bit would be set when HW clears the DRAM_MANUAL_0[MODE_REGISTER_READ] bit to indicate MRR completion. 0x0: NO_TIMEOUT (No Timeout(default)) 0x1: TIMEOUT (Timeout)

0x0044D074 BIMC_S_DDR0_SHKE_DRAM_STATUS**Type:** R**Clock:** HCLK**Reset State:** 0x00000000

SHKE Dram status based on SW trigger in DRAM_MANUAL_0 CSR control.

BIMC_S_DDR0_SHKE_DRAM_STATUS

Bits	Name	Description
2	WDOG_SELF_RFSH	This is the DRAM self refresh status corresponding to the watchdog reset event received. If set, it indicates that the memory controller is currently in self refresh state due to a watchdog reset event and hence requires a DDR_MANUAL_0[EXIT_SELF_REFRESH] trigger to exit from the forced self refresh state. 0x0: NOT_IN_SELF_REFRESH (Not In Self Refresh(default)) 0x1: IN_SELF_REFRESH (In Self Refresh)
1	SW_SELF_RFSH	This is the DRAM self refresh status corresponding to the SW triggered command via DRAM_MANUAL_0[ENTER_SELF_REFRESH_IDLE/_ASAP] and DDR_MANUAL_0[EXIT_SELF_REFRESH] CSR. If set, it indicates that the memory controller is currently in SW triggered self refresh state and hence requires a DDR_MANUAL_0[EXIT_SELF_REFRESH] trigger to exit from the SW forced self refresh state. 0x0: NOT_IN_SELF_REFRESH (Not In Self Refresh(default)) 0x1: IN_SELF_REFRESH (In Self Refresh)

BIMC_S_DDR0_SHKE_DRAM_STATUS (cont.)

Bits	Name	Description
0	STALL	This is the DRAM stall status corresponding to the SW triggered command via DRAM_MANUAL_0[STALL_CMDS] and DDR_MANUAL_0[RESUME_CMDS] CSR. If set, it indicates that the memory controller is currently in SW triggered stalled state and hence requires a DDR_MANUAL_0[RESUME_CMDS] trigger to exit from the SW forced stall state. 0x0: NOT_STALLED (Not Stalled(default)) 0x1: STALLED (Stalled)

0x0044D0A0 BIMC_S_DDR0_SHKE_PERIODIC_MRR**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

SHKE Periodic MRR Command.

This is typically used to generate periodic LPDDR2/3 MRR commands to monitor the temperature of the memory device and appropriately adjust the refresh rate depending on the SDRAM device temperature.

BIMC_S_DDR0_SHKE_PERIODIC_MRR

Bits	Name	Description
27:20	MRR_ADDR	Specify which mode register to access. For LPDDR2/3 devices, this would typically be set to 0x04.
17:16	MRR_RANK_SEL	SW: RW, HW: R Specify DRAM rank to which the command goes. The HW will only send the command to a rank which is initialized. 0x0: NOT_VALID (Not valid (default)) 0x1: RANK_0_ONLY (Rank 0 only) 0x2: RANK_1_ONLY (Rank 1 only) 0x3: BOTH_RANKS_0_AND_1 (Both ranks 0 and 1)

BIMC_S_DDR0_SHKE_PERIODIC_MRR (cont.)

Bits	Name	Description
15:0	MRR_INTERVAL	SW: RW, HW: R Number of timer clock edges between periodic MRR samples. This interval is maintained by the controller on a per-rank basis. The timer clock typically runs at 32 kHz, hence providing a range of 10-2050ms periodic interval. The minimum supported value is ?d4 for 1 rank systems and ?d13 for 2 rank systems. 0x0: DISABLED (Disabled (default)) 0x100: TIME_OF_10MS (Time of 10ms) 0x200: TIME_OF_20MS (Time of 20ms) 0x400: TIME_OF_30MS (Time of 30ms) 0x800: TIME_OF_60MS (Time of 60ms) 0x1000: TIME_OF_130MS (Time of 130ms) 0x2000: TIME_OF_260MS (Time of 260ms) 0x4000: TIME_OF_510MS (Time of 510ms) 0x8000: TIME_OF_1030MS (Time of 1030ms) 0xFFFF: TIME_OF_2050MS (Time of 2050ms)

0x0044D0B0 BIMC_S_DDR0_SHKE_PERIODIC_MRR_RDATA_RANK1_U**Type:** R**Clock:** HCLK**Reset State:** 0x00000000

SHKE Periodic Mode Register Read (Upper 32 bits) from Memory Rank1. The data in this register is valid only for 64-bit memory devices. (does not apply for LPDDR2/3 devices).

BIMC_S_DDR0_SHKE_PERIODIC_MRR_RDATA_RANK1_U

Bits	Name	Description
31:0	MREG_DATA	This is the read data corresponding to the Mode Register Read command issued to rank1 due to periodic mode register reads enabled via PERIODIC_MRR CSR. The data captured represents the upper 32 bits for the first beat of read data for 64bit memory devices.

0x0044D0B4 BIMC_S_DDR0_SHKE_PERIODIC_MRR_RDATA_RANK1_L**Type:** R**Clock:** HCLK**Reset State:** 0x00000000

SHKE Periodic Mode Register Read (Lower 32 bits) from Memory Rank1. The data in this register is valid for both 64-bit & 32-bit memory devices.

BIMC_S_DDR0_SHKE_PERIODIC_MRR_RDATA_RANK1_L

Bits	Name	Description
31:0	MREG_DATA	This is the read data corresponding to the Mode Register Read command issued to rank1 due to periodic mode register reads enabled via PERIODIC_MRR CSR. The data captured represents the lower 32 bits for the first beat of read data for 64bit memory devices or the only 32 bits of data for the first beat of read data for 32 bit memory devices.

0x0044D0B8 BIMC_S_DDR0_SHKE_PERIODIC_MRR_RDATA_RANK0_U**Type:** R**Clock:** HCLK**Reset State:** 0x00000000

SHKE Periodic Mode Register Read (Upper 32 bits) from Memory Rank0. The data in this register is valid only for 64-bit memory devices. (does not apply for LPDDR2/3 devices)

BIMC_S_DDR0_SHKE_PERIODIC_MRR_RDATA_RANK0_U

Bits	Name	Description
31:0	MREG_DATA	This is the read data corresponding to the Mode Register Read command issued to rank0 due to periodic mode register reads enabled via PERIODIC_MRR CSR. The data captured represents the upper 32 bits for the first beat of read data for 64bit memory devices.

0x0044D0BC BIMC_S_DDR0_SHKE_PERIODIC_MRR_RDATA_RANK0_L**Type:** R**Clock:** HCLK**Reset State:** 0x00000000

SHKE Periodic Mode Register Read (Lower 32 bits) from Memory Rank0. The data in this register is valid for both 64-bit & 32-bit memory devices.

BIMC_S_DDR0_SHKE_PERIODIC_MRR_RDATA_RANK0_L

Bits	Name	Description
31:0	MREG_DATA	This is the read data corresponding to the Mode Register Read command issued to rank0 due to periodic mode register reads enabled via PERIODIC_MRR CSR. The data captured represents the lower 32 bits for the first beat of read data for 64bit memory devices or the only 32 bits of data for the first beat of read data for 32 bit memory devices.

0x0044D0D0 BIMC_S_DDR0_SHKE_PERIODIC_ZQCAL**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

This register provides control for periodic ZQCalibration commands to be issued to the SDRAM memory device.

BIMC_S_DDR0_SHKE_PERIODIC_ZQCAL

Bits	Name	Description
29:28	RANK_SEL	SW: RW, HW: R Specify DRAM rank to which the command goes. The HW will only send the command to a rank which is initialized. 0x0: NOT_VALID (Not valid (default)) 0x1: RANK_0_ONLY (Rank 0 only) 0x2: RANK_1_ONLY (Rank 1 only) 0x3: BOTH_RANKS_0_AND_1 (Both ranks 0 and 1)
15:0	ZQCAL_INTERVAL	SW: RW, HW: R Number of timer clock edges between periodic ZQ calibrations. This interval is maintained by the controller on a per-rank basis. The timer clock typically runs at 32kHz, hence providing a range of 10-2050ms periodic interval. The minimum supported value is ?d4 for 1 rank systems and ?d13 for 2 rank systems. 0x0: DISABLED (Disabled (default)) 0x100: TIME_OF_10MS (Time of 10ms) 0x200: TIME_OF_20MS (Time of 20ms) 0x400: TIME_OF_30MS (Time of 30ms) 0x800: TIME_OF_60MS (Time of 60ms) 0x1000: TIME_OF_130MS (Time of 130ms) 0x2000: TIME_OF_260MS (Time of 260ms) 0x4000: TIME_OF_510MS (Time of 510ms) 0x8000: TIME_OF_1030MS (Time of 1030ms) 0xFFFF: TIME_OF_2050MS (Time of 2050ms)

0x0044D0F0 BIMC_S_DDR0_SHKE_AUTO_REFRESH_CNTL**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

This register provides the controls for auto refresh functionality for the SDRAM devices.

BIMC_S_DDR0_SHKE_AUTO_REFRESH_CNTL

Bits	Name	Description
29	AUTO_RFSH_ENABLE_RANK1	SW: RW, HW: R Auto Refresh Enable for Rank1 0x0: DISABLE (Disable (default)) 0x1: ENABLE (Enable)
28	AUTO_RFSH_ENABLE_RANK0	SW: RW, HW: R Auto Refresh Enable for Rank0 0x0: DISABLE (Disable (default)) 0x1: ENABLE (Enable)
25:24	AUTO_REFRESH_MODE	SW: RW, HW: R Specifies the auto refresh mode. Per-bank refresh mode is only supported for LPDDR2/3 8-bank devices. 0x0: ALL_BANK_REFRESH (All Bank Refresh (default)) 0x1: PER_BANK_REFRESH (Per Bank Refresh)
23:22	AUTO_SWITCH_ALL_BANK_MODE	SW: RW, HW: R If configured in per-bank refresh mode, this setting controls whether HW automatically switches to all-bank refresh mode when a certain high temperature of the memory device is detected which requires a faster refresh rate. When enabled, HW automatically switches back to the configured per-bank refresh mode when the device temperature returns below the threshold specified. When set to '10', HW will automatically switch to all-bank refresh mode when a TREFI multiplier of 0.25 is received and it switches back to per-bank refresh mode when a multiplier greater than 0.25 is received (eg. 0.5 or 1). It is illegal to change this CSR setting after the rank has been initialized. 0x0: DISABLE (Disable (default)) 0x1: TREFI_0_5X_OR_TREFI_0_25X (TREFI_0.5X_OR_TREFI_0.25X) 0x2: TREFI_0_25X (TREFI_0.25X) 0x3: NOT_VALID (Not Valid)
16	DISABLE_AUTO_REFRESH_TEMP_UPDATE	SW: RW, HW: R Disables the refresh rate updates based on SRR based DRAM temperature updates. It is illegal to change this CSR setting after the rank has been initialized. 0x0: ENABLE (Enable (default)) 0x1: DISABLE (Disable)
9:0	TREFI	SW: RW, HW: R If SHKE_AUTO_RFSH_CNTL_1[TREFI] is non-zero, then this field is used as TREFI value for rank0. If SHKE_AUTO_RFSH_CNTL_1[TREFI] is zero, then this field is used as TREFI value for both ranks. Number of clock cycle delay (in t _{cxo} clock cycles) between all-bank auto-refresh commands. This is required to be set to the ?desired t _{TREFI} ? -1 t _{cxo} cycles. It is illegal to change this CSR setting after the rank has been initialized.

0x0044D0F4 BIMC_S_DDR0_SHKE_SELF_REFRESH_CNTL**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

This register provides the controls for HW activity based self refresh.

BIMC_S_DDR0_SHKE_SELF_REFRESH_CNTL

Bits	Name	Description
29	HW_SELF_RFSH_ENABLE_RANK1	SW: RW, HW: R Enables HW to look for no activity for rank1 and trigger an idle timer specified based on the value of HW_SELF_RFSH_TIMER. If there is no activity still and the timer expires, the HW will initiate entry to self refresh mode. 0x0: DISABLE (Disable (default)) 0x1: ENABLE (Enable)
28	HW_SELF_RFSH_ENABLE_RANK0	SW: RW, HW: R Enables HW to look for no activity for rank0 and trigger an idle timer specified based on the value of HW_SELF_RFSH_TIMER. If there is no activity still and the timer expires, the HW will initiate entry to self refresh mode. 0x0: DISABLE (Disable (default)) 0x1: ENABLE (Enable)
23:12	HW_SELF_RFSH_IDLE_TIMER	SW: RW, HW: R Count idle tctx cycles before entering self refresh. The timer is based on XO clock frequency. Valid only when HW_SELF_RFSH_ENABLE_RANK* or CONCURRENT_SELF_RFSH_EN is set. The minimum supported value is 4.

BIMC_S_DDR0_SHKE_SELF_REFRESH_CNTL (cont.)

Bits	Name	Description
8	CONCURRENT_SELF_REFRESH_EN	SW: RW, HW: R Concurrent Self Refresh Mode. This field is only valid if both the memory ranks are populated with DRAM devices and if only one or none of the individual self refresh enable bits for the ranks are set. (Currently only support this field can be set if and only if the individual self refresh enable bits for the ranks are both off) If the individual self refresh enable bits for both ranks (above) are off and this field is set (only supported mode), then self refresh is only entered when both ranks are idle and the idle timer has expired for both ranks. The above is a recommended setting for PCDDR3 (or LPDDR3) for dual rank ODT configurations wherein the inactive rank is selected to provide termination for a read/write to the other rank. If the concurrent self refresh enable bit is set and only the hardware self refresh enable bit for rank1 is set (not supported), then the controller would enter self refresh for rank1 entirely based on rank1 traffic, however the self refresh enter for rank0 would happen only if rank1 is in self refresh and rank0's idle timer has expired. Rank0 will come out of self refresh if either of the ranks go non-idle. This setting is recommended for LPDDR3 (external) dual rank devices wherein only 1 ODT pin exists. 0x0: DISABLE (Disable (default)) 0x1: ENABLE (Enable)

0x0044D100 BIMC_S_DDR0_SHKE_AUTO_REFRESH_CNTL_1**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

This register provides the controls for auto refresh functionality for Rank1 for the SDRAM devices.

BIMC_S_DDR0_SHKE_AUTO_REFRESH_CNTL_1

Bits	Name	Description
9:0	TREFI	SW: RW, HW: R This field is only for Rank1 refresh rate. If the contents of this field are zero, then SHKE_AUTO_REFRESH_CNTL[TREFI] is used as the initial value for both ranks, else SHKE_AUTO_REFRESH_CNTL[TREFI] is used as the initial value for rank0 and this field is used as the initial value for rank1. Number of clock cycle delay (in t _{cxo} clock cycles) between all-bank auto-refresh commands. This is required to be set to the ?desired tREFI? -1 t _{cxo} cycles. It is illegal to change this CSR setting after the rank has been initialized.

0x0044D104 BIMC_S_DDR0_SHKE_AUTO_REFRESH_CNTL_2**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

This register provides the programmable delay for updating the refresh rate based on SDRAM's device temperature. This register is NOT SUPPORTED if retention based power collapse is used.

BIMC_S_DDR0_SHKE_AUTO_REFRESH_CNTL_2

Bits	Name	Description
31:16	RFSH_UPDATE_TIMER	SW: RW, HW: R Number of timer clock edges to wait before updating the refresh rate. This is only for the refresh rates selected in DLY_RFSH_CONDN below. Typically, SW would want to set this to the amount of time it takes to service a high temperature interrupt from DPE indicating high temperature is reached and hence the min. frequency of operation needs to be increased for the design to perform data commands in addition to keeping up with the auto refresh requirements of the device. A value of 0 indicates that the timer is disabled. The minimum supported value for this timer is ?h3. Typically, the value programmed in this timer would be far less than the value in PERIODIC_MRR[MRR_INTERVAL].

BIMC_S_DDR0_SHKE_AUTO_REFRESH_CNTL_2 (cont.)

Bits	Name	Description
0	DLY_RFSH_CONDN	SW: RW, HW: R If the timer above is valid (non-zero), then this CSR field determines at what requested refresh rates should the timer be enabled. 0x0: TREFI_0_5X_OR_LESS (TREFI_0.5X_OR_LESS) 0x1: TREFI_0_25X (TREFI_0.25X)

3 Peripheral Configuration NOC

0x00500000 PCNOC_OBS_ID_COREID

Type: R

Reset State: 0x3D716D0D

PCNOC_OBS_ID_COREID

Bits	Name	Description
31:8	CORECHECKSUM	Field containing a checksum of the parameters of the IP.
7:0	CORETYPEID	Field identifying the type of IP.

0x00500004 PCNOC_OBS_ID_REVISIONID

Type: R

Reset State: 0x01222B00

PCNOC_OBS_ID_REVISIONID

Bits	Name	Description
31:8	FLEXNOCID	Field containing the build revision of the software used to generate the IP HDL code.
7:0	USERID	Field containing a user defined value, not used anywhere inside the IP itself.

0x00500008 PCNOC_OBS_FAULTEN

Type: RW

Reset State: 0x00000000

PCNOC_OBS_FAULTEN

Bits	Name	Description
0	FAULTEN	Enable Fault output

0x0050000C PCNOC_OBS_ERRVLD**Type:** R**Reset State:** Undefined**PCNOC_OBS_ERRVLD**

Bits	Name	Description
0	ERRVLD	Error logged Valid

0x00500010 PCNOC_OBS_ERRCLR**Type:** RW**Reset State:** 0x00000000**PCNOC_OBS_ERRCLR**

Bits	Name	Description
0	ERRCLR	Clr ErrVld status

0x00500014 PCNOC_OBS_ERRLOG0**Type:** R**Reset State:** 0b1XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX

Header: Lock, Opcode, Len1, ErrCode values

PCNOC_OBS_ERRLOG0

Bits	Name	Description
31	FORMAT	Format of ErrLog0 register
25:16	LEN1	Header: Len1 value
10:8	ERRCODE	Header: Error Code value
4:1	OPC	Header: Opcode value
0	LOCK	Header: Lock bit value

0x00500018 PCNOC_OBS_ERRLOG1**Type:** R**Reset State:** Undefined

PCNOC_OBS_ERRLOG1

Bits	Name	Description
29:0	ERRLOG1	Header: Routeld lsb value

0x00500020 PCNOC_OBS_ERRLOG3**Type:** R**Reset State:** Undefined**PCNOC_OBS_ERRLOG3**

Bits	Name	Description
31:0	ERRLOG3	Header: Addr lsb value

0x00500024 PCNOC_OBS_ERRLOG4**Type:** R**Reset State:** Undefined**PCNOC_OBS_ERRLOG4**

Bits	Name	Description
0	ERRLOG4	Header: Addr msb value

0x00500028 PCNOC_OBS_ERRLOG5**Type:** R**Reset State:** Undefined**PCNOC_OBS_ERRLOG5**

Bits	Name	Description
15:0	ERRLOG5	Header: User lsb value

00x00501080 PCNOC_QXM_SDC1_TFILT_ID_COREID**Type:** R**Reset State:** 0xFE931609

PCNOC_QXM_SDC1_TFILT_ID_COREID

Bits	Name	Description
31:8	CORECHECKSUM	Field containing a checksum of the parameters of the IP.
7:0	CORETYPEID	Field identifying the type of IP.

0x00501084 PCNOC_QXM_SDC1_TFILT_ID_REVISIONID**Type:** R**Reset State:** 0x01222B00**PCNOC_QXM_SDC1_TFILT_ID_REVISIONID**

Bits	Name	Description
31:8	FLEXNOCID	Field containing the build revision of the software used to generate the IP HDL code.
7:0	USERID	Field containing a user defined value, not used anywhere inside the IP itself.

0x00501088 PCNOC_QXM_SDC1_TFILT_MODE**Type:** RW**Reset State:** 0x00000000**PCNOC_QXM_SDC1_TFILT_MODE**

Bits	Name	Description
0	MODE	Register Mode is a 1-bit register that sets the filtering mode as follows: handshake Mode = 0 or latency Mode = 1.

0x0050108C PCNOC_QXM_SDC1_TFILT_ADDRBASE_LOW**Type:** RW**Reset State:** 0x00000000**PCNOC_QXM_SDC1_TFILT_ADDRBASE_LOW**

Bits	Name	Description
31:0	ADDRBASE_LOW	Address base LSB register.

0x00501090 PCNOC_QXM_SDC1_TFILT_ADDRBASE_HIGH**Type:** RW**Reset State:** 0x00000000**PCNOC_QXM_SDC1_TFILT_ADDRBASE_HIGH**

Bits	Name	Description
0	ADDRBASE_HIGH	Address base MSB register.

0x00501094 PCNOC_QXM_SDC1_TFILT_ADDRWINDOWSIZE**Type:** RW**Reset State:** 0x00000000**PCNOC_QXM_SDC1_TFILT_ADDRWINDOWSIZE**

Bits	Name	Description
5:0	ADDRWINDOWSIZE	Register AddrWindowSize contains the encoded address mask used to filter packets: the effective Mask value is equal to $\sim(2^{**}\text{AddrWindowSize} - 1)$. A packet is a candidate when $\text{ReqInfo.Addr} \& \text{AddrMask} = \text{AddrBase} \& \text{AddrMask}$.

0x00501098 PCNOC_QXM_SDC1_TFILT_SRCIDBASE**Type:** RW**Reset State:** 0x00000000**PCNOC_QXM_SDC1_TFILT_SRCIDBASE**

Bits	Name	Description
14:0	SRCIDBASE	Reserved, null-width register. Register SrcIdBase is only available for specific designs, and contains the source ID base used to filter packets. The number of bits is set with parameter wSrcId.

0x0050109C PCNOC_QXM_SDC1_TFILT_SRCIDMASK**Type:** RW**Reset State:** 0x00000000

PCNOC_QXM_SDC1_TFILT_SRCIDMASK

Bits	Name	Description
14:0	SRCIDMASK	Reserved, null-width register. Register SrcIdMask is only available for specific designs, and contains the source ID mask used to filter packets. The number of bits is set with parameter wSrcId. A packet is a candidate when ReqInfo.SrcId + SrcIdMask = SrcIdBase + SrcIdMask.

0x005010A0 PCNOC_QXM_SDC1_TFILT_OPCODE**Type:** RW**Reset State:** 0x00000000**PCNOC_QXM_SDC1_TFILT_OPCODE**

Bits	Name	Description
1	WREN	When set to 1, selects WR requests.
0	RDEN	When set to 1, selects RD requests.

0x005010A4 PCNOC_QXM_SDC1_TFILT_USERBASE**Type:** RW**Reset State:** 0x00000000**PCNOC_QXM_SDC1_TFILT_USERBASE**

Bits	Name	Description
15:0	USERBASE	This register contains the User base used to filter requests.

0x005010A8 PCNOC_QXM_SDC1_TFILT_USERMASK**Type:** RW**Reset State:** 0x00000000**PCNOC_QXM_SDC1_TFILT_USERMASK**

Bits	Name	Description
15:0	USERMASK	This register contains the User mask used to filter requests.

0x00501100 PCNOC_QXM_SDC2_TFILT_ID_COREID**Type:** R**Reset State:** 0x1B00AF09**PCNOC_QXM_SDC2_TFILT_ID_COREID**

Bits	Name	Description
31:8	CORECHECKSUM	Field containing a checksum of the parameters of the IP.
7:0	CORETYPEID	Field identifying the type of IP.

0x00501104 PCNOC_QXM_SDC2_TFILT_ID_REVISIONID**Type:** R**Reset State:** 0x01222B00**PCNOC_QXM_SDC2_TFILT_ID_REVISIONID**

Bits	Name	Description
31:8	FLEXNOCID	Field containing the build revision of the software used to generate the IP HDL code.
7:0	USERID	Field containing a user defined value, not used anywhere inside the IP itself.

0x00501108 PCNOC_QXM_SDC2_TFILT_MODE**Type:** RW**Reset State:** 0x00000000**PCNOC_QXM_SDC2_TFILT_MODE**

Bits	Name	Description
0	MODE	Register Mode is a 1-bit register that sets the filtering mode as follows: handshake Mode = 0 or latency Mode = 1.

0x0050110C PCNOC_QXM_SDC2_TFILT_ADDRBASE_LOW**Type:** RW**Reset State:** 0x00000000**PCNOC_QXM_SDC2_TFILT_ADDRBASE_LOW**

Bits	Name	Description
31:0	ADDRBASE_LOW	Address base LSB register.

0x00501110 PCNOC_QXM_SDC2_TFILT_ADDRBASE_HIGH**Type:** RW**Reset State:** 0x00000000**PCNOC_QXM_SDC2_TFILT_ADDRBASE_HIGH**

Bits	Name	Description
0	ADDRBASE_HIGH	Address base MSB register.

0x00501114 PCNOC_QXM_SDC2_TFILT_ADDRWINDOWSIZE**Type:** RW**Reset State:** 0x00000000**PCNOC_QXM_SDC2_TFILT_ADDRWINDOWSIZE**

Bits	Name	Description
5:0	ADDRWINDOWSIZE	Register AddrWindowSize contains the encoded address mask used to filter packets: the effective Mask value is equal to $\sim(2^{**}\text{AddrWindowSize} - 1)$. A packet is a candidate when $\text{ReqInfo.Addr} \& \text{AddrMask} = \text{AddrBase} \& \text{AddrMask}$.

0x00501118 PCNOC_QXM_SDC2_TFILT_SRCIDBASE**Type:** RW**Reset State:** 0x00000000**PCNOC_QXM_SDC2_TFILT_SRCIDBASE**

Bits	Name	Description
14:0	SRCIDBASE	Reserved, null-width register. Register SrcIdBase is only available for specific designs, and contains the source ID base used to filter packets. The number of bits is set with parameter wSrcId.

0x0050111C PCNOC_QXM_SDC2_TFILT_SRCIDMASK**Type:** RW**Reset State:** 0x00000000

PCNOC_QXM_SDC2_TFILT_SRCIDMASK

Bits	Name	Description
14:0	SRCIDMASK	Reserved, null-width register. Register SrcIdMask is only available for specific designs, and contains the source ID mask used to filter packets. The number of bits is set with parameter wSrcId. A packet is a candidate when ReqInfo.SrcId + SrcIdMask = SrcIdBase + SrcIdMask.

0x00501120 PCNOC_QXM_SDC2_TFILT_OPCODE**Type:** RW**Reset State:** 0x00000000**PCNOC_QXM_SDC2_TFILT_OPCODE**

Bits	Name	Description
1	WREN	When set to 1, selects WR requests.
0	RDEN	When set to 1, selects RD requests.

0x00501124 PCNOC_QXM_SDC2_TFILT_USERBASE**Type:** RW**Reset State:** 0x00000000**PCNOC_QXM_SDC2_TFILT_USERBASE**

Bits	Name	Description
15:0	USERBASE	This register contains the User base used to filter requests.

0x00501128 PCNOC_QXM_SDC2_TFILT_USERMASK**Type:** RW**Reset State:** 0x00000000**PCNOC_QXM_SDC2_TFILT_USERMASK**

Bits	Name	Description
15:0	USERMASK	This register contains the User mask used to filter requests.

0x00501180 PCNOC_QHM2_TFILT_ID_COREID**Type:** R**Reset State:** 0xEC07D309**PCNOC_QHM2_TFILT_ID_COREID**

Bits	Name	Description
31:8	CORECHECKSUM	Field containing a checksum of the parameters of the IP.
7:0	CORETYPEID	Field identifying the type of IP.

0x00501184 PCNOC_QHM2_TFILT_ID_REVISIONID**Type:** R**Reset State:** 0x01222B00**PCNOC_QHM2_TFILT_ID_REVISIONID**

Bits	Name	Description
31:8	FLEXNOCID	Field containing the build revision of the software used to generate the IP HDL code.
7:0	USERID	Field containing a user defined value, not used anywhere inside the IP itself.

0x00501188 PCNOC_QHM2_TFILT_MODE**Type:** RW**Reset State:** 0x00000000**PCNOC_QHM2_TFILT_MODE**

Bits	Name	Description
0	MODE	Register Mode is a 1-bit register that sets the filtering mode as follows: handshake Mode = 0 or latency Mode = 1.

0x0050118C PCNOC_QHM2_TFILT_ADDRBASE_LOW**Type:** RW**Reset State:** 0x00000000**PCNOC_QHM2_TFILT_ADDRBASE_LOW**

Bits	Name	Description
31:0	ADDRBASE_LOW	Address base LSB register.

0x00501194 PCNOC_QHM2_TFILT_ADDRWINDOWSIZE**Type:** RW**Reset State:** 0x00000000**PCNOC_QHM2_TFILT_ADDRWINDOWSIZE**

Bits	Name	Description
5:0	ADDRWINDOWSIZE	Register AddrWindowSize contains the encoded address mask used to filter packets: the effective Mask value is equal to $\sim(2^{**}\text{AddrWindowSize} - 1)$. A packet is a candidate when $\text{ReqInfo.Addr} \& \text{AddrMask} = \text{AddrBase} \& \text{AddrMask}$.

0x00501198 PCNOC_QHM2_TFILT_SRCIDBASE**Type:** RW**Reset State:** 0x00000000**PCNOC_QHM2_TFILT_SRCIDBASE**

Bits	Name	Description
14:0	SRCIDBASE	Reserved, null-width register. Register SrcIdBase is only available for specific designs, and contains the source ID base used to filter packets. The number of bits is set with parameter wSrcId.

0x0050119C PCNOC_QHM2_TFILT_SRCIDMASK**Type:** RW**Reset State:** 0x00000000**PCNOC_QHM2_TFILT_SRCIDMASK**

Bits	Name	Description
14:0	SRCIDMASK	Reserved, null-width register. Register SrcIdMask is only available for specific designs, and contains the source ID mask used to filter packets. The number of bits is set with parameter wSrcId. A packet is a candidate when $\text{ReqInfo.SrcId} + \text{SrcIdMask} = \text{SrcIdBase} + \text{SrcIdMask}$.

0x005011A0 PCNOC_QHM2_TFILT_OPCODE**Type:** RW**Reset State:** 0x00000000

PCNOC_QHM2_TFILT_OPCODE

Bits	Name	Description
1	WREN	When set to 1, selects WR requests.
0	RDEN	When set to 1, selects RD requests.

0x005011A4 PCNOC_QHM2_TFILT_USERBASE**Type:** RW**Reset State:** 0x00000000**PCNOC_QHM2_TFILT_USERBASE**

Bits	Name	Description
15:0	USERBASE	This register contains the User base used to filter requests.

0x005011A8 PCNOC_QHM2_TFILT_USERMASK**Type:** RW**Reset State:** 0x00000000**PCNOC_QHM2_TFILT_USERMASK**

Bits	Name	Description
15:0	USERMASK	This register contains the User mask used to filter requests.

0x00501200 PCNOC_QHM1_TFILT_ID_COREID**Type:** R**Reset State:** 0x318A7609**PCNOC_QHM1_TFILT_ID_COREID**

Bits	Name	Description
31:8	CORECHECKSUM	Field containing a checksum of the parameters of the IP.
7:0	CORETYPEID	Field identifying the type of IP.

0x00501204 PCNOC_QHM1_TFILT_ID_REVISIONID**Type:** R**Reset State:** 0x01222B00

PCNOC_QHM1_TFILT_ID_REVISIONID

Bits	Name	Description
31:8	FLEXNOCID	Field containing the build revision of the software used to generate the IP HDL code.
7:0	USERID	Field containing a user defined value, not used anywhere inside the IP itself.

0x00501208 PCNOC_QHM1_TFILT_MODE**Type:** RW**Reset State:** 0x00000000**PCNOC_QHM1_TFILT_MODE**

Bits	Name	Description
0	MODE	Register Mode is a 1-bit register that sets the filtering mode as follows: handshake Mode = 0 or latency Mode = 1.

0x0050120C PCNOC_QHM1_TFILT_ADDRBASE_LOW**Type:** RW**Reset State:** 0x00000000**PCNOC_QHM1_TFILT_ADDRBASE_LOW**

Bits	Name	Description
31:0	ADDRBASE_LOW	Address base LSB register.

0x00501214 PCNOC_QHM1_TFILT_ADDRWINDOWSIZE**Type:** RW**Reset State:** 0x00000000**PCNOC_QHM1_TFILT_ADDRWINDOWSIZE**

Bits	Name	Description
5:0	ADDRWINDOWSIZE	Register AddrWindowSize contains the encoded address mask used to filter packets: the effective Mask value is equal to $\sim(2^{**}\text{AddrWindowSize} - 1)$. A packet is a candidate when $\text{ReqInfo.Addr} \& \text{AddrMask} = \text{AddrBase} \& \text{AddrMask}$.

0x00501218 PCNOC_QHM1_TFILT_SRCIDBASE**Type:** RW**Reset State:** 0x00000000**PCNOC_QHM1_TFILT_SRCIDBASE**

Bits	Name	Description
14:0	SRCIDBASE	Reserved, null-width register. Register SrcIdBase is only available for specific designs, and contains the source ID base used to filter packets. The number of bits is set with parameter wSrcId.

0x0050121C PCNOC_QHM1_TFILT_SRCIDMASK**Type:** RW**Reset State:** 0x00000000**PCNOC_QHM1_TFILT_SRCIDMASK**

Bits	Name	Description
14:0	SRCIDMASK	Reserved, null-width register. Register SrcIdMask is only available for specific designs, and contains the source ID mask used to filter packets. The number of bits is set with parameter wSrcId. A packet is a candidate when $\text{ReqInfo.SrcId} + \text{SrcIdMask} = \text{SrcIdBase} + \text{SrcIdMask}$.

0x00501220 PCNOC_QHM1_TFILT_OPCODE**Type:** RW**Reset State:** 0x00000000**PCNOC_QHM1_TFILT_OPCODE**

Bits	Name	Description
1	WREN	When set to 1, selects WR requests.
0	RDEN	When set to 1, selects RD requests.

0x00501224 PCNOC_QHM1_TFILT_USERBASE**Type:** RW**Reset State:** 0x00000000

PCNOC_QHM1_TFILT_USERBASE

Bits	Name	Description
15:0	USERBASE	This register contains the User base used to filter requests.

0x00501228 PCNOC_QHM1_TFILT_USERMASK**Type:** RW**Reset State:** 0x00000000**PCNOC_QHM1_TFILT_USERMASK**

Bits	Name	Description
15:0	USERMASK	This register contains the User mask used to filter requests.

0x00501280 PCNOC_QXM_SNOC_TFILT_ID_COREID**Type:** R**Reset State:** 0x214A1A09**PCNOC_QXM_SNOC_TFILT_ID_COREID**

Bits	Name	Description
31:8	CORECHECKSUM	Field containing a checksum of the parameters of the IP.
7:0	CORETYPEID	Field identifying the type of IP.

0x00501284 PCNOC_QXM_SNOC_TFILT_ID_REVISIONID**Type:** R**Reset State:** 0x01222B00**PCNOC_QXM_SNOC_TFILT_ID_REVISIONID**

Bits	Name	Description
31:8	FLEXNOCID	Field containing the build revision of the software used to generate the IP HDL code.
7:0	USERID	Field containing a user defined value, not used anywhere inside the IP itself.

0x00501288 PCNOC_QXM_SNOCTFILT_MODE**Type:** RW**Reset State:** 0x00000000**PCNOC_QXM_SNOCTFILT_MODE**

Bits	Name	Description
0	MODE	Register Mode is a 1-bit register that sets the filtering mode as follows: handshake Mode = 0 or latency Mode = 1.

0x0050128C PCNOC_QXM_SNOCTFILT_ADDRBASE_LOW**Type:** RW**Reset State:** 0x00000000**PCNOC_QXM_SNOCTFILT_ADDRBASE_LOW**

Bits	Name	Description
31:0	ADDRBASE_LOW	Address base LSB register.

0x00501290 PCNOC_QXM_SNOCTFILT_ADDRBASE_HIGH**Type:** RW**Reset State:** 0x00000000**PCNOC_QXM_SNOCTFILT_ADDRBASE_HIGH**

Bits	Name	Description
0	ADDRBASE_HIGH	Address base MSB register.

0x00501294 PCNOC_QXM_SNOCTFILT_ADDRWINDOWSIZE**Type:** RW**Reset State:** 0x00000000**PCNOC_QXM_SNOCTFILT_ADDRWINDOWSIZE**

Bits	Name	Description
5:0	ADDRWINDOWSIZE	Register AddrWindowSize contains the encoded address mask used to filter packets: the effective Mask value is equal to $\sim(2^{**}\text{AddrWindowSize} - 1)$. A packet is a candidate when $\text{ReqInfo.Addr} \& \text{AddrMask} = \text{AddrBase} \& \text{AddrMask}$.

0x00501298 PCNOC_QXM_SNOG_TFILT_SRCIDBASE**Type:** RW**Reset State:** 0x00000000**PCNOC_QXM_SNOG_TFILT_SRCIDBASE**

Bits	Name	Description
14:0	SRCIDBASE	Reserved, null-width register. Register SrcIdBase is only available for specific designs, and contains the source ID base used to filter packets. The number of bits is set with parameter wSrcId.

0x0050129C PCNOC_QXM_SNOG_TFILT_SRCIDMASK**Type:** RW**Reset State:** 0x00000000**PCNOC_QXM_SNOG_TFILT_SRCIDMASK**

Bits	Name	Description
14:0	SRCIDMASK	Reserved, null-width register. Register SrcIdMask is only available for specific designs, and contains the source ID mask used to filter packets. The number of bits is set with parameter wSrcId. A packet is a candidate when $\text{ReqInfo.SrcId} + \text{SrcIdMask} = \text{SrcIdBase} + \text{SrcIdMask}$.

0x005012A0 PCNOC_QXM_SNOG_TFILT_OPCODE**Type:** RW**Reset State:** 0x00000000**PCNOC_QXM_SNOG_TFILT_OPCODE**

Bits	Name	Description
1	WREN	When set to 1, selects WR requests.
0	RDEN	When set to 1, selects RD requests.

0x005012A4 PCNOC_QXM_SNOG_TFILT_USERBASE**Type:** RW**Reset State:** 0x00000000

PCNOC_QXM_SNOG_TFILT_USERBASE

Bits	Name	Description
15:0	USERBASE	This register contains the User base used to filter requests.

0x005012A8 PCNOC_QXM_SNOG_TFILT_USERMASK**Type:** RW**Reset State:** 0x00000000**PCNOC_QXM_SNOG_TFILT_USERMASK**

Bits	Name	Description
15:0	USERMASK	This register contains the User mask used to filter requests.

0x00502000 PCNOC_TPRB0_TPROF_ID_COREID**Type:** R**Reset State:** 0xD1B7540A**PCNOC_TPRB0_TPROF_ID_COREID**

Bits	Name	Description
31:8	CORECHECKSUM	Field containing a checksum of the parameters of the IP.
7:0	CORETYPEID	Field identifying the type of IP.

0x00502004 PCNOC_TPRB0_TPROF_ID_REVISIONID**Type:** R**Reset State:** 0x01222B00**PCNOC_TPRB0_TPROF_ID_REVISIONID**

Bits	Name	Description
31:8	FLEXNOCID	Field containing the build revision of the software used to generate the IP HDL code.
7:0	USERID	Field containing a user defined value, not used anywhere inside the IP itself.

0x00502008 PCNOC_TPRB0_TPROF_EN**Type:** RW**Reset State:** 0x00000000**PCNOC_TPRB0_TPROF_EN**

Bits	Name	Description
0	EN	Register En is a 1-bit register that enables the transaction probe counter unit.

0x0050200C PCNOC_TPRB0_TPROF_MODE**Type:** RW**Reset State:** 0x00000000**PCNOC_TPRB0_TPROF_MODE**

Bits	Name	Description
1:0	MODE	Register Mode sets the counting mode per observed port. Each bit per observation port defines the incrementing mode. (Mode = 0 for Delay, Mode = 1 for Pending)

0x00502010 PCNOC_TPRB0_TPROF_OBSERVEDSEL_0**Type:** RW**Reset State:** 0x00000000**PCNOC_TPRB0_TPROF_OBSERVEDSEL_0**

Bits	Name	Description
2:0	OBSERVEDSEL_0	Register ObservedSel_i selects the port to be observed: 0.

0x00502014 PCNOC_TPRB0_TPROF_OBSERVEDSEL_1**Type:** RW**Reset State:** 0x00000000**PCNOC_TPRB0_TPROF_OBSERVEDSEL_1**

Bits	Name	Description
2:0	OBSERVEDSEL_1	Register ObservedSel_i selects the port to be observed: 1.

0x00502020 PCNOC_TPRB0_TPROF_NTENURELINES_0**Type:** RW**Reset State:** 0x00000000**PCNOC_TPRB0_TPROF_NTENURELINES_0**

Bits	Name	Description
2:0	NTENURELINES_0	Register nTenureLine_i indicates the number of transaction probe lines allocated to each observed port: 0.

0x0050202C PCNOC_TPRB0_TPROF_THRESHOLDS_0_0**Type:** RW**Reset State:** 0x00000000**PCNOC_TPRB0_TPROF_THRESHOLDS_0_0**

Bits	Name	Description
3:0	THRESHOLDS_0_0	Register Threshold_i_j contains the threshold index '0' that allows computation of threshold values.

0x00502030 PCNOC_TPRB0_TPROF_THRESHOLDS_0_1**Type:** RW**Reset State:** 0x00000000**PCNOC_TPRB0_TPROF_THRESHOLDS_0_1**

Bits	Name	Description
3:0	THRESHOLDS_0_1	Register Threshold_i_j contains the threshold index '1' that allows computation of threshold values.

0x00502034 PCNOC_TPRB0_TPROF_THRESHOLDS_0_2**Type:** RW**Reset State:** 0x00000000**PCNOC_TPRB0_TPROF_THRESHOLDS_0_2**

Bits	Name	Description
3:0	THRESHOLDS_0_2	Register Threshold_i_j contains the threshold index '2' that allows computation of threshold values.

0x0050203C PCNOC_TPRB0_TPROF_THRESHOLDS_1_0**Type:** RW**Reset State:** 0x00000000**PCNOC_TPRB0_TPROF_THRESHOLDS_1_0**

Bits	Name	Description
3:0	THRESHOLDS_1_0	Register Threshold_i_j contains the threshold index '0' that allows computation of threshold values.

0x00502040 PCNOC_TPRB0_TPROF_THRESHOLDS_1_1**Type:** RW**Reset State:** 0x00000000**PCNOC_TPRB0_TPROF_THRESHOLDS_1_1**

Bits	Name	Description
3:0	THRESHOLDS_1_1	Register Threshold_i_j contains the threshold index '1' that allows computation of threshold values.

0x00502044 PCNOC_TPRB0_TPROF_THRESHOLDS_1_2**Type:** RW**Reset State:** 0x00000000**PCNOC_TPRB0_TPROF_THRESHOLDS_1_2**

Bits	Name	Description
3:0	THRESHOLDS_1_2	Register Threshold_i_j contains the threshold index '2' that allows computation of threshold values.

0x0050206C PCNOC_TPRB0_TPROF_OVERFLOWSTATUS**Type:** R**Reset State:** Undefined**PCNOC_TPRB0_TPROF_OVERFLOWSTATUS**

Bits	Name	Description
1:0	OVERFLOWSTATUS	Register OverflowStatus is a status register that handles the overflow status on each observed port. OverflowStatus = nObservable. Bit 0 gives the status of observed port 0.

0x00502070 PCNOC_TPRB0_TPROF_OVERFLOWRESET**Type:** RW**Reset State:** 0x00000000**PCNOC_TPRB0_TPROF_OVERFLOWRESET**

Bits	Name	Description
1:0	OVERFLOWRESET	Register OverflowReset is a pulse register that clears overflow status bits per observed port on each write access. OverflowReset = nObservable. Writing 0x2 clears the overflow status of observed port 1.

0x00502074 PCNOC_TPRB0_TPROF_PENDINGEVENTMODE**Type:** RW**Reset State:** 0x00000000**PCNOC_TPRB0_TPROF_PENDINGEVENTMODE**

Bits	Name	Description
0	PENDINGEVENTMODE	Register pendingEventMode is a 1-bit register that configures the pending event mode. When set to 0 (CYCLE), and when register mode is set to PENDING, the pending event is generated on each cycle when the counter is greater than zero. When set to 1 (STOP) the pending event is generated on each stop event.

0x00502078 PCNOC_TPRB0_TPROF_PRESCALER**Type:** RW**Reset State:** 0x00000000**PCNOC_TPRB0_TPROF_PRESCALER**

Bits	Name	Description
7:0	PRESCALER	8Register Prescaler is an-bit pre-scaling register that accepts any pre-scaling value between 1 (default) and 256. If set to 0, pre-scaling is disabled. If set to any other supported value 'n', the threshold counter value is divided by (n + 1).

0x00503000 PCNOC_EPRB0_SPROF_ID_COREID**Type:** R**Reset State:** 0x2E2A2E06

PCNOC_EPRB0_SPROF_ID_COREID

Bits	Name	Description
31:8	CORECHECKSUM	Field containing a checksum of the parameters of the IP.
7:0	CORETYPEID	Field identifying the type of IP.

0x00503004 PCNOC_EPRB0_SPROF_ID_REVISIONID**Type:** R**Reset State:** 0x01222B00**PCNOC_EPRB0_SPROF_ID_REVISIONID**

Bits	Name	Description
31:8	FLEXNOCID	Field containing the build revision of the software used to generate the IP HDL code.
7:0	USERID	Field containing a user defined value, not used anywhere inside the IP itself.

0x00503008 PCNOC_EPRB0_SPROF_MAINCTL**Type:** RW**Reset State:** 0x00000000

Register MainCtl contains probe global control bits. The register has seven bit fields:

PCNOC_EPRB0_SPROF_MAINCTL

Bits	Name	Description
7	FILTBYTEALWAYSCHAINABLEEN	When set to 0, Filter Byte / ByteEn Cnt will be connected to all statistics counters where odd Stat count are not chainable. When set to 1, Filter Byte / ByteEn Cnt will be connected only to chainable stat counter (even stat counter).
6	INTRUSIVEMODE	When set to 1, register field IntrusiveMode enables trace operation in Intrusive flow-control mode. When set to 0, the register enables trace operation in Overflow flow-control mode
5	STATCONDDUMP	When set, register field StatCondDump enables the dump of a statistics frame to the range of counter values set for registers StatAlarmMin, StatAlarmMax, and AlarmMode. This field also renders register StatAlarmStatus inoperative. When parameter statisticsCounterAlarm is set to False, the StatCondDump register bit is reserved.
4	ALARMEN	When set, register field AlarmEn enables the probe to collect alarm-related information. When the register field bit is null, both TraceAlarm and StatAlarm outputs are driven to 0.

PCNOC_EPRB0_SPROF_MAINCTL (cont.)

Bits	Name	Description
3	STATEN	When set to 1, register field StatEn enables statistics profiling. The probe sendS statistics results to the output for signal ObsTx. All statistics counters are cleared when the StatEn bit goes from 0 to 1. When set to 0, counters are disabled.
2	PAYLOADEN	Register field PayloadEn, when set to 1, enables traces to contain headers and payload. When set to 0, only headers are reported.
1	TRACEEN	Register field TraceEn enables the probe to send filtered packets (Trace) on the ObsTx observation output.
0	ERREN	Register field ErrEn enables the probe to send on the ObsTx output any packet with Error status, independently of filtering mechanisms, thus constituting a simple supplementary global filter.

0x0050300C PCNOC_EPRB0_SPROF_CFGCTL**Type:** RW**Reset State:** 0b000000000000000000000000000000X0**PCNOC_EPRB0_SPROF_CFGCTL**

Bits	Name	Type	Description
1	ACTIVE	R	
0	GLOBALEN	RW	

0x00503014 PCNOC_EPRB0_SPROF_FILTERLUT**Type:** RW**Reset State:** 0x00000000**PCNOC_EPRB0_SPROF_FILTERLUT**

Bits	Name	Description
3:0	FILTERLUT	Register FilterLut contains a look-up table that is used to combine filter outputs in order to trace packets. Packet tracing is enabled when the FilterLut bit of index (FNout ... F0out) is equal to 1. The number of bits in register FilterLut is determined by the setting for parameter nPort, calculated as 2**nPort. When parameter nFilter is set to None or 0, FilterLut is reserved.

0x00503018 PCNOC_EPRB0_SPROF_TRACEALARMEN**Type:** RW**Reset State:** 0x00000000

PCNOC_EPRB0_SPROF_TRACEALARMEN

Bits	Name	Description
2:0	TRACEALARMEN	Register TraceAlarmEn controls which lookup table or filter can set the TraceAlarm signal output once the trace alarm status is set. The number of bits in register TraceAlarmEn is determined by the value set for parameter nFilter + 1. Bit nFilter controls the lookup table output, and bits nFilter:0 control the corresponding filter output. When parameter nFilter is set to None, TraceAlarmEn is reserved.

0x0050301C PCNOC_EPRB0_SPROF_TRACEALARMSTATUS**Type:** R**Reset State:** Undefined**PCNOC_EPRB0_SPROF_TRACEALARMSTATUS**

Bits	Name	Description
2:0	TRACEALARMSTATUS	Register TraceAlarmStatus is a read-only register that indicates which lookup table or filter has been matched by a packet, independently of register TraceAlarmEn bit configuration. The number of bits in TraceAlarmStatus is determined by the value set for parameter nFilter + 1. When nFilter is set to None, TraceAlarmStatus is reserved.

0x00503020 PCNOC_EPRB0_SPROF_TRACEALARMCLR**Type:** RW**Reset State:** 0x00000000**PCNOC_EPRB0_SPROF_TRACEALARMCLR**

Bits	Name	Description
2:0	TRACEALARMCLR	Register TraceAlarmClr is a write-only register. A write operation to this register clears the bits in register TraceAlarmStatus, which are set by this write. The number of bits in register TraceAlarmClr is determined by the value set for parameter nFilter + 1. When nFilter is set to None, TraceAlarmClr is reserved.

0x00503024 PCNOC_EPRB0_SPROF_STATPERIOD**Type:** RW**Reset State:** 0x00000000

PCNOC_EPRB0_SPROF_STATPERIOD

Bits	Name	Description
4:0	STATPERIOD	Register StatPeriod is a 5-bit register that sets a period, within a range of 2-8 megacycles, during which statistics are collected before being dumped automatically. Setting the register effectively enables automatic mode operation for statistics collection. Note: The configured period is also used for statistics alarm generation. The period is calculated with the formula: $N_Cycle = 2^{**min(23, StatPeriod)}$ When register StatPeriod is set to its default value 0, automatic dump mode is disabled, and register StatGo is activated for manual mode operation. Note: When parameter nStatisticsCounter is set to 0, statistics collection is disabled, with the consequence that all statistics collection registers, including StatPeriod, are reserved.

0x00503028 PCNOC_EPRB0_SPROF_STATGO**Type:** RW**Reset State:** 0x00000000**PCNOC_EPRB0_SPROF_STATGO**

Bits	Name	Description
0	STATGO	Writing to 1-bit pulse register StatGo generates a statistics dump. The register is active when statistics collection operates in its manual mode, that is, when register StatPeriod is set to 0.

0x0050302C PCNOC_EPRB0_SPROF_STATALARMMIN**Type:** RW**Reset State:** 0x00000000**PCNOC_EPRB0_SPROF_STATALARMMIN**

Bits	Name	Description
31:0	STATALARMMIN	Register StatAlarmMin contains the minimum count value used in statistics alarm comparisons. The number of bits is equal to twice the value set for parameter wStatisticsCounter. When parameter statisticsCounterAlarm is set to False, StatAlarmMin is reserved.

0x00503030 PCNOC_EPRB0_SPROF_STATALARMMAX**Type:** RW**Reset State:** 0x00000000

PCNOC_EPRB0_SPROF_STATALARMMAX

Bits	Name	Description
31:0	STATALARMMAX	Register StatAlarmMax contains the maximum count value used in statistics alarm comparisons. The number of bits is equal to twice the value set for parameter wStatisticsCounter. When parameter statisticsCounterAlarm is set to False, StatAlarmMax is reserved.

0x00503034 PCNOC_EPRB0_SPROF_STATALARMSSTATUS**Type:** R**Reset State:** Undefined**PCNOC_EPRB0_SPROF_STATALARMSSTATUS**

Bits	Name	Description
0	STATALARMSSTATUS	Register StatAlarmStatus is a read-only 1-bit register indicating that at least one statistics counter has exceeded the programmed values for registers StatAlarmMin or StatAlarmMax. Output signal StatAlarm is equal to the values stored in register MainCtl fields StatAlarmStatus and AlarmEn. When parameter statisticsCounterAlarm is set to False, StatAlarmStatus is reserved.

0x00503038 PCNOC_EPRB0_SPROF_STATALARMCCLR**Type:** RW**Reset State:** 0x00000000**PCNOC_EPRB0_SPROF_STATALARMCCLR**

Bits	Name	Description
0	STATALARMCCLR	Register StatAlarmClr is a 1-bit write-only register. A write operation to this register clears the register StatAlarmStatus bit. When parameter statisticsCounterAlarm is set to False, StatAlarmClr is reserved.

0x00503044 PCNOC_EPRB0_SPROF_FILTERS_0_ROUTEIDBASE**Type:** RW**Reset State:** 0x00000000**PCNOC_EPRB0_SPROF_FILTERS_0_ROUTEIDBASE**

Bits	Name	Description
29:0	FILTERS_0_ROUTEIDBASE	Register RouteIdBase contains the RouteId-IsbFilterRouteId bits base used to filter packets.

0x00503048 PCNOC_EPRB0_SPROF_FILTERS_0_ROUTEIDMASK**Type:** RW**Reset State:** 0x00000000**PCNOC_EPRB0_SPROF_FILTERS_0_ROUTEIDMASK**

Bits	Name	Description
29:0	FILTERS_0_ROUTEIDMASK	Register RouteIdMask contains the RouteId-IsbFilterRouteId mask used to filter packets. A packet is candidate when packet.RouteId >> IsbFilterRouteId & RouteIdMask = RouteIdBase & RouteIdMask.

0x0050304C PCNOC_EPRB0_SPROF_FILTERS_0_ADDRBASE_LOW**Type:** RW**Reset State:** 0x00000000**PCNOC_EPRB0_SPROF_FILTERS_0_ADDRBASE_LOW**

Bits	Name	Description
31:0	FILTERS_0_ADDRBASE_LO W	Address LSB register.

0x00503050 PCNOC_EPRB0_SPROF_FILTERS_0_ADDRBASE_HIGH**Type:** RW**Reset State:** 0x00000000**PCNOC_EPRB0_SPROF_FILTERS_0_ADDRBASE_HIGH**

Bits	Name	Description
0	FILTERS_0_ADDRBASE_HI GH	Address MSB register.

0x00503054 PCNOC_EPRB0_SPROF_FILTERS_0_WINDOWSIZE**Type:** RW**Reset State:** 0x00000000

PCNOC_EPRB0_SPROF_FILTERS_0_WINDOWSIZE

Bits	Name	Description
5:0	FILTERS_0_WINDOWSIZE	Register WindowSize contains the encoded address mask used to filter packets. The effective Mask value is equal to $\sim(2^{\max(\text{WindowSize}, \text{packet.Len})} - 1)$. A packet is candidate when $\text{packet.Addr} \& \text{Mask} = \text{AddrBase} \& \text{Mask}$. This allows filtering of packets having an intersection with the AddrBase/WindowSize burst aligned region, even if the region is smaller than the packet.

0x00503060 PCNOC_EPRB0_SPROF_FILTERS_0_OPCODE**Type:** RW**Reset State:** 0x00000000**PCNOC_EPRB0_SPROF_FILTERS_0_OPCODE**

Bits	Name	Description
3	URGEN	Selects URG packets (urgency).
2	LOCKEN	Selects RDX-WR, RDL, WRC and Linked sequence.
1	WREN	Selects WR packets.
0	RDEN	Selects RD packets.

0x00503064 PCNOC_EPRB0_SPROF_FILTERS_0_STATUS**Type:** RW**Reset State:** 0x00000000

Register Status is 2-bit register that selects candidate packets based on packet status.

PCNOC_EPRB0_SPROF_FILTERS_0_STATUS

Bits	Name	Description
1	RSPEN	Selects RSP and FAIL-CONT status packets.
0	REQEN	Selects REQ status packets.

0x00503068 PCNOC_EPRB0_SPROF_FILTERS_0_LENGTH**Type:** RW**Reset State:** 0x00000000

PCNOC_EPRB0_SPROF_FILTERS_0_LENGTH

Bits	Name	Description
3:0	FILTERS_0_LENGTH	Register Length is 4-bit register that selects candidate packets if their number of bytes is less than or equal to 2^{Length} .

0x0050306C PCNOC_EPRB0_SPROF_FILTERS_0_URGENCY**Type:** RW**Reset State:** 0x00000000**PCNOC_EPRB0_SPROF_FILTERS_0_URGENCY**

Bits	Name	Description
1:0	FILTERS_0_URGENCY	Register Urgency contains the minimum urgency level used to filter packets. A packet is a candidate when Urgency is greater than or equal to $2^{(\text{Urgency}-1)}$.

0x00503080 PCNOC_EPRB0_SPROF_FILTERS_1_ROUTEIDBASE**Type:** RW**Reset State:** 0x00000000**PCNOC_EPRB0_SPROF_FILTERS_1_ROUTEIDBASE**

Bits	Name	Description
29:0	FILTERS_1_ROUTEIDBASE	Register RouteIdBase contains the RouteId-IsbFilterRouteId bits base used to filter packets.

0x00503084 PCNOC_EPRB0_SPROF_FILTERS_1_ROUTEIDMASK**Type:** RW**Reset State:** 0x00000000**PCNOC_EPRB0_SPROF_FILTERS_1_ROUTEIDMASK**

Bits	Name	Description
29:0	FILTERS_1_ROUTEIDMASK	Register RouteIdMask contains the RouteId-IsbFilterRouteId mask used to filter packets. A packet is candidate when $\text{packet.RouteId} \gg \text{IsbFilterRouteId} \ \& \ \text{RouteIdMask} = \text{RouteIdBase} \ \& \ \text{RouteIdMask}$.

0x00503088 PCNOC_EPRB0_SPROF_FILTERS_1_ADDRBASE_LOW**Type:** RW**Reset State:** 0x00000000**PCNOC_EPRB0_SPROF_FILTERS_1_ADDRBASE_LOW**

Bits	Name	Description
31:0	FILTERS_1_ADDRBASE_LO W	Address LSB register.

0x0050308C PCNOC_EPRB0_SPROF_FILTERS_1_ADDRBASE_HIGH**Type:** RW**Reset State:** 0x00000000**PCNOC_EPRB0_SPROF_FILTERS_1_ADDRBASE_HIGH**

Bits	Name	Description
0	FILTERS_1_ADDRBASE_HI GH	Address MSB register.

0x00503090 PCNOC_EPRB0_SPROF_FILTERS_1_WINDOWSIZE**Type:** RW**Reset State:** 0x00000000**PCNOC_EPRB0_SPROF_FILTERS_1_WINDOWSIZE**

Bits	Name	Description
5:0	FILTERS_1_WINDOWSIZE	Register WindowSize contains the encoded address mask used to filter packets. The effective Mask value is equal to $\sim(2^{\max(\text{WindowSize}, \text{packet.Len})} - 1)$. A packet is candidate when $\text{packet.Addr} \& \text{Mask} = \text{AddrBase} \& \text{Mask}$. This allows filtering of packets having an intersection with the AddrBase/WindowSize burst aligned region, even if the region is smaller than the packet.

0x0050309C PCNOC_EPRB0_SPROF_FILTERS_1_OPCODE**Type:** RW**Reset State:** 0x00000000**PCNOC_EPRB0_SPROF_FILTERS_1_OPCODE**

Bits	Name	Description
3	URGEN	Selects URG packets (urgency).

PCNOC_EPRB0_SPROF_FILTERS_1_OPCODE (cont.)

Bits	Name	Description
2	LOCKEN	Selects RDX-WR, RDL, WRC and Linked sequence.
1	WREN	Selects WR packets.
0	RDEN	Selects RD packets.

0x005030A0 PCNOC_EPRB0_SPROF_FILTERS_1_STATUS**Type:** RW**Reset State:** 0x00000000

Register Status is 2-bit register that selects candidate packets based on packet status.

PCNOC_EPRB0_SPROF_FILTERS_1_STATUS

Bits	Name	Description
1	RSPEN	Selects RSP and FAIL-CONT status packets.
0	REQEN	Selects REQ status packets.

0x005030A4 PCNOC_EPRB0_SPROF_FILTERS_1_LENGTH**Type:** RW**Reset State:** 0x00000000**PCNOC_EPRB0_SPROF_FILTERS_1_LENGTH**

Bits	Name	Description
3:0	FILTERS_1_LENGTH	Register Length is 4-bit register that selects candidate packets if their number of bytes is less than or equal to 2^{**}Length .

0x005030A8 PCNOC_EPRB0_SPROF_FILTERS_1_URGENCY**Type:** RW**Reset State:** 0x00000000**PCNOC_EPRB0_SPROF_FILTERS_1_URGENCY**

Bits	Name	Description
1:0	FILTERS_1_URGENCY	Register Urgency contains the minimum urgency level used to filter packets. A packet is a candidate when Urgency is greater than or equal to $2^{**}(\text{Urgency}-1)$.

0x00503138 PCNOC_EPRB0_SPROF_COUNTERS_0_SRC**Type:** RW**Reset State:** 0x00000000**PCNOC_EPRB0_SPROF_COUNTERS_0_SRC**

Bits	Name	Description
4:0	INTEVENT	Internal packet event

0x0050313C PCNOC_EPRB0_SPROF_COUNTERS_0_ALARMMODE**Type:** RW**Reset State:** 0x00000000**PCNOC_EPRB0_SPROF_COUNTERS_0_ALARMMODE**

Bits	Name	Description
1:0	COUNTERS_0_ALARMMODE	Register AlarmMode is a 2-bit register that is present when parameter statisticsCounterAlarm is set to True. The register defines the statistics-alarm behavior of the counter.

0x00503140 PCNOC_EPRB0_SPROF_COUNTERS_0_VAL**Type:** R**Reset State:** Undefined**PCNOC_EPRB0_SPROF_COUNTERS_0_VAL**

Bits	Name	Description
15:0	COUNTERS_0_VAL	Register Val is a read-only register that is always present. The register contains the statistics counter value either pending StatAlarm output, or when statistics collection is suspended subsequent to triggers or signal statSuspend.

0x0050314C PCNOC_EPRB0_SPROF_COUNTERS_1_SRC**Type:** RW**Reset State:** 0x00000000**PCNOC_EPRB0_SPROF_COUNTERS_1_SRC**

Bits	Name	Description
4:0	INTEVENT	Internal packet event

0x00503150 PCNOC_EPRB0_SPROF_COUNTERS_1_ALARMMODE**Type:** RW**Reset State:** 0x00000000**PCNOC_EPRB0_SPROF_COUNTERS_1_ALARMMODE**

Bits	Name	Description
1:0	COUNTERS_1_ALARMMODE	Register AlarmMode is a 2-bit register that is present when parameter statisticsCounterAlarm is set to True. The register defines the statistics-alarm behavior of the counter.

0x00503154 PCNOC_EPRB0_SPROF_COUNTERS_1_VAL**Type:** R**Reset State:** Undefined**PCNOC_EPRB0_SPROF_COUNTERS_1_VAL**

Bits	Name	Description
15:0	COUNTERS_1_VAL	Register Val is a read-only register that is always present. The register contains the statistics counter value either pending StatAlarm output, or when statistics collection is suspended subsequent to triggers or signal statSuspend.

0x00503160 PCNOC_EPRB0_SPROF_COUNTERS_2_SRC**Type:** RW**Reset State:** 0x00000000**PCNOC_EPRB0_SPROF_COUNTERS_2_SRC**

Bits	Name	Description
4:0	INTEVENT	Internal packet event

0x00503164 PCNOC_EPRB0_SPROF_COUNTERS_2_ALARMMODE**Type:** RW**Reset State:** 0x00000000**PCNOC_EPRB0_SPROF_COUNTERS_2_ALARMMODE**

Bits	Name	Description
1:0	COUNTERS_2_ALARMMODE	Register AlarmMode is a 2-bit register that is present when parameter statisticsCounterAlarm is set to True. The register defines the statistics-alarm behavior of the counter.

0x00503168 PCNOC_EPRB0_SPROF_COUNTERS_2_VAL**Type:** R**Reset State:** Undefined**PCNOC_EPRB0_SPROF_COUNTERS_2_VAL**

Bits	Name	Description
15:0	COUNTERS_2_VAL	Register Val is a read-only register that is always present. The register contains the statistics counter value either pending StatAlarm output, or when statistics collection is suspended subsequent to triggers or signal statSuspend.

0x00503174 PCNOC_EPRB0_SPROF_COUNTERS_3_SRC**Type:** RW**Reset State:** 0x00000000**PCNOC_EPRB0_SPROF_COUNTERS_3_SRC**

Bits	Name	Description
4:0	INTEVENT	Internal packet event

0x00503178 PCNOC_EPRB0_SPROF_COUNTERS_3_ALARMMODE**Type:** RW**Reset State:** 0x00000000**PCNOC_EPRB0_SPROF_COUNTERS_3_ALARMMODE**

Bits	Name	Description
1:0	COUNTERS_3_ALARMMODE	Register AlarmMode is a 2-bit register that is present when parameter statisticsCounterAlarm is set to True. The register defines the statistics-alarm behavior of the counter.

0x0050317C PCNOC_EPRB0_SPROF_COUNTERS_3_VAL**Type:** R**Reset State:** Undefined

PCNOC_EPRB0_SPROF_COUNTERS_3_VAL

Bits	Name	Description
15:0	COUNTERS_3_VAL	Register Val is a read-only register that is always present. The register contains the statistics counter value either pending StatAlarm output, or when statistics collection is suspended subsequent to triggers or signal statSuspend.

0x00503188 PCNOC_EPRB0_SPROF_COUNTERS_4_SRC**Type:** RW**Reset State:** 0x00000000**PCNOC_EPRB0_SPROF_COUNTERS_4_SRC**

Bits	Name	Description
4:0	INTEVENT	Internal packet event

0x0050318C PCNOC_EPRB0_SPROF_COUNTERS_4_ALARMMODE**Type:** RW**Reset State:** 0x00000000**PCNOC_EPRB0_SPROF_COUNTERS_4_ALARMMODE**

Bits	Name	Description
1:0	COUNTERS_4_ALARMMODE	Register AlarmMode is a 2-bit register that is present when parameter statisticsCounterAlarm is set to True. The register defines the statistics-alarm behavior of the counter.

0x00503190 PCNOC_EPRB0_SPROF_COUNTERS_4_VAL**Type:** R**Reset State:** Undefined**PCNOC_EPRB0_SPROF_COUNTERS_4_VAL**

Bits	Name	Description
15:0	COUNTERS_4_VAL	Register Val is a read-only register that is always present. The register contains the statistics counter value either pending StatAlarm output, or when statistics collection is suspended subsequent to triggers or signal statSuspend.

0x0050319C PCNOC_EPRB0_SPROF_COUNTERS_5_SRC**Type:** RW**Reset State:** 0x00000000**PCNOC_EPRB0_SPROF_COUNTERS_5_SRC**

Bits	Name	Description
4:0	INTEVENT	Internal packet event

0x005031A0 PCNOC_EPRB0_SPROF_COUNTERS_5_ALARMMODE**Type:** RW**Reset State:** 0x00000000**PCNOC_EPRB0_SPROF_COUNTERS_5_ALARMMODE**

Bits	Name	Description
1:0	COUNTERS_5_ALARMMODE	Register AlarmMode is a 2-bit register that is present when parameter statisticsCounterAlarm is set to True. The register defines the statistics-alarm behavior of the counter.

0x005031A4 PCNOC_EPRB0_SPROF_COUNTERS_5_VAL**Type:** R**Reset State:** Undefined**PCNOC_EPRB0_SPROF_COUNTERS_5_VAL**

Bits	Name	Description
15:0	COUNTERS_5_VAL	Register Val is a read-only register that is always present. The register contains the statistics counter value either pending StatAlarm output, or when statistics collection is suspended subsequent to triggers or signal statSuspend.

0x005031B0 PCNOC_EPRB0_SPROF_COUNTERS_6_SRC**Type:** RW**Reset State:** 0x00000000**PCNOC_EPRB0_SPROF_COUNTERS_6_SRC**

Bits	Name	Description
4:0	INTEVENT	Internal packet event

0x005031B4 PCNOC_EPRB0_SPROF_COUNTERS_6_ALARMMODE**Type:** RW**Reset State:** 0x00000000**PCNOC_EPRB0_SPROF_COUNTERS_6_ALARMMODE**

Bits	Name	Description
1:0	COUNTERS_6_ALARMMODE	Register AlarmMode is a 2-bit register that is present when parameter statisticsCounterAlarm is set to True. The register defines the statistics-alarm behavior of the counter.

0x005031B8 PCNOC_EPRB0_SPROF_COUNTERS_6_VAL**Type:** R**Reset State:** Undefined**PCNOC_EPRB0_SPROF_COUNTERS_6_VAL**

Bits	Name	Description
15:0	COUNTERS_6_VAL	Register Val is a read-only register that is always present. The register contains the statistics counter value either pending StatAlarm output, or when statistics collection is suspended subsequent to triggers or signal statSuspend.

0x005031C4 PCNOC_EPRB0_SPROF_COUNTERS_7_SRC**Type:** RW**Reset State:** 0x00000000**PCNOC_EPRB0_SPROF_COUNTERS_7_SRC**

Bits	Name	Description
4:0	INTEVENT	Internal packet event

0x005031C8 PCNOC_EPRB0_SPROF_COUNTERS_7_ALARMMODE**Type:** RW**Reset State:** 0x00000000**PCNOC_EPRB0_SPROF_COUNTERS_7_ALARMMODE**

Bits	Name	Description
1:0	COUNTERS_7_ALARMMODE	Register AlarmMode is a 2-bit register that is present when parameter statisticsCounterAlarm is set to True. The register defines the statistics-alarm behavior of the counter.

0x005031CC PCNOC_EPRB0_SPROF_COUNTERS_7_VAL**Type:** R**Reset State:** Undefined**PCNOC_EPRB0_SPROF_COUNTERS_7_VAL**

Bits	Name	Description
15:0	COUNTERS_7_VAL	Register Val is a read-only register that is always present. The register contains the statistics counter value either pending StatAlarm output, or when statistics collection is suspended subsequent to triggers or signal statSuspend.

0x00504000 PCNOC_TPRB0_SPROF_ID_COREID**Type:** R**Reset State:** 0x26712606**PCNOC_TPRB0_SPROF_ID_COREID**

Bits	Name	Description
31:8	CORECHECKSUM	Field containing a checksum of the parameters of the IP.
7:0	CORETYPEID	Field identifying the type of IP.

0x00504004 PCNOC_TPRB0_SPROF_ID_REVISIONID**Type:** R**Reset State:** 0x01222B00**PCNOC_TPRB0_SPROF_ID_REVISIONID**

Bits	Name	Description
31:8	FLEXNOCID	Field containing the build revision of the software used to generate the IP HDL code.
7:0	USERID	Field containing a user defined value, not used anywhere inside the IP itself.

0x00504008 PCNOC_TPRB0_SPROF_MAINCTL**Type:** RW**Reset State:** 0x00000000

Register MainCtl contains probe global control bits. The register has seven bit fields:

PCNOC_TPRB0_SPROF_MAINCTL

Bits	Name	Type	Description
7	FILTBYTEALWAYSCHAINABLEEN	RW	When set to 0, Filter Byte / ByteEn Cnt will be connected to all statistics counters where odd Stat count are not chainable. When set to 1, Filter Byte / ByteEn Cnt will be connected only to chainable stat counter (even stat counter).
6	INTRUSIVEMODE	R	When set to 1, register field IntrusiveMode enables trace operation in Intrusive flow-control mode. When set to 0, the register enables trace operation in Overflow flow-control mode
5	STATCONDDUMP	RW	When set, register field StatCondDump enables the dump of a statistics frame to the range of counter values set for registers StatAlarmMin, StatAlarmMax, and AlarmMode. This field also renders register StatAlarmStatus inoperative. When parameter statisticsCounterAlarm is set to False, the StatCondDump register bit is reserved.
4	ALARMEN	RW	When set, register field AlarmEn enables the probe to collect alarm-related information. When the register field bit is null, both TraceAlarm and StatAlarm outputs are driven to 0.
3	STATEN	RW	When set to 1, register field StatEn enables statistics profiling. The probe sendS statistics results to the output for signal ObsTx. All statistics counters are cleared when the StatEn bit goes from 0 to 1. When set to 0, counters are disabled.
2	PAYLOADEN	RW	Register field PayloadEn, when set to 1, enables traces to contain headers and payload. When set to 0, only headers are reported.
1	TRACEEN	RW	Register field TraceEn enables the probe to send filtered packets (Trace) on the ObsTx observation output.
0	ERREN	RW	Register field ErrEn enables the probe to send on the ObsTx output any packet with Error status, independently of filtering mechanisms, thus constituting a simple supplementary global filter.

0x0050400C PCNOC_TPRB0_SPROF_CFGCTL**Type:** RW**Reset State:** 0b000000000000000000000000000000X0**PCNOC_TPRB0_SPROF_CFGCTL**

Bits	Name	Type	Description
1	ACTIVE	R	
0	GLOBALEN	RW	

0x00504024 PCNOC_TPRB0_SPROF_STATPERIOD**Type:** RW**Reset State:** 0x00000000**PCNOC_TPRB0_SPROF_STATPERIOD**

Bits	Name	Description
4:0	STATPERIOD	Register StatPeriod is a 5-bit register that sets a period, within a range of 2-8 megacycles, during which statistics are collected before being dumped automatically. Setting the register effectively enables automatic mode operation for statistics collection. Note: The configured period is also used for statistics alarm generation. The period is calculated with the formula: $N_Cycle = 2^{min(23, StatPeriod)}$ When register StatPeriod is set to its default value 0, automatic dump mode is disabled, and register StatGo is activated for manual mode operation. Note: When parameter nStatisticsCounter is set to 0, statistics collection is disabled, with the consequence that all statistics collection registers, including StatPeriod, are reserved.

0x00504028 PCNOC_TPRB0_SPROF_STATGO**Type:** RW**Reset State:** 0x00000000**PCNOC_TPRB0_SPROF_STATGO**

Bits	Name	Description
0	STATGO	Writing to 1-bit pulse register StatGo generates a statistics dump. The register is active when statistics collection operates in its manual mode, that is, when register StatPeriod is set to 0.

0x0050402C PCNOC_TPRB0_SPROF_STATALARMMIN**Type:** RW**Reset State:** 0x00000000**PCNOC_TPRB0_SPROF_STATALARMMIN**

Bits	Name	Description
31:0	STATALARMMIN	Register StatAlarmMin contains the minimum count value used in statistics alarm comparisons. The number of bits is equal to twice the value set for parameter wStatisticsCounter. When parameter statisticsCounterAlarm is set to False, StatAlarmMin is reserved.

0x00504030 PCNOC_TPRB0_SPROF_STATALARMMAX**Type:** RW**Reset State:** 0x00000000**PCNOC_TPRB0_SPROF_STATALARMMAX**

Bits	Name	Description
31:0	STATALARMMAX	Register StatAlarmMax contains the maximum count value used in statistics alarm comparisons. The number of bits is equal to twice the value set for parameter wStatisticsCounter. When parameter statisticsCounterAlarm is set to False, StatAlarmMax is reserved.

0x00504034 PCNOC_TPRB0_SPROF_STATALARMSSTATUS**Type:** R**Reset State:** Undefined**PCNOC_TPRB0_SPROF_STATALARMSSTATUS**

Bits	Name	Description
0	STATALARMSSTATUS	Register StatAlarmStatus is a read-only 1-bit register indicating that at least one statistics counter has exceeded the programmed values for registers StatAlarmMin or StatAlarmMax. Output signal StatAlarm is equal to the values stored in register MainCtl fields StatAlarmStatus and AlarmEn. When parameter statisticsCounterAlarm is set to False, StatAlarmStatus is reserved.

0x00504038 PCNOC_TPRB0_SPROF_STATALARMCCLR**Type:** RW**Reset State:** 0x00000000**PCNOC_TPRB0_SPROF_STATALARMCCLR**

Bits	Name	Description
0	STATALARMCCLR	Register StatAlarmClr is a 1-bit write-only register. A write operation to this register clears the register StatAlarmStatus bit. When parameter statisticsCounterAlarm is set to False, StatAlarmClr is reserved.

0x00504138 PCNOC_TPRB0_SPROF_COUNTERS_0_SRC**Type:** RW**Reset State:** 0x00000000

PCNOC_TPRB0_SPROF_COUNTERS_0_SRC

Bits	Name	Description
5	EXTEVENT	When set to 1 counts the cycles where ExtEvent[IntEvent] = 1. It exists when nExtEvent > 0.
4:0	INTEVENT	Internal packet event

0x0050413C PCNOC_TPRB0_SPROF_COUNTERS_0_ALARMMODE**Type:** RW**Reset State:** 0x00000000**PCNOC_TPRB0_SPROF_COUNTERS_0_ALARMMODE**

Bits	Name	Description
1:0	COUNTERS_0_ALARMMODE	Register AlarmMode is a 2-bit register that is present when parameter statisticsCounterAlarm is set to True. The register defines the statistics-alarm behavior of the counter.

0x00504140 PCNOC_TPRB0_SPROF_COUNTERS_0_VAL**Type:** R**Reset State:** Undefined**PCNOC_TPRB0_SPROF_COUNTERS_0_VAL**

Bits	Name	Description
15:0	COUNTERS_0_VAL	Register Val is a read-only register that is always present. The register contains the statistics counter value either pending StatAlarm output, or when statistics collection is suspended subsequent to triggers or signal statSuspend.

0x0050414C PCNOC_TPRB0_SPROF_COUNTERS_1_SRC**Type:** RW**Reset State:** 0x00000000**PCNOC_TPRB0_SPROF_COUNTERS_1_SRC**

Bits	Name	Description
5	EXTEVENT	When set to 1 counts the cycles where ExtEvent[IntEvent] = 1. It exists when nExtEvent > 0.
4:0	INTEVENT	Internal packet event

0x00504150 PCNOC_TPRB0_SPROF_COUNTERS_1_ALARMMODE**Type:** RW**Reset State:** 0x00000000**PCNOC_TPRB0_SPROF_COUNTERS_1_ALARMMODE**

Bits	Name	Description
1:0	COUNTERS_1_ALARMMODE	Register AlarmMode is a 2-bit register that is present when parameter statisticsCounterAlarm is set to True. The register defines the statistics-alarm behavior of the counter.

0x00504154 PCNOC_TPRB0_SPROF_COUNTERS_1_VAL**Type:** R**Reset State:** Undefined**PCNOC_TPRB0_SPROF_COUNTERS_1_VAL**

Bits	Name	Description
15:0	COUNTERS_1_VAL	Register Val is a read-only register that is always present. The register contains the statistics counter value either pending StatAlarm output, or when statistics collection is suspended subsequent to triggers or signal statSuspend.

0x00504160 PCNOC_TPRB0_SPROF_COUNTERS_2_SRC**Type:** RW**Reset State:** 0x00000000**PCNOC_TPRB0_SPROF_COUNTERS_2_SRC**

Bits	Name	Description
5	EXTEVENT	When set to 1 counts the cycles where ExtEvent[IntEvent] = 1. It exists when nExtEvent > 0.
4:0	INTEVENT	Internal packet event

0x00504164 PCNOC_TPRB0_SPROF_COUNTERS_2_ALARMMODE**Type:** RW**Reset State:** 0x00000000

PCNOC_TPRB0_SPROF_COUNTERS_2_ALARMMODE

Bits	Name	Description
1:0	COUNTERS_2_ALARMMODE	Register AlarmMode is a 2-bit register that is present when parameter statisticsCounterAlarm is set to True. The register defines the statistics-alarm behavior of the counter.

0x00504168 PCNOC_TPRB0_SPROF_COUNTERS_2_VAL**Type:** R**Reset State:** Undefined**PCNOC_TPRB0_SPROF_COUNTERS_2_VAL**

Bits	Name	Description
15:0	COUNTERS_2_VAL	Register Val is a read-only register that is always present. The register contains the statistics counter value either pending StatAlarm output, or when statistics collection is suspended subsequent to triggers or signal statSuspend.

0x00504174 PCNOC_TPRB0_SPROF_COUNTERS_3_SRC**Type:** RW**Reset State:** 0x00000000**PCNOC_TPRB0_SPROF_COUNTERS_3_SRC**

Bits	Name	Description
5	EXTEVENT	When set to 1 counts the cycles where ExtEvent[IntEvent] = 1. It exists when nExtEvent > 0.
4:0	INTEVENT	Internal packet event

0x00504178 PCNOC_TPRB0_SPROF_COUNTERS_3_ALARMMODE**Type:** RW**Reset State:** 0x00000000**PCNOC_TPRB0_SPROF_COUNTERS_3_ALARMMODE**

Bits	Name	Description
1:0	COUNTERS_3_ALARMMODE	Register AlarmMode is a 2-bit register that is present when parameter statisticsCounterAlarm is set to True. The register defines the statistics-alarm behavior of the counter.

0x0050417C PCNOC_TPRB0_SPROF_COUNTERS_3_VAL**Type:** R**Reset State:** Undefined**PCNOC_TPRB0_SPROF_COUNTERS_3_VAL**

Bits	Name	Description
15:0	COUNTERS_3_VAL	Register Val is a read-only register that is always present. The register contains the statistics counter value either pending StatAlarm output, or when statistics collection is suspended subsequent to triggers or signal statSuspend.

0x00504188 PCNOC_TPRB0_SPROF_COUNTERS_4_SRC**Type:** RW**Reset State:** 0x00000000**PCNOC_TPRB0_SPROF_COUNTERS_4_SRC**

Bits	Name	Description
5	EXTEVENT	When set to 1 counts the cycles where ExtEvent[IntEvent] = 1. It exists when nExtEvent > 0.
4:0	INTEVENT	Internal packet event

0x0050418C PCNOC_TPRB0_SPROF_COUNTERS_4_ALARMMODE**Type:** RW**Reset State:** 0x00000000**PCNOC_TPRB0_SPROF_COUNTERS_4_ALARMMODE**

Bits	Name	Description
1:0	COUNTERS_4_ALARMMODE	Register AlarmMode is a 2-bit register that is present when parameter statisticsCounterAlarm is set to True. The register defines the statistics-alarm behavior of the counter.

0x00504190 PCNOC_TPRB0_SPROF_COUNTERS_4_VAL**Type:** R**Reset State:** Undefined

PCNOC_TPRB0_SPROF_COUNTERS_4_VAL

Bits	Name	Description
15:0	COUNTERS_4_VAL	Register Val is a read-only register that is always present. The register contains the statistics counter value either pending StatAlarm output, or when statistics collection is suspended subsequent to triggers or signal statSuspend.

0x0050419C PCNOC_TPRB0_SPROF_COUNTERS_5_SRC**Type:** RW**Reset State:** 0x00000000**PCNOC_TPRB0_SPROF_COUNTERS_5_SRC**

Bits	Name	Description
5	EXTEVENT	When set to 1 counts the cycles where ExtEvent[IntEvent] = 1. It exists when nExtEvent > 0.
4:0	INTEVENT	Internal packet event

0x005041A0 PCNOC_TPRB0_SPROF_COUNTERS_5_ALARMMODE**Type:** RW**Reset State:** 0x00000000**PCNOC_TPRB0_SPROF_COUNTERS_5_ALARMMODE**

Bits	Name	Description
1:0	COUNTERS_5_ALARMMODE	Register AlarmMode is a 2-bit register that is present when parameter statisticsCounterAlarm is set to True. The register defines the statistics-alarm behavior of the counter.

0x005041A4 PCNOC_TPRB0_SPROF_COUNTERS_5_VAL**Type:** R**Reset State:** Undefined**PCNOC_TPRB0_SPROF_COUNTERS_5_VAL**

Bits	Name	Description
15:0	COUNTERS_5_VAL	Register Val is a read-only register that is always present. The register contains the statistics counter value either pending StatAlarm output, or when statistics collection is suspended subsequent to triggers or signal statSuspend.

0x005041B0 PCNOC_TPRB0_SPROF_COUNTERS_6_SRC**Type:** RW**Reset State:** 0x00000000**PCNOC_TPRB0_SPROF_COUNTERS_6_SRC**

Bits	Name	Description
5	EXTEVENT	When set to 1 counts the cycles where ExtEvent[IntEvent] = 1. It exists when nExtEvent > 0.
4:0	INTEVENT	Internal packet event

0x005041B4 PCNOC_TPRB0_SPROF_COUNTERS_6_ALARMMODE**Type:** RW**Reset State:** 0x00000000**PCNOC_TPRB0_SPROF_COUNTERS_6_ALARMMODE**

Bits	Name	Description
1:0	COUNTERS_6_ALARMMODE	Register AlarmMode is a 2-bit register that is present when parameter statisticsCounterAlarm is set to True. The register defines the statistics-alarm behavior of the counter.

0x005041B8 PCNOC_TPRB0_SPROF_COUNTERS_6_VAL**Type:** R**Reset State:** Undefined**PCNOC_TPRB0_SPROF_COUNTERS_6_VAL**

Bits	Name	Description
15:0	COUNTERS_6_VAL	Register Val is a read-only register that is always present. The register contains the statistics counter value either pending StatAlarm output, or when statistics collection is suspended subsequent to triggers or signal statSuspend.

0x005041C4 PCNOC_TPRB0_SPROF_COUNTERS_7_SRC**Type:** RW**Reset State:** 0x00000000

PCNOC_TPRB0_SPROF_COUNTERS_7_SRC

Bits	Name	Description
5	EXTEVENT	When set to 1 counts the cycles where ExtEvent[IntEvent] = 1. It exists when nExtEvent > 0.
4:0	INTEVENT	Internal packet event

0x005041C8 PCNOC_TPRB0_SPROF_COUNTERS_7_ALARMMODE**Type:** RW**Reset State:** 0x00000000**PCNOC_TPRB0_SPROF_COUNTERS_7_ALARMMODE**

Bits	Name	Description
1:0	COUNTERS_7_ALARMMODE	Register AlarmMode is a 2-bit register that is present when parameter statisticsCounterAlarm is set to True. The register defines the statistics-alarm behavior of the counter.

0x005041CC PCNOC_TPRB0_SPROF_COUNTERS_7_VAL**Type:** R**Reset State:** Undefined**PCNOC_TPRB0_SPROF_COUNTERS_7_VAL**

Bits	Name	Description
15:0	COUNTERS_7_VAL	Register Val is a read-only register that is always present. The register contains the statistics counter value either pending StatAlarm output, or when statistics collection is suspended subsequent to triggers or signal statSuspend.

0x005041D8 PCNOC_TPRB0_SPROF_COUNTERS_8_SRC**Type:** RW**Reset State:** 0x00000000**PCNOC_TPRB0_SPROF_COUNTERS_8_SRC**

Bits	Name	Description
5	EXTEVENT	When set to 1 counts the cycles where ExtEvent[IntEvent] = 1. It exists when nExtEvent > 0.
4:0	INTEVENT	Internal packet event

0x005041DC PCNOC_TPRB0_SPROF_COUNTERS_8_ALARMMODE**Type:** RW**Reset State:** 0x00000000**PCNOC_TPRB0_SPROF_COUNTERS_8_ALARMMODE**

Bits	Name	Description
1:0	COUNTERS_8_ALARMMODE	Register AlarmMode is a 2-bit register that is present when parameter statisticsCounterAlarm is set to True. The register defines the statistics-alarm behavior of the counter.

0x005041E0 PCNOC_TPRB0_SPROF_COUNTERS_8_VAL**Type:** R**Reset State:** Undefined**PCNOC_TPRB0_SPROF_COUNTERS_8_VAL**

Bits	Name	Description
15:0	COUNTERS_8_VAL	Register Val is a read-only register that is always present. The register contains the statistics counter value either pending StatAlarm output, or when statistics collection is suspended subsequent to triggers or signal statSuspend.

0x005041EC PCNOC_TPRB0_SPROF_COUNTERS_9_SRC**Type:** RW**Reset State:** 0x00000000**PCNOC_TPRB0_SPROF_COUNTERS_9_SRC**

Bits	Name	Description
5	EXTEVENT	When set to 1 counts the cycles where ExtEvent[IntEvent] = 1. It exists when nExtEvent > 0.
4:0	INTEVENT	Internal packet event

0x005041F0 PCNOC_TPRB0_SPROF_COUNTERS_9_ALARMMODE**Type:** RW**Reset State:** 0x00000000

PCNOC_TPRB0_SPROF_COUNTERS_9_ALARMMODE

Bits	Name	Description
1:0	COUNTERS_9_ALARMMODE	Register AlarmMode is a 2-bit register that is present when parameter statisticsCounterAlarm is set to True. The register defines the statistics-alarm behavior of the counter.

0x005041F4 PCNOC_TPRB0_SPROF_COUNTERS_9_VAL**Type:** R**Reset State:** Undefined**PCNOC_TPRB0_SPROF_COUNTERS_9_VAL**

Bits	Name	Description
15:0	COUNTERS_9_VAL	Register Val is a read-only register that is always present. The register contains the statistics counter value either pending StatAlarm output, or when statistics collection is suspended subsequent to triggers or signal statSuspend.

0x00504200 PCNOC_TPRB0_SPROF_COUNTERS_10_SRC**Type:** RW**Reset State:** 0x00000000**PCNOC_TPRB0_SPROF_COUNTERS_10_SRC**

Bits	Name	Description
5	EXTEVENT	When set to 1 counts the cycles where ExtEvent[IntEvent] = 1. It exists when nExtEvent > 0.
4:0	INTEVENT	Internal packet event

0x00504204 PCNOC_TPRB0_SPROF_COUNTERS_10_ALARMMODE**Type:** RW**Reset State:** 0x00000000**PCNOC_TPRB0_SPROF_COUNTERS_10_ALARMMODE**

Bits	Name	Description
1:0	COUNTERS_10_ALARMMODE	Register AlarmMode is a 2-bit register that is present when parameter statisticsCounterAlarm is set to True. The register defines the statistics-alarm behavior of the counter.

0x00504208 PCNOC_TPRB0_SPROF_COUNTERS_10_VAL**Type:** R**Reset State:** Undefined**PCNOC_TPRB0_SPROF_COUNTERS_10_VAL**

Bits	Name	Description
15:0	COUNTERS_10_VAL	Register Val is a read-only register that is always present. The register contains the statistics counter value either pending StatAlarm output, or when statistics collection is suspended subsequent to triggers or signal statSuspend.

0x00504214 PCNOC_TPRB0_SPROF_COUNTERS_11_SRC**Type:** RW**Reset State:** 0x00000000**PCNOC_TPRB0_SPROF_COUNTERS_11_SRC**

Bits	Name	Description
5	EXTEVENT	When set to 1 counts the cycles where ExtEvent[IntEvent] = 1. It exists when nExtEvent > 0.
4:0	INTEVENT	Internal packet event

0x00504218 PCNOC_TPRB0_SPROF_COUNTERS_11_ALARMMODE**Type:** RW**Reset State:** 0x00000000**PCNOC_TPRB0_SPROF_COUNTERS_11_ALARMMODE**

Bits	Name	Description
1:0	COUNTERS_11_ALARMMODE	Register AlarmMode is a 2-bit register that is present when parameter statisticsCounterAlarm is set to True. The register defines the statistics-alarm behavior of the counter.

0x0050421C PCNOC_TPRB0_SPROF_COUNTERS_11_VAL**Type:** R**Reset State:** Undefined

PCNOC_TPRB0_SPROF_COUNTERS_11_VAL

Bits	Name	Description
15:0	COUNTERS_11_VAL	Register Val is a read-only register that is always present. The register contains the statistics counter value either pending StatAlarm output, or when statistics collection is suspended subsequent to triggers or signal statSuspend.

0x00504228 PCNOC_TPRB0_SPROF_COUNTERS_12_SRC**Type:** RW**Reset State:** 0x00000000**PCNOC_TPRB0_SPROF_COUNTERS_12_SRC**

Bits	Name	Description
5	EXTEVENT	When set to 1 counts the cycles where ExtEvent[IntEvent] = 1. It exists when nExtEvent > 0.
4:0	INTEVENT	Internal packet event

0x0050422C PCNOC_TPRB0_SPROF_COUNTERS_12_ALARMMODE**Type:** RW**Reset State:** 0x00000000**PCNOC_TPRB0_SPROF_COUNTERS_12_ALARMMODE**

Bits	Name	Description
1:0	COUNTERS_12_ALARMMODE	Register AlarmMode is a 2-bit register that is present when parameter statisticsCounterAlarm is set to True. The register defines the statistics-alarm behavior of the counter.

0x00504230 PCNOC_TPRB0_SPROF_COUNTERS_12_VAL**Type:** R**Reset State:** Undefined**PCNOC_TPRB0_SPROF_COUNTERS_12_VAL**

Bits	Name	Description
15:0	COUNTERS_12_VAL	Register Val is a read-only register that is always present. The register contains the statistics counter value either pending StatAlarm output, or when statistics collection is suspended subsequent to triggers or signal statSuspend.

0x0050423C PCNOC_TPRB0_SPROF_COUNTERS_13_SRC**Type:** RW**Reset State:** 0x00000000**PCNOC_TPRB0_SPROF_COUNTERS_13_SRC**

Bits	Name	Description
5	EXTEVENT	When set to 1 counts the cycles where ExtEvent[IntEvent] = 1. It exists when nExtEvent > 0.
4:0	INTEVENT	Internal packet event

0x00504240 PCNOC_TPRB0_SPROF_COUNTERS_13_ALARMMODE**Type:** RW**Reset State:** 0x00000000**PCNOC_TPRB0_SPROF_COUNTERS_13_ALARMMODE**

Bits	Name	Description
1:0	COUNTERS_13_ALARMMODE	Register AlarmMode is a 2-bit register that is present when parameter statisticsCounterAlarm is set to True. The register defines the statistics-alarm behavior of the counter.

0x00504244 PCNOC_TPRB0_SPROF_COUNTERS_13_VAL**Type:** R**Reset State:** Undefined**PCNOC_TPRB0_SPROF_COUNTERS_13_VAL**

Bits	Name	Description
15:0	COUNTERS_13_VAL	Register Val is a read-only register that is always present. The register contains the statistics counter value either pending StatAlarm output, or when statistics collection is suspended subsequent to triggers or signal statSuspend.

0x00504250 PCNOC_TPRB0_SPROF_COUNTERS_14_SRC**Type:** RW**Reset State:** 0x00000000

PCNOC_TPRB0_SPROF_COUNTERS_14_SRC

Bits	Name	Description
5	EXTEVENT	When set to 1 counts the cycles where ExtEvent[IntEvent] = 1. It exists when nExtEvent > 0.
4:0	INTEVENT	Internal packet event

0x00504254 PCNOC_TPRB0_SPROF_COUNTERS_14_ALARMMODE**Type:** RW**Reset State:** 0x00000000**PCNOC_TPRB0_SPROF_COUNTERS_14_ALARMMODE**

Bits	Name	Description
1:0	COUNTERS_14_ALARMMODE	Register AlarmMode is a 2-bit register that is present when parameter statisticsCounterAlarm is set to True. The register defines the statistics-alarm behavior of the counter.

0x00504258 PCNOC_TPRB0_SPROF_COUNTERS_14_VAL**Type:** R**Reset State:** Undefined**PCNOC_TPRB0_SPROF_COUNTERS_14_VAL**

Bits	Name	Description
15:0	COUNTERS_14_VAL	Register Val is a read-only register that is always present. The register contains the statistics counter value either pending StatAlarm output, or when statistics collection is suspended subsequent to triggers or signal statSuspend.

0x00504264 PCNOC_TPRB0_SPROF_COUNTERS_15_SRC**Type:** RW**Reset State:** 0x00000000**PCNOC_TPRB0_SPROF_COUNTERS_15_SRC**

Bits	Name	Description
5	EXTEVENT	When set to 1 counts the cycles where ExtEvent[IntEvent] = 1. It exists when nExtEvent > 0.
4:0	INTEVENT	Internal packet event

0x00504268 PCNOC_TPRB0_SPROF_COUNTERS_15_ALARMMODE**Type:** RW**Reset State:** 0x00000000**PCNOC_TPRB0_SPROF_COUNTERS_15_ALARMMODE**

Bits	Name	Description
1:0	COUNTERS_15_ALARMMODE	Register AlarmMode is a 2-bit register that is present when parameter statisticsCounterAlarm is set to True. The register defines the statistics-alarm behavior of the counter.

0x0050426C PCNOC_TPRB0_SPROF_COUNTERS_15_VAL**Type:** R**Reset State:** Undefined**PCNOC_TPRB0_SPROF_COUNTERS_15_VAL**

Bits	Name	Description
15:0	COUNTERS_15_VAL	Register Val is a read-only register that is always present. The register contains the statistics counter value either pending StatAlarm output, or when statistics collection is suspended subsequent to triggers or signal statSuspend.

0x0050C000 PCNOC_QHM1_QOS_ID_COREID**Type:** R**Reset State:** 0xB88D1704**PCNOC_QHM1_QOS_ID_COREID**

Bits	Name	Description
31:8	CORECHECKSUM	Field containing a checksum of the parameters of the IP.
7:0	CORETYPEID	Field identifying the type of IP.

0x0050C004 PCNOC_QHM1_QOS_ID_REVISIONID**Type:** R**Reset State:** 0x01222B00

PCNOC_QHM1_QOS_ID_REVISIONID

Bits	Name	Description
31:8	FLEXNOCID	Field containing the build revision of the software used to generate the IP HDL code.
7:0	USERID	Field containing a user defined value, not used anywhere inside the IP itself.

0x0050C008 PCNOC_QHM1_QOS_PRIORITY**Type:** RW**Reset State:** 0x00000000

Priority register.

PCNOC_QHM1_QOS_PRIORITY

Bits	Name	Description
3:2	P1	In Regulator mode, defines the HIGH hurry level. In Fixed/Limiter mode, defines the Urgency level for READ transactions.
1:0	P0	In Regulator mode, defines the LOW hurry level. In Fixed/Limiter mode, defines the Urgency level for WRITE transactions.

0x0050C00C PCNOC_QHM1_QOS_MODE**Type:** RW**Reset State:** 0x00000000**PCNOC_QHM1_QOS_MODE**

Bits	Name	Description
1:0	MODE	Functional Mode: 0=Fixed, 1=Limiter, 2=Bypass, 3=Regulator.

0x0050C018 PCNOC_QHM1_QOS_EXTCONTROL**Type:** RW**Reset State:** 0x00000000

External inputs control.

PCNOC_QHM1_QOS_EXTCONTROL

Bits	Name	Description
2	INTCLKEN	Replace the External reference by the local clock.
1	EXTTHREN	ExtThr input controls Low/High priority instead of bandwidth threshold.
0	SOCKETQOSEN	Combines the Socket QoS with Regulator QoS.

0x0050D000 PCNOC_QHM2_QOS_ID_COREID**Type:** R**Reset State:** 0x5F073304**PCNOC_QHM2_QOS_ID_COREID**

Bits	Name	Description
31:8	CORECHECKSUM	Field containing a checksum of the parameters of the IP.
7:0	CORETYPEID	Field identifying the type of IP.

0x0050D004 PCNOC_QHM2_QOS_ID_REVISIONID**Type:** R**Reset State:** 0x01222B00**PCNOC_QHM2_QOS_ID_REVISIONID**

Bits	Name	Description
31:8	FLEXNOCID	Field containing the build revision of the software used to generate the IP HDL code.
7:0	USERID	Field containing a user defined value, not used anywhere inside the IP itself.

0x0050D008 PCNOC_QHM2_QOS_PRIORITY**Type:** RW**Reset State:** 0x00000000

Priority register.

PCNOC_QHM2_QOS_PRIORITY

Bits	Name	Description
3:2	P1	In Regulator mode, defines the HIGH hurry level. In Fixed/Limiter mode, defines the Urgency level for READ transactions.
1:0	P0	In Regulator mode, defines the LOW hurry level. In Fixed/Limiter mode, defines the Urgency level for WRITE transactions.

0x0050D00C PCNOC_QHM2_QOS_MODE**Type:** RW**Reset State:** 0x00000000**PCNOC_QHM2_QOS_MODE**

Bits	Name	Description
1:0	MODE	Functional Mode: 0=Fixed, 1=Limiter, 2=Bypass, 3=Regulator.

0x0050D018 PCNOC_QHM2_QOS_EXTCONTROL**Type:** RW**Reset State:** 0x00000000

External inputs control.

PCNOC_QHM2_QOS_EXTCONTROL

Bits	Name	Description
2	INTCLKEN	Replace the External reference by the local clock.
1	EXTTHREN	ExtThr input controls Low/High priority instead of bandwidth threshold.
0	SOCKETQOSEN	Combines the Socket QoS with Regulator QoS.

0x0050E000 PCNOC_SDC1_QOS_ID_COREID**Type:** R**Reset State:** 0xE4C38F04**PCNOC_SDC1_QOS_ID_COREID**

Bits	Name	Description
31:8	CORECHECKSUM	Field containing a checksum of the parameters of the IP.

PCNOC_SDC1_QOS_ID_COREID (cont.)

Bits	Name	Description
7:0	CORETYPEID	Field identifying the type of IP.

0x0050E004 PCNOC_SDC1_QOS_ID_REVISIONID**Type:** R**Reset State:** 0x01222B00**PCNOC_SDC1_QOS_ID_REVISIONID**

Bits	Name	Description
31:8	FLEXNOCID	Field containing the build revision of the software used to generate the IP HDL code.
7:0	USERID	Field containing a user defined value, not used anywhere inside the IP itself.

0x0050E008 PCNOC_SDC1_QOS_PRIORITY**Type:** RW**Reset State:** 0x00000000

Priority register.

PCNOC_SDC1_QOS_PRIORITY

Bits	Name	Description
3:2	P1	In Regulator mode, defines the HIGH hurry level. In Fixed/Limiter mode, defines the Urgency level for READ transactions.
1:0	P0	In Regulator mode, defines the LOW hurry level. In Fixed/Limiter mode, defines the Urgency level for WRITE transactions.

0x0050E00C PCNOC_SDC1_QOS_MODE**Type:** RW**Reset State:** 0x00000000**PCNOC_SDC1_QOS_MODE**

Bits	Name	Description
1:0	MODE	Functional Mode: 0=Fixed, 1=Limiter, 2=Bypass, 3=Regulator.

0x0050E018 PCNOC_SDC1_QOS_EXTCONTROL**Type:** RW**Reset State:** 0x00000000

External inputs control.

PCNOC_SDC1_QOS_EXTCONTROL

Bits	Name	Description
2	INTCLKEN	Replace the External reference by the local clock.
1	EXTTHREN	ExtThr input controls Low/High priority instead of bandwidth threshold.
0	SOCKETQOSEN	Combines the Socket QoS with Regulator QoS.

0x0050F000 PCNOC_SDC2_QOS_ID_COREID**Type:** R**Reset State:** 0xBE565A04**PCNOC_SDC2_QOS_ID_COREID**

Bits	Name	Description
31:8	CORECHECKSUM	Field containing a checksum of the parameters of the IP.
7:0	CORETYPEID	Field identifying the type of IP.

0x0050F004 PCNOC_SDC2_QOS_ID_REVISIONID**Type:** R**Reset State:** 0x01222B00**PCNOC_SDC2_QOS_ID_REVISIONID**

Bits	Name	Description
31:8	FLEXNOCID	Field containing the build revision of the software used to generate the IP HDL code.
7:0	USERID	Field containing a user defined value, not used anywhere inside the IP itself.

0x0050F008 PCNOC_SDC2_QOS_PRIORITY**Type:** RW**Reset State:** 0x00000000

Priority register.

PCNOC_SDC2_QOS_PRIORITY

Bits	Name	Description
3:2	P1	In Regulator mode, defines the HIGH hurry level. In Fixed/Limiter mode, defines the Urgency level for READ transactions.
1:0	P0	In Regulator mode, defines the LOW hurry level. In Fixed/Limiter mode, defines the Urgency level for WRITE transactions.

0x0050F00C PCNOC_SDC2_QOS_MODE

Type: RW

Reset State: 0x00000000

PCNOC_SDC2_QOS_MODE

Bits	Name	Description
1:0	MODE	Functional Mode: 0=Fixed, 1=Limiter, 2=Bypass, 3=Regulator.

0x0050F018 PCNOC_SDC2_QOS_EXTCONTROL

Type: RW

Reset State: 0x00000000

External inputs control.

PCNOC_SDC2_QOS_EXTCONTROL

Bits	Name	Description
2	INTCLKEN	Replace the External reference by the local clock.
1	EXTTHREN	ExtThr input controls Low/High priority instead of bandwidth threshold.
0	SOCKETQOSEN	Combines the Socket QoS with Regulator QoS.

0x00510000 PCNOC_SNOQ_QOS_ID_COREID

Type: R

Reset State: 0xE0779004

PCNOC_SNOC_QOS_ID_COREID

Bits	Name	Description
31:8	CORECHECKSUM	Field containing a checksum of the parameters of the IP.
7:0	CORETYPEID	Field identifying the type of IP.

0x00510004 PCNOC_SNOC_QOS_ID_REVISIONID**Type:** R**Reset State:** 0x01222B00**PCNOC_SNOC_QOS_ID_REVISIONID**

Bits	Name	Description
31:8	FLEXNOCID	Field containing the build revision of the software used to generate the IP HDL code.
7:0	USERID	Field containing a user defined value, not used anywhere inside the IP itself.

0x00510008 PCNOC_SNOC_QOS_PRIORITY**Type:** RW**Reset State:** 0x00000000

Priority register.

PCNOC_SNOC_QOS_PRIORITY

Bits	Name	Description
3:2	P1	In Regulator mode, defines the HIGH hurry level. In Fixed/Limiter mode, defines the Urgency level for READ transactions.
1:0	P0	In Regulator mode, defines the LOW hurry level. In Fixed/Limiter mode, defines the Urgency level for WRITE transactions.

0x0051000C PCNOC_SNOC_QOS_MODE**Type:** RW**Reset State:** 0x00000000**PCNOC_SNOC_QOS_MODE**

Bits	Name	Description
1:0	MODE	Functional Mode: 0=Fixed, 1=Limiter, 2=Bypass, 3=Regulator.

0x00510018 PCNOC_SNOC_QOS_EXTCONTROL**Type:** RW**Reset State:** 0x00000000

External inputs control.

PCNOC_SNOC_QOS_EXTCONTROL

Bits	Name	Description
2	INTCLKEN	Replace the External reference by the local clock.
1	EXTTHREN	ExtThr input controls Low/High priority instead of bandwidth threshold.
0	SOCKETQOSEN	Combines the Socket QoS with Regulator QoS.

3.1 PCNOC_0_BUS_TIMEOUT**0x00023004 PCNOC_0_ABT_INST_ID****Type:** RW**Clock:** AHB Clock**Reset State:** 0x00000000**PCNOC_0_ABT_INST_ID**

Bits	Name	Description
31:0	INSTID	SW configurable field to provide unique instance number for each of the AHB Timeout Slaves in the chip.

0x00023008 PCNOC_0_ABT_NUM_SLAVES**Type:** R**Clock:** AHB Clock**Reset State:** 0x00000005**PCNOC_0_ABT_NUM_SLAVES**

Bits	Name	Description
5:0	NUMSLAVES	Specifies the number of slaves on the AHB bus excluding the Timeout Slave. Valid values are from 1 to 32.

0x0002300C PCNOC_0_ABT_TIMER_LOADVAL**Type:** RW**Clock:** AHB Clock**Reset State:** 0x000000FF**PCNOC_0_ABT_TIMER_LOADVAL**

Bits	Name	Description
7:0	LOADVAL	Load the timer with this value, when the timer is in load mode, i.e, MODE field in ABT_TIMER_MODE is 0. Note: LOADVAL of 0 is not valid.

0x00023010 PCNOC_0_ABT_TIMER_MODE**Type:** RW**Clock:** AHB Clock**Reset State:** 0x00000000**PCNOC_0_ABT_TIMER_MODE**

Bits	Name	Description
0	MODE	If this field is 0, the timer is in load mode and the LOADVAL field value in ABT_TIMER_LOADVAL is loaded into the timer. If this field is 1, the timer is in run mode.

0x00023014 PCNOC_0_ABT_INTR_STATUS**Type:** R**Clock:** AHB Clock**Reset State:** 0x00000000**PCNOC_0_ABT_INTR_STATUS**

Bits	Name	Description
0	INTRSTATUS	Specifies the timeout interrupt status irrespective of interrupt enable.

0x00023018 PCNOC_0_ABT_INTR_CLEAR**Type:** W**Clock:** AHB Clock**Reset State:** 0x00000000

PCNOC_0_ABT_INTR_CLEAR

Bits	Name	Description
0	INTRCLEAR	Clears the pending interrupt.

0x0002301C PCNOC_0_ABT_INTR_ENABLE**Type:** RW**Clock:** AHB Clock**Reset State:** 0x00000000**PCNOC_0_ABT_INTR_ENABLE**

Bits	Name	Description
0	INTRENABLE	Enable an interrupt generation on timeout event.

0x00023020 PCNOC_0_ABT_SYND_VALID**Type:** R**Clock:** AHB Clock**Reset State:** 0x00000000**PCNOC_0_ABT_SYND_VALID**

Bits	Name	Description
0	SYNDVALID	Set when the information in other syndrome registers is valid.

0x00023024 PCNOC_0_ABT_SYND_CLEAR**Type:** W**Clock:** AHB Clock**Reset State:** 0x00000000**PCNOC_0_ABT_SYND_CLEAR**

Bits	Name	Description
0	SYNDCLEAR	Clears the syndrome valid bit and syndrome information (ABT_SYND_ID, ABT_SYND_ADDR, ABT_SYND_HREADY).

0x00023028 PCNOC_0_ABT_SYND_ID

Type: R
Clock: AHB Clock
Reset State: 0x00000000

PCNOC_0_ABT_SYND_ID

Bits	Name	Description
15:13	BID	Bus ID of the transaction that caused bus hang.
12:8	PID	Port ID of the transaction that caused bus hang.
7:0	MID	Master ID of the transaction that caused bus hang.

0x0002302C PCNOC_0_ABT_SYND_ADDR0

Type: R
Clock: AHB Clock
Reset State: 0x00000000

PCNOC_0_ABT_SYND_ADDR0

Bits	Name	Description
31:0	SYNDADDR0	Lower 32-bit address field of the transaction that caused bus hang.

0x00023030 PCNOC_0_ABT_SYND_ADDR1

Type: R
Clock: AHB Clock
Reset State: 0x00000000

PCNOC_0_ABT_SYND_ADDR1

Bits	Name	Description
31:0	SYNDADDR1	Upper 32-bit address field of the transaction that caused bus hang. Reserved for future use.

0x00023034 PCNOC_0_ABT_SYND_HREADY

Type: R
Clock: AHB Clock
Reset State: 0xFFFFFFFF

PCNOC_0_ABT_SYND_HREADY

Bits	Name	Description
31:0	SYNDHREADY	Contains the status of Hready for each slave on the bus during the timeout event. The valid bit fields are [ABT_NUM_SLAVES-1:0]. Each valid bit field corresponds to a physical slave on the bus.

3.2 PCNOC_1_BUS_TIMEOUT**0x00024004 PCNOC_1_ABT_INST_ID****Type:** RW**Clock:** AHB Clock**Reset State:** 0x00000000**PCNOC_1_ABT_INST_ID**

Bits	Name	Description
31:0	INSTID	SW configurable field to provide unique instance number for each of the AHB Timeout Slaves in the chip.

0x00024008 PCNOC_1_ABT_NUM_SLAVES**Type:** R**Clock:** AHB Clock**Reset State:** 0x00000005**PCNOC_1_ABT_NUM_SLAVES**

Bits	Name	Description
5:0	NUMSLAVES	Specifies the number of slaves on the AHB bus excluding the Timeout Slave. Valid values are from 1 to 32.

0x0002400C PCNOC_1_ABT_TIMER_LOADVAL**Type:** RW**Clock:** AHB Clock**Reset State:** 0x000000FF

PCNOC_1_AB_TTIMER_LOADVAL

Bits	Name	Description
7:0	LOADVAL	Load the timer with this value, when the timer is in load mode, i.e, MODE field in AB_TTIMER_MODE is 0. Note: LOADVAL of 0 is not valid.

0x00024010 PCNOC_1_AB_TTIMER_MODE**Type:** RW**Clock:** AHB Clock**Reset State:** 0x00000000**PCNOC_1_AB_TTIMER_MODE**

Bits	Name	Description
0	MODE	If this field is 0, the timer is in load mode and the LOADVAL field value in AB_TTIMER_LOADVAL is loaded into the timer. If this field is 1, the timer is in run mode.

0x00024014 PCNOC_1_AB_TINTR_STATUS**Type:** R**Clock:** AHB Clock**Reset State:** 0x00000000**PCNOC_1_AB_TINTR_STATUS**

Bits	Name	Description
0	INTRSTATUS	Specifies the timeout interrupt status irrespective of interrupt enable.

0x00024018 PCNOC_1_AB_TINTR_CLEAR**Type:** W**Clock:** AHB Clock**Reset State:** 0x00000000**PCNOC_1_AB_TINTR_CLEAR**

Bits	Name	Description
0	INTRCLEAR	Clears the pending interrupt.

0x0002401C PCNOC_1_ABT_INTR_ENABLE**Type:** RW**Clock:** AHB Clock**Reset State:** 0x00000000**PCNOC_1_ABT_INTR_ENABLE**

Bits	Name	Description
0	INTRENABLE	Enable an interrupt generation on timeout event.

0x00024020 PCNOC_1_ABT_SYND_VALID**Type:** R**Clock:** AHB Clock**Reset State:** 0x00000000**PCNOC_1_ABT_SYND_VALID**

Bits	Name	Description
0	SYNDVALID	Set when the information in other syndrome registers is valid.

0x00024024 PCNOC_1_ABT_SYND_CLEAR**Type:** W**Clock:** AHB Clock**Reset State:** 0x00000000**PCNOC_1_ABT_SYND_CLEAR**

Bits	Name	Description
0	SYNDCLEAR	Clears the syndrome valid bit and syndrome information (ABT_SYND_ID, ABT_SYND_ADDR, ABT_SYND_HREADY).

0x00024028 PCNOC_1_ABT_SYND_ID**Type:** R**Clock:** AHB Clock**Reset State:** 0x00000000**PCNOC_1_ABT_SYND_ID**

Bits	Name	Description
15:13	BID	Bus ID of the transaction that caused bus hang.

PCNOC_1_ABT_SYND_ID (cont.)

Bits	Name	Description
12:8	PID	Port ID of the transaction that caused bus hang.
7:0	MID	Master ID of the transaction that caused bus hang.

0x0002402C PCNOC_1_ABT_SYND_ADDR0**Type:** R**Clock:** AHB Clock**Reset State:** 0x00000000**PCNOC_1_ABT_SYND_ADDR0**

Bits	Name	Description
31:0	SYNDADDR0	Lower 32-bit address field of the transaction that caused bus hang.

0x00024030 PCNOC_1_ABT_SYND_ADDR1**Type:** R**Clock:** AHB Clock**Reset State:** 0x00000000**PCNOC_1_ABT_SYND_ADDR1**

Bits	Name	Description
31:0	SYNDADDR1	Upper 32-bit address field of the transaction that caused bus hang. Reserved for future use.

0x00024034 PCNOC_1_ABT_SYND_HREADY**Type:** R**Clock:** AHB Clock**Reset State:** 0xFFFFFFFF**PCNOC_1_ABT_SYND_HREADY**

Bits	Name	Description
31:0	SYNDHREADY	Contains the status of Hready for each slave on the bus during the timeout event. The valid bit fields are [ABT_NUM_SLAVES-1:0]. Each valid bit field corresponds to a physical slave on the bus.

3.3 PCNOC_2_BUS_TIMEOUT

0x00025004 PCNOC_2_ABT_INST_ID

Type: RW

Clock: AHB Clock

Reset State: 0x00000000

PCNOC_2_ABT_INST_ID

Bits	Name	Description
31:0	INSTID	SW configurable field to provide unique instance number for each of the AHB Timeout Slaves in the chip.

0x00025008 PCNOC_2_ABT_NUM_SLAVES

Type: R

Clock: AHB Clock

Reset State: 0x00000005

PCNOC_2_ABT_NUM_SLAVES

Bits	Name	Description
5:0	NUMSLAVES	Specifies the number of slaves on the AHB bus excluding the Timeout Slave. Valid values are from 1 to 32.

0x0002500C PCNOC_2_ABT_TIMER_LOADVAL

Type: RW

Clock: AHB Clock

Reset State: 0x000000FF

PCNOC_2_ABT_TIMER_LOADVAL

Bits	Name	Description
7:0	LOADVAL	Load the timer with this value, when the timer is in load mode, i.e, MODE field in ABT_TIMER_MODE is 0. Note: LOADVAL of 0 is not valid.

0x00025010 PCNOC_2_ABT_TIMER_MODE

Type: RW

Clock: AHB Clock

Reset State: 0x00000000

PCNOC_2_AB_TTIMER_MODE

Bits	Name	Description
0	MODE	If this field is 0, the timer is in load mode and the LOADVAL field value in ABT_TIMER_LOADVAL is loaded into the timer. If this field is 1, the timer is in run mode.

0x00025014 PCNOC_2_AB_TINTR_STATUS**Type:** R**Clock:** AHB Clock**Reset State:** 0x00000000**PCNOC_2_AB_TINTR_STATUS**

Bits	Name	Description
0	INTRSTATUS	Specifies the timeout interrupt status irrespective of interrupt enable.

0x00025018 PCNOC_2_AB_TINTR_CLEAR**Type:** W**Clock:** AHB Clock**Reset State:** 0x00000000**PCNOC_2_AB_TINTR_CLEAR**

Bits	Name	Description
0	INTRCLEAR	Clears the pending interrupt.

0x0002501C PCNOC_2_AB_TINTR_ENABLE**Type:** RW**Clock:** AHB Clock**Reset State:** 0x00000000**PCNOC_2_AB_TINTR_ENABLE**

Bits	Name	Description
0	INTRENABLE	Enable an interrupt generation on timeout event.

0x00025020 PCNOC_2_ABT_SYND_VALID

Type: R
Clock: AHB Clock
Reset State: 0x00000000

PCNOC_2_ABT_SYND_VALID

Bits	Name	Description
0	SYNDVALID	Set when the information in other syndrome registers is valid.

0x00025024 PCNOC_2_ABT_SYND_CLEAR

Type: W
Clock: AHB Clock
Reset State: 0x00000000

PCNOC_2_ABT_SYND_CLEAR

Bits	Name	Description
0	SYNDCLEAR	Clears the syndrome valid bit and syndrome information (ABT_SYND_ID, ABT_SYND_ADDR, ABT_SYND_HREADY).

0x00025028 PCNOC_2_ABT_SYND_ID

Type: R
Clock: AHB Clock
Reset State: 0x00000000

PCNOC_2_ABT_SYND_ID

Bits	Name	Description
15:13	BID	Bus ID of the transaction that caused bus hang.
12:8	PID	Port ID of the transaction that caused bus hang.
7:0	MID	Master ID of the transaction that caused bus hang.

0x0002502C PCNOC_2_ABT_SYND_ADDR0

Type: R
Clock: AHB Clock
Reset State: 0x00000000

PCNOC_2_AB_T_SYND_ADDR0

Bits	Name	Description
31:0	SYNDADDR0	Lower 32-bit address field of the transaction that caused bus hang.

0x00025030 PCNOC_2_AB_T_SYND_ADDR1**Type:** R**Clock:** AHB Clock**Reset State:** 0x00000000**PCNOC_2_AB_T_SYND_ADDR1**

Bits	Name	Description
31:0	SYNDADDR1	Upper 32-bit address field of the transaction that caused bus hang. Reserved for future use.

0x00025034 PCNOC_2_AB_T_SYND_HREADY**Type:** R**Clock:** AHB Clock**Reset State:** 0xFFFFFFFF**PCNOC_2_AB_T_SYND_HREADY**

Bits	Name	Description
31:0	SYNDHREADY	Contains the status of Hready for each slave on the bus during the timeout event. The valid bit fields are [ABT_NUM_SLAVES-1:0]. Each valid bit field corresponds to a physical slave on the bus.

3.4 PCNOC_3_BUS_TIMEOUT**0x00026004 PCNOC_3_AB_T_INST_ID****Type:** RW**Clock:** AHB Clock**Reset State:** 0x00000000**PCNOC_3_AB_T_INST_ID**

Bits	Name	Description
31:0	INSTID	SW configurable field to provide unique instance number for each of the AHB Timeout Slaves in the chip.

0x00026008 PCNOC_3_ABT_NUM_SLAVES

Type: R
Clock: AHB Clock
Reset State: 0x00000005

PCNOC_3_ABT_NUM_SLAVES

Bits	Name	Description
5:0	NUMSLAVES	Specifies the number of slaves on the AHB bus excluding the Timeout Slave. Valid values are from 1 to 32.

0x0002600C PCNOC_3_ABT_TIMER_LOADVAL

Type: RW
Clock: AHB Clock
Reset State: 0x000000FF

PCNOC_3_ABT_TIMER_LOADVAL

Bits	Name	Description
7:0	LOADVAL	Load the timer with this value, when the timer is in load mode, i.e, MODE field in ABT_TIMER_MODE is 0. Note: LOADVAL of 0 is not valid.

0x00026010 PCNOC_3_ABT_TIMER_MODE

Type: RW
Clock: AHB Clock
Reset State: 0x00000000

PCNOC_3_ABT_TIMER_MODE

Bits	Name	Description
0	MODE	If this field is 0, the timer is in load mode and the LOADVAL field value in ABT_TIMER_LOADVAL is loaded into the timer. If this field is 1, the timer is in run mode.

0x00026014 PCNOC_3_ABT_INTR_STATUS

Type: R
Clock: AHB Clock
Reset State: 0x00000000

PCNOC_3_ABT_INTR_STATUS

Bits	Name	Description
0	INTRSTATUS	Specifies the timeout interrupt status irrespective of interrupt enable.

0x00026018 PCNOC_3_ABT_INTR_CLEAR**Type:** W**Clock:** AHB Clock**Reset State:** 0x00000000**PCNOC_3_ABT_INTR_CLEAR**

Bits	Name	Description
0	INTRCLEAR	Clears the pending interrupt.

0x0002601C PCNOC_3_ABT_INTR_ENABLE**Type:** RW**Clock:** AHB Clock**Reset State:** 0x00000000**PCNOC_3_ABT_INTR_ENABLE**

Bits	Name	Description
0	INTRENABLE	Enable an interrupt generation on timeout event.

0x00026020 PCNOC_3_ABT_SYND_VALID**Type:** R**Clock:** AHB Clock**Reset State:** 0x00000000**PCNOC_3_ABT_SYND_VALID**

Bits	Name	Description
0	SYNDVALID	Set when the information in other syndrome registers is valid.

0x00026024 PCNOC_3_AB_T_SYND_CLEAR**Type:** W**Clock:** AHB Clock**Reset State:** 0x00000000**PCNOC_3_AB_T_SYND_CLEAR**

Bits	Name	Description
0	SYNDCLEAR	Clears the syndrome valid bit and syndrome information (ABT_SYND_ID, ABT_SYND_ADDR, ABT_SYND_HREADY).

0x00026028 PCNOC_3_AB_T_SYND_ID**Type:** R**Clock:** AHB Clock**Reset State:** 0x00000000**PCNOC_3_AB_T_SYND_ID**

Bits	Name	Description
15:13	BID	Bus ID of the transaction that caused bus hang.
12:8	PID	Port ID of the transaction that caused bus hang.
7:0	MID	Master ID of the transaction that caused bus hang.

0x0002602C PCNOC_3_AB_T_SYND_ADDR0**Type:** R**Clock:** AHB Clock**Reset State:** 0x00000000**PCNOC_3_AB_T_SYND_ADDR0**

Bits	Name	Description
31:0	SYNDADDR0	Lower 32-bit address field of the transaction that caused bus hang.

0x00026030 PCNOC_3_AB_T_SYND_ADDR1**Type:** R**Clock:** AHB Clock**Reset State:** 0x00000000

PCNOC_3_ABT_SYND_ADDR1

Bits	Name	Description
31:0	SYNDADDR1	Upper 32-bit address field of the transaction that caused bus hang. Reserved for future use.

0x00026034 PCNOC_3_ABT_SYND_HREADY**Type:** R**Clock:** AHB Clock**Reset State:** 0xFFFFFFFF**PCNOC_3_ABT_SYND_HREADY**

Bits	Name	Description
31:0	SYNDHREADY	Contains the status of Hready for each slave on the bus during the timeout event. The valid bit fields are [ABT_NUM_SLAVES-1:0]. Each valid bit field corresponds to a physical slave on the bus.

3.5 PCNOC_4_BUS_TIMEOUT**0x00027004 PCNOC_4_ABT_INST_ID****Type:** RW**Clock:** AHB Clock**Reset State:** 0x00000000**PCNOC_4_ABT_INST_ID**

Bits	Name	Description
31:0	INSTID	SW configurable field to provide unique instance number for each of the AHB Timeout Slaves in the chip.

0x00027008 PCNOC_4_ABT_NUM_SLAVES**Type:** R**Clock:** AHB Clock**Reset State:** 0x00000003**PCNOC_4_ABT_NUM_SLAVES**

Bits	Name	Description
5:0	NUMSLAVES	Specifies the number of slaves on the AHB bus excluding the Timeout Slave. Valid values are from 1 to 32.

0x0002700C PCNOC_4_AB_TTIMER_LOADVAL**Type:** RW**Clock:** AHB Clock**Reset State:** 0x000000FF**PCNOC_4_AB_TTIMER_LOADVAL**

Bits	Name	Description
7:0	LOADVAL	Load the timer with this value, when the timer is in load mode, i.e, MODE field in AB_TTIMER_MODE is 0. Note: LOADVAL of 0 is not valid.

0x00027010 PCNOC_4_AB_TTIMER_MODE**Type:** RW**Clock:** AHB Clock**Reset State:** 0x00000000**PCNOC_4_AB_TTIMER_MODE**

Bits	Name	Description
0	MODE	If this field is 0, the timer is in load mode and the LOADVAL field value in AB_TTIMER_LOADVAL is loaded into the timer. If this field is 1, the timer is in run mode.

0x00027014 PCNOC_4_AB_TINTR_STATUS**Type:** R**Clock:** AHB Clock**Reset State:** 0x00000000**PCNOC_4_AB_TINTR_STATUS**

Bits	Name	Description
0	INTRSTATUS	Specifies the timeout interrupt status irrespective of interrupt enable.

0x00027018 PCNOC_4_AB_TINTR_CLEAR**Type:** W**Clock:** AHB Clock**Reset State:** 0x00000000

PCNOC_4_ABT_INTR_CLEAR

Bits	Name	Description
0	INTRCLEAR	Clears the pending interrupt.

0x0002701C PCNOC_4_ABT_INTR_ENABLE**Type:** RW**Clock:** AHB Clock**Reset State:** 0x00000000**PCNOC_4_ABT_INTR_ENABLE**

Bits	Name	Description
0	INTRENABLE	Enable an interrupt generation on timeout event.

0x00027020 PCNOC_4_ABT_SYND_VALID**Type:** R**Clock:** AHB Clock**Reset State:** 0x00000000**PCNOC_4_ABT_SYND_VALID**

Bits	Name	Description
0	SYNDVALID	Set when the information in other syndrome registers is valid.

0x00027024 PCNOC_4_ABT_SYND_CLEAR**Type:** W**Clock:** AHB Clock**Reset State:** 0x00000000**PCNOC_4_ABT_SYND_CLEAR**

Bits	Name	Description
0	SYNDCLEAR	Clears the syndrome valid bit and syndrome information (ABT_SYND_ID, ABT_SYND_ADDR, ABT_SYND_HREADY).

0x00027028 PCNOC_4_AB_T_SYND_ID

Type: R
Clock: AHB Clock
Reset State: 0x00000000

PCNOC_4_AB_T_SYND_ID

Bits	Name	Description
15:13	BID	Bus ID of the transaction that caused bus hang.
12:8	PID	Port ID of the transaction that caused bus hang.
7:0	MID	Master ID of the transaction that caused bus hang.

0x0002702C PCNOC_4_AB_T_SYND_ADDR0

Type: R
Clock: AHB Clock
Reset State: 0x00000000

PCNOC_4_AB_T_SYND_ADDR0

Bits	Name	Description
31:0	SYNDADDR0	Lower 32-bit address field of the transaction that caused bus hang.

0x00027030 PCNOC_4_AB_T_SYND_ADDR1

Type: R
Clock: AHB Clock
Reset State: 0x00000000

PCNOC_4_AB_T_SYND_ADDR1

Bits	Name	Description
31:0	SYNDADDR1	Upper 32-bit address field of the transaction that caused bus hang. Reserved for future use.

0x00027034 PCNOC_4_AB_T_SYND_HREADY

Type: R
Clock: AHB Clock
Reset State: 0xFFFFFFFF

PCNOC_4_ABT_SYND_HREADY

Bits	Name	Description
31:0	SYNDHREADY	Contains the status of Hready for each slave on the bus during the timeout event. The valid bit fields are [ABT_NUM_SLAVES-1:0]. Each valid bit field corresponds to a physical slave on the bus.

3.6 PCNOC_5_BUS_TIMEOUT

0x00028004 PCNOC_5_ABT_INST_ID

Type: RW

Clock: AHB Clock

Reset State: 0x00000000

PCNOC_5_ABT_INST_ID

Bits	Name	Description
31:0	INSTID	SW configurable field to provide unique instance number for each of the AHB Timeout Slaves in the chip.

0x00028008 PCNOC_5_ABT_NUM_SLAVES

Type: R

Clock: AHB Clock

Reset State: 0x00000001

PCNOC_5_ABT_NUM_SLAVES

Bits	Name	Description
5:0	NUMSLAVES	Specifies the number of slaves on the AHB bus excluding the Timeout Slave. Valid values are from 1 to 32.

0x0002800C PCNOC_5_ABT_TIMER_LOADVAL

Type: RW

Clock: AHB Clock

Reset State: 0x000000FF

PCNOC_5_ABT_TIMER_LOADVAL

Bits	Name	Description
7:0	LOADVAL	Load the timer with this value, when the timer is in load mode, i.e, MODE field in ABT_TIMER_MODE is 0. Note: LOADVAL of 0 is not valid.

0x00028010 PCNOC_5_ABT_TIMER_MODE

Type: RW

Clock: AHB Clock

Reset State: 0x00000000

PCNOC_5_AB_TTIMER_MODE

Bits	Name	Description
0	MODE	If this field is 0, the timer is in load mode and the LOADVAL field value in AB_TTIMER_LOADVAL is loaded into the timer. If this field is 1, the timer is in run mode.

0x00028014 PCNOC_5_AB_TINTR_STATUS**Type:** R**Clock:** AHB Clock**Reset State:** 0x00000000**PCNOC_5_AB_TINTR_STATUS**

Bits	Name	Description
0	INTRSTATUS	Specifies the timeout interrupt status irrespective of interrupt enable.

0x00028018 PCNOC_5_AB_TINTR_CLEAR**Type:** W**Clock:** AHB Clock**Reset State:** 0x00000000**PCNOC_5_AB_TINTR_CLEAR**

Bits	Name	Description
0	INTRCLEAR	Clears the pending interrupt.

0x0002801C PCNOC_5_AB_TINTR_ENABLE**Type:** RW**Clock:** AHB Clock**Reset State:** 0x00000000**PCNOC_5_AB_TINTR_ENABLE**

Bits	Name	Description
0	INTRENABLE	Enable an interrupt generation on timeout event.

0x00028020 PCNOC_5_AB_T_SYND_VALID

Type: R
Clock: AHB Clock
Reset State: 0x00000000

PCNOC_5_AB_T_SYND_VALID

Bits	Name	Description
0	SYNDVALID	Set when the information in other syndrome registers is valid.

0x00028024 PCNOC_5_AB_T_SYND_CLEAR

Type: W
Clock: AHB Clock
Reset State: 0x00000000

PCNOC_5_AB_T_SYND_CLEAR

Bits	Name	Description
0	SYNDCLEAR	Clears the syndrome valid bit and syndrome information (AB_T_SYND_ID, AB_T_SYND_ADDR, AB_T_SYND_HREADY).

0x00028028 PCNOC_5_AB_T_SYND_ID

Type: R
Clock: AHB Clock
Reset State: 0x00000000

PCNOC_5_AB_T_SYND_ID

Bits	Name	Description
15:13	BID	Bus ID of the transaction that caused bus hang.
12:8	PID	Port ID of the transaction that caused bus hang.
7:0	MID	Master ID of the transaction that caused bus hang.

0x0002802C PCNOC_5_AB_T_SYND_ADDR0

Type: R
Clock: AHB Clock
Reset State: 0x00000000

PCNOC_5_ABT_SYND_ADDR0

Bits	Name	Description
31:0	SYNDADDR0	Lower 32-bit address field of the transaction that caused bus hang.

0x00028030 PCNOC_5_ABT_SYND_ADDR1**Type:** R**Clock:** AHB Clock**Reset State:** 0x00000000**PCNOC_5_ABT_SYND_ADDR1**

Bits	Name	Description
31:0	SYNDADDR1	Upper 32-bit address field of the transaction that caused bus hang. Reserved for future use.

0x00028034 PCNOC_5_ABT_SYND_HREADY**Type:** R**Clock:** AHB Clock**Reset State:** 0xFFFFFFFF**PCNOC_5_ABT_SYND_HREADY**

Bits	Name	Description
31:0	SYNDHREADY	Contains the status of Hready for each slave on the bus during the timeout event. The valid bit fields are [ABT_NUM_SLAVES-1:0]. Each valid bit field corresponds to a physical slave on the bus.

3.7 PCNOC_6_BUS_TIMEOUT**0x00029004 PCNOC_6_AB_TINST_ID****Type:** RW**Clock:** AHB Clock**Reset State:** 0x00000000**PCNOC_6_AB_TINST_ID**

Bits	Name	Description
31:0	INSTID	SW configurable field to provide unique instance number for each of the AHB Timeout Slaves in the chip.

0x00029008 PCNOC_6_ABT_NUM_SLAVES

Type: R
Clock: AHB Clock
Reset State: 0x00000002

PCNOC_6_ABT_NUM_SLAVES

Bits	Name	Description
5:0	NUMSLAVES	Specifies the number of slaves on the AHB bus excluding the Timeout Slave. Valid values are from 1 to 32.

0x0002900C PCNOC_6_ABT_TIMER_LOADVAL

Type: RW
Clock: AHB Clock
Reset State: 0x000000FF

PCNOC_6_ABT_TIMER_LOADVAL

Bits	Name	Description
7:0	LOADVAL	Load the timer with this value, when the timer is in load mode, i.e, MODE field in ABT_TIMER_MODE is 0. Note: LOADVAL of 0 is not valid.

0x00029010 PCNOC_6_ABT_TIMER_MODE

Type: RW
Clock: AHB Clock
Reset State: 0x00000000

PCNOC_6_ABT_TIMER_MODE

Bits	Name	Description
0	MODE	If this field is 0, the timer is in load mode and the LOADVAL field value in ABT_TIMER_LOADVAL is loaded into the timer. If this field is 1, the timer is in run mode.

0x00029014 PCNOC_6_ABT_INTR_STATUS

Type: R
Clock: AHB Clock
Reset State: 0x00000000

PCNOC_6_ABT_INTR_STATUS

Bits	Name	Description
0	INTRSTATUS	Specifies the timeout interrupt status irrespective of interrupt enable.

0x00029018 PCNOC_6_ABT_INTR_CLEAR**Type:** W**Clock:** AHB Clock**Reset State:** 0x00000000**PCNOC_6_ABT_INTR_CLEAR**

Bits	Name	Description
0	INTRCLEAR	Clears the pending interrupt.

0x0002901C PCNOC_6_ABT_INTR_ENABLE**Type:** RW**Clock:** AHB Clock**Reset State:** 0x00000000**PCNOC_6_ABT_INTR_ENABLE**

Bits	Name	Description
0	INTRENABLE	Enable an interrupt generation on timeout event.

0x00029020 PCNOC_6_ABT_SYND_VALID**Type:** R**Clock:** AHB Clock**Reset State:** 0x00000000**PCNOC_6_ABT_SYND_VALID**

Bits	Name	Description
0	SYNDVALID	Set when the information in other syndrome registers is valid.

0x00029024 PCNOC_6_AB_T_SYND_CLEAR**Type:** W**Clock:** AHB Clock**Reset State:** 0x00000000**PCNOC_6_AB_T_SYND_CLEAR**

Bits	Name	Description
0	SYNDCLEAR	Clears the syndrome valid bit and syndrome information (ABT_SYND_ID, ABT_SYND_ADDR, ABT_SYND_HREADY).

0x00029028 PCNOC_6_AB_T_SYND_ID**Type:** R**Clock:** AHB Clock**Reset State:** 0x00000000**PCNOC_6_AB_T_SYND_ID**

Bits	Name	Description
15:13	BID	Bus ID of the transaction that caused bus hang.
12:8	PID	Port ID of the transaction that caused bus hang.
7:0	MID	Master ID of the transaction that caused bus hang.

0x0002902C PCNOC_6_AB_T_SYND_ADDR0**Type:** R**Clock:** AHB Clock**Reset State:** 0x00000000**PCNOC_6_AB_T_SYND_ADDR0**

Bits	Name	Description
31:0	SYNDADDR0	Lower 32-bit address field of the transaction that caused bus hang.

0x00029030 PCNOC_6_AB_T_SYND_ADDR1**Type:** R**Clock:** AHB Clock**Reset State:** 0x00000000

PCNOC_6_AB_T_SYND_ADDR1

Bits	Name	Description
31:0	SYNDADDR1	Upper 32-bit address field of the transaction that caused bus hang. Reserved for future use.

0x00029034 PCNOC_6_AB_T_SYND_HREADY**Type:** R**Clock:** AHB Clock**Reset State:** 0xFFFFFFFF**PCNOC_6_AB_T_SYND_HREADY**

Bits	Name	Description
31:0	SYNDHREADY	Contains the status of Hready for each slave on the bus during the timeout event. The valid bit fields are [ABT_NUM_SLAVES-1:0]. Each valid bit field corresponds to a physical slave on the bus.

3.8 PCNOC_7_BUS_TIMEOUT**0x0002A004 PCNOC_7_AB_T_INST_ID****Type:** RW**Clock:** AHB Clock**Reset State:** 0x00000000**PCNOC_7_AB_T_INST_ID**

Bits	Name	Description
31:0	INSTID	SW configurable field to provide unique instance number for each of the AHB Timeout Slaves in the chip.

0x0002A008 PCNOC_7_AB_T_NUM_SLAVES**Type:** R**Clock:** AHB Clock**Reset State:** 0x00000004**PCNOC_7_AB_T_NUM_SLAVES**

Bits	Name	Description
5:0	NUMSLAVES	Specifies the number of slaves on the AHB bus excluding the Timeout Slave. Valid values are from 1 to 32.

0x0002A00C PCNOC_7_ABT_TIMER_LOADVAL**Type:** RW**Clock:** AHB Clock**Reset State:** 0x000000FF**PCNOC_7_ABT_TIMER_LOADVAL**

Bits	Name	Description
7:0	LOADVAL	Load the timer with this value, when the timer is in load mode, i.e, MODE field in ABT_TIMER_MODE is 0. Note: LOADVAL of 0 is not valid.

0x0002A010 PCNOC_7_ABT_TIMER_MODE**Type:** RW**Clock:** AHB Clock**Reset State:** 0x00000000**PCNOC_7_ABT_TIMER_MODE**

Bits	Name	Description
0	MODE	If this field is 0, the timer is in load mode and the LOADVAL field value in ABT_TIMER_LOADVAL is loaded into the timer. If this field is 1, the timer is in run mode.

0x0002A014 PCNOC_7_ABT_INTR_STATUS**Type:** R**Clock:** AHB Clock**Reset State:** 0x00000000**PCNOC_7_ABT_INTR_STATUS**

Bits	Name	Description
0	INTRSTATUS	Specifies the timeout interrupt status irrespective of interrupt enable.

0x0002A018 PCNOC_7_ABT_INTR_CLEAR**Type:** W**Clock:** AHB Clock**Reset State:** 0x00000000

PCNOC_7_ABT_INTR_CLEAR

Bits	Name	Description
0	INTRCLEAR	Clears the pending interrupt.

0x0002A01C PCNOC_7_ABT_INTR_ENABLE**Type:** RW**Clock:** AHB Clock**Reset State:** 0x00000000**PCNOC_7_ABT_INTR_ENABLE**

Bits	Name	Description
0	INTRENABLE	Enable an interrupt generation on timeout event.

0x0002A020 PCNOC_7_ABT_SYND_VALID**Type:** R**Clock:** AHB Clock**Reset State:** 0x00000000**PCNOC_7_ABT_SYND_VALID**

Bits	Name	Description
0	SYNDVALID	Set when the information in other syndrome registers is valid.

0x0002A024 PCNOC_7_ABT_SYND_CLEAR**Type:** W**Clock:** AHB Clock**Reset State:** 0x00000000**PCNOC_7_ABT_SYND_CLEAR**

Bits	Name	Description
0	SYNDCLEAR	Clears the syndrome valid bit and syndrome information (ABT_SYND_ID, ABT_SYND_ADDR, ABT_SYND_HREADY).

0x0002A028 PCNOC_7_AB_T_SYND_ID

Type: R
Clock: AHB Clock
Reset State: 0x00000000

PCNOC_7_AB_T_SYND_ID

Bits	Name	Description
15:13	BID	Bus ID of the transaction that caused bus hang.
12:8	PID	Port ID of the transaction that caused bus hang.
7:0	MID	Master ID of the transaction that caused bus hang.

0x0002A02C PCNOC_7_AB_T_SYND_ADDR0

Type: R
Clock: AHB Clock
Reset State: 0x00000000

PCNOC_7_AB_T_SYND_ADDR0

Bits	Name	Description
31:0	SYNDADDR0	Lower 32-bit address field of the transaction that caused bus hang.

0x0002A030 PCNOC_7_AB_T_SYND_ADDR1

Type: R
Clock: AHB Clock
Reset State: 0x00000000

PCNOC_7_AB_T_SYND_ADDR1

Bits	Name	Description
31:0	SYNDADDR1	Upper 32-bit address field of the transaction that caused bus hang. Reserved for future use.

0x0002A034 PCNOC_7_AB_T_SYND_HREADY

Type: R
Clock: AHB Clock
Reset State: 0xFFFFFFFF

PCNOC_7_ABT_SYND_HREADY

Bits	Name	Description
31:0	SYNDHREADY	Contains the status of Hready for each slave on the bus during the timeout event. The valid bit fields are [ABT_NUM_SLAVES-1:0]. Each valid bit field corresponds to a physical slave on the bus.

3.9 PCNOC_8_BUS_TIMEOUT**0x0002B004 PCNOC_8_ABT_INST_ID****Type:** RW**Clock:** AHB Clock**Reset State:** 0x00000000**PCNOC_8_ABT_INST_ID**

Bits	Name	Description
31:0	INSTID	SW configurable field to provide unique instance number for each of the AHB Timeout Slaves in the chip.

0x0002B008 PCNOC_8_ABT_NUM_SLAVES**Type:** R**Clock:** AHB Clock**Reset State:** 0x00000003**PCNOC_8_ABT_NUM_SLAVES**

Bits	Name	Description
5:0	NUMSLAVES	Specifies the number of slaves on the AHB bus excluding the Timeout Slave. Valid values are from 1 to 32.

0x0002B00C PCNOC_8_ABT_TIMER_LOADVAL**Type:** RW**Clock:** AHB Clock**Reset State:** 0x000000FF

PCNOC_8_ABT_TIMER_LOADVAL

Bits	Name	Description
7:0	LOADVAL	Load the timer with this value, when the timer is in load mode, i.e, MODE field in ABT_TIMER_MODE is 0. Note: LOADVAL of 0 is not valid.

0x0002B010 PCNOC_8_ABT_TIMER_MODE**Type:** RW**Clock:** AHB Clock**Reset State:** 0x00000000**PCNOC_8_ABT_TIMER_MODE**

Bits	Name	Description
0	MODE	If this field is 0, the timer is in load mode and the LOADVAL field value in ABT_TIMER_LOADVAL is loaded into the timer. If this field is 1, the timer is in run mode.

0x0002B014 PCNOC_8_ABT_INTR_STATUS**Type:** R**Clock:** AHB Clock**Reset State:** 0x00000000**PCNOC_8_ABT_INTR_STATUS**

Bits	Name	Description
0	INTRSTATUS	Specifies the timeout interrupt status irrespective of interrupt enable.

0x0002B018 PCNOC_8_ABT_INTR_CLEAR**Type:** W**Clock:** AHB Clock**Reset State:** 0x00000000**PCNOC_8_ABT_INTR_CLEAR**

Bits	Name	Description
0	INTRCLEAR	Clears the pending interrupt.

0x0002B01C PCNOC_8_ABT_INTR_ENABLE**Type:** RW**Clock:** AHB Clock**Reset State:** 0x00000000**PCNOC_8_ABT_INTR_ENABLE**

Bits	Name	Description
0	INTRENABLE	Enable an interrupt generation on timeout event.

0x0002B020 PCNOC_8_ABT_SYND_VALID**Type:** R**Clock:** AHB Clock**Reset State:** 0x00000000**PCNOC_8_ABT_SYND_VALID**

Bits	Name	Description
0	SYNDVALID	Set when the information in other syndrome registers is valid.

0x0002B024 PCNOC_8_ABT_SYND_CLEAR**Type:** W**Clock:** AHB Clock**Reset State:** 0x00000000**PCNOC_8_ABT_SYND_CLEAR**

Bits	Name	Description
0	SYNDCLEAR	Clears the syndrome valid bit and syndrome information (ABT_SYND_ID, ABT_SYND_ADDR, ABT_SYND_HREADY).

0x0002B028 PCNOC_8_ABT_SYND_ID**Type:** R**Clock:** AHB Clock**Reset State:** 0x00000000**PCNOC_8_ABT_SYND_ID**

Bits	Name	Description
15:13	BID	Bus ID of the transaction that caused bus hang.

PCNOC_8_ABT_SYND_ID (cont.)

Bits	Name	Description
12:8	PID	Port ID of the transaction that caused bus hang.
7:0	MID	Master ID of the transaction that caused bus hang.

0x0002B02C PCNOC_8_ABT_SYND_ADDR0**Type:** R**Clock:** AHB Clock**Reset State:** 0x00000000**PCNOC_8_ABT_SYND_ADDR0**

Bits	Name	Description
31:0	SYNDADDR0	Lower 32-bit address field of the transaction that caused bus hang.

0x0002B030 PCNOC_8_ABT_SYND_ADDR1**Type:** R**Clock:** AHB Clock**Reset State:** 0x00000000**PCNOC_8_ABT_SYND_ADDR1**

Bits	Name	Description
31:0	SYNDADDR1	Upper 32-bit address field of the transaction that caused bus hang. Reserved for future use.

0x0002B034 PCNOC_8_ABT_SYND_HREADY**Type:** R**Clock:** AHB Clock**Reset State:** 0xFFFFFFFF**PCNOC_8_ABT_SYND_HREADY**

Bits	Name	Description
31:0	SYNDHREADY	Contains the status of Hready for each slave on the bus during the timeout event. The valid bit fields are [ABT_NUM_SLAVES-1:0]. Each valid bit field corresponds to a physical slave on the bus.

3.10 PCNOC_9_BUS_TIMEOUT

0x0002C004 PCNOC_9_ABT_INST_ID

Type: RW

Clock: AHB Clock

Reset State: 0x00000000

PCNOC_9_ABT_INST_ID

Bits	Name	Description
31:0	INSTID	SW configurable field to provide unique instance number for each of the AHB Timeout Slaves in the chip.

0x0002C008 PCNOC_9_ABT_NUM_SLAVES

Type: R

Clock: AHB Clock

Reset State: 0x00000003

PCNOC_9_ABT_NUM_SLAVES

Bits	Name	Description
5:0	NUMSLAVES	Specifies the number of slaves on the AHB bus excluding the Timeout Slave. Valid values are from 1 to 32.

0x0002C00C PCNOC_9_ABT_TIMER_LOADVAL

Type: RW

Clock: AHB Clock

Reset State: 0x000000FF

PCNOC_9_ABT_TIMER_LOADVAL

Bits	Name	Description
7:0	LOADVAL	Load the timer with this value, when the timer is in load mode, i.e, MODE field in ABT_TIMER_MODE is 0. Note: LOADVAL of 0 is not valid.

0x0002C010 PCNOC_9_ABT_TIMER_MODE

Type: RW

Clock: AHB Clock

Reset State: 0x00000000

PCNOC_9_AB_TTIMER_MODE

Bits	Name	Description
0	MODE	If this field is 0, the timer is in load mode and the LOADVAL field value in ABT_TIMER_LOADVAL is loaded into the timer. If this field is 1, the timer is in run mode.

0x0002C014 PCNOC_9_AB_TINTR_STATUS**Type:** R**Clock:** AHB Clock**Reset State:** 0x00000000**PCNOC_9_AB_TINTR_STATUS**

Bits	Name	Description
0	INTRSTATUS	Specifies the timeout interrupt status irrespective of interrupt enable.

0x0002C018 PCNOC_9_AB_TINTR_CLEAR**Type:** W**Clock:** AHB Clock**Reset State:** 0x00000000**PCNOC_9_AB_TINTR_CLEAR**

Bits	Name	Description
0	INTRCLEAR	Clears the pending interrupt.

0x0002C01C PCNOC_9_AB_TINTR_ENABLE**Type:** RW**Clock:** AHB Clock**Reset State:** 0x00000000**PCNOC_9_AB_TINTR_ENABLE**

Bits	Name	Description
0	INTRENABLE	Enable an interrupt generation on timeout event.

0x0002C020 PCNOC_9_AB_T_SYND_VALID

Type: R
Clock: AHB Clock
Reset State: 0x00000000

PCNOC_9_AB_T_SYND_VALID

Bits	Name	Description
0	SYNDVALID	Set when the information in other syndrome registers is valid.

0x0002C024 PCNOC_9_AB_T_SYND_CLEAR

Type: W
Clock: AHB Clock
Reset State: 0x00000000

PCNOC_9_AB_T_SYND_CLEAR

Bits	Name	Description
0	SYNDCLEAR	Clears the syndrome valid bit and syndrome information (ABT_SYND_ID, ABT_SYND_ADDR, ABT_SYND_HREADY).

0x0002C028 PCNOC_9_AB_T_SYND_ID

Type: R
Clock: AHB Clock
Reset State: 0x00000000

PCNOC_9_AB_T_SYND_ID

Bits	Name	Description
15:13	BID	Bus ID of the transaction that caused bus hang.
12:8	PID	Port ID of the transaction that caused bus hang.
7:0	MID	Master ID of the transaction that caused bus hang.

0x0002C02C PCNOC_9_AB_T_SYND_ADDR0

Type: R
Clock: AHB Clock
Reset State: 0x00000000

PCNOC_9_AB_T_SYND_ADDR0

Bits	Name	Description
31:0	SYNDADDR0	Lower 32-bit address field of the transaction that caused bus hang.

0x0002C030 PCNOC_9_AB_T_SYND_ADDR1**Type:** R**Clock:** AHB Clock**Reset State:** 0x00000000**PCNOC_9_AB_T_SYND_ADDR1**

Bits	Name	Description
31:0	SYNDADDR1	Upper 32-bit address field of the transaction that caused bus hang. Reserved for future use.

0x0002C034 PCNOC_9_AB_T_SYND_HREADY**Type:** R**Clock:** AHB Clock**Reset State:** 0xFFFFFFFF**PCNOC_9_AB_T_SYND_HREADY**

Bits	Name	Description
31:0	SYNDHREADY	Contains the status of Hready for each slave on the bus during the timeout event. The valid bit fields are [ABT_NUM_SLAVES-1:0]. Each valid bit field corresponds to a physical slave on the bus.

4 Clock Controller

4.1 GCC_GLOBAL_CLK_CTL_REG

0x01821000 GCC_GPLL0_MODE

Type: RW

Clock: CLK

Reset State: 0x00000000

GPLL0_MODE register configures and controls GPLL0.

GCC_GPLL0_MODE

Bits	Name	Description
21	PLL_VOTE_FSM_RESET	Resets PLL voting FSM 0:Deasserts Reset to PLL voting FSM 1:Resets PLL voting FSM
20	PLL_VOTE_FSM_ENA	Enable PLL voting FSM 0:Disabled 1:Enabled
19:14	PLL_BIAS_COUNT	Sets PLL bias count of PLL voting FSM. The count specifies X * 16 PLL reference clock cycles to wait for biasing the PLL during bring up.
13:8	PLL_LOCK_COUNT	Sets PLL lock time of PLL voting FSM. The count specifies the number of BIAS wait time interval to wait before determining PLL is locked. If lock count is Y, it will wait Y * PLL_BIAS_COUNT * 16 PLL reference clock cycles before declaring PLL is locked. If this value is set to 0, the lock_det output of the PLL is used instead
3	RESERVED	Reserved
2	PLL_RESET_N	This bitfield is used for reset of digital logic in PLL. This includes the MND counters and integer divider
1	PLL_BYPASSNL	Specify the PLL bypass mode 0:Bypass PLL 1:Use PLL

GCC_GPLL0_MODE (cont.)

Bits	Name	Description
0	PLL_OUTCTRL	Specify the PLL output control 0:Disable PLL output and save power 1:Activate PLL outputs

0x01821004 GCC_GPLL0_L_VAL**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

Following equation is used to calculate the GPLL0 output frequency

$$VCO = REF * [L + (M/N)]$$

Where R represents the pre-divider of 2 and P represents post-divider

The value of GPLL0 register is the given value of L for that equation.

GCC_GPLL0_L_VAL

Bits	Name	Description
7:0	PLL_L	This bitfield contains 8-bit L value. This specifies the PLL VCO feedback divide ratio.

0x01821008 GCC_GPLL0_M_VAL**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

Following equation is used to calculate the GPLL0 output frequency

$$VCO = REF * [L + (M/N)]$$

Where R represents the pre-divider of 2 and P represents post-divider

The value of GPLL0 register is the given value of M for that equation.

GCC_GPLL0_M_VAL

Bits	Name	Description
18:0	PLL_M	This bitfield contains 19-bit M value. This specifies the PLL M-value for frac-N accumulator.

0x0182100C GCC_GPLL0_N_VAL**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

Following equation is used to calculate the GPLL0 output frequency

$$VCO = REF * [L + (M/N)]$$

Where R represents the pre-divider of 2 and P represents post-divider

The value of GPLL0 register is the given value of N for that equation.

GCC_GPLL0_N_VAL

Bits	Name	Description
18:0	PLL_N	This bitfield contains 19-bit N value. This specifies the PLL N-value for frac-N accumulator.

0x01821010 GCC_GPLL0_USER_CTL**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

This register contains register bits for specifying GPLL0 user configuration settings.

GCC_GPLL0_USER_CTL

Bits	Name	Description
31:25	RESERVE_BITS31_25	
24	MN_EN	PLL fractional mode control. 0:Disable 1:Enable
23:21	RESERVE_BITS23_21	
20	VCO_SEL	VCO selection 1: Select low freq VCO (100-500 Mhz) 0: Select high-freq VCO (400-1400 Mhz)
19:13	RESERVE_BITS19_13	
12	PRE_DIV_RATIO	Clk_ref pre-divide ratio. 0: 1 1: 2
11:10	RESERVE_BITS11_10	

GCC_GPLL0_USER_CTL (cont.)

Bits	Name	Description
9:8	POST_DIV_RATIO	Output clock Post divide ration. 00: 1 01: 2 10: Reserved 11: 4
7	OUTPUT_INV	Output invert state. 0: Output invert disable 1: Output invert enable
6:5	RESERVED	Reserved
4	RESERVED	Reserved
3	PLLOUT_LV_EARLY	Enable or disable of pllout_lv_early output. 0: Disable 1: Enable
2	PLLOUT_LV_BIST	Enable or disable of pllout_lv_bist output. 0: Disable 1: Enable
1	PLLOUT_LV_AUX	Enable or disable of pllout_lv_aux output. 0: Disable 1: Enable
0	PLLOUT_LV_MAIN	Enable or disable of pllout_lv_main output. 0: Disable 1: Enable

0x01821014 GCC_GPLL0_CONFIG_CTL**Type:** RW**Clock:** CLK**Reset State:** 0x00031000

This register contains register bits for specifying GPLL0 configuration control settings.

GCC_GPLL0_CONFIG_CTL

Bits	Name	Description
31:30	RESERVE_BITS31_30	
29	FORCE_PFD_UP	Force PFD Up (force_PFD<1>). 0 Normal operation (default). 1 Force PFD UP -> 1

GCC_GPLL0_CONFIG_CTL (cont.)

Bits	Name	Description
28	FORCE_PFD_DOWN	Force PFD down (force_PFD<0>). 0 Normal operation (default). 1 Force PFD DN -> 1
27:26	NMOSC_FREQ_CTRL	Noise measurement oscillator frequency control. 00: Highest (2.8 GHz at TT / 1.05V / 40°C) (default) 01: 1/2 highest 10: 1/3 highest 11: 1/4 highest
25:24	PFD_DZSEL	PFD reset pulse width adjustment (PFD_dzsel<1:0>). 00 1x delay (default) 01: 2x delay 10: 3x delay 11: 4x delay
23	NMOSC_EN	Noise measurement oscillator control (nmosc_en) 0: Disable (default) 1: Enable
22	RESERVE_BIT22	
21:20	ICP_DIV	Charge pump bias mirror multiplication factor(icp_div<1:0>). 00: 1 (default) 01: 3/2 10: 7/4 11: 9/4
19:18	IREG_DIV	Regulator bias mirror multiplication factor (ireg_div<1:0>). 00: 1 (default) 01: 7/5 10: 9/5 11: 11/5
17:16	CUSEL	VCO desoupling cap unit size (cusel <1:0>). 00: 125 fF 01: 250 fF 10: 375 fF 11: 500 fF (default)
15	REF_MODE	Reference circuit mode (ref_mode). 0: Bandgap Mode: Seed currents source from bandgap circuit and Nwell refernce circuit active. (default) 1: Supply-independent mode: Seed currents sourced from supply-independent circuit and Nwell reference pulled to VDDA.
14	RESERVE_BIT14	

GCC_GPLL0_CONFIG_CTL (cont.)

Bits	Name	Description
13:12	CFG_LOCKDET	Lock detector phase error threshold (cfg_lockdet<1:0>). 00: 2pi/2 01: 2pi/15 (default) 10: 2pi/23 11: 2pi/31
11	FORCE_ISEED	Seed current control (force_iseed). 0: Turn off seed currents when clock is detected. (default) 1: Keep seed currents on at all times.
10:0	RESERVE_BITS10_0	

0x0182101C GCC_GPLL0_STATUS**Type:** R**Clock:** CLK**Reset State:** 0x00000000

GPLL0 Status register.

GCC_GPLL0_STATUS

Bits	Name	Description
17	PLL_ACTIVE_FLAG	pll_active_flag from PLL voting FSM. It indicates when FSM has enabled the PLL and PLL should be locked.
16	PLL_LOCK_DET	Lock Detect signal from PLL.
15:0	PLL_D	Status output straight from PLL. Bit 0 is clock detect status. Bit 15:1 is shorted to VSSD.

0x01820000 GCC_GPLL1_MODE**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

GPLL1_MODE register configures and controls GPLL1.

GCC_GPLL1_MODE

Bits	Name	Description
21	PLL_VOTE_FSM_RESET	Resets PLL voting FSM 0:Deasserts Reset to PLL voting FSM 1:Resets PLL voting FSM
20	PLL_VOTE_FSM_ENA	Enable PLL voting FSM 0:Disabled 1:Enabled
19:14	PLL_BIAS_COUNT	Sets PLL bias count of PLL voting FSM. The count specifies X * 16 PLL reference clock cycles to wait for biasing the PLL during bring up.
13:8	PLL_LOCK_COUNT	Sets PLL lock time of PLL voting FSM. The count specifies the number of BIAS wait time interval to wait before determining PLL is locked. If lock count is Y, it will wait Y * PLL_BIAS_COUNT * 16 PLL reference clock cycles before declaring PLL is locked. If this value is set to 0, the lock_det output of the PLL is used instead
3	RESERVED	Reserved
2	PLL_RESET_N	This bitfield is used for reset of digital logic in PLL. This includes the MND counters and integer divider
1	PLL_BYPASSNL	Specify the PLL bypass mode 0:Bypass PLL 1:Use PLL
0	PLL_OUTCTRL	Specify the PLL output control 0:Disable PLL output and save power 1:Activate PLL outputs

0x01820004 GCC_GPLL1_L_VAL**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

Following equation is used to calculate the GPLL1 output frequency

$$VCO = REF * [L + (M/N)]$$

Where R represents the pre-divider of 2 and P represents post-divider

The value of GPLL1 register is the given value of L for that equation.

GCC_GPLL1_L_VAL

Bits	Name	Description
7:0	PLL_L	This bitfield contains 8-bit L value. This specifies the PLL VCO feedback divide ratio.

0x01820008 GCC_GPLL1_M_VAL**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

Following equation is used to calculate the GPLL1 output frequency

$$VCO = REF * [L + (M/N)]$$

Where R represents the pre-divider of 2 and P represents post-divider

The value of GPLL1 register is the given value of M for that equation.

GCC_GPLL1_M_VAL

Bits	Name	Description
18:0	PLL_M	This bitfield contains 19-bit M value. This specifies the PLL M-value for frac-N accumulator.

0x0182000C GCC_GPLL1_N_VAL**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

Following equation is used to calculate the GPLL1 output frequency

$$VCO = REF * [L + (M/N)]$$

Where R represents the pre-divider of 2 and P represents post-divider

The value of GPLL1 register is the given value of N for that equation.

GCC_GPLL1_N_VAL

Bits	Name	Description
18:0	PLL_N	This bitfield contains 19-bit N value. This specifies the PLL N-value for frac-N accumulator.

0x01820010 GCC_GPLL1_USER_CTL**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

This register contains register bits for specifying GPLL1 user configuration settings.

GCC_GPLL1_USER_CTL

Bits	Name	Description
31:25	RESERVE_BITS31_25	
24	MN_EN	PLL fractional mode control. 0: Disable 1: Enable
23:21	RESERVE_BITS23_21	
20	VCO_SEL	VCO selection 1: Select low freq VCO (100-500 Mhz) 0: Select high-freq VCO (400-1400 Mhz)
19:13	RESERVE_BITS19_13	
12	PRE_DIV_RATIO	Clk_ref pre-divide ratio. 0: 1 1: 2
11:10	RESERVE_BITS11_10	
9:8	POST_DIV_RATIO	Output clock Post divide ration. 00: 1 01: 2 10: Reserved 11: 4
7	OUTPUT_INV	Output invert state. 0: Output invert disable 1: Output invert enable
6:5	RESERVE_BITS6_5	
4	RESERVED	Reserved
3	PLLOUT_LV_EARLY	Enable or disable of pllout_lv_early output. 0: Disable 1: Enable
2	PLLOUT_LV_BIST	Enable or disable of pllout_lv_bist output. 0: Disable 1: Enable
1	PLLOUT_LV_AUX	Enable or disable of pllout_lv_aux output. 0: Disable 1: Enable

GCC_GPLL1_USER_CTL (cont.)

Bits	Name	Description
0	PLLOUT_LV_MAIN	Enable or disable of pllout_lv_main output. 0: Disable 1: Enable

0x01820014 GCC_GPLL1_CONFIG_CTL**Type:** RW**Clock:** CLK**Reset State:** 0x00031000

This register contains register bits for specifying GPLL1 configuration control settings.

GCC_GPLL1_CONFIG_CTL

Bits	Name	Description
31:30	RESERVE_BITS31_30	
29	FORCE_PFD_UP	Force PFD Up (force_PFD<1>). 0 Normal operation (default). 1 Force PFD UP -> 1
28	FORCE_PFD_DOWN	Force PFD down (force_PFD<0>). 0 Normal operation (default). 1 Force PFD DN -> 1
27:26	NMOSC_FREQ_CTRL	Noise measurement oscillator frequency control. 00: Highest (2.8 GHz at TT / 1.05V / 40°C) (default) 01: 1/2 highest 10: 1/3 highest 11: 1/4 highest
25:24	PFD_DZSEL	PFD reset pulse width adjustment (PFD_dzsel<1:0>). 00 1x delay (default) 01: 2x delay 10: 3x delay 11: 4x delay
23	NMOSC_EN	Noise measurement oscillator control (nmosc_en) 0: Disable (default) 1: Enable
22	RESERVE_BIT22	
21:20	ICP_DIV	Charge pump bias mirror multiplication factor(icp_div<1:0>). 00: 1 (default) 01: 3/2 10: 7/4 11: 9/4

GCC_GPLL1_CONFIG_CTL (cont.)

Bits	Name	Description
19:18	IREG_DIV	Regulator bias mirror multiplication factor (ireg_div<1:0>). 00: 1 (default) 01: 7/5 10: 9/5 11: 11/5
17:16	CUSEL	VCO desoupling cap unit size (cusel <1:0>). 00: 125 fF 01: 250 fF 10: 375 fF 11: 500 fF (default)
15	REF_MODE	Reference circuit mode (ref_mode). 0: Bandgap Mode: Seed currents source from bandgap circuit and Nwell reference circuit active. (default) 1: Supply-independent mode: Seed currents sourced from supply-independent circuit and Nwell reference pulled to VDDA.
14	RESERVE_BIT14	
13:12	CFG_LOCKDET	Lock detector phase error threshold (cfg_lockdet<1:0>). 00: $2\pi/2$ 01: $2\pi/15$ (default) 10: $2\pi/23$ 11: $2\pi/31$
11	FORCE_ISEED	Seed current control (force_iseed). 0: Turn off seed currents when clock is detected. (default) 1: Keep seed currents on at all times.
10:0	RESERVE_BITS10_0	

0x0182001C GCC_GPLL1_STATUS**Type:** R**Clock:** CLK**Reset State:** 0x00000000

GPLL1 Status register.

GCC_GPLL1_STATUS

Bits	Name	Description
17	PLL_ACTIVE_FLAG	pll_active_flag from PLL voting FSM. It indicates when FSM has enabled the PLL and PLL should be locked.
16	PLL_LOCK_DET	Lock Detect signal from PLL.

GCC_GPLL1_STATUS (cont.)

Bits	Name	Description
15:0	PLL_D	Status output straight from PLL. Bit 0 is clock detect status. Bit 15:1 is shorted to VSSD.

0x0184A000 GCC_GPLL2_MODE**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

GPLL2_MODE register configures and controls GPLL2.

GCC_GPLL2_MODE

Bits	Name	Description
21	PLL_VOTE_FSM_RESET	Resets PLL voting FSM 0:Deasserts Reset to PLL voting FSM 1:Resets PLL voting FSM
20	PLL_VOTE_FSM_ENA	Enable PLL voting FSM 0:Disabled 1:Enabled
19:14	PLL_BIAS_COUNT	Sets PLL bias count of PLL voting FSM. The count specifies X * 16 PLL reference clock cycles to wait for biasing the PLL during bring up.
13:8	PLL_LOCK_COUNT	Sets PLL lock time of PLL voting FSM. The count specifies the number of BIAS wait time interval to wait before determining PLL is locked. If lock count is Y, it will wait Y * PLL_BIAS_COUNT * 16 PLL reference clock cycles before declaring PLL is locked. If this value is set to 0, the lock_det output of the PLL is used instead
3	RESERVED	Reserved
2	PLL_RESET_N	This bitfield is used for reset of digital logic in PLL. This includes the MND counters and integer divider
1	PLL_BYPASSNL	Specify the PLL bypass mode 0:Bypass PLL 1:Use PLL
0	PLL_OUTCTRL	Specify the PLL output control 0:Disable PLL output and save power 1:Activate PLL outputs

0x0184A004 GCC_GPLL2_L_VAL**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

Following equation is used to calculate the GPLL2 output frequency

$$VCO = REF * [L + (M/N)]$$

Where R represents the pre-divider of 2 and P represents post-divider

The value of GPLL2 register is the given value of L for that equation.

GCC_GPLL2_L_VAL

Bits	Name	Description
7:0	PLL_L	This bitfield contains 8-bit L value. This specifies the PLL VCO feedback divide ratio.

0x0184A008 GCC_GPLL2_M_VAL**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

Following equation is used to calculate the GPLL2 output frequency

$$VCO = REF * [L + (M/N)]$$

Where R represents the pre-divider of 2 and P represents post-divider

The value of GPLL2 register is the given value of M for that equation.

GCC_GPLL2_M_VAL

Bits	Name	Description
18:0	PLL_M	This bitfield contains 19-bit M value. This specifies the PLL M-value for frac-N accumulator.

0x0184A00C GCC_GPLL2_N_VAL**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

Following equation is used to calculate the GPLL2 output frequency

$$VCO = REF * [L + (M/N)]$$

Where R represents the pre-divider of 2 and P represents post-divider

The value of GPLL2 register is the given value of N for that equation.

GCC_GPLL2_N_VAL

Bits	Name	Description
18:0	PLL_N	This bitfield contains 19-bit N value. This specifies the PLL N-value for frac-N accumulator.

0x0184A010 GCC_GPLL2_USER_CTL

Type: RW

Clock: CLK

Reset State: 0x00000000

This register contains register bits for specifying GPLL2 user configuration settings.

GCC_GPLL2_USER_CTL

Bits	Name	Description
31:25	RESERVE_BITS31_25	
24	MN_EN	PLL fractional mode control. 0: Disable 1: Enable
23:21	RESERVE_BITS23_21	
20	VCO_SEL	VCO selection 1: Select low freq VCO (100-500 Mhz) 0: Select high-freq VCO (400-1400 Mhz)
19:13	RESERVE_BITS19_13	
12	PRE_DIV_RATIO	Clk_ref pre-divide ratio. 0: 1 1: 2
11:10	RESERVE_BITS11_10	
9:8	POST_DIV_RATIO	Output clock Post divide ration. 00: 1 01: 2 10: Reserved 11: 4
7	OUTPUT_INV	Output invert state. 0: Output invert disable 1: Output invert enable

GCC_GPLL2_USER_CTL (cont.)

Bits	Name	Description
6:5	RESERVE_BITS6_5	
4	RESERVED	Reserved
3	PLLOUT_LV_EARLY	Enable or disable of pllout_lv_early output. 0: Disable 1: Enable
2	PLLOUT_LV_BIST	Enable or disable of pllout_lv_bist output. 0: Disable 1: Enable
1	PLLOUT_LV_AUX	Enable or disable of pllout_lv_aux output. 0: Disable 1: Enable
0	PLLOUT_LV_MAIN	Enable or disable of pllout_lv_main output. 0: Disable 1: Enable

0x0184A014 GCC_GPLL2_CONFIG_CTL**Type:** RW**Clock:** CLK**Reset State:** 0x00031000

This register contains register bits for specifying GPLL2 configuration control settings.

GCC_GPLL2_CONFIG_CTL

Bits	Name	Description
31:30	RESERVE_BITS31_30	
29	FORCE_PFD_UP	Force PFD Up (force_PFD<1>). 0 Normal operation (default). 1 Force PFD UP -> 1
28	FORCE_PFD_DOWN	Force PFD down (force_PFD<0>). 0 Normal operation (default). 1 Force PFD DN -> 1
27:26	NMOSC_FREQ_CTRL	Noise measurement oscillator frequency control. 00: Highest (2.8 GHz at TT / 1.05V / 40°C) (default) 01: 1/2 highest 10: 1/3 highest 11: 1/4 highest

GCC_GPLL2_CONFIG_CTL (cont.)

Bits	Name	Description
25:24	PFD_DZSEL	PFD reset pulse width adjustment (PFD_dzsel<1:0>). 00 1x delay (default) 01: 2x delay 10: 3x delay 11: 4x delay
23	NMOSC_EN	Noise measurement oscillator control (nmosc_en) 0: Disable (default) 1: Enable
22	RESERVE_BIT22	
21:20	ICP_DIV	Charge pump bias mirror multiplication factor (icp_div<1:0>). 00: 1 (default) 01: 3/2 10: 7/4 11: 9/4
19:18	IREG_DIV	Regulator bias mirror multiplication factor (ireg_div<1:0>). 00: 1 (default) 01: 7/5 10: 9/5 11: 11/5
17:16	CUSEL	VCO desoupling cap unit size (cusel <1:0>). 00: 125 fF 01: 250 fF 10: 375 fF 11: 500 fF (default)
15	REF_MODE	Reference circuit mode (ref_mode). 0: Bandgap Mode: Seed currents source from bandgap circuit and Nwell reference circuit active. (default) 1: Supply-independent mode: Seed currents sourced from supply-independent circuit and Nwell reference pulled to VDDA.
14	RESERVE_BIT14	
13:12	CFG_LOCKDET	Lock detector phase error threshold (cfg_lockdet<1:0>). 00: 2pi/2 01: 2pi/15 (default) 10: 2pi/23 11: 2pi/31
11	FORCE_ISEED	Seed current control (force_iseed). 0: Turn off seed currents when clock is detected. (default) 1: Keep seed currents on at all times.
10:0	RESERVE_BITS10_0	

0x0184A01C GCC_GPLL2_STATUS

Type: R
Clock: CLK
Reset State: 0x00000000

GPLL2 Status register.

GCC_GPLL2_STATUS

Bits	Name	Description
17	PLL_ACTIVE_FLAG	pll_active_flag from PLL voting FSM. It indicates when FSM has enabled the PLL and PLL should be locked.
16	PLL_LOCK_DET	Lock Detect signal from PLL.
15:0	PLL_D	Status output straight from PLL. Bit 0 is clock detect status. Bit 15:1 is shorted to VSSD.

0x01823000 GCC_BIMC_PLL_MODE

Type: RW
Clock: CLK
Reset State: 0x00000000

BIMC_PLL_MODE register configures and controls BIMC_PLL.

GCC_BIMC_PLL_MODE

Bits	Name	Description
21	PLL_VOTE_FSM_RESET	Resets PLL voting FSM 0:Deasserts Reset to PLL voting FSM 1:Resets PLL voting FSM
20	PLL_VOTE_FSM_ENA	Enable PLL voting FSM 0:Disabled 1:Enabled
19:14	PLL_BIAS_COUNT	Sets PLL bias count of PLL voting FSM. The count specifies X * 16 PLL reference clock cycles to wait for biasing the PLL during bring up.
13:8	PLL_LOCK_COUNT	Sets PLL lock time of PLL voting FSM. The count specifies the number of BIAS wait time interval to wait before determining PLL is locked. If lock count is Y, it will wait Y * PLL_BIAS_COUNT * 16 PLL reference clock cycles before declaring PLL is locked. If this value is set to 0, the lock_det output of the PLL is used instead
3	RESERVED	Reserved
2	PLL_RESET_N	This bitfield is used for reset of digital logic in PLL. This includes the MND counters and integer divider

GCC_BIMC_PLL_MODE (cont.)

Bits	Name	Description
1	PLL_BYPASSNL	Specify the PLL bypass mode 0:Bypass PLL 1:Use PLL
0	PLL_OUTCTRL	Specify the PLL output control 0:Disable PLL output and save power 1:Activate PLL outputs

0x01823004 GCC_BIMC_PLL_L_VAL**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

Following equation is used to calculate the BIMC_PLL output frequency

$$VCO = REF * [L + (M/N)]$$

Where R represents the pre-divider of 2 and P represents post-divider

The value of BIMC_PLL register is the given value of L for that equation.

GCC_BIMC_PLL_L_VAL

Bits	Name	Description
7:0	PLL_L	This bitfield contains 8-bit L value. This specifies the PLL VCO feedback divide ratio.

0x01823008 GCC_BIMC_PLL_M_VAL**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

Following equation is used to calculate the BIMC_PLL output frequency

$$VCO = REF * [L + (M/N)]$$

Where R represents the pre-divider of 2 and P represents post-divider

The value of BIMC_PLL register is the given value of M for that equation.

GCC_BIMC_PLL_M_VAL

Bits	Name	Description
18:0	PLL_M	This bitfield contains 19-bit M value. This specifies the PLL M-value for frac-N accumulator.

0x0182300C GCC_BIMC_PLL_N_VAL**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

Following equation is used to calculate the BIMC_PLL output frequency

$$VCO = REF * [L + (M/N)]$$

Where R represents the pre-divider of 2 and P represents post-divider

The value of BIMC_PLL register is the given value of N for that equation.

GCC_BIMC_PLL_N_VAL

Bits	Name	Description
18:0	PLL_N	This bitfield contains 19-bit N value. This specifies the PLL N-value for frac-N accumulator.

0x01823010 GCC_BIMC_PLL_USER_CTL**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

This register contains register bits for specifying BIMC_PLL user configuration settings.

GCC_BIMC_PLL_USER_CTL

Bits	Name	Description
31:25	RESERVE_BITS31_25	
24	MN_EN	PLL fractional mode control. 0:Disable 1:Enable
23:21	RESERVE_BITS23_21	
20	VCO_SEL	VCO selection 1: Select low freq VCO (100-500 Mhz) 0: Select high-freq VCO (400-1400 Mhz)

GCC_BIMC_PLL_USER_CTL (cont.)

Bits	Name	Description
19:13	RESERVE_BITS19_13	
12	PRE_DIV_RATIO	Clk_ref pre-divide ratio. 0: 1 1: 2
11:10	RESERVE_BITS11_10	
9:8	POST_DIV_RATIO	Output clock Post divide ration. 00: 1 01: 2 10: Reserved 11: 4
7	OUTPUT_INV	Output invert state. 0: Output invert disable 1: Output invert enable
6:5	RESERVE_BITS6_5	
4	RESERVED	Reserved
3	PLLOUT_LV_EARLY	Enable or disable of pllout_lv_early output. 0: Disable 1: Enable
2	PLLOUT_LV_BIST	Enable or disable of pllout_lv_bist output. 0: Disable 1: Enable
1	PLLOUT_LV_AUX	Enable or disable of pllout_lv_aux output. 0: Disable 1: Enable
0	PLLOUT_LV_MAIN	Enable or disable of pllout_lv_main output. 0: Disable 1: Enable

0x01823014 GCC_BIMC_PLL_CONFIG_CTL**Type:** RW**Clock:** CLK**Reset State:** 0x00031000

This register contains register bits for specifying BIMC_PLL configuration control settings.

GCC_BIMC_PLL_CONFIG_CTL

Bits	Name	Description
31:30	RESERVE_BITS31_30	

GCC_BIMC_PLL_CONFIG_CTL (cont.)

Bits	Name	Description
29	FORCE_PFD_UP	Force PFD Up (force_PFD<1>). 0 Normal operation (default). 1 Force PFD UP -> 1
28	FORCE_PFD_DOWN	Force PFD down (force_PFD<0>). 0 Normal operation (default). 1 Force PFD DN -> 1
27:26	NMOSC_FREQ_CTRL	Noise measurement oscillator frequency control. 00: Highest (2.8 GHz at TT / 1.05V / 40°C) (default) 01: 1/2 highest 10: 1/3 highest 11: 1/4 highest
25:24	PFD_DZSEL	PFD reset pulse width adjustment (PFD_dzsel<1:0>). 00 1x delay (default) 01: 2x delay 10: 3x delay 11: 4x delay
23	NMOSC_EN	Noise measurement oscillator control (nmosc_en) 0: Disable (default) 1: Enable
22	RESERVE_BIT22	
21:20	ICP_DIV	Charge pump bias mirror multiplication factor(icp_div<1:0>). 00: 1 (default) 01: 3/2 10: 7/4 11: 9/4
19:18	IREG_DIV	Regulator bias mirror multiplication factor (ireg_div<1:0>). 00: 1 (default) 01: 7/5 10: 9/5 11: 11/5
17:16	CUSEL	VCO decoupling cap unit size (cusel <1:0>). 00: 125 fF 01: 250 fF 10: 375 fF 11: 500 fF (default)
15	REF_MODE	Reference circuit mode (ref_mode). 0: Bandgap Mode: Seed currents source from bandgap circuit and Nwell reference circuit active. (default) 1: Supply-independent mode: Seed currents sourced from supply-independent circuit and Nwell reference pulled to VDDA.
14	RESERVE_BIT14	

GCC_BIMC_PLL_CONFIG_CTL (cont.)

Bits	Name	Description
13:12	CFG_LOCKDET	Lock detector phase error threshold (cfg_lockdet<1:0>). 00: 2pi/2 01: 2pi/15 (default) 10: 2pi/23 11: 2pi/31
11	FORCE_ISEED	Seed current control (force_iseed). 0: Turn off seed currents when clock is detected. (default) 1: Keep seed currents on at all times.
10:0	RESERVE_BITS10_0	

0x0182301C GCC_BIMC_PLL_STATUS**Type:** R**Clock:** CLK**Reset State:** 0x00000000

BIMC_PLL Status register.

GCC_BIMC_PLL_STATUS

Bits	Name	Description
17	PLL_ACTIVE_FLAG	pll_active_flag from PLL voting FSM. It indicates when FSM has enabled the PLL and PLL should be locked.
16	PLL_LOCK_DET	Lock Detect signal from PLL.
15:0	PLL_D	Status output straight from PLL. Bit 0 is clock detect status. Bit 15:1 is shorted to VSSD.

0x01826004 GCC_SYSTEM_NOC_BFDCD_CMD_RCGR**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

SYSTEM NOC root clock generator command register. It allows for SW to trigger the configuration data

GCC_SYSTEM_NOC_BFDCD_CMD_RCGR

Bits	Name	Type	Description
31	ROOT_OFF	R	This bitfield provides indication on status of the RCG; Mainly for SW to determine if the generator is actively producing clock 1: Clock Off 0: Clock On; Generator is actively producing the clock
4	DIRTY_CFG_RCGR	R	Indicates change in CFG_RCGR register configuration
1	ROOT_EN	RW	Enable Clock Generator output. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
0	UPDATE	RW	Update to the new programmed configuration. It should be cleared upon switching to new configuration 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x01826008 GCC_SYSTEM_NOC_BFDCD_CFG_RCGR**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

SYSTEM NOC root clock generator configuration register. It allows for SW to configure Root Clock Generation Macro

GCC_SYSTEM_NOC_BFDCD_CFG_RCGR

Bits	Name	Description
10:8	SRC_SEL	This bitfield selects clock source that will be used to source the RCG output 0x0: SRC0 (CXO.) 0x1: SRC1 (GPLL0_MAIN.) 0x2: SRC2 (BIMC_PLL_AUX.) 0x3: SRC3 (Unused(Grounded).) 0x4: SRC4 (Unused(Grounded).) 0x5: SRC5 (Unused(Grounded).) 0x6: SRC6 (Unused(Grounded).) 0x7: SRC7 (Reserved)

GCC_SYSTEM_NOC_BFDCD_CFG_RCGR (cont.)

Bits	Name	Description
4:0	SRC_DIV	This field defines the integer and half integer portion of the divider setting of the RCG. The value specifies is $2N-1$ where N is the divide ratio. For example, if the half integer divide ration is 2.5, the src_div should be set at 00100. 0x0: BYPASS (Bypsss.) 0x1: DIV1 (Divide by 1.) 0x2: DIV1_5 (Divide by 1.5.) 0x3: DIV2 (Divide by 2.) 0x4: DIV2_5 (Divide by 2.5.) 0x5: DIV3 (Divide by 3.) 0x6: DIV3_5 (Divide by 3.5.) 0x7: DIV4 (Divide by 4.) 0x8: DIV4_5 (Divide by 4.5.) 0x9: DIV5 (Divide by 5.) 0xA: DIV5_5 (Divide by 5.5.) 0xB: DIV6 (Divide by 6.) 0xC: DIV6_5 (Divide by 6.5.) 0xD: DIV7 (Divide by 7.) 0xE: DIV7_5 (Divide by 7.5.) 0xF: DIV8 (Divide by 8.) 0x10: DIV8_5 (Divide by 8.5.) 0x11: DIV9 (Divide by 9.) 0x12: DIV9_5 (Divide by 9.5.) 0x13: DIV10 (Divide by 10.) 0x14: DIV10_5 (Divide by 10.5.) 0x15: DIV11 (Divide by 11.) 0x16: DIV11_5 (Divide by 11.5.) 0x17: DIV12 (Divide by 12.) 0x18: DIV12_5 (Divide by 12.5.) 0x19: DIV13 (Divide by 13.) 0x1A: DIV13_5 (Divide by 13.5.) 0x1B: DIV14 (Divide by 14.) 0x1C: DIV14_5 (Divide by 14.5.) 0x1D: DIV15 (Divide by 15.) 0x1E: DIV15_5 (Divide by 15.5.) 0x1F: DIV16 (Divide by 16.)

0x01827000 GCC_PCNOB_BFDCD_CMD_RCGR**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

PC NOC root clock generator command register. It allows for SW to trigger the configuration data

GCC_PCNOC_BFDCD_CMD_RCGR

Bits	Name	Type	Description
31	ROOT_OFF	R	This bitfield provides indication on status of the RCG; Mainly for SW to determine if the generator is actively producing clock 1: Clock Off 0: Clock On; Generator is actively producing the clock
4	DIRTY_CFG_RCGR	R	Indicates change in CFG_RCGR register configuration
1	ROOT_EN	RW	Enable Clock Generator output. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
0	UPDATE	RW	Update to the new programmed configuration. It should be cleared upon switching to new configuration 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x01827004 GCC_PCNOC_BFDCD_CFG_RCGR

Type: RW

Clock: CLK

Reset State: 0x00000000

PC NOC root clock generator configuration register. It allows for SW to configure Root Clock Generation Macro

GCC_PCNOC_BFDCD_CFG_RCGR

Bits	Name	Description
10:8	SRC_SEL	This bitfield selects clock source that will be used to source the RCG output 0x0: SRC0 (CXO.) 0x1: SRC1 (GPLLO_MAIN.) 0x2: SRC2 (BIMC_PLL_AUX.) 0x3: SRC3 (Unused(Grounded).) 0x4: SRC4 (Unused(Grounded).) 0x5: SRC5 (Unused(Grounded).) 0x6: SRC6 (Unused(Grounded).) 0x7: Reserved

GCC_PCNOG_BFDCD_CFG_RCGR (cont.)

Bits	Name	Description
4:0	SRC_DIV	This field defines the integer and half integer portion of the divider setting of the RCG. The value specifies is $2N-1$ where N is the divide ratio. For example, if the half integer divide ration is 2.5, the src_div should be set at 00100. 0x0: BYPASS (Bypsss.) 0x1: DIV1 (Divide by 1.) 0x2: DIV1_5 (Divide by 1.5.) 0x3: DIV2 (Divide by 2.) 0x4: DIV2_5 (Divide by 2.5.) 0x5: DIV3 (Divide by 3.) 0x6: DIV3_5 (Divide by 3.5.) 0x7: DIV4 (Divide by 4.) 0x8: DIV4_5 (Divide by 4.5.) 0x9: DIV5 (Divide by 5.) 0xA: DIV5_5 (Divide by 5.5.) 0xB: DIV6 (Divide by 6.) 0xC: DIV6_5 (Divide by 6.5.) 0xD: DIV7 (Divide by 7.) 0xE: DIV7_5 (Divide by 7.5.) 0xF: DIV8 (Divide by 8.) 0x10: DIV8_5 (Divide by 8.5.) 0x11: DIV9 (Divide by 9.) 0x12: DIV9_5 (Divide by 9.5.) 0x13: DIV10 (Divide by 10.) 0x14: DIV10_5 (Divide by 10.5.) 0x15: DIV11 (Divide by 11.) 0x16: DIV11_5 (Divide by 11.5.) 0x17: DIV12 (Divide by 12.) 0x18: DIV12_5 (Divide by 12.5.) 0x19: DIV13 (Divide by 13.) 0x1A: DIV13_5 (Divide by 13.5.) 0x1B: DIV14 (Divide by 14.) 0x1C: DIV14_5 (Divide by 14.5.) 0x1D: DIV15 (Divide by 15.) 0x1E: DIV15_5 (Divide by 15.5.) 0x1F: DIV16 (Divide by 16.)

0x01826020 GCC_SYS_NOC_AXI_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x00000001

System NOC AXI clock branch control register. It allows SW control of CBC soft macro

GCC_SYS_NOC_AXI_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	This bitfield indicated if the clock branch is halted 0:Clock On 1:Clock Off
0	CLK_ENABLE	RW	This bitfield indicated if the clock branch is enabled. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x01826024 GCC_SYS_NOC_QDSS_STM_AXI_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

System NOC QDSS STM AXI clock branch control register. It allows SW control of CBC soft macro

GCC_SYS_NOC_QDSS_STM_AXI_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	This bitfield indicated if the clock branch is halted 0:Clock On 1:Clock Off
0	CLK_ENABLE	RW	This bitfield indicated if the clock branch is enabled. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x01826028 GCC_SYS_NOC_APSS_AHB_CBCR**Type:** R**Clock:** CLK**Reset State:** 0x00000000

System NOC APSS AHB clock branch control register. It allows SW control of CBC soft macro

GCC_SYS_NOC_APSS_AHB_CBCR

Bits	Name	Description
31	CLK_OFF	This bitfield indicated if the clock branch is halted 0:Clock On 1:Clock Off

0x0182602C GCC_SNOC_PCNOC_AHB_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x00000001

System NOC Periph NOC AHB clock branch control register. It allows SW control of CBC soft macro

GCC_SNOC_PCNOC_AHB_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	This bitfield indicated if the clock branch is halted 0:Clock On 1:Clock Off
0	CLK_ENABLE	RW	This bitfield indicated if the clock branch is enabled. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x01827018 GCC_PCNOC_BCR**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

CONFIG NOC block control register

GCC_PCNOC_BCR

Bits	Name	Description
0	BLK_ARES	This bitfield controls asynchronous software reset for the block. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x0182701C GCC_PCNOC_AHB_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x00000001

Config NOC AHB clock branch control register. It allows SW control of CBC soft macro. Note that during mission mode, the CBC control by this register should

not be disabled. Therefore, this register should not be touched.

GCC_PCNOC_AHB_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	This bitfield indicated if the clock branch is halted 0:Clock On 1:Clock Off
0	CLK_ENABLE	RW	This bitfield indicated if the clock branch is enabled. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x01827020 GCC_PCNOC_DDR_CFG_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

Config NOC DDR config clock branch control register. It allows SW control of CBC soft macro

GCC_PCNOC_DDR_CFG_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	This bitfield indicated if the clock branch is halted 0:Clock On 1:Clock Off
0	CLK_ENABLE	RW	This bitfield indicated if the clock branch is enabled. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x01827024 GCC_PCNOC_RPM_AHB_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x00000001

Config NOC RPM AHB clock branch control register. It allows SW control of CBC soft macro

GCC_PCNOC_RPM_AHB_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	This bitfield indicated if the clock branch is halted 0:Clock On 1:Clock Off

GCC_PCNOG_RPM_AHB_CBCR (cont.)

Bits	Name	Type	Description
0	CLK_ENABLE	RW	This bitfield indicated if the clock branch is enabled. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x0180E000 GCC_IMEM_BCR**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

IMEM block control register

GCC_IMEM_BCR

Bits	Name	Description
0	BLK_ARES	This bitfield controls asynchronous software reset for the block. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x0180E004 GCC_IMEM_AXI_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x1000CFF0

IMEM AXI clock branch control register. It allows SW control of CBC soft macro

GCC_IMEM_AXI_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	This bitfield indicated if the clock branch is halted 0:Clock On 1:Clock Off
30:28	NOC_HANDSHAKE_FSM_STATUS	R	This bitfield is used to check the FSM state
15	NOC_HANDSHAKE_FSM_ENABLE	RW	This bitfield enables the FSM to use so that slaveready and socketcon handshake is used prior to affecting the enable of the clock branch cell. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

GCC_IMEM_AXI_CBCR (cont.)

Bits	Name	Type	Description
14	FORCE_MEM_CORE_ON	RW	This bit force the core on signal to remain active during halt state of the clk. 0x1: FORCE_ENABLE (Force-on Mode.) 0x0: FORCE_DISABLE (Normal Mode force disabled.)
13	FORCE_MEM_PERIPH_ON	RW	This bit force the periph_on signal to remain active during halt state of the clk. 0x1: FORCE_ENABLE (Force-on Mode.) 0x0: FORCE_DISABLE (Normal Mode force disabled.)
12	FORCE_MEM_PERIPH_OFF	RW	This bit force the periph_off signal to remain active during halt state of the clk. 0x1: FORCE_ENABLE (Force-on Mode.) 0x0: FORCE_DISABLE (Normal Mode force disabled.)
11:8	WAKEUP	RW	This bitfield specifies the number of clock cycles it takes for the clock to turn back on. 0x0: CLOCK0 (Zero cycle) 0x1: CLOCK1 (One cycle) 0x2: CLOCK2 (Two cycles) 0x3: CLOCK3 (Three cycles) 0x4: CLOCK4 (Four cycles) 0x5: CLOCK5 (Five cycles) 0x6: CLOCK6 (Six cycles) 0x7: CLOCK7 (Seven cycles) 0x8: CLOCK8 (Eight cycles) 0x9: CLOCK9 (Nine cycles) 0xA: CLOCK10 (Ten cycles) 0xB: CLOCK11 (Eleven cycles) 0xC: CLOCK12 (Twelve cycles) 0xD: CLOCK13 (Thirteen cycles) 0xE: CLOCK14 (Fourteen cycles) 0xF: CLOCK15 (Fifteen cycles)

GCC_IMEM_AXI_CBCR (cont.)

Bits	Name	Type	Description
7:4	SLEEP	RW	This bitfield specifies the number of clock cycles till complete sleep where headswitch of the corresponding memories will be opened. 0x0: CLOCK0 (Zero cycle) 0x1: CLOCK1 (One cycle) 0x2: CLOCK2 (Two cycles) 0x3: CLOCK3 (Three cycles) 0x4: CLOCK4 (Four cycles) 0x5: CLOCK5 (Five cycles) 0x6: CLOCK6 (Six cycles) 0x7: CLOCK7 (Seven cycles) 0x8: CLOCK8 (Eight cycles) 0x9: CLOCK9 (Nine cycles) 0xA: CLOCK10 (Ten cycles) 0xB: CLOCK11 (Eleven cycles) 0xC: CLOCK12 (Twelve cycles) 0xD: CLOCK13 (Thirteen cycles) 0xE: CLOCK14 (Fourteen cycles) 0xF: CLOCK15 (Fifteen cycles)

0x0180E008 GCC_IMEM_CFG_AHB_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x80008000

IMEM config AHB clock branch control register. It allows SW control of CBC soft macro

GCC_IMEM_CFG_AHB_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	This bitfield indicated if the clock branch is halted 0:Clock On 1:Clock Off
30:28	NOC_HANDSHAKE_FSM_STATUS	R	This bitfield is used to check the FSM state
15	NOC_HANDSHAKE_FSM_ENABLE	RW	This bitfield enables the FSM to use so that slaverready and socketcon hankshake is used prior to affecting the enable of the clock branch cell. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
0	CLK_ENABLE	RW	This bitfield indicated if the clock branch is enabled. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x0184B000 GCC_MMSS_BCR**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

MMSS block control register

GCC_MMSS_BCR

Bits	Name	Description
1	GMEM_AXI_ARES	This bitfield controls asynchronous software reset for the gmem axi clock alone. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
0	BLK_ARES	This bitfield controls asynchronous software reset for the block. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x0185903C GCC_GMEM_SYS_NOC_AXI_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x80008000

GMEM System NOC AXI clock branch control register. It allows SW control of CBC soft macro

GCC_GMEM_SYS_NOC_AXI_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	This bitfield indicated if the clock branch is halted 0:Clock On 1:Clock Off
30:28	NOC_HANDSHAKE_FSM_STATUS	R	This bitfield is used to check the FSM state
15	NOC_HANDSHAKE_FSM_ENABLE	RW	This bitfield enables the FSM to use so that slaveready and socketcon hankshake is used prior to affecting the enable of the clock branch cell. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x01831024 GCC_BIMC_GFX_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x80004FF0

MMSS GFX clock branch control register. It allows SW control of CBC soft macro

GCC_BIMC_GFX_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	This bitfield indicated if the clock branch is halted 0:Clock On 1:Clock Off
14	FORCE_MEM_CORE_ON	RW	This bit force the core on signal to remain active during halt state of the clk. 0x1: FORCE_ENABLE (Force-on Mode.) 0x0: FORCE_DISABLE (Normal Mode force disabled.)
13	FORCE_MEM_PERIPH_ON	RW	This bit force the periph_on signal to remain active during halt state of the clk. 0x1: FORCE_ENABLE (Force-on Mode.) 0x0: FORCE_DISABLE (Normal Mode force disabled.)
12	FORCE_MEM_PERIPH_OFF	RW	This bit force the periph_off signal to remain active during halt state of the clk. 0x1: FORCE_ENABLE (Force-on Mode.) 0x0: FORCE_DISABLE (Normal Mode force disabled.)
11:8	WAKEUP	RW	This bitfield specifies the number of clock cycles it takes for the clock to turn back on. 0x0: CLOCK0 (Zero cycle) 0x1: CLOCK1 (One cycle) 0x2: CLOCK2 (Two cycles) 0x3: CLOCK3 (Three cycles) 0x4: CLOCK4 (Four cycles) 0x5: CLOCK5 (Five cycles) 0x6: CLOCK6 (Six cycles) 0x7: CLOCK7 (Seven cycles) 0x8: CLOCK8 (Eight cycles) 0x9: CLOCK9 (Nine cycles) 0xA: CLOCK10 (Ten cycles) 0xB: CLOCK11 (Eleven cycles) 0xC: CLOCK12 (Twelve cycles) 0xD: CLOCK13 (Thirteen cycles) 0xE: CLOCK14 (Fourteen cycles) 0xF: CLOCK15 (Fifteen cycles)

GCC_BIMC_GFX_CBCR (cont.)

Bits	Name	Type	Description
7:4	SLEEP	RW	This bitfield specifies the number of clock cycles till complete sleep where headswitch of the corresponding memories will be opened. 0x0: CLOCK0 (Zero cycle) 0x1: CLOCK1 (One cycle) 0x2: CLOCK2 (Two cycles) 0x3: CLOCK3 (Three cycles) 0x4: CLOCK4 (Four cycles) 0x5: CLOCK5 (Five cycles) 0x6: CLOCK6 (Six cycles) 0x7: CLOCK7 (Seven cycles) 0x8: CLOCK8 (Eight cycles) 0x9: CLOCK9 (Nine cycles) 0xA: CLOCK10 (Ten cycles) 0xB: CLOCK11 (Eleven cycles) 0xC: CLOCK12 (Twelve cycles) 0xD: CLOCK13 (Thirteen cycles) 0xE: CLOCK14 (Fourteen cycles) 0xF: CLOCK15 (Fifteen cycles)
0	CLK_ENABLE	RW	This bitfield indicated if the clock branch is enabled. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x01849000 GCC_MSS_CFG_AHB_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x80008000

MSS config AHB clock branch control register. It allows SW control of CBC soft macro

GCC_MSS_CFG_AHB_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	This bitfield indicated if the clock branch is halted 0:Clock On 1:Clock Off
30:28	NOC_HANDSHAKE_FSM_STATUS	R	This bitfield is used to check the FSM state
15	NOC_HANDSHAKE_FSM_ENABLE	RW	This bitfield enables the FSM to use so that slaveready and socketcon hankshake is used prior to affecting the enable of the clock branch cell. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

GCC_MSS_CFG_AHB_CBCR (cont.)

Bits	Name	Type	Description
0	CLK_ENABLE	RW	This bitfield indicated if the clock branch is enabled. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x01849004 GCC_MSS_Q6_BIMC_AXI_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

MSS Q6 BIMC AXI clock branch control register. It allows SW control of CBC soft macro

GCC_MSS_Q6_BIMC_AXI_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	This bitfield indicated if the clock branch is halted 0:Clock On 1:Clock Off
1	HW_CTL	RW	This bit is used to enable hardware dynamic gating of the clock branch. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
0	CLK_ENABLE	RW	0x1: ENABLE 0x0: DISABLE

0x01829000 GCC_QDSS_BCR**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

QDSS block control register

GCC_QDSS_BCR

Bits	Name	Description
0	BLK_ARES	This bitfield controls asynchronous software reset for the block. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x01829004 GCC_QDSS_DAP_AHB_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

QDSS DAP AHB clock branch control register. It allows SW control of CBC soft macro

GCC_QDSS_DAP_AHB_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	This bitfield indicated if the clock branch is halted 0:Clock On 1:Clock Off
0	CLK_ENABLE	RW	This bitfield indicated if the clock branch is enabled. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x01829008 GCC_QDSS_CFG_AHB_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x80008000

QDSS config AHB clock branch control register. It allows SW control of CBC soft macro

GCC_QDSS_CFG_AHB_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	This bitfield indicated if the clock branch is halted 0:Clock On 1:Clock Off
30:28	NOC_HANDSHAKE_FSM_STATUS	R	This bitfield is used to check the FSM state
15	NOC_HANDSHAKE_FSM_ENABLE	RW	This bitfield enables the FSM to use so that slaveready and socketcon handshake is used prior to affecting the enable of the clock branch cell. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
0	CLK_ENABLE	RW	This bitfield indicated if the clock branch is enabled. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x01829028 GCC_QDSS_ETR_USB_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

QDSS ETR USB clock branch control register. It allows SW control of CBC soft macro

GCC_QDSS_ETR_USB_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	This bitfield indicated if the clock branch is halted 0:Clock On 1:Clock Off
0	CLK_ENABLE	RW	This bitfield indicated if the clock branch is enabled. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x0182902C GCC_QDSS_STM_CMD_RCGR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

QDSS STM root clock generator command register. It allows for SW to trigger the configuration data

GCC_QDSS_STM_CMD_RCGR

Bits	Name	Type	Description
31	ROOT_OFF	R	This bitfield provides indication on status of the RCG; Mainly for SW to determine if the generator is actively producing clock 1: Clock Off 0: Clock On; Generator is actively producing the clock
4	DIRTY_CFG_RCGR	R	Indicates change in CFG_RCGR register configuration
1	ROOT_EN	RW	Enable Clock Generator output. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
0	UPDATE	RW	Update to the new programmed configuration. It should be cleared upon switching to new configuration 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x01829030 GCC_QDSS_STM_CFG_RCGR**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

QDSS STM root clock generator configuration register. It allows for SW to configure Root Clock Generation Macro

GCC_QDSS_STM_CFG_RCGR

Bits	Name	Description
10:8	SRC_SEL	This bitfield selects clock source that will be used to source the RCG output 0x0: SRC0 (CXO.) 0x1: SRC1 (GPLL0_MAIN.) 0x2: SRC2 (Unused(Grounded).) 0x3: SRC3 (Unused(Grounded).) 0x4: SRC4 (Unused(Grounded).) 0x5: SRC5 (Unused(Grounded).) 0x6: SRC6 (Unused(Grounded).) 0x7: Reserved

GCC_QDSS_STM_CFG_RCGR (cont.)

Bits	Name	Description
4:0	SRC_DIV	This field defines the integer and half integer portion of the divider setting of the RCG. The value specifies is $2N-1$ where N is the divide ratio. For example, if the half integer divide ration is 2.5, the src_div should be set at 00100. 0x0: BYPASS (Bypsss.) 0x1: DIV1 (Divide by 1.) 0x2: DIV1_5 (Divide by 1.5.) 0x3: DIV2 (Divide by 2.) 0x4: DIV2_5 (Divide by 2.5.) 0x5: DIV3 (Divide by 3.) 0x6: DIV3_5 (Divide by 3.5.) 0x7: DIV4 (Divide by 4.) 0x8: DIV4_5 (Divide by 4.5.) 0x9: DIV5 (Divide by 5.) 0xA: DIV5_5 (Divide by 5.5.) 0xB: DIV6 (Divide by 6.) 0xC: DIV6_5 (Divide by 6.5.) 0xD: DIV7 (Divide by 7.) 0xE: DIV7_5 (Divide by 7.5.) 0xF: DIV8 (Divide by 8.) 0x10: DIV8_5 (Divide by 8.5.) 0x11: DIV9 (Divide by 9.) 0x12: DIV9_5 (Divide by 9.5.) 0x13: DIV10 (Divide by 10.) 0x14: DIV10_5 (Divide by 10.5.) 0x15: DIV11 (Divide by 11.) 0x16: DIV11_5 (Divide by 11.5.) 0x17: DIV12 (Divide by 12.) 0x18: DIV12_5 (Divide by 12.5.) 0x19: DIV13 (Divide by 13.) 0x1A: DIV13_5 (Divide by 13.5.) 0x1B: DIV14 (Divide by 14.) 0x1C: DIV14_5 (Divide by 14.5.) 0x1D: DIV15 (Divide by 15.) 0x1E: DIV15_5 (Divide by 15.5.) 0x1F: DIV16 (Divide by 16.)

0x01829044 GCC_QDSS_STM_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x80008000

QDSS STM clock branch control register. It allows SW control of CBC soft macro

GCC_QDSS_STM_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	This bitfield indicated if the clock branch is halted 0:Clock On 1:Clock Off
30:28	NOC_HANDSHAKE_FSM_STATUS	R	This bitfield is used to check the FSM state
15	NOC_HANDSHAKE_FSM_ENABLE	RW	This bitfield enables the FSM to use so that slaveready and socketcon hankshake is used prior to affecting the enable of the clock branch cell. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
0	CLK_ENABLE	RW	This bitfield indicated if the clock branch is enabled. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x01829048 GCC_QDSS_TRACECLKIN_CMD_RCGR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

QDSS TRACECLKIN root clock generator command register. It allows for SW to trigger the configuration data

GCC_QDSS_TRACECLKIN_CMD_RCGR

Bits	Name	Type	Description
31	ROOT_OFF	R	This bitfield provides indication on status of the RCG; Mainly for SW to determine if the generator is actively producing clock 1: Clock Off 0: Clock On; Generator is actively producing the clock
4	DIRTY_CFG_RCGR	R	Indicates change in CFG_RCGR register configuration
1	ROOT_EN	RW	Enable Clock Generator output. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
0	UPDATE	RW	Update to the new programmed configuration. It should be cleared upon switching to new configuration 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x0182904C GCC_QDSS_TRACECLKIN_CFG_RCGR**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

QDSS TRACECLKIN root clock generator configuration register. It allows for SW to configure Root Clock Generation Macro

GCC_QDSS_TRACECLKIN_CFG_RCGR

Bits	Name	Description
10:8	SRC_SEL	This bitfield selects clock source that will be used to source the RCG output 0x0: SRC0 (CXO.) 0x1: SRC1 (GPLL1_MAIN.) 0x2: SRC2 (GPLL2_AUX.) 0x3: SRC3 (GPLL0_AUX.) 0x4: SRC4 (Unused(Grounded).) 0x5: SRC5 (Unused(Grounded).) 0x6: SRC6 (Unused(Grounded).) 0x7: Reserved

GCC_QDSS_TRACECLKIN_CFG_RCGR (cont.)

Bits	Name	Description
4:0	SRC_DIV	This field defines the integer and half integer portion of the divider setting of the RCG. The value specifies is $2N-1$ where N is the divide ratio. For example, if the half integer divide ration is 2.5, the src_div should be set at 00100. 0x0: BYPASS (Bypsss.) 0x1: DIV1 (Divide by 1.) 0x2: DIV1_5 (Divide by 1.5.) 0x3: DIV2 (Divide by 2.) 0x4: DIV2_5 (Divide by 2.5.) 0x5: DIV3 (Divide by 3.) 0x6: DIV3_5 (Divide by 3.5.) 0x7: DIV4 (Divide by 4.) 0x8: DIV4_5 (Divide by 4.5.) 0x9: DIV5 (Divide by 5.) 0xA: DIV5_5 (Divide by 5.5.) 0xB: DIV6 (Divide by 6.) 0xC: DIV6_5 (Divide by 6.5.) 0xD: DIV7 (Divide by 7.) 0xE: DIV7_5 (Divide by 7.5.) 0xF: DIV8 (Divide by 8.) 0x10: DIV8_5 (Divide by 8.5.) 0x11: DIV9 (Divide by 9.) 0x12: DIV9_5 (Divide by 9.5.) 0x13: DIV10 (Divide by 10.) 0x14: DIV10_5 (Divide by 10.5.) 0x15: DIV11 (Divide by 11.) 0x16: DIV11_5 (Divide by 11.5.) 0x17: DIV12 (Divide by 12.) 0x18: DIV12_5 (Divide by 12.5.) 0x19: DIV13 (Divide by 13.) 0x1A: DIV13_5 (Divide by 13.5.) 0x1B: DIV14 (Divide by 14.) 0x1C: DIV14_5 (Divide by 14.5.) 0x1D: DIV15 (Divide by 15.) 0x1E: DIV15_5 (Divide by 15.5.) 0x1F: DIV16 (Divide by 16.)

0x01829060 GCC_QDSS_TRACECLKIN_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

QDSS TRACECLKIN clock branch control register. It allows SW control of CBC soft macro

GCC_QDSS_TRACECLKIN_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	This bitfield indicated if the clock branch is halted 0:Clock On 1:Clock Off
0	CLK_ENABLE	RW	This bitfield indicated if the clock branch is enabled. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x01829064 GCC_QDSS_TSCTR_CMD_RCGR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

QDSS TSCTR root clock generator command register. It allows for SW to trigger the configuration data

GCC_QDSS_TSCTR_CMD_RCGR

Bits	Name	Type	Description
31	ROOT_OFF	R	This bitfield provides indication on status of the RCG; Mainly for SW to determine if the generator is actively producing clock 1: Clock Off 0: Clock On; Generator is actively producing the clock
4	DIRTY_CFG_RCGR	R	Indicates change in CFG_RCGR register configuration
1	ROOT_EN	RW	Enable Clock Generator output. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
0	UPDATE	RW	Update to the new programmed configuration. It should be cleared upon switching to new configuration 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x01829068 GCC_QDSS_TSCTR_CFG_RCGR**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

QDSS TSCTR root clock generator configuration register. It allows for SW to configure Root Clock Generation Macro

GCC_QDSS_TSCTR_CFG_RCGR

Bits	Name	Description
10:8	SRC_SEL	This bitfield selects clock source that will be used to source the RCG output 0x0: SRC0 (CXO.) 0x1: SRC1 (GPLL1_MAIN.) 0x2: SRC2 (GPLL2_AUX.) 0x3: SRC3 (GPLL0_AUX.) 0x4: SRC4 (Unused(Grounded).) 0x5: SRC5 (Unused(Grounded).) 0x6: SRC6 (Unused(Grounded).) 0x7: Reserved

GCC_QDSS_TSCTR_CFG_RCGR (cont.)

Bits	Name	Description
4:0	SRC_DIV	This field defines the integer and half integer portion of the divider setting of the RCG. The value specifies is $2N-1$ where N is the divide ratio. For example, if the half integer divide ration is 2.5, the src_div should be set at 00100. 0x0: BYPASS (Bypsss.) 0x1: DIV1 (Divide by 1.) 0x2: DIV1_5 (Divide by 1.5.) 0x3: DIV2 (Divide by 2.) 0x4: DIV2_5 (Divide by 2.5.) 0x5: DIV3 (Divide by 3.) 0x6: DIV3_5 (Divide by 3.5.) 0x7: DIV4 (Divide by 4.) 0x8: DIV4_5 (Divide by 4.5.) 0x9: DIV5 (Divide by 5.) 0xA: DIV5_5 (Divide by 5.5.) 0xB: DIV6 (Divide by 6.) 0xC: DIV6_5 (Divide by 6.5.) 0xD: DIV7 (Divide by 7.) 0xE: DIV7_5 (Divide by 7.5.) 0xF: DIV8 (Divide by 8.) 0x10: DIV8_5 (Divide by 8.5.) 0x11: DIV9 (Divide by 9.) 0x12: DIV9_5 (Divide by 9.5.) 0x13: DIV10 (Divide by 10.) 0x14: DIV10_5 (Divide by 10.5.) 0x15: DIV11 (Divide by 11.) 0x16: DIV11_5 (Divide by 11.5.) 0x17: DIV12 (Divide by 12.) 0x18: DIV12_5 (Divide by 12.5.) 0x19: DIV13 (Divide by 13.) 0x1A: DIV13_5 (Divide by 13.5.) 0x1B: DIV14 (Divide by 14.) 0x1C: DIV14_5 (Divide by 14.5.) 0x1D: DIV15 (Divide by 15.) 0x1E: DIV15_5 (Divide by 15.5.) 0x1F: DIV16 (Divide by 16.)

0x0182907C GCC_QDSS_TSCTR_DIV2_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

QDSS TSCTR DIV2 clock branch control register. It allows SW control of CBC soft macro

GCC_QDSS_TSCTR_DIV2_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	This bitfield indicated if the clock branch is halted 0:Clock On 1:Clock Off
0	CLK_ENABLE	RW	This bitfield indicated if the clock branch is enabled. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x01829080 GCC_QDSS_TSCTR_DIV3_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

QDSS TSCTR DIV3 clock branch control register. It allows SW control of CBC soft macro

GCC_QDSS_TSCTR_DIV3_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	This bitfield indicated if the clock branch is halted 0:Clock On 1:Clock Off
0	CLK_ENABLE	RW	This bitfield indicated if the clock branch is enabled. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x01829084 GCC_QDSS_DAP_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

QDSS DAP clock branch control register. It allows SW control of CBC soft macro

GCC_QDSS_DAP_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	This bitfield indicated if the clock branch is halted 0:Clock On 1:Clock Off

GCC_QDSS_DAP_CBCR (cont.)

Bits	Name	Type	Description
0	CLK_ENABLE	RW	This bitfield indicated if the clock branch is enabled. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x01829088 GCC_QDSS_TSCTR_DIV4_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

QDSS TSCTR DIV4 clock branch control register. It allows SW control of CBC soft macro

GCC_QDSS_TSCTR_DIV4_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	This bitfield indicated if the clock branch is halted 0:Clock On 1:Clock Off
0	CLK_ENABLE	RW	This bitfield indicated if the clock branch is enabled. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x0182908C GCC_QDSS_TSCTR_DIV8_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

QDSS TSCTR DIV8 clock branch control register. It allows SW control of CBC soft macro

GCC_QDSS_TSCTR_DIV8_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	This bitfield indicated if the clock branch is halted 0:Clock On 1:Clock Off
0	CLK_ENABLE	RW	This bitfield indicated if the clock branch is enabled. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x01829090 GCC_QDSS_TSCTR_DIV16_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

QDSS TSCTR DIV16 clock branch control register. It allows SW control of CBC soft macro

GCC_QDSS_TSCTR_DIV16_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	This bitfield indicated if the clock branch is halted 0:Clock On 1:Clock Off
0	CLK_ENABLE	RW	This bitfield indicated if the clock branch is enabled. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x01830000 GCC_GCC_SLEEP_CMD_RCGR**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

GCC SLEEP root clock generator command register. It allows for SW to trigger the configuration data

GCC_GCC_SLEEP_CMD_RCGR

Bits	Name	Type	Description
31	ROOT_OFF	R	This bitfield provides indication on status of the RCG; Mainly for SW to determine if the generator is actively producing clock 1: Clock Off 0: Clock On; Generator is actively producing the clock
1	ROOT_EN	RW	Enable Clock Generator output. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x01841000 GCC_USB_HS_BCR**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

USB HS block control register

GCC_USB_HS_BCR

Bits	Name	Description
0	BLK_ARES	This bitfield controls asynchronous software reset for the block. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x01841004 GCC_USB_HS_SYSTEM_CBCR

Type: RW

Clock: CLK

Reset State: 0x80004FF0

USB HS SYSTEM clock branch control register. It allows SW control of CBC soft macro

GCC_USB_HS_SYSTEM_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	This bitfield indicated if the clock branch is halted 0:Clock On 1:Clock Off
14	FORCE_MEM_CORE_ON	RW	This bit force the core on signal to remain active during halt state of the clk. 0x1: FORCE_ENABLE (Force-on Mode.) 0x0: FORCE_DISABLE (Normal Mode force disabled.)
13	FORCE_MEM_PERIPH_ON	RW	This bit force the periph_on signal to remain active during halt state of the clk. 0x1: FORCE_ENABLE (Force-on Mode.) 0x0: FORCE_DISABLE (Normal Mode force disabled.)
12	FORCE_MEM_PERIPH_OFF	RW	This bit force the periph_off signal to remain active during halt state of the clk. 0x1: FORCE_ENABLE (Force-on Mode.) 0x0: FORCE_DISABLE (Normal Mode force disabled.)

GCC_USB_HS_SYSTEM_CBCR (cont.)

Bits	Name	Type	Description
11:8	WAKEUP	RW	This bitfield specifies the number of clock cycles it takes for the clock to turn back on. 0x0: CLOCK0 (Zero cycle) 0x1: CLOCK1 (One cycle) 0x2: CLOCK2 (Two cycles) 0x3: CLOCK3 (Three cycles) 0x4: CLOCK4 (Four cycles) 0x5: CLOCK5 (Five cycles) 0x6: CLOCK6 (Six cycles) 0x7: CLOCK7 (Seven cycles) 0x8: CLOCK8 (Eight cycles) 0x9: CLOCK9 (Nine cycles) 0xA: CLOCK10 (Ten cycles) 0xB: CLOCK11 (Eleven cycles) 0xC: CLOCK12 (Twelve cycles) 0xD: CLOCK13 (Thirteen cycles) 0xE: CLOCK14 (Fourteen cycles) 0xF: CLOCK15 (Fifteen cycles)
7:4	SLEEP	RW	This bitfield specifies the number of clock cycles till complete sleep where headswitch of the corresponding memories will be opened. 0x0: CLOCK0 (Zero cycle) 0x1: CLOCK1 (One cycle) 0x2: CLOCK2 (Two cycles) 0x3: CLOCK3 (Three cycles) 0x4: CLOCK4 (Four cycles) 0x5: CLOCK5 (Five cycles) 0x6: CLOCK6 (Six cycles) 0x7: CLOCK7 (Seven cycles) 0x8: CLOCK8 (Eight cycles) 0x9: CLOCK9 (Nine cycles) 0xA: CLOCK10 (Ten cycles) 0xB: CLOCK11 (Eleven cycles) 0xC: CLOCK12 (Twelve cycles) 0xD: CLOCK13 (Thirteen cycles) 0xE: CLOCK14 (Fourteen cycles) 0xF: CLOCK15 (Fifteen cycles)
0	CLK_ENABLE	RW	This bitfield indicated if the clock branch is enabled. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x01841008 GCC_USB_HS_AHB_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x8000CFF0

USB HS AHB clock branch control register. It allows SW control of CBC soft macro

GCC_USB_HS_AHB_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	This bitfield indicated if the clock branch is halted 0:Clock On 1:Clock Off
30:28	NOC_HANDSHAKE_FSM_STATUS	R	This bitfield is used to check the FSM state
15	NOC_HANDSHAKE_FSM_ENABLE	RW	This bitfield enables the FSM to use so that slaveready and socketcon handshake is used prior to affecting the enable of the clock branch cell. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
14	FORCE_MEM_CORE_ON	RW	This bit force the core on signal to remain active during halt state of the clk. 0x1: FORCE_ENABLE (Force-on Mode.) 0x0: FORCE_DISABLE (Normal Mode force disabled.)
13	FORCE_MEM_PERIPH_ON	RW	This bit force the periph_on signal to remain active during halt state of the clk. 0x1: FORCE_ENABLE (Force-on Mode.) 0x0: FORCE_DISABLE (Normal Mode force disabled.)
12	FORCE_MEM_PERIPH_OFF	RW	This bit force the periph_off signal to remain active during halt state of the clk. 0x1: FORCE_ENABLE (Force-on Mode.) 0x0: FORCE_DISABLE (Normal Mode force disabled.)

GCC_USB_HS_AHB_CBCR (cont.)

Bits	Name	Type	Description
11:8	WAKEUP	RW	This bitfield specifies the number of clock cycles it takes for the clock to turn back on. 0x0: CLOCK0 (Zero cycle) 0x1: CLOCK1 (One cycle) 0x2: CLOCK2 (Two cycles) 0x3: CLOCK3 (Three cycles) 0x4: CLOCK4 (Four cycles) 0x5: CLOCK5 (Five cycles) 0x6: CLOCK6 (Six cycles) 0x7: CLOCK7 (Seven cycles) 0x8: CLOCK8 (Eight cycles) 0x9: CLOCK9 (Nine cycles) 0xA: CLOCK10 (Ten cycles) 0xB: CLOCK11 (Eleven cycles) 0xC: CLOCK12 (Twelve cycles) 0xD: CLOCK13 (Thirteen cycles) 0xE: CLOCK14 (Fourteen cycles) 0xF: CLOCK15 (Fifteen cycles)
7:4	SLEEP	RW	This bitfield specifies the number of clock cycles till complete sleep where headswitch of the corresponding memories will be opened. 0x0: CLOCK0 (Zero cycle) 0x1: CLOCK1 (One cycle) 0x2: CLOCK2 (Two cycles) 0x3: CLOCK3 (Three cycles) 0x4: CLOCK4 (Four cycles) 0x5: CLOCK5 (Five cycles) 0x6: CLOCK6 (Six cycles) 0x7: CLOCK7 (Seven cycles) 0x8: CLOCK8 (Eight cycles) 0x9: CLOCK9 (Nine cycles) 0xA: CLOCK10 (Ten cycles) 0xB: CLOCK11 (Eleven cycles) 0xC: CLOCK12 (Twelve cycles) 0xD: CLOCK13 (Thirteen cycles) 0xE: CLOCK14 (Fourteen cycles) 0xF: CLOCK15 (Fifteen cycles)
0	CLK_ENABLE	RW	This bitfield indicated if the clock branch is enabled. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x0184100C GCC_USB_HS_INACTIVITY_TIMERS_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

USB HS INACTIVITY TIMERS clock branch control register. It allows SW control of CBC soft macro

GCC_USB_HS_INACTIVITY_TIMERS_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	This bitfield indicated if the clock branch is halted 0:Clock On 1:Clock Off
0	CLK_ENABLE	RW	This bitfield indicated if the clock branch is enabled. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x01841010 GCC_USB_HS_SYSTEM_CMD_RCGR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

USB HS SYSTEM root clock generator command register. It allows for SW to trigger the configuration data

GCC_USB_HS_SYSTEM_CMD_RCGR

Bits	Name	Type	Description
31	ROOT_OFF	R	This bitfield provides indication on status of the RCG; Mainly for SW to determine if the generator is actively producing clock 1: Clock Off 0: Clock On; Generator is actively producing the clock
4	DIRTY_CFG_RCGR	R	Indicates change in CFG_RCGR register configuration
1	ROOT_EN	RW	Enable Clock Generator output. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
0	UPDATE	RW	Update to the new programmed configuration. It should be cleared upon switching to new configuration 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x01841014 GCC_USB_HS_SYSTEM_CFG_RCGR**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

USB HS SYSTEM root clock generator configuration register. It allows for SW to configure Root Clock Generation Macro

GCC_USB_HS_SYSTEM_CFG_RCGR

Bits	Name	Description
10:8	SRC_SEL	This bitfield selects clock source that will be used to source the RCG output 0x0: SRC0 (CXO.) 0x1: SRC1 (GPLL0_MAIN.) 0x2: SRC2 (Unused(Grounded).) 0x3: SRC3 (Unused(Grounded).) 0x4: SRC4 (Unused(Grounded).) 0x5: SRC5 (Unused(Grounded).) 0x6: SRC6 (Unused(Grounded).) 0x7: Reserved

GCC_USB_HS_SYSTEM_CFG_RCGR (cont.)

Bits	Name	Description
4:0	SRC_DIV	This field defines the integer and half integer portion of the divider setting of the RCG. The value specifies is $2N-1$ where N is the divide ratio. For example, if the half integer divide ration is 2.5, the src_div should be set at 00100. 0x0: BYPASS (Bypsss.) 0x1: DIV1 (Divide by 1.) 0x2: DIV1_5 (Divide by 1.5.) 0x3: DIV2 (Divide by 2.) 0x4: DIV2_5 (Divide by 2.5.) 0x5: DIV3 (Divide by 3.) 0x6: DIV3_5 (Divide by 3.5.) 0x7: DIV4 (Divide by 4.) 0x8: DIV4_5 (Divide by 4.5.) 0x9: DIV5 (Divide by 5.) 0xA: DIV5_5 (Divide by 5.5.) 0xB: DIV6 (Divide by 6.) 0xC: DIV6_5 (Divide by 6.5.) 0xD: DIV7 (Divide by 7.) 0xE: DIV7_5 (Divide by 7.5.) 0xF: DIV8 (Divide by 8.) 0x10: DIV8_5 (Divide by 8.5.) 0x11: DIV9 (Divide by 9.) 0x12: DIV9_5 (Divide by 9.5.) 0x13: DIV10 (Divide by 10.) 0x14: DIV10_5 (Divide by 10.5.) 0x15: DIV11 (Divide by 11.) 0x16: DIV11_5 (Divide by 11.5.) 0x17: DIV12 (Divide by 12.) 0x18: DIV12_5 (Divide by 12.5.) 0x19: DIV13 (Divide by 13.) 0x1A: DIV13_5 (Divide by 13.5.) 0x1B: DIV14 (Divide by 14.) 0x1C: DIV14_5 (Divide by 14.5.) 0x1D: DIV15 (Divide by 15.) 0x1E: DIV15_5 (Divide by 15.5.) 0x1F: DIV16 (Divide by 16.)

0x01841028 GCC_USB2A_PHY_BCR**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

USB2A PHY block control register

GCC_USB2A_PHY_BCR

Bits	Name	Description
0	BLK_ARES	This bitfield controls asynchronous software reset for the block. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x0184102C GCC_USB2A_PHY_SLEEP_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

USB 2A PHY sleep clock control register

GCC_USB2A_PHY_SLEEP_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	This bitfield indicated if the clock branch is halted 0:Clock On 1:Clock Off
0	CLK_ENABLE	RW	This bitfield indicated if the clock branch is enabled. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x01842000 GCC_SDCC1_BCR**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

SDCC1 block control register

GCC_SDCC1_BCR

Bits	Name	Description
0	BLK_ARES	This bitfield controls asynchronous software reset for the block. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x01842004 GCC_SDCC1_APPS_CMD_RCGR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

SDCC1 application root clock generator command register. It allows for SW to trigger the configuration data

GCC_SDCC1_APPS_CMD_RCGR

Bits	Name	Type	Description
31	ROOT_OFF	R	This bitfield provides indication on status of the RCG; Mainly for SW to determine if the generator is actively producing clock 1: Clock Off 0: Clock On; Generator is actively producing the clock
7	DIRTY_D	R	Indicates change in D register configuration
6	DIRTY_M	R	Indicates change in M register configuration
5	DIRTY_N	R	Indicates change in N register configuration
4	DIRTY_CFG_RCGR	R	Indicates change in CFG_RCGR register configuration
1	ROOT_EN	RW	Enable Clock Generator output. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
0	UPDATE	RW	Update to the new programmed configuration. It should be cleared upon switching to new configuration 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x01842008 GCC_SDCC1_APPS_CFG_RCGR**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

SDCC1 application root clock generator configuration register. It allows for SW to configure Root Clock Generation Macro

GCC_SDCC1_APPS_CFG_RCGR

Bits	Name	Description
13:12	MODE	This bitfield provides SW selection of MN counter mode. 0x3: SINGLE_EDGE (Single-edge.) 0x2: DUAL_EDGE (Dual-edge.) 0x1: SWALLOW (Swallow.) 0x0: BYPASS (Bypass.)
10:8	SRC_SEL	This bitfield selects clock source that will be used to source the RCG output 0x0: SRC0 (CXO.) 0x1: SRC1 (GPLL0_MAIN.) 0x2: SRC2 (Unused(Grounded).) 0x3: SRC3 (Unused(Grounded).) 0x4: SRC4 (Unused(Grounded).) 0x5: SRC5 (Unused(Grounded).) 0x6: SRC6 (Unused(Grounded).) 0x7: Reserved

GCC_SDCC1_APPS_CFG_RCGR (cont.)

Bits	Name	Description
4:0	SRC_DIV	This field defines the integer and half integer portion of the divider setting of the RCG. The value specifies is $2N-1$ where N is the divide ratio. For example, if the half integer divide ration is 2.5, the src_div should be set at 00100. 0x0: BYPASS (Bypsss.) 0x1: DIV1 (Divide by 1.) 0x2: DIV1_5 (Divide by 1.5.) 0x3: DIV2 (Divide by 2.) 0x4: DIV2_5 (Divide by 2.5.) 0x5: DIV3 (Divide by 3.) 0x6: DIV3_5 (Divide by 3.5.) 0x7: DIV4 (Divide by 4.) 0x8: DIV4_5 (Divide by 4.5.) 0x9: DIV5 (Divide by 5.) 0xA: DIV5_5 (Divide by 5.5.) 0xB: DIV6 (Divide by 6.) 0xC: DIV6_5 (Divide by 6.5.) 0xD: DIV7 (Divide by 7.) 0xE: DIV7_5 (Divide by 7.5.) 0xF: DIV8 (Divide by 8.) 0x10: DIV8_5 (Divide by 8.5.) 0x11: DIV9 (Divide by 9.) 0x12: DIV9_5 (Divide by 9.5.) 0x13: DIV10 (Divide by 10.) 0x14: DIV10_5 (Divide by 10.5.) 0x15: DIV11 (Divide by 11.) 0x16: DIV11_5 (Divide by 11.5.) 0x17: DIV12 (Divide by 12.) 0x18: DIV12_5 (Divide by 12.5.) 0x19: DIV13 (Divide by 13.) 0x1A: DIV13_5 (Divide by 13.5.) 0x1B: DIV14 (Divide by 14.) 0x1C: DIV14_5 (Divide by 14.5.) 0x1D: DIV15 (Divide by 15.) 0x1E: DIV15_5 (Divide by 15.5.) 0x1F: DIV16 (Divide by 16.)

0x0184200C GCC_SDCC1_APPS_M**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

SDCC1 application root clock generator M value register. This register houses the M value for MND divider.

GCC_SDCC1_APPS_M

Bits	Name	Description
7:0	M	This bitfield represents M Value of the M/N:D Counter

0x01842010 GCC_SDCC1_APPS_N

Type: RW

Clock: CLK

Reset State: 0x00000000

SDCC1 application root clock generator N value register. This register houses the NOT(N-M) value for MND divider.

GCC_SDCC1_APPS_N

Bits	Name	Description
7:0	NOT_N_MINUS_M	This bitfield represents NOT(N-M) Value of the M/N:D Counter

0x01842014 GCC_SDCC1_APPS_D

Type: RW

Clock: CLK

Reset State: 0x00000000

SDCC1 application root clock generator D value register. This register houses the NOT 2D value for MND divider.

GCC_SDCC1_APPS_D

Bits	Name	Description
7:0	NOT_2D	This bitfield represents NOT(2*D) Value of the M/N:D Counter

0x01842018 GCC_SDCC1_APPS_CBCR

Type: RW

Clock: CLK

Reset State: 0x80004FF0

SDCC1 application clock branch control register. It allows SW control of CBC soft macro

GCC_SDCC1_APPS_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	This bitfield indicated if the clock branch is halted 0:Clock On 1:Clock Off
14	FORCE_MEM_CORE_ON	RW	This bit force the core on signal to remain active during halt state of the clk. 0x1: FORCE_ENABLE (Force-on Mode.) 0x0: FORCE_DISABLE (Normal Mode force disabled.)
13	FORCE_MEM_PERIPH_ON	RW	This bit force the periph_on signal to remain active during halt state of the clk. 0x1: FORCE_ENABLE (Force-on Mode.) 0x0: FORCE_DISABLE (Normal Mode force disabled.)
12	FORCE_MEM_PERIPH_OFF	RW	This bit force the periph_off signal to remain active during halt state of the clk. 0x1: FORCE_ENABLE (Force-on Mode.) 0x0: FORCE_DISABLE (Normal Mode force disabled.)
11:8	WAKEUP	RW	This bitfield specifies the number of clock cycles it takes for the clock to turn back on. 0x0: CLOCK0 (Zero cycle) 0x1: CLOCK1 (One cycle) 0x2: CLOCK2 (Two cycles) 0x3: CLOCK3 (Three cycles) 0x4: CLOCK4 (Four cycles) 0x5: CLOCK5 (Five cycles) 0x6: CLOCK6 (Six cycles) 0x7: CLOCK7 (Seven cycles) 0x8: CLOCK8 (Eight cycles) 0x9: CLOCK9 (Nine cycles) 0xA: CLOCK10 (Ten cycles) 0xB: CLOCK11 (Eleven cycles) 0xC: CLOCK12 (Twelve cycles) 0xD: CLOCK13 (Thirteen cycles) 0xE: CLOCK14 (Fourteen cycles) 0xF: CLOCK15 (Fifteen cycles)

GCC_SDCC1_APPS_CBCR (cont.)

Bits	Name	Type	Description
7:4	SLEEP	RW	This bitfield specifies the number of clock cycles till complete sleep where headswitch of the corresponding memories will be opened. 0x0: CLOCK0 (Zero cycle) 0x1: CLOCK1 (One cycle) 0x2: CLOCK2 (Two cycles) 0x3: CLOCK3 (Three cycles) 0x4: CLOCK4 (Four cycles) 0x5: CLOCK5 (Five cycles) 0x6: CLOCK6 (Six cycles) 0x7: CLOCK7 (Seven cycles) 0x8: CLOCK8 (Eight cycles) 0x9: CLOCK9 (Nine cycles) 0xA: CLOCK10 (Ten cycles) 0xB: CLOCK11 (Eleven cycles) 0xC: CLOCK12 (Twelve cycles) 0xD: CLOCK13 (Thirteen cycles) 0xE: CLOCK14 (Fourteen cycles) 0xF: CLOCK15 (Fifteen cycles)
0	CLK_ENABLE	RW	This bitfield indicated if the clock branch is enabled. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x0184201C GCC_SDCC1_AHB_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x8000CFF0

SDCC1 AHB clock branch control register. It allows SW control of CBC soft macro

GCC_SDCC1_AHB_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	This bitfield indicated if the clock branch is halted 0:Clock On 1:Clock Off
30:28	NOC_HANDSHAKE_FSM_STATUS	R	This bitfield is used to check the FSM state
15	NOC_HANDSHAKE_FSM_ENABLE	RW	This bitfield enables the FSM to use so that slave ready and socketcon handshake is used prior to affecting the enable of the clock branch cell. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

GCC_SDCC1_AHB_CBCR (cont.)

Bits	Name	Type	Description
14	FORCE_MEM_CORE_ON	RW	This bit force the core on signal to remain active during halt state of the clk. 0x1: FORCE_ENABLE (Force-on Mode.) 0x0: FORCE_DISABLE (Normal Mode force disabled.)
13	FORCE_MEM_PERIPH_ON	RW	This bit force the periph_on signal to remain active during halt state of the clk. 0x1: FORCE_ENABLE (Force-on Mode.) 0x0: FORCE_DISABLE (Normal Mode force disabled.)
12	FORCE_MEM_PERIPH_OFF	RW	This bit force the periph_off signal to remain active during halt state of the clk. 0x1: FORCE_ENABLE (Force-on Mode.) 0x0: FORCE_DISABLE (Normal Mode force disabled.)
11:8	WAKEUP	RW	This bitfield specifies the number of clock cycles it takes for the clock to turn back on. 0x0: CLOCK0 (Zero cycle) 0x1: CLOCK1 (One cycle) 0x2: CLOCK2 (Two cycles) 0x3: CLOCK3 (Three cycles) 0x4: CLOCK4 (Four cycles) 0x5: CLOCK5 (Five cycles) 0x6: CLOCK6 (Six cycles) 0x7: CLOCK7 (Seven cycles) 0x8: CLOCK8 (Eight cycles) 0x9: CLOCK9 (Nine cycles) 0xA: CLOCK10 (Ten cycles) 0xB: CLOCK11 (Eleven cycles) 0xC: CLOCK12 (Twelve cycles) 0xD: CLOCK13 (Thirteen cycles) 0xE: CLOCK14 (Fourteen cycles) 0xF: CLOCK15 (Fifteen cycles)

GCC_SDCC1_AHB_CBCR (cont.)

Bits	Name	Type	Description
7:4	SLEEP	RW	This bitfield specifies the number of clock cycles till complete sleep where headswitch of the corresponding memories will be opened. 0x0: CLOCK0 (Zero cycle) 0x1: CLOCK1 (One cycle) 0x2: CLOCK2 (Two cycles) 0x3: CLOCK3 (Three cycles) 0x4: CLOCK4 (Four cycles) 0x5: CLOCK5 (Five cycles) 0x6: CLOCK6 (Six cycles) 0x7: CLOCK7 (Seven cycles) 0x8: CLOCK8 (Eight cycles) 0x9: CLOCK9 (Nine cycles) 0xA: CLOCK10 (Ten cycles) 0xB: CLOCK11 (Eleven cycles) 0xC: CLOCK12 (Twelve cycles) 0xD: CLOCK13 (Thirteen cycles) 0xE: CLOCK14 (Fourteen cycles) 0xF: CLOCK15 (Fifteen cycles)
0	CLK_ENABLE	RW	This bitfield indicated if the clock branch is enabled. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x01842020 GCC_SDCC1_MISC**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

SDCC1 misc register

GCC_SDCC1_MISC

Bits	Name	Description
0	CDC_ARES	This bitfield controls asynchronous software reset for the CDC block. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x01843000 GCC_SDCC2_BCR**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

SDCC2 block control register

GCC_SDCC2_BCR

Bits	Name	Description
0	BLK_ARES	This bitfield controls asynchronous software reset for the block. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x01843004 GCC_SDCC2_APPS_CMD_RCGR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

SDCC2 application root clock generator command register. It allows for SW to trigger the configuration data

GCC_SDCC2_APPS_CMD_RCGR

Bits	Name	Type	Description
31	ROOT_OFF	R	This bitfield provides indication on status of the RCG; Mainly for SW to determine if the generator is actively producing clock 1: Clock Off 0: Clock On; Generator is actively producing the clock
7	DIRTY_D	R	Indicates change in D register configuration
6	DIRTY_M	R	Indicates change in M register configuration
5	DIRTY_N	R	Indicates change in N register configuration
4	DIRTY_CFG_RCGR	R	Indicates change in CFG_RCGR register configuration
1	ROOT_EN	RW	Enable Clock Generator output. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
0	UPDATE	RW	Update to the new programmed configuration. It should be cleared upon switching to new configuration 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x01843008 GCC_SDCC2_APPS_CFG_RCGR**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

SDCC2 application root clock generator configuration register. It allows for SW to configure Root Clock Generation Macro

GCC_SDCC2_APPS_CFG_RCGR

Bits	Name	Description
13:12	MODE	This bitfield provides SW selection of MN counter mode. 0x3: SINGLE_EDGE (Single-edge.) 0x2: DUAL_EDGE (Dual-edge.) 0x1: SWALLOW (Swallow.) 0x0: BYPASS (Bypass.)
10:8	SRC_SEL	This bitfield selects clock source that will be used to source the RCG output 0x0: SRC0 (CXO.) 0x1: SRC1 (GPLLO_MAIN.) 0x2: SRC2 (Unused(Grounded).) 0x3: SRC3 (Unused(Grounded).) 0x4: SRC4 (Unused(Grounded).) 0x5: SRC5 (Unused(Grounded).) 0x6: SRC6 (Unused(Grounded).) 0x7: Reserved

GCC_SDCC2_APPS_CFG_RCGR (cont.)

Bits	Name	Description
4:0	SRC_DIV	This field defines the integer and half integer portion of the divider setting of the RCG. The value specifies is $2N-1$ where N is the divide ratio. For example, if the half integer divide ration is 2.5, the src_div should be set at 00100. 0x0: BYPASS (Bypsss.) 0x1: DIV1 (Divide by 1.) 0x2: DIV1_5 (Divide by 1.5.) 0x3: DIV2 (Divide by 2.) 0x4: DIV2_5 (Divide by 2.5.) 0x5: DIV3 (Divide by 3.) 0x6: DIV3_5 (Divide by 3.5.) 0x7: DIV4 (Divide by 4.) 0x8: DIV4_5 (Divide by 4.5.) 0x9: DIV5 (Divide by 5.) 0xA: DIV5_5 (Divide by 5.5.) 0xB: DIV6 (Divide by 6.) 0xC: DIV6_5 (Divide by 6.5.) 0xD: DIV7 (Divide by 7.) 0xE: DIV7_5 (Divide by 7.5.) 0xF: DIV8 (Divide by 8.) 0x10: DIV8_5 (Divide by 8.5.) 0x11: DIV9 (Divide by 9.) 0x12: DIV9_5 (Divide by 9.5.) 0x13: DIV10 (Divide by 10.) 0x14: DIV10_5 (Divide by 10.5.) 0x15: DIV11 (Divide by 11.) 0x16: DIV11_5 (Divide by 11.5.) 0x17: DIV12 (Divide by 12.) 0x18: DIV12_5 (Divide by 12.5.) 0x19: DIV13 (Divide by 13.) 0x1A: DIV13_5 (Divide by 13.5.) 0x1B: DIV14 (Divide by 14.) 0x1C: DIV14_5 (Divide by 14.5.) 0x1D: DIV15 (Divide by 15.) 0x1E: DIV15_5 (Divide by 15.5.) 0x1F: DIV16 (Divide by 16.)

0x0184300C GCC_SDCC2_APPS_M**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

SDCC2 application root clock generator M value register. This register houses the M value for MND divider.

GCC_SDCC2_APPS_M

Bits	Name	Description
7:0	M	This bitfield represents M Value of the M/N:D Counter

0x01843010 GCC_SDCC2_APPS_N

Type: RW

Clock: CLK

Reset State: 0x00000000

SDCC2 application root clock generator N value register. This register houses the NOT(N-M) value for MND divider.

GCC_SDCC2_APPS_N

Bits	Name	Description
7:0	NOT_N_MINUS_M	This bitfield represents NOT(N-M) Value of the M/N:D Counter

0x01843014 GCC_SDCC2_APPS_D

Type: RW

Clock: CLK

Reset State: 0x00000000

SDCC2 application root clock generator D value register. This register houses the NOT 2D value for MND divider.

GCC_SDCC2_APPS_D

Bits	Name	Description
7:0	NOT_2D	This bitfield represents NOT(2*D) Value of the M/N:D Counter

0x01843018 GCC_SDCC2_APPS_CBCR

Type: RW

Clock: CLK

Reset State: 0x80004FF0

SDCC2 application clock branch control register. It allows SW control of CBC soft macro

GCC_SDCC2_APPS_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	This bitfield indicated if the clock branch is halted 0:Clock On 1:Clock Off
14	FORCE_MEM_CORE_ON	RW	This bit force the core on signal to remain active during halt state of the clk. 0x1: FORCE_ENABLE (Force-on Mode.) 0x0: FORCE_DISABLE (Normal Mode force disabled.)
13	FORCE_MEM_PERIPH_ON	RW	This bit force the periph_on signal to remain active during halt state of the clk. 0x1: FORCE_ENABLE (Force-on Mode.) 0x0: FORCE_DISABLE (Normal Mode force disabled.)
12	FORCE_MEM_PERIPH_OFF	RW	This bit force the periph_off signal to remain active during halt state of the clk. 0x1: FORCE_ENABLE (Force-on Mode.) 0x0: FORCE_DISABLE (Normal Mode force disabled.)
11:8	WAKEUP	RW	This bitfield specifies the number of clock cycles it takes for the clock to turn back on. 0x0: CLOCK0 (Zero cycle) 0x1: CLOCK1 (One cycle) 0x2: CLOCK2 (Two cycles) 0x3: CLOCK3 (Three cycles) 0x4: CLOCK4 (Four cycles) 0x5: CLOCK5 (Five cycles) 0x6: CLOCK6 (Six cycles) 0x7: CLOCK7 (Seven cycles) 0x8: CLOCK8 (Eight cycles) 0x9: CLOCK9 (Nine cycles) 0xA: CLOCK10 (Ten cycles) 0xB: CLOCK11 (Eleven cycles) 0xC: CLOCK12 (Twelve cycles) 0xD: CLOCK13 (Thirteen cycles) 0xE: CLOCK14 (Fourteen cycles) 0xF: CLOCK15 (Fifteen cycles)

GCC_SDCC2_APPS_CBCR (cont.)

Bits	Name	Type	Description
7:4	SLEEP	RW	This bitfield specifies the number of clock cycles till complete sleep where headswitch of the corresponding memories will be opened. 0x0: CLOCK0 (Zero cycle) 0x1: CLOCK1 (One cycle) 0x2: CLOCK2 (Two cycles) 0x3: CLOCK3 (Three cycles) 0x4: CLOCK4 (Four cycles) 0x5: CLOCK5 (Five cycles) 0x6: CLOCK6 (Six cycles) 0x7: CLOCK7 (Seven cycles) 0x8: CLOCK8 (Eight cycles) 0x9: CLOCK9 (Nine cycles) 0xA: CLOCK10 (Ten cycles) 0xB: CLOCK11 (Eleven cycles) 0xC: CLOCK12 (Twelve cycles) 0xD: CLOCK13 (Thirteen cycles) 0xE: CLOCK14 (Fourteen cycles) 0xF: CLOCK15 (Fifteen cycles)
0	CLK_ENABLE	RW	This bitfield indicated if the clock branch is enabled. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x0184301C GCC_SDCC2_AHB_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x8000CFF0

SDCC2 AHB clock branch control register. It allows SW control of CBC soft macro

GCC_SDCC2_AHB_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	This bitfield indicated if the clock branch is halted 0:Clock On 1:Clock Off
30:28	NOC_HANDSHAKE_FSM_STATUS	R	This bitfield is used to check the FSM state
15	NOC_HANDSHAKE_FSM_ENABLE	RW	This bitfield enables the FSM to use so that slave ready and socketcon handshake is used prior to affecting the enable of the clock branch cell. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

GCC_SDCC2_AHB_CBCR (cont.)

Bits	Name	Type	Description
14	FORCE_MEM_CORE_ON	RW	This bit force the core on signal to remain active during halt state of the clk. 0x1: FORCE_ENABLE (Force-on Mode.) 0x0: FORCE_DISABLE (Normal Mode force disabled.)
13	FORCE_MEM_PERIPH_ON	RW	This bit force the periph_on signal to remain active during halt state of the clk. 0x1: FORCE_ENABLE (Force-on Mode.) 0x0: FORCE_DISABLE (Normal Mode force disabled.)
12	FORCE_MEM_PERIPH_OFF	RW	This bit force the periph_off signal to remain active during halt state of the clk. 0x1: FORCE_ENABLE (Force-on Mode.) 0x0: FORCE_DISABLE (Normal Mode force disabled.)
11:8	WAKEUP	RW	This bitfield specifies the number of clock cycles it takes for the clock to turn back on. 0x0: CLOCK0 (Zero cycle) 0x1: CLOCK1 (One cycle) 0x2: CLOCK2 (Two cycles) 0x3: CLOCK3 (Three cycles) 0x4: CLOCK4 (Four cycles) 0x5: CLOCK5 (Five cycles) 0x6: CLOCK6 (Six cycles) 0x7: CLOCK7 (Seven cycles) 0x8: CLOCK8 (Eight cycles) 0x9: CLOCK9 (Nine cycles) 0xA: CLOCK10 (Ten cycles) 0xB: CLOCK11 (Eleven cycles) 0xC: CLOCK12 (Twelve cycles) 0xD: CLOCK13 (Thirteen cycles) 0xE: CLOCK14 (Fourteen cycles) 0xF: CLOCK15 (Fifteen cycles)

GCC_SDCC2_AHB_CBCR (cont.)

Bits	Name	Type	Description
7:4	SLEEP	RW	This bitfield specifies the number of clock cycles till complete sleep where headswitch of the corresponding memories will be opened. 0x0: CLOCK0 (Zero cycle) 0x1: CLOCK1 (One cycle) 0x2: CLOCK2 (Two cycles) 0x3: CLOCK3 (Three cycles) 0x4: CLOCK4 (Four cycles) 0x5: CLOCK5 (Five cycles) 0x6: CLOCK6 (Six cycles) 0x7: CLOCK7 (Seven cycles) 0x8: CLOCK8 (Eight cycles) 0x9: CLOCK9 (Nine cycles) 0xA: CLOCK10 (Ten cycles) 0xB: CLOCK11 (Eleven cycles) 0xC: CLOCK12 (Twelve cycles) 0xD: CLOCK13 (Thirteen cycles) 0xE: CLOCK14 (Fourteen cycles) 0xF: CLOCK15 (Fifteen cycles)
0	CLK_ENABLE	RW	This bitfield indicated if the clock branch is enabled. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x01801000 GCC_BLSP1_BCR**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

BLSP1 block control register

GCC_BLSP1_BCR

Bits	Name	Description
0	BLK_ARES	This bitfield controls asynchronous software reset for the block. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x01801004 GCC_BLSP1_SLEEP_CBCR**Type:** R**Clock:** CLK**Reset State:** 0x80000000

BLSP1 SLEEP clock branch control register. It allows SW control of CBC soft macro

GCC_BLSP1_SLEEP_CBCR

Bits	Name	Description
31	CLK_OFF	This bitfield indicated if the clock branch is halted 0:Clock On 1:Clock Off

0x01801008 GCC_BLSP1_AHB_CBCR

Type: RW

Clock: CLK

Reset State: 0x8000CFF0

BLSP1 AHB clock branch control register. It allows SW control of CBC soft macro

GCC_BLSP1_AHB_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	This bitfield indicated if the clock branch is halted 0:Clock On 1:Clock Off
30:28	NOC_HANDSHAKE_FSM_STATUS	R	This bitfield is used to check the FSM state
15	NOC_HANDSHAKE_FSM_ENABLE	RW	This bitfield enables the FSM to use so that slave ready and socketcon handshake is used prior to affecting the enable of the clock branch cell. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
14	FORCE_MEM_CORE_ON	RW	This bit force the core on signal to remain active during halt state of the clk. 0x1: FORCE_ENABLE (Force-on Mode.) 0x0: FORCE_DISABLE (Normal Mode force disabled.)
13	FORCE_MEM_PERIPH_ON	RW	This bit force the periph_on signal to remain active during halt state of the clk. 0x1: FORCE_ENABLE (Force-on Mode.) 0x0: FORCE_DISABLE (Normal Mode force disabled.)
12	FORCE_MEM_PERIPH_OFF	RW	This bit force the periph_off signal to remain active during halt state of the clk. 0x1: FORCE_ENABLE (Force-on Mode.) 0x0: FORCE_DISABLE (Normal Mode force disabled.)

GCC_BLSP1_AHB_CBCR (cont.)

Bits	Name	Type	Description
11:8	WAKEUP	RW	This bitfield specifies the number of clock cycles it takes for the clock to turn back on. 0x0: CLOCK0 (Zero cycle) 0x1: CLOCK1 (One cycle) 0x2: CLOCK2 (Two cycles) 0x3: CLOCK3 (Three cycles) 0x4: CLOCK4 (Four cycles) 0x5: CLOCK5 (Five cycles) 0x6: CLOCK6 (Six cycles) 0x7: CLOCK7 (Seven cycles) 0x8: CLOCK8 (Eight cycles) 0x9: CLOCK9 (Nine cycles) 0xA: CLOCK10 (Ten cycles) 0xB: CLOCK11 (Eleven cycles) 0xC: CLOCK12 (Twelve cycles) 0xD: CLOCK13 (Thirteen cycles) 0xE: CLOCK14 (Fourteen cycles) 0xF: CLOCK15 (Fifteen cycles)
7:4	SLEEP	RW	This bitfield specifies the number of clock cycles till complete sleep where headswitch of the corresponding memories will be opened. 0x0: CLOCK0 (Zero cycle) 0x1: CLOCK1 (One cycle) 0x2: CLOCK2 (Two cycles) 0x3: CLOCK3 (Three cycles) 0x4: CLOCK4 (Four cycles) 0x5: CLOCK5 (Five cycles) 0x6: CLOCK6 (Six cycles) 0x7: CLOCK7 (Seven cycles) 0x8: CLOCK8 (Eight cycles) 0x9: CLOCK9 (Nine cycles) 0xA: CLOCK10 (Ten cycles) 0xB: CLOCK11 (Eleven cycles) 0xC: CLOCK12 (Twelve cycles) 0xD: CLOCK13 (Thirteen cycles) 0xE: CLOCK14 (Fourteen cycles) 0xF: CLOCK15 (Fifteen cycles)

0x01802000 GCC_BLSP1_QUP1_BCR**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

BLSP1 QUP1 block control register

GCC_BLSP1_QUP1_BCR

Bits	Name	Description
0	BLK_ARES	This bitfield controls asynchronous software reset for the block. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x01802004 GCC_BLSP1_QUP1_SPI_APPS_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

BLSP1 QUP1 SPI application clock branch control register. It allows SW control of CBC soft macro

GCC_BLSP1_QUP1_SPI_APPS_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	This bitfield indicated if the clock branch is halted 0:Clock On 1:Clock Off
0	CLK_ENABLE	RW	This bitfield indicated if the clock branch is enabled. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x01802008 GCC_BLSP1_QUP1_I2C_APPS_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

BLSP1 QUP1 I2C application clock branch control register. It allows SW control of CBC soft macro

GCC_BLSP1_QUP1_I2C_APPS_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	This bitfield indicated if the clock branch is halted 0:Clock On 1:Clock Off
0	CLK_ENABLE	RW	This bitfield indicated if the clock branch is enabled. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x0180200C GCC_BLSP1_QUP1_I2C_APPS_CMD_RCGR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

BLSP1 QUP1 I2C application root clock generator command register. It allows for SW to trigger the configuration data

GCC_BLSP1_QUP1_I2C_APPS_CMD_RCGR

Bits	Name	Type	Description
31	ROOT_OFF	R	This bitfield provides indication on status of the RCG; Mainly for SW to determine if the generator is actively producing clock 1: Clock Off 0: Clock On; Generator is actively producing the clock
4	DIRTY_CFG_RCGR	R	Indicates change in CFG_RCGR register configuration
1	ROOT_EN	RW	Enable Clock Generator output. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
0	UPDATE	RW	Update to the new programmed configuration. It should be cleared upon switching to new configuration 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x01802010 GCC_BLSP1_QUP1_I2C_APPS_CFG_RCGR**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

BLSP1 QUP1 I2C application root clock generator configuration register. It allows for SW to configure Root Clock Generation Macro

GCC_BLSP1_QUP1_I2C_APPS_CFG_RCGR

Bits	Name	Description
10:8	SRC_SEL	This bitfield selects clock source that will be used to source the RCG output 0x0: SRC0 (CXO.) 0x1: SRC1 (GPLL0_MAIN.) 0x2: SRC2 (Unused(Grounded).) 0x3: SRC3 (Unused(Grounded).) 0x4: SRC4 (Unused(Grounded).) 0x5: SRC5 (Unused(Grounded).) 0x6: SRC6 (Unused(Grounded).) 0x7: Reserved

GCC_BLSP1_QUP1_I2C_APPS_CFG_RCGR (cont.)

Bits	Name	Description
4:0	SRC_DIV	This field defines the integer and half integer portion of the divider setting of the RCGR. The value specifies is $2N-1$ where N is the divide ratio. For example, if the half integer divide ration is 2.5, the src_div should be set at 00100. 0x0: BYPASS (Bypsss.) 0x1: DIV1 (Divide by 1.) 0x2: DIV1_5 (Divide by 1.5.) 0x3: DIV2 (Divide by 2.) 0x4: DIV2_5 (Divide by 2.5.) 0x5: DIV3 (Divide by 3.) 0x6: DIV3_5 (Divide by 3.5.) 0x7: DIV4 (Divide by 4.) 0x8: DIV4_5 (Divide by 4.5.) 0x9: DIV5 (Divide by 5.) 0xA: DIV5_5 (Divide by 5.5.) 0xB: DIV6 (Divide by 6.) 0xC: DIV6_5 (Divide by 6.5.) 0xD: DIV7 (Divide by 7.) 0xE: DIV7_5 (Divide by 7.5.) 0xF: DIV8 (Divide by 8.) 0x10: DIV8_5 (Divide by 8.5.) 0x11: DIV9 (Divide by 9.) 0x12: DIV9_5 (Divide by 9.5.) 0x13: DIV10 (Divide by 10.) 0x14: DIV10_5 (Divide by 10.5.) 0x15: DIV11 (Divide by 11.) 0x16: DIV11_5 (Divide by 11.5.) 0x17: DIV12 (Divide by 12.) 0x18: DIV12_5 (Divide by 12.5.) 0x19: DIV13 (Divide by 13.) 0x1A: DIV13_5 (Divide by 13.5.) 0x1B: DIV14 (Divide by 14.) 0x1C: DIV14_5 (Divide by 14.5.) 0x1D: DIV15 (Divide by 15.) 0x1E: DIV15_5 (Divide by 15.5.) 0x1F: DIV16 (Divide by 16.)

0x01803000 GCC_BLSP1_QUP2_I2C_APPS_CMD_RCGR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

BLSP1 QUP2 I2C application root clock generator command register. It allows for SW to trigger the configuration data

GCC_BLSP1_QUP2_I2C_APPS_CMD_RCGR

Bits	Name	Type	Description
31	ROOT_OFF	R	This bitfield provides indication on status of the RCG; Mainly for SW to determine if the generator is actively producing clock 1: Clock Off 0: Clock On; Generator is actively producing the clock
4	DIRTY_CFG_RCGR	R	Indicates change in CFG_RCGR register configuration
1	ROOT_EN	RW	Enable Clock Generator output. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
0	UPDATE	RW	Update to the new programmed configuration. It should be cleared upon switching to new configuration 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x01803004 GCC_BLSP1_QUP2_I2C_APPS_CFG_RCGR

Type: RW

Clock: CLK

Reset State: 0x00000000

BLSP1 QUP2 I2C application root clock generator configuration register. It allows for SW to configure Root Clock Generation Macro

GCC_BLSP1_QUP2_I2C_APPS_CFG_RCGR

Bits	Name	Description
10:8	SRC_SEL	This bitfield selects clock source that will be used to source the RCG output 0x0: SRC0 (CXO.) 0x1: SRC1 (GPLL0_MAIN.) 0x2: SRC2 (Unused(Grounded).) 0x3: SRC3 (Unused(Grounded).) 0x4: SRC4 (Unused(Grounded).) 0x5: SRC5 (Unused(Grounded).) 0x6: SRC6 (Unused(Grounded).) 0x7: Reserved

GCC_BLSP1_QUP2_I2C_APPS_CFG_RCGR (cont.)

Bits	Name	Description
4:0	SRC_DIV	This field defines the integer and half integer portion of the divider setting of the RCGR. The value specifies is $2N-1$ where N is the divide ratio. For example, if the half integer divide ration is 2.5, the src_div should be set at 00100. 0x0: BYPASS (Bypsss.) 0x1: DIV1 (Divide by 1.) 0x2: DIV1_5 (Divide by 1.5.) 0x3: DIV2 (Divide by 2.) 0x4: DIV2_5 (Divide by 2.5.) 0x5: DIV3 (Divide by 3.) 0x6: DIV3_5 (Divide by 3.5.) 0x7: DIV4 (Divide by 4.) 0x8: DIV4_5 (Divide by 4.5.) 0x9: DIV5 (Divide by 5.) 0xA: DIV5_5 (Divide by 5.5.) 0xB: DIV6 (Divide by 6.) 0xC: DIV6_5 (Divide by 6.5.) 0xD: DIV7 (Divide by 7.) 0xE: DIV7_5 (Divide by 7.5.) 0xF: DIV8 (Divide by 8.) 0x10: DIV8_5 (Divide by 8.5.) 0x11: DIV9 (Divide by 9.) 0x12: DIV9_5 (Divide by 9.5.) 0x13: DIV10 (Divide by 10.) 0x14: DIV10_5 (Divide by 10.5.) 0x15: DIV11 (Divide by 11.) 0x16: DIV11_5 (Divide by 11.5.) 0x17: DIV12 (Divide by 12.) 0x18: DIV12_5 (Divide by 12.5.) 0x19: DIV13 (Divide by 13.) 0x1A: DIV13_5 (Divide by 13.5.) 0x1B: DIV14 (Divide by 14.) 0x1C: DIV14_5 (Divide by 14.5.) 0x1D: DIV15 (Divide by 15.) 0x1E: DIV15_5 (Divide by 15.5.) 0x1F: DIV16 (Divide by 16.)

0x01804000 GCC_BLSP1_QUP3_I2C_APPS_CMD_RCGR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

BLSP1 QUP3 I2C application root clock generator command register. It allows for SW to trigger the configuration data

GCC_BLSP1_QUP3_I2C_APPS_CMD_RCGR

Bits	Name	Type	Description
31	ROOT_OFF	R	This bitfield provides indication on status of the RCG; Mainly for SW to determine if the generator is actively producing clock 1: Clock Off 0: Clock On; Generator is actively producing the clock
4	DIRTY_CFG_RCGR	R	Indicates change in CFG_RCGR register configuration
1	ROOT_EN	RW	Enable Clock Generator output. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
0	UPDATE	RW	Update to the new programmed configuration. It should be cleared upon switching to new configuration 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x01804004 GCC_BLSP1_QUP3_I2C_APPS_CFG_RCGR

Type: RW

Clock: CLK

Reset State: 0x00000000

BLSP1 QUP3 I2C application root clock generator configuration register. It allows for SW to configure Root Clock Generation Macro

GCC_BLSP1_QUP3_I2C_APPS_CFG_RCGR

Bits	Name	Description
10:8	SRC_SEL	This bitfield selects clock source that will be used to source the RCG output 0x0: SRC0 (CXO.) 0x1: SRC1 (GPLL0_MAIN.) 0x2: SRC2 (Unused(Grounded).) 0x3: SRC3 (Unused(Grounded).) 0x4: SRC4 (Unused(Grounded).) 0x5: SRC5 (Unused(Grounded).) 0x6: SRC6 (Unused(Grounded).) 0x7: Reserved

GCC_BLSP1_QUP3_I2C_APPS_CFG_RCGR (cont.)

Bits	Name	Description
4:0	SRC_DIV	This field defines the integer and half integer portion of the divider setting of the RCGR. The value specifies is $2N-1$ where N is the divide ratio. For example, if the half integer divide ration is 2.5, the src_div should be set at 00100. 0x0: BYPASS (Bypsss.) 0x1: DIV1 (Divide by 1.) 0x2: DIV1_5 (Divide by 1.5.) 0x3: DIV2 (Divide by 2.) 0x4: DIV2_5 (Divide by 2.5.) 0x5: DIV3 (Divide by 3.) 0x6: DIV3_5 (Divide by 3.5.) 0x7: DIV4 (Divide by 4.) 0x8: DIV4_5 (Divide by 4.5.) 0x9: DIV5 (Divide by 5.) 0xA: DIV5_5 (Divide by 5.5.) 0xB: DIV6 (Divide by 6.) 0xC: DIV6_5 (Divide by 6.5.) 0xD: DIV7 (Divide by 7.) 0xE: DIV7_5 (Divide by 7.5.) 0xF: DIV8 (Divide by 8.) 0x10: DIV8_5 (Divide by 8.5.) 0x11: DIV9 (Divide by 9.) 0x12: DIV9_5 (Divide by 9.5.) 0x13: DIV10 (Divide by 10.) 0x14: DIV10_5 (Divide by 10.5.) 0x15: DIV11 (Divide by 11.) 0x16: DIV11_5 (Divide by 11.5.) 0x17: DIV12 (Divide by 12.) 0x18: DIV12_5 (Divide by 12.5.) 0x19: DIV13 (Divide by 13.) 0x1A: DIV13_5 (Divide by 13.5.) 0x1B: DIV14 (Divide by 14.) 0x1C: DIV14_5 (Divide by 14.5.) 0x1D: DIV15 (Divide by 15.) 0x1E: DIV15_5 (Divide by 15.5.) 0x1F: DIV16 (Divide by 16.)

0x01805000 GCC_BLSP1_QUP4_I2C_APPS_CMD_RCGR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

BLSP1 QUP4 I2C application root clock generator command register. It allows for SW to trigger the configuration data

GCC_BLSP1_QUP4_I2C_APPS_CMD_RCGR

Bits	Name	Type	Description
31	ROOT_OFF	R	This bitfield provides indication on status of the RCG; Mainly for SW to determine if the generator is actively producing clock 1: Clock Off 0: Clock On; Generator is actively producing the clock
4	DIRTY_CFG_RCGR	R	Indicates change in CFG_RCGR register configuration
1	ROOT_EN	RW	Enable Clock Generator output. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
0	UPDATE	RW	Update to the new programmed configuration. It should be cleared upon switching to new configuration 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x01805004 GCC_BLSP1_QUP4_I2C_APPS_CFG_RCGR

Type: RW

Clock: CLK

Reset State: 0x00000000

BLSP1 QUP4 I2C application root clock generator configuration register. It allows for SW to configure Root Clock Generation Macro

GCC_BLSP1_QUP4_I2C_APPS_CFG_RCGR

Bits	Name	Description
10:8	SRC_SEL	This bitfield selects clock source that will be used to source the RCG output 0x0: SRC0 (CXO.) 0x1: SRC1 (GPLL0_MAIN.) 0x2: SRC2 (Unused(Grounded).) 0x3: SRC3 (Unused(Grounded).) 0x4: SRC4 (Unused(Grounded).) 0x5: SRC5 (Unused(Grounded).) 0x6: SRC6 (Unused(Grounded).) 0x7: Reserved

GCC_BLSP1_QUP4_I2C_APPS_CFG_RCGR (cont.)

Bits	Name	Description
4:0	SRC_DIV	This field defines the integer and half integer portion of the divider setting of the RCGR. The value specifies is $2N-1$ where N is the divide ratio. For example, if the half integer divide ration is 2.5, the src_div should be set at 00100. 0x0: BYPASS (Bypsss.) 0x1: DIV1 (Divide by 1.) 0x2: DIV1_5 (Divide by 1.5.) 0x3: DIV2 (Divide by 2.) 0x4: DIV2_5 (Divide by 2.5.) 0x5: DIV3 (Divide by 3.) 0x6: DIV3_5 (Divide by 3.5.) 0x7: DIV4 (Divide by 4.) 0x8: DIV4_5 (Divide by 4.5.) 0x9: DIV5 (Divide by 5.) 0xA: DIV5_5 (Divide by 5.5.) 0xB: DIV6 (Divide by 6.) 0xC: DIV6_5 (Divide by 6.5.) 0xD: DIV7 (Divide by 7.) 0xE: DIV7_5 (Divide by 7.5.) 0xF: DIV8 (Divide by 8.) 0x10: DIV8_5 (Divide by 8.5.) 0x11: DIV9 (Divide by 9.) 0x12: DIV9_5 (Divide by 9.5.) 0x13: DIV10 (Divide by 10.) 0x14: DIV10_5 (Divide by 10.5.) 0x15: DIV11 (Divide by 11.) 0x16: DIV11_5 (Divide by 11.5.) 0x17: DIV12 (Divide by 12.) 0x18: DIV12_5 (Divide by 12.5.) 0x19: DIV13 (Divide by 13.) 0x1A: DIV13_5 (Divide by 13.5.) 0x1B: DIV14 (Divide by 14.) 0x1C: DIV14_5 (Divide by 14.5.) 0x1D: DIV15 (Divide by 15.) 0x1E: DIV15_5 (Divide by 15.5.) 0x1F: DIV16 (Divide by 16.)

0x01806000 GCC_BLSP1_QUP5_I2C_APPS_CMD_RCGR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

BLSP1 QUP5 I2C application root clock generator command register. It allows for SW to trigger the configuration data

GCC_BLSP1_QUP5_I2C_APPS_CMD_RCGR

Bits	Name	Type	Description
31	ROOT_OFF	R	This bitfield provides indication on status of the RCG; Mainly for SW to determine if the generator is actively producing clock 1: Clock Off 0: Clock On; Generator is actively producing the clock
4	DIRTY_CFG_RCGR	R	Indicates change in CFG_RCGR register configuration
1	ROOT_EN	RW	Enable Clock Generator output. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
0	UPDATE	RW	Update to the new programmed configuration. It should be cleared upon switching to new configuration 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x01806004 GCC_BLSP1_QUP5_I2C_APPS_CFG_RCGR

Type: RW

Clock: CLK

Reset State: 0x00000000

BLSP1 QUP5 I2C application root clock generator configuration register. It allows for SW to configure Root Clock Generation Macro

GCC_BLSP1_QUP5_I2C_APPS_CFG_RCGR

Bits	Name	Description
10:8	SRC_SEL	This bitfield selects clock source that will be used to source the RCG output 0x0: SRC0 (CXO.) 0x1: SRC1 (GPLL0_MAIN.) 0x2: SRC2 (Unused(Grounded).) 0x3: SRC3 (Unused(Grounded).) 0x4: SRC4 (Unused(Grounded).) 0x5: SRC5 (Unused(Grounded).) 0x6: SRC6 (Unused(Grounded).) 0x7: Reserved

GCC_BLSP1_QUP5_I2C_APPS_CFG_RCGR (cont.)

Bits	Name	Description
4:0	SRC_DIV	This field defines the integer and half integer portion of the divider setting of the RCGR. The value specifies is $2N-1$ where N is the divide ratio. For example, if the half integer divide ration is 2.5, the src_div should be set at 00100. 0x0: BYPASS (Bypsss.) 0x1: DIV1 (Divide by 1.) 0x2: DIV1_5 (Divide by 1.5.) 0x3: DIV2 (Divide by 2.) 0x4: DIV2_5 (Divide by 2.5.) 0x5: DIV3 (Divide by 3.) 0x6: DIV3_5 (Divide by 3.5.) 0x7: DIV4 (Divide by 4.) 0x8: DIV4_5 (Divide by 4.5.) 0x9: DIV5 (Divide by 5.) 0xA: DIV5_5 (Divide by 5.5.) 0xB: DIV6 (Divide by 6.) 0xC: DIV6_5 (Divide by 6.5.) 0xD: DIV7 (Divide by 7.) 0xE: DIV7_5 (Divide by 7.5.) 0xF: DIV8 (Divide by 8.) 0x10: DIV8_5 (Divide by 8.5.) 0x11: DIV9 (Divide by 9.) 0x12: DIV9_5 (Divide by 9.5.) 0x13: DIV10 (Divide by 10.) 0x14: DIV10_5 (Divide by 10.5.) 0x15: DIV11 (Divide by 11.) 0x16: DIV11_5 (Divide by 11.5.) 0x17: DIV12 (Divide by 12.) 0x18: DIV12_5 (Divide by 12.5.) 0x19: DIV13 (Divide by 13.) 0x1A: DIV13_5 (Divide by 13.5.) 0x1B: DIV14 (Divide by 14.) 0x1C: DIV14_5 (Divide by 14.5.) 0x1D: DIV15 (Divide by 15.) 0x1E: DIV15_5 (Divide by 15.5.) 0x1F: DIV16 (Divide by 16.)

0x01807000 GCC_BLSP1_QUP6_I2C_APPS_CMD_RCGR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

BLSP1 QUP6 I2C application root clock generator command register. It allows for SW to trigger the configuration data

GCC_BLSP1_QUP6_I2C_APPS_CMD_RCGR

Bits	Name	Type	Description
31	ROOT_OFF	R	This bitfield provides indication on status of the RCG; Mainly for SW to determine if the generator is actively producing clock 1: Clock Off 0: Clock On; Generator is actively producing the clock
4	DIRTY_CFG_RCGR	R	Indicates change in CFG_RCGR register configuration
1	ROOT_EN	RW	Enable Clock Generator output. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
0	UPDATE	RW	Update to the new programmed configuration. It should be cleared upon switching to new configuration 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x01807004 GCC_BLSP1_QUP6_I2C_APPS_CFG_RCGR

Type: RW

Clock: CLK

Reset State: 0x00000000

BLSP1 QUP6 I2C application root clock generator configuration register. It allows for SW to configure Root Clock Generation Macro

GCC_BLSP1_QUP6_I2C_APPS_CFG_RCGR

Bits	Name	Description
10:8	SRC_SEL	This bitfield selects clock source that will be used to source the RCG output 0x0: SRC0 (CXO.) 0x1: SRC1 (GPLL0_MAIN.) 0x2: SRC2 (Unused(Grounded).) 0x3: SRC3 (Unused(Grounded).) 0x4: SRC4 (Unused(Grounded).) 0x5: SRC5 (Unused(Grounded).) 0x6: SRC6 (Unused(Grounded).) 0x7: Reserved

GCC_BLSP1_QUP6_I2C_APPS_CFG_RCGR (cont.)

Bits	Name	Description
4:0	SRC_DIV	This field defines the integer and half integer portion of the divider setting of the RCGR. The value specifies is $2N-1$ where N is the divide ratio. For example, if the half integer divide ration is 2.5, the src_div should be set at 00100. 0x0: BYPASS (Bypsss.) 0x1: DIV1 (Divide by 1.) 0x2: DIV1_5 (Divide by 1.5.) 0x3: DIV2 (Divide by 2.) 0x4: DIV2_5 (Divide by 2.5.) 0x5: DIV3 (Divide by 3.) 0x6: DIV3_5 (Divide by 3.5.) 0x7: DIV4 (Divide by 4.) 0x8: DIV4_5 (Divide by 4.5.) 0x9: DIV5 (Divide by 5.) 0xA: DIV5_5 (Divide by 5.5.) 0xB: DIV6 (Divide by 6.) 0xC: DIV6_5 (Divide by 6.5.) 0xD: DIV7 (Divide by 7.) 0xE: DIV7_5 (Divide by 7.5.) 0xF: DIV8 (Divide by 8.) 0x10: DIV8_5 (Divide by 8.5.) 0x11: DIV9 (Divide by 9.) 0x12: DIV9_5 (Divide by 9.5.) 0x13: DIV10 (Divide by 10.) 0x14: DIV10_5 (Divide by 10.5.) 0x15: DIV11 (Divide by 11.) 0x16: DIV11_5 (Divide by 11.5.) 0x17: DIV12 (Divide by 12.) 0x18: DIV12_5 (Divide by 12.5.) 0x19: DIV13 (Divide by 13.) 0x1A: DIV13_5 (Divide by 13.5.) 0x1B: DIV14 (Divide by 14.) 0x1C: DIV14_5 (Divide by 14.5.) 0x1D: DIV15 (Divide by 15.) 0x1E: DIV15_5 (Divide by 15.5.) 0x1F: DIV16 (Divide by 16.)

0x01802024 GCC_BLSP1_QUP1_SPI_APPS_CMD_RCGR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

BLSP1 QUP1 SPI application root clock generator command register. It allows for SW to trigger the configuration data

GCC_BLSP1_QUP1_SPI_APPS_CMD_RCGR

Bits	Name	Type	Description
31	ROOT_OFF	R	This bitfield provides indication on status of the RCG; Mainly for SW to determine if the generator is actively producing clock 1: Clock Off 0: Clock On; Generator is actively producing the clock
7	DIRTY_D	R	Indicates change in D register configuration
6	DIRTY_M	R	Indicates change in M register configuration
5	DIRTY_N	R	Indicates change in N register configuration
4	DIRTY_CFG_RCGR	R	Indicates change in CFG_RCGR register configuration
1	ROOT_EN	RW	Enable Clock Generator output. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
0	UPDATE	RW	Update to the new programmed configuration. It should be cleared upon switching to new configuration 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x01802028 GCC_BLSP1_QUP1_SPI_APPS_CFG_RCGR

Type: RW

Clock: CLK

Reset State: 0x00000000

BLSP1 QUP1 SPI application root clock generator configuration register. It allows for SW to configure Root Clock Generation Macro

GCC_BLSP1_QUP1_SPI_APPS_CFG_RCGR

Bits	Name	Description
13:12	MODE	This bitfield provides SW selection of MN counter mode. 0x3: SINGLE_EDGE (Single-edge.) 0x2: DUAL_EDGE (Dual-edge.) 0x1: SWALLOW (Swallow.) 0x0: BYPASS (Bypass.)

GCC_BLSP1_QUP1_SPI_APPS_CFG_RCGR (cont.)

Bits	Name	Description
10:8	SRC_SEL	This bitfield selects clock source that will be used to source the RCG output 0x0: SRC0 (CXO.) 0x1: SRC1 (GPLL0_MAIN.) 0x2: SRC2 (Unused(Grounded).) 0x3: SRC3 (Unused(Grounded).) 0x4: SRC4 (Unused(Grounded).) 0x5: SRC5 (Unused(Grounded).) 0x6: SRC6 (Unused(Grounded).) 0x7: Reserved

GCC_BLSP1_QUP1_SPI_APPS_CFG_RCGR (cont.)

Bits	Name	Description
4:0	SRC_DIV	This field defines the integer and half integer portion of the divider setting of the RCGR. The value specifies is $2N-1$ where N is the divide ratio. For example, if the half integer divide ration is 2.5, the src_div should be set at 00100. 0x0: BYPASS (Bypsss.) 0x1: DIV1 (Divide by 1.) 0x2: DIV1_5 (Divide by 1.5.) 0x3: DIV2 (Divide by 2.) 0x4: DIV2_5 (Divide by 2.5.) 0x5: DIV3 (Divide by 3.) 0x6: DIV3_5 (Divide by 3.5.) 0x7: DIV4 (Divide by 4.) 0x8: DIV4_5 (Divide by 4.5.) 0x9: DIV5 (Divide by 5.) 0xA: DIV5_5 (Divide by 5.5.) 0xB: DIV6 (Divide by 6.) 0xC: DIV6_5 (Divide by 6.5.) 0xD: DIV7 (Divide by 7.) 0xE: DIV7_5 (Divide by 7.5.) 0xF: DIV8 (Divide by 8.) 0x10: DIV8_5 (Divide by 8.5.) 0x11: DIV9 (Divide by 9.) 0x12: DIV9_5 (Divide by 9.5.) 0x13: DIV10 (Divide by 10.) 0x14: DIV10_5 (Divide by 10.5.) 0x15: DIV11 (Divide by 11.) 0x16: DIV11_5 (Divide by 11.5.) 0x17: DIV12 (Divide by 12.) 0x18: DIV12_5 (Divide by 12.5.) 0x19: DIV13 (Divide by 13.) 0x1A: DIV13_5 (Divide by 13.5.) 0x1B: DIV14 (Divide by 14.) 0x1C: DIV14_5 (Divide by 14.5.) 0x1D: DIV15 (Divide by 15.) 0x1E: DIV15_5 (Divide by 15.5.) 0x1F: DIV16 (Divide by 16.)

0x0180202C GCC_BLSP1_QUP1_SPI_APPS_M**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

BLSP1 QUP1 SPI application root clock generator M value register. This register houses the M value for MND divider.

GCC_BLSP1_QUP1_SPI_APPS_M

Bits	Name	Description
7:0	M	This bitfield represents M Value of the M/N:D Counter

0x01802030 GCC_BLSP1_QUP1_SPI_APPS_N

Type: RW

Clock: CLK

Reset State: 0x00000000

BLSP1 QUP1 SPI application root clock generator N value register. This register houses the NOT(N-M) value for MND divider.

GCC_BLSP1_QUP1_SPI_APPS_N

Bits	Name	Description
7:0	NOT_N_MINUS_M	This bitfield represents NOT(N-M) Value of the M/N:D Counter

0x01802034 GCC_BLSP1_QUP1_SPI_APPS_D

Type: RW

Clock: CLK

Reset State: 0x00000000

BLSP1 QUP1 SPI application root clock generator D value register. This register houses the NOT 2D value for MND divider.

GCC_BLSP1_QUP1_SPI_APPS_D

Bits	Name	Description
7:0	NOT_2D	This bitfield represents NOT(2*D) Value of the M/N:D Counter

0x01802038 GCC_BLSP1_UART1_BCR

Type: RW

Clock: CLK

Reset State: 0x00000000

BLSP1 UART1 block control register

GCC_BLSP1_UART1_BCR

Bits	Name	Description
0	BLK_ARES	This bitfield controls asynchronous software reset for the block. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x0180203C GCC_BLSP1_UART1_APPS_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

BLSP1 UART1 application clock branch control register. It allows SW control of CBC soft macro

GCC_BLSP1_UART1_APPS_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	This bitfield indicated if the clock branch is halted 0:Clock On 1:Clock Off
0	CLK_ENABLE	RW	This bitfield indicated if the clock branch is enabled. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x01802040 GCC_BLSP1_UART1_SIM_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

BLSP1 UART1 SIM clock branch control register. It allows SW control of CBC soft macro

GCC_BLSP1_UART1_SIM_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	This bitfield indicated if the clock branch is halted 0:Clock On 1:Clock Off
0	CLK_ENABLE	RW	This bitfield indicated if the clock branch is enabled. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x01802044 GCC_BLSP1_UART1_APPS_CMD_RCGR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

BLSP1 UART1 application root clock generator command register. It allows for SW to trigger the configuration data

GCC_BLSP1_UART1_APPS_CMD_RCGR

Bits	Name	Type	Description
31	ROOT_OFF	R	This bitfield provides indication on status of the RCG; Mainly for SW to determine if the generator is actively producing clock 1: Clock Off 0: Clock On; Generator is actively producing the clock
7	DIRTY_D	R	Indicates change in D register configuration
6	DIRTY_M	R	Indicates change in M register configuration
5	DIRTY_N	R	Indicates change in N register configuration
4	DIRTY_CFG_RCGR	R	Indicates change in CFG_RCGR register configuration
1	ROOT_EN	RW	Enable Clock Generator output. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
0	UPDATE	RW	Update to the new programmed configuration. It should be cleared upon switching to new configuration 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x01802048 GCC_BLSP1_UART1_APPS_CFG_RCGR**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

BLSP1 UART1 application root clock generator configuration register. It allows for SW to configure Root Clock Generation Macro

GCC_BLSP1_UART1_APPS_CFG_RCGR

Bits	Name	Description
13:12	MODE	This bitfield provides SW selection of MN counter mode. 0x3: SINGLE_EDGE (Single-edge.) 0x2: DUAL_EDGE (Dual-edge.) 0x1: SWALLOW (Swallow.) 0x0: BYPASS (Bypass.)
10:8	SRC_SEL	This bitfield selects clock source that will be used to source the RCG output 0x0: SRC0 (CXO.) 0x1: SRC1 (GPLL0_MAIN.) 0x2: SRC2 (Unused(Grounded).) 0x3: SRC3 (Unused(Grounded).) 0x4: SRC4 (Unused(Grounded).) 0x5: SRC5 (Unused(Grounded).) 0x6: SRC6 (Unused(Grounded).) 0x7: Reserved

GCC_BLSP1_UART1_APPS_CFG_RCGR (cont.)

Bits	Name	Description
4:0	SRC_DIV	This field defines the integer and half integer portion of the divider setting of the RCGR. The value specifies is $2N-1$ where N is the divide ratio. For example, if the half integer divide ration is 2.5, the src_div should be set at 00100. 0x0: BYPASS (Bypsss.) 0x1: DIV1 (Divide by 1.) 0x2: DIV1_5 (Divide by 1.5.) 0x3: DIV2 (Divide by 2.) 0x4: DIV2_5 (Divide by 2.5.) 0x5: DIV3 (Divide by 3.) 0x6: DIV3_5 (Divide by 3.5.) 0x7: DIV4 (Divide by 4.) 0x8: DIV4_5 (Divide by 4.5.) 0x9: DIV5 (Divide by 5.) 0xA: DIV5_5 (Divide by 5.5.) 0xB: DIV6 (Divide by 6.) 0xC: DIV6_5 (Divide by 6.5.) 0xD: DIV7 (Divide by 7.) 0xE: DIV7_5 (Divide by 7.5.) 0xF: DIV8 (Divide by 8.) 0x10: DIV8_5 (Divide by 8.5.) 0x11: DIV9 (Divide by 9.) 0x12: DIV9_5 (Divide by 9.5.) 0x13: DIV10 (Divide by 10.) 0x14: DIV10_5 (Divide by 10.5.) 0x15: DIV11 (Divide by 11.) 0x16: DIV11_5 (Divide by 11.5.) 0x17: DIV12 (Divide by 12.) 0x18: DIV12_5 (Divide by 12.5.) 0x19: DIV13 (Divide by 13.) 0x1A: DIV13_5 (Divide by 13.5.) 0x1B: DIV14 (Divide by 14.) 0x1C: DIV14_5 (Divide by 14.5.) 0x1D: DIV15 (Divide by 15.) 0x1E: DIV15_5 (Divide by 15.5.) 0x1F: DIV16 (Divide by 16.)

0x0180204C GCC_BLSP1_UART1_APPS_M**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

BLSP1 UART1 application root clock generator M value register. This register houses the M value for MND divider.

GCC_BLSP1_UART1_APPS_M

Bits	Name	Description
15:0	M	This bitfield represents M Value of the M/N:D Counter

0x01802050 GCC_BLSP1_UART1_APPS_N

Type: RW

Clock: CLK

Reset State: 0x00000000

BLSP1 UART1 application root clock generator N value register. This register houses the NOT(N-M) value for MND divider.

GCC_BLSP1_UART1_APPS_N

Bits	Name	Description
15:0	NOT_N_MINUS_M	This bitfield represents NOT(N-M) Value of the M/N:D Counter

0x01802054 GCC_BLSP1_UART1_APPS_D

Type: RW

Clock: CLK

Reset State: 0x00000000

BLSP1 UART1 application root clock generator D value register. This register houses the NOT 2D value for MND divider.

GCC_BLSP1_UART1_APPS_D

Bits	Name	Description
15:0	NOT_2D	This bitfield represents NOT(2*D) Value of the M/N:D Counter

0x01803008 GCC_BLSP1_QUP2_BCR

Type: RW

Clock: CLK

Reset State: 0x00000000

BLSP1 QUP2 block control register

GCC_BLSP1_QUP2_BCR

Bits	Name	Description
0	BLK_ARES	This bitfield controls asynchronous software reset for the block. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x0180300C GCC_BLSP1_QUP2_SPI_APPS_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

BLSP1 QUP2 SPI application clock branch control register. It allows SW control of CBC soft macro

GCC_BLSP1_QUP2_SPI_APPS_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	This bitfield indicated if the clock branch is halted 0:Clock On 1:Clock Off
0	CLK_ENABLE	RW	This bitfield indicated if the clock branch is enabled. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x01803010 GCC_BLSP1_QUP2_I2C_APPS_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

BLSP1 QUP2 I2C application clock branch control register. It allows SW control of CBC soft macro

GCC_BLSP1_QUP2_I2C_APPS_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	This bitfield indicated if the clock branch is halted 0:Clock On 1:Clock Off
0	CLK_ENABLE	RW	This bitfield indicated if the clock branch is enabled. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x01803014 GCC_BLSP1_QUP2_SPI_APPS_CMD_RCGR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

BLSP1 QUP2 SPI application root clock generator command register. It allows for SW to trigger the configuration data

GCC_BLSP1_QUP2_SPI_APPS_CMD_RCGR

Bits	Name	Type	Description
31	ROOT_OFF	R	This bitfield provides indication on status of the RCG; Mainly for SW to determine if the generator is actively producing clock 1: Clock Off 0: Clock On; Generator is actively producing the clock
7	DIRTY_D	R	Indicates change in D register configuration
6	DIRTY_M	R	Indicates change in M register configuration
5	DIRTY_N	R	Indicates change in N register configuration
4	DIRTY_CFG_RCGR	R	Indicates change in CFG_RCGR register configuration
1	ROOT_EN	RW	Enable Clock Generator output. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
0	UPDATE	RW	Update to the new programmed configuration. It should be cleared upon switching to new configuration 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x01803018 GCC_BLSP1_QUP2_SPI_APPS_CFG_RCGR**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

BLSP1 QUP2 SPI application root clock generator configuration register. It allows for SW to configure Root Clock Generation Macro

GCC_BLSP1_QUP2_SPI_APPS_CFG_RCGR

Bits	Name	Description
13:12	MODE	This bitfield provides SW selection of MN counter mode. 0x3: SINGLE_EDGE (Single-edge.) 0x2: DUAL_EDGE (Dual-edge.) 0x1: SWALLOW (Swallow.) 0x0: BYPASS (Bypass.)
10:8	SRC_SEL	This bitfield selects clock source that will be used to source the RCG output 0x0: SRC0 (CXO.) 0x1: SRC1 (GPLL0_MAIN.) 0x2: SRC2 (Unused(Grounded).) 0x3: SRC3 (Unused(Grounded).) 0x4: SRC4 (Unused(Grounded).) 0x5: SRC5 (Unused(Grounded).) 0x6: SRC6 (Unused(Grounded).) 0x7: Reserved

GCC_BLSP1_QUP2_SPI_APPS_CFG_RCGR (cont.)

Bits	Name	Description
4:0	SRC_DIV	This field defines the integer and half integer portion of the divider setting of the RCGR. The value specifies is $2N-1$ where N is the divide ratio. For example, if the half integer divide ration is 2.5, the src_div should be set at 00100. 0x0: BYPASS (Bypsss.) 0x1: DIV1 (Divide by 1.) 0x2: DIV1_5 (Divide by 1.5.) 0x3: DIV2 (Divide by 2.) 0x4: DIV2_5 (Divide by 2.5.) 0x5: DIV3 (Divide by 3.) 0x6: DIV3_5 (Divide by 3.5.) 0x7: DIV4 (Divide by 4.) 0x8: DIV4_5 (Divide by 4.5.) 0x9: DIV5 (Divide by 5.) 0xA: DIV5_5 (Divide by 5.5.) 0xB: DIV6 (Divide by 6.) 0xC: DIV6_5 (Divide by 6.5.) 0xD: DIV7 (Divide by 7.) 0xE: DIV7_5 (Divide by 7.5.) 0xF: DIV8 (Divide by 8.) 0x10: DIV8_5 (Divide by 8.5.) 0x11: DIV9 (Divide by 9.) 0x12: DIV9_5 (Divide by 9.5.) 0x13: DIV10 (Divide by 10.) 0x14: DIV10_5 (Divide by 10.5.) 0x15: DIV11 (Divide by 11.) 0x16: DIV11_5 (Divide by 11.5.) 0x17: DIV12 (Divide by 12.) 0x18: DIV12_5 (Divide by 12.5.) 0x19: DIV13 (Divide by 13.) 0x1A: DIV13_5 (Divide by 13.5.) 0x1B: DIV14 (Divide by 14.) 0x1C: DIV14_5 (Divide by 14.5.) 0x1D: DIV15 (Divide by 15.) 0x1E: DIV15_5 (Divide by 15.5.) 0x1F: DIV16 (Divide by 16.)

0x0180301C GCC_BLSP1_QUP2_SPI_APPS_M**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

BLSP1 QUP2 SPI application root clock generator M value register. This register houses the M value for MND divider.

GCC_BLSP1_QUP2_SPI_APPS_M

Bits	Name	Description
7:0	M	This bitfield represents M Value of the M/N:D Counter

0x01803020 GCC_BLSP1_QUP2_SPI_APPS_N

Type: RW

Clock: CLK

Reset State: 0x00000000

BLSP1 QUP2 SPI application root clock generator N value register. This register houses the NOT(N-M) value for MND divider.

GCC_BLSP1_QUP2_SPI_APPS_N

Bits	Name	Description
7:0	NOT_N_MINUS_M	This bitfield represents NOT(N-M) Value of the M/N:D Counter

0x01803024 GCC_BLSP1_QUP2_SPI_APPS_D

Type: RW

Clock: CLK

Reset State: 0x00000000

BLSP1 QUP2 SPI application root clock generator D value register. This register houses the NOT 2D value for MND divider.

GCC_BLSP1_QUP2_SPI_APPS_D

Bits	Name	Description
7:0	NOT_2D	This bitfield represents NOT(2*D) Value of the M/N:D Counter

0x01803028 GCC_BLSP1_UART2_BCR

Type: RW

Clock: CLK

Reset State: 0x00000000

BLSP1 UART2 block control register

GCC_BLSP1_UART2_BCR

Bits	Name	Description
0	BLK_ARES	This bitfield controls asynchronous software reset for the block. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x0180302C GCC_BLSP1_UART2_APPS_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

BLSP1 UART2 application clock branch control register. It allows SW control of CBC soft macro

GCC_BLSP1_UART2_APPS_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	This bitfield indicated if the clock branch is halted 0:Clock On 1:Clock Off
0	CLK_ENABLE	RW	This bitfield indicated if the clock branch is enabled. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x01803030 GCC_BLSP1_UART2_SIM_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

BLSP1 UART2 SIM clock branch control register. It allows SW control of CBC soft macro

GCC_BLSP1_UART2_SIM_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	This bitfield indicated if the clock branch is halted 0:Clock On 1:Clock Off
0	CLK_ENABLE	RW	This bitfield indicated if the clock branch is enabled. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x01803034 GCC_BLSP1_UART2_APPS_CMD_RCGR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

BLSP1 UART2 application root clock generator command register. It allows for SW to trigger the configuration data

GCC_BLSP1_UART2_APPS_CMD_RCGR

Bits	Name	Type	Description
31	ROOT_OFF	R	This bitfield provides indication on status of the RCG; Mainly for SW to determine if the generator is actively producing clock 1: Clock Off 0: Clock On; Generator is actively producing the clock
7	DIRTY_D	R	Indicates change in D register configuration
6	DIRTY_M	R	Indicates change in M register configuration
5	DIRTY_N	R	Indicates change in N register configuration
4	DIRTY_CFG_RCGR	R	Indicates change in CFG_RCGR register configuration
1	ROOT_EN	RW	Enable Clock Generator output. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
0	UPDATE	RW	Update to the new programmed configuration. It should be cleared upon switching to new configuration 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x01803038 GCC_BLSP1_UART2_APPS_CFG_RCGR**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

BLSP1 UART2 application root clock generator configuration register. It allows for SW to configure Root Clock Generation Macro

GCC_BLSP1_UART2_APPS_CFG_RCGR

Bits	Name	Description
13:12	MODE	This bitfield provides SW selection of MN counter mode. 0x3: SINGLE_EDGE (Single-edge.) 0x2: DUAL_EDGE (Dual-edge.) 0x1: SWALLOW (Swallow.) 0x0: BYPASS (Bypass.)
10:8	SRC_SEL	This bitfield selects clock source that will be used to source the RCG output 0x0: SRC0 (CXO.) 0x1: SRC1 (GPLL0_MAIN.) 0x2: SRC2 (Unused(Grounded).) 0x3: SRC3 (Unused(Grounded).) 0x4: SRC4 (Unused(Grounded).) 0x5: SRC5 (Unused(Grounded).) 0x6: SRC6 (Unused(Grounded).) 0x7: Reserved

GCC_BLSP1_UART2_APPS_CFG_RCGR (cont.)

Bits	Name	Description
4:0	SRC_DIV	This field defines the integer and half integer portion of the divider setting of the RCGR. The value specifies is $2N-1$ where N is the divide ratio. For example, if the half integer divide ration is 2.5, the src_div should be set at 00100. 0x0: BYPASS (Bypsss.) 0x1: DIV1 (Divide by 1.) 0x2: DIV1_5 (Divide by 1.5.) 0x3: DIV2 (Divide by 2.) 0x4: DIV2_5 (Divide by 2.5.) 0x5: DIV3 (Divide by 3.) 0x6: DIV3_5 (Divide by 3.5.) 0x7: DIV4 (Divide by 4.) 0x8: DIV4_5 (Divide by 4.5.) 0x9: DIV5 (Divide by 5.) 0xA: DIV5_5 (Divide by 5.5.) 0xB: DIV6 (Divide by 6.) 0xC: DIV6_5 (Divide by 6.5.) 0xD: DIV7 (Divide by 7.) 0xE: DIV7_5 (Divide by 7.5.) 0xF: DIV8 (Divide by 8.) 0x10: DIV8_5 (Divide by 8.5.) 0x11: DIV9 (Divide by 9.) 0x12: DIV9_5 (Divide by 9.5.) 0x13: DIV10 (Divide by 10.) 0x14: DIV10_5 (Divide by 10.5.) 0x15: DIV11 (Divide by 11.) 0x16: DIV11_5 (Divide by 11.5.) 0x17: DIV12 (Divide by 12.) 0x18: DIV12_5 (Divide by 12.5.) 0x19: DIV13 (Divide by 13.) 0x1A: DIV13_5 (Divide by 13.5.) 0x1B: DIV14 (Divide by 14.) 0x1C: DIV14_5 (Divide by 14.5.) 0x1D: DIV15 (Divide by 15.) 0x1E: DIV15_5 (Divide by 15.5.) 0x1F: DIV16 (Divide by 16.)

0x0180303C GCC_BLSP1_UART2_APPS_M**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

BLSP1 UART2 application root clock generator M value register. This register houses the M value for MND divider.

GCC_BLSP1_UART2_APPS_M

Bits	Name	Description
15:0	M	This bitfield represents M Value of the M/N:D Counter

0x01803040 GCC_BLSP1_UART2_APPS_N

Type: RW

Clock: CLK

Reset State: 0x00000000

BLSP1 UART2 application root clock generator N value register. This register houses the NOT(N-M) value for MND divider.

GCC_BLSP1_UART2_APPS_N

Bits	Name	Description
15:0	NOT_N_MINUS_M	This bitfield represents NOT(N-M) Value of the M/N:D Counter

0x01803044 GCC_BLSP1_UART2_APPS_D

Type: RW

Clock: CLK

Reset State: 0x00000000

BLSP1 UART2 application root clock generator D value register. This register houses the NOT 2D value for MND divider.

GCC_BLSP1_UART2_APPS_D

Bits	Name	Description
15:0	NOT_2D	This bitfield represents NOT(2*D) Value of the M/N:D Counter

0x01804018 GCC_BLSP1_QUP3_BCR

Type: RW

Clock: CLK

Reset State: 0x00000000

BLSP1 QUP3 block control register

GCC_BLSP1_QUP3_BCR

Bits	Name	Description
0	BLK_ARES	This bitfield controls asynchronous software reset for the block. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x0180401C GCC_BLSP1_QUP3_SPI_APPS_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

BLSP1 QUP3 SPI application clock branch control register. It allows SW control of CBC soft macro

GCC_BLSP1_QUP3_SPI_APPS_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	This bitfield indicated if the clock branch is halted 0:Clock On 1:Clock Off
0	CLK_ENABLE	RW	This bitfield indicated if the clock branch is enabled. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x01804020 GCC_BLSP1_QUP3_I2C_APPS_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

BLSP1 QUP3 I2C application clock branch control register. It allows SW control of CBC soft macro

GCC_BLSP1_QUP3_I2C_APPS_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	This bitfield indicated if the clock branch is halted 0:Clock On 1:Clock Off
0	CLK_ENABLE	RW	This bitfield indicated if the clock branch is enabled. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x01804024 GCC_BLSP1_QUP3_SPI_APPS_CMD_RCGR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

BLSP1 QUP3 SPI application root clock generator command register. It allows for SW to trigger the configuration data

GCC_BLSP1_QUP3_SPI_APPS_CMD_RCGR

Bits	Name	Type	Description
31	ROOT_OFF	R	This bitfield provides indication on status of the RCG; Mainly for SW to determine if the generator is actively producing clock 1: Clock Off 0: Clock On; Generator is actively producing the clock
7	DIRTY_D	R	Indicates change in D register configuration
6	DIRTY_M	R	Indicates change in M register configuration
5	DIRTY_N	R	Indicates change in N register configuration
4	DIRTY_CFG_RCGR	R	Indicates change in CFG_RCGR register configuration
1	ROOT_EN	RW	Enable Clock Generator output. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
0	UPDATE	RW	Update to the new programmed configuration. It should be cleared upon switching to new configuration 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x01804028 GCC_BLSP1_QUP3_SPI_APPS_CFG_RCGR**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

BLSP1 QUP3 SPI application root clock generator configuration register. It allows for SW to configure Root Clock Generation Macro

GCC_BLSP1_QUP3_SPI_APPS_CFG_RCGR

Bits	Name	Description
13:12	MODE	This bitfield provides SW selection of MN counter mode. 0x3: SINGLE_EDGE (Single-edge.) 0x2: DUAL_EDGE (Dual-edge.) 0x1: SWALLOW (Swallow.) 0x0: BYPASS (Bypass.)
10:8	SRC_SEL	This bitfield selects clock source that will be used to source the RCG output 0x0: SRC0 (CXO.) 0x1: SRC1 (GPLL0_MAIN.) 0x2: SRC2 (Unused(Grounded).) 0x3: SRC3 (Unused(Grounded).) 0x4: SRC4 (Unused(Grounded).) 0x5: SRC5 (Unused(Grounded).) 0x6: SRC6 (Unused(Grounded).) 0x7: Reserved

GCC_BLSP1_QUP3_SPI_APPS_CFG_RCGR (cont.)

Bits	Name	Description
4:0	SRC_DIV	This field defines the integer and half integer portion of the divider setting of the RCGR. The value specifies is $2N-1$ where N is the divide ratio. For example, if the half integer divide ration is 2.5, the src_div should be set at 00100. 0x0: BYPASS (Bypsss.) 0x1: DIV1 (Divide by 1.) 0x2: DIV1_5 (Divide by 1.5.) 0x3: DIV2 (Divide by 2.) 0x4: DIV2_5 (Divide by 2.5.) 0x5: DIV3 (Divide by 3.) 0x6: DIV3_5 (Divide by 3.5.) 0x7: DIV4 (Divide by 4.) 0x8: DIV4_5 (Divide by 4.5.) 0x9: DIV5 (Divide by 5.) 0xA: DIV5_5 (Divide by 5.5.) 0xB: DIV6 (Divide by 6.) 0xC: DIV6_5 (Divide by 6.5.) 0xD: DIV7 (Divide by 7.) 0xE: DIV7_5 (Divide by 7.5.) 0xF: DIV8 (Divide by 8.) 0x10: DIV8_5 (Divide by 8.5.) 0x11: DIV9 (Divide by 9.) 0x12: DIV9_5 (Divide by 9.5.) 0x13: DIV10 (Divide by 10.) 0x14: DIV10_5 (Divide by 10.5.) 0x15: DIV11 (Divide by 11.) 0x16: DIV11_5 (Divide by 11.5.) 0x17: DIV12 (Divide by 12.) 0x18: DIV12_5 (Divide by 12.5.) 0x19: DIV13 (Divide by 13.) 0x1A: DIV13_5 (Divide by 13.5.) 0x1B: DIV14 (Divide by 14.) 0x1C: DIV14_5 (Divide by 14.5.) 0x1D: DIV15 (Divide by 15.) 0x1E: DIV15_5 (Divide by 15.5.) 0x1F: DIV16 (Divide by 16.)

0x0180402C GCC_BLSP1_QUP3_SPI_APPS_M**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

BLSP1 QUP3 SPI application root clock generator M value register. This register houses the M value for MND divider.

GCC_BLSP1_QUP3_SPI_APPS_M

Bits	Name	Description
7:0	M	This bitfield represents M Value of the M/N:D Counter

0x01804030 GCC_BLSP1_QUP3_SPI_APPS_N

Type: RW

Clock: CLK

Reset State: 0x00000000

BLSP1 QUP3 SPI application root clock generator N value register. This register houses the NOT(N-M) value for MND divider.

GCC_BLSP1_QUP3_SPI_APPS_N

Bits	Name	Description
7:0	NOT_N_MINUS_M	This bitfield represents NOT(N-M) Value of the M/N:D Counter

0x01804034 GCC_BLSP1_QUP3_SPI_APPS_D

Type: RW

Clock: CLK

Reset State: 0x00000000

BLSP1 QUP3 SPI application root clock generator D value register. This register houses the NOT 2D value for MND divider.

GCC_BLSP1_QUP3_SPI_APPS_D

Bits	Name	Description
7:0	NOT_2D	This bitfield represents NOT(2*D) Value of the M/N:D Counter

0x01805018 GCC_BLSP1_QUP4_BCR

Type: RW

Clock: CLK

Reset State: 0x00000000

BLSP1 QUP4 block control register

GCC_BLSP1_QUP4_BCR

Bits	Name	Description
0	BLK_ARES	This bitfield controls asynchronous software reset for the block. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x0180501C GCC_BLSP1_QUP4_SPI_APPS_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

BLSP1 QUP4 SPI application clock branch control register. It allows SW control of CBC soft macro

GCC_BLSP1_QUP4_SPI_APPS_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	This bitfield indicated if the clock branch is halted 0:Clock On 1:Clock Off
0	CLK_ENABLE	RW	This bitfield indicated if the clock branch is enabled. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x01805020 GCC_BLSP1_QUP4_I2C_APPS_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

BLSP1 QUP4 I2C application clock branch control register. It allows SW control of CBC soft macro

GCC_BLSP1_QUP4_I2C_APPS_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	This bitfield indicated if the clock branch is halted 0:Clock On 1:Clock Off
0	CLK_ENABLE	RW	This bitfield indicated if the clock branch is enabled. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x01805024 GCC_BLSP1_QUP4_SPI_APPS_CMD_RCGR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

BLSP1 QUP4 SPI application root clock generator command register. It allows for SW to trigger the configuration data

GCC_BLSP1_QUP4_SPI_APPS_CMD_RCGR

Bits	Name	Type	Description
31	ROOT_OFF	R	This bitfield provides indication on status of the RCG; Mainly for SW to determine if the generator is actively producing clock 1: Clock Off 0: Clock On; Generator is actively producing the clock
7	DIRTY_D	R	Indicates change in D register configuration
6	DIRTY_M	R	Indicates change in M register configuration
5	DIRTY_N	R	Indicates change in N register configuration
4	DIRTY_CFG_RCGR	R	Indicates change in CFG_RCGR register configuration
1	ROOT_EN	RW	Enable Clock Generator output. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
0	UPDATE	RW	Update to the new programmed configuration. It should be cleared upon switching to new configuration 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x01805028 GCC_BLSP1_QUP4_SPI_APPS_CFG_RCGR**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

BLSP1 QUP4 SPI application root clock generator configuration register. It allows for SW to configure Root Clock Generation Macro

GCC_BLSP1_QUP4_SPI_APPS_CFG_RCGR

Bits	Name	Description
13:12	MODE	This bitfield provides SW selection of MN counter mode. 0x3: SINGLE_EDGE (Single-edge.) 0x2: DUAL_EDGE (Dual-edge.) 0x1: SWALLOW (Swallow.) 0x0: BYPASS (Bypass.)
10:8	SRC_SEL	This bitfield selects clock source that will be used to source the RCG output 0x0: SRC0 (CXO.) 0x1: SRC1 (GPLL0_MAIN.) 0x2: SRC2 (Unused(Grounded).) 0x3: SRC3 (Unused(Grounded).) 0x4: SRC4 (Unused(Grounded).) 0x5: SRC5 (Unused(Grounded).) 0x6: SRC6 (Unused(Grounded).) 0x7: Reserved

GCC_BLSP1_QUP4_SPI_APPS_CFG_RCGR (cont.)

Bits	Name	Description
4:0	SRC_DIV	This field defines the integer and half integer portion of the divider setting of the RCGR. The value specifies is $2N-1$ where N is the divide ratio. For example, if the half integer divide ration is 2.5, the src_div should be set at 00100. 0x0: BYPASS (Bypsss.) 0x1: DIV1 (Divide by 1.) 0x2: DIV1_5 (Divide by 1.5.) 0x3: DIV2 (Divide by 2.) 0x4: DIV2_5 (Divide by 2.5.) 0x5: DIV3 (Divide by 3.) 0x6: DIV3_5 (Divide by 3.5.) 0x7: DIV4 (Divide by 4.) 0x8: DIV4_5 (Divide by 4.5.) 0x9: DIV5 (Divide by 5.) 0xA: DIV5_5 (Divide by 5.5.) 0xB: DIV6 (Divide by 6.) 0xC: DIV6_5 (Divide by 6.5.) 0xD: DIV7 (Divide by 7.) 0xE: DIV7_5 (Divide by 7.5.) 0xF: DIV8 (Divide by 8.) 0x10: DIV8_5 (Divide by 8.5.) 0x11: DIV9 (Divide by 9.) 0x12: DIV9_5 (Divide by 9.5.) 0x13: DIV10 (Divide by 10.) 0x14: DIV10_5 (Divide by 10.5.) 0x15: DIV11 (Divide by 11.) 0x16: DIV11_5 (Divide by 11.5.) 0x17: DIV12 (Divide by 12.) 0x18: DIV12_5 (Divide by 12.5.) 0x19: DIV13 (Divide by 13.) 0x1A: DIV13_5 (Divide by 13.5.) 0x1B: DIV14 (Divide by 14.) 0x1C: DIV14_5 (Divide by 14.5.) 0x1D: DIV15 (Divide by 15.) 0x1E: DIV15_5 (Divide by 15.5.) 0x1F: DIV16 (Divide by 16.)

0x0180502C GCC_BLSP1_QUP4_SPI_APPS_M**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

BLSP1 QUP4 SPI application root clock generator M value register. This register houses the M value for MND divider.

GCC_BLSP1_QUP4_SPI_APPS_M

Bits	Name	Description
7:0	M	This bitfield represents M Value of the M/N:D Counter

0x01805030 GCC_BLSP1_QUP4_SPI_APPS_N

Type: RW

Clock: CLK

Reset State: 0x00000000

BLSP1 QUP4 SPI application root clock generator N value register. This register houses the NOT(N-M) value for MND divider.

GCC_BLSP1_QUP4_SPI_APPS_N

Bits	Name	Description
7:0	NOT_N_MINUS_M	This bitfield represents NOT(N-M) Value of the M/N:D Counter

0x01805034 GCC_BLSP1_QUP4_SPI_APPS_D

Type: RW

Clock: CLK

Reset State: 0x00000000

BLSP1 QUP4 SPI application root clock generator D value register. This register houses the NOT 2D value for MND divider.

GCC_BLSP1_QUP4_SPI_APPS_D

Bits	Name	Description
7:0	NOT_2D	This bitfield represents NOT(2*D) Value of the M/N:D Counter

0x0182E020 GCC_SPMI_PCNOA_AHB_CBCR

Type: RW

Clock: CLK

Reset State: 0x10008000

GCC_SPMI_PCNOA_AHB_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	
30:28	NOC_HANDSHAKE_FSM_STATUS	R	
15	NOC_HANDSHAKE_FSM_ENABLE	RW	0x1: ENABLE 0x0: DISABLE

0x0182F01C GCC_SPDM_PCNOA_CY_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

SPDM Periph NOC cycle counter clock branch control register. It allows SW control of CBC soft macro

GCC_SPDM_PCNOA_CY_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	This bitfield indicated if the clock branch is halted 0:Clock On 1:Clock Off
0	CLK_ENABLE	RW	This bitfield indicated if the clock branch is enabled. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x01831020 GCC_BIMC_APSS_AXI_CBCR**Type:** R**Clock:** CLK**Reset State:** 0x00000000

BIMC APSS AXI clock branch control register. It allows SW control of CBC soft macro

GCC_BIMC_APSS_AXI_CBCR

Bits	Name	Description
31	CLK_OFF	This bitfield indicated if the clock branch is halted 0:Clock On 1:Clock Off

0x0183201C GCC_DDR_DIM_CFG_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x80008000

DDR DIM config clock branch control register. It allows SW control of CBC soft macro

GCC_DDR_DIM_CFG_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	This bitfield indicated if the clock branch is halted 0:Clock On 1:Clock Off
30:28	NOC_HANDSHAKE_FSM_STATUS	R	This bitfield is used to check the FSM state
15	NOC_HANDSHAKE_FSM_ENABLE	RW	This bitfield enables the FSM to use so that slaveready and socketcon handshake is used prior to affecting the enable of the clock branch cell. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
0	CLK_ENABLE	RW	This bitfield indicated if the clock branch is enabled. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x0181207C GCC_APSS_TCU_CMD_RCGR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

APSS TCU Aysnc Mode root clock generator command register. It allows for SW to trigger the configuration data

GCC_APSS_TCU_CMD_RCGR

Bits	Name	Type	Description
31	ROOT_OFF	R	This bitfield provides indication on status of the RCG; Mainly for SW to determine if the generator is actively producing clock 1: Clock Off 0: Clock On; Generator is actively producing the clock
4	DIRTY_CFG_RCGR	R	Indicates change in CFG_RCGR register configuration

GCC_APSS_TCU_CMD_RCGR (cont.)

Bits	Name	Type	Description
1	ROOT_EN	RW	Enable Clock Generator output. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
0	UPDATE	RW	Update to the new programmed configuration. It should be cleared upon switching to new configuration 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x01812080 GCC_APSS_TCU_CFG_RCGR**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

APSS TCU Async Mode BIMC GPU root clock generator configuration register. It allows for SW to configure Root Clock Generation Macro

GCC_APSS_TCU_CFG_RCGR

Bits	Name	Description
10:8	SRC_SEL	This bitfield selects clock source that will be used to source the RCG output 0x0: SRC0 (CXO.) 0x1: SRC1 (GPLL0_MAIN) 0x2: SRC2 (BIMC_PLL_AUX.) 0x3: SRC3 (GPLL1_AUX.) 0x4: SRC4 (Unused(Grounded).) 0x5: SRC5 (Unused(Grounded).) 0x6: SRC6 (Unused(Grounded).) 0x7: Reserved

GCC_APSS_TCU_CFG_RCGR (cont.)

Bits	Name	Description
4:0	SRC_DIV	This field defines the integer and half integer portion of the divider setting of the RCG. The value specifies is $2N-1$ where N is the divide ratio. For example, if the half integer divide ration is 2.5, the src_div should be set at 00100. 0x0: BYPASS (Bypsss.) 0x1: DIV1 (Divide by 1.) 0x2: DIV1_5 (Divide by 1.5.) 0x3: DIV2 (Divide by 2.) 0x4: DIV2_5 (Divide by 2.5.) 0x5: DIV3 (Divide by 3.) 0x6: DIV3_5 (Divide by 3.5.) 0x7: DIV4 (Divide by 4.) 0x8: DIV4_5 (Divide by 4.5.) 0x9: DIV5 (Divide by 5.) 0xA: DIV5_5 (Divide by 5.5.) 0xB: DIV6 (Divide by 6.) 0xC: DIV6_5 (Divide by 6.5.) 0xD: DIV7 (Divide by 7.) 0xE: DIV7_5 (Divide by 7.5.) 0xF: DIV8 (Divide by 8.) 0x10: DIV8_5 (Divide by 8.5.) 0x11: DIV9 (Divide by 9.) 0x12: DIV9_5 (Divide by 9.5.) 0x13: DIV10 (Divide by 10.) 0x14: DIV10_5 (Divide by 10.5.) 0x15: DIV11 (Divide by 11.) 0x16: DIV11_5 (Divide by 11.5.) 0x17: DIV12 (Divide by 12.) 0x18: DIV12_5 (Divide by 12.5.) 0x19: DIV13 (Divide by 13.) 0x1A: DIV13_5 (Divide by 13.5.) 0x1B: DIV14 (Divide by 14.) 0x1C: DIV14_5 (Divide by 14.5.) 0x1D: DIV15 (Divide by 15.) 0x1E: DIV15_5 (Divide by 15.5.) 0x1F: DIV16 (Divide by 16.)

0x01831028 GCC_BIMC_GPU_CMD_RCGR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

BIMC GPU root clock generator command register. It allows for SW to trigger the configuration data

GCC_BIMC_GPU_CMD_RCGR

Bits	Name	Type	Description
31	ROOT_OFF	R	This bitfield provides indication on status of the RCG; Mainly for SW to determine if the generator is actively producing clock 1: Clock Off 0: Clock On; Generator is actively producing the clock
4	DIRTY_CFG_RCGR	R	Indicates change in CFG_RCGR register configuration
1	ROOT_EN	RW	Enable Clock Generator output. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
0	UPDATE	RW	Update to the new programmed configuration. It should be cleared upon switching to new configuration 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x0183102C GCC_BIMC_GPU_CFG_RCGR

Type: RW

Clock: CLK

Reset State: 0x00000000

BIMC GPU root clock generator configuration register. It allows for SW to configure Root Clock Generation Macro

GCC_BIMC_GPU_CFG_RCGR

Bits	Name	Description
10:8	SRC_SEL	This bitfield selects clock source that will be used to source the RCG output 0x0: SRC0 (CXO.) 0x1: SRC1 (GPLL0_MAIN) 0x2: SRC2 (BIMC_PLL_AUX.) 0x3: SRC3 (GPLL1_AUX.) 0x4: SRC4 (Unused(Grounded).) 0x5: SRC5 (Unused(Grounded).) 0x6: SRC6 (Unused(Grounded).) 0x7: Reserved

GCC_BIMC_GPU_CFG_RCGR (cont.)

Bits	Name	Description
4:0	SRC_DIV	This field defines the integer and half integer portion of the divider setting of the RCG. The value specifies is $2N-1$ where N is the divide ratio. For example, if the half integer divide ration is 2.5, the src_div should be set at 00100. 0x0: BYPASS (Bypsss.) 0x1: DIV1 (Divide by 1.) 0x2: DIV1_5 (Divide by 1.5.) 0x3: DIV2 (Divide by 2.) 0x4: DIV2_5 (Divide by 2.5.) 0x5: DIV3 (Divide by 3.) 0x6: DIV3_5 (Divide by 3.5.) 0x7: DIV4 (Divide by 4.) 0x8: DIV4_5 (Divide by 4.5.) 0x9: DIV5 (Divide by 5.) 0xA: DIV5_5 (Divide by 5.5.) 0xB: DIV6 (Divide by 6.) 0xC: DIV6_5 (Divide by 6.5.) 0xD: DIV7 (Divide by 7.) 0xE: DIV7_5 (Divide by 7.5.) 0xF: DIV8 (Divide by 8.) 0x10: DIV8_5 (Divide by 8.5.) 0x11: DIV9 (Divide by 9.) 0x12: DIV9_5 (Divide by 9.5.) 0x13: DIV10 (Divide by 10.) 0x14: DIV10_5 (Divide by 10.5.) 0x15: DIV11 (Divide by 11.) 0x16: DIV11_5 (Divide by 11.5.) 0x17: DIV12 (Divide by 12.) 0x18: DIV12_5 (Divide by 12.5.) 0x19: DIV13 (Divide by 13.) 0x1A: DIV13_5 (Divide by 13.5.) 0x1B: DIV14 (Divide by 14.) 0x1C: DIV14_5 (Divide by 14.5.) 0x1D: DIV15 (Divide by 15.) 0x1E: DIV15_5 (Divide by 15.5.) 0x1F: DIV16 (Divide by 16.)

0x01831040 GCC_BIMC_GPU_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

BIMC GPU clock control register

GCC_BIMC_GPU_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	This bitfield indicated if the clock branch is halted 0:Clock On 1:Clock Off
0	CLK_ENABLE	RW	This bitfield indicated if the clock branch is enabled. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x01831044 GCC_BIMC_TCU_CBCR**Type:** R**Clock:** CLK**Reset State:** 0x00000000

BIMC clock for BIMC-TCU Interface, clock branch control register. It allows SW control of CBC soft macro

GCC_BIMC_TCU_CBCR

Bits	Name	Description
31	CLK_OFF	This bitfield indicated if the clock branch is halted 0:Clock On 1:Clock Off

0x0181C000 GCC_ULTAUDIO_PCNO_C_MPORT_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

ULTAUDIO SYS_NOC MPORT clock branch control register. It allows SW control of CBC soft macro

GCC_ULTAUDIO_PCNO_C_MPORT_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	This bitfield indicated if the clock branch is halted 0:Clock On 1:Clock Off

GCC_ULTAUDIO_PCNOG_MPORT_CBCR (cont.)

Bits	Name	Type	Description
1	HW_CTL	RW	This bit is used to enable hardware dynamic gating of the clock branch. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
0	CLK_ENABLE	RW	This bitfield indicated if the clock branch is enabled. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x0181C004 GCC_ULTAUDIO_PCNOG_SWAY_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x80008000

ULTAUDIO SYS_NOG SWAY clock branch control register. It allows SW control of CBC soft macro

GCC_ULTAUDIO_PCNOG_SWAY_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	This bitfield indicated if the clock branch is halted 0:Clock On 1:Clock Off
30:28	NOC_HANDSHAKE_FSM_STATUS	R	This bitfield is used to check the FSM state
15	NOC_HANDSHAKE_FSM_ENABLE	RW	This bitfield enables the FSM to use so that slave ready and socketcon handshake is used prior to affecting the enable of the clock branch cell. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
0	CLK_ENABLE	RW	This bitfield indicated if the clock branch is enabled. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x0181C0B4 GCC_ULT_AUDIO_BCR**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

ULT_AUDIO block control register

GCC_ULT_AUDIO_BCR

Bits	Name	Description
0	BLK_ARES	This bitfield controls asynchronous software reset for the block. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x01846000 GCC_APSS_AHB_CMD_RCGR**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

APSS AHB root clock generator command register. It allows for SW to trigger the configuration data

GCC_APSS_AHB_CMD_RCGR

Bits	Name	Type	Description
31	ROOT_OFF	R	This bitfield provides indication on status of the RCG; Mainly for SW to determine if the generator is actively producing clock 1: Clock Off 0: Clock On; Generator is actively producing the clock
4	DIRTY_CFG_RCGR	R	Indicates change in CFG_RCGR register configuration
1	ROOT_EN	RW	Enable Clock Generator output. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
0	UPDATE	RW	Update to the new programmed configuration. It should be cleared upon switching to new configuration 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x01846004 GCC_APSS_AHB_CFG_RCGR**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

APSS AHB root clock generator configuration register. It allows for SW to configure Root Clock Generation Macro

GCC_APSS_AHB_CFG_RCGR

Bits	Name	Description
10:8	SRC_SEL	This bitfield selects clock source that will be used to source the RCG output 0x0: SRC0 (CXO.) 0x1: SRC1 (GPLL0_MAIN.) 0x2: SRC2 (Unused(Grounded).) 0x3: SRC3 (Unused(Grounded).) 0x4: SRC4 (Unused(Grounded).) 0x5: SRC5 (Unused(Grounded).) 0x6: SRC6 (Unused(Grounded).) 0x7: Reserved

GCC_APSS_AHB_CFG_RCGR (cont.)

Bits	Name	Description
4:0	SRC_DIV	This field defines the integer and half integer portion of the divider setting of the RCGR. The value specifies is $2N-1$ where N is the divide ratio. For example, if the half integer divide ration is 2.5, the src_div should be set at 00100. 0x0: BYPASS (Bypsss.) 0x1: DIV1 (Divide by 1.) 0x2: DIV1_5 (Divide by 1.5.) 0x3: DIV2 (Divide by 2.) 0x4: DIV2_5 (Divide by 2.5.) 0x5: DIV3 (Divide by 3.) 0x6: DIV3_5 (Divide by 3.5.) 0x7: DIV4 (Divide by 4.) 0x8: DIV4_5 (Divide by 4.5.) 0x9: DIV5 (Divide by 5.) 0xA: DIV5_5 (Divide by 5.5.) 0xB: DIV6 (Divide by 6.) 0xC: DIV6_5 (Divide by 6.5.) 0xD: DIV7 (Divide by 7.) 0xE: DIV7_5 (Divide by 7.5.) 0xF: DIV8 (Divide by 8.) 0x10: DIV8_5 (Divide by 8.5.) 0x11: DIV9 (Divide by 9.) 0x12: DIV9_5 (Divide by 9.5.) 0x13: DIV10 (Divide by 10.) 0x14: DIV10_5 (Divide by 10.5.) 0x15: DIV11 (Divide by 11.) 0x16: DIV11_5 (Divide by 11.5.) 0x17: DIV12 (Divide by 12.) 0x18: DIV12_5 (Divide by 12.5.) 0x19: DIV13 (Divide by 13.) 0x1A: DIV13_5 (Divide by 13.5.) 0x1B: DIV14 (Divide by 14.) 0x1C: DIV14_5 (Divide by 14.5.) 0x1D: DIV15 (Divide by 15.) 0x1E: DIV15_5 (Divide by 15.5.) 0x1F: DIV16 (Divide by 16.)

0x01846018 GCC_APSS_AHB_MISC**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

APSS AHB clock control misc configuration register. It allows for SW to configure clock dividers used to generate the APSS AHB clock

GCC_APSS_AHB_MISC

Bits	Name	Description
7:4	APSS_AHB_CLK_AUTO_SCALE_DIV	This bitfield is used to configure the div ratio when auto scaling is enable. 0x0: DIV1 (Divide by 1.) 0x1: DIV2 (Divide by 2.) 0x2: DIV3 (Divide by 3.) 0x3: DIV4 (Divide by 4.) 0x4: DIV5 (Divide by 5.) 0x5: DIV6 (Divide by 6.) 0x6: DIV7 (Divide by 7.) 0x7: DIV8 (Divide by 8.) 0x8: DIV9 (Divide by 9.) 0x9: DIV10 (Divide by 10.) 0xA: DIV11 (Divide by 11.) 0xB: DIV12 (Divide by 12.) 0xC: DIV13 (Divide by 13.) 0xD: DIV14 (Divide by 14.) 0xE: DIV15 (Divide by 15.) 0xF: DIV16 (Divide by 16.)
0	APSS_AHB_CLK_AUTO_SCALE_DIS	This bit controls whether RPM clock automatic clock scaling can be disabled. 0x1: SCALE_DISABLE (Auto APSS AHB clock scaling is disable.) 0x0: SCALE_NOT_DISABLE (Auto APSS AHB clock scaling is enable.)

0x0184601C GCC_APSS_AHB_CBCR

Type: RW

Clock: CLK

Reset State: 0x10008000

APSS AHB clock branch control register. It allows SW control of CBC soft macro

GCC_APSS_AHB_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	This bitfield indicated if the clock branch is halted 0:Clock On 1:Clock Off
30:28	NOC_HANDSHAKE_FSM_STATUS	R	This bitfield is used to check the FSM state

GCC_APSS_AHB_CBCR (cont.)

Bits	Name	Type	Description
15	NOC_HANDSHAKE_FSM_EN	RW	This bitfield enables the FSM to use so that slaverdy and socketcon hankshake is used prior to affecting the enable of the clock branch cell. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x01846020 GCC_APSS_AXI_CBCR**Type:** R**Clock:** CLK**Reset State:** 0x00000000

APSS AXI clock branch control register. It allows SW control of CBC soft macro

GCC_APSS_AXI_CBCR

Bits	Name	Description
31	CLK_OFF	This bitfield indicated if the clock branch is halted 0:Clock On 1:Clock Off

0x01848000 GCC_PCNOC_BUS_TIMEOUT0_BCR**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

PC NoC BUS time out 0 block control register

GCC_PCNOC_BUS_TIMEOUT0_BCR

Bits	Name	Description
0	BLK_ARES	This bitfield controls asynchronous software reset for the block. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x01848004 GCC_PCNOB_BUS_TIMEOUT0_AHB_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

PC NoC BUS time out 0 AHB clock branch control register. It allows SW control of CBC soft macro

GCC_PCNOB_BUS_TIMEOUT0_AHB_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	This bitfield indicated if the clock branch is halted 0:Clock On 1:Clock Off
0	CLK_ENABLE	RW	This bitfield indicated if the clock branch is enabled. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x01848008 GCC_PCNOB_BUS_TIMEOUT1_BCR**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

PC NoC BUS time out 1 block control register

GCC_PCNOB_BUS_TIMEOUT1_BCR

Bits	Name	Description
0	BLK_ARES	This bitfield controls asynchronous software reset for the block. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x0184800C GCC_PCNOB_BUS_TIMEOUT1_AHB_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

PC NoC BUS time out 1 AHB clock branch control register. It allows SW control of CBC soft macro

GCC_PCNOB_BUS_TIMEOUT1_AHB_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	This bitfield indicated if the clock branch is halted 0:Clock On 1:Clock Off
0	CLK_ENABLE	RW	This bitfield indicated if the clock branch is enabled. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x01848010 GCC_PCNOB_BUS_TIMEOUT2_BCR**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

PC NoC BUS time out 2 block control register

GCC_PCNOB_BUS_TIMEOUT2_BCR

Bits	Name	Description
0	BLK_ARES	This bitfield controls asynchronous software reset for the block. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x01848014 GCC_PCNOB_BUS_TIMEOUT2_AHB_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

PC NoC BUS time out 2 AHB clock branch control register. It allows SW control of CBC soft macro

GCC_PCNOB_BUS_TIMEOUT2_AHB_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	This bitfield indicated if the clock branch is halted 0:Clock On 1:Clock Off
0	CLK_ENABLE	RW	This bitfield indicated if the clock branch is enabled. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x01848018 GCC_PCNOB_BUS_TIMEOUT3_BCR**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

PC NoC BUS time out 3 block control register

GCC_PCNOB_BUS_TIMEOUT3_BCR

Bits	Name	Description
0	BLK_ARES	This bitfield controls asynchronous software reset for the block. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x0184801C GCC_PCNOB_BUS_TIMEOUT3_AHB_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

PC NoC BUS time out 3 AHB clock branch control register. It allows SW control of CBC soft macro

GCC_PCNOB_BUS_TIMEOUT3_AHB_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	This bitfield indicated if the clock branch is halted 0:Clock On 1:Clock Off
0	CLK_ENABLE	RW	This bitfield indicated if the clock branch is enabled. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x01848020 GCC_PCNOB_BUS_TIMEOUT4_BCR**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

PC NoC BUS time out 4 block control register

GCC_PCNOC_BUS_TIMEOUT4_BCR

Bits	Name	Description
0	BLK_ARES	This bitfield controls asynchronous software reset for the block. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x01848024 GCC_PCNOC_BUS_TIMEOUT4_AHB_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

PC NoC BUS time out 4 AHB clock branch control register. It allows SW control of CBC soft macro

GCC_PCNOC_BUS_TIMEOUT4_AHB_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	This bitfield indicated if the clock branch is halted 0:Clock On 1:Clock Off
0	CLK_ENABLE	RW	This bitfield indicated if the clock branch is enabled. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x01848028 GCC_PCNOC_BUS_TIMEOUT5_BCR**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

PC NoC BUS time out 5 block control register

GCC_PCNOC_BUS_TIMEOUT5_BCR

Bits	Name	Description
0	BLK_ARES	This bitfield controls asynchronous software reset for the block. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x0184802C GCC_PCNOB_BUS_TIMEOUT5_AHB_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

PC NoC BUS time out 5 AHB clock branch control register. It allows SW control of CBC soft macro

GCC_PCNOB_BUS_TIMEOUT5_AHB_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	This bitfield indicated if the clock branch is halted 0:Clock On 1:Clock Off
0	CLK_ENABLE	RW	This bitfield indicated if the clock branch is enabled. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x01848030 GCC_PCNOB_BUS_TIMEOUT6_BCR**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

PC NoC BUS time out 6 block control register

GCC_PCNOB_BUS_TIMEOUT6_BCR

Bits	Name	Description
0	BLK_ARES	This bitfield controls asynchronous software reset for the block. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x01848034 GCC_PCNOB_BUS_TIMEOUT6_AHB_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

PC NoC BUS time out 6 AHB clock branch control register. It allows SW control of CBC soft macro

GCC_PCNOC_BUS_TIMEOUT6_AHB_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	This bitfield indicated if the clock branch is halted 0:Clock On 1:Clock Off
0	CLK_ENABLE	RW	This bitfield indicated if the clock branch is enabled. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x01848038 GCC_PCNOC_BUS_TIMEOUT7_BCR**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

PC NoC BUS time out 7 block control register

GCC_PCNOC_BUS_TIMEOUT7_BCR

Bits	Name	Description
0	BLK_ARES	This bitfield controls asynchronous software reset for the block. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x0184803C GCC_PCNOC_BUS_TIMEOUT7_AHB_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

PC NoC BUS time out 7 AHB clock branch control register. It allows SW control of CBC soft macro

GCC_PCNOC_BUS_TIMEOUT7_AHB_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	This bitfield indicated if the clock branch is halted 0:Clock On 1:Clock Off
0	CLK_ENABLE	RW	This bitfield indicated if the clock branch is enabled. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x01848040 GCC_PCNOB_BUS_TIMEOUT8_BCR**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

PC NoC BUS time out 8 block control register

GCC_PCNOB_BUS_TIMEOUT8_BCR

Bits	Name	Description
0	BLK_ARES	This bitfield controls asynchronous software reset for the block. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x01848044 GCC_PCNOB_BUS_TIMEOUT8_AHB_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

PC NoC BUS time out 8 AHB clock branch control register. It allows SW control of CBC soft macro

GCC_PCNOB_BUS_TIMEOUT8_AHB_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	This bitfield indicated if the clock branch is halted 0:Clock On 1:Clock Off
0	CLK_ENABLE	RW	This bitfield indicated if the clock branch is enabled. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x01848048 GCC_PCNOB_BUS_TIMEOUT9_BCR**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

PC NoC BUS time out 9 block control register

GCC_PCNOC_BUS_TIMEOUT9_BCR

Bits	Name	Description
0	BLK_ARES	This bitfield controls asynchronous software reset for the block. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x0184804C GCC_PCNOC_BUS_TIMEOUT9_AHB_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

PC NoC BUS time out 9 AHB clock branch control register. It allows SW control of CBC soft macro

GCC_PCNOC_BUS_TIMEOUT9_AHB_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	This bitfield indicated if the clock branch is halted 0:Clock On 1:Clock Off
0	CLK_ENABLE	RW	This bitfield indicated if the clock branch is enabled. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x0181700C GCC_PCNOC_MPU_CFG_AHB_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x80008000

PC NOC MPU config AHB clock branch control register. It allows SW control of CBC soft macro

GCC_PCNOC_MPU_CFG_AHB_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	This bitfield indicated if the clock branch is halted 0:Clock On 1:Clock Off
30:28	NOC_HANDSHAKE_FSM_STATUS	R	This bitfield is used to check the FSM state

GCC_PCNOG_MPU_CFG_AHB_CBCR (cont.)

Bits	Name	Type	Description
15	NOC_HANDSHAKE_FSM_EN	RW	This bitfield enables the FSM to use so that slaverdy and socketcon hankshake is used prior to affecting the enable of the clock branch cell. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
0	CLK_ENABLE	RW	This bitfield indicated if the clock branch is enabled. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x01812010 GCC_GFX_TBU_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x80004FF0

GFX TBU clock branch control register. It allows SW control of CBC soft macro. The CBC enable for CRYPTO clock will be managed by clock voting registers.

GCC_GFX_TBU_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	This bitfield indicated if the clock branch is halted 0:Clock On 1:Clock Off
14	FORCE_MEM_CORE_ON	RW	This bit force the core on signal to remain active during halt state of the clk. 0x1: FORCE_ENABLE (Force-on Mode.) 0x0: FORCE_DISABLE (Normal Mode force disabled.)
13	FORCE_MEM_PERIPH_ON	RW	This bit force the periph_on signal to remain active during halt state of the clk. 0x1: FORCE_ENABLE (Force-on Mode.) 0x0: FORCE_DISABLE (Normal Mode force disabled.)
12	FORCE_MEM_PERIPH_OFF	RW	This bit force the periph_off signal to remain active during halt state of the clk. 0x1: FORCE_ENABLE (Force-on Mode.) 0x0: FORCE_DISABLE (Normal Mode force disabled.)

GCC_GFX_TBU_CBCR (cont.)

Bits	Name	Type	Description
11:8	WAKEUP	RW	This bitfield specifies the number of clock cycles it takes for the clock to turn back on. 0x0: CLOCK0 (Zero cycle) 0x1: CLOCK1 (One cycle) 0x2: CLOCK2 (Two cycles) 0x3: CLOCK3 (Three cycles) 0x4: CLOCK4 (Four cycles) 0x5: CLOCK5 (Five cycles) 0x6: CLOCK6 (Six cycles) 0x7: CLOCK7 (Seven cycles) 0x8: CLOCK8 (Eight cycles) 0x9: CLOCK9 (Nine cycles) 0xA: CLOCK10 (Ten cycles) 0xB: CLOCK11 (Eleven cycles) 0xC: CLOCK12 (Twelve cycles) 0xD: CLOCK13 (Thirteen cycles) 0xE: CLOCK14 (Fourteen cycles) 0xF: CLOCK15 (Fifteen cycles)
7:4	SLEEP	RW	This bitfield specifies the number of clock cycles till complete sleep where headswitch of the corresponding memories will be opened. 0x0: CLOCK0 (Zero cycle) 0x1: CLOCK1 (One cycle) 0x2: CLOCK2 (Two cycles) 0x3: CLOCK3 (Three cycles) 0x4: CLOCK4 (Four cycles) 0x5: CLOCK5 (Five cycles) 0x6: CLOCK6 (Six cycles) 0x7: CLOCK7 (Seven cycles) 0x8: CLOCK8 (Eight cycles) 0x9: CLOCK9 (Nine cycles) 0xA: CLOCK10 (Ten cycles) 0xB: CLOCK11 (Eleven cycles) 0xC: CLOCK12 (Twelve cycles) 0xD: CLOCK13 (Thirteen cycles) 0xE: CLOCK14 (Fourteen cycles) 0xF: CLOCK15 (Fifteen cycles)

0x01812014 GCC_VENUS_TBU_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x80004FF0

VENUS TBU clock branch control register. It allows SW control of CBC soft macro. The CBC enable for CRYPTO clock will be managed by clock voting registers.

GCC_VENUS_TBU_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	This bitfield indicated if the clock branch is halted 0:Clock On 1:Clock Off
14	FORCE_MEM_CORE_ON	RW	This bit force the core on signal to remain active during halt state of the clk. 0x1: FORCE_ENABLE (Force-on Mode.) 0x0: FORCE_DISABLE (Normal Mode force disabled.)
13	FORCE_MEM_PERIPH_ON	RW	This bit force the periph_on signal to remain active during halt state of the clk. 0x1: FORCE_ENABLE (Force-on Mode.) 0x0: FORCE_DISABLE (Normal Mode force disabled.)
12	FORCE_MEM_PERIPH_OFF	RW	This bit force the periph_off signal to remain active during halt state of the clk. 0x1: FORCE_ENABLE (Force-on Mode.) 0x0: FORCE_DISABLE (Normal Mode force disabled.)
11:8	WAKEUP	RW	This bitfield specifies the number of clock cycles it takes for the clock to turn back on. 0x0: CLOCK0 (Zero cycle) 0x1: CLOCK1 (One cycle) 0x2: CLOCK2 (Two cycles) 0x3: CLOCK3 (Three cycles) 0x4: CLOCK4 (Four cycles) 0x5: CLOCK5 (Five cycles) 0x6: CLOCK6 (Six cycles) 0x7: CLOCK7 (Seven cycles) 0x8: CLOCK8 (Eight cycles) 0x9: CLOCK9 (Nine cycles) 0xA: CLOCK10 (Ten cycles) 0xB: CLOCK11 (Eleven cycles) 0xC: CLOCK12 (Twelve cycles) 0xD: CLOCK13 (Thirteen cycles) 0xE: CLOCK14 (Fourteen cycles) 0xF: CLOCK15 (Fifteen cycles)

GCC_VENUS_TBU_CBCR (cont.)

Bits	Name	Type	Description
7:4	SLEEP	RW	This bitfield specifies the number of clock cycles till complete sleep where headswitch of the corresponding memories will be opened. 0x0: CLOCK0 (Zero cycle) 0x1: CLOCK1 (One cycle) 0x2: CLOCK2 (Two cycles) 0x3: CLOCK3 (Three cycles) 0x4: CLOCK4 (Four cycles) 0x5: CLOCK5 (Five cycles) 0x6: CLOCK6 (Six cycles) 0x7: CLOCK7 (Seven cycles) 0x8: CLOCK8 (Eight cycles) 0x9: CLOCK9 (Nine cycles) 0xA: CLOCK10 (Ten cycles) 0xB: CLOCK11 (Eleven cycles) 0xC: CLOCK12 (Twelve cycles) 0xD: CLOCK13 (Thirteen cycles) 0xE: CLOCK14 (Fourteen cycles) 0xF: CLOCK15 (Fifteen cycles)

0x01812098 GCC_APSS_TCU_ASYNC_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x00004FF0

APSS TCU Async Mode clock branch control register. It allows SW control of CBC soft macro.

GCC_APSS_TCU_ASYNC_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	This bitfield indicates if the clock branch is halted. This is default ON clock only if APSS TCU - BIMC async mode is enabled. 0:Clock On 1:Clock Off
14	FORCE_MEM_CORE_ON	RW	This bit force the core on signal to remain active during halt state of the clk. 0x1: FORCE_ENABLE (Force-on Mode.) 0x0: FORCE_DISABLE (Normal Mode force disabled.)
13	FORCE_MEM_PERIPH_ON	RW	This bit force the periph_on signal to remain active during halt state of the clk. 0x1: FORCE_ENABLE (Force-on Mode.) 0x0: FORCE_DISABLE (Normal Mode force disabled.)

GCC_APSS_TCU_ASYNC_CBCR (cont.)

Bits	Name	Type	Description
12	FORCE_MEM_PERIPH_OF F	RW	This bit force the periph_off signal to remain active during halt state of the clk. 0x1: FORCE_ENABLE (Force-on Mode.) 0x0: FORCE_DISABLE (Normal Mode force disabled.)
11:8	WAKEUP	RW	This bitfield specifies the number of clock cycles it takes for the clock to turn back on. 0x0: CLOCK0 (Zero cycle) 0x1: CLOCK1 (One cycle) 0x2: CLOCK2 (Two cycles) 0x3: CLOCK3 (Three cycles) 0x4: CLOCK4 (Four cycles) 0x5: CLOCK5 (Five cycles) 0x6: CLOCK6 (Six cycles) 0x7: CLOCK7 (Seven cycles) 0x8: CLOCK8 (Eight cycles) 0x9: CLOCK9 (Nine cycles) 0xA: CLOCK10 (Ten cycles) 0xB: CLOCK11 (Eleven cycles) 0xC: CLOCK12 (Twelve cycles) 0xD: CLOCK13 (Thirteen cycles) 0xE: CLOCK14 (Fourteen cycles) 0xF: CLOCK15 (Fifteen cycles)
7:4	SLEEP	RW	This bitfield specifies the number of clock cycles till complete sleep where headswitch of the corresponding memories will be opened. 0x0: CLOCK0 (Zero cycle) 0x1: CLOCK1 (One cycle) 0x2: CLOCK2 (Two cycles) 0x3: CLOCK3 (Three cycles) 0x4: CLOCK4 (Four cycles) 0x5: CLOCK5 (Five cycles) 0x6: CLOCK6 (Six cycles) 0x7: CLOCK7 (Seven cycles) 0x8: CLOCK8 (Eight cycles) 0x9: CLOCK9 (Nine cycles) 0xA: CLOCK10 (Ten cycles) 0xB: CLOCK11 (Eleven cycles) 0xC: CLOCK12 (Twelve cycles) 0xD: CLOCK13 (Thirteen cycles) 0xE: CLOCK14 (Fourteen cycles) 0xF: CLOCK15 (Fifteen cycles)

0x01812018 GCC_APSS_TCU_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x00004FF0

APSS TCU clock branch control register. It allows SW control of CBC soft macro. The CBC enable for CRYPTO clock will be managed by clock voting registers.

GCC_APSS_TCU_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	This bitfield indicated if the clock branch is halted 0:Clock On 1:Clock Off
14	FORCE_MEM_CORE_ON	RW	This bit force the core on signal to remain active during halt state of the clk. 0x1: FORCE_ENABLE (Force-on Mode.) 0x0: FORCE_DISABLE (Normal Mode force disabled.)
13	FORCE_MEM_PERIPH_ON	RW	This bit force the periph_on signal to remain active during halt state of the clk. 0x1: FORCE_ENABLE (Force-on Mode.) 0x0: FORCE_DISABLE (Normal Mode force disabled.)
12	FORCE_MEM_PERIPH_OFF	RW	This bit force the periph_off signal to remain active during halt state of the clk. 0x1: FORCE_ENABLE (Force-on Mode.) 0x0: FORCE_DISABLE (Normal Mode force disabled.)
11:8	WAKEUP	RW	This bitfield specifies the number of clock cycles it takes for the clock to turn back on. 0x0: CLOCK0 (Zero cycle) 0x1: CLOCK1 (One cycle) 0x2: CLOCK2 (Two cycles) 0x3: CLOCK3 (Three cycles) 0x4: CLOCK4 (Four cycles) 0x5: CLOCK5 (Five cycles) 0x6: CLOCK6 (Six cycles) 0x7: CLOCK7 (Seven cycles) 0x8: CLOCK8 (Eight cycles) 0x9: CLOCK9 (Nine cycles) 0xA: CLOCK10 (Ten cycles) 0xB: CLOCK11 (Eleven cycles) 0xC: CLOCK12 (Twelve cycles) 0xD: CLOCK13 (Thirteen cycles) 0xE: CLOCK14 (Fourteen cycles) 0xF: CLOCK15 (Fifteen cycles)

GCC_APSS_TCU_CBCR (cont.)

Bits	Name	Type	Description
7:4	SLEEP	RW	This bitfield specifies the number of clock cycles till complete sleep where headswitch of the corresponding memories will be opened. 0x0: CLOCK0 (Zero cycle) 0x1: CLOCK1 (One cycle) 0x2: CLOCK2 (Two cycles) 0x3: CLOCK3 (Three cycles) 0x4: CLOCK4 (Four cycles) 0x5: CLOCK5 (Five cycles) 0x6: CLOCK6 (Six cycles) 0x7: CLOCK7 (Seven cycles) 0x8: CLOCK8 (Eight cycles) 0x9: CLOCK9 (Nine cycles) 0xA: CLOCK10 (Ten cycles) 0xB: CLOCK11 (Eleven cycles) 0xC: CLOCK12 (Twelve cycles) 0xD: CLOCK13 (Thirteen cycles) 0xE: CLOCK14 (Fourteen cycles) 0xF: CLOCK15 (Fifteen cycles)

0x0181201C GCC_MDP_TBU_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x80004FF0

MDP TBU clock branch control register. It allows SW control of CBC soft macro. The CBC enable for CRYPTO clock will be managed by clock voting registers.

GCC_MDP_TBU_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	This bitfield indicated if the clock branch is halted 0:Clock On 1:Clock Off
14	FORCE_MEM_CORE_ON	RW	This bit force the core on signal to remain active during halt state of the clk. 0x1: FORCE_ENABLE (Force-on Mode.) 0x0: FORCE_DISABLE (Normal Mode force disabled.)
13	FORCE_MEM_PERIPH_ON	RW	This bit force the periph_on signal to remain active during halt state of the clk. 0x1: FORCE_ENABLE (Force-on Mode.) 0x0: FORCE_DISABLE (Normal Mode force disabled.)

GCC_MDP_TBU_CBCR (cont.)

Bits	Name	Type	Description
12	FORCE_MEM_PERIPH_OF F	RW	This bit force the periph_off signal to remain active during halt state of the clk. 0x1: FORCE_ENABLE (Force-on Mode.) 0x0: FORCE_DISABLE (Normal Mode force disabled.)
11:8	WAKEUP	RW	This bitfield specifies the number of clock cycles it takes for the clock to turn back on. 0x0: CLOCK0 (Zero cycle) 0x1: CLOCK1 (One cycle) 0x2: CLOCK2 (Two cycles) 0x3: CLOCK3 (Three cycles) 0x4: CLOCK4 (Four cycles) 0x5: CLOCK5 (Five cycles) 0x6: CLOCK6 (Six cycles) 0x7: CLOCK7 (Seven cycles) 0x8: CLOCK8 (Eight cycles) 0x9: CLOCK9 (Nine cycles) 0xA: CLOCK10 (Ten cycles) 0xB: CLOCK11 (Eleven cycles) 0xC: CLOCK12 (Twelve cycles) 0xD: CLOCK13 (Thirteen cycles) 0xE: CLOCK14 (Fourteen cycles) 0xF: CLOCK15 (Fifteen cycles)
7:4	SLEEP	RW	This bitfield specifies the number of clock cycles till complete sleep where headswitch of the corresponding memories will be opened. 0x0: CLOCK0 (Zero cycle) 0x1: CLOCK1 (One cycle) 0x2: CLOCK2 (Two cycles) 0x3: CLOCK3 (Three cycles) 0x4: CLOCK4 (Four cycles) 0x5: CLOCK5 (Five cycles) 0x6: CLOCK6 (Six cycles) 0x7: CLOCK7 (Seven cycles) 0x8: CLOCK8 (Eight cycles) 0x9: CLOCK9 (Nine cycles) 0xA: CLOCK10 (Ten cycles) 0xB: CLOCK11 (Eleven cycles) 0xC: CLOCK12 (Twelve cycles) 0xD: CLOCK13 (Thirteen cycles) 0xE: CLOCK14 (Fourteen cycles) 0xF: CLOCK15 (Fifteen cycles)

0x01812020 GCC_GFX_TCU_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x00004FF0

GFX TBU clock branch control register. It allows SW control of CBC soft macro. The CBC enable for CRYPTO clock will be managed by clock voting registers.

GCC_GFX_TCU_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	This bitfield indicated if the clock branch is halted 0:Clock On 1:Clock Off
14	FORCE_MEM_CORE_ON	RW	This bit force the core on signal to remain active during halt state of the clk. 0x1: FORCE_ENABLE (Force-on Mode.) 0x0: FORCE_DISABLE (Normal Mode force disabled.)
13	FORCE_MEM_PERIPH_ON	RW	This bit force the periph_on signal to remain active during halt state of the clk. 0x1: FORCE_ENABLE (Force-on Mode.) 0x0: FORCE_DISABLE (Normal Mode force disabled.)
12	FORCE_MEM_PERIPH_OFF	RW	This bit force the periph_off signal to remain active during halt state of the clk. 0x1: FORCE_ENABLE (Force-on Mode.) 0x0: FORCE_DISABLE (Normal Mode force disabled.)
11:8	WAKEUP	RW	This bitfield specifies the number of clock cycles it takes for the clock to turn back on. 0x0: CLOCK0 (Zero cycle) 0x1: CLOCK1 (One cycle) 0x2: CLOCK2 (Two cycles) 0x3: CLOCK3 (Three cycles) 0x4: CLOCK4 (Four cycles) 0x5: CLOCK5 (Five cycles) 0x6: CLOCK6 (Six cycles) 0x7: CLOCK7 (Seven cycles) 0x8: CLOCK8 (Eight cycles) 0x9: CLOCK9 (Nine cycles) 0xA: CLOCK10 (Ten cycles) 0xB: CLOCK11 (Eleven cycles) 0xC: CLOCK12 (Twelve cycles) 0xD: CLOCK13 (Thirteen cycles) 0xE: CLOCK14 (Fourteen cycles) 0xF: CLOCK15 (Fifteen cycles)

GCC_GFX_TCU_CBCR (cont.)

Bits	Name	Type	Description
7:4	SLEEP	RW	This bitfield specifies the number of clock cycles till complete sleep where headswitch of the corresponding memories will be opened. 0x0: CLOCK0 (Zero cycle) 0x1: CLOCK1 (One cycle) 0x2: CLOCK2 (Two cycles) 0x3: CLOCK3 (Three cycles) 0x4: CLOCK4 (Four cycles) 0x5: CLOCK5 (Five cycles) 0x6: CLOCK6 (Six cycles) 0x7: CLOCK7 (Seven cycles) 0x8: CLOCK8 (Eight cycles) 0x9: CLOCK9 (Nine cycles) 0xA: CLOCK10 (Ten cycles) 0xB: CLOCK11 (Eleven cycles) 0xC: CLOCK12 (Twelve cycles) 0xD: CLOCK13 (Thirteen cycles) 0xE: CLOCK14 (Fourteen cycles) 0xF: CLOCK15 (Fifteen cycles)

0x01812024 GCC_MSS_TBU_AXI_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x80004FF0

MSS TBU AXI clock branch control register. It allows SW control of CBC soft macro. The CBC enable for CRYPTO clock will be managed by clock voting registers.

GCC_MSS_TBU_AXI_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	This bitfield indicated if the clock branch is halted 0:Clock On 1:Clock Off
14	FORCE_MEM_CORE_ON	RW	This bit force the core on signal to remain active during halt state of the clk. 0x1: FORCE_ENABLE (Force-on Mode.) 0x0: FORCE_DISABLE (Normal Mode force disabled.)
13	FORCE_MEM_PERIPH_ON	RW	This bit force the periph_on signal to remain active during halt state of the clk. 0x1: FORCE_ENABLE (Force-on Mode.) 0x0: FORCE_DISABLE (Normal Mode force disabled.)

GCC_MSS_TBU_AXI_CBCR (cont.)

Bits	Name	Type	Description
12	FORCE_MEM_PERIPH_OF F	RW	This bit force the periph_off signal to remain active during halt state of the clk. 0x1: FORCE_ENABLE (Force-on Mode.) 0x0: FORCE_DISABLE (Normal Mode force disabled.)
11:8	WAKEUP	RW	This bitfield specifies the number of clock cycles it takes for the clock to turn back on. 0x0: CLOCK0 (Zero cycle) 0x1: CLOCK1 (One cycle) 0x2: CLOCK2 (Two cycles) 0x3: CLOCK3 (Three cycles) 0x4: CLOCK4 (Four cycles) 0x5: CLOCK5 (Five cycles) 0x6: CLOCK6 (Six cycles) 0x7: CLOCK7 (Seven cycles) 0x8: CLOCK8 (Eight cycles) 0x9: CLOCK9 (Nine cycles) 0xA: CLOCK10 (Ten cycles) 0xB: CLOCK11 (Eleven cycles) 0xC: CLOCK12 (Twelve cycles) 0xD: CLOCK13 (Thirteen cycles) 0xE: CLOCK14 (Fourteen cycles) 0xF: CLOCK15 (Fifteen cycles)
7:4	SLEEP	RW	This bitfield specifies the number of clock cycles till complete sleep where headswitch of the corresponding memories will be opened. 0x0: CLOCK0 (Zero cycle) 0x1: CLOCK1 (One cycle) 0x2: CLOCK2 (Two cycles) 0x3: CLOCK3 (Three cycles) 0x4: CLOCK4 (Four cycles) 0x5: CLOCK5 (Five cycles) 0x6: CLOCK6 (Six cycles) 0x7: CLOCK7 (Seven cycles) 0x8: CLOCK8 (Eight cycles) 0x9: CLOCK9 (Nine cycles) 0xA: CLOCK10 (Ten cycles) 0xB: CLOCK11 (Eleven cycles) 0xC: CLOCK12 (Twelve cycles) 0xD: CLOCK13 (Thirteen cycles) 0xE: CLOCK14 (Fourteen cycles) 0xF: CLOCK15 (Fifteen cycles)

0x01812028 GCC_MSS_TBU_GSS_AXI_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x80004FF0

MSS TBU GSS Axi clock branch control register. It allows SW control of CBC soft macro. The CBC enable for CRYPTO clock will be managed by clock voting registers.

GCC_MSS_TBU_GSS_AXI_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	This bitfield indicated if the clock branch is halted 0:Clock On 1:Clock Off
14	FORCE_MEM_CORE_ON	RW	This bit force the core on signal to remain active during halt state of the clk. 0x1: FORCE_ENABLE (Force-on Mode.) 0x0: FORCE_DISABLE (Normal Mode force disabled.)
13	FORCE_MEM_PERIPH_ON	RW	This bit force the periph_on signal to remain active during halt state of the clk. 0x1: FORCE_ENABLE (Force-on Mode.) 0x0: FORCE_DISABLE (Normal Mode force disabled.)
12	FORCE_MEM_PERIPH_OFF	RW	This bit force the periph_off signal to remain active during halt state of the clk. 0x1: FORCE_ENABLE (Force-on Mode.) 0x0: FORCE_DISABLE (Normal Mode force disabled.)
11:8	WAKEUP	RW	This bitfield specifies the number of clock cycles it takes for the clock to turn back on. 0x0: CLOCK0 (Zero cycle) 0x1: CLOCK1 (One cycle) 0x2: CLOCK2 (Two cycles) 0x3: CLOCK3 (Three cycles) 0x4: CLOCK4 (Four cycles) 0x5: CLOCK5 (Five cycles) 0x6: CLOCK6 (Six cycles) 0x7: CLOCK7 (Seven cycles) 0x8: CLOCK8 (Eight cycles) 0x9: CLOCK9 (Nine cycles) 0xA: CLOCK10 (Ten cycles) 0xB: CLOCK11 (Eleven cycles) 0xC: CLOCK12 (Twelve cycles) 0xD: CLOCK13 (Thirteen cycles) 0xE: CLOCK14 (Fourteen cycles) 0xF: CLOCK15 (Fifteen cycles)

GCC_MSS_TBU_GSS_AXI_CBCR (cont.)

Bits	Name	Type	Description
7:4	SLEEP	RW	This bitfield specifies the number of clock cycles till complete sleep where headswitch of the corresponding memories will be opened. 0x0: CLOCK0 (Zero cycle) 0x1: CLOCK1 (One cycle) 0x2: CLOCK2 (Two cycles) 0x3: CLOCK3 (Three cycles) 0x4: CLOCK4 (Four cycles) 0x5: CLOCK5 (Five cycles) 0x6: CLOCK6 (Six cycles) 0x7: CLOCK7 (Seven cycles) 0x8: CLOCK8 (Eight cycles) 0x9: CLOCK9 (Nine cycles) 0xA: CLOCK10 (Ten cycles) 0xB: CLOCK11 (Eleven cycles) 0xC: CLOCK12 (Twelve cycles) 0xD: CLOCK13 (Thirteen cycles) 0xE: CLOCK14 (Fourteen cycles) 0xF: CLOCK15 (Fifteen cycles)

0x0181202C GCC_MSS_TBU_Q6_AXI_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x80004FF0

MSS TBU Q6 AXI clock branch control register. It allows SW control of CBC soft macro. The CBC enable for CRYPTO clock will be managed by clock voting registers.

GCC_MSS_TBU_Q6_AXI_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	This bitfield indicated if the clock branch is halted 0:Clock On 1:Clock Off
14	FORCE_MEM_CORE_ON	RW	This bit force the core on signal to remain active during halt state of the clk. 0x1: FORCE_ENABLE (Force-on Mode.) 0x0: FORCE_DISABLE (Normal Mode force disabled.)
13	FORCE_MEM_PERIPH_ON	RW	This bit force the periph_on signal to remain active during halt state of the clk. 0x1: FORCE_ENABLE (Force-on Mode.) 0x0: FORCE_DISABLE (Normal Mode force disabled.)

GCC_MSS_TBU_Q6_AXI_CBCR (cont.)

Bits	Name	Type	Description
12	FORCE_MEM_PERIPH_OF F	RW	This bit force the periph_off signal to remain active during halt state of the clk. 0x1: FORCE_ENABLE (Force-on Mode.) 0x0: FORCE_DISABLE (Normal Mode force disabled.)
11:8	WAKEUP	RW	This bitfield specifies the number of clock cycles it takes for the clock to turn back on. 0x0: CLOCK0 (Zero cycle) 0x1: CLOCK1 (One cycle) 0x2: CLOCK2 (Two cycles) 0x3: CLOCK3 (Three cycles) 0x4: CLOCK4 (Four cycles) 0x5: CLOCK5 (Five cycles) 0x6: CLOCK6 (Six cycles) 0x7: CLOCK7 (Seven cycles) 0x8: CLOCK8 (Eight cycles) 0x9: CLOCK9 (Nine cycles) 0xA: CLOCK10 (Ten cycles) 0xB: CLOCK11 (Eleven cycles) 0xC: CLOCK12 (Twelve cycles) 0xD: CLOCK13 (Thirteen cycles) 0xE: CLOCK14 (Fourteen cycles) 0xF: CLOCK15 (Fifteen cycles)
7:4	SLEEP	RW	This bitfield specifies the number of clock cycles till complete sleep where headswitch of the corresponding memories will be opened. 0x0: CLOCK0 (Zero cycle) 0x1: CLOCK1 (One cycle) 0x2: CLOCK2 (Two cycles) 0x3: CLOCK3 (Three cycles) 0x4: CLOCK4 (Four cycles) 0x5: CLOCK5 (Five cycles) 0x6: CLOCK6 (Six cycles) 0x7: CLOCK7 (Seven cycles) 0x8: CLOCK8 (Eight cycles) 0x9: CLOCK9 (Nine cycles) 0xA: CLOCK10 (Ten cycles) 0xB: CLOCK11 (Eleven cycles) 0xC: CLOCK12 (Twelve cycles) 0xD: CLOCK13 (Thirteen cycles) 0xE: CLOCK14 (Fourteen cycles) 0xF: CLOCK15 (Fifteen cycles)

0x01812030 GCC_PCNOCTBU_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x00004FF0

PCNOCTBU clock branch control register. It allows SW control of CBC soft macro

GCC_PCNOCTBU_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	This bitfield indicated if the clock branch is halted 0:Clock On 1:Clock Off
14	FORCE_MEM_CORE_ON	RW	This bit force the core on signal to remain active during halt state of the clk. 0x1: FORCE_ENABLE (Force-on Mode.) 0x0: FORCE_DISABLE (Normal Mode force disabled.)
13	FORCE_MEM_PERIPH_ON	RW	This bit force the periph_on signal to remain active during halt state of the clk. 0x1: FORCE_ENABLE (Force-on Mode.) 0x0: FORCE_DISABLE (Normal Mode force disabled.)
12	FORCE_MEM_PERIPH_OFF	RW	This bit force the periph_off signal to remain active during halt state of the clk. 0x1: FORCE_ENABLE (Force-on Mode.) 0x0: FORCE_DISABLE (Normal Mode force disabled.)
11:8	WAKEUP	RW	This bitfield specifies the number of clock cycles it takes for the clock to turn back on. 0x0: CLOCK0 (Zero cycle) 0x1: CLOCK1 (One cycle) 0x2: CLOCK2 (Two cycles) 0x3: CLOCK3 (Three cycles) 0x4: CLOCK4 (Four cycles) 0x5: CLOCK5 (Five cycles) 0x6: CLOCK6 (Six cycles) 0x7: CLOCK7 (Seven cycles) 0x8: CLOCK8 (Eight cycles) 0x9: CLOCK9 (Nine cycles) 0xA: CLOCK10 (Ten cycles) 0xB: CLOCK11 (Eleven cycles) 0xC: CLOCK12 (Twelve cycles) 0xD: CLOCK13 (Thirteen cycles) 0xE: CLOCK14 (Fourteen cycles) 0xF: CLOCK15 (Fifteen cycles)

GCC_PCNOG_TBU_CBCR (cont.)

Bits	Name	Type	Description
7:4	SLEEP	RW	This bitfield specifies the number of clock cycles till complete sleep where headswitch of the corresponding memories will be opened. 0x0: CLOCK0 (Zero cycle) 0x1: CLOCK1 (One cycle) 0x2: CLOCK2 (Two cycles) 0x3: CLOCK3 (Three cycles) 0x4: CLOCK4 (Four cycles) 0x5: CLOCK5 (Five cycles) 0x6: CLOCK6 (Six cycles) 0x7: CLOCK7 (Seven cycles) 0x8: CLOCK8 (Eight cycles) 0x9: CLOCK9 (Nine cycles) 0xA: CLOCK10 (Ten cycles) 0xB: CLOCK11 (Eleven cycles) 0xC: CLOCK12 (Twelve cycles) 0xD: CLOCK13 (Thirteen cycles) 0xE: CLOCK14 (Fourteen cycles) 0xF: CLOCK15 (Fifteen cycles)

0x01812048 GCC_PRONTO_TBU_CBCR**Type:** R**Clock:** CLK**Reset State:** 0x80000000

PCNOG TBU clock branch control register. It allows SW control of CBC soft macro

GCC_PRONTO_TBU_CBCR

Bits	Name	Description
31	CLK_OFF	This bitfield indicated if the clock branch is halted 0:Clock On 1:Clock Off

0x01812034 GCC_JPEG_TBU_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x80004FF0

JPEG TBU clock branch control register. It allows SW control of CBC soft macro

GCC_JPEG_TBU_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	This bitfield indicated if the clock branch is halted 0:Clock On 1:Clock Off
14	FORCE_MEM_CORE_ON	RW	This bit force the core on signal to remain active during halt state of the clk. 0x1: FORCE_ENABLE (Force-on Mode.) 0x0: FORCE_DISABLE (Normal Mode force disabled.)
13	FORCE_MEM_PERIPH_ON	RW	This bit force the periph_on signal to remain active during halt state of the clk. 0x1: FORCE_ENABLE (Force-on Mode.) 0x0: FORCE_DISABLE (Normal Mode force disabled.)
12	FORCE_MEM_PERIPH_OFF	RW	This bit force the periph_off signal to remain active during halt state of the clk. 0x1: FORCE_ENABLE (Force-on Mode.) 0x0: FORCE_DISABLE (Normal Mode force disabled.)
11:8	WAKEUP	RW	This bitfield specifies the number of clock cycles it takes for the clock to turn back on. 0x0: CLOCK0 (Zero cycle) 0x1: CLOCK1 (One cycle) 0x2: CLOCK2 (Two cycles) 0x3: CLOCK3 (Three cycles) 0x4: CLOCK4 (Four cycles) 0x5: CLOCK5 (Five cycles) 0x6: CLOCK6 (Six cycles) 0x7: CLOCK7 (Seven cycles) 0x8: CLOCK8 (Eight cycles) 0x9: CLOCK9 (Nine cycles) 0xA: CLOCK10 (Ten cycles) 0xB: CLOCK11 (Eleven cycles) 0xC: CLOCK12 (Twelve cycles) 0xD: CLOCK13 (Thirteen cycles) 0xE: CLOCK14 (Fourteen cycles) 0xF: CLOCK15 (Fifteen cycles)

GCC_JPEG_TBU_CBCR (cont.)

Bits	Name	Type	Description
7:4	SLEEP	RW	This bitfield specifies the number of clock cycles till complete sleep where headswitch of the corresponding memories will be opened. 0x0: CLOCK0 (Zero cycle) 0x1: CLOCK1 (One cycle) 0x2: CLOCK2 (Two cycles) 0x3: CLOCK3 (Three cycles) 0x4: CLOCK4 (Four cycles) 0x5: CLOCK5 (Five cycles) 0x6: CLOCK6 (Six cycles) 0x7: CLOCK7 (Seven cycles) 0x8: CLOCK8 (Eight cycles) 0x9: CLOCK9 (Nine cycles) 0xA: CLOCK10 (Ten cycles) 0xB: CLOCK11 (Eleven cycles) 0xC: CLOCK12 (Twelve cycles) 0xD: CLOCK13 (Thirteen cycles) 0xE: CLOCK14 (Fourteen cycles) 0xF: CLOCK15 (Fifteen cycles)

0x01812038 GCC_SMMU_CFG_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x80008000

SMMU CFG clock branch control register. It allows SW control of CBC soft macro

GCC_SMMU_CFG_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	This bitfield indicated if the clock branch is halted 0:Clock On 1:Clock Off
30:28	NOC_HANDSHAKE_FSM_STATUS	R	This bitfield is used to check the FSM state
15	NOC_HANDSHAKE_FSM_ENABLE	RW	This bitfield enables the FSM to use so that slaveredy and socketcon hankshake is used prior to affecting the enable of the clock branch cell. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x0181203C GCC_VFE_TBU_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x80004FF0

VFE TBU clock branch control register. It allows SW control of CBC soft macro

GCC_VFE_TBU_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	This bitfield indicated if the clock branch is halted 0:Clock On 1:Clock Off
14	FORCE_MEM_CORE_ON	RW	This bit force the core on signal to remain active during halt state of the clk. 0x1: FORCE_ENABLE (Force-on Mode.) 0x0: FORCE_DISABLE (Normal Mode force disabled.)
13	FORCE_MEM_PERIPH_ON	RW	This bit force the periph_on signal to remain active during halt state of the clk. 0x1: FORCE_ENABLE (Force-on Mode.) 0x0: FORCE_DISABLE (Normal Mode force disabled.)
12	FORCE_MEM_PERIPH_OFF	RW	This bit force the periph_off signal to remain active during halt state of the clk. 0x1: FORCE_ENABLE (Force-on Mode.) 0x0: FORCE_DISABLE (Normal Mode force disabled.)
11:8	WAKEUP	RW	This bitfield specifies the number of clock cycles it takes for the clock to turn back on. 0x0: CLOCK0 (Zero cycle) 0x1: CLOCK1 (One cycle) 0x2: CLOCK2 (Two cycles) 0x3: CLOCK3 (Three cycles) 0x4: CLOCK4 (Four cycles) 0x5: CLOCK5 (Five cycles) 0x6: CLOCK6 (Six cycles) 0x7: CLOCK7 (Seven cycles) 0x8: CLOCK8 (Eight cycles) 0x9: CLOCK9 (Nine cycles) 0xA: CLOCK10 (Ten cycles) 0xB: CLOCK11 (Eleven cycles) 0xC: CLOCK12 (Twelve cycles) 0xD: CLOCK13 (Thirteen cycles) 0xE: CLOCK14 (Fourteen cycles) 0xF: CLOCK15 (Fifteen cycles)

GCC_VFE_TBU_CBCR (cont.)

Bits	Name	Type	Description
7:4	SLEEP	RW	This bitfield specifies the number of clock cycles till complete sleep where headswitch of the corresponding memories will be opened. 0x0: CLOCK0 (Zero cycle) 0x1: CLOCK1 (One cycle) 0x2: CLOCK2 (Two cycles) 0x3: CLOCK3 (Three cycles) 0x4: CLOCK4 (Four cycles) 0x5: CLOCK5 (Five cycles) 0x6: CLOCK6 (Six cycles) 0x7: CLOCK7 (Seven cycles) 0x8: CLOCK8 (Eight cycles) 0x9: CLOCK9 (Nine cycles) 0xA: CLOCK10 (Ten cycles) 0xB: CLOCK11 (Eleven cycles) 0xC: CLOCK12 (Twelve cycles) 0xD: CLOCK13 (Thirteen cycles) 0xE: CLOCK14 (Fourteen cycles) 0xF: CLOCK15 (Fifteen cycles)

0x01812044 GCC_GTCU_AHB_CBCR**Type:** R**Clock:** CLK**Reset State:** 0x80000000

GTCU AHB clock branch control register. It allows SW control of CBC soft macro

GCC_GTCU_AHB_CBCR

Bits	Name	Description
31	CLK_OFF	This bitfield indicated if the clock branch is halted 0:Clock On 1:Clock Off

0x01812094 GCC_GTCU_AHB_BRIDGE_CBCR**Type:** R**Clock:** CLK**Reset State:** 0x80000000

GTCU AHB2AHB Bridge clock branch control register. It allows SW control of CBC soft macro

GCC_GTCU_AHB_BRIDGE_CBCR

Bits	Name	Description
31	CLK_OFF	This bitfield indicated if the clock branch is halted 0:Clock On 1:Clock Off

0x0187C004 GCC_SMMU_CATS_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

SMMU CATS64 clock control register

GCC_SMMU_CATS_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	This bitfield indicated if the clock branch is halted 0:Clock On 1:Clock Off
0	CLK_ENABLE	RW	This bitfield indicated if the clock branch is enabled. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x0187C000 GCC_SMMU_CATS_BCR**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

SMMU CATS Block control register

GCC_SMMU_CATS_BCR

Bits	Name	Description
0	BLK_ARES	This is for reset for the CATS 64 and CATS 128 Clock 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x01812000 GCC_SMMU_BCR**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

SMMU Blocks control register

GCC_SMMU_BCR

Bits	Name	Description
0	BLK_ARES	This can be used to disable the clocks of the SMMU blocks. Each TBU/TCU and CFG Clock has its own BCR, hence this need not be used for reset 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x01812050 GCC_APSS_TCU_BCR**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

APSS TCU Blocks control register

GCC_APSS_TCU_BCR

Bits	Name	Description
0	BLK_ARES	This bitfield controls asynchronous software reset for the block. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x01862000 GCC_MDP_TBU_BCR**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

MDP TBU Blocks control register

GCC_MDP_TBU_BCR

Bits	Name	Description
0	BLK_ARES	This bitfield controls asynchronous software reset for the block. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x01863000 GCC_GFX_TBU_BCR**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

GFX TBU Blocks control register

GCC_GFX_TBU_BCR

Bits	Name	Description
0	BLK_ARES	This bitfield controls asynchronous software reset for the block. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x01864000 GCC_GFX_TCU_BCR**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

GFX TCU Blocks control register

GCC_GFX_TCU_BCR

Bits	Name	Description
0	BLK_ARES	This bitfield controls asynchronous software reset for the block. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x01865000 GCC_MSS_TBU_AXI_BCR**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

MSS TBU Blocks control register

GCC_MSS_TBU_AXI_BCR

Bits	Name	Description
0	BLK_ARES	This bitfield controls asynchronous software reset for the block. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x01867000 GCC_MSS_TBU_Q6_AXI_BCR**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

MSS Q6 TBU Blocks control register

GCC_MSS_TBU_Q6_AXI_BCR

Bits	Name	Description
0	BLK_ARES	This bitfield controls asynchronous software reset for the block. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x01866000 GCC_MSS_TBU_GSS_AXI_BCR**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

MSS GSS TBU Blocks control register

GCC_MSS_TBU_GSS_AXI_BCR

Bits	Name	Description
0	BLK_ARES	This bitfield controls asynchronous software reset for the block. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x01868000 GCC_GTCU_AHB_BCR**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

GTCU AHB Blocks control register

GCC_GTCU_AHB_BCR

Bits	Name	Description
0	BLK_ARES	This bitfield controls asynchronous software reset for the block. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x01869000 GCC_SMMU_CFG_BCR**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

SMMU CFG Clock Blocks control register

GCC_SMMU_CFG_BCR

Bits	Name	Description
0	BLK_ARES	This bitfield controls asynchronous software reset for the block. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x0186A000 GCC_VFE_TBU_BCR**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

VFE TBU Blocks control register

GCC_VFE_TBU_BCR

Bits	Name	Description
0	BLK_ARES	This bitfield controls asynchronous software reset for the block. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x0186B000 GCC_VENUS_TBU_BCR**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

VENUS TBU Blocks control register

GCC_VENUS_TBU_BCR

Bits	Name	Description
0	BLK_ARES	This bitfield controls asynchronous software reset for the block. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x0186C000 GCC_JPEG_TBU_BCR

Type: RW

Clock: CLK

Reset State: 0x00000000

JPEG TBU Blocks control register

GCC_JPEG_TBU_BCR

Bits	Name	Description
0	BLK_ARES	This bitfield controls asynchronous software reset for the block. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x0186D000 GCC_PRONTO_TBU_BCR

Type: RW

Clock: CLK

Reset State: 0x00000000

PRONTO TBU Blocks control register

GCC_PRONTO_TBU_BCR

Bits	Name	Description
0	BLK_ARES	This bitfield controls asynchronous software reset for the block. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x01812058 GCC_PCNOCTBU_BCR**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

PCNOCTBU Blocks control register

GCC_PCNOCTBU_BCR

Bits	Name	Description
0	BLK_ARES	This bitfield controls asynchronous software reset for the block. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x0183600C GCC_RPM_SMMU_CLOCK_BRANCH_ENA_VOTE**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

RPM SMMU clock enable voting register for votable clocks.

GCC_RPM_SMMU_CLOCK_BRANCH_ENA_VOTE

Bits	Name	Description
13	GTCU_AHB_CLK_ENA	Enable GTCU_AHB Clock 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
12	SMMU_CFG_CLK_ENA	Enable SMMU_CFG Clock 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
11	PCNOCTBU_CLK_ENA	Enable PCNOCTBU Clock 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
10	JPEG_TBU_CLK_ENA	Enable JPEG_TBU Clock 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
9	VFE_TBU_CLK_ENA	Enable VFE_TBU Clock 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
8	MSS_TBU_Q6_AXI_CLK_ENA	Enable MSS_Q6_TBU Clock 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

GCC_RPM_SMMU_CLOCK_BRANCH_ENA_VOTE (cont.)

Bits	Name	Description
7	MSS_TBU_GSS_AXI_CLK_ENA	Enable MSS_GSS_TBU Clock 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
6	MSS_TBU_AXI_CLK_ENA	Enable MSS_AXI_TBU Clock 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
5	VENUS_TBU_CLK_ENA	Enable VENUS_TBU Clock 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
4	MDP_TBU_CLK_ENA	Enable MDP_TBU Clock 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
3	GFX_TBU_CLK_ENA	Enable GFX_TBU Clock 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
2	GFX_TCU_CLK_ENA	Enable GFX_TCU Clock 0x0: ENABLE (In this bit of this register, there is a need to write '0' to enable the clock.) 0x1: DISABLE (This bit got merged into a flop tray and to have 0 as Disable we had to add a Flop which CL asked to avoid.)
1	APSS_TCU_CLK_ENA	Enable APSS_TCU Clock 0x0: ENABLE (This is the same case as above bit. '0' enables the clock and reason is same.) 0x1: DISABLE (This bit also got merged into a flop tray and to avoid Flop ECO we have '1' as Disable.)
0	PRONTO_TBU_CLK_ENA	Enable PRONTO_TBU clock 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x01836008 GCC_RPM_CLOCK_SLEEP_ENA_VOTE**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

RPM clock sleep voting register for votable clocks.

GCC_RPM_CLOCK_SLEEP_ENA_VOTE

Bits	Name	Description
17	IMEM_AXI_CLK_SLEEP_ENA	Enable for allowing sleeping of IMEM AXI clock 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
16	SYS_NOC_APSS_AHB_CLK_SLEEP_ENA	Enable for allowing sleeping sys_noc_apss_ahb_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
15	BIMC_APSS_AXI_CLK_SLEEP_ENA	Enable for allowing sleeping bimc_apss_axi_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
14	APSS_AHB_CLK_SLEEP_ENA	Enable for allowing sleeping apss_ahb_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
13	APSS_AXI_CLK_SLEEP_ENA	Enable for allowing sleeping apss_axi_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
12	MPM_AHB_CLK_SLEEP_ENA	Enable for allowing sleeping mpm_ahb_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
11	GMEM_SYS_NOC_AXI_CLK_SLEEP_ENA	Enable for allowing sleeping gmem_sys_noc_axi_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
10	BLSP1_AHB_CLK_SLEEP_ENA	Enable for allowing sleeping blsp1_ahb_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
9	BLSP1_SLEEP_CLK_SLEEP_ENA	Enable for allowing sleeping blsp1_sleep_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
8	PRNG_AHB_CLK_SLEEP_ENA	Enable for allowing sleeping prng_ahb_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
7	BOOT_ROM_AHB_CLK_SLEEP_ENA	Enable for allowing sleeping boot_rom_ahb_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
6	MSG_RAM_AHB_CLK_SLEEP_ENA	Enable for allowing sleeping msg_ram_ahb_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
5	TLMM_AHB_CLK_SLEEP_ENA	Enable for allowing sleeping tlmm_ahb_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

GCC_RPM_CLOCK_SLEEP_ENA_VOTE (cont.)

Bits	Name	Description
4	TLMM_CLK_SLEEP_ENA	Enable for allowing sleeping tmm_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
3	SPMI_PCNOA_AHB_CLK_SLEEP_ENA	0x1: ENABLE 0x0: DISABLE
2	CRYPTO_CLK_SLEEP_ENA	Enable for allowing sleeping crypto_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
1	CRYPTO_AXI_CLK_SLEEP_ENA	Enable for allowing sleeping crypto_axi_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
0	CRYPTO_AHB_CLK_SLEEP_ENA	Enable for allowing sleeping crypto_ahb_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x01836010 GCC_RPM_SMMU_CLOCK_SLEEP_ENA_VOTE**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

RPM SMMU clock sleep voting register for votable clocks.

GCC_RPM_SMMU_CLOCK_SLEEP_ENA_VOTE

Bits	Name	Description
13	GTCU_AHB_CLK_SLEEP_ENA	Enable for allowing sleeping of GTCU_AHB Clock 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
12	SMMU_CFG_CLK_SLEEP_ENA	Enable for allowing sleeping of SMMU_CFG Clock 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
11	PCNOC_TBU_CLK_SLEEP_ENA	Enable for allowing sleeping of PCNOC_TBU Clock 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
10	JPEG_TBU_CLK_SLEEP_ENA	Enable for allowing sleeping of JPEG_TBU Clock 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
9	VFE_TBU_CLK_SLEEP_ENA	Enable for allowing sleeping of VFE_TBU Clock 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

GCC_RPM_SMMU_CLOCK_SLEEP_ENA_VOTE (cont.)

Bits	Name	Description
8	MSS_TBU_Q6_AXI_CLK_SLEEP_ENA	Enable for allowing sleeping of MSS_Q6_TBU Clock 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
7	MSS_TBU_GSS_AXI_CLK_SLEEP_ENA	Enable for allowing sleeping of MSS_GSS_TBU Clock 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
6	MSS_TBU_AXI_CLK_SLEEP_ENA	Enable for allowing sleeping of MSS_AXI_TBU Clock 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
5	VENUS_TBU_CLK_SLEEP_ENA	Enable for allowing sleeping of VENUS_TBU Clock 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
4	MDP_TBU_CLK_SLEEP_ENA	Enable for allowing sleeping of MDP_TBU Clock 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
3	GFX_TBU_CLK_SLEEP_ENA	Enable for allowing sleeping of GFX_TBU Clock 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
2	GFX_TCU_CLK_SLEEP_ENA	Enable for allowing sleeping of GFX_TCU Clock 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
1	APSS_TCU_CLK_SLEEP_ENA	Enable for allowing sleeping of APSS_TCU Clock 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
0	PRONTO_TBU_CLK_SLEEP_ENA	Enable for allowing sleeping of PRONTO_TBU Clock 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x01845000 GCC_APCS_GPLL_ENA_VOTE**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

APCS GPLL enable voting register.

GCC_APCS_GPLL_ENA_VOTE

Bits	Name	Description
3	BIMC_PLL	Enable BIMC_PLL 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
2	GPLL2	Enable GPLL2 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
1	GPLL1	Enable GPLL1 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
0	GPLL0	Enable GPLL0 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x01845004 GCC_APCS_CLOCK_BRANCH_ENA_VOTE**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

APCS clock enable voting register for votable clocks.

GCC_APCS_CLOCK_BRANCH_ENA_VOTE

Bits	Name	Description
17	IMEM_AXI_CLK_ENA	Enable imem AXI clock 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
16	SYS_NOC_APSS_AHB_CLK_ENA	Enable sys_noc_apss_ahb_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
15	BIMC_APSS_AXI_CLK_ENA	Enable bimc_apss_axi_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
14	APSS_AHB_CLK_ENA	Enable apss_ahb_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
13	APSS_AXI_CLK_ENA	Enable apss_axi_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

GCC_APCS_CLOCK_BRANCH_ENA_VOTE (cont.)

Bits	Name	Description
12	MPM_AHB_CLK_ENA	Enable mpm_ahb_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
11	GMEM_SYS_NOC_AXI_CLK_ENA	Enable gmem_sys_noc_axi_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
10	BLSP1_AHB_CLK_ENA	Enable blsp1_ahb_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
9	BLSP1_SLEEP_CLK_ENA	Enable blsp1_sleep_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
8	PRNG_AHB_CLK_ENA	Enable prng_ahb_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
7	BOOT_ROM_AHB_CLK_ENA	Enable boot_rom_ahb_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
6	MSG_RAM_AHB_CLK_ENA	Enable msg_ram_ahb_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
5	TLMM_AHB_CLK_ENA	Enable tlmm_ahb_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
4	TLMM_CLK_ENA	Enable tlmm_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
3	SPMI_PCNOC_AHB_CLK_ENA	0x1: ENABLE 0x0: DISABLE
2	CRYPTO_CLK_ENA	Enable crypto_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
1	CRYPTO_AXI_CLK_ENA	Enable crypto_axi_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
0	CRYPTO_AHB_CLK_ENA	Enable crypto_ahb_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x0184500C GCC_APCS_SMMU_CLOCK_BRANCH_ENA_VOTE**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

APCS SMMU clock enable voting register for votable clocks.

GCC_APCS_SMMU_CLOCK_BRANCH_ENA_VOTE

Bits	Name	Description
13	GTCU_AHB_CLK_ENA	Enable GTCU_AHB CLock 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
12	SMMU_CFG_CLK_ENA	Enable SMMU_CFG CLock 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
11	PCNOC_TBU_CLK_ENA	Enable PCNOC_TBU Clock 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
10	JPEG_TBU_CLK_ENA	Enable JPEG_TBU Clock 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
9	VFE_TBU_CLK_ENA	Enable VFE_TBU Clock 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
8	MSS_TBU_Q6_AXI_CLK_ENA	Enable MSS_Q6_TBU Clock 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
7	MSS_TBU_GSS_AXI_CLK_ENA	Enable MSS_GSS_TBU Clock 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
6	MSS_TBU_AXI_CLK_ENA	Enable MSS_AXI_TBU Clock 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
5	VENUS_TBU_CLK_ENA	Enable VENUS_TBU Clock 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
4	MDP_TBU_CLK_ENA	Enable MDP_TBU Clock 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
3	GFX_TBU_CLK_ENA	Enable GFX_TBU Clock 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

GCC_APCS_SMMU_CLOCK_BRANCH_ENA_VOTE (cont.)

Bits	Name	Description
2	GFX_TCU_CLK_ENA	Enable GFX_TCU Clock 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
1	APSS_TCU_CLK_ENA	Enable APSS_TCU Clock 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
0	PRONTO_TBU_CLK_ENA	Enable PRONTO_TBU clock 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x01845008 GCC_APCS_CLOCK_SLEEP_ENA_VOTE**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

APCS clock sleep voting register for votable clocks.

GCC_APCS_CLOCK_SLEEP_ENA_VOTE

Bits	Name	Description
17	IMEM_AXI_CLK_SLEEP_ENA	Enable for allowing sleeping of IMEM AXI clock 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
16	SYS_NOC_APSS_AHB_CLK_SLEEP_ENA	Enable for allowing sleeping sys_noc_apss_ahb_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
15	BIMC_APSS_AXI_CLK_SLEEP_ENA	Enable for allowing sleeping bimc_apss_axi_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
14	APSS_AHB_CLK_SLEEP_ENA	Enable for allowing sleeping apss_ahb_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
13	APSS_AXI_CLK_SLEEP_ENA	Enable for allowing sleeping apss_axi_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
12	MPM_AHB_CLK_SLEEP_ENA	Enable for allowing sleeping mpm_ahb_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

GCC_APCS_CLOCK_SLEEP_ENA_VOTE (cont.)

Bits	Name	Description
11	GMEM_SYS_NOC_AXI_CLK_SLEEP_ENA	Enable for allowing sleeping gmem_sys_noc_axi_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
10	BLSP1_AHB_CLK_SLEEP_ENA	Enable for allowing sleeping blsp1_ahb_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
9	BLSP1_SLEEP_CLK_SLEEP_ENA	Enable for allowing sleeping blsp1_sleep_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
8	PRNG_AHB_CLK_SLEEP_ENA	Enable for allowing sleeping prng_ahb_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
7	BOOT_ROM_AHB_CLK_SLEEP_ENA	Enable for allowing sleeping boot_rom_ahb_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
6	MSG_RAM_AHB_CLK_SLEEP_ENA	Enable for allowing sleeping msg_ram_ahb_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
5	TLMM_AHB_CLK_SLEEP_ENA	Enable for allowing sleeping tlmm_ahb_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
4	TLMM_CLK_SLEEP_ENA	Enable for allowing sleeping tlmm_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
3	SPMI_PCNOB_AHB_CLK_SLEEP_ENA	0x1: ENABLE 0x0: DISABLE
2	CRYPTO_CLK_SLEEP_ENA	Enable for allowing sleeping crypto_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
1	CRYPTO_AXI_CLK_SLEEP_ENA	Enable for allowing sleeping crypto_axi_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
0	CRYPTO_AHB_CLK_SLEEP_ENA	Enable for allowing sleeping crypto_ahb_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x01845010 GCC_APCS_SMMU_CLOCK_SLEEP_ENA_VOTE**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

APCS SMMU clock sleep voting register for votable clocks.

GCC_APCS_SMMU_CLOCK_SLEEP_ENA_VOTE

Bits	Name	Description
13	GTCU_AHB_CLK_SLEEP_ENA	Enable for allowing sleeping of GTCU_AHB Clock 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
12	SMMU_CFG_CLK_SLEEP_ENA	Enable for allowing sleeping of SMMU_CFG Clock 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
11	PCNOC_TBU_CLK_SLEEP_ENA	Enable for allowing sleeping of PCNOC_TBU Clock 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
10	JPEG_TBU_CLK_SLEEP_ENA	Enable for allowing sleeping of JPEG_TBU Clock 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
9	VFE_TBU_CLK_SLEEP_ENA	Enable for allowing sleeping of VFE_TBU Clock 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
8	MSS_TBU_Q6_AXI_CLK_SLEEP_ENA	Enable for allowing sleeping of MSS_Q6_TBU Clock 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
7	MSS_TBU_GSS_AXI_CLK_SLEEP_ENA	Enable for allowing sleeping of MSS_GSS_TBU Clock 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
6	MSS_TBU_AXI_CLK_SLEEP_ENA	Enable for allowing sleeping of MSS_AXI_TBU Clock 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
5	VENUS_TBU_CLK_SLEEP_ENA	Enable for allowing sleeping of VENUS_TBU Clock 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
4	MDP_TBU_CLK_SLEEP_ENA	Enable for allowing sleeping of MDP_TBU Clock 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
3	GFX_TBU_CLK_SLEEP_ENA	Enable for allowing sleeping of GFX_TBU Clock 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

GCC_APCS_SMMU_CLOCK_SLEEP_ENA_VOTE (cont.)

Bits	Name	Description
2	GFX_TCU_CLK_SLEEP_ENA	Enable for allowing sleeping of GFX_TCU Clock 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
1	APSS_TCU_CLK_SLEEP_ENA	Enable for allowing sleeping of APSS_TCU Clock 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
0	PRONTO_TBU_CLK_SLEEP_ENA	Enable for allowing sleeping of PRONTO_TBU Clock 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x01813010 GCC_APCS_TZ_GPLL_ENA_VOTE**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

APCS Trust zone GPLL enable voting register.

GCC_APCS_TZ_GPLL_ENA_VOTE

Bits	Name	Description
3	BIMC_PLL	Enable BIMC_PLL 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
2	GPLL2	Enable GPLL2 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
1	GPLL1	Enable GPLL1 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
0	GPLL0	Enable GPLL0 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x01813014 GCC_APCS_TZ_CLOCK_BRANCH_ENA_VOTE**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

APCS Trust Zone clock enable voting register for votable clocks.

GCC_APCS_TZ_CLOCK_BRANCH_ENA_VOTE

Bits	Name	Description
17	IMEM_AXI_CLK_ENA	Enable imem AXI clock 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
16	SYS_NOC_APSS_AHB_CLK_ENA	Enable sys_noc_apss_ahb_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
15	BIMC_APSS_AXI_CLK_ENA	Enable bimc_apss_axi_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
14	APSS_AHB_CLK_ENA	Enable apss_ahb_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
13	APSS_AXI_CLK_ENA	Enable apss_axi_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
12	MPM_AHB_CLK_ENA	Enable mpm_ahb_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
11	GMEM_SYS_NOC_AXI_CLK_ENA	Enable gmem_sys_noc_axi_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
10	BLSP1_AHB_CLK_ENA	Enable blsp1_ahb_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
9	BLSP1_SLEEP_CLK_ENA	Enable blsp1_sleep_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
8	PRNG_AHB_CLK_ENA	Enable prng_ahb_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
7	BOOT_ROM_AHB_CLK_ENA	Enable boot_rom_ahb_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
6	MSG_RAM_AHB_CLK_ENA	Enable msg_ram_ahb_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
5	TLMM_AHB_CLK_ENA	Enable tlmm_ahb_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

GCC_APCS_TZ_CLOCK_BRANCH_ENA_VOTE (cont.)

Bits	Name	Description
4	TLMM_CLK_ENA	Enable tlmm_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
3	SPMI_PCNOC_AHB_CLK_ENA	0x1: ENABLE 0x0: DISABLE
2	CRYPTO_CLK_ENA	Enable crypto_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
1	CRYPTO_AXI_CLK_ENA	Enable crypto_axi_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
0	CRYPTO_AHB_CLK_ENA	Enable crypto_ahb_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x01813020 GCC_APCS_TZ_SMMU_CLOCK_BRANCH_ENA_VOTE**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

APCS Trust Zone SMMU clock enable voting register for votable clocks.

GCC_APCS_TZ_SMMU_CLOCK_BRANCH_ENA_VOTE

Bits	Name	Description
13	GTCU_AHB_CLK_ENA	Enable GTCU_AHB CLock 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
12	SMMU_CFG_CLK_ENA	Enable SMMU_CFG CLock 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
11	PCNOC_TBU_CLK_ENA	Enable PCNOC_TBU Clock 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
10	JPEG_TBU_CLK_ENA	Enable JPEG_TBU Clock 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
9	VFE_TBU_CLK_ENA	Enable VFE_TBU Clock 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

GCC_APCS_TZ_SMMU_CLOCK_BRANCH_ENA_VOTE (cont.)

Bits	Name	Description
8	MSS_TBU_Q6_AXI_CLK_ENA	Enable MSS_Q6_TBU Clock 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
7	MSS_TBU_GSS_AXI_CLK_ENA	Enable MSS_GSS_TBU Clock 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
6	MSS_TBU_AXI_CLK_ENA	Enable MSS_AXI_TBU Clock 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
5	VENUS_TBU_CLK_ENA	Enable VENUS_TBU Clock 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
4	MDP_TBU_CLK_ENA	Enable MDP_TBU Clock 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
3	GFX_TBU_CLK_ENA	Enable GFX_TBU Clock 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
2	GFX_TCU_CLK_ENA	Enable GFX_TCU Clock 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
1	APSS_TCU_CLK_ENA	Enable APSS_TCU Clock 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
0	PRONTO_TBU_CLK_ENA	Enable PRONTO_TBU clock 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x01813018 GCC_APCS_TZ_CLOCK_SLEEP_ENA_VOTE**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

APCS Trust Zone clock sleep voting register for votable clocks.

GCC_APCS_TZ_CLOCK_SLEEP_ENA_VOTE

Bits	Name	Description
17	IMEM_AXI_CLK_SLEEP_ENA	Enable for allowing sleeping of IMEM AXI clock 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
16	SYS_NOC_APSS_AHB_CLK_SLEEP_ENA	Enable for allowing sleeping sys_noc_apss_ahb_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
15	BIMC_APSS_AXI_CLK_SLEEP_ENA	Enable for allowing sleeping bimc_apss_axi_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
14	APSS_AHB_CLK_SLEEP_ENA	Enable for allowing sleeping apss_ahb_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
13	APSS_AXI_CLK_SLEEP_ENA	Enable for allowing sleeping apss_axi_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
12	MPM_AHB_CLK_SLEEP_ENA	Enable for allowing sleeping mpm_ahb_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
11	GMEM_SYS_NOC_AXI_CLK_SLEEP_ENA	Enable for allowing sleeping gmem_sys_noc_axi_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
10	BLSP1_AHB_CLK_SLEEP_ENA	Enable for allowing sleeping blsp1_ahb_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
9	BLSP1_SLEEP_CLK_SLEEP_ENA	Enable for allowing sleeping blsp1_sleep_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
8	PRNG_AHB_CLK_SLEEP_ENA	Enable for allowing sleeping prng_ahb_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
7	BOOT_ROM_AHB_CLK_SLEEP_ENA	Enable for allowing sleeping boot_rom_ahb_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
6	MSG_RAM_AHB_CLK_SLEEP_ENA	Enable for allowing sleeping msg_ram_ahb_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
5	TLMM_AHB_CLK_SLEEP_ENA	Enable for allowing sleeping tlmm_ahb_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

GCC_APCS_TZ_CLOCK_SLEEP_ENA_VOTE (cont.)

Bits	Name	Description
4	TLMM_CLK_SLEEP_ENA	Enable for allowing sleeping tmm_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
3	SPMI_PCNOA_AHB_CLK_SLEEP_ENA	0x1: ENABLE 0x0: DISABLE
2	CRYPTO_CLK_SLEEP_ENA	Enable for allowing sleeping crypto_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
1	CRYPTO_AXI_CLK_SLEEP_ENA	Enable for allowing sleeping crypto_axi_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
0	CRYPTO_AHB_CLK_SLEEP_ENA	Enable for allowing sleeping crypto_ahb_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x01813024 GCC_APCS_TZ_SMMU_CLOCK_SLEEP_ENA_VOTE**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

APCS Trust Zone SMMU clock sleep voting register for votable clocks.

GCC_APCS_TZ_SMMU_CLOCK_SLEEP_ENA_VOTE

Bits	Name	Description
13	GTCU_AHB_CLK_SLEEP_ENA	Enable for allowing sleeping of GTCU_AHB Clock 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
12	SMMU_CFG_CLK_SLEEP_ENA	Enable for allowing sleeping of SMMU_CFG Clock 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
11	PCNOC_TBU_CLK_SLEEP_ENA	Enable for allowing sleeping of PCNOC_TBU Clock 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
10	JPEG_TBU_CLK_SLEEP_ENA	Enable for allowing sleeping of JPEG_TBU Clock 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
9	VFE_TBU_CLK_SLEEP_ENA	Enable for allowing sleeping of VFE_TBU Clock 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

GCC_APCS_TZ_SMMU_CLOCK_SLEEP_ENA_VOTE (cont.)

Bits	Name	Description
8	MSS_TBU_Q6_AXI_CLK_SLEEP_ENA	Enable for allowing sleeping of MSS_Q6_TBU Clock 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
7	MSS_TBU_GSS_AXI_CLK_SLEEP_ENA	Enable for allowing sleeping of MSS_GSS_TBU Clock 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
6	MSS_TBU_AXI_CLK_SLEEP_ENA	Enable for allowing sleeping of MSS_AXI_TBU Clock 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
5	VENUS_TBU_CLK_SLEEP_ENA	Enable for allowing sleeping of VENUS_TBU Clock 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
4	MDP_TBU_CLK_SLEEP_ENA	Enable for allowing sleeping of MDP_TBU Clock 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
3	GFX_TBU_CLK_SLEEP_ENA	Enable for allowing sleeping of GFX_TBU Clock 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
2	GFX_TCU_CLK_SLEEP_ENA	Enable for allowing sleeping of GFX_TCU Clock 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
1	APSS_TCU_CLK_SLEEP_ENA	Enable for allowing sleeping of APSS_TCU Clock 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
0	PRONTO_TBU_CLK_SLEEP_ENA	Enable for allowing sleeping of PRONTO_TBU Clock 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x01861000 GCC_APCS_HYP_GPLL_ENA_VOTE**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

APCS HyperVisor GPLL enable voting register.

GCC_APCS_HYP_GPLL_ENA_VOTE

Bits	Name	Description
3	BIMC_PLL	Enable BIMC_PLL 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
2	GPLL2	Enable GPLL2 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
1	GPLL1	Enable GPLL1 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
0	GPLL0	Enable GPLL0 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x01861004 GCC_APCS_HYP_CLOCK_BRANCH_ENA_VOTE**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

APCS Hypervisor clock enable voting register for votable clocks.

GCC_APCS_HYP_CLOCK_BRANCH_ENA_VOTE

Bits	Name	Description
17	IMEM_AXI_CLK_ENA	Enable imem AXI clock 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
16	SYS_NOC_APSS_AHB_CLK_ENA	Enable sys_noc_apss_ahb_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
15	BIMC_APSS_AXI_CLK_ENA	Enable bimc_apss_axi_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
14	APSS_AHB_CLK_ENA	Enable apss_ahb_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
13	APSS_AXI_CLK_ENA	Enable apss_axi_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

GCC_APCS_HYP_CLOCK_BRANCH_ENA_VOTE (cont.)

Bits	Name	Description
12	MPM_AHB_CLK_ENA	Enable mpm_ahb_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
11	GMEM_SYS_NOC_AXI_CLK_ENA	Enable gmem_sys_noc_axi_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
10	BLSP1_AHB_CLK_ENA	Enable blsp1_ahb_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
9	BLSP1_SLEEP_CLK_ENA	Enable blsp1_sleep_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
8	PRNG_AHB_CLK_ENA	Enable prng_ahb_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
7	BOOT_ROM_AHB_CLK_ENA	Enable boot_rom_ahb_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
6	MSG_RAM_AHB_CLK_ENA	Enable msg_ram_ahb_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
5	TLMM_AHB_CLK_ENA	Enable tlmm_ahb_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
4	TLMM_CLK_ENA	Enable tlmm_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
3	SPMI_PCNOC_AHB_CLK_ENA	0x1: ENABLE 0x0: DISABLE
2	CRYPTO_CLK_ENA	Enable crypto_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
1	CRYPTO_AXI_CLK_ENA	Enable crypto_axi_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
0	CRYPTO_AHB_CLK_ENA	Enable crypto_ahb_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x0186100C GCC_APCS_HYP_SMMU_CLOCK_BRANCH_ENA_VOTE**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

APCS Hypervisor SMMU clock enable voting register for votable clocks.

GCC_APCS_HYP_SMMU_CLOCK_BRANCH_ENA_VOTE

Bits	Name	Description
13	GTCU_AHB_CLK_ENA	Enable GTCU_AHB CLock 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
12	SMMU_CFG_CLK_ENA	Enable SMMU_CFG CLock 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
11	PCNOC_TBU_CLK_ENA	Enable PCNOC_TBU Clock 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
10	JPEG_TBU_CLK_ENA	Enable JPEG_TBU Clock 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
9	VFE_TBU_CLK_ENA	Enable VFE_TBU Clock 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
8	MSS_TBU_Q6_AXI_CLK_ENA	Enable MSS_Q6_TBU Clock 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
7	MSS_TBU_GSS_AXI_CLK_ENA	Enable MSS_GSS_TBU Clock 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
6	MSS_TBU_AXI_CLK_ENA	Enable MSS_AXI_TBU Clock 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
5	VENUS_TBU_CLK_ENA	Enable VENUS_TBU Clock 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
4	MDP_TBU_CLK_ENA	Enable MDP_TBU Clock 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
3	GFX_TBU_CLK_ENA	Enable GFX_TBU Clock 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

GCC_APCS_HYP_SMMU_CLOCK_BRANCH_ENA_VOTE (cont.)

Bits	Name	Description
2	GFX_TCU_CLK_ENA	Enable GFX_TCU Clock 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
1	APSS_TCU_CLK_ENA	Enable APSS_TCU Clock 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
0	PRONTO_TBU_CLK_ENA	Enable PRONTO_TBU clock 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x01861008 GCC_APCS_HYP_CLOCK_SLEEP_ENA_VOTE**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

APCS HyperVisor clock sleep voting register for votable clocks.

GCC_APCS_HYP_CLOCK_SLEEP_ENA_VOTE

Bits	Name	Description
17	IMEM_AXI_CLK_SLEEP_ENA	Enable for allowing sleeping of IMEM AXI clock 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
16	SYS_NOC_APSS_AHB_CLK_SLEEP_ENA	Enable for allowing sleeping sys_noc_apss_ahb_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
15	BIMC_APSS_AXI_CLK_SLEEP_ENA	Enable for allowing sleeping bimc_apss_axi_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
14	APSS_AHB_CLK_SLEEP_ENA	Enable for allowing sleeping apss_ahb_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
13	APSS_AXI_CLK_SLEEP_ENA	Enable for allowing sleeping apss_axi_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
12	MPM_AHB_CLK_SLEEP_ENA	Enable for allowing sleeping mpm_ahb_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

GCC_APCS_HYP_CLOCK_SLEEP_ENA_VOTE (cont.)

Bits	Name	Description
11	GMEM_SYS_NOC_AXI_CLK_SLEEP_ENA	Enable for allowing sleeping gmem_sys_noc_axi_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
10	BLSP1_AHB_CLK_SLEEP_ENA	Enable for allowing sleeping blsp1_ahb_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
9	BLSP1_SLEEP_CLK_SLEEP_ENA	Enable for allowing sleeping blsp1_sleep_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
8	PRNG_AHB_CLK_SLEEP_ENA	Enable for allowing sleeping prng_ahb_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
7	BOOT_ROM_AHB_CLK_SLEEP_ENA	Enable for allowing sleeping boot_rom_ahb_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
6	MSG_RAM_AHB_CLK_SLEEP_ENA	Enable for allowing sleeping msg_ram_ahb_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
5	TLMM_AHB_CLK_SLEEP_ENA	Enable for allowing sleeping tlmm_ahb_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
4	TLMM_CLK_SLEEP_ENA	Enable for allowing sleeping tlmm_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
3	SPMI_PCNOC_AHB_CLK_SLEEP_ENA	0x1: ENABLE 0x0: DISABLE
2	CRYPTO_CLK_SLEEP_ENA	Enable for allowing sleeping crypto_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
1	CRYPTO_AXI_CLK_SLEEP_ENA	Enable for allowing sleeping crypto_axi_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
0	CRYPTO_AHB_CLK_SLEEP_ENA	Enable for allowing sleeping crypto_ahb_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x01861010 GCC_APCS_HYP_SMMU_CLOCK_SLEEP_ENA_VOTE**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

APCS HyperVisor SMMU clock sleep voting register for votable clocks.

GCC_APCS_HYP_SMMU_CLOCK_SLEEP_ENA_VOTE

Bits	Name	Description
13	GTCU_AHB_CLK_SLEEP_ENA	Enable for allowing sleeping of GTCU_AHB Clock 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
12	SMMU_CFG_CLK_SLEEP_ENA	Enable for allowing sleeping of SMMU_CFG Clock 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
11	PCNOC_TBU_CLK_SLEEP_ENA	Enable for allowing sleeping of PCNOC_TBU Clock 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
10	JPEG_TBU_CLK_SLEEP_ENA	Enable for allowing sleeping of JPEG_TBU Clock 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
9	VFE_TBU_CLK_SLEEP_ENA	Enable for allowing sleeping of VFE_TBU Clock 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
8	MSS_TBU_Q6_AXI_CLK_SLEEP_ENA	Enable for allowing sleeping of MSS_Q6_TBU Clock 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
7	MSS_TBU_GSS_AXI_CLK_SLEEP_ENA	Enable for allowing sleeping of MSS_GSS_TBU Clock 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
6	MSS_TBU_AXI_CLK_SLEEP_ENA	Enable for allowing sleeping of MSS_AXI_TBU Clock 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
5	VENUS_TBU_CLK_SLEEP_ENA	Enable for allowing sleeping of VENUS_TBU Clock 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
4	MDP_TBU_CLK_SLEEP_ENA	Enable for allowing sleeping of MDP_TBU Clock 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
3	GFX_TBU_CLK_SLEEP_ENA	Enable for allowing sleeping of GFX_TBU Clock 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

GCC_APCS_HYP_SMMU_CLOCK_SLEEP_ENA_VOTE (cont.)

Bits	Name	Description
2	GFX_TCU_CLK_SLEEP_ENA	Enable for allowing sleeping of GFX_TCU Clock 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
1	APSS_TCU_CLK_SLEEP_ENA	Enable for allowing sleeping of APSS_TCU Clock 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
0	PRONTO_TBU_CLK_SLEEP_ENA	Enable for allowing sleeping of PRONTO_TBU Clock 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x0181B000 GCC_MSS_Q6_GPLL_ENA_VOTE**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

MSS Q6 GPLL enable voting register.

GCC_MSS_Q6_GPLL_ENA_VOTE

Bits	Name	Description
3	BIMC_PLL	Enable BIMC_PLL 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
2	GPLL2	Enable GPLL2 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
1	GPLL1	Enable GPLL1 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
0	GPLL0	Enable GPLL0 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x0181B004 GCC_MSS_Q6_CLOCK_BRANCH_ENA_VOTE**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

MSS Q6 clock enable voting register for votable clocks.

GCC_MSS_Q6_CLOCK_BRANCH_ENA_VOTE

Bits	Name	Description
17	IMEM_AXI_CLK_ENA	Enable imem AXI clock 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
16	SYS_NOC_APSS_AHB_CLK_ENA	Enable sys_noc_apss_ahb_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
15	BIMC_APSS_AXI_CLK_ENA	Enable bimc_apss_axi_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
14	APSS_AHB_CLK_ENA	Enable apss_ahb_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
13	APSS_AXI_CLK_ENA	Enable apss_axi_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
12	MPM_AHB_CLK_ENA	Enable mpm_ahb_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
11	GMEM_SYS_NOC_AXI_CLK_ENA	Enable gmem_sys_noc_axi_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
10	BLSP1_AHB_CLK_ENA	Enable blsp1_ahb_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
9	BLSP1_SLEEP_CLK_ENA	Enable blsp1_sleep_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
8	PRNG_AHB_CLK_ENA	Enable prng_ahb_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
7	BOOT_ROM_AHB_CLK_ENA	Enable boot_rom_ahb_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
6	MSG_RAM_AHB_CLK_ENA	Enable msg_ram_ahb_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
5	TLMM_AHB_CLK_ENA	Enable tlmm_ahb_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

GCC_MSS_Q6_CLOCK_BRANCH_ENA_VOTE (cont.)

Bits	Name	Description
4	TLMM_CLK_ENA	Enable tlmm_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
3	SPMI_PCNOC_AHB_CLK_ENA	0x1: ENABLE 0x0: DISABLE
2	CRYPTO_CLK_ENA	Enable crypto_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
1	CRYPTO_AXI_CLK_ENA	Enable crypto_axi_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
0	CRYPTO_AHB_CLK_ENA	Enable crypto_ahb_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x0181B008 GCC_MSS_Q6_CLOCK_SLEEP_ENA_VOTE**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

MSS Q6 clock sleep voting register for votable clocks.

GCC_MSS_Q6_CLOCK_SLEEP_ENA_VOTE

Bits	Name	Description
17	IMEM_AXI_CLK_SLEEP_ENA	Enable for allowing sleeping of IMEM AXI clock 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
16	SYS_NOC_APSS_AHB_CLK_SLEEP_ENA	Enable for allowing sleeping sys_noc_apss_ahb_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
15	BIMC_APSS_AXI_CLK_SLEEP_ENA	Enable for allowing sleeping bimc_apss_axi_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
14	APSS_AHB_CLK_SLEEP_ENA	Enable for allowing sleeping apss_ahb_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
13	APSS_AXI_CLK_SLEEP_ENA	Enable for allowing sleeping apss_axi_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

GCC_MSS_Q6_CLOCK_SLEEP_ENA_VOTE (cont.)

Bits	Name	Description
12	MPM_AHB_CLK_SLEEP_ENA	Enable for allowing sleeping mpm_ahb_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
11	GMEM_SYS_NOC_AXI_CLK_SLEEP_ENA	Enable for allowing sleeping gmem_sys_noc_axi_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
10	BLSP1_AHB_CLK_SLEEP_ENA	Enable for allowing sleeping blsp1_ahb_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
9	BLSP1_SLEEP_CLK_SLEEP_ENA	Enable for allowing sleeping blsp1_sleep_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
8	PRNG_AHB_CLK_SLEEP_ENA	Enable for allowing sleeping prng_ahb_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
7	BOOT_ROM_AHB_CLK_SLEEP_ENA	Enable for allowing sleeping boot_rom_ahb_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
6	MSG_RAM_AHB_CLK_SLEEP_ENA	Enable for allowing sleeping msg_ram_ahb_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
5	TLMM_AHB_CLK_SLEEP_ENA	Enable for allowing sleeping tlmm_ahb_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
4	TLMM_CLK_SLEEP_ENA	Enable for allowing sleeping tlmm_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
3	SPMI_PCNOC_AHB_CLK_SLEEP_ENA	0x1: ENABLE 0x0: DISABLE
2	CRYPTO_CLK_SLEEP_ENA	Enable for allowing sleeping crypto_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
1	CRYPTO_AXI_CLK_SLEEP_ENA	Enable for allowing sleeping crypto_axi_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
0	CRYPTO_AHB_CLK_SLEEP_ENA	Enable for allowing sleeping crypto_ahb_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x01870000 GCC_WCSS_GPLL_ENA_VOTE**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

WCSS GPLL enable voting register.

GCC_WCSS_GPLL_ENA_VOTE

Bits	Name	Description
3	BIMC_PLL	Enable BIMC_PLL 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
2	GPLL2	Enable GPLL2 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
1	GPLL1	Enable GPLL1 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
0	GPLL0	Enable GPLL0 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x01870004 GCC_WCSS_CLOCK_BRANCH_ENA_VOTE**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

WCSS clock enable voting register for votable clocks.

GCC_WCSS_CLOCK_BRANCH_ENA_VOTE

Bits	Name	Description
17	IMEM_AXI_CLK_ENA	Enable imem AXI clock 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
16	SYS_NOC_APSS_AHB_CLK_ENA	Enable sys_noc_apss_ahb_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
15	BIMC_APSS_AXI_CLK_ENA	Enable bimc_apss_axi_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

GCC_WCSS_CLOCK_BRANCH_ENA_VOTE (cont.)

Bits	Name	Description
14	APSS_AHB_CLK_ENA	Enable apss_ahb_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
13	APSS_AXI_CLK_ENA	Enable apss_axi_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
12	MPM_AHB_CLK_ENA	Enable mpm_ahb_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
11	GMEM_SYS_NOC_AXI_CLK_ENA	Enable gmem_sys_noc_axi_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
10	BLSP1_AHB_CLK_ENA	Enable blsp1_ahb_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
9	BLSP1_SLEEP_CLK_ENA	Enable blsp1_sleep_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
8	PRNG_AHB_CLK_ENA	Enable prng_ahb_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
7	BOOT_ROM_AHB_CLK_ENA	Enable boot_rom_ahb_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
6	MSG_RAM_AHB_CLK_ENA	Enable msg_ram_ahb_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
5	TLMM_AHB_CLK_ENA	Enable tlmm_ahb_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
4	TLMM_CLK_ENA	Enable tlmm_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
3	SPMI_PCNOB_AHB_CLK_ENA	0x1: ENABLE 0x0: DISABLE
2	CRYPTO_CLK_ENA	Enable crypto_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

GCC_WCSS_CLOCK_BRANCH_ENA_VOTE (cont.)

Bits	Name	Description
1	CRYPTO_AXI_CLK_ENA	Enable crypto_axi_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
0	CRYPTO_AHB_CLK_ENA	Enable crypto_ahb_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x01870008 GCC_WCSS_CLOCK_SLEEP_ENA_VOTE**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

WCSS clock sleep voting register for votable clocks.

GCC_WCSS_CLOCK_SLEEP_ENA_VOTE

Bits	Name	Description
17	IMEM_AXI_CLK_SLEEP_ENA	Enable for allowing sleeping of IMEM AXI clock 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
16	SYS_NOC_APSS_AHB_CLK_SLEEP_ENA	Enable for allowing sleeping sys_noc_apss_ahb_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
15	BIMC_APSS_AXI_CLK_SLEEP_ENA	Enable for allowing sleeping bimc_apss_axi_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
14	APSS_AHB_CLK_SLEEP_ENA	Enable for allowing sleeping apss_ahb_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
13	APSS_AXI_CLK_SLEEP_ENA	Enable for allowing sleeping apss_axi_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
12	MPM_AHB_CLK_SLEEP_ENA	Enable for allowing sleeping mpm_ahb_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
11	GMEM_SYS_NOC_AXI_CLK_SLEEP_ENA	Enable for allowing sleeping gmem_sys_noc_axi_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

GCC_WCSS_CLOCK_SLEEP_ENA_VOTE (cont.)

Bits	Name	Description
10	BLSP1_AHB_CLK_SLEEP_ENA	Enable for allowing sleeping blsp1_ahb_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
9	BLSP1_SLEEP_CLK_SLEEP_ENA	Enable for allowing sleeping blsp1_sleep_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
8	PRNG_AHB_CLK_SLEEP_ENA	Enable for allowing sleeping prng_ahb_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
7	BOOT_ROM_AHB_CLK_SLEEP_ENA	Enable for allowing sleeping boot_rom_ahb_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
6	MSG_RAM_AHB_CLK_SLEEP_ENA	Enable for allowing sleeping msg_ram_ahb_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
5	TLMM_AHB_CLK_SLEEP_ENA	Enable for allowing sleeping tlmm_ahb_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
4	TLMM_CLK_SLEEP_ENA	Enable for allowing sleeping tlmm_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
3	SPMI_PCNOB_AHB_CLK_SLEEP_ENA	0x1: ENABLE 0x0: DISABLE
2	CRYPTO_CLK_SLEEP_ENA	Enable for allowing sleeping crypto_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
1	CRYPTO_AXI_CLK_SLEEP_ENA	Enable for allowing sleeping crypto_axi_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
0	CRYPTO_AHB_CLK_SLEEP_ENA	Enable for allowing sleeping crypto_ahb_clk 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x01810004 GCC_VENUS_RESTART**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

VENUS_RESTART register allows SW to restart (reset) Venus

GCC_VENUS_RESTART

Bits	Name	Description
0	VENUS_RESTART	Allows restart (reset) to VENUS. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x01811000 GCC_WCSS_RESTART**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

WCSS_RESTART register allows SW to restart (reset) Wireless Sub-system

GCC_WCSS_RESTART

Bits	Name	Description
0	WCSS_RESTART	Allows restart (reset) to WCSS. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x01815008 GCC_RESET_STATUS**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

RESET STATUS Captures reset status for various sources.

GCC_RESET_STATUS

Bits	Name	Description
5	SECURE_WDOG_EXPIRE_STATUS	Set when secure wdog expire happens. 1: Secure wdog expired
4	PMIC_ABNORMAL_RESIN_STATUS	Set when abnormal resin occurs. 1: abnormal resin triggered
3	TSENSE_RESET_STATUS	Set when temperature sensor activates. 1: tsensor triggered
2	SRST_STATUS	Set high when any SRST triggered. 1: srst triggered
1:0	RESERVED	RESERVED

0x0181500C GCC_SW_SRST**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

SW SRST allows system reset for debugger.

GCC_SW_SRST

Bits	Name	Description
0	SW_SRST	set to reset system. 0x1: ENABLE (Software System reset enable.) 0x0: DISABLE (Software System reset disable.)

0x0181301C GCC_PROC_HALT**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

PROC HALT allows halting processor for debugger.

GCC_PROC_HALT

Bits	Name	Description
0	PROC_HALT	Clear to release halt to processor. 0x1: HALT (Processor halt.) 0x0: NOT_HALT (Processor not halt.)

0x01874004 GCC_CLOCK_FRQ_MEASURE_CTL**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

The clock frequency measure control register holds the value to load in the XO DIV4 counter. It also contains the enable to turn on both up and down counters for measure frequency for the corresponding debug clock.

NOTE The XO DIV4 counter use the gcc_xo_div4_clk. One must enable the gcc_xo_div4_clk by programming GCC_XO_DIV4_CBCR.

GCC_CLOCK_FRQ_MEASURE_CTL

Bits	Name	Description
20	CNT_EN	This bitfield is used to enable both the Up and Down counter. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
19:0	XO_DIV4_TERM_CNT	After being loaded, this value is decremented and used to compare in order to figure out the frequency of the clock being measured. $\text{Freq}(\text{measure clock}) = \text{Freq}(\text{XO_DIV4}) * (\text{MEASURE_CNT} + 1.5) / (\text{XO_DIV4_TERM_CNT} + 3.5)$

0x01874008 GCC_CLOCK_FRQ_MEASURE_STATUS**Type:** R**Clock:** CLK**Reset State:** 0x00000000

This register is read only register which contains the status of the measurement counter and its value.

It is used to figure out the frequency of the clock being measured at the end.

GCC_CLOCK_FRQ_MEASURE_STATUS

Bits	Name	Description
25	XO_DIV4_CNT_DONE	This bitfield is automatically set (1) when the XO_DIV4 counter terminates. It can be polled by the microprocessor to determine when the value of the measurement counter is valid
24:0	MEASURE_CNT	These read only register bits hold the value stored in the measurement counter, which is used to figure out the frequency of the clock being measured. $\text{Freq}(\text{measure clock}) = \text{Freq}(\text{XO_DIV4}) * (\text{MEASURE_CNT} + 1.5) / (\text{XO_DIV4_TERM_CNT} + 3.5)$

0x01808000 GCC_GP1_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

GP1 clock branch control register. It allows SW control of CBC soft macro

GCC_GP1_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	This bitfield indicated if the clock branch is halted 0:Clock On 1:Clock Off
0	CLK_ENABLE	RW	This bitfield indicated if the clock branch is enabled. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x01808004 GCC_GP1_CMD_RCGR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

GP1 root clock generator command register. It allows for SW to trigger the configuration data

GCC_GP1_CMD_RCGR

Bits	Name	Type	Description
31	ROOT_OFF	R	This bitfield provides indication on status of the RCG; Mainly for SW to determine if the generator is actively producing clock 1: Clock Off 0: Clock On; Generator is actively producing the clock
7	DIRTY_D	R	Indicates change in D register configuration
6	DIRTY_M	R	Indicates change in M register configuration
5	DIRTY_N	R	Indicates change in N register configuration
4	DIRTY_CFG_RCGR	R	Indicates change in CFG_RCGR register configuration
1	ROOT_EN	RW	Enable Clock Generator output. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
0	UPDATE	RW	Update to the new programmed configuration. It should be cleared upon switching to new configuration 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x01808008 GCC_GP1_CFG_RCGR**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

GP1 root clock generator configuration register. It allows for SW to configure Root Clock Generation Macro

GCC_GP1_CFG_RCGR

Bits	Name	Description
13:12	MODE	This bitfield provides SW selection of MN counter mode. 0x3: SINGLE_EDGE (Single-edge.) 0x2: DUAL_EDGE (Dual-edge.) 0x1: SWALLOW (Swallow.) 0x0: BYPASS (Bypass.)
10:8	SRC_SEL	This bitfield selects clock source that will be used to source the RCG output 0x0: SRC0 (CXO.) 0x1: SRC1 (GPLL0_MAIN.) 0x2: SRC2 (Unused(Grounded).) 0x3: SRC3 (Unused(Grounded).) 0x4: SRC4 (Unused(Grounded).) 0x5: SRC5 (Unused(Grounded).) 0x6: SRC6 (Unused(Grounded).) 0x7: SRC7 (Unused(Grounded).)

GCC_GP1_CFG_RCGR (cont.)

Bits	Name	Description
4:0	SRC_DIV	This field defines the integer and half integer portion of the divider setting of the RCG. The value specifies is $2N-1$ where N is the divide ratio. For example, if the half integer divide ration is 2.5, the src_div should be set at 00100. 0x0: BYPASS (Bypsss.) 0x1: DIV1 (Divide by 1.) 0x2: DIV1_5 (Divide by 1.5.) 0x3: DIV2 (Divide by 2.) 0x4: DIV2_5 (Divide by 2.5.) 0x5: DIV3 (Divide by 3.) 0x6: DIV3_5 (Divide by 3.5.) 0x7: DIV4 (Divide by 4.) 0x8: DIV4_5 (Divide by 4.5.) 0x9: DIV5 (Divide by 5.) 0xA: DIV5_5 (Divide by 5.5.) 0xB: DIV6 (Divide by 6.) 0xC: DIV6_5 (Divide by 6.5.) 0xD: DIV7 (Divide by 7.) 0xE: DIV7_5 (Divide by 7.5.) 0xF: DIV8 (Divide by 8.) 0x10: DIV8_5 (Divide by 8.5.) 0x11: DIV9 (Divide by 9.) 0x12: DIV9_5 (Divide by 9.5.) 0x13: DIV10 (Divide by 10.) 0x14: DIV10_5 (Divide by 10.5.) 0x15: DIV11 (Divide by 11.) 0x16: DIV11_5 (Divide by 11.5.) 0x17: DIV12 (Divide by 12.) 0x18: DIV12_5 (Divide by 12.5.) 0x19: DIV13 (Divide by 13.) 0x1A: DIV13_5 (Divide by 13.5.) 0x1B: DIV14 (Divide by 14.) 0x1C: DIV14_5 (Divide by 14.5.) 0x1D: DIV15 (Divide by 15.) 0x1E: DIV15_5 (Divide by 15.5.) 0x1F: DIV16 (Divide by 16.)

0x0180800C GCC_GP1_M**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

GP1 root clock generator M value register. This register houses the M value for MND divider.

GCC_GP1_M

Bits	Name	Description
7:0	M	This bitfield represents M Value of the M/N:D Counter

0x01808010 GCC_GP1_N**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

GP1 root clock generator N value register. This register houses the NOT(N-M) value for MND divider.

GCC_GP1_N

Bits	Name	Description
7:0	NOT_N_MINUS_M	This bitfield represents NOT(N-M) Value of the M/N:D Counter

0x01808014 GCC_GP1_D**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

GP1 root clock generator D value register. This register houses the NOT 2D value for MND divider.

GCC_GP1_D

Bits	Name	Description
7:0	NOT_2D	This bitfield represents NOT(2*D) Value of the M/N:D Counter

0x01809000 GCC_GP2_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

GP2 clock branch control register. It allows SW control of CBC soft macro

GCC_GP2_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	This bitfield indicated if the clock branch is halted 0:Clock On 1:Clock Off
0	CLK_ENABLE	RW	This bitfield indicated if the clock branch is enabled. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x01809004 GCC_GP2_CMD_RCGR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

GP2 root clock generator command register. It allows for SW to trigger the configuration data

GCC_GP2_CMD_RCGR

Bits	Name	Type	Description
31	ROOT_OFF	R	This bitfield provides indication on status of the RCG; Mainly for SW to determine if the generator is actively producing clock 1: Clock Off 0: Clock On; Generator is actively producing the clock
7	DIRTY_D	R	Indicates change in D register configuration
6	DIRTY_M	R	Indicates change in M register configuration
5	DIRTY_N	R	Indicates change in N register configuration
4	DIRTY_CFG_RCGR	R	Indicates change in CFG_RCGR register configuration
1	ROOT_EN	RW	Enable Clock Generator output. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
0	UPDATE	RW	Update to the new programmed configuration. It should be cleared upon switching to new configuration 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x01809008 GCC_GP2_CFG_RCGR**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

GP2 root clock generator configuration register. It allows for SW to configure Root Clock Generation Macro

GCC_GP2_CFG_RCGR

Bits	Name	Description
13:12	MODE	This bitfield provides SW selection of MN counter mode. 0x3: SINGLE_EDGE (Single-edge.) 0x2: DUAL_EDGE (Dual-edge.) 0x1: SWALLOW (Swallow.) 0x0: BYPASS (Bypass.)
10:8	SRC_SEL	This bitfield selects clock source that will be used to source the RCG output 0x0: SRC0 (CXO.) 0x1: SRC1 (GPLL0_MAIN.) 0x2: SRC2 (Unused(Grounded).) 0x3: SRC3 (Unused(Grounded).) 0x4: SRC4 (Unused(Grounded).) 0x5: SRC5 (Unused(Grounded).) 0x6: SRC6 (Unused(Grounded).) 0x7: SRC7 (Unused(Grounded).)

GCC_GP2_CFG_RCGR (cont.)

Bits	Name	Description
4:0	SRC_DIV	This field defines the integer and half integer portion of the divider setting of the RCG. The value specifies is $2N-1$ where N is the divide ratio. For example, if the half integer divide ration is 2.5, the src_div should be set at 00100. 0x0: BYPASS (Bypsss.) 0x1: DIV1 (Divide by 1.) 0x2: DIV1_5 (Divide by 1.5.) 0x3: DIV2 (Divide by 2.) 0x4: DIV2_5 (Divide by 2.5.) 0x5: DIV3 (Divide by 3.) 0x6: DIV3_5 (Divide by 3.5.) 0x7: DIV4 (Divide by 4.) 0x8: DIV4_5 (Divide by 4.5.) 0x9: DIV5 (Divide by 5.) 0xA: DIV5_5 (Divide by 5.5.) 0xB: DIV6 (Divide by 6.) 0xC: DIV6_5 (Divide by 6.5.) 0xD: DIV7 (Divide by 7.) 0xE: DIV7_5 (Divide by 7.5.) 0xF: DIV8 (Divide by 8.) 0x10: DIV8_5 (Divide by 8.5.) 0x11: DIV9 (Divide by 9.) 0x12: DIV9_5 (Divide by 9.5.) 0x13: DIV10 (Divide by 10.) 0x14: DIV10_5 (Divide by 10.5.) 0x15: DIV11 (Divide by 11.) 0x16: DIV11_5 (Divide by 11.5.) 0x17: DIV12 (Divide by 12.) 0x18: DIV12_5 (Divide by 12.5.) 0x19: DIV13 (Divide by 13.) 0x1A: DIV13_5 (Divide by 13.5.) 0x1B: DIV14 (Divide by 14.) 0x1C: DIV14_5 (Divide by 14.5.) 0x1D: DIV15 (Divide by 15.) 0x1E: DIV15_5 (Divide by 15.5.) 0x1F: DIV16 (Divide by 16.)

0x0180900C GCC_GP2_M**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

GP2 root clock generator M value register. This register houses the M value for MND divider.

GCC_GP2_M

Bits	Name	Description
7:0	M	This bitfield represents M Value of the M/N:D Counter

0x01809010 GCC_GP2_N**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

GP2 root clock generator N value register. This register houses the NOT(N-M) value for MND divider.

GCC_GP2_N

Bits	Name	Description
7:0	NOT_N_MINUS_M	This bitfield represents NOT(N-M) Value of the M/N:D Counter

0x01809014 GCC_GP2_D**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

GP2 root clock generator D value register. This register houses the NOT 2D value for MND divider.

GCC_GP2_D

Bits	Name	Description
7:0	NOT_2D	This bitfield represents NOT(2*D) Value of the M/N:D Counter

0x0180A000 GCC_GP3_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

GP3 clock branch control register. It allows SW control of CBC soft macro

GCC_GP3_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	This bitfield indicated if the clock branch is halted 0:Clock On 1:Clock Off
0	CLK_ENABLE	RW	This bitfield indicated if the clock branch is enabled. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x0180A004 GCC_GP3_CMD_RCGR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

GP3 root clock generator command register. It allows for SW to trigger the configuration data

GCC_GP3_CMD_RCGR

Bits	Name	Type	Description
31	ROOT_OFF	R	This bitfield provides indication on status of the RCG; Mainly for SW to determine if the generator is actively producing clock 1: Clock Off 0: Clock On; Generator is actively producing the clock
7	DIRTY_D	R	Indicates change in D register configuration
6	DIRTY_M	R	Indicates change in M register configuration
5	DIRTY_N	R	Indicates change in N register configuration
4	DIRTY_CFG_RCGR	R	Indicates change in CFG_RCGR register configuration
1	ROOT_EN	RW	Enable Clock Generator output. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
0	UPDATE	RW	Update to the new programmed configuration. It should be cleared upon switching to new configuration 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x0180A008 GCC_GP3_CFG_RCGR**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

GP3 root clock generator configuration register. It allows for SW to configure Root Clock Generation Macro

GCC_GP3_CFG_RCGR

Bits	Name	Description
13:12	MODE	This bitfield provides SW selection of MN counter mode. 0x3: SINGLE_EDGE (Single-edge.) 0x2: DUAL_EDGE (Dual-edge.) 0x1: SWALLOW (Swallow.) 0x0: BYPASS (Bypass.)
10:8	SRC_SEL	This bitfield selects clock source that will be used to source the RCG output 0x0: SRC0 (CXO.) 0x1: SRC1 (GPLL0_MAIN.) 0x2: SRC2 (Unused(Grounded).) 0x3: SRC3 (Unused(Grounded).) 0x4: SRC4 (Unused(Grounded).) 0x5: SRC5 (Unused(Grounded).) 0x6: SRC6 (Unused(Grounded).) 0x7: SRC7 (Unused(Grounded).)

GCC_GP3_CFG_RCGR (cont.)

Bits	Name	Description
4:0	SRC_DIV	This field defines the integer and half integer portion of the divider setting of the RCG. The value specifies is $2N-1$ where N is the divide ratio. For example, if the half integer divide ration is 2.5, the src_div should be set at 00100. 0x0: BYPASS (Bypsss.) 0x1: DIV1 (Divide by 1.) 0x2: DIV1_5 (Divide by 1.5.) 0x3: DIV2 (Divide by 2.) 0x4: DIV2_5 (Divide by 2.5.) 0x5: DIV3 (Divide by 3.) 0x6: DIV3_5 (Divide by 3.5.) 0x7: DIV4 (Divide by 4.) 0x8: DIV4_5 (Divide by 4.5.) 0x9: DIV5 (Divide by 5.) 0xA: DIV5_5 (Divide by 5.5.) 0xB: DIV6 (Divide by 6.) 0xC: DIV6_5 (Divide by 6.5.) 0xD: DIV7 (Divide by 7.) 0xE: DIV7_5 (Divide by 7.5.) 0xF: DIV8 (Divide by 8.) 0x10: DIV8_5 (Divide by 8.5.) 0x11: DIV9 (Divide by 9.) 0x12: DIV9_5 (Divide by 9.5.) 0x13: DIV10 (Divide by 10.) 0x14: DIV10_5 (Divide by 10.5.) 0x15: DIV11 (Divide by 11.) 0x16: DIV11_5 (Divide by 11.5.) 0x17: DIV12 (Divide by 12.) 0x18: DIV12_5 (Divide by 12.5.) 0x19: DIV13 (Divide by 13.) 0x1A: DIV13_5 (Divide by 13.5.) 0x1B: DIV14 (Divide by 14.) 0x1C: DIV14_5 (Divide by 14.5.) 0x1D: DIV15 (Divide by 15.) 0x1E: DIV15_5 (Divide by 15.5.) 0x1F: DIV16 (Divide by 16.)

0x0180A00C GCC_GP3_M**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

GP3 root clock generator M value register. This register houses the M value for MND divider.

GCC_GP3_M

Bits	Name	Description
7:0	M	This bitfield represents M Value of the M/N:D Counter

0x0180A010 GCC_GP3_N**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

GP3 root clock generator N value register. This register houses the NOT(N-M) value for MND divider.

GCC_GP3_N

Bits	Name	Description
7:0	NOT_N_MINUS_M	This bitfield represents NOT(N-M) Value of the M/N:D Counter

0x0180A014 GCC_GP3_D**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

GP3 root clock generator D value register. This register houses the NOT 2D value for MND divider.

GCC_GP3_D

Bits	Name	Description
7:0	NOT_2D	This bitfield represents NOT(2*D) Value of the M/N:D Counter

0x01860000 GCC_APSS_MISC**Type:** RW**Clock:** CLK**Reset State:** 0x00000003

This register contains 3 bits which are described below.

GCC_APSS_MISC

Bits	Name	Description
2	AUX2_CLK_ENABLE	To enable AUX2 clock for APSS, from GPLL1. This bitfield indicated if the clock branch is enabled. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
1	AUX1_CLK_ENABLE	To enable AUX1 clock for APSS, from GPLL0. This bitfield indicated if the clock branch is enabled. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
0	RPM_RESET_REMOVAL	This bitfield should be programmed to remove RPM Core from Reset. 0x1: RESET (RPM COre is in Reset.) 0x0: RELEASE (RPM COre reset is released.)

0x0187F000 GCC_TIC_MODE_APSS_BOOT**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

This register contains one bit which when set, will allow APSS to BOOT in TIC Mode.

GCC_TIC_MODE_APSS_BOOT

Bits	Name	Description
0	APSS_BOOT_IN_TIC_MODE	This bitfield should be programmed for any use case of TIC needing to boot APSS in TIC Mode. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x01840000 GCC_USB_BOOT_CLOCK_CTL**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

This register contains one bit which when set, will turn on all clocks necessary to access USB 2.0 during boot code.

GCC_USB_BOOT_CLOCK_CTL

Bits	Name	Description
0	CLK_ENABLE	This bitfield indicated if the clock branch is enabled. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x01835000 GCC_RAW_SLEEP_CLK_CTRL**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

This register contains one bit which when set, will turn off the gating feature of raw sleep clock when XO clock is disabled at the XO pad by MPM.

GCC_RAW_SLEEP_CLK_CTRL

Bits	Name	Description
0	GATING_DISABLE	This bitfield controls the disable to the raw sleep clock gating feature. 0x1: DISABLE (Sleep clock gating is disabled so that when XO clock is disabled at pad, raw sleep clock will keep running.) 0x0: ENABLE (Sleep clock gating is enabled so that when XO clock is disabled at pad, raw sleep clock will stop running at well.)

0x01800008 GCC_GCC_SPARE0_REG**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

This register is spare register protected by RPU

GCC_GCC_SPARE0_REG

Bits	Name	Description
31:0	SPARE_BITS	32 bits spare bits

0x0180000C GCC_GCC_SPARE1_REG**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

This register is spare register protected by RPU

GCC_GCC_SPARE1_REG

Bits	Name	Description
31:0	SPARE_BITS	32 bits spare bits

0x0187E000 GCC_GCC_SPARE2_REG**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

This register is spare register not protected by RPU

GCC_GCC_SPARE2_REG

Bits	Name	Description
31:0	SPARE_BITS	32 bits spare bits

0x0187E004 GCC_GCC_SPARE3_REG**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

This register is spare register not protected by RPU

GCC_GCC_SPARE3_REG

Bits	Name	Description
31:0	SPARE_BITS	32 bits spare bits

0x0182A000 GCC_DCD_BCR**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

DCD block control register

GCC_DCD_BCR

Bits	Name	Description
0	BLK_ARES	This bitfield controls asynchronous software reset for the block. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x0182A004 GCC_DCD_XO_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x00000001

DCD XO clock branch control register. It allows SW control of DCD XO CBC soft macro

GCC_DCD_XO_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	This bitfield indicated if the clock branch is halted 0:Clock On 1:Clock Off
0	CLK_ENABLE	RW	This bitfield indicated if the clock branch is enabled. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x0182A008 GCC_SNOC_DCD_CONFIG**Type:** RW**Clock:** CLK**Reset State:** 0x00004045

This register contains register bits for specifying DCD user configuration settings.

GCC_SNOC_DCD_CONFIG

Bits	Name	Description
31	DCD_ENABLE	Enable bit for DCD
14:0	ALLOWED_DIV	Allowed divide values for DCD

0x0182A00C GCC_SNOC_DCD_HYSTERESIS_CNT**Type:** RW**Clock:** CLK**Reset State:** 0x00100080

This register contains register bits for specifying DCD user configuration settings.

GCC_SNOC_DCD_HYSTERESIS_CNT

Bits	Name	Description
31:12	FIRST_CNT	Hysteresis count on first divide on DCD
11:0	NEXT_CNT	Hysteresis count on succeeding divides on DCD

0x0182A010 GCC_PCNOCD_DCD_CONFIG**Type:** RW**Clock:** CLK**Reset State:** 0x00004045

This register contains register bits for specifying DCD user configuration settings.

GCC_PCNOCD_DCD_CONFIG

Bits	Name	Description
31	DCD_ENABLE	Enable bit for DCD
14:0	ALLOWED_DIV	Allowed divide values for DCD

0x0182A014 GCC_PCNOCD_DCD_HYSTERESIS_CNT**Type:** RW**Clock:** CLK**Reset State:** 0x00100080

This register contains register bits for specifying DCD user configuration settings.

GCC_PCNOG_DCD_HYSTERESIS_CNT

Bits	Name	Description
31:12	FIRST_CNT	Hysteresis count on first divide on DCD
11:0	NEXT_CNT	Hysteresis count on succeeding divides on DCD

0x0182F024 GCC_MM_SPDM_BCR**Type:** RW**Clock:** CLK**Reset State:** 0x00000000**GCC_MM_SPDM_BCR**

Bits	Name	Description
0	BLK_ARES	

0x0182F028 GCC_SPDM_JPEG0_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

PSCBC Registers for MM_SPDM_JPEG0

GCC_SPDM_JPEG0_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	Branch Clock Off 0: Clock is ON 1: Clock is OFF
0	CLK_ENABLE		SW Clock Enable Bit 0: Clock Disabled 1: Clock Enabled

0x0182F02C GCC_SPDM_MDP_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

PSCBC Registers for MM_SPDM_MDP

GCC_SPDM_MDP_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	Branch Clock Off 0: Clock is ON 1: Clock is OFF
0	CLK_ENABLE		SW Clock Enable Bit 0: Clock Disabled 1: Clock Enabled

0x0182F034 GCC_SPDM_VCODEC0_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

PSCBC Registers for MM_SPDM_VCODEC0

GCC_SPDM_VCODEC0_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	Branch Clock Off 0: Clock is ON 1: Clock is OFF
0	CLK_ENABLE		SW Clock Enable Bit 0: Clock Disabled 1: Clock Enabled

0x0182F038 GCC_SPDM_VFE0_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

PSCBC Registers for MM_SPDM_VFE0

GCC_SPDM_VFE0_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	Branch Clock Off 0: Clock is ON 1: Clock is OFF

GCC_SPDM_VFE0_CBCR (cont.)

Bits	Name	Type	Description
0	CLK_ENABLE		SW Clock Enable Bit 0: Clock Disabled 1: Clock Enabled

0x0182F03C GCC_SPDM_GFX3D_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

PSCBC Registers for MM_SPDM_GFX3D

GCC_SPDM_GFX3D_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	Branch Clock Off 0: Clock is ON 1: Clock is OFF
0	CLK_ENABLE		SW Clock Enable Bit 0: Clock Disabled 1: Clock Enabled

0x0182F040 GCC_SPDM_AHB_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

PSCBC Registers for MM_SPDM_AHB

GCC_SPDM_AHB_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	Branch Clock Off 0: Clock is ON 1: Clock is OFF
0	CLK_ENABLE		SW Clock Enable Bit 0: Clock Disabled 1: Clock Enabled

0x0182F044 GCC_SPDM_PCLK0_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

PSCBC Registers for MM_SPDM_PCLK0

GCC_SPDM_PCLK0_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	Branch Clock Off 0: Clock is ON 1: Clock is OFF
0	CLK_ENABLE		SW Clock Enable Bit 0: Clock Disabled 1: Clock Enabled

0x0182F048 GCC_SPDM_CSI0_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

PSCBC Registers for MM_SPDM_CSI0

GCC_SPDM_CSI0_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	Branch Clock Off 0: Clock is ON 1: Clock is OFF
0	CLK_ENABLE		SW Clock Enable Bit 0: Clock Disabled 1: Clock Enabled

0x0184C000 GCC_VCODEC0_CMD_RCGR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

Root Clock Generator Command Register VCODEC0

GCC_VCODEC0_CMD_RCGR

Bits	Name	Type	Description
31	ROOT_OFF	R	The ROOT_OFF status allows SW to determine if the generator is actively producing a clock 0: Active 1: Not Active
7	DIRTY_D	R	Dirty bit for the D register. This bit is active if the D register is written, and will clear after the update bit asserts. 0: Not Active 1: Active
6	DIRTY_N	R	Dirty bit for the N register. This bit is active if the N register is written, and will clear after the update bit asserts. 0: Not Active 1: Active
5	DIRTY_M	R	Dirty bit for the M register. This bit is active if the M register is written, and will clear after the update bit asserts. 0: Not Active 1: Active
4	DIRTY_CFG_RCGR	R	Dirty bit for the CFG_RCGR register. This bit is active if the CFG_RCGR register is written, and will clear after the update bit asserts. 0: Not Active 1: Active
1	ROOT_EN		The ROOT_EN bit controls the output of the clock generator. This enable may be Ore with a HW clock enable and then sent to the RCG. 0: Disabled 1: Enabled
0	UPDATE		This UPDATE bit is SW and HW modified. SW sets the bit when the configuration values are programmed, and when they have taken effect, then HW clears the bit to 0. 0: Not Active 1: Active

0x0184C004 GCC_VCODEC0_CFG_RCGR**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

Clock Generator Config Register for VCODEC0

GCC_VCODEC0_CFG_RCGR

Bits	Name	Description
13:12	MODE	mode 00: bypass mode 01: swallow mode 10: dual edge mode 11: single edge mode
10:8	SRC_SEL	Root Source Select 000: cxo 001: GPLLO_MAIN 010: gnd 011: gnd 100: gnd 101: gnd 110: gnd 111: Reserved

GCC_VCODEC0_CFG_RCGR (cont.)

Bits	Name	Description
4:0	SRC_DIV	The SRC_DIV bits define the integer and half integer portion of the divider setting. 00000: Div 1 00001: Div 1 00010: Div 1.5 00011: Div 2 00100: Div 2.5 00101: Div 3 00110: Div 3.5 00111: Div 4 01000: Div 4.5 01001: Div 5 01010: Div 5.5 01011: Div 6 01100: Div 6.5 01101: Div 7 01110: Div 7.5 01111: Div 8 10000: Div 8.5 10001: Div 9 10010: Div 9.5 10011: Div 10 10100: Div 10.5 10101: Div 11 10110: Div 11.5 10111: Div 12 11000: Div 12.5 11001: Div 13 11010: Div 13.5 11011: Div 14 11100: Div 14.5 11101: Div 15 11110: Div 15.5 11111: Div 16

0x0184C008 GCC_VCODEC0_M**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

Program M for the MND divider, where division ration is $F_{out} = F_{in} \cdot (M / N)$

GCC_VCODEC0_M

Bits	Name	Description
7:0	M	Program M

0x0184C00C GCC_VCODEC0_N**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

Program NOT_N_MINUS_M (1's complement of N-M); The N value derived is used for the MND divider, where division ration is $F_{out} = F_{in} \cdot (M / N)$

GCC_VCODEC0_N

Bits	Name	Description
7:0	NOT_N_MINUS_M	Program NOT_N_MINUS_M

0x0184C010 GCC_VCODEC0_D**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

Program NOT_2D (1's complement of 2D); The D value derived is used for the MND divider, where the duty cycle is D/N

GCC_VCODEC0_D

Bits	Name	Description
7:0	NOT_2D	Program NOT_2D

0x0184C014 GCC_VENUS0_BCR**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

BCR Registers for VENUS0

GCC_VENUS0_BCR

Bits	Name	Description
0	BLK_ARES	The block asynchronous reset bit, when set, the block clocks are disabled and the ares_out signals go active. 0: Not Active 1: Active

0x0184C018 GCC_VENUS0_GDSCR**Type:** RW**Clock:** CLK**Reset State:** 0x00222001

GDS Registers for VENUS0

GCC_VENUS0_GDSCR

Bits	Name	Type	Description
31	PWR_ON	R	PWR State 0: Power collapsed 1: Power on
30:27	GDSC_STATE	R	GDS FSM State 0000: PowerOff 0001: AssertAresAndRampUp_en_few 0010: DeassertAres 0011: RampUp_en_rest 0100: PowerOn 0101: UnclampIO 0110: NoRetainFF 0111: RestoreFF 1000: NA 1001: NA 1010: NA 1011: NA 1100: DisableClk 1101: RetainFF 1110: NA 1111: ClampIOAndSaveFF

GCC_VENUS0_GDSCR (cont.)

Bits	Name	Type	Description
23:20	EN_REST_WAIT		Wait cntr for the remaining PD switches to power on 0000: cntr value is cxc x 1 cycle 0001: cntr value is cxc x 2 cycle 0010: cntr value is cxc x 4 cycle [default] 0011: cntr value is cxc x 8 cycle 0100: cntr value is cxc x 16 cycle 0101: cntr value is cxc x 32 cycle 0110: cntr value is cxc x 64 cycle 0111: cntr value is cxc x 128 cycle 1000: cntr value is cxc x 256 cycle 1001: cntr value is cxc x 512 cycle 1010: cntr value is cxc x 1024 cycle 1011: cntr value is cxc x 2048 cycle 1100: cntr value is cxc x 2048 cycle 1101: cntr value is cxc x 2048 cycle 1110: cntr value is cxc x 2048 cycle 1111: cntr value is cxc x 2048 cycle
19:16	EN_FEW_WAIT		Wait cntr for the first few PD switches to power on 0000: cntr value is cxc x 1 cycle 0001: cntr value is cxc x 2 cycle 0010: cntr value is cxc x 4 cycle [default] 0011: cntr value is cxc x 8 cycle 0100: cntr value is cxc x 16 cycle 0101: cntr value is cxc x 32 cycle 0110: cntr value is cxc x 64 cycle 0111: cntr value is cxc x 128 cycle 1000: cntr value is cxc x 256 cycle 1001: cntr value is cxc x 512 cycle 1010: cntr value is cxc x 1024 cycle 1011: cntr value is cxc x 2048 cycle 1100: cntr value is cxc x 2048 cycle 1101: cntr value is cxc x 2048 cycle 1110: cntr value is cxc x 2048 cycle 1111: cntr value is cxc x 2048 cycle

GCC_VENUS0_GDSCR (cont.)

Bits	Name	Type	Description
15:12	CLK_DIS_WAIT		Wait while all clocks in this PD are turned off 0000: cntr value is cxo x 1 cycle 0001: cntr value is cxo x 2 cycle 0010: cntr value is cxo x 4 cycle [default] 0011: cntr value is cxo x 8 cycle 0100: cntr value is cxo x 16 cycle 0101: cntr value is cxo x 32 cycle 0110: cntr value is cxo x 64 cycle 0111: cntr value is cxo x 128 cycle 1000: cntr value is cxo x 256 cycle 1001: cntr value is cxo x 512 cycle 1010: cntr value is cxo x 1024 cycle 1011: cntr value is cxo x 2048 cycle 1100: cntr value is cxo x 2048 cycle 1101: cntr value is cxo x 2048 cycle 1110: cntr value is cxo x 2048 cycle 1111: cntr value is cxo x 2048 cycle
10	RESTORE		Restore 0: Not Active 1: Active
9	SAVE		Save 0: Not Active 1: Active
8	RETAIN		Retain 0: Not Active 1: Active
7	EN_REST		Enable Rest 0: Not Active 1: Active
6	EN_FEW		Enable Few 0: Not Active 1: Active
5	CLAMP_IO		Clamp IO 0: Not Active 1: Active
4	CLK_DISABLE		Clk Disable 0: Not Active 1: Active
3	PD_ARES		Reset 0: Not Active 1: Active

GCC_VENUS0_GDSCR (cont.)

Bits	Name	Type	Description
2	SW_OVERRIDE		SW Override 0: GDSC sequence based on HW FSM 1: GDSC sequence based on SW
1	HW_CONTROL		HW Control 0: Power off by HW trigger 1: Power on by HW trigger
0	SW_COLLAPSE		SW Collapse 0: Power on through SW 1: Power off through SW, default

0x0184C01C GCC_VENUS0_VCODEC0_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x80004FF0

PSCBC Registers for VENUS0_VCODEC0

GCC_VENUS0_VCODEC0_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	Branch Clock Off 0: Clock is ON 1: Clock is OFF
14	FORCE_MEM_CORE_ON		Force Mem Core bits to be on 0: Memory Core can be powered down by the clock branch slp control 1: Force Memory Core to be powered on (default)
13	FORCE_MEM_PERIPH_ON		Force Mem Periph logics to be on 0: Memory Periphery logic can be powered down by the clock branch slp control (default) 1: Force Memory Periphery logic to be powerd on
12	FORCE_MEM_PERIPH_OFF		Force Mem Periph logics to be off 0: Memory Periphery logic is powered on by the clock branch slp control (default) 1: Force Memory Periphery logic to be powerd off

GCC_VENUS0_VCODEC0_CBCR (cont.)

Bits	Name	Type	Description
11:8	WAKEUP		When the clock is enabled, the memory control signal will be turned on immediately, but the clock will remain gated until WAKEUP clock cycles later. 0000: Number of clock cycle 1 0001: Number of clock cycle 2 0010: Number of clock cycle 3 0011: Number of clock cycle 4 0100: Number of clock cycle 5 0101: Number of clock cycle 6 0110: Number of clock cycle 7 1000: Number of clock cycle 8 1001: Number of clock cycle 9 1010: Number of clock cycle 10 1011: Number of clock cycle 11 1100: Number of clock cycle 12 1101: Number of clock cycle 13 1110: Number of clock cycle 14 1111: Number of clock cycle 15
7:4	SLEEP		When the clock is disabled, the clock will be gated immediately, but the memory control signals will not be turned off until a delay of SLEEP clock cycles. 0000: Number of clock cycle 1 0001: Number of clock cycle 2 0010: Number of clock cycle 3 0011: Number of clock cycle 4 0100: Number of clock cycle 5 0101: Number of clock cycle 6 0110: Number of clock cycle 7 1000: Number of clock cycle 8 1001: Number of clock cycle 9 1010: Number of clock cycle 10 1011: Number of clock cycle 11 1100: Number of clock cycle 12 1101: Number of clock cycle 13 1110: Number of clock cycle 14 1111: Number of clock cycle 15
0	CLK_ENABLE		SW Clock Enable Bit 0: Clock Disabled 1: Clock Enabled

0x0184C020 GCC_VENUS0_AHB_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x80008000

PSCBC Registers for VENUS0_AHB

GCC_VENUS0_AHB_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	Branch Clock Off 0: Clock is ON 1: Clock is OFF
30:28	NOC_HANDSHAKE_FSM_STATE	R	FSM State 000: OFFS = Off 001: WFON = Wait For Clk On 010: SRIU = Slave Ready is Up 011: WFSL = Wait for Socketcon Low 100: WOFF = Wait for Clk Off 101: Not Used 110: Not Used 111: Not Used
15	NOC_HANDSHAKE_FSM_EN		SlaveReady and Socket Connect FSM Enable; Default is 1 0: SW Clock Enabled 1: FSM Clock Enabled
0	CLK_ENABLE		SW Clock Enable Bit 0: Clock Disabled. 1: Clock Enabled.

0x0184C024 GCC_VENUS0_AXI_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x80004FF0

PSCBC Registers for VENUS0_AXI

GCC_VENUS0_AXI_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	Branch Clock Off 0: Clock is ON 1: Clock is OFF

GCC_VENUS0_AXI_CBCR (cont.)

Bits	Name	Type	Description
14	FORCE_MEM_CORE_ON		Force Mem Core bits to be on 0: Memory Core can be powered down by the clock branch slp control 1: Force Memory Core to be powered on (default)
13	FORCE_MEM_PERIPH_ON		Force Mem Periph logics to be on 0: Memory Periphery logic can be powered down by the clock branch slp control (default) 1: Force Memory Periphery logic to be powered on
12	FORCE_MEM_PERIPH_OFF		Force Mem Periph logics to be off 0: Memory Periphery logic is powered on by the clock branch slp control (default) 1: Force Memory Periphery logic to be powerd off
11:8	WAKEUP		When the clock is enabled, the memory control signal will be turned on immediately, but the clock will remain gated until WAKEUP clock cycles later. 0000: Number of clock cycle 1 0001: Number of clock cycle 2 0010: Number of clock cycle 3 0011: Number of clock cycle 4 0100: Number of clock cycle 5 0101: Number of clock cycle 6 0110: Number of clock cycle 7 1000: Number of clock cycle 8 1001: Number of clock cycle 9 1010: Number of clock cycle 10 1011: Number of clock cycle 11 1100: Number of clock cycle 12 1101: Number of clock cycle 13 1110: Number of clock cycle 14 1111: Number of clock cycle 15

GCC_VENUS0_AXI_CBCR (cont.)

Bits	Name	Type	Description
7:4	SLEEP		When the clock is disabled, the clock will be gated immediately, but the memory control signals will not be turned off until a delay of SLEEP clock cycles. 0000: Number of clock cycle 1 0001: Number of clock cycle 2 0010: Number of clock cycle 3 0011: Number of clock cycle 4 0100: Number of clock cycle 5 0101: Number of clock cycle 6 0110: Number of clock cycle 7 1000: Number of clock cycle 8 1001: Number of clock cycle 9 1010: Number of clock cycle 10 1011: Number of clock cycle 11 1100: Number of clock cycle 12 1101: Number of clock cycle 13 1110: Number of clock cycle 14 1111: Number of clock cycle 15
0	CLK_ENABLE		SW Clock Enable Bit 0: Clock Disabled 1: Clock Enabled

0x0184D000 GCC_PCLK0_CMD_RCGR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

Root Clock Generator Command Register PCLK0

GCC_PCLK0_CMD_RCGR

Bits	Name	Type	Description
31	ROOT_OFF	R	The ROOT_OFF status allows SW to determine if the generator is actively producing a clock 0: Active 1: Not Active
7	DIRTY_D	R	Dirty bit for the D register. This bit is active if the D register is written, and will clear after the update bit asserts. 0: Not Active 1: Active

GCC_PCLK0_CMD_RCGR (cont.)

Bits	Name	Type	Description
6	DIRTY_N	R	Dirty bit for the N register. This bit is active if the N register is written, and will clear after the update bit asserts. 0: Not Active 1: Active
5	DIRTY_M	R	Dirty bit for the M register. This bit is active if the M register is written, and will clear after the update bit asserts. 0: Not Active 1: Active
4	DIRTY_CFG_RCGR	R	Dirty bit for the CFG_RCGR register. This bit is active if the CFG_RCGR register is written, and will clear after the update bit asserts. 0: Not Active 1: Active
1	ROOT_EN		The ROOT_EN bit controls the output of the clock generator. This enable may be ORed with a HW clock enable and then sent to the RCG. 0: Disabled 1: Enabled
0	UPDATE		This UPDATE bit is SW and HW modified. SW sets the bit when the configuration values are programmed, and when they have taken effect, then HW clears the bit to 0. 0: Not Active 1: Active

0x0184D004 GCC_PCLK0_CFG_RCGR**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

Clock Generator Config Register for PCLK0

GCC_PCLK0_CFG_RCGR

Bits	Name	Description
13:12	MODE	mode 00: bypass mode 01: swallow mode 10: dual edge mode 11: single edge mode

GCC_PCLK0_CFG_RCGR (cont.)

Bits	Name	Description
10:8	SRC_SEL	Root Source Select 000: cxo 001: dsi0_phy_pll_out_dsiclk 010: GPLL0_AUX 011: gnd 100: gnd 101: gnd 110: gnd 111: Reserved
4:0	SRC_DIV	The SRC_DIV bits define the integer and half integer portion of the divider setting. 00000: Div 1 00001: Div 1 00010: Div 1.5 00011: Div 2 00100: Div 2.5 00101: Div 3 00110: Div 3.5 00111: Div 4 01000: Div 4.5 01001: Div 5 01010: Div 5.5 01011: Div 6 01100: Div 6.5 01101: Div 7 01110: Div 7.5 01111: Div 8 10000: Div 8.5 10001: Div 9 10010: Div 9.5 10011: Div 10 10100: Div 10.5 10101: Div 11 10110: Div 11.5 10111: Div 12 11000: Div 12.5 11001: Div 13 11010: Div 13.5 11011: Div 14 11100: Div 14.5 11101: Div 15 11110: Div 15.5 11111: Div 16

0x0184D008 GCC_PCLK0_M**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

Program M for the MND divider, where division ration is $F_{out} = F_{in} \cdot (M / N)$

GCC_PCLK0_M

Bits	Name	Description
7:0	M	Program M

0x0184D00C GCC_PCLK0_N**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

Program NOT_N_MINUS_M (1's complement of N-M); The N value derived is used for the MND divider, where division ration is $F_{out} = F_{in} \cdot (M / N)$

GCC_PCLK0_N

Bits	Name	Description
7:0	NOT_N_MINUS_M	Program NOT_N_MINUS_M

0x0184D010 GCC_PCLK0_D**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

Program NOT_2D (1's complement of 2D); The D value derived is used for the MND divider, where the duty cycle is D/N

GCC_PCLK0_D

Bits	Name	Description
7:0	NOT_2D	Program NOT_2D

0x0184D014 GCC_MDP_CMD_RCGR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

Root Clock Generator Command Register MDP

GCC_MDP_CMD_RCGR

Bits	Name	Type	Description
31	ROOT_OFF	R	The ROOT_OFF status allows SW to determine if the generator is actively producing a clock 0: Active 1: Not Active
4	DIRTY_CFG_RCGR	R	Dirty bit for the CFG_RCGR register. This bit is active if the CFG_RCGR register is written, and will clear after the update bit asserts. 0: Not Active 1: Active
1	ROOT_EN		The ROOT_EN bit controls the output of the clock generator. This enable may be ORed with a HW clock enable and then sent to the RCG. 0: Disabled 1: Enabled
0	UPDATE		This UPDATE bit is SW and HW modified. SW sets the bit when the configuration values are programmed, and when they have taken effect, then HW clears the bit to 0. 0: Not Active 1: Active

0x0184D018 GCC_MDP_CFG_RCGR**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

Clock Generator Config Register for MDP

GCC_MDP_CFG_RCGR

Bits	Name	Description
10:8	SRC_SEL	Root Source Select 000: cxo 001: GPLLO_MAIN 010: dsi0_phy_pll_out_dsiclk 011: gnd 100: gnd 101: gnd 110: gnd 111: Reserved
4:0	SRC_DIV	The SRC_DIV bits define the integer and half integer portion of the divider setting. 00000: Div 1 00001: Div 1 00010: Div 1.5 00011: Div 2 00100: Div 2.5 00101: Div 3 00110: Div 3.5 00111: Div 4 01000: Div 4.5 01001: Div 5 01010: Div 5.5 01011: Div 6 01100: Div 6.5 01101: Div 7 01110: Div 7.5 01111: Div 8 10000: Div 8.5 10001: Div 9 10010: Div 9.5 10011: Div 10 10100: Div 10.5 10101: Div 11 10110: Div 11.5 10111: Div 12 11000: Div 12.5 11001: Div 13 11010: Div 13.5 11011: Div 14 11100: Div 14.5 11101: Div 15 11110: Div 15.5 11111: Div 16

0x0184D02C GCC_VSYNC_CMD_RCGR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

Root Clock Generator Command Register VSYNC

GCC_VSYNC_CMD_RCGR

Bits	Name	Type	Description
31	ROOT_OFF	R	The ROOT_OFF status allows SW to determine if the generator is actively producing a clock 0: Active 1: Not Active
4	DIRTY_CFG_RCGR	R	Dirty bit for the CFG_RCGR register. This bit is active if the CFG_RCGR register is written, and will clear after the update bit asserts. 0: Not Active 1: Active
1	ROOT_EN		The ROOT_EN bit controls the output of the clock generator. This enable may be ORed with a HW clock enable and then sent to the RCG. 0: Disabled 1: Enabled
0	UPDATE		This UPDATE bit is SW and HW modified. SW sets the bit when the configuration values are programmed, and when they have taken effect, then HW clears the bit to 0. 0: Not Active 1: Active

0x0184D030 GCC_VSYNC_CFG_RCGR**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

Clock Generator Config Register for VSYNC

GCC_VSYNC_CFG_RCGR

Bits	Name	Description
10:8	SRC_SEL	Root Source Select 000: cxo 001: gnd 010: GPLL0_AUX 011: gnd 100: gnd 101: gnd 110: gnd 111: Reserved
4:0	SRC_DIV	The SRC_DIV bits define the integer and half integer portion of the divider setting. 00000: Div 1 00001: Div 1 00010: Div 1.5 00011: Div 2 00100: Div 2.5 00101: Div 3 00110: Div 3.5 00111: Div 4 01000: Div 4.5 01001: Div 5 01010: Div 5.5 01011: Div 6 01100: Div 6.5 01101: Div 7 01110: Div 7.5 01111: Div 8 10000: Div 8.5 10001: Div 9 10010: Div 9.5 10011: Div 10 10100: Div 10.5 10101: Div 11 10110: Div 11.5 10111: Div 12 11000: Div 12.5 11001: Div 13 11010: Div 13.5 11011: Div 14 11100: Div 14.5 11101: Div 15 11110: Div 15.5 11111: Div 16

0x0184D044 GCC_BYTE0_CMD_RCGR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

Root Clock Generator Command Register BYTE0

GCC_BYTE0_CMD_RCGR

Bits	Name	Type	Description
31	ROOT_OFF	R	The ROOT_OFF status allows SW to determine if the generator is actively producing a clock 0: Active 1: Not Active
4	DIRTY_CFG_RCGR	R	Dirty bit for the CFG_RCGR register. This bit is active if the CFG_RCGR register is written, and will clear after the update bit asserts. 0: Not Active 1: Active
1	ROOT_EN		The ROOT_EN bit controls the output of the clock generator. This enable may be ORed with a HW clock enable and then sent to the RCG. 0: Disabled 1: Enabled
0	UPDATE		This UPDATE bit is SW and HW modified. SW sets the bit when the configuration values are programmed, and when they have taken effect, then HW clears the bit to 0. 0: Not Active 1: Active

0x0184D048 GCC_BYTE0_CFG_RCGR**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

Clock Generator Config Register for BYTE0

GCC_BYTE0_CFG_RCGR

Bits	Name	Description
10:8	SRC_SEL	Root Source Select 000: cxo 001: dsi0_phy_pll_out_byteclk 010: GPLL0_AUX 011: gnd 100: gnd 101: gnd 110: gnd 111: Reserved
4:0	SRC_DIV	The SRC_DIV bits define the integer and half integer portion of the divider setting. 00000: Div 1 00001: Div 1 00010: Div 1.5 00011: Div 2 00100: Div 2.5 00101: Div 3 00110: Div 3.5 00111: Div 4 01000: Div 4.5 01001: Div 5 01010: Div 5.5 01011: Div 6 01100: Div 6.5 01101: Div 7 01110: Div 7.5 01111: Div 8 10000: Div 8.5 10001: Div 9 10010: Div 9.5 10011: Div 10 10100: Div 10.5 10101: Div 11 10110: Div 11.5 10111: Div 12 11000: Div 12.5 11001: Div 13 11010: Div 13.5 11011: Div 14 11100: Div 14.5 11101: Div 15 11110: Div 15.5 11111: Div 16

0x0184D05C GCC_ESC0_CMD_RCGR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

Root Clock Generator Command Register ESC0

GCC_ESC0_CMD_RCGR

Bits	Name	Type	Description
31	ROOT_OFF	R	The ROOT_OFF status allows SW to determine if the generator is actively producing a clock 0: Active 1: Not Active
4	DIRTY_CFG_RCGR	R	Dirty bit for the CFG_RCGR register. This bit is active if the CFG_RCGR register is written, and will clear after the update bit asserts. 0: Not Active 1: Active
1	ROOT_EN		The ROOT_EN bit controls the output of the clock generator. This enable may be ORed with a HW clock enable and then sent to the RCG. 0: Disabled 1: Enabled
0	UPDATE		This UPDATE bit is SW and HW modified. SW sets the bit when the configuration values are programmed, and when they have taken effect, then HW clears the bit to 0. 0: Not Active 1: Active

0x0184D060 GCC_ESC0_CFG_RCGR**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

Clock Generator Config Register for ESC0

GCC_ESC0_CFG_RCGR

Bits	Name	Description
10:8	SRC_SEL	Root Source Select 000: cxo 001: gnd 010: dsi0_phy_pll_out_byteclk 011: gnd 100: gnd 101: gnd 110: gnd 111: Reserved
4:0	SRC_DIV	The SRC_DIV bits define the integer and half integer portion of the divider setting. 00000: Div 1 00001: Div 1 00010: Div 1.5 00011: Div 2 00100: Div 2.5 00101: Div 3 00110: Div 3.5 00111: Div 4 01000: Div 4.5 01001: Div 5 01010: Div 5.5 01011: Div 6 01100: Div 6.5 01101: Div 7 01110: Div 7.5 01111: Div 8 10000: Div 8.5 10001: Div 9 10010: Div 9.5 10011: Div 10 10100: Div 10.5 10101: Div 11 10110: Div 11.5 10111: Div 12 11000: Div 12.5 11001: Div 13 11010: Div 13.5 11011: Div 14 11100: Div 14.5 11101: Div 15 11110: Div 15.5 11111: Div 16

0x0184D074 GCC_MDSS_BCR**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

BCR Registers for MDSS

GCC_MDSS_BCR

Bits	Name	Description
0	BLK_ARES	The block asynchronous reset bit, when set, the block clocks are disabled and the ares_out signals go active. 0: Not Active 1: Active

0x0184D078 GCC_MDSS_GDSCR**Type:** RW**Clock:** CLK**Reset State:** 0x00222001

GDS Registers for MDSS

GCC_MDSS_GDSCR

Bits	Name	Type	Description
31	PWR_ON	R	PWR State 0: Power collapsed 1: Power on

GCC_MDSS_GDSCR (cont.)

Bits	Name	Type	Description
30:27	GDSC_STATE	R	GDS FSM State 0000: PowerOff 0001: AssertAresAndRampUp_en_few 0010: DeassertAres 0011: RampUp_en_rest 0100: PowerOn 0101: UnclampIO 0110: NoRetainFF 0111: RestoreFF 1000: NA 1001: NA 1010: NA 1011: NA 1100: DisableClk 1101: RetainFF 1110: NA 1111: ClampIOAndSaveFF
23:20	EN_REST_WAIT		Wait cntr for the remaining PD switches to power on 0000: cntr value is cxc x 1 cycle 0001: cntr value is cxc x 2 cycle 0010: cntr value is cxc x 4 cycle [default] 0011: cntr value is cxc x 8 cycle 0100: cntr value is cxc x 16 cycle 0101: cntr value is cxc x 32 cycle 0110: cntr value is cxc x 64 cycle 0111: cntr value is cxc x 128 cycle 1000: cntr value is cxc x 256 cycle 1001: cntr value is cxc x 512 cycle 1010: cntr value is cxc x 1024 cycle 1011: cntr value is cxc x 2048 cycle 1100: cntr value is cxc x 2048 cycle 1101: cntr value is cxc x 2048 cycle 1110: cntr value is cxc x 2048 cycle 1111: cntr value is cxc x 2048 cycle

GCC_MDSS_GDSCR (cont.)

Bits	Name	Type	Description
19:16	EN_FEW_WAIT		Wait cntr for the first few PD switches to power on 0000: cntr value is cxc x 1 cycle 0001: cntr value is cxc x 2 cycle 0010: cntr value is cxc x 4 cycle [default] 0011: cntr value is cxc x 8 cycle 0100: cntr value is cxc x 16 cycle 0101: cntr value is cxc x 32 cycle 0110: cntr value is cxc x 64 cycle 0111: cntr value is cxc x 128 cycle 1000: cntr value is cxc x 256 cycle 1001: cntr value is cxc x 512 cycle 1010: cntr value is cxc x 1024 cycle 1011: cntr value is cxc x 2048 cycle 1100: cntr value is cxc x 2048 cycle 1101: cntr value is cxc x 2048 cycle 1110: cntr value is cxc x 2048 cycle 1111: cntr value is cxc x 2048 cycle
15:12	CLK_DIS_WAIT		Wait while all clocks in this PD are turned off 0000: cntr value is cxc x 1 cycle 0001: cntr value is cxc x 2 cycle 0010: cntr value is cxc x 4 cycle [default] 0011: cntr value is cxc x 8 cycle 0100: cntr value is cxc x 16 cycle 0101: cntr value is cxc x 32 cycle 0110: cntr value is cxc x 64 cycle 0111: cntr value is cxc x 128 cycle 1000: cntr value is cxc x 256 cycle 1001: cntr value is cxc x 512 cycle 1010: cntr value is cxc x 1024 cycle 1011: cntr value is cxc x 2048 cycle 1100: cntr value is cxc x 2048 cycle 1101: cntr value is cxc x 2048 cycle 1110: cntr value is cxc x 2048 cycle 1111: cntr value is cxc x 2048 cycle
10	RESTORE		Restore 0: Not Active 1: Active
9	SAVE		Save 0: Not Active 1: Active
8	RETAIN		Retain 0: Not Active 1: Active

GCC_MDSS_GDSCR (cont.)

Bits	Name	Type	Description
7	EN_REST		Enable Rest 0: Not Active 1: Active
6	EN_FEW		Enable Few 0: Not Active 1: Active
5	CLAMP_IO		Clamp IO 0: Not Active 1: Active
4	CLK_DISABLE		Clk Disable 0: Not Active 1: Active
3	PD_ARES		Reset 0: Not Active 1: Active
2	SW_OVERRIDE		SW Override 0: GDSC sequence based on HW FSM 1: GDSC sequence based on SW
1	HW_CONTROL		HW Control 0: Power off by HW trigger 1: Power on by HW trigger
0	SW_COLLAPSE		SW Collapse 0: Power on through SW 1: Power off through SW, default

0x0184D07C GCC_MDSS_AHB_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x80008000

PSCBC Registers for MDSS_AHB

GCC_MDSS_AHB_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	Branch Clock Off 0: Clock is ON 1: Clock is OFF

GCC_MDSS_AHB_CBCR (cont.)

Bits	Name	Type	Description
30:28	NOC_HANDSHAKE_FSM_STATE	R	FSM State 000: OFFS = Off 001: WFON = Wait For Clk On 010: SRIU = Slave Ready is Up 011: WFSL = Wait for Socketcon Low 100: WOFF = Wait for Clk Off 101: Not Used 110: Not Used 111: Not Used
15	NOC_HANDSHAKE_FSM_EN		SlaveReady and Socket Connect FSM Enable; Default is 1 0: SW Clock Enabled 1: FSM Clock Enabled
0	CLK_ENABLE		SW Clock Enable Bit 0: Clock Disabled. 1: Clock Enabled.

0x0184D080 GCC_MDSS_AXI_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x80004FF0

PSCBC Registers for MDSS_AXI

GCC_MDSS_AXI_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	Branch Clock Off 0: Clock is ON 1: Clock is OFF
14	FORCE_MEM_CORE_ON		Force Mem Core bits to be on 0: Memory Core can be powered down by the clock branch slp control 1: Force Memory Core to be powered on (default)
13	FORCE_MEM_PERIPH_ON		Force Mem Periph logics to be on 0: Memory Periphery logic can be powered down by the clock branch slp control (default) 1: Force Memory Periphery logic to be powerd on
12	FORCE_MEM_PERIPH_OFF		Force Mem Periph logics to be off 0: Memory Periphery logic is powered on by the clock branch slp control (default) 1: Force Memory Periphery logic to be powerd off

GCC_MDSS_AXI_CBCR (cont.)

Bits	Name	Type	Description
11:8	WAKEUP		When the clock is enabled, the memory control signal will be turned on immediately, but the clock will remain gated until WAKEUP clock cycles later. 0000: Number of clock cycle 1 0001: Number of clock cycle 2 0010: Number of clock cycle 3 0011: Number of clock cycle 4 0100: Number of clock cycle 5 0101: Number of clock cycle 6 0110: Number of clock cycle 7 1000: Number of clock cycle 8 1001: Number of clock cycle 9 1010: Number of clock cycle 10 1011: Number of clock cycle 11 1100: Number of clock cycle 12 1101: Number of clock cycle 13 1110: Number of clock cycle 14 1111: Number of clock cycle 15
7:4	SLEEP		When the clock is disabled, the clock will be gated immediately, but the memory control signals will not be turned off until a delay of SLEEP clock cycles. 0000: Number of clock cycle 1 0001: Number of clock cycle 2 0010: Number of clock cycle 3 0011: Number of clock cycle 4 0100: Number of clock cycle 5 0101: Number of clock cycle 6 0110: Number of clock cycle 7 1000: Number of clock cycle 8 1001: Number of clock cycle 9 1010: Number of clock cycle 10 1011: Number of clock cycle 11 1100: Number of clock cycle 12 1101: Number of clock cycle 13 1110: Number of clock cycle 14 1111: Number of clock cycle 15
0	CLK_ENABLE		SW Clock Enable Bit 0: Clock Disabled 1: Clock Enabled

0x0184D084 GCC_MDSS_PCLK0_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

PSCBC Registers for MDSS_PCLK0

GCC_MDSS_PCLK0_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	Branch Clock Off 0: Clock is ON 1: Clock is OFF
0	CLK_ENABLE		SW Clock Enable Bit 0: Clock Disabled 1: Clock Enabled

0x0184D088 GCC_MDSS_MDP_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x80004FF0

PSCBC Registers for MDSS_MDP

GCC_MDSS_MDP_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	Branch Clock Off 0: Clock is ON 1: Clock is OFF
14	FORCE_MEM_CORE_ON		Force Mem Core bits to be on 0: Memory Core can be powered down by the clock branch slp control 1: Force Memory Core to be powered on (default)
13	FORCE_MEM_PERIPH_ON		Force Mem Periph logics to be on 0: Memory Periphery logic can be powered down by the clock branch slp control (default) 1: Force Memory Periphery logic to be powerd on
12	FORCE_MEM_PERIPH_OF F		Force Mem Periph logics to be off 0: Memory Periphery logic is powered on by the clock branch slp control (default) 1: Force Memory Periphery logic to be powerd off

GCC_MDSS_MDP_CBCR (cont.)

Bits	Name	Type	Description
11:8	WAKEUP		When the clock is enabled, the memory control signal will be turned on immediately, but the clock will remain gated until WAKEUP clock cycles later. 0000: Number of clock cycle 1 0001: Number of clock cycle 2 0010: Number of clock cycle 3 0011: Number of clock cycle 4 0100: Number of clock cycle 5 0101: Number of clock cycle 6 0110: Number of clock cycle 7 1000: Number of clock cycle 8 1001: Number of clock cycle 9 1010: Number of clock cycle 10 1011: Number of clock cycle 11 1100: Number of clock cycle 12 1101: Number of clock cycle 13 1110: Number of clock cycle 14 1111: Number of clock cycle 15
7:4	SLEEP		When the clock is disabled, the clock will be gated immediately, but the memory control signals will not be turned off until a delay of SLEEP clock cycles. 0000: Number of clock cycle 1 0001: Number of clock cycle 2 0010: Number of clock cycle 3 0011: Number of clock cycle 4 0100: Number of clock cycle 5 0101: Number of clock cycle 6 0110: Number of clock cycle 7 1000: Number of clock cycle 8 1001: Number of clock cycle 9 1010: Number of clock cycle 10 1011: Number of clock cycle 11 1100: Number of clock cycle 12 1101: Number of clock cycle 13 1110: Number of clock cycle 14 1111: Number of clock cycle 15
0	CLK_ENABLE		SW Clock Enable Bit 0: Clock Disabled 1: Clock Enabled

0x0184D090 GCC_MDSS_VSYNC_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

PSCBC Registers for MDSS_VSYNC

GCC_MDSS_VSYNC_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	Branch Clock Off 0: Clock is ON 1: Clock is OFF
0	CLK_ENABLE		SW Clock Enable Bit 0: Clock Disabled 1: Clock Enabled

0x0184D094 GCC_MDSS_BYTE0_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

PSCBC Registers for MDSS_BYTE0

GCC_MDSS_BYTE0_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	Branch Clock Off 0: Clock is ON 1: Clock is OFF
0	CLK_ENABLE		SW Clock Enable Bit 0: Clock Disabled 1: Clock Enabled

0x0184D098 GCC_MDSS_ESC0_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

PSCBC Registers for MDSS_ESC0

GCC_MDSS_ESC0_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	Branch Clock Off 0: Clock is ON 1: Clock is OFF
0	CLK_ENABLE		SW Clock Enable Bit 0: Clock Disabled 1: Clock Enabled

0x0184E000 GCC_CSI0PHYTIMER_CMD_RCGR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

Root Clock Generator Command Register CSI0PHYTIMER

GCC_CSI0PHYTIMER_CMD_RCGR

Bits	Name	Type	Description
31	ROOT_OFF	R	The ROOT_OFF status allows SW to determine if the generator is actively producing a clock 0: Active 1: Not Active
4	DIRTY_CFG_RCGR	R	Dirty bit for the CFG_RCGR register. This bit is active if the CFG_RCGR register is written, and will clear after the update bit asserts. 0: Not Active 1: Active
1	ROOT_EN		The ROOT_EN bit controls the output of the clock generator. This enable may be ORed with a HW clock enable and then sent to the RCG. 0: Disabled 1: Enabled
0	UPDATE		This UPDATE bit is SW and HW modified. SW sets the bit when the configuration values are programmed, and when they have taken effect, then HW clears the bit to 0. 0: Not Active 1: Active

0x0184E004 GCC_CSI0PHYTIMER_CFG_RCGR**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

Clock Generator Config Register for CSI0PHYTIMER

GCC_CSI0PHYTIMER_CFG_RCGR

Bits	Name	Description
10:8	SRC_SEL	Root Source Select 000: cxo 001: GPLL0_MAIN 010: GPLL1_AUX 011: gnd 100: gnd 101: gnd 110: gnd 111: Reserved

GCC_CSI0PHYTIMER_CFG_RCGR (cont.)

Bits	Name	Description
4:0	SRC_DIV	The SRC_DIV bits define the integer and half integer portion of the divider setting. 00000: Div 1 00001: Div 1 00010: Div 1.5 00011: Div 2 00100: Div 2.5 00101: Div 3 00110: Div 3.5 00111: Div 4 01000: Div 4.5 01001: Div 5 01010: Div 5.5 01011: Div 6 01100: Div 6.5 01101: Div 7 01110: Div 7.5 01111: Div 8 10000: Div 8.5 10001: Div 9 10010: Div 9.5 10011: Div 10 10100: Div 10.5 10101: Div 11 10110: Div 11.5 10111: Div 12 11000: Div 12.5 11001: Div 13 11010: Div 13.5 11011: Div 14 11100: Div 14.5 11101: Div 15 11110: Div 15.5 11111: Div 16

0x0184E018 GCC_CAMSS_PHY0_BCR**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

BCR Registers for CAMSS_PHY0

GCC_CAMSS_PHY0_BCR

Bits	Name	Description
0	BLK_ARES	The block asynchronous reset bit, when set, the block clocks are disabled and the ares_out signals go active. 0: Not Active 1: Active

0x0184E01C GCC_CAMSS_PHY0_CSI0PHYTIMER_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

PSCBC Registers for CAMSS_PHY0_CSI0PHYTIMER

GCC_CAMSS_PHY0_CSI0PHYTIMER_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	Branch Clock Off 0: Clock is ON 1: Clock is OFF
0	CLK_ENABLE		SW Clock Enable Bit 0: Clock Disabled 1: Clock Enabled

0x0184F000 GCC_CSI1PHYTIMER_CMD_RCGR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

Root Clock Generator Command Register CSI1PHYTIMER

GCC_CSI1PHYTIMER_CMD_RCGR

Bits	Name	Type	Description
31	ROOT_OFF	R	The ROOT_OFF status allows SW to determine if the generator is actively producing a clock 0: Active 1: Not Active

GCC_CSI1PHYTIMER_CMD_RCGR (cont.)

Bits	Name	Type	Description
4	DIRTY_CFG_RCGR	R	Dirty bit for the CFG_RCGR register. This bit is active if the CFG_RCGR register is written, and will clear after the update bit asserts. 0: Not Active 1: Active
1	ROOT_EN		The ROOT_EN bit controls the output of the clock generator. This enable may be ORed with a HW clock enable and then sent to the RCG. 0: Disabled 1: Enabled
0	UPDATE		This UPDATE bit is SW and HW modified. SW sets the bit when the configuration values are programmed, and when they have taken effect, then HW clears the bit to 0. 0: Not Active 1: Active

0x0184F004 GCC_CSI1PHYTIMER_CFG_RCGR**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

Clock Generator Config Register for CSI1PHYTIMER

GCC_CSI1PHYTIMER_CFG_RCGR

Bits	Name	Description
10:8	SRC_SEL	Root Source Select 000: cxo 001: GPLL0_MAIN 010: GPLL1_AUX 011: gnd 100: gnd 101: gnd 110: gnd 111: Reserved

GCC_CSI1PHYTIMER_CFG_RCGR (cont.)

Bits	Name	Description
4:0	SRC_DIV	The SRC_DIV bits define the integer and half integer portion of the divider setting. 00000: Div 1 00001: Div 1 00010: Div 1.5 00011: Div 2 00100: Div 2.5 00101: Div 3 00110: Div 3.5 00111: Div 4 01000: Div 4.5 01001: Div 5 01010: Div 5.5 01011: Div 6 01100: Div 6.5 01101: Div 7 01110: Div 7.5 01111: Div 8 10000: Div 8.5 10001: Div 9 10010: Div 9.5 10011: Div 10 10100: Div 10.5 10101: Div 11 10110: Div 11.5 10111: Div 12 11000: Div 12.5 11001: Div 13 11010: Div 13.5 11011: Div 14 11100: Div 14.5 11101: Div 15 11110: Div 15.5 11111: Div 16

0x0184F018 GCC_CAMSS_PHY1_BCR**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

BCR Registers for CAMSS_PHY1

GCC_CAMSS_PHY1_BCR

Bits	Name	Description
0	BLK_ARES	The block asynchronous reset bit, when set, the block clocks are disabled and the ares_out signals go active. 0: Not Active 1: Active

0x0184F01C GCC_CAMSS_PHY1_CSI1PHYTIMER_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

PSCBC Registers for CAMSS_PHY1_CSI1PHYTIMER

GCC_CAMSS_PHY1_CSI1PHYTIMER_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	Branch Clock Off 0: Clock is ON 1: Clock is OFF
0	CLK_ENABLE		SW Clock Enable Bit 0: Clock Disabled 1: Clock Enabled

0x0184E020 GCC_CSI0_CMD_RCGR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

Root Clock Generator Command Register CSI0

GCC_CSI0_CMD_RCGR

Bits	Name	Type	Description
31	ROOT_OFF	R	The ROOT_OFF status allows SW to determine if the generator is actively producing a clock 0: Active 1: Not Active

GCC_CSI0_CMD_RCGR (cont.)

Bits	Name	Type	Description
4	DIRTY_CFG_RCGR	R	Dirty bit for the CFG_RCGR register. This bit is active if the CFG_RCGR register is written, and will clear after the update bit asserts. 0: Not Active 1: Active
1	ROOT_EN		The ROOT_EN bit controls the output of the clock generator. This enable may be ORed with a HW clock enable and then sent to the RCG. 0: Disabled 1: Enabled
0	UPDATE		This UPDATE bit is SW and HW modified. SW sets the bit when the configuration values are programmed, and when they have taken effect, then HW clears the bit to 0. 0: Not Active 1: Active

0x0184E024 GCC_CSI0_CFG_RCGR**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

Clock Generator Config Register for CSI0

GCC_CSI0_CFG_RCGR

Bits	Name	Description
10:8	SRC_SEL	Root Source Select 000: cxo 001: GPLLO_MAIN 010: gnd 011: gnd 100: gnd 101: gnd 110: gnd 111: Reserved

GCC_CSI0_CFG_RCGR (cont.)

Bits	Name	Description
4:0	SRC_DIV	The SRC_DIV bits define the integer and half integer portion of the divider setting. 00000: Div 1 00001: Div 1 00010: Div 1.5 00011: Div 2 00100: Div 2.5 00101: Div 3 00110: Div 3.5 00111: Div 4 01000: Div 4.5 01001: Div 5 01010: Div 5.5 01011: Div 6 01100: Div 6.5 01101: Div 7 01110: Div 7.5 01111: Div 8 10000: Div 8.5 10001: Div 9 10010: Div 9.5 10011: Div 10 10100: Div 10.5 10101: Div 11 10110: Div 11.5 10111: Div 12 11000: Div 12.5 11001: Div 13 11010: Div 13.5 11011: Div 14 11100: Div 14.5 11101: Div 15 11110: Div 15.5 11111: Div 16

0x0184E038 GCC_CAMSS_CSI0_BCR**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

BCR Registers for CAMSS_CSI0

GCC_CAMSS_CSI0_BCR

Bits	Name	Description
0	BLK_ARES	The block asynchronous reset bit, when set, the block clocks are disabled and the ares_out signals go active. 0: Not Active 1: Active

0x0184E03C GCC_CAMSS_CSI0_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

PSCBC Registers for CAMSS_CSI0

GCC_CAMSS_CSI0_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	Branch Clock Off 0: Clock is ON 1: Clock is OFF
0	CLK_ENABLE		SW Clock Enable Bit 0: Clock Disabled 1: Clock Enabled

0x0184E040 GCC_CAMSS_CSI0_AHB_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

PSCBC Registers for CAMSS_CSI0_AHB

GCC_CAMSS_CSI0_AHB_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	Branch Clock Off 0: Clock is ON 1: Clock is OFF
0	CLK_ENABLE		SW Clock Enable Bit 0: Clock Disabled 1: Clock Enabled

0x0184E044 GCC_CAMSS_CSI0PHY_BCR**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

BCR Registers for CAMSS_CSI0PHY

GCC_CAMSS_CSI0PHY_BCR

Bits	Name	Description
0	BLK_ARES	The block asynchronous reset bit, when set, the block clocks are disabled and the ares_out signals go active. 0: Not Active 1: Active

0x0184E048 GCC_CAMSS_CSI0PHY_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

PSCBC Registers for CAMSS_CSI0PHY

GCC_CAMSS_CSI0PHY_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	Branch Clock Off 0: Clock is ON 1: Clock is OFF
0	CLK_ENABLE		SW Clock Enable Bit 0: Clock Disabled 1: Clock Enabled

0x0184E04C GCC_CAMSS_CSI0RDI_BCR**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

BCR Registers for CAMSS_CSI0RDI

GCC_CAMSS_CSI0RDI_BCR

Bits	Name	Description
0	BLK_ARES	The block asynchronous reset bit, when set, the block clocks are disabled and the ares_out signals go active. 0: Not Active 1: Active

0x0184E050 GCC_CAMSS_CSI0RDI_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

PSCBC Registers for CAMSS_CSI0RDI

GCC_CAMSS_CSI0RDI_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	Branch Clock Off 0: Clock is ON 1: Clock is OFF
0	CLK_ENABLE		SW Clock Enable Bit 0: Clock Disabled 1: Clock Enabled

0x0184E054 GCC_CAMSS_CSI0PIX_BCR**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

BCR Registers for CAMSS_CSI0PIX

GCC_CAMSS_CSI0PIX_BCR

Bits	Name	Description
0	BLK_ARES	The block asynchronous reset bit, when set, the block clocks are disabled and the ares_out signals go active. 0: Not Active 1: Active

0x0184E058 GCC_CAMSS_CSI0PIX_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

PSCBC Registers for CAMSS_CSI0PIX

GCC_CAMSS_CSI0PIX_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	Branch Clock Off 0: Clock is ON 1: Clock is OFF
0	CLK_ENABLE		SW Clock Enable Bit 0: Clock Disabled 1: Clock Enabled

0x0184F020 GCC_CSI1_CMD_RCGR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

Root Clock Generator Command Register CSI1

GCC_CSI1_CMD_RCGR

Bits	Name	Type	Description
31	ROOT_OFF	R	The ROOT_OFF status allows SW to determine if the generator is actively producing a clock 0: Active 1: Not Active
4	DIRTY_CFG_RCGR	R	Dirty bit for the CFG_RCGR register. This bit is active if the CFG_RCGR register is written, and will clear after the update bit asserts. 0: Not Active 1: Active
1	ROOT_EN		The ROOT_EN bit controls the output of the clock generator. This enable may be ORed with a HW clock enable and then sent to the RCG. 0: Disabled 1: Enabled

GCC_CSI1_CMD_RCGR (cont.)

Bits	Name	Type	Description
0	UPDATE		This UPDATE bit is SW and HW modified. SW sets the bit when the configuration values are programmed, and when they have taken effect, then HW clears the bit to 0. 0: Not Active 1: Active

0x0184F024 GCC_CSI1_CFG_RCGR**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

Clock Generator Config Register for CSI1

GCC_CSI1_CFG_RCGR

Bits	Name	Description
10:8	SRC_SEL	Root Source Select 000: cxo 001: GPLL0_MAIN 010: gnd 011: gnd 100: gnd 101: gnd 110: gnd 111: Reserved

GCC_CSI1_CFG_RCGR (cont.)

Bits	Name	Description
4:0	SRC_DIV	The SRC_DIV bits define the integer and half integer portion of the divider setting. 00000: Div 1 00001: Div 1 00010: Div 1.5 00011: Div 2 00100: Div 2.5 00101: Div 3 00110: Div 3.5 00111: Div 4 01000: Div 4.5 01001: Div 5 01010: Div 5.5 01011: Div 6 01100: Div 6.5 01101: Div 7 01110: Div 7.5 01111: Div 8 10000: Div 8.5 10001: Div 9 10010: Div 9.5 10011: Div 10 10100: Div 10.5 10101: Div 11 10110: Div 11.5 10111: Div 12 11000: Div 12.5 11001: Div 13 11010: Div 13.5 11011: Div 14 11100: Div 14.5 11101: Div 15 11110: Div 15.5 11111: Div 16

0x0184F038 GCC_CAMSS_CSI1_BCR**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

BCR Registers for CAMSS_CSI1

GCC_CAMSS_CSI1_BCR

Bits	Name	Description
0	BLK_ARES	The block asynchronous reset bit, when set, the block clocks are disabled and the ares_out signals go active. 0: Not Active 1: Active

0x0184F03C GCC_CAMSS_CSI1_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

PSCBC Registers for CAMSS_CSI1

GCC_CAMSS_CSI1_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	Branch Clock Off 0: Clock is ON 1: Clock is OFF
0	CLK_ENABLE		SW Clock Enable Bit 0: Clock Disabled 1: Clock Enabled

0x0184F040 GCC_CAMSS_CSI1_AHB_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

PSCBC Registers for CAMSS_CSI1_AHB

GCC_CAMSS_CSI1_AHB_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	Branch Clock Off 0: Clock is ON 1: Clock is OFF
0	CLK_ENABLE		SW Clock Enable Bit 0: Clock Disabled 1: Clock Enabled

0x0184F044 GCC_CAMSS_CSI1PHY_BCR**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

BCR Registers for CAMSS_CSI1PHY

GCC_CAMSS_CSI1PHY_BCR

Bits	Name	Description
0	BLK_ARES	The block asynchronous reset bit, when set, the block clocks are disabled and the ares_out signals go active. 0: Not Active 1: Active

0x0184F048 GCC_CAMSS_CSI1PHY_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

PSCBC Registers for CAMSS_CSI1PHY

GCC_CAMSS_CSI1PHY_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	Branch Clock Off 0: Clock is ON 1: Clock is OFF
0	CLK_ENABLE		SW Clock Enable Bit 0: Clock Disabled 1: Clock Enabled

0x0184F04C GCC_CAMSS_CSI1RDI_BCR**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

BCR Registers for CAMSS_CSI1RDI

GCC_CAMSS_CSI1RDI_BCR

Bits	Name	Description
0	BLK_ARES	The block asynchronous reset bit, when set, the block clocks are disabled and the ares_out signals go active. 0: Not Active 1: Active

0x0184F050 GCC_CAMSS_CSI1RDI_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

PSCBC Registers for CAMSS_CSI1RDI

GCC_CAMSS_CSI1RDI_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	Branch Clock Off 0: Clock is ON 1: Clock is OFF
0	CLK_ENABLE		SW Clock Enable Bit 0: Clock Disabled 1: Clock Enabled

0x0184F054 GCC_CAMSS_CSI1PIX_BCR**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

BCR Registers for CAMSS_CSI1PIX

GCC_CAMSS_CSI1PIX_BCR

Bits	Name	Description
0	BLK_ARES	The block asynchronous reset bit, when set, the block clocks are disabled and the ares_out signals go active. 0: Not Active 1: Active

0x0184F058 GCC_CAMSS_CSI1PIX_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

PSCBC Registers for CAMSS_CSI1PIX

GCC_CAMSS_CSI1PIX_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	Branch Clock Off 0: Clock is ON 1: Clock is OFF
0	CLK_ENABLE		SW Clock Enable Bit 0: Clock Disabled 1: Clock Enabled

0x01850000 GCC_CAMSS_ISPIF_BCR**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

BCR Registers for CAMSS_ISPIF

GCC_CAMSS_ISPIF_BCR

Bits	Name	Description
0	BLK_ARES	The block asynchronous reset bit, when set, the block clocks are disabled and the ares_out signals go active. 0: Not Active 1: Active

0x01850004 GCC_CAMSS_ISPIF_AHB_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

PSCBC Registers for CAMSS_ISPIF_AHB

GCC_CAMSS_ISPIF_AHB_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	Branch Clock Off 0: Clock is ON 1: Clock is OFF
0	CLK_ENABLE		SW Clock Enable Bit 0: Clock Disabled 1: Clock Enabled

0x01851000 GCC_CCI_CMD_RCGR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

Root Clock Generator Command Register CCI

GCC_CCI_CMD_RCGR

Bits	Name	Type	Description
31	ROOT_OFF	R	The ROOT_OFF status allows SW to determine if the generator is actively producing a clock 0: Active 1: Not Active
7	DIRTY_D	R	Dirty bit for the D register. This bit is active if the D register is written, and will clear after the update bit asserts. 0: Not Active 1: Active
6	DIRTY_N	R	Dirty bit for the N register. This bit is active if the N register is written, and will clear after the update bit asserts. 0: Not Active 1: Active
5	DIRTY_M	R	Dirty bit for the M register. This bit is active if the M register is written, and will clear after the update bit asserts. 0: Not Active 1: Active
4	DIRTY_CFG_RCGR	R	Dirty bit for the CFG_RCGR register. This bit is active if the CFG_RCGR register is written, and will clear after the update bit asserts. 0: Not Active 1: Active

GCC_CCI_CMD_RCGR (cont.)

Bits	Name	Type	Description
1	ROOT_EN		The ROOT_EN bit controls the output of the clock generator. This enable may be ORed with a HW clock enable and then sent to the RCG. 0: Disabled 1: Enabled
0	UPDATE		This UPDATE bit is SW and HW modified. SW sets the bit when the configuration values are programmed, and when they have taken effect, then HW clears the bit to 0. 0: Not Active 1: Active

0x01851004 GCC_CCI_CFG_RCGR**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

Clock Generator Config Register for CCI

GCC_CCI_CFG_RCGR

Bits	Name	Description
13:12	MODE	mode 00: bypass mode 01: swallow mode 10: dual edge mode 11: single edge mode
10:8	SRC_SEL	Root Source Select; Note gcc_gp1l1 is a backup 480Mhz source and the PLL may not be on. Only request SW to turn on if other sources cannot generate the required clock. 000: cxo 001: gnd 010: GPLLO_AUX 011: gnd 100: gnd 101: gnd 110: gnd 111: Reserved

GCC_CCI_CFG_RCGR (cont.)

Bits	Name	Description
4:0	SRC_DIV	The SRC_DIV bits define the integer and half integer portion of the divider setting. 00000: Div 1 00001: Div 1 00010: Div 1.5 00011: Div 2 00100: Div 2.5 00101: Div 3 00110: Div 3.5 00111: Div 4 01000: Div 4.5 01001: Div 5 01010: Div 5.5 01011: Div 6 01100: Div 6.5 01101: Div 7 01110: Div 7.5 01111: Div 8 10000: Div 8.5 10001: Div 9 10010: Div 9.5 10011: Div 10 10100: Div 10.5 10101: Div 11 10110: Div 11.5 10111: Div 12 11000: Div 12.5 11001: Div 13 11010: Div 13.5 11011: Div 14 11100: Div 14.5 11101: Div 15 11110: Div 15.5 11111: Div 16

0x01851008 GCC_CCI_M**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

Program M for the MND divider, where division ration is $F_{out} = F_{in} \cdot (M / N)$

GCC_CCI_M

Bits	Name	Description
7:0	M	Program M

0x0185100C GCC_CCI_N**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

Program NOT_N_MINUS_M (1's complement of N-M); The N value derived is used for the MND divider, where division ration is $F_{out} = F_{in} \cdot (M / N)$

GCC_CCI_N

Bits	Name	Description
7:0	NOT_N_MINUS_M	Program NOT_N_MINUS_M

0x01851010 GCC_CCI_D**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

Program NOT_2D (1's complement of 2D); The D value derived is used for the MND divider, where the duty cycle is D/N

GCC_CCI_D

Bits	Name	Description
7:0	NOT_2D	Program NOT_2D

0x01851014 GCC_CAMSS_CCI_BCR**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

BCR Registers for CAMSS_CCI

GCC_CAMSS_CCI_BCR

Bits	Name	Description
0	BLK_ARES	The block asynchronous reset bit, when set, the block clocks are disabled and the ares_out signals go active. 0: Not Active 1: Active

0x01851018 GCC_CAMSS_CCI_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

PSCBC Registers for CAMSS_CCI

GCC_CAMSS_CCI_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	Branch Clock Off 0: Clock is ON 1: Clock is OFF
0	CLK_ENABLE		SW Clock Enable Bit 0: Clock Disabled 1: Clock Enabled

0x0185101C GCC_CAMSS_CCI_AHB_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

PSCBC Registers for CAMSS_CCI_AHB

GCC_CAMSS_CCI_AHB_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	Branch Clock Off 0: Clock is ON 1: Clock is OFF
0	CLK_ENABLE		SW Clock Enable Bit 0: Clock Disabled 1: Clock Enabled

0x01852000 GCC_MCLK0_CMD_RCGR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

Root Clock Generator Command Register MCLK0

GCC_MCLK0_CMD_RCGR

Bits	Name	Type	Description
31	ROOT_OFF	R	The ROOT_OFF status allows SW to determine if the generator is actively producing a clock 0: Active 1: Not Active
7	DIRTY_D	R	Dirty bit for the D register. This bit is active if the D register is written, and will clear after the update bit asserts. 0: Not Active 1: Active
6	DIRTY_N	R	Dirty bit for the N register. This bit is active if the N register is written, and will clear after the update bit asserts. 0: Not Active 1: Active
5	DIRTY_M	R	Dirty bit for the M register. This bit is active if the M register is written, and will clear after the update bit asserts. 0: Not Active 1: Active
4	DIRTY_CFG_RCGR	R	Dirty bit for the CFG_RCGR register. This bit is active if the CFG_RCGR register is written, and will clear after the update bit asserts. 0: Not Active 1: Active
1	ROOT_EN		The ROOT_EN bit controls the output of the clock generator. This enable may be ORed with a HW clock enable and then sent to the RCG. 0: Disabled 1: Enabled
0	UPDATE		This UPDATE bit is SW and HW modified. SW sets the bit when the configuration values are programmed, and when they have taken effect, then HW clears the bit to 0. 0: Not Active 1: Active

0x01852004 GCC_MCLK0_CFG_RCGR**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

Clock Generator Config Register for MCLK0

GCC_MCLK0_CFG_RCGR

Bits	Name	Description
13:12	MODE	mode 00: bypass mode 01: swallow mode 10: dual edge mode 11: single edge mode
10:8	SRC_SEL	Root Source Select; Note gcc_gpll1 is a backup 480Mhz source and the PLL may not be on. Only request SW to turn on if other sources cannot generate the required clock. 000: cxo 001: GPLL0_MAIN 010: GPLL1_AUX 011: gnd 100: gnd 101: gnd 110: core_pi_sleep_clk 111: Reserved

GCC_MCLK0_CFG_RCGR (cont.)

Bits	Name	Description
4:0	SRC_DIV	The SRC_DIV bits define the integer and half integer portion of the divider setting. 00000: Div 1 00001: Div 1 00010: Div 1.5 00011: Div 2 00100: Div 2.5 00101: Div 3 00110: Div 3.5 00111: Div 4 01000: Div 4.5 01001: Div 5 01010: Div 5.5 01011: Div 6 01100: Div 6.5 01101: Div 7 01110: Div 7.5 01111: Div 8 10000: Div 8.5 10001: Div 9 10010: Div 9.5 10011: Div 10 10100: Div 10.5 10101: Div 11 10110: Div 11.5 10111: Div 12 11000: Div 12.5 11001: Div 13 11010: Div 13.5 11011: Div 14 11100: Div 14.5 11101: Div 15 11110: Div 15.5 11111: Div 16

0x01852008 GCC_MCLK0_M**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

Program M for the MND divider, where division ration is $F_{out} = F_{in} \cdot (M / N)$

GCC_MCLK0_M

Bits	Name	Description
7:0	M	Program M

0x0185200C GCC_MCLK0_N**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

Program NOT_N_MINUS_M (1's complement of N-M); The N value derived is used for the MND divider, where division ration is $F_{out} = F_{in} \cdot (M / N)$

GCC_MCLK0_N

Bits	Name	Description
7:0	NOT_N_MINUS_M	Program NOT_N_MINUS_M

0x01852010 GCC_MCLK0_D**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

Program NOT_2D (1's complement of 2D); The D value derived is used for the MND divider, where the duty cycle is D/N

GCC_MCLK0_D

Bits	Name	Description
7:0	NOT_2D	Program NOT_2D

0x01852014 GCC_CAMSS_MCLK0_BCR**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

BCR Registers for CAMSS_MCLK0

GCC_CAMSS_MCLK0_BCR

Bits	Name	Description
0	BLK_ARES	The block asynchronous reset bit, when set, the block clocks are disabled and the ares_out signals go active. 0: Not Active 1: Active

0x01852018 GCC_CAMSS_MCLK0_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

PSCBC Registers for CAMSS_MCLK0

GCC_CAMSS_MCLK0_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	Branch Clock Off 0: Clock is ON 1: Clock is OFF
0	CLK_ENABLE		SW Clock Enable Bit 0: Clock Disabled 1: Clock Enabled

0x01853000 GCC_MCLK1_CMD_RCGR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

Root Clock Generator Command Register MCLK1

GCC_MCLK1_CMD_RCGR

Bits	Name	Type	Description
31	ROOT_OFF	R	The ROOT_OFF status allows SW to determine if the generator is actively producing a clock 0: Active 1: Not Active

GCC_MCLK1_CMD_RCGR (cont.)

Bits	Name	Type	Description
7	DIRTY_D	R	Dirty bit for the D register. This bit is active if the D register is written, and will clear after the update bit asserts. 0: Not Active 1: Active
6	DIRTY_N	R	Dirty bit for the N register. This bit is active if the N register is written, and will clear after the update bit asserts. 0: Not Active 1: Active
5	DIRTY_M	R	Dirty bit for the M register. This bit is active if the M register is written, and will clear after the update bit asserts. 0: Not Active 1: Active
4	DIRTY_CFG_RCGR	R	Dirty bit for the CFG_RCGR register. This bit is active if the CFG_RCGR register is written, and will clear after the update bit asserts. 0: Not Active 1: Active
1	ROOT_EN		The ROOT_EN bit controls the output of the clock generator. This enable may be ORed with a HW clock enable and then sent to the RCG. 0: Disabled 1: Enabled
0	UPDATE		This UPDATE bit is SW and HW modified. SW sets the bit when the configuration values are programmed, and when they have taken effect, then HW clears the bit to 0. 0: Not Active 1: Active

0x01853004 GCC_MCLK1_CFG_RCGR**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

Clock Generator Config Register for MCLK1

GCC_MCLK1_CFG_RCGR

Bits	Name	Description
13:12	MODE	mode 00: bypass mode 01: swallow mode 10: dual edge mode 11: single edge mode
10:8	SRC_SEL	Root Source Select; Note gcc_gpll1 is a backup 480Mhz source and the PLL may not be on. Only request SW to turn on if other sources cannot generate the required clock. 000: cxo 001: GPLL0_MAIN 010: GPLL1_AUX 011: gnd 100: gnd 101: gnd 110: core_pi_sleep_clk 111: Reserved

GCC_MCLK1_CFG_RCGR (cont.)

Bits	Name	Description
4:0	SRC_DIV	The SRC_DIV bits define the integer and half integer portion of the divider setting. 00000: Div 1 00001: Div 1 00010: Div 1.5 00011: Div 2 00100: Div 2.5 00101: Div 3 00110: Div 3.5 00111: Div 4 01000: Div 4.5 01001: Div 5 01010: Div 5.5 01011: Div 6 01100: Div 6.5 01101: Div 7 01110: Div 7.5 01111: Div 8 10000: Div 8.5 10001: Div 9 10010: Div 9.5 10011: Div 10 10100: Div 10.5 10101: Div 11 10110: Div 11.5 10111: Div 12 11000: Div 12.5 11001: Div 13 11010: Div 13.5 11011: Div 14 11100: Div 14.5 11101: Div 15 11110: Div 15.5 11111: Div 16

0x01853008 GCC_MCLK1_M**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

Program M for the MND divider, where division ration is $F_{out} = F_{in} \cdot (M / N)$

GCC_MCLK1_M

Bits	Name	Description
7:0	M	Program M

0x0185300C GCC_MCLK1_N**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

Program NOT_N_MINUS_M (1's complement of N-M); The N value derived is used for the MND divider, where division ration is $F_{out} = F_{in} \cdot (M / N)$

GCC_MCLK1_N

Bits	Name	Description
7:0	NOT_N_MINUS_M	Program NOT_N_MINUS_M

0x01853010 GCC_MCLK1_D**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

Program NOT_2D (1's complement of 2D); The D value derived is used for the MND divider, where the duty cycle is D/N

GCC_MCLK1_D

Bits	Name	Description
7:0	NOT_2D	Program NOT_2D

0x01853014 GCC_CAMSS_MCLK1_BCR**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

BCR Registers for CAMSS_MCLK1

GCC_CAMSS_MCLK1_BCR

Bits	Name	Description
0	BLK_ARES	The block asynchronous reset bit, when set, the block clocks are disabled and the ares_out signals go active. 0: Not Active 1: Active

0x01853018 GCC_CAMSS_MCLK1_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

PSCBC Registers for CAMSS_MCLK1

GCC_CAMSS_MCLK1_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	Branch Clock Off 0: Clock is ON 1: Clock is OFF
0	CLK_ENABLE		SW Clock Enable Bit 0: Clock Disabled 1: Clock Enabled

0x01854000 GCC_MM_GP0_CMD_RCGR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

Root Clock Generator Command Register GP0

GCC_MM_GP0_CMD_RCGR

Bits	Name	Type	Description
31	ROOT_OFF	R	The ROOT_OFF status allows SW to determine if the generator is actively producing a clock 0: Active 1: Not Active

GCC_MM_GP0_CMD_RCGR (cont.)

Bits	Name	Type	Description
7	DIRTY_D	R	Dirty bit for the D register. This bit is active if the D register is written, and will clear after the update bit asserts. 0: Not Active 1: Active
6	DIRTY_N	R	Dirty bit for the N register. This bit is active if the N register is written, and will clear after the update bit asserts. 0: Not Active 1: Active
5	DIRTY_M	R	Dirty bit for the M register. This bit is active if the M register is written, and will clear after the update bit asserts. 0: Not Active 1: Active
4	DIRTY_CFG_RCGR	R	Dirty bit for the CFG_RCGR register. This bit is active if the CFG_RCGR register is written, and will clear after the update bit asserts. 0: Not Active 1: Active
1	ROOT_EN		The ROOT_EN bit controls the output of the clock generator. This enable may be ORed with a HW clock enable and then sent to the RCG. 0: Disabled 1: Enabled
0	UPDATE		This UPDATE bit is SW and HW modified. SW sets the bit when the configuration values are programmed, and when they have taken effect, then HW clears the bit to 0. 0: Not Active 1: Active

0x01854004 GCC_MM_GP0_CFG_RCGR**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

Clock Generator Config Register for GP0

GCC_MM_GP0_CFG_RCGR

Bits	Name	Description
13:12	MODE	mode 00: bypass mode 01: swallow mode 10: dual edge mode 11: single edge mode
10:8	SRC_SEL	Root Source Select; Note gcc_gpll1 is a backup 480Mhz source and the PLL may not be on. Only request SW to turn on if other sources cannot generate the required clock. 000: cxo 001: GPLL0_MAIN 010: GPLL1_AUX 011: gnd 100: gnd 101: gnd 110: core_pi_sleep_clk 111: Reserved

GCC_MM_GP0_CFG_RCGR (cont.)

Bits	Name	Description
4:0	SRC_DIV	The SRC_DIV bits define the integer and half integer portion of the divider setting. 00000: Div 1 00001: Div 1 00010: Div 1.5 00011: Div 2 00100: Div 2.5 00101: Div 3 00110: Div 3.5 00111: Div 4 01000: Div 4.5 01001: Div 5 01010: Div 5.5 01011: Div 6 01100: Div 6.5 01101: Div 7 01110: Div 7.5 01111: Div 8 10000: Div 8.5 10001: Div 9 10010: Div 9.5 10011: Div 10 10100: Div 10.5 10101: Div 11 10110: Div 11.5 10111: Div 12 11000: Div 12.5 11001: Div 13 11010: Div 13.5 11011: Div 14 11100: Div 14.5 11101: Div 15 11110: Div 15.5 11111: Div 16

0x01854008 GCC_MM_GP0_M**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

Program M for the MND divider, where division ration is $F_{out} = F_{in} \cdot (M / N)$

GCC_MM_GP0_M

Bits	Name	Description
7:0	M	Program M

0x0185400C GCC_MM_GP0_N**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

Program NOT_N_MINUS_M (1's complement of N-M); The N value derived is used for the MND divider, where division ration is $F_{out} = F_{in} \cdot (M / N)$

GCC_MM_GP0_N

Bits	Name	Description
7:0	NOT_N_MINUS_M	Program NOT_N_MINUS_M

0x01854010 GCC_MM_GP0_D**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

Program NOT_2D (1's complement of 2D); The D value derived is used for the MND divider, where the duty cycle is D/N

GCC_MM_GP0_D

Bits	Name	Description
7:0	NOT_2D	Program NOT_2D

0x01854014 GCC_CAMSS_GP0_BCR**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

BCR Registers for CAMSS_GP0

GCC_CAMSS_GP0_BCR

Bits	Name	Description
0	BLK_ARES	The block asynchronous reset bit, when set, the block clocks are disabled and the ares_out signals go active. 0: Not Active 1: Active

0x01854018 GCC_CAMSS_GP0_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

PSCBC Registers for CAMSS_GP0

GCC_CAMSS_GP0_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	Branch Clock Off 0: Clock is ON 1: Clock is OFF
0	CLK_ENABLE		SW Clock Enable Bit 0: Clock Disabled 1: Clock Enabled

0x01855000 GCC_MM_GP1_CMD_RCGR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

Root Clock Generator Command Register GP1

GCC_MM_GP1_CMD_RCGR

Bits	Name	Type	Description
31	ROOT_OFF	R	The ROOT_OFF status allows SW to determine if the generator is actively producing a clock 0: Active 1: Not Active

GCC_MM_GP1_CMD_RCGR (cont.)

Bits	Name	Type	Description
7	DIRTY_D	R	Dirty bit for the D register. This bit is active if the D register is written, and will clear after the update bit asserts. 0: Not Active 1: Active
6	DIRTY_N	R	Dirty bit for the N register. This bit is active if the N register is written, and will clear after the update bit asserts. 0: Not Active 1: Active
5	DIRTY_M	R	Dirty bit for the M register. This bit is active if the M register is written, and will clear after the update bit asserts. 0: Not Active 1: Active
4	DIRTY_CFG_RCGR	R	Dirty bit for the CFG_RCGR register. This bit is active if the CFG_RCGR register is written, and will clear after the update bit asserts. 0: Not Active 1: Active
1	ROOT_EN		The ROOT_EN bit controls the output of the clock generator. This enable may be ORed with a HW clock enable and then sent to the RCG. 0: Disabled 1: Enabled
0	UPDATE		This UPDATE bit is SW and HW modified. SW sets the bit when the configuration values are programmed, and when they have taken effect, then HW clears the bit to 0. 0: Not Active 1: Active

0x01855004 GCC_MM_GP1_CFG_RCGR**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

Clock Generator Config Register for GP1

GCC_MM_GP1_CFG_RCGR

Bits	Name	Description
13:12	MODE	mode 00: bypass mode 01: swallow mode 10: dual edge mode 11: single edge mode
10:8	SRC_SEL	Root Source Select; Note gcc_gpll1 is a backup 480Mhz source and the PLL may not be on. Only request SW to turn on if other sources cannot generate the required clock. 000: cxo 001: GPLL0_MAIN 010: GPLL1_AUX 011: gnd 100: gnd 101: gnd 110: core_pi_sleep_clk 111: Reserved

GCC_MM_GP1_CFG_RCGR (cont.)

Bits	Name	Description
4:0	SRC_DIV	The SRC_DIV bits define the integer and half integer portion of the divider setting. 00000: Div 1 00001: Div 1 00010: Div 1.5 00011: Div 2 00100: Div 2.5 00101: Div 3 00110: Div 3.5 00111: Div 4 01000: Div 4.5 01001: Div 5 01010: Div 5.5 01011: Div 6 01100: Div 6.5 01101: Div 7 01110: Div 7.5 01111: Div 8 10000: Div 8.5 10001: Div 9 10010: Div 9.5 10011: Div 10 10100: Div 10.5 10101: Div 11 10110: Div 11.5 10111: Div 12 11000: Div 12.5 11001: Div 13 11010: Div 13.5 11011: Div 14 11100: Div 14.5 11101: Div 15 11110: Div 15.5 11111: Div 16

0x01855008 GCC_MM_GP1_M**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

Program M for the MND divider, where division ration is $F_{out} = F_{in} \cdot (M / N)$

GCC_MM_GP1_M

Bits	Name	Description
7:0	M	Program M

0x0185500C GCC_MM_GP1_N**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

Program NOT_N_MINUS_M (1's complement of N-M); The N value derived is used for the MND divider, where division ration is $F_{out} = F_{in} \cdot (M / N)$

GCC_MM_GP1_N

Bits	Name	Description
7:0	NOT_N_MINUS_M	Program NOT_N_MINUS_M

0x01855010 GCC_MM_GP1_D**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

Program NOT_2D (1's complement of 2D); The D value derived is used for the MND divider, where the duty cycle is D/N

GCC_MM_GP1_D

Bits	Name	Description
7:0	NOT_2D	Program NOT_2D

0x01855014 GCC_CAMSS_GP1_BCR**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

BCR Registers for CAMSS_GP1

GCC_CAMSS_GP1_BCR

Bits	Name	Description
0	BLK_ARES	The block asynchronous reset bit, when set, the block clocks are disabled and the ares_out signals go active. 0: Not Active 1: Active

0x01855018 GCC_CAMSS_GP1_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

PSCBC Registers for CAMSS_GP1

GCC_CAMSS_GP1_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	Branch Clock Off 0: Clock is ON 1: Clock is OFF
0	CLK_ENABLE		SW Clock Enable Bit 0: Clock Disabled 1: Clock Enabled

0x01856000 GCC_CAMSS_TOP_BCR**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

BCR Registers for CAMSS_TOP

GCC_CAMSS_TOP_BCR

Bits	Name	Description
0	BLK_ARES	The block asynchronous reset bit, when set, the block clocks are disabled and the ares_out signals go active. 0: Not Active 1: Active

0x0185A014 GCC_CAMSS_AHB_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

PSCBC Registers for CAMSS_TOP_AHB

GCC_CAMSS_AHB_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	Branch Clock Off 0: Clock is ON 1: Clock is OFF
0	CLK_ENABLE		SW Clock Enable Bit 0: Clock Disabled. 1: Clock Enabled.

0x0185A018 GCC_CAMSS_AHB_BCR**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

BCR Registers for CAMSS_AHB

GCC_CAMSS_AHB_BCR

Bits	Name	Description
0	BLK_ARES	The block asynchronous reset bit, when set, the block clocks are disabled and the ares_out signals go active. 0: Not Active 1: Active

0x01856004 GCC_CAMSS_TOP_AHB_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x80008000

PSCBC Registers for CAMSS_AHB

GCC_CAMSS_TOP_AHB_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	Branch Clock Off 0: Clock is ON 1: Clock is OFF
30:28	NOC_HANDSHAKE_FSM_STATE	R	FSM State 000: OFFS = Off 001: WFON = Wait For Clk On 010: SRIU = Slave Ready is Up 011: WFSL = Wait for Socketcon Low 100: WOFF = Wait for Clk Off 101: Not Used 110: Not Used 111: Not Used
15	NOC_HANDSHAKE_FSM_EN		SlaveReady and Socket Connect FSM Enable; Default is 1 0: SW Clock Enabled 1: FSM Clock Enabled
0	CLK_ENABLE		SW Clock Enable Bit 0: Clock Disabled. 1: Clock Enabled.

0x01856008 GCC_CAMSS_MICRO_BCR**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

BCR Registers for CAMSS_MICRO

GCC_CAMSS_MICRO_BCR

Bits	Name	Description
0	BLK_ARES	The block asynchronous reset bit, when set, the block clocks are disabled and the ares_out signals go active. 0: Not Active 1: Active

0x0185600C GCC_CAMSS_MICRO_AHB_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x80004FF0

PSCBC Registers for CAMSS_MICRO_AHB

GCC_CAMSS_MICRO_AHB_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	Branch Clock Off 0: Clock is ON 1: Clock is OFF
14	FORCE_MEM_CORE_ON		Force Mem Core bits to be on 0: Memory Core can be powered down by the clock branch slp control 1: Force Memory Core to be powered on (default)
13	FORCE_MEM_PERIPH_ON		Force Mem Periph logics to be on 0: Memory Periphery logic can be powered down by the clock branch slp control (default) 1: Force Memory Periphery logic to be powerd on
12	FORCE_MEM_PERIPH_OF F		Force Mem Periph logics to be off 0: Memory Periphery logic is powered on by the clock branch slp control (default) 1: Force Memory Periphery logic to be powerd off
11:8	WAKEUP		When the clock is enabled, the memory control signal will be turned on immediately, but the clock will remain gated until WAKEUP clock cycles later. 0000: Number of clock cycle 1 0001: Number of clock cycle 2 0010: Number of clock cycle 3 0011: Number of clock cycle 4 0100: Number of clock cycle 5 0101: Number of clock cycle 6 0110: Number of clock cycle 7 1000: Number of clock cycle 8 1001: Number of clock cycle 9 1010: Number of clock cycle 10 1011: Number of clock cycle 11 1100: Number of clock cycle 12 1101: Number of clock cycle 13 1110: Number of clock cycle 14 1111: Number of clock cycle 15

GCC_CAMSS_MICRO_AHB_CBCR (cont.)

Bits	Name	Type	Description
7:4	SLEEP		When the clock is disabled, the clock will be gated immediately, but the memory control signals will not be turned off until a delay of SLEEP clock cycles. 0000: Number of clock cycle 1 0001: Number of clock cycle 2 0010: Number of clock cycle 3 0011: Number of clock cycle 4 0100: Number of clock cycle 5 0101: Number of clock cycle 6 0110: Number of clock cycle 7 1000: Number of clock cycle 8 1001: Number of clock cycle 9 1010: Number of clock cycle 10 1011: Number of clock cycle 11 1100: Number of clock cycle 12 1101: Number of clock cycle 13 1110: Number of clock cycle 14 1111: Number of clock cycle 15
0	CLK_ENABLE		SW Clock Enable Bit 0: Clock Disabled 1: Clock Enabled

0x01857000 GCC_JPEG0_CMD_RCGR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

Root Clock Generator Command Register JPEG0

GCC_JPEG0_CMD_RCGR

Bits	Name	Type	Description
31	ROOT_OFF	R	The ROOT_OFF status allows SW to determine if the generator is actively producing a clock 0: Active 1: Not Active
4	DIRTY_CFG_RCGR	R	Dirty bit for the CFG_RCGR register. This bit is active if the CFG_RCGR register is written, and will clear after the update bit asserts. 0: Not Active 1: Active

GCC_JPEG0_CMD_RCGR (cont.)

Bits	Name	Type	Description
1	ROOT_EN		The ROOT_EN bit controls the output of the clock generator. This enable may be ORed with a HW clock enable and then sent to the RCG. 0: Disabled 1: Enabled
0	UPDATE		This UPDATE bit is SW and HW modified. SW sets the bit when the configuration values are programmed, and when they have taken effect, then HW clears the bit to 0. 0: Not Active 1: Active

0x01857004 GCC_JPEG0_CFG_RCGR**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

Clock Generator Config Register for JPEG0

GCC_JPEG0_CFG_RCGR

Bits	Name	Description
10:8	SRC_SEL	Root Source Select 000: cxo 001: GPLL0_MAIN 010: gnd 011: gnd 100: gnd 101: gnd 110: gnd 111: Reserved

GCC_JPEG0_CFG_RCGR (cont.)

Bits	Name	Description
4:0	SRC_DIV	The SRC_DIV bits define the integer and half integer portion of the divider setting. 00000: Div 1 00001: Div 1 00010: Div 1.5 00011: Div 2 00100: Div 2.5 00101: Div 3 00110: Div 3.5 00111: Div 4 01000: Div 4.5 01001: Div 5 01010: Div 5.5 01011: Div 6 01100: Div 6.5 01101: Div 7 01110: Div 7.5 01111: Div 8 10000: Div 8.5 10001: Div 9 10010: Div 9.5 10011: Div 10 10100: Div 10.5 10101: Div 11 10110: Div 11.5 10111: Div 12 11000: Div 12.5 11001: Div 13 11010: Div 13.5 11011: Div 14 11100: Div 14.5 11101: Div 15 11110: Div 15.5 11111: Div 16

0x01857018 GCC_CAMSS_JPEG_BCR**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

BCR Registers for CAMSS_JPEG

GCC_CAMSS_JPEG_BCR

Bits	Name	Description
0	BLK_ARES	The block asynchronous reset bit, when set, the block clocks are disabled and the ares_out signals go active. 0: Not Active 1: Active

0x0185701C GCC_CAMSS_JPEG_GDSCR**Type:** RW**Clock:** CLK**Reset State:** 0x00222001

GDS Registers for CAMSS_JPEG

GCC_CAMSS_JPEG_GDSCR

Bits	Name	Type	Description
31	PWR_ON	R	PWR State 0: Power collapsed 1: Power on
30:27	GDSC_STATE	R	GDS FSM State 0000: PowerOff 0001: AssertAresAndRampUp_en_few 0010: DeassertAres 0011: RampUp_en_rest 0100: PowerOn 0101: UnclampIO 0110: NoRetainFF 0111: RestoreFF 1000: NA 1001: NA 1010: NA 1011: NA 1100: DisableClk 1101: RetainFF 1110: NA 1111: ClampIOAndSaveFF

GCC_CAMSS_JPEG_GDSCR (cont.)

Bits	Name	Type	Description
23:20	EN_REST_WAIT		Wait cntr for the remaining PD switches to power on 0000: cntr value is cxc x 1 cycle 0001: cntr value is cxc x 2 cycle 0010: cntr value is cxc x 4 cycle [default] 0011: cntr value is cxc x 8 cycle 0100: cntr value is cxc x 16 cycle 0101: cntr value is cxc x 32 cycle 0110: cntr value is cxc x 64 cycle 0111: cntr value is cxc x 128 cycle 1000: cntr value is cxc x 256 cycle 1001: cntr value is cxc x 512 cycle 1010: cntr value is cxc x 1024 cycle 1011: cntr value is cxc x 2048 cycle 1100: cntr value is cxc x 2048 cycle 1101: cntr value is cxc x 2048 cycle 1110: cntr value is cxc x 2048 cycle 1111: cntr value is cxc x 2048 cycle
19:16	EN_FEW_WAIT		Wait cntr for the first few PD switches to power on 0000: cntr value is cxc x 1 cycle 0001: cntr value is cxc x 2 cycle 0010: cntr value is cxc x 4 cycle [default] 0011: cntr value is cxc x 8 cycle 0100: cntr value is cxc x 16 cycle 0101: cntr value is cxc x 32 cycle 0110: cntr value is cxc x 64 cycle 0111: cntr value is cxc x 128 cycle 1000: cntr value is cxc x 256 cycle 1001: cntr value is cxc x 512 cycle 1010: cntr value is cxc x 1024 cycle 1011: cntr value is cxc x 2048 cycle 1100: cntr value is cxc x 2048 cycle 1101: cntr value is cxc x 2048 cycle 1110: cntr value is cxc x 2048 cycle 1111: cntr value is cxc x 2048 cycle

GCC_CAMSS_JPEG_GDSCR (cont.)

Bits	Name	Type	Description
15:12	CLK_DIS_WAIT		Wait while all clocks in this PD are turned off 0000: cntr value is cxo x 1 cycle 0001: cntr value is cxo x 2 cycle 0010: cntr value is cxo x 4 cycle [default] 0011: cntr value is cxo x 8 cycle 0100: cntr value is cxo x 16 cycle 0101: cntr value is cxo x 32 cycle 0110: cntr value is cxo x 64 cycle 0111: cntr value is cxo x 128 cycle 1000: cntr value is cxo x 256 cycle 1001: cntr value is cxo x 512 cycle 1010: cntr value is cxo x 1024 cycle 1011: cntr value is cxo x 2048 cycle 1100: cntr value is cxo x 2048 cycle 1101: cntr value is cxo x 2048 cycle 1110: cntr value is cxo x 2048 cycle 1111: cntr value is cxo x 2048 cycle
10	RESTORE		Restore 0: Not Active 1: Active
9	SAVE		Save 0: Not Active 1: Active
8	RETAIN		Retain 0: Not Active 1: Active
7	EN_REST		Enable Rest 0: Not Active 1: Active
6	EN_FEW		Enable Few 0: Not Active 1: Active
5	CLAMP_IO		Clamp IO 0: Not Active 1: Active
4	CLK_DISABLE		Clk Disable 0: Not Active 1: Active
3	PD_ARES		Reset 0: Not Active 1: Active

GCC_CAMSS_JPEG_GDSCR (cont.)

Bits	Name	Type	Description
2	SW_OVERRIDE		SW Override 0: GDSC sequence based on HW FSM 1: GDSC sequence based on SW
1	HW_CONTROL		HW Control 0: Power off by HW trigger 1: Power on by HW trigger
0	SW_COLLAPSE		SW Collapse 0: Power on through SW 1: Power off through SW, default

0x01857020 GCC_CAMSS_JPEG0_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x80004FF0

PSCBC Registers for CAMSS_JPEG0

GCC_CAMSS_JPEG0_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	Branch Clock Off 0: Clock is ON 1: Clock is OFF
14	FORCE_MEM_CORE_ON		Force Mem Core bits to be on 0: Memory Core can be powered down by the clock branch slp control 1: Force Memory Core to be powered on (default)
13	FORCE_MEM_PERIPH_ON		Force Mem Periph logics to be on 0: Memory Periphery logic can be powered down by the clock branch slp control (default) 1: Force Memory Periphery logic to be powerd on
12	FORCE_MEM_PERIPH_OFF		Force Mem Periph logics to be off 0: Memory Periphery logic is powered on by the clock branch slp control (default) 1: Force Memory Periphery logic to be powerd off

GCC_CAMSS_JPEG0_CBCR (cont.)

Bits	Name	Type	Description
11:8	WAKEUP		When the clock is enabled, the memory control signal will be turned on immediately, but the clock will remain gated until WAKEUP clock cycles later. 0000: Number of clock cycle 1 0001: Number of clock cycle 2 0010: Number of clock cycle 3 0011: Number of clock cycle 4 0100: Number of clock cycle 5 0101: Number of clock cycle 6 0110: Number of clock cycle 7 1000: Number of clock cycle 8 1001: Number of clock cycle 9 1010: Number of clock cycle 10 1011: Number of clock cycle 11 1100: Number of clock cycle 12 1101: Number of clock cycle 13 1110: Number of clock cycle 14 1111: Number of clock cycle 15
7:4	SLEEP		When the clock is disabled, the clock will be gated immediately, but the memory control signals will not be turned off until a delay of SLEEP clock cycles. 0000: Number of clock cycle 1 0001: Number of clock cycle 2 0010: Number of clock cycle 3 0011: Number of clock cycle 4 0100: Number of clock cycle 5 0101: Number of clock cycle 6 0110: Number of clock cycle 7 1000: Number of clock cycle 8 1001: Number of clock cycle 9 1010: Number of clock cycle 10 1011: Number of clock cycle 11 1100: Number of clock cycle 12 1101: Number of clock cycle 13 1110: Number of clock cycle 14 1111: Number of clock cycle 15
0	CLK_ENABLE		SW Clock Enable Bit 0: Clock Disabled 1: Clock Enabled

0x01857024 GCC_CAMSS_JPEG_AHB_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

PSCBC Registers for CAMSS_JPEG_AHB

GCC_CAMSS_JPEG_AHB_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	Branch Clock Off 0: Clock is ON 1: Clock is OFF
0	CLK_ENABLE		SW Clock Enable Bit 0: Clock Disabled 1: Clock Enabled

0x01857028 GCC_CAMSS_JPEG_AXI_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x80004FF0

PSCBC Registers for CAMSS_JPEG_AXI

GCC_CAMSS_JPEG_AXI_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	Branch Clock Off 0: Clock is ON 1: Clock is OFF
14	FORCE_MEM_CORE_ON		Force Mem Core bits to be on 0: Memory Core can be powered down by the clock branch slp control 1: Force Memory Core to be powered on (default)
13	FORCE_MEM_PERIPH_ON		Force Mem Periph logics to be on 0: Memory Periphery logic can be powered down by the clock branch slp control (default) 1: Force Memory Periphery logic to be powerd on
12	FORCE_MEM_PERIPH_OF F		Force Mem Periph logics to be off 0: Memory Periphery logic is powered on by the clock branch slp control (default) 1: Force Memory Periphery logic to be powerd off

GCC_CAMSS_JPEG_AXI_CBCR (cont.)

Bits	Name	Type	Description
11:8	WAKEUP		When the clock is enabled, the memory control signal will be turned on immediately, but the clock will remain gated until WAKEUP clock cycles later. 0000: Number of clock cycle 1 0001: Number of clock cycle 2 0010: Number of clock cycle 3 0011: Number of clock cycle 4 0100: Number of clock cycle 5 0101: Number of clock cycle 6 0110: Number of clock cycle 7 1000: Number of clock cycle 8 1001: Number of clock cycle 9 1010: Number of clock cycle 10 1011: Number of clock cycle 11 1100: Number of clock cycle 12 1101: Number of clock cycle 13 1110: Number of clock cycle 14 1111: Number of clock cycle 15
7:4	SLEEP		When the clock is disabled, the clock will be gated immediately, but the memory control signals will not be turned off until a delay of SLEEP clock cycles. 0000: Number of clock cycle 1 0001: Number of clock cycle 2 0010: Number of clock cycle 3 0011: Number of clock cycle 4 0100: Number of clock cycle 5 0101: Number of clock cycle 6 0110: Number of clock cycle 7 1000: Number of clock cycle 8 1001: Number of clock cycle 9 1010: Number of clock cycle 10 1011: Number of clock cycle 11 1100: Number of clock cycle 12 1101: Number of clock cycle 13 1110: Number of clock cycle 14 1111: Number of clock cycle 15
0	CLK_ENABLE		SW Clock Enable Bit 0: Clock Disabled 1: Clock Enabled

0x01858000 GCC_VFE0_CMD_RCGR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

Root Clock Generator Command Register VFE0

GCC_VFE0_CMD_RCGR

Bits	Name	Type	Description
31	ROOT_OFF	R	The ROOT_OFF status allows SW to determine if the generator is actively producing a clock 0: Active 1: Not Active
4	DIRTY_CFG_RCGR	R	Dirty bit for the CFG_RCGR register. This bit is active if the CFG_RCGR register is written, and will clear after the update bit asserts. 0: Not Active 1: Active
1	ROOT_EN		The ROOT_EN bit controls the output of the clock generator. This enable may be ORed with a HW clock enable and then sent to the RCG. 0: Disabled 1: Enabled
0	UPDATE		This UPDATE bit is SW and HW modified. SW sets the bit when the configuration values are programmed, and when they have taken effect, then HW clears the bit to 0. 0: Not Active 1: Active

0x01858004 GCC_VFE0_CFG_RCGR**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

Clock Generator Config Register for VFE0

GCC_VFE0_CFG_RCGR

Bits	Name	Description
10:8	SRC_SEL	Root Source Select 000: cxo 001: GPLL0_MAIN 010: GPLL2_MAIN 011: gnd 100: gnd 101: gnd 110: gnd 111: Reserved
4:0	SRC_DIV	The SRC_DIV bits define the integer and half integer portion of the divider setting. 00000: Div 1 00001: Div 1 00010: Div 1.5 00011: Div 2 00100: Div 2.5 00101: Div 3 00110: Div 3.5 00111: Div 4 01000: Div 4.5 01001: Div 5 01010: Div 5.5 01011: Div 6 01100: Div 6.5 01101: Div 7 01110: Div 7.5 01111: Div 8 10000: Div 8.5 10001: Div 9 10010: Div 9.5 10011: Div 10 10100: Div 10.5 10101: Div 11 10110: Div 11.5 10111: Div 12 11000: Div 12.5 11001: Div 13 11010: Div 13.5 11011: Div 14 11100: Div 14.5 11101: Div 15 11110: Div 15.5 11111: Div 16

0x01858018 GCC_CPP_CMD_RCGR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

Root Clock Generator Command Register CPP

GCC_CPP_CMD_RCGR

Bits	Name	Type	Description
31	ROOT_OFF	R	The ROOT_OFF status allows SW to determine if the generator is actively producing a clock 0: Active 1: Not Active
4	DIRTY_CFG_RCGR	R	Dirty bit for the CFG_RCGR register. This bit is active if the CFG_RCGR register is written, and will clear after the update bit asserts. 0: Not Active 1: Active
1	ROOT_EN		The ROOT_EN bit controls the output of the clock generator. This enable may be ORed with a HW clock enable and then sent to the RCG. 0: Disabled 1: Enabled
0	UPDATE		This UPDATE bit is SW and HW modified. SW sets the bit when the configuration values are programmed, and when they have taken effect, then HW clears the bit to 0. 0: Not Active 1: Active

0x0185801C GCC_CPP_CFG_RCGR**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

Clock Generator Config Register for CPP

GCC_CPP_CFG_RCGR

Bits	Name	Description
10:8	SRC_SEL	Root Source Select 000: cxo 001: GPLL0_MAIN 010: GPLL2_MAIN 011: gnd 100: gnd 101: gnd 110: gnd 111: Reserved
4:0	SRC_DIV	The SRC_DIV bits define the integer and half integer portion of the divider setting. 00000: Div 1 00001: Div 1 00010: Div 1.5 00011: Div 2 00100: Div 2.5 00101: Div 3 00110: Div 3.5 00111: Div 4 01000: Div 4.5 01001: Div 5 01010: Div 5.5 01011: Div 6 01100: Div 6.5 01101: Div 7 01110: Div 7.5 01111: Div 8 10000: Div 8.5 10001: Div 9 10010: Div 9.5 10011: Div 10 10100: Div 10.5 10101: Div 11 10110: Div 11.5 10111: Div 12 11000: Div 12.5 11001: Div 13 11010: Div 13.5 11011: Div 14 11100: Div 14.5 11101: Div 15 11110: Div 15.5 11111: Div 16

0x01858030 GCC_CAMSS_VFE_BCR**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

BCR Registers for CAMSS_VFE

GCC_CAMSS_VFE_BCR

Bits	Name	Description
0	BLK_ARES	The block asynchronous reset bit, when set, the block clocks are disabled and the ares_out signals go active. 0: Not Active 1: Active

0x01858034 GCC_CAMSS_VFE_GDSCR**Type:** RW**Clock:** CLK**Reset State:** 0x00222001

GDS Registers for CAMSS_VFE

GCC_CAMSS_VFE_GDSCR

Bits	Name	Type	Description
31	PWR_ON	R	PWR State 0: Power collapsed 1: Power on

GCC_CAMSS_VFE_GDSCR (cont.)

Bits	Name	Type	Description
30:27	GDSC_STATE	R	GDS FSM State 0000: PowerOff 0001: AssertAresAndRampUp_en_few 0010: DeassertAres 0011: RampUp_en_rest 0100: PowerOn 0101: UnclampIO 0110: NoRetainFF 0111: RestoreFF 1000: NA 1001: NA 1010: NA 1011: NA 1100: DisableClk 1101: RetainFF 1110: NA 1111: ClampIOAndSaveFF
23:20	EN_REST_WAIT		Wait cntr for the remaining PD switches to power on 0000: cntr value is cxc x 1 cycle 0001: cntr value is cxc x 2 cycle 0010: cntr value is cxc x 4 cycle [default] 0011: cntr value is cxc x 8 cycle 0100: cntr value is cxc x 16 cycle 0101: cntr value is cxc x 32 cycle 0110: cntr value is cxc x 64 cycle 0111: cntr value is cxc x 128 cycle 1000: cntr value is cxc x 256 cycle 1001: cntr value is cxc x 512 cycle 1010: cntr value is cxc x 1024 cycle 1011: cntr value is cxc x 2048 cycle 1100: cntr value is cxc x 2048 cycle 1101: cntr value is cxc x 2048 cycle 1110: cntr value is cxc x 2048 cycle 1111: cntr value is cxc x 2048 cycle

GCC_CAMSS_VFE_GDSCR (cont.)

Bits	Name	Type	Description
19:16	EN_FEW_WAIT		Wait cntr for the first few PD switches to power on 0000: cntr value is cxc x 1 cycle 0001: cntr value is cxc x 2 cycle 0010: cntr value is cxc x 4 cycle [default] 0011: cntr value is cxc x 8 cycle 0100: cntr value is cxc x 16 cycle 0101: cntr value is cxc x 32 cycle 0110: cntr value is cxc x 64 cycle 0111: cntr value is cxc x 128 cycle 1000: cntr value is cxc x 256 cycle 1001: cntr value is cxc x 512 cycle 1010: cntr value is cxc x 1024 cycle 1011: cntr value is cxc x 2048 cycle 1100: cntr value is cxc x 2048 cycle 1101: cntr value is cxc x 2048 cycle 1110: cntr value is cxc x 2048 cycle 1111: cntr value is cxc x 2048 cycle
15:12	CLK_DIS_WAIT		Wait while all clocks in this PD are turned off 0000: cntr value is cxc x 1 cycle 0001: cntr value is cxc x 2 cycle 0010: cntr value is cxc x 4 cycle [default] 0011: cntr value is cxc x 8 cycle 0100: cntr value is cxc x 16 cycle 0101: cntr value is cxc x 32 cycle 0110: cntr value is cxc x 64 cycle 0111: cntr value is cxc x 128 cycle 1000: cntr value is cxc x 256 cycle 1001: cntr value is cxc x 512 cycle 1010: cntr value is cxc x 1024 cycle 1011: cntr value is cxc x 2048 cycle 1100: cntr value is cxc x 2048 cycle 1101: cntr value is cxc x 2048 cycle 1110: cntr value is cxc x 2048 cycle 1111: cntr value is cxc x 2048 cycle
10	RESTORE		Restore 0: Not Active 1: Active
9	SAVE		Save 0: Not Active 1: Active
8	RETAIN		Retain 0: Not Active 1: Active

GCC_CAMSS_VFE_GDSCR (cont.)

Bits	Name	Type	Description
7	EN_REST		Enable Rest 0: Not Active 1: Active
6	EN_FEW		Enable Few 0: Not Active 1: Active
5	CLAMP_IO		Clamp IO 0: Not Active 1: Active
4	CLK_DISABLE		Clk Disable 0: Not Active 1: Active
3	PD_ARES		Reset 0: Not Active 1: Active
2	SW_OVERRIDE		SW Override 0: GDSC sequence based on HW FSM 1: GDSC sequence based on SW
1	HW_CONTROL		HW Control 0: Power off by HW trigger 1: Power on by HW trigger
0	SW_COLLAPSE		SW Collapse 0: Power on through SW 1: Power off through SW, default

0x01858038 GCC_CAMSS_VFE0_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x80004FF0

PSCBC Registers for CAMSS_VFE0

GCC_CAMSS_VFE0_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	Branch Clock Off 0: Clock is ON 1: Clock is OFF

GCC_CAMSS_VFE0_CBCR (cont.)

Bits	Name	Type	Description
14	FORCE_MEM_CORE_ON		Force Mem Core bits to be on 0: Memory Core can be powered down by the clock branch slp control 1: Force Memory Core to be powered on (default)
13	FORCE_MEM_PERIPH_ON		Force Mem Periph logics to be on 0: Memory Periphery logic can be powered down by the clock branch slp control (default) 1: Force Memory Periphery logic to be powered on
12	FORCE_MEM_PERIPH_OFF		Force Mem Periph logics to be off 0: Memory Periphery logic is powered on by the clock branch slp control (default) 1: Force Memory Periphery logic to be powered off
11:8	WAKEUP		When the clock is enabled, the memory control signal will be turned on immediately, but the clock will remain gated until WAKEUP clock cycles later. 0000: Number of clock cycle 1 0001: Number of clock cycle 2 0010: Number of clock cycle 3 0011: Number of clock cycle 4 0100: Number of clock cycle 5 0101: Number of clock cycle 6 0110: Number of clock cycle 7 1000: Number of clock cycle 8 1001: Number of clock cycle 9 1010: Number of clock cycle 10 1011: Number of clock cycle 11 1100: Number of clock cycle 12 1101: Number of clock cycle 13 1110: Number of clock cycle 14 1111: Number of clock cycle 15

GCC_CAMSS_VFE0_CBCR (cont.)

Bits	Name	Type	Description
7:4	SLEEP		When the clock is disabled, the clock will be gated immediately, but the memory control signals will not be turned off until a delay of SLEEP clock cycles. 0000: Number of clock cycle 1 0001: Number of clock cycle 2 0010: Number of clock cycle 3 0011: Number of clock cycle 4 0100: Number of clock cycle 5 0101: Number of clock cycle 6 0110: Number of clock cycle 7 1000: Number of clock cycle 8 1001: Number of clock cycle 9 1010: Number of clock cycle 10 1011: Number of clock cycle 11 1100: Number of clock cycle 12 1101: Number of clock cycle 13 1110: Number of clock cycle 14 1111: Number of clock cycle 15
0	CLK_ENABLE		SW Clock Enable Bit 0: Clock Disabled 1: Clock Enabled

0x0185803C GCC_CAMSS_CPP_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x80004FF0

PSCBC Registers for CAMSS_VFE_CPP

GCC_CAMSS_CPP_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	Branch Clock Off 0: Clock is ON 1: Clock is OFF
14	FORCE_MEM_CORE_ON		Force Mem Core bits to be on 0: Memory Core can be powered down by the clock branch slp control 1: Force Memory Core to be powered on (default)
13	FORCE_MEM_PERIPH_ON		Force Mem Periph logics to be on 0: Memory Periphery logic can be powered down by the clock branch slp control (default) 1: Force Memory Periphery logic to be powerd on

GCC_CAMSS_CPP_CBCR (cont.)

Bits	Name	Type	Description
12	FORCE_MEM_PERIPH_OF F		Force Mem Periph logics to be off 0: Memory Periphery logic is powered on by the clock branch slp control (default) 1: Force Memory Periphery logic to be powerd off
11:8	WAKEUP		When the clock is enabled, the memory control signal will be turned on immediately, but the clock will remain gated until WAKEUP clock cycles later. 0000: Number of clock cycle 1 0001: Number of clock cycle 2 0010: Number of clock cycle 3 0011: Number of clock cycle 4 0100: Number of clock cycle 5 0101: Number of clock cycle 6 0110: Number of clock cycle 7 1000: Number of clock cycle 8 1001: Number of clock cycle 9 1010: Number of clock cycle 10 1011: Number of clock cycle 11 1100: Number of clock cycle 12 1101: Number of clock cycle 13 1110: Number of clock cycle 14 1111: Number of clock cycle 15
7:4	SLEEP		When the clock is disabled, the clock will be gated immediately, but the memory control signals will not be turned off until a delay of SLEEP clock cycles. 0000: Number of clock cycle 1 0001: Number of clock cycle 2 0010: Number of clock cycle 3 0011: Number of clock cycle 4 0100: Number of clock cycle 5 0101: Number of clock cycle 6 0110: Number of clock cycle 7 1000: Number of clock cycle 8 1001: Number of clock cycle 9 1010: Number of clock cycle 10 1011: Number of clock cycle 11 1100: Number of clock cycle 12 1101: Number of clock cycle 13 1110: Number of clock cycle 14 1111: Number of clock cycle 15
0	CLK_ENABLE		SW Clock Enable Bit 0: Clock Disabled 1: Clock Enabled

0x01858040 GCC_CAMSS_CPP_AHB_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

PSCBC Registers for CAMSS_VFE_CPP_AHB

GCC_CAMSS_CPP_AHB_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	Branch Clock Off 0: Clock is ON 1: Clock is OFF
0	CLK_ENABLE		SW Clock Enable Bit 0: Clock Disabled 1: Clock Enabled

0x01858044 GCC_CAMSS_VFE_AHB_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

PSCBC Registers for CAMSS_VFE_AHB

GCC_CAMSS_VFE_AHB_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	Branch Clock Off 0: Clock is ON 1: Clock is OFF
0	CLK_ENABLE		SW Clock Enable Bit 0: Clock Disabled 1: Clock Enabled

0x01858048 GCC_CAMSS_VFE_AXI_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x80004FF0

PSCBC Registers for CAMSS_VFE_AXI

GCC_CAMSS_VFE_AXI_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	Branch Clock Off 0: Clock is ON 1: Clock is OFF
14	FORCE_MEM_CORE_ON		Force Mem Core bits to be on 0: Memory Core can be powered down by the clock branch slp control 1: Force Memory Core to be powered on (default)
13	FORCE_MEM_PERIPH_ON		Force Mem Periph logics to be on 0: Memory Periphery logic can be powered down by the clock branch slp control (default) 1: Force Memory Periphery logic to be powerd on
12	FORCE_MEM_PERIPH_OF F		Force Mem Periph logics to be off 0: Memory Periphery logic is powered on by the clock branch slp control (default) 1: Force Memory Periphery logic to be powerd off
11:8	WAKEUP		When the clock is enabled, the memory control signal will be turned on immediately, but the clock will remain gated until WAKEUP clock cycles later. 0000: Number of clock cycle 1 0001: Number of clock cycle 2 0010: Number of clock cycle 3 0011: Number of clock cycle 4 0100: Number of clock cycle 5 0101: Number of clock cycle 6 0110: Number of clock cycle 7 1000: Number of clock cycle 8 1001: Number of clock cycle 9 1010: Number of clock cycle 10 1011: Number of clock cycle 11 1100: Number of clock cycle 12 1101: Number of clock cycle 13 1110: Number of clock cycle 14 1111: Number of clock cycle 15

GCC_CAMSS_VFE_AXI_CBCR (cont.)

Bits	Name	Type	Description
7:4	SLEEP		When the clock is disabled, the clock will be gated immediately, but the memory control signals will not be turned off until a delay of SLEEP clock cycles. 0000: Number of clock cycle 1 0001: Number of clock cycle 2 0010: Number of clock cycle 3 0011: Number of clock cycle 4 0100: Number of clock cycle 5 0101: Number of clock cycle 6 0110: Number of clock cycle 7 1000: Number of clock cycle 8 1001: Number of clock cycle 9 1010: Number of clock cycle 10 1011: Number of clock cycle 11 1100: Number of clock cycle 12 1101: Number of clock cycle 13 1110: Number of clock cycle 14 1111: Number of clock cycle 15
0	CLK_ENABLE		SW Clock Enable Bit 0: Clock Disabled 1: Clock Enabled

0x0185804C GCC_CAMSS_CSI_VFE0_BCR**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

BCR Registers for CAMSS_CSI_VFE0

GCC_CAMSS_CSI_VFE0_BCR

Bits	Name	Description
0	BLK_ARES	The block asynchronous reset bit, when set, the block clocks are disabled and the ares_out signals go active. 0: Not Active 1: Active

0x01858050 GCC_CAMSS_CSI_VFE0_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x80004FF0

PSCBC Registers for CAMSS_CSI_VFE0

GCC_CAMSS_CSI_VFE0_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	Branch Clock Off 0: Clock is ON 1: Clock is OFF
14	FORCE_MEM_CORE_ON		Force Mem Core bits to be on 0: Memory Core can be powered down by the clock branch slp control 1: Force Memory Core to be powered on (default)
13	FORCE_MEM_PERIPH_ON		Force Mem Periph logics to be on 0: Memory Periphery logic can be powered down by the clock branch slp control (default) 1: Force Memory Periphery logic to be powerd on
12	FORCE_MEM_PERIPH_OF F		Force Mem Periph logics to be off 0: Memory Periphery logic is powered on by the clock branch slp control (default) 1: Force Memory Periphery logic to be powerd off
11:8	WAKEUP		When the clock is enabled, the memory control signal will be turned on immediately, but the clock will remain gated until WAKEUP clock cycles later. 0000: Number of clock cycle 1 0001: Number of clock cycle 2 0010: Number of clock cycle 3 0011: Number of clock cycle 4 0100: Number of clock cycle 5 0101: Number of clock cycle 6 0110: Number of clock cycle 7 1000: Number of clock cycle 8 1001: Number of clock cycle 9 1010: Number of clock cycle 10 1011: Number of clock cycle 11 1100: Number of clock cycle 12 1101: Number of clock cycle 13 1110: Number of clock cycle 14 1111: Number of clock cycle 15

GCC_CAMSS_CSI_VFE0_CBCR (cont.)

Bits	Name	Type	Description
7:4	SLEEP		When the clock is disabled, the clock will be gated immediately, but the memory control signals will not be turned off until a delay of SLEEP clock cycles. 0000: Number of clock cycle 1 0001: Number of clock cycle 2 0010: Number of clock cycle 3 0011: Number of clock cycle 4 0100: Number of clock cycle 5 0101: Number of clock cycle 6 0110: Number of clock cycle 7 1000: Number of clock cycle 8 1001: Number of clock cycle 9 1010: Number of clock cycle 10 1011: Number of clock cycle 11 1100: Number of clock cycle 12 1101: Number of clock cycle 13 1110: Number of clock cycle 14 1111: Number of clock cycle 15
0	CLK_ENABLE		SW Clock Enable Bit 0: Clock Disabled 1: Clock Enabled

0x01859000 GCC_GFX3D_CMD_RCGR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

Root Clock Generator Command Register GFX3D

GCC_GFX3D_CMD_RCGR

Bits	Name	Type	Description
31	ROOT_OFF	R	The ROOT_OFF status allows SW to determine if the generator is actively producing a clock 0: Active 1: Not Active
4	DIRTY_CFG_RCGR	R	Dirty bit for the CFG_RCGR register. This bit is active if the CFG_RCGR register is written, and will clear after the update bit asserts. 0: Not Active 1: Active

GCC_GFX3D_CMD_RCGR (cont.)

Bits	Name	Type	Description
1	ROOT_EN		The ROOT_EN bit controls the output of the clock generator. This enable may be ORed with a HW clock enable and then sent to the RCG. 0: Disabled 1: Enabled
0	UPDATE		This UPDATE bit is SW and HW modified. SW sets the bit when the configuration values are programmed, and when they have taken effect, then HW clears the bit to 0. 0: Not Active 1: Active

0x01859004 GCC_GFX3D_CFG_RCGR**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

Clock Generator Config Register for GFX3D

GCC_GFX3D_CFG_RCGR

Bits	Name	Description
10:8	SRC_SEL	Root Source Select 000: cxo 001: GPLL1_MAIN 010: GPLL2_AUX 011: GPLL0_AUX 100: gnd 101: gnd 110: gnd 111: Reserved

GCC_GFX3D_CFG_RCGR (cont.)

Bits	Name	Description
4:0	SRC_DIV	The SRC_DIV bits define the integer and half integer portion of the divider setting. 00000: Div 1 00001: Div 1 00010: Div 1.5 00011: Div 2 00100: Div 2.5 00101: Div 3 00110: Div 3.5 00111: Div 4 01000: Div 4.5 01001: Div 5 01010: Div 5.5 01011: Div 6 01100: Div 6.5 01101: Div 7 01110: Div 7.5 01111: Div 8 10000: Div 8.5 10001: Div 9 10010: Div 9.5 10011: Div 10 10100: Div 10.5 10101: Div 11 10110: Div 11.5 10111: Div 12 11000: Div 12.5 11001: Div 13 11010: Div 13.5 11011: Div 14 11100: Div 14.5 11101: Div 15 11110: Div 15.5 11111: Div 16

0x01859018 GCC_OXILI_BCR**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

BCR Registers for OXILI

GCC_OXILI_BCR

Bits	Name	Description
0	BLK_ARES	The block asynchronous reset bit, when set, the block clocks are disabled and the ares_out signals go active. 0: Not Active 1: Active

0x0185901C GCC_OXILI_GDSCR**Type:** RW**Clock:** CLK**Reset State:** 0x00222001

GDS Registers for OXILI

GCC_OXILI_GDSCR

Bits	Name	Type	Description
31	PWR_ON	R	PWR State 0: Power collapsed 1: Power on
30:27	GDSC_STATE	R	GDS FSM State 0000: PowerOff 0001: AssertAresAndRampUp_en_few 0010: DeassertAres 0011: RampUp_en_rest 0100: PowerOn 0101: UnclampIO 0110: NoRetainFF 0111: RestoreFF 1000: NA 1001: NA 1010: NA 1011: NA 1100: DisableClk 1101: RetainFF 1110: NA 1111: ClampIOAndSaveFF

GCC_OXILI_GDSCR (cont.)

Bits	Name	Type	Description
23:20	EN_REST_WAIT		Wait cntr for the remaining PD switches to power on 0000: cntr value is cxc x 1 cycle 0001: cntr value is cxc x 2 cycle 0010: cntr value is cxc x 4 cycle [default] 0011: cntr value is cxc x 8 cycle 0100: cntr value is cxc x 16 cycle 0101: cntr value is cxc x 32 cycle 0110: cntr value is cxc x 64 cycle 0111: cntr value is cxc x 128 cycle 1000: cntr value is cxc x 256 cycle 1001: cntr value is cxc x 512 cycle 1010: cntr value is cxc x 1024 cycle 1011: cntr value is cxc x 2048 cycle 1100: cntr value is cxc x 2048 cycle 1101: cntr value is cxc x 2048 cycle 1110: cntr value is cxc x 2048 cycle 1111: cntr value is cxc x 2048 cycle
19:16	EN_FEW_WAIT		Wait cntr for the first few PD switches to power on 0000: cntr value is cxc x 1 cycle 0001: cntr value is cxc x 2 cycle 0010: cntr value is cxc x 4 cycle [default] 0011: cntr value is cxc x 8 cycle 0100: cntr value is cxc x 16 cycle 0101: cntr value is cxc x 32 cycle 0110: cntr value is cxc x 64 cycle 0111: cntr value is cxc x 128 cycle 1000: cntr value is cxc x 256 cycle 1001: cntr value is cxc x 512 cycle 1010: cntr value is cxc x 1024 cycle 1011: cntr value is cxc x 2048 cycle 1100: cntr value is cxc x 2048 cycle 1101: cntr value is cxc x 2048 cycle 1110: cntr value is cxc x 2048 cycle 1111: cntr value is cxc x 2048 cycle

GCC_OXILI_GDSCR (cont.)

Bits	Name	Type	Description
15:12	CLK_DIS_WAIT		Wait while all clocks in this PD are turned off 0000: cntr value is cxo x 1 cycle 0001: cntr value is cxo x 2 cycle 0010: cntr value is cxo x 4 cycle [default] 0011: cntr value is cxo x 8 cycle 0100: cntr value is cxo x 16 cycle 0101: cntr value is cxo x 32 cycle 0110: cntr value is cxo x 64 cycle 0111: cntr value is cxo x 128 cycle 1000: cntr value is cxo x 256 cycle 1001: cntr value is cxo x 512 cycle 1010: cntr value is cxo x 1024 cycle 1011: cntr value is cxo x 2048 cycle 1100: cntr value is cxo x 2048 cycle 1101: cntr value is cxo x 2048 cycle 1110: cntr value is cxo x 2048 cycle 1111: cntr value is cxo x 2048 cycle
10	RESTORE		Restore 0: Not Active 1: Active
9	SAVE		Save 0: Not Active 1: Active
8	RETAIN		Retain 0: Not Active 1: Active
7	EN_REST		Enable Rest 0: Not Active 1: Active
6	EN_FEW		Enable Few 0: Not Active 1: Active
5	CLAMP_IO		Clamp IO 0: Not Active 1: Active
4	CLK_DISABLE		Clk Disable 0: Not Active 1: Active
3	PD_ARES		Reset 0: Not Active 1: Active

GCC_OXILI_GDSCR (cont.)

Bits	Name	Type	Description
2	SW_OVERRIDE		SW Override 0: GDSC sequence based on HW FSM 1: GDSC sequence based on SW
1	HW_CONTROL		HW Control 0: Power off by HW trigger 1: Power on by HW trigger
0	SW_COLLAPSE		SW Collapse 0: Power on through SW 1: Power off through SW, default

0x01859020 GCC_OXILI_GFX3D_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x80004FF0

PSCBC Registers for OXILI_GFX3D

GCC_OXILI_GFX3D_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	Branch Clock Off 0: Clock is ON 1: Clock is OFF
14	FORCE_MEM_CORE_ON		Force Mem Core bits to be on 0: Memory Core can be powered down by the clock branch slp control 1: Force Memory Core to be powered on (default)
13	FORCE_MEM_PERIPH_ON		Force Mem Periph logics to be on 0: Memory Periphery logic can be powered down by the clock branch slp control (default) 1: Force Memory Periphery logic to be powerd on
12	FORCE_MEM_PERIPH_OFF		Force Mem Periph logics to be off 0: Memory Periphery logic is powered on by the clock branch slp control (default) 1: Force Memory Periphery logic to be powerd off

GCC_OXILI_GFX3D_CBCR (cont.)

Bits	Name	Type	Description
11:8	WAKEUP		When the clock is enabled, the memory control signal will be turned on immediately, but the clock will remain gated until WAKEUP clock cycles later. 0000: Number of clock cycle 1 0001: Number of clock cycle 2 0010: Number of clock cycle 3 0011: Number of clock cycle 4 0100: Number of clock cycle 5 0101: Number of clock cycle 6 0110: Number of clock cycle 7 1000: Number of clock cycle 8 1001: Number of clock cycle 9 1010: Number of clock cycle 10 1011: Number of clock cycle 11 1100: Number of clock cycle 12 1101: Number of clock cycle 13 1110: Number of clock cycle 14 1111: Number of clock cycle 15
7:4	SLEEP		When the clock is disabled, the clock will be gated immediately, but the memory control signals will not be turned off until a delay of SLEEP clock cycles. 0000: Number of clock cycle 1 0001: Number of clock cycle 2 0010: Number of clock cycle 3 0011: Number of clock cycle 4 0100: Number of clock cycle 5 0101: Number of clock cycle 6 0110: Number of clock cycle 7 1000: Number of clock cycle 8 1001: Number of clock cycle 9 1010: Number of clock cycle 10 1011: Number of clock cycle 11 1100: Number of clock cycle 12 1101: Number of clock cycle 13 1110: Number of clock cycle 14 1111: Number of clock cycle 15
0	CLK_ENABLE		SW Clock Enable Bit 0: Clock Disabled 1: Clock Enabled

0x01859024 GCC_OXILI_GMEM_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x80004FF0

PSCBC Registers for OXILI_GMEM

GCC_OXILI_GMEM_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	Branch Clock Off 0: Clock is ON 1: Clock is OFF
14	FORCE_MEM_CORE_ON		Force Mem Core bits to be on 0: Memory Core can be powered down by the clock branch slp control 1: Force Memory Core to be powered on (default)
13	FORCE_MEM_PERIPH_ON		Force Mem Periph logics to be on 0: Memory Periphery logic can be powered down by the clock branch slp control (default) 1: Force Memory Periphery logic to be powerd on
12	FORCE_MEM_PERIPH_OF F		Force Mem Periph logics to be off 0: Memory Periphery logic is powered on by the clock branch slp control (default) 1: Force Memory Periphery logic to be powerd off
11:8	WAKEUP		When the clock is enabled, the memory control signal will be turned on immediately, but the clock will remain gated until WAKEUP clock cycles later. 0000: Number of clock cycle 1 0001: Number of clock cycle 2 0010: Number of clock cycle 3 0011: Number of clock cycle 4 0100: Number of clock cycle 5 0101: Number of clock cycle 6 0110: Number of clock cycle 7 1000: Number of clock cycle 8 1001: Number of clock cycle 9 1010: Number of clock cycle 10 1011: Number of clock cycle 11 1100: Number of clock cycle 12 1101: Number of clock cycle 13 1110: Number of clock cycle 14 1111: Number of clock cycle 15

GCC_OXILI_GMEM_CBCR (cont.)

Bits	Name	Type	Description
7:4	SLEEP		When the clock is disabled, the clock will be gated immediately, but the memory control signals will not be turned off until a delay of SLEEP clock cycles. 0000: Number of clock cycle 1 0001: Number of clock cycle 2 0010: Number of clock cycle 3 0011: Number of clock cycle 4 0100: Number of clock cycle 5 0101: Number of clock cycle 6 0110: Number of clock cycle 7 1000: Number of clock cycle 8 1001: Number of clock cycle 9 1010: Number of clock cycle 10 1011: Number of clock cycle 11 1100: Number of clock cycle 12 1101: Number of clock cycle 13 1110: Number of clock cycle 14 1111: Number of clock cycle 15
0	CLK_ENABLE		SW Clock Enable Bit 0: Clock Disabled 1: Clock Enabled

0x01859028 GCC_OXILI_AHB_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x80008000

CBC Registers for OXILI_AHB

GCC_OXILI_AHB_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	Branch Clock Off 0: Clock is ON 1: Clock is OFF

GCC_OXILI_AHB_CBCR (cont.)

Bits	Name	Type	Description
30:28	NOC_HANDSHAKE_FSM_STATE	R	FSM State 000: OFFS = Off 001: WFON = Wait For Clk On 010: SRIU = Slave Ready is Up 011: WFSL = Wait for Socketcon Low 100: WOFF = Wait for Clk Off 101: Not Used 110: Not Used 111: Not Used
15	NOC_HANDSHAKE_FSM_EN		SlaveReady and Socket Connect FSM Enable; Default is 1 0: SW Clock Enabled 1: FSM Clock Enabled
0	CLK_ENABLE		SW Clock Enable Bit 0: Clock Disabled. 1: Clock Enabled.

0x0185902C GCC_GMEM_BCR**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

BCR Registers for GMEM

GCC_GMEM_BCR

Bits	Name	Description
0	BLK_ARES	The block asynchronous reset bit, when set, the block clocks are disabled and the ares_out signals go active. 0: Not Active 1: Active

0x0185A000 GCC_CAMSS_AHB_CMD_RCGR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

Root Clock Generator Command Register CAMSS_AHB

GCC_CAMSS_AHB_CMD_RCGR

Bits	Name	Type	Description
31	ROOT_OFF	R	The ROOT_OFF status allows SW to determine if the generator is actively producing a clock 0: Active 1: Not Active
7	DIRTY_D	R	Dirty bit for the D register. This bit is active if the D register is written, and will clear after the update bit asserts. 0: Not Active 1: Active
6	DIRTY_N	R	Dirty bit for the N register. This bit is active if the N register is written, and will clear after the update bit asserts. 0: Not Active 1: Active
5	DIRTY_M	R	Dirty bit for the M register. This bit is active if the M register is written, and will clear after the update bit asserts. 0: Not Active 1: Active
4	DIRTY_CFG_RCGR	R	Dirty bit for the CFG_RCGR register. This bit is active if the CFG_RCGR register is written, and will clear after the update bit asserts. 0: Not Active 1: Active
1	ROOT_EN		The ROOT_EN bit controls the output of the clock generator. This enable may be ORed with a HW clock enable and then sent to the RCG. 0: Disabled 1: Enabled
0	UPDATE		This UPDATE bit is SW and HW modified. SW sets the bit when the configuration values are programmed, and when they have taken effect, then HW clears the bit to 0. 0: Not Active 1: Active

0x0185A004 GCC_CAMSS_AHB_CFG_RCGR**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

Clock Generator Config Register for CAMSS_AHB

GCC_CAMSS_AHB_CFG_RCGR

Bits	Name	Description
13:12	MODE	mode 00: bypass mode 01: swallow mode 10: dual edge mode 11: single edge mode
10:8	SRC_SEL	Root Source Select 000: cxo 001: GPLLO_MAIN 010: gnd 011: gnd 100: gnd 101: gnd 110: gnd 111: Reserved

GCC_CAMSS_AHB_CFG_RCGR (cont.)

Bits	Name	Description
4:0	SRC_DIV	The SRC_DIV bits define the integer and half integer portion of the divider setting. 00000: Div 1 00001: Div 1 00010: Div 1.5 00011: Div 2 00100: Div 2.5 00101: Div 3 00110: Div 3.5 00111: Div 4 01000: Div 4.5 01001: Div 5 01010: Div 5.5 01011: Div 6 01100: Div 6.5 01101: Div 7 01110: Div 7.5 01111: Div 8 10000: Div 8.5 10001: Div 9 10010: Div 9.5 10011: Div 10 10100: Div 10.5 10101: Div 11 10110: Div 11.5 10111: Div 12 11000: Div 12.5 11001: Div 13 11010: Div 13.5 11011: Div 14 11100: Div 14.5 11101: Div 15 11110: Div 15.5 11111: Div 16

0x0185A008 GCC_CAMSS_AHB_M**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

Program M for the MND divider, where division ration is $F_{out} = F_{in} \cdot (M / N)$

GCC_CAMSS_AHB_M

Bits	Name	Description
7:0	M	Program M

0x0185A00C GCC_CAMSS_AHB_N**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

Program NOT_N_MINUS_M (1's complement of N-M); The N value derived is used for the MND divider, where division ration is $F_{out} = F_{in} \cdot (M / N)$

GCC_CAMSS_AHB_N

Bits	Name	Description
7:0	NOT_N_MINUS_M	Program NOT_N_MINUS_M

0x0185A010 GCC_CAMSS_AHB_D**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

Program NOT_2D (1's complement of 2D); The D value derived is used for the MND divider, where the duty cycle is D/N

GCC_CAMSS_AHB_D

Bits	Name	Description
7:0	NOT_2D	Program NOT_2D

0x0185B00C GCC_GX_DOMAIN_MISC**Type:** RW**Clock:** CLK**Reset State:** 0x00001101

Misc registers for Oxili Ocmem in GX rail

GCC_GX_DOMAIN_MISC

Bits	Name	Description
12	GMEM_FIFO_HALT	
8	OXILI_FIFO_HALT	Refer to ECO QCTDD00250650. This bit is not used for fifo halt; it is used as clamp mem in gfx domain. 0: Not Active 1: Active
4	OXILI_GMEM_RESET	
0	OXILI_GMEM_CLAMP_IO	

0x0181C008 GCC_AUDIO_CORE_BCR**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

BCR Registers for AUDIO_CORE

GCC_AUDIO_CORE_BCR

Bits	Name	Type	Description
31	DFD_STATUS	R	DFD Status 0: Not Active 1: Active
2	FORCE_RESET		Force Reset 0: Not Active 1: Active
1	DFD_EN		DFD Enable 0: Not Active 1: Active
0	BLK_ARES		The block asynchronous reset bit, when set, the clocks are disabled and the ares_out signals go active. 0: Not Active 1: Active

0x0181C054 GCC_ULTAUDIO_LPAIF_PRI_I2S_CMD_RCGR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

RCG Command Register

GCC_ULTAUDIO_LPAIF_PRI_I2S_CMD_RCGR

Bits	Name	Type	Description
31	ROOT_OFF	R	READ ONLY. Indicates if the RCG is actively producing a clock. 0: clock off 1: clock on
7	DIRTY_D	R	READ ONLY. Indicates change in the D register configuration.
6	DIRTY_N	R	READ ONLY. Indicates change in the N register configuration.
5	DIRTY_M	R	READ ONLY. Indicates change in the M register configuration.
4	DIRTY_CFG_RCGR	R	READ ONLY. Indicates change in the CFG_RCGR register configuration.
1	ROOT_EN		RCG enable: 0: disabled 1: enabled
0	UPDATE		Write 1 to this bit to update the RCG with a new programmed configuration. This bit is cleared by hardware.

0x0181C058 GCC_ULTAUDIO_LPAIF_PRI_I2S_CFG_RCGR**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

RCG Configuration Register

GCC_ULTAUDIO_LPAIF_PRI_I2S_CFG_RCGR

Bits	Name	Description
13:12	MODE	Specifies the MN counter mode: 0: bypass mode 1: swallow mode 2: dual edge mode 3: single edge mode

GCC_ULTAUDIO_LPAIF_PRI_I2S_CFG_RCGR (cont.)

Bits	Name	Description
10:8	SRC_SEL	Specifies which clock source is output from the RCG. 0: CXO 1: GPLL1_MAIN 2: PRI_I2S_EXT_CLK_IN 3: no clock (ground) 4: no clock (ground) 5: no clock (ground) 6: sleep_clk 7: Reserved
4:0	SRC_DIV	Specifies the integer and half integer portion of the divider setting of the RCG. The value specified is $2N-1$ where N is the divide ratio. For example, if the half integer divide ratio is 2.5, the SRC_DIV value should be set at 0x04. 0: Bypass 1: Div-1 2: Div-1.5 3: Div-2 4: Div-2.5 5: Div-3 6: Div-3.5 7: Div-4 8: Div-4.5 9: Div-5 10: Div-5.5 11: Div-6 12: Div-6.5 13: Div-7 14: Div-7.5 15: Div-8 16: Div-8.5 17: Div-9 18: Div-9.5 19: Div-10 20: Div-10.5 21: Div-11 22: Div-11.5 23: Div-12 24: Div-12.5 25: Div-13 26: Div-13.5 27: Div-14 28: Div-14.5 29: Div-15 30: Div-15.5 31: Div-16

0x0181C05C GCC_ULTAUDIO_LPAIF_PRI_I2S_M**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

RCG M Register

GCC_ULTAUDIO_LPAIF_PRI_I2S_M

Bits	Name	Description
7:0	M	Specifies the M value of the M/N:D counter.

0x0181C060 GCC_ULTAUDIO_LPAIF_PRI_I2S_N**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

RCG N Register

GCC_ULTAUDIO_LPAIF_PRI_I2S_N

Bits	Name	Description
7:0	NOT_N_MINUS_M	Specifies the NOT(N-M) value of the M/N:D counter.

0x0181C064 GCC_ULTAUDIO_LPAIF_PRI_I2S_D**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

RCG D Register

GCC_ULTAUDIO_LPAIF_PRI_I2S_D

Bits	Name	Description
7:0	NOT_2D	Specifies the NOT(2*D) value of the M/N:D counter.

0x0181C068 GCC_ULTAUDIO_LPAIF_PRI_I2S_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

CBC Registers

GCC_ULTAUDIO_LPAIF_PRI_I2S_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	READ ONLY This bit indicates if the clock branch is halted: - 0:Clock On - 1:Clock Off
1	HW_CTL		Clock gating control: 0 = software controlled 1 = hardware controlled
0	CLK_ENABLE		Asynchronous clock enable.

0x0181C06C GCC_ULTAUDIO_LPAIF_SEC_I2S_CMD_RCGR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

RCG Command Register

GCC_ULTAUDIO_LPAIF_SEC_I2S_CMD_RCGR

Bits	Name	Type	Description
31	ROOT_OFF	R	READ ONLY. Indicates if the RCG is actively producing a clock. 0: clock off 1: clock on
7	DIRTY_D	R	READ ONLY. Indicates change in the D register configuration.
6	DIRTY_N	R	READ ONLY. Indicates change in the N register configuration.
5	DIRTY_M	R	READ ONLY. Indicates change in the M register configuration.
4	DIRTY_CFG_RCGR	R	READ ONLY. Indicates change in the CFG_RCGR register configuration.

GCC_ULTAUDIO_LPAIF_SEC_I2S_CMD_RCGR (cont.)

Bits	Name	Type	Description
1	ROOT_EN		RCG enable: 0: disabled 1: enabled
0	UPDATE		Write 1 to this bit to update the RCG with a new programmed configuration. This bit is cleared by hardware.

0x0181C070 GCC_ULTAUDIO_LPAIF_SEC_I2S_CFG_RCGR**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

RCG Configuration Register

GCC_ULTAUDIO_LPAIF_SEC_I2S_CFG_RCGR

Bits	Name	Description
13:12	MODE	Specifies the MN counter mode: 0: bypass mode 1: swallow mode 2: dual edge mode 3: single edge mode
10:8	SRC_SEL	Specifies which clock source is output from the RCG. 0: CXO 1: GPLL1_MAIN 2: SEC_I2S_EXT_CLK_IN 3: no clock (ground) 4: no clock (ground) 5: no clock (ground) 6: sleep_clk 7: Reserved

GCC_ULTAUDIO_LPAIF_SEC_I2S_CFG_RCGR (cont.)

Bits	Name	Description
4:0	SRC_DIV	Specifies the integer and half integer portion of the divider setting of the RCG. The value specified is 2N-1 where N is the divide ratio. For example, if the half integer divide ratio is 2.5, the SRC_DIV value should be set at 0x04. 0: Bypass 1: Div-1 2: Div-1.5 3: Div-2 4: Div-2.5 5: Div-3 6: Div-3.5 7: Div-4 8: Div-4.5 9: Div-5 10: Div-5.5 11: Div-6 12: Div-6.5 13: Div-7 14: Div-7.5 15: Div-8 16: Div-8.5 17: Div-9 18: Div-9.5 19: Div-10 20: Div-10.5 21: Div-11 22: Div-11.5 23: Div-12 24: Div-12.5 25: Div-13 26: Div-13.5 27: Div-14 28: Div-14.5 29: Div-15 30: Div-15.5 31: Div-16

0x0181C074 GCC_ULTAUDIO_LPAIF_SEC_I2S_M**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

RCG M Register

GCC_ULTAUDIO_LPAIF_SEC_I2S_M

Bits	Name	Description
7:0	M	Specifies the M value of the M/N:D counter.

0x0181C078 GCC_ULTAUDIO_LPAIF_SEC_I2S_N**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

RCG N Register

GCC_ULTAUDIO_LPAIF_SEC_I2S_N

Bits	Name	Description
7:0	NOT_N_MINUS_M	Specifies the NOT(N-M) value of the M/N:D counter.

0x0181C07C GCC_ULTAUDIO_LPAIF_SEC_I2S_D**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

RCG D Register

GCC_ULTAUDIO_LPAIF_SEC_I2S_D

Bits	Name	Description
7:0	NOT_2D	Specifies the NOT(2*D) value of the M/N:D counter.

0x0181C080 GCC_ULTAUDIO_LPAIF_SEC_I2S_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

CBC Registers

GCC_ULTAUDIO_LPAIF_SEC_I2S_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	READ ONLY This bit indicates if the clock branch is halted: - 0:Clock On - 1:Clock Off
1	HW_CTL		Clock gating control: 0 = software controlled 1 = hardware controlled
0	CLK_ENABLE		Asynchronous clock enable.

0x0181C034 GCC_ULTAUDIO_XO_CMD_RCGR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

RCG Command Register

GCC_ULTAUDIO_XO_CMD_RCGR

Bits	Name	Type	Description
31	ROOT_OFF	R	READ ONLY. Indicates if the RCG is actively producing a clock. 0: clock off 1: clock on
7	DIRTY_D	R	READ ONLY. Indicates change in the D register configuration.
6	DIRTY_N	R	READ ONLY. Indicates change in the N register configuration.
5	DIRTY_M	R	READ ONLY. Indicates change in the M register configuration.
4	DIRTY_CFG_RCGR	R	READ ONLY. Indicates change in the CFG_RCGR register configuration.
1	ROOT_EN		RCG enable: 0: disabled 1: enabled
0	UPDATE		Write 1 to this bit to update the RCG with a new programmed configuration. This bit is cleared by hardware.

0x0181C038 GCC_ULTAUDIO_XO_CFG_RCGR**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

RCG Configuration Register

GCC_ULTAUDIO_XO_CFG_RCGR

Bits	Name	Description
13:12	MODE	Specifies the MN counter mode: 0: bypass mode 1: swallow mode 2: dual edge mode 3: single edge mode
10:8	SRC_SEL	Specifies which clock source is output from the RCG. 0: CXO 1: no clock (ground) 2: no clock (ground) 3: no clock (ground) 4: no clock (ground) 5: no clock (ground) 6: sleep_clk 7: Reserved

GCC_ULTAUDIO_XO_CFG_RCGR (cont.)

Bits	Name	Description
4:0	SRC_DIV	Specifies the integer and half integer portion of the divider setting of the RCG. The value specified is $2N-1$ where N is the divide ratio. For example, if the half integer divide ratio is 2.5, the SRC_DIV value should be set at 0x04. 0: Bypass 1: Div-1 2: Div-1.5 3: Div-2 4: Div-2.5 5: Div-3 6: Div-3.5 7: Div-4 8: Div-4.5 9: Div-5 10: Div-5.5 11: Div-6 12: Div-6.5 13: Div-7 14: Div-7.5 15: Div-8 16: Div-8.5 17: Div-9 18: Div-9.5 19: Div-10 20: Div-10.5 21: Div-11 22: Div-11.5 23: Div-12 24: Div-12.5 25: Div-13 26: Div-13.5 27: Div-14 28: Div-14.5 29: Div-15 30: Div-15.5 31: Div-16

0x0181C010 GCC_ULTAUDIO_AHBFABRIC_CMD_RCGR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

RCG Command Register

GCC_ULTAUDIO_AHBFABRIC_CMD_RCGR

Bits	Name	Type	Description
31	ROOT_OFF	R	READ ONLY. Indicates if the RCG is actively producing a clock. 0: clock off 1: clock on
7	DIRTY_D	R	READ ONLY. Indicates change in the D register configuration.
6	DIRTY_N	R	READ ONLY. Indicates change in the N register configuration.
5	DIRTY_M	R	READ ONLY. Indicates change in the M register configuration.
4	DIRTY_CFG_RCGR	R	READ ONLY. Indicates change in the CFG_RCGR register configuration.
1	ROOT_EN		RCG enable: 0: disabled 1: enabled
0	UPDATE		Write 1 to this bit to update the RCG with a new programmed configuration. This bit is cleared by hardware.

0x0181C014 GCC_ULTAUDIO_AHBFABRIC_CFG_RCGR**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

RCG Configuration Register

GCC_ULTAUDIO_AHBFABRIC_CFG_RCGR

Bits	Name	Description
13:12	MODE	Specifies the MN counter mode: 0: bypass mode 1: swallow mode 2: dual edge mode 3: single edge mode

GCC_ULTAUDIO_AHBFABRIC_CFG_RCGR (cont.)

Bits	Name	Description
10:8	SRC_SEL	Specifies which clock source is output from the RCG. 0: CXO 1: GPLL0_MAIN 2: GPLL1_AUX 3: no clock (ground) 4: no clock (ground) 5: no clock (ground) 6: sleep_clk 7: Reserved
4:0	SRC_DIV	Specifies the integer and half integer portion of the divider setting of the RCG. The value specified is 2N-1 where N is the divide ratio. For example, if the half integer divide ratio is 2.5, the SRC_DIV value should be set at 0x04. 0: Bypass 1: Div-1 2: Div-1.5 3: Div-2 4: Div-2.5 5: Div-3 6: Div-3.5 7: Div-4 8: Div-4.5 9: Div-5 10: Div-5.5 11: Div-6 12: Div-6.5 13: Div-7 14: Div-7.5 15: Div-8 16: Div-8.5 17: Div-9 18: Div-9.5 19: Div-10 20: Div-10.5 21: Div-11 22: Div-11.5 23: Div-12 24: Div-12.5 25: Div-13 26: Div-13.5 27: Div-14 28: Div-14.5 29: Div-15 30: Div-15.5 31: Div-16

0x0181C018 GCC_ULTAUDIO_AHBFABRIC_M**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

RCG M Register

GCC_ULTAUDIO_AHBFABRIC_M

Bits	Name	Description
7:0	M	Specifies the M value of the M/N:D counter.

0x0181C01C GCC_ULTAUDIO_AHBFABRIC_N**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

RCG N Register

GCC_ULTAUDIO_AHBFABRIC_N

Bits	Name	Description
7:0	NOT_N_MINUS_M	Specifies the NOT(N-M) value of the M/N:D counter.

0x0181C020 GCC_ULTAUDIO_AHBFABRIC_D**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

RCG D Register

GCC_ULTAUDIO_AHBFABRIC_D

Bits	Name	Description
7:0	NOT_2D	Specifies the NOT(2*D) value of the M/N:D counter.

0x0181C024 GCC_ULTAUDIO_AHBFBABRIC_IXFABRIC_LPM_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x80004FF0

CBC Registers

GCC_ULTAUDIO_AHBFBABRIC_IXFABRIC_LPM_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	READ ONLY This bit indicates if the clock branch is halted: - 0:Clock On - 1:Clock Off
14	FORCE_MEM_CORE_ON		Control bit to force memory core on: 1 = memory core forced on, 0 = not in force
13	FORCE_MEM_PERIPH_ON		Control to force memory peripheral on: 1 = memory peripheral forced on, 0 = not in force This bit has priority over up_force_periph_off.
12	FORCE_MEM_PERIPH_OFF		Control bit to force memory peripheral off: 1 = memory peripheral force off, 0 = not in force This bit is effectively only if up_force_periph_on = 0.
11:8	WAKEUP		Specifies the latency between memory turning on and clock turning on during wakeup process
7:4	SLEEP		Specifies the latency between clock going off and memory going off during sleep process.
1	HW_CTL		Clock gating control: 0 = software controlled 1 = hardware controlled
0	CLK_ENABLE		Asynchronous clock enable.

0x0181C028 GCC_ULTAUDIO_AHBFBABRIC_IXFABRIC_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

CBC Registers

GCC_ULTAUDIO_AHBFABRIC_IXFABRIC_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	READ ONLY This bit indicates if the clock branch is halted: - 0:Clock On - 1:Clock Off
1	HW_CTL		Clock gating control: 0 = software controlled 1 = hardware controlled
0	CLK_ENABLE		Asynchronous clock enable.

0x0181C04C RESERVED**0x0181C050 GCC_ULTAUDIO_STC_XO_CBCR****Type:** RW**Clock:** CLK**Reset State:** 0x80000000

CBC Registers

GCC_ULTAUDIO_STC_XO_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	READ ONLY This bit indicates if the clock branch is halted: - 0:Clock On - 1:Clock Off
1	HW_CTL		Clock gating control: 0 = software controlled 1 = hardware controlled
0	CLK_ENABLE		Asynchronous clock enable.

0x0181C030 GCC_ULTAUDIO_AHBFABRIC_EFABRIC_SPDM_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

CBC Registers

GCC_ULTAUDIO_AHBFABRIC_EFABRIC_SPDM_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	READ ONLY This bit indicates if the clock branch is halted: - 0:Clock On - 1:Clock Off
1	HW_CTL		Clock gating control: 0 = software controlled 1 = hardware controlled
0	CLK_ENABLE		Asynchronous clock enable.

0x0181C09C GCC_CODEC_DIGCODEC_CMD_RCGR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

RCG Command Register

GCC_CODEC_DIGCODEC_CMD_RCGR

Bits	Name	Type	Description
31	ROOT_OFF	R	READ ONLY. Indicates if the RCG is actively producing a clock. 0: clock off 1: clock on
7	DIRTY_D	R	READ ONLY. Indicates change in the D register configuration.
6	DIRTY_N	R	READ ONLY. Indicates change in the N register configuration.
5	DIRTY_M	R	READ ONLY. Indicates change in the M register configuration.
4	DIRTY_CFG_RCGR	R	READ ONLY. Indicates change in the CFG_RCGR register configuration.
1	ROOT_EN		RCG enable: 0: disabled 1: enabled
0	UPDATE		Write 1 to this bit to update the RCG with a new programmed configuration. This bit is cleared by hardware.

0x0181C0A0 GCC_CODEC_DIGCODEC_CFG_RCGR**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

RCG Configuration Register

GCC_CODEC_DIGCODEC_CFG_RCGR

Bits	Name	Description
13:12	MODE	Specifies the MN counter mode: 0: bypass mode 1: swallow mode 2: dual edge mode 3: single edge mode
10:8	SRC_SEL	Specifies which clock source is output from the RCG. 0: CXO 1: GPLL1_MAIN 2: CODEC_MCLK_EXT_CLK_IN 3: no clock (ground) 4: no clock (ground) 5: no clock (ground) 6: sleep_clk 7: Reserved

GCC_CODEC_DIGCODEC_CFG_RCGR (cont.)

Bits	Name	Description
4:0	SRC_DIV	Specifies the integer and half integer portion of the divider setting of the RCG. The value specified is 2N-1 where N is the divide ratio. For example, if the half integer divide ratio is 2.5, the SRC_DIV value should be set at 0x04. 0: Bypass 1: Div-1 2: Div-1.5 3: Div-2 4: Div-2.5 5: Div-3 6: Div-3.5 7: Div-4 8: Div-4.5 9: Div-5 10: Div-5.5 11: Div-6 12: Div-6.5 13: Div-7 14: Div-7.5 15: Div-8 16: Div-8.5 17: Div-9 18: Div-9.5 19: Div-10 20: Div-10.5 21: Div-11 22: Div-11.5 23: Div-12 24: Div-12.5 25: Div-13 26: Div-13.5 27: Div-14 28: Div-14.5 29: Div-15 30: Div-15.5 31: Div-16

0x0181C0A4 GCC_CODEC_DIGCODEC_M**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

RCG M Register

GCC_CODEC_DIGCODEC_M

Bits	Name	Description
7:0	M	Specifies the M value of the M/N:D counter.

0x0181C0A8 GCC_CODEC_DIGCODEC_N**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

RCG N Register

GCC_CODEC_DIGCODEC_N

Bits	Name	Description
7:0	NOT_N_MINUS_M	Specifies the NOT(N-M) value of the M/N:D counter.

0x0181C0AC GCC_CODEC_DIGCODEC_D**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

RCG D Register

GCC_CODEC_DIGCODEC_D

Bits	Name	Description
7:0	NOT_2D	Specifies the NOT(2*D) value of the M/N:D counter.

0x0181C0B0 GCC_CODEC_DIGCODEC_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

CBC Registers

GCC_CODEC_DIGCODEC_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	READ ONLY This bit indicates if the clock branch is halted: - 0:Clock On - 1:Clock Off
1	HW_CTL		Clock gating control: 0 = software controlled 1 = hardware controlled
0	CLK_ENABLE		Asynchronous clock enable.

0x0181C084 GCC_ULTAUDIO_LPAIF_AUX_I2S_CMD_RCGR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

RCG Command Register

GCC_ULTAUDIO_LPAIF_AUX_I2S_CMD_RCGR

Bits	Name	Type	Description
31	ROOT_OFF	R	READ ONLY. Indicates if the RCG is actively producing a clock. 0: clock off 1: clock on
7	DIRTY_D	R	READ ONLY. Indicates change in the D register configuration.
6	DIRTY_N	R	READ ONLY. Indicates change in the N register configuration.
5	DIRTY_M	R	READ ONLY. Indicates change in the M register configuration.
4	DIRTY_CFG_RCGR	R	READ ONLY. Indicates change in the CFG_RCGR register configuration.
1	ROOT_EN		RCG enable: 0: disabled 1: enabled
0	UPDATE		Write 1 to this bit to update the RCG with a new programmed configuration. This bit is cleared by hardware.

0x0181C088 GCC_ULTAUDIO_LPAIF_AUX_I2S_CFG_RCGR**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

RCG Configuration Register

GCC_ULTAUDIO_LPAIF_AUX_I2S_CFG_RCGR

Bits	Name	Description
13:12	MODE	Specifies the MN counter mode: 0: bypass mode 1: swallow mode 2: dual edge mode 3: single edge mode
10:8	SRC_SEL	Specifies which clock source is output from the RCG. 0: CXO 1: GPLL1_MAIN 2: SEC_I2S_EXT_CLK_IN 3: no clock (ground) 4: no clock (ground) 5: no clock (ground) 6: sleep_clk 7: Reserved

GCC_ULTAUDIO_LPAIF_AUX_I2S_CFG_RCGR (cont.)

Bits	Name	Description
4:0	SRC_DIV	Specifies the integer and half integer portion of the divider setting of the RCG. The value specified is 2N-1 where N is the divide ratio. For example, if the half integer divide ratio is 2.5, the SRC_DIV value should be set at 0x04. 0: Bypass 1: Div-1 2: Div-1.5 3: Div-2 4: Div-2.5 5: Div-3 6: Div-3.5 7: Div-4 8: Div-4.5 9: Div-5 10: Div-5.5 11: Div-6 12: Div-6.5 13: Div-7 14: Div-7.5 15: Div-8 16: Div-8.5 17: Div-9 18: Div-9.5 19: Div-10 20: Div-10.5 21: Div-11 22: Div-11.5 23: Div-12 24: Div-12.5 25: Div-13 26: Div-13.5 27: Div-14 28: Div-14.5 29: Div-15 30: Div-15.5 31: Div-16

0x0181C08C GCC_ULTAUDIO_LPAIF_AUX_I2S_M**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

RCG M Register

GCC_ULTAUDIO_LPAIF_AUX_I2S_M

Bits	Name	Description
7:0	M	Specifies the M value of the M/N:D counter.

0x0181C090 GCC_ULTAUDIO_LPAIF_AUX_I2S_N**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

RCG N Register

GCC_ULTAUDIO_LPAIF_AUX_I2S_N

Bits	Name	Description
7:0	NOT_N_MINUS_M	Specifies the NOT(N-M) value of the M/N:D counter.

0x0181C094 GCC_ULTAUDIO_LPAIF_AUX_I2S_D**Type:** RW**Clock:** CLK**Reset State:** 0x00000000

RCG D Register

GCC_ULTAUDIO_LPAIF_AUX_I2S_D

Bits	Name	Description
7:0	NOT_2D	Specifies the NOT(2*D) value of the M/N:D counter.

0x0181C098 GCC_ULTAUDIO_LPAIF_AUX_I2S_CBCR**Type:** RW**Clock:** CLK**Reset State:** 0x80000000

CBC Registers

GCC_ULTAUDIO_LPAIF_AUX_I2S_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	READ ONLY This bit indicates if the clock branch is halted: - 0:Clock On - 1:Clock Off
1	HW_CTL		Clock gating control: 0 = software controlled 1 = hardware controlled
0	CLK_ENABLE		Asynchronous clock enable.

5 Core Top CSR

5.1 TCSR_TCSR_MUTEX

TCSR mutex

5.1.1 TCSR_TCSR_MUTEX

TCSR mutex

**0x01905000+ TCSR_MUTEX_REG_n, n=[0..31]
0x1000*n**

Type: RW

Clock: AHB_CLK

Reset State: 0x00000000

The MUTEX_REGn register specifies the information regarding the access status of each of the 32 sections of memory, which is denoted by n. Register contents of 0 imply that the respective section of memory is free to use, and therefore unlocked. Non-zero contents would imply that the respective section of memory is in use, and therefore locked

TCSR_MUTEX_REG_n

Bits	Name	Description
7:0	MUTEX	Mutex register data

5.2 TCSR_TCSR_REGS

TCSR internal

5.2.1 TCSR_TCSR_REGS

TCSR internal

0x01939800 TCSR_MUTEX_REG_RESET**Type:** RW**Clock:** AHB_CLK**Reset State:** 0x00000000

The MUTEX_REG_RESET register resets the MUTEX_REG_%n% blocks. Software has to assert the reset in order to clear the MUTEX_REGn. The reset bit needs to be de-asserted before SW can write to Mutex registers

TCSR_MUTEX_REG_RESET

Bits	Name	Description
0	MUTEX_RESET	0: deasserts reset 1: asserts reset

0x0193B07C TCSR_TCSR_CLK_EN**Type:** RW**Clock:** AHB_CLK**Reset State:** 0x00000001

This register controls HW clock gating logics in core_top_csr block

TCSR_TCSR_CLK_EN

Bits	Name	Description
0	TCSR_CLK_EN	0: clock gating logic is enabled. clock to TCSR is turned off when there is no AHB transactions. 1: Clock gating logic is disabled. clock to TCSR is always on

0x0193C000 TCSR_TIMEOUT_SLAVE_GLB_EN**Type:** RW**Clock:** AHB_CLK**Reset State:** 0x00000000

This register controls all the timeout slaves in the chip

TCSR_TIMEOUT_SLAVE_GLB_EN

Bits	Name	Description
0	TIMEOUT_SLAVE_GLB_EN	Enable bit to globally enable or disable all the timeout slaves in the chip 0: Disable 1: enable

**0x0193D000+ TCSR_TZ_WONCE_n, n=[0..15]
0x04*n****Type:** RW**Clock:** AHB_CLK**Reset State:** 0x00000000

Trust Zone Write Once registers

TCSR_TZ_WONCE_n

Bits	Name	Description
31:0	TZ_WONCE_ADDRESS	Write once registers used by TZ to pass address. This register is reset only on chip POR and can be written only once after POR

0x0193D100 TCSR_BOOT_MISC_DETECT**Type:** RW**Clock:** AHB_CLK**Reset State:** 0x00000000

Boot download Cookie register

TCSR_BOOT_MISC_DETECT

Bits	Name	Description
31:0	BOOT_MISC_DETECT	32-bit register used by Apps PBL to check cookie and enter DLOAD mode. This register is reset only at chip POR and retains value across all other resets

0x0193E000 TCSR_MMSS_RPM_IRQ_EN**Type:** RW**Clock:** AHB_CLK**Reset State:** 0x00000000

RPM INTERRUPT ENABLE

TCSR_MMSS_RPM_IRQ_EN

Bits	Name	Description
24	RPM_IRQ_EN_VENUS0_IRQ	RPM IRQ Enable for venus0_irq from venus0 0: disable 1: enable
23	RPM_IRQ_EN_CAMSS_IRQ10	RPM IRQ Enable for camss_irq10 from camss 0: disable 1: enable
22	RPM_IRQ_EN_CAMSS_IRQ11	RPM IRQ Enable for camss_irq11 from camss 0: disable 1: enable
21	RPM_IRQ_EN_CAMSS_IRQ12	RPM IRQ Enable for camss_irq12 from camss 0: disable 1: enable
20	RPM_IRQ_EN_MDSS_IRQ	RPM IRQ Enable for mdss_irq from mdss 0: disable 1: enable
16	RPM_IRQ_EN_GC_SYS_IRQ3	RPM IRQ Enable for o_gc_sys_irq3 from oxili 0: disable 1: enable
15	RPM_IRQ_EN_GC_SYS_IRQ2	RPM IRQ Enable for o_gc_sys_irq2 from oxili 0: disable 1: enable
14	RPM_IRQ_EN_GC_SYS_IRQ1	RPM IRQ Enable for o_gc_sys_irq1 from oxili 0: disable 1: enable
13	RPM_IRQ_EN_GC_SYS_IRQ0	RPM IRQ Enable for o_gc_sys_irq0 from oxili 0: disable 1: enable
12	RPM_IRQ_EN_CAMSS_IRQ0	RPM IRQ Enable for camss_irq0 from camss 0: disable 1: enable
11	RPM_IRQ_EN_CAMSS_IRQ1	RPM IRQ Enable for camss_irq1 from camss 0: disable 1: enable
10	RPM_IRQ_EN_CAMSS_IRQ2	RPM IRQ Enable for camss_irq2 from camss 0: disable 1: enable

TCSR_MMSS_RPM_IRQ_EN (cont.)

Bits	Name	Description
9	RPM_IRQ_EN_CAMSS_IRQ 3	RPM IRQ Clear for camss_irq3 from camss 0: disable 1: enable
8	RPM_IRQ_EN_CAMSS_IRQ 4	RPM IRQ Enable for camss_irq4 from camss 0: disable 1: enable
7	RPM_IRQ_EN_CAMSS_IRQ 5	RPM IRQ Enable for camss_irq5 from camss 0: disable 1: enable
6	RPM_IRQ_EN_CAMSS_IRQ 6	RPM IRQ Enable for camss_irq6 from camss 0: disable 1: enable
5	RPM_IRQ_EN_CAMSS_IRQ 7	RPM IRQ Enable for camss_irq7 from camss 0: disable 1: enable
4	RPM_IRQ_EN_CAMSS_IRQ 8	RPM IRQ Enable for camss_irq8 from camss 0: disable 1: enable
3	RPM_IRQ_EN_CAMSS_IRQ 9	RPM IRQ Enable for camss_irq9 from camss 0: disable 1: enable
2	RPM_IRQ_EN_CSIPHY_0_I RQ	RPM IRQ Enable for csiphy_0_irq from csiphy 0: disable 1: enable
1	RPM_IRQ_EN_CSIPHY_1_I RQ	RPM IRQ Enable for csiphy_1_irq from csiphy 0: disable 1: enable

0x0193E004 TCSR_MMSS_RPM_IRQ_CLEAR**Type:** RW**Clock:** AHB_CLK**Reset State:** 0x00000000

RPM INTERRUPT CLEAR FOR IRQs

TCSR_MMSS_RPM_IRQ_CLEAR

Bits	Name	Description
24	RPM_IRQ_CLEAR_VENUS0_IRQ	RPM IRQ clear for venus0_irq from venus0 0: not clear 1: clear
23	RPM_IRQ_CLEAR_CAMSS_IRQ10	RPM IRQ Clear for camss_irq10 from camss 0: not clear 1: clear
22	RPM_IRQ_CLEAR_CAMSS_IRQ11	RPM IRQ Clear for camss_irq11 from camss 0: not clear 1: clear
21	RPM_IRQ_CLEAR_CAMSS_IRQ12	RPM IRQ Clear for camss_irq12 from camss 0: not clear 1: clear
20	RPM_IRQ_CLEAR_MDSS_IRQ	RPM IRQ clear for mdss_irq from mdss 0: not clear 1: clear
16	RPM_IRQ_CLEAR_GC_SYS_IRQ3	RPM IRQ clear for o_gc_sys_irq3 from oxili 0: not clear 1: clear
15	RPM_IRQ_CLEAR_GC_SYS_IRQ2	RPM IRQ clear for o_gc_sys_irq2 from oxili 0: not clear 1: clear
14	RPM_IRQ_CLEAR_GC_SYS_IRQ1	RPM IRQ clear for o_gc_sys_irq1 from oxili 0: not clear 1: clear
13	RPM_IRQ_CLEAR_GC_SYS_IRQ0	RPM IRQ clear for o_gc_sys_irq0 from oxili 0: not clear 1: clear
12	RPM_IRQ_CLEAR_CAMSS_IRQ0	RPM IRQ Clear for camss_irq0 from camss 0: not clear 1: clear
11	RPM_IRQ_CLEAR_CAMSS_IRQ1	RPM IRQ Clear for camss_irq1 from camss 0: not clear 1: clear
10	RPM_IRQ_CLEAR_CAMSS_IRQ2	RPM IRQ Clear for camss_irq2 from camss 0: not clear 1: clear
9	RPM_IRQ_CLEAR_CAMSS_IRQ3	RPM IRQ Clear for camss_irq3 from camss 0: not clear 1: clear

TCSR_MMSS_RPM_IRQ_CLEAR (cont.)

Bits	Name	Description
8	RPM_IRQ_CLEAR_CAMSS_IRQ4	RPM IRQ Clear for camss_irq4 from camss 0: not clear 1: clear
7	RPM_IRQ_CLEAR_CAMSS_IRQ5	RPM IRQ Clear for camss_irq5 from camss 0: not clear 1: clear
6	RPM_IRQ_CLEAR_CAMSS_IRQ6	RPM IRQ Clear for camss_irq6 from camss 0: not clear 1: clear
5	RPM_IRQ_CLEAR_CAMSS_IRQ7	RPM IRQ Clear for camss_irq7 from camss 0: not clear 1: clear
4	RPM_IRQ_CLEAR_CAMSS_IRQ8	RPM IRQ Clear for camss_irq8 from camss 0: not clear 1: clear
3	RPM_IRQ_CLEAR_CAMSS_IRQ9	RPM IRQ Clear for camss_irq9 from camss 0: not clear 1: clear
2	RPM_IRQ_CLEAR_CSIPHY_0_IRQ	RPM IRQ Clear for csiphy_0_irq from csiphy 0: not clear 1: clear
1	RPM_IRQ_CLEAR_CSIPHY_1_IRQ	RPM IRQ Clear for csiphy_1_irq from csiphy 0: not clear 1: clear

0x0193E008 TCSR_MMSS_RPM_IRQ_STATUS**Type:** R**Clock:** AHB_CLK**Reset State:** 0x00000000

RPM INTERRUPT STATUS

TCSR_MMSS_RPM_IRQ_STATUS

Bits	Name	Description
24	RPM_IRQ_STATUS_VENUS_0_IRQ	RPM IRQ Status for venus0_irq from venus0
23	RPM_IRQ_STATUS_CAMSS_IRQ10	RPM IRQ Status for camss_irq10 from camss

TCSR_MMSS_RPM_IRQ_STATUS (cont.)

Bits	Name	Description
22	RPM_IRQ_STATUS_CAMSS_IRQ11	RPM IRQ Status for camss_irq11 from camss
21	RPM_IRQ_STATUS_CAMSS_IRQ12	RPM IRQ Status for camss_irq12 from camss
20	RPM_IRQ_STATUS_MDSS_IRQ	RPM IRQ Status for mdss_irq from mdss
16	RPM_IRQ_STATUS_GC_SY S_IRQ3	RPM IRQ Status for o_gc_sys_irq3 from oxili
15	RPM_IRQ_STATUS_GC_SY S_IRQ2	RPM IRQ Status for o_gc_sys_irq2 from oxili
14	RPM_IRQ_STATUS_GC_SY S_IRQ1	RPM IRQ Status for o_gc_sys_irq1 from oxili
13	RPM_IRQ_STATUS_GC_SY S_IRQ0	RPM IRQ Status for o_gc_sys_irq0 from oxili
12	RPM_IRQ_STATUS_CAMSS_IRQ0	RPM IRQ Status for camss_irq0 from camss
11	RPM_IRQ_STATUS_CAMSS_IRQ1	RPM IRQ Status for camss_irq1 from camss
10	RPM_IRQ_STATUS_CAMSS_IRQ2	RPM IRQ Status for camss_irq2 from camss
9	RPM_IRQ_STATUS_CAMSS_IRQ3	RPM IRQ Status for camss_irq3 from camss
8	RPM_IRQ_STATUS_CAMSS_IRQ4	RPM IRQ Status for camss_irq4 from camss
7	RPM_IRQ_STATUS_CAMSS_IRQ5	RPM IRQ Status for camss_irq5 from camss
6	RPM_IRQ_STATUS_CAMSS_IRQ6	RPM IRQ Status for camss_irq6 from camss
5	RPM_IRQ_STATUS_CAMSS_IRQ7	RPM IRQ Status for camss_irq7 from camss
4	RPM_IRQ_STATUS_CAMSS_IRQ8	RPM IRQ Status for camss_irq8 from camss
3	RPM_IRQ_STATUS_CAMSS_IRQ9	RPM IRQ Status for camss_irq9 from camss
2	RPM_IRQ_STATUS_CSIPHY_0_IRQ	RPM IRQ Status for csiphy_0_irq from csiphy
1	RPM_IRQ_STATUS_CSIPHY_1_IRQ	RPM IRQ Status for csiphy_1_irq from csiphy

0x0193E010 TCSR_MMSS_OXILI_GC_SYS_AHB_STATUS**Type:** R**Clock:** AHB_CLK**Reset State:** 0x00000000

Read Only registers for Oxili debug purpose

TCSR_MMSS_OXILI_GC_SYS_AHB_STATUS

Bits	Name	Description
31:0	OXILI_GC_SYS_AHB_STATUS	GC_SYS_AHB_STATUS from Oxili [31] Reserved [30:28] Status of the ahb request signals [27] Reserved [26:24] Status of the ahb grant signals [23] Indicates transfer from CP PFP is split [22] Indicates transfer from CP ME is split [21:20] Gives the master id for the transfer 0 - HOST master 1 - CP ME master 2 - CP PFP master [19:16] Gives the slave id for the transfer 0 - MISC 1 - VBIF 2 - PC 3 - CP 4 - RBP 5 - UCHE 6 - HLSQ 7 - TSE 8 - VFD 9 - VPC 10 - VSC 11 - Reserved [15] Indicates there is transfer in progress [14] Indicates a transfer was a write transfer [13] Reserved [12] Indicates a current transfer is being stalled [11] Indicates a transfer completed [10] Indicates a split transfer on last transfer [09] Indicates a error transfer on last transfer [08] Reserved [07:00] This is the higher 8 bits of the transfer timeout counter

0x0193E014 TCSR_MMSS_OXILI_CMD_REG**Type:** RW**Clock:** AHB_CLK**Reset State:** 0x00000000

Command registers for Oxili debug purpose

TCSR_MMSS_OXILI_CMD_REG

Bits	Name	Description
3:0	OXILI_CMD	Command registers for Oxili debug purpose [0] Release the bus that is waiting for a transfer to complete. The bus will return an ERROR response. This is for debug purpose. If another transfer is performed to the slave that is hanging, the bus could hang again. [1] Reset the transfer timeout counter [3:2] unused

0x0193E020 TCSR_MMSS_DSI_ULP_CLAMP_CTRL**Type:** RW**Clock:** AHB_CLK**Reset State:** 0x00000000

DSI ULP CLAMP CONTROLLER

TCSR_MMSS_DSI_ULP_CLAMP_CTRL

Bits	Name	Description
15	DSI0_CLAMP_EN	DSI0 CLAMP Enable
9	DSI0_CLKLN_EN	DSI0 CLKLN Enable
8	DSI0_CLKLN_ULPS_REQUEST	DSI0 CLKLN ULPS Request
7	DSI0_DLN0_EN	DSI0 DLN0 Enable
6	DSI0_DLN0_ULPS_REQUEST	DSI0 DLN0 ULPS Request
5	DSI0_DLN1_EN	DSI0 DLN1 Enable
4	DSI0_DLN1_ULPS_REQUEST	DSI0 DLN1 ULPS Request
3	DSI0_DLN2_EN	DSI0 DLN2 Enable
2	DSI0_DLN2_ULPS_REQUEST	DSI0 DLN2 ULPS Request
1	DSI0_DLN3_EN	DSI0 DLN3 Enable

TCSR_MMSS_DSI_ULP_CLAMP_CTRL (cont.)

Bits	Name	Description
0	DSI0_DLN3_ULPS_REQUE ST	DSI0 DLN3 ULPS Request

0x0193E024 TCSR_MMSS_DSI_PHYRESET_CTRL**Type:** RW**Clock:** AHB_CLK**Reset State:** 0x00000000

DSI PHY RESET CONTROL REGISTER

TCSR_MMSS_DSI_PHYRESET_CTRL

Bits	Name	Description
0	DSI_PHYRESET_CTRL	When set to 0, no effect, when set to 1 DSI PHY will not be reset by mmcc_mdss_ahb_ares

0x0193E030 RESERVED**0x0193E034 RESERVED****0x0193E038 TCSR_MMSS_IMEM_RAM_CONFIG****Type:** RW**Clock:** AHB_CLK**Reset State:** 0x00000000

IMEM RAM Configuration

TCSR_MMSS_IMEM_RAM_CONFIG

Bits	Name	Description
2	EX_RAM_CLK_EN	Not Used in A305 design
1:0	EX_RAM_CONFIG	External ram access configuration 00: No External ram allocated to AXI 10: All external ram allocated to AXI Note: EX_RAM_CONFIG[1] is used as the selector for glitch free Mux

0x0193F000 TCSR_TBU_BYPASS_ENABLE**Type:** RW**Clock:** AHB_CLK**Reset State:** 0x00000000

TBU bypass enable register

TCSR_TBU_BYPASS_ENABLE

Bits	Name	Description
9:0	TBU_BYPASS_ENABLE	TBU bypass enable register for SMMU.

0x01941000 TCSR_QPDI_DISABLE_CFG**Type:** RW**Clock:** AHB_CLK**Reset State:** 0x00000001

Qualcomm PMIC debug interface disable

TCSR_QPDI_DISABLE_CFG

Bits	Name	Description
0	QPDI_DISABLE_CFG	QPDI disable bit. 1=Disable 0=Enable.

0x01942080 TCSR_COMPILER_VDDCX_ACC_0**Type:** RW**Clock:** AHB_CLK**Reset State:** 0x00000000

Compiler RAM ACC value of VDD CX power domain

TCSR_COMPILER_VDDCX_ACC_0

Bits	Name	Description
31:26	COMPILER_HDHPLLSP127	ACC[6:1] bits for HDHPLLSP127. ACC[0] & ACC[7] are derived from mode selected by MEM_ACC_SEL_VDDCX
25:20	COMPILER_HPLVRF	ACC[6:1] bits for HPLVRF(2-port). ACC[0] & ACC[7] are derived from mode selected by MEM_ACC_SEL_VDDCX

TCSR_COMPILER_VDDCX_ACC_0 (cont.)

Bits	Name	Description
19:16	COMPILER_LLRF240_1	ACC[3:0] bits for LLRF240(1-port)
15:12	COMPILER_STDSP155	ACC[3:0] bits for STDSP155
11:8	COMPILER_LLRF240_2	ACC[3:0] bits for LLRF240(2-port)
7:4	COMPILER_LLDP155	ACC[3:0] bits for LLPDP155
3:0	COMPILER_LLSP155	ACC[3:0] bits for LLSP155/UDHLLSP155

0x01942084 TCSR_COMPILER_VDDCX_ACC_1**Type:** RW**Clock:** AHB_CLK**Reset State:** 0x00000000

Complier RAM ACC value of VDD CX power domain

TCSR_COMPILER_VDDCX_ACC_1

Bits	Name	Description
3:0	COMPILER_UHDLLPDP155	ACC[3:0] bits for UHDLLPDP155

0x01942090 TCSR_CUSTOM_VDDCX_ACC_0**Type:** RW**Clock:** AHB_CLK**Reset State:** 0x00000000

Custom RAM ACC value of VDD CX power domain

TCSR_CUSTOM_VDDCX_ACC_0

Bits	Name	Description
31:30	CUSTOM_UHPSP127_L	ACC[2:1] bits for UHPSP127. This is partial field. Rest of the bits of this field[6:3] are in next register(CUSTOM_VDDCX_ACC_1). ACC[0] & ACC[7] are derived from mode selected by MEM_ACC_SEL_VDDCX
29:24	CUSTOM_UHPSP155_OTHERS	ACC[6:1] bits for UHPSP155 for others. ACC[0] & ACC[7] are derived from mode selected by MEM_ACC_SEL_VDDCX
23:18	CUSTOM_UHPSP155_DATA	ACC[6:1] bits for UHPSP155 for L1 data. ACC[0] & ACC[7] are derived from mode selected by MEM_ACC_SEL_VDDCX
17:12	CUSTOM_8P_RF	ACC[6:1] bits for 8P RF. ACC[0] & ACC[7] are derived from mode selected by MEM_ACC_SEL_VDDCX

TCSR_CUSTOM_VDDCX_ACC_0 (cont.)

Bits	Name	Description
11:6	CUSTOM_6P_RF	ACC[6:1] bits for 6P RF. ACC[0] & ACC[7] are derived from mode selected by MEM_ACC_SEL_VDDCX
5:0	CUSTOM_HPLVF240_BYPASS	ACC[6:1] bits for HPLVF240_bypass. ACC[0] & ACC[7] are derived from mode selected by MEM_ACC_SEL_VDDCX

0x01942094 TCSR_CUSTOM_VDDCX_ACC_1**Type:** RW**Clock:** AHB_CLK**Reset State:** 0x00000000

Custom RAM ACC value of VDD CX power domain

TCSR_CUSTOM_VDDCX_ACC_1

Bits	Name	Description
21:16	CUSTOM_VHPLLSP127	ACC[6:1] bits for VHPLLSP127. ACC[0] & ACC[7] are derived from mode selected by MEM_ACC_SEL_VDDCX
15:10	CUSTOM_HPLLSP155	ACC[6:1] bits for HPLLSP155. ACC[0] & ACC[7] are derived from mode selected by MEM_ACC_SEL_VDDCX
9:4	CUSTOM_HPLLSP127	ACC[6:1] bits for HPLLSP127. ACC[0] & ACC[7] are derived from mode selected by MEM_ACC_SEL_VDDCX
3:0	CUSTOM_UHPSP127_H	ACC[6:3] bits for UHPSP127. This is partial field. Rest of the bits of this field[2:1] are in previous register(CUSTOM_VDDCX_ACC_0). ACC[0] & ACC[7] are derived from mode selected by MEM_ACC_SEL_VDDCX

0x01942100 TCSR_MEM_ACC_SEL_VDDCX**Type:** RW**Clock:** AHB_CLK**Reset State:** 0x00000001

This register controls ACC mode of VDD CX power domain

TCSR_MEM_ACC_SEL_VDDCX

Bits	Name	Description
1:0	MEM_ACC_SEL_VDDCX	2'b01: Nominal 2'b11: Turbo 2'b00: SVS 2'b10: Reserved

0x01946000 TCSR_MEM_ACC_SEL_APC**Type:** RW**Clock:** AHB_CLK**Reset State:** 0x00000101

This register controls ACC mode of APC L1 and L2 power domain

TCSR_MEM_ACC_SEL_APC

Bits	Name	Description
9:8	MEM_ACC_SEL_APC_L2	Mode control only for APC L2 memories 2'b01: Nominal 2'b11: Turbo 2'b00: SVS 2'b10: Reserved
1:0	MEM_ACC_SEL_APC_L1	Mode control only for APC L1 memories 2'b01: Nominal 2'b11: Turbo 2'b00: SVS 2'b10: Reserved

0x01946004 TCSR_CUSTOM_APC_ACC**Type:** RW**Clock:** AHB_CLK**Reset State:** 0x00000000

Custom RAM ACC value of APC power domain

TCSR_CUSTOM_APC_ACC

Bits	Name	Description
11:6	CUSTOM_UHPSP155_APC_L2	ACC[6:1] bits for UHPSP155 for L2. ACC[0] & ACC[7] are derived from mode selected by MEM_ACC_SEL_APC_L2
5:0	CUSTOM_UHPSP155_APC_L1	ACC[6:1] bits for UHPSP155 for L1. ACC[0] & ACC[7] are derived from mode selected by MEM_ACC_SEL_APC_L1

0x01947000 TCSR_SYS_POWER_CTRL**Type:** RW**Clock:** AHB_CLK**Reset State:** 0x00000000

indicate status of MPM power controls

TCSR_SYS_POWER_CTRL

Bits	Name	Description
15:0	SYS_POWER_CTRL	[15:11] reserved [10:8] USB PORT TYPE 3'b000 = SDP (500mA HID Grant) 3'b001 = DCP 3'b010 = CDP 3'b011 = ACA-Dock 3'b100 = ACA-A 3'b101 = ACA-B 3'b110 = ACA-C 3'b111 = None/Unconfigured ABOVE_WEAK_TH 1'b0 = VBAT is below the weak (bootable) threshold 1'b1 = VBAT is above the weak (bootable) threshold [6:5] CHG_PATH 2'b00 = None (PMIC is powered from battery, no valid charger attached) 2'b01 = USB path charger attached 2'b10 = DCIN path charger attached 2'b11 = Both USB and DCIN path chargers are attached [4] SKIP 1'b0 = Standard PMIC dual stage PON sequence, primary sequence first 1'b1 = Single stage PON sequence, includes primary followed by secondary sequence without MSM PMIC PON Handshake required. [3] FLCB_PMIC 1'b0 = PMIC is non-FLCB compatible or 3rd party PMIC 1'b1 = PMIC is FLCB compatible [2] BATTERY_PRESENT (passed from PMIC PBL_ACCESS1 register) 1'b0 = No Battery 1'b1 = Battery is present in the system [1:0] Reserved

0x01947004 TCSR_USB_CORE_ID**Type:** RW**Clock:** AHB_CLK**Reset State:** 0x00000000

This register indicate the USB core ID which is used for download during boot

TCSR_USB_CORE_ID

Bits	Name	Description
1:0	USB_CORE_ID	2'b00 = core 0 2'b01 = core 1 2'b10 = core 2 2'b11 = core 3

0x01948000 TCSR_LDO_SLEEP_CTRL**Type:** RW**Clock:** AHB_CLK**Reset State:** 0x00000000

Control the LDO sleep

TCSR_LDO_SLEEP_CTRL

Bits	Name	Description
0	LDO_SLEEP	0 = LDO is on 1 = LDO is off; This state is not considered available. It would potentially cause reliability damage

0x01948004 TCSR_LDO_UPDATE_STATE_CTRL**Type:** RW**Clock:** AHB_CLK**Reset State:** 0x00000000

Control the LDO control update

TCSR_LDO_UPDATE_STATE_CTRL

Bits	Name	Description
0	LDO_UPDATE_STATE	0 = the LDO latches are not enabled 1 = LDO Latches are enabled to change state.

0x01948008 TCSR_LDO_OBIAS_CTRL**Type:** RW**Clock:** AHB_CLK**Reset State:** 0x00000000

Control the LDO output bias

TCSR_LDO_OBIAS_CTRL

Bits	Name	Description
0	LDO_OBIAS_ON	0 = output stage has no internal current sinking 1 = LDO internally sinks 10uA from Ppass

0x0194800C TCSR_LDO_VREF_CONFIG**Type:** RW**Clock:** AHB_CLK**Reset State:** 0x00000007

Adjust the vref voltage that goes from the SCALER to the REG

TCSR_LDO_VREF_CONFIG

Bits	Name	Description
3:0	LDO_VREF_CONFIG	$vref = 0.8625 + (-0.0125 * vref_config<3:0>)$; 0000 mapped to 1000 internally. Effectively 0000 => 1000

0x01948010 TCSR_LDO_IB_CONFIG**Type:** RW**Clock:** AHB_CLK**Reset State:** 0x00000002

Adjust the internal ibias current that goes from the SCALER to Regulator

TCSR_LDO_IB_CONFIG

Bits	Name	Description
2:0	LDO_IB_CONFIG	$ireg = 0.25\mu A + 0.25\mu A * ldo_ib_config<2:0>$; 000 mapped to 100 internally

0x01948014 TCSR_LDO_BGC_CONFIG**Type:** RW**Clock:** AHB_CLK**Reset State:** 0x00000000**TCSR_LDO_BGC_CONFIG**

Bits	Name	Description
2:0	LDO_BGC	LDO_BGC[0] = 0: power down local reference LDO_BGC[0] = 1: power up local reference LDO_BGC[1]: reserved LDO_BGC[2]: reserved

0x01948018 TCSR_LDO_VREF_CTRL**Type:** RW**Clock:** AHB_CLK**Reset State:** 0x00000000

LDO Vref select

TCSR_LDO_VREF_CTRL

Bits	Name	Description
16	LDO_VREF_SEL_OVR	0 = selects LDO_VREF_SEL_MPM 1 = selects LDO_VREF_SW
0	LDO_VREF_SEL_SW	0 = selects PMIC 1.25V reference 1 = selects local BAND gap reference

0x0194801C TCSR_LDO_LD_EN**Type:** RW**Clock:** AHB_CLK**Reset State:** 0x00000000

Control the LDO load

TCSR_LDO_LD_EN

Bits	Name	Description
31	LDO_LD_EN	LDO_LD_EN = 0: disables the test load at LDO output LDO_LD_EN = 1: Enable the test load

0x01948020 TCSR_LDO_LD_CTRL**Type:** RW**Clock:** AHB_CLK**Reset State:** 0x00000000

Control the LDO load current

TCSR_LDO_LD_CTRL

Bits	Name	Description
23:16	LDO_LD_MSB	8'b0 = no load from bigger current branch oad current = $0.5\text{mA} * \text{ldo001_ld_lsb}<0> + 0.5\text{mA} * \text{ldo001_ld_lsb}<1> + 0.5\text{mA} * \text{ldo001_ld_lsb}<2> + 0.5\text{mA} * \text{ldo001_ld_lsb}<3> + 1\text{mA} * \text{ldo001_ld_lsb}<4> + 1\text{mA} * \text{ldo001_ld_lsb}<5> + 1\text{mA} * \text{ldo001_ld_lsb}<6> + 1\text{mA} * \text{ldo001_ld_lsb}<7>$
7:0	LDO_LD_LSB	8'b0 = no load from smaller current branch Load current = $0.125\text{mA} * \text{ldo001_ld_lsb}<0> + 0.125\text{mA} * \text{ldo001_ld_lsb}<1> + 0.125\text{mA} * \text{ldo001_ld_lsb}<2> + 0.125\text{mA} * \text{ldo001_ld_lsb}<3> + 0.25\text{mA} * \text{ldo001_ld_lsb}<4> + 0.25\text{mA} * \text{ldo001_ld_lsb}<5> + 0.25\text{mA} * \text{ldo001_ld_lsb}<6> + 0.25\text{mA} * \text{ldo001_ld_lsb}<7>$

0x01948024 TCSR_LDO_OSC_RESETB**Type:** RW**Clock:** AHB_CLK**Reset State:** 0x00000000

Control the LDO Oscillator

TCSR_LDO_OSC_RESETB

Bits	Name	Description
31	LDO_OSC_RESETB	0 = disables the noise measure oscillator 1 = enables the noise measure oscillator

0x01948028 TCSR_LDO_OSC_CTRL**Type:** RW**Clock:** AHB_CLK**Reset State:** 0x00000000

Control the LDO Oscillator

TCSR_LDO_OSC_CTRL

Bits	Name	Description
1:0	LDO_OSC_CTRL	00 = Noise measurement oscillator frequency control Highest (2.8 GHz at TT / 1.05V / 40°C) 01 = 1/2 highest 10 = 1/3 highest 11 = 1/4 highest

0x0194802C TCSR_LDO_DFT_EN_CTRL**Type:** RW**Clock:** AHB_CLK**Reset State:** 0x00000000

Control the LDO in DFT mode

TCSR_LDO_DFT_EN_CTRL

Bits	Name	Description
31	LDO_DFT_EN	0 = Reserved 1 = Reserved

0x01948030 TCSR_LDO_DFT_CTRL**Type:** RW**Clock:** AHB_CLK**Reset State:** 0x00000000

Control the LDO in DFT bus output selection

TCSR_LDO_DFT_CTRL

Bits	Name	Description
2:0	LDO_DFT_CONFIG	3'b000: Reserved 3'b001: Reserved 3'b010: Reserved 3'b011: Reserved 3'b100: Reserved 3'b101: Reserved 3'b110: Reserved 3'b111: Reserved

0x0194A000 TCSR_SPDM_CNT_CLK_CTRL**Type:** RW**Clock:** AHB_CLK**Reset State:** 0x00000000

This register controls Mux selects for cycle clock multiplexors to select one clock from two possible inputs, cy_clk_a and cy_clk_b. There is one mux per cycle counter and each mux select is 1-bit wide

TCSR_SPDM_CNT_CLK_CTRL

Bits	Name	Description
15:0	SPDM_CNT_CLK_MUX_SEL	cycle clock mux sel bit n is for mux n

0x0194A004 TCSR_SPDM_DLY_FIFO_EN**Type:** RW**Clock:** AHB_CLK**Reset State:** 0x00000000

This register enables delay fifo for each offline 4bit port input. Delay fifos are used to transfer offline data between the source and the SPDM core pseudo-synchronously

TCSR_SPDM_DLY_FIFO_EN

Bits	Name	Description
31:0	SPDM_DLY_FIFO_EN	each bit corresponds to delay for fifo used for offline port Bit[0] corresponds to delay for fifo used for offline port #0 bit value 0: Disable 1: Enable

0x0194A008 TCSR_SPDM_STG1_MUX_SEL**Type:** RW**Clock:** AHB_CLK**Reset State:** 0x00000000

This register controls Mux selects for stage 1 muxes to select 8 primary ports to monitor 32 offline ports. Each stage 1 mux is 2-bits wide and selects 1 of 4 offline ports

TCSR_SPDM_STG1_MUX_SEL

Bits	Name	Description
15:0	SPDM_STG1_MUX_SEL	Bits 31:16 Reserved_31_16 15:14 spdm_stg1_mux_sel7 selects between Port #19, Port #23, Port #27, Port #31 to create Primary Port #7 13:12 spdm_stg1_mux_sel6 selects between Port #18, Port #22, Port #26, Port #30 to create Primary Port #6 11:10 spdm_stg1_mux_sel5 selects between Port #17, Port #21, Port #25, Port #29 to create Primary Port #5 9:8 spdm_stg1_mux_sel4 selects between Port #16, Port #20, Port #24, Port #28 to create Primary Port #4 7:6 spdm_stg1_mux_sel3 selects between Port #3, Port #7, Port #11, Port #15 to create Primary Port #3 5:4 spdm_stg1_mux_sel2 selects between Port #2, Port #6, Port #10, Port #14 to create Primary Port #2 3:2 spdm_stg1_mux_sel1 selects between Port #1, Port #5, Port #9, Port #13 to create Primary Port #1 1:0 spdm_stg1_mux_sel0 selects between Port #0, Port #4, Port #8, Port #12 to create Primary Port #0

0x0194A00C TCSR_SPDM_STG2_A_MUX_SEL**Type:** RW**Clock:** AHB_CLK**Reset State:** 0x00000000

This register controls Mux selects for stage 3 multiplexors to select one bit from each port (selected in stage 2 mux) for monitoring in an offline monitor. There is one mux per offline monitor and each stage 3 mux select is 2-bits wide

TCSR_SPDM_STG2_A_MUX_SEL

Bits	Name	Description
31:0	SPDM_STG2_A_MUX_SEL	Bits 31:30 spdm_stg2_a_mux_sel12 selects between Primary Ports #0-#3 for monitoring in offline monitor #15 29:28 spdm_stg2_a_mux_sel12 selects between Primary Ports #0-#3 for monitoring in offline monitor #14 27:26 spdm_stg2_a_mux_sel12 selects between Primary Ports #0-#3 for monitoring in offline monitor #13 25:24 spdm_stg2_a_mux_sel12 selects between Primary Ports #0-#3 for monitoring in offline monitor #12 23:22 spdm_stg2_a_mux_sel11 selects between Primary Ports #0-#3 for monitoring in offline monitor #11 21:20 spdm_stg2_a_mux_sel10 selects between Primary Ports #0-#3 for monitoring in offline monitor #10 19:18 spdm_stg2_a_mux_sel9 selects between Primary Ports #0-#3 for monitoring in offline monitor #9 17:16 spdm_stg2_a_mux_sel8 selects between Primary Ports #0-#3 for monitoring in offline monitor #8 15:14 spdm_stg2_a_mux_sel7 selects between Primary Ports #0-#3 for monitoring in offline monitor #7 13:12 spdm_stg2_a_mux_sel6 selects between Primary Ports #0-#3 for monitoring in offline monitor #6 11:10 spdm_stg2_a_mux_sel5 selects between Primary Ports #0-#3 for monitoring in offline monitor #5 9:8 spdm_stg2_a_mux_sel4 selects between Primary Ports #0-#3 for monitoring in offline monitor #4 7:6 spdm_stg2_a_mux_sel3 selects between Primary Ports #0-#3 for monitoring in offline monitor #3 5:4 spdm_stg2_a_mux_sel2 selects between Primary Ports #0-#3 for monitoring in offline monitor #2 3:2 spdm_stg2_a_mux_sel1 selects between Primary Ports #0-#3 for monitoring in offline monitor #1 1:0 spdm_stg2_a_mux_sel0

TCSR_SPDM_STG2_A_MUX_SEL (cont.)

Bits	Name	Description
		selects between Primary Ports #0-#3 for monitoring in offline monitor #0

0x0194A010 TCSR_SPDM_STG2_B_MUX_SEL**Type:** RW**Clock:** AHB_CLK**Reset State:** 0x00000000

This register controls Mux selects for stage 2 muxes to select one of 4 primary ports (selected in stage 1 mux) for monitoring in an offline monitor. There is one mux per offline monitor and each stage 2 mux select is 2-bits wide

TCSR_SPDM_STG2_B_MUX_SEL

Bits	Name	Description
31:0	SPDM_STG2_B_MUX_SEL	Bits 31:30 spdm_stg2_b_mux_sel31 selects between Primary Ports #4-#7 for monitoring in offline monitor #31 29:28 spdm_stg2_b_mux_sel30 selects between Primary Ports #4-#7 for monitoring in offline monitor #30 27:26 spdm_stg2_b_mux_sel29 selects between Primary Ports #4-#7 for monitoring in offline monitor #29 25:24 spdm_stg2_b_mux_sel28 selects between Primary Ports #4-#7 for monitoring in offline monitor #28 23:22 spdm_stg2_b_mux_sel27 selects between Primary Ports #4-#7 for monitoring in offline monitor #27 21:20 spdm_stg2_b_mux_sel26 selects between Primary Ports #4-#7 for monitoring in offline monitor #26 19:18 spdm_stg2_b_mux_sel25 selects between Primary Ports #4-#7 for monitoring in offline monitor #25 17:16 spdm_stg2_b_mux_sel24 selects between Primary Ports #4-#7 for monitoring in offline monitor #24 15:14 spdm_stg2_b_mux_sel23 selects between Primary Ports #4-#7 for monitoring in offline monitor #23 13:12 spdm_stg2_b_mux_sel22 selects between Primary Ports #4-#7 for monitoring in offline monitor #22 11:10 spdm_stg2_b_mux_sel21 selects between Primary Ports #4-#7 for monitoring in offline monitor #21 9:8 spdm_stg2_b_mux_sel20 selects between Primary Ports #4-#7 for monitoring in offline monitor #20 7:6 spdm_stg2_b_mux_sel19 selects between Primary Ports #4-#7 for monitoring in offline monitor #19 5:4 spdm_stg2_b_mux_sel18 selects between Primary Ports #4-#7 for monitoring in offline monitor #18 3:2 spdm_stg2_b_mux_sel17 selects between Primary Ports #4-#7 for monitoring in offline monitor #17 1:0 spdm_stg2_b_mux_sel16

TCSR_SPDM_STG2_B_MUX_SEL (cont.)

Bits	Name	Description
		selects between Primary Ports #4-#7 for monitoring in offline monitor #16

0x0194A014 TCSR_SPDM_STG3_A_MUX_SEL**Type:** RW**Clock:** AHB_CLK**Reset State:** 0x00000000

This register controls Mux selects for stage 3 multiplexors to select one bit from each port (selected in stage 2 mux) for monitoring in an offline monitor. There is one mux per offline monitor and each stage 3 mux select is 2-bits wide

TCSR_SPDM_STG3_A_MUX_SEL

Bits	Name	Description
31:0	SPDM_STG3_A_MUX_SEL	Bits 31:30 spdm_stg3_a_mux_sel15 selects one of 4-bits from Stage 2 Mux #15 for monitoring in offline monitor #15 29:28 spdm_stg3_a_mux_sel14 selects one of 4-bits from Stage 2 Mux #14 for monitoring in offline monitor #14 27:26 spdm_stg3_a_mux_sel13 selects one of 4-bits from Stage 2 Mux #13 for monitoring in offline monitor #13 25:24 spdm_stg3_a_mux_sel12 selects one of 4-bits from Stage 2 Mux #12 for monitoring in offline monitor #12 23:22 spdm_stg3_a_mux_sel11 selects one of 4-bits from Stage 2 Mux #11 for monitoring in offline monitor #11 21:20 spdm_stg3_a_mux_sel10 selects one of 4-bits from Stage 2 Mux #10 for monitoring in offline monitor #10 19:18 spdm_stg3_a_mux_sel9 selects one of 4-bits from Stage 2 Mux #9 for monitoring in offline monitor #9 17:16 spdm_stg3_a_mux_sel8 selects one of 4-bits from Stage 2 Mux #8 for monitoring in offline monitor #8 15:14 spdm_stg3_a_mux_sel7 selects one of 4-bits from Stage 2 Mux #7 for monitoring in offline monitor #7 13:12 spdm_stg3_a_mux_sel6 selects one of 4-bits from Stage 2 Mux #6 for monitoring in offline monitor #6 11:10 spdm_stg3_a_mux_sel5 selects one of 4-bits from Stage 2 Mux #5 for monitoring in offline monitor #5 9:8 spdm_stg3_a_mux_sel4 selects one of 4-bits from Stage 2 Mux #4 for monitoring in offline monitor #4 7:6 spdm_stg3_a_mux_sel3 selects one of 4-bits from Stage 2 Mux #3 for monitoring in offline monitor #3 5:4 spdm_stg3_a_mux_sel2 selects one of 4-bits from Stage 2 Mux #2 for monitoring in offline monitor #2 3:2 spdm_stg3_a_mux_sel1 selects one of 4-bits from Stage 2 Mux #1 for monitoring in offline monitor #1 1:0 spdm_stg3_a_mux_sel0

TCSR_SPDM_STG3_A_MUX_SEL (cont.)

Bits	Name	Description
		selects one of 4-bits from Stage 2 Mux #0 for monitoring in offline monitor #0

0x0194A018 TCSR_SPDM_STG3_B_MUX_SEL**Type:** RW**Clock:** AHB_CLK**Reset State:** 0x00000000

This register controls Mux selects for stage 3 multiplexors to select one bit from each port (selected in stage 2 mux) for monitoring in an offline monitor. There is one mux per offline monitor and each stage 3 mux select is 2-bits wide.

TCSR_SPDM_STG3_B_MUX_SEL

Bits	Name	Description
31:0	SPDM_STG3_B_MUX_SEL	Bits 31:30 spdm_stg3_a_mux_sel15 selects one of 4-bits from Stage 2 Mux #15 for monitoring in offline monitor #15 29:28 spdm_stg3_a_mux_sel14 selects one of 4-bits from Stage 2 Mux #14 for monitoring in offline monitor #14 27:26 spdm_stg3_a_mux_sel13 selects one of 4-bits from Stage 2 Mux #13 for monitoring in offline monitor #13 25:24 spdm_stg3_a_mux_sel12 selects one of 4-bits from Stage 2 Mux #12 for monitoring in offline monitor #12 23:22 spdm_stg3_a_mux_sel11 selects one of 4-bits from Stage 2 Mux #11 for monitoring in offline monitor #11 21:20 spdm_stg3_a_mux_sel10 selects one of 4-bits from Stage 2 Mux #10 for monitoring in offline monitor #10 19:18 spdm_stg3_a_mux_sel9 selects one of 4-bits from Stage 2 Mux #9 for monitoring in offline monitor #9 17:16 spdm_stg3_a_mux_sel8 selects one of 4-bits from Stage 2 Mux #8 for monitoring in offline monitor #8 15:14 spdm_stg3_a_mux_sel7 selects one of 4-bits from Stage 2 Mux #7 for monitoring in offline monitor #7 13:12 spdm_stg3_a_mux_sel6 selects one of 4-bits from Stage 2 Mux #6 for monitoring in offline monitor #6 11:10 spdm_stg3_a_mux_sel5 selects one of 4-bits from Stage 2 Mux #5 for monitoring in offline monitor #5 9:8 spdm_stg3_a_mux_sel4 selects one of 4-bits from Stage 2 Mux #4 for monitoring in offline monitor #4 7:6 spdm_stg3_a_mux_sel3 selects one of 4-bits from Stage 2 Mux #3 for monitoring in offline monitor #3 5:4 spdm_stg3_a_mux_sel2 selects one of 4-bits from Stage 2 Mux #2 for monitoring in offline monitor #2 3:2 spdm_stg3_a_mux_sel1 selects one of 4-bits from Stage 2 Mux #1 for monitoring in offline monitor #1 1:0 spdm_stg3_a_mux_sel0

TCSR_SPDM_STG3_B_MUX_SEL (cont.)

Bits	Name	Description
		selects one of 4-bits from Stage 2 Mux #0 for monitoring in offline monitor #0

0x0194B000 RESERVED**0x0194B080+ TCSR_PHSS_UART_PRONTO_INT_SEL_n, n=[0..1]
0x10*n****Type:** RW**Clock:** AHB_CLK**Reset State:** 0x00000000

Each register select which PHSS interrupt goes to one bit interrupt output. register %n% controls phss_uart_pronto_int[%n%]. the enabled interrupts ORed together to generate output.

TCSR_PHSS_UART_PRONTO_INT_SEL_n

Bits	Name	Description
1	PRONTO_BLSP1_UART_2_IRQ_ENABLE	1 = enabled. 0 = disabled
0	PRONTO_BLSP1_UART_1_IRQ_ENABLE	1 = enabled. 0 = disabled

**0x0194B0C0+ TCSR_PHSS_UART_MSS_INT_SEL_n, n=[0..1]
0x10*n****Type:** RW**Clock:** AHB_CLK**Reset State:** 0x00000000

Each register select which PHSS interrupt goes to one bit interrupt output to MSS. register %n% controls phss_uart_mss_int[%n%]. the enabled interrupts ORed together to generate output

TCSR_PHSS_UART_MSS_INT_SEL_n

Bits	Name	Description
1	MSS_BLSP1_UART_2_IRQ_ENABLE	1 = enabled. 0 = disabled
0	MSS_BLSP1_UART_1_IRQ_ENABLE	1 = enabled. 0 = disabled

**0x0194B100+ TCSR_PHSS_QUP_PRONTO_INT_SEL_n, n=[0..1]
0x10*n**
Type: RW**Clock:** AHB_CLK**Reset State:** 0x00000000

select which PHSS interrupt goes to one bit interrupt output to PRONTO. register %n% controls phss_qup_pronto_int[%n%]. the enabled interrupts ORed together to generate output

TCSR_PHSS_QUP_PRONTO_INT_SEL_n

Bits	Name	Description
5	PRONTO_BLSP1_QUP_6_I RQ_ENABLE	1 = enabled. 0 = disabled
4	PRONTO_BLSP1_QUP_5_I RQ_ENABLE	1 = enabled. 0 = disabled
3	PRONTO_BLSP1_QUP_4_I RQ_ENABLE	1 = enabled. 0 = disabled
2	PRONTO_BLSP1_QUP_3_I RQ_ENABLE	1 = enabled. 0 = disabled
1	PRONTO_BLSP1_QUP_2_I RQ_ENABLE	1 = enabled. 0 = disabled
0	PRONTO_BLSP1_QUP_1_I RQ_ENABLE	1 = enabled. 0 = disabled

**0x0194B140+ TCSR_PHSS_QUP_MSS_INT_SEL_n, n=[0..1]
0x10*n**
Type: RW**Clock:** AHB_CLK**Reset State:** 0x00000000

Each register select which PHSS interrupt goes to one bit interrupt output to MSS. register %n% controls phss_qup_mss_int[%n%]. the enabled interrupts ORed together to generate output

TCSR_PHSS_QUP_MSS_INT_SEL_n

Bits	Name	Description
5	MSS_BLSP1_QUP_6_IRQ_ ENABLE	1 = enabled. 0 = disabled
4	MSS_BLSP1_QUP_5_IRQ_ ENABLE	1 = enabled. 0 = disabled
3	MSS_BLSP1_QUP_4_IRQ_ ENABLE	1 = enabled. 0 = disabled

TCSR_PHSS_QUP_MSS_INT_SEL_n (cont.)

Bits	Name	Description
2	MSS_BLSP1_QUP_3_IRQ_ENABLE	1 = enabled. 0 = disabled
1	MSS_BLSP1_QUP_2_IRQ_ENABLE	1 = enabled. 0 = disabled
0	MSS_BLSP1_QUP_1_IRQ_ENABLE	1 = enabled. 0 = disabled

0x0194D020 TCSR_TIMEOUT_INTR_STATUS**Type:** R**Clock:** AHB_CLK**Reset State:** 0x00000000

This is interrupt status register

TCSR_TIMEOUT_INTR_STATUS

Bits	Name	Description
22	ULTAUDIO_AHBI_TIMEOUT_IRQ	Status of interrupt 1 = interrupt asserted. 0 = interrupt deasserted
20	MSS_CONFIG_TIMEOUT_IRQ	Status of interrupt 1 = interrupt asserted. 0 = interrupt deasserted
14	SNOC_S1_TIMEOUT_IRQ	Status of interrupt 1 = interrupt asserted. 0 = interrupt deasserted
13	SNOC_S0_TIMEOUT_IRQ	Status of interrupt 1 = interrupt asserted. 0 = interrupt deasserted
9	PCNOC_S9_TIMEOUT_IRQ	Status of interrupt 1 = interrupt asserted. 0 = interrupt deasserted
8	PCNOC_S8_TIMEOUT_IRQ	Status of interrupt 1 = interrupt asserted. 0 = interrupt deasserted
7	PCNOC_S7_TIMEOUT_IRQ	Status of interrupt 1 = interrupt asserted. 0 = interrupt deasserted

TCSR_TIMEOUT_INTR_STATUS (cont.)

Bits	Name	Description
6	PCNOC_S6_TIMEOUT_IRQ	Status of interrupt 1 = interrupt asserted. 0 = interrupt deasserted
5	PCNOC_S5_TIMEOUT_IRQ	Status of interrupt 1 = interrupt asserted. 0 = interrupt deasserted
4	PCNOC_S4_TIMEOUT_IRQ	Status of interrupt 1 = interrupt asserted. 0 = interrupt deasserted
3	PCNOC_S3_TIMEOUT_IRQ	Status of interrupt 1 = interrupt asserted. 0 = interrupt deasserted
2	PCNOC_S2_TIMEOUT_IRQ	Status of interrupt 1 = interrupt asserted. 0 = interrupt deasserted
1	PCNOC_S1_TIMEOUT_IRQ	Status of interrupt 1 = interrupt asserted. 0 = interrupt deasserted
0	PCNOC_S0_TIMEOUT_IRQ	Status of interrupt 1 = interrupt asserted. 0 = interrupt deasserted

0x0194D030 TCSR_TIMEOUT_INTR_RPM_ENABLE**Type:** RW**Clock:** AHB_CLK**Reset State:** 0x00000000

This is interrupt enable register for RPM. Set a bit to 1 enables the corresponding interrupt to propagate

TCSR_TIMEOUT_INTR_RPM_ENABLE

Bits	Name	Description
22	ULTAUDIO_AHBI_TIMEOUT_IRQ_RPM_ENABLE	1 = interrupt enabled. 0 = interrupt disabled
20	MSS_CONFIG_TIMEOUT_IRQ_RPM_ENABLE	1 = interrupt enabled. 0 = interrupt disabled
14	SNOC_S1_TIMEOUT_IRQ_RPM_ENABLE	1 = interrupt enabled. 0 = interrupt disabled

TCSR_TIMEOUT_INTR_RPM_ENABLE (cont.)

Bits	Name	Description
13	SNOC_S0_TIMEOUT_IRQ_RPM_ENABLE	1 = interrupt enabled. 0 = interrupt disabled
9	PCNOC_S9_TIMEOUT_IRQ_RPM_ENABLE	1 = interrupt enabled. 0 = interrupt disabled
8	PCNOC_S8_TIMEOUT_IRQ_RPM_ENABLE	1 = interrupt enabled. 0 = interrupt disabled
7	PCNOC_S7_TIMEOUT_IRQ_RPM_ENABLE	1 = interrupt enabled. 0 = interrupt disabled
6	PCNOC_S6_TIMEOUT_IRQ_RPM_ENABLE	1 = interrupt enabled. 0 = interrupt disabled
5	PCNOC_S5_TIMEOUT_IRQ_RPM_ENABLE	1 = interrupt enabled. 0 = interrupt disabled
4	PCNOC_S4_TIMEOUT_IRQ_RPM_ENABLE	1 = interrupt enabled. 0 = interrupt disabled
3	PCNOC_S3_TIMEOUT_IRQ_RPM_ENABLE	1 = interrupt enabled. 0 = interrupt disabled
2	PCNOC_S2_TIMEOUT_IRQ_RPM_ENABLE	1 = interrupt enabled. 0 = interrupt disabled
1	PCNOC_S1_TIMEOUT_IRQ_RPM_ENABLE	1 = interrupt enabled. 0 = interrupt disabled
0	PCNOC_S0_TIMEOUT_IRQ_RPM_ENABLE	1 = interrupt enabled. 0 = interrupt disabled

0x0194D040 TCSR_TIMEOUT_INTR_APSS_ENABLE**Type:** RW**Clock:** AHB_CLK**Reset State:** 0x00000000

This is interrupt enable register for APSS

TCSR_TIMEOUT_INTR_APSS_ENABLE

Bits	Name	Description
22	ULTAUDIO_AHBI_TIMEOUT_IRQ_APSS_ENABLE	1 = interrupt enabled. 0 = interrupt disabled
20	MSS_CONFIG_TIMEOUT_IRQ_APSS_ENABLE	1 = interrupt enabled. 0 = interrupt disabled

TCSR_TIMEOUT_INTR_APSS_ENABLE (cont.)

Bits	Name	Description
14	SNOC_S1_TIMEOUT_IRQ_APSS_ENABLE	1 = interrupt enabled. 0 = interrupt disabled
13	SNOC_S0_TIMEOUT_IRQ_APSS_ENABLE	1 = interrupt enabled. 0 = interrupt disabled
9	PCNOC_S9_TIMEOUT_IRQ_APSS_ENABLE	1 = interrupt enabled. 0 = interrupt disabled
8	PCNOC_S8_TIMEOUT_IRQ_APSS_ENABLE	1 = interrupt enabled. 0 = interrupt disabled
7	PCNOC_S7_TIMEOUT_IRQ_APSS_ENABLE	1 = interrupt enabled. 0 = interrupt disabled
6	PCNOC_S6_TIMEOUT_IRQ_APSS_ENABLE	1 = interrupt enabled. 0 = interrupt disabled
5	PCNOC_S5_TIMEOUT_IRQ_APSS_ENABLE	1 = interrupt enabled. 0 = interrupt disabled
4	PCNOC_S4_TIMEOUT_IRQ_APSS_ENABLE	1 = interrupt enabled. 0 = interrupt disabled
3	PCNOC_S3_TIMEOUT_IRQ_APSS_ENABLE	1 = interrupt enabled. 0 = interrupt disabled
2	PCNOC_S2_TIMEOUT_IRQ_APSS_ENABLE	1 = interrupt enabled. 0 = interrupt disabled
1	PCNOC_S1_TIMEOUT_IRQ_APSS_ENABLE	1 = interrupt enabled. 0 = interrupt disabled
0	PCNOC_S0_TIMEOUT_IRQ_APSS_ENABLE	1 = interrupt enabled. 0 = interrupt disabled

0x0194D070 TCSR_PRONTO_BUS_TIMEOUT_RESET**Type:** RW**Clock:** AHB_CLK**Reset State:** 0x00000000

This is SW reset register for the u_ahb_bus_timeout_snoc_s3 for Pronto

TCSR_PRONTO_BUS_TIMEOUT_RESET

Bits	Name	Description
0	WCSS_BUS_TIMEOUT_NO_C_SOFT_RESET	1 = reset asserted. 0 = reset de-asserted

0x0194F000 TCSR_MSSQ6_HALTREQ**Type:** RW**Clock:** AHB_CLK**Reset State:** 0x00000000

AXI bus Halt request

TCSR_MSSQ6_HALTREQ

Bits	Name	Description
0	MSSQ6_HALTREQ	AXI bus Halt request

0x0194F004 TCSR_MSSQ6_HALTACK**Type:** R**Clock:** AHB_CLK**Reset State:** 0x00000000

AXI bus Halt request acknowledge

TCSR_MSSQ6_HALTACK

Bits	Name	Description
0	MSSQ6_HALTACK	AXI bus Halt request acknowledge

0x0194F008 TCSR_MSSQ6_MASTER_IDLE**Type:** R**Clock:** AHB_CLK**Reset State:** 0x00000001

AXI bus is idle

TCSR_MSSQ6_MASTER_IDLE

Bits	Name	Description
0	MSSQ6_MASTER_IDLE	AXI bus is idle

0x0194F010 TCSR_MSSQ6_POWER_STATE**Type:** R**Clock:** AHB_CLK**Reset State:** 0x00000000

indicate power state of QDSP6 in MSS

TCSR_MSSQ6_POWER_STATE

Bits	Name	Description
0	MSSQ6_POWER_STATE	1 = power is on 0 = power is off

0x01951000 TCSR_MSS_NC_HALTREQ**Type:** RW**Clock:** AHB_CLK**Reset State:** 0x00000000

AXI bus Halt request

TCSR_MSS_NC_HALTREQ

Bits	Name	Description
0	MSS_NC_HALTREQ	AXI bus Halt request

0x01951004 TCSR_MSS_NC_HALTACK**Type:** R**Clock:** AHB_CLK**Reset State:** 0x00000000

AXI bus Halt request acknowledge

TCSR_MSS_NC_HALTACK

Bits	Name	Description
0	MSS_NC_HALTACK	AXI bus Halt request acknowledge

0x01951008 TCSR_MSS_NC_MASTER_IDLE**Type:** R**Clock:** AHB_CLK**Reset State:** 0x00000001

AXI bus is idle

TCSR_MSS_NC_MASTER_IDLE

Bits	Name	Description
0	MSS_NC_MASTER_IDLE	AXI bus is idle

0x01953000 TCSR_SDC1_HALTREQ**Type:** RW**Clock:** AHB_CLK**Reset State:** 0x00000000

AXI bus Halt request

TCSR_SDC1_HALTREQ

Bits	Name	Description
0	SDC1_HALTREQ	AXI bus Halt request

0x01953004 TCSR_SDC1_HALTACK**Type:** R**Clock:** AHB_CLK**Reset State:** 0x00000000

AXI bus Halt request acknowledge

TCSR_SDC1_HALTACK

Bits	Name	Description
0	SDC1_HALTACK	AXI bus Halt request acknowledge

0x01953008 TCSR_SDC1_MASTER_IDLE**Type:** R**Clock:** AHB_CLK**Reset State:** 0x00000001

AXI bus is idle

TCSR_SDC1_MASTER_IDLE

Bits	Name	Description
0	SDC1_MASTER_IDLE	AXI bus is idle

0x01954000 TCSR_SDC2_HALTREQ**Type:** RW**Clock:** AHB_CLK**Reset State:** 0x00000000

AXI bus Halt request

TCSR_SDC2_HALTREQ

Bits	Name	Description
0	SDC2_HALTREQ	AXI bus Halt request

0x01954004 TCSR_SDC2_HALTACK**Type:** R**Clock:** AHB_CLK**Reset State:** 0x00000000

AXI bus Halt request acknowledge

TCSR_SDC2_HALTACK

Bits	Name	Description
0	SDC2_HALTACK	AXI bus Halt request acknowledge

0x01954008 TCSR_SDC2_MASTER_IDLE**Type:** R**Clock:** AHB_CLK**Reset State:** 0x00000001

AXI bus is idle

TCSR_SDC2_MASTER_IDLE

Bits	Name	Description
0	SDC2_MASTER_IDLE	AXI bus is idle

0x01955000 TCSR_ULT_AUDIO_CORE_ABORT_REQ**Type:** RW**Clock:** AHB_CLK**Reset State:** 0x00000000

ULT audio SSR abort request

TCSR_ULT_AUDIO_CORE_ABORT_REQ

Bits	Name	Description
0	ULT_AUDIO_CORE_ABORT_REQ	ULTaudio abort request for Sub System Reset

0x01955004 TCSR_ULT_AUDIO_CORE_ABORT_ACK**Type:** R**Clock:** AHB_CLK**Reset State:** 0x00000000

ULTaudio SSR abort request acknowledge

TCSR_ULT_AUDIO_CORE_ABORT_ACK

Bits	Name	Description
0	ULT_AUDIO_CORE_ABORT_ACK	ULTaudio abort request acknowledge for Sub System Reset

0x01957000 TCSR_PNOC_SNOC_MEMTYPE**Type:** RW**Clock:** AHB_CLK**Reset State:** 0x0000000A

Mux for selecting pnoc to snoc amemtype, shared and innershared

TCSR_PNOC_SNOC_MEMTYPE

Bits	Name	Description
8	MEM_TYPE_SEL	- 0 = Amemtype value selected from PNOC. - 1 = Amemtype value selected from register phss_pnoc_snoc_memtype.
4	INNERSHARED_VALUE	Ainnershared value for MEM_TYPE_SEL = 1
3	SHARED_VALUE	Ashared value for MEM_TYPE_SEL = 1
2:0	MEM_TYPE_VALUE	Amemtype value for MEM_TYPE_SEL = 1

6 DMA Engine for Hardware Retention BIMC Wrapper

6.1 DEHR_BIMC

DEHR Configuration

6.1.1 DEHR_BIMC_DEHR

DEHR Configuration

0x004B2010 DEHR_BIMC_DEHR_CGC_CFG

Type: RW

Clock: dehr_cfg_clk

Reset State: 0x00000000

DEHR_BIMC_DEHR_CGC_CFG

Bits	Name	Description
0	SW_CGC_DISABLE	When set, it disables the low power feature (no gated clock).

0x004B2104 DEHR_BIMC_DEHR_MSA_CFG

Type: R

Clock: dehr_cfg_clk

Reset State: 0x00000000

This register provides the status of DEHR_ENA and DEHR_LOCK bits of DEHR_SEC_CFG register owned by MSA

DEHR_BIMC_DEHR_MSA_CFG

Bits	Name	Description
4	MSA_DEHR_ENA	When read, returns the DEHR_ENA bit value in DEHR_SEC_CFG register
0	MSA_DEHR_LOCK	When read, returns the DEHR_LOCK bit value in DEHR_SEC_CFG register

0x004B2140 DEHR_BIMC_DEHR_RAM_CFG**Type:** RW**Clock:** dehr_cfg_clk**Reset State:** 0x00000000**DEHR_BIMC_DEHR_RAM_CFG**

Bits	Name	Description
15:0	CFG_BASE_ADDR	Defines the RAM base address for configuration data.

0x004B2200 DEHR_BIMC_DEHR_DMA_STATUS**Type:** RW**Clock:** dehr_cfg_clk**Reset State:** 0x00000000**DEHR_BIMC_DEHR_DMA_STATUS**

Bits	Name	Type	Description
20	RESTORE_IN_PROGRESS	R	When set, it tells us it is in the middle of the Restore operation. It is automatically cleared when RESTORE_DONE bit is set.
16	RESTORE_DONE	RW	The Restore operation is done. It is automatically cleared when next Restore operation starts.
4	SAVE_IN_PROGRESS	R	When set, it tells us it is in the middle of Save operation. This bit is automatically cleared when SAVE_DONE bit is set.
0	SAVE_DONE	RW	The Save Operation is done. It is automatically cleared when next Save operation starts.

0x004B2404 DEHR_BIMC_DEHR_SW_DMA_CFG**Type:** RW**Clock:** dehr_cfg_clk**Reset State:** 0x00000000

The register can be used to manually start Save or Restore operation

DEHR_BIMC_DEHR_SW_DMA_CFG

Bits	Name	Description
8	SW_RESTORE_REQ	When DBG_ENA is set, set this bit to 1 will manually start the Restore operation.
0	SW_SAVE_REQ	When DBG_ENA is set, set this bit to 1 will manually start the Save operation.

0x004B3000 DEHR_BIMC_DEHR_SEC_CFG**Type:** RW**Clock:** dehr_cfg_clk**Reset State:** 0x00000000**DEHR_BIMC_DEHR_SEC_CFG**

Bits	Name	Description
4	DEHR_ENA	When set, DEHR gets the authentication, and is able to do save/restore operations. It cannot be set unless the DEHR_LOCK is also set. It can be cleared at any time
0	DEHR_LOCK	When set, it is not allowed to update the configuration of the DEHR

7 EBI1 Physical Layer Configuration

7.1 DIM_C00_DDRPHY_CA

0x00480000 DIM_C00_DIM_CA_TOP_CFG

Type: RW

Clock: HCLK

Reset State: 0x00000031

The DIM_CA_TOP_CFG register configures the following:

DIM_CA_TOP_CFG

DIM_C00_DIM_CA_TOP_CFG

Bits	Name	Description
24	CA_LB_MODE_SEL	Selects external or internal loop back mode 1'b1: internal loop back mode 1'b0: external loop back mode (default)
23	CA_LB_MODE_EN	Enables external or internal loop back mode 1'b1: loop back mode enabled 1'b0: loop back mode disabled (default)
22	RESERVED	Reserved
21	MISR_DATA_CLK_SEL	selects b/w MISR data or SEED and Loopback clock or FF clock 1'b1: reserved 1'b0: seed from csr ff clk (default)
20	ILOCAL_CTLR_SEL	Select source of PCNT/NCNT/PNCNT_VALID 1'b1: external 1'b0: internal (default)
16	SDR_MODE_EN	1'b1: Enables SDR mode on address bus 1'b0: Enables DDR mode on address bus (default)
8	RESERVED	Reserved

DIM_C00_DIM_CA_TOP_CFG (cont.)

Bits	Name	Description
5	WR_PIPE_EXTEND1	Write pipe extend select (when hp_mode = 1'b1) 1'b1: with pipe extend (default) 1'b0: without pipe extend
4	WR_PIPE_EXTEND0	Write pipe extend select (when hp_mode = 1'b0) 1'b1: with pipe extend (default) 1'b0: without pipe extend
0	WR_CLK_SEL	Write clock select 1'b1: With write CDC (default) 1'b0: Bypass write CDC, write clock from cdiv2 phi2 output

0x00480010 DIM_C00_DIM_CA_PAD_CFG0**Type:** RW**Clock:** HCLK**Reset State:** 0xE0222240

The DIM_CA_PAD_CFG0 register configures the following:

DIM_CA_PAD_CFG0

DIM_C00_DIM_CA_PAD_CFG0

Bits	Name	Description
31	CA_DDR_MODE1	Mode pin for DDR (when hp_mode = 1'b1) 1'b1: DDR2 modes (default) 1'b0: DDR1 modes
30	CA_DDR_MODE0	Mode pin for DDR (when hp_mode = 1'b0) 1'b1: DDR2 modes (default) 1'b0: DDR1 modes
29	CA_LV_MODE	Mode pin for high/low voltage regime
28	CA_ODT_ENA	Enable bit for on-die termination
27:26	CA_ODT	Impedance control bit settings for on-die termination
25:24	CA_PULL	Input pull control 2'b00: no pull (default) 2'b01: pull down 2'b10: keeper 2'b11: pull up
21:20	CA_NSLEW	Slew rate control bits for output path NMOS
17:16	CA_PSLEW	Slew rate control bits for output path PMOS

DIM_C00_DIM_CA_PAD_CFG0 (cont.)

Bits	Name	Description
13:12	CA_NRXDEL	Delay control bits to increase strength of NMOS input drive
9:8	CA_PRXDEL	Delay control bits to increase strength of PMOS input driver
7	CA_VM_SHIFT_ENA	VM (voltage at output of on-die termination) shift enable bit
6:4	CA_ROUT	Impedance control bit settings for output driver
2:0	CA_DCC	Duty cycle correction bits for output path

0x00480014 DIM_C00_DIM_CA_PAD_CFG1**Type:** RW**Clock:** HCLK**Reset State:** 0xE0220440

The DIM_CA_PAD_CFG1 register configures the following:

DIM_CA_PAD_CFG1

DIM_C00_DIM_CA_PAD_CFG1

Bits	Name	Description
31	CK_DDR_MODE1	Mode select for DDR (when hp_mode = 1'b1) 1'b1: DDR2 modes (default) 1'b0: DDR1 modes
30	CK_DDR_MODE0	Mode select for DDR (when hp_mode = 1'b0) 1'b1: DDR2 modes (default) 1'b0: DDR1 modes
29	CK_LV_MODE	Mode select for high/low voltage regime
28	CK_CMFB_ENA	Common mode feedback loop enable
27	CK_ODT_ENA1	Enable bit for on-die termination (when hp_mode = 1'b1)
26	CK_ODT_ENA	Enable bit for on-die termination (when hp_mode = 1'b0)
25:24	CK_ODT	Impedance control bit settings for on-die termination
21:20	CK_NSLEW	Slew rate control bits for output path NMOS
17:16	CK_PSLEW	Slew rate control bits for output path PMOS
13	CK_CUR_MODE1	Current/Voltage mode selection (when hp_mode = 1'b1) 1'b1: Current mode 1'b0: Voltage mode (default)
12	CK_CUR_MODE0	Current/Voltage mode selection (when hp_mode = 1'b0) 1'b1: Current mode 1'b0: Voltage mode (default)

DIM_C00_DIM_CA_PAD_CFG1 (cont.)

Bits	Name	Description
10:8	CK_I_DRV	Control bit settings for bias current
7	CK_VM_SHIFT_ENA	VM (voltage at output of on-die termination) shift enable bit
6:4	CK_ROUT	Impedance control bit settings for output driver
2:0	CK_DCC	Duty cycle correction bits for output path

0x00480018 DIM_C00_DIM_CA_PAD_CFG2**Type:** RW**Clock:** HCLK**Reset State:** 0x1000000A

The DIM_CA_PAD_CFG2 register configures the following:

DIM_CA_PAD_CFG2

DIM_C00_DIM_CA_PAD_CFG2

Bits	Name	Description
29	VREF_LDO_ENA	Enable VREF pad LDO Must be 0 for x2 core as there is no internal LDO
28	VREF_PASSGATE_ENA	Enable pass gate from padsig to vref_int pin (default 1)
27:24	VREF_SP_OUT	Unused
21:18	VREF_SP_IN	Unused
17	VREF_BYPASS_ENA	Enable bypass gate from resistive divider to vref_int pin
16	VREF_RDIV_ENA	Enable resistive divider for VREF voltage control
13:8	VREF_PULLDN_SPEED_CNTL	Unused
5:0	VREF_LEVEL_CNTL	Control over the VREF voltage level through resistive divider

0x0048001C DIM_C00_DIM_CA_PAD_CFG3**Type:** RW**Clock:** HCLK**Reset State:** 0x30300000

The DIM_CA_PAD_CFG3 register configures the following:

DIM_CA_PAD_CFG3

DIM_C00_DIM_CA_PAD_CFG3

Bits	Name	Description
31:30	CS_N_IE	
29:28	CS_N_OE	
25	CK_IE	
24	CK_OE	
23:22	CKE_IE	
21:20	CKE_OE	
19:10	CA_IE	
9:0	CA_OE	

0x00480020 DIM_C00_DIM_CA_PAD_CFG4**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

The DIM_CA_PAD_CFG4 register configures the following:

DIM_CA_PAD_CFG4

DIM_C00_DIM_CA_PAD_CFG4

Bits	Name	Description
31:30	CS_N_OE_DYN_ENA	Enable dynamic control of CS_N OE 1'b1: Dynamic control enabled and CS_N_OE_DYN[n] bits are OR-ed with common OE from controller 1'b0: (default) Dynamic control disabled, CS_N OE gated by CA_PAD_CFG3 bits Note: This register should not be set in external loopback mode since it will change the IE control source.
29:28	CS_N_OE_DYN	Dynamic OE control for each of the CS_N outputs, OR-ed with common controller OE.
23:22	CKE_OE_DYN_ENA	Enable dynamic control of CKE OE 1'b1: Dynamic control enabled and CKE_OE_DYN[n] bits are OR-ed with common OE from controller 1'b0: (default) Dynamic control disabled, CKE OE gated by CA_PAD_CFG3 bits
21:20	CKE_OE_DYN	Dynamic OE control for each of the CA outputs, OR-ed with common controller OE.

DIM_C00_DIM_CA_PAD_CFG4 (cont.)

Bits	Name	Description
19:10	CA_OE_DYN_ENA	Enable dynamic control of CA OE 1'b1: Dynamic control enabled and CA_OE_DYN[n] bits are OR-ed with common OE from controller 1'b0: (default) Dynamic control disabled, CA OE gated by CA_PAD_CFG3 bits
9:0	CA_OE_DYN	Dynamic OE control for each of the CA outputs, OR-ed with common controller OE.

0x00480024 DIM_C00_DDRPHY_CA_MISR_CFG**Type:** RW**Clock:** HCLK**Reset State:** 0x872C56BD

The DDRPHY_CA_MISR_CFG register configures the following:

DDRPHY_CA_MISR_CFG

DIM_C00_DDRPHY_CA_MISR_CFG

Bits	Name	Description
31:0	SEED_VALUE	Loopback MISR seed

0x00480028 DIM_C00_DDRPHY_CA_MISR_EVEN_STATUS**Type:** R**Clock:** HCLK**Reset State:** 0x00000000

The DDRPHY_CA_MISR_EVEN_STATUS register configures the following:

DDRPHY_CA_MISR_EVEN_STATUS

DIM_C00_DDRPHY_CA_MISR_EVEN_STATUS

Bits	Name	Description
31:0	SIGNATURE	Even clock Loopback MISR signature

0x0048002C DIM_C00_DDRPHY_CA_MISR_ODD_STATUS**Type:** R**Clock:** HCLK**Reset State:** 0x00000000

The DDRPHY_CA_MISR_ODD_STATUS register configures the following:

DDRPHY_CA_MISR_ODD_STATUS

DIM_C00_DDRPHY_CA_MISR_ODD_STATUS

Bits	Name	Description
31:0	SIGNATURE	Odd Clock Loopback MISR signature

0x00480030 DIM_C00_DIM_CA_CDC_CTLR_CFG0**Type:** RW**Clock:** HCLK**Reset State:** 0x02000000

The DIM_CA_CDC_CTLR_CFG0 register configures the following:

DIM_CA_CDC_CTLR_CFG0

DIM_C00_DIM_CA_CDC_CTLR_CFG0

Bits	Name	Description
25	ERROR_CODE_UPDATE_EN	1'b1: Enable overwriting (default) the error code for subsequent calibrations. 1'b0: Disable overwriting the error code during subsequent calibrations
24	TRACKING_CAL_ENA	1'b1: Relationship between subunits and oscillator count is learnt and used for tracking calibration. 1'b0: Periodic calibration triggers cause full recalibration
23:20	OSC_COUNT_ERR_TOLERANCE	Magnitude of the difference between oscillator count and target count when the algorithm assumes convergence and stops further search.
19	TRACK_CALIB_MODE	1'b0: Tracking calibration uses previous Target_count value. 1'b1 Accurate mode: Target_count is recomputed at the start of tracking calibration
18	FULL_DELAY	1'b0: unitstep and subunit caps are from calibration or SW settings 1'b1: All unit step and subunit caps are selected for the delay arrays.
17	HW_AUTOCAL_ENA	1'b1: Enables periodic auto calibration 1'b0: Periodic calibration is disabled

DIM_C00_DIM_CA_CDC_CTLR_CFG0 (cont.)

Bits	Name	Description
16	SW_TRIG_FULL_CALIB	A low to high transition on this register bit triggers the CDC calibration statemachine to initiate full calibration of the standby delay array. Also resets tracking mode for the delay arrays.
13:12	OSC_PRE_DIV	The factor by which the raw oscillator frequency is divided before being given to the counting logic. 2'b00 Divide by 1 2'b01 Divide by 2 2'b10 Divide by 4 2'b11 Divide by 8
10:0	TMUX_CHAR	Cchar is a characterized constant equal to the ratio of output mux offset delay to average unit step delay. This register is programmed by multiplying the constant with 1024 and rounding to the nearest integer. E.g. if Cchar = 0.67, $RO(0.67 * 1024) = 686$, hence program TMUX_CHAR = 11'h2AE

0x00480034 DIM_C00_DIM_CA_CDC_CTLR_CFG1**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

The DIM_CA_CDC_CTLR_CFG1 register configures the following:

DIM_CA_CDC_CTLR_CFG1

DIM_C00_DIM_CA_CDC_CTLR_CFG1

Bits	Name	Description
26:24	OSC_COUNT_DELAY	Number of oscillator cycles from the end of reference gate signal to assertion of oscillation counter done. Legal values are from 0 to 4.
23:20	STANDBY_DELAY	This is the number of reference clock cycles from when a newly calibrated delay array is made active to when the current active delay array is put into standby mode. 4'b0: Disables the counter. Hardware waits for explicit acknowledgement of delay array switch from CDC. Others: Explicit acknowledgement mode is disabled and the counter is used
18:16	DEL_MODE_DELAY	Number of reference clock cycles between putting a newly calibrated delay array in delay mode to selecting it as the active delay array. The minimum allowed value is 1
14:12	OSC_MODE_DELAY	Number of reference clock cycles between putting a delay array in oscillator mode to starting the oscillation counter. This delay is allowed for the oscillations to stabilize through the delay array. The minimum allowed value is 1

DIM_C00_DIM_CA_CDC_CTLR_CFG1 (cont.)

Bits	Name	Description
10:8	DECODER_DELAY	Number of reference clock cycles allowed for unitstep and subunit counts to be encoded into the unit and finecap select lines on the DDA. The minimum allowed value is 1.
6:4	DIVIDER_DELAY	Number of reference clock cycles allowed for arithmetic division operations. The minimum allowed value is 1
2:0	MULTIPLIER_DELAY	Number of reference clock cycles allowed for arithmetic multiplication operations. The minimum allowed value is 1

0x00480038 DIM_C00_DIM_CA_CDC_CAL_TIMER_CFG0**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

The DIM_CA_CDC_CAL_TIMER_CFG0 register configures the following:

DIM_CA_CDC_CAL_TIMER_CFG0

DIM_C00_DIM_CA_CDC_CAL_TIMER_CFG0

Bits	Name	Description
24	INVALID_TIMER_ENA	1'b1: Enable the timer counting number of missed calibration intervals after which the CDC is invalid. 1'b0: Timer disabled A low to high transition on this register bit loads the timer with programmed value below.
23:20	INVALID_TIMER_VAL	Number of missed calibration intervals after which CDC invalid state is asserted.
16	TIMER_ENA	1'b1: Enables 32 kHz sleep clock based periodic autocalibration timer. 1'b0: Timer disabled A low to high transition on this register bit loads the timer with programmed value below.
15:0	TIMER_VAL	Count value for periodic autocalibration interval based on sleep clock.

0x0048003C DIM_C00_DIM_CA_CDC_CAL_TIMER_CFG1**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

The DIM_CA_CDC_CAL_TIMER_CFG1 register configures the following:

DIM_CA_CDC_CAL_TIMER_CFG1

DIM_C00_DIM_CA_CDC_CAL_TIMER_CFG1

Bits	Name	Description
12	FF_TIMER_ENA	1'b1: Enables reference clock based periodic autocalibration timer. 1'b0: Timer disabled A low to high transition on this register bit loads the timer with programmed value below.
9:0	FF_TIMER_VAL	Count value for periodic autocalibration interval based on reference clock.

0x00480040 DIM_C00_DIM_CA_CDC_REFCOUNT_CFG

Type: RW

Clock: HCLK

Reset State: 0x00000000

The DIM_CA_CDC_REFCOUNT_CFG register configures the following:

DIM_CA_CDC_REFCOUNT_CFG

DIM_C00_DIM_CA_CDC_REFCOUNT_CFG

Bits	Name	Description
31:16	TREF	Reference clock period programmed in pico seconds.
13:8	CCAL_REF_COUNT	Number of reference clock cycles used for oscillation counts during coarse calibration. Maximum supported value is 59.
5:0	FCAL_REF_COUNT	Number of reference clock cycles used for oscillation counts during fine calibration. Maximum supported value is 59

0x00480044 DIM_C00_DIM_CA_CDC_COARSE_CAL_CFG

Type: RW

Clock: HCLK

Reset State: 0x00000000

The register DIM_CA_CDC_COARSE_CAL_CFG configures the following:

DIM_CA_CDC_COARSE_CAL_CFG

DIM_C00_DIM_CA_CDC_COARSE_CAL_CFG

Bits	Name	Description
12:8	CCAL_SUBUNIT_CAPS	Subunit count needed for an effective unit step count of 32 during coarse calibration. This value is also used by software oscillator mode.
5:0	CCAL_UNITSTEPS	unitstep count needed for an effective unit step count of 32 during coarse calibration.

0x00480048 DIM_C00_DIM_CA_CDC_RSVD_CFG**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

The DIM_CA_CDC_RSVD_CFG register configures the following:

DIM_CA_CDC_RSVD_CFG

DIM_C00_DIM_CA_CDC_RSVD_CFG

Bits	Name	Description
15:0	TEMP_FIELD	Reserved register

0x0048004C DIM_C00_DIM_CA_CDC_OFFSET_CFG**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

The DIM_CA_CDC_OFFSET_CFG register configures the following:

DIM_CA_CDC_OFFSET_CFG

DIM_C00_DIM_CA_CDC_OFFSET_CFG

Bits	Name	Description
20	SUBUNIT_OFFSET_MODE	1'b1: The programmed offset is directly loaded as subunit count 1'b0: Programmed offset is added/subtracted from the subunit count
19	SUBUNIT_OFFSET_SIGN	1'b1: Offset is subtracted from the subunit count 1'b0: Offset is added to the subunit count
16:12	SUBUNIT_OFFSET	Software subunit offset

DIM_C00_DIM_CA_CDC_OFFSET_CFG (cont.)

Bits	Name	Description
8	UNIT_OFFSET_MODE	1'b1: The programmed offset is directly loaded as unit step count 1'b0: Programmed offset is added/subtracted from the unit step count
7	UNIT_OFFSET_SIGN	1'b1: Offset is subtracted from the unitstep count 1'b0: Offset is added to the unitstep count
5:0	UNIT_OFFSET	Software unitstep offset

0x00480050 DIM_C00_DIM_CA_CDC_DELAY_CFG**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

The DIM_CA_CDC_DELAY_CFG register configures the following:

DIM_CA_CDC_DELAY_CFG

DIM_C00_DIM_CA_CDC_DELAY_CFG

Bits	Name	Description
28	TARGET_COUNT_ENA	1'b1: Enable software specified Target_count for calibration instead of value calculated by hardware. 1'b0: Hardware computed Target_count used.
27:16	TARGET_COUNT	Software specified Target_count to which the statemachine calibrates
11:0	DELAY_VAL	Delay required from CDC programmed in pico seconds. This value has to be greater than the minimum delay supported by DDA Tmin.

0x00480054 DIM_C00_DIM_CA_CDC_SW_MODE_CFG**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

The DIM_CA_CDC_SW_MODE_CFG register configures the following:

DIM_CA_CDC_SW_MODE_CFG

DIM_C00_DIM_CA_CDC_SW_MODE_CFG

Bits	Name	Description
3	SW_DEL_MODE	A low to high transition on this register bit puts the delay array selected by SW_DA_SEL into delay mode and connects it to the DDA output. This register has to remain set for the duration of SW controlled delay mode.
2	SW_OSC_MODE	Reserved register.
1	SW_DA_SEL	Selects the delay array inside DDA during software delay or oscillator mode.
0	SW_LOAD	1'b1: Enable loading of the software offsets from the offset configuration register. This register has to remain set for the duration of SW mode operation.

0x0048005C DIM_C00_DIM_CA_CDC_SW_OVRD_CFG**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000**DIM_C00_DIM_CA_CDC_SW_OVRD_CFG**

Bits	Name	Description
16	SW_REF_GATE	Write 1'b1 to start the reference window. Write 1'b0 to end the reference window. Number of reference cycle in the window will be reported in SW_REF_GATE_COUNT in CDC_STATUS0
11	SW_OVRD_DA1_OSC_EN	In software override mode, 1'b1 written to this bit puts delay array 1 in oscillator-mode
10	SW_OVRD_DA1_IN_EN	In software override mode, 1'b1 written to this bit puts delay array 1 in delay-mode
9	SW_OVRD_DA0_OSC_EN	In software override mode, 1'b1 written to this bit puts delay array 0 in oscillator-mode
8	SW_OVRD_DA0_IN_EN	In software override mode, 1'b1 written to this bit puts delay array 0 in delay-mode
3	SW_OVRD_CDC_COUNTEN R_RST	1'b1 written to this bit resets the oscillation counter. Follow this with 1'b0 before setting reference gate.
2	SW_OVRD_LOAD_DA_SEL	In software override mode, this bit select the delay array to which unit and subunit offset values from CDC_OFFSET_CFG register will be loaded.
1	SW_OVRD_ACTV_DA_SEL	In software override mode, this register selects the delay array connected to the CDC output
0	SW_OVRD_ENA	1'b1 written to this bit puts the CDC interface in software override mode where the below register bits control the CDC. 1'b0: CDC is under hardware control

0x00480070 DIM_C00_DIM_CA_CDC_STATUS0**Type:** R**Clock:** HCLK**Reset State:** 0x0000000C

The DIM_CA_CDC_STATUS0 register configures the following:

DIM_CA_CDC_STATUS0

DIM_C00_DIM_CA_CDC_STATUS0

Bits	Name	Description
26:24	CDC_ERROR_CODE	Error code for internal errors: 3'b000: No error 3'b001: Programmed delay less than Tmux 3'b010: Programmed delay less than Tmin 3'b011: Unit-step overflow: Overflow during fine calibration. Desired delay cannot be realized even when all unit-steps are selected. 3'b100: Unit-step underflow 3'b101: Unit-step overflow: Overflow during coarse calibration. Estimated unit steps needed to realize the desired delay are greater than the total number of unit-steps in design. Note: The CDC_ERROR_CODE is not a sticky register and is updated after each CDC calibration.
23:16	SW_REF_GATE_COUNT	Number of reference clock cycles in the software generated oscillation gate window doing SW override mode
15:4	OSC_COUNT	Current value of oscillator count
3	CURR_SEL_DA	This bit indicates the current active delay array connected to the output. 1'b0: DA0 is selected 1'b1: DA1 is selected
2	CTLR_SM_IDLE	1'b1 Indicates that the calibration state machine is idle 1'b0 indicates calibration statemachine is busy
1	OSC_DONE	This status register is set after oscillations count is finished by CDC. It gets reset when a new oscillator count is started.
0	CALIBRATION_DONE	1'b1 indicates that calibration is done and CURR_SUBUNIT_STEPS, CURR_UNIT_STEPS status registers are valid.

0x00480074 DIM_C00_DIM_CA_CDC_STATUS1**Type:** R**Clock:** HCLK**Reset State:** 0x00000000

The DIM_CA_CDC_STATUS1 register configures the following:

DIM_CA_CDC_STATUS1

DIM_C00_DIM_CA_CDC_STATUS1

Bits	Name	Description
27:16	CURR_DELAY_VALUE	HW Computed value of current delay after calibration by the state machine. This register is valid if CALIBRATION_DONE status bit is set.
11:0	CURR_TMUX_DELAY	Current value of output mux offset delay computed by HW.

0x00480078 DIM_C00_DIM_CA_CDC_STATUS2

Type: R

Clock: HCLK

Reset State: 0x10331033

The DIM_CA_CDC_STATUS2 register configures the following:

DIM_CA_CDC_STATUS2

DIM_C00_DIM_CA_CDC_STATUS2

Bits	Name	Description
28:24	DA1_SUBUNITS	Current value of subunit cap settings applied to delay array 1
21:16	DA1_UNITSTEPS	Current value of unit steps applied to delay array 1
12:8	DA0_SUBUNITS	Current value of subunit cap settings applied to delay array 0
5:0	DA0_UNITSTEPS	Current value of unit steps applied to delay array 0

0x0048007C DIM_C00_DIM_CA_CDC_STATUS3

Type: R

Clock: HCLK

Reset State: 0x00000000

The DIM_CA_CDC_STATUS3 register configures the following:

DIM_CA_CDC_STATUS3

DIM_C00_DIM_CA_CDC_STATUS3

Bits	Name	Description
31:24	NUM_OF_OSC_ITER	Number of oscillator iterations before convergence was achieved during tracking

DIM_C00_DIM_CA_CDC_STATUS3 (cont.)

Bits	Name	Description
23:12	COUNT_ERROR	Difference between target and actual oscillator counts at the start of tracking calibration
11:0	CURR_TARGET_COUNT	HW computed value of current Target_count used by the calibration state machine.

0x004800E0 DIM_C00_DIM_CA_IOC_CTLR_CFG**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

The DIM_CA_IOC_CTLR_CFG register configures the following:

DIM_CA_IOC_CTLR_CFG

DIM_C00_DIM_CA_IOC_CTLR_CFG

Bits	Name	Description
31	CAL_NOW	SW: RW Set this bit to 1'b1 will cause the IO calibration starts immediately. 0x0: NO_OP (no-op) 0x1: START_IOCAL_IMMEDIATELY (start IOCal immediately This bit has to be cleared after the calibration has been done.)
30	IO_CAL_AUTO	Periodic auto calibration mode. Writing a '1' to this bit will trigger periodic auto calibration with the period specified in IOC_CTLR_TIMER_CFG register. If '0', it disables the timer based on sleep clock. Note that this does not impact the timer based on fixed frequency clock (ffclk).
29	IO_CAL_FF_TIMER_EN	Fixed Frequency Timer mode. Writing a '1' to this bit will set the timer running off tcxo clock. If '0', it disables the timer based on fixed frequency clock.
28	IO_CAL_BANDGAP_DYN_CTRL	Enable dynamic control of the bandgap element: This low-power feature is only available on x2 core. Bit reserved for x0/x1 cores. 1'b1: Enabled - bandgap element turned on only during IO calibration or when current mode is enabled. 1'b0: Disabled (default) - bandgap element turned on/off statically based on BANDHGAP_ENA0/1 bits.
25	SW_FFCLK_ON	Writing a '1' to this field will turn on the fixed frequency (xo) clock on signal
24	LV_MODE	SW: RW Enable/Disable low-voltage mode (MIF2 pad only) 0x0: ENUM_1_8V (1.8V) 0x1: NON_1_8V (non-1.8V)

DIM_C00_DIM_CA_IOC_CTLR_CFG (cont.)

Bits	Name	Description
20:16	MARGIN_LOAD	SW: RW If the difference between the current IOCal result is greater than the last result by the number specified here, then the IOCal controller will request value update to DDR controller. 0x0: ALWAYS_UPDATE (always update)
13:12	IMP_SEL	SW: RW Select bits to choose which impedance to calibrate to
10	PN_SEL_CA	SW: RW Enables loading of HW calibrated values for ca pads. Writing a '1' to this bit would let the IOCAL to request traffic stall on ca pads when it is time to update the pads with the new value.
9	PN_SEL_DATA	SW: RW Enables loading of HW calibrated values for dq/dqs pads. Writing a '1' to this bit would let the IOCAL to request traffic stall on data pads when it is time to update the pads with the new value.
8	CAL_USE_LAST	SW: RW Select the initial value to start IO calibration 0x0: START_FROM_A_FIXED_VALUES_SPECIFIED_IN_IOC_CTLR_PNCNT_CFG (start from a fixed values specified in IOC_CTLR_PNCNT_CFG (default)) 0x1: START_FROM_PREVIOUS_IOCAL_PNCNT_RESULTS (start from previous IOCal PNCNT results)
6:4	SAMPLE_POINT	SW: RW Specify number of samples per measure point 0x0: ENUM_1 (1) 0x1: ENUM_3 (3) 0x2: ENUM_5 (5) 0x3: ENUM_7 (7) 0x4: ENUM_9 (9) 0x5: ENUM_11 (11) 0x6: ENUM_13 (13) 0x7: ENUM_15 (15)
3	DDR_MODE1	
2	DDR_MODE0	
1	BANDGAP_ENA1	SW: RW Pin to enable bandgap independent of diff_ctl 0x0: DISABLE (disable) 0x1: ENABLE (enable)
0	BANDGAP_ENA0	SW: RW Pin to enable bandgap independent of diff_ctl 0x0: DISABLE (disable) 0x1: ENABLE (enable)

0x004800E4 DIM_C00_DIM_CA_IOC_CTLR_PNCNT_CFG**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

The DIM_CA_IOC_CTLR_CFG register configures the following:

This register contains the initial value for the next calibration to start from. Software can set the hardware to either start calibration from a fixed value or from previous results. If software chooses to use fixed value by setting CAN_USE_LAST in register IOC_CTLR_CFG to be 0, then the values in this register will be used.

DIM_CA_IOC_CTLR_CFG

DIM_C00_DIM_CA_IOC_CTLR_PNCNT_CFG

Bits	Name	Description
12:8	NCNT_INIT_CSR	SW: RW Starting PCNT value used in calibration. Only take effect when CAN_USE_LAST is asserted
4:0	PCNT_INIT_CSR	SW: RW Starting NCNT value used in calibration. Only take effect when CAN_USE_LAST is asserted

0x004800E8 DIM_C00_DIM_CA_IOC_CTLR_TIMER_CFG**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

The DIM_CA_IOC_CTLR_TIMER_CFG register configures the following:

DIM_CA_IOC_CTLR_TIMER_CFG

DIM_C00_DIM_CA_IOC_CTLR_TIMER_CFG

Bits	Name	Description
31:16	TIMER_PERIOD	SW: RW Recalibration Period. The period is measured in timer clock cycles. Typically it's a 32kHz clock. The minimum period that can be programmed is 3. 0,1,2: Invalid values
15:0	FF_TIMER_PERIOD	SW: RW Recalibration Period for the timer running of xo clock. 0x0: INVALID (Invalid)

0x004800EC DIM_C00_DIM_CA_IOC_CTLR_TIMER_STATUS**Type:** R**Clock:** HCLK**Reset State:** 0x00000000

The DIM_CA_IOC_CTLR_TIMER_STATUS register configures the following:

DIM_CA_IOC_CTLR_TIMER_STATUS

DIM_C00_DIM_CA_IOC_CTLR_TIMER_STATUS

Bits	Name	Description
15:0	TIMER_STATUS	Current Auto Calibration Timer value. As this register is written in sleep clock domain and read in xo clock domain and no hardware synchronization in place, It is required by the software to read this register 4 times to get the correct value.

0x004800F4 DIM_C00_DIM_CA_IOC_CTLR_STATUS**Type:** R**Clock:** HCLK**Reset State:** 0x00011010

The DIM_CA_IOC_CTLR_STATUS register configures the following:

DIM_CA_IOC_CTLR_STATUS

DIM_C00_DIM_CA_IOC_CTLR_STATUS

Bits	Name	Description
31	INIT_ILOCAL_DONE	SW: R The very first IO Calibration is finished This bit is sticky. once it becomes 1'b1 until software writes it back to 0. 0x0: INIT_CAL_NEVER_DONE (Init-cal never done) 0x1: INIT_CAL_FINISHED (Init-cal finished)
18	ILOCAL_DONE_D	SW: R IO Calibration is finished 0x0: IN_PROGRESS (in progress) 0x1: FINISHED (finished)
17	ILOCAL_BUSY	SW: R Status of calibration State machine 0x0: IDLE (idle) 0x1: BUSY (busy)

DIM_C00_DIM_CA_IOC_CTLR_STATUS (cont.)

Bits	Name	Description
16	SYNC_COMP	SW: R comp value from IOCal pad
12:8	NCNT_HOLD	SW: R Current NCNT value used
4:0	PCNT_HOLD	SW: R Current PCNT value used

0x00480100 DIM_C00_DIM_CA_CA_IOC_SLV_CFG**Type:** RW**Clock:** HCLK**Reset State:** 0x10100000

The DIM_CA_CA_IOC_SLV_CFG register configures the following:

This register specifies the overriding pcnt and ncnt values. Overriding mode is controlled by PNCNT_HW_LOAD_EN bit in this register.

DIM_CA_CA_IOC_SLV_CFG

DIM_C00_DIM_CA_CA_IOC_SLV_CFG

Bits	Name	Description
31	PNCNT_HW_LOAD_EN	Select PCNT/NCNT Source for pads. This is the mux control that chooses either auto calibrated value or the software override value. 0x0: OVERRIDE (override (default)) 0x1: AUTO_CAL (auto cal)
28:24	NCNT_SW_VAL	SW: RW NCNT Override Value for the pads
20:16	PCNT_SW_VAL	SW: RW PCNT Override Value for the pads
15	NCNT_OFFSET_SIGN	controls if the NOFFSET is added for subtracted on top of calibration result 0x0: ADD (Add) 0x1: SUBTRACT (Subtract)
12:8	NCNT_SW_OFFSET	SW: RW The offset applied on top of NCNT value from calibration before it gets sent to IO pads 0x0: COUNT_0 (count 0(default))

DIM_C00_DIM_CA_CA_IOC_SLV_CFG (cont.)

Bits	Name	Description
7	PCNT_OFFSET_SIGN	controls if the POFFSET is added for subtracted on top of calibration result 0x0: ADD (Add) 0x1: SUBTRACT (Subtract)
4:0	PCNT_SW_OFFSET	SW: RW The offset applied on top of PCNT value from calibration before it gets sent to IO pads 0x0: COUNT_0 (count 0(default))

0x00480104 DIM_C00_DIM_CA_CA_IOC_SLV_STATUS**Type:** R**Clock:** HCLK**Reset State:** 0x00001010

The DIM_CA_CA_IOC_SLV_STATUS register configures the following:

DIM_CA_CA_IOC_SLV_STATUS

DIM_C00_DIM_CA_CA_IOC_SLV_STATUS

Bits	Name	Description
12:8	PAD_NCNT	The final NCNT value that is applied to the pads.
4:0	PAD_PCNT	The final PCNT value that is applied to the pads.

0x00480110 DIM_C00_DIM_CA_CK_IOC_SLV_CFG**Type:** RW**Clock:** HCLK**Reset State:** 0x10100000

The DIM_CA_CK_IOC_SLV_CFG register configures the following:

DIM_CA_CK_IOC_SLV_CFG

DIM_C00_DIM_CA_CK_IOC_SLV_CFG

Bits	Name	Description
31	PNCNT_HW_LOAD_EN	Select PCNT/NCNT Source for pads. This is the mux control that chooses either auto calibrated value or the software override value. 0x0: OVERRIDE (override (default)) 0x1: AUTO_CAL (auto cal)

DIM_C00_DIM_CA_CK_IOC_SLV_CFG (cont.)

Bits	Name	Description
28:24	NCNT_SW_VAL	SW: RW NCNT Override Value for the pads
20:16	PCNT_SW_VAL	SW: RW PCNT Override Value for the pads
15	NCNT_OFFSET_SIGN	controls if the NOFFSET is added for subtracted on top of calibration result 0x0: ADD (Add) 0x1: SUBTRACT (Subtract)
12:8	NCNT_SW_OFFSET	SW: RW The offset applied on top of NCNT value from calibration before it gets sent to IO pads 0x0: COUNT_0 (count 0(default))
7	PCNT_OFFSET_SIGN	controls if the POFFSET is added for subtracted on top of calibration result 0x0: ADD (Add) 0x1: SUBTRACT (Subtract)
4:0	PCNT_SW_OFFSET	SW: RW The offset applied on top of PCNT value from calibration before it gets sent to IO pads 0x0: COUNT_0 (count 0(default))

0x00480114 DIM_C00_DIM_CA_CK_IOC_SLV_STATUS**Type:** R**Clock:** HCLK**Reset State:** 0x00001010

The DIM_CA_CK_IOC_SLV_STATUS register configures the following:

DIM_CA_CK_IOC_SLV_STATUS

DIM_C00_DIM_CA_CK_IOC_SLV_STATUS

Bits	Name	Description
12:8	PAD_NCNT	The final NCNT value that is applied to the pads.
4:0	PAD_PCNT	The final PCNT value that is applied to the pads.

7.2 DIM_D00_DDRPHY_DQ

0x00480800 DIM_D00_DIM_DQ_TOP_CFG

Type: RW

Clock: HCLK

Reset State: 0x00000031

The DIM_DQ_TOP_CFG register configures the following:

DIM_DQ_TOP_CFG

DIM_D00_DIM_DQ_TOP_CFG

Bits	Name	Description
30	DQ_LB_MODE_SEL	Selects external or internal loop back mode 1'b1: internal loop back mode 1'b0: external loop back mode (default)
29	DQ_LB_MODE_EN	Enables external or internal loop back mode 1'b1: loop back mode enabled 1'b0: loop back mode disabled (default)
28	RESERVED	Reserved
27	MISR_DATA_CLK_SEL	selects b/w MISR data or SEED and Loopback clock or FF clock 1'b1: from dte 1'b0: seed from csr ff clk (default)
26	CDC_LDO_EN	Enablement of CDC LDO 1'b1: Enabled 1'b0: Disabled LDO and power provided from switches (default)
25	CDC_SWITCH_RC_EN	Enablement of CDC power RC (LPF) switch 1'b1: Enabled 1'b0: Disabled (default)
24	CDC_SWITCH_BYPASS_OF F	Enablement of CDC power bypass switch 1'b1: Disabled 1'b0: Enabled (default)
16	RCW_EN	Enablement of the Read Capture Window 1'b1: Enable the RCW 1'b0: Disabled (default)
8	RESERVED	Reserved
5	WR_PIPE_EXTEND1	Write pipe extend select (when hp_mode = 1'b1) 1'b1: with pipe extend (default) 1'b0: without pipe extend
4	WR_PIPE_EXTEND0	Write pipe extend select (when hp_mode = 1'b0) 1'b1: with pipe extend (default) 1'b0: without pipe extend

DIM_D00_DIM_DQ_TOP_CFG (cont.)

Bits	Name	Description
0	WR_CLK_SEL	Write clock select 1'b1: With write CDC (default) 1'b0: Bypass write CDC, write clock from cdiv2 phi2 output

0x00480810 DIM_D00_DIM_DQ_PAD_CFG0**Type:** RW**Clock:** HCLK**Reset State:** 0xE0222240

The DIM_DQ_PAD_CFG0 register configures the following:

DIM_DQ_PAD_CFG0

DIM_D00_DIM_DQ_PAD_CFG0

Bits	Name	Description
31	DQ_DDR_MODE1	Mode pin for DDR (when hp_mode = 1'b1) 1'b1: DDR2 modes (default) 1'b0: DDR1 modes
30	DQ_DDR_MODE0	Mode pin for DDR (when hp_mode = 1'b0) 1'b1: DDR2 modes (default) 1'b0: DDR1 modes
29	DQ_LV_MODE	Mode select for high/low voltage regime
28	DQ_ODT_ENA	Unused - ODT controlled by PAD_CFG3
27:26	DQ_ODT	Unused - ODT controlled by PAD_CFG3
25:24	DQ_PULL_B	Input pull control
21:20	DQ_NSLEW	Slew rate control bits for output path NMOS
17:16	DQ_PSLEW	Slew rate control bits for output path PMOS
13:12	DQ_NRXDEL	Delay control bits to increase strength of NMOS input drive
9:8	DQ_PRXDEL	Delay control bits to increase strength of PMOS input driver
7	DQ_VM_SHIFT_ENA	VM (voltage at output of on-die termination) shift enable bit
6:4	DQ_ROUT	Impedance control bit settings for output driver
2:0	DQ_DCC	Duty cycle correction bits for output path

0x00480814 DIM_D00_DIM_DQ_PAD_CFG1**Type:** RW**Clock:** HCLK**Reset State:** 0xE0222240

The DIM_DQ_PAD_CFG1 register configures the following:

DIM_DQ_PAD_CFG1

DIM_D00_DIM_DQ_PAD_CFG1

Bits	Name	Description
31	DQS_DDR_MODE1	Mode pin for DDR (when hp_mode = 1'b1) 1'b1: DDR2 modes (default) 1'b0: DDR1 modes
30	DQS_DDR_MODE0	Mode pin for DDR (when hp_mode = 1'b0) 1'b1: DDR2 modes (default) 1'b0: DDR1 modes
29	DQS_LV_MODE	Mode select for high/low voltage regime
28	DQS_ODT_ENA	Unused - ODT controlled by PAD_CFG3
27:26	DQS_ODT	Unused - ODT controlled by PAD_CFG3
25:24	DQS_PULL	Input pull control 2'b00: no pull (default) 2'b01: pull down 2'b10: keeper 2'b11: pull up
21:20	DQS_NSLEW	Slew rate control bits for output path NMOS
17:16	DQS_PSLEW	Slew rate control bits for output path PMOS
13:12	DQS_NRXDEL	Delay control bits to increase strength of NMOS input drive
9:8	DQS_PRXDEL	Delay control bits to increase strength of PMOS input drive
7	DQS_VM_SHIFT_ENA	VM (voltage at output of on-die termination) shift enable bit
6:4	DQS_ROUT	Impedance control bit settings for output driver
2:0	DQS_DCC	Duty cycle correction bits for output path

0x00480818 DIM_D00_DIM_DQ_PAD_CFG2**Type:** RW**Clock:** HCLK**Reset State:** 0x1000000A

The DIM_DQ_PAD_CFG2 register configures the following:

DIM_DQ_PAD_CFG2

DIM_D00_DIM_DQ_PAD_CFG2

Bits	Name	Description
29	VREF_LDO_ENA	Enable VREF pad LDO Must be 0 for x2 core as there is no internal LDO
28	VREF_PASSGATE_ENA	Enable pass gate from padsig to vref_int pin (default 1)
27:24	VREF_SP_OUT	Unused
21:18	VREF_SP_IN	Unused
17	VREF_BYPASS_ENA	Enable bypass gate from resistive divider to vref_int pin
16	VREF_RDIV_ENA	Enable resistive divider for VREF voltage control
13:8	VREF_PULLDN_SPEED_CN TL	Unused
5:0	VREF_LEVEL_CNTL	Control over the VREF voltage level through resistive divider

0x0048081C DIM_D00_DIM_DQ_DQ_SMT_STATUS

Type: R

Clock: HCLK

Reset State: 0x00000000

The DIM_DQ_DQ_SMT_STATUS register reflects the status of SMT signals from DQ pads intended for CA training:

DIM_DQ_DQ_SMT_STATUS

DIM_D00_DIM_DQ_DQ_SMT_STATUS

Bits	Name	Description
7:0	DIM_DQ_DQ_SMT_STATUS	Status of SMT pins from DQ pads

0x00480820 DIM_D00_DIM_DQ_PAD_CFG3

Type: RW

Clock: HCLK

Reset State: 0x1000FF11

The DIM_DQ_PAD_CFG3 register configures the following:

DIM_DQ_PAD_CFG3

DIM_D00_DIM_DQ_PAD_CFG3

Bits	Name	Description
28	DQS_DIFF_MODE	DQS output mode control 1'b1: differential output (default) 1'b0: single-ended output; _n output is HIZ
27	RCW_ODT_ENA1	Enable ODT for RCW pad (when hp_mode=1)
26	RCW_ODT_ENA0	Enable ODT for RCW pad (when hp_mode=0)
25:24	RCW_ODT	Impedance control for on-die termination on RCW pad
23	DQ_ODT_ENA1	Enable ODT for DQ pads (when hp_mode=1)
22	DQ_ODT_ENA0	Enable ODT for DQ pads (when hp_mode=0)
21:20	DQ_ODT	Impedance control for on-die termination on DQ pads
19	DQS_ODT_ENA1	Enable ODT for DQS pad (when hp_mode=1)
18	DQS_ODT_ENA0	Enable ODT for DQS pad (when hp_mode=0)
17:16	DQS_ODT	Impedance control for on-die termination on DQS pad
15:8	DQ_IE_OE	Enable both input receiver and output driver for all DQ pads
5	RCW_IE_OE	Enable both input receiver and output driver for RCW pad
4	DQS_IE_OE	Enable both input receiver and output driver for DQS pad
1	DM_IE	Input received enable for DM pad
0	DM_OE	Output driver enable for DM pad

0x00480824 DIM_D00_DIM_DQ_PAD_CFG4**Type:** RW**Clock:** HCLK**Reset State:** 0xE0222240

The DIM_DQ_PAD_CFG4 register configures the following:

DIM_DQ_PAD_CFG4

DIM_D00_DIM_DQ_PAD_CFG4

Bits	Name	Description
31	RCW_DDR_MODE1	Mode pin for DDR (when hp_mode = 1'b1) 1'b1: DDR2 modes (default) 1'b0: DDR1 modes
30	RCW_DDR_MODE0	Mode pin for DDR (when hp_mode = 1'b0) 1'b1: DDR2 modes (default) 1'b0: DDR1 modes

DIM_D00_DIM_DQ_PAD_CFG4 (cont.)

Bits	Name	Description
29	RCW_LV_MODE	Mode select for high/low voltage regime
25:24	RCW_PULL_B	Input pull control
21:20	RCW_NSLEW	Slew rate control bits for output path NMOS
17:16	RCW_PSLEW	Slew rate control bits for output path PMOS
13:12	RCW_NRXDEL	Delay control bits to increase strength of NMOS input drive
9:8	RCW_PRXDEL	Delay control bits to increase strength of PMOS input driver
7	RCW_VM_SHIFT_ENA	VM (voltage at output of on-die termination) shift enable bit
6:4	RCW_ROUT	Impedance control bit settings for output driver
2:0	RCW_DCC	Duty cycle correction bits for output path

0x0048080C DIM_D00_DDRPHY_DQ_MISR_CFG**Type:** RW**Clock:** HCLK**Reset State:** 0x872C56BD

The DDRPHY_DQ_MISR_CFG register configures the following:

DDRPHY_DQ_MISR_CFG

DIM_D00_DDRPHY_DQ_MISR_CFG

Bits	Name	Description
31:0	SEED_VALUE	Loopback MISR seed

0x00480828 DIM_D00_DDRPHY_DQ_MISR_EVEN_STATUS**Type:** R**Clock:** HCLK**Reset State:** 0x00000000

The DDRPHY_DQ_MISR_EVEN_STATUS register configures the following:

DDRPHY_DQ_MISR_EVEN_STATUS

DIM_D00_DDRPHY_DQ_MISR_EVEN_STATUS

Bits	Name	Description
31:0	SIGNATURE	Even clock Loopback MISR signature

0x0048082C DIM_D00_DDRPHY_DQ_MISR_ODD_STATUS**Type:** R**Clock:** HCLK**Reset State:** 0x00000000

The DDRPHY_DQ_MISR_ODD_STATUS register configures the following:

DDRPHY_DQ_MISR_ODD_STATUS

DIM_D00_DDRPHY_DQ_MISR_ODD_STATUS

Bits	Name	Description
31:0	SIGNATURE	Odd Clock Loopback MISR signature

0x00480830 DIM_D00_DIM_DQ_CDC_CTLR_CFG0**Type:** RW**Clock:** HCLK**Reset State:** 0x02000000

The DIM_DQ_CDC_CTLR_CFG0 register configures the following:

DIM_DQ_CDC_CTLR_CFG0

DIM_D00_DIM_DQ_CDC_CTLR_CFG0

Bits	Name	Description
26	ERROR_CODE_UPDATE_EN	1'b1: Enable overwriting (default) the error code for subsequent calibrations. 1'b0: Disable overwriting the error code during subsequent calibrations
25	STAGGER_CAL_ENA	1'b1: Stagger calibration of write and read CDCs once after the other to reduce peak voltage drop. 1'b0: Both write and read CDCs calibrated simultaneously This bit is introduced on the x2 core, so not present on x0 and x1 versions of DIM/PHY.
24	TRACKING_CAL_ENA	1'b1: Relationship between subunits and oscillator count is learnt and used for tracking calibration. 1'b0: Periodic calibration triggers cause full recalibration
23:20	OSC_COUNT_ERR_TOLERANCE	Magnitude of the difference between oscillator count and target count when the algorithm assumes convergence and stops further search.
19	TRACK_CALIB_MODE	1'b0: Tracking calibration uses previous Target_count value. 1'b1 Accurate mode: Target_count is recomputed at the start of tracking calibration

DIM_D00_DIM_DQ_CDC_CTLR_CFG0 (cont.)

Bits	Name	Description
18	FULL_DELAY	1'b0: unitstep and subunit caps are from calibration or SW settings 1'b1: All unit step and subunit caps are selected for the delay arrays.
17	HW_AUTOCAL_ENA	1'b1: Enables periodic auto calibration 1'b0: Periodic calibration is disabled
16	SW_TRIG_FULL_CALIB	A low to high transition on this register bit triggers the CDC calibration statemachine to initiate full calibration of the standby delay array. Also resets tracking mode for the delay arrays.
13:12	OSC_PRE_DIV	The factor by which the raw oscillator frequency is divided before being given to the counting logic. 2'b00 Divide by 1 2'b01 Divide by 2 2'b10 Divide by 4 2'b11 Divide by 8
10:0	TMUX_CHAR	Cchar is a characterized constant equal to the ratio of output mux offset delay to average unit step delay. This register is programmed by multiplying the constant with 1024 and rounding to the nearest integer. E.g. if Cchar = 0.67, $RO(0.67 * 1024) = 686$, hence program TMUX_CHAR = 11'h2AE

0x00480834 DIM_D00_DIM_DQ_CDC_CTLR_CFG1**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

The DIM_DQ_CDC_CTLR_CFG1 register configures the following:

DIM_DQ_CDC_CTLR_CFG1

DIM_D00_DIM_DQ_CDC_CTLR_CFG1

Bits	Name	Description
26:24	OSC_COUNT_DELAY	Number of oscillator cycles from the end of reference gate signal to assertion of oscillation counter done. Legal values are from 0 to 4.
23:20	STANDBY_DELAY	This is the number of reference clock cycles from when a newly calibrated delay array is made active to when the current active delay array is put into standby mode. 4'b0: Disables the counter. Hardware waits for explicit acknowledgement of delay array switch from CDC. Others: Explicit acknowledgement mode is disabled and the counter is used

DIM_D00_DIM_DQ_CDC_CTLR_CFG1 (cont.)

Bits	Name	Description
18:16	DEL_MODE_DELAY	Number of reference clock cycles between putting a newly calibrated delay array in delay mode to selecting it as the active delay array. The minimum allowed value is 1
14:12	OSC_MODE_DELAY	Number of reference clock cycles between putting a delay array in oscillator mode to starting the oscillation counter. This delay is allowed for the oscillations to stabilize through the delay array. The minimum allowed value is 1
10:8	DECODER_DELAY	Number of reference clock cycles allowed for unitstep and subunit counts to be encoded into the unit and finecap select lines on the DDA. The minimum allowed value is 1.
6:4	DIVIDER_DELAY	Number of reference clock cycles allowed for arithmetic division operations. The minimum allowed value is 1
2:0	MULTIPLIER_DELAY	Number of reference clock cycles allowed for arithmetic multiplication operations. The minimum allowed value is 1

0x00480838 DIM_D00_DIM_DQ_CDC_CAL_TIMER_CFG0**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

The DIM_DQ_CDC_CAL_TIMER_CFG0 register configures the following:

DIM_DQ_CDC_CAL_TIMER_CFG0

DIM_D00_DIM_DQ_CDC_CAL_TIMER_CFG0

Bits	Name	Description
24	INVALID_TIMER_ENA	1'b1: Enable the timer counting number of missed calibration intervals after which the CDC is invalid. 1'b0: Timer disabled A low to high transition on this register bit loads the timer with programmed value below.
23:20	INVALID_TIMER_VAL	Number of missed calibration intervals after which CDC invalid state is asserted.
16	TIMER_ENA	1'b1: Enables 32 kHz sleep clock based periodic autocalibration timer. 1'b0: Timer disabled A low to high transition on this register bit loads the timer with programmed value below.
15:0	TIMER_VAL	Count value for periodic autocalibration interval based on sleep clock.

0x0048083C DIM_D00_DIM_DQ_CDC_CAL_TIMER_CFG1**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

The DIM_DQ_CDC_CAL_TIMER_CFG1 register configures the following:

DIM_DQ_CDC_CAL_TIMER_CFG1

DIM_D00_DIM_DQ_CDC_CAL_TIMER_CFG1

Bits	Name	Description
12	FF_TIMER_ENA	1'b1: Enables reference clock based periodic autocalibration timer. 1'b0: Timer disabled A low to high transition on this register bit loads the timer with programmed value below.
9:0	FF_TIMER_VAL	Count value for periodic autocalibration interval based on reference clock.

0x00480840 DIM_D00_DIM_DQ_CDC_REFCOUNT_CFG**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

The DIM_DQ_CDC_REFCOUNT_CFG register configures the following:

DIM_DQ_CDC_REFCOUNT_CFG

DIM_D00_DIM_DQ_CDC_REFCOUNT_CFG

Bits	Name	Description
31:16	TREF	Reference clock period programmed in pico seconds.
13:8	CCAL_REF_COUNT	Number of reference clock cycles used for oscillation counts during coarse calibration. Maximum supported value is 59.
5:0	FCAL_REF_COUNT	Number of reference clock cycles used for oscillation counts during fine calibration. Maximum supported value is 59

0x00480844 DIM_D00_DIM_DQ_CDC_COARSE_CAL_CFG**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

The register DIM_DQ_CDC_COARSE_CAL_CFG configures the following:

DIM_DQ_CDC_COARSE_CAL_CFG

DIM_D00_DIM_DQ_CDC_COARSE_CAL_CFG

Bits	Name	Description
12:8	CCAL_SUBUNIT_CAPS	Subunit count needed for an effective unit step count of 32 during coarse calibration. This value is also used by software oscillator mode.
5:0	CCAL_UNITSTEPS	unitstep count needed for an effective unit step count of 32 during coarse calibration.

0x00480848 DIM_D00_DIM_DQ_CDC_RSVD_CFG

Type: RW**Clock:** HCLK**Reset State:** 0x00000000

The DIM_DQ_CDC_RSVD_CFG register configures the following:

DIM_DQ_CDC_RSVD_CFG

DIM_D00_DIM_DQ_CDC_RSVD_CFG

Bits	Name	Description
15:0	TEMP_FIELD	Reserved register

0x0048084C DIM_D00_DIM_DQ_RD_CDC_OFFSET_CFG

Type: RW**Clock:** HCLK**Reset State:** 0x00000000

The DIM_DQ_RD_CDC_OFFSET_CFG register configures the following:

DIM_DQ_RD_CDC_OFFSET_CFG

DIM_D00_DIM_DQ_RD_CDC_OFFSET_CFG

Bits	Name	Description
20	SUBUNIT_OFFSET_MODE	1'b1: The programmed offset is directly loaded as subunit count 1'b0: Programmed offset is added/subtracted from the subunit count
19	SUBUNIT_OFFSET_SIGN	1'b1: Offset is subtracted from the subunit count 1'b0: Offset is added to the subunit count
16:12	SUBUNIT_OFFSET	Software subunit offset

DIM_D00_DIM_DQ_RD_CDC_OFFSET_CFG (cont.)

Bits	Name	Description
8	UNIT_OFFSET_MODE	1'b1: The programmed offset is directly loaded as unit step count 1'b0: Programmed offset is added/subtracted from the unit step count
7	UNIT_OFFSET_SIGN	1'b1: Offset is subtracted from the unitstep count 1'b0: Offset is added to the unitstep count
5:0	UNIT_OFFSET	Software unitstep offset

0x00480850 DIM_D00_DIM_DQ_RD_CDC_DELAY_CFG**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

The DIM_DQ_RD_CDC_DELAY_CFG register configures the following:

DIM_DQ_RD_CDC_DELAY_CFG

DIM_D00_DIM_DQ_RD_CDC_DELAY_CFG

Bits	Name	Description
28	TARGET_COUNT_ENA	1'b1: Enable software specified Target_count for calibration instead of value calculated by hardware. 1'b0: Hardware computed Target_count used.
27:16	TARGET_COUNT	Software specified Target_count to which the statemachine calibrates
11:0	DELAY_VAL	Delay required from CDC programmed in pico seconds. This value has to be greater than the minimum delay supported by DDA Tmin.

0x00480854 DIM_D00_DIM_DQ_RD_CDC_SW_MODE_CFG**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

The DIM_DQ_RD_CDC_SW_MODE_CFG register configures the following:

DIM_DQ_RD_CDC_SW_MODE_CFG

DIM_D00_DIM_DQ_RD_CDC_SW_MODE_CFG

Bits	Name	Description
3	SW_DEL_MODE	A low to high transition on this register bit puts the delay array selected by SW_DA_SEL into delay mode and connects it to the DDA output. This register has to remain set for the duration of SW controlled delay mode.
2	SW_OSC_MODE	Reserved register.
1	SW_DA_SEL	Selects the delay array inside DDA during software delay or oscillator mode.
0	SW_LOAD	1'b1: Enable loading of the software offsets from the offset configuration register. This register has to remain set for the duration of SW mode operation.

0x0048085C DIM_D00_DIM_DQ_RD_CDC_SW_OVRD_CFG**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

The DIM_DQ_RD_CDC_SW_OVRD_CFG register configures the following:

DIM_DQ_RD_CDC_SW_OVRD_CFG

DIM_D00_DIM_DQ_RD_CDC_SW_OVRD_CFG

Bits	Name	Description
16	SW_REF_GATE	Write 1'b1 to start the reference window. Write 1'b0 to end the reference window. Number of reference cycle in the window will be reported in SW_REF_GATE_COUNT in CDC_STATUS0
11	SW_OVRD_DA1_OSC_EN	In software override mode, 1'b1 written to this bit puts delay array 1 in oscillator-mode
10	SW_OVRD_DA1_IN_EN	In software override mode, 1'b1 written to this bit puts delay array 1 in delay-mode
9	SW_OVRD_DA0_OSC_EN	In software override mode, 1'b1 written to this bit puts delay array 0 in oscillator-mode
8	SW_OVRD_DA0_IN_EN	In software override mode, 1'b1 written to this bit puts delay array 0 in delay-mode
3	SW_OVRD_CDC_COUNTERR_RST	1'b1 written to this bit resets the oscillation counter. Follow this with 1'b0 before setting reference gate.
2	SW_OVRD_LOAD_DA_SEL	In software override mode, this bit select the delay array to which unit and subunit offset values from CDC_OFFSET_CFG register will be loaded.

DIM_D00_DIM_DQ_RD_CDC_SW_OVRD_CFG (cont.)

Bits	Name	Description
1	SW_OVRD_ACTV_DA_SEL	In software override mode, this register selects the delay array connected to the CDC output
0	SW_OVRD_ENA	1'b1 written to this bit puts the CDC interface in software override mode where the below register bits control the CDC. 1'b0: CDC is under hardware control

0x00480860 DIM_D00_DIM_DQ_RD_CDC_SLAVE_DDA_CFG**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

The DIM_DQ_RD_CDC_SLAVE_DDA_CFG register configures the following:

DIM_DQ_RD_CDC_SLAVE_DDA_CFG

DIM_D00_DIM_DQ_RD_CDC_SLAVE_DDA_CFG

Bits	Name	Description
17	SLAV_DDA_OFFSET_MODE	1'b1: The programmed offset is directly loaded as unit-step count 1'b0: Programmed offset is added/subtracted from the unit-step count
16	SLAV_DDA_OFFSET_SIGN	1'b1: Offset is subtracted from the unit-step count. This subtraction feature is not supported on x0 and x1 PHY cores (bit RESERVED). Supported on the x2 core version. 1'b0: Offset is added to the unit-step count
15:12	SLAVE_DDA_OFFSET	unit-step offset for slave DDA.
10:0	SLAVE_DDA_DELAY	Delay required from slave DDA programmed in pico seconds.

0x00480870 DIM_D00_DIM_DQ_RD_CDC_STATUS0**Type:** R**Clock:** HCLK**Reset State:** 0x0000000C

The DIM_DQ_RD_CDC_STATUS0 register configures the following:

DIM_DQ_RD_CDC_STATUS0

DIM_D00_DIM_DQ_RD_CDC_STATUS0

Bits	Name	Description
26:24	CDC_ERROR_CODE	Error code for internal errors: 3'b000: No error 3'b001: Programmed delay less than Tmux 3'b010: Programmed delay less than Tmin 3'b011: Unit-step overflow: Overflow during fine calibration. Desired delay cannot be realized even when all unit-steps are selected. 3'b100: Unit-step underflow 3'b101: Unit-step overflow: Overflow during coarse calibration. Estimated unit steps needed to realize the desired delay are greater than the total number of unit-steps in design. Note: The CDC_ERROR_CODE is not a sticky register and is updated after each CDC calibration.
23:16	SW_REF_GATE_COUNT	Number of reference clock cycles in the software generated oscillation gate window doing SW override mode
15:4	OSC_COUNT	Current value of oscillator count
3	CURR_SEL_DA	This bit indicates the current active delay array connected to the output. 1'b0: DA0 is selected 1'b1: DA1 is selected
2	CTLR_SM_IDLE	1'b1 Indicates that the calibration state machine is idle 1'b0 indicates calibration statemachine is busy
1	OSC_DONE	This status register is set after oscillations count is finished by CDC. It gets reset when a new oscillator count is started.
0	CALIBRATION_DONE	1'b1 indicates that calibration is done and CURR_SUBUNIT_STEPS, CURR_UNIT_STEPS status registers are valid.

0x00480874 DIM_D00_DIM_DQ_RD_CDC_STATUS1**Type:** R**Clock:** HCLK**Reset State:** 0x00000000

The DIM_DQ_RD_CDC_STATUS1 register configures the following:

DIM_DQ_RD_CDC_STATUS1

DIM_D00_DIM_DQ_RD_CDC_STATUS1

Bits	Name	Description
27:16	CURR_DELAY_VALUE	HW Computed value of current delay after calibration by the state machine. This register is valid if CALIBRATION_DONE status bit is set.
11:0	CURR_TMUX_DELAY	Current value of output mux offset delay computed by HW.

0x00480878 DIM_D00_DIM_DQ_RD_CDC_STATUS2**Type:** R**Clock:** HCLK**Reset State:** 0x10331033

The DIM_DQ_RD_CDC_STATUS2 register configures the following:

DIM_DQ_RD_CDC_STATUS2

DIM_D00_DIM_DQ_RD_CDC_STATUS2

Bits	Name	Description
28:24	DA1_SUBUNITS	Current value of subunit cap settings applied to delay array 1
21:16	DA1_UNITSTEPS	Current value of unit steps applied to delay array 1
12:8	DA0_SUBUNITS	Current value of subunit cap settings applied to delay array 0
5:0	DA0_UNITSTEPS	Current value of unit steps applied to delay array 0

0x0048087C DIM_D00_DIM_DQ_RD_CDC_STATUS3**Type:** R**Clock:** HCLK**Reset State:** 0x00000000

The DIM_DQ_RD_CDC_STATUS3 register configures the following:

DIM_DQ_RD_CDC_STATUS3

DIM_D00_DIM_DQ_RD_CDC_STATUS3

Bits	Name	Description
31:24	NUM_OF_OSC_ITER	Number of oscillator iterations before convergence was achieved during tracking
23:12	COUNT_ERROR	Difference between target and actual oscillator counts at the start of tracking calibration

DIM_D00_DIM_DQ_RD_CDC_STATUS3 (cont.)

Bits	Name	Description
11:0	CURR_TARGET_COUNT	HW computed value of current Target_count used by the calibration state machine.

0x00480880 DIM_D00_DIM_DQ_RD_CDC_STATUS4**Type:** R**Clock:** HCLK**Reset State:** 0x000000FF

The DIM_DQ_RD_CDC_STATUS4 register configures the following:

DIM_DQ_RD_CDC_STATUS4

DIM_D00_DIM_DQ_RD_CDC_STATUS4

Bits	Name	Description
7:4	SLAVE_DDA_DA1_TAPS	Number if unit taps applied to delay array 1 of slave DDA
3:0	SLAVE_DDA_DA0_TAPS	Number if unit taps applied to delay array 0 of slave DDA

0x004808AC DIM_D00_DIM_DQ_WR_CDC_OFFSET_CFG**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

The DIM_DQ_WR_CDC_OFFSET_CFG register configures the following:

DIM_DQ_WR_CDC_OFFSET_CFG

DIM_D00_DIM_DQ_WR_CDC_OFFSET_CFG

Bits	Name	Description
20	SUBUNIT_OFFSET_MODE	1'b1: The programmed offset is directly loaded as subunit count 1'b0: Programmed offset is added/subtracted from the subunit count
19	SUBUNIT_OFFSET_SIGN	1'b1: Offset is subtracted from the subunit count 1'b0: Offset is added to the subunit count
16:12	SUBUNIT_OFFSET	Software subunit offset
8	UNIT_OFFSET_MODE	1'b1: The programmed offset is directly loaded as unit step count 1'b0: Programmed offset is added/subtracted from the unit step count

DIM_D00_DIM_DQ_WR_CDC_OFFSET_CFG (cont.)

Bits	Name	Description
7	UNIT_OFFSET_SIGN	1'b1: Offset is subtracted from the unitstep count 1'b0: Offset is added to the unitstep count
5:0	UNIT_OFFSET	Software unitstep offset

0x004808B0 DIM_D00_DIM_DQ_WR_CDC_DELAY_CFG**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

The DIM_DQ_WR_CDC_DELAY_CFG register configures the following:

DIM_DQ_WR_CDC_DELAY_CFG

DIM_D00_DIM_DQ_WR_CDC_DELAY_CFG

Bits	Name	Description
28	TARGET_COUNT_ENA	1'b1: Enable software specified Target_count for calibration instead of value calculated by hardware. 1'b0: Hardware computed Target_count used.
27:16	TARGET_COUNT	Software specified Target_count to which the statemachine calibrates
11:0	DELAY_VAL	Delay required from CDC programmed in pico seconds. This value has to be greater than the minimum delay supported by DDA Tmin.

0x004808B4 DIM_D00_DIM_DQ_WR_CDC_SW_MODE_CFG**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

The DIM_DQ_WR_CDC_SW_MODE_CFG register configures the following:

DIM_DQ_WR_CDC_SW_MODE_CFG

DIM_D00_DIM_DQ_WR_CDC_SW_MODE_CFG

Bits	Name	Description
3	SW_DEL_MODE	A low to high transition on this register bit puts the delay array selected by SW_DA_SEL into delay mode and connects it to the DDA output. This register has to remain set for the duration of SW controlled delay mode.

DIM_D00_DIM_DQ_WR_CDC_SW_MODE_CFG (cont.)

Bits	Name	Description
2	SW_OSC_MODE	Reserved register.
1	SW_DA_SEL	Selects the delay array inside DDA during software delay or oscillator mode.
0	SW_LOAD	1'b1: Enable loading of the software offsets from the offset configuration register. This register has to remain set for the duration of SW mode operation.

0x004808BC DIM_D00_DIM_DQ_WR_CDC_SW_OVRD_CFG**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

The DIM_DQ_WR_CDC_SW_OVRD_CFG register configures the following:

DIM_DQ_WR_CDC_SW_OVRD_CFG

DIM_D00_DIM_DQ_WR_CDC_SW_OVRD_CFG

Bits	Name	Description
16	SW_REF_GATE	Write 1'b1 to start the reference window. Write 1'b0 to end the reference window. Number of reference cycle in the window will be reported in SW_REF_GATE_COUNT in CDC_STATUS0
11	SW_OVRD_DA1_OSC_EN	In software override mode, 1'b1 written to this bit puts delay array 1 in oscillator-mode
10	SW_OVRD_DA1_IN_EN	In software override mode, 1'b1 written to this bit puts delay array 1 in delay-mode
9	SW_OVRD_DA0_OSC_EN	In software override mode, 1'b1 written to this bit puts delay array 0 in oscillator-mode
8	SW_OVRD_DA0_IN_EN	In software override mode, 1'b1 written to this bit puts delay array 0 in delay-mode
3	SW_OVRD_CDC_COUNTERR_RST	1'b1 written to this bit resets the oscillation counter. Follow this with 1'b0 before setting reference gate.
2	SW_OVRD_LOAD_DA_SEL	In software override mode, this bit select the delay array to which unit and subunit offset values from CDC_OFFSET_CFG register will be loaded.
1	SW_OVRD_ACTV_DA_SEL	In software override mode, this register selects the delay array connected to the CDC output
0	SW_OVRD_ENA	1'b1 written to this bit puts the CDC interface in software override mode where the below register bits control the CDC. 1'b0: CDC is under hardware control

0x004808D0 DIM_D00_DIM_DQ_WR_CDC_STATUS0**Type:** R**Clock:** HCLK**Reset State:** 0x0000000C

The DIM_DQ_WR_CDC_STATUS0 register configures the following:

DIM_DQ_WR_CDC_STATUS0

DIM_D00_DIM_DQ_WR_CDC_STATUS0

Bits	Name	Description
26:24	CDC_ERROR_CODE	Error code for internal errors: 3'b000: No error 3'b001: Programmed delay less than Tmux 3'b010: Programmed delay less than Tmin 3'b011: Unit-step overflow: Overflow during fine calibration. Desired delay cannot be realized even when all unit-steps are selected. 3'b100: Unit-step underflow 3'b101: Unit-step overflow: Overflow during coarse calibration. Estimated unit steps needed to realize the desired delay are greater than the total number of unit-steps in design. Note: The CDC_ERROR_CODE is not a sticky register and is updated after each CDC calibration.
23:16	SW_REF_GATE_COUNT	Number of reference clock cycles in the software generated oscillation gate window during SW override mode
15:4	OSC_COUNT	Current value of oscillator count
3	CURR_SEL_DA	This bit indicates the current active delay array connected to the output. 1'b0: DA0 is selected 1'b1: DA1 is selected
2	CTLR_SM_IDLE	1'b1 Indicates that the calibration state machine is idle 1'b0 indicates calibration statemachine is busy
1	OSC_DONE	This status register is set after oscillations count is finished by CDC. It gets reset when a new oscillator count is started.
0	CALIBRATION_DONE	1'b1 indicates that calibration is done and CURR_SUBUNIT_STEPS, CURR_UNIT_STEPS status registers are valid.

0x004808D4 DIM_D00_DIM_DQ_WR_CDC_STATUS1**Type:** R**Clock:** HCLK**Reset State:** 0x00000000

The DIM_DQ_WR_CDC_STATUS1 register configures the following:

DIM_DQ_WR_CDC_STATUS1

DIM_D00_DIM_DQ_WR_CDC_STATUS1

Bits	Name	Description
27:16	CURR_DELAY_VALUE	HW Computed value of current delay after calibration by the state machine. This register is valid if CALIBRATION_DONE status bit is set.
11:0	CURR_TMUX_DELAY	Current value of output mux offset delay computed by HW.

0x004808D8 DIM_D00_DIM_DQ_WR_CDC_STATUS2

Type: R

Clock: HCLK

Reset State: 0x10331033

The DIM_DQ_WR_CDC_STATUS2 register configures the following:

DIM_DQ_WR_CDC_STATUS2

DIM_D00_DIM_DQ_WR_CDC_STATUS2

Bits	Name	Description
28:24	DA1_SUBUNITS	Current value of subunit cap settings applied to delay array 1
21:16	DA1_UNITSTEPS	Current value of unit steps applied to delay array 1
12:8	DA0_SUBUNITS	Current value of subunit cap settings applied to delay array 0
5:0	DA0_UNITSTEPS	Current value of unit steps applied to delay array 0

0x004808DC DIM_D00_DIM_DQ_WR_CDC_STATUS3

Type: R

Clock: HCLK

Reset State: 0x00000000

The DIM_DQ_WR_CDC_STATUS3 register configures the following:

DIM_DQ_WR_CDC_STATUS3

DIM_D00_DIM_DQ_WR_CDC_STATUS3

Bits	Name	Description
31:24	NUM_OF_OSC_ITER	Number of oscillator iterations before convergence was achieved during tracking

DIM_D00_DIM_DQ_WR_CDC_STATUS3 (cont.)

Bits	Name	Description
23:12	COUNT_ERROR	Difference between target and actual oscillator counts at the start of tracking calibration
11:0	CURR_TARGET_COUNT	HW computed value of current Target_count used by the calibration state machine.

0x00480900 DIM_D00_DIM_DQ_DQ_IOC_SLV_CFG**Type:** RW**Clock:** HCLK**Reset State:** 0x10100000

The DIM_DQ_DQ_IOC_SLV_CFG register configures the following:

DIM_DQ_DQ_IOC_SLV_CFG

DIM_D00_DIM_DQ_DQ_IOC_SLV_CFG

Bits	Name	Description
31	PNCNT_HW_LOAD_EN	
28:24	NCNT_SW_VAL	
20:16	PCNT_SW_VAL	
15	NCNT_OFFSET_SIGN	
12:8	NCNT_SW_OFFSET	
7	PCNT_OFFSET_SIGN	
4:0	PCNT_SW_OFFSET	

0x00480904 DIM_D00_DIM_DQ_DQ_IOC_SLV_STATUS**Type:** R**Clock:** HCLK**Reset State:** 0x00001010

The DIM_DQ_DQ_IOC_SLV_STATUS register configures the following:

DIM_CA_CA_IOC_SLV_STATUS

DIM_D00_DIM_DQ_DQ_IOC_SLV_STATUS

Bits	Name	Description
12:8	PAD_NCNT	Impedance calibration bit settings for NMOS

DIM_D00_DIM_DQ_DQ_IOC_SLV_STATUS (cont.)

Bits	Name	Description
4:0	PAD_PCNT	Impedance calibration bit settings for PMOS

0x00480910 DIM_D00_DIM_DQ_DQS_IOC_SLV_CFG**Type:** RW**Clock:** HCLK**Reset State:** 0x10100000

The DIM_DQ_DQS_IOC_SLV_CFG register configures the following:

DIM_DQ_DQS_IOC_SLV_CFG

DIM_D00_DIM_DQ_DQS_IOC_SLV_CFG

Bits	Name	Description
31	PNCNT_HW_LOAD_EN	
28:24	NCNT_SW_VAL	
20:16	PCNT_SW_VAL	
15	NCNT_OFFSET_SIGN	
12:8	NCNT_SW_OFFSET	
7	PCNT_OFFSET_SIGN	
4:0	PCNT_SW_OFFSET	

0x00480914 DIM_D00_DIM_DQ_DQS_IOC_SLV_STATUS**Type:** R**Clock:** HCLK**Reset State:** 0x00001010

The DIM_DQ_DQS_IOC_SLV_STATUS register configures the following:

DIM_DQ_DQS_IOC_SLV_STATUS

DIM_D00_DIM_DQ_DQS_IOC_SLV_STATUS

Bits	Name	Description
12:8	PAD_NCNT	Impedance calibration bit settings for NMOS
4:0	PAD_PCNT	Impedance calibration bit settings for PMOS

7.3 DIM_D01_DDRPHY_DQ

0x00481000 DIM_D01_DIM_DQ_TOP_CFG

Type: RW

Clock: HCLK

Reset State: 0x00000031

The DIM_DQ_TOP_CFG register configures the following:

DIM_DQ_TOP_CFG

DIM_D01_DIM_DQ_TOP_CFG

Bits	Name	Description
30	DQ_LB_MODE_SEL	Selects external or internal loop back mode 1'b1: internal loop back mode 1'b0: external loop back mode (default)
29	DQ_LB_MODE_EN	Enables external or internal loop back mode 1'b1: loop back mode enabled 1'b0: loop back mode disabled (default)
28	RESERVED	Reserved
27	MISR_DATA_CLK_SEL	selects b/w MISR data or SEED and Loopback clock or FF clock 1'b1: from dte 1'b0: seed from csr ff clk (default)
26	CDC_LDO_EN	Enablement of CDC LDO 1'b1: Enabled 1'b0: Disabled LDO and power provided from switches (default)
25	CDC_SWITCH_RC_EN	Enablement of CDC power RC (LPF) switch 1'b1: Enabled 1'b0: Disabled (default)
24	CDC_SWITCH_BYPASS_OFF	Enablement of CDC power bypass switch 1'b1: Disabled 1'b0: Enabled (default)
16	RCW_EN	Enablement of the Read Capture Window 1'b1: Enable the RCW 1'b0: Disabled (default)
8	RESERVED	Reserved
5	WR_PIPE_EXTEND1	Write pipe extend select (when hp_mode = 1'b1) 1'b1: with pipe extend (default) 1'b0: without pipe extend
4	WR_PIPE_EXTEND0	Write pipe extend select (when hp_mode = 1'b0) 1'b1: with pipe extend (default) 1'b0: without pipe extend

DIM_D01_DIM_DQ_TOP_CFG (cont.)

Bits	Name	Description
0	WR_CLK_SEL	Write clock select 1'b1: With write CDC (default) 1'b0: Bypass write CDC, write clock from cdiv2 phi2 output

0x00481010 DIM_D01_DIM_DQ_PAD_CFG0**Type:** RW**Clock:** HCLK**Reset State:** 0xE0222240

The DIM_DQ_PAD_CFG0 register configures the following:

DIM_DQ_PAD_CFG0

DIM_D01_DIM_DQ_PAD_CFG0

Bits	Name	Description
31	DQ_DDR_MODE1	Mode pin for DDR (when hp_mode = 1'b1) 1'b1: DDR2 modes (default) 1'b0: DDR1 modes
30	DQ_DDR_MODE0	Mode pin for DDR (when hp_mode = 1'b0) 1'b1: DDR2 modes (default) 1'b0: DDR1 modes
29	DQ_LV_MODE	Mode select for high/low voltage regime
28	DQ_ODT_ENA	Unused - ODT controlled by PAD_CFG3
27:26	DQ_ODT	Unused - ODT controlled by PAD_CFG3
25:24	DQ_PULL_B	Input pull control
21:20	DQ_NSLEW	Slew rate control bits for output path NMOS
17:16	DQ_PSLEW	Slew rate control bits for output path PMOS
13:12	DQ_NRXDEL	Delay control bits to increase strength of NMOS input drive
9:8	DQ_PRXDEL	Delay control bits to increase strength of PMOS input driver
7	DQ_VM_SHIFT_ENA	VM (voltage at output of on-die termination) shift enable bit
6:4	DQ_ROUT	Impedance control bit settings for output driver
2:0	DQ_DCC	Duty cycle correction bits for output path

0x00481014 DIM_D01_DIM_DQ_PAD_CFG1**Type:** RW**Clock:** HCLK**Reset State:** 0xE0222240

The DIM_DQ_PAD_CFG1 register configures the following:

DIM_DQ_PAD_CFG1

DIM_D01_DIM_DQ_PAD_CFG1

Bits	Name	Description
31	DQS_DDR_MODE1	Mode pin for DDR (when hp_mode = 1'b1) 1'b1: DDR2 modes (default) 1'b0: DDR1 modes
30	DQS_DDR_MODE0	Mode pin for DDR (when hp_mode = 1'b0) 1'b1: DDR2 modes (default) 1'b0: DDR1 modes
29	DQS_LV_MODE	Mode select for high/low voltage regime
28	DQS_ODT_ENA	Unused - ODT controlled by PAD_CFG3
27:26	DQS_ODT	Unused - ODT controlled by PAD_CFG3
25:24	DQS_PULL	Input pull control 2'b00: no pull (default) 2'b01: pull down 2'b10: keeper 2'b11: pull up
21:20	DQS_NSLEW	Slew rate control bits for output path NMOS
17:16	DQS_PSLEW	Slew rate control bits for output path PMOS
13:12	DQS_NRXDEL	Delay control bits to increase strength of NMOS input drive
9:8	DQS_PRXDEL	Delay control bits to increase strength of PMOS input drive
7	DQS_VM_SHIFT_ENA	VM (voltage at output of on-die termination) shift enable bit
6:4	DQS_ROUT	Impedance control bit settings for output driver
2:0	DQS_DCC	Duty cycle correction bits for output path

0x00481018 DIM_D01_DIM_DQ_PAD_CFG2**Type:** RW**Clock:** HCLK**Reset State:** 0x1000000A

The DIM_DQ_PAD_CFG2 register configures the following:

DIM_DQ_PAD_CFG2

DIM_D01_DIM_DQ_PAD_CFG2

Bits	Name	Description
29	VREF_LDO_ENA	Enable VREF pad LDO Must be 0 for x2 core as there is no internal LDO
28	VREF_PASSGATE_ENA	Enable pass gate from padsig to vref_int pin (default 1)
27:24	VREF_SP_OUT	Unused
21:18	VREF_SP_IN	Unused
17	VREF_BYPASS_ENA	Enable bypass gate from resistive divider to vref_int pin
16	VREF_RDIV_ENA	Enable resistive divider for VREF voltage control
13:8	VREF_PULLDN_SPEED_CN TL	Unused
5:0	VREF_LEVEL_CNTL	Control over the VREF voltage level through resistive divider

0x0048101C DIM_D01_DIM_DQ_DQ_SMT_STATUS

Type: R

Clock: HCLK

Reset State: 0x00000000

The DIM_DQ_DQ_SMT_STATUS register reflects the status of SMT signals from DQ pads intended for CA training:

DIM_DQ_DQ_SMT_STATUS

DIM_D01_DIM_DQ_DQ_SMT_STATUS

Bits	Name	Description
7:0	DIM_DQ_DQ_SMT_STATUS	Status of SMT pins from DQ pads

0x00481020 DIM_D01_DIM_DQ_PAD_CFG3

Type: RW

Clock: HCLK

Reset State: 0x1000FF11

The DIM_DQ_PAD_CFG3 register configures the following:

DIM_DQ_PAD_CFG3

DIM_D01_DIM_DQ_PAD_CFG3

Bits	Name	Description
28	DQS_DIFF_MODE	DQS output mode control 1'b1: differential output (default) 1'b0: single-ended output; _n output is HIZ
27	RCW_ODT_ENA1	Enable ODT for RCW pad (when hp_mode=1)
26	RCW_ODT_ENA0	Enable ODT for RCW pad (when hp_mode=0)
25:24	RCW_ODT	Impedance control for on-die termination on RCW pad
23	DQ_ODT_ENA1	Enable ODT for DQ pads (when hp_mode=1)
22	DQ_ODT_ENA0	Enable ODT for DQ pads (when hp_mode=0)
21:20	DQ_ODT	Impedance control for on-die termination on DQ pads
19	DQS_ODT_ENA1	Enable ODT for DQS pad (when hp_mode=1)
18	DQS_ODT_ENA0	Enable ODT for DQS pad (when hp_mode=0)
17:16	DQS_ODT	Impedance control for on-die termination on DQS pad
15:8	DQ_IE_OE	Enable both input receiver and output driver for all DQ pads
5	RCW_IE_OE	Enable both input receiver and output driver for RCW pad
4	DQS_IE_OE	Enable both input receiver and output driver for DQS pad
1	DM_IE	Input received enable for DM pad
0	DM_OE	Output driver enable for DM pad

0x00481024 DIM_D01_DIM_DQ_PAD_CFG4**Type:** RW**Clock:** HCLK**Reset State:** 0xE0222240

The DIM_DQ_PAD_CFG4 register configures the following:

DIM_DQ_PAD_CFG4

DIM_D01_DIM_DQ_PAD_CFG4

Bits	Name	Description
31	RCW_DDR_MODE1	Mode pin for DDR (when hp_mode = 1'b1) 1'b1: DDR2 modes (default) 1'b0: DDR1 modes
30	RCW_DDR_MODE0	Mode pin for DDR (when hp_mode = 1'b0) 1'b1: DDR2 modes (default) 1'b0: DDR1 modes

DIM_D01_DIM_DQ_PAD_CFG4 (cont.)

Bits	Name	Description
29	RCW_LV_MODE	Mode select for high/low voltage regime
25:24	RCW_PULL_B	Input pull control
21:20	RCW_NSLEW	Slew rate control bits for output path NMOS
17:16	RCW_PSLEW	Slew rate control bits for output path PMOS
13:12	RCW_NRXDEL	Delay control bits to increase strength of NMOS input drive
9:8	RCW_PRXDEL	Delay control bits to increase strength of PMOS input driver
7	RCW_VM_SHIFT_ENA	VM (voltage at output of on-die termination) shift enable bit
6:4	RCW_ROUT	Impedance control bit settings for output driver
2:0	RCW_DCC	Duty cycle correction bits for output path

0x0048100C DIM_D01_DDRPHY_DQ_MISR_CFG**Type:** RW**Clock:** HCLK**Reset State:** 0x872C56BD

The DDRPHY_DQ_MISR_CFG register configures the following:

DDRPHY_DQ_MISR_CFG

DIM_D01_DDRPHY_DQ_MISR_CFG

Bits	Name	Description
31:0	SEED_VALUE	Loopback MISR seed

0x00481028 DIM_D01_DDRPHY_DQ_MISR_EVEN_STATUS**Type:** R**Clock:** HCLK**Reset State:** 0x00000000

The DDRPHY_DQ_MISR_EVEN_STATUS register configures the following:

DDRPHY_DQ_MISR_EVEN_STATUS

DIM_D01_DDRPHY_DQ_MISR_EVEN_STATUS

Bits	Name	Description
31:0	SIGNATURE	Even clock Loopback MISR signature

0x0048102C DIM_D01_DDRPHY_DQ_MISR_ODD_STATUS**Type:** R**Clock:** HCLK**Reset State:** 0x00000000

The DDRPHY_DQ_MISR_ODD_STATUS register configures the following:

DDRPHY_DQ_MISR_ODD_STATUS

DIM_D01_DDRPHY_DQ_MISR_ODD_STATUS

Bits	Name	Description
31:0	SIGNATURE	Odd Clock Loopback MISR signature

0x00481030 DIM_D01_DIM_DQ_CDC_CTLR_CFG0**Type:** RW**Clock:** HCLK**Reset State:** 0x02000000

The DIM_DQ_CDC_CTLR_CFG0 register configures the following:

DIM_DQ_CDC_CTLR_CFG0

DIM_D01_DIM_DQ_CDC_CTLR_CFG0

Bits	Name	Description
26	ERROR_CODE_UPDATE_EN	1'b1: Enable overwriting (default) the error code for subsequent calibrations. 1'b0: Disable overwriting the error code during subsequent calibrations
25	STAGGER_CAL_ENA	1'b1: Stagger calibration of write and read CDCs once after the other to reduce peak voltage drop. 1'b0: Both write and read CDCs calibrated simultaneously This bit is introduced on the x2 core, so not present on x0 and x1 versions of DIM/PHY.
24	TRACKING_CAL_ENA	1'b1: Relationship between subunits and oscillator count is learnt and used for tracking calibration. 1'b0: Periodic calibration triggers cause full recalibration
23:20	OSC_COUNT_ERR_TOLERANCE	Magnitude of the difference between oscillator count and target count when the algorithm assumes convergence and stops further search.
19	TRACK_CALIB_MODE	1'b0: Tracking calibration uses previous Target_count value. 1'b1 Accurate mode: Target_count is recomputed at the start of tracking calibration

DIM_D01_DIM_DQ_CDC_CTLR_CFG0 (cont.)

Bits	Name	Description
18	FULL_DELAY	1'b0: unitstep and subunit caps are from calibration or SW settings 1'b1: All unit step and subunit caps are selected for the delay arrays.
17	HW_AUTOCAL_ENA	1'b1: Enables periodic auto calibration 1'b0: Periodic calibration is disabled
16	SW_TRIG_FULL_CALIB	A low to high transition on this register bit triggers the CDC calibration statemachine to initiate full calibration of the standby delay array. Also resets tracking mode for the delay arrays.
13:12	OSC_PRE_DIV	The factor by which the raw oscillator frequency is divided before being given to the counting logic. 2'b00 Divide by 1 2'b01 Divide by 2 2'b10 Divide by 4 2'b11 Divide by 8
10:0	TMUX_CHAR	Cchar is a characterized constant equal to the ratio of output mux offset delay to average unit step delay. This register is programmed by multiplying the constant with 1024 and rounding to the nearest integer. E.g. if Cchar = 0.67, $RO(0.67 * 1024) = 686$, hence program TMUX_CHAR = 11'h2AE

0x00481034 DIM_D01_DIM_DQ_CDC_CTLR_CFG1**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

The DIM_DQ_CDC_CTLR_CFG1 register configures the following:

DIM_DQ_CDC_CTLR_CFG1

DIM_D01_DIM_DQ_CDC_CTLR_CFG1

Bits	Name	Description
26:24	OSC_COUNT_DELAY	Number of oscillator cycles from the end of reference gate signal to assertion of oscillation counter done. Legal values are from 0 to 4.
23:20	STANDBY_DELAY	This is the number of reference clock cycles from when a newly calibrated delay array is made active to when the current active delay array is put into standby mode. 4'b0: Disables the counter. Hardware waits for explicit acknowledgement of delay array switch from CDC. Others: Explicit acknowledgement mode is disabled and the counter is used

DIM_D01_DIM_DQ_CDC_CTLR_CFG1 (cont.)

Bits	Name	Description
18:16	DEL_MODE_DELAY	Number of reference clock cycles between putting a newly calibrated delay array in delay mode to selecting it as the active delay array. The minimum allowed value is 1
14:12	OSC_MODE_DELAY	Number of reference clock cycles between putting a delay array in oscillator mode to starting the oscillation counter. This delay is allowed for the oscillations to stabilize through the delay array. The minimum allowed value is 1
10:8	DECODER_DELAY	Number of reference clock cycles allowed for unitstep and subunit counts to be encoded into the unit and finecap select lines on the DDA. The minimum allowed value is 1.
6:4	DIVIDER_DELAY	Number of reference clock cycles allowed for arithmetic division operations. The minimum allowed value is 1
2:0	MULTIPLIER_DELAY	Number of reference clock cycles allowed for arithmetic multiplication operations. The minimum allowed value is 1

0x00481038 DIM_D01_DIM_DQ_CDC_CAL_TIMER_CFG0**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

The DIM_DQ_CDC_CAL_TIMER_CFG0 register configures the following:

DIM_DQ_CDC_CAL_TIMER_CFG0

DIM_D01_DIM_DQ_CDC_CAL_TIMER_CFG0

Bits	Name	Description
24	INVALID_TIMER_ENA	1'b1: Enable the timer counting number of missed calibration intervals after which the CDC is invalid. 1'b0: Timer disabled A low to high transition on this register bit loads the timer with programmed value below.
23:20	INVALID_TIMER_VAL	Number of missed calibration intervals after which CDC invalid state is asserted.
16	TIMER_ENA	1'b1: Enables 32 kHz sleep clock based periodic autocalibration timer. 1'b0: Timer disabled A low to high transition on this register bit loads the timer with programmed value below.
15:0	TIMER_VAL	Count value for periodic autocalibration interval based on sleep clock.

0x0048103C DIM_D01_DIM_DQ_CDC_CAL_TIMER_CFG1**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

The DIM_DQ_CDC_CAL_TIMER_CFG1 register configures the following:

DIM_DQ_CDC_CAL_TIMER_CFG1

DIM_D01_DIM_DQ_CDC_CAL_TIMER_CFG1

Bits	Name	Description
12	FF_TIMER_ENA	1'b1: Enables reference clock based periodic autocalibration timer. 1'b0: Timer disabled A low to high transition on this register bit loads the timer with programmed value below.
9:0	FF_TIMER_VAL	Count value for periodic autocalibration interval based on reference clock.

0x00481040 DIM_D01_DIM_DQ_CDC_REFCOUNT_CFG**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

The DIM_DQ_CDC_REFCOUNT_CFG register configures the following:

DIM_DQ_CDC_REFCOUNT_CFG

DIM_D01_DIM_DQ_CDC_REFCOUNT_CFG

Bits	Name	Description
31:16	TREF	Reference clock period programmed in pico seconds.
13:8	CCAL_REF_COUNT	Number of reference clock cycles used for oscillation counts during coarse calibration. Maximum supported value is 59.
5:0	FCAL_REF_COUNT	Number of reference clock cycles used for oscillation counts during fine calibration. Maximum supported value is 59

0x00481044 DIM_D01_DIM_DQ_CDC_COARSE_CAL_CFG**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

The register DIM_DQ_CDC_COARSE_CAL_CFG configures the following:

DIM_DQ_CDC_COARSE_CAL_CFG

DIM_D01_DIM_DQ_CDC_COARSE_CAL_CFG

Bits	Name	Description
12:8	CCAL_SUBUNIT_CAPS	Subunit count needed for an effective unit step count of 32 during coarse calibration. This value is also used by software oscillator mode.
5:0	CCAL_UNITSTEPS	unitstep count needed for an effective unit step count of 32 during coarse calibration.

0x00481048 DIM_D01_DIM_DQ_CDC_RSVD_CFG

Type: RW**Clock:** HCLK**Reset State:** 0x00000000

The DIM_DQ_CDC_RSVD_CFG register configures the following:

DIM_DQ_CDC_RSVD_CFG

DIM_D01_DIM_DQ_CDC_RSVD_CFG

Bits	Name	Description
15:0	TEMP_FIELD	Reserved register

0x0048104C DIM_D01_DIM_DQ_RD_CDC_OFFSET_CFG

Type: RW**Clock:** HCLK**Reset State:** 0x00000000

The DIM_DQ_RD_CDC_OFFSET_CFG register configures the following:

DIM_DQ_RD_CDC_OFFSET_CFG

DIM_D01_DIM_DQ_RD_CDC_OFFSET_CFG

Bits	Name	Description
20	SUBUNIT_OFFSET_MODE	1'b1: The programmed offset is directly loaded as subunit count 1'b0: Programmed offset is added/subtracted from the subunit count
19	SUBUNIT_OFFSET_SIGN	1'b1: Offset is subtracted from the subunit count 1'b0: Offset is added to the subunit count
16:12	SUBUNIT_OFFSET	Software subunit offset

DIM_D01_DIM_DQ_RD_CDC_OFFSET_CFG (cont.)

Bits	Name	Description
8	UNIT_OFFSET_MODE	1'b1: The programmed offset is directly loaded as unit step count 1'b0: Programmed offset is added/subtracted from the unit step count
7	UNIT_OFFSET_SIGN	1'b1: Offset is subtracted from the unitstep count 1'b0: Offset is added to the unitstep count
5:0	UNIT_OFFSET	Software unitstep offset

0x00481050 DIM_D01_DIM_DQ_RD_CDC_DELAY_CFG**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

The DIM_DQ_RD_CDC_DELAY_CFG register configures the following:

DIM_DQ_RD_CDC_DELAY_CFG

DIM_D01_DIM_DQ_RD_CDC_DELAY_CFG

Bits	Name	Description
28	TARGET_COUNT_ENA	1'b1: Enable software specified Target_count for calibration instead of value calculated by hardware. 1'b0: Hardware computed Target_count used.
27:16	TARGET_COUNT	Software specified Target_count to which the statemachine calibrates
11:0	DELAY_VAL	Delay required from CDC programmed in pico seconds. This value has to be greater than the minimum delay supported by DDA Tmin.

0x00481054 DIM_D01_DIM_DQ_RD_CDC_SW_MODE_CFG**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

The DIM_DQ_RD_CDC_SW_MODE_CFG register configures the following:

DIM_DQ_RD_CDC_SW_MODE_CFG

DIM_D01_DIM_DQ_RD_CDC_SW_MODE_CFG

Bits	Name	Description
3	SW_DEL_MODE	A low to high transition on this register bit puts the delay array selected by SW_DA_SEL into delay mode and connects it to the DDA output. This register has to remain set for the duration of SW controlled delay mode.
2	SW_OSC_MODE	Reserved register.
1	SW_DA_SEL	Selects the delay array inside DDA during software delay or oscillator mode.
0	SW_LOAD	1'b1: Enable loading of the software offsets from the offset configuration register. This register has to remain set for the duration of SW mode operation.

0x0048105C DIM_D01_DIM_DQ_RD_CDC_SW_OVRD_CFG**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

The DIM_DQ_RD_CDC_SW_OVRD_CFG register configures the following:

DIM_DQ_RD_CDC_SW_OVRD_CFG

DIM_D01_DIM_DQ_RD_CDC_SW_OVRD_CFG

Bits	Name	Description
16	SW_REF_GATE	Write 1'b1 to start the reference window. Write 1'b0 to end the reference window. Number of reference cycle in the window will be reported in SW_REF_GATE_COUNT in CDC_STATUS0
11	SW_OVRD_DA1_OSC_EN	In software override mode, 1'b1 written to this bit puts delay array 1 in oscillator-mode
10	SW_OVRD_DA1_IN_EN	In software override mode, 1'b1 written to this bit puts delay array 1 in delay-mode
9	SW_OVRD_DA0_OSC_EN	In software override mode, 1'b1 written to this bit puts delay array 0 in oscillator-mode
8	SW_OVRD_DA0_IN_EN	In software override mode, 1'b1 written to this bit puts delay array 0 in delay-mode
3	SW_OVRD_CDC_COUNTEN R_RST	1'b1 written to this bit resets the oscillation counter. Follow this with 1'b0 before setting reference gate.
2	SW_OVRD_LOAD_DA_SEL	In software override mode, this bit select the delay array to which unit and subunit offset values from CDC_OFFSET_CFG register will be loaded.

DIM_D01_DIM_DQ_RD_CDC_SW_OVRD_CFG (cont.)

Bits	Name	Description
1	SW_OVRD_ACTV_DA_SEL	In software override mode, this register selects the delay array connected to the CDC output
0	SW_OVRD_ENA	1'b1 written to this bit puts the CDC interface in software override mode where the below register bits control the CDC. 1'b0: CDC is under hardware control

0x00481060 DIM_D01_DIM_DQ_RD_CDC_SLAVE_DDA_CFG**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

The DIM_DQ_RD_CDC_SLAVE_DDA_CFG register configures the following:

DIM_DQ_RD_CDC_SLAVE_DDA_CFG

DIM_D01_DIM_DQ_RD_CDC_SLAVE_DDA_CFG

Bits	Name	Description
17	SLAV_DDA_OFFSET_MODE	1'b1: The programmed offset is directly loaded as unit-step count 1'b0: Programmed offset is added/subtracted from the unit-step count
16	SLAV_DDA_OFFSET_SIGN	1'b1: Offset is subtracted from the unit-step count. This subtraction feature is not supported on x0 and x1 PHY cores (bit RESERVED). Supported on the x2 core version. 1'b0: Offset is added to the unit-step count
15:12	SLAVE_DDA_OFFSET	unit-step offset for slave DDA.
10:0	SLAVE_DDA_DELAY	Delay required from slave DDA programmed in pico seconds.

0x00481070 DIM_D01_DIM_DQ_RD_CDC_STATUS0**Type:** R**Clock:** HCLK**Reset State:** 0x0000000C

The DIM_DQ_RD_CDC_STATUS0 register configures the following:

DIM_DQ_RD_CDC_STATUS0

DIM_D01_DIM_DQ_RD_CDC_STATUS0

Bits	Name	Description
26:24	CDC_ERROR_CODE	Error code for internal errors: 3'b000: No error 3'b001: Programmed delay less than Tmux 3'b010: Programmed delay less than Tmin 3'b011: Unit-step overflow: Overflow during fine calibration. Desired delay cannot be realized even when all unit-steps are selected. 3'b100: Unit-step underflow 3'b101: Unit-step overflow: Overflow during coarse calibration. Estimated unit steps needed to realize the desired delay are greater than the total number of unit-steps in design. Note: The CDC_ERROR_CODE is not a sticky register and is updated after each CDC calibration.
23:16	SW_REF_GATE_COUNT	Number of reference clock cycles in the software generated oscillation gate window doing SW override mode
15:4	OSC_COUNT	Current value of oscillator count
3	CURR_SEL_DA	This bit indicates the current active delay array connected to the output. 1'b0: DA0 is selected 1'b1: DA1 is selected
2	CTLR_SM_IDLE	1'b1 Indicates that the calibration state machine is idle 1'b0 indicates calibration statemachine is busy
1	OSC_DONE	This status register is set after oscillations count is finished by CDC. It gets reset when a new oscillator count is started.
0	CALIBRATION_DONE	1'b1 indicates that calibration is done and CURR_SUBUNIT_STEPS, CURR_UNIT_STEPS status registers are valid.

0x00481074 DIM_D01_DIM_DQ_RD_CDC_STATUS1**Type:** R**Clock:** HCLK**Reset State:** 0x00000000

The DIM_DQ_RD_CDC_STATUS1 register configures the following:

DIM_DQ_RD_CDC_STATUS1

DIM_D01_DIM_DQ_RD_CDC_STATUS1

Bits	Name	Description
27:16	CURR_DELAY_VALUE	HW Computed value of current delay after calibration by the state machine. This register is valid if CALIBRATION_DONE status bit is set.
11:0	CURR_TMUX_DELAY	Current value of output mux offset delay computed by HW.

0x00481078 DIM_D01_DIM_DQ_RD_CDC_STATUS2**Type:** R**Clock:** HCLK**Reset State:** 0x10331033

The DIM_DQ_RD_CDC_STATUS2 register configures the following:

DIM_DQ_RD_CDC_STATUS2

DIM_D01_DIM_DQ_RD_CDC_STATUS2

Bits	Name	Description
28:24	DA1_SUBUNITS	Current value of subunit cap settings applied to delay array 1
21:16	DA1_UNITSTEPS	Current value of unit steps applied to delay array 1
12:8	DA0_SUBUNITS	Current value of subunit cap settings applied to delay array 0
5:0	DA0_UNITSTEPS	Current value of unit steps applied to delay array 0

0x0048107C DIM_D01_DIM_DQ_RD_CDC_STATUS3**Type:** R**Clock:** HCLK**Reset State:** 0x00000000

The DIM_DQ_RD_CDC_STATUS3 register configures the following:

DIM_DQ_RD_CDC_STATUS3

DIM_D01_DIM_DQ_RD_CDC_STATUS3

Bits	Name	Description
31:24	NUM_OF_OSC_ITER	Number of oscillator iterations before convergence was achieved during tracking
23:12	COUNT_ERROR	Difference between target and actual oscillator counts at the start of tracking calibration

DIM_D01_DIM_DQ_RD_CDC_STATUS3 (cont.)

Bits	Name	Description
11:0	CURR_TARGET_COUNT	HW computed value of current Target_count used by the calibration state machine.

0x00481080 DIM_D01_DIM_DQ_RD_CDC_STATUS4**Type:** R**Clock:** HCLK**Reset State:** 0x000000FF

The DIM_DQ_RD_CDC_STATUS4 register configures the following:

DIM_DQ_RD_CDC_STATUS4

DIM_D01_DIM_DQ_RD_CDC_STATUS4

Bits	Name	Description
7:4	SLAVE_DDA_DA1_TAPS	Number if unit taps applied to delay array 1 of slave DDA
3:0	SLAVE_DDA_DA0_TAPS	Number if unit taps applied to delay array 0 of slave DDA

0x004810AC DIM_D01_DIM_DQ_WR_CDC_OFFSET_CFG**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

The DIM_DQ_WR_CDC_OFFSET_CFG register configures the following:

DIM_DQ_WR_CDC_OFFSET_CFG

DIM_D01_DIM_DQ_WR_CDC_OFFSET_CFG

Bits	Name	Description
20	SUBUNIT_OFFSET_MODE	1'b1: The programmed offset is directly loaded as subunit count 1'b0: Programmed offset is added/subtracted from the subunit count
19	SUBUNIT_OFFSET_SIGN	1'b1: Offset is subtracted from the subunit count 1'b0: Offset is added to the subunit count
16:12	SUBUNIT_OFFSET	Software subunit offset
8	UNIT_OFFSET_MODE	1'b1: The programmed offset is directly loaded as unit step count 1'b0: Programmed offset is added/subtracted from the unit step count

DIM_D01_DIM_DQ_WR_CDC_OFFSET_CFG (cont.)

Bits	Name	Description
7	UNIT_OFFSET_SIGN	1'b1: Offset is subtracted from the unitstep count 1'b0: Offset is added to the unitstep count
5:0	UNIT_OFFSET	Software unitstep offset

0x004810B0 DIM_D01_DIM_DQ_WR_CDC_DELAY_CFG**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

The DIM_DQ_WR_CDC_DELAY_CFG register configures the following:

DIM_DQ_WR_CDC_DELAY_CFG

DIM_D01_DIM_DQ_WR_CDC_DELAY_CFG

Bits	Name	Description
28	TARGET_COUNT_ENA	1'b1: Enable software specified Target_count for calibration instead of value calculated by hardware. 1'b0: Hardware computed Target_count used.
27:16	TARGET_COUNT	Software specified Target_count to which the statemachine calibrates
11:0	DELAY_VAL	Delay required from CDC programmed in pico seconds. This value has to be greater than the minimum delay supported by DDA Tmin.

0x004810B4 DIM_D01_DIM_DQ_WR_CDC_SW_MODE_CFG**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

The DIM_DQ_WR_CDC_SW_MODE_CFG register configures the following:

DIM_DQ_WR_CDC_SW_MODE_CFG

DIM_D01_DIM_DQ_WR_CDC_SW_MODE_CFG

Bits	Name	Description
3	SW_DEL_MODE	A low to high transition on this register bit puts the delay array selected by SW_DA_SEL into delay mode and connects it to the DDA output. This register has to remain set for the duration of SW controlled delay mode.

DIM_D01_DIM_DQ_WR_CDC_SW_MODE_CFG (cont.)

Bits	Name	Description
2	SW_OSC_MODE	Reserved register.
1	SW_DA_SEL	Selects the delay array inside DDA during software delay or oscillator mode.
0	SW_LOAD	1'b1: Enable loading of the software offsets from the offset configuration register. This register has to remain set for the duration of SW mode operation.

0x004810BC DIM_D01_DIM_DQ_WR_CDC_SW_OVRD_CFG**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

The DIM_DQ_WR_CDC_SW_OVRD_CFG register configures the following:

DIM_DQ_WR_CDC_SW_OVRD_CFG

DIM_D01_DIM_DQ_WR_CDC_SW_OVRD_CFG

Bits	Name	Description
16	SW_REF_GATE	Write 1'b1 to start the reference window. Write 1'b0 to end the reference window. Number of reference cycle in the window will be reported in SW_REF_GATE_COUNT in CDC_STATUS0
11	SW_OVRD_DA1_OSC_EN	In software override mode, 1'b1 written to this bit puts delay array 1 in oscillator-mode
10	SW_OVRD_DA1_IN_EN	In software override mode, 1'b1 written to this bit puts delay array 1 in delay-mode
9	SW_OVRD_DA0_OSC_EN	In software override mode, 1'b1 written to this bit puts delay array 0 in oscillator-mode
8	SW_OVRD_DA0_IN_EN	In software override mode, 1'b1 written to this bit puts delay array 0 in delay-mode
3	SW_OVRD_CDC_COUNTERR_RST	1'b1 written to this bit resets the oscillation counter. Follow this with 1'b0 before setting reference gate.
2	SW_OVRD_LOAD_DA_SEL	In software override mode, this bit select the delay array to which unit and subunit offset values from CDC_OFFSET_CFG register will be loaded.
1	SW_OVRD_ACTV_DA_SEL	In software override mode, this register selects the delay array connected to the CDC output
0	SW_OVRD_ENA	1'b1 written to this bit puts the CDC interface in software override mode where the below register bits control the CDC. 1'b0: CDC is under hardware control

0x004810D0 DIM_D01_DIM_DQ_WR_CDC_STATUS0**Type:** R**Clock:** HCLK**Reset State:** 0x0000000C

The DIM_DQ_WR_CDC_STATUS0 register configures the following:

DIM_DQ_WR_CDC_STATUS0

DIM_D01_DIM_DQ_WR_CDC_STATUS0

Bits	Name	Description
26:24	CDC_ERROR_CODE	Error code for internal errors: 3'b000: No error 3'b001: Programmed delay less than Tmux 3'b010: Programmed delay less than Tmin 3'b011: Unit-step overflow: Overflow during fine calibration. Desired delay cannot be realized even when all unit-steps are selected. 3'b100: Unit-step underflow 3'b101: Unit-step overflow: Overflow during coarse calibration. Estimated unit steps needed to realize the desired delay are greater than the total number of unit-steps in design. Note: The CDC_ERROR_CODE is not a sticky register and is updated after each CDC calibration.
23:16	SW_REF_GATE_COUNT	Number of reference clock cycles in the software generated oscillation gate window doing SW override mode
15:4	OSC_COUNT	Current value of oscillator count
3	CURR_SEL_DA	This bit indicates the current active delay array connected to the output. 1'b0: DA0 is selected 1'b1: DA1 is selected
2	CTLR_SM_IDLE	1'b1 Indicates that the calibration state machine is idle 1'b0 indicates calibration statemachine is busy
1	OSC_DONE	This status register is set after oscillations count is finished by CDC. It gets reset when a new oscillator count is started.
0	CALIBRATION_DONE	1'b1 indicates that calibration is done and CURR_SUBUNIT_STEPS, CURR_UNIT_STEPS status registers are valid.

0x004810D4 DIM_D01_DIM_DQ_WR_CDC_STATUS1**Type:** R**Clock:** HCLK**Reset State:** 0x00000000

The DIM_DQ_WR_CDC_STATUS1 register configures the following:

DIM_DQ_WR_CDC_STATUS1

DIM_D01_DIM_DQ_WR_CDC_STATUS1

Bits	Name	Description
27:16	CURR_DELAY_VALUE	HW Computed value of current delay after calibration by the state machine. This register is valid if CALIBRATION_DONE status bit is set.
11:0	CURR_TMUX_DELAY	Current value of output mux offset delay computed by HW.

0x004810D8 DIM_D01_DIM_DQ_WR_CDC_STATUS2

Type: R

Clock: HCLK

Reset State: 0x10331033

The DIM_DQ_WR_CDC_STATUS2 register configures the following:

DIM_DQ_WR_CDC_STATUS2

DIM_D01_DIM_DQ_WR_CDC_STATUS2

Bits	Name	Description
28:24	DA1_SUBUNITS	Current value of subunit cap settings applied to delay array 1
21:16	DA1_UNITSTEPS	Current value of unit steps applied to delay array 1
12:8	DA0_SUBUNITS	Current value of subunit cap settings applied to delay array 0
5:0	DA0_UNITSTEPS	Current value of unit steps applied to delay array 0

0x004810DC DIM_D01_DIM_DQ_WR_CDC_STATUS3

Type: R

Clock: HCLK

Reset State: 0x00000000

The DIM_DQ_WR_CDC_STATUS3 register configures the following:

DIM_DQ_WR_CDC_STATUS3

DIM_D01_DIM_DQ_WR_CDC_STATUS3

Bits	Name	Description
31:24	NUM_OF_OSC_ITER	Number of oscillator iterations before convergence was achieved during tracking

DIM_D01_DIM_DQ_WR_CDC_STATUS3 (cont.)

Bits	Name	Description
23:12	COUNT_ERROR	Difference between target and actual oscillator counts at the start of tracking calibration
11:0	CURR_TARGET_COUNT	HW computed value of current Target_count used by the calibration state machine.

0x00481100 DIM_D01_DIM_DQ_DQ_IOC_SLV_CFG**Type:** RW**Clock:** HCLK**Reset State:** 0x10100000

The DIM_DQ_DQ_IOC_SLV_CFG register configures the following:

DIM_DQ_DQ_IOC_SLV_CFG

DIM_D01_DIM_DQ_DQ_IOC_SLV_CFG

Bits	Name	Description
31	PNCNT_HW_LOAD_EN	
28:24	NCNT_SW_VAL	
20:16	PCNT_SW_VAL	
15	NCNT_OFFSET_SIGN	
12:8	NCNT_SW_OFFSET	
7	PCNT_OFFSET_SIGN	
4:0	PCNT_SW_OFFSET	

0x00481104 DIM_D01_DIM_DQ_DQ_IOC_SLV_STATUS**Type:** R**Clock:** HCLK**Reset State:** 0x00001010

The DIM_DQ_DQ_IOC_SLV_STATUS register configures the following:

DIM_CA_CA_IOC_SLV_STATUS

DIM_D01_DIM_DQ_DQ_IOC_SLV_STATUS

Bits	Name	Description
12:8	PAD_NCNT	Impedance calibration bit settings for NMOS

DIM_D01_DIM_DQ_DQ_IOC_SLV_STATUS (cont.)

Bits	Name	Description
4:0	PAD_PCNT	Impedance calibration bit settings for PMOS

0x00481110 DIM_D01_DIM_DQ_DQS_IOC_SLV_CFG**Type:** RW**Clock:** HCLK**Reset State:** 0x10100000

The DIM_DQ_DQS_IOC_SLV_CFG register configures the following:

DIM_DQ_DQS_IOC_SLV_CFG

DIM_D01_DIM_DQ_DQS_IOC_SLV_CFG

Bits	Name	Description
31	PNCNT_HW_LOAD_EN	
28:24	NCNT_SW_VAL	
20:16	PCNT_SW_VAL	
15	NCNT_OFFSET_SIGN	
12:8	NCNT_SW_OFFSET	
7	PCNT_OFFSET_SIGN	
4:0	PCNT_SW_OFFSET	

0x00481114 DIM_D01_DIM_DQ_DQS_IOC_SLV_STATUS**Type:** R**Clock:** HCLK**Reset State:** 0x00001010

The DIM_DQ_DQS_IOC_SLV_STATUS register configures the following:

DIM_DQ_DQS_IOC_SLV_STATUS

DIM_D01_DIM_DQ_DQS_IOC_SLV_STATUS

Bits	Name	Description
12:8	PAD_NCNT	Impedance calibration bit settings for NMOS
4:0	PAD_PCNT	Impedance calibration bit settings for PMOS

7.4 DIM_D02_DDRPHY_DQ

0x00481800 DIM_D02_DIM_DQ_TOP_CFG

Type: RW

Clock: HCLK

Reset State: 0x00000031

The DIM_DQ_TOP_CFG register configures the following:

DIM_DQ_TOP_CFG

DIM_D02_DIM_DQ_TOP_CFG

Bits	Name	Description
30	DQ_LB_MODE_SEL	Selects external or internal loop back mode 1'b1: internal loop back mode 1'b0: external loop back mode (default)
29	DQ_LB_MODE_EN	Enables external or internal loop back mode 1'b1: loop back mode enabled 1'b0: loop back mode disabled (default)
28	RESERVED	Reserved
27	MISR_DATA_CLK_SEL	selects b/w MISR data or SEED and Loopback clock or FF clock 1'b1: from dte 1'b0: seed from csr ff clk (default)
26	CDC_LDO_EN	Enablement of CDC LDO 1'b1: Enabled 1'b0: Disabled LDO and power provided from switches (default)
25	CDC_SWITCH_RC_EN	Enablement of CDC power RC (LPF) switch 1'b1: Enabled 1'b0: Disabled (default)
24	CDC_SWITCH_BYPASS_OFF	Enablement of CDC power bypass switch 1'b1: Disabled 1'b0: Enabled (default)
16	RCW_EN	Enablement of the Read Capture Window 1'b1: Enable the RCW 1'b0: Disabled (default)
8	RESERVED	Reserved
5	WR_PIPE_EXTEND1	Write pipe extend select (when hp_mode = 1'b1) 1'b1: with pipe extend (default) 1'b0: without pipe extend
4	WR_PIPE_EXTEND0	Write pipe extend select (when hp_mode = 1'b0) 1'b1: with pipe extend (default) 1'b0: without pipe extend

DIM_D02_DIM_DQ_TOP_CFG (cont.)

Bits	Name	Description
0	WR_CLK_SEL	Write clock select 1'b1: With write CDC (default) 1'b0: Bypass write CDC, write clock from cdiv2 phi2 output

0x00481810 DIM_D02_DIM_DQ_PAD_CFG0**Type:** RW**Clock:** HCLK**Reset State:** 0xE0222240

The DIM_DQ_PAD_CFG0 register configures the following:

DIM_DQ_PAD_CFG0

DIM_D02_DIM_DQ_PAD_CFG0

Bits	Name	Description
31	DQ_DDR_MODE1	Mode pin for DDR (when hp_mode = 1'b1) 1'b1: DDR2 modes (default) 1'b0: DDR1 modes
30	DQ_DDR_MODE0	Mode pin for DDR (when hp_mode = 1'b0) 1'b1: DDR2 modes (default) 1'b0: DDR1 modes
29	DQ_LV_MODE	Mode select for high/low voltage regime
28	DQ_ODT_ENA	Unused - ODT controlled by PAD_CFG3
27:26	DQ_ODT	Unused - ODT controlled by PAD_CFG3
25:24	DQ_PULL_B	Input pull control
21:20	DQ_NSLEW	Slew rate control bits for output path NMOS
17:16	DQ_PSLEW	Slew rate control bits for output path PMOS
13:12	DQ_NRXDEL	Delay control bits to increase strength of NMOS input drive
9:8	DQ_PRXDEL	Delay control bits to increase strength of PMOS input driver
7	DQ_VM_SHIFT_ENA	VM (voltage at output of on-die termination) shift enable bit
6:4	DQ_ROUT	Impedance control bit settings for output driver
2:0	DQ_DCC	Duty cycle correction bits for output path

0x00481814 DIM_D02_DIM_DQ_PAD_CFG1**Type:** RW**Clock:** HCLK**Reset State:** 0xE0222240

The DIM_DQ_PAD_CFG1 register configures the following:

DIM_DQ_PAD_CFG1

DIM_D02_DIM_DQ_PAD_CFG1

Bits	Name	Description
31	DQS_DDR_MODE1	Mode pin for DDR (when hp_mode = 1'b1) 1'b1: DDR2 modes (default) 1'b0: DDR1 modes
30	DQS_DDR_MODE0	Mode pin for DDR (when hp_mode = 1'b0) 1'b1: DDR2 modes (default) 1'b0: DDR1 modes
29	DQS_LV_MODE	Mode select for high/low voltage regime
28	DQS_ODT_ENA	Unused - ODT controlled by PAD_CFG3
27:26	DQS_ODT	Unused - ODT controlled by PAD_CFG3
25:24	DQS_PULL	Input pull control 2'b00: no pull (default) 2'b01: pull down 2'b10: keeper 2'b11: pull up
21:20	DQS_NSLEW	Slew rate control bits for output path NMOS
17:16	DQS_PSLEW	Slew rate control bits for output path PMOS
13:12	DQS_NRXDEL	Delay control bits to increase strength of NMOS input drive
9:8	DQS_PRXDEL	Delay control bits to increase strength of PMOS input drive
7	DQS_VM_SHIFT_ENA	VM (voltage at output of on-die termination) shift enable bit
6:4	DQS_ROUT	Impedance control bit settings for output driver
2:0	DQS_DCC	Duty cycle correction bits for output path

0x00481818 DIM_D02_DIM_DQ_PAD_CFG2**Type:** RW**Clock:** HCLK**Reset State:** 0x1000000A

The DIM_DQ_PAD_CFG2 register configures the following:

DIM_DQ_PAD_CFG2

DIM_D02_DIM_DQ_PAD_CFG2

Bits	Name	Description
29	VREF_LDO_ENA	Enable VREF pad LDO Must be 0 for x2 core as there is no internal LDO
28	VREF_PASSGATE_ENA	Enable pass gate from padsig to vref_int pin (default 1)
27:24	VREF_SP_OUT	Unused
21:18	VREF_SP_IN	Unused
17	VREF_BYPASS_ENA	Enable bypass gate from resistive divider to vref_int pin
16	VREF_RDIV_ENA	Enable resistive divider for VREF voltage control
13:8	VREF_PULLDN_SPEED_CN TL	Unused
5:0	VREF_LEVEL_CNTL	Control over the VREF voltage level through resistive divider

0x0048181C DIM_D02_DIM_DQ_DQ_SMT_STATUS

Type: R

Clock: HCLK

Reset State: 0x00000000

The DIM_DQ_DQ_SMT_STATUS register reflects the status of SMT signals from DQ pads intended for CA training:

DIM_DQ_DQ_SMT_STATUS

DIM_D02_DIM_DQ_DQ_SMT_STATUS

Bits	Name	Description
7:0	DIM_DQ_DQ_SMT_STATUS	Status of SMT pins from DQ pads

0x00481820 DIM_D02_DIM_DQ_PAD_CFG3

Type: RW

Clock: HCLK

Reset State: 0x1000FF11

The DIM_DQ_PAD_CFG3 register configures the following:

DIM_DQ_PAD_CFG3

DIM_D02_DIM_DQ_PAD_CFG3

Bits	Name	Description
28	DQS_DIFF_MODE	DQS output mode control 1'b1: differential output (default) 1'b0: single-ended output; _n output is HIZ
27	RCW_ODT_ENA1	Enable ODT for RCW pad (when hp_mode=1)
26	RCW_ODT_ENA0	Enable ODT for RCW pad (when hp_mode=0)
25:24	RCW_ODT	Impedance control for on-die termination on RCW pad
23	DQ_ODT_ENA1	Enable ODT for DQ pads (when hp_mode=1)
22	DQ_ODT_ENA0	Enable ODT for DQ pads (when hp_mode=0)
21:20	DQ_ODT	Impedance control for on-die termination on DQ pads
19	DQS_ODT_ENA1	Enable ODT for DQS pad (when hp_mode=1)
18	DQS_ODT_ENA0	Enable ODT for DQS pad (when hp_mode=0)
17:16	DQS_ODT	Impedance control for on-die termination on DQS pad
15:8	DQ_IE_OE	Enable both input receiver and output driver for all DQ pads
5	RCW_IE_OE	Enable both input receiver and output driver for RCW pad
4	DQS_IE_OE	Enable both input receiver and output driver for DQS pad
1	DM_IE	Input received enable for DM pad
0	DM_OE	Output driver enable for DM pad

0x00481824 DIM_D02_DIM_DQ_PAD_CFG4**Type:** RW**Clock:** HCLK**Reset State:** 0xE0222240

The DIM_DQ_PAD_CFG4 register configures the following:

DIM_DQ_PAD_CFG4

DIM_D02_DIM_DQ_PAD_CFG4

Bits	Name	Description
31	RCW_DDR_MODE1	Mode pin for DDR (when hp_mode = 1'b1) 1'b1: DDR2 modes (default) 1'b0: DDR1 modes
30	RCW_DDR_MODE0	Mode pin for DDR (when hp_mode = 1'b0) 1'b1: DDR2 modes (default) 1'b0: DDR1 modes

DIM_D02_DIM_DQ_PAD_CFG4 (cont.)

Bits	Name	Description
29	RCW_LV_MODE	Mode select for high/low voltage regime
25:24	RCW_PULL_B	Input pull control
21:20	RCW_NSLEW	Slew rate control bits for output path NMOS
17:16	RCW_PSLEW	Slew rate control bits for output path PMOS
13:12	RCW_NRXDEL	Delay control bits to increase strength of NMOS input drive
9:8	RCW_PRXDEL	Delay control bits to increase strength of PMOS input driver
7	RCW_VM_SHIFT_ENA	VM (voltage at output of on-die termination) shift enable bit
6:4	RCW_ROUT	Impedance control bit settings for output driver
2:0	RCW_DCC	Duty cycle correction bits for output path

0x0048180C DIM_D02_DDRPHY_DQ_MISR_CFG**Type:** RW**Clock:** HCLK**Reset State:** 0x872C56BD

The DDRPHY_DQ_MISR_CFG register configures the following:

DDRPHY_DQ_MISR_CFG

DIM_D02_DDRPHY_DQ_MISR_CFG

Bits	Name	Description
31:0	SEED_VALUE	Loopback MISR seed

0x00481828 DIM_D02_DDRPHY_DQ_MISR_EVEN_STATUS**Type:** R**Clock:** HCLK**Reset State:** 0x00000000

The DDRPHY_DQ_MISR_EVEN_STATUS register configures the following:

DDRPHY_DQ_MISR_EVEN_STATUS

DIM_D02_DDRPHY_DQ_MISR_EVEN_STATUS

Bits	Name	Description
31:0	SIGNATURE	Even clock Loopback MISR signature

0x0048182C DIM_D02_DDRPHY_DQ_MISR_ODD_STATUS**Type:** R**Clock:** HCLK**Reset State:** 0x00000000

The DDRPHY_DQ_MISR_ODD_STATUS register configures the following:

DDRPHY_DQ_MISR_ODD_STATUS

DIM_D02_DDRPHY_DQ_MISR_ODD_STATUS

Bits	Name	Description
31:0	SIGNATURE	Odd Clock Loopback MISR signature

0x00481830 DIM_D02_DIM_DQ_CDC_CTLR_CFG0**Type:** RW**Clock:** HCLK**Reset State:** 0x02000000

The DIM_DQ_CDC_CTLR_CFG0 register configures the following:

DIM_DQ_CDC_CTLR_CFG0

DIM_D02_DIM_DQ_CDC_CTLR_CFG0

Bits	Name	Description
26	ERROR_CODE_UPDATE_EN	1'b1: Enable overwriting (default) the error code for subsequent calibrations. 1'b0: Disable overwriting the error code during subsequent calibrations
25	STAGGER_CAL_ENA	1'b1: Stagger calibration of write and read CDCs once after the other to reduce peak voltage drop. 1'b0: Both write and read CDCs calibrated simultaneously This bit is introduced on the x2 core, so not present on x0 and x1 versions of DIM/PHY.
24	TRACKING_CAL_ENA	1'b1: Relationship between subunits and oscillator count is learnt and used for tracking calibration. 1'b0: Periodic calibration triggers cause full recalibration
23:20	OSC_COUNT_ERR_TOLERANCE	Magnitude of the difference between oscillator count and target count when the algorithm assumes convergence and stops further search.
19	TRACK_CALIB_MODE	1'b0: Tracking calibration uses previous Target_count value. 1'b1 Accurate mode: Target_count is recomputed at the start of tracking calibration

DIM_D02_DIM_DQ_CDC_CTLR_CFG0 (cont.)

Bits	Name	Description
18	FULL_DELAY	1'b0: unitstep and subunit caps are from calibration or SW settings 1'b1: All unit step and subunit caps are selected for the delay arrays.
17	HW_AUTOCAL_ENA	1'b1: Enables periodic auto calibration 1'b0: Periodic calibration is disabled
16	SW_TRIG_FULL_CALIB	A low to high transition on this register bit triggers the CDC calibration statemachine to initiate full calibration of the standby delay array. Also resets tracking mode for the delay arrays.
13:12	OSC_PRE_DIV	The factor by which the raw oscillator frequency is divided before being given to the counting logic. 2'b00 Divide by 1 2'b01 Divide by 2 2'b10 Divide by 4 2'b11 Divide by 8
10:0	TMUX_CHAR	Cchar is a characterized constant equal to the ratio of output mux offset delay to average unit step delay. This register is programmed by multiplying the constant with 1024 and rounding to the nearest integer. E.g. if Cchar = 0.67, $RO(0.67 * 1024) = 686$, hence program TMUX_CHAR = 11'h2AE

0x00481834 DIM_D02_DIM_DQ_CDC_CTLR_CFG1**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

The DIM_DQ_CDC_CTLR_CFG1 register configures the following:

DIM_DQ_CDC_CTLR_CFG1

DIM_D02_DIM_DQ_CDC_CTLR_CFG1

Bits	Name	Description
26:24	OSC_COUNT_DELAY	Number of oscillator cycles from the end of reference gate signal to assertion of oscillation counter done. Legal values are from 0 to 4.
23:20	STANDBY_DELAY	This is the number of reference clock cycles from when a newly calibrated delay array is made active to when the current active delay array is put into standby mode. 4'b0: Disables the counter. Hardware waits for explicit acknowledgement of delay array switch from CDC. Others: Explicit acknowledgement mode is disabled and the counter is used

DIM_D02_DIM_DQ_CDC_CTLR_CFG1 (cont.)

Bits	Name	Description
18:16	DEL_MODE_DELAY	Number of reference clock cycles between putting a newly calibrated delay array in delay mode to selecting it as the active delay array. The minimum allowed value is 1
14:12	OSC_MODE_DELAY	Number of reference clock cycles between putting a delay array in oscillator mode to starting the oscillation counter. This delay is allowed for the oscillations to stabilize through the delay array. The minimum allowed value is 1
10:8	DECODER_DELAY	Number of reference clock cycles allowed for unitstep and subunit counts to be encoded into the unit and finecap select lines on the DDA. The minimum allowed value is 1.
6:4	DIVIDER_DELAY	Number of reference clock cycles allowed for arithmetic division operations. The minimum allowed value is 1
2:0	MULTIPLIER_DELAY	Number of reference clock cycles allowed for arithmetic multiplication operations. The minimum allowed value is 1

0x00481838 DIM_D02_DIM_DQ_CDC_CAL_TIMER_CFG0**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

The DIM_DQ_CDC_CAL_TIMER_CFG0 register configures the following:

DIM_DQ_CDC_CAL_TIMER_CFG0

DIM_D02_DIM_DQ_CDC_CAL_TIMER_CFG0

Bits	Name	Description
24	INVALID_TIMER_ENA	1'b1: Enable the timer counting number of missed calibration intervals after which the CDC is invalid. 1'b0: Timer disabled A low to high transition on this register bit loads the timer with programmed value below.
23:20	INVALID_TIMER_VAL	Number of missed calibration intervals after which CDC invalid state is asserted.
16	TIMER_ENA	1'b1: Enables 32 kHz sleep clock based periodic autocalibration timer. 1'b0: Timer disabled A low to high transition on this register bit loads the timer with programmed value below.
15:0	TIMER_VAL	Count value for periodic autocalibration interval based on sleep clock.

0x0048183C DIM_DQ_DIM_DQ_CDC_CAL_TIMER_CFG1**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

The DIM_DQ_CDC_CAL_TIMER_CFG1 register configures the following:

DIM_DQ_CDC_CAL_TIMER_CFG1

DIM_DQ_DIM_DQ_CDC_CAL_TIMER_CFG1

Bits	Name	Description
12	FF_TIMER_ENA	1'b1: Enables reference clock based periodic autocalibration timer. 1'b0: Timer disabled A low to high transition on this register bit loads the timer with programmed value below.
9:0	FF_TIMER_VAL	Count value for periodic autocalibration interval based on reference clock.

0x00481840 DIM_DQ_DIM_DQ_CDC_REFCOUNT_CFG**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

The DIM_DQ_CDC_REFCOUNT_CFG register configures the following:

DIM_DQ_CDC_REFCOUNT_CFG

DIM_DQ_DIM_DQ_CDC_REFCOUNT_CFG

Bits	Name	Description
31:16	TREF	Reference clock period programmed in pico seconds.
13:8	CCAL_REF_COUNT	Number of reference clock cycles used for oscillation counts during coarse calibration. Maximum supported value is 59.
5:0	FCAL_REF_COUNT	Number of reference clock cycles used for oscillation counts during fine calibration. Maximum supported value is 59

0x00481844 DIM_DQ_DIM_DQ_CDC_COARSE_CAL_CFG**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

The register DIM_DQ_CDC_COARSE_CAL_CFG configures the following:

DIM_DQ_CDC_COARSE_CAL_CFG

DIM_D02_DIM_DQ_CDC_COARSE_CAL_CFG

Bits	Name	Description
12:8	CCAL_SUBUNIT_CAPS	Subunit count needed for an effective unit step count of 32 during coarse calibration. This value is also used by software oscillator mode.
5:0	CCAL_UNITSTEPS	unitstep count needed for an effective unit step count of 32 during coarse calibration.

0x00481848 DIM_D02_DIM_DQ_CDC_RSVD_CFG

Type: RW

Clock: HCLK

Reset State: 0x00000000

The DIM_DQ_CDC_RSVD_CFG register configures the following:

DIM_DQ_CDC_RSVD_CFG

DIM_D02_DIM_DQ_CDC_RSVD_CFG

Bits	Name	Description
15:0	TEMP_FIELD	Reserved register

0x0048184C DIM_D02_DIM_DQ_RD_CDC_OFFSET_CFG

Type: RW

Clock: HCLK

Reset State: 0x00000000

The DIM_DQ_RD_CDC_OFFSET_CFG register configures the following:

DIM_DQ_RD_CDC_OFFSET_CFG

DIM_D02_DIM_DQ_RD_CDC_OFFSET_CFG

Bits	Name	Description
20	SUBUNIT_OFFSET_MODE	1'b1: The programmed offset is directly loaded as subunit count 1'b0: Programmed offset is added/subtracted from the subunit count
19	SUBUNIT_OFFSET_SIGN	1'b1: Offset is subtracted from the subunit count 1'b0: Offset is added to the subunit count
16:12	SUBUNIT_OFFSET	Software subunit offset

DIM_D02_DIM_DQ_RD_CDC_OFFSET_CFG (cont.)

Bits	Name	Description
8	UNIT_OFFSET_MODE	1'b1: The programmed offset is directly loaded as unit step count 1'b0: Programmed offset is added/subtracted from the unit step count
7	UNIT_OFFSET_SIGN	1'b1: Offset is subtracted from the unitstep count 1'b0: Offset is added to the unitstep count
5:0	UNIT_OFFSET	Software unitstep offset

0x00481850 DIM_D02_DIM_DQ_RD_CDC_DELAY_CFG**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

The DIM_DQ_RD_CDC_DELAY_CFG register configures the following:

DIM_DQ_RD_CDC_DELAY_CFG

DIM_D02_DIM_DQ_RD_CDC_DELAY_CFG

Bits	Name	Description
28	TARGET_COUNT_ENA	1'b1: Enable software specified Target_count for calibration instead of value calculated by hardware. 1'b0: Hardware computed Target_count used.
27:16	TARGET_COUNT	Software specified Target_count to which the statemachine calibrates
11:0	DELAY_VAL	Delay required from CDC programmed in pico seconds. This value has to be greater than the minimum delay supported by DDA Tmin.

0x00481854 DIM_D02_DIM_DQ_RD_CDC_SW_MODE_CFG**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

The DIM_DQ_RD_CDC_SW_MODE_CFG register configures the following:

DIM_DQ_RD_CDC_SW_MODE_CFG

DIM_D02_DIM_DQ_RD_CDC_SW_MODE_CFG

Bits	Name	Description
3	SW_DEL_MODE	A low to high transition on this register bit puts the delay array selected by SW_DA_SEL into delay mode and connects it to the DDA output. This register has to remain set for the duration of SW controlled delay mode.
2	SW_OSC_MODE	Reserved register.
1	SW_DA_SEL	Selects the delay array inside DDA during software delay or oscillator mode.
0	SW_LOAD	1'b1: Enable loading of the software offsets from the offset configuration register. This register has to remain set for the duration of SW mode operation.

0x0048185C DIM_D02_DIM_DQ_RD_CDC_SW_OVRD_CFG**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

The DIM_DQ_RD_CDC_SW_OVRD_CFG register configures the following:

DIM_DQ_RD_CDC_SW_OVRD_CFG

DIM_D02_DIM_DQ_RD_CDC_SW_OVRD_CFG

Bits	Name	Description
16	SW_REF_GATE	Write 1'b1 to start the reference window. Write 1'b0 to end the reference window. Number of reference cycle in the window will be reported in SW_REF_GATE_COUNT in CDC_STATUS0
11	SW_OVRD_DA1_OSC_EN	In software override mode, 1'b1 written to this bit puts delay array 1 in oscillator-mode
10	SW_OVRD_DA1_IN_EN	In software override mode, 1'b1 written to this bit puts delay array 1 in delay-mode
9	SW_OVRD_DA0_OSC_EN	In software override mode, 1'b1 written to this bit puts delay array 0 in oscillator-mode
8	SW_OVRD_DA0_IN_EN	In software override mode, 1'b1 written to this bit puts delay array 0 in delay-mode
3	SW_OVRD_CDC_COUNTERR_RST	1'b1 written to this bit resets the oscillation counter. Follow this with 1'b0 before setting reference gate.
2	SW_OVRD_LOAD_DA_SEL	In software override mode, this bit select the delay array to which unit and subunit offset values from CDC_OFFSET_CFG register will be loaded.

DIM_D02_DIM_DQ_RD_CDC_SW_OVRD_CFG (cont.)

Bits	Name	Description
1	SW_OVRD_ACTV_DA_SEL	In software override mode, this register selects the delay array connected to the CDC output
0	SW_OVRD_ENA	1'b1 written to this bit puts the CDC interface in software override mode where the below register bits control the CDC. 1'b0: CDC is under hardware control

0x00481860 DIM_D02_DIM_DQ_RD_CDC_SLAVE_DDA_CFG**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

The DIM_DQ_RD_CDC_SLAVE_DDA_CFG register configures the following:

DIM_DQ_RD_CDC_SLAVE_DDA_CFG

DIM_D02_DIM_DQ_RD_CDC_SLAVE_DDA_CFG

Bits	Name	Description
17	SLAV_DDA_OFFSET_MODE	1'b1: The programmed offset is directly loaded as unit-step count 1'b0: Programmed offset is added/subtracted from the unit-step count
16	SLAV_DDA_OFFSET_SIGN	1'b1: Offset is subtracted from the unit-step count. This subtraction feature is not supported on x0 and x1 PHY cores (bit RESERVED). Supported on the x2 core version. 1'b0: Offset is added to the unit-step count
15:12	SLAVE_DDA_OFFSET	unit-step offset for slave DDA.
10:0	SLAVE_DDA_DELAY	Delay required from slave DDA programmed in pico seconds.

0x00481870 DIM_D02_DIM_DQ_RD_CDC_STATUS0**Type:** R**Clock:** HCLK**Reset State:** 0x0000000C

The DIM_DQ_RD_CDC_STATUS0 register configures the following:

DIM_DQ_RD_CDC_STATUS0

DIM_D02_DIM_DQ_RD_CDC_STATUS0

Bits	Name	Description
26:24	CDC_ERROR_CODE	Error code for internal errors: 3'b000: No error 3'b001: Programmed delay less than Tmux 3'b010: Programmed delay less than Tmin 3'b011: Unit-step overflow: Overflow during fine calibration. Desired delay cannot be realized even when all unit-steps are selected. 3'b100: Unit-step underflow 3'b101: Unit-step overflow: Overflow during coarse calibration. Estimated unit steps needed to realize the desired delay are greater than the total number of unit-steps in design. Note: The CDC_ERROR_CODE is not a sticky register and is updated after each CDC calibration.
23:16	SW_REF_GATE_COUNT	Number of reference clock cycles in the software generated oscillation gate window doing SW override mode
15:4	OSC_COUNT	Current value of oscillator count
3	CURR_SEL_DA	This bit indicates the current active delay array connected to the output. 1'b0: DA0 is selected 1'b1: DA1 is selected
2	CTLR_SM_IDLE	1'b1 Indicates that the calibration state machine is idle 1'b0 indicates calibration statemachine is busy
1	OSC_DONE	This status register is set after oscillations count is finished by CDC. It gets reset when a new oscillator count is started.
0	CALIBRATION_DONE	1'b1 indicates that calibration is done and CURR_SUBUNIT_STEPS, CURR_UNIT_STEPS status registers are valid.

0x00481874 DIM_D02_DIM_DQ_RD_CDC_STATUS1**Type:** R**Clock:** HCLK**Reset State:** 0x00000000

The DIM_DQ_RD_CDC_STATUS1 register configures the following:

DIM_DQ_RD_CDC_STATUS1

DIM_D02_DIM_DQ_RD_CDC_STATUS1

Bits	Name	Description
27:16	CURR_DELAY_VALUE	HW Computed value of current delay after calibration by the state machine. This register is valid if CALIBRATION_DONE status bit is set.
11:0	CURR_TMUX_DELAY	Current value of output mux offset delay computed by HW.

0x00481878 DIM_D02_DIM_DQ_RD_CDC_STATUS2**Type:** R**Clock:** HCLK**Reset State:** 0x10331033

The DIM_DQ_RD_CDC_STATUS2 register configures the following:

DIM_DQ_RD_CDC_STATUS2

DIM_D02_DIM_DQ_RD_CDC_STATUS2

Bits	Name	Description
28:24	DA1_SUBUNITS	Current value of subunit cap settings applied to delay array 1
21:16	DA1_UNITSTEPS	Current value of unit steps applied to delay array 1
12:8	DA0_SUBUNITS	Current value of subunit cap settings applied to delay array 0
5:0	DA0_UNITSTEPS	Current value of unit steps applied to delay array 0

0x0048187C DIM_D02_DIM_DQ_RD_CDC_STATUS3**Type:** R**Clock:** HCLK**Reset State:** 0x00000000

The DIM_DQ_RD_CDC_STATUS3 register configures the following:

DIM_DQ_RD_CDC_STATUS3

DIM_D02_DIM_DQ_RD_CDC_STATUS3

Bits	Name	Description
31:24	NUM_OF_OSC_ITER	Number of oscillator iterations before convergence was achieved during tracking
23:12	COUNT_ERROR	Difference between target and actual oscillator counts at the start of tracking calibration

DIM_D02_DIM_DQ_RD_CDC_STATUS3 (cont.)

Bits	Name	Description
11:0	CURR_TARGET_COUNT	HW computed value of current Target_count used by the calibration state machine.

0x00481880 DIM_D02_DIM_DQ_RD_CDC_STATUS4**Type:** R**Clock:** HCLK**Reset State:** 0x000000FF

The DIM_DQ_RD_CDC_STATUS4 register configures the following:

DIM_DQ_RD_CDC_STATUS4

DIM_D02_DIM_DQ_RD_CDC_STATUS4

Bits	Name	Description
7:4	SLAVE_DDA_DA1_TAPS	Number if unit taps applied to delay array 1 of slave DDA
3:0	SLAVE_DDA_DA0_TAPS	Number if unit taps applied to delay array 0 of slave DDA

0x004818AC DIM_D02_DIM_DQ_WR_CDC_OFFSET_CFG**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

The DIM_DQ_WR_CDC_OFFSET_CFG register configures the following:

DIM_DQ_WR_CDC_OFFSET_CFG

DIM_D02_DIM_DQ_WR_CDC_OFFSET_CFG

Bits	Name	Description
20	SUBUNIT_OFFSET_MODE	1'b1: The programmed offset is directly loaded as subunit count 1'b0: Programmed offset is added/subtracted from the subunit count
19	SUBUNIT_OFFSET_SIGN	1'b1: Offset is subtracted from the subunit count 1'b0: Offset is added to the subunit count
16:12	SUBUNIT_OFFSET	Software subunit offset
8	UNIT_OFFSET_MODE	1'b1: The programmed offset is directly loaded as unit step count 1'b0: Programmed offset is added/subtracted from the unit step count

DIM_D02_DIM_DQ_WR_CDC_OFFSET_CFG (cont.)

Bits	Name	Description
7	UNIT_OFFSET_SIGN	1'b1: Offset is subtracted from the unitstep count 1'b0: Offset is added to the unitstep count
5:0	UNIT_OFFSET	Software unitstep offset

0x004818B0 DIM_D02_DIM_DQ_WR_CDC_DELAY_CFG**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

The DIM_DQ_WR_CDC_DELAY_CFG register configures the following:

DIM_DQ_WR_CDC_DELAY_CFG

DIM_D02_DIM_DQ_WR_CDC_DELAY_CFG

Bits	Name	Description
28	TARGET_COUNT_ENA	1'b1: Enable software specified Target_count for calibration instead of value calculated by hardware. 1'b0: Hardware computed Target_count used.
27:16	TARGET_COUNT	Software specified Target_count to which the statemachine calibrates
11:0	DELAY_VAL	Delay required from CDC programmed in pico seconds. This value has to be greater than the minimum delay supported by DDA Tmin.

0x004818B4 DIM_D02_DIM_DQ_WR_CDC_SW_MODE_CFG**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

The DIM_DQ_WR_CDC_SW_MODE_CFG register configures the following:

DIM_DQ_WR_CDC_SW_MODE_CFG

DIM_D02_DIM_DQ_WR_CDC_SW_MODE_CFG

Bits	Name	Description
3	SW_DEL_MODE	A low to high transition on this register bit puts the delay array selected by SW_DA_SEL into delay mode and connects it to the DDA output. This register has to remain set for the duration of SW controlled delay mode.

DIM_D02_DIM_DQ_WR_CDC_SW_MODE_CFG (cont.)

Bits	Name	Description
2	SW_OSC_MODE	Reserved register.
1	SW_DA_SEL	Selects the delay array inside DDA during software delay or oscillator mode.
0	SW_LOAD	1'b1: Enable loading of the software offsets from the offset configuration register. This register has to remain set for the duration of SW mode operation.

0x004818BC DIM_D02_DIM_DQ_WR_CDC_SW_OVRD_CFG**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

The DIM_DQ_WR_CDC_SW_OVRD_CFG register configures the following:

DIM_DQ_WR_CDC_SW_OVRD_CFG

DIM_D02_DIM_DQ_WR_CDC_SW_OVRD_CFG

Bits	Name	Description
16	SW_REF_GATE	Write 1'b1 to start the reference window. Write 1'b0 to end the reference window. Number of reference cycle in the window will be reported in SW_REF_GATE_COUNT in CDC_STATUS0
11	SW_OVRD_DA1_OSC_EN	In software override mode, 1'b1 written to this bit puts delay array 1 in oscillator-mode
10	SW_OVRD_DA1_IN_EN	In software override mode, 1'b1 written to this bit puts delay array 1 in delay-mode
9	SW_OVRD_DA0_OSC_EN	In software override mode, 1'b1 written to this bit puts delay array 0 in oscillator-mode
8	SW_OVRD_DA0_IN_EN	In software override mode, 1'b1 written to this bit puts delay array 0 in delay-mode
3	SW_OVRD_CDC_COUNTERR_RST	1'b1 written to this bit resets the oscillation counter. Follow this with 1'b0 before setting reference gate.
2	SW_OVRD_LOAD_DA_SEL	In software override mode, this bit select the delay array to which unit and subunit offset values from CDC_OFFSET_CFG register will be loaded.
1	SW_OVRD_ACTV_DA_SEL	In software override mode, this register selects the delay array connected to the CDC output
0	SW_OVRD_ENA	1'b1 written to this bit puts the CDC interface in software override mode where the below register bits control the CDC. 1'b0: CDC is under hardware control

0x004818D0 DIM_D02_DIM_DQ_WR_CDC_STATUS0**Type:** R**Clock:** HCLK**Reset State:** 0x0000000C

The DIM_DQ_WR_CDC_STATUS0 register configures the following:

DIM_DQ_WR_CDC_STATUS0

DIM_D02_DIM_DQ_WR_CDC_STATUS0

Bits	Name	Description
26:24	CDC_ERROR_CODE	Error code for internal errors: 3'b000: No error 3'b001: Programmed delay less than Tmux 3'b010: Programmed delay less than Tmin 3'b011: Unit-step overflow: Overflow during fine calibration. Desired delay cannot be realized even when all unit-steps are selected. 3'b100: Unit-step underflow 3'b101: Unit-step overflow: Overflow during coarse calibration. Estimated unit steps needed to realize the desired delay are greater than the total number of unit-steps in design. Note: The CDC_ERROR_CODE is not a sticky register and is updated after each CDC calibration.
23:16	SW_REF_GATE_COUNT	Number of reference clock cycles in the software generated oscillation gate window doing SW override mode
15:4	OSC_COUNT	Current value of oscillator count
3	CURR_SEL_DA	This bit indicates the current active delay array connected to the output. 1'b0: DA0 is selected 1'b1: DA1 is selected
2	CTLR_SM_IDLE	1'b1 Indicates that the calibration state machine is idle 1'b0 indicates calibration statemachine is busy
1	OSC_DONE	This status register is set after oscillations count is finished by CDC. It gets reset when a new oscillator count is started.
0	CALIBRATION_DONE	1'b1 indicates that calibration is done and CURR_SUBUNIT_STEPS, CURR_UNIT_STEPS status registers are valid.

0x004818D4 DIM_D02_DIM_DQ_WR_CDC_STATUS1**Type:** R**Clock:** HCLK**Reset State:** 0x00000000

The DIM_DQ_WR_CDC_STATUS1 register configures the following:

DIM_DQ_WR_CDC_STATUS1

DIM_D02_DIM_DQ_WR_CDC_STATUS1

Bits	Name	Description
27:16	CURR_DELAY_VALUE	HW Computed value of current delay after calibration by the state machine. This register is valid if CALIBRATION_DONE status bit is set.
11:0	CURR_TMUX_DELAY	Current value of output mux offset delay computed by HW.

0x004818D8 DIM_D02_DIM_DQ_WR_CDC_STATUS2

Type: R

Clock: HCLK

Reset State: 0x10331033

The DIM_DQ_WR_CDC_STATUS2 register configures the following:

DIM_DQ_WR_CDC_STATUS2

DIM_D02_DIM_DQ_WR_CDC_STATUS2

Bits	Name	Description
28:24	DA1_SUBUNITS	Current value of subunit cap settings applied to delay array 1
21:16	DA1_UNITSTEPS	Current value of unit steps applied to delay array 1
12:8	DA0_SUBUNITS	Current value of subunit cap settings applied to delay array 0
5:0	DA0_UNITSTEPS	Current value of unit steps applied to delay array 0

0x004818DC DIM_D02_DIM_DQ_WR_CDC_STATUS3

Type: R

Clock: HCLK

Reset State: 0x00000000

The DIM_DQ_WR_CDC_STATUS3 register configures the following:

DIM_DQ_WR_CDC_STATUS3

DIM_D02_DIM_DQ_WR_CDC_STATUS3

Bits	Name	Description
31:24	NUM_OF_OSC_ITER	Number of oscillator iterations before convergence was achieved during tracking

DIM_D02_DIM_DQ_WR_CDC_STATUS3 (cont.)

Bits	Name	Description
23:12	COUNT_ERROR	Difference between target and actual oscillator counts at the start of tracking calibration
11:0	CURR_TARGET_COUNT	HW computed value of current Target_count used by the calibration state machine.

0x00481900 DIM_D02_DIM_DQ_DQ_IOC_SLV_CFG**Type:** RW**Clock:** HCLK**Reset State:** 0x10100000

The DIM_DQ_DQ_IOC_SLV_CFG register configures the following:

DIM_DQ_DQ_IOC_SLV_CFG

DIM_D02_DIM_DQ_DQ_IOC_SLV_CFG

Bits	Name	Description
31	PNCNT_HW_LOAD_EN	
28:24	NCNT_SW_VAL	
20:16	PCNT_SW_VAL	
15	NCNT_OFFSET_SIGN	
12:8	NCNT_SW_OFFSET	
7	PCNT_OFFSET_SIGN	
4:0	PCNT_SW_OFFSET	

0x00481904 DIM_D02_DIM_DQ_DQ_IOC_SLV_STATUS**Type:** R**Clock:** HCLK**Reset State:** 0x00001010

The DIM_DQ_DQ_IOC_SLV_STATUS register configures the following:

DIM_CA_CA_IOC_SLV_STATUS

DIM_D02_DIM_DQ_DQ_IOC_SLV_STATUS

Bits	Name	Description
12:8	PAD_NCNT	Impedance calibration bit settings for NMOS

DIM_D02_DIM_DQ_DQ_IOC_SLV_STATUS (cont.)

Bits	Name	Description
4:0	PAD_PCNT	Impedance calibration bit settings for PMOS

0x00481910 DIM_D02_DIM_DQ_DQS_IOC_SLV_CFG**Type:** RW**Clock:** HCLK**Reset State:** 0x10100000

The DIM_DQ_DQS_IOC_SLV_CFG register configures the following:

DIM_DQ_DQS_IOC_SLV_CFG

DIM_D02_DIM_DQ_DQS_IOC_SLV_CFG

Bits	Name	Description
31	PNCNT_HW_LOAD_EN	
28:24	NCNT_SW_VAL	
20:16	PCNT_SW_VAL	
15	NCNT_OFFSET_SIGN	
12:8	NCNT_SW_OFFSET	
7	PCNT_OFFSET_SIGN	
4:0	PCNT_SW_OFFSET	

0x00481914 DIM_D02_DIM_DQ_DQS_IOC_SLV_STATUS**Type:** R**Clock:** HCLK**Reset State:** 0x00001010

The DIM_DQ_DQS_IOC_SLV_STATUS register configures the following:

DIM_DQ_DQS_IOC_SLV_STATUS

DIM_D02_DIM_DQ_DQS_IOC_SLV_STATUS

Bits	Name	Description
12:8	PAD_NCNT	Impedance calibration bit settings for NMOS
4:0	PAD_PCNT	Impedance calibration bit settings for PMOS

7.5 DIM_D03_DDRPHY_DQ

0x00482000 DIM_D03_DIM_DQ_TOP_CFG

Type: RW

Clock: HCLK

Reset State: 0x00000031

The DIM_DQ_TOP_CFG register configures the following:

DIM_DQ_TOP_CFG

DIM_D03_DIM_DQ_TOP_CFG

Bits	Name	Description
30	DQ_LB_MODE_SEL	Selects external or internal loop back mode 1'b1: internal loop back mode 1'b0: external loop back mode (default)
29	DQ_LB_MODE_EN	Enables external or internal loop back mode 1'b1: loop back mode enabled 1'b0: loop back mode disabled (default)
28	RESERVED	Reserved
27	MISR_DATA_CLK_SEL	selects b/w MISR data or SEED and Loopback clock or FF clock 1'b1: reserved 1'b0: seed from csr ff clk (default)
26	CDC_LDO_EN	Enablement of CDC LDO 1'b1: Enabled 1'b0: Disabled LDO and power provided from switches (default)
25	CDC_SWITCH_RC_EN	Enablement of CDC power RC (LPF) switch 1'b1: Enabled 1'b0: Disabled (default)
24	CDC_SWITCH_BYPASS_OFF	Enablement of CDC power bypass switch 1'b1: Disabled 1'b0: Enabled (default)
16	RCW_EN	Enablement of the Read Capture Window 1'b1: Enable the RCW 1'b0: Disabled (default)
8	RESERVED	Reserved
5	WR_PIPE_EXTEND1	Write pipe extend select (when hp_mode = 1'b1) 1'b1: with pipe extend (default) 1'b0: without pipe extend
4	WR_PIPE_EXTEND0	Write pipe extend select (when hp_mode = 1'b0) 1'b1: with pipe extend (default) 1'b0: without pipe extend

DIM_D03_DIM_DQ_TOP_CFG (cont.)

Bits	Name	Description
0	WR_CLK_SEL	Write clock select 1'b1: With write CDC (default) 1'b0: Bypass write CDC, write clock from cdiv2 phi2 output

0x00482010 DIM_D03_DIM_DQ_PAD_CFG0**Type:** RW**Clock:** HCLK**Reset State:** 0xE0222240

The DIM_DQ_PAD_CFG0 register configures the following:

DIM_DQ_PAD_CFG0

DIM_D03_DIM_DQ_PAD_CFG0

Bits	Name	Description
31	DQ_DDR_MODE1	Mode pin for DDR (when hp_mode = 1'b1) 1'b1: DDR2 modes (default) 1'b0: DDR1 modes
30	DQ_DDR_MODE0	Mode pin for DDR (when hp_mode = 1'b0) 1'b1: DDR2 modes (default) 1'b0: DDR1 modes
29	DQ_LV_MODE	Mode select for high/low voltage regime
28	DQ_ODT_ENA	Unused - ODT controlled by PAD_CFG3
27:26	DQ_ODT	Unused - ODT controlled by PAD_CFG3
25:24	DQ_PULL_B	Input pull control
21:20	DQ_NSLEW	Slew rate control bits for output path NMOS
17:16	DQ_PSLEW	Slew rate control bits for output path PMOS
13:12	DQ_NRXDEL	Delay control bits to increase strength of NMOS input drive
9:8	DQ_PRXDEL	Delay control bits to increase strength of PMOS input driver
7	DQ_VM_SHIFT_ENA	VM (voltage at output of on-die termination) shift enable bit
6:4	DQ_ROUT	Impedance control bit settings for output driver
2:0	DQ_DCC	Duty cycle correction bits for output path

0x00482014 DIM_D03_DIM_DQ_PAD_CFG1**Type:** RW**Clock:** HCLK**Reset State:** 0xE0222240

The DIM_DQ_PAD_CFG1 register configures the following:

DIM_DQ_PAD_CFG1

DIM_D03_DIM_DQ_PAD_CFG1

Bits	Name	Description
31	DQS_DDR_MODE1	Mode pin for DDR (when hp_mode = 1'b1) 1'b1: DDR2 modes (default) 1'b0: DDR1 modes
30	DQS_DDR_MODE0	Mode pin for DDR (when hp_mode = 1'b0) 1'b1: DDR2 modes (default) 1'b0: DDR1 modes
29	DQS_LV_MODE	Mode select for high/low voltage regime
28	DQS_ODT_ENA	Unused - ODT controlled by PAD_CFG3
27:26	DQS_ODT	Unused - ODT controlled by PAD_CFG3
25:24	DQS_PULL	Input pull control 2'b00: no pull (default) 2'b01: pull down 2'b10: keeper 2'b11: pull up
21:20	DQS_NSLEW	Slew rate control bits for output path NMOS
17:16	DQS_PSLEW	Slew rate control bits for output path PMOS
13:12	DQS_NRXDEL	Delay control bits to increase strength of NMOS input drive
9:8	DQS_PRXDEL	Delay control bits to increase strength of PMOS input drive
7	DQS_VM_SHIFT_ENA	VM (voltage at output of on-die termination) shift enable bit
6:4	DQS_ROUT	Impedance control bit settings for output driver
2:0	DQS_DCC	Duty cycle correction bits for output path

0x00482018 DIM_D03_DIM_DQ_PAD_CFG2**Type:** RW**Clock:** HCLK**Reset State:** 0x1000000A

The DIM_DQ_PAD_CFG2 register configures the following:

DIM_DQ_PAD_CFG2

DIM_D03_DIM_DQ_PAD_CFG2

Bits	Name	Description
29	VREF_LDO_ENA	Enable VREF pad LDO Must be 0 for x2 core as there is no internal LDO
28	VREF_PASSGATE_ENA	Enable pass gate from padsig to vref_int pin (default 1)
27:24	VREF_SP_OUT	Unused
21:18	VREF_SP_IN	Unused
17	VREF_BYPASS_ENA	Enable bypass gate from resistive divider to vref_int pin
16	VREF_RDIV_ENA	Enable resistive divider for VREF voltage control
13:8	VREF_PULLDN_SPEED_CN TL	Unused
5:0	VREF_LEVEL_CNTL	Control over the VREF voltage level through resistive divider

0x0048201C DIM_D03_DIM_DQ_DQ_SMT_STATUS

Type: R

Clock: HCLK

Reset State: 0x00000000

The DIM_DQ_DQ_SMT_STATUS register reflects the status of SMT signals from DQ pads intended for CA training:

DIM_DQ_DQ_SMT_STATUS

DIM_D03_DIM_DQ_DQ_SMT_STATUS

Bits	Name	Description
7:0	DIM_DQ_DQ_SMT_STATUS	Status of SMT pins from DQ pads

0x00482020 DIM_D03_DIM_DQ_PAD_CFG3

Type: RW

Clock: HCLK

Reset State: 0x1000FF11

The DIM_DQ_PAD_CFG3 register configures the following:

DIM_DQ_PAD_CFG3

DIM_D03_DIM_DQ_PAD_CFG3

Bits	Name	Description
28	DQS_DIFF_MODE	DQS output mode control 1'b1: differential output (default) 1'b0: single-ended output; _n output is HIZ
27	RCW_ODT_ENA1	Enable ODT for RCW pad (when hp_mode=1)
26	RCW_ODT_ENA0	Enable ODT for RCW pad (when hp_mode=0)
25:24	RCW_ODT	Impedance control for on-die termination on RCW pad
23	DQ_ODT_ENA1	Enable ODT for DQ pads (when hp_mode=1)
22	DQ_ODT_ENA0	Enable ODT for DQ pads (when hp_mode=0)
21:20	DQ_ODT	Impedance control for on-die termination on DQ pads
19	DQS_ODT_ENA1	Enable ODT for DQS pad (when hp_mode=1)
18	DQS_ODT_ENA0	Enable ODT for DQS pad (when hp_mode=0)
17:16	DQS_ODT	Impedance control for on-die termination on DQS pad
15:8	DQ_IE_OE	Enable both input receiver and output driver for all DQ pads
5	RCW_IE_OE	Enable both input receiver and output driver for RCW pad
4	DQS_IE_OE	Enable both input receiver and output driver for DQS pad
1	DM_IE	Input received enable for DM pad
0	DM_OE	Output driver enable for DM pad

0x00482024 DIM_D03_DIM_DQ_PAD_CFG4**Type:** RW**Clock:** HCLK**Reset State:** 0xE0222240

The DIM_DQ_PAD_CFG4 register configures the following:

DIM_DQ_PAD_CFG4

DIM_D03_DIM_DQ_PAD_CFG4

Bits	Name	Description
31	RCW_DDR_MODE1	Mode pin for DDR (when hp_mode = 1'b1) 1'b1: DDR2 modes (default) 1'b0: DDR1 modes
30	RCW_DDR_MODE0	Mode pin for DDR (when hp_mode = 1'b0) 1'b1: DDR2 modes (default) 1'b0: DDR1 modes

DIM_D03_DIM_DQ_PAD_CFG4 (cont.)

Bits	Name	Description
29	RCW_LV_MODE	Mode select for high/low voltage regime
25:24	RCW_PULL_B	Input pull control
21:20	RCW_NSLEW	Slew rate control bits for output path NMOS
17:16	RCW_PSLEW	Slew rate control bits for output path PMOS
13:12	RCW_NRXDEL	Delay control bits to increase strength of NMOS input drive
9:8	RCW_PRXDEL	Delay control bits to increase strength of PMOS input driver
7	RCW_VM_SHIFT_ENA	VM (voltage at output of on-die termination) shift enable bit
6:4	RCW_ROUT	Impedance control bit settings for output driver
2:0	RCW_DCC	Duty cycle correction bits for output path

0x0048200C DIM_D03_DDRPHY_DQ_MISR_CFG**Type:** RW**Clock:** HCLK**Reset State:** 0x872C56BD

The DDRPHY_DQ_MISR_CFG register configures the following:

DDRPHY_DQ_MISR_CFG

DIM_D03_DDRPHY_DQ_MISR_CFG

Bits	Name	Description
31:0	SEED_VALUE	Loopback MISR seed

0x00482028 DIM_D03_DDRPHY_DQ_MISR_EVEN_STATUS**Type:** R**Clock:** HCLK**Reset State:** 0x00000000

The DDRPHY_DQ_MISR_EVEN_STATUS register configures the following:

DDRPHY_DQ_MISR_EVEN_STATUS

DIM_D03_DDRPHY_DQ_MISR_EVEN_STATUS

Bits	Name	Description
31:0	SIGNATURE	Even clock Loopback MISR signature

0x0048202C DIM_D03_DDRPHY_DQ_MISR_ODD_STATUS**Type:** R**Clock:** HCLK**Reset State:** 0x00000000

The DDRPHY_DQ_MISR_ODD_STATUS register configures the following:

DDRPHY_DQ_MISR_ODD_STATUS

DIM_D03_DDRPHY_DQ_MISR_ODD_STATUS

Bits	Name	Description
31:0	SIGNATURE	Odd Clock Loopback MISR signature

0x00482030 DIM_D03_DIM_DQ_CDC_CTLR_CFG0**Type:** RW**Clock:** HCLK**Reset State:** 0x02000000

The DIM_DQ_CDC_CTLR_CFG0 register configures the following:

DIM_DQ_CDC_CTLR_CFG0

DIM_D03_DIM_DQ_CDC_CTLR_CFG0

Bits	Name	Description
26	ERROR_CODE_UPDATE_EN	1'b1: Enable overwriting (default) the error code for subsequent calibrations. 1'b0: Disable overwriting the error code during subsequent calibrations
25	STAGGER_CAL_ENA	1'b1: Stagger calibration of write and read CDCs once after the other to reduce peak voltage drop. 1'b0: Both write and read CDCs calibrated simultaneously This bit is introduced on the x2 core, so not present on x0 and x1 versions of DIM/PHY.
24	TRACKING_CAL_ENA	1'b1: Relationship between subunits and oscillator count is learnt and used for tracking calibration. 1'b0: Periodic calibration triggers cause full recalibration
23:20	OSC_COUNT_ERR_TOLERANCE	Magnitude of the difference between oscillator count and target count when the algorithm assumes convergence and stops further search.
19	TRACK_CALIB_MODE	1'b0: Tracking calibration uses previous Target_count value. 1'b1 Accurate mode: Target_count is recomputed at the start of tracking calibration

DIM_D03_DIM_DQ_CDC_CTLR_CFG0 (cont.)

Bits	Name	Description
18	FULL_DELAY	1'b0: unitstep and subunit caps are from calibration or SW settings 1'b1: All unit step and subunit caps are selected for the delay arrays.
17	HW_AUTOCAL_ENA	1'b1: Enables periodic auto calibration 1'b0: Periodic calibration is disabled
16	SW_TRIG_FULL_CALIB	A low to high transition on this register bit triggers the CDC calibration statemachine to initiate full calibration of the standby delay array. Also resets tracking mode for the delay arrays.
13:12	OSC_PRE_DIV	The factor by which the raw oscillator frequency is divided before being given to the counting logic. 2'b00 Divide by 1 2'b01 Divide by 2 2'b10 Divide by 4 2'b11 Divide by 8
10:0	TMUX_CHAR	Cchar is a characterized constant equal to the ratio of output mux offset delay to average unit step delay. This register is programmed by multiplying the constant with 1024 and rounding to the nearest integer. E.g. if Cchar = 0.67, $RO(0.67 * 1024) = 686$, hence program TMUX_CHAR = 11'h2AE

0x00482034 DIM_D03_DIM_DQ_CDC_CTLR_CFG1**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

The DIM_DQ_CDC_CTLR_CFG1 register configures the following:

DIM_DQ_CDC_CTLR_CFG1

DIM_D03_DIM_DQ_CDC_CTLR_CFG1

Bits	Name	Description
26:24	OSC_COUNT_DELAY	Number of oscillator cycles from the end of reference gate signal to assertion of oscillation counter done. Legal values are from 0 to 4.
23:20	STANDBY_DELAY	This is the number of reference clock cycles from when a newly calibrated delay array is made active to when the current active delay array is put into standby mode. 4'b0: Disables the counter. Hardware waits for explicit acknowledgement of delay array switch from CDC. Others: Explicit acknowledgement mode is disabled and the counter is used

DIM_D03_DIM_DQ_CDC_CTLR_CFG1 (cont.)

Bits	Name	Description
18:16	DEL_MODE_DELAY	Number of reference clock cycles between putting a newly calibrated delay array in delay mode to selecting it as the active delay array. The minimum allowed value is 1
14:12	OSC_MODE_DELAY	Number of reference clock cycles between putting a delay array in oscillator mode to starting the oscillation counter. This delay is allowed for the oscillations to stabilize through the delay array. The minimum allowed value is 1
10:8	DECODER_DELAY	Number of reference clock cycles allowed for unitstep and subunit counts to be encoded into the unit and finecap select lines on the DDA. The minimum allowed value is 1.
6:4	DIVIDER_DELAY	Number of reference clock cycles allowed for arithmetic division operations. The minimum allowed value is 1
2:0	MULTIPLIER_DELAY	Number of reference clock cycles allowed for arithmetic multiplication operations. The minimum allowed value is 1

0x00482038 DIM_D03_DIM_DQ_CDC_CAL_TIMER_CFG0**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

The DIM_DQ_CDC_CAL_TIMER_CFG0 register configures the following:

DIM_DQ_CDC_CAL_TIMER_CFG0

DIM_D03_DIM_DQ_CDC_CAL_TIMER_CFG0

Bits	Name	Description
24	INVALID_TIMER_ENA	1'b1: Enable the timer counting number of missed calibration intervals after which the CDC is invalid. 1'b0: Timer disabled A low to high transition on this register bit loads the timer with programmed value below.
23:20	INVALID_TIMER_VAL	Number of missed calibration intervals after which CDC invalid state is asserted.
16	TIMER_ENA	1'b1: Enables 32 kHz sleep clock based periodic autocalibration timer. 1'b0: Timer disabled A low to high transition on this register bit loads the timer with programmed value below.
15:0	TIMER_VAL	Count value for periodic autocalibration interval based on sleep clock.

0x0048203C DIM_D03_DIM_DQ_CDC_CAL_TIMER_CFG1**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

The DIM_DQ_CDC_CAL_TIMER_CFG1 register configures the following:

DIM_DQ_CDC_CAL_TIMER_CFG1

DIM_D03_DIM_DQ_CDC_CAL_TIMER_CFG1

Bits	Name	Description
12	FF_TIMER_ENA	1'b1: Enables reference clock based periodic autocalibration timer. 1'b0: Timer disabled A low to high transition on this register bit loads the timer with programmed value below.
9:0	FF_TIMER_VAL	Count value for periodic autocalibration interval based on reference clock.

0x00482040 DIM_D03_DIM_DQ_CDC_REFCOUNT_CFG**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

The DIM_DQ_CDC_REFCOUNT_CFG register configures the following:

DIM_DQ_CDC_REFCOUNT_CFG

DIM_D03_DIM_DQ_CDC_REFCOUNT_CFG

Bits	Name	Description
31:16	TREF	Reference clock period programmed in pico seconds.
13:8	CCAL_REF_COUNT	Number of reference clock cycles used for oscillation counts during coarse calibration. Maximum supported value is 59.
5:0	FCAL_REF_COUNT	Number of reference clock cycles used for oscillation counts during fine calibration. Maximum supported value is 59

0x00482044 DIM_D03_DIM_DQ_CDC_COARSE_CAL_CFG**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

The register DIM_DQ_CDC_COARSE_CAL_CFG configures the following:

DIM_DQ_CDC_COARSE_CAL_CFG

DIM_D03_DIM_DQ_CDC_COARSE_CAL_CFG

Bits	Name	Description
12:8	CCAL_SUBUNIT_CAPS	Subunit count needed for an effective unit step count of 32 during coarse calibration. This value is also used by software oscillator mode.
5:0	CCAL_UNITSTEPS	unitstep count needed for an effective unit step count of 32 during coarse calibration.

0x00482048 DIM_D03_DIM_DQ_CDC_RSVD_CFG

Type: RW**Clock:** HCLK**Reset State:** 0x00000000

The DIM_DQ_CDC_RSVD_CFG register configures the following:

DIM_DQ_CDC_RSVD_CFG

DIM_D03_DIM_DQ_CDC_RSVD_CFG

Bits	Name	Description
15:0	TEMP_FIELD	Reserved register

0x0048204C DIM_D03_DIM_DQ_RD_CDC_OFFSET_CFG

Type: RW**Clock:** HCLK**Reset State:** 0x00000000

The DIM_DQ_RD_CDC_OFFSET_CFG register configures the following:

DIM_DQ_RD_CDC_OFFSET_CFG

DIM_D03_DIM_DQ_RD_CDC_OFFSET_CFG

Bits	Name	Description
20	SUBUNIT_OFFSET_MODE	1'b1: The programmed offset is directly loaded as subunit count 1'b0: Programmed offset is added/subtracted from the subunit count
19	SUBUNIT_OFFSET_SIGN	1'b1: Offset is subtracted from the subunit count 1'b0: Offset is added to the subunit count
16:12	SUBUNIT_OFFSET	Software subunit offset

DIM_D03_DIM_DQ_RD_CDC_OFFSET_CFG (cont.)

Bits	Name	Description
8	UNIT_OFFSET_MODE	1'b1: The programmed offset is directly loaded as unit step count 1'b0: Programmed offset is added/subtracted from the unit step count
7	UNIT_OFFSET_SIGN	1'b1: Offset is subtracted from the unitstep count 1'b0: Offset is added to the unitstep count
5:0	UNIT_OFFSET	Software unitstep offset

0x00482050 DIM_D03_DIM_DQ_RD_CDC_DELAY_CFG**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

The DIM_DQ_RD_CDC_DELAY_CFG register configures the following:

DIM_DQ_RD_CDC_DELAY_CFG

DIM_D03_DIM_DQ_RD_CDC_DELAY_CFG

Bits	Name	Description
28	TARGET_COUNT_ENA	1'b1: Enable software specified Target_count for calibration instead of value calculated by hardware. 1'b0: Hardware computed Target_count used.
27:16	TARGET_COUNT	Software specified Target_count to which the statemachine calibrates
11:0	DELAY_VAL	Delay required from CDC programmed in pico seconds. This value has to be greater than the minimum delay supported by DDA Tmin.

0x00482054 DIM_D03_DIM_DQ_RD_CDC_SW_MODE_CFG**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

The DIM_DQ_RD_CDC_SW_MODE_CFG register configures the following:

DIM_DQ_RD_CDC_SW_MODE_CFG

DIM_D03_DIM_DQ_RD_CDC_SW_MODE_CFG

Bits	Name	Description
3	SW_DEL_MODE	A low to high transition on this register bit puts the delay array selected by SW_DA_SEL into delay mode and connects it to the DDA output. This register has to remain set for the duration of SW controlled delay mode.
2	SW_OSC_MODE	Reserved register.
1	SW_DA_SEL	Selects the delay array inside DDA during software delay or oscillator mode.
0	SW_LOAD	1'b1: Enable loading of the software offsets from the offset configuration register. This register has to remain set for the duration of SW mode operation.

0x0048205C DIM_D03_DIM_DQ_RD_CDC_SW_OVRD_CFG**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

The DIM_DQ_RD_CDC_SW_OVRD_CFG register configures the following:

DIM_DQ_RD_CDC_SW_OVRD_CFG

DIM_D03_DIM_DQ_RD_CDC_SW_OVRD_CFG

Bits	Name	Description
16	SW_REF_GATE	Write 1'b1 to start the reference window. Write 1'b0 to end the reference window. Number of reference cycle in the window will be reported in SW_REF_GATE_COUNT in CDC_STATUS0
11	SW_OVRD_DA1_OSC_EN	In software override mode, 1'b1 written to this bit puts delay array 1 in oscillator-mode
10	SW_OVRD_DA1_IN_EN	In software override mode, 1'b1 written to this bit puts delay array 1 in delay-mode
9	SW_OVRD_DA0_OSC_EN	In software override mode, 1'b1 written to this bit puts delay array 0 in oscillator-mode
8	SW_OVRD_DA0_IN_EN	In software override mode, 1'b1 written to this bit puts delay array 0 in delay-mode
3	SW_OVRD_CDC_COUNTEN R_RST	1'b1 written to this bit resets the oscillation counter. Follow this with 1'b0 before setting reference gate.
2	SW_OVRD_LOAD_DA_SEL	In software override mode, this bit select the delay array to which unit and subunit offset values from CDC_OFFSET_CFG register will be loaded.

DIM_D03_DIM_DQ_RD_CDC_SW_OVRD_CFG (cont.)

Bits	Name	Description
1	SW_OVRD_ACTV_DA_SEL	In software override mode, this register selects the delay array connected to the CDC output
0	SW_OVRD_ENA	1'b1 written to this bit puts the CDC interface in software override mode where the below register bits control the CDC. 1'b0: CDC is under hardware control

0x00482060 DIM_D03_DIM_DQ_RD_CDC_SLAVE_DDA_CFG**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

The DIM_DQ_RD_CDC_SLAVE_DDA_CFG register configures the following:

DIM_DQ_RD_CDC_SLAVE_DDA_CFG

DIM_D03_DIM_DQ_RD_CDC_SLAVE_DDA_CFG

Bits	Name	Description
17	SLAV_DDA_OFFSET_MODE	1'b1: The programmed offset is directly loaded as unit-step count 1'b0: Programmed offset is added/subtracted from the unit-step count
16	SLAV_DDA_OFFSET_SIGN	1'b1: Offset is subtracted from the unit-step count. This subtraction feature is not supported on x0 and x1 PHY cores (bit RESERVED). Supported on the x2 core version. 1'b0: Offset is added to the unit-step count
15:12	SLAVE_DDA_OFFSET	unit-step offset for slave DDA.
10:0	SLAVE_DDA_DELAY	Delay required from slave DDA programmed in pico seconds.

0x00482070 DIM_D03_DIM_DQ_RD_CDC_STATUS0**Type:** R**Clock:** HCLK**Reset State:** 0x0000000C

The DIM_DQ_RD_CDC_STATUS0 register configures the following:

DIM_DQ_RD_CDC_STATUS0

DIM_D03_DIM_DQ_RD_CDC_STATUS0

Bits	Name	Description
26:24	CDC_ERROR_CODE	Error code for internal errors: 3'b000: No error 3'b001: Programmed delay less than Tmux 3'b010: Programmed delay less than Tmin 3'b011: Unit-step overflow: Overflow during fine calibration. Desired delay cannot be realized even when all unit-steps are selected. 3'b100: Unit-step underflow 3'b101: Unit-step overflow: Overflow during coarse calibration. Estimated unit steps needed to realize the desired delay are greater than the total number of unit-steps in design. Note: The CDC_ERROR_CODE is not a sticky register and is updated after each CDC calibration.
23:16	SW_REF_GATE_COUNT	Number of reference clock cycles in the software generated oscillation gate window doing SW override mode
15:4	OSC_COUNT	Current value of oscillator count
3	CURR_SEL_DA	This bit indicates the current active delay array connected to the output. 1'b0: DA0 is selected 1'b1: DA1 is selected
2	CTLR_SM_IDLE	1'b1 Indicates that the calibration state machine is idle 1'b0 indicates calibration statemachine is busy
1	OSC_DONE	This status register is set after oscillations count is finished by CDC. It gets reset when a new oscillator count is started.
0	CALIBRATION_DONE	1'b1 indicates that calibration is done and CURR_SUBUNIT_STEPS, CURR_UNIT_STEPS status registers are valid.

0x00482074 DIM_D03_DIM_DQ_RD_CDC_STATUS1**Type:** R**Clock:** HCLK**Reset State:** 0x00000000

The DIM_DQ_RD_CDC_STATUS1 register configures the following:

DIM_DQ_RD_CDC_STATUS1

DIM_D03_DIM_DQ_RD_CDC_STATUS1

Bits	Name	Description
27:16	CURR_DELAY_VALUE	HW Computed value of current delay after calibration by the state machine. This register is valid if CALIBRATION_DONE status bit is set.
11:0	CURR_TMUX_DELAY	Current value of output mux offset delay computed by HW.

0x00482078 DIM_D03_DIM_DQ_RD_CDC_STATUS2**Type:** R**Clock:** HCLK**Reset State:** 0x10331033

The DIM_DQ_RD_CDC_STATUS2 register configures the following:

DIM_DQ_RD_CDC_STATUS2

DIM_D03_DIM_DQ_RD_CDC_STATUS2

Bits	Name	Description
28:24	DA1_SUBUNITS	Current value of subunit cap settings applied to delay array 1
21:16	DA1_UNITSTEPS	Current value of unit steps applied to delay array 1
12:8	DA0_SUBUNITS	Current value of subunit cap settings applied to delay array 0
5:0	DA0_UNITSTEPS	Current value of unit steps applied to delay array 0

0x0048207C DIM_D03_DIM_DQ_RD_CDC_STATUS3**Type:** R**Clock:** HCLK**Reset State:** 0x00000000

The DIM_DQ_RD_CDC_STATUS3 register configures the following:

DIM_DQ_RD_CDC_STATUS3

DIM_D03_DIM_DQ_RD_CDC_STATUS3

Bits	Name	Description
31:24	NUM_OF_OSC_ITER	Number of oscillator iterations before convergence was achieved during tracking
23:12	COUNT_ERROR	Difference between target and actual oscillator counts at the start of tracking calibration

DIM_D03_DIM_DQ_RD_CDC_STATUS3 (cont.)

Bits	Name	Description
11:0	CURR_TARGET_COUNT	HW computed value of current Target_count used by the calibration state machine.

0x00482080 DIM_D03_DIM_DQ_RD_CDC_STATUS4**Type:** R**Clock:** HCLK**Reset State:** 0x000000FF

The DIM_DQ_RD_CDC_STATUS4 register configures the following:

DIM_DQ_RD_CDC_STATUS4

DIM_D03_DIM_DQ_RD_CDC_STATUS4

Bits	Name	Description
7:4	SLAVE_DDA_DA1_TAPS	Number if unit taps applied to delay array 1 of slave DDA
3:0	SLAVE_DDA_DA0_TAPS	Number if unit taps applied to delay array 0 of slave DDA

0x004820AC DIM_D03_DIM_DQ_WR_CDC_OFFSET_CFG**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

The DIM_DQ_WR_CDC_OFFSET_CFG register configures the following:

DIM_DQ_WR_CDC_OFFSET_CFG

DIM_D03_DIM_DQ_WR_CDC_OFFSET_CFG

Bits	Name	Description
20	SUBUNIT_OFFSET_MODE	1'b1: The programmed offset is directly loaded as subunit count 1'b0: Programmed offset is added/subtracted from the subunit count
19	SUBUNIT_OFFSET_SIGN	1'b1: Offset is subtracted from the subunit count 1'b0: Offset is added to the subunit count
16:12	SUBUNIT_OFFSET	Software subunit offset
8	UNIT_OFFSET_MODE	1'b1: The programmed offset is directly loaded as unit step count 1'b0: Programmed offset is added/subtracted from the unit step count

DIM_D03_DIM_DQ_WR_CDC_OFFSET_CFG (cont.)

Bits	Name	Description
7	UNIT_OFFSET_SIGN	1'b1: Offset is subtracted from the unitstep count 1'b0: Offset is added to the unitstep count
5:0	UNIT_OFFSET	Software unitstep offset

0x004820B0 DIM_D03_DIM_DQ_WR_CDC_DELAY_CFG**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

The DIM_DQ_WR_CDC_DELAY_CFG register configures the following:

DIM_DQ_WR_CDC_DELAY_CFG

DIM_D03_DIM_DQ_WR_CDC_DELAY_CFG

Bits	Name	Description
28	TARGET_COUNT_ENA	1'b1: Enable software specified Target_count for calibration instead of value calculated by hardware. 1'b0: Hardware computed Target_count used.
27:16	TARGET_COUNT	Software specified Target_count to which the statemachine calibrates
11:0	DELAY_VAL	Delay required from CDC programmed in pico seconds. This value has to be greater than the minimum delay supported by DDA Tmin.

0x004820B4 DIM_D03_DIM_DQ_WR_CDC_SW_MODE_CFG**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

The DIM_DQ_WR_CDC_SW_MODE_CFG register configures the following:

DIM_DQ_WR_CDC_SW_MODE_CFG

DIM_D03_DIM_DQ_WR_CDC_SW_MODE_CFG

Bits	Name	Description
3	SW_DEL_MODE	A low to high transition on this register bit puts the delay array selected by SW_DA_SEL into delay mode and connects it to the DDA output. This register has to remain set for the duration of SW controlled delay mode.

DIM_D03_DIM_DQ_WR_CDC_SW_MODE_CFG (cont.)

Bits	Name	Description
2	SW_OSC_MODE	Reserved register.
1	SW_DA_SEL	Selects the delay array inside DDA during software delay or oscillator mode.
0	SW_LOAD	1'b1: Enable loading of the software offsets from the offset configuration register. This register has to remain set for the duration of SW mode operation.

0x004820BC DIM_D03_DIM_DQ_WR_CDC_SW_OVRD_CFG**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

The DIM_DQ_WR_CDC_SW_OVRD_CFG register configures the following:

DIM_DQ_WR_CDC_SW_OVRD_CFG

DIM_D03_DIM_DQ_WR_CDC_SW_OVRD_CFG

Bits	Name	Description
16	SW_REF_GATE	Write 1'b1 to start the reference window. Write 1'b0 to end the reference window. Number of reference cycle in the window will be reported in SW_REF_GATE_COUNT in CDC_STATUS0
11	SW_OVRD_DA1_OSC_EN	In software override mode, 1'b1 written to this bit puts delay array 1 in oscillator-mode
10	SW_OVRD_DA1_IN_EN	In software override mode, 1'b1 written to this bit puts delay array 1 in delay-mode
9	SW_OVRD_DA0_OSC_EN	In software override mode, 1'b1 written to this bit puts delay array 0 in oscillator-mode
8	SW_OVRD_DA0_IN_EN	In software override mode, 1'b1 written to this bit puts delay array 0 in delay-mode
3	SW_OVRD_CDC_COUNTERR_RST	1'b1 written to this bit resets the oscillation counter. Follow this with 1'b0 before setting reference gate.
2	SW_OVRD_LOAD_DA_SEL	In software override mode, this bit select the delay array to which unit and subunit offset values from CDC_OFFSET_CFG register will be loaded.
1	SW_OVRD_ACTV_DA_SEL	In software override mode, this register selects the delay array connected to the CDC output
0	SW_OVRD_ENA	1'b1 written to this bit puts the CDC interface in software override mode where the below register bits control the CDC. 1'b0: CDC is under hardware control

0x004820D0 DIM_D03_DIM_DQ_WR_CDC_STATUS0**Type:** R**Clock:** HCLK**Reset State:** 0x0000000C

The DIM_DQ_WR_CDC_STATUS0 register configures the following:

DIM_DQ_WR_CDC_STATUS0

DIM_D03_DIM_DQ_WR_CDC_STATUS0

Bits	Name	Description
26:24	CDC_ERROR_CODE	Error code for internal errors: 3'b000: No error 3'b001: Programmed delay less than Tmux 3'b010: Programmed delay less than Tmin 3'b011: Unit-step overflow: Overflow during fine calibration. Desired delay cannot be realized even when all unit-steps are selected. 3'b100: Unit-step underflow 3'b101: Unit-step overflow: Overflow during coarse calibration. Estimated unit steps needed to realize the desired delay are greater than the total number of unit-steps in design. Note: The CDC_ERROR_CODE is not a sticky register and is updated after each CDC calibration.
23:16	SW_REF_GATE_COUNT	Number of reference clock cycles in the software generated oscillation gate window doing SW override mode
15:4	OSC_COUNT	Current value of oscillator count
3	CURR_SEL_DA	This bit indicates the current active delay array connected to the output. 1'b0: DA0 is selected 1'b1: DA1 is selected
2	CTLR_SM_IDLE	1'b1 Indicates that the calibration state machine is idle 1'b0 indicates calibration statemachine is busy
1	OSC_DONE	This status register is set after oscillations count is finished by CDC. It gets reset when a new oscillator count is started.
0	CALIBRATION_DONE	1'b1 indicates that calibration is done and CURR_SUBUNIT_STEPS, CURR_UNIT_STEPS status registers are valid.

0x004820D4 DIM_D03_DIM_DQ_WR_CDC_STATUS1**Type:** R**Clock:** HCLK**Reset State:** 0x00000000

The DIM_DQ_WR_CDC_STATUS1 register configures the following:

DIM_DQ_WR_CDC_STATUS1

DIM_D03_DIM_DQ_WR_CDC_STATUS1

Bits	Name	Description
27:16	CURR_DELAY_VALUE	HW Computed value of current delay after calibration by the state machine. This register is valid if CALIBRATION_DONE status bit is set.
11:0	CURR_TMUX_DELAY	Current value of output mux offset delay computed by HW.

0x004820D8 DIM_D03_DIM_DQ_WR_CDC_STATUS2

Type: R

Clock: HCLK

Reset State: 0x10331033

The DIM_DQ_WR_CDC_STATUS2 register configures the following:

DIM_DQ_WR_CDC_STATUS2

DIM_D03_DIM_DQ_WR_CDC_STATUS2

Bits	Name	Description
28:24	DA1_SUBUNITS	Current value of subunit cap settings applied to delay array 1
21:16	DA1_UNITSTEPS	Current value of unit steps applied to delay array 1
12:8	DA0_SUBUNITS	Current value of subunit cap settings applied to delay array 0
5:0	DA0_UNITSTEPS	Current value of unit steps applied to delay array 0

0x004820DC DIM_D03_DIM_DQ_WR_CDC_STATUS3

Type: R

Clock: HCLK

Reset State: 0x00000000

The DIM_DQ_WR_CDC_STATUS3 register configures the following:

DIM_DQ_WR_CDC_STATUS3

DIM_D03_DIM_DQ_WR_CDC_STATUS3

Bits	Name	Description
31:24	NUM_OF_OSC_ITER	Number of oscillator iterations before convergence was achieved during tracking

DIM_D03_DIM_DQ_WR_CDC_STATUS3 (cont.)

Bits	Name	Description
23:12	COUNT_ERROR	Difference between target and actual oscillator counts at the start of tracking calibration
11:0	CURR_TARGET_COUNT	HW computed value of current Target_count used by the calibration state machine.

0x00482100 DIM_D03_DIM_DQ_DQ_IOC_SLV_CFG**Type:** RW**Clock:** HCLK**Reset State:** 0x10100000

The DIM_DQ_DQ_IOC_SLV_CFG register configures the following:

DIM_DQ_DQ_IOC_SLV_CFG

DIM_D03_DIM_DQ_DQ_IOC_SLV_CFG

Bits	Name	Description
31	PNCNT_HW_LOAD_EN	
28:24	NCNT_SW_VAL	
20:16	PCNT_SW_VAL	
15	NCNT_OFFSET_SIGN	
12:8	NCNT_SW_OFFSET	
7	PCNT_OFFSET_SIGN	
4:0	PCNT_SW_OFFSET	

0x00482104 DIM_D03_DIM_DQ_DQ_IOC_SLV_STATUS**Type:** R**Clock:** HCLK**Reset State:** 0x00001010

The DIM_DQ_DQ_IOC_SLV_STATUS register configures the following:

DIM_CA_CA_IOC_SLV_STATUS

DIM_D03_DIM_DQ_DQ_IOC_SLV_STATUS

Bits	Name	Description
12:8	PAD_NCNT	Impedance calibration bit settings for NMOS

DIM_D03_DIM_DQ_DQ_IOC_SLV_STATUS (cont.)

Bits	Name	Description
4:0	PAD_PCNT	Impedance calibration bit settings for PMOS

0x00482110 DIM_D03_DIM_DQ_DQS_IOC_SLV_CFG**Type:** RW**Clock:** HCLK**Reset State:** 0x10100000

The DIM_DQ_DQS_IOC_SLV_CFG register configures the following:

DIM_DQ_DQS_IOC_SLV_CFG

DIM_D03_DIM_DQ_DQS_IOC_SLV_CFG

Bits	Name	Description
31	PNCNT_HW_LOAD_EN	
28:24	NCNT_SW_VAL	
20:16	PCNT_SW_VAL	
15	NCNT_OFFSET_SIGN	
12:8	NCNT_SW_OFFSET	
7	PCNT_OFFSET_SIGN	
4:0	PCNT_SW_OFFSET	

0x00482114 DIM_D03_DIM_DQ_DQS_IOC_SLV_STATUS**Type:** R**Clock:** HCLK**Reset State:** 0x00001010

The DIM_DQ_DQS_IOC_SLV_STATUS register configures the following:

DIM_DQ_DQS_IOC_SLV_STATUS

DIM_D03_DIM_DQ_DQS_IOC_SLV_STATUS

Bits	Name	Description
12:8	PAD_NCNT	Impedance calibration bit settings for NMOS
4:0	PAD_PCNT	Impedance calibration bit settings for PMOS

7.6 EBI1_AHB2PHY

AHB2PHY

7.6.1 EBI1_AHB2PHY

AHB2PHY

0x0049F008 EBI1_AHB2PHY_BROADCAST_EN_CFG_LOWER

Type: RW

Clock: hclk

Reset State: 0x00000000

AHB2PHY_BROADCAST_EN_CFG_LOWER register configures the PHY chip select lower broadcast enables.

EBI1_AHB2PHY_BROADCAST_EN_CFG_LOWER

Bits	Name	Description
31:0	BROADCAST_EN_LOWER	ENABLE(1) DISABLE(0) bit[0]: broadcast for the chip select 0 (o_phy_cs[0]) bit[1]: broadcast for the chip select 1 (o_phy_cs[1]) bit[2]: broadcast for the chip select 2 (o_phy_cs[2]) bit[31]: broadcast for the chip select (o_phy_cs[31])

0x0049F00C EBI1_AHB2PHY_BROADCAST_EN_CFG_UPPER

Type: RW

Clock: hclk

Reset State: 0x00000000

AHB2PHY_BROADCAST_EN_CFG_UPPER register configures the PHY chip select upper broadcast enables.

EBI1_AHB2PHY_BROADCAST_EN_CFG_UPPER

Bits	Name	Description
29:0	BROADCAST_EN_UPPER	ENABLE(1) DISABLE(0) bit[0]: broadcast for the chip select 0 (o_phy_cs[32]) bit[1]: broadcast for the chip select 1 (o_phy_cs[33]) bit[2]: broadcast for the chip select 2 (o_phy_cs[34]) bit[29]: broadcast for the chip select (o_phy_cs[61])

0x0049F010 EBI1_AHB2PHY_TOP_CFG**Type:** RW**Clock:** hclk**Reset State:** 0x00000000

AHB2PHY_TOP_CFG register configures AHB2PHY PHY wait states.

EBI1_AHB2PHY_TOP_CFG

Bits	Name	Description
5:4	WRITE_WAIT_STATES	0x0: NO_WAIT_STATES 0x1: ONE_WAIT_STATE 0x2: TWO_WAIT_STATES 0x3: THREE_WAIT_STATES
1:0	READ_WAIT_STATES	0x0: NO_WAIT_STATES 0x1: ONE_WAIT_STATE 0x2: TWO_WAIT_STATES 0x3: THREE_WAIT_STATES

7.7 EBI1_AHB2PHY_BROADCAST

AHB2PHY Broadcast Address Space

7.7.1 EBI1_AHB2PHY_BROADCAST

AHB2PHY Broadcast Address Space

**0x0049F800+ EBI1_AHB2PHY_BROADCAST_ADDRESS_SPACE_n, n=[0..255]
0x4*n****Type:** RW**Clock:** hclk**Reset State:** 0x00000000

A write or read access to this space activates all PHY chip selects that have their broadcast enable bit set to a 1.

EBI1_AHB2PHY_BROADCAST_ADDRESS_SPACE_n

Bits	Name	Description
31:0	BROADCAST_SPACE	The entire 1 kB address space for this slave chip select is dedicated to the Broadcast function. If a write or read access is performed in this address space the chip selects of all PHY slaves that have been enabled will be active during the access. A PHY slave is enabled for broadcast by setting the BROADCAST_EN bit in the AHB2PHY_BROADCAST_EN_CFG register. Each PHY slave has a dedicated bit in this register.

8 A53SS

8.1 APCS_QTMR_AC

QTMR AC Block

8.1.1 APCS_QTMR_AC

QTMR AC Block

0x0B020000 APCS_QTMR_AC_CNTFRQ

Type: RW

Clock: SYS_AHB_CLK

Reset State: 0x00000000

Security Treatment: Restricted

Counter Frequency register. Counter frequency in ticks per second

APCS_QTMR_AC_CNTFRQ

Bits	Name	Description
31:0	CNTFRQ	

0x0B020004 APCS_QTMR_AC_CNTSR

Type: RW

Clock: SYS_AHB_CLK

Reset State: 0x00000000

Security Treatment: Restricted

The Counter Secure Register (CNTSR) contains a bit for Secure access to each timer frame, including both views of the timer frame and the access to the associated CNTACR and CNTVOFF registers in the Access Control region.

APCS_QTMR_AC_CNTSR

Bits	Name	Description
6:0	NSN	Non-secure access to frame N. N lies from 0 to \$NUM_FRAME-1 (# of configured frame) 0x0: SECURE_ONLY (Secure access only - non-secure access is RAZ/WI) 0x1: SECURE_OR_NONSECURE (Secure or Non-secure access)

0x0B020008 APCS_QTMR_AC_CNTTID**Type:** R**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** Both

The Counter Timer ID (CNTTID) is 32-bit register that identifies the number of timer frames implemented, which frame have virtual offsets and timers, and which frames have second views

APCS_QTMR_AC_CNTTID

Bits	Name	Description
31:28	F7_CFG	Frame 7 configuration (same definition as Frame 0)
27:24	F6_CFG	Frame 6 configuration (same definition as Frame 0)
23:20	F5_CFG	Frame 5 configuration (same definition as Frame 0)
19:16	F4_CFG	Frame 4 configuration (same definition as Frame 0)
15:12	F3_CFG	Frame 3 configuration (same definition as Frame 0)
11:8	F2_CFG	Frame 2 configuration (same definition as Frame 0)
7:4	F1_CFG	Frame 1 configuration (same definition as Frame 0)
3:0	F0_CFG	Frame 0 configuration 0x3: RSVD (Reserved (Read as zero)) 0x2: FPLO (Frame has second view) 0x1: FVI (Virtual frame implemented) 0x0: FI (Frame implemented)

**0x0B020040+ APCS_QTMR_AC_CNTACR_n, n=[0..6]
0x4*n****Type:** RW**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTSR[n]

The Counter Access Control Register (CNTACR) provide access control to the individual elements of timer frame n, including the second view of that timer frame n

APCS_QTMR_AC_CNTACR_n

Bits	Name	Description
5	RWPT	Read/Write access to CNTP_*register 0x0: ACCESS_DENIED (Access denied) 0x1: ACCESS_ALLOWED (Access allowed)
4	RWVT	Read/Write access to CNTV_*register 0x0: ACCESS_DENIED (Access denied) 0x1: ACCESS_ALLOWED (Access allowed)
3	RVOFF	Read access to CNTVOFF register 0x0: ACCESS_DENIED (Access denied) 0x1: ACCESS_ALLOWED (Access allowed)
2	RFRQ	Read access to CNTFRQ register 0x0: ACCESS_DENIED (Access denied) 0x1: ACCESS_ALLOWED (Access allowed)
1	RPVCT	Read access to CNTVCT register 0x0: ACCESS_DENIED (Access denied) 0x1: ACCESS_ALLOWED (Access allowed)
0	RPCT	Read access to CNTPCT register 0x0: ACCESS_DENIED (Access denied) 0x1: ACCESS_ALLOWED (Access allowed)

0x0B020080+ APCS_QTMR_AC_CNTVOFF_LO_n, n=[0..6] 0x8*n

Type: RW

Clock: SYS_AHB_CLK

Reset State: 0x00000000

Security Treatment: Controlled by QTMR_AC_CNTSR[n]

The lower 32 bits of Virtual Offset register

APCS_QTMR_AC_CNTVOFF_LO_n

Bits	Name	Description
31:0	CNTVOFF_LO	Lower bits [31:0] of virtual Offset for Frame n

0x0B020084+ APCS_QTMR_AC_CNTVOFF_HI_n, n=[0..6] 0x8*n

Type: RW

Clock: SYS_AHB_CLK

Reset State: 0x00000000

Security Treatment: Controlled by QTMR_AC_CNTSR[n]

The upper 32 bits of Virtual Offset register

APCS_QTMR_AC_CNTVOFF_HI_n

Bits	Name	Description
23:0	CNTVOFF_HI	Higher bits [31:0] of virtual Offset for Frame n

8.2 APCS_F0_QTMR_V1

QTMR V1 Block

8.2.1 APCS_F0_QTMR_V1

QTMR V1 Block

0x0B021000 APCS_F0_QTMR_V1_Cntpct_Lo

Type: R

Clock: SYS_AHB_CLK

Reset State: 0x00000000

Security Treatment: Controlled by QTMR_AC_CNTACR_n[RPCT]

The lower 32 bits of Physical Counter

APCS_F0_QTMR_V1_Cntpct_Lo

Bits	Name	Description
31:0	CNTPCT_LO	Lower bits [31:0] of Physical Counter

0x0B021004 APCS_F0_QTMR_V1_CNTPCT_HI**Type:** R**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RPCT]

The upper 32 bits of Physical Counter

APCS_F0_QTMR_V1_CNTPCT_HI

Bits	Name	Description
23:0	CNTPCT_HI	Higher bits [31:0] of Physical Counter

0x0B021008 APCS_F0_QTMR_V1_CNTVCT_LO**Type:** R**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RVCT]

The lower 32 bits of Virtual Counter

APCS_F0_QTMR_V1_CNTVCT_LO

Bits	Name	Description
31:0	CNTVCT_LO	Lower bits [31:0] of Virtual Counter

0x0B02100C APCS_F0_QTMR_V1_CNTVCT_HI**Type:** R**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RVCT]

The upper 32 bits of Virtual Counter

APCS_F0_QTMR_V1_CNTVCT_HI

Bits	Name	Description
23:0	CNTVCT_HI	Higher bits [31:0] of Virtual Counter

0x0B021010 APCS_F0_QTMR_V1_CNTFRQ**Type:** R**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RFRQ]

Read QTMR_AC_CNTFRQ register

APCS_F0_QTMR_V1_CNTFRQ

Bits	Name	Description
31:0	CNTFRQ	Counter frequency in ticks per second

0x0B021014 APCS_F0_QTMR_V1_CNTPL0ACR**Type:** RW**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** Both

The Second View Counter Access Control Register (CNTPL0ACR) provide second view access control to the individual elements of timer frame n, except CNTVOFF and CNTPL0ACR registers

APCS_F0_QTMR_V1_CNTPL0ACR

Bits	Name	Description
9	PL0CTEN	Second view read/write access to CNTP_* register
8	PL0VTEN	Second view read/write access to CNTV_* register
1	PL0VCTEN	Second view read access to CNTVCT and CNTFRQ register
0	PL0PCTEN	Second view read access to CNTPCT and CNTFRQ register

0x0B021018 APCS_F0_QTMR_V1_CNTVOFF_LO**Type:** R**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RVOFF]

Read QTMR_AC_CNTVOFF_LO_n register

APCS_F0_QTMR_V1_CNTVOFF_LO

Bits	Name	Description
31:0	CNTVOFF_LO	Lower bits [31:0] of virtual Offset for Frame

0x0B02101C APCS_F0_QTMR_V1_CNTVOFF_HI**Type:** R**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RVOFF]

Read QTMR_AC_CNTVOFF_HI_n register

APCS_F0_QTMR_V1_CNTVOFF_HI

Bits	Name	Description
23:0	CNTVOFF_HI	Higher bits [31:0] of virtual Offset for Frame

0x0B021020 APCS_F0_QTMR_V1_CNTP_CVAL_LO**Type:** RW**Clock:** SYS_TMR_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RWPT]

The Physical Timer Compare Value register is compare against the CNTPCT counter. The interrupt signal is asserted when the CNTPCT counter is greater than equal to CNTP_CVAL

APCS_F0_QTMR_V1_CNTP_CVAL_LO

Bits	Name	Description
31:0	CNTP_CVAL_LO	Lower bits [31:0] of compare value for interrupt for Frame

0x0B021024 APCS_F0_QTMR_V1_CNTP_CVAL_HI**Type:** RW**Clock:** SYS_TMR_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RWPT]

The Physical Timer Compare Value register is compare against the CNTPCT counter. The interrupt signal is asserted when the CNTPCT counter is greater than equal to CNTP_CVAL

APCS_F0_QTMR_V1_CNTP_CVAL_HI

Bits	Name	Description
23:0	CNTP_CVAL_HI	Higher bits [31:0] of compare value for interrupt for Frame

0x0B021028 APCS_F0_QTMR_V1_CNTP_TVAL**Type:** RW**Clock:** SYS_TMR_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RWPT]

An alternative 32 bit TVAL view to provide read or write CVAL register.

The TVAL reads:

 $TVAL = (CVAL - CNTP)$

The TVAL write CVAL with:

 $CVAL = (CNTP + \text{signExtend}(TVAL))$ **APCS_F0_QTMR_V1_CNTP_TVAL**

Bits	Name	Description
31:0	CNTP_TVAL	Timer Value in tick per second for Frame

0x0B02102C APCS_F0_QTMR_V1_CNTP_CTL**Type:** RW**Clock:** SYS_TMR_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RWPT]

The Physical Timer Control Register which can disable the timer and can mask its interrupt. If the timer is disabled, interrupt are masked, but the timer value continue to count down.

APCS_F0_QTMR_V1_CNTP_CTL

Bits	Name	Type	Description
2	ISTAT	R	Interrupt Status before masking (Read Only) 0x0: INTERRUPT_NOT_PENDING 0x1: INTERRUPT_PENDING

APCS_F0_QTMR_V1_CNTP_CTL (cont.)

Bits	Name	Type	Description
1	IMSK	RW	Mask Interrupt 0x0: UNMASK_INTERRUPT 0x1: MASK_INTERRUPT
0	EN	RW	Enable Timer 0x0: DISABLED 0x1: ENABLED

0x0B021030 APCS_F0_QTMR_V1_CNTV_CVAL_LO**Type:** RW**Clock:** SYS_TMR_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RWVT]

The Virtual Timer Compare Value register is compare against the CNTVCT counter. The interrupt signal is asserted when the CNTVCT counter is greater than equal to CNTV_CVAL. This register is implemented if FVI (Virtual Frame implemented) for the Frame n is set in QTMR_AC_CNTTID register.

APCS_F0_QTMR_V1_CNTV_CVAL_LO

Bits	Name	Description
31:0	CNTV_CVAL_LO	Lower bits [31:0] of compare value for interrupt for Frame n

0x0B021034 APCS_F0_QTMR_V1_CNTV_CVAL_HI**Type:** RW**Clock:** SYS_TMR_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RWVT]

The Virtual Timer Compare Value register is compare against the CNTVCT counter. The interrupt signal is asserted when the CNTVCT counter is greater than equal to CNTV_CVAL. This register is implemented if FVI (Virtual Frame implemented) for the Frame n is set in QTMR_AC_CNTTID register.

APCS_F0_QTMR_V1_CNTV_CVAL_HI

Bits	Name	Description
23:0	CNTV_CVAL_HI	Higher bits [31:0] of compare value for interrupt for Frame n

0x0B021038 APCS_F0_QTMR_V1_CNTV_TVAL**Type:** RW**Clock:** SYS_TMR_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RWVT]

An alternative 32 bit TVAL view to provide read or write CVAL register.

The TVAL reads:

$$TVAL = (CVAL - (CNTP - CNTVOFF))$$

The TVAL write CVAL with:

$$CVAL = (CNTP - CNTVOFF) + \text{signExtend}(TVAL)$$

This register is implemented if FVI (Virtual Frame implemented) for the Frame n is set in QTMR_AC_CNTTID register.

APCS_F0_QTMR_V1_CNTV_TVAL

Bits	Name	Description
31:0	CNTV_TVAL	Timer Value in tick per second for Frame

0x0B02103C APCS_F0_QTMR_V1_CNTV_CTL**Type:** RW**Clock:** SYS_TMR_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RWVT]

The Virtual Timer Control Register which can disable the timer and can mask its interrupt. If the timer is disabled, interrupt are masked, but the timer value continue to count down.

This register is implemented if FVI (Virtual Frame implemented) for the Frame n is set in QTMR_AC_CNTTID register.

APCS_F0_QTMR_V1_CNTV_CTL

Bits	Name	Type	Description
2	ISTAT	R	Interrupt Status before masking (Read Only) 0x0: INTERRUPT_NOT_PENDING 0x1: INTERRUPT_PENDING

APCS_F0_QTMR_V1_CNTV_CTL (cont.)

Bits	Name	Type	Description
1	IMSK	RW	Mask Interrupt 0x0: UNMASK_INTERRUPT 0x1: MASK_INTERRUPT
0	EN	RW	Enable Timer 0x0: DISABLED 0x1: ENABLED

8.3 APCS_F0_QTMR_V2

QTMR V2 Block

8.3.1 APCS_F0_QTMR_V2

QTMR V2 Block

0x0B022000 APCS_F0_QTMR_V2_CNTPCT_LO**Type:** R**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by TMR_V1_CNTPL0ACR[PL0PCTEN]

The lower 32 bits of Physical Counter

APCS_F0_QTMR_V2_CNTPCT_LO

Bits	Name	Description
31:0	CNTPCT_LO	Lower bits [31:0] of Physical Counter

0x0B022004 APCS_F0_QTMR_V2_CNTPCT_HI**Type:** R**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by TMR_V1_CNTPL0ACR[PL0PCTEN]

The upper 32 bits of Physical Counter

APCS_F0_QTMR_V2_CNTPCT_HI

Bits	Name	Description
23:0	CNTPCT_HI	Higher bits [31:0] of Physical Counter

0x0B022008 APCS_F0_QTMR_V2_CNTVCT_LO**Type:** R**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by TMR_V1_CNTPL0ACR[PL0VCTEN]

The lower 32 bits of Physical Counter

APCS_F0_QTMR_V2_CNTVCT_LO

Bits	Name	Description
31:0	CNTVCT_LO	Lower bits [31:0] of Virtual Counter

0x0B02200C APCS_F0_QTMR_V2_CNTVCT_HI**Type:** R**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by TMR_V1_CNTPL0ACR[PL0VCTEN]

The upper 32 bits of Physical Counter

APCS_F0_QTMR_V2_CNTVCT_HI

Bits	Name	Description
23:0	CNTVCT_HI	Higher bits [31:0] of Virtual Counter

0x0B022010 APCS_F0_QTMR_V2_CNTFRQ**Type:** R**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by TMR_V1_CNTPL0ACR[PL0PCTEN] or TMR_V1_CNTPL0ACR[PL0VCTEN]

Read QTMR_AC_CNTFRQ register.

APCS_F0_QTMR_V2_CNTFRQ

Bits	Name	Description
31:0	CNTFRQ	Counter frequency in ticks per second

0x0B022020 APCS_F0_QTMR_V2_CNTP_CVAL_LO**Type:** RW**Clock:** SYS_TMR_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by TMR_V1_CNTPL0ACR[PL0PTEN]

The Physical Timer Compare Value register is compare against the CNTPCT counter. The interrupt signal is asserted when the CNTPCT counter is greater than equal to CNTP_CVAL

APCS_F0_QTMR_V2_CNTP_CVAL_LO

Bits	Name	Description
31:0	CNTP_CVAL_LO	Higher bits [31:0] of compare value for interrupt for Frame

0x0B022024 APCS_F0_QTMR_V2_CNTP_CVAL_HI**Type:** RW**Clock:** SYS_TMR_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by TMR_V1_CNTPL0ACR[PL0PTEN]

The Physical Timer Compare Value register is compare against the CNTPCT counter. The interrupt signal is asserted when the CNTPCT counter is greater than equal to CNTP_CVAL

APCS_F0_QTMR_V2_CNTP_CVAL_HI

Bits	Name	Description
23:0	CNTP_CVAL_HI	Higher bits [31:0] of compare value for interrupt for Frame

0x0B022028 APCS_F0_QTMR_V2_CNTP_TVAL**Type:** RW**Clock:** SYS_TMR_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by TMR_V1_CNTPL0ACR[PL0PTEN]

An alternative 32 bit TVAL view to provide read or write CVAL register.

The TVAL reads:

$$TVAL = (CVAL - CNTP)$$

The TVAL write CVAL with:

$$CVAL = (CNTP + \text{signExtend}(TVAL))$$

APCS_F0_QTMR_V2_CNTP_TVAL

Bits	Name	Description
31:0	CNTP_TVAL	Timer Value in tick per second for Frame

0x0B02202C APCS_F0_QTMR_V2_CNTP_CTL

Type: RW

Clock: SYS_TMR_CLK

Reset State: 0x00000000

Security Treatment: Controlled by TMR_V1_CNTPL0ACR[PL0PTEN]

The Physical Timer Control Register which can disable the timer and can mask its interrupt. If the timer is disabled, interrupt are masked, but the timer value continue to count down.

APCS_F0_QTMR_V2_CNTP_CTL

Bits	Name	Type	Description
2	ISTAT	R	Interrupt Status before masking (Read Only) 0x0: INTERRUPT_NOT_PENDING 0x1: INTERRUPT_PENDING
1	IMSK	RW	Mask Interrupt 0x0: UNMASK_INTERRUPT 0x1: MASK_INTERRUPT
0	EN	RW	Enable Timer 0x0: DISABLED 0x1: ENABLED

0x0B022030 APCS_F0_QTMR_V2_CNTV_CVAL_LO

Type: RW

Clock: SYS_TMR_CLK

Reset State: 0x00000000

Security Treatment: Controlled by TMR_V1_CNTPL0ACR[PL0VTEN]

The Virtual Timer Compare Value register is compare against the CNTVCT counter. The interrupt

signal is asserted when the CNTVCT counter is greater than equal to CNTV_CVAL. This register is implemented if FVI (Virtual Frame implemented) for the Frame n is set in

QTMR_AC_CNTTID register

APCS_F0_QTMR_V2_CNTV_CVAL_LO

Bits	Name	Description
31:0	CNTV_CVAL_LO	Lower bits [31:0] of compare value for interrupt for Frame n

0x0B022034 APCS_F0_QTMR_V2_CNTV_CVAL_HI

Type: RW

Clock: SYS_TMR_CLK

Reset State: 0x00000000

Security Treatment: Controlled by TMR_V1_CNTPL0ACR[PL0VTEN]

The Virtual Timer Compare Value register is compare against the CNTVCT counter. The interrupt signal is asserted when the CNTVCT counter is greater than equal to CNTV_CVAL. This register is implemented if FVI (Virtual Frame implemented) for the Frame n is set in QTMR_AC_CNTTID register.

APCS_F0_QTMR_V2_CNTV_CVAL_HI

Bits	Name	Description
23:0	CNTV_CVAL_HI	Higher bits [31:0] of compare value for interrupt for Frame n

0x0B022038 APCS_F0_QTMR_V2_CNTV_TVAL

Type: RW

Clock: SYS_TMR_CLK

Reset State: 0x00000000

Security Treatment: Controlled by TMR_V1_CNTPL0ACR[PL0VTEN]

An alternative 32 bit TVAL view to provide read or write CVAL register.

The TVAL reads:

$$TVAL = (CVAL - (CNTP - CNTVOFF))$$

The TVAL write CVAL with:

$$CVAL = (CNTP - CNTVOFF) + \text{signExtend}(TVAL)$$

This register is implemented if FVI (Virtual Frame implemented) for the Frame n is set in QTMR_AC_CNTTID register.

APCS_F0_QTMR_V2_CNTV_TVAL

Bits	Name	Description
31:0	CNTV_TVAL	Timer Value in tick per second for Frame

0x0B02203C APCS_F0_QTMR_V2_CNTV_CTL**Type:** RW**Clock:** SYS_TMR_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by TMR_V1_CNTPL0ACR[PL0VTEN]

The Virtual Timer Control Register which can disable the timer and can mask its interrupt. If the timer is disabled, interrupt are masked, but the timer value continue to count down.

This register is implemented if FVI (Virtual Frame implemented) for the Frame n is set in QTMR_AC_CNTTID register.

APCS_F0_QTMR_V2_CNTV_CTL

Bits	Name	Type	Description
2	ISTAT	R	Interrupt Status before masking (Read Only) 0x0: INTERRUPT_NOT_PENDING 0x1: INTERRUPT_PENDING
1	IMSK	RW	Mask Interrupt 0x0: UNMASK_INTERRUPT 0x1: MASK_INTERRUPT
0	EN	RW	Enable Timer 0x0: DISABLED 0x1: ENABLED

8.4 APCS_F1_QTMR_V1

QTMR V1 Block

8.4.1 APCS_F1_QTMR_V1

QTMR V1 Block

0x0B023000 APCS_F1_QTMR_V1_CNTPCT_LO**Type:** R**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RPCT]

The lower 32 bits of Physical Counter

APCS_F1_QTMR_V1_CNTPCT_LO

Bits	Name	Description
31:0	CNTPCT_LO	Lower bits [31:0] of Physical Counter

0x0B023004 APCS_F1_QTMR_V1_CNTPCT_HI**Type:** R**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RPCT]

The upper 32 bits of Physical Counter

APCS_F1_QTMR_V1_CNTPCT_HI

Bits	Name	Description
23:0	CNTPCT_HI	Higher bits [31:0] of Physical Counter

0x0B023008 APCS_F1_QTMR_V1_CNTVCT_LO**Type:** R**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RVCT]

The lower 32 bits of Virtual Counter

APCS_F1_QTMR_V1_CNTVCT_LO

Bits	Name	Description
31:0	CNTVCT_LO	Lower bits [31:0] of Virtual Counter

0x0B02300C APCS_F1_QTMR_V1_CNTVCT_HI**Type:** R**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RVCT]

The upper 32 bits of Virtual Counter

APCS_F1_QTMR_V1_CNTVCT_HI

Bits	Name	Description
23:0	CNTVCT_HI	Higher bits [31:0] of Virtual Counter

0x0B023010 APCS_F1_QTMR_V1_CNTFRQ**Type:** R**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RFRQ]

Read QTMR_AC_CNTFRQ register

APCS_F1_QTMR_V1_CNTFRQ

Bits	Name	Description
31:0	CNTFRQ	Counter frequency in ticks per second

0x0B023014 APCS_F1_QTMR_V1_CNTPL0ACR**Type:** RW**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** Both

The Second View Counter Access Control Register (CNTPL0ACR) provide second view access control to the individual elements of timer frame n, except CNTVOFF and CNTPL0ACR registers

APCS_F1_QTMR_V1_CNTPL0ACR

Bits	Name	Description
9	PL0CTEN	Second view read/write access to CNTP_* register
8	PL0VTEN	Second view read/write access to CNTV_* register
1	PL0VCTEN	Second view read access to CNTVCT and CNTFRQ register

APCS_F1_QTMR_V1_CNTPL0ACR (cont.)

Bits	Name	Description
0	PL0PCTEN	Second view read access to CNTPCT and CNTFRQ register

0x0B023018 APCS_F1_QTMR_V1_CNTVOFF_LO**Type:** R**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RVOFF]

Read QTMR_AC_CNTVOFF_LO_n register

APCS_F1_QTMR_V1_CNTVOFF_LO

Bits	Name	Description
31:0	CNTVOFF_LO	Lower bits [31:0] of virtual Offset for Frame

0x0B02301C APCS_F1_QTMR_V1_CNTVOFF_HI**Type:** R**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RVOFF]

Read QTMR_AC_CNTVOFF_HI_n register

APCS_F1_QTMR_V1_CNTVOFF_HI

Bits	Name	Description
23:0	CNTVOFF_HI	Higher bits [31:0] of virtual Offset for Frame

0x0B023020 APCS_F1_QTMR_V1_CNTP_CVAL_LO**Type:** RW**Clock:** SYS_TMR_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RWPT]

The Physical Timer Compare Value register is compare against the CNTPCT counter. The interrupt signal is asserted when the CNTPCT counter is greater than equal to CNTP_CVAL

APCS_F1_QTMR_V1_CNTP_CVAL_LO

Bits	Name	Description
31:0	CNTP_CVAL_LO	Lower bits [31:0] of compare value for interrupt for Frame

0x0B023024 APCS_F1_QTMR_V1_CNTP_CVAL_HI**Type:** RW**Clock:** SYS_TMR_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RWPT]

The Physical Timer Compare Value register is compare against the CNTPCT counter. The interrupt signal is asserted when the CNTPCT counter is greater than equal to CNTP_CVAL

APCS_F1_QTMR_V1_CNTP_CVAL_HI

Bits	Name	Description
23:0	CNTP_CVAL_HI	Higher bits [31:0] of compare value for interrupt for Frame

0x0B023028 APCS_F1_QTMR_V1_CNTP_TVAL**Type:** RW**Clock:** SYS_TMR_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RWPT]

An alternative 32 bit TVAL view to provide read or write CVAL register.

The TVAL reads:

$$TVAL = (CVAL - CNTP)$$

The TVAL write CVAL with:

$$CVAL = (CNTP + \text{signExtend}(TVAL))$$

APCS_F1_QTMR_V1_CNTP_TVAL

Bits	Name	Description
31:0	CNTP_TVAL	Timer Value in tick per second for Frame

0x0B02302C APCS_F1_QTMR_V1_CNTP_CTL**Type:** RW**Clock:** SYS_TMR_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RWPT]

The Physical Timer Control Register which can disable the timer and can mask its interrupt. If the timer is disabled, interrupt are masked, but the timer value continue to count down.

APCS_F1_QTMR_V1_CNTP_CTL

Bits	Name	Type	Description
2	ISTAT	R	Interrupt Status before masking (Read Only) 0x0: INTERRUPT_NOT_PENDING 0x1: INTERRUPT_PENDING
1	IMSK	RW	Mask Interrupt 0x0: UNMASK_INTERRUPT 0x1: MASK_INTERRUPT
0	EN	RW	Enable Timer 0x0: DISABLED 0x1: ENABLED

0x0B023030 APCS_F1_QTMR_V1_CNTV_CVAL_LO**Type:** RW**Clock:** SYS_TMR_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RWVT]

The Virtual Timer Compare Value register is compare against the CNTVCT counter. The interrupt signal is asserted when the CNTVCT counter is greater than equal to CNTV_CVAL. This register is implemented if FVI (Virtual Frame implemented) for the Frame n is set in QTMR_AC_CNTTID register.

APCS_F1_QTMR_V1_CNTV_CVAL_LO

Bits	Name	Description
31:0	CNTV_CVAL_LO	Lower bits [31:0] of compare value for interrupt for Frame n

0x0B023034 APCS_F1_QTMR_V1_CNTV_CVAL_HI**Type:** RW**Clock:** SYS_TMR_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RWVT]

The Virtual Timer Compare Value register is compare against the CNTVCT counter. The interrupt signal is asserted when the CNTVCT counter is greater than equal to CNTV_CVAL. This register is implemented if FVI (Virtual Frame implemented) for the Frame n is set in QTMR_AC_CNTTID register.

APCS_F1_QTMR_V1_CNTV_CVAL_HI

Bits	Name	Description
23:0	CNTV_CVAL_HI	Higher bits [31:0] of compare value for interrupt for Frame n

0x0B023038 APCS_F1_QTMR_V1_CNTV_TVAL**Type:** RW**Clock:** SYS_TMR_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RWVT]

An alternative 32 bit TVAL view to provide read or write CVAL register.

The TVAL reads:

$$TVAL = (CVAL - (CNTP - CNTVOFF))$$

The TVAL write CVAL with:

$$CVAL = (CNTP - CNTVOFF) + \text{signExtend}(TVAL)$$

This register is implemented if FVI (Virtual Frame implemented) for the Frame n is set in QTMR_AC_CNTTID register.

APCS_F1_QTMR_V1_CNTV_TVAL

Bits	Name	Description
31:0	CNTV_TVAL	Timer Value in tick per second for Frame

0x0B02303C APCS_F1_QTMR_V1_CNTV_CTL**Type:** RW**Clock:** SYS_TMR_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RWVT]

The Virtual Timer Control Register which can disable the timer and can mask its interrupt. If the timer is disabled, interrupt are masked, but the timer value continue to count down.

This register is implemented if FVI (Virtual Frame implemented) for the Frame n is set in QTMR_AC_CNTTID register.

APCS_F1_QTMR_V1_CNTV_CTL

Bits	Name	Type	Description
2	ISTAT	R	Interrupt Status before masking (Read Only) 0x0: INTERRUPT_NOT_PENDING 0x1: INTERRUPT_PENDING
1	IMSK	RW	Mask Interrupt 0x0: UNMASK_INTERRUPT 0x1: MASK_INTERRUPT
0	EN	RW	Enable Timer 0x0: DISABLED 0x1: ENABLED

8.5 APCS_F2_QTMR_V1

QTMR V1 Block

8.5.1 APCS_F2_QTMR_V1

QTMR V1 Block

0x0B024000 APCS_F2_QTMR_V1_CNTPCT_LO**Type:** R**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RPCT]

The lower 32 bits of Physical Counter

APCS_F2_QTMR_V1_CNTPCT_LO

Bits	Name	Description
31:0	CNTPCT_LO	Lower bits [31:0] of Physical Counter

0x0B024004 APCS_F2_QTMR_V1_CNTPCT_HI**Type:** R**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RPCT]

The upper 32 bits of Physical Counter

APCS_F2_QTMR_V1_CNTPCT_HI

Bits	Name	Description
23:0	CNTPCT_HI	Higher bits [31:0] of Physical Counter

0x0B024008 APCS_F2_QTMR_V1_CNTVCT_LO**Type:** R**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RVCT]

The lower 32 bits of Virtual Counter

APCS_F2_QTMR_V1_CNTVCT_LO

Bits	Name	Description
31:0	CNTVCT_LO	Lower bits [31:0] of Virtual Counter

0x0B02400C APCS_F2_QTMR_V1_CNTVCT_HI**Type:** R**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RVCT]

The upper 32 bits of Virtual Counter

APCS_F2_QTMR_V1_CNTVCT_HI

Bits	Name	Description
23:0	CNTVCT_HI	Higher bits [31:0] of Virtual Counter

0x0B024010 APCS_F2_QTMR_V1_CNTFRQ**Type:** R**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RFRQ]

Read QTMR_AC_CNTFRQ register

APCS_F2_QTMR_V1_CNTFRQ

Bits	Name	Description
31:0	CNTFRQ	Counter frequency in ticks per second

0x0B024014 APCS_F2_QTMR_V1_CNTPL0ACR**Type:** RW**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** Both

The Second View Counter Access Control Register (CNTPL0ACR) provide second view access control to the individual elements of timer frame n, except CNTVOFF and CNTPL0ACR registers

APCS_F2_QTMR_V1_CNTPL0ACR

Bits	Name	Description
9	PL0CTEN	Second view read/write access to CNTP_* register
8	PL0VTEN	Second view read/write access to CNTV_* register
1	PL0VCTEN	Second view read access to CNTVCT and CNTFRQ register
0	PL0PCTEN	Second view read access to CNTPCT and CNTFRQ register

0x0B024018 APCS_F2_QTMR_V1_CNTVOFF_LO**Type:** R**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RVOFF]

Read QTMR_AC_CNTVOFF_LO_n register

APCS_F2_QTMR_V1_CNTVOFF_LO

Bits	Name	Description
31:0	CNTVOFF_LO	Lower bits [31:0] of virtual Offset for Frame

0x0B02401C APCS_F2_QTMR_V1_CNTVOFF_HI**Type:** R**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RVOFF]

Read QTMR_AC_CNTVOFF_HI_n register

APCS_F2_QTMR_V1_CNTVOFF_HI

Bits	Name	Description
23:0	CNTVOFF_HI	Higher bits [31:0] of virtual Offset for Frame

0x0B024020 APCS_F2_QTMR_V1_CNTP_CVAL_LO**Type:** RW**Clock:** SYS_TMR_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RWPT]

The Physical Timer Compare Value register is compare against the CNTPCT counter. The interrupt signal is asserted when the CNTPCT counter is greater than equal to CNTP_CVAL

APCS_F2_QTMR_V1_CNTP_CVAL_LO

Bits	Name	Description
31:0	CNTP_CVAL_LO	Lower bits [31:0] of compare value for interrupt for Frame

0x0B024024 APCS_F2_QTMR_V1_CNTP_CVAL_HI**Type:** RW**Clock:** SYS_TMR_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RWPT]

The Physical Timer Compare Value register is compare against the CNTPCT counter. The interrupt signal is asserted when the CNTPCT counter is greater than equal to CNTP_CVAL

APCS_F2_QTMR_V1_CNTP_CVAL_HI

Bits	Name	Description
23:0	CNTP_CVAL_HI	Higher bits [31:0] of compare value for interrupt for Frame

0x0B024028 APCS_F2_QTMR_V1_CNTP_TVAL**Type:** RW**Clock:** SYS_TMR_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RWPT]

An alternative 32 bit TVAL view to provide read or write CVAL register.

The TVAL reads:

$$TVAL = (CVAL - CNTP)$$

The TVAL write CVAL with:

$$CVAL = (CNTP + \text{signExtend}(TVAL))$$
APCS_F2_QTMR_V1_CNTP_TVAL

Bits	Name	Description
31:0	CNTP_TVAL	Timer Value in tick per second for Frame

0x0B02402C APCS_F2_QTMR_V1_CNTP_CTL**Type:** RW**Clock:** SYS_TMR_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RWPT]

The Physical Timer Control Register which can disable the timer and can mask its interrupt. If the timer is disabled, interrupt are masked, but the timer value continue to count down.

APCS_F2_QTMR_V1_CNTP_CTL

Bits	Name	Type	Description
2	ISTAT	R	Interrupt Status before masking (Read Only) 0x0: INTERRUPT_NOT_PENDING 0x1: INTERRUPT_PENDING
1	IMSK	RW	Mask Interrupt 0x0: UNMASK_INTERRUPT 0x1: MASK_INTERRUPT
0	EN	RW	Enable Timer 0x0: DISABLED 0x1: ENABLED

0x0B024030 APCS_F2_QTMR_V1_CNTV_CVAL_LO**Type:** RW**Clock:** SYS_TMR_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RWVT]

The Virtual Timer Compare Value register is compare against the CNTVCT counter. The interrupt signal is asserted when the CNTVCT counter is greater than equal to CNTV_CVAL. This register is implemented if FVI (Virtual Frame implemented) for the Frame n is set in QTMR_AC_CNTTID register.

APCS_F2_QTMR_V1_CNTV_CVAL_LO

Bits	Name	Description
31:0	CNTV_CVAL_LO	Lower bits [31:0] of compare value for interrupt for Frame n

0x0B024034 APCS_F2_QTMR_V1_CNTV_CVAL_HI**Type:** RW**Clock:** SYS_TMR_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RWVT]

The Virtual Timer Compare Value register is compare against the CNTVCT counter. The interrupt signal is asserted when the CNTVCT counter is greater than equal to CNTV_CVAL. This register is implemented if FVI (Virtual Frame implemented) for the Frame n is set in QTMR_AC_CNTTID register.

APCS_F2_QTMR_V1_CNTV_CVAL_HI

Bits	Name	Description
23:0	CNTV_CVAL_HI	Higher bits [31:0] of compare value for interrupt for Frame n

0x0B024038 APCS_F2_QTMR_V1_CNTV_TVAL**Type:** RW**Clock:** SYS_TMR_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RWVT]

An alternative 32 bit TVAL view to provide read or write CVAL register.

The TVAL reads:

$$\text{TVAL} = (\text{CVAL} - (\text{CNTP} - \text{CNTVOFF}))$$

The TVAL write CVAL with:

$$\text{CVAL} = (\text{CNTP} - \text{CNTVOFF}) + \text{signExtend}(\text{TVAL})$$

This register is implemented if FVI (Virtual Frame implemented) for the Frame n is set in QTMR_AC_CNTTID register.

APCS_F2_QTMR_V1_CNTV_TVAL

Bits	Name	Description
31:0	CNTV_TVAL	Timer Value in tick per second for Frame

0x0B02403C APCS_F2_QTMR_V1_CNTV_CTL**Type:** RW**Clock:** SYS_TMR_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RWVT]

The Virtual Timer Control Register which can disable the timer and can mask its interrupt. If the timer is disabled, interrupt are masked, but the timer value continue to count down.

This register is implemented if FVI (Virtual Frame implemented) for the Frame n is set in QTMR_AC_CNTTID register.

APCS_F2_QTMR_V1_CNTV_CTL

Bits	Name	Type	Description
2	ISTAT	R	Interrupt Status before masking (Read Only) 0x0: INTERRUPT_NOT_PENDING 0x1: INTERRUPT_PENDING
1	IMSK	RW	Mask Interrupt 0x0: UNMASK_INTERRUPT 0x1: MASK_INTERRUPT
0	EN	RW	Enable Timer 0x0: DISABLED 0x1: ENABLED

8.6 APCS_F3_QTMR_V1

QTMR V1 Block

8.6.1 APCS_F3_QTMR_V1

QTMR V1 Block

0x0B025000 APCS_F3_QTMR_V1_Cntpct_Lo**Type:** R**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RPCT]

The lower 32 bits of Physical Counter

APCS_F3_QTMR_V1_Cntpct_Lo

Bits	Name	Description
31:0	CNTPCT_LO	Lower bits [31:0] of Physical Counter

0x0B025004 APCS_F3_QTMR_V1_Cntpct_Hi**Type:** R**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RPCT]

The upper 32 bits of Physical Counter

APCS_F3_QTMR_V1_Cntpct_HI

Bits	Name	Description
23:0	CNTPCT_HI	Higher bits [31:0] of Physical Counter

0x0B025008 APCS_F3_QTMR_V1_Cntvct_LO

Type: R

Clock: SYS_AHB_CLK

Reset State: 0x00000000

Security Treatment: Controlled by QTMR_AC_CNTACR_n[RVCT]

The lower 32 bits of Virtual Counter

APCS_F3_QTMR_V1_Cntvct_LO

Bits	Name	Description
31:0	CNTVCT_LO	Lower bits [31:0] of Virtual Counter

0x0B02500C APCS_F3_QTMR_V1_Cntvct_HI

Type: R

Clock: SYS_AHB_CLK

Reset State: 0x00000000

Security Treatment: Controlled by QTMR_AC_CNTACR_n[RVCT]

The upper 32 bits of Virtual Counter

APCS_F3_QTMR_V1_Cntvct_HI

Bits	Name	Description
23:0	CNTVCT_HI	Higher bits [31:0] of Virtual Counter

0x0B025010 APCS_F3_QTMR_V1_Cntfrq

Type: R

Clock: SYS_AHB_CLK

Reset State: 0x00000000

Security Treatment: Controlled by QTMR_AC_CNTACR_n[RFRQ]

Read QTMR_AC_CNTFRQ register

APCS_F3_QTMR_V1_CNTFRQ

Bits	Name	Description
31:0	CNTFRQ	Counter frequency in ticks per second

0x0B025014 APCS_F3_QTMR_V1_CNTPL0ACR**Type:** RW**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** Both

The Second View Counter Access Control Register (CNTPL0ACR) provide second view access control to the individual elements of timer frame n, except CNTVOFF and CNTPL0ACR registers

APCS_F3_QTMR_V1_CNTPL0ACR

Bits	Name	Description
9	PL0CTEN	Second view read/write access to CNTP_* register
8	PL0VTEN	Second view read/write access to CNTV_* register
1	PL0VCTEN	Second view read access to CNTVCT and CNTFRQ register
0	PL0PCTEN	Second view read access to CNTPCT and CNTFRQ register

0x0B025018 APCS_F3_QTMR_V1_CNTVOFF_LO**Type:** R**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RVOFF]

Read QTMR_AC_CNTVOFF_LO_n register

APCS_F3_QTMR_V1_CNTVOFF_LO

Bits	Name	Description
31:0	CNTVOFF_LO	Lower bits [31:0] of virtual Offset for Frame

0x0B02501C APCS_F3_QTMR_V1_CNTVOFF_HI**Type:** R**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RVOFF]

Read QTMR_AC_CNTVOFF_HI_n register

APCS_F3_QTMR_V1_CNTVOFF_HI

Bits	Name	Description
23:0	CNTVOFF_HI	Higher bits [31:0] of virtual Offset for Frame

0x0B025020 APCS_F3_QTMR_V1_CNTP_CVAL_LO**Type:** RW**Clock:** SYS_TMR_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RWPT]

The Physical Timer Compare Value register is compare against the CNTPCT counter. The interrupt signal is asserted when the CNTPCT counter is greater than equal to CNTP_CVAL

APCS_F3_QTMR_V1_CNTP_CVAL_LO

Bits	Name	Description
31:0	CNTP_CVAL_LO	Lower bits [31:0] of compare value for interrupt for Frame

0x0B025024 APCS_F3_QTMR_V1_CNTP_CVAL_HI**Type:** RW**Clock:** SYS_TMR_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RWPT]

The Physical Timer Compare Value register is compare against the CNTPCT counter. The interrupt signal is asserted when the CNTPCT counter is greater than equal to CNTP_CVAL

APCS_F3_QTMR_V1_CNTP_CVAL_HI

Bits	Name	Description
23:0	CNTP_CVAL_HI	Higher bits [31:0] of compare value for interrupt for Frame

0x0B025028 APCS_F3_QTMR_V1_CNTP_TVAL**Type:** RW**Clock:** SYS_TMR_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RWPT]

An alternative 32 bit TVAL view to provide read or write CVAL register.

The TVAL reads:

$$TVAL = (CVAL - CNTP)$$

The TVAL write CVAL with:

$$CVAL = (CNTP + \text{signExtend}(TVAL))$$
APCS_F3_QTMR_V1_CNTP_TVAL

Bits	Name	Description
31:0	CNTP_TVAL	Timer Value in tick per second for Frame

0x0B02502C APCS_F3_QTMR_V1_CNTP_CTL**Type:** RW**Clock:** SYS_TMR_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RWPT]

The Physical Timer Control Register which can disable the timer and can mask its interrupt. If the timer is disabled, interrupt are masked, but the timer value continue to count down.

APCS_F3_QTMR_V1_CNTP_CTL

Bits	Name	Type	Description
2	ISTAT	R	Interrupt Status before masking (Read Only) 0x0: INTERRUPT_NOT_PENDING 0x1: INTERRUPT_PENDING
1	IMSK	RW	Mask Interrupt 0x0: UNMASK_INTERRUPT 0x1: MASK_INTERRUPT
0	EN	RW	Enable Timer 0x0: DISABLED 0x1: ENABLED

0x0B025030 APCS_F3_QTMR_V1_CNTV_CVAL_LO**Type:** RW**Clock:** SYS_TMR_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RWVT]

The Virtual Timer Compare Value register is compare against the CNTVCT counter. The interrupt signal is asserted when the CNTVCT counter is greater than equal to CNTV_CVAL. This register is implemented if FVI (Virtual Frame implemented) for the Frame n is set in

QTMR_AC_CNTTID register.

APCS_F3_QTMR_V1_CNTV_CVAL_LO

Bits	Name	Description
31:0	CNTV_CVAL_LO	Lower bits [31:0] of compare value for interrupt for Frame n

0x0B025034 APCS_F3_QTMR_V1_CNTV_CVAL_HI**Type:** RW**Clock:** SYS_TMR_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RWVT]

The Virtual Timer Compare Value register is compare against the CNTVCT counter. The interrupt signal is asserted when the CNTVCT counter is greater than equal to CNTV_CVAL. This register is implemented if FVI (Virtual Frame implemented) for the Frame n is set in

QTMR_AC_CNTTID register.

APCS_F3_QTMR_V1_CNTV_CVAL_HI

Bits	Name	Description
23:0	CNTV_CVAL_HI	Higher bits [31:0] of compare value for interrupt for Frame n

0x0B025038 APCS_F3_QTMR_V1_CNTV_TVAL**Type:** RW**Clock:** SYS_TMR_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RWVT]

An alternative 32 bit TVAL view to provide read or write CVAL register.

The TVAL reads:

$$TVAL = (CVAL - (CNTP - CNTVOFF))$$

The TVAL write CVAL with:

$$CVAL = (CNTP - CNTVOFF) + \text{signExtend}(TVAL)$$

This register is implemented if FVI (Virtual Frame implemented) for the Frame n is set in QTMR_AC_CNTTID register.

APCS_F3_QTMR_V1_CNTV_TVAL

Bits	Name	Description
31:0	CNTV_TVAL	Timer Value in tick per second for Frame

0x0B02503C APCS_F3_QTMR_V1_CNTV_CTL

Type: RW

Clock: SYS_TMR_CLK

Reset State: 0x00000000

Security Treatment: Controlled by QTMR_AC_CNTACR_n[RWVT]

The Virtual Timer Control Register which can disable the timer and can mask its interrupt. If the timer is disabled, interrupt are masked, but the timer value continue to count down.

This register is implemented if FVI (Virtual Frame implemented) for the Frame n is set in QTMR_AC_CNTTID register.

APCS_F3_QTMR_V1_CNTV_CTL

Bits	Name	Type	Description
2	ISTAT	R	Interrupt Status before masking (Read Only) 0x0: INTERRUPT_NOT_PENDING 0x1: INTERRUPT_PENDING
1	IMSK	RW	Mask Interrupt 0x0: UNMASK_INTERRUPT 0x1: MASK_INTERRUPT
0	EN	RW	Enable Timer 0x0: DISABLED 0x1: ENABLED

8.7 APCS_F4_QTMR_V1

APCS_F4_QTMR_V1

QTMR V1 Block

8.7.1 APCS_F4_QTMR_V1

QTMR V1 Block

0x0B026000 APCS_F4_QTMR_V1_CNTPCT_LO

Type: R

Clock: SYS_AHB_CLK

Reset State: 0x00000000

Security Treatment: Controlled by QTMR_AC_CNTACR_n[RPCT]

The lower 32 bits of Physical Counter

APCS_F4_QTMR_V1_CNTPCT_LO

Bits	Name	Description
31:0	CNTPCT_LO	Lower bits [31:0] of Physical Counter

0x0B026004 APCS_F4_QTMR_V1_CNTPCT_HI

Type: R

Clock: SYS_AHB_CLK

Reset State: 0x00000000

Security Treatment: Controlled by QTMR_AC_CNTACR_n[RPCT]

The upper 32 bits of Physical Counter

APCS_F4_QTMR_V1_CNTPCT_HI

Bits	Name	Description
23:0	CNTPCT_HI	Higher bits [31:0] of Physical Counter

0x0B026008 APCS_F4_QTMR_V1_CNTVCT_LO**Type:** R**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RVCT]

The lower 32 bits of Virtual Counter

APCS_F4_QTMR_V1_CNTVCT_LO

Bits	Name	Description
31:0	CNTVCT_LO	Lower bits [31:0] of Virtual Counter

0x0B02600C APCS_F4_QTMR_V1_CNTVCT_HI**Type:** R**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RVCT]

The upper 32 bits of Virtual Counter

APCS_F4_QTMR_V1_CNTVCT_HI

Bits	Name	Description
23:0	CNTVCT_HI	Higher bits [31:0] of Virtual Counter

0x0B026010 APCS_F4_QTMR_V1_CNTFRQ**Type:** R**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RFRQ]

Read QTMR_AC_CNTFRQ register

APCS_F4_QTMR_V1_CNTFRQ

Bits	Name	Description
31:0	CNTFRQ	Counter frequency in ticks per second

0x0B026014 APCS_F4_QTMR_V1_CNTPL0ACR**Type:** RW**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** Both

The Second View Counter Access Control Register (CNTPL0ACR) provide second view access control to the individual elements of timer frame n, except CNTVOFF and CNTPL0ACR registers

APCS_F4_QTMR_V1_CNTPL0ACR

Bits	Name	Description
9	PL0CTEN	Second view read/write access to CNTP_* register
8	PL0VTEN	Second view read/write access to CNTV_* register
1	PL0VCTEN	Second view read access to CNTVCT and CNTFRQ register
0	PL0PCTEN	Second view read access to CNTPCT and CNTFRQ register

0x0B026018 APCS_F4_QTMR_V1_CNTVOFF_LO**Type:** R**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RVOFF]

Read QTMR_AC_CNTVOFF_LO_n register

APCS_F4_QTMR_V1_CNTVOFF_LO

Bits	Name	Description
31:0	CNTVOFF_LO	Lower bits [31:0] of virtual Offset for Frame

0x0B02601C APCS_F4_QTMR_V1_CNTVOFF_HI**Type:** R**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RVOFF]

Read QTMR_AC_CNTVOFF_HI_n register

APCS_F4_QTMR_V1_CNTVOFF_HI

Bits	Name	Description
23:0	CNTVOFF_HI	Higher bits [31:0] of virtual Offset for Frame

0x0B026020 APCS_F4_QTMR_V1_CNTP_CVAL_LO**Type:** RW**Clock:** SYS_TMR_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RWPT]

The Physical Timer Compare Value register is compare against the CNTPCT counter. The interrupt signal is asserted when the CNTPCT counter is greater than equal to CNTP_CVAL

APCS_F4_QTMR_V1_CNTP_CVAL_LO

Bits	Name	Description
31:0	CNTP_CVAL_LO	Lower bits [31:0] of compare value for interrupt for Frame

0x0B026024 APCS_F4_QTMR_V1_CNTP_CVAL_HI**Type:** RW**Clock:** SYS_TMR_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RWPT]

The Physical Timer Compare Value register is compare against the CNTPCT counter. The interrupt signal is asserted when the CNTPCT counter is greater than equal to CNTP_CVAL

APCS_F4_QTMR_V1_CNTP_CVAL_HI

Bits	Name	Description
23:0	CNTP_CVAL_HI	Higher bits [31:0] of compare value for interrupt for Frame

0x0B026028 APCS_F4_QTMR_V1_CNTP_TVAL**Type:** RW**Clock:** SYS_TMR_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RWPT]

An alternative 32 bit TVAL view to provide read or write CVAL register.

The TVAL reads:

$$TVAL = (CVAL - CNTP)$$

The TVAL write CVAL with:

$$CVAL = (CNTP + \text{signExtend}(TVAL))$$

APCS_F4_QTMR_V1_CNTP_TVAL

Bits	Name	Description
31:0	CNTP_TVAL	Timer Value in tick per second for Frame

0x0B02602C APCS_F4_QTMR_V1_CNTP_CTL

Type: RW

Clock: SYS_TMR_CLK

Reset State: 0x00000000

Security Treatment: Controlled by QTMR_AC_CNTACR_n[RWPT]

The Physical Timer Control Register which can disable the timer and can mask its interrupt. If the timer is disabled, interrupt are masked, but the timer value continue to count down.

APCS_F4_QTMR_V1_CNTP_CTL

Bits	Name	Type	Description
2	ISTAT	R	Interrupt Status before masking (Read Only) 0x0: INTERRUPT_NOT_PENDING 0x1: INTERRUPT_PENDING
1	IMSK	RW	Mask Interrupt 0x0: UNMASK_INTERRUPT 0x1: MASK_INTERRUPT
0	EN	RW	Enable Timer 0x0: DISABLED 0x1: ENABLED

0x0B026030 APCS_F4_QTMR_V1_CNTV_CVAL_LO

Type: RW

Clock: SYS_TMR_CLK

Reset State: 0x00000000

Security Treatment: Controlled by QTMR_AC_CNTACR_n[RWVT]

The Virtual Timer Compare Value register is compare against the CNTVCT counter. The interrupt

signal is asserted when the CNTVCT counter is greater than equal to CNTV_CVAL. This register is implemented if FVI (Virtual Frame implemented) for the Frame n is set in QTMR_AC_CNTTID register.

APCS_F4_QTMR_V1_CNTV_CVAL_LO

Bits	Name	Description
31:0	CNTV_CVAL_LO	Lower bits [31:0] of compare value for interrupt for Frame n

0x0B026034 APCS_F4_QTMR_V1_CNTV_CVAL_HI

Type: RW
Clock: SYS_TMR_CLK
Reset State: 0x00000000

Security Treatment: Controlled by QTMR_AC_CNTACR_n[RWVT]

The Virtual Timer Compare Value register is compare against the CNTVCT counter. The interrupt signal is asserted when the CNTVCT counter is greater than equal to CNTV_CVAL. This register is implemented if FVI (Virtual Frame implemented) for the Frame n is set in QTMR_AC_CNTTID register.

APCS_F4_QTMR_V1_CNTV_CVAL_HI

Bits	Name	Description
23:0	CNTV_CVAL_HI	Higher bits [31:0] of compare value for interrupt for Frame n

0x0B026038 APCS_F4_QTMR_V1_CNTV_TVAL

Type: RW
Clock: SYS_TMR_CLK
Reset State: 0x00000000

Security Treatment: Controlled by QTMR_AC_CNTACR_n[RWVT]

An alternative 32 bit TVAL view to provide read or write CVAL register.

The TVAL reads:

$$TVAL = (CVAL - (CNTP - CNTVOFF))$$

The TVAL write CVAL with:

$$CVAL = (CNTP - CNTVOFF) + \text{signExtend}(TVAL)$$

This register is implemented if FVI (Virtual Frame implemented) for the Frame n is set in

QTMR_AC_CNTTID register.

APCS_F4_QTMR_V1_CNTV_TVAL

Bits	Name	Description
31:0	CNTV_TVAL	Timer Value in tick per second for Frame

0x0B02603C APCS_F4_QTMR_V1_CNTV_CTL

Type: RW

Clock: SYS_TMR_CLK

Reset State: 0x00000000

Security Treatment: Controlled by QTMR_AC_CNTACR_n[RWVT]

The Virtual Timer Control Register which can disable the timer and can mask its interrupt. If the timer is disabled, interrupt are masked, but the timer value continue to count down.

This register is implemented if FVI (Virtual Frame implemented) for the Frame n is set in QTMR_AC_CNTTID register.

APCS_F4_QTMR_V1_CNTV_CTL

Bits	Name	Type	Description
2	ISTAT	R	Interrupt Status before masking (Read Only) 0x0: INTERRUPT_NOT_PENDING 0x1: INTERRUPT_PENDING
1	IMSK	RW	Mask Interrupt 0x0: UNMASK_INTERRUPT 0x1: MASK_INTERRUPT
0	EN	RW	Enable Timer 0x0: DISABLED 0x1: ENABLED

8.8 APCS_F5_QTMR_V1

QTMR V1 Block

8.8.1 APCS_F5_QTMR_V1

QTMR V1 Block

0x0B027000 APCS_F5_QTMR_V1_CNTPCT_LO**Type:** R**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RPCT]

The lower 32 bits of Physical Counter

APCS_F5_QTMR_V1_CNTPCT_LO

Bits	Name	Description
31:0	CNTPCT_LO	Lower bits [31:0] of Physical Counter

0x0B027004 APCS_F5_QTMR_V1_CNTPCT_HI**Type:** R**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RPCT]

The upper 32 bits of Physical Counter

APCS_F5_QTMR_V1_CNTPCT_HI

Bits	Name	Description
23:0	CNTPCT_HI	Higher bits [31:0] of Physical Counter

0x0B027008 APCS_F5_QTMR_V1_CNTVCT_LO**Type:** R**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RVCT]

The lower 32 bits of Virtual Counter

APCS_F5_QTMR_V1_CNTVCT_LO

Bits	Name	Description
31:0	CNTVCT_LO	Lower bits [31:0] of Virtual Counter

0x0B02700C APCS_F5_QTMR_V1_CNTVCT_HI**Type:** R**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RVCT]

The upper 32 bits of Virtual Counter

APCS_F5_QTMR_V1_CNTVCT_HI

Bits	Name	Description
23:0	CNTVCT_HI	Higher bits [31:0] of Virtual Counter

0x0B027010 APCS_F5_QTMR_V1_CNTFRQ**Type:** R**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RFRQ]

Read QTMR_AC_CNTFRQ register

APCS_F5_QTMR_V1_CNTFRQ

Bits	Name	Description
31:0	CNTFRQ	Counter frequency in ticks per second

0x0B027014 APCS_F5_QTMR_V1_CNTPL0ACR**Type:** RW**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** Both

The Second View Counter Access Control Register (CNTPL0ACR) provide second view access control to the individual elements of timer frame n, except CNTVOFF and CNTPL0ACR registers

APCS_F5_QTMR_V1_CNTPL0ACR

Bits	Name	Description
9	PL0CTEN	Second view read/write access to CNTP_* register
8	PL0VTEN	Second view read/write access to CNTV_* register
1	PL0VCTEN	Second view read access to CNTVCT and CNTFRQ register

APCS_F5_QTMR_V1_CNTPL0ACR (cont.)

Bits	Name	Description
0	PL0PCTEN	Second view read access to CNTPCT and CNTFRQ register

0x0B027018 APCS_F5_QTMR_V1_CNTVOFF_LO**Type:** R**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RVOFF]

Read QTMR_AC_CNTVOFF_LO_n register

APCS_F5_QTMR_V1_CNTVOFF_LO

Bits	Name	Description
31:0	CNTVOFF_LO	Lower bits [31:0] of virtual Offset for Frame

0x0B02701C APCS_F5_QTMR_V1_CNTVOFF_HI**Type:** R**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RVOFF]

Read QTMR_AC_CNTVOFF_HI_n register

APCS_F5_QTMR_V1_CNTVOFF_HI

Bits	Name	Description
23:0	CNTVOFF_HI	Higher bits [31:0] of virtual Offset for Frame

0x0B027020 APCS_F5_QTMR_V1_CNTP_CVAL_LO**Type:** RW**Clock:** SYS_TMR_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RWPT]

The Physical Timer Compare Value register is compare against the CNTPCT counter. The interrupt signal is asserted when the CNTPCT counter is greater than equal to CNTP_CVAL

APCS_F5_QTMR_V1_CNTP_CVAL_LO

Bits	Name	Description
31:0	CNTP_CVAL_LO	Lower bits [31:0] of compare value for interrupt for Frame

0x0B027024 APCS_F5_QTMR_V1_CNTP_CVAL_HI**Type:** RW**Clock:** SYS_TMR_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RWPT]

The Physical Timer Compare Value register is compare against the CNTPCT counter. The interrupt signal is asserted when the CNTPCT counter is greater than equal to CNTP_CVAL

APCS_F5_QTMR_V1_CNTP_CVAL_HI

Bits	Name	Description
23:0	CNTP_CVAL_HI	Higher bits [31:0] of compare value for interrupt for Frame

0x0B027028 APCS_F5_QTMR_V1_CNTP_TVAL**Type:** RW**Clock:** SYS_TMR_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RWPT]

An alternative 32 bit TVAL view to provide read or write CVAL register.

The TVAL reads:

$$TVAL = (CVAL - CNTP)$$

The TVAL write CVAL with:

$$CVAL = (CNTP + \text{signExtend}(TVAL))$$

APCS_F5_QTMR_V1_CNTP_TVAL

Bits	Name	Description
31:0	CNTP_TVAL	Timer Value in tick per second for Frame

0x0B02702C APCS_F5_QTMR_V1_CNTP_CTL**Type:** RW**Clock:** SYS_TMR_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RWPT]

The Physical Timer Control Register which can disable the timer and can mask its interrupt. If the timer is disabled, interrupt are masked, but the timer value continue to count down.

APCS_F5_QTMR_V1_CNTP_CTL

Bits	Name	Type	Description
2	ISTAT	R	Interrupt Status before masking (Read Only) 0x0: INTERRUPT_NOT_PENDING 0x1: INTERRUPT_PENDING
1	IMSK	RW	Mask Interrupt 0x0: UNMASK_INTERRUPT 0x1: MASK_INTERRUPT
0	EN	RW	Enable Timer 0x0: DISABLED 0x1: ENABLED

0x0B027030 APCS_F5_QTMR_V1_CNTV_CVAL_LO**Type:** RW**Clock:** SYS_TMR_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RWVT]

The Virtual Timer Compare Value register is compare against the CNTVCT counter. The interrupt signal is asserted when the CNTVCT counter is greater than equal to CNTV_CVAL. This register is implemented if FVI (Virtual Frame implemented) for the Frame n is set in QTMR_AC_CNTTID register.

APCS_F5_QTMR_V1_CNTV_CVAL_LO

Bits	Name	Description
31:0	CNTV_CVAL_LO	Lower bits [31:0] of compare value for interrupt for Frame n

0x0B027034 APCS_F5_QTMR_V1_CNTV_CVAL_HI**Type:** RW**Clock:** SYS_TMR_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RWVT]

The Virtual Timer Compare Value register is compare against the CNTVCT counter. The interrupt signal is asserted when the CNTVCT counter is greater than equal to CNTV_CVAL. This register is implemented if FVI (Virtual Frame implemented) for the Frame n is set in

QTMR_AC_CNTTID register.

APCS_F5_QTMR_V1_CNTV_CVAL_HI

Bits	Name	Description
23:0	CNTV_CVAL_HI	Higher bits [31:0] of compare value for interrupt for Frame n

0x0B027038 APCS_F5_QTMR_V1_CNTV_TVAL**Type:** RW**Clock:** SYS_TMR_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RWVT]

An alternative 32 bit TVAL view to provide read or write CVAL register.

The TVAL reads:

$$TVAL = (CVAL - (CNTP - CNTVOFF))$$

The TVAL write CVAL with:

$$CVAL = (CNTP - CNTVOFF) + \text{signExtend}(TVAL)$$

This register is implemented if FVI (Virtual Frame implemented) for the Frame n is set in

QTMR_AC_CNTTID register.

APCS_F5_QTMR_V1_CNTV_TVAL

Bits	Name	Description
31:0	CNTV_TVAL	Timer Value in tick per second for Frame

0x0B02703C APCS_F5_QTMR_V1_CNTV_CTL**Type:** RW**Clock:** SYS_TMR_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RWVT]

The Virtual Timer Control Register which can disable the timer and can mask its interrupt. If the timer is disabled, interrupt are masked, but the timer value continue to count down.

This register is implemented if FVI (Virtual Frame implemented) for the Frame n is set in QTMR_AC_CNTTID register.

APCS_F5_QTMR_V1_CNTV_CTL

Bits	Name	Type	Description
2	ISTAT	R	Interrupt Status before masking (Read Only) 0x0: INTERRUPT_NOT_PENDING 0x1: INTERRUPT_PENDING
1	IMSK	RW	Mask Interrupt 0x0: UNMASK_INTERRUPT 0x1: MASK_INTERRUPT
0	EN	RW	Enable Timer 0x0: DISABLED 0x1: ENABLED

8.9 APCS_F6_QTMR_V1

QTMR V1 Block

8.9.1 APCS_F6_QTMR_V1

QTMR V1 Block

0x0B028000 APCS_F6_QTMR_V1_CNTPCT_LO**Type:** R**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RPCT]

The lower 32 bits of Physical Counter

APCS_F6_QTMR_V1_CNTPCT_LO

Bits	Name	Description
31:0	CNTPCT_LO	Lower bits [31:0] of Physical Counter

0x0B028004 APCS_F6_QTMR_V1_CNTPCT_HI**Type:** R**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RPCT]

The upper 32 bits of Physical Counter

APCS_F6_QTMR_V1_CNTPCT_HI

Bits	Name	Description
23:0	CNTPCT_HI	Higher bits [31:0] of Physical Counter

0x0B028008 APCS_F6_QTMR_V1_CNTVCT_LO**Type:** R**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RVCT]

The lower 32 bits of Virtual Counter

APCS_F6_QTMR_V1_CNTVCT_LO

Bits	Name	Description
31:0	CNTVCT_LO	Lower bits [31:0] of Virtual Counter

0x0B02800C APCS_F6_QTMR_V1_CNTVCT_HI**Type:** R**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RVCT]

The upper 32 bits of Virtual Counter

APCS_F6_QTMR_V1_CNTVCT_HI

Bits	Name	Description
23:0	CNTVCT_HI	Higher bits [31:0] of Virtual Counter

0x0B028010 APCS_F6_QTMR_V1_CNTFRQ**Type:** R**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RFRQ]

Read QTMR_AC_CNTFRQ register

APCS_F6_QTMR_V1_CNTFRQ

Bits	Name	Description
31:0	CNTFRQ	Counter frequency in ticks per second

0x0B028014 APCS_F6_QTMR_V1_CNTPL0ACR**Type:** RW**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** Both

The Second View Counter Access Control Register (CNTPL0ACR) provide second view access control to the individual elements of timer frame n, except CNTVOFF and CNTPL0ACR registers

APCS_F6_QTMR_V1_CNTPL0ACR

Bits	Name	Description
9	PL0CTEN	Second view read/write access to CNTP_* register
8	PL0VTEN	Second view read/write access to CNTV_* register
1	PL0VCTEN	Second view read access to CNTVCT and CNTFRQ register
0	PL0PCTEN	Second view read access to CNTPCT and CNTFRQ register

0x0B028018 APCS_F6_QTMR_V1_CNTVOFF_LO**Type:** R**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RVOFF]

Read QTMR_AC_CNTVOFF_LO_n register

APCS_F6_QTMR_V1_CNTVOFF_LO

Bits	Name	Description
31:0	CNTVOFF_LO	Lower bits [31:0] of virtual Offset for Frame

0x0B02801C APCS_F6_QTMR_V1_CNTVOFF_HI**Type:** R**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RVOFF]

Read QTMR_AC_CNTVOFF_HI_n register

APCS_F6_QTMR_V1_CNTVOFF_HI

Bits	Name	Description
23:0	CNTVOFF_HI	Higher bits [31:0] of virtual Offset for Frame

0x0B028020 APCS_F6_QTMR_V1_CNTP_CVAL_LO**Type:** RW**Clock:** SYS_TMR_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RWPT]

The Physical Timer Compare Value register is compare against the CNTPCT counter. The interrupt signal is asserted when the CNTPCT counter is greater than equal to CNTP_CVAL

APCS_F6_QTMR_V1_CNTP_CVAL_LO

Bits	Name	Description
31:0	CNTP_CVAL_LO	Lower bits [31:0] of compare value for interrupt for Frame

0x0B028024 APCS_F6_QTMR_V1_CNTP_CVAL_HI**Type:** RW**Clock:** SYS_TMR_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RWPT]

The Physical Timer Compare Value register is compare against the CNTPCT counter. The interrupt signal is asserted when the CNTPCT counter is greater than equal to CNTP_CVAL

APCS_F6_QTMR_V1_CNTP_CVAL_HI

Bits	Name	Description
23:0	CNTP_CVAL_HI	Higher bits [31:0] of compare value for interrupt for Frame

0x0B028028 APCS_F6_QTMR_V1_CNTP_TVAL**Type:** RW**Clock:** SYS_TMR_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RWPT]

An alternative 32 bit TVAL view to provide read or write CVAL register.

The TVAL reads:

$$TVAL = (CVAL - CNTP)$$

The TVAL write CVAL with:

$$CVAL = (CNTP + \text{signExtend}(TVAL))$$
APCS_F6_QTMR_V1_CNTP_TVAL

Bits	Name	Description
31:0	CNTP_TVAL	Timer Value in tick per second for Frame

0x0B02802C APCS_F6_QTMR_V1_CNTP_CTL**Type:** RW**Clock:** SYS_TMR_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RWPT]

The Physical Timer Control Register which can disable the timer and can mask its interrupt. If the timer is disabled, interrupt are masked, but the timer value continue to count down.

APCS_F6_QTMR_V1_CNTP_CTL

Bits	Name	Type	Description
2	ISTAT	R	Interrupt Status before masking (Read Only) 0x0: INTERRUPT_NOT_PENDING 0x1: INTERRUPT_PENDING
1	IMSK	RW	Mask Interrupt 0x0: UNMASK_INTERRUPT 0x1: MASK_INTERRUPT
0	EN	RW	Enable Timer 0x0: DISABLED 0x1: ENABLED

0x0B028030 APCS_F6_QTMR_V1_CNTV_CVAL_LO**Type:** RW**Clock:** SYS_TMR_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RWVT]

The Virtual Timer Compare Value register is compare against the CNTVCT counter. The interrupt signal is asserted when the CNTVCT counter is greater than equal to CNTV_CVAL. This register is implemented if FVI (Virtual Frame implemented) for the Frame n is set in QTMR_AC_CNTTID register.

APCS_F6_QTMR_V1_CNTV_CVAL_LO

Bits	Name	Description
31:0	CNTV_CVAL_LO	Lower bits [31:0] of compare value for interrupt for Frame n

0x0B028034 APCS_F6_QTMR_V1_CNTV_CVAL_HI**Type:** RW**Clock:** SYS_TMR_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RWVT]

The Virtual Timer Compare Value register is compare against the CNTVCT counter. The interrupt signal is asserted when the CNTVCT counter is greater than equal to CNTV_CVAL. This register is implemented if FVI (Virtual Frame implemented) for the Frame n is set in

QTMR_AC_CNTTID register.

APCS_F6_QTMR_V1_CNTV_CVAL_HI

Bits	Name	Description
23:0	CNTV_CVAL_HI	Higher bits [31:0] of compare value for interrupt for Frame n

0x0B028038 APCS_F6_QTMR_V1_CNTV_TVAL

Type: RW

Clock: SYS_TMR_CLK

Reset State: 0x00000000

Security Treatment: Controlled by QTMR_AC_CNTACR_n[RWVT]

An alternative 32 bit TVAL view to provide read or write CVAL register.

The TVAL reads:

$$TVAL = (CVAL - (CNTP - CNTVOFF))$$

The TVAL write CVAL with:

$$CVAL = (CNTP - CNTVOFF) + \text{signExtend}(TVAL)$$

This register is implemented if FVI (Virtual Frame implemented) for the Frame n is set in

QTMR_AC_CNTTID register.

APCS_F6_QTMR_V1_CNTV_TVAL

Bits	Name	Description
31:0	CNTV_TVAL	Timer Value in tick per second for Frame

0x0B02803C APCS_F6_QTMR_V1_CNTV_CTL

Type: RW

Clock: SYS_TMR_CLK

Reset State: 0x00000000

Security Treatment: Controlled by QTMR_AC_CNTACR_n[RWVT]

The Virtual Timer Control Register which can disable the timer and can mask its interrupt. If the timer is disabled, interrupt are masked, but the timer value continue to count down.

This register is implemented if FVI (Virtual Frame implemented) for the Frame n is set in

QTMR_AC_CNTTID register.

APCS_F6_QTMR_V1_CNTV_CTL

Bits	Name	Type	Description
2	ISTAT	R	Interrupt Status before masking (Read Only) 0x0: INTERRUPT_NOT_PENDING 0x1: INTERRUPT_PENDING
1	IMSK	RW	Mask Interrupt 0x0: UNMASK_INTERRUPT 0x1: MASK_INTERRUPT
0	EN	RW	Enable Timer 0x0: DISABLED 0x1: ENABLED

8.10 APCS_A53SS_HYP**0x0B01A004 APCS_VMM_IPC_INTERRUPT****Type:** W**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** None**APCS_VMM_IPC_INTERRUPT**

Bits	Name	Description
31:28	SPARE_IPC_31_28	Spare
27:24	SPARE_IPC_27_24	Spare
23:20	SPARE_IPC_23_20	Spare
19:16	SPARE_IPC_19_16	Spare
15:12	SPARE_IPC_15_12	Spare
11:8	LPASS_ADSP_IPC	Interrupts to LPASS ADSP
7:4	SPARE_IPC_7_4	Spare
3:0	RPM_IPC	Interrupts to RPM

0x0B01A010 APCS_L2_SPM_VOTE_HYP**Type:** RW**Clock:** SYS_AHB_CLK**Reset State:** 0xC000FFFF**Security Treatment:** None

Hypervisor Vote for L2 SPM functionality

APCS_L2_SPM_VOTE_HYP

Bits	Name	Description
31	PMIC_HANDSHAKE_EN	Hypervisor Vote for L2 SPM <-> PMIC Handshake. If not set, SPM cannot initiate any requests to PMIC
30	RPM_HANDSHAKE_EN	Hypervisor Vote for L2 SPM <-> RPM Handshake. If not set, SPM cannot initiate handshake with RPM
15:0	PWR_CTL_EN	Hypervisor Vote for L2 SPM PWR_CTL bits. If not set, PWR_CTL bit is forced LOW

0x0B01A020 APCS_BANKED_SPM_VOTE_HYP

Type: RW

Clock: SYS_AHB_CLK

Reset State: 0xC0000000

Security Treatment: None

Hypervisor Vote for Core SPM functionality

APCS_BANKED_SPM_VOTE_HYP

Bits	Name	Description
31	PMIC_HANDSHAKE_EN	Hypervisor Vote for Core SPM <-> PMIC Handshake. If not set, SPM cannot initiate any requests to PMIC
30	RPM_HANDSHAKE_EN	Hypervisor Vote for Core SPM <-> RPM Handshake. If not set, SPM cannot initiate handshake with RPM

0x0B01A030 APCS_ALIAS0_SPM_VOTE_HYP

Type: RW

Clock: SYS_AHB_CLK

Reset State: 0xC0000000

Security Treatment: None

Hypervisor Vote for Core SPM functionality

APCS_ALIAS0_SPM_VOTE_HYP

Bits	Name	Description
31	PMIC_HANDSHAKE_EN	Hypervisor Vote for Core SPM <-> PMIC Handshake. If not set, SPM cannot initiate any requests to PMIC

APCS_ALIAS0_SPM_VOTE_HYP (cont.)

Bits	Name	Description
30	RPM_HANDSHAKE_EN	Hypervisor Vote for Core SPM <-> RPM Handshake. If not set, SPM cannot initiate handshake with RPM

0x0B01A034 APCS_ALIAS1_SPM_VOTE_HYP**Type:** RW**Clock:** SYS_AHB_CLK**Reset State:** 0xC0000000**Security Treatment:** None

Hypervisor Vote for Core SPM functionality

APCS_ALIAS1_SPM_VOTE_HYP

Bits	Name	Description
31	PMIC_HANDSHAKE_EN	Hypervisor Vote for Core SPM <-> PMIC Handshake. If not set, SPM cannot initiate any requests to PMIC
30	RPM_HANDSHAKE_EN	Hypervisor Vote for Core SPM <-> RPM Handshake. If not set, SPM cannot initiate handshake with RPM

0x0B01A038 APCS_ALIAS2_SPM_VOTE_HYP**Type:** RW**Clock:** SYS_AHB_CLK**Reset State:** 0xC0000000**Security Treatment:** None

Hypervisor Vote for Core SPM functionality

APCS_ALIAS2_SPM_VOTE_HYP

Bits	Name	Description
31	PMIC_HANDSHAKE_EN	Hypervisor Vote for Core SPM <-> PMIC Handshake. If not set, SPM cannot initiate any requests to PMIC
30	RPM_HANDSHAKE_EN	Hypervisor Vote for Core SPM <-> RPM Handshake. If not set, SPM cannot initiate handshake with RPM

0x0B01A03C APCS_ALIAS3_SPM_VOTE_HYP**Type:** RW**Clock:** SYS_AHB_CLK**Reset State:** 0xC0000000**Security Treatment:** None

Hypervisor Vote for Core SPM functionality

APCS_ALIAS3_SPM_VOTE_HYP

Bits	Name	Description
31	PMIC_HANDSHAKE_EN	Hypervisor Vote for Core SPM <-> PMIC Handshake. If not set, SPM cannot initiate any requests to PMIC
30	RPM_HANDSHAKE_EN	Hypervisor Vote for Core SPM <-> RPM Handshake. If not set, SPM cannot initiate handshake with RPM

0x0B01811C RESERVED**0x0B018120 RESERVED****0x0B018124 RESERVED****8.11 APCS_KPSS_WDT****0x0B017000 APCS_WDOG_SECURE****Type:** RW**Clock:** SYS_AHB_CLK**Reset State:** 0x00000001**Security Treatment:** Restricted

The WDOG_SECURE register is used to enable or disable non-secure bus transactions. The bus **security** status is signaled by APROTNS pin. When APROTNS is set to '0' the state of the APCS_WDTSECURE register is not considered - a secure transaction is always allowed. When APROTNS is set to '1', the APCS_WDTSECURE register's SECURE bit must be '1' for access to be granted. If the SECURE bit is set to '0', the non-secure (APROTNS = '1') transactions are denied. When access is denied, a write is ignored, and a read will return '0'. No error is signaled.

When reading the APCS_WDTSECURE register, non secure agents will always read '0'. Secure agents will see the actual value of the register. The same is true for the registers or bits each bit in the APCS_WDTSECURE register correspond to.

APCS_WDOG_SECURE

Bits	Name	Description
0	WDT	0x1: NSEC (NSEC) 0x0: SEC (SEC) Controls security treatment for the Watch Dog Timer registers: WDOG_RESET, WDOG_CTL, WDOG_STATUS, WDOG_BARK_TIME, WDOG_BITE_TIME, WDOG_TEST_LOAD, STATUS, WDOG_TEST_LOAD, WDOG_TEST)

0x0B017004 APCS_WDOG_RESET**Type:** W**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** .Controlled by WDOG_SECURE [WDT].

The WDOG_RESET register resets the watch dog counter. In normal operation, the microprocessor should periodically write this command register to reset the watch dog counter. This command register also disables the watchdog freeze

APCS_WDOG_RESET

Bits	Name	Description
0	RESET	This bit generates the WDT_STB during the non-sleep mode. A pulse is generated on WDT_STB when this bit is written with a ?1

0x0B017008 APCS_WDOG_CTL**Type:** W**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** .Controlled by WDOG_SECURE [WDT].

The WDOG_CTL register controls when the watch dog timer is enabled.

APCS_WDOG_CTL

Bits	Name	Description
1	UNMASKED_INT_ENABLE	Enable unmasked interrupt (IRQ or FIQ) to wake up the wdog timer: 0x1: ENABLE (Enable) 0x0: DISABLE (Disable)

APCS_WDOG_CTL (cont.)

Bits	Name	Description
0	ENABLE	Enable wdog timer: 0x1: ENABLE (Enable) 0x0: DISABLE (Disable)

0x0B01700C APCS_WDOG_STATUS**Type:** R**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** .Controlled by WDOG_SECURE [WDT].

The WDOG_STATUS register is the watchdog status register.

APCS_WDOG_STATUS

Bits	Name	Description
31	FROZEN	This bit indicates whether the or not wdog timer is frozen at count 0. 0x1: YES (frozen) 0x0: NO (not frozen)
20:1	COUNT	Counter value [19:0] of the watch dog counter. Note: The sleep counter value is sampled using the AHB clk. Multiple reads are required to determine the value (assuming AHB clk is much faster than sleep clock).
0	EXPIRED_STATUS	0x1: YES (expired (wdog timer has expired)) 0x0: NO (not expired (wdog timer has not expired))

0x0B017010 APCS_WDOG_BARK_TIME**Type:** RW**Clock:** SYS_AHB_CLK**Reset State:** 0x0007FFFF**Security Treatment:** .Controlled by WDOG_SECURE [WDT].

The WDOG_BARK_TIME register defines the time in sleep_clk when to assert the watchdog ('bark') interrupts. The value can be between 0 and 12787 (the time when the watchdog times out). The default value is 0x3FFF, but this reset value is not readable through the AHB interface. Note that this value is larger than the watchdog timeout of 0x31F3, effectively disabling the watchdog ('bark') interrupt.

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APCS_WDOG_BARK_TIME

Bits	Name	Description
19:0	DATA	The WDOG counter value on which to trigger the bark interrupt.

0x0B017014 APCS_WDOG_BITE_TIME**Type:** RW**Clock:** SYS_AHB_CLK**Reset State:** 0x000FFFFFF**Security Treatment:** .Controlled by WDOG_SECURE [WDT].

The WDOG_BITE_TIME register defines the time in sleep_clk when to assert the watchdog ('bark') interrupts. The value can be between 0 and 12787 (the time when the watchdog times out). The default value is 0x3FFF, but this reset value is not readable through the AHB interface. Note that this value is larger than the watchdog timeout of 0x31F3, effectively disabling the watchdog ('bark') interrupt.

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APCS_WDOG_BITE_TIME

Bits	Name	Description
19:0	DATA	The WDOG bite expiration time [19:0].

8.12 APCS_SH_KPSS_PLL**0x0B016000 APCS_SH_PLL_MODE****Type:** RW**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by APCS_PLLSECURE[PLL_CTL].**APCS_SH_PLL_MODE**

Bits	Name	Description
3	RESERVED	RESERVED
2	RESET_N	
1	BYPASSNL	
0	OUTCTRL	

0x0B016004 APCS_SH_PLL_L_VAL**Type:** RW**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by APCS_PLLSECURE[PLL_CTL].**APCS_SH_PLL_L_VAL**

Bits	Name	Description
7:0	L_VAL	PLL VCO feedback division ratio

0x0B016008 APCS_SH_PLL_M_VAL**Type:** RW**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by APCS_PLLSECURE[PLL_CTL].**APCS_SH_PLL_M_VAL**

Bits	Name	Description
18:0	M_VAL	PLL VCO fraction divide numerator

0x0B01600C APCS_SH_PLL_N_VAL**Type:** RW**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by APCS_PLLSECURE[PLL_CTL].**APCS_SH_PLL_N_VAL**

Bits	Name	Description
18:0	N_VAL	PLL VCO fraction divide denominator

0x0B016010 APCS_SH_PLL_USER_CTL**Type:** RW**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by APCS_PLLSECURE[PLL_CTL].

APCS_SH_PLL_USER_CTL

Bits	Name	Description
31:25	RESERVE_BITS31_25	RESERVED
24	MN_EN	PLL fractional mode control (mn_en)
23:13	RESERVE_BITS23_13	RESERVED
12	PRE_DIV_RATIO	PLL pre-divider control(prediv2_en)
11:10	RESERVE_BITS11_10	RESERVED
9:8	POST_DIV_RATIO	PLL post-divider control (postdiv_ctl<1:0>)
7	OUTPUT_INV	Output clock polarity (inv_output)
6:5	RESERVE_BITS6_5	RESERVED
4	RESERVED	Reserved
3	PLLOUT_LV_EARLY	pllout_lv_early enable (lvearly_en)
2	PLLOUT_LV_AUX2	pllout_lv_aux2 enable (lvaux2_en)
1	PLLOUT_LV_AUX	pllout_lv_aux enable (lvaux_en)
0	PLLOUT_LV_MAIN	pllout_lv_main enable (lvmain_en)

0x0B016014 APCS_SH_PLL_CONFIG_CTL**Type:** RW**Clock:** SYS_AHB_CLK**Reset State:** 0x40014665**Security Treatment:** Controlled by APCS_PLLSECURE[PLL_CTL].**APCS_SH_PLL_CONFIG_CTL**

Bits	Name	Description
31:0	SR2_PLL_LP_CONFIG_CTL_31_0	Ref SR2_PLL_LP Datasheet, used with hpm wrapper

0x0B01601C APCS_SH_PLL_STATUS**Type:** R**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by APCS_PLLSECURE[PLL_CTL].

APCS_SH_PLL_STATUS

Bits	Name	Description
16	LOCK_DET	PLL Lock Detect Output
15:0	STATUS	PLL Status Register

0x0B016030 APCS_SH_PLL_SECURE**Type:** RW**Clock:** SYS_AHB_CLK**Reset State:** 0x00000001**Security Treatment:** Restricted

The APCS_PLLSECURE register is used to enable or disable non-secure bus transactions. The bus **security** status is signaled by A53SS's APROTNS pin. When APROTNS is set to '0' the state of the APCS_PLLSECURE register is not considered - a secure transaction is always allowed. When APROTNS is set to '1', the APCS_PLLSECURE register's SECURE bit must be '1' for access to be granted. If the SECURE bit is set to '0', the non-secure (APROTNS = '1') transactions are denied. When access is denied, a write is ignored, and a read will return '0'. No error is signaled.

When reading the APCS_PLLSECURE register, non secure agents will always read '0'. Secure agents will see the actual value of the register. The same is true for the registers or bits each bit in the APCS_PLLSECURE register correspond to.

APCS_SH_PLL_SECURE

Bits	Name	Description
0	PLL_CTL	0x1: NSEC (NSEC) 0x0: SEC (SEC) Controls security treatment for the PLL control registers: APCS_PLL_MODE, APCS_PLL_CONFIG_CTL, APCS_PLL_L_VAL, APCS_PLL_M_VAL, APCS_PLL_N_VAL, APCS_PLL_DROOP_CTL, APCS_PLL_TEST_CTL, APCS_PLL_STATUS)

8.13 APCS_L2_SAW2**0x0B012000 APCS_L2_SAW2_SECURE****Type:** RW**Clock:** sys_ref_clk**Reset State:** 0x00000007**Security Treatment:** Restricted

The SAW2_SECURE register is used to enable or disable non-secure bus transactions. The bus **security** status is signaled by ahb_saw_hprotns pin. When ahb_saw_hprotns is set to '0' the state of the SAW2_SECURE register is not considered - a secure transaction is always allowed. When ahb_saw_hprotns is set to '1', the SAW2_SECURE register **security** treatment bit of that register must be '1' for access to be granted. If the **security** treatment bit of the register is set to '0', the non-secure (ahb_saw_hprotns = '1') transactions are denied. When access is denied, a write is ignored, and a read will return '0'. No error is signaled.

When reading the SAW2_SECURE register, non secure agents will always read '0'. Secure agents will see the actual value of the register. The reset value is controlled by CFGNSINIT input pin.

APCS_L2_SAW2_SECURE

Bits	Name	Description
2	SAW_CTL	0x1: NSEC (NSEC) 0x0: SEC (SEC) Controls security treatment for SAW2 registers: SAW2_ID, SAW2_CFG, SAW2_STS_0, SAW2_STS_1, SAW2_RST, SAW2_VERSION)
1	PWR_CTL	0x1: NSEC (NSEC) 0x0: SEC (SEC) Controls security treatment for SPM registers: SAW2_SPM_CTL, SAW2_SPM_PMIC_DLY, SAW2_SPM_PMIC_DATA_0, SAW2_SPM_PMIC_DATA_1, SAW2_SPM_SLP_SEQ_ENTRY_n)
0	VLT_CTL	0x1: NSEC (NSEC) 0x0: SEC (SEC)

0x0B012004 APCS_L2_SAW2_ID

Type: R

Clock: sys_ref_clk

Reset State: 0x2010086F

Security Treatment: Controlled by SAW2_SECURE [SAW_CTL].

This read only register reports the revision and parameter information for the SAW2 core.

APCS_L2_SAW2_ID

Bits	Name	Description
31:24	NUM_SPM_ENTRY	SAW2 parameter: Indicates number of SAW2_SPM_SLP_SEQ_ENTRY register implemented. Value can range from 1 - 128
21:16	NUM_PWR_CTL	SAW2 parameter: Indicates number of power control implemented. Value can range from 4 - 32
12:8	PMIC_DATA_WIDTH	SAW2 parameter: Indicate PMIC data bus width Values can range from 8-16

APCS_L2_SAW2_ID (cont.)

Bits	Name	Description
6:4	NUM_PMIC_DATA	SAW2 parameter: Indicates number of SAW2_SPM_PMIC_DATA_n register implemented. Value can range from 1-7
3	CFG_NS_ACCESS	SAW2 parameter: Reset value of SAW2_SECURE register
2	PMIC_ARB_INTF	SAW2 parameter: Indicates PMIC Arbiter Interface function is implemented
1	RESERVED	Reserved
0	SPM_PRESENT	SAW2 parameter: Indicates SPM function is implemented

0x0B012008 APCS_L2_SAW2_CFG**Type:** RW**Clock:** sys_ref_clk**Reset State:** 0x00000000**Security Treatment:** Controlled by SAW2_SECURE [SAW_CTL].**APCS_L2_SAW2_CFG**

Bits	Name	Description
9	RESERVED	Reserved
8	FRC_REF_CLK_ON	Chicken bit to force saw_sys_ref_clk_on_req ON.
4:0	CLK_DIV	Divider ratio for clock. This is used to generate timer tick for the timer. Timer tick is asserted every (CLK_DIV + 1) sys_ref_clk period.

0x0B01200C APCS_L2_SAW2_SPM_STS**Type:** R**Clock:** sys_ref_clk**Reset State:** 0x00000000**Security Treatment:** Controlled by SAW2_SECURE [SAW_CTL].

This read only register provides SW with SAW2 status.

APCS_L2_SAW2_SPM_STS

Bits	Name	Description
31:16	CURR_PWR_CTL	Power Control Outputs.
15	SHTDWN_REQ	This bit reflects the shutdown request from the SPM(spm_rpm_shutdown_req) to RPM.

APCS_L2_SAW2_SPM_STS (cont.)

Bits	Name	Description
14	SHTDWN_ACK	This bit reflects the shutdown acknowledgement from the RPM(rpm_spm_shutdown_ack) to SPM.
13	BRNGUP_REQ	This bit reflects the bringup request from the SPM(spm_rpm_bringup_req) to RPM.
12	BRNGUP_ACK	This bit reflects the bringup acknowledgement from the RPM(rpm_spm_bringup_ack) to SPM.
11:10	RPM_STATE	State of the RPM FSM: 0x0: RUN (RUN(waiting for SPM request)) 0x1: STDNACK (STDNACK (waiting for shutdown ACK from RPM)) 0x2: BGUPACK (BGUPACK (waiting for bringup ACK from RPM)) 0x3: WAKEUP (WAKEUP (waiting for wakeup interrupt))
8:0	SPM_CMD_ADDR	Last SPM command executed.

0x0B012010 RESERVED**0x0B012014 APCS_L2_SAW2_PMIC_STS****Type:** R**Clock:** sys_ref_clk**Reset State:** 0x00000000**Security Treatment:** Controlled by SAW2_SECURE [SAW_CTL].

This read only register provides SW with SAW2 status.

APCS_L2_SAW2_PMIC_STS

Bits	Name	Description
29:20	RESERVED	Reserved
18	CURR_PMIC_SIZE	Last PMIC VLVL Size sent to the PMIC Arbiter (and the PMIC).
17:16	PMIC_STATE	State of the PMIC FSM: 0x0: IDLE (IDLE) 0x1: DONE (DONE (waiting for DONE form PMIC Arb before transitions back to IDLE)) 0x2: DELAY (DELAY (waiting for delay count termination before transitions back to IDLE)) 0x3: DONEBAR (DONEBAR (waiting for DONE to deassert before another pending request is processed))
7:0	CURR_PMIC_DATA	Last PMIC DATA sent to the PMIC Arbiter (and the PMIC)

0x0B012018 RESERVED**0x0B01201C APCS_L2_SAW2_VCTL****Type:** RW**Clock:** sys_ref_clk**Reset State:** 0x00000000**Security Treatment:** Controlled by SAW2_SECURE [VLT_CTL].

Though this register is read/writable, it also causes a command pulse to the PMIC FSM. Writing this register results in a transaction to the PMIC with SAW2_VCTL being sent to the PMIC. SAW2 support both 8901 and 8058 regulator.

APCS_L2_SAW2_VCTL

Bits	Name	Description
20	SIZE	PMIC Data Size
18:16	ADR_IDX	PMIC Arbiter Address Index. Drive the saw_pmic_addr_idx output port
7:0	PMIC_DATA	PMIC DATA sent to the PMIC Arbiter.

0x0B012020 RESERVED**0x0B012024 RESERVED****0x0B012030 APCS_L2_SAW2_SPM_CTL****Type:** RW**Clock:** sys_ref_clk**Reset State:** 0x00000000**Security Treatment:** Controlled by SAW2_SECURE [PWR_CTL].

The SAW2_SPM_CTL register is used to control the subsystem power management system. This are parameters that controls the operation of SPM FSM.

APCS_L2_SAW2_SPM_CTL

Bits	Name	Description
31:24	EVENT_SYNC	SPM Event input configuration. There is an individual bit for each sys_spm_event input
17	SLP_CMD_MODE	Sleep Command Mode
16	SPM_SYS_PC_MODE	Indication that we're in power collapse mode
12:4	SPM_START_ADR	Start address for the SPM sequence.

APCS_L2_SAW2_SPM_CTL (cont.)

Bits	Name	Description
3	ISAR	Inhibit Start Address Reset 0x0: END_OF_PROGRAM_RESET_THE_SPM_START_ADR_TO_ZERO (End of program reset the SPM_START_ADR to zero.) 0x1: INHIBIT_END_OF_PROGRAM_TO_RESET_SPM_START_ADR (Inhibit End of program to reset SPM_START_ADR)
2:1	WAKEUP_CONFIG	Wakeup Configuration 0x0: SYS_SPM_WAKEUP (sys_spm_wakeup) 0x1: SYS_SPM_WAKEUP_OR_NOT_CPU_SPM_WAIT_REQ (sys_spm_wakeup or !cpu_spm_wait_req) 0x2: SYS_SPM_WAKEUP_OR_RISING_EDGE_OF_SYS_SPM_DBG_NOPWRDWN (sys_spm_wakeup or rising edge of sys_spm_dbg_nopwrdsn) 0x3: SYS_SPM_WAKEUP_OR_NOT_CPU_SPM_WAIT_REQ_OR_RISING_EDGE_OF_SYS_SPM_DBG_NOPWRDWN (sys_spm_wakeup or !cpu_spm_wait_req or rising edge of sys_spm_dbg_nopwrdsn)
0	SPM_EN	SPM En.

0x0B012034 APCS_L2_SAW2_SPM_DLY**Type:** RW**Clock:** sys_ref_clk**Reset State:** 0x00000000**Security Treatment:** Controlled by SAW2_SECURE [PWR_CTL].

SPM Delay. Controls the time between SPM_SLP_SEQ SPM Delay and the next SPM_SLP_SEQ command.

APCS_L2_SAW2_SPM_DLY

Bits	Name	Description
29:20	DLY3	SPM Delay. Controls the time between SPM_SLP_SEQ SPM Delay and the next SPM_SLP_SEQ command. Time range: 52ns to 1.7ms
19:10	DLY2	SPM Delay. Controls the time between SPM_SLP_SEQ SPM Delay and the next SPM_SLP_SEQ command. Time range: 52ns to 1.7ms
9:0	DLY1	SPM Delay. Controls the time between SPM_SLP_SEQ SPM Delay and the next SPM_SLP_SEQ command. Time range: 52ns to 1.7ms

0x0B012038 APCS_L2_SAW2_SPM_STS2**Type:** R**Clock:** sys_ref_clk**Reset State:** 0x00000000**Security Treatment:** Controlled by SAW2_SECURE [SAW_CTL].

This read only register provides SW with SPM power control status.

APCS_L2_SAW2_SPM_STS2

Bits	Name	Description
15:0	CURR_PWR_CTL	Power control Outputs

**0x0B012400+ APCS_L2_SAW2_SPM_SLP_SEQ_ENTRY_n, n=[0..31]
0x4*n****Type:** RW**Clock:** sys_ref_clk**Reset State:** 0x00000000**Security Treatment:** Controlled by SAW2_SECURE [PWR_CTL].

The SAW2_SPM_SLP_SEQ_ENTRY_n is an array used to sequence through the steps during various power mode. The register width is defined by CLOG_PWR_CTL parameter.

APCS_L2_SAW2_SPM_SLP_SEQ_ENTRY_n

Bits	Name	Description
29:24	CMD3	SPM PWR CMD3
21:16	CMD2	SPM PWR CMD2
13:8	CMD1	SPM PWR CMD1
5:0	CMD0	SPM PWR CMD0

**0x0B012040+ APCS_L2_SAW2_SPM_PMIC_DATA_n, n=[0..5]
0x4*n****Type:** RW**Clock:** sys_ref_clk**Reset State:** 0x00000000**Security Treatment:** Controlled by SAW2_SECURE [PWR_CTL].

This register provide the PMIC data values for SPM FSM PMIC transaction.

APCS_L2_SAW2_SPM_PMIC_DATA_n

Bits	Name	Description
26:24	DLY	PMIC Delay. Controls the time between SPM_SLP_SEQ PMIC command with SAW2_SPM_PMIC_DATA_n and the next SPM_SLP_SEQ command. Time range: 13.33us to 1.7ms
20	SIZE	PMIC DATA size 0x0: ENUM_ONE_BYTE 0x1: ENUM_TWO_BYTE
18:16	ADR_IDX	PMIC Arbiter Address Index. Drive the saw_pmic_addr_idx output port.
7:0	DATA	PMIC DATA sent to the PMIC Arbiter

8.14 APCS_BANKED_SAW2

0x0B009000 APCS_BANKED_SAW2_SECURE**Type:** RW**Clock:** sys_ref_clk**Reset State:** 0x00000000**Security Treatment:** Restricted

The SAW2_SECURE register is used to enable or disable non-secure bus transactions. The bus **security** status is signaled by ahb_saw_hprotns pin. When ahb_saw_hprotns is set to '0' the state of the SAW2_SECURE register is not considered - a secure transaction is always allowed. When ahb_saw_hprotns is set to '1', the SAW2_SECURE register **security** treatment bit of that register must be '1' for access to be granted. If the **security** treatment bit of the register is set to '0', the non-secure (ahb_saw_hprotns = '1') transactions are denied. When access is denied, a write is ignored, and a read will return '0'. No error is signaled.

When reading the SAW2_SECURE register, non secure agents will always read '0'. Secure agents will see the actual value of the register. The reset value is controlled by CFGNSINIT input pin.

APCS_BANKED_SAW2_SECURE

Bits	Name	Description
2	SAW_CTL	0x1: NSEC (NSEC) 0x0: SEC (SEC) Controls security treatment for SAW2 registers: SAW2_ID, SAW2_CFG, SAW2_STS_0, SAW2_STS_1, SAW2_RST, SAW2_VERSION)

APCS_BANKED_SAW2_SECURE (cont.)

Bits	Name	Description
1	PWR_CTL	0x1: NSEC (NSEC) 0x0: SEC (SEC) Controls security treatment for SPM registers: SAW2_SPM_CTL, SAW2_SPM_PMIC_DLY, SAW2_SPM_PMIC_DATA_0, SAW2_SPM_PMIC_DATA_1, SAW2_SPM_SLP_SEQ_ENTRY_n)
0	VLT_CTL	0x1: NSEC (NSEC) 0x0: SEC (SEC)

0x0B009004 APCS_BANKED_SAW2_ID**Type:** R**Clock:** sys_ref_clk**Reset State:** 0x120F0861**Security Treatment:** Controlled by SAW2_SECURE [SAW_CTL].

This read only register reports the revision and parameter information for the SAW2 core.

APCS_BANKED_SAW2_ID

Bits	Name	Description
31:24	NUM_SPM_ENTRY	SAW2 parameter: Indicates number of SAW2_SPM_SLP_SEQ_ENTRY register implemented. Value can range from 1 - 128
21:16	NUM_PWR_CTL	SAW2 parameter: Indicates number of power control implemented. Value can range from 4 - 32
12:8	PMIC_DATA_WIDTH	SAW2 parameter: Indicate PMIC data bus width Values can range from 8-16
6:4	NUM_PMIC_DATA	SAW2 parameter: Indicates number of SAW2_SPM_PMIC_DATA_n register implemented. Value can range from 1-7
3	CFG_NS_ACCESS	SAW2 parameter: Reset value of SAW2_SECURE register
2	PMIC_ARB_INTF	SAW2 parameter: Indicates PMIC Arbiter Interface function is implemented
1	RESERVED	Reserved
0	SPM_PRESENT	SAW2 parameter: Indicates SPM function is implemented

0x0B009008 APCS_BANKED_SAW2_CFG**Type:** RW**Clock:** sys_ref_clk**Reset State:** 0x00000000**Security Treatment:** Controlled by SAW2_SECURE [SAW_CTL].**APCS_BANKED_SAW2_CFG**

Bits	Name	Description
9	RESERVED	Reserved
8	FRC_REF_CLK_ON	Chicken bit to force saw_sys_ref_clk_on_req ON.
4:0	CLK_DIV	Divider ratio for clock. This is used to generate timer tick for the timer. Timer tick is asserted every (CLK_DIV + 1) sys_ref_clk period.

0x0B00900C APCS_BANKED_SAW2_SPM_STS**Type:** R**Clock:** sys_ref_clk**Reset State:** 0x00000000**Security Treatment:** Controlled by SAW2_SECURE [SAW_CTL].

This read only register provides SW with SAW2 status.

APCS_BANKED_SAW2_SPM_STS

Bits	Name	Description
31:16	CURR_PWR_CTL	Power Control Outputs.
15	SHTDWN_REQ	This bit reflects the shutdown request from the SPM(spm_rpm_shutdown_req) to RPM.
14	SHTDWN_ACK	This bit reflects the shutdown acknowledgement from the RPM(rpm_spm_shutdown_ack) to SPM.
13	BRNGUP_REQ	This bit reflects the bringup request from the SPM(spm_rpm_bringup_req) to RPM.
12	BRNGUP_ACK	This bit reflects the bringup acknowledgement from the RPM(rpm_spm_bringup_ack) to SPM.
11:10	RPM_STATE	State of the RPM FSM: 0x0: RUN (RUN(waiting for SPM request)) 0x1: STDNACK (STDNACK (waiting for shutdown ACK from RPM)) 0x2: BGUPACK (BGUPACK (waiting for bringup ACK from RPM)) 0x3: WAKEUP (WAKEUP (waiting for wakeup interrupt))
8:0	SPM_CMD_ADDR	Last SPM command executed.

0x0B009010 RESERVED**0x0B009014 APCS_BANKED_SAW2_PMIC_STS****Type:** R**Clock:** sys_ref_clk**Reset State:** 0x00000000**Security Treatment:** Controlled by SAW2_SECURE [SAW_CTL].

This read only register provides SW with SAW2 status.

APCS_BANKED_SAW2_PMIC_STS

Bits	Name	Description
29:20	RESERVED	Reserved
18	CURR_PMIC_SIZE	Last PMIC VLVL Size sent to the PMIC Arbiter (and the PMIC).
17:16	PMIC_STATE	State of the PMIC FSM: 0x0: IDLE (IDLE) 0x1: DONE (DONE (waiting for DONE form PMIC Arb before transitions back to IDLE)) 0x2: DELAY (DELAY (waiting for delay count termination before transitions back to IDLE)) 0x3: DONEBAR (DONEBAR (waiting for DONE to deassert before another pending request is processed))
7:0	CURR_PMIC_DATA	Last PMIC DATA sent to the PMIC Arbiter (and the PMIC)

0x0B009018 APCS_BANKED_SAW2_RST**Type:** RW**Clock:** sys_ref_clk**Reset State:** 0x00000000**Security Treatment:** Controlled by SAW2_SECURE [SAW_CTL].

The SAW2_RST register is used to reset the SAW logic. This register clear itself. It does not reset any SAW2 CSR's.

APCS_BANKED_SAW2_RST

Bits	Name	Description
0	RESERVED	Reserved

0x0B009030 APCS_BANKED_SAW2_SPM_CTL**Type:** RW**Clock:** sys_ref_clk**Reset State:** 0x00000000**Security Treatment:** Controlled by SAW2_SECURE [PWR_CTL].

The SAW2_SPM_CTL register is used to control the subsystem power management system. This are parameters that controls the operation of SPM FSM.

APCS_BANKED_SAW2_SPM_CTL

Bits	Name	Description
31:24	EVENT_SYNC	SPM Event input configuration. There is an individual bit for each sys_spm_event input
17	SLP_CMD_MODE	Sleep Command M0de
16	SPM_SYS_PC_MODE	Indication that we're in power collapse mode
12:4	SPM_START_ADR	Start address for the SPM sequence.
3	ISAR	Inhibit Start Address Reset 0x0: END_OF_PROGRAM_RESET_THE_SPM_START_ADR_TO_ZERO (End of program reset the SPM_START_ADR to zero.) 0x1: INHIBIT_END_OF_PROGRAM_TO_RESET_SPM_START_ADR (Inhibit End of program to reset SPM_START_ADR)
2:1	WAKEUP_CONFIG	Wakeup Configuration 0x0: SYS_SPM_WAKEUP (sys_spm_wakeup) 0x1: SYS_SPM_WAKEUP_OR_NOT_CPU_SPM_WAIT_REQ (sys_spm_wakeup or !cpu_spm_wait_req) 0x2: SYS_SPM_WAKEUP_OR_RISING_EDGE_OF_SYS_SPM_DBG_NOPWRDWN (sys_spm_wakeup or rising edge of sys_spm_dbg_nopwrdsn) 0x3: SYS_SPM_WAKEUP_OR_NOT_CPU_SPM_WAIT_REQ_OR_RISING_EDGE_OF_SYS_SPM_DBG_NOPWRDWN (sys_spm_wakeup or !cpu_spm_wait_req or rising edge of sys_spm_dbg_nopwrdsn)
0	SPM_EN	SPM En.

0x0B009034 APCS_BANKED_SAW2_SPM_DLY**Type:** RW**Clock:** sys_ref_clk**Reset State:** 0x00000000**Security Treatment:** Controlled by SAW2_SECURE [PWR_CTL].

SPM Delay. Controls the time between SPM_SLP_SEQ SPM Delay and the next SPM_SLP_SEQ command.

APCS_BANKED_SAW2_SPM_DLY

Bits	Name	Description
29:20	DLY3	SPM Delay. Controls the time between SPM_SLP_SEQ SPM Delay and the next SPM_SLP_SEQ command. Time range: 52ns to 1.7ms
19:10	DLY2	SPM Delay. Controls the time between SPM_SLP_SEQ SPM Delay and the next SPM_SLP_SEQ command. Time range: 52ns to 1.7ms
9:0	DLY1	SPM Delay. Controls the time between SPM_SLP_SEQ SPM Delay and the next SPM_SLP_SEQ command. Time range: 52ns to 1.7ms

0x0B009038 APCS_BANKED_SAW2_SPM_STS2

Type: R

Clock: sys_ref_clk

Reset State: 0x00000000

Security Treatment: Controlled by SAW2_SECURE [SAW_CTL].

This read only register provides SW with SPM power control status.

APCS_BANKED_SAW2_SPM_STS2

Bits	Name	Description
14:0	CURR_PWR_CTL	Power control Outputs

0x0B009400+ APCS_BANKED_SAW2_SPM_SLP_SEQ_ENTRY_n, n=[0..17] 0x4*n

Type: RW

Clock: sys_ref_clk

Reset State: 0x00000000

Security Treatment: Controlled by SAW2_SECURE [PWR_CTL].

The SAW2_SPM_SLP_SEQ_ENTRY_n is an array used to sequence through the steps during various power mode. The register width is defined by CLOG_PWR_CTL parameter.

APCS_BANKED_SAW2_SPM_SLP_SEQ_ENTRY_n

Bits	Name	Description
29:24	CMD3	SPM PWR CMD3
21:16	CMD2	SPM PWR CMD2

APCS_BANKED_SAW2_SPM_SLP_SEQ_ENTRY_n (cont.)

Bits	Name	Description
13:8	CMD1	SPM PWR CMD1
5:0	CMD0	SPM PWR CMD0

8.15 APCS_ALIAS0_SAW2

0x0B089000 APCS_ALIAS0_SAW2_SECURE**Type:** RW**Clock:** sys_ref_clk**Reset State:** 0x00000000**Security Treatment:** Restricted

The SAW2_SECURE register is used to enable or disable non-secure bus transactions. The bus **security** status is signaled by ahb_saw_hprotns pin. When ahb_saw_hprotns is set to '0' the state of the SAW2_SECURE register is not considered - a secure transaction is always allowed. When ahb_saw_hprotns is set to '1', the SAW2_SECURE register **security** treatment bit of that register must be '1' for access to be granted. If the **security** treatment bit of the register is set to '0', the non-secure (ahb_saw_hprotns = '1') transactions are denied. When access is denied, a write is ignored, and a read will return '0'. No error is signaled.

When reading the SAW2_SECURE register, non secure agents will always read '0'. Secure agents will see the actual value of the register. The reset value is controlled by CFGNSINIT input pin.

APCS_ALIAS0_SAW2_SECURE

Bits	Name	Description
2	SAW_CTL	0x1: NSEC (NSEC) 0x0: SEC (SEC) Controls security treatment for SAW2 registers: SAW2_ID, SAW2_CFG, SAW2_STS_0, SAW2_STS_1, SAW2_RST, SAW2_VERSION)
1	PWR_CTL	0x1: NSEC (NSEC) 0x0: SEC (SEC) Controls security treatment for SPM registers: SAW2_SPM_CTL, SAW2_SPM_PMIC_DLY, SAW2_SPM_PMIC_DATA_0, SAW2_SPM_PMIC_DATA_1, SAW2_SPM_SLP_SEQ_ENTRY_n)
0	VLT_CTL	0x1: NSEC (NSEC) 0x0: SEC (SEC)

0x0B089004 APCS_ALIAS0_SAW2_ID**Type:** R**Clock:** sys_ref_clk**Reset State:** 0x120F0861**Security Treatment:** Controlled by SAW2_SECURE [SAW_CTL].

This read only register reports the revision and parameter information for the SAW2 core.

APCS_ALIAS0_SAW2_ID

Bits	Name	Description
31:24	NUM_SPM_ENTRY	SAW2 parameter: Indicates number of SAW2_SPM_SLP_SEQ_ENTRY register implemented. Value can range from 1 - 128
21:16	NUM_PWR_CTL	SAW2 parameter: Indicates number of power control implemented. Value can range from 4 - 32
12:8	PMIC_DATA_WIDTH	SAW2 parameter: Indicate PMIC data bus width Values can range from 8-16
6:4	NUM_PMIC_DATA	SAW2 parameter: Indicates number of SAW2_SPM_PMIC_DATA_n register implemented. Value can range from 1-7
3	CFG_NS_ACCESS	SAW2 parameter: Reset value of SAW2_SECURE register
2	PMIC_ARB_INTF	SAW2 parameter: Indicates PMIC Arbiter Interface function is implemented
1	RESERVED	Reserved
0	SPM_PRESENT	SAW2 parameter: Indicates SPM function is implemented

0x0B089008 APCS_ALIAS0_SAW2_CFG**Type:** RW**Clock:** sys_ref_clk**Reset State:** 0x00000000**Security Treatment:** Controlled by SAW2_SECURE [SAW_CTL].**APCS_ALIAS0_SAW2_CFG**

Bits	Name	Description
9	RESERVED	Reserved
8	FRC_REF_CLK_ON	Chicken bit to force saw_sys_ref_clk_on_req ON.
4:0	CLK_DIV	Divider ratio for clock. This is used to generate timer tick for the timer. Timer tick is asserted every (CLK_DIV + 1) sys_ref_clk period.

0x0B08900C APCS_ALIAS0_SAW2_SPM_STS**Type:** R**Clock:** sys_ref_clk**Reset State:** 0x00000000**Security Treatment:** Controlled by SAW2_SECURE [SAW_CTL].

This read only register provides SW with SAW2 status.

APCS_ALIAS0_SAW2_SPM_STS

Bits	Name	Description
31:16	CURR_PWR_CTL	Power Control Outputs.
15	SHTDWN_REQ	This bit reflects the shutdown request from the SPM(spm_rpm_shutdown_req) to RPM.
14	SHTDWN_ACK	This bit reflects the shutdown acknowledgement from the RPM(rpm_spm_shutdown_ack) to SPM.
13	BRNGUP_REQ	This bit reflects the bringup request from the SPM(spm_rpm_bringup_req) to RPM.
12	BRNGUP_ACK	This bit reflects the bringup acknowledgement from the RPM(rpm_spm_bringup_ack) to SPM.
11:10	RPM_STATE	State of the RPM FSM: 0x0: RUN (RUN(waiting for SPM request)) 0x1: STDNACK (STDNACK (waiting for shutdown ACK from RPM)) 0x2: BGUPACK (BGUPACK (waiting for bringup ACK from RPM)) 0x3: WAKEUP (WAKEUP (waiting for wakeup interrupt))
8:0	SPM_CMD_ADDR	Last SPM command executed.

0x0B089010 RESERVED**0x0B089014 APCS_ALIAS0_SAW2_PMIC_STS****Type:** R**Clock:** sys_ref_clk**Reset State:** 0x00000000**Security Treatment:** Controlled by SAW2_SECURE [SAW_CTL].

This read only register provides SW with SAW2 status.

APCS_ALIAS0_SAW2_PMIC_STS

Bits	Name	Description
29:20	RESERVED	Reserved

APCS_ALIAS0_SAW2_PMIC_STS (cont.)

Bits	Name	Description
18	CURR_PMIC_SIZE	Last PMIC VLVL Size sent to the PMIC Arbiter (and the PMIC).
17:16	PMIC_STATE	State of the PMIC FSM: 0x0: IDLE (IDLE) 0x1: DONE (DONE (waiting for DONE form PMIC Arb before transitions back to IDLE)) 0x2: DELAY (DELAY (waiting for delay count termination before transitions back to IDLE)) 0x3: DONEBAR (DONEBAR (waiting for DONE to deassert before another pending request is processed))
7:0	CURR_PMIC_DATA	Last PMIC DATA sent to the PMIC Arbiter (and the PMIC)

0x0B089018 APCS_ALIAS0_SAW2_RST**Type:** RW**Clock:** sys_ref_clk**Reset State:** 0x00000000**Security Treatment:** Controlled by SAW2_SECURE [SAW_CTL].

The SAW2_RST register is used to reset the SAW logic. This register clear itself. It does not reset any SAW2 CSR's.

APCS_ALIAS0_SAW2_RST

Bits	Name	Description
0	RST	Reset SPM FSM and control registers.

0x0B089030 APCS_ALIAS0_SAW2_SPM_CTL**Type:** RW**Clock:** sys_ref_clk**Reset State:** 0x00000000**Security Treatment:** Controlled by SAW2_SECURE [PWR_CTL].

The SAW2_SPM_CTL register is used to control the subsystem power management system. This are parameters that controls the operation of SPM FSM.

APCS_ALIAS0_SAW2_SPM_CTL

Bits	Name	Description
31:24	EVENT_SYNC	SPM Event input configuration. There is an individual bit for each sys_spm_event input
17	SLP_CMD_MODE	Sleep Command M0de

APCS_ALIAS0_SAW2_SPM_CTL (cont.)

Bits	Name	Description
16	SPM_SYS_PC_MODE	Indication that we're in power collapse mode
12:4	SPM_START_ADR	Start address for the SPM sequence.
3	ISAR	Inhibit Start Address Reset 0x0: END_OF_PROGRAM_RESET_THE_SPM_START_ADR_TO_ZERO (End of program reset the SPM_START_ADR to zero.) 0x1: INHIBIT_END_OF_PROGRAM_TO_RESET_SPM_START_ADR (Inhibit End of program to reset SPM_START_ADR)
2:1	WAKEUP_CONFIG	Wakeup Configuration 0x0: SYS_SPM_WAKEUP (sys_spm_wakeup) 0x1: SYS_SPM_WAKEUP_OR_NOT_CPU_SPM_WAIT_REQ (sys_spm_wakeup or !cpu_spm_wait_req) 0x2: SYS_SPM_WAKEUP_OR_RISING_EDGE_OF_SYS_SPM_DBG_NOPWRDWN (sys_spm_wakeup or rising edge of sys_spm_dbg_nopwrdsn) 0x3: SYS_SPM_WAKEUP_OR_NOT_CPU_SPM_WAIT_REQ_OR_RISING_EDGE_OF_SYS_SPM_DBG_NOPWRDWN (sys_spm_wakeup or !cpu_spm_wait_req or rising edge of sys_spm_dbg_nopwrdsn)
0	SPM_EN	SPM En.

0x0B089034 APCS_ALIAS0_SAW2_SPM_DLY**Type:** RW**Clock:** sys_ref_clk**Reset State:** 0x00000000**Security Treatment:** Controlled by SAW2_SECURE [PWR_CTL].

SPM Delay. Controls the time between SPM_SLP_SEQ SPM Delay and the next SPM_SLP_SEQ command.

APCS_ALIAS0_SAW2_SPM_DLY

Bits	Name	Description
29:20	DLY3	SPM Delay. Controls the time between SPM_SLP_SEQ SPM Delay and the next SPM_SLP_SEQ command. Time range: 52ns to 1.7ms
19:10	DLY2	SPM Delay. Controls the time between SPM_SLP_SEQ SPM Delay and the next SPM_SLP_SEQ command. Time range: 52ns to 1.7ms

APCS_ALIAS0_SAW2_SPM_DLY (cont.)

Bits	Name	Description
9:0	DLY1	SPM Delay. Controls the time between SPM_SLP_SEQ SPM Delay and the next SPM_SLP_SEQ command. Time range: 52ns to 1.7ms

0x0B089038 APCS_ALIAS0_SAW2_SPM_STS2**Type:** R**Clock:** sys_ref_clk**Reset State:** 0x00000000**Security Treatment:** Controlled by SAW2_SECURE [SAW_CTL].

This read only register provides SW with SPM power control status.

APCS_ALIAS0_SAW2_SPM_STS2

Bits	Name	Description
14:0	CURR_PWR_CTL	Power control Outputs

**0x0B089400+ APCS_ALIAS0_SAW2_SPM_SLP_SEQ_ENTRY_n, n=[0..17]
0x4*n****Type:** RW**Clock:** sys_ref_clk**Reset State:** 0x00000000**Security Treatment:** Controlled by SAW2_SECURE [PWR_CTL].

The SAW2_SPM_SLP_SEQ_ENTRY_n is an array used to sequence through the steps during various power mode. The register width is defined by CLOG_PWR_CTL parameter.

APCS_ALIAS0_SAW2_SPM_SLP_SEQ_ENTRY_n

Bits	Name	Description
29:24	CMD3	SPM PWR CMD3
21:16	CMD2	SPM PWR CMD2
13:8	CMD1	SPM PWR CMD1
5:0	CMD0	SPM PWR CMD0

8.16 APCS_ALIAS1_SAW2

0x0B099000 APCS_ALIAS1_SAW2_SECURE

Type: RW

Clock: sys_ref_clk

Reset State: 0x00000000

Security Treatment: Restricted

The SAW2_SECURE register is used to enable or disable non-secure bus transactions. The bus **security** status is signaled by ahb_saw_hprotns pin. When ahb_saw_hprotns is set to '0' the state of the SAW2_SECURE register is not considered - a secure transaction is always allowed. When ahb_saw_hprotns is set to '1', the SAW2_SECURE register **security** treatment bit of that register must be '1' for access to be granted. If the **security** treatment bit of the register is set to '0', the non-secure (ahb_saw_hprotns = '1') transactions are denied. When access is denied, a write is ignored, and a read will return '0'. No error is signaled.

When reading the SAW2_SECURE register, non secure agents will always read '0'. Secure agents will see the actual value of the register. The reset value is controlled by CFGNSINIT input pin.

APCS_ALIAS1_SAW2_SECURE

Bits	Name	Description
2	SAW_CTL	0x1: NSEC (NSEC) 0x0: SEC (SEC) Controls security treatment for SAW2 registers: SAW2_ID, SAW2_CFG, SAW2_STS_0, SAW2_STS_1, SAW2_RST, SAW2_VERSION)
1	PWR_CTL	0x1: NSEC (NSEC) 0x0: SEC (SEC) Controls security treatment for SPM registers: SAW2_SPM_CTL, SAW2_SPM_PMIC_DLY, SAW2_SPM_PMIC_DATA_0, SAW2_SPM_PMIC_DATA_1, SAW2_SPM_SLP_SEQ_ENTRY_n)
0	VLT_CTL	0x1: NSEC (NSEC) 0x0: SEC (SEC)

0x0B099004 APCS_ALIAS1_SAW2_ID

Type: R

Clock: sys_ref_clk

Reset State: 0x120F0861

Security Treatment: Controlled by SAW2_SECURE [SAW_CTL].

This read only register reports the revision and parameter information for the SAW2 core.

APCS_ALIAS1_SAW2_ID

Bits	Name	Description
31:24	NUM_SPM_ENTRY	SAW2 parameter: Indicates number of SAW2_SPM_SLP_SEQ_ENTRY register implemented. Value can range from 1 - 128
21:16	NUM_PWR_CTL	SAW2 parameter: Indicates number of power control implemented. Value can range from 4 - 32
12:8	PMIC_DATA_WIDTH	SAW2 parameter: Indicate PMIC data bus width Values can range from 8-16
6:4	NUM_PMIC_DATA	SAW2 parameter: Indicates number of SAW2_SPM_PMIC_DATA_n register implemented. Value can range from 1-7
3	CFG_NS_ACCESS	SAW2 parameter: Reset value of SAW2_SECURE register
2	PMIC_ARB_INTF	SAW2 parameter: Indicates PMIC Arbiter Interface function is implemented
1	RESERVED	Reserved
0	SPM_PRESENT	SAW2 parameter: Indicates SPM function is implemented

0x0B099008 APCS_ALIAS1_SAW2_CFG**Type:** RW**Clock:** sys_ref_clk**Reset State:** 0x00000000**Security Treatment:** Controlled by SAW2_SECURE [SAW_CTL].**APCS_ALIAS1_SAW2_CFG**

Bits	Name	Description
9	RESERVED	Reserved
8	FRC_REF_CLK_ON	Chicken bit to force saw_sys_ref_clk_on_req ON.
4:0	CLK_DIV	Divider ratio for clock. This is used to generate timer tick for the timer. Timer tick is asserted every (CLK_DIV + 1) sys_ref_clk period.

0x0B09900C APCS_ALIAS1_SAW2_SPM_STS**Type:** R**Clock:** sys_ref_clk**Reset State:** 0x00000000**Security Treatment:** Controlled by SAW2_SECURE [SAW_CTL].

This read only register provides SW with SAW2 status.

APCS_ALIAS1_SAW2_SPM_STS

Bits	Name	Description
31:16	CURR_PWR_CTL	Power Control Outputs.
15	SHTDWN_REQ	This bit reflects the shutdown request from the SPM(spm_rpm_shutdown_req) to RPM.
14	SHTDWN_ACK	This bit reflects the shutdown acknowledgement from the RPM(rpm_spm_shutdown_ack) to SPM.
13	BRNGUP_REQ	This bit reflects the bringup request from the SPM(spm_rpm_bringup_req) to RPM.
12	BRNGUP_ACK	This bit reflects the bringup acknowledgement from the RPM(rpm_spm_bringup_ack) to SPM.
11:10	RPM_STATE	State of the RPM FSM: 0x0: RUN (RUN(waiting for SPM request)) 0x1: STDNACK (STDNACK (waiting for shutdown ACK from RPM)) 0x2: BGUPACK (BGUPACK (waiting for bringup ACK from RPM)) 0x3: WAKEUP (WAKEUP (waiting for wakeup interrupt))
8:0	SPM_CMD_ADDR	Last SPM command executed.

0x0B099010 RESERVED**0x0B099014 APCS_ALIAS1_SAW2_PMIC_STS****Type:** R**Clock:** sys_ref_clk**Reset State:** 0x00000000**Security Treatment:** Controlled by SAW2_SECURE [SAW_CTL].

This read only register provides SW with SAW2 status.

APCS_ALIAS1_SAW2_PMIC_STS

Bits	Name	Description
29:20	RESERVED	Reserved
18	CURR_PMIC_SIZE	Last PMIC VLVL Size sent to the PMIC Arbiter (and the PMIC).
17:16	PMIC_STATE	State of the PMIC FSM: 0x0: IDLE (Reserved) 0x1: DONE (DONE (waiting for DONE form PMIC Arb before transitions back to IDLE)) 0x2: DELAY (DELAY (waiting for delay count termination before transitions back to IDLE)) 0x3: DONEBAR (DONEBAR (waiting for DONE to deassert before another pending request is processed))

APCS_ALIAS1_SAW2_PMIC_STS (cont.)

Bits	Name	Description
7:0	CURR_PMIC_DATA	Last PMIC DATA sent to the PMIC Arbiter (and the PMIC)

0x0B099018 APCS_ALIAS1_SAW2_RST**Type:** RW**Clock:** sys_ref_clk**Reset State:** 0x00000000**Security Treatment:** Controlled by SAW2_SECURE [SAW_CTL].

The SAW2_RST register is used to reset the SAW logic. This register clear itself. It does not reset any SAW2 CSR's.

APCS_ALIAS1_SAW2_RST

Bits	Name	Description
0	RST	Reset SPM FSM and control registers.

0x0B099030 APCS_ALIAS1_SAW2_SPM_CTL**Type:** RW**Clock:** sys_ref_clk**Reset State:** 0x00000000**Security Treatment:** Controlled by SAW2_SECURE [PWR_CTL].

The SAW2_SPM_CTL register is used to control the subsystem power management system. This are parameters that controls the operation of SPM FSM.

APCS_ALIAS1_SAW2_SPM_CTL

Bits	Name	Description
31:24	EVENT_SYNC	SPM Event input configuration. There is an individual bit for each sys_spm_event input
17	SLP_CMD_MODE	Sleep Command M0de
16	SPM_SYS_PC_MODE	Indication that we're in power collapse mode
12:4	SPM_START_ADR	Start address for the SPM sequence.
3	ISAR	Inhibit Start Address Reset 0x0: END_OF_PROGRAM_RESET_THE_SPM_START_ADR_TO_ZERO (End of program reset the SPM_START_ADR to zero.) 0x1: INHIBIT_END_OF_PROGRAM_TO_RESET_SPM_START_ADR (Inhibit End of program to reset SPM_START_ADR)

APCS_ALIAS1_SAW2_SPM_CTL (cont.)

Bits	Name	Description
2:1	WAKEUP_CONFIG	Wakeup Configuration 0x0: SYS_SPM_WAKEUP (sys_spm_wakeup) 0x1: SYS_SPM_WAKEUP_OR_NOT_CPU_SPM_WAIT_REQ (sys_spm_wakeup or !cpu_spm_wait_req) 0x2: SYS_SPM_WAKEUP_OR_RISING_EDGE_OF_SYS_SPM_DBG_NOPWRDWN (sys_spm_wakeup or rising edge of sys_spm_dbg_nopwrdsn) 0x3: SYS_SPM_WAKEUP_OR_NOT_CPU_SPM_WAIT_REQ_OR_RISING_EDGE_OF_SYS_SPM_DBG_NOPWRDWN (sys_spm_wakeup or !cpu_spm_wait_req or rising edge of sys_spm_dbg_nopwrdsn)
0	SPM_EN	SPM En.

0x0B099034 APCS_ALIAS1_SAW2_SPM_DLY**Type:** RW**Clock:** sys_ref_clk**Reset State:** 0x00000000**Security Treatment:** Controlled by SAW2_SECURE [PWR_CTL].

SPM Delay. Controls the time between SPM_SLP_SEQ SPM Delay and the next SPM_SLP_SEQ command.

APCS_ALIAS1_SAW2_SPM_DLY

Bits	Name	Description
29:20	DLY3	SPM Delay. Controls the time between SPM_SLP_SEQ SPM Delay and the next SPM_SLP_SEQ command. Time range: 52ns to 1.7ms
19:10	DLY2	SPM Delay. Controls the time between SPM_SLP_SEQ SPM Delay and the next SPM_SLP_SEQ command. Time range: 52ns to 1.7ms
9:0	DLY1	SPM Delay. Controls the time between SPM_SLP_SEQ SPM Delay and the next SPM_SLP_SEQ command. Time range: 52ns to 1.7ms

0x0B099038 APCS_ALIAS1_SAW2_SPM_STS2**Type:** R**Clock:** sys_ref_clk**Reset State:** 0x00000000**Security Treatment:** Controlled by SAW2_SECURE [SAW_CTL].

This read only register provides SW with SPM power control status.

APCS_ALIAS1_SAW2_SPM_STS2

Bits	Name	Description
14:0	CURR_PWR_CTL	Power control Outputs

0x0B099400+ APCS_ALIAS1_SAW2_SPM_SLP_SEQ_ENTRY_n, n=[0..17] 0x4*n

Type: RW

Clock: sys_ref_clk

Reset State: 0x00000000

Security Treatment: Controlled by SAW2_SECURE [PWR_CTL].

The SAW2_SPM_SLP_SEQ_ENTRY_n is an array used to sequence through the steps during various power mode. The register width is defined by CLOG_PWR_CTL parameter.

APCS_ALIAS1_SAW2_SPM_SLP_SEQ_ENTRY_n

Bits	Name	Description
29:24	CMD3	SPM PWR CMD3
21:16	CMD2	SPM PWR CMD2
13:8	CMD1	SPM PWR CMD1
5:0	CMD0	SPM PWR CMD0

8.17 APCS_ALIAS2_SAW2

0x0B0A9000 APCS_ALIAS2_SAW2_SECURE

Type: RW

Clock: sys_ref_clk

Reset State: 0x00000000

Security Treatment: Restricted

The SAW2_SECURE register is used to enable or disable non-secure bus transactions. The bus **security** status is signaled by ahb_saw_hprotns pin. When ahb_saw_hprotns is set to '0' the state of the SAW2_SECURE register is not considered - a secure transaction is always allowed. When ahb_saw_hprotns is set to '1', the SAW2_SECURE register **security** treatment bit of that register must be '1' for access to be granted. If the **security** treatment bit of the register is set to '0', the non-secure (ahb_saw_hprotns = '1') transactions are denied. When access is denied, a write is ignored, and a read will return '0'. No error is signaled.

When reading the SAW2_SECURE register, non secure agents will always read '0'. Secure agents will see the actual value of the register. The reset value is controlled by CFGNSINIT input pin.

APCS_ALIAS2_SAW2_SECURE

Bits	Name	Description
2	SAW_CTL	0x1: NSEC (NSEC) 0x0: SEC (SEC) Controls security treatment for SAW2 registers: SAW2_ID, SAW2_CFG, SAW2_STS_0, SAW2_STS_1, SAW2_RST, SAW2_VERSION)
1	PWR_CTL	0x1: NSEC (NSEC) 0x0: SEC (SEC) Controls security treatment for SPM registers: SAW2_SPM_CTL, SAW2_SPM_PMIC_DLY, SAW2_SPM_PMIC_DATA_0, SAW2_SPM_PMIC_DATA_1, SAW2_SPM_SLP_SEQ_ENTRY_n)
0	VLT_CTL	0x1: NSEC (NSEC) 0x0: SEC (SEC)

0x0B0A9004 APCS_ALIAS2_SAW2_ID

Type: R

Clock: sys_ref_clk

Reset State: 0x120F0861

Security Treatment: Controlled by SAW2_SECURE [SAW_CTL].

This read only register reports the revision and parameter information for the SAW2 core.

APCS_ALIAS2_SAW2_ID

Bits	Name	Description
31:24	NUM_SPM_ENTRY	SAW2 parameter: Indicates number of SAW2_SPM_SLP_SEQ_ENTRY register implemented. Value can range from 1 - 128
21:16	NUM_PWR_CTL	SAW2 parameter: Indicates number of power control implemented. Value can range from 4 - 32
12:8	PMIC_DATA_WIDTH	SAW2 parameter: Indicate PMIC data bus width Values can range from 8-16
6:4	NUM_PMIC_DATA	SAW2 parameter: Indicates number of SAW2_SPM_PMIC_DATA_n register implemented. Value can range from 1-7
3	CFG_NS_ACCESS	SAW2 parameter: Reset value of SAW2_SECURE register
2	PMIC_ARB_INTF	SAW2 parameter: Indicates PMIC Arbiter Interface function is implemented
1	RESERVED	Reserved

APCS_ALIAS2_SAW2_ID (cont.)

Bits	Name	Description
0	SPM_PRESENT	SAW2 parameter: Indicates SPM function is implemented

0x0B0A9008 APCS_ALIAS2_SAW2_CFG**Type:** RW**Clock:** sys_ref_clk**Reset State:** 0x00000000**Security Treatment:** Controlled by SAW2_SECURE [SAW_CTL].**APCS_ALIAS2_SAW2_CFG**

Bits	Name	Description
9	RESERVED	Reserved
8	FRC_REF_CLK_ON	Chicken bit to force saw_sys_ref_clk_on_req ON.
4:0	CLK_DIV	Divider ratio for clock. This is used to generate timer tick for the timer. Timer tick is asserted every (CLK_DIV + 1) sys_ref_clk period.

0x0B0A900C APCS_ALIAS2_SAW2_SPM_STS**Type:** R**Clock:** sys_ref_clk**Reset State:** 0x00000000**Security Treatment:** Controlled by SAW2_SECURE [SAW_CTL].

This read only register provides SW with SAW2 status.

APCS_ALIAS2_SAW2_SPM_STS

Bits	Name	Description
31:16	CURR_PWR_CTL	Power Control Outputs.
15	SHTDWN_REQ	This bit reflects the shutdown request from the SPM(spm_rpm_shutdown_req) to RPM.
14	SHTDWN_ACK	This bit reflects the shutdown acknowledgement from the RPM(rpm_spm_shutdown_ack) to SPM.
13	BRNGUP_REQ	This bit reflects the bringup request from the SPM(spm_rpm_bringup_req) to RPM.
12	BRNGUP_ACK	This bit reflects the bringup acknowledgement from the RPM(rpm_spm_bringup_ack) to SPM.

APCS_ALIAS2_SAW2_SPM_STS (cont.)

Bits	Name	Description
11:10	RPM_STATE	State of the RPM FSM: 0x0: RUN (RUN(waiting for SPM request)) 0x1: STDNACK (STDNACK (waiting for shutdown ACK from RPM)) 0x2: BGUPACK (BGUPACK (waiting for bringup ACK from RPM)) 0x3: WAKEUP (WAKEUP (waiting for wakeup interrupt))
8:0	SPM_CMD_ADDR	Last SPM command executed.

0x0B0A9010 RESERVED**0x0B0A9014 APCS_ALIAS2_SAW2_PMIC_STS****Type:** R**Clock:** sys_ref_clk**Reset State:** 0x00000000**Security Treatment:** Controlled by SAW2_SECURE [SAW_CTL].

This read only register provides SW with SAW2 status.

APCS_ALIAS2_SAW2_PMIC_STS

Bits	Name	Description
29:20	RESERVED	Reserved
18	CURR_PMIC_SIZE	Last PMIC VLVL Size sent to the PMIC Arbiter (and the PMIC).
17:16	PMIC_STATE	State of the PMIC FSM: 0x0: IDLE (Reserved) 0x1: DONE (DONE (waiting for DONE form PMIC Arb before transitions back to IDLE)) 0x2: DELAY (DELAY (waiting for delay count termination before transitions back to IDLE)) 0x3: DONEBAR (DONEBAR (waiting for DONE to deassert before another pending request is processed))
7:0	CURR_PMIC_DATA	Last PMIC DATA sent to the PMIC Arbiter (and the PMIC)

0x0B0A9018 APCS_ALIAS2_SAW2_RST**Type:** RW**Clock:** sys_ref_clk**Reset State:** 0x00000000**Security Treatment:** Controlled by SAW2_SECURE [SAW_CTL].

The SAW2_RST register is used to reset the SAW logic. This register clear itself. It does not reset any SAW2 CSR's.

APCS_ALIAS2_SAW2_RST

Bits	Name	Description
0	RST	Reset SPM FSM and control registers.

0x0B0A9030 APCS_ALIAS2_SAW2_SPM_CTL

Type: RW

Clock: sys_ref_clk

Reset State: 0x00000000

Security Treatment: Controlled by SAW2_SECURE [PWR_CTL].

The SAW2_SPM_CTL register is used to control the subsystem power management system. This are parameters that controls the operation of SPM FSM.

APCS_ALIAS2_SAW2_SPM_CTL

Bits	Name	Description
31:24	EVENT_SYNC	SPM Event input configuration. There is an individual bit for each sys_spm_event input
17	SLP_CMD_MODE	Sleep Command Mode
16	SPM_SYS_PC_MODE	Indication that we're in power collapse mode
12:4	SPM_START_ADR	Start address for the SPM sequence.
3	ISAR	Inhibit Start Address Reset 0x0: END_OF_PROGRAM_RESET_THE_SPM_START_ADR_TO_ZERO (End of program reset the SPM_START_ADR to zero.) 0x1: INHIBIT_END_OF_PROGRAM_TO_RESET_SPM_START_ADR (Inhibit End of program to reset SPM_START_ADR)
2:1	WAKEUP_CONFIG	Wakeup Configuration 0x0: SYS_SPM_WAKEUP (sys_spm_wakeup) 0x1: SYS_SPM_WAKEUP_OR_NOT_CPU_SPM_WAIT_REQ (sys_spm_wakeup or !cpu_spm_wait_req) 0x2: SYS_SPM_WAKEUP_OR_RISING_EDGE_OF_SYS_SPM_DBG_NOPWRDWN (sys_spm_wakeup or rising edge of sys_spm_dbg_nopwrdsn) 0x3: SYS_SPM_WAKEUP_OR_NOT_CPU_SPM_WAIT_REQ_OR_RISING_EDGE_OF_SYS_SPM_DBG_NOPWRDWN (sys_spm_wakeup or !cpu_spm_wait_req or rising edge of sys_spm_dbg_nopwrdsn)
0	SPM_EN	SPM En.

0x0B0A9034 APCS_ALIAS2_SAW2_SPM_DLY**Type:** RW**Clock:** sys_ref_clk**Reset State:** 0x00000000**Security Treatment:** Controlled by SAW2_SECURE [PWR_CTL].

SPM Delay. Controls the time between SPM_SLP_SEQ SPM Delay and the next SPM_SLP_SEQ command.

APCS_ALIAS2_SAW2_SPM_DLY

Bits	Name	Description
29:20	DLY3	SPM Delay. Controls the time between SPM_SLP_SEQ SPM Delay and the next SPM_SLP_SEQ command. Time range: 52ns to 1.7ms
19:10	DLY2	SPM Delay. Controls the time between SPM_SLP_SEQ SPM Delay and the next SPM_SLP_SEQ command. Time range: 52ns to 1.7ms
9:0	DLY1	SPM Delay. Controls the time between SPM_SLP_SEQ SPM Delay and the next SPM_SLP_SEQ command. Time range: 52ns to 1.7ms

0x0B0A9038 APCS_ALIAS2_SAW2_SPM_STS2**Type:** R**Clock:** sys_ref_clk**Reset State:** 0x00000000**Security Treatment:** Controlled by SAW2_SECURE [SAW_CTL].

This read only register provides SW with SPM power control status.

APCS_ALIAS2_SAW2_SPM_STS2

Bits	Name	Description
14:0	CURR_PWR_CTL	Power control Outputs

**0x0B0A9400+ APCS_ALIAS2_SAW2_SPM_SLP_SEQ_ENTRY_n, n=[0..17]
0x4*n****Type:** RW**Clock:** sys_ref_clk**Reset State:** 0x00000000**Security Treatment:** Controlled by SAW2_SECURE [PWR_CTL].

The SAW2_SPM_SLP_SEQ_ENTRY_n is an array used to sequence through the steps during various power mode. The register width is defined by CLOG_PWR_CTL parameter.

APCS_ALIAS2_SAW2_SPM_SLP_SEQ_ENTRY_n

Bits	Name	Description
29:24	CMD3	SPM PWR CMD3
21:16	CMD2	SPM PWR CMD2
13:8	CMD1	SPM PWR CMD1
5:0	CMD0	SPM PWR CMD0

8.18 APCS_ALIAS3_SAW2

0x0B0B9000 APCS_ALIAS3_SAW2_SECURE**Type:** RW**Clock:** sys_ref_clk**Reset State:** 0x00000000**Security Treatment:** Restricted

The SAW2_SECURE register is used to enable or disable non-secure bus transactions. The bus **security** status is signaled by ahb_saw_hprotns pin. When ahb_saw_hprotns is set to '0' the state of the SAW2_SECURE register is not considered - a secure transaction is always allowed. When ahb_saw_hprotns is set to '1', the SAW2_SECURE register **security** treatment bit of that register must be '1' for access to be granted. If the **security** treatment bit of the register is set to '0', the non-secure (ahb_saw_hprotns = '1') transactions are denied. When access is denied, a write is ignored, and a read will return '0'. No error is signaled.

When reading the SAW2_SECURE register, non secure agents will always read '0'. Secure agents will see the actual value of the register. The reset value is controlled by CFGNSINIT input pin.

APCS_ALIAS3_SAW2_SECURE

Bits	Name	Description
2	SAW_CTL	0x1: NSEC (NSEC) 0x0: SEC (SEC) Controls security treatment for SAW2 registers: SAW2_ID, SAW2_CFG, SAW2_STS_0, SAW2_STS_1, SAW2_RST, SAW2_VERSION)
1	PWR_CTL	0x1: NSEC (NSEC) 0x0: SEC (SEC) Controls security treatment for SPM registers: SAW2_SPM_CTL, SAW2_SPM_PMIC_DLY, SAW2_SPM_PMIC_DATA_0, SAW2_SPM_PMIC_DATA_1, SAW2_SPM_SLP_SEQ_ENTRY_n)
0	VLT_CTL	0x1: NSEC (NSEC) 0x0: SEC (SEC)

0x0B0B9004 APCS_ALIAS3_SAW2_ID**Type:** R**Clock:** sys_ref_clk**Reset State:** 0x120F0861**Security Treatment:** Controlled by SAW2_SECURE [SAW_CTL].

This read only register reports the revision and parameter information for the SAW2 core.

APCS_ALIAS3_SAW2_ID

Bits	Name	Description
31:24	NUM_SPM_ENTRY	SAW2 parameter: Indicates number of SAW2_SPM_SLP_SEQ_ENTRY register implemented. Value can range from 1 - 128
21:16	NUM_PWR_CTL	SAW2 parameter: Indicates number of power control implemented. Value can range from 4 - 32
12:8	PMIC_DATA_WIDTH	SAW2 parameter: Indicate PMIC data bus width Values can range from 8-16
6:4	NUM_PMIC_DATA	SAW2 parameter: Indicates number of SAW2_SPM_PMIC_DATA_n register implemented. Value can range from 1-7
3	CFG_NS_ACCESS	SAW2 parameter: Reset value of SAW2_SECURE register
2	PMIC_ARB_INTF	SAW2 parameter: Indicates PMIC Arbiter Interface function is implemented
1	RESERVED	Reserved
0	SPM_PRESENT	SAW2 parameter: Indicates SPM function is implemented

0x0B0B9008 APCS_ALIAS3_SAW2_CFG**Type:** RW**Clock:** sys_ref_clk**Reset State:** 0x00000000**Security Treatment:** Controlled by SAW2_SECURE [SAW_CTL].**APCS_ALIAS3_SAW2_CFG**

Bits	Name	Description
9	RESERVED	Reserved
8	FRC_REF_CLK_ON	Chicken bit to force saw_sys_ref_clk_on_req ON.
4:0	CLK_DIV	Divider ratio for clock. This is used to generate timer tick for the timer. Timer tick is asserted every (CLK_DIV + 1) sys_ref_clk period.

0x0B0B900C APCS_ALIAS3_SAW2_SPM_STS**Type:** R**Clock:** sys_ref_clk**Reset State:** 0x00000000**Security Treatment:** Controlled by SAW2_SECURE [SAW_CTL].

This read only register provides SW with SAW2 status.

APCS_ALIAS3_SAW2_SPM_STS

Bits	Name	Description
31:16	CURR_PWR_CTL	Power Control Outputs.
15	SHTDWN_REQ	This bit reflects the shutdown request from the SPM(spm_rpm_shutdown_req) to RPM.
14	SHTDWN_ACK	This bit reflects the shutdown acknowledgement from the RPM(rpm_spm_shutdown_ack) to SPM.
13	BRNGUP_REQ	This bit reflects the bringup request from the SPM(spm_rpm_bringup_req) to RPM.
12	BRNGUP_ACK	This bit reflects the bringup acknowledgement from the RPM(rpm_spm_bringup_ack) to SPM.
11:10	RPM_STATE	State of the RPM FSM: 0x0: RUN (RUN(waiting for SPM request)) 0x1: STDNACK (STDNACK (waiting for shutdown ACK from RPM)) 0x2: BGUPACK (BGUPACK (waiting for bringup ACK from RPM)) 0x3: WAKEUP (WAKEUP (waiting for wakeup interrupt))
8:0	SPM_CMD_ADDR	Last SPM command executed.

0x0B0B9010 RESERVED**0x0B0B9014 APCS_ALIAS3_SAW2_PMIC_STS****Type:** R**Clock:** sys_ref_clk**Reset State:** 0x00000000**Security Treatment:** Controlled by SAW2_SECURE [SAW_CTL].

This read only register provides SW with SAW2 status.

APCS_ALIAS3_SAW2_PMIC_STS

Bits	Name	Description
29:20	RESERVED	Reserved

APCS_ALIAS3_SAW2_PMIC_STS (cont.)

Bits	Name	Description
18	CURR_PMIC_SIZE	Last PMIC VLVL Size sent to the PMIC Arbiter (and the PMIC).
17:16	PMIC_STATE	State of the PMIC FSM: 0x0: IDLE (IDLE (Reserved)) 0x1: DONE (DONE (waiting for DONE form PMIC Arb before transitions back to IDLE)) 0x2: DELAY (DELAY (waiting for delay count termination before transitions back to IDLE)) 0x3: DONEBAR (DONEBAR (waiting for DONE to deassert before another pending request is processed))
7:0	CURR_PMIC_DATA	Last PMIC DATA sent to the PMIC Arbiter (and the PMIC)

0x0B0B9018 APCS_ALIAS3_SAW2_RST**Type:** RW**Clock:** sys_ref_clk**Reset State:** 0x00000000**Security Treatment:** Controlled by SAW2_SECURE [SAW_CTL].

The SAW2_RST register is used to reset the SAW logic. This register clear itself. It does not reset any SAW2 CSR's.

APCS_ALIAS3_SAW2_RST

Bits	Name	Description
0	RST	Reset SPM FSM and control registers.

0x0B0B9030 APCS_ALIAS3_SAW2_SPM_CTL**Type:** RW**Clock:** sys_ref_clk**Reset State:** 0x00000000**Security Treatment:** Controlled by SAW2_SECURE [PWR_CTL].

The SAW2_SPM_CTL register is used to control the subsystem power management system. This are parameters that controls the operation of SPM FSM.

APCS_ALIAS3_SAW2_SPM_CTL

Bits	Name	Description
31:24	EVENT_SYNC	SPM Event input configuration. There is an individual bit for each sys_spm_event input
17	SLP_CMD_MODE	Sleep Command M0de

APCS_ALIAS3_SAW2_SPM_CTL (cont.)

Bits	Name	Description
16	SPM_SYS_PC_MODE	Indication that we're in power collapse mode
12:4	SPM_START_ADR	Start address for the SPM sequence.
3	ISAR	Inhibit Start Address Reset 0x0: END_OF_PROGRAM_RESET_THE_SPM_START_ADR_TO_ZERO (End of program reset the SPM_START_ADR to zero.) 0x1: INHIBIT_END_OF_PROGRAM_TO_RESET_SPM_START_ADR (Inhibit End of program to reset SPM_START_ADR)
2:1	WAKEUP_CONFIG	Wakeup Configuration 0x0: SYS_SPM_WAKEUP (sys_spm_wakeup) 0x1: SYS_SPM_WAKEUP_OR_NOT_CPU_SPM_WAIT_REQ (sys_spm_wakeup or !cpu_spm_wait_req) 0x2: SYS_SPM_WAKEUP_OR_RISING_EDGE_OF_SYS_SPM_DBG_NOPWRDWN (sys_spm_wakeup or rising edge of sys_spm_dbg_nopwrdsn) 0x3: SYS_SPM_WAKEUP_OR_NOT_CPU_SPM_WAIT_REQ_OR_RISING_EDGE_OF_SYS_SPM_DBG_NOPWRDWN (sys_spm_wakeup or !cpu_spm_wait_req or rising edge of sys_spm_dbg_nopwrdsn)
0	SPM_EN	SPM En.

0x0B0B9034 APCS_ALIAS3_SAW2_SPM_DLY**Type:** RW**Clock:** sys_ref_clk**Reset State:** 0x00000000**Security Treatment:** Controlled by SAW2_SECURE [PWR_CTL].

SPM Delay. Controls the time between SPM_SLP_SEQ SPM Delay and the next SPM_SLP_SEQ command.

APCS_ALIAS3_SAW2_SPM_DLY

Bits	Name	Description
29:20	DLY3	SPM Delay. Controls the time between SPM_SLP_SEQ SPM Delay and the next SPM_SLP_SEQ command. Time range: 52ns to 1.7ms
19:10	DLY2	SPM Delay. Controls the time between SPM_SLP_SEQ SPM Delay and the next SPM_SLP_SEQ command. Time range: 52ns to 1.7ms

APCS_ALIAS3_SAW2_SPM_DLY (cont.)

Bits	Name	Description
9:0	DLY1	SPM Delay. Controls the time between SPM_SLP_SEQ SPM Delay and the next SPM_SLP_SEQ command. Time range: 52ns to 1.7ms

0x0B0B9038 APCS_ALIAS3_SAW2_SPM_STS2**Type:** R**Clock:** sys_ref_clk**Reset State:** 0x00000000**Security Treatment:** Controlled by SAW2_SECURE [SAW_CTL].

This read only register provides SW with SPM power control status.

APCS_ALIAS3_SAW2_SPM_STS2

Bits	Name	Description
14:0	CURR_PWR_CTL	Power control Outputs

**0x0B0B9400+ APCS_ALIAS3_SAW2_SPM_SLP_SEQ_ENTRY_n, n=[0..17]
0x4*n****Type:** RW**Clock:** sys_ref_clk**Reset State:** 0x00000000**Security Treatment:** Controlled by SAW2_SECURE [PWR_CTL].

The SAW2_SPM_SLP_SEQ_ENTRY_n is an array used to sequence through the steps during various power mode. The register width is defined by CLOG_PWR_CTL parameter.

APCS_ALIAS3_SAW2_SPM_SLP_SEQ_ENTRY_n

Bits	Name	Description
29:24	CMD3	SPM PWR CMD3
21:16	CMD2	SPM PWR CMD2
13:8	CMD1	SPM PWR CMD1
5:0	CMD0	SPM PWR CMD0

8.19 APCS_KPSS_GLB

0x0B011000 APCS_GLB_SECURE

Type: RW

Clock: SYS_AHB_CLK

Reset State: 0x0000001F

Security Treatment: Restricted

The GLB_SECURE register is used to enable or disable non-secure bus transactions. The bus **security** status is signaled by A53SS's APROTNS pin. When APROTNS is set to '0' the state of the GLB_SECURE register is not considered - a secure transaction is always allowed. When APROTNS is set to '1', the GLB_SECURE register's SECURE bit must be '1' for access to be granted. If the SECURE bit is set to '0', the non-secure (APROTNS = '1') transactions are denied. When access is denied, a write is ignored, and a read will return '0'. No error is signaled.

When reading the GLB_SECURE register, non secure agents will always read '0'. Secure agents will see the actual value of the register. The same is true for the registers or bits each bit in the GLB_SECURE register correspond to.

APCS_GLB_SECURE

Bits	Name	Description
4	VOTE	Secure bit for SPM Voting by HLOS 0x1: NSEC (NSEC) 0x0: SEC (SEC) Controls security treatment for SPM Vote registers: L2_SPM_VOTE_HLOS)
3	ACP	Secure bit for ACP related CSRs 0x1: NSEC (NSEC) 0x0: SEC (SEC) Controls security treatment for ACP related registers: AINACTS_CNT, MISC_PWR_CTL, MISC_PWR_STS)
2	RESERVED	Reserved
1	PWR	0x1: NSEC (NSEC) 0x0: SEC (SEC) Controls security treatment for the global sleep control registers: L2_PWR_CTL, L2_PWR_STATUS, DBG_PWR_CTL)
0	STS	0x1: NSEC (NSEC) 0x0: SEC (SEC) Controls security treatment for the global clock control registers: PWR_CTL_OVERRIDE, GLB_CLK_STATUS, APCS_L2_AUX_CLK_SEL)

0x0B011004 APCS_TZ_IPC_INTERRUPT**Type:** W**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** Restricted

The TZ_IPC_INTERRUPT register is used by trustzone software to send Inter-process communication interrupt to RPM

APCS_TZ_IPC_INTERRUPT

Bits	Name	Description
23:20	TZ_SPARE_IPC	TZ Spare IPC Interrupt
2:0	RPM_TZ_IPC	TZ Interrupt to RPM

0x0B011008 APCS_IPC_INTERRUPT**Type:** W**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** None

Inter-process communication signals from A53 to other masters.

APCS_IPC_INTERRUPT

Bits	Name	Description
23:20	SPARE_IPC	Spare Interrupt
19:16	WCN_IPC	Interrupt to WConnect (LPASS).
15:12	MSS_IPC	Interrupt to GPS
11:8	ADSP_IPC	Interrupt to ADSP (LPASS).
2:0	RPM_HLOS_IPC	HLOS Interrupt to RPM.

0x0B01100C APCS_PWR_CTL_OVERRIDE**Type:** RW**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by GLB_SECURE [PWR].

The PWR_CTL_OVERRIDE register is used to enable the global APCS clocks. This is used to workaround hardware dynamic clock gating problems.

APCS_PWR_CTL_OVERRIDE

Bits	Name	Description
22	PRESETDBG	When Set, this bit drives PRESETDBGn signal ZERO
21:18	L1_RAM_COLLAPSE_DIS	When Set, this bit will override the L1 RAM collapse operation by SPM. Instead the L1 rams will be retained in the case where APC voltage is retained
17	SCU_RAM_COLLAPSE_DIS	When Set, this bit will override the SCU RAM collapse operation by SPM. Instead the SCU rams will be retained in the case where APC voltage is retained
16	L2_RET_SLP_DIS	When set, this bit will take L2 array out of retention mode. This bit needs to be set before the vdd_mx voltage is lowered. The L2 data will be lost if vdd_mx is lowered to a voltage level such that the internal array voltage is below the retention voltage.
4:0	CLK_EN	Clock enables for global APCS clocks. This is only to workaround hardware dynamic clock gating problems. 0x1: ALWAYS_ENABLED (Always enabled) 0x0: DYNAMICALLY_ENABLED_BY_HARDWARE (dynamically enabled by hardware) [5]= SYS_apcsL2Aux_Clk [4] = SYS_apccL2Pll_Clk [3] = SYS_apcsPiSleep_Clk [2] = SYS_apcsRef_Clk [1] = SYS_apcsAhb_Clk [0] = SYS_apcsQsb_Clk)

0x0B011010 APCS_GLB_CLK_STATUS**Type:** R**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by GLB_SECURE [STS].

The GLB_CLK_STATUS register is used to monitor the output of the clock gating cells.

APCS_GLB_CLK_STATUS

Bits	Name	Description
19:16	QGIC_WAKEUP	Wakeup output of QGIC

APCS_GLB_CLK_STATUS (cont.)

Bits	Name	Description
4:0	STS	Status for global APCS clocks 0x1: TOGGLING (toggling,) 0x0: HALTED (halted [5]= APCS_sysL2AuxClkOnReq [4] = APCS_sysApcsPllClkOnReq [3] = APCS_sysSleepClkOnReq [2] = APCS_sysRefClkOnReq [1] = APCS_sysAhbClkOnReq [0] = APCS_sysQsbClkOnReq)

0x0B011014 APCS_L2_PWR_CTL**Type:** RW**Clock:** SYS_AHB_CLK**Reset State:** 0x0000010F**Security Treatment:** Controlled by GLB_SECURE [PWR].

The L2_PWR_CTL register controls the L2 Global Distributed Head Switch and L2 array head switches.

APCS_L2_PWR_CTL

Bits	Name	Description
28	PMIC_APC_ON	Report that PMIC APC supply is ON
25:16	L2_HS_CNT	Count value for headswitched L2 logic
13	L2_RET_SLP	
12	SYS_RESET	Setting this bit asserts the SYS_apccSys_Areset_N input to the A53SS CPU, causing a warm system reset (all powered non-debug logic will be reset) to the CPU core without resetting subsystem functions such as timers, clock control, etc.
11	L2_SLEEP_STATE	Report L2 logic is in sleep state
10	L2_HS_RST	When set, this bit asserts reset to the headswitched L2 logic
9	L2_HS_EN	When set, this bit closes the switches for the headswitched L2 logic, putting it into an operational state
8	L2_HS_CLAMP	When set, this bit clamps the outputs from the headswitched L2 logic into the unswitched domain This bit is ORed with SAW outputs before passing to the CPU(resets to 1).

APCS_L2_PWR_CTL (cont.)

Bits	Name	Description
3:0	L2_ARRAY_HS	When set, these bits closes the array head switches for 512 KB sections of the L2 cache array. When clear (default reset value), array head switch is open, resulting a lower cache array leakage when data does not need to be retained in the array. This bits is ORed with SAW outputs before passing to the CPU

0x0B011018 APCS_L2_PWR_STATUS**Type:** R**Clock:** SYS_AHB_CLK**Reset State:** 0x03013100**Security Treatment:** Controlled by GLB_SECURE [STS].

The L2_PWR_STATUS register contains the status bits necessary to support an MP with foot switched L2.

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APCS_L2_PWR_STATUS

Bits	Name	Description
28	APC_SUPPLY_ON	This bit reflect if PMIC APC SUPPLY is ON or OFF
27	SAW_SLP_ACK	This bit reflects the status of APCS_apccL2SleepAck signal
26	DBG_PWRUP_REQ	This bit reflects the status of SYS_apcsDbgPwrUpReq
25	L2_CLK_IDLE	This bit reflects the status of the APCC_sysL2ClkIdle signal
24	L2_SLV_IDLE	This bit reflects the status of the APCC_qsbSlvIdle signal
20:16	L2_HW_EVENT	This bit reflects the status of APCS_sysL2HWEvents[4:0] signals
13	DBG_RST_STS	This bit reflects the status of APCS_apccDbg_Areset_N signal
12	L2_HS_RST_STS	This bit reflects the status of APCS_apccL2swvd_Areset_N signal
10	L2_RET_SLP_STS	This bit reflects the status of APCS_apccSwCtlL2RetentionSleep signal
9	L2_HS_STS	Status for headswitched L2 logic
8	L2_CLMP_STS	This bit reflects the status of APCS_apccClampL2swvd signal
3:0	L2_ARRAY_HS_STS	This bit reflects the status of APCS_apccSwCtlL2ArrayCollapse signal

0x0B011030 APCS_IDR**Type:** R**Clock:** SYS_AHB_CLK**Reset State:** 0x00404744**Security Treatment:** Controlled by GLB_SECURE [STS].

This read only register reports the PARAMETER information for the A53SS core.

APCS_IDR

Bits	Name	Description
26:25	CFG_APCPLL	Indicate APC PLL FUSE value
24	QDSSGEN	Indicate the QDSS version configuration value
23:21	CFG_HBID	Indicate the A53SS HBID configuration value
20:16	CFG_HPID	Indicate the A53SS HPID configuration value
15:12	L2_SIZE	Indicate number of L2 Data Array implemented. Each L2 Data Array Size is 128 KB (for 1MB L2) or 64 KB (for 512 KB L2)
11:8	NUM_FRAME	Indicate number of QTMR physical frame implemented
7:4	VARIANT	Reserved for A53SS
3:0	NUM_CPU	Indicates number of Cortex-A53 CPU implemented.

0x0B011034 APCS_XO_CBCR**Type:** RW**Clock:** SYS_AHB_CLK**Reset State:** 0x00000001**Security Treatment:** Controlled by GLB_SECURE [PWR].

This register provides software control for A53SS XO clock gating.

APCS_XO_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	Read Only. This bit indicates if the clock branch is halted..
1	HW_CTL	RW	This bit indicates dynamic clock gating is supported.
0	SW_CTL	RW	Clock enables for A53SS XO clock.

0x0B011038 APCS_SLEEP_CBCR**Type:** RW**Clock:** SYS_AHB_CLK**Reset State:** 0x00000001**Security Treatment:** Controlled by GLB_SECURE [PWR].

This register provides software control for A53SS SLEEP clock gating.

APCS_SLEEP_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	Read Only. This bit indicates if the clock branch is halted..
1	HW_CTL	RW	This bit indicates dynamic clock gating is supported.
0	SW_CTL	RW	Clock enables for A53SS SLEEP clock.

0x0B011050 APCS_CMD_RCGR**Type:** RW**Clock:** SYS_AHB_CLK**Reset State:** 0x80000002**Security Treatment:** Controlled by GLB_SECURE [PWR].

HM_RCG Command Register. It allows for SW to trigger the configuration data

APCS_CMD_RCGR

Bits	Name	Type	Description
31	ROOT_OFF	R	This bitfield provides indication on status of the RCG; Mainly for SW to determine if the generator is actively producing clock 1: Clock Off 0: Clock On; Generator is actively producing the clock
4	DIRTY_CFG_RCGR	R	Indicates change in CFG_RCGR register configuration
1	ROOT_EN	RW	Enable Clock Generator output. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
0	UPDATE	RW	Update to the new programmed configuration. It should be cleared upon switching to new configuration 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x0B011054 APCS_CFG_RCGR**Type:** RW**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by GLB_SECURE [PWR].

HM_RCG Configuration Register. It allows for SW to configure Root Clock Generation Macro

APCS_CFG_RCGR

Bits	Name	Description
10:8	SRC_SEL	This bitfield selects clock source that will be used to source the RCG output 0x0: SRC0 (refRoot_Clk.) 0x1: SRC1 (SYS_apcsAux_Clk_0.) 0x2: SRC2 (SYS_apcsAux_Clk_1.) 0x3: SRC3 (SYS_apcsAux_Clk_2.) 0x4: SRC4 (SYS_apcsAux_Clk_3.) 0x5: SRC5 (APCN_apcPll_Clk.) 0x6: SRC6 (SYS_apcsQsb_Clk.) 0x7: reserved

APCS_CFG_RCGR (cont.)

Bits	Name	Description
4:0	SRC_DIV	This field defines the integer and half integer portion of the divider setting of the RCG. The value specifies is $2N-1$ where N is the divide ratio. For example, if the half integer divide ration is 2.5, the src_div should be set at 00100. 0x0: BYPASS (Bypsss.) 0x1: DIV1 (Divide by 1.) 0x2: DIV1_5 (Divide by 1.5.) 0x3: DIV2 (Divide by 2.) 0x4: DIV2_5 (Divide by 2.5.) 0x5: DIV3 (Divide by 3.) 0x6: DIV3_5 (Divide by 3.5.) 0x7: DIV4 (Divide by 4.) 0x8: DIV4_5 (Divide by 4.5.) 0x9: DIV5 (Divide by 5.) 0xA: DIV5_5 (Divide by 5.5.) 0xB: DIV6 (Divide by 6.) 0xC: DIV6_5 (Divide by 6.5.) 0xD: DIV7 (Divide by 7.) 0xE: DIV7_5 (Divide by 7.5.) 0xF: DIV8 (Divide by 8.) 0x10: DIV8_5 (Divide by 8.5.) 0x11: DIV9 (Divide by 9.) 0x12: DIV9_5 (Divide by 9.5.) 0x13: DIV10 (Divide by 10.) 0x14: DIV10_5 (Divide by 10.5.) 0x15: DIV11 (Divide by 11.) 0x16: DIV11_5 (Divide by 11.5.) 0x17: DIV12 (Divide by 12.) 0x18: DIV12_5 (Divide by 12.5.) 0x19: DIV13 (Divide by 13.) 0x1A: DIV13_5 (Divide by 13.5.) 0x1B: DIV14 (Divide by 14.) 0x1C: DIV14_5 (Divide by 14.5.) 0x1D: DIV15 (Divide by 15.) 0x1E: DIV15_5 (Divide by 15.5.) 0x1F: DIV16 (Divide by 16.)

0x0B011058 APCS_CORE_CBCR**Type:** RW**Clock:** SYS_AHB_CLK**Reset State:** 0x80000000**Security Treatment:** Controlled by GLB_SECURE [PWR].

CBCR Register. It allows SW control of CBC soft macro

APCS_CORE_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	This bitfield indicated if the clock branch is halted 0:Clock On 1:Clock Off
1	HW_CTL	RW	This bit is used to enable hardware dynamic gating of the clock branch. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)
0	CLK_ENABLE	RW	This bitfield indicated if the clock branch is enabled. 0x1: ENABLE (Enable.) 0x0: DISABLE (Disable.)

0x0B01106C APCS_A53_CFG_STS

Type: RW

Clock: SYS_AHB_CLK

Reset State: 0x00000002

Security Treatment: Controlled by GLB_SECURE [PWR].

A53_CFG_STS Register

APCS_A53_CFG_STS

Bits	Name	Description
31:24	CLUSTERIDAFF1	Value read in the Cluster ID Affinity Level 1field, MPIDR bits [15:8], of the CP15 MPDIR register. These pins are only sampled during reset of the processor.
23:16	CLUSTERIDAFF2	Value read in the Cluster ID Affinity Level 2field, MPIDR bits [23:16], of the CP15 MPDIR register. These pins are only sampled during reset of the processor.
1	ACINACTM	
0	CFGSDISABLE	

0x0B011074 APCS_AUX_CLK_ON

Type: RW

Clock: SYS_AHB_CLK

Reset State: 0x00000000

Security Treatment: Controlled by GLB_SECURE [PWR].

AUX_CLK_ON Register

APCS_AUX_CLK_ON

Bits	Name	Description
3:0	AUX_CLK_ON_REQ	

0x0B011078 APCS_QSB_LP

Type: RW**Clock:** SYS_AHB_CLK**Reset State:** 0x00000001**Security Treatment:** Controlled by GLB_SECURE [PWR].

QSB_LP Clock on Req Register

APCS_QSB_LP

Bits	Name	Description
6:1	QSB_LP_CNTR	No of Core clock cycle to wait before clock_on req is pulled low, once bus goes idle
0	QSB_LP_OVERRIDE_BIT	Override bit to disable LP functionality, once set qsb_clock_on is always one

0x0B011100 APCS_APPS_BOOTFSM_CTL

Type: RW**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** none

BOOTFSM Override register. This register can be used to go to particular states. This register can override state only if DBGEN is zero.

APCS_APPS_BOOTFSM_CTL

Bits	Name	Description
7:4	STATE_OVERRIDE_VAL	STATE Value that need to be updated
0	STATE_OVERRIDE	update state with STATE_OVERRIDE_VAL. Edge is sampled by HW.

0x0B011104 APCS_APPS_BOOTFSM_STATUS**Type:** R**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** none

BOOTFSM status register. This register can be used to check the current states as well as some relevant signal status. This will reflect the current status of signals.

APCS_APPS_BOOTFSM_STATUS

Bits	Name	Description
31:0	BOOTFSM_STATUS	Status will be updated whenever there is a state change.

0x0B011110 APCS_L2_SPM_AUX_STARTADDR**Type:** RW**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** Restricted

AUX Start Address for L2 SPM

APCS_L2_SPM_AUX_STARTADDR

Bits	Name	Description
16	AUX_STARTADDR_SEL	Select Auxiliary start address for L2SPM
8:0	AUX_STARTADDR	Auxiliary start address for L2SPM

0x0B011200 APCS_L2_SPM_VOTE_TZ**Type:** RW**Clock:** SYS_AHB_CLK**Reset State:** 0xC000FFFF**Security Treatment:** Restricted

TZ Vote for SPM functionality

APCS_L2_SPM_VOTE_TZ

Bits	Name	Description
31	PMIC_HANDSHAKE_EN	TZ Vote for SPM <-> PMIC Handshake. If not set, SPM cannot initiate any requests from PMIC

APCS_L2_SPM_VOTE_TZ (cont.)

Bits	Name	Description
30	RPM_HANDSHAKE_EN	TZ Vote for SPM <-> RPM Handshake. If not set, SPM cannot initiate handshake with RPM
15:0	PWR_CTL_EN	TZ Vote for SPM PWR_CTL bits. If not set, PWR_CTL bit is forced LOW

0x0B011204 APCS_L2_SPM_VOTE_HLOS**Type:** RW**Clock:** SYS_AHB_CLK**Reset State:** 0xC000FFFF**Security Treatment:** Controlled by GLB_SECURE [VOTE].

HLOS Vote for SPM functionality

APCS_L2_SPM_VOTE_HLOS

Bits	Name	Description
31	PMIC_HANDSHAKE_EN	HLOS Vote for SPM <-> PMIC Handshake. If not set, SPM cannot initiate any requests from PMIC
30	RPM_HANDSHAKE_EN	HLOS Vote for SPM <-> RPM Handshake. If not set, SPM cannot initiate handshake with RPM
15:0	PWR_CTL_EN	HLOS Vote for SPM PWR_CTL bits. If not set, PWR_CTL bit is forced LOW

0x0B011208 APCS_L2_SPM_VOTE_INT_STATUS**Type:** R**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** None

Interrupt generated due to contention in Voting

APCS_L2_SPM_VOTE_INT_STATUS

Bits	Name	Description
31	PMIC_HANDSHAKE_INT_STATUS	Set when there is a contention in Voting from TZ/HYP/HLOS
30	RPM_HANDSHAKE_INT_STATUS	Set when there is a contention in Voting from TZ/HYP/HLOS
15:0	PWR_CTL_INT_STATUS	Set when there is a contention in Voting from TZ/HYP/HLOS

0x0B01120C APCS_L2_SPM_VOTE_INT_CLEAR**Type:** W**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** None

Interrupt Clear register

APCS_L2_SPM_VOTE_INT_CLEAR

Bits	Name	Description
31	PMIC_HANDSHAKE_INT_CLEAR	Self Clear register to reset the interrupt generated due to contention
30	RPM_HANDSHAKE_INT_CLEAR	Self Clear register to reset the interrupt generated due to contention
15:0	PWR_CTL_INT_CLEAR	Self Clear register to reset the interrupt generated due to contention

0x0B011220 APCS_AINACTS_CNT**Type:** RW**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by GLB_SECURE [ACP].

AINACTS Count register

APCS_AINACTS_CNT

Bits	Name	Type	Description
31	AINACTS_CNT_WIP	R	AINACTS_CNT update in progress. The count value needs to be written into a register in XO clock domain. Any updates the count value while this bit is set HIGH shall be ignored
23:0	AINACTS_CNT_VAL	RW	No. of XO (19.2Mhz) cycles to wait before setting AINACTS once ACP bus goes idle

0x0B011228 APCS_ACP_L2_PM_STATUS**Type:** R**Clock:** SYS_AHB_CLK**Reset State:** 0x00040040**Security Treatment:** Controlled by GLB_SECURE [STS]

ACP/L2 Power Management Controller Status register

APCS_ACP_L2_PM_STATUS

Bits	Name	Description
31	QSB_CLK_ON_REQ	reflects the value of signal qsb_clk_on_req
30	WSTARTS	reflects the value of signal wstarts
29:24	WR_REQ_CNT	reflects the value of signal wr_req_cnt
23	ALL_CORE_IN_STBL_LPM	reflects the value of signal all_core_in_stbl_lpm
22	L2SPM_WAIT_REQ	reflects the value of signal l2spm_wait_req
21	L2SPM_TIMER_TRIG_ACK	reflects the value of signal l2spm_timer_trig_ack
20	L2SPM_TIMER_TRIG	reflects the value of signal l2spm_timer_trig
19	CBC_EN_OVERRIDE	reflects the value of signal cbc_en_override
18	MASK_ACP_XFERS	reflects the value of signal mask_acp_xfers
17:16	STATE	reflects the state of acp_l2_pm_ctl FSM
15	L2SPM_PC_MODE_QSB	reflects the state of signal l2spm_pc_mode_qsb
14	AINACTS_CNT_WR_ACK	reflects the state of signal ainacts_cnt_wr_ack
13	ACP_STALL	reflects the state of signal acp_stall: When this is asserted all the Request channels AR/AW/W are Masked and A53SS will not serve any new transfers
12	L2FLUSHDONE	reflects the state of signal HW signal L2FLUSHDONE. Note this reflects the state of CORTEXA53 pin L2FLUSHDONE. This is latched on to CSR L2FlushDone
11	L2FLUSHREQ	reflects the state of signal HW signal L2FLUSHREQ. Note it is generated when L2FlushReq is programmed and acp transfers are stalled. This gets deasserted on receiving L2FLUSHDONE from A53
10	TIMER_EXP	reflects the value of signal timer_exp
9	TIMER_EN	reflects the value of signal timer_en
8	TIMER_TRIG_ACK_QSB	reflects the value of signal timer_trig_ack_qsb
7	TIMER_TRIG	reflects the value of signal timer_trig
6	AINACTS	reflects the value of signal ainacts
5:0	XFER_CNT	reflects the value of outstanding transfer count on ACP bus

0x0B011244 APCS_QCHANNEL_CPU_NEON**Type:** RW**Clock:** SYS_AHB_CLK**Reset State:** 0x000F000F**Security Treatment:** Restricted

Qchannel handshake for CPU/NEON

APCS_QCHANNEL_CPU_NEON

Bits	Name	Type	Description
31:28	NEONQACTIVE	R	Q channel interface for NEON
27:24	NEONQDENY	R	Q channel interface for NEON
23:20	NEONQACCEPTN	R	Q channel interface for NEON
19:16	NEONQREQN	RW	Q channel interface for NEON
15:12	CPUQACTIVE	R	Q channel interface for CPU
11:8	CPUQDENY	R	Q channel interface for CPU
7:4	CPUQACCEPTN	R	Q channel interface for CPU
3:0	CPUQREQN	RW	Q channel interface for CPU

0x0B011248 APCS_QCHANNEL_L2**Type:** RW**Clock:** SYS_AHB_CLK**Reset State:** 0x00000001**Security Treatment:** Restricted

Qchannel handshake for L2

APCS_QCHANNEL_L2

Bits	Name	Type	Description
3	L2QACTIVE	R	Q channel interface for L2
2	L2QDENY	R	Q channel interface for L2
1	L2QACCEPTN	R	Q channel interface for L2
0	L2QREQN	RW	Q channel interface for L2

0x0B01122C APCS_MISC_PWR_CTL**Type:** RW**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by GLB_SECURE [ACP].

Miscellaneous Power Control Register

APCS_MISC_PWR_CTL

Bits	Name	Description
8	CLR_EX_MON_REQ	Reserved/Not used
4	ACP_STALL_REQ	Stall transactions on ACP channel
0	L2_FLUSH_REQ	L2 Flush Request

0x0B011230 APCS_MISC_PWR_STS**Type:** R**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by GLB_SECURE [ACP].

Miscellaneous Power Status Register

APCS_MISC_PWR_STS

Bits	Name	Description
8	CLR_EX_MON_ACK	Reserved/Not used
4	ACP_STALL_ACK	ACP transactions stalled
0	L2_FLUSH_DONE	L2 FLush Complete acknowledge

0x0B011240 APCS_CORE_HS_STATE**Type:** R**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** None

Power collapse info for each of the core just before watchdog expiry

APCS_CORE_HS_STATE

Bits	Name	Description
3:0	HS_STATE	Head switch state for each of the core latched just before watch dog expiry. The state retains its value across watch dog reset 0x1: SET (Core was powered On before watch dog expiry) 0x0: CLEAR (Core was powered Off before watch dog expiry)

8.20 APCS_KPSS_CFG**0x0B010000 APCS_CFG_SECURE****Type:** RW**Clock:** SYS_AHB_CLK**Reset State:** 0x00000001**Security Treatment:** Restricted

The CFG_SECURE register is used to enable or disable non-secure bus transactions. The bus **security** status is signaled by A53SS's APROTNS pin. When APROTNS is set to '0' the state of the CFG_SECURE register is not considered - a secure transaction is always allowed. When APROTNS is set to '1', the CFG_SECURE register's SECURE bit must be '1' for access to be granted. If the SECURE bit is set to '0', the non-secure (APROTNS = '1') transactions are denied. When access is denied, a write is ignored, and a read will return '0'. No error is signaled.

When reading the CFG_SECURE register, non secure agents will always read '0'. Secure agents will see the actual value of the register. The same is true for the registers or bits each bit in the APCS_GCCSECURE register correspond to.

APCS_CFG_SECURE

Bits	Name	Description
0	GLB_CTL	0x1: NSEC (NSEC) 0x0: SEC (SEC) Controls security treatment for the global control registers: START_ADDR.)

0x0B010004 APCS_BOOT_START_ADDR_SEC**Type:** RW**Clock:** SYS_AHB_CLK**Reset State:** 0x00100005**Security Treatment:** Controlled by CFG_SECURE[CFG_CTL].

The BOOT_START_ADDR_SEC register is used to determine the address to boot from in secure mode. It resets to the value on SYS_apcsCFGRSTADDR[31:16]. Reset by SYS_apcsSYSPor_Ares|SYS_apcsSys_Ares

APCS_BOOT_START_ADDR_SEC

Bits	Name	Description
31:16	START_ADDR	Start address for the A53
2	BOOT_128KB_EN	128 KB BOOT enable
1	VINITHI	0 = starts exceptions vectors at address 0x00000000 1 = starts exception vectors at address 0xFFFF0000
0	REMAP_EN	Enable remapping

0x0B010008 APCS_BOOT_START_ADDR_NSEC**Type:** RW**Clock:** SYS_AHB_CLK**Reset State:** 0x00100000**Security Treatment:** Secure and Nonsecure access

The BOOT_START_ADDR_NSEC register is used to determine the address to boot from in non-secure mode. It resets to the value on SYS_apcsCFGRSTADDR[31:16]. Reset by SYS_apcsSYSPor_Ares|SYS_apcsSys_Ares

APCS_BOOT_START_ADDR_NSEC

Bits	Name	Type	Description
31:16	START_ADDR	RW	Start address for the A53
2	BOOT_128KB_EN	RW	128 KB BOOT enable
1	VINITHI	R	This is RO field and returns the copy of BOOT_START_ADDR_SEC VINITHI value
0	REMAP_EN	RW	Enable remapping

0x0B01000C APCS_AA64NAA32_REG**Type:** RW**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** Secure**APCS_AA64NAA32_REG**

Bits	Name	Description
0	AA64NAA32	

0x0B010010 APCS_TCM_START_ADDR**Type:** RW**Clock:** SYS_AHB_CLK**Reset State:** 0x08000000**Security Treatment:** Controlled by CFG_SECURE[CFG_CTL].

The TCM_REDIRECT register gives out the address([31:19]) for comparison to decide if the transaction goes to TCM while in boot, TCM_REDIRECT[0] is an enable pin for the TCM_REDIRECT Logic

APCS_TCM_START_ADDR

Bits	Name	Description
31:19	TCM_START_ADDR	TCM address for the A53
1	TCM_REDIRECT_EN_1	Enable Both L2 and TCM
0	TCM_REDIRECT_EN_0	Enable TCM_REDIRECT

8.21 APCS_BANKED_KPSS_ACS**0x0B008000 APCS_BANKED_APC_SECURE****Type:** RW**Clock:** SYS_AHB_CLK**Reset State:** 0x0000000F**Security Treatment:** Restricted

The APC_SECURE register is used to enable or disable non-secure bus transactions. The bus **security** status is signaled by A53SS's APROTNS pin. When APROTNS is set to '0' the state of the APC_SECURE register is not considered - a secure transaction is always allowed. When APROTNS is set to '1', the APC_SECURE register's SECURE bit must be '1' for access to be granted. If the SECURE bit is set to '0', the non-secure (APROTNS = '1') transactions are denied. When access is denied, a write is ignored, and a read will return '0'. No error is signaled.

When reading the APC_SECURE register, non secure agents will always read '0'. Secure agents will see the actual value of the register. The same is true for the registers or bits each bit in the APC_SECURE register correspond to.

APCS_BANKED_APC_SECURE

Bits	Name	Description
3	VOTE	Secure bit for SPM Voting by HLOS 0x1: NSEC (NSEC) 0x0: SEC (SEC) Controls security treatment for SPM Vote registers: SPM_VOTE_HLOS)
2	RESERVED	Reserved
1	CLK_CTL	0x1: NSEC (NSEC) 0x0: SEC (SEC) Controls security treatment for the sleep control registers: APC_PWR_STATUS, APC_PWR_GATE_STATUS)
0	SLP_CTL	0x1: NSEC (NSEC) 0x0: SEC (SEC) Controls security treatment for the clock control registers: CPU_PWR_CTL, CPU_TRGTD_DBG_RST, APC_PWR_GATE_CTL, APC_A53_CFG_STS)

0x0B008004 APCS_BANKED_CPU_PWR_CTL**Type:** RW**Clock:** SYS_AHB_CLK**Reset State:** 0x00000088**Security Treatment:** Controlled by APC_SECURE [SLP_CTL].

The CPU_PWR_CTL register is used to control the clamps, reset and array head switches.

APCS_BANKED_CPU_PWR_CTL

Bits	Name	Description
16	CLK_EN	Clock enables for CORE clock. This is only to workaround hardware dynamic clock gating problems. 0x1: ALWAYS_ENABLED (Always enabled) 0x0: DYNAMICALLY_ENABLED_BY_HARDWARE (dynamically enabled by hardware [20:18]= SYS_apcNAux_Clk[3:1] [17] = SYS_apcsRef_Clk [16] = SYS_apcNPll_Clk)
10	CORE_SLEEP_STATE	Report that the CPU is in sleep state
7	CORE_PWRD_UP	Report that the CPU is powered up.

APCS_BANKED_CPU_PWR_CTL (cont.)

Bits	Name	Description
6	GATE_CLK	When set, this bit forces the CPU clock to be gated off. This control is intended to permit external masters such as the RPM to gate off the CPU clock during sleep modes, when power control is being done by an external master instead of the SPM. The output of this register bit is ORed with a clock gating control output from the SPM, so that if either the SPM asserts the control, or the RPM sets this bit, the CPU clock will be gated off. When set, this bit gate the CPU clock that must be asserted during sleep state. This bit is ORed with a SAW output before passing to the CPU core.
5	COREPOR_RST	When set, this bit asserts corepor areset to the CPU core.
4	CORE_RST	When set, this bit asserts core areset to the COPU core.
3	CORE_MEM_HS	When cleared, this bit opens a head switch to put the Core Memories into a low-leakage, non-data-retaining state. This bit is ORed with a SAW output before passing to the CPU core.
2	L1_RST_DIS	When set, this bit disables L1 initialization during low power mode exit
1	CORE_MEM_CLAMP	When set this bit clamps the output from the headswitch Core Memories. This bit is OR'ed with saw outputs before passing to the Core Memories
0	CLAMP	When set, this bit asserts the clamp signal that must be asserted prior to collapsing CPU voltage, vdd_apc0. This bit is ORed with SAW output before passing to the CPU core.

0x0B008040 APCS_BANKED_APC_A53_CFG_STS**Type:** R**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by APC_SECURE [SLP_CTL].**APCS_BANKED_APC_A53_CFG_STS**

Bits	Name	Description
28	SMPNAMP	

0x0B008008 APCS_BANKED_APC_PWR_STATUS**Type:** R**Clock:** SYS_AHB_CLK**Reset State:** 0x00040700**Security Treatment:** Controlled by APC_SECURE [CLK_CTL].

The APC_PWR_STATUS register is used to monitor the output of the clock gating cells.

APCS_BANKED_APC_PWR_STATUS

Bits	Name	Description
27	SAW_SLP_ACK	This bit reflects the status of APCN_apccWaitAck signal
25	CORE_NO_PWR_DWN	This bit reflect the status of the APCN_dbgCoreNoPwrDwn bits.
24	CORE_PWRUP_REQ	This bit reflect the status of the APCN_dbgCorePwrUpReq bits.
20:16	HW_EVENT	This bit reflect the status of APCN_sysHwEvent[4:0] bits.
13	RET_SLP_ACK	This bit reflect the status of APCN_sysRetSlpAck bit. L2 assert the signal when it see the APCN_sysRetSlpReq and CPU snoop FIFO is quieced. L2 de-assert this signal after APCN_sysRetSlpReq is de-asserted and all snoop invalidation is completed.
12	RET_SLP_REQ	This bit reflect the status of APCN_apccRetSlpReq bit
10	TRGTD_DBG_RST	This bit reflect the status of APCN_apcTargetedDbg_Areset_N bit
9	CORE_RST	This bit reflect the status of APCN_apccCore_Areset_N bit
8	COREPOR_RST	This bit reflect the status of APCN_apccCorePor_Areset_N bit
7	GDHS_STS	Status for APC block headswitch
6	CORE_MEM_HS_STS	
5	CLAMP	This bit reflect the status of the SYS_apcNClampApcNvd bit.
4	FRC_CLK_OFF	This bit reflect the status of APCN_apccForceClocksOff bit
3	AHB_CLK	This bit reflect the status of APCN_ahbClkOnReq bit
2	REF_CLK	This bit reflect the status of APCN_refClkOnReq bit
1	CORE_AUX_CLK	This bits reflect the status of the APCS_sysApcNAuxClkOnReq bits.
0	CORE_PLL_CLK	This bit reflect the status of the APCS_sysApcsPllClkOnReq bit.

0x0B008014 APCS_BANKED_APC_PWR_GATE_CTL

Type: RW

Clock: SYS_AHB_CLK

Reset State: 0x10000001

Security Treatment: Controlled by APC_SECURE [SLP_CTL].

APC_PWR_GATE_CTL is used to control the GDHS switch for CPU core.

APCS_BANKED_APC_PWR_GATE_CTL

Bits	Name	Description
31:24	GDHS_CNT	Delay count between assertion of en_few and en_rest control of CPU GDHS.
0	GDHS_EN	When set, this bit closes the block head switches for the CPU logic putting it into an operational state.

0x0B008030 APCS_BANKED_APC_PWR_GATE_STATUS**Type:** R**Clock:** SYS_AHB_CLK**Reset State:** 0x00010000**Security Treatment:** Controlled by APC_SECURE [CLK_CTL].

This register is used to monitor the GDHS power control input of the A53 CPU

APCS_BANKED_APC_PWR_GATE_STATUS

Bits	Name	Description
16	IDLE	This bit reflects if power gate sequence is switching between different power modes
4:1	EN_REST	This bits reflect the status of the EnRest[3:0] bits
0	EN_FEW	This bit reflect the status of the EnFew bit.

0x0B008044 APCS_BANKED_SPM_AUX_STARTADDR**Type:** RW**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** Restricted

AUX Start Address for CPU SPM

APCS_BANKED_SPM_AUX_STARTADDR

Bits	Name	Description
16	AUX_STARTADDR_SEL	Select Auxillary start address for CPU SPM
8:0	AUX_STARTADDR	Auxillary start address for CPU SPM

0x0B008050 APCS_BANKED_SPM_VOTE_TZ**Type:** RW**Clock:** SYS_AHB_CLK**Reset State:** 0xC0007FFF**Security Treatment:** Restricted

TZ Vote for SPM functionality

APCS_BANKED_SPM_VOTE_TZ

Bits	Name	Description
31	PMIC_HANDSHAKE_EN	TZ Vote for SPM <-> PMIC Handshake. If not set, SPM cannot initiate any requests to PMIC
30	RPM_HANDSHAKE_EN	TZ Vote for SPM <-> RPM Handshake. If not set, SPM cannot initiate handshake with RPM
14:0	PWR_CTL_EN	TZ Vote for SPM PWR_CTL bits. If not set, PWR_CTL bit is forced LOW

0x0B008054 APCS_BANKED_SPM_VOTE_HLOS**Type:** RW**Clock:** SYS_AHB_CLK**Reset State:** 0xC0007FFF**Security Treatment:** Controlled by APC_SECURE [VOTE]

HLOS Vote for SPM functionality

APCS_BANKED_SPM_VOTE_HLOS

Bits	Name	Description
31	PMIC_HANDSHAKE_EN	HLOS Vote for SPM <-> PMIC Handshake. If not set, SPM cannot initiate any requests to PMIC
30	RPM_HANDSHAKE_EN	HLOS Vote for SPM <-> RPM Handshake. If not set, SPM cannot initiate handshake with RPM
14:0	PWR_CTL_EN	HLOS Vote for SPM PWR_CTL bits. If not set, PWR_CTL bit is forced LOW

0x0B008058 APCS_BANKED_SPM_VOTE_INT_STATUS**Type:** R**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** None

Interrupt generated due to contention in Voting

APCS_BANKED_SPM_VOTE_INT_STATUS

Bits	Name	Description
31	PMIC_HANDSHAKE_INT_STATUS	Set when there is a contention in Voting from TZ/HYP/HLOS
30	RPM_HANDSHAKE_INT_STATUS	Set when there is a contention in Voting from TZ/HYP/HLOS
14:0	PWR_CTL_INT_STATUS	Set when there is a contention in Voting from TZ/HYP/HLOS

0x0B00805C APCS_BANKED_SPM_VOTE_INT_CLEAR

Type: W

Clock: SYS_AHB_CLK

Reset State: 0x00000000

Security Treatment: None

Interrupt Clear register

APCS_BANKED_SPM_VOTE_INT_CLEAR

Bits	Name	Description
31	PMIC_HANDSHAKE_INT_CLEAR	Self Clear register to reset the interrupt generated due to contention
30	RPM_HANDSHAKE_INT_CLEAR	Self Clear register to reset the interrupt generated due to contention
14:0	PWR_CTL_INT_CLEAR	Self Clear register to reset the interrupt generated due to contention

8.22 APCS_ALIAS0_KPSS_ACS

0x0B088000 APCS_ALIAS0_APC_SECURE

Type: RW

Clock: SYS_AHB_CLK

Reset State: 0x0000000F

Security Treatment: Restricted

The APC_SECURE register is used to enable or disable non-secure bus transactions. The bus **security** status is signaled by A53SS's APROTNS pin. When APROTNS is set to '0' the state of the APC_SECURE register is not considered - a secure transaction is always allowed. When APROTNS is set to '1', the APC_SECURE register's SECURE bit must be '1' for access to be

granted. If the SECURE bit is set to '0', the non-secure (APROTNS = '1') transactions are denied. When access is denied, a write is ignored, and a read will return '0'. No error is signaled.

When reading the APC_SECURE register, non secure agents will always read '0'. Secure agents will see the actual value of the register. The same is true for the registers or bits each bit in the APC_SECURE register correspond to.

APCS_ALIAS0_APC_SECURE

Bits	Name	Description
3	VOTE	Secure bit for SPM Voting by HLOS 0x1: NSEC (NSEC) 0x0: SEC (SEC) Controls security treatment for SPM Vote registers: SPM_VOTE_HLOS)
2	RESERVED	Reserved
1	CLK_CTL	0x1: NSEC (NSEC) 0x0: SEC (SEC) Controls security treatment for the sleep control registers: APC_PWR_STATUS, APC_PWR_GATE_STATUS)
0	SLP_CTL	0x1: NSEC (NSEC) 0x0: SEC (SEC) Controls security treatment for the clock control registers: CPU_PWR_CTL, CPU_TRGTD_DBG_RST, APC_PWR_GATE_CTL, APC_A53_CFG_STS)

0x0B088004 APCS_ALIAS0_CPU_PWR_CTL

Type: RW

Clock: SYS_AHB_CLK

Reset State: 0x00000088

Security Treatment: Controlled by APC_SECURE [SLP_CTL].

The CPU_PWR_CTL register is used to control the clamps, reset and array head switches.

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APCS_ALIAS0_CPU_PWR_CTL

Bits	Name	Description
16	CLK_EN	Clock enables for CORE clock. This is only to workaround hardware dynamic clock gating problems. 0x1: ALWAYS_ENABLED (Always enabled) 0x0: DYNAMICALLY_ENABLED_BY_HARDWARE (dynamically enabled by hardware) [20:18]= SYS_apcNAux_Clk[3:1] [17] = SYS_apcsRef_Clk [16] = SYS_apcNPII_Clk)
10	CORE_SLEEP_STATE	Report that the CPU is in sleep state
7	CORE_PWRD_UP	Report that the CPU is powered up.
6	GATE_CLK	When set, this bit forces the CPU clock to be gated off. This control is intended to permit external masters such as the RPM to gate off the CPU clock during sleep modes, when power control is being done by an external master instead of the SPM. The output of this register bit is ORed with a clock gating control output from the SPM, so that if either the SPM asserts the control, or the RPM sets this bit, the CPU clock will be gated off. When set, this bit gate the CPU clock that must be asserted during sleep state. This bit is ORed with a SAW output before passing to the CPU core.
5	COREPOR_RST	When set, this bit asserts corepor areset to the CPU core.
4	CORE_RST	When set, this bit asserts core areset to the COPU core.
3	CORE_MEM_HS	When cleared, this bit opens a head switch to put the Core Memories into a low-leakage, non-data-retaining state. This bit is ORed with a SAW output before passing to the CPU core.
2	L1_RST_DIS	When set, this bit disables L1 initialization during low power mode exit
1	CORE_MEM_CLAMP	When set this bit clamps the output from the headswitch Core Memories. This bit is OR'ed with saw outputs before passing to the Core Memories
0	CLAMP	When set, this bit asserts the clamp signal that must be asserted prior to collapsing CPU voltage, vdd_apc0. This bit is ORed with SAW output before passing to the CPU core.

0x0B088040 APCS_ALIAS0_APC_A53_CFG_STS**Type:** R**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by APC_SECURE [SLP_CTL].

APCS_ALIAS0_APC_A53_CFG_STS

Bits	Name	Description
28	SMPNAMP	

0x0B088008 APCS_ALIAS0_APC_PWR_STATUS**Type:** R**Clock:** SYS_AHB_CLK**Reset State:** 0x00040700**Security Treatment:** Controlled by APC_SECURE [CLK_CTL].

The APC_PWR_STATUS register is used to monitor the output of the clock gating cells.

APCS_ALIAS0_APC_PWR_STATUS

Bits	Name	Description
27	SAW_SLP_ACK	This bit reflects the status of APCN_apccWaitAck signal
25	CORE_NO_PWR_DWN	This bit reflect the status of the APCN_dbgCoreNoPwrDwn bits.
24	CORE_PWRUP_REQ	This bit reflect the status of the APCN_dbgCorePwrUpReq bits.
20:16	HW_EVENT	This bit reflect the status of APCN_sysHwEvent[4:0] bits.
13	RET_SLP_ACK	This bit reflect the status of APCN_sysRetSlpAck bit. L2 assert the signal when it see the APCN_sysRetSlpReq and CPU snoop FIFO is quieced. L2 de-assert this signal after APCN_sysRetSlpReq is de-asserted and all snoop invalidation is completed.
12	RET_SLP_REQ	This bit reflect the status of APCN_apccRetSlpReq bit
10	TRGTD_DBG_RST	This bit reflect the status of APCN_apcTargetedDbg_Areset_N bit
9	CORE_RST	This bit reflect the status of APCN_apccCore_Areset_N bit
8	COREPOR_RST	This bit reflect the status of APCN_apccCorePor_Areset_N bit
7	GDHS_STS	Status for APC block headswitch
6	CORE_MEM_HS_STS	
5	CLAMP	This bit reflect the status of the SYS_apcNClampApcNvd bit.
4	FRC_CLK_OFF	This bit reflect the status of APCN_apccForceClocksOff bit
3	AHB_CLK	This bit reflect the status of APCN_ahbClkOnReq bit
2	REF_CLK	This bit reflect the status of APCN_refClkOnReq bit
1	CORE_AUX_CLK	This bits reflect the status of the APCS_sysApcNAuxClkOnReq bits.
0	CORE_PLL_CLK	This bit reflect the status of the APCS_sysApcsPllClkOnReq bit.

0x0B088014 APCS_ALIAS0_APC_PWR_GATE_CTL**Type:** RW**Clock:** SYS_AHB_CLK**Reset State:** 0x10000001**Security Treatment:** Controlled by APC_SECURE [SLP_CTL].

APC_PWR_GATE_CTL is used to control the GDHS switch for CPU core.

APCS_ALIAS0_APC_PWR_GATE_CTL

Bits	Name	Description
31:24	GDHS_CNT	Delay count between assertion of en_few and en_rest control of CPU GDHS.
0	GDHS_EN	When set, this bit closes the block head switches for the CPU logic putting it into an operational state.

0x0B088030 APCS_ALIAS0_APC_PWR_GATE_STATUS**Type:** R**Clock:** SYS_AHB_CLK**Reset State:** 0x00010000**Security Treatment:** Controlled by APC_SECURE [CLK_CTL].

This register is used to monitor the GDHS power control input of the A53 CPU

APCS_ALIAS0_APC_PWR_GATE_STATUS

Bits	Name	Description
16	IDLE	This bit reflects if power gate sequence is switching between different power modes
4:1	EN_REST	This bits reflect the status of the EnRest[3:0] bits
0	EN_FEW	This bit reflect the status of the EnFew bit.

0x0B088044 APCS_ALIAS0_SPM_AUX_STARTADDR**Type:** RW**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** Restricted

AUX Start Address for CPU SPM

APCS_ALIAS0_SPM_AUX_STARTADDR

Bits	Name	Description
16	AUX_STARTADDR_SEL	Select Auxillary start address for CPU SPM
8:0	AUX_STARTADDR	Auxillary start address for CPU SPM

0x0B088050 APCS_ALIAS0_SPM_VOTE_TZ**Type:** RW**Clock:** SYS_AHB_CLK**Reset State:** 0xC0007FFF**Security Treatment:** Restricted

TZ Vote for SPM functionality

APCS_ALIAS0_SPM_VOTE_TZ

Bits	Name	Description
31	PMIC_HANDSHAKE_EN	TZ Vote for SPM <-> PMIC Handshake. If not set, SPM cannot initiate any requests to PMIC
30	RPM_HANDSHAKE_EN	TZ Vote for SPM <-> RPM Handshake. If not set, SPM cannot initiate handshake with RPM
14:0	PWR_CTL_EN	TZ Vote for SPM PWR_CTL bits. If not set, PWR_CTL bit is forced LOW

0x0B088054 APCS_ALIAS0_SPM_VOTE_HLOS**Type:** RW**Clock:** SYS_AHB_CLK**Reset State:** 0xC0007FFF**Security Treatment:** Controlled by APC_SECURE [VOTE]

HLOS Vote for SPM functionality

APCS_ALIAS0_SPM_VOTE_HLOS

Bits	Name	Description
31	PMIC_HANDSHAKE_EN	HLOS Vote for SPM <-> PMIC Handshake. If not set, SPM cannot initiate any requests to PMIC
30	RPM_HANDSHAKE_EN	HLOS Vote for SPM <-> RPM Handshake. If not set, SPM cannot initiate handshake with RPM
14:0	PWR_CTL_EN	HLOS Vote for SPM PWR_CTL bits. If not set, PWR_CTL bit is forced LOW

0x0B088058 APCS_ALIAS0_SPM_VOTE_INT_STATUS**Type:** R**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** None

Interrupt generated due to contention in Voting

APCS_ALIAS0_SPM_VOTE_INT_STATUS

Bits	Name	Description
31	PMIC_HANDSHAKE_INT_STATUS	Set when there is a contention in Voting from TZ/HYP/HLOS
30	RPM_HANDSHAKE_INT_STATUS	Set when there is a contention in Voting from TZ/HYP/HLOS
14:0	PWR_CTL_INT_STATUS	Set when there is a contention in Voting from TZ/HYP/HLOS

0x0B08805C APCS_ALIAS0_SPM_VOTE_INT_CLEAR**Type:** W**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** None

Interrupt Clear register

APCS_ALIAS0_SPM_VOTE_INT_CLEAR

Bits	Name	Description
31	PMIC_HANDSHAKE_INT_CLEAR	Self Clear register to reset the interrupt generated due to contention
30	RPM_HANDSHAKE_INT_CLEAR	Self Clear register to reset the interrupt generated due to contention
14:0	PWR_CTL_INT_CLEAR	Self Clear register to reset the interrupt generated due to contention

8.23 APCS_ALIAS1_KPSS_ACS

0x0B098000 APCS_ALIAS1_APC_SECURE

Type: RW

Clock: SYS_AHB_CLK

Reset State: 0x0000000F

Security Treatment: Restricted

The APC_SECURE register is used to enable or disable non-secure bus transactions. The bus **security** status is signaled by A53SS's APROTNS pin. When APROTNS is set to '0' the state of the APC_SECURE register is not considered - a secure transaction is always allowed. When APROTNS is set to '1', the APC_SECURE register's SECURE bit must be '1' for access to be granted. If the SECURE bit is set to '0', the non-secure (APROTNS = '1') transactions are denied. When access is denied, a write is ignored, and a read will return '0'. No error is signaled.

When reading the APC_SECURE register, non secure agents will always read '0'. Secure agents will see the actual value of the register. The same is true for the registers or bits each bit in the APC_SECURE register correspond to.

APCS_ALIAS1_APC_SECURE

Bits	Name	Description
3	VOTE	Secure bit for SPM Voting by HLOS 0x1: NSEC (NSEC) 0x0: SEC (SEC) Controls security treatment for SPM Vote registers: SPM_VOTE_HLOS)
2	RESERVED	Reserved
1	CLK_CTL	0x1: NSEC (NSEC) 0x0: SEC (SEC) Controls security treatment for the sleep control registers: APC_PWR_STATUS, APC_PWR_GATE_STATUS)
0	SLP_CTL	0x1: NSEC (NSEC) 0x0: SEC (SEC) Controls security treatment for the clock control registers: CPU_PWR_CTL, CPU_TRGTD_DBG_RST, APC_PWR_GATE_CTL, APC_A53_CFG_STS)

0x0B098004 APCS_ALIAS1_CPU_PWR_CTL

Type: RW

Clock: SYS_AHB_CLK

Reset State: 0x00000088

Security Treatment: Controlled by APC_SECURE [SLP_CTL].

The CPU_PWR_CTL register is used to control the clamps, reset and array head switches.

APCS_ALIAS1_CPU_PWR_CTL

Bits	Name	Description
16	CLK_EN	Clock enables for CORE clock. This is only to workaround hardware dynamic clock gating problems. 0x1: ALWAYS_ENABLED (Always enabled) 0x0: DYNAMICALLY_ENABLED_BY_HARDWARE (dynamically enabled by hardware) [20:18]= SYS_apcNAux_Clk[3:1] [17] = SYS_apcsRef_Clk [16] = SYS_apcNPIL_Clk)
10	CORE_SLEEP_STATE	Report that the CPU is in sleep state
7	CORE_PWRD_UP	Report that the CPU is powered up.
6	GATE_CLK	When set, this bit forces the CPU clock to be gated off. This control is intended to permit external masters such as the RPM to gate off the CPU clock during sleep modes, when power control is being done by an external master instead of the SPM. The output of this register bit is ORed with a clock gating control output from the SPM, so that if either the SPM asserts the control, or the RPM sets this bit, the CPU clock will be gated off. When set, this bit gate the CPU clock that must be asserted during sleep state. This bit is ORed with a SAW output before passing to the CPU core.
5	COREPOR_RST	When set, this bit asserts corepor areset to the CPU core.
4	CORE_RST	When set, this bit asserts core areset to the COPU core.
3	CORE_MEM_HS	When cleared, this bit opens a head switch to put the Core Memories into a low-leakage, non-data-retaining state. This bit is ORed with a SAW output before passing to the CPU core.
2	L1_RST_DIS	When set, this bit disables L1 initialization during low power mode exit
1	CORE_MEM_CLAMP	When set this bit clamps the output from the headswitch Core Memories. This bit is OR'ed with saw outputs before passing to the Core Memories
0	CLAMP	When set, this bit asserts the clamp signal that must be asserted prior to collapsing CPU voltage, vdd_apc0. This bit is ORed with SAW output before passing to the CPU core.

0x0B098040 APCS_ALIAS1_APC_A53_CFG_STS**Type:** R**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by APC_SECURE [SLP_CTL].

APCS_ALIAS1_APC_A53_CFG_STS

Bits	Name	Description
28	SMPNAMP	

0x0B098008 APCS_ALIAS1_APC_PWR_STATUS**Type:** R**Clock:** SYS_AHB_CLK**Reset State:** 0x00040700**Security Treatment:** Controlled by APC_SECURE [CLK_CTL].

The APC_PWR_STATUS register is used to monitor the output of the clock gating cells.

APCS_ALIAS1_APC_PWR_STATUS

Bits	Name	Description
27	SAW_SLP_ACK	This bit reflects the status of APCN_apccWaitAck signal
25	CORE_NO_PWR_DWN	This bit reflect the status of the APCN_dbgCoreNoPwrDwn bits.
24	CORE_PWRUP_REQ	This bit reflect the status of the APCN_dbgCorePwrUpReq bits.
20:16	HW_EVENT	This bit reflect the status of APCN_sysHwEvent[4:0] bits.
13	RET_SLP_ACK	This bit reflect the status of APCN_sysRetSlpAck bit. L2 assert the signal when it see the APCN_sysRetSlpReq and CPU snoop FIFO is quieced. L2 de-assert this signal after APCN_sysRetSlpReq is de-asserted and all snoop invalidation is completed.
12	RET_SLP_REQ	This bit reflect the status of APCN_apccRetSlpReq bit
10	TRGTD_DBG_RST	This bit reflect the status of APCN_apcTargetedDbg_Areset_N bit
9	CORE_RST	This bit reflect the status of APCN_apccCore_Areset_N bit
8	COREPOR_RST	This bit reflect the status of APCN_apccCorePor_Areset_N bit
7	GDHS_STS	Status for APC block headswitch
6	CORE_MEM_HS_STS	
5	CLAMP	This bit reflect the status of the SYS_apcNClampApcNvd bit.
4	FRC_CLK_OFF	This bit reflect the status of APCN_apccForceClocksOff bit
3	AHB_CLK	This bit reflect the status of APCN_ahbClkOnReq bit
2	REF_CLK	This bit reflect the status of APCN_refClkOnReq bit
1	CORE_AUX_CLK	This bits reflect the status of the APCS_sysApcNAuxClkOnReq bits.
0	CORE_PLL_CLK	This bit reflect the status of the APCS_sysApcsPllClkOnReq bit.

0x0B098014 APCS_ALIAS1_APC_PWR_GATE_CTL**Type:** RW**Clock:** SYS_AHB_CLK**Reset State:** 0x10000001**Security Treatment:** Controlled by APC_SECURE [SLP_CTL].

APC_PWR_GATE_CTL is used to control the GDHS switch for CPU core.

APCS_ALIAS1_APC_PWR_GATE_CTL

Bits	Name	Description
31:24	GDHS_CNT	Delay count between assertion of en_few and en_rest control of CPU GDHS.
0	GDHS_EN	When set, this bit closes the block head switches for the CPU logic putting it into an operational state.

0x0B098030 APCS_ALIAS1_APC_PWR_GATE_STATUS**Type:** R**Clock:** SYS_AHB_CLK**Reset State:** 0x00010000**Security Treatment:** Controlled by APC_SECURE [CLK_CTL].

This register is used to monitor the GDHS power control input of the A53 CPU

APCS_ALIAS1_APC_PWR_GATE_STATUS

Bits	Name	Description
16	IDLE	This bit reflects if power gate sequence is switching between different power modes
4:1	EN_REST	This bits reflect the status of the EnRest[3:0] bits
0	EN_FEW	This bit reflect the status of the EnFew bit.

0x0B098044 APCS_ALIAS1_SPM_AUX_STARTADDR**Type:** RW**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** Restricted

AUX Start Address for CPU SPM

APCS_ALIAS1_SPM_AUX_STARTADDR

Bits	Name	Description
16	AUX_STARTADDR_SEL	Select Auxillary start address for CPU SPM
8:0	AUX_STARTADDR	Auxillary start address for CPU SPM

0x0B098050 APCS_ALIAS1_SPM_VOTE_TZ**Type:** RW**Clock:** SYS_AHB_CLK**Reset State:** 0xC0007FFF**Security Treatment:** Restricted

TZ Vote for SPM functionality

APCS_ALIAS1_SPM_VOTE_TZ

Bits	Name	Description
31	PMIC_HANDSHAKE_EN	TZ Vote for SPM <-> PMIC Handshake. If not set, SPM cannot initiate any requests to PMIC
30	RPM_HANDSHAKE_EN	TZ Vote for SPM <-> RPM Handshake. If not set, SPM cannot initiate handshake with RPM
14:0	PWR_CTL_EN	TZ Vote for SPM PWR_CTL bits. If not set, PWR_CTL bit is forced LOW

0x0B098054 APCS_ALIAS1_SPM_VOTE_HLOS**Type:** RW**Clock:** SYS_AHB_CLK**Reset State:** 0xC0007FFF**Security Treatment:** Controlled by APC_SECURE [VOTE]

HLOS Vote for SPM functionality

APCS_ALIAS1_SPM_VOTE_HLOS

Bits	Name	Description
31	PMIC_HANDSHAKE_EN	HLOS Vote for SPM <-> PMIC Handshake. If not set, SPM cannot initiate any requests to PMIC
30	RPM_HANDSHAKE_EN	HLOS Vote for SPM <-> RPM Handshake. If not set, SPM cannot initiate handshake with RPM
14:0	PWR_CTL_EN	HLOS Vote for SPM PWR_CTL bits. If not set, PWR_CTL bit is forced LOW

0x0B098058 APCS_ALIAS1_SPM_VOTE_INT_STATUS**Type:** R**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** None

Interrupt generated due to contention in Voting

APCS_ALIAS1_SPM_VOTE_INT_STATUS

Bits	Name	Description
31	PMIC_HANDSHAKE_INT_STATUS	Set when there is a contention in Voting from TZ/HYP/HLOS
30	RPM_HANDSHAKE_INT_STATUS	Set when there is a contention in Voting from TZ/HYP/HLOS
14:0	PWR_CTL_INT_STATUS	Set when there is a contention in Voting from TZ/HYP/HLOS

0x0B09805C APCS_ALIAS1_SPM_VOTE_INT_CLEAR**Type:** W**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** None

Interrupt Clear register

APCS_ALIAS1_SPM_VOTE_INT_CLEAR

Bits	Name	Description
31	PMIC_HANDSHAKE_INT_CLEAR	Self Clear register to reset the interrupt generated due to contention
30	RPM_HANDSHAKE_INT_CLEAR	Self Clear register to reset the interrupt generated due to contention
14:0	PWR_CTL_INT_CLEAR	Self Clear register to reset the interrupt generated due to contention

8.24 APCS_ALIAS2_KPSS_ACS

0x0B0A8000 APCS_ALIAS2_APC_SECURE

Type: RW

Clock: SYS_AHB_CLK

Reset State: 0x0000000F

Security Treatment: Restricted

The APC_SECURE register is used to enable or disable non-secure bus transactions. The bus **security** status is signaled by A53SS's APROTNS pin. When APROTNS is set to '0' the state of the APC_SECURE register is not considered - a secure transaction is always allowed. When APROTNS is set to '1', the APC_SECURE register's SECURE bit must be '1' for access to be granted. If the SECURE bit is set to '0', the non-secure (APROTNS = '1') transactions are denied. When access is denied, a write is ignored, and a read will return '0'. No error is signaled.

When reading the APC_SECURE register, non secure agents will always read '0'. Secure agents will see the actual value of the register. The same is true for the registers or bits each bit in the APC_SECURE register correspond to.

APCS_ALIAS2_APC_SECURE

Bits	Name	Description
3	VOTE	Secure bit for SPM Voting by HLOS 0x1: NSEC (NSEC) 0x0: SEC (SEC) Controls security treatment for SPM Vote registers: SPM_VOTE_HLOS)
2	RESERVED	Reserved
1	CLK_CTL	0x1: NSEC (NSEC) 0x0: SEC (SEC) Controls security treatment for the sleep control registers: APC_PWR_STATUS, APC_PWR_GATE_STATUS)
0	SLP_CTL	0x1: NSEC (NSEC) 0x0: SEC (SEC) Controls security treatment for the clock control registers: CPU_PWR_CTL, CPU_TRGTD_DBG_RST, APC_PWR_GATE_CTL, APC_A53_CFG_STS)

0x0B0A8004 APCS_ALIAS2_CPU_PWR_CTL

Type: RW

Clock: SYS_AHB_CLK

Reset State: 0x00000088

Security Treatment: Controlled by APC_SECURE [SLP_CTL].

The CPU_PWR_CTL register is used to control the clamps, reset and array head switches.

APCS_ALIAS2_CPU_PWR_CTL

Bits	Name	Description
16	CLK_EN	Clock enables for CORE clock. This is only to workaround hardware dynamic clock gating problems. 0x1: ALWAYS_ENABLED (Always enabled) 0x0: DYNAMICALLY_ENABLED_BY_HARDWARE (dynamically enabled by hardware) [20:18]= SYS_apcNAux_Clk[3:1] [17] = SYS_apcsRef_Clk [16] = SYS_apcNPIL_Clk)
10	CORE_SLEEP_STATE	Report that the CPU is in sleep state
7	CORE_PWRD_UP	Report that the CPU is powered up.
6	GATE_CLK	When set, this bit forces the CPU clock to be gated off. This control is intended to permit external masters such as the RPM to gate off the CPU clock during sleep modes, when power control is being done by an external master instead of the SPM. The output of this register bit is ORed with a clock gating control output from the SPM, so that if either the SPM asserts the control, or the RPM sets this bit, the CPU clock will be gated off. When set, this bit gate the CPU clock that must be asserted during sleep state. This bit is ORed with a SAW output before passing to the CPU core.
5	COREPOR_RST	When set, this bit asserts corepor areset to the CPU core.
4	CORE_RST	When set, this bit asserts core areset to the COPU core.
3	CORE_MEM_HS	When cleared, this bit opens a head switch to put the Core Memories into a low-leakage, non-data-retaining state. This bit is ORed with a SAW output before passing to the CPU core.
2	L1_RST_DIS	When set, this bit disables L1 initialization during low power mode exit
1	CORE_MEM_CLAMP	When set this bit clamps the output from the headswitch Core Memories. This bit is OR'ed with saw outputs before passing to the Core Memories
0	CLAMP	When set, this bit asserts the clamp signal that must be asserted prior to collapsing CPU voltage, vdd_apc0. This bit is ORed with SAW output before passing to the CPU core.

0x0B0A8040 APCS_ALIAS2_APC_A53_CFG_STS**Type:** R**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by APC_SECURE [SLP_CTL].

APCS_ALIAS2_APC_A53_CFG_STS

Bits	Name	Description
28	SMPNAMP	

0x0B0A8008 APCS_ALIAS2_APC_PWR_STATUS**Type:** R**Clock:** SYS_AHB_CLK**Reset State:** 0x00040700**Security Treatment:** Controlled by APC_SECURE [CLK_CTL].

The APC_PWR_STATUS register is used to monitor the output of the clock gating cells.

APCS_ALIAS2_APC_PWR_STATUS

Bits	Name	Description
27	SAW_SLP_ACK	This bit reflects the status of APCN_apccWaitAck signal
25	CORE_NO_PWR_DWN	This bit reflect the status of the APCN_dbgCoreNoPwrDwn bits.
24	CORE_PWRUP_REQ	This bit reflect the status of the APCN_dbgCorePwrUpReq bits.
20:16	HW_EVENT	This bit reflect the status of APCN_sysHwEvent[4:0] bits.
13	RET_SLP_ACK	This bit reflect the status of APCN_sysRetSlpAck bit. L2 assert the signal when it see the APCN_sysRetSlpReq and CPU snoop FIFO is quieced. L2 de-assert this signal after APCN_sysRetSlpReq is de-asserted and all snoop invalidation is completed.
12	RET_SLP_REQ	This bit reflect the status of APCN_apccRetSlpReq bit
10	TRGTD_DBG_RST	This bit reflect the status of APCN_apcTargetedDbg_Areset_N bit
9	CORE_RST	This bit reflect the status of APCN_apccCore_Areset_N bit
8	COREPOR_RST	This bit reflect the status of APCN_apccCorePor_Areset_N bit
7	GDHS_STS	Status for APC block headswitch
6	CORE_MEM_HS_STS	
5	CLAMP	This bit reflect the status of the SYS_apcNClampApcNvd bit.
4	FRC_CLK_OFF	This bit reflect the status of APCN_apccForceClocksOff bit
3	AHB_CLK	This bit reflect the status of APCN_ahbClkOnReq bit
2	REF_CLK	This bit reflect the status of APCN_refClkOnReq bit
1	CORE_AUX_CLK	This bits reflect the status of the APCS_sysApcNAuxClkOnReq bits.
0	CORE_PLL_CLK	This bit reflect the status of the APCS_sysApcsPllClkOnReq bit.

0x0B0A8014 APCS_ALIAS2_APC_PWR_GATE_CTL**Type:** RW**Clock:** SYS_AHB_CLK**Reset State:** 0x10000001**Security Treatment:** Controlled by APC_SECURE [SLP_CTL].

APC_PWR_GATE_CTL is used to control the GDHS switch for CPU core.

APCS_ALIAS2_APC_PWR_GATE_CTL

Bits	Name	Description
31:24	GDHS_CNT	Delay count between assertion of en_few and en_rest control of CPU GDHS.
0	GDHS_EN	When set, this bit closes the block head switches for the CPU logic putting it into an operational state.

0x0B0A8030 APCS_ALIAS2_APC_PWR_GATE_STATUS**Type:** R**Clock:** SYS_AHB_CLK**Reset State:** 0x00010000**Security Treatment:** Controlled by APC_SECURE [CLK_CTL].

This register is used to monitor the GDHS power control input of the A53 CPU

APCS_ALIAS2_APC_PWR_GATE_STATUS

Bits	Name	Description
16	IDLE	This bit reflects if power gate sequence is switching between different power modes
4:1	EN_REST	This bits reflect the status of the EnRest[3:0] bits
0	EN_FEW	This bit reflect the status of the EnFew bit.

0x0B0A8044 APCS_ALIAS2_SPM_AUX_STARTADDR**Type:** RW**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** Restricted

AUX Start Address for CPU SPM

APCS_ALIAS2_SPM_AUX_STARTADDR

Bits	Name	Description
16	AUX_STARTADDR_SEL	Select Auxillary start address for CPU SPM
8:0	AUX_STARTADDR	Auxillary start address for CPU SPM

0x0B0A8050 APCS_ALIAS2_SPM_VOTE_TZ**Type:** RW**Clock:** SYS_AHB_CLK**Reset State:** 0xC0007FFF**Security Treatment:** Restricted

TZ Vote for SPM functionality

APCS_ALIAS2_SPM_VOTE_TZ

Bits	Name	Description
31	PMIC_HANDSHAKE_EN	TZ Vote for SPM <-> PMIC Handshake. If not set, SPM cannot initiate any requests to PMIC
30	RPM_HANDSHAKE_EN	TZ Vote for SPM <-> RPM Handshake. If not set, SPM cannot initiate handshake with RPM
14:0	PWR_CTL_EN	TZ Vote for SPM PWR_CTL bits. If not set, PWR_CTL bit is forced LOW

0x0B0A8054 APCS_ALIAS2_SPM_VOTE_HLOS**Type:** RW**Clock:** SYS_AHB_CLK**Reset State:** 0xC0007FFF**Security Treatment:** Controlled by APC_SECURE [VOTE]

HLOS Vote for SPM functionality

APCS_ALIAS2_SPM_VOTE_HLOS

Bits	Name	Description
31	PMIC_HANDSHAKE_EN	HLOS Vote for SPM <-> PMIC Handshake. If not set, SPM cannot initiate any requests to PMIC
30	RPM_HANDSHAKE_EN	HLOS Vote for SPM <-> RPM Handshake. If not set, SPM cannot initiate handshake with RPM
14:0	PWR_CTL_EN	HLOS Vote for SPM PWR_CTL bits. If not set, PWR_CTL bit is forced LOW

0x0B0A8058 APCS_ALIAS2_SPM_VOTE_INT_STATUS**Type:** R**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** None

Interrupt generated due to contention in Voting

APCS_ALIAS2_SPM_VOTE_INT_STATUS

Bits	Name	Description
31	PMIC_HANDSHAKE_INT_STATUS	Set when there is a contention in Voting from TZ/HYP/HLOS
30	RPM_HANDSHAKE_INT_STATUS	Set when there is a contention in Voting from TZ/HYP/HLOS
14:0	PWR_CTL_INT_STATUS	Set when there is a contention in Voting from TZ/HYP/HLOS

0x0B0A805C APCS_ALIAS2_SPM_VOTE_INT_CLEAR**Type:** W**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** None

Interrupt Clear register

APCS_ALIAS2_SPM_VOTE_INT_CLEAR

Bits	Name	Description
31	PMIC_HANDSHAKE_INT_CLEAR	Self Clear register to reset the interrupt generated due to contention
30	RPM_HANDSHAKE_INT_CLEAR	Self Clear register to reset the interrupt generated due to contention
14:0	PWR_CTL_INT_CLEAR	Self Clear register to reset the interrupt generated due to contention

8.25 APCS_ALIAS3_KPSS_ACS

0x0B0B8000 APCS_ALIAS3_APC_SECURE

Type: RW

Clock: SYS_AHB_CLK

Reset State: 0x0000000F

Security Treatment: Restricted

The APC_SECURE register is used to enable or disable non-secure bus transactions. The bus **security** status is signaled by A53SS's APROTNS pin. When APROTNS is set to '0' the state of the APC_SECURE register is not considered - a secure transaction is always allowed. When APROTNS is set to '1', the APC_SECURE register's SECURE bit must be '1' for access to be granted. If the SECURE bit is set to '0', the non-secure (APROTNS = '1') transactions are denied. When access is denied, a write is ignored, and a read will return '0'. No error is signaled.

When reading the APC_SECURE register, non secure agents will always read '0'. Secure agents will see the actual value of the register. The same is true for the registers or bits each bit in the APC_SECURE register correspond to.

APCS_ALIAS3_APC_SECURE

Bits	Name	Description
3	VOTE	Secure bit for SPM Voting by HLOS 0x1: NSEC (NSEC) 0x0: SEC (SEC) Controls security treatment for SPM Vote registers: SPM_VOTE_HLOS)
2	RESERVED	Reserved
1	CLK_CTL	0x1: NSEC (NSEC) 0x0: SEC (SEC) Controls security treatment for the sleep control registers: APC_PWR_STATUS, APC_PWR_GATE_STATUS)
0	SLP_CTL	0x1: NSEC (NSEC) 0x0: SEC (SEC) Controls security treatment for the clock control registers: CPU_PWR_CTL, CPU_TRGTD_DBG_RST, APC_PWR_GATE_CTL, APC_A53_CFG_STS)

0x0B0B8004 APCS_ALIAS3_CPU_PWR_CTL

Type: RW

Clock: SYS_AHB_CLK

Reset State: 0x00000088

Security Treatment: Controlled by APC_SECURE [SLP_CTL].

The CPU_PWR_CTL register is used to control the clamps, reset and array head switches.

APCS_ALIAS3_CPU_PWR_CTL

Bits	Name	Description
16	CLK_EN	Clock enables for CORE clock. This is only to workaround hardware dynamic clock gating problems. 0x1: ALWAYS_ENABLED (Always enabled) 0x0: DYNAMICALLY_ENABLED_BY_HARDWARE (dynamically enabled by hardware) [20:18]= SYS_apcNAux_Clk[3:1] [17] = SYS_apcsRef_Clk [16] = SYS_apcNPIL_Clk)
10	CORE_SLEEP_STATE	Report that the CPU is in sleep state
7	CORE_PWRD_UP	Report that the CPU is powered up.
6	GATE_CLK	When set, this bit forces the CPU clock to be gated off. This control is intended to permit external masters such as the RPM to gate off the CPU clock during sleep modes, when power control is being done by an external master instead of the SPM. The output of this register bit is ORed with a clock gating control output from the SPM, so that if either the SPM asserts the control, or the RPM sets this bit, the CPU clock will be gated off. When set, this bit gate the CPU clock that must be asserted during sleep state. This bit is ORed with a SAW output before passing to the CPU core.
5	COREPOR_RST	When set, this bit asserts corepor areset to the CPU core.
4	CORE_RST	When set, this bit asserts core areset to the COPU core.
3	CORE_MEM_HS	When cleared, this bit opens a head switch to put the Core Memories into a low-leakage, non-data-retaining state. This bit is ORed with a SAW output before passing to the CPU core.
2	L1_RST_DIS	When set, this bit disables L1 initialization during low power mode exit
1	CORE_MEM_CLAMP	When set this bit clamps the output from the headswitch Core Memories. This bit is OR'ed with saw outputs before passing to the Core Memories
0	CLAMP	When set, this bit asserts the clamp signal that must be asserted prior to collapsing CPU voltage, vdd_apc0. This bit is ORed with SAW output before passing to the CPU core.

0x0B0B8040 APCS_ALIAS3_APC_A53_CFG_STS**Type:** R**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by APC_SECURE [SLP_CTL].

APCS_ALIAS3_APC_A53_CFG_STS

Bits	Name	Description
28	SMPNAMP	

0x0B0B8008 APCS_ALIAS3_APC_PWR_STATUS**Type:** R**Clock:** SYS_AHB_CLK**Reset State:** 0x00040700**Security Treatment:** Controlled by APC_SECURE [CLK_CTL].

The APC_PWR_STATUS register is used to monitor the output of the clock gating cells.

APCS_ALIAS3_APC_PWR_STATUS

Bits	Name	Description
27	SAW_SLP_ACK	This bit reflects the status of APCN_apccWaitAck signal
25	CORE_NO_PWR_DWN	This bit reflect the status of the APCN_dbgCoreNoPwrDwn bits.
24	CORE_PWRUP_REQ	This bit reflect the status of the APCN_dbgCorePwrUpReq bits.
20:16	HW_EVENT	This bit reflect the status of APCN_sysHwEvent[4:0] bits.
13	RET_SLP_ACK	This bit reflect the status of APCN_sysRetSlpAck bit. L2 assert the signal when it see the APCN_sysRetSlpReq and CPU snoop FIFO is quieced. L2 de-assert this signal after APCN_sysRetSlpReq is de-asserted and all snoop invalidation is completed.
12	RET_SLP_REQ	This bit reflect the status of APCN_apccRetSlpReq bit
10	TRGTD_DBG_RST	This bit reflect the status of APCN_apcTargetedDbg_Areset_N bit
9	CORE_RST	This bit reflect the status of APCN_apccCore_Areset_N bit
8	COREPOR_RST	This bit reflect the status of APCN_apccCorePor_Areset_N bit
7	GDHS_STS	Status for APC block headswitch
6	CORE_MEM_HS_STS	
5	CLAMP	This bit reflect the status of the SYS_apcNClampApcNvd bit.
4	FRC_CLK_OFF	This bit reflect the status of APCN_apccForceClocksOff bit
3	AHB_CLK	This bit reflect the status of APCN_ahbClkOnReq bit
2	REF_CLK	This bit reflect the status of APCN_refClkOnReq bit
1	CORE_AUX_CLK	This bits reflect the status of the APCS_sysApcNAuxClkOnReq bits.
0	CORE_PLL_CLK	This bit reflect the status of the APCS_sysApcsPllClkOnReq bit.

0x0B0B8014 APCS_ALIAS3_APC_PWR_GATE_CTL**Type:** RW**Clock:** SYS_AHB_CLK**Reset State:** 0x10000001**Security Treatment:** Controlled by APC_SECURE [SLP_CTL].

APC_PWR_GATE_CTL is used to control the GDHS switch for CPU core.

APCS_ALIAS3_APC_PWR_GATE_CTL

Bits	Name	Description
31:24	GDHS_CNT	Delay count between assertion of en_few and en_rest control of CPU GDHS.
0	GDHS_EN	When set, this bit closes the block head switches for the CPU logic putting it into an operational state.

0x0B0B8030 APCS_ALIAS3_APC_PWR_GATE_STATUS**Type:** R**Clock:** SYS_AHB_CLK**Reset State:** 0x00010000**Security Treatment:** Controlled by APC_SECURE [CLK_CTL].

This register is used to monitor the GDHS power control input of the A53 CPU

APCS_ALIAS3_APC_PWR_GATE_STATUS

Bits	Name	Description
16	IDLE	This bit reflects if power gate sequence is switching between different power modes
4:1	EN_REST	This bits reflect the status of the EnRest[3:0] bits
0	EN_FEW	This bit reflect the status of the EnFew bit.

0x0B0B8044 APCS_ALIAS3_SPM_AUX_STARTADDR**Type:** RW**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** Restricted

AUX Start Address for CPU SPM

APCS_ALIAS3_SPM_AUX_STARTADDR

Bits	Name	Description
16	AUX_STARTADDR_SEL	Select Auxillary start address for CPU SPM
8:0	AUX_STARTADDR	Auxillary start address for CPU SPM

0x0B0B8050 APCS_ALIAS3_SPM_VOTE_TZ**Type:** RW**Clock:** SYS_AHB_CLK**Reset State:** 0xC0007FFF**Security Treatment:** Restricted

TZ Vote for SPM functionality

APCS_ALIAS3_SPM_VOTE_TZ

Bits	Name	Description
31	PMIC_HANDSHAKE_EN	TZ Vote for SPM <-> PMIC Handshake. If not set, SPM cannot initiate any requests to PMIC
30	RPM_HANDSHAKE_EN	TZ Vote for SPM <-> RPM Handshake. If not set, SPM cannot initiate handshake with RPM
14:0	PWR_CTL_EN	TZ Vote for SPM PWR_CTL bits. If not set, PWR_CTL bit is forced LOW

0x0B0B8054 APCS_ALIAS3_SPM_VOTE_HLOS**Type:** RW**Clock:** SYS_AHB_CLK**Reset State:** 0xC0007FFF**Security Treatment:** Controlled by APC_SECURE [VOTE]

HLOS Vote for SPM functionality

APCS_ALIAS3_SPM_VOTE_HLOS

Bits	Name	Description
31	PMIC_HANDSHAKE_EN	HLOS Vote for SPM <-> PMIC Handshake. If not set, SPM cannot initiate any requests to PMIC
30	RPM_HANDSHAKE_EN	HLOS Vote for SPM <-> RPM Handshake. If not set, SPM cannot initiate handshake with RPM
14:0	PWR_CTL_EN	HLOS Vote for SPM PWR_CTL bits. If not set, PWR_CTL bit is forced LOW

0x0B0B8058 APCS_ALIAS3_SPM_VOTE_INT_STATUS**Type:** R**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** None

Interrupt generated due to contention in Voting

APCS_ALIAS3_SPM_VOTE_INT_STATUS

Bits	Name	Description
31	PMIC_HANDSHAKE_INT_STATUS	Set when there is a contention in Voting from TZ/HYP/HLOS
30	RPM_HANDSHAKE_INT_STATUS	Set when there is a contention in Voting from TZ/HYP/HLOS
14:0	PWR_CTL_INT_STATUS	Set when there is a contention in Voting from TZ/HYP/HLOS

0x0B0B805C APCS_ALIAS3_SPM_VOTE_INT_CLEAR**Type:** W**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** None

Interrupt Clear register

APCS_ALIAS3_SPM_VOTE_INT_CLEAR

Bits	Name	Description
31	PMIC_HANDSHAKE_INT_CLEAR	Self Clear register to reset the interrupt generated due to contention
30	RPM_HANDSHAKE_INT_CLEAR	Self Clear register to reset the interrupt generated due to contention
14:0	PWR_CTL_INT_CLEAR	Self Clear register to reset the interrupt generated due to contention

8.26 APCS_QGIC2

**0x0B000080+ APCS_GICD_ISRn, n=[0..8]
0x4*n**
Type: RW**Clock:** CLK**Reset State:** 0x00000000**Security Treatment:** Restricted

Each bit in an GICD_ISR register controls the **security** state of an interrupt, to be either secure or non-secure. You can only access these registers with secure read or secure write accesses.

NOTE The GICD_ISR register at address offset 0x0080 is banked, that is there is one per CPU Interface. It is only accessible from the designated CPU and controls that CPU's PPI and SGI interrupts. The remaining registers control the SPI interrupts.

APCS_GICD_ISRn

Bits	Name	Description
31:0	INT_NS	0x0: SEC (SEC (Assigns INTID N to the Secure state.)) 0x1: NS (NS (Assigns INTID N to the Non-secure state.))

**0x0B000100+ APCS_GICD_ISENABLERn, n=[0..8]
0x4*n**
Type: RW**Clock:** CLK**Reset State:** 0x00000000**Security Treatment:** NS-int Dependant

Each bit in an GICD_ISENABLER register controls the enabling of an interrupt.

Reading this register returns the currently enabled interrupts subject to NS-int Dependant access rules.

NOTE The GICD_ISENABLER register at address offset 0x0100 is banked, that is there is one per CPU Interface. It is only accessible from the designated CPU and controls that CPU's PPI and SGI interrupts. The remaining registers control the SPI interrupts.

APCS_GICD_ISENABLERn

Bits	Name	Description
31:0	INT	0x0: NA (NA (No effect. Use GICD_ICENABLER register.)) 0x1: SET (SET (Enables INTID N.))

**0x0B000180+ APCS_GICD_ICENABLERn, n=[0..8]
0x4*n****Type:** RW**Clock:** CLK**Reset State:** 0x00000000**Security Treatment:** NS-int Dependant

Each bit in an GICD_ICENABLER register controls the disabling of an interrupt.

Reading this register returns the currently enabled interrupts subject to NS-int Dependant access rules.

NOTE The GICD_ICENABLER register at address offset 0x0180 is banked, that is there is one per CPU Interface. It is only accessible from the designated CPU and controls that CPU's PPI and SGI interrupts. The remaining registers control the SPI interrupts.

APCS_GICD_ICENABLERn

Bits	Name	Description
31:0	INT	0x0: NA (NA (No effect. Use GICD_ISENABLER register.)) 0x1: CLR (CLR (Disables INTID N.))

**0x0B000200+ APCS_GICD_ISPENDRn, n=[0..8]
0x4*n****Type:** RW**Clock:** CLK**Reset State:** 0x00000000**Security Treatment:** NS-int Dependant

Each bit in the GICD_ISPENDR register provides the pending status of an interrupt. Writing a 1 to any of these bits will transition the corresponding interrupt to the Pending (or Active and Pending) state

NOTE The GICD_ISPENDR register at address offset 0x0200 is repeated once per CPU Interface. It is only accessible from the designated CPU and controls that CPU's PPI interrupts. SGI interrupt bits are read-only, however the distributor updates these using the GICD_SGIR register. The remaining registers control the SPI interrupts.

APCS_GICD_ISPENDRn

Bits	Name	Description
31:0	INT	0x0: NA (NA (No effect. Use GICD_ICPENDR register.)) 0x1: SET (SET (Sets INTID N to the Pending state.))

**0x0B000280+ APCS_GICD_ICPENDRn, n=[0..8]
0x4*n****Type:** RW**Clock:** CLK**Reset State:** 0x00000000**Security Treatment:** NS-int Dependant

Each bit in the GICD_ICPENDR register provides the pending status of an interrupt. Writing a 1 to any of these bits will transition the corresponding interrupt out of the Pending (or Active and Pending) state

NOTE The GICD_ICPENDR register at address offset 0x0280 is banked, that is there is one per CPU Interface. It is only accessible from the designated CPU and controls that CPU's PPI interrupts. SGI interrupt bits are read-only, however the distributor updates these using the GICD_SGIR register. The remaining registers control the SPI interrupts.

APCS_GICD_ICPENDRn

Bits	Name	Description
31:0	INT	0x0: NA (NA (No effect. Use GICD_ISPENDR register.)) 0x1: CLR (CLR (Clears INTID N. from the pending state.))

**0x0B000300+ APCS_GICD_ISACTIVERn, n=[0..8]
0x4*n****Type:** RW**Clock:** CLK**Reset State:** 0x00000000**Security Treatment:** NS-int Dependant

Each bit in the GICD_ISACTIVER register provides the active status of an interrupt. Writing a 1 to any of these bits will transition the corresponding interrupt to the Active (or Active and Pending) state

NOTE The GICD_ISACTIVER register at address offset 0x0300 is repeated once per CPU Interface. It is only accessible from the designated CPU and provides information for that CPU's PPI and SGI interrupts. The remaining registers control the SPI interrupts.

APCS_GICD_ISACTIVERn

Bits	Name	Description
31:0	INT	0x0: DO_NOTHING (DO NOTHING) 0x1: SET (SET)

**0x0B000380+ APCS_GICD_ICACTIVER_n, n=[0..8]
0x4*n****Type:** RW**Clock:** CLK**Reset State:** 0x00000000**Security Treatment:** NS-int Dependant

Each bit in the GICD_ICACTIVER register provides the active status of an interrupt. Writing a 1 to any of these bits will transition the corresponding interrupt out of the Active (or Active and Pending) state

NOTE The GICD_ICACTIVER register at address offset 0x0300 is repeated once per CPU Interface. It is only accessible from the designated CPU and provides information for that CPU's PPI and SGI interrupts. The remaining registers control the SPI interrupts.

APCS_GICD_ICACTIVER_n

Bits	Name	Description
31:0	INT	0x1: CLR (CLR) 0x0: DO_NOTHING (DO NOTHING)

**0x0B000400+ APCS_GICD_IPRIORITYR_n, n=[0..71]
0x4*n****Type:** RW**Clock:** CLK**Reset State:** 0x00000000**Security Treatment:** NS-int Dependant

Each field in the GICD_IPRIORITYR registers controls the priority level of an interrupt.

NOTE The GICD_IPRIORITYR registers at address offsets 0x0400 to 0x41C are repeated once per CPU Interface. These are only accessible from the designated CPU and control the priority levels for that CPU's PPI and SGI interrupts. The remaining registers control the SPI interrupt priority levels.

NOTE This register is byte addressable.

The GICD_IPRIORITYR register behaves differently depending on the **security** setting of interrupt associated with the address written (NS-int) and the **security** of the bus transaction (NS-prot). Figure 26-1 shows how these differences affect different types of access.

Figure 26-1 Summary of the secure and non-secure views of interrupt priority

APCS_GICD_IPRIORITYR_n

Bits	Name	Description
31:27	INT3	
23:19	INT2	

APCS_GICD_IPRIORITYRn (cont.)

Bits	Name	Description
15:11	INT1	
7:3	INT0	All bits cleared (0x0) is the highest priority. All bits set (5'b11111) is the lowest priority. For non-secure write access, the MSB is always set (1). For non-secure read access, the MSB is always clear (0). Non-secure entities can't use the highest priorities without secure software setting such a priority. The highest priorities are typically reserved for secure software. On the other hand, secure software can, if needed, write and read the MSB. Secure entities have access to all priorities. Secure interrupts can use priority numbers 5'b10000-5'b11111 (the lower half of the priority spectrum) if needed. Due to strict 'less than' comparisons used in the CPU interface, setting an interrupt's priority to 5'b11111 effectively disables that interrupt. The number of priorities actually supported is 32. Software can determine this number by writing 0xFF and reading back the result.

**0x0B000800+ APCS_GICD_ITARGETSRn, n=[0..71]
0x4*n****Type:** RW**Clock:** CLK**Reset State:** 0x00000000**Security Treatment:** NS-int Dependant

Each field in the GICD_ITARGETSR registers controls the destination CPU(s) of an interrupt.

NOTE The GICD_ITARGETSR registers at address offsets 0x0800 to 0x81C are not actually implemented. (PPI interrupts, being private, only have one destination CPU. SGI interrupts are generated by the GICD_SGIR register only and do not need the GICD_ITARGETSR.) The remaining registers control the SPI interrupt destinations.

NOTE This register is byte addressable.

APCS_GICD_ITARGETSRn

Bits	Name	Description
31:24	INT3	
23:16	INT2	
15:8	INT1	

APCS_GICD_ITARGETSRn (cont.)

Bits	Name	Description
7:0	INT0	Each bit represents one of 8 CPUs. Bit 0 represents the zeroth CPU, bit 7 represents the seventh CPU. Setting this field to 0 will disable the interrupt as no CPU will see it set. For INTIDs < 32 (the SGI and PPI), reads return the bit corresponding to the CPU performing the read (i.e. CPU 0 reads back 0x01010101). Writes are ignored. The number of CPUs actually supported is determined by NUM_CPU generic. Software can determine this number by writing 0xFF and reading back the result.

**0x0B000C00+APCS_GICD_ICFGRn, n=[0..17]
0x4*n****Type:** RW**Clock:** CLK**Reset State:** 0x55555555**Security Treatment:** NS-int Dependant

Each field in the GICD_ICFGR registers allows:

- control of type of an SPI:
- level-sensitive
- edge-sensitive
- reading the type of PPIs and SGIs

NOTE: The GICD_ICFGR registers at address offsets 0x0C00 to 0xC04 are repeated once per CPU Interface. These are only accessible from the designated CPU and allow read access for that CPU's PPI and SGI interrupts. The remaining registers control the SPI interrupts.

NOTE: PPI(0) is the maintenance interrupt and must be configured as level type.

NOTE: This register is not byte addressable, interrupts should be disabled before configuring their type.

NOTE: Reset state for GICD_ICFGR0 is 0xaaaaaaaa.

APCS_GICD_ICFGRn

Bits	Name	Description
31:30	INT15	
29:28	INT14	
27:26	INT13	
25:24	INT12	
23:22	INT11	

APCS_GICD_ICFGRn (cont.)

Bits	Name	Description
21:20	INT10	
19:18	INT9	
17:16	INT8	
15:14	INT7	
13:12	INT6	
11:10	INT5	
9:8	INT4	
7:6	INT3	
5:4	INT2	
3:2	INT1	
1:0	INT0	0x0: LVL_N_TO_N (LVL N to N) 0x1: LVL_1_TO_N (LVL 1 to N) 0x2: EDGE_N_TO_N (EDGE N to N) 0x3: EDGE_1_TO_N (EDGE 1 to N) These bits behave differently for each of the three interrupt types as follows: - For SGI, read back as 0b10 - SGI interrupts are edge-sensitive and use the N-N model. - For PPI/SPI, read/writable as 0b01 or 0b11 - PPI/SPI interrupts can be edge or level sensitive and use the 1-N model.)

**0x0B000F10+ APCS_GICD_CPENDSGIRn, n=[0..3]
 0x4*n**
Type: RW**Clock:** CLK**Reset State:** 0x00000000**Security Treatment:** Banked

Each field in the GICD_CPENDSGIR register, when written as 1, clears the corresponding pending bit of the source CPU for the corresponding SGI.

NOTE If the SGI is marked secure, only a secure access can access that SGI's pending bits.

NOTE This register is byte addressable.

APCS_GICD_CPENDSGIRn

Bits	Name	Description
31:24	SGI3	

APCS_GICD_CPENDSGIRn (cont.)

Bits	Name	Description
23:16	SGI2	
15:8	SGI1	
7:0	SGI0	Each bit represents one of 8 CPUs. Bit 0 represents the zeroth CPU, bit 7 represents the seventh CPU. Writing a 1 to any bit clears the pending status bit from the CPU to the SGI.

**0x0B000F20+ APCS_GICD_SPENDSGIRn, n=[0..3]
0x4*n****Type:** RW**Clock:** CLK**Reset State:** 0x00000000**Security Treatment:** Banked

Each field in the GICD_SPENDSGIR register, when written as 1, sets the corresponding pending bit of the source CPU for the corresponding SGI.

NOTE If the SGI is marked secure, only a secure access can access that SGI's pending bits.

NOTE This register is byte addressable.

APCS_GICD_SPENDSGIRn

Bits	Name	Description
31:24	SGI3	
23:16	SGI2	
15:8	SGI1	
7:0	SGI0	Each bit represents one of 8 CPUs. Bit 0 represents the zeroth CPU, bit 7 represents the seventh CPU. Writing a 1 to any bit sets the pending status bit from the CPU to the SGI.

**0x0B001100+ APCS_GICH_LRn, n=[0..3]
0x4*n****Type:** RW**Clock:** CLK**Reset State:** 0x00000000**Security Treatment:** Common

The GICH_LRn provides access to each of the list registers in the Hypervisor Interface.

APCS_GICH_LRn

Bits	Name	Description
31	HW	Indicates that this is a Hardware interrupt. When this bit is set, bits 19:10 contain the Physical interrupt number, and when this interrupt is subject to an EOI request, a Deactivate Interrupt request will be sent to the Interrupt Distributor.
30	GRP	Indicates that this interrupt should be treated as Group1. This means that the interrupt will present as an IRQ if GICV_CTLR[EnableG1] bit is set. If the GRP bit is clear, the interrupt is put into Group0 and will present as either IRQ or FIQ depending on the setting of the GICV_CTLR[FIQen] but only if GICV_CTLR[EnableG0] bit is also set. This bit is also used with GICV_CTLR[CBPR] to determine which Binary Point register is used in GICV for preemption calculations.
29:28	STATE	0x0: IDL (IDL (INVALID)) 0x1: PNA (PNA (PENDING, NOT ACTIVE)) 0x2: A (A (ACTIVE)) 0x3: PA (PA (PENDING and ACTIVE)) NOTE: for hardware interrupts, the PENDING+ACTIVE state is held in the physical Distributor rather than the VCPUIF. The PENDING+ACTIVE state (11) should only be used for software originated interrupts (Virtual Devices) or SGIs.)
27:23	PRI	Indicates the priority of this interrupt. This is converted to a traditional 8 bit QGIC priority value by appending three zero bits.
19:10	PHY_ID	The physical ID of the interrupt (which SGI/PPI/SPI is it?). When the HW bit is set this field indicates the physical interrupt ID, which is forwarded to the Distributor when this interrupt is subject to an EOI request. See below for details when the HW bit is clear. If the Physical ID is between 16 and 31 inclusive (PPI space), the Deactivation will apply to the PPI associated with the same physical CPU as the VCPUIF making the request. If the Physical ID is between 0 and 15 inclusive (SGI space), the Deactivation will still apply to SGI associated with the same physical CPU as the VCPUIF making the request but it is expected that the HW bit and Physical ID will NOT be used for SGIs.
9:0	VIRTL_ID	Indicates the Virtual ID. This ID is returned to the Guest OS when this interrupt is acknowledged via the GICV_IAR (Interrupt Acknowledge register).

9 OCIMEM Wrapper CSR

9.1 OCIMEM_CSR

Description of OCIMEM Control and Status Register Space.

9.1.1 OCIMEM_CSR

OCIMEM Core Control Registers

Description of OCIMEM Control and Status Register Space.

0x0005000C OCIMEM_GEN_STAT

Type: R

Reset State: 0x00000000

OCIMEM_GEN_STAT

Bits	Name	Description
1	AXI_ERR_SLEEP	Access to memory in sleep mode error. This field set on the first AXI error and is cleared by writing OCIMEM_INTC_CLR.AXI_ERR_INT_CLR
0	AXI_ERR_DECODE	Address decode error on an AXI port. This field set on the first AXI error and is cleared by writing OCIMEM_INTC_CLR.AXI_ERR_INT_CLR

0x00050010 OCIMEM_INTC_CLR

Type: RW

Reset State: 0x00000000

OCIMEM Interrupt Control. Writing a '1' to any bit field clears the associated interrupt. Writing follow-up '0s' is not required

OCIMEM_INTC_CLR

Bits	Name	Description
0	AXI_ERR_INT_CLR	Clear AXI error interrupt

0x00050014 OCIMEM_INTC_MASK**Type:** RW**Reset State:** 0x00000001

OCIMEM Interrupt Mask. This register is used to mask (disable) interrupt sources.

OCIMEM_INTC_MASK

Bits	Name	Description
0	AXI_ERR_INT_MSK	AXI error interrupt mask bit

0x00050018 OCIMEM_INTC_STAT**Type:** R**Reset State:** 0x00000000

OCIMEM Interrupt Status register.

OCIMEM_INTC_STAT

Bits	Name	Description
0	AXI_ERR_INT_STAT	AXI Address decode error. Query OCIMEM_GEN_STAT register for additional details on error. Clear by writing OCIMEM_INTC_CLR.AXI_ERR_INT_CLR

0x0005001C OCIMEM_OSW_STATUS**Type:** R**Reset State:** 0x00000001

OCIMEM OSW Status Register

OCIMEM_OSW_STATUS

Bits	Name	Description
5	OSW_RRESP_FIFO_FULL	Indicate whether FIFO is full. 0 - NOT FULL 1 - FULL
4	OSW_WRESP_FIFO_FULL	Indicate whether FIFO is full. 0 - NOT FULL 1 - FULL
3	OSW_RDDATA_FIFO_FULL	Indicate whether FIFO is full. 0 - NOT FULL 1 - FULL
2	OSW_WRDATA_FIFO_FULL	It indicates whether the Write Data FIFO within OSW0 is currently FULL or not.
1	OSW_CMDFIFO_FULL	It indicates whether the Command FIFO within OSW0 is currently FULL or not.
0	OSW_STAT_IDLE	It indicates whether OSW0 is currently IDLE or BUSY.

0x00050034 OCIMEM_PSCGC_TIMERS**Type:** RW**Reset State:** 0x00000F0F

This register sets the timers that ensure that there is sufficient margin between the memory clocks turning off/on and the memory sleep signals begin asserted/de-asserted

OCIMEM_PSCGC_TIMERS

Bits	Name	Description
11:8	TO_SLEEP_COUNTER	4 bit count down, must be 20ns
3:0	WAKEUP_COUNTER	4 bit count down, must be 20ns

0x00050038 OCIMEM_PSCGC_STAT**Type:** R**Reset State:** 0x00000000

This register provides the clk-off status of each memory cgc controller.

OCIMEM_PSCGC_STAT

Bits	Name	Description
0	M0_PSCGC_CLKOFF	Indicates clock-off status for the associated memory macro

0x0005003C OCIMEM_PSCGC_M0_M7_CTL**Type:** RW**Reset State:** 0x00000002

Memory Banks 0 to 7 PSCGC Power Control. This register is used to perform fine-grain sleep control over OCIMEM memory macros.

OCIMEM_PSCGC_M0_M7_CTL

Bits	Name	Description
2:0	M0_PSCGC_CTL	Controls the inputs {async_clk_halt, up_force_core_on, up_force_periph_on} for the 3 macros in Bank 0 in Elan. These macros are put to sleep/waken together. bit [2 1 0] 0 0 0 = norm_passthru - upstream sleep-controller is in control (default) 0 1 0 = norm_force_core_on - memory core is forced on 0 0 1 = norm_force_periph_on - peripheral controller is forced on 0 1 1 = norm_force_all_on - peripheral controller and core are forced on 1 1 0 = slp_ret - sleep with retention, periph off 1 1 1 = slp_ret_periph_on - sleep with retention, periph on 1 0 0 = slp_nret - sleep without retention other = Not supported Note: 1. typical sequence from normal to sleep with retention state sequence (one prgm write): norm_passthru -> slp_ret 2. typical sequence from sleep with retention to normal state sequence (two prgm writes): slp_ret -> norm_force_core_on -> norm_passthru 3. typical sequence from normal to sleep no retention state sequence (one prgm write): norm_passthru -> slp_nret 4. typical sequence from sleep no retention to normal state sequence (one prgm write): slp_nret -> norm_passthru

0x00050060 OCIMEM_ERR_ADDRESS**Type:** R**Reset State:** 0x00000000

The OCIMEM_ERR_ADDRESS register contains the lower (from NOC slave port) address of the request that caused the AXI bus error. This register will be updated by hardware upon the detection of an AXI address decode or sleep error. Once set, the register will be locked until bit OCIMEM_INTC_CLR.AXI_ERR_INT_CLR is written to clear the error.

OCIMEM_ERR_ADDRESS

Bits	Name	Description
31:0	ERROR_ADDRESS	Lower (System NOC Slave port) address of the request that caused the bus error. For Elan 2 MB is reserved so MSB is bit 20:3 are significant

0x00050064 OCIMEM_AXI_ERR_SYNDROME

Type: R

Reset State: 0x00000000

The OCIMEM_AXI_ERR_SYNDROME register provides various data about the request that caused the AXI bus error. This register will be updated by hardware upon the detection of an address decode or sleep error. Once set, the register will be locked until bit OCIMEM_INTC_CLR.AXI_ERR_INT_CLR is written to clear the error.

OCIMEM_AXI_ERR_SYNDROME

Bits	Name	Description
31:29	ERROR_ABID	The Bus ID of the request that caused the bus error.
28:24	ERROR_APID	The Port ID of the request that caused the bus error.
23:16	ERROR_AMID	The Master ID of the request that caused the bus error.
14:8	ERROR_ATID	The Transfer ID of the request that caused the bus error
5	ERROR_ABURST	The ABURST of the request that caused the bus error
4	ERROR_AWRITE	The awrite of the request that caused bus error
3:0	ERROR_ALEN	The burst length of the request that caused bus error

10 WCSS

10.1 WCSS_BT_SYSCCLK

10.1.1 WCSS_A_BT_SYSCCLK_BT_SYSCCLK

0x0A440050 WCSS_A_BT_SYSCCLK_CMPR_EN

Type: RW
Clock: PCLK
Reset State: 0x00000000

WCSS_A_BT_SYSCCLK_CMPR_EN

Bits	Name	Description
1	CMPR_INT_STAT_1	Comparator Interrupt Status
0	CMPR_EN_0	0x1: Enable comparator interrupt

0x0A440054 WCSS_A_BT_SYSCCLK_CMPR_INTGR

Type: RW
Clock: PCLK
Reset State: 0x00000000

WCSS_A_BT_SYSCCLK_CMPR_INTGR

Bits	Name	Description
27:0	CMPR_INTGR_27_0	Integer part of the comparator

0x0A440058 WCSS_A_BT_SYSCCLK_CMPR_FRACT

Type: RW
Clock: PCLK
Reset State: 0x00000000

WCSS_A_BT_SYSCLK_CMPR_FRACT

Bits	Name	Description
12:0	CMPR_FRACT_12_0	Fractional part of the comparator

0x0A440070 WCSS_A_BT_SYSCLK_COUNTVAL**Type:** RW**Clock:** PCLK**Reset State:** 0x0000FA00**WCSS_A_BT_SYSCLK_COUNTVAL**

Bits	Name	Description
16:0	COUNTVAL_16_0	Average number of 16 MHz cycles in an LPO clock cycle. A read returns the last calculated lpo_countval. A write forces a new lpo_countval to be loaded. Bit 16:7 is the integer portion of lpo_countval. Bit 6:0 is the fractional portion of lpo_countval.

0x0A440100 WCSS_A_BT_SYSCLK_CONTROL**Type:** RW**Clock:** PCLK**Reset State:** 0x00000000

Calibration Control

WCSS_A_BT_SYSCLK_CONTROL

Bits	Name	Description
2	CONTROL_2	0x0: selects calibration length of 64. 0x1: selects calibration length of 128.
1:0	CONTROL_1_0	0x0: Calibrator disabled. 0x1: Calibrate once when issued. 0x2: Calibrate when issued, repeat calibration after exiting deep sleep. 0x3: Calibrate continuously.

0x0A440104 WCSS_A_BT_SYSCCLK_COMPLETE**Type:** RW**Clock:** PCLK**Reset State:** 0x00000000**WCSS_A_BT_SYSCCLK_COMPLETE**

Bits	Name	Description
0	COMPLETE_0	Single sticky bit that is set when the calibration is complete. This bit can be set or cleared by writing to this register.

0x0A440120 WCSS_A_BT_SYSCCLK_CLKN_CAPTURE_CTL**Type:** RW**Clock:** PCLK**Reset State:** 0x00000000**WCSS_A_BT_SYSCCLK_CLKN_CAPTURE_CTL**

Bits	Name	Description
2	CLKN_CAPTURE_BUSY	Indicates hardware is enabled to generate CLKN_PULSE_32K and pulse has not completed 0x0: Idle 0x1: Busy
1	CLKN_PULSE_32K	One 32K768 pulse status
0	CLKN_CAPTURE_PULSE_EN	Write 1 to enable the generation of a pulse on the next edge of the 32k768 sleep clock. Bit is self clearing.

0x0A440124 WCSS_A_BT_SYSCCLK_CLKN_CAPTURE_INTEGER_VAL**Type:** R**Clock:** PCLK**Reset State:** 0x00000000**WCSS_A_BT_SYSCCLK_CLKN_CAPTURE_INTEGER_VAL**

Bits	Name	Description
27:0	CLKN_INTEGER	integer value captured by CLKN_PULSE_32K.

0x0A440128 WCSS_A_BT_SYSCLK_CLKN_CAPTURE_FRACT_VAL

Type: R
Clock: PCLK
Reset State: 0x00000000

WCSS_A_BT_SYSCLK_CLKN_CAPTURE_FRACT_VAL

Bits	Name	Description
12:0	CLKN_FRACT	fractional value captured by CLKN_PULSE_32K.

0x0A440130 WCSS_A_BT_SYSCLK_MASTER_MEM_ADDR

Type: RW
Clock: PCLK
Reset State: 0x0000FB29

WCSS_A_BT_SYSCLK_MASTER_MEM_ADDR

Bits	Name	Description
15:0	MASTER_ADDR	Upper 16 bits of address (31:16) used for LC accesses to AHB address space. LC sources the lower AHB addresses bits (15:0).

0x0A440134 WCSS_A_BT_SYSCLK_SLAVE_BT_ADDR

Type: RW
Clock: PCLK
Reset State: 0x0000FB00

WCSS_A_BT_SYSCLK_SLAVE_BT_ADDR

Bits	Name	Description
15:7	SLAVE_ADDR	Upper 9 bits of address (31:23) used for LC accesses to crypto (through AHB).
6:0	RESERVED_X134	

0x0A4400A0 WCSS_A_BT_SYSCLK_PVC_SSB12_INTR

Type: RW
Clock: PCLK
Reset State: 0x00000008

Raw status

WCSS_A_BT_SYSCLK_PVC_SSB12_INTR

Bits	Name	Description
8	PVC_RESET	0x1: reset all PVC hardware (FIFO, PVC, SSB1). 0x0: no reset.
7	FLUSH_PVC_FIFO	0x1: Flush PVC FIFO. 0x0: no flush.
6	ENABLE_PVC_BUSY_INTR	0x1: enable PVC_BUSY condition to activate SSB12_INTR interrupt line. 0x0: disabled.
5	PVC_BUSY	0x1: PVC SSB12 hardware port is busy. 0x0: Software performs SSB12 reads/writes when this bit is unset, to allow PVC traffic priority.
4	SSB12_INT	0x1: SSB12_INT_ENABLE[0] is set and SSB12_RD_STATUS[27] is set.
3	PVC_FIFO_EMPTY	Raw PVC_FIFO_EMPTY status
2	ENABLE_PVC_FIFO_EMPTY_INTR	0x1: enable PVC_FIFO_EMPTY condition to activate SSB12_INTR interrupt line.
1	PVC_FIFO_FULL	Raw PVC_FIFO_FULL status.
0	ENABLE_PVC_FIFO_FULL_INTR	0x1: enable PVC_FIFO_FULL condition to activate SSB12_INTR interrupt line.

10.2 WCSS_BT**10.2.1 WCSS_A_BT_BT_LINK_CTRL****0x0A3FF000 WCSS_A_BT_INT_0_INTERRUPT_CONTROL****Type:** RW**Clock:** PCLK**Reset State:** 0x00000000**WCSS_A_BT_INT_0_INTERRUPT_CONTROL**

Bits	Name	Description
0	INT0EN	Main enable for Int0 0x0: Disable 0x1: Enable, allows unmasked events to generate an interrupt

0x0A3FF004 WCSS_A_BT_INT_0_BTC_INTERRUPT_MASK**Type:** RW**Clock:** PCLK**Reset State:** 0x00000000

These registers do not affect the raw status interrupt register.

WCSS_A_BT_INT_0_BTC_INTERRUPT_MASK

Bits	Name	Description
9	MASKRXABORT	Reserved - rxabort does not generate interrupt independent of mask setting
8	MASKRSSI	Reserved - rssi does not generate interrupt independent of mask setting
7	MASKFS	0x0: Masked 0x1: Unmasked
6	MASKTXABORT	0x0: Masked 0x1: Unmasked
5	MASKTX	0x0: Masked 0x1: Unmasked
4	MASKLC	0x0: Masked 0x1: Unmasked
3	MASKHALF	0x0: Masked 0x1: Unmasked
2	MASKRX	0x0: Masked 0x1: Unmasked
1	MASKSLOT	0x0: Masked 0x1: Unmasked
0	MASKPRE	0x0: Masked 0x1: Unmasked

0x0A3FF008 WCSS_A_BT_INT_0_BTC_INTERRUPT_STATUS_AND_ACKNOWLEDGE**Type:** RW**Clock:** PCLK**Reset State:** 0x0000000A

The Int0En and corresponding mask registers do not affect the raw status interrupt register.

WCSS_A_BT_INT_0_BTC_INTERRUPT_STATUS_AND_ACKNOWLEDGE

Bits	Name	Description
9	STATRXABORT	Raw Rx abort status 0x0: No abort 0x1: abort
8	STATRSSI	RSSI capture from serial interface status 0x0: RSSI not captured 0x1: RSSI captured
7	STATFS	Raw coex_frame_sync interrupt status 0x0: No interrupt 0x1: Interrupt is pending
6	STATTXABORT	Raw Tx abort interrupt status 0x0: No interrupt 0x1: Interrupt is pending
5	STATTX	Raw Tx interrupt status 0x0: No interrupt 0x1: Interrupt is pending
4	STATLC	Raw LC interrupt status 0x0: No interrupt 0x1: Interrupt is pending
3	STATHALF	Raw pre-halfslot interrupt status 0x0: No interrupt 0x1: Interrupt is pending
2	STATRX	Raw Rx interrupt status 0x0: No interrupt 0x1: Interrupt is pending
1	STATSLOT	Raw slot-timer interrupt status 0x0: No interrupt 0x1: Interrupt is pending
0	STATPRE	Raw preslot interrupt status 0x0: No interrupt 0x1: Interrupt is pending

0x0A3FF00C WCSS_A_BT_INT_0_BTC_PRE SLOT_FILTER**Type:** RW**Clock:** PCLK**Reset State:** 0x00000000

Int0 preslot filter control and status; shared down counter that is loaded or cleared by writing to this address decrements every preslot timeout until the down counter reaches zero

WCSS_A_BT_INT_0_BTC_PRE SLOT_FILTER

Bits	Name	Description
7:0	PRESLOTFLTR	Range is 0 to 0xFF (255) preslot time-outs Write : starts the filter by loading the number of consecutive preslot time-outs to be filtered; writing a zero immediately disables the filter. Read : returns the current number of preslots still to be filtered; zero indicates that the filter is no longer active.

0x0A3FF010 WCSS_A_BT_INT_0_BTC_SLOT_BURST_TIMER**Type:** RW**Clock:** PCLK**Reset State:** 0x00000000

Int0 programmable slot timer; generates an interrupt at a specific time during a slot.

WCSS_A_BT_INT_0_BTC_SLOT_BURST_TIMER

Bits	Name	Description
13	HALFSLOT	0x0: Use half slot 0 0x1: Use half slot 1
12:0	SLOTTIMER	Indicates at which fractional bt clock location to fire the interrupt Time during slot=[312.5 usec * SlotTimer(13)]+[62.5 nsec * SlotTimer(12:0)] Fractional range=0 to 0x1387 (4999) resolution = 62.5 nsec

0x0A3FF014 WCSS_A_BT_BLUETOOTH_CLOCK_N_COMPARATOR_INTERRUPT_CONTROL_STATUS**Type:** RW**Clock:** PCLK**Reset State:** 0x00000000

ClkCmpEn does not affect this register's raw status.

WCSS_A_BT_BLUETOOTH_CLOCK_N_COMPARATOR_INTERRUPT_CONTROL_STATUS

Bits	Name	Description
5	CLKCMPINT2	Raw clock N comparator interrupt status. 0x0: No interrupt 0x1: interrupt is pending

WCSS_A_BT_BLUETOOTH_CLOCK_N_COMPARATOR_INTERRUPT_CONTROL_STATUS

Bits	Name	Description
4	CLKCMPEN2	Clock N comparator interrupt enable 0x0: Disables interrupt and clears interrupt status ClkCmplInt = 0 0x1: Enables interrupt
3	CLKCMPINT1	Raw clock N comparator interrupt status 0x0: No interrupt 0x1: interrupt is pending
2	CLKCMPEN1	Clock N comparator interrupt enable 0x0: Disables interrupt and clears interrupt status ClkCmplInt = 0 0x1: Enables interrupt
1	CLKCMPINT0	Raw clock N comparator interrupt status 0x0: No interrupt 0x1: interrupt is pending
0	CLKCMPEN0	Clock N comparator interrupt enable 0x0: Disables interrupt and clears interrupt status ClkCmplInt = 0 0x1: Enables interrupt

0x0A3FF018 WCSS_A_BT_CAPTURED_BT_CLKN_INTEGER_PORTION**Type:** R**Clock:** PCLK**Reset State:** 0x00000000**WCSS_A_BT_CAPTURED_BT_CLKN_INTEGER_PORTION**

Bits	Name	Description
27:0	BTCLKN_40_13	Bits 40:13 of the clkn captured instantly when a Tx or Rx hard align

0x0A3FF01C WCSS_A_BT_CAPTURED_BT_CLKN_FRACTIONAL_PORTION**Type:** R**Clock:** PCLK**Reset State:** 0x00000000**WCSS_A_BT_CAPTURED_BT_CLKN_FRACTIONAL_PORTION**

Bits	Name	Description
12:0	BTCLKN_12_0	Bits 12:0 of the clkn captured instantly when a Tx or Rx hard align

0x0A3FF020 WCSS_A_BT_CAPTURED_BT_CLKA_INTEGER_PORTION

Type: R
Clock: PCLK
Reset State: 0x00000000

WCSS_A_BT_CAPTURED_BT_CLKA_INTEGER_PORTION

Bits	Name	Description
27:0	BTCLKA_40_13	Bits 40:13 of the local ClkA captured instantly when a Tx or Rx hard align

0x0A3FF024 WCSS_A_BT_CAPTURED_BT_CLKA_FRACTIONAL_PORTION

Type: R
Clock: PCLK
Reset State: 0x00000000

WCSS_A_BT_CAPTURED_BT_CLKA_FRACTIONAL_PORTION

Bits	Name	Description
12:0	BTCLKA_12_0	Bits 12:0 of the local ClkA captured instantly when a Tx or Rx hard align

0x0A3FF028 WCSS_A_BT_BT_CLOCK_CONTROL

Type: RW
Clock: PCLK
Reset State: 0x00000000

This register does not apply to LE.

WCSS_A_BT_BT_CLOCK_CONTROL

Bits	Name	Description
2	CLKNLDSEL	Select edge detector circuit to use for coex_frame_sync 0x0: use qct_edge_detect component 0x1: use edge detector from register delays
1	CLKNLDEN	Enable load of BT CLKN on a rising edge of LTE coex_frame_sync 0x0: Disabled 0x1: Load BT ClkN value into LTE Frame Sync Captured ClkN registers on rising edge of coex_frame_sync

WCSS_A_BT_BT_CLOCK_CONTROL (cont.)

Bits	Name	Description
0	LDFHS	Load clk(27:2) from the last received FHS packet into the Bt ClkA(27:2) counter; bit(1:0) of Bt ClkA remains unchanged 0x0: No change 0x1: Load FHS clk 27:2 into Bt ClkA27:2 fractional bits stall during the load

0x0A3FF02C WCSS_A_BT_BT_CLOCK_ALU_COMMAND**Type:** RW**Clock:** PCLK**Reset State:** 0x00000000

This is the Bluetooth clock ALU command register and issues (writes) the ALU command as programmed in the individual registers. It does not handle long LE packets because of the MOD 5000 operation inside the ALU.

WCSS_A_BT_BT_CLOCK_ALU_COMMAND

Bits	Name	Description
18:16	OP_0	Selects input 0 of the Bluetooth clock ALU 0x0: Zeroes 0x1: bt_clk operand register 0x2: bt_clkn 0x3: bt_clka 0x4: bt_clk accumulator register 0x5: bt_clk_cap and zeroes
14:12	OP_1	Selects input 1 of the Bluetooth clock ALU 0x0: Zeroes 0x1: bt_clk operand register 0x2: bt_clkn 0x3: bt_clka 0x4: bt_clk accumulator register 0x5: bt_clk_cap and zeroes
8	OP	Executed by the Bluetooth clock ALU 0x0: Add OP_0 + OP_1 0x1: Subtract OP_0 - OP_1
6	LR	Load accumulator register with entire ALU result; not affected by update fields. 0x0: Do not capture alu result in accumulator register 0x1: Update the integer IntgrAcc and fractional FracAcc accumulator registers with the ALU result for read-back or accumulation or both

WCSS_A_BT_BT_CLOCK_ALU_COMMAND (cont.)

Bits	Name	Description
5	LN	Load ClkN counter with the entire or partial ALU result. 0x0: Do not capture ALU result in Bt ClkN 0x1: Update the entire or partial integer and fractional Bt ClkN counters with the alu result according to the update fields. Fractional bits stall during the load.
4	LA	Load ClkA counter with the entire or partial ALU result. 0x0: Do not capture ALU result in Bt ClkA 0x1: Update the entire or partial integer and fractional Bt ClkA counters with the ALU result according to update fields. Fractional bits stall during load
3	UU	Update upper Bt Clk field allows bit(27:2) of Bt ClkN or Bt ClkA, or both, to be updated when an LN or LA, or both, ALU command is issued. 0x0: Do not update bit 27:2 0x1: Update bit 27:2
2	U1	Update Bt Clk (1) field (U1) allows bit(1) of Bt ClkN and/or Bt ClkA to be updated when a LN and/or LA alu command is issued. 0x0: Do not update bit 1 0x1: Update bit 1
1	U0	Update Bt Clk (0) field allows bit(0) of Bt ClkN and/or Bt ClkA to be updated when a LN and/or LA alu command is issued. 0x0: Do not update bit 0 0x1: Update bit 0
0	UF	Update fractional Bt Clk field allows the 13-bit fractional portion of Bt ClkN and/or Bt ClkA to be updated when an LN and/or LA alu command is issued. 0x0: Do not update fractional bits 0x1: Update fractional bits

0x0A3FF030 WCSS_A_BT_BT_CLOCK_ALU_INTEGER_OPERAND**Type:** RW**Clock:** PCLK**Reset State:** 0x00000000

WCSS_A_BT_BT_CLOCK_ALU_INTEGER_OPERAND

Bits	Name	Description
27:0	INTGROPRND	Integer portion of the Bluetooth clock ALU scratchpad operand; selected as an input to the ALU by OP_0 and OP_1; does not apply to LE. Range is 0 to 0xFFFFFFFF halfslots Resolution is 312.5 usec

0x0A3FF034 WCSS_A_BT_BT_CLOCK_ALU_FRACTIONAL_OPERAND**Type:** RW**Clock:** PCLK**Reset State:** 0x00000000**WCSS_A_BT_BT_CLOCK_ALU_FRACTIONAL_OPERAND**

Bits	Name	Description
12:0	FRACOPRND	Fractional portion of the Bluetooth clock ALU scratchpad operand; selected as an input to the ALU by OP_0 and OP_1; does not apply to LE. Range = 0 to 0x1387 (4999) Resolution = 62.5 nsec

0x0A3FF038 WCSS_A_BT_BT_CLOCK_ALU_INTEGER_ACCUMULATOR**Type:** RW**Clock:** PCLK**Reset State:** 0x00000000**WCSS_A_BT_BT_CLOCK_ALU_INTEGER_ACCUMULATOR**

Bits	Name	Description
27:0	INTGRACC	Integer portion of the Bluetooth clock ALU accumulator register; shared register that is selected as an input to the ALU by OP_0 and OP_1; can also be loaded with the result of an ALU operation when the LR command bit is set; does not apply to LE. Range = 0 to 0xFFFFFFFF halfslots Resolution = 312.5 usec

0x0A3FF03C WCSS_A_BT_BT_CLOCK_ALU_FRACTIONAL_ACCUMULATOR**Type:** RW**Clock:** PCLK**Reset State:** 0x00000000**WCSS_A_BT_BT_CLOCK_ALU_FRACTIONAL_ACCUMULATOR**

Bits	Name	Description
12:0	FRACACC	Fractional portion of the Bluetooth clock ALU accumulator register; shared register that is selected as an input to the ALU by OP_0 and OP_1 can also be loaded with the result of an ALU operation when the LR command bit is set; does not apply to LE. Range = 0 to 0x1387 (4999) Resolution = 62.5 nsec

0x0A3FF040 WCSS_A_BT_LTE_FRAME_SYNC_CAPTURED_BT_CLKN_INTEGER_PORTION**Type:** R**Clock:** PCLK**Reset State:** 0x00000000**WCSS_A_BT_LTE_FRAME_SYNC_CAPTURED_BT_CLKN_INTEGER_PORTION**

Bits	Name	Description
27:0	BTCLKN_40_13	Bits 40:13 of the ClkN captured at the rising edge of coex_frame_sync

0x0A3FF044 WCSS_A_BT_LTE_FRAME_SYNC_CAPTURED_BT_CLKN_FRACTIONAL_PORTION**Type:** R**Clock:** PCLK**Reset State:** 0x00000000**WCSS_A_BT_LTE_FRAME_SYNC_CAPTURED_BT_CLKN_FRACTIONAL_PORTION**

Bits	Name	Description
12:0	BTCLKN_12_0	Bits 12:0 of the ClkN captured at the rising edge of coex_frame_sync

0x0A3FF048 WCSS_A_BT_ENCRYPTION_CONTROL**Type:** RW**Clock:** PCLK**Reset State:** 0x00000000**WCSS_A_BT_ENCRYPTION_CONTROL**

Bits	Name	Description
3	MIC_EN	For encryption and decryption when MIC is disabled, software should program the Crypto CNTR mode: 0x0: Disable MIC (for Crypto) 0x1: Enable MIC (for Crypto)
2	CRYPTO_EN	0x0: Disable Crypto 0x1: Enable Crypto for encryption, decryption
1	BRENCREN	Enables encryption for broadcast packets during transmit and receive; a broadcast packet has an all-zero am_addr field in the packet header; shall be set to 0 for LE. 0x0: Disable broadcast encryption 0x1: Enable broadcast encryption, except for FHS packets
0	ENCREN	Enables encryption for non-broadcast packets during transmit and receive; a non-broadcast packet has a non-zero am_addr field in the packet header; applies to BR/EDR and LE. 0x0: Disable non-broadcast encryption. 0x1: Enable non-broadcast encryption, except for FHS packets

0x0A3FF04C WCSS_A_BT_ENCRYPTION_K_C_KEY_0_LE_SESSION_KEY_0**Type:** RW**Clock:** PCLK**Reset State:** 0x00000000**WCSS_A_BT_ENCRYPTION_K_C_KEY_0_LE_SESSION_KEY_0**

Bits	Name	Description
31:0	ENCRKEY_31_0	KC encryption key is 128 bits long, using 4 consecutive words; loaded into the encryption engine during blending for transmit and receive.

0x0A3FF050 WCSS_A_BT_ENCRYPTION_K_C_KEY_1_LE_SESSION_KEY_1**Type:** RW**Clock:** PCLK**Reset State:** 0x00000000**WCSS_A_BT_ENCRYPTION_K_C_KEY_1_LE_SESSION_KEY_1**

Bits	Name	Description
31:0	ENCRKEY_63_32	See ENCRYPTION_K_C_KEY_0_LE_SESSION_KEY_0.

0x0A3FF054 WCSS_A_BT_ENCRYPTION_K_C_KEY_2_LE_SESSION_KEY_2**Type:** RW**Clock:** PCLK**Reset State:** 0x00000000**WCSS_A_BT_ENCRYPTION_K_C_KEY_2_LE_SESSION_KEY_2**

Bits	Name	Description
31:0	ENCRKEY_95_64	See ENCRYPTION_K_C_KEY_0_LE_SESSION_KEY_0.

0x0A3FF058 WCSS_A_BT_ENCRYPTION_K_C_KEY_3_LE_SESSION_KEY_3**Type:** RW**Clock:** PCLK**Reset State:** 0x00000000**WCSS_A_BT_ENCRYPTION_K_C_KEY_3_LE_SESSION_KEY_3**

Bits	Name	Description
31:0	ENCRKEY_127_96	See ENCRYPTION_K_C_KEY_0_LE_SESSION_KEY_0.

0x0A3FF05C WCSS_A_BT_CAPTURED_UAP**Type:** R**Clock:** PCLK**Reset State:** 0x00000000**WCSS_A_BT_CAPTURED_UAP**

Bits	Name	Description
7:0	UAPCAP	UAP field of the BdAddr(31:24) that was captured by the link controller; used to initialize the HEC and CRC LFSRs for the Tx and Rx data paths.

0x0A3FF060 WCSS_A_BT_BT_DEVICE_ADDRESS_0**Type:** RW**Clock:** PCLK**Reset State:** 0x00000000**WCSS_A_BT_BT_DEVICE_ADDRESS_0**

Bits	Name	Description
31:0	BDADDR_31_0	Bluetooth device address, which is 48 bits long using two consecutive words; used directly by the hop kernel and the encryption engine. UAP portion, BdAddr(31:24), is captured and frozen at the TxCmdTo/RxCmdTo; used to initialize the Tx and Rx HEC and CRC LFSRs For LE, the BdAddr field is used to initialize the CRC LFSR input BdAddr(23:0) used as the 24-bit CRC initialization value for the LE CRC BdAddr(47:24) are unused in LE

0x0A3FF064 WCSS_A_BT_BT_DEVICE_ADDRESS_1**Type:** RW**Clock:** PCLK**Reset State:** 0x00000000**WCSS_A_BT_BT_DEVICE_ADDRESS_1**

Bits	Name	Description
15:0	BDADDR_47_32	See BT_DEVICE_ADDRESS_0

0x0A3FF068 WCSS_A_BT_HOP_CONTROL**Type:** RW**Clock:** PCLK**Reset State:** 0x00000000**WCSS_A_BT_HOP_CONTROL**

Bits	Name	Description
7	AFHEN	Enables adaptive frequency hopping; enables the same channel mechanism and the substitution of a valid channel for a bad channel; shall be set to 0 for LE. 0x0: Disable 0x1: Enable

WCSS_A_BT_HOP_CONTROL (cont.)

Bits	Name	Description
6	OFFSEONLY	Controls the use of HopOffset when generating the hop frequency (HopFreq) for the radio synthesizer; shall be set to 1 for LE (software must program the hop frequency.) 0x0: Add HopOffset and the calculated channel to generate HopFreq; normal 0x1: Use HopOffset directly to generate HopFreq; testing/LE functional
4	FRZCLK	Controls portions of the captured Bluetooth clock when an LC command recaptures the selected Bluetooth clock (Capt0 = 1 or Capt1 = 1); shall be set to 0 for LE. 0x0: Update entire captured Bluetooth clock with the selected Bluetooth clock + 1 0x1: Freeze bits 16:12, 4:2, 0 of the captured Bluetooth clock; update rest of the captured Bluetooth clock with the selected Bluetooth clock + 1
3	PGTRAIN	Selects the page train (and koffset) to be used when the hop kernel calculates a new channel. 0x0: Use A train koffset = 24 0x1: Use B train koffset = 8
2:0	HOPTYPE	Selects the hopping sequence to be used by the hop kernel when calculating a channel; Shall be set to 000 for LE. 0x0: Page scan/inquiry scan, X00 0x1: Page/inquiry, X01 0x2: Slave page response/inquiry response 0x6: Master page response/inquiry response 0x3: Connection, X11

0x0A3FF06C WCSS_A_BT_HOP_N_COUNT**Type:** RW**Clock:** PCLK**Reset State:** 0x00000000

WCSS_A_BT_HOP_N_COUNT

Bits	Name	Description
4:0	HOPNCNT	N count to be used by the hop kernel when calculating a channel; used by the hop kernel only when HopType = master page response/inquiry response or slave page response / inquiry response; Shall be set to 0 for LE.

0x0A3FF070 WCSS_A_BT_HOP_OFFSET**Type:** RW**Clock:** PCLK**Reset State:** 0x00000000**WCSS_A_BT_HOP_OFFSET**

Bits	Name	Description
6:0	HOPOFFSET	Hop offset register, which should be set to 2 in normal mode (OffsetOnly = 0); is added to the calculated channel (0 to 78) to produce a valid frequency offset (2 to 80 Mhz). In BT test mode (OffsetOnly = 1), used directly, therefore it should be set to a valid frequency offset within the specified range Valid range = 0x02 (2) to 0x50 (80) Mhz; values out of range are not restricted.

0x0A3FF074 WCSS_A_BT_HOP_COEXISTENCE_LOW_CHANNEL**Type:** RW**Clock:** PCLK**Reset State:** 0x00000000**WCSS_A_BT_HOP_COEXISTENCE_LOW_CHANNEL**

Bits	Name	Description
6:0	LOWCHAN	Lower limit of the 802.11x range of channels (see Coexistence Command). Range = 0 to 0x4E (78)

0x0A3FF078 WCSS_A_BT_HOP_COEXISTENCE_HIGH_CHANNEL**Type:** RW**Clock:** PCLK**Reset State:** 0x00000000**WCSS_A_BT_HOP_COEXISTENCE_HIGH_CHANNEL**

Bits	Name	Description
6:0	HIGHCHAN	Upper limit of the 802.11x range of channels (see Coexistence command). Range = 0 to 0x4E (78)

0x0A3FF07C WCSS_A_BT_HOP_AFH_CHANNEL_MAP_0**Type:** RW**Clock:** PCLK**Reset State:** 0x00000000

AFH channel map, which is 79 bits long; using 3 consecutive words; shall be set to 0 for LE. The hop kernel indexes this channel map only when adaptive frequency hopping is enabled (AfhEn = 1). Each bit location represents a corresponding channel (bit 0 is channel 0, etc).

A particular channel can be avoided by setting its corresponding bit location to indicate a bad channel. When the hop kernel calculates a channel that is deemed bad according to this channel map, it substitutes a valid AFH channel.

WCSS_A_BT_HOP_AFH_CHANNEL_MAP_0

Bits	Name	Description
31:0	AFHMAP_31_0	Each bit location represents a corresponding channel (bit 0 is channel 0, etc). 0: Indicates a bad channel to be avoided. 1: Indicates a valid channel

0x0A3FF080 WCSS_A_BT_HOP_AFH_CHANNEL_MAP_1**Type:** RW**Clock:** PCLK**Reset State:** 0x00000000**WCSS_A_BT_HOP_AFH_CHANNEL_MAP_1**

Bits	Name	Description
31:0	AFHMAP_63_32	See HOP_AFH_CHANNEL_MAP_0

0x0A3FF084 WCSS_A_BT_HOP_AFH_CHANNEL_MAP_2

Type: RW
Clock: PCLK
Reset State: 0x00000000

WCSS_A_BT_HOP_AFH_CHANNEL_MAP_2

Bits	Name	Description
14:0	AFHMAP_78_64	See HOP_AFH_CHANNEL_MAP_0

0x0A3FF088 WCSS_A_BT_HOP_HALF_SLOT_0_BCA_CHANNEL

Type: RW
Clock: PCLK
Reset State: 0x00000000

WCSS_A_BT_HOP_HALF_SLOT_0_BCA_CHANNEL

Bits	Name	Description
6:0	BCA0CHAN	Channel that is added to HopOffset when a power burst measurement (bad channel assessment) LC command is issued for halfslot 0 (see Cmd0). Valid range = 0 to 0x4E (78); values out of range are not restricted.

0x0A3FF08C WCSS_A_BT_HOP_HALF_SLOT_1_BCA_CHANNEL

Type: RW
Clock: PCLK
Reset State: 0x00000000

WCSS_A_BT_HOP_HALF_SLOT_1_BCA_CHANNEL

Bits	Name	Description
6:0	BCA1CHAN	See HOP_HALF_SLOT_0_BCA_CHANNEL

0x0A3FF090 WCSS_A_BT_HOP_FREQUENCY

Type: R
Clock: PCLK
Reset State: 0x00000000

Hop frequency generated for the radio synthesizer; the high-level priority scheme that shows how the other hop registers are used to generate the hop frequency:

HopFreq <= HopOffset when (OffsetOnly = 1) else

HopOffset + Bca0Chan when (servicing Cmd0 = RxBca) else

HopOffset + Bca1Chan when (servicing Cmd1 = RxBca) else

HopOffset + calculated channel

WCSS_A_BT_HOP_FREQUENCY

Bits	Name	Description
6:0	HOPFREQ	Valid range = 0x02 (2) to 0x50 (80) Mhz. Values out of range are not restricted.

0x0A3FF094 WCSS_A_BT_LE_CHANNEL_INDEX

Type: RW

Clock: PCLK

Reset State: 0x00000000

Used as the seed to the LE whitening LFSR; not the same value as the hop frequency; this field is written once per LE event because the frequency of all packets in a LE event are the same; not used for BR/EDR packets.

WCSS_A_BT_LE_CHANNEL_INDEX

Bits	Name	Description
5:0	LECHANNELINDEX	Valid range = 0 to 39; values out of range are not restricted.

0x0A3FF098 WCSS_A_BT_HOP_FREQUENCY_1

Type: R

Clock: PCLK

Reset State: 0x00000000

Hop frequency from previous packet HOP_FREQUENCY is captured in HOP_FREQUENCY_1 at the beginning of LC RM_TUNING state

WCSS_A_BT_HOP_FREQUENCY_1

Bits	Name	Description
6:0	HOPFREQ	Valid range = 0x02 (2) to 0x50 (80) Mhz. Values out of range are not restricted.

0x0A3FF0A0 WCSS_A_BT_LC_COMMAND_HALF_SLOT_0**Type:** RW**Clock:** PCLK**Reset State:** 0x00000000

Link controller command register for halfslot 0. For write, issues the link control command as programmed in the individual registers; processed at the TxCmdTo/RxCmdTo when the selected Bluetooth clock bit(0) = 1; these time-outs precede the first halfslot. For read, returns the last command that was issued.

WCSS_A_BT_LC_COMMAND_HALF_SLOT_0

Bits	Name	Description
7	MODMODE	Selects between BR/EDR and LE for the next link controller command. 0x0: BR/EDR 0x1: LE
6	CAPT0	Capture field that controls the capturing of the selected Bluetooth clock + 1 and the UAP portion of the BdAddr when this command is issued; capturing takes place at the TxCmdTo/RxCmdTo preceding the first halfslot. 0x0: Disable capturing 0x1: Enable capturing
5:4	CALC0	Calculate field that controls whether the hop calculator runs when this command is issued; decision to run the hop calculator occurs at the TxCmdTo/RxCmdTo preceding the first halfslot. AfhEn = 0: Hop calculator runs when Cmd0 Idle AfhEn = 1: Hop calculator runs when Cmd0 Idle and bt_clk(1) = 1 0x0: Link controller automatically runs the hop calculator 0x1: Force the hop calculator to run regardless of Cmd0 0x2: Force disabling of the hop calculator regardless of Cmd0

WCSS_A_BT_LC_COMMAND_HALF_SLOT_0 (cont.)

Bits	Name	Description
3:0	CMD0	Command field that controls what activity will take place during the first halfslot. 0x0: Idle do not Tx or Rx 0x1: Tx 0x2: Rx data 0x3: Rx data X scrambler initialized with hop calculator X input 0x4: Rx ID 0x6: Rx scan data 0x7: Rx scan data X scrambler initialized with hop calculator X input 0x8: Rx scan ID 0x9: Rx BCA bad channel assessment power burst measurement 0xA: Continuous Tx test mode configured by transmit buffer 0xB: Continuous Rx Test Mod_1 1 Mbps data 0xC: Continuous Rx Test Mod_2 2 Mbps data 0xD: Continuous Rx Test Mod_3 3 Mbps data 0xE: Tx X scrambler initialized with hop calculator X input

0x0A3FF0A4 WCSS_A_BT_LC_COMMAND_HALF_SLOT_1**Type:** RW**Clock:** PCLK**Reset State:** 0x00000000**WCSS_A_BT_LC_COMMAND_HALF_SLOT_1**

Bits	Name	Description
7	MODMODE	Selects between BR/EDR and LE for the next link controller command. 0x0: BR/EDR 0x1: LE
6	CAPT1	Capture field that controls the capturing of the selected Bluetooth clock + 1 and the UAP portion of the BdAddr when this command is issued; capturing takes place at the TxCmdTo/RxCmdTo preceding the first halfslot. 0x0: Disable capturing 0x1: Enable capturing

WCSS_A_BT_LC_COMMAND_HALF_SLOT_1 (cont.)

Bits	Name	Description
5:4	CALC1	Calculate field that controls whether the hop calculator runs when this command is issued; decision to run the hop calculator occurs at the TxCmdTo/RxCmdTo preceding the first halfslot. AfhEn = 0: Hop calculator runs when Cmd0 Idle AfhEn = 1: Hop calculator runs when Cmd0 Idle and bt_clk(1) = 1 0x0: Link controller automatically runs the hop calculator 0x1: Force the hop calculator to run regardless of Cmd0 0x2: Force disabling of the hop calculator regardless of Cmd0
3:0	CMD1	Command field that controls what activity will take place during the first halfslot. 0x0: Idle do not Tx or Rx 0x1: Tx 0x2: Rx data 0x3: Rx data X scrambler initialized with hop calculator X input 0x4: Rx ID 0x6: Rx scan data 0x7: Rx scan data X scrambler initialized with hop calculator X input 0x8: Rx scan ID 0x9: Rx BCA bad channel assessment power burst measurement 0xA: Continuous Tx test mode configured by transmit buffer 0xB: Continuous Rx Test Mod_1 1 Mbps data 0xC: Continuous Rx Test Mod_2 2 Mbps data 0xD: Continuous Rx Test Mod_3 3 Mbps data 0xE: Tx X scrambler initialized with hop calculator X input

0x0A3FF0A8 WCSS_A_BT_LC_ABORT_ISSUE_AND_STATUS**Type:** RW**Clock:** PCLK**Reset State:** 0x0000000F**WCSS_A_BT_LC_ABORT_ISSUE_AND_STATUS**

Bits	Name	Description
6	COEXLATE	Coexistence command timing; late window is from the coexistence command time-out to the BtClk(0) rollover. 0x0: Coexistence command was issued on time 0x1: Coexistence command was issued late

WCSS_A_BT_LC_ABORT_ISSUE_AND_STATUS (cont.)

Bits	Name	Description
5	LC1LATE	LC command halfslot 1 timing; late window is from the appropriate Tx, Rx, or last command time-out to the BtClk rollover that marks halfslot 1. 0x0: LC command halfslot 1 was issued on time 0x1: LC command halfslot 1 was issued late
4	LC0LATE	LC command halfslot 0 timing; late window is from the appropriate Tx, Rx, or last command time-out to the BtClk rollover that marks halfslot 0. 0x0: LC command halfslot 0 was issued on time 0x1: LC command halfslot 0 was issued late
3	HOPDONE	Hop calculator state machine status 0x0: Hop calculator state machine is calculating a frequency not done 0x1: Hop calculator state machine is not calculating a frequency done
2	COEXDONE	Coexistence state machine status 0x0: Coexistence state machine is processing a command not done 0x1: Coexistence state machine is not processing a command done
1	LCDONE	Link controller state machine status 0x0: Link controller state machine is processing a command not done 0x1: Link controller state machine is not processing a command done
0	LCABORT	Link controller abort control and status Read: 0x0: link controller abort is still in progress (not done) 0x1: link controller abort has been completed (done) Write: 0x0: No change 0x1: Issue an abort to the link controller

0x0A3FF0AC WCSS_A_BT_LL_R_TRIGGER_REPETITION**Type:** RW**Clock:** PCLK**Reset State:** 0x00000000

WCSS_A_BT_LL_R_TRIGGER_REPETITION

Bits	Name	Description
13:0	REPEAT	Desired repetition of trigger AC, in unit of 32usec. Minimum value is 2 since at least a sync0 and a sync1 will be sent.

0x0A3FF0B0 WCSS_A_BT_RX_CONTROL**Type:** RW**Clock:** PCLK**Reset State:** 0x00000000**WCSS_A_BT_RX_CONTROL**

Bits	Name	Description
6	CDWINEN	Enables the carrier detect window 0x0: Disable 0x1: Enable
5	HECALL	Controls the checking of the HEC bits. To save power the receiver should be disabled immediately upon detecting a HEC error. However, the entire HEC should be received before aborting if software needs an accurate number of 1/3 FEC errors that occurred during the packet header. 0x0: Check the HEC on the fly as the bits are received and abort the receive immediately if a HEC mismatch is detected 0x1: Receive all the HEC bits before checking for a possible HEC mismatch
4	HEC1SLOT	Controls the receive abort of 1-slot packets due to an AM address mismatch. In some cases, it is vital to receive the entire packet header and check the HEC before aborting due to an AM address mismatch. For example: A slave device may not transmit during an Sco frame if the received packet (from the master) was intended for another device. To detect this, the slave device must check the AM address and verify the HEC before deciding to transmit. 0x0: Immediately abort receiving a 1-slot packet due to an AM address mismatch 0x1: Always check the HEC before aborting the receive of a 1-slot packet due to an AM address mismatch

WCSS_A_BT_RX_CONTROL (cont.)

Bits	Name	Description
3	RXAMADIS	Disables the receive AM address filter 0x0: Enable the AM address filter; only packets with a valid AM address in the base AM address array to be received; receive of packets with an invalid AM address are aborted and the AmAddrBad status is set. 0x1: Disable the AM address filter; packets with any AM address are received and the AmAddrBad status is never set
2	WININTEN	Enables asserting of the Rx interrupt for a receive window time-out; receive window time-out is generated when servicing any Rx LC command, except Rx Bca 0x0: Prevents a receive window time-out from assert Rx interrupt 0x1: Enables a receive window time-out to assert Rx interrupt
1	HDRALIGNEN	Enables slot alignment for receiving non-ID packets; enables capturing of the frame timing error FrameErr right before slot alignment. Slot alignment occurs when a received packet header is deemed valid - when the HEC matches and the number of 1/3 FEC packet header errors is below the specified threshold; the following Bluetooth clock adjustment is made: [Bt ClkA = integer captured clock(27:0) & fractional RxHdrAlign(12:0)] 0x0: Disable non-ID packet slot alignment 0x1: Enable non-ID packet slot alignment on a valid packet header; first capture the FrameErr before slot alignment
0	ACDALIGNEN	Enables slot alignment for receiving ID packets. Slot alignment occurs when the receiver detects a valid access code - when the modem correlator matches the sync word; the following Bluetooth clock adjustment is made: [Bt ClkA = integer Bt ClkA(27:2) & 00 & fractional RxAcAlign(12:0)] 0x0: Disable ID packet slot alignment 0x1: Enable ID packet slot alignment on a valid access code detect

0x0A3FF0B4 WCSS_A_BT_RX_NORMAL_APERTURE_WINDOW**Type:** RW**Clock:** PCLK**Reset State:** 0x00000000

Normal aperture receive window duration, which is opened only after receiving 70 new symbols when these LC commands are being serviced:

Rx data Rx data X Rx ID

WCSS_A_BT_RX_NORMAL_APERTURE_WINDOW

Bits	Name	Description
5:0	ACDWIN	Range = 0 to 0x3F (63) usec 0x0: Value of zero is used to open an infinite receive window

0x0A3FF0B8 WCSS_A_BT_RX_WIDE_APERTURE_WINDOW**Type:** RW**Clock:** PCLK**Reset State:** 0x00000000

Wide aperture receive window duration, which is opened only after receiving 70 new symbols when these LC commands are being serviced:

Rx scan data

Rx scan data X

Rx scan ID

WCSS_A_BT_RX_WIDE_APERTURE_WINDOW

Bits	Name	Description
21:0	SCANWIN	Range = 0 to 0x3FFFFFF (4,194,303) usec 0x0: Value of zero is used to open an infinite receive window

0x0A3FF0BC WCSS_A_BT_RX_BCA_WINDOW**Type:** RW**Clock:** PCLK**Reset State:** 0x00000000

Bad channel assessment receive window duration, which is opened immediately when the Rx Bca LC command is being serviced; software must disable the modem correlator prior to running BCA.

WCSS_A_BT_RX_BCA_WINDOW

Bits	Name	Description
8:0	BCAWIN	Range = 0 to 0x1FF (511) usec 0x0: Value of zero is used to open an infinite receive window 0x1: Value of one is used to open an infinite receive window

0x0A3FF0C0 WCSS_A_BT_SW_WHITEN_SEED

Type: RW

Clock: PCLK

Reset State: 0x00000000

Whitening seed and enable for LLR trigger

WCSS_A_BT_SW_WHITEN_SEED

Bits	Name	Description
7:1	SEED	7 bit seed
0	EN	enable signal to use the seed to initialize LFSR.

0x0A3FF0C4 WCSS_A_BT_RX_ACL_HEADER_THRESHOLD

Type: RW

Clock: PCLK

Reset State: 0x00000000

Threshold of 1/3 FEC errors allowed in an ACL packet header; does not apply to LE. If the number of 1/3 FEC errors in an ACL packet header exceeds this threshold, the receive is aborted and the HdrFecBad status is set

WCSS_A_BT_RX_ACL_HEADER_THRESHOLD

Bits	Name	Description
4:0	ACLTHRESH	Threshold of 18 or larger prevents receive from aborting Range = 0 to 0x1F (31) errors

0x0A3FF0C8 WCSS_A_BT_RX_SCO_HEADER_THRESHOLD**Type:** RW**Clock:** PCLK**Reset State:** 0x00000000

Threshold of 1/3 FEC errors allowed in a SCO packet header; does not apply to LE. If the number of 1/3 FEC errors in a SCO packet header exceeds this threshold, the receive is aborted and the HdrFecBad status is set

WCSS_A_BT_RX_SCO_HEADER_THRESHOLD

Bits	Name	Description
4:0	SCOTHRESH	Threshold of 18 or larger prevents the receive from aborting Range = 0 to 0x1F (31) errors

0x0A3FF0CC WCSS_A_BT_HOP_OFFSET_1**Type:** RW**Clock:** PCLK**Reset State:** 0x00000000**WCSS_A_BT_HOP_OFFSET_1**

Bits	Name	Description
6:0	HOPOFFSET	Hop offset register for the 2nd half-slot. Added in for LLR.

0x0A3FF0D0 WCSS_A_BT_RX_SYNC_WORD_0**Type:** RW**Clock:** PCLK**Reset State:** 0x00000000**WCSS_A_BT_RX_SYNC_WORD_0**

Bits	Name	Description
31:0	RX_SYNC_WORD_0	Bit [31:0] of the sync_word

0x0A3FF0D4 WCSS_A_BT_RX_SYNC_WORD_1**Type:** RW**Clock:** PCLK**Reset State:** 0x00000000

WCSS_A_BT_RX_SYNC_WORD_1

Bits	Name	Description
31:0	RX_SYNC_WORD_1	Bit [63:32] of the sync_word

0x0A3FF0D8 WCSS_A_BT_RX_SCO_PAYLOAD_LENGTH**Type:** RW**Clock:** PCLK**Reset State:** 0x00000000

Payload byte length that is used to receive Sco or eSco packets; Before an Sco or eSco packet is to be received, this length field must be set to the expected payload byte length, which is predefined for Sco packets but can vary, as negotiated, for eSco packets Dv packets do not use this length field; does not apply to LE

WCSS_A_BT_RX_SCO_PAYLOAD_LENGTH

Bits	Name	Description
9:0	RXSCOLNGTH	Range = 0 to 0x3FF (1023) bytes

0x0A3FF0DC WCSS_A_BT_RX_AM_ADDRESS_ARRAY_POINTER**Type:** RW**Clock:** PCLK**Reset State:** 0x00000000

Pointer to the word-aligned starting address of the receive base AM address array, located in internal RAM; not a pointer to a link-list; does not apply to LE. Receive logic reads from base AM address array offset by the received base AM address for every packet type that has a packet header

WCSS_A_BT_RX_AM_ADDRESS_ARRAY_POINTER

Bits	Name	Description
15:2	RXAMAPTR	Range = 0 to 16k words (the RAM may be sized smaller)

0x0A3FF0E0 WCSS_A_BT_RX_BUFFER_SCO_POINTER**Type:** RW**Clock:** PCLK**Reset State:** 0x00000000

Pointer to the starting address of the Sco receive buffer link-list, which with its corresponding buffer are located in the internal RAM; receive logic accesses this link-list and writes to its corresponding buffer while receiving a Sco or eSco packet with a non-zero payload byte length; (voice portion of Dv, Hv1, Hv2, Hv3, Hv4, Hv5, 2-Hv3, 2-Hv5, 3-Hv1, 3-Hv3, 3-Hv5); does not apply to LE

WCSS_A_BT_RX_BUFFER_SCO_POINTER

Bits	Name	Description
15:2	RXSCOPTR	Range = 0 to 16k words (the RAM may be sized smaller)

0x0A3FF0E4 WCSS_A_BT_RX_BUFFER_ACL_POINTER

Type: RW

Clock: PCLK

Reset State: 0x00000000

Pointer to the starting address of the Acl (non-Lmp) receive buffer link-list, which with its corresponding buffer are located in the internal RAM; Receive logic accesses this link-list and writes to its corresponding buffer while receiving a non-Lmp Acl packet with a non-zero payload byte length (data portion of non-LMP Dv, non-Lmp Dm1, Dm3, Dm5, Dh1, Dh3, Dh5, 2-Dh1, 2-Dh3, 2-Dh5, 3-Dh1, 3-Dh3, 3-Dh5) does not apply to LE

WCSS_A_BT_RX_BUFFER_ACL_POINTER

Bits	Name	Description
15:2	RXACLPTR	Range = 0 to 16k words (the RAM may be sized smaller)

0x0A3FF0E8 WCSS_A_BT_RX_BUFFER_LMP_POINTER

Type: RW

Clock: PCLK

Reset State: 0x00000000

Pointer to the starting address of the Lmp receive buffer link-list, which with its corresponding buffer is located in the internal RAM; does not apply to LE. Receive logic accesses this link-list and writes to its corresponding buffer while receiving a Lmp packet with a non-zero payload byte length (data portion of Lmp Dv, Lmp Dm1)

WCSS_A_BT_RX_BUFFER_LMP_POINTER

Bits	Name	Description
15:2	RXLMPPTR	Range = 0 to 16k words (the Ram may be sized smaller)

0x0A3FF0EC WCSS_A_BT_RX_BUFFER_CONTROL_POINTER**Type:** RW**Clock:** PCLK**Reset State:** 0x00000000

Pointer to the starting address of the control receive buffer link-list, which with its corresponding buffer are located in the internal RAM; does not apply to LE. Receive logic will access this link-list and write to its corresponding buffer while receiving a Control packet with a non-zero payload byte length (FHS, Aux)

WCSS_A_BT_RX_BUFFER_CONTROL_POINTER

Bits	Name	Description
15:2	RXCNTLPTR	Range = 0 to 16k words (the Ram may be sized smaller)

0x0A3FF0F0 WCSS_A_BT_RX_BUFFER_LE_POINTER**Type:** RW**Clock:** PCLK**Reset State:** 0x00000000

Pointer to the starting address of the LE receive buffer link-list, which with its corresponding buffer is located in the internal RAM. Receive logic will access this link-list and write to its corresponding buffer while receiving a LE packet with a non-zero payload byte length (data portion of all LE data packets)

WCSS_A_BT_RX_BUFFER_LE_POINTER

Bits	Name	Description
15:2	RXLELPTR	Range = 0 to 16k words (the Ram may be sized smaller)

0x0A3FF0F4 WCSS_A_BT_RX_LOW_POWER_CONTROL**Type:** RW**Clock:** PCLK**Reset State:** 0x00000000**WCSS_A_BT_RX_LOW_POWER_CONTROL**

Bits	Name	Description
4	LPPSBISTDONE	Once software has read the 1 result, the value is cleared automatically 0x0: BIST not done 0x1: BIST complete

WCSS_A_BT_RX_LOW_POWER_CONTROL (cont.)

Bits	Name	Description
3	LPPSBISTSTART	Initially, should be written with a 1 for start of the test prior to a half slot boundary. The first BIST period is synchronized to the halfslot; every subsequent period is fully under software control, which must also program LPPSBISTEn, LPPSBISTTimer and LowPwrEn. Must be toggled between BIST periods to start the next test 0x0: No action 0x1: Start LPPS BIST
2	LPPSBISTEN	0x0: Disable LPPS BIST 0x1: Enable LPPS BIST
1	EDABORTEN	Energy detect abort enable 0x0: Leave receiver on during duration of the scan window; hardware shuts receiver off only at the window time out interrupt; software still has option to write an LC abort during this period 0x1: Immediately turn the receiver off after the first energy detect interrupt is received
0	LOWPWREN	Low power mode enable, which enables the energy detect block in the radio/modem and also controls related logic in the link controller 0x0: Disable low power mode 0x1: Enable low power mode

0x0A3FF0FC WCSS_A_BT_SYNC_WORD**Type:** RW**Clock:** PCLK**Reset State:** 0x00000000

Programmable wait time for Sync Word detection. The enable is self cleared by HW after SW sets it.

WCSS_A_BT_SYNC_WORD

Bits	Name	Description
7:1	DELAY	In unit of usec, default to 32. SW should tune it for LLR RX
0	DELAY_SEL	self clearing after both halfslot commands are done

0x0A3FF100 WCSS_A_BT_RX_COMPOSITE_STATUS**Type:** RW**Clock:** PCLK**Reset State:** 0x00000000

Composite receive status register: Write, a write of any value to this address clears the StatValid status register Read, returns the status of the current or last receive operation

WCSS_A_BT_RX_COMPOSITE_STATUS

Bits	Name	Description
8	RXABORT	Indicates Rx abort status 0x0: coexistence rx abort has not been encountered 0x1: The receive operation has been aborted by coexistence
7	STATVALID	Indicates that the receive status is valid. Software clears this status bits by writing any value to this particular address. Hardware sets this status bit at the beginning of the receive operation when all the receive status bits are cleared 0x0: The receive status registers has not been updated by hardware 0x1: The receive status registers has been updated with the latest status
6	RXBUFERR	Receive buffer link-list overflow status (disabled by RxBufDis); always cleared at the beginning of the receive operation 0x0: Null pointer has not been encountered 0x1: The receive operation was aborted due to a Null Pointer that was read from one of the receive link-lists
5	AMADDRBAD	AM address filter status (disabled by RxAmaDis), which is always cleared at the beginning of the receive operation; does not apply to LE 0x0: Invalid AM address has not been encountered 0x1: Receive operation has been aborted for an invalid AM address; PktHdr contains new AM address and packet index fields

WCSS_A_BT_RX_COMPOSITE_STATUS (cont.)

Bits	Name	Description
4	ACDVALID	EnergyDet; operates in conjunction with the low power mode enable: always cleared at the beginning of the receive operation LowPwrEn = 0 indicates that a valid access code has been detected; 0x0: A valid access code has not been detected 0x1: A valid access code has been detected while the receive window was open and energy was detected controlled by AcdCarrier LowPwrEn = 1 indicates that energy has been detected; During a receive the value is held until software has read it even if there are more energy detection events. 0x0: No energy detected 0x1: Packet detector has detected energy
3	RXHALFSLOT	Indicates which LC halfslot command (Cmd0, Cmd1) was being serviced when the receive status bits were generated; updated at the beginning of the receive operation 0x0: Receive status is generated by Cmd0 first halfslot 0x1: Receive status is generated by Cmd1 second halfslot
2	PYLDBAD	Composite receive payload status; PyldFecBad, CrcBad, and LengthBad status bits are OR-ed together to indicate that the receive operation was aborted due to an error during the payload; does not apply to LE
1	PKTHDRBAD	Composite receive packet header status; ExtFecBad, HdrFecBad, and HecBad status bits are OR-ed together to indicate that the receive operation was aborted due to an error during the packet/extended header
0	WINTO	(Duplicated) Receive window time-out status (disabled by an infinite receive window duration); always cleared at the beginning of the receive operation 0x0: Receive window has not timed out 0x1: Receive operation ended due to a receive window time-out; valid access code was not detected or the Bca window timed out

0x0A3FF104 WCSS_A_BT_RX_DETAILED_BAD_STATUS**Type:** R**Clock:** PCLK**Reset State:** 0x00000000

WCSS_A_BT_RX_DETAILED_BAD_STATUS

Bits	Name	Description
7	PYLD FECBAD	Payload 2/3 FEC uncorrectable double error checker status (disabled by RxFecDis) always cleared at the beginning of the receive operation; does not apply to LE 0x0: An uncorrectable 2/3 FEC error has not been encountered during the payload 0x1: The receive operation has been aborted due to a 2/3 FEC uncorrectable double error in the payload
6	CRCBAD	CRC checker status (disabled by Rx_crcDis); always cleared at the beginning of the receive operation 0x0: A CRC mismatch has not been encountered 0x1: The receive operation has been aborted for a CRC mismatch; timing of the CRC check is based on the Hec_crcCapt control register
5	LENGTHBAD	Payload length checker status (disabled by RxLnghDis); always cleared at the beginning of the receive operation 0x0: Payload byte length is less-than-or-equal to the allowed maximum 0x1: Receive operation has been aborted for a payload byte length that exceeds the maximum for that particular packet type; payload length from the receive payload header is used; however, for Sco and eSco packets the programmed expected payload length RxScoLngh is used
3	HDR FECBAD	Packet header 1/3 FEC error checker status (disabled by a large threshold); always cleared at the beginning of the receive operation; does not apply to LE. 0x0: Number of 1/3 FEC errors during the packet header is within limits 0x1: Receive operation has been aborted because the number of 1/3 FEC errors during the packet header has exceeded the AclThresh or ScoThresh threshold; this check occurs at the end of the packet header and may not occur if an HEC error checked on the fly aborts the receive operation early
2	HECBAD	HEC checker status (disabled by RxHecDis); always cleared at the beginning of the receive operation; does not apply to LE. 0x0: HEC mismatch has not been encountered 0x1: Receive operation has been aborted due to a HEC mismatch; timing of the HEC check is based on the HecAll and Hec_crcCapt control registers

WCSS_A_BT_RX_DETAILED_BAD_STATUS (cont.)

Bits	Name	Description
1	AMADDRBAD	(Duplicated) AM address filter status (disabled by RxAmaDis); does not apply to LE
0	WINTO	(Duplicated) Receive window time-out status (disabled by an infinite receive window duration)

0x0A3FF108 WCSS_A_BT_RX_FEC_AND_DONE_STATUS**Type:** R**Clock:** PCLK**Reset State:** 0x00000010**WCSS_A_BT_RX_FEC_AND_DONE_STATUS**

Bits	Name	Description
6	CRCGOOD	CRC checker status (forced by RxCrcDis); always cleared at the beginning of the receive operation. 0x0: CRC has not been checked or a mismatch was encountered 0x1: 16-bit CRC field was received without any mismatches
5	HECGOOD	HEC checker status (forced by RxHecDis); always cleared at the beginning of the receive operation; does not apply to LE 0x0: HEC has not been checked or a mismatch was encountered 0x1: 8-bit HEC field was received without any mismatches
4	RXPKTDONE	Status of the receive logic; rising edge of this status bit marks the end of a receive operation. Software can use this rising edge when receiving a 60 byte 2-Hv3 with CRC to set the preslot time-out; currently the longest per-slot packet (excluding extended headers) 0x0: Receive logic is busy with a receive operation 0x1: Receive logic is not busy with a receive operation
3	FECHDRFLAG	Status of packet header 1/3 FEC error counter (FecHdrCnt); this flag and counter is not disabled by a large AclThresh or ScoThresh threshold; does not apply to LE 0x0: FecHdrCnt counter is equal to zero 0x1: FecHdrCnt counter is not equal to zero
1	FECCORFLAG	Status of payload 1/3 or 2/3 FEC single error corrected counter (FecCorCnt); this flag and counter is not disabled with PyldFecDis; does not apply to LE. 0x0: FecCorCnt counter is equal to zero 0x1: FecCorCnt counter is not equal to zero

WCSS_A_BT_RX_FEC_AND_DONE_STATUS (cont.)

Bits	Name	Description
0	FECDETFLAG	Status of payload 2/3 FEC uncorrectable double error detected counter (FecDetCnt); this flag and counter is not disabled with PyldFecDis; does not apply to LE. 0x0: FecDetCnt counter is equal to zero 0x1: FecDetCnt counter is not equal to zero

0x0A3FF10C WCSS_A_BT_RX_FRAME_TIMING_ERROR**Type:** R**Clock:** PCLK**Reset State:** 0x00000000

Receive frame timing error (updating is disabled by HdrAlignEn); when a valid packet header of a non-ID packet is detected, this register is updated with the current fractional bits of Bt ClkA right before hard-aligning the entire Bt ClkA

WCSS_A_BT_RX_FRAME_TIMING_ERROR

Bits	Name	Description
13:0	FRAMEERR	BR/EDR, range = 0 to 0x1387 (4999); resolution = 62.5 nsec LE, range = 0 to 0x3FFF (16383); resolution = 62.5 nsec; LE Usage TBD

0x0A3FF11C WCSS_A_BT_RX_PACKET_HEADER_ERROR_COUNT**Type:** R**Clock:** PCLK**Reset State:** 0x00000000

Packet header 1/3 FEC error counter (flagged by FecHdrFlag); Always cleared at the beginning of the receive operation does not apply to LE

WCSS_A_BT_RX_PACKET_HEADER_ERROR_COUNT

Bits	Name	Description
4:0	FECHDRCNT	Range = 0 to 0x12 (18) bit errors (limited by the packet header length)

0x0A3FF124 WCSS_A_BT_RX_PAYLOAD_2_3_FEC_ERROR_DETECTED_COUNT

Type: R
Clock: PCLK
Reset State: 0x00000000

Payload 2/3 FEC uncorrectable double error detected counter (flagged by FecDetFlag); Always cleared at the beginning of the receive operation does not apply to LE

WCSS_A_BT_RX_PAYLOAD_2_3_FEC_ERROR_DETECTED_COUNT

Bits	Name	Description
7:0	FECDETCNT	Range = 0 to 0xFF (255) bit errors

0x0A3FF128 WCSS_A_BT_RX_PAYLOAD_FEC_ERROR_CORRECTED_COUNT

Type: R
Clock: PCLK
Reset State: 0x00000000

Payload 1/3 or 2/3 FEC single error corrected counter (flagged by FecCorFlag); Always cleared at the beginning of the receive operation does not apply to LE

WCSS_A_BT_RX_PAYLOAD_FEC_ERROR_CORRECTED_COUNT

Bits	Name	Description
7:0	FECCORCNT	Range = 0 to 0xFF (255) bit errors

0x0A3FF12C WCSS_A_BT_RX_ENERGY_DETECT_COUNT

Type: R
Clock: PCLK
Reset State: 0x00000000

Copy of the 16 LSBS of the Rx wide aperture window, captured when the energy detect block found a strong signal Value held until software has read it, then it may be updated by hardware Software must set the LowPwrEn bit in the Rx low power control register before this metric becomes active Cleared at the beginning of a receive operation

WCSS_A_BT_RX_ENERGY_DETECT_COUNT

Bits	Name	Description
15:0	EDCNT	Count value has a resolution of 1 us

0x0A3FF134 WCSS_A_BT_RX_LE_PAYLOAD_HEADER_DATA**Type:** R**Clock:** PCLK**Reset State:** 0x00000000

LE payload byte length from the last receive operation; always cleared at the beginning of the receive operation

WCSS_A_BT_RX_LE_PAYLOAD_HEADER_DATA

Bits	Name	Description
25:24	B_1	LE header from the last receive operation
23:16	LELENGTH	The LE payload byte length from the last receive operation
7:0	LEHDR	The LE header from the last receive operation

0x0A3FF138 WCSS_A_BT_RX_PACKET_HEADER_DATA**Type:** R**Clock:** PCLK**Reset State:** 0x00007F00

Combination of the packet header bits of the last receive operation; aligned to match the format of the packet header in the transmit buffer

WCSS_A_BT_RX_PACKET_HEADER_DATA

Bits	Name	Description
18	SEQN	Sequence number from the packet header of the last receive operation; always cleared at the beginning of the receive operation
17	ARQN	Acknowledge indication from the packet header of the last receive operation; always cleared at the beginning of the receive operation
16	FLOW1	Flow control from the packet header of the last receive operation; always cleared at the beginning of the receive operation
14:8	PKTINDEX	Packet index decoded from the last receive operation PktIndex(6): 0 PktIndex(5): 0 PktIndex(4): Sco_ptt and/or Acl_ptt field read from the base am address array PktIndex(3:0): Type code initially forced by the LC command being serviced or later updated from the received packet header

WCSS_A_BT_RX_PACKET_HEADER_DATA (cont.)

Bits	Name	Description
2:0	BASEAMA	Active member address from the packet header of the last receive operation; always cleared at the beginning of the receive operation

0x0A3FF13C WCSS_A_BT_RX_PAYLOAD_HEADER_DATA**Type:** R**Clock:** PCLK**Reset State:** 0x00000000

Combination of the payload header bits of the last receive operation These bits are aligned to match the format of the payload header in the transmit buffer

WCSS_A_BT_RX_PAYLOAD_HEADER_DATA

Bits	Name	Description
25:16	PYDLNGTH	The payload byte length from the last receive operation The RxScoLngth is loaded into this register when receiving Sco or eSco packets. A length of 18 bytes is loaded into this register when receiving FHS packets. For all other packets the length field from the payload header is loaded into this register. always cleared at the beginning of the receive operation
11:8	UNDEF	Undefined field from a 2-byte payload header of the last receive operation. Undef(0) bit has been defined as the msb (10th bit) of the payload byte length for enhanced rate Acl packets (2-Dh1, 2-Dh3, 2-Dh5, 3-Dh1, 3-Dh3, and 3-Dh5) If one of these packets are received then Undef(0) = 0 and PyldLngth indicates the correct payload byte length always cleared at the beginning of the receive operation
2	FLOW2	Flow control from the payload header of the last receive operation always cleared at the beginning of the receive operation
1:0	LCH	Logical channel indicator from the payload header of the last receive operation always cleared at the beginning of the receive operation

0x0A3FF144 WCSS_A_BT_RX_PACKET_HEADER_HEC

Type: R
Clock: PCLK
Reset State: 0x00000000

HEC field from the packet header of the last receive operation; always cleared at the beginning of the receive operation; The HecCrcCapt control register must be set for the entire HEC field to be received does not apply to LE

WCSS_A_BT_RX_PACKET_HEADER_HEC

Bits	Name	Description
7:0	PKTHDRHEC	

0x0A3FF148 WCSS_A_BT_RX_PAYLOAD_CRC

Type: R
Clock: PCLK
Reset State: 0x00000000

CRC field from the payload of the last receive operation; always cleared at the beginning of the receive operation. HecCrcCapt control register must be set for the entire CRC field to be received

WCSS_A_BT_RX_PAYLOAD_CRC

Bits	Name	Description
23:0	PYLDCRC	For BR/EDR, the CRC is the lower 16 bits For LE, the CRC is 23 bits

0x0A3FF14C WCSS_A_BT_TX_SCO_PAYLOAD_LENGTH

Type: RW
Clock: PCLK
Reset State: 0x00000000

Payload byte length that is used to transmit Sco or eSco packets; DV packets do not use this length field; Before a Sco or eSco packet is transmitted, this length field must be set to the correct payload byte length Payload byte length is predefined for Sco packets but can vary, as negotiated, for eSco packets does not apply to LE

WCSS_A_BT_TX_SCO_PAYLOAD_LENGTH

Bits	Name	Description
9:0	TXSCOLNGTH	Range = 0 to 0x3FF (1023) bytes

0x0A3FF150 WCSS_A_BT_TX_BUFFER_POINTER**Type:** RW**Clock:** PCLK**Reset State:** 0x00000000

Pointer to the starting address of the transmit buffer link-list, which with its corresponding buffer are located in the internal RAM. Transmit logic accesses this link-list and read from its corresponding buffer while transmitting any packet or continuous pattern. As a safety feature, software can write a null pointer to the first entry of the transmit link-list or simply set this pointer to zero. After software configures the transmit data portion of the link-list then the null pointer can be replaced with a valid entry that points the first word of the transmit buffer. This method prevents old data from being transmitted if software does not meet their deadlines; also more robust than inverting the CRC field with TxInvCrc.

WCSS_A_BT_TX_BUFFER_POINTER

Bits	Name	Description
15:2	TXBUFPTR	Range = 0 to 16k words (the RAM may be sized smaller)

0x0A3FF158 WCSS_A_BT_TX_STATUS**Type:** R**Clock:** PCLK**Reset State:** 0x00000000

Transmit buffer link-list overflow status; the transmit logic is allowed to request one extra read from the transmit buffer after the null pointer has been read from the transmit buffer link-list; always cleared at the beginning of the transmit operation.

WCSS_A_BT_TX_STATUS

Bits	Name	Description
1	TXBUFERR	0x0: Null pointer has not been encountered, or only one extra read has been requested by the transmit logic 0x1: Transmit logic has requested two more reads from the transmit buffer since the null pointer was read from the transmit buffer link-list

WCSS_A_BT_TX_STATUS (cont.)

Bits	Name	Description
0	TXABORT	(Duplicated) Transmit abort status When channel avoidance coexistence is enabled this bit is updated after the hop calculator has completed when servicing a Tx LC command When packet traffic arbitration coexistence is enabled, this bit is updated when the Tx_Confirm input indicates an abort 0x0: Transmit operation was not aborted 0x1: Transmit operation was aborted because this Bluetooth device has low priority BtPriority = 0, and the calculated channel fell within the range of 802.11 channels LowChan calculated channel HighChan

0x0A3FF15C WCSS_A_BT_COEXISTENCE_RF_ACTIVE_DELAY_TIMER**Type:** RW**Clock:** PCLK**Reset State:** 0x00000000

Delays the falling edge of the external rf_active signal by the programmed value.

WCSS_A_BT_COEXISTENCE_RF_ACTIVE_DELAY_TIMER

Bits	Name	Description
6:0	COEXRFDELAY	Disabled if set to zero; only applies to Tx and is ignored during Rx Range = 0 to 0x7F (max 127 us, 1us increments)

0x0A3FF160 WCSS_A_BT_SOFTWARE_CAPTURE_CALCULATE_COMMAND**Type:** RW**Clock:** PCLK**Reset State:** 0x00000000**WCSS_A_BT_SOFTWARE_CAPTURE_CALCULATE_COMMAND**

Bits	Name	Description
2	SWCAPT	Capture field that controls the capture of the selected Bluetooth clock + 1 and the UAP portion of the BdAddr when this command is issued 0x0: Do not capture the clock and UAP 0x1: Capture the clock and UAP

WCSS_A_BT_SOFTWARE_CAPTURE_CALCULATE_COMMAND (cont.)

Bits	Name	Description
1:0	SWCALC	Calculate field that controls whether the hop calculator runs when this command is issued; bits do not match Cmd0 and Cmd1. AfhEn = 0: Hop calculator runs. AfhEn = 1: Hop calculator runs when bt_clk(1) = 1. 0x0: Do not run the hop cal_1 0x1: Force the hop calculator when this command is issued 0x2: Do not run the hop cal_2 0x3: Automatically runs the hop calculator

0x0A3FF164 WCSS_A_BT_COEXISTENCE_COMMAND**Type:** RW**Clock:** PCLK**Reset State:** 0x00000000

Coexistence command register:

For writes, issues the coexistence command as programmed in the individual registers, and processed at the CoexCmdTo For reads, returns the last command that was issued.

WCSS_A_BT_COEXISTENCE_COMMAND

Bits	Name	Description
15	QTABTRESEND	Indicates the status of the QTA Bt bit 8, RESEND 0x0: Last WLAN hardware state resend not requested 0x1: Last WLAN hardware state resend requested
14	QTABTMODE	Controls QTA Bt character bit 7, MODE 0x0: Receive slot 0x1: Transmit slot
13	QTABTMOREDATA	Controls QTA Bt character bit 6, MORE_DATA. 0x0: ACL buffers empty 0x1: ACL buffers still contain data
12:10	QTABTPRIORITY	Controls QTA Bt character bits 5:3, PRIORITY 0x0: Priority 0 0x1: Priority 1 0x2: Priority 2 0x3: Priority 3 0x4: Priority 4 0x5: Priority 5 0x6: Priority 6 0x7: Priority 7
9	QTABTAIESYNC	Controls QTA Bt character bit 1, AIE_SYNC 0x0: Even anchor 0x1: Odd anchor

WCSS_A_BT_COEXISTENCE_COMMAND (cont.)

Bits	Name	Description
8	QTABTACTIVE	Controls QTA BT character bit 2, BT_ACTIVE ignored if QtaBypass bit is 0; generated by link controller hardware. 0x0: BT_ACTIVE = 0 0x1: BT_ACTIVE = 1
7	STOPSLOT2	Controls the capturing of the selected Bluetooth clock + 1 and the UAP portion of the BdAddr when this command is issued; occurs at the CoexCmdTo; not buffered and must be maintained until CoexStatTo. 0x0: Disable capturing 0x1: Enable capturing
6	COEXCAPT	Controls the behavior during the second slot when using the CoexCmd = 01,10 sequence (i.e., Coex is in the Slot2_Mode state) 0x0: Allow Slot2_Mode exit only if a link controller command is present in half slot 1 0x1: Allow Slot2_Mode exit after the next Tx or Rx command timeout expires; coex waits for the link controller to finish processing
5:4	COEXCALC	Controls whether the hop calculator runs when this command is issued; determination to run the hop calculator occurs at the CoexCmdTo; not buffered and must be maintained until CoexStatTo AfhEn = 0: Hop calculator runs AfhEn = 1: Hop calculator runs when bt_clk(1) = 1 0x0: Automatically runs the hop calculator 0x1: Force the hop calculator to run 0x2: Force disabling of the hop calculator
3	COEXPRI	Controls the priority signaling of the external coexistence interface and the internal channel avoidance mechanism; not buffered and must be maintained until CoexStatTo. 0x0: Low priority 0x1: High priority
2	COEXMODE	Controls the Tx/Rx signaling of the external coexistence interface; not buffered and must be maintained until CoexStatTo 0x0: Receive 0x1: Transmit

WCSS_A_BT_COEXISTENCE_COMMAND (cont.)

Bits	Name	Description
1:0	COEXCMD	Controls the activity on the external coexistence interface 0x0: Stop activity on the interface immediately but wait for hop calculator to finish 0x1: Start slot 1 activity, then hold the interface signals until a slot 2 command is issued; state machine must be in idle when issuing this command 0x2: Start and finish slot 2 activity; state machine should already be waiting for a slot 2 command 0x3: Start and finish slot 1 activity; state machine must be in Idle when issuing this command

0x0A3FF168 WCSS_A_BT_COEXISTENCE_CONTROL**Type:** RW**Clock:** PCLK**Reset State:** 0x00000000**WCSS_A_BT_COEXISTENCE_CONTROL**

Bits	Name	Description
15	PTAABORTRX	Enable pta_rx_confx to abort Rx packets PTA interface uses the same signal (tx_confx) for both tx and rx 0x0: Disable 0x1: Enable
13	CXMABORTRXEN	Enable cxm_rx_grant to abort Rx packets 0x0: Disable 0x1: Enable
12	CXMABORTTXEN	Enable cxm_tx_grant to abort Tx packets 0x0: Disable 0x1: Enable
11	QTAABORTRXEN	Enable qta_rx_confx to abort Rx packets 0x0: Disable 0x1: Enable
10	QTAABORTTXEN	Enable qta_tx_confx to abort Tx packets 0x0: Disable 0x1: Enable
9	QTABYPASS	Controls QTA protocol timing; when set, the QtaBtn bits are forwarded to QTA transmit UART immediately, bypassing PTA timing logic 0x0: QTA Bt timing under hardware control 0x1: QTA Bt hardware timing bypassed

WCSS_A_BT_COEXISTENCE_CONTROL (cont.)

Bits	Name	Description
7	PTATXHIPRI	Packet traffic arbitration; controls the high priority transmit feature 0x0: Disable; a transmit aborted according to the PtaAbortTx and PtaAbortTxon control registers 0x1: Enable; Tx_Confx input ignored for an entire frame when a high priority transmit command has been issued
6	PTAABORTTXON	Controls a transmit abort that is already in progress (when tx_on=1) 0x0: Disable 0x1: Enable; abort occurs only if PtaAbortTx=1 and CoexEn=1 and Tx_Confx = 1
5	PTAABORTTX	Enable pta_tx_confx to abort Tx packets 0x0: Disable 0x1: Enable
4	PTAEN	Enables the generation of the external interface outputs: Rf_Active, Status and Freq 0x0: Disable 0x1: Enable
3	CAVPRIRX	Channel avoidance; controls the activity of the Bt_Priority output during a high priority receive that occurs on an 802.11x channel 0x0: A high priority Receive does not affect Bt_Priority 0x1: Set Bt_Priority for a high priority Receive that is in the 802.11x band
2	CAVPRITX	Controls the activity of the Bt_Priority output during a high priority transmit that occurs on an 802.11x channel 0x0: High priority transmit does not affect Bt_Priority 0x1: Set Bt_Priority for a high priority transmit
1	CAVABORTTX	Controls the aborting of a transmit 0x0: Disable the transmit abort 0x1: Enable the transmit abort; occurs only if CoexEn = 1 and CoexPri = 0, and HopAbort = 1 on 802.11x channel
0	CAVEN	Enables the generation of the external interface output Bt_Priority 0x0: Disable 0x1: Enable

0x0A3FF16C WCSS_A_BT_COEXISTENCE_ENABLE_AND_POLARITY**Type:** RW**Clock:** PCLK**Reset State:** 0x00000000

WCSS_A_BT_COEXISTENCE_ENABLE_AND_POLARITY

Bits	Name	Description
8	POLCXMGRANT	Sets the polarity of the CxM grant input 0x0: Normal; high input indicates that the Bluetooth should be prevented 0x1: Inverted; low input indicates that the Bluetooth should be prevented
7	COEXEN	Enables the clock to the coexistence logic 0x0: Disable by gating the clock 0x1: Enable by ungating the clock
6	QTAEN	Enables the QTA output signals. CoexEn must be active for QTA 0x0: Disable QTA 0x1: Enable QTA
5	POLCHDATA	Sets the polarity of the coexistence interface Chan_Data input 0x0: Normal; high input prevents Bt_Priority from updating 0x1: Inverted; low input prevents Bt_Priority from updating
4	POLBTPRI	Sets the polarity of the coexistence interface Bt_Priority output 0x0: Normal; high output indicates a high priority Bluetooth event 0x1: Inverted; low output indicates a high priority Bluetooth event
3	POLRFACT	Sets the polarity of the coexistence interface Rf_Active output 0x0: Normal; high output indicates that Bluetooth is active 0x1: Inverted; low output indicates that Bluetooth is active
2	POLSTAT	Sets the polarity of the coexistence interface Status output 0x0: Normal; high output indicates Bluetooth high priority or Bluetooth Tx 0x1: Inverted; low output indicates Bluetooth high priority or Bluetooth Tx
1	POLFREQ	Sets the polarity of the coexistence interface Freq output 0x0: Normal; high output indicates a frequency overlap 0x1: Inverted; low output indicates a frequency overlap
0	POLTXCNFX	Sets the polarity of the coexistence interface Tx_Confx input 0x0: Normal; high input indicates that the Bluetooth should be prevented 0x1: Inverted; low input indicates that the Bluetooth should be prevented

0x0A3FF170 WCSS_A_BT_COEXISTENCE_STATUS**Type:** R**Clock:** PCLK**Reset State:** 0x00000080

WCSS_A_BT_COEXISTENCE_STATUS

Bits	Name	Description
15	QTAWLRESEND	Indicates the status of the QTA WLAN bit 8 0x0: Last Bluetooth hardware state resend not requested 0x1: Last Bluetooth hardware state resend requested
14	QTAWL7	Reserved. Indicates the status of the QTA WLAN bit 7
13	QTAWL6	Reserved. Indicates the status of the QTA WLAN bit 6
12	QTAWL5	Reserved. Indicates the status of the QTA WLAN bit 5
11	QTAWL4	Reserved. Indicates the status of the QTA WLAN bit 4
10	QTAWL3	Reserved. Indicates the status of the QTA WLAN bit 3
9	QTAWL2	Reserved. Indicates the status of the QTA WLAN bit 2
8	QTAWLCNFX	Indicates the status of the QTA WLAN bit 1. Tx_confirm input signal as seen by the link controller in QTA mode. 0x0: Low signal is being received 0x1: High signal is being received
7	HOPABORT	Indicates the status of the calculated hop channel (LowChan channel HighChan) 0x0: Channel is outside of the 802.11x channel band 0x1: Channel is within the 802.11x channel band; does not mean Tx abort occurred
6	TXABORT	(Duplicated) Transmit abort status
5	COEXCHDATA	Indicates the level of the coexistence Chan_Data input signal as seen by the Bluetooth controller. Pin muxing must be set appropriately to connect this signal to a pin 0x0: Low signal is being received 0x1: High signal is being received
4	COEXBTPRI	Indicates the level of the coexistence Bt_Priority output signal as the Bluetooth controller generates it. Pin muxing must be set appropriately to drive this level to a pin 0x0: Low signal is being generated 0x1: High signal is being generated
3	COEXRFACT	Indicates the level of the coexistence Rf_Active output signal as the Bluetooth controller generates it Pin muxing must be set appropriately to drive this level to a pin 0x0: Low signal is being generated 0x1: High signal is being generated

WCSS_A_BT_COEXISTENCE_STATUS (cont.)

Bits	Name	Description
2	COEXSTAT	Indicates the level of the coexistence Status output signal as the Bluetooth controller generates it Pin muxing must be set appropriately to drive this level to a pin 0x0: Low signal is being generated 0x1: High signal is being generated
1	COEXFREQ	Indicates the level of the coexistence Freq output signal as the Bluetooth controller generates it Pin muxing must be set appropriately to drive this level to a pin 0x0: Low signal is being generated 0x1: High signal is being generated
0	COEXTXCNFX	Indicates the level of the coexistence Tx_Confx output signal as seen by the Bluetooth controller Pin muxing must be set appropriately to drive this level to a pin 0x0: Low signal is being received 0x1: High signal is being received

0x0A3FF184 WCSS_A_BT_HARDWARE_VERIFICATION_CONTROL_1**Type:** RW**Clock:** PCLK**Reset State:** 0x00000000**WCSS_A_BT_HARDWARE_VERIFICATION_CONTROL_1**

Bits	Name	Description
2	BBIF_ENABLE	Enables BBIF interface when set to 1
1	PKTRPTEN	Enables the repeated packet Tx or Rx test mode; the number of packets must also be programmed using PktRptNum register
0	RESERVED	reserved

0x0A3FF188 WCSS_A_BT_HARDWARE_VERIFICATION_CONTROL_2**Type:** RW**Clock:** PCLK**Reset State:** 0x00000000

WCSS_A_BT_HARDWARE_VERIFICATION_CONTROL_2

Bits	Name	Description
5:4	LCCONT	Enables the continuous execution of LC commands by the link controller 0x0: Single overlaid execution on halfslot boundaries; normal 0x1: Continuous overlaid execution on halfslot boundaries; for testing 0x2: Continuous ping-ponged execution on slot boundaries; for testing 0x3: Continuous externally triggered execution on halfslot boundaries
3:2	BTCLKDIS	Disables the Bluetooth clock counters; can be used to save power when the device is not in use. Bt ClkA: Running freely (hard-align allowed) Integer portion is frozen (hard-align disabled) Fractional portion is running freely (allow hard-align) Integer portion is running freely (hard-align disabled) Fractional portion is running freely (allow hard-align) Integer portion is frozen (hard-align disabled) Fractional portion is frozen (allow hard-align) 0x0: Bt ClkN is running fre_1; normal 0x1: Bt ClkN is running fre_2; testing 0x2: Bt ClkN integer and fractional portion is frozen; used for BERT testing 0x3: Bt ClkN integer and fractional portion is frozen; testing
1	BERTPRBS	Selects the PRBS sequence to be used when BertPacket = 1 0x0: PRBS-9 pattern initialized with all ones 0x1: PRBS-15 pattern initialized with all ones
0	BTCLKSEL	Selects the integer and fractional Bluetooth clock that is used by the burst timers and link controller; also selects the input to the captured Bluetooth clock that is distributed to the hop calculator, encryption engine, Tx scramble, and Rx descramble 0x0: Select Bt ClkA 0x1: Select Bt ClkN

0x0A3FF18C WCSS_A_BT_HARDWARE_VERIFICATION_CONTROL_3**Type:** RW**Clock:** PCLK**Reset State:** 0x00000000

Manual control over RM_CMD[2:0]

WCSS_A_BT_HARDWARE_VERIFICATION_CONTROL_3

Bits	Name	Description
2:0	RMIF_FORCE	Write: sets RM_CMD[2:0] = RMif_Force[2:0] Read: returns the current value of RM_CMD[2:0]

0x0A3FF190 WCSS_A_BT_HARDWARE_VERIFICATION_CLEAR_STATE_MACHINES**Type:** RW**Clock:** PCLK**Reset State:** 0x00000000**WCSS_A_BT_HARDWARE_VERIFICATION_CLEAR_STATE_MACHINES**

Bits	Name	Description
6	CLRCOEXFSM	Synchronous clear for the coexistence finite state machine 0x0: No change 0x1: Force the coexistence FSM and pending command back to idle
5	CLRBUFFSM	Synchronous clear for the buffer control finite state machine 0x0: No change 0x1: Force the buffer control FSM idle
4	CLRBBIFFSM	Synchronous clear for the BBIF finite state machine 0x0: No change 0x1: Force the BBIF FSM to idle
3	CLRHOPFSM	Synchronous clear for the hop calculator and modulo divider finite state machines 0x0: No change 0x1: Force the hop calculator and modulo divider FSM to idle; all other state machine outputs dividend, divisor, hop channel are forced to zero
2	CLRLCFSM	Synchronous clear for the LC finite state machine and its pending commands 0x0: No change 0x1: Force the LC FSM and pending commands to idle
1	CLRTXFSM	Synchronous clear for the Tx control finite state machine 0x0: No change 0x1: Force the Tx control FSM to idle
0	CLRRXFSM	Synchronous clear for the Rx control finite state machine and the Rx scan window 0x0: No change 0x1: Force the Rx control FSM to idle; disable Rx scan window

0x0A3FF194 WCSS_A_BT_HARDWARE_VERIFICATION_STATUS_1**Type:** R**Clock:** PCLK**Reset State:** 0x00000000

Number of packet header bit errors (after 1/3 FEC decode) received during hardware verification modes.

Accumulates the number of packet header mismatches during continuous hardware verification receive bursts Expected packet header is programmed into EncrKey(31:0) during test mode Once this accumulator reaches its maximum value it does not roll over PyldCrc register is reused during hardware verification modes to generate this result

WCSS_A_BT_HARDWARE_VERIFICATION_STATUS_1

Bits	Name	Description
15:0	RXHDRERR	Range = 0 to 0xFFFF (65,535) bit errors

0x0A3FF198 WCSS_A_BT_HARDWARE_VERIFICATION_STATUS_2

Type: R

Clock: PCLK

Reset State: 0x00000000

Number of payload bit errors received during hardware verification modes Accumulates the number of Prbs-9 or Prbs-15 mismatches during a continuous receive test or during continuous hardware verification receive bursts Once this accumulator reaches its maximum value it does not roll over FecCorCnt and FecDetCnt registers are reused during hardware verification modes to generate this result.

WCSS_A_BT_HARDWARE_VERIFICATION_STATUS_2

Bits	Name	Description
15:0	RXPYLDERR	Range = 0 to 0xFFFF (65,535) bit errors

0x0A3FF1A0 WCSS_A_BT_PACKET_CONTROL

Type: RW

Clock: PCLK

Reset State: 0x00000000

WCSS_A_BT_PACKET_CONTROL

Bits	Name	Description
6	CMDABORTDIS	Disables the automatic aborting of late LC and coexistence commands 0x0: Enable 0x1: Disable

WCSS_A_BT_PACKET_CONTROL (cont.)

Bits	Name	Description
5	HECCRCCAPT	Controls the checking of the HEC and CRC bits for BR/EDR and the CRC bits for LE; reading the received HEC and CRC field requires that this bit is set 0x0: Timing of the HEC check is based on the HecAll control register; check the CRC on the fly as the bits are received and abort the receive immediately if a CRC mismatch is detected 0x1: Receive all the HEC and CRC bits before checking for possible mismatches
4	SCRDIS	Disables the scrambling of the packet header, extended header, and payload for both transmit and receive operations 0x0: Enable scrambling of packet header and payload 0x1: Disable all scrambling
2	SCO_CRCEN	Enables the appending (Tx) and checking (Rx) of the CRC at the end of the payload of the following Sco packets: (Hv3, Hv4, Hv5, 2-Hv3, 2-Hv5, 3-Hv1, 3-Hv3, 3-Hv5) 0x0: Enable 0x1: Disable
1	AUX1FECEN	Enables the encoding (Tx) and checking (Rx) of the 2/3 FEC coding during the payload portion of an Aux1 packet 0x0: Enable 0x1: Disable
0	AUX1CRCEN	Enables the appending (Tx) and checking (Rx) of the CRC at the end of the payload portion of an Aux1 packet 0x0: Enable 0x1: Disable

0x0A3FF1A8 WCSS_A_BT_LE_CONTROL_2**Type:** RW**Clock:** PCLK**Reset State:** 0x00000000**WCSS_A_BT_LE_CONTROL_2**

Bits	Name	Description
0	LESCANABORTMODE	Controls how a scan is aborted during LE mode 0x0: Stop scan only upon receiving a packet with a good CRC 0x1: Stop scan only upon receiving a packet with a good or bad CRC

0x0A3FF1AC WCSS_A_BT_BLUETOOTH_CLOCK_N_INTEGER_COMPARE_VALUE_0**Type:** RW**Clock:** PCLK**Reset State:** 0x00000000**WCSS_A_BT_BLUETOOTH_CLOCK_N_INTEGER_COMPARE_VALUE_0**

Bits	Name	Description
27:0	CLKCMPINTGR_27_0	Integer compare value for Bluetooth clock N comparator interrupt

0x0A3FF1B0 WCSS_A_BT_BLUETOOTH_CLOCK_N_FRACTIONAL_COMPARE_VALUE_0**Type:** RW**Clock:** PCLK**Reset State:** 0x00000000**WCSS_A_BT_BLUETOOTH_CLOCK_N_FRACTIONAL_COMPARE_VALUE_0**

Bits	Name	Description
12:0	CLKCMPFRACT_12_0	Fractional compare value for Bluetooth clock N comparator interrupt

0x0A3FF1B4 WCSS_A_BT_BLUETOOTH_CLOCK_N_INTEGER_COMPARE_VALUE_1**Type:** RW**Clock:** PCLK**Reset State:** 0x00000000**WCSS_A_BT_BLUETOOTH_CLOCK_N_INTEGER_COMPARE_VALUE_1**

Bits	Name	Description
27:0	CLKCMPINTGR_27_0	Integer compare value for Bluetooth clock N comparator interrupt

0x0A3FF1B8 WCSS_A_BT_BLUETOOTH_CLOCK_N_FRACTIONAL_COMPARE_VALUE_1**Type:** RW**Clock:** PCLK**Reset State:** 0x00000000**WCSS_A_BT_BLUETOOTH_CLOCK_N_FRACTIONAL_COMPARE_VALUE_1**

Bits	Name	Description
12:0	CLKCMPFRACT_12_0	Fractional compare value for Bluetooth clock N comparator interrupt

0x0A3FF1BC WCSS_A_BT_BURST_COEXISTENCE_COMMAND**Type:** RW**Clock:** PCLK**Reset State:** 0x00000000

Compare value for generating the coexistence command burst timer (CoexCmdTo); coexistence state machine runs the hop calculator and starts the coexistence signaling at this burst timer if a coexistence command has been issued CoexCmdTo must occur at least 3.5 usec before the TxCmdTo and the RxCmdTo to prevent hop calculator contention (unless Calc0, Calc1 = CoexCalc) SlotTimer(13) determines which halfslot (0 or 1) to use; SlotTimer(12:0) indicates which fractional Bluetooth clock location is to fire the interrupt

WCSS_A_BT_BURST_COEXISTENCE_COMMAND

Bits	Name	Description
13:0	BRSTCOEXCMD	Time during slot = [312.5 usec * SlotTimer(13)] + [62.5 nsec * SlotTimer(12:0)] Fractional range = 0 to 0x1387 (4999); resolution = 62.5 nsec

0x0A3FF1C0 WCSS_A_BT_BURST_COEXISTENCE_STATUS**Type:** RW**Clock:** PCLK**Reset State:** 0x00000000

Compare value for generating the coexistence command burst timer (CoexStatTo); coexistence state machine continues the coexistence signaling at this burst timer SlotTimer(12:0) indicates which fractional Bluetooth clock location is to fire the interrupt.

WCSS_A_BT_BURST_COEXISTENCE_STATUS

Bits	Name	Description
13:0	BRSTCOEXSTAT	Time during slot = [312.5 usec * SlotTimer(13)] + [62.5 nsec * SlotTimer(12:0)] Fractional range = 0 to 0x1387 (4999); resolution = 62.5 nsec

0x0A3FF1C4 WCSS_A_BT_BURST_PRE SLOT**Type:** RW**Clock:** PCLK**Reset State:** 0x00000000

Fractional compare bits for generating the preslot burst timer (PreslotTo) and the prehalfslot burst timer (PrehalfTo); can generate an Int0 at a specific time during a slot. Preslot burst timer is

generated during the second halfslot ($bt_clk(0) = 1$) Prehalfslot burst timer is generated during the first halfslot ($bt_clk(0) = 0$).

WCSS_A_BT_BURST_PRESLOT

Bits	Name	Description
12:0	BRSTPRESLOT	PreslotTo = Time before first halfslot = 312.5 usec - [62.5 nsec * BrstPreslot] PrehalfTo = Time before second halfslot = 312.5 usec - [62.5 nsec * BrstPreslot] Range = 0 to 0x1387 (4999); resolution = 62.5 nsec

0x0A3FF1C8 WCSS_A_BT_BURST_TX_COMMAND

Type: RW

Clock: PCLK

Reset State: 0x00000000

Fractional compare bits for generating the Tx command burst timer (TxCmdTo); link controller starts the transmit process at this burst timer if a Tx LC command has been issued; Tx command burst timer is generated every halfslot.

WCSS_A_BT_BURST_TX_COMMAND

Bits	Name	Description
12:0	BRSTTXCMD	TxCmdTo = Time before halfslot = 312.5 usec - [62.5 nsec * BrstTxCmd] Range = 0 to 0x1387 (4999); resolution = 62.5 nsec

0x0A3FF1CC WCSS_A_BT_BURST_TX_HOP_STROBE

Type: RW

Clock: PCLK

Reset State: 0x00000000

Fractional compare bits for generating the Tx hop strobe burst timer (TxHstbTo); link controller continues the transmit process at this burst timer by sending the Hop_Strobe_Tx RM command to the RM; Tx hop strobe burst timer is generated every halfslot.

WCSS_A_BT_BURST_TX_HOP_STROBE

Bits	Name	Description
12:0	BRSTTXHSTB	TxHstbTo = Time before halfslot = 312.5 usec - [62.5 nsec * BrstTxHstb] Range = 0 to 0x1387 (4999); resolution = 62.5 nsec

0x0A3FF1D0 WCSS_A_BT_BURST_TX_ON**Type:** RW**Clock:** PCLK**Reset State:** 0x00000000

Fractional compare bits for generating the Tx on burst timer (TxOnTo); link controller continues the transmit process at this burst timer by activating the internal Tx On signal, which enables the baseband transmit logic and gated clock to activate Tx Enable with valid Tx Data exactly 3 usec after TxOnTo; Tx On burst timer is generated every halfslot.

WCSS_A_BT_BURST_TX_ON

Bits	Name	Description
12:0	BRSTTXON	TxOnTo = Time before halfslot = 312.5 usec - [62.5 nsec * BrstTxOn] Range = 0 to 0x1387 (4999); resolution = 62.5 nsec

0x0A3FF1D4 WCSS_A_BT_BURST_RX_COMMAND**Type:** RW**Clock:** PCLK**Reset State:** 0x00000000

Fractional compare bits for generating the Rx command burst timer (RxCmdTo); link controller starts the receive process at this burst timer if an Rx LC command has been issued; Rx command burst timer is generated every halfslot.

WCSS_A_BT_BURST_RX_COMMAND

Bits	Name	Description
12:0	BRSTRXCMD	RxCmdTo = Time before halfslot = 312.5 usec - [62.5 nsec * BrstRxCmd] Range = 0 to 0x1387 (4999); resolution = 62.5 nsec

0x0A3FF1D8 WCSS_A_BT_BURST_RX_HOP_STROBE**Type:** RW**Clock:** PCLK**Reset State:** 0x00000000

Fractional compare bits for generating the Rx hop strobe burst timer (RxHstbTo); link controller continues the receive process at this burst timer by sending the Hop_Strobe_Rx RM command to the RM; Rx hop strobe burst timer is generated every halfslot.

WCSS_A_BT_BURST_RX_HOP_STROBE

Bits	Name	Description
12:0	BRSTRXHSTB	RxHstbTo = Time before halfslot = 312.5 usec - [62.5 nsec * BrstRxHstb] Range = 0 to 0x1387 (4999); resolution = 62.5 nsec

0x0A3FF1DC WCSS_A_BT_BURST_RX_ON**Type:** RW**Clock:** PCLK**Reset State:** 0x00000000

Fractional compare bits for generating the Rx On burst timer (RxOnTo); link controller continues the receive process at this burst timer by activating the internal Rx On signal, which enables the baseband receive logic and their gated clocks; Rx On burst timer is generated every halfslot.

WCSS_A_BT_BURST_RX_ON

Bits	Name	Description
12:0	BRSTRXON	RxOnTo = Time before halfslot = 312.5 usec - [62.5 nsec * BrstRxOn] Range = 0 to 0x1387 (4999); resolution = 62.5 nsec

0x0A3FF1E0 WCSS_A_BT_RX_ACCESS_CODE_DETECT_HARD_ALIGN**Type:** RW**Clock:** PCLK**Reset State:** 0x00000000

Fractional value for hard aligning on ID packets (enabled by AcdAlignEn); does not apply for LE.

WCSS_A_BT_RX_ACCESS_CODE_DETECT_HARD_ALIGN

Bits	Name	Description
12:0	RXACDALIGN	Time before halfslot = 312.5 usec - [62.5 nsec * [RxAccdAlign +1]] Range = 0 to 0x1387 (4999); resolution = 62.5 nsec

0x0A3FF1E4 WCSS_A_BT_RX_PACKET_HEADER_HARD_ALIGN**Type:** RW**Clock:** PCLK**Reset State:** 0x00000000

Fractional value for hard aligning on non-ID packets BR/EDR packets (enabled by HdrAlignEn)

WCSS_A_BT_RX_PACKET_HEADER_HARD_ALIGN

Bits	Name	Description
13:0	RXHDRALIGN	Time before halfslot = 312.5 usec - [62.5 nsec * [RxHdrAlign +1]] Range = 0 to 0x1387 (4999); resolution = 62.5 nsec

0x0A3FF1E8 WCSS_A_BT_TX_GUARD_SYNC_LENGTH

Type: RW

Clock: PCLK

Reset State: 0x00000000

Baseband transmit guard sync gap length for enhanced rate packets; during this time the RM switches to a modulator that may have a different latency.

WCSS_A_BT_TX_GUARD_SYNC_LENGTH

Bits	Name	Description
4:0	TXGSLNGTH	Range = 1 to 0x1F (31) usec

0x0A3FF1EC WCSS_A_BT_SOFTWARE_INTERRUPT_LATENCY

Type: RW

Clock: PCLK

Reset State: 0x00000000

Estimated latency of software interrupt service routine; setting this value to zero causes the Rx interrupt to fire exactly at the end of a packet; a non-zero value causes the Rx interrupt to fire SwIntLat usec earlier.

WCSS_A_BT_SOFTWARE_INTERRUPT_LATENCY

Bits	Name	Description
4:0	SWINTLAT	Range = 0 to 0x1D (29) usec

0x0A3FF1F0 WCSS_A_BT_RX_DEMODULATOR_DELAY

Type: RW

Clock: PCLK

Reset State: 0x00000000

Baseband receive demodulator delay; the receive logic issues the Rm_Idle, RM command RXDMDDLY usec before the last data symbol is sampled ID packets, 0-byte, 1-byte Aux1 (no FEC or CRC), and 0-byte 8DPSK (3Mbps) packets are not affected.

WCSS_A_BT_RX_DEMODULATOR_DELAY

Bits	Name	Description
3:0	RXDMDDLY	For basic rate packets, range = 0 to 0xC (12) usec For enhanced data rate packets, range = 0 to 0x5 (5) usec

0x0A3FF1F4 WCSS_A_BT_TX_PACKET_HEADER_HARD_ALIGN

Type: RW

Clock: PCLK

Reset State: 0x00000000

Fractional value for hard aligning on LE packets (enabled by TxHdrAlignEn) not used for BR/EDR packets When the alignment occurs, the fractional counter rollover value is forced to [Payload length*8*16) + TxHdrAlign] Value of the counter is forced to 56*16.

WCSS_A_BT_TX_PACKET_HEADER_HARD_ALIGN

Bits	Name	Description
13:0	TXHDRALIGN	Range = 0 to 0x3FFF (16383) resolution = 62.5 nsec

0x0A3FF1FC WCSS_A_BT_BT_TO_LPASS_AUDIO

Type: RW

Clock: PCLK

Reset State: 0x00000000

WCSS_A_BT_BT_TO_LPASS_AUDIO

Bits	Name	Description
0	BTLAPSSAUDIORXINTR	bt_lpass_audio_rx_intr signal to LPASS

0x0A3FF200 WCSS_A_BT_LC_RX_AGC_RSSI_OUT_0

Type: R

Clock: PCLK

Reset State: 0x00000000

WCSS_A_BT_LC_RX_AGC_RSSI_OUT_0

Bits	Name	Description
24	CORR_VALID	correlation valid
17:16	LNA_AGC_GAIN	LNA AGC gain for current packet 0x0: High 0x1: Nominal 0x3: Low
15:0	AGC_RSSI	Signal amplitude estimate for current packet

0x0A3FF204 WCSS_A_BT_LC_RX_AGC_RSSI_OUT_1

Type: R
Clock: PCLK
Reset State: 0x00000000

WCSS_A_BT_LC_RX_AGC_RSSI_OUT_1

Bits	Name	Description
24	CORR_VALID	correlation valid
17:16	LNA_AGC_GAIN	LNA AGC Gain for previous packet 0x0: High 0x1: Nominal 0x3: Low
15:0	AGC_RSSI	Signal amplitude estimate for previous packet

0x0A3FF208 WCSS_A_BT_LC_SIF_CTRL

Type: RW
Clock: PCLK
Reset State: 0x00000000

WCSS_A_BT_LC_SIF_CTRL

Bits	Name	Description
0	LC_SIF_SW_RESET	Soft reset of two-wire serial interface which is combined with asynchronous reset_n during scan_mode, this reset has no effect

0x0A3FF320 WCSS_A_BT_MWS_CTRL**Type:** RW**Clock:** PCLK**Reset State:** 0x00000000

Control the MWS TX ON signal

WCSS_A_BT_MWS_CTRL

Bits	Name	Description
16	ENABLE	Enable the BT_TX_ON.
15:8	TRAIL_DLY_CNT	DELAY the trailing edge of BT_TX_ON by the cnt value, in us
7:0	LEAD_DLY_CNT	DELAY the leading edge of BT_TX_ON by the cnt value, in us

0x0A3FF348 WCSS_A_BT_INT_0_LE_BTC_SLOT_BURST_TIMER**Type:** RW**Clock:** PCLK**Reset State:** 0x00000000

LE Interrupt 0 programmable slot timer; generates an interrupt at a specific time during a slot; the timeout value is compared directly with no offset Time before halfslot = 312.5 usec - [62.5 nsec * [SlotTimer]].

WCSS_A_BT_INT_0_LE_BTC_SLOT_BURST_TIMER

Bits	Name	Description
13:0	SLOTTIMER	Range = 0 to 0x1387 (4999); resolution = 62.5 nsec

0x0A3FF34C WCSS_A_BT_LE_BURST_COEXISTENCE_COMMAND**Type:** RW**Clock:** PCLK**Reset State:** 0x00000000

Compare value for generating the coexistence command burst timer (CoexCmdTo); the coexistence state machine runs the hop calculator and starts the coexistence signaling at this burst timer if a coexistence command has been issued CoexCmdTo must occur at least 3.5 usec before the TxCmdTo and the RxCmdTo to prevent hop calculator contention (unless Calc0, Calc1 = CoexCalc) Time before halfslot = 312.5 usec - [62.5 nsec * [BrstCoexCmd]]

WCSS_A_BT_LE_BURST_COEXISTENCE_COMMAND

Bits	Name	Description
13:0	BRSTCOEXCMD	Range = 0 to 0x1387 (4999); resolution = 62.5 nsec

0x0A3FF350 WCSS_A_BT_LE_BURST_COEXISTENCE_STATUS**Type:** RW**Clock:** PCLK**Reset State:** 0x00000000

Compare value for generating the coexistence command burst timer (CoexStatTo); the coexistence state machine continues the coexistence signaling at this burst timer Time before halfslot = 312.5 usec - [62.5 nsec * [BrstCoexStat]].

WCSS_A_BT_LE_BURST_COEXISTENCE_STATUS

Bits	Name	Description
13:0	BRSTCOEXSTAT	Range = 0 to 0x1387 (4999); resolution = 62.5 nsec

0x0A3FF354 WCSS_A_BT_LE_BURST_PRESLOT**Type:** RW**Clock:** PCLK**Reset State:** 0x00000000

Fractional compare bits for generating the preslot burst timer (PreslotTo) and the prehalfslot burst timer (PrehalfTo) for LE, these are the same these burst timers can generate an Int0 at a specific time during a slot. Time before halfslot = 312.5 usec - [62.5 nsec * [BrstPreSlot]]

WCSS_A_BT_LE_BURST_PRESLOT

Bits	Name	Description
13:0	BRSTPRESLOT	Range = 0 to 0x1387 (4999); resolution = 62.5 nsec

0x0A3FF358 WCSS_A_BT_LE_BURST_TX_COMMAND**Type:** RW**Clock:** PCLK**Reset State:** 0x00000000

Fractional compare bits for generating the Tx command burst timer (TxCmdTo); link controller starts the transmit process at this burst timer if a Tx LC command has been issued Tx command burst timer is generated every slot Time before halfslot = 312.5 usec - [62.5 nsec * [BrstTxCmd]].

WCSS_A_BT_LE_BURST_TX_COMMAND

Bits	Name	Description
13:0	BRSTTXCMD	Range = 0 to 0x1387 (4999) resolution = 62.5 nsec

0x0A3FF35C WCSS_A_BT_LE_BURST_TX_HOP_STROBE**Type:** RW**Clock:** PCLK**Reset State:** 0x00000000

Fractional compare bits for generating the Tx hop strobe burst timer (TxHstbTo); link controller continues the transmit process at this burst timer by sending the Hop_Strobe_Tx RM command to the RM Tx hop strobe burst timer is generated every slot. Time before halfslot = 312.5 usec - [62.5 nsec * [BrstTxHstb]].

WCSS_A_BT_LE_BURST_TX_HOP_STROBE

Bits	Name	Description
13:0	BRSTTXHSTB	Range = 0 to 0x1387 (4999); resolution = 62.5 nsec

0x0A3FF360 WCSS_A_BT_LE_BURST_TX_ON**Type:** RW**Clock:** PCLK**Reset State:** 0x00000000

Fractional compare bits for generating the Tx on burst timer (TxOnTo) link controller continues the transmit process at this burst timer by activating the internal Tx On signal, which enables the baseband transmit logic and gated clock that activates Tx Enable with valid Tx Data exactly 3 usec after TxOnTo Tx On burst timer is generated every slot. Time before halfslot = 312.5 usec - [62.5 nsec * [TxOnTo]].

WCSS_A_BT_LE_BURST_TX_ON

Bits	Name	Description
13:0	BRSTTXON	Range = 0 to 0x1387 (4999); resolution = 62.5 nsec

0x0A3FF364 WCSS_A_BT_LE_BURST_RX_COMMAND**Type:** RW**Clock:** PCLK**Reset State:** 0x00000000

Fractional compare bits for generating the Rx command burst timer (RxCmdTo); link controller starts the receive process at this burst timer if a Rx LC command has been issued Rx command burst timer is generated every slot. Time before halfslot = 312.5 usec - [62.5 nsec * [BrstRxCmd]]

WCSS_A_BT_LE_BURST_RX_COMMAND

Bits	Name	Description
13:0	BRSTRXCMD	Range = 0 to 0x1387 (4999); resolution = 62.5 nsec

0x0A3FF368 WCSS_A_BT_LE_BURST_RX_HOP_STROBE**Type:** RW**Clock:** PCLK**Reset State:** 0x00000000

Fractional compare bits for generating the Rx hop strobe burst timer (RxHstbTo); link controller continues the receive process at this burst timer by sending the Hop_Strobe_Rx rm command to the Rm Rx hop strobe burst timer is generated every slot Time before halfslot = 312.5 usec - [62.5 nsec * [BrstRxHstb]]

WCSS_A_BT_LE_BURST_RX_HOP_STROBE

Bits	Name	Description
13:0	BRSTRXHSTB	Range = 0 to 0x1387 (4999); resolution = 62.5 nsec

0x0A3FF36C WCSS_A_BT_LE_BURST_RX_ON**Type:** RW**Clock:** PCLK**Reset State:** 0x00000000

Fractional compare bits for generating the Rx on burst timer (RxOnTo) link controller continues the receive process at this burst timer by activating the internal Rx On signal, which enables the baseband receive logic and their gated clocks Rx On burst timer is generated every slot Time before halfslot = 312.5 usec - [62.5 nsec * [BrstRxOn]]

WCSS_A_BT_LE_BURST_RX_ON

Bits	Name	Description
13:0	BRSTRXON	Range = 0 to 0x1387 (4999); resolution = 62.5 nsec

0x0A3FF370 WCSS_A_BT_PACKET_REPEAT_NUMBER**Type:** RW**Clock:** PCLK**Reset State:** 0x00000000

Number of packets to send or receive during continuous test mode if the programmed value is read prior to issuing an LC halfslot command, the value is unchanged; the next LC halfslot command initiates packet processing and the counter decrements must set the PktRptEn bit to enable this register. When the test is complete, reading this register should return a zero; a nonzero value indicates that not all packets were processed. Readback values do not reflect the programmed value.

WCSS_A_BT_PACKET_REPEAT_NUMBER

Bits	Name	Description
15:0	PKTRPTNUM	Range = 0 to 0xFFFF

0x0A3FF374 WCSS_A_BT_GOOD_CRC_COUNTER**Type:** R**Clock:** PCLK**Reset State:** 0x00000000

Number of good CRC values received during a repeated Rx test value is reset to zero when a PktRptNum is written. Must set the PktRptEn bit to enable this register.

WCSS_A_BT_GOOD_CRC_COUNTER

Bits	Name	Description
15:0	GOODCRCCTR	Range = 0 to 0xFFFF

0x0A3FF378 WCSS_A_BT_LE_RX_HARD_ALIGN**Type:** RW**Clock:** PCLK**Reset State:** 0x00000000

Fractional value for hard aligning on LE packets (enabled by HdrAlignEn); when the alignment occurs, the fractional counter rollover value is forced to $[(\text{Payload length} * 8 * 16) + \text{RxHdrAlign}]$ value of the counter is forced to $56 * 16$ Time before slot = $[(\text{Len} * 8 * 16) + \text{RxHdrAlign}] * 62.5\text{ns}$

WCSS_A_BT_LE_RX_HARD_ALIGN

Bits	Name	Description
13:0	RXHDRALIGN	Range = 0 to 0x3FFF (16383); resolution = 62.5 nsec

0x0A3FF37C WCSS_A_BT_LE_CONTROL

Type: RW

Clock: PCLK

Reset State: 0x00000000

WCSS_A_BT_LE_CONTROL

Bits	Name	Description
7	RESETFRACCLK	Writing a 1 to this bit forces an immediate alignment of the Bluetooth clock; causes a fractional clock rollover (reset to zero); increments CLKA 0x0: No effect 0x1: Reset fractional clock
6	SKIPCALC	Skip hop calculation 0x0: Process hop calculation even if it is not needed 0x1: Skip hop calculation state; HopStrobe burst timer is ignored
5	CMD0EN	Force LC command 0 processing 0x0: Use legacy command 0/1; processing based on Bluetooth clock 0x1: Always use command 0
4	LEMODMODE	LE modulation mode 0x0: Modulation mod is GFSK BR 0x1: Modulation mode is GMSK LE
3	LEMODBYP	Bypass LE modulation to the modem 0x0: Do not bypass; modulation is GMSK only during LE mode 0x1: Set modulation mode according to LEModMode bit

WCSS_A_BT_LE_CONTROL (cont.)

Bits	Name	Description
2	TXHDRALIGNEN	Enables slot alignment for LE Tx packets; slot alignment occurs when a LE Tx payload length has been captured; Bluetooth clock adjustments: [CLKA fractional counter rollover value = Payload length*8*16 + TxHdrAlign] Even if enabled, the fractional clock is initialized to 4999 at the beginning of a Tx operation and changes only if a packet is sent Preslot burst timer max value is also set at the same time the clock adjustment is made 0x0: Disable LE Tx header slot alignment 0x1: Enable LE Tx header slot alignment when a valid payload length has been captured
1	LECORRMODE	LE correlation mode 0x0: Correlate against 32-bit sync word default 0x1: Correlate against 32-bit sync word and 4 MSBs of preamble
0	LEEN	LE timer enable; setting can be written at any time and be latched at the next Bluetooth clock A rollover; burst timers from the corresponding domain only be in effect after the rollover 0x0: Enable Bluetooth time domain 0x1: Enable LE time domain

0x0A3FF380 WCSS_A_BT_LE_HOP_OFFSET**Type:** RW**Clock:** PCLK**Reset State:** 0x00000000

To be used with LE and with setting OffsetOnly = 1 sent directly from the LC to the modem and represents the actual frequency to be tuned; hardware does not calculate the hop frequency for LE; this value is not the LE channel index

WCSS_A_BT_LE_HOP_OFFSET

Bits	Name	Description
6:0	HOPOFFSET	Valid range = 0x02 (2) to 0x50 (80) Mhz values out of range are not restricted

0x0A3FF388 WCSS_A_BT_BLUETOOTH_CLOCK_N_INTEGER_COMPARE_VALUE_2**Type:** RW**Clock:** PCLK**Reset State:** 0x00000000**WCSS_A_BT_BLUETOOTH_CLOCK_N_INTEGER_COMPARE_VALUE_2**

Bits	Name	Description
27:0	CLKCMPINTGR_27_0	Integer compare value for Bluetooth Clock N Comparator output drives Tx audio interrupt to LPASS

0x0A3FF38C WCSS_A_BT_BLUETOOTH_CLOCK_N_FRACTIONAL_COMPARE_VALUE_2**Type:** RW**Clock:** PCLK**Reset State:** 0x00000000**WCSS_A_BT_BLUETOOTH_CLOCK_N_FRACTIONAL_COMPARE_VALUE_2**

Bits	Name	Description
12:0	CLKCMPFRACT_12_0	Fractional compare value for Bluetooth Clock N Comparator output drives Tx audio interrupt to LPASS

00x0A3FF39 WCSS_A_BT_ERROR_STATUS_REGISTER**C****Type:** R**Clock:** PCLK**Reset State:** 0x00000000**WCSS_A_BT_ERROR_STATUS_REGISTER**

Bits	Name	Description
3	ERROR_CRPT_WR	Indicates link controller error access when gam_cmd_busy 0x1: btc_w_req_cc error from crypto_ahb_ctrl
2	ERROR_CRPT_RD	Indicates link controller error access when gam_cmd_busy 0x1: btc_r_req_cc error from crypto_ahb_ctrl
1	ERROR_CTRL_WR	Indicates link controller error access when gam_cmd_busy 0x1: btc_w_req error from buffer controller
0	ERROR_CTRL_RD	Indicates link controller error access when gam_cmd_busy 0x1: btc_r_req error from buffer controller

0x0A3FF3A0 WCSS_A_BT_CLOCK_GATING**Type:** RW**Clock:** PCLK**Reset State:** 0x00000001**WCSS_A_BT_CLOCK_GATING**

Bits	Name	Description
2	CRYPTO_SW_RESET	Resets Crypto block 0x0: Normal 0x1: Reset
1	SW_RESET	Resets link controller including Crypto interface logic (both AHB and 32M domain) 0x0: Normal 0x1: Reset
0	CRYPTO_CLK_GATING_EN	Enables clock gating for Crypto4 block 0x0: Disable 0x1: Enable

0x0A3FF3A4 WCSS_A_BT_GAM_GAS_CONFIG**Type:** RW**Clock:** PCLK**Reset State:** 0x0000000D

Configuration of the GAM/GAS bridges

WCSS_A_BT_GAM_GAS_CONFIG

Bits	Name	Description
25	GAM_CLK_GATING_DISABLE	0x0: allows hardware to clock gate for low power 0x1: forces clock on
24	GAS_CLK_GATING_DISABLE	0x0: allows hardware to clock gate for low power 0x1: forces clock on
23:20	GAM_CFG_BUSY_LIMIT	The maximum number of clock cycles the GAM should wait for the application to either present data to the GAM during writes or to deassert the read stall signal during reads, before the GAM will release the AHB.
19:12	GAS_CFG_HSPLIT_RESP_LIMIT	This configuration value specifies the maximum number of clocks that a master may be split for, before it will be unsplit. Units are in 1024 clock cycles

WCSS_A_BT_GAM_GAS_CONFIG (cont.)

Bits	Name	Description
11:4	GAS_CFG_HSPLIT_MST_RESP_LIMIT	This configuration value specifies the maximum number of clk cycles that a master has once it is unsplit, to return and complete the transaction. Units are in 1024 clock cycles
3:0	GAS_CFG_HREADY_LIMIT	This configuration value specifies the maximum number of clk cycles that the GAS may hold HREADY low for. If this limit is reached, the GAS will issue a split to the master and proceed to the SUSPENDED state (cmd_status=SUSPENDED)

10.3 WCSS_A_QTA_BT_BT_QTA_UART**0x0A4C0000 WCSS_A_QTA_BT_INTR****Type:** RW**Reset State:** 0x00000000

Interrupt status register.

WCSS_A_QTA_BT_INTR

Bits	Name	Description
26	RX_FIFO_UFLOW_INTR	Receive FIFO Underflow interrupt status bit. Software must write a 1 to this bit to clear it.
25	RX_FIFO_OFLOW_INTR	Receive FIFO Overflow interrupt status bit. Software must write a 1 to this bit to clear it.
24	RX_FIFO_RDY_INTR	Receive FIFO Ready interrupt status bit: Indicates when the Receive FIFO is ready to be drained of software characters. This interrupt is set every clock cycle while the FIFO fill level is greater than the Receive FIFO Trigger Level (rx_trig_level). Software must write a 1 to this bit to clear it.
17	TX_FIFO_OFLOW_INTR	Transmit FIFO Overflow interrupt status bit. Software must write a 1 to this bit to clear it.
16	TX_FIFO_RDY_INTR	Transmit FIFO Ready interrupt status bit: Indicates when the Transmit FIFO is ready to accept more software characters for transmission. This interrupt is set every clock cycle while the FIFO fill level is less than or equal to the Transmit FIFO Trigger Level (tx_trig_level). Software must write a 1 to this bit to clear it.
8	RX_HW_CHAR_INTR	Received Hardware Character interrupt status bit. Software must write a 1 to this bit to clear it.
1	PARITY_ERR_INTR	UART Parity Error interrupt status bit. Software must write a 1 to this bit to clear it.

WCSS_A_QTA_BT_INTR (cont.)

Bits	Name	Description
0	FRAME_ERR_INTR	UART Frame Error interrupt status bit. Software must write a 1 to this bit to clear it.

0x0A4C0004 WCSS_A_QTA_BT_INTR_EN**Type:** RW**Reset State:** 0x00000000

Interrupt enable register.

WCSS_A_QTA_BT_INTR_EN

Bits	Name	Description
26	RX_FIFO_UFLOW_INTR_EN	Interrupt enable for the Receive FIFO Underflow interrupt.
25	RX_FIFO_OFLOW_INTR_EN	Interrupt enable for the Receive FIFO Overflow interrupt.
24	RX_FIFO_RDY_INTR_EN	Interrupt enable for the Receive FIFO Ready interrupt.
17	TX_FIFO_OFLOW_INTR_EN	Interrupt enable for the Transmit FIFO Overflow interrupt.
16	TX_FIFO_RDY_INTR_EN	Interrupt enable for the Transmit FIFO Ready interrupt.
8	RX_HW_CHAR_INTR_EN	Interrupt enable for the Received Hardware Character interrupt.
1	PARITY_ERR_INTR_EN	Interrupt enable for the Parity Error interrupt.
0	FRAME_ERR_INTR_EN	Interrupt enable for the Frame Error interrupt.

0x0A4C0008 WCSS_A_QTA_BT_UART_CTRL**Type:** RW**Reset State:** 0x03280060

UART control register.

WCSS_A_QTA_BT_UART_CTRL

Bits	Name	Description
31:24	BAUD_RATE_NUMERATOR	Sets the numerator for the ratio of the system clock frequency to the baud rate.
23:16	BAUD_RATE_DENOMINATOR	Sets the denominator for the ratio of the system clock frequency to the baud rate.
6	TX_INV_EN	When set, enables the UART's serial TX output inverter.

WCSS_A_QTA_BT_UART_CTRL (cont.)

Bits	Name	Description
5	RX_INV_EN	When set, enables the UART's serial RX input inverter.
4	POWER_DOWN	When set, puts the UART into power-down mode. The UART transmitter will complete its current transmission and then disable all further transmissions. The UART receiver will discard the current character being received and then ignore all further receptions.
3	PARITY_EN	Enables parity bit generation in the UART transmitter, and parity checking in the UART receiver.
2	PARITY_ODD	When parity is enabled, this bit specifies whether the parity is odd or even. When 1, the parity is odd; when 0, it is even.
1:0	NUM_STOPS	Specifies the number of stop bits (1-4) at the end of each character. The number of stop bits per character = num_stops + 1.

0x0A4C000C WCSS_A_QTA_BT_HW_CTRL**Type:** RW**Reset State:** 0x00000014

Hardware layer control register.

WCSS_A_QTA_BT_HW_CTRL

Bits	Name	Type	Description
31:24	LAST_RX_HW_CHAR	R	This field captures the last received hardware character that was passed on to the Signaling layer.
15	TX_BUFFER_STATE	R	State of the 1-character-deep Hardware TX Buffer. When 1, the buffer is full; when 0, the buffer is empty.
14:8	TX_BUFFER_CONTENTS	R	This field captures the hardware character presently stored in the TX Buffer. Resend bit excluded as only non-Resend-Requests are stored in this buffer.
7	RESEND_REQ_FLAG	R	Indicates the present state of the Resend Request flag. When set, it indicates that a Resend Request is pending transmission.
4	EXCLUSIVE_RESEND_EN	RW	Exclusive Resend Enable: When set, the QTA module will treat Signal State bits within Resend Request characters as invalid. When clear, the Signal State bits are valid regardless of whether the character is a Resend Request.
3	RESEND_REQ_PASSTHRU_EN	RW	Resend Request Pass-Through Enable: When set, the QTA module will pass received Resend Request characters through to the Signaling layer for processing. NB: This feature is unsupported for the WLAN version of the QTA module, because the Signaling interface for the WLAN version does not support Resend signaling.

WCSS_A_QTA_BT_HW_CTRL (cont.)

Bits	Name	Type	Description
2	RESEND_RESP_EN	RW	Resend Response Enable: When this bit is set, the QTA module will automatically respond to received Resend Requests by resending its last (non-Resend-Request) hardware character. Otherwise, the QTA module will not respond to Resend Requests received from the peer device.
1	RETRANSMIT	W	Retransmit TX Buffer contents: Writing a 1 to this bit causes the last (non-Resend-Request) hardware character to be re-transmitted. This bit is self-clearing.
0	RESEND_REQUEST	W	Send a Resend Request: Writing a 1 to this bit causes a Resend Request character to be queued for transmission. This bit is self-clearing.

0x0A4C0010 WCSS_A_QTA_BT_HW_TX_BUFFER_MODIFY**Type:** W**Reset State:** 0x00000000

TX Buffer modify register.

WCSS_A_QTA_BT_HW_TX_BUFFER_MODIFY

Bits	Name	Description
6:0	DATA	Data to be written to the TX Buffer. Writing to this register causes the contents of the TX Buffer to be modified to the value of the data field, and causes the TX buffer state to be set to full.

0x0A4C0014 WCSS_A_QTA_BT_SW_CTRL**Type:** RW**Reset State:** 0x00000000

Software layer control register.

WCSS_A_QTA_BT_SW_CTRL

Bits	Name	Type	Description
29:24	RX_FILL	R	Indicates the number of software characters presently queued in the Receive FIFO.
21	RX_FLUSH	W	Receive FIFO Flush: Writing a 1 to this bit flushes the contents of the Receive FIFO. This bit is self-clearing.
20:16	RX_TRIG_LEVEL	RW	Receive FIFO Trigger Level. See the rx_fifo_rdy_int interrupt status register description.

WCSS_A_QTA_BT_SW_CTRL (cont.)

Bits	Name	Type	Description
13:8	TX_FILL	R	Indicates the number of software characters presently queued in the Transmit FIFO.
5	TX_FLUSH	W	Transmit FIFO Flush: Writing a 1 to this bit flushes the contents of the Transmit FIFO. This bit is self-clearing.
4:0	TX_TRIG_LEVEL	RW	Transmit FIFO Trigger Level. See the tx_fifo_rdy_int interrupt status register description.

0x0A4C0018 WCSS_A_QTA_BT_SW_TX_FIFO_IN**Type:** W**Reset State:** 0x00000000

Software Transmit FIFO write data register.

WCSS_A_QTA_BT_SW_TX_FIFO_IN

Bits	Name	Description
7:0	DATA	Software character to be transmitted.

0x0A4C001C WCSS_A_QTA_BT_SW_RX_FIFO_OUT**Type:** R**Reset State:** 0x00000000

Software Receive FIFO read data register.

WCSS_A_QTA_BT_SW_RX_FIFO_OUT

Bits	Name	Description
7:0	DATA	Software character received.

10.4 WCSS_BT_SSBi2_CFG**10.4.1 WCSS_A_BT_SSBi2_CFG_BT_SSBi2_CFG****0x0A4BFFEC WCSS_A_BT_SSBi2_CFG_SSBi2_PVC_FIFO****Type:** W**Clock:** PCLK**Reset State:** 0x00000000

WCSS_A_BT_SSB12_CFG_SSB12_PVC_FIFO

Bits	Name	Description
12:8	PVC_INDEX	
7:0	PVC_DATA	Sub-Section 2.2.1: Crystal calibration Stop Parsing at Section 2.3: Link controller registers Start Parsing at Section 1.2: SSB12 Configuration Micro ARM registers

10.5 WCSS_A_SSB12_CFG_SSB12_CFG**0x0A4A0000 WCSS_A_SSB12_CFG_SSB12_CTL****Type:** RW**Clock:** AHB bus clock**Reset State:** 0x00083005

Specifies various generic SSB12 control information. Bits 19:13 comprising of fields SSB12_DATA_PDEN, FCLK_MODE, FCLK_SW_OVERRIDE and NUM_FCLK are either asynchronously reset or set. This is because they control the signals going directly to a pad. Remaining bits 12:0 are synchronously reset or set.

WCSS_A_SSB12_CFG_SSB12_CTL

Bits	Name	Description
19:16	NUM_FCLK	Indicates a programmable number of active forwarding clock cycles following a SSB12 'transaction done' for a corresponding SSB12 request. Unless there is a new SSB12 transaction request, the forwarding clock is shut off after toggling NUM_FCLK cycles at the PMIC SSB12 clock frequency. The suggested setting (also the default value) is 8 clock cycles.
15	FCLK_SW_OVERRIDE	This is a software override bit to control the CGC for forwarding clock. Set this bit to 1 to allow the SSB12 clock to continue as a forwarding clock through the CGC (provided FLCK_MODE = 1). This override bit is provided to allow continuous toggling clock to SSB12 slave irrespective of how the HW logic controls the CGC. Set this bit to 0 to enable pure HW control on the forwarding clock CGC.
14	FCLK_MODE	Set this bit to 1 to activate forwarding clock mode. When this bit is 0, the FCLK_SW_OVERRIDE field has no meaning. Out of reset, the mode is set to 0 indicating that the SSB12 slave does not work off the forwarding clock. When FCLK_MODE = 0; no forwarding clock. When FCLK_MODE = 1; FCLK_SW_OVERRIDE = 1; forwarding clock always toggles. When FCLK_MODE = 1; FCLK_SW_OVERRIDE = 0; forwarding clock is controlled by HW logic.
13	SSB12_DATA_PDEN	Pulldown enable for SSB12_DATA pad.

WCSS_A_SSB_I2_CFG_SSB_I2_CTL (cont.)

Bits	Name	Description
12	PA_CGC_CTRL	This is to control the SSB_I2 clock going to this PMIC arbiter. Set this bit to 1 when you want to allow the clock to flow through the CGC. Set this bit to 0 to shut off clock to PMIC arbiter. Out of reset, this bit is set to 1.
10:8	SSB_I2_ADDR_WIDTH	Specifies what is the addressing mode in SSB_I2 mode. This field is only applicable in SSB_I2 mode and not the old SSB_I1 mode. 0x0: ENUM_8_BIT_ADDRESSING_MODE (8-bit addressing mode) 0x1: ENUM_9_BIT_ADDRESSING_MODE_AND_SO_ON (9-bit addressing mode and so on)
7	SSB_I2_MODE	Set this bit to 1 when SW wants to switch to new SSB_I2 mode. Set this is 0 when in old SSB_I1 mode.
5:4	SEL_RD_DATA	Specifies which RD_DATA value to select out of four possible combinations. This signal allows the returned data from the slave to be captured inside the master, 1, 2, or 3 symbols late in order to accommodate a large range of routing delay. Based on this field, the entire read transaction length will increase by SEL_RD_DATA number of symbols.
3:2	SSB_I2_DATA_DEL	Specifies the number of 1/2 clock periods, SSB_I2_DATA will be delayed before sampling inside.
1:0	IDLE_SYMS	The master will ensure there are at least this number of idle symbol periods in between the accesses. The idle time is inserted at the end of the read or write transaction. This supported values are always between 1 - 3.

0x0A4A0004 WCSS_A_SSB_I2_CFG_SSB_I2_INT_ENABLE**Type:** RW**Clock:** AHB bus clock**Reset State:** 0x00000000

Provides control on whether the TRANSACTION-DONE interrupts can be generated or not (per SW master).

WCSS_A_SSB_I2_CFG_SSB_I2_INT_ENABLE

Bits	Name	Description
6:0	ENABLE_SSB_I2_INT	SSB_I2 transaction done interrupts for each SW master are generated only when the corresponding bits are set. The Interrupt mapping corresponds to the entries in the VIT. For e.g. if the VMID in VIT index 0 corresponds to the SW master RPM and VMID in index 1 of the VIT corresponds to SW master ARM 11, the bit 0 in the register will enable the interrupt corresponding to RPM, bit 1 will enable interrupt corresponding to ARM11.

0x0A4A0008 WCSS_A_SSBi2_CFG_SSBi2_PRIORITIES**Type:** RW**Clock:** AHB bus clock**Reset State:** 0x00000000

Specifies the priorities for SSBi2 arbiter ports. The SSBi2 block supports up to maximum of 8 ports for arbitration. Each port has 3 bits to denote its priority. The bits are synchronously reset.

CAUTION Since the reset state value of SSBi2_PRIORITIES is 0, only SW requests (hard-coded to port #0) can get through. SW has to program this register to allow hardware requests through the SSBi2 arbiter.

WCSS_A_SSBi2_CFG_SSBi2_PRIORITIES

Bits	Name	Description
11:9	PRIORITY3	Specifies the port number of host, with a priority that is 1 less than PRIORITY2.
8:6	PRIORITY2	Specifies the port number of host, with a priority that is 1 less than PRIORITY1.
5:3	PRIORITY1	Specifies the port number of host, with a priority that is 1 less than PRIORITY0.
2:0	PRIORITY0	Specifies the port number of the highest priority host.

0x0A4A000C WCSS_A_SSBi2_CFG_SSBi2_BANKS_2_3_CTL**Type:** RW**Clock:** AHB bus clock**Reset State:** 0xFFFFFFFF

This register specifies whether an access is allowed to banks 2 and 3 of PMIC. The access control to the banks 2 and 3 of PMIC is set per VMID. Bit 0 corresponds to access control for VMID 0 and bit 31 corresponds to access control for VMID 31.

WCSS_A_SSBi2_CFG_SSBi2_BANKS_2_3_CTL

Bits	Name	Description
31:0	ACCESS_CONTROL	Set each of the 32 bits to 1 if the master corresponding to that VMID wishes to initiate SW requests to banks 2 and 3 of PMIC.

0x0A4A0010 WCSS_A_SSBi2_CFG_SSBi2_ILLEGAL_STATUS**Type:** R**Clock:** AHB bus clock**Reset State:** 0x00000000

Returns status information of the software master that tried to access a SSBI2 command micro register, and failed **security** through the VIT table.

WCSS_A_SSBI2_CFG_SSBI2_ILLEGAL_STATUS

Bits	Name	Description
5:1	ILLEGAL_VMID	Indicates the VMID of the master that performed the most recent illegal access. The value in this field is only meaningful when the ILLEGAL_ACCESS bit is set. On software reads, the value is retained unless another illegal access with a different VMID happens. In that case, this field will have the VMID corresponding to the most recent illegal master.
0	ILLEGAL_ACCESS	When set (1) indicates an illegal access. An illegal access is defined as an access that does not find a matching VMID in the SSBI2_VITn registers. This bit is cleared on a software read provided there is no other illegal access in the same clock cycle that a software read happens.

**0x0A4A0020+WCSS_A_SSBI2_CFG_SSBI2_VITn, n=[0..6]
0x4*n**

Type: RW

Clock: AHB bus clock

Reset State: 0x00000000

This is the VMID index table(VIT) used for enabling concurrent SSBI2 SW requests among multiple SW masters. Secure entities are expected to assign a VMID for the SW master corresponding to an index. The SSBI2_VIT0 has the VMID associated with index 0, SSBI2_VIT1 has the VMID associated with index 1 and so on.

CAUTION By default, only the SW master with a VMID of '0' has access to registers in SSBI2 command micro space. SROT/RPM has to populate this table before valid SW masters can initiate SSBI2 SW transactions.

SSBI2_VITn, n=[0..6]

WCSS_A_SSBI2_CFG_SSBI2_VITn

Bits	Name	Description
4:0	VMID	VMID corresponding to index n.

0x0A4A00A0 WCSS_A_SSBI2_CFG_SSBI2_PVC_CTL

Type: RW

Clock: AHB bus clock

Reset State: 0x00000000

This register provides the global enable/disable for the PVC interface module, per PVC port enable/disable as well as captures PVC interface busy status.

WARNING In a PMIC arbiter wrapper system having more than 1 PMIC arbiter module; if a PVC port(x) is enabled in one of the PMIC arbiters, the corresponding PVC_port(x) in all the other PMIC arbiters shall be disabled. If multiple PVC_port(x)'s enabled simultaneously, the hardware does not guarantee correct operation.

WCSS_A_SSB_I2_CFG_SSB_I2_PVC_CTL

Bits	Name	Description
8	PVC_INTF_BUSY	This is a status bit. When set (1) indicates that the PVC interface module is busy (meaning there is atleast one PVC transaction being serviced). Busy line goes high when PVC interface detects a new request on one of the PVC ports and stays high until it samples a done from SSB_I2 arbiter. Writes to this bit will not have any effect and will be ignored.
3	PVC_PORT3_EN	Set this bit to 1 to enable PVC port 3. PVC port 3 maps to SMPSS CPU1 SAW.
2	PVC_PORT2_EN	Set this bit to 1 to enable PVC port 2. PVC port 2 maps to SMPSS CPU 0 SAW.
1	PVC_PORT1_EN	Set this bit to 1 to enable PVC port 1. PVC port 1 maps to LPASS Q6 SAW.
0	PVC_INTF_EN	Set this bit to 1 to enable PVC interface module.

0x0A4A00A4 WCSS_A_SSB_I2_CFG_SSB_I2_PVC_TABLEn, n=[0..31] +0x4*n

Type: RW

Clock: AHB bus clock

Reset State: 0x00000000

This set of registers indicates the full PVC table corresponding to all the PVC ports. For e.g.: there are 4 entries per PVC port(set by RTL generic) and there are 3 PVC ports (set by RTL generic) in 12 entries. Each entry is 15-bit address to the SSB_I2 slave registers. This indirect addressing from the PVC ports to the SSB_I2 slave registers provides a way to enforce **security** on hardware SSB_I2 requests.

NOTE Though this current document shows 12 entries, the number of entries should scale based on the GENERIC settings for the number of PVC ports and the address index width. Also though 15-bit addresses are supported, 10-bit addressing is all that is expected for current PMICs.

SSB_I2_PVC_TABLEn, n=[0...11]

WCSS_A_SSBi2_CFG_SSBi2_PVC_TABLEn

Bits	Name	Description
14:0	ADDR	15-bit address of the SSBi2 slave registers.

10.6 WCSS_A_SSBi2_CMD_SSBi2_CMD**0x0A480000 WCSS_A_SSBi2_CMD_SSBi2_CMD****Type:** W**Clock:** AHB bus clock**Reset State:** 0x00000000

The SW masters write to this register to initiate either a read or a write SSBi2 transaction. Through this register, the SW can address up to 32 kB of SSBi2 slave register space. However the PMICs only house a maximum of 1024 registers and hence the values on the most significant 5-bits of the REG_ADDR_14_8 field are ignored. The REG_DATA field is only applicable for write transaction and is ignored for reads. The read data returned back from the SSBi2 slave is directed to the SSBi2_RD_STATUS register. Bit 23 is left as a reserved bit to enable slave address to grow to 16-bits in the future.

WCSS_A_SSBi2_CMD_SSBi2_CMD

Bits	Name	Description
24	RDWRN	Denotes whether the command to perform is a register write (0) or read (1).
22:16	REG_ADDR_14_8	Specifies the upper 7 bits, 14:8, of SSBi slave register address.
15:8	REG_ADDR_7_0	Lower 8 bits, 7:0, of SSBi slave register address.
7:0	REG_DATA	Write data for a slave register.

0x0A480004 WCSS_A_SSBi2_CMD_SSBi2_RD_STATUS**Type:** R**Clock:** AHB bus clock**Reset State:** 0x00000000

This register provides the read data as well as the status of the transaction. This register also returns back information associated with the transaction such as the SSBi2 slave register address and the transaction access type. SW is expected to poll TRANS_DONE and TRANS_DENIED until either of these bits are set to 1.

- TRANS_DONE = 0, TRANS_DENIED = 1; indicates to SW that the current transaction was rejected because the master did not have permissions to access the corresponding SSBi2 slave register.

- TRANS_DONE = 1, TRANS_DENIED = 0; indicates to SW that the current transaction was completed successfully. For writes, it means that the write data was sent out on the SSBI wire and for reads, it means that the read data is returned back and is now in SSBI2_RD_STATUS[REG_DATA]. At this point, the SW is said to have completed one transaction.
- Both TRANS_DONE and TRANS_DENIED being either 0 or 1 does not happen and is not a valid combination after a transaction has been initiated.

WCSS_A_SSBI2_CMD_SSBI2_RD_STATUS

Bits	Name	Description
27	TRANS_DONE	When set (1), indicates that the SSBI2 has completed the transaction. If this was a write request, the write has completed. If this was a read request, the read data has returned back.. NOTE This bit is cleared on read.
26	TRANS_DENIED	When set (1), indicates that a SSBI2 transaction was denied because the master did not have access to the particular SSBI2 slave SW register to where the access was intended. For PM8058 banks 2 and 3, this bit is set based on the settings in SSBI2_BANKS_2_3_CTL register that blocked the transaction. NOTE This bit is cleared on read.
25	TRANS_PROG	When set (1), indicates that a SSBI2 transaction is in progress. This signal will be set for the duration from REQ assertion until DONE assertion.
24	RDWRN	Returns the access type of transaction. A value of 1 implies a read transaction and a value of 0 implies a write transaction.
22:8	REG_ADDR	Returns the address to which the SSBI2 transaction was performed.
7:0	REG_DATA	Returned read register data. NOTE The read data is not cleared on reads, however will return all zeroes for a SW master whose transaction was denied or was an illegal SW master.

10.7 WCSS_A_PMU_PMU**0x0A21B000 RESERVED****0x0A21B008 WCSS_A_PMU_TOP_AHB_CBCR****Type:** RW**Reset State:** 0x00000001

cbcr for top ahb clk

WCSS_A_PMU_TOP_AHB_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	clock off
19:16	CLK_DIV	RW	clock enable
14	FORCE_MEM_CORE_ON	RW	force_mem_core_on
13	FORCE_MEM_PERIPH_ON	RW	force_mem_periph_on
12	FORCE_MEM_PERIPH_OFF	RW	force_mem_periph_off
11:8	WAKEUP	RW	wakeup
7:4	SLEEP	RW	sleep
1	HW_CTL	RW	hw control enable
0	CLK_ENABLE	RW	clock enable

00x0A21B010 WCSS_A_PMU_TOP_SLP_CBCR**Type:** RW**Reset State:** 0x00000001

cbr for top slp clk

WCSS_A_PMU_TOP_SLP_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	clock off
19:16	CLK_DIV	RW	clock enable
14	FORCE_MEM_CORE_ON	RW	force_mem_core_on
13	FORCE_MEM_PERIPH_ON	RW	force_mem_periph_on
12	FORCE_MEM_PERIPH_OFF	RW	force_mem_periph_off
11:8	WAKEUP	RW	wakeup
7:4	SLEEP	RW	sleep
1	HW_CTL	RW	hw control enable
0	CLK_ENABLE	RW	clock enable

0x0A21B014 WCSS_A_PMU_TOP_XO_CBCR**Type:** RW**Reset State:** 0x00000001

cbr for top xo clk

WCSS_A_PMU_TOP_XO_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	clock off
19:16	CLK_DIV	RW	clock enable
14	FORCE_MEM_CORE_ON	RW	force_mem_core_on
13	FORCE_MEM_PERIPH_ON	RW	force_mem_periph_on
12	FORCE_MEM_PERIPH_OFF	RW	force_mem_periph_off
11:8	WAKEUP	RW	wakeup
7:4	SLEEP	RW	sleep
1	HW_CTL	RW	hw control enable
0	CLK_ENABLE	RW	clock enable

0x0A21B018 WCSS_A_PMU_AXI_M_CBCR

Type: RW

Reset State: 0x00000001

cbr for o_wcss_axi_m_clk

WCSS_A_PMU_AXI_M_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	clock off
1	HW_CTL	RW	hw control enable
0	CLK_ENABLE	RW	clock enable

0x0A21B020 WCSS_A_PMU_COMMON_SS_BCR

Type: RW

Reset State: 0x00000000

bcr for common_ss

WCSS_A_PMU_COMMON_SS_BCR

Bits	Name	Type	Description
31	DFD_STATUS	R	dfd enable

WCSS_A_PMU_COMMON_SS_BCR (cont.)

Bits	Name	Type	Description
1	DFD_EN	RW	dfd enable
0	BLK_ARES	RW	asynchronous reset to Pronto top modules

0x0A21B024 WCSS_A_PMU_COMMON_GDSCR**Type:** RW**Reset State:** 0x00222001

gdscr for common_ss

WCSS_A_PMU_COMMON_GDSCR

Bits	Name	Type	Description
31	PD_OFF	R	
30:27	GDSC_STATE	R	
23:20	EN_REST_WAIT	RW	
19:16	EN_FEW_WAIT	RW	
15:12	CLK_DIS_WAIT	RW	
10	RESTORE	RW	
9	SAVE	RW	
8	RETAIN	RW	
7	EN_REST	RW	
6	EN_FEW	RW	
5	CLAMP_IO	RW	
4	CLK_DISABLE	RW	
3	PD_ARES	RW	
2	SW_OVERRIDE	RW	
1	HW_CONTROL	RW	collapse control is from HW.
0	SW_COLLAPSE	RW	sw collapse the GDHS.

0x0A21B02C WCSS_A_PMU_COMMON_TRC_CBCR**Type:** RW**Reset State:** 0x80000000

cbcr for common trc clk

WCSS_A_PMU_COMMON_TRC_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	clock off
19:16	CLK_DIV	RW	clock enable
14	FORCE_MEM_CORE_ON	RW	force_mem_core_on
13	FORCE_MEM_PERIPH_ON	RW	force_mem_periph_on
12	FORCE_MEM_PERIPH_OFF	RW	force_mem_periph_off
11:8	WAKEUP	RW	wakeup
7:4	SLEEP	RW	sleep
1	HW_CTL	RW	hw control enable
0	CLK_ENABLE	RW	clock enable

0x0A21B030 WCSS_A_PMU_COMMON_CPU_CBCR**Type:** RW**Reset State:** 0x80000FF0

cbr for common cpu clk

WCSS_A_PMU_COMMON_CPU_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	clock off
19:16	CLK_DIV	RW	clock enable
14	FORCE_MEM_CORE_ON	RW	force_mem_core_on
13	FORCE_MEM_PERIPH_ON	RW	force_mem_periph_on
12	FORCE_MEM_PERIPH_OFF	RW	force_mem_periph_off
11:8	WAKEUP	RW	wakeup
7:4	SLEEP	RW	sleep
1	HW_CTL	RW	hw control enable
0	CLK_ENABLE	RW	clock enable

0x0A21B034 WCSS_A_PMU_COMMON_AHB_CBCR**Type:** RW**Reset State:** 0x80000FF0

cbr for common ahb clk

WCSS_A_PMU_COMMON_AHB_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	clock off
19:16	CLK_DIV	RW	clock enable
14	FORCE_MEM_CORE_ON	RW	force_mem_core_on
13	FORCE_MEM_PERIPH_ON	RW	force_mem_periph_on
12	FORCE_MEM_PERIPH_OFF	RW	force_mem_periph_off
11:8	WAKEUP	RW	wakeup
7:4	SLEEP	RW	sleep
1	HW_CTL	RW	hw control enable
0	CLK_ENABLE	RW	clock enable

0x0A21B038 WCSS_A_PMU_COMMON_BT_32M_CBCR

Type: RW

Reset State: 0x80000000

cbr for common bt_32m clk

WCSS_A_PMU_COMMON_BT_32M_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	clock off
19:16	CLK_DIV	RW	clock enable
14	FORCE_MEM_CORE_ON	RW	force_mem_core_on
13	FORCE_MEM_PERIPH_ON	RW	force_mem_periph_on
12	FORCE_MEM_PERIPH_OFF	RW	force_mem_periph_off
11:8	WAKEUP	RW	wakeup
7:4	SLEEP	RW	sleep
1	HW_CTL	RW	hw control enable
0	CLK_ENABLE	RW	clock enable

0x0A21B03C WCSS_A_PMU_COMMON_BT_FM_19M_CBCR**Type:** RW**Reset State:** 0x80000000

cbr for common bt_fm_19m clk

WCSS_A_PMU_COMMON_BT_FM_19M_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	clock off
19:16	CLK_DIV	RW	clock enable
14	FORCE_MEM_CORE_ON	RW	force_mem_core_on
13	FORCE_MEM_PERIPH_ON	RW	force_mem_periph_on
12	FORCE_MEM_PERIPH_OFF	RW	force_mem_periph_off
11:8	WAKEUP	RW	wakeup
7:4	SLEEP	RW	sleep
1	HW_CTL	RW	hw control enable
0	CLK_ENABLE	RW	clock enable

0x0A21B040 WCSS_A_PMU_COMMON_SLP_CBCR**Type:** RW**Reset State:** 0x80000000

cbr for common slp clk

WCSS_A_PMU_COMMON_SLP_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	clock off
19:16	CLK_DIV	RW	clock enable
14	FORCE_MEM_CORE_ON	RW	force_mem_core_on
13	FORCE_MEM_PERIPH_ON	RW	force_mem_periph_on
12	FORCE_MEM_PERIPH_OFF	RW	force_mem_periph_off
11:8	WAKEUP	RW	wakeup
7:4	SLEEP	RW	sleep
1	HW_CTL	RW	hw control enable
0	CLK_ENABLE	RW	clock enable

0x0A21B044 WCSS_A_PMU_COMMON_XO_CBCR**Type:** RW**Reset State:** 0x00000001

cbr for common xo clk

WCSS_A_PMU_COMMON_XO_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	clock off
19:16	CLK_DIV	RW	clock enable
14	FORCE_MEM_CORE_ON	RW	force_mem_core_on
13	FORCE_MEM_PERIPH_ON	RW	force_mem_periph_on
12	FORCE_MEM_PERIPH_OFF	RW	force_mem_periph_off
11:8	WAKEUP	RW	wakeup
7:4	SLEEP	RW	sleep
1	HW_CTL	RW	hw control enable
0	CLK_ENABLE	RW	clock enable

0x0A21B050 WCSS_A_PMU_WLAN_BCR**Type:** RW**Reset State:** 0x00000000

bcr for wlan

WCSS_A_PMU_WLAN_BCR

Bits	Name	Type	Description
31	DFD_STATUS	R	dfd enable
1	DFD_EN	RW	dfd enable
0	BLK_ARES	RW	asynchronous reset to Pronto top modules

0x0A21B054 WCSS_A_PMU_WLAN_GDSCR**Type:** RW**Reset State:** 0x00222001

gdscr for wlan_ss

WCSS_A_PMU_WLAN_GDSCR

Bits	Name	Type	Description
31	PD_OFF	R	
30:27	GDSC_STATE	R	
23:20	EN_REST_WAIT	RW	
19:16	EN_FEW_WAIT	RW	
15:12	CLK_DIS_WAIT	RW	
10	RESTORE	RW	
9	SAVE	RW	
8	RETAIN	RW	
7	EN_REST	RW	
6	EN_FEW	RW	
5	CLAMP_IO	RW	
4	CLK_DISABLE	RW	
3	PD_ARES	RW	
2	SW_OVERRIDE	RW	
1	HW_CONTROL	RW	collapse control is from HW.
0	SW_COLLAPSE	RW	sw collapse the GDHS.

0x0A21B058 WCSS_A_PMU_WLAN_RFA_CAL_CBCR**Type:** RW**Reset State:** 0x80000000

cbr for wlan_rfa_cal_clk

WCSS_A_PMU_WLAN_RFA_CAL_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	clock off
17:16	CLK_DIV	RW	clock division ration is clk_div + 1
14	FORCE_MEM_CORE_ON	RW	force_mem_core_on
13	FORCE_MEM_PERIPH_ON	RW	force_mem_periph_on
12	FORCE_MEM_PERIPH_OFF	RW	force_mem_periph_off
11:8	WAKEUP	RW	wakeup
7:4	SLEEP	RW	sleep

WCSS_A_PMU_WLAN_RFA_CAL_CBCR (cont.)

Bits	Name	Type	Description
1	HW_CTL	RW	hw control enable
0	CLK_ENABLE	RW	clock enable

0x0A21B05C WCSS_A_PMU_WLAN_PHY_MAIN1_FAST_CBCR**Type:** RW**Reset State:** 0x80000000

cbr for wlan phy_main1_fast clk

WCSS_A_PMU_WLAN_PHY_MAIN1_FAST_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	clock off
19:16	CLK_DIV	RW	clock enable
14	FORCE_MEM_CORE_ON	RW	force_mem_core_on
13	FORCE_MEM_PERIPH_ON	RW	force_mem_periph_on
12	FORCE_MEM_PERIPH_OFF	RW	force_mem_periph_off
11:8	WAKEUP	RW	wakeup
7:4	SLEEP	RW	sleep
1	HW_CTL	RW	hw control enable
0	CLK_ENABLE	RW	clock enable

0x0A21B060 WCSS_A_PMU_WLAN_PHY_MAIN1_TX_CBCR**Type:** RW**Reset State:** 0x80000FF0

cbr for wlan phy_main1 clk

WCSS_A_PMU_WLAN_PHY_MAIN1_TX_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	clock off
19:16	CLK_DIV	RW	clock enable
14	FORCE_MEM_CORE_ON	RW	force_mem_core_on
13	FORCE_MEM_PERIPH_ON	RW	force_mem_periph_on

WCSS_A_PMU_WLAN_PHY_MAIN1_TX_CBCR (cont.)

Bits	Name	Type	Description
12	FORCE_MEM_PERIPH_OFF	RW	force_mem_periph_off
11:8	WAKEUP	RW	wakeup
7:4	SLEEP	RW	sleep
1	HW_CTL	RW	hw control enable
0	CLK_ENABLE	RW	clock enable

0x0A21B064 WCSS_A_PMU_WLAN_PHY_MAIN1_RX_CBCR**Type:** RW**Reset State:** 0x80000FF0

cbr for wlan phy_main1 clk

WCSS_A_PMU_WLAN_PHY_MAIN1_RX_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	clock off
19:16	CLK_DIV	RW	clock enable
14	FORCE_MEM_CORE_ON	RW	force_mem_core_on
13	FORCE_MEM_PERIPH_ON	RW	force_mem_periph_on
12	FORCE_MEM_PERIPH_OFF	RW	force_mem_periph_off
11:8	WAKEUP	RW	wakeup
7:4	SLEEP	RW	sleep
1	HW_CTL	RW	hw control enable
0	CLK_ENABLE	RW	clock enable

0x0A21B068 WCSS_A_PMU_WLAN_PHY_MAIN2_CBCR**Type:** RW**Reset State:** 0x80000FF0

cbr for wlan phy_main2 clk

WCSS_A_PMU_WLAN_PHY_MAIN2_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	clock off

WCSS_A_PMU_WLAN_PHY_MAIN2_CBCR (cont.)

Bits	Name	Type	Description
19:16	CLK_DIV	RW	clock enable
14	FORCE_MEM_CORE_ON	RW	force_mem_core_on
13	FORCE_MEM_PERIPH_ON	RW	force_mem_periph_on
12	FORCE_MEM_PERIPH_OFF	RW	force_mem_periph_off
11:8	WAKEUP	RW	wakeup
7:4	SLEEP	RW	sleep
1	HW_CTL	RW	hw control enable
0	CLK_ENABLE	RW	clock enable

0x0A21B070 WCSS_A_PMU_WLAN_PHY_DAC_CBCR**Type:** RW**Reset State:** 0x80000000

cbcr for wlan phy_dac clk

WCSS_A_PMU_WLAN_PHY_DAC_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	clock off
19:16	CLK_DIV	RW	clock enable
14	FORCE_MEM_CORE_ON	RW	force_mem_core_on
13	FORCE_MEM_PERIPH_ON	RW	force_mem_periph_on
12	FORCE_MEM_PERIPH_OFF	RW	force_mem_periph_off
11:8	WAKEUP	RW	wakeup
7:4	SLEEP	RW	sleep
1	HW_CTL	RW	hw control enable
0	CLK_ENABLE	RW	clock enable

0x0A21B074 WCSS_A_PMU_WLAN_AHB_CBCR**Type:** RW**Reset State:** 0x80000FF0

cbcr for wlan ahb clk

WCSS_A_PMU_WLAN_AHB_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	clock off
19:16	CLK_DIV	RW	clock enable
14	FORCE_MEM_CORE_ON	RW	force_mem_core_on
13	FORCE_MEM_PERIPH_ON	RW	force_mem_periph_on
12	FORCE_MEM_PERIPH_OFF	RW	force_mem_periph_off
11:8	WAKEUP	RW	wakeup
7:4	SLEEP	RW	sleep
1	HW_CTL	RW	hw control enable
0	CLK_ENABLE	RW	clock enable

0x0A21B078 WCSS_A_PMU_WLAN_PHY_ADC_CBCR**Type:** RW**Reset State:** 0x80000000

cbr for wlan phy_adc clk

WCSS_A_PMU_WLAN_PHY_ADC_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	clock off
19:16	CLK_DIV	RW	clock enable
14	FORCE_MEM_CORE_ON	RW	force_mem_core_on
13	FORCE_MEM_PERIPH_ON	RW	force_mem_periph_on
12	FORCE_MEM_PERIPH_OFF	RW	force_mem_periph_off
11:8	WAKEUP	RW	wakeup
7:4	SLEEP	RW	sleep
1	HW_CTL	RW	hw control enable
0	CLK_ENABLE	RW	clock enable

0x0A21B07C WCSS_A_PMU_WLAN_RFIF_CBCR**Type:** RW**Reset State:** 0x80000000

cbr for wlan rfif clk

WCSS_A_PMU_WLAN_RFIF_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	clock off
19:16	CLK_DIV	RW	clock enable
14	FORCE_MEM_CORE_ON	RW	force_mem_core_on
13	FORCE_MEM_PERIPH_ON	RW	force_mem_periph_on
12	FORCE_MEM_PERIPH_OFF	RW	force_mem_periph_off
11:8	WAKEUP	RW	wakeup
7:4	SLEEP	RW	sleep
1	HW_CTL	RW	hw control enable
0	CLK_ENABLE	RW	clock enable

0x0A21B080 WCSS_A_PMU_ADC_240M_CBCR

Type: RW

Reset State: 0x80000000

cbr for adc 240m clk

WCSS_A_PMU_ADC_240M_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	clock off
19:16	CLK_DIV	RW	clock enable
14	FORCE_MEM_CORE_ON	RW	force_mem_core_on
13	FORCE_MEM_PERIPH_ON	RW	force_mem_periph_on
12	FORCE_MEM_PERIPH_OFF	RW	force_mem_periph_off
11:8	WAKEUP	RW	wakeup
7:4	SLEEP	RW	sleep
1	HW_CTL	RW	hw control enable
0	CLK_ENABLE	RW	clock enable

0x0A21B084 WCSS_A_PMU_DAC_SM_CBCR**Type:** RW**Reset State:** 0x80000000

cbr for dac_sm_clk

WCSS_A_PMU_DAC_SM_CBCR

Bits	Name	Type	Description
31	CLK_OFF	R	clock off
17:16	CLK_DIV	RW	clock division ration is clk_div + 1
14	FORCE_MEM_CORE_ON	RW	force_mem_core_on
13	FORCE_MEM_PERIPH_ON	RW	force_mem_periph_on
12	FORCE_MEM_PERIPH_OFF	RW	force_mem_periph_off
11:8	WAKEUP	RW	wakeup
7:4	SLEEP	RW	sleep
1	HW_CTL	RW	hw control enable
0	CLK_ENABLE	RW	clock enable

0x0A21B0A0 WCSS_A_PMU_PHY_FAST_CMD_RCGR**Type:** RW**Reset State:** 0x00000000

cmd_cgr for phy_fast clk src

WCSS_A_PMU_PHY_FAST_CMD_RCGR

Bits	Name	Type	Description
31	ROOT_OFF	R	root clock off
4	DIRTY_CFG_RCGR	R	dirty bit for cfg_rcgr. When cfg_rcgr is written the bit is set to 1 and cleared when update bit is set.
1	ROOT_EN	RW	root_en
0	UPDATE	RW	update configuration and status configuration and status

0x0A21B0A4 WCSS_A_PMU_PHY_FAST_CFG_RCGR**Type:** RW**Reset State:** 0x00000000

cfg_cgr for phy_fast clk src

WCSS_A_PMU_PHY_FAST_CFG_RCGR

Bits	Name	Description
10:8	SRC_SEL	src_sel
4:0	SRC_DIV	src_div. division ration = (src_div+1)/2

0x0A21B0C0 WCSS_A_PMU_PHY_CMD_RCGR

Type: RW

Reset State: 0x80000000

cmd_cgr for phy clk src

WCSS_A_PMU_PHY_CMD_RCGR

Bits	Name	Type	Description
31	ROOT_OFF	R	root clock off
7	DIRTY_D	R	dirty bit for d. When d is written the bit is set to 1 and cleared when update bit is set.
6	DIRTY_N	R	dirty bit for n. When n is written the bit is set to 1 and cleared when update bit is set.
5	DIRTY_M	R	dirty bit for m. When m is written the bit is set to 1 and cleared when update bit is set.
4	DIRTY_CFG_RCGR	R	dirty bit for cfg_rcgr. When cfg_rcgr is written the bit is set to 1 and cleared when update bit is set.
1	ROOT_EN	RW	root_en
0	UPDATE	RW	update configuration and status configuration and status

0x0A21B0C4 WCSS_A_PMU_PHY_CFG_RCGR

Type: RW

Reset State: 0x00000000

cfg_cgr for phy clk src

WCSS_A_PMU_PHY_CFG_RCGR

Bits	Name	Description
13:12	MODE	mnd mode. 00 - bypass, 11 - mnd
10:8	SRC_SEL	src_sel

WCSS_A_PMU_PHY_CFG_RCGR (cont.)

Bits	Name	Description
4:0	SRC_DIV	src_div

0x0A21B0C8 WCSS_A_PMU_PHY_M**Type:** RW**Reset State:** 0x00000000

m for phy clk src

WCSS_A_PMU_PHY_M

Bits	Name	Description
7:0	M	m for mnd_rcg

0x0A21B0CC WCSS_A_PMU_PHY_N**Type:** RW**Reset State:** 0x00000000

not_n_minus_m for phy clk src

WCSS_A_PMU_PHY_N

Bits	Name	Description
7:0	NOT_N_MINUS_M	not_n_minus_m for mnd_rcg

0x0A21B0D0 WCSS_A_PMU_PHY_D**Type:** RW**Reset State:** 0x00000000

not_2d for phy clk src

WCSS_A_PMU_PHY_D

Bits	Name	Description
7:0	NOT_2D	not_2d for mnd_rcg

0x0A21B0E0 WCSS_A_PMU_ADCDAC_CMD_RCGR**Type:** RW**Reset State:** 0x00000000

cmd_cgr for adcdac clk src

WCSS_A_PMU_ADCDAC_CMD_RCGR

Bits	Name	Type	Description
31	ROOT_OFF	R	root clock off
7	DIRTY_D	R	dirty bit for d. When d is written the bit is set to 1 and cleared when update bit is set.
6	DIRTY_N	R	dirty bit for n. When n is written the bit is set to 1 and cleared when update bit is set.
5	DIRTY_M	R	dirty bit for m. When m is written the bit is set to 1 and cleared when update bit is set.
4	DIRTY_CFG_RCGR	R	dirty bit for cfg_rcgr. When cfg_rcgr is written the bit is set to 1 and cleared when update bit is set.
1	ROOT_EN	RW	root_en
0	UPDATE	RW	update configuration and status

0x0A21B0E4 WCSS_A_PMU_ADCDAC_CFG_RCGR**Type:** RW**Reset State:** 0x00000000

cfg_cgr for adcdac clk src

WCSS_A_PMU_ADCDAC_CFG_RCGR

Bits	Name	Description
13:12	MODE	mnd mode. 00 - bypass, 11 - mnd
10:8	SRC_SEL	src_sel
4:0	SRC_DIV	src_div

0x0A21B0E8 WCSS_A_PMU_ADCDAC_M**Type:** RW**Reset State:** 0x00000000

m for adcdac clk src

WCSS_A_PMU_ADCDAC_M

Bits	Name	Description
7:0	M	m for mnd_rcg

0x0A21B0EC WCSS_A_PMU_ADCDAC_N**Type:** RW**Reset State:** 0x00000000

not_n_minus_m for adcdac clk src

WCSS_A_PMU_ADCDAC_N

Bits	Name	Description
7:0	NOT_N_MINUS_M	not_n_minus_m for mnd_rcg

0x0A21B0F0 WCSS_A_PMU_ADCDAC_D**Type:** RW**Reset State:** 0x00000000

not_2d for adcdac clk src

WCSS_A_PMU_ADCDAC_D

Bits	Name	Description
7:0	NOT_2D	not_2d for mnd_rcg

0x0A21B100 WCSS_A_PMU_RFIF_CMD_RCGR**Type:** RW**Reset State:** 0x00000000

cmd_cgr for rfif clk src

WCSS_A_PMU_RFIF_CMD_RCGR

Bits	Name	Type	Description
31	ROOT_OFF	R	root clock off
7	DIRTY_D	R	dirty bit for d. When d is written the bit is set to 1 and cleared when update bit is set.

WCSS_A_PMU_RFIF_CMD_RCGR (cont.)

Bits	Name	Type	Description
6	DIRTY_N	R	dirty bit for n. When n is written the bit is set to 1 and cleared when update bit is set.
5	DIRTY_M	R	dirty bit for m. When m is written the bit is set to 1 and cleared when update bit is set.
4	DIRTY_CFG_RCGR	R	dirty bit for cfg_rcgr. When cfg_rcgr is written the bit is set to 1 and cleared when update bit is set.
1	ROOT_EN	RW	root_en
0	UPDATE	RW	update configuration and status

0x0A21B104 WCSS_A_PMU_RFIF_CFG_RCGR**Type:** RW**Reset State:** 0x00000000

cfg_cgr for rfif clk src

WCSS_A_PMU_RFIF_CFG_RCGR

Bits	Name	Description
13:12	MODE	mnd mode. 00 - bypass, 11 - mnd
10:8	SRC_SEL	src_sel
4:0	SRC_DIV	src_div

0x0A21B108 WCSS_A_PMU_RFIF_M**Type:** RW**Reset State:** 0x00000000

m for rfif clk src

WCSS_A_PMU_RFIF_M

Bits	Name	Description
7:0	M	m for mnd_rcg

0x0A21B10C WCSS_A_PMU_RFIF_N**Type:** RW**Reset State:** 0x00000000

not_n_minus_m for rfif clk src

WCSS_A_PMU_RFIF_N

Bits	Name	Description
7:0	NOT_N_MINUS_M	not_n_minus_m for mnd_rcg

0x0A21B110 WCSS_A_PMU_RFIF_D

Type: RW

Reset State: 0x00000000

not_2d for rfif clk src

WCSS_A_PMU_RFIF_D

Bits	Name	Description
7:0	NOT_2D	not_2d for mnd_rcg

0x0A21B120 WCSS_A_PMU_CPU_AHB_CMD_RCGR

Type: RW

Reset State: 0x00000002

cmd_cgr for cpu_ahb clk src

WCSS_A_PMU_CPU_AHB_CMD_RCGR

Bits	Name	Type	Description
31	ROOT_OFF	R	root clock off
4	DIRTY_CFG_RCGR	R	dirty bit for cfg_rcgr. When cfg_rcgr is written the bit is set to 1 and cleared when update bit is set.
1	ROOT_EN	RW	root_en
0	UPDATE	RW	update configuration and status

0x0A21B124 WCSS_A_PMU_CPU_AHB_CFG_RCGR

Type: RW

Reset State: 0x00000000

cfg_cgr for cpu_ahb clk src

WCSS_A_PMU_CPU_AHB_CFG_RCGR

Bits	Name	Description
10:8	SRC_SEL	src_sel
4:0	SRC_DIV	src_div

0x0A21B134 WCSS_A_PMU_CPU_AHB_MISC**Type:** RW**Reset State:** 0x00000001

misc for cpu_ahb clk src

WCSS_A_PMU_CPU_AHB_MISC

Bits	Name	Description
3:0	SRC_DIV	src_div

0x0A21B140 WCSS_A_PMU_BT_32M_CMD_RCGR**Type:** RW**Reset State:** 0x00000000

cmd_cgr for bt_32m clk src

WCSS_A_PMU_BT_32M_CMD_RCGR

Bits	Name	Type	Description
31	ROOT_OFF	R	root clock off
4	DIRTY_CFG_RCGR	R	dirty bit for cfg_rcgr. When cfg_rcgr is written the bit is set to 1 and cleared when update bit is set.
1	ROOT_EN	RW	root_en
0	UPDATE	RW	update configuration and status

0x0A21B144 WCSS_A_PMU_BT_32M_CFG_RCGR**Type:** RW**Reset State:** 0x00000000

cfg_cgr for bt_32m clk src

WCSS_A_PMU_BT_32M_CFG_RCGR

Bits	Name	Description
10:8	SRC_SEL	src_sel
4:0	SRC_DIV	src_div

0x0A21B160 WCSS_A_PMU_BT_FM_19M_CMD_RCGR**Type:** RW**Reset State:** 0x00000000

cmd_cgr for bt_fm_19m clk src

WCSS_A_PMU_BT_FM_19M_CMD_RCGR

Bits	Name	Type	Description
31	ROOT_OFF	R	root clock off
7	DIRTY_D	R	dirty bit for d. When d is written the bit is set to 1 and cleared when update bit is set.
6	DIRTY_N	R	dirty bit for n. When n is written the bit is set to 1 and cleared when update bit is set.
5	DIRTY_M	R	dirty bit for m. When m is written the bit is set to 1 and cleared when update bit is set.
4	DIRTY_CFG_RCGR	R	dirty bit for cfg_rcgr. When cfg_rcgr is written the bit is set to 1 and cleared when update bit is set.
1	ROOT_EN	RW	root_en
0	UPDATE	RW	update configuration and status

0x0A21B164 WCSS_A_PMU_BT_FM_19M_CFG_RCGR**Type:** RW**Reset State:** 0x00000000

cfg_cgr for bt_fm_19m clk src

WCSS_A_PMU_BT_FM_19M_CFG_RCGR

Bits	Name	Description
13:12	MODE	mnd mode. 00 - bypass, 11 - mnd
10:8	SRC_SEL	src_sel
4:0	SRC_DIV	src_div

0x0A21B180 WCSS_A_PMU_SLP_CMD_RCGR**Type:** RW**Reset State:** 0x00000002

cmd_cgr for slp clk src

WCSS_A_PMU_SLP_CMD_RCGR

Bits	Name	Type	Description
31	ROOT_OFF	R	root clock off
4	DIRTY_CFG_RCGR	R	dirty bit for cfg_rcgr. When cfg_rcgr is written the bit is set to 1 and cleared when update bit is set.
1	ROOT_EN	RW	root_en
0	UPDATE	RW	update configuration and status

0x0A21B184 WCSS_A_PMU_SLP_CFG_RCGR**Type:** RW**Reset State:** 0x00000600

cfg_cgr for slp clk src

WCSS_A_PMU_SLP_CFG_RCGR

Bits	Name	Description
10:8	SRC_SEL	src_sel
4:0	SRC_DIV	src_div

0x0A21B1A0 WCSS_A_PMU_XO_CMD_RCGR**Type:** RW**Reset State:** 0x00000002

cmd_cgr for xo clk src

WCSS_A_PMU_XO_CMD_RCGR

Bits	Name	Type	Description
31	ROOT_OFF	R	root clock off
4	DIRTY_CFG_RCGR	R	dirty bit for cfg_rcgr. When cfg_rcgr is written the bit is set to 1 and cleared when update bit is set.

WCSS_A_PMU_XO_CMD_RCGR (cont.)

Bits	Name	Type	Description
1	ROOT_EN	RW	root_en
0	UPDATE	RW	update configuration and status

0x0A21B1A4 WCSS_A_PMU_XO_CFG_RCGR**Type:** RW**Reset State:** 0x00000000

cfg_cgr for xo clk src

WCSS_A_PMU_XO_CFG_RCGR

Bits	Name	Description
10:8	SRC_SEL	src_sel
4:0	SRC_DIV	src_div

0x0A21B1C0 WCSS_A_PMU_PLL_MODE**Type:** RW**Reset State:** 0x00000000

PLL config_ctl

WCSS_A_PMU_PLL_MODE

Bits	Name	Description
5:4	REF_XO_SELECT	ref_clk source select. 0 : 19.2MHz XO 1 : 24MHz from IRIS
3	RESERVED	reserved
2	PLL_RESET_N	reset_n input
1	PLL_BYPASSNL	bypassnl input
0	PLL_OUTCTRL	enable input

0x0A21B1C4 WCSS_A_PMU_PLL_L_VAL**Type:** RW**Reset State:** 0x00000032

PLL L value

WCSS_A_PMU_PLL_L_VAL

Bits	Name	Description
7:0	L	PLL L value

0x0A21B1C8 WCSS_A_PMU_PLL_M_VAL**Type:** RW**Reset State:** 0x00000000

PLL M value

WCSS_A_PMU_PLL_M_VAL

Bits	Name	Description
18:0	M	PLL M value

0x0A21B1CC WCSS_A_PMU_PLL_N_VAL**Type:** RW**Reset State:** 0x00000000

PLL N value

WCSS_A_PMU_PLL_N_VAL

Bits	Name	Description
18:0	N	PLL M value

0x0A21B1D0 WCSS_A_PMU_PLL_USER_CTL**Type:** RW**Reset State:** 0x00000000

PLL user_ctl

WCSS_A_PMU_PLL_USER_CTL

Bits	Name	Description
31:0	PLL_USER_CTL	PLL test_ctl input

0x0A21B1D4 WCSS_A_PMU_PLL_CONFIG_CTL**Type:** RW**Reset State:** 0x00141200

PLL config_ctl

WCSS_A_PMU_PLL_CONFIG_CTL

Bits	Name	Description
31:0	PLL_CONFIG_CTL	PLL config_ctl input

0x0A21B1DC WCSS_A_PMU_PLL_STATUS**Type:** R**Reset State:** 0x00000000

pll status

WCSS_A_PMU_PLL_STATUS

Bits	Name	Description
16	PLL_LOCK_DET	pll lock_det output
15:0	PLL_STATUS	pll status output

0x0A21C004 WCSS_A_PMU_CFG**Type:** RW**Reset State:** 0x00400000

Configuration .

WCSS_A_PMU_CFG

Bits	Name	Type	Description
22	BW_CONFIG_VALID_BYPASS	RW	When this bit is set, bw_config_tx_valid, bw_config_rx_valid are not used, and PHY takes bw_config_rx and bw_config_tx as static signal, and doesn't synchronize them. Provided for riva backward compatibility.
21	BW_CONFIG_RX_VALID	RW	RX BW configuration valid. After changing bw_config_rx, SW should write '1' to this bit until PHY synchronizes bw_config_rx to proper clock domain. SW should clear this bit once PHY has done synchronization, usually after PHY configuration is complete.

WCSS_A_PMU_CFG (cont.)

Bits	Name	Type	Description
20	BW_CONFIG_TX_VALID	RW	TX BW configuration valid. After changing bw_config_tx, SW should write '1' to this bit until PHY synchronizes bw_config_tx to proper clock domain. SW should clear this bit once PHY has done synchronization, usually after PHY configuration is complete.
19:18	BW_CONFIG_RX	RW	RX BW configuration
17:16	BW_CONFIG_TX	RW	TX BW configuration
10	IRIS_REG_READ_STS	R	IRIS register read command status. 1 : in progress 0 : done or idle
9	IRIS_REG_READ	RW	send iris register read command to IRIS
8	IRIS_RESET_STS	R	IRIS reset command status. 1 : in progress 0 : done or idle
7	IRIS_RESET	RW	send reset command to IRIS
6	IRIS_XO_CFG_STS	R	IRIS XO configuration status. 1 : in progress 0 : done or idle
5	GC_BUS_MUX_SEL	RW	1 : GC bus is controlled by top 0 : GC bus is controlled by WLAN PHY
4	IRIS_XO_EN	RW	1 : enable XO during configuration 0 : disable XO during configuration
3	IRIS_XO_CFG	RW	start IRIS XO configuration
2:1	IRIS_XO_MODE	RW	IRIS XO mode. 2'b11 : 48MHz XO 2'b01 : 38.4MHz TCXO 2'b00 : 19.2MHz TCXO
0	WARM_BOOT	RW	warm/cold boot indication.

0x0A21C008 WCSS_A_PMU_OVRD_EN**Type:** RW**Reset State:** 0x00000002

enable override of controls

WCSS_A_PMU_OVRD_EN

Bits	Name	Description
1	CCU_PMU_DISABLE_CLK_GATING_OVRD_EN	override ccu_pmu_disable_clk_gating for PMU
0	DUMMY_OVRD_EN	override disable_clk_gating for PMU

0x0A21C00C WCSS_A_PMU_OVRD_VAL**Type:** RW**Reset State:** 0x00000002

override value of controls

WCSS_A_PMU_OVRD_VAL

Bits	Name	Description
1	CCU_PMU_DISABLE_CLK_GATING_OVRD_VAL	override value of ccu_pmu_disable_clk_gating for PMU
0	DUMMY_OVRD_VAL	override value of disable_clk_gating for PMU

0x0A21C010 WCSS_A_PMU_SLP_TMR_CTL**Type:** RW**Reset State:** 0x00000000

control for sleep timer

WCSS_A_PMU_SLP_TMR_CTL

Bits	Name	Description
8	ENABLE_SLP_TMR_CLK	enable clocks to slp_tmr
7	CLK_INT_CLR	Clear slp_clk_int.
6	TMR_EN_INT_CLR	Clear tmr_en_int.
5	CLK_INT_EN	Enable interrupt at the next rising edge of sleep clock. Due to synchronization, it may not be the very next rising edge, but could be 1~2 sleep clocks later.
4	TMR_EN_INT_EN	Enable interrupt when sleep timer is enabled. Interrupt will be asserted at the rising edge of sleep clock when sleep timer starts to increase or decrease.
3	TMR_ADD	add slp_tmr_val_wr[31:0] to the current sleep timer value.
2	TMR_LD	load sleep timer with slp_tmr_val_wr[31:0].
1	TMR_DEC	1 : Decrement sleep timer when tmr_en is set. 0 : increment sleep timer when tmr_en is set.
0	TMR_EN	Enable sleep timer increment or decrement.

0x0A21C014 WCSS_A_PMU_SLP_TMR_STS**Type:** R**Reset State:** 0x00000000

Status of sleep timer.

WCSS_A_PMU_SLP_TMR_STS

Bits	Name	Description
2	CLK_INT	Status of slp_clk_int.
1	TMR_EN_INT	Status of tmr_en_int.
0	TOGGLE	This bit toggles at the every rising edge of sleep clock.

0x0A21C018 WCSS_A_PMU_SLP_TMR_VAL_WR**Type:** RW**Reset State:** 0x00000000

sleep timer value to be written

WCSS_A_PMU_SLP_TMR_VAL_WR

Bits	Name	Description
31:0	SLP_TMR_WR	sleep timer value to be written .

0x0A21C01C WCSS_A_PMU_SLP_TMR_VAL**Type:** R**Reset State:** 0x00000000

current sleep timer value

WCSS_A_PMU_SLP_TMR_VAL

Bits	Name	Description
31:0	SLP_TMR	current sleep timer value

0x0A21C020 WCSS_A_PMU_WLAN_SLP_TMR_CTL**Type:** RW**Reset State:** 0x00000000

WLAN control of sleep timer

WCSS_A_PMU_WLAN_SLP_TMR_CTL

Bits	Name	Description
2	WLAN_SLP_TMR_INT_CLR	clear sleep timer interrupt for WLAN.
1	WLAN_SLP_TMR_INT_EN	enable sleep timer interrupt for WLAN.
0	WLAN_SLP_TMR_EN	enable sleep timer for WLAN.

0x0A21C024 WCSS_A_PMU_WLAN_SLP_TMR_STS

Type: R

Reset State: 0x00000000

sleep timer status for WLAN

WCSS_A_PMU_WLAN_SLP_TMR_STS

Bits	Name	Description
1	WLAN_SLP_TMR_INT	status of WLAN sleep timer interrupt
0	WLAN_SLP_TMR_INT_RAW	raw status of WLAN sleep timer interrupt

0x0A21C028 WCSS_A_PMU_WLAN_SLP_TMR_EXP

Type: RW

Reset State: 0x00000000

sleep timer expiration value for WLAN

WCSS_A_PMU_WLAN_SLP_TMR_EXP

Bits	Name	Description
31:0	WLAN_SLP_TMR_EXP	sleep timer expiration value for WLAN.

0x0A21C02C WCSS_A_PMU_BT_SLP_TMR_CTL

Type: RW

Reset State: 0x00000000

BT control of sleep timer

WCSS_A_PMU_BT_SLP_TMR_CTL

Bits	Name	Description
2	BT_SLP_TMR_INT_CLR	clear sleep timer interrupt for BT.
1	BT_SLP_TMR_INT_EN	enable sleep timer interrupt for BT.
0	BT_SLP_TMR_EN	enable sleep timer for BT.

0x0A21C030 WCSS_A_PMU_BT_SLP_TMR_STS**Type:** R**Reset State:** 0x00000000

sleep timer status for BT

WCSS_A_PMU_BT_SLP_TMR_STS

Bits	Name	Description
1	BT_SLP_TMR_INT	status of BT sleep timer interrupt
0	BT_SLP_TMR_INT_RAW	raw status of BT sleep timer interrupt

0x0A21C034 WCSS_A_PMU_BT_SLP_TMR_EXP**Type:** RW**Reset State:** 0x00000000

sleep timer expiration value for BT

WCSS_A_PMU_BT_SLP_TMR_EXP

Bits	Name	Description
31:0	BT_SLP_TMR_EXP	sleep timer expiration value for BT.

0x0A21C038 WCSS_A_PMU_PWR_CNTL_ADCDAC**Type:** RW**Reset State:** 0x00000000

power control to ADC/DAC block

WCSS_A_PMU_PWR_CNTL_ADCDAC

Bits	Name	Description
1	IDDQ_N_ADC	Freeze IO from ADC.
0	SHUTDOWN_N_ADC	shutdown ADC regulator.

0x0A21C03C WCSS_A_PMU_WMAC_CSR**Type:** RW**Reset State:** 0x00000600

control and status to WMAC modules

WCSS_A_PMU_WMAC_CSR

Bits	Name	Type	Description
31:24	RXP_RSSI_ANT1	R	RXP rssi ant1
23:16	RXP_RSSI_ANT0	R	RXP rssi ant0
10	MTU_TSF_TIMER_EN_SYNC	R	TSF timer enable synchronized to the rising edge of sleep clock and synchronized back to AHB clock. TSF timer will be enabled using this signal.
9	MTU_TSF_TIMER_EN	RW	TSF timer enable for TSF timer restoration. This signal will be synchronized to the rising edge of sleep clock and synchronized back to AHB clock to actually enable TSF timer.
8:7	RXP_WAKEUP_REASON	R	RXP wake up reason. May need to move it to common SS.
4	TPE_CONSIDER_PWRON_BT	RW	TPE consider poweron BT
3	MCU_QTA_VALID	RW	MCU qta_valid
2	MCU_QTA_SEL	RW	MCU qta_sel
1	MCU_GAS_CLKGATE_DISABLE	RW	clock gate disable to MCU GAS
0	MCU_CLKGATE_DISABLE	RW	clock gate disable to MCU

0x0A21C040 WCSS_A_PMU_COMMON_CSR**Type:** RW**Reset State:** 0x00000000

This register controls clock and soft reset of common SS and Pronto top

WCSS_A_PMU_COMMON_CSR

Bits	Name	Description
3	PMU_QGIC2_INVERT_LTE_ACTIVE	when set to 1, qgic2 will expect negedge of lte_active interrupt as active state
2	PMU_QTMR_NS_SELECT	TBD
1	PMU_QGIC2_NS_SELECT	TBD
0	A2XB_CFG_EN	Set to enable A2XB bridge

0x0A21C044 WCSS_A_PMU_ROOT_CLK_ENABLE**Type:** RW**Reset State:** 0x000007F7

This register controls root clock enable to common SS and Pronto top

WCSS_A_PMU_ROOT_CLK_ENABLE

Bits	Name	Description
10	PMU_SLP_ROOT_CLK_ENABLE	slp root clock enable to PMU module
9	PMU_XO_ROOT_CLK_ENABLE	XO root clock enable to PMU module
8	QGIC2_ROOT_CLK_ENABLE	root clock enable to QGIC2 module
7	QTIMER_AHB_ROOT_CLK_ENABLE	AHB root clock enable to QTIMER module
6	QTIMER_XO_ROOT_CLK_ENABLE	XO root clock enable to QTIMER module
5	AOA2AB_ROOT_CLK_ENABLE	root clock enable to AOA2AB module
4	AOAHB_ROOT_CLK_ENABLE	root clock enable to AOAHB module
2	ECAHB_ROOT_CLK_ENABLE	root clock enable to ECAHB module
1	RESERVED	Reserved
0	CRCM_CCU_ROOT_CLK_ENABLE	root clock enable to CCU module

0x0A21C048 WCSS_A_PMU_CLKGATE_DISABLE**Type:** RW**Reset State:** 0x000000B6**WCSS_A_PMU_CLKGATE_DISABLE**

Bits	Name	Description
14	PMU_SLP_TMR_CLKGATE_DISABLE	clock gate disable to PMU SLP_TMR module

WCSS_A_PMU_CLKGATE_DISABLE (cont.)

Bits	Name	Description
13	PMU_IRIS_CFG_CLKGATE_DISABLE	clock gate disable to PMU IRIS_CFG module
12	PMU_WDOG_CLKGATE_DISABLE	clock gate disable to PMU WDOG module
11	QGIC2_CLKGATE_DISABLE	clock gate disable to QGIC2 module
10	QTIMER_AHB_CLKGATE_DISABLE	AHB clock gate disable to QTIMER module
9	WLAN_GLOBAL_CLK_GATE_DISABLE	global clock gate disable for wlan
8	COMMON_GLOBAL_CLK_GATE_DISABLE	global clock gate disable for common_ss
7	PMU_XO_CLKGATE_DISABLE	XO clock gate disable to PMU module
6	QTIMER_XO_CLKGATE_DISABLE	XO clock gate disable to QTIMER module
5	AOA2AB_CLKGATE_DISABLE	clock gate disable to AOA2AB module
4	AOAHB_CLKGATE_DISABLE	clock gate disable to AOAHB module
2	ECAHB_CLKGATE_DISABLE	clock gate disable to ECAHB module
1	RESERVED	Reserved
0	CCU_CLKGATE_DISABLE	clock gate disable to CCU module

0x0A21C04C WCSS_A_PMU_SOFT_RESET**Type:** RW**Reset State:** 0x00000400**WCSS_A_PMU_SOFT_RESET**

Bits	Name	Type	Description
10	CRCM_CCPU_SOFT_RESET	RW	soft reset to ARM9. ARM9 will be kept in reset until SW clears it during cold boot up. This is to make sure clk_ccpu and clk_ahb and hclken are all stabilized when reset to ARM9 is released.
9	PMU_SLP_SOFT_RESET	RW	soft reset to sleep clock related logic in PMU including sleep timer.
8	PMU_TCXO_SOFT_RESET	RW	soft reset to tcxo related logic in PMU including SAW2 and GDS controller.
7	QGIC2_SOFT_RESET	RW	soft reset to QGIC2 module
6	QTIMER_SOFT_RESET	RW	soft reset to QTIMER module
5	AOA2AB_SOFT_RESET	W	soft reset to AOA2AB module. It is self-clearing, and 1-clock pulse will be generated. Assertion happens after 4 clock cycles from writing to this register.
4	AOAHB_SOFT_RESET	W	soft reset to AOAHB module. It is self-clearing, and 1-clock pulse will be generated. Assertion happens after 4 clock cycles from writing to this register.

WCSS_A_PMU_SOFT_RESET (cont.)

Bits	Name	Type	Description
2	ECAHB_SOFT_RESET	W	soft reset to ECAHB module. It is self-clearing, and 1-clock pulse will be generated. Assertion happens after 4 clock cycles from writing to this register.
1	MCU_SOFT_RESET	RW	soft reset to MCU module
0	CRCM_CCU_SOFT_RESET	RW	soft reset to CCU module

0x0A21C060 WCSS_A_PMU_WLAN_POWER_MEASURE**Type:** RW**Reset State:** 0x00000000

wlan_power_measure_mask control: Used to generate a masked vector

WCSS_A_PMU_WLAN_POWER_MEASURE

Bits	Name	Description
12:8	WLAN_POWER_MEASURE_MATCH	bit match for wlan_power_tbusl
4:0	WLAN_POWER_MEASURE_MASK	bit mask for wlan_power_tbusl

0x0A21C064 WCSS_A_PMU_WCSS_IPC**Type:** W**Reset State:** 0x00000000

Interrupt output other subsystems.

WCSS_A_PMU_WCSS_IPC

Bits	Name	Description
23	WCSS_SPARE_SW_INTR_3	o_wcss_spare_sw_intr_3
22	WCSS_SPARE_SW_INTR_2	o_wcss_spare_sw_intr_2
21	WCSS_SPARE_SW_INTR_1	o_wcss_spare_sw_intr_1
20	WCSS_SPARE_SW_INTR_0	o_wcss_spare_sw_intr_0
15	WCSS_MSS_SW_INTR_3	o_wcss_mss_sw_intr_3
14	WCSS_MSS_SW_INTR_2	o_wcss_mss_sw_intr_2
13	WCSS_MSS_SW_INTR_1	o_wcss_mss_sw_intr_1
12	WCSS_MSS_SW_INTR_0	o_wcss_mss_sw_intr_0
11	WCSS_LPASS_SMD_HI	o_wcss_lpass_smd_hi

WCSS_A_PMU_WCSS_IPC (cont.)

Bits	Name	Description
10	WCSS_LPASS_SMD_MED	o_wcss_lpass_smd_med
9	WCSS_LPASS_SMD_LO	o_wcss_lpass_smd_lo
8	WCSS_LPASS_MBOX_INTR	o_wcss_lpass_mbox_intr
7	WCSS_APSS_SMD_HI	o_wcss_apss_smd_hi
6	WCSS_APSS_SMD_MED	o_wcss_apss_smd_med
5	WCSS_APSS_SMD_LO	o_wcss_apss_smd_lo
4	WCSS_APSS_SMSM_IRQ	o_wcss_apss_smsm_irq
2	WCSS_RPM_GP_LOW_IRQ	o_wcss_rpm_gp_low_irq
1	WCSS_RPM_GP_MED_IRQ	o_wcss_rpm_gp_med_irq
0	WCSS_RPM_GP_HI_IRQ	o_wcss_rpm_gp_hi_irq

0x0A21C068 WCSS_A_PMU_WDOG_CTL**Type:** RW**Reset State:** 0x00000000

control register for watchdog timer

WCSS_A_PMU_WDOG_CTL

Bits	Name	Type	Description
4	RESERVED		reserved
3	WDOG_UNMASKED_INT_ENABLE	RW	unmasked IRQ or FIQ will enable wdog timer
2	WDOG_ENABLE	RW	enables wdog timer.
1	WDOG_FREEZE	W	freezes the watchdog timer at count 0. The microprocessor should write this register before it goes to power-on sleep to avoid a watchdog timeout during sleep.
0	WDOG_RESET	W	resets the watch dog counter. In normal operation, the microprocessor should periodically write the register to reset watch dog. This bit also disables watchdog freeze.

0x0A21C06C WCSS_A_PMU_WDOG_STS**Type:** R**Reset State:** 0x00000000

watchdog status register

WCSS_A_PMU_WDOG_STS

Bits	Name	Description
3	RESERVED	reserved
2	WDOG_CNT_RESET_STATUS	show the status of wdog_res signal for test.
1	WDOG_FROZEN	1 : watchdog timer is frozen. 0 : watchdog timer is not frozen.
0	WDOG_EXPIRED_STATUS	1 : watchdog timer has expired. 0 : watchdog timer has not expired.

0x0A21C070 WCSS_A_PMU_WDOG_COUNT

Type: R

Reset State: 0x00000000

watchdog count register

WCSS_A_PMU_WDOG_COUNT

Bits	Name	Description
20:0	WDOG_COUNT	counter value of watchdog counter. Multiple reads are required to determine the value (assuming ahb clock is much faster than sleep clock)

0x0A21C074 WCSS_A_PMU_WDOG_BITE_SECURE

Type: W

Reset State: 0x00000000

this register needs to be written first to unlock write access to WDOG_BITE_TIME register.

WCSS_A_PMU_WDOG_BITE_SECURE

Bits	Name	Description
31:0	WDOG_BITE_SECURE	when this register is written to 0xA1A602E7, WDOG_BITE_TIME can up updated. Once WDOG_BITE_TIME is updated, write access to WDOG_BITE_TIME is blocked until this register is again written to the same pattern.

0x0A21C078 WCSS_A_PMU_WDOG_BARK_TIME**Type:** RW**Reset State:** 0x0003FFFF

defines the time in sleep_clk when to assert the wdog bark interrupt. The value can be between 1 and 0x1fffff (the time when the watchdog times out). The default value is 0x3ffff.

WCSS_A_PMU_WDOG_BARK_TIME

Bits	Name	Type	Description
31	SYNC_STATUS	R	when set(1) wdog bark time is synchronizing to sleep_clk. Data value is not guaranteed until this bit is clear(0).
20:0	WDOG_BARK_TIME	RW	wdog counter vale on which to trigger the bark interrupt

0x0A21C07C WCSS_A_PMU_WDOG_BITE_TIME**Type:** RW**Reset State:** 0x0007FFFF

defines the time in sleep_clk when to assert the wdog bite interrupt. The value can be between 1 and 21'h1ffff (the time when the watchdog times out). The default value is 0x7ffff.

WCSS_A_PMU_WDOG_BITE_TIME

Bits	Name	Type	Description
31	SYNC_STATUS	R	when set(1) wdog bark time is synchronizing to sleep_clk. Data value is not guaranteed until this bit is clear(0).
20:0	WDOG_BITE_TIME	RW	wdog counter vale on which to trigger the bite interrupt. wdog_bite_time should be greater than wdog_bark_time.

0x0A21C08C WCSS_A_PMU_AOA2AB_CTL**Type:** RW**Reset State:** 0x00000000

aoa2ab control register

WCSS_A_PMU_AOA2AB_CTL

Bits	Name	Description
20	AOA2AB_AHB_ERR_CLEAR	clear AOA2AB_AHB_ERR_ADDR
19:16	AOA2AB_CFG_BUSY_LIMIT	a2ab input config signal

WCSS_A_PMU_AOA2AB_CTL (cont.)

Bits	Name	Description
15:0	AOA2AB_TIMEOUT_VALUE	a2ab transaction timeout time in units of 0x1fff clock cycles. 0x0 means timeout timer is disabled.

0x0A21C090 WCSS_A_PMU_AOA2AB_WRAP_ADDR**Type:** R**Reset State:** 0x00000000

aoa2ab wrap address register

WCSS_A_PMU_AOA2AB_WRAP_ADDR

Bits	Name	Description
31:0	AOA2AB_WRAP_ADDR	aoa2ab wrap address.

0x0A21C094 WCSS_A_PMU_AOA2AB_AHB_ERR_ADDR**Type:** R**Reset State:** 0x00000000

aoa2ab ahb error address

WCSS_A_PMU_AOA2AB_AHB_ERR_ADDR

Bits	Name	Description
31:0	AOA2AB_AHB_ERR_ADDR	aoa2ab ahb error address. Cleared by writing to AOA2AB_CTL register

0x0A21C09C WCSS_A_PMU_DAC_CAL_CTL**Type:** RW**Reset State:** 0x00000000

control DAC calibration logic

WCSS_A_PMU_DAC_CAL_CTL

Bits	Name	Description
0	DAC_SM_SOFT_RESET	soft reset to DAC calibration state machine

0x0A21C0A0 WCSS_A_PMU_BT_SW_EVENT**Type:** RW**Reset State:** 0x00000000

scratch register that BT SW can write to indicate specific SW event happening

WCSS_A_PMU_BT_SW_EVENT

Bits	Name	Description
31:0	BT_SW_EVENT	Each bit indicates a specific SW event occurring.

0x0A21C124 WCSS_A_PMU_IRIS_XO_SETTLE_TIME**Type:** RW**Reset State:** 0x00000000

IRIS XO settle timer expiration value

WCSS_A_PMU_IRIS_XO_SETTLE_TIME

Bits	Name	Description
31:0	IRIS_XO_SETTLE_TIME	the expiration value of iris_xo_settle_timer in 19.2MHz tick.

0x0A21C128 WCSS_A_PMU_IRIS_XO_SETTLE_TIMER_CSR**Type:** RW**Reset State:** 0x00000000

IRIS XO settle timer control and status

WCSS_A_PMU_IRIS_XO_SETTLE_TIMER_CSR

Bits	Name	Type	Description
1	IRIS_XO_SETTLE_TIMER_EXP_CLR	W	When written 1, iris_xo_settle_timer_exp will be cleared to 0.
0	IRIS_XO_SETTLE_TIMER_EXP	R	the status of iris_xo_settle_timer expiration. The timer will start running when SPM enables it, and will assert iris_xo_settle_timer_exp when the value matches with iris_xo_settle_time register above.

0x0A21C12C WCSS_A_PMU_IRIS_XO_TIMER**Type:** R**Reset State:** 0x00000000

current value of iris_xo_timer

WCSS_A_PMU_IRIS_XO_TIMER

Bits	Name	Description
31:0	IRIS_XO_TIMER	The register shows the current value of iris xo timer. There is no synchronization done, so SW should ignore lower bits changing. It is only debug purpose.

0x0A21C130 WCSS_A_PMU_COMMON_SS_MEM_SLP_CTL**Type:** RW**Reset State:** 0x00000000

individual memory sleep control for common_ss. If SW wants to put common_ss memories in different sleep mode, SW should program this register, and keep COMMON_AHB_CBCR's force_mem_core_on=0, force_mem_periph_on=0.

WCSS_A_PMU_COMMON_SS_MEM_SLP_CTL

Bits	Name	Description
7	PATCH_RAM_FORCE_MEM_PERIPH_ON	when this bit is set to 1, ROM patch memory periph is forced to be on.
6	PATCH_RAM_FORCE_MEM_CORE_ON	when this bit is set to 1, ROM patch memory core is forced to be on. This bit is used to have ROM patch memory in retention mode.
5	CCPU_FORCE_MEM_PERIPH_ON	when this bit is set to 1, ccpu memory periph is forced to be on excpe the ROM patch memory.
4	CCPU_FORCE_MEM_CORE_ON	when this bit is set to 1, ccpu memory core is forced to be on excpe the ROM patch memory. This bit is used to have ccpu in retention mode.
3	DXE_FORCE_MEM_PERIPH_ON	when this bit is set to 1, dxe memory periph is forced to be on.
2	DXE_FORCE_MEM_CORE_ON	when this bit is set to 1, dxe memory core is forced to be on. This bit is used to have dxe in retention mode.
1	CMEM_FORCE_MEM_PERIPH_ON	when this bit is set to 1, cmem memory periph is forced to be on.
0	CMEM_FORCE_MEM_CORE_ON	when this bit is set to 1, cmem memory core is forced to be on. This bit is used to have cMEM in retention mode.

0x0A21C134 WCSS_A_PMU_IRIS_REG_READ**Type:** RW**Reset State:** 0x00000000

address and data for IRIS register read command

WCSS_A_PMU_IRIS_REG_READ

Bits	Name	Type	Description
31:16	IRIS_REG_RDATA	R	read data returned for iris_reg_addr. It is valid only when CFG.iris_reg_read_sts=0.
10:0	IRIS_REG_ADDR	RW	address of IRIS register to be read. After setting the address, CFG.iris_reg_read should be set to '1' to initiate the command through 5-wire interface.

0x0A21C138 WCSS_A_PMU_WLAN_COEX_CTL**Type:** RW**Reset State:** 0x00000000

Control register to generate wlan_tx_on in 5G/2.5G mode for Coex.

WCSS_A_PMU_WLAN_COEX_CTL

Bits	Name	Description
2	WLAN_IN_2G_TX_ON_EN	TX Enable for wlan in 2G
1	WLAN_IN_5G_TX_ON_EN	TX Enable for wlan in 5G
0	WLAN_IN_5G_MODE	wlan in 5G mode

0x0A21D000 WCSS_A_PMU_CCPU_CTL**Type:** RW**Reset State:** 0x00000000

This register represents Controls for CCPU.

WCSS_A_PMU_CCPU_CTL

Bits	Name	Type	Description
4	CCPU_STANDBYWFI	R	The status standbywfi signal from arm9.

WCSS_A_PMU_CCPU_CTL (cont.)

Bits	Name	Type	Description
2	CCPU_REMAP_EN	RW	Set to '1' to enable the boot address region to be re-mapped to a different 64K region in the memory map specified in ccpu_remap_addr reg.
1	CCPU_INSERT_WAIT_FOR_INTERRUPT_CMD_EN	RW	When set, cCPU logic returns a "wait for interrupt" command at boot address.
0	CCPU_HIGH_INTERRUPT_VECTOR_EN	RW	1: When set, cCPU fetches instructions from 0xFFFF_0000 0: When clear, cCPU fetches instructions from 0x0

0x0A21D004 WCSS_A_PMU_CCPU_BOOT_REMAP_ADDR**Type:** RW**Reset State:** 0x00000000

The base address for memory re-map the 64K boot address region.

WCSS_A_PMU_CCPU_BOOT_REMAP_ADDR

Bits	Name	Description
15:0	CCPU_BOOT_REMAP_ADDR	Base address of the 64K boot window onto the DAHB.

0x0A21D008 WCSS_A_PMU_CMEM_ACCESS_CTL**Type:** RW**Reset State:** 0x00000000

This register control the access to cmem address

WCSS_A_PMU_CMEM_ACCESS_CTL

Bits	Name	Description
0	ENABLE_FULL_CMEM_ACCESS	when set to '1', the lower 64KB address of CMEM will be accessible by all masters. when set to '0', ARM9 is the only master that can access the lower 64KB region of CMEM. Access from other masters will be returned by error response.

10.8 WCSS_A_AOAHB_AOAHB

0x0A21A000 WCSS_A_AOAHB_AOAHB_AOWBR_PL

Type: RW

Reset State: 0x00000001

aoahb Arbiter arbitration priority level for aowbr master. In the aoahb arbitration scheme the master with highest priority level always wins. Should there be more than one master with the same priority then the winning master will be selected using a round-robin algorithm.

WCSS_A_AOAHB_AOAHB_AOWBR_PL

Bits	Name	Description
3:0	PRIORITY	Arbitration priority for aowbr master. The lowest priority level is 1 and the highest priority level is 15. A priority level of 0 has a special significance, it means that the master is disabled and stops participating in the arbitration scheme. Note that the aoahb arbiter will prevent a master from disabling itself.

0x0A21A004 WCSS_A_AOAHB_AOAHB_AOA2AB_PL

Type: RW

Reset State: 0x00000001

aoahb Arbiter arbitration priority level for aoa2ab master. In the aoahb arbitration scheme the master with highest priority level always wins. Should there be more than one master with the same priority then the winning master will be selected using a round-robin algorithm.

WCSS_A_AOAHB_AOAHB_AOA2AB_PL

Bits	Name	Description
3:0	PRIORITY	Arbitration priority for aoa2ab master. The lowest priority level is 1 and the highest priority level is 15. A priority level of 0 has a special significance, it means that the master is disabled and stops participating in the arbitration scheme. Note that the aoahb arbiter will prevent a master from disabling itself.

0x0A21A048 WCSS_A_AOAHB_AOAHB_DFLT_MST

Type: RW

Reset State: 0x00000000

This register identifies the default master for the aoahb.

WCSS_A_AOAHB_AOAHB_DFLT_MST

Bits	Name	Description
3:0	DFT_MST	Default Master ID number register. The default master is the master that is granted by the bus when no master has requested ownership. NOTE: Writing anything beyond the 4 R/W bits will cause the entire write to be discarded. 0x0: DMY (Grant ownership to aoahb dummy master) 0x1: AOWBR (Grant ownership to aoahb aowbr master) 0x2: AOA2AB (Grant ownership to aoahb aoa2ab master)

10.9 WCSS_A_SAW2_SAW2**0x0A219000 WCSS_A_SAW2_SAW2_SECURE****Type:** RW**Clock:** sys_ref_clk**Reset State:** 0x00000000**Security Treatment:** Restricted

The SAW2_SECURE register is used to enable or disable non-secure bus transactions. The bus **security** status is signaled by ahb_saw_hprotns pin. When ahb_saw_hprotns is set to '0' the state of the SAW2_SECURE register is not considered - a secure transaction is always allowed. When ahb_saw_hprotns is set to '1', the SAW2_SECURE register **security** treatment bit of that register must be '1' for access to be granted. If the **security** treatment bit of the register is set to '0', the non-secure (ahb_saw_hprotns = '1') transactions are denied. When access is denied, a write is ignored, and a read will return '0'. No error is signaled.

When reading the SAW2_SECURE register, non secure agents will always read '0'. Secure agents will see the actual value of the register. The reset value is controlled by CFGNSINIT input pin.

WCSS_A_SAW2_SAW2_SECURE

Bits	Name	Description
2	SAW_CTL	0x1: NSEC (NSEC) 0x0: SEC (SEC) Controls security treatment for SAW2 registers: SAW2_ID, SAW2_CFG, SAW2_STS_0, SAW2_STS_1, SAW2_RST, SAW2_VERSION)
1	PWR_CTL	0x1: NSEC (NSEC) 0x0: SEC (SEC) Controls security treatment for SPM registers: SAW2_SPM_CTL, SAW2_SPM_PMIC_DLY, SAW2_SPM_PMIC_DATA_0, SAW2_SPM_PMIC_DATA_1, SAW2_SPM_SLP_SEQ_ENTRY_n)
0	VLT_CTL	0x1: NSEC (NSEC) 0x0: SEC (SEC)

0x0A219004 WCSS_A_SAW2_SAW2_ID**Type:** R**Clock:** sys_ref_clk**Reset State:** 0x200B0811**Security Treatment:** Controlled by SAW2_SECURE [SAW_CTL].

This read only register reports the revision and parameter information for the SAW2 core.

WCSS_A_SAW2_SAW2_ID

Bits	Name	Description
31:24	NUM_SPM_ENTRY	SAW2 parameter: Indicates number of SAW2_SPM_SLP_SEQ_ENTRY register implemented. Value can range from 1 - 128
21:16	NUM_PWR_CTL	SAW2 parameter: Indicates number of power control implemented. Value can range from 4 - 32
12:8	PMIC_DATA_WIDTH	SAW2 parameter: Indicate PMIC data bus width Values can range from 8-16
6:4	NUM_PMIC_DATA	SAW2 parameter: Indicates number of SAW2_SPM_PMIC_DATA_n register implemented. Value can range from 1-7
3	CFG_NS_ACCESS	SAW2 parameter: Reset value of SAW2_SECURE register
2	PMIC_ARB_INTF	SAW2 parameter: Indicates PMIC Arbiter Interface function is implemented
1	RESERVED	Reserved
0	SPM_PRESENT	SAW2 parameter: Indicates SPM function is implemented

0x0A219008 WCSS_A_SAW2_SAW2_CFG**Type:** RW**Clock:** sys_ref_clk**Reset State:** 0x00000000**Security Treatment:** Controlled by SAW2_SECURE [SAW_CTL].**WCSS_A_SAW2_SAW2_CFG**

Bits	Name	Description
9	RESERVED	reserved
8	FRC_REF_CLK_ON	Chicken bit to force saw_sys_ref_clk_on_req ON.
4:0	CLK_DIV	Divider ratio for clock. This is used to generate timer tick for the timer. Timer tick is asserted every (CLK_DIV + 1) sys_ref_clk period.

0x0A21900C WCSS_A_SAW2_SAW2_SPM_STS**Type:** R**Clock:** sys_ref_clk**Reset State:** 0x00000000**Security Treatment:** Controlled by SAW2_SECURE [SAW_CTL].

This read only register provides SW with SAW2 status.

WCSS_A_SAW2_SAW2_SPM_STS

Bits	Name	Description
31:16	CURR_PWR_CTL	Power Control Outputs.
15	SHTDWN_REQ	This bit reflects the shutdown request from the SPM(spm_rpm_shutdown_req) to RPM.
14	SHTDWN_ACK	This bit reflects the shutdown acknowledgement from the RPM(rpm_spm_shutdown_ack) to SPM.
13	BRNGUP_REQ	This bit reflects the bringup request from the SPM(spm_rpm_bringup_req) to RPM.
12	BRNGUP_ACK	This bit reflects the bringup acknowledgement from the RPM(rpm_spm_bringup_ack) to SPM.
11:10	RPM_STATE	State of the RPM FSM: 0x0: RUN (RUN(waiting for SPM request)) 0x1: STDNACK (STDNACK (waiting for shutdown ACK from RPM)) 0x2: BGUPACK (BGUPACK (waiting for bringup ACK from RPM)) 0x3: WAKEUP (WAKEUP (waiting for wakeup interrupt))
8:0	SPM_CMD_ADDR	Last SPM command executed.

0x0A219010 RESERVED**0x0A219014 WCSS_A_SAW2_SAW2_PMIC_STS****Type:** R**Clock:** sys_ref_clk**Reset State:** 0x00000000**Security Treatment:** Controlled by SAW2_SECURE [SAW_CTL].

This read only register provides SW with SAW2 status.

WCSS_A_SAW2_SAW2_PMIC_STS

Bits	Name	Description
29:20	RESERVED	Reserved

WCSS_A_SAW2_SAW2_PMIC_STS (cont.)

Bits	Name	Description
18	CURR_PMIC_SIZE	Last PMIC VLVL Size sent to the PMIC Arbiter (and the PMIC).
17:16	PMIC_STATE	State of the PMIC FSM: 0x0: IDLE (IDLE) 0x1: DONE (DONE (waiting for DONE form PMIC Arb before transitions back to IDLE)) 0x2: DELAY (DELAY (waiting for delay count termination before transitions back to IDLE)) 0x3: DONEBAR (DONEBAR (waiting for DONE to deassert before another pending request is processed))
7:0	CURR_PMIC_DATA	Last PMIC DATA sent to the PMIC Arbiter (and the PMIC)

0x0A219018 WCSS_A_SAW2_SAW2_RST**Type:** RW**Clock:** sys_ref_clk**Reset State:** 0x00000000**Security Treatment:** Controlled by SAW2_SECURE [SAW_CTL].

The SAW2_RST register is used to reset the SAW logic. This register clear itself. It does not reset any SAW2 CSR's.

WCSS_A_SAW2_SAW2_RST

Bits	Name	Description
0	RESERVED	Reserved

0x0A219030 WCSS_A_SAW2_SAW2_SPM_CTL**Type:** RW**Clock:** sys_ref_clk**Reset State:** 0x00000000**Security Treatment:** Controlled by SAW2_SECURE [PWR_CTL].

The SAW2_SPM_CTL register is used to control the subsystem power management system. This are parameters that controls the operation of SPM FSM.

WCSS_A_SAW2_SAW2_SPM_CTL

Bits	Name	Description
31:24	EVENT_SYNC	SPM Event input configuration. There is an individual bit for each sys_spm_event input
17	SLP_CMD_MODE	Sleep Command M0de

WCSS_A_SAW2_SAW2_SPM_CTL (cont.)

Bits	Name	Description
16	SPM_SYS_PC_MODE	Indication that we're in power collapse mode
12:4	SPM_START_ADR	Start address for the SPM sequence.
3	ISAR	Inhibit Start Address Reset 0x0: END_OF_PROGRAM_RESET_THE_SPM_START_ADR_TO_ZERO (End of program reset the SPM_START_ADR to zero.) 0x1: INHIBIT_END_OF_PROGRAM_TO_RESET_SPM_START_ADR (Inhibit End of program to reset SPM_START_ADR)
2:1	WAKEUP_CONFIG	Wakeup Configuration 0x0: SYS_SPM_WAKEUP (sys_spm_wakeup) 0x1: SYS_SPM_WAKEUP_OR_NOT_CPU_SPM_WAIT_REQ (sys_spm_wakeup or !cpu_spm_wait_req) 0x2: SYS_SPM_WAKEUP_OR_RISING_EDGE_OF_SYS_SPM_DBG_NOPWRDWN (sys_spm_wakeup or rising edge of sys_spm_dbg_nopwrdsn) 0x3: SYS_SPM_WAKEUP_OR_NOT_CPU_SPM_WAIT_REQ_OR_RISING_EDGE_OF_SYS_SPM_DBG_NOPWRDWN (sys_spm_wakeup or !cpu_spm_wait_req or rising edge of sys_spm_dbg_nopwrdsn)
0	SPM_EN	SPM En.

0x0A219034 WCSS_A_SAW2_SAW2_SPM_DLY**Type:** RW**Clock:** sys_ref_clk**Reset State:** 0x00000000**Security Treatment:** Controlled by SAW2_SECURE [PWR_CTL].

SPM Delay. Controls the time between SPM_SLP_SEQ SPM Delay and the next SPM_SLP_SEQ command.

WCSS_A_SAW2_SAW2_SPM_DLY

Bits	Name	Description
29:20	DLY3	SPM Delay. Controls the time between SPM_SLP_SEQ SPM Delay and the next SPM_SLP_SEQ command. Time range: 52ns to 1.7ms
19:10	DLY2	SPM Delay. Controls the time between SPM_SLP_SEQ SPM Delay and the next SPM_SLP_SEQ command. Time range: 52ns to 1.7ms

WCSS_A_SAW2_SAW2_SPM_DLY (cont.)

Bits	Name	Description
9:0	DLY1	SPM Delay. Controls the time between SPM_SLP_SEQ SPM Delay and the next SPM_SLP_SEQ command. Time range: 52ns to 1.7ms

0x0A219038 WCSS_A_SAW2_SAW2_SPM_STS2**Type:** R**Clock:** sys_ref_clk**Reset State:** 0x00000000**Security Treatment:** Controlled by SAW2_SECURE [SAW_CTL].

This read only register provides SW with SPM power control status.

WCSS_A_SAW2_SAW2_SPM_STS2

Bits	Name	Description
10:0	CURR_PWR_CTL	Power control Outputs

**0x0A219400+ WCSS_A_SAW2_SAW2_SPM_SLP_SEQ_ENTRY_n, n=[0..31]
0x4*n****Type:** RW**Clock:** sys_ref_clk**Reset State:** 0x00000000**Security Treatment:** Controlled by SAW2_SECURE [PWR_CTL].

The SAW2_SPM_SLP_SEQ_ENTRY_n is an array used to sequence through the steps during various power mode. The register width is defined by CLOG_PWR_CTL parameter.

WCSS_A_SAW2_SAW2_SPM_SLP_SEQ_ENTRY_n

Bits	Name	Description
31:24	CMD3	SPM PWR CMD3
23:16	CMD2	SPM PWR CMD2
15:8	CMD1	SPM PWR CMD1
7:0	CMD0	SPM PWR CMD0

10.10 WCSS_QTMR_AC

QTMR AC Block

10.10.1 WCSS_A_QTMR_AC_QTMR_AC

QTMR AC Block

0x0A211000 WCSS_A_QTMR_AC_QTMR_AC_CNTFRQ

Type: RW

Clock: SYS_AHB_CLK

Reset State: 0x00000000

Security Treatment: Restricted

Counter Frequency register. Counter frequency in ticks per second

WCSS_A_QTMR_AC_QTMR_AC_CNTFRQ

Bits	Name	Description
31:0	CNTFRQ	

0x0A211004 WCSS_A_QTMR_AC_QTMR_AC_CNTSR

Type: RW

Clock: SYS_AHB_CLK

Reset State: 0x00000000

Security Treatment: Restricted

The Counter Secure Register (CNTSR) contains a bit for Secure access to each timer frame, including both views of the timer frame and the access to the associated CNTACR and CNTVOFF registers in the Access Control region.

WCSS_A_QTMR_AC_QTMR_AC_CNTSR

Bits	Name	Description
4:0	NSN	Non-secure access to frame N. N lies from 0 to \$NUM_FRAME-1 (# of configured frame) 0x0: SECURE_ONLY (Secure access only - non-secure access is RAZ/WI) 0x1: SECURE_OR_NONSECURE (Secure or Non-secure access)

0x0A211008 WCSS_A_QTMR_AC_QTMR_AC_CNTTID

Type: R

Clock: SYS_AHB_CLK

Reset State: 0x00000000

Security Treatment: Both

The Counter Timer ID (CNTTID) is 32-bit register that identifies the number of timer frames implemented, which frame have virtual offsets and timers, and which frames have second views.

WCSS_A_QTMR_AC_QTMR_AC_CNTTID

Bits	Name	Description
31:28	F7_CFG	Frame 7 configuration (same definition as Frame 0)
27:24	F6_CFG	Frame 6 configuration (same definition as Frame 0)
23:20	F5_CFG	Frame 5 configuration (same definition as Frame 0)
19:16	F4_CFG	Frame 4 configuration (same definition as Frame 0)
15:12	F3_CFG	Frame 3 configuration (same definition as Frame 0)
11:8	F2_CFG	Frame 2 configuration (same definition as Frame 0)
7:4	F1_CFG	Frame 1 configuration (same definition as Frame 0)
3:0	F0_CFG	Frame 0 configuration 0x3: RSVD (Reserved (Read as zero)) 0x2: FPLO (Frame has second view) 0x1: FVI (Virtual frame implemented) 0x0: FI (Frame implemented)

0x0A211040+ WCSS_A_QTMR_AC_QTMR_AC_CNTACR_n, n=[0..4] 0x4*n

Type: RW

Clock: SYS_AHB_CLK

Reset State: 0x00000000

Security Treatment: Controlled by QTMR_AC_CNTSR[n] The Counter Access Control Register (CNTACR) provide access control to the individual elements of timer frame n, including the second view of that timer frame n.

WCSS_A_QTMR_AC_QTMR_AC_CNTACR_n

Bits	Name	Description
5	RWPT	Read/Write access to CNTP_*register 0x0: ACCESS_DENIED (Access denied) 0x1: ACCESS_ALLOWED (Access allowed)
4	RWVT	Read/Write access to CNTV_*register 0x0: ACCESS_DENIED (Access denied) 0x1: ACCESS_ALLOWED (Access allowed)
3	RVOFF	Read access to CNTVOFF register 0x0: ACCESS_DENIED (Access denied) 0x1: ACCESS_ALLOWED (Access allowed)
2	RFRQ	Read access to CNTFRQ register 0x0: ACCESS_DENIED (Access denied) 0x1: ACCESS_ALLOWED (Access allowed)

WCSS_A_QTMR_AC_QTMR_AC_CNTACR_n (cont.)

Bits	Name	Description
1	RPVCT	Read access to CNTVCT register 0x0: ACCESS_DENIED (Access denied) 0x1: ACCESS_ALLOWED (Access allowed)
0	RPCT	Read access to CNTPCT register 0x0: ACCESS_DENIED (Access denied) 0x1: ACCESS_ALLOWED (Access allowed)

**0x0A211080+ WCSS_A_QTMR_AC_QTMR_AC_CNTVOFF_LO_n, n=[0..4]
0x8*n****Type:** RW**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTSR[n]

The lower 32 bits of Virtual Offset register

WCSS_A_QTMR_AC_QTMR_AC_CNTVOFF_LO_n

Bits	Name	Description
31:0	CNTVOFF_LO	Lower bits [31:0] of virtual Offset for Frame n

**0x0A211084+ WCSS_A_QTMR_AC_QTMR_AC_CNTVOFF_HI_n, n=[0..4]
0x8*n****Type:** RW**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTSR[n]

The upper 32 bits of Virtual Offset register

WCSS_A_QTMR_AC_QTMR_AC_CNTVOFF_HI_n

Bits	Name	Description
23:0	CNTVOFF_HI	Higher bits [31:0] of virtual Offset for Frame n

0x0A211FC0 WCSS_A_QTMR_AC_QTMR_AC_CFG**Type:** RW**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** Both

This register is used for global qtmr controls

WCSS_A_QTMR_AC_QTMR_AC_CFG

Bits	Name	Description
0	RESERVED	reserved

10.11 WCSS_QTMR_V1_T0

QTMR V1 Block

10.11.1 WCSS_A_QTMR_V1_T0_QTMR_V1

QTMR V1 Block

0x0A212000 WCSS_A_QTMR_V1_T0_QTMR_V1_Cntpct_Lo**Type:** R**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RPCT]

The lower 32 bits of Physical Counter

WCSS_A_QTMR_V1_T0_QTMR_V1_Cntpct_Lo

Bits	Name	Description
31:0	CNTPCT_LO	Lower bits [31:0] of Physical Counter

0x0A212004 WCSS_A_QTMR_V1_T0_QTMR_V1_Cntpct_Hi**Type:** R**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RPCT]

The upper 32 bits of Physical Counter

WCSS_A_QTMR_V1_T0_QTMR_V1_Cntpct_HI

Bits	Name	Description
23:0	CNTPCT_HI	Higher bits [31:0] of Physical Counter

0x0A212008 WCSS_A_QTMR_V1_T0_QTMR_V1_Cntvct_LO**Type:** R**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RVCT]

The lower 32 bits of Virtual Counter

WCSS_A_QTMR_V1_T0_QTMR_V1_Cntvct_LO

Bits	Name	Description
31:0	CNTVCT_LO	Lower bits [31:0] of Virtual Counter

0x0A21200C WCSS_A_QTMR_V1_T0_QTMR_V1_Cntvct_HI**Type:** R**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RVCT]

The upper 32 bits of Virtual Counter

WCSS_A_QTMR_V1_T0_QTMR_V1_Cntvct_HI

Bits	Name	Description
23:0	CNTVCT_HI	Higher bits [31:0] of Virtual Counter

0x0A212010 WCSS_A_QTMR_V1_T0_QTMR_V1_Cntfrq**Type:** R**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RFRQ]

Read QTMR_AC_CNTFRQ register

WCSS_A_QTMR_V1_T0_QTMR_V1_CNTFRQ

Bits	Name	Description
31:0	CNTFRQ	Counter frequency in ticks per second

0x0A212014 WCSS_A_QTMR_V1_T0_QTMR_V1_CNTPL0ACR**Type:** RW**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** Both

The Second View Counter Access Control Register (CNTPL0ACR) provide second view access control to the individual elements of timer frame n, except CNTVOFF and CNTPL0ACR registers

WCSS_A_QTMR_V1_T0_QTMR_V1_CNTPL0ACR

Bits	Name	Description
9	PL0CTEN	Second view read/write access to CNTP_* register
8	PL0VTEN	Second view read/write access to CNTV_* register
1	PL0VCTEN	Second view read access to CNTVCT and CNTFRQ register
0	PL0PCTEN	Second view read access to CNTPCT and CNTFRQ register

0x0A212018 WCSS_A_QTMR_V1_T0_QTMR_V1_CNTVOFF_LO**Type:** R**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RVOFF]

Read QTMR_AC_CNTVOFF_LO_n register

WCSS_A_QTMR_V1_T0_QTMR_V1_CNTVOFF_LO

Bits	Name	Description
31:0	CNTVOFF_LO	Lower bits [31:0] of virtual Offset for Frame

0x0A21201C WCSS_A_QTMR_V1_T0_QTMR_V1_CNTVOFF_HI**Type:** R**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RVOFF]

Read QTMR_AC_CNTVOFF_HI_n register

WCSS_A_QTMR_V1_T0_QTMR_V1_CNTVOFF_HI

Bits	Name	Description
23:0	CNTVOFF_HI	Higher bits [31:0] of virtual Offset for Frame

0x0A212020 WCSS_A_QTMR_V1_T0_QTMR_V1_CNTP_CVAL_LO**Type:** RW**Clock:** SYS_TMR_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RWPT]

The Physical Timer Compare Value register is compare against the CNTPCT counter. The interrupt signal is asserted when the CNTPCT counter is greater than equal to CNTP_CVAL

WCSS_A_QTMR_V1_T0_QTMR_V1_CNTP_CVAL_LO

Bits	Name	Description
31:0	CNTP_CVAL_LO	Lower bits [31:0] of compare value for interrupt for Frame

0x0A212024 WCSS_A_QTMR_V1_T0_QTMR_V1_CNTP_CVAL_HI**Type:** RW**Clock:** SYS_TMR_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RWPT]

The Physical Timer Compare Value register is compare against the CNTPCT counter. The interrupt signal is asserted when the CNTPCT counter is greater than equal to CNTP_CVAL

WCSS_A_QTMR_V1_T0_QTMR_V1_CNTP_CVAL_HI

Bits	Name	Description
23:0	CNTP_CVAL_HI	Higher bits [31:0] of compare value for interrupt for Frame

0x0A212028 WCSS_A_QTMR_V1_T0_QTMR_V1_CNTP_TVAL**Type:** RW**Clock:** SYS_TMR_CLK**Reset State:** 0x00000000

Security Treatment: Controlled by QTMR_AC_CNTACR_n[RWPT] An alternative 32 bit TVAL view to provide read or write CVAL register.

The TVAL reads:

$$TVAL = (CVAL - CNTP)$$

The TVAL write CVAL with:

$$CVAL = (CNTP + \text{signExtend}(TVAL))$$
WCSS_A_QTMR_V1_T0_QTMR_V1_CNTP_TVAL

Bits	Name	Description
31:0	CNTP_TVAL	Timer Value in tick per second for Frame

0x0A21202C WCSS_A_QTMR_V1_T0_QTMR_V1_CNTP_CTL**Type:** RW**Clock:** SYS_TMR_CLK**Reset State:** 0x00000000

Security Treatment: Controlled by QTMR_AC_CNTACR_n[RWPT] The Physical Timer Control Register which can disable the timer and can mask its interrupt. If the timer is disabled, interrupt are masked, but the timer value continue to count down.

WCSS_A_QTMR_V1_T0_QTMR_V1_CNTP_CTL

Bits	Name	Type	Description
2	ISTAT	R	Interrupt Status before masking (Read Only) 0x0: INTERRUPT_NOT_PENDING 0x1: INTERRUPT_PENDING
1	IMSK	RW	Mask Interrupt 0x0: UNMASK_INTERRUPT 0x1: MASK_INTERRUPT
0	EN	RW	Enable Timer 0x0: DISABLED 0x1: ENABLED

0x0A212030 WCSS_A_QTMR_V1_T0_QTMR_V1_CNTV_CVAL_LO**Type:** RW**Clock:** SYS_TMR_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RWVT]

The Virtual Timer Compare Value register is compare against the CNTVCT counter. The interrupt signal is asserted when the CNTVCT counter is greater than equal to CNTV_CVAL. This register is implemented if FVI (Virtual Frame implemented) for the Frame n is set in QTMR_AC_CNTTID register.

WCSS_A_QTMR_V1_T0_QTMR_V1_CNTV_CVAL_LO

Bits	Name	Description
31:0	CNTV_CVAL_LO	Lower bits [31:0] of compare value for interrupt for Frame n

0x0A212034 WCSS_A_QTMR_V1_T0_QTMR_V1_CNTV_CVAL_HI**Type:** RW**Clock:** SYS_TMR_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RWVT]

The Virtual Timer Compare Value register is compare against the CNTVCT counter. The interrupt signal is asserted when the CNTVCT counter is greater than equal to CNTV_CVAL. This register is implemented if FVI (Virtual Frame implemented) for the Frame n is set in QTMR_AC_CNTTID register.

WCSS_A_QTMR_V1_T0_QTMR_V1_CNTV_CVAL_HI

Bits	Name	Description
23:0	CNTV_CVAL_HI	Higher bits [31:0] of compare value for interrupt for Frame n

0x0A212038 WCSS_A_QTMR_V1_T0_QTMR_V1_CNTV_TVAL**Type:** RW**Clock:** SYS_TMR_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RWVT]

An alternative 32 bit TVAL view to provide read or write CVAL register. The TVAL reads: $TVAL = (CVAL - (CNTP - CNTVOFF))$ The TVAL write CVAL with:

$$CVAL = (CNTP - CNTVOFF) + \text{signExtend}(TVAL)$$

This register is implemented if FVI (Virtual Frame implemented) for the Frame n is set in QTMR_AC_CNTTID register.

WCSS_A_QTMR_V1_T0_QTMR_V1_CNTV_TVAL

Bits	Name	Description
31:0	CNTV_TVAL	Timer Value in tick per second for Frame

0x0A21203C WCSS_A_QTMR_V1_T0_QTMR_V1_CNTV_CTL

Type: RW

Clock: SYS_TMR_CLK

Reset State: 0x00000000

Security Treatment: Controlled by QTMR_AC_CNTACR_n[RWVT]

The Virtual Timer Control Register which can disable the timer and can mask its interrupt. If the timer is disabled, interrupt are masked, but the timer value continue to count down. This register is implemented if FVI (Virtual Frame implemented) for the Frame n is set in QTMR_AC_CNTTID register.

WCSS_A_QTMR_V1_T0_QTMR_V1_CNTV_CTL

Bits	Name	Type	Description
2	ISTAT	R	Interrupt Status before masking (Read Only) 0x0: INTERRUPT_NOT_PENDING 0x1: INTERRUPT_PENDING
1	IMSK	RW	Mask Interrupt 0x0: UNMASK_INTERRUPT 0x1: MASK_INTERRUPT
0	EN	RW	Enable Timer 0x0: DISABLED 0x1: ENABLED

10.12 WCSS_QTMR_V1_T1

QTMR V1 Block

10.12.1 WCSS_A_QTMR_V1_T1_QTMR_V1

QTMR V1 Block

0x0A213000 WCSS_A_QTMR_V1_T1_QTMR_V1_CNTPCT_LO**Type:** R**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RPCT]

The lower 32 bits of Physical Counter

WCSS_A_QTMR_V1_T1_QTMR_V1_CNTPCT_LO

Bits	Name	Description
31:0	CNTPCT_LO	Lower bits [31:0] of Physical Counter

0x0A213004 WCSS_A_QTMR_V1_T1_QTMR_V1_CNTPCT_HI**Type:** R**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RPCT]

The upper 32 bits of Physical Counter

WCSS_A_QTMR_V1_T1_QTMR_V1_CNTPCT_HI

Bits	Name	Description
23:0	CNTPCT_HI	Higher bits [31:0] of Physical Counter

0x0A213008 WCSS_A_QTMR_V1_T1_QTMR_V1_CNTVCT_LO**Type:** R**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RVCT]

The lower 32 bits of Virtual Counter

WCSS_A_QTMR_V1_T1_QTMR_V1_CNTVCT_LO

Bits	Name	Description
31:0	CNTVCT_LO	Lower bits [31:0] of Virtual Counter

0x0A21300C WCSS_A_QTMR_V1_T1_QTMR_V1_CNTVCT_HI**Type:** R**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RVCT]

The upper 32 bits of Virtual Counter

WCSS_A_QTMR_V1_T1_QTMR_V1_CNTVCT_HI

Bits	Name	Description
23:0	CNTVCT_HI	Higher bits [31:0] of Virtual Counter

0x0A213010 WCSS_A_QTMR_V1_T1_QTMR_V1_CNTFRQ**Type:** R**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RFRQ]

Read QTMR_AC_CNTFRQ register

WCSS_A_QTMR_V1_T1_QTMR_V1_CNTFRQ

Bits	Name	Description
31:0	CNTFRQ	Counter frequency in ticks per second

0x0A213014 WCSS_A_QTMR_V1_T1_QTMR_V1_CNTPL0ACR**Type:** RW**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** Both

The Second View Counter Access Control Register (CNTPL0ACR) provide second view access control to the individual elements of timer frame n, except CNTVOFF and CNTPL0ACR registers

WCSS_A_QTMR_V1_T1_QTMR_V1_CNTPL0ACR

Bits	Name	Description
9	PL0CTEN	Second view read/write access to CNT*_* register
8	PL0VTEN	Second view read/write access to CNTV_* register
1	PL0VCTEN	Second view read access to CNTVCT and CNTFRQ register

WCSS_A_QTMR_V1_T1_QTMR_V1_CNTPL0ACR (cont.)

Bits	Name	Description
0	PL0PCTEN	Second view read access to CNTPCT and CNTFRQ register

0x0A213018 WCSS_A_QTMR_V1_T1_QTMR_V1_CNTVOFF_LO**Type:** R**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RVOFF]

Read QTMR_AC_CNTVOFF_LO_n register

WCSS_A_QTMR_V1_T1_QTMR_V1_CNTVOFF_LO

Bits	Name	Description
31:0	CNTVOFF_LO	Lower bits [31:0] of virtual Offset for Frame

0x0A21301C WCSS_A_QTMR_V1_T1_QTMR_V1_CNTVOFF_HI**Type:** R**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RVOFF]

Read QTMR_AC_CNTVOFF_HI_n register

WCSS_A_QTMR_V1_T1_QTMR_V1_CNTVOFF_HI

Bits	Name	Description
23:0	CNTVOFF_HI	Higher bits [31:0] of virtual Offset for Frame

0x0A213020 WCSS_A_QTMR_V1_T1_QTMR_V1_CNTP_CVAL_LO**Type:** RW**Clock:** SYS_TMR_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RWPT]

The Physical Timer Compare Value register is compare against the CNTPCT counter. The interrupt signal is asserted when the CNTPCT counter is greater than equal to CNTP_CVAL

WCSS_A_QTMR_V1_T1_QTMR_V1_CNTP_CVAL_LO

Bits	Name	Description
31:0	CNTP_CVAL_LO	Lower bits [31:0] of compare value for interrupt for Frame

0x0A213024 WCSS_A_QTMR_V1_T1_QTMR_V1_CNTP_CVAL_HI**Type:** RW**Clock:** SYS_TMR_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RWPT]

The Physical Timer Compare Value register is compare against the CNTPCT counter. The interrupt signal is asserted when the CNTPCT counter is greater than equal to CNTP_CVAL

WCSS_A_QTMR_V1_T1_QTMR_V1_CNTP_CVAL_HI

Bits	Name	Description
23:0	CNTP_CVAL_HI	Higher bits [31:0] of compare value for interrupt for Frame

0x0A213028 WCSS_A_QTMR_V1_T1_QTMR_V1_CNTP_TVAL**Type:** RW**Clock:** SYS_TMR_CLK**Reset State:** 0x00000000

Security Treatment: Controlled by QTMR_AC_CNTACR_n[RWPT] An alternative 32 bit TVAL view to provide read or write CVAL register.

The TVAL reads:

$$TVAL = (CVAL - CNTP)$$

The TVAL write CVAL with:

$$CVAL = (CNTP + \text{signExtend}(TVAL))$$

WCSS_A_QTMR_V1_T1_QTMR_V1_CNTP_TVAL

Bits	Name	Description
31:0	CNTP_TVAL	Timer Value in tick per second for Frame

0x0A21302C WCSS_A_QTMR_V1_T1_QTMR_V1_CNTP_CTL**Type:** RW**Clock:** SYS_TMR_CLK**Reset State:** 0x00000000

Security Treatment: Controlled by QTMR_AC_CNTACR_n[RWPT] The Physical Timer Control Register which can disable the timer and can mask its interrupt. If the timer is disabled, interrupt are masked, but the timer value continue to count down.

WCSS_A_QTMR_V1_T1_QTMR_V1_CNTP_CTL

Bits	Name	Type	Description
2	ISTAT	R	Interrupt Status before masking (Read Only) 0x0: INTERRUPT_NOT_PENDING 0x1: INTERRUPT_PENDING
1	IMSK	RW	Mask Interrupt 0x0: UNMASK_INTERRUPT 0x1: MASK_INTERRUPT
0	EN	RW	Enable Timer 0x0: DISABLED 0x1: ENABLED

0x0A213030 WCSS_A_QTMR_V1_T1_QTMR_V1_CNTV_CVAL_LO**Type:** RW**Clock:** SYS_TMR_CLK**Reset State:** 0x00000000

Security Treatment: Controlled by QTMR_AC_CNTACR_n[RWVT]

The Virtual Timer Compare Value register is compare against the CNTVCT counter. The interrupt signal is asserted when the CNTVCT counter is greater than equal to CNTV_CVAL. This register is implemented if FVI (Virtual Frame implemented) for the Frame n is set in QTMR_AC_CNTTID register.

WCSS_A_QTMR_V1_T1_QTMR_V1_CNTV_CVAL_LO

Bits	Name	Description
31:0	CNTV_CVAL_LO	Lower bits [31:0] of compare value for interrupt for Frame n

0x0A213034 WCSS_A_QTMR_V1_T1_QTMR_V1_CNTV_CVAL_HI**Type:** RW**Clock:** SYS_TMR_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RWVT]

The Virtual Timer Compare Value register is compare against the CNTVCT counter. The interrupt signal is asserted when the CNTVCT counter is greater than equal to CNTV_CVAL. This register is implemented if FVI (Virtual Frame implemented) for the Frame n is set in QTMR_AC_CNTTID register.

WCSS_A_QTMR_V1_T1_QTMR_V1_CNTV_CVAL_HI

Bits	Name	Description
23:0	CNTV_CVAL_HI	Higher bits [31:0] of compare value for interrupt for Frame n

0x0A213038 WCSS_A_QTMR_V1_T1_QTMR_V1_CNTV_TVAL**Type:** RW**Clock:** SYS_TMR_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RWVT]

An alternative 32 bit TVAL view to provide read or write CVAL register. The TVAL reads: $TVAL = (CVAL - (CNTP - CNTVOFF))$. The TVAL write CVAL with: $CVAL = (CNTP - CNTVOFF) + \text{signExtend}(TVAL)$

This register is implemented if FVI (Virtual Frame implemented) for the Frame n is set in QTMR_AC_CNTTID register.

WCSS_A_QTMR_V1_T1_QTMR_V1_CNTV_TVAL

Bits	Name	Description
31:0	CNTV_TVAL	Timer Value in tick per second for Frame

0x0A21303C WCSS_A_QTMR_V1_T1_QTMR_V1_CNTV_CTL**Type:** RW**Clock:** SYS_TMR_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RWVT]

The Virtual Timer Control Register which can disable the timer and can mask its interrupt. If the timer is disabled, interrupt are masked, but the timer value continue to count down.

This register is implemented if FVI (Virtual Frame implemented) for the Frame n is set in QTMR_AC_CNTTID register.

WCSS_A_QTMR_V1_T1_QTMR_V1_CNTV_CTL

Bits	Name	Type	Description
2	ISTAT	R	Interrupt Status before masking (Read Only) 0x0: INTERRUPT_NOT_PENDING 0x1: INTERRUPT_PENDING
1	IMSK	RW	Mask Interrupt 0x0: UNMASK_INTERRUPT 0x1: MASK_INTERRUPT
0	EN	RW	Enable Timer 0x0: DISABLED 0x1: ENABLED

10.13 WCSS_QTMR_V1_T2

QTMR V1 Block

10.13.1 WCSS_A_QTMR_V1_T2_QTMR_V1

QTMR V1 Block

0x0A214000 WCSS_A_QTMR_V1_T2_QTMR_V1_Cntpct_LO

Type: R

Clock: SYS_AHB_CLK

Reset State: 0x00000000

Security Treatment: Controlled by QTMR_AC_CNTACR_n[RPCT]

The lower 32 bits of Physical Counter

WCSS_A_QTMR_V1_T2_QTMR_V1_Cntpct_LO

Bits	Name	Description
31:0	CNTPCT_LO	Lower bits [31:0] of Physical Counter

0x0A214004 WCSS_A_QTMR_V1_T2_QTMR_V1_Cntpct_HI

Type: R

Clock: SYS_AHB_CLK

Reset State: 0x00000000

Security Treatment: Controlled by QTMR_AC_CNTACR_n[RPCT]

The upper 32 bits of Physical Counter

WCSS_A_QTMR_V1_T2_QTMR_V1_Cntpct_HI

Bits	Name	Description
23:0	CNTPCT_HI	Higher bits [31:0] of Physical Counter

0x0A214008 WCSS_A_QTMR_V1_T2_QTMR_V1_Cntvct_LO**Type:** R**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RVCT]

The lower 32 bits of Virtual Counter

WCSS_A_QTMR_V1_T2_QTMR_V1_Cntvct_LO

Bits	Name	Description
31:0	CNTVCT_LO	Lower bits [31:0] of Virtual Counter

0x0A21400C WCSS_A_QTMR_V1_T2_QTMR_V1_Cntvct_HI**Type:** R**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RVCT]

The upper 32 bits of Virtual Counter

WCSS_A_QTMR_V1_T2_QTMR_V1_Cntvct_HI

Bits	Name	Description
23:0	CNTVCT_HI	Higher bits [31:0] of Virtual Counter

0x0A214010 WCSS_A_QTMR_V1_T2_QTMR_V1_Cntfrq**Type:** R**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RFRQ]

Read QTMR_AC_CNTFRQ register

WCSS_A_QTMR_V1_T2_QTMR_V1_CNTFRQ

Bits	Name	Description
31:0	CNTFRQ	Counter frequency in ticks per second

0x0A214014 WCSS_A_QTMR_V1_T2_QTMR_V1_CNTPL0ACR**Type:** RW**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** Both

The Second View Counter Access Control Register (CNTPL0ACR) provide second view access control to the individual elements of timer frame n, except CNTVOFF and CNTPL0ACR registers

WCSS_A_QTMR_V1_T2_QTMR_V1_CNTPL0ACR

Bits	Name	Description
9	PL0CTEN	Second view read/write access to CNTP_* register
8	PL0VTEN	Second view read/write access to CNTV_* register
1	PL0VCTEN	Second view read access to CNTVCT and CNTFRQ register
0	PL0PCTEN	Second view read access to CNTPCT and CNTFRQ register

0x0A214018 WCSS_A_QTMR_V1_T2_QTMR_V1_CNTVOFF_LO**Type:** R**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RVOFF]

Read QTMR_AC_CNTVOFF_LO_n register

WCSS_A_QTMR_V1_T2_QTMR_V1_CNTVOFF_LO

Bits	Name	Description
31:0	CNTVOFF_LO	Lower bits [31:0] of virtual Offset for Frame

0x0A21401C WCSS_A_QTMR_V1_T2_QTMR_V1_CNTVOFF_HI**Type:** R**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RVOFF]

Read QTMR_AC_CNTVOFF_HI_n register

WCSS_A_QTMR_V1_T2_QTMR_V1_CNTVOFF_HI

Bits	Name	Description
23:0	CNTVOFF_HI	Higher bits [31:0] of virtual Offset for Frame

0x0A214020 WCSS_A_QTMR_V1_T2_QTMR_V1_CNTP_CVAL_LO**Type:** RW**Clock:** SYS_TMR_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RWPT]

The Physical Timer Compare Value register is compare against the CNTPCT counter. The interrupt signal is asserted when the CNTPCT counter is greater than equal to CNTP_CVAL

WCSS_A_QTMR_V1_T2_QTMR_V1_CNTP_CVAL_LO

Bits	Name	Description
31:0	CNTP_CVAL_LO	Lower bits [31:0] of compare value for interrupt for Frame

0x0A214024 WCSS_A_QTMR_V1_T2_QTMR_V1_CNTP_CVAL_HI**Type:** RW**Clock:** SYS_TMR_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RWPT]

The Physical Timer Compare Value register is compare against the CNTPCT counter. The interrupt signal is asserted when the CNTPCT counter is greater than equal to CNTP_CVAL

WCSS_A_QTMR_V1_T2_QTMR_V1_CNTP_CVAL_HI

Bits	Name	Description
23:0	CNTP_CVAL_HI	Higher bits [31:0] of compare value for interrupt for Frame

0x0A214028 WCSS_A_QTMR_V1_T2_QTMR_V1_CNTP_TVAL**Type:** RW**Clock:** SYS_TMR_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RWPT]

An alternative 32 bit TVAL view to provide read or write CVAL register.

The TVAL reads:

TVAL = (CVAL - CNTP)

The TVAL write CVAL with:

CVAL = (CNTP + signExtend(TVAL))

WCSS_A_QTMR_V1_T2_QTMR_V1_CNTP_TVAL

Bits	Name	Description
31:0	CNTP_TVAL	Timer Value in tick per second for Frame

0x0A21402C WCSS_A_QTMR_V1_T2_QTMR_V1_CNTP_CTL**Type:** RW**Clock:** SYS_TMR_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RWPT]

The Physical Timer Control Register which can disable the timer and can mask its interrupt. If the timer is disabled, interrupt are masked, but the timer value continue to count down.

WCSS_A_QTMR_V1_T2_QTMR_V1_CNTP_CTL

Bits	Name	Type	Description
2	ISTAT	R	Interrupt Status before masking (Read Only) 0x0: INTERRUPT_NOT_PENDING 0x1: INTERRUPT_PENDING
1	IMSK	RW	Mask Interrupt 0x0: UNMASK_INTERRUPT 0x1: MASK_INTERRUPT
0	EN	RW	Enable Timer 0x0: DISABLED 0x1: ENABLED

0x0A214030 WCSS_A_QTMR_V1_T2_QTMR_V1_CNTV_CVAL_LO**Type:** RW**Clock:** SYS_TMR_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RWVT]

The Virtual Timer Compare Value register is compare against the CNTVCT counter. The interrupt signal is asserted when the CNTVCT counter is greater than equal to CNTV_CVAL. This register is implemented if FVI (Virtual Frame implemented) for the Frame n is set in QTMR_AC_CNTTID register.

WCSS_A_QTMR_V1_T2_QTMR_V1_CNTV_CVAL_LO

Bits	Name	Description
31:0	CNTV_CVAL_LO	Lower bits [31:0] of compare value for interrupt for Frame n

0x0A214034 WCSS_A_QTMR_V1_T2_QTMR_V1_CNTV_CVAL_HI**Type:** RW**Clock:** SYS_TMR_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RWVT]

The Virtual Timer Compare Value register is compare against the CNTVCT counter. The interrupt signal is asserted when the CNTVCT counter is greater than equal to CNTV_CVAL. This register is implemented if FVI (Virtual Frame implemented) for the Frame n is set in QTMR_AC_CNTTID register.

WCSS_A_QTMR_V1_T2_QTMR_V1_CNTV_CVAL_HI

Bits	Name	Description
23:0	CNTV_CVAL_HI	Higher bits [31:0] of compare value for interrupt for Frame n

0x0A214038 WCSS_A_QTMR_V1_T2_QTMR_V1_CNTV_TVAL**Type:** RW**Clock:** SYS_TMR_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RWVT]

An alternative 32 bit TVAL view to provide read or write CVAL register.

The TVAL reads:

$$TVAL = (CVAL - (CNTP - CNTVOFF))$$

The TVAL write CVAL with:

$$CVAL = (CNTP - CNTVOFF) + \text{signExtend}(TVAL)$$

This register is implemented if FVI (Virtual Frame implemented) for the Frame n is set in QTMR_AC_CNTTID register.

WCSS_A_QTMR_V1_T2_QTMR_V1_CNTV_TVAL

Bits	Name	Description
31:0	CNTV_TVAL	Timer Value in tick per second for Frame

0x0A21403C WCSS_A_QTMR_V1_T2_QTMR_V1_CNTV_CTL

Type: RW

Clock: SYS_TMR_CLK

Reset State: 0x00000000

Security Treatment: Controlled by QTMR_AC_CNTACR_n[RWVT]

The Virtual Timer Control Register which can disable the timer and can mask its interrupt. If the timer is disabled, interrupt are masked, but the timer value continue to count down. This register is implemented if FVI (Virtual Frame implemented) for the Frame n is set in QTMR_AC_CNTTID register.

WCSS_A_QTMR_V1_T2_QTMR_V1_CNTV_CTL

Bits	Name	Type	Description
2	ISTAT	R	Interrupt Status before masking (Read Only) 0x0: INTERRUPT_NOT_PENDING 0x1: INTERRUPT_PENDING
1	IMSK	RW	Mask Interrupt 0x0: UNMASK_INTERRUPT 0x1: MASK_INTERRUPT
0	EN	RW	Enable Timer 0x0: DISABLED 0x1: ENABLED

10.14 WCSS_QTMR_V1_T3

QTMR V1 Block

10.14.1 WCSS_A_QTMR_V1_T3_QTMR_V1

QTMR V1 Block

0x0A215000 WCSS_A_QTMR_V1_T3_QTMR_V1_CNTPCT_LO**Type:** R**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RPCT]

The lower 32 bits of Physical Counter

WCSS_A_QTMR_V1_T3_QTMR_V1_CNTPCT_LO

Bits	Name	Description
31:0	CNTPCT_LO	Lower bits [31:0] of Physical Counter

0x0A215004 WCSS_A_QTMR_V1_T3_QTMR_V1_CNTPCT_HI**Type:** R**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RPCT]

The upper 32 bits of Physical Counter

WCSS_A_QTMR_V1_T3_QTMR_V1_CNTPCT_HI

Bits	Name	Description
23:0	CNTPCT_HI	Higher bits [31:0] of Physical Counter

0x0A215008 WCSS_A_QTMR_V1_T3_QTMR_V1_CNTVCT_LO**Type:** R**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RVCT]

The lower 32 bits of Virtual Counter

WCSS_A_QTMR_V1_T3_QTMR_V1_CNTVCT_LO

Bits	Name	Description
31:0	CNTVCT_LO	Lower bits [31:0] of Virtual Counter

0x0A21500C WCSS_A_QTMR_V1_T3_QTMR_V1_CNTVCT_HI**Type:** R**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RVCT]

The upper 32 bits of Virtual Counter

WCSS_A_QTMR_V1_T3_QTMR_V1_CNTVCT_HI

Bits	Name	Description
23:0	CNTVCT_HI	Higher bits [31:0] of Virtual Counter

0x0A215010 WCSS_A_QTMR_V1_T3_QTMR_V1_CNTFRQ**Type:** R**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RFRQ]

Read QTMR_AC_CNTFRQ register

WCSS_A_QTMR_V1_T3_QTMR_V1_CNTFRQ

Bits	Name	Description
31:0	CNTFRQ	Counter frequency in ticks per second

0x0A215014 WCSS_A_QTMR_V1_T3_QTMR_V1_CNTPL0ACR**Type:** RW**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** Both

The Second View Counter Access Control Register (CNTPL0ACR) provide second view access control to the individual elements of timer frame n, except CNTVOFF and CNTPL0ACR registers

WCSS_A_QTMR_V1_T3_QTMR_V1_CNTPL0ACR

Bits	Name	Description
9	PL0CTEN	Second view read/write access to CNTP_* register
8	PL0VTEN	Second view read/write access to CNTV_* register
1	PL0VCTEN	Second view read access to CNTVCT and CNTFRQ register

WCSS_A_QTMR_V1_T3_QTMR_V1_CNTPL0ACR (cont.)

Bits	Name	Description
0	PL0PCTEN	Second view read access to CNTPCT and CNTFRQ register

0x0A215018 WCSS_A_QTMR_V1_T3_QTMR_V1_CNTVOFF_LO**Type:** R**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RVOFF]

Read QTMR_AC_CNTVOFF_LO_n register

WCSS_A_QTMR_V1_T3_QTMR_V1_CNTVOFF_LO

Bits	Name	Description
31:0	CNTVOFF_LO	Lower bits [31:0] of virtual Offset for Frame

0x0A21501C WCSS_A_QTMR_V1_T3_QTMR_V1_CNTVOFF_HI**Type:** R**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RVOFF]

Read QTMR_AC_CNTVOFF_HI_n register

WCSS_A_QTMR_V1_T3_QTMR_V1_CNTVOFF_HI

Bits	Name	Description
23:0	CNTVOFF_HI	Higher bits [31:0] of virtual Offset for Frame

0x0A215020 WCSS_A_QTMR_V1_T3_QTMR_V1_CNTP_CVAL_LO**Type:** RW**Clock:** SYS_TMR_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RWPT]

The Physical Timer Compare Value register is compare against the CNTPCT counter. The interrupt signal is asserted when the CNTPCT counter is greater than equal to CNTP_CVAL

WCSS_A_QTMR_V1_T3_QTMR_V1_CNTP_CVAL_LO

Bits	Name	Description
31:0	CNTP_CVAL_LO	Lower bits [31:0] of compare value for interrupt for Frame

0x0A215024 WCSS_A_QTMR_V1_T3_QTMR_V1_CNTP_CVAL_HI**Type:** RW**Clock:** SYS_TMR_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RWPT]

The Physical Timer Compare Value register is compare against the CNTPCT counter. The interrupt signal is asserted when the CNTPCT counter is greater than equal to CNTP_CVAL

WCSS_A_QTMR_V1_T3_QTMR_V1_CNTP_CVAL_HI

Bits	Name	Description
23:0	CNTP_CVAL_HI	Higher bits [31:0] of compare value for interrupt for Frame

0x0A215028 WCSS_A_QTMR_V1_T3_QTMR_V1_CNTP_TVAL**Type:** RW**Clock:** SYS_TMR_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RWPT]

An alternative 32 bit TVAL view to provide read or write CVAL register.

The TVAL reads:

$$TVAL = (CVAL - CNTP)$$

The TVAL write CVAL with:

$$CVAL = (CNTP + \text{signExtend}(TVAL))$$

WCSS_A_QTMR_V1_T3_QTMR_V1_CNTP_TVAL

Bits	Name	Description
31:0	CNTP_TVAL	Timer Value in tick per second for Frame

0x0A21502C WCSS_A_QTMR_V1_T3_QTMR_V1_CNTP_CTL**Type:** RW**Clock:** SYS_TMR_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RWPT]

The Physical Timer Control Register which can disable the timer and can mask its interrupt. If the timer is disabled, interrupt are masked, but the timer value continue to count down.

WCSS_A_QTMR_V1_T3_QTMR_V1_CNTP_CTL

Bits	Name	Type	Description
2	ISTAT	R	Interrupt Status before masking (Read Only) 0x0: INTERRUPT_NOT_PENDING 0x1: INTERRUPT_PENDING
1	IMSK	RW	Mask Interrupt 0x0: UNMASK_INTERRUPT 0x1: MASK_INTERRUPT
0	EN	RW	Enable Timer 0x0: DISABLED 0x1: ENABLED

0x0A215030 WCSS_A_QTMR_V1_T3_QTMR_V1_CNTV_CVAL_LO**Type:** RW**Clock:** SYS_TMR_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RWVT]

The Virtual Timer Compare Value register is compare against the CNTVCT counter. The interrupt signal is asserted when the CNTVCT counter is greater than equal to CNTV_CVAL. This register is implemented if FVI (Virtual Frame implemented) for the Frame n is set in QTMR_AC_CNTTID register.

WCSS_A_QTMR_V1_T3_QTMR_V1_CNTV_CVAL_LO

Bits	Name	Description
31:0	CNTV_CVAL_LO	Lower bits [31:0] of compare value for interrupt for Frame n

0x0A215034 WCSS_A_QTMR_V1_T3_QTMR_V1_CNTV_CVAL_HI**Type:** RW**Clock:** SYS_TMR_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RWVT]

The Virtual Timer Compare Value register is compare against the CNTVCT counter. The interrupt signal is asserted when the CNTVCT counter is greater than equal to CNTV_CVAL. This register is implemented if FVI (Virtual Frame implemented) for the Frame n is set in QTMR_AC_CNTTID register.

WCSS_A_QTMR_V1_T3_QTMR_V1_CNTV_CVAL_HI

Bits	Name	Description
23:0	CNTV_CVAL_HI	Higher bits [31:0] of compare value for interrupt for Frame n

0x0A215038 WCSS_A_QTMR_V1_T3_QTMR_V1_CNTV_TVAL**Type:** RW**Clock:** SYS_TMR_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RWVT]

An alternative 32 bit TVAL view to provide read or write CVAL register.

The TVAL reads: $TVAL = (CVAL - (CNTP - CNTVOFF))$

The TVAL write CVAL with: $CVAL = (CNTP - CNTVOFF) + \text{signExtend}(TVAL)$

This register is implemented if FVI (Virtual Frame implemented) for the Frame n is set in QTMR_AC_CNTTID register.

WCSS_A_QTMR_V1_T3_QTMR_V1_CNTV_TVAL

Bits	Name	Description
31:0	CNTV_TVAL	Timer Value in tick per second for Frame

0x0A21503C WCSS_A_QTMR_V1_T3_QTMR_V1_CNTV_CTL**Type:** RW**Clock:** SYS_TMR_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RWVT]

The Virtual Timer Control Register which can disable the timer and can mask its interrupt. If the timer is disabled, interrupt are masked, but the timer value continue to count down. This register is

implemented if FVI (Virtual Frame implemented) for the Frame n is set in QTMR_AC_CNTTID register.

WCSS_A_QTMR_V1_T3_QTMR_V1_CNTV_CTL

Bits	Name	Type	Description
2	ISTAT	R	Interrupt Status before masking (Read Only) 0x0: INTERRUPT_NOT_PENDING 0x1: INTERRUPT_PENDING
1	IMSK	RW	Mask Interrupt 0x0: UNMASK_INTERRUPT 0x1: MASK_INTERRUPT
0	EN	RW	Enable Timer 0x0: DISABLED 0x1: ENABLED

10.15 WCSS_QTMR_V1_T4

QTMR V1 Block

10.15.1 WCSS_A_QTMR_V1_T4_QTMR_V1

QTMR V1 Block

0x0A216000 WCSS_A_QTMR_V1_T4_QTMR_V1_Cntpct_LO

Type: R

Clock: SYS_AHB_CLK

Reset State: 0x00000000

Security Treatment: Controlled by QTMR_AC_CNTACR_n[RPCT]

The lower 32 bits of Physical Counter

WCSS_A_QTMR_V1_T4_QTMR_V1_Cntpct_LO

Bits	Name	Description
31:0	CNTPCT_LO	Lower bits [31:0] of Physical Counter

0x0A216004 WCSS_A_QTMR_V1_T4_QTMR_V1_Cntpct_HI

Type: R

Clock: SYS_AHB_CLK

Reset State: 0x00000000

Security Treatment: Controlled by QTMR_AC_CNTACR_n[RPCT]

The upper 32 bits of Physical Counter

WCSS_A_QTMR_V1_T4_QTMR_V1_Cntpct_HI

Bits	Name	Description
23:0	CNTPCT_HI	Higher bits [31:0] of Physical Counter

0x0A216008 WCSS_A_QTMR_V1_T4_QTMR_V1_Cntvct_LO**Type:** R**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RVCT]

The lower 32 bits of Virtual Counter

WCSS_A_QTMR_V1_T4_QTMR_V1_Cntvct_LO

Bits	Name	Description
31:0	CNTVCT_LO	Lower bits [31:0] of Virtual Counter

0x0A21600C WCSS_A_QTMR_V1_T4_QTMR_V1_Cntvct_HI**Type:** R**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RVCT]

The upper 32 bits of Virtual Counter

WCSS_A_QTMR_V1_T4_QTMR_V1_Cntvct_HI

Bits	Name	Description
23:0	CNTVCT_HI	Higher bits [31:0] of Virtual Counter

0x0A216010 WCSS_A_QTMR_V1_T4_QTMR_V1_Cntfrq**Type:** R**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RFRQ]

Read QTMR_AC_CNTFRQ register

WCSS_A_QTMR_V1_T4_QTMR_V1_CNTFRQ

Bits	Name	Description
31:0	CNTFRQ	Counter frequency in ticks per second

0x0A216014 WCSS_A_QTMR_V1_T4_QTMR_V1_CNTPL0ACR**Type:** RW**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** Both

The Second View Counter Access Control Register (CNTPL0ACR) provide second view access control to the individual elements of timer frame n, except CNTVOFF and CNTPL0ACR registers

WCSS_A_QTMR_V1_T4_QTMR_V1_CNTPL0ACR

Bits	Name	Description
9	PL0CTEN	Second view read/write access to CNTP_* register
8	PL0VTEN	Second view read/write access to CNTV_* register
1	PL0VCTEN	Second view read access to CNTVCT and CNTFRQ register
0	PL0PCTEN	Second view read access to CNTPCT and CNTFRQ register

0x0A216018 WCSS_A_QTMR_V1_T4_QTMR_V1_CNTVOFF_LO**Type:** R**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RVOFF]

Read QTMR_AC_CNTVOFF_LO_n register

WCSS_A_QTMR_V1_T4_QTMR_V1_CNTVOFF_LO

Bits	Name	Description
31:0	CNTVOFF_LO	Lower bits [31:0] of virtual Offset for Frame

0x0A21601C WCSS_A_QTMR_V1_T4_QTMR_V1_CNTVOFF_HI**Type:** R**Clock:** SYS_AHB_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RVOFF]

Read QTMR_AC_CNTVOFF_HI_n register

WCSS_A_QTMR_V1_T4_QTMR_V1_CNTVOFF_HI

Bits	Name	Description
23:0	CNTVOFF_HI	Higher bits [31:0] of virtual Offset for Frame

0x0A216020 WCSS_A_QTMR_V1_T4_QTMR_V1_CNTP_CVAL_LO**Type:** RW**Clock:** SYS_TMR_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RWPT]

The Physical Timer Compare Value register is compare against the CNTPCT counter. The interrupt signal is asserted when the CNTPCT counter is greater than equal to CNTP_CVAL

WCSS_A_QTMR_V1_T4_QTMR_V1_CNTP_CVAL_LO

Bits	Name	Description
31:0	CNTP_CVAL_LO	Lower bits [31:0] of compare value for interrupt for Frame

0x0A216024 WCSS_A_QTMR_V1_T4_QTMR_V1_CNTP_CVAL_HI**Type:** RW**Clock:** SYS_TMR_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RWPT]

The Physical Timer Compare Value register is compare against the CNTPCT counter. The interrupt signal is asserted when the CNTPCT counter is greater than equal to CNTP_CVAL

WCSS_A_QTMR_V1_T4_QTMR_V1_CNTP_CVAL_HI

Bits	Name	Description
23:0	CNTP_CVAL_HI	Higher bits [31:0] of compare value for interrupt for Frame

0x0A216028 WCSS_A_QTMR_V1_T4_QTMR_V1_CNTP_TVAL**Type:** RW**Clock:** SYS_TMR_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RWPT]

An alternative 32 bit TVAL view to provide read or write CVAL register.

The TVAL reads: $TVAL = (CVAL - CNTP)$ The TVAL write CVAL with: $CVAL = (CNTP + \text{signExtend}(TVAL))$

WCSS_A_QTMR_V1_T4_QTMR_V1_CNTP_TVAL

Bits	Name	Description
31:0	CNTP_TVAL	Timer Value in tick per second for Frame

0x0A21602C WCSS_A_QTMR_V1_T4_QTMR_V1_CNTP_CTL**Type:** RW**Clock:** SYS_TMR_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RWPT]

The Physical Timer Control Register which can disable the timer and can mask its interrupt. If the timer is disabled, interrupt are masked, but the timer value continue to count down.

WCSS_A_QTMR_V1_T4_QTMR_V1_CNTP_CTL

Bits	Name	Type	Description
2	ISTAT	R	Interrupt Status before masking (Read Only) 0x0: INTERRUPT_NOT_PENDING 0x1: INTERRUPT_PENDING
1	IMSK	RW	Mask Interrupt 0x0: UNMASK_INTERRUPT 0x1: MASK_INTERRUPT
0	EN	RW	Enable Timer 0x0: DISABLED 0x1: ENABLED

0x0A216030 WCSS_A_QTMR_V1_T4_QTMR_V1_CNTV_CVAL_LO**Type:** RW**Clock:** SYS_TMR_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RWVT]

The Virtual Timer Compare Value register is compare against the CNTVCT counter. The interrupt signal is asserted when the CNTVCT counter is greater than equal to CNTV_CVAL. This register is implemented if FVI (Virtual Frame implemented) for the Frame n is set in QTMR_AC_CNTTID register.

WCSS_A_QTMR_V1_T4_QTMR_V1_CNTV_CVAL_LO

Bits	Name	Description
31:0	CNTV_CVAL_LO	Lower bits [31:0] of compare value for interrupt for Frame n

0x0A216034 WCSS_A_QTMR_V1_T4_QTMR_V1_CNTV_CVAL_HI**Type:** RW**Clock:** SYS_TMR_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RWVT]

The Virtual Timer Compare Value register is compare against the CNTVCT counter. The interrupt signal is asserted when the CNTVCT counter is greater than equal to CNTV_CVAL. This register is implemented if FVI (Virtual Frame implemented) for the Frame n is set in QTMR_AC_CNTTID register.

WCSS_A_QTMR_V1_T4_QTMR_V1_CNTV_CVAL_HI

Bits	Name	Description
23:0	CNTV_CVAL_HI	Higher bits [31:0] of compare value for interrupt for Frame n

0x0A216038 WCSS_A_QTMR_V1_T4_QTMR_V1_CNTV_TVAL**Type:** RW**Clock:** SYS_TMR_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RWVT] An alternative 32 bit TVAL view to provide read or write CVAL register. The TVAL reads:

$TVAL = (CVAL - (CNTP - CNTVOFF))$ The TVAL write CVAL with: $CVAL = (CNTP - CNTVOFF) + \text{signExtend}(TVAL)$ This register is implemented if FVI (Virtual Frame implemented) for the Frame n is set in QTMR_AC_CNTTID register.

WCSS_A_QTMR_V1_T4_QTMR_V1_CNTV_TVAL

Bits	Name	Description
31:0	CNTV_TVAL	Timer Value in tick per second for Frame

0x0A21603C WCSS_A_QTMR_V1_T4_QTMR_V1_CNTV_CTL**Type:** RW**Clock:** SYS_TMR_CLK**Reset State:** 0x00000000**Security Treatment:** Controlled by QTMR_AC_CNTACR_n[RWVT]

The Virtual Timer Control Register which can disable the timer and can mask its interrupt. If the timer is disabled, interrupt are masked, but the timer value continue to count down. This register is implemented if FVI (Virtual Frame implemented) for the Frame n is set in QTMR_AC_CNTTID register.

WCSS_A_QTMR_V1_T4_QTMR_V1_CNTV_CTL

Bits	Name	Type	Description
2	ISTAT	R	Interrupt Status before masking (Read Only) 0x0: INTERRUPT_NOT_PENDING 0x1: INTERRUPT_PENDING
1	IMSK	RW	Mask Interrupt 0x0: UNMASK_INTERRUPT 0x1: MASK_INTERRUPT
0	EN	RW	Enable Timer 0x0: DISABLED 0x1: ENABLED

10.16 WCSS_A_QGIC2_QGIC2**0x0A20A000 WCSS_A_QGIC2_GICD_CTLR****Type:** RW**Clock:** CLK**Reset State:** 0x00000000**Security Treatment:** Banked

The GICD_CTLR register controls if the Distributor responds to external interrupt stimulus changes. Non-secure access Distributor provides access to the enable_ns register in bit 0.

Secure access Distributor provides access to the enable_s register in bit 0, and the enable_ns register in bit 1.

WCSS_A_QGIC2_GICD_CTLR

Bits	Name	Description
1	ENABLE_NS	This bit is an alias of the enable_ns bit. This bit is only usable by Secure software. 0x0: CLR (CLR) 0x1: SET (SET)
0	ENABLE	This bit is the enable bit for both Secure and Non-secure software. Secure software accesses enable_s at this location. Non-secure software accesses enable_ns. 0x0: CLR (CLR) 0x1: SET (SET)

0x0A20A004 WCSS_A_QGIC2_GICD_TYPER**Type:** R**Clock:** CLK**Reset State:** Undefined**Security Treatment:** Common

The GICD_TYPER register provides information about the configuration of the QGIC2.

WCSS_A_QGIC2_GICD_TYPER

Bits	Name	Description
15:11	LSPI	Returns the number of Lockable Shared Peripheral Interrupts (LSPIs) that the QGIC2 contains. QGIC2 does not support lockable SPIs, always reads back 0x0.
10	TZ	TrustZone support. QGIC2 supports TrustZone, always reads back 0x1.
9:8	RESERVED_BITS9_TO_8	
7:5	CPU_NUM	Returns the number of CPU interfaces that the GIC provides. The GIC provides either: 0b 000 = one CPU interface 0b 001 = two CPU interfaces 0b 010 = three CPU interfaces 0b 011 = four CPU interfaces 0b 100 = five CPU interfaces 0b 101 = six CPU interfaces 0b 110 = seven CPU interfaces 0b 111 = eight CPU interfaces.

WCSS_A_QGIC2_GICD_TYPER (cont.)

Bits	Name	Description
4:0	IT_LINES	Returns the number of INTIDs, to the nearest 32, that the Distributor provides. Read as: 0b 00000 = the Distributor is configured for up to 32 INTIDs 0b 00001 = the Distributor is configured for up to 64 INTIDs 0b 00010 = the Distributor is configured for up to 96 INTIDs 0b 00011 = the Distributor is configured for up to 128 INTIDs 0b 11110 = the Distributor is configured for up to 992 INTIDs 0b 11111 = the Distributor is configured for up to 1020 INTIDs.

0x0A20A008 WCSS_A_QGIC2_GICD_IIDR**Type:** R**Clock:** CLK**Reset State:** 0x00001070**Security Treatment:** Common

The GICD_IIDR register provides the implementor of the Distributor and the revision of the Distributor.

WCSS_A_QGIC2_GICD_IIDR

Bits	Name	Description
15:12	REVISION	Returns the revision number of the Distributor, 0x1
11:0	IMPLEMENTOR	Returns Qualcomm's JEP106 ID number, 0b 01110000

0x0A20A020 WCSS_A_QGIC2_GICD_ANSACR**Type:** RW**Clock:** CLK**Reset State:** 0x00000000**Security Treatment:** Restricted

The auxiliary non-secure access control register (GICD_ANSACR) is used to control non-secure access to GICD_CGCR.

WCSS_A_QGIC2_GICD_ANSACR

Bits	Name	Description
0	GICD_CGCR	0x0: SEC (SEC (Dis-allows non-secure access.)) 0x1: NS (NS (Allows non-secure access.))

0x0A20A024 WCSS_A_QGIC2_GICD_CGCR**Type:** RW**Clock:** CLK**Reset State:** 0x00000000**Security Treatment:** Restricted

The clock gate control register (GICD_CGCR) is used to control localized clock gating over-ride logic. Setting bits in GICD_CGCR disables the corresponding local clock gating logic. The local clock gating logic normally turns off a local (small subset) clock tree automatically if there is no need for the clock. If any error is found in the gating logic, these bits can be used to over-ride it.

WCSS_A_QGIC2_GICD_CGCR

Bits	Name	Description
31:17	RESERVED_31_TO_17	
16	TOP	0x1: DISABLE (DISABLE) 0x0: ENABLE (ENABLE) Controls the top level clock gate (which gates entire QGIC2.)
15:4	RESERVED_15_TO_4	
3	DI_SGI_STATE	0x1: DISABLE (DISABLE) 0x0: ENABLE (ENABLE) Controls the clock gate for Distributor interrupt state data base registers for SGI interrupts.)
2	DI_PPI_SPI_STATE	0x1: DISABLE (DISABLE) 0x0: ENABLE (ENABLE) Controls the clock gate for Distributor interrupt state data base registers for PPI and SPI interrupts.)
1	DI_DEMET	0x1: DISABLE (DISABLE) 0x0: ENABLE (ENABLE) Controls the clock gate for Distributor interrupt input demet registers.)
0	DI_RD	0x1: DISABLE (DISABLE) 0x0: ENABLE (ENABLE) Controls the clock gate for Distributor CRIF Read Data.)

**0x0A20A080+WCSS_A_QGIC2_GICD_ISRn, n=[0..8]
0x4*n****Type:** RW**Clock:** CLK**Reset State:** 0x00000000**Security Treatment:** Restricted

Each bit in an GICD_ISR register controls the **security** state of an interrupt, to be either secure or non-secure. You can only access these registers with secure read or secure write accesses.

NOTE The GICD_ISR register at address offset 0x0080 is banked, that is there is one per CPU Interface. It is only accessible from the designated CPU and controls that CPU's PPI and SGI interrupts. The remaining registers control the SPI interrupts.

WCSS_A_QGIC2_GICD_ISRn

Bits	Name	Description
31:0	INT_NS	0x0: SEC (SEC (Assigns INTID N to the Secure state.)) 0x1: NS (NS (Assigns INTID N to the Non-secure state.))

0x0A20A100+WCSS_A_QGIC2_GICD_ISENABLERn, n=[0..8] 0x4*n

Type: RW

Clock: CLK

Reset State: 0x00000000

Security Treatment: NS-int Dependant

Each bit in an GICD_ISENABLER register controls the enabling of an interrupt.

Reading this register returns the currently enabled interrupts subject to NS-int Dependant access rules.

NOTE The GICD_ISENABLER register at address offset 0x0100 is banked, that is there is one per CPU Interface. It is only accessible from the designated CPU and controls that CPU's PPI and SGI interrupts. The remaining registers control the SPI interrupts.

WCSS_A_QGIC2_GICD_ISENABLERn

Bits	Name	Description
31:0	INT	0x0: NA (NA (No effect. Use GICD_ICENABLER register.)) 0x1: SET (SET (Enables INTID N.))

0x0A20A180+WCSS_A_QGIC2_GICD_ICENABLERn, n=[0..8] 0x4*n

Type: RW

Clock: CLK

Reset State: 0x00000000

Security Treatment: NS-int Dependant

Each bit in an GICD_ICENABLER register controls the disabling of an interrupt.

Reading this register returns the currently enabled interrupts subject to NS-int Dependant access rules.

NOTE The GICD_ICENABLER register at address offset 0x0180 is banked, that is there is one per CPU Interface. It is only accessible from the designated CPU and controls that CPU's PPI and SGI interrupts. The remaining registers control the SPI interrupts.

WCSS_A_QGIC2_GICD_ICENABLERn

Bits	Name	Description
31:0	INT	0x0: NA (NA (No effect. Use GICD_ISENABLER register.)) 0x1: CLR (CLR (Disables INTID N.))

0x0A20A200+WCSS_A_QGIC2_GICD_ISPENDRn, n=[0..8]
0x4*n
Type: RW**Clock:** CLK**Reset State:** 0x00000000**Security Treatment:** NS-int Dependant

Each bit in the GICD_ISPENDR register provides the pending status of an interrupt. Writing a 1 to any of these bits will transition the corresponding interrupt to the Pending (or Active and Pending) state

NOTE The GICD_ISPENDR register at address offset 0x0200 is repeated once per CPU Interface. It is only accessible from the designated CPU and controls that CPU's PPI interrupts. SGI interrupt bits are read-only, however the distributor updates these using the GICD_SGIR register. The remaining registers control the SPI interrupts.

WCSS_A_QGIC2_GICD_ISPENDRn

Bits	Name	Description
31:0	INT	0x0: NA (NA (No effect. Use GICD_ICPENDR register.)) 0x1: SET (SET (Sets INTID N to the Pending state.))

0x0A20A280+WCSS_A_QGIC2_GICD_ICPENDRn, n=[0..8]
0x4*n
Type: RW**Clock:** CLK**Reset State:** 0x00000000**Security Treatment:** NS-int Dependant

Each bit in the GICD_ICPENDR register provides the pending status of an interrupt. Writing a 1 to any of these bits will transition the corresponding interrupt out of the Pending (or Active and Pending) state

NOTE The GICD_ICPENDR register at address offset 0x0280 is banked, that is there is one per CPU Interface. It is only accessible from the designated CPU and controls that CPU's PPI interrupts. SGI interrupt bits are read-only, however the distributor updates these using the GICD_SGIR register. The remaining registers control the SPI interrupts.

WCSS_A_QGIC2_GICD_ICPENDRn

Bits	Name	Description
31:0	INT	0x0: NA (NA (No effect. Use GICD_ISPENDR register.)) 0x1: CLR (CLR (Clears INTID N. from the pending state.))

0x0A20A300+WCSS_A_QGIC2_GICD_ISACTIVERn, n=[0..8]**0x4*n****Type:** RW**Clock:** CLK**Reset State:** 0x00000000**Security Treatment:** NS-int Dependant

Each bit in the GICD_ISACTIVER register provides the active status of an interrupt. Writing a 1 to any of these bits will transition the corresponding interrupt to the Active (or Active and Pending) state

NOTE The GICD_ISACTIVER register at address offset 0x0300 is repeated once per CPU Interface. It is only accessible from the designated CPU and provides information for that CPU's PPI and SGI interrupts. The remaining registers control the SPI interrupts.

WCSS_A_QGIC2_GICD_ISACTIVERn

Bits	Name	Description
31:0	INT	0x0: DO_NOTHING (DO NOTHING) 0x1: SET (SET)

0x0A20A380+WCSS_A_QGIC2_GICD_ICACTIVERn, n=[0..8]**0x4*n****Type:** RW**Clock:** CLK**Reset State:** 0x00000000**Security Treatment:** NS-int Dependant

Each bit in the GICD_ICACTIVER register provides the active status of an interrupt. Writing a 1 to any of these bits will transition the corresponding interrupt out of the Active (or Active and Pending) state

NOTE The GICD_ICACTIVER register at address offset 0x0300 is repeated once per CPU Interface. It is only accessible from the designated CPU and provides information for that CPU's PPI and SGI interrupts. The remaining registers control the SPI interrupts.

WCSS_A_QGIC2_GICD_ICACTIVERn

Bits	Name	Description
31:0	INT	0x1: CLR (CLR) 0x0: DO_NOTHING (DO NOTHING)

0x0A20A400+WCSS_A_QGIC2_GICD_IPRIORITYRn, n=[0..71]**0x4*n****Type:** RW**Clock:** CLK**Reset State:** 0x00000000**Security Treatment:** NS-int Dependant

Each field in the GICD_IPRIORITYR registers controls the priority level of an interrupt.

NOTE The GICD_IPRIORITYR registers at address offsets 0x0400 to 0x41C are repeated once per CPU Interface. These are only accessible from the designated CPU and control the priority levels for that CPU's PPI and SGI interrupts. The remaining registers control the SPI interrupt priority levels.

NOTE This register is byte addressable.

The GICD_IPRIORITYR register behaves differently depending on the **security** setting of interrupt associated with the address written (NS-int) and the **security** of the bus transaction (NS-prot). Figure 26-1 shows how these differences affect different types of access.

Figure 26-1 Summary of the secure and non-secure views of interrupt priority

WCSS_A_QGIC2_GICD_IPRIORITYRn

Bits	Name	Description
31:27	INT3	
23:19	INT2	
15:11	INT1	

WCSS_A_QGIC2_GICD_IPRIORITYRn (cont.)

Bits	Name	Description
7:3	INT0	All bits cleared (0x0) is the highest priority. All bits set (5'b11111) is the lowest priority. For non-secure write access, the MSB is always set (1). For non-secure read access, the MSB is always clear (0). Non-secure entities can't use the highest priorities without secure software setting such a priority. The highest priorities are typically reserved for secure software. On the other hand, secure software can, if needed, write and read the MSB. Secure entities have access to all priorities. Secure interrupts can use priority numbers 5'b10000-5'b11111 (the lower half of the priority spectrum) if needed. Due to strict 'less than' comparisons used in the CPU interface, setting an interrupt's priority to 5'b11111 effectively disables that interrupt. The number of priorities actually supported is 32. Software can determine this number by writing 0xFF and reading back the result.

**0x0A20A800+WCSS_A_QGIC2_GICD_ITARGETSRn, n=[0..71]
 0x4*n**
Type: RW**Clock:** CLK**Reset State:** 0x00000000**Security Treatment:** NS-int Dependant

Each field in the GICD_ITARGETSR registers controls the destination CPU(s) of an interrupt.

NOTE The GICD_ITARGETSR registers at address offsets 0x0800 to 0x81C are not actually implemented. (PPI interrupts, being private, only have one destination CPU. SGI interrupts are generated by the GICD_SGIR register only and do not need the GICD_ITARGETSR.) The remaining registers control the SPI interrupt destinations.

NOTE This register is byte addressable.

WCSS_A_QGIC2_GICD_ITARGETSRn

Bits	Name	Description
31:24	INT3	
23:16	INT2	
15:8	INT1	

WCSS_A_QGIC2_GICD_ITARGETSRn (cont.)

Bits	Name	Description
7:0	INT0	Each bit represents one of 8 CPUs. Bit 0 represents the zeroth CPU, bit 7 represents the seventh CPU. Setting this field to 0 will disable the interrupt as no CPU will see it set. For INTIDs < 32 (the SGI and PPI), reads return the bit corresponding to the CPU performing the read (i.e. CPU 0 reads back 0x01010101). Writes are ignored. The number of CPUs actually supported is determined by NUM_CPU generic. Software can determine this number by writing 0xFF and reading back the result.

**0x0A20AC00 WCSS_A_QGIC2_GICD_ICFGRn, n=[0..17]
+0x4*n****Type:** RW**Clock:** CLK**Reset State:** 0x55555555**Security Treatment:** NS-int Dependant

Each field in the GICD_ICFGR registers allows:

- control of type of an SPI:
- level-sensitive
- edge-sensitive
- reading the type of PPIs and SGIs

NOTE The GICD_ICFGR registers at address offsets 0x0C00 to 0xC04 are repeated once per CPU Interface. These are only accessible from the designated CPU and allow read access for that CPU's PPI and SGI interrupts. The remaining registers control the SPI interrupts.

NOTE PPI(0) is the maintenance interrupt and must be configured as level type.

NOTE This register is not byte addressable, interrupts should be disabled before configuring their type.

NOTE Reset state for GICD_ICFGR0 is 0xaaaaaaaa.

WCSS_A_QGIC2_GICD_ICFGRn

Bits	Name	Description
31:30	INT15	
29:28	INT14	
27:26	INT13	
25:24	INT12	
23:22	INT11	

WCSS_A_QGIC2_GICD_ICFGRn (cont.)

Bits	Name	Description
21:20	INT10	
19:18	INT9	
17:16	INT8	
15:14	INT7	
13:12	INT6	
11:10	INT5	
9:8	INT4	
7:6	INT3	
5:4	INT2	
3:2	INT1	
1:0	INT0	0x0: LVL_N_TO_N (LVL N to N) 0x1: LVL_1_TO_N (LVL 1 to N) 0x2: EDGE_N_TO_N (EDGE N to N) 0x3: EDGE_1_TO_N (EDGE 1 to N) These bits behave differently for each of the three interrupt types as follows: - For SGI, read back as 0b10 - SGI interrupts are edge-sensitive and use the N-N model. - For PPI/SPI, read/writable as 0b01 or 0b11 - PPI/SPI interrupts can be edge or level sensitive and use the 1-N model.)

0x0A20AF00 WCSS_A_QGIC2_GICD_SGIR**Type:** W**Clock:** CLK**Reset State:** 0x00000000**Security Treatment:** Banked

The GICD_SGIR register provides the method by which software may cause interrupts to become pending for designated target CPU(s). The QGIC2 maintains an internal database of all outstanding software interrupt requests pended to each CPU Interface and rotates through them in Round Robin fashion. Signaling the same interrupt a second time before it is serviced will result in only one interrupt from the source CPU to the target CPU(s).

WCSS_A_QGIC2_GICD_SGIR

Bits	Name	Description
31:26	RESERVED_BITS31_TO_26	

WCSS_A_QGIC2_GICD_SGIR (cont.)

Bits	Name	Description
25:24	T_FILTER	Target Filter 0x0: LIST (LIST (Use the T_LIST as is)) 0x1: OTHERS (OTHERS (Send to all CPUs accept the CPU MID making the request, ignoring T_LIST)) 0x2: MID (MID (Send to only the CPU MID making the request, ignoring T_LIST)) 0x3: NA (NA (Reserved))
23:16	T_LIST	Target List. Each bit set represents a CPU target for the INT_ID, subject to the T_FILTER field. 0x1: CPU0 (CPU0) 0x2: CPU1 (CPU1) 0x4: CPU2 (CPU2) 0x8: CPU3 (CPU3) 0x10: CPU4 (CPU4) 0x20: CPU5 (CPU5) 0x40: CPU6 (CPU6) 0x80: CPU7 (CPU7)
15	SATT	Security ATtribute. This bit is only programmable by secure Software. 0x0: SECURE (SECURE (Secure interrupt is issued)) 0x1: NONSECURE (NONSECURE (Non-secure interrupt is issued).)
14:4	RESERVED_BITS14_TO_4	
3:0	INT_ID	The INT ID number (0-15) of the SGI to be set as pending. See Table 26-3 SGI pending truth table for security (SATT vs. NS-prot) on page 14 for information on the conditions that allow an SGI to be set pending.

**0x0A20AF10+WCSS_A_QGIC2_GICD_CPENDSGIRn, n=[0..3]
0x4*n****Type:** RW**Clock:** CLK**Reset State:** 0x00000000**Security Treatment:** Banked

Each field in the GICD_CPENDSGIR register, when written as 1, clears the corresponding pending bit of the source CPU for the corresponding SGI.

NOTE If the SGI is marked secure, only a secure access can access that SGI's pending bits.

NOTE This register is byte addressable.

WCSS_A_QGIC2_GICD_CPENDSGIRn

Bits	Name	Description
31:24	SGI3	
23:16	SGI2	
15:8	SGI1	
7:0	SGI0	Each bit represents one of 8 CPUs. Bit 0 represents the zeroth CPU, bit 7 represents the seventh CPU. Writing a 1 to any bit clears the pending status bit from the CPU to the SGI.

**0x0A20AF20+WCSS_A_QGIC2_GICD_SPENDSGIRn, n=[0..3]
0x4*n****Type:** RW**Clock:** CLK**Reset State:** 0x00000000**Security Treatment:** Banked

Each field in the GICD_SPENDSGIR register, when written as 1, sets the corresponding pending bit of the source CPU for the corresponding SGI.

NOTE If the SGI is marked secure, only a secure access can access that SGI's pending bits.

NOTE This register is byte addressable.

WCSS_A_QGIC2_GICD_SPENDSGIRn

Bits	Name	Description
31:24	SGI3	
23:16	SGI2	
15:8	SGI1	
7:0	SGI0	Each bit represents one of 8 CPUs. Bit 0 represents the zeroth CPU, bit 7 represents the seventh CPU. Writing a 1 to any bit sets the pending status bit from the CPU to the SGI.

0x0A20AFD0 WCSS_A_QGIC2_GICD_PIDR0**Type:** R**Clock:** CLK**Reset State:** 0x00000090**Security Treatment:** Common

The GICD_PIDR0 provides access to QGIC2 peripheral identification.

WCSS_A_QGIC2_GICD_PIDR0

Bits	Name	Description
7:0	PART_NUM	Returns lower byte of the QGIC2 part number 0xB390, 0x90

0x0A20AFD4 WCSS_A_QGIC2_GICD_PIDR1**Type:** R**Clock:** CLK**Reset State:** 0x00000003**Security Treatment:** Common

The GICD_PIDR1 provides access to QGIC2 peripheral identification.

WCSS_A_QGIC2_GICD_PIDR1

Bits	Name	Description
7:4	DESIGNER	Returns Qualcomm JEDEC JEP code bits 3:0, 0x0.
3:0	PART_NUM	Returns upper nibble of the QGIC2 part number 0x390, 0x3

0x0A20AFD8 WCSS_A_QGIC2_GICD_PIDR2**Type:** R**Clock:** CLK**Reset State:** 0x0000002F**Security Treatment:** Common

The GICD_PIDR2 provides access to QGIC2 peripheral identification.

WCSS_A_QGIC2_GICD_PIDR2

Bits	Name	Description
7:4	ARCH_VERSION	Returns the QGIC2's architecture revision number, 0x2
3	USES_JEP_CODE	Reads 0x1.
2:0	DESIGNER	Returns Qualcomm JEDEC JEP code bits 6:4, 0x7.

0x0A20AFDC WCSS_A_QGIC2_GICD_PIDR3**Type:** R**Clock:** CLK**Reset State:** 0x00000000**Security Treatment:** Common

The GICD_PIDR3 provides access to QGIC2 peripheral identification.

WCSS_A_QGIC2_GICD_PIDR3

Bits	Name	Description
7:4	REVISION	Returns 0x0.

0x0A20AFE0 WCSS_A_QGIC2_GICD_PIDR4

Type: R

Clock: CLK

Reset State: 0x00000000

Security Treatment: Common

The GICD_PIDR4 provides access to QGIC2 peripheral identification.

WCSS_A_QGIC2_GICD_PIDR4

Bits	Name	Description
3:0	DESIGNER	Qualcomm JEDEC JEP code bits 10:7, 0x0

0x0A20AFE4 WCSS_A_QGIC2_GICD_PIDR5

Type: R

Clock: CLK

Reset State: 0x00000000

Security Treatment: Common

The GICD_PIDR5 provides access to QGIC2 peripheral identification.

WCSS_A_QGIC2_GICD_PIDR5

Bits	Name	Description
7:0	RESRVD_BY_ARM	Reserved in the specification for future use.

0x0A20AFE8 WCSS_A_QGIC2_GICD_PIDR6

Type: R

Clock: CLK

Reset State: 0x00000000

Security Treatment: Common

The GICD_PIDR6 provides access to QGIC2 peripheral identification.

WCSS_A_QGIC2_GICD_PIDR6

Bits	Name	Description
7:0	RESRVD_BY_ARM	Reserved in the specification for future use.

0x0A20AFEC WCSS_A_QGIC2_GICD_PIDR7**Type:** R**Clock:** CLK**Reset State:** 0x00000000**Security Treatment:** Common

The GICD_PIDR7 provides access to QGIC2 peripheral identification.

WCSS_A_QGIC2_GICD_PIDR7

Bits	Name	Description
7:0	RESRVD_BY_ARM	Reserved in the specification for future use.

0x0A20AFF0 WCSS_A_QGIC2_GICD_CIDR0**Type:** R**Clock:** CLK**Reset State:** 0x0000000D**Security Treatment:** Common

The GICD_CIDR0 provides access to QGIC2 component identification.

WCSS_A_QGIC2_GICD_CIDR0

Bits	Name	Description
7:0	COMP_ID_0	Reads back as 0x0D.

0x0A20AFF4 WCSS_A_QGIC2_GICD_CIDR1**Type:** R**Clock:** CLK**Reset State:** 0x000000F0**Security Treatment:** Common

The GICD_CIDR1 provides access to QGIC2 component identification.

WCSS_A_QGIC2_GICD_CIDR1

Bits	Name	Description
7:0	COMP_ID_1	Reads back as 0xF0.

0x0A20AFF8 WCSS_A_QGIC2_GICD_CIDR2**Type:** R**Clock:** CLK**Reset State:** 0x00000005**Security Treatment:** Common

The GICD_CIDR2 provides access to QGIC2 component identification.

WCSS_A_QGIC2_GICD_CIDR2

Bits	Name	Description
7:0	COMP_ID_2	Reads back as 0x05.

0x0A20AFFC WCSS_A_QGIC2_GICD_CIDR3**Type:** R**Clock:** CLK**Reset State:** 0x000000B1**Security Treatment:** Common

The GICD_CIDR3 provides access to QGIC2 component identification.

WCSS_A_QGIC2_GICD_CIDR3

Bits	Name	Description
7:0	COMP_ID_3	Reads back as 0xB1.

0x0A20B000 WCSS_A_QGIC2_GICH_HCR**Type:** RW**Clock:** CLK**Reset State:** 0x00000000**Security Treatment:** Common

The GICH_HCR configures the QGIC2 Hypervisor interface.

WCSS_A_QGIC2_GICH_HCR

Bits	Name	Description
31:27	EOICOUNT	Counts the number of EOIs which have been received without a corresponding entry being found among the list registers. Incremented automatically by the CPU Interface when such an EOI is received. EOIs which do not clear a bit in the Active Priorities register do not cause an increment. The maintenance interrupt is asserted whenever this field is non-zero and the SKIDIE bit is set (see below).
7	VDG1IE	If this bit is set, the maintenance interrupt will be asserted whenever the GICV_CTLR[EnableG1] is cleared. 0x0: CLR (CLR) 0x1: SET (SET)
6	VEG1IE	If this bit is set, the maintenance interrupt will be asserted whenever the GICV_CTLR[EnableG1] is set. 0x0: CLR (CLR) 0x1: SET (SET)
5	VDG0IE	If this bit is set, the maintenance interrupt will be asserted whenever the GICV_CTLR[EnableG0] is cleared. 0x0: CLR (CLR) 0x1: SET (SET)
4	VEG0IE	If this bit is set, the maintenance interrupt will be asserted whenever the GICV_CTLR[EnableG0] is set. 0x0: CLR (CLR) 0x1: SET (SET)
3	NPIE	No Pending Interrupt Maintenance Interrupt Enable. If this bit is set, the maintenance interrupt will be asserted whenever no list registers contain an interrupt in PENDING state. 0x0: CLR (CLR) 0x1: SET (SET)
2	SKIDIE	SKID Maintenance Interrupt Enable. If this bit is set, the maintenance interrupt will be asserted whenever GICH_HCR[EOICount] is non-zero. 0x0: CLR (CLR) 0x1: SET (SET)
1	UIE	Underflow Maintenance Interrupt Enable. If this bit is set, the maintenance interrupt will be asserted whenever zero or one of the list entries are marked as valid. 0x0: CLR (CLR) 0x1: SET (SET)
0	EN	Enable or Disable maintenance and by proxy Virtual CPU interface Interrupts 0x0: CLR (CLR) 0x1: SET (SET)

0x0A20B004 WCSS_A_QGIC2_GICH_VTR**Type:** R**Clock:** CLK**Reset State:** 0x90000003**Security Treatment:** Common

The GICH_VTR provides information about the configuration of the Hypervisor Interface (virtual GIC type register).

WCSS_A_QGIC2_GICH_VTR

Bits	Name	Description
31:29	PRIBITS	Indicates the number of priority bits implemented, minus one. For this version of the specification the only acceptable value is 100, indicating 5 bits of priority (32 levels)
28:26	PREBITS	Indicates the number of preemption bits implemented, minus one. For this version of the specification the only acceptable value is 100, indicating 5 bits of preemption (32 levels)
5:0	LISTREGS	Indicates the number of implemented List Registers, minus one. QGIC2 has 4 list registers. This is set to 0x3

0x0A20B008 WCSS_A_QGIC2_GICH_VMCR**Type:** RW**Clock:** CLK**Reset State:** 0x004C0000**Security Treatment:** Common

The GICH_VMCR contains read/write aliases of all the Virtual Machine accessible state (aliases from the GICV_* registers). This alias allows the Hypervisor to easily save and restore this state with a single read or write, and without needing to map in the Virtual Machine registers.

Note: Various bits of this register include the terms group 0 or group 1 in their names. This is purely for compatibility with the physical GIC interface and does not pertain to actual TrustZone security.

WCSS_A_QGIC2_GICH_VMCR

Bits	Name	Description
31:27	VMPMR	Alias of the GICV_PMR bits 7:3 in GICV
23:21	VMG0BP	Alias of the group0 GICV_BPR bit in GICV
20:18	VMG1BP	Alias of the group1 GICV_ABPR bit in GICV
9	VEM	Alias of the GICV_CTLR[EOIMode] bit in GICV
4	VMGBPR	Alias of the GICV_CTLR[GBPR] bit in GICV

WCSS_A_QGIC2_GICH_VMCR (cont.)

Bits	Name	Description
3	VMFIQEN	Alias of the GICV_CTLR[FIQen] bit in QGICV.
2	VMACKCTL	Alias of the GICV_CTLR[ActCt] bit in QGICV.
1	VMENABLE_G1	Alias of the GICV_CTLR[ENABLE_G1] bit in QGICV.
0	VMENABLE_G0	Alias of the GICV_CTLR[ENABLE_G0] bit in QGICV.

0x0A20B010 WCSS_A_QGIC2_GICH_MISR**Type:** R**Clock:** CLK**Reset State:** 0x00000000**Security Treatment:** Common

The GICH_MISR is the Maintenance interrupt syndrome register in the Hypervisor Interface. This register indicates which Maintenance Interrupts are currently asserted. This takes into account the individual Maintenance Interrupt Enable bits in GICH_HCR. However, the global enable in GICH_HCR (EN) is not factored in to these bits. Therefore the maintenance interrupt is asserted if and only if at least one bit is set in this register and the EN bit in GICH_HCR is set.

WCSS_A_QGIC2_GICH_MISR

Bits	Name	Description
7	VDNSI	
6	VENSI	
5	VDSI	
4	VESI	
3	NPI	
2	SKIDI	
1	UI	
0	EI	

0x0A20B020 WCSS_A_QGIC2_GICH_EISR**Type:** R**Clock:** CLK**Reset State:** 0x00000000**Security Treatment:** Common

The GICH_EISR provides EOI interrupt status in the Hypervisor Interface - which Software interrupts in the LRn registers that have EI set and are currently in the Invalid state.

WCSS_A_QGIC2_GICH_EISR

Bits	Name	Description
3:0	LR	0x0: NOT_EOI (NOT_EOI) 0x1: EOI (EOI)

0x0A20B030 WCSS_A_QGIC2_GICH_ELRSR**Type:** R**Clock:** CLK**Reset State:** 0x0000000F**Security Treatment:** Common

The GICH_ELRSR provides Empty List Register Status in the Hypervisor Interface.

WCSS_A_QGIC2_GICH_ELRSR

Bits	Name	Description
3:0	LR	0x0: NOT_EMPTY (NOT_EMPTY) 0x1: EMPTY (EMPTY)

0x0A20B0F0 WCSS_A_QGIC2_GICH_APR**Type:** RW**Clock:** CLK**Reset State:** 0x00000000**Security Treatment:** Common

The GICH_APR provides information about the priorities currently active in the Hypervisor Interface.

WCSS_A_QGIC2_GICH_APR

Bits	Name	Description
31:0	PRI	0x0: INACTIVE (INACTIVE) 0x1: ACTIVE (ACTIVE)

**0x0A20B100+WCSS_A_QGIC2_GICH_LRn, n=[0..3]
0x4*n****Type:** RW**Clock:** CLK**Reset State:** 0x00000000**Security Treatment:** Common

The GICH_LRn provides access to each of the list registers in the Hypervisor Interface.

WCSS_A_QGIC2_GICH_LRn

Bits	Name	Description
31	HW	Indicates that this is a Hardware interrupt. When this bit is set, bits 19:10 contain the Physical interrupt number, and when this interrupt is subject to an EOI request, a Deactivate Interrupt request will be sent to the Interrupt Distributor.
30	GRP	Indicates that this interrupt should be treated as Group1. This means that the interrupt will present as an IRQ if GICV_CTLR[EnableG1] bit is set. If the GRP bit is clear, the interrupt is put into Group0 and will present as either IRQ or FIQ depending on the setting of the GICV_CTLR[FIQen] but only if GICV_CTLR[EnableG0] bit is also set. This bit is also used with GICV_CTLR[CBPR] to determine which Binary Point register is used in GICV for preemption calculations.
29:28	STATE	0x0: IDL (IDL (INVALID)) 0x1: PNA (PNA (PENDING, NOT ACTIVE)) 0x2: A (A (ACTIVE)) 0x3: PA (PA (PENDING and ACTIVE)) NOTE: for hardware interrupts, the PENDING+ACTIVE state is held in the physical Distributor rather than the VCPUIF. The PENDING+ACTIVE state (11) should only be used for software originated interrupts (Virtual Devices) or SGIs.)
27:23	PRI	Indicates the priority of this interrupt. This is converted to a traditional 8 bit QGIC priority value by appending three zero bits.
19:10	PHY_ID	The physical ID of the interrupt (which SGI/PPI/SPI is it?). When the HW bit is set this field indicates the physical interrupt ID, which is forwarded to the Distributor when this interrupt is subject to an EOI request. See below for details when the HW bit is clear. If the Physical ID is between 16 and 31 inclusive (PPI space), the Deactivation will apply to the PPI associated with the same physical CPU as the VCPUIF making the request. If the Physical ID is between 0 and 15 inclusive (SGI space), the Deactivation will still apply to SGI associated with the same physical CPU as the VCPUIF making the request but it is expected that the HW bit and Physical ID will NOT be used for SGIs.
9:0	VIRTL_ID	Indicates the Virtual ID. This ID is returned to the Guest OS when this interrupt is acknowledged via the GICV_IAR (Interrupt Acknowledge register).

0x0A20B120 WCSS_A_QGIC2_GICH_SW_LR**Type:** RW**Clock:** CLK**Reset State:** 0x00000000**Security Treatment:** Common

This register is not a real physical register or address in the design. It is placed here in the Software Manual to describe the alternate meanings for bits 19:10 in the LRn registers. Adding it here allows SW to pick up defines for these alternate meanings and use them in LRn accesses.

The HW uses these alternate definitions when bit 31 (HW) is set to 0 - indicating a (a pure virtual interrupt) written directly to this Hypervisor interface or an interrupt generated due to an SGI that originated in the Distributor. Bits 31:20 and 9:0 retain the normal GICH_LRn meanings.

WCSS_A_QGIC2_GICH_SW_LR

Bits	Name	Description
19	EI	Indicates that this interrupt should trigger an EOI Interrupt. When this bit is set, a maintenance interrupt will be asserted whenever this interrupt becomes INVALID.
12:10	CPUID	If this interrupt is an SGI, indicates the requesting CPU ID. This will appear in the relevant field of the VM Interrupt Acknowledge register. If the interrupt is not an SGI, software must set this field to 0.

0x0A20C000 WCSS_A_QGIC2_GICC_CTLR**Type:** RW**Clock:** CLK**Reset State:** 0x00000000**Security Treatment:** Banked

The GICC_CTLR configures the QGIC2 CPU interface.

Non-secure access Access is granted only to bits 0, 5,6 and 9.

Secure access Access is granted to all register bits. Non-secure bits 0,5,6, and 9 are aliased into bits 1, 7, 8, and 10 for access by secure software.Modifying these aliased bits will modify the non-secure settings. Bits 0, 5, 6, 9 are physically banked into Secure and Non-secure space.

WCSS_A_QGIC2_GICC_CTLR

Bits	Name	Description
10	EOIMODENS	For secure software, a read/writable alias of Non-Secure EOImode.

WCSS_A_QGIC2_GICC_CTLR (cont.)

Bits	Name	Description
9	EOIMODE	For secure software, controls the behavior of secure access to GICC_EOI and GICC_DIR. For non-secure software, controls the behavior of non-secure access to GICC_EOI and GICC_DIR. 0x0: PD_AND_DI (PD_AND_DI (Both Priority Drop and Deactivate Interrupt, GICC_DIR does not function)) 0x1: PD (PD(GICC_EOI performs Priority Drop, GICC_DIR performs deactivate interrupt function))
8	IRQBYPDISABLENS	For secure software, a read/writable alias of Non-secure version of IRQBypDisable.
7	FIQBYPDISABLENS	For secure software, a read/writable alias of Non-secure version of FIQBypDisable
6	IRQBYPDISABLE	For Non-secure software, when CPU Interface is disabled, prevents the legacy bypass IRQ signal from being forwarded to the CPU (if IRQ is under Non-secure control). For secure software, prevents bypass if FIQ is under secure control. 0x0: CLR (CLR) 0x1: SET (SET)
5	FIQBYPDISABLE	For Non-secure software, when CPU Interface is disabled, prevents the legacy bypass FIQ signal from being forwarded to the CPU (if FIQ is under Non-secure control). For secure software, prevents bypass if FIQ is under secure control. 0x0: CLR (CLR) 0x1: SET (SET)
4	SBPR	Controls which binary point register is used to calculate preemption. 0x0: BANKED (BANKED (secure interrupts use the secure Binary Point Register, non-secure interrupts use the non-secure Binary Point Register)) 0x1: RESTRICTED (RESTRICTED (all interrupts use the secure Binary Point Register))
3	S_DEST	Controls destination of secure interrupts. 0x0: IRQ (IRQ) 0x1: FIQ (FIQ)

WCSS_A_QGIC2_GICC_CTLR (cont.)

Bits	Name	Description
2	ACKCTL	Controls the side effect behavior of a secure read request to the Interrupt Acknowledge Register in the case where the highest priority pending interrupt is non-secure. If AckCtl is set to 0, a secure read request to the Interrupt Acknowledge Register returns an Interrupt ID value of 1022, and the read request does not cause the interrupt to be acknowledged (that is the Pending status of the interrupt remains unchanged). If AckCtl is set to 1, a secure read request to the Interrupt Acknowledge Register returns the Interrupt ID value of the non-secure interrupt, and causes the interrupt to be acknowledged (that is the interrupt becomes Active, or Active and Pending). 0x0: DISABLE_ACK_OF_NS_PENDING (DISABLE ACK OF NS PENDING) 0x1: ENABLE_ACK_OF_NS_PENDING (ENABLE ACK OF NS PENDING)
1	ENABLE_NS	For secure software, a read/writable alias of non secure software's enable bit (bit 0 of this same register in non secure space). 0x0: CLR (CLR) 0x1: SET (SET)
0	ENABLE	For non-secure software, this bit enables/disables non secure interrupts. For secure software, this bit enables/disables secure interrupts. 0x0: CLR (CLR) 0x1: SET (SET)

0x0A20C004 WCSS_A_QGIC2_GICC_PMR**Type:** RW**Clock:** CLK**Reset State:** 0x00000000**Security Treatment:** Banked

The GICC_PMR register configures the QGIC2 CPU interface Priority Mask. The Priority Mask can be used to limit the interrupts that can cause an interrupt request to the CPU based on priority levels.

Non-secure access Write access is granted only if bit 7 is 0x1, writes are ignored otherwise. Reads are always granted, however, bit 7 is always set in the data returned. Bit 0 is always zero.

Secure access Access is granted to all register bits.

WCSS_A_QGIC2_GICC_PMR

Bits	Name	Description
7:3	LEVEL	Set the Priority Mask Level. The CPU interface asserts an interrupt request to CPU if the priority of the highest Pending interrupt sent by the interrupt Distributor is strictly higher than at least the mask set in Priority Mask Register.

0x0A20C008 WCSS_A_QGIC2_GICC_BPR**Type:** RW**Clock:** CLK**Reset State:** 0x00000002**Security Treatment:** Banked

The GICC_BPR register configures the QGIC2 CPU interface Binary Point. The Binary Point is used to limit interrupts that can cause an interrupt request to the CPU based on the interrupt's priority and the interrupt priority that currently being serviced by the CPU (if there is one). If the CPU is not servicing an interrupt, the Binary Point register is not used.

Non-secure access Provides access to the non-secure Binary Point register.

Secure access Provides access to the secure Binary Point register. Secure software may access the non secure Binary Point register at offset 0x001C

WCSS_A_QGIC2_GICC_BPR

Bits	Name	Description
2:0	VAL	The VAL setting is used to determine the priority bits used for preemption according to Table 26-6.

0x0A20C00C WCSS_A_QGIC2_GICC_IAR**Type:** R**Clock:** CLK**Reset State:** 0x000003FF**Security Treatment:** Banked

The GICC_IAR register is used by the CPU to obtain the ID of the interrupt which caused the assertion if IRQn or FIQn.

Performing this read has the side effect of causing the Distributor to change the interrupt from the Pending state to the Active or Active and Pending state according the description below:

Using three bits defined by AHB bus **security** state (1=non-secure), **security** of the HPPI(1=non-secure) and the value of GICC_CTLR[AckCtl]:

00x - respond with the true interrupt number, transitioning to active (secure request to secure interrupt, ACKCTL is don't care)

010 - respond with 1022 (secure request to non-secure interrupt but ACKCTL=0 disabling secure handling of non-secure interrupts)

011 - respond with the true interrupt number, transitioning to active (secure request to non-secure interrupt with ACKCTL=1)

10x - respond with 1023 (non-secure request to secure interrupt)

11x - respond with the true interrupt number, transitioning to active (non-secure request to non-secure interrupt)

WCSS_A_QGIC2_GICC_IAR

Bits	Name	Description
12:10	CPU_ID	CPU Source ID value is only valid if the interrupt is a Software Interrupt, in which case it contains the ID of the CPU which requested the Interrupt. For PPIs and SPIs the CPU source ID field is read as zero and should be ignored.
9:0	INT_ID	Interrupt Identifier. The value returned in this field is dependent on the security state of the access to the register, the security state of any outstanding interrupts, the value of GICC_CTLR[AckCtl] as defined in the description above.

0x0A20C010 WCSS_A_QGIC2_GICC_EOIR

Type: W

Clock: CLK

Reset State: 0x00000000

Security Treatment: Banked

Unlike QGIC1, GICC_EOIR (End Of Interrupt) register has two possible behaviors. By default it acts as it did in QGIC1, that is, it can be written to indicate when software has finished handling an interrupt.

In QGIC2, if GICC_CTLR[EOImode] is set (1), writing this register has the 'priority drop' (PD) effect. The PD effect is when the active priority bit for the highest priority active interrupt is cleared. The key difference, when EOImode is one, is that the Distributor is not notified that this happened - that is, the interrupt itself does not become inactive.

When EOImode is zero, writing to this register will, in addition to the PD effect described above, set the interrupt to Inactive or Pending (if prior to writing to this register, the interrupt was both Active and Pending it becomes just pending) in the Distributor.

In either case, the value written must be the interrupt ID, and the CPU Source ID for SGIs, of the interrupt that is being completed.

The **security** status of the write to this register (NS-prot) must match the **security** status (NS-int) of the interrupt ID, according to Table 26-7 for the interrupt to be successfully cleared.

WCSS_A_QGIC2_GICC_EOIR

Bits	Name	Description
12:10	CPU_ID	CPU Source ID value is only valid if the interrupt is a Software Interrupt, in which case it must match the ID of the CPU which requested the Interrupt. As described in Table 26-7, the CPU Source ID must match an active interrupt in order for the GICC_EOIR to take effect. The CPU_ID is written must match that delivered to software when GICC_IAR is read. For PPIs and SPIs the CPU source ID field is ignored.
9:0	INT_ID	Interrupt Identifier. The value must match the interrupt ID that software received when reading the GICC_IAR register. If the security settings match (see Table 26-7) a write will cause the priority drop effect and if EOImode=0, the Active state in the Distributor for the corresponding interrupt will also be cleared.

0x0A20C014 WCSS_A_QGIC2_GICC_RPR

Type: R

Clock: CLK

Reset State: 0x000000F8

Security Treatment: Banked

The GICC_RPR (Running Priority) register provides access to the highest priority of all the Active interrupts on this CPU. The priority value returned is sensitive to the **security** status (NS-int) of the interrupt currently running and the NS-prot

Non-secure access If the currently interrupt is secure (NS-int = 0), and a non-secure read of the Running Priority register is attempted (NS-prot = 1), then the value 0 (maximum priority) is returned.

Secure access Returns the true value in the register.

WCSS_A_QGIC2_GICC_RPR

Bits	Name	Description
7:3	VAL	Running Priority

0x0A20C018 WCSS_A_QGIC2_GICC_HPIR

Type: R

Clock: CLK

Reset State: 0x000003FF

Security Treatment: Banked

The GICC_HPPIR register provides access to the highest priority pending interrupt to the CPU Interface.

NOTE Interrupts that are Active and Pending in the Distributor are not considered candidates to become the highest priority pending interrupt.

Using three bits defined by AHB bus **security** state (1=non-secure), **security** of the HPPI(1=non-secure) and the value of GICC_CTLR[AckCtl]:

00x - respond with the true interrupt number (secure request to secure interrupt, ACKCTL is don't care)

010 - respond with 1022 (secure request to non-secure interrupt but ACKCTL=0 disabling secure handling of non-secure interrupts)

011 - respond with the true interrupt number (secure request to non-secure interrupt with ACKCTL=1)

10x - respond with 1023 (non-secure request to secure interrupt)

11x - respond with the true interrupt number (non-secure request to non-secure interrupt).

WCSS_A_QGIC2_GICC_HPPIR

Bits	Name	Description
12:10	CPU_ID	CPU Source ID value is only valid if the interrupt is a Software Interrupt, in which case it contains the ID of the CPU which requested the Interrupt. For PPIs and SPIs the CPU source ID field is read as zero and should be ignored.
9:0	INT_ID	Interrupt Identifier. The value returned in this field is dependent on the security state of the access to the register, and the security state of any outstanding interrupts, as described above.

0x0A20C01C WCSS_A_QGIC2_GICC_ABPR

Type: RW

Clock: CLK

Reset State: 0x00000003

Security Treatment: Restricted

The GICC_ABPR register provides a copy of the non-secure binary point register for use by secure software. See the definition of the GICC_BPR register for details on the use of this register.

Non-secure access No access.

Secure access Access granted, updates are reflected to non-secure software at the GICC_BPR register at address offset 0x08.

WCSS_A_QGIC2_GICC_ABPR

Bits	Name	Description
2:0	VAL	Same as the GICC_BPR register - affects only non-secure interrupts.

0x0A20C020 WCSS_A_QGIC2_GICC_AIAR**Type:** R**Clock:** CLK**Reset State:** 0x000003FF**Security Treatment:** Restricted

The GICC_AIAR register is used by the CPU to obtain the ID of the interrupt which caused the assertion if IRQn or FIQn. This is a secure only register used to read non-secure interrupts by Secure Software.

Performing this read has the side effect of causing the Distributor to change the interrupt from the Pending state to the Active or Active and Pending state.

Non-secure access No access. Read as zero.

Secure access See description below.

Based on the 2 bits defined by the AHB **Security** of the request (1=non-secure) and the **security** of the interrupt (1=non-secure):

00 - respond with 1023 (secure request that behaves as if it is a non-secure request, to secure interrupt)

01 - respond with the true interrupt number, transitioning to active (secure request that behaves as if it is a non-secure request, to non-secure interrupt)

1x - RAZ/WI (non-secure request cannot access this register)

WCSS_A_QGIC2_GICC_AIAR

Bits	Name	Description
12:10	CPU_ID	CPU Source ID value is only valid if the interrupt is a Software Interrupt, in which case it contains the ID of the CPU which requested the Interrupt. For PPIs and SPIs the CPU source ID field is read as zero and should be ignored.
9:0	INT_ID	Interrupt Identifier.

0x0A20C024 WCSS_A_QGIC2_GICC_AEOIR**Type:** W**Clock:** CLK**Reset State:** 0x00000000**Security Treatment:** Restricted

A secure write to this register has an identical effect to a non-secure write to GICC_EOIR, with the exception of the EOImode bit; the value of the secure EOImode bit (bit 9) in GICC_CTLR is used to determine whether the interrupt is deactivated or just priority dropped rather than the non-secure EOImode bit (bit 10).

This register must be used by secure software to signal completion of non-secure interrupts when GICC_CTLR AckCtl bit is set to 0; GICC_EOIR cannot be used for this purpose since it would affect GICC_APRn rather than GICC_NSAPRn.

Non-secure access ignored

Secure access signal completion of non-secure interrupt

WCSS_A_QGIC2_GICC_AEOIR

Bits	Name	Description
12:10	CPU_ID	CPU Source ID value is only valid if the interrupt is a Software Interrupt, in which case it must match the ID of the CPU which requested the Interrupt. The CPU Source ID must match an active interrupt in order for the GICC_AEOIR to take effect. The CPU_ID is written must match that delivered to software when GICC_IAR is read. For PPIs and SPIs the CPU source ID field is ignored.
9:0	INT_ID	Interrupt Identifier. The value must match the interrupt ID that software received when reading the GICC_IAR register. Interrupt Identifier. A write will cause the priority drop effect and if EOImode=0, the Active state in the Distributor for the corresponding interrupt will also be cleared.

0x0A20C028 WCSS_A_QGIC2_GICC_AHPPIR**Type:** R**Clock:** CLK**Reset State:** 0x000003FF**Security Treatment:** Restricted

This is a Secure-Only register.

A read from this register has an identical effect to a Non-Secure read from GICC_HPPIR. It can therefore be used to check for a pending Non-Secure interrupt without the possibility of returning a Secure Interrupt ID value.

Interrupts that are Active and Pending in the Distributor are not considered candidates to become the highest priority pending interrupt.

Non-secure access No access. Read as zero.

Secure access See description below.

NOTE

Using three bits defined by AHB bus **security** state (1=non-secure), **security** of the HPPI(1=non-secure) and the value of GICC_CTLR[AckCtl]:

00x - respond with 1023 (hppi is a secure interrupt)

01x - respond with the true interrupt number (hppi is a non-secure interrupt)

1xx - RAZ/WI (non-secure request cannot access this register).

WCSS_A_QGIC2_GICC_AHPPIR

Bits	Name	Description
12:10	CPU_ID	CPU Source ID value is only valid if the interrupt is a Software Interrupt, in which case it contains the ID of the CPU which requested the Interrupt. For PPIs and SPIs the CPU source ID field is read as zero and should be ignored.
9:0	INT_ID	Interrupt Identifier. The value returned in this field is dependent on the security state of the access to the register, and the security state of any outstanding interrupts, as described above.

0x0A20C0D0 WCSS_A_QGIC2_GICC_APR

Type: RW

Clock: CLK

Reset State: 0x00000000

Security Treatment: Banked

Each bit in the GICC_APR register provides the active status of priorities in the CPU Interface. Non-secure access Bits 15-0 represent priorities 31 to 16 of the non-secure active priorities. Secure access Bits 31-0 represent priorities 31 to 0 of the active secure priorities.

WCSS_A_QGIC2_GICC_APR

Bits	Name	Description
31:0	PRI	0x1: A (A (ACTIVE)) 0x0: NA (NA (NOT ACTIVE))

0x0A20C0E0 WCSS_A_QGIC2_GICC_NSAPR**Type:** RW**Clock:** CLK**Reset State:** 0x00000000**Security Treatment:** Restricted

Each bit in the GICC_NSAPR register provides the active status of non-secure priorities in the CPU Interface. Non-secure access None - write ignore, read-as-zero. Secure access Bits 31-0 represent priorities 31 to 0 of the active non-secure priorities.

WCSS_A_QGIC2_GICC_NSAPR

Bits	Name	Description
31:0	PRI	0x1: A (A (ACTIVE)) 0x0: NA (NA (NOT ACTIVE))

0x0A20C0FC WCSS_A_QGIC2_GICC_IIDR**Type:** R**Clock:** CLK**Reset State:** 0x00020070**Security Treatment:** Common

The GICC_IIDR register provides information about the QGIC2 CPU Interface version and device implementation information.

WCSS_A_QGIC2_GICC_IIDR

Bits	Name	Description
31:20	PART_NUM	Part Number: 0x0
19:16	ARCH_VERSION	Architecture Version : 0x2
15:12	REVISION	Revision number: 0x0
11:0	IMPLEMENTOR	Bits[11:8] contain the implementor's JEP106 continuation code, 0x0 Bit[7] is always 0 Bits[6:0] contain bits [6:0] of the implementor's JEP106 code,

0x0A20D000 WCSS_A_QGIC2_GICC_DIR**Type:** W**Clock:** CLK**Reset State:** 0x00000000**Security Treatment:** Banked

The GICC_DIR register provides the mechanism by which Software can Deactivate a specific interrupt.

Secure writes can deactivate secure and non-secure interrupts.

Non-secure writes can only deactivate non-secure interrupts.

Note that unlike GICC_EOIR and GICC_IAR, there is no restriction of the Deactivation of Non-Secure interrupts with Secure accesses. The AckCtl bit does not affect the behaviour of GICC_DIR. This is because unlike GICC_IAR there is no chance of the access affecting an interrupt from an unexpected **security** state, and unlike GICC_EOIR there is no effect on GICC_APRn or GICC_NSAPRn, so there is no need to indicate which register should be affected.

WCSS_A_QGIC2_GICC_DIR

Bits	Name	Description
12:10	CPU_ID	CPU Source ID value is only valid if the interrupt is a Software Interrupt, in which case it contains the ID of the CPU which requested the Interrupt. For PPIs and SPIs the CPU source ID field is read as zero and should be ignored.
9:0	INT_ID	Interrupt Identifier. INT_ID written should match value returned from GICC_IAR read.

0x0A20E000 WCSS_A_QGIC2_GICV_CTLR

Type: RW

Clock: CLK

Reset State: 0x00000000

Security Treatment: Common

The GICV_CTLR configures the QGIC2 Virtual CPU interface.

WCSS_A_QGIC2_GICV_CTLR

Bits	Name	Description
9	EOIMODE	Controls the behavior of Non-secure access to GICV_EOI and GICV_DIR 0x0: PD_AND_DI (PD_AND_DI (Both Priority Drop and Deactivate Interrupt, GICV_DIR does not function)) 0x1: PD (PD(GICV_EOI performs Priority Drop, GICV_DIR performs deactivate interrupt function))
4	GBPR	Controls which binary point register is used to calculate preemption, GICV_BPR or the GICV_ABPR. 0x0: BANKED (BANKED (Group 0 interrupts use the Group 0 Binary Point Register, Group 1 interrupts use the Group 1 Binary Point Register)) 0x1: G0 (G0 (all interrupts use the Group 0 Binary Point Register))

WCSS_A_QGIC2_GICV_CTLR (cont.)

Bits	Name	Description
3	FIQEN	Controls destination of secure interrupts. 0x0: IRQ (IRQ) 0x1: FIQ (FIQ)
2	ACKCTL	Controls whether reads of the VM Interrupt Acknowledge Register can acknowledge interrupts where the NS bit is set in the List Register. 0x0: NO_IAR_4G0 (NO_IAR_4G0 (On a read of GICV_IAR, the highest priority unmasked PENDING interrupt is acknowledged only if the NS bit is clear in the corresponding List Register. If the NS bit in that List Register is set, the read will return 1022 and the interrupt can only be acknowledge by a read to the VM Aliased Interrupt Acknowledge Register.)) 0x1: IAR_4G0 (IAR_4G0 (On a read of GICV_IAR, the highest priority unmasked PENDING interrupt is always acknowledged regardless of the NS bit setting in the corresponding List Register.))
1	ENABLE_G1	This bit enables Group 1 interrupts. 0x0: CLR (CLR) 0x1: SET (SET)
0	ENABLE_G0	This bit enables Group 0 interrupts. 0x0: CLR (CLR) 0x1: SET (SET)

0x0A20E004 WCSS_A_QGIC2_GICV_PMR**Type:** RW**Clock:** CLK**Reset State:** 0x00000000**Security Treatment:** Common

The GICV_PMR register configures the QGIC2 Virtual CPU interface Priority Mask. The Priority Mask can be used to limit the interrupts that can cause an interrupt request to the CPU based on priority levels.

WCSS_A_QGIC2_GICV_PMR

Bits	Name	Description
7:3	LEVEL	Set the Priority Mask Level. The CPU interface asserts an interrupt request to CPU if the priority of the highest Pending interrupt sent by the interrupt Distributor is strictly higher than at least the mask set in Priority Mask Register.

0x0A20E008 WCSS_A_QGIC2_GICV_BPR**Type:** RW**Clock:** CLK**Reset State:** 0x00000002**Security Treatment:** Common

The GICV_BPR register configures the QGIC2 Virtual CPU interface Binary Point. The Binary Point is used to limit interrupts that can cause an interrupt request to the CPU based on the interrupt's priority and the interrupt priority that currently being serviced by the CPU (if there is one). If the CPU is not servicing an interrupt, the Binary Point register is not used.

WCSS_A_QGIC2_GICV_BPR

Bits	Name	Description
2:0	VAL	The VAL setting is used to determine the priority bits used for preemption according to Table 26-6.

0x0A20E00C WCSS_A_QGIC2_GICV_IAR**Type:** R**Clock:** CLK**Reset State:** 0x000003FF**Security Treatment:** Common

The GICV_IAR register is used by the CPU to obtain the ID of the interrupt which caused the assertion if IRQn or FIQn.

Performing this read has the side effect of causing the GICH to change the interrupt from the Pending state to the Active or Active and Pending state according to the rules below:

Using the 2 bits defined by **Security** of the HPPI(1=non-secure) and the value of GICV_CTLR[AckCtl]:

0x - respond with the true interrupt number, transitioning to active (ignore APROTNS and consider as a secure request to secure interrupt, ACKCTL is don't care)

10 - respond with 1022 (ignore APROTNS and consider as secure request to non-secure interrupt but ACKCTL=0 disabling secure handling of non-secure interrupts)

11 - respond with the true interrupt number, transitioning to active (ignore APROTNS and consider as secure request to non-secure interrupt with ACKCTL=1)

WCSS_A_QGIC2_GICV_IAR

Bits	Name	Description
12:10	CPU_ID	CPU Source ID value is only valid if the interrupt is a Software Interrupt, in which case it contains the ID of the CPU which requested the Interrupt. For PPIs and SPIs the CPU source ID field is read as zero and should be ignored.
9:0	INT_ID	Interrupt Identifier - a virtual interrupt number determined by the Hypervisor SW.

0x0A20E010 WCSS_A_QGIC2_GICV_EOIR**Type:** W**Clock:** CLK**Reset State:** 0x00000000**Security Treatment:** Common

GICV_EOIR (End Of Interrupt) register has two possible behaviors. By default it can be written to to indicate when software has finished handling an interrupt.

If GICV_CTLR[EOImode] is set (1), writing this register has the 'priority drop' (PD) effect. The PD effect is when the active priority bit for the highest priority active interrupt is cleared. The key difference, when EOImode is (1), is the interrupt itself does not become inactive.

When EOImode is (0), writing to this register will, in addition to the PD effect described above, set the interrupt to Inactive or Pending (if prior to writing to this register, the interrupt was both Active and Pending it becomes just pending). However, if GICV_CTLR[EOImode] is (0), then the CPU_ID and INT_ID must match an active interrupt for a state change to occur.

In either case, the value written must be the interrupt ID, and the CPU Source ID for SGIs, of the interrupt that is being completed.

WCSS_A_QGIC2_GICV_EOIR

Bits	Name	Description
12:10	CPU_ID	CPU Source ID value is only valid if the interrupt is a Software Interrupt, in which case it must match the ID of the CPU which requested the Interrupt. The CPU_ID is written must match that delivered to software when GICV_IAR is read. For PPIs and SPIs the CPU source ID field is ignored.
9:0	INT_ID	Interrupt Identifier. The value must match the interrupt ID that software received when reading the GICV_IAR register.

0x0A20E014 WCSS_A_QGIC2_GICV_RPR**Type:** R**Clock:** CLK**Reset State:** 0x000000F8**Security Treatment:** Common

The GICV_RPR (Running Priority) register provides access to the highest priority of all the Active interrupts on this Virtual CPU.

WCSS_A_QGIC2_GICV_RPR

Bits	Name	Description
7:3	VAL	Running Priority

0x0A20E018 WCSS_A_QGIC2_GICV_HPPIR**Type:** R**Clock:** CLK**Reset State:** 0x000003FF**Security Treatment:** Common

The GICV_HPPIR register provides access to the highest priority pending interrupt to the Virtual CPU Interface. If no interrupt is pending, responds with 0x3FF.

NOTE Interrupts that are Active and Pending in the GICH (hypervisor) are not considered candidates to be come the highest priority pending interrupt.

Using 2 bits defined by the value of GICV_CTLR[AckCtl] and the NS bit of the HPPI(1=non-secure):

1x - respond with the virtual interrupt number

00 - respond with the virtual interrupt number

01 - respond with 1022

WCSS_A_QGIC2_GICV_HPPIR

Bits	Name	Description
12:10	CPU_ID	CPU Source ID value is only valid if the interrupt is a Software Interrupt, in which case it contains the ID of the CPU which requested the Interrupt. For PPIs and SPIs the CPU source ID field is read as zero and should be ignored.
9:0	INT_ID	Interrupt Identifier.

0x0A20E01C WCSS_A_QGIC2_GICV_ABPR**Type:** RW**Clock:** CLK**Reset State:** 0x00000003**Security Treatment:** Common

The GICV_ABPR register provides access to the non-secure binary point register used by non-secure interrupts when GICV_CTRL[GBPR]=0. See the definition of the GICV_BPR register for details on the use of this register.

WCSS_A_QGIC2_GICV_ABPR

Bits	Name	Description
2:0	VAL	Same as the GICV_BPR register - affects only non-secure interrupts.

0x0A20E020 WCSS_A_QGIC2_GICV_AIAR**Type:** R**Clock:** CLK**Reset State:** 0x000003FF**Security Treatment:** Common

The GICV_AIAR register is used by the CPU to obtain the ID of the interrupt which caused the assertion if IRQn or FIQn.

Performing this read has the side effect of causing the GICH to change the interrupt from the Pending state to the Active or Active and Pending state according to the rules below:

Using the **security** state of the pending interrupt:

- 0 - respond with 1023

- 1 - respond with the true interrupt number, transitioning to active

WCSS_A_QGIC2_GICV_AIAR

Bits	Name	Description
12:10	CPU_ID	CPU Source ID value is only valid if the interrupt is a Software Interrupt, in which case it contains the ID of the CPU which requested the Interrupt. For PPIs and SPIs the CPU source ID field is read as zero and should be ignored.
9:0	INT_ID	Interrupt Identifier - a virtual interrupt number determined by the Hypervisor SW.

0x0A20E024 WCSS_A_QGIC2_GICV_AEOIR**Type:** W**Clock:** CLK**Reset State:** 0x00000000**Security Treatment:** Common

This register presents the same programmers model as GICC_AEOIR.

The effect of writing to this register is identical to the effect of writing the same value to the GICV_EOIR register. This alias is provided for compatibility with the Physical CPU Interface, where the two registers have a different effect due to the provision of separate Secure and Non-Secure Active Priority Registers.

WCSS_A_QGIC2_GICV_AEOIR

Bits	Name	Description
12:10	CPU_ID	CPU Source ID value is only valid if the interrupt is a Software Interrupt, in which case it must match the ID of the CPU which requested the Interrupt. The CPU_ID is written must match that delivered to software when GICV_IAR is read. For PPIs and SPIs the CPU source ID field is ignored.
9:0	INT_ID	Interrupt Identifier. The value must match the interrupt ID that software received when reading the GICV_IAR register.

0x0A20E028 WCSS_A_QGIC2_GICV_AHPPIR**Type:** R**Clock:** CLK**Reset State:** 0x000003FF**Security Treatment:** Common

This register presents the same programmers model as GICC_AHPPIR.

This register returns the interrupt ID of the highest priority PENDING interrupt if it has an NS value of 1. Reading this interrupt has no side effects and simply returns the highest priority PENDING interrupt among the list registers or a status code according to the following table:

Using the NS bit of the HPPI(1=non-secure):

0 - 1023

1 - respond with the virtual interrupt number

Note that in some cases the returned value may not be strictly accurate if interrupts are currently paged out in software.

WCSS_A_QGIC2_GICV_AHPPIR

Bits	Name	Description
12:10	CPU_ID	CPU Source ID value is only valid if the interrupt is a Software Interrupt, in which case it contains the ID of the CPU which requested the Interrupt. For PPIs and SPIs the CPU source ID field is read as zero and should be ignored.
9:0	INT_ID	Interrupt Identifier.

0x0A20E0D0 WCSS_A_QGIC2_GICV_APR**Type:** RW**Clock:** CLK**Reset State:** 0x00000000**Security Treatment:** Common

Each bit in the GICV_APR is an alias of GICH_APR. The data and access control are one and the same.

WCSS_A_QGIC2_GICV_APR

Bits	Name	Description
31:0	PRI	0x1: A (A (ACTIVE)) 0x0: NA (NA (NOT ACTIVE))

0x0A20E0FC WCSS_A_QGIC2_GICV_IIDR**Type:** R**Clock:** CLK**Reset State:** 0x00020070**Security Treatment:** Common

The GICV_IIDR register provides information about the QGIC2 Virtual CPU Interface version and device implementor information.

WCSS_A_QGIC2_GICV_IIDR

Bits	Name	Description
31:20	PART_NUM	Part Number: 0x390
19:16	ARCH_VERSION	Architecture Version : 0x2
15:12	REVISION	Revision number: 0x0

WCSS_A_QGIC2_GICV_IIDR (cont.)

Bits	Name	Description
11:0	IMPLEMENTOR	Bits[11:8] contain the implementor's JEP106 continuation code, 0x0 Bit[7] is always 0 Bits[6:0] contain bits [6:0] of the implementor's JEP106 code,

0x0A20F000 WCSS_A_QGIC2_GICV_DIR**Type:** W**Clock:** CLK**Reset State:** 0x00000000**Security Treatment:** Banked

The GICV_DIR register provides the mechanism by which Software can Deactivate a specific interrupt.

WCSS_A_QGIC2_GICV_DIR

Bits	Name	Description
12:10	CPU_ID	CPU Source ID value is only valid if the interrupt is a Software Interrupt, in which case it contains the ID of the CPU which requested the Interrupt.
9:0	INT_ID	Interrupt Identifier. The value should match the INT_ID returned from the read of GICV_IAR.

10.17 WCSS_A_LTE_CXM_LTE_CXM**0x0A208000 WCSS_A_LTE_CXM_CXM_WLAN_TX_GRANT_SEL****Type:** RW**Reset State:** 0x00000000

CxM_WLAN_TX_Grant_sel

WCSS_A_LTE_CXM_CXM_WLAN_TX_GRANT_SEL

Bits	Name	Description
0	CXM_WLAN_TX_GRANT_SEL	TX Grant Select for Wlan. 1: Select CxM Wlan Tx Grant, 0: Select QTA tx grant.

0x0A208004 WCSS_A_LTE_CXM_CXM_WLAN_RX_GRANT_SEL**Type:** RW**Reset State:** 0x00000000

CxM_WLAN_RX_Grant_sel

WCSS_A_LTE_CXM_CXM_WLAN_RX_GRANT_SEL

Bits	Name	Description
0	CXM_WLAN_RX_GRANT_SEL	RX Grant Select for Wlan. 1: Select CxM Wlan Rx Grant, 0: Select QTA rx grant.

0x0A208008 WCSS_A_LTE_CXM_CXM_WLAN_TX_RX_GRANT_EN**Type:** RW**Reset State:** 0x0000000F

CxM_TX_RX_Grant_Enable

WCSS_A_LTE_CXM_CXM_WLAN_TX_RX_GRANT_EN

Bits	Name	Description
3	CXM_WLAN_RX_GRANT_BM_OVERRIDE_VALUE	Override Value of WLAN RX Grant.
2	CXM_WLAN_RX_GRANT_BM_OVERRIDE_ENABLE	Enable override of WLAN RX Grant.
1	CXM_WLAN_TX_GRANT_BM_OVERRIDE_VALUE	Override Value of WLAN TX Grant.
0	CXM_WLAN_TX_GRANT_BM_OVERRIDE_ENABLE	Enable override of WLAN TX Grant.

0x0A20800C WCSS_A_LTE_CXM_CXM_TIMER_COUNTER**Type:** R**Reset State:** 0x00000000

CxM_Timer_Counter

WCSS_A_LTE_CXM_CXM_TIMER_COUNTER

Bits	Name	Description
31:0	TIMER_COUNTER	Counter running at 19.2 MHz

0x0A208014 WCSS_A_LTE_CXM_CXM_SW_CAPTURE_CLK**Type:** RW**Reset State:** 0x00000000

CXM_SW_Capture_CLK

WCSS_A_LTE_CXM_CXM_SW_CAPTURE_CLK

Bits	Name	Description
0	CXM_SW_CAPTURE_CLK	0->1 triggers sync timing.

0x0A208018 WCSS_A_LTE_CXM_CXM_SW_CAPTURE_SEL**Type:** RW**Reset State:** 0x00000000

CXM_SW_Capture_SEL

WCSS_A_LTE_CXM_CXM_SW_CAPTURE_SEL

Bits	Name	Description
0	CXM_SW_CAPTURE_SEL	0: Select SW_Capture_CLK to trigger timing sync 1: Select LET_Frame_Sync_Test_In to trigger timing sync.

0x0A20801C WCSS_A_LTE_CXM_LTE_CXM_COUNTER_CAPTURED**Type:** R**Reset State:** 0x00000000

CxM_Counter_Captured

WCSS_A_LTE_CXM_LTE_CXM_COUNTER_CAPTURED

Bits	Name	Description
31:0	CXM_COUNTER_CAPTURED	Value captured from CxM_Timer_Counter when timing sync is triggered.

0x0A208020 WCSS_A_LTE_CXM_CXM_CP_INTERRUPT_CTRL**Type:** RW**Reset State:** 0x00000000

CxM_Interrupt_Ctrl

WCSS_A_LTE_CXM_CXM_CP_INTERRUPT_CTRL

Bits	Name	Description
3:0	CXM_INTERRUPT_CTRL	Bit_i controls the mode of strobe i, i= 0,1,2,3. 0: indicates the strobe is one-shot, 1: indicates the strobe is periodic.

0x0A208024 WCSS_A_LTE_CXM_CXM_CP_INTERRUPT_MASK_N**Type:** RW**Reset State:** 0x00000000

CxM_Interrupt_Mask_n

WCSS_A_LTE_CXM_CXM_CP_INTERRUPT_MASK_N

Bits	Name	Description
3:0	CXM_INTERRUPT_MASK_N	Bit_i controls the mode of strobe i, i= 0,1,2,3. 0: strobe is disabled 1: strobe is enabled

0x0A208028 WCSS_A_LTE_CXM_CXM_CP_INTERRUPT_RESET_N**Type:** RW**Reset State:** 0x00000000

CxM_Interrupt_Reset_n

WCSS_A_LTE_CXM_CXM_CP_INTERRUPT_RESET_N

Bits	Name	Description
3:0	CXM_INTERRUPT_RESET_N	Bit_i resets strobe i, i= 0,1,2,3. 0 indicates reset, i.e., no further strobes are generated after reset. For one-shot interrupt, it is self cleared to zero. For periodic interrupt, it is not self-cleared (CxM-SW can reset it).

0x0A20802C WCSS_A_LTE_CXM_CXM_CP_INTERRUPT_STATUS**Type:** R**Reset State:** 0x00000000

CxM_Interrupt_Status

WCSS_A_LTE_CXM_CXM_CP_INTERRUPT_STATUS

Bits	Name	Description
3:0	CXM_INTERRUPT_STATUS	0x1: indicates the CxM_Interrupt is generated by strobe 0 0x2: indicates the CxM_Interrupt is generated by strobe 1 0x4: indicates the CxM_Interrupt is generated by strobe 2 0x8: indicates the CxM_Interrupt is generated by strobe 3 This register is readable by CxM-SW. This register is self-cleared.

0x0A208030 WCSS_A_LTE_CXM_CXM_WLAN_TX_GRANT_ENABLE1**Type:** RW**Reset State:** 0x00000000

CxM_WLAN_TX_Grant_Enable1

WCSS_A_LTE_CXM_CXM_WLAN_TX_GRANT_ENABLE1

Bits	Name	Description
0	CXM_WLAN_TX_GRANT_ENABLE1	This register is used together with CxM_Timer_Counter CxM_WLAN_TX_Grant_Threshold1, and CxM_WLAN_TX_Grant_Value1

0x0A208034 WCSS_A_LTE_CXM_CXM_WLAN_TX_GRANT_THRESHOLD1**Type:** RW**Reset State:** 0x00000000

CxM_WLAN_TX_Grant_Threshold1

WCSS_A_LTE_CXM_CXM_WLAN_TX_GRANT_THRESHOLD1

Bits	Name	Description
31:0	CXM_WLAN_TX_GRANT_THRESHOLD1	This register is used together with CxM_Timer_Counter, CxM_WLAN_TX_Grant_Enable1, and CxM_WLAN_TX_Grant_Value1.

0x0A208038 WCSS_A_LTE_CXM_CXM_WLAN_TX_GRANT_VALUE1**Type:** RW**Reset State:** 0x00000000

CxM_WLAN_TX_Grant_Value1

WCSS_A_LTE_CXM_CXM_WLAN_TX_GRANT_VALUE1

Bits	Name	Description
0	CXM_WLAN_TX_GRANT_VALUE1	This register is used together with CxM_Timer_Counter, CxM_WLAN_TX_Grant_Enable1, and CxM_WLAN_TX_Grant_Threshold1.

0x0A20803C WCSS_A_LTE_CXM_CXM_WLAN_TX_GRANT_ENABLE2**Type:** RW**Reset State:** 0x00000000

CxM_WLAN_TX_Grant_Enable2

WCSS_A_LTE_CXM_CXM_WLAN_TX_GRANT_ENABLE2

Bits	Name	Description
0	CXM_WLAN_TX_GRANT_VALUE2	This register is used together with CxM_Timer_Counter, CxM_WLAN_TX_Grant_Enable2, and CxM_WLAN_TX_Grant_Threshold2.

0x0A208040 WCSS_A_LTE_CXM_CXM_WLAN_TX_GRANT_THRESHOLD2**Type:** RW**Reset State:** 0x00000000

CxM_WLAN_TX_Grant_Threshold2

WCSS_A_LTE_CXM_CXM_WLAN_TX_GRANT_THRESHOLD2

Bits	Name	Description
31:0	CXM_WLAN_TX_GRANT_THRESHOLD2	This register is used together with CxM_Timer_Counter, CxM_WLAN_TX_Grant_Enable2, and CxM_WLAN_TX_Grant_Value2.

0x0A208044 WCSS_A_LTE_CXM_CXM_WLAN_TX_GRANT_VALUE2**Type:** RW**Reset State:** 0x00000000

CxM_WLAN_TX_Grant_Value2

WCSS_A_LTE_CXM_CXM_WLAN_TX_GRANT_VALUE2

Bits	Name	Description
0	CXM_WLAN_TX_GRANT_VALUE2	This register is used together with CxM_Timer_Counter, CxM_WLAN_TX_Grant_Enable1, and CxM_WLAN_TX_Grant_Threshold2.

0x0A208048 WCSS_A_LTE_CXM_CXM_WLAN_RX_GRANT_ENABLE1**Type:** RW**Reset State:** 0x00000000

CxM_WLAN_RX_Grant_Enable1

WCSS_A_LTE_CXM_CXM_WLAN_RX_GRANT_ENABLE1

Bits	Name	Description
0	CXM_WLAN_RX_GRANT_ENABLE1	This register is used together with CxM_Timer_Counter CxM_WLAN_RX_Grant_Threshold1, and CxM_WLAN_RX_Grant_Value1

0x0A20804C WCSS_A_LTE_CXM_CXM_WLAN_RX_GRANT_THRESHOLD1**Type:** RW**Reset State:** 0x00000000

CxM_WLAN_RX_Grant_Threshold1

WCSS_A_LTE_CXM_CXM_WLAN_RX_GRANT_THRESHOLD1

Bits	Name	Description
31:0	CXM_WLAN_RX_GRANT_THRESHOLD1	This register is used together with CxM_Timer_Counter, CxM_WLAN_RX_Grant_Enable1, and CxM_WLAN_RX_Grant_Value1.

0x0A208050 WCSS_A_LTE_CXM_CXM_WLAN_RX_GRANT_VALUE1**Type:** RW**Reset State:** 0x00000000

CxM_WLAN_RX_Grant_Value1

WCSS_A_LTE_CXM_CXM_WLAN_RX_GRANT_VALUE1

Bits	Name	Description
0	CXM_WLAN_RX_GRANT_VALUE1	This register is used together with CxM_Timer_Counter, CxM_WLAN_RX_Grant_Enable1, and CxM_WLAN_RX_Grant_Threshold1.

0x0A208054 WCSS_A_LTE_CXM_CXM_WLAN_RX_GRANT_ENABLE2**Type:** RW**Reset State:** 0x00000000

CxM_WLAN_RX_Grant_Enable2

WCSS_A_LTE_CXM_CXM_WLAN_RX_GRANT_ENABLE2

Bits	Name	Description
0	CXM_WLAN_RX_GRANT_ENABLE2	This register is used together with CxM_Timer_Counter CxM_WLAN_RX_Grant_Threshold2, and CxM_WLAN_RX_Grant_Value2

0x0A208058 WCSS_A_LTE_CXM_CXM_WLAN_RX_GRANT_THRESHOLD2**Type:** RW**Reset State:** 0x00000000

CxM_WLAN_RX_Grant_Threshold2

WCSS_A_LTE_CXM_CXM_WLAN_RX_GRANT_THRESHOLD2

Bits	Name	Description
31:0	CXM_WLAN_RX_GRANT_THRESHOLD2	This register is used together with CxM_Timer_Counter, CxM_WLAN_RX_Grant_Enable2, and CxM_WLAN_RX_Grant_Value2.

0x0A20805C WCSS_A_LTE_CXM_CXM_WLAN_RX_GRANT_VALUE2**Type:** RW**Reset State:** 0x00000000

CxM_WLAN_RX_Grant_Value2

WCSS_A_LTE_CXM_CXM_WLAN_RX_GRANT_VALUE2

Bits	Name	Description
0	CXM_WLAN_RX_GRANT_VALUE2	This register is used together with CxM_Timer_Counter, CxM_WLAN_RX_Grant_Enable2, and CxM_WLAN_RX_Grant_Threshold2.

0x0A208060 WCSS_A_LTE_CXM_CXM_BT_TX_GRANT_ENABLE1**Type:** RW**Reset State:** 0x00000000

CxM_BT_TX_Grant_Enable1

WCSS_A_LTE_CXM_CXM_BT_TX_GRANT_ENABLE1

Bits	Name	Description
0	CXM_BT_TX_GRANT_ENABLE1	This register is used together with CxM_Timer_Counter, CxM_BT_TX_Grant_Threshold1, and CxM_BT_TX_Grant_Value1.

0x0A208064 WCSS_A_LTE_CXM_CXM_BT_TX_GRANT_THRESHOLD1**Type:** RW**Reset State:** 0x00000000

CxM_BT_TX_Grant_Threshold1

WCSS_A_LTE_CXM_CXM_BT_TX_GRANT_THRESHOLD1

Bits	Name	Description
31:0	CXM_BT_TX_GRANT_THRESHOLD1	This register is used together with CxM_Timer_Counter, CxM_BT_TX_Grant_Enable1, and CxM_BT_TX_Grant_Value1.

0x0A208068 WCSS_A_LTE_CXM_CXM_BT_TX_GRANT_VALUE1**Type:** RW**Reset State:** 0x00000000

CxM_BT_TX_Grant_Value1

WCSS_A_LTE_CXM_CXM_BT_TX_GRANT_VALUE1

Bits	Name	Description
0	CXM_BT_TX_GRANT_VALU E1	This register is used together with CxM_Timer_Counter, CxM_BT_TX_Grant_Enable1, and CxM_BT_TX_Grant_Threshold1.

0x0A20806C WCSS_A_LTE_CXM_CXM_BT_TX_GRANT_ENABLE2**Type:** RW**Reset State:** 0x00000000

CxM_BT_TX_Grant_Enable2

WCSS_A_LTE_CXM_CXM_BT_TX_GRANT_ENABLE2

Bits	Name	Description
0	CXM_BT_TX_GRANT_VALU E2	This register is used together with CxM_Timer_Counter, CxM_BT_TX_Grant_Enable2, and CxM_BT_TX_Grant_Threshold2.

0x0A208070 WCSS_A_LTE_CXM_CXM_BT_TX_GRANT_THRESHOLD2**Type:** RW**Reset State:** 0x00000000

CxM_BT_TX_Grant_Threshold2

WCSS_A_LTE_CXM_CXM_BT_TX_GRANT_THRESHOLD2

Bits	Name	Description
31:0	CXM_BT_TX_GRANT_THR ESHOLD2	This register is used together with CxM_Timer_Counter, CxM_BT_TX_Grant_Enable2, and CxM_BT_TX_Grant_Value2.

0x0A208074 WCSS_A_LTE_CXM_CXM_BT_TX_GRANT_VALUE2**Type:** RW**Reset State:** 0x00000000

CxM_BT_TX_Grant_Value2

WCSS_A_LTE_CXM_CXM_BT_TX_GRANT_VALUE2

Bits	Name	Description
0	CXM_BT_TX_GRANT_VALU E2	This register is used together with CxM_Timer_Counter, CxM_BT_TX_Grant_Enable1, and CxM_BT_TX_Grant_Threshold2.

0x0A208078 WCSS_A_LTE_CXM_CXM_BT_RX_GRANT_ENABLE1**Type:** RW**Reset State:** 0x00000000

CxM_BT_RX_Grant_Enable1

WCSS_A_LTE_CXM_CXM_BT_RX_GRANT_ENABLE1

Bits	Name	Description
0	CXM_BT_RX_GRANT_ENA BLE1	This register is used together with CxM_Timer_Counter CxM_BT_RX_Grant_Threshold1, and CxM_BT_RX_Grant_Value1

0x0A20807C WCSS_A_LTE_CXM_CXM_BT_RX_GRANT_THRESHOLD1**Type:** RW**Reset State:** 0x00000000

CxM_BT_RX_Grant_Threshold1

WCSS_A_LTE_CXM_CXM_BT_RX_GRANT_THRESHOLD1

Bits	Name	Description
31:0	CXM_BT_RX_GRANT_THR ESHOLD1	This register is used together with CxM_Timer_Counter, CxM_BT_RX_Grant_Enable1, and CxM_BT_RX_Grant_Value1.

0x0A208080 WCSS_A_LTE_CXM_CXM_BT_RX_GRANT_VALUE1**Type:** RW**Reset State:** 0x00000000

CxM_BT_RX_Grant_Value1

WCSS_A_LTE_CXM_CXM_BT_RX_GRANT_VALUE1

Bits	Name	Description
0	CXM_BT_RX_GRANT_VAL UE1	This register is used together with CxM_Timer_Counter, CxM_BT_RX_Grant_Enable1, and CxM_BT_RX_Grant_Threshold1.

0x0A208084 WCSS_A_LTE_CXM_CXM_BT_RX_GRANT_ENABLE2**Type:** RW**Reset State:** 0x00000000

CxM_BT_RX_Grant_Enable2

WCSS_A_LTE_CXM_CXM_BT_RX_GRANT_ENABLE2

Bits	Name	Description
0	CXM_BT_RX_GRANT_ENA BLE2	This register is used together with CxM_Timer_Counter CxM_BT_RX_Grant_Threshold2, and CxM_BT_RX_Grant_Value2

0x0A208088 WCSS_A_LTE_CXM_CXM_BT_RX_GRANT_THRESHOLD2**Type:** RW**Reset State:** 0x00000000

CxM_BT_RX_Grant_Threshold2

WCSS_A_LTE_CXM_CXM_BT_RX_GRANT_THRESHOLD2

Bits	Name	Description
31:0	CXM_BT_RX_GRANT_THR ESHOLD2	This register is used together with CxM_Timer_Counter, CxM_BT_RX_Grant_Enable2, and CxM_BT_RX_Grant_Value2.

0x0A20808C WCSS_A_LTE_CXM_CXM_BT_RX_GRANT_VALUE2**Type:** RW**Reset State:** 0x00000000

CxM_BT_RX_Grant_Value2

WCSS_A_LTE_CXM_CXM_BT_RX_GRANT_VALUE2

Bits	Name	Description
0	CXM_BT_RX_GRANT_VAL UE2	This register is used together with CxM_Timer_Counter, CxM_BT_RX_Grant_Enable2, and CxM_BT_RX_Grant_Threshold2.

0x0A208090 WCSS_A_LTE_CXM_CXM_BT_RX_TX_GRANT_SEL**Type:** RW**Reset State:** 0x00000001

CxM_BT_RX_Grant_sel

WCSS_A_LTE_CXM_CXM_BT_RX_TX_GRANT_SEL

Bits	Name	Description
2	CXM_BT_TX_GRANT_SEL	0: Cxm_BT_TX_Grant goes to the mux output; 1: WLAN_ACTIVE goes to the mux output S/R by CxM-SW.
1:0	CXM_BT_RX_GRANT_SEL	00: CxM_BT_RX_Grant_bm OR'ed with wlan_tx_en reaches the mux output. 01: CxM_BT_RX_Grant_bm OR'ed with mtu_mcu_tx_rx_busy reaches the mux output. 02: CxM_BT_RX_Grant_bm reaches the mux output. 03: not allowed S/R by CxM-SW.

0x0A208094 WCSS_A_LTE_CXM_CXM_BT_TX_RX_GRANT_EN**Type:** RW**Reset State:** 0x00000005

CxM_BT_TX_RX_Grant_en

WCSS_A_LTE_CXM_CXM_BT_TX_RX_GRANT_EN

Bits	Name	Description
3	CXM_BT_RX_GRANT_BM_ OVERWRITE_VALUE	Override Value for BT RX Grant.
2	CXM_BT_RX_GRANT_BM_ OVERWRITE_ENABLE	Enable grant override for BT RX Grant.
1	CXM_BT_TX_GRANT_BM_ OVERWRITE_VALUE	Override Value for BT TX Grant.
0	CXM_BT_TX_GRANT_BM_ OVERWRITE_ENABLE	Enable grant override for BT TX Grant.

0x0A208098 WCSS_A_LTE_CXM_CXM_CP_INTERRUPT_STROBE_0_START_COUNT**Type:** RW**Reset State:** 0x00000000

CxM_Interrupt_Strobe_0_Start_Count

WCSS_A_LTE_CXM_CXM_CP_INTERRUPT_STROBE_0_START_COUNT

Bits	Name	Description
31:0	CXM_INTERRUPT_STROBE_0_START_COUNT	When Strobe i is in one-shot mode, stobe i shall generate a one-shot pulse when the CxM_Timer_Counter is equal to CxM_Interrupt_Strobe_i_Start_Count . When Strobe i is in periodic mode, stobe i shall generate a periodic pulse that starts when the CxM_Timer_Counter is equal to CxM_Interrupt_Strobe_i_Start_Count , which the periodicity defined by the following register. S/R by CxM-SW.

0x0A20809C WCSS_A_LTE_CXM_CXM_CP_INTERRUPT_STROBE_1_START_COUNT**Type:** RW**Reset State:** 0x00000000

CxM_Interrupt_Strobe_1_Start_Count

WCSS_A_LTE_CXM_CXM_CP_INTERRUPT_STROBE_1_START_COUNT

Bits	Name	Description
31:0	CXM_INTERRUPT_STROBE_1_START_COUNT	When Strobe i is in one-shot mode, stobe i shall generate a one-shot pulse when the CxM_Timer_Counter is equal to CxM_Interrupt_Strobe_i_Start_Count . When Strobe i is in periodic mode, stobe i shall generate a periodic pulse that starts when the CxM_Timer_Counter is equal to CxM_Interrupt_Strobe_i_Start_Count , which the periodicity defined by the following register. S/R by CxM-SW.

0x0A2080A0 WCSS_A_LTE_CXM_CXM_CP_INTERRUPT_STROBE_2_START_COUNT**Type:** RW**Reset State:** 0x00000000

CxM_Interrupt_Strobe_2_Start_Count

WCSS_A_LTE_CXM_CXM_CP_INTERRUPT_STROBE_2_START_COUNT

Bits	Name	Description
31:0	CXM_INTERRUPT_STOBE_2_START_COUNT	When Strobe i is in one-shot mode, stobe i shall generate a one-shot pulse when the CxM_Timer_Counter is equal to CxM_Interrupt_Stobe_i_Start_Count . When Strobe i is in periodic mode, stobe i shall generate a periodic pulse that starts when the CxM_Timer_Counter is equal to CxM_Interrupt_Stobe_i_Start_Count, which the periodicity defined by the following register. S/R by CxM-SW.

0x0A2080A4 WCSS_A_LTE_CXM_CXM_CP_INTERRUPT_STROBE_3_START_COUNT**Type:** RW**Reset State:** 0x00000000

CxM_Interrupt_Stobe_3_Start_Count

WCSS_A_LTE_CXM_CXM_CP_INTERRUPT_STROBE_3_START_COUNT

Bits	Name	Description
31:0	CXM_INTERRUPT_STOBE_3_START_COUNT	When Strobe i is in one-shot mode, stobe i shall generate a one-shot pulse when the CxM_Timer_Counter is equal to CxM_Interrupt_Stobe_i_Start_Count . When Strobe i is in periodic mode, stobe i shall generate a periodic pulse that starts when the CxM_Timer_Counter is equal to CxM_Interrupt_Stobe_i_Start_Count , which the periodicity defined by the following register. S/R by CxM-SW.

0x0A2080A8 WCSS_A_LTE_CXM_CXM_CP_INTERRUPT_STROBE_0_PERIODICITY**Type:** RW**Reset State:** 0x00000000

CxM_Interrupt_Stobe_0_Periodicity

WCSS_A_LTE_CXM_CXM_CP_INTERRUPT_STROBE_0_PERIODICITY

Bits	Name	Description
23:0	CXM_INTERRUPT_STOBE_0_PERIODICITY	When Strobe i is in periodic mode, stobe i should generate a strobe with periodicity defined by CxM_Interrupt_Stobe_i_Periodicity. The same clock as the CxM-Timer is used here. S/R by CxM-SW.

0x0A2080AC WCSS_A_LTE_CXM_CXM_CP_INTERRUPT_STROBE_1_PERIODICITY**Type:** RW**Reset State:** 0x00000000

CxM_Interrupt_Strobe_1_Periodicity

WCSS_A_LTE_CXM_CXM_CP_INTERRUPT_STROBE_1_PERIODICITY

Bits	Name	Description
23:0	CXM_INTERRUPT_STROBE_1_PERIODICITY	When Strobe i is in periodic mode, stobe i should generate a strobe with periodicity defined by CxM_Interrupt_Strobe_i_Periodicity. The same clock as the CxM-Timer is used here. S/R by CxM-SW.

0x0A2080B0 WCSS_A_LTE_CXM_CXM_CP_INTERRUPT_STROBE_2_PERIODICITY**Type:** RW**Reset State:** 0x00000000

CxM_Interrupt_Strobe_2_Periodicity

WCSS_A_LTE_CXM_CXM_CP_INTERRUPT_STROBE_2_PERIODICITY

Bits	Name	Description
23:0	CXM_INTERRUPT_STROBE_2_PERIODICITY	When Strobe i is in periodic mode, stobe i should generate a strobe with periodicity defined by CxM_Interrupt_Strobe_i_Periodicity. The same clock as the CxM-Timer is used here. S/R by CxM-SW.

0x0A2080B4 WCSS_A_LTE_CXM_CXM_CP_INTERRUPT_STROBE_3_PERIODICITY**Type:** RW**Reset State:** 0x00000000

CxM_Interrupt_Strobe_3_Periodicity

WCSS_A_LTE_CXM_CXM_CP_INTERRUPT_STROBE_3_PERIODICITY

Bits	Name	Description
23:0	CXM_INTERRUPT_STROBE_3_PERIODICITY	When Strobe i is in periodic mode, stobe i should generate a strobe with periodicity defined by CxM_Interrupt_Strobe_i_Periodicity. The same clock as the CxM-Timer is used here. S/R by CxM-SW.

0x0A2080B8 WCSS_A_LTE_CXM_CXM_WCN_PRIORITY_ENABLES**Type:** RW**Reset State:** 0x00000010

Following registers are related to WCN_PRIORITY

WCSS_A_LTE_CXM_CXM_WCN_PRIORITY_ENABLES

Bits	Name	Description
6	CXM_FRAME_SYNC_SEL	CxM_frame_sync_sel
5	CXM_WCN_PRIORITY_OVE RWRITE_VALUE	CxM_WCN_Priority_overwrite_value
4	CXM_WCN_PRIORITY_OVE RWRITE_ENABLE	CxM_WCN_Priority_overwrite_enable
3	CXM_WCN_PRIORITY_VAL UE2	CxM_WCN_Priority_Value2
2	CXM_WCN_PRIORITY_ENA BLE2	CxM_WCN_Priority_Enable2
1	CXM_WCN_PRIORITY_VAL UE1	CxM_WCN_Priority_Value1
0	CXM_WCN_PRIORITY_ENA BLE1	CxM_WCN_Priority_Enable1

0x0A2080BC WCSS_A_LTE_CXM_CXM_WCN_PRIORITY_THRESHOLD1**Type:** RW**Reset State:** 0x00000000

CXM_WCN_Priority_Threshold1

WCSS_A_LTE_CXM_CXM_WCN_PRIORITY_THRESHOLD1

Bits	Name	Description
31:0	CXM_WCN_PRIORITY_THR ESHOLD1	This register is used together with CxM_WCN_Priority_Enable1, CxM_WCN_Priority_Enable2, CxM_WCN_Priority_Value1, CxM_WCN_Priority_Value2, etc.

0x0A2080C0 WCSS_A_LTE_CXM_CXM_WCN_PRIORITY_THRESHOLD2**Type:** RW**Reset State:** 0x00000000

CXM_WCN_Priority_Threshold2

WCSS_A_LTE_CXM_CXM_WCN_PRIORITY_THRESHOLD2

Bits	Name	Description
31:0	CXM_WCN_PRIORITY_THR ESHOLD2	This register is used together with CxM_WCN_Priority_Enable1, CxM_WCN_Priority_Enable2, CxM_WCN_Priority_Value1, CxM_WCN_Priority_Value2, etc.

0x0A2080C4 WCSS_A_LTE_CXM_CXM_LTE_ACTIVE_FRAME_SYNC_VALUE**Type:** R**Reset State:** 0x00000000

This register "polls" the frame_sync input and LTE_active input values.

WCSS_A_LTE_CXM_CXM_LTE_ACTIVE_FRAME_SYNC_VALUE

Bits	Name	Description
1	I_RIVA_LTE_ACTIVE	This register captures the i_riva_lte_active input signal value.
0	I_RIVA_LTE_FRAME_SYNC	This register captures the i_riva_lte_frame_sync input signal value.

10.18 WCSS_A_SPDM_AHBTM**0x0A203000 WCSS_A_SPDM_AHBTM_CFG****Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

The AHBTM_CFG register configures the following:

WCSS_A_SPDM_AHBTM_CFG

Bits	Name	Description
0	CGC_CLK_ENA	0x0: NO_CLOCK_RUNNING (No clock running) 0x1: CLOCK_ENABLED (Clock enabled)

**0x0A203020+ WCSS_A_SPDM_AHBTM_Cn_CFG0, n=[0..7]
0x20*n****Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

The AHBTM_Cn_CFG0 register configures the following:

AHBTM_Cn_CFG0, n=[0,7]]

WCSS_A_SPDM_AHBTM_Cn_CFG0

Bits	Name	Description
31:28	PORT_SELECT	Identifies which AHB port the monitor should be observing

0x0A203024+ WCSS_A_SPDM_AHBTM_Cn_CFG1, n=[0..7] 0x20*n

Type: RW

Clock: HCLK

Reset State: 0x00000000

The AHBTM_Cn_CFG1 register configures the following:

AHBTM_Cn_CFG1, n=[0,7] (Continued)

WCSS_A_SPDM_AHBTM_Cn_CFG1

Bits	Name	Description
20	SEQ_ADDR_ENA	Setting (1) this bit enables the Sequential Address state checking in the enable logic of the counter. On reset this bit is initialized to 0.
18	HMASTLOCK_ENA	Setting (1) this bit enables the HMASTLOCK state checking in the enable logic of the counter. On reset this bit is initialized to 0.
17	HWRITE_ENA	Setting (1) this bit enables the HWRITE state checking in the enable logic of the counter. On reset this bit is initialized to 0.
16	HREADY_ENA	Setting (1) this bit enables the HREADY state checking in the enable logic of the counter. On reset this bit is initialized to 0.
15	HBURST_ENA	Setting (1) this bit enables the HBURST state checking in the enable logic of the counter. On reset this bit is initialized to 0.
14	HTRANS_ENA	Setting (1) this bit enables the HTRANS state checking in the enable logic of the counter. On reset this bit is initialized to 0.
13	HSIZE_ENA	Setting (1) this bit enables the HSIZE state checking in the enable logic of the counter. On reset this bit is initialized to 0.
12	HRESP_ENA	Setting (1) this bit enables the HRESP state checking in the enable logic of the counter. On reset this bit is initialized to 0.

WCSS_A_SPDM_AHBTM_Cn_CFG1 (cont.)

Bits	Name	Description
9	HPROT_ENA	Setting (1) this bit enables the HPROT state checking in the enable logic of the counter. On reset this bit is initialized to 0.
8	HSEL_ENA	Setting (1) this bit enables the HSEL state checking in the enable logic of the counter. On reset this bit is initialized to 0.
4	HADDR_ENA	Setting (1) this bit enables the HADDR state checking in the enable logic of the counter. On reset this bit is initialized to 0.
1	MSTR_GNT_ENA	Setting (1) this bit enables the MSTR_GNTstate checking in the enable logic of the counter. On reset this bit is initialized to 0.
0	ARB_LAT_ENA	Setting (1) this bit enables the ARB_LAT state checking in the enable logic of the counter. On reset this bit is initialized to 0.

**0x0A203028+ WCSS_A_SPDM_AHBTM_Cn_CFG2, n=[0..7]
0x20*n****Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

The AHBTM_Cn_CFG2 register configures the following:

AHBTM_Cn_CFG2, n=[0,7]

WCSS_A_SPDM_AHBTM_Cn_CFG2

Bits	Name	Description
8	HMASTLOCK_VAL	Value used in the match circuit.
4	HWRITE_VAL	Value used in the match circuit.
0	HREADY_VAL	Value used in the match circuit.

**0x0A20302C+ WCSS_A_SPDM_AHBTM_Cn_CFG3, n=[0..7]
0x20*n****Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

The AHBTM_Cn_CFG3 register configures the following:

AHBTM_Cn_CFG3, n=[0,7]

WCSS_A_SPDM_AHBTM_Cn_CFG3

Bits	Name	Description
30:28	HBURST_MASK	Setting (1) these bits masks off the corresponding bit in the VAL register prior to being used in the match circuit.
27	HBURST_INV	Setting (1) this bit inverts the match circuit output to provide the ability to disable the counter when a match occurs.
26:24	HBURST_VAL	Value used in the match circuit.
22:20	HTRANS_MASK	Setting (1) these bits masks off the corresponding bit in the VAL register prior to being used in the match circuit.
19	HTRANS_INV	Setting (1) this bit inverts the match circuit output to provide the ability to disable the counter when a match occurs.
18:16	HTRANS_VAL	Value used in the match circuit.
14:12	HSIZE_MASK	Setting (1) these bits masks off the corresponding bit in the VAL register prior to being used in the match circuit.
11	HSIZE_INV	Setting (1) this bit inverts the match circuit output to provide the ability to disable the counter when a match occurs.
10:8	HSIZE_VAL	Value used in the match circuit.
6:4	HRESP_MASK	Setting (1) these bits masks off the corresponding bit in the VAL register prior to being used in the match circuit.
3	HRESP_INV	Setting (1) this bit inverts the match circuit output to provide the ability to disable the counter when a match occurs.
2:0	HRESP_VAL	Value used in the match circuit.

**0x0A203030+ WCSS_A_SPDM_AHBTM_Cn_CFG4, n=[0..7]
0x20*n**
Type: RW**Clock:** HCLK**Reset State:** 0x00000000

The AHBTM_Cn_CFG4 register configures the following:

AHBTM_Cn_CFG4, n=[0,7]

WCSS_A_SPDM_AHBTM_Cn_CFG4

Bits	Name	Description
27:24	HPROT_MASK	Setting (1) this bit masks off the corresponding bit in the VAL register prior to being used in the match circuit.
20	HPROT_INV	Setting (1) this bit inverts the match circuit output to provide the ability to disable the counter when a match occurs.
19:16	HPROT_VAL	Value used in the match circuit.

WCSS_A_SPDM_AHBTM_Cn_CFG4 (cont.)

Bits	Name	Description
11:4	HSEL_MASK	Setting (1) this bit masks off the corresponding bit in the VAL register prior to being used in the match circuit.
0	HSEL_VAL	Value used in the match circuit.

**0x0A203034+ WCSS_A_SPDM_AHBTM_Cn_CFG5, n=[0..7]
0x20*n****Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

The AHBTM_Cn_CFG5 register configures the following:

AHBTM_Cn_CFG5, n=[0,7]

WCSS_A_SPDM_AHBTM_Cn_CFG5

Bits	Name	Description
28	HADDR_INV	Setting (1) this bit inverts the match circuit output to provide the ability to disable the counter when a match occurs.
27:8	HADDR_VAL	Value used in the match circuit.
7:0	HADDR_MASK	Setting (1) this bit masks off the corresponding bit (7->15...0->8) in the VAL register prior to being used in the match circuit.

**0x0A203038+ WCSS_A_SPDM_AHBTM_Cn_CFG6, n=[0..7]
0x20*n****Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

The AHBTM_Cn_CFG6 register configures the following:

AHBTM_Cn_CFG6, n=[0,7]

WCSS_A_SPDM_AHBTM_Cn_CFG6

Bits	Name	Description
27:20	MSTR_GNT_MASK	Setting (1) these bits masks off the corresponding bits being used in the match circuit. When set = 0xFF this disables the MSTR_GNT state checking from the enable logic of the counter
16	MSTR_GNT_VAL	Value used in the match circuit.

WCSS_A_SPDM_AHBTM_Cn_CFG6 (cont.)

Bits	Name	Description
11:4	ARB_LAT_MASK	Setting (1) these bits masks off the corresponding bits being used in the match circuit. When set = 0xFF this disables the ARB_LAT state checking from the enable logic of the counter
0	ARB_LAT_VAL	Value used in the match circuit.

10.19 WCSS_A_DXE_0_DXE_0**0x0A202000 WCSS_A_DXE_0_DMA_CSR****Type:** RW**Reset State:** 0x00002728

The DMA Control and status Register (DMA_CSR) is used for configuration settings that, in general, apply to all the DMA channels.

WCSS_A_DXE_0_DMA_CSR

Bits	Name	Type	Description
31:24	RESERVED	R	Reserved. Writes ignored, reads will return 0.
23:16	SPARE	RW	Spare bits to be used as necessary for possible ECO bug fixes.
15	H2H_SYNC_EN	RW	When set, DXE will wait for confirmation from MIF that H2H written data has made it all the way out to memory before continuing with the transfers. This will slow things down but guarantee that the data is always consistent no matter what MIF port is used to write it or read it back. When clear, as long as the data is read from the same MIF port from which it was written, MIF will guarantee that data coherency without the need to wait.
14	PAUSED	R	This status bit will be set when as a result of setting the pause request bit (PAUSE), the state machine has completed any ongoing chunk transfer, released the AHB interfaces, and entered the pause state. This will bit be cleared when as a result of clearing the pause request bit (PAUSE), the state machine has gone back to processing any outstanding transfers or else it has gone back to the IDLE state to wait for new transfer requests.
13	ECTR_EN	RW	Event Counter Enable. When set the event counters will be enabled.
12:8	B2H_TSTMP_OFF	RW	Timestamp Offset. This bitfield specifies the offset (multiple of 4 bytes) to be used for writing the start and end timestamps in the BD for B2H type transfers

WCSS_A_DXE_0_DMA_CSR (cont.)

Bits	Name	Type	Description
7:3	H2B_TSTMP_OFF	RW	Timestamp Offset. This bitfield specifies the offset (multiple of 4 bytes) to be used for writing the start and end timestamps in the BD for H2B transfers.
2	TSTMP_EN	RW	Timestamp Enable. When set, this bit enables the writing of start and end timestamps for H2B and B2H transfers. The timestamps will be written in the BD at the offset specified by the setting of the TMSTMP_OFF bitfield as two consecutive 32-bit starting with the start timestamp. When this bit is cleared no timestamps will be written to the BD.
1	PAUSE	RW	Pause the DMA engine. The engine will stop as soon as the next chunk transfer is complete so no data will be lost. It will also release the AHB interfaces. No progress will be made until software clears this bit.
0	EN	RW	Enables the DMA engine

0x0A202004 WCSS_A_DXE_0_DMA_ENCH**Type:** RW**Reset State:** 0x00000000

This register is write-only and provides a convenient way for software to enable one or more channels with a single register write. When writing to this register, each data bit set HIGH causes the corresponding CH_CSR[EN] bit in the channel to be set thereby enabling the channel. Data bits that are low have no effect in their corresponding channel CSR[EN] bits.

WCSS_A_DXE_0_DMA_ENCH

Bits	Name	Type	Description
31:8	RSVD	R	Reserved. Writes ignored, reads return 0
7	CH7_EN	W	Enable channel 7.
6	CH6_EN	W	Enable channel 6.
5	CH5_EN	W	Enable channel 5.
4	CH4_EN	W	Enable channel 4.
3	CH3_EN	W	Enable channel 3.
2	CH2_EN	W	Enable channel 2.
1	CH1_EN	W	Enable channel 1.
0	CH0_EN	W	Enable channel 0.

0x0A202008 WCSS_A_DXE_0_DMA_CH_EN**Type:** R**Reset State:** 0x00000000

This register is read-only and indicates which DMA channels are enabled. Each bit in this register reflects the status of the corresponding EN bit in each of the channels' CHn_CSRs. As such, used in combination with DMA_DONE, it provides a convenient way for software to determine which channels are available for a new DMA operation.

WCSS_A_DXE_0_DMA_CH_EN

Bits	Name	Description
31:8	RSVD	Reserved. Writes ignored, reads return 0
7	CH7_EN	Enable (EN) bit status for DMA Channel 7 in CH7_CSR.
6	CH6_EN	Enable (EN) bit status for DMA Channel 6 in CH6_CSR.
5	CH5_EN	Enable (EN) bit status for DMA Channel 5 in CH5_CSR.
4	CH4_EN	Enable (EN) bit status for DMA Channel 4 in CH4_CSR.
3	CH3_EN	Enable (EN) bit status for DMA Channel 3 in CH3_CSR.
2	CH2_EN	Enable (EN) bit status for DMA Channel 2 in CH2_CSR.
1	CH1_EN	Enable (EN) bit status for DMA Channel 1 in CH1_CSR.
0	CH0_EN	Enable (EN) bit status for DMA Channel 0 in CH0_CSR.

0x0A20200C WCSS_A_DXE_0_DMA_CH_DONE**Type:** R**Reset State:** 0x00000000

This register is read-only and indicates which DMA channels are done with their programmed transfer, each bit in this register reflects the status of the corresponding DONE bit in each of the channels' CHn_CSRs. As such, it provides a convenient way for software to determine which channels have finished their programmed DMA operation.

WCSS_A_DXE_0_DMA_CH_DONE

Bits	Name	Description
31:8	RSVD	Reserved. Writes ignored, reads return 0
7	CH7_DONE	Done (DONE) bit for DMA Channel 7 in CH7_CSR.
6	CH6_DONE	Done (DONE) bit for DMA Channel 6 in CH6_CSR.
5	CH5_DONE	Done (DONE) bit for DMA Channel 5 in CH5_CSR.
4	CH4_DONE	Done (DONE) bit for DMA Channel 4 in CH4_CSR.

WCSS_A_DXE_0_DMA_CH_DONE (cont.)

Bits	Name	Description
3	CH3_DONE	Done (DONE) bit for DMA Channel 3 in CH3_CSR.
2	CH2_DONE	Done (DONE) bit for DMA Channel 2 in CH2_CSR.
1	CH1_DONE	Done (DONE) bit for DMA Channel 1 in CH1_CSR.
0	CH0_DONE	Done (DONE) bit for DMA Channel 0 in CH0_CSR.

0x0A202010 WCSS_A_DXE_0_DMA_CH_ERR**Type:** R**Reset State:** 0x00000000

This register is read-only and indicates which DMA channels have errors, each bit in this register reflects the status of the corresponding ERR bit in each of the channels' CHn_CSRs. As such, it provides a convenient way for software to determine which channels have incurred errors during operation.

WCSS_A_DXE_0_DMA_CH_ERR

Bits	Name	Description
31:8	RSVD	Reserved. Writes ignored, reads return 0
7	CH7_ERR	Error (ERR) bit for DMA Channel 7 in CH7_CSR.
6	CH6_ERR	Error (ERR) bit for DMA Channel 6 in CH6_CSR.
5	CH5_ERR	Error (ERR) bit for DMA Channel 5 in CH5_CSR.
4	CH4_ERR	Error (ERR) bit for DMA Channel 4 in CH4_CSR.
3	CH3_ERR	Error (ERR) bit for DMA Channel 3 in CH3_CSR.
2	CH2_ERR	Error (ERR) bit for DMA Channel 2 in CH2_CSR.
1	CH1_ERR	Error (ERR) bit for DMA Channel 1 in CH1_CSR.
0	CH0_ERR	Error (ERR) bit for DMA Channel 0 in CH0_CSR.

0x0A202014 WCSS_A_DXE_0_DMA_CH_STOP**Type:** R**Reset State:** 0x00000000

This register is read-only and indicates which DMA channels have been stopped by software request, each bit in this register reflects the status of the corresponding STOP bit in each of the channels' CHn_CSRs. As such, it provides a convenient way for software to determine which channels have incurred stops during operation.

WCSS_A_DXE_0_DMA_CH_STOP

Bits	Name	Description
31:8	RSVD	Reserved. Writes ignored, reads return 0
7	CH7_STOP	Stop (STOP) bit for DMA Channel 7 in CH7_CSR.
6	CH6_STOP	Stop (STOP) bit for DMA Channel 6 in CH6_CSR.
5	CH5_STOP	Stop (STOP) bit for DMA Channel 5 in CH5_CSR.
4	CH4_STOP	Stop (STOP) bit for DMA Channel 4 in CH4_CSR.
3	CH3_STOP	Stop (STOP) bit for DMA Channel 3 in CH3_CSR.
2	CH2_STOP	Stop (STOP) bit for DMA Channel 2 in CH2_CSR.
1	CH1_STOP	Stop (STOP) bit for DMA Channel 1 in CH1_CSR.
0	CH0_STOP	Stop (STOP) bit for DMA Channel 0 in CH0_CSR.

0x0A202018 WCSS_A_DXE_0_INT_MSK**Type:** RW**Reset State:** 0x00000000

The Interrupt Mask register defines the functionality of DXE's all-channel combined interrupt output. A bit set to one in the mask enables the generation of the interrupt for that source channel, a zero disables the generation of an interrupt.

WCSS_A_DXE_0_INT_MSK

Bits	Name	Type	Description
31:8	RSVD	R	Reserved. Writes ignored, reads return 0
7	CH7_INTEN	RW	Interrupt Enable. Enable DMA Channel 7 Interrupts.
6	CH6_INTEN	RW	Interrupt Enable. Enable DMA Channel 6 Interrupts.
5	CH5_INTEN	RW	Interrupt Enable. Enable DMA Channel 5 Interrupts.
4	CH4_INTEN	RW	Interrupt Enable. Enable DMA Channel 4 Interrupts.
3	CH3_INTEN	RW	Interrupt Enable. Enable DMA Channel 3 Interrupts.
2	CH2_INTEN	RW	Interrupt Enable. Enable DMA Channel 2 Interrupts.
1	CH1_INTEN	RW	Interrupt Enable. Enable DMA Channel 1 Interrupts.
0	CH0_INTEN	RW	Interrupt Enable. Enable DMA Channel 0 Interrupts.

0x0A20201C WCSS_A_DXE_0_INT_SRC_MSKD**Type:** R**Reset State:** 0x00000000

This read-only register is the bitwise AND of INT_SRC_RAW with INT_MSK. If any of its bits is a one, the DXE will generate an interrupt request to the interrupt controller.

WCSS_A_DXE_0_INT_SRC_MSKD

Bits	Name	Description
31:8	RSVD	Reserved. Writes ignored, reads return 0
7	CH7_INT	ED, Done or Error interrupt request by CH7 after masking with INT_MSK[7].
6	CH6_INT	ED, Done or Error interrupt request by CH6 after masking with INT_MSK[6].
5	CH5_INT	ED, Done or Error interrupt request by CH5 after masking with INT_MSK[5].
4	CH4_INT	ED, Done or Error interrupt request by CH4 after masking with INT_MSK[4].
3	CH3_INT	ED, Done or Error interrupt request by CH3 after masking with INT_MSK[3].
2	CH2_INT	ED, Done or Error interrupt request by CH2 after masking with INT_MSK[2].
1	CH1_INT	ED, Done or Error interrupt request by CH1 after masking with INT_MSK[1].
0	CH0_INT	ED, Done or Error interrupt request by CH0 after masking with INT_MSK[0].

0x0A202020 WCSS_A_DXE_0_INT_SRC_RAW**Type:** R**Reset State:** 0x00000000

This read-only register combines the sources of all types of error interrupt requests: descriptor, done and error types. It is basically the bitwise OR of INT_ED_SRC, INT_DONE_SRC and INT_ERR_SRC

WCSS_A_DXE_0_INT_SRC_RAW

Bits	Name	Description
31:8	RSVD	Reserved. Writes ignored, reads return 0
7	CH7_INT	ED, Done or Error interrupt requested by CH7.

WCSS_A_DXE_0_INT_SRC_RAW (cont.)

Bits	Name	Description
6	CH6_INT	ED, Done or Error interrupt requested by CH6.
5	CH5_INT	ED, Done or Error interrupt requested by CH5.
4	CH4_INT	ED, Done or Error interrupt requested by CH4.
3	CH3_INT	ED, Done or Error interrupt requested by CH3.
2	CH2_INT	ED, Done or Error interrupt requested by CH2.
1	CH1_INT	ED, Done or Error interrupt requested by CH1.
0	CH0_INT	ED Done or Error interrupt requested by CH0.

0x0A202024 WCSS_A_DXE_0_INT_ED_SRC**Type:** R**Reset State:** 0x00000000

This read-only register combines the sources of all external descriptor requested interrupts. It is the "raw" unmasked value from the channel registers.

WCSS_A_DXE_0_INT_ED_SRC

Bits	Name	Description
31:8	RSVD	Reserved. Writes ignored, reads return 0
7	CH7_ED_INT	External descriptor interrupt requested by CH7.
6	CH6_ED_INT	External descriptor interrupt requested by CH6.
5	CH5_ED_INT	External descriptor interrupt requested by CH5.
4	CH4_ED_INT	External descriptor interrupt requested by CH4.
3	CH3_ED_INT	External descriptor interrupt requested by CH3.
2	CH2_ED_INT	External descriptor interrupt requested by CH2.
1	CH1_ED_INT	External descriptor interrupt requested by CH1.
0	CH0_ED_INT	External descriptor interrupt requested by CH0.

0x0A202028 WCSS_A_DXE_0_INT_DONE_SRC**Type:** R**Reset State:** 0x00000000

This read-only register combines the sources of all done interrupt requests. It is the "raw" unmasked value from the channel registers.

WCSS_A_DXE_0_INT_DONE_SRC

Bits	Name	Description
31:8	RSVD	Reserved. Writes ignored, reads return 0
7	CH7_DONE_INT	Done interrupt requested by CH7.
6	CH6_DONE_INT	Done interrupt requested by CH6.
5	CH5_DONE_INT	Done interrupt requested by CH5.
4	CH4_DONE_INT	Done interrupt requested by CH4.
3	CH3_DONE_INT	Done interrupt requested by CH3.
2	CH2_DONE_INT	Done interrupt requested by CH2.
1	CH1_DONE_INT	Done interrupt requested by CH1.
0	CH0_DONE_INT	Done interrupt requested by CH0.

0x0A20202C WCSS_A_DXE_0_INT_ERR_SRC**Type:** R**Reset State:** 0x00000000

This read-only register combines the sources of all error interrupt requests. It is the "raw" unmasked value from the channel registers.

WCSS_A_DXE_0_INT_ERR_SRC

Bits	Name	Description
31:8	RSVD	Reserved. Writes ignored, reads return 0
7	CH7_ERR_INT	Error interrupt requested by CH7.
6	CH6_ERR_INT	Error interrupt requested by CH6.
5	CH5_ERR_INT	Error interrupt requested by CH5.
4	CH4_ERR_INT	Error interrupt requested by CH4.
3	CH3_ERR_INT	Error interrupt requested by CH3.
2	CH2_ERR_INT	Error interrupt requested by CH2.
1	CH1_ERR_INT	Error interrupt requested by CH1.
0	CH0_ERR_INT	Error interrupt requested by CH0.

0x0A202030 WCSS_A_DXE_0_INT_CLR**Type:** RW**Reset State:** 0x00000000

This register is write-only and is used to clear interrupt request from either all or specific individual channels. When writing to this register, each data bit set HIGH causes the corresponding bits in the channel CSR to be cleared. All ED, DONE and ERROR source will be cleared at once. Data bits that are low have no effect on the corresponding bit in the channel CSR register.

WCSS_A_DXE_0_INT_CLR

Bits	Name	Type	Description
31:8	RSVD	R	Reserved. Writes ignored, reads return 0
7	CH7_INT_CLR	W	Clear all interrupt sources in CH7.
6	CH6_INT_CLR	W	Clear all interrupt sources in CH6.
5	CH5_INT_CLR	W	Clear all interrupt sources in CH5.
4	CH4_INT_CLR	W	Clear all interrupt sources in CH4.
3	CH3_INT_CLR	W	Clear all interrupt sources in CH3.
2	CH2_INT_CLR	W	Clear all interrupt sources in CH2.
1	CH1_INT_CLR	W	Clear all interrupt sources in CH1.
0	CH0_INT_CLR	W	Clear all interrupt sources in CH0.

0x0A202034 WCSS_A_DXE_0_INT_ED_CLR**Type:** RW**Reset State:** 0x00000000

This register is write-only and is used to clear external descriptor interrupt requests simultaneously from either all or specific individual channels. When writing to this register, each data bit set HIGH causes the corresponding bits in the channel CSR to be cleared. Data bits that are low have no effect on the corresponding bit in the channel CSR register.

WCSS_A_DXE_0_INT_ED_CLR

Bits	Name	Type	Description
31:8	RSVD	R	Reserved. Writes ignored, reads return 0
7	CH7_INT_ED_CLR	W	Clear ED interrupt source in CH7.
6	CH6_INT_ED_CLR	W	Clear ED interrupt source in CH6.
5	CH5_INT_ED_CLR	W	Clear ED interrupt source in CH5.

WCSS_A_DXE_0_INT_ED_CLR (cont.)

Bits	Name	Type	Description
4	CH4_INT_ED_CLR	W	Clear ED interrupt source in CH4.
3	CH3_INT_ED_CLR	W	Clear ED interrupt source in CH3.
2	CH2_INT_ED_CLR	W	Clear ED interrupt source in CH2.
1	CH1_INT_ED_CLR	W	Clear ED interrupt source in CH1.
0	CH0_INT_ED_CLR	W	Clear ED interrupt source in CH0.

0x0A202038 WCSS_A_DXE_0_INT_DONE_CLR**Type:** RW**Reset State:** 0x00000000

This register is write-only and is used to clear DONE interrupt requests simultaneously from either all or specific individual channels. When writing to this register, each data bit set HIGH causes the corresponding bits in the channel CSR to be cleared. Data bits that are low have no effect on the corresponding bit in the channel CSR register.

WCSS_A_DXE_0_INT_DONE_CLR

Bits	Name	Type	Description
31:8	RSVD	R	Reserved. Writes ignored, reads return 0
7	CH7_INT_DONE_CLR	W	Clear DONE interrupt source in CH7.
6	CH6_INT_DONE_CLR	W	Clear DONE interrupt source in CH6.
5	CH5_INT_DONE_CLR	W	Clear DONE interrupt source in CH5.
4	CH4_INT_DONE_CLR	W	Clear DONE interrupt source in CH4.
3	CH3_INT_DONE_CLR	W	Clear DONE interrupt source in CH3.
2	CH2_INT_DONE_CLR	W	Clear DONE interrupt source in CH2.
1	CH1_INT_DONE_CLR	W	Clear DONE interrupt source in CH1.
0	CH0_INT_DONE_CLR	W	Clear DONE interrupt source in CH0.

0x0A20203C WCSS_A_DXE_0_INT_ERR_CLR**Type:** RW**Reset State:** 0x00000000

This register is write-only and is used to clear ERROR interrupt requests simultaneously from either all or specific individual channels. When writing to this register, each data bit set HIGH causes the corresponding bits in the channel CSR to be cleared. Data bits that are low have no effect on the corresponding bit in the channel CSR register.

WCSS_A_DXE_0_INT_ERR_CLR

Bits	Name	Type	Description
31:8	RSVD	R	Reserved. Writes ignored, reads return 0
7	CH7_INT_ERR_CLR	W	Clear ERROR interrupt source in CH7.
6	CH6_INT_ERR_CLR	W	Clear ERROR interrupt source in CH6.
5	CH5_INT_ERR_CLR	W	Clear ERROR interrupt source in CH5.
4	CH4_INT_ERR_CLR	W	Clear ERROR interrupt source in CH4.
3	CH3_INT_ERR_CLR	W	Clear ERROR interrupt source in CH3.
2	CH2_INT_ERR_CLR	W	Clear ERROR interrupt source in CH2.
1	CH1_INT_ERR_CLR	W	Clear ERROR interrupt source in CH1.
0	CH0_INT_ERR_CLR	W	Clear ERROR interrupt source in CH0.

0x0A202040 WCSS_A_DXE_0_DMA_CH_PRES**Type:** R**Reset State:** 0x000000FF

This register is read-only and indicates which DMA channels exist in the hardware.

WCSS_A_DXE_0_DMA_CH_PRES

Bits	Name	Description
31	CH31_PRES	Channel 31 Present. When high signifies that the channel exists in this implementation of the HW.
30	CH30_PRES	Channel 30 Present. When high signifies that the channel exists in this implementation of the HW.
29	CH29_PRES	Channel 29 Present. When high signifies that the channel exists in this implementation of the HW.
28	CH28_PRES	Channel 28 Present. When high signifies that the channel exists in this implementation of the HW.
27	CH27_PRES	Channel 27 Present. When high signifies that the channel exists in this implementation of the HW.
26	CH26_PRES	Channel 26 Present. When high signifies that the channel exists in this implementation of the HW.
25	CH25_PRES	Channel 25 Present. When high signifies that the channel exists in this implementation of the HW.
24	CH24_PRES	Channel 24 Present. When high signifies that the channel exists in this implementation of the HW.
23	CH23_PRES	Channel 23 Present. When high signifies that the channel exists in this implementation of the HW.

WCSS_A_DXE_0_DMA_CH_PRES (cont.)

Bits	Name	Description
22	CH22_PRES	Channel 22 Present. When high signifies that the channel exists in this implementation of the HW.
21	CH21_PRES	Channel 21 Present. When high signifies that the channel exists in this implementation of the HW.
20	CH20_PRES	Channel 20 Present. When high signifies that the channel exists in this implementation of the HW.
19	CH19_PRES	Channel 19 Present. When high signifies that the channel exists in this implementation of the HW.
18	CH18_PRES	Channel 18 Present. When high signifies that the channel exists in this implementation of the HW.
17	CH17_PRES	Channel 17 Present. When high signifies that the channel exists in this implementation of the HW.
16	CH16_PRES	Channel 16 Present. When high signifies that the channel exists in this implementation of the HW.
15	CH15_PRES	Channel 15 Present. When high signifies that the channel exists in this implementation of the HW.
14	CH14_PRES	Channel 14 Present. When high signifies that the channel exists in this implementation of the HW.
13	CH13_PRES	Channel 13 Present. When high signifies that the channel exists in this implementation of the HW.
12	CH12_PRES	Channel 12 Present. When high signifies that the channel exists in this implementation of the HW.
11	CH11_PRES	Channel 11 Present. When high signifies that the channel exists in this implementation of the HW.
10	CH10_PRES	Channel 10 Present. When high signifies that the channel exists in this implementation of the HW.
9	CH9_PRES	Channel 9 Present. When high signifies that the channel exists in this implementation of the HW.
8	CH8_PRES	Channel 8 Present. When high signifies that the channel exists in this implementation of the HW.
7	CH7_PRES	Channel 7 Present. When high signifies that the channel exists in this implementation of the HW.
6	CH6_PRES	Channel 6 Present. When high signifies that the channel exists in this implementation of the HW.
5	CH5_PRES	Channel 5 Present. When high signifies that the channel exists in this implementation of the HW.
4	CH4_PRES	Channel 4 Present. When high signifies that the channel exists in this implementation of the HW.
3	CH3_PRES	Channel 3 Present. When high signifies that the channel exists in this implementation of the HW.

WCSS_A_DXE_0_DMA_CH_PRES (cont.)

Bits	Name	Description
2	CH2_PRES	Channel 2 Present. When high signifies that the channel exists in this implementation of the HW.
1	CH1_PRES	Channel 1 Present. When high signifies that the channel exists in this implementation of the HW.
0	CH0_PRES	Channel 0 Present. When high signifies that the channel exists in this implementation of the HW.

0x0A202050 WCSS_A_DXE_0_PF_BD**Type:** R**Reset State:** 0x00000000

This read-only register contains the index of any prefetched BD index. The index is only meaningful if the accompanying valid bit is set.

WCSS_A_DXE_0_PF_BD

Bits	Name	Description
31:17	RESERVED	Reserved. Writes ignored, reads will return 0.
16	VALID	Valid. When set indicates that the INDEX bitfield contains a valid BD index pointer.
15:0	INDEX	BD Index.

0x0A202054 WCSS_A_DXE_0_PF_PDU**Type:** R**Reset State:** 0x00000000

This read-only register contains the index of any prefetched PDU index. The index is only meaningful if the accompanying valid bit is set.

WCSS_A_DXE_0_PF_PDU

Bits	Name	Description
31:17	RESERVED	Reserved. Reads will return 0.
16	VALID	Valid. When set indicates that the INDEX bitfield contains a valid PDU index pointer.
15:0	INDEX	PDU Index.

0x0A202058 WCSS_A_DXE_0_REVISION**Type:** R**Reset State:** 0x0000001D

This register's value represents the DXE RTL revision present in the hardware.

WCSS_A_DXE_0_REVISION

Bits	Name	Description
31:0	REVID	Revision ID. Corresponds to the label number of DXE (revision of .info file)

0x0A20205C WCSS_A_DXE_0_BMU_SB_QDAT_AVAIL**Type:** R**Reset State:** 0x00000000

This read-only register provides the status of BMU WQ dataavailable sideband signals as seen by DXE.

WCSS_A_DXE_0_BMU_SB_QDAT_AVAIL

Bits	Name	Description
31:25	RSVD1	Reserved.
24:2	QDAT_AV	BMU WQ Data Aavailable.
1:0	RSVD	Reserved. BMU WQs 1 and 0 are notavailable for general use.

0x0A202060 WCSS_A_DXE_0_BMU_SB_PDU_AVAIL**Type:** R**Reset State:** 0x00000000

This read-only register provides the status of BMU PDUavailable sideband signals as seen by DXE.

WCSS_A_DXE_0_BMU_SB_PDU_AVAIL

Bits	Name	Description
31:11	RSVD1	Reserved.
10:2	PDU_AV	BMU BD/PDU Aavailable.
1:0	RSVD	Reserved. BMU thresholds 0 and 1 are notavailable for general use.

0x0A202064 WCSS_A_DXE_0_STATE**Type:** R**Reset State:** 0x00000000

This read-only register provides key DXE state machine status to help debug.

WCSS_A_DXE_0_STATE

Bits	Name	Description
31	RX_SPAVAIL	Space isavailable in one or more of the UIF RX endpoints
30	TX_MGMT_DAVAIL	Data isavailable in the UIF TX management endpoint
29	TX_DAT_DAVAIL	Data isavailable in the UIF TX data endpoint
28	WQ_DAVAIL	Data isavailable in selected BMU WQ
27	PDU_AVAIL	PDUs areavailable for current threshold
26:25	XFER_IDX	Id of the transfer size source that turned out to be the minimum and therefore picked as size of current transfer
24:17	XFER_SZ	Calculated max transfer size from minimum of chunk size and space or dataavailable in either source PDU, destination PDU, or UIF endpoint.
16:12	SEL_CH	Currently Selected Channel.
11:10	PFSM	State of prefetch engine state machine
9:8	DFSM	State of DPDMA state machine
7:6	SFSM	State of SPDMA state machine

WCSS_A_DXE_0_STATE (cont.)

Bits	Name	Description
5:0	MFSM	State of main state machine 0x0: IDLE (Idle state) 0x1: H2H_START (Preparing to start a H2H type transfer) 0x2: H2H_RD (Read transfer) 0x3: H2H_WR (Write transfer) 0x4: H2H_UPDATE (Update channel register values) 0x5: LD_DESC (Load external descriptor) 0x6: CKEDVAL (Check descriptor valid bit) 0x7: CLREDVAL (Clear descriptor valid bit) 0x8: WB_CTRL (Update CH_CTRL with value in DESC_CTRL word of descriptor) 0x9: WB_SZ (Update CH_SZ with value in DESC_SZ word of descriptor) 0xA: WB_SADR (Update CH_SADR with value in DESC_SADRL and DESC_SADRH words of descriptor) 0xB: WB_DADR (Update CH_DADR with value in DESC_DADRL and DESC_DADRH words of descriptor) 0xC: WB_DESC (Update CH_DESC with value in DESC_NXTL and DESC_NXTH words of descriptor) 0xD: PAUSE (State machine paused due to SW pause request) 0xE: B2H_START (Preparing to start a B2H type transfer) 0xF: B2H_POPWQ (Popping BMU work queue) 0x10: B2H_GET_BD_INFO (Read in head, tail, mpdu length, num pdus, etc) 0x11: B2H_RD_BD (Reading BD data contents from BD index) 0x12: B2H_WR_BD (Writing out BD contents) 0x13: B2H_RD_PDU (Read transfer) 0x14: B2H_WR_PDU (Write transfer) 0x15: B2H_NXT_PDU (Get index of next PDU in the chain) 0x16: B2H_UPNCK_PDU (Check that next PDU index is valid) 0x17: B2H_WR_TSTMP (Update start and end timestamps for B2H transfers) 0x18: B2H_REL_PKT (Release BD and linked PDUs) 0x19: H2B_START (Preparing to start a H2B type transfer) 0x1A: H2B_GET_PDU (Obtain a new BD or PDU index) 0x1B: H2B_RD_PDU (Read transfer) 0x1C: H2B_WR_PDU (Write transfer) 0x1D: H2B_LNK_PDU (Link current PDU with new PDU) 0x1E: H2B_UPDT_BD1 (Prepare to update BD fields)

WCSS_A_DXE_0_STATE (cont.)

Bits	Name	Description
		0x1F: H2B_UPDT_BD2 (Update head, tail, pdu_count, etc fields of BD) 0x20: H2B_PUSHWQ (Push BMU work queue) 0x21: ABORT (SW requested transfer abort) 0x22: ERROR (Error detected in transfer) 0x23: H2B_WR_TSTMP (Update start and end timestamps for H2B transfers) 0x24: WR_TOT_XFER_LEN (write total xfer length to SIF at the start of every new H2H or B2H transfer) 0x25: H2B_BD_TMPLT_RD (Read BD template data) 0x26: H2B_BD_TMPLT_WR (Write BD template data to allocated BD)

0x0A202068 WCSS_A_DXE_0_ARB_CH_MSK**Type:** R**Reset State:** 0x00000000

This register is read-only and indicates which enabled DMA channels have been masked from active arbitration by some blocking condition

WCSS_A_DXE_0_ARB_CH_MSK

Bits	Name	Description
31:8	RSVD	Reserved. Reads return 0
7	CH7_MSKD	When set indicates that CH7 is masked by the arbiter and is therefore currently not actively participating in the arbitration because of some blocking condition that would prevent it from making progress.
6	CH6_MSKD	When set indicates that CH6 is masked by the arbiter and is therefore currently not actively participating in the arbitration because of some blocking condition that would prevent it from making progress.
5	CH5_MSKD	When set indicates that CH5 is masked by the arbiter and is therefore currently not actively participating in the arbitration because of some blocking condition that would prevent it from making progress.
4	CH4_MSKD	When set indicates that CH4 is masked by the arbiter and is therefore currently not actively participating in the arbitration because of some blocking condition that would prevent it from making progress.
3	CH3_MSKD	When set indicates that CH3 is masked by the arbiter and is therefore currently not actively participating in the arbitration because of some blocking condition that would prevent it from making progress.

WCSS_A_DXE_0_ARB_CH_MSK (cont.)

Bits	Name	Description
2	CH2_MSKD	When set indicates that CH2 is masked by the arbiter and is therefore currently not actively participating in the arbitration because of some blocking condition that would prevent it from making progress.
1	CH1_MSKD	When set indicates that CH1 is masked by the arbiter and is therefore currently not actively participating in the arbitration because of some blocking condition that would prevent it from making progress.
0	CH0_MSKD	When set indicates that CH0 is masked by the arbiter and is therefore currently not actively participating in the arbitration because of some blocking condition that would prevent it from making progress.

0x0A20206C WCSS_A_DXE_0_UIF_SB**Type:** R**Reset State:** 0x00000000

This read-only register provides the status of important UIF sideband signals

WCSS_A_DXE_0_UIF_SB

Bits	Name	Description
31:22	RSVD	Reserved. Writes ignored, reads will return 0.
21:14	RX_SPACE_REM_DAT_REM	Provides the number of dwords that can be written to the active Rx endpoint
13	RX4_SPAVAIL	When high means that there is spaceavailable in the Rx1 endpoint
12	RX3_SPAVAIL	When high means that there is spaceavailable in the Rx1 endpoint
11	RX2_SPAVAIL	When high means that there is spaceavailable in the Rx1 endpoint
10	RX1_SPAVAIL	When high means that there is spaceavailable in the Rx1 endpoint
9:2	TX_DAT_REM	Provides the number of dwordsavailable in the active Tx endpoint
1	TX_MGMT_DAVAIL	When high means that there is dataavailable in the Tx management endpoint
0	TX_DAT_DAVAIL	When high means that there is dataavailable in the Tx data endpoint

0x0A202070 WCSS_A_DXE_0_TIMESTAMP**Type:** R**Reset State:** 0x00000046

This read-only register provides the current value of the MTU timestamp

WCSS_A_DXE_0_TIMESTAMP

Bits	Name	Description
31:0	TSTMP	

0x0A202074 WCSS_A_DXE_0_ARB_CH_MSK_CLR

Type: RW

Reset State: 0x00000000

This write-only register provides for a way to forcibly unmask currently masked channels and return them to active participation in the arbitration

WCSS_A_DXE_0_ARB_CH_MSK_CLR

Bits	Name	Type	Description
31:8	RSVD	R	Reserved. Writes ignored. Reads return 0
7	CH7_MSK_CLR	W	Writing a one clears the arbiter mask for channel 7 there by returning the channel to active arbitration. Reads always return 0.
6	CH6_MSK_CLR	W	Writing a one clears the arbiter mask for channel 6 there by returning the channel to active arbitration. Reads always return 0.
5	CH5_MSK_CLR	W	Writing a one clears the arbiter mask for channel 5 there by returning the channel to active arbitration. Reads always return 0.
4	CH4_MSK_CLR	W	Writing a one clears the arbiter mask for channel 4 there by returning the channel to active arbitration. Reads always return 0.
3	CH3_MSK_CLR	W	Writing a one clears the arbiter mask for channel 3 there by returning the channel to active arbitration. Reads always return 0.
2	CH2_MSK_CLR	W	Writing a one clears the arbiter mask for channel 2 there by returning the channel to active arbitration. Reads always return 0.
1	CH1_MSK_CLR	W	Writing a one clears the arbiter mask for channel 1 there by returning the channel to active arbitration. Reads always return 0.
0	CH0_MSK_CLR	W	Writing a one clears the arbiter mask for channel 0 there by returning the channel to active arbitration. Reads always return 0.

0x0A202078 WCSS_A_DXE_0_CTR_PRES**Type:** R**Reset State:** 0x000000FF

This register is read-only and indicates how many counters actually exist in this implementation of the hardware.

WCSS_A_DXE_0_CTR_PRES

Bits	Name	Description
31	CTR31_PRES	Counter 31 Present. When high signifies that the counter exists in this implementation of the HW.
30	CTR30_PRES	Counter 30 Present. When high signifies that the counter exists in this implementation of the HW.
29	CTR29_PRES	Counter 29 Present. When high signifies that the counter exists in this implementation of the HW.
28	CTR28_PRES	Counter 28 Present. When high signifies that the counter exists in this implementation of the HW.
27	CTR27_PRES	Counter 27 Present. When high signifies that the counter exists in this implementation of the HW.
26	CTR26_PRES	Counter 26 Present. When high signifies that the counter exists in this implementation of the HW.
25	CTR25_PRES	Counter 25 Present. When high signifies that the counter exists in this implementation of the HW.
24	CTR24_PRES	Counter 24 Present. When high signifies that the counter exists in this implementation of the HW.
23	CTR23_PRES	Counter 23 Present. When high signifies that the counter exists in this implementation of the HW.
22	CTR22_PRES	Counter 22 Present. When high signifies that the counter exists in this implementation of the HW.
21	CTR21_PRES	Counter 21 Present. When high signifies that the counter exists in this implementation of the HW.
20	CTR20_PRES	Counter 20 Present. When high signifies that the counter exists in this implementation of the HW.
19	CTR19_PRES	Counter 19 Present. When high signifies that the counter exists in this implementation of the HW.
18	CTR18_PRES	Counter 18 Present. When high signifies that the counter exists in this implementation of the HW.
17	CTR17_PRES	Counter 17 Present. When high signifies that the counter exists in this implementation of the HW.
16	CTR16_PRES	Counter 16 Present. When high signifies that the counter exists in this implementation of the HW.

WCSS_A_DXE_0_CTR_PRES (cont.)

Bits	Name	Description
15	CTR15_PRES	Counter 15 Present. When high signifies that the counter exists in this implementation of the HW.
14	CTR14_PRES	Counter 14 Present. When high signifies that the counter exists in this implementation of the HW.
13	CTR13_PRES	Counter 13 Present. When high signifies that the counter exists in this implementation of the HW.
12	CTR12_PRES	Counter 12 Present. When high signifies that the counter exists in this implementation of the HW.
11	CTR11_PRES	Counter 11 Present. When high signifies that the counter exists in this implementation of the HW.
10	CTR10_PRES	Counter 10 Present. When high signifies that the counter exists in this implementation of the HW.
9	CTR9_PRES	Counter 9 Present. When high signifies that the counter exists in this implementation of the HW.
8	CTR8_PRES	Counter 8 Present. When high signifies that the counter exists in this implementation of the HW.
7	CTR7_PRES	Counter 7 Present. When high signifies that the counter exists in this implementation of the HW.
6	CTR6_PRES	Counter 6 Present. When high signifies that the counter exists in this implementation of the HW.
5	CTR5_PRES	Counter 5 Present. When high signifies that the counter exists in this implementation of the HW.
4	CTR4_PRES	Counter 4 Present. When high signifies that the counter exists in this implementation of the HW.
3	CTR3_PRES	Counter 3 Present. When high signifies that the counter exists in this implementation of the HW.
2	CTR2_PRES	Counter 2 Present. When high signifies that the counter exists in this implementation of the HW.
1	CTR1_PRES	Counter 1 Present. When high signifies that the counter exists in this implementation of the HW.
0	CTR0_PRES	Counter 0 Present. When high signifies that the counter exists in this implementation of the HW.

0x0A20207C WCSS_A_DXE_0_CTR_CLR**Type:** RW**Reset State:** 0x00000000

This register is write-only and is used to reset the count from one or more specific event counters. When writing to this register, each data bit set HIGH causes the count value in the corresponding counter to be cleared. Data bits that are low have no effect on the corresponding counter.

WCSS_A_DXE_0_CTR_CLR

Bits	Name	Type	Description
31:8	RSVD	R	Reserved. Writes ignored, reads return 0
7	CTR7_CLR	W	Zero the count value in CTR7.
6	CTR6_CLR	W	Zero the count value in CTR6.
5	CTR5_CLR	W	Zero the count value in CTR5.
4	CTR4_CLR	W	Zero the count value in CTR4.
3	CTR3_CLR	W	Zero the count value in CTR3.
2	CTR2_CLR	W	Zero the count value in CTR2.
1	CTR1_CLR	W	Zero the count value in CTR1.
0	CTR0_CLR	W	Zero the count value in CTR0.

0x0A202080 WCSS_A_DXE_0_BD_TMPLTL**Type:** RW**Reset State:** 0x00000000

The BD template base address high register holds the low bits of the base address for the BD templates store. DXE supports up to 4 templates and automatically calculates the address of the selected template based on the base address and the values programmed in CHn_CTRL[BDT_IDX] as follows: $BD_TMPLT_ADDR = BD_TMPLT_BASE + (128 * CHn_CTRL[BDT_IDX])$

WCSS_A_DXE_0_BD_TMPLTL

Bits	Name	Description
31:0	BASEL	BD Template Base Low Address.

0x0A202084 WCSS_A_DXE_0_BD_TMPLTH**Type:** RW**Reset State:** 0x00000000

The BD template base address high register holds the high bits of the base address for the BD templates store. See the description of CH_SADRL for more information.

WCSS_A_DXE_0_BD_TMPLTH

Bits	Name	Type	Description
31:2	RSVD	R	Reserved. Writes ignored. Reads always return zeroes.

WCSS_A_DXE_0_BD_TMPLTH (cont.)

Bits	Name	Type	Description
1:0	BASEH	RW	Base Address High. This bitfield holds bits 33 and 32 of the BD template base address. With the current address mapping in Virgo these high order bits determine where the transfer is routed and whether the lower 32 bits of the address (the ones held in BD_TMPLTL) go untranslated through the target interface or if they use the interface's address translation windows. This applies only to PCI and PCIe interfaces. Since the BD templates should in almost all cases be stored in internal chip memory (MIF) this bitfield should almost always be set to zero. See the bitfield encoding descriptions for an explanation. 0x0: ADDR_XTLN (If address in PIF or XIF address space then address goes through translation in PIF or XIF) 0x1: PIF_NO_XTLN (Always maps to PIF and full 32-bits in CH_SADRL goes through PIF untranslated) 0x2: XIF_NO_XTLN (Always maps to XIF and full 32-bits in CH_SADRL goes through XIF untranslated) 0x3: RSVD (Reserved)

0x0A202088 WCSS_A_DXE_0_BDT_DAT_START**Type:** RW**Reset State:** 0x00000040

The BD template data start offset register holds the offset used to start writing payload data when the DXE is in H2B mode and the BD template feature has been enabled. In this mode, at the start of every new packet transfer, DXE will copy only enough bytes from the BD template to fill the space between byte 0 of the BD and the data start offset. Once this much of the BD template has been transferred, DXE will start reading data from the programmed source address and writing it into the BD starting at the programmed offset. Due to limitations of other modules in the chip, it's imperative that the value in this register allow for any header data to fit completely within the BD of the packet. That is, the total of adding the value contained in the BD template field "MPDU Header length" to the value in this register needs to be less than or equal to 127 in order to avoid problems. Further, it is necessary that the value of this register match the "MPDU Header Offset" field programmed in the BD template. DXE makes no attempt to check this is really the case.

WCSS_A_DXE_0_BDT_DAT_START

Bits	Name	Type	Description
31:7	RSVD	R	Reserved. Writes ignored. Reads always return zeroes.
6:0	OFFSET	RW	Data start offset.

0x0A202200 WCSS_A_DXE_0_COUNTER0**Type:** RW**Reset State:** 0x00000000

This register can be configured to count whole packets in the H2B or B2H direction or to count the number of transfers (descriptors) for H2H transactions. For now this counter is hardwired to always count all transfers completing on channel 0, but will eventually be able to count events happening on any channel and also only count transfers with the INC bit set in the CH_CSR (Not implemented yet)

The value gets cleared at reset and the counter can also be cleared or preset to any starting value by directly writing to it. As a convenience to the programmer the CTR_CLR register has been provided to quickly clear the value in one or more of the event counters.

WCSS_A_DXE_0_COUNTER0

Bits	Name	Type	Description
31:24	RSVD	R	
23:0	CNT	RW	

0x0A202204 WCSS_A_DXE_0_COUNTER1**Type:** RW**Reset State:** 0x00000000

This register can be configured to count whole packets in the H2B or B2H direction or to count the number of transfers (descriptors) for H2H transactions. For now this counter is hardwired to always count all transfers completing on channel 1, but will eventually be able to count events happening on any channel and also only count transfers with the INC bit set in the CH_CSR (Not implemented yet)

The value gets cleared at reset and the counter can also be cleared or preset to any starting value by directly writing to it. As a convenience to the programmer the CTR_CLR register has been provided to quickly clear the value in one or more of the event counters.

WCSS_A_DXE_0_COUNTER1

Bits	Name	Type	Description
31:24	RSVD	R	
23:0	CNT	RW	

0x0A202208 WCSS_A_DXE_0_COUNTER2**Type:** RW**Reset State:** 0x00000000

This register can be configured to count whole packets in the H2B or B2H direction or to count the number of transfers (descriptors) for H2H transactions. For now this counter is hardwired to always count all transfers completing on channel 2, but will eventually be able to count events happening on any channel and also only count transfers with the INC bit set in the CH_CSR (Not implemented yet)

The value gets cleared at reset and the counter can also be cleared or preset to any starting value by directly writing to it. As a convenience to the programmer the CTR_CLR register has been provided to quickly clear the value in one or more of the event counters.

WCSS_A_DXE_0_COUNTER2

Bits	Name	Type	Description
31:24	RSVD	R	
23:0	CNT	RW	

0x0A20220C WCSS_A_DXE_0_COUNTER3**Type:** RW**Reset State:** 0x00000000

This register can be configured to count whole packets in the H2B or B2H direction or to count the number of transfers (descriptors) for H2H transactions. For now this counter is hardwired to always count all transfers completing on channel 3, but will eventually be able to count events happening on any channel and also only count transfers with the INC bit set in the CH_CSR (Not implemented yet)

The value gets cleared at reset and the counter can also be cleared or preset to any starting value by directly writing to it. As a convenience to the programmer the CTR_CLR register has been provided to quickly clear the value in one or more of the event counters.

WCSS_A_DXE_0_COUNTER3

Bits	Name	Type	Description
31:24	RSVD	R	
23:0	CNT	RW	

0x0A202210 WCSS_A_DXE_0_COUNTER4**Type:** RW**Reset State:** 0x00000000

This register can be configured to count whole packets in the H2B or B2H direction or to count the number of transfers (descriptors) for H2H transactions. For now this counter is hardwired to always count all transfers completing on channel 4, but will eventually be able to count events happening on any channel and also only count transfers with the INC bit set in the CH_CSR (Not implemented yet)

The value gets cleared at reset and the counter can also be cleared or preset to any starting value by directly writing to it. As a convenience to the programmer the CTR_CLR register has been provided to quickly clear the value in one or more of the event counters.

WCSS_A_DXE_0_COUNTER4

Bits	Name	Type	Description
31:24	RSVD	R	
23:0	CNT	RW	

0x0A202214 WCSS_A_DXE_0_COUNTER5**Type:** RW**Reset State:** 0x00000000

This register can be configured to count whole packets in the H2B or B2H direction or to count the number of transfers (descriptors) for H2H transactions. For now this counter is hardwired to always count all transfers completing on channel 5, but will eventually be able to count events happening on any channel and also only count transfers with the INC bit set in the CH_CSR (Not implemented yet)

The value gets cleared at reset and the counter can also be cleared or preset to any starting value by directly writing to it. As a convenience to the programmer the CTR_CLR register has been provided to quickly clear the value in one or more of the event counters.

WCSS_A_DXE_0_COUNTER5

Bits	Name	Type	Description
31:24	RSVD	R	
23:0	CNT	RW	

0x0A202218 WCSS_A_DXE_0_COUNTER6**Type:** RW**Reset State:** 0x00000000

This register can be configured to count whole packets in the H2B or B2H direction or to count the number of transfers (descriptors) for H2H transactions. For now this counter is hardwired to always count all transfers completing on channel 6, but will eventually be able to count events happening on any channel and also only count transfers with the INC bit set in the CH_CSR (Not implemented yet)

The value gets cleared at reset and the counter can also be cleared or preset to any starting value by directly writing to it. As a convenience to the programmer the CTR_CLR register has been provided to quickly clear the value in one or more of the event counters.

WCSS_A_DXE_0_COUNTER6

Bits	Name	Type	Description
31:24	RSVD	R	
23:0	CNT	RW	

0x0A20221C WCSS_A_DXE_0_COUNTER7**Type:** RW**Reset State:** 0x00000000

This register can be configured to count whole packets in the H2B or B2H direction or to count the number of transfers (descriptors) for H2H transactions. For now this counter is hardwired to always count all transfers completing on channel 7, but will eventually be able to count events happening on any channel and also only count transfers with the INC bit set in the CH_CSR (Not implemented yet)

The value gets cleared at reset and the counter can also be cleared or preset to any starting value by directly writing to it. As a convenience to the programmer the CTR_CLR register has been provided to quickly clear the value in one or more of the event counters.

WCSS_A_DXE_0_COUNTER7

Bits	Name	Type	Description
31:24	RSVD	R	
23:0	CNT	RW	

0x0A202400 WCSS_A_DXE_0_CH0_CTRL**Type:** RW**Reset State:** 0x10000708

Serves to configure channel 0 operation

WCSS_A_DXE_0_CH0_CTRL

Bits	Name	Type	Description
31	SWAP	RW	Endian Byte Swap Enable. When high instructs DXE to perform an endian swap of the incoming byte stream. The corresponding bitfield in the DXE descriptor control word is located at DXE_DESC_CTRL[20]
30:29	BDT_IDX	RW	BD Template Index. When XTYPE=H2B and BDH=1, DXE supports creating a BD by using template data from one of up to 4 different BD templates. The BD template base store is pointed at by the DXE BD_TMPLT(H/L) registers and the address used to read the template data is calculated as follows: $BD_TMPLT_ADDR = BD_TMPLT_BASE + (128 * CH_CTRL[BDT_IDX])$ The corresponding bitfield in the control word of the external descriptor is mapped to bits[19:18]
28	DFMT	RW	Descriptor Format. This bitfield is used to select between long and short descriptor formats. When using the short descriptor format, the high part of the source, destination and descriptor addresses will be taken from the SADDRH_DFLT, DADDRH_DFLT or DESCH_DFLT registers respectively. Default is to use the long descriptor format 0x0: SHORT (Short descriptor format (5 dwords)) 0x1: LONG (Long descriptor format (8 dwords))
27	ABORT	RW	Writing a one to this bit will cause the channel to stop its current transfer and set the ERR bit and ERR_CODE fields in CH_STATUS. Note that if the channel is currently disabled or if it is about to be disabled because it has just finished its programmed transfer, then the abort request will be ignored and the error bit and error codes won't be set. If the channel is currently masked by the arbiter then setting this bit will forcibly unmask it so that the transfer can be aborted the next time the channel is selected by the internal arbiter. The abort request is automatically cleared by hw once the channel has aborted the transfer. Also note that since the DXE will not attempt to do any cleanup when a transfer is aborted, software might need to do so. For example, for H2B transfers, SW should manually release the BD and any PDUs currently in use by the partial transfer or else a BD/PDU leak will result.

WCSS_A_DXE_0_CH0_CTRL (cont.)

Bits	Name	Type	Description
26	ENDIANNESS	RW	Specifies the endianness of the master generating the data. Affects alignment of valid bytes for non-dword aligned reads/writes. The corresponding field in the DXE descriptor control word is located in DXE_DESC_CTRL[21] 0x0: BIGEND (System is Big-Endian) 0x1: LTLEND (System is Little-Endian)
25:22	CTR_SEL	RW	Counter Select. If the event counters are enabled (DMA_CSR[ECTR_EN]=1) then this bitfield selects the counter that will be incremented at the end of the transfer. By default the value in this bitfield is the same as the channel number so that after reset channel 0 increments counter 0, channel 1 increments counter 1 and so on.
21	EDVEN	RW	When set, the DMA engine will clear the VAL bit in the DESC_CSR word of the current descriptor once it has completed processing it. This is useful for implementing a descriptor "ring" (see the description of the VALID bit in Table 3.1)
20	EDEN	RW	External Descriptor Enable. Use External Descriptor Linked List
19	INE_DONE	RW	Enable Channel Interrupt when Channel is Done. The corresponding interrupt source bit is INT_DONE
18	INE_ERR	RW	Enable Channel Interrupt on Errors. The corresponding interrupt source bit is INT_ERR
17	INE_ED	RW	Enable External Descriptor Interrupt. When this bit is set, an interrupt will be generated after processing a descriptor item that has the INT bit set in its DESC_CTRL word (see Table 3.1). The corresponding interrupt source bit is INT_ED
16	STOP	R	Channel processing stopped due to a request in the last processed linked list descriptor (see the STOP bit in the definition of the DESC_CSR word in Table 3.1). This bit will be automatically cleared when the channel is restarted.
15:13	PRIO	RW	This field sets the channel priority. 7: Highest Priority. 0: Lowest Priority.

WCSS_A_DXE_0_CH0_CTRL (cont.)

Bits	Name	Type	Description
12:9	BTHLD_SEL	RW	BMU Threshold Select. This value tells DXE which of the BMU BD/PDU thresholds to check before issuing BMU commands that reserve, or get BDs or PDUs. It also specifies the module index value that will be used to issue any required BMU commands (see BMU spec for details). The value of this bitfield is only meaningful for transfer types that involve the BMU and therefore its value will be ignored for H2H transfers. 0x0: RSVD0 (Reserved) 0x1: RSVD1 (Reserved) 0x2: THLD2 (Use BD/PDU threshold 2) 0x3: THLD3 (Use BD/PDU threshold 3) 0x4: THLD4 (Use BD/PDU threshold 4) 0x5: THLD5 (Use BD/PDU threshold 5) 0x6: THLD6 (Use BD/PDU threshold 6) 0x7: THLD7 (Use BD/PDU threshold 7) 0x8: THLD8 (Use BD/PDU threshold 8) 0x9: THLD9 (Use BD/PDU threshold 9) 0xA: THLD10 (Use BD/PDU threshold 10) 0xB: RSVD11 (Reserved) 0xC: RSVD12 (Reserved) 0xD: RSVD13 (Reserved) 0xE: RSVD14 (Reserved) 0xF: RSVD15 (Reserved)
8	PDU_REL	RW	0: Don't release BD and PDUs when done. 1: Release BD and PDUs when done. 0x0: KEEP (Don) 0x1: RELEASE (Release BD and payload PDUs)

WCSS_A_DXE_0_CH0_CTRL (cont.)

Bits	Name	Type	Description
7	PIQ	RW	next descriptor Pointer address Is Queue. When set the source of the descriptor address is interpreted as a queue ID instead of just the address of a flat memory location. The only valid source queues for descriptors are the UIF Tx data and Tx management endpoints. Any other IDs will result the channel aborting the transfer with an invalid source queue error. Therefore, when this bit is set, SW must ensure that the CH_DESCCL[QID] bitfield is set to a valid queue ID and that the CH_DESCCL[BASE] bitfield is set to the base address of the source UIF endpoint. This base address is formed by taking the base address for the UIF block in the SAHB bus which can be found in the Nova address map document and then adding the internal UIF offset for the correct source endpoint. This offsets can be obtained from the UIF specification document. For convenience, the current internal UIF offsets are provided in the description of the QID bitfield in the CH_DESCCL register description. 0x0: FLAT (Source address maps to regular flat memory) 0x1: QUEUE (Source address maps to FIFO type memory (UIF Tx endpoints))
6	DIQ	RW	Destination Is Queue. When set the destination address is interpreted as a queue ID instead of just the address of a flat memory location. The queue could be either a BMU queue, one of the UIF RX endpoints or and EIF outbound FIFO. When this bit is set, SW must ensure that the DADR[QID] bitfield is set to a valid queue ID and that the DADR[BASE] bitfield is set to the base address of the destination queue for UIF or EIF, or to zero for a BMU queue. endpoint. 0x0: FLAT (Destination address maps to regular flat memory) 0x1: QUEUE (Destination address maps to a BMU queue or FIFO type memory)
5	SIQ	RW	Source Is Queue. When set the source address is interpreted as a queue ID instead of just the address of a flat memory location. The queue could be either a BMU queue, one of the UIF TX endpoints or and EIF inbound FIFO. When this bit is set, SW must ensure that the SADR[QID] bitfield is set to a valid queue ID and that the SADR[BASE] bitfield is set to the base address of the source queue for UIF or EIF, or to zero for a BMU queue. endpoint. 0x0: FLAT (Source address maps to regular flat memory) 0x1: QUEUE (Source address maps to a BMU queue or FIFO type memory)

WCSS_A_DXE_0_CH0_CTRL (cont.)

Bits	Name	Type	Description
4	BDH	RW	BD Handling bit. The behavior depends on the setting of XTYPE. If XTYPE is: H2B: Then if BDH is: 0: Use first 128 bytes of packet as BD. 1: Create BD from BD template (Up to 4 supported. Also see description for BDT_IDX). B2H: Then if BDH is: 0: Insert BD in front of payload. 1: Discard BD. For XTYPE values of H2H and B2B the value of this bit is ignored.
3	EOP	RW	End of Packet bit. For H2B transfers marks the end of a packet Note that when not using external descriptors DXE cannot support the gather functionality and therefore the EOP bit must be set. For XTYPE values of H2H, B2B and B2H the value of this bit is ignored.
2:1	XTYPE	RW	Transfer Type. Specifies the type of transfer to perform: 00: H2H. Host space to Host space transfer. 01: B2B. BMU space to BMU space transfer. 10: H2B. Host space to BMU translation. 11: B2H. BMU space to Host space translation. For cross-space transfers (H2B and B2H) the DMA engine will also perform the required data format translation. The format translation behavior can be modified by the setting of the BDH and PAD bits. 0x0: H2H (Host space to Host space transfer) 0x1: B2B (BMU space to BMU space transfer) 0x2: H2B (Host space to BMU space transfer) 0x3: B2H (BMU space to Host space transfer)

WCSS_A_DXE_0_CH0_CTRL (cont.)

Bits	Name	Type	Description
0	EN	RW	Channel Enable or Restart. Writing this bit with a one either starts a new DMA transfer according to the parameters programmed in the channel's configuration registers and clears the DONE bit or, if the channel is already enabled but the STOP bit is set, then it restarts channel processing and also clears the STOP bit. This bit is automatically cleared when a single transfer completes or, if the channel is configured to use linked list descriptors, when the last transfer in the descriptor linked list completes (DESC_NXT is NULL or/and DESC_CSR[VALID]=0). The channel is also disabled and this bit cleared if a channel error is encountered. Reading this bit indicates whether a channel is currently enabled or disabled: 0: Channel Disabled. 1: Channel Enabled. The Channel Enable bit status can also be found by reading the DMA_ENCH register which aggregates the EN bits of all channels. Note: Software must take care to not disable the channel while it is busy or data could be lost.

0x0A202404 WCSS_A_DXE_0_CH0_STATUS**Type:** RW**Reset State:** 0x00000000

Serves to report the status of the programmed channel operation.

WCSS_A_DXE_0_CH0_STATUS

Bits	Name	Type	Description
31:16	RSVD	R	Reserved. Reads always return 0. Writes are ignored.
15	INT_DONE	R	Interrupt Source: Asserted when the channel has finished all processing.
14	INT_ERR	R	Interrupt Source: Asserted when a channel error has occurred. Masked by the INE_ERR
13	INT_ED	R	Interrupt Source. Descriptor Done and INT requested
12	ABORT_REQ	R	Abort Request. Software has requested that the current operation be aborted. However, if CH_STATUS[EN] is still set then the channel has registered the request but is not yet in a position to abort the transfer "safely". Once the channel finishes its current operation it will abort the transfer and set the error flag and error code bitfields accordingly.

WCSS_A_DXE_0_CH0_STATUS (cont.)

Bits	Name	Type	Description
11	STOP_REQ	R	Stop Request. Software has requested that the channel stop its operation after the current descriptor has been processed. Once this happens the STOPD bitfield will be set to one and the channel will be disabled.
10:6	ERR_CODE	R	Used to identify the source or cause of an error. 0x0: NONE (No Errors) 0x1: SAHB_ERR (DXE received Error response while performing a transfer on the SAHB) 0x2: H2H_RD_BUS_ERR (SAHB bus error encountered during H2H read operation) 0x3: H2H_WR_BUS_ERR (SAHB bus error encountered during H2H write operation) 0x4: PRG_INV_XTYPE (B2B not yet supported) 0x5: BERR_POPWQ (BMU error on POP WQ command) 0x6: BERR_PUSHWQ (BMU error on PUSH WQ command) 0x7: BERR_RLSS (BMU error on Release BD and attached PDUs command) 0x8: BERR_GETPDU (BMU error on Get PDU command) 0x9: PRG_INV_WQ (Programming error: Unsupported BMU Work Queue) 0xA: PRG_INV_H2H_SRC_QID (Programming error: Invalid H2H source queue ID) 0xB: PRG_INV_H2H_DST_QID (Programming error: Invalid H2H destination queue ID) 0xC: PRG_INV_B2H_SRC_QID (Programming error: Invalid B2H BMU source work queue ID) 0xD: PRG_INV_B2H_DST_QID (Programming error: Invalid B2H destination queue ID) 0xE: PRG_INV_B2H_SRC_IDX (Programming error: Invalid B2H source BD index) 0xF: PRG_INV_H2B_SRC_QID (Programming error: Invalid H2B source queue ID) 0x10: PRG_INV_H2B_DST_QID (Programming error: Invalid H2B BMU destination work queue ID) 0x11: PRG_INV_H2B_DST_IDX (Programming error: Invalid H2B destination BD index) 0x12: PRG_INV_H2B_SZ (Programming error: Invalid H2B transfer size of 0 when BDH=1) 0x13: PRG_INV_SADR (Programming error: Invalid Source Address) 0x14: PRG_INV_DADR (Programming error: Invalid Destination Address) 0x15: PRG_INV_EDADR (Programming error: Invalid Descriptor Address) 0x16: PRG_INV_SRC_WQID (Programming error: Invalid Destination WQ ID)

WCSS_A_DXE_0_CH0_STATUS (cont.)

Bits	Name	Type	Description
			0x17: PRG_INV_DST_WQID (Programming error: Invalid Destination WQ ID) 0x18: PRG_XTYPE_MSM TCH (Programming error: XTYPE mismatch during H2B gather transfer) 0x19: PKT_ERR (Malformed packet) 0x1A: ABORT (Aborted transfer due to SW request) 0x1B: PDU_CNT_OVFL (PDU count field overflow)
5	ERR	RW	DMA channel stopped due to error. To clear the error status it is necessary to write a 1 into this bit position.
4	STOPD	R	Channel Stopped. The channel has stopped further processing and disabled itself due to a stop request in the last processed descriptor.
3	MSKD	R	Channel Masked. The channel has been automatically masked out from arbitration due to a blocking condition.
2	DONE	R	DMA channel Done. This bit will be automatically cleared the next time the channel is enabled.
1	BUSY	R	Channel busy. High when the channel is enabled and currently active (won internal arbitration)
0	EN	R	Channel Enabled. Copy of CH_CTRL[EN].

0x0A202408 WCSS_A_DXE_0_CH0_SZ**Type:** RW**Reset State:** 0x00000000

The transfer size register specifies the total, remaining, and burst or 'chunk' transfer sizes for each channel. The maximum total transfer size is 16K-1 bytes. The chunk size control can be useful to control the relative bandwidth used by each channel as the dma engine will change channels, release the AHB bus, and re-arbitrate for access after transferring chunk size bytes.

WCSS_A_DXE_0_CH0_SZ

Bits	Name	Type	Description
31:28	CHK_SZ	RW	Chunk transfer size. Specifies the number of bytes (in multiples of 8) to be transferred at one given time (not implying they are buffered but that they will be transferred for each start event in one bus request cycle). If chunk size is zero the DMA engine will always perform maximum chunk size transfers. Maximum chunk size is 128 bytes.
27:14	REM_SZ	R	Remaining transfer size. Specifies the number of bytes that remain to be transferred to complete the current operation.

WCSS_A_DXE_0_CH0_SZ (cont.)

Bits	Name	Type	Description
13:0	TOT_SZ	RW	Total Transfer Size. Specifies the number of bytes to be transferred. Maximum total transfer size is (4K-1) bytes. Note: This field is unnecessary and will be ignored for B2H or B2B transfers, when the source of the transfer is in BMU space (either BMU queue or direct pointer to BD). This is because the size of the transfer will be the packet length encoded in the BD at offset 0x0c bits [11:0].

0x0A20240C WCSS_A_DXE_0_CH0_SADRL**Type:** RW**Reset State:** 0x00000000

The channel source address register specifies either the source address in flat internal or external memory, or else the BMU queue ID, UIF TX endpoint ID, or EIF inbound FIFO ID for the transfer. The register must be programmed directly by software before the channel is enabled, then, once the DMA channel is enabled, the register is updated after each atomic chunk transfer (read+write) with the number of bytes transferred. (except when using a UIF/EIF queue) and also after loading a new descriptor. When using flat memory as the source, addresses can have arbitrary byte-alignment. When using a BMU queue, UIF endpoint or EIF fifo as a source, the software is responsible for programming a valid queue ID and correct base address for the queue. The layout of bitfields in this register depends on the setting of the CH_CSR[SIQ] bit. When CH_CSR[SIQ] is clear then the register has no bitfields and the full 32-bits are interpreted as the physical address of a location in flat memory, and when it is set this register has two bitfields: The SQID bitfield used to specify the queue ID, and the BASE bitfield used to provide the base address for the UIF or EIF source endpoint/FIFO.

For B2H or B2B transfers, when the source is a BMU Queue ID (CH_CSR[SIQ] = 1 and SWQID matches BMU), the DXE will issue a BMU POPWQ command to pop the specified queue and obtain the BD index of the source packet. Else, when the source is an address (CH_CSR[SIQ] = 0), the DXE will interpret the address as the index of the packet BD and will add it to the BMU_BASE to obtain the physical address to use to fetch it directly from memory.

For H2B or H2H transfers, if the source is a UIF endpoint or EIF FIFO (CH_CSR[SIQ] = 1 and SQID matches UIF/EIF), SW should program the SAHB base address of the UIF source endpoint in the BASE bitfield and the queue ID to match in the SWQID bitfield, then the DXE will use the base address given to generate the physical address and read the data over the SAHB bus and also use the appropriate sideband signals as specified by the SQID field to communicate with the selected UIF endpoint. SW should make sure that the BASE and SQID fields are consistent and refer to the same endpoint or else hangs or unpredictable behavior might occur. Else, when the source is just a location in flat memory space, (CH_CSR[SIQ] = 0), the bitfields lose their special meaning and DXE will just use the full 32-bits programmed into the register as the physical address to use to fetch the data. For this case, SW should make sure that the provided address maps to the intended source interface (Internal memory, USB, PCI, PCIe) by conforming to the address map specified for the SAHB bus.

To summarize, for B2H and B2B transfers the address must conform to the following format:

For index of BD mode (CH_CSR[SIQ] = 0):

ADDR[31:7] = BMU_BASE + BD_INDEX[15:0]

ADDR[6:0] = 7'd0

For BMU Queue ID mode (CH_CSR[SIQ] = 1 & SQID matching one of the BMU queues):

ADDR[31:7] = Any value will do but zero preferred

ADDR[6:0] = Any BMU Queue ID as specified in description of SQID field

For H2B and H2H transfers the address must conform to the following format:

For sources in flat memory space (CH_CSR[SIQ] = 0):

ADDR[31:0] = address mapping to source interface

For UIF RX endpoints or EIF FIFOs (CH_CSR[SIQ] = 1 & QID matching UIF or EIF):

ADDR[31:7] = UIF RX endpoint base or EIF inbound FIFO base

ADDR[6:0] = Queue ID matching UIF or EIF as specified in description of SQID field

WCSS_A_DXE_0_CH0_SADRL

Bits	Name	Description
31:7	BASE	Base Address. Valid only when the CH_CSR[SIQ] bit is set. For this case, this bitfield provides the base address of the source UIF TX endpoint or EIF inbound FIFO. Consult the appropriate interface uarch for the offset within the interface's address space. When CH_CSR[SIQ] bit is not set this field loses its special meaning and is interpreted only as part of the full 32-bit source address.

WCSS_A_DXE_0_CH0_SADRL (cont.)

Bits	Name	Description
6:0	SQID	Source Queue ID. Valid only when the CH_CSR[SIQ] bit is set. For this case, this bitfield provides the ID for the source BMU, UIF or EIF queue to be used for the transfer. When CH_CSR[SIQ] bit is not set this field loses its special meaning and is interpreted only as part of the full 32-bit source address. 0x0: UIF_TX_DATA_EP (UIF TX Data Endpoint) 0x1: UIF_TX_MGMT_EP (UIF TX Management Endpoint) 0x2: BMU_2 (BMU Work Queue 2) 0x3: BMU_3 (BMU Work Queue 3) 0x4: BMU_4 (BMU Work Queue 4) 0x5: BMU_5 (BMU Work Queue 5) 0x6: BMU_6 (BMU Work Queue 6) 0x7: BMU_7 (BMU Work Queue 7) 0x8: BMU_8 (BMU Work Queue 8) 0x9: BMU_9 (BMU Work Queue 9) 0xA: BMU_10 (BMU Work Queue 10) 0xB: BMU_11 (BMU Work Queue 11) 0xC: BMU_12 (BMU Work Queue 12) 0xD: BMU_13 (BMU Work Queue 13) 0xE: BMU_14 (BMU Work Queue 14) 0xF: BMU_15 (BMU Work Queue 15) 0x10: BMU_16 (BMU Work Queue 16) 0x11: BMU_17 (BMU Work Queue 17) 0x12: BMU_18 (BMU Work Queue 18) 0x13: BMU_19 (BMU Work Queue 19) 0x14: BMU_20 (BMU Work Queue 20) 0x15: BMU_21 (BMU Work Queue 21) 0x16: BMU_22 (BMU Work Queue 22) 0x17: BMU_23 (BMU Work Queue 23) 0x18: BMU_24 (BMU Work Queue 24) 0x19: RSVD_25 (Reserved) 0x1A: RSVD_26 (Reserved) 0x1B: RSVD_27 (Reserved) 0x1C: RSVD_28 (Reserved) 0x1D: RSVD_29 (Reserved) 0x1E: RSVD_30 (Reserved) 0x1F: RSVD_31 (Reserved)

0x0A202410 WCSS_A_DXE_0_CH0_SADRH**Type:** RW**Reset State:** 0x00000000

The channel source address high register holds the high bits of the source address for a transfer. Refer to the description of CH_SADRL for more info.

WCSS_A_DXE_0_CH0_SADRH

Bits	Name	Type	Description
31:2	RSVD	R	Reserved. Writes ignored. Reads always return zeroes.
1:0	ADDRH	RW	For taurus, this register holds bits 33 and 32 of the address. With the current address mapping these high order bits determine where the transfer is routed and whether the lower 32 bits of the address (the ones held in CH_SADRL) go untranslated through PIF or XIF or if they use the PIF and XIF address translation windows. See the bitfield encoding descriptions for an explanation. 0x0: ADDR_XTLN (If address in PIF or XIF address space then address goes through translation in PIF or XIF) 0x1: PIF_NO_XTLN (Always maps to PIF and full 32-bits in CH_SADRL goes through PIF untranslated) 0x2: XIF_NO_XTLN (Always maps to XIF and full 32-bits in CH_SADRL goes through XIF untranslated) 0x3: RSVD (Reserved)

0x0A202414 WCSS_A_DXE_0_CH0_DADRL**Type:** RW**Reset State:** 0x00000000

The channel destination address register specifies either the destination address in flat internal or external memory, or else the BMU queue ID, UIF RX endpoint ID, or EIF outbound FIFO ID for the transfer. The register must be programmed directly by software before the channel is enabled, then, once the DMA channel is enabled, the register is updated after each atomic chunk transfer (read+write) with the number of bytes transferred. (except when using a UIF/EIF queue) and also after loading a new descriptor. When using flat memory as the destination, addresses can have arbitrary byte-alignment. When using a BMU queue, UIF endpoint or EIF fifo as a destination, the software is responsible for programming a valid queue ID and correct base address for the queue. The layout of bitfields in this register depends on the setting of the CH_CSR[DIQ] bit. When this bit is clear then the register has no bitfields and the full 32-bits are interpreted as the physical address of a location in flat memory, and when the bit is set this register has two bitfields: The DQID bitfield used to specify the queue ID, and the BASE bitfield used to provide the base address for the UIF or EIF destination endpoint/FIFO.

For H2B or B2B transfers, when using a BMU queue as the destination (CH_CSR[DIQ] = 1 and DQID matches BMU), the DXE will push the index of the BD of the packet into the specified queue at the end of the packet transfer. Therefore, software is responsible for programming a valid BMU Queue ID.

Else, when the destination is an index (CH_CSR[DIQ] = 0), the BD will be written to the address obtained by adding the provided index to the BMU_BASE, again, the software must ensure that said address is legal and known to BMU. That is, software must preallocate the BD (get the BD from BMU via GET BD/PDU cmd) for the BD before starting the transfer. For B2H or H2H transfers, if the destination is a UIF endpoint or EIF FIFO (CH_CSR[DIQ] = 1 and DQID matches UIF/EIF), SW should program the SAHB base address of the UIF destination endpoint in the

BASE bitfield and the queue ID to match in the DQID bitfield, then the DXE will use the base address given to generate the physical address and write the data over the SAHB bus and also use the appropriate sideband signals as specified by the DQID field to communicate with the selected UIF endpoint. SW should make sure that the BASE and DQID fields are consistent and refer to the same endpoint else hangs or unpredictable behavior might occur. On the other hand, when the destination is just a location in flat memory space, (CH_CSR[DIQ] = 0), be it internal or external, the bitfields lose their special meaning and DXE will just use the full 32-bits programmed into the register as the address to write the data. For this case, SW should make sure that the address maps to the intended destination interface (Internal memory, USB, PCI, PCIe) by conforming to the address map specified for the SAHB bus. To summarize, for H2B and B2B transfers the address must conform to the following format:

For index of BD mode (CH_CSR[DIQ] = 0):

ADDR[31:7] = BMU_BASE + BD_INDEX[15:0]

ADDR[6:0] = 7'd0

For BMU Queue ID mode (CH_CSR[DIQ] = 1 & DQID matching one of the BMU queues):

ADDR[31:5] = Any value will do but zero preferred

ADDR[4:0] = Any BMU Queue ID as specified in description of QID field

For B2H and H2H transfers the address must conform to the following format:

For destinations in flat memory space (CH_CSR[DIQ] = 0): ADDR[31:0] = address mapping to destination interface

For UIF RX endpoints or EIF FIFOs (CH_CSR[DIQ] = 1 & DQID matching UIF or EIF):

ADDR[31:5] = UIF RX endpoint base or EIF inbound FIFO base

ADDR[4:0] = Queue ID matching UIF or EIF as specified in description of DQID field

WCSS_A_DXE_0_CH0_DADRL

Bits	Name	Description
31:7	BASE	Base Address. Valid only when the CH_CSR[DIQ] bit is set. For this case, this bitfield provides the base address of the destination UIF RX endpoint or EIF outbound FIFO (for BMU SW may use any value can be used but zero is suggested). Consult the appropriate interface uarch for the offset within the interface's address space. When CH_CSR[SIQ] bit is not set this field loses its special meaning and is interpreted only as part of the full 32-bit source address.

WCSS_A_DXE_0_CH0_DADRL (cont.)

Bits	Name	Description
6:0	DQID	Destination Queue ID. Valid only when the CH_CSR[SIQ] bit is set. For this case, this bitfield provides the ID for the destination BMU WQ, UIF RX endpoint or EIF outbound queue to be used for the transfer. When CH_CSR[SIQ] bit is not set this field loses its special meaning and is interpreted only as part of the full 32-bit source address. 0x0: RSVD_0 (Reserved) 0x1: RSVD_1 (Reserved) 0x2: BMU_2 (BMU Work Queue 2) 0x3: BMU_3 (BMU Work Queue 3) 0x4: BMU_4 (BMU Work Queue 4) 0x5: BMU_5 (BMU Work Queue 5) 0x6: BMU_6 (BMU Work Queue 6) 0x7: BMU_7 (BMU Work Queue 7) 0x8: BMU_8 (BMU Work Queue 8) 0x9: BMU_9 (BMU Work Queue 9) 0xA: BMU_10 (BMU Work Queue 10) 0xB: BMU_11 (BMU Work Queue 11) 0xC: BMU_12 (BMU Work Queue 12) 0xD: BMU_13 (BMU Work Queue 13) 0xE: BMU_14 (BMU Work Queue 14) 0xF: BMU_15 (BMU Work Queue 15) 0x10: BMU_16 (BMU Work Queue 16) 0x11: BMU_17 (BMU Work Queue 17) 0x12: BMU_18 (BMU Work Queue 18) 0x13: BMU_19 (BMU Work Queue 19) 0x14: BMU_20 (BMU Work Queue 20) 0x15: BMU_21 (BMU Work Queue 21) 0x16: BMU_22 (BMU Work Queue 22) 0x17: BMU_23 (BMU Work Queue 23) 0x18: BMU_24 (BMU Work Queue 24) 0x19: BMU_25 (BMU Work Queue 25) 0x1A: BMU_26 (BMU Work Queue 26) 0x1B: BMU_SINK_WQ (BMU_SINK (WQ255)) 0x1C: UIF_RX_1_EP (UIF RX 1 endpoint) 0x1D: UIF_RX_2_EP (UIF RX 2 endpoint) 0x1E: UIF_RX_3_EP (UIF RX 3 endpoint) 0x1F: UIF_RX_4_EP (UIF RX 4 endpoint)

0x0A202418 WCSS_A_DXE_0_CH0_DADRH**Type:** RW**Reset State:** 0x00000000

The channel destination address high register holds the high bits of the destination address for a transfer. See the description of CH_DADRL for more info.

WCSS_A_DXE_0_CH0_DADRH

Bits	Name	Type	Description
31:2	RSVD	R	Reserved. Writes ignored. Reads always return zeroes.
1:0	ADDRH	RW	This register holds bits 33 and 32 of the address. With the current address mapping these high order bits determine where the transfer is routed and whether the lower 32 bits of the address (the ones held in CH_DADRL) go untranslated through PIF or XIF or if they use the PIF and XIF address translation windows. See the bitfield encoding descriptions for an explanation. 0x0: ADDR_XTLN (If address in PIF or XIF address space then address goes through translation in PIF or XIF) 0x1: PIF_NO_XTLN (Always maps to PIF and full 32-bits in CH_DADRL goes through PIF untranslated) 0x2: XIF_NO_XTLN (Always maps to XIF and full 32-bits in CH_DADRL goes through XIF untranslated) 0x3: RSVD (Reserved. Maps to default slave. Reads return error, writes are ignored)

0x0A20241C WCSS_A_DXE_0_CH0_DESCL

Type: RW

Reset State: 0x00000000

The Channel 0 Linked List Descriptor Pointer register contains the 32 lower bits of the address of the next descriptor to be loaded in the linked list. This address must be dword (32-bit) aligned (addr[1:0]=0 if CH_CTRL[PIQ]=0). Each complete transfer descriptor consists of 7 32-bit values arranged contiguously in memory. The value of this register will be overwritten with the address of the start of the next descriptor set in the list after the current descriptor has been fetched. The meaning of the value in this register depends on the value of the CH_CTRL[PIQ] bitfield similar to the way in which SADRL and DADRL are depending in SIQ and DIQ respectively. For CH_CTRL[PIQ]=0 then this register has no bitfields and the value is just interpreted as an address in flat memory space. When CH_CTRL[PIQ] is set then the descriptor will be read from FIFO type memory in UIF and the register has 2 bitfields: BASE, and SQID. BASE specifies the base address of the UIF Tx queue (derived from the start of the UIF address space in SAHB plus the internal UIF offset).

WCSS_A_DXE_0_CH0_DESCL

Bits	Name	Description
31:7	BASE	This bitfield only exists and has meaning when the descriptor needs to be read from a UIF TX queue (and therefore CH_CTRL[PIQ]=1), then it serves to specify the base address of the UIF endpoint to use as the source of the descriptor. Refer to the Nova address map document and to the UIF architecture document for guaranteed correct values to use here. For convenience but without a guarantee of accuracy, the base value to use for both the Tx data and management endpoints is the following: base = UIF base + TD_and_TM_offset = 0x0E02_5000 + x100 = 0x0E02_5100 >> 7 = 0x1C_04A2
6:0	SQID	This bitfield only exists and has meaning when the descriptor needs to be read from a UIF TX queue (and therefore CH_CTRL[PIQ]=1), then it serves to specify which of the two UIF TX queues to use. Note that in this case SW also needs to program the correct base address of the source UIF endpoint in the BASE bitfield. 0x0: UIF_TX_DATA_EP (Descriptor to be read from UIF TX Data Endpoint) 0x1: UIF_TX_MGMT_EP (Descriptor to be read from UIF TX Management Endpoint)

0x0A202420 WCSS_A_DXE_0_CH0_DESCH**Type:** RW**Reset State:** 0x00000000

The channel destination address high register holds the high bits of the the address of the next descriptor to be loaded in the linked list.

WCSS_A_DXE_0_CH0_DESCH

Bits	Name	Type	Description
31:2	RSVD	R	Reserved. Writes ignored. Reads always return zeroes.

WCSS_A_DXE_0_CH0_DESCH (cont.)

Bits	Name	Type	Description
1:0	ADDRH	RW	For taurus, this register holds bits 33 and 32 of the address. With the current address mapping these high order bits determine where the transfer is routed and whether the lower 32 bits of the address (the ones held in CH_DADRL) go untranslated through PIF or XIF or if they use the PIF and XIF address translation windows. See the bitfield encoding descriptions for an explanation. Note that if the source of the descriptor is either of the UIF endpoints (CH_CTRL[PIQ]=1) then the value in ADDRH has to be 0 or else the descriptor read could end up mapping to the XIF or PIF. 0x0: ADDR_XTLN (If address in PIF or XIF address space then address goes through translation in PIF or XIF) 0x1: PIF_NO_XTLN (Always maps to PIF and full 32-bits in CH_DADRL goes through PIF untranslated) 0x2: XIF_NO_XTLN (Always maps to XIF and full 32-bits in CH_DADRL goes through XIF untranslated) 0x3: RSVD (Reserved. Maps to default slave. Reads return error, writes are ignored)

0x0A202424 WCSS_A_DXE_0_CH0_LST_DESCL**Type:** R**Reset State:** 0x00000000

The read-only Channel 0 Current/last Linked List Descriptor Pointer register contains the low 32-bits of the address of the descriptor currently being processed or of the last descriptor processed if the channel is disabled.

The meaning of the value and bitfields with regards to CH_CTRL[PIQ] is the same as was explained for the CH_DESCL register.

WCSS_A_DXE_0_CH0_LST_DESCL

Bits	Name	Description
31:7	BASE	See description of CH_DESCL[BASE].
6:0	SQID	See description of CH_DESCL[SQID]. 0x0: UIF_TX_DATA_EP (Descriptor to be read from UIF TX Data Endpoint) 0x1: UIF_TX_MGMT_EP (Descriptor to be read from UIF TX Management Endpoint)

0x0A202428 WCSS_A_DXE_0_CH0_LST_DESCH**Type:** R**Reset State:** 0x00000000

The channel destination address high register holds the high bits of the the address of the next descriptor to be loaded in the linked list. Note that The meaning of the value and bitfields is the same as was explained for the CH_DESCL register.

WCSS_A_DXE_0_CH0_LST_DESCH

Bits	Name	Description
31:2	RSVD	Reserved. Writes ignored. Reads always return zeroes.
1:0	ADDRH	For taurus, this register holds bits 33 and 32 of the address.

0x0A20242C WCSS_A_DXE_0_CH0_BD**Type:** R**Reset State:** 0x00000000**WCSS_A_DXE_0_CH0_BD**

Bits	Name	Description
31:24	RSVD	Reserved. Always returns 0 when read.
23	VAL	Valid. BD has been loaded and is valid.
22:16	PDU_CNT	PDU count of packet.
15:0	IDX	Index of BD. To get actual physical address must be added to BD/PDU base.

0x0A202430 WCSS_A_DXE_0_CH0_HEAD**Type:** R**Reset State:** 0x00000000

Address of head of packet.

WCSS_A_DXE_0_CH0_HEAD

Bits	Name	Description
31:27	RSVD	Reserved.
26	DESCVAL	Descriptor valid. Signifies descriptor has already been loaded. Used to prevent reloading of a descriptor.

WCSS_A_DXE_0_CH0_HEAD (cont.)

Bits	Name	Description
25:21	DWQID	Destination WQ ID. Original BMU, UIF or EIF destination queue. Valid only when CH_CSR[DIQ]=1
20:16	SWQID	Source WQ ID. Original BMU, UIF or EIF source queue. Valid only when CH_CSR[SIQ]=1
15:0	IDX	Index of Head. Must be added to BD/PDU base for actual physical address.

0x0A202434 WCSS_A_DXE_0_CH0_TAIL**Type:** R**Reset State:** 0x00000000

Address of tail of packet.

WCSS_A_DXE_0_CH0_TAIL

Bits	Name	Description
31:16	RSVD	Reserved.
15:0	IDX	Index of Tail. Must be added to BD/PDU base for actual physical address

0x0A202438 WCSS_A_DXE_0_CH0_PDU**Type:** R**Reset State:** 0x00000000

PDU info register.

All its fields are working fields that get modified as the programmed operation progresses.

WCSS_A_DXE_0_CH0_PDU

Bits	Name	Description
31	RSVD	Reserved. Always returns 0 when read.
30:24	PDU_CNT	Current PDU count.
23:8	IDX	Index of current PDU. Must be added to BD/PDU base for actual physical address
7:0	FST_OFF	First PDU offset.

0x0A20243C WCSS_A_DXE_0_CH0_TSTMP**Type:** R**Reset State:** 0x00000000

Contains last sampled value of the system timestamp

WCSS_A_DXE_0_CH0_TSTMP

Bits	Name	Description
31:0	VALUE	

0x0A202440 WCSS_A_DXE_0_CH1_CTRL**Type:** RW**Reset State:** 0x10400708

Serves to configure channel 1 operation

WCSS_A_DXE_0_CH1_CTRL

Bits	Name	Type	Description
31	SWAP	RW	Endian Byte Swap Enable. When high instructs DXE to perform an endian swap of the incoming byte stream. The corresponding bitfield in the DXE descriptor control word is located at DXE_DESC_CTRL[20]
30:29	BDT_IDX	RW	BD Template Index. When XTYPE=H2B and BDH=1, DXE supports creating a BD by using template data from one of up to 4 different BD templates. The BD template base store is pointed at by the DXE BD_TMPLT(H/L) registers and the address used to read the template data is calculated as follows: $BD_TMPLT_ADDR = BD_TMPLT_BASE + (128 * CH_CTRL[BDT_IDX])$ The corresponding bitfield in the control word of the external descriptor is mapped to bits[19:18]
28	DFMT	RW	Descriptor Format. This bitfield is used to select between long and short descriptor formats. When using the short descriptor format, the high part of the source, destination and descriptor addresses will be taken from the SADDRH_DFLT, DADDRH_DFLT or DESCH_DFLT registers respectively. Default is to use the long descriptor format 0x0: SHORT (Short descriptor format (5 dwords)) 0x1: LONG (Long descriptor format (8 dwords))

WCSS_A_DXE_0_CH1_CTRL (cont.)

Bits	Name	Type	Description
27	ABORT	RW	Writing a one to this bit will cause the channel to stop its current transfer and set the ERR bit and ERR_CODE fields in CH_STATUS. Note that if the channel is currently disabled or if it is about to be disabled because it has just finished its programmed transfer, then the abort request will be ignored and the error bit and error codes won't be set. If the channel is currently masked by the arbiter then setting this bit will forcibly unmask it so that the transfer can be aborted the next time the channel is selected by the internal arbiter. The abort request is automatically cleared by hw once the channel has aborted the transfer. Also note that since the DXE will not attempt to do any cleanup when a transfer is aborted, software might need to do so. For example, for H2B transfers, SW should manually release the BD and any PDUs currently in use by the partial transfer or else a BD/PDU leak will result.
26	ENDIANNESS	RW	Specifies the endianness of the master generating the data. Affects alignment of valid bytes for non-dword aligned reads/writes. The corresponding field in the DXE descriptor control word is located in DXE_DESC_CTRL[21] 0x0: BIGEND (System is Big-Endian) 0x1: LTLEND (System is Little-Endian)
25:22	CTR_SEL	RW	Counter Select. If the event counters are enabled (DMA_CSR[ECTR_EN]=1) then this bitfield selects the counter that will be incremented at the end of the transfer. By default the value in this bitfield is the same as the channel number so that after reset channel 0 increments counter 0, channel 1 increments counter 1 and so on.
21	EDVEN	RW	When set, the DMA engine will clear the VAL bit in the DESC_CSR word of the current descriptor once it has completed processing it. This is useful for implementing a descriptor "ring" (see the description of the VALID bit in Table 3.1)
20	EDEN	RW	External Descriptor Enable. Use External Descriptor Linked List
19	INE_DONE	RW	Enable Channel Interrupt when Channel is Done. The corresponding interrupt source bit is INT_DONE
18	INE_ERR	RW	Enable Channel Interrupt on Errors. The corresponding interrupt source bit is INT_ERR
17	INE_ED	RW	Enable External Descriptor Interrupt. When this bit is set, an interrupt will be generated after processing a descriptor item that has the INT bit set in its DESC_CTRL word (see Table 3.1). The corresponding interrupt source bit is INT_ED

WCSS_A_DXE_0_CH1_CTRL (cont.)

Bits	Name	Type	Description
16	STOP	R	Channel processing stopped due to a request in the last processed linked list descriptor (see the STOP bit in the definition of the DESC_CSR word in Table 3.1). This bit will be automatically cleared when the channel is restarted.
15:13	PRIO	RW	This field sets the channel priority. 7: Highest Priority. 0: Lowest Priority.
12:9	BTHLD_SEL	RW	BMU Threshold Select. This value tells DXE which of the BMU BD/PDU thresholds to check before issuing BMU commands that reserve, or get BDs or PDUs. It also specifies the module index value that will be used to issue any required BMU commands (see BMU spec for details). The value of this bitfield is only meaningful for transfer types that involve the BMU and therefore its value will be ignored for H2H transfers. 0x0: RSVD0 (Reserved) 0x1: RSVD1 (Reserved) 0x2: THLD2 (Use BD/PDU threshold 2) 0x3: THLD3 (Use BD/PDU threshold 3) 0x4: THLD4 (Use BD/PDU threshold 4) 0x5: THLD5 (Use BD/PDU threshold 5) 0x6: THLD6 (Use BD/PDU threshold 6) 0x7: THLD7 (Use BD/PDU threshold 7) 0x8: THLD8 (Use BD/PDU threshold 8) 0x9: THLD9 (Use BD/PDU threshold 9) 0xA: THLD10 (Use BD/PDU threshold 10) 0xB: RSVD11 (Reserved) 0xC: RSVD12 (Reserved) 0xD: RSVD13 (Reserved) 0xE: RSVD14 (Reserved) 0xF: RSVD15 (Reserved)
8	PDU_REL	RW	0: Don't release BD and PDUs when done. 1: Release BD and PDUs when done. 0x0: KEEP (Don) 0x1: RELEASE (Release BD and payload PDUs)

WCSS_A_DXE_0_CH1_CTRL (cont.)

Bits	Name	Type	Description
7	PIQ	RW	next descriptor Pointer address Is Queue. When set the source of the descriptor address is interpreted as a queue ID instead of just the address of a flat memory location. The only valid source queues for descriptors are the UIF Tx data and Tx management endpoints. Any other IDs will result the channel aborting the transfer with an invalid source queue error. Therefore, when this bit is set, SW must ensure that the CH_DESCCL[QID] bitfield is set to a valid queue ID and that the CH_DESCCL[BASE] bitfield is set to the base address of the source UIF endpoint. This base address is formed by taking the base address for the UIF block in the SAHB bus which can be found in the Nova address map document and then adding the internal UIF offset for the correct source endpoint. This offsets can be obtained from the UIF specification document. For convenience, the current internal UIF offsets are provided in the description of the QID bitfield in the CH_DESCCL register description. 0x0: FLAT (Source address maps to regular flat memory) 0x1: QUEUE (Source address maps to FIFO type memory (UIF Tx endpoints))
6	DIQ	RW	Destination Is Queue. When set the destination address is interpreted as a queue ID instead of just the address of a flat memory location. The queue could be either a BMU queue, one of the UIF RX endpoints or and EIF outbound FIFO. When this bit is set, SW must ensure that the DADR[QID] bitfield is set to a valid queue ID and that the DADR[BASE] bitfield is set to the base address of the destination queue for UIF or EIF, or to zero for a BMU queue. endpoint. 0x0: FLAT (Destination address maps to regular flat memory) 0x1: QUEUE (Destination address maps to a BMU queue or FIFO type memory)
5	SIQ	RW	Source Is Queue. When set the source address is interpreted as a queue ID instead of just the address of a flat memory location. The queue could be either a BMU queue, one of the UIF TX endpoints or and EIF inbound FIFO. When this bit is set, SW must ensure that the SADR[QID] bitfield is set to a valid queue ID and that the SADR[BASE] bitfield is set to the base address of the source queue for UIF or EIF, or to zero for a BMU queue. endpoint. 0x0: FLAT (Source address maps to regular flat memory) 0x1: QUEUE (Source address maps to a BMU queue or FIFO type memory)

WCSS_A_DXE_0_CH1_CTRL (cont.)

Bits	Name	Type	Description
4	BDH	RW	BD Handling bit. The behavior depends on the setting of XTYPE. If XTYPE is: H2B: Then if BDH is: 0: Use first 128 bytes of packet as BD. 1: Create BD from BD template (Up to 4 supported. Also see desc for BDT_IDX). B2H: Then if BDH is: 0: Insert BD in front of payload. 1: Discard BD. For XTYPE values of H2H and B2B the value of this bit is ignored.
3	EOP	RW	End of Packet bit. For H2B transfers marks the end of a packet Note that when not using external descriptors DXE cannot support the gather functionality and therefore the EOP bit must be set. For XTYPE values of H2H, B2B and B2H the value of this bit is ignored.
2:1	XTYPE	RW	Transfer Type. Specifies the type of transfer to perform: 00: H2H. Host space to Host space transfer. 01: B2B. BMU space to BMU space transfer. 10: H2B. Host space to BMU translation. 11: B2H. BMU space to Host space translation. For cross-space transfers (H2B and B2H) the DMA engine will also perform the required data format translation. The format translation behavior can be modified by the setting of the BDH and PAD bits. 0x0: H2H (Host space to Host space transfer) 0x1: B2B (BMU space to BMU space transfer) 0x2: H2B (Host space to BMU space transfer) 0x3: B2H (BMU space to Host space transfer)

WCSS_A_DXE_0_CH1_CTRL (cont.)

Bits	Name	Type	Description
0	EN	RW	Channel Enable or Restart. Writing this bit with a one either starts a new DMA transfer according to the parameters programmed in the channel's configuration registers and clears the DONE bit or, if the channel is already enabled but the STOP bit is set, then it restarts channel processing and also clears the STOP bit. This bit is automatically cleared when a single transfer completes or, if the channel is configured to use linked list descriptors, when the last transfer in the descriptor linked list completes (DESC_NXT is NULL or/and DESC_CSR[VALID]=0). The channel is also disabled and this bit cleared if a channel error is encountered. Reading this bit indicates whether a channel is currently enabled or disabled: 0: Channel Disabled. 1: Channel Enabled. The Channel Enable bit status can also be found by reading the DMA_ENCH register which aggregates the EN bits of all channels. Note: Software must take care to not disable the channel while it is busy or data could be lost.

0x0A202444 WCSS_A_DXE_0_CH1_STATUS**Type:** RW**Reset State:** 0x00000000

Serves to report the status of the programmed channel operation.

WCSS_A_DXE_0_CH1_STATUS

Bits	Name	Type	Description
31:16	RSVD	R	Reserved. Reads always return 0. Writes are ignored.
15	INT_DONE	R	Interrupt Source: Asserted when the channel has finished all processing.
14	INT_ERR	R	Interrupt Source: Asserted when a channel error has occurred. Masked by the INE_ERR
13	INT_ED	R	Interrupt Source. Descriptor Done and INT requested
12	ABORT_REQ	R	Abort Request. Software has requested that the current operation be aborted. However, if CH_STATUS[EN] is still set then the channel has registered the request but is not yet in a position to abort the transfer "safely". Once the channel finishes its current operation it will abort the transfer and set the error flag and error code bitfields accordingly.

WCSS_A_DXE_0_CH1_STATUS (cont.)

Bits	Name	Type	Description
11	STOP_REQ	R	Stop Request. Software has requested that the channel stop its operation after the current descriptor has been processed. Once this happens the STOPD bitfield will be set to one and the channel will be disabled.
10:6	ERR_CODE	R	Used to identify the source or cause of an error. 0x0: NONE (No Errors) 0x1: SAHB_ERR (DXE received Error response while performing a transfer on the SAHB) 0x2: H2H_RD_BUS_ERR (SAHB bus error encountered during H2H read operation) 0x3: H2H_WR_BUS_ERR (SAHB bus error encountered during H2H write operation) 0x4: PRG_INV_XTYPE (B2B not yet supported) 0x5: BERR_POPWQ (BMU error on POP WQ command) 0x6: BERR_PUSHWQ (BMU error on PUSH WQ command) 0x7: BERR_RLSS (BMU error on Release BD and attached PDUs command) 0x8: BERR_GETPDU (BMU error on Get PDU command) 0x9: PRG_INV_WQ (Programming error: Unsupported BMU Work Queue) 0xA: PRG_INV_H2H_SRC_QID (Programming error: Invalid H2H source queue ID) 0xB: PRG_INV_H2H_DST_QID (Programming error: Invalid H2H destination queue ID) 0xC: PRG_INV_B2H_SRC_QID (Programming error: Invalid B2H BMU source work queue ID) 0xD: PRG_INV_B2H_DST_QID (Programming error: Invalid B2H destination queue ID) 0xE: PRG_INV_B2H_SRC_IDX (Programming error: Invalid B2H source BD index) 0xF: PRG_INV_H2B_SRC_QID (Programming error: Invalid H2B source queue ID) 0x10: PRG_INV_H2B_DST_QID (Programming error: Invalid H2B BMU destination work queue ID) 0x11: PRG_INV_H2B_DST_IDX (Programming error: Invalid H2B destination BD index) 0x12: PRG_INV_H2B_SZ (Programming error: Invalid H2B transfer size of 0 when BDH=1) 0x13: PRG_INV_SADR (Programming error: Invalid Source Address) 0x14: PRG_INV_DADR (Programming error: Invalid Destination Address) 0x15: PRG_INV_EDADR (Programming error: Invalid Descriptor Address) 0x16: PRG_INV_SRC_WQID (Programming error: Invalid Destination WQ ID)

WCSS_A_DXE_0_CH1_STATUS (cont.)

Bits	Name	Type	Description
			0x17: PRG_INV_DST_WQID (Programming error: Invalid Destination WQ ID) 0x18: PRG_XTYPE_MSM TCH (Programming error: XTYPE mismatch during H2B gather transfer) 0x19: PKT_ERR (Malformed packet) 0x1A: ABORT (Aborted transfer due to SW request) 0x1B: PDU_CNT_OVFL (PDU count field overflow)
5	ERR	RW	DMA channel stopped due to error. To clear the error status it is necessary to write a 1 into this bit position.
4	STOPD	R	Channel Stopped. The channel has stopped further processing and disabled itself due to a stop request in the last processed descriptor.
3	MSKD	R	Channel Masked. The channel has been automatically masked out from arbitration due to a blocking condition.
2	DONE	R	DMA channel Done. This bit will be automatically cleared the next time the channel is enabled.
1	BUSY	R	Channel busy. High when the channel is enabled and currently active (won internal arbitration)
0	EN	R	Channel Enabled. Copy of CH_CTRL[EN].

0x0A202448 WCSS_A_DXE_0_CH1_SZ**Type:** RW**Reset State:** 0x00000000

The transfer size register specifies the total, remaining, and burst or 'chunk' transfer sizes for each channel. The maximum total transfer size is 16K-1 bytes. The chunk size control can be useful to control the relative bandwidth used by each channel as the dma engine will change channels, release the AHB bus, and rearbitrate for access after transferring chunk size bytes.

WCSS_A_DXE_0_CH1_SZ

Bits	Name	Type	Description
31:28	CHK_SZ	RW	Chunk transfer size. Specifies the number of bytes (in multiples of 8) to be transferred at one given time (not implying they are buffered but that they will be transferred for each start event in one bus request cycle). If chunk size is zero the DMA engine will always perform maximum chunk size transfers. Maximum chunk size is 128 bytes.
27:14	REM_SZ	R	Remaining transfer size. Specifies the number of bytes that remain to be transferred to complete the current operation.

WCSS_A_DXE_0_CH1_SZ (cont.)

Bits	Name	Type	Description
13:0	TOT_SZ	RW	Total Transfer Size. Specifies the number of bytes to be transferred. Maximum total transfer size is (4K-1) bytes. Note: This field is unnecessary and will be ignored for B2H or B2B transfers, when the source of the transfer is in BMU space (either BMU queue or direct pointer to BD). This is because the size of the transfer will be the packet length encoded in the BD at offset 0x0c bits [11:0].

0x0A20244C WCSS_A_DXE_0_CH1_SADRL**Type:** RW**Reset State:** 0x00000000

The channel source address register specifies either the source address in flat internal or external memory, or else the BMU queue ID, UIF TX endpoint ID, or EIF inbound FIFO ID for the transfer. The register must be programmed directly by software before the channel is enabled, then, once the DMA channel is enabled, the register is updated after each atomic chunk transfer (read+write) with the number of bytes transferred. (except when using a UIF/EIF queue) and also after loading a new descriptor. When using flat memory as the source, addresses can have arbitrary byte-alignment. When using a BMU queue, UIF endpoint or EIF fifo as a source, the software is responsible for programming a valid queue ID and correct base address for the queue. The layout of bitfields in this register depends on the setting of the CH_CSR[SIQ] bit. When CH_CSR[SIQ] is clear then the register has no bitfields and the full 32-bits are interpreted as the physical address of a location in flat memory, and when it is set this register has two bitfields: The SQID bitfield used to specify the queue ID, and the BASE bitfield used to provide the base address for the UIF or EIF source endpoint/FIFO.

For B2H or B2B transfers, when the source is a BMU Queue ID (CH_CSR[SIQ] = 1 and SWQID matches BMU), the DXE will issue a BMU POPWQ command to pop the specified queue and obtain the BD index of the source packet. Else, when the source is an address (CH_CSR[SIQ] = 0), the DXE will interpret the address as the index of the packet BD and will add it to the BMU_BASE to obtain the physical address to use to fetch it directly from memory.

For H2B or H2H transfers, if the source is a UIF endpoint or EIF FIFO (CH_CSR[SIQ] = 1 and SQID matches UIF/EIF), SW should program the SAHB base address of the UIF source endpoint in the BASE bitfield and the queue ID to match in the SWQID bitfield, then the DXE will use the base address given to generate the physical address and read the data over the SAHB bus and also use the appropriate sideband signals as specified by the SQID field to communicate with the selected UIF endpoint. SW should make sure that the BASE and SQID fields are consistent and refer to the same endpoint or else hangs or unpredictable behavior might occur. Else, when the source is just a location in flat memory space, (CH_CSR[SIQ] = 0), the bitfields lose their special meaning and DXE will just use the full 32-bits programmed into the register as the physical address to use to fetch the data. For this case, SW should make sure that the provided address maps to the intended source interface (Internal memory, USB, PCI, PCIe) by conforming to the address map specified for the SAHB bus.

To summarize, for B2H and B2B transfers the address must conform to the following format:

For index of BD mode (CH_CSR[SIQ] = 0):

ADDR[31:7] = BMU_BASE + BD_INDEX[15:0]

ADDR[6:0] = 7'd0

For BMU Queue ID mode (CH_CSR[SIQ] = 1 & SQID matching one of the BMU queues):

ADDR[31:7] = Any value will do but zero preferred

ADDR[6:0] = Any BMU Queue ID as specified in description of SQID field

For H2B and H2H transfers the address must conform to the following format:

For sources in flat memory space (CH_CSR[SIQ] = 0): ADDR[31:0] = address mapping to source interface

For UIF RX endpoints or EIF FIFOs (CH_CSR[SIQ] = 1 & QID matching UIF or EIF):

ADDR[31:7] = UIF RX endpoint base or EIF inbound FIFO base

ADDR[6:0] = Queue ID matching UIF or EIF as specified in description of SQID field

WCSS_A_DXE_0_CH1_SADRL

Bits	Name	Description
31:7	BASE	Base Address. Valid only when the CH_CSR[SIQ] bit is set. For this case, this bitfield provides the base address of the source UIF TX endpoint or EIF inbound FIFO. Consult the appropriate interface uarch for the offset within the interface's address space. When CH_CSR[SIQ] bit is not set this field loses its special meaning and is interpreted only as part of the full 32-bit source address.

WCSS_A_DXE_0_CH1_SADRL (cont.)

Bits	Name	Description
6:0	SQID	Source Queue ID. Valid only when the CH_CSR[SIQ] bit is set. For this case, this bitfield provides the ID for the source BMU, UIF or EIF queue to be used for the transfer. When CH_CSR[SIQ] bit is not set this field loses its special meaning and is interpreted only as part of the full 32-bit source address. 0x0: UIF_TX_DATA_EP (UIF TX Data Endpoint) 0x1: UIF_TX_MGMT_EP (UIF TX Management Endpoint) 0x2: BMU_2 (BMU Work Queue 2) 0x3: BMU_3 (BMU Work Queue 3) 0x4: BMU_4 (BMU Work Queue 4) 0x5: BMU_5 (BMU Work Queue 5) 0x6: BMU_6 (BMU Work Queue 6) 0x7: BMU_7 (BMU Work Queue 7) 0x8: BMU_8 (BMU Work Queue 8) 0x9: BMU_9 (BMU Work Queue 9) 0xA: BMU_10 (BMU Work Queue 10) 0xB: BMU_11 (BMU Work Queue 11) 0xC: BMU_12 (BMU Work Queue 12) 0xD: BMU_13 (BMU Work Queue 13) 0xE: BMU_14 (BMU Work Queue 14) 0xF: BMU_15 (BMU Work Queue 15) 0x10: BMU_16 (BMU Work Queue 16) 0x11: BMU_17 (BMU Work Queue 17) 0x12: BMU_18 (BMU Work Queue 18) 0x13: BMU_19 (BMU Work Queue 19) 0x14: BMU_20 (BMU Work Queue 20) 0x15: BMU_21 (BMU Work Queue 21) 0x16: BMU_22 (BMU Work Queue 22) 0x17: BMU_23 (BMU Work Queue 23) 0x18: BMU_24 (BMU Work Queue 24) 0x19: RSVD_25 (Reserved) 0x1A: RSVD_26 (Reserved) 0x1B: RSVD_27 (Reserved) 0x1C: RSVD_28 (Reserved) 0x1D: RSVD_29 (Reserved) 0x1E: RSVD_30 (Reserved) 0x1F: RSVD_31 (Reserved)

0x0A202450 WCSS_A_DXE_0_CH1_SADRH**Type:** RW**Reset State:** 0x00000000

The channel source address high register holds the high bits of the source address for a transfer. See the description of CH_SADRL for more info.

WCSS_A_DXE_0_CH1_SADRH

Bits	Name	Type	Description
31:2	RSVD	R	Reserved. Writes ignored. Reads always return zeroes.
1:0	ADDRH	RW	For taurus, this register holds bits 33 and 32 of the address. With the current address mapping these high order bits determine where the transfer is routed and whether the lower 32 bits of the address (the ones held in CH_SADRL) go untranslated through PIF or XIF or if they use the PIF and XIF address translation windows. See the bitfield encoding descriptions for an explanation. 0x0: ADDR_XTLN (If address in PIF or XIF address space then address goes through translation in PIF or XIF) 0x1: PIF_NO_XTLN (Always maps to PIF and full 32-bits in CH_SADRL goes through PIF untranslated) 0x2: XIF_NO_XTLN (Always maps to XIF and full 32-bits in CH_SADRL goes through XIF untranslated) 0x3: RSVD (Reserved)

0x0A202454 WCSS_A_DXE_0_CH1_DADRL**Type:** RW**Reset State:** 0x00000000

The channel destination address register specifies either the destination address in flat internal or external memory, or else the BMU queue ID, UIF RX endpoint ID, or EIF outbound FIFO ID for the transfer. The register must be programmed directly by software before the channel is enabled, then, once the DMA channel is enabled, the register is updated after each atomic chunk transfer (read+write) with the number of bytes transferred. (except when using a UIF/EIF queue) and also after loading a new descriptor. When using flat memory as the destination, addresses can have arbitrary byte-alignment. When using a BMU queue, UIF endpoint or EIF fifo as a destination, the software is responsible for programming a valid queue ID and correct base address for the queue. The layout of bitfields in this register depends on the setting of the CH_CSR[DIQ] bit. When this bit is clear then the register has no bitfields and the full 32-bits are interpreted as the physical address of a location in flat memory, and when the bit is set this register has two bitfields: The DQID bitfield used to specify the queue ID, and the BASE bitfield used to provide the base address for the UIF or EIF destination endpoint/FIFO.

For H2B or B2B transfers, when using a BMU queue as the destination (CH_CSR[DIQ] = 1 and DQID matches BMU), the DXE will push the index of the BD of the packet into the specified queue at the end of the packet transfer. Therefore, software is responsible for programming a valid BMU Queue ID.

Else, when the destination is an index (CH_CSR[DIQ] = 0), the BD will be written to the address obtained by adding the provided index to the BMU_BASE, again, the software must ensure that said address is legal and known to BMU. That is, software must preallocate the BD (get the BD from BMU via GET BD/PDU cmd) for the BD before starting the transfer. For B2H or H2H transfers, if the destination is a UIF endpoint or EIF FIFO (CH_CSR[DIQ] = 1 and DQID matches UIF/EIF), SW should program the SAHB base address of the UIF destination endpoint in the

BASE bitfield and the queue ID to match in the DQID bitfield, then the DXE will use the base address given to generate the physical address and write the data over the SAHB bus and also use the appropriate sideband signals as specified by the DQID field to communicate with the selected UIF endpoint. SW should make sure that the BASE and DQID fields are consistent and refer to the same endpoint else hangs or unpredictable behavior might occur. On the other hand, when the destination is just a location in flat memory space, (CH_CSR[DIQ] = 0), be it internal or external, the bitfields lose their special meaning and DXE will just use the full 32-bits programmed into the register as the address to write the data. For this case, SW should make sure that the address maps to the intended destination interface (Internal memory, USB, PCI, PCIe) by conforming to the address map specified for the SAHB bus. To summarize, for H2B and B2B transfers the address must conform to the following format:

For index of BD mode (CH_CSR[DIQ] = 0): ADDR[31:7] = BMU_BASE + BD_INDEX[15:0]

ADDR[6:0] = 7'd0

For BMU Queue ID mode (CH_CSR[DIQ] = 1 & DQID matching one of the BMU queues):

ADDR[31:5] = Any value will do but zero preferred

ADDR[4:0] = Any BMU Queue ID as specified in description of QID field

For B2H and H2H transfers the address must conform to the following format:

For destinations in flat memory space (CH_CSR[DIQ] = 0): ADDR[31:0] = address mapping to destination interface

For UIF RX endpoints or EIF FIFOs (CH_CSR[DIQ] = 1 & DQID matching UIF or EIF):

ADDR[31:5] = UIF RX endpoint base or EIF inbound FIFO base

ADDR[4:0] = Queue ID matching UIF or EIF as specified in description of DQID field

WCSS_A_DXE_0_CH1_DADRL

Bits	Name	Description
31:7	BASE	Base Address. Valid only when the CH_CSR[DIQ] bit is set. For this case, this bitfield provides the base address of the destination UIF RX endpoint or EIF outbound FIFO (for BMU SW may use any value can be used but zero is suggested). Consult the appropriate interface uarch for the offset within the interface's address space. When CH_CSR[SIQ] bit is not set this field loses its special meaning and is interpreted only as part of the full 32-bit source address.

WCSS_A_DXE_0_CH1_DADRL (cont.)

Bits	Name	Description
6:0	DQID	Destination Queue ID. Valid only when the CH_CSR[SIQ] bit is set. For this case, this bitfield provides the ID for the destination BMU WQ, UIF RX endpoint or EIF outbound queue to be used for the transfer. When CH_CSR[SIQ] bit is not set this field loses its special meaning and is interpreted only as part of the full 32-bit source address. 0x0: RSVD_0 (Reserved) 0x1: RSVD_1 (Reserved) 0x2: BMU_2 (BMU Work Queue 2) 0x3: BMU_3 (BMU Work Queue 3) 0x4: BMU_4 (BMU Work Queue 4) 0x5: BMU_5 (BMU Work Queue 5) 0x6: BMU_6 (BMU Work Queue 6) 0x7: BMU_7 (BMU Work Queue 7) 0x8: BMU_8 (BMU Work Queue 8) 0x9: BMU_9 (BMU Work Queue 9) 0xA: BMU_10 (BMU Work Queue 10) 0xB: BMU_11 (BMU Work Queue 11) 0xC: BMU_12 (BMU Work Queue 12) 0xD: BMU_13 (BMU Work Queue 13) 0xE: BMU_14 (BMU Work Queue 14) 0xF: BMU_15 (BMU Work Queue 15) 0x10: BMU_16 (BMU Work Queue 16) 0x11: BMU_17 (BMU Work Queue 17) 0x12: BMU_18 (BMU Work Queue 18) 0x13: BMU_19 (BMU Work Queue 19) 0x14: BMU_20 (BMU Work Queue 20) 0x15: BMU_21 (BMU Work Queue 21) 0x16: BMU_22 (BMU Work Queue 22) 0x17: BMU_23 (BMU Work Queue 23) 0x18: BMU_24 (BMU Work Queue 24) 0x19: BMU_25 (BMU Work Queue 25) 0x1A: BMU_26 (BMU Work Queue 26) 0x1B: BMU_SINK_WQ (BMU_SINK (WQ255)) 0x1C: UIF_RX_1_EP (UIF RX 1 endpoint) 0x1D: UIF_RX_2_EP (UIF RX 2 endpoint) 0x1E: UIF_RX_3_EP (UIF RX 3 endpoint) 0x1F: UIF_RX_4_EP (UIF RX 4 endpoint)

0x0A202458 WCSS_A_DXE_0_CH1_DADRH**Type:** RW**Reset State:** 0x00000000

The channel destination address high register holds the high bits of the destination address for a transfer. See the description of CH_DADRL for more info.

WCSS_A_DXE_0_CH1_DADRH

Bits	Name	Type	Description
31:2	RSVD	R	Reserved. Writes ignored. Reads always return zeroes.
1:0	ADDRH	RW	This register holds bits 33 and 32 of the address. With the current address mapping these high order bits determine where the transfer is routed and whether the lower 32 bits of the address (the ones held in CH_DADRL) go untranslated through PIF or XIF or if they use the PIF and XIF address translation windows. See the bitfield encoding descriptions for an explanation. 0x0: ADDR_XTLN (If address in PIF or XIF address space then address goes through translation in PIF or XIF) 0x1: PIF_NO_XTLN (Always maps to PIF and full 32-bits in CH_DADRL goes through PIF untranslated) 0x2: XIF_NO_XTLN (Always maps to XIF and full 32-bits in CH_DADRL goes through XIF untranslated) 0x3: RSVD (Reserved. Maps to default slave. Reads return error, writes are ignored)

0x0A20245C WCSS_A_DXE_0_CH1_DESCL

Type: RW

Reset State: 0x00000000

The Channel 1 Linked List Descriptor Pointer register contains the 32 lower bits of the address of the next descriptor to be loaded in the linked list. This address must be dword (32-bit) aligned (addr[1:0]=0 if CH_CTRL[PIQ]=0). Each complete transfer descriptor consists of 7 32-bit values arranged contiguously in memory. The value of this register will be overwritten with the address of the start of the next descriptor set in the list after the current descriptor has been fetched. The meaning of the value in this register depends on the value of the CH_CTRL[PIQ] bitfield similar to the way in which SADRL and DADRL are depending in SIQ and DIQ respectively. For CH_CTRL[PIQ]=0 then this register has no bitfields and the value is just interpreted as an address in flat memory space. When CH_CTRL[PIQ] is set then the descriptor will be read from FIFO type memory in UIF and the register has 2 bitfields: BASE, and SQID. BASE specifies the base address of the UIF Tx queue (derived from the start of the UIF address space in SAHB plus the internal UIF offset).

WCSS_A_DXE_0_CH1_DESCL

Bits	Name	Description
31:7	BASE	This bitfield only exists and has meaning when the descriptor needs to be read from a UIF TX queue (and therefore CH_CTRL[PIQ]=1), then it serves to specify the base address of the UIF endpoint to use as the source of the descriptor. Refer to the Nova address map document and to the UIF architecture document for guaranteed correct values to use here. For convenience but without a guarantee of accuracy, the base value to use for both the Tx data and management endpoints is the following: base = UIF base + TD_and_TM_offset = 0x0E02_5000 + x100 = 0x0E02_5100>>7 = 0x1C_04A2
6:0	SQID	This bitfield only exists and has meaning when the descriptor needs to be read from a UIF TX queue (and therefore CH_CTRL[PIQ]=1), then it serves to specify which of the two UIF TX queues to use. Note that in this case SW also needs to program the correct base address of the source UIF endpoint in the BASE bitfield. 0x0: UIF_TX_DATA_EP (Descriptor to be read from UIF TX Data Endpoint) 0x1: UIF_TX_MGMT_EP (Descriptor to be read from UIF TX Management Endpoint)

0x0A202460 WCSS_A_DXE_0_CH1_DESCH**Type:** RW**Reset State:** 0x00000000

The channel destination address high register holds the high bits of the the address of the next descriptor to be loaded in the linked list.

WCSS_A_DXE_0_CH1_DESCH

Bits	Name	Type	Description
31:2	RSVD	R	Reserved. Writes ignored. Reads always return zeroes.

WCSS_A_DXE_0_CH1_DESCH (cont.)

Bits	Name	Type	Description
1:0	ADDRH	RW	For taurus, this register holds bits 33 and 32 of the address. With the current address mapping these high order bits determine where the transfer is routed and whether the lower 32 bits of the address (the ones held in CH_DADRL) go untranslated through PIF or XIF or if they use the PIF and XIF address translation windows. See the bitfield encoding descriptions for an explanation. Note that if the source of the descriptor is either of the UIF endpoints (CH_CTRL[PIQ]=1) then the value in ADDRH has to be 0 or else the descriptor read could end up mapping to the XIF or PIF. 0x0: ADDR_XTLN (If address in PIF or XIF address space then address goes through translation in PIF or XIF) 0x1: PIF_NO_XTLN (Always maps to PIF and full 32-bits in CH_DADRL goes through PIF untranslated) 0x2: XIF_NO_XTLN (Always maps to XIF and full 32-bits in CH_DADRL goes through XIF untranslated) 0x3: RSVD (Reserved. Maps to default slave. Reads return error, writes are ignored)

0x0A202464 WCSS_A_DXE_0_CH1_LST_DESCL**Type:** R**Reset State:** 0x00000000

The read-only Channel 1 Current/last Linked List Descriptor Pointer register contains the low 32-bits of the address of the descriptor currently being processed or of the last descriptor processed if the channel is disabled.

The meaning of the value and bitfields with regards to CH_CTRL[PIQ] is the same as was explained for the CH_DESCL register.

WCSS_A_DXE_0_CH1_LST_DESCL

Bits	Name	Description
31:7	BASE	See description of CH_DESCL[BASE].
6:0	SQID	See description of CH_DESCL[SQID]. 0x0: UIF_TX_DATA_EP (Descriptor to be read from UIF TX Data Endpoint) 0x1: UIF_TX_MGMT_EP (Descriptor to be read from UIF TX Management Endpoint)

0x0A202468 WCSS_A_DXE_0_CH1_LST_DESCH**Type:** R**Reset State:** 0x00000000

The channel destination address high register holds the high bits of the the address of the next descriptor to be loaded in the linked list. Note that The meaning of the value and bitfields is the same as was explained for the CH_DESCL register.

WCSS_A_DXE_0_CH1_LST_DESCH

Bits	Name	Description
31:2	RSVD	Reserved. Writes ignored. Reads always return zeroes.
1:0	ADDRH	For taurus, this register holds bits 33 and 32 of the address.

0x0A20246C WCSS_A_DXE_0_CH1_BD**Type:** R**Reset State:** 0x00000000**WCSS_A_DXE_0_CH1_BD**

Bits	Name	Description
31:24	RSVD	Reserved. Always returns 0 when read.
23	VAL	Valid. BD has been loaded and is valid.
22:16	PDU_CNT	PDU count of packet.
15:0	IDX	Index of BD. To get actual physical address must be added to BD/PDU base.

0x0A202470 WCSS_A_DXE_0_CH1_HEAD**Type:** R**Reset State:** 0x00000000

Address of head of packet.

WCSS_A_DXE_0_CH1_HEAD

Bits	Name	Description
31:27	RSVD	Reserved.
26	DESCVAL	Descriptor valid. Signifies descriptor has already been loaded. Used to prevent reloading of a descriptor.

WCSS_A_DXE_0_CH1_HEAD (cont.)

Bits	Name	Description
25:21	DWQID	Destination WQ ID. Original BMU, UIF or EIF destination queue. Valid only when CH_CSR[DIQ]=1
20:16	SWQID	Source WQ ID. Original BMU, UIF or EIF source queue. Valid only when CH_CSR[SIQ]=1
15:0	IDX	Index of Head. Must be added to BD/PDU base for actual physical address.

0x0A202474 WCSS_A_DXE_0_CH1_TAIL**Type:** R**Reset State:** 0x00000000

Address of tail of packet.

WCSS_A_DXE_0_CH1_TAIL

Bits	Name	Description
31:16	RSVD	Reserved.
15:0	IDX	Index of Tail. Must be added to BD/PDU base for actual physical address

0x0A202478 WCSS_A_DXE_0_CH1_PDU**Type:** R**Reset State:** 0x00000000

PDU info register.

All its fields are working fields that get modified as the programmed operation progresses.

WCSS_A_DXE_0_CH1_PDU

Bits	Name	Description
31	RSVD	Reserved. Always returns 0 when read.
30:24	PDU_CNT	Current PDU count.
23:8	IDX	Index of current PDU. Must be added to BD/PDU base for actual physical address
7:0	FST_OFF	First PDU offset.

0x0A20247C WCSS_A_DXE_0_CH1_TSTMP**Type:** R**Reset State:** 0x00000000

Contains last sampled value of the system timestamp

WCSS_A_DXE_0_CH1_TSTMP

Bits	Name	Description
31:0	VALUE	

0x0A202480 WCSS_A_DXE_0_CH2_CTRL**Type:** RW**Reset State:** 0x10800708

Serves to configure channel 2 operation

WCSS_A_DXE_0_CH2_CTRL

Bits	Name	Type	Description
31	SWAP	RW	Endian Byte Swap Enable. When high instructs DXE to perform an endian swap of the incoming byte stream. The corresponding bitfield in the DXE descriptor control word is located at DXE_DESC_CTRL[20]
30:29	BDT_IDX	RW	BD Template Index. When XTYPE=H2B and BDH=1, DXE supports creating a BD by using template data from one of up to 4 different BD templates. The BD template base store is pointed at by the DXE BD_TMPLT(H/L) registers and the address used to read the template data is calculated as follows: $BD_TMPLT_ADDR = BD_TMPLT_BASE + (128 * CH_CTRL[BDT_IDX])$ The corresponding bitfield in the control word of the external descriptor is mapped to bits[19:18]
28	DFMT	RW	Descriptor Format. This bitfield is used to select between long and short descriptor formats. When using the short descriptor format, the high part of the source, destination and descriptor addresses will be taken from the SADDRH_DFLT, DADDRH_DFLT or DESCH_DFLT registers respectively. Default is to use the long descriptor format 0x0: SHORT (Short descriptor format (5 dwords)) 0x1: LONG (Long descriptor format (8 dwords))

WCSS_A_DXE_0_CH2_CTRL (cont.)

Bits	Name	Type	Description
27	ABORT	RW	Writing a one to this bit will cause the channel to stop its current transfer and set the ERR bit and ERR_CODE fields in CH_STATUS. Note that if the channel is currently disabled or if it is about to be disabled because it has just finished its programmed transfer, then the abort request will be ignored and the error bit and error codes won't be set. If the channel is currently masked by the arbiter then setting this bit will forcibly unmask it so that the transfer can be aborted the next time the channel is selected by the internal arbiter. The abort request is automatically cleared by hw once the channel has aborted the transfer. Also note that since the DXE will not attempt to do any cleanup when a transfer is aborted, software might need to do so. For example, for H2B transfers, SW should manually release the BD and any PDUs currently in use by the partial transfer or else a BD/PDU leak will result.
26	ENDIANNESS	RW	Specifies the endianness of the master generating the data. Affects alignment of valid bytes for non-dword aligned reads/writes. The corresponding field in the DXE descriptor control word is located in DXE_DESC_CTRL[21] 0x0: BIGEND (System is Big-Endian) 0x1: LTLEND (System is Little-Endian)
25:22	CTR_SEL	RW	Counter Select. If the event counters are enabled (DMA_CSR[ECTR_EN]=1) then this bitfield selects the counter that will be incremented at the end of the transfer. By default the value in this bitfield is the same as the channel number so that after reset channel 0 increments counter 0, channel 1 increments counter 1 and so on.
21	EDVEN	RW	When set, the DMA engine will clear the VAL bit in the DESC_CSR word of the current descriptor once it has completed processing it. This is useful for implementing a descriptor "ring" (see the description of the VALID bit in Table 3.1)
20	EDEN	RW	External Descriptor Enable. Use External Descriptor Linked List
19	INE_DONE	RW	Enable Channel Interrupt when Channel is Done. The corresponding interrupt source bit is INT_DONE
18	INE_ERR	RW	Enable Channel Interrupt on Errors. The corresponding interrupt source bit is INT_ERR
17	INE_ED	RW	Enable External Descriptor Interrupt. When this bit is set, an interrupt will be generated after processing a descriptor item that has the INT bit set in its DESC_CTRL word (see Table 3.1). The corresponding interrupt source bit is INT_ED

WCSS_A_DXE_0_CH2_CTRL (cont.)

Bits	Name	Type	Description
16	STOP	R	Channel processing stopped due to a request in the last processed linked list descriptor (see the STOP bit in the definition of the DESC_CSR word in Table 3.1). This bit will be automatically cleared when the channel is restarted.
15:13	PRIO	RW	This field sets the channel priority. 7: Highest Priority. 0: Lowest Priority.
12:9	BTHLD_SEL	RW	BMU Threshold Select. This value tells DXE which of the BMU BD/PDU thresholds to check before issuing BMU commands that reserve, or get BDs or PDUs. It also specifies the module index value that will be used to issue any required BMU commands (see BMU spec for details). The value of this bitfield is only meaningful for transfer types that involve the BMU and therefore its value will be ignored for H2H transfers. 0x0: RSVD0 (Reserved) 0x1: RSVD1 (Reserved) 0x2: THLD2 (Use BD/PDU threshold 2) 0x3: THLD3 (Use BD/PDU threshold 3) 0x4: THLD4 (Use BD/PDU threshold 4) 0x5: THLD5 (Use BD/PDU threshold 5) 0x6: THLD6 (Use BD/PDU threshold 6) 0x7: THLD7 (Use BD/PDU threshold 7) 0x8: THLD8 (Use BD/PDU threshold 8) 0x9: THLD9 (Use BD/PDU threshold 9) 0xA: THLD10 (Use BD/PDU threshold 10) 0xB: RSVD11 (Reserved) 0xC: RSVD12 (Reserved) 0xD: RSVD13 (Reserved) 0xE: RSVD14 (Reserved) 0xF: RSVD15 (Reserved)
8	PDU_REL	RW	0: Don't release BD and PDUs when done. 1: Release BD and PDUs when done. 0x0: KEEP (Don) 0x1: RELEASE (Release BD and payload PDUs)

WCSS_A_DXE_0_CH2_CTRL (cont.)

Bits	Name	Type	Description
7	PIQ	RW	next descriptor Pointer address Is Queue. When set the source of the descriptor address is interpreted as a queue ID instead of just the address of a flat memory location. The only valid source queues for descriptors are the UIF Tx data and Tx management endpoints. Any other IDs will result the channel aborting the transfer with an invalid source queue error. Therefore, when this bit is set, SW must ensure that the CH_DESCCL[QID] bitfield is set to a valid queue ID and that the CH_DESCCL[BASE] bitfield is set to the base address of the source UIF endpoint. This base address is formed by taking the base address for the UIF block in the SAHB bus which can be found in the Nova address map document and then adding the internal UIF offset for the correct source endpoint. This offsets can be obtained from the UIF specification document. For convenience, the current internal UIF offsets are provided in the description of the QID bitfield in the CH_DESCCL register description. 0x0: FLAT (Source address maps to regular flat memory) 0x1: QUEUE (Source address maps to FIFO type memory (UIF Tx endpoints))
6	DIQ	RW	Destination Is Queue. When set the destination address is interpreted as a queue ID instead of just the address of a flat memory location. The queue could be either a BMU queue, one of the UIF RX endpoints or and EIF outbound FIFO. When this bit is set, SW must ensure that the DADR[QID] bitfield is set to a valid queue ID and that the DADR[BASE] bitfield is set to the base address of the destination queue for UIF or EIF, or to zero for a BMU queue. endpoint. 0x0: FLAT (Destination address maps to regular flat memory) 0x1: QUEUE (Destination address maps to a BMU queue or FIFO type memory)
5	SIQ	RW	Source Is Queue. When set the source address is interpreted as a queue ID instead of just the address of a flat memory location. The queue could be either a BMU queue, one of the UIF TX endpoints or and EIF inbound FIFO. When this bit is set, SW must ensure that the SADR[QID] bitfield is set to a valid queue ID and that the SADR[BASE] bitfield is set to the base address of the source queue for UIF or EIF, or to zero for a BMU queue. endpoint. 0x0: FLAT (Source address maps to regular flat memory) 0x1: QUEUE (Source address maps to a BMU queue or FIFO type memory)

WCSS_A_DXE_0_CH2_CTRL (cont.)

Bits	Name	Type	Description
4	BDH	RW	BD Handling bit. The behavior depends on the setting of XTYPE. If XTYPE is: H2B: Then if BDH is: 0: Use first 128 bytes of packet as BD. 1: Create BD from BD template (Up to 4 supported. Also see desc for BDT_IDX). B2H: Then if BDH is: 0: Insert BD in front of payload. 1: Discard BD. For XTYPE values of H2H and B2B the value of this bit is ignored.
3	EOP	RW	End of Packet bit. For H2B transfers marks the end of a packet Note that when not using external descriptors DXE cannot support the gather functionality and therefore the EOP bit must be set. For XTYPE values of H2H, B2B and B2H the value of this bit is ignored.
2:1	XTYPE	RW	Transfer Type. Specifies the type of transfer to perform: 00: H2H. Host space to Host space transfer. 01: B2B. BMU space to BMU space transfer. 10: H2B. Host space to BMU translation. 11: B2H. BMU space to Host space translation. For cross-space transfers (H2B and B2H) the DMA engine will also perform the required data format translation. The format translation behavior can be modified by the setting of the BDH and PAD bits. 0x0: H2H (Host space to Host space transfer) 0x1: B2B (BMU space to BMU space transfer) 0x2: H2B (Host space to BMU space transfer) 0x3: B2H (BMU space to Host space transfer)

WCSS_A_DXE_0_CH2_CTRL (cont.)

Bits	Name	Type	Description
0	EN	RW	Channel Enable or Restart. Writing this bit with a one either starts a new DMA transfer according to the parameters programmed in the channel's configuration registers and clears the DONE bit or, if the channel is already enabled but the STOP bit is set, then it restarts channel processing and also clears the STOP bit. This bit is automatically cleared when a single transfer completes or, if the channel is configured to use linked list descriptors, when the last transfer in the descriptor linked list completes (DESC_NXT is NULL or/and DESC_CSR[VALID]=0). The channel is also disabled and this bit cleared if a channel error is encountered. Reading this bit indicates whether a channel is currently enabled or disabled: 0: Channel Disabled. 1: Channel Enabled. The Channel Enable bit status can also be found by reading the DMA_ENCH register which aggregates the EN bits of all channels. Note: Software must take care to not disable the channel while it is busy or data could be lost.

0x0A202484 WCSS_A_DXE_0_CH2_STATUS**Type:** RW**Reset State:** 0x00000000

Serves to report the status of the programmed channel operation.

WCSS_A_DXE_0_CH2_STATUS

Bits	Name	Type	Description
31:16	RSVD	R	Reserved. Reads always return 0. Writes are ignored.
15	INT_DONE	R	Interrupt Source: Asserted when the channel has finished all processing.
14	INT_ERR	R	Interrupt Source: Asserted when a channel error has occurred. Masked by the INE_ERR
13	INT_ED	R	Interrupt Source. Descriptor Done and INT requested
12	ABORT_REQ	R	Abort Request. Software has requested that the current operation be aborted. However, if CH_STATUS[EN] is still set then the channel has registered the request but is not yet in a position to abort the transfer "safely". Once the channel finishes its current operation it will abort the transfer and set the error flag and error code bitfields accordingly.

WCSS_A_DXE_0_CH2_STATUS (cont.)

Bits	Name	Type	Description
11	STOP_REQ	R	Stop Request. Software has requested that the channel stop its operation after the current descriptor has been processed. Once this happens the STOPD bitfield will be set to one and the channel will be disabled.
10:6	ERR_CODE	R	Used to identify the source or cause of an error. 0x0: NONE (No Errors) 0x1: SAHB_ERR (DXE received Error response while performing a transfer on the SAHB) 0x2: H2H_RD_BUS_ERR (SAHB bus error encountered during H2H read operation) 0x3: H2H_WR_BUS_ERR (SAHB bus error encountered during H2H write operation) 0x4: PRG_INV_XTYPE (B2B not yet supported) 0x5: BERR_POPWQ (BMU error on POP WQ command) 0x6: BERR_PUSHWQ (BMU error on PUSH WQ command) 0x7: BERR_RLSS (BMU error on Release BD and attached PDUs command) 0x8: BERR_GETPDU (BMU error on Get PDU command) 0x9: PRG_INV_WQ (Programming error: Unsupported BMU Work Queue) 0xA: PRG_INV_H2H_SRC_QID (Programming error: Invalid H2H source queue ID) 0xB: PRG_INV_H2H_DST_QID (Programming error: Invalid H2H destination queue ID) 0xC: PRG_INV_B2H_SRC_QID (Programming error: Invalid B2H BMU source work queue ID) 0xD: PRG_INV_B2H_DST_QID (Programming error: Invalid B2H destination queue ID) 0xE: PRG_INV_B2H_SRC_IDX (Programming error: Invalid B2H source BD index) 0xF: PRG_INV_H2B_SRC_QID (Programming error: Invalid H2B source queue ID) 0x10: PRG_INV_H2B_DST_QID (Programming error: Invalid H2B BMU destination work queue ID) 0x11: PRG_INV_H2B_DST_IDX (Programming error: Invalid H2B destination BD index) 0x12: PRG_INV_H2B_SZ (Programming error: Invalid H2B transfer size of 0 when BDH=1) 0x13: PRG_INV_SADR (Programming error: Invalid Source Address) 0x14: PRG_INV_DADR (Programming error: Invalid Destination Address) 0x15: PRG_INV_EDADR (Programming error: Invalid Descriptor Address) 0x16: PRG_INV_SRC_WQID (Programming error: Invalid Destination WQ ID)

WCSS_A_DXE_0_CH2_STATUS (cont.)

Bits	Name	Type	Description
			0x17: PRG_INV_DST_WQID (Programming error: Invalid Destination WQ ID) 0x18: PRG_XTYPE_MSM TCH (Programming error: XTYPE mismatch during H2B gather transfer) 0x19: PKT_ERR (Malformed packet) 0x1A: ABORT (Aborted transfer due to SW request) 0x1B: PDU_CNT_OVFL (PDU count field overflow)
5	ERR	RW	DMA channel stopped due to error. To clear the error status it is necessary to write a 1 into this bit position.
4	STOPD	R	Channel Stopped. The channel has stopped further processing and disabled itself due to a stop request in the last processed descriptor.
3	MSKD	R	Channel Masked. The channel has been automatically masked out from arbitration due to a blocking condition.
2	DONE	R	DMA channel Done. This bit will be automatically cleared the next time the channel is enabled.
1	BUSY	R	Channel busy. High when the channel is enabled and currently active (won internal arbitration)
0	EN	R	Channel Enabled. Copy of CH_CTRL[EN].

0x0A202488 WCSS_A_DXE_0_CH2_SZ**Type:** RW**Reset State:** 0x00000000

The transfer size register specifies the total, remaining, and burst or 'chunk' transfer sizes for each channel. The maximum total transfer size is 16K-1 bytes. The chunk size control can be useful to control the relative bandwidth used by each channel as the dma engine will change channels, release the AHB bus, and rearbitrate for access after transferring chunk size bytes.

WCSS_A_DXE_0_CH2_SZ

Bits	Name	Type	Description
31:28	CHK_SZ	RW	Chunk transfer size. Specifies the number of bytes (in multiples of 8) to be transferred at one given time (not implying they are buffered but that they will be transferred for each start event in one bus request cycle). If chunk size is zero the DMA engine will always perform maximum chunk size transfers. Maximum chunk size is 128 bytes.
27:14	REM_SZ	R	Remaining transfer size. Specifies the number of bytes that remain to be transferred to complete the current operation.

WCSS_A_DXE_0_CH2_SZ (cont.)

Bits	Name	Type	Description
13:0	TOT_SZ	RW	Total Transfer Size. Specifies the number of bytes to be transferred. Maximum total transfer size is (4K-1) bytes. Note: This field is unnecessary and will be ignored for B2H or B2B transfers, when the source of the transfer is in BMU space (either BMU queue or direct pointer to BD). This is because the size of the transfer will be the packet length encoded in the BD at offset 0x0c bits [11:0].

0x0A20248C WCSS_A_DXE_0_CH2_SADRL**Type:** RW**Reset State:** 0x00000000

The channel source address register specifies either the source address in flat internal or external memory, or else the BMU queue ID, UIF TX endpoint ID, or EIF inbound FIFO ID for the transfer. The register must be programmed directly by software before the channel is enabled, then, once the DMA channel is enabled, the register is updated after each atomic chunk transfer (read+write) with the number of bytes transferred. (except when using a UIF/EIF queue) and also after loading a new descriptor. When using flat memory as the source, addresses can have arbitrary byte-alignment. When using a BMU queue, UIF endpoint or EIF fifo as a source, the software is responsible for programming a valid queue ID and correct base address for the queue. The layout of bitfields in this register depends on the setting of the CH_CSR[SIQ] bit. When CH_CSR[SIQ] is clear then the register has no bitfields and the full 32-bits are interpreted as the physical address of a location in flat memory, and when it is set this register has two bitfields: The SQID bitfield used to specify the queue ID, and the BASE bitfield used to provide the base address for the UIF or EIF source endpoint/FIFO.

For B2H or B2B transfers, when the source is a BMU Queue ID (CH_CSR[SIQ] = 1 and SWQID matches BMU), the DXE will issue a BMU POPWQ command to pop the specified queue and obtain the BD index of the source packet. Else, when the source is an address (CH_CSR[SIQ] = 0), the DXE will interpret the address as the index of the packet BD and will add it to the BMU_BASE to obtain the physical address to use to fetch it directly from memory.

For H2B or H2H transfers, if the source is a UIF endpoint or EIF FIFO (CH_CSR[SIQ] = 1 and SQID matches UIF/EIF), SW should program the SAHB base address of the UIF source endpoint in the BASE bitfield and the queue ID to match in the SWQID bitfield, then the DXE will use the base address given to generate the physical address and read the data over the SAHB bus and also use the appropriate sideband signals as specified by the SQID field to communicate with the selected UIF endpoint. SW should make sure that the BASE and SQID fields are consistent and refer to the same endpoint or else hangs or unpredictable behavior might occur. Else, when the source is just a location in flat memory space, (CH_CSR[SIQ] = 0), the bitfields lose their special meaning and DXE will just use the full 32-bits programmed into the register as the physical address to use to fetch the data. For this case, SW should make sure that the provided address maps to the intended source interface (Internal memory, USB, PCI, PCIe) by conforming to the address map specified for the SAHB bus.

To summarize, for B2H and B2B transfers the address must conform to the following format:

For index of BD mode (CH_CSR[SIQ] = 0):

ADDR[31:7] = BMU_BASE + BD_INDEX[15:0]

ADDR[6:0] = 7'd0

For BMU Queue ID mode (CH_CSR[SIQ] = 1 & SQID matching one of the BMU queues):

ADDR[31:7] = Any value will do but zero preferred

ADDR[6:0] = Any BMU Queue ID as specified in description of SQID field

For H2B and H2H transfers the address must conform to the following format:

For sources in flat memory space (CH_CSR[SIQ] = 0): ADDR[31:0] = address mapping to source interface

For UIF RX endpoints or EIF FIFOs (CH_CSR[SIQ] = 1 & QID matching UIF or EIF):

ADDR[31:7] = UIF RX endpoint base or EIF inbound FIFO base

ADDR[6:0] = Queue ID matching UIF or EIF as specified in description of SQID field

WCSS_A_DXE_0_CH2_SADRL

Bits	Name	Description
31:7	BASE	Base Address. Valid only when the CH_CSR[SIQ] bit is set. For this case, this bitfield provides the base address of the source UIF TX endpoint or EIF inbound FIFO. Consult the appropriate interface uarch for the offset within the interface's address space. When CH_CSR[SIQ] bit is not set this field loses its special meaning and is interpreted only as part of the full 32-bit source address.

WCSS_A_DXE_0_CH2_SADRL (cont.)

Bits	Name	Description
6:0	SQID	Source Queue ID. Valid only when the CH_CSR[SIQ] bit is set. For this case, this bitfield provides the ID for the source BMU, UIF or EIF queue to be used for the transfer. When CH_CSR[SIQ] bit is not set this field loses its special meaning and is interpreted only as part of the full 32-bit source address. 0x0: UIF_TX_DATA_EP (UIF TX Data Endpoint) 0x1: UIF_TX_MGMT_EP (UIF TX Management Endpoint) 0x2: BMU_2 (BMU Work Queue 2) 0x3: BMU_3 (BMU Work Queue 3) 0x4: BMU_4 (BMU Work Queue 4) 0x5: BMU_5 (BMU Work Queue 5) 0x6: BMU_6 (BMU Work Queue 6) 0x7: BMU_7 (BMU Work Queue 7) 0x8: BMU_8 (BMU Work Queue 8) 0x9: BMU_9 (BMU Work Queue 9) 0xA: BMU_10 (BMU Work Queue 10) 0xB: BMU_11 (BMU Work Queue 11) 0xC: BMU_12 (BMU Work Queue 12) 0xD: BMU_13 (BMU Work Queue 13) 0xE: BMU_14 (BMU Work Queue 14) 0xF: BMU_15 (BMU Work Queue 15) 0x10: BMU_16 (BMU Work Queue 16) 0x11: BMU_17 (BMU Work Queue 17) 0x12: BMU_18 (BMU Work Queue 18) 0x13: BMU_19 (BMU Work Queue 19) 0x14: BMU_20 (BMU Work Queue 20) 0x15: BMU_21 (BMU Work Queue 21) 0x16: BMU_22 (BMU Work Queue 22) 0x17: BMU_23 (BMU Work Queue 23) 0x18: BMU_24 (BMU Work Queue 24) 0x19: RSVD_25 (Reserved) 0x1A: RSVD_26 (Reserved) 0x1B: RSVD_27 (Reserved) 0x1C: RSVD_28 (Reserved) 0x1D: RSVD_29 (Reserved) 0x1E: RSVD_30 (Reserved) 0x1F: RSVD_31 (Reserved)

0x0A202490 WCSS_A_DXE_0_CH2_SADRH**Type:** RW**Reset State:** 0x00000000

The channel source address high register holds the high bits of the source address for a transfer. See the description of CH_SADRL for more info.

WCSS_A_DXE_0_CH2_SADRH

Bits	Name	Type	Description
31:2	RSVD	R	Reserved. Writes ignored. Reads always return zeroes.
1:0	ADDRH	RW	For taurus, this register holds bits 33 and 32 of the address. With the current address mapping these high order bits determine where the transfer is routed and whether the lower 32 bits of the address (the ones held in CH_SADRL) go untranslated through PIF or XIF or if they use the PIF and XIF address translation windows. See the bitfield encoding descriptions for an explanation. 0x0: ADDR_XTLN (If address in PIF or XIF address space then address goes through translation in PIF or XIF) 0x1: PIF_NO_XTLN (Always maps to PIF and full 32-bits in CH_SADRL goes through PIF untranslated) 0x2: XIF_NO_XTLN (Always maps to XIF and full 32-bits in CH_SADRL goes through XIF untranslated) 0x3: RSVD (Reserved)

0x0A202494 WCSS_A_DXE_0_CH2_DADRL**Type:** RW**Reset State:** 0x00000000

The channel destination address register specifies either the destination address in flat internal or external memory, or else the BMU queue ID, UIF RX endpoint ID, or EIF outbound FIFO ID for the transfer. The register must be programmed directly by software before the channel is enabled, then, once the DMA channel is enabled, the register is updated after each atomic chunk transfer (read+write) with the number of bytes transferred. (except when using a UIF/EIF queue) and also after loading a new descriptor. When using flat memory as the destination, addresses can have arbitrary byte-alignment. When using a BMU queue, UIF endpoint or EIF fifo as a destination, the software is responsible for programming a valid queue ID and correct base address for the queue. The layout of bitfields in this register depends on the setting of the CH_CSR[DIQ] bit. When this bit is clear then the register has no bitfields and the full 32-bits are interpreted as the physical address of a location in flat memory, and when the bit is set this register has two bitfields: The DQID bitfield used to specify the queue ID, and the BASE bitfield used to provide the base address for the UIF or EIF destination endpoint/FIFO.

For H2B or B2B transfers, when using a BMU queue as the destination (CH_CSR[DIQ] = 1 and DQID matches BMU), the DXE will push the index of the BD of the packet into the specified queue at the end of the packet transfer. Therefore, software is responsible for programming a valid BMU Queue ID.

Else, when the destination is an index (CH_CSR[DIQ] = 0), the BD will be written to the address obtained by adding the provided index to the BMU_BASE, again, the software must ensure that said address is legal and known to BMU. That is, software must preallocate the BD (get the BD from BMU via GET BD/PDU cmd) for the BD before starting the transfer. For B2H or H2H transfers, if the destination is a UIF endpoint or EIF FIFO (CH_CSR[DIQ] = 1 and DQID matches UIF/EIF), SW should program the SAHB base address of the UIF destination endpoint in the

BASE bitfield and the queue ID to match in the DQID bitfield, then the DXE will use the base address given to generate the physical address and write the data over the SAHB bus and also use the appropriate sideband signals as specified by the DQID field to communicate with the selected UIF endpoint. SW should make sure that the BASE and DQID fields are consistent and refer to the same endpoint else hangs or unpredictable behavior might occur. On the other hand, when the destination is just a location in flat memory space, (CH_CSR[DIQ] = 0), be it internal or external, the bitfields lose their special meaning and DXE will just use the full 32-bits programmed into the register as the address to write the data. For this case, SW should make sure that the address maps to the intended destination interface (Internal memory, USB, PCI, PCIe) by conforming to the address map specified for the SAHB bus. To summarize, for H2B and B2B transfers the address must conform to the following format:

For index of BD mode (CH_CSR[DIQ] = 0): ADDR[31:7] = BMU_BASE + BD_INDEX[15:0]

ADDR[6:0] = 7'd0

For BMU Queue ID mode (CH_CSR[DIQ] = 1 & DQID matching one of the BMU queues):

ADDR[31:5] = Any value will do but zero preferred

ADDR[4:0] = Any BMU Queue ID as specified in description of QID field

For B2H and H2H transfers the address must conform to the following format:

For destinations in flat memory space (CH_CSR[DIQ] = 0): ADDR[31:0] = address mapping to destination interface

For UIF RX endpoints or EIF FIFOs (CH_CSR[DIQ] = 1 & DQID matching UIF or EIF):

ADDR[31:5] = UIF RX endpoint base or EIF inbound FIFO base

ADDR[4:0] = Queue ID matching UIF or EIF as specified in description of DQID field

WCSS_A_DXE_0_CH2_DADRL

Bits	Name	Description
31:7	BASE	Base Address. Valid only when the CH_CSR[DIQ] bit is set. For this case, this bitfield provides the base address of the destination UIF RX endpoint or EIF outbound FIFO (for BMU SW may use any value can be used but zero is suggested). Consult the appropriate interface uarch for the offset within the interface's address space. When CH_CSR[SIQ] bit is not set this field loses its special meaning and is interpreted only as part of the full 32-bit source address.

WCSS_A_DXE_0_CH2_DADRL (cont.)

Bits	Name	Description
6:0	DQID	Destination Queue ID. Valid only when the CH_CSR[SIQ] bit is set. For this case, this bitfield provides the ID for the destination BMU WQ, UIF RX endpoint or EIF outbound queue to be used for the transfer. When CH_CSR[SIQ] bit is not set this field loses its special meaning and is interpreted only as part of the full 32-bit source address. 0x0: RSVD_0 (Reserved) 0x1: RSVD_1 (Reserved) 0x2: BMU_2 (BMU Work Queue 2) 0x3: BMU_3 (BMU Work Queue 3) 0x4: BMU_4 (BMU Work Queue 4) 0x5: BMU_5 (BMU Work Queue 5) 0x6: BMU_6 (BMU Work Queue 6) 0x7: BMU_7 (BMU Work Queue 7) 0x8: BMU_8 (BMU Work Queue 8) 0x9: BMU_9 (BMU Work Queue 9) 0xA: BMU_10 (BMU Work Queue 10) 0xB: BMU_11 (BMU Work Queue 11) 0xC: BMU_12 (BMU Work Queue 12) 0xD: BMU_13 (BMU Work Queue 13) 0xE: BMU_14 (BMU Work Queue 14) 0xF: BMU_15 (BMU Work Queue 15) 0x10: BMU_16 (BMU Work Queue 16) 0x11: BMU_17 (BMU Work Queue 17) 0x12: BMU_18 (BMU Work Queue 18) 0x13: BMU_19 (BMU Work Queue 19) 0x14: BMU_20 (BMU Work Queue 20) 0x15: BMU_21 (BMU Work Queue 21) 0x16: BMU_22 (BMU Work Queue 22) 0x17: BMU_23 (BMU Work Queue 23) 0x18: BMU_24 (BMU Work Queue 24) 0x19: BMU_25 (BMU Work Queue 25) 0x1A: BMU_26 (BMU Work Queue 26) 0x1B: BMU_SINK_WQ (BMU_SINK (WQ255)) 0x1C: UIF_RX_1_EP (UIF RX 1 endpoint) 0x1D: UIF_RX_2_EP (UIF RX 2 endpoint) 0x1E: UIF_RX_3_EP (UIF RX 3 endpoint) 0x1F: UIF_RX_4_EP (UIF RX 4 endpoint)

0x0A202498 WCSS_A_DXE_0_CH2_DADRH**Type:** RW**Reset State:** 0x00000000

The channel destination address high register holds the high bits of the destination address for a transfer. See the description of CH_DADRL for more info.

WCSS_A_DXE_0_CH2_DADRH

Bits	Name	Type	Description
31:2	RSVD	R	Reserved. Writes ignored. Reads always return zeroes.
1:0	ADDRH	RW	This register holds bits 33 and 32 of the address. With the current address mapping these high order bits determine where the transfer is routed and whether the lower 32 bits of the address (the ones held in CH_DADRL) go untranslated through PIF or XIF or if they use the PIF and XIF address translation windows. See the bitfield encoding descriptions for an explanation. 0x0: ADDR_XTLN (If address in PIF or XIF address space then address goes through translation in PIF or XIF) 0x1: PIF_NO_XTLN (Always maps to PIF and full 32-bits in CH_DADRL goes through PIF untranslated) 0x2: XIF_NO_XTLN (Always maps to XIF and full 32-bits in CH_DADRL goes through XIF untranslated) 0x3: RSVD (Reserved. Maps to default slave. Reads return error, writes are ignored)

0x0A20249C WCSS_A_DXE_0_CH2_DESCL

Type: RW

Reset State: 0x00000000

The Channel 2 Linked List Descriptor Pointer register contains the 32 lower bits of the address of the next descriptor to be loaded in the linked list. This address must be dword (32-bit) aligned (addr[1:0]=0 if CH_CTRL[PIQ]=0). Each complete transfer descriptor consists of 7 32-bit values arranged contiguously in memory. The value of this register will be overwritten with the address of the start of the next descriptor set in the list after the current descriptor has been fetched. The meaning of the value in this register depends on the value of the CH_CTRL[PIQ] bitfield similar to the way in which SADRL and DADRL are depending in SIQ and DIQ respectively. For CH_CTRL[PIQ]=0 then this register has no bitfields and the value is just interpreted as an address in flat memory space. When CH_CTRL[PIQ] is set then the descriptor will be read from FIFO type memory in UIF and the register has 2 bitfields: BASE, and SQID. BASE specifies the base address of the UIF Tx queue (derived from the start of the UIF address space in SAHB plus the internal UIF offset).

WCSS_A_DXE_0_CH2_DESCL

Bits	Name	Description
31:7	BASE	This bitfield only exists and has meaning when the descriptor needs to be read from a UIF TX queue (and therefore CH_CTRL[PIQ]=1), then it serves to specify the base address of the UIF endpoint to use as the source of the descriptor. Refer to the Nova address map document and to the UIF architecture document for guaranteed correct values to use here. For convenience but without a guarantee of accuracy, the base value to use for both the Tx data and management endpoints is the following: $\text{base} = \text{UIF base} + \text{TD_and_TM_offset} = 0x0E02_5000 + x100 = 0x0E02_5100 \gg 7 = 0x1C_04A2$
6:0	SQID	This bitfield only exists and has meaning when the descriptor needs to be read from a UIF TX queue (and therefore CH_CTRL[PIQ]=1), then it serves to specify which of the two UIF TX queues to use. Note that in this case SW also needs to program the correct base address of the source UIF endpoint in the BASE bitfield. 0x0: UIF_TX_DATA_EP (Descriptor to be read from UIF TX Data Endpoint) 0x1: UIF_TX_MGMT_EP (Descriptor to be read from UIF TX Management Endpoint)

0x0A2024A0 WCSS_A_DXE_0_CH2_DESCH**Type:** RW**Reset State:** 0x00000000

The channel destination address high register holds the high bits of the the address of the next descriptor to be loaded in the linked list.

WCSS_A_DXE_0_CH2_DESCH

Bits	Name	Type	Description
31:2	RSVD	R	Reserved. Writes ignored. Reads always return zeroes.

WCSS_A_DXE_0_CH2_DESCH (cont.)

Bits	Name	Type	Description
1:0	ADDRH	RW	For taurus, this register holds bits 33 and 32 of the address. With the current address mapping these high order bits determine where the transfer is routed and whether the lower 32 bits of the address (the ones held in CH_DADRL) go untranslated through PIF or XIF or if they use the PIF and XIF address translation windows. See the bitfield encoding descriptions for an explanation. Note that if the source of the descriptor is either of the UIF endpoints (CH_CTRL[PIQ]=1) then the value in ADDRH has to be 0 or else the descriptor read could end up mapping to the XIF or PIF. 0x0: ADDR_XTLN (If address in PIF or XIF address space then address goes through translation in PIF or XIF) 0x1: PIF_NO_XTLN (Always maps to PIF and full 32-bits in CH_DADRL goes through PIF untranslated) 0x2: XIF_NO_XTLN (Always maps to XIF and full 32-bits in CH_DADRL goes through XIF untranslated) 0x3: RSVD (Reserved. Maps to default slave. Reads return error, writes are ignored)

0x0A2024A4 WCSS_A_DXE_0_CH2_LST_DESCL**Type:** R**Reset State:** 0x00000000

The read-only Channel 2 Current/last Linked List Descriptor Pointer register contains the low 32-bits of the address of the descriptor currently being processed or of the last descriptor processed if the channel is disabled.

The meaning of the value and bitfields with regards to CH_CTRL[PIQ] is the same as was explained for the CH_DESCL register.

WCSS_A_DXE_0_CH2_LST_DESCL

Bits	Name	Description
31:7	BASE	See description of CH_DESCL[BASE].
6:0	SQID	See description of CH_DESCL[SQID]. 0x0: UIF_TX_DATA_EP (Descriptor to be read from UIF TX Data Endpoint) 0x1: UIF_TX_MGMT_EP (Descriptor to be read from UIF TX Management Endpoint)

0x0A2024A8 WCSS_A_DXE_0_CH2_LST_DESCH**Type:** R**Reset State:** 0x00000000

The channel destination address high register holds the high bits of the the address of the next descriptor to be loaded in the linked list. Note that The meaning of the value and bitfields is the same as was explained for the CH_DESCL register.

WCSS_A_DXE_0_CH2_LST_DESCH

Bits	Name	Description
31:2	RSVD	Reserved. Writes ignored. Reads always return zeroes.
1:0	ADDRH	For taurus, this register holds bits 33 and 32 of the address.

0x0A2024AC WCSS_A_DXE_0_CH2_BD**Type:** R**Reset State:** 0x00000000**WCSS_A_DXE_0_CH2_BD**

Bits	Name	Description
31:24	RSVD	Reserved. Always returns 0 when read.
23	VAL	Valid. BD has been loaded and is valid.
22:16	PDU_CNT	PDU count of packet.
15:0	IDX	Index of BD. To get actual physical address must be added to BD/PDU base.

0x0A2024B0 WCSS_A_DXE_0_CH2_HEAD**Type:** R**Reset State:** 0x00000000

Address of head of packet.

WCSS_A_DXE_0_CH2_HEAD

Bits	Name	Description
31:27	RSVD	Reserved.
26	DESCVAL	Descriptor valid. Signifies descriptor has already been loaded. Used to prevent reloading of a descriptor.

WCSS_A_DXE_0_CH2_HEAD (cont.)

Bits	Name	Description
25:21	DWQID	Destination WQ ID. Original BMU, UIF or EIF destination queue. Valid only when CH_CSR[DIQ]=1
20:16	SWQID	Source WQ ID. Original BMU, UIF or EIF source queue. Valid only when CH_CSR[SIQ]=1
15:0	IDX	Index of Head. Must be added to BD/PDU base for actual physical address.

0x0A2024B4 WCSS_A_DXE_0_CH2_TAIL**Type:** R**Reset State:** 0x00000000

Address of tail of packet.

WCSS_A_DXE_0_CH2_TAIL

Bits	Name	Description
31:16	RSVD	Reserved.
15:0	IDX	Index of Tail. Must be added to BD/PDU base for actual physical address

0x0A2024B8 WCSS_A_DXE_0_CH2_PDU**Type:** R**Reset State:** 0x00000000

PDU info register.

All its fields are working fields that get modified as the programmed operation progresses.

WCSS_A_DXE_0_CH2_PDU

Bits	Name	Description
31	RSVD	Reserved. Always returns 0 when read.
30:24	PDU_CNT	Current PDU count.
23:8	IDX	Index of current PDU. Must be added to BD/PDU base for actual physical address
7:0	FST_OFF	First PDU offset.

0x0A2024BC WCSS_A_DXE_0_CH2_TSTMP**Type:** R**Reset State:** 0x00000000

Contains last sampled value of the system timestamp

WCSS_A_DXE_0_CH2_TSTMP

Bits	Name	Description
31:0	VALUE	

0x0A2024C0 WCSS_A_DXE_0_CH3_CTRL**Type:** RW**Reset State:** 0x10C00708

Serves to configure channel 3 operation

WCSS_A_DXE_0_CH3_CTRL

Bits	Name	Type	Description
31	SWAP	RW	Endian Byte Swap Enable. When high instructs DXE to perform an endian swap of the incoming byte stream. The corresponding bitfield in the DXE descriptor control word is located at DXE_DESC_CTRL[20]
30:29	BDT_IDX	RW	BD Template Index. When XTYPE=H2B and BDH=1, DXE supports creating a BD by using template data from one of up to 4 different BD templates. The BD template base store is pointed at by the DXE BD_TMPLT(H/L) registers and the address used to read the template data is calculated as follows: $BD_TMPLT_ADDR = BD_TMPLT_BASE + (128 * CH_CTRL[BDT_IDX])$ The corresponding bitfield in the control word of the external descriptor is mapped to bits[19:18]
28	DFMT	RW	Descriptor Format. This bitfield is used to select between long and short descriptor formats. When using the short descriptor format, the high part of the source, destination and descriptor addresses will be taken from the SADDRH_DFLT, DADDRH_DFLT or DESCH_DFLT registers respectively. Default is to use the long descriptor format 0x0: SHORT (Short descriptor format (5 dwords)) 0x1: LONG (Long descriptor format (8 dwords))

WCSS_A_DXE_0_CH3_CTRL (cont.)

Bits	Name	Type	Description
27	ABORT	RW	Writing a one to this bit will cause the channel to stop its current transfer and set the ERR bit and ERR_CODE fields in CH_STATUS. Note that if the channel is currently disabled or if it is about to be disabled because it has just finished its programmed transfer, then the abort request will be ignored and the error bit and error codes won't be set. If the channel is currently masked by the arbiter then setting this bit will forcibly unmask it so that the transfer can be aborted the next time the channel is selected by the internal arbiter. The abort request is automatically cleared by hw once the channel has aborted the transfer. Also note that since the DXE will not attempt to do any cleanup when a transfer is aborted, software might need to do so. For example, for H2B transfers, SW should manually release the BD and any PDUs currently in use by the partial transfer or else a BD/PDU leak will result.
26	ENDIANNESS	RW	Specifies the endianness of the master generating the data. Affects alignment of valid bytes for non-dword aligned reads/writes. The corresponding field in the DXE descriptor control word is located in DXE_DESC_CTRL[21] 0x0: BIGEND (System is Big-Endian) 0x1: LTLEND (System is Little-Endian)
25:22	CTR_SEL	RW	Counter Select. If the event counters are enabled (DMA_CSR[ECTR_EN]=1) then this bitfield selects the counter that will be incremented at the end of the transfer. By default the value in this bitfield is the same as the channel number so that after reset channel 0 increments counter 0, channel 1 increments counter 1 and so on.
21	EDVEN	RW	When set, the DMA engine will clear the VAL bit in the DESC_CSR word of the current descriptor once it has completed processing it. This is useful for implementing a descriptor "ring" (see the description of the VALID bit in Table 3.1)
20	EDEN	RW	External Descriptor Enable. Use External Descriptor Linked List
19	INE_DONE	RW	Enable Channel Interrupt when Channel is Done. The corresponding interrupt source bit is INT_DONE
18	INE_ERR	RW	Enable Channel Interrupt on Errors. The corresponding interrupt source bit is INT_ERR
17	INE_ED	RW	Enable External Descriptor Interrupt. When this bit is set, an interrupt will be generated after processing a descriptor item that has the INT bit set in its DESC_CTRL word (see Table 3.1). The corresponding interrupt source bit is INT_ED

WCSS_A_DXE_0_CH3_CTRL (cont.)

Bits	Name	Type	Description
16	STOP	R	Channel processing stopped due to a request in the last processed linked list descriptor (see the STOP bit in the definition of the DESC_CSR word in Table 3.1). This bit will be automatically cleared when the channel is restarted.
15:13	PRIO	RW	This field sets the channel priority. 7: Highest Priority. 0: Lowest Priority.
12:9	BTHLD_SEL	RW	BMU Threshold Select. This value tells DXE which of the BMU BD/PDU thresholds to check before issuing BMU commands that reserve, or get BDs or PDUs. It also specifies the module index value that will be used to issue any required BMU commands (see BMU spec for details). The value of this bitfield is only meaningful for transfer types that involve the BMU and therefore its value will be ignored for H2H transfers. 0x0: RSVD0 (Reserved) 0x1: RSVD1 (Reserved) 0x2: THLD2 (Use BD/PDU threshold 2) 0x3: THLD3 (Use BD/PDU threshold 3) 0x4: THLD4 (Use BD/PDU threshold 4) 0x5: THLD5 (Use BD/PDU threshold 5) 0x6: THLD6 (Use BD/PDU threshold 6) 0x7: THLD7 (Use BD/PDU threshold 7) 0x8: THLD8 (Use BD/PDU threshold 8) 0x9: THLD9 (Use BD/PDU threshold 9) 0xA: THLD10 (Use BD/PDU threshold 10) 0xB: RSVD11 (Reserved) 0xC: RSVD12 (Reserved) 0xD: RSVD13 (Reserved) 0xE: RSVD14 (Reserved) 0xF: RSVD15 (Reserved)
8	PDU_REL	RW	0: Don't release BD and PDUs when done. 1: Release BD and PDUs when done. 0x0: KEEP (Don) 0x1: RELEASE (Release BD and payload PDUs)

WCSS_A_DXE_0_CH3_CTRL (cont.)

Bits	Name	Type	Description
7	PIQ	RW	next descriptor Pointer address Is Queue. When set the source of the descriptor address is interpreted as a queue ID instead of just the address of a flat memory location. The only valid source queues for descriptors are the UIF Tx data and Tx management endpoints. Any other IDs will result the channel aborting the transfer with an invalid source queue error. Therefore, when this bit is set, SW must ensure that the CH_DESCCL[QID] bitfield is set to a valid queue ID and that the CH_DESCCL[BASE] bitfield is set to the base address of the source UIF endpoint. This base address is formed by taking the base address for the UIF block in the SAHB bus which can be found in the Nova address map document and then adding the internal UIF offset for the correct source endpoint. This offsets can be obtained from the UIF specification document. For convenience, the current internal UIF offsets are provided in the description of the QID bitfield in the CH_DESCCL register description. 0x0: FLAT (Source address maps to regular flat memory) 0x1: QUEUE (Source address maps to FIFO type memory (UIF Tx endpoints))
6	DIQ	RW	Destination Is Queue. When set the destination address is interpreted as a queue ID instead of just the address of a flat memory location. The queue could be either a BMU queue, one of the UIF RX endpoints or and EIF outbound FIFO. When this bit is set, SW must ensure that the DADR[QID] bitfield is set to a valid queue ID and that the DADR[BASE] bitfield is set to the base address of the destination queue for UIF or EIF, or to zero for a BMU queue. endpoint. 0x0: FLAT (Destination address maps to regular flat memory) 0x1: QUEUE (Destination address maps to a BMU queue or FIFO type memory)
5	SIQ	RW	Source Is Queue. When set the source address is interpreted as a queue ID instead of just the address of a flat memory location. The queue could be either a BMU queue, one of the UIF TX endpoints or and EIF inbound FIFO. When this bit is set, SW must ensure that the SADR[QID] bitfield is set to a valid queue ID and that the SADR[BASE] bitfield is set to the base address of the source queue for UIF or EIF, or to zero for a BMU queue. endpoint. 0x0: FLAT (Source address maps to regular flat memory) 0x1: QUEUE (Source address maps to a BMU queue or FIFO type memory)

WCSS_A_DXE_0_CH3_CTRL (cont.)

Bits	Name	Type	Description
4	BDH	RW	BD Handling bit. The behavior depends on the setting of XTYPE. If XTYPE is: H2B: Then if BDH is: 0: Use first 128 bytes of packet as BD. 1: Create BD from BD template (Up to 4 supported. Also see desc for BDT_IDX). B2H: Then if BDH is: 0: Insert BD in front of payload. 1: Discard BD. For XTYPE values of H2H and B2B the value of this bit is ignored.
3	EOP	RW	End of Packet bit. For H2B transfers marks the end of a packet Note that when not using external descriptors DXE cannot support the gather functionality and therefore the EOP bit must be set. For XTYPE values of H2H, B2B and B2H the value of this bit is ignored.
2:1	XTYPE	RW	Transfer Type. Specifies the type of transfer to perform: 00: H2H. Host space to Host space transfer. 01: B2B. BMU space to BMU space transfer. 10: H2B. Host space to BMU translation. 11: B2H. BMU space to Host space translation. For cross-space transfers (H2B and B2H) the DMA engine will also perform the required data format translation. The format translation behavior can be modified by the setting of the BDH and PAD bits. 0x0: H2H (Host space to Host space transfer) 0x1: B2B (BMU space to BMU space transfer) 0x2: H2B (Host space to BMU space transfer) 0x3: B2H (BMU space to Host space transfer)

WCSS_A_DXE_0_CH3_CTRL (cont.)

Bits	Name	Type	Description
0	EN	RW	Channel Enable or Restart. Writing this bit with a one either starts a new DMA transfer according to the parameters programmed in the channel's configuration registers and clears the DONE bit or, if the channel is already enabled but the STOP bit is set, then it restarts channel processing and also clears the STOP bit. This bit is automatically cleared when a single transfer completes or, if the channel is configured to use linked list descriptors, when the last transfer in the descriptor linked list completes (DESC_NXT is NULL or/and DESC_CSR[VALID]=0). The channel is also disabled and this bit cleared if a channel error is encountered. Reading this bit indicates whether a channel is currently enabled or disabled: 0: Channel Disabled. 1: Channel Enabled. The Channel Enable bit status can also be found by reading the DMA_ENCH register which aggregates the EN bits of all channels. Note: Software must take care to not disable the channel while it is busy or data could be lost.

0x0A2024C4 WCSS_A_DXE_0_CH3_STATUS**Type:** RW**Reset State:** 0x00000000

Serves to report the status of the programmed channel operation.

WCSS_A_DXE_0_CH3_STATUS

Bits	Name	Type	Description
31:16	RSVD	R	Reserved. Reads always return 0. Writes are ignored.
15	INT_DONE	R	Interrupt Source: Asserted when the channel has finished all processing.
14	INT_ERR	R	Interrupt Source: Asserted when a channel error has occurred. Masked by the INE_ERR
13	INT_ED	R	Interrupt Source. Descriptor Done and INT requested
12	ABORT_REQ	R	Abort Request. Software has requested that the current operation be aborted. However, if CH_STATUS[EN] is still set then the channel has registered the request but is not yet in a position to abort the transfer "safely". Once the channel finishes its current operation it will abort the transfer and set the error flag and error code bitfields accordingly.

WCSS_A_DXE_0_CH3_STATUS (cont.)

Bits	Name	Type	Description
11	STOP_REQ	R	Stop Request. Software has requested that the channel stop its operation after the current descriptor has been processed. Once this happens the STOPD bitfield will be set to one and the channel will be disabled.
10:6	ERR_CODE	R	Used to identify the source or cause of an error. 0x0: NONE (No Errors) 0x1: SAHB_ERR (DXE received Error response while performing a transfer on the SAHB) 0x2: H2H_RD_BUS_ERR (SAHB bus error encountered during H2H read operation) 0x3: H2H_WR_BUS_ERR (SAHB bus error encountered during H2H write operation) 0x4: PRG_INV_XTYPE (B2B not yet supported) 0x5: BERR_POPWQ (BMU error on POP WQ command) 0x6: BERR_PUSHWQ (BMU error on PUSH WQ command) 0x7: BERR_RLSS (BMU error on Release BD and attached PDUs command) 0x8: BERR_GETPDU (BMU error on Get PDU command) 0x9: PRG_INV_WQ (Programming error: Unsupported BMU Work Queue) 0xA: PRG_INV_H2H_SRC_QID (Programming error: Invalid H2H source queue ID) 0xB: PRG_INV_H2H_DST_QID (Programming error: Invalid H2H destination queue ID) 0xC: PRG_INV_B2H_SRC_QID (Programming error: Invalid B2H BMU source work queue ID) 0xD: PRG_INV_B2H_DST_QID (Programming error: Invalid B2H destination queue ID) 0xE: PRG_INV_B2H_SRC_IDX (Programming error: Invalid B2H source BD index) 0xF: PRG_INV_H2B_SRC_QID (Programming error: Invalid H2B source queue ID) 0x10: PRG_INV_H2B_DST_QID (Programming error: Invalid H2B BMU destination work queue ID) 0x11: PRG_INV_H2B_DST_IDX (Programming error: Invalid H2B destination BD index) 0x12: PRG_INV_H2B_SZ (Programming error: Invalid H2B transfer size of 0 when BDH=1) 0x13: PRG_INV_SADR (Programming error: Invalid Source Address) 0x14: PRG_INV_DADR (Programming error: Invalid Destination Address) 0x15: PRG_INV_EDADR (Programming error: Invalid Descriptor Address) 0x16: PRG_INV_SRC_WQID (Programming error: Invalid Destination WQ ID)

WCSS_A_DXE_0_CH3_STATUS (cont.)

Bits	Name	Type	Description
			0x17: PRG_INV_DST_WQID (Programming error: Invalid Destination WQ ID) 0x18: PRG_XTYPE_MSM TCH (Programming error: XTYPE mismatch during H2B gather transfer) 0x19: PKT_ERR (Malformed packet) 0x1A: ABORT (Aborted transfer due to SW request) 0x1B: PDU_CNT_OVFL (PDU count field overflow)
5	ERR	RW	DMA channel stopped due to error. To clear the error status it is necessary to write a 1 into this bit position.
4	STOPD	R	Channel Stopped. The channel has stopped further processing and disabled itself due to a stop request in the last processed descriptor.
3	MSKD	R	Channel Masked. The channel has been automatically masked out from arbitration due to a blocking condition.
2	DONE	R	DMA channel Done. This bit will be automatically cleared the next time the channel is enabled.
1	BUSY	R	Channel busy. High when the channel is enabled and currently active (won internal arbitration)
0	EN	R	Channel Enabled. Copy of CH_CTRL[EN].

0x0A2024C8 WCSS_A_DXE_0_CH3_SZ**Type:** RW**Reset State:** 0x00000000

The transfer size register specifies the total, remaining, and burst or 'chunk' transfer sizes for each channel. The maximum total transfer size is 16K-1 bytes. The chunk size control can be useful to control the relative bandwidth used by each channel as the dma engine will change channels, release the AHB bus, and re-arbitrate for access after transferring chunk size bytes.

WCSS_A_DXE_0_CH3_SZ

Bits	Name	Type	Description
31:28	CHK_SZ	RW	Chunk transfer size. Specifies the number of bytes (in multiples of 8) to be transferred at one given time (not implying they are buffered but that they will be transferred for each start event in one bus request cycle). If chunk size is zero the DMA engine will always perform maximum chunk size transfers. Maximum chunk size is 128 bytes.
27:14	REM_SZ	R	Remaining transfer size. Specifies the number of bytes that remain to be transferred to complete the current operation.

WCSS_A_DXE_0_CH3_SZ (cont.)

Bits	Name	Type	Description
13:0	TOT_SZ	RW	Total Transfer Size. Specifies the number of bytes to be transferred. Maximum total transfer size is (4K-1) bytes. Note: This field is unnecessary and will be ignored for B2H or B2B transfers, when the source of the transfer is in BMU space (either BMU queue or direct pointer to BD). This is because the size of the transfer will be the packet length encoded in the BD at offset 0x0c bits [11:0].

0x0A2024CC WCSS_A_DXE_0_CH3_SADRL**Type:** RW**Reset State:** 0x00000000

The channel source address register specifies either the source address in flat internal or external memory, or else the BMU queue ID, UIF TX endpoint ID, or EIF inbound FIFO ID for the transfer. The register must be programmed directly by software before the channel is enabled, then, once the DMA channel is enabled, the register is updated after each atomic chunk transfer (read+write) with the number of bytes transferred. (except when using a UIF/EIF queue) and also after loading a new descriptor. When using flat memory as the source, addresses can have arbitrary byte-alignment. When using a BMU queue, UIF endpoint or EIF fifo as a source, the software is responsible for programming a valid queue ID and correct base address for the queue. The layout of bitfields in this register depends on the setting of the CH_CSR[SIQ] bit. When CH_CSR[SIQ] is clear then the register has no bitfields and the full 32-bits are interpreted as the physical address of a location in flat memory, and when it is set this register has two bitfields: The SQID bitfield used to specify the queue ID, and the BASE bitfield used to provide the base address for the UIF or EIF source endpoint/FIFO.

For B2H or B2B transfers, when the source is a BMU Queue ID (CH_CSR[SIQ] = 1 and SWQID matches BMU), the DXE will issue a BMU POPWQ command to pop the specified queue and obtain the BD index of the source packet. Else, when the source is an address (CH_CSR[SIQ] = 0), the DXE will interpret the address as the index of the packet BD and will add it to the BMU_BASE to obtain the physical address to use to fetch it directly from memory.

For H2B or H2H transfers, if the source is a UIF endpoint or EIF FIFO (CH_CSR[SIQ] = 1 and SQID matches UIF/EIF), SW should program the SAHB base address of the UIF source endpoint in the BASE bitfield and the queue ID to match in the SWQID bitfield, then the DXE will use the base address given to generate the physical address and read the data over the SAHB bus and also use the appropriate sideband signals as specified by the SQID field to communicate with the selected UIF endpoint. SW should make sure that the BASE and SQID fields are consistent and refer to the same endpoint or else hangs or unpredictable behavior might occur. Else, when the source is just a location in flat memory space, (CH_CSR[SIQ] = 0), the bitfields lose their special meaning and DXE will just use the full 32-bits programmed into the register as the physical address to use to fetch the data. For this case, SW should make sure that the provided address maps to the intended source interface (Internal memory, USB, PCI, PCIe) by conforming to the address map specified for the SAHB bus.

To summarize, for B2H and B2B transfers the address must conform to the following format:

For index of BD mode (CH_CSR[SIQ] = 0):

ADDR[31:7] = BMU_BASE + BD_INDEX[15:0]

ADDR[6:0] = 7'd0

For BMU Queue ID mode (CH_CSR[SIQ] = 1 & SQID matching one of the BMU queues):

ADDR[31:7] = Any value will do but zero preferred

ADDR[6:0] = Any BMU Queue ID as specified in description of SQID field

For H2B and H2H transfers the address must conform to the following format:

For sources in flat memory space (CH_CSR[SIQ] = 0): ADDR[31:0] = address mapping to source interface

For UIF RX endpoints or EIF FIFOs (CH_CSR[SIQ] = 1 & QID matching UIF or EIF):

ADDR[31:7] = UIF RX endpoint base or EIF inbound FIFO base

ADDR[6:0] = Queue ID matching UIF or EIF as specified in description of SQID field

WCSS_A_DXE_0_CH3_SADRL

Bits	Name	Description
31:7	BASE	Base Address. Valid only when the CH_CSR[SIQ] bit is set. For this case, this bitfield provides the base address of the source UIF TX endpoint or EIF inbound FIFO. Consult the appropriate interface uarch for the offset within the interface's address space. When CH_CSR[SIQ] bit is not set this field loses its special meaning and is interpreted only as part of the full 32-bit source address.

WCSS_A_DXE_0_CH3_SADRL (cont.)

Bits	Name	Description
6:0	SQID	Source Queue ID. Valid only when the CH_CSR[SIQ] bit is set. For this case, this bitfield provides the ID for the source BMU, UIF or EIF queue to be used for the transfer. When CH_CSR[SIQ] bit is not set this field loses its special meaning and is interpreted only as part of the full 32-bit source address. 0x0: UIF_TX_DATA_EP (UIF TX Data Endpoint) 0x1: UIF_TX_MGMT_EP (UIF TX Management Endpoint) 0x2: BMU_2 (BMU Work Queue 2) 0x3: BMU_3 (BMU Work Queue 3) 0x4: BMU_4 (BMU Work Queue 4) 0x5: BMU_5 (BMU Work Queue 5) 0x6: BMU_6 (BMU Work Queue 6) 0x7: BMU_7 (BMU Work Queue 7) 0x8: BMU_8 (BMU Work Queue 8) 0x9: BMU_9 (BMU Work Queue 9) 0xA: BMU_10 (BMU Work Queue 10) 0xB: BMU_11 (BMU Work Queue 11) 0xC: BMU_12 (BMU Work Queue 12) 0xD: BMU_13 (BMU Work Queue 13) 0xE: BMU_14 (BMU Work Queue 14) 0xF: BMU_15 (BMU Work Queue 15) 0x10: BMU_16 (BMU Work Queue 16) 0x11: BMU_17 (BMU Work Queue 17) 0x12: BMU_18 (BMU Work Queue 18) 0x13: BMU_19 (BMU Work Queue 19) 0x14: BMU_20 (BMU Work Queue 20) 0x15: BMU_21 (BMU Work Queue 21) 0x16: BMU_22 (BMU Work Queue 22) 0x17: BMU_23 (BMU Work Queue 23) 0x18: BMU_24 (BMU Work Queue 24) 0x19: RSVD_25 (Reserved) 0x1A: RSVD_26 (Reserved) 0x1B: RSVD_27 (Reserved) 0x1C: RSVD_28 (Reserved) 0x1D: RSVD_29 (Reserved) 0x1E: RSVD_30 (Reserved) 0x1F: RSVD_31 (Reserved)

0x0A2024D0 WCSS_A_DXE_0_CH3_SADRH**Type:** RW**Reset State:** 0x00000000

The channel source address high register holds the high bits of the source address for a transfer. See the description of CH_SADRL for more info.

WCSS_A_DXE_0_CH3_SADRH

Bits	Name	Type	Description
31:2	RSVD	R	Reserved. Writes ignored. Reads always return zeroes.
1:0	ADDRH	RW	For taurus, this register holds bits 33 and 32 of the address. With the current address mapping these high order bits determine where the transfer is routed and whether the lower 32 bits of the address (the ones held in CH_SADRL) go untranslated through PIF or XIF or if they use the PIF and XIF address translation windows. See the bitfield encoding descriptions for an explanation. 0x0: ADDR_XTLN (If address in PIF or XIF address space then address goes through translation in PIF or XIF) 0x1: PIF_NO_XTLN (Always maps to PIF and full 32-bits in CH_SADRL goes through PIF untranslated) 0x2: XIF_NO_XTLN (Always maps to XIF and full 32-bits in CH_SADRL goes through XIF untranslated) 0x3: RSVD (Reserved)

0x0A2024D4 WCSS_A_DXE_0_CH3_DADRL**Type:** RW**Reset State:** 0x00000000

The channel destination address register specifies either the destination address in flat internal or external memory, or else the BMU queue ID, UIF RX endpoint ID, or EIF outbound FIFO ID for the transfer. The register must be programmed directly by software before the channel is enabled, then, once the DMA channel is enabled, the register is updated after each atomic chunk transfer (read+write) with the number of bytes transferred. (except when using a UIF/EIF queue) and also after loading a new descriptor. When using flat memory as the destination, addresses can have arbitrary byte-alignment. When using a BMU queue, UIF endpoint or EIF fifo as a destination, the software is responsible for programming a valid queue ID and correct base address for the queue. The layout of bitfields in this register depends on the setting of the CH_CSR[DIQ] bit. When this bit is clear then the register has no bitfields and the full 32-bits are interpreted as the physical address of a location in flat memory, and when the bit is set this register has two bitfields: The DQID bitfield used to specify the queue ID, and the BASE bitfield used to provide the base address for the UIF or EIF destination endpoint/FIFO.

For H2B or B2B transfers, when using a BMU queue as the destination (CH_CSR[DIQ] = 1 and DQID matches BMU), the DXE will push the index of the BD of the packet into the specified queue at the end of the packet transfer. Therefore, software is responsible for programming a valid BMU Queue ID.

Else, when the destination is an index (CH_CSR[DIQ] = 0), the BD will be written to the address obtained by adding the provided index to the BMU_BASE, again, the software must ensure that said address is legal and known to BMU. That is, software must preallocate the BD (get the BD from BMU via GET BD/PDU cmd) for the BD before starting the transfer. For B2H or H2H transfers, if the destination is a UIF endpoint or EIF FIFO (CH_CSR[DIQ] = 1 and DQID matches UIF/EIF), SW should program the SAHB base address of the UIF destination endpoint in the

BASE bitfield and the queue ID to match in the DQID bitfield, then the DXE will use the base address given to generate the physical address and write the data over the SAHB bus and also use the appropriate sideband signals as specified by the DQID field to communicate with the selected UIF endpoint. SW should make sure that the BASE and DQID fields are consistent and refer to the same endpoint else hangs or unpredictable behavior might occur. On the other hand, when the destination is just a location in flat memory space, (CH_CSR[DIQ] = 0), be it internal or external, the bitfields lose their special meaning and DXE will just use the full 32-bits programmed into the register as the address to write the data. For this case, SW should make sure that the address maps to the intended destination interface (Internal memory, USB, PCI, PCIe) by conforming to the address map specified for the SAHB bus. To summarize, for H2B and B2B transfers the address must conform to the following format:

For index of BD mode (CH_CSR[DIQ] = 0): ADDR[31:7] = BMU_BASE + BD_INDEX[15:0]

ADDR[6:0] = 7'd0

For BMU Queue ID mode (CH_CSR[DIQ] = 1 & DQID matching one of the BMU queues):

ADDR[31:5] = Any value will do but zero preferred

ADDR[4:0] = Any BMU Queue ID as specified in description of QID field

For B2H and H2H transfers the address must conform to the following format:

For destinations in flat memory space (CH_CSR[DIQ] = 0): ADDR[31:0] = address mapping to destination interface

For UIF RX endpoints or EIF FIFOs (CH_CSR[DIQ] = 1 & DQID matching UIF or EIF):

ADDR[31:5] = UIF RX endpoint base or EIF inbound FIFO base

ADDR[4:0] = Queue ID matching UIF or EIF as specified in description of DQID field

WCSS_A_DXE_0_CH3_DADRL

Bits	Name	Description
31:7	BASE	Base Address. Valid only when the CH_CSR[DIQ] bit is set. For this case, this bitfield provides the base address of the destination UIF RX endpoint or EIF outbound FIFO (for BMU SW may use any value can be used but zero is suggested). Consult the appropriate interface uarch for the offset within the interface's address space. When CH_CSR[SIQ] bit is not set this field loses its special meaning and is interpreted only as part of the full 32-bit source address.

WCSS_A_DXE_0_CH3_DADRL (cont.)

Bits	Name	Description
6:0	DQID	Destination Queue ID. Valid only when the CH_CSR[SIQ] bit is set. For this case, this bitfield provides the ID for the destination BMU WQ, UIF RX endpoint or EIF outbound queue to be used for the transfer. When CH_CSR[SIQ] bit is not set this field loses its special meaning and is interpreted only as part of the full 32-bit source address. 0x0: RSVD_0 (Reserved) 0x1: RSVD_1 (Reserved) 0x2: BMU_2 (BMU Work Queue 2) 0x3: BMU_3 (BMU Work Queue 3) 0x4: BMU_4 (BMU Work Queue 4) 0x5: BMU_5 (BMU Work Queue 5) 0x6: BMU_6 (BMU Work Queue 6) 0x7: BMU_7 (BMU Work Queue 7) 0x8: BMU_8 (BMU Work Queue 8) 0x9: BMU_9 (BMU Work Queue 9) 0xA: BMU_10 (BMU Work Queue 10) 0xB: BMU_11 (BMU Work Queue 11) 0xC: BMU_12 (BMU Work Queue 12) 0xD: BMU_13 (BMU Work Queue 13) 0xE: BMU_14 (BMU Work Queue 14) 0xF: BMU_15 (BMU Work Queue 15) 0x10: BMU_16 (BMU Work Queue 16) 0x11: BMU_17 (BMU Work Queue 17) 0x12: BMU_18 (BMU Work Queue 18) 0x13: BMU_19 (BMU Work Queue 19) 0x14: BMU_20 (BMU Work Queue 20) 0x15: BMU_21 (BMU Work Queue 21) 0x16: BMU_22 (BMU Work Queue 22) 0x17: BMU_23 (BMU Work Queue 23) 0x18: BMU_24 (BMU Work Queue 24) 0x19: BMU_25 (BMU Work Queue 25) 0x1A: BMU_26 (BMU Work Queue 26) 0x1B: BMU_SINK_WQ (BMU_SINK (WQ255)) 0x1C: UIF_RX_1_EP (UIF RX 1 endpoint) 0x1D: UIF_RX_2_EP (UIF RX 2 endpoint) 0x1E: UIF_RX_3_EP (UIF RX 3 endpoint) 0x1F: UIF_RX_4_EP (UIF RX 4 endpoint)

0x0A2024D8 WCSS_A_DXE_0_CH3_DADRH**Type:** RW**Reset State:** 0x00000000

The channel destination address high register holds the high bits of the destination address for a transfer. See the description of CH_DADRL for more info.

WCSS_A_DXE_0_CH3_DADRH

Bits	Name	Type	Description
31:2	RSVD	R	Reserved. Writes ignored. Reads always return zeroes.
1:0	ADDRH	RW	This register holds bits 33 and 32 of the address. With the current address mapping these high order bits determine where the transfer is routed and whether the lower 32 bits of the address (the ones held in CH_DADRL) go untranslated through PIF or XIF or if they use the PIF and XIF address translation windows. See the bitfield encoding descriptions for an explanation. 0x0: ADDR_XTLN (If address in PIF or XIF address space then address goes through translation in PIF or XIF) 0x1: PIF_NO_XTLN (Always maps to PIF and full 32-bits in CH_DADRL goes through PIF untranslated) 0x2: XIF_NO_XTLN (Always maps to XIF and full 32-bits in CH_DADRL goes through XIF untranslated) 0x3: RSVD (Reserved. Maps to default slave. Reads return error, writes are ignored)

0x0A2024DC WCSS_A_DXE_0_CH3_DESCL

Type: RW

Reset State: 0x00000000

The Channel 3 Linked List Descriptor Pointer register contains the 32 lower bits of the address of the next descriptor to be loaded in the linked list. This address must be dword (32-bit) aligned (addr[1:0]=0 if CH_CTRL[PIQ]=0). Each complete transfer descriptor consists of 7 32-bit values arranged contiguously in memory. The value of this register will be overwritten with the address of the start of the next descriptor set in the list after the current descriptor has been fetched. The meaning of the value in this register depends on the value of the CH_CTRL[PIQ] bitfield similar to the way in which SADRL and DADRL are depending in SIQ and DIQ respectively. For CH_CTRL[PIQ]=0 then this register has no bitfields and the value is just interpreted as an address in flat memory space. When CH_CTRL[PIQ] is set then the descriptor will be read from FIFO type memory in UIF and the register has 2 bitfields: BASE, and SQID. BASE specifies the base address of the UIF Tx queue (derived from the start of the UIF address space in SAHB plus the internal UIF offset).

WCSS_A_DXE_0_CH3_DESCL

Bits	Name	Description
31:7	BASE	This bitfield only exists and has meaning when the descriptor needs to be read from a UIF TX queue (and therefore CH_CTRL[PIQ]=1), then it serves to specify the base address of the UIF endpoint to use as the source of the descriptor. Refer to the Nova address map document and to the UIF architecture document for guaranteed correct values to use here. For convenience but without a guarantee of accuracy, the base value to use for both the Tx data and management endpoints is the following: $\text{base} = \text{UIF base} + \text{TD_and_TM_offset} = 0x0E02_5000 + x100 = 0x0E02_5100 \gg 7 = 0x1C_04A2$
6:0	SQID	This bitfield only exists and has meaning when the descriptor needs to be read from a UIF TX queue (and therefore CH_CTRL[PIQ]=1), then it serves to specify which of the two UIF TX queues to use. Note that in this case SW also needs to program the correct base address of the source UIF endpoint in the BASE bitfield. 0x0: UIF_TX_DATA_EP (Descriptor to be read from UIF TX Data Endpoint) 0x1: UIF_TX_MGMT_EP (Descriptor to be read from UIF TX Management Endpoint)

0x0A2024E0 WCSS_A_DXE_0_CH3_DESCH**Type:** RW**Reset State:** 0x00000000

The channel destination address high register holds the high bits of the the address of the next descriptor to be loaded in the linked list.

WCSS_A_DXE_0_CH3_DESCH

Bits	Name	Type	Description
31:2	RSVD	R	Reserved. Writes ignored. Reads always return zeroes.

WCSS_A_DXE_0_CH3_DESCH (cont.)

Bits	Name	Type	Description
1:0	ADDRH	RW	For taurus, this register holds bits 33 and 32 of the address. With the current address mapping these high order bits determine where the transfer is routed and whether the lower 32 bits of the address (the ones held in CH_DADRL) go untranslated through PIF or XIF or if they use the PIF and XIF address translation windows. See the bitfield encoding descriptions for an explanation. Note that if the source of the descriptor is either of the UIF endpoints (CH_CTRL[PIQ]=1) then the value in ADDRH has to be 0 or else the descriptor read could end up mapping to the XIF or PIF. 0x0: ADDR_XTLN (If address in PIF or XIF address space then address goes through translation in PIF or XIF) 0x1: PIF_NO_XTLN (Always maps to PIF and full 32-bits in CH_DADRL goes through PIF untranslated) 0x2: XIF_NO_XTLN (Always maps to XIF and full 32-bits in CH_DADRL goes through XIF untranslated) 0x3: RSVD (Reserved. Maps to default slave. Reads return error, writes are ignored)

0x0A2024E4 WCSS_A_DXE_0_CH3_LST_DESCL**Type:** R**Reset State:** 0x00000000

The read-only Channel 3 Current/last Linked List Descriptor Pointer register contains the low 32-bits of the address of the descriptor currently being processed or of the last descriptor processed if the channel is disabled.

The meaning of the value and bitfields with regards to CH_CTRL[PIQ] is the same as was explained for the CH_DESCL register.

WCSS_A_DXE_0_CH3_LST_DESCL

Bits	Name	Description
31:7	BASE	See description of CH_DESCL[BASE].
6:0	SQID	See description of CH_DESCL[SQID]. 0x0: UIF_TX_DATA_EP (Descriptor to be read from UIF TX Data Endpoint) 0x1: UIF_TX_MGMT_EP (Descriptor to be read from UIF TX Management Endpoint)

0x0A2024E8 WCSS_A_DXE_0_CH3_LST_DESCH**Type:** R**Reset State:** 0x00000000

The channel destination address high register holds the high bits of the the address of the next descriptor to be loaded in the linked list. Note that The meaning of the value and bitfields is the same as was explained for the CH_DESCL register.

WCSS_A_DXE_0_CH3_LST_DESCH

Bits	Name	Description
31:2	RSVD	Reserved. Writes ignored. Reads always return zeroes.
1:0	ADDRH	For taurus, this register holds bits 33 and 32 of the address.

0x0A2024EC WCSS_A_DXE_0_CH3_BD**Type:** R**Reset State:** 0x00000000**WCSS_A_DXE_0_CH3_BD**

Bits	Name	Description
31:24	RSVD	Reserved. Always returns 0 when read.
23	VAL	Valid. BD has been loaded and is valid.
22:16	PDU_CNT	PDU count of packet.
15:0	IDX	Index of BD. To get actual physical address must be added to BD/PDU base.

0x0A2024F0 WCSS_A_DXE_0_CH3_HEAD**Type:** R**Reset State:** 0x00000000

Address of head of packet.

WCSS_A_DXE_0_CH3_HEAD

Bits	Name	Description
31:27	RSVD	Reserved.
26	DESCVAL	Descriptor valid. Signifies descriptor has already been loaded. Used to prevent reloading of a descriptor.

WCSS_A_DXE_0_CH3_HEAD (cont.)

Bits	Name	Description
25:21	DWQID	Destination WQ ID. Original BMU, UIF or EIF destination queue. Valid only when CH_CSR[DIQ]=1
20:16	SWQID	Source WQ ID. Original BMU, UIF or EIF source queue. Valid only when CH_CSR[SIQ]=1
15:0	IDX	Index of Head. Must be added to BD/PDU base for actual physical address.

0x0A2024F4 WCSS_A_DXE_0_CH3_TAIL**Type:** R**Reset State:** 0x00000000

Address of tail of packet.

WCSS_A_DXE_0_CH3_TAIL

Bits	Name	Description
31:16	RSVD	Reserved.
15:0	IDX	Index of Tail. Must be added to BD/PDU base for actual physical address

0x0A2024F8 WCSS_A_DXE_0_CH3_PDU**Type:** R**Reset State:** 0x00000000

PDU info register.

All its fields are working fields that get modified as the programmed operation progresses.

WCSS_A_DXE_0_CH3_PDU

Bits	Name	Description
31	RSVD	Reserved. Always returns 0 when read.
30:24	PDU_CNT	Current PDU count.
23:8	IDX	Index of current PDU. Must be added to BD/PDU base for actual physical address
7:0	FST_OFF	First PDU offset.

0x0A2024FC WCSS_A_DXE_0_CH3_TSTMP**Type:** R**Reset State:** 0x00000000

Contains last sampled value of the system timestamp

WCSS_A_DXE_0_CH3_TSTMP

Bits	Name	Description
31:0	VALUE	

0x0A202500 WCSS_A_DXE_0_CH4_CTRL**Type:** RW**Reset State:** 0x11000708

Serves to configure channel 4 operation

WCSS_A_DXE_0_CH4_CTRL

Bits	Name	Type	Description
31	SWAP	RW	Endian Byte Swap Enable. When high instructs DXE to perform an endian swap of the incoming byte stream. The corresponding bitfield in the DXE descriptor control word is located at DXE_DESC_CTRL[20]
30:29	BDT_IDX	RW	BD Template Index. When XTYPE=H2B and BDH=1, DXE supports creating a BD by using template data from one of up to 4 different BD templates. The BD template base store is pointed at by the DXE BD_TMPLT(H/L) registers and the address used to read the template data is calculated as follows: $BD_TMPLT_ADDR = BD_TMPLT_BASE + (128 * CH_CTRL[BDT_IDX])$ The corresponding bitfield in the control word of the external descriptor is mapped to bits[19:18]
28	DFMT	RW	Descriptor Format. This bitfield is used to select between long and short descriptor formats. When using the short descriptor format, the high part of the source, destination and descriptor addresses will be taken from the SADDRH_DFLT, DADDRH_DFLT or DESCH_DFLT registers respectively. Default is to use the long descriptor format 0x0: SHORT (Short descriptor format (5 dwords)) 0x1: LONG (Long descriptor format (8 dwords))

WCSS_A_DXE_0_CH4_CTRL (cont.)

Bits	Name	Type	Description
27	ABORT	RW	Writing a one to this bit will cause the channel to stop its current transfer and set the ERR bit and ERR_CODE fields in CH_STATUS. Note that if the channel is currently disabled or if it is about to be disabled because it has just finished its programmed transfer, then the abort request will be ignored and the error bit and error codes won't be set. If the channel is currently masked by the arbiter then setting this bit will forcibly unmask it so that the transfer can be aborted the next time the channel is selected by the internal arbiter. The abort request is automatically cleared by hw once the channel has aborted the transfer. Also note that since the DXE will not attempt to do any cleanup when a transfer is aborted, software might need to do so. For example, for H2B transfers, SW should manually release the BD and any PDUs currently in use by the partial transfer or else a BD/PDU leak will result.
26	ENDIANNESS	RW	Specifies the endianness of the master generating the data. Affects alignment of valid bytes for non-dword aligned reads/writes. The corresponding field in the DXE descriptor control word is located in DXE_DESC_CTRL[21] 0x0: BIGEND (System is Big-Endian) 0x1: LTLEND (System is Little-Endian)
25:22	CTR_SEL	RW	Counter Select. If the event counters are enabled (DMA_CSR[ECTR_EN]=1) then this bitfield selects the counter that will be incremented at the end of the transfer. By default the value in this bitfield is the same as the channel number so that after reset channel 0 increments counter 0, channel 1 increments counter 1 and so on.
21	EDVEN	RW	When set, the DMA engine will clear the VAL bit in the DESC_CSR word of the current descriptor once it has completed processing it. This is useful for implementing a descriptor "ring" (see the description of the VALID bit in Table 3.1)
20	EDEN	RW	External Descriptor Enable. Use External Descriptor Linked List
19	INE_DONE	RW	Enable Channel Interrupt when Channel is Done. The corresponding interrupt source bit is INT_DONE
18	INE_ERR	RW	Enable Channel Interrupt on Errors. The corresponding interrupt source bit is INT_ERR
17	INE_ED	RW	Enable External Descriptor Interrupt. When this bit is set, an interrupt will be generated after processing a descriptor item that has the INT bit set in its DESC_CTRL word (see Table 3.1). The corresponding interrupt source bit is INT_ED

WCSS_A_DXE_0_CH4_CTRL (cont.)

Bits	Name	Type	Description
16	STOP	R	Channel processing stopped due to a request in the last processed linked list descriptor (see the STOP bit in the definition of the DESC_CSR word in Table 3.1). This bit will be automatically cleared when the channel is restarted.
15:13	PRIO	RW	This field sets the channel priority. 7: Highest Priority. 0: Lowest Priority.
12:9	BTHLD_SEL	RW	BMU Threshold Select. This value tells DXE which of the BMU BD/PDU thresholds to check before issuing BMU commands that reserve, or get BDs or PDUs. It also specifies the module index value that will be used to issue any required BMU commands (see BMU spec for details). The value of this bitfield is only meaningful for transfer types that involve the BMU and therefore its value will be ignored for H2H transfers. 0x0: RSVD0 (Reserved) 0x1: RSVD1 (Reserved) 0x2: THLD2 (Use BD/PDU threshold 2) 0x3: THLD3 (Use BD/PDU threshold 3) 0x4: THLD4 (Use BD/PDU threshold 4) 0x5: THLD5 (Use BD/PDU threshold 5) 0x6: THLD6 (Use BD/PDU threshold 6) 0x7: THLD7 (Use BD/PDU threshold 7) 0x8: THLD8 (Use BD/PDU threshold 8) 0x9: THLD9 (Use BD/PDU threshold 9) 0xA: THLD10 (Use BD/PDU threshold 10) 0xB: RSVD11 (Reserved) 0xC: RSVD12 (Reserved) 0xD: RSVD13 (Reserved) 0xE: RSVD14 (Reserved) 0xF: RSVD15 (Reserved)
8	PDU_REL	RW	0: Don't release BD and PDUs when done. 1: Release BD and PDUs when done. 0x0: KEEP (Don) 0x1: RELEASE (Release BD and payload PDUs)

WCSS_A_DXE_0_CH4_CTRL (cont.)

Bits	Name	Type	Description
7	PIQ	RW	next descriptor Pointer address Is Queue. When set the source of the descriptor address is interpreted as a queue ID instead of just the address of a flat memory location. The only valid source queues for descriptors are the UIF Tx data and Tx management endpoints. Any other IDs will result the channel aborting the transfer with an invalid source queue error. Therefore, when this bit is set, SW must ensure that the CH_DESCCL[QID] bitfield is set to a valid queue ID and that the CH_DESCCL[BASE] bitfield is set to the base address of the source UIF endpoint. This base address is formed by taking the base address for the UIF block in the SAHB bus which can be found in the Nova address map document and then adding the internal UIF offset for the correct source endpoint. This offsets can be obtained from the UIF specification document. For convenience, the current internal UIF offsets are provided in the description of the QID bitfield in the CH_DESCCL register description. 0x0: FLAT (Source address maps to regular flat memory) 0x1: QUEUE (Source address maps to FIFO type memory (UIF Tx endpoints))
6	DIQ	RW	Destination Is Queue. When set the destination address is interpreted as a queue ID instead of just the address of a flat memory location. The queue could be either a BMU queue, one of the UIF RX endpoints or and EIF outbound FIFO. When this bit is set, SW must ensure that the DADR[QID] bitfield is set to a valid queue ID and that the DADR[BASE] bitfield is set to the base address of the destination queue for UIF or EIF, or to zero for a BMU queue. endpoint. 0x0: FLAT (Destination address maps to regular flat memory) 0x1: QUEUE (Destination address maps to a BMU queue or FIFO type memory)
5	SIQ	RW	Source Is Queue. When set the source address is interpreted as a queue ID instead of just the address of a flat memory location. The queue could be either a BMU queue, one of the UIF TX endpoints or and EIF inbound FIFO. When this bit is set, SW must ensure that the SADR[QID] bitfield is set to a valid queue ID and that the SADR[BASE] bitfield is set to the base address of the source queue for UIF or EIF, or to zero for a BMU queue. endpoint. 0x0: FLAT (Source address maps to regular flat memory) 0x1: QUEUE (Source address maps to a BMU queue or FIFO type memory)

WCSS_A_DXE_0_CH4_CTRL (cont.)

Bits	Name	Type	Description
4	BDH	RW	BD Handling bit. The behavior depends on the setting of XTYPE. If XTYPE is: H2B: Then if BDH is: 0: Use first 128 bytes of packet as BD. 1: Create BD from BD template (Up to 4 supported. Also see desc for BDT_IDX). B2H: Then if BDH is: 0: Insert BD in front of payload. 1: Discard BD. For XTYPE values of H2H and B2B the value of this bit is ignored.
3	EOP	RW	End of Packet bit. For H2B transfers marks the end of a packet Note that when not using external descriptors DXE cannot support the gather functionality and therefore the EOP bit must be set. For XTYPE values of H2H, B2B and B2H the value of this bit is ignored.
2:1	XTYPE	RW	Transfer Type. Specifies the type of transfer to perform: 00: H2H. Host space to Host space transfer. 01: B2B. BMU space to BMU space transfer. 10: H2B. Host space to BMU translation. 11: B2H. BMU space to Host space translation. For cross-space transfers (H2B and B2H) the DMA engine will also perform the required data format translation. The format translation behavior can be modified by the setting of the BDH and PAD bits. 0x0: H2H (Host space to Host space transfer) 0x1: B2B (BMU space to BMU space transfer) 0x2: H2B (Host space to BMU space transfer) 0x3: B2H (BMU space to Host space transfer)

WCSS_A_DXE_0_CH4_CTRL (cont.)

Bits	Name	Type	Description
0	EN	RW	Channel Enable or Restart. Writing this bit with a one either starts a new DMA transfer according to the parameters programmed in the channel's configuration registers and clears the DONE bit or, if the channel is already enabled but the STOP bit is set, then it restarts channel processing and also clears the STOP bit. This bit is automatically cleared when a single transfer completes or, if the channel is configured to use linked list descriptors, when the last transfer in the descriptor linked list completes (DESC_NXT is NULL or/and DESC_CSR[VALID]=0). The channel is also disabled and this bit cleared if a channel error is encountered. Reading this bit indicates whether a channel is currently enabled or disabled: 0: Channel Disabled. 1: Channel Enabled. The Channel Enable bit status can also be found by reading the DMA_ENCH register which aggregates the EN bits of all channels. Note: Software must take care to not disable the channel while it is busy or data could be lost.

0x0A202504 WCSS_A_DXE_0_CH4_STATUS**Type:** RW**Reset State:** 0x00000000

Serves to report the status of the programmed channel operation.

WCSS_A_DXE_0_CH4_STATUS

Bits	Name	Type	Description
31:16	RSVD	R	Reserved. Reads always return 0. Writes are ignored.
15	INT_DONE	R	Interrupt Source: Asserted when the channel has finished all processing.
14	INT_ERR	R	Interrupt Source: Asserted when a channel error has occurred. Masked by the INE_ERR
13	INT_ED	R	Interrupt Source. Descriptor Done and INT requested
12	ABORT_REQ	R	Abort Request. Software has requested that the current operation be aborted. However, if CH_STATUS[EN] is still set then the channel has registered the request but is not yet in a position to abort the transfer "safely". Once the channel finishes its current operation it will abort the transfer and set the error flag and error code bitfields accordingly.

WCSS_A_DXE_0_CH4_STATUS (cont.)

Bits	Name	Type	Description
11	STOP_REQ	R	Stop Request. Software has requested that the channel stop its operation after the current descriptor has been processed. Once this happens the STOPD bitfield will be set to one and the channel will be disabled.
10:6	ERR_CODE	R	Used to identify the source or cause of an error. 0x0: NONE (No Errors) 0x1: SAHB_ERR (DXE received Error response while performing a transfer on the SAHB) 0x2: H2H_RD_BUS_ERR (SAHB bus error encountered during H2H read operation) 0x3: H2H_WR_BUS_ERR (SAHB bus error encountered during H2H write operation) 0x4: PRG_INV_XTYPE (B2B not yet supported) 0x5: BERR_POPWQ (BMU error on POP WQ command) 0x6: BERR_PUSHWQ (BMU error on PUSH WQ command) 0x7: BERR_RLSS (BMU error on Release BD and attached PDUs command) 0x8: BERR_GETPDU (BMU error on Get PDU command) 0x9: PRG_INV_WQ (Programming error: Unsupported BMU Work Queue) 0xA: PRG_INV_H2H_SRC_QID (Programming error: Invalid H2H source queue ID) 0xB: PRG_INV_H2H_DST_QID (Programming error: Invalid H2H destination queue ID) 0xC: PRG_INV_B2H_SRC_QID (Programming error: Invalid B2H BMU source work queue ID) 0xD: PRG_INV_B2H_DST_QID (Programming error: Invalid B2H destination queue ID) 0xE: PRG_INV_B2H_SRC_IDX (Programming error: Invalid B2H source BD index) 0xF: PRG_INV_H2B_SRC_QID (Programming error: Invalid H2B source queue ID) 0x10: PRG_INV_H2B_DST_QID (Programming error: Invalid H2B BMU destination work queue ID) 0x11: PRG_INV_H2B_DST_IDX (Programming error: Invalid H2B destination BD index) 0x12: PRG_INV_H2B_SZ (Programming error: Invalid H2B transfer size of 0 when BDH=1) 0x13: PRG_INV_SADR (Programming error: Invalid Source Address) 0x14: PRG_INV_DADR (Programming error: Invalid Destination Address) 0x15: PRG_INV_EDADR (Programming error: Invalid Descriptor Address) 0x16: PRG_INV_SRC_WQID (Programming error: Invalid Destination WQ ID)

WCSS_A_DXE_0_CH4_STATUS (cont.)

Bits	Name	Type	Description
			0x17: PRG_INV_DST_WQID (Programming error: Invalid Destination WQ ID) 0x18: PRG_XTYPE_MSM TCH (Programming error: XTYPE mismatch during H2B gather transfer) 0x19: PKT_ERR (Malformed packet) 0x1A: ABORT (Aborted transfer due to SW request) 0x1B: PDU_CNT_OVFL (PDU count field overflow)
5	ERR	RW	DMA channel stopped due to error. To clear the error status it is necessary to write a 1 into this bit position.
4	STOPD	R	Channel Stopped. The channel has stopped further processing and disabled itself due to a stop request in the last processed descriptor.
3	MSKD	R	Channel Masked. The channel has been automatically masked out from arbitration due to a blocking condition.
2	DONE	R	DMA channel Done. This bit will be automatically cleared the next time the channel is enabled.
1	BUSY	R	Channel busy. High when the channel is enabled and currently active (won internal arbitration)
0	EN	R	Channel Enabled. Copy of CH_CTRL[EN].

0x0A202508 WCSS_A_DXE_0_CH4_SZ**Type:** RW**Reset State:** 0x00000000

The transfer size register specifies the total, remaining, and burst or 'chunk' transfer sizes for each channel. The maximum total transfer size is 16K-1 bytes. The chunk size control can be useful to control the relative bandwidth used by each channel as the dma engine will change channels, release the AHB bus, and rearbitrate for access after transferring chunk size bytes.

WCSS_A_DXE_0_CH4_SZ

Bits	Name	Type	Description
31:28	CHK_SZ	RW	Chunk transfer size. Specifies the number of bytes (in multiples of 8) to be transferred at one given time (not implying they are buffered but that they will be transferred for each start event in one bus request cycle). If chunk size is zero the DMA engine will always perform maximum chunk size transfers. Maximum chunk size is 128 bytes.
27:14	REM_SZ	R	Remaining transfer size. Specifies the number of bytes that remain to be transferred to complete the current operation.

WCSS_A_DXE_0_CH4_SZ (cont.)

Bits	Name	Type	Description
13:0	TOT_SZ	RW	Total Transfer Size. Specifies the number of bytes to be transferred. Maximum total transfer size is (4K-1) bytes. Note: This field is unnecessary and will be ignored for B2H or B2B transfers, when the source of the transfer is in BMU space (either BMU queue or direct pointer to BD). This is because the size of the transfer will be the packet length encoded in the BD at offset 0x0c bits [11:0].

0x0A20250C WCSS_A_DXE_0_CH4_SADRL**Type:** RW**Reset State:** 0x00000000

The channel source address register specifies either the source address in flat internal or external memory, or else the BMU queue ID, UIF TX endpoint ID, or EIF inbound FIFO ID for the transfer. The register must be programmed directly by software before the channel is enabled, then, once the DMA channel is enabled, the register is updated after each atomic chunk transfer (read+write) with the number of bytes transferred. (except when using a UIF/EIF queue) and also after loading a new descriptor. When using flat memory as the source, addresses can have arbitrary byte-alignment. When using a BMU queue, UIF endpoint or EIF fifo as a source, the software is responsible for programming a valid queue ID and correct base address for the queue. The layout of bitfields in this register depends on the setting of the CH_CSR[SIQ] bit. When CH_CSR[SIQ] is clear then the register has no bitfields and the full 32-bits are interpreted as the physical address of a location in flat memory, and when it is set this register has two bitfields: The SQID bitfield used to specify the queue ID, and the BASE bitfield used to provide the base address for the UIF or EIF source endpoint/FIFO.

For B2H or B2B transfers, when the source is a BMU Queue ID (CH_CSR[SIQ] = 1 and SWQID matches BMU), the DXE will issue a BMU POPWQ command to pop the specified queue and obtain the BD index of the source packet. Else, when the source is an address (CH_CSR[SIQ] = 0), the DXE will interpret the address as the index of the packet BD and will add it to the BMU_BASE to obtain the physical address to use to fetch it directly from memory.

For H2B or H2H transfers, if the source is a UIF endpoint or EIF FIFO (CH_CSR[SIQ] = 1 and SQID matches UIF/EIF), SW should program the SAHB base address of the UIF source endpoint in the BASE bitfield and the queue ID to match in the SWQID bitfield, then the DXE will use the base address given to generate the physical address and read the data over the SAHB bus and also use the appropriate sideband signals as specified by the SQID field to communicate with the selected UIF endpoint. SW should make sure that the BASE and SQID fields are consistent and refer to the same endpoint or else hangs or unpredictable behavior might occur. Else, when the source is just a location in flat memory space, (CH_CSR[SIQ] = 0), the bitfields lose their special meaning and DXE will just use the full 32-bits programmed into the register as the physical address to use to fetch the data. For this case, SW should make sure that the provided address maps to the intended source interface (Internal memory, USB, PCI, PCIe) by conforming to the address map specified for the SAHB bus.

To summarize, for B2H and B2B transfers the address must conform to the following format:

For index of BD mode (CH_CSR[SIQ] = 0):

ADDR[31:7] = BMU_BASE + BD_INDEX[15:0]

ADDR[6:0] = 7'd0

For BMU Queue ID mode (CH_CSR[SIQ] = 1 & SQID matching one of the BMU queues):

ADDR[31:7] = Any value will do but zero preferred

ADDR[6:0] = Any BMU Queue ID as specified in description of SQID field

For H2B and H2H transfers the address must conform to the following format:

For sources in flat memory space (CH_CSR[SIQ] = 0): ADDR[31:0] = address mapping to source interface

For UIF RX endpoints or EIF FIFOs (CH_CSR[SIQ] = 1 & QID matching UIF or EIF):

ADDR[31:7] = UIF RX endpoint base or EIF inbound FIFO base

ADDR[6:0] = Queue ID matching UIF or EIF as specified in description of SQID field

WCSS_A_DXE_0_CH4_SADRL

Bits	Name	Description
31:7	BASE	Base Address. Valid only when the CH_CSR[SIQ] bit is set. For this case, this bitfield provides the base address of the source UIF TX endpoint or EIF inbound FIFO. Consult the appropriate interface uarch for the offset within the interface's address space. When CH_CSR[SIQ] bit is not set this field loses its special meaning and is interpreted only as part of the full 32-bit source address.

WCSS_A_DXE_0_CH4_SADRL (cont.)

Bits	Name	Description
6:0	SQID	Source Queue ID. Valid only when the CH_CSR[SIQ] bit is set. For this case, this bitfield provides the ID for the source BMU, UIF or EIF queue to be used for the transfer. When CH_CSR[SIQ] bit is not set this field loses its special meaning and is interpreted only as part of the full 32-bit source address. 0x0: UIF_TX_DATA_EP (UIF TX Data Endpoint) 0x1: UIF_TX_MGMT_EP (UIF TX Management Endpoint) 0x2: BMU_2 (BMU Work Queue 2) 0x3: BMU_3 (BMU Work Queue 3) 0x4: BMU_4 (BMU Work Queue 4) 0x5: BMU_5 (BMU Work Queue 5) 0x6: BMU_6 (BMU Work Queue 6) 0x7: BMU_7 (BMU Work Queue 7) 0x8: BMU_8 (BMU Work Queue 8) 0x9: BMU_9 (BMU Work Queue 9) 0xA: BMU_10 (BMU Work Queue 10) 0xB: BMU_11 (BMU Work Queue 11) 0xC: BMU_12 (BMU Work Queue 12) 0xD: BMU_13 (BMU Work Queue 13) 0xE: BMU_14 (BMU Work Queue 14) 0xF: BMU_15 (BMU Work Queue 15) 0x10: BMU_16 (BMU Work Queue 16) 0x11: BMU_17 (BMU Work Queue 17) 0x12: BMU_18 (BMU Work Queue 18) 0x13: BMU_19 (BMU Work Queue 19) 0x14: BMU_20 (BMU Work Queue 20) 0x15: BMU_21 (BMU Work Queue 21) 0x16: BMU_22 (BMU Work Queue 22) 0x17: BMU_23 (BMU Work Queue 23) 0x18: BMU_24 (BMU Work Queue 24) 0x19: RSVD_25 (Reserved) 0x1A: RSVD_26 (Reserved) 0x1B: RSVD_27 (Reserved) 0x1C: RSVD_28 (Reserved) 0x1D: RSVD_29 (Reserved) 0x1E: RSVD_30 (Reserved) 0x1F: RSVD_31 (Reserved)

0x0A202510 WCSS_A_DXE_0_CH4_SADRH**Type:** RW**Reset State:** 0x00000000

The channel source address high register holds the high bits of the source address for a transfer. See the description of CH_SADRL for more info.

WCSS_A_DXE_0_CH4_SADRH

Bits	Name	Type	Description
31:2	RSVD	R	Reserved. Writes ignored. Reads always return zeroes.
1:0	ADDRH	RW	For taurus, this register holds bits 33 and 32 of the address. With the current address mapping these high order bits determine where the transfer is routed and whether the lower 32 bits of the address (the ones held in CH_SADRL) go untranslated through PIF or XIF or if they use the PIF and XIF address translation windows. See the bitfield encoding descriptions for an explanation. 0x0: ADDR_XTLN (If address in PIF or XIF address space then address goes through translation in PIF or XIF) 0x1: PIF_NO_XTLN (Always maps to PIF and full 32-bits in CH_SADRL goes through PIF untranslated) 0x2: XIF_NO_XTLN (Always maps to XIF and full 32-bits in CH_SADRL goes through XIF untranslated) 0x3: RSVD (Reserved)

0x0A202514 WCSS_A_DXE_0_CH4_DADRL**Type:** RW**Reset State:** 0x00000000

The channel destination address register specifies either the destination address in flat internal or external memory, or else the BMU queue ID, UIF RX endpoint ID, or EIF outbound FIFO ID for the transfer. The register must be programmed directly by software before the channel is enabled, then, once the DMA channel is enabled, the register is updated after each atomic chunk transfer (read+write) with the number of bytes transferred. (except when using a UIF/EIF queue) and also after loading a new descriptor. When using flat memory as the destination, addresses can have arbitrary byte-alignment. When using a BMU queue, UIF endpoint or EIF fifo as a destination, the software is responsible for programming a valid queue ID and correct base address for the queue. The layout of bitfields in this register depends on the setting of the CH_CSR[DIQ] bit. When this bit is clear then the register has no bitfields and the full 32-bits are interpreted as the physical address of a location in flat memory, and when the bit is set this register has two bitfields: The DQID bitfield used to specify the queue ID, and the BASE bitfield used to provide the base address for the UIF or EIF destination endpoint/FIFO.

For H2B or B2B transfers, when using a BMU queue as the destination (CH_CSR[DIQ] = 1 and DQID matches BMU), the DXE will push the index of the BD of the packet into the specified queue at the end of the packet transfer. Therefore, software is responsible for programming a valid BMU Queue ID.

Else, when the destination is an index (CH_CSR[DIQ] = 0), the BD will be written to the address obtained by adding the provided index to the BMU_BASE, again, the software must ensure that said address is legal and known to BMU. That is, software must preallocate the BD (get the BD from BMU via GET BD/PDU cmd) for the BD before starting the transfer. For B2H or H2H transfers, if the destination is a UIF endpoint or EIF FIFO (CH_CSR[DIQ] = 1 and DQID matches UIF/EIF), SW should program the SAHB base address of the UIF destination endpoint in the

BASE bitfield and the queue ID to match in the DQID bitfield, then the DXE will use the base address given to generate the physical address and write the data over the SAHB bus and also use the appropriate sideband signals as specified by the DQID field to communicate with the selected UIF endpoint. SW should make sure that the BASE and DQID fields are consistent and refer to the same endpoint else hangs or unpredictable behavior might occur. On the other hand, when the destination is just a location in flat memory space, (CH_CSR[DIQ] = 0), be it internal or external, the bitfields lose their special meaning and DXE will just use the full 32-bits programmed into the register as the address to write the data. For this case, SW should make sure that the address maps to the intended destination interface (Internal memory, USB, PCI, PCIe) by conforming to the address map specified for the SAHB bus. To summarize, for H2B and B2B transfers the address must conform to the following format:

For index of BD mode (CH_CSR[DIQ] = 0): ADDR[31:7] = BMU_BASE + BD_INDEX[15:0]

ADDR[6:0] = 7'd0

For BMU Queue ID mode (CH_CSR[DIQ] = 1 & DQID matching one of the BMU queues):

ADDR[31:5] = Any value will do but zero preferred

ADDR[4:0] = Any BMU Queue ID as specified in description of QID field

For B2H and H2H transfers the address must conform to the following format:

For destinations in flat memory space (CH_CSR[DIQ] = 0): ADDR[31:0] = address mapping to destination interface

For UIF RX endpoints or EIF FIFOs (CH_CSR[DIQ] = 1 & DQID matching UIF or EIF):

ADDR[31:5] = UIF RX endpoint base or EIF inbound FIFO base

ADDR[4:0] = Queue ID matching UIF or EIF as specified in description of DQID field

WCSS_A_DXE_0_CH4_DADRL

Bits	Name	Description
31:7	BASE	Base Address. Valid only when the CH_CSR[DIQ] bit is set. For this case, this bitfield provides the base address of the destination UIF RX endpoint or EIF outbound FIFO (for BMU SW may use any value can be used but zero is suggested). Consult the appropriate interface uarch for the offset within the interface's address space. When CH_CSR[SIQ] bit is not set this field loses its special meaning and is interpreted only as part of the full 32-bit source address.

WCSS_A_DXE_0_CH4_DADRL (cont.)

Bits	Name	Description
6:0	DQID	Destination Queue ID. Valid only when the CH_CSR[SIQ] bit is set. For this case, this bitfield provides the ID for the destination BMU WQ, UIF RX endpoint or EIF outbound queue to be used for the transfer. When CH_CSR[SIQ] bit is not set this field loses its special meaning and is interpreted only as part of the full 32-bit source address. 0x0: RSVD_0 (Reserved) 0x1: RSVD_1 (Reserved) 0x2: BMU_2 (BMU Work Queue 2) 0x3: BMU_3 (BMU Work Queue 3) 0x4: BMU_4 (BMU Work Queue 4) 0x5: BMU_5 (BMU Work Queue 5) 0x6: BMU_6 (BMU Work Queue 6) 0x7: BMU_7 (BMU Work Queue 7) 0x8: BMU_8 (BMU Work Queue 8) 0x9: BMU_9 (BMU Work Queue 9) 0xA: BMU_10 (BMU Work Queue 10) 0xB: BMU_11 (BMU Work Queue 11) 0xC: BMU_12 (BMU Work Queue 12) 0xD: BMU_13 (BMU Work Queue 13) 0xE: BMU_14 (BMU Work Queue 14) 0xF: BMU_15 (BMU Work Queue 15) 0x10: BMU_16 (BMU Work Queue 16) 0x11: BMU_17 (BMU Work Queue 17) 0x12: BMU_18 (BMU Work Queue 18) 0x13: BMU_19 (BMU Work Queue 19) 0x14: BMU_20 (BMU Work Queue 20) 0x15: BMU_21 (BMU Work Queue 21) 0x16: BMU_22 (BMU Work Queue 22) 0x17: BMU_23 (BMU Work Queue 23) 0x18: BMU_24 (BMU Work Queue 24) 0x19: BMU_25 (BMU Work Queue 25) 0x1A: BMU_26 (BMU Work Queue 26) 0x1B: BMU_SINK_WQ (BMU_SINK (WQ255)) 0x1C: UIF_RX_1_EP (UIF RX 1 endpoint) 0x1D: UIF_RX_2_EP (UIF RX 2 endpoint) 0x1E: UIF_RX_3_EP (UIF RX 3 endpoint) 0x1F: UIF_RX_4_EP (UIF RX 4 endpoint)

0x0A202518 WCSS_A_DXE_0_CH4_DADRH**Type:** RW**Reset State:** 0x00000000

The channel destination address high register holds the high bits of the destination address for a transfer. See the description of CH_DADRL for more info.

WCSS_A_DXE_0_CH4_DADRH

Bits	Name	Type	Description
31:2	RSVD	R	Reserved. Writes ignored. Reads always return zeroes.
1:0	ADDRH	RW	This register holds bits 33 and 32 of the address. With the current address mapping these high order bits determine where the transfer is routed and whether the lower 32 bits of the address (the ones held in CH_DADRL) go untranslated through PIF or XIF or if they use the PIF and XIF address translation windows. See the bitfield encoding descriptions for an explanation. 0x0: ADDR_XTLN (If address in PIF or XIF address space then address goes through translation in PIF or XIF) 0x1: PIF_NO_XTLN (Always maps to PIF and full 32-bits in CH_DADRL goes through PIF untranslated) 0x2: XIF_NO_XTLN (Always maps to XIF and full 32-bits in CH_DADRL goes through XIF untranslated) 0x3: RSVD (Reserved. Maps to default slave. Reads return error, writes are ignored)

0x0A20251C WCSS_A_DXE_0_CH4_DESCL

Type: RW

Reset State: 0x00000000

The Channel 4 Linked List Descriptor Pointer register contains the 32 lower bits of the address of the next descriptor to be loaded in the linked list. This address must be dword (32-bit) aligned (addr[1:0]=0 if CH_CTRL[PIQ]=0). Each complete transfer descriptor consists of 7 32-bit values arranged contiguously in memory. The value of this register will be overwritten with the address of the start of the next descriptor set in the list after the current descriptor has been fetched. The meaning of the value in this register depends on the value of the CH_CTRL[PIQ] bitfield similar to the way in which SADRL and DADRL are depending in SIQ and DIQ respectively. For CH_CTRL[PIQ]=0 then this register has no bitfields and the value is just interpreted as an address in flat memory space. When CH_CTRL[PIQ] is set then the descriptor will be read from FIFO type memory in UIF and the register has 2 bitfields: BASE, and SQID. BASE specifies the base address of the UIF Tx queue (derived from the start of the UIF address space in SAHB plus the internal UIF offset).

WCSS_A_DXE_0_CH4_DESCL

Bits	Name	Description
31:7	BASE	This bitfield only exists and has meaning when the descriptor needs to be read from a UIF TX queue (and therefore CH_CTRL[PIQ]=1), then it serves to specify the base address of the UIF endpoint to use as the source of the descriptor. Refer to the Nova address map document and to the UIF architecture document for guaranteed correct values to use here. For convenience but without a guarantee of accuracy, the base value to use for both the Tx data and management endpoints is the following: $\text{base} = \text{UIF base} + \text{TD_and_TM_offset} = 0x0E02_5000 + x100 = 0x0E02_5100 \gg 7 = 0x1C_04A2$
6:0	SQID	This bitfield only exists and has meaning when the descriptor needs to be read from a UIF TX queue (and therefore CH_CTRL[PIQ]=1), then it serves to specify which of the two UIF TX queues to use. Note that in this case SW also needs to program the correct base address of the source UIF endpoint in the BASE bitfield. 0x0: UIF_TX_DATA_EP (Descriptor to be read from UIF TX Data Endpoint) 0x1: UIF_TX_MGMT_EP (Descriptor to be read from UIF TX Management Endpoint)

0x0A202520 WCSS_A_DXE_0_CH4_DESCH**Type:** RW**Reset State:** 0x00000000

The channel destination address high register holds the high bits of the the address of the next descriptor to be loaded in the linked list.

WCSS_A_DXE_0_CH4_DESCH

Bits	Name	Type	Description
31:2	RSVD	R	Reserved. Writes ignored. Reads always return zeroes.

WCSS_A_DXE_0_CH4_DESCH (cont.)

Bits	Name	Type	Description
1:0	ADDRH	RW	For taurus, this register holds bits 33 and 32 of the address. With the current address mapping these high order bits determine where the transfer is routed and whether the lower 32 bits of the address (the ones held in CH_DADRL) go untranslated through PIF or XIF or if they use the PIF and XIF address translation windows. See the bitfield encoding descriptions for an explanation. Note that if the source of the descriptor is either of the UIF endpoints (CH_CTRL[PIQ]=1) then the value in ADDRH has to be 0 or else the descriptor read could end up mapping to the XIF or PIF. 0x0: ADDR_XTLN (If address in PIF or XIF address space then address goes through translation in PIF or XIF) 0x1: PIF_NO_XTLN (Always maps to PIF and full 32-bits in CH_DADRL goes through PIF untranslated) 0x2: XIF_NO_XTLN (Always maps to XIF and full 32-bits in CH_DADRL goes through XIF untranslated) 0x3: RSVD (Reserved. Maps to default slave. Reads return error, writes are ignored)

0x0A202524 WCSS_A_DXE_0_CH4_LST_DESCL**Type:** R**Reset State:** 0x00000000

The read-only Channel 4 Current/last Linked List Descriptor Pointer register contains the low 32-bits of the address of the descriptor currently being processed or of the last descriptor processed if the channel is disabled.

The meaning of the value and bitfields with regards to CH_CTRL[PIQ] is the same as was explained for the CH_DESCL register.

WCSS_A_DXE_0_CH4_LST_DESCL

Bits	Name	Description
31:7	BASE	See description of CH_DESCL[BASE].
6:0	SQID	See description of CH_DESCL[SQID]. 0x0: UIF_TX_DATA_EP (Descriptor to be read from UIF TX Data Endpoint) 0x1: UIF_TX_MGMT_EP (Descriptor to be read from UIF TX Management Endpoint)

0x0A202528 WCSS_A_DXE_0_CH4_LST_DESCH**Type:** R**Reset State:** 0x00000000

The channel destination address high register holds the high bits of the the address of the next descriptor to be loaded in the linked list. Note that The meaning of the value and bitfields is the same as was explained for the CH_DESCL register.

WCSS_A_DXE_0_CH4_LST_DESCH

Bits	Name	Description
31:2	RSVD	Reserved. Writes ignored. Reads always return zeroes.
1:0	ADDRH	For taurus, this register holds bits 33 and 32 of the address.

0x0A20252C WCSS_A_DXE_0_CH4_BD**Type:** R**Reset State:** 0x00000000**WCSS_A_DXE_0_CH4_BD**

Bits	Name	Description
31:24	RSVD	Reserved. Always returns 0 when read.
23	VAL	Valid. BD has been loaded and is valid.
22:16	PDU_CNT	PDU count of packet.
15:0	IDX	Index of BD. To get actual physical address must be added to BD/PDU base.

0x0A202530 WCSS_A_DXE_0_CH4_HEAD**Type:** R**Reset State:** 0x00000000

Address of head of packet.

WCSS_A_DXE_0_CH4_HEAD

Bits	Name	Description
31:27	RSVD	Reserved.
26	DESCVAL	Descriptor valid. Signifies descriptor has already been loaded. Used to prevent reloading of a descriptor.

WCSS_A_DXE_0_CH4_HEAD (cont.)

Bits	Name	Description
25:21	DWQID	Destination WQ ID. Original BMU, UIF or EIF destination queue. Valid only when CH_CSR[DIQ]=1
20:16	SWQID	Source WQ ID. Original BMU, UIF or EIF source queue. Valid only when CH_CSR[SIQ]=1
15:0	IDX	Index of Head. Must be added to BD/PDU base for actual physical address.

0x0A202534 WCSS_A_DXE_0_CH4_TAIL**Type:** R**Reset State:** 0x00000000

Address of tail of packet.

WCSS_A_DXE_0_CH4_TAIL

Bits	Name	Description
31:16	RSVD	Reserved.
15:0	IDX	Index of Tail. Must be added to BD/PDU base for actual physical address

0x0A202538 WCSS_A_DXE_0_CH4_PDU**Type:** R**Reset State:** 0x00000000

PDU info register.

All its fields are working fields that get modified as the programmed operation progresses.

WCSS_A_DXE_0_CH4_PDU

Bits	Name	Description
31	RSVD	Reserved. Always returns 0 when read.
30:24	PDU_CNT	Current PDU count.
23:8	IDX	Index of current PDU. Must be added to BD/PDU base for actual physical address
7:0	FST_OFF	First PDU offset.

0x0A20253C WCSS_A_DXE_0_CH4_TSTMP**Type:** R**Reset State:** 0x00000000

Contains last sampled value of the system timestamp

WCSS_A_DXE_0_CH4_TSTMP

Bits	Name	Description
31:0	VALUE	

0x0A202540 WCSS_A_DXE_0_CH5_CTRL**Type:** RW**Reset State:** 0x11400708

Serves to configure channel 5 operation

WCSS_A_DXE_0_CH5_CTRL

Bits	Name	Type	Description
31	SWAP	RW	Endian Byte Swap Enable. When high instructs DXE to perform an endian swap of the incoming byte stream. The corresponding bitfield in the DXE descriptor control word is located at DXE_DESC_CTRL[20]
30:29	BDT_IDX	RW	BD Template Index. When XTYPE=H2B and BDH=1, DXE supports creating a BD by using template data from one of up to 4 different BD templates. The BD template base store is pointed at by the DXE BD_TMPLT(H/L) registers and the address used to read the template data is calculated as follows: $BD_TMPLT_ADDR = BD_TMPLT_BASE + (128 * CH_CTRL[BDT_IDX])$ The corresponding bitfield in the control word of the external descriptor is mapped to bits[19:18]
28	DFMT	RW	Descriptor Format. This bitfield is used to select between long and short descriptor formats. When using the short descriptor format, the high part of the source, destination and descriptor addresses will be taken from the SADDRH_DFLT, DADDRH_DFLT or DESCH_DFLT registers respectively. Default is to use the long descriptor format 0x0: SHORT (Short descriptor format (5 dwords)) 0x1: LONG (Long descriptor format (8 dwords))

WCSS_A_DXE_0_CH5_CTRL (cont.)

Bits	Name	Type	Description
27	ABORT	RW	Writing a one to this bit will cause the channel to stop its current transfer and set the ERR bit and ERR_CODE fields in CH_STATUS. Note that if the channel is currently disabled or if it is about to be disabled because it has just finished its programmed transfer, then the abort request will be ignored and the error bit and error codes won't be set. If the channel is currently masked by the arbiter then setting this bit will forcibly unmask it so that the transfer can be aborted the next time the channel is selected by the internal arbiter. The abort request is automatically cleared by hw once the channel has aborted the transfer. Also note that since the DXE will not attempt to do any cleanup when a transfer is aborted, software might need to do so. For example, for H2B transfers, SW should manually release the BD and any PDUs currently in use by the partial transfer or else a BD/PDU leak will result.
26	ENDIANNESS	RW	Specifies the endianness of the master generating the data. Affects alignment of valid bytes for non-dword aligned reads/writes. The corresponding field in the DXE descriptor control word is located in DXE_DESC_CTRL[21] 0x0: BIGEND (System is Big-Endian) 0x1: LTLEND (System is Little-Endian)
25:22	CTR_SEL	RW	Counter Select. If the event counters are enabled (DMA_CSR[ECTR_EN]=1) then this bitfield selects the counter that will be incremented at the end of the transfer. By default the value in this bitfield is the same as the channel number so that after reset channel 0 increments counter 0, channel 1 increments counter 1 and so on.
21	EDVEN	RW	When set, the DMA engine will clear the VAL bit in the DESC_CSR word of the current descriptor once it has completed processing it. This is useful for implementing a descriptor "ring" (see the description of the VALID bit in Table 3.1)
20	EDEN	RW	External Descriptor Enable. Use External Descriptor Linked List
19	INE_DONE	RW	Enable Channel Interrupt when Channel is Done. The corresponding interrupt source bit is INT_DONE
18	INE_ERR	RW	Enable Channel Interrupt on Errors. The corresponding interrupt source bit is INT_ERR
17	INE_ED	RW	Enable External Descriptor Interrupt. When this bit is set, an interrupt will be generated after processing a descriptor item that has the INT bit set in its DESC_CTRL word (see Table 3.1). The corresponding interrupt source bit is INT_ED

WCSS_A_DXE_0_CH5_CTRL (cont.)

Bits	Name	Type	Description
16	STOP	R	Channel processing stopped due to a request in the last processed linked list descriptor (see the STOP bit in the definition of the DESC_CSR word in Table 3.1). This bit will be automatically cleared when the channel is restarted.
15:13	PRIO	RW	This field sets the channel priority. 7: Highest Priority. 0: Lowest Priority.
12:9	BTHLD_SEL	RW	BMU Threshold Select. This value tells DXE which of the BMU BD/PDU thresholds to check before issuing BMU commands that reserve, or get BDs or PDUs. It also specifies the module index value that will be used to issue any required BMU commands (see BMU spec for details). The value of this bitfield is only meaningful for transfer types that involve the BMU and therefore its value will be ignored for H2H transfers. 0x0: RSVD0 (Reserved) 0x1: RSVD1 (Reserved) 0x2: THLD2 (Use BD/PDU threshold 2) 0x3: THLD3 (Use BD/PDU threshold 3) 0x4: THLD4 (Use BD/PDU threshold 4) 0x5: THLD5 (Use BD/PDU threshold 5) 0x6: THLD6 (Use BD/PDU threshold 6) 0x7: THLD7 (Use BD/PDU threshold 7) 0x8: THLD8 (Use BD/PDU threshold 8) 0x9: THLD9 (Use BD/PDU threshold 9) 0xA: THLD10 (Use BD/PDU threshold 10) 0xB: RSVD11 (Reserved) 0xC: RSVD12 (Reserved) 0xD: RSVD13 (Reserved) 0xE: RSVD14 (Reserved) 0xF: RSVD15 (Reserved)
8	PDU_REL	RW	0: Don't release BD and PDUs when done. 1: Release BD and PDUs when done. 0x0: KEEP (Don) 0x1: RELEASE (Release BD and payload PDUs)

WCSS_A_DXE_0_CH5_CTRL (cont.)

Bits	Name	Type	Description
7	PIQ	RW	next descriptor Pointer address Is Queue. When set the source of the descriptor address is interpreted as a queue ID instead of just the address of a flat memory location. The only valid source queues for descriptors are the UIF Tx data and Tx management endpoints. Any other IDs will result the channel aborting the transfer with an invalid source queue error. Therefore, when this bit is set, SW must ensure that the CH_DESCCL[QID] bitfield is set to a valid queue ID and that the CH_DESCCL[BASE] bitfield is set to the base address of the source UIF endpoint. This base address is formed by taking the base address for the UIF block in the SAHB bus which can be found in the Nova address map document and then adding the internal UIF offset for the correct source endpoint. This offsets can be obtained from the UIF specification document. For convenience, the current internal UIF offsets are provided in the description of the QID bitfield in the CH_DESCCL register description. 0x0: FLAT (Source address maps to regular flat memory) 0x1: QUEUE (Source address maps to FIFO type memory (UIF Tx endpoints))
6	DIQ	RW	Destination Is Queue. When set the destination address is interpreted as a queue ID instead of just the address of a flat memory location. The queue could be either a BMU queue, one of the UIF RX endpoints or and EIF outbound FIFO. When this bit is set, SW must ensure that the DADR[QID] bitfield is set to a valid queue ID and that the DADR[BASE] bitfield is set to the base address of the destination queue for UIF or EIF, or to zero for a BMU queue. endpoint. 0x0: FLAT (Destination address maps to regular flat memory) 0x1: QUEUE (Destination address maps to a BMU queue or FIFO type memory)
5	SIQ	RW	Source Is Queue. When set the source address is interpreted as a queue ID instead of just the address of a flat memory location. The queue could be either a BMU queue, one of the UIF TX endpoints or and EIF inbound FIFO. When this bit is set, SW must ensure that the SADR[QID] bitfield is set to a valid queue ID and that the SADR[BASE] bitfield is set to the base address of the source queue for UIF or EIF, or to zero for a BMU queue. endpoint. 0x0: FLAT (Source address maps to regular flat memory) 0x1: QUEUE (Source address maps to a BMU queue or FIFO type memory)

WCSS_A_DXE_0_CH5_CTRL (cont.)

Bits	Name	Type	Description
4	BDH	RW	BD Handling bit. The behavior depends on the setting of XTYPE. If XTYPE is: H2B: Then if BDH is: 0: Use first 128 bytes of packet as BD. 1: Create BD from BD template (Up to 4 supported. Also see desc for BDT_IDX). B2H: Then if BDH is: 0: Insert BD in front of payload. 1: Discard BD. For XTYPE values of H2H and B2B the value of this bit is ignored.
3	EOP	RW	End of Packet bit. For H2B transfers marks the end of a packet Note that when not using external descriptors DXE cannot support the gather functionality and therefore the EOP bit must be set. For XTYPE values of H2H, B2B and B2H the value of this bit is ignored.
2:1	XTYPE	RW	Transfer Type. Specifies the type of transfer to perform: 00: H2H. Host space to Host space transfer. 01: B2B. BMU space to BMU space transfer. 10: H2B. Host space to BMU translation. 11: B2H. BMU space to Host space translation. For cross-space transfers (H2B and B2H) the DMA engine will also perform the required data format translation. The format translation behavior can be modified by the setting of the BDH and PAD bits. 0x0: H2H (Host space to Host space transfer) 0x1: B2B (BMU space to BMU space transfer) 0x2: H2B (Host space to BMU space transfer) 0x3: B2H (BMU space to Host space transfer)

WCSS_A_DXE_0_CH5_CTRL (cont.)

Bits	Name	Type	Description
0	EN	RW	Channel Enable or Restart. Writing this bit with a one either starts a new DMA transfer according to the parameters programmed in the channel's configuration registers and clears the DONE bit or, if the channel is already enabled but the STOP bit is set, then it restarts channel processing and also clears the STOP bit. This bit is automatically cleared when a single transfer completes or, if the channel is configured to use linked list descriptors, when the last transfer in the descriptor linked list completes (DESC_NXT is NULL or/and DESC_CSR[VALID]=0). The channel is also disabled and this bit cleared if a channel error is encountered. Reading this bit indicates whether a channel is currently enabled or disabled: 0: Channel Disabled. 1: Channel Enabled. The Channel Enable bit status can also be found by reading the DMA_ENCH register which aggregates the EN bits of all channels. Note: Software must take care to not disable the channel while it is busy or data could be lost.

0x0A202544 WCSS_A_DXE_0_CH5_STATUS**Type:** RW**Reset State:** 0x00000000

Serves to report the status of the programmed channel operation.

WCSS_A_DXE_0_CH5_STATUS

Bits	Name	Type	Description
31:16	RSVD	R	Reserved. Reads always return 0. Writes are ignored.
15	INT_DONE	R	Interrupt Source: Asserted when the channel has finished all processing.
14	INT_ERR	R	Interrupt Source: Asserted when a channel error has occurred. Masked by the INE_ERR
13	INT_ED	R	Interrupt Source. Descriptor Done and INT requested
12	ABORT_REQ	R	Abort Request. Software has requested that the current operation be aborted. However, if CH_STATUS[EN] is still set then the channel has registered the request but is not yet in a position to abort the transfer "safely". Once the channel finishes its current operation it will abort the transfer and set the error flag and error code bitfields accordingly.

WCSS_A_DXE_0_CH5_STATUS (cont.)

Bits	Name	Type	Description
11	STOP_REQ	R	Stop Request. Software has requested that the channel stop its operation after the current descriptor has been processed. Once this happens the STOPD bitfield will be set to one and the channel will be disabled.
10:6	ERR_CODE	R	Used to identify the source or cause of an error. 0x0: NONE (No Errors) 0x1: SAHB_ERR (DXE received Error response while performing a transfer on the SAHB) 0x2: H2H_RD_BUS_ERR (SAHB bus error encountered during H2H read operation) 0x3: H2H_WR_BUS_ERR (SAHB bus error encountered during H2H write operation) 0x4: PRG_INV_XTYPE (B2B not yet supported) 0x5: BERR_POPWQ (BMU error on POP WQ command) 0x6: BERR_PUSHWQ (BMU error on PUSH WQ command) 0x7: BERR_RLSS (BMU error on Release BD and attached PDUs command) 0x8: BERR_GETPDU (BMU error on Get PDU command) 0x9: PRG_INV_WQ (Programming error: Unsupported BMU Work Queue) 0xA: PRG_INV_H2H_SRC_QID (Programming error: Invalid H2H source queue ID) 0xB: PRG_INV_H2H_DST_QID (Programming error: Invalid H2H destination queue ID) 0xC: PRG_INV_B2H_SRC_QID (Programming error: Invalid B2H BMU source work queue ID) 0xD: PRG_INV_B2H_DST_QID (Programming error: Invalid B2H destination queue ID) 0xE: PRG_INV_B2H_SRC_IDX (Programming error: Invalid B2H source BD index) 0xF: PRG_INV_H2B_SRC_QID (Programming error: Invalid H2B source queue ID) 0x10: PRG_INV_H2B_DST_QID (Programming error: Invalid H2B BMU destination work queue ID) 0x11: PRG_INV_H2B_DST_IDX (Programming error: Invalid H2B destination BD index) 0x12: PRG_INV_H2B_SZ (Programming error: Invalid H2B transfer size of 0 when BDH=1) 0x13: PRG_INV_SADR (Programming error: Invalid Source Address) 0x14: PRG_INV_DADR (Programming error: Invalid Destination Address) 0x15: PRG_INV_EDADR (Programming error: Invalid Descriptor Address) 0x16: PRG_INV_SRC_WQID (Programming error: Invalid Destination WQ ID)

WCSS_A_DXE_0_CH5_STATUS (cont.)

Bits	Name	Type	Description
			0x17: PRG_INV_DST_WQID (Programming error: Invalid Destination WQ ID) 0x18: PRG_XTYPE_MSM TCH (Programming error: XTYPE mismatch during H2B gather transfer) 0x19: PKT_ERR (Malformed packet) 0x1A: ABORT (Aborted transfer due to SW request) 0x1B: PDU_CNT_OVFL (PDU count field overflow)
5	ERR	RW	DMA channel stopped due to error. To clear the error status it is necessary to write a 1 into this bit position.
4	STOPD	R	Channel Stopped. The channel has stopped further processing and disabled itself due to a stop request in the last processed descriptor.
3	MSKD	R	Channel Masked. The channel has been automatically masked out from arbitration due to a blocking condition.
2	DONE	R	DMA channel Done. This bit will be automatically cleared the next time the channel is enabled.
1	BUSY	R	Channel busy. High when the channel is enabled and currently active (won internal arbitration)
0	EN	R	Channel Enabled. Copy of CH_CTRL[EN].

0x0A202548 WCSS_A_DXE_0_CH5_SZ**Type:** RW**Reset State:** 0x00000000

The transfer size register specifies the total, remaining, and burst or 'chunk' transfer sizes for each channel. The maximum total transfer size is 16K-1 bytes. The chunk size control can be useful to control the relative bandwidth used by each channel as the dma engine will change channels, release the AHB bus, and re-arbitrate for access after transferring chunk size bytes.

WCSS_A_DXE_0_CH5_SZ

Bits	Name	Type	Description
31:28	CHK_SZ	RW	Chunk transfer size. Specifies the number of bytes (in multiples of 8) to be transferred at one given time (not implying they are buffered but that they will be transferred for each start event in one bus request cycle). If chunk size is zero the DMA engine will always perform maximum chunk size transfers. Maximum chunk size is 128 bytes.
27:14	REM_SZ	R	Remaining transfer size. Specifies the number of bytes that remain to be transferred to complete the current operation.

WCSS_A_DXE_0_CH5_SZ (cont.)

Bits	Name	Type	Description
13:0	TOT_SZ	RW	Total Transfer Size. Specifies the number of bytes to be transferred. Maximum total transfer size is (4K-1) bytes. Note: This field is unnecessary and will be ignored for B2H or B2B transfers, when the source of the transfer is in BMU space (either BMU queue or direct pointer to BD). This is because the size of the transfer will be the packet length encoded in the BD at offset 0x0c bits [11:0].

0x0A20254C WCSS_A_DXE_0_CH5_SADRL**Type:** RW**Reset State:** 0x00000000

The channel source address register specifies either the source address in flat internal or external memory, or else the BMU queue ID, UIF TX endpoint ID, or EIF inbound FIFO ID for the transfer. The register must be programmed directly by software before the channel is enabled, then, once the DMA channel is enabled, the register is updated after each atomic chunk transfer (read+write) with the number of bytes transferred. (except when using a UIF/EIF queue) and also after loading a new descriptor. When using flat memory as the source, addresses can have arbitrary byte-alignment. When using a BMU queue, UIF endpoint or EIF fifo as a source, the software is responsible for programming a valid queue ID and correct base address for the queue. The layout of bitfields in this register depends on the setting of the CH_CSR[SIQ] bit. When CH_CSR[SIQ] is clear then the register has no bitfields and the full 32-bits are interpreted as the physical address of a location in flat memory, and when it is set this register has two bitfields: The SQID bitfield used to specify the queue ID, and the BASE bitfield used to provide the base address for the UIF or EIF source endpoint/FIFO.

For B2H or B2B transfers, when the source is a BMU Queue ID (CH_CSR[SIQ] = 1 and SWQID matches BMU), the DXE will issue a BMU POPWQ command to pop the specified queue and obtain the BD index of the source packet. Else, when the source is an address (CH_CSR[SIQ] = 0), the DXE will interpret the address as the index of the packet BD and will add it to the BMU_BASE to obtain the physical address to use to fetch it directly from memory.

For H2B or H2H transfers, if the source is a UIF endpoint or EIF FIFO (CH_CSR[SIQ] = 1 and SQID matches UIF/EIF), SW should program the SAHB base address of the UIF source endpoint in the BASE bitfield and the queue ID to match in the SWQID bitfield, then the DXE will use the base address given to generate the physical address and read the data over the SAHB bus and also use the appropriate sideband signals as specified by the SQID field to communicate with the selected UIF endpoint. SW should make sure that the BASE and SQID fields are consistent and refer to the same endpoint or else hangs or unpredictable behavior might occur. Else, when the source is just a location in flat memory space, (CH_CSR[SIQ] = 0), the bitfields lose their special meaning and DXE will just use the full 32-bits programmed into the register as the physical address to use to fetch the data. For this case, SW should make sure that the provided address maps to the intended source interface (Internal memory, USB, PCI, PCIe) by conforming to the address map specified for the SAHB bus.

To summarize, for B2H and B2B transfers the address must conform to the following format:

For index of BD mode (CH_CSR[SIQ] = 0):

ADDR[31:7] = BMU_BASE + BD_INDEX[15:0]

ADDR[6:0] = 7'd0

For BMU Queue ID mode (CH_CSR[SIQ] = 1 & SQID matching one of the BMU queues):

ADDR[31:7] = Any value will do but zero preferred

ADDR[6:0] = Any BMU Queue ID as specified in description of SQID field

For H2B and H2H transfers the address must conform to the following format:

For sources in flat memory space (CH_CSR[SIQ] = 0): ADDR[31:0] = address mapping to source interface

For UIF RX endpoints or EIF FIFOs (CH_CSR[SIQ] = 1 & QID matching UIF or EIF):

ADDR[31:7] = UIF RX endpoint base or EIF inbound FIFO base

ADDR[6:0] = Queue ID matching UIF or EIF as specified in description of SQID field

WCSS_A_DXE_0_CH5_SADRL

Bits	Name	Description
31:7	BASE	Base Address. Valid only when the CH_CSR[SIQ] bit is set. For this case, this bitfield provides the base address of the source UIF TX endpoint or EIF inbound FIFO. Consult the appropriate interface uarch for the offset within the interface's address space. When CH_CSR[SIQ] bit is not set this field loses its special meaning and is interpreted only as part of the full 32-bit source address.

WCSS_A_DXE_0_CH5_SADRL (cont.)

Bits	Name	Description
6:0	SQID	Source Queue ID. Valid only when the CH_CSR[SIQ] bit is set. For this case, this bitfield provides the ID for the source BMU, UIF or EIF queue to be used for the transfer. When CH_CSR[SIQ] bit is not set this field loses its special meaning and is interpreted only as part of the full 32-bit source address. 0x0: UIF_TX_DATA_EP (UIF TX Data Endpoint) 0x1: UIF_TX_MGMT_EP (UIF TX Management Endpoint) 0x2: BMU_2 (BMU Work Queue 2) 0x3: BMU_3 (BMU Work Queue 3) 0x4: BMU_4 (BMU Work Queue 4) 0x5: BMU_5 (BMU Work Queue 5) 0x6: BMU_6 (BMU Work Queue 6) 0x7: BMU_7 (BMU Work Queue 7) 0x8: BMU_8 (BMU Work Queue 8) 0x9: BMU_9 (BMU Work Queue 9) 0xA: BMU_10 (BMU Work Queue 10) 0xB: BMU_11 (BMU Work Queue 11) 0xC: BMU_12 (BMU Work Queue 12) 0xD: BMU_13 (BMU Work Queue 13) 0xE: BMU_14 (BMU Work Queue 14) 0xF: BMU_15 (BMU Work Queue 15) 0x10: BMU_16 (BMU Work Queue 16) 0x11: BMU_17 (BMU Work Queue 17) 0x12: BMU_18 (BMU Work Queue 18) 0x13: BMU_19 (BMU Work Queue 19) 0x14: BMU_20 (BMU Work Queue 20) 0x15: BMU_21 (BMU Work Queue 21) 0x16: BMU_22 (BMU Work Queue 22) 0x17: BMU_23 (BMU Work Queue 23) 0x18: BMU_24 (BMU Work Queue 24) 0x19: RSVD_25 (Reserved) 0x1A: RSVD_26 (Reserved) 0x1B: RSVD_27 (Reserved) 0x1C: RSVD_28 (Reserved) 0x1D: RSVD_29 (Reserved) 0x1E: RSVD_30 (Reserved) 0x1F: RSVD_31 (Reserved)

0x0A202550 WCSS_A_DXE_0_CH5_SADRH**Type:** RW**Reset State:** 0x00000000

The channel source address high register holds the high bits of the source address for a transfer. See the description of CH_SADRL for more info.

WCSS_A_DXE_0_CH5_SADRH

Bits	Name	Type	Description
31:2	RSVD	R	Reserved. Writes ignored. Reads always return zeroes.
1:0	ADDRH	RW	For taurus, this register holds bits 33 and 32 of the address. With the current address mapping these high order bits determine where the transfer is routed and whether the lower 32 bits of the address (the ones held in CH_SADRL) go untranslated through PIF or XIF or if they use the PIF and XIF address translation windows. See the bitfield encoding descriptions for an explanation. 0x0: ADDR_XTLN (If address in PIF or XIF address space then address goes through translation in PIF or XIF) 0x1: PIF_NO_XTLN (Always maps to PIF and full 32-bits in CH_SADRL goes through PIF untranslated) 0x2: XIF_NO_XTLN (Always maps to XIF and full 32-bits in CH_SADRL goes through XIF untranslated) 0x3: RSVD (Reserved)

0x0A202554 WCSS_A_DXE_0_CH5_DADRL**Type:** RW**Reset State:** 0x00000000

The channel destination address register specifies either the destination address in flat internal or external memory, or else the BMU queue ID, UIF RX endpoint ID, or EIF outbound FIFO ID for the transfer. The register must be programmed directly by software before the channel is enabled, then, once the DMA channel is enabled, the register is updated after each atomic chunk transfer (read+write) with the number of bytes transferred. (except when using a UIF/EIF queue) and also after loading a new descriptor. When using flat memory as the destination, addresses can have arbitrary byte-alignment. When using a BMU queue, UIF endpoint or EIF fifo as a destination, the software is responsible for programming a valid queue ID and correct base address for the queue. The layout of bitfields in this register depends on the setting of the CH_CSR[DIQ] bit. When this bit is clear then the register has no bitfields and the full 32-bits are interpreted as the physical address of a location in flat memory, and when the bit is set this register has two bitfields: The DQID bitfield used to specify the queue ID, and the BASE bitfield used to provide the base address for the UIF or EIF destination endpoint/FIFO.

For H2B or B2B transfers, when using a BMU queue as the destination (CH_CSR[DIQ] = 1 and DQID matches BMU), the DXE will push the index of the BD of the packet into the specified queue at the end of the packet transfer. Therefore, software is responsible for programming a valid BMU Queue ID.

Else, when the destination is an index (CH_CSR[DIQ] = 0), the BD will be written to the address obtained by adding the provided index to the BMU_BASE, again, the software must ensure that said address is legal and known to BMU. That is, software must preallocate the BD (get the BD from BMU via GET BD/PDU cmd) for the BD before starting the transfer. For B2H or H2H transfers, if the destination is a UIF endpoint or EIF FIFO (CH_CSR[DIQ] = 1 and DQID matches UIF/EIF), SW should program the SAHB base address of the UIF destination endpoint in the

BASE bitfield and the queue ID to match in the DQID bitfield, then the DXE will use the base address given to generate the physical address and write the data over the SAHB bus and also use the appropriate sideband signals as specified by the DQID field to communicate with the selected UIF endpoint. SW should make sure that the BASE and DQID fields are consistent and refer to the same endpoint else hangs or unpredictable behavior might occur. On the other hand, when the destination is just a location in flat memory space, (CH_CSR[DIQ] = 0), be it internal or external, the bitfields lose their special meaning and DXE will just use the full 32-bits programmed into the register as the address to write the data. For this case, SW should make sure that the address maps to the intended destination interface (Internal memory, USB, PCI, PCIe) by conforming to the address map specified for the SAHB bus. To summarize, for H2B and B2B transfers the address must conform to the following format:

For index of BD mode (CH_CSR[DIQ] = 0): ADDR[31:7] = BMU_BASE + BD_INDEX[15:0]

ADDR[6:0] = 7'd0

For BMU Queue ID mode (CH_CSR[DIQ] = 1 & DQID matching one of the BMU queues):

ADDR[31:5] = Any value will do but zero preferred

ADDR[4:0] = Any BMU Queue ID as specified in description of QID field

For B2H and H2H transfers the address must conform to the following format:

For destinations in flat memory space (CH_CSR[DIQ] = 0): ADDR[31:0] = address mapping to destination interface

For UIF RX endpoints or EIF FIFOs (CH_CSR[DIQ] = 1 & DQID matching UIF or EIF):

ADDR[31:5] = UIF RX endpoint base or EIF inbound FIFO base

ADDR[4:0] = Queue ID matching UIF or EIF as specified in description of DQID field

WCSS_A_DXE_0_CH5_DADRL

Bits	Name	Description
31:7	BASE	Base Address. Valid only when the CH_CSR[DIQ] bit is set. For this case, this bitfield provides the base address of the destination UIF RX endpoint or EIF outbound FIFO (for BMU SW may use any value can be used but zero is suggested). Consult the appropriate interface uarch for the offset within the interface's address space. When CH_CSR[SIQ] bit is not set this field loses its special meaning and is interpreted only as part of the full 32-bit source address.

WCSS_A_DXE_0_CH5_DADRL (cont.)

Bits	Name	Description
6:0	DQID	Destination Queue ID. Valid only when the CH_CSR[SIQ] bit is set. For this case, this bitfield provides the ID for the destination BMU WQ, UIF RX endpoint or EIF outbound queue to be used for the transfer. When CH_CSR[SIQ] bit is not set this field loses its special meaning and is interpreted only as part of the full 32-bit source address. 0x0: RSVD_0 (Reserved) 0x1: RSVD_1 (Reserved) 0x2: BMU_2 (BMU Work Queue 2) 0x3: BMU_3 (BMU Work Queue 3) 0x4: BMU_4 (BMU Work Queue 4) 0x5: BMU_5 (BMU Work Queue 5) 0x6: BMU_6 (BMU Work Queue 6) 0x7: BMU_7 (BMU Work Queue 7) 0x8: BMU_8 (BMU Work Queue 8) 0x9: BMU_9 (BMU Work Queue 9) 0xA: BMU_10 (BMU Work Queue 10) 0xB: BMU_11 (BMU Work Queue 11) 0xC: BMU_12 (BMU Work Queue 12) 0xD: BMU_13 (BMU Work Queue 13) 0xE: BMU_14 (BMU Work Queue 14) 0xF: BMU_15 (BMU Work Queue 15) 0x10: BMU_16 (BMU Work Queue 16) 0x11: BMU_17 (BMU Work Queue 17) 0x12: BMU_18 (BMU Work Queue 18) 0x13: BMU_19 (BMU Work Queue 19) 0x14: BMU_20 (BMU Work Queue 20) 0x15: BMU_21 (BMU Work Queue 21) 0x16: BMU_22 (BMU Work Queue 22) 0x17: BMU_23 (BMU Work Queue 23) 0x18: BMU_24 (BMU Work Queue 24) 0x19: BMU_25 (BMU Work Queue 25) 0x1A: BMU_26 (BMU Work Queue 26) 0x1B: BMU_SINK_WQ (BMU_SINK (WQ255)) 0x1C: UIF_RX_1_EP (UIF RX 1 endpoint) 0x1D: UIF_RX_2_EP (UIF RX 2 endpoint) 0x1E: UIF_RX_3_EP (UIF RX 3 endpoint) 0x1F: UIF_RX_4_EP (UIF RX 4 endpoint)

0x0A202558 WCSS_A_DXE_0_CH5_DADRH**Type:** RW**Reset State:** 0x00000000

The channel destination address high register holds the high bits of the destination address for a transfer. See the description of CH_DADRL for more info.

WCSS_A_DXE_0_CH5_DADRH

Bits	Name	Type	Description
31:2	RSVD	R	Reserved. Writes ignored. Reads always return zeroes.
1:0	ADDRH	RW	This register holds bits 33 and 32 of the address. With the current address mapping these high order bits determine where the transfer is routed and whether the lower 32 bits of the address (the ones held in CH_DADRL) go untranslated through PIF or XIF or if they use the PIF and XIF address translation windows. See the bitfield encoding descriptions for an explanation. 0x0: ADDR_XTLN (If address in PIF or XIF address space then address goes through translation in PIF or XIF) 0x1: PIF_NO_XTLN (Always maps to PIF and full 32-bits in CH_DADRL goes through PIF untranslated) 0x2: XIF_NO_XTLN (Always maps to XIF and full 32-bits in CH_DADRL goes through XIF untranslated) 0x3: RSVD (Reserved. Maps to default slave. Reads return error, writes are ignored)

0x0A20255C WCSS_A_DXE_0_CH5_DESCL

Type: RW

Reset State: 0x00000000

The Channel 5 Linked List Descriptor Pointer register contains the 32 lower bits of the address of the next descriptor to be loaded in the linked list. This address must be dword (32-bit) aligned (addr[1:0]=0 if CH_CTRL[PIQ]=0). Each complete transfer descriptor consists of 7 32-bit values arranged contiguously in memory. The value of this register will be overwritten with the address of the start of the next descriptor set in the list after the current descriptor has been fetched. The meaning of the value in this register depends on the value of the CH_CTRL[PIQ] bitfield similar to the way in which SADRL and DADRL are depending in SIQ and DIQ respectively. For CH_CTRL[PIQ]=0 then this register has no bitfields and the value is just interpreted as an address in flat memory space. When CH_CTRL[PIQ] is set then the descriptor will be read from FIFO type memory in UIF and the register has 2 bitfields: BASE, and SQID. BASE specifies the base address of the UIF Tx queue (derived from the start of the UIF address space in SAHB plus the internal UIF offset).

WCSS_A_DXE_0_CH5_DESCL

Bits	Name	Description
31:7	BASE	This bitfield only exists and has meaning when the descriptor needs to be read from a UIF TX queue (and therefore CH_CTRL[PIQ]=1), then it serves to specify the base address of the UIF endpoint to use as the source of the descriptor. Refer to the Nova address map document and to the UIF architecture document for guaranteed correct values to use here. For convenience but without a guarantee of accuracy, the base value to use for both the Tx data and management endpoints is the following: base = UIF base + TD_and_TM_offset = 0x0E02_5000 + x100 = 0x0E02_5100 >> 7 = 0x1C_04A2
6:0	SQID	This bitfield only exists and has meaning when the descriptor needs to be read from a UIF TX queue (and therefore CH_CTRL[PIQ]=1), then it serves to specify which of the two UIF TX queues to use. Note that in this case SW also needs to program the correct base address of the source UIF endpoint in the BASE bitfield. 0x0: UIF_TX_DATA_EP (Descriptor to be read from UIF TX Data Endpoint) 0x1: UIF_TX_MGMT_EP (Descriptor to be read from UIF TX Management Endpoint)

0x0A202560 WCSS_A_DXE_0_CH5_DESCH**Type:** RW**Reset State:** 0x00000000

The channel destination address high register holds the high bits of the the address of the next descriptor to be loaded in the linked list.

WCSS_A_DXE_0_CH5_DESCH

Bits	Name	Type	Description
31:2	RSVD	R	Reserved. Writes ignored. Reads always return zeroes.

WCSS_A_DXE_0_CH5_DESCH (cont.)

Bits	Name	Type	Description
1:0	ADDRH	RW	For taurus, this register holds bits 33 and 32 of the address. With the current address mapping these high order bits determine where the transfer is routed and whether the lower 32 bits of the address (the ones held in CH_DADRL) go untranslated through PIF or XIF or if they use the PIF and XIF address translation windows. See the bitfield encoding descriptions for an explanation. Note that if the source of the descriptor is either of the UIF endpoints (CH_CTRL[PIQ]=1) then the value in ADDRH has to be 0 or else the descriptor read could end up mapping to the XIF or PIF. 0x0: ADDR_XTLN (If address in PIF or XIF address space then address goes through translation in PIF or XIF) 0x1: PIF_NO_XTLN (Always maps to PIF and full 32-bits in CH_DADRL goes through PIF untranslated) 0x2: XIF_NO_XTLN (Always maps to XIF and full 32-bits in CH_DADRL goes through XIF untranslated) 0x3: RSVD (Reserved. Maps to default slave. Reads return error, writes are ignored)

0x0A202564 WCSS_A_DXE_0_CH5_LST_DESCL**Type:** R**Reset State:** 0x00000000

The read-only Channel 5 Current/last Linked List Descriptor Pointer register contains the low 32-bits of the address of the descriptor currently being processed or of the last descriptor processed if the channel is disabled.

The meaning of the value and bitfields with regards to CH_CTRL[PIQ] is the same as was explained for the CH_DESCL register.

WCSS_A_DXE_0_CH5_LST_DESCL

Bits	Name	Description
31:7	BASE	See description of CH_DESCL[BASE].
6:0	SQID	See description of CH_DESCL[SQID]. 0x0: UIF_TX_DATA_EP (Descriptor to be read from UIF TX Data Endpoint) 0x1: UIF_TX_MGMT_EP (Descriptor to be read from UIF TX Management Endpoint)

0x0A202568 WCSS_A_DXE_0_CH5_LST_DESCH**Type:** R**Reset State:** 0x00000000

The channel destination address high register holds the high bits of the the address of the next descriptor to be loaded in the linked list. Note that The meaning of the value and bitfields is the same as was explained for the CH_DESCL register.

WCSS_A_DXE_0_CH5_LST_DESCH

Bits	Name	Description
31:2	RSVD	Reserved. Writes ignored. Reads always return zeroes.
1:0	ADDRH	For taurus, this register holds bits 33 and 32 of the address.

0x0A20256C WCSS_A_DXE_0_CH5_BD**Type:** R**Reset State:** 0x00000000**WCSS_A_DXE_0_CH5_BD**

Bits	Name	Description
31:24	RSVD	Reserved. Always returns 0 when read.
23	VAL	Valid. BD has been loaded and is valid.
22:16	PDU_CNT	PDU count of packet.
15:0	IDX	Index of BD. To get actual physical address must be added to BD/PDU base.

0x0A202570 WCSS_A_DXE_0_CH5_HEAD**Type:** R**Reset State:** 0x00000000

Address of head of packet.

WCSS_A_DXE_0_CH5_HEAD

Bits	Name	Description
31:27	RSVD	Reserved.
26	DESCVAL	Descriptor valid. Signifies descriptor has already been loaded. Used to prevent reloading of a descriptor.

WCSS_A_DXE_0_CH5_HEAD (cont.)

Bits	Name	Description
25:21	DWQID	Destination WQ ID. Original BMU, UIF or EIF destination queue. Valid only when CH_CSR[DIQ]=1
20:16	SWQID	Source WQ ID. Original BMU, UIF or EIF source queue. Valid only when CH_CSR[SIQ]=1
15:0	IDX	Index of Head. Must be added to BD/PDU base for actual physical address.

0x0A202574 WCSS_A_DXE_0_CH5_TAIL**Type:** R**Reset State:** 0x00000000

Address of tail of packet.

WCSS_A_DXE_0_CH5_TAIL

Bits	Name	Description
31:16	RSVD	Reserved.
15:0	IDX	Index of Tail. Must be added to BD/PDU base for actual physical address

0x0A202578 WCSS_A_DXE_0_CH5_PDU**Type:** R**Reset State:** 0x00000000

PDU info register.

All its fields are working fields that get modified as the programmed operation progresses.

WCSS_A_DXE_0_CH5_PDU

Bits	Name	Description
31	RSVD	Reserved. Always returns 0 when read.
30:24	PDU_CNT	Current PDU count.
23:8	IDX	Index of current PDU. Must be added to BD/PDU base for actual physical address
7:0	FST_OFF	First PDU offset.

0x0A20257C WCSS_A_DXE_0_CH5_TSTMP**Type:** R**Reset State:** 0x00000000

Contains last sampled value of the system timestamp

WCSS_A_DXE_0_CH5_TSTMP

Bits	Name	Description
31:0	VALUE	

0x0A202580 WCSS_A_DXE_0_CH6_CTRL**Type:** RW**Reset State:** 0x11800708

Serves to configure channel 6 operation

WCSS_A_DXE_0_CH6_CTRL

Bits	Name	Type	Description
31	SWAP	RW	Endian Byte Swap Enable. When high instructs DXE to perform an endian swap of the incoming byte stream. The corresponding bitfield in the DXE descriptor control word is located at DXE_DESC_CTRL[20]
30:29	BDT_IDX	RW	BD Template Index. When XTYPE=H2B and BDH=1, DXE supports creating a BD by using template data from one of up to 4 different BD templates. The BD template base store is pointed at by the DXE BD_TMPLT(H/L) registers and the address used to read the template data is calculated as follows: $BD_TMPLT_ADDR = BD_TMPLT_BASE + (128 * CH_CTRL[BDT_IDX])$ The corresponding bitfield in the control word of the external descriptor is mapped to bits[19:18]
28	DFMT	RW	Descriptor Format. This bitfield is used to select between long and short descriptor formats. When using the short descriptor format, the high part of the source, destination and descriptor addresses will be taken from the SADDRH_DFLT, DADDRH_DFLT or DESCH_DFLT registers respectively. Default is to use the long descriptor format 0x0: SHORT (Short descriptor format (5 dwords)) 0x1: LONG (Long descriptor format (8 dwords))

WCSS_A_DXE_0_CH6_CTRL (cont.)

Bits	Name	Type	Description
27	ABORT	RW	Writing a one to this bit will cause the channel to stop its current transfer and set the ERR bit and ERR_CODE fields in CH_STATUS. Note that if the channel is currently disabled or if it is about to be disabled because it has just finished its programmed transfer, then the abort request will be ignored and the error bit and error codes won't be set. If the channel is currently masked by the arbiter then setting this bit will forcibly unmask it so that the transfer can be aborted the next time the channel is selected by the internal arbiter. The abort request is automatically cleared by hw once the channel has aborted the transfer. Also note that since the DXE will not attempt to do any cleanup when a transfer is aborted, software might need to do so. For example, for H2B transfers, SW should manually release the BD and any PDUs currently in use by the partial transfer or else a BD/PDU leak will result.
26	ENDIANNESS	RW	Specifies the endianness of the master generating the data. Affects alignment of valid bytes for non-dword aligned reads/writes. The corresponding field in the DXE descriptor control word is located in DXE_DESC_CTRL[21] 0x0: BIGEND (System is Big-Endian) 0x1: LTLEND (System is Little-Endian)
25:22	CTR_SEL	RW	Counter Select. If the event counters are enabled (DMA_CSR[ECTR_EN]=1) then this bitfield selects the counter that will be incremented at the end of the transfer. By default the value in this bitfield is the same as the channel number so that after reset channel 0 increments counter 0, channel 1 increments counter 1 and so on.
21	EDVEN	RW	When set, the DMA engine will clear the VAL bit in the DESC_CSR word of the current descriptor once it has completed processing it. This is useful for implementing a descriptor "ring" (see the description of the VALID bit in Table 3.1)
20	EDEN	RW	External Descriptor Enable. Use External Descriptor Linked List
19	INE_DONE	RW	Enable Channel Interrupt when Channel is Done. The corresponding interrupt source bit is INT_DONE
18	INE_ERR	RW	Enable Channel Interrupt on Errors. The corresponding interrupt source bit is INT_ERR
17	INE_ED	RW	Enable External Descriptor Interrupt. When this bit is set, an interrupt will be generated after processing a descriptor item that has the INT bit set in its DESC_CTRL word (see Table 3.1). The corresponding interrupt source bit is INT_ED

WCSS_A_DXE_0_CH6_CTRL (cont.)

Bits	Name	Type	Description
16	STOP	R	Channel processing stopped due to a request in the last processed linked list descriptor (see the STOP bit in the definition of the DESC_CSR word in Table 3.1). This bit will be automatically cleared when the channel is restarted.
15:13	PRIO	RW	This field sets the channel priority. 7: Highest Priority. 0: Lowest Priority.
12:9	BTHLD_SEL	RW	BMU Threshold Select. This value tells DXE which of the BMU BD/PDU thresholds to check before issuing BMU commands that reserve, or get BDs or PDUs. It also specifies the module index value that will be used to issue any required BMU commands (see BMU spec for details). The value of this bitfield is only meaningful for transfer types that involve the BMU and therefore its value will be ignored for H2H transfers. 0x0: RSVD0 (Reserved) 0x1: RSVD1 (Reserved) 0x2: THLD2 (Use BD/PDU threshold 2) 0x3: THLD3 (Use BD/PDU threshold 3) 0x4: THLD4 (Use BD/PDU threshold 4) 0x5: THLD5 (Use BD/PDU threshold 5) 0x6: THLD6 (Use BD/PDU threshold 6) 0x7: THLD7 (Use BD/PDU threshold 7) 0x8: THLD8 (Use BD/PDU threshold 8) 0x9: THLD9 (Use BD/PDU threshold 9) 0xA: THLD10 (Use BD/PDU threshold 10) 0xB: RSVD11 (Reserved) 0xC: RSVD12 (Reserved) 0xD: RSVD13 (Reserved) 0xE: RSVD14 (Reserved) 0xF: RSVD15 (Reserved)
8	PDU_REL	RW	0: Don't release BD and PDUs when done. 1: Release BD and PDUs when done. 0x0: KEEP (Don) 0x1: RELEASE (Release BD and payload PDUs)

WCSS_A_DXE_0_CH6_CTRL (cont.)

Bits	Name	Type	Description
7	PIQ	RW	next descriptor Pointer address Is Queue. When set the source of the descriptor address is interpreted as a queue ID instead of just the address of a flat memory location. The only valid source queues for descriptors are the UIF Tx data and Tx management endpoints. Any other IDs will result the channel aborting the transfer with an invalid source queue error. Therefore, when this bit is set, SW must ensure that the CH_DESCCL[QID] bitfield is set to a valid queue ID and that the CH_DESCCL[BASE] bitfield is set to the base address of the source UIF endpoint. This base address is formed by taking the base address for the UIF block in the SAHB bus which can be found in the Nova address map document and then adding the internal UIF offset for the correct source endpoint. This offsets can be obtained from the UIF specification document. For convenience, the current internal UIF offsets are provided in the description of the QID bitfield in the CH_DESCCL register description. 0x0: FLAT (Source address maps to regular flat memory) 0x1: QUEUE (Source address maps to FIFO type memory (UIF Tx endpoints))
6	DIQ	RW	Destination Is Queue. When set the destination address is interpreted as a queue ID instead of just the address of a flat memory location. The queue could be either a BMU queue, one of the UIF RX endpoints or and EIF outbound FIFO. When this bit is set, SW must ensure that the DADR[QID] bitfield is set to a valid queue ID and that the DADR[BASE] bitfield is set to the base address of the destination queue for UIF or EIF, or to zero for a BMU queue. endpoint. 0x0: FLAT (Destination address maps to regular flat memory) 0x1: QUEUE (Destination address maps to a BMU queue or FIFO type memory)
5	SIQ	RW	Source Is Queue. When set the source address is interpreted as a queue ID instead of just the address of a flat memory location. The queue could be either a BMU queue, one of the UIF TX endpoints or and EIF inbound FIFO. When this bit is set, SW must ensure that the SADR[QID] bitfield is set to a valid queue ID and that the SADR[BASE] bitfield is set to the base address of the source queue for UIF or EIF, or to zero for a BMU queue. endpoint. 0x0: FLAT (Source address maps to regular flat memory) 0x1: QUEUE (Source address maps to a BMU queue or FIFO type memory)

WCSS_A_DXE_0_CH6_CTRL (cont.)

Bits	Name	Type	Description
4	BDH	RW	BD Handling bit. The behavior depends on the setting of XTYPE. If XTYPE is: H2B: Then if BDH is: 0: Use first 128 bytes of packet as BD. 1: Create BD from BD template (Up to 4 supported. Also see desc for BDT_IDX). B2H: Then if BDH is: 0: Insert BD in front of payload. 1: Discard BD. For XTYPE values of H2H and B2B the value of this bit is ignored.
3	EOP	RW	End of Packet bit. For H2B transfers marks the end of a packet Note that when not using external descriptors DXE cannot support the gather functionality and therefore the EOP bit must be set. For XTYPE values of H2H, B2B and B2H the value of this bit is ignored.
2:1	XTYPE	RW	Transfer Type. Specifies the type of transfer to perform: 00: H2H. Host space to Host space transfer. 01: B2B. BMU space to BMU space transfer. 10: H2B. Host space to BMU translation. 11: B2H. BMU space to Host space translation. For cross-space transfers (H2B and B2H) the DMA engine will also perform the required data format translation. The format translation behavior can be modified by the setting of the BDH and PAD bits. 0x0: H2H (Host space to Host space transfer) 0x1: B2B (BMU space to BMU space transfer) 0x2: H2B (Host space to BMU space transfer) 0x3: B2H (BMU space to Host space transfer)

WCSS_A_DXE_0_CH6_CTRL (cont.)

Bits	Name	Type	Description
0	EN	RW	Channel Enable or Restart. Writing this bit with a one either starts a new DMA transfer according to the parameters programmed in the channel's configuration registers and clears the DONE bit or, if the channel is already enabled but the STOP bit is set, then it restarts channel processing and also clears the STOP bit. This bit is automatically cleared when a single transfer completes or, if the channel is configured to use linked list descriptors, when the last transfer in the descriptor linked list completes (DESC_NXT is NULL or/and DESC_CSR[VALID]=0). The channel is also disabled and this bit cleared if a channel error is encountered. Reading this bit indicates whether a channel is currently enabled or disabled: 0: Channel Disabled. 1: Channel Enabled. The Channel Enable bit status can also be found by reading the DMA_ENCH register which aggregates the EN bits of all channels. Note: Software must take care to not disable the channel while it is busy or data could be lost.

0x0A202584 WCSS_A_DXE_0_CH6_STATUS**Type:** RW**Reset State:** 0x00000000

Serves to report the status of the programmed channel operation.

WCSS_A_DXE_0_CH6_STATUS

Bits	Name	Type	Description
31:16	RSVD	R	Reserved. Reads always return 0. Writes are ignored.
15	INT_DONE	R	Interrupt Source: Asserted when the channel has finished all processing.
14	INT_ERR	R	Interrupt Source: Asserted when a channel error has occurred. Masked by the INE_ERR
13	INT_ED	R	Interrupt Source. Descriptor Done and INT requested
12	ABORT_REQ	R	Abort Request. Software has requested that the current operation be aborted. However, if CH_STATUS[EN] is still set then the channel has registered the request but is not yet in a position to abort the transfer "safely". Once the channel finishes its current operation it will abort the transfer and set the error flag and error code bitfields accordingly.

WCSS_A_DXE_0_CH6_STATUS (cont.)

Bits	Name	Type	Description
11	STOP_REQ	R	Stop Request. Software has requested that the channel stop its operation after the current descriptor has been processed. Once this happens the STOPD bitfield will be set to one and the channel will be disabled.
10:6	ERR_CODE	R	Used to identify the source or cause of an error. 0x0: NONE (No Errors) 0x1: SAHB_ERR (DXE received Error response while performing a transfer on the SAHB) 0x2: H2H_RD_BUS_ERR (SAHB bus error encountered during H2H read operation) 0x3: H2H_WR_BUS_ERR (SAHB bus error encountered during H2H write operation) 0x4: PRG_INV_XTYPE (B2B not yet supported) 0x5: BERR_POPWQ (BMU error on POP WQ command) 0x6: BERR_PUSHWQ (BMU error on PUSH WQ command) 0x7: BERR_RLSS (BMU error on Release BD and attached PDUs command) 0x8: BERR_GETPDU (BMU error on Get PDU command) 0x9: PRG_INV_WQ (Programming error: Unsupported BMU Work Queue) 0xA: PRG_INV_H2H_SRC_QID (Programming error: Invalid H2H source queue ID) 0xB: PRG_INV_H2H_DST_QID (Programming error: Invalid H2H destination queue ID) 0xC: PRG_INV_B2H_SRC_QID (Programming error: Invalid B2H BMU source work queue ID) 0xD: PRG_INV_B2H_DST_QID (Programming error: Invalid B2H destination queue ID) 0xE: PRG_INV_B2H_SRC_IDX (Programming error: Invalid B2H source BD index) 0xF: PRG_INV_H2B_SRC_QID (Programming error: Invalid H2B source queue ID) 0x10: PRG_INV_H2B_DST_QID (Programming error: Invalid H2B BMU destination work queue ID) 0x11: PRG_INV_H2B_DST_IDX (Programming error: Invalid H2B destination BD index) 0x12: PRG_INV_H2B_SZ (Programming error: Invalid H2B transfer size of 0 when BDH=1) 0x13: PRG_INV_SADR (Programming error: Invalid Source Address) 0x14: PRG_INV_DADR (Programming error: Invalid Destination Address) 0x15: PRG_INV_EDADR (Programming error: Invalid Descriptor Address) 0x16: PRG_INV_SRC_WQID (Programming error: Invalid Destination WQ ID)

WCSS_A_DXE_0_CH6_STATUS (cont.)

Bits	Name	Type	Description
			0x17: PRG_INV_DST_WQID (Programming error: Invalid Destination WQ ID) 0x18: PRG_XTYPE_MSM TCH (Programming error: XTYPE mismatch during H2B gather transfer) 0x19: PKT_ERR (Malformed packet) 0x1A: ABORT (Aborted transfer due to SW request) 0x1B: PDU_CNT_OVFL (PDU count field overflow)
5	ERR	RW	DMA channel stopped due to error. To clear the error status it is necessary to write a 1 into this bit position.
4	STOPD	R	Channel Stopped. The channel has stopped further processing and disabled itself due to a stop request in the last processed descriptor.
3	MSKD	R	Channel Masked. The channel has been automatically masked out from arbitration due to a blocking condition.
2	DONE	R	DMA channel Done. This bit will be automatically cleared the next time the channel is enabled.
1	BUSY	R	Channel busy. High when the channel is enabled and currently active (won internal arbitration)
0	EN	R	Channel Enabled. Copy of CH_CTRL[EN].

0x0A202588 WCSS_A_DXE_0_CH6_SZ**Type:** RW**Reset State:** 0x00000000

The transfer size register specifies the total, remaining, and burst or 'chunk' transfer sizes for each channel. The maximum total transfer size is 16K-1 bytes. The chunk size control can be useful to control the relative bandwidth used by each channel as the dma engine will change channels, release the AHB bus, and rearbitrate for access after transferring chunk size bytes.

WCSS_A_DXE_0_CH6_SZ

Bits	Name	Type	Description
31:28	CHK_SZ	RW	Chunk transfer size. Specifies the number of bytes (in multiples of 8) to be transferred at one given time (not implying they are buffered but that they will be transferred for each start event in one bus request cycle). If chunk size is zero the DMA engine will always perform maximum chunk size transfers. Maximum chunk size is 128 bytes.
27:14	REM_SZ	R	Remaining transfer size. Specifies the number of bytes that remain to be transferred to complete the current operation.

WCSS_A_DXE_0_CH6_SZ (cont.)

Bits	Name	Type	Description
13:0	TOT_SZ	RW	Total Transfer Size. Specifies the number of bytes to be transferred. Maximum total transfer size is (4K-1) bytes. Note: This field is unnecessary and will be ignored for B2H or B2B transfers, when the source of the transfer is in BMU space (either BMU queue or direct pointer to BD). This is because the size of the transfer will be the packet length encoded in the BD at offset 0x0c bits [11:0].

0x0A20258C WCSS_A_DXE_0_CH6_SADRL**Type:** RW**Reset State:** 0x00000000

The channel source address register specifies either the source address in flat internal or external memory, or else the BMU queue ID, UIF TX endpoint ID, or EIF inbound FIFO ID for the transfer. The register must be programmed directly by software before the channel is enabled, then, once the DMA channel is enabled, the register is updated after each atomic chunk transfer (read+write) with the number of bytes transferred. (except when using a UIF/EIF queue) and also after loading a new descriptor. When using flat memory as the source, addresses can have arbitrary byte-alignment. When using a BMU queue, UIF endpoint or EIF fifo as a source, the software is responsible for programming a valid queue ID and correct base address for the queue. The layout of bitfields in this register depends on the setting of the CH_CSR[SIQ] bit. When CH_CSR[SIQ] is clear then the register has no bitfields and the full 32-bits are interpreted as the physical address of a location in flat memory, and when it is set this register has two bitfields: The SQID bitfield used to specify the queue ID, and the BASE bitfield used to provide the base address for the UIF or EIF source endpoint/FIFO.

For B2H or B2B transfers, when the source is a BMU Queue ID (CH_CSR[SIQ] = 1 and SWQID matches BMU), the DXE will issue a BMU POPWQ command to pop the specified queue and obtain the BD index of the source packet. Else, when the source is an address (CH_CSR[SIQ] = 0), the DXE will interpret the address as the index of the packet BD and will add it to the BMU_BASE to obtain the physical address to use to fetch it directly from memory.

For H2B or H2H transfers, if the source is a UIF endpoint or EIF FIFO (CH_CSR[SIQ] = 1 and SQID matches UIF/EIF), SW should program the SAHB base address of the UIF source endpoint in the BASE bitfield and the queue ID to match in the SWQID bitfield, then the DXE will use the base address given to generate the physical address and read the data over the SAHB bus and also use the appropriate sideband signals as specified by the SQID field to communicate with the selected UIF endpoint. SW should make sure that the BASE and SQID fields are consistent and refer to the same endpoint or else hangs or unpredictable behavior might occur. Else, when the source is just a location in flat memory space, (CH_CSR[SIQ] = 0), the bitfields lose their special meaning and DXE will just use the full 32-bits programmed into the register as the physical address to use to fetch the data. For this case, SW should make sure that the provided address maps to the intended source interface (Internal memory, USB, PCI, PCIe) by conforming to the address map specified for the SAHB bus.

To summarize, for B2H and B2B transfers the address must conform to the following format:

For index of BD mode (CH_CSR[SIQ] = 0):

ADDR[31:7] = BMU_BASE + BD_INDEX[15:0]

ADDR[6:0] = 7'd0

For BMU Queue ID mode (CH_CSR[SIQ] = 1 & SQID matching one of the BMU queues):

ADDR[31:7] = Any value will do but zero preferred

ADDR[6:0] = Any BMU Queue ID as specified in description of SQID field

For H2B and H2H transfers the address must conform to the following format:

For sources in flat memory space (CH_CSR[SIQ] = 0): ADDR[31:0] = address mapping to source interface

For UIF RX endpoints or EIF FIFOs (CH_CSR[SIQ] = 1 & QID matching UIF or EIF):

ADDR[31:7] = UIF RX endpoint base or EIF inbound FIFO base

ADDR[6:0] = Queue ID matching UIF or EIF as specified in description of SQID field

WCSS_A_DXE_0_CH6_SADRL

Bits	Name	Description
31:7	BASE	Base Address. Valid only when the CH_CSR[SIQ] bit is set. For this case, this bitfield provides the base address of the source UIF TX endpoint or EIF inbound FIFO. Consult the appropriate interface uarch for the offset within the interface's address space. When CH_CSR[SIQ] bit is not set this field loses its special meaning and is interpreted only as part of the full 32-bit source address.

WCSS_A_DXE_0_CH6_SADRL (cont.)

Bits	Name	Description
6:0	SQID	Source Queue ID. Valid only when the CH_CSR[SIQ] bit is set. For this case, this bitfield provides the ID for the source BMU, UIF or EIF queue to be used for the transfer. When CH_CSR[SIQ] bit is not set this field loses its special meaning and is interpreted only as part of the full 32-bit source address. 0x0: UIF_TX_DATA_EP (UIF TX Data Endpoint) 0x1: UIF_TX_MGMT_EP (UIF TX Management Endpoint) 0x2: BMU_2 (BMU Work Queue 2) 0x3: BMU_3 (BMU Work Queue 3) 0x4: BMU_4 (BMU Work Queue 4) 0x5: BMU_5 (BMU Work Queue 5) 0x6: BMU_6 (BMU Work Queue 6) 0x7: BMU_7 (BMU Work Queue 7) 0x8: BMU_8 (BMU Work Queue 8) 0x9: BMU_9 (BMU Work Queue 9) 0xA: BMU_10 (BMU Work Queue 10) 0xB: BMU_11 (BMU Work Queue 11) 0xC: BMU_12 (BMU Work Queue 12) 0xD: BMU_13 (BMU Work Queue 13) 0xE: BMU_14 (BMU Work Queue 14) 0xF: BMU_15 (BMU Work Queue 15) 0x10: BMU_16 (BMU Work Queue 16) 0x11: BMU_17 (BMU Work Queue 17) 0x12: BMU_18 (BMU Work Queue 18) 0x13: BMU_19 (BMU Work Queue 19) 0x14: BMU_20 (BMU Work Queue 20) 0x15: BMU_21 (BMU Work Queue 21) 0x16: BMU_22 (BMU Work Queue 22) 0x17: BMU_23 (BMU Work Queue 23) 0x18: BMU_24 (BMU Work Queue 24) 0x19: RSVD_25 (Reserved) 0x1A: RSVD_26 (Reserved) 0x1B: RSVD_27 (Reserved) 0x1C: RSVD_28 (Reserved) 0x1D: RSVD_29 (Reserved) 0x1E: RSVD_30 (Reserved) 0x1F: RSVD_31 (Reserved)

0x0A202590 WCSS_A_DXE_0_CH6_SADRH**Type:** RW**Reset State:** 0x00000000

The channel source address high register holds the high bits of the source address for a transfer. See the description of CH_SADRL for more info.

WCSS_A_DXE_0_CH6_SADRH

Bits	Name	Type	Description
31:2	RSVD	R	Reserved. Writes ignored. Reads always return zeroes.
1:0	ADDRH	RW	For taurus, this register holds bits 33 and 32 of the address. With the current address mapping these high order bits determine where the transfer is routed and whether the lower 32 bits of the address (the ones held in CH_SADRL) go untranslated through PIF or XIF or if they use the PIF and XIF address translation windows. See the bitfield encoding descriptions for an explanation. 0x0: ADDR_XTLN (If address in PIF or XIF address space then address goes through translation in PIF or XIF) 0x1: PIF_NO_XTLN (Always maps to PIF and full 32-bits in CH_SADRL goes through PIF untranslated) 0x2: XIF_NO_XTLN (Always maps to XIF and full 32-bits in CH_SADRL goes through XIF untranslated) 0x3: RSVD (Reserved)

0x0A202594 WCSS_A_DXE_0_CH6_DADRL**Type:** RW**Reset State:** 0x00000000

The channel destination address register specifies either the destination address in flat internal or external memory, or else the BMU queue ID, UIF RX endpoint ID, or EIF outbound FIFO ID for the transfer. The register must be programmed directly by software before the channel is enabled, then, once the DMA channel is enabled, the register is updated after each atomic chunk transfer (read+write) with the number of bytes transferred. (except when using a UIF/EIF queue) and also after loading a new descriptor. When using flat memory as the destination, addresses can have arbitrary byte-alignment. When using a BMU queue, UIF endpoint or EIF fifo as a destination, the software is responsible for programming a valid queue ID and correct base address for the queue. The layout of bitfields in this register depends on the setting of the CH_CSR[DIQ] bit. When this bit is clear then the register has no bitfields and the full 32-bits are interpreted as the physical address of a location in flat memory, and when the bit is set this register has two bitfields: The DQID bitfield used to specify the queue ID, and the BASE bitfield used to provide the base address for the UIF or EIF destination endpoint/FIFO.

For H2B or B2B transfers, when using a BMU queue as the destination (CH_CSR[DIQ] = 1 and DQID matches BMU), the DXE will push the index of the BD of the packet into the specified queue at the end of the packet transfer. Therefore, software is responsible for programming a valid BMU Queue ID.

Else, when the destination is an index (CH_CSR[DIQ] = 0), the BD will be written to the address obtained by adding the provided index to the BMU_BASE, again, the software must ensure that said address is legal and known to BMU. That is, software must preallocate the BD (get the BD from BMU via GET BD/PDU cmd) for the BD before starting the transfer. For B2H or H2H transfers, if the destination is a UIF endpoint or EIF FIFO (CH_CSR[DIQ] = 1 and DQID matches UIF/EIF), SW should program the SAHB base address of the UIF destination endpoint in the

BASE bitfield and the queue ID to match in the DQID bitfield, then the DXE will use the base address given to generate the physical address and write the data over the SAHB bus and also use the appropriate sideband signals as specified by the DQID field to communicate with the selected UIF endpoint. SW should make sure that the BASE and DQID fields are consistent and refer to the same endpoint else hangs or unpredictable behavior might occur. On the other hand, when the destination is just a location in flat memory space, (CH_CSR[DIQ] = 0), be it internal or external, the bitfields lose their special meaning and DXE will just use the full 32-bits programmed into the register as the address to write the data. For this case, SW should make sure that the address maps to the intended destination interface (Internal memory, USB, PCI, PCIe) by conforming to the address map specified for the SAHB bus. To summarize, for H2B and B2B transfers the address must conform to the following format:

For index of BD mode (CH_CSR[DIQ] = 0): ADDR[31:7] = BMU_BASE + BD_INDEX[15:0]

ADDR[6:0] = 7'd0

For BMU Queue ID mode (CH_CSR[DIQ] = 1 & DQID matching one of the BMU queues):

ADDR[31:5] = Any value will do but zero preferred

ADDR[4:0] = Any BMU Queue ID as specified in description of QID field

For B2H and H2H transfers the address must conform to the following format:

For destinations in flat memory space (CH_CSR[DIQ] = 0): ADDR[31:0] = address mapping to destination interface

For UIF RX endpoints or EIF FIFOs (CH_CSR[DIQ] = 1 & DQID matching UIF or EIF):

ADDR[31:5] = UIF RX endpoint base or EIF inbound FIFO base

ADDR[4:0] = Queue ID matching UIF or EIF as specified in description of DQID field

WCSS_A_DXE_0_CH6_DADRL

Bits	Name	Description
31:7	BASE	Base Address. Valid only when the CH_CSR[DIQ] bit is set. For this case, this bitfield provides the base address of the destination UIF RX endpoint or EIF outbound FIFO (for BMU SW may use any value can be used but zero is suggested). Consult the appropriate interface uarch for the offset within the interface's address space. When CH_CSR[SIQ] bit is not set this field loses its special meaning and is interpreted only as part of the full 32-bit source address.

WCSS_A_DXE_0_CH6_DADRL (cont.)

Bits	Name	Description
6:0	DQID	Destination Queue ID. Valid only when the CH_CSR[SIQ] bit is set. For this case, this bitfield provides the ID for the destination BMU WQ, UIF RX endpoint or EIF outbound queue to be used for the transfer. When CH_CSR[SIQ] bit is not set this field loses its special meaning and is interpreted only as part of the full 32-bit source address. 0x0: RSVD_0 (Reserved) 0x1: RSVD_1 (Reserved) 0x2: BMU_2 (BMU Work Queue 2) 0x3: BMU_3 (BMU Work Queue 3) 0x4: BMU_4 (BMU Work Queue 4) 0x5: BMU_5 (BMU Work Queue 5) 0x6: BMU_6 (BMU Work Queue 6) 0x7: BMU_7 (BMU Work Queue 7) 0x8: BMU_8 (BMU Work Queue 8) 0x9: BMU_9 (BMU Work Queue 9) 0xA: BMU_10 (BMU Work Queue 10) 0xB: BMU_11 (BMU Work Queue 11) 0xC: BMU_12 (BMU Work Queue 12) 0xD: BMU_13 (BMU Work Queue 13) 0xE: BMU_14 (BMU Work Queue 14) 0xF: BMU_15 (BMU Work Queue 15) 0x10: BMU_16 (BMU Work Queue 16) 0x11: BMU_17 (BMU Work Queue 17) 0x12: BMU_18 (BMU Work Queue 18) 0x13: BMU_19 (BMU Work Queue 19) 0x14: BMU_20 (BMU Work Queue 20) 0x15: BMU_21 (BMU Work Queue 21) 0x16: BMU_22 (BMU Work Queue 22) 0x17: BMU_23 (BMU Work Queue 23) 0x18: BMU_24 (BMU Work Queue 24) 0x19: BMU_25 (BMU Work Queue 25) 0x1A: BMU_26 (BMU Work Queue 26) 0x1B: BMU_SINK_WQ (BMU_SINK (WQ255)) 0x1C: UIF_RX_1_EP (UIF RX 1 endpoint) 0x1D: UIF_RX_2_EP (UIF RX 2 endpoint) 0x1E: UIF_RX_3_EP (UIF RX 3 endpoint) 0x1F: UIF_RX_4_EP (UIF RX 4 endpoint)

0x0A202598 WCSS_A_DXE_0_CH6_DADRH**Type:** RW**Reset State:** 0x00000000

The channel destination address high register holds the high bits of the destination address for a transfer. See the description of CH_DADRL for more info.

WCSS_A_DXE_0_CH6_DADRH

Bits	Name	Type	Description
31:2	RSVD	R	Reserved. Writes ignored. Reads always return zeroes.
1:0	ADDRH	RW	This register holds bits 33 and 32 of the address. With the current address mapping these high order bits determine where the transfer is routed and whether the lower 32 bits of the address (the ones held in CH_DADRL) go untranslated through PIF or XIF or if they use the PIF and XIF address translation windows. See the bitfield encoding descriptions for an explanation. 0x0: ADDR_XTLN (If address in PIF or XIF address space then address goes through translation in PIF or XIF) 0x1: PIF_NO_XTLN (Always maps to PIF and full 32-bits in CH_DADRL goes through PIF untranslated) 0x2: XIF_NO_XTLN (Always maps to XIF and full 32-bits in CH_DADRL goes through XIF untranslated) 0x3: RSVD (Reserved. Maps to default slave. Reads return error, writes are ignored)

0x0A20259C WCSS_A_DXE_0_CH6_DESCL

Type: RW

Reset State: 0x00000000

The Channel 6 Linked List Descriptor Pointer register contains the 32 lower bits of the address of the next descriptor to be loaded in the linked list. This address must be dword (32-bit) aligned (addr[1:0]=0 if CH_CTRL[PIQ]=0). Each complete transfer descriptor consists of 7 32-bit values arranged contiguously in memory. The value of this register will be overwritten with the address of the start of the next descriptor set in the list after the current descriptor has been fetched. The meaning of the value in this register depends on the value of the CH_CTRL[PIQ] bitfield similar to the way in which SADRL and DADRL are depending in SIQ and DIQ respectively. For CH_CTRL[PIQ]=0 then this register has no bitfields and the value is just interpreted as an address in flat memory space. When CH_CTRL[PIQ] is set then the descriptor will be read from FIFO type memory in UIF and the register has 2 bitfields: BASE, and SQID. BASE specifies the base address of the UIF Tx queue (derived from the start of the UIF address space in SAHB plus the internal UIF offset).

WCSS_A_DXE_0_CH6_DESCL

Bits	Name	Description
31:7	BASE	This bitfield only exists and has meaning when the descriptor needs to be read from a UIF TX queue (and therefore CH_CTRL[PIQ]=1), then it serves to specify the base address of the UIF endpoint to use as the source of the descriptor. Refer to the Nova address map document and to the UIF architecture document for guaranteed correct values to use here. For convenience but without a guarantee of accuracy, the base value to use for both the Tx data and management endpoints is the following: $\text{base} = \text{UIF base} + \text{TD_and_TM_offset} = 0x0E02_5000 + x100 = 0x0E02_5100 \gg 7 = 0x1C_04A2$
6:0	SQID	This bitfield only exists and has meaning when the descriptor needs to be read from a UIF TX queue (and therefore CH_CTRL[PIQ]=1), then it serves to specify which of the two UIF TX queues to use. Note that in this case SW also needs to program the correct base address of the source UIF endpoint in the BASE bitfield. 0x0: UIF_TX_DATA_EP (Descriptor to be read from UIF TX Data Endpoint) 0x1: UIF_TX_MGMT_EP (Descriptor to be read from UIF TX Management Endpoint)

0x0A2025A0 WCSS_A_DXE_0_CH6_DESCH**Type:** RW**Reset State:** 0x00000000

The channel destination address high register holds the high bits of the the address of the next descriptor to be loaded in the linked list.

WCSS_A_DXE_0_CH6_DESCH

Bits	Name	Type	Description
31:2	RSVD	R	Reserved. Writes ignored. Reads always return zeroes.

WCSS_A_DXE_0_CH6_DESCH (cont.)

Bits	Name	Type	Description
1:0	ADDRH	RW	For taurus, this register holds bits 33 and 32 of the address. With the current address mapping these high order bits determine where the transfer is routed and whether the lower 32 bits of the address (the ones held in CH_DADRL) go untranslated through PIF or XIF or if they use the PIF and XIF address translation windows. See the bitfield encoding descriptions for an explanation. Note that if the source of the descriptor is either of the UIF endpoints (CH_CTRL[PIQ]=1) then the value in ADDRH has to be 0 or else the descriptor read could end up mapping to the XIF or PIF. 0x0: ADDR_XTLN (If address in PIF or XIF address space then address goes through translation in PIF or XIF) 0x1: PIF_NO_XTLN (Always maps to PIF and full 32-bits in CH_DADRL goes through PIF untranslated) 0x2: XIF_NO_XTLN (Always maps to XIF and full 32-bits in CH_DADRL goes through XIF untranslated) 0x3: RSVD (Reserved. Maps to default slave. Reads return error, writes are ignored)

0x0A2025A4 WCSS_A_DXE_0_CH6_LST_DESCL**Type:** R**Reset State:** 0x00000000

The read-only Channel 6 Current/last Linked List Descriptor Pointer register contains the low 32-bits of the address of the descriptor currently being processed or of the last descriptor processed if the channel is disabled.

The meaning of the value and bitfields with regards to CH_CTRL[PIQ] is the same as was explained for the CH_DESCL register.

WCSS_A_DXE_0_CH6_LST_DESCL

Bits	Name	Description
31:7	BASE	See description of CH_DESCL[BASE].
6:0	SQID	See description of CH_DESCL[SQID]. 0x0: UIF_TX_DATA_EP (Descriptor to be read from UIF TX Data Endpoint) 0x1: UIF_TX_MGMT_EP (Descriptor to be read from UIF TX Management Endpoint)

0x0A2025A8 WCSS_A_DXE_0_CH6_LST_DESCH**Type:** R**Reset State:** 0x00000000

The channel destination address high register holds the high bits of the the address of the next descriptor to be loaded in the linked list. Note that The meaning of the value and bitfields is the same as was explained for the CH_DESCL register.

WCSS_A_DXE_0_CH6_LST_DESCH

Bits	Name	Description
31:2	RSVD	Reserved. Writes ignored. Reads always return zeroes.
1:0	ADDRH	For taurus, this register holds bits 33 and 32 of the address.

0x0A2025AC WCSS_A_DXE_0_CH6_BD**Type:** R**Reset State:** 0x00000000**WCSS_A_DXE_0_CH6_BD**

Bits	Name	Description
31:24	RSVD	Reserved. Always returns 0 when read.
23	VAL	Valid. BD has been loaded and is valid.
22:16	PDU_CNT	PDU count of packet.
15:0	IDX	Index of BD. To get actual physical address must be added to BD/PDU base.

0x0A2025B0 WCSS_A_DXE_0_CH6_HEAD**Type:** R**Reset State:** 0x00000000

Address of head of packet.

WCSS_A_DXE_0_CH6_HEAD

Bits	Name	Description
31:27	RSVD	Reserved.
26	DESCVAL	Descriptor valid. Signifies descriptor has already been loaded. Used to prevent reloading of a descriptor.

WCSS_A_DXE_0_CH6_HEAD (cont.)

Bits	Name	Description
25:21	DWQID	Destination WQ ID. Original BMU, UIF or EIF destination queue. Valid only when CH_CSR[DIQ]=1
20:16	SWQID	Source WQ ID. Original BMU, UIF or EIF source queue. Valid only when CH_CSR[SIQ]=1
15:0	IDX	Index of Head. Must be added to BD/PDU base for actual physical address.

0x0A2025B4 WCSS_A_DXE_0_CH6_TAIL**Type:** R**Reset State:** 0x00000000

Address of tail of packet.

WCSS_A_DXE_0_CH6_TAIL

Bits	Name	Description
31:16	RSVD	Reserved.
15:0	IDX	Index of Tail. Must be added to BD/PDU base for actual physical address

0x0A2025B8 WCSS_A_DXE_0_CH6_PDU**Type:** R**Reset State:** 0x00000000

PDU info register.

All its fields are working fields that get modified as the programmed operation progresses.

WCSS_A_DXE_0_CH6_PDU

Bits	Name	Description
31	RSVD	Reserved. Always returns 0 when read.
30:24	PDU_CNT	Current PDU count.
23:8	IDX	Index of current PDU. Must be added to BD/PDU base for actual physical address
7:0	FST_OFF	First PDU offset.

0x0A2025BC WCSS_A_DXE_0_CH6_TSTMP**Type:** R**Reset State:** 0x00000000

Contains last sampled value of the system timestamp

WCSS_A_DXE_0_CH6_TSTMP

Bits	Name	Description
31:0	VALUE	

0x0A2025C0 WCSS_A_DXE_0_CH7_CTRL**Type:** RW**Reset State:** 0x11C00708

Serves to configure channel 7 operation

WCSS_A_DXE_0_CH7_CTRL

Bits	Name	Type	Description
31	SWAP	RW	Endian Byte Swap Enable. When high instructs DXE to perform an endian swap of the incoming byte stream. The corresponding bitfield in the DXE descriptor control word is located at DXE_DESC_CTRL[20]
30:29	BDT_IDX	RW	BD Template Index. When XTYPE=H2B and BDH=1, DXE supports creating a BD by using template data from one of up to 4 different BD templates. The BD template base store is pointed at by the DXE BD_TMPLT(H/L) registers and the address used to read the template data is calculated as follows: $BD_TMPLT_ADDR = BD_TMPLT_BASE + (128 * CH_CTRL[BDT_IDX])$ The corresponding bitfield in the control word of the external descriptor is mapped to bits[19:18]
28	DFMT	RW	Descriptor Format. This bitfield is used to select between long and short descriptor formats. When using the short descriptor format, the high part of the source, destination and descriptor addresses will be taken from the SADDRH_DFLT, DADDRH_DFLT or DESCH_DFLT registers respectively. Default is to use the long descriptor format 0x0: SHORT (Short descriptor format (5 dwords)) 0x1: LONG (Long descriptor format (8 dwords))

WCSS_A_DXE_0_CH7_CTRL (cont.)

Bits	Name	Type	Description
27	ABORT	RW	Writing a one to this bit will cause the channel to stop its current transfer and set the ERR bit and ERR_CODE fields in CH_STATUS. Note that if the channel is currently disabled or if it is about to be disabled because it has just finished its programmed transfer, then the abort request will be ignored and the error bit and error codes won't be set. If the channel is currently masked by the arbiter then setting this bit will forcibly unmask it so that the transfer can be aborted the next time the channel is selected by the internal arbiter. The abort request is automatically cleared by hw once the channel has aborted the transfer. Also note that since the DXE will not attempt to do any cleanup when a transfer is aborted, software might need to do so. For example, for H2B transfers, SW should manually release the BD and any PDUs currently in use by the partial transfer or else a BD/PDU leak will result.
26	ENDIANNESS	RW	Specifies the endianness of the master generating the data. Affects alignment of valid bytes for non-dword aligned reads/writes. The corresponding field in the DXE descriptor control word is located in DXE_DESC_CTRL[21] 0x0: BIGEND (System is Big-Endian) 0x1: LTLEND (System is Little-Endian)
25:22	CTR_SEL	RW	Counter Select. If the event counters are enabled (DMA_CSR[ECTR_EN]=1) then this bitfield selects the counter that will be incremented at the end of the transfer. By default the value in this bitfield is the same as the channel number so that after reset channel 0 increments counter 0, channel 1 increments counter 1 and so on.
21	EDVEN	RW	When set, the DMA engine will clear the VAL bit in the DESC_CSR word of the current descriptor once it has completed processing it. This is useful for implementing a descriptor "ring" (see the description of the VALID bit in Table 3.1)
20	EDEN	RW	External Descriptor Enable. Use External Descriptor Linked List
19	INE_DONE	RW	Enable Channel Interrupt when Channel is Done. The corresponding interrupt source bit is INT_DONE
18	INE_ERR	RW	Enable Channel Interrupt on Errors. The corresponding interrupt source bit is INT_ERR
17	INE_ED	RW	Enable External Descriptor Interrupt. When this bit is set, an interrupt will be generated after processing a descriptor item that has the INT bit set in its DESC_CTRL word (see Table 3.1). The corresponding interrupt source bit is INT_ED

WCSS_A_DXE_0_CH7_CTRL (cont.)

Bits	Name	Type	Description
16	STOP	R	Channel processing stopped due to a request in the last processed linked list descriptor (see the STOP bit in the definition of the DESC_CSR word in Table 3.1). This bit will be automatically cleared when the channel is restarted.
15:13	PRIO	RW	This field sets the channel priority. 7: Highest Priority. 0: Lowest Priority.
12:9	BTHLD_SEL	RW	BMU Threshold Select. This value tells DXE which of the BMU BD/PDU thresholds to check before issuing BMU commands that reserve, or get BDs or PDUs. It also specifies the module index value that will be used to issue any required BMU commands (see BMU spec for details). The value of this bitfield is only meaningful for transfer types that involve the BMU and therefore its value will be ignored for H2H transfers. 0x0: RSVD0 (Reserved) 0x1: RSVD1 (Reserved) 0x2: THLD2 (Use BD/PDU threshold 2) 0x3: THLD3 (Use BD/PDU threshold 3) 0x4: THLD4 (Use BD/PDU threshold 4) 0x5: THLD5 (Use BD/PDU threshold 5) 0x6: THLD6 (Use BD/PDU threshold 6) 0x7: THLD7 (Use BD/PDU threshold 7) 0x8: THLD8 (Use BD/PDU threshold 8) 0x9: THLD9 (Use BD/PDU threshold 9) 0xA: THLD10 (Use BD/PDU threshold 10) 0xB: RSVD11 (Reserved) 0xC: RSVD12 (Reserved) 0xD: RSVD13 (Reserved) 0xE: RSVD14 (Reserved) 0xF: RSVD15 (Reserved)
8	PDU_REL	RW	0: Don't release BD and PDUs when done. 1: Release BD and PDUs when done. 0x0: KEEP (Don) 0x1: RELEASE (Release BD and payload PDUs)

WCSS_A_DXE_0_CH7_CTRL (cont.)

Bits	Name	Type	Description
7	PIQ	RW	next descriptor Pointer address Is Queue. When set the source of the descriptor address is interpreted as a queue ID instead of just the address of a flat memory location. The only valid source queues for descriptors are the UIF Tx data and Tx management endpoints. Any other IDs will result the channel aborting the transfer with an invalid source queue error. Therefore, when this bit is set, SW must ensure that the CH_DESCCL[QID] bitfield is set to a valid queue ID and that the CH_DESCCL[BASE] bitfield is set to the base address of the source UIF endpoint. This base address is formed by taking the base address for the UIF block in the SAHB bus which can be found in the Nova address map document and then adding the internal UIF offset for the correct source endpoint. This offsets can be obtained from the UIF specification document. For convenience, the current internal UIF offsets are provided in the description of the QID bitfield in the CH_DESCCL register description. 0x0: FLAT (Source address maps to regular flat memory) 0x1: QUEUE (Source address maps to FIFO type memory (UIF Tx endpoints))
6	DIQ	RW	Destination Is Queue. When set the destination address is interpreted as a queue ID instead of just the address of a flat memory location. The queue could be either a BMU queue, one of the UIF RX endpoints or and EIF outbound FIFO. When this bit is set, SW must ensure that the DADR[QID] bitfield is set to a valid queue ID and that the DADR[BASE] bitfield is set to the base address of the destination queue for UIF or EIF, or to zero for a BMU queue. endpoint. 0x0: FLAT (Destination address maps to regular flat memory) 0x1: QUEUE (Destination address maps to a BMU queue or FIFO type memory)
5	SIQ	RW	Source Is Queue. When set the source address is interpreted as a queue ID instead of just the address of a flat memory location. The queue could be either a BMU queue, one of the UIF TX endpoints or and EIF inbound FIFO. When this bit is set, SW must ensure that the SADR[QID] bitfield is set to a valid queue ID and that the SADR[BASE] bitfield is set to the base address of the source queue for UIF or EIF, or to zero for a BMU queue. endpoint. 0x0: FLAT (Source address maps to regular flat memory) 0x1: QUEUE (Source address maps to a BMU queue or FIFO type memory)

WCSS_A_DXE_0_CH7_CTRL (cont.)

Bits	Name	Type	Description
4	BDH	RW	BD Handling bit. The behavior depends on the setting of XTYPE. If XTYPE is: H2B: Then if BDH is: 0: Use first 128 bytes of packet as BD. 1: Create BD from BD template (Up to 4 supported. Also see desc for BDT_IDX). B2H: Then if BDH is: 0: Insert BD in front of payload. 1: Discard BD. For XTYPE values of H2H and B2B the value of this bit is ignored.
3	EOP	RW	End of Packet bit. For H2B transfers marks the end of a packet Note that when not using external descriptors DXE cannot support the gather functionality and therefore the EOP bit must be set. For XTYPE values of H2H, B2B and B2H the value of this bit is ignored.
2:1	XTYPE	RW	Transfer Type. Specifies the type of transfer to perform: 00: H2H. Host space to Host space transfer. 01: B2B. BMU space to BMU space transfer. 10: H2B. Host space to BMU translation. 11: B2H. BMU space to Host space translation. For cross-space transfers (H2B and B2H) the DMA engine will also perform the required data format translation. The format translation behavior can be modified by the setting of the BDH and PAD bits. 0x0: H2H (Host space to Host space transfer) 0x1: B2B (BMU space to BMU space transfer) 0x2: H2B (Host space to BMU space transfer) 0x3: B2H (BMU space to Host space transfer)

WCSS_A_DXE_0_CH7_CTRL (cont.)

Bits	Name	Type	Description
0	EN	RW	Channel Enable or Restart. Writing this bit with a one either starts a new DMA transfer according to the parameters programmed in the channel's configuration registers and clears the DONE bit or, if the channel is already enabled but the STOP bit is set, then it restarts channel processing and also clears the STOP bit. This bit is automatically cleared when a single transfer completes or, if the channel is configured to use linked list descriptors, when the last transfer in the descriptor linked list completes (DESC_NXT is NULL or/and DESC_CSR[VALID]=0). The channel is also disabled and this bit cleared if a channel error is encountered. Reading this bit indicates whether a channel is currently enabled or disabled: 0: Channel Disabled. 1: Channel Enabled. The Channel Enable bit status can also be found by reading the DMA_ENCH register which aggregates the EN bits of all channels. Note: Software must take care to not disable the channel while it is busy or data could be lost.

0x0A2025C4 WCSS_A_DXE_0_CH7_STATUS**Type:** RW**Reset State:** 0x00000000

Serves to report the status of the programmed channel operation.

WCSS_A_DXE_0_CH7_STATUS

Bits	Name	Type	Description
31:16	RSVD	R	Reserved. Reads always return 0. Writes are ignored.
15	INT_DONE	R	Interrupt Source: Asserted when the channel has finished all processing.
14	INT_ERR	R	Interrupt Source: Asserted when a channel error has occurred. Masked by the INE_ERR
13	INT_ED	R	Interrupt Source. Descriptor Done and INT requested
12	ABORT_REQ	R	Abort Request. Software has requested that the current operation be aborted. However, if CH_STATUS[EN] is still set then the channel has registered the request but is not yet in a position to abort the transfer "safely". Once the channel finishes its current operation it will abort the transfer and set the error flag and error code bitfields accordingly.

WCSS_A_DXE_0_CH7_STATUS (cont.)

Bits	Name	Type	Description
11	STOP_REQ	R	Stop Request. Software has requested that the channel stop its operation after the current descriptor has been processed. Once this happens the STOPD bitfield will be set to one and the channel will be disabled.
10:6	ERR_CODE	R	Used to identify the source or cause of an error. 0x0: NONE (No Errors) 0x1: SAHB_ERR (DXE received Error response while performing a transfer on the SAHB) 0x2: H2H_RD_BUS_ERR (SAHB bus error encountered during H2H read operation) 0x3: H2H_WR_BUS_ERR (SAHB bus error encountered during H2H write operation) 0x4: PRG_INV_XTYPE (B2B not yet supported) 0x5: BERR_POPWQ (BMU error on POP WQ command) 0x6: BERR_PUSHWQ (BMU error on PUSH WQ command) 0x7: BERR_RLSS (BMU error on Release BD and attached PDUs command) 0x8: BERR_GETPDU (BMU error on Get PDU command) 0x9: PRG_INV_WQ (Programming error: Unsupported BMU Work Queue) 0xA: PRG_INV_H2H_SRC_QID (Programming error: Invalid H2H source queue ID) 0xB: PRG_INV_H2H_DST_QID (Programming error: Invalid H2H destination queue ID) 0xC: PRG_INV_B2H_SRC_QID (Programming error: Invalid B2H BMU source work queue ID) 0xD: PRG_INV_B2H_DST_QID (Programming error: Invalid B2H destination queue ID) 0xE: PRG_INV_B2H_SRC_IDX (Programming error: Invalid B2H source BD index) 0xF: PRG_INV_H2B_SRC_QID (Programming error: Invalid H2B source queue ID) 0x10: PRG_INV_H2B_DST_QID (Programming error: Invalid H2B BMU destination work queue ID) 0x11: PRG_INV_H2B_DST_IDX (Programming error: Invalid H2B destination BD index) 0x12: PRG_INV_H2B_SZ (Programming error: Invalid H2B transfer size of 0 when BDH=1) 0x13: PRG_INV_SADR (Programming error: Invalid Source Address) 0x14: PRG_INV_DADR (Programming error: Invalid Destination Address) 0x15: PRG_INV_EDADR (Programming error: Invalid Descriptor Address) 0x16: PRG_INV_SRC_WQID (Programming error: Invalid Destination WQ ID)

WCSS_A_DXE_0_CH7_STATUS (cont.)

Bits	Name	Type	Description
			0x17: PRG_INV_DST_WQID (Programming error: Invalid Destination WQ ID) 0x18: PRG_XTYPE_MSM TCH (Programming error: XTYPE mismatch during H2B gather transfer) 0x19: PKT_ERR (Malformed packet) 0x1A: ABORT (Aborted transfer due to SW request) 0x1B: PDU_CNT_OVFL (PDU count field overflow)
5	ERR	RW	DMA channel stopped due to error. To clear the error status it is necessary to write a 1 into this bit position.
4	STOPD	R	Channel Stopped. The channel has stopped further processing and disabled itself due to a stop request in the last processed descriptor.
3	MSKD	R	Channel Masked. The channel has been automatically masked out from arbitration due to a blocking condition.
2	DONE	R	DMA channel Done. This bit will be automatically cleared the next time the channel is enabled.
1	BUSY	R	Channel busy. High when the channel is enabled and currently active (won internal arbitration)
0	EN	R	Channel Enabled. Copy of CH_CTRL[EN].

0x0A2025C8 WCSS_A_DXE_0_CH7_SZ**Type:** RW**Reset State:** 0x00000000

The transfer size register specifies the total, remaining, and burst or 'chunk' transfer sizes for each channel. The maximum total transfer size is 16K-1 bytes. The chunk size control can be useful to control the relative bandwidth used by each channel as the dma engine will change channels, release the AHB bus, and re-arbitrate for access after transferring chunk size bytes.

WCSS_A_DXE_0_CH7_SZ

Bits	Name	Type	Description
31:28	CHK_SZ	RW	Chunk transfer size. Specifies the number of bytes (in multiples of 8) to be transferred at one given time (not implying they are buffered but that they will be transferred for each start event in one bus request cycle). If chunk size is zero the DMA engine will always perform maximum chunk size transfers. Maximum chunk size is 128 bytes.
27:14	REM_SZ	R	Remaining transfer size. Specifies the number of bytes that remain to be transferred to complete the current operation.

WCSS_A_DXE_0_CH7_SZ (cont.)

Bits	Name	Type	Description
13:0	TOT_SZ	RW	Total Transfer Size. Specifies the number of bytes to be transferred. Maximum total transfer size is (4K-1) bytes. Note: This field is unnecessary and will be ignored for B2H or B2B transfers, when the source of the transfer is in BMU space (either BMU queue or direct pointer to BD). This is because the size of the transfer will be the packet length encoded in the BD at offset 0x0c bits [11:0].

0x0A2025CC WCSS_A_DXE_0_CH7_SADRL**Type:** RW**Reset State:** 0x00000000

The channel source address register specifies either the source address in flat internal or external memory, or else the BMU queue ID, UIF TX endpoint ID, or EIF inbound FIFO ID for the transfer. The register must be programmed directly by software before the channel is enabled, then, once the DMA channel is enabled, the register is updated after each atomic chunk transfer (read+write) with the number of bytes transferred. (except when using a UIF/EIF queue) and also after loading a new descriptor. When using flat memory as the source, addresses can have arbitrary byte-alignment. When using a BMU queue, UIF endpoint or EIF fifo as a source, the software is responsible for programming a valid queue ID and correct base address for the queue. The layout of bitfields in this register depends on the setting of the CH_CSR[SIQ] bit. When CH_CSR[SIQ] is clear then the register has no bitfields and the full 32-bits are interpreted as the physical address of a location in flat memory, and when it is set this register has two bitfields: The SQID bitfield used to specify the queue ID, and the BASE bitfield used to provide the base address for the UIF or EIF source endpoint/FIFO.

For B2H or B2B transfers, when the source is a BMU Queue ID (CH_CSR[SIQ] = 1 and SWQID matches BMU), the DXE will issue a BMU POPWQ command to pop the specified queue and obtain the BD index of the source packet. Else, when the source is an address (CH_CSR[SIQ] = 0), the DXE will interpret the address as the index of the packet BD and will add it to the BMU_BASE to obtain the physical addresses to use to fetch it directly from memory.

For H2B or H2H transfers, if the source is a UIF endpoint or EIF FIFO (CH_CSR[SIQ] = 1 and SQID matches UIF/EIF), SW should program the SAHB base address of the UIF source endpoint in the BASE bitfield and the queue ID to match in the SWQID bitfield, then the DXE will use the base address given to generate the physical address and read the data over the SAHB bus and also use the appropriate sideband signals as specified by the SQID field to communicate with the selected UIF endpoint. SW should make sure that the BASE and SQID fields are consistent and refer to the same endpoint or else hangs or unpredictable behavior might occur. Else, when the source is just a location in flat memory space, (CH_CSR[SIQ] = 0), the bitfields lose their special meaning and DXE will just use the full 32-bits programmed into the register as the physical address to use to fetch the data. For this case, SW should make sure that the provided address maps to the intended source interface (Internal memory, USB, PCI, PCIe) by conforming to the address map specified for the SAHB bus.

To summarize, for B2H and B2B transfers the address must conform to the following format:

For index of BD mode (CH_CSR[SIQ] = 0):

ADDR[31:7] = BMU_BASE + BD_INDEX[15:0]

ADDR[6:0] = 7'd0

For BMU Queue ID mode (CH_CSR[SIQ] = 1 & SQID matching one of the BMU queues):

ADDR[31:7] = Any value will do but zero preferred

ADDR[6:0] = Any BMU Queue ID as specified in description of SQID field

For H2B and H2H transfers the address must conform to the following format:

For sources in flat memory space (CH_CSR[SIQ] = 0): ADDR[31:0] = address mapping to source interface

For UIF RX endpoints or EIF FIFOs (CH_CSR[SIQ] = 1 & QID matching UIF or EIF):

ADDR[31:7] = UIF RX endpoint base or EIF inbound FIFO base

ADDR[6:0] = Queue ID matching UIF or EIF as specified in description of SQID field

WCSS_A_DXE_0_CH7_SADRL

Bits	Name	Description
31:7	BASE	Base Address. Valid only when the CH_CSR[SIQ] bit is set. For this case, this bitfield provides the base address of the source UIF TX endpoint or EIF inbound FIFO. Consult the appropriate interface uarch for the offset within the interface's address space. When CH_CSR[SIQ] bit is not set this field loses its special meaning and is interpreted only as part of the full 32-bit source address.

WCSS_A_DXE_0_CH7_SADRL (cont.)

Bits	Name	Description
6:0	SQID	Source Queue ID. Valid only when the CH_CSR[SIQ] bit is set. For this case, this bitfield provides the ID for the source BMU, UIF or EIF queue to be used for the transfer. When CH_CSR[SIQ] bit is not set this field loses its special meaning and is interpreted only as part of the full 32-bit source address. 0x0: UIF_TX_DATA_EP (UIF TX Data Endpoint) 0x1: UIF_TX_MGMT_EP (UIF TX Management Endpoint) 0x2: BMU_2 (BMU Work Queue 2) 0x3: BMU_3 (BMU Work Queue 3) 0x4: BMU_4 (BMU Work Queue 4) 0x5: BMU_5 (BMU Work Queue 5) 0x6: BMU_6 (BMU Work Queue 6) 0x7: BMU_7 (BMU Work Queue 7) 0x8: BMU_8 (BMU Work Queue 8) 0x9: BMU_9 (BMU Work Queue 9) 0xA: BMU_10 (BMU Work Queue 10) 0xB: BMU_11 (BMU Work Queue 11) 0xC: BMU_12 (BMU Work Queue 12) 0xD: BMU_13 (BMU Work Queue 13) 0xE: BMU_14 (BMU Work Queue 14) 0xF: BMU_15 (BMU Work Queue 15) 0x10: BMU_16 (BMU Work Queue 16) 0x11: BMU_17 (BMU Work Queue 17) 0x12: BMU_18 (BMU Work Queue 18) 0x13: BMU_19 (BMU Work Queue 19) 0x14: BMU_20 (BMU Work Queue 20) 0x15: BMU_21 (BMU Work Queue 21) 0x16: BMU_22 (BMU Work Queue 22) 0x17: BMU_23 (BMU Work Queue 23) 0x18: BMU_24 (BMU Work Queue 24) 0x19: RSVD_25 (Reserved) 0x1A: RSVD_26 (Reserved) 0x1B: RSVD_27 (Reserved) 0x1C: RSVD_28 (Reserved) 0x1D: RSVD_29 (Reserved) 0x1E: RSVD_30 (Reserved) 0x1F: RSVD_31 (Reserved)

0x0A2025D0 WCSS_A_DXE_0_CH7_SADRH**Type:** RW**Reset State:** 0x00000000

The channel source address high register holds the high bits of the source address for a transfer. See the description of CH_SADRL for more info.

WCSS_A_DXE_0_CH7_SADRH

Bits	Name	Type	Description
31:2	RSVD	R	Reserved. Writes ignored. Reads always return zeroes.
1:0	ADDRH	RW	For taurus, this register holds bits 33 and 32 of the address. With the current address mapping these high order bits determine where the transfer is routed and whether the lower 32 bits of the address (the ones held in CH_SADRL) go untranslated through PIF or XIF or if they use the PIF and XIF address translation windows. See the bitfield encoding descriptions for an explanation. 0x0: ADDR_XTLN (If address in PIF or XIF address space then address goes through translation in PIF or XIF) 0x1: PIF_NO_XTLN (Always maps to PIF and full 32-bits in CH_SADRL goes through PIF untranslated) 0x2: XIF_NO_XTLN (Always maps to XIF and full 32-bits in CH_SADRL goes through XIF untranslated) 0x3: RSVD (Reserved)

0x0A2025D4 WCSS_A_DXE_0_CH7_DADRL**Type:** RW**Reset State:** 0x00000000

The channel destination address register specifies either the destination address in flat internal or external memory, or else the BMU queue ID, UIF RX endpoint ID, or EIF outbound FIFO ID for the transfer. The register must be programmed directly by software before the channel is enabled, then, once the DMA channel is enabled, the register is updated after each atomic chunk transfer (read+write) with the number of bytes transferred. (except when using a UIF/EIF queue) and also after loading a new descriptor. When using flat memory as the destination, addresses can have arbitrary byte-alignment. When using a BMU queue, UIF endpoint or EIF fifo as a destination, the software is responsible for programming a valid queue ID and correct base address for the queue. The layout of bitfields in this register depends on the setting of the CH_CSR[DIQ] bit. When this bit is clear then the register has no bitfields and the full 32-bits are interpreted as the physical address of a location in flat memory, and when the bit is set this register has two bitfields: The DQID bitfield used to specify the queue ID, and the BASE bitfield used to provide the base address for the UIF or EIF destination endpoint/FIFO.

For H2B or B2B transfers, when using a BMU queue as the destination (CH_CSR[DIQ] = 1 and DQID matches BMU), the DXE will push the index of the BD of the packet into the specified queue at the end of the packet transfer. Therefore, software is responsible for programming a valid BMU Queue ID.

Else, when the destination is an index (CH_CSR[DIQ] = 0), the BD will be written to the address obtained by adding the provided index to the BMU_BASE, again, the software must ensure that said address is legal and known to BMU. That is, software must preallocate the BD (get the BD from BMU via GET BD/PDU cmd) for the BD before starting the transfer. For B2H or H2H transfers, if the destination is a UIF endpoint or EIF FIFO (CH_CSR[DIQ] = 1 and DQID matches UIF/EIF), SW should program the SAHB base address of the UIF destination endpoint in the

BASE bitfield and the queue ID to match in the DQID bitfield, then the DXE will use the base address given to generate the physical address and write the data over the SAHB bus and also use the appropriate sideband signals as specified by the DQID field to communicate with the selected UIF endpoint. SW should make sure that the BASE and DQID fields are consistent and refer to the same endpoint else hangs or unpredictable behavior might occur. On the other hand, when the destination is just a location in flat memory space, (CH_CSR[DIQ] = 0), be it internal or external, the bitfields lose their special meaning and DXE will just use the full 32-bits programmed into the register as the address to write the data. For this case, SW should make sure that the address maps to the intended destination interface (Internal memory, USB, PCI, PCIe) by conforming to the address map specified for the SAHB bus. To summarize, for H2B and B2B transfers the address must conform to the following format:

For index of BD mode (CH_CSR[DIQ] = 0): ADDR[31:7] = BMU_BASE + BD_INDEX[15:0]

ADDR[6:0] = 7'd0

For BMU Queue ID mode (CH_CSR[DIQ] = 1 & DQID matching one of the BMU queues):

ADDR[31:5] = Any value will do but zero preferred

ADDR[4:0] = Any BMU Queue ID as specified in description of QID field

For B2H and H2H transfers the address must conform to the following format:

For destinations in flat memory space (CH_CSR[DIQ] = 0): ADDR[31:0] = address mapping to destination interface

For UIF RX endpoints or EIF FIFOs (CH_CSR[DIQ] = 1 & DQID matching UIF or EIF):

ADDR[31:5] = UIF RX endpoint base or EIF inbound FIFO base

ADDR[4:0] = Queue ID matching UIF or EIF as specified in description of DQID field

WCSS_A_DXE_0_CH7_DADRL

Bits	Name	Description
31:7	BASE	Base Address. Valid only when the CH_CSR[DIQ] bit is set. For this case, this bitfield provides the base address of the destination UIF RX endpoint or EIF outbound FIFO (for BMU SW may use any value can be used but zero is suggested). Consult the appropriate interface uarch for the offset within the interface's address space. When CH_CSR[SIQ] bit is not set this field loses its special meaning and is interpreted only as part of the full 32-bit source address.

WCSS_A_DXE_0_CH7_DADRL (cont.)

Bits	Name	Description
6:0	DQID	Destination Queue ID. Valid only when the CH_CSR[SIQ] bit is set. For this case, this bitfield provides the ID for the destination BMU WQ, UIF RX endpoint or EIF outbound queue to be used for the transfer. When CH_CSR[SIQ] bit is not set this field loses its special meaning and is interpreted only as part of the full 32-bit source address. 0x0: RSVD_0 (Reserved) 0x1: RSVD_1 (Reserved) 0x2: BMU_2 (BMU Work Queue 2) 0x3: BMU_3 (BMU Work Queue 3) 0x4: BMU_4 (BMU Work Queue 4) 0x5: BMU_5 (BMU Work Queue 5) 0x6: BMU_6 (BMU Work Queue 6) 0x7: BMU_7 (BMU Work Queue 7) 0x8: BMU_8 (BMU Work Queue 8) 0x9: BMU_9 (BMU Work Queue 9) 0xA: BMU_10 (BMU Work Queue 10) 0xB: BMU_11 (BMU Work Queue 11) 0xC: BMU_12 (BMU Work Queue 12) 0xD: BMU_13 (BMU Work Queue 13) 0xE: BMU_14 (BMU Work Queue 14) 0xF: BMU_15 (BMU Work Queue 15) 0x10: BMU_16 (BMU Work Queue 16) 0x11: BMU_17 (BMU Work Queue 17) 0x12: BMU_18 (BMU Work Queue 18) 0x13: BMU_19 (BMU Work Queue 19) 0x14: BMU_20 (BMU Work Queue 20) 0x15: BMU_21 (BMU Work Queue 21) 0x16: BMU_22 (BMU Work Queue 22) 0x17: BMU_23 (BMU Work Queue 23) 0x18: BMU_24 (BMU Work Queue 24) 0x19: BMU_25 (BMU Work Queue 25) 0x1A: BMU_26 (BMU Work Queue 26) 0x1B: BMU_SINK_WQ (BMU_SINK (WQ255)) 0x1C: UIF_RX_1_EP (UIF RX 1 endpoint) 0x1D: UIF_RX_2_EP (UIF RX 2 endpoint) 0x1E: UIF_RX_3_EP (UIF RX 3 endpoint) 0x1F: UIF_RX_4_EP (UIF RX 4 endpoint)

0x0A2025D8 WCSS_A_DXE_0_CH7_DADRH**Type:** RW**Reset State:** 0x00000000

The channel destination address high register holds the high bits of the destination address for a transfer. See the description of CH_DADRL for more info.

WCSS_A_DXE_0_CH7_DADRH

Bits	Name	Type	Description
31:2	RSVD	R	Reserved. Writes ignored. Reads always return zeroes.
1:0	ADDRH	RW	This register holds bits 33 and 32 of the address. With the current address mapping these high order bits determine where the transfer is routed and whether the lower 32 bits of the address (the ones held in CH_DADRL) go untranslated through PIF or XIF or if they use the PIF and XIF address translation windows. See the bitfield encoding descriptions for an explanation. 0x0: ADDR_XTLN (If address in PIF or XIF address space then address goes through translation in PIF or XIF) 0x1: PIF_NO_XTLN (Always maps to PIF and full 32-bits in CH_DADRL goes through PIF untranslated) 0x2: XIF_NO_XTLN (Always maps to XIF and full 32-bits in CH_DADRL goes through XIF untranslated) 0x3: RSVD (Reserved. Maps to default slave. Reads return error, writes are ignored)

0x0A2025DC WCSS_A_DXE_0_CH7_DESCL

Type: RW

Reset State: 0x00000000

The Channel 7 Linked List Descriptor Pointer register contains the 32 lower bits of the address of the next descriptor to be loaded in the linked list. This address must be dword (32-bit) aligned (addr[1:0]=0 if CH_CTRL[PIQ]=0). Each complete transfer descriptor consists of 7 32-bit values arranged contiguously in memory. The value of this register will be overwritten with the address of the start of the next descriptor set in the list after the current descriptor has been fetched. The meaning of the value in this register depends on the value of the CH_CTRL[PIQ] bitfield similar to the way in which SADRL and DADRL are depending in SIQ and DIQ respectively. For CH_CTRL[PIQ]=0 then this register has no bitfields and the value is just interpreted as an address in flat memory space. When CH_CTRL[PIQ] is set then the descriptor will be read from FIFO type memory in UIF and the register has 2 bitfields: BASE, and SQID. BASE specifies the base address of the UIF Tx queue (derived from the start of the UIF address space in SAHB plus the internal UIF offset).

WCSS_A_DXE_0_CH7_DESCL

Bits	Name	Description
31:7	BASE	This bitfield only exists and has meaning when the descriptor needs to be read from a UIF TX queue (and therefore CH_CTRL[PIQ]=1), then it serves to specify the base address of the UIF endpoint to use as the source of the descriptor. Refer to the Nova address map document and to the UIF architecture document for guaranteed correct values to use here. For convenience but without a guarantee of accuracy, the base value to use for both the Tx data and management endpoints is the following: $\text{base} = \text{UIF base} + \text{TD_and_TM_offset} = 0x0E02_5000 + x100 = 0x0E02_5100 \gg 7 = 0x1C_04A2$
6:0	SQID	This bitfield only exists and has meaning when the descriptor needs to be read from a UIF TX queue (and therefore CH_CTRL[PIQ]=1), then it serves to specify which of the two UIF TX queues to use. Note that in this case SW also needs to program the correct base address of the source UIF endpoint in the BASE bitfield. 0x0: UIF_TX_DATA_EP (Descriptor to be read from UIF TX Data Endpoint) 0x1: UIF_TX_MGMT_EP (Descriptor to be read from UIF TX Management Endpoint)

0x0A2025E0 WCSS_A_DXE_0_CH7_DESCH**Type:** RW**Reset State:** 0x00000000

The channel destination address high register holds the high bits of the address of the next descriptor to be loaded in the linked list.

WCSS_A_DXE_0_CH7_DESCH

Bits	Name	Type	Description
31:2	RSVD	R	Reserved. Writes ignored. Reads always return zeros.

WCSS_A_DXE_0_CH7_DESCH (cont.)

Bits	Name	Type	Description
1:0	ADDRH	RW	For taurus, this register holds bits 33 and 32 of the address. With the current address mapping these high order bits determine where the transfer is routed and whether the lower 32 bits of the address (the ones held in CH_DADRL) go untranslated through PIF or XIF or if they use the PIF and XIF address translation windows. See the bitfield encoding descriptions for an explanation. Note that if the source of the descriptor is either of the UIF endpoints (CH_CTRL[PIQ]=1) then the value in ADDRH has to be 0 or else the descriptor read could end up mapping to the XIF or PIF. 0x0: ADDR_XTLN (If address in PIF or XIF address space then address goes through translation in PIF or XIF) 0x1: PIF_NO_XTLN (Always maps to PIF and full 32-bits in CH_DADRL goes through PIF untranslated) 0x2: XIF_NO_XTLN (Always maps to XIF and full 32-bits in CH_DADRL goes through XIF untranslated) 0x3: RSVD (Reserved. Maps to default slave. Reads return error, writes are ignored)

0x0A2025E4 WCSS_A_DXE_0_CH7_LST_DESCL**Type:** R**Reset State:** 0x00000000

The read-only Channel 7 Current/last Linked List Descriptor Pointer register contains the low 32-bits of the address of the descriptor currently being processed or of the last descriptor processed if the channel is disabled.

The meaning of the value and bitfields with regards to CH_CTRL[PIQ] is the same as was explained for the CH_DESCL register.

WCSS_A_DXE_0_CH7_LST_DESCL

Bits	Name	Description
31:7	BASE	See description of CH_DESCL[BASE].
6:0	SQID	See description of CH_DESCL[SQID]. 0x0: UIF_TX_DATA_EP (Descriptor to be read from UIF TX Data Endpoint) 0x1: UIF_TX_MGMT_EP (Descriptor to be read from UIF TX Management Endpoint)

0x0A2025E8 WCSS_A_DXE_0_CH7_LST_DESCH**Type:** R**Reset State:** 0x00000000

The channel destination address high register holds the high bits of the address of the next descriptor to be loaded in the linked list. Note that The meaning of the value and bitfields is the same as was explained for the CH_DESCL register.

WCSS_A_DXE_0_CH7_LST_DESCH

Bits	Name	Description
31:2	RSVD	Reserved. Writes ignored. Reads always return zeroes.
1:0	ADDRH	For taurus, this register holds bits 33 and 32 of the address.

0x0A2025EC WCSS_A_DXE_0_CH7_BD**Type:** R**Reset State:** 0x00000000**WCSS_A_DXE_0_CH7_BD**

Bits	Name	Description
31:24	RSVD	Reserved. Always returns 0 when read.
23	VAL	Valid. BD has been loaded and is valid.
22:16	PDU_CNT	PDU count of packet.
15:0	IDX	Index of BD. To get actual physical address must be added to BD/PDU base.

0x0A2025F0 WCSS_A_DXE_0_CH7_HEAD**Type:** R**Reset State:** 0x00000000

Address of head of packet.

WCSS_A_DXE_0_CH7_HEAD

Bits	Name	Description
31:27	RSVD	Reserved.
26	DESCVAL	Descriptor valid. Signifies descriptor has already been loaded. Used to prevent reloading of a descriptor.

WCSS_A_DXE_0_CH7_HEAD (cont.)

Bits	Name	Description
25:21	DWQID	Destination WQ ID. Original BMU, UIF or EIF destination queue. Valid only when CH_CSR[DIQ]=1
20:16	SWQID	Source WQ ID. Original BMU, UIF or EIF source queue. Valid only when CH_CSR[SIQ]=1
15:0	IDX	Index of Head. Must be added to BD/PDU base for actual physical address.

0x0A2025F4 WCSS_A_DXE_0_CH7_TAIL**Type:** R**Reset State:** 0x00000000

Address of tail of packet.

WCSS_A_DXE_0_CH7_TAIL

Bits	Name	Description
31:16	RSVD	Reserved.
15:0	IDX	Index of Tail. Must be added to BD/PDU base for actual physical address

0x0A2025F8 WCSS_A_DXE_0_CH7_PDU**Type:** R**Reset State:** 0x00000000

PDU info register.

All its fields are working fields that get modified as the programmed operation progresses.

WCSS_A_DXE_0_CH7_PDU

Bits	Name	Description
31	RSVD	Reserved. Always returns 0 when read.
30:24	PDU_CNT	Current PDU count.
23:8	IDX	Index of current PDU. Must be added to BD/PDU base for actual physical address
7:0	FST_OFF	First PDU offset.

0x0A2025FC WCSS_A_DXE_0_CH7_TSTMP**Type:** R**Reset State:** 0x00000000

Contains last sampled value of the system timestamp

WCSS_A_DXE_0_CH7_TSTMP

Bits	Name	Description
31:0	VALUE	

10.20 WCSS_A_CMEM_CMEM**0x0A201000 WCSS_A_CMEM_CMEM_INT****Type:** R**Reset State:** 0x00000000

Interrupts in CMEM.

WCSS_A_CMEM_CMEM_INT

Bits	Name	Description
5	CMEM_ACPU_INVALID_RA_DDR_INT	Invalid read address access from CPU port interrupt.
4	CMEM_ACPU_INVALID_WA_DDR_INT	Invalid write address access from CPU port interrupt.
3	CMEM_DAHB_INVALID_RA_DDR_INT	Invalid address read access from DAHB port interrupt.
2	CMEM_DAHB_INVALID_WA_DDR_INT	Invalid address write access from DAHB port interrupt.
1	CMEM_CAHB_INVALID_RA_DDR_INT	Invalid address read access from CAHB port interrupt.
0	CMEM_CAHB_INVALID_WA_DDR_INT	Invalid address write access from CAHB port interrupt.

0x0A201004 WCSS_A_CMEM_CMEM_INT_EN**Type:** RW**Reset State:** 0x00000000

Enable assertion of interrupt

WCSS_A_CMEM_CMEM_INT_EN

Bits	Name	Description
5	CMEM_ACPU_INVALID_RA_DDR_INT_EN	Enable invalid read address access from CPU interrupt.
4	CMEM_ACPU_INVALID_WA_DDR_INT_EN	Enable invalid write address access from CPU interrupt.
3	CMEM_DAHB_INVALID_RA_DDR_INT_EN	Enable invalid address read access from DAHB interrupt.
2	CMEM_DAHB_INVALID_WA_DDR_INT_EN	Enable invalid address write access from DAHB interrupt.
1	CMEM_CAHB_INVALID_RA_DDR_INT_EN	Enable invalid address read access from CAHB interrupt.
0	CMEM_CAHB_INVALID_WA_DDR_INT_EN	Enable invalid address write access from CAHB interrupt.

0x0A201008 WCSS_A_CMEM_CMEM_INT_CLR**Type:** RW**Reset State:** 0x00000000

Clear the interrupts status bit.

WCSS_A_CMEM_CMEM_INT_CLR

Bits	Name	Description
5	CMEM_ACPU_INVALID_RA_DDR_INT_CLR	Clear invalid read address access from CPU interrupt.
4	CMEM_ACPU_INVALID_WA_DDR_INT_CLR	Clear invalid write address access from CPU interrupt.
3	CMEM_DAHB_INVALID_RA_DDR_INT_CLR	Clear invalid address read access from DAHB interrupt.
2	CMEM_DAHB_INVALID_WA_DDR_INT_CLR	Clear invalid address write access from DAHB interrupt.
1	CMEM_CAHB_INVALID_RA_DDR_INT_CLR	Clear invalid address read access from CAHB interrupt.
0	CMEM_CAHB_INVALID_WA_DDR_INT_CLR	Clear invalid address write access from CAHB interrupt.

0x0A20100C WCSS_A_CMEM_CMEM_MEM_CFG**Type:** RW**Reset State:** 0x00780000

Memory rdOnly

WCSS_A_CMEM_CMEM_MEM_CFG

Bits	Name	Type	Description
31:16	CMEM_LRAM_KBYTES	R	Memory configuration
3:0	MEM_CFG	RW	

0x0A201010 WCSS_A_CMEM_CMEM_TRANSBLK_CODE**Type:** RW**Reset State:** 0x00000000

Unlock the access to transblk_start_addr, transblk_end_addr and transblk_port_en registers by writing 32'hA1A602E7

WCSS_A_CMEM_CMEM_TRANSBLK_CODE

Bits	Name	Description
31:0	CMEM_TRANSBLK_CODE	Transaction blocking write-access code.

0x0A201014 WCSS_A_CMEM_CMEM_TRANSBLK_START_ADDR**Type:** RW**Reset State:** 0x00000000

Starting address of transfer block region in memory.

WCSS_A_CMEM_CMEM_TRANSBLK_START_ADDR

Bits	Name	Description
31:10	CMEM_TRANSBLK_START_ADDR	starting address of transfer block region in kbytes

0x0A201018 WCSS_A_CMEM_CMEM_TRANSBLK_END_ADDR**Type:** RW**Reset State:** 0x00000000

Ending address of transfer block region in memory.

WCSS_A_CMEM_CMEM_TRANSBLK_END_ADDR

Bits	Name	Description
31:10	CMEM_TRANSBLK_END_A DDR	ending address of transfer block region in kbytes

0x0A20101C WCSS_A_CMEM_CMEM_TRANSBLK_PORT_EN**Type:** RW**Reset State:** 0x00000000

transfer block enable for each port.

WCSS_A_CMEM_CMEM_TRANSBLK_PORT_EN

Bits	Name	Description
23:20	CMEM_TRANSBLK_MSTID1	second AHB master ID blocked from accessing blocked region. valid only when cmem_transblk_mstid1_vld is set to 1.
19:16	CMEM_TRANSBLK_MSTID0	first AHB master ID blocked from accessing blocked region. valid only when cmem_transblk_mstid0_vld is set to 1.
3	CMEM_TRANSBLK_MSTID1_VLD	cmem_transblk_mstid1 is valid
2	CMEM_TRANSBLK_MSTID0_VLD	cmem_transblk_mstid0 is valid
1:0	CMEM_TRANSBLK_PORT_EN	transfer block enable for each port

0x0A201020 WCSS_A_CMEM_CMEM_ARB_POLICY**Type:** RW**Reset State:** 0x00000000

Arbitration policy when two AHB ports are active

WCSS_A_CMEM_CMEM_ARB_POLICY

Bits	Name	Description
24:16	CMEM_CAHB_PRTY	CAHB port priority for each 9 SRAM block.Bit n corresponds to SRAM_n.
9:1	CMEM_LRAM_ARB_POLICY	Arbitration for each 9 SRAM block. Bit n corresponds to SRAM_n.
0	CMEM_AHB_ARB_POLICY	Arbitration policy. 0x0: PRR (Prioritized Round-Robin) 0x1: Reserved

0x0A201024 WCSS_A_CMEM_CMEM_PRIORITY**Type:** RW**Reset State:** 0x00000000

Priority setting of dahb, and cpu.

WCSS_A_CMEM_CMEM_PRIORITY

Bits	Name	Description
24:16	CMEM_ACPU_PRTY	CPU port priority for each 9 SRAM block.Bit n corresponds to SRAM_n.
8:0	CMEM_DAHB_PRTY	DAHB port priority for each 9 SRAM block.Bit n corresponds to SRAM_n.

0x0A20102C WCSS_A_CMEM_CMEM_INVALID_HISTORY**Type:** R**Reset State:** 0x00000000

history of up to first 4 invalid address access

WCSS_A_CMEM_CMEM_INVALID_HISTORY

Bits	Name	Description
31:28	ACC_TYPE3	access type for the forth invalid access. bit[0] - read or write. 0 : read, 1 : write bit[1] - access is from CAHB bit[2] - access is from DAHB bit[3] - access is from CPU
27:24	MST_ID3	master ID that the forth invalid access has occurred if the access is from CAHB or DAHB
23:20	ACC_TYPE2	access type for the third invalid access. bit[0] - read or write. 0 : read, 1 : write bit[1] - access is from CAHB bit[2] - access is from DAHB bit[3] - access is from CPU
19:16	MST_ID2	master ID that the third invalid access has occurred if the access is from CAHB or DAHB
15:12	ACC_TYPE1	access type for the second invalid access. bit[0] - read or write. 0 : read, 1 : write bit[1] - access is from CAHB bit[2] - access is from DAHB bit[3] - access is from CPU
11:8	MST_ID1	master ID that the second invalid access has occurred if the access is from CAHB or DAHB
7:4	ACC_TYPE0	access type for the first invalid access. bit[0] - read or write. 0 : read, 1 : write bit[1] - access is from CAHB bit[2] - access is from DAHB bit[3] - access is from CPU
3:0	MST_ID0	master ID that the first invalid access has occurred if the access is from CAHB or DAHB

0x0A201030 WCSS_A_CMEM_CMEM_INVALID_ADDR0**Type:** R**Reset State:** 0x00000000

address that the first invalid access has happened.

WCSS_A_CMEM_CMEM_INVALID_ADDR0

Bits	Name	Description
31:0	CMEM_INVALID_ADDR0	address that the first invalid access has happened.

0x0A201034 WCSS_A_CMEM_CMEM_INVALID_ADDR1**Type:** R**Reset State:** 0x00000000

address that the second invalid access has happened.

WCSS_A_CMEM_CMEM_INVALID_ADDR1

Bits	Name	Description
31:0	CMEM_INVALID_ADDR1	address that the second invalid access has happened.

0x0A201038 WCSS_A_CMEM_CMEM_INVALID_ADDR2**Type:** R**Reset State:** 0x00000000

address that the third invalid access has happened.

WCSS_A_CMEM_CMEM_INVALID_ADDR2

Bits	Name	Description
31:0	CMEM_INVALID_ADDR2	address that the third invalid access has happened.

0x0A20103C WCSS_A_CMEM_CMEM_INVALID_ADDR3**Type:** R**Reset State:** 0x00000000

address that the forth invalid access has happened.

WCSS_A_CMEM_CMEM_INVALID_ADDR3

Bits	Name	Description
31:0	CMEM_INVALID_ADDR3	address that the forth invalid access has happened.

0x0A201040 WCSS_A_CMEM_CMEM_INVALID_CTL**Type:** RW**Reset State:** 0x00000000

control register for clearing invalid history and enabling error response.

WCSS_A_CMEM_CMEM_INVALID_CTL

Bits	Name	Description
1	CMEM_BUS_ERROR_ON_INVALID	when enabled, AHB error will be returned when invalid access occurs.
0	CMEM_INVALID_CLR	clear invalid history including id, type and address

0x0A201048 WCSS_A_CMEM_CMEM_CNT_CTRL

Type: RW

Reset State: 0x00000000

This register controls the cmem counters

WCSS_A_CMEM_CMEM_CNT_CTRL

Bits	Name	Description
1	CMEM_CNT_CLR	Clear the cmem counters to 0.
0	CMEM_CNT_EN	Enable the cmem counters to increment.

0x0A20104C WCSS_A_CMEM_CMEM_CAHB_BUSY_CNT

Type: R

Reset State: 0x00000000

32-bit of cmem_cahb_busy_cnt

WCSS_A_CMEM_CMEM_CAHB_BUSY_CNT

Bits	Name	Description
31:0	CMEM_CAHB_BUSY_CNT	Number of AHB clock cycle while CAHB port is busy.

0x0A201050 WCSS_A_CMEM_CMEM_CAHB_IDLE_CNT**Type:** R**Reset State:** 0x00000000

32-bit of cmem_cahb_idle_cnt

WCSS_A_CMEM_CMEM_CAHB_IDLE_CNT

Bits	Name	Description
31:0	CMEM_CAHB_IDLE_CNT	Number of AHB clock cycle while CAHB port is idle.

0x0A201054 WCSS_A_CMEM_CMEM_CAHB_ACTIVE_CNT**Type:** R**Reset State:** 0x00000000

32-bit of cmem_cahb_active_cnt

WCSS_A_CMEM_CMEM_CAHB_ACTIVE_CNT

Bits	Name	Description
31:0	CMEM_CAHB_ACTIVE_CNT	Number of AHB clock cycle while CAHB port is busy and active.

0x0A201058 WCSS_A_CMEM_CMEM_DAHB_BUSY_CNT**Type:** R**Reset State:** 0x00000000

32-bit of cmem_dahb_busy_cnt

WCSS_A_CMEM_CMEM_DAHB_BUSY_CNT

Bits	Name	Description
31:0	CMEM_DAHB_BUSY_CNT	Number of AHB clock cycle while DAHB port is busy.

0x0A20105C WCSS_A_CMEM_CMEM_DAHB_IDLE_CNT**Type:** R**Reset State:** 0x00000000

32-bit of cmem_dahb_idle_cnt

WCSS_A_CMEM_CMEM_DAHB_IDLE_CNT

Bits	Name	Description
31:0	CMEM_DAHB_IDLE_CNT	Number of AHB clock cycle while DAHB port is idle.

0x0A201060 WCSS_A_CMEM_CMEM_DAHB_ACTIVE_CNT

Type: R

Reset State: 0x00000000

32-bit of cmem_dahb_active_cnt

WCSS_A_CMEM_CMEM_DAHB_ACTIVE_CNT

Bits	Name	Description
31:0	CMEM_DAHB_ACTIVE_CNT	Number of AHB clock cycle while DAHB port is busy and active.

0x0A201064 WCSS_A_CMEM_CMEM_ACPU_BUSY_CNT

Type: R

Reset State: 0x00000000

32-bit of cmem_acpu_busy_cnt

WCSS_A_CMEM_CMEM_ACPU_BUSY_CNT

Bits	Name	Description
31:0	CMEM_ACPU_BUSY_CNT	Number of ACPU clock cycle while CPU memory port is busy.

0x0A201068 WCSS_A_CMEM_CMEM_ACPU_IDLE_CNT

Type: R

Reset State: 0x00000000

32-bit of cmem_acpu_idle_cnt

WCSS_A_CMEM_CMEM_ACPU_IDLE_CNT

Bits	Name	Description
31:0	CMEM_ACPU_IDLE_CNT	Number of ACPU clock cycle while CPU memory port is busy.

0x0A20106C WCSS_A_CMEM_CMEM_ACPU_ACTIVE_CNT**Type:** R**Reset State:** 0x00000000

32-bit of cmem_acpu_active_cnt

WCSS_A_CMEM_CMEM_ACPU_ACTIVE_CNT

Bits	Name	Description
31:0	CMEM_ACPU_ACTIVE_CNT	Number of ACPU clock cycle while CPU memory port is busy and active.

0x0A201078 WCSS_A_CMEM_CMEM_CLK_GATE**Type:** RW**Reset State:** 0x00010104

disable control for top cmem clock gates

WCSS_A_CMEM_CMEM_CLK_GATE

Bits	Name	Type	Description
26	CPU_MEM_CLK_DISABLE	RW	when set, cpu clock to the memories will be blocked. The bit should be set to 1 before changing mem_cfg to avoid clock glitch to memories.
25	TRC_MEM_CLK_DISABLE	RW	when set, trc clock to the memories will be blocked. The bit should be set to 1 before changing mem_cfg to avoid clock glitch to memories.
24	PHYDBG_MEM_CLK_DISABLE	RW	when set, phydbg clock to the memories will be blocked. The bit should be set to 1 before changing mem_cfg to avoid clock glitch to memories.
23:16	GAS_BUSY_MSK	RW	number of cycles that gas_busy enables the clock after it is de-asserted.

WCSS_A_CMEM_CMEM_CLK_GATE (cont.)

Bits	Name	Type	Description
15:8	AHB_BUSY_MSK	RW	number of cycles that ahb_busy enables the clock after it is de-asserted.
7	DAHB_ABORT_HAPPENED	R	indication of abort happened.
6	DAHB_ABORT_CLR	RW	clear abort_happened
5	CAHB_ABORT_HAPPENED	R	indication of abort happened.
4	CAHB_ABORT_CLR	RW	clear abort_happened
3	CMEM_RD_CLK_GATE_DISABLE	RW	Disable read path clock gate.
2	CMEM_MEM_CLK_GATE_DISABLE	RW	Disable top clock gate for memory clock.
1	CMEM_ACPU_CLK_GATE_DISABLE	RW	Disable top clock gate for 40/80MHz ACPU clock.
0	CMEM_AHB_CLK_GATE_DISABLE	RW	Disable top clock gate for 40MHz AHB clock.

0x0A20107C WCSS_A_CMEM_CMEM_CAHB_RX_CTL**Type:** RW**Reset State:** 0x00000000

various control for CAHB rx fifo and data path

WCSS_A_CMEM_CMEM_CAHB_RX_CTL

Bits	Name	Description
2	CMEM_RX_CLR_ON_CMD_EN	when the bit is set, rx path is cleared when new transaction starts.
1	CMEM_RX_CLR	clear the RX FIFO and the read data path
0	CMEM_RX_FIFO_CLR	clear the RX FIFO

0x0A201080 WCSS_A_CMEM_CMEM_DAHB_RX_CTL**Type:** RW**Reset State:** 0x00000000

various control for DAHB rx fifo and data path

WCSS_A_CMEM_CMEM_DAHB_RX_CTL

Bits	Name	Description
2	CMEM_RX_CLR_ON_CMD_EN	when the bit is set, rx path is cleared when new transaction starts.
1	CMEM_RX_CLR	clear the RX FIFO and the read data path
0	CMEM_RX_FIFO_CLR	clear the RX FIFO

10.21 WCSS_A_CCAHB_CCAHB**0x0A200000 WCSS_A_CCAHB_CCAHB_BT_PL****Type:** RW**Reset State:** 0x00000005

ccahb Arbiter arbitration priority level for bt master. In the ccahb arbitration scheme the master with highest priority level always wins. Should there be more than one master with the same priority then the winning master will be selected using a round-robin algorithm.

WCSS_A_CCAHB_CCAHB_BT_PL

Bits	Name	Description
3:0	PRIORITY	Arbitration priority for bt master. The lowest priority level is 1 and the highest priority level is 15. A priority level of 0 has a special significance, it means that the master is disabled and stops participating in the arbitration scheme. Note that the ccahb arbiter will prevent a master from disabling itself.

0x0A200004 WCSS_A_CCAHB_CCAHB_FM_PL**Type:** RW**Reset State:** 0x00000005

ccahb Arbiter arbitration priority level for fm master. In the ccahb arbitration scheme the master with highest priority level always wins. Should there be more than one master with the same priority then the winning master will be selected using a round-robin algorithm.

WCSS_A_CCAHB_CCAHB_FM_PL

Bits	Name	Description
3:0	PRIORITY	Arbitration priority for fm master. The lowest priority level is 1 and the highest priority level is 15. A priority level of 0 has a special significance, it means that the master is disabled and stops participating in the arbitration scheme. Note that the ccahb arbiter will prevent a master from disabling itself.

0x0A200008 WCSS_A_CCAHB_CCAHB_RRI_PL**Type:** RW**Reset State:** 0x00000001

ccahb Arbiter arbitration priority level for rri master. In the ccahb arbitration scheme the master with highest priority level always wins. Should there be more than one master with the same priority then the winning master will be selected using a round-robin algorithm.

WCSS_A_CCAHB_CCAHB_RRI_PL

Bits	Name	Description
3:0	PRIORITY	Arbitration priority for rri master. The lowest priority level is 1 and the highest priority level is 15. A priority level of 0 has a special significance, it means that the master is disabled and stops participating in the arbitration scheme. Note that the ccahb arbiter will prevent a master from disabling itself.

0x0A20000C WCSS_A_CCAHB_CCAHB_DXE_PL**Type:** RW**Reset State:** 0x00000001

ccahb Arbiter arbitration priority level for dxm master. In the ccahb arbitration scheme the master with highest priority level always wins. Should there be more than one master with the same priority then the winning master will be selected using a round-robin algorithm.

WCSS_A_CCAHB_CCAHB_DXE_PL

Bits	Name	Description
3:0	PRIORITY	Arbitration priority for dxm master. The lowest priority level is 1 and the highest priority level is 15. A priority level of 0 has a special significance, it means that the master is disabled and stops participating in the arbitration scheme. Note that the ccahb arbiter will prevent a master from disabling itself.

0x0A200010 WCSS_A_CCAHB_CCAHB_CCPU_PL**Type:** RW**Reset State:** 0x00000004

ccahb Arbiter arbitration priority level for ccpu master. In the ccahb arbitration scheme the master with highest priority level always wins. Should there be more than one master with the same priority then the winning master will be selected using a round-robin algorithm.

WCSS_A_CCAHB_CCAHB_CCPU_PL

Bits	Name	Description
3:0	PRIORITY	Arbitration priority for ccpu master. The lowest priority level is 1 and the highest priority level is 15. A priority level of 0 has a special significance, it means that the master is disabled and stops participating in the arbitration scheme. Note that the ccahb arbiter will prevent a master from disabling itself.

0x0A200014 WCSS_A_CCAHB_CCAHB_A2AB_PL**Type:** RW**Reset State:** 0x00000001

ccahb Arbiter arbitration priority level for a2ab master. In the ccahb arbitration scheme the master with highest priority level always wins. Should there be more than one master with the same priority then the winning master will be selected using a round-robin algorithm.

WCSS_A_CCAHB_CCAHB_A2AB_PL

Bits	Name	Description
3:0	PRIORITY	Arbitration priority for a2ab master. The lowest priority level is 1 and the highest priority level is 15. A priority level of 0 has a special significance, it means that the master is disabled and stops participating in the arbitration scheme. Note that the ccahb arbiter will prevent a master from disabling itself.

0x0A200018 WCSS_A_CCAHB_CCAHB_CBR_PL**Type:** RW**Reset State:** 0x00000005

ccahb Arbiter arbitration priority level for cbr master. In the ccahb arbitration scheme the master with highest priority level always wins. Should there be more than one master with the same priority then the winning master will be selected using a round-robin algorithm.

WCSS_A_CCAHB_CCAHB_CBR_PL

Bits	Name	Description
3:0	PRIORITY	Arbitration priority for cbr master. The lowest priority level is 1 and the highest priority level is 15. A priority level of 0 has a special significance, it means that the master is disabled and stops participating in the arbitration scheme. Note that the ccahb arbiter will prevent a master from disabling itself.

0x0A200048 WCSS_A_CCAHB_CCAHB_DFLT_MST**Type:** RW**Reset State:** 0x00000000

This register identifies the default master for the ccahb

WCSS_A_CCAHB_CCAHB_DFLT_MST

Bits	Name	Description
3:0	DFT_MST	Default Master ID number register. The default master is the master that is granted by the bus when no master has requested ownership. NOTE: Writing anything beyond the 4 R/W bits will cause the entire write to be discarded. 0x0: DMY (Grant ownership to ccahb dummy master) 0x1: BT (Grant ownership to ccahb bt master) 0x2: FM (Grant ownership to ccahb fm master) 0x3: RRI (Grant ownership to ccahb rri master) 0x4: DXE (Grant ownership to ccahb dxe master) 0x5: CCPU (Grant ownership to ccahb ccpu master) 0x6: A2AB (Grant ownership to ccahb a2ab master) 0x7: CBR (Grant ownership to ccahb cbr master)

10.22 WCSS_A_FM_SSB_I2_CFG_SSB_I2_CFG**0x0A1E0000 WCSS_A_FM_SSB_I2_CFG_SSB_I2_CTL****Type:** RW**Clock:** AHB bus clock**Reset State:** 0x00083005

Specifies various generic SSB_I2 control information. Bits 19:13 comprising of fields SSB_I2_DATA_PDEN, FCLK_MODE, FCLK_SW_OVERRIDE and NUM_FCLK are either

asynchronously reset or set. This is because they control the signals going directly to a pad. Remaining bits 12:0 are synchronously reset or set.

WCSS_A_FM_SSB12_CFG_SSB12_CTL

Bits	Name	Description
19:16	NUM_FCLK	Indicates a programmable number of active forwarding clock cycles following a SSB12 'transaction done' for a corresponding SSB12 request. Unless there is a new SSB12 transaction request, the forwarding clock is shut off after toggling NUM_FCLK cycles at the PMIC SSB12 clock frequency. The suggested setting (also the default value) is 8 clock cycles.
15	FCLK_SW_OVERRIDE	This is a software override bit to control the CGC for forwarding clock. Set this bit to 1 to allow the SSB12 clock to continue as a forwarding clock through the CGC (provided FCLK_MODE = 1). This override bit is provided to allow continuous toggling clock to SSB12 slave irrespective of how the HW logic controls the CGC. Set this bit to 0 to enable pure HW control on the forwarding clock CGC.
14	FCLK_MODE	Set this bit to 1 to activate forwarding clock mode. When this bit is 0, the FCLK_SW_OVERRIDE field has no meaning. Out of reset, the mode is set to 0 indicating that the SSB12 slave does not work off the forwarding clock. When FCLK_MODE = 0; no forwarding clock. When FCLK_MODE = 1; FCLK_SW_OVERRIDE = 1; forwarding clock always toggles. When FCLK_MODE = 1; FCLK_SW_OVERRIDE = 0; forwarding clock is controlled by HW logic.
13	SSB12_DATA_PDEN	Pulldown enable for SSB12_DATA pad.
12	PA_CGC_CTRL	This is to control the SSB12 clock going to this PMIC arbiter. Set this bit to 1 when you want to allow the clock to flow through the CGC. Set this bit to 0 to shut off clock to PMIC arbiter. Out of reset, this bit is set to 1.
10:8	SSB12_ADDR_WIDTH	Specifies what is the addressing mode in SSB12 mode. This field is only applicable in SSB12 mode and not the old SSB11 mode. 0x0: ENUM_8_BIT_ADDRESSING_MODE (8-bit addressing mode) 0x1: ENUM_9_BIT_ADDRESSING_MODE_AND_SO_ON (9-bit addressing mode and so on)
7	SSB12_MODE	Set this bit to 1 when SW wants to switch to new SSB12 mode. Set this to 0 when in old SSB11 mode.
5:4	SEL_RD_DATA	Specifies which RD_DATA value to select out of four possible combinations. This signal allows the returned data from the slave to be captured inside the master, 1, 2, or 3 symbols late in order to accommodate a large range of routing delay. Based on this field, the entire read transaction length will increase by SEL_RD_DATA number of symbols.
3:2	SSB12_DATA_DEL	Specifies the number of 1/2 clock periods, SSB12_DATA will be delayed before sampling inside.

WCSS_A_FM_SSBi2_CFG_SSBi2_CTL (cont.)

Bits	Name	Description
1:0	IDLE_SYMS	The master will ensure there are at least this number of idle symbol periods in between the accesses. The idle time is inserted at the end of the read or write transaction. This supported values are always between 1 - 3.

0x0A1E0004 WCSS_A_FM_SSBi2_CFG_SSBi2_INT_ENABLE**Type:** RW**Clock:** AHB bus clock**Reset State:** 0x00000000

Provides control on whether the TRANSACTION-DONE interrupts can be generated or not (per SW master).

WCSS_A_FM_SSBi2_CFG_SSBi2_INT_ENABLE

Bits	Name	Description
6:0	ENABLE_SSBi2_INT	SSBi2 transaction done interrupts for each SW master are generated only when the corresponding bits are set. The Interrupt mapping corresponds to the entries in the VIT . For e.g. if the VMID in VIT index 0 corresponds to the SW master RPM and VMID in index 1 of the VIT corresponds to SW master ARM 11, the bit 0 in the register will enable the interrupt corresponding to RPM , bit 1 will enable interrupt corresponding to ARM11.

0x0A1E0008 WCSS_A_FM_SSBi2_CFG_SSBi2_PRIORITIES**Type:** RW**Clock:** AHB bus clock**Reset State:** 0x00000000

Specifies the priorities for SSBi2 arbiter ports. The SSBi2 block supports up to maximum of 8 ports for arbitration. Each port has 3 bits to denote its priority. The bits are synchronously reset.

CAUTION Since the reset state value of SSBi2_PRIORITIES is 0, only SW requests (hard-coded to port #0) can get through. SW has to program this register to allow hardware requests through the SSBi2 arbiter.

WCSS_A_FM_SSBi2_CFG_SSBi2_PRIORITIES

Bits	Name	Description
11:9	PRIORITY3	Specifies the port number of host, with a priority that is 1 less then PRIORITY2.

WCSS_A_FM_SSB_I2_CFG_SSB_I2_PRIORITIES (cont.)

Bits	Name	Description
8:6	PRIORITY2	Specifies the port number of host, with a priority that is 1 less than PRIORITY1.
5:3	PRIORITY1	Specifies the port number of host, with a priority that is 1 less than PRIORITY0.
2:0	PRIORITY0	Specifies the port number of the highest priority host.

0x0A1E000C WCSS_A_FM_SSB_I2_CFG_SSB_I2_BANKS_2_3_CTL**Type:** RW**Clock:** AHB bus clock**Reset State:** 0xFFFFFFFF

This register specifies whether an access is allowed to banks 2 and 3 of PMIC. The access control to the banks 2 and 3 of PMIC is set per VMID. Bit 0 corresponds to access control for VMID 0 and bit 31 corresponds to access control for VMID 31.

WCSS_A_FM_SSB_I2_CFG_SSB_I2_BANKS_2_3_CTL

Bits	Name	Description
31:0	ACCESS_CONTROL	Set each of the 32 bits to 1 if the master corresponding to that VMID wishes to initiate SW requests to banks 2 and 3 of PMIC.

0x0A1E0010 WCSS_A_FM_SSB_I2_CFG_SSB_I2_ILLEGAL_STATUS**Type:** R**Clock:** AHB bus clock**Reset State:** 0x00000000

Returns status information of the software master that tried to access a SSB_I2 command micro register, and failed **security** through the VIT table.

WCSS_A_FM_SSB_I2_CFG_SSB_I2_ILLEGAL_STATUS

Bits	Name	Description
5:1	ILLEGAL_VMID	Indicates the VMID of the master that performed the most recent illegal access. The value in this field is only meaningful when the ILLEGAL_ACCESS bit is set. On software reads, the value is retained unless another illegal access with a different VMID happens. In that case, this field will have the VMID corresponding to the most recent illegal master.

WCSS_A_FM_SSBi2_CFG_SSBi2_ILLEGAL_STATUS (cont.)

Bits	Name	Description
0	ILLEGAL_ACCESS	When set (1) indicates an illegal access. An illegal access is defined as an access that does not find a matching VMID in the SSBi2_VITn registers. This bit is cleared on a software read provided there is no other illegal access in the same clock cycle that a software read happens.

0x0A1E0020+ WCSS_A_FM_SSBi2_CFG_SSBi2_VITn, n=[0..6]**0x4*n****Type:** RW**Clock:** AHB bus clock**Reset State:** 0x00000000

This is the VMID index table (VIT) used for enabling concurrent SSBi2 SW requests among multiple SW masters. Secure entities are expected to assign a VMID for the SW master corresponding to an index. The SSBi2_VIT0 has the VMID associated with index 0, SSBi2_VIT1 has the VMID associated with index 1 and so on.

CAUTION By default, only the SW master with a VMID of '0' has access to registers in SSBi2 command micro space. SROT/RPM has to populate this table before valid SW masters can initiate SSBi2 SW transactions.

SSBi2_VITn, n=[0..6]

WCSS_A_FM_SSBi2_CFG_SSBi2_VITn

Bits	Name	Description
4:0	VMID	VMID corresponding to index n.

0x0A1E00A0 WCSS_A_FM_SSBi2_CFG_SSBi2_PVC_CTL**Type:** RW**Clock:** AHB bus clock**Reset State:** 0x00000000

This register provides the global enable/disable for the PVC interface module, per PVC port enable/disable as well as captures PVC interface busy status.

WARNING In a PMIC arbiter wrapper system having more than 1 PMIC arbiter module; if a PVC port(x) is enabled in one of the PMIC arbiters, the corresponding PVC_port(x) in all the other PMIC arbiters shall be disabled. If multiple PVC_port(x)'s enabled simultaneously, the hardware does not guarantee correct operation.

WCSS_A_FM_SSBi2_CFG_SSBi2_PVC_CTL

Bits	Name	Description
8	PVC_INTF_BUSY	This is a status bit. When set (1) indicates that the PVC interface module is busy (meaning there is atleast one PVC transaction being serviced). Busy line goes high when PVC interface detects a new request on one of the PVC ports and stays high until it samples a done from SSBi2 arbiter. Writes to this bit will not have any effect and will be ignored.
3	PVC_PORT3_EN	Set this bit to 1 to enable PVC port 3. PVC port 3 maps to SMPSS CPU1 SAW.
2	PVC_PORT2_EN	Set this bit to 1 to enable PVC port 2. PVC port 2 maps to SMPSS CPU 0 SAW.
1	PVC_PORT1_EN	Set this bit to 1 to enable PVC port 1. PVC port 1 maps to LPASS Q6 SAW.
0	PVC_INTF_EN	Set this bit to 1 to enable PVC interface module.

**0x0A1E00A4 WCSS_A_FM_SSBi2_CFG_SSBi2_PVC_TABLEn, n=[0..31]
+0x4*n****Type:** RW**Clock:** AHB bus clock**Reset State:** 0x00000000

This set of registers indicates the full PVC table corresponding to all the PVC ports. For e.g.: there are 4 entries per PVC port(set by RTL generic) and there are 3 PVC ports (set by RTL generic). Each entry is 15-bit address to the SSBi2 slave registers. This indirect addressing from the PVC ports to the SSBi2 slave registers provides a way to enforce **security** on hardware SSBi2 requests.

NOTE Though this current document shows 12 entries, the number of entries should scale based on the GENERIC settings for the number of PVC ports and the address index width. Also though 15-bit addresses are supported, 10-bit addressing is all that is expected for current PMICs.

SSBi2_PVC_TABLEn, n=[0...11]

WCSS_A_FM_SSBi2_CFG_SSBi2_PVC_TABLEn

Bits	Name	Description
14:0	ADDR	15-bit address of the SSBi2 slave registers.

10.23 WCSS_A_FM_SSBi2_CMD_SSBi2_CMD

0x0A1C0000 WCSS_A_FM_SSBi2_CMD_SSBi2_CMD

Type: W

Clock: AHB bus clock

Reset State: 0x00000000

The SW masters write to this register to initiate either a read or a write SSBi2 transaction. Through this register, the SW can address up to 32 kB of SSBi2 slave register space. However the PMICs only house a maximum of 1024 registers and hence the values on the most significant 5-bits of the REG_ADDR_14_8 field are ignored. The REG_DATA field is only applicable for write transaction and is ignored for reads. The read data returned back from the SSBi2 slave is directed to the SSBi2_RD_STATUS register. Bit 23 is left as a reserved bit to enable slave address to grow to 16-bits in the future.

WCSS_A_FM_SSBi2_CMD_SSBi2_CMD

Bits	Name	Description
24	RDWRN	Denotes whether the command to perform is a register write (0) or read (1).
22:16	REG_ADDR_14_8	Specifies the upper 7 bits, 14:8, of SSBi slave register address.
15:8	REG_ADDR_7_0	Lower 8 bits, 7:0, of SSBi slave register address.
7:0	REG_DATA	Write data for a slave register.

0x0A1C0004 WCSS_A_FM_SSBi2_CMD_SSBi2_RD_STATUS

Type: R

Clock: AHB bus clock

Reset State: 0x00000000

This register provides the read data as well as the status of the transaction. This register also returns back information associated with the transaction such as the SSBi2 slave register address and the transaction access type. SW is expected to poll TRANS_DONE and TRANS_DENIED until either of these bits are set to 1.

- TRANS_DONE = 0, TRANS_DENIED = 1; indicates to SW that the current transaction was rejected because the master did not have permissions to access the corresponding SSBi2 slave register.
- TRANS_DONE = 1, TRANS_DENIED = 0; indicates to SW that the current transaction was completed successfully. For writes, it means that the write data was sent out on the SSBi wire and for reads, it means that the read data is returned back and is now in SSBi2_RD_STATUS[REG_DATA]. At this point, the SW is said to have completed one transaction.

- Both TRANS_DONE and TRANS_DENIED being either 0 or 1 does not happen and is not a valid combination after a transaction has been initiated.

WCSS_A_FM_SSB_I2_CMD_SSB_I2_RD_STATUS

Bits	Name	Description
27	TRANS_DONE	When set (1), indicates that the SSB_I2 has completed the transaction. If this was a write request, the write has completed. If this was a read request, the read data has returned back.. NOTE This bit is cleared on read.
26	TRANS_DENIED	When set (1), indicates that a SSB_I2 transaction was denied because the master did not have access to the particular SSB_I2 slave SW register to where the access was intended.
25	TRANS_PROG	When set (1), indicates that a SSB_I2 transaction is in progress. This signal will be set for the duration from REQ assertion until DONE assertion.
24	RDWRN	Returns the access type of transaction. A value of 1 implies a read transaction and a value of 0 implies a write transaction.
22:8	REG_ADDR	Returns the address to which the SSB_I2 transaction was performed.
7:0	REG_DATA	Returned read register data. NOTE The read data is not cleared on reads, however will return all zeroes for a SW master whose transaction was denied or was an illegal SW master.

10.24 WCSS_A_FM_RIVA_FM**0x0A180000 WCSS_A_FM_TX_CONTROL****Type:** RW**Clock:** APB**Reset State:** 0x00000000

The TX_EN register is to control Riva-FM TX functionality.

WCSS_A_FM_TX_CONTROL

Bits	Name	Description
11:10	AUDIO_IN_SHIFT_CTRL	Used for rounding the left and right audio input. The TX_AUDIO_IN_SHIFT_CTRL register is used for controlling the rounding operation of the audio input data stream. 0x0: ROUND_2 (2 bit round) 0x1: ROUND_3 (3 bit round) 0x2: ROUND_4 (4 bit round) 0x3: ROUND_5 (5 bit round)

WCSS_A_FM_TX_CONTROL (cont.)

Bits	Name	Description
9:5	PRE_EMPH_CONST	0x13 for 50u pre-emph 0x1D for 75u pre-emph The PRE_EMPH_CONST is to set the TX pre-emphasis value.
4:3	DC_BLOCK	# of bits to truncate from the DC blocking accumulator. The TX_DC_BLOCK register is used to select the number of bits for dc blocking feedback. The DC Blocking accumulation registers are AUDIO_D+9 bits wide, and the 2-bit DC_BLOCK_SEL input is used to select 1-of-4 truncation bit sizes (no-feedback, 9-bit, 10-bit and 11-bit truncation), for selectable AUDIO_D-2-bit-width feedback data 0x0: DISABLE (no DC blocking) 0x1: TRUNC_9 (truncate 9 bit from feedback) 0x2: TRUNC_10 (truncate 10 bit from feedback) 0x3: TRUNC_11 (truncate 11 bit from feedback)
2	PRE_EMPH_EN	This is a debugging mode register only. For normal operation should be set to 1 at all times. 0x1: ENABLE (ENABLE (Enables pre-emphasis)) 0x0: DISABLE (DISABLE (Disables pre-emphasis))
1	STEREO	0x1: STEREO (STEREO (Enables Tx Stereo Mode)) 0x0: MONO (MONO (The TX_STEREO register is to enable the stereo mode. Incase of disable the stereo mode mono modulation should be increased. Mono mode modulation mono should be increased.))
0	TX_EN	Active high 0x1: ENABLE (ENABLE (Enables TX. For proper operation, this register should be set as the last programmed register for TX.)) 0x0: DISABLE (DISABLE (Disables TX and gates off all clocks, except for Tx Registers.))

0x0A180004 WCSS_A_FM_TX_MODULATION**Type:** RW**Clock:** APB**Reset State:** 0x00000000

The TX_MOD register is to set the modulation percentage for Mono, Stereo, RDS, and Pilot.

WCSS_A_FM_TX_MODULATION

Bits	Name	Description
31:24	MOD_RDS	Modulation multiplier for RDS, 0x80=9.5% Value = (% modulation/9.5%) * 128
23:16	MOD_PILOT	Modulation multiplier for 19 kHz pilot, 0x80=9.5% Value = (% modulation/9.5%) * 128

WCSS_A_FM_TX_MODULATION (cont.)

Bits	Name	Description
15:8	MOD_STEREO	Modulation multiplier for L-R, 0x80=42.75% Values = (% modulation/42.75%) * 128
7:0	MOD_MONO	Modulation multiplier for L+R, 0x80=42.75% Values = (% modulation/42.75%) * 128 the modulation percentage for Mono

0x0A180008 WCSS_A_FM_TX_AUDIO_LIMIT**Type:** RW**Clock:** APB**Reset State:** 0x00000000**WCSS_A_FM_TX_AUDIO_LIMIT**

Bits	Name	Description
27:14	AUDIO_LIMIT_2	The TX_AUDIO_LIMIT_2 field is used for controlling Audio limiting (saturation) level. This register should be set to 0x1B58 = 7000 for normal operation. This register is used to saturate the audio output of the 4x up-sampler (48 kHz to 192 kHz up-sampler)
13:0	AUDIO_LIMIT_1	The TX_AUDIO_LIMIT_1 fields used for controlling Audio limiting (saturation) level. This register should be set to 0x1B58 = 7000 for normal operation. This register is used to saturate the audio input to the 4x up-sampler (48 kHz to 192 kHz up-sampler). Value = % Modulation * (214)

0x0A18000C WCSS_A_FM_TX_RDS_CONTROL**Type:** RW**Clock:** APB**Reset State:** 0x00000000

The TX_RDS_EN register is used to enable and control the RDS functionalists.

WCSS_A_FM_TX_RDS_CONTROL

Bits	Name	Description
1	INV_RDS_SYMBOL	0x1: INVERT (INVERT (Invert RDS digital waveform data)) 0x0: NORMAL (NORMAL (Normal RDS digital waveform)) The TX_INV_RDS_SYMBOL register is used to invert RDS digital waveform. This is a debugging mode only register
0	RDS_EN	0x1: ENABLE (ENABLE (Enables RDS)) 0x0: DISABLE (DISABLE (Disables RDS))

0x0A180010 WCSS_A_FM_TX_RDS_BLK_DATA_A_B**Type:** RW**Clock:** APB**Reset State:** 0x00000000

The TX_RDS_BLK_DATA_A register is used to hold 16-bit RDS block A.

The TX_RDS_BLK_DATA_B register is used to hold 16-bit RDS block B

WCSS_A_FM_TX_RDS_BLK_DATA_A_B

Bits	Name	Description
31:16	RDS_BLK_DATA_A	RDS block data A
15:0	RDS_BLK_DATA_B	RDS block data B

0x0A180014 WCSS_A_FM_TX_RDS_BLK_DATA_C_D**Type:** RW**Clock:** APB**Reset State:** 0x00000000

The TX_RDS_BLK_DATA_C register is used to hold 16-bit RDS block C.

The TX_RDS_BLK_DATA_D register is used to hold 16-bit RDS block D.

WCSS_A_FM_TX_RDS_BLK_DATA_C_D

Bits	Name	Description
31:16	RDS_BLK_DATA_C	RDS block data C
15:0	RDS_BLK_DATA_D	RDS block data D

0x0A180018 WCSS_A_FM_TX_SOFT_RESET**Type:** RW**Clock:** APB**Reset State:** 0x00000000

The TX_SOFT_RESET register controls resetting the Tx data path under SW control.

WCSS_A_FM_TX_SOFT_RESET

Bits	Name	Description
0	SOFT_RESET	0x0: OPERATION (OPERATION) 0x1: RESET (SW can write this field to 1 to reset the entire Tx Data path. TX Registers are not affected. SW must, however, write a 0 back to this register to bring the Tx Data path out of reset.)

0x0A18001C WCSS_A_FM_TX_TMUX_SEL**Type:** RW**Clock:** APB**Reset State:** 0x00000000

The TX_TMUX_SEL register is to control the Tx testmux to bring out internal signals on core-level pins for debug.

WCSS_A_FM_TX_TMUX_SEL

Bits	Name	Description
4:0	TMUX_CNTL	Controls the Tx testmux to bring out internal signals on core-level pins for debug. Refer 2.1 Tx Testmux for a table of signals selected on different select values.

0x0A190000 WCSS_A_FM_RX_CONTROL**Type:** RW**Clock:** APB**Reset State:** 0x00000000

The RX_CNTRL register is control main functions in Riva-FM RX.

WCSS_A_FM_RX_CONTROL

Bits	Name	Description
0	RX_EN	0x0: DISABLE (DISABLE) 0x1: ENABLE (When 1, the RX modem is enabled. Clocks are enabled, and the logic is enabled for the RX, except for RX_RDS. There register RX_RDS_EN is used to enable RX_RDS separately.)

0x0A190004 WCSS_A_FM_RX_SOFT_RESET**Type:** RW**Clock:** APB**Reset State:** 0x00000000

The RX_SOFT_RESET register controls resetting the RX data path under SW control.

WCSS_A_FM_RX_SOFT_RESET

Bits	Name	Description
0	SOFT_RESET	0x0: OPERATION (OPERATION) 0x1: RESET (SW can write this field to 1 to reset the entire RX Data path including RDS. RX Registers are not affected. SW must, however, write a 0 back to this register to bring FM_RX out of reset.)

0x0A190024 WCSS_A_FM_RX_TMUX_SEL**Type:** RW**Clock:** APB**Reset State:** 0x00000000

The RX_TMUX_SEL register selects among the different internal signals that can be brought out on the testmux output for debug.

WCSS_A_FM_RX_TMUX_SEL

Bits	Name	Description
11:6	TMUX_CNTL_1	select 16 bit debug for bit[31:16] of RX debug mux. Selects combinations of signals to make visible for debug. Refer 2.2 RX Testmux for a table of signals selected on different select values.
5:0	TMUX_CNTL_0	select 16 bit debug for bit [15:0] of RX debug mux. Selects combinations of signals to make visible for debug. RX Testmux for a table of signals selected on different select values.

0x0A190028 WCSS_A_FM_RX_DEMOD_GLITCH_THRESHOLD**Type:** RW**Clock:** APB**Reset State:** 0x00000000

The RX_DEMOD_GLITCH_THRESHOLD register is used to control the value to compare against by FM DEMOD phase to determine if a phase glitch was detected.

WCSS_A_FM_RX_DEMOD_GLITCH_THRESHOLD

Bits	Name	Description
14:0	DEMOD_GLITCH_THRESH_OLD	The values computed for FM DEMOD Phase are compared against this value. If the FM DEMOD Phase exceeds this value, a phase glitch is detected

0x0A19002C WCSS_A_FM_RX_GLITCH_STOP_search_CNT**Type:** RW**Clock:** APB**Reset State:** 0x00000000

The RX_GLITCH_STOP_search_CNT register is used to control the end of glitch search window in 240 kHz clock counts.

WCSS_A_FM_RX_GLITCH_STOP_search_CNT

Bits	Name	Description
6:0	STOP_SEARCH_CNT	End of glitch search window in 240 kHz clock count.

0x0A190030 WCSS_A_FM_RX_GLITCH_START_search_CNT**Type:** RW**Clock:** APB**Reset State:** 0x00000000

The RX_GLITCH_START_search_CNT register is used to control the start of glitch search window in 240 kHz clock counts.

WCSS_A_FM_RX_GLITCH_START_search_CNT

Bits	Name	Description
6:0	GLITCH_START_SEARCH_CNT	Start of glitch search window in 240 kHz clock count.

0x0A190034 WCSS_A_FM_RX_GLITCH_BLANK_CNT_TH**Type:** RW**Clock:** APB**Reset State:** 0x00000000

The RX_GLITCH_BLANK_CNT_TH register is used to control the threshold used to limit the duration of phase compensation.

WCSS_A_FM_RX_GLITCH_BLANK_CNT_TH

Bits	Name	Description
3:0	BLANK_CNT_TH	Threshold used to limit the duration of phase compensation. This register should be set to any value larger than 1. setting it to 1 is equivalent to set it to 0.

0x0A190038 WCSS_A_FM_RX_PLL_BANK_CHG_STB**Type:** RW**Clock:** APB**Reset State:** 0x00000000

The RX_PLL_BANK_CHG_STB register is used to control the change in PLL bank.

WCSS_A_FM_RX_PLL_BANK_CHG_STB

Bits	Name	Description
0	PLL_BANK_CHG_STB	0x0: NOT_START (NOT_START) 0x1: START (When 1, indicates that the radio has performed a bank change. SW must first write a 1, followed by writing a 0 to produce a pulse to HW. This register is used by FM DEMOD for Phase Glitch Detection.)

0x0A19003C WCSS_A_FM_RX_PHASE_COMP_BYPASS**Type:** RW**Clock:** APB**Reset State:** 0x00000000

The RX_PHASE_COMP_BYPASS register is used to bypass the FM DEMOD Phase Compensation.

WCSS_A_FM_RX_PHASE_COMP_BYPASS

Bits	Name	Description
0	PHASE_COMP_BYPASS	0x0: ENABLE (ENABLE) 0x1: BYPASS (When set to 1, the FM DEMOD would bypass phase compensation operation.)

0x0A190040 WCSS_A_FM_RX_MPX_DCC_CTRL**Type:** RW**Clock:** APB**Reset State:** 0x00000000

to control the functionality of MPX DCC

The RX_MPX_DCC_BYPASS register is used to bypass the MPX_DCC module.

The RX_MPX_DCC_SEL register is used to control the Symmetric Rounding of second Symmetric Saturation in the MPX_DCC data path.

WCSS_A_FM_RX_MPX_DCC_CTRL

Bits	Name	Description
2:1	MPX_DCC_SEL	Selects how many bits to Symmetric Round the result of second Symmetric Saturation in the MPX_DCC data path. This register directly controls the M parameter passed to SROUND function used in the design. 0x0: ROUND_3 (Round by 3-bits) 0x1: ROUND_4 (Round by 4-bits) 0x2: ROUND_5 (Round by 5-bits)
0	MPX_DCC_BYPASS	0x0: ENABLE (ENABLE) 0x1: BYPASS (When 1, the MPX_DCC module is bypassed)

0x0A190044 WCSS_A_FM_RX_MPX_DCC_Z_DELAY_REGISTER**Type:** RW**Clock:** APB**Reset State:** 0x00000000

The RX_MPX_DCC_Z_M1 field is used to hold the value of the 20-bit delay element in MPX_DCC.

WCSS_A_FM_RX_MPX_DCC_Z_DELAY_REGISTER

Bits	Name	Description
19:0	MPX_DCC_Z_M1	There is a 20 bit delay element that is part of the normal data path, so it will be updated by the HW. Also, on SW reset, the hooks are provided for the SW to update this register

0x0A190048 WCSS_A_FM_RX_PILOT_DET_THRESHOLD**Type:** RW**Clock:** APB**Reset State:** 0x00000000

The RX_PDET_HI_THRESHOLD field is used to control the pilot amplitude threshold for turning pilot detection flag on.

The RX_PDET_LO_THRESHOLD field is used to control the pilot amplitude threshold for turning pilot detection flag off.

WCSS_A_FM_RX_PILOT_DET_THRESHOLD

Bits	Name	Description
19:10	PDET_LO_THRESHOLD	Pilot amplitude threshold for turning pilot detection flag from on to off.
9:0	PDET_HI_THRESHOLD	Pilot amplitude threshold for turning pilot detection flag from off to on. PDET_HI_THRESHOLD should be higher in value than PDET_LO_THRESHOLD. Setting this register to lower than PDET_LO_THRESHOLD could cause unstable Pilot detection for stereo channels.

0x0A19004C WCSS_A_FM_RX_DELTA_R57K_PHASE_ADJUST**Type:** RW**Clock:** APB**Reset State:** 0x00000000

This register is used to control the phase adjustment for the R57I and R57Q subcarrier to account for RDS BPF group delay.

WCSS_A_FM_RX_DELTA_R57K_PHASE_ADJUST

Bits	Name	Description
12:0	DELTA_IDX3	Phase adjustment for the R57I and R57Q sub carrier to account for RDS BPF group delay.

0x0A190050 WCSS_A_FM_RX_BLEND_WEIGHT**Type:** RW**Clock:** APB**Reset State:** 0x00000000

The RX_BLEND_WEIGHT register is used to control the blending weight.

WCSS_A_FM_RX_BLEND_WEIGHT

Bits	Name	Description
9:0	BLEND_WEIGHT	Blending weight determined by the audio control unit in SW. should have max vlaue of 0x280 for best stable operation. any higher value than 0x280 could cause audio saturation issues may lead to audio quality

0x0A190054 WCSS_A_FM_RX_SOFTMUTE_WEIGHT**Type:** RW**Clock:** APB**Reset State:** 0x00000000

The RX_SOFTMUTE_WEIGHT register is used to control the soft mute weight.

WCSS_A_FM_RX_SOFTMUTE_WEIGHT

Bits	Name	Description
8:0	SOFTMUTE_WEIGHT	Soft mute weight determined by the audio control unit in SW

0x0A190058 WCSS_A_FM_RX_BLEND_SOFTMUTE_LPF_M_SEL**Type:** RW**Clock:** APB**Reset State:** 0x00000000

The RX_BLEND_SOFTMUTE_LPF_M_SEL register is used to select the blending weight LPF and soft mute LPF.

WCSS_A_FM_RX_BLEND_SOFTMUTE_LPF_M_SEL

Bits	Name	Description
5:4	SF_WGHT_M	Soft mute LPF M select
1:0	BLEND_WGHT_M	Blending weight LPF M select

0x0A19005C WCSS_A_FM_RX_MS_DECODER_CNTL**Type:** RW**Clock:** APB**Reset State:** 0x00000000

The RX_MS_DECODER_CNTL register is used to control various aspects of MS_DECODER.

WCSS_A_FM_RX_MS_DECODER_CNTL

Bits	Name	Description
5	R_HARD_MUTE	0x0: NO_MUTE (NO_MUTE 10b: MUTE when set to 1 by SW, turn on hard mute for the right channel)
4	L_HARD_MUTE	0x0: NO_MUTE (NO_MUTE) 0x1: MUTE (When set to 1 by SW, turn on hard mute for the left channel.)
1	DE_EMPS_TC	De-emphasis filter time constant select. 0x0: TC_50US (TC_50US) 0x1: TC_75US (TC_75US)
0	MONO_DEC	0x0: NORMAL (NORMAL) 0x1: FORCE_MONO (FORCE_MONO when set to 1, force mono decode.)

0x0A190060 WCSS_A_FM_RX_MS_DECODER_STEREO_ON**Type:** R**Clock:** APB**Reset State:** 0x00000000**WCSS_A_FM_RX_MS_DECODER_STEREO_ON**

Bits	Name	Description
0	STEREO_ON	a read-only bit for SW (output from mono-stereo decoder)

0x0A190064 WCSS_A_FM_RX_RDS_CTRL**Type:** RW**Clock:** APB**Reset State:** 0x00000000

The RX_RDS_EN register is used to enable the RX_RDS block, IQ Analyzer and also to soft reset the RX_RDS block.

WCSS_A_FM_RX_RDS_CTRL

Bits	Name	Description
2	RDS_SOFT_RESET	0x0: OPERATION (OPERATION) 0x1: RESET (RESET SW can write this field to 1 to reset the entire RX RDS. SW must, however, write a 0 back to this field to bring RX_RDS out of reset.)
1	IQ_AN_EN	0x0: DISABLE (DISABLE) 0x1: ENABLE (ENABLE When 1, enables IQ signal analyzer.)

WCSS_A_FM_RX_RDS_CTRL (cont.)

Bits	Name	Description
0	RDS_EN	0x0: DISABLE (DISABLE) 0x1: ENABLE (ENABLE When 1, enables RDS reception.)

0x0A190068 WCSS_A_FM_RX_RDS_IQ_DET_CTRL**Type:** RW**Clock:** APB**Reset State:** 0x00000000

The RX_RDS_IQ_INT_DUR register is used to control the IQ signal detector integration duration in bit intervals.

The RX_RDS_IQ_DET_THRESH register is used to control the IQ signal detector threshold for qualifying SIG_DET and SIG_NOT_DET conditions

WCSS_A_FM_RX_RDS_IQ_DET_CTRL

Bits	Name	Description
21:4	IQ_DET_THRESH	IQ_DET_THRESH
3:0	IQ_INT_DUR	IQ signal detector integration duration in bit intervals. The value specified should be one less than the number of bit intervals desired.

0x0A19006C WCSS_A_FM_RX_RDS_ACQ_MODE**Type:** RW**Clock:** APB**Reset State:** 0x00000000

The RX_RDS_ACQ_MODE register is used to specify the averaging duration for acquisition of RDS signal.

WCSS_A_FM_RX_RDS_ACQ_MODE

Bits	Name	Description
1:0	ACQ_MODE	Identifies Bit Synchronizer averaging duration for acquisition of RDS signal. 0x0: AVG_16 (4, 8, 16-bit averaging stepped acquisition.) 0x1: AVG_32 (8, 16, 32-bit averaging stepped acquisition.) 0x2: AVG_64 (16, 32, 64-bit averaging stepped acquisition)

0x0A190070 WCSS_A_FM_RX_RDS_TRK_MODE**Type:** RW**Clock:** APB**Reset State:** 0x00000000

The RX_RDS_TRK_MODE register is used to specify averaging duration for RDS signal tracking.

WCSS_A_FM_RX_RDS_TRK_MODE

Bits	Name	Description
1:0	TRK_MODE	Identifies Bit Synchronizer averaging duration for tracking of RDS signal. 0x0: AVG_32 (32-bit averaging tracking, valid only with ACQ_MODE = 0) 0x1: AVG_64 (64-bit averaging tracking, valid only with ACQ_MODE = 0 or 1) 0x2: AVG_128 (128-bit averaging tracking)

0x0A190074 WCSS_A_FM_RX_RDS_STROBE_MARGIN**Type:** RW**Clock:** APB**Reset State:** 0x00000000

The RX_RDS_STROBE_MARGIN register is used to specify the number of consecutive negative values that are required prior to a positive value when detecting the negative-to-positive transition of R19 for generating the C19 signal.

WCSS_A_FM_RX_RDS_STROBE_MARGIN

Bits	Name	Description
1:0	STROBE_MARGIN	Specifies the number of consecutive negative values that are required prior to a positive value when detecting the negative-to-positive transition of R19 for generating the C19 signal (19 kHz strobe). 0d0: NEGATIVE_1 (1 negative value) 0d1: NEGATIVE_2 (2 negative values) 0d2: NEGATIVE_3 (3 negative values) 0d3: NEGATIVE_4 (4 negative values)

0x0A190078 WCSS_A_FM_RX_RDS_MF_LPF_CNT**Type:** RW**Clock:** APB**Reset State:** 0x00000000

The RX_RDS_MF_LPF_CNT register is used to control the delay count to allow matched filter and low pass filter to settle.

WCSS_A_FM_RX_RDS_MF_LPF_CNT

Bits	Name	Description
7:4	MF_CNT	Delay count to allow matched filter to settle.
3:0	LPF_CNT	Delay count for allowing RDS BPF, down converter and LPF to settle. Only values 1 to 15 are valid.

0x0A19007C WCSS_A_FM_RX_RDS_IQ_AN_INT_DUR_SEL**Type:** RW**Clock:** APB**Reset State:** 0x00000000

The RX_RDS_IQ_AN_INT_DUR_SEL register is used to select the IQ signal analyzer integration duration.

WCSS_A_FM_RX_RDS_IQ_AN_INT_DUR_SEL

Bits	Name	Description
2:0	IQ_AN_INT_DUR_SEL	Selects IQ signal analyzer integration duration.

0x0A190080 WCSS_A_FM_RX_RDS_IQ_ANALYZER_SELECT**Type:** RW**Clock:** APB**Reset State:** 0x00000000

1-bit register to enable IQ analyzer. Both IQ detector and analyzer are running regardless of the setting of this bit. This register is to select which block controls the RDS PHY state machine.

This register controls which block to do the following

- Select the IQ arm
- Start the match filters

Both blocks can generate interrupts. IQ analyzer generates energy level for both channels regardless of the setting of this bit.

WCSS_A_FM_RX_RDS_IQ_ANALYZER_SELECT

Bits	Name	Description
1	IQ_ANA_SIG_DET_SEL	0x0: SELECT_SIG_DET_FROM_IQ_DETECTOR (select sig det from IQ detector) 0x1: SELECT_SIG_DET_FROM_IQ_ANALYZER (select sig det from IQ analyzer - this is legacy register should set as bit 0)

0x0A190084 WCSS_A_FM_RX_RDS_IQ_AN_DET_THRESH**Type:** RW**Clock:** APB**Reset State:** 0x00000000

The RX_RDS_IQ_AN_DET_THRESH register is used to control the IQ signal analyzer threshold for qualifying IQ_AN_SIG_DET condition.

WCSS_A_FM_RX_RDS_IQ_AN_DET_THRESH

Bits	Name	Description
13:0	IQ_AN_DET_THRESH	LSBs of IQ signal analyzer threshold for qualifying IQ_AN_SIG_DET condition.

0x0A190088 WCSS_A_FM_RX_RDS_RECEIVE_MODE_SELECT**Type:** RW**Clock:** APB**Reset State:** 0x00000000

This register is to set the RDS receive mode. Two modes are supported by FM65 2.0

- Coherent mode, 19K pilot and 57K RDS tone are locked
- Non Coherent mode, 19K pilot and 57K RDS tone are not locked, also this mode is used in case of missing 19k pilot or mono mode with RDS.

WCSS_A_FM_RX_RDS_RECEIVE_MODE_SELECT

Bits	Name	Description
0	RDS_MODE_SELECT	0x0: COHERENT (COHERENT (Coherent mode legacy mode)) 0x1: NON_COHERENT (NON_COHERENT (Non Coherent))

0x0A19008C WCSS_A_FM_RX_RDS_STATUS**Type:** R**Clock:** APB**Reset State:** 0x00000000

The RX_RDS_STATUS register is used to indicate to SW several status bits.

WCSS_A_FM_RX_RDS_STATUS

Bits	Name	Description
2	IQ_AN_SIG_DET	0x0: NOT_DETECTED (NOT_DETECTED) 0x1: DETECTED (DETECTED Indicates (when 1) that the IQ analyzer detected a signal that meets or exceeds IQ_AN_DET_THRESH value.)
1	IQ_AN_SELECT	0x0: I_ARM (I_ARM) 0x1: Q_ARM (Q_ARM Indicates whether the IQ signal analyzer selected I (when 0) or Q (when 1) arm of the RDS signal.)
0	IQ_DET_SELECT	0x0: I_ARM (I_ARM) 0x1: Q_ARM (Q_ARM Indicates whether the IQ detector selected I (when 0) or Q (when 1) arm of the RDS signal.)

0x0A190090 WCSS_A_FM_RX_RDS_IQ_BUFFER**Type:** R**Clock:** APB**Reset State:** 0x00000000

The RX_RDS_I_BUFFER field is used to indicate to SW the I arm signal level measurement results from IQ Signal Analyzer.

The RX_RDS_Q_BUFFER field is used to indicate to SW the Q arm signal level measurement results from IQ Signal Analyzer.

WCSS_A_FM_RX_RDS_IQ_BUFFER

Bits	Name	Description
27:14	RX_RDS_Q_BUFFER	RDS Q arm signal level measurement results from I/Q Signal Analyzer.
13:0	RX_RDS_I_BUFFER	LSBs of RDS I arm signal level measurement results from I/Q Signal Analyzer

0x0A190094 WCSS_A_FM_RX_RDS_TIME_TRACKING_THRESHOLD**Type:** RW**Clock:** APB**Reset State:** 0x00000000**WCSS_A_FM_RX_RDS_TIME_TRACKING_THRESHOLD**

Bits	Name	Description
9:5	DROP_THRESHOLD	Peak index difference threshold to drop bit. This is 5 bit signed value. This register is used only in non-coherent mode. The register is used to set the peak index difference threshold to drop bits. If the current peak index minus previous peak index is less than RX_RDS_DROP_BIT_THRESHOLD value the current received bit will be dropped
4:0	REPEAT_THRESHOLD	Peak index difference threshold to add bit. This is 5 bit signed value. This register is used only in non-coherent mode. The register is used to set the peak index difference threshold to add bits. So if the current peak index minus previous index is greater than RX_RDS_REPEAT_BIT_THRESHOLD value the current received bit will be repeated.

0x0A190098 WCSS_A_FM_RX_RDS_PEAK_INDEX**Type:** R**Clock:** APB**Reset State:** 0x00000000

16 bit RDS peak index output from the bit synchronizer. In case of detection of repeat/drop demod bit inside the rds physical layer this register will reflect the peak indexes that cause the repeat/drop to happen. This register stores the last 4 peak indexes that cause repeat/drop to happen. This register could be considered as 4 deep FIFO for the peak index but only store values at the event of adding or dropping bits

WCSS_A_FM_RX_RDS_PEAK_INDEX

Bits	Name	Description
15:0	RDS_PEAK_INDEX	bit 15 - 12 PEAK_INDEX_3 same as PEAK_INDEX_1 bit 11 - 8 PEAK_INDEX_2 same as PEAK_INDEX_0 bit 7 - 4 PEAK_INDEX_1 bit 3 - 0 PEAK_INDEX_0

0x0A19009C WCSS_A_FM_RX_RDS_DL_CONFIG**Type:** RW**Clock:** APB**Reset State:** 0x00000000

The RX_RDS_DL_CONFIG register is used to configure SYSTEM_TYPE, C_CP_TYPE and C_CP_ADAPT for RDS Data Link.

WCSS_A_FM_RX_RDS_DL_CONFIG

Bits	Name	Description
2	C_CP_ADAPT	Identifies whether the DL receiver will adaptive identify Type A or Type B reception (C or Cp Type offsets for third block of groups). 0x0: NO_ADAPT (NO_ADAPT) 0x1: ADAPT (ADAPT)
1	C_CP_TYPE	Identifies Type A or Type B RDS reception to be used by the DL receiver. 0x0: A_TYPE (A_TYPE (Type A, C offset for third block of group)) 0x1: B_TYPE (B_TYPE (Type B, C offset for third block of group))
0	SYSTEM_TYPE	0x0: RBDS (RBDS) 0x1: RDS (RDS Configures receiver for RDS (if set to 1) or RBDS (if set to 0) reception.)

0x0A1900A0 WCSS_A_FM_RX_RDS_T1Q1E**Type:** RW**Clock:** APB**Reset State:** 0x00000000

The RX_RDS_T1Q1E register is used to control the exponential representation of the threshold value for loss of synchronization in acquisition mode.

WCSS_A_FM_RX_RDS_T1Q1E

Bits	Name	Description
3:0	T1Q1E	Exponential representation of the threshold value for loss of synchronization in acquisition mode.

0x0A1900A4 WCSS_A_FM_RX_RDS_T1Q2E**Type:** RW**Clock:** APB**Reset State:** 0x00000000

The RX_RDS_T1Q2E register is used to control the exponential representation of the threshold value for loss of synchronization in tracking mode.

WCSS_A_FM_RX_RDS_T1Q2E

Bits	Name	Description
3:0	T1Q2E	Exponential representation of the threshold value for loss of synchronization in tracking mode.

0x0A1900A8 WCSS_A_FM_RX_RDS_BLOCK_DATA_0_1

Type: R

Clock: APB

Reset State: 0x00000000

The RX_RDS_BLOCK_DATA_0 field is used to store the received RDS BLOCK DATA 0.

The RX_RDS_BLOCK_DATA_1 fields used to store the received RDS BLOCK DATA 1

WCSS_A_FM_RX_RDS_BLOCK_DATA_0_1

Bits	Name	Description
31:16	RDS_BLOCK_DATA_0	RDS receive block data 0
15:0	RDS_BLOCK_DATA_1	RDS receive block data 1

0x0A1900AC WCSS_A_FM_RX_RDS_BLOCK_DATA_2_3

Type: R

Clock: APB

Reset State: 0x00000000

The RX_RDS_BLOCK_DATA_2 field is used to store the received RDS BLOCK DATA 2.

The RX_RDS_BLOCK_DATA_3 field is used to store the received RDS BLOCK DATA 3

WCSS_A_FM_RX_RDS_BLOCK_DATA_2_3

Bits	Name	Description
31:16	RDS_BLOCK_DATA_2	RDS receive block data 2,
15:0	RDS_BLOCK_DATA_3	RDS receive block data 3

0x0A1900B0 WCSS_A_FM_RX_RDS_BLOCK_STATUS

Type: R
Clock: APB
Reset State: 0x00000000

The RX_RDS_BLOCK_STATUS_0 field is used to store the status associated with RDS BLOCK DATA 0.

The RX_RDS_BLOCK_STATUS_1 field is used to store the status associated with RDS BLOCK DATA 1.

The RX_RDS_BLOCK_STATUS_2 field is used to store the status associated with RDS BLOCK DATA 2.

The RX_RDS_BLOCK_STATUS_3 field is used to store the status associated with RDS BLOCK DATA 3.

WCSS_A_FM_RX_RDS_BLOCK_STATUS

Bits	Name	Description
30:24	RDS_BLOCK_STATUS_0	RDS receive data block 0 status
22:16	RDS_BLOCK_STATUS_1	RDS receive data block 1 status
14:8	RDS_BLOCK_STATUS_2	RDS receive data block 2 status
6:0	RDS_BLOCK_STATUS_3	RDS receive data block 3 status

0x0A1900B4 WCSS_A_FM_RX_RDS_BLOCK_DATA_RAW_0

Type: R
Clock: APB
Reset State: 0x00000000

The RX_RDS_BLOCK_DATA_RAW_0 register is used to store the raw received RDS BLOCK DATA 0 and BLOCK OFFSET 0 before any correction is done.

WCSS_A_FM_RX_RDS_BLOCK_DATA_RAW_0

Bits	Name	Description
25:16	RDS_BLOCK_OFFSET_0	RDS receive raw block offset 0,
15:0	RDS_BLOCK_DATA_RAW_0	RDS receive raw block data 0,

0x0A1900B8 WCSS_A_FM_RX_RDS_BLOCK_DATA_RAW_1

Type: R
Clock: APB
Reset State: 0x00000000

The RX_RDS_BLOCK_DATA_RAW_1 register is used to store the raw received RDS BLOCK DATA 1 and BLOCK OFFSET 1 before any correction is done.

WCSS_A_FM_RX_RDS_BLOCK_DATA_RAW_1

Bits	Name	Description
25:16	RDS_BLOCK_OFFSET_1	RDS receive raw block offset 1,
15:0	RDS_BLOCK_DATA_RAW_1	RDS receive raw block data 1,

0x0A1900BC WCSS_A_FM_RX_RDS_BLOCK_DATA_RAW_2

Type: R
Clock: APB
Reset State: 0x00000000

The RX_RDS_BLOCK_DATA_RAW_2 register is used to store the raw received RDS BLOCK DATA 2 and BLOCK OFFSET 2 before any correction is done.

WCSS_A_FM_RX_RDS_BLOCK_DATA_RAW_2

Bits	Name	Description
25:16	RDS_BLOCK_OFFSET_2	RDS receive raw block offset 2,
15:0	RDS_BLOCK_DATA_RAW_2	RDS receive raw block data 2,

0x0A1900C0 WCSS_A_FM_RX_RDS_BLOCK_DATA_RAW_3

Type: R
Clock: APB
Reset State: 0x00000000

The RX_RDS_BLOCK_DATA_RAW_3 register is used to store the raw received RDS BLOCK DATA 3 and BLOCK OFFSET 3 before any correction is done.

WCSS_A_FM_RX_RDS_BLOCK_DATA_RAW_3

Bits	Name	Description
25:16	RDS_BLOCK_OFFSET_3	RDS receive raw block offset 3,
15:0	RDS_BLOCK_DATA_RAW_3	RDS receive raw block data 3,

0x0A1900C4 WCSS_A_FM_RX_RDS_TMUX_SEL**Type:** RW**Clock:** APB**Reset State:** 0x00000000

The RX_RDS_TMUX_CNTL register is used to select one of several debug selections available for bringing out internal signals on testmux for lab debug.

WCSS_A_FM_RX_RDS_TMUX_SEL

Bits	Name	Description
18:0	RESERVED	reserved

0x0A1900C8 WCSS_A_FM_RX_BLENDING_READBACK**Type:** R**Clock:** APB**Reset State:** 0x00000000

10-bit read only debug register for blending value. This register should read back the same value as RX_BLEND_WEIGHT after settling time.

WCSS_A_FM_RX_BLENDING_READBACK

Bits	Name	Description
9:0	BLEND_WEIGHT_RB	Blending

0x0A1900CC WCSS_A_FM_RX_MUTE_READBACK**Type:** R**Clock:** APB**Reset State:** 0x00000000

9 bit debugging read back register for reading back the soft mute read. This register should read back the same value as RX_SOFTMUTE_WEIGHT after conversion time.

WCSS_A_FM_RX_MUTE_READBACK

Bits	Name	Description
8:0	SOFT_MUTE_WEIGHT_RB	For RX_SOFTMUTE_WEIGHT

0x0A1900D0 WCSS_A_FM_RX_PLL_FREERUN_EN**Type:** RW**Clock:** APB**Reset State:** 0x00000000

1-bit to enable PLL free run mode. In this mode the PLL generates 19k tone regardless of the existence of pilot tone from the received channel.

WCSS_A_FM_RX_PLL_FREERUN_EN

Bits	Name	Description
0	FREE_RUN_ENABLE	0x0: DISABLE (DISABLE (disable free run)) 0x1: ENABLE (ENABLE (enable free run, should only be used with Mono RX stations))

0x0A1900D4 WCSS_A_FM_RX_ROT_CTRL_0**Type:** RW**Clock:** APB**Reset State:** 0x00000000

This register is to control the spur rotator and removal block.

The idea behind the spur removal block is to rotate the signal such that the spur will move to DC. The Then you remove a DC value representing the level of the spur. There are two options for that DC value. The first is to use a DC tracking filter to estimate it. The second is to use a predetermined value in spur_comp_i/q registers.

To use this register, the ROT3_SEL bit must be managed properly. The first write to this register must set ROT3_SEL=1, to select new rotator and to activate the fields defined in this register. Subsequent writes must be used to set the other fields in this register.

For legacy mode, set ROT3_SEL=0 and the register fields will be as defined as in legacy register RX_ROT_CTRL, where the field definitions are different than in RX_ROT_CTRL_0.

WCSS_A_FM_RX_ROT_CTRL_0

Bits	Name	Description
31	ROT3_SEL	Select for legacy(LUT-based) or new(CORDIC-based) rotator. 0x0: LEGACY (One Spur Removal block with a LUT based rotator and 1kHz resolution) 0x1: CORDIC (Up to 3 Spur Removal blocks with CORDIC based Rotators and 0.228 Hz resolution in 240kHz sampling rate)

WCSS_A_FM_RX_ROT_CTRL_0 (cont.)

Bits	Name	Description
27	BYPASS_IN_G1G2G3	When Legacy_based Rotator is selected, this field is located at bit 15. 0x0: NORMAL (NORMAL) 0x1: BYPASS_IN_G1G2G3 (BYPASS_IN_G1G2G3 (bypass in any non zero gain state))
26:7	ROT_FREQ	Rotation Frequency. When Legacy_based Rotator is selected, this field is located at bits [14:7] and has 1kHz resolution. When CORDIC_based Rotator is selected, the 20 bits [26:7] represents the spur frequency normalized by the sampling rate (240kHz). The resolution is 0.228 Hz.
6	FORCE_DC_TRACK	0x0: USE_DC (USE_DC (enable built in DC tracking logic in spur rotator)) 0x1: USE_SPUR (USE_SPUR (use spur_comp_I/Q registers value as dc level to be subtracted from signal after rotation))
5	SPUR_DC_DELAY	0x0: DELAY_2048 (DELAY_2048 (enable dc tracking circuit after 2048 240kHz cycles)) 0x1: DELAY_4096 (DELAY_4096 (enable dc tracking circuit after 4096 240kHz cycles))
4	SPUR_1K_ROT_ENABLE	0x0: DISABLE (DISABLE (disable and bypass spur rotator)) 0x1: ENABLE (ENABLE (enable spur rotator))
3:2	SPUR_DC_IIR_RND	Set the rounding position for DC tracking. 0x0: ROUND_3 (Recommended) 0x1: ROUND_4 0x2: ROUND_5 0x3: ROUND_6
1:0	SPUR_INT_SETTING	Set the DC tracking integration period. Number of cycles below in 240kHz period 0x0: INTEGRATE_256 (Recommended) 0x1: INTEGRATE_512 0x2: INTEGRATE_1024 0x3: INTEGRATE_2048

0x0A1900D8 WCSS_A_FM_SPUR_COMP_FIXED_VALUE_0**Type:** RW**Clock:** APB**Reset State:** 0x00000000

16 bit register for Spur comp value. The value programmed into this register will be subtracted from the I or Q arm after rotation. For details please refer to the HLD and system docs.

WCSS_A_FM_SPUR_COMP_FIXED_VALUE_0

Bits	Name	Description
31:16	SPUR_COMP_Q	for spur comp Q 16 bit register for Spur comp value. The value programmed into this register will be subtracted from the Q arm after rotation. For details please refer to the HLD and system docs
15:0	SPUR_COMP_I	or spur comp I

0x0A1900E0 WCSS_A_FM_DEMOD_DC_IF_OFFSET**Type:** RW**Clock:** APB**Reset State:** 0x00000000

This is 16bit register contains fm_demod 16 bit dc offset. This register should be calculated based on the rotator freq. This register is used by fm_demod value to subtract FM DC offset after demodulation

$$\text{DEMOD_DC_IF_OFFSET} = (429 * (-\text{rot_freq} + 100)) / 4$$

WCSS_A_FM_DEMOD_DC_IF_OFFSET

Bits	Name	Description
15:0	DEMOD_DC_IF_OFFSET	DEMOD_DC_IF_OFFSET Note this register should be set according to ROT_FREQ setting. The max rotation freq supported by the design is 200KHz.

0x0A1900E4 WCSS_A_FM_SPUR_COMP_DC_VALUE_RD_BACK_0**Type:** R**Clock:** APB**Reset State:** 0x00000000

2x12 bit register for spur DC estimation read back value. this value could be used to estimate spur power.

WCSS_A_FM_SPUR_COMP_DC_VALUE_RD_BACK_0

Bits	Name	Description
27:16	SPUR_DC_COMP_Q	dc value estimation for Q
11:0	SPUR_DC_COMP_I	dc value estimation for I

0x0A190118 WCSS_A_FM_SPUR_TRACK_INPUT_SEL**Type:** RW**Clock:** APB**Reset State:** 0x00000000

This selects which spur removal stage the spur tracking block takes input from.

WCSS_A_FM_SPUR_TRACK_INPUT_SEL

Bits	Name	Description
1:0	SEL	00: first spur removal; 01: second spur removal; 10: third spur removal

0x0A19011C WCSS_A_FM_SPUR_TRACK_CMD_STATUS**Type:** R**Clock:** APB**Reset State:** 0x00000001

Write to this address will trigger spur tracking operation, and clearing the status bit. When spur tracking operation is done, status bit will be set to 1.

WCSS_A_FM_SPUR_TRACK_CMD_STATUS

Bits	Name	Description
0	DONE	Spur tracking done

0x0A190120 WCSS_A_FM_RX_ROT_CTRL_1**Type:** RW**Clock:** APB**Reset State:** 0x00000000

This register is to control the second spur removal block.

The idea behind the spur removal block is to rotate the signal such that the spur will move to DC. Then you remove a DC value representing the level of the spur. There are two options for that DC value. The first is to use a DC tracking filter to estimate it. The second is to use a predetermined value in spur_comp_i/q registers.

The second spur removal block should only be enabled when first spur removal block is enabled.

WCSS_A_FM_RX_ROT_CTRL_1

Bits	Name	Description
27	BYPASS_IN_G1G2G3	0x0: NORMAL (NORMAL) 0x1: BYPASS_IN_G1G2G3 (BYPASS_IN_G1G2G3 (bypass in any non zero gain state))
26:7	ROT_FREQ	Rotation Frequency. Available only when CORDIC_based Rotator is selected. The 20 bits [26:7] represents the spur frequency normalized by the sampling rate (240kHz). The resolution is 0.228 Hz.
6	FORCE_DC_TRACK	0x0: USE_DC (USE_DC (enable built in DC tracking logic in spur rotator)) 0x1: USE_SPUR (USE_SPUR (use spur_comp_I/Q registers value as dc level to be subtracted from signal after rotation))
5	SPUR_DC_DELAY	0x0: DELAY_2048 (DELAY_2048 (enable dc tracking circuit after 2048 240kHz cycles)) 0x1: DELAY_4096 (DELAY_4096 (enable dc tracking circuit after 4096 240kHz cycles))
4	SPUR_1K_ROT_ENABLE	0x0: DISABLE (DISABLE (disable and bypass spur rotator)) 0x1: ENABLE (ENABLE (enable spur rotator))
3:2	SPUR_DC_IIR_RND	Set the rounding position for DC tracking. 0x0: ROUND_3 (Recommended) 0x1: ROUND_4 0x2: ROUND_5 0x3: ROUND_6
1:0	SPUR_INT_SETTING	Set the DC tracking integration period. Number of cycles below in 240kHz period. 0x0: INTEGRATE_256 (Recommended) 0x1: INTEGRATE_512 0x2: INTEGRATE_1024 0x3: INTEGRATE_2048

0x0A190124 WCSS_A_FM_SPUR_COMP_FIXED_VALUE_1**Type:** RW**Clock:** APB**Reset State:** 0x00000000

16 bit register for Spur comp value. The value programmed into this register will be subtracted from the I or Q arm after rotation. For details please refer to the HLD and system docs.

WCSS_A_FM_SPUR_COMP_FIXED_VALUE_1

Bits	Name	Description
27:16	SPUR_COMP_Q	for spur comp Q 16 bit register for Spur comp value. The value programmed into this register will be subtracted from the Q arm after rotation. For details please refer to the HLD and system docs
11:0	SPUR_COMP_I	or spur comp I

0x0A190128 WCSS_A_FM_SPUR_COMP_DC_VALUE_RD_BACK_1**Type:** R**Clock:** APB**Reset State:** 0x00000000

2x12 bit register for spur DC estimation read back value. this value could be used to estimate spur power.

WCSS_A_FM_SPUR_COMP_DC_VALUE_RD_BACK_1

Bits	Name	Description
27:16	SPUR_DC_COMP_Q	dc value estimation for Q
11:0	SPUR_DC_COMP_I	dc value estimation for I

0x0A19012C WCSS_A_FM_RX_ROT_CTRL_2**Type:** RW**Clock:** APB**Reset State:** 0x00000000

This register is to control the third spur removal block.

The idea behind the spur removal block is to rotate the signal such that the spur will move to DC. The Then you remove a DC value representing the level of the spur. There are two options for that DC value. The first is to use a DC tracking filter to estimate it. The second is to use a predetermined value in spur_comp_i/q registers.

The third spur removal block should only be enabled when the first 2 spur removal blocks are enabled.

WCSS_A_FM_RX_ROT_CTRL_2

Bits	Name	Description
27	BYPASS_IN_G1G2G3	0x0: NORMAL (NORMAL) 0x1: BYPASS_IN_G1G2G3 (BYPASS_IN_G1G2G3 (bypass in any non zero gain state))
26:7	ROT_FREQ	Rotation Frequency. Available only when CORDIC_based Rotator is selected. The 20 bits [26:7] represents the spur frequency normalized by the sampling rate (240kHz). The resolution is 0.228 Hz.
6	FORCE_DC_TRACK	0x0: USE_DC (USE_DC (enable built in DC tracking logic in spur rotator)) 0x1: USE_SPUR (USE_SPUR (use spur_comp_I/Q registers value as dc level to be subtracted from signal after rotation))
5	SPUR_DC_DELAY	0x0: DELAY_2048 (DELAY_2048 (enable dc tracking circuit after 2048 240kHz cycles)) 0x1: DELAY_4096 (DELAY_4096 (enable dc tracking circuit after 4096 240kHz cycles))
4	SPUR_1K_ROT_ENABLE	0x0: DISABLE (DISABLE (disable and bypass spur rotator)) 0x1: ENABLE (ENABLE (enable spur rotator))
3:2	SPUR_DC_IIR_RND	Set the rounding position for DC tracking. 0x0: ROUND_3 (Recommended) 0x1: ROUND_4 0x2: ROUND_5 0x3: ROUND_6
1:0	SPUR_INT_SETTING	Set the DC tracking integration period. Number of cycles below in 240kHz period. 0x0: INTEGRATE_256 (Recommended) 0x1: INTEGRATE_512 0x2: INTEGRATE_1024 0x3: INTEGRATE_2048

0x0A190130 WCSS_A_FM_SPUR_COMP_FIXED_VALUE_2**Type:** RW**Clock:** APB**Reset State:** 0x00000000

16 bit register for Spur comp value. The value programmed into this register will be subtracted from the I or Q arm after rotation. For details refer to the HLD and system docs.

WCSS_A_FM_SPUR_COMP_FIXED_VALUE_2

Bits	Name	Description
27:16	SPUR_COMP_Q	for spur comp Q 16 bit register for Spur comp value. The value programmed into this register will be subtracted from the Q arm after rotation. For details refer to the HLD and system docs
11:0	SPUR_COMP_I	or spur comp I

0x0A190134 WCSS_A_FM_SPUR_COMP_DC_VALUE_RD_BACK_2**Type:** R**Clock:** APB**Reset State:** 0x00000000

2x12 bit register for spur DC estimation read back value. this value could be used to estimate spur power.

WCSS_A_FM_SPUR_COMP_DC_VALUE_RD_BACK_2

Bits	Name	Description
27:16	SPUR_DC_COMP_Q	dc value estimation for Q
11:0	SPUR_DC_COMP_I	dc value estimation for I

0x0A1A0000 WCSS_A_FM_GAS_GAM_CTRL**Type:** RW**Clock:** HCLK**Reset State:** 0x0000000D

GAS/GAM control register

WCSS_A_FM_GAS_GAM_CTRL

Bits	Name	Description
25	DISABLE_GAM_CLK_GATE	0x0: ENABLE (ENABLE (enable GAM clock gating)) 0x1: DISABLE (DISABLE (disable GAM clock gating))
24	DISABLE_GAS_CLK_GATE	0x0: ENABLE (ENABLE (enable GAS clock gating)) 0x1: DISABLE (DISABLE (disable GAS clock gating))
23:20	GAM_CFG_BUSY_LIMIT	The maximum number of clock cycles the GAM should wait for the application to either present data to the GAM during writes or to deassert the read stall signal during reads, before the GAM will release the AHB

WCSS_A_FM_GAS_GAM_CTRL (cont.)

Bits	Name	Description
19:12	CFG_HSPLIT_RESP_LIMIT	GAS HSPLIT limit, should be set to 0
11:4	CFG_SPLIT_MST_RESP_LIMIT	GAS split, should be set to 0
3:0	CFG_HREADY_LIMIT	GAS slave H Ready limit. should be set to zero for Prima 1.0

0x0A1A0004 WCSS_A_FM_FM_INTR_EVENT**Type:** R**Clock:** APB**Reset State:** 0x00000000

Reading this register will clear all the status bits

WCSS_A_FM_FM_INTR_EVENT

Bits	Name	Description
19	SLV_BYTE_EN_ERROR	slave byte enable error, if byte enable is not all 1. FM slave interface assume all access are 32 bit registers write
18	SSBI_INTR	caused by FM SSBI read/write completion
17	AHB_ACESS_ERROR	caused by FM is not able to read or write AHB
16	UNDER_FLOW	this bit is set if FM TX mode is enabled and BUFFER_TAIL is equal BUFFER_HEAD
15	OVER_FLOW	this bit is set if FM RX mode is enabled and BUFFER_HEAD is equal BUFFER_TAIL
14	AUDIO_SAMPLES_RDY	interrupt event caused by n-samples of FM audio are written or read from cMEM
13	TX_RDS_INTR	Interrupt event caused by RDS buffer is empty. SW should write the entire group (4 blocks) to prevent under flow. The TX_RDS_INTR field holds the interrupt status event bits for all RDS interrupt sources in TX. This bit is cleared on a read by SW.
12	PILOT_DET_DEASSERT_INTR	Interrupt event caused by signal PILOT_DET de-asserted after previously having been asserted
11	PILOT_DET_INTR	Interrupt event caused by signal PILOT_DET asserted
10	IQ_AN_SEGMENT_DONE_INTR	Interrupt event caused by completion of integration of a segment by IQ signal analyzer.
9	C_CP_CHANGE_INTR	Interrupt event caused by change from Type A to Type B reception and vice-versa.
8	RDS_SYNCH_LOSS_INTR	Interrupt event caused by loss of group synchronization.
7	RDS_NOT_SYNCH_INTR	Interrupt event caused by failure to achieve group synchronization.

WCSS_A_FM_FM_INTR_EVENT (cont.)

Bits	Name	Description
6	RDS_SYNCH_INTR	Interrupt event caused by achievement of group synchronization.
5	SIG_NOT_DET_INTR	Interrupt event caused by failure to detect signal
4	SIG_DET_INTR	Interrupt event caused by detection of signal when it meets or exceeds the RX_RDS_IQ_DET_THRESH.
3	RDS_BLOCK_3_INTR	Interrupt event caused by valid data in block register 3
2	RDS_BLOCK_2_INTR	Interrupt event caused by valid data in block register 2
1	RDS_BLOCK_1_INTR	Interrupt event caused by valid data in block register 1
0	RDS_BLOCK_0_INTR	Interrupt event caused by valid data in block register 0

0x0A1A0008 WCSS_A_FM_FM_INTR_ENABLE**Type:** RW**Clock:** APB**Reset State:** 0x00000000

This register has individual interrupt enable bits for each interrupt event. When an FM_INTR_EVENT is 1 and the corresponding INTR_ENABLE bit is set, an interrupt is generated to SW.

WCSS_A_FM_FM_INTR_ENABLE

Bits	Name	Description
19	SLV_BYTE_EN_ERROR_EN	0x0: DISBALE (DISABLE) 0x1: ENABLE (ENABLE slave byte enable error, if byte enable is not all 1. FM slave interface assume all access are 32 bit registers write)
18	SSBI_INTR_EN	0x0: DISABLE (DISABLE) 0x1: ENABLE (ENABLE caused by FM SSBI read/write completion)
17	AHB_ACESS_ERROR_EN	0x0: DISABLE (DISABLE) 0x1: ENABLE (ENABLE caused by FM is not able to read or write AHB)
16	UNDER_FLOW_EN	0x0: DISABLE (DISABLE) 0x1: ENABLE (ENABLE caused by FM TX is reading from empty buffer)
15	OVER_FLOW_EN	0x0: DISABLE (DISABLE) 0x1: ENABLE (ENABLE caused by FM RX is writing to full buffer)
14	AUDIO_SAMPLES_RDY_EN	0x0: DISABLE (DISABLE) 0x1: ENABLE (ENABLE interrupt event caused by n-samples of FM audio are written or read from cMEM)

WCSS_A_FM_FM_INTR_ENABLE (cont.)

Bits	Name	Description
13	TX_RDS_INTR_EN	0x0: DISABLE (DISABLE) 0x1: ENABLE (ENABLE Interrupt event caused by RDS buffer is empty. SW should write the entire group (4 blocks) to prevent under flow. The TX_RDS_INTR field holds the interrupt status event bits for all RDS interrupt sources in TX. This bit is cleared on a read by SW.)
12	PILOT_DET_DEASSERT_INTR_EN	0x0: DISABLE (DISABLE) 0x1: ENABLE (ENABLE Interrupt event caused by signal PILOT_DET de-asserted after previously having been asserted)
11	PILOT_DET_INTR_EN	0x0: DISABLE (DISABLE) 0x1: ENABLE (ENABLE Interrupt event caused by signal PILOT_DET asserted)
10	IQ_AN_SEGMENT_DONE_INTR_EN	0x0: DISABLE (DISABLE) 0x1: ENABLE (ENABLE Interrupt event caused by completion of integration of a segment by IQ signal analyzer.)
9	C_CP_CHANGE_INTR_EN	0x0: DISABLE (DISABLE) 0x1: ENABLE (ENABLE Interrupt event caused by change from Type A to Type B reception and vice-versa.)
8	RDS_SYNCH_LOSS_INTR_EN	0x0: DISABLE (DISABLE) 0x1: ENABLE (ENABLE Interrupt event caused by loss of group synchronization.)
7	RDS_NOT_SYNCH_INTR_EN	0x0: DISABLE (DISABLE) 0x1: ENABLE (ENABLE Interrupt event caused by failure to achieve group synchronization.)
6	RDS_SYNCH_INTR_EN	0x0: DISABLE (DISABLE) 0x1: ENABLE (ENABLE Interrupt event caused by achievement of group synchronization.)
5	SIG_NOT_DET_INTR_EN	0x0: DISABLE (DISABLE) 0x1: ENABLE (ENABLE Interrupt event caused by failure to detect signal)
4	SIG_DET_INTR_EN	0x0: DISABLE (DISABLE) 0x1: ENABLE (ENABLE Interrupt event caused by detection of signal when it meets or exceeds the RX_RDS_IQ_DET_THRESH.)
3	RDS_BLOCK_3_INTR_EN	0x0: DISABLE (DISABLE) 0x1: ENABLE (ENABLE Interrupt event caused by valid data in block register 3)
2	RDS_BLOCK_2_INTR_EN	0x0: DISABLE (DISABLE) 0x1: ENABLE (ENABLE Interrupt event caused by valid data in block register 2)
1	RDS_BLOCK_1_INTR_EN	0x0: DISABLE (DISABLE) 0x1: ENABLE (ENABLE Interrupt event caused by valid data in block register 1)

WCSS_A_FM_FM_INTR_ENABLE (cont.)

Bits	Name	Description
0	RDS_BLOCK_0_INTR_EN	0x0: DISABLE (DISABLE) 0x1: ENABLE (ENABLE Interrupt event caused by valid data in block register 0)

0x0A1A000C WCSS_A_FM_FM_BUFFER_START_ADDRESS**Type:** RW**Clock:** APB**Reset State:** 0x00000000

This register is to set the FM buffer inside cMEM start address. Note that bit 0&1 are set to 0

WCSS_A_FM_FM_BUFFER_START_ADDRESS

Bits	Name	Description
31:2	BUFFER_START	Start address of FM buffer inside Riva cMEM

0x0A1A0010 WCSS_A_FM_FM_BUFFER_LENGTH**Type:** RW**Clock:** APB**Reset State:** 0x00000000

This register is to set the FM buffer length inside cMEM.

WCSS_A_FM_FM_BUFFER_LENGTH

Bits	Name	Description
14:0	BUFFER_LENGTH	buffer length in dwords

0x0A1A0014 WCSS_A_FM_FM_BUFFER_HEAD**Type:** RW**Clock:** APB**Reset State:** 0x00000000

This register is to set the buffer head.

WCSS_A_FM_FM_BUFFER_HEAD

Bits	Name	Description
14:0	BUFFER_HEAD	buffer head relative to start in dwords. for FM RX this register is updated by FM HW for FM TX this register should be updated by Audio driver

0x0A1A0018 WCSS_A_FM_FM_BUFFER_TAIL**Type:** RW**Clock:** APB**Reset State:** 0x00000000

This register is to set the buffer tail

WCSS_A_FM_FM_BUFFER_TAIL

Bits	Name	Description
14:0	BUFFER_TAIL	buffer tail relative to start in dwords for FM TX this register is updated by FM HW for FM RX this register should be updated by Audio driver

0x0A1A001C WCSS_A_FM_AUDIO_INTR_CONTROL**Type:** RW**Clock:** APB**Reset State:** 0x00000000

This register is to set the number of FM audio samples to generate interrupt. also this register is to select pulse or level audio interrupts to LPASS.

WCSS_A_FM_AUDIO_INTR_CONTROL

Bits	Name	Description
18	PULSE_LEVEL_SEL	0x0: LEVEL (LEVEL (active high, will be cleared by setting CLR_AUDIO_INTR)) 0x1: PULSE (PULSE (1 us interrupt pulse to LPASS is generated))
17	CLR_AUDIO_SAMPLE_COUNTER	0x0: RUNNING (RUNNING) 0x1: REST (REST toggle between 0 and 1 are required to reset the audio samples counters)
14:0	AUDIO_INTR_INTERVAL	number of audio samples to generates interrupts to the audio sub system

0x0A1A0020 WCSS_A_FM_CLR_AUDIO_LEVEL_INTR

Type: R
Clock: APB
Reset State: 0x00000000

Read this register will cause LPASS interrupt to clear if it is set to level.

WCSS_A_FM_CLR_AUDIO_LEVEL_INTR

Bits	Name	Description
0	CLR_AUDIO_INTR	Reading this register will clear the LPASS interrupts

0x0A1A0024 WCSS_A_FM_AUDIO_SAMPLES_COUNTER_VFR_SNAP_READBACK

Type: R
Clock: APB
Reset State: 0x00000000

Snap of FM samples counter based on the pos edge of VFR signal.

WCSS_A_FM_AUDIO_SAMPLES_COUNTER_VFR_SNAP_READBACK

Bits	Name	Description
31:0	AUDIO_SAMPLES_VFR_COUNT	sampled AUDIO_SAMPLES_COUNT based on VFR

0x0A1A0028 WCSS_A_FM_AUDIO_SAMPLES_COUNTER_READBACK

Type: R
Clock: APB
Reset State: 0x00000000

counts the number of FM samples, this is free running counter, it roll over to 0.

WCSS_A_FM_AUDIO_SAMPLES_COUNTER_READBACK

Bits	Name	Description
31:0	AUDIO_SAMPLES_COUNT	number of audio samples. this register is used to check for sample-slipping in the audio sub system

0x0A1A002C WCSS_A_FM_SW_RESET**Type:** RW**Clock:** APB**Reset State:** 0x00000000

Software reset

WCSS_A_FM_SW_RESET

Bits	Name	Description
2	SLV_SW_RESET	0x0: NORMAL (NORMAL (normal mode)) 0x1: RESET (RESET (reset Slave block) slave will not recover from this reset until power up)
1	MSTR_SW_RESET	0x0: NORMAL (NORMAL (normal)) 0x1: RESET (RESET (reset slave block))
0	SDI_SW_RESET	0x0: NORMAL (NORMAL (normal mode)) 0x1: RESET (RESET (reset serial data interface))

0x0A1A0034 WCSS_A_FM_SDI_CTRL**Type:** RW**Clock:** APB**Reset State:** 0x00000000

Serial Data Interface control register

WCSS_A_FM_SDI_CTRL

Bits	Name	Description
0	SDI_EN	0x0: DISABLE (DISABLE (disable SDI)) 0x1: ENABLE (ENABLE (Enable SDI))

0x0A1A0038 WCSS_A_FM_SCRAMBLER_CTRL**Type:** RW**Clock:** APB**Reset State:** 0x00000000

.Control the scrambler/descrambler block.

WCSS_A_FM_SCRAMBLER_CTRL

Bits	Name	Description
2	SCRM_TYPE	0x0: ADDITIVE (ADDITIVE) 0x1: MULTIPLICATIVE (MULTIPLICATIVE)
1	SCRM_DESCRM_SEL	0x0: SCRAMBLER_ENABLED (SCRAMBLER_ENABLED) 0x1: DE_SCRAMBLER_ENABLED (DE_SCRAMBLER_ENABLED)
0	SCRM_EN	0x1: RUNNING (RUNNING) 0x0: BYPASS (BYPASS)

0x0A1A003C WCSS_A_FM_SCRAMBLER_POLY**Type:** RW**Clock:** APB**Reset State:** 0x00000000

.bit [16:0] of Scrambler poly.

WCSS_A_FM_SCRAMBLER_POLY

Bits	Name	Description
16:0	SCRM_POLY	Specify LFSR polynomial

0x0A1A0040 WCSS_A_FM_FM_INTR_CTRL**Type:** RW**Clock:** APB**Reset State:** 0x00000000

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WCSS_A_FM_FM_INTR_CTRL

Bits	Name	Description
0	FM_INTR_PULSE_LVL_SEL	0x0: PULSE (PULSE) 0x1: LEVEL (LEVEL)

0x0A1A0044 WCSS_A_FM_AUDIO_DATA_CONTROL**Type:** RW**Clock:** APB**Reset State:** 0x00000000

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WCSS_A_FM_AUDIO_DATA_CONTROL

Bits	Name	Description
1	SWAP_AUDIO_BYTES	0x0: NORMAL (NORMAL (NO Swap)) 0x1: SWAP (SWAP (swap bytes for 16 bit left and right audio channels))
0	SWAP_AUDIO	0x0: NORMAL (NORMAL (Left right)) 0x1: SWAP (SWAP (right left))

0x0A1A0090 WCSS_A_FM_SPUR_TRK_START_ADDRESS**Type:** RW**Clock:** APB**Reset State:** 0x00000000

This register is to set the buffer starting address for storing spur tracking data. Note that bit 0&1 are set to 0. The buffer length is 256 words.

WCSS_A_FM_SPUR_TRK_START_ADDRESS

Bits	Name	Description
31:2	BUFFER_START	Start address of spur tracking buffer inside cMEM

0x0A1A0094 WCSS_A_FM_RX_AHB_ACCESS_ERROR_STATUS**Type:** R**Clock:** APB**Reset State:** 0x00000000

When interrupt event RX_AHB_ACCESS_ERROR happens, this register shows the source of the error.

WCSS_A_FM_RX_AHB_ACCESS_ERROR_STATUS

Bits	Name	Description
1	SPUR_TRK_ERROR	error happened when writing to spur tracking buffer inside cMEM

WCSS_A_FM_RX_AHB_ACCESS_ERROR_STATUS (cont.)

Bits	Name	Description
0	RX_AUDIO_ERROR	error happened when writing to audio data buffer inside cMEM

10.25 WCSS_A_A2XB_A2XB**0x0A100400 WCSS_A_A2XB_A2XB_CFG****Type:** RW**Reset State:** 0x00000020

This register's value configures general A2XB features.

WCSS_A_A2XB_A2XB_CFG

Bits	Name	Type	Description
21:16	MSTR_SPLIT_TIMEOUT	R	Shows the split timeout status of each of the CDAHB masters (one bit per master). When set signifies that the master has been split for longer than the timeout limit and therefore has seen a split timeout. The bits are sticky so that if more than one timeout happen the new timed out masters get OR-ed in with the previous value.
15:10	MSTR_SPLIT_STAT	R	Shows the split status of each of the CDAHB masters (one bit per master). When set signifies that the master has been split by A2XB. There are four reasons for a master being split: 1. Command Queue is full so request could not be accepted 2. Write data Queue is full so write data could not be accepted 3. Read request is pending and A2XB is waiting on AXI read completion 4. A write by another master is pending due to <2> so a new write command cannot be accepted
9:2	SPLIT_LIMIT	RW	configures the split timeout limit in the AHB slave in thousands of cycles. If a master is split for more than the configured limit a split timeout error interrupt will be generated.

WCSS_A_A2XB_A2XB_CFG (cont.)

Bits	Name	Type	Description
1	BUSY_OFF	RW	Purely a debug feature. A bug in the busy generation logic has the potential to hang every master in Riva. Should this type of bug be encountered, setting this bit to 1 will ensure that busy is always low, thereby preventing masters from getting hung waiting for completion of writes. Note that this will likely cause data coherency issues between masters.
0	ENABLE	R	Set to enable the A2XB bridge functionality. This register is actually on the PMU and this read only bit just reflects the status of the corresponding PMU register. When set to zero all rd/wr requests not destined to internal A2XB registers will generate an AHB error response.

0x0A100404 WCSS_A_A2XB_INT_MSK**Type:** RW**Reset State:** 0x0000000F

The value of this register is used to mask interrupt sources.

WCSS_A_A2XB_INT_MSK

Bits	Name	Description
3	A2XB_AXI_HALT_REQ_SET_INT_MSK	When clear it masks off the a2xb_axi_halt_req_set interrupt request.
2	A2XB_SPLIT_TIMEOUT_ERR_INT_MSK	When clear it masks off the a2xb_split_timeout_err interrupt request.
1	A2XB_RD_ERR_INT_MSK	When clear it masks off the a2xb_rd_err interrupt request.
0	A2XB_WR_ERR_INT_MSK	When clear it masks off the a2xb_wr_err interrupt request.

0x0A100408 WCSS_A_A2XB_INT_SRC_MSKD**Type:** R**Reset State:** 0x00000000

The value of this register is the masked interrupt source. It is the logical AND of the INT_SRC with INT_MSK

WCSS_A_A2XB_INT_SRC_MSKD

Bits	Name	Description
3	A2XB_AXI_HALT_REQ_SET_INT_MSKD	This field is the masked a2xb_axi_halt_req_set_int
2	A2XB_SPLIT_TIMEOUT_ERR_INT_MSKD	This field is the masked a2xb_split_timeout_err_int
1	A2XB_RD_ERR_INT_MSKD	This field is the masked a2xb_rd_err_int
0	A2XB_WR_ERR_INT_MSKD	This field is the masked a2xb_wr_err_int

0x0A10040C WCSS_A_A2XB_INT_SRC**Type:** R**Reset State:** 0x00000000

The value of this register is the interrupt source.

WCSS_A_A2XB_INT_SRC

Bits	Name	Description
3	A2XB_AXI_HALT_REQ_SET_INT	This field is only set when A2XB receives a halt request from the AXI NOC (visible through halt[haltreq_external]=1) It is NOT set for internally driven halt requests (halt[haltreq_internal]=1) Upon receiving the halt request, the A2XB will complete all pending AXI transfers but will from then on return error responses to any AHB transaction destined to the AXI master port. Local access to A2XB CSRs will still work normally.
2	A2XB_SPLIT_TIMEOUT_ERR_INT	This field is set when the A2XB GAS detects that a master has been split for longer than the timeout limit. This is a fatal error that will require at least a full A2XB reset to recover. On occurrence, A2XB will immediately unsplit any split masters, will flush all its buffers, will discard any data incoming on its AXI port side and will from then on return error responses to any AHB transaction destined to the fabric. Local access to A2XB CSRs will still work normally.
1	A2XB_RD_ERR_INT	This field is set when A2XB receives an error response from the AXI fabric on any data beat of a read transaction. Data following the first beat with an error and belonging to the same transfer will be discarded and when the AHB master returns to complete its read it will be able to complete only as many beats as completed without an AXI error (if any) and will then receive an AHB error response on the beat that incurred the AXI error.

WCSS_A_A2XB_INT_SRC (cont.)

Bits	Name	Description
0	A2XB_WR_ERR_INT	This field is set when A2XB receives an error response from the AXI fabric for a write transaction. Since the AHB protocol provides no way to notify the originating master of errors incurred on posted writes, this interrupt will be the only indication that a write error occurred.

0x0A100410 WCSS_A_A2XB_INT_CLR**Type:** W**Reset State:** 0x00000000

The value of this register is used to clear interrupt sources.

WCSS_A_A2XB_INT_CLR

Bits	Name	Description
3	A2XB_AXI_HALT_REQ_SET_INT_CLR	This field when set will clear the halt request set interrupt indication. This only clears the interrupt request, A2XB reset will be required to return to normal operation.
2	A2XB_SPLIT_TIMEOUT_ERR_INT_CLR	This field when set will clear the split timeout error interrupt. This only clears the interrupt request, A2XB reset will be required to return to normal operation.
1	AXI_RD_ERR_INT_CLR	This field when set will clear the AXI read error interrupt.
0	AXI_WR_ERR_INT_CLR	This field when set will clear the AXI write error interrupt.

0x0A10041C WCSS_A_A2XB_ERR_INFO**Type:** R**Reset State:** 0x00000000

Records which AHB master IDs have received AXI write or read error responses for any of their transfers. Once set, the bit can only be cleared by resetting the A2XB.

WCSS_A_A2XB_ERR_INFO

Bits	Name	Description
11	MID_6_RD_ERR	Set if AHB MID 6 got an AXI read error response in any of its transfers.
10	MID_5_RD_ERR	Set if AHB MID 5 got an AXI read error response in any of its transfers.
9	MID_4_RD_ERR	Set if AHB MID 4 got an AXI read error response in any of its transfers.
8	MID_3_RD_ERR	Set if AHB MID 3 got an AXI read error response in any of its transfers.
7	MID_2_RD_ERR	Set if AHB MID 2 got an AXI read error response in any of its transfers.
6	MID_1_RD_ERR	Set if AHB MID 1 got an AXI read error response in any of its transfers.
5	MID_6_WR_ERR	Set if AHB MID 6 got an AXI write error response in any of its transfers.
4	MID_5_WR_ERR	Set if AHB MID 5 got an AXI write error response in any of its transfers.
3	MID_4_WR_ERR	Set if AHB MID 4 got an AXI write error response in any of its transfers.
2	MID_3_WR_ERR	Set if AHB MID 3 got an AXI write error response in any of its transfers.
1	MID_2_WR_ERR	Set if AHB MID 2 got an AXI write error response in any of its transfers.
0	MID_1_WR_ERR	Set if AHB MID 1 got an AXI write error response in any of its transfers.

0x0A100420 WCSS_A_A2XB_REVISION**Type:** R**Reset State:** 0x00000045

This register's value represents the A2XB RTL revision present in the hardware.

WCSS_A_A2XB_REVISION

Bits	Name	Description
31:0	REVID	Revision ID. Corresponds to the label number of A2XB (revision of .info file)

0x0A100428 WCSS_A_A2XB_HALT**Type:** RW**Reset State:** 0x00000008

This register contains control and status bits related to the haltreq/haltack interface.

WCSS_A_A2XB_HALT

Bits	Name	Type	Description
3	MASTER_IDLE	R	This bit is only set when there are no pending AXI transfers so the AXI master interface is idle.
2	HALTACK	R	Reflects the status of the haltack WCSS output signal. This bit can be set if either internal or external halt requests are asserted and only after the interface has completed all pending AXI transfers and is idle.
1	HALTREQ_EXTERNAL	R	Reflects the status of the external haltreq input signal to the WCSS subsystem. The triggered behavior when set is identical to the one described for the haltreq_internal bit.
0	HALTREQ_INTERNAL	RW	Internal haltreq request. When set this bit triggers the haltreq/haltack protocol. When set, the A2XB interface will finish all pending AXI transfers but will not accept any new AXI-bound requests from any master returning an AHB error response instead. Once set this bit is set it can only be cleared by resetting the A2XB interface. Accesses to A2XB csr registers are not affected by this bit.

0x0A10042C WCSS_A_A2XB_AXI_SB_CFG**Type:** RW**Reset State:** 0x000000DB

This register serves to configure the values used to drive the AXI sideband signals that are not used within WCSS and for which otherwise constant values would be driven.

WCSS_A_A2XB_AXI_SB_CFG

Bits	Name	Description
8:6	AMEMTYPE_HW_MASTER S	This bitfield contains the value that will be used to drive the AXI amemtype[2:0] signal for ALL transfers originated by (non-cpu) all other hardware masters. HW modules only access packet memory in DDR and so its default value should not be changed.
5:3	AMEMTYPE_CPU_BURST	This bitfield contains the value that will be used to drive the AXI amemtype[2:0] signal for burst transfers originated by the internal WCSS cpu, such accesses are always used for cacheable regions (HPROT[3]=1)
2:0	AMEMTYPE_CPU_SINGLE	This bitfield contains the value that will be used to drive the AXI amemtype[2:0] signal for single beat transfers originated by the internal WCSS cpu. The concern was that using a memtype of NORMAL to access locations with potential sideeffects (FIFOs, registers) could result in corruption in certain scenarios. This field provides a way to use a type of strongly-ordered or device for this type of transfers. This is because cacheable access always use burst transfers while non-cacheable accesses will always be mapped to single transfers. So, HPROT[3:2]=01 (non-cacheable, bufferable) which is roughly equivalent to AMEMTYPE=DEVICE, can be inferred when the bridge sees a cpu initiated single-beat 32-bit transfer.<CR> Also, Note that HPROT[3:2]=01 (non-cacheable, bufferable), even if available to the bridge and used to automatically set amemtype=DEVICE, would not guarantee the requirement of the NOC team that a DEVICE type never go to the DDR. This is because DDR memory locations shared or used by WCSS ARM9 SW to communicate with other processors need to be marked as non-cacheable because no mechanism exists to synchronize the Pronto ARM9 caches when the contents of the shared memory are changed by other processors in the system. I really don't see a way to simultaneously guarantee the dual requirement that CSR accesses always use DEVICE while DDR accesses always use NORMAL without building a non-portable (if hardwired) or expensive (if programmable) linkage of the WCSS subsystem to a target MSM 's memory map simply to map to the required amemtype. This is not a failure of the WCSS subsystem but rather a consequence of the system using non-standard ARM signals which our stock ARM9 CPU obviously does not support

10.26 WCSS_A_ADU_ADU

0x0A082800 WCSS_A_ADU_CONTROL

Type: RW

Reset State: 0x0BFF0002

Contains configuration information for the adu

WCSS_A_ADU_CONTROL

Bits	Name	Type	Description
31:24	NEXT_WQ	RW	This field indicates the WQ that the ADU will push the frames to after they have been processed. This field is only used when 'push_wq_selection_mode' is set to '1'.
23:16	WOW_WQ	RW	This field indicates the WQ that the ADU will push the frames during WOW mode. This field is only used when 'push_wq_selection_mode' is set to '1'.
15	REG_REINIT_ERR_STATE_CFG	RW	When set, ADU reg reinit FSM will go back to idle instead of staying in error state.
13	REG_REINIT_GOTO_IDLE	RW	When set, ADU reg reinit FSM (BPS, MIMO) will go to idle.
8	CLEAR_TIMEOUT_CNT	W	Writing a '1' will reset the timeout counter and clear timeout error interrupt. Always read 0.
7	DISABLE_LENGTH_CHECK	RW	When set, the ADU will not check the subframe length error
6	PUSH_WQ_SELECTION_WOW_MODE	RW	This field controls where the ADU finds the index for the WQ to which the processed frames need to be pushed to in WOW mode. This bit should be set (a 0->1 transition) prior to WOW mode to select the WOW_WQ. Upon the assertion of dpu_wakeuphost, ADU will push to the normal WQ. 0: ADU will push the frame to the WQ that is selected by the push_wq_selection_mode 1: ADU will push the frame to the WQ that is indicate in the 'WOW_WQ' field in this register
5	ADU_ERROR_EXIT	W	Writing a '1' will clear the interrupts and ADU will go back to IDLE from ERROR_STATE. Always read 0

WCSS_A_ADU_CONTROL (cont.)

Bits	Name	Type	Description
4	ADU_BYPASS_OPTION_ENABLE	RW	When set, ADU will immediately forward the frame and NOT check if the frame is AMSDU
3	CLEAR_COUNTERS	RW	When set, the ADU packet counters are cleared To re-enable the counters, this bit needs to be cleared again
2	DISABLE_HT_FIELD_DETECTION	RW	When set, the ADU will NOT check for the presence of the HT control field
1	PUSH_WQ_SELECTION_MODE	RW	This field controls where the ADU finds the index for the WQ to which the processed frames need to be pushed to. 0: ADU will push the frame to the WQ that is indicated in the 'ADU/DPU routing flag' in the BD descriptor. 1: ADU will push the frame to the WQ that is indicate in the 'Next WQ' field in this register
0	ADU_ENABLE	RW	When set, the ADU is enabled. When the adu gets disabled in the middle of a transfer, it will finish the ongoing transfer.

0x0A082804 WCSS_A_ADU_INTERRUPT_STATUS**Type:** RW**Reset State:** 0x00000000

Contains status information on what interrupt has been generated Note that the interrupt is only passed on to the mCPU when the error interrupt has been enabled in the 'ADU interrupt enable' register.

WCSS_A_ADU_INTERRUPT_STATUS

Bits	Name	Type	Description
15:14	RESERVED		The two bits are reserved.
13	ADU_POLL_CMD_ERROR	R	When set, adu poll cmd reached retry timeout error occurred
12	UMA_WRONG_WQ_ERROR	R	When set, adu uma tx push to its own WQ error occurred
11	TIMEOUT_ERROR	R	When set, timeout error occurred

WCSS_A_ADU_INTERRUPT_STATUS (cont.)

Bits	Name	Type	Description
10	WRONG_WQ_ERROR	R	When set, adu push to its own WQ error occurred
9	INCORRECT_LINK_ERROR	R	When set, an incorrect BD/PDU link error occurred
8	BAD_BD_TYPE_ERROR	R	When set, a bad bd type error occurred
7	GAM_WRITE_ERROR	R	When set, a gam write error occurred
6	SW_REG_REINIT_DONE	R	When set, the sw triggered reg reinit is done
5	DELIMITER_LENGTH_NOT_AVAIL_ERROR	R	When set, the adu was not able to extract the delimiter length from the frame
4	GAM_READ_ERROR	R	When set, the gam reported a read error
3	GBI_PUSH_ERROR	R	When set, the gbi reported an error while a push operation was ongoing
2	MPDU_HEADER_NOT_IN_BD_ERROR	R	When set, the mpdu header was not located fully within the BD
1	GBI_POP_ZERO_ERROR	R	When set, the gbi returned a NULL BD pointer when the BMU was indicating DA in its WQ.
0	GBI_POP_ERROR	R	When set, the gbi reported an error while a pop operation was ongoing

0x0A082808 WCSS_A_ADU_INTERRUPT_ENABLE**Type:** RW**Reset State:** 0x00000000

Contains the enable bits for which interrupt has to be generated

WCSS_A_ADU_INTERRUPT_ENABLE

Bits	Name	Description
15:14	ADU_RESERVED	The two bits are reserved

WCSS_A_ADU_INTERRUPT_ENABLE (cont.)

Bits	Name	Description
13	ADU_POLL_CMD_ERROR_ENABLE	When set, adu poll cmd reach retry timeout error will generate error interrupt
12	UMA_WRONG_WQ_ERROR_ENABLE	When set, adu uma tx push its own WQ error will generate error interrupt
11	TIMEOUT_ERROR_ENABLE	When set, timeout error will generate error interrupt
10	WRONG_WQ_ERROR_ENABLE	When set, adu push its own WQ error will generate error interrupt
9	INCORRECT_LINK_ERROR_ENABLE	When set, an incorrect BD/PDU link error will generate error interrupt
8	BAD_BD_TYPE_ERROR_ENABLE	When set, a bad bd type error will generate error interrupt
7	GAM_WRITE_ERROR_ENABLE	When set, a gam write error will generate an adu error interrupt
6	SW_REG_REINIT_DONE_ENABLE	When set, when the sw triggered reg reinit is done the interrupt will be generated
5	DELIMITER_LENGTH_NOT_AVAIL_ERROR_ENABLE	When set, and the adu was not able to extract the delimiter length from the frame an adu error interrupt will be generated
4	GAM_READ_ERROR_ENABLE	When set, and the gam reported a read error an adu error interrupt will be generated
3	GBI_PUSH_ERROR_ENABLE	When set, and the gbi reported an error while a push operation was ongoing, an adu error interrupt will be generated
2	MPDU_HEADER_NOT_IN_BD_ERROR_ENABLE	When set, and the mpdu header was not located fully within the BD an adu error interrupt will be generated
1	GBI_POP_ZERO_ERROR_ENABLE	When set, and the gbi returned a NULL BD pointer when the BMU was indicating DA in its WQ, an adu error interrupt will be generated
0	GBI_POP_ERROR_ENABLE	When set, and the gbi reported an error while a pop operation was ongoing, an adu error interrupt will be generated

0x0A082810 WCSS_A_ADU_ADU_STATUS**Type:** R**Reset State:** 0x00000000

Provides status information about the ADU

WCSS_A_ADU_ADU_STATUS

Bits	Name	Description
28:24	ENCODED_ADU_REINIT_FSM	The FSM status of adu reinit
23	ADU_SM_STATE_MSB	The MSB of statemachine state of the ADU controller
22	BMU_UMA_TX_WQ_DA	The status of the bmu_uma_tx_wq_da
21	ENCODED_GAS_INTERFACE_SM_STATE	The status of the gas interface.
20	UMA_TX_STATUS	The status of the uma tx. 0 means uma tx is idle
19:15	UMA_TX_SM_STATE	The statemachine state of the uma tx
13	ADU_MULTIPLE_RX_UPDATE_ONGOING	shows the status of adu_multiple_rx_update_ongoing
12:8	ADU_SM_STATE	The 5LSB of statemachine state of the ADU controller
5	ADU_FIXED_WQ_PUSH_MODE_WOW_ENABLED	Shows the status of adu_fixed_wq_push_mode_wow_enabled signal. It's set to 1 when control[6] is set to 1, and clear to 0 when control[6] is cleared or when dpu_wakuphost assert
4	DPU_WAKEUPHOST	Shows the status of dpu_wakuphost
3	ADU_STATUS	When set, the ADU controller is NOT in it's IDLE state
2	ADU_MCU_ERR_INT	shows the status of the ADU error interrupt signal going to the mcu
1	DATA_AVAIL	shows the data_aavailable signal from the bmu
0	BD_PDU_AVAIL	shows the bd_pdu_avail signal from the bmu

0x0A082814 RESERVED**0x0A082818 RESERVED****0x0A08281C RESERVED****0x0A082820 WCSS_A_ADU_ADU_COUNTERS1****Type:** R**Reset State:** 0x00000000

gives information on the number of frames processed

WCSS_A_ADU_ADU_COUNTERS1

Bits	Name	Description
23:16	NUMBER_OF_AMSDU_FRAMES_PROCESSED	Indicates the number of amsdu frames processed by the ADU
7:0	NUMBER_OF_FRAMES_POPPED	Indicates the number of frames that were popped by the ADU

0x0A082824 WCSS_A_ADU_ADU_COUNTERS2**Type:** R**Reset State:** 0x00000000

gives information on the number of frames processed

WCSS_A_ADU_ADU_COUNTERS2

Bits	Name	Description
23:16	NUMBER_OF_NON_AMSDU_FRAMES_PROCESSED	Indicates the number of non amsdu frames processed by the ADU
7:0	NUMBER_OF_BYPASS_FRAMES_PROCESSED	Indicates the number of frames for which the 'ADU bypass' bit in the BD was set

0x0A082828 WCSS_A_ADU_ADU_COUNTERS3**Type:** R**Reset State:** 0x00000000

gives information on the number of frames processed

WCSS_A_ADU_ADU_COUNTERS3

Bits	Name	Description
23:16	NR_OF_ERROR_FRAMES	Indicates the number of error frames which has been pushed to error wq.
7:0	NR_OF_MISCONFIGURED_AMSDU_FRAMES	Indicates the number of amsdu frames processed that were not correctly configured

0x0A08282C WCSS_A_ADU_CONTROL2**Type:** RW**Reset State:** 0x03FFF92A

Contains configuration information for the adu

WCSS_A_ADU_CONTROL2

Bits	Name	Description
31:26	SPARE_BIT	Spare register bits for ECOs
25:12	TIMEOUT_CNT	This field specify the timeout value in unit of 1 us when adu fsm doesn't return to IDLE.
11:0	MAX_LENGTH	This field specify the max subframe length is allowed.

0x0A082830 WCSS_A_ADU_TIMEOUT_READ**Type:** R**Reset State:** 0x00000000

Contains timeout counter information for the adu

WCSS_A_ADU_TIMEOUT_READ

Bits	Name	Description
19:0	TIMEOUT_CNT_REG	This field read the timeout counter value

0x0A08283C WCSS_A_ADU_UMA_CONFIG**Type:** RW**Reset State:** 0xA0106000

This register is to config the UMA function

WCSS_A_ADU_UMA_CONFIG

Bits	Name	Description
31	UMA_TX_MCUSEBC_EN	When set, mc address will use bc addr for da search.
30:26	UMA_TX_QID_FOR_NONQOS	The Queueid to fill the BD when the packet is non QoS frame.
25	UMA_TX_EATRACT_TID	When set, uma tx extract the tid from 802.1Q/IP header.
24	UMA_RX_LLC_EN	When set and bit 21 is set, enables LLC removal.
23	UMA_RX_CHECKSUM_EN	When set and bit 21 is set, enables the TCP checksum validation in RX direction
22	UMA_RX_FT_EN	When set and bit 21 is set, enables the frame translation in RX direction
21	UMA_RX_EN	When set, enables the data path enhancement in RX direction. When clear, UMA RX simply forward the frame to the specified WQ.
20	UMA_TX_PUSH_WQ_SEL	When set, UMA will push the frame to the WQ specified in bit [19:12] When clear, UMA will push the frame to the WQ specified in the routing flag in BD
19:12	UMA_TX_PUSH_WQ	Specify the WQ that UMA should push to in TX direction if bit 20 is set.
11:4	TX_DEFAULT_STAID	The default descriptor index used for frame translation in TX direction when there's no DA match in the search table.
3	UMA_TX_LLC_EN	When set and bit 0 is set, enables LLC insertion in TX direction
2	UMA_TX_CHECKSUM_EN	When set and bit 0 is set, enables TCO checksum computation in TX direction

WCSS_A_ADU_UMA_CONFIG (cont.)

Bits	Name	Description
1	UMA_TX_FT_EN	When set and bit 0 is set, enables frame translation in TX direction.
0	UMA_TX_EN	When set, enables data path enhancement in TX direction. When clear, UMA simply forward the incoming frame to the specified WQ.

0x0A082840 WCSS_A_ADU_UMA_CTRL**Type:** RW**Reset State:** 0x0000FF03

This register is to control the UMA function

WCSS_A_ADU_UMA_CTRL

Bits	Name	Type	Description
31:24	UMA_TX_AP_WQ	RW	The WQ UMA will push when there's no match.
23:16	UMA_RX_MC_WQ	RW	The WQ UMA will push when DA is a mc/bc address.
15:8	UMA_RX_ERROR_WQ	RW	The Error WQ UMA will forward when exception occurs, such as payload size less than 8 bytes.
7	UMA_RX_MCWQ_EN	RW	When set, enable UMA RX push the frame to uma_rx_mc_wq when DA is a mc/bc address
6	UMA_TX_APWQ_EN	RW	When set, enable UMA TX push the frame to uma_tx_ap_wq when there's no match
5	UMA_DISABLE_MIN_PAYLOAD_CHECK	RW	When set, disable the min payload check
4	UMA_COUNTER_CLEAR	W	Write a '1' to clear the UMA TX Counters. Always read '0'.
3	UMA_TX_QID_FROM_BD	RW	When set, UMA won't update the queue id in BD. When clear, queue id will be a 1-1 mapping from tid(QoS) or register (non-QoS)
2	UMA_TX_ERROR_EXIT	W	Write a '1' to clear the UMA TX interrupts and UMA TX FSM goes back to IDLE. Always read '0'.

WCSS_A_ADU_UMA_CTRL (cont.)

Bits	Name	Type	Description
1	UMA_RX_ENABLE	RW	When set, UMA RX is enabled. When clear, UMA RX is off. adu_ctrl will push the processed frame to the specified WQ.
0	UMA_TX_ENABLE	RW	When set, UMA TX is enabled. When clear, UMA TX is off. No frame will be popped from BMU.

0x0A082844 WCSS_A_ADU_UMA_DESP_TABLE_ADDR**Type:** RW**Reset State:** 0x00000000

This register contains the base address of uma's descriptor table

WCSS_A_ADU_UMA_DESP_TABLE_ADDR

Bits	Name	Description
31:0	UMA_DESP_TABLE_BASE_ADDR	Base address of UMA's descriptor table

0x0A082848 WCSS_A_ADU_UMA_TX_ETHERTYPE_TABLE1**Type:** RW**Reset State:** 0x00000000

This register contains 2 entries of the ethernet table for uma tx

WCSS_A_ADU_UMA_TX_ETHERTYPE_TABLE1

Bits	Name	Description
31:16	UMA_TX_ETHERTYPE_ENTRY2	One of the entries of uma tx ethernet table
15:0	UMA_TX_ETHERTYPE_ENTRY1	One of the entries of uma tx ethernet table

0x0A08284C WCSS_A_ADU_UMA_TX_ETHERTYPE_TABLE2**Type:** RW**Reset State:** 0x00000000

This register contains 2 entries of the ethertype table for uma tx

WCSS_A_ADU_UMA_TX_ETHERTYPE_TABLE2

Bits	Name	Description
31:16	UMA_TX_ETHERTYPE_ENTRY4	One of the entries of uma tx ethertype table
15:0	UMA_TX_ETHERTYPE_ENTRY3	One of the entries of uma tx ethertype table

0x0A082850 WCSS_A_ADU_UMA_RX_ETHERTYPE_TABLE1

Type: RW

Reset State: 0x00000000

This register contains 2 entries of the ethertype table for uma rx

WCSS_A_ADU_UMA_RX_ETHERTYPE_TABLE1

Bits	Name	Description
31:16	UMA_RX_ETHERTYPE_ENTRY2	One of the entries of uma rx ethertype table
15:0	UMA_RX_ETHERTYPE_ENTRY1	One of the entries of uma rx ethertype table

0x0A082854 WCSS_A_ADU_UMA_RX_ETHERTYPE_TABLE2

Type: RW

Reset State: 0x00000000

This register contains 2 entries of the ethertype table for uma rx

WCSS_A_ADU_UMA_RX_ETHERTYPE_TABLE2

Bits	Name	Description
31:16	UMA_RX_ETHERTYPE_ENTRY4	One of the entries of uma rx ethertype table
15:0	UMA_RX_ETHERTYPE_ENTRY3	One of the entries of uma rx ethertype table

0x0A082858 WCSS_A_ADU_UMA_TX_COUNTERS1**Type:** R**Reset State:** 0x00000000

This register contains counters information of uma tx

WCSS_A_ADU_UMA_TX_COUNTERS1

Bits	Name	Description
23:16	UMA_TX_PUSHED_COUNTER	Counters of the frame uma tx pushed
7:0	UMA_TX_POPPED_COUNTER	Counters of the frame uma tx popped

0x0A08285C WCSS_A_ADU_UMA_TX_COUNTERS2**Type:** R**Reset State:** 0x00000000

This register contains counters information of uma tx

WCSS_A_ADU_UMA_TX_COUNTERS2

Bits	Name	Description
23:16	UMA_TX_ERROR_COUNTER	Counters of the frames forwarded to error wq by UMA TX
7:0	UMA_TX_FT_COUNTER	Counters of the frame translated by uma tx

0x0A082860 WCSS_A_ADU_UMA_LLC_COUNTERS**Type:** R**Reset State:** 0x00000000

This register contains counters information of uma llc insertion/removal

WCSS_A_ADU_UMA_LLC_COUNTERS

Bits	Name	Description
23:16	UMA_RX_LLC_COUNTER	Counters of the LLC removal done by uma rx
7:0	UMA_TX_LLC_COUNTER	Counters of the LLC insertion done by uma tx

0x0A082864 WCSS_A_ADU_UMA_INDEX_TABLE_WDATA_U**Type:** RW**Reset State:** 0x00000000

This register contains the upper bits of the entry of index table

WCSS_A_ADU_UMA_INDEX_TABLE_WDATA_U

Bits	Name	Description
23:16	UMA_SEARCHTABLE_INDEX	index (in unit of 3 words) to the descriptor table
15:0	UMA_SEARCHTABLE_WDATA_U	Upper bits of the entry of index table

0x0A082868 WCSS_A_ADU_UMA_INDEX_TABLE_WDATA_L**Type:** RW**Reset State:** 0x00000000

This register contains the lower bits of the entry of index table

WCSS_A_ADU_UMA_INDEX_TABLE_WDATA_L

Bits	Name	Description
31:0	UMA_SEARCHTABLE_WDATA_L	Lower bits of the 1st entry of index table

0x0A08286C WCSS_A_ADU_UMA_INDEX_TABLE_ADDR_CTRL**Type:** RW**Reset State:** 0x00000000

This register contains the address of index table the data to be written to/read from

WCSS_A_ADU_UMA_INDEX_TABLE_ADDR_CTRL

Bits	Name	Type	Description
8	UMA_SEARCHTABLE_RW	W	Always read '0' Write '1' to write the data {uma_index_data_u, uma_index_data_l} to the address uma_index_addr Write '0' to load the contents of the index table addressed by uma_index_addr to uma_index_table_data_l[u] registers.
7:0	UMA_SEARCHTABLE_ADDR	RW	address of the index table to be written to/read from

0x0A082870 WCSS_A_ADU_UMA_INDEX_TABLE_RDATA_U**Type:** R**Reset State:** 0x00000000

This register contains the upper bits of the entry of index table for read

WCSS_A_ADU_UMA_INDEX_TABLE_RDATA_U

Bits	Name	Description
23:16	UMA_SEARCHTABLE_RDATA_INDEX	Index of the index table for read
15:0	UMA_SEARCHTABLE_RDATA_U	Upper bits of the entry of index table for read

0x0A082874 WCSS_A_ADU_UMA_INDEX_TABLE_RDATA_L**Type:** R**Reset State:** 0x00000000

This register contains the lower bits of the entry of index table for read

WCSS_A_ADU_UMA_INDEX_TABLE_RDATA_L

Bits	Name	Description
31:0	UMA_SEARCHTABLE_RDATA_L	Lower bits of the 1st entry of index table for read

0x0A082878 WCSS_A_ADU_UMA_TX_DEFAULT_WMACADDR_U**Type:** RW**Reset State:** 0x00000000

This register contains the upper bits of the default wmac addr to be used when there's no DA match

WCSS_A_ADU_UMA_TX_DEFAULT_WMACADDR_U

Bits	Name	Description
15:0	UMA_DEFAULT_WMACADDR_U	Upper bits of the default wmac addr

0x0A08287C WCSS_A_ADU_UMA_TX_DEFAULT_WMACADDR_L**Type:** RW**Reset State:** 0x00000000

This register contains the lower bits of the default wmac addr to be used when there's no DA match

WCSS_A_ADU_UMA_TX_DEFAULT_WMACADDR_L

Bits	Name	Description
31:0	UMA_DEFAULT_WMACADDR_L	Lower bits of the default wmac addr

0x0A082880 WCSS_A_ADU_UMA_CTRL2**Type:** RW**Reset State:** 0x001104FF

This register is to control the UMA function

WCSS_A_ADU_UMA_CTRL2

Bits	Name	Description
31:24	ADU_FORWARD_AMSDU_WQ	When adu_amsdu_en (bit20) is cleared to 0, the amsdu pkt will be pushed to this WQ.
20	ADU_AMSDU_EN	When set, adu will deaggregate the amsdu frame into MSDUs. When disable, the amsdu pkt will be forward to the adu_forward_amsdu WQ (bit[31:24]).
19	UMA_TX_HTC_EN	When set, htc field insertion will be enabled. When the bit in sta descriptor is set, the htc field will be inserted into wmac frame header.
18	UMA_TX_BD_EN	When set, uma tx will fill queueid(based on the configuration), sta index, dpu index, dpu sig in the BD when there's no frame translation. Ignored when FT is enabled
17	WOW_WAKEUP_SEL	When set, adu will check the BD for wow wakeup indication. When clear, adu will check the sideband signal from dpu for wow wakeup.
16	RX_PRIORITY_ROUTING_WQ_EN	When set, adu will push to the rx_priority_routing_wq when the rx dxe priority routing bit is set in BD
15:8	RX_PRIORITY_ROUTING_WQ	The WQ adu will push to when the rx dxe priority routing bit is set in BD
7:0	UMA_TX_ERROR_WQ	The WQ uma tx will push the frame when there's error

0x0A082884 WCSS_A_ADU_UMA_TX_COUNTERS3**Type:** R**Reset State:** 0x00000000

This register contains counters information of uma tx

WCSS_A_ADU_UMA_TX_COUNTERS3

Bits	Name	Description
23:16	UMA_TX_SYNC_WQ_COUNTER	Counters of the uma tx push to sync wq
7:0	UMA_TX_BYPASS_COUNTER	Counters of the uma tx bypass

0x0A0828B8 WCSS_A_ADU_UMA_TX_WQ_QID_CTRL**Type:** RW**Reset State:** 0x00000000

This register contains the control signals to enable/disable whether or not checking the threshold value for mcpu/btqm wq push

WCSS_A_ADU_UMA_TX_WQ_QID_CTRL

Bits	Name	Description
24	UMA_TX_BD_FILL_EN	When set, UMA TX fill the extra fields in BD.

0x0A0828C0 WCSS_A_ADU_ADU_POLL_CMD_ERROR_ADDR**Type:** R**Reset State:** 0x00000000

This register contains the poll cmd address when it reaches retry timeout error

WCSS_A_ADU_ADU_POLL_CMD_ERROR_ADDR

Bits	Name	Description
31:0	ADU_POLL_CMD_ERROR_ADDR	The poll cmd address when adu_poll_cmd_error is generated

0x0A0828C4 WCSS_A_ADU_ADU_SW_REG_REINIT_ADDR**Type:** RW**Reset State:** 0x00000000

This register contains the sw triggered reg reinit table address

WCSS_A_ADU_ADU_SW_REG_REINIT_ADDR

Bits	Name	Description
31:2	SW_REG_REINIT_ADDR	The sw triggered reg reinit table address. always word address

0x0A0828C8 WCSS_A_ADU_ADU_SW_REG_REINIT_TRIGGER**Type:** R**Reset State:** 0x00000000

This register to trigger the reg reinit

WCSS_A_ADU_ADU_SW_REG_REINIT_TRIGGER

Bits	Name	Description
0	SW_REG_REINIT_P	Write '1' to trigger the reg reinit. Always read 0.

0x0A0828CC WCSS_A_ADU_UMA_TX_HTC_HT**Type:** RW**Reset State:** 0x0000FFC0

This register contains the htc field in ht format to be filled in frame translation

WCSS_A_ADU_UMA_TX_HTC_HT

Bits	Name	Description
31:0	UMA_TX_HTC_HT	htc field in ht format to be filled in frame translation

0x0A0828D0 WCSS_A_ADU_UMA_TX_HTC_VHT**Type:** RW**Reset State:** 0x0000FFE1

This register contains the htc field in vht format to be filled in frame translation

WCSS_A_ADU_UMA_TX_HTC_VHT

Bits	Name	Description
31:0	UMA_TX_HTC_VHT	htc field in vht format to be filled in frame translation

0x0A0828D4 WCSS_A_ADU_UMA_TX_TID_TO_QID**Type:** RW**Reset State:** 0x76543210

This register contains the tid to queueid mapping

WCSS_A_ADU_UMA_TX_TID_TO_QID

Bits	Name	Description
31:28	QUEUEID_FOR_TID7	queueid for tid 7
27:24	QUEUEID_FOR_TID6	queueid for tid 6
23:20	QUEUEID_FOR_TID5	queueid for tid 5
19:16	QUEUEID_FOR_TID4	queueid for tid 4
15:12	QUEUEID_FOR_TID3	queueid for tid 3
11:8	QUEUEID_FOR_TID2	queueid for tid 2
7:4	QUEUEID_FOR_TID1	queueid for tid 1
3:0	QUEUEID_FOR_TID0	queueid for tid 0

0x0A0828D8 WCSS_A_ADU_UMA_TX_DEFAULT_WMACADDR_U_1**Type:** RW**Reset State:** 0x00000000

This register contains the upper bits of the default wmac addr to be used for BSSID1 when there's no DA match

WCSS_A_ADU_UMA_TX_DEFAULT_WMACADDR_U_1

Bits	Name	Description
15:0	UMA_DEFAULT_WMACADDR_U	Upper bits of the default wmac addr for BSSID1

0x0A0828DC WCSS_A_ADU_UMA_TX_DEFAULT_WMACADDR_L_1**Type:** RW**Reset State:** 0x00000000

This register contains the lower bits of the default wmac addr to be used for BSSID1 when there's no DA match

WCSS_A_ADU_UMA_TX_DEFAULT_WMACADDR_L_1

Bits	Name	Description
31:0	UMA_DEFAULT_WMACADDR_L	Lower bits of the default wmac addr for BSSID1

0x0A0828E0 WCSS_A_ADU_UMA_TX_DEFAULT_WMACADDR_U_2**Type:** RW**Reset State:** 0x00000000

This register contains the upper bits of the default wmac addr to be used for BSSID2 when there's no DA match

WCSS_A_ADU_UMA_TX_DEFAULT_WMACADDR_U_2

Bits	Name	Description
15:0	UMA_DEFAULT_WMACADDR_U	Upper bits of the default wmac addr for BSSID2

0x0A0828E4 WCSS_A_ADU_UMA_TX_DEFAULT_WMACADDR_L_2**Type:** RW**Reset State:** 0x00000000

This register contains the lower bits of the default wmac addr to be used for BSSID2 when there's no DA match

WCSS_A_ADU_UMA_TX_DEFAULT_WMACADDR_L_2

Bits	Name	Description
31:0	UMA_DEFAULT_WMACADDR_L	Lower bits of the default wmac addr for BSSID2

0x0A0828E8 WCSS_A_ADU_UMA_TX_DEFAULT_WMACADDR_U_3**Type:** RW**Reset State:** 0x00000000

This register contains the upper bits of the default wmac addr to be used for BSSID3 when there's no DA match

WCSS_A_ADU_UMA_TX_DEFAULT_WMACADDR_U_3

Bits	Name	Description
15:0	UMA_DEFAULT_WMACADDR_U	Upper bits of the default wmac addr for BSSID3

0x0A0828EC WCSS_A_ADU_UMA_TX_DEFAULT_WMACADDR_L_3**Type:** RW**Reset State:** 0x00000000

This register contains the lower bits of the default wmac addr to be used for BSSID3 when there's no DA match

WCSS_A_ADU_UMA_TX_DEFAULT_WMACADDR_L_3

Bits	Name	Description
31:0	UMA_DEFAULT_WMACADDR_L	Lower bits of the default wmac addr for BSSID3

0x0A0828F0 WCSS_A_ADU_UMA_TX_DEFAULT_STAID**Type:** RW**Reset State:** 0x00000000

This register contains the default staid for bssid1-3 for descriptor

WCSS_A_ADU_UMA_TX_DEFAULT_STAID

Bits	Name	Description
23:16	TX_DEFAULT_STAID_3	The default descriptor index of bssid3 used for frame translation in TX direction when there's no DA match in the search table.
15:8	TX_DEFAULT_STAID_2	The default descriptor index of bssid2 used for frame translation in TX direction when there's no DA match in the search table.
7:0	TX_DEFAULT_STAID_1	The default descriptor index of bssid1 used for frame translation in TX direction when there's no DA match in the search table..

0x0A0828F4 WCSS_A_ADU_ADU_REINIT_CONFIG**Type:** RW**Reset State:** 0x00000A00

This register configs the additional info of the adu reinit command to access external addr space

WCSS_A_ADU_ADU_REINIT_CONFIG

Bits	Name	Type	Description
15:8	I_WCSS_CFG_BASE_ADD RESS	R	The base address of the internal address space
7:0	SW_REG_REINIT_CMD_EX T_BASE_ADDR	RW	When the feature is enabled, the upper bits of the register address when bit24 of the command line is set to 1

10.27 WCSS_A_AGC_AGC**0x0A013C00 WCSS_A_AGC_AGC_RESET****Type:** RW**Reset State:** 0x0001

Soft reset of the AGC state machine.

WCSS_A_AGC_AGC_RESET

Bits	Name	Description
0	RESET	Reset bit. 0x0: NORMAL (Normal operation) 0x1: RESET (Reset AGC state machine)

0x0A013C04 WCSS_A_AGC_DIS_MODE**Type:** RW**Reset State:** 0x0000

Disables detection of OFDM (11a,11g)/OFDM-MIMO, Barker/CCK (11b) packets or double bandwidth 11a/MIMO packets. Useful to reduce misdetect probability in a pure 11a or 11b environment.

WCSS_A_AGC_DIS_MODE

Bits	Name	Description
3	DISABLE_11AC80	Disable quadruple bandwidth bit for 11ac80 MHz packets. 0x0: ENABLE (Enable 11ac 80MHz detection.) 0x1: DISABLE (Disable 11ac 80MHz packet detection.)
2	DISABLE_11N40	Disable double bandwidth bit for 11n40 MHz packets. 0x0: ENABLE (Enable 11n 40MHz detection.) 0x1: DISABLE (Disable 11n 40MHz packet detection.)
1	DISABLE_11B	Disable 11b bit. 0x0: ENABLE (Enable 11b packet detection.) 0x1: DISABLE (Disable 11b packet detection.)
0	DISABLE_11AG	Disable 11a/g bit. 0x0: ENABLE (Enable 11a/g packet detection.) 0x1: DISABLE (Disable 11a/g packet detection.)

0x0A013C08 WCSS_A_AGC_RX_OVERRIDE**Type:** RW**Reset State:** 0x0000

This register allows the EN_RX (Receive Enable) outputs of the BBIC, the standby input and the powerdown inputs of the built-in ADC to be software- controlled. These outputs are used to enable the AD converters and RF ASICs for debugging and calibration purposes. The power down bits are used to put all of the ADCs in power down mode (for instance in beacon power save mode) or part of its ADCs to be powered down (for instance a product not using all of the antenna).

WCSS_A_AGC_RX_OVERRIDE

Bits	Name	Description
4	ENRX_VAL	Enable values for receive chain 0 to 0. For these values to take effect, the corresponding override bits should be set.
0	STBY_VAL	ADC standby values for receive chain 0 to 0. Note that the standby values are asserted low, and de-asserted high. For these values to take effect, the corresponding override bits should be set.

0x0A013C0C WCSS_A_AGC_PHY_LOOPBACK**Type:** RW**Reset State:** 0x0008

Configures support for PHY loopback.

WCSS_A_AGC_PHY_LOOPBACK

Bits	Name	Description
3	DIG_LOOPBACK	Enables the loopback digitally.
2	PMI_EN	Enables loop backed data to go out the PMI to the MAC.
1:0	MODE	Sets the PHY loopback mode. PHY loopback can be enabled statically, where it is always in loopback mode, or dynamically where it only enters loopback mode when we're transmitting. 0x0: DISABLED (Disabled) 0x1: STATIC_EN (Always enabled) 0x2: DYN_EN (Only enabled during packet transmission)

0x0A013C10 WCSS_A_AGC_CONFIG_XBAR**Type:** RW**Reset State:** 0x0000

Sets the configuration for the digital / analog crossbar. This will configure how a particular analog receive chain is mapped to a particular digital RX processing chain. In addition, it will also configure how a particular digital TX processing chain is connected to a particular analog transmit chain. An analog chain includes the baseband analog + the corresponding RF chain. Although there are separate controls for txxbar and rxxbar, for normal mode of operation, they should be set the same. Refer to the System Engineering Specification on the edge portal for the configuration and connectivity of xbar. Note that no analog chain other than analog chain 2 can feed the digital chain 2 and hence there is no xbar functionality on digital chain 2

WCSS_A_AGC_CONFIG_XBAR

Bits	Name	Description
15	PHYDBG_SEL	PHYDBG input select. Used for playing PHYDBG data into RX ADC path.
14	IQSWAP_EN	Enables swapping of the I and Q rails of the ADC.

0x0A013C14 WCSS_A_AGC_CW_DETECT**Type:** RW**Reset State:** 0x000B

At packet detection, the PHY detects narrowband interference to avoid false alarms caused by narrowband interference. This type of interference could arise as the result of LO leakage, LO interference of other IEEE802.11 devices in the vicinity of the receiver and other, intentional radiators, such as Bluetooth devices. of factors the explanation of which is beyond the scope of this guide. When CW is found the packet detection is ignored since it is most likely that the packet detectors triggered on the CW interference. The CW frequency is used to set a CW notchfilter

inside the RXFIR to cancel out that interferer in the future, until a different CW interferer is found or until a timer expires.

WCSS_A_AGC_CW_DETECT

Bits	Name	Description
7	TXLO_DIS	Disables the use of the Rxfir with a TX LO notch. 0x0: ENABLE (Enable TX LO Notch Filter.) 0x1: DISABLE (Disable TX LO Notch Filter.)
6	CW_EN_RXB_DATA	Enables CW notch processing when the AGC is in states 9 or 11. 0x0: DISABLE (Disable CW Notch Filter when processing RXB data.) 0x1: ENABLE (Enable CW Notch Filter when processing RXB data.)
5	CW_EN_RXB	Enables CW notch processing when the AGC is in states 2,3 or 7. 0x0: DISABLE (Disable CW Notch Filter when processing RXB preamble.) 0x1: ENABLE (Enable CW Notch Filter when processing RXB preamble.)
4	CW_EN_RXA_DATA	Enables CW notch processing when the AGC is in states 8 or 10. 0x0: DISABLE (Disable CW Notch Filter when processing RXA data.) 0x1: ENABLE (Enable CW Notch Filter when processing RXA data.)
3	CW_EN_RXA	Enables CW notch processing when the AGC is in states 2,3 or 6. 0x0: DISABLE (Disable CW Notch Filter when processing RXA preamble.) 0x1: ENABLE (Enable CW Notch Filter when processing RXA preamble.)
2	DC_DIS	Disables the use of the Rxfir with a DC notch. Note that this bitfield should be set in conjunction with the RACTL's rxfir_scaling_coeff register. 0x0: ENABLE (Enable DC Notch Filter.) 0x1: DISABLE (Disable DC Notch Filter.)
1	RADAR_DETECT_DIS	0x1: DISABLE (Disable use of CW detector during radar detection.) 0x0: ENABLE (Enable use of CW detector during radar detection .)
0	PKT_DETECT_DIS	0x1: DISABLE (Disable use of CW detects to abort packet detects.) 0x0: ENABLE (Enable use of CW detects to abort packet detects. .)

0x0A013C18 WCSS_A_AGC_CW_NOTCH_TIMEOUT

Type: RW

Reset State: 0x0064

The CW notch filter applies a notch to the receiver input signal at a frequency as measured by the CW detector. The notch is applied until the CW detector passes a different value to the notch filter, or until the notch filter's built-in timer expires. When this timer expires the input signal bypasses the notch filter altogether. The timeout value of the timer is defined in this register.

WCSS_A_AGC_CW_NOTCH_TIMEOUT

Bits	Name	Description
8	DIS	Setting this field to 1 is equivalent to setting an infinite timeout value. When set, the cw notch index will remain the same until is set to a different value.
7:0	VAL	Timeout value. This value is multiplied by 2^{19} to get the number of 80 MHz clock cycles that make up the timeout time. 0x1: MINIMUM ($100 * 2^{19} = 0.65$ seconds) 0x64: DEFAULT ($1 * 2^{19} = 6.5$ milliseconds) 0xFF: MAXIMUM ($255 * 2^{19} = 1.67$ seconds)

0x0A013C1C WCSS_A_AGC_GAIN_MODE

Type: RW

Reset State: 0x0000

WCSS_A_AGC_GAIN_MODE

Bits	Name	Description
1	MULT_STEP	This bit defines if only a single COARSE_STEP or SAT_STEP is made or if multiple steps are possible. 0x0: SING (Single sat/coarse gain step) 0x1: MULT (Multiple sat/coarse gain steps)
0	USE_RXFIR_PWR	Setting this field to 1 will cause the gain computation to come from the power measured in the RXFIR power detector instead of the ADC power detector.

0x0A013C24 WCSS_A_AGC_GAINSET0

Type: RW

Reset State: 0x0000

Allows to break AGC gain setting loop and manually override AGC setting of chain 0. For debugging purposes.

WCSS_A_AGC_GAINSET0

Bits	Name	Description
7	OVERRIDE	Override bit. 0x0: NORMAL (automatic gain control) 0x1: OVERRIDE (manual gain setting)
6:0	GAIN	Gain setting.

0x0A013C28 WCSS_A_AGC_GAINSET_WRITE**Type:** W**Reset State:** 0x0000

When a 1 is written to this register, the gain registers of the RFIC(s) are loaded with the values stored in bit [3:0] of registers GAINSET0. 0, provided that the override bit [4] these registers is set. At the moment that these registers are written, the gain that is programmed in the RFIC receivers will remain at its previously programmed value. Writing to this register enforces a write of the gain values to the RFIC gain registers, which would otherwise happen only when the AGC decides to change gain when a packet is received or a strong signal clips the RF frontend, or when the AGC is reset (through RX_RESET signal, SLEEP signal, packet transmission or by writing to AGC_RESET register. On the chains on which the override bit is not set, the current gain remains unchanged. This register is provided for debugging purposes. This register cannot be read back

WCSS_A_AGC_GAINSET_WRITE

Bits	Name	Description
0	OVERRIDE	Gain write pulse bit. 0x0: NORMAL (Idle) 0x1: OVERRIDE (Write gain settings to the RFIC)

0x0A013C2C WCSS_A_AGC_STAT_CTRL**Type:** RW**Reset State:** 0x0003

Control for AGC specific stat info

WCSS_A_AGC_STAT_CTRL

Bits	Name	Description
1	HTSTS_RSSI_EN	Enables RSSI measurements on (V)HT-STS symbol if there is one. Otherwise it will only use RSSI from the L-LTF.
0	RSSI_TYPE	0x0: ADC (Output ADC RSSI) 0x1: RXFIR (Output RXFIR RSSI)

0x0A013C30 WCSS_A_AGC_SUBBAND_CONFIG**Type:** RW**Reset State:** 0x0000

Sets subband configuration.

WCSS_A_AGC_SUBBAND_CONFIG

Bits	Name	Description
3:2	STG3_SUBBAND	Selects the subband location for the stage 3 RxFIR filter. 0x0: LOWER 0x1: UPPER 0x2: CENTER 0x3: RESERVED
1:0	STG2_SUBBAND	Selects the subband location for the stage 2 RxFIR filter. 0x0: LOWER 0x1: UPPER 0x2: CENTER 0x3: RESERVED

0x0A013C34 WCSS_A_AGC_CHANNEL_FREQ**Type:** RW**Reset State:** 0x096C

This stores the channel frequency in MHz is stored used to calculate resampler offset for 11b operation. This register needs to be reprogrammed every time the MAC switches to a different RF channel. Width is 13 bits, so 11b mode can also operate in 5 GHz band, if the need arises.

WCSS_A_AGC_CHANNEL_FREQ

Bits	Name	Description
12:0	FREQ	Channel frequency 0x96C: CH1 (Channel 1 (2412 MHz)) 0x971: CH2 (Channel 2 (2417 MHz)) 0x976: CH3 (Channel 3 (2422 MHz)) 0x97B: CH4 (Channel 4 (2427 MHz)) 0x980: CH5 (Channel 5 (2432 MHz)) 0x985: CH6 (Channel 6 (2437 MHz)) 0x98A: CH7 (Channel 7 (2442 MHz)) 0x98F: CH8 (Channel 8 (2447 MHz)) 0x994: CH9 (Channel 9 (2452 MHz)) 0x999: CH10 (Channel 10 (2457 MHz)) 0x99E: CH11 (Channel 11 (2462 MHz)) 0x9A3: CH12 (Channel 12 (2467 MHz)) 0x9A8: CH13 (Channel 13 (2472 MHz)) 0x9B4: CH14 (Channel 14 (2484 MHz))

0x0A013C38 WCSS_A_AGC_N_ACTIVE**Type:** RW**Reset State:** 0x0001

Specifies number of receive chains used to receive a packet.

WCSS_A_AGC_N_ACTIVE

Bits	Name	Description
0	NUMBER	Receive chains.

0x0A013C3C WCSS_A_AGC_N_LISTEN**Type:** RW**Reset State:** 0x0011

Specifies number of receive chains used to listen for a packet.

WCSS_A_AGC_N_LISTEN

Bits	Name	Description
4	RF_ANT_EN	If number of listen chains is 0, then this specifies which RF antennas to enable during listen mode.
0	NUMBER	Listen chains.

0x0A013C40 WCSS_A_AGC_N_MEASURE**Type:** RW**Reset State:** 0x0001

Specifies number of receive chains that is switched on to measure power on.

WCSS_A_AGC_N_MEASURE

Bits	Name	Description
0	NUMBER	Measure chains.

0x0A013C44 WCSS_A_AGC_N_CAPTURE**Type:** RW**Reset State:** 0x0011

Specifies number of receive chains used to capture a packet.

WCSS_A_AGC_N_CAPTURE

Bits	Name	Description
4	RF_ANT_EN	If number of capture chains is 0, then this specifies which RF antennas to enable during capture mode.
0	NUMBER	Receive chains.

0x0A013C48 WCSS_A_AGC_CCA_MODE**Type:** RW**Reset State:** 0x0B6D

Determines how the CCA signal from the PHY to the MAC is generated.

WCSS_A_AGC_CCA_MODE

Bits	Name	Description
12	FORCE_TX_CCA_IDLE	Setting to 1 forces the Tx CCA to be idle all the time. For debug purposes only.

WCSS_A_AGC_CCA_MODE (cont.)

Bits	Name	Description
11:9	SB3	Clear Channel Assessment Mode for channel subband 3 0x0: CLEAR 0x1: BUSY 0x2: GI 0x3: ED_OR_GI 0x4: GI_OR_CS_AND_CD 0x5: ED_OR_GI_AND_CS_AND_CD
8:6	SB2	Clear Channel Assessment Mode for channel subband 2 0x0: CLEAR 0x1: BUSY 0x2: GI 0x3: ED_OR_GI 0x4: GI_OR_CS_AND_CD 0x5: ED_OR_GI_AND_CS_AND_CD
5:3	SB1	Clear Channel Assessment Mode for channel subband 1 0x0: CLEAR 0x1: BUSY 0x2: GI 0x3: ED_OR_GI 0x4: GI_OR_CS_AND_CD 0x5: ED_OR_GI_AND_CS_AND_CD
2:0	SB0	Clear Channel Assessment Mode for channel subband 0 0x0: CLEAR 0x1: BUSY 0x2: GI 0x3: ED_OR_GI 0x4: GI_OR_CS_AND_CD 0x5: ED_OR_GI_AND_CS_AND_CD

0x0A013C4C WCSS_A_AGC_CCA_GI_DET_CFG**Type:** RW**Reset State:** 0x017D**WCSS_A_AGC_CCA_GI_DET_CFG**

Bits	Name	Description
9:1	TH_LGI	
0	DIS	Disable for the GI-based cca detector.

0x0A013C50 WCSS_A_AGC_TH_CD20**Type:** RW**Reset State:** 0x0000

If the 11a or 11b packet correlator detects a packet and the RSSI on receive Chain 0 exceeds this threshold, then the carrier detect flag is set. Depending on the value of the CCA mode register this determines the behavior of the CCA signal from the PHY to the MAC. Value 8'hff is used to disable this check.

WCSS_A_AGC_TH_CD20

Bits	Name	Description
6:0	TH	CD Threshold. Default value (0 = -?? dB below max level).

0x0A013C54 WCSS_A_AGC_TH_CD40**Type:** RW**Reset State:** 0x0000

If the 11a or 11b packet correlator detects a packet and the RSSI on receive Chain 0 exceeds this threshold, then the carrier detect flag is set. Depending on the value of the CCA mode register this determines the behavior of the CCA signal from the PHY to the MAC. Value 8'hff is used to disable this check.

WCSS_A_AGC_TH_CD40

Bits	Name	Description
6:0	TH	CD Threshold. Default value (0 = -?? dB below max level).

0x0A013C58 WCSS_A_AGC_TH_CD80**Type:** RW**Reset State:** 0x0000

If the 11a or 11b packet correlator detects a packet and the RSSI on receive Chain 0 exceeds this threshold, then the carrier detect flag is set. Depending on the value of the CCA mode register this determines the behavior of the CCA signal from the PHY to the MAC. Value 8'hff is used to disable this check.

WCSS_A_AGC_TH_CD80

Bits	Name	Description
6:0	TH	CD Threshold. Default value (0 = -?? dB below max level).

0x0A013C5C WCSS_A_AGC_TH_CS20**Type:** RW**Reset State:** 0x0012

If the 11a or 11b packet correlator detects a packet and the RSSI on receive Chain 0 exceeds the carrier detect threshold, and the received energy also exceeds the carrier sense threshold, then the carrier sense flag is set. Depending on the value of the CCA mode register this determines the behavior of the CCA signal from the PHY to the MAC. Value 8'hff is used to disable this check.

WCSS_A_AGC_TH_CS20

Bits	Name	Description
6:0	TH	CS Threshold Default value (-82 dBm)

0x0A013C60 WCSS_A_AGC_TH_CS40**Type:** RW**Reset State:** 0x0012

If the 11a or 11b packet correlator detects a packet and the RSSI on receive Chain 0 exceeds the carrier detect threshold, and the received energy also exceeds the carrier sense threshold, then the carrier sense flag is set. Depending on the value of the CCA mode register this determines the behavior of the CCA signal from the PHY to the MAC. Value 8'hff is used to disable this check.

WCSS_A_AGC_TH_CS40

Bits	Name	Description
6:0	TH	CS Threshold Default value (-82 dBm)

0x0A013C64 WCSS_A_AGC_TH_CS80**Type:** RW**Reset State:** 0x0012

If the 11a or 11b packet correlator detects a packet and the RSSI on receive Chain 0 exceeds the carrier detect threshold, and the received energy also exceeds the carrier sense threshold, then the carrier sense flag is set. Depending on the value of the CCA mode register this determines the behavior of the CCA signal from the PHY to the MAC. Value 8'hff is used to disable this check.

WCSS_A_AGC_TH_CS80

Bits	Name	Description
6:0	TH	CS Threshold Default value (-82 dBm)

0x0A013C68 WCSS_A_AGC_TH_EDET**Type:** RW**Reset State:** 0x2626

If the RSSI on receive chain 0 exceeds this threshold then the energy detect flag is set. Depending on the value of the CCA mode register this determines the behavior of the CCA signal from the PHY to the MAC. Value 8'hff is used to disable this check.

WCSS_A_AGC_TH_EDET

Bits	Name	Description
14:8	TH20	ED Threshold. Default value (-62 dBm)
6:0	TH40	ED Threshold. Default value (-62 dBm)

0x0A013C6C WCSS_A_AGC_D_AGC_A**Type:** RW**Reset State:** 0x0000

This latency is the sum of three components. The ADC latency, the latency through the FIR and CAL module and the latency for AGC 11a processing, which is determined by interference locator processing latency. Ranges from 0 to 1.575 micro seconds.

WCSS_A_AGC_D_AGC_A

Bits	Name	Description
5:0	DELAY	Propagation time in 80/40MHz dynamic clocks.

0x0A013C70 WCSS_A_AGC_D_AGC_B**Type:** RW**Reset State:** 0x0006

This latency is the sum of two components. The ADC latency, and the latency through the FIR and CAL module. Ranges from 0 to 1.575 micro seconds.

WCSS_A_AGC_D_AGC_B

Bits	Name	Description
5:0	DELAY	Propagation time in 50ns units. Default value is 6, corresponding to 300ns.

0x0A013C74 WCSS_A_AGC_RSSI_OFFSET0**Type:** RW**Reset State:** 0x003A

This register holds an offset value (determined at manufacturing time) that is used to translate rssi values to dBm values as measured at the antenna input. The same offset is used to correct both the ADC and FIR RSSI.

WCSS_A_AGC_RSSI_OFFSET0

Bits	Name	Description
7:0	OFFSET	Offset is a 2cc encoded number that is added to the rssi computed by the power module.

0x0A013C78 WCSS_A_AGC_RSSI_BT_OFFSET0**Type:** RW**Reset State:** 0x0000

When a Bluetooth transmission takes place the RSSI on one of the receive antennae is reduced. This register is used to compensate for that attenuation change so that the reported RSSI is the true dBm number at the antenna.

WCSS_A_AGC_RSSI_BT_OFFSET0

Bits	Name	Description
7:0	OFFSET	Offset is a 2cc encoded number that is added to the rssi computed by the power module.

0x0A013C7C WCSS_A_AGC_CLIP_COUNT**Type:** W**Reset State:** 0x00000000

Debug Counter. Counts transitions between CORR state and GAIN_SETTLE state due to ADC clip events.

WCSS_A_AGC_CLIP_COUNT

Bits	Name	Description
15:0	COUNT	Counter output. This counter is reset by writing a 0 to it.

0x0A013C80 WCSS_A_AGC_SAT_COUNT**Type:** W**Reset State:** 0x00000000

Debug Counter. Counts transitions between CORR state and GAIN_SETTLE state due to RF sat events.

WCSS_A_AGC_SAT_COUNT

Bits	Name	Description
15:0	COUNT	Counter output. This counter is reset by writing a 0 to it.

0x0A013C84 WCSS_A_AGC_PIP_COUNT**Type:** W**Reset State:** 0x00000000

Debug Counter. Counts the number of packet in packet detection (power_jum_det during some of the states)

WCSS_A_AGC_PIP_COUNT

Bits	Name	Description
7:0	COUNT	Counter output. This counter is reset by writing a 0 to it.

0x0A013C88 WCSS_A_AGC_RXA_NOT_AVALID_COUNT**Type:** W**Reset State:** 0x00000000

BTCF not_avalid signal

WCSS_A_AGC_RXA_NOT_AVALID_COUNT

Bits	Name	Description
15:0	COUNT	Counter output. This counter is reset by writing a 0 to it.

0x0A013C8C WCSS_A_AGC_RXA_HDR_INVALID_COUNT**Type:** W**Reset State:** 0x00000000

RXA header invalid counter.

WCSS_A_AGC_RXA_HDR_INVALID_COUNT

Bits	Name	Description
15:0	COUNT	Counter output. This counter is reset by writing a 0 to it.

0x0A013C90 WCSS_A_AGC_RXB_NO_B_PKT_DET_COUNT**Type:** W**Reset State:** 0x00000000**WCSS_A_AGC_RXB_NO_B_PKT_DET_COUNT**

Bits	Name	Description
15:0	COUNT	Counter output. This counter is reset by writing a 0 to it.

0x0A013C94 WCSS_A_AGC_RXB_INV_HDR_COUNT**Type:** W**Reset State:** 0x00000000**WCSS_A_AGC_RXB_INV_HDR_COUNT**

Bits	Name	Description
15:0	COUNT	Counter output. This counter is reset by writing a 0 to it.

0x0A013C98 WCSS_A_AGC_RXB_ILL_HDR_COUNT**Type:** W**Reset State:** 0x00000000**WCSS_A_AGC_RXB_ILL_HDR_COUNT**

Bits	Name	Description
15:0	COUNT	Counter output. This counter is reset by writing a 0 to it.

0x0A013C9C WCSS_A_AGC_COARSE_DCO_UPDATE_COUNTER**Type:** W**Reset State:** 0x00000000

WCSS_A_AGC_COARSE_DCO_UPDATE_COUNTER

Bits	Name	Description
7:0	VALUE	Coarse DCO measurement update counter.

0x0A013CA0 WCSS_A_AGC_FALSE_AC_ABORT_COUNT**Type:** W**Reset State:** 0x00000000

Counts number of aborts of legacy packets that occur due to clip events in potential VHT-STS.

WCSS_A_AGC_FALSE_AC_ABORT_COUNT

Bits	Name	Description
7:0	COUNT	Counter output. This counter is reset by writing a 0 to it.

0x0A013CA4 WCSS_A_AGC_LATE_AC_DET_COUNT**Type:** W**Reset State:** 0x00000000

Counts number of late 11AC detects that cause missed VHT-STS gain commands.

WCSS_A_AGC_LATE_AC_DET_COUNT

Bits	Name	Description
7:0	COUNT	Counter output. This counter is reset by writing a 0 to it.

0x0A013CA8 WCSS_A_AGC_CCA_PKTDET_STATUS**Type:** R**Reset State:** 0x00000000

Samples the current value of the CCA and pktdet flags

WCSS_A_AGC_CCA_PKTDET_STATUS

Bits	Name	Description
6	LSIG_TXOP	
5	EARLY_PKTDET_N	
4	PKTDET_N	

WCSS_A_AGC_CCA_PKTDET_STATUS (cont.)

Bits	Name	Description
3:0	CCA	

0x0A013CAC WCSS_A_AGC_SAMPLE_ADC_I0**Type:** R**Reset State:** 0x00000000**WCSS_A_AGC_SAMPLE_ADC_I0**

Bits	Name	Description
11	OVFL	Overflow bit. 0x0: NO_OVFL (No overflow) 0x1: OVFL (Overflow)
10:0	ADC	ADC output, offset binary encoded ADC Chain 0 i output.

0x0A013CB0 WCSS_A_AGC_SAMPLE_ADC_Q0**Type:** R**Reset State:** 0x00000000**WCSS_A_AGC_SAMPLE_ADC_Q0**

Bits	Name	Description
11	OVFL	Overflow bit. 0x0: NO_OVFL (No overflow) 0x1: OVFL (Overflow)
10:0	ADC	ADC output, offset binary encoded ADC Chain 0 q output.

0x0A013CB4 WCSS_A_AGC_ADC_RSSI0**Type:** R**Reset State:** 0x00000000

Read out RSSI measured right after ADC0 while not receiving a packet. Invalid bit indicates that RSSI is not current, since a packet is being received. The value is the RSSI stored just before the AGC detected the packet.

WCSS_A_AGC_ADC_RSSI0

Bits	Name	Description
8	NAV_BUSY	NAV status which helps software to allow/ignore the sampled rssi value for 11k measurements. 0x0: IDLE (NAV idle) 0x1: BUSY (NAV busy)
7	INVALID	RSSI0 not current 0x0: VALID (RSSI valid on antenna 0) 0x1: INVALID (RSSI invalid on antenna 0)
6:0	RSSI	RSSI measurement on Chain 0 ranging from 0 (lowest RSSI) to 113 (highest RSSI), measured in dB.

0x0A013CB8 WCSS_A_AGC_FIR_RSSI0**Type:** R**Reset State:** 0x00000000

Read out RSSI measured after FIR while not receiving a packet. Invalid bit indicates that RSSI is not current, since a packet is being received. The value is the RSSI stored just before the AGC detected the packet.

WCSS_A_AGC_FIR_RSSI0

Bits	Name	Description
8	NAV_BUSY	NAV status which helps software to allow/ignore the sampled rssi value for 11k measurements. 0x0: IDLE (NAV idle) 0x1: BUSY (NAV busy)
7	INVALID	RSSI0 not current 0x0: VALID (RSSI valid on antenna 0) 0x1: INVALID (RSSI invalid on antenna 0)
6:0	RSSI	RSSI measurement on Chain 0 ranging from 0 (lowest RSSI) to 113 (highest RSSI), measured in dB.

0x0A013CBC WCSS_A_AGC_COARSE_DCO_CHAIN0_REAL**Type:** R**Reset State:** 0x0000

This register stores the measured DC offset value from the AGC_POWER module if the value exceeds the th_dco_cal parameter during CORR state and (rx_gain == INIT_GAIN). The DC offset is measured from the ADC data with the LSB truncated.

WCSS_A_AGC_COARSE_DCO_CHAIN0_REAL

Bits	Name	Description
9:0	VALUE	Value of DC offset on the real component of chain 0 measured by the AGC_POWER module

0x0A013CC0 WCSS_A_AGC_COARSE_DCO_CHAIN0_IMAG**Type:** R**Reset State:** 0x0000

This register stores the measured DC offset value from the AGC_POWER module if the value exceeds the th_dco_cal parameter during CORR state and (rx_gain == INIT_GAIN). The DC offset is measured from the ADC data with the LSB truncated.

WCSS_A_AGC_COARSE_DCO_CHAIN0_IMAG

Bits	Name	Description
9:0	VALUE	Value of DC offset on the image component of chain 0 measured by the AGC_POWER module

0x0A013CC4 WCSS_A_AGC_INIT_GAIN**Type:** RW**Reset State:** 0x0048

Initial gain setting in 1dB steps.

WCSS_A_AGC_INIT_GAIN

Bits	Name	Description
6:0	GAIN	Gain setting in 1dB steps.

0x0A013CC8 WCSS_A_AGC_MAX_GAIN**Type:** RW**Reset State:** 0x0048

maximum gain that can be commanded to RF ASIC.

WCSS_A_AGC_MAX_GAIN

Bits	Name	Description
6:0	GAIN	For RFIC operation the 7 bits denote 1 dB steps.

0x0A013CCC WCSS_A_AGC_GAIN_OFFSET0**Type:** RW**Reset State:** 0x0000

receiver gain offset in 1dB steps per chain

WCSS_A_AGC_GAIN_OFFSET0

Bits	Name	Description
4:0	VAL	

0x0A013CD0 WCSS_A_AGC_TR_SW_GAIN_THRESH0**Type:** RW**Reset State:** 0x0000

The T/R switch is driven to transmit mode whenever the commanded gain index is below this threshold. Setting this to 0 effectively disables this.

WCSS_A_AGC_TR_SW_GAIN_THRESH0

Bits	Name	Description
6:0	GAIN	

0x0A013CD4 WCSS_A_AGC_SAT_STEP**Type:** RW**Reset State:** 0x0072

The sat_step field is used to specify the step used to gain the RF Frontend down if the RF is saturated. The same value is used to step up the gain if the signal energy is below TH_NO_SIGNAL. This gain step is specified in multiples of 2dB from 0 dB to 128 dB. The enable field is used to switch this functionality on or off.

WCSS_A_AGC_SAT_STEP

Bits	Name	Description
7	IGNORE_ADC_CLIP	Normally, a sat step is only taken if a digital (or ADC) clip is also seen at the same time, but if you set this bit to 1, then only the RF sat flag is used.
6	SAT_STEP_EN	Allow AGC to make gain step when in Vega mode RF frontend is in saturation. 0x0: DISABLE (Saturated gain step disabled) 0x1: ENABLE (Saturated gain step enabled)
5:0	SAT_STEP	Coarse gain step. Default value is 25, or 50dB step for Vega mode

0x0A013CD8 WCSS_A_AGC_COARSE_STEP_HT**Type:** RW**Reset State:** 0x0514

Coarse gain step. Used to step gain down if one chain's ADCs clip. Used to step up if the signal energy is below TH_NO_SIGNAL. Specified in multiples of 4dB steps, from 0 dB to 60 dB.

WCSS_A_AGC_COARSE_STEP_HT

Bits	Name	Description
14	VHT_STS_FINE_GAIN_SET_DIS	Enable VHT-STS fine gain setting 0x1: DISABLE (Disable VHT-STS fine gain setting.) 0x0: ENABLE (Enable VHT-STS fine gain setting)
13	VHT_STS_COARSE_GAIN_SET_DIS	Enable VHT-STS coarse gain setting 0x1: DISABLE (Disable VHT-STS coarse gain setting.) 0x0: ENABLE (Enable VHT-STS coarse gain setting)
12	HT_STS_GAIN_SET_DIS	Enable HT-STS gain setting 0x1: DISABLE (Disable HT-STS gain setting.) 0x0: ENABLE (Enable HT-STS gain setting)
11:6	VHTSTS_STEP	Coarse gain step for 2nd STF in mixed mode packets.
5:0	HTSTS_STEP	Coarse gain step.

0x0A013CDC WCSS_A_AGC_COARSE_STEP**Type:** RW**Reset State:** 0x0059

Coarse gain step. Used to step gain down if one chain's ADCs clip. Used to step up if the signal energy is below TH_NO_SIGNAL. Specified in multiples of 4dB steps, from 0 dB to 60 dB.

WCSS_A_AGC_COARSE_STEP

Bits	Name	Description
6	COARSE_STEP_EN	Allow AGC to make coarse gain step when digital clip is detected. 0x0: DISABLE (Coarse gain step disabled) 0x1: ENABLE (Coarse gain step enabled)
5:0	COARSE_STEP	Coarse gain step.

0x0A013CE0 WCSS_A_AGC_MAX_POWER_DIFF**Type:** RW**Reset State:** 0x0004

The value in this register specifies maximum power difference. If the received power on all any antenna is this much less than the power on the strongest antenna, then the AGC limits the gain for the chain with the less power to that corresponding with a power of the strongest power minus the value in this register. This threshold avoids gaining up receive chains with weak signal and thereby blowing up noise. antenna.

WCSS_A_AGC_MAX_POWER_DIFF

Bits	Name	Description
3:0	DIFFERENCE	Maximum power difference in 4 dB steps. From 0 .. 14*4=56 dB

0x0A013CE4 WCSS_A_AGC_TARGET_BO_ACTIVE**Type:** RW**Reset State:** 0x0020

Target Backoff for Listen chains. The gain is set this amount of dBs below the gain that would fully saturate the receiver. The backoff is specified in 0.5 dB steps.

WCSS_A_AGC_TARGET_BO_ACTIVE

Bits	Name	Description
5:0	BACK_OFF	Backoff value. Default backoff is 32, or 16 dB.

0x0A013CE8 WCSS_A_AGC_TARGET_BO_INACTIVE**Type:** RW**Reset State:** 0x0024

Target Backoff for Receive chains, that are not listen chains. The gain is set this amount of dBs below the gain that would fully saturate the receiver. This backoff is typically higher than that of listen antennas to accommodate the unknown upfading on the inactive receive chains. The backoff is specified in 0.5 dB steps.

WCSS_A_AGC_TARGET_BO_INACTIVE

Bits	Name	Description
5:0	BACK_OFF	Backoff value. Default backoff is 36, i.e. 18 dB.

0x0A013CEC WCSS_A_AGC_TH_CLIP_HIGH

Type: RW

Reset State: 0x0734

If the output of any of the ADCs is greater than this value the clip flag of the corresponding ADC is asserted.

WCSS_A_AGC_TH_CLIP_HIGH

Bits	Name	Description
10:0	THRESHOLD	Clip threshold in offset binary. Example: if the 2's complement threshold is 400(0x190), then the 11-bit offset binary code is $400 + 1024 = 1424$, +800 corresponds to 1824

0x0A013CF0 WCSS_A_AGC_TH_CLIP_LOW

Type: RW

Reset State: 0x00CC

If the output of any of the ADCs is less than this value the clip flag of the corresponding ADC is asserted.

WCSS_A_AGC_TH_CLIP_LOW

Bits	Name	Description
10:0	THRESHOLD	Clip threshold in offset binary. Example: if the 2's complement threshold is -400, then the 11-bit offset binary code is $-400 + 1024 = 624$. -800 corresponds to 224

0x0A013CF4 WCSS_A_AGC_TH_SIGNAL_HIGH

Type: RW

Reset State: 0x00B4

AGC compares power with this threshold. If power is higher, a coarse gain reduction is done. In steps of 16384 (2^{14}).

WCSS_A_AGC_TH_SIGNAL_HIGH

Bits	Name	Description
7:0	THRESHOLD	Power threshold. Default value 110 (1802240).

0x0A013CF8 WCSS_A_AGC_TH_SIGNAL_LOW

Type: RW

Reset State: 0x0046

AGC compares power with this threshold. If power is lower, a coarse gain reduction is done if the ADCs clip. Specified in steps of 1024 (2^{10}).

WCSS_A_AGC_TH_SIGNAL_LOW

Bits	Name	Description
7:0	THRESHOLD	Power threshold. Default value 60 (61440).

0x0A013CFC WCSS_A_AGC_TH_HTSTS_CLIP

Type: RW

Reset State: 0x006E

Minimum ADC power during HT-STTS needed for clip detection.

WCSS_A_AGC_TH_HTSTS_CLIP

Bits	Name	Description
7:0	THRESHOLD	Power threshold. Default value 110 (1802240).

0x0A013D00 WCSS_A_AGC_TH_VHTSTS_CLIP

Type: RW

Reset State: 0x006E

Minimum ADC power during VHT-STTS needed for clip detection.

WCSS_A_AGC_TH_VHTSTS_CLIP

Bits	Name	Description
7:0	THRESHOLD	Power threshold. Default value 110 (1802240).

0x0A013D04 WCSS_A_AGC_TH_POWER_CLIP**Type:** RW**Reset State:** 0x003F

Used to do extra gain reduction if an antenna is clipping, but there is no time to do a coarse step.

WCSS_A_AGC_TH_POWER_CLIP

Bits	Name	Description
5:0	THLD	Power clip threshold.

0x0A013D08 WCSS_A_AGC_SAT_DELAY**Type:** RW**Reset State:** 0x0010**WCSS_A_AGC_SAT_DELAY**

Bits	Name	Description
5:0	TIME	

0x0A013D0C WCSS_A_AGC_CLIP_WINDOW**Type:** RW**Reset State:** 0x0000**WCSS_A_AGC_CLIP_WINDOW**

Bits	Name	Description
11:6	THRESHOLD	Threshold for the minimum number of clips needed in the window for a true clip to be detected.
5:0	TIME	Window size in 80MHz clocks that clips are monitored.

0x0A013D10 WCSS_A_AGC_TH_NO_SIGNAL**Type:** RW**Reset State:** 0x0001

Decision Threshold for absence of signal energy, which will result in the AGC stepping up the gain, if it is not already at its maximum setting.

WCSS_A_AGC_TH_NO_SIGNAL

Bits	Name	Description
7:0	THRESHOLD	Detection threshold. Default value 1.

0x0A013D14 WCSS_A_AGC_TH_DCO_CAL**Type:** RW**Reset State:** 0x000A

This register holds the RSSI threshold to trigger capture of the DCO value measured by AGC_POWER during CORR state

WCSS_A_AGC_TH_DCO_CAL

Bits	Name	Description
6:0	THLD	Coarse DCO measurement threshold.

0x0A013D18 WCSS_A_AGC_DET_DATA_SHIFT**Type:** RW**Reset State:** 0x0005

Right shift the data by this amount

WCSS_A_AGC_DET_DATA_SHIFT

Bits	Name	Description
3:2	DET11A	Right shift the det11a detector input by this amount
1:0	DET11B	Right shift the det11b detector input by this amount

0x0A013D1C WCSS_A_AGC_TH_RXB_PWR_HIGH**Type:** RW**Reset State:** 0x0005

This threshold is used to by the AGC to set the digital AGC shift value for det11b and RXB.

WCSS_A_AGC_TH_RXB_PWR_HIGH

Bits	Name	Description
5:0	THLD	11B Digital AGC power setting threshold.

0x0A013D20 WCSS_A_AGC_TH_CW_LOW

Type: RW

Reset State: 0x0020

The CW detect threshold is set to this value if non of the RSSI values exceeds the th_cw_rssi threshold.

WCSS_A_AGC_TH_CW_LOW

Bits	Name	Description
7:0	THRESHOLD	CW detect threshold.

0x0A013D24 WCSS_A_AGC_TH_CW_HIGH

Type: RW

Reset State: 0x00C8

The CW interference detect threshold is set to this value if one or more of the RSSI values exceeds the th_cw_rssi threshold.

WCSS_A_AGC_TH_CW_HIGH

Bits	Name	Description
7:0	THRESHOLD	CW detect threshold.

0x0A013D28 WCSS_A_AGC_TH_CW_RSSI

Type: RW

Reset State: 0x000E

If one of the RSSI values exceeds this threshold, the cd detect threshold is set to th_cw_high. Otherwise the CW detect threshold is set to the_cw_low. The CW detect threshold is updated when an RXA packet or an RXB packet is received.

WCSS_A_AGC_TH_CW_RSSI

Bits	Name	Description
6:0	THRESHOLD	RSSI threshold.

0x0A013D2C WCSS_A_AGC_CW_MODE_DELAY**Type:** RW**Reset State:** 0x7D00

Specifies various delays for several operating modes of the CW detector

WCSS_A_AGC_CW_MODE_DELAY

Bits	Name	Description
15:8	RADAR_DET_DELAY	Delay before starting cs_det after a pulse detect

0x0A013D30 WCSS_A_AGC_TH_D0_A**Type:** RW**Reset State:** 0x0087

This registers sets the correlation threshold above which the AGC decides an 802.11a packet is being received.

WCSS_A_AGC_TH_D0_A

Bits	Name	Description
7:0	THRESHOLD	

0x0A013D34 WCSS_A_AGC_TH_D0_A_LOW**Type:** RW**Reset State:** 0x0078

This registers sets the correlation threshold above which the AGC decides an 802.11a packet with high SNR is being received. The purpose of this register is to have a more sensitive and hence faster packet detection, since high SNR signals require two coarse gain steps, which are time consuming.

WCSS_A_AGC_TH_D0_A_LOW

Bits	Name	Description
7:0	THRESHOLD	

0x0A013D38 WCSS_A_AGC_TH_D0_B**Type:** RW**Reset State:** 0x008C

This registers sets the correlation threshold above which the AGC decides an 802.11b packet is being received. It can be set lower than the threshold for 11a packets since there is a longer time interval to correlate over.

WCSS_A_AGC_TH_D0_B

Bits	Name	Description
7:0	THRESHOLD	

0x0A013D3C WCSS_A_AGC_TH_D0_B_TF_EST**Type:** RW**Reset State:** 0x0050

This registers sets the correlation threshold above which the AGC decides an 802.11b packet is being received during the timing and frequency estimation state. This is a reconfirmation test which reduces the false alarm rate of the AGC for 11b packets. It is generally set lower than the initial threshold for 11b packets since the correlator is to allow margin for a low correlation value which can occur at the moment the correlator output is sampled (at the end of the 10 us tf(time and frequency)-estimation interval).

WCSS_A_AGC_TH_D0_B_TF_EST

Bits	Name	Description
7:0	THLD	

0x0A013D40 WCSS_A_AGC_TH_D0_11N**Type:** RW**Reset State:** 0x005A

This registers sets the correlation threshold above which the AGC decides an 802.11n packet is being received

WCSS_A_AGC_TH_D0_11N

Bits	Name	Description
7:0	VAL	

0x0A013D44 WCSS_A_AGC_TH_D0_11N_LOW**Type:** RW**Reset State:** 0x0078

This registers sets the correlation threshold above which the AGC decides an 802.11n packet is being received

WCSS_A_AGC_TH_D0_11N_LOW

Bits	Name	Description
7:0	VAL	

0x0A013D48 WCSS_A_AGC_TH_D0_11AC**Type:** RW**Reset State:** 0x0064

This registers sets the correlation threshold above which the AGC decides an 802.11ac packet is being received

WCSS_A_AGC_TH_D0_11AC

Bits	Name	Description
7:0	VAL	

0x0A013D4C WCSS_A_AGC_TH_D0_11AC_LOW**Type:** RW**Reset State:** 0x005A

This registers sets the correlation threshold above which the AGC decides an 802.11ac packet is being received

WCSS_A_AGC_TH_D0_11AC_LOW

Bits	Name	Description
7:0	VAL	

0x0A013D50 WCSS_A_AGC_TH_MAXCORBA**Type:** RW**Reset State:** 0x0000

This registers sets the cross check correlation threshold above which the AGC decides between 802.11b vs 802.11a packet is being received

WCSS_A_AGC_TH_MAXCORBA

Bits	Name	Description
7:0	VAL	

0x0A013D54 WCSS_A_AGC_TH_MAXCOR40**Type:** RW**Reset State:** 0x00FF

This registers sets the cross check correlation threshold above which the AGC decides between 802.11n40 vs 802.11a packet is being received

WCSS_A_AGC_TH_MAXCOR40

Bits	Name	Description
7:0	VAL	

0x0A013D58 WCSS_A_AGC_TH_MAXCORA40**Type:** RW**Reset State:** 0x0050

This registers sets the cross check correlation threshold above which the AGC decides between 802.11a and a 802.11n packet is being received

WCSS_A_AGC_TH_MAXCORA40

Bits	Name	Description
7:0	VAL	

0x0A013D5C WCSS_A_AGC_TH_MAXCOR20**Type:** RW**Reset State:** 0x00FF

This registers sets the cross check correlation threshold above which the AGC decides between 802.11ac vs 802.11a packet is being received

WCSS_A_AGC_TH_MAXCOR20

Bits	Name	Description
10	DIS	
9:0	VAL	

0x0A013D60 WCSS_A_AGC_TH_MAXCORA80

Type: RW

Reset State: 0x001E

This registers sets the cross check correlation threshold above which the AGC decides between 802.11a and a 802.11n packet is being received

WCSS_A_AGC_TH_MAXCORA80

Bits	Name	Description
7:0	VAL	

0x0A013D64 WCSS_A_AGC_TH_MAXCORAB

Type: RW

Reset State: 0x0005

For a positive 802.11b packet detection, not only should the auto correlation value divided by the signal power exceed TH_D0_A (or TH_D0_A_LOW in high SNR), it should also exceed that of the 802.11b bandwidth packet detector multiplied by the value in this register, divided by 256. The default fraction of the double bandwidth detector output that the 802.11b detector should exceed is 0.5 (128/256).

WCSS_A_AGC_TH_MAXCORAB

Bits	Name	Description
7:0	THRESHOLD	

0x0A013D68 WCSS_A_AGC_C11B

Type: RW

Reset State: 0x00E2

Averaging Parameter for correlation metric of 11b detector. Allowed values are 200-255

WCSS_A_AGC_C11B

Bits	Name	Description
7:0	VAL	

0x0A013D6C WCSS_A_AGC_C11BP**Type:** RW**Reset State:** 0x00FD

Averaging Parameter for power metric of 11b detector. Allowed values are 200-255

WCSS_A_AGC_C11BP

Bits	Name	Description
7:0	VAL	

0x0A013D70 WCSS_A_AGC_TH_CAP_A**Type:** RW**Reset State:** 0x0032

Detection threshold for capture event when reception of current 11a packet has been aborted. Value 0 is disables this check.

WCSS_A_AGC_TH_CAP_A

Bits	Name	Description
7:0	LEVEL_JUMP	Delta signal level. Default value 50 (7dB).

0x0A013D74 WCSS_A_AGC_TH_CAP_B**Type:** RW**Reset State:** 0x0032

Detection threshold for capture event when reception of current 11b packet has been aborted. Value 0 is disables this check.

WCSS_A_AGC_TH_CAP_B

Bits	Name	Description
7:0	LEVEL_JUMP	Delta signal level. Default value 50 (7dB).

0x0A013D78 WCSS_A_AGC_TH_CAP_RXA**Type:** RW**Reset State:** 0x0032

Detection threshold for capture event when 11a packet is being received. Value 0 disables this check.

WCSS_A_AGC_TH_CAP_RXA

Bits	Name	Description
7:0	LEVEL_JUMP	Delta signal level. Default value 50 (7dB).

0x0A013D7C WCSS_A_AGC_TH_CAP_RXB**Type:** RW**Reset State:** 0x0032

Detection threshold for capture event when 11b packet is being received. Value 0 disables this check.

WCSS_A_AGC_TH_CAP_RXB

Bits	Name	Description
7:0	LEVEL_JUMP	Delta signal level. Default value 50 (7dB).

0x0A013D80 WCSS_A_AGC_TH_PIP_PWR_JUMP**Type:** RW**Reset State:** 0x0000

Threshold for power jump to detect for a packet-in-packet case

WCSS_A_AGC_TH_PIP_PWR_JUMP

Bits	Name	Description
7:0	VAL	

0x0A013D84 WCSS_A_AGC_TH_EOP_LVL_DROP**Type:** RW**Reset State:** 0x0010

Threshold for signal level drop to detect end of packet in the case of a corrupted length field.

WCSS_A_AGC_TH_EOP_LVL_DROP

Bits	Name	Description
7:0	EOP_LVL_DROP	Delta signal level Default value 16 (12dB).

0x0A013D88 WCSS_A_AGC_TH_CORR_DROP**Type:** RW**Reset State:** 0x0000

AGC compares power with this threshold. If power is lower, in correlation state then an 802.11a trigger is ignored.

WCSS_A_AGC_TH_CORR_DROP

Bits	Name	Description
7:0	THRESHOLD	Power drop threshold. Default value 0 (i.e. disabled).

0x0A013D8C WCSS_A_AGC_CORR_TIME**Type:** RW**Reset State:** 0x01E0

This register specifies the time, in increments of 25ns, which the AGC can spend correlating if an energy jump has been detected in the AGC state.

WCSS_A_AGC_CORR_TIME

Bits	Name	Description
11:0	TIME	Correlation Time.

0x0A013D90 WCSS_A_AGC_ALL_RX_ON_TIME**Type:** RW**Reset State:** 0x0019

Time to leave all antennas on after a set_all_rx command. Units are in 0.8us steps.

WCSS_A_AGC_ALL_RX_ON_TIME

Bits	Name	Description
5:0	DELAY	

0x0A013D94 WCSS_A_AGC_TRIGGER_DELAY**Type:** RW**Reset State:** 0x0014

Detector triggers are ignored for a time specified by this register after a set_all_rx command. This enables the detectors to build up a reliable power estimate.

WCSS_A_AGC_TRIGGER_DELAY

Bits	Name	Description
4:0	TIME	Delay in 100ns increments.

0x0A013D98 WCSS_A_AGC_PWR_NOTCH_ALPHA1**Type:** RW**Reset State:** 0x012C**WCSS_A_AGC_PWR_NOTCH_ALPHA1**

Bits	Name	Description
8:0	ALPHA	

0x0A013D9C WCSS_A_AGC_PWR_NOTCH_ALPHA2**Type:** RW**Reset State:** 0x01FF**WCSS_A_AGC_PWR_NOTCH_ALPHA2**

Bits	Name	Description
8:0	ALPHA	

0x0A013DA0 WCSS_A_AGC_PWR_NOTCH_T_ALPHA**Type:** RW**Reset State:** 0x0008**WCSS_A_AGC_PWR_NOTCH_T_ALPHA**

Bits	Name	Description
9	BYPASS	0x0: ENABLE (Enable pwr DC notch filter) 0x1: BYPASS (Bypass pwr DC notch filter)
8:0	TIME	

0x0A013DA4 WCSS_A_AGC_D_TX_SETTLE**Type:** RW**Reset State:** 0x0028

Ranges from 0 to 12.775 micro seconds.

WCSS_A_AGC_D_TX_SETTLE

Bits	Name	Description
8:0	DELAY	Propagation time in 25ns units.

0x0A013DA8 WCSS_A_AGC_D_UNSATURATED**Type:** RW**Reset State:** 0x0010

Gain settling delay for RF RX non-saturated RX FE. Settling time in increments of 25 ns (40 MHz samples) Ranges from 0 to 1.575 micro seconds.

WCSS_A_AGC_D_UNSATURATED

Bits	Name	Description
5:0	DELAY	Settling time in 25ns units.

0x0A013DAC WCSS_A_AGC_D_SATURATED**Type:** RW**Reset State:** 0x0010

Gain settling delay for RF RX saturated RX FE. Settling time in increments of 25 ns (40 MHz samples) Ranges from 0 to 1.575 micro seconds.

WCSS_A_AGC_D_SATURATED

Bits	Name	Description
5:0	DELAY	Settling time in 25ns units.

0x0A013DB0 WCSS_A_AGC_D_FIRANDCAL**Type:** RW**Reset State:** 0x0010

RXFIR and CAL propagation delay. Ranges from 0 to 1.575 micro seconds.

WCSS_A_AGC_D_FIRANDCAL

Bits	Name	Description
5:0	DELAY	Propagation time in 25ns units. Default value is 14, corresponding to 350 ns.

0x0A013DB4 WCSS_A_AGC_D_HT_STS_COARSE_MEAS**Type:** RW**Reset State:** 0x0028**WCSS_A_AGC_D_HT_STS_COARSE_MEAS**

Bits	Name	Description
7:0	TIME	Settling time in 25ns units.

0x0A013DB8 WCSS_A_AGC_D_HT_STS_FINE_MEAS**Type:** RW**Reset State:** 0x0064**WCSS_A_AGC_D_HT_STS_FINE_MEAS**

Bits	Name	Description
7:0	TIME	Settling time in 25ns units.

0x0A013DBC WCSS_A_AGC_D_VHT_STS_COARSE_MEAS**Type:** RW**Reset State:** 0x0028**WCSS_A_AGC_D_VHT_STS_COARSE_MEAS**

Bits	Name	Description
7:0	TIME	Settling time in 25ns units.

0x0A013DC0 WCSS_A_AGC_D_VHT_STS_FINE_MEAS**Type:** RW**Reset State:** 0x0066

WCSS_A_AGC_D_VHT_STS_FINE_MEAS

Bits	Name	Description
7:0	TIME	Settling time in 25ns units.

0x0A013DC4 WCSS_A_AGC_D_VHT_DET_TIME**Type:** RW**Reset State:** 0x008E**WCSS_A_AGC_D_VHT_DET_TIME**

Bits	Name	Description
7:0	TIME	Max time in 25ns units that VHT detection is allowed from the start of the VHT-STS.

0x0A013DC8 WCSS_A_AGC_SIFS_TIME_A**Type:** RW**Reset State:** 0x0028

This register specifies the time in 25 ns increments, that the receiver is insensitive after the reception of an IEEE802.11a packet. This value may be set to zero which increases false detection probability of ACK packets, but which allows receptions of packets with a non-standard interframe spacing.

WCSS_A_AGC_SIFS_TIME_A

Bits	Name	Description
9:0	SIFS_A	SIFS time. Default value is 40, or 1 micro seconds (16 us as per 11a standard - 4us CCA hangover time). 0x0: MIN (0 ns, minimum value.) 0x1: STEP (25 ns, step size value.) 0x3FF: MAX (25.57 us, maximum value.)

0x0A013DCC WCSS_A_AGC_SIFS_TIME_B**Type:** RW**Reset State:** 0x0028

This register specifies the time in 25 ns increments, that the receiver is insensitive after the reception of an IEEE802.11b packet. This value may be set to zero which increases false detection probability of ACK packet, but which allows receptions of packets with a non-standard interframe spacing.

WCSS_A_AGC_SIFS_TIME_B

Bits	Name	Description
9:0	SIFS_B	SIFS time. Default value is 40, or 1 micro seconds. 0x0: MIN (0 ns, minimum value.) 0x1: STEP (25 ns, step size value.) 0x3FF: MAX (25.57 us, maximum value.)

0x0A013DD0 WCSS_A_AGC_MAX_RX_TIME_A**Type:** RW**Reset State:** 0x01AD

Max receive time of an 11a packet in 12.8 us increments. Used to prevent lockup of the AGC state machine when it is unable to determine end of packet in the case of a corrupted length field when no end of packet level drop is detected.

WCSS_A_AGC_MAX_RX_TIME_A

Bits	Name	Description
11:0	MAX_RX_TIME	Max RX time. Default value $20+4096*8/6=5482$ (6 Mbit/s packet of 4096 bytes plus OFDM preamble) and $429*12.8=5491.2$)

0x0A013DD4 WCSS_A_AGC_MAX_RX_TIME_B**Type:** RW**Reset State:** 0x0A10

Max receive time of an 11b packet in 25.6 us increments. Used to prevent lockup of the AGC state machine when it is unable to determine end of packet in the case of a corrupted length field when no end of packet level drop is detected.

WCSS_A_AGC_MAX_RX_TIME_B

Bits	Name	Description
11:0	MAX_RX_TIME	Max RX time. Default value $192+4096*8=32960$ (1 Mbit/s packet of 4096 bytes plus long preamble) and $2576*12.8=32973$)

0x0A013DD8 WCSS_A_AGC_RTT_CFG**Type:** RW**Reset State:** 0x0000

RTT configuration register.

WCSS_A_AGC_RTT_CFG

Bits	Name	Description
4	ALWAYS_VAL_RX2TX	For debugging purposes only, this eliminates the need for the WMAC to validate the t3-t2 measurement.
3	MEAS_SIFS_EN	Enables reporting the SIFS time instead of delta between packet arrivals as required in 11v. This is for compatibility with the old RTT.
2	REF_TIMER_EN	Enables the reference timer used for absolute time measurements. This should always be enabled if RTT is enabled.
1:0	MODE	Sets the RTT mode of operation. 0x0: OFF 0x1: TX2RX 0x2: RX2TX 0x3: RESERVED

0x0A013DDC WCSS_A_AGC_RTT_SIFS_WINDOW**Type:** RW**Reset State:** 0xC878

Sets the min/max SIFS time that is allowed. Anything outside this window will be regarded as an invalid RTT measurement.

WCSS_A_AGC_RTT_SIFS_WINDOW

Bits	Name	Description
15:8	MAX_TIMEOUT	In 100ns steps.
7:0	MIN_TIMEOUT	In 100ns steps.

0x0A013DE0 WCSS_A_AGC_RTT_OFFSET_20**Type:** RW**Reset State:** 0x0000

RTT offset calibration value for 20 MHz packets

WCSS_A_AGC_RTT_OFFSET_20

Bits	Name	Description
15:8	RX_OFFSET	Signed number in 12.5ns steps.
7:0	TX_OFFSET	Unsigned number in 12.5ns steps.

0x0A013DE4 WCSS_A_AGC_RTT_OFFSET_40**Type:** RW**Reset State:** 0x0000

RTT offset calibration value for 40 MHz packets

WCSS_A_AGC_RTT_OFFSET_40

Bits	Name	Description
15:8	RX_OFFSET	Signed number in 12.5ns steps.
7:0	TX_OFFSET	Unsigned number in 12.5ns steps.

0x0A013DE8 WCSS_A_AGC_RTT_OFFSET_80**Type:** RW**Reset State:** 0x0000

RTT offset calibration value for 80 MHz packets

WCSS_A_AGC_RTT_OFFSET_80

Bits	Name	Description
15:8	RX_OFFSET	Signed number in 12.5ns steps.
7:0	TX_OFFSET	Unsigned number in 12.5ns steps.

0x0A013DEC WCSS_A_AGC_RTT_VALID_COUNT**Type:** W**Reset State:** 0x00000000**WCSS_A_AGC_RTT_VALID_COUNT**

Bits	Name	Description
7:0	COUNT	Counts number of RTT valids.

0x0A013DF0 WCSS_A_AGC_RTT_DELTA_MEAS**Type:** R**Reset State:** 0x00000000

WCSS_A_AGC_RTT_DELTA_MEAS

Bits	Name	Description
13:0	TIME	In 12.5ns steps.

0x0A013DF4 WCSS_A_AGC_RTT_REF_TIME_MEAS_LS16**Type:** R**Reset State:** 0x00000000**WCSS_A_AGC_RTT_REF_TIME_MEAS_LS16**

Bits	Name	Description
15:0	TIME	In 12.5ns steps.

0x0A013DF8 WCSS_A_AGC_RTT_REF_TIME_MEAS_MS16**Type:** R**Reset State:** 0x00000000**WCSS_A_AGC_RTT_REF_TIME_MEAS_MS16**

Bits	Name	Description
15:0	TIME	In 12.5ns steps.

0x0A013DFC WCSS_A_AGC_RDET_RSSI_THRESHOLD**Type:** RW**Reset State:** 0x007F

If the maximum of the N_LISTEN RSSI inputs as calculated by the power block in the AGC exceeds the value in the up threshold field of this register, a start of pulse is flagged to the pulse train detector. When the max RSSI sinks below the value of the down threshold field, the end of pulse is flagged. If both thresholds are set to 0, the analog pulse detector (in the Mars RF ASIC) is used instead. The start of the pulse is detected on the rising edge of this signal. The end of the pulse is detected on the falling edge.

WCSS_A_AGC_RDET_RSSI_THRESHOLD

Bits	Name	Description
13:7	DET_THLD_DN	Pulse detection threshold.
6:0	DET_THLD_UP	Pulse detection threshold.

0x0A013E00 WCSS_A_AGC_RDET_MIN_IAT_WRITE**Type:** RW**Reset State:** 0x6F7C

This register defines the maximum inter-arrival time between radar pulses, measured in multiples of 128 samples. Pulse that arrive after the previous pulse within a time less than the value in this register are ignored by the radar pulse train detector. If set equal to max_iat, then this register has no effect on the operation of the radar detector. This value gets written to the appropriate train detector selected by det_sel, with the write itself enabled using det_write (see register rdet_reset).

WCSS_A_AGC_RDET_MIN_IAT_WRITE

Bits	Name	Description
15:0	MIN_IAT	Minimum Inter-arrival time. default is 0.499 ms (radar type 3). 0x1: STEP_SIZE (6.4 us)

0x0A013E04 WCSS_A_AGC_RDET_MAX_IAT_WRITE**Type:** RW**Reset State:** 0x01BF

This register defines the maximum inter-arrival time between radar pulses, measured in multiples of 128 samples. If this time is exceeded, then the radar detector assumes the pulse train has ended and checks whether more than min_pulse_train_count1 pulses have been counted. This value gets written to the appropriate train detector selected by det_sel, with the write itself enabled using det_write (see register rdet_reset).

WCSS_A_AGC_RDET_MAX_IAT_WRITE

Bits	Name	Description
15:0	MAX_IAT	Maximum Inter-arrival time. Default is 3.072 (ms radar type 1). 0x1: STEP_SIZE (6.4 us)

0x0A013E08 WCSS_A_AGC_RDET_PULSE_WIDTH_MARGIN_WRITE**Type:** RW**Reset State:** 0x0005

When the first pulse of a train arrives, its duration is stored. Subsequent pulses' durations are compared to this first one. If a pulse of a different length is encountered, the pulse train ends. The comparison uses a margin as defined by this register. The check performed is if $\text{abs}(\text{current_pulse_duration} - \text{first_pulse_duration}) \leq \text{this register}$ then the current pulse does not break the train, and the pulse counter is increased. This value gets written to the appropriate train detector selected by det_sel, with the write itself enabled using det_write (see register rdet_reset).

WCSS_A_AGC_RDET_PULSE_WIDTH_MARGIN_WRITE

Bits	Name	Description
7:0	PW_MRGN	Pulse width margin.

0x0A013E0C WCSS_A_AGC_RDET_MIN_PULSE_TRAIN_COUNT1_WRITE**Type:** RW**Reset State:** 0x0004

If a radar pulse train ends because the maximum inter-arrival time has expired, the current number of pulses counted is compared with this register. If the number of pulses is greater than or equal to this register's contents, then the radar detection flag is raised. This value gets written to the appropriate train detector selected by `det_sel`, with the write itself enabled using `det_write` (see register `rdet_reset`).

WCSS_A_AGC_RDET_MIN_PULSE_TRAIN_COUNT1_WRITE

Bits	Name	Description
15:0	CNT_THLD	Pulse count threshold.

0x0A013E10 WCSS_A_AGC_RDET_MIN_PULSE_TRAIN_COUNT2_WRITE**Type:** RW**Reset State:** 0x0004

If a radar pulse train ends because a pulse of different duration as the first pulse in the train is detected, the current number of pulses counted is compared with this register. If the number of pulses is greater than or equal to this register's contents, then the radar detection flag is raised. This value gets written to the appropriate train detector selected by `det_sel`, with the write itself enabled using `det_write` (see register `rdet_reset`).

WCSS_A_AGC_RDET_MIN_PULSE_TRAIN_COUNT2_WRITE

Bits	Name	Description
15:0	CNT_THLD	Pulse count threshold.

0x0A013E14 WCSS_A_AGC_RDET_MIN_PULSE_TRAIN_COUNT3_WRITE**Type:** RW**Reset State:** 0x0004

If a radar a pulse is detected the number of pulses counted is increased. If the number of pulses counted is greater than or equal to this register's contents, then the radar detection flag is raised.

This value gets written to the appropriate train detector selected by `det_sel`, with the write itself enabled using `det_write` (see register `rdet_reset`).

WCSS_A_AGC_RDET_MIN_PULSE_TRAIN_COUNT3_WRITE

Bits	Name	Description
15:0	CNT_THLD	Pulse count threshold.

0x0A013E18 WCSS_A_AGC_RDET_MIN_PULSE_TRAIN_COUNT4_WRITE

Type: RW

Reset State: 0x0000

If a pulse arrives before the minimum inter-arrival time has expired the current number of pulses counted is compared with this register. If the number of pulses is greater than or equal to this register's contents, then the radar detection flag is raised. Special values are provided to simply ignore radar pulses which arrive within the minimum arrival time or to reset the pulse counter without comparing the current pulse count with this register. This value gets written to the appropriate train detector selected by `det_sel`, with the write itself enabled using `det_write` (see register `rdet_reset`).

WCSS_A_AGC_RDET_MIN_PULSE_TRAIN_COUNT4_WRITE

Bits	Name	Description
15:0	CNT_THLD	Pulse count threshold. Default is 0 0x0: NO_RESET (Ignore pulse, do not reset pulse counter) 0xFFFF: NO_CHECK (reset pulse counter, do not compare)

0x0A013E1C WCSS_A_AGC_RDET_MIN_RADAR_PULSE_WIDTH_WRITE

Type: RW

Reset State: 0x0012

Minimum radar pulse width, specified in samples. When the end of a pulse is detected, its duration is compared to the contents of this registers. When the pulse duration is less than or equal to this register's content, the pulse is ignored. This mechanism is to ensure that short glitches (for instance ripple on a pulse which is just on the `rss_i_threshold` level) or a rapidly switching clip detect signal are ignored. This value gets written to the appropriate train detector selected by `det_sel`, with the write itself enabled using `det_write` (see register `rdet_reset`).

WCSS_A_AGC_RDET_MIN_RADAR_PULSE_WIDTH_WRITE

Bits	Name	Description
15:0	MIN_PW	Minimum pulse width.

0x0A013E20 WCSS_A_AGC_RDET_MAX_RADAR_PULSE_WIDTH_WRITE**Type:** RW**Reset State:** 0x0016

Maximum radar pulse width, specified in samples. When the end of a pulse is detected, its duration is compared to the contents of this registers. When the pulse duration is greater than or equal to the this register's content, the pulse is ignored. This mechanism is to ensure that packet transmissions (typically of much longer duration as radar pulses) are not counted as radar pulses. Setting this register to 0 will disable the radar detector. This value gets written to the appropriate train detector selected by `det_sel`, with the write itself enabled using `det_write` (see register `rdet_reset`).

WCSS_A_AGC_RDET_MAX_RADAR_PULSE_WIDTH_WRITE

Bits	Name	Description
15:0	MAX_PW	Maximum pulse width.

0x0A013E24 WCSS_A_AGC_RDET_MIN_IAT_READ**Type:** R**Reset State:** 0x00000000

This register defines the maximum inter-arrival time between radar pulses, measured in multiples of 128 samples. Pulse that arrive after the previous pulse within a time less than the value in this register are ignored by the radar pulse train detector. If set equal to `max_iat`, then this value has no effect on the operation of the radar detector. This value is read from the appropriate train detector selected by `det_sel` (see register `rdet_reset`).

WCSS_A_AGC_RDET_MIN_IAT_READ

Bits	Name	Description
15:0	MIN_IAT	Minimum Inter-arrival time. default is 0.499 ms (radar type 3). 0x1: STEP_SIZE (6.4 us)

0x0A013E28 WCSS_A_AGC_RDET_MAX_IAT_READ**Type:** R**Reset State:** 0x00000000

This register defines the maximum inter-arrival time between radar pulses, measured in multiples of 128 samples. If this time is exceeded, then the radar detector assumes the pulse train has ended and checks whether more than `min_pulse_train_count1` pulses have been counted. This value is read from the appropriate train detector selected by `det_sel` (see register `rdet_reset`).

WCSS_A_AGC_RDET_MAX_IAT_READ

Bits	Name	Description
15:0	MAX_IAT	Maximum Inter-arrival time. Default is 3.072 (ms radar type 1). 0x1: STEP_SIZE (6.4 us)

0x0A013E2C WCSS_A_AGC_RDET_PULSE_WIDTH_MARGIN_READ**Type:** R**Reset State:** 0x00000000

When the first pulse of a train arrives, its duration is stored. Subsequent pulses' durations are compared to this first one. If a pulse of a different length is encountered, the pulse train ends. The comparison uses a margin as defined by this register. The check performed is if $\text{abs}(\text{current_pulse_duration} - \text{first_pulse_duration}) \leq \text{this register}$ then the current pulse does not break the train, and the pulse counter is increased. This value is read from the appropriate train detector selected by `det_sel` (see register `rdet_reset`).

WCSS_A_AGC_RDET_PULSE_WIDTH_MARGIN_READ

Bits	Name	Description
7:0	PW_MRGN	Pulse width margin.

0x0A013E30 WCSS_A_AGC_RDET_MIN_PULSE_TRAIN_COUNT1_READ**Type:** R**Reset State:** 0x00000000

If a radar pulse train ends because the maximum inter-arrival time has expired, the current number of pulses counted is compared with this register. If the number of pulses is greater than or equal to this register's contents, then the radar detection flag is raised. This value is read from the appropriate train detector selected by `det_sel` (see register `rdet_reset`).

WCSS_A_AGC_RDET_MIN_PULSE_TRAIN_COUNT1_READ

Bits	Name	Description
15:0	CNT_THLD	Pulse count threshold.

0x0A013E34 WCSS_A_AGC_RDET_MIN_PULSE_TRAIN_COUNT2_READ**Type:** R**Reset State:** 0x00000000

If a radar pulse train ends because a pulse of different duration as the first pulse in the train is detected, the current number of pulses counted is compared with this register. If the number of pulses is greater than or equal to this register's contents, then the radar detection flag is raised. This value is read from the appropriate train detector selected by det_sel (see register rdet_reset).

WCSS_A_AGC_RDET_MIN_PULSE_TRAIN_COUNT2_READ

Bits	Name	Description
15:0	CNT_THLD	Pulse count threshold.

0x0A013E38 WCSS_A_AGC_RDET_MIN_PULSE_TRAIN_COUNT3_READ

Type: R

Reset State: 0x00000000

If a radar a pulse is detected the number of pulses counted is increased. If the number of pulses counted is greater than or equal to this register's contents, then the radar detection flag is raised. This value is read from the appropriate train detector selected by det_sel (see register rdet_reset).

WCSS_A_AGC_RDET_MIN_PULSE_TRAIN_COUNT3_READ

Bits	Name	Description
15:0	CNT_THLD	Pulse count threshold.

0x0A013E3C WCSS_A_AGC_RDET_MIN_PULSE_TRAIN_COUNT4_READ

Type: R

Reset State: 0x00000000

If a pulse arrives before the minimum inter-arrival time has expired the current number of pulses counted is compared with this register. If the number of pulses is greater than or equal to this register's contents, then the radar detection flag is raised. Special values are provided to simply ignore radar pulses which arrive within the minimum arrival time or to reset the pulse counter without comparing the current pulse count with this register. This value is read from the appropriate train detector selected by det_sel (see register rdet_reset).

WCSS_A_AGC_RDET_MIN_PULSE_TRAIN_COUNT4_READ

Bits	Name	Description
15:0	CNT_THLD	Pulse count threshold. Default is 0 0x0: NO_RESET (Ignore pulse, do not reset pulse counter) 0xFFFF: NO_CHECK (reset pulse counter, do not compare)

0x0A013E40 WCSS_A_AGC_RDET_MIN_RADAR_PULSE_WIDTH_READ**Type:** R**Reset State:** 0x00000000

Minimum radar pulse width, specified in samples. When the end of a pulse is detected, its duration is compared to the contents of this registers. When the pulse duration is less than or equal to the this register's content, the pulse is ignored. This mechanism is to ensure that short glitches (for instance ripple on a pulse which is just on the rssi_threshold level) or a rapidly switching clip detect signal are ignored. This value is read from the appropriate train detector selected by det_sel (see register rdet_reset).

WCSS_A_AGC_RDET_MIN_RADAR_PULSE_WIDTH_READ

Bits	Name	Description
15:0	MIN_PW	Minimum pulse width.

0x0A013E44 WCSS_A_AGC_RDET_MAX_RADAR_PULSE_WIDTH_READ**Type:** R**Reset State:** 0x00000000

Maximum radar pulse width, specified in samples. When the end of a pulse is detected, its duration is compared to the contents of this registers. When the pulse duration is greater than or equal to the this register's content, the pulse is ignored. This mechanism is to ensure that packet transmissions (typically of much longer duration as radar pulses) are not counted as radar pulses. Setting this register to 0 will disable the radar detector. This value is read from the appropriate train detector selected by det_sel (see register rdet_reset).

WCSS_A_AGC_RDET_MAX_RADAR_PULSE_WIDTH_READ

Bits	Name	Description
15:0	MAX_PW	Maximum pulse width.

0x0A013E48 WCSS_A_AGC_RDET_RESET**Type:** RW**Reset State:** 0x08FF

Reset the radar detector to its initial state by writing a 1 followed by a zero to bit 0 in this register. Writing a 1 will disable the radar detector. The radar detector is disabled by default.

WCSS_A_AGC_RDET_RESET

Bits	Name	Description
14	DET_WRITE	Train detector APB write bit. 0x0: DIS (Disable APB write to train detectors) 0x1: EN (Enable APB write to train detectors)
13:11	RSSI_MODE	Configure RSSI combination mode. 0x0: RSSI0 (Use chain 0 rssi) 0x1: MAX_RSSI (Use max rssi of n_listen chains) 0x2: MIN_RSSI (Use min rssi of n_listen chains)
10:8	DET_SEL	Select train detector to program. 0x7: SEL7 (program train detector 7) 0x6: SEL6 (program train detector 6) 0x5: SEL5 (program train detector 5) 0x4: SEL4 (program train detector 4) 0x3: SEL3 (program train detector 3) 0x2: SEL2 (program train detector 2) 0x1: SEL1 (program train detector 1) 0x0: SEL0 (program train detector 0)
7:0	DET_RST	8'b0xxxxxx -> enable7 (Enable train detector 7) 8'b1xxxxxx -> disable7 (Disable train detector 7) 8'bx0xxxxx -> enable6 (Enable train detector 6) 8'bx1xxxxx -> disable6 (Disable train detector 6) 8'bxx0xxxx -> enable5 (Enable train detector 5) 8'bxx1xxxx -> disable5 (Disable train detector 5) 8'bx0xxxx -> enable4 (Enable train detector 4) 8'bx1xxxx -> disable4 (Disable train detector 4) 8'bxxx0xxx -> enable3 (Enable train detector 3) 8'bxxx1xxx -> disable3 (Disable train detector 3) 8'bxxx0xx -> enable2 (Enable train detector 2) 8'bxxx1xx -> disable2 (Disable train detector 2) 8'bxxxx0x -> enable1 (Enable train detector 1) 8'bxxxx1x -> disable1 (Disable train detector 1) 8'bxxxxx0 -> enable0 (Enable train detector and rest of radar detector) 8'bxxxxx1 -> disable0 (Disable train detector and rest of radar detector) Enable/Disable of radar detector and train detector instantiations.

0x0A013E4C WCSS_A_AGC_RDET_RADAR_DETECTED**Type:** R**Reset State:** 0x00000000

When this bit is set a radar pulse train has been detected. This bit must be reset by the radar detection (interrupt) services routine by writing a '1' into the dflag_rst register.

WCSS_A_AGC_RDET_RADAR_DETECTED

Bits	Name	Description
5:3	TRIGGER_DET	Number of detector instance that triggered the radar detector.
2:1	TRIGGER_SRC	Trigger source. For debugging purposes this field display the reason why the radar detector triggered. 0x1: C1 (Pulse count exceeds min_pulse_count1) 0x2: C2 (Pulse count exceeds min_pulse_count2) 0x3: C3 (Pulse count exceeds min_pulse_count3) 0x0: C4 (Pulse count exceeds min_pulse_count4)
0	PULSE_DET	Detection bit. 0x0: IDEL (No pulse train detected) 0x1: PDET (Pulse train detected)

0x0A013E50 WCSS_A_AGC_RDET_PULSEWIDTH**Type:** R**Reset State:** 0x00000000

This register specifies the width of the first pulse in the detected pulse train. All other pulses in the train have an equal width, to within the pulse_width_margin.

WCSS_A_AGC_RDET_PULSEWIDTH

Bits	Name	Description
15:0	PULSE_WIDTH	Pulse width.

0x0A013E54 WCSS_A_AGC_RDET_NUM_DETECTED**Type:** R**Reset State:** 0x00000000

This register reports the number of pulses counted by the radar detection algorithm.

WCSS_A_AGC_RDET_NUM_DETECTED

Bits	Name	Description
15:0	NUM_DET	Number of pulses counted.

0x0A013E58 WCSS_A_AGC_RDET_MEASUREMENT_DELAY**Type:** RW**Reset State:** 0x0020

This register configures the time interval (measured in rssi samples) after which the radar detector measures rssi level after it returns to the CORR state after a gain set due to a clip event. It can not evaluate RSSI immediately since the RSSI circuit comes out of reset when the CORR state is entered. Therefore it takes time for the RSSI to ramp up (about 32 samples). The time programmed in this register should be larger than the RSSI rampup time.

WCSS_A_AGC_RDET_MEASUREMENT_DELAY

Bits	Name	Description
8	CORR_FLAG_BYPASS	Correlation flag function bypass. If asserted, the correlation flag function in the RDET pulse detector (used to delay transition of the main state machine from IDLE to CLIP_S1) will be bypassed. Default setting = 1'b0.
7:0	TIME	Measurement delay.

0x0A013E5C WCSS_A_AGC_RDET_FLAG_RESET**Type:** W**Reset State:** 0x0000

This register contains 2 bits that control the operation of the radar detector.

WCSS_A_AGC_RDET_FLAG_RESET

Bits	Name	Description
0	DFLAG_RST	Reset detector flag. While the detector flag is set further radar detects are ignored and the num_detected and pulse width registers are not updated. By writing a 1 to this register, the detector flag is reset and will be set when the next radar burst is detected. 0x0: NORMAL (Normal) 0x1: RESET (Reset detector flag)

0x0A013E60 WCSS_A_AGC_RSSI_THRESHOLD_CLIPPED**Type:** RW**Reset State:** 0x0000

RSSI threshold for clipped pulses

WCSS_A_AGC_RSSI_THRESHOLD_CLIPPED

Bits	Name	Description
13:7	DET_THLD_DN	Pulse detection threshold.
6:0	DET_THLD_UP	Pulse detection threshold.

0x0A013E64 WCSS_A_AGC_CW_PULSE_WIDTH**Type:** RW**Reset State:** 0x0019

When the radar detector's pulse detector detects a radar pulse it monitors the CW detect flag. When this flag is set the pulse detector sets a local flag called `cw_detected`. When at the end of the radar pulse, the `cw_detected` flag is still not set, the in the pulse is discounted as a radar pulse (which should be narrow-band) and the `pulse_cancel` output is asserted. Some pulses are too short to perform a cw detection on. The cw detection time is the sum of the sample data gathering time (which is programmed in CSR `cw_radar_delay`) and the fft and cw detect calculation time. If the pulse duration is less than the number in this register, then the radar detector's pulse detector assumes that the detected pulse is a radar pulse. The radar detector's train detectors may still reject is because it is not in between the minimum and maximum radar pulse width.

WCSS_A_AGC_CW_PULSE_WIDTH

Bits	Name	Description
7:0	VAL	Duration of data gathering and calculation time for cw detection, specified in multiples of 0.8 us (16 20 Mhz sample).

0x0A013E68 WCSS_A_AGC_CW_RADAR**Type:** RW**Reset State:** 0x007D

This register configures if and with what delay the radar detector can start the CW detector.

WCSS_A_AGC_CW_RADAR

Bits	Name	Description
9	DET_DIS	Disables control of the CW detector by the radar detector.
8:0	DELAY	Amount of delay in 40 MHz samples between start pulse of the radar detector and the time that the CW detector starts calculating.

0x0A013E6C WCSS_A_AGC_ACI_DETECT_TH**Type:** RW**Reset State:** 0x0148

When detecting a radar pulse, the radar detector's pulse detector calculates the difference between the total (pre FIR) RSSI and the in-band (post FIR) RSSI. If this difference exceeds the threshold programmed in this register, the aci_detect flag is asserted. If, during a radar pulse, the ratio of the total time that this flag is asserted and the pulse's duration exceeds the second threshold specified in this register, then the radar detector decides that this pulse was in the adjacent channel. The radar detector will then ignore this pulse. This mechanism helps to avoid false triggers on packet traffic, interference and radar pulses in the adjacent channel. This type of false triggers would have the adverse effect that the radar detector would vacate a clean channel because of a noisy neighbor.

WCSS_A_AGC_ACI_DETECT_TH

Bits	Name	Description
12	DETECT_DIS	0x1: DISABLE (Disable use of ACI detector during radar detection.) 0x0: ENABLE (Enable use of ACI detector during radar detection.)
11:4	TH_RSSI_ACI	This register field specifies the RSSI delta which the pre and post FIR RSSI should differ by, to be considered adjacent channel power.
3:0	ACI_FRACTION	This register field specifies the fractional threshold that the pre and post FIR RSSI difference should exceed to be counted as ACI (Adjacent Channel Interference). This threshold is specified in integer multiples of 1/16-ths. For example, a value of 8 means that the RSSI should differ by more than th_rss_aci for more than 50% of the duration of the detected radar pulse for it to be ruled ACI.

0x0A013E70 WCSS_A_AGC_TRANSITION_COUNT_GLOBAL_EN**Type:** RW**Reset State:** 0x0001

This is a global enable/disable for all transition counters. This is useful for freezing the counters at the same time instant before reading or writing to them.

WCSS_A_AGC_TRANSITION_COUNT_GLOBAL_EN

Bits	Name	Description
0	ENABLE	

0x0A013E74 WCSS_A_AGC_TRANSITION_COUNT0_EN**Type:** RW**Reset State:** 0x0000

Control word for state transition counter 0. This counter can be triggered on any transition between any 2 AGC states.

WCSS_A_AGC_TRANSITION_COUNT0_EN

Bits	Name	Description
10	ENABLE	Enables counter.
9:5	NEXT_STATE	State in which transition ends.
4:0	CURRENT_STATE	State from which transition originates.

0x0A013E78 WCSS_A_AGC_TRANSITION_COUNT1_EN**Type:** RW**Reset State:** 0x0000

Control word for state transition counter 1. This counter can be triggered on any transition between any 2 AGC states.

WCSS_A_AGC_TRANSITION_COUNT1_EN

Bits	Name	Description
10	ENABLE	Enables counter.
9:5	NEXT_STATE	State in which transition ends.
4:0	CURRENT_STATE	State from which transition originates.

0x0A013E7C WCSS_A_AGC_TRANSITION_COUNT2_EN**Type:** RW**Reset State:** 0x0000

Control word for state transition counter 2. This counter can be triggered on any transition between any 2 AGC states.

WCSS_A_AGC_TRANSITION_COUNT2_EN

Bits	Name	Description
10	ENABLE	Enables counter.
9:5	NEXT_STATE	State in which transition ends.

WCSS_A_AGC_TRANSITION_COUNT2_EN (cont.)

Bits	Name	Description
4:0	CURRENT_STATE	State from which transition originates.

0x0A013E80 WCSS_A_AGC_TRANSITION_COUNT3_EN**Type:** RW**Reset State:** 0x0000

Control word for state transition counter 3. This counter can be triggered on any transition between any 2 AGC states.

WCSS_A_AGC_TRANSITION_COUNT3_EN

Bits	Name	Description
10	ENABLE	Enables counter.
9:5	NEXT_STATE	State in which transition ends.
4:0	CURRENT_STATE	State from which transition originates.

0x0A013E84 WCSS_A_AGC_TRANSITION_COUNT4_EN**Type:** RW**Reset State:** 0x0000

Control word for state transition counter 4. This counter can be triggered on any transition between any 2 AGC states.

WCSS_A_AGC_TRANSITION_COUNT4_EN

Bits	Name	Description
10	ENABLE	Enables counter.
9:5	NEXT_STATE	State in which transition ends.
4:0	CURRENT_STATE	State from which transition originates.

0x0A013E88 WCSS_A_AGC_TRANSITION_COUNT5_EN**Type:** RW**Reset State:** 0x0000

Control word for state transition counter 5. This counter can be triggered on any transition between any 2 AGC states.

WCSS_A_AGC_TRANSITION_COUNT5_EN

Bits	Name	Description
10	ENABLE	Enables counter.
9:5	NEXT_STATE	State in which transition ends.
4:0	CURRENT_STATE	State from which transition originates.

0x0A013E8C WCSS_A_AGC_TRANSITION_COUNT6_EN**Type:** RW**Reset State:** 0x0000

Control word for state transition counter 6. This counter can be triggered on any transition between any 2 AGC states.

WCSS_A_AGC_TRANSITION_COUNT6_EN

Bits	Name	Description
10	ENABLE	Enables counter.
9:5	NEXT_STATE	State in which transition ends.
4:0	CURRENT_STATE	State from which transition originates.

0x0A013E90 WCSS_A_AGC_TRANSITION_COUNT7_EN**Type:** RW**Reset State:** 0x0000

Control word for state transition counter 7. This counter can be triggered on any transition between any 2 AGC states.

WCSS_A_AGC_TRANSITION_COUNT7_EN

Bits	Name	Description
10	ENABLE	Enables counter.
9:5	NEXT_STATE	State in which transition ends.
4:0	CURRENT_STATE	State from which transition originates.

0x0A013E94 WCSS_A_AGC_TRANSITION_COUNT8_EN**Type:** RW**Reset State:** 0x0000

Control word for state transition counter 8. This counter can be triggered on any transition between any 2 AGC states.

WCSS_A_AGC_TRANSITION_COUNT8_EN

Bits	Name	Description
10	ENABLE	Enables counter.
9:5	NEXT_STATE	State in which transition ends.
4:0	CURRENT_STATE	State from which transition originates.

0x0A013E98 WCSS_A_AGC_TRANSITION_COUNT9_EN**Type:** RW**Reset State:** 0x0000

Control word for state transition counter 9. This counter can be triggered on any transition between any 2 AGC states.

WCSS_A_AGC_TRANSITION_COUNT9_EN

Bits	Name	Description
10	ENABLE	Enables counter.
9:5	NEXT_STATE	State in which transition ends.
4:0	CURRENT_STATE	State from which transition originates.

0x0A013E9C WCSS_A_AGC_TRANSITION_COUNT10_EN**Type:** RW**Reset State:** 0x0000

Control word for state transition counter 10. This counter can be triggered on any transition between any 2 AGC states.

WCSS_A_AGC_TRANSITION_COUNT10_EN

Bits	Name	Description
10	ENABLE	Enables counter.
9:5	NEXT_STATE	State in which transition ends.

WCSS_A_AGC_TRANSITION_COUNT10_EN (cont.)

Bits	Name	Description
4:0	CURRENT_STATE	State from which transition originates.

0x0A013EA0 WCSS_A_AGC_TRANSITION_COUNT11_EN**Type:** RW**Reset State:** 0x0000

Control word for state transition counter 11. This counter can be triggered on any transition between any 2 AGC states.

WCSS_A_AGC_TRANSITION_COUNT11_EN

Bits	Name	Description
10	ENABLE	Enables counter.
9:5	NEXT_STATE	State in which transition ends.
4:0	CURRENT_STATE	State from which transition originates.

0x0A013EA4 WCSS_A_AGC_TRANSITION_COUNT0**Type:** W**Reset State:** 0x00000000

Debug Counter. Counts transitions between arbitrary states of the AGC state machine.

WCSS_A_AGC_TRANSITION_COUNT0

Bits	Name	Description
15:0	COUNT	Counter output. This counter is reset by writing a 0 to it.

0x0A013EA8 WCSS_A_AGC_TRANSITION_COUNT1**Type:** W**Reset State:** 0x00000000

Debug Counter. Counts transitions between arbitrary states of the AGC state machine.

WCSS_A_AGC_TRANSITION_COUNT1

Bits	Name	Description
15:0	COUNT	Counter output. This counter is reset by writing a 0 to it.

0x0A013EAC WCSS_A_AGC_TRANSITION_COUNT2**Type:** W**Reset State:** 0x00000000

Debug Counter. Counts transitions between arbitrary states of the AGC state machine.

WCSS_A_AGC_TRANSITION_COUNT2

Bits	Name	Description
15:0	COUNT	Counter output. This counter is reset by writing a 0 to it.

0x0A013EB0 WCSS_A_AGC_TRANSITION_COUNT3**Type:** W**Reset State:** 0x00000000

Debug Counter. Counts transitions between arbitrary states of the AGC state machine.

WCSS_A_AGC_TRANSITION_COUNT3

Bits	Name	Description
15:0	COUNT	Counter output. This counter is reset by writing a 0 to it.

0x0A013EB4 WCSS_A_AGC_TRANSITION_COUNT4**Type:** W**Reset State:** 0x00000000

Debug Counter. Counts transitions between arbitrary states of the AGC state machine.

WCSS_A_AGC_TRANSITION_COUNT4

Bits	Name	Description
15:0	COUNT	Counter output. This counter is reset by writing a 0 to it.

0x0A013EB8 WCSS_A_AGC_TRANSITION_COUNT5**Type:** W**Reset State:** 0x00000000

Debug Counter. Counts transitions between arbitrary states of the AGC state machine.

WCSS_A_AGC_TRANSITION_COUNT5

Bits	Name	Description
15:0	COUNT	Counter output. This counter is reset by writing a 0 to it.

0x0A013EBC WCSS_A_AGC_TRANSITION_COUNT6**Type:** W**Reset State:** 0x00000000

Debug Counter. Counts transitions between arbitrary states of the AGC state machine.

WCSS_A_AGC_TRANSITION_COUNT6

Bits	Name	Description
15:0	COUNT	Counter output. This counter is reset by writing a 0 to it.

0x0A013EC0 WCSS_A_AGC_TRANSITION_COUNT7**Type:** W**Reset State:** 0x00000000

Debug Counter. Counts transitions between arbitrary states of the AGC state machine.

WCSS_A_AGC_TRANSITION_COUNT7

Bits	Name	Description
15:0	COUNT	Counter output. This counter is reset by writing a 0 to it.

0x0A013EC4 WCSS_A_AGC_TRANSITION_COUNT8**Type:** W**Reset State:** 0x00000000

Debug Counter. Counts transitions between arbitrary states of the AGC state machine.

WCSS_A_AGC_TRANSITION_COUNT8

Bits	Name	Description
15:0	COUNT	Counter output. This counter is reset by writing a 0 to it.

0x0A013EC8 WCSS_A_AGC_TRANSITION_COUNT9**Type:** W**Reset State:** 0x00000000

Debug Counter. Counts transitions between arbitrary states of the AGC state machine.

WCSS_A_AGC_TRANSITION_COUNT9

Bits	Name	Description
15:0	COUNT	Counter output. This counter is reset by writing a 0 to it.

0x0A013ECC WCSS_A_AGC_TRANSITION_COUNT10**Type:** W**Reset State:** 0x00000000

Debug Counter. Counts transitions between arbitrary states of the AGC state machine.

WCSS_A_AGC_TRANSITION_COUNT10

Bits	Name	Description
15:0	COUNT	Counter output. This counter is reset by writing a 0 to it.

0x0A013ED0 WCSS_A_AGC_TRANSITION_COUNT11**Type:** W**Reset State:** 0x00000000

Debug Counter. Counts transitions between arbitrary states of the AGC state machine.

WCSS_A_AGC_TRANSITION_COUNT11

Bits	Name	Description
15:0	COUNT	Counter output. This counter is reset by writing a 0 to it.

0x0A013ED4 WCSS_A_AGC_NOISE_HIST_CTRL**Type:** RW**Reset State:** 0x0001

Controls various modes of operation of the noise histogram module. It can be reset and stopped, it can be read out and reset, it can read out without being reset, and it can be read out, reset and stopped.

WCSS_A_AGC_NOISE_HIST_CTRL

Bits	Name	Description
2	READ_AND_CLEAR	When this bit is set and the read_pulse bit is written to, the current noise histogram bin counter values are latched into the rssi status registers, the noise histogram counters are reset and noise histogram collection continues.
1	READOUT	When this bit is set and the read_pulse bit is written to, the current noise histogram bin counter values are latched into to the rssi status registers. Noise histogram collection continues.
0	RESET	When this bit is set the current noise histogram bin counter values are written to the rssi status registers, the noise histogram collection is halted and the counters are reset.

0x0A013ED8 WCSS_A_AGC_NOISE_HIST_BIN0_MSW**Type:** R**Reset State:** 0x00000000

Counts RSSI samples within IPI (idle power indicator) level 0. Most significant 10 bits.

WCSS_A_AGC_NOISE_HIST_BIN0_MSW

Bits	Name	Description
9:0	IPI_COUNT	IPI level counter value.

0x0A013EDC WCSS_A_AGC_NOISE_HIST_BIN0_LSW**Type:** R**Reset State:** 0x00000000

Counts RSSI samples within IPI (idle power indicator) level 0. Least significant 16 bits.

WCSS_A_AGC_NOISE_HIST_BIN0_LSW

Bits	Name	Description
15:0	IPI_COUNT	IPI level counter value.

0x0A013EE0 WCSS_A_AGC_NOISE_HIST_BIN1_MSW**Type:** R**Reset State:** 0x00000000

Counts RSSI samples within IPI (idle power indicator) level 1. Most significant 10 bits.

WCSS_A_AGC_NOISE_HIST_BIN1_MSW

Bits	Name	Description
9:0	IPI_COUNT	IPI level counter value.

0x0A013EE4 WCSS_A_AGC_NOISE_HIST_BIN1_LSW**Type:** R**Reset State:** 0x00000000

Counts RSSI samples within IPI (idle power indicator) level 1. Least significant 16 bits.

WCSS_A_AGC_NOISE_HIST_BIN1_LSW

Bits	Name	Description
15:0	IPI_COUNT	IPI level counter value.

0x0A013EE8 WCSS_A_AGC_NOISE_HIST_BIN2_MSW**Type:** R**Reset State:** 0x00000000

Counts RSSI samples within IPI (idle power indicator) level 2. Most significant 10 bits.

WCSS_A_AGC_NOISE_HIST_BIN2_MSW

Bits	Name	Description
9:0	IPI_COUNT	IPI level counter value.

0x0A013EEC WCSS_A_AGC_NOISE_HIST_BIN2_LSW**Type:** R**Reset State:** 0x00000000

Counts RSSI samples within IPI (idle power indicator) level 2. Least significant 16 bits.

WCSS_A_AGC_NOISE_HIST_BIN2_LSW

Bits	Name	Description
15:0	IPI_COUNT	IPI level counter value.

0x0A013EF0 WCSS_A_AGC_NOISE_HIST_BIN3_MSW**Type:** R**Reset State:** 0x00000000

Counts RSSI samples within IPI (idle power indicator) level 3. Most significant 10 bits.

WCSS_A_AGC_NOISE_HIST_BIN3_MSW

Bits	Name	Description
9:0	IPI_COUNT	IPI level counter value.

0x0A013EF4 WCSS_A_AGC_NOISE_HIST_BIN3_LSW**Type:** R**Reset State:** 0x00000000

Counts RSSI samples within IPI (idle power indicator) level 3. Least significant 16 bits.

WCSS_A_AGC_NOISE_HIST_BIN3_LSW

Bits	Name	Description
15:0	IPI_COUNT	IPI level counter value.

0x0A013EF8 WCSS_A_AGC_NOISE_HIST_BIN4_MSW**Type:** R**Reset State:** 0x00000000

Counts RSSI samples within IPI (idle power indicator) level 4. Most significant 10 bits.

WCSS_A_AGC_NOISE_HIST_BIN4_MSW

Bits	Name	Description
9:0	IPI_COUNT	IPI level counter value.

0x0A013EFC WCSS_A_AGC_NOISE_HIST_BIN4_LSW**Type:** R**Reset State:** 0x00000000

Counts RSSI samples within IPI (idle power indicator) level 4. Least significant 16 bits.

WCSS_A_AGC_NOISE_HIST_BIN4_LSW

Bits	Name	Description
15:0	IPI_COUNT	IPI level counter value.

0x0A013F00 WCSS_A_AGC_NOISE_HIST_BIN5_MSW**Type:** R**Reset State:** 0x00000000

Counts RSSI samples within IPI (idle power indicator) level 5. Most significant 10 bits.

WCSS_A_AGC_NOISE_HIST_BIN5_MSW

Bits	Name	Description
9:0	IPI_COUNT	IPI level counter value.

0x0A013F04 WCSS_A_AGC_NOISE_HIST_BIN5_LSW**Type:** R**Reset State:** 0x00000000

Counts RSSI samples within IPI (idle power indicator) level 5. Least significant 16 bits.

WCSS_A_AGC_NOISE_HIST_BIN5_LSW

Bits	Name	Description
15:0	IPI_COUNT	IPI level counter value.

0x0A013F08 WCSS_A_AGC_NOISE_HIST_BIN6_MSW**Type:** R**Reset State:** 0x00000000

Counts RSSI samples within IPI (idle power indicator) level 6. Most significant 10 bits.

WCSS_A_AGC_NOISE_HIST_BIN6_MSW

Bits	Name	Description
9:0	IPI_COUNT	IPI level counter value.

0x0A013F0C WCSS_A_AGC_NOISE_HIST_BIN6_LSW**Type:** R**Reset State:** 0x00000000

Counts RSSI samples within IPI (idle power indicator) level 6. Least significant 16 bits.

WCSS_A_AGC_NOISE_HIST_BIN6_LSW

Bits	Name	Description
15:0	IPI_COUNT	IPI level counter value.

0x0A013F10 WCSS_A_AGC_NOISE_HIST_BIN7_MSW**Type:** R**Reset State:** 0x00000000

Counts RSSI samples within IPI (idle power indicator) level 7. Most significant 10 bits.

WCSS_A_AGC_NOISE_HIST_BIN7_MSW

Bits	Name	Description
9:0	IPI_COUNT	IPI level counter value.

0x0A013F14 WCSS_A_AGC_NOISE_HIST_BIN7_LSW**Type:** R**Reset State:** 0x00000000

Counts RSSI samples within IPI (idle power indicator) level 7. Least significant 16 bits.

WCSS_A_AGC_NOISE_HIST_BIN7_LSW

Bits	Name	Description
15:0	IPI_COUNT	IPI level counter value.

0x0A013F18 WCSS_A_AGC_NOISE_HIST_BIN8_MSW**Type:** R**Reset State:** 0x00000000

Counts RSSI samples within IPI (idle power indicator) level 8. Most significant 10 bits.

WCSS_A_AGC_NOISE_HIST_BIN8_MSW

Bits	Name	Description
9:0	IPI_COUNT	IPI level counter value.

0x0A013F1C WCSS_A_AGC_NOISE_HIST_BIN8_LSW**Type:** R**Reset State:** 0x00000000

Counts RSSI samples within IPI (idle power indicator) level 8. Least significant 16 bits.

WCSS_A_AGC_NOISE_HIST_BIN8_LSW

Bits	Name	Description
15:0	IPI_COUNT	IPI level counter value.

0x0A013F20 WCSS_A_AGC_NOISE_HIST_BIN9_MSW**Type:** R**Reset State:** 0x00000000

Counts RSSI samples within IPI (idle power indicator) level 9. Most significant 10 bits.

WCSS_A_AGC_NOISE_HIST_BIN9_MSW

Bits	Name	Description
9:0	IPI_COUNT	IPI level counter value.

0x0A013F24 WCSS_A_AGC_NOISE_HIST_BIN9_LSW**Type:** R**Reset State:** 0x00000000

Counts RSSI samples within IPI (idle power indicator) level 9. Least significant 16 bits.

WCSS_A_AGC_NOISE_HIST_BIN9_LSW

Bits	Name	Description
15:0	IPI_COUNT	IPI level counter value.

0x0A013F28 WCSS_A_AGC_NOISE_HIST_BIN10_MSW**Type:** R**Reset State:** 0x00000000

Counts RSSI samples within IPI (idle power indicator) level 10. Most significant 10 bits.

WCSS_A_AGC_NOISE_HIST_BIN10_MSW

Bits	Name	Description
9:0	IPI_COUNT	IPI level counter value.

0x0A013F2C WCSS_A_AGC_NOISE_HIST_BIN10_LSW**Type:** R**Reset State:** 0x00000000

Counts RSSI samples within IPI (idle power indicator) level 10. Least significant 16 bits.

WCSS_A_AGC_NOISE_HIST_BIN10_LSW

Bits	Name	Description
15:0	IPI_COUNT	IPI level counter value.

0x0A013F30 WCSS_A_AGC_NOISE_HIST_BUSY_MSW**Type:** R**Reset State:** 0x00000000

Counts microseconds during noise histogram collection during which no noise is measured because a packet is transmitted, received or NAV is busy. Most significant 10 bits.

WCSS_A_AGC_NOISE_HIST_BUSY_MSW

Bits	Name	Description
9:0	COUNT	IPI level counter value.

0x0A013F34 WCSS_A_AGC_NOISE_HIST_BUSY_LSW**Type:** R**Reset State:** 0x00000000

Counts microseconds during noise histogram collection during which no noise is measured because a packet is transmitted, received or NAV is busy. Least significant 16 bits.

WCSS_A_AGC_NOISE_HIST_BUSY_LSW

Bits	Name	Description
15:0	COUNT	IPI level counter value.

0x0A013F38 WCSS_A_AGC_WATCHDOG_MASK**Type:** RW**Reset State:** 0x0001

This register is used to mask the watchdog-driven reset of the AGC state machine. When the reset is not masked, an expiry of the AGC watchdog timer will set the timeout state register, generate an interrupt and reset the AGC state machine. When the reset is masked, an expiry of the AGC watchdog timer will set the timeout state register, generate an interrupt, but will not reset the AGC state machine. This functionality is intended for debugging purposes when the AGC state machine needs to remain in the hanging state so the various conditions that may have lead to the hangup can be studied before they are automatically reset by the watchdog timer. registerComment

WCSS_A_AGC_WATCHDOG_MASK

Bits	Name	Description
0	MASK	0x0: UNMASKED (watchdog-driven reset enabled) 0x1: MASKED (watchdog-driven reset disabled)

0x0A013F3C WCSS_A_AGC_TIMEOUT_STATE**Type:** R**Reset State:** 0x00000000

AGC state in which watchdog timer expired.

WCSS_A_AGC_TIMEOUT_STATE

Bits	Name	Description
4:0	STATE	AGC State.

0x0A013F40 WCSS_A_AGC_AGC_WATCHDOG_TIMEOUT_WAIT_DCO_A20**Type:** RW**Reset State:** 0x0000

This register configures the watchdog timer for the WAIT_DCO_A state of the AGC state machine. The watchdog timer is reset when the AGC state machine transitions from one state to another. When its value exceeds the threshold programmed in this register it resets the AGC state

machine as well as the RXA and RXB modules. A value of zero disables the watchdog timer in this state.

WCSS_A_AGC_AGC_WATCHDOG_TIMEOUT_WAIT_DCO_A20

Bits	Name	Description
7:0	TIME	Timeout delay. This value is multiplied by 2 to arrive at the number of 40 MHz clock cycles after which the watchdog timer expires. This results in a maximum delay of $255 * 2 * 12.5 \text{ ns} = 6.375 \text{ us}$.

0x0A013F44 WCSS_A_AGC_AGC_WATCHDOG_TIMEOUT_WAIT_DCO_A40

Type: RW

Reset State: 0x0000

This register configures the watchdog timer for the WAIT_DCO_A40 state of the AGC state machine. The watchdog timer is reset when the AGC state machine transitions from one state to another. When its value exceeds the threshold programmed in this register it resets the AGC state machine as well as the RXA and RXB modules. A value of zero disables the watchdog timer in this state.

WCSS_A_AGC_AGC_WATCHDOG_TIMEOUT_WAIT_DCO_A40

Bits	Name	Description
7:0	TIME	Timeout delay. This value is multiplied by 2 to arrive at the number of 40 MHz clock cycles after which the watchdog timer expires. This results in a maximum delay of $255 * 2 * 12.5 \text{ ns} = 6.375 \text{ us}$.

0x0A013F48 WCSS_A_AGC_AGC_WATCHDOG_TIMEOUT_WAIT_DCO_A80

Type: RW

Reset State: 0x0000

This register configures the watchdog timer for the WAIT_DCO_A80 state of the AGC state machine. The watchdog timer is reset when the AGC state machine transitions from one state to another. When its value exceeds the threshold programmed in this register it resets the AGC state machine as well as the RXA and RXB modules. A value of zero disables the watchdog timer in this state.

WCSS_A_AGC_AGC_WATCHDOG_TIMEOUT_WAIT_DCO_A80

Bits	Name	Description
7:0	TIME	Timeout delay. This value is multiplied by 2 to arrive at the number of 40 MHz clock cycles after which the watchdog timer expires. This results in a maximum delay of $255 * 2 * 12.5 \text{ ns} = 6.375 \text{ us}$.

0x0A013F4C WCSS_A_AGC_AGC_WATCHDOG_TIMEOUT_WAIT_DCO_B**Type:** RW**Reset State:** 0x0000

This register configures the watchdog timer for the WAIT_DCO_B state of the AGC state machine. The watchdog timer is reset when the AGC state machine transitions from one state to another. When its value exceeds the threshold programmed in this register it resets the AGC state machine as well as the RXA and RXB modules. A value of zero disables the watchdog timer in this state.

WCSS_A_AGC_AGC_WATCHDOG_TIMEOUT_WAIT_DCO_B

Bits	Name	Description
7:0	TIME	Timeout delay. This value is multiplied by 2 to arrive at the number of 40 MHz clock cycles after which the watchdog timer expires. This results in a maximum delay of $255 * 2 * 12.5 \text{ ns} = 6.375 \text{ us}$.

0x0A013F50 WCSS_A_AGC_AGC_WATCHDOG_TIMEOUT_WAIT_A**Type:** RW**Reset State:** 0x0000

This register configures the watchdog timer for the WAIT_A state of the AGC state machine. The watchdog timer is reset when the AGC state machine transitions from one state to another. When its value exceeds the threshold programmed in this register it resets the AGC state machine as well as the RXA and RXB modules. A value of zero disables the watchdog timer in this state.

WCSS_A_AGC_AGC_WATCHDOG_TIMEOUT_WAIT_A

Bits	Name	Description
7:0	TIME	Timeout delay. This value is multiplied by 8 (2^3) to arrive at the number of 40 MHz clock cycles after which the watchdog timer expires. This results in a maximum delay of $255 * 2^3 * 25 \text{ ns} = 51 \text{ us}$.

0x0A013F54 WCSS_A_AGC_AGC_WATCHDOG_TIMEOUT_RXA_HDR**Type:** RW**Reset State:** 0x0000

This register configures the watchdog timer for the RXA_HDR state of the AGC state machine. The watchdog timer is reset when the AGC state machine transitions from one state to another. When its value exceeds the threshold programmed in this register it resets the AGC state machine as well as the RXA and RXB modules. A value of zero disables the watchdog timer in this state.

WCSS_A_AGC_AGC_WATCHDOG_TIMEOUT_RXA_HDR

Bits	Name	Description
7:0	TIME	Timeout delay. This value is multiplied by 8 (2^3) to arrive at the number of 40 MHz clock cycles after which the watchdog timer expires. This results in a maximum delay of $255 * 2^3 * 25 \text{ ns} = 51 \text{ us}$.

0x0A013F58 WCSS_A_AGC_AGC_WATCHDOG_TIMEOUT_HT_STS_COARSE**Type:** RW**Reset State:** 0x0000

This register configures the watchdog timer for the HT_STS_COARSE state of the AGC state machine. The watchdog timer is reset when the AGC state machine transitions from one state to another. When its value exceeds the threshold programmed in this register it resets the AGC state machine as well as the RXA and RXB modules. A value of zero disables the watchdog timer in this state.

WCSS_A_AGC_AGC_WATCHDOG_TIMEOUT_HT_STS_COARSE

Bits	Name	Description
7:0	TIME	Timeout delay. This value is multiplied by 2 to arrive at the number of 40 MHz clock cycles after which the watchdog timer expires. This results in a maximum delay of $255 * 2 * 12.5 \text{ ns} = 6.375 \text{ us}$.

0x0A013F5C WCSS_A_AGC_AGC_WATCHDOG_TIMEOUT_HT_STS_FINE**Type:** RW**Reset State:** 0x0000

This register configures the watchdog timer for the HT_STS_FINE state of the AGC state machine. The watchdog timer is reset when the AGC state machine transitions from one state to another. When its value exceeds the threshold programmed in this register it resets the AGC state machine as well as the RXA and RXB modules. A value of zero disables the watchdog timer in this state.

WCSS_A_AGC_AGC_WATCHDOG_TIMEOUT_HT_STS_FINE

Bits	Name	Description
7:0	TIME	Timeout delay. This value is multiplied by 2 to arrive at the number of 40 MHz clock cycles after which the watchdog timer expires. This results in a maximum delay of $255 * 2 * 12.5 \text{ ns} = 6.375 \text{ us}$.

0x0A013F60 WCSS_A_AGC_AGC_WATCHDOG_TIMEOUT_VHT_STS_COARSE**Type:** RW**Reset State:** 0x0000

This register configures the watchdog timer for the VHT_STS_COARSE state of the AGC state machine. The watchdog timer is reset when the AGC state machine transitions from one state to another. When its value exceeds the threshold programmed in this register it resets the AGC state machine as well as the RXA and RXB modules. A value of zero disables the watchdog timer in this state.

WCSS_A_AGC_AGC_WATCHDOG_TIMEOUT_VHT_STS_COARSE

Bits	Name	Description
7:0	TIME	Timeout delay. This value is multiplied by 2 to arrive at the number of 40 MHz clock cycles after which the watchdog timer expires. This results in a maximum delay of $255 * 2 * 12.5 \text{ ns} = 6.375 \text{ us}$.

0x0A013F64 WCSS_A_AGC_AGC_WATCHDOG_TIMEOUT_VHT_STS_FINE**Type:** RW**Reset State:** 0x0000

This register configures the watchdog timer for the VHT_STS_FINE state of the AGC state machine. The watchdog timer is reset when the AGC state machine transitions from one state to another. When its value exceeds the threshold programmed in this register it resets the AGC state machine as well as the RXA and RXB modules. A value of zero disables the watchdog timer in this state.

WCSS_A_AGC_AGC_WATCHDOG_TIMEOUT_VHT_STS_FINE

Bits	Name	Description
7:0	TIME	Timeout delay. This value is multiplied by 2 to arrive at the number of 40 MHz clock cycles after which the watchdog timer expires. This results in a maximum delay of $255 * 2 * 12.5 \text{ ns} = 6.375 \text{ us}$.

0x0A013F68 WCSS_A_AGC_AGC_WATCHDOG_TIMEOUT_RXA_DATA**Type:** RW**Reset State:** 0x0000

This register configures the watchdog timer for the RXA_DATA state of the AGC state machine. The watchdog timer is reset when the AGC state machine transitions from one state to another. When its value exceeds the threshold programmed in this register it resets the AGC state machine as well as the RXA and RXB modules. A value of zero disables the watchdog timer in this state.

WCSS_A_AGC_AGC_WATCHDOG_TIMEOUT_RXA_DATA

Bits	Name	Description
7:0	TIME	Timeout delay. This value is multiplied by 16384 (2^{14}) to arrive at the number of 40 MHz clock cycles after which the watchdog timer expires. This results in a maximum delay of $255 * 2^{14} * 25 \text{ ns} = 104 \text{ ms}$.

0x0A013F6C WCSS_A_AGC_AGC_WATCHDOG_TIMEOUT_CAP_A**Type:** RW**Reset State:** 0x0000

This register configures the watchdog timer for the CAP_A state of the AGC state machine. The watchdog timer is reset when the AGC state machine transitions from one state to another. When its value exceeds the threshold programmed in this register it resets the AGC state machine as well as the RXA and RXB modules. A value of zero disables the watchdog timer in this state.

WCSS_A_AGC_AGC_WATCHDOG_TIMEOUT_CAP_A

Bits	Name	Description
7:0	TIME	Timeout delay. This value is multiplied by 16384 (2^{14}) to arrive at the number of 40 MHz clock cycles after which the watchdog timer expires. This results in a maximum delay of $255 * 2^{14} * 25 \text{ ns} = 104 \text{ ms}$.

0x0A013F70 WCSS_A_AGC_AGC_WATCHDOG_TIMEOUT_TF_EST**Type:** RW**Reset State:** 0x0000

This register configures the watchdog timer for the TF_EST state of the AGC state machine. The watchdog timer is reset when the AGC state machine transitions from one state to another. When its value exceeds the threshold programmed in this register it resets the AGC state machine as well as the RXA and RXB modules. A value of zero disables the watchdog timer in this state.

WCSS_A_AGC_AGC_WATCHDOG_TIMEOUT_TF_EST

Bits	Name	Description
7:0	TIME	Timeout delay. This value is multiplied by 4 (2^2) to arrive at the number of 40 MHz clock cycles after which the watchdog timer expires. This results in a maximum delay of $255 * 2^2 * 25 \text{ ns} = 25.5 \text{ us}$.

0x0A013F74 WCSS_A_AGC_AGC_WATCHDOG_TIMEOUT_WAIT_B**Type:** RW**Reset State:** 0x0000

This register configures the watchdog timer for the WAIT_B state of the AGC state machine. The watchdog timer is reset when the AGC state machine transitions from one state to another. When its value exceeds the threshold programmed in this register it resets the AGC state machine as well as the RXA and RXB modules. A value of zero disables the watchdog timer in this state.

WCSS_A_AGC_AGC_WATCHDOG_TIMEOUT_WAIT_B

Bits	Name	Description
7:0	TIME	Timeout delay. This value is multiplied by 64 (2^6) to arrive at the number of 40 MHz clock cycles after which the watchdog timer expires. This results in a maximum delay of $255 * 2^6 * 25 \text{ ns} = 408 \text{ us}$.

0x0A013F78 WCSS_A_AGC_AGC_WATCHDOG_TIMEOUT_RXB_HDR**Type:** RW**Reset State:** 0x007D

This register configures the watchdog timer for the RXB_HDR state of the AGC state machine. The watchdog timer is reset when the AGC state machine transitions from one state to another. When its value exceeds the threshold programmed in this register it resets the AGC state machine as well as the RXA and RXB modules. A value of zero disables the watchdog timer in this state.

WCSS_A_AGC_AGC_WATCHDOG_TIMEOUT_RXB_HDR

Bits	Name	Description
7:0	TIME	Timeout delay. This value is multiplied by 64 (2^6) to arrive at the number of 40 MHz clock cycles after which the watchdog timer expires. This results in a maximum delay of $255 * 2^6 * 25 \text{ ns} = 408 \text{ us}$.

0x0A013F7C WCSS_A_AGC_AGC_WATCHDOG_TIMEOUT_RXB_DATA**Type:** RW**Reset State:** 0x0000

This register configures the watchdog timer for the RXB_DATA state of the AGC state machine. The watchdog timer is reset when the AGC state machine transitions from one state to another. When its value exceeds the threshold programmed in this register it resets the AGC state machine as well as the RXA and RXB modules. A value of zero disables the watchdog timer in this state.

WCSS_A_AGC_AGC_WATCHDOG_TIMEOUT_RXB_DATA

Bits	Name	Description
7:0	TIME	Timeout delay. This value is multiplied by 16384 (2^{14}) to arrive at the number of 40 MHz clock cycles after which the watchdog timer expires. This results in a maximum delay of $255 * 2^{14} * 25 \text{ ns} = 104 \text{ ms}$.

0x0A013F80 WCSS_A_AGC_AGC_WATCHDOG_TIMEOUT_CAP_B**Type:** RW**Reset State:** 0x0000

This register configures the watchdog timer for the CAP_B state of the AGC state machine. The watchdog timer is reset when the AGC state machine transitions from one state to another. When its value exceeds the threshold programmed in this register it resets the AGC state machine as well as the RXA and RXB modules. A value of zero disables the watchdog timer in this state.

WCSS_A_AGC_AGC_WATCHDOG_TIMEOUT_CAP_B

Bits	Name	Description
7:0	TIME	Timeout delay. This value is multiplied by 16384 (2^{14}) to arrive at the number of 40 MHz clock cycles after which the watchdog timer expires. This results in a maximum delay of $255 * 2^{14} * 25 \text{ ns} = 104 \text{ ms}$.

0x0A013F84 WCSS_A_AGC_AGC_WATCHDOG_TIMEOUT_WAIT_SIFS**Type:** RW**Reset State:** 0x0000

This register configures the watchdog timer for the WAIT_SIFS state of the AGC state machine. The watchdog timer is reset when the AGC state machine transitions from one state to another. When its value exceeds the threshold programmed in this register it resets the AGC state machine as well as the RXA and RXB modules. A value of zero disables the watchdog timer in this state.

WCSS_A_AGC_AGC_WATCHDOG_TIMEOUT_WAIT_SIFS

Bits	Name	Description
7:0	TIME	Timeout delay. This value is multiplied by 8 (2^3) to arrive at the number of 40 MHz clock cycles after which the watchdog timer expires. This results in a maximum delay of $255 * 2^3 * 25 \text{ ns} = 51 \text{ us}$.

0x0A013F88 WCSS_A_AGC_CCA0_WATCHDOG_TIMEOUT**Type:** RW**Reset State:** 0x0000

This register sets the maximum time CCA0 can be busy before a timeout condition is detected. Setting to 0 disables this watchdog check.

WCSS_A_AGC_CCA0_WATCHDOG_TIMEOUT

Bits	Name	Description
15:0	TIME	Timeout value in 102.4us steps. Setting to 0 disables the timeout.

0x0A013F8C WCSS_A_AGC_CCA1_WATCHDOG_TIMEOUT**Type:** RW**Reset State:** 0x0000

This register sets the maximum time CCA1 can be busy before a timeout condition is detected. Setting to 0 disables this watchdog check.

WCSS_A_AGC_CCA1_WATCHDOG_TIMEOUT

Bits	Name	Description
15:0	TIME	Timeout value in 102.4us steps. Setting to 0 disables the timeout.

0x0A013F90 WCSS_A_AGC_CCA2_WATCHDOG_TIMEOUT**Type:** RW**Reset State:** 0x0000

This register sets the maximum time CCA2 or CCA3 can be busy before a timeout condition is detected. Setting to 0 disables this watchdog check.

WCSS_A_AGC_CCA2_WATCHDOG_TIMEOUT

Bits	Name	Description
15:0	TIME	Timeout value in 102.4us steps. Setting to 0 disables the timeout.

0x0A013F94 WCSS_A_AGC_CCA0_WATCHDOG_EXPIRE_COUNT**Type:** W**Reset State:** 0x00000000

CCA0 watchdog expire count.

WCSS_A_AGC_CCA0_WATCHDOG_EXPIRE_COUNT

Bits	Name	Description
7:0	COUNT	

0x0A013F98 WCSS_A_AGC_CCA1_WATCHDOG_EXPIRE_COUNT

Type: W

Reset State: 0x00000000

CCA1 watchdog expire count.

WCSS_A_AGC_CCA1_WATCHDOG_EXPIRE_COUNT

Bits	Name	Description
7:0	COUNT	

0x0A013F9C WCSS_A_AGC_CCA2_WATCHDOG_EXPIRE_COUNT

Type: W

Reset State: 0x00000000

CCA2 watchdog expire count.

WCSS_A_AGC_CCA2_WATCHDOG_EXPIRE_COUNT

Bits	Name	Description
7:0	COUNT	

0x0A013FA0 WCSS_A_AGC_RXFIR_CONFIG

Type: RW

Reset State: 0x0000

Contains RxFIR configuration settings.

WCSS_A_AGC_RXFIR_CONFIG

Bits	Name	Description
0	DEC_BY_4	Enables decimation-by-4 in the stage 1 RxFIR instead of decimation-by-3 in the normal functional mode.

0x0A013FA4 WCSS_A_AGC_VHTSIGA_P_AID1**Type:** RW**Reset State:** 0x0000**WCSS_A_AGC_VHTSIGA_P_AID1**

Bits	Name	Description
8:0	VALUE	Use for filtering 11ac SU packet

0x0A013FA8 WCSS_A_AGC_VHTSIGA_P_AID2**Type:** RW**Reset State:** 0x0000**WCSS_A_AGC_VHTSIGA_P_AID2**

Bits	Name	Description
8:0	VALUE	Use for filtering 11ac SU packet

0x0A013FAC WCSS_A_AGC_VHTSIGA_P_AID3**Type:** RW**Reset State:** 0x0000**WCSS_A_AGC_VHTSIGA_P_AID3**

Bits	Name	Description
8:0	VALUE	Use for filtering 11ac SU packet

0x0A013FB0 WCSS_A_AGC_VHTSIGA_P_AID4**Type:** RW**Reset State:** 0x0000**WCSS_A_AGC_VHTSIGA_P_AID4**

Bits	Name	Description
8:0	VALUE	Use for filtering 11ac SU packet

0x0A013FB4 WCSS_A_AGC_VHTSIGA_P_AID5**Type:** RW**Reset State:** 0x0000**WCSS_A_AGC_VHTSIGA_P_AID5**

Bits	Name	Description
8:0	VALUE	Use for filtering 11ac SU packet

0x0A013FB8 WCSS_A_AGC_VHTSIGA_BWENCODING**Type:** RW**Reset State:** 0x01AC

Contains programming for bandwidth

WCSS_A_AGC_VHTSIGA_BWENCODING

Bits	Name	Description
8:6	BW80	Fully specified field to indicate 20M mode, 40M mode and 80M mode.
5:3	BW40	
2:0	BW20	

0x0A013FBC WCSS_A_AGC_VHTSIGA_GID1**Type:** RW**Reset State:** 0x003F

Four programmable GID values can be designated as SU (single user) for the receiver. If the receiver finds that the incoming GID matches any of the four GID values programmed in this and the following register, then it will be designated as an SU packet.

WCSS_A_AGC_VHTSIGA_GID1

Bits	Name	Description
11:6	GID2	Second programmable SU GID.
5:0	GID1	First programmable SU GID.

0x0A013FC0 WCSS_A_AGC_VHTSIGA_GID2**Type:** RW**Reset State:** 0x0000

Four programmable GID values can be designated as SU (single user) for the receiver. If the receiver finds that the incoming GID matches any of the four GID values programmed in this and the preceding register, then it will be designated as an SU packet.

WCSS_A_AGC_VHTSIGA_GID2

Bits	Name	Description
11:6	GID4	Fourth programmable SU GID.
5:0	GID3	Third programmable SU GID.

0x0A013FC4 WCSS_A_AGC_CCA_GI_DET_CFG_LGI**Type:** RW**Reset State:** 0xA59B**WCSS_A_AGC_CCA_GI_DET_CFG_LGI**

Bits	Name	Description
15:8	TIMER_MAX_END	This register holds the upper limit on the timer (running off 40MHz clock) by when the subsequent ccadet-threshold crossing is expected. Ideally this value is 160 implying 4 usec.
7:0	TIMER_MIN_END	This register holds the lower limit on the timer (running off 40MHz clock) by when the subsequent ccadet-threshold crossing is expected. Ideally this value is 160 implying 4 usec.

0x0A013FC8 WCSS_A_AGC_CCA_GI_DET_CFG_SGI**Type:** RW**Reset State:** 0x958B**WCSS_A_AGC_CCA_GI_DET_CFG_SGI**

Bits	Name	Description
15:8	TIMER_MAX_END	This register holds the upper limit on the timer (running off 40MHz clock) by when the subsequent ccadet-threshold crossing is expected. Ideally this value is 144 implying 3.6 usec.
7:0	TIMER_MIN_END	This register holds the lower limit on the timer (running off 40MHz clock) by when the subsequent ccadet-threshold crossing is expected. Ideally this value is 144 implying 3.6 usec.

0x0A013FCC WCSS_A_AGC_CCA_GI_DET_CFG1**Type:** RW**Reset State:** 0x00BE**WCSS_A_AGC_CCA_GI_DET_CFG1**

Bits	Name	Description
8:0	TH_SGI	

10.28 WCSS_A_BBANA_BBANA**0x0A014000 WCSS_A_BBANA_ADC_CTRL****Type:** RW**Reset State:** 0x0000

This set of registers are used to control ADC operations.

WCSS_A_BBANA_ADC_CTRL

Bits	Name	Description
9	OFFSET_MEAS	Enable ADCs to do offset calibration.
8:7	INTLV_CTRL	Used to control ADC interleaving.
6	ADC_PWR_DOWN_N	Power down I & Q reference, clock and output (active low). Not currently used. 0x0: PWR_DOWN (Disables both I & Q, reference, clock and output) 0x1: ENABLED (ADC I & Q, reference, clock and outputs enabled)
5	OUTPUT_DISABLE_Q	Output disable control for ADC Q data rail. 0x0: NORMAL (Normal ADC output level) 0x1: MID (Mid level AC output)
4	EN_Q_OVERRIDE	Enable override control for ADC Q data rail. 0x0: NO_OVERRIDE (Signal in bit 3 is not used to control ADC enable signal) 0x1: OVERRIDE (Signal in bit 3 is used to control ADC enable signal)
3	EN_Q	Enable ADC for Q data rail. 0x0: DISABLED (ADC Q data rail disabled) 0x1: ENABLED (ADC Q data rail enabled)
2	OUTPUT_DISABLE_I	Output disable control for ADC I data rail. 0x0: NORMAL (Normal ADC output level) 0x1: MID (Mid level AC output)

WCSS_A_BBANA_ADC_CTRL (cont.)

Bits	Name	Description
1	EN_I_OVERRIDE	Enable override control for ADC I data rail. 0x0: NO_OVERRIDE (Signal in bit 0 is not used to control ADC enable signal) 0x1: OVERRIDE (Signal in bit 0 is used to control ADC enable signal)
0	EN_I	Enable ADC for I data rail. 0x0: DISABLED (ADC I data rail disabled) 0x1: ENABLED (ADC I data rail enabled)

0x0A014004 WCSS_A_BBANA_ADC_CLK_CTRL**Type:** RW**Reset State:** 0x0000

This set of registers are used for controlling the ADC clock settings.

WCSS_A_BBANA_ADC_CLK_CTRL

Bits	Name	Description
5:4	ADJ_DUTY_CYCLE_PW	Setting used to adjust the duty cycle pulse width
3	DUTY_CYCLE_EN	Setting used to enable/disable the duty cycle circuit 0x0: DISABLED (Duty cycle circuit is disabled) 0x1: ENABLED (Duty cycle circuit is enabled)
2	CLKEDGE_SEL	Setting used to select the edge of the clock to be used by the ADCs 0x0: POSEDGE (Positive edge of clock is used) 0x1: NEGEDGE (Negative edge of clock is used)
1	CLKSRC_SEL	Setting used to select the clock source to the ADCs
0	DISABLE_CLK	Used to enable/disable clock to ADCs. 0x0: ENABLED (Clock to the ADCs is enabled) 0x1: DISABLED (Clock to the ADCs is disabled)

0x0A014008 WCSS_A_BBANA_ADC_REF_CTRL**Type:** RW**Reset State:** 0x0000

This set of registers are used for controlling the reference circuit settings.

WCSS_A_BBANA_ADC_REF_CTRL

Bits	Name	Description
6	REF_PWRDOWN	Power down the reference.
5	VDD_SEL	Switch the reference to VDD.
4	BIAS_SEL	Selects the source of the reference bias.
3:2	OUT_BIAS	Used to adjust the bias current of the reference generator.
1:0	V_ADJUST	Register value used to adjust the ADC reference voltage.

0x0A01400C WCSS_A_BBANA_ADC_CONV_ADJ**Type:** RW**Reset State:** 0x0000

This set of registers are used for adjusting various conversion settings.

WCSS_A_BBANA_ADC_CONV_ADJ

Bits	Name	Description
8:7	INP_FILT_POLE	Register value used to modify the input filter pole frequency.
6:5	CONV_DLY	Adjust the conversion delay.
4:3	COMM_BIAS	Adjust the common mode bias.
2:0	REG_ADJUST	Register value used to adjust the regulator.

0x0A014014 WCSS_A_BBANA_ADC_SPARE3**Type:** RW**Reset State:** 0x0000

This register contains the third set of ADC spare bits.

WCSS_A_BBANA_ADC_SPARE3

Bits	Name	Description
7:4	SPARE3_UNUSED	Register value holding unused (unmapped) ADC spare3 bits.
3	CONV_CTRL	Conversion control register. 0x0: CONV_UNDEF_0 (Conversion control 0 - unused) 0x1: CONV_UNDEF_1 (Conversion control 1 - unused)

WCSS_A_BBANA_ADC_SPARE3 (cont.)

Bits	Name	Description
2:0	FILT_CTRL	ADC filter control register. 0x0: FILT_UNDEF_0 (filter ctrl undefined 0) 0x1: FILT_UNDEF_1 (filter ctrl undefined 1) 0x2: FILT_UNDEF_2 (filter ctrl undefined 2) 0x3: FILT_UNDEF_3 (filter ctrl undefined 3) 0x4: FILT_UNDEF_4 (filter ctrl undefined 4) 0x5: FILT_UNDEF_5 (filter ctrl undefined 5) 0x6: FILT_UNDEF_6 (filter ctrl undefined 6) 0x7: FILT_UNDEF_7 (filter ctrl undefined 7)

0x0A014018 WCSS_A_BBANA_DAC_CTRL**Type:** RW**Reset State:** 0x0000

This set of registers are used to control DAC operations.

WCSS_A_BBANA_DAC_CTRL

Bits	Name	Description
12:10	DAC_CFG1	Register for DAC configuration settings.
9	CLKEDGE_SEL	Setting used to select the edge of the clock to be used by the DACs 0x0: POSEDGE (Positive edge of clock is used to register data) 0x1: NEGEDGE (Negative edge of clock is used to register data)
8	INPUT_STANDBY_EN_CTRL	Note: This bit is no longer used. Control of TXAIF input signal on to DAC enable/standby signal. 0x0: ENABLE_DRIVEN (TXAIF standby signal drives DAC enable signal directly. The DAC standby is driven by the standby override registers.) 0x1: STANDBY_DRIVEN (TXAIF standby signal drives DAC standby signal directly. The DAC enable is driven by the enable override registers.)
7	STANDBY_Q_OVERRIDE	Standby override control for DAC Q data rail. 0x0: NO_OVERRIDE (Signal in bit 6 is not used to control DAC standby status) 0x1: OVERRIDE (Signal in bit 6 is used to control DAC standby status)
6	STANDBY_Q	Standby control for DAC Q data rail. 0x0: NORMAL (DAC in normal mode) 0x1: MID (Place DAC in standby)

WCSS_A_BBANA_DAC_CTRL (cont.)

Bits	Name	Description
5	EN_Q_OVERRIDE	Enable override control for DAC Q data rail. 0x0: NO_OVERRIDE (Signal in bit 4 is not used to control DAC enable signal) 0x1: OVERRIDE (Signal in bit 4 is used to control DAC enable signal)
4	EN_Q	Enable DAC for Q data rail. 0x0: DISABLED (DAC Q data rail disabled) 0x1: ENABLED (DAC Q data rail enabled)
3	STANDBY_I_OVERRIDE	Standby override control for DAC I data rail. 0x0: NO_OVERRIDE (Signal in bit 2 is not used to control DAC standby status) 0x1: OVERRIDE (Signal in bit 2 is used to control DAC standby status)
2	STANDBY_I	Standby control for DAC I data rail. 0x0: NORMAL (DAC in normal mode) 0x1: STANDBY (Place DAC in standby)
1	EN_I_OVERRIDE	Enable override control for DAC I data rail. 0x0: NO_OVERRIDE (Signal in bit 0 is not used to control DAC enable signal) 0x1: OVERRIDE (Signal in bit 0 is used to control DAC enable signal)
0	EN_I	Enable DAC for I data rail. 0x0: DISABLED (DAC I data rail disabled) 0x1: ENABLED (DAC I data rail enabled)

0x0A01401C WCSS_A_BBANA_DAC_BIAS**Type:** RW**Reset State:** 0x0000

This set of registers are used to control the DAC bias.

WCSS_A_BBANA_DAC_BIAS

Bits	Name	Description
6:0	TX_BIAS	Select output current value.

0x0A014024 WCSS_A_BBANA_DAC_CFG**Type:** RW**Reset State:** 0x0000

This set of registers are used to control 19-bit extended DAC configuration.

WCSS_A_BBANA_DAC_CFG

Bits	Name	Description
15:0	DAC_CFG0	Register for DAC lower 16-bit configuration settings.

0x0A014028 WCSS_A_BBANA_DAC_CAL_CTRL

Type: RW

Reset State: 0x0000

This set of registers are used to control DAC calibration

WCSS_A_BBANA_DAC_CAL_CTRL

Bits	Name	Description
14	CAL_EN	Enable the calibration state machine
13:8	SM_DIN	Data to write in CALDAC memory for write operation
7	SM_WR	Write signal for CALDAC memory
6:0	SM_ADDR	CALDAC memory address lines

0x0A01402C WCSS_A_BBANA_DAC_CAL_CFG

Type: RW

Reset State: 0x0000

This set of registers are used to set DAC calibration configuration

WCSS_A_BBANA_DAC_CAL_CFG

Bits	Name	Description
15:0	CAL_CFG	CSM configuration bits

0x0A014030 WCSS_A_BBANA_DAC_CAL_STATUS

Type: R

Reset State: 0x0000

This set of registers capture the status of the DAC calibration

WCSS_A_BBANA_DAC_CAL_STATUS

Bits	Name	Description
7	CAL_ERR	Calibration error
6	CAL_BUSY	Calibration in progress
5:0	SM_DOUT	Data read from CALDAC memory for read operation

10.29 WCSS_A_BMU_BMU**0x0A080000 WCSS_A_BMU_CONTROL****Type:** RW**Reset State:** 0x03FE03FE

Sets the valid range of BD and PDU indexes in the BMU.

WCSS_A_BMU_CONTROL

Bits	Name	Description
31:16	MAX_PDU_INDEX_NR	Typically the MAX PDU index is greater then the MAC BD index. In that case BDs can not be used as PDUs and vice versa. When MAX_BD_Index and MAX_PDU_index are programmed to the same value, the BMU is set in a special mode: Mixed mode. Each PDU can be used as a BD or each BD used as a PDU. At initialization the PDU idle list does not need to be created, just the BD idle list. When PDUs are requested, they will be taken from the BD idle list. During operation BD and PDU idle lists will both be maintained, but all released PDUs will automatically get moved back to the BD idle list. In this mode, the MAX BD and PDU index number can not be greater then the MAX BD index number that the HW allows. Note that before the WQs get enabled, this value must be programmed accordingly to the amount of packet memoryavailable. This is similar to have to program the bd_pbu_base_address in the MCU. In Volans, the BMU should be configured to only operate in Mixed Mode.
10:0	MAX_BD_INDEX_NR	Virgo can maximum support 2K BDs. The maximum value that can therefor be programmed is 2047. Where as Libra and Volans can support only 1023 BDs Note that before the WQs get enabled, this value must be programmed accordingly to the amount of packet memoryavailable. This is similar to have to program the bd_pbu_base_address in the MCU.

0x0A080004 WCSS_A_BMU_ERR_INTR_STATUS**Type:** R**Reset State:** 0x00000000

This register indicates the BMU Error Interrupt Status. When one of these errors occur, the status will always reflect this. The Err_Intr_ENABLE register is used to indicate which error bits will cause an interrupt to be generated. Writing a 1'b1 to a particular bit will clear the error status bit. Most of these errors are generated when the BMU receives a command. In this case the command will be stored in the 'err_int_addr' and 'err_int_wdata' registers.

WCSS_A_BMU_ERR_INTR_STATUS

Bits	Name	Description
31	BMU_ACCESS_DIRECTION	When one of the error indications below gets set, this bit will indicate if this happened on a read or write BMU access. 1'b1 indicates write access 1'b0 indicates read access. On a write access, register err_int_wdata will contain the write data.
30	BMU_INTERRUPT_STATUS	Status of the BMU err interrupt status signal going to the mcu. This is the OR of all interrupts enabled in this register and bmu_idle_bd_pdu_threshold_interrupt_status bit from register: bmu_idle_bd_pdu_status.
28	BTQM_ERR	This is the OR of all the enabled bits from btqm_err_status register. Set when the btqm detects an error or some other operational related interrupt in the btqm_err_status register. The details on which btqm error happened can be found in register 'btqm_err_int_status'.
27	DOUBLE_RELEASE_PDU_ERR	Error set when head or tail of a list of linked PDUs that is being released is equal to the head or tail pointer of the idle PDU list. Note that there can be a delay between the release command and the detection of this problem due to buffering.
26	DOUBLE_RELEASE_BD_ERR	Error set when head or tail of a list of linked BDs that is being released is equal to the head or tail pointer of the idle BD list. Note that there can be a delay between the release command and the detection of this problem due to buffering.
25	DOUBLE_PUSH_ERR	Error set when a push command contains a BD index equal to the head or tail pointer of the WQ it is pushed to.
24	RELEASE_FRAG_BD_INDEX_ERR	When BMU starts releasing defragmentation BD, if a fragment contains invalid BD index this error will be set.
23	MULTIPLE_READY_ERR	Error set when multiple modules in the bmu indicate ready. That should not happen.
22	RELEASE_INTEGRITY_CHECK_PDU_LINK_ERR	Error set when on a release command, the integrity check failed. The pdu links were incorrect.
21	PUSH_INTEGRITY_CHECK_PDU_LINK_ERR	Error set when on a push bd command, the integrity check failed. The pdu links were incorrect.
20	PUSH_INTEGRITY_CHECK_HEAD_TAIL_PDU_ERR	Error set when on a push bd command, the integrity check failed. The head/tail/entries part in the BD was incorrect.
19	PUSH_INTEGRITY_CHECK_PREV_PDU_ERR	Error set when on a push bd command, the integrity check failed. The prev. pdu pointer in the BD was incorrect.
18	PREV_PDU_INDEX_ERR	Error set when on a release of BD with linked PDUs, and the A-MSDU feature is enabled and the prev PDU pointer has an invalid index.

WCSS_A_BMU_ERR_INTR_STATUS (cont.)

Bits	Name	Description
17	NR_OF_PDU_OVERFLOW_ERR	Error set when on a release of PDUs, the number of idle PDUs would gone over the theoretical amount as determined by the MAX BD index and MAX PDU index from the BMU control register
16	NR_OF_BD_OVERFLOW_ERR	Error set when on a release of BDs, the number of idle BDs would gone over the theoretical amount as determined by the MAX BD index from the BMU control register
15	WRGAM_ERR	Set when the write GAM indicated an internal error
14	RDGAM_ERR	Set when the read GAM indicated an internal error
13	TIME_OUT_ERR	Set when the BMU internally detected is has been hanging There is a timeout counter in the BMU, which tracks the length of each transaction. The timeout count value can be programmed in register: bmu_watchdog_timeout_time For this error the 'err_int_addr' and 'err_int_wdata' registers are updated so it is recorded which bmu command caused this error.
12	IDLE_PDU_LIST_TAIL_ERR	The pdu idle list is corrupted. In the PDU idle list the PDU prefetch unit has read the PDU pointer index from the one but last PDU in the PDU idle list. It was expected that this PDU index pointer points to the PDU indicated in the idle_pdu_tail register. This was however not the case. THIS is a serious error as it indicates that the idle PDU list is corrupted.
11	IDLE_PDU_LIST_CNT_ERR	The pdu idle list is corrupted. In the PDU idle list the PDU prefetch unit has found an PDU who was pointing to the PDU tail index, which would indicate that the end of the idle PDU list has been reached. But this does not correspond with the idle PDU count in the idle_pdu_count register. This error indicates that more PDUs were expected to be in the idle PDU list. THIS is a serious error as it indicates that the idle PDU list is corrupted.
10	PDU_INDEX_FROM_BD_ERR	When a BD with linked PDUs was released, the BMU detected that in the BD the number of linked PDUs was non zero, but the head or tail PDU pointer index in the BD was greater then the MAX PDU index range. This is an indication that the Idle PDU list is broken and is a serious error.
9	RELEASE_FIFO_FULL_WARNING	Set when the BMU release fifo got full. This will not break BMU operation, but still should not occur as it can effect latency on the AHB bus. This however is minimized due to the split transaction generated by the BMU GAS if the release takes too long.
8	BD_LINKED_LIST_ERR	Set when in the bd idle list a pointer to a bd index was found that is outside of the valid bd index ranges. This is a serious error and the only way to recover is to soft reset the BMU or generate a bmu module reset
7	PDU_LINKED_LIST_ERR	Set when in the pdu idle list a pointer to a pdu index was found that is outside of the valid pdu index ranges. This is a serious error and the only way to recover is to soft reset the BMU or generate a bmu module reset

WCSS_A_BMU_ERR_INTR_STATUS (cont.)

Bits	Name	Description
6	OUT_OF_MODULE_RANGE_ERR	Set when a BMU command with module index that was out of allowed range was used The valid module index is less than or equal to BMU_HIGHEST_VALID_MASTER_INDEX For this error the 'err_int_addr' and 'err_int_wdata' registers are updated so it is recorded which bmu command caused this error.
5	OUT_OF_PDU_RANGE	Set when a BMU command with PDU index that was out of allowed range was used For this error the 'err_int_addr' and 'err_int_wdata' registers are updated so it is recorded which bmu command caused this error.
4	OUT_OF_BD_RANGE	Set when a BMU command with BD index that was out of allowed range was used For this error the 'err_int_addr' and 'err_int_wdata' registers are updated so it is recorded which bmu command caused this error.
3	UNDEFINED_CMD	Set when an undefined BMU command was given to the BMU For this error the 'err_int_addr' and 'err_int_wdata' registers are updated so it is recorded which bmu command caused this error.
2	WQ_INVALID	Set when an unenabled or undefined WQ was used in a BMU command In Volans, the valid WQs are "2 to 16 and 23 to 26" For this error the 'err_int_addr' and 'err_int_wdata' registers are updated so it is recorded which bmu command caused this error.
1	WQ_FULL	This is set with push command when the WQ entries are greater/equal to highest_valid_bd_index value. This shouldn't happen in the system For this error the 'err_int_addr' and 'err_int_wdata' registers are updated so it is recorded which bmu command caused this error.
0	NOT_ENOUGH_BD_PDU_WARNING	Set when a request for BD_PDUs came in when they were not available. The modules should be able to handle this and resubmit their request at a later time. This is not an error, but is given for informational purposes

0x0A080008 WCSS_A_BMU_ERR_INTR_ENABLE**Type:** RW**Reset State:** 0x00000000

This register is used as BMU Interrupt enable Mask. By default all the interrupts are disabled For descriptions of each error interrupt, see register: Err_Intr_STATUS

WCSS_A_BMU_ERR_INTR_ENABLE

Bits	Name	Description
28	BTQM_ERR_ENABLE	
27	DOUBLE_RELEASE_PDU_ERR_ENABLE	

WCSS_A_BMU_ERR_INTR_ENABLE (cont.)

Bits	Name	Description
26	DOUBLE_RELEASE_BD_ERR_ENABLE	
25	DOUBLE_PUSH_ERR_ENABLE	
24	RELEASE_FRAG_BD_INDEX_ERR_ENABLE	
23	MULTIPLE_READY_ERR_ENABLE	
22	RELEASE_INTEGRITY_CHECK_PDU_LINK_ERR_ENABLE	
21	PUSH_INTEGRITY_CHECK_PDU_LINK_ERR_ENABLE	
20	PUSH_INTEGRITY_CHECK_HEAD_TAIL_PDU_ERR_ENABLE	
19	PUSH_INTEGRITY_CHECK_PREV_PDU_ERR_ENABLE	
18	PREV_PDU_INDEX_ERR_ENABLE	
17	NR_OF_PDU_OVERFLOW_ERR_ENABLE	
16	NR_OF_BD_OVERFLOW_ERR_ENABLE	
15	WRGAM_ERR_ENABLE	
14	RDGAM_ERR_ENABLE	
13	TIMEOUT_ERR_ENABLE	
12	IDLE_PDU_LIST_TAIL_ERR_ENABLE	
11	IDLE_PDU_LIST_CNT_ERR_ENABLE	
10	PDU_INDEX_FROM_BD_ERR_ENABLE	
9	RELEASE_FIFO_FULL_WARNING_ENABLE	
8	BD_LINKED_LIST_ERR_ENABLE	
7	PDU_LINKED_LIST_ERR_ENABLE	
6	OUT_OF_MODULE_RANGE_ERR_ENABLE	

WCSS_A_BMU_ERR_INTR_ENABLE (cont.)

Bits	Name	Description
5	OUT_OF_PDU_RANGE_ERR_ENABLE	
4	OUT_OF_BD_RANGE_ERR_ENABLE	
3	UNDEFINED_CMD_ERR_ENABLE	
2	WQ_INVALID_ERR_ENABLE	
1	WQ_FULL_ERR_ENABLE	
0	NOT_ENOUGH_BD_PDU_WARNING_ENABLE	

0x0A08000C WCSS_A_BMU_ERR_INT_ADDR**Type:** R**Reset State:** 0x00000000

This register contains the BMU access address that caused the last error interrupt is generated This information is for debug purpose only

WCSS_A_BMU_ERR_INT_ADDR

Bits	Name	Description
31:0	BMU_ERR_ACCESS_ADDRESS	

0x0A080010 WCSS_A_BMU_ERR_INT_WDATA**Type:** R**Reset State:** 0x00000000

This register contains the BMU access wdata that caused the last error interrupt is generated. This information is for debug purpose only The data in here is only valid when 'BMU_access_direction' in register Err_Intr_STATUS is set.

WCSS_A_BMU_ERR_INT_WDATA

Bits	Name	Description
31:0	BMU_ERR_ACCESS_WDATA	

0x0A080014 WCSS_A_BMU_BD_PDU_THRESHOLD0**Type:** RW**Reset State:** 0x00000035

Controls the bd_pdu_aavailable[0] signal to RXP. If the available free BDs or PDUs are less or equal to these programmed values, the bd_pdu_aavailable[0] signal to the RXP will not be set, causing that master to stall.

WCSS_A_BMU_BD_PDU_THRESHOLD0

Bits	Name	Description
31:16	PDU_THRESHOLD_0	
10:0	BD_THRESHOLD_0	

0x0A080018 WCSS_A_BMU_BD_PDU_THRESHOLD1**Type:** RW**Reset State:** 0x00000001

Controls the bd_pdu_aavailable[1] signal to DPU. If the available free BDs or PDUs are less or equal to these programmed values, the bd_pdu_aavailable[1] signal to the DPU will not be set, causing that master to stall. The SW programmers guide will have guidance on how to program these values.

WCSS_A_BMU_BD_PDU_THRESHOLD1

Bits	Name	Description
31:16	PDU_THRESHOLD_1	
10:0	BD_THRESHOLD_1	

0x0A08001C WCSS_A_BMU_BD_PDU_THRESHOLD2**Type:** RW**Reset State:** 0x00000001

Controls the bd_pdu_aavailable[2] signal to DPU. If the available free BDs or PDUs are less or equal to these programmed values, the bd_pdu_aavailable[2] signal to the DPU will not be set, causing that master to stall. The SW programmers guide will have guidance on how to program these values.

WCSS_A_BMU_BD_PDU_THRESHOLD2

Bits	Name	Description
31:16	PDU_THRESHOLD_2	
10:0	BD_THRESHOLD_2	

0x0A080020 WCSS_A_BMU_BD_PDU_THRESHOLD3**Type:** RW**Reset State:** 0x00000001

Controls the bd_pdu_aavailable[3]. The assignment/usage of this signal is programmable. It could be used by SW or used by the DXE. If the available free BDs or PDUs are less or equal to these programmed values, the bd_pdu_aavailable[3] signal will not be set, typically preventing that new BDs and/or PDUs get reserved or requested. The SW programmers guide will have guidance on how to program these values.

WCSS_A_BMU_BD_PDU_THRESHOLD3

Bits	Name	Description
31:16	PDU_THRESHOLD_3	
10:0	BD_THRESHOLD_3	

0x0A080024 WCSS_A_BMU_BD_PDU_THRESHOLD4**Type:** RW**Reset State:** 0x00000028

Controls the bd_pdu_aavailable[4]. The assignment/usage of this signal is programmable. It could be used by SW or used by the DXE. If the available free BDs or PDUs are less or equal to these programmed values, the bd_pdu_aavailable[4] signal will not be set, typically preventing that new BDs and/or PDUs get reserved or requested. The SW programmers guide will have guidance on how to program these values.

WCSS_A_BMU_BD_PDU_THRESHOLD4

Bits	Name	Description
31:16	PDU_THRESHOLD_4	
10:0	BD_THRESHOLD_4	

0x0A080028 WCSS_A_BMU_BD_PDU_THRESHOLD5**Type:** RW**Reset State:** 0x00000028

Controls the bd_pdu_aavailable[5]. The assignment/usage of this signal is programmable. It could be used by SW or used by the DXE If the available free BDs or PDUs are less or equal to these programmed values, the bd_pdu_aavailable[5] signal will not be set, typically preventing that new BDs and/or PDUs get reserved or requested. The SW programmers guide will have guidance on how to program these values.

WCSS_A_BMU_BD_PDU_THRESHOLD5

Bits	Name	Description
31:16	PDU_THRESHOLD_5	
10:0	BD_THRESHOLD_5	

0x0A08002C WCSS_A_BMU_BD_PDU_THRESHOLD6**Type:** RW**Reset State:** 0x00000001

Controls the bd_pdu_aavailable[6]. The assignment/usage of this signal is programmable. It could be used by SW or used by the DXE If the available free BDs or PDUs are less or equal to these programmed values, the bd_pdu_aavailable[6] signal will not be set, typically preventing that new BDs and/or PDUs get reserved or requested. The SW programmers guide will have guidance on how to program these values.

WCSS_A_BMU_BD_PDU_THRESHOLD6

Bits	Name	Description
31:16	PDU_THRESHOLD_6	
10:0	BD_THRESHOLD_6	

0x0A080030 WCSS_A_BMU_BD_PDU_THRESHOLD7**Type:** RW**Reset State:** 0x00000029

Controls the bd_pdu_aavailable[7]. The assignment/usage of this signal is programmable. It could be used by SW or used by the DXE If the available free BDs or PDUs are less or equal to these programmed values, the bd_pdu_aavailable[7] signal will not be set, typically preventing that new BDs and/or PDUs get reserved or requested. The SW programmers guide will have guidance on how to program these values.

WCSS_A_BMU_BD_PDU_THRESHOLD7

Bits	Name	Description
31:16	PDU_THRESHOLD_7	
10:0	BD_THRESHOLD_7	

0x0A080034 WCSS_A_BMU_BD_PDU_THRESHOLD8**Type:** RW**Reset State:** 0x00000029

Controls the bd_pdu_aavailable[8]. The assignment/usage of this signal is programmable. It could be used by SW or used by the DXE. If the available free BDs or PDUs are less or equal to these programmed values, the bd_pdu_aavailable[8] signal will not be set, typically preventing that new BDs and/or PDUs get reserved or requested. The SW programmers guide will have guidance on how to program these values.

WCSS_A_BMU_BD_PDU_THRESHOLD8

Bits	Name	Description
31:16	PDU_THRESHOLD_8	
10:0	BD_THRESHOLD_8	

0x0A080038 WCSS_A_BMU_BD_PDU_THRESHOLD9**Type:** RW**Reset State:** 0x000003FF

Controls the bd_pdu_aavailable[9]. The assignment/usage of this signal is programmable. It could be used by SW or used by the DXE. If the available free BDs or PDUs are less or equal to these programmed values, the bd_pdu_aavailable[9] signal will not be set, typically preventing that new BDs and/or PDUs get reserved or requested. The SW programmers guide will have guidance on how to program these values.

WCSS_A_BMU_BD_PDU_THRESHOLD9

Bits	Name	Description
31:16	PDU_THRESHOLD_9	
10:0	BD_THRESHOLD_9	

0x0A08003C WCSS_A_BMU_BD_PDU_THRESHOLD10**Type:** RW**Reset State:** 0x000003FF

Controls the bd_pdu_aavailable[10]. The assignment/usage of this signal is programmable. It could be used by SW or used by the DXE If the available free BDs or PDUs are less or equal to these programmed values, the bd_pdu_aavailable[10] signal will not be set, typically preventing that new BDs and/or PDUs get reserved or requested. The SW programmers guide will have guidance on how to program these values.

WCSS_A_BMU_BD_PDU_THRESHOLD10

Bits	Name	Description
31:16	PDU_THRESHOLD_10	
10:0	BD_THRESHOLD_10	

0x0A080040 WCSS_A_BMU_BD_PDU_RESERVED0**Type:** R**Reset State:** 0x00000000

Indicates how many BDs or PDUs have been reserved by the RXP A module can use the BMU command Idle BD and PDU reservation request to have the BMU set aside a number of BDs and PDUs for that module. These BDs and PDUs will only be set aside if this does not violate the 'BD and PDU threshold values that were set aside for this master. This register gives inside into how many BDs and PDUs have been set aside to this master. This number will decrement each time the module gets a BD or PDU.

WCSS_A_BMU_BD_PDU_RESERVED0

Bits	Name	Description
31:16	NUMBER_RESERVED_PDU	
10:0	NUMBER_RESERVED_BD	

0x0A080044 WCSS_A_BMU_BD_PDU_RESERVED1**Type:** R**Reset State:** 0x00000000

Indicates how many BDs or PDUs have been reserved by the DPU A module can use the BMU command Idle BD and PDU reservation request to have the BMU set aside a number of BDs and PDUs for that module. These BDs and PDUs will only be set aside if this does not violate the 'BD and PDU threshold values that were set aside for this master. This register gives inside into how

many BDs and PDUs have been set aside to this master. This number will decrement each time the module gets a BD or PDU.

WCSS_A_BMU_BD_PDU_RESERVED1

Bits	Name	Description
31:16	NUMBER_RESERVED_PDU	
10:0	NUMBER_RESERVED_BD	

0x0A080048 WCSS_A_BMU_BD_PDU_RESERVED2

Type: R

Reset State: 0x00000000

Indicates how many BDs or PDUs have been reserved by master 2 (software assignable) A module can use the BMU command Idle BD and PDU reservation request to have the BMU set aside a number of BDs and PDUs for that module. These BDs and PDUs will only be set aside if this does not violate the 'BD and PDU threshold values that were set aside for this master. This register gives insight into how many BDs and PDUs have been set aside to this master. This number will decrement each time the module gets a BD or PDU. Note that this can be used as a mechanism by SW to set a number of PDUs and BDs aside for a particular DXE channel.

WCSS_A_BMU_BD_PDU_RESERVED2

Bits	Name	Description
31:16	NUMBER_RESERVED_PDU	
10:0	NUMBER_RESERVED_BD	

0x0A08004C WCSS_A_BMU_BD_PDU_RESERVED3

Type: R

Reset State: 0x00000000

Indicates how many BDs or PDUs have been reserved by master 3 (software assignable) A module can use the BMU command Idle BD and PDU reservation request to have the BMU set aside a number of BDs and PDUs for that module. These BDs and PDUs will only be set aside if this does not violate the 'BD and PDU threshold values that were set aside for this master. This register gives insight into how many BDs and PDUs have been set aside to this master. This number will decrement each time the module gets a BD or PDU. Note that this can be used as a mechanism by SW to set a number of PDUs and BDs aside for a particular DXE channel.

WCSS_A_BMU_BD_PDU_RESERVED3

Bits	Name	Description
31:16	NUMBER_RESERVED_PDU	
10:0	NUMBER_RESERVED_BD	

0x0A080050 WCSS_A_BMU_BD_PDU_RESERVED4**Type:** R**Reset State:** 0x00000000

Indicates how many BDs or PDUs have been reserved by master 4 (software assignable). A module can use the BMU command Idle BD and PDU reservation request to have the BMU set aside a number of BDs and PDUs for that module. These BDs and PDUs will only be set aside if this does not violate the 'BD and PDU threshold values that were set aside for this master. This register gives insight into how many BDs and PDUs have been set aside to this master. This number will decrement each time the module gets a BD or PDU. Note that this can be used as a mechanism by SW to set a number of PDUs and BDs aside for a particular DXE channel.

WCSS_A_BMU_BD_PDU_RESERVED4

Bits	Name	Description
31:16	NUMBER_RESERVED_PDU	
10:0	NUMBER_RESERVED_BD	

0x0A080054 WCSS_A_BMU_BD_PDU_RESERVED5**Type:** R**Reset State:** 0x00000000

Indicates how many BDs or PDUs have been reserved by master 5 (software assignable). A module can use the BMU command Idle BD and PDU reservation request to have the BMU set aside a number of BDs and PDUs for that module. These BDs and PDUs will only be set aside if this does not violate the 'BD and PDU threshold values that were set aside for this master. This register gives insight into how many BDs and PDUs have been set aside to this master. This number will decrement each time the module gets a BD or PDU. Note that this can be used as a mechanism by SW to set a number of PDUs and BDs aside for a particular DXE channel.

WCSS_A_BMU_BD_PDU_RESERVED5

Bits	Name	Description
31:16	NUMBER_RESERVED_PDU	
10:0	NUMBER_RESERVED_BD	

0x0A080058 WCSS_A_BMU_BD_PDU_RESERVED6**Type:** R**Reset State:** 0x00000000

Indicates how many BDs or PDUs have been reserved by master 6 (software assignable) A module can use the BMU command Idle BD and PDU reservation request to have the BMU set aside a number of BDs and PDUs for that module. These BDs and PDUs will only be set aside if this does not violate the 'BD and PDU threshold values that were set aside for this master. This register gives insight into how many BDs and PDUs have been set aside to this master. This number will decrement each time the module gets a BD or PDU. Note that this can be used as a mechanism by SW to set a number of PDUs and BDs aside for a particular DXE channel.

WCSS_A_BMU_BD_PDU_RESERVED6

Bits	Name	Description
31:16	NUMBER_RESERVED_PDU	
10:0	NUMBER_RESERVED_BD	

0x0A08005C WCSS_A_BMU_BD_PDU_RESERVED7**Type:** R**Reset State:** 0x00000000

Indicates how many BDs or PDUs have been reserved by master 7 (software assignable) A module can use the BMU command Idle BD and PDU reservation request to have the BMU set aside a number of BDs and PDUs for that module. These BDs and PDUs will only be set aside if this does not violate the 'BD and PDU threshold values that were set aside for this master. This register gives insight into how many BDs and PDUs have been set aside to this master. This number will decrement each time the module gets a BD or PDU. Note that this can be used as a mechanism by SW to set a number of PDUs and BDs aside for a particular DXE channel.

WCSS_A_BMU_BD_PDU_RESERVED7

Bits	Name	Description
31:16	NUMBER_RESERVED_PDU	
10:0	NUMBER_RESERVED_BD	

0x0A080060 WCSS_A_BMU_BD_PDU_RESERVED8**Type:** R**Reset State:** 0x00000000

Indicates how many BDs or PDUs have been reserved by master 8 (software assignable) A module can use the BMU command Idle BD and PDU reservation request to have the BMU set

aside a number of BDs and PDUs for that module. These BDs and PDUs will only be set aside if this does not violate the 'BD and PDU threshold values that were set aside for this master. This register gives inside into how many BDs and PDUs have been set aside to this master. This number will decrement each time the module gets a BD or PDU. Note that this can be used as a mechanism by SW to set a number of PDUs and BDs aside for a particular DXE channel.

WCSS_A_BMU_BD_PDU_RESERVED8

Bits	Name	Description
31:16	NUMBER_RESERVED_PDU	
10:0	NUMBER_RESERVED_BD	

0x0A080064 WCSS_A_BMU_BD_PDU_RESERVED9

Type: R

Reset State: 0x00000000

Indicates how many BDs or PDUs have been reserved by master 9 (software assignable) A module can use the BMU command Idle BD and PDU reservation request to have the BMU set aside a number of BDs and PDUs for that module. These BDs and PDUs will only be set aside if this does not violate the 'BD and PDU threshold values that were set aside for this master. This register gives inside into how many BDs and PDUs have been set aside to this master. This number will decrement each time the module gets a BD or PDU. Note that this can be used as a mechanism by SW to set a number of PDUs and BDs aside for a particular DXE channel.

WCSS_A_BMU_BD_PDU_RESERVED9

Bits	Name	Description
31:16	NUMBER_RESERVED_PDU	
10:0	NUMBER_RESERVED_BD	

0x0A080068 WCSS_A_BMU_BD_PDU_RESERVED10

Type: R

Reset State: 0x00000000

Indicates how many BDs or PDUs have been reserved by master 10 (software assignable) A module can use the BMU command Idle BD and PDU reservation request to have the BMU set aside a number of BDs and PDUs for that module. These BDs and PDUs will only be set aside if this does not violate the 'BD and PDU threshold values that were set aside for this master. This register gives inside into how many BDs and PDUs have been set aside to this master. This number will decrement each time the module gets a BD or PDU. Note that this can be used as a mechanism by SW to set a number of PDUs and BDs aside for a particular DXE channel.

WCSS_A_BMU_BD_PDU_RESERVED10

Bits	Name	Description
31:16	NUMBER_RESERVED_PDU	
10:0	NUMBER_RESERVED_BD	

0x0A08006C WCSS_A_BMU_WQ_ENABLE**Type:** RW**Reset State:** 0x00000000

Controls which WQ will be enabled. These WQs should be enabled during initialization, after the BD and PDU idle lists have been created and all head and tail pointers for the to be enabled WQs have been configured. The number of WQs to be enabled need to be decided at that time. Once enabled, a WQ should never be disabled again, as this can cause BD and PDU leaks. Bit 0 corresponds to WQ0, Bit 26 corresponds to WQ 26.

WCSS_A_BMU_WQ_ENABLE

Bits	Name	Description
27:0	WQ_ENABLE	WQ0: Assigned for Idle BD from DDR/Ext Memory. WQ1: Assigned for Idle PDU WQ2: Assigned for mCPU RX WQ WQ3: Can be assigned to DPU, DXE or mCPU WQ4: Can be assigned to DPU, DXE or mCPU WQ5: Can be assigned to DPU, DXE or mCPU WQ6: Can be assigned to DPU, DXE or mCPU WQ7: Can be assigned to DPU, DXE or mCPU WQ8: Can be assigned to DPU, DXE or mCPU WQ9: Can be assigned to DPU, DXE or mCPU WQ10: Can be assigned to DPU, DXE or mCPU WQ11: Can be assigned to DXE or mCPU WQ12: Can be assigned to DXE or mCPU WQ13: Can be assigned to DXE or mCPU WQ14: Can be assigned to DXE or mCPU WQ15: Can be assigned to DXE or mCPU WQ16: Can be assigned to DXE or mCPU WQ17: Can be assigned to DXE or mCPU WQ18: Can be assigned to mCPU WQ19: Can be assigned to mCPU WQ20: Can be assigned to mCPU WQ21: Can be assigned to mCPU WQ22: Can be assigned to mCPU WQ23: Can be assigned to mCPU WQ24: Can be assigned to mCPU, or ADU WQ25: High priority entrance WQ for the BTQM (BMU Transmit Queue Manager) WQ26: Low priority entrance WQ for the BTQM (BMU Transmit Queue Manager) WQ27: Assigned for Idle BD from cMem/Local Memory.

0x0A080070 WCSS_A_BMU_DPU_WQ_ASSIGNMENT**Type:** RW**Reset State:** 0x00000809

WQs 3 to 10 can be assigned to the DPU. This register controls if they are assigned to the DPU and if so, in which direction: TX or RX By default, WQ 3 is enabled and assigned to the TX side of the DPU By default, WQ 4 is enabled and assigned to the RX side of the DPU

WCSS_A_BMU_DPU_WQ_ASSIGNMENT

Bits	Name	Description
15	WQ10_DPU_ASSIGNMENT_DIRECTION	Only valid when WQ 10 is assigned to the DPU. When set, WQ 10 is assigned to the TX side of the DPU When NOT set, WQ 10 is assigned to the RX side of the DPU
14	WQ9_DPU_ASSIGNMENT_DIRECTION	Only valid when WQ 9 is assigned to the DPU. When set, and WQ 9 is assigned to the TX side of the DPU When NOT set, WQ 9 is assigned to the RX side of the DPU
13	WQ8_DPU_ASSIGNMENT_DIRECTION	Only valid when WQ 8 is assigned to the DPU. When set, and WQ 8 is assigned to the TX side of the DPU When NOT set, WQ 8 is assigned to the RX side of the DPU
12	WQ7_DPU_ASSIGNMENT_DIRECTION	Only valid when WQ 7 is assigned to the DPU. When set, and WQ 7 is assigned to the TX side of the DPU When NOT set, WQ 7 is assigned to the RX side of the DPU
11	WQ6_DPU_ASSIGNMENT_DIRECTION	Only valid when WQ 6 is assigned to the DPU. When set, and WQ 6 is assigned to the TX side of the DPU When NOT set, WQ 6 is assigned to the RX side of the DPU
10	WQ5_DPU_ASSIGNMENT_DIRECTION	Only valid when WQ 5 is assigned to the DPU. When set, and WQ 5 is assigned to the TX side of the DPU When NOT set, WQ 5 is assigned to the RX side of the DPU
9	WQ4_DPU_ASSIGNMENT_DIRECTION	Only valid when WQ 4 is assigned to the DPU. When set, and WQ 4 is assigned to the TX side of the DPU When NOT set, WQ 4 is assigned to the RX side of the DPU
8	WQ3_DPU_ASSIGNMENT_DIRECTION	Only valid when WQ 3 is assigned to the DPU. When set, and WQ 3 is assigned to the TX side of the DPU When NOT set, WQ 3 is assigned to the RX side of the DPU
7	WQ10_DPU_ASSIGNMENT_ENABLE	When set, WQ 10 is assigned to the DPU
6	WQ9_DPU_ASSIGNMENT_ENABLE	When set, WQ 9 is assigned to the DPU
5	WQ8_DPU_ASSIGNMENT_ENABLE	When set, WQ 8 is assigned to the DPU
4	WQ7_DPU_ASSIGNMENT_ENABLE	When set, WQ 7 is assigned to the DPU
3	WQ6_DPU_ASSIGNMENT_ENABLE	When set, WQ 6 is assigned to the DPU
2	WQ5_DPU_ASSIGNMENT_ENABLE	When set, WQ 5 is assigned to the DPU
1	WQ4_DPU_ASSIGNMENT_ENABLE	When set, WQ 4 is assigned to the DPU

WCSS_A_BMU_DPU_WQ_ASSIGNMENT (cont.)

Bits	Name	Description
0	WQ3_DPU_ASSIGNMENT_ENABLE	When set, WQ 3 is assigned to the DPU

0x0A080080 WCSS_A_BMU_GAS_CONFIGURATION**Type:** RW**Reset State:** 0x000D0000

These are the configuration bits for the gas The default values should be OK This register is provided as a backup

WCSS_A_BMU_GAS_CONFIGURATION

Bits	Name	Description
19:16	GAS_CFG_HREADY_LIMIT	
15:8	GAS_CFG_HSPLIT_RESP_LIMIT	
7:0	GAS_CFG_HSPLIT_MST_R ESP_LIMIT	

0x0A080084 WCSS_A_BMU_AVAILABLE_BD_PDU_AFTER_RSV**Type:** R**Reset State:** 0x00000000

This register can be used to find out how many idle PDUs and BDs are actually available from the BD and PDU idle list when the number of reserved BDs and PDUs for all the different modules have been taken into account.

WCSS_A_BMU_AVAILABLE_BD_PDU_AFTER_RSV

Bits	Name	Description
31:16	AVAILABLE_PDUS	
15:0	AVAILABLE_BDS	

0x0A080088 WCSS_A_BMU_CONTROL2**Type:** RW**Reset State:** 0x00000001

This register can be used to control bmu A-MSDU support BD/PDU integrity checking by the BMU To enable rxp_bmu_dxe_datalength for DXE wq byte count To enable enhanced reserve all command To enable priority based BD/PDU reservation

WCSS_A_BMU_CONTROL2

Bits	Name	Description
31	FREEZE_ONE_ERROR_LESS_STRICKED_ENABLE	When bit 'freeze_one_error' is set, and this bit is set, and an error is detected in the BMU, the BMU will freeze, but in a little less rigorous way then when this bit is not set. Get commands will still go through, and the BD/PDUavailable signals to the RXP and DPU will not be masked off. This will prevent DPU and RXP getting into some error state.
7	BLOCK_RXP_PUSH_INTEGRITY_CHECK	When set, and 'push_data_integrity_check' is also set, the push integrity will be performed on all WQs except WQ 2. That is the WQ that RXP is pushing to.
6	FREEZE_ONE_ERROR_ENABLE	When set, and an error is detected in the BMU, the BMU will freeze. That is, release commands are thrown away, get bd/dpu commands will not be successful anymore, and reservation request will return unsuccessful. All wq signals and bd/pduavailable signals will be set to low.
5	RELEASE_INTEGRITY_CHECK	When set, and a release command is received by the BMU, the BMU will check the integrity of the BD and linked PDUs. If a problem is found with the linked DPU pointers, an error is reported and the frame is not pushed into the WQ, so that the contents remainsavailable for debugging
4	PUSH_DATA_INTEGRITY_CHECK	When set, and a push command is received by the BMU, the BMU will check the integrity of the BD and linked PDUs. If a problem is found with the BD/DPU pointers, an error is reported and the frame is not pushed into the WQ, so that the contents remainsavailable for debugging The index of the BD that caused the problem can be read in register: bmu_push_integrity_err_bd_index
3	RXP_BMU_DXE_DATALEN_ENABLE	When set, any packet push from Rxp to Dxe, rxp_bmu_dxe_dataLEN is taken to calculate dxe data length
2	ENHANCED_RESERVE_ALL_ENABLE	When set, reserve all command checks for TX memory threshold if tx_threshold is greater than than BTQM memory size then it reserves min of (tx_threshold - BTQM memory size, idle Packet memory size - requested module threshold)
1	PRIORITY_BASED_RES_ENABLE	When set, enables priority based BD/PDU reservation command
0	AMSDU_RELEASE_SUPPORT_ENABLE	When set, the A-MSDU support for releasing PDUs from BDs is enabled. This means that the BMU will evaluate the setting of the PREV PDU index field in the BD. If this field is zero, the PDUs are release according to the head/tail and PDU count linked to the BD. When the PREV PDU index field is unequal to zero, the BMU will use this index as the 'tail' index of the list of PDUs to release. The amount of PDUs to release is set equal to PDU count minus 1.

0x0A08008C WCSS_A_BMU_DISABLE_WQ_DA**Type:** RW**Reset State:** 0x00000000

This register can be used to disable the wq daavailable signals. In that case, modules will think that their WQ is empty.

WCSS_A_BMU_DISABLE_WQ_DA

Bits	Name	Description
26:2	DISABLE_WQ_DA	Bit 2 corresponds with WQ 2, bit 3 with WQ 3 etc. When set, the wq daavailable signal to a WQ will always indicate that the WQ is empty. This overwrites any real amount of frames buffered in a WQ.

0x0A080090 WCSS_A_BMU_DISABLE_BD_PDU_AVAIL**Type:** RW**Reset State:** 0x00000000

This register can be used to disable the bd_pduavailable signals. In that case, modules will think that no more BDs or PDUs areavailable, and will stall their operation

WCSS_A_BMU_DISABLE_BD_PDU_AVAIL

Bits	Name	Description
10:0	DISABLE_BD_PDU_AVAIL	Bit 0 corresponds with master 0, bit 1 with master 1 etc. When set, the bd_pduavailable signal to a master will always indicate that no BDs and PDUs areavailable. This overwrites any real amount of BDs or PDUs beingavailable for that module.

0x0A080094 WCSS_A_BMU_LAST_FOUR_POP_TRACE_CONTROL**Type:** RW**Reset State:** 0x00000000

This register controls the enabling of the last 4 pop command tracing control. The POP command tracing can be enabled for one specific (programmable) WQ. It also provides insight into how many times a pop command was issues for the specified WQ

WCSS_A_BMU_LAST_FOUR_POP_TRACE_CONTROL

Bits	Name	Type	Description
31:16	POP_CMD_COUNT	R	The number of times a POP command was issued on the specified WQ.

WCSS_A_BMU_LAST_FOUR_POP_TRACE_CONTROL (cont.)

Bits	Name	Type	Description
8	POP_TRACE_ENABLE	RW	When set, the pop tracing functionality for the WQ as specified in field 'pop_trace_wq_index' is enabled. The BD indexes returned for the last 4 pop commands that were issued for the specified WQ, are captured in the registers: pop_cmd_trace_bd_index0, pop_cmd_trace_bd_index1, pop_cmd_trace_bd_index2 and pop_cmd_trace_bd_index3. The index in the pop_cmd_trace_bd_index0 is the BD index returned with the most recent pop command. Whenever the 'pop_trace_wq_index' field gets changed the pop trace functionality first needs to be disabled. That will clear the 4 stored indexes and the pop counter.
4:0	POP_TRACE_WQ_INDEX	RW	The WQ index for which the pop commands need to be traced

0x0A080098 WCSS_A_BMU_POP_CMD_TRACE_BD_INDEX0**Type:** R**Reset State:** 0x00000000

This register contains a BD index returned as the result of a POP command on a specific WQ. See the description for the 'last_four_pop_trace_control' register for additional details.

WCSS_A_BMU_POP_CMD_TRACE_BD_INDEX0

Bits	Name	Description
11:0	POP_BD_INDEX0	BD index

0x0A08009C WCSS_A_BMU_POP_CMD_TRACE_BD_INDEX1**Type:** R**Reset State:** 0x00000000

This register contains a BD index returned as the result of a POP command on a specific WQ. See the description for the 'last_four_pop_trace_control' register for additional details.

WCSS_A_BMU_POP_CMD_TRACE_BD_INDEX1

Bits	Name	Description
11:0	POP_BD_INDEX1	BD index

0x0A0800A0 WCSS_A_BMU_POP_CMD_TRACE_BD_INDEX2**Type:** R**Reset State:** 0x00000000

This register contains a BD index returned as the result of a POP command on a specific WQ. See the description for the 'last_four_pop_trace_control' register for additional details.

WCSS_A_BMU_POP_CMD_TRACE_BD_INDEX2

Bits	Name	Description
11:0	POP_BD_INDEX2	BD index

0x0A0800A4 WCSS_A_BMU_POP_CMD_TRACE_BD_INDEX3**Type:** R**Reset State:** 0x00000000

This register contains a BD index returned as the result of a POP command on a specific WQ. See the description for the 'last_four_pop_trace_control' register for additional details.

WCSS_A_BMU_POP_CMD_TRACE_BD_INDEX3

Bits	Name	Description
11:0	POP_BD_INDEX3	BD index

0x0A0800AC WCSS_A_BMU_BMU_INTERNAL_MEM_TRACE_ADDR**Type:** R**Reset State:** 0x00000000

This register Indicates the head index pointer of where the last BMU command data was written to BMU internal memory

WCSS_A_BMU_BMU_INTERNAL_MEM_TRACE_ADDR

Bits	Name	Description
10:0	BMU_INTERNAL_MEM_INDEX	internal BMU memory address that will be used next to store the internal BMU tracing BMU commands and data. In units of BD index.

0x0A0800B0 WCSS_A_BMU_RELEASE_ERROR_DETAILS**Type:** R**Reset State:** 0x00000000

When a release error is reported, this register contains PDU index values of current & previous PDUs that caused release error This is debug information

WCSS_A_BMU_RELEASE_ERROR_DETAILS

Bits	Name	Description
31:0	RELEASE_ERROR_DETAILS	Bits 31:16 will indicate the previous PDU index Bits 15:0 will indicate the current PDU index

0x0A0800B4 WCSS_A_BMU_RELEASE_ERROR_DETAILS_LOW**Type:** R**Reset State:** 0x00000000

When a release error is reported, this register contains debug information

WCSS_A_BMU_RELEASE_ERROR_DETAILS_LOW

Bits	Name	Description
23:0	RELEASE_ERROR_DETAILS_LOW	Bit 23: when set, indicates a release of PDU linked list is ongoing when NOT set, bit 22 indicates the ongoing command: Bit 22: when NOT set, indicates a release of BD-PDU command is ongoing when set, indicates a release is ongoing based on a push to the sink WQ: FF That means, this can be a BD-PDU release or a release based on a defrag BD. The BMU will first investigate what the BD type is and process is accordingly. Bits [15:0]: when bit 23 is set: PDU head index of the PDU linked list when bit 23 is NOT set: BD index

0x0A0800B8 WCSS_A_BMU_RELEASE_ERROR_DETAILS_HIGH**Type:** R**Reset State:** 0x00000000

When a release error is reported, this register contains debug information It only contains valid info when bit 23 of register release_error_details_low is set

WCSS_A_BMU_RELEASE_ERROR_DETAILS_HIGH

Bits	Name	Description
23:0	RELEASE_ERROR_DETAIL_S_HIGH	Bits [23:16]: number of PDUs in the PDU linked list being released Bits [15:0]: PDU tail index of the PDU linked list

0x0A0800BC WCSS_A_BMU_RELEASE_FIFO_ENTRIES**Type:** R**Reset State:** 0x00000000

The register provides insight into how many entries in the release fifo have been used

WCSS_A_BMU_RELEASE_FIFO_ENTRIES

Bits	Name	Description
15:8	NR_PDU_LINKS_CHECKED	When a release (integrity) check error is reported, this field can help debug the problem This field will indicate how many PDUs the integrity check has gone down the PDU linked list that is connected to the BD
5:0	NR_USED_REL_FIFO_ENTRIES	

0x0A0800C0 WCSS_A_BMU_LAST_BD_PREFETCH_HEAD**Type:** R**Reset State:** 0x00000000

The register provides insight info what the last used bd prefetch head was. This information can be usefull in case a linked BD list error is reported

WCSS_A_BMU_LAST_BD_PREFETCH_HEAD

Bits	Name	Description
26:16	REMAINING_AVAILABLE_BD_FOR_PREFETCH_CNT	
10:0	PREFETCH_HEAD_BD_INDEX	

0x0A0800C4 WCSS_A_BMU_BD_PREFETCH_HEAD_BD_LINK**Type:** R**Reset State:** 0x00000000

The register provides insight into which BD index was the bd prefetch head was pointing to This information can be usefull in case a linked BD list error is reported

WCSS_A_BMU_BD_PREFETCH_HEAD_BD_LINK

Bits	Name	Description
10:0	NEXT_PREFETCH_HEAD_BD_INDEX	

0x0A0800C8 WCSS_A_BMU_BMU_PUSH_INTEGRITY_ERR_BD_INDEX

Type: R

Reset State: 0x00000000

This register indicates the bd index of the BD that was pushed into a WQ, but the integrity check on the BD with linked PDUs failed

WCSS_A_BMU_BMU_PUSH_INTEGRITY_ERR_BD_INDEX

Bits	Name	Description
10:0	INTEGRITY_ERR_BD_INDEX	BD index.

0x0A0800CC WCSS_A_BMU_BMU_WATCHDOG_TIMEOUT_TIME

Type: RW

Reset State: 0x0000FFFF

This register indicates the timeout in nr of clock cycles when an ahb access has started but does not get finished

WCSS_A_BMU_BMU_WATCHDOG_TIMEOUT_TIME

Bits	Name	Description
19:0	BMU_WATCHDOG_TIMEOUT_TIME	timeout time in nr of clk cycles. When set to zero, the watchdog functionality is disabled

0x0A0800D0 WCSS_A_BMU_BMU_SM_STATES_ON_ERROR_LOW

Type: R

Reset State: 0x00000000

This register provides a snapshot of the bmu statemachines when an error is reported.

WCSS_A_BMU_BMU_SM_STATES_ON_ERROR_LOW

Bits	Name	Description
31:0	BMU_SM_STATES_ON_ERROR_LOW	[31:28] = encoded_release_pdu_ctrl_sm_state[3:0]; [27:25] = encoded_release_bd_ctrl_sm_state[2:0]; [24:22] = encoded_release_ctrl_sm_state[2:0]; [21:19] = encoded_pdu_prefetch_mgr_sm_state[2:0]; [18:15] = encoded_bd_prefetch_mgr_sm_state[3:0]; [14:12] = encoded_bd_pdu_mgr_sm_state[2:0]; [11:8] = encoded_integrity_check_state[3:0]; [7:4] = {1'b0, encoded_wq_mgr_state[2:0]}; [3:0] = encoded_gas_interface_sm_state;

0x0A0800D4 WCSS_A_BMU_BMU_SM_STATES_ON_ERROR_HIGH**Type:** R**Reset State:** 0x00000000

This register provides a snapshot of the bmu statemachines when an error is reported.

WCSS_A_BMU_BMU_SM_STATES_ON_ERROR_HIGH

Bits	Name	Description
31	ERROR_CAPTURED	when set, an error state has been captured
12:0	BMU_SM_STATES_ON_ERROR_HIGH	[12] = bmu_mcu_err_int; [11] = freeze_bmu_state; [10:4] = test_bus_locally_prefetched_pdu_count[BMU_HIGH_BIT_PREFETCH_MEM + 1:0]; [3:1] = encoded_prefetch_fifo_ctrl_sm_state[2:0]; [0] = encoded_release_pdu_ctrl_sm_state[4];

0x0A0800D8 WCSS_A_BMU_BMU_PREFETCH_ERROR_DETAILS**Type:** R**Reset State:** 0x00000000

This register provides insight to the error that was reported on an reported prefetch error

WCSS_A_BMU_BMU_PREFETCH_ERROR_DETAILS

Bits	Name	Description
31:0	BMU_PREFETCH_ERROR_DETAILS	rdgam_prefetch_pdu_rdata_valid [15:0] = used_pdu_head_index_reg [31:16] = rdgam_prefetch_pdu_rdata[15:0] tail_of_pdu_list_reached_based_on_tail_pointer [15:0] = used_pdu_head_index_reg; [23:16] = locally_prefetched_pdu_count_reg; [31:24] = pdu_count_aavailable_for_prefetch[7:0]; next_tail_of_pdu_list_reached_based_on_pdu_count [15:0] = used_pdu_head_index_reg; [31:16] = local_pdu_head_index_reg;

0x0A0800DC WCSS_A_BMU_CONTROL3**Type:** RW**Reset State:** 0x00000000

This register can be used to control details for the bmu build in BD/PDU integrity checking. The enabling of the overall integrity check is controlled by bits in the control2 register. This register allows that check to tune in into specific WQs (for the push command check) or specific masters (for the release check).

WCSS_A_BMU_CONTROL3

Bits	Name	Description
15	PUSH_INTEGRITY_FOR_SPECIFIC_MASTER_SELECTED	When set, the push WQ integrity check will only be performed on pushes to the WQ identified by "push_integrity_check_wq_nr".
12:8	PUSH_INTEGRITY_CHECK_WQ_NR	This is the number of the WQ for which the BD/PDUs pushed into this WQ need to be checked for their integrity.
7	RELEASE_INTEGRITY_FOR_SPECIFIC_MASTER_SELECTED	When set, the release BD/PDU integrity check will only be performed for the master identified by "release_integrity_check_master_id".
5:0	RELEASE_INTEGRITY_CHECK_MASTER_ID	This is the id of the master from which the releases need to be checked for its integrity.

0x0A0800E0 WCSS_A_BMU_CONTROL4**Type:** RW**Reset State:** 0x00000F28

This register can be used to control the generation of the 'priority' signal to the DAHB arbiter who passes this on to the MIF. This priority setting will help BMU getting faster 'prioritized' access to the MIF memory over other (SYS, mCPU) interfaces to the MIF.

WCSS_A_BMU_CONTROL4

Bits	Name	Type	Description
31	PREFETCH_FIFO_ACCESS_PRIORITY_SET	R	When set, the prefetch fifo priority signal is set, resulting in the MIF giving priority to the DAHB accesses.
30	RELEASE_FIFO_ACCESS_PRIORITY_SET	R	When set, the release fifo priority signal is set, resulting in the MIF giving priority to the DAHB accesses.
29	PDU_PREFETCH_ACCESS_PRIORITY_SET	R	When set, the pdu based prefetch priority signal is set, resulting in the MIF giving priority to the DAHB accesses.
28	BD_PREFETCH_ACCESS_PRIORITY_SET	R	When set, the bd based prefetch priority signal is set, resulting in the MIF giving priority to the DAHB accesses.
22:16	NR_OF_PREFETCH_ENTRIES	R	The number of entries in the PDU prefetch fifo
11:8	PREFETCH_FIFO_ACCESS_PRIORITY_THRESHOLD	RW	The 'priority' signal is set when the number of used PDU entries in the prefetch fifo gets below this threshold, and there are still PDUs left that can be prefetched When BMU is operating in the special mode where every 128 byte memory segment can be BD or PDU, the amount of the 'priority' signal is set when the prefetched BDs are below the threshold, and the PDU prefetch fifo has less than this threshold number of used entries.
6:0	RELEASE_FIFO_ACCESS_PRIORITY_THRESHOLD	RW	The 'priority signal is set when the release fifo gets filled above this threshold

0x0A0800E4 WCSS_A_BMU_DPU_RSV_STATUS**Type:** RW**Reset State:** 0x00000000

This register can be used to observe and clear pending reservation commands that were issued by the dpu to the BMU. When a reservation command from the DPU is not granted, the nr of BDs and PDUs that were requested are remembered by the BMU. The BMU uses this number, together with the programmed BD/PDU threshold to control the generation of the bmu_pdu_bd_pdu_avail_for_rsv_req signals for the DPU.

WCSS_A_BMU_DPU_RSV_STATUS

Bits	Name	Type	Description
31	BMU_PDU_RX_BD_PDU_AVAILABLE_CLEAR	RW	When set, will clear the pending DPU rx reservation
30	BMU_PDU_RX_BD_PDU_AVAILABLE_FOR_RSV_REQ	R	Status of the bmu_pdu_rx_bd_pdu_avail_for_rsv_req signal to the dpu
28:24	DPU_RX_PENDING_BD_RSV_REQ	R	The number of bds the dpu rx has requested

WCSS_A_BMU_DPU_RSV_STATUS (cont.)

Bits	Name	Type	Description
23:16	DPU_RX_PENDING_PDU_RSV_REQ	R	The number of pdus the dpu rx has requested
15	BMU_PDU_TX_BD_PDU_AVAIL_CLEAR	RW	When set, will clear the pending DPU tx reservation
14	BMU_PDU_TX_BD_PDU_AVAIL_FOR_RSV_REQ	R	Status of the bmu_pdu_tx_bd_pdu_avail_for_rsv_req signal to the dpu
12:8	DPU_TX_PENDING_BD_RSVD_REQ	R	The number of bds the dpu tx has requested
7:0	DPU_TX_PENDING_PDU_RSV_REQ	R	The number of pdus the dpu tx has requested

0x0A0800F0 WCSS_A_BMU_BTQM_CONTROL1**Type:** RW**Reset State:** 0x00000000

controls BTQM (BMU Transmit Queue Manager) features

WCSS_A_BMU_BTQM_CONTROL1

Bits	Name	Type	Description
31	FAST_STAMEM_INIT_STATUS	R	When set, the fast sta-mem initialization is ongoing
28	FAST_STA_MEM_INIT	RW	Writing a 1 to this register bit, the BTQM will initialize the STA mem to all zeros, which means all staids are disabled. It is safest to immediately clear this bit after it was set.
27	BTQM_COUNTERS_CLEAR	R	Writing a 1 to this register bit will clear the btqm related counters. This register auto clears.
26	CFG_MASK_OFF_APSD_SESSION_ENABLE	RW	When set, the 'get_info_apspd_session_ongoing' signal will never be set.
25	CFG_FIXED_BCN_QUEUE_ID_MAPPING	RW	When set, the beacon queueid 0 will always be mapped to BO_DA0, beacon queueid 1 is mapped to BO_DA1, etc. When not set, the same mapping as for the NON beacon STAids is used.
24	CFG_CORRECTED_ARBITRATION_ENABLED	RW	When set, the arbitration logic treats the lowest STAid that has data among all STAids that have data for the same queueid, with the same priority as all the others STAids. When NOT set, the lowest STAid that has data among all STAids that have data for the same queueid, might get less opportunity to transmit

WCSS_A_BMU_BTQM_CONTROL1 (cont.)

Bits	Name	Type	Description
23	TPE_STATS_BUSY_FLAG_SET_ON_FEEDBACK	RW	When set, the ampdu stats busy signal will be set when a TPE feedback signal is received. It will only get cleared when a BA bitmap has been received and has been fully processed
22	APSD_TRIGGER_PM_BIT_SETTING_ENABLE	RW	When set, an incoming apsd frame that has an TID that is of the trigger category, is always considered to be setting the power save status of the STA, irrespective of the pm bit in the frame.
21	PSPOLL_PM_BIT_SETTING_ENABLE	RW	When set, an incoming pspoll frame is always considered to be setting the power save status of the STA, irrespective of the pm bit in the frame
20	APSD_TRIGGER_PM_BIT_MASKING_ENABLE	RW	When set, the PM bit from the received APSD trigger frame is masked off and not interpreted to (properly) indicate the power state of the transmitters STA
19	PSPOLL_PM_BIT_MASKING_ENABLE	RW	When set, the PM bit from the received PS-Poll frame is masked off and not interpreted to (properly) indicate the power state of the transmitters STA
18	REGISTER_APSD_TRIGGER_WHEN_DATA_AVAILABLE	RW	When set, and QoS Null transmission is NOT enabled as response to an APSD trigger, the APSD trigger will only be registered when there is data in any of the delivery enabled queues
17	REGISTER_PSPOLL_TRIGGER_WHEN_DATA_AVAILABLE	RW	When set, and Data Null transmission is NOT enabled as response to a PSPoll, the pspoll trigger will only be registered when there is data in a transmit queue
16	ADVANCED_MORE_DATA_BIT_GENERATION_ENABLED	RW	When set, data frames that are pushed into a transmit queue from which a transmission is happening, will be taken into account for the more data bit generation.
15	COMBINED_SEARCH_ENABLED	RW	When set, BEACON STA and NON BEACON sta search arbitration will happen in parallel as long as NO NON BEACON sta is in power save mode
13	QOS_NULL_TID_SOURCE_IS_TRIGGER_TID	RW	When set, and a qos null frame is transmitted, the TID inserted in the qos null frame is set to the tid field of the trigger frame received. When Not set, the TID inserted in the QoS Null frame will be based on the U-APSD queue with the highest queueid number.
12	SW_PROCESSING_OF_PSPOLL_DATA_NULL_ENABLED	RW	When set, and a valid PSPOLL is received, the reception will be registered in the STA descriptor, so that SW can find out this trigger was received and the start the handling of it.
11	SW_PROCESSING_OF_APSD_QOS_NULL_ENABLED	RW	When set, and a valid APSD trigger is received, the trigger will be registered in the STA descriptor, so that SW can find out this trigger was received and the start the handling of it.
10	RANDOM_PSPOLL_APSD_TRIGGER_PRIORITY_SCHEDULE_ENABLED	RW	When set, and a valid pspoll or APSD trigger is received, the STA has a 50% chance to be put close to the start of the transmit queue.

WCSS_A_BMU_BTQM_CONTROL1 (cont.)

Bits	Name	Type	Description
9	PSPOLL_APSD_TRIGGER_PRIORITY_SCHEDULING_ENABLE	RW	When set, and a valid pspoll or APSD trigger is received, the STA is put at close to the start of the transmit queue
8	BD_PUSH_OWNERSHIP_CHECK_ENABLE	RW	When set, the ownership check on all the push commands gets enabled
7	BD_OWNERSHIP_ASSIGNMENT_DISABLE	RW	When set, the BD_PDU manager module will stop updating the BD ownership assignment when a BD is taken out of the BD idle list and handed over to an other module. Disabling this feature reduces the overhead on the BD SRAM memory
6	BD_OWNERSHIP_CHECK_ENABLE	RW	When set, the will start assigning and tracking who owns a BD. There are 4 owners: 0: BD idle list 1: External module 2: WQ managed by the BMU 3: BTQM transmit WQ
5	APSD_AMPDU_BA_SUPPORT_ENABLED	RW	When set, the BTQM can handle BA policy transmissions while APSD is enabled. Frames can be aggregated in an AMPDU.
4	BTQM_RXP_PWR_INTERFACE_ENABLE	RW	When set, the BMU interface to the RXP related to the power state processing of a station processing is enabled
3	BTQM_RXP_BA_INTERFACE_ENABLE	RW	When set, the BMU interface to the RXP related to the Block Ack bitmap processing is enabled
2	BTQM_TPE_INTERFACE_ENABLE	RW	When set, the BMU interface to the TPE is enabled
1	BTQM_QUEUEING_CTRL_LP_ENABLE	RW	When set, the btqm will start popping frames out of the low priority BTQM entrance WQ 26 and pushing them into the Transmit WQs
0	BTQM_QUEUEING_CTRL_HP_ENABLE	RW	When set, the btqm will start popping frames out of the high priority BTQM entrance WQ 25 and pushing them into the Transmit WQs

0x0A0800F4 WCSS_A_BMU_BTQM_CONTROL2**Type:** RW**Reset State:** 0x00000000

controls BTQM (BMU Transmit Queue Manager) features

WCSS_A_BMU_BTQM_CONTROL2

Bits	Name	Description
31	CFG_BA_PARTIAL_BITMAP_UPDATE	When set, and an AMPDU session, if BA is received without sending BAR(Partial bitmap) and received BA SSN is greater than current SSN the frames are not dropped.

WCSS_A_BMU_BTQM_CONTROL2 (cont.)

Bits	Name	Description
30	CFG_SW_OWNERSHIP_HA S_PRIORITY	When set, the sw access should be getting higher priority over the other ownership request accesses
29	CFG_SENT_QOS_NULL_AF TER_EOSP_WITH_RETRY_ DROP	When set, and an AMPDU session exchange with the TPE is ongoing for which the EOSP gets set, and any frame for which the EOSP was set gets dropped because the max retry count is reached, the BMU will set the internal' QoS data sent flag, so that when no BA is received in which frames get acknowledged, only one more QoS Null frame will be sent, and not any more data frames.
28	CFG_DATA_NULL_SENT_HI GH_PRIORITY_ENABLE	When set, and a data null frame has been sent, but has not yet been acknowledged, the frame will be resent with high priority.
27	CFG_QOS_NULL_SENT_HI GH_PRIORITY_ENABLE	When set, and a qos null frame has been sent, but has not yet been acknowledged, the frame will be resent with high priority.
26	CFG_AMPDU_APSD_MISSI NG_EOSP_HANDLING_EN ABLE	When set, more frames get acknowledged by an BA then the Remaining frame count and NO EOSP was sent (which should not have happened), the BTQM will force a QOS NULL to be sent next, with the EOSP bit set.
25	CFG_END_APSD_AFTER_B A_APSD_EOSP_SENT	When set, and an APSD session with BA is ongoing and an EOSP was set, and the BA gets received, the APSD session will be ended independent of how many frames were acknowledged
24	CFG_SEND_QOSNULL_AF TER_BA_APSD_EOSP_SEN T	When set, and an APSD session with BA is ongoing and an EOSP was set, and the BA gets received, a QoS null will be sent if the BA is not confirming all frames to be ACKED.
10:0	BTQM_QUEUEID_QUEUE_ ENABLE	When set, the corresponding queueid transmission is enabled. Bit zero corresponds with queueid0 This register allows high level blocking of all the transmission of frames out of a particular queueid

0x0A0800F8 WCSS_A_BMU_BTQM_STATUS**Type:** R**Reset State:** 0x00000000

This register indicates the status of several BTQM features

WCSS_A_BMU_BTQM_STATUS

Bits	Name	Description
3	BTQM_TPE_INT_STATUS	When set, the BTQM TPE interface is active providing the TPE queue related information.
2	BTQM_ARBITER_STATUS	When set, the BTQM arbiter is active searching for the next STA to transmit a new frame from
1	BTQM_BA_HANDLER_STAT US	When set, the BTQM Block Ack handler is processing a Block ACK bitmap

WCSS_A_BMU_BTQM_STATUS (cont.)

Bits	Name	Description
0	BTQM_QUEUE_CTRL_STATUS	When set, the BTQM queue controller is active popping a frame out of one of the BTQM entrance WQs and pushing them into the transmit WQs

0x0A0800FC WCSS_A_BMU_BTQM_STA_ENABLE_DISABLE_CONTROL**Type:** RW**Reset State:** 0x00000000

Controls the enable-ing or disable-ing of a STA Whenever a value is written into this register, the BTQM will process the STA settings that came with it. A status bit will indicate if the BTQM is still processing an 'update' command. The next update comand should not be issued till the status bit has been cleared. Note that this can sometimes take quite a while. If for example just before a STA-disable command, a transmit exchange sequence with that STA had started, that transmit exchange sequence will be allowed to finish first. This is done to prevent abnormal abortion of sequences.

WCSS_A_BMU_BTQM_STA_ENABLE_DISABLE_CONTROL

Bits	Name	Type	Description
31	STA_UPDATE_STATUS	R	When set to one, BTQM is processing the last STAid configuration command written to this register.
10:8	STA_CONFIGURATION_COMMAND	RW	000: Enable the transmit queues, but do NOT enable transmission 001: Enable the transmit queues, AND enable transmission 010: Disable the transmit queues AND disable transmission 011: Disable the transmit queues AND disable transmission AND cleanup any data left in the transmit queues 100: Clear the fields related to control frames (ps-poll bit, apsd trigger bit, data null sent bit, qos null sent bit, remaining apsd frame count and eosp sent bit) 101: Disable transmission, but leave transmit queues enabled AND clear the fields related to control frames (ps-poll bit, apsd trigger bit, data null sent bit, qos null sent bit, remaining apsd frame count and eosp sent bit) 110: Disable transmission, but leave transmit queues enabled
3:0	STAID	RW	The STAid for which the STA operation is intended.

0x0A080100 WCSS_A_BMU_TRANSMIT_QUEUE_ACCESS_CONTROL1**Type:** RW**Reset State:** 0x00000000

Selects the queue to access

WCSS_A_BMU_TRANSMIT_QUEUE_ACCESS_CONTROL1

Bits	Name	Description
11:8	QUEUEID	
3:0	STAID	

0x0A080104 WCSS_A_BMU_TRANSMIT_QUEUE_ACCESS_CONTROL2**Type:** RW**Reset State:** 0x00000000

Part 1 of the queue information to write back to the bmu Note that this format corresponds to the first word of the TX queue info in MIF memory

WCSS_A_BMU_TRANSMIT_QUEUE_ACCESS_CONTROL2

Bits	Name	Description
31	PREV_TX_DROP_SETTING	
30	BAR_REQ_SETTING	
10:0	NR_OF_FRAMES_IN_QUEUE	

0x0A080108 WCSS_A_BMU_TRANSMIT_QUEUE_ACCESS_CONTROL3**Type:** RW**Reset State:** 0x00000000

Part 2 of the queue information to write back to the bmu Note that this format corresponds to the second word of the TX queue info in MIF memory

WCSS_A_BMU_TRANSMIT_QUEUE_ACCESS_CONTROL3

Bits	Name	Description
26:16	QUEUE_HEAD_BD_INDEX	
10:0	QUEUE_TAIL_BD_INDEX	

0x0A08010C WCSS_A_BMU_TRANSMIT_QUEUE_ACCESS_CONTROL4**Type:** RW**Reset State:** 0x80000000

controls what kind of access will happen on the transmit queue

WCSS_A_BMU_TRANSMIT_QUEUE_ACCESS_CONTROL4

Bits	Name	Type	Description
31	ACCESS_DONE	R	When set, the queue access control action has finished.
0	READ_WRITE_CONTROL	RW	Only valid when 'set_frame_drop_indication' is NOT set. When 'read_write_control' is set to 1, a transmit queue read access is initialized. The read result will be available in registers 'transmit_queue_access_control5' and 6, after the 'access done' bit is set. When set to 0, a transmit queue write access is initialized. The data that is present in the registers 'transmit_queue_access_control1' and 2, is written back to the transmit queue maintained by the BTQM.

0x0A080110 WCSS_A_BMU_TRANSMIT_QUEUE_ACCESS_CONTROL5**Type:** R**Reset State:** 0x00000000

Part 1 of the queue information read from the bmu. Only valid after the 'access done' bit is set in 'transmit_queue_access_control4'. Note that this format corresponds to the first word of the TX queue info in MIF memory.

WCSS_A_BMU_TRANSMIT_QUEUE_ACCESS_CONTROL5

Bits	Name	Description
31	PREV_TX_DROP_SETTING	
30	BAR_REQ_SETTING	
10:0	NR_OF_FRAMES_IN_QUEUE	

0x0A080114 WCSS_A_BMU_TRANSMIT_QUEUE_ACCESS_CONTROL6**Type:** R**Reset State:** 0x00000000

Part 2 of the queue information read from the bmu. Only valid after the 'access done' bit is set in 'transmit_queue_access_control4'. Note that this format corresponds to the second word of the TX queue info in MIF memory.

WCSS_A_BMU_TRANSMIT_QUEUE_ACCESS_CONTROL6

Bits	Name	Description
26:16	QUEUE_HEAD_BD_INDEX	

WCSS_A_BMU_TRANSMIT_QUEUE_ACCESS_CONTROL6 (cont.)

Bits	Name	Description
10:0	QUEUE_TAIL_BD_INDEX	

0x0A080118 WCSS_A_BMU_BD_INDEX_INFO_ACCESS_CONTROL1**Type:** RW**Reset State:** 0x00000000

Selects the bd index to access

WCSS_A_BMU_BD_INDEX_INFO_ACCESS_CONTROL1

Bits	Name	Type	Description
31	BD_INDEX_INFO_ACCESS_STATUS	R	When set, the BD index memory access is ongoing. The read data will not be valid until this bit is cleared. No new bd index command should be given when this bit is set.
16	BD_INDEX_UPDATE	RW	When set, the BD index info in BMU memory will be updated with the data given in registers 'bd_index_info_access_control2 - 4' When NOT set, the BD index info will be read from the BMU memory and will be made available in registers: 'bd_index_info_access_control5 - 7'
10:0	BD_INDEX	RW	The bd index for which the BD index info needs to be read or written

0x0A08011C WCSS_A_BMU_BD_INDEX_INFO_ACCESS_CONTROL2**Type:** RW**Reset State:** 0x00000000

Part 1 of the bd index information to write back to the bmu

WCSS_A_BMU_BD_INDEX_INFO_ACCESS_CONTROL2

Bits	Name	Description
31:0	BD_INDEX_INFO_0	

0x0A080120 WCSS_A_BMU_BD_INDEX_INFO_ACCESS_CONTROL3**Type:** RW**Reset State:** 0x00000000

Part 2 of the bd index information to write back to the bmu

WCSS_A_BMU_BD_INDEX_INFO_ACCESS_CONTROL3

Bits	Name	Description
31:0	BD_INDEX_INFO_1	

0x0A080124 WCSS_A_BMU_BD_INDEX_INFO_ACCESS_CONTROL4**Type:** RW**Reset State:** 0x00000000

Part 3 of the bd index information to write back to the bmu

WCSS_A_BMU_BD_INDEX_INFO_ACCESS_CONTROL4

Bits	Name	Description
5:0	BD_INDEX_INFO_2	

0x0A080128 WCSS_A_BMU_BD_INDEX_INFO_ACCESS_CONTROL5**Type:** R**Reset State:** 0x00000000

Part 1 of the bd index information read from the bmu. Only valid after the 'access done' bit is set in 'bd_index_info_access_control5'

WCSS_A_BMU_BD_INDEX_INFO_ACCESS_CONTROL5

Bits	Name	Description
31:0	BD_INDEX_INFO_0	

0x0A08012C WCSS_A_BMU_BD_INDEX_INFO_ACCESS_CONTROL6**Type:** R**Reset State:** 0x00000000

Part 2 of the bd index information read from the bmu. Only valid after the 'access done' bit is set in 'bd_index_info_access_control5'

WCSS_A_BMU_BD_INDEX_INFO_ACCESS_CONTROL6

Bits	Name	Description
31:0	BD_INDEX_INFO_1	

0x0A080130 WCSS_A_BMU_BD_INDEX_INFO_ACCESS_CONTROL7**Type:** R**Reset State:** 0x00000000

Part 3 of the bd index information read from the bmu. Only valid after the 'access done' bit is set in 'bd_index_info_access_control5'

WCSS_A_BMU_BD_INDEX_INFO_ACCESS_CONTROL7

Bits	Name	Description
5:0	BD_INDEX_INFO_2	

0x0A080134 WCSS_A_BMU_BAR_AND_FRAME_DROP_CONTROL**Type:** RW**Reset State:** 0x00000000

controls bar indication and frame drop indication This operation complete is indicated by setting bit-31 of transmit_queue_access_control4 register

WCSS_A_BMU_BAR_AND_FRAME_DROP_CONTROL

Bits	Name	Description
2	SET_QUEUE_BAR_INDICATION	When set to 1, the BTQM will force the 'BAR request' setting in the indicated transmit queue to be set. This feature should be used when a Block ack session is to be initialized, and with the 'BAR request' setting set, the TPE will be forced to first transmit a BAR frame.
1	SET_FRAME_DROP_INDICATION	When set to 1, the BTQM will force the 'drop frame' setting in the indicated transmit queue to be set.

0x0A080138 WCSS_A_BMU_TX_QUEUE_STAID_QUEUEID_CONFIG**Type:** RW**Reset State:** 0x00000000

controls the valid STAid and queueid ranges

WCSS_A_BMU_TX_QUEUE_STAID_QUEUEID_CONFIG

Bits	Name	Description
19:16	MAX_VALID_QUEUEID	The maximum allowed queueid. When a frame is pushed to a transmit WQ for a queueid higher then the programmed value, an error indication will be generated.
11:8	MAX_VALID_STAID	The maximum allowed staId. When a frame is pushed to a transmit WQ for a STAid higher then the programmed value, an error indication will be generated.
3:0	NR_OF_BEACON_STAIDS	The number of staids set aside for beacon handling The STAids reserved start from 0. A number of 5 programmed here, means STAids 0, 1, 2, 3, and 4 are set aside for beacon transmission

0x0A08013C WCSS_A_BMU_ENHANCED_TRACING_CONTROL1**Type:** RW**Reset State:** 0x00000000

Control for the enhance gen5/6 HW tracing capabilities

WCSS_A_BMU_ENHANCED_TRACING_CONTROL1

Bits	Name	Description
31	SW_BMU_HWTRACE_ENABLE	Implemented: When set, tracing is enabled for SW data
30	BTQM_ARBITER_SEARCH_HWTRACE_ENABLE	Implemented: When set, tracing is enabled for btqm next STA/qid transmission selection
29	BTQM_PWR_STATE_HWTRACE_ENABLE	Implemented: When set, tracing is enabled for receive frames pwr state handling
28	TPE_BMU_HWTRACE_ENABLE	Implemented: When set, tracing is enabled for signals from the TPE module
27	GET_BD_PDU_TRACE_FOR_PDU_ENABLE	Implemented: When set, get bd and pdu command tracing is enabled for GET commands that request at least one PDU
26	NAV_UPDATE_TRACE_ENABLE	Implemented: When set, NAV updated (setting or clearing) are registered
25	PHYTX_IDLE_TRACE_ENABLE	Implemented: When set, phytx idle level changes are registered
24	RXP_PKDET_N_TRACE_ENABLE	Implemented: When set, pktdet_n level changes are registered
23	RXP_SECONDARY_CCA_TRACE_ENABLE	Implemented: When set, secondary cca level changes are registered
22	RXP_PRIMARY_CCA_TRACE_ENABLE	Implemented: When set, primary cca level changes are registered

WCSS_A_BMU_ENHANCED_TRACING_CONTROL1 (cont.)

Bits	Name	Description
21	RXP_RPE_BITMAP_UPDAT E_TRACE_ENABLE	Implemented: When set, rxp - rpe bitmap update interaction is registered
20	RXP_RPE_DUPL_DETECT_ TRACE_ENABLE	Implemented: When set, rxp - rpe dupl detect interaction is registered
19	BTQM_QUEUEING_CTRL	Implemented: When set, popping of frames by the BTQM transmit queue controller of frames pushed into the BTQM entrance WQs are registered
18	TPE_FEEDBACK_CMD_TR ACE_ENABLE	Implemented: When set, tpe feedback info is registered
17	TPE_GET_BDIDX_CMD_TR ACE_ENABLE	Implemented: When set, tpe bd index info command are registered
16	TPE_GET_LENGTH_CMD_ TRACE_ENABLE	Implemented: When set, tpe get length info command are registered
15	TPE_GET_INFO_CMD_TRA CE_ENABLE	Implemented: When set, tpe get info command are registered
14	DETAILED_BD_IDLE_LIST_ PUSH_HW_TRACE_ENABL E	Implemented: When set, every push to the bd idle list is registered
13	BTQM_QUEUEING_DROP_ FRAME_TRACE_ENABLE	Implemented: When set, frame drops by the BTQM queue controller tracing is enabled
12	RELEASE_TPE_INT_FRAM E_TRACE_ENABLE	Implemented: When set, release frames from tpe interface tracing is enabled
11	RELEASE_DISABLED_STA_ QUEUE_TRACE_ENABLE	Implemented: When set, release disabled_sta queue command tracing is enabled
10	RELEASE_BA_DROP_TRA CE_ENABLE	When set, release ba_drop command tracing is enabled
9	RELEASE_SINK_CMD_TRA CE_ENABLE	Implemented: When set, release sink command tracing is enabled
8	RELEASE_BD_PDU_CMD_ TRACE_ENABLE	Implemented: When set, release bd_pdu command tracing is enabled
7	RELEASE_PDU_CMD_TRA CE_ENABLE	Implemented: When set, release pdu command tracing is enabled
6	RELEASE_BD_CMD_TRAC E_ENABLE	Implemented: When set, release bd command tracing is enabled
5	BD_PDU_RESERVATION_T RACE_ENABLE	Implemented: When set, bd pdu reservation command tracing is enabled
4	GET_BD_PDU_TRACE_FO R_BD_ENABLE	Implemented: When set, get bd and pdu command tracing is enabled for GET commands that request at least one BD
3	PUSH_CMD_TRACE_ENAB LE	Implemented: When set, push command tracing is enabled

WCSS_A_BMU_ENHANCED_TRACING_CONTROL1 (cont.)

Bits	Name	Description
2	POP_CMD_TRACE_ENABLE	Implemented: When set, pop command tracing is enabled
1	TRACE_FREEZE_ON_ERR	Implemented: When set, the tracing memory will freeze when the bmu detects an internal error for which an BMU error interrupt was enabled.
0	ENHANCED_TRACING_ENABLE	Implemented: When set, all legacy (gen 4) modes of BMU HW-TRACING are disabled, and the new mode is enabled. In this mode all legacy tracing control bit settings are discarded

0x0A080140 WCSS_A_BMU_BTQM_QUEUE_INFO_BASE_ADDR**Type:** RW**Reset State:** 0x00000000

sets the memory address location where the BTQM is going to find the transmit queues

WCSS_A_BMU_BTQM_QUEUE_INFO_BASE_ADDR

Bits	Name	Description
31:2	ADDRESS	The memory address where the BTQM is going to find the transmit queues

0x0A080144 WCSS_A_BMU_QOS_QUEUEID_MAPPING1**Type:** RW**Reset State:** 0x000000FF

indicates to the BMU which queueids contain QoS frames

WCSS_A_BMU_QOS_QUEUEID_MAPPING1

Bits	Name	Description
10:0	QOS_SETTING	When a bit is set, the corresponding queueid will contain QoS frames

0x0A080148 WCSS_A_BMU_QOS_QUEUEID_MAPPING2**Type:** RW**Reset State:** 0x76543210

indicates to the BMU what QoS TID frames are located in a queue. This setting is only valid for a queueid, if the corresponding 'qos_setting' bit for that queue is set.

WCSS_A_BMU_QOS_QUEUEID_MAPPING2

Bits	Name	Description
31:28	TID_FOR_QUEUEID7	
27:24	TID_FOR_QUEUEID6	
23:20	TID_FOR_QUEUEID5	
19:16	TID_FOR_QUEUEID4	
15:12	TID_FOR_QUEUEID3	
11:8	TID_FOR_QUEUEID2	
7:4	TID_FOR_QUEUEID1	
3:0	TID_FOR_QUEUEID0	

0x0A08014C WCSS_A_BMU_QOS_QUEUEID_MAPPING3

Type: RW

Reset State: 0x00000000

indicates to the BMU what QoS TID frames are located in a queue. This setting is only valid for a queueid, if the corresponding 'qos_setting' bit for that queue is set.

WCSS_A_BMU_QOS_QUEUEID_MAPPING3

Bits	Name	Description
11:8	TID_FOR_QUEUEID10	
7:4	TID_FOR_QUEUEID9	
3:0	TID_FOR_QUEUEID8	

0x0A080154 WCSS_A_BMU_QUEUEID_BO_MAPPING1

Type: RW

Reset State: 0x44557667

indicates to the BMU how queueid are mapped to BO engines

WCSS_A_BMU_QUEUEID_BO_MAPPING1

Bits	Name	Description
30:28	BO_FOR_QUEUEID7	

WCSS_A_BMU_QUEUEID_BO_MAPPING1 (cont.)

Bits	Name	Description
26:24	BO_FOR_QUEUEID6	
22:20	BO_FOR_QUEUEID5	
18:16	BO_FOR_QUEUEID4	
14:12	BO_FOR_QUEUEID3	
10:8	BO_FOR_QUEUEID2	
6:4	BO_FOR_QUEUEID1	
2:0	BO_FOR_QUEUEID0	

0x0A080158 WCSS_A_BMU_QUEUEID_BO_MAPPING2**Type:** RW**Reset State:** 0x00000032

indicates to the BMU how queueid are mapped to BO engines

WCSS_A_BMU_QUEUEID_BO_MAPPING2

Bits	Name	Description
10:8	BO_FOR_QUEUEID10	
6:4	BO_FOR_QUEUEID9	
2:0	BO_FOR_QUEUEID8	

0x0A080160 WCSS_A_BMU_FAST_TRACE_READING_CONTROL**Type:** RW**Reset State:** 0x00000000

Indicates the start bd index address from which fast trace reading should start programming this field set the trace reading head ptr automatically back to this programmed index

WCSS_A_BMU_FAST_TRACE_READING_CONTROL

Bits	Name	Type	Description
26:16	CURRENT_FAST_TRACE_BD_INDEX_ADDRESS	R	The current bd index used for the next bd index read. This value gets automatically updated to the 'fast_trace_start_bd_index_address' field when that is reprogrammed. This value also gets automatically updated to point to the trace table entry when 'fast_trace_reading_data2' register is read. The hardware takes care of 'wrapping around', when the end of the trace memory has been reached.
10:0	FAST_TRACE_START_BD_INDEX_ADDRESS	RW	

0x0A080164 WCSS_A_BMU_FAST_TRACE_READING_DATA0**Type:** R**Reset State:** 0x00000000

returns the lower 32 bits of the bd index data of the bd index indicated in field:
current_fast_trace_bd_index_address

WCSS_A_BMU_FAST_TRACE_READING_DATA0

Bits	Name	Description
31:0	TRACE_DATA0	The lower 32 bits of the trace data

0x0A080168 WCSS_A_BMU_FAST_TRACE_READING_DATA1**Type:** R**Reset State:** 0x00000000

returns the middle 32 bits of the bd index data of the bd index indicated in field:
current_fast_trace_bd_index_address.

WCSS_A_BMU_FAST_TRACE_READING_DATA1

Bits	Name	Description
31:0	TRACE_DATA1	The upper X bits of the trace data

0x0A08016C WCSS_A_BMU_FAST_TRACE_READING_DATA2**Type:** R**Reset State:** 0x00000000

returns the upper X bits of the bd index data of the bd index indicated in field:
current_fast_trace_bd_index_address. Reading from this register, automatically updates the
'current_fast_trace_bd_index_address' field to point to the next entry in the trace table (exact width
of this field: TBD)

WCSS_A_BMU_FAST_TRACE_READING_DATA2

Bits	Name	Description
31:0	TRACE_DATA2	The upper X bits of the trace data

0x0A080170 WCSS_A_BMU_FAST_BD_LINK_SETUP_CONTROL

Type: RW

Reset State: 0x00000000

Initiates fast automated linking of BDs

WCSS_A_BMU_FAST_BD_LINK_SETUP_CONTROL

Bits	Name	Type	Description
31	CREATE_BD_LINKED_LIST	RW	When writing a 1 to this field, the BMU will start linking all the BDs indicated by the 'bd_start_index' and 'bd_end_index' fields.
30	BD_LINK_LIST_CREATION_STATUS	R	When set, the BMU is still busy creating the BD linked list
26:16	BD_END_INDEX	RW	The index of the last BD in the BD linked list to be created by HW
10:0	BD_START_INDEX	RW	The index of the first BD in the BD linked list to be created by HW

0x0A080174 WCSS_A_BMU_STA_CONFIG_STATUS1

Type: RW

Reset State: 0x00000000

Selects the STAid for which the STA parameters need to be read Whenever this register is written, the hardware will start reading or writing the sta parameters related to this staid

WCSS_A_BMU_STA_CONFIG_STATUS1

Bits	Name	Type	Description
31	READ_WRITE_STATUS	R	When set, the hardware is busy retrieving the station information for the last time this register got programmed

WCSS_A_BMU_STA_CONFIG_STATUS1 (cont.)

Bits	Name	Type	Description
8	STA_CONFIG_UPDATE	RW	When NOT set, the hardware will read the sta_id settings When set, the hardware will write back the sta_id settings as programmed in the 'sta_config_status2' and 'sta_config_status3' registers.
3:0	STAID	RW	The STAID for which the station parameters need to be read

0x0A080178 WCSS_A_BMU_STA_CONFIG_STATUS2**Type:** RW**Reset State:** 0x00000000

Provides the STA information corresponding to the 'sta_id' programmed in the 'sta_config_status1' register. This register only has valid contents when the 'read_write_status' bit in the 'sta_config_status1' register is NOT set.

WCSS_A_BMU_STA_CONFIG_STATUS2

Bits	Name	Description
22	EOSP_SENT_STATUS	When set to one, the BTQM has sent a QoS frame for which the EOSP bit in the QoS field was set. This field is used by the BTQM to track the sending of this indicating in case APSD in combination with AMPDU aggregation is used.
21	QOS_NULL_SENT_ONCE_STATUS	When set to one, the BTQM has sent a QoS Null frame. This field is used to be able to indicate that the previous transmission was not successful, and the next time around the QoS Null will be resent one more time.
20	DATA_NULL_SENT_ONCE_STATUS	When set to one, the BTQM has sent a Data Null frame. This field is used to be able to indicate that the previous transmission was not successful, and the next time around the Data Null will be resent one more time.
18:16	CURRENT_REMAINING_TX_FRAME_CNT	This value will be taken as the 'current remaining tx frame count'. Typically this value should be set to zero, and access here is only provided for backup or testing purposes. When set to a non zero value, it will only be used when also 'current_apsd_trig_received_setting' has been set, and the 'sta_u_apsd_enable' has been set.
14	CURRENT_APSD_TRIG_RECEIVED_SETTING	When set to one, the BTQM will assume that an apsd trigger frame has just been received.
13	CURRENT_PS_POLL_RECEIVED_SETTING	When set to one, the BTQM will assume that a ps_poll frame has just been received.
12	CURRENT_PWR_STATE_SETTING	When set to one, the BTQM will assume that the STA is in power save mode.

WCSS_A_BMU_STA_CONFIG_STATUS2 (cont.)

Bits	Name	Description
10:8	INITIAL_REMAINING_TX_FRAME_CNT	After receiving an APSD trigger frame, the value in this register will represent the number of response frames that will be sent to the STA. A value of 7 indicates, to drain the complete transmit queue
6	QOS_NULL_RESP_ENABLE	When set, and an apsd trigger is received and all APSD delivery enabled queues are empty, the BTQM will indicate to the TPE to transmit a QoS null frame
5	U_DATA_NULL_RESP_ENABLE	When set, and a ps-poll apsd trigger is received and all the (non APSD deliver enabled) queues are empty, the BTQM will indicate to the TPE to transmit a Data null frame
4	STA_U_APSD_TRACK_ENABLE	When set to one, the BTQM will the reception and handling of APSD trigger frames
1	STA_TX_ENABLE	When set, transmission for this sta is enabled
0	STA_ENABLE	When set, the STAid is enabled

0x0A08017C WCSS_A_BMU_STA_CONFIG_STATUS3**Type:** RW**Reset State:** 0x00000000

Provides the STA information corresponding to the 'sta_id' programmed in the 'sta_config_status1' register. This register only has valid contents when the 'read_write_status' bit in the 'sta_config_status1' register is NOT set.

WCSS_A_BMU_STA_CONFIG_STATUS3

Bits	Name	Description
31:16	DELIVERY_ENABLED_TCID_QUEUES	When set, the corresponding TCID is part of the delivery enabled queues. Bit 0 corresponds with TCID 0
15:0	TRIGGER_ENABLED_TCID	When set, and a QoS Data frame of the corresponding TCID is received, the frame will be interpreted as an APSD trigger frame

0x0A080180 WCSS_A_BMU_STA_CONFIG_STATUS4**Type:** RW**Reset State:** 0x00000000

Provides the STA information corresponding to the 'sta_id' programmed in the 'sta_config_status1' register. This register only has valid contents when the 'read_write_status' bit in the 'sta_config_status1' register is NOT set.

WCSS_A_BMU_STA_CONFIG_STATUS4

Bits	Name	Description
10:0	DATA_AVAILABLE	When set, the corresponding queueid queue contains frames Bit 0, corresponds to queueid 0 Bit 1, corresponds to queueid 1 ...

0x0A080184 WCSS_A_BMU_BTQM_ERR_STATUS**Type:** R**Reset State:** 0x00000000

Provides status information on BTQM internal errors

WCSS_A_BMU_BTQM_ERR_STATUS

Bits	Name	Description
27	BTQM_POLL_STATUS_TIMER_INT_STATUS	
26	STA_CONFIG_DONE_INT_STATUS	
25	BTQM_SW_QUEUE_ACCESS_DONE_INT_STATUS	
24	STA_CONTROL_UPDATE_DONE_INT_STATUS	
23	LMEM_BD_PUSHED_TO_BTQM_QUEUE_ERR	
22	BTQM_BA_SEQ_NRX_SEQ_NRX_ERR	
21	BTQM_FRAME_DATA_AVAILABLE_ERR	
20	BTQM_FRAME_TIMEOUT_ERR	
19	RELEASE_BTQM_LINK_BD_INDEX_ERR	
18	SET_PREV_LAST_FRAME_NULL_BD_INDEX_ERR	
17	SET_SECOND_FRAME_NULL_BD_INDEX_ERR	
16	RELEASE_BD_PDU_CMD_OWNERSHIP_ERR	
15	RELEASE_FRAG_BD_CMD_OWNERSHIP_ERR	

WCSS_A_BMU_BTQM_ERR_STATUS (cont.)

Bits	Name	Description
14	RELEASE_PDU_TO_BD_C MD_OWNERSHIP_ERR	
13	BD_PDU_MGR_OWNERSHI P_ERR	
12	BTQM_TPE_INT_OWNER SHIP_ERR	
11	BTQM_QUEUE_CTRL_OW NERSHIP_ERR	
10	BA_HANDLER_OWNERSHI P_ERR	
9	RELEASE_BD_OWNERSHI P_ERR	
8	PUSH_BD_OWNERSHIP_E RR	
7	EMPTY_TX_QUEUE_PREF ETCHED_WARNING	
6	CONTROL_FRAME_ERR	
5	FRAME_FORMAT_ERR	
4	INVALID_BD_PARAMETER S_ERR	
3	UNEXPECTED_ARBITER_R ESULT_WARNING	
2	BROKEN_TRANSMIT_QUE UE_LINK_ERR	
1	TPE_INTERACTION_ERR	
0	LINKED_LIST_ERR	

0x0A080188 WCSS_A_BMU_BTQM_ERR_ENABLE**Type:** RW**Reset State:** 0x005FFF7F

Enables for the BTQM error or warning interrupt generation

WCSS_A_BMU_BTQM_ERR_ENABLE

Bits	Name	Description
27	BTQM_POLL_STATUS_TIM ER_INT_ENABLE	

WCSS_A_BMU_BTQM_ERR_ENABLE (cont.)

Bits	Name	Description
26	STA_CONFIG_DONE_INT_ENABLE	
25	BTQM_SW_QUEUE_ACCE SS_DONE_INT_ENABLE	
24	STA_CONTROL_UPDATE_ DONE_INT_ENABLE	
23	LMEM_BD_PUSHED_TO_B TQM_QUEUE_ERR_ENABL E	
22	BTQM_BA_SEQ_NR_GE_T X_SEQ_NR_ERR_ENABLE	
21	BTQM_FRAME_DATA_AVAI L_ERR_ENABLE	
20	BTQM_FRAME_TIMEOUT_ ERR_ENABLE	
19	RELEASE_BTQM_LINK_BD _INDEX_ERR_ENABLE	
18	SET_PREV_LAST_FRAME_ NULL_BD_INDEX_ERR_EN ABLE	
17	SET_SECOND_FRAME_NU LL_BD_INDEX_ERR_ENAB LE	
16	RELEASE_BD_PDU_CMD_ OWNERSHIP_ERR_ENABL E	
15	RELEASE_FRAG_BD_CMD _OWNERSHIP_ERR_ENAB LE	
14	RELEASE_PDU_TO_BD_C MD_OWNERSHIP_ERR_EN ABLE	
13	BD_PDU_MGR_OWNERSHI P_ERR_ENABLE	
12	BTQM_TPE_INT_OWNER SHIP_ERR_ENABLE	
11	BTQM_QUEUE_CTRL_OW NERSHIP_ERR_ENABLE	
10	BA_HANDLER_OWNERSHI P_ERR_ENABLE	
9	RELEASE_BD_OWNERSHI P_ERR_ENABLE	

WCSS_A_BMU_BTQM_ERR_ENABLE (cont.)

Bits	Name	Description
8	PUSH_BD_OWNERSHIP_ERR_ENABLE	
7	EMPTY_TX_QUEUE_PREFETCHED_WARNING	
6	CONTROL_FRAME_ERR_ENABLE	
5	FRAME_FORMAT_ERR_ENABLE	
4	INVALID_BD_PARAMETERS_ERR_ENABLE	
3	UNEXPECTED_ARBITER_RESULT_WARNING	
2	BROKEN_TRANSMIT_QUEUE_LINK_ERR_ENABLE	
1	TPE_INTERACTION_ERR_ENABLE	
0	LINKED_LIST_ERR_ENABLE	

0x0A08018C WCSS_A_BMU_BTQM_ERR_STATE**Type:** R**Reset State:** 0x00000000

Provides the BTQM statemachine states when a BMU error interrupt is generated

WCSS_A_BMU_BTQM_ERR_STATE

Bits	Name	Description
29:0	BTQM_STATES	Snapshot of the btqm statemachines states when a BMU error interrupt is generated

0x0A080190 WCSS_A_BMU_BTQM_ERR_STATE2**Type:** R**Reset State:** 0x00000000

Provides part 2 of the BTQM statemachine states when a BMU error interrupt is generated

WCSS_A_BMU_BTQM_ERR_STATE2

Bits	Name	Description
19:0	BTQM_STATES	Snapshot of the btqm statemachines states when a BMU error interrupt is generated

0x0A080194 WCSS_A_BMU_BTQM_TPE_INT_STATS**Type:** R**Reset State:** 0x00000000

Provides status information on the bmu_tpe interface

WCSS_A_BMU_BTQM_TPE_INT_STATS

Bits	Name	Description
19	TPE_INTERACTION_ERROR_STATUS	When set, a TPE interface interaction error occurred
18:16	TPE_INTERACTION_ERROR_CODE	indicates the first tpe interface interaction error that happened
15:8	BO_DATA_AVAIL	status of the dataavailable signals to the TPE
4:0	TPE_INT_SM_STATE	Represent the statemachine state

0x0A0801B4 WCSS_A_BMU_RETRY_LIMIT_DROP_CNT**Type:** R**Reset State:** 0x00000000

Provides the number of frames that were dropped by the BTQM because the retry limit was reached.

WCSS_A_BMU_RETRY_LIMIT_DROP_CNT

Bits	Name	Description
23:0	RETRY_LIMIT_DROP_CNT	

0x0A0801B8 WCSS_A_BMU_TPE_DROP_CNT**Type:** R**Reset State:** 0x00000000

Provides the number of frames that were dropped by the BTQM because the TPE gave a 'DROP feedback' command

WCSS_A_BMU_TPE_DROP_CNT

Bits	Name	Description
23:0	TPE_DROP_CNT	

0x0A0801BC WCSS_A_BMU_BA_HANDLER_ACKED_DROP_CNT**Type:** R**Reset State:** 0x00000000

Provides the number of frames that were dropped by the BTQM BA handler because the frame reception got acknowledge by the BA bitmap

WCSS_A_BMU_BA_HANDLER_ACKED_DROP_CNT

Bits	Name	Description
23:0	BA_HANDLER_ACKED_DROP_CNT	

0x0A0801C0 WCSS_A_BMU_BA_HANDLER_WINDOW_MOVED_DROP_CNT**Type:** R**Reset State:** 0x00000000

Provides the number of frames that were dropped by the BTQM BA handler because the BA bitmap SSN moved on beyond the frame Sequence number still in the queue

WCSS_A_BMU_BA_HANDLER_WINDOW_MOVED_DROP_CNT

Bits	Name	Description
23:0	BA_HANDLER_WINDOW_MOVED_DROP_CNT	

0x0A0801C4 WCSS_A_BMU_GET_INFO_CNT**Type:** R**Reset State:** 0x00000000

Provides the number times the BTQM received a 'get_info' command from the TPE

WCSS_A_BMU_GET_INFO_CNT

Bits	Name	Description
23:0	GET_INFO_CNT	

0x0A0801C8 WCSS_A_BMU_GET_BD_INDEX_CNT**Type:** R**Reset State:** 0x00000000

Provides the number times the BTQM received a 'get_bd_index' command from the TPE

WCSS_A_BMU_GET_BD_INDEX_CNT

Bits	Name	Description
23:0	GET_BD_INDEX_CNT	

0x0A0801CC WCSS_A_BMU_PWR_SAVE_STA_CNT**Type:** RW**Reset State:** 0x00000000

The number of STAs that are in power save mode & bit map for power save stations

WCSS_A_BMU_PWR_SAVE_STA_CNT

Bits	Name	Description
31:16	POWER_SAVE_STAS_BIT_MAP	
8:0	PWR_SAVE_STA_CNT	

0x0A0801D0 WCSS_A_BMU_BTQM_FRAME_CNT**Type:** R**Reset State:** 0x00000000

The number of frames currently managed by the btqm

WCSS_A_BMU_BTQM_FRAME_CNT

Bits	Name	Description
10:0	BTQM_FRAME_CNT	

0x0A0801D4 WCSS_A_BMU_BTQM_TIMEOUT_CHECK_CTRL**Type:** RW**Reset State:** 0x00000000

The number of clk cycles the BTQM is allowed to empty all btqm transmit queues measured from the point where the last frame got pushed into the BTQM. When this number is exceeded, an BTQM timeout error interrupt is generated. A value of zero disables this check.

WCSS_A_BMU_BTQM_TIMEOUT_CHECK_CTRL

Bits	Name	Description
31:0	BTQM_TIMEOUT_CLK_CYC LE_CNT	

0x0A0801D8 WCSS_A_BMU_PWR_STATE_CHANGE_INT_CONTROL**Type:** RW**Reset State:** 0x00000000

Provides interrupt control and status of STA power state changes.

WCSS_A_BMU_PWR_STATE_CHANGE_INT_CONTROL

Bits	Name	Type	Description
31	PWR_STATE_CHANGE_INT _AUTO_CLEAR_ENABLE	RW	
30	PWR_STATE_CHANGE_INT _OVERFLOW	RW	
29	PWR_STATE_STATUS2	R	
28	PWR_STATE_STATUS1	R	
27	PWR_STATE_STATUS0	R	
26	PWR_STATE_ENTRY2_VAL ID	RW	
25	PWR_STATE_ENTRY1_VAL ID	RW	
24	PWR_STATE_ENTRY0_VAL ID	RW	
23:16	STAID_PWR_INT2	R	
15:8	STAID_PWR_INT1	R	
7:0	STAID_PWR_INT0	R	

0x0A0801DC WCSS_A_BMU_BTQM_INTERRUPT_STATUS**Type:** R**Reset State:** 0x00000000

This register indicates the status of BTQM interrupts. When one of the corresponding events occur, the status will reflect this. An interrupt is only generated if the interrupt is also enabled (see next register).

WCSS_A_BMU_BTQM_INTERRUPT_STATUS

Bits	Name	Description
2	PWRSTATE_DROP_INTERRUPT	When set, the maximum number of frames in a queue for an STA in power save mode has been reached.
1	BTQM_QUEUES_EMPTY_INTERRUPT	When set, the BTQM queues have become empty.
0	PWRSTATE_CHANGE_INTERRUPT	When set, an STA has indicated a power state change.

0x0A0801E0 WCSS_A_BMU_BTQM_INTERRUPT_ENABLE**Type:** RW**Reset State:** 0x00000000

This register controls the enables of BTQM interrupts.

WCSS_A_BMU_BTQM_INTERRUPT_ENABLE

Bits	Name	Description
2	PWRSTATE_DROP_INTERRUPT_ENABLE	When set, an interrupt will be generated when the maximum number of frames in a queue for an STA in power save mode has been reached.
1	BTQM_QUEUES_EMPTY_INTERRUPT_ENABLE	When set, an interrupt will be generated when the BTQM queues have become empty.
0	PWRSTATE_CHANGE_INTERRUPT_ENABLE	When set, an interrupt will be generated when an STA has indicated a power state change.

0x0A0801E4 WCSS_A_BMU_PWR_SAVE_DROP_CTRL**Type:** RW**Reset State:** 0x00000000

This register controls dropping of frames for an STA in power save mode.

WCSS_A_BMU_PWR_SAVE_DROP_CTRL

Bits	Name	Description
8	PWR_SAVE_DROP_ENABLE	When set, and the number of frames in a queue for a STA is larger than this 'drop_threshold' number, the frame will be dropped.
7:0	PWR_SAVE_DROP_THRES HOLD	When set, an interrupt will be generated when an STA has indicated a power state change

0x0A0801E8 WCSS_A_BMU_PWR_SAVE_DROP_INT_CONTROL**Type:** RW**Reset State:** 0x00000000

Provides interrupt control and status for the power save frame DROP interrupt generations (For when BTQM starts dropping frames for an STA in power save mode because the max threshold of frames is reached.)

WCSS_A_BMU_PWR_SAVE_DROP_INT_CONTROL

Bits	Name	Type	Description
31	DROP_INT_AUTO_CLEAR_ENABLE	RW	
30	DROP_INT_OVERFLOW	RW	
29	DROP_ENTRY1_VALID	RW	
28	DROP_ENTRY0_VALID	RW	
25:21	DROP_QUEUEID_PWR_INT1	R	
20:16	DROP_QUEUEID_PWR_INT0	R	
15:8	DROP_STAID_PWR_INT1	R	
7:0	DROP_STAID_PWR_INT0	R	

0x0A0801EC WCSS_A_BMU_BMU_BACKUP_CONTROL**Type:** RW**Reset State:** 0x00000000

This register controls backup functionality build into the BMU

WCSS_A_BMU_BMU_BACKUP_CONTROL

Bits	Name	Description
4	SW_TRIGGERED_REARBITRAION_ENABLE	When set, BTQM arbiter to force a re-arbitration for the STAs It removes the current selected STAs from all the prefetched queues, and then re-start the arbitration Once rearbitration is done this bit will be cleared
3	CFG_CLEAR_DA_ON_FINDING_EMPTY_QUEUE_ENABLE	When set, and an empty queue is prefetched, the dataavailable signal corresponding to that STA/queueid will be cleared
2	STA_DA_AVAIL_BACKUP_MODE_ENABLED	When set, on every frame pushed into a queue, all sta daavailable settings will be set
1	ARBITRATION_BACKUP_MODE_ENABLED	When set, on every frame pushed into a queue, all queue daavailable and min/max values will be set again
0	RESERVED	Do NOT set

0x0A0801F0 WCSS_A_BMU_BMU_QUEUE_DELAY_ACCESS_CONTROL**Type:** RW**Reset State:** 0x0A0A0000

This register to control the access of Queue/Transmit delay parameters

WCSS_A_BMU_BMU_QUEUE_DELAY_ACCESS_CONTROL

Bits	Name	Description
28:24	QUEUE_DELAY_RANGE_FIELD_VALUE	Queue delay Bin 0 Range field value for first bin
20:16	TRANSMIT_DELAY_RANGE_FIELD_VALUE	Transmit delay Bin 0 Range field value for first bin
8	READ_MODIFY_WRITE_ZERO	Control to access Queue/Transmit Delay parameters 0 : Data read only 1 : After reading the value, data is initialized to zero
7	QUEUE_TRANSMIT_DELAY	Control to access Queue/Transmit Delay parameters 0 : Queue Delay 1 : Transmit Delay
6:3	TID_INDEX	TID access control
2:0	BIN_INDEX	Bin access control 000 : Bin-0 001 : Bin-1 . . 101 : Bin-5

0x0A0801F4 WCSS_A_BMU_QUEUE_TRANSMIT_DELAY_VALUE**Type:** R**Reset State:** 0x00000000

Queue/Transmit delay value, and this register access is controlled through "bmu_queue_delay_access_control" register

WCSS_A_BMU_QUEUE_TRANSMIT_DELAY_VALUE

Bits	Name	Description
31:0	QUEUE_TRANSMIT_DELAY_VALUE	

0x0A0801F8 WCSS_A_BMU_DXE_RX_WQ_INDEX

Type: RW

Reset State: 0x0000040B

DXE rx WQ index where the received packed are being pushed. BMU keeps track of byte cnt for this WQ for SIF-DXE interface

WCSS_A_BMU_DXE_RX_WQ_INDEX

Bits	Name	Description
15:8	DXE_RX_PRIORITY_WQ_INDEX	DXE high WQ index for high priority traffic
7:0	DXE_RX_WQ_INDEX	DXE WQ index for normal traffic

0x0A0801FC WCSS_A_BMU_BMU_SIF_DATA_LENGTH

Type: R

Reset State: 0x00000000

Number bytes pushed to DXE Rx WQ

WCSS_A_BMU_BMU_SIF_DATA_LENGTH

Bits	Name	Description
16	DXE_WQ_DATA_UNDERFLOW	When this bit is set, SIF transfer byte count is more than DXE WQ data
15:0	BMU_SIF_DATA_LENGTH	Data accumulated in DXE WQ

0x0A080200 WCSS_A_BMU_SW_BMU_DXE_DATA_LENGTH

Type: RW

Reset State: 0x00000000

Number bytes pushed to DXE Rx WQ

WCSS_A_BMU_SW_BMU_DXE_DATA_LENGTH

Bits	Name	Description
15:0	SW_BMU_DXE_DATA_LENGTH	Data length to be added to DXE Rx WQ count

0x0A080208 WCSS_A_BMU_BTQM_PACKET_MEMORY_UNIT_SIZE**Type:** RW**Reset State:** 0x00000001

BTQM reports consumed memory in btqm_packet_memory_unit_size*128-bytes

WCSS_A_BMU_BTQM_PACKET_MEMORY_UNIT_SIZE

Bits	Name	Description
3:0	BTQM_PACKET_MEMORY_UNIT_SIZE	

0x0A080210 WCSS_A_BMU_BTQM_TX_THRESHOLD**Type:** RW**Reset State:** 0x00000080

Controls the memory resources for Tx traffic If the Tx memory usage is greater than equal to this value Any request for BD/PDUs from reserve all command is ignored

WCSS_A_BMU_BTQM_TX_THRESHOLD

Bits	Name	Description
10:0	BTQM_MEM_THRESHOLD	

0x0A080214 WCSS_A_BMU_BTQM_STAID_3_TO_0_MEM_STATUS**Type:** R**Reset State:** 0x00000000

BTQM memory status for STAid3 to STAid0. It is multiples of 128bytes. This needs to be multiplied by setting in "btqm_packet_memory_unit_size" register.

WCSS_A_BMU_BTQM_STAID_3_TO_0_MEM_STATUS

Bits	Name	Description
31:24	STAID3	
23:16	STAID2	
15:8	STAID1	
7:0	STAID0	

0x0A080218 WCSS_A_BMU_BTQM_STAID_7_TO_4_MEM_STATUS**Type:** R**Reset State:** 0x00000000

BTQM memory status for STAid7 to STAid4. It is multiples of 128bytes. This needs to be multiplied by setting in "btqm_packet_memory_unit_size" register.

WCSS_A_BMU_BTQM_STAID_7_TO_4_MEM_STATUS

Bits	Name	Description
31:24	STAID7	
23:16	STAID6	
15:8	STAID5	
7:0	STAID4	

0x0A08021C WCSS_A_BMU_BTQM_PKT_MEMORY_STATUS**Type:** R**Reset State:** 0x00000000

BTQM combined memory status for all the stations.

WCSS_A_BMU_BTQM_PKT_MEMORY_STATUS

Bits	Name	Description
10:0	BTQM_PKT_MEMORY_STATUS	

0x0A080220 WCSS_A_BMU_HP_RX_MEMORY_UPPER_LIMIT**Type:** RW**Reset State:** 0x000000E8

Controls the memory resources for Rx high priority RX traffic. If the Rx memory usage is greater than equal to this value Any request for BD/PDUs from Rx modules is ignored

WCSS_A_BMU_HP_RX_MEMORY_UPPER_LIMIT

Bits	Name	Description
10:0	HP_RX_MEMORY_UPPER_LIMIT	

0x0A080224 WCSS_A_BMU_LP_RX_MEMORY_UPPER_LIMIT**Type:** RW**Reset State:** 0x000000D1

Controls the memory resources for Rx high priority RX traffic. If the Rx memory usage is greater than equal to this value Any request for BD/PDUs from Rx modules is ignored

WCSS_A_BMU_LP_RX_MEMORY_UPPER_LIMIT

Bits	Name	Description
10:0	LP_RX_MEMORY_UPPER_LIMIT	

0x0A080228 WCSS_A_BMU_BTQM_STAID_1_TO_0_MEM_THRESHOLD**Type:** RW**Reset State:** 0x00000000

BTQM memory threshold for STAid1 to STAid0. It is multiples of 128bytes. BMU generates an interrupt when STAx memory usage is less than this threshold value

WCSS_A_BMU_BTQM_STAID_1_TO_0_MEM_THRESHOLD

Bits	Name	Description
26:16	STAID1	
10:0	STAID0	

0x0A08022C WCSS_A_BMU_BTQM_STAID_3_TO_2_MEM_THRESHOLD**Type:** RW**Reset State:** 0x00000000

BTQM memory threshold for STAid3 to STAid2. It is multiples of 128bytes. BMU generates an interrupt when STAx memory usage is less than this threshold value

WCSS_A_BMU_BTQM_STAID_3_TO_2_MEM_THRESHOLD

Bits	Name	Description
26:16	STAID3	
10:0	STAID2	

0x0A080230 WCSS_A_BMU_BTQM_STAID_5_TO_4_MEM_THRESHOLD**Type:** RW**Reset State:** 0x00000000

BTQM memory threshold for STAid5 to STAid4. It is multiples of 128bytes. BMU generates an interrupt when STAx memory usage is less than this threshold value

WCSS_A_BMU_BTQM_STAID_5_TO_4_MEM_THRESHOLD

Bits	Name	Description
26:16	STAID5	
10:0	STAID4	

0x0A080234 WCSS_A_BMU_BTQM_STAID_7_TO_6_MEM_THRESHOLD**Type:** RW**Reset State:** 0x00000000

BTQM memory threshold for STAid7 to STAid6. It is multiples of 128bytes. BMU generates an interrupt when STAx memory usage is less than this threshold value

WCSS_A_BMU_BTQM_STAID_7_TO_6_MEM_THRESHOLD

Bits	Name	Description
26:16	STAID7	
10:0	STAID6	

0x0A080238 WCSS_A_BMU_BTQM_STAID_15_TO_0_MEM_USAGE_STATUS**Type:** RW**Reset State:** 0x00000000

BTQM memory status for STAid15 to STAid0. When the memory usage for STAidX is less than the threshold value the bit will be set

WCSS_A_BMU_BTQM_STAID_15_TO_0_MEM_USAGE_STATUS

Bits	Name	Type	Description
31	BMU_MCU_QUEUE_USAGE_THRESHOLD_INTERRUPT_STATUS	RW	Writing one to this register bit will clear the interrupt
30	BMU_MCU_QUEUE_USAGE_THRESHOLD_INTERRUPT_ENABLE	RW	When this bit is set, bmu_mcu_queue_usage_threshold_interrupt is enabled
15:0	STAID	R	

0x0A08023C WCSS_A_BMU_POLL_STATUS_TIMER_THRESHOLD**Type:** RW**Reset State:** 0x0000FFFF

BTQM poll status threshold, this value represents no.of clock cycles BTQM generates an interrupt when the delay exceeds this value

WCSS_A_BMU_POLL_STATUS_TIMER_THRESHOLD

Bits	Name	Description
15:0	POLL_STATUS_TIMER_THRESHOLD	

0x0A080240 WCSS_A_BMU_BMU_IDLE_BD_PDU_STATUS**Type:** RW**Reset State:** 0x00000000

BMU keeps track of no.of idle bd/pdus in the system. When this idle bd/pdu number reaches the programmed threshold value it generates an interrupt

WCSS_A_BMU_BMU_IDLE_BD_PDU_STATUS

Bits	Name	Description
31	BMU_IDLE_BD_PDU_THRESHOLD_INTERRUPT_STATUS	Writing one to this register bit will clear the interrupt
15	BMU_IDLE_BD_PDU_THRESHOLD_INTERRUPT_ENABLE	When this bit is set, bmu_idle_bd_pdu_threshold_interrupt is enabled
10:0	BMU_IDLE_BD_PDU_THRESHOLD	

0x0A080244 WCSS_A_BMU_BTQM_STAID_11_TO_8_MEM_STATUS**Type:** R**Reset State:** 0x00000000

BTQM memory status for STAid11 to STAid8. It is multiples of 128bytes. This needs to be multiplied by setting in "btqm_packet_memory_unit_size" register.

WCSS_A_BMU_BTQM_STAID_11_TO_8_MEM_STATUS

Bits	Name	Description
31:24	STAID11	
23:16	STAID10	
15:8	STAID9	
7:0	STAID8	

0x0A080248 WCSS_A_BMU_BTQM_STAID_15_TO_12_MEM_STATUS**Type:** R**Reset State:** 0x00000000

BTQM memory status for STAid15 to STAid12. It is multiples of 128bytes. This needs to be multiplied by setting in "btqm_packet_memory_unit_size" register.

WCSS_A_BMU_BTQM_STAID_15_TO_12_MEM_STATUS

Bits	Name	Description
31:24	STAID15	
23:16	STAID14	
15:8	STAID13	

WCSS_A_BMU_BTQM_STAID_15_TO_12_MEM_STATUS (cont.)

Bits	Name	Description
7:0	STAID12	

0x0A08024C WCSS_A_BMU_BTQM_STAID_9_TO_8_MEM_THRESHOLD**Type:** RW**Reset State:** 0x00000000

BTQM memory threshold for STAid9 to STAid8. It is multiples of 128bytes. BMU generates an interrupt when STAx memory usage is less than this threshold value

WCSS_A_BMU_BTQM_STAID_9_TO_8_MEM_THRESHOLD

Bits	Name	Description
26:16	STAID9	
10:0	STAID8	

0x0A080250 WCSS_A_BMU_BTQM_STAID_11_TO_10_MEM_THRESHOLD**Type:** RW**Reset State:** 0x00000000

BTQM memory threshold for STAid11 to STAid10. It is multiples of 128bytes. BMU generates an interrupt when STAx memory usage is less than this threshold value

WCSS_A_BMU_BTQM_STAID_11_TO_10_MEM_THRESHOLD

Bits	Name	Description
26:16	STAID11	
10:0	STAID10	

0x0A080254 WCSS_A_BMU_BTQM_STAID_13_TO_12_MEM_THRESHOLD**Type:** RW**Reset State:** 0x00000000

BTQM memory threshold for STAid13 to STAid12. It is multiples of 128bytes. BMU generates an interrupt when STAx memory usage is less than this threshold value

WCSS_A_BMU_BTQM_STAID_13_TO_12_MEM_THRESHOLD

Bits	Name	Description
26:16	STAID13	
10:0	STAID12	

0x0A080258 WCSS_A_BMU_BTQM_STAID_15_TO_14_MEM_THRESHOLD**Type:** RW**Reset State:** 0x00000000

BTQM memory threshold for STAid15 to STAid14. It is multiples of 128bytes. BMU generates an interrupt when STAx memory usage is less than this threshold value

WCSS_A_BMU_BTQM_STAID_15_TO_14_MEM_THRESHOLD

Bits	Name	Description
26:16	STAID15	
10:0	STAID14	

0x0A08025C WCSS_A_BMU_BMU_PKT_MEMORY_CONFIG_MODE**Type:** RW**Reset State:** 0x00000000

To configure each module operational mode i.e to use local/external memory as packet memory to get BD/PDUs 00 : EXternal Memory Only -> Defined master gets BD/PDUs from External (DDR) memory only 01 : Local Memory Only -> Defined master gets BD/PDUs from Local Memory(cMem) only 10 : Local/Ext -> Defined master gets from cMem and whenavailable BD/PDUs falls below the threshold level BMU gives out BD/PDUs for DDR memory 11 : Reserved

WCSS_A_BMU_BMU_PKT_MEMORY_CONFIG_MODE

Bits	Name	Description
21:20	MASTER10	To configure Master10 module operational mode
19:18	MASTER9	To configure Master9 module operational mode
17:16	MASTER8	To configure Master8 module operational mode
15:14	MASTER7	To configure Master7 module operational mode
13:12	MASTER6	To configure Master6 module operational mode
11:10	MASTER5	To configure Master5 module operational mode

WCSS_A_BMU_BMU_PKT_MEMORY_CONFIG_MODE (cont.)

Bits	Name	Description
9:8	MASTER4	To configure Master4 module operational mode
7:6	MASTER3	To configure Master3 module operational mode
5:4	MASTER2	To configure Master2 module operational mode
3:2	MASTER1	To configure Master1 module operational mode
1:0	MASTER0	To configure Master0 module operational mode

0x0A080260 WCSS_A_BMU_AVAILABLE_BD_PDU_AFTER_RSV_IN_LOCAL_MEM**Type:** R**Reset State:** 0x00000000

This register can be used to find out how many idle BD/PDUs are actually available (in Local memory) from the BD and PDU idle list when the number of reserved BDs and PDUs for all the different modules have been taken into account.

WCSS_A_BMU_AVAILABLE_BD_PDU_AFTER_RSV_IN_LOCAL_MEM

Bits	Name	Description
15:0	AVAILABLE_BD_PDUS_IN_LOCAL_MEMORY	

0x0A080264 WCSS_A_BMU_BLOCK_DPU_RX_THRESHOLD**Type:** RW**Reset State:** 0x00000000

If at certain moment no flat memory space is available anymore, DXE RX will stop pulling out frames from cMEM. cMEM can now fill up with frame pulled by DPU RX out of DDR, to prevent Tx block, this threshold is used block the data available signal to DPU_Rx.

WCSS_A_BMU_BLOCK_DPU_RX_THRESHOLD

Bits	Name	Description
10:0	BLOCK_DPU_RX_THRESHOLD	

0x0A080268 WCSS_A_BMU_ADU_RX_LOW_THRESHOLD**Type:** RW**Reset State:** 0x00000000

ADU RX can consume cMEM BDs in case of A-MSDU de-aggregation, ADU cMEM threshold should have the smallest value in Receive path BMU will switch to this threshold value when there is a block in Rx-path

WCSS_A_BMU_ADU_RX_LOW_THRESHOLD

Bits	Name	Description
10:0	ADU_RX_LOW_THRESHOLD	

0x0A08026C WCSS_A_BMU_DXE_TX_LOW_THRESHOLD**Type:** RW**Reset State:** 0x00000000

when DXE RX WQ is not empty (and RX path might be stalling), increase the memory amount available for TX data (=> Decrease threshold) When RX is stalled BMU will switch to this threshold

WCSS_A_BMU_DXE_TX_LOW_THRESHOLD

Bits	Name	Description
10:0	DXE_TX_LOW_THRESHOLD	

0x0A080270 WCSS_A_BMU_DXE_RX_DA_SELECTION**Type:** RW**Reset State:** 0x00000810

Data available signal that goes all the way from BMU to DXE to prevent Rx blocking, These are the bits to configure assigned DXE WQs. By default WQ-4 & WQ-11 assumed as DXE Rx WQs

WCSS_A_BMU_DXE_RX_DA_SELECTION

Bits	Name	Description
24:2	DXE_RX_DA_SELECTION	

10.30 WCSS_A_BTCF_BTCF

0x0A014800 WCSS_A_BTCF_BTCF_CONFIG

Type: RW

Reset State: 0x0EDD

Programs the offsets that will be added to the computed values

WCSS_A_BTCF_BTCF_CONFIG

Bits	Name	Description
11:8	FFT_TOFFSET256	signed 2's complement number which is used to adjust the timing offset for the 256-point FFT start time
7:4	FFT_TOFFSET128	signed 2's complement number which is used to adjust the timing offset for the 128-point FFT start time
3:0	FFT_TOFFSET64	signed 2's complement number which is used to adjust the timing offset for the 64-point FFT start time

0x0A014804 WCSS_A_BTCF_BTCF_DET_RATIO

Type: RW

Reset State: 0x0005

Detection thresholds for OFDM packets.

WCSS_A_BTCF_BTCF_DET_RATIO

Bits	Name	Description
2:0	VALUE	The alpha value used to compute the pkt_valid flag. unsigned number. A combination of pkt_det_ratio and pkt_det_dshift determine the 11a detection threshold.

0x0A014808 WCSS_A_BTCF_BTCF_DET_DSHIFT

Type: RW

Reset State: 0x0001

Detection shift values for OFDM packets.

WCSS_A_BTCF_BTCF_DET_DSHIFT

Bits	Name	Description
2:0	VALUE	The shift value used to compute the pkt_valid flag. unsigned number

0x0A01480C WCSS_A_BTCF_BTCFMIN**Type:** R**Reset State:** 0x00000000

Stores the last computed btcf_min value

WCSS_A_BTCF_BTCFMIN

Bits	Name	Description
14:0	BTCF	The last latched btcf value when newmin was asserted. This is denoted as Bmin(t) in the uarch.

0x0A014810 WCSS_A_BTCF_PWRMIN**Type:** R**Reset State:** 0x00000000

Stores the last computed pow(t) value corresponding to btcf_min

WCSS_A_BTCF_PWRMIN

Bits	Name	Description
15:0	POW	The last latched pow(t) value when newmin was asserted

0x0A014814 WCSS_A_BTCF_OFFSETMIN**Type:** R**Reset State:** 0x00000000

Stores the last computed offset counter value corresponding to btcf_min

WCSS_A_BTCF_OFFSETMIN

Bits	Name	Description
7:0	COUNT	The last latched offset count when newmin was asserted. This provides the count value from when the first valid btcf value was computed. This value added with fft_toffset gives the deltata to the TDC module

10.31 WCSS_A_CAHB_CAHB**0x0A082C00 WCSS_A_CAHB_CAHB_RXP_PL****Type:** RW**Reset State:** 0x00000005

cahb Arbiter arbitration priority level for rxp master. In the cahb arbitration scheme the master with highest priority level always wins. Should there be more than one master with the same priority then the winning master will be selected using a round-robin algorithm.

WCSS_A_CAHB_CAHB_RXP_PL

Bits	Name	Description
3:0	PRIORITY	Arbitration priority for rxp master. The lowest priority level is 1 and the highest priority level is 15. A priority level of 0 has a special significance, it means that the master is disabled and stops participating in the arbitration scheme. Note that the cahb arbiter will prevent a master from disabling itself.

0x0A082C04 WCSS_A_CAHB_CAHB_TPE_PL**Type:** RW**Reset State:** 0x00000005

cahb Arbiter arbitration priority level for tpe master. In the cahb arbitration scheme the master with highest priority level always wins. Should there be more than one master with the same priority then the winning master will be selected using a round-robin algorithm.

WCSS_A_CAHB_CAHB_TPE_PL

Bits	Name	Description
3:0	PRIORITY	Arbitration priority for tpe master. The lowest priority level is 1 and the highest priority level is 15. A priority level of 0 has a special significance, it means that the master is disabled and stops participating in the arbitration scheme. Note that the cahb arbiter will prevent a master from disabling itself.

0x0A082C08 WCSS_A_CAHB_CAHB_RPE_PL**Type:** RW**Reset State:** 0x00000005

cahb Arbiter arbitration priority level for rpe master. In the cahb arbitration scheme the master with highest priority level always wins. Should there be more than one master with the same priority then the winning master will be selected using a round-robin algorithm.

WCSS_A_CAHB_CAHB_RPE_PL

Bits	Name	Description
3:0	PRIORITY	Arbitration priority for rpe master. The lowest priority level is 1 and the highest priority level is 15. A priority level of 0 has a special significance, it means that the master is disabled and stops participating in the arbitration scheme. Note that the cahb arbiter will prevent a master from disabling itself.

0x0A082C0C WCSS_A_CAHB_CAHB_RPEM_PL**Type:** RW**Reset State:** 0x00000005

cahb Arbiter arbitration priority level for rpem master. In the cahb arbitration scheme the master with highest priority level always wins. Should there be more than one master with the same priority then the winning master will be selected using a round-robin algorithm.

WCSS_A_CAHB_CAHB_RPEM_PL

Bits	Name	Description
3:0	PRIORITY	Arbitration priority for rpem master. The lowest priority level is 1 and the highest priority level is 15. A priority level of 0 has a special significance, it means that the master is disabled and stops participating in the arbitration scheme. Note that the cahb arbiter will prevent a master from disabling itself.

0x0A082C10 WCSS_A_CAHB_CAHB_DPU_PL**Type:** RW**Reset State:** 0x00000001

cahb Arbiter arbitration priority level for dpu master. In the cahb arbitration scheme the master with highest priority level always wins. Should there be more than one master with the same priority then the winning master will be selected using a round-robin algorithm.

WCSS_A_CAHB_CAHB_DPU_PL

Bits	Name	Description
3:0	PRIORITY	Arbitration priority for dpu master. The lowest priority level is 1 and the highest priority level is 15. A priority level of 0 has a special significance, it means that the master is disabled and stops participating in the arbitration scheme. Note that the cahb arbiter will prevent a master from disabling itself.

0x0A082C14 WCSS_A_CAHB_CAHB_ADU_PL**Type:** RW**Reset State:** 0x00000001

cahb Arbiter arbitration priority level for adu master. In the cahb arbitration scheme the master with highest priority level always wins. Should there be more than one master with the same priority then the winning master will be selected using a round-robin algorithm.

WCSS_A_CAHB_CAHB_ADU_PL

Bits	Name	Description
3:0	PRIORITY	Arbitration priority for adu master. The lowest priority level is 1 and the highest priority level is 15. A priority level of 0 has a special significance, it means that the master is disabled and stops participating in the arbitration scheme. Note that the cahb arbiter will prevent a master from disabling itself.

0x0A082C18 WCSS_A_CAHB_CAHB_CBR_PL**Type:** RW**Reset State:** 0x00000004

cahb Arbiter arbitration priority level for cbr master. In the cahb arbitration scheme the master with highest priority level always wins. Should there be more than one master with the same priority then the winning master will be selected using a round-robin algorithm.

WCSS_A_CAHB_CAHB_CBR_PL

Bits	Name	Description
3:0	PRIORITY	Arbitration priority for cbr master. The lowest priority level is 1 and the highest priority level is 15. A priority level of 0 has a special significance, it means that the master is disabled and stops participating in the arbitration scheme. Note that the cahb arbiter will prevent a master from disabling itself.

0x0A082C1C WCSS_A_CAHB_CAHB_CSU_PL**Type:** RW**Reset State:** 0x00000001

cahb Arbiter arbitration priority level for csu master. In the cahb arbitration scheme the master with highest priority level always wins. Should there be more than one master with the same priority then the winning master will be selected using a round-robin algorithm.

WCSS_A_CAHB_CAHB_CSU_PL

Bits	Name	Description
3:0	PRIORITY	Arbitration priority for csu master. The lowest priority level is 1 and the highest priority level is 15. A priority level of 0 has a special significance, it means that the master is disabled and stops participating in the arbitration scheme. Note that the cahb arbiter will prevent a master from disabling itself.

0x0A082C48 WCSS_A_CAHB_CAHB_DFLT_MST**Type:** RW**Reset State:** 0x00000000

This register identifies the default master for the cahb

WCSS_A_CAHB_CAHB_DFLT_MST

Bits	Name	Description
3:0	DFT_MST	Default Master ID number register. The default master is the master that is granted by the bus when no master has requested ownership. NOTE: Writing anything beyond the 4 R/W bits will cause the entire write to be discarded. 0x0: DMY (Grant ownership to cahb dummy master) 0x1: RXP (Grant ownership to cahb rxp master) 0x2: TPE (Grant ownership to cahb tpe master) 0x3: RPE (Grant ownership to cahb rpe master) 0x4: RPEM (Grant ownership to cahb rpem master) 0x5: DPU (Grant ownership to cahb dpu master) 0x6: ADU (Grant ownership to cahb adu master) 0x7: CBR (Grant ownership to cahb cbr master) 0x8: CSU (Grant ownership to cahb csu master)

10.32 WCSS_A_CAL_CAL

0x0A008000 WCSS_A_CAL_CALMODE

Type: RW

Reset State: 0x00000000

Determines the calibration mode and generates calibration enables during these modes. The Enable_Measure bit in the MSB of the register is used to enable the measurement of the value. For instance, in DC offset initial calibration mode, the CPU writes to the DC offset DAC in the RF ASIC through the SPI interface. The CPU then has to command the DC offset calibration module in the PHY to measure the DC offset (at the ADC output) after the new DAC output (in the RF ASIC) can be observed at the ADC output (typically after 0.5 us after the DAC is written to in the RF ASIC). Hence, once the CPU writes to the SPI interface, it then waits for a specific period of time (1 us) after the SPI write completes, and then asserts this bit to command the DC offset module to start the measurement. Once the measurement is done (as indicated by the CAL_STATUS) register, this bit has to be disabled so as to setup for the next measurement.

WCSS_A_CAL_CALMODE

Bits	Name	Description
2:0	MODE	Sets the calibration mode value. 3'b101 to 3'b111 are unused. 0x0: NORMAL (Normal Mode, No calibration being done) 0x1: INITDCCAL (DC Offset Initial Calibration Mode. In this mode we correct for the DC offset in all of the receive chains (3x2) by writing to the DC offset correction DAC in the respective RF ICs. This offset measurement is done serially (one receive chain at a time).) 0x2: RESDCCAL (Digital/Residual DC Offset Calibration Mode. In this mode we measure the DC offset (to account for any drift in DC offset due to temperature and/or gain settings) and correct for it digitally at the output of the ADCs for all the receive chains (3x2). This offset measurement is done in parallel.) 0x3: IQCAL (I/Q Amplitude Phase Imbalance Calibration Mode. In this mode we measure the amplitude and phase imbalance between the I and Q for each receive chain) 0x4: TXLOCAL (TX LO Leakage Measurement mode. In this mode we measure the transmit chain local oscillator leakage by connecting the transmit path to the receive path and measuring the DC offset introduced by the transmit path at the output of the ADCs in the receive chain. The transmit DACs are then adjusted to minimize this measured DC offset)

0x0A008004 WCSS_A_CAL_MEASURE

Type: W

Reset State: 0x0000

By writing to this register, in calibration modes other than normal mode, a measure pulse is pulsed to the calibration module. Originally, this was a level signal called Enable_Measure. With the wr_Pulse register type being supported, it was decided to add this, so that software does not have to disable and then enable for the cal module to detect a pulse

WCSS_A_CAL_MEASURE

Bits	Name	Description
0	CMD	Measure write pulse bit which enables the measurement for the specified calibration mode in the cal_mode register 0x0: NORMAL (Idle) 0x1: MEASURE (Measure pulse to calibration controller)

0x0A008008 WCSS_A_CAL_CHAIN_SEL

Type: RW

Reset State: 0x00000000

Determines which transmit/receive channel is being calibrated. If Bit 2 is 0, then the I channel of the transmit/receive chain is selected. If Bit 2 is 1, then the Q chain of the transmit/receive chain is selected. CS1 and CS0 determine which of the receive chains 0,1,2 or the transmit chains 0,1 are being calibrated as shown in table below.

WCSS_A_CAL_CHAIN_SEL

Bits	Name	Description
2	IQSEL	Selects I/Q inputs for the inputs specified in chain 0x0: ISEL (I input Select) 0x1: QSEL (Q input Select)
1:0	CHAIN	Selects receive chain output 0x0: CHAIN0 (Chain 0 Select) 0x1: CHAIN1 (Chain 1 Select) 0x2: CHAIN2 (Chain 2 Select) 0x3: CHAIN3 (Chain 3 Select)

0x0A00800C WCSS_A_CAL_LENGTH_INIT

Type: RW

Reset State: 0x03FF

The CAL_LENGTH registers are used to determine what length should the calibration module average over. If it is desirable to average over N samples, the CPU should write N-1 to this register where N is always an integer number that represents some power of 2.

WCSS_A_CAL_LENGTH_INIT

Bits	Name	Description
11:0	LENGTH	

0x0A008010 WCSS_A_CAL_LENGTH_RSA**Type:** RW**Reset State:** 0x0001

The CAL_LENGTH registers are used to determine what length should the calibration module average over. If it is desirable to average over N samples, the CPU should write N-1 to this register where N is always an integer number that represents some power of 2.

WCSS_A_CAL_LENGTH_RSA

Bits	Name	Description
11:0	LENGTH	

0x0A008014 WCSS_A_CAL_LENGTH_RSB**Type:** RW**Reset State:** 0x00FF

The CAL_LENGTH registers are used to determine what length should the calibration module average over. If it is desirable to average over N samples, the CPU should write N-1 to this register where N is always an integer number that represents some power of 2.

WCSS_A_CAL_LENGTH_RSB

Bits	Name	Description
11:0	LENGTH	

0x0A008018 WCSS_A_CAL_PIE_FFT**Type:** RW**Reset State:** 0x0000

Size of FFT for IQ phase imbalance measurement.

WCSS_A_CAL_PIE_FFT

Bits	Name	Description
1:0	SIZE	0x0: PTS64 (FFT size = 64 points) 0x1: PTS128 (FFT size = 128 points) 0x2: PTS256 (FFT size = 256 points)

0x0A00801C WCSS_A_CAL_DCO_THRESHOLD**Type:** RW**Reset State:** 0x004C

Threshold value for the digital DC offset measured on a per-packet basis. If the measured digital DC offset is more than this threshold, the DC offset module sets the CAL_ATTENTION register to request CPU to recalibrate. Default is 150 mV (= 76 LSBs of a 10-bit ADC with a -1V to +1V swing) = 0x4C

WCSS_A_CAL_DCO_THRESHOLD

Bits	Name	Description
10:0	VALUE	

0x0A008020 WCSS_A_CAL_DCTIMER1**Type:** RW**Reset State:** 0x2C24

When the DC offset module is commanded to measure the digital DC offset either by the CPU or the AGC module (the AGC module does so on every packet), the DC offset module terminates all the RFICs inputs and waits for the time (number for 80MHz clocks) specified in the CAL_DC_TIMER1 register. This allows for the terminated input to be observed at the ADC outputs. This time is typically in the order of 0.5 us = 20 40MHz clocks = 40 80 MHz clocks = 0x28 - 1 POR. Note that this number should always be an odd number (to be a multiple of 40 MHz clocks)

WCSS_A_CAL_DCTIMER1

Bits	Name	Description
15:8	VALUE_RSB	
7:0	VALUE_RSA	

0x0A008024 WCSS_A_CAL_DCTIMER2**Type:** RW**Reset State:** 0x2000

When the DC offset module is commanded to measure the digital DC offset either by the CPU or the AGC module (the AGC module does so on every packet), the DC offset module terminates all the RFICs inputs and waits for the time (number for 80MHz clocks) specified in the CAL_DC_TIMER1 register. This allows for the terminated input to be observed at the ADC outputs. This time is typically in the order of $0.5 \mu s = 20 \text{ 40MHz} = 40 \text{ 80 MHz clocks} = 0x28-1$. Now when the DC offset measurement is done, the DC offset module switches the RF frontend back in. After switching the RF FE in, the DC offset module waits for CAL_DC_TIMER2 time before it can assert a data valid signal to the rest of the blocks. The reason there are two different timer registers for turning on and turning off the RF FE is so as to support any asymmetry in turning on/turning off the RF FE. If this saves clocks, then it allows data to be sampled earlier which may be particularly useful in per-packet calibration. Presently, the default value is 0.5us.

WCSS_A_CAL_DCTIMER2

Bits	Name	Description
15:8	VALUE_RSB	
7:0	VALUE_RSA	

0x0A008028 WCSS_A_CAL_DCOFFSET**Type:** R**Reset State:** 0x0000

Measured DC offset, lower order bits. This register gives the measured DC offset (the output of the ADC of the selected chain) when the calibration mode is in DC Offset Initial Calibration and/or Digital DC offset calibration mode. This is a 11-bit 2's complement value. The CPU reads this value when it reads the CAL_STATUS to be done. The value read corresponds to the value in CHAIN_SEL.

WCSS_A_CAL_DCOFFSET

Bits	Name	Description
10:0	DCOFFSET	

0x0A00802C WCSS_A_CAL_PWRDCOFFSET**Type:** R**Reset State:** 0x0000

Measured power of DC offset, lower order bits. This register gives the measured power of the input signal (the output of the ADC of the selected chain) when the calibration mode is in DC Offset Initial Calibration and/or Digital DC offset calibration mode. This is a 11-bit 2's complement value. The CPU reads this value when it reads the CAL_STATUS to be done. The value read corresponds to the value in CHAIN_SEL.

WCSS_A_CAL_PWRDCOFFSET

Bits	Name	Description
15:0	POWER	

0x0A008030 WCSS_A_CAL_CLR_DCOFFSET

Type: W

Reset State: 0x0000

Usually the last measured DC offset is held for the next packet as well until a new dco measurement is commanded by the AGC. Since, DC offset can change with time, when the AGC comes out of reset (like after beacon power save modes), software may decide to clear the DC offset register.

WCSS_A_CAL_CLR_DCOFFSET

Bits	Name	Description
0	CMD	Clear write pulse bit which clears the latched DC offset values 0x0: NORMAL (Idle) 0x1: MEASURE (Clear command pulse to DC Offset module)

0x0A008034 WCSS_A_CAL_CHN0_DCO_OVR

Type: RW

Reset State: 0x0000

Used to override the measured DC offset for chain 0

WCSS_A_CAL_CHN0_DCO_OVR

Bits	Name	Description
15:8	Q0	
7:0	I0	

0x0A008038 WCSS_A_CAL_STATUS**Type:** R**Reset State:** 0x0000

Indicates the calibration status for the current mode

WCSS_A_CAL_STATUS

Bits	Name	Description
15:14	STATUS	Only busy, idle and done are implemented 0x0: IDLE (Idle) 0x1: DONE (Done) 0x2: BUSY (Busy) 0x3: ERR (Error)

0x0A00803C WCSS_A_CAL_OVERRIDE**Type:** RW**Reset State:** 0x0000

Sets the test mode for calibration

WCSS_A_CAL_OVERRIDE

Bits	Name	Description
0	CHN0_DC	DCO bypass bit 0x0: NORMAL (DCO Correction is performed) 0x1: BYPASS (DCO Correction is bypassed)

0x0A008040 WCSS_A_CAL_CORR_ENABLE**Type:** RW**Reset State:** 0x0002

This register provides packet-type specific enables for DCO correction

WCSS_A_CAL_CORR_ENABLE

Bits	Name	Description
1	EN_11B	
0	EN_OFDM	

0x0A008044 WCSS_A_CAL_AIEST**Type:** R**Reset State:** 0x0000

When CAL_MODE is set to I/Q amplitude imbalance, this register gives the amplitude imbalance for that tone (CAL_IQ_TONENUM reg) for the current gain setting in AGC_GAIN_SET register. This is a 10-bit value. The CPU reads this value when it reads the CAL_STATUS to be done. The value read corresponds to the value in CHAIN_SEL.

WCSS_A_CAL_AIEST

Bits	Name	Description
8:0	AIEST	

0x0A008048 WCSS_A_CAL_IQ_CORR_MODE**Type:** RW**Reset State:** 0x0005

This register is used for controlling the corrector modes.

WCSS_A_CAL_IQ_CORR_MODE

Bits	Name	Description
3:1	C0_THROUGH_TAP_SEL	Allows for selecting the specific tap location as the through tap component for the I/Q correctors. Default selection is the centre tap location. Non-listed encodings default to centre tap selection. 0x0: C0_IN_DATA_TAP (Through tap selects input data (no delay)) 0x1: C0_IN_DATA_DLY1 (Through tap selects input data delayed by 1 cycle) 0x2: C0_IN_DATA_DLY2 (Through tap selects input data delayed by 2 cycles (centre tap)) 0x3: C0_IN_DATA_DLY3 (Through tap selects input data delayed by 3 cycles) 0x4: C0_IN_DATA_DLY4 (Through tap selects input data (final tap))
0	BYPASS	When set(1), the iq corr coefficients are set to 0. When 0, iq corr coefficients are read from IQ_CORR_COEFF memory. This register was added to purely save test time and sim issues so the memories don't have to be slammed at the start of the sim

0x0A008050 WCSS_A_CAL_FORCE_TR_SWITCH**Type:** RW**Reset State:** 0x0000

Using this register, the T/R switches may be set to TX mode during DCO calibration. The register provides this function to be enabled in any of the three following categories - coarse (periodic) cal, OFDM packet residual CAL and 11b residual CAL.

WCSS_A_CAL_FORCE_TR_SWITCH

Bits	Name	Description
2	EN_11B	
1	EN_OFDM	
0	EN_COARSE	

0x0A008054 WCSS_A_CAL_AIEST_DISABLE**Type:** RW**Reset State:** 0x0000

Using this register, the IQ amplitude imbalance estimate may be suppressed during IQ phase imbalance calculation.

WCSS_A_CAL_AIEST_DISABLE

Bits	Name	Description
0	FLAG	

0x0A008058 WCSS_A_CAL_BARAM_ACCESS**Type:** RW**Reset State:** 0x0000

This register controls which data stream has access to the BARAM memory. When set, the BARAM is only accessible by the host, not by internal logic.

WCSS_A_CAL_BARAM_ACCESS

Bits	Name	Description
0	APB	

0x0A008800+ WCSS_A_CAL_BARAMn_0, n=[0..255]**0x8*n****Type:** RW**Reset State:** 0x0000

Read write interface to the burst average RAM in the PIE module

WCSS_A_CAL_BARAMn_0

Bits	Name	Description
15:0	IAVG	

0x0A008804+ WCSS_A_CAL_BARAMn_1, n=[0..255]**0x8*n****Type:** RW**Reset State:** 0x0000

Read write interface to the burst average RAM in the PIE module

WCSS_A_CAL_BARAMn_1

Bits	Name	Description
15:0	QAVG	

0x0A009000+ WCSS_A_CAL_IQ_CORR_COEFF_MEMn_0, n=[0..39]**0x20*n****Type:** RW**Reset State:** 0x0000

Contains RX IQ imbalance correction coefficients for each RF gain setting

WCSS_A_CAL_IQ_CORR_COEFF_MEMn_0

Bits	Name	Description
15:9	C0_I1_0	2's complement IQ imbalance correction coefficient for I-rail tap1 on chain 0
8:0	C0_I0	2's complement IQ imbalance correction coefficient for I-rail tap0 on chain 0

0x0A009004+ WCSS_A_CAL_IQ_CORR_COEFF_MEMn_1, n=[0..39]**0x20*n****Type:** RW**Reset State:** 0x0000

Contains RX IQ imbalance correction coefficients for each RF gain setting

WCSS_A_CAL_IQ_CORR_COEFF_MEMn_1

Bits	Name	Description
15:11	C0_I3_0	2's complement IQ imbalance correction coefficient for I-rail tap3 on chain 0
10:2	C0_I2	2's complement IQ imbalance correction coefficient for I-rail tap2 on chain 0
1:0	C0_I1_1	2's complement IQ imbalance correction coefficient for I-rail tap1 on chain 0

0x0A009008+ WCSS_A_CAL_IQ_CORR_COEFF_MEMn_2, n=[0..39] 0x20*n

Type: RW

Reset State: 0x0000

Contains RX IQ imbalance correction coefficients for each RF gain setting

WCSS_A_CAL_IQ_CORR_COEFF_MEMn_2

Bits	Name	Description
15:13	C0_Q0_0	2's complement IQ imbalance correction coefficient for Q-rail tap0 on chain 0
12:4	C0_I4	2's complement IQ imbalance correction coefficient for I-rail tap4 on chain 0
3:0	C0_I3_1	2's complement IQ imbalance correction coefficient for I-rail tap3 on chain 0

0x0A00900C+ WCSS_A_CAL_IQ_CORR_COEFF_MEMn_3, n=[0..39] 0x20*n

Type: RW

Reset State: 0x0000

Contains RX IQ imbalance correction coefficients for each RF gain setting

WCSS_A_CAL_IQ_CORR_COEFF_MEMn_3

Bits	Name	Description
15	C0_Q2_0	2's complement IQ imbalance correction coefficient for Q-rail tap2 on chain 0
14:6	C0_Q1	2's complement IQ imbalance correction coefficient for Q-rail tap1 on chain 0

WCSS_A_CAL_IQ_CORR_COEFF_MEMn_3 (cont.)

Bits	Name	Description
5:0	C0_Q0_1	2's complement IQ imbalance correction coefficient for Q-rail tap0 on chain 0

**0x0A009010+ WCSS_A_CAL_IQ_CORR_COEFF_MEMn_4, n=[0..39]
0x20*n****Type:** RW**Reset State:** 0x0000

Contains RX IQ imbalance correction coefficients for each RF gain setting

WCSS_A_CAL_IQ_CORR_COEFF_MEMn_4

Bits	Name	Description
15:8	C0_Q3_0	2's complement IQ imbalance correction coefficient for Q-rail tap3 on chain 0
7:0	C0_Q2_1	2's complement IQ imbalance correction coefficient for Q-rail tap2 on chain 0

**0x0A009014+ WCSS_A_CAL_IQ_CORR_COEFF_MEMn_5, n=[0..39]
0x20*n****Type:** RW**Reset State:** 0x0000

Contains RX IQ imbalance correction coefficients for each RF gain setting

WCSS_A_CAL_IQ_CORR_COEFF_MEMn_5

Bits	Name	Description
9:1	C0_Q4	2's complement IQ imbalance correction coefficient for Q-rail tap4 on chain 0
0	C0_Q3_1	2's complement IQ imbalance correction coefficient for Q-rail tap3 on chain 0

10.33 WCSS_A_CCU_CCU**0x0A204000 WCSS_A_CCU_CCU_ENABLE_CLK****Type:** RW**Reset State:** 0x00017FFF

This register enables the clock to Common SS modules. If set, the clock is enabled.

WCSS_A_CCU_CCU_ENABLE_CLK

Bits	Name	Description
16	AOWBR_ROOT_CLK_ENABLE	Enable clock for aowbr
15	RESERVED	reserved
14	PMU_TCXO_ENABLE_CLK	Enable clock for pmu_tcxo [UNUSED]
13	LTE_ENABLE_CLK	Enable clock for lte
12	FM_ENABLE_CLK	Enable clock for fm
11	BT_ENABLE_CLK	Enable clock for bt
10	TIC_ENABLE_CLK	Enable clock for tic
9	ROM_ENABLE_CLK	Enable clock for rom
8	SPDM_ENABLE_CLK	Enable clock for spdm
7	CDAHB_ENABLE_CLK	Enable clock for cda hb
6	CCAHB_ENABLE_CLK	Enable clock for cca hb
5	A2XB_ENABLE_CLK	Enable clock for a2xb
4	A2AB_ENABLE_CLK	Enable clock for a2ab
3	RRI_ENABLE_CLK	Enable clock for rri
2	DXE_ENABLE_CLK	Enable clock for dxe
1	CMEM_ENABLE_CLK	Enable clock for cmem
0	CCPU_ENABLE_CLK	Enable clock for ccpu

0x0A204004 WCSS_A_CCU_CCU_DISABLE_CLK_GATING**Type:** RW**Reset State:** 0x0002C5F0

Controls clock gating in the module. If set, clock gating to the module and in the module is disabled.

WCSS_A_CCU_CCU_DISABLE_CLK_GATING

Bits	Name	Description
17	AOWBR_CLKGATE_DISABLE	Disable clock gating for aowbr
16	RESERVED	reserved
15	PMU_TCXO_DISABLE_CLK_GATING	Disable clk gating for pmu_tcxo [UNUSED]

WCSS_A_CCU_CCU_DISABLE_CLK_GATING (cont.)

Bits	Name	Description
14	PMU_DISABLE_CLK_GATING	Disable clk gating for pmu
13	LTE_DISABLE_CLK_GATING	Disable clk gating for lte
12	FM_DISABLE_CLK_GATING	Disable clk gating for fm
11	BT_DISABLE_CLK_GATING	Disable clk gating for bt
10	TIC_DISABLE_CLK_GATING	Disable clk gating for tic
9	AXI_FABRIC_DISABLE_CLK_GATING	Disable clk gating for AXI Fabric clock
8	SPDM_DISABLE_CLK_GATING	Disable clk gating for spdm
7	CDAHB_DISABLE_CLK_GATING	Disable clk gating for cdaHB
6	CCAHB_DISABLE_CLK_GATING	Disable clk gating for ccaHB
5	A2XB_DISABLE_CLK_GATING	Disable clk gating for a2xb
4	A2AB_DISABLE_CLK_GATING	Disable clk gating for a2ab
3	RRI_DISABLE_CLK_GATING	Disable clk gating for rri
2	DXE_DISABLE_CLK_GATING	Disable clk gating for dxe
1	CMEM_DISABLE_CLK_GATING	Disable clk gating for cmem
0	CCPU_DISABLE_CLK_GATING	Disable clk gating for ccpu

0x0A204008 WCSS_A_CCU_CCU_GCU_DISABLE_CLK_GATING**Type:** RW**Reset State:** 0x000040F0

Controls clock gating in the module. If set, clock gating to the module and in the module is disabled.

WCSS_A_CCU_CCU_GCU_DISABLE_CLK_GATING

Bits	Name	Description
14	AOWBR_GCU_DISABLE_CLK_GATING	Disable GCU clk gating for aowbr
13	LTE_GCU_DISABLE_CLK_GATING	Disable GCU clk gating for lte
12	FM_GCU_DISABLE_CLK_GATING	Disable GCU clk gating for fm
11	BT_GCU_DISABLE_CLK_GATING	Disable GCU clk gating for bt
10	TIC_GCU_DISABLE_CLK_GATING	Disable GCU clk gating for tic
9	ROM_GCU_DISABLE_CLK_GATING	Disable GCU clk gating for rom
8	SPDM_GCU_DISABLE_CLK_GATING	Disable GCU clk gating for spdm
7	CDAHB_GCU_DISABLE_CLK_GATING	Disable GCU clk gating for cdaHB
6	CCAHB_GCU_DISABLE_CLK_GATING	Disable GCU clk gating for ccaHB
5	A2XB_GCU_DISABLE_CLK_GATING	Disable GCU clk gating for a2xb
4	A2AB_GCU_DISABLE_CLK_GATING	Disable GCU clk gating for a2ab
3	RRI_GCU_DISABLE_CLK_GATING	Disable GCU clk gating for rri
2	DXE_GCU_DISABLE_CLK_GATING	Disable GCU clk gating for dxe
1	CMEM_GCU_DISABLE_CLK_GATING	Disable GCU clk gating for cmem
0	CCPU_GCU_DISABLE_CLK_GATING	Disable GCU clk gating for ccpu

0x0A20400C WCSS_A_CCU_CCU_CRCM_CONTROL**Type:** RW**Reset State:** 0x00000000

This register represents CRCM control signals

WCSS_A_CCU_CCU_CRCM_CONTROL

Bits	Name	Description
2	WMAC_SYNC_RST_CLK_EN_OVERRIDE	wmac_sync_rst_clk_en_override to CRCM
1	CMEM_MEM_CLK_ENABLE_D	cmem_mem_clk_enabled to CMEM
0	RESERVED	reserved

0x0A204010 WCSS_A_CCU_CCU_SOFT_RESET**Type:** RW**Reset State:** 0x00000000

This register controls Soft Reset control of Common SS modules. Set high to reset.

WCSS_A_CCU_CCU_SOFT_RESET

Bits	Name	Description
18	AOWBR_SOFT_RESET	Soft Reset for aowbr. Asserts the reset when set. Needs to be cleared by SW.
17	RESERVED	reserved
16	SPARE_16	Unused.
15	SPARE_15	Unused.
14	PMU_AHB_SOFT_RESET	Soft Reset for PMU. Asserts the reset when set. Needs to be cleared by SW.
13	LTE_SOFT_RESET	Soft Reset for LTE. Asserts the reset when set. Needs to be cleared by SW.
12	FM_SOFT_RESET	Soft Reset for FM. Asserts the reset when set. Needs to be cleared by SW.
11	BT_SOFT_RESET	Soft Reset for FM. Asserts the reset when set. Needs to be cleared by SW.
10	TIC_SOFT_RESET	Soft Reset for TIC. Asserts the reset when set. Needs to be cleared by SW.
9	ROM_SOFT_RESET	Soft Reset for ROM. Asserts the reset when set. Needs to be cleared by SW.
8	SPDM_SOFT_RESET	Soft Reset for SPDM. Asserts the reset when set. Needs to be cleared by SW.
7	CDAHB_SOFT_RESET	Soft Reset for CDAHB. Asserts the reset when set. Needs to be cleared by SW.
6	CCAHB_SOFT_RESET	Soft Reset for CCHB. Asserts the reset after 4 clock cycles. Self-clearing

WCSS_A_CCU_CCU_SOFT_RESET (cont.)

Bits	Name	Description
5	A2XB_SOFT_RESET	Soft Reset for A2XB. Asserts the reset when set. Needs to be cleared by SW.
4	A2AB_SOFT_RESET	Soft Reset for A2AB. Asserts the reset after 4 clock cycles. Self-clearing
3	RRI_SOFT_RESET	Soft Reset for RRI. Asserts the reset when set. Needs to be cleared by SW.
2	DXE_SOFT_RESET	Soft Reset for DXE. Asserts the reset when set. Needs to be cleared by SW.
1	CMEM_SOFT_RESET	Soft Reset for CMEM. Asserts the reset when set. Needs to be cleared by SW.
0	CCPU_SOFT_RESET	Soft Reset for CCPU. Asserts the reset when set. Needs to be cleared by SW.

0x0A205000 WCSS_A_CCU_CCU_ROM_BASE_ADDR1**Type:** RW**Reset State:** 0x00000000

Base Address for ROM in Dwords; Left-Shift by 2-bits to get the byte address

WCSS_A_CCU_CCU_ROM_BASE_ADDR1

Bits	Name	Description
27:21	BA3	patch3_base_addr
20:14	BA2	patch2_base_addr
13:7	BA1	patch1_base_addr
6:0	BA0	patch0_base_addr

0x0A205004 WCSS_A_CCU_CCU_ROM_BASE_ADDR2**Type:** RW**Reset State:** 0x00000000

Base Address for ROM in Dwords; Left-Shift by 2-bits to get the byte address

WCSS_A_CCU_CCU_ROM_BASE_ADDR2

Bits	Name	Description
27:21	BA7	patch7_base_addr

WCSS_A_CCU_CCU_ROM_BASE_ADDR2 (cont.)

Bits	Name	Description
20:14	BA6	patch6_base_addr
13:7	BA5	patch5_base_addr
6:0	BA4	patch4_base_addr

0x0A205008 WCSS_A_CCU_CCU_ROM_START_ADDR1**Type:** RW**Reset State:** 0x00000000

Start Address for ROM in Dwords; Left-Shift by 2-bits to get the byte address

WCSS_A_CCU_CCU_ROM_START_ADDR1

Bits	Name	Description
31:16	SA1	patch1_start_addr
15:0	SA0	patch0_start_addr

0x0A20500C WCSS_A_CCU_CCU_ROM_START_ADDR2**Type:** RW**Reset State:** 0x00000000

Start Address for ROM in Dwords; Left-Shift by 2-bits to get the byte address

WCSS_A_CCU_CCU_ROM_START_ADDR2

Bits	Name	Description
31:16	SA3	patch3_start_addr
15:0	SA2	patch2_start_addr

0x0A205010 WCSS_A_CCU_CCU_ROM_START_ADDR3**Type:** RW**Reset State:** 0x00000000

Start Address for ROM in Dwords; Left-Shift by 2-bits to get the byte address

WCSS_A_CCU_CCU_ROM_START_ADDR3

Bits	Name	Description
31:16	SA5	patch5_start_addr
15:0	SA4	patch4_start_addr

0x0A205014 WCSS_A_CCU_CCU_ROM_START_ADDR4**Type:** RW**Reset State:** 0x00000000

Start Address for ROM in Dwords; Left-Shift by 2-bits to get the byte address

WCSS_A_CCU_CCU_ROM_START_ADDR4

Bits	Name	Description
31:16	SA7	patch7_start_addr
15:0	SA6	patch6_start_addr

0x0A205018 WCSS_A_CCU_CCU_ROM_END_ADDR1**Type:** RW**Reset State:** 0x00000000

End Address for ROM in Dwords; Left-Shift by 2-bits to get the byte address

WCSS_A_CCU_CCU_ROM_END_ADDR1

Bits	Name	Description
31:16	EA1	patch1_end_addr
15:0	EA0	patch0_end_addr

0x0A20501C WCSS_A_CCU_CCU_ROM_END_ADDR2**Type:** RW**Reset State:** 0x00000000

End Address for ROM in Dwords; Left-Shift by 2-bits to get the byte address

WCSS_A_CCU_CCU_ROM_END_ADDR2

Bits	Name	Description
31:16	EA3	patch3_end_addr
15:0	EA2	patch2_end_addr

0x0A205020 WCSS_A_CCU_CCU_ROM_END_ADDR3**Type:** RW**Reset State:** 0x00000000

End Address for ROM in Dwords; Left-Shift by 2-bits to get the byte address

WCSS_A_CCU_CCU_ROM_END_ADDR3

Bits	Name	Description
31:16	EA5	patch5_end_addr
15:0	EA4	patch4_end_addr

0x0A205024 WCSS_A_CCU_CCU_ROM_END_ADDR4**Type:** RW**Reset State:** 0x00000000

End Address for ROM in Dwords; Left-Shift by 2-bits to get the byte address

WCSS_A_CCU_CCU_ROM_END_ADDR4

Bits	Name	Description
31:16	EA7	patch7_end_addr
15:0	EA6	patch6_end_addr

0x0A205038 WCSS_A_CCU_CCU_RRI_SW_REG_REINIT**Type:** W**Reset State:** 0x00000000

CCU_RRI_SW_REG_REINIT

WCSS_A_CCU_CCU_RRI_SW_REG_REINIT

Bits	Name	Description
0	RRI_SW_REG_REINIT_P	Write a one to initiate the reg-reinit process. Write only

0x0A20503C WCSS_A_CCU_CCU_RRI_SW_REG_REINIT_ADDR**Type:** RW**Reset State:** 0x00000000

CCU_RRI_SW_REG_REINIT_ADDR

WCSS_A_CCU_CCU_RRI_SW_REG_REINIT_ADDR

Bits	Name	Description
31:2	RRI_SW_REG_REINIT_ADDR	The starting (word) address of the sw triggered register re-configuration table.

0x0A205040 WCSS_A_CCU_CCU_RRI_POLL_CMD_ERR_ADDR**Type:** R**Reset State:** 0x00000000

CCU_RRI_POLL_CMD_ERR_ADDR

WCSS_A_CCU_CCU_RRI_POLL_CMD_ERR_ADDR

Bits	Name	Description
31:2	RRI_POLL_CMD_ERR_ADDR	the poll cmd address when it reaches retry timeout error. Read Only

0x0A205044 WCSS_A_CCU_CCU_RRI_MISC_CFG**Type:** RW**Reset State:** 0x00000000

CCU_RRI_MISC_CFG

WCSS_A_CCU_CCU_RRI_MISC_CFG

Bits	Name	Type	Description
27:20	CCU_RRI_SW_REG_REINIT_CMD_EXT_BASE_ADDR	RW	For RRI to config the external base addr if RRI should go out of pronto addr space

WCSS_A_CCU_CCU_RRI_MISC_CFG (cont.)

Bits	Name	Type	Description
19:15	RRI_CCU_ENCODED_FSM	R	rri_ccu_encoded_fsm, Read Only
14:11	RESERVED	RW	reserved
10	CCU_RRI_GOTO_IDLE	RW	When set, force RRI FSM to go back to idle. Default to 0
9	CCU_RRI_ERR_STATE_CFG	RW	When set, RRI FSM will continue to finish the process when error occurs. When cleared, RRI FSM will go to error state and SW has to write to the ccu_rri_err_exit bit to clear the error. Default to 0.
8	CCU_RRI_ERR_EXIT	RW	Write 1 for RRI FSM to go back to IDLE from error state. Always read 0.
7:0	CCU_RRI_ONE_USEC_LIMIT	RW	Value in number of clock cycles to generate the 1usec pulse.

0x0A20504C WCSS_A_CCU_CCU_MISC_CTL**Type:** RW**Reset State:** 0x00012000

CCU Controller related misc control.

WCSS_A_CCU_CCU_MISC_CTL

Bits	Name	Description
16	CCU_CCPU_A9SS_JTAG_SYNC_EN	ccu_ccpu_a9ss_jtag_sync_en, disable to bypass the sync logic on JTAG signals in A9SS if needed
15:8	CCU_CCPU_SPM_ACK_EXTEND_COUNT_VAL	Count in AHB Clocks to extend the gating of irq/fiq from the posedge of spm_ack
7	CCU_CCPU_DISABLE_SPM_ACK_LOGIC	When set it disables the spm_ack extend logic implemented in CCPU for Pronto SS
6	CCU_CCPU_WRAP8_DISABLE	When set it disables the wrap8 logic implemented in CCPU for Riva 2.0
5	CCPU_CMEM_TIMEOUT_CHK_EN	CCPU CMEM Timeout check enable
4	CCPU_STATUS_COUNT_RESET	Signal to reset CCPU Status count
3	CCPU_STATUS_COUNT_ENABLE	Signal to enable CCPU Status count
2	ITCM_ENABLE	0: ITCM interface on ARM9 disabled 1: ITCM interface on ARM9 enabled
1	GAS2CRIF_WR_CLK_GATE_DISABLE	0: QGICs CRIF wr_clk gating not disabled 1: QGICs CRIF wr_clk gating disabled

WCSS_A_CCU_CCU_MISC_CTL (cont.)

Bits	Name	Description
0	GAS2CRIF_NS_SELECT	0: QGICs CRIF accesses are Secure 1: QGICs CRIF accesses are Non-Secure

0x0A205050 WCSS_A_CCU_CCU_CCPU_STALL_COUNT**Type:** R**Reset State:** 0x00000000

CCPU Stall counter value

WCSS_A_CCU_CCU_CCPU_STALL_COUNT

Bits	Name	Description
31:0	CCPU_STALL_COUNT	CCPU Stall Counter value

0x0A205054 WCSS_A_CCU_CCU_CCPU_ACTIVE_COUNT**Type:** R**Reset State:** 0x00000000

CCPU Active counter value.

WCSS_A_CCU_CCU_CCPU_ACTIVE_COUNT

Bits	Name	Description
31:0	CCPU_ACTIVE_COUNT	CCPU Active Counter value

0x0A205058 WCSS_A_CCU_CCU_CCPU_INVALID_ADDR**Type:** R**Reset State:** 0x00000000

CCPU Last Invalid Addr.

WCSS_A_CCU_CCU_CCPU_INVALID_ADDR

Bits	Name	Description
31:0	CCPU_INVALID_ADDR	CCPU Last Invalid Addr

0x0A20505C WCSS_A_CCU_CCU_CCPU_LAST_ADDR0**Type:** R**Reset State:** 0x00000000

CCPU Last Address 0. The last 3 addresses the ARM9 has requested over the memory bus

WCSS_A_CCU_CCU_CCPU_LAST_ADDR0

Bits	Name	Description
31:0	CCPU_LAST_ADDR0	CCPU Last Address 0

0x0A205060 WCSS_A_CCU_CCU_CCPU_LAST_ADDR1**Type:** R**Reset State:** 0x00000000

CCPU Last Address 1. The last 3 addresses the ARM9 has requested over the memory bus

WCSS_A_CCU_CCU_CCPU_LAST_ADDR1

Bits	Name	Description
31:0	CCPU_LAST_ADDR1	CCPU Last Address 1

0x0A205064 WCSS_A_CCU_CCU_CCPU_LAST_ADDR2**Type:** R**Reset State:** 0x00000000

CCPU Last Address 2. The last 3 addresses the ARM9 has requested over the memory bus

WCSS_A_CCU_CCU_CCPU_LAST_ADDR2

Bits	Name	Description
31:0	CCPU_LAST_ADDR2	CCPU Last Address 2

0x0A205068 WCSS_A_CCU_CCU_A2AB_ERR_ADDR**Type:** R**Reset State:** 0x00000000

A2AB error address register.

WCSS_A_CCU_CCU_A2AB_ERR_ADDR

Bits	Name	Description
31:0	A2AB_CCU_AHB_ERR_ADDR	a2ab_ccu_ahb_err_addr, can be cleared by writing to A2AB_CTL register

0x0A20506C WCSS_A_CCU_CCU_A2AB_CTL**Type:** RW**Reset State:** 0x00000000

A2AB control register.

WCSS_A_CCU_CCU_A2AB_CTL

Bits	Name	Description
0	A2AB_AHB_ERR_CLEAR	Write 1 to clear CCU_A2AB_ERR_ADDR register and generate ccu_a2ab_ahb_err_clear signal. SW needs to clear this bit to de-assert the clear signal.

0x0A205070 WCSS_A_CCU_CCU_A2AB_CFG**Type:** RW**Reset State:** 0x00000000

A2AB config register.

WCSS_A_CCU_CCU_A2AB_CFG

Bits	Name	Description
3:0	CCU_A2AB_CFG_HREADY_LIMIT	A2AB input config signal

0x0A205074 WCSS_A_CCU_CCU_WRAP_ADDR**Type:** RW**Reset State:** 0x00000000

A2AB wrap access address register.

WCSS_A_CCU_CCU_WRAP_ADDR

Bits	Name	Description
31:0	A2AB_WRAP_ADDR	a2ab_wrap_addr, Last wrap access address

0x0A205078 WCSS_A_CCU_CCU_AOWBR_ERR_ADDR**Type:** R**Reset State:** 0x00000000

AOWBR error address register.

WCSS_A_CCU_CCU_AOWBR_ERR_ADDR

Bits	Name	Description
31:0	AOWBR_CCU_AHB_ERR_A DDR	aowbr_ccu_ahb_err_addr

0x0A206000 WCSS_A_CCU_CCU_CCPU_TO_MSM_MB_INT_STAT**Type:** R**Reset State:** 0x00000000register indicating for which mailboxes an interrupt to the **msm** is pending**WCSS_A_CCU_CCU_CCPU_TO_MSM_MB_INT_STAT**

Bits	Name	Description
3:0	MBX_INT_STATUS	Each bit stands for the corresponding mailbox. When set, an interrupt from that Mailbox is pending

0x0A206004 WCSS_A_CCU_CCU_MSM_TO_CCPU_MB_INT_STAT**Type:** R**Reset State:** 0x00000000

register indicating for which mailboxes an interrupt to the cCPU is pending

WCSS_A_CCU_CCU_MSM_TO_CCPU_MB_INT_STAT

Bits	Name	Description
3:0	MBX_INT_STATUS	Each bit stands for the corresponding mailbox. When set, an interrupt from that Mailbox is pending

0x0A206008 WCSS_A_CCU_CCU_MB0_CONTROL**Type:** RW**Reset State:** 0x00000000

General purpose Mailbox register, which can be used to generate interrupts to the ccpu or to the **msm**.

WCSS_A_CCU_CCU_MB0_CONTROL

Bits	Name	Description
31:2	MAILBOX_VALUE	A 30 bit value from sender to receiver
1	INT_DIRECTION	Interrupt direction: When set to zero: cCPU to msm When set to one: msm to cCPU Hard Coded to Zero: ReadOnly
0	MB_RESET	When set, all to this MB related parameters, including the counters, are cleared.

0x0A20600C WCSS_A_CCU_CCU_MB0_CONTROL_COUNTERS**Type:** R**Reset State:** 0x00000000

Access counters for tracking the usage and handling of the MB interrupt

WCSS_A_CCU_CCU_MB0_CONTROL_COUNTERS

Bits	Name	Description
18:16	WRITE_CNT	A 3 bit value indicating the number of times a none reset write operation has been performed on the mb0_control register. This value gets cleared on a mb0 reset.
2:0	READ_CNT	A 3 bit value indicating the number of times a read operation has been performed on the control register of this mailbox This value gets cleared on a MB reset.

0x0A206010 WCSS_A_CCU_CCU_MB1_CONTROL**Type:** RW**Reset State:** 0x00000000

General purpose Mailbox register, which can be used to generate interrupts to the ccpu or to the **msm**.

WCSS_A_CCU_CCU_MB1_CONTROL

Bits	Name	Description
31:2	MAILBOX_VALUE	A 30 bit value from sender to receiver
1	INT_DIRECTION	Interrupt direction: When set to zero: cCPU to msm When set to one: msm to cCPU Hard Coded to Zero: ReadOnly
0	MB_RESET	When set, all to this MB related parameters, including the counters, are cleared.

0x0A206014 WCSS_A_CCU_CCU_MB1_CONTROL_COUNTERS**Type:** R**Reset State:** 0x00000000

Access counters for tracking the usage and handling of the MB interrupt

WCSS_A_CCU_CCU_MB1_CONTROL_COUNTERS

Bits	Name	Description
18:16	WRITE_CNT	A 3 bit value indicating the number of times a none reset write operation has been performed on the mb1_control register. This value gets cleared on a mb1 reset.
2:0	READ_CNT	A 3 bit value indicating the number of times a read operation has been performed on the control register of this mailbox This value gets cleared on a MB reset

0x0A206018 WCSS_A_CCU_CCU_MB2_CONTROL**Type:** RW**Reset State:** 0x00000000General purpose Mailbox register, which can be used to generate interrupts to the ccpu or to the **msm**.**WCSS_A_CCU_CCU_MB2_CONTROL**

Bits	Name	Description
31:2	MAILBOX_VALUE	A 30 bit value from sender to receiver
1	INT_DIRECTION	Interrupt direction: When set to zero: cCPU to msm When set to one: msm to cCPU Hard Coded to Zero: ReadOnly
0	MB_RESET	When set, all to this MB related parameters, including the counters, are cleared.

0x0A20601C WCSS_A_CCU_CCU_MB2_CONTROL_COUNTERS**Type:** R**Reset State:** 0x00000000

Access counters for tracking the usage and handling of the MB interrupt

WCSS_A_CCU_CCU_MB2_CONTROL_COUNTERS

Bits	Name	Description
18:16	WRITE_CNT	A 3 bit value indicating the number of times a none reset write operation has been performed on the mb2_control register. This value gets cleared on a mb2 reset.
2:0	READ_CNT	A 3 bit value indicating the number of times a read operation has been performed on the control register of this mailbox This value gets cleared on a MB reset

0x0A206020 WCSS_A_CCU_CCU_MB3_CONTROL**Type:** RW**Reset State:** 0x00000000

General purpose Mailbox register, which can be used to generate interrupts to the ccpu or to the **msm**.

WCSS_A_CCU_CCU_MB3_CONTROL

Bits	Name	Description
31:2	MAILBOX_VALUE	A 30 bit value from sender to receiver
1	INT_DIRECTION	Interrupt direction: When set to zero: cCPU to msm When set to one: msm to cCPU Hard Coded to Zero: ReadOnly
0	MB_RESET	When set, all to this MB related parameters, including the counters, are cleared.

0x0A206024 WCSS_A_CCU_CCU_MB3_CONTROL_COUNTERS**Type:** R**Reset State:** 0x00000000

Access counters for tracking the usage and handling of the MB interrupt

WCSS_A_CCU_CCU_MB3_CONTROL_COUNTERS

Bits	Name	Description
18:16	WRITE_CNT	A 3 bit value indicating the number of times a none reset write operation has been performed on the mb3_control register. This value gets cleared on a mb3 reset.
2:0	READ_CNT	A 3 bit value indicating the number of times a read operation has been performed on the control register of this mailbox This value gets cleared on a MB reset

0x0A206028 WCSS_A_CCU_CCU_MB4_CONTROL**Type:** RW**Reset State:** 0x00000002

General purpose Mailbox register, which can be used to generate interrupts to the ccpu or to the **msm**.

WCSS_A_CCU_CCU_MB4_CONTROL

Bits	Name	Description
31:2	MAILBOX_VALUE	A 30 bit value from sender to receiver
1	INT_DIRECTION	Interrupt direction: When set to zero: cCPU to msm When set to one: msm to cCPU Hard Coded to One: ReadOnly
0	MB_RESET	When set, all to this MB related parameters, including the counters, are cleared.

0x0A20602C WCSS_A_CCU_CCU_MB4_CONTROL_COUNTERS**Type:** R**Reset State:** 0x00000000

Access counters for tracking the usage and handling of the MB interrupt

WCSS_A_CCU_CCU_MB4_CONTROL_COUNTERS

Bits	Name	Description
18:16	WRITE_CNT	A 3 bit value indicating the number of times a none reset write operation has been performed on the mb4_control register. This value gets cleared on a mb4 reset.
2:0	READ_CNT	A 3 bit value indicating the number of times a read operation has been performed on the control register of this mailbox This value gets cleared on a MB reset

0x0A206030 WCSS_A_CCU_CCU_MB5_CONTROL**Type:** RW**Reset State:** 0x00000002

General purpose Mailbox register, which can be used to generate interrupts to the ccpu or to the **msm**.

WCSS_A_CCU_CCU_MB5_CONTROL

Bits	Name	Description
31:2	MAILBOX_VALUE	A 30 bit value from sender to receiver
1	INT_DIRECTION	Interrupt direction: When set to zero: cCPU to msm When set to one: msm to cCPU Hard Coded to One: ReadOnly
0	MB_RESET	When set, all to this MB related parameters, including the counters, are cleared.

0x0A206034 WCSS_A_CCU_CCU_MB5_CONTROL_COUNTERS**Type:** R**Reset State:** 0x00000000

Access counters for tracking the usage and handling of the MB interrupt

WCSS_A_CCU_CCU_MB5_CONTROL_COUNTERS

Bits	Name	Description
18:16	WRITE_CNT	A 3 bit value indicating the number of times a none reset write operation has been performed on the mb5_control register. This value gets cleared on a mb5 reset.
2:0	READ_CNT	A 3 bit value indicating the number of times a read operation has been performed on the control register of this mailbox This value gets cleared on a MB reset

0x0A206038 WCSS_A_CCU_CCU_MB6_CONTROL**Type:** RW**Reset State:** 0x00000002

General purpose Mailbox register, which can be used to generate interrupts to the ccpu or to the **msm**.

WCSS_A_CCU_CCU_MB6_CONTROL

Bits	Name	Description
31:2	MAILBOX_VALUE	A 30 bit value from sender to receiver
1	INT_DIRECTION	Interrupt direction: When set to zero: cCPU to msm When set to one: msm to cCPU Hard Coded to One: ReadOnly
0	MB_RESET	When set, all to this MB related parameters, including the counters, are cleared.

0x0A20603C WCSS_A_CCU_CCU_MB6_CONTROL_COUNTERS**Type:** R**Reset State:** 0x00000000

Access counters for tracking the usage and handling of the MB interrupt

WCSS_A_CCU_CCU_MB6_CONTROL_COUNTERS

Bits	Name	Description
18:16	WRITE_CNT	A 3 bit value indicating the number of times a none reset write operation has been performed on the mb6_control register. This value gets cleared on a mb6 reset.
2:0	READ_CNT	A 3 bit value indicating the number of times a read operation has been performed on the control register of this mailbox This value gets cleared on a MB reset

0x0A206040 WCSS_A_CCU_CCU_MB7_CONTROL**Type:** RW**Reset State:** 0x00000002

General purpose Mailbox register, which can be used to generate interrupts to the ccpu or to the **msm**.

WCSS_A_CCU_CCU_MB7_CONTROL

Bits	Name	Description
31:2	MAILBOX_VALUE	A 30 bit value from sender to receiver
1	INT_DIRECTION	Interrupt direction: When set to zero: cCPU to msm When set to one: msm to cCPU Hard Coded to One: ReadOnly
0	MB_RESET	When set, all to this MB related parameters, including the counters, are cleared.

0x0A206044 WCSS_A_CCU_CCU_MB7_CONTROL_COUNTERS**Type:** R**Reset State:** 0x00000000

Access counters for tracking the usage and handling of the MB interrupt

WCSS_A_CCU_CCU_MB7_CONTROL_COUNTERS

Bits	Name	Description
18:16	WRITE_CNT	A 3 bit value indicating the number of times a none reset write operation has been performed on the mb7_control register. This value gets cleared on a mb7 reset.
2:0	READ_CNT	A 3 bit value indicating the number of times a read operation has been performed on the control register of this mailbox This value gets cleared on a MB reset

0x0A2060B0 WCSS_A_CCU_CCU_MUTEX0**Type:** RW**Reset State:** 0x00000101

General purpose semaphore and/or mutex register

WCSS_A_CCU_CCU_MUTEX0

Bits	Name	Type	Description
31	RESET	RW	When set, both Current Count and MaxCount fields get initialized with the value written to the MAXcount field. This bit will not get set on a write operation.
15:8	MAXCOUNT	RW	The Max count of semaphoresavailable. Can only be set when the Reset bit is written with value 1.
7:0	CURRENTCOUNT	R	Current count of the semaphore. On a read, the current value of the Semaphore is returned and if the CurrentCount > 0, the CurrentCount value is decreased by one. On a write, the current value of the Semaphore is increased by one if the CurrentCount < MaxCount

0x0A2060B4 WCSS_A_CCU_CCU_MUTEX1**Type:** RW**Reset State:** 0x00000101

General purpose semaphore and/or mutex register

WCSS_A_CCU_CCU_MUTEX1

Bits	Name	Type	Description
31	RESET	RW	When set, both Current Count and MaxCount fields get initialized with the value written to the MAXcount field. This bit will not get set on a write operation.
15:8	MAXCOUNT	RW	The Max count of semaphoresavailable. Can only be set when the Reset bit is written with value 1.
7:0	CURRENTCOUNT	R	Current count of the semaphore. On a read, the current value of the Semaphore is returned and if the CurrentCount > 0, the CurrentCount value is decreased by one. On a write, the current value of the Semaphore is increased by one if the CurrentCount < MaxCount

0x0A2060B8 WCSS_A_CCU_CCU_MUTEX2**Type:** RW**Reset State:** 0x00000101

General purpose semaphore and/or mutex register

WCSS_A_CCU_CCU_MUTEX2

Bits	Name	Type	Description
31	RESET	RW	When set, both Current Count and MaxCount fields get initialized with the value written to the MAXcount field. This bit will not get set on a write operation.
15:8	MAXCOUNT	RW	The Max count of semaphoresavailable. Can only be set when the Reset bit is written with value 1.
7:0	CURRENTCOUNT	R	Current count of the semaphore. On a read, the current value of the Semaphore is returned and if the CurrentCount > 0, the CurrentCount value is decreased by one. On a write, the current value of the Semaphore is increased by one if the CurrentCount < MaxCount

0x0A2060BC WCSS_A_CCU_CCU_MUTEX3**Type:** RW**Reset State:** 0x00000101

General purpose semaphore and/or mutex register

WCSS_A_CCU_CCU_MUTEX3

Bits	Name	Type	Description
31	RESET	RW	When set, both Current Count and MaxCount fields get initialized with the value written to the MAXcount field. This bit will not get set on a write operation.
15:8	MAXCOUNT	RW	The Max count of semaphoresavailable. Can only be set when the Reset bit is written with value 1.
7:0	CURRENTCOUNT	R	Current count of the semaphore. On a read, the current value of the Semaphore is returned and if the CurrentCount > 0, the CurrentCount value is decreased by one. On a write, the current value of the Semaphore is increased by one if the CurrentCount < MaxCount

0x0A2060C0 WCSS_A_CCU_CCU_MUTEX4**Type:** RW**Reset State:** 0x00000101

General purpose semaphore and/or mutex register

WCSS_A_CCU_CCU_MUTEX4

Bits	Name	Type	Description
31	RESET	RW	When set, both Current Count and MaxCount fields get initialized with the value written to the MAXcount field. This bit will not get set on a write operation.
15:8	MAXCOUNT	RW	The Max count of semaphoresavailable. Can only be set when the Reset bit is written with value 1.
7:0	CURRENTCOUNT	R	Current count of the semaphore. On a read, the current value of the Semaphore is returned and if the CurrentCount > 0, the CurrentCount value is decreased by one. On a write, the current value of the Semaphore is increased by one if the CurrentCount < MaxCount

0x0A2060C4 WCSS_A_CCU_CCU_MUTEX5**Type:** RW**Reset State:** 0x00000101

General purpose semaphore and/or mutex register

WCSS_A_CCU_CCU_MUTEX5

Bits	Name	Type	Description
31	RESET	RW	When set, both Current Count and MaxCount fields get initialized with the value written to the MAXcount field. This bit will not get set on a write operation.
15:8	MAXCOUNT	RW	The Max count of semaphoresavailable. Can only be set when the Reset bit is written with value 1.
7:0	CURRENTCOUNT	R	Current count of the semaphore. On a read, the current value of the Semaphore is returned and if the CurrentCount > 0, the CurrentCount value is decreased by one. On a write, the current value of the Semaphore is increased by one if the CurrentCount < MaxCount

0x0A2060C8 WCSS_A_CCU_CCU_MUTEX6**Type:** RW**Reset State:** 0x00000101

General purpose semaphore and/or mutex register

WCSS_A_CCU_CCU_MUTEX6

Bits	Name	Type	Description
31	RESET	RW	When set, both Current Count and MaxCount fields get initialized with the value written to the MAXcount field. This bit will not get set on a write operation.
15:8	MAXCOUNT	RW	The Max count of semaphoresavailable. Can only be set when the Reset bit is written with value 1.
7:0	CURRENTCOUNT	R	Current count of the semaphore. On a read, the current value of the Semaphore is returned and if the CurrentCount > 0, the CurrentCount value is decreased by one. On a write, the current value of the Semaphore is increased by one if the CurrentCount < MaxCount

0x0A2060CC WCSS_A_CCU_CCU_MUTEX7**Type:** RW**Reset State:** 0x00000101

General purpose semaphore and/or mutex register

WCSS_A_CCU_CCU_MUTEX7

Bits	Name	Type	Description
31	RESET	RW	When set, both Current Count and MaxCount fields get initialized with the value written to the MAXcount field. This bit will not get set on a write operation.
15:8	MAXCOUNT	RW	The Max count of semaphores available. Can only be set when the Reset bit is written with value 1.
7:0	CURRENTCOUNT	R	Current count of the semaphore. On a read, the current value of the Semaphore is returned and if the CurrentCount > 0, the CurrentCount value is decreased by one. On a write, the current value of the Semaphore is increased by one if the CurrentCount < MaxCount

0x0A20507C WCSS_A_CCU_CCU_TRACE_CONTROL1**Type:** RW**Reset State:** 0x00000000

Trace function enable/disable control

WCSS_A_CCU_CCU_TRACE_CONTROL1

Bits	Name	Description
1	CCU_TRC_BUF_CFG	0 = captured samples write is blocked when buffer is full, 1 = overwrite mode enabled
0	CCU_TRC_ENABLE	When set, the trace function is enabled

0x0A205080 WCSS_A_CCU_CCU_TRACE_CONTROL2**Type:** RW**Reset State:** 0x7FFF0000

Trace function sample storage location in cMEM control

WCSS_A_CCU_CCU_TRACE_CONTROL2

Bits	Name	Description
30:16	CCU_TRC_END_ADDR	trace buffer cMEM ending address
14:0	CCU_TRC_START_ADDR	trace buffer cMEM starting address

0x0A205084 WCSS_A_CCU_CCU_TRC_WFT_CAPTURE_CTRL**Type:** RW**Reset State:** 0x00000000

Wait for Trigger capture control

WCSS_A_CCU_CCU_TRC_WFT_CAPTURE_CTRL

Bits	Name	Description
31:30	CCU_WFT_CAPTURE_MODE	00 = no capture of samples, 01 = capture samples every clock, 10 = capture only when signals selected by capture mask have changed states, 11 = reserved
29:0	CCU_WFT_CAPTURE_MASK	Wait For trigger capture mask

0x0A205088 WCSS_A_CCU_CCU_TRC_WFT_STATE_CTRL**Type:** RW**Reset State:** 0x00000000

Wait for Trigger state control

WCSS_A_CCU_CCU_TRC_WFT_STATE_CTRL

Bits	Name	Description
31:16	CCU_MAX_CAPTURE_COUNT	Maximum number of samples to be captured after entering this state. A value of 0 indicates unlimited samples
15:5	CCU_SPARE2	Reserved
4:0	CCU_TS_TRANSITION_ENABLE	These enables indicate to which other TSs the trigger statemachine is allowed to jump when the corresponding trigger is seen. When multiple enabled triggers come in at the same time, the lowest TS gets priority Bit 0: When set, transition to TS1 enabled when TS1 trigger is seen Bit 1: When set, transition to TS2 enabled when TS2 trigger is seen Bit 2: When set, transition to TS3 enabled when TS3 trigger is seen Bit 3: When set, transition to TS4 enabled when TS4 trigger is seen Bit 4: When set, transition to TS5 enabled when TS5 trigger is seen

0x0A20508C WCSS_A_CCU_CCU_TRC_TS1_CAPTURE_CTRL**Type:** RW**Reset State:** 0x00000000

Trigger State 1 capture control

WCSS_A_CCU_CCU_TRC_TS1_CAPTURE_CTRL

Bits	Name	Description
31:30	CCU_TS1_CAPTURE_MODE	00 = no capture of samples, 01 = capture samples every clock, 10 = capture only when signals selected by capture mask have changed states, 11 = reserved
29:0	CCU_TS1_CAPTURE_MASK	TS1 capture mask

0x0A205090 WCSS_A_CCU_CCU_TRC_TS1_STATE_CTRL**Type:** RW**Reset State:** 0x00000000

Trigger State 1 state control

WCSS_A_CCU_CCU_TRC_TS1_STATE_CTRL

Bits	Name	Description
31:16	CCU_MAX_CAPTURE_COUNT	Maximum number of samples to be captured after entering this state. A value of 0 indicates unlimited samples
15:5	CCU_SPARE2	Reserved
4:0	CCU_TS_TRANSITION_ENABLE	These enables indicate to which other TSs the trigger statemachine is allowed to jump when the corresponding trigger is seen. When multiple enabled triggers come in at the same time, the lowest TS gets priority Bit 0: When set, transition to TS1 enabled when TS1 trigger is seen Bit 1: When set, transition to TS2 enabled when TS2 trigger is seen Bit 2: When set, transition to TS3 enabled when TS3 trigger is seen Bit 3: When set, transition to TS4 enabled when TS4 trigger is seen Bit 4: When set, transition to TS5 enabled when TS5 trigger is seen

0x0A205094 WCSS_A_CCU_CCU_TRC_TS1_TRIGGER_MASK**Type:** RW**Reset State:** 0x00000000

TS1 Trigger Mask control

WCSS_A_CCU_CCU_TRC_TS1_TRIGGER_MASK

Bits	Name	Description
29:0	CCU_TRIGGER_MASK	All bits set to 1 in this mask, selects the signals in the test bus for which a trigger value is defined. All bits set to 0 indicate testbus signals that are don't cares, and will not be used for this trigger.

0x0A205098 WCSS_A_CCU_CCU_TRC_TS1_TRIGGER_VALUE**Type:** RW**Reset State:** 0x00000000

TS1 Trigger value control

WCSS_A_CCU_CCU_TRC_TS1_TRIGGER_VALUE

Bits	Name	Description
29:0	CCU_TRIGGER_VALUE	For each bit that is a 1 in the ts1 trigger mask, the value in of the corresponding bit in this register indicates the trigger value of the testbus signal.

0x0A20509C WCSS_A_CCU_CCU_TRC_TS2_CAPTURE_CTRL**Type:** RW**Reset State:** 0x00000000

Trigger State 2 capture control

WCSS_A_CCU_CCU_TRC_TS2_CAPTURE_CTRL

Bits	Name	Description
31:30	CCU_TS2_CAPTURE_MODE	00 = no capture of samples, 01 = capture samples every clock, 10 = capture only when signals selected by capture mask have changed states, 11 = reserved
29:0	CCU_TS2_CAPTURE_MASK	TS1 capture mask

0x0A2050A0 WCSS_A_CCU_CCU_TRC_TS2_STATE_CTRL**Type:** RW**Reset State:** 0x00000000

Trigger State 2 state control

WCSS_A_CCU_CCU_TRC_TS2_STATE_CTRL

Bits	Name	Description
31:16	CCU_MAX_CAPTURE_COUNT	Maximum number of samples to be captured after entering this state. A value of 0 indicates unlimited samples
15:5	CCU_SPARE2	Reserved

WCSS_A_CCU_CCU_TRC_TS2_STATE_CTRL (cont.)

Bits	Name	Description
4:0	CCU_TS_TRANSITION_ENABLE	These enables indicate to which other TSs the trigger statemachine is allowed to jump when the corresponding trigger is seen. When multiple enabled triggers come in at the same time, the lowest TS gets priority Bit 0: When set, transition to TS1 enabled when TS1 trigger is seen Bit 1: When set, transition to TS2 enabled when TS2 trigger is seen Bit 2: When set, transition to TS3 enabled when TS3 trigger is seen Bit 3: When set, transition to TS4 enabled when TS4 trigger is seen Bit 4: When set, transition to TS5 enabled when TS5 trigger is seen

0x0A2050A4 WCSS_A_CCU_CCU_TRC_TS2_TRIGGER_MASK**Type:** RW**Reset State:** 0x00000000

TS2 Trigger Mask control

WCSS_A_CCU_CCU_TRC_TS2_TRIGGER_MASK

Bits	Name	Description
29:0	CCU_TRIGGER_MASK	All bits set to 1 in this mask, selects the signals in the test bus for which a trigger value is defined. All bits set to 0 indicate testbus signals that are don't cares, and will not be used for this trigger.

0x0A2050A8 WCSS_A_CCU_CCU_TRC_TS2_TRIGGER_VALUE**Type:** RW**Reset State:** 0x00000000

TS2 Trigger value control

WCSS_A_CCU_CCU_TRC_TS2_TRIGGER_VALUE

Bits	Name	Description
29:0	CCU_TRIGGER_VALUE	For each bit that is a 1 in the ts2 trigger mask, the value in of the corresponding bit in this register indicates the trigger value of the testbus signal.

0x0A2050AC WCSS_A_CCU_CCU_TRC_TS3_CAPTURE_CTRL**Type:** RW**Reset State:** 0x00000000

Trigger State 3 capture control

WCSS_A_CCU_CCU_TRC_TS3_CAPTURE_CTRL

Bits	Name	Description
31:30	CCU_TS3_CAPTURE_MODE	00 = no capture of samples, 01 = capture samples every clock, 10 = capture only when signals selected by capture mask have changed states, 11 = reserved
29:0	CCU_TS3_CAPTURE_MASK	TS1 capture mask

0x0A2050B0 WCSS_A_CCU_CCU_TRC_TS3_STATE_CTRL**Type:** RW**Reset State:** 0x00000000

Trigger State 3 state control

WCSS_A_CCU_CCU_TRC_TS3_STATE_CTRL

Bits	Name	Description
31:16	CCU_MAX_CAPTURE_COUNT	Maximum number of samples to be captured after entering this state. A value of 0 indicates unlimited samples
15:5	CCU_SPARE2	Reserved
4:0	CCU_TS_TRANSITION_ENABLE	These enables indicate to which other TSs the trigger statemachine is allowed to jump when the corresponding trigger is seen. When multiple enabled triggers come in at the same time, the lowest TS gets priority Bit 1: When set, transition to TS1 enabled when TS1 trigger is seen Bit 2: When set, transition to TS2 enabled when TS2 trigger is seen Bit 3: When set, transition to TS3 enabled when TS3 trigger is seen Bit 4: When set, transition to TS4 enabled when TS4 trigger is seen Bit 5: When set, transition to TS5 enabled when TS5 trigger is seen

0x0A2050B4 WCSS_A_CCU_CCU_TRC_TS3_TRIGGER_MASK**Type:** RW**Reset State:** 0x00000000

TS3 Trigger Mask control

WCSS_A_CCU_CCU_TRC_TS3_TRIGGER_MASK

Bits	Name	Description
29:0	CCU_TRIGGER_MASK	All bits set to 1 in this mask, selects the signals in the test bus for which a trigger value is defined. All bits set to 0 indicate testbus signals that are don't cares, and will not be used for this trigger.

0x0A2050B8 WCSS_A_CCU_CCU_TRC_TS3_TRIGGER_VALUE**Type:** RW**Reset State:** 0x00000000

TS3 Trigger value control

WCSS_A_CCU_CCU_TRC_TS3_TRIGGER_VALUE

Bits	Name	Description
29:0	CCU_TRIGGER_VALUE	For each bit that is a 1 in the ts3 trigger mask, the value in of the corresponding bit in this register indicates the trigger value of the testbus signal.

0x0A2050BC WCSS_A_CCU_CCU_TRC_TS4_CAPTURE_CTRL**Type:** RW**Reset State:** 0x00000000

Trigger State 4 capture control

WCSS_A_CCU_CCU_TRC_TS4_CAPTURE_CTRL

Bits	Name	Description
31:30	CCU_TS4_CAPTURE_MODE	00 = no capture of samples, 01 = capture samples every clock, 10 = capture only when signals selected by capture mask have changed states, 11 = reserved
29:0	CCU_TS4_CAPTURE_MASK	TS1 capture mask

0x0A2050C0 WCSS_A_CCU_CCU_TRC_TS4_STATE_CTRL**Type:** RW**Reset State:** 0x00000000

Trigger State 4 state control

WCSS_A_CCU_CCU_TRC_TS4_STATE_CTRL

Bits	Name	Description
31:16	CCU_MAX_CAPTURE_COUNT	Maximum number of samples to be captured after entering this state. A value of 0 indicates unlimited samples
15:5	CCU_SPARE2	Reserved

WCSS_A_CCU_CCU_TRC_TS4_STATE_CTRL (cont.)

Bits	Name	Description
4:0	CCU_TS_TRANSITION_EN ABLE	These enables indicate to which other TSs the trigger statemachine is allowed to jump when the corresponding trigger is seen. When multiple enabled triggers come in at the same time, the lowest TS gets priority Bit 1: When set, transition to TS1 enabled when TS1 trigger is seen Bit 2: When set, transition to TS2 enabled when TS2 trigger is seen Bit 3: When set, transition to TS3 enabled when TS3 trigger is seen Bit 4: When set, transition to TS4 enabled when TS4 trigger is seen Bit 5: When set, transition to TS5 enabled when TS5 trigger is seen

0x0A2050C4 WCSS_A_CCU_CCU_TRC_TS4_TRIGGER_MASK**Type:** RW**Reset State:** 0x00000000

TS4 Trigger Mask control

WCSS_A_CCU_CCU_TRC_TS4_TRIGGER_MASK

Bits	Name	Description
29:0	CCU_TRIGGER_MASK	All bits set to 1 in this mask, selects the signals in the test bus for which a trigger value is defined. All bits set to 0 indicate testbus signals that are don't cares, and will not be used for this trigger.

0x0A2050C8 WCSS_A_CCU_CCU_TRC_TS4_TRIGGER_VALUE**Type:** RW**Reset State:** 0x00000000

TS4 Trigger value control

WCSS_A_CCU_CCU_TRC_TS4_TRIGGER_VALUE

Bits	Name	Description
29:0	CCU_TRIGGER_VALUE	For each bit that is a 1 in the ts4 trigger mask, the value in of the corresponding bit in this register indicates the trigger value of the testbus signal.

0x0A2050CC WCSS_A_CCU_CCU_TRC_TS5_CAPTURE_CTRL**Type:** RW**Reset State:** 0x00000000

Trigger State 5 capture control

WCSS_A_CCU_CCU_TRC_TS5_CAPTURE_CTRL

Bits	Name	Description
31:30	CCU_TS5_CAPTURE_MODE	00 = no capture of samples, 01 = capture samples every clock, 10 = capture only when signals selected by capture mask have changed states, 11 = reserved
29:0	CCU_TS5_CAPTURE_MASK	TS1 capture mask

0x0A2050D0 WCSS_A_CCU_CCU_TRC_TS5_STATE_CTRL**Type:** RW**Reset State:** 0x00000000

Trigger State 5 state control

WCSS_A_CCU_CCU_TRC_TS5_STATE_CTRL

Bits	Name	Description
31:16	CCU_MAX_CAPTURE_COUNT	Maximum number of samples to be captured after entering this state. A value of 0 indicates unlimited samples
15:5	CCU_SPARE2	Reserved
4:0	CCU_TS_TRANSITION_ENABLE	These enables indicate to which other TSs the trigger statemachine is allowed to jump when the corresponding trigger is seen. When multiple enabled triggers come in at the same time, the lowest TS gets priority Bit 0: When set, transition to TS1 enabled when TS1 trigger is seen Bit 1: When set, transition to TS2 enabled when TS2 trigger is seen Bit 2: When set, transition to TS3 enabled when TS3 trigger is seen Bit 3: When set, transition to TS4 enabled when TS4 trigger is seen Bit 4: When set, transition to TS5 enabled when TS5 trigger is seen

0x0A2050D4 WCSS_A_CCU_CCU_TRC_TS5_TRIGGER_MASK**Type:** RW**Reset State:** 0x00000000

TS5 Trigger Mask control

WCSS_A_CCU_CCU_TRC_TS5_TRIGGER_MASK

Bits	Name	Description
29:0	CCU_TRIGGER_MASK	All bits set to 1 in this mask, selects the signals in the test bus for which a trigger value is defined. All bits set to 0 indicate testbus signals that are don't cares, and will not be used for this trigger.

0x0A2050D8 WCSS_A_CCU_CCU_TRC_TS5_TRIGGER_VALUE**Type:** RW**Reset State:** 0x00000000

TS5 Trigger value control

WCSS_A_CCU_CCU_TRC_TS5_TRIGGER_VALUE

Bits	Name	Description
29:0	CCU_TRIGGER_VALUE	For each bit that is a 1 in the ts5 trigger mask, the value in of the corresponding bit in this register indicates the trigger value of the testbus signal.

0x0A2050DC WCSS_A_CCU_CCU_DXE_INT_SELECT**Type:** RW**Reset State:** 0x00000000

Select the DXE interrupts to generate the xfer done and rx data avail apss interrupts

WCSS_A_CCU_CCU_DXE_INT_SELECT

Bits	Name	Description
23:16	RX_DATA_AVAIL_DXE_CHANNEL_SELECT	Bit 16: when set, dxe channel 0 interrupt is ORed into pronto_apps_wlan_rx_data_avail interrupt. Bit 17: when set, dxe channel 1 interrupt is ORed into pronto_apps_wlan_rx_data_avail interrupt. Bit 18: when set, dxe channel 2 interrupt is ORed into pronto_apps_wlan_rx_data_avail interrupt. Bit 19: when set, dxe channel 3 interrupt is ORed into pronto_apps_wlan_rx_data_avail interrupt. Bit 20: when set, dxe channel 4 interrupt is ORed into pronto_apps_wlan_rx_data_avail interrupt. Bit 21: when set, dxe channel 5 interrupt is ORed into pronto_apps_wlan_rx_data_avail interrupt. Bit 22: when set, dxe channel 6 interrupt is ORed into pronto_apps_wlan_rx_data_avail interrupt. Bit 23: when set, dxe channel 7 interrupt is ORed into pronto_apps_wlan_rx_data_avail interrupt.

WCSS_A_CCU_CCU_DXE_INT_SELECT (cont.)

Bits	Name	Description
7:0	XFER_DONE_DXE_CHANN EL_SELECT	Bit 0: when set, dxe channel 0 interrupt is ORed into pronto_apps_wlan_data_xfer_done interrupt. Bit 1: when set, dxe channel 1 interrupt is ORed into pronto_apps_wlan_data_xfer_done interrupt. Bit 2: when set, dxe channel 2 interrupt is ORed into pronto_apps_wlan_data_xfer_done interrupt. Bit 3: when set, dxe channel 3 interrupt is ORed into pronto_apps_wlan_data_xfer_done interrupt. Bit 4: when set, dxe channel 4 interrupt is ORed into pronto_apps_wlan_data_xfer_done interrupt. Bit 5: when set, dxe channel 5 interrupt is ORed into pronto_apps_wlan_data_xfer_done interrupt. Bit 6: when set, dxe channel 6 interrupt is ORed into pronto_apps_wlan_data_xfer_done interrupt. Bit 7: when set, dxe channel 7 interrupt is ORed into pronto_apps_wlan_data_xfer_done interrupt.

0x0A2050E0 WCSS_A_CCU_CCU_DXE_INT_SELECT_STATUS**Type:** R**Reset State:** 0x00000000

The pending DXE interrupts that generate the xfer done and rx data avail apss interrupts

WCSS_A_CCU_CCU_DXE_INT_SELECT_STATUS

Bits	Name	Description
23:16	RX_DATA_AVAIL_DXE_CHA NNEL_STATUS	Bit 16: when set, dxe channel 0 interrupt is setting pronto_apps_wlan_rx_data_avail interrupt. Bit 17: when set, dxe channel 1 interrupt is setting pronto_apps_wlan_rx_data_avail interrupt. Bit 18: when set, dxe channel 2 interrupt is setting pronto_apps_wlan_rx_data_avail interrupt. Bit 19: when set, dxe channel 3 interrupt is setting pronto_apps_wlan_rx_data_avail interrupt. Bit 20: when set, dxe channel 4 interrupt is setting pronto_apps_wlan_rx_data_avail interrupt. Bit 21: when set, dxe channel 5 interrupt is setting pronto_apps_wlan_rx_data_avail interrupt. Bit 22: when set, dxe channel 6 interrupt is setting pronto_apps_wlan_rx_data_avail interrupt. Bit 23: when set, dxe channel 7 interrupt is setting pronto_apps_wlan_rx_data_avail interrupt.

WCSS_A_CCU_CCU_DXE_INT_SELECT_STATUS (cont.)

Bits	Name	Description
7:0	XFER_DONE_DXE_CHANNEL_STATUS	Bit 0: when set, dxe channel 0 interrupt is setting pronto_apps_wlan_data_xfer_done interrupt. Bit 1: when set, dxe channel 1 interrupt is setting pronto_apps_wlan_data_xfer_done interrupt. Bit 2: when set, dxe channel 2 interrupt is setting pronto_apps_wlan_data_xfer_done interrupt. Bit 3: when set, dxe channel 3 interrupt is setting pronto_apps_wlan_data_xfer_done interrupt. Bit 4: when set, dxe channel 4 interrupt is setting pronto_apps_wlan_data_xfer_done interrupt. Bit 5: when set, dxe channel 5 interrupt is setting pronto_apps_wlan_data_xfer_done interrupt. Bit 6: when set, dxe channel 6 interrupt is setting pronto_apps_wlan_data_xfer_done interrupt. Bit 7: when set, dxe channel 7 interrupt is setting pronto_apps_wlan_data_xfer_done interrupt.

0x0A2050E4 WCSS_A_CCU_CCU_ASIC_INT_ENABLE**Type:** RW**Reset State:** 0x00000000

The interrupt enables for generating apss interrupt: pronto_apss_asic_interrupt

WCSS_A_CCU_CCU_ASIC_INT_ENABLE

Bits	Name	Description
23	DXE_CHANNEL7_ENABLE	When set, dxe channel 7 interrupt will set pronto_apss_asic_intr interrupt.
22	DXE_CHANNEL6_ENABLE	When set, dxe channel 6 interrupt will set pronto_apss_asic_intr interrupt.
21	DXE_CHANNEL5_ENABLE	When set, dxe channel 5 interrupt will set pronto_apss_asic_intr interrupt.
20	DXE_CHANNEL4_ENABLE	When set, dxe channel 4 interrupt will set pronto_apss_asic_intr interrupt.
19	DXE_CHANNEL3_ENABLE	When set, dxe channel 3 interrupt will set pronto_apss_asic_intr interrupt.
18	DXE_CHANNEL2_ENABLE	When set, dxe channel 2 interrupt will set pronto_apss_asic_intr interrupt.
17	DXE_CHANNEL1_ENABLE	When set, dxe channel 1 interrupt will set pronto_apss_asic_intr interrupt.
16	DXE_CHANNEL0_ENABLE	When set, dxe channel 0 interrupt will set pronto_apss_asic_intr interrupt.
15	AOWBR_TIMEOUT_INTR_EN	when set, either AOWBR read or write timeout will set pronto_apss_asic_intr interrupt.
14	AOWBR_AHB_ERR_INTR_EN	when set, either AOWBR read or write AHB error will set pronto_apss_asic_intr interrupt.

WCSS_A_CCU_CCU_ASIC_INT_ENABLE (cont.)

Bits	Name	Description
13:12	CCU_SPARE	Reserved
11	MAC_HOST_INTR_EN	When set, it enables the MAC-HOST interrupt (generated from MCU register: MAC_HOST_INT_(MASKED)_STATUS)
10	ADU_ERR_WQ10_WQ13_D AVAIL_INTR_EN	When set, it enables the ADU error WQ10 or WQ13 dataavailable interrupt.
9	DPU_ERR_WQ8_WQ9_DAV AIL_INTR_EN	When set, it enables the DPU error WQ8 or WQ9 dataavailable interrupt.
8	MTU_TIMER_3_INTR_EN	When set, it enables the MTU timer 3 timeout interrupt.
7	MTU_TIMER_2_INTR_EN	When set, it enables the MTU timer 2 timeout interrupt.
6:3	SYS_MB_INTR_EN	When each bit is set, it enables the corresponding SYS MB interrupt. Bit3: MCU MB0 Bit4: MCU MB1, etc
2	DPU_MIC_ERR_INTR_EN	When set, it enables the DPU MIC error interrupt.
1	WATCHDOG_INTR_2_EN	When set, it enables the watchdog timer 2 event interrupt.
0	CCU_SPARE2	Reserved

0x0A2050E8 WCSS_A_CCU_CCU_ASIC_INT_STATUS**Type:** R**Reset State:** 0x00000000

The (enable masked) status of all inputs generating apss interrupt: pronto_apss_asic_interrupt

WCSS_A_CCU_CCU_ASIC_INT_STATUS

Bits	Name	Description
23	DXE_CHANNEL7_STATUS	When set, dxe channel 7 interrupt will set pronto_apss_asic_intr interrupt.
22	DXE_CHANNEL6_STATUS	When set, dxe channel 6 interrupt will set pronto_apss_asic_intr interrupt.
21	DXE_CHANNEL5_STATUS	When set, dxe channel 5 interrupt will set pronto_apss_asic_intr interrupt.
20	DXE_CHANNEL4_STATUS	When set, dxe channel 4 interrupt will set pronto_apss_asic_intr interrupt.
19	DXE_CHANNEL3_STATUS	When set, dxe channel 3 interrupt will set pronto_apss_asic_intr interrupt.
18	DXE_CHANNEL2_STATUS	When set, dxe channel 2 interrupt will set pronto_apss_asic_intr interrupt.
17	DXE_CHANNEL1_STATUS	When set, dxe channel 1 interrupt will set pronto_apss_asic_intr interrupt.

WCSS_A_CCU_CCU_ASIC_INT_STATUS (cont.)

Bits	Name	Description
16	DXE_CHANNEL0_STATUS	When set, dxe channel 0 interrupt will set pronto_apss_asic_intr interrupt.
15	AOWBR_TIMEOUT_INTR	When set, AOWBR read or write transaction timeout will set pronto_apss_asic_intr interrupt. This status bit will be cleared when AOWBR ahb_err_timeout_flags register is read.
14	AOWBR_AHB_ERR_INTR	When set, AOWBR read or write AHB error will set pronto_apss_asic_intr interrupt. This status bit will be cleared when AOWBR ahb_err_timeout_flags register is read.
11	MAC_HOST_INTR	When set, the MAC-HOST interrupt (generated from MCU register: MAC_HOST_INT_(MASKED)_STATUS) is pending
10	ADU_ERR_WQ10_WQ13_DATAVAIL_INTR	When set, it indicates that the ADU error WQ10 or WQ13 dataavailable interrupt occurs.
9	DPU_ERR_WQ8_WQ9_DATAVAIL_INTR	When set, it indicates that the DPU error WQ8 or WQ9 dataavailable interrupt occurs.
8	MTU_TIMER_3_INTR	When set, it indicates that the MTU timer 3 timeout interrupt occurs.
7	MTU_TIMER_2_INTR	When set, it indicates that the MTU timer 2 timeout interrupt occurs.
6:3	SYS_MB_INTR	When each bit is set, it indicates that the corresponding SYS MB interrupt occurs.
2	DPU_MIC_ERR_INTR	When set, it indicates that a DPU MIC error interrupt occurs.
1	WATCHDOG_2_EVENT_INTERRUPT	When set, it indicates that the watchdog timer 2 event interrupt occurs.

0x0A2050EC RESERVED**0x0A2050F0 RESERVED****0x0A205100 WCSS_A_CCU_CCU_TRC_STATUS****Type:** R**Reset State:** 0x00000000

CCU_TRC_STATUS

WCSS_A_CCU_CCU_TRC_STATUS

Bits	Name	Description
6:2	TRC_CCU_TRIG_EVENT	trc_ccu_trig_event
1	TRC_CCU_BUF_FULL	trc_ccu_buf_full status

WCSS_A_CCU_CCU_TRC_STATUS (cont.)

Bits	Name	Description
0	TRC_CCU_BUF_INIT_DONE	trc_ccu_buf_init_done status

0x0A205104 WCSS_A_CCU_CCU_TRC_TRIG_SAMPLE**Type:** R**Reset State:** 0x00000000

CCU_TRC_TRIG_SAMPLE

WCSS_A_CCU_CCU_TRC_TRIG_SAMPLE

Bits	Name	Description
29:0	TRC_CCU_TRIG_SAMPLE	trc_ccu_trig_sample

0x0A205108 WCSS_A_CCU_CCU_TRC_RD_ADDR**Type:** R**Reset State:** 0x00000000

CCU_TRC_RD_ADDR

WCSS_A_CCU_CCU_TRC_RD_ADDR

Bits	Name	Description
30:16	TRC_CCU_BUF_RD_END_ADDR	trc_ccu_buf_rd_end_addr
14:0	TRC_CCU_BUF_RD_START_ADDR	trc_ccu_buf_rd_start_addr

0x0A20510C WCSS_A_CCU_SLP_CALIB_CYCLES_REG**Type:** RW**Reset State:** 0x00000020

slp_calib_cycles_reg

WCSS_A_CCU_SLP_CALIB_CYCLES_REG

Bits	Name	Description
31:0	SLP_CYCLES_PER_CALIB_REG	Number of SLP clock cycles to calibrate

0x0A205110 WCSS_A_CCU_SLP_CALIB_CTL_REG**Type:** RW**Reset State:** 0x00000000

slp_calib_ctl_reg

WCSS_A_CCU_SLP_CALIB_CTL_REG

Bits	Name	Description
1	SLP_CALIB_DONE_CLR	Clear slp_calib_done flag. This bit is self clearing upon the flag being cleared.
0	SLP_CALIB_EN	SLP calib enable

0x0A205114 WCSS_A_CCU_SLP_CALIB_DONE_REG**Type:** R**Reset State:** 0x00000000

slp_calib_done_reg

WCSS_A_CCU_SLP_CALIB_DONE_REG

Bits	Name	Description
0	SLP_CALIB_DONE_AHB	slp_calib_done_ahb flag, can be cleared by writing to the slp_calib_ctl_reg

0x0A205118 WCSS_A_CCU_SLP_CALIB_CNT_AHB_PER_SLP_REG**Type:** R**Reset State:** 0x00000000

slp_calib_cnt_ahb_per_slp_reg

WCSS_A_CCU_SLP_CALIB_CNT_AHB_PER_SLP_REG

Bits	Name	Description
31:0	CNT_AHB_CYCLES_PER_SLP_CALIB	Final cnt_ahb_cycles_per_slp_calib

0x0A20511C WCSS_A_CCU_AOWBR_TIMEOUT_ADDR_REG**Type:** R**Reset State:** 0x00000000

This register recorded the start address of either a write or read transaction that CCAHB master has issued to AOWBR and has resulted in suspend timeout.

WCSS_A_CCU_AOWBR_TIMEOUT_ADDR_REG

Bits	Name	Description
31:0	TIMEOUT_START_ADDR	The start address of read or write transaction that resulted in suspend timeout.

0x0A205120 WCSS_A_CCU_AOWBR_ERR_TIMEOUT_FLAGS_REG**Type:** R**Reset State:** 0x00000000

This register recorded either the bus error flag of a read or write transaction that AOAHB slave returned an ERROR response. This register also recorded the suspend timeout flag of a read or write transaction that has been issued by a CCAHB master.

WCSS_A_CCU_AOWBR_ERR_TIMEOUT_FLAGS_REG

Bits	Name	Description
3	CCAHB_RD_TIMEOUT	Read transaction suspend timeout flag.
2	CCAHB_WR_TIMEOUT	Write transaction suspend timeout flag.
1	AOAHB_RD_AHB_ERR	AOAHB slave returned read beat bus error.
0	AOAHB_WR_AHB_ERR	AOAHB slave returned write beat bus error.

10.34 WCSS_A_CCU_UART_CCU_UART

0x0A207000 WCSS_A_CCU_UART_UART_RBR

Type: R

Reset State: 0x00000000

Receive Buffer Register

WCSS_A_CCU_UART_UART_RBR

Bits	Name	Description
7:0	VALUE	

0x0A207004 WCSS_A_CCU_UART_UART_DLH

Type: RW

Reset State: 0x00000000

Divisor Latch High Register

WCSS_A_CCU_UART_UART_DLH

Bits	Name	Description
7:0	VALUE	

0x0A207008 WCSS_A_CCU_UART_UART_IIR

Type: R

Reset State: 0x00000000

Interrupt Identify Register

WCSS_A_CCU_UART_UART_IIR

Bits	Name	Description
7:6	FEN	
3:0	ID	

0x0A20700C WCSS_A_CCU_UART_UART_LCR**Type:** RW**Reset State:** 0x00000000

Line Control Register

WCSS_A_CCU_UART_UART_LCR

Bits	Name	Description
7	DLAB	
6	BREAK	
4	EPS	
3	PEN	
2	STOP	
1:0	DLS	

0x0A207010 WCSS_A_CCU_UART_UART_MCR**Type:** RW**Reset State:** 0x00000000

Modem Control Register

WCSS_A_CCU_UART_UART_MCR

Bits	Name	Description
5	AFCE	
4	LOOPBACK	
3	OUT2	
2	OUT1	
1	RTS	
0	DTR	

0x0A207014 WCSS_A_CCU_UART_UART_LSR**Type:** R**Reset State:** 0x00000060

Line Status Register

WCSS_A_CCU_UART_UART_LSR

Bits	Name	Description
7	RFE	
6	TEMPT	
5	THRE	
4	BI	
3	FE	
2	PE	
1	OE	
0	DR	

0x0A207018 WCSS_A_CCU_UART_UART_MSR**Type:** R**Reset State:** 0x00000011

Modem Status Register

WCSS_A_CCU_UART_UART_MSR

Bits	Name	Description
7	DCD	
6	RI	
5	DSR	
4	CTS	
3	DDCD	
2	TERI	
1	DDSR	
0	DCTS	

0x0A20701C WCSS_A_CCU_UART_UART_SCRATCH**Type:** RW**Reset State:** 0x00000000

Scratch Pad Register

WCSS_A_CCU_UART_UART_SCRATCH

Bits	Name	Description
7:0	SCRATCH	

0x0A207070 WCSS_A_CCU_UART_UART_FAR**Type:** R**Reset State:** 0x00000000

FIFO Access Register

WCSS_A_CCU_UART_UART_FAR

Bits	Name	Description
0	FAR	

0x0A20707C WCSS_A_CCU_UART_UART_USR**Type:** R**Reset State:** 0x00000000

UART Status Register

WCSS_A_CCU_UART_UART_USR

Bits	Name	Description
0	BUSY	

0x0A2070A4 WCSS_A_CCU_UART_UART_HTX**Type:** RW**Reset State:** 0x00000000

Halt TX Register

WCSS_A_CCU_UART_UART_HTX

Bits	Name	Description
0	HALT	

0x0A2070A8 WCSS_A_CCU_UART_UART_DMASA**Type:** R**Reset State:** 0x00000000

DMA Software Acknowledge Register

WCSS_A_CCU_UART_UART_DMASA

Bits	Name	Description
0	ACK	

10.35 WCSS_A_CDAH_B_CDAH_B**0x0A100000 WCSS_A_CDAH_B_CDAH_B_RRI_PL****Type:** RW**Reset State:** 0x00000002

cdahb Arbiter arbitration priority level for rri master. In the cdahb arbitration scheme the master with highest priority level always wins. Should there be more than one master with the same priority then the winning master will be selected using a round-robin algorithm.

WCSS_A_CDAH_B_CDAH_B_RRI_PL

Bits	Name	Description
3:0	PRIORITY	Arbitration priority for rri master. The lowest priority level is 1 and the highest priority level is 15. A priority level of 0 has a special significance, it means that the master is disabled and stops participating in the arbitration scheme. Note that the cdahb arbiter will prevent a master from disabling itself.

0x0A100004 WCSS_A_CDAH_B_CDAH_B_FD_B_R_PL**Type:** RW**Reset State:** 0x00000004

cdahb Arbiter arbitration priority level for fdbr master. In the cdahb arbitration scheme the master with highest priority level always wins. Should there be more than one master with the same priority then the winning master will be selected using a round-robin algorithm.

WCSS_A_CDAHB_CDAHB_FDBR_PL

Bits	Name	Description
3:0	PRIORITY	Arbitration priority for fdbbr master. The lowest priority level is 1 and the highest priority level is 15. A priority level of 0 has a special significance, it means that the master is disabled and stops participating in the arbitration scheme. Note that the cdaHB arbiter will prevent a master from disabling itself.

0x0A100008 WCSS_A_CDAHB_CDAHB_CCPU_PL**Type:** RW**Reset State:** 0x00000005

cdahb Arbiter arbitration priority level for ccpu master. In the cdahb arbitration scheme the master with highest priority level always wins. Should there be more than one master with the same priority then the winning master will be selected using a round-robin algorithm.

WCSS_A_CDAHB_CDAHB_CCPU_PL

Bits	Name	Description
3:0	PRIORITY	Arbitration priority for ccpu master. The lowest priority level is 1 and the highest priority level is 15. A priority level of 0 has a special significance, it means that the master is disabled and stops participating in the arbitration scheme. Note that the cdaHB arbiter will prevent a master from disabling itself.

0x0A10000C WCSS_A_CDAHB_CDAHB_DBR_PL**Type:** RW**Reset State:** 0x00000003

cdahb Arbiter arbitration priority level for dbr master. In the cdahb arbitration scheme the master with highest priority level always wins. Should there be more than one master with the same priority then the winning master will be selected using a round-robin algorithm.

WCSS_A_CDAHB_CDAHB_DBR_PL

Bits	Name	Description
3:0	PRIORITY	Arbitration priority for dbr master. The lowest priority level is 1 and the highest priority level is 15. A priority level of 0 has a special significance, it means that the master is disabled and stops participating in the arbitration scheme. Note that the cdaHB arbiter will prevent a master from disabling itself.

0x0A100010 WCSS_A_CDAHB_CDAHB_A2AB_PL**Type:** RW**Reset State:** 0x00000001

cdahb Arbiter arbitration priority level for a2ab master. In the cdahb arbitration scheme the master with highest priority level always wins. Should there be more than one master with the same priority then the winning master will be selected using a round-robin algorithm.

WCSS_A_CDAHB_CDAHB_A2AB_PL

Bits	Name	Description
3:0	PRIORITY	Arbitration priority for a2ab master. The lowest priority level is 1 and the highest priority level is 15. A priority level of 0 has a special significance, it means that the master is disabled and stops participating in the arbitration scheme. Note that the cdahb arbiter will prevent a master from disabling itself.

0x0A100014 WCSS_A_CDAHB_CDAHB_DXE_PL**Type:** RW**Reset State:** 0x00000002

cdahb Arbiter arbitration priority level for dxm master. In the cdahb arbitration scheme the master with highest priority level always wins. Should there be more than one master with the same priority then the winning master will be selected using a round-robin algorithm.

WCSS_A_CDAHB_CDAHB_DXE_PL

Bits	Name	Description
3:0	PRIORITY	Arbitration priority for dxm master. The lowest priority level is 1 and the highest priority level is 15. A priority level of 0 has a special significance, it means that the master is disabled and stops participating in the arbitration scheme. Note that the cdahb arbiter will prevent a master from disabling itself.

0x0A100048 WCSS_A_CDAHB_CDAHB_DFLT_MST**Type:** RW**Reset State:** 0x00000000

This register identifies the default master for the cdahb

WCSS_A_CDAH_B_CDAH_DFLT_MST

Bits	Name	Description
3:0	DFT_MST	Default Master ID number register. The default master is the master that is granted by the bus when no master has requested ownership. NOTE: Writing anything beyond the 4 R/W bits will cause the entire write to be discarded. 0x0: DMY (Grant ownership to cdahb dummy master) 0x1: RRI (Grant ownership to cdahb rri master) 0x2: FDBR (Grant ownership to cdahb fdbr master) 0x3: CCPU (Grant ownership to cdahb ccpu master) 0x4: DBR (Grant ownership to cdahb dbr master) 0x5: A2AB (Grant ownership to cdahb a2ab master) 0x6: DXE (Grant ownership to cdahb dxe master)

10.36 WCSS_A_CFG_CFG_WCSS**0x0A141000 WCSS_A_CFG_CSI_TEMPLATE_DW0****Type:** RW**Reset State:** 0x00000000

this register contains first dword of the CSI template

WCSS_A_CFG_CSI_TEMPLATE_DW0

Bits	Name	Description
31:0	CFG_TEMPLATE_DW0	First dword of template is stored in register since memory is not big enough. SO this register contains first dword of the template, this is filled by SW. Refer CSI template in to bd_datastructure document for the contents of this dword

0x0A141004 WCSS_A_CFG_CSI_TEMPLATE_MEM**Type:** RW**Reset State:** 0x00000000

Registers 000-810 give access to 528 locations of the CSI template that is stored in the local memory refer to BD_Data_structure.doc for template format

WCSS_A_CFG_CSI_TEMPLATE_MEM

Bits	Name	Description
31:0	CFG_TEMPLATE_DATA	Check bit 32 of the config register, if it is zero then YOu can either read from any of the locations or write to any of the locations.

0x0A141844 WCSS_A_CFG_CFG_STALL_TIMEOUT**Type:** RW**Reset State:** 0x00000A00

Stall timeout functionality - allows CFG to return to idle if PMI interface stalls.

WCSS_A_CFG_CFG_STALL_TIMEOUT

Bits	Name	Description
16	STALL_TIMEOUT_EN	Enable stall timeout functionality on the MPI.
11:0	STALL_TIMEOUT_TIME	Number of cycles of no activity during PMI transaction before stall timeout occurs.

0x0A141848 WCSS_A_CFG_CFG_CONFIG**Type:** RW**Reset State:** 0x00150000

gives some configurations for CFG

WCSS_A_CFG_CFG_CONFIG

Bits	Name	Type	Description
31	CSIPROC_REGACCESS_FAIL	RW	when set indicates template is being updates internally hence read or write access failed this needs to be cleared by SW
30	CSIPROC_BUSY	R	when set indicates template is being updates internally
29	CSIPROC_TEMPLATE_READY	R	when set indicates template is ready
28:24	LOCAL_CLKGATE_DIS	RW	local clock gate disable for all clk gates in cfg bit0 - gas/config modules bit1 - async fifo bit2 - phy_rxif bit3 - csi_proc bit4 - memif
23:16	CFG_11AC_VHCB_CATEGORY	RW	VHTCB frame is a management Action frame which has category field which is not decided in IEEE spec hence it is made programmable Program VHTCB category field here

WCSS_A_CFG_CFG_CONFIG (cont.)

Bits	Name	Type	Description
12:11	RESERVED		reserved
10:8	RESERVED		reserved
3	SW_TEMPLATE_READY	RW	This is done only for debug purpose. SW can first write to bit 2 in this register to ignore PHY transfers, then write the VHT-CB frame or CSI frame to memory when it is finished creating template, write 1'b1 to bit 3 indicating to TPE frame is ready for transfer
2	IGNORE_PHY_XFERS_EN	RW	when this bit is set any phy transfer is ignored and only SW will do the writes, this is for debug purpose.
1	SOFT_PMI_ABORT_EN	RW	Soft PMI abort enable. this is basically to create phy abort thro' sw in this case need to make sure stall timeout is enabled
0	CFG_CSI_EN	RW	Enables the CSI functionality, by default it is disabled

0x0A14184C WCSS_A_CFG_STALL_TIMEOUT_PHY_ABORT_CNT**Type:** R**Reset State:** 0x00000000

Number of times stall timeout occurs with PHY interface. this register also has PHY abort cntnts

WCSS_A_CFG_STALL_TIMEOUT_PHY_ABORT_CNT

Bits	Name	Description
27:24	FIFOFULL_CNT	counts number of times fifo full happened
23:16	PHY_CLKSTART_CNT	counts number of clock starts
11:8	PHY_ABORT_CNT	PHY Abort Packet Cnt.
3:0	STALL_TIMEOUT_COUNT	Number of times the PMI interface encounters a stall timeout

0x0A141858 WCSS_A_CFG_CFG_SM_STATE**Type:** R**Reset State:** 0x00010001

this register gives the state of the two CFG state machines

WCSS_A_CFG_CFG_SM_STATE

Bits	Name	Description
21:16	RXIF_PHY_STATE	RX interface to phy state machine state
8:0	CSIPROC_STATE	CSI processing state machine state

0x0A14185C WCSS_A_CFG_CFG_TPE_INFO_CNT**Type:** R**Reset State:** 0x00000000

this register gives the counts number of tpe_info_p pulses

WCSS_A_CFG_CFG_TPE_INFO_CNT

Bits	Name	Description
15:0	CFG_TPE_INFO_P_CNT	counter for number of tpe_cfg_info_p pulses

10.37 WCSS_A_DAHB_DAHB**0x0A120000 WCSS_A_DAHB_DAHB_DPUW_PL****Type:** RW**Reset State:** 0x00000004

dahb Arbiter arbitration priority level for dpuw master. In the dahb arbitration scheme the master with highest priority level always wins. Should there be more than one master with the same priority then the winning master will be selected using a round-robin algorithm.

WCSS_A_DAHB_DAHB_DPUW_PL

Bits	Name	Description
3:0	PRIORITY	Arbitration priority for dpuw master. The lowest priority level is 1 and the highest priority level is 15. A priority level of 0 has a special significance, it means that the master is disabled and stops participating in the arbitration scheme. Note that the dahb arbiter will prevent a master from disabling itself.

0x0A120004 WCSS_A_DAHB_DAHB_DPUR_PL**Type:** RW**Reset State:** 0x00000004

dahb Arbiter arbitration priority level for dpur master. In the dahb arbitration scheme the master with highest priority level always wins. Should there be more than one master with the same priority then the winning master will be selected using a round-robin algorithm.

WCSS_A_DAHB_DAHB_DPUR_PL

Bits	Name	Description
3:0	PRIORITY	Arbitration priority for dpur master. The lowest priority level is 1 and the highest priority level is 15. A priority level of 0 has a special significance, it means that the master is disabled and stops participating in the arbitration scheme. Note that the dahb arbiter will prevent a master from disabling itself.

0x0A120008 WCSS_A_DAHB_DAHB_BMUW_PL

Type: RW

Reset State: 0x00000005

dahb Arbiter arbitration priority level for bmuw master. In the dahb arbitration scheme the master with highest priority level always wins. Should there be more than one master with the same priority then the winning master will be selected using a round-robin algorithm.

WCSS_A_DAHB_DAHB_BMUW_PL

Bits	Name	Description
3:0	PRIORITY	Arbitration priority for bmuw master. The lowest priority level is 1 and the highest priority level is 15. A priority level of 0 has a special significance, it means that the master is disabled and stops participating in the arbitration scheme. Note that the dahb arbiter will prevent a master from disabling itself.

0x0A12000C WCSS_A_DAHB_DAHB_BMUR_PL

Type: RW

Reset State: 0x00000005

dahb Arbiter arbitration priority level for bmur master. In the dahb arbitration scheme the master with highest priority level always wins. Should there be more than one master with the same priority then the winning master will be selected using a round-robin algorithm.

WCSS_A_DAHB_DAHB_BMUR_PL

Bits	Name	Description
3:0	PRIORITY	Arbitration priority for bmur master. The lowest priority level is 1 and the highest priority level is 15. A priority level of 0 has a special significance, it means that the master is disabled and stops participating in the arbitration scheme. Note that the dahb arbiter will prevent a master from disabling itself.

0x0A120010 WCSS_A_DAHB_DAHB_ADU_PL**Type:** RW**Reset State:** 0x00000002

dahb Arbiter arbitration priority level for adu master. In the dahb arbitration scheme the master with highest priority level always wins. Should there be more than one master with the same priority then the winning master will be selected using a round-robin algorithm.

WCSS_A_DAHB_DAHB_ADU_PL

Bits	Name	Description
3:0	PRIORITY	Arbitration priority for adu master. The lowest priority level is 1 and the highest priority level is 15. A priority level of 0 has a special significance, it means that the master is disabled and stops participating in the arbitration scheme. Note that the dahb arbiter will prevent a master from disabling itself.

0x0A120014 WCSS_A_DAHB_DAHB_DBR_PL**Type:** RW**Reset State:** 0x00000005

dahb Arbiter arbitration priority level for dbr master. In the dahb arbitration scheme the master with highest priority level always wins. Should there be more than one master with the same priority then the winning master will be selected using a round-robin algorithm.

WCSS_A_DAHB_DAHB_DBR_PL

Bits	Name	Description
3:0	PRIORITY	Arbitration priority for dbr master. The lowest priority level is 1 and the highest priority level is 15. A priority level of 0 has a special significance, it means that the master is disabled and stops participating in the arbitration scheme. Note that the dahb arbiter will prevent a master from disabling itself.

0x0A120018 WCSS_A_DAHB_DAHB_CSU_PL**Type:** RW**Reset State:** 0x00000004

dahb Arbiter arbitration priority level for csu master. In the dahb arbitration scheme the master with highest priority level always wins. Should there be more than one master with the same priority then the winning master will be selected using a round-robin algorithm.

WCSS_A_DAHB_DAHB_CSU_PL

Bits	Name	Description
3:0	PRIORITY	Arbitration priority for csu master. The lowest priority level is 1 and the highest priority level is 15. A priority level of 0 has a special significance, it means that the master is disabled and stops participating in the arbitration scheme. Note that the dahb arbiter will prevent a master from disabling itself.

0x0A120048 WCSS_A_DAHB_DAHB_DFLT_MST**Type:** RW**Reset State:** 0x00000000

This register identifies the default master for the dahb

WCSS_A_DAHB_DAHB_DFLT_MST

Bits	Name	Description
3:0	DFT_MST	Default Master ID number register. The default master is the master that is granted by the bus when no master has requested ownership. NOTE: Writing anything beyond the 4 R/W bits will cause the entire write to be discarded. 0x0: DMY (Grant ownership to dahb dummy master) 0x1: DPUW (Grant ownership to dahb dpuw master) 0x2: DPUR (Grant ownership to dahb dpur master) 0x3: BMUW (Grant ownership to dahb bmuw master) 0x4: BMUR (Grant ownership to dahb bmur master) 0x5: ADU (Grant ownership to dahb adu master) 0x6: DBR (Grant ownership to dahb dbr master) 0x7: CSU (Grant ownership to dahb csu master)

10.38 WCSS_A_DPU_DPU

0x0A081800 WCSS_A_DPU_DPU_TXPKTCOUNT

Type: RW

Reset State: 0x00000000

Total TX packets handled by the PDU

WCSS_A_DPU_DPU_TXPKTCOUNT

Bits	Name	Description
15:0	DD_TXPKTCOUNT	Total TX packets handled by the PDU

0x0A081804 WCSS_A_DPU_DPU_RXPKTCOUNT

Type: RW

Reset State: 0x00000000

Total RX packets handled by the PDU

WCSS_A_DPU_DPU_RXPKTCOUNT

Bits	Name	Description
15:0	DD_RXPKTCOUNT	Total RX packets handled by the PDU

0x0A08180C WCSS_A_DPU_DPU_BASE_ADDR

Type: RW

Reset State: 0x00010000

Base address of the DPU data structure

WCSS_A_DPU_DPU_BASE_ADDR

Bits	Name	Description
31:0	DD_BASE	Base address of the DPU data structure

0x0A081810 WCSS_A_DPU_DPU_KEYBASE_ADDR

Type: RW

Reset State: 0x00011000

Base address of the Key data structure

WCSS_A_DPU_DPU_KEYBASE_ADDR

Bits	Name	Description
31:0	DD_KEYBASE	Base address of the key data structure

0x0A081814 WCSS_A_DPU_DPU_MICKEYBASE_ADDR

Type: RW

Reset State: 0x00012000

Base address of the TKIP Mic Key data structure

WCSS_A_DPU_DPU_MICKEYBASE_ADDR

Bits	Name	Description
31:0	DD_MICKEYBASE	Base address of the TKIP MIC key data structure

0x0A081818 WCSS_A_DPU_DPU_RCBASE_ADDR

Type: RW

Reset State: 0x00013000

Base address of the Reply_counter data structure

WCSS_A_DPU_DPU_RCBASE_ADDR

Bits	Name	Description
31:0	DD_RCBASE	Base address of the reply-counter data structure

0x0A08181C WCSS_A_DPU_DPU_CONTROL

Type: RW

Reset State: 0x00000006

Holds DPU based configuration fields

WCSS_A_DPU_DPU_CONTROL

Bits	Name	Type	Description
31	SP_CLKGATE_DISABLE	RW	Disable clock gating for Security block 0: Clock gating is enabled 1: Clock gating is disabled

WCSS_A_DPU_DPU_CONTROL (cont.)

Bits	Name	Type	Description
30	WEP_CLKGATE_DISABLE	RW	Disable clock gating for RC4 engine 0: Clock gating is enabled 1: Clock gating is disabled
29	TKIP_CLKGATE_DISABLE	RW	Disable clock gating for TKIP block 0: Clock gating is enabled 1: Clock gating is disabled
28	AES_CLKGATE_DISABLE	RW	Disable clock gating for AES engine 0: Clock gating is enabled 1: Clock gating is disabled
27	FRAG_CLKGATE_DISABLE	RW	Disable clock gating for fragmentation/defragmentation block 0: Clock gating is enabled 1: Clock gating is disabled
26	PDUBD_STALL	R	DPU is stalled as no PDU/BD's are available and the DPU requires more. 0: DPU is not stalled 1: DPU is stalled
25	IGR_BUSY	R	DPU Egress State-machine is Busy indication. 0: Ingress SM is IDLE. 1: Ingress SM is BUSY.
24	EGR_BUSY	R	DPU Egress State-machine is Busy indication. 0: Egress SM is IDLE. 1: Egress SM is BUSY.
23	SP_BUSY	R	Encryption/Decryption engine is Busy indication. 0: SP is IDLE. 1: SP is BUSY.
18	WAPI_BUSY	R	WAPI Encryption/Decryption engine is Busy indication. 0: WAPI is IDLE. 1: WAPI is BUSY.
17	SP_RC4_BUSY	R	RC4 Encryption/Decryption engine is Busy indication. 0: SP is IDLE. 1: SP is BUSY.
16	SP_AES_BUSY	R	AES Encryption/Decryption engine is Busy indication. 0: SP is IDLE. 1: SP is BUSY.
15	WAPI_CLKGATE_DISABLE	RW	Disable clock gating for WAPI engine 0: Clock gating is enabled 1: Clock gating is disabled
14	CSR_CLKGATE_DISABLE	RW	Disable clock gating for config reg module 0: Clock gating is enabled 1: Clock gating is disabled
13	WAPI_REVERSE_IV_ENDIAN_RX	RW	This bit reverses the endianness of WAPI replay counter in rx direction
12	INCL_PN_IN_BD_EN	RW	By enabling this bit, DPU writes the packet number in BD(0x38) when replay check is disabled
11	WAPI_REVERSE_IV_ENDIAN_TX	RW	This bit reverses the endianness of WAPI replay counter in tx direction
10	WAPI_QOS_EN	RW	This bit enables wapi QoS field to calculate MIC value
9	RHS_WINCHK_EN	RW	This bit enables right hand side window check
8	WQ_TX_RSV_EN	RW	This bit enables reservations on for TX WQ's
7	WQ_RX_RSV_EN	RW	This bit enables reservations on for RX WQ's
6	DROP_EQ_LEN	RW	Setting this bit drops all envelope only received packets 0: Removes the envelope and passes zero byte packet to Host 1: Drops all envelope only received packets

WCSS_A_DPU_DPU_CONTROL (cont.)

Bits	Name	Type	Description
5	ROUTING_FLAG_EN	RW	Setting this bit overrides the BD routing flag 0: Packet will be routed to the value set in BD 1: Packet will be routed to the value set in routing flag register
4	COPY_FRAME_EN	RW	Setting this bit enables Copying of incoming local memory frame to external memory, i This is needed only when there is no encryption/fragmentation 0: No copying 1: Copy incoming local memory BD/PDUs to External memory BD/PDUs
3	SEQUENCE_NO_INS_EN	RW	Setting this bit enables inserting of DPU generated sequence no in frame control 0: Sequence no from Host is passed on 1: Inserts Sequence number generated by DPU
2	RX_TAG_CHK_EN	RW	Setting this bit enables BD TAG checking against the DPU TAG for RX packets 0: Don't check BD and DD tag on RX 1: Do check BD and DD tag on RX
1	TX_TAG_CHK_EN	RW	Setting this bit enables BD TAG checking against the DPU TAG for TX packets 0: Don't check BD and DD tag on TX 1: Do check BD and DD tag on TX
0	WQ_ARB_MODE	RW	Setting this bit selects the arbitration scheme of the Work Queues (WQ). 0: Arbitration priority is ordered RX0->RX7 followed by TX0->TX7 i.e. RX WQ's have higher priority than TX WQ's 1: Arbitration priority is ordered RX0,TX0,RX1,TX1,RX2,TX2,etc i.e. RX WQ 0 has higher priority than TX WQ 0 which has higher priority than RX WQ 1 and so on.

0x0A081820 WCSS_A_DPU_DPU_INTERRUPT_MASK**Type:** RW**Reset State:** 0x00000000

Enables/Disables the DPU interrupts

WCSS_A_DPU_DPU_INTERRUPT_MASK

Bits	Name	Description
31	EGR_WR_ERROR	Setting this bit enables the EGR State Machine Error interrupt. The interrupt occurs when the DPU egress state-machine enters the error state. Such errors are triggered by errors on the Write AHB bus, which should not happen. The DPU does not handle these errors and may result in a DPU lock up. 0: Disabled 1: Enabled

WCSS_A_DPU_DPU_INTERRUPT_MASK (cont.)

Bits	Name	Description
30	EGR_GBI_ERROR	Setting this bit enables the EGR State Machine Error interrupt. The interrupt occurs when the DPU egress state-machine enters the error state. Such errors are triggered by errors on the Command AHB bus, which should not happen. The DPU does not handle these errors and may result in a DPU lock up. 0: Disabled 1: Enabled
29	IGR_ERROR	Setting this bit enables the IGR State Machine Error interrupt. The interrupt occurs when the DPU ingress state-machine enters the error state. Such errors are triggered by errors on the Read AHB bus, which should not happen. The DPU does not handle these errors and may result in a DPU lock up. 0: Disabled 1: Enabled
28	PDUBD_STALL	Setting this bit enables the PDU/BD Stalled interrupt. The interrupt occurs when the DPU requires BD/PDU's but none are available. 0: Disabled 1: Enabled
27	WATCHDOG_TIMER	Setting this bit enables the Watchdog Timer interrupt. The interrupt occurs when the Watchdog timer reaches its programmable threshold. 0: Disabled 1: Enabled
24	DPU_WAKEUPHOST_EN	Setting this bit enables the wake-up host interrupt. In WoWL mode whenever it detects magic packet the interrupt bit is set 0: Disabled 1: Enabled
23	MIC_ERR_EN	Setting this bit enables the MIC error interrupt. The interrupt occurs when a TKIP MIC is detected as errored. 0: Disabled 1: Enabled
20:0	PKT_ERR_EN	The state of each bit enables/disables an interrupt for the corresponding Packet Error Code. The interrupt occurs when a packet Error is detected. 0: Disabled (per-bit) 1: Enabled (per-bit)

0x0A081824 WCSS_A_DPU_DPU_INTERRUPT_STATUS**Type:** R**Reset State:** 0x00000000

Returns the status of the DPU interrupts

WCSS_A_DPU_DPU_INTERRUPT_STATUS

Bits	Name	Description
31	EGR_WR_ERROR	This bit indicates the status of the EGR State Machine Error interrupt. The event occurs when the DPU egress state-machine enters the error state. Such errors are triggered by errors on the Write AHB bus which, should not happen. The DPU does not handle these errors and may result in a DPU lock up. 0: Disabled 1: Enabled

WCSS_A_DPU_DPU_INTERRUPT_STATUS (cont.)

Bits	Name	Description
30	EGR_GBI_ERROR	This bit indicates the status of the EGR State Machine Error interrupt. The event occurs when the DPU egress state-machine enters the error state. Such errors are triggered by errors on the Command AHB bus which, should not happen. The DPU does not handle these errors and may result in a DPU lock up. 0: Disabled 1: Enabled
29	IGR_ERROR	This bit indicates the status of the IGR State Machine Error interrupt. The event occurs when the DPU ingress state-machine enters the error state. Such errors are triggered by errors on the Read AHB bus, which should not happen. The DPU does not handle these errors and may result in a DPU lock up. 0: Disabled 1: Enabled
28	PDUBD_STALL	This bit indicates the status of the PDU/BD Stalled interrupt. The event occurs when the DPU requires BD/PDU's but none are available. 0: Disabled 1: Enabled
27	WATCHDOG_TIMER	This bit indicates the status of the Watchdog Timer event. The event occurs when the Watchdog timer reaches its programmable threshold. 0: Disabled 1: Enabled
23	MIC_ERR	This bit indicates the status of the MIC error event. The event occurs when a TKIP MIC is detected as errored. 0: Disabled 1: Enabled
20:0	PKT_ERR	The state of each bit enables/disables an interrupt for the corresponding Packet Error Code. The interrupt occurs when a packet Error is detected. 0: Disabled (per-bit) 1: Enabled (per-bit) 0x2: DPU_BAD_TAG_ERR (BD / DD Tag mis-match) 0x4: DPU_BAD_BD_ERR (BD corruption) 0x20: DPU_FRAG_CNT_ERR (DPU - Max Frag Count) 0x40: DPU_BAD_MIC_ERR (SP - TKIP MIC Error) 0x100: DPU_BAD_DECRYPT_ERR (SP - Decryption failed) 0x200: DPU_ENV_ONLY_ERR (SP - Minimum length packet) 0x400: DPU_ENV_PART_ERR (SP - Less than Minimum length packet) 0x800: DPU_ZERO_LEN_ERR (SP - Zero length packet) 0x1000: DPU_BAD_EXTIV_ERR (SP - ExtIV not set (AES or TKIP)) 0x2000: DPU_BAD_KID_ERR (SP - RX Key ID mis-match) 0x4000: DPU_BAD_TSC1_ERR (SP - WEP SEED Mis-match in TKIP) 0x8000: DPU_UNPROT_ERR (SP - Un-protected packet on encrypted channel) 0x10000: DPU_PROT_ERR (SP - Protected packet on un-encrypted channel) 0x20000: DPU_BAD_RC_ERR (SP - Replay Counter mis-match) 0x40000: DPU_RC_WRAP_ERR (SP - Replay Counter Overflow) 0x80000: DPU_STALL_FLUSH (DPU - STALL and flushed) 0x100000: DPU_BD_CHNG_ERR (DPU - BD changed during fragmentation)

0x0A081828 WCSS_A_DPU_DPU_BIP_MMIE**Type:** RW**Reset State:** 0x00001000

This is the Management MIC Information Element This contains Element ID and Length

WCSS_A_DPU_DPU_BIP_MMIE

Bits	Name	Description
15:8	LENGTH	Default this value is 16, This is Length Information
7:0	ELEMENT_ID	Default this value is 0, This is MMIE Element ID

0x0A081848 WCSS_A_DPU_DPU_WATCHDOG**Type:** RW**Reset State:** 0xFFFF0000

This value represents the DPU watch dog timer and controls. The watchdog self resets to 0 when the DPU is idle and increments by 1 at 128 micro-second intervals when the DPU is not idle.

NOTE: The Watchdog does not stop/change the operation of the DPU.

WCSS_A_DPU_DPU_WATCHDOG

Bits	Name	Type	Description
31:16	WATCHDOG_THRES	RW	This is the value that if reached the DPU will issue a timeout interrupt. Writing to this register resets the watchdog timer
15:0	WATCHDOG_VALUE	R	This is the current value of the DPU watchdog timer.

0x0A08184C WCSS_A_DPU_DPU_BD_IN_COUNTS**Type:** R**Reset State:** 0x00000000

This register provides debug counts of BD's into the DPU

WCSS_A_DPU_DPU_BD_IN_COUNTS

Bits	Name	Description
19:16	BD_REQUEST	This is the count of BD's requested by the DPU.
3:0	BD_POP	This is the count of BD's popped by the DPU.

0x0A081850 WCSS_A_DPU_DPU_BD_OUT_COUNTS**Type:** R**Reset State:** 0x00000000

This register provides debug counts of BD's out of the DPU

WCSS_A_DPU_DPU_BD_OUT_COUNTS

Bits	Name	Description
19:16	BD_RELEASE	This is the count of BD's released by the DPU.
3:0	BD_PUSH	This is the count of BD's pushed by the DPU.

0x0A081854 WCSS_A_DPU_DPU_WQ_3_RESERVE**Type:** RW**Reset State:** 0x00000110

This register is used to define the number of BD/PDU's the DPU reserves before processing a TX/RX 3 WQ. If the reservation fails the TX/RX 3 to 10 WQ's are disabled from selection until there enough BD/PDU's to satisfy the reservation request. The BD/PDU reservation values must be less than TX/RX 4 WQ reservation values.

WCSS_A_DPU_DPU_WQ_3_RESERVE

Bits	Name	Description
12:8	BD_CNT	This the number of BD's the DPU will reserve before processing a WQ 3.
6:0	PDU_CNT	This the number of PDU's the DPU will reserve before processing a WQ 3.

0x0A08185C WCSS_A_DPU_DPU_WQ_5_RESERVE**Type:** RW**Reset State:** 0x00000110

This register is used to define the number of BD/PDU's the DPU reserves before processing a TX/RX 5 WQ. If the reservation fails the TX/RX 5 to 10 WQ's are disabled from selection until there enough BD/PDU's to satisfy the reservation request. The BD/PDU reservation values must be less than TX/RX 6 WQ reservation values.

WCSS_A_DPU_DPU_WQ_5_RESERVE

Bits	Name	Description
12:8	BD_CNT	This the number of BD's the DPU will reserve before processing a WQ 5.
6:0	PDU_CNT	This the number of PDU's the DPU will reserve before processing a WQ 5.

0x0A081860 WCSS_A_DPU_DPU_WQ_6_RESERVE**Type:** RW**Reset State:** 0x00000110

This register is used to define the number of BD/PDU's the DPU reserves before processing a TX/RX 6 WQ. If the reservation fails the TX/RX 6 to 10 WQ's are disabled from selection until there enough BD/PDU's to satisfy the reservation request. The BD/PDU reservation values must be less than TX/RX 7 WQ reservation values.

WCSS_A_DPU_DPU_WQ_6_RESERVE

Bits	Name	Description
12:8	BD_CNT	This the number of BD's the DPU will reserve before processing a WQ 6.
6:0	PDU_CNT	This the number of PDU's the DPU will reserve before processing a WQ 6.

0x0A081864 WCSS_A_DPU_DPU_WQ_7_RESERVE**Type:** RW**Reset State:** 0x00000110

This register is used to define the number of BD/PDU's the DPU reserves before processing a TX/RX 4 WQ. If the reservation fails the TX/RX 7 to 10 WQ's are disabled from selection until there enough BD/PDU's to satisfy the reservation request. The BD/PDU reservation values must be less than TX/RX 8 WQ reservation values.

WCSS_A_DPU_DPU_WQ_7_RESERVE

Bits	Name	Description
12:8	BD_CNT	This the number of BD's the DPU will reserve before processing a WQ 7.
6:0	PDU_CNT	This the number of PDU's the DPU will reserve before processing a WQ 7.

0x0A081874 WCSS_A_DPU_DPU_ERROR_WQ**Type:** RW**Reset State:** 0x0000FF08

This register is used to define two Work Queues to which error packets can be routed. By default, both of these error work queues are set to 0xFF, which is intended to be a 'junk' work queue where data is discarded.

WCSS_A_DPU_DPU_ERROR_WQ

Bits	Name	Description
15:8	ERR_WQ_B	This indicates the work queue number for work queue B.
7:0	ERR_WQ_A	This indicates the work queue number for work queue A.

0x0A081878 WCSS_A_DPU_DPU_ERROR_WQ_SELECT**Type:** RW**Reset State:** 0x00000000

This register is used to select the work queue routing for error packet with particular error codes. Each error code is represented by a single bit in this register; for example bit 3 is used for selecting routing of packets with error code 3. A set bit will direct error packets with that error code to work queue B. A cleared bit will direct error packets with that error code to work queue A. By default all errors are routed to error work queue A.

WCSS_A_DPU_DPU_ERROR_WQ_SELECT

Bits	Name	Description
18:1	ERR_WQ_SELECT	If bit n is set in this field then error packets with error code n are sent to error work queue B; if the bit is clear then they are sent to work queue A. All bit numbers other than valid error codes are reserved and should be left zero.

0x0A08187C WCSS_A_DPU_DPU_ERROR_WQ_COUNT**Type:** RW**Reset State:** 0x00000000

This register contains counters for errors pushed to each error work queue. The counts are cleared when it is written.

WCSS_A_DPU_DPU_ERROR_WQ_COUNT

Bits	Name	Description
19:16	ERR_WQ_B_CNT	Count of error packets that have been routed to error work queue B.
3:0	ERR_WQ_A_CNT	Count of error packets that have been routed to error work queue A.

0x0A081880 WCSS_A_DPU_DPU_FRAG_COUNT**Type:** RW**Reset State:** 0x00000000

This register contains counters for fragments received and transmitted. The counts are cleared when it is written.

WCSS_A_DPU_DPU_FRAG_COUNT

Bits	Name	Description
31:16	RX_FRAG_CNT	This is a count of fragments de-fragmented by the DPU. Packets that pass through the DPU without being de-fragmented do not increment this count.
15:0	TX_FRAG_CNT	This is a count of fragments created by the DPU. Packets that pass through the DPU without being fragmented do not increment this count.

0x0A0818E4 WCSS_A_DPU_DPU_WAPIMICKEYBASE_ADDR**Type:** RW**Reset State:** 0x00012000

Base address of the WAPI Mic Key data structure

WCSS_A_DPU_DPU_WAPIMICKEYBASE_ADDR

Bits	Name	Description
31:0	DD_WAPIMICKEYBASE	Base address of the WAPI MIC key data structure

0x0A0818E8 WCSS_A_DPU_DPU_BUG_FIX**Type:** RW**Reset State:** 0x00000000

This register holds bug fix controls.

WCSS_A_DPU_DPU_BUG_FIX

Bits	Name	Description
2	TKIP_PRIORITY_SEL	Selects the method by which the PRIORITY field is generated for the TKIP MIC calculations. 0: from the BD TID field. PRIORITY = {4'h0,BD_TID} 1: from the QC field. PRIORITY = (QC_VALID) ? QC[7:0] : 8'h0

0x0A0818EC WCSS_A_DPU_DPU_AES_MASK

Type: RW

Reset State: 0x80000000

This register holds AES AAD Mask controls.

WCSS_A_DPU_DPU_AES_MASK

Bits	Name	Description
31	FRAME_CONTROL_MASK_15	Selects the masking control for Frame Control bit 15 0: Frame Control bit 15 set to 0 1: Frame Control bit 15 set to value in MAC Header
7	QOS_MASK_7	Selects the masking control for QOS bit 7 0: QOS bit 7 set to 0 1: QOS bit 7 set to value in MAC Header

0x0A0818F0 WCSS_A_DPU_DPU_TKIP_MASK

Type: RW

Reset State: 0x00000000

This register holds TKIP MIC Mask controls.

WCSS_A_DPU_DPU_TKIP_MASK

Bits	Name	Description
7	PRIORITY_MASK_7	Selects the masking control for PRORITY bit 7 0: PRIORITY bit 7 set to 0 1: PRIORITY bit 7 set to value in MAC Header

0x0A0818F4 WCSS_A_DPU_DPU_SEQUENCE_NO

Type: RW

Reset State: 0x00000000

This register holds Sequence number.

WCSS_A_DPU_DPU_SEQUENCE_NO

Bits	Name	Description
11:0	SEQUENCE_NO	DPU generated Sequence Number

0x0A0818F8 WCSS_A_DPU_DPU_ROUTING_FLAG**Type:** RW**Reset State:** 0x0000FF18

This register holds the Routing Flag Value.

WCSS_A_DPU_DPU_ROUTING_FLAG

Bits	Name	Description
15:8	WOW_ROUTING_FLAG	Routing flag to rout the packets in WOW mode (This is controlled by wow_routing_flag_en)
7:0	ROUTING_FLAG	Routing flag to rout the packets in WOW mode (This controlled by wow_en)

0x0A0818FC WCSS_A_DPU_DPU_HOST_WAKEUP_STATUS_MAGICBD**Type:** R**Reset State:** 0x00000000

This register holds host wakeup reason code and magic BD index Value

WCSS_A_DPU_DPU_HOST_WAKEUP_STATUS_MAGICBD

Bits	Name	Description
31:16	MAGIC_PACKET_BD_INDEX	Bd index value for the Magic packet
15	HOST_WAKEUP_STATUS	Host wake-up status
6:0	HOST_WAKEUP_REASON_CODE	Host wake-up reason code

0x0A081900 WCSS_A_DPU_DPU_WOW_CONFIG**Type:** RW**Reset State:** 0x00000000

Holds WOW based configuration fields

WCSS_A_DPU_DPU_WOW_CONFIG

Bits	Name	Description
7	WOW_ROUTING_FLAG_EN	When this bit is enabled 0: 1:
6	WOW_PKTERR_EN	When this bit is enabled 0: 1:
5	WOW_CHANSWITCH_EN	When this bit is set wow block will also wake up the host if channel switch frame is received 0: 1: When this bit is set wow block will also wake up the host if channel switch frame is received
4	WOW_DEAUTH_EN	When this bit is set wow block will also wake up the host if deauthentication packet is received 0: 1: When this bit is set wow block will also wake up the host if deauthentication packet is received
3	WOW_DISASS_EN	When this bit is set wow block will also wake up the host if disassociation packet is received 0: 1: When this bit is set wow block will also wake up the host if disassociation packet is received
2:1	WOW_MODE	four wow modes from the register data that is programmed by SW 00: Magic packet 01: Client only(reserved) 10: AP-Assist(reserved) 11: AP
0	WOW_EN	Setting this bit enables/disables the wow mode. 0: WOW blockdisabled 1: WOW block enable

0x0A081904 WCSS_A_DPU_DPU_WOW_TIMERS**Type:** RW**Reset State:** 0x00000000

Link down Timer :The value programmed indicates the threshold for consecutive missed beacons. If the timer reaches this value host is woken up. Wake-up Timer : Time out timer if an wakeup even has not occurred and this timer expires we wakeup the host

WCSS_A_DPU_DPU_WOW_TIMERS

Bits	Name	Description
31:8	WAKEUP	
5:0	LINK_DOWN	

0x0A081908 WCSS_A_DPU_DPU_WOW_MAGIC_PACKET_HI**Type:** RW**Reset State:** 0x00000000

This is 48bit mac address that is written by software, this is the pattern that is used for magic packet. It can be any pattern also

WCSS_A_DPU_DPU_WOW_MAGIC_PACKET_HI

Bits	Name	Description
15:0	MAGIC_PACKET_47_TO_32	This is 48bit mac address that is written by software

0x0A08190C WCSS_A_DPU_DPU_WOW_MAGIC_PACKET_LO**Type:** RW**Reset State:** 0x00000000

This is 48bit mac address that is written by software, this is the pattern that is used for magic packet. It can be any pattern also

WCSS_A_DPU_DPU_WOW_MAGIC_PACKET_LO

Bits	Name	Description
31:0	MAGIC_PACKET_31_TO_0	This is 48bit mac address that is written by software

0x0A081910 WCSS_A_DPU_WOW_FOUND_MAGIC_COUNTER**Type:** R**Reset State:** 0x00000000

Number of packets received after magic packet has been detected Note That these packets will be pushed to DXE WQ including magic packet, which is something you can verify

WCSS_A_DPU_WOW_FOUND_MAGIC_COUNTER

Bits	Name	Description
31:0	VALUE	

0x0A081914 WCSS_A_DPU_WOW_WAITING4MAGIC_COUNTER**Type:** R**Reset State:** 0x00000000

Number of packets received before we detect a magic packet These packets are pushed to drop WQ

WCSS_A_DPU_WOW_WAITING4MAGIC_COUNTER

Bits	Name	Description
31:0	VALUE	

0x0A081918 WCSS_A_DPU_WAPI_STA_KEY_INDEX_VALUES**Type:** R**Reset State:** 0x00000000

Station key index values

WCSS_A_DPU_WAPI_STA_KEY_INDEX_VALUES

Bits	Name	Description
31:0	VALUE	

0x0A081920 WCSS_A_DPU_WAPI_STA0_KEY_INDEXES**Type:** RW**Reset State:** 0x00000000

This register holds wapi key indexes to map key Ids

WCSS_A_DPU_WAPI_STA0_KEY_INDEXES

Bits	Name	Description
31:24	KEY_INDEX3	Keyindex value to map keyid3
23:16	KEY_INDEX2	Keyindex value to map keyid2
15:8	KEY_INDEX1	Keyindex value to map keyid1
7:0	KEY_INDEX0	Keyindex value to map keyid0

0x0A081924 WCSS_A_DPU_WAPI_STA1_KEY_INDEXES**Type:** RW**Reset State:** 0x00000000

This register holds wapi key indexes to map key Ids

WCSS_A_DPU_WAPI_STA1_KEY_INDEXES

Bits	Name	Description
31:24	KEY_INDEX3	Keyindex value to map keyid3
23:16	KEY_INDEX2	Keyindex value to map keyid2
15:8	KEY_INDEX1	Keyindex value to map keyid1
7:0	KEY_INDEX0	Keyindex value to map keyid0

0x0A081928 WCSS_A_DPU_WAPI_STA2_KEY_INDEXES**Type:** RW**Reset State:** 0x00000000

This register holds wapi key indexes to map key Ids

WCSS_A_DPU_WAPI_STA2_KEY_INDEXES

Bits	Name	Description
31:24	KEY_INDEX3	Keyindex value to map keyid3
23:16	KEY_INDEX2	Keyindex value to map keyid2
15:8	KEY_INDEX1	Keyindex value to map keyid1
7:0	KEY_INDEX0	Keyindex value to map keyid0

0x0A08192C WCSS_A_DPU_WAPI_STA3_KEY_INDEXES**Type:** RW**Reset State:** 0x00000000

This register holds wapi key indexes to map key Ids

WCSS_A_DPU_WAPI_STA3_KEY_INDEXES

Bits	Name	Description
31:24	KEY_INDEX3	Keyindex value to map keyid3
23:16	KEY_INDEX2	Keyindex value to map keyid2
15:8	KEY_INDEX1	Keyindex value to map keyid1
7:0	KEY_INDEX0	Keyindex value to map keyid0

0x0A081930 WCSS_A_DPU_WAPI_STA4_KEY_INDEXES**Type:** RW**Reset State:** 0x00000000

This register holds wapi key indexes to map key Ids

WCSS_A_DPU_WAPI_STA4_KEY_INDEXES

Bits	Name	Description
31:24	KEY_INDEX3	Keyindex value to map keyid3
23:16	KEY_INDEX2	Keyindex value to map keyid2
15:8	KEY_INDEX1	Keyindex value to map keyid1
7:0	KEY_INDEX0	Keyindex value to map keyid0

0x0A081934 WCSS_A_DPU_WAPI_STA5_KEY_INDEXES**Type:** RW**Reset State:** 0x00000000

This register holds wapi key indexes to map key Ids

WCSS_A_DPU_WAPI_STA5_KEY_INDEXES

Bits	Name	Description
31:24	KEY_INDEX3	Keyindex value to map keyid3
23:16	KEY_INDEX2	Keyindex value to map keyid2
15:8	KEY_INDEX1	Keyindex value to map keyid1
7:0	KEY_INDEX0	Keyindex value to map keyid0

0x0A081938 WCSS_A_DPU_WAPI_STA6_KEY_INDEXES**Type:** RW**Reset State:** 0x00000000

This register holds wapi key indexes to map key Ids

WCSS_A_DPU_WAPI_STA6_KEY_INDEXES

Bits	Name	Description
31:24	KEY_INDEX3	Keyindex value to map keyid3
23:16	KEY_INDEX2	Keyindex value to map keyid2

WCSS_A_DPU_WAPI_STA6_KEY_INDEXES (cont.)

Bits	Name	Description
15:8	KEY_INDEX1	Keyindex value to map keyid1
7:0	KEY_INDEX0	Keyindex value to map keyid0

0x0A08193C WCSS_A_DPU_WAPI_STA7_KEY_INDEXES**Type:** RW**Reset State:** 0x00000000

This register holds wapi key indexes to map key Ids

WCSS_A_DPU_WAPI_STA7_KEY_INDEXES

Bits	Name	Description
31:24	KEY_INDEX3	Keyindex value to map keyid3
23:16	KEY_INDEX2	Keyindex value to map keyid2
15:8	KEY_INDEX1	Keyindex value to map keyid1
7:0	KEY_INDEX0	Keyindex value to map keyid0

0x0A081940 WCSS_A_DPU_WAPI_STA8_KEY_INDEXES**Type:** RW**Reset State:** 0x00000000

This register holds wapi key indexes to map key Ids

WCSS_A_DPU_WAPI_STA8_KEY_INDEXES

Bits	Name	Description
31:24	KEY_INDEX3	Keyindex value to map keyid3
23:16	KEY_INDEX2	Keyindex value to map keyid2
15:8	KEY_INDEX1	Keyindex value to map keyid1
7:0	KEY_INDEX0	Keyindex value to map keyid0

0x0A081944 WCSS_A_DPU_WAPI_STA9_KEY_INDEXES**Type:** RW**Reset State:** 0x00000000

This register holds wapi key indexes to map key Ids

WCSS_A_DPU_WAPI_STA9_KEY_INDEXES

Bits	Name	Description
31:24	KEY_INDEX3	Keyindex value to map keyid3
23:16	KEY_INDEX2	Keyindex value to map keyid2
15:8	KEY_INDEX1	Keyindex value to map keyid1
7:0	KEY_INDEX0	Keyindex value to map keyid0

0x0A081948 WCSS_A_DPU_WAPI_STA10_KEY_INDEXES

Type: RW

Reset State: 0x00000000

This register holds wapi key indexes to map key Ids

WCSS_A_DPU_WAPI_STA10_KEY_INDEXES

Bits	Name	Description
31:24	KEY_INDEX3	Keyindex value to map keyid3
23:16	KEY_INDEX2	Keyindex value to map keyid2
15:8	KEY_INDEX1	Keyindex value to map keyid1
7:0	KEY_INDEX0	Keyindex value to map keyid0

0x0A08194C WCSS_A_DPU_WAPI_STA11_KEY_INDEXES

Type: RW

Reset State: 0x00000000

This register holds wapi key indexes to map key Ids

WCSS_A_DPU_WAPI_STA11_KEY_INDEXES

Bits	Name	Description
31:24	KEY_INDEX3	Keyindex value to map keyid3
23:16	KEY_INDEX2	Keyindex value to map keyid2
15:8	KEY_INDEX1	Keyindex value to map keyid1
7:0	KEY_INDEX0	Keyindex value to map keyid0

0x0A081950 WCSS_A_DPU_WAPI_STA12_KEY_INDEXES**Type:** RW**Reset State:** 0x00000000

This register holds wapi key indexes to map key Ids

WCSS_A_DPU_WAPI_STA12_KEY_INDEXES

Bits	Name	Description
31:24	KEY_INDEX3	Keyindex value to map keyid3
23:16	KEY_INDEX2	Keyindex value to map keyid2
15:8	KEY_INDEX1	Keyindex value to map keyid1
7:0	KEY_INDEX0	Keyindex value to map keyid0

0x0A081954 WCSS_A_DPU_WAPI_STA13_KEY_INDEXES**Type:** RW**Reset State:** 0x00000000

This register holds wapi key indexes to map key Ids

WCSS_A_DPU_WAPI_STA13_KEY_INDEXES

Bits	Name	Description
31:24	KEY_INDEX3	Keyindex value to map keyid3
23:16	KEY_INDEX2	Keyindex value to map keyid2
15:8	KEY_INDEX1	Keyindex value to map keyid1
7:0	KEY_INDEX0	Keyindex value to map keyid0

0x0A081958 WCSS_A_DPU_WAPI_STA14_KEY_INDEXES**Type:** RW**Reset State:** 0x00000000

This register holds wapi key indexes to map key Ids

WCSS_A_DPU_WAPI_STA14_KEY_INDEXES

Bits	Name	Description
31:24	KEY_INDEX3	Keyindex value to map keyid3
23:16	KEY_INDEX2	Keyindex value to map keyid2

WCSS_A_DPU_WAPI_STA14_KEY_INDEXES (cont.)

Bits	Name	Description
15:8	KEY_INDEX1	Keyindex value to map keyid1
7:0	KEY_INDEX0	Keyindex value to map keyid0

0x0A08195C WCSS_A_DPU_WAPI_STA15_KEY_INDEXES**Type:** RW**Reset State:** 0x00000000

This register holds wapi key indexes to map key Ids

WCSS_A_DPU_WAPI_STA15_KEY_INDEXES

Bits	Name	Description
31:24	KEY_INDEX3	Keyindex value to map keyid3
23:16	KEY_INDEX2	Keyindex value to map keyid2
15:8	KEY_INDEX1	Keyindex value to map keyid1
7:0	KEY_INDEX0	Keyindex value to map keyid0

0x0A081964 WCSS_A_DPU_WAPI_PN_CONFIG**Type:** RW**Reset State:** 0x00000064

This register configures WAPI packet number even/Odd check in Rx path and PN increment by one/two in Tx path Bit Field Definitions 00 : Sequential 01 : Odd 10 : Even

WCSS_A_DPU_WAPI_PN_CONFIG

Bits	Name	Description
15:14	CASE7	{uc_bc, to_bs, from_ds} = 3'b111
13:12	CASE6	{uc_bc, to_bs, from_ds} = 3'b110
11:10	CASE5	{uc_bc, to_bs, from_ds} = 3'b101
9:8	CASE4	{uc_bc, to_bs, from_ds} = 3'b100
7:6	CASE3	{uc_bc, to_bs, from_ds} = 3'b011
5:4	CASE2	{uc_bc, to_bs, from_ds} = 3'b010
3:2	CASE1	{uc_bc, to_bs, from_ds} = 3'b001
1:0	CASE0	{uc_bc, to_bs, from_ds} = 3'b000

10.39 WCSS_A_FDAHB_FDAHB

0x0A140000 WCSS_A_FDAHB_FDAHB_RXP_PL

Type: RW

Reset State: 0x00000004

fdahb Arbiter arbitration priority level for rxp master. In the fdahb arbitration scheme the master with highest priority level always wins. Should there be more than one master with the same priority then the winning master will be selected using a round-robin algorithm.

WCSS_A_FDAHB_FDAHB_RXP_PL

Bits	Name	Description
3:0	PRIORITY	Arbitration priority for rxp master. The lowest priority level is 1 and the highest priority level is 15. A priority level of 0 has a special significance, it means that the master is disabled and stops participating in the arbitration scheme. Note that the fdahb arbiter will prevent a master from disabling itself.

0x0A140004 WCSS_A_FDAHB_FDAHB_TXP_PL

Type: RW

Reset State: 0x00000004

fdahb Arbiter arbitration priority level for txp master. In the fdahb arbitration scheme the master with highest priority level always wins. Should there be more than one master with the same priority then the winning master will be selected using a round-robin algorithm.

WCSS_A_FDAHB_FDAHB_TXP_PL

Bits	Name	Description
3:0	PRIORITY	Arbitration priority for txp master. The lowest priority level is 1 and the highest priority level is 15. A priority level of 0 has a special significance, it means that the master is disabled and stops participating in the arbitration scheme. Note that the fdahb arbiter will prevent a master from disabling itself.

0x0A140008 WCSS_A_FDAHB_FDAHB_TPE_PL

Type: RW

Reset State: 0x00000004

fdahb Arbiter arbitration priority level for tpe master. In the fdahb arbitration scheme the master with highest priority level always wins. Should there be more than one master with the same priority then the winning master will be selected using a round-robin algorithm.

WCSS_A_FDAH_B_FDAH_B_TPE_PL

Bits	Name	Description
3:0	PRIORITY	Arbitration priority for tpe master. The lowest priority level is 1 and the highest priority level is 15. A priority level of 0 has a special significance, it means that the master is disabled and stops participating in the arbitration scheme. Note that the fdahb arbiter will prevent a master from disabling itself.

0x0A14000C WCSS_A_FDAH_B_FDAH_B_FDBR_PL

Type: RW

Reset State: 0x00000005

fdahb Arbiter arbitration priority level for fdbr master. In the fdahb arbitration scheme the master with highest priority level always wins. Should there be more than one master with the same priority then the winning master will be selected using a round-robin algorithm.

WCSS_A_FDAH_B_FDAH_B_FDBR_PL

Bits	Name	Description
3:0	PRIORITY	Arbitration priority for fdbr master. The lowest priority level is 1 and the highest priority level is 15. A priority level of 0 has a special significance, it means that the master is disabled and stops participating in the arbitration scheme. Note that the fdahb arbiter will prevent a master from disabling itself.

0x0A140010 WCSS_A_FDAH_B_FDAH_B_RPE_PL

Type: RW

Reset State: 0x00000004

fdahb Arbiter arbitration priority level for rpe master. In the fdahb arbitration scheme the master with highest priority level always wins. Should there be more than one master with the same priority then the winning master will be selected using a round-robin algorithm.

WCSS_A_FDAHB_FDAHB_RPE_PL

Bits	Name	Description
3:0	PRIORITY	Arbitration priority for rpe master. The lowest priority level is 1 and the highest priority level is 15. A priority level of 0 has a special significance, it means that the master is disabled and stops participating in the arbitration scheme. Note that the fdahb arbiter will prevent a master from disabling itself.

0x0A140048 WCSS_A_FDAHB_FDAHB_DFLT_MST**Type:** RW**Reset State:** 0x00000000

This register identifies the default master for the fdahb

WCSS_A_FDAHB_FDAHB_DFLT_MST

Bits	Name	Description
3:0	DFT_MST	Default Master ID number register. The default master is the master that is granted by the bus when no master has requested ownership. NOTE: Writing anything beyond the 4 R/W bits will cause the entire write to be discarded. 0x0: DMY (Grant ownership to fdahb dummy master) 0x1: RXP (Grant ownership to fdahb rxp master) 0x2: TXP (Grant ownership to fdahb txp master) 0x3: TPE (Grant ownership to fdahb tpe master) 0x4: FDBR (Grant ownership to fdahb fdbr master) 0x5: RPE (Grant ownership to fdahb rpe master)

10.40 WCSS_A_FFT_FFT

0x0A00C000+WCSS_A_FFT_FFT_RAM_0n_0, n=[0..255]
0x8*n

Type: RW**Reset State:** 0x0000

Ping-Pong FFT Reorder RAM shared by all receive chains.

WCSS_A_FFT_FFT_RAM_0n_0

Bits	Name	Description
15:12	IMAG_DATA0_0	12-bit imagee data for Rx antenna 0
11:0	REAL_DATA0	12-bit real data for Rx antenna 0

0x0A00C004+WCSS_A_FFT_FFT_RAM_0n_1, n=[0..255]
0x8*n
Type: RW**Reset State:** 0x0000

Ping-Pong FFT Reorder RAM shared by all receive chains.

WCSS_A_FFT_FFT_RAM_0n_1

Bits	Name	Description
7:0	IMAG_DATA0_1	12-bit imagee data for Rx antenna 0

0x0A00C800+WCSS_A_FFT_FFT_RAM_1n_0, n=[0..255]
0x8*n
Type: RW**Reset State:** 0x0000

Ping-Pong FFT Reorder RAM shared by all receive chains.

WCSS_A_FFT_FFT_RAM_1n_0

Bits	Name	Description
15:12	IMAG_DATA0_0	12-bit imagee data for Rx antenna 0
11:0	REAL_DATA0	12-bit real data for Rx antenna 0

0x0A00C804+WCSS_A_FFT_FFT_RAM_1n_1, n=[0..255]
0x8*n
Type: RW**Reset State:** 0x0000

Ping-Pong FFT Reorder RAM shared by all receive chains.

WCSS_A_FFT_FFT_RAM_1n_1

Bits	Name	Description
7:0	IMAG_DATA0_1	12-bit imagee data for Rx antenna 0

0x0A00D000 WCSS_A_FFT_APB_ACCESS**Type:** RW**Reset State:** 0x0000

This register controls APB access of the FFT Reorder RAM.

WCSS_A_FFT_APB_ACCESS

Bits	Name	Description
0	APB_MODE	APB access enable bit - internal memory may be accessed only when this bit is asserted

0x0A00D004 WCSS_A_FFT_FFT_TOGGLE**Type:** R**Reset State:** 0x00000000

This register is a 1-bit counter that is updated at the end of each FFT block.

WCSS_A_FFT_FFT_TOGGLE

Bits	Name	Description
0	COUNT	FFT block count

10.41 WCSS_A_MCU_MCU**0x0A080C00 WCSS_A_MCU_BD_PDU_BASE_ADDR****Type:** RW**Reset State:** 0x00000000

Sets the memory base address of the BDs and PDUs in external memory.

WCSS_A_MCU_BD_PDU_BASE_ADDR

Bits	Name	Description
31:7	BD_PDU_BASE_ADDRESS	Base address of the BDs and PDUs in external memory. Note that the lowest 8 bits can not be programmed as these bits will always have to be zero. BDs and PDUs are in units of 128 bytes.

0x0A080C04 WCSS_A_MCU_MLC_CONTROL**Type:** RW**Reset State:** 0x00330000

MAC Loop back Control register

WCSS_A_MCU_MLC_CONTROL

Bits	Name	Description
31	MLC_LOOPBACK_ENABLE	When set to 1'b1, the MLC loopback functionality is enabled
30	MLC_SNOOPING_ENABLE	When set to 1'b1, the MLC snooping functionality is enabled
29	MLC_ADDRESS_SWAP_ENABLE	When set to 1'b1, the MLC will swap address 1 and address2 of the frame received from the TXP
23:16	BYTE_FORWARDING_INTERVAL	The number of 160MHz clock cycles in between each data byte requested from the TXP and forwarded to the RXP. A value of 8'd3 corresponds with 315 Mbps. A value of 8'd7 corresponds with 100 mbps. This setting is only used in MLC loopback functionality mode, not snooping mode. bits [17:16] are read only and are always tied to 2'b11 since this is the minimum value that you can program. It is tied to this value preventing user from programming it anything other

0x0A080C0C WCSS_A_MCU_DBR_STATUS**Type:** RW**Reset State:** 0x00000000

Status register of the DBR

WCSS_A_MCU_DBR_STATUS

Bits	Name	Description
5	DBR_GAM_ERROR	When set to 1'b1, the gam inside the DBR reported an error to overcome some rtl issue, please write to this bit to clear DBR_incorrect_length_error
4	DBR_INCORRECT_LENGTH_ERROR	When set to 1'b1, the DBR reported an error: a module tried an access that was larger then 1 word to overcome some rtl issue, please write to this bit to clear DBR_GAM_error

0x0A080C10 WCSS_A_MCU_MCPU_TO_HOST_MB_INT_STAT**Type:** R**Reset State:** 0x00000000

register indicating for which mailboxes an interrupt to the host is pending

WCSS_A_MCU_MCPU_TO_HOST_MB_INT_STAT

Bits	Name	Description
15:0	MBX_INT_STATUS	Each bit stands for the corresponding mailbox. When set, an interrupt from that Mailbox is pending

0x0A080C14 WCSS_A_MCU_HOST_TO_MCPU_MB_INT_STAT**Type:** R**Reset State:** 0x00000000

register indicating for which mailboxes an interrupt to the mCPU is pending

WCSS_A_MCU_HOST_TO_MCPU_MB_INT_STAT

Bits	Name	Description
15:0	MBX_INT_STATUS	Each bit stands for the corresponding mailbox. When set, an interrupt from that Mailbox is pending

0x0A080C18 WCSS_A_MCU_MB0_CONTROL**Type:** RW**Reset State:** 0x00000000

General purpose Mailbox register, which can be used to generate interrupts to the mcpu or to the host.

WCSS_A_MCU_MB0_CONTROL

Bits	Name	Type	Description
31:2	MAILBOX_VALUE	RW	A 30 bit value from sender to receiver
1	INT_DIRECTION	RW	Interrupt direction: When set to zero: mCPU to host When set to one: host to mCPU
0	MB_RESET	R	When set, all to this MB related parameters, including the counters, are cleared.

0x0A080C1C WCSS_A_MCU_MB0_CONTROL_COUNTERS**Type:** R**Reset State:** 0x00000000

Access counters for tracking the usage and handling of the MB interrupt

WCSS_A_MCU_MB0_CONTROL_COUNTERS

Bits	Name	Description
18:16	WRITE_CNT	A 3 bit value indicating the number of times a none reset write operation has been performed on the mb0_control register. This value gets cleared on a mb0 reset.

WCSS_A_MCU_MB0_CONTROL_COUNTERS (cont.)

Bits	Name	Description
2:0	READ_CNT	A 3 bit value indicating the number of times a read operation has been performed on the control register of this mailbox This value gets cleared on a MB reset.

0x0A080C20 WCSS_A_MCU_MB1_CONTROL**Type:** RW**Reset State:** 0x00000000

General purpose Mailbox register, which can be used to generate interrupts to the mcpu or to the host.

WCSS_A_MCU_MB1_CONTROL

Bits	Name	Type	Description
31:2	MAILBOX_VALUE	RW	A 30 bit value from sender to receiver
1	INT_DIRECTION	RW	Interrupt direction: When set to zero: mCPU to host When set to one: host to mCPU
0	MB_RESET	R	When set, all to this MB related parameters, including the counters, are cleared.

0x0A080C24 WCSS_A_MCU_MB1_CONTROL_COUNTERS**Type:** R**Reset State:** 0x00000000

Access counters for tracking the usage and handling of the MB interrupt

WCSS_A_MCU_MB1_CONTROL_COUNTERS

Bits	Name	Description
18:16	WRITE_CNT	A 3 bit value indicating the number of times a none reset write operation has been performed on the mb1_control register. This value gets cleared on a mb1 reset.
2:0	READ_CNT	A 3 bit value indicating the number of times a read operation has been performed on the control register of this mailbox This value gets cleared on a MB reset

0x0A080C28 WCSS_A_MCU_MB2_CONTROL**Type:** RW**Reset State:** 0x00000000

General purpose Mailbox register, which can be used to generate interrupts to the mcpu or to the host.

WCSS_A_MCU_MB2_CONTROL

Bits	Name	Type	Description
31:2	MAILBOX_VALUE	RW	A 30 bit value from sender to receiver
1	INT_DIRECTION	RW	Interrupt direction: When set to zero: mCPU to host When set to one: host to mCPU
0	MB_RESET	R	When set, all to this MB related parameters, including the counters, are cleared.

0x0A080C2C WCSS_A_MCU_MB2_CONTROL_COUNTERS**Type:** R**Reset State:** 0x00000000

Access counters for tracking the usage and handling of the MB interrupt

WCSS_A_MCU_MB2_CONTROL_COUNTERS

Bits	Name	Description
18:16	WRITE_CNT	A 3 bit value indicating the number of times a none reset write operation has been performed on the mb2_control register. This value gets cleared on a mb2 reset.
2:0	READ_CNT	A 3 bit value indicating the number of times a read operation has been performed on the control register of this mailbox This value gets cleared on a MB reset

0x0A080C30 WCSS_A_MCU_MB3_CONTROL**Type:** RW**Reset State:** 0x00000000

General purpose Mailbox register, which can be used to generate interrupts to the mcpu or to the host.

WCSS_A_MCU_MB3_CONTROL

Bits	Name	Type	Description
31:2	MAILBOX_VALUE	RW	A 30 bit value from sender to receiver
1	INT_DIRECTION	RW	Interrupt direction: When set to zero: mCPU to host When set to one: host to mCPU
0	MB_RESET	R	When set, all to this MB related parameters, including the counters, are cleared.

0x0A080C34 WCSS_A_MCU_MB3_CONTROL_COUNTERS**Type:** R**Reset State:** 0x00000000

Access counters for tracking the usage and handling of the MB interrupt

WCSS_A_MCU_MB3_CONTROL_COUNTERS

Bits	Name	Description
18:16	WRITE_CNT	A 3 bit value indicating the number of times a none reset write operation has been performed on the mb3_control register. This value gets cleared on a mb3 reset.
2:0	READ_CNT	A 3 bit value indicating the number of times a read operation has been performed on the control register of this mailbox This value gets cleared on a MB reset

0x0A080C98 WCSS_A_MCU_MUTEX0**Type:** R**Reset State:** 0x00000101

General purpose semaphore and/or mutex register

WCSS_A_MCU_MUTEX0

Bits	Name	Description
31	RESET	When set, both Current Count and MaxCount fields get initialized with the value written to the MAXcount field. This bit will not get set on a write operation.
15:8	MAXCOUNT	THE Max count of semaphoresavailable. Can only be set when the Reset bit is written with value 1.

WCSS_A_MCU_MUTEX0 (cont.)

Bits	Name	Description
7:0	CURRENTCOUNT	Current count of the semaphore. On a read, the current value of the Semaphore is returned and if the CurrentCount > 0, the CurrentCount value is decreased by one. On a write, the current value of the Semaphore is increased by one if the CurrentCount < MaxCount

0x0A080C9C WCSS_A_MCU_MUTEX1**Type:** R**Reset State:** 0x00000101

General purpose semaphore and/or mutex register

WCSS_A_MCU_MUTEX1

Bits	Name	Description
31	RESET	When set, both Current Count and MaxCount fields get initialized with the value written to the MAXcount field. This bit will not get set on a write operation.
15:8	MAXCOUNT	The Max count of semaphoresavailable. Can only be set when the Reset bit is written with value 1.
7:0	CURRENTCOUNT	Current count of the semaphore. On a read, the current value of the Semaphore is returned and if the CurrentCount > 0, the CurrentCount value is decreased by one. On a write, the current value of the Semaphore is increased by one if the CurrentCount < MaxCount

0x0A080CA0 WCSS_A_MCU_MUTEX2**Type:** R**Reset State:** 0x00000101

General purpose semaphore and/or mutex register

WCSS_A_MCU_MUTEX2

Bits	Name	Description
31	RESET	When set, both Current Count and MaxCount fields get initialized with the value written to the MAXcount field. This bit will not get set on a write operation.
15:8	MAXCOUNT	The Max count of semaphoresavailable. Can only be set when the Reset bit is written with value 1.

WCSS_A_MCU_MUTEX2 (cont.)

Bits	Name	Description
7:0	CURRENTCOUNT	Current count of the semaphore. On a read, the current value of the Semaphore is returned and if the CurrentCount > 0, the CurrentCount value is decreased by one. On a write, the current value of the Semaphore is increased by one if the CurrentCount < MaxCount

0x0A080CA4 WCSS_A_MCU_MUTEX3**Type:** R**Reset State:** 0x00000101

General purpose semaphore and/or mutex register

WCSS_A_MCU_MUTEX3

Bits	Name	Description
31	RESET	When set, both Current Count and MaxCount fields get initialized with the value written to the MAXcount field. This bit will not get set on a write operation.
15:8	MAXCOUNT	The Max count of semaphoresavailable. Can only be set when the Reset bit is written with value 1.
7:0	CURRENTCOUNT	Current count of the semaphore. On a read, the current value of the Semaphore is returned and if the CurrentCount > 0, the CurrentCount value is decreased by one. On a write, the current value of the Semaphore is increased by one if the CurrentCount < MaxCount

0x0A080CB8 WCSS_A_MCU_SOFT_RESET**Type:** RW**Reset State:** 0x00000000

Control to soft reset MAC modules. For mCPU reset control, refer to register: MCPU_reset_control

WCSS_A_MCU_SOFT_RESET

Bits	Name	Type	Description
31	TXP_PHYTX_ABORT	RW	When set, the abort signal to the PHYTX is raised.
30	CAHB_SOFT_RESET_DOW NCNT_EN	RW	When set, the cahb_soft_reset_downcount starts to count for 63 cycles then reset the mrcm_cahb
20	CSU_SOFT_RESET	RW	When set, the csu is held in soft reset

WCSS_A_MCU_SOFT_RESET (cont.)

Bits	Name	Type	Description
19	FDAHB_SOFT_RESET	RW	When set, the fdahb is held in soft reset
18	FDBR_SOFT_RESET	RW	When set, the fdbr is held in soft reset
17	CBR_SOFT_RESET	W	write 1 to soft reset cbr. This is self-clear
16	QTA_SOFT_RESET	RW	When set, the QTA is held in soft reset
15	CFG_SOFT_RESET	RW	When set, the CFG is held in soft reset
14	PHY_SOFT_RESET	RW	When set, the PHY is held in soft reset
13	ADU_SOFT_RESET	RW	When set, the ADU is held in soft reset
12	RPE_SOFT_RESET	RW	When set, the RPE is held in soft reset
11	TPE_SOFT_RESET	RW	When set, the TPE is held in soft reset
10	DAHB_SOFT_RESET	RW	When set, the DAHB is held in soft reset
9	MCU_SOFT_RESET	RW	When set, the MCU is held in soft reset
8	PHY_AHB_2_APB_SOFT_RESET	RW	When set, the PHY AHB to APB interface is held in soft reset
7	DPU_SOFT_RESET	RW	When set, the DPU is held in soft reset
6	BMU_SOFT_RESET	RW	When set, the BMU is held in soft reset
5	MTU_SOFT_RESET	RW	When set, the MTU (MAC Timing Unit) is held in soft reset When set, the RXP is held in soft reset
3	TXP_SOFT_RESET	RW	When set, the TXP is held in soft reset
2	DBR_SOFT_RESET	RW	Reserved
0	MLC_SOFT_RESET	RW	When set, the MLC (MAC Loopback Controller) is held in soft reset

0x0A080CBC WCSS_A_MCU_MAC_CLK_GATING_ENABLE**Type:** RW**Reset State:** 0x047F8DF9

Control to enable the dynamic clock gating functionality in the MAC modules

WCSS_A_MCU_MAC_CLK_GATING_ENABLE

Bits	Name	Description
31:27	SPARE	spare register bits
26	MCU_CSU_CLK_GATING_ENABLE	When set, the csu dynamic clock gating function is enabled
25	MCU_FDAHB_CLK_GATING_ENABLE	When set, the fdahb dynamic clock gating function is enabled

WCSS_A_MCU_MAC_CLK_GATING_ENABLE (cont.)

Bits	Name	Description
24	MCU_FDBR_CLK_GATING_ENABLE	When set, the fdbr dynamic clock gating function is enabled
23	MCU_CBR_CLK_GATING_ENABLE	When set, the cbr dynamic clock gating function is enabled
22	MCU_CFG_CLK_GATING_ENABLE	When set, the cfg dynamic clock gating function is enabled
21	MCU_GAS_CLK_GATING_ENABLE	When set, the MCU gas dynamic clock gating function is enabled
20	QTA_CLK_GATING_ENABLE	When set, the QTA dynamic clock gating function is enabled
19	ARM_CLK_GATING_ENABLE	When set, the ARM dynamic clock gating function is enabled
18	BTC_CLK_GATING_ENABLE	When set, the BTC dynamic clock gating function is enabled
17	ADU_CLK_GATING_ENABLE	When set, the ADU dynamic clock gating function is enabled
16	TPE_CLK_GATING_ENABLE	When set, the TPE dynamic clock gating function is enabled
15	RPE_CLK_GATING_ENABLE	When set, the RPE dynamic clock gating function is enabled
14	DPU_HISTBUF_CLK_ENABLE	This bit is a little bit different since 0-> the clks to hist buff are enabled 1-> the clks are disabled write zero to enable the clock
13	DAHB_CLK_GATING_ENABLE	When set, the DAHB bus module dynamic clock gating function is enabled
12	CAHB_CLK_GATING_ENABLE	When set, the CAHB module dynamic clock gating function is enabled
11	MCU_RESERVED1	RESERVED
10	MIF_CLK_GATING_ENABLE	Reserved
9	MCU_CLK_GATING_ENABLE	When set, the MCU dynamic clock gating function is enabled
8	PHY_AHB_2_APB_CLK_GATING_ENABLE	When set, the PHY AHB to APB interface dynamic clock gating function is enabled
7	DPU_CLK_GATING_ENABLE	When set, the DPU dynamic clock gating function is enabled
6	BMU_CLK_GATING_ENABLE	When set, the BMU dynamic clock gating function is enabled
5	MTU_CLK_GATING_ENABLE	When set, the MTU (MAC Timing Unit) dynamic clock gating function is enabled
4	RXP_CLK_GATING_ENABLE	When set, the RXP dynamic clock gating function is enabled

WCSS_A_MCU_MAC_CLK_GATING_ENABLE (cont.)

Bits	Name	Description
3	TXP_CLK_GATING_ENABLE	When set, the TXP dynamic clock gating function is enabled
2	DBR_CLK_GATING_ENABLE	When set, the DBR dynamic clock gating function is enabled
1	RESERVED2	RESERVED
0	MLC_CLK_GATING_ENABLE	When set, the MLC (MAC Loopback Controller) dynamic clock gating function is enabled

0x0A080CD4 WCSS_A_MCU_MAC_MODULES_ROOT_CLK_ENABLES**Type:** RW**Reset State:** 0x0007FFFD

Control to enable the root clocks for each of the MAC modules

WCSS_A_MCU_MAC_MODULES_ROOT_CLK_ENABLES

Bits	Name	Description
18	CSU_ROOT_CLK_ENABLE	When set, the csu root clock is enabled
17	FDAHB_ROOT_CLK_ENABLE	When set, the fdahb root clock is enabled
16	FDBR_ROOT_CLK_ENABLE	When set, the fdbr root clock is enabled
15	CBR_ROOT_CLK_ENABLE	When set, the cbr root clock is enabled
14	CFG_ROOT_CLK_ENABLE	When set, the CFG root clock is enabled
13	RTCK_ROOT_CLK_ENABLE	Reserved.
12	ADU_ROOT_CLK_ENABLE	When set, the ADU root clock is enabled
11	RPE_ROOT_CLK_ENABLE	When set, the RPE root clock is enabled
10	TPE_ROOT_CLK_ENABLE	When set, the TPE root clock is enabled
9	MCPU_ROOT_CLK_ENABLE	When set, the MCPU root clock is enabled
8	PHY_AHB_2_APB_ROOT_CLK_ENABLE	When set, the PHY AHB to APB interface root clock is enabled
7	DPU_ROOT_CLK_ENABLE	When set, the DPU root clock is enabled
6	BMU_ROOT_CLK_ENABLE	When set, the BMU root clock is enabled
5	MTU_ROOT_CLK_ENABLE	When set, the MTU (MAC Timing Unit) root clock is enabled

WCSS_A_MCU_MAC_MODULES_ROOT_CLK_ENABLES (cont.)

Bits	Name	Description
4	RXP_ROOT_CLK_ENABLE	When set, the RXP root clock is enabled
3	TXP_ROOT_CLK_ENABLE	When set, the TXP root clock is enabled
2	DBR_ROOT_CLK_ENABLE	When set, the DBR root clock is enabled
0	MLC_ROOT_CLK_ENABLE	When set, the MLC (MAC Loopback Controller) root clock is enabled

0x0A080CD8 WCSS_A_MCU_MAC_MODULES_GOTO_IDLE_CONTROL**Type:** RW**Reset State:** 0x00000000

The statemachines in the MAC modules can be forced to go back to idle by setting these signals. This resembles somewhat the softreset functionality, except that the configuration registers maintain their settings. It is therefore less intrusive than a soft reset, and might achieve restoring the modules into a known state

WCSS_A_MCU_MAC_MODULES_GOTO_IDLE_CONTROL

Bits	Name	Description
13	CSU_GOTO_IDLE	When set, the CSU HW state machines will return to idle
12	CFG_GOTO_IDLE	When set, the CFG HW state machines will return to idle
11	ADU_GOTO_IDLE	When set, the ADU HW state machines will return to idle
10	TPE_GOTO_IDLE	When set, the TPE HW state machines will return to idle
9	RPE_GOTO_IDLE	When set, the RPE HW state machines will return to idle
8	PHY_AHB_2_APB_GOTO_IDLE	When set, the PHY AHB to APB interface HW state machines will return to idle
7	DPU_GOTO_IDLE	When set, the DPU HW state machines will return to idle
6	BMU_GOTO_IDLE	When set, the BMU HW state machines will return to idle
5	MTU_GOTO_IDLE	When set, the MTU (MAC Timing Unit) HW state machines will return to idle When set, the RXP HW state machines will return to idle
2	DBR_GOTO_IDLE	Reserved
0	MLC_GOTO_IDLE	When set, the MLC (MAC Loopback Controller) HW state machines will return to idle

0x0A080CEC WCSS_A_MCU_APB2PHY_STATUS**Type:** RW**Reset State:** 0x00000000

Status register of the apb interface to the PHY

WCSS_A_MCU_APB2PHY_STATUS

Bits	Name	Description
1	AHB2PHY_MCU_TIMEOUT_ERROR1	When set to 1'b1, a timeout has happened in the apb2phy interface
0	AHB2PHY_MCU_TIMEOUT_ERROR2	When set to 1'b1, a timeout has happened in the apb2phy interface

0x0A080CF0 WCSS_A_MCU_MLC_STATUS**Type:** R**Reset State:** 0x00000000

Status register of the MLC

WCSS_A_MCU_MLC_STATUS

Bits	Name	Description
31:28	MLC_CONTROLLER_STATUS	State that the MLC controller is in: 0x0: idle 0xf:0x1: MLC active
27:24	NR_OF_RXP_STARTS	The number of txp_mlc_start pulses received
23:22	NR_OF_TXP_MLC_ABORTS	The number of txp_mlc_aborts received. This number freezes on 2'b11.
21:20	NR_OF_MLC_ENDS	The number of command bytes passed on to the RXP
19:12	NR_OF_COMMAND_BYTES	The number of command bytes passed on to the RXP
11:0	NR_OF_DATA_BYTES	The number of data bytes and statistics bytes passed on to the RXP

0x0A080CF4 WCSS_A_MCU_MCU_RESERVED**Type:** RW**Reset State:** 0x00000000

These bits have been reserved for future use

WCSS_A_MCU_MCU_RESERVED

Bits	Name	Description
15:1	MCU_RESERVED	Spare register bits read writable
0	PMU_AGC_SLEEP	pmu_agc_sleep

0x0A080D00 WCSS_A_MCU_RF_WARMUP_CONTROL**Type:** RW**Reset State:** 0x00000000

This register has controls for txp for rf warmup

WCSS_A_MCU_RF_WARMUP_CONTROL

Bits	Name	Type	Description
2	TXP_MCU_RF_WARMUP	R	This is a read only signal. It is basically the status of the of the warmup signal that is given by the TXP to the PHY
1	MCU_TXP_RF_WARMUP	RW	Reserved. Not used.
0	MCU_TXP_RF_WARMUP_END	RW	Reserved. Not used.

0x0A080D04 WCSS_A_MCU_BTC**Type:** RW**Reset State:** 0x00000040

This register controls the operation of the BTC function. This register also masks the various conditions that this device will be reporting 'ACTIVE' to the Bluetooth device. Setting a bit to "1" enables the condition, "0" masks(disables) the condition.

WCSS_A_MCU_BTC

Bits	Name	Type	Description
23	MTU_BTC_TX_RX_BUSY	R	Gives status of busy signal going to BTC after the override
22	MTU_MCU_TX_RX_BUSY	R	Gives status of busy signal coming from MTU
21	BTC_GPIO_INTEL_WLAN_ACTIVE	R	Reserved. 2-wire is not supported
20	BTC_BSR_WLAN_ACTIVE	R	Gives status of wlan active signal going to the pad
19	BTC_BLUETOOTH_ACTIVE	R	Gives status input bluetooth active signal
18	BTC_WLAN_ACTIVE	R	Gives status of wlan active signal from logic (before override)

WCSS_A_MCU_BTC (cont.)

Bits	Name	Type	Description
17	BTC_INTEL_WLAN_ACTIVE	R	Reserved. 2-wire is not supported
16	CHANNEL_DATA_DONE	RW	Reserved. 2-wire is not supported
12	BTC_WLAN_ACTIVE_SW_SEL	RW	When '1', this bit selects override for the wlan active signal
11	BTC_WLAN_ACTIVE_OVERRIDE	RW	When '1', this bit overrides the wlan active signal from logic
10	BTC_INTEL_WLAN_ACTIVE_OVERRIDE	RW	Reserved. 2-wire is not supported
9	BTC_INTEL_WLAN_ACTIVE_SW_SEL	RW	Reserved. 2-wire is not supported
8	BTC_TX_RX_BUSY_SW_SEL	RW	Reserved
7	BTC_TX_RX_BUSY_OVERRIDE	RW	Reserved
6	BT_ACTIVE_EN	RW	When '1', this bit enables the sensing of the Bluetooth devices activity on the shared transmission channel for both Dell and Intel modes of operation.
5	BTC_MODE	RW	Reserved. 2-wire is not supported
4	CHANNEL_DATA_WR_EN	RW	Reserved. 2-wire is not supported
3:0	CHANNEL_DATA	RW	Reserved. 2-wire is not supported

0x0A080D08 WCSS_A_MCU_RF_ON_OFF_CONTROL**Type:** R**Reset State:** 0x00000000

This register reflects the status of radio on/off HW pin

WCSS_A_MCU_RF_ON_OFF_CONTROL

Bits	Name	Description
0	RD_ON_OFF_HWPIN_STAT	1'b0 : radio OFF 1'b1 : radio ON

0x0A080D94 RESERVED**0x0A080E10 WCSS_A_MCU_DABH_MIF_PRIORITY_GENERATION_CONTROL****Type:** RW**Reset State:** 0x00000000

controls the way the 'priority signal from the DAHB to the MIF is generated

WCSS_A_MCU_DABH_MIF_PRIORITY_GENERATION_CONTROL

Bits	Name	Description
1:0	DABH_MIF_PRIORITY_MODE	00: txp, rxp bus request based, BMU priority request based 01: txp, rxp and bmu bus request based (when BMU priority requested) AND mif txp/rxp/bmu busy based 10: same as 01, but with extended bus request version 11: request starts on a rising edge of the busrequest from txp/rxp/bmu(when bmu priority requested) and ends at the falling edge of the txp/rxp/bmu mif busy indication

0x0A080E14 WCSS_A_MCU_BTC_CTRL**Type:** RW**Reset State:** 0x0006019F

This register controls the BTC function in 3/4 wire mode.

WCSS_A_MCU_BTC_CTRL

Bits	Name	Type	Description
29:27	SCO_RESERVE_SLOT	RW	To setup the sco/esco reserve slot in terms of slot
26	FORCE_SYNC_TO_SCO	RW	A 0->1 transition will force pta to sync to the bt_active when the counter reaches 0 within the programmed boundary as sco start. SW has to clear this bit.
25	STATS_EN	RW	When set, enable the states counters
24	ESCO_WAIT_FOR_RETX_EN	RW	When set, the wlan will not be allow to transmit until the esco retransmission slot is cleared
23:21	ESCO_RETRANSMIT_SLOT	RW	To setup the esco retransmission slot in terms of slot
20:16	SCO_PERIOD	RW	To setup the sco period in terms of slot
15	PERIOD_INT_CLEAR	W	Reserved. Not required any more
14	PRIORITY_INT_CLEAR	W	Reserved. Not required any more
13	BT_SIMUL_TX_NONOVERLAP	RW	Reserved. No simultaneous transmission
12	BT_SIMUL_TX_ALL	RW	Reserved. No simultaneous transmission

WCSS_A_MCU_BT_CTRL (cont.)

Bits	Name	Type	Description
11	PERIOD_INT_DISABLE	RW	Reserved. Not required any more
10	PRIORITY_INT_DISABLE	RW	Reserved. Not required any more
9:8	BT_3_4WIRE_MODE	RW	Reserved. Only 3-wire is supported
7	BT_DELL_PRIORITY	RW	Reserved. 2-wire is not supported
6	BT_SVO_MODE	RW	Set this bit if BT is in SVO mode. 0x0: NOT_IN_SVO (BT is not in SVO mode.) 0x1: IN_SVO (BT is in SVO mode.)
5	BT_SCO_MODE	RW	Set this bit if BT is in SCO mode. 0x0: NOT_IN_SCO (BT is not in SCO mode.) 0x1: IN_SCO (BT is in SCO mode.)
4	BT_STATUS_POLARITY	RW	Set this bit if BT is operating in 3/4 wire protocol. 0x0: TX_LOW (bt_priority is low when BT TX) 0x1: TX_HIGH (bt_priority is high when BT TX)
3	WLAN_ACTIVE_POLARITY	RW	To configure the polarity of wlan_active signal from BT. 0x0: ACTIVE_LOW (wlan_active is low when WLAN is busy.) 0x1: ACTIVE_HIGH (wlan_active is high when WLAN is busy.)
2	BT_FREQ_POLARITY	RW	Reserved. 4-wire is not supported
1	BT_PRIORITY_POLARITY	RW	To configure the polarity of bt_priority signal from BT 0x0: ACTIVE_LOW (bt_priority is low when BT is in high priority) 0x1: ACTIVE_HIGH (bt_priority is high when BT is in high priority)
0	BT_ACTIVE_POLARITY	RW	To configure the polarity of bt_active signal from BT 0x0: ACTIVE_LOW (bt_active is low to indicate BT traffic) 0x1: ACTIVE_HIGH (bt_active is high to indicate BT traffic)

0x0A080E18 WCSS_A_MCU_BT_CTRL2**Type:** RW**Reset State:** 0x01003200

This register is to setup the counter value in 3/4 wire mode.

WCSS_A_MCU_BT_CTRL2

Bits	Name	Description
31:24	OFFSET_UNCERTANTY	To setup the allowed offset value for period validation

WCSS_A_MCU_BTC_CTRL2 (cont.)

Bits	Name	Description
23	BEFORE_LAG	Reserved 0x0: CNT_LAG (early_end_cnt is after bt_active assertion) 0x1: CNT_PRIORI (early_end_cnt is before bt_active assertion)
15:8	EARLY_END_CNT	To setup the number of cycles for wlan to stop to transmit after bt_active assertion.(keep the name of the bit field in case any TB might use it for whatever purpose). Note this value should be larger than t2 to avoid wrong priority bt traffic request signal sent to TPE.
7:0	LATE_START_CNT	To setup the number of cycles for wlan to start to transmit after bt_active deassertion.

0x0A080E1C WCSS_A_MCU_BTC_CTRL3**Type:** RW**Reset State:** 0x00014196

This register is to setup the timing spec in 3/4 wire mode.

WCSS_A_MCU_BTC_CTRL3

Bits	Name	Description
17:12	BTC_T3	To setup the T3 number (default to 20us).
11:8	BTC_T2	To setup the T2 number (default to 1us).
7:0	BTC_T1	To setup the T1 number (default to 150us).

0x0A080E20 WCSS_A_MCU_BTC_STATUS**Type:** RW**Reset State:** 0x010EA500

This register contains the status of internal signals.

WCSS_A_MCU_BTC_STATUS

Bits	Name	Type	Description
31	WAIT_FOR_ESCO_DONE	R	The status of the esco_bt_is_done signal. If it's in esco mode, and the BTC_CTRL[24] is set, the signal will be asserted until the esco retransmission is done. Otherwise this signal will be 0.
30	BTC_MODE	R	When '1', BT is in tx in current slot. When '0' and bt_active is high, BT is in rx in current slot

WCSS_A_MCU_BTC_STATUS (cont.)

Bits	Name	Type	Description
29:24	SCO_SLOT_CNT	R	The sco slot counter value
23	SCO_SYNC_FLAG	R	To indicate whether the sco period is locked(1) or not(0)
22:8	BT_AVAIL_CNT	RW	The sco period counter which s/w can write to change the value (The big field name is kept in case any mcs relies on it)
7	MASTER_SLAVE_STATUS	R	To indicate whether it's master(1) or slave(0) This is only valid in sco/esco mode.
6	BT_PERIOD_STATUS	R	The status of the bt_period_int signal which validates the bt mode and period (Always read 0 since the interrupt is removed)
5	BT_PRIORITY_STATUS	R	The status of the bt_priority_int signal signalwhich validates the bt mode and priority (Always read 0 since the interrupt is removed)
4	BT_PRIORITY_TO_TPE	R	The status of the bt_priority signal to TPE
3	BT_TRAFFIC_REQUEST	R	The status of the bt_traffic_request signal to TPE
2	BT_FREQ	R	The status of the incoming bt_freq signal (Always read 0 since we only use 3 wire protocol)
1	BT_PRIORITY	R	The status of the incoming bt_priority signal
0	BT_ACTIVE_S	R	The status of the sync-ed bt_active signal

0x0A080E24 WCSS_A_MCU_BTC_STATUS2**Type:** R**Reset State:** 0x00007FFF

This register contains the status of internal signals.

WCSS_A_MCU_BTC_STATUS2

Bits	Name	Description
31:28	CNT_3750US	Reserved
27:24	S_STATE	The encoded state machine status
23:16	CNT_T2_T3	The status of the t2_t3 counter
14:0	BT_AVAIL_CNT	The down counter to next bt-active assertion in sco/esco mode to TPE

0x0A080E2C RESERVED**0x0A080E30 RESERVED****0x0A080E38 WCSS_A_MCU_FIQ_EN****Type:** RW**Reset State:** 0x00000000

This register contains the acpu fiq enable signals

WCSS_A_MCU_FIQ_EN

Bits	Name	Description
31	SW_1_FIQ_EN	When set, sw 1 interrupt is enabled.
30	RXP_ERR_FIQ_EN	When set, rxp error interrupt is enabled.
29	QTA_FIQ_EN	When set, qta interrupt is enabled.
28	MTU_TIMER7_EN	When set, mtu timer7/tsf_timer7 interrupt is enabled. See register 'MTU_TIMER_INTERRUPT_SOURCE' for timer selection
27	MTU_TIMER6_EN	When set, mtu timer6/tsf_timer6 interrupt is enabled. See register 'MTU_TIMER_INTERRUPT_SOURCE' for timer selection
26	PHY_HIF_FIQ_EN	When set, phy hif interrupt is enabled.
25	WQ_DATA_AVAIL_FIQ_EN	When set, wq dataavailable interrupt is enabled.
24	PHY_FIQ_IRQ_EN	When set, phy irq interrupt is enabled.
23	PHY_FIQ_FIQ_EN	When set, phy fiq interrupt is enabled.
22	TPE_FIQ_EN	When set, tpe_mcu_int interrupt is enabled.
21	MIF_FIQ_EN	When set, mif interrupt is enabled.
20	BT_INACTIVE_FIQ_EN	When set, the bt inactive interrupt is enabled.
19	BT_ACTIVE_FIQ_EN	When set, the bt active interrupt is enabled.
18	MTU_TIMER5_EN	When set, mtu timer5/tsf_timer5 interrupt is enabled. See register 'MTU_TIMER_INTERRUPT_SOURCE' for timer selection
17	ADU_FIQ_EN	When set, adu interrupt is enabled.
16	MTU_TIMER4_EN	When set, mtu timer4/tsf_timer4 interrupt is enabled. See register 'MTU_TIMER_INTERRUPT_SOURCE' for timer selection
15	RPE_FIQ_EN	When set, rpe interrupt is enabled.
13	COMBINED_FIQ_EN	When set, the combined txp/dpu/dbr/ahb2phy interrupt is enabled. See combined_int related registers for detail
12	RXP_FIQ_EN	When set, rxp beacon related interrupt is enabled.
7	HOST_TO_ACPU_MB3_FIQ_EN	When set, host to acpu MB3 interrupt is enabled.

WCSS_A_MCU_FIQ_EN (cont.)

Bits	Name	Description
6	HOST_TO_ACPU_MB2_FIQ_EN	When set, host to acpu MB2 interrupt is enabled.
5	HOST_TO_ACPU_MB1_FIQ_EN	When set, host to acpu MB1 interrupt is enabled.
4	HOST_TO_ACPU_MB0_FIQ_EN	When set, host to acpu MB0 interrupt is enabled.
3	MTU_FIQ_EN	When set, combined mtu interrupt is enabled. See mtu_int related registers for detail
2	SW_0_FIQ_EN	When set, sw 0 interrupt is enabled.
1	SCU_UART_FIQ_EN	When set, uart interrupt is enabled.
0	MTU_WD_1_FIQ_EN	When set, mtu watchdog timer (warning) interrupt is enabled.

0x0A080E3C WCSS_A_MCU_FIQ_STATUS**Type:** R**Reset State:** 0x00000000

This register contains the acpu fiq raw status signals

WCSS_A_MCU_FIQ_STATUS

Bits	Name	Description
31	SW_1_FIQ_STATUS	sw 1 interrupt status.
30	RXP_ERR_FIQ_STATUS	rxp error interrupt status.
29	QTA_FIQ_STATUS	qta interrupt status.
28	MTU_TIMER7_STATUS	When set, mtu timer7/tsf_timer7 interrupt status See register 'MTU_TIMER_INTERRUPT_SOURCE' for timer selection
27	MTU_TIMER6_STATUS	When set, mtu timer6/tsf_timer6 interrupt status See register 'MTU_TIMER_INTERRUPT_SOURCE' for timer selection
26	PHY_HIF_FIQ_STATUS	phy hif interrupt status.
25	WQ_DATA_AVAIL_FIQ_STATUS	wq dataavailable interrupt status.
24	PHY_FIQ_IRQ_STATUS	phy irq interrupt status.
23	PHY_FIQ_FIQ_STATUS	phy fiq interrupt status.
22	TPE_FIQ_STATUS	tpe_mcu_int interrupt status.
21	MIF_FIQ_STATUS	mif interrupt status.
20	BT_INACTIVE_FIQ_STATUS	bt inactive interrupt status.

WCSS_A_MCU_FIQ_STATUS (cont.)

Bits	Name	Description
19	BT_ACTIVE_FIQ_STATUS	bt active interrupt status.
18	MTU_TIMER5_STATUS	When set, mtu timer5/tsf_timer5 interrupt status See register 'MTU_TIMER_INTERRUPT_SOURCE' for timer selection
17	ADU_FIQ_STATUS	adu interrupt status.
16	MTU_TIMER4_STATUS	When set, mtu timer4/tsf_timer4 interrupt status See register 'MTU_TIMER_INTERRUPT_SOURCE' for timer selection
15	RPE_FIQ_STATUS	rpe interrupt status.
13	COMBINED_FIQ_STATUS	the combined txp/dpu/dbr/ahb2phy interrupt status. See combined_int related registers for detail
12	RXP_FIQ_STATUS	rxp beacon related interrupt status.
7	HOST_TO_ACPU_MB3_FIQ_STATUS	host to acpu MB3 interrupt status.
6	HOST_TO_ACPU_MB2_FIQ_STATUS	host to acpu MB2 interrupt status.
5	HOST_TO_ACPU_MB1_FIQ_STATUS	host to acpu MB1 interrupt status.
4	HOST_TO_ACPU_MB0_FIQ_STATUS	host to acpu MB0 interrupt status.
3	MTU_FIQ_STATUS	combined mtu interrupt status. See mtu_int related registers for detail
2	SW_0_FIQ_STATUS	sw 0 interrupt status.
1	SCU_UART_FIQ_STATUS	uart interrupt status.
0	MTU_WD_1_FIQ_STATUS	mtu watchdog timer (warning) interrupt status.

0x0A080E40 WCSS_A_MCU_FIQ_MASKED_STATUS**Type:** R**Reset State:** 0x00000000

This register contains the acpu fiq masked status

WCSS_A_MCU_FIQ_MASKED_STATUS

Bits	Name	Description
31	SW_1_FIQ_MASKED_STATUS	sw 1 interrupt masked status.
30	RXP_ERR_FIQ_MASKED_STATUS	rxp error interrupt masked status.

WCSS_A_MCU_FIQ_MASKED_STATUS (cont.)

Bits	Name	Description
29	QTA_FIQ_MASKED_STATUS	qta interrupt masked status.
28	MTU_TIMER7_MASKED_STATUS	When set, mtu timer7/tsf_timer7 masked status See register 'MTU_TIMER_INTERRUPT_SOURCE' for timer selection
27	MTU_TIMER6_MASKED_STATUS	When set, mtu timer6/tsf_timer6 masked status See register 'MTU_TIMER_INTERRUPT_SOURCE' for timer selection
26	PHY_HIF_FIQ_MASKED_STATUS	phy hif interrupt masked status.
25	WQ_DATA_AVAIL_FIQ_MASKED_STATUS	wq dataavailable interrupt masked status.
24	PHY_FIQ_IRQ_MASKED_STATUS	phy irq interrupt masked status.
23	PHY_FIQ_FIQ_MASKED_STATUS	phy fiq interrupt masked status.
22	TPE_FIQ_MASKED_STATUS	tpe_mcu_int interrupt masked status.
21	MIF_FIQ_MASKED_STATUS	mif interrupt masked status.
20	BT_INACTIVE_FIQ_MASKED_STATUS	bt inactive interrupt masked status.
19	BT_ACTIVE_FIQ_MASKED_STATUS	bt active interrupt masked status.
18	MTU_TIMER5_MASKED_STATUS	When set, mtu timer5/tsf_timer5 masked status See register 'MTU_TIMER_INTERRUPT_SOURCE' for timer selection
17	ADU_FIQ_MASKED_STATUS	adu interrupt masked status.
16	MTU_TIMER4_MASKED_STATUS	When set, mtu timer4/tsf_timer4 masked status See register 'MTU_TIMER_INTERRUPT_SOURCE' for timer selection
15	RPE_FIQ_MASKED_STATUS	rpe interrupt masked status.
13	COMBINED_FIQ_MASKED_STATUS	the combined txp/dpu/dbr/ahb2phy interrupt masked status. See combined_int related registers for detail
12	RXP_FIQ_MASKED_STATUS	rxp beacon related interrupt masked status.
11	DXE_MXU_CHAIN_2_FIQ_MASKED_STATUS	dx channel 2 interrupt masked status.
10	DXE_MXU_CHAIN_1_FIQ_MASKED_STATUS	dx channel 1 interrupt masked status.
9	DXE_MXU_CHAIN_0_FIQ_MASKED_STATUS	dx channel 0 interrupt masked status.
8	PMU_PWR_UP_FIQ_MASKED_STATUS	pmu_pwr_up interrupt masked status.

WCSS_A_MCU_FIQ_MASKED_STATUS (cont.)

Bits	Name	Description
7	HOST_TO_ACPU_MB3_FIQ_MASKED_STATUS	host to acpu MB3 interrupt masked status.
6	HOST_TO_ACPU_MB2_FIQ_MASKED_STATUS	host to acpu MB2 interrupt masked status.
5	HOST_TO_ACPU_MB1_FIQ_MASKED_STATUS	host to acpu MB1 interrupt masked status.
4	HOST_TO_ACPU_MB0_FIQ_MASKED_STATUS	host to acpu MB0 interrupt masked status.
3	MTU_FIQ_MASKED_STATUS	combined mtu interrupt masked status. See mtu_int related registers for detail
2	SW_0_FIQ_MASKED_STATUS	sw 0 interrupt masked status.
1	SCU_UART_FIQ_MASKED_STATUS	uart interrupt masked status.
0	MTU_WD_1_FIQ_MASKED_STATUS	mtu watchdog timer (warning) interrupt masked status.

0x0A080E44 WCSS_A_MCU_FIQ_CLEAR**Type:** W**Reset State:** 0x00000000

This register contains the acpu fiq clear signals

WCSS_A_MCU_FIQ_CLEAR

Bits	Name	Description
31	SW_1_FIQ_CLEAR	When set, sw 1 interrupt is cleared.
30	RXP_ERR_FIQ_CLEAR	When set, rxp error interrupt is cleared.
29	QTA_FIQ_CLEAR	When set, qta interrupt is cleared.
28	MTU_TIMER7_CLEAR	When set, mtu timer7/tsf_timer7 interrupt is cleared
27	MTU_TIMER6_CLEAR	When set, mtu timer6/tsf_timer6 interrupt is cleared
26	PHY_HIF_FIQ_CLEAR	When set, phy hif interrupt is cleared.
25	WQ_DATA_AVAIL_FIQ_CLEAR	When set, wq dataavailable interrupt is cleared.
24	PHY_FIQ_IRQ_CLEAR	When set, phy irq interrupt is cleared.
23	PHY_FIQ_FIQ_CLEAR	When set, phy fiq interrupt is cleared.
22	TPE_FIQ_CLEAR	When set, tpe_mcu_int interrupt is cleared.

WCSS_A_MCU_FIQ_CLEAR (cont.)

Bits	Name	Description
21	MIF_FIQ_CLEAR	When set, mif interrupt is cleared.
20	BT_INACTIVE_FIQ_CLEAR	When set, bt inactive interrupt is cleared.
19	BT_ACTIVE_FIQ_CLEAR	When set, bt active interrupt is cleared.
18	MTU_TIMER5_CLEAR	When set, mtu timer5/tsf_timer5 interrupt is cleared
17	ADU_FIQ_CLEAR	When set, adu interrupt is cleared.
16	MTU_TIMER4_CLEAR	When set, mtu timer4/tsf_timer4 interrupt is cleared
15	RPE_FIQ_CLEAR	When set, rpe interrupt is cleared.
13	COMBINED_FIQ_CLEAR	When set, the combined txp/dpu/dbt/ahb2phy interrupt is cleared. See combined_int related registers for detail
12	RXP_FIQ_CLEAR	When set, rxp beacon related interrupt is cleared.
7	HOST_TO_ACPU_MB3_FIQ_CLEAR	When set, host to acpu MB3 interrupt is cleared.
6	HOST_TO_ACPU_MB2_FIQ_CLEAR	When set, host to acpu MB2 interrupt is cleared.
5	HOST_TO_ACPU_MB1_FIQ_CLEAR	When set, host to acpu MB1 interrupt is cleared.
4	HOST_TO_ACPU_MB0_FIQ_CLEAR	When set, host to acpu MB0 interrupt is cleared.
3	MTU_FIQ_CLEAR	When set, combined mtu interrupt is cleared. See mtu_int related registers for detail
2	SW_0_FIQ_CLEAR	When set, sw 0 interrupt is cleared.
1	SCU_UART_FIQ_CLEAR	When set, uart interrupt is cleared.
0	MTU_WD_1_FIQ_CLEAR	When set, mtu watchdog timer (warning) interrupt is cleared.

0x0A080E48 WCSS_A_MCU_IRQ_EN**Type:** RW**Reset State:** 0x00000000

This register contains the acpu irq enable signals

WCSS_A_MCU_IRQ_EN

Bits	Name	Description
31	SW_1_IRQ_EN	When set, sw 1 interrupt is enabled.
30	RXP_ERR_IRQ_EN	When set, rxp error interrupt is enabled.
29	QTA_IRQ_EN	When set, qta interrupt is enabled.

WCSS_A_MCU_IRQ_EN (cont.)

Bits	Name	Description
28	MTU_TIMER7_EN	When set, mtu timer7/tsf_timer7 interrupt is enabled. See register 'MTU_TIMER_INTERRUPT_SOURCE' for timer selection
27	MTU_TIMER6_EN	When set, mtu timer6/tsf_timer6 interrupt is enabled. See register 'MTU_TIMER_INTERRUPT_SOURCE' for timer selection
26	PHY_HIF_IRQ_EN	When set, phy hif interrupt is enabled.
25	WQ_DATA_AVAIL_IRQ_EN	When set, wq dataavailable interrupt is enabled.
24	PHY_IRQ_IRQ_EN	When set, phy irq interrupt is enabled.
23	PHY_FIQ_IRQ_EN	When set, phy fiq interrupt is enabled.
22	TPE_IRQ_EN	When set, tpe_mcu_int interrupt is enabled.
21	MIF_IRQ_EN	When set, mif interrupt is enabled.
20	BT_INACTIVE_IRQ_EN	When set, bt inactive interrupt is enabled.
19	BT_ACTIVE_IRQ_EN	When set, bt active interrupt is enabled.
18	MTU_TIMER5_EN	When set, mtu timer5/tsf_timer5 interrupt is enabled. See register 'MTU_TIMER_INTERRUPT_SOURCE' for timer selection
17	ADU_IRQ_EN	When set, adu interrupt is enabled.
16	MTU_TIMER4_EN	When set, mtu timer4/tsf_timer4 interrupt is enabled. See register 'MTU_TIMER_INTERRUPT_SOURCE' for timer selection
15	RPE_IRQ_EN	When set, rpe interrupt is enabled.
13	COMBINED_IRQ_EN	When set, the combined txp/dpu/dbr/ahb2phy interrupt is enabled. See combined_int related registers for detail
12	RXP_IRQ_EN	When set, rxp beacon related interrupt is enabled.
7	HOST_TO_ACPU_MB3_IRQ_EN	When set, host to acpu MB3 interrupt is enabled.
6	HOST_TO_ACPU_MB2_IRQ_EN	When set, host to acpu MB2 interrupt is enabled.
5	HOST_TO_ACPU_MB1_IRQ_EN	When set, host to acpu MB1 interrupt is enabled.
4	HOST_TO_ACPU_MB0_IRQ_EN	When set, host to acpu MB0 interrupt is enabled.
3	MTU_IRQ_EN	When set, combined mtu interrupt is enabled. See mtu_int related registers for detail
2	SW_0_IRQ_EN	When set, sw 0 interrupt is enabled.
1	SCU_UART_IRQ_EN	When set, uart interrupt is enabled.
0	MTU_WD_1_IRQ_EN	When set, mtu watchdog timer (warning) interrupt is enabled.

0x0A080E4C WCSS_A_MCU_IRQ_STATUS**Type:** R**Reset State:** 0x00000000

This register contains the acpu irq raw status signals

WCSS_A_MCU_IRQ_STATUS

Bits	Name	Description
31	SW_1_IRQ_STATUS	sw 1 interrupt status.
30	RXP_ERR_IRQ_STATUS	rxp error interrupt status.
29	QTA_IRQ_STATUS	qta interrupt status.
28	MTU_TIMER7_STATUS	When set, mtu timer7/tsf_timer7 interrupt status See register 'MTU_TIMER_INTERRUPT_SOURCE' for timer selection
27	MTU_TIMER6_STATUS	When set, mtu timer6/tsf_timer6 interrupt status See register 'MTU_TIMER_INTERRUPT_SOURCE' for timer selection
26	PHY_HIF_IRQ_STATUS	phy hif interrupt status.
25	WQ_DATA_AVAIL_IRQ_STATUS	wq dataavailable interrupt status.
24	PHY_IRQ_IRQ_STATUS	phy irq interrupt status.
23	PHY_FIQ_IRQ_STATUS	phy fiq interrupt status.
22	TPE_IRQ_STATUS	tpe_mcu_int interrupt status.
21	MIF_IRQ_STATUS	mif interrupt status.
20	BT_INACTIVE_IRQ_STATUS	bt inactive interrupt status.
19	BT_ACTIVE_IRQ_STATUS	bt_active interrupt status.
18	MTU_TIMER5_STATUS	When set, mtu timer5/tsf_timer5 interrupt status See register 'MTU_TIMER_INTERRUPT_SOURCE' for timer selection
17	ADU_IRQ_STATUS	adu interrupt status.
16	MTU_TIMER4_STATUS	When set, mtu timer4/tsf_timer4 interrupt status See register 'MTU_TIMER_INTERRUPT_SOURCE' for timer selection
15	RPE_IRQ_STATUS	rpe interrupt status.
13	COMBINED_IRQ_STATUS	the combined txp/dpu/dbr/ahb2phy interrupt status. See combined_int related registers for detail
12	RXP_IRQ_STATUS	rxp beacon related interrupt status.
7	HOST_TO_ACPU_MB3_IRQ_STATUS	host to acpu MB3 interrupt status.
6	HOST_TO_ACPU_MB2_IRQ_STATUS	host to acpu MB2 interrupt status.

WCSS_A_MCU_IRQ_STATUS (cont.)

Bits	Name	Description
5	HOST_TO_ACPU_MB1_IRQ_STATUS	host to acpu MB1 interrupt status.
4	HOST_TO_ACPU_MB0_IRQ_STATUS	host to acpu MB0 interrupt status.
3	MTU_IRQ_STATUS	combined mtu interrupt status. See mtu_int related registers for detail
2	SW_0_IRQ_STATUS	sw 0 interrupt status.
1	SCU_UART_IRQ_STATUS	uart interrupt status.
0	MTU_WD_1_IRQ_STATUS	mtu watchdog timer (warning) interrupt status.

0x0A080E50 WCSS_A_MCU_IRQ_MASKED_STATUS**Type:** R**Reset State:** 0x00000000

This register contains the acpu irq masked status

WCSS_A_MCU_IRQ_MASKED_STATUS

Bits	Name	Description
31	SW_1_IRQ_MASKED_STATUS	sw 1 interrupt masked status.
30	RXP_ERR_IRQ_MASKED_STATUS	rxp error interrupt masked status.
29	QTA_IRQ_MASKED_STATUS	qta interrupt masked status.
28	MTU_TIMER7_MASKED_STATUS	When set, mtu timer7/tsf_timer7 masked status See register 'MTU_TIMER_INTERRUPT_SOURCE' for timer selection
27	MTU_TIMER6_MASKED_STATUS	When set, mtu timer6/tsf_timer6 masked status See register 'MTU_TIMER_INTERRUPT_SOURCE' for timer selection
26	PHY_HIF_IRQ_MASKED_STATUS	phy hif interrupt masked status.
25	WQ_DATA_AVAIL_IRQ_MASKED_STATUS	wq dataavailable interrupt masked status.
24	PHY_IRQ_IRQ_MASKED_STATUS	phy irq interrupt masked status.
23	PHY_FIQ_IRQ_MASKED_STATUS	phy fiq interrupt masked status.
22	TPE_IRQ_MASKED_STATUS	tpe_mcu_int interrupt masked status.

WCSS_A_MCU_IRQ_MASKED_STATUS (cont.)

Bits	Name	Description
21	MIF_IRQ_MASKED_STATUS	mif interrupt masked status.
20	BT_INACTIVE_IRQ_MASKED_STATUS	bt inactive interrupt masked status.
19	BT_ACTIVE_IRQ_MASKED_STATUS	bt_active interrupt masked status.
18	MTU_TIMER5_MASKED_STATUS	When set, mtu timer5/tsf_timer5 masked status See register 'MTU_TIMER_INTERRUPT_SOURCE' for timer selection
17	ADU_IRQ_MASKED_STATUS	adu interrupt masked status.
16	MTU_TIMER4_MASKED_STATUS	When set, mtu timer4/tsf_timer4 masked status See register 'MTU_TIMER_INTERRUPT_SOURCE' for timer selection
15	RPE_IRQ_MASKED_STATUS	rpe interrupt masked status.
13	COMBINED_IRQ_MASKED_STATUS	the combined txp/dpu/dbt/ahb2phy interrupt masked status. See combined_int related registers for detail
12	RXP_IRQ_MASKED_STATUS	rxp beacon related interrupt masked status.
7	HOST_TO_ACPU_MB3_IRQ_MASKED_STATUS	host to acpu MB3 interrupt masked status.
6	HOST_TO_ACPU_MB2_IRQ_MASKED_STATUS	host to acpu MB2 interrupt masked status.
5	HOST_TO_ACPU_MB1_IRQ_MASKED_STATUS	host to acpu MB1 interrupt masked status.
4	HOST_TO_ACPU_MB0_IRQ_MASKED_STATUS	host to acpu MB0 interrupt masked status.
3	MTU_IRQ_MASKED_STATUS	combined mtu interrupt masked status. See mtu_int related registers for detail
2	SW_0_IRQ_MASKED_STATUS	sw 0 interrupt masked status.
1	SCU_UART_IRQ_MASKED_STATUS	uart interrupt masked status.
0	MTU_WD_1_IRQ_MASKED_STATUS	mtu watchdog timer (warning) interrupt masked status.

0x0A080E54 WCSS_A_MCU_IRQ_CLEAR**Type:** W**Reset State:** 0x00000000

This register contains the acpu irq clear signals

WCSS_A_MCU_IRQ_CLEAR

Bits	Name	Description
31	SW_1_IRQ_CLEAR	When set, sw 1 interrupt is cleared.
30	RXP_ERR_IRQ_CLEAR	When set, rxp error interrupt is cleared.
29	QTA_IRQ_CLEAR	When set, qta interrupt is cleared.
28	MTU_TIMER7_CLEAR	When set, mtu timer7/tsf_timer7 interrupt is cleared
27	MTU_TIMER6_CLEAR	When set, mtu timer6/tsf_timer6 interrupt is cleared
26	PHY_HIF_IRQ_CLEAR	When set, phy hif interrupt is cleared.
25	WQ_DATA_AVAIL_IRQ_CLEAR	When set, wq dataavailable interrupt is cleared.
24	PHY_IRQ_IRQ_CLEAR	When set, phy irq interrupt is cleared.
23	PHY_FIQ_IRQ_CLEAR	When set, phy fiq interrupt is cleared.
22	TPE_IRQ_CLEAR	When set, tpe_mcu_int interrupt is cleared.
21	MIF_IRQ_CLEAR	When set, mif interrupt is cleared.
20	BT_INACTIVE_IRQ_CLEAR	When set, bt inactive interrupt is cleared.
19	BT_ACTIVE_IRQ_CLEAR	When set, bt active interrupt is cleared.
18	MTU_TIMER5_CLEAR	When set, mtu timer5/tsf_timer5 interrupt is cleared
17	ADU_IRQ_CLEAR	When set, adu interrupt is cleared.
16	MTU_TIMER4_CLEAR	When set, mtu timer4/tsf_timer4 interrupt is cleared
15	RPE_IRQ_CLEAR	When set, rpe interrupt is cleared.
13	COMBINED_IRQ_CLEAR	When set, the combined txp/dpu/dbt/ahb2phy interrupt is cleared. See combined_int related registers for detail
12	RXP_IRQ_CLEAR	When set, rxp beacon related interrupt is cleared.
7	HOST_TO_ACPU_MB3_IRQ_CLEAR	When set, host to acpu MB3 interrupt is cleared.
6	HOST_TO_ACPU_MB2_IRQ_CLEAR	When set, host to acpu MB2 interrupt is cleared.
5	HOST_TO_ACPU_MB1_IRQ_CLEAR	When set, host to acpu MB1 interrupt is cleared.
4	HOST_TO_ACPU_MB0_IRQ_CLEAR	When set, host to acpu MB0 interrupt is cleared.
3	MTU_IRQ_CLEAR	When set, combined mtu interrupt is cleared. See mtu_int related registers for detail
2	SW_0_IRQ_CLEAR	When set, sw 0 interrupt is cleared.
1	SCU_UART_IRQ_CLEAR	When set, uart interrupt is cleared.
0	MTU_WD_1_IRQ_CLEAR	When set, mtu watchdog timer (warning) interrupt is cleared.

0x0A080E58 WCSS_A_MCU_MTU_INT_EN**Type:** RW**Reset State:** 0x00000000

This register contains the mtu interrupt enable signals

WCSS_A_MCU_MTU_INT_EN

Bits	Name	Description
14	MTU_MCU_TX_BOUNDARY_INT_EN	When set, mtu-tpe transmit boundary interrupt is enabled
13	MTU_ERR_EN	When set, mtu error interrupt is enabled
12	WD_ENABLE_DISABLE_ERROR_EN	When set, mtu wd_enable_disable_error interrupt is enabled
11	WD_PROTECTION_ERROR_EN	When set, mtu wd_protection_error interrupt is enabled
10	RESP_TIMEOUT_MISSING_PKT_PUSH_EN	When set, mtu resp_timeout_missing_pkt_push interrupt is enabled
9	RESP_TIMEOUT_MISSING_PKTDET_EN	When set, mtu resp_timeout_missing_pktdet interrupt is enabled
8	RESP_TIMEOUT_MISSING_EARLY_PKTDET_EN	When set, mtu resp_timeout_missing_early_pktdet interrupt is enabled
7	MTU_PIFS_TO_EN	When set, mtu pifs timeout interrupt is enabled
6	MTU_SIFS_TO_EN	When set, mtu sifs timeout interrupt is enabled
5	MTU_BKOF1_EN	When set, mtu bkof1 interrupt is enabled
4	MTU_BKOF0_EN	When set, mtu bkof0 interrupt is enabled
3	MTU_TIMER3_EN	When set, mtu timer3/tsf_timer3 interrupt is enabled See register 'MTU_TIMER_INTERRUPT_SOURCE' for timer selection
2	MTU_TIMER2_EN	When set, mtu timer2/tsf_timer2 interrupt is enabled See register 'MTU_TIMER_INTERRUPT_SOURCE' for timer selection
1	MTU_TIMER1_EN	When set, mtu timer1/tsf_timer1 interrupt is enabled See register 'MTU_TIMER_INTERRUPT_SOURCE' for timer selection
0	MTU_TIMER0_EN	When set, mtu timer0/tsf_timer0 interrupt is enabled See register 'MTU_TIMER_INTERRUPT_SOURCE' for timer selection

0x0A080E5C WCSS_A_MCU_MTU_INT_STATUS**Type:** R**Reset State:** 0x00000000

This register contains the mtu interrupt raw status signals

WCSS_A_MCU_MTU_INT_STATUS

Bits	Name	Description
14	MTU_MCU_TX_BOUNDARY_INT_STATUS	mtu-tpe transmit boundary interrupt status
13	MTU_ERR_STATUS	mtu error interrupt status
12	WD_ENABLE_DISABLE_ERROR_STATUS	mtu wd_enable_disable_error interrupt status
11	WD_PROTECTION_ERROR_STATUS	mtu wd_protection_error interrupt status
10	RESP_TIMEOUT_MISSING_PKT_PUSH_STATUS	mtu resp_timeout_missing_pkt_push interrupt status
9	RESP_TIMEOUT_MISSING_PKTDET_STATUS	mtu resp_timeout_missing_pktdet interrupt status
8	RESP_TIMEOUT_MISSING_EARLY_PKTDET_STATUS	mtu resp_timeout_missing_early_pktdet interrupt status
7	MTU_PIFS_TO_STATUS	mtu pifs timeout interrupt status
6	MTU_SIFS_TO_STATUS	mtu sifs timeout interrupt status
5	MTU_BKOF1_STATUS	mtu bkof1 interrupt status
4	MTU_BKOF0_STATUS	mtu bkof0 interrupt status
3	MTU_TIMER3_STATUS	mtu timer3/tsf timer3 interrupt status
2	MTU_TIMER2_STATUS	mtu timer2/tsf timer2 interrupt status
1	MTU_TIMER1_STATUS	mtu timer1/tsf timer1 interrupt status
0	MTU_TIMER0_STATUS	mtu timer0/tsf timer0 interrupt status

0x0A080E60 WCSS_A_MCU_MTU_INT_MASKED_STATUS**Type:** R**Reset State:** 0x00000000

This register contains the mtu interrupt masked status signals

WCSS_A_MCU_MTU_INT_MASKED_STATUS

Bits	Name	Description
14	MTU_MCU_TX_BOUNDARY_INT_MASK_STATUS	mtu-tpe transmit boundary interrupt masked status
13	MTU_ERR_MASK_STATUS	mtu error interrupt masked status
12	WD_ENABLE_DISABLE_ERROR_MASK_STATUS	mtu wd_enable_disable_error interrupt masked status

WCSS_A_MCU_MTU_INT_MASKED_STATUS (cont.)

Bits	Name	Description
11	WD_PROTECTION_ERROR_MASK_STATUS	mtu wd_protection_error interrupt masked status
10	RESP_TIMEOUT_MISSING_PKT_PUSH_MASK_STATUS	mtu resp_timeout_missing_pkt_push interrupt masked status
9	RESP_TIMEOUT_MISSING_PKTDET_MASK_STATUS	mtu resp_timeout_missing_pktdet interrupt masked status
8	RESP_TIMEOUT_MISSING_EARLY_PKTDET_MASK_STATUS	mtu resp_timeout_missing_early_pktdet interrupt masked status
7	MTU_PIFS_TO_MASK_STATUS	mtu pifs timeout interrupt masked status
6	MTU_SIFS_TO_MASK_STATUS	mtu sifs timeout interrupt masked status
5	MTU_BKOF1_MASK_STATUS	mtu bkof1 interrupt masked status
4	MTU_BKOF0_MASK_STATUS	mtu bkof0 interrupt masked status
3	MTU_TIMER3_MASK_STATUS	mtu timer3/tsf timer3 interrupt masked status
2	MTU_TIMER2_MASK_STATUS	mtu timer2/tsf timer2 interrupt masked status
1	MTU_TIMER1_MASK_STATUS	mtu timer1/tsf timer1 interrupt masked status
0	MTU_TIMER0_MASK_STATUS	mtu timer0/tsf timer0 interrupt masked status

0x0A080E64 WCSS_A_MCU_MTU_INT_CLEAR**Type:** W**Reset State:** 0x00000000

This register contains the mtu interrupt clear signals

WCSS_A_MCU_MTU_INT_CLEAR

Bits	Name	Description
14	MTU_MCU_TX_BOUNDARY_INT_CLEAR	When set, mtu-tpe transmit boundary interrupt is cleared
13	MTU_ERR_CLEAR	When set, mtu error interrupt is cleared
12	WD_ENABLE_DISABLE_ERROR_CLEAR	When set, mtu wd_enable_disable_error interrupt is cleared

WCSS_A_MCU_MTU_INT_CLEAR (cont.)

Bits	Name	Description
11	WD_PROTECTION_ERROR_CLEAR	When set, mtu wd_protection_error interrupt is cleared
10	RESP_TIMEOUT_MISSING_PKT_PUSH_CLEAR	When set, mtu resp_timeout_missing_pkt_push interrupt is cleared
9	RESP_TIMEOUT_MISSING_PKTDET_CLEAR	When set, mtu resp_timeout_missing_pktdet interrupt is cleared
8	RESP_TIMEOUT_MISSING_EARLY_PKTDET_CLEAR	When set, mtu resp_timeout_missing_early_pktdet interrupt is cleared
7	MTU_PIFS_TO_CLEAR	When set, mtu pifs timeout interrupt is cleared
6	MTU_SIFS_TO_CLEAR	When set, mtu sifs timeout interrupt is cleared
5	MTU_BKOF1_CLEAR	When set, mtu bkof1 interrupt is cleared
4	MTU_BKOF0_CLEAR	When set, mtu bkof0 interrupt is cleared
3	MTU_TIMER3_CLEAR	When set, mtu timer3/tsf timer3 interrupt is cleared
2	MTU_TIMER2_CLEAR	When set, mtu timer2/tsf timer2 interrupt is cleared
1	MTU_TIMER1_CLEAR	When set, mtu timer1/tsf timer1 interrupt is cleared
0	MTU_TIMER0_CLEAR	When set, mtu timer0/tsf timer0 interrupt is cleared

0x0A080E68 WCSS_A_MCU_COMBINED_INT_EN**Type:** RW**Reset State:** 0x00000000

This register contains the combined interrupt enable signals

WCSS_A_MCU_COMBINED_INT_EN

Bits	Name	Description
24	CSU_MCU_INT_EN	When set, the csu error interrupt is enabled
23	WLAN_BRIDGE_AHB_ERR_INT_EN	When set, the wlan bridge ahb error interrupt is enabled
21	BTC_PATTERN_DET_INT_EN	When set, btc_pattern_det_int_p interrupt is enabled
20	BTC_SCO_INT_EN	When set, btc_sco_int_p interrupt is enabled
19	TPE_MCU_WRONG_SSN_OR_TID_ERROR_P_EN	When set, tpe_mcu_wrong_ssn_or_tid_error_p interrupt is enabled
18	TPE_MCU_BD_BASED_TX_INT_1_P_EN	When set, tpe_mcu_bd_based_tx_int_1_p interrupt is enabled

WCSS_A_MCU_COMBINED_INT_EN (cont.)

Bits	Name	Description
17	TPE_MCU_BD_BASED_TX_INT_0_P_EN	When set, tpe_mcu_bd_based_tx_int_0_p interrupt is enabled
16	TPE_MCU_ERROR_VAILD_P_EN	When set, tpe_mcu_error_vaild_p interrupt is enabled
15	TPE_MCU_GOT_ACK_FOR_BACK_P_EN	When set, tpe_mcu_got_ack_for_back_p interrupt is enabled
14	BMU_MCU_PWRSTATE_DROP_INT_EN	When set, bmu_mcu_pwrstate_drop_interrupt is enabled
13	BMU_MCU_PWRSTATE_CHANGE_INT_EN	When set, bmu_mcu_pwrstate_change_interrupt is enabled
12	BMU_MCU_BTQM_QUEUE_EMPTY_INTERRUPT_EN	When set, bmu_mcu_btqm_queues_empty_interrupt interrupt is enabled
11	BMU_MCU_ERR_INT_EN	When set, bmu_mcu_err_int interrupt is enabled
10	AHB2PHY_MCU_TIMEOUT_ERROR2_EN	When set, ahb2phy_mcu_timeout_error2 interrupt is enabled
9	AHB2PHY_MCU_TIMEOUT_ERROR1_EN	When set, ahb2phy_mcu_timeout_error1 interrupt is enabled
8	DBR_MCU_GAM_ERR_EN	When set, dbr_mcu_gam_err interrupt is enabled
7	DBR_MCU_INCORRECT_LENGTH_ERR_EN	When set, dbr_mcu_incorrect_length_err interrupt is enabled
6	DPU_MCU_MICERR_P_EN	When set, dpu_mcu_micerr_p interrupt is enabled
5	DPU_MCU_ERRINTR_P_EN	When set, dpu_mcu_errintr_p interrupt is enabled
4	BMU_MCU_QUEUE_USAGE_THRESHOLD_INTERRUPT_EN	When set, bmu_mcu_queue_usage_threshold_interrupt is enabled
3	ACPU_MCU_MIF_TIMEOUT_INT_P_EN	When set, acpu_mcu_mif_timeout_int_p interrupt is enabled
2	ACPU_MCU_AHB_ERR_INT_P_EN	When set, acpu_mcu_ahb_err_int_p interrupt is enabled
1	TXP_MCU_TXP_TIMEOUT_INT_P_EN	When set, txp_mcu_txp_timeout_int_p interrupt is enabled
0	TXP_MCU_ERR_INT_EN	When set, txp_mcu_err_int interrupt is enabled

0x0A080E6C WCSS_A_MCU_COMBINED_INT_STATUS**Type:** R**Reset State:** 0x00000000

This register contains the combined interrupt raw status signals

WCSS_A_MCU_COMBINED_INT_STATUS

Bits	Name	Description
24	CSU_MCU_INT_STATUS	csu error interrupt status
23	WLAN_BRIDGE_AHB_ERR_INT_STATUS	wlan bridge ahb error interrupt status
21	BTC_PATTERN_DET_INT_P_STATUS	btc_pattern_det_int_p interrupt status
20	BTC_SCO_INT_P_STATUS	btc_sco_int_p interrupt status
19	TPE_MCU_WRONG_SSN_OR_TID_ERROR_P_STATUS	tpe_mcu_wrong_ssn_or_tid_error_p interrupt status
18	TPE_MCU_BD_BASED_TX_INT_1_P_STATUS	tpe_mcu_bd_based_tx_int_1_p interrupt status
17	TPE_MCU_BD_BASED_TX_INT_0_P_STATUS	tpe_mcu_bd_based_tx_int_0_p interrupt status
16	TPE_MCU_ERROR_VAILD_P_STATUS	tpe_mcu_error_vaild_p interrupt status
15	TPE_MCU_GOT_ACK_FOR_BACK_P_STATUS	tpe_mcu_got_ack_for_back_p interrupt status
14	BMU_MCU_PWRSTATE_DROP_INT_STATUS	bmu_mcu_pwrstate_drop_interrupt status
13	BMU_MCU_PWRSTATE_CHANGE_INT_STATUS	bmu_mcu_pwrstate_change_interrupt status
12	BMU_MCU_BTQM_QUEUE_EMPTY_INTERRUPT_STATUS	bmu_mcu_btqm_queues_empty_interrupt interrupt status
11	BMU_MCU_ERR_INT_STATUS	bmu_mcu_err_int interrupt status
10	AHB2PHY_MCU_TIMEOUT_ERROR2_STATUS	ahb2phy_mcu_timeout_error2 interrupt status
9	AHB2PHY_MCU_TIMEOUT_ERROR1_STATUS	ahb2phy_mcu_timeout_error1 interrupt status
8	DBR_MCU_GAM_ERR_STATUS	dbr_mcu_gam_err interrupt status
7	DBR_MCU_INCORRECT_LENGTH_ERR_STATUS	dbr_mcu_incorrect_length_err interrupt status
6	DPU_MCU_MICERR_P_STATUS	dpu_mcu_micerr_p interrupt status
5	DPU_MCU_ERRINTR_P_STATUS	dpu_mcu_errintr_p interrupt status
4	BMU_MCU_QUEUE_USAGE_THRESHOLD_INTERRUPT_STATUS	bmu_mcu_queue_usage_threshold_interrupt interrupt status

WCSS_A_MCU_COMBINED_INT_STATUS (cont.)

Bits	Name	Description
3	ACPU_MCU_MIF_TIMEOUT_INT_P_STATUS	acpu_mcu_mif_timeout_int_p interrupt status
2	ACPU_MCU_AHB_ERR_INT_P_STATUS	acpu_mcu_ahb_err_int_p interrupt status
1	TXP_MCU_TXP_TIMEOUT_INT_P_STATUS	txp_mcu_txp_timeout_int_p interrupt status
0	TXP_MCU_ERR_INT_STATUS	txp_mcu_err_int interrupt status

0x0A080E70 WCSS_A_MCU_COMBINED_INT_MASKED_STATUS**Type:** R**Reset State:** 0x00000000

This register contains the combined interrupt masked status signals

WCSS_A_MCU_COMBINED_INT_MASKED_STATUS

Bits	Name	Description
24	CSU_MCU_ERR_INT_MASK_STATUS	csu error interrupt masked status
23	WLAN_BRIDGE_AHB_ERR_INT_MASK_STATUS	wlan bridge ahb error interrupt masked status
21	BTC_PATTERN_DET_INT_P_MASK_STATUS	btc_pattern_det_int_p interrupt masked status
20	BTC_SCO_INT_P_MASK_STATUS	btc_sco_int_p interrupt masked status
19	TPE_MCU_WRONG_SSN_OR_TID_ERROR_P_MASK_STATUS	tpe_mcu_wrong_ssn_or_tid_error_p interrupt masked status
18	TPE_MCU_BD_BASED_TX_INT_1_P_MASK_STATUS	tpe_mcu_bd_based_tx_int_1_p interrupt masked status
17	TPE_MCU_BD_BASED_TX_INT_0_P_MASK_STATUS	tpe_mcu_bd_based_tx_int_0_p interrupt masked status
16	TPE_MCU_ERROR_VAILD_P_MASK_STATUS	tpe_mcu_error_vaild_p interrupt masked status
15	TPE_MCU_GOT_ACK_FOR_BACK_P_MASK_STATUS	tpe_mcu_got_ack_for_back_p interrupt masked status
14	BMU_MCU_PWRSTATE_DROP_INT_MASK_STATUS	bmu_mcu_pwrstate_drop_interrupt masked status
13	BMU_MCU_PWRSTATE_CHANGE_INT_MASK_STATUS	bmu_mcu_pwrstate_change_interrupt masked status

WCSS_A_MCU_COMBINED_INT_MASKED_STATUS (cont.)

Bits	Name	Description
12	BMU_MCU_BTQM_QUEUE_S_EMPTY_INTERRUPT_MASK_STATUS	bmu_mcu_btqm_queues_empty_interrupt interrupt masked status
11	BMU_MCU_ERR_INT_MASK_STATUS	bmu_mcu_err_int interrupt masked status
10	AHB2PHY_MCU_TIMEOUT_ERROR2_MASK_STATUS	ahb2phy_mcu_timeout_error2 interrupt masked status
9	AHB2PHY_MCU_TIMEOUT_ERROR1_MASK_STATUS	ahb2phy_mcu_timeout_error1 interrupt masked status
8	DBR_MCU_GAM_ERR_MASK_STATUS	dbr_mcu_gam_err interrupt masked status
7	DBR_MCU_INCORRECT_LENGTH_ERR_MASK_STATUS	dbr_mcu_incorrect_length_err interrupt masked status
6	DPU_MCU_MICERR_P_MASK_STATUS	dpu_mcu_micerr_p interrupt masked status
5	DPU_MCU_ERRINTR_P_MASK_STATUS	dpu_mcu_errintr_p interrupt masked status
4	BMU_MCU_QUEUE_USAGE_THRESHOLD_INTERRUPT_MASK_STATUS	bmu_mcu_queue_usage_threshold_interrupt interrupt masked status
3	ACPU_MCU_MIF_TIMEOUT_INT_P_MASK_STATUS	acpu_mcu_mif_timeout_int_p interrupt masked status
2	ACPU_MCU_AHB_ERR_INT_P_MASK_STATUS	acpu_mcu_ahb_err_int_p interrupt masked status
1	TXP_MCU_TXP_TIMEOUT_INT_P_MASK_STATUS	txp_mcu_txp_timeout_int_p interrupt masked status
0	TXP_MCU_ERR_INT_MASK_STATUS	txp_mcu_err_int interrupt masked status

0x0A080E74 WCSS_A_MCU_COMBINED_INT_CLEAR**Type:** RW**Reset State:** 0x00000000

This register contains the combined interrupt clear signals

WCSS_A_MCU_COMBINED_INT_CLEAR

Bits	Name	Type	Description
24	CSU_MCU_ERR_INT_CLEAR	RW	When set, csu error interrupt is cleared

WCSS_A_MCU_COMBINED_INT_CLEAR (cont.)

Bits	Name	Type	Description
23	WLAN_BRIDGE_AHB_ERR_INT_CLEAR	RW	When set, wlan bridge ahb error interrupt is cleared
21	BTC_PATTERN_DET_INT_CLEAR	RW	When set, btc_pattern_det_int_p interrupt is cleared
20	BTC_SCO_INT_CLEAR	RW	When set, btc_sco_int_p interrupt is cleared
19	TPE_MCU_WRONG_SSN_OR_TID_ERROR_P_CLEAR	W	When set, tpe_mcu_wrong_ssn_or_tid_error_p interrupt is cleared
18	TPE_MCU_BD_BASED_TX_INT_1_P_CLEAR	W	When set, tpe_mcu_bd_based_tx_int_1_p interrupt is cleared
17	TPE_MCU_BD_BASED_TX_INT_0_P_CLEAR	W	When set, tpe_mcu_bd_based_tx_int_0_p interrupt is cleared
16	TPE_MCU_ERROR_VAILD_P_CLEAR	W	When set, tpe_mcu_error_vaild_p interrupt is cleared
15	TPE_MCU_GOT_ACK_FOR_BACK_P_CLEAR	W	When set, tpe_mcu_got_ack_for_back_p interrupt is cleared
14	BMU_MCU_PWRSTATE_DROP_INT_CLEAR	W	When set, bmu_mcu_pwrstate_drop_interrupt is cleared
13	BMU_MCU_PWRSTATE_CHANGE_INT_CLEAR	W	When set, bmu_mcu_pwrstate_change_interrupt is cleared
12	BMU_MCU_BTQM_QUEUE_EMPTY_INTERRUPT_CLEAR	W	When set, bmu_mcu_btqm_queues_empty_interrupt interrupt is cleared
11	BMU_MCU_ERR_INT_CLEAR	W	When set, bmu_mcu_err_int interrupt is cleared
10	AHB2PHY_MCU_TIMEOUT_ERROR2_CLEAR	W	When set, ahb2phy_mcu_timeout_error2 interrupt is cleared
9	AHB2PHY_MCU_TIMEOUT_ERROR1_CLEAR	W	When set, ahb2phy_mcu_timeout_error1 interrupt is cleared
8	DBR_MCU_GAM_ERR_CLEAR	W	When set, dbr_mcu_gam_err interrupt is cleared
7	DBR_MCU_INCORRECT_LENGTH_ERR_CLEAR	W	When set, dbr_mcu_incorrect_length_err interrupt is cleared
6	DPU_MCU_MICERR_P_CLEAR	W	When set, dpu_mcu_micerr_p interrupt is cleared
5	DPU_MCU_ERRINTR_P_CLEAR	W	When set, dpu_mcu_errintr_p interrupt is cleared
4	BMU_MCU_QUEUE_USAGE_THRESHOLD_INTERRUPT_CLEAR	W	When set, bmu_mcu_queue_usage_threshold_interrupt_clear interrupt is cleared
3	ACPU_MCU_MIF_TIMEOUT_INT_P_CLEAR	W	When set, acpu_mcu_mif_timeout_int_p interrupt is cleared

WCSS_A_MCU_COMBINED_INT_CLEAR (cont.)

Bits	Name	Type	Description
2	ACPU_MCU_AHB_ERR_INT_P_CLEAR	W	When set, acpu_mcu_ahb_err_int_p interrupt is cleared
1	TXP_MCU_TXP_TIMEOUT_INT_P_CLEAR	W	When set, txp_mcu_txp_timeout_int_p interrupt is cleared
0	TXP_MCU_ERR_INT_CLEAR	W	When set, txp_mcu_err_int interrupt is cleared

0x0A080E78 WCSS_A_MCU_BMU_WQ_INT_EN**Type:** RW**Reset State:** 0x00000000

This register contains the bmu wq dataavailable interrupt enable signals

WCSS_A_MCU_BMU_WQ_INT_EN

Bits	Name	Description
24:2	BMU_WQ_INT_EN	BMU WQ 2-24 dataavailable

0x0A080E7C WCSS_A_MCU_BMU_WQ_INT_STATUS**Type:** R**Reset State:** 0x00000000

This register contains the bmu wq dataavailable interrupt raw status signals

WCSS_A_MCU_BMU_WQ_INT_STATUS

Bits	Name	Description
24:2	BMU_WQ_INT_STATUS	BMU WQ 2-24 dataavailable

0x0A080E80 WCSS_A_MCU_BMU_WQ_INT_MASKED_STATUS**Type:** R**Reset State:** 0x00000000

This register contains the bmu wq dataavailable interrupt masked status signals

WCSS_A_MCU_BMU_WQ_INT_MASKED_STATUS

Bits	Name	Description
24:2	BMU_WQ_INT_MASKED_STATUS	BMU WQ 2-24 dataavailable

0x0A080E84 WCSS_A_MCU_BMU_WQ_INT_CLEAR**Type:** W**Reset State:** 0x00000000

This register contains the bmu wq dataavailable interrupt clear signals

WCSS_A_MCU_BMU_WQ_INT_CLEAR

Bits	Name	Description
24:2	BMU_WQ_INT_CLEAR	BMU WQ 2-24 dataavailable

0x0A080E98 WCSS_A_MCU_SW_INT_0**Type:** RW**Reset State:** 0x00000000

This register contains the sw interrupt signals

WCSS_A_MCU_SW_INT_0

Bits	Name	Description
0	COMBINED_INT_EN	SW generated interrupt

0x0A080E9C WCSS_A_MCU_SW_INT_1**Type:** RW**Reset State:** 0x00000000

This register contains the sw interrupt signals

WCSS_A_MCU_SW_INT_1

Bits	Name	Description
0	COMBINED_INT_EN	SW generated interrupt1

0x0A080EA0 WCSS_A_MCU_MCU_HOST_INT_EN**Type:** RW**Reset State:** 0x00000000

This register contains the mcu_host interrupt enable signals

WCSS_A_MCU_MCU_HOST_INT_EN

Bits	Name	Description
11	RESERVE	This bit is reserved.
10	ADU_ERR_WQ_DA_AVAIL_EN	When set, adu_err_wq_da_avail interrupt is enabled
9	DPU_ERR_WQ_DA_AVAIL_EN	When set, dpu_err_wq_da_avail interrupt is enabled
8	MTU_TIMER3_EN	When set, mtu timer3/tsf_timer3 interrupt is enabled. See register 'MTU_TIMER_INTERRUPT_SOURCE' for timer selection
7	MTU_TIMER2_EN	When set, mtu timer2/tsf_timer2 interrupt is enabled. See register 'MTU_TIMER_INTERRUPT_SOURCE' for timer selection
6	MB3_EN	When set, mb3 interrupt is enabled
5	MB2_EN	When set, mb2 interrupt is enabled
4	MB1_EN	When set, mb1 interrupt is enabled
3	MB0_EN	When set, mb0 interrupt is enabled
2	DPU_MIC_ERROR_EN	When set, dpu mic error interrupt is enabled
1	MTU_WATCHDOG_EN	When set, mtu watchdog interrupt is enabled

0x0A080EA4 WCSS_A_MCU_MCU_HOST_INT_STATUS**Type:** R**Reset State:** 0x00000000

This register contains the mcu_host interrupt raw status signals

WCSS_A_MCU_MCU_HOST_INT_STATUS

Bits	Name	Description
11	RESERVE	This bit is reserve
10	ADU_ERR_WQ_DA_AVAIL_STATUS	adu_err_wq_da_avail status
9	DPU_ERR_WQ_DA_AVAIL_STATUS	dpu_err_wq_da_avail status

WCSS_A_MCU_MCU_HOST_INT_STATUS (cont.)

Bits	Name	Description
8	MTU_TIMER3_STATUS	mtu timer3/tsf_timer3 status See register 'MTU_TIMER_INTERRUPT_SOURCE' for timer selection
7	MTU_TIMER2_STATUS	mtu timer2/tsf_timer2 status See register 'MTU_TIMER_INTERRUPT_SOURCE' for timer selection
6	MB3_STATUS	mb3 status
5	MB2_STATUS	mb2 status
4	MB1_STATUS	mb1 status
3	MB0_STATUS	mb0 status
2	DPU_MIC_ERROR_STATUS	dpu mic error status
1	MTU_WATCHDOG_STATUS	mtu watchdog status

0x0A080EA8 WCSS_A_MCU_MCU_HOST_INT_MASKED_STATUS**Type:** R**Reset State:** 0x00000000

This register contains the mcu_host interrupt masked status signals

WCSS_A_MCU_MCU_HOST_INT_MASKED_STATUS

Bits	Name	Description
11	RESERVE	This bit is reserve
10	ADU_ERR_WQ_DA_AVAIL_MASK_STATUS	adu_err_wq_da_avail masked status
9	DPU_ERR_WQ_DA_AVAIL_MASK_STATUS	dpu_err_wq_da_avail masked status
8	MTU_TIMER3_MASK_STAT US	mtu timer2/tsf_timer2 masked status See register 'MTU_TIMER_INTERRUPT_SOURCE' for timer selection
7	MTU_TIMER2_MASK_STAT US	mtu timer2/tsf_timer2 masked status See register 'MTU_TIMER_INTERRUPT_SOURCE' for timer selection
6	MB3_MASK_STATUS	mb3 masked status
5	MB2_MASK_STATUS	mb2 masked status
4	MB1_MASK_STATUS	mb1 masked status
3	MB0_MASK_STATUS	mb0 masked status
2	DPU_MIC_ERROR_MASK_ STATUS	dpu mic error masked status
1	MTU_WATCHDOG_MASK_ STATUS	mtu watchdog masked status

0x0A080EAC WCSS_A_MCU_MCU_HOST_INT_CLEAR**Type:** W**Reset State:** 0x00000000

This register contains the mcu_host interrupt clear signals

WCSS_A_MCU_MCU_HOST_INT_CLEAR

Bits	Name	Description
11	RESERVE	This bit is reserve for future usage
10	ADU_ERR_WQ_DA_AVAIL_CLEAR	When set, clear adu_err_wq_da_avail interrupt
9	DPU_ERR_WQ_DA_AVAIL_CLEAR	When set, clear dpu_err_wq_da_avail interrupt
8	MTU_TIMER3_CLEAR	When set, clear mtu timer3/tsf_timer3 interrupt
7	MTU_TIMER2_CLEAR	When set, clear mtu timer2/tsf_timer2 interrupt
6	MB3_CLEAR	When set, clear mb3 interrupt
5	MB2_CLEAR	When set, clear mb2 interrupt
4	MB1_CLEAR	When set, clear mb1 interrupt
3	MB0_CLEAR	When set, clear mb0 interrupt
2	DPU_MIC_ERROR_CLEAR	When set, clear dpu mic error interrupt
1	MTU_WATCHDOG_CLEAR	When set, clear mtu watchdog interrupt

0x0A080EB0 WCSS_A_MCU_MCU_PMU_INFO**Type:** RW**Reset State:** 0x00000000

This register contains the RF register reinitialization is done information

WCSS_A_MCU_MCU_PMU_INFO

Bits	Name	Description
0	MCU_PMU_RF_REINIT_DONE	SW writes to this register to tell pmu that RF reg reinit is done.

0x0A080EB4 WCSS_A_MCU_MCU_WMAC_STATUS**Type:** R**Reset State:** 0x00000000

This register contains the wmac modules status

WCSS_A_MCU_MCU_WMAC_STATUS

Bits	Name	Description
11	CSU_MCU_STATUS	When set, csu is busy.
10	ACPU_MCU_STATUS	When set, acpu is busy.
9	SIF_MCU_STATUS	When set, sif is busy.
8	BTQM_MCU_STATUS	When set, btqm queue is not empty.
7	BMU_MCU_STATUS	When set, bmu wq is not empty.
6	TPE_MCU_STATUS	When set, tpe is busy.
5	TXP_MCU_STATUS	When set, txp is busy.
4	DXE_MCU_STATUS	When set, dxe is busy.
3	ADU_MCU_STATUS	When set, adu is busy.
2	DPU_MCU_STATUS	When set, dpu is busy.
1	RPE_MCU_STATUS	When set, rpe is busy.
0	RXP_MCU_STATUS	When set, rxp is busy.

0x0A080EB8 WCSS_A_MCU_MAC_HOST_INT_EN

Type: RW

Reset State: 0x00000000

This register contains the mac to host interrupt enable signals. The bit order is the same as those in the IRQ/FIQ register which are used by acpu, this is dedicated for host.

WCSS_A_MCU_MAC_HOST_INT_EN

Bits	Name	Description
31	SW_1_INT_EN	When set, sw 1 interrupt is enabled.
30	RXP_ERR_INT_EN	When set, rxp error interrupt is enabled.
29	QTA_INT_EN	When set, qta interrupt is enabled.
28	MTU_TIMER7_EN	When set, mtu timer7/tsf_timer7 interrupt is enabled. See register 'MTU_TIMER_INTERRUPT_SOURCE' for timer selection
27	MTU_TIMER6_EN	When set, mtu timer6/tsf_timer6 interrupt is enabled. See register 'MTU_TIMER_INTERRUPT_SOURCE' for timer selection
26	PHY_HIF_INT_EN	When set, phy hif interrupt is enabled.
25	WQ_DATA_AVAIL_INT_EN	When set, wq dataavailable interrupt is enabled.
24	PHY_IRQ_INT_EN	When set, phy irq interrupt is enabled.

WCSS_A_MCU_MAC_HOST_INT_EN (cont.)

Bits	Name	Description
23	PHY_FIQ_INT_EN	When set, phy fiq interrupt is enabled.
22	TPE_INT_EN	When set, tpe_mcu_int interrupt is enabled.
21	MIF_INT_EN	When set, mif interrupt is enabled.
20	BT_INACTIVE_INT_EN	When set, bt_inactive interrupt is enabled.
19	BT_ACTIVE_INT_EN	When set, bt active interrupt is enabled.
18	MTU_TIMER5_EN	When set, mtu timer5/tsf_timer5 interrupt is enabled. See register 'MTU_TIMER_INTERRUPT_SOURCE' for timer selection
17	ADU_INT_EN	When set, adu interrupt is enabled.
16	MTU_TIMER4_EN	When set, mtu timer4/tsf_timer4 interrupt is enabled. See register 'MTU_TIMER_INTERRUPT_SOURCE' for timer selection
15	RPE_INT_EN	When set, rpe interrupt is enabled.
13	COMBINED_INT_EN	When set, the combined txp/dpu/dbt/ahb2phy interrupt is enabled. See combined_int related registers for detail
12	RXP_INT_EN	When set, rxp beacon related interrupt is enabled.
7	ACPU_TO_HOST_MB3_INT_EN	When set, acpu to host MB3 interrupt is enabled.
6	ACPU_TO_HOST_MB2_INT_EN	When set, acpu to host MB2 interrupt is enabled.
5	ACPU_TO_HOST_MB1_INT_EN	When set, acpu to host MB1 interrupt is enabled.
4	ACPU_TO_HOST_MB0_INT_EN	When set, acpu to host MB0 interrupt is enabled.
3	MTU_INT_EN	When set, combined mtu interrupt is enabled. See mtu_int related registers for detail
2	SW_0_INT_EN	When set, sw 0 interrupt is enabled.
1	SCU_UART_INT_EN	When set, uart interrupt is enabled.
0	MTU_WD_1_INT_EN	When set, mtu watchdog timer (warning) interrupt is enabled.

0x0A080EBC WCSS_A_MCU_MAC_HOST_INT_STATUS**Type:** R**Reset State:** 0x00000000

This register contains the mac to host interrupt raw status signals. The bit order is the same as those in the IRQ/FIQ register which are used by acpu, this is dedicated for host.

WCSS_A_MCU_MAC_HOST_INT_STATUS

Bits	Name	Description
31	SW_1_INT_STATUS	sw 1 interrupt status.
30	RXP_ERR_INT_STATUS	rxp error interrupt status.
29	QTA_INT_STATUS	qta interrupt status
28	MTU_TIMER7_STATUS	When set, mtu timer7/tsf_timer7 interrupt status See register 'MTU_TIMER_INTERRUPT_SOURCE' for timer selection
27	MTU_TIMER6_STATUS	When set, mtu timer6/tsf_timer6 interrupt status See register 'MTU_TIMER_INTERRUPT_SOURCE' for timer selection
26	PHY_HIF_INT_STATUS	phy hif interrupt status.
25	WQ_DATA_AVAIL_INT_STATUS	wq dataavailable interrupt status.
24	PHY_IRQ_INT_STATUS	phy irq interrupt status.
23	PHY_FIQ_INT_STATUS	phy fiq interrupt status.
22	TPE_INT_STATUS	tpe_mcu_int interrupt status.
21	MIF_INT_STATUS	mif interrupt status.
20	BT_INACTIVE_INT_STATUS	bt inactive interrupt status.
19	BT_ACTIVE_INT_STATUS	bt active interrupt status.
18	MTU_TIMER5_STATUS	When set, mtu timer5/tsf_timer5 interrupt status See register 'MTU_TIMER_INTERRUPT_SOURCE' for timer selection
17	ADU_INT_STATUS	adu interrupt status.
16	MTU_TIMER4_STATUS	When set, mtu timer4/tsf_timer4 interrupt status See register 'MTU_TIMER_INTERRUPT_SOURCE' for timer selection
15	RPE_INT_STATUS	rpe interrupt status.
13	COMBINED_INT_STATUS	the combined txp/dpu/dbr/ahb2phy interrupt status. See combined_int related registers for detail
12	RXP_INT_STATUS	rxp beacon related interrupt status.
7	ACPU_TO_HOST_MB3_INT_STATUS	acpu to host MB3 interrupt status.
6	ACPU_TO_HOST_MB2_INT_STATUS	acpu to host MB2 interrupt status.
5	ACPU_TO_HOST_MB1_INT_STATUS	acpu to host MB1 interrupt status.
4	ACPU_TO_HOST_MB0_INT_STATUS	acpu to host MB0 interrupt status.
3	MTU_INT_STATUS	combined mtu interrupt status. See mtu_int related registers for detail
2	SW_0_INT_STATUS	sw 0 interrupt status.

WCSS_A_MCU_MAC_HOST_INT_STATUS (cont.)

Bits	Name	Description
1	SCU_UART_INT_STATUS	uart interrupt status.
0	MTU_WD_1_INT_STATUS	mtu watchdog timer (warning) interrupt status.

0x0A080EC0 WCSS_A_MCU_MAC_HOST_INT_MASKED_STATUS**Type:** R**Reset State:** 0x00000000

This register contains the mac to host interrupt masked status. The bit order is the same as those in the IRQ/FIQ register which are used by acpu, this is dedicated for host.

WCSS_A_MCU_MAC_HOST_INT_MASKED_STATUS

Bits	Name	Description
31	SW_1_INT_MASKED_STATUS	sw 1 interrupt masked status.
30	RXP_ERR_INT_MASKED_STATUS	rxp error interrupt masked status.
29	QTA_INT_MASKED_STATUS	qta interrupt masked status
28	MTU_TIMER7_MASKED_STATUS	When set, mtu timer7/tsf_timer7 masked status See register 'MTU_TIMER_INTERRUPT_SOURCE' for timer selection
27	MTU_TIMER6_MASKED_STATUS	When set, mtu timer6/tsf_timer6 masked status See register 'MTU_TIMER_INTERRUPT_SOURCE' for timer selection
26	PHY_HIF_INT_MASKED_STATUS	phy hif interrupt masked status.
25	WQ_DATA_AVAIL_INT_MASKED_STATUS	wq dataavailable interrupt masked status.
24	PHY_IRQ_INT_MASKED_STATUS	phy irq interrupt masked status.
23	PHY_FIQ_INT_MASKED_STATUS	phy fiq interrupt masked status.
22	TPE_INT_MASKED_STATUS	tpe_mcu_int interrupt masked status.
21	MIF_INT_MASKED_STATUS	mif interrupt masked status.
20	BT_INACTIVE_INT_MASKED_STATUS	bt inactive interrupt masked status.
19	BT_ACTIVE_INT_MASKED_STATUS	bt active interrupt masked status.

WCSS_A_MCU_MAC_HOST_INT_MASKED_STATUS (cont.)

Bits	Name	Description
18	MTU_TIMER5_MASKED_STATUS	When set, mtu timer5/tsf_timer5 masked status See register 'MTU_TIMER_INTERRUPT_SOURCE' for timer selection
17	ADU_INT_MASKED_STATUS	adu interrupt masked status.
16	MTU_TIMER4_MASKED_STATUS	When set, mtu timer4/tsf_timer4 masked status See register 'MTU_TIMER_INTERRUPT_SOURCE' for timer selection
15	RPE_INT_MASKED_STATUS	rpe interrupt masked status.
13	COMBINED_INT_MASKED_STATUS	the combined txp/dpu/dbr/ahb2phy interrupt masked status. See combined_int related registers for detail
12	RXP_INT_MASKED_STATUS	rxp beacon related interrupt masked status.
7	ACPU_TO_HOST_MB3_INT_MASKED_STATUS	acpu to host MB3 interrupt masked status.
6	ACPU_TO_HOST_MB2_INT_MASKED_STATUS	acpu to host MB2 interrupt masked status.
5	ACPU_TO_HOST_MB1_INT_MASKED_STATUS	acpu to host MB1 interrupt masked status.
4	ACPU_TO_HOST_MB0_INT_MASKED_STATUS	acpu to host MB0 interrupt masked status.
3	MTU_INT_MASKED_STATUS	combined mtu interrupt masked status. See mtu_int related registers for detail
2	SW_0_INT_MASKED_STATUS	sw 0 interrupt masked status.
1	SCU_UART_INT_MASKED_STATUS	uart interrupt masked status.
0	MTU_WD_1_INT_MASKED_STATUS	mtu watchdog timer (warning) interrupt masked status.

0x0A080EC4 WCSS_A_MCU_MAC_HOST_INT_CLEAR**Type:** W**Reset State:** 0x00000000

This register contains the mac to host interrupt clear signals. The bit order is the same as those in the IRQ/FIQ register which are used by acpu, this is dedicated for host.

WCSS_A_MCU_MAC_HOST_INT_CLEAR

Bits	Name	Description
31	SW_1_INT_CLEAR	When set, sw 1 interrupt is cleared.

WCSS_A_MCU_MAC_HOST_INT_CLEAR (cont.)

Bits	Name	Description
30	RXP_ERR_INT_CLEAR	When set, rxp error interrupt is cleared.
29	QTA_INT_CLEAR	When set, qta interrupt is cleared.
28	MTU_TIMER7_CLEAR	When set, mtu timer7/tsf_timer7 interrupt is cleared
27	MTU_TIMER6_CLEAR	When set, mtu timer6/tsf_timer6 interrupt is cleared
26	PHY_HIF_INT_CLEAR	When set, phy hif interrupt is cleared.
25	WQ_DATA_AVAIL_INT_CLEAR	When set, wq dataavailable interrupt is cleared.
24	PHY_IRQ_INT_CLEAR	When set, phy irq interrupt is cleared.
23	PHY_FIQ_INT_CLEAR	When set, phy fiq interrupt is cleared.
22	TPE_INT_CLEAR	When set, tpe_mcu_int interrupt is cleared.
21	MIF_INT_CLEAR	When set, mif interrupt is cleared.
20	BT_INACTIVE_INT_CLEAR	When set, bt inactive interrupt is cleared.
19	BT_ACTIVE_INT_CLEAR	When set, bt active interrupt is cleared.
18	MTU_TIMER5_CLEAR	When set, mtu timer5/tsf_timer5 interrupt is cleared
17	ADU_INT_CLEAR	When set, adu interrupt is cleared.
16	MTU_TIMER4_CLEAR	When set, mtu timer4/tsf_timer4 interrupt is cleared
15	RPE_INT_CLEAR	When set, rpe interrupt is cleared.
13	COMBINED_INT_CLEAR	When set, the combined txp/dpu/dbr/ahb2phy interrupt is cleared. See combined_int related registers for detail
12	RXP_INT_CLEAR	When set, rxp beacon related interrupt is cleared.
7	ACPU_TO_HOST_MB3_INT_CLEAR	When set, acpu to host MB3 interrupt is cleared.
6	ACPU_TO_HOST_MB2_INT_CLEAR	When set, acpu to host MB2 interrupt is cleared.
5	ACPU_TO_HOST_MB1_INT_CLEAR	When set, acpu to host MB1 interrupt is cleared.
4	ACPU_TO_HOST_MB0_INT_CLEAR	When set, acpu to host MB0 interrupt is cleared.
3	MTU_INT_CLEAR	When set, combined mtu interrupt is cleared. See mtu_int related registers for detail
2	SW_0_INT_CLEAR	When set, sw 0 interrupt is cleared.
1	SCU_UART_INT_CLEAR	When set, uart interrupt is cleared.
0	MTU_WD_1_INT_CLEAR	When set, mtu watchdog timer (warning) interrupt is cleared.

0x0A080EC8 WCSS_A_MCU_MTU_HOST_INT_EN**Type:** RW**Reset State:** 0x00000000

This register contains the mtu interrupt enable signals

WCSS_A_MCU_MTU_HOST_INT_EN

Bits	Name	Description
14	MTU_MCU_TX_BOUNDARY_INT_HOSTINT_EN	When set, mtu_mcu_tx_boundary_int interrupt is enabled
13	MTU_ERR_HOSTINT_EN	When set, mtu error interrupt is enabled
12	WD_ENABLE_DISABLE_ERROR_HOSTINT_EN	When set, mtu wd_enable_disable_error interrupt is enabled
11	WD_PROTECTION_ERROR_HOSTINT_EN	When set, mtu wd_protection_error interrupt is enabled
10	RESP_TIMEOUT_MISSING_PKT_PUSH_HOSTINT_EN	When set, mtu resp_timeout_missing_pkt_push interrupt is enabled
9	RESP_TIMEOUT_MISSING_PKTDET_HOSTINT_EN	When set, mtu resp_timeout_missing_pktdet interrupt is enabled
8	RESP_TIMEOUT_MISSING_EARLY_PKTDET_HOSTINT_EN	When set, mtu resp_timeout_missing_early_pktdet interrupt is enabled
7	MTU_PIFS_TO_HOSTINT_EN	When set, mtu pifs timeout interrupt is enabled
6	MTU_SIFS_TO_HOSTINT_EN	When set, mtu sifs timeout interrupt is enabled
5	MTU_BKOF1_HOSTINT_EN	When set, mtu bkof1 interrupt is enabled
4	MTU_BKOF0_HOSTINT_EN	When set, mtu bkof0 interrupt is enabled
3	MTU_TIMER3_HOSTINT_EN	When set, mtu timer3/tsf_timer3 interrupt is enabled See register 'MTU_TIMER_INTERRUPT_SOURCE' for timer selection
2	MTU_TIMER2_HOSTINT_EN	When set, mtu timer2/tsf_timer2 interrupt is enabled See register 'MTU_TIMER_INTERRUPT_SOURCE' for timer selection
1	MTU_TIMER1_HOSTINT_EN	When set, mtu timer1/tsf_timer1 interrupt is enabled See register 'MTU_TIMER_INTERRUPT_SOURCE' for timer selection
0	MTU_TIMER0_HOSTINT_EN	When set, mtu timer0/tsf_timer0 interrupt is enabled See register 'MTU_TIMER_INTERRUPT_SOURCE' for timer selection

0x0A080ECC WCSS_A_MCU_MTU_HOST_INT_MASKED_STATUS**Type:** R**Reset State:** 0x00000000

This register contains the mtu interrupt masked status signals

WCSS_A_MCU_MTU_HOST_INT_MASKED_STATUS

Bits	Name	Description
14	MTU_MCU_TX_BOUNDARY_INT_HOSTINT_MASK_STATUS	tu_mcu_tx_boundary_int interrupt masked status
13	MTU_ERR_HOSTINT_MASK_STATUS	mtu error interrupt masked status
12	WD_ENABLE_DISABLE_ERROR_HOSTINT_MASK_STATUS	mtu wd_enable_disable_error interrupt masked status
11	WD_PROTECTION_ERROR_HOSTINT_MASK_STATUS	mtu wd_protection_error interrupt masked status
10	RESP_TIMEOUT_MISSING_PKT_PUSH_HOSTINT_MASK_STATUS	mtu resp_timeout_missing_pkt_push interrupt masked status
9	RESP_TIMEOUT_MISSING_PKTDET_HOSTINT_MASK_STATUS	mtu resp_timeout_missing_pktdet interrupt masked status
8	RESP_TIMEOUT_MISSING_EARLY_PKTDET_HOSTINT_MASK_STATUS	mtu resp_timeout_missing_early_pktdet interrupt masked status
7	MTU_PIFS_TO_HOSTINT_MASK_STATUS	mtu pifs timeout interrupt masked status
6	MTU_SIFS_TO_HOSTINT_MASK_STATUS	mtu sifs timeout interrupt masked status
5	MTU_BKOF1_HOSTINT_MASK_STATUS	mtu bkof1 interrupt masked status
4	MTU_BKOF0_HOSTINT_MASK_STATUS	mtu bkof0 interrupt masked status
3	MTU_TIMER3_HOSTINT_MASK_STATUS	mtu timer3/tsf_timer3 interrupt masked status See register 'MTU_TIMER_INTERRUPT_SOURCE' for timer selection
2	MTU_TIMER2_HOSTINT_MASK_STATUS	mtu timer2/tsf_timer2 interrupt masked status See register 'MTU_TIMER_INTERRUPT_SOURCE' for timer selection
1	MTU_TIMER1_HOSTINT_MASK_STATUS	mtu timer1/tsf_timer1 interrupt masked status See register 'MTU_TIMER_INTERRUPT_SOURCE' for timer selection
0	MTU_TIMER0_HOSTINT_MASK_STATUS	mtu timer0/tsf_timer0 interrupt masked status See register 'MTU_TIMER_INTERRUPT_SOURCE' for timer selection

0x0A080ED0 WCSS_A_MCU_COMBINED_HOST_INT_EN**Type:** RW**Reset State:** 0x00000000

This register contains the combined interrupt enable signals

WCSS_A_MCU_COMBINED_HOST_INT_EN

Bits	Name	Description
24	CSU_MCU_ERR_INT_HOST_INT_EN	When set, csu error interrupt is enabled
23	WLAN_BRIDGE_AHB_ERR_INT_HOSTINT_EN	When set, wlan bridge ahb error interrupt is enabled
21	BTC_PATTERN_DET_INT_HOSTINT_EN	When set, btc_pattern_det_int_p interrupt is enabled
20	BTC_SCO_INT_HOSTINT_EN	When set, btc_sco_int_p interrupt is enabled
19	TPE_MCU_WRONG_SSN_OR_TID_ERROR_P_HOSTINT_EN	When set, tpe_mcu_wrong_ssn_or_tid_error_p interrupt is enabled
18	TPE_MCU_BD_BASED_TX_INT_1_P_HOSTINT_EN	When set, tpe_mcu_bd_based_tx_int_1_p interrupt is enabled
17	TPE_MCU_BD_BASED_TX_INT_0_P_HOSTINT_EN	When set, tpe_mcu_bd_based_tx_int_0_p interrupt is enabled
16	TPE_MCU_ERROR_VAILD_P_HOSTINT_EN	When set, tpe_mcu_error_vaild_p interrupt is enabled
15	TPE_MCU_GOT_ACK_FOR_BACK_P_HOSTINT_EN	When set, tpe_mcu_got_ack_for_back_p interrupt is enabled
14	BMU_MCU_PWRSTATE_DROP_INT_EN	When set, bmu_mcu_pwrstate_drop_interrupt interrupt is enabled
13	BMU_MCU_PWRSTATE_CHANGE_INT_EN	When set, bmu_mcu_pwrstate_change_interrupt interrupt is enabled
12	BMU_MCU_BTQM_QUEUE_EMPTY_INTERRUPT_HOSTINT_EN	When set, bmu_mcu_btqm_queues_empty_interrupt interrupt is enabled
11	BMU_MCU_ERR_INT_HOSTINT_EN	When set, bmu_mcu_err_int interrupt is enabled
10	AHB2PHY_MCU_TIMEOUT_ERROR2_HOSTINT_EN	When set, ahb2phy_mcu_timeout_error2 interrupt is enabled
9	AHB2PHY_MCU_TIMEOUT_ERROR1_HOSTINT_EN	When set, ahb2phy_mcu_timeout_error1 interrupt is enabled
8	DBR_MCU_GAM_ERR_HOSTINT_EN	When set, dbr_mcu_gam_err interrupt is enabled

WCSS_A_MCU_COMBINED_HOST_INT_EN (cont.)

Bits	Name	Description
7	DBR_MCU_INCORRECT_LENGTH_ERR_HOSTINT_EN	When set, dbr_mcu_incorrect_length_err interrupt is enabled
6	DPU_MCU_MICERR_P_HOSTINT_EN	When set, dpu_mcu_micerr_p interrupt is enabled
5	DPU_MCU_ERRINTR_P_HOSTINT_EN	When set, dpu_mcu_errintr_p interrupt is enabled
4	BMU_MCU_QUEUE_USAGE_THRESHOLD_INTERRUPT_EN	When set, bmu_mcu_queue_usage_threshold_interrupt is enabled
3	ACPU_MCU_MIF_TIMEOUT_INT_P_HOSTINT_EN	When set, acpu_mcu_mif_timeout_int_p interrupt is enabled
2	ACPU_MCU_AHB_ERR_INT_P_HOSTINT_EN	When set, acpu_mcu_ahb_err_int_p interrupt is enabled
1	TXP_MCU_TXP_TIMEOUT_INT_P_HOSTINT_EN	When set, txp_mcu_txp_timeout_int_p interrupt is enabled
0	TXP_MCU_ERR_INT_HOSTINT_EN	When set, txp_mcu_err_int interrupt is enabled

0x0A080ED4 WCSS_A_MCU_COMBINED_HOST_INT_MASKED_STATUS**Type:** R**Reset State:** 0x00000000

This register contains the combined interrupt masked status signals

WCSS_A_MCU_COMBINED_HOST_INT_MASKED_STATUS

Bits	Name	Description
24	CSU_MCU_ERR_INT_HOSTINT_MASK_STATUS	csu error interrupt masked status
23	WLAN_BRIDGE_AHB_ERR_INT_HOSTINT_MASK_STATUS	wlan bridge ahb error interrupt masked status
21	BTC_PATTERN_DET_INT_P_HOSTINT_MASK_STATUS	btc_pattern_det_int_p interrupt masked status
20	BTC_SCO_INT_P_HOSTINT_MASK_STATUS	btc_sco_int_p interrupt masked status
19	TPE_MCU_WRONG_SSN_OR_TID_ERROR_P_HOSTINT_MASK_STATUS	tpe_mcu_wrong_ssn_or_tid_error_p interrupt masked status

WCSS_A_MCU_COMBINED_HOST_INT_MASKED_STATUS (cont.)

Bits	Name	Description
18	TPE_MCU_BD_BASED_TX_INT_1_P_HOSTINT_MASK_STATUS	tpe_mcu_bd_based_tx_int_1_p interrupt masked status
17	TPE_MCU_BD_BASED_TX_INT_0_P_HOSTINT_MASK_STATUS	tpe_mcu_bd_based_tx_int_0_p interrupt masked status
16	TPE_MCU_ERROR_VAILD_P_HOSTINT_MASK_STATUS	tpe_mcu_error_vaild_p interrupt masked status
15	TPE_MCU_GOT_ACK_FOR_BACK_P_HOSTINT_MASK_STATUS	tpe_mcu_got_ack_for_back_p interrupt masked status
14	BMU_MCU_PWRSTATE_DROP_INT_STATUS	bmu_mcu_pwrstate_drop_interrupt masked status
13	BMU_MCU_PWRSTATE_CHANGE_INT_STATUS	bmu_mcu_pwrstate_change_interrupt masked status
12	BMU_MCU_BTQM_QUEUE_EMPTY_INTERRUPT_HOSTINT_MASK_STATUS	bmu_mcu_btqm_queues_empty_interrupt interrupt masked status
11	BMU_MCU_ERR_INT_HOSTINT_MASK_STATUS	bmu_mcu_err_int interrupt masked status
10	AHB2PHY_MCU_TIMEOUT_ERROR2_HOSTINT_MASK_STATUS	ahb2phy_mcu_timeout_error2 interrupt masked status
9	AHB2PHY_MCU_TIMEOUT_ERROR1_HOSTINT_MASK_STATUS	ahb2phy_mcu_timeout_error1 interrupt masked status
8	DBR_MCU_GAM_ERR_HOSTINT_MASK_STATUS	dbr_mcu_gam_err interrupt masked status
7	DBR_MCU_INCORRECT_LENGTH_ERR_HOSTINT_MASK_STATUS	dbr_mcu_incorrect_length_err interrupt masked status
6	DPU_MCU_MICERR_P_HOSTINT_MASK_STATUS	dpu_mcu_micerr_p interrupt masked status
5	DPU_MCU_ERRINTR_P_HOSTINT_MASK_STATUS	dpu_mcu_errintr_p interrupt masked status
4	BMU_MCU_QUEUE_USAGE_THRESHOLD_INTERRUPT_STATUS	bmu_mcu_queue_usage_threshold_interrupt interrupt status
3	ACPU_MCU_MIF_TIMEOUT_INT_P_HOSTINT_MASK_STATUS	acpu_mcu_mif_timeout_int_p interrupt masked status

WCSS_A_MCU_COMBINED_HOST_INT_MASKED_STATUS (cont.)

Bits	Name	Description
2	ACPU_MCU_AHB_ERR_INT_P_HOSTINT_MASK_STATUS	acpu_mcu_ahb_err_int_p interrupt masked status
1	TXP_MCU_TXP_TIMEOUT_INT_P_HOSTINT_MASK_STATUS	txp_mcu_txp_timeout_int_p interrupt masked status
0	TXP_MCU_ERR_INT_HOST_INT_MASK_STATUS	txp_mcu_err_int interrupt masked status

0x0A080ED8 WCSS_A_MCU_BMU_WQ_HOST_INT_EN**Type:** RW**Reset State:** 0x00000000

This register contains the bmu wq dataavailable interrupt enable signals

WCSS_A_MCU_BMU_WQ_HOST_INT_EN

Bits	Name	Description
24:2	BMU_WQ_HOST_INT_EN	BMU WQ 2-24 data available

0x0A080EDC WCSS_A_MCU_BMU_WQ_HOST_INT_MASKED_STATUS**Type:** R**Reset State:** 0x00000000

This register contains the bmu wq dataavailable interrupt masked status signals

WCSS_A_MCU_BMU_WQ_HOST_INT_MASKED_STATUS

Bits	Name	Description
24:2	BMU_WQ_HOST_INT_MASKED_STATUS	BMU WQ 2-24 data available

0x0A080EE0 WCSS_A_MCU_MTU_TIMER_INTERRUPT_SOURCE**Type:** RW**Reset State:** 0x00000000

This register controls which interrupt, TSF based or timer based, interrupt from MTU is selected

WCSS_A_MCU_MTU_TIMER_INTERRUPT_SOURCE

Bits	Name	Description
7	TSF_TIMER7_INT_SELECT	When set, in all interrupt related registers indicating mtu timer7/tsf_timer7 interrupt, the interrupt source is set to TSF. When NOT set, the interrupt source is set to mtu timer 7
6	TSF_TIMER6_INT_SELECT	When set, in all interrupt related registers indicating mtu timer6/tsf_timer6 interrupt, the interrupt source is set to TSF. When NOT set, the interrupt source is set to mtu timer 6
5	TSF_TIMER5_INT_SELECT	When set, in all interrupt related registers indicating mtu timer5/tsf_timer5 interrupt, the interrupt source is set to TSF. When NOT set, the interrupt source is set to mtu timer 5
4	TSF_TIMER4_INT_SELECT	When set, in all interrupt related registers indicating mtu timer4/tsf_timer4 interrupt, the interrupt source is set to TSF. When NOT set, the interrupt source is set to mtu timer 4
3	TSF_TIMER3_INT_SELECT	When set, in all interrupt related registers indicating mtu timer3/tsf_timer3 interrupt, the interrupt source is set to TSF. When NOT set, the interrupt source is set to mtu timer 3
2	TSF_TIMER2_INT_SELECT	When set, in all interrupt related registers indicating mtu timer2/tsf_timer2 interrupt, the interrupt source is set to TSF. When NOT set, the interrupt source is set to mtu timer 2
1	TSF_TIMER1_INT_SELECT	When set, in all interrupt related registers indicating mtu timer1/tsf_timer1 interrupt, the interrupt source is set to TSF. When NOT set, the interrupt source is set to mtu timer 1
0	TSF_TIMER0_INT_SELECT	When set, in all interrupt related registers indicating mtu timer0/tsf_timer0 interrupt, the interrupt source is set to TSF. When NOT set, the interrupt source is set to mtu timer 0

0x0A080F00 WCSS_A_MCU_BTC_CTRL4**Type:** RW**Reset State:** 0x00000000

This register is to setup the pta.

WCSS_A_MCU_BTC_CTRL4

Bits	Name	Description
15:0	SCO_INTERRUPT_OFFSET	The btc_sco_int will be trigger when the sco period counter matches the value When MSB is '1', the interrupt is to be triggered the programmed time after the bt_active assertion When MSB is '0', the interrupt is to be triggered the programmed time before the bt_active assertion

0x0A080F04 WCSS_A_MCU_BTC_PATTERN_DET1**Type:** RW**Reset State:** 0x00000000

This register contains the info for pattern detection

WCSS_A_MCU_BTC_PATTERN_DET1

Bits	Name	Description
15:0	DATA	The mask bits for pattern detection. The LSB should be 1.

0x0A080F08 WCSS_A_MCU_BTC_PATTERN_DET2**Type:** RW**Reset State:** 0x00000000

This register contains the set up for pattern detection

WCSS_A_MCU_BTC_PATTERN_DET2

Bits	Name	Description
31:16	BT_PRIORITY_PATTERN_DET	The bt_priority pattern for pattern detection
15:0	BT_ACTIVE_PATTERN_DET	The bt_active pattern for pattern detection

0x0A080F0C WCSS_A_MCU_BTC_PATTERN_HISTORY**Type:** R**Reset State:** 0x00000000

This register contains the history of bt_active and bt_priority

WCSS_A_MCU_BTC_PATTERN_HISTORY

Bits	Name	Description
31:16	BT_PRIORITY_HISTORY	The bt_priority history
15:0	BT_ACTIVE_HISTORY	The bt_active history

0x0A080F10 WCSS_A_MCU_BTC_HI_GRANT**Type:** RW**Reset State:** 0x00000000

This register countans the stats of bt high priority grant

WCSS_A_MCU_BTC_HI_GRANT

Bits	Name	Description
31:20	BT_HI_PRI_GRANT_CNT	BT high priority grant counts
19:0	BT_HI_PRI_GRANT_TIME	BT high priority grant time in microsecond

0x0A080F14 WCSS_A_MCU_BTC_LOW_GRANT**Type:** RW**Reset State:** 0x00000000

This register contains the stats of bt log priority grant

WCSS_A_MCU_BTC_LOW_GRANT

Bits	Name	Description
31:20	BT_LOW_PRI_GRANT_CNT	BT low priority grant counts
19:0	BT_LOW_PRI_GRANT_TIME	BT low priority grant time in microsecond

0x0A080F18 WCSS_A_MCU_BTC_REQ_CNT**Type:** RW**Reset State:** 0x00000000

This register is to setup the pta.

WCSS_A_MCU_BTC_REQ_CNT

Bits	Name	Description
23:12	BT_LO_PRI_REQ_CNT	BT low priority requests.
11:0	BT_HI_PRI_REQ_CNT	BT high priority requests.

0x0A080F1C WCSS_A_MCU_MRCM_CLK_GATE_DISABLE**Type:** RW**Reset State:** 0x00003883

This register controls the dynamic clock gating of the wmac modules in mrcm

WCSS_A_MCU_MRCM_CLK_GATE_DISABLE

Bits	Name	Description
14	MCU_MRCM_CSU_GCU_CLK_GATE_DISABLE	When set, disable csu dynamic clock gating
13	MCU_MRCM_FDAH_B_GCU_CLK_GATE_DISABLE	When set, disable fdahb dynamic clock gating
12	MCU_MRCM_FDBR_GCU_CLK_GATE_DISABLE	When set, disable fdbr dynamic clock gating
11	MCU_MRCM_CBR_GCU_CLK_GATE_DISABLE	When set, disable cbr dynamic clock gating
10	MCU_MRCM_TXP_GCU_CLK_GATE_DISABLE	When set, disable txp dynamic clock gating
9	MCU_MRCM_CFG_GCU_CLK_GATE_DISABLE	When set, disable cfg dynamic clock gating
8	MCU_MRCM_RXP_GCU_CLK_GATE_DISABLE	When set, disable rxp dynamic clock gating
7	MCU_MRCM_DBR_GCU_CLK_GATE_DISABLE	When set, disable dbr dynamic clock gating
6	MCU_MRCM_BMU_GCU_CLK_GATE_DISABLE	When set, disable bmu dynamic clock gating
5	MCU_MRCM_DPU_GCU_CLK_GATE_DISABLE	When set, disable dpu dynamic clock gating
4	MCU_MRCM_ADU_GCU_CLK_GATE_DISABLE	When set, disable adu dynamic clock gating
3	MCU_MRCM_TPE_GCU_CLK_GATE_DISABLE	When set, disable tpe dynamic clock gating
2	MCU_MRCM_RPE_GCU_CLK_GATE_DISABLE	When set, disable rpe dynamic clock gating
1	MCU_MRCM_CAHB_GCU_CLK_GATE_DISABLE	When set, disable cahb dynamic clock gating
0	MCU_MRCM_DAHB_GCU_CLK_GATE_DISABLE	When set, disable dahb dynamic clock gating

0x0A080F20 WCSS_A_MCU_TXP_BUSY_CNT**Type:** RW**Reset State:** 0x00000000

This register controls the counter counting on the time when txp is busy

WCSS_A_MCU_TXP_BUSY_CNT

Bits	Name	Description
23:0	TXP_BUSY_CNT	The time (in unit of us) when txp is busy

0x0A080F24 WCSS_A_MCU_WLAN_BRIDGE_CTRL**Type:** RW**Reset State:** 0x00000000

This register contains the control signal to the wlan bridges

WCSS_A_MCU_WLAN_BRIDGE_CTRL

Bits	Name	Description
23:20	MCU_FDBR_CDAHB2FDAHB_TBUS_SEL	The testbus control signal for fdbr cdahb2fdahb
19:16	MCU_FDBR_FDAHB2CDAHB_TBUS_SEL	The testbus control signal for fdbr fdahb2cdahb
15:12	MCU_DBR_CDAHB2DAHB_TBUS_SEL	The testbus control signal for dbr dahb2cdahb
11:8	MCU_DBR_DAHB2CDAHB_TBUS_SEL	The testbus control signal for dbr dahb2cdahb
7:4	MCU_CBR_CCAHB2CAHB_TBUS_SEL	The testbus control signal for cbr ccahb2cahb
3:0	MCU_CBR_CAHB2CCAHB_TBUS_SEL	The testbus control signal for cbr cahb2ccahb

0x0A080F30 WCSS_A_MCU_BD_PDU_BASE_ADDR_LOCAL**Type:** RW**Reset State:** 0x00000000

Sets the memory base address of the BDs and PDUs in the local memory.

WCSS_A_MCU_BD_PDU_BASE_ADDR_LOCAL

Bits	Name	Description
31:7	BD_PDU_BASE_ADDRESS	Base address of the BDs and PDUs in the local memory. Note that the lowest 8 bits can not be programmed as these bits will always have to be zero. BDs and PDUs are in units of 128 bytes.

0x0A080F34 WCSS_A_MCU_CMEM_BDINDEX_THRESHOLD**Type:** RW**Reset State:** 0x00000000

The bd/pdu threshold in local memory.

WCSS_A_MCU_CMEM_BDINDEX_THRESHOLD

Bits	Name	Description
15:0	CMEM_BDINDEX_THRESH OLD	If the bd/pdu index is less than this value, the pkt is in local memory.

0x0A080F38 WCSS_A_MCU_CSU_CONFIGURATION**Type:** RW**Reset State:** 0x00000000

This register configures the CSU module operations.

WCSS_A_MCU_CSU_CONFIGURATION

Bits	Name	Description
31:28	MCU_CSU_SPARE	CSU spare configuration register.
27:24	RESERVED	reserved
23	MCU_CSU_RX_PRIORITY_SEL	Enable higher priority for serving RX bmu_csu_rx_wq_da.
22	MCU_CSU_RX_COUNTER_CLEAR	Enable RX UDP Checksum logic.
21	MCU_CSU_RX_EN	Enable RX UDP Checksum logic.
18	MCU_CSU_RX_ERROR_EX IT	Enable RX UDP Checksum logic.
17	MCU_CSU_RX_UDP_CHEC KSUM_VALIDATE_EN	Enable RX UDP Checksum logic.
16	MCU_CSU_RX_TCP_CHEC KSUM_VALIDATE_EN	Enable RX TCP Checksum logic.

WCSS_A_MCU_CSU_CONFIGURATION (cont.)

Bits	Name	Description
15	MCU_CSU_RX_IP_CHECKSUM_VALIDATE_EN	Enable RX IP Checksum logic.
14	MCU_CSU_RX_MODULE_EN	Enable CSU RX module
13	MCU_CSU_TX_COUNTER_CLEAR	Enable clearing of all status counters.
12	MCU_CSU_TX_EN	Enable TX FSM logic.
9	MCU_CSU_TX_ERROR_EXIT	Enable exiting of CSU TX Error FSM logic.
8	MCU_CSU_TX_UDP_CHECKSUM_GEN_EN	Enable CSU TX UDP Checksum logic.
7	MCU_CSU_TX_TCP_CHECKSUM_GEN_EN	Enable CSU TX TCP Checksum logic.
6	MCU_CSU_TX_IP_CHECKSUM_GEN_EN	Enable CSU TX IP Checksum logic.
5	MCU_CSU_TX_MODULE_EN	Enable CSU TX logic.
4	MCU_CSU_VLAN_ETHER_EN	Enables VLAN Tag EtherType processing
3	MCU_CSU_IPV6_ETHER_EN	Enables IPv6 EtherType processing
2	MCU_CSU_IPV4_ETHER_EN	Enables IPv4 EtherType processing

0x0A080F3C WCSS_A_MCU_MCU_CSU_NEXTHDR_BITMAP_0**Type:** RW**Reset State:** 0x00000000

This register allows software to define Next Header for IPv6

WCSS_A_MCU_MCU_CSU_NEXTHDR_BITMAP_0

Bits	Name	Description
31:0	MCU_CSU_NEXTHDR_BITMAP_0	This register stores Next Header decoded values.

0x0A080F40 WCSS_A_MCU_MCU_CSU_NEXTHDR_BITMAP_1**Type:** RW**Reset State:** 0x00000000

This register allows software to define Next Header for IPv6

WCSS_A_MCU_MCU_CSU_NEXTHDR_BITMAP_1

Bits	Name	Description
31:0	MCU_CSU_NEXTHDR_BITMAP_1	This register stores Next Header decoded values.

0x0A080F44 WCSS_A_MCU_MCU_CSU_NEXTHDR_BITMAP_2**Type:** RW**Reset State:** 0x00000000

This register allows software to define Next Header for IPv6

WCSS_A_MCU_MCU_CSU_NEXTHDR_BITMAP_2

Bits	Name	Description
31:0	MCU_CSU_NEXTHDR_BITMAP_2	This register stores Next Header decoded values.

0x0A080F48 WCSS_A_MCU_MCU_CSU_NEXTHDR_BITMAP_3**Type:** RW**Reset State:** 0x00000000

This register allows software to define Next Header for IPv6

WCSS_A_MCU_MCU_CSU_NEXTHDR_BITMAP_3

Bits	Name	Description
31:0	MCU_CSU_NEXTHDR_BITMAP_3	This register stores Next Header decoded values.

0x0A080F4C WCSS_A_MCU_MCU_CSU_NEXTHDR_BITMAP_4**Type:** RW**Reset State:** 0x00000000

This register allows software to define Next Header for IPv6

WCSS_A_MCU_MCU_CSU_NEXTHDR_BITMAP_4

Bits	Name	Description
31:0	MCU_CSU_NEXTHDR_BIT MAP_4	This register stores Next Header decoded values.

0x0A080F50 WCSS_A_MCU_MCU_CSU_NEXTHDR_BITMAP_5

Type: RW

Reset State: 0x00000000

This register allows software to define Next Header for IPv6

WCSS_A_MCU_MCU_CSU_NEXTHDR_BITMAP_5

Bits	Name	Description
31:0	MCU_CSU_NEXTHDR_BIT MAP_5	This register stores Next Header decoded values.

0x0A080F54 WCSS_A_MCU_MCU_CSU_NEXTHDR_BITMAP_6

Type: RW

Reset State: 0x00000000

This register allows software to define Next Header for IPv6

WCSS_A_MCU_MCU_CSU_NEXTHDR_BITMAP_6

Bits	Name	Description
31:0	MCU_CSU_NEXTHDR_BIT MAP_6	This register stores Next Header decoded values.

0x0A080F58 WCSS_A_MCU_MCU_CSU_NEXTHDR_BITMAP_7

Type: RW

Reset State: 0x00000000

This register allows software to define Next Header for IPv6

WCSS_A_MCU_MCU_CSU_NEXTHDR_BITMAP_7

Bits	Name	Description
31:0	MCU_CSU_NEXTHDR_BITMAP_7	This register stores Next Header decoded values.

0x0A080F5C WCSS_A_MCU_MCU_CSU_TX_WORK_QUEUES**Type:** RW**Reset State:** 0x00150006

This register contains CSU work queues

WCSS_A_MCU_MCU_CSU_TX_WORK_QUEUES

Bits	Name	Description
23:16	MCU_CSU_TX_POP_WQ_INDEX	CSU TX Pop Work Queue Index
15:8	MCU_CSU_TX_ERROR_WQ	CSU TX Error Push Work Queue Index
7:0	MCU_CSU_TX_PUSH_WQ	CSU TX Push Work Queue Index

0x0A080F6C WCSS_A_MCU_MCU_CSU_RX_WORK_QUEUES**Type:** RW**Reset State:** 0x0416000B

This register contains CSU work queues

WCSS_A_MCU_MCU_CSU_RX_WORK_QUEUES

Bits	Name	Description
31:24	MCU_CSU_RX_HPR_PUSH_WQ	CSU RX High Priority Push Work Queue
23:16	MCU_CSU_RX_POP_WQ_INDEX	CSU RX Pop Work Queue Index
15:8	MCU_CSU_RX_ERROR_WQ	CSU RX Error Push Work Queue Index
7:0	MCU_CSU_RX_PUSH_WQ	CSU RX Push Work Queue Index

0x0A080F7C WCSS_A_MCU_MCU_CSU_INTERRUPT**Type:** RW**Reset State:** 0x00000000

This register allows software to see number of error conditions

WCSS_A_MCU_MCU_CSU_INTERRUPT

Bits	Name	Type	Description
15:8	CSU_MCU_INT_STATUS	R	This register records the causes of interrupts.
7:0	MCU_CSU_INT_MASK	RW	This register is used to program interrupt masks.

0x0A080F80 WCSS_A_MCU_MCU_CBR_CCAHB_ERR_ADDR**Type:** R**Reset State:** 0x00000000

This register records the error beat address in either a read or write transaction on CCAHB when CBR detects CCAHB slave error response.

WCSS_A_MCU_MCU_CBR_CCAHB_ERR_ADDR

Bits	Name	Description
31:0	ERROR_BEAT_ADDRESS	The address of a write or read beat that resulted in error response sent by CCAHB slave. This register will not be cleared when it is read.

0x0A080F84 WCSS_A_MCU_MCU_CBR_CAHB_ERR_ADDR**Type:** R**Reset State:** 0x00000000

This register records the error beat address in either a read or write transaction on CAHB when CBR detects CAHB slave error response.

WCSS_A_MCU_MCU_CBR_CAHB_ERR_ADDR

Bits	Name	Description
31:0	ERROR_BEAT_ADDRESS	The address of a write or read beat that resulted in error response sent by CAHB slave. This register will not be cleared when it is read.

0x0A080F88 WCSS_A_MCU_MCU_CBR_CCAHB_TIMEOUT_ADDR**Type:** R**Reset State:** 0x00000000

This register recorded the start address of a read or write transaction originated from CCAHB that encountered a suspend timeout.

WCSS_A_MCU_MCU_CBR_CCAHB_TIMEOUT_ADDR

Bits	Name	Description
31:0	START_ADDRESS	The start address of a write or read transaction originated from a CCAHB master that resulted in suspend timeout. This register will not be cleared when it is read.

0x0A080F8C WCSS_A_MCU_MCU_CBR_CAHB_TIMEOUT_ADDR**Type:** R**Reset State:** 0x00000000

This register recorded the start address of a read or write transaction originated from CAHB that encountered a suspend timeout.

WCSS_A_MCU_MCU_CBR_CAHB_TIMEOUT_ADDR

Bits	Name	Description
31:0	START_ADDRESS	The start address of a write or read transaction originated from a CAHB master that resulted in suspend timeout. This register will not be cleared when it is read.

0x0A080F90 WCSS_A_MCU_MCU_DBR_CDAHB_ERR_ADDR**Type:** R**Reset State:** 0x00000000

This register records the error beat address in either a read or write transaction on CDAHB when DBR detects CDAHB slave error response.

WCSS_A_MCU_MCU_DBR_CDAHB_ERR_ADDR

Bits	Name	Description
31:0	ERROR_BEAT_ADDRESS	The address of a write or read beat that resulted in error response sent by CDAHB slave. This register will not be cleared when it is read.

0x0A080F94 WCSS_A_MCU_MCU_DBR_DAHB_ERR_ADDR**Type:** R**Reset State:** 0x00000000

This register records the error beat address in either a read or write transaction on DAHB when DBR detects DAHB slave error response.

WCSS_A_MCU_MCU_DBR_DAHB_ERR_ADDR

Bits	Name	Description
31:0	ERROR_BEAT_ADDRESS	The address of a write or read beat that resulted in error response sent by DAHB slave. This register will not be cleared when it is read.

0x0A080F98 WCSS_A_MCU_MCU_DBR_CDAHBTIMEOUT_ADDR**Type:** R**Reset State:** 0x00000000

This register recorded the start address of a read or write transaction originated from CDAHBT that encountered a suspend timeout at DBR.

WCSS_A_MCU_MCU_DBR_CDAHBTIMEOUT_ADDR

Bits	Name	Description
31:0	START_ADDRESS	The start address of a write or read transaction originated from a CDAHBT master that resulted in suspend timeout. This register will not be cleared when it is read.

0x0A080F9C WCSS_A_MCU_MCU_DBR_DAHBTIMEOUT_ADDR**Type:** R**Reset State:** 0x00000000

This register recorded the start address of a read or write transaction originated from DAHB that encountered a suspend timeout at DBR.

WCSS_A_MCU_MCU_DBR_DAHBTIMEOUT_ADDR

Bits	Name	Description
31:0	START_ADDRESS	The start address of a write or read transaction originated from a DAHB master that resulted in suspend timeout. This register will not be cleared when it is read.

0x0A080FA0 WCSS_A_MCU_MCU_FDBR_CDAHB_ERR_ADDR**Type:** R**Reset State:** 0x00000000

This register records the error beat address in either a read or write transaction on CDAHB when FDBR detects CDAHB slave error response.

WCSS_A_MCU_MCU_FDBR_CDAHB_ERR_ADDR

Bits	Name	Description
31:0	ERROR_BEAT_ADDRESS	The address of a write or read beat that resulted in error response sent by CDAHB slave. This register will not be cleared when it is read.

0x0A080FA4 WCSS_A_MCU_MCU_FDBR_FDAHB_ERR_ADDR**Type:** R**Reset State:** 0x00000000

This register records the error beat address in either a read or write transaction on FDAHB when FDBR detects DAHB slave error response.

WCSS_A_MCU_MCU_FDBR_FDAHB_ERR_ADDR

Bits	Name	Description
31:0	ERROR_BEAT_ADDRESS	The address of a write or read beat that resulted in error response sent by FDAHB slave. This register will not be cleared when it is read.

0x0A080FA8 WCSS_A_MCU_MCU_FDBR_CDAHB_TIMEOUT_ADDR**Type:** R**Reset State:** 0x00000000

This register recorded the start address of a read or write transaction originated from CDAHB that encountered a suspend timeout at FDBR.

WCSS_A_MCU_MCU_FDBR_CDAHB_TIMEOUT_ADDR

Bits	Name	Description
31:0	START_ADDRESS	The start address of a write or read transaction originated from a CDAHB master that resulted in suspend timeout. This register will not be cleared when it is read.

0x0A080FAC WCSS_A_MCU_MCU_FDBR_FDAHB_TIMEOUT_ADDR**Type:** R**Reset State:** 0x00000000

This register recorded the start address of a read or write transaction originated from FDAHB that encountered a suspend timeout at FDBR.

WCSS_A_MCU_MCU_FDBR_FDAHB_TIMEOUT_ADDR

Bits	Name	Description
31:0	START_ADDRESS	The start address of a write or read transaction originated from a FDAHB master that resulted in suspend timeout. This register will not be cleared when it is read.

0x0A080FB0 WCSS_A_MCU_MCU_WLAN_BRIDGE_ERR_TIMEOUT_SRC**Type:** R**Reset State:** 0x00000000

This register records all CBR, DBR and FDBR AHB error pulses as well as read and write transactions suspend timeout pulses. When this register is read, its content will be cleared.

WCSS_A_MCU_MCU_WLAN_BRIDGE_ERR_TIMEOUT_SRC

Bits	Name	Description
23	FDBR_MCU_FDAHB_RD_TIMEOUT_P	FDBR FDAHB to CDAHB read transaction suspend timeout.
22	FDBR_MCU_FDAHB_WR_TIMEOUT_P	FDBR FDAHB to CDAHB write transaction suspend timeout.
21	FDBR_MCU_CDAHB_RD_TIMEOUT_P	FDBR CDAHB to FDAHB read transaction suspend timeout.
20	FDBR_MCU_CDAHB_WR_TIMEOUT_P	FDBR CDAHB to FDAHB write transaction suspend timeout.
19	FDBR_MCU_FDAHB_RD_AHB_ERR_P	FDBR detects FDAHB slave read transaction error response.
18	FDBR_MCU_FDAHB_WR_AHB_ERR_P	FDBR detects FDAHB slave write transaction error response.
17	FDBR_MCU_CDAHB_RD_AHB_ERR_P	FDBR detects CDAHB slave read transaction error response.
16	FDBR_MCU_CDAHB_WR_AHB_ERR_P	FDBR detects CDAHB slave write transaction error response.
15	DBR_MCU_DAHB_RD_TIMEOUT_P	DBR DAHB to CDAHB read transaction suspend timeout.

WCSS_A_MCU_MCU_WLAN_BRIDGE_ERR_TIMEOUT_SRC (cont.)

Bits	Name	Description
14	DBR_MCU_DAHB_WR_TIM EOUT_P	DBR DAHB to CDAHB write transaction suspend timeout.
13	DBR_MCU_CDAHB_RD_T IMEOUT_P	DBR CDAHB to DAHB read transaction suspend timeout.
12	DBR_MCU_CDAHB_WR_T IMEOUT_P	DBR CDAHB to DAHB write transaction suspend timeout.
11	DBR_MCU_DAHB_RD_A HB_ERR_P	DBR detects DAHB slave read transaction error response.
10	DBR_MCU_DAHB_WR_A HB_ERR_P	DBR detects DAHB slave write transaction error response.
9	DBR_MCU_CDAHB_RD_A HB_ERR_P	DBR detects CDAHB slave read transaction error response.
8	DBR_MCU_CDAHB_WR_A HB_ERR_P	DBR detects CDAHB slave write transaction error response.
7	CBR_MCU_CAHB_RD_T IMEOUT_P	CBR CAHB to CCAHB read transaction suspend timeout.
6	CBR_MCU_CAHB_WR_T IMEOUT_P	CBR CAHB to CCAHB write transaction suspend timeout.
5	CBR_MCU_CCAHB_RD_T IMEOUT_P	CBR CCAHB to CAHB read transaction suspend timeout.
4	CBR_MCU_CCAHB_WR_T IMEOUT_P	CBR CCAHB to CAHB write transaction suspend timeout.
3	CBR_MCU_CAHB_RD_A HB_ERR_P	CBR detects CAHB slave read transaction error response.
2	CBR_MCU_CAHB_WR_A HB_ERR_P	CBR detects CAHB slave write transaction error response.
1	CBR_MCU_CCAHB_RD_A HB_ERR_P	CBR detects CCAHB slave read transaction error response.
0	CBR_MCU_CCAHB_WR_A HB_ERR_P	CBR detects CCAHB slave write transaction error response.

10.42 WCSS_A_MPI_MPI**0x0A013400 WCSS_A_MPI_WATCHDOG_MAX_COMMAND1****Type:** RW**Reset State:** 0x0000

Allowed time for state machine to remain in COMMAND1 state

WCSS_A_MPI_WATCHDOG_MAX_COMMAND1

Bits	Name	Description
15:0	CYCLES	Allowed period of time in this state = (Register value * 128 * 12.5)ns. The state machine automatically adjusts for 20/40/80MHz bandwidth settings. Setting a value of 0 disables the watchdog.

0x0A013404 WCSS_A_MPI_WATCHDOG_MAX_COMMAND2**Type:** RW**Reset State:** 0x0000

Allowed time for state machine to remain in COMMAND2 state

WCSS_A_MPI_WATCHDOG_MAX_COMMAND2

Bits	Name	Description
15:0	CYCLES	Allowed period of time in this state = (Register value * 128 * 12.5)ns. The state machine automatically adjusts for 20/40/80MHz bandwidth settings. Setting a value of 0 disables the watchdog.

0x0A013408 WCSS_A_MPI_WATCHDOG_MAX_WAIT**Type:** RW**Reset State:** 0x0000

Allowed time for state machine to remain in WAIT state

WCSS_A_MPI_WATCHDOG_MAX_WAIT

Bits	Name	Description
15:0	CYCLES	Allowed period of time in this state = (Register value * 128 * 12.5)ns. The state machine automatically adjusts for 20/40/80MHz bandwidth settings. Setting a value of 0 disables the watchdog.

0x0A01340C WCSS_A_MPI_WATCHDOG_MAX_TRANSFER**Type:** RW**Reset State:** 0x0000

Allowed time between request or data_valid pulses in TRANSFER state

WCSS_A_MPI_WATCHDOG_MAX_TRANSFER

Bits	Name	Description
15:0	CYCLES	Allowed period of time in this state = (Register value * 128 * 12.5)ns. The state machine automatically adjusts for 20/40/80MHz bandwidth settings. Setting a value of 0 disables the watchdog.

0x0A013410 WCSS_A_MPI_MAX_REQAORB**Type:** RW**Reset State:** 0x0000

Holds the maximum number of txa or txb requests per packet that should be allowed during transmit. Default value of 0 disables the reporting of max_request_exceeded violation

WCSS_A_MPI_MAX_REQAORB

Bits	Name	Description
15:0	VALUE	

0x0A013414 WCSS_A_MPI_COMMAND2_DELAY**Type:** RW**Reset State:** 0x026C

Delay between end of first 4 bytes of COMMAND and start of rest of COMMAND. The state machine automatically adjusts for 20/40/80MHz bandwidth settings.

WCSS_A_MPI_COMMAND2_DELAY

Bits	Name	Description
15:0	VALUE	

0x0A013418 WCSS_A_MPI_MPI_ENABLE**Type:** RW**Reset State:** 0x0000

This register must be set to 0x1 for the MPI to respond to TXP signals. This register is used to disable the MPI during MAC loopback tests.

WCSS_A_MPI_MPI_ENABLE

Bits	Name	Description
0	VALUE	

0x0A01341C WCSS_A_MPI_TXP_MPI_START**Type:** W**Reset State:** 0x00000000

Counts number of TXP_MPI_START events received by TX Mac-PHY Interface module.

WCSS_A_MPI_TXP_MPI_START

Bits	Name	Description
15:0	COUNT	TXP_MPI_START event counter.

0x0A013420 WCSS_A_MPI_MPI_TXP_REQ**Type:** W**Reset State:** 0x00000000

Counts number of MPI_TXP_REQ generated by TX MAC-PHY Interface module.

WCSS_A_MPI_MPI_TXP_REQ

Bits	Name	Description
15:0	COUNT	MPI_TXP_REQ event counter.

0x0A013424 WCSS_A_MPI_TXA_MPI_DATA_REQ**Type:** W**Reset State:** 0x00000000

Counts number of TXA_MPI_DATA_REQ events received by TX Mac-PHY Interface module.

WCSS_A_MPI_TXA_MPI_DATA_REQ

Bits	Name	Description
15:0	COUNT	TXA_MPI_DATA_REQ event counter.

0x0A013428 WCSS_A_MPI_TXB_MPI_DATA_REQ**Type:** W**Reset State:** 0x00000000

Counts number of TXB_MPI_DATA_REQ events received by TX Mac-PHY Interface module.

WCSS_A_MPI_TXB_MPI_DATA_REQ

Bits	Name	Description
15:0	COUNT	TXB_MPI_DATA_REQ event counter.

0x0A01342C WCSS_A_MPI_TXP_MPI_DATA_VAL**Type:** W**Reset State:** 0x00000000

Counts number of TXP_MPI_DATA_VAL pulses received.

WCSS_A_MPI_TXP_MPI_DATA_VAL

Bits	Name	Description
15:0	COUNT	TXP_MPI_DATA_VAL event counter.

0x0A013430 WCSS_A_MPI_MPI_TXA_DATA_VAL**Type:** W**Reset State:** 0x00000000

Counts number of MPI_TXA_DATA_VAL generated by TX MAC-PHY Interface module.

WCSS_A_MPI_MPI_TXA_DATA_VAL

Bits	Name	Description
15:0	COUNT	MPI_TXA_DATA_VAL event counter.

0x0A013434 WCSS_A_MPI_MPI_TXB_DATA_VAL**Type:** W**Reset State:** 0x00000000

Counts number of MPI_TXB_DATA_VAL generated by TX MAC-PHY Interface module.

WCSS_A_MPI_MPI_TXB_DATA_VAL

Bits	Name	Description
15:0	COUNT	MPI_TXB_DATA_VAL event counter.

0x0A013438 WCSS_A_MPI_MPI_TXA_LASTBYTE**Type:** W**Reset State:** 0x00000000

Counts number of MPI_TXA_LASTBYTE generated by TX MAC-PHY Interface module.

WCSS_A_MPI_MPI_TXA_LASTBYTE

Bits	Name	Description
15:0	COUNT	MPI_TXA_LASTBYTE event counter.

0x0A01343C WCSS_A_MPI_MPI_TXA_CW_LASTBYTE**Type:** W**Reset State:** 0x00000000

Counts number of MPI_TXA_CW_LASTBYTE generated by TX MAC-PHY Interface module.

WCSS_A_MPI_MPI_TXA_CW_LASTBYTE

Bits	Name	Description
15:0	COUNT	MPI_TXA_CW_LASTBYTE event counter.

0x0A013440 WCSS_A_MPI_MPI_TXA_CW_MAX_PARITY**Type:** W**Reset State:** 0x00000000

Counts number of MPI_TXA_CW_MAX_PARITY generated by TX MAC-PHY Interface module.

WCSS_A_MPI_MPI_TXA_CW_MAX_PARITY

Bits	Name	Description
15:0	COUNT	MPI_TXA_CW_MAX_PARITY event counter.

0x0A013444 WCSS_A_MPI_MPI_STATE**Type:** R**Reset State:** 0x00000000

Stores the last 4 MPI states.

WCSS_A_MPI_MPI_STATE

Bits	Name	Description
15:0	VALUE	Bit[15:12] = Previous MPI state (N-3) Bit[11:8] = Previous MPI state (N-2) Bit[7:4] = Previous MPI state (N-1) Bit[3:0] = Current MPI state

0x0A013448 WCSS_A_MPI_STATUS_FLAGS**Type:** R**Reset State:** 0x00000000

Indicates current state of the IDLE and RF warmup flags.

WCSS_A_MPI_STATUS_FLAGS

Bits	Name	Description
1	RF_WARMUP	Bit[0] = PHY idle flag - MAC may transmit when this flag is set Bit[1] = RF warmup flag - indicates that MAC has asserted the RF warm-up flag
0	IDLE	

0x0A01344C WCSS_A_MPI_REQ_ALARM**Type:** W**Reset State:** 0x00000000

Counts number of REQ_ALARM abort pulses generated by TX MAC-PHY Interface module (when too many MPI requests were issued or if contiguous request pulses received).

WCSS_A_MPI_REQ_ALARM

Bits	Name	Description
15:0	COUNT	

0x0A013450 WCSS_A_MPI_WATCHDOG_ALARM**Type:** W**Reset State:** 0x00000000

Counts number of WATCHDOG_ALARM abort pulses generated by TX MAC-PHY Interface module.

WCSS_A_MPI_WATCHDOG_ALARM

Bits	Name	Description
15:0	COUNT	

0x0A013454 WCSS_A_MPI_COMMAND_ALARM**Type:** W**Reset State:** 0x00000000

Counts number of times that a command word was received when data was expected by the TX MAC-PHY Interface module and vice versa.

WCSS_A_MPI_COMMAND_ALARM

Bits	Name	Description
15:0	COUNT	

0x0A013458 WCSS_A_MPI_TXP_ABORT**Type:** W**Reset State:** 0x00000000

Counts number of times that an abort was received from the TXP.

WCSS_A_MPI_TXP_ABORT

Bits	Name	Description
15:0	COUNT	

0x0A01345C WCSS_A_MPI_TXCTL_ABORT**Type:** W**Reset State:** 0x00000000

Counts number of times that an abort was received from the TXCTL.

WCSS_A_MPI_TXCTL_ABORT

Bits	Name	Description
15:0	COUNT	

10.43 WCSS_A_MTU_MTU**0x0A081400 WCSS_A_MTU_DIFS_LIMIT_0TO3****Type:** RW**Reset State:** 0x1C1C2525

AIFS limit register for 0 to 3 back off engines. The value is in micro seconds Though the name is DIFS, it is used for both DIFS,AIFS. It is up to s/w to select the value

WCSS_A_MTU_DIFS_LIMIT_0TO3

Bits	Name	Description
31:24	SW_MTU_DIFS_LIMIT_3	AIFS value in micro seconds for the back-off engine 3
23:16	SW_MTU_DIFS_LIMIT_2	AIFS value in micro seconds for the back-off engine 2
15:8	SW_MTU_DIFS_LIMIT_1	AIFS value in micro seconds for the back-off engine 1
7:0	SW_MTU_DIFS_LIMIT_0	AIFS value in micro seconds for the back-off engine 0

0x0A081404 WCSS_A_MTU_DIFS_LIMIT_4TO7**Type:** RW**Reset State:** 0x1F431616

AIFS limit register for 4 to 7 back off engines. The value is in micro seconds Though the name is DIFS, it is used for both DIFS,AIFS. It is up to s/w to select the value

WCSS_A_MTU_DIFS_LIMIT_4TO7

Bits	Name	Description
31:24	SW_MTU_DIFS_LIMIT_7	AIFS value in micro seconds for the back-off engine 7
23:16	SW_MTU_DIFS_LIMIT_6	AIFS value in micro seconds for the back-off engine 6
15:8	SW_MTU_DIFS_LIMIT_5	AIFS value in micro seconds for the back-off engine 5
7:0	SW_MTU_DIFS_LIMIT_4	AIFS value in micro seconds for the back-off engine 4

0x0A081408 WCSS_A_MTU_EIFS_PIFS_SLOT_LIMIT**Type:** RW**Reset State:** 0x005A1309

This register contains the limits for eifs, pifs and the slot_limit for data and bcn

WCSS_A_MTU_EIFS_PIFS_SLOT_LIMIT

Bits	Name	Description
31:16	SW_MTU_EIFS_LIMIT	EIFS value in micro seconds
15:8	SW_PIFS_LIMIT	PIFS value in micro seconds
7:0	SW_MTU_SLOT_LIMIT	slot limit value in micro seconds for all the back-off engines

0x0A08140C WCSS_A_MTU_SW_MTU_BCN_SLOT_USEC_SIFS_LIMIT**Type:** RW**Reset State:** 0x20007714

this register contains the limits for beacon slot usec pulse generation sifs

WCSS_A_MTU_SW_MTU_BCN_SLOT_USEC_SIFS_LIMIT

Bits	Name	Description
31:24	SW_MTU_EARLY_PKT_DET_MISS_LIMIT	sw_mtu_early_pkt_det_miss_limit-- this is used in respons_to(ack_to) logic if the system is expecting a response and if the early_pkt_det is HIGH for this limit, an interrupt is generated (old name is cca_miss_limit)
15:8	SW_MTU_ONE_USEC_LIMIT	limit value to be programmed to get one usec pulse. Basically this is number of clk cycles. this value is loaded into a counter and keep getting decremented at every clk.
7:0	SW_MTU_BCN_SLOT_LIMIT	slot limit value in micro seconds for all the back-off engines

0x0A081410 WCSS_A_MTU_SW_MTU_MISC_LIMITS**Type:** RW**Reset State:** 0x0C20091E

This register contains the limits for one usec puls and for programmable pulse generation the programmable pulse is sued by the timers in one of the modes and the sw delay limits

WCSS_A_MTU_SW_MTU_MISC_LIMITS

Bits	Name	Description
31:24	SW_MTU_PUSH_MISS_LIMIT	Limit to control the missing_push interrupt. If pkt_det goes high and we don't get a packet from the rxp before this limit, the interrupt will go.
21:16	SW_MTU_TXP_DELAY_LIMIT	TXP is indicated earlier indication of transmission of a pkt. The number in this register indicates the delay between the indication to s/w and the corresponding flag set that goes to txp The actual delay is the number programmed multiplied by the sw_mtu_txp_delay_pulse_limit multiplied by 8ns
11:8	SW_MTU_TXP_DELAY_PULSE_LIMIT	this number dictates the resolution of the delay between the s/w interrupt and the corresponding txp flag
7:0	TIMER_RESOLUTION_LIMIT	A pulse is generated at the end of 'this limit' number of 120mhz pulses. This pulse represents the resolution in timers.

0x0A081414 WCSS_A_MTU_NAV_CNT**Type:** RW**Reset State:** 0x000000FF

This register contains the nav value NAV is updated by hw as well as by the s/w. s/w has higher priority over the h/w updates

WCSS_A_MTU_NAV_CNT

Bits	Name	Description
31:0	NAV_CNT	This register contains the nav value. There has to some default. So I kept ff The value is in micro seconds

0x0A081418 WCSS_A_MTU_TSF_TIMER_LO**Type:** RW**Reset State:** 0x00000000

This register contains the lower 32 bits of the tsf timer

WCSS_A_MTU_TSF_TIMER_LO

Bits	Name	Description
31:0	TSF_TIMER_LO	This register contains the lower 32 bits of tsf timer

0x0A08141C WCSS_A_MTU_TSF_TIMER_HI**Type:** RW**Reset State:** 0x00000000

This register contains the upper 32 bits of the tsf timer

WCSS_A_MTU_TSF_TIMER_HI

Bits	Name	Description
31:0	TSF_TIMER_HI	This register contains the upper 32 bits of tsf timer

0x0A081420 WCSS_A_MTU_TIMER_0**Type:** RW**Reset State:** 0x00FFFFFF

This register contains the timer 0 value. This timer is rd/writable,up/down Based on the configuration, this can be used either for up or down counter and generates an interrupt on reaching Zero in down mode. in up mode interrupt is generated when the timer reaches the sw_match_register_0.

WCSS_A_MTU_TIMER_0

Bits	Name	Description
23:0	TIMER_0	This register contains the timer 0 value. This timer is rd/writable,up/down

0x0A081424 WCSS_A_MTU_TIMER_1**Type:** RW**Reset State:** 0x00FFFFFF

This register contains the timer 1 value. This timer is rd/writable,up/down Based on the configuration, this can be used either for up or down counter and generates an interrupt on reaching Zero in down mode. in up mode interrupt is generated when the timer reaches the sw_match_register_1.

WCSS_A_MTU_TIMER_1

Bits	Name	Description
23:0	TIMER_1	This register contains the timer 1 value. This timer is rd/writable,up/down

0x0A081428 WCSS_A_MTU_TIMER_2**Type:** RW**Reset State:** 0x00FFFFFF

This register contains the timer 2 value. This timer is rd/writable,up/down Based on the configuration, this can be used either for up or down counter and generates an interrupt on reaching Zero in down mode. in up mode interrupt is generated when the timer reaches the sw_match_register_2.

WCSS_A_MTU_TIMER_2

Bits	Name	Description
23:0	TIMER_2	This register contains the timer 2 value. This timer is rd/writable,up/down

0x0A08142C WCSS_A_MTU_TIMER_3**Type:** RW**Reset State:** 0x00FFFFFF

This register contains the timer 3 value. This timer is rd/writable,up/down Based on the configuration, this can be used either for up or down counter and generates an interrupt on reaching Zero in down mode. in up mode interrupt is generated when the timer reaches the sw_match_register_3.

WCSS_A_MTU_TIMER_3

Bits	Name	Description
23:0	TIMER_3	This register contains the timer 3 value. This timer is rd/writable,up/down

0x0A081430 WCSS_A_MTU_TIMER_4**Type:** RW**Reset State:** 0x00FFFFFF

This register contains the timer 4 value. This timer is rd/writable,up/down Based on the configuration, this can be used either for up or down counter and generates an interrupt on reaching Zero in down mode. in up mode interrupt is generated when the timer reaches the sw_match_register_4.

WCSS_A_MTU_TIMER_4

Bits	Name	Description
23:0	TIMER_4	This register contains the timer 4 value. This timer is rd/writable,up/down

0x0A081434 WCSS_A_MTU_TIMER_5**Type:** RW**Reset State:** 0x00FFFFFF

This register contains the timer 5 value. This timer is rd/writable,up/down Based on the configuration, this can be used either for up or down counter and generates an interrupt on reaching Zero in down mode. in up mode interrupt is generated when the timer reaches the sw_match_register_5.

WCSS_A_MTU_TIMER_5

Bits	Name	Description
23:0	TIMER_5	This register contains the timer 5 value. This timer is rd/writable,up/down

0x0A081438 WCSS_A_MTU_TIMER_6**Type:** RW**Reset State:** 0x00FFFFFF

This register contains the timer 6 value. This timer is rd/writable,up/down Based on the configuration, this can be used either for up or down counter and generates an interrupt on reaching Zero in down mode. in up mode interrupt is generated when the timer reaches the sw_match_register_6.

WCSS_A_MTU_TIMER_6

Bits	Name	Description
23:0	TIMER_6	This register contains the timer 6 value. This timer is rd/writable,up/down

0x0A08143C WCSS_A_MTU_TIMER_7**Type:** RW**Reset State:** 0x00FFFFFF

This register contains the timer 7 value. This timer is rd/writable,up/down Based on the configuration, this can be used either for up or down counter and generates an interrupt on reaching Zero in down mode. in up mode interrupt is generated when the timer reaches the sw_match_register_7.

WCSS_A_MTU_TIMER_7

Bits	Name	Description
23:0	TIMER_7	This register contains the timer 7 value. This timer is rd/writable,up/down

0x0A081440 WCSS_A_MTU_TIMER_CONTROL

Type: RW

Reset State: 0x00000000

This register contains all the controls for the timers.

WCSS_A_MTU_TIMER_CONTROL

Bits	Name	Description
31:16	SW_MTU_BASIC_UNIT_SELECT	2 bit for each timer is provided. bit [1:0] for timer 0, bit [3:2] for timer 1... To make these timers more flexible, 3 basic units are provided. if the control =0, the timer will be stopped (this is the only way to disable the counters if the control =01, timer is ++ at every clk if the control =10, timer is ++ at every one us if the control =11, timer is ++ at every programmable pulse. Basically in this mode, the period of the pulse is also controlled by s/w. Note that this programmable pulse is only one for all the timers
11:0	SW_MTU_TIMER_UP_DOWN_CNTRL	1 bit for each timer is provided. bit 0 for timer 0, bit 1 for timer 1... This controls the up/down functionality of the timers. 1-> up, 0->down.

0x0A081444 WCSS_A_MTU_BKOF_CNT_0

Type: RW

Reset State: 0x00000003

This register contains the bkof_cnt of bkof 0. The units are slot number. In one of the modes, sw can control this counter and run the back offs with this value. But this is normal way of operating. In this mode, sw needs to take care of the random number generation also

WCSS_A_MTU_BKOF_CNT_0

Bits	Name	Description
15:0	BKOF_CNT_0	This register contains the bkof_counter value for the back-off 0 In the sw controlled mode, (sw_mtu_use_sw_bkof_counter = 1), sw will have to write this register before starting the back-off. if sw_mtu_use_sw_bkof_counter =0, hw generates this value = (lfsr_data & sw_mtu_cw_reg)

0x0A081448 WCSS_A_MTU_BKOF_CNT_1**Type:** RW**Reset State:** 0x00000004

This register contains the bkof_cnt of bkof 1. In one of the modes, sw can control this counter and run the back offs with this value. But this is normal way of operating In this mode, sw needs to take care of the random number generation also

WCSS_A_MTU_BKOF_CNT_1

Bits	Name	Description
15:0	BKOF_CNT_1	This register contains the bkof_counter value for the back-off 1 In the sw controlled mode, (sw_mtu_use_sw_bkof_counter = 1), sw will have to write this register before starting the back-off. if sw_mtu_use_sw_bkof_counter =0, hw generates this value = (lfsr_data & sw_mtu_cw_reg)

0x0A08144C WCSS_A_MTU_BKOF_CNT_2**Type:** RW**Reset State:** 0x00000007

This register contains the bkof_cnt of bkof 2. In one of the modes, sw can control this counter and run the back offs with this value. But this is normal way of operating In this mode, sw needs to take care of the random number generation also

WCSS_A_MTU_BKOF_CNT_2

Bits	Name	Description
15:0	BKOF_CNT_2	This register contains the bkof_counter value for the back-off 2 In the sw controlled mode, (sw_mtu_use_sw_bkof_counter = 1), sw will have to write this register before starting the back-off. if sw_mtu_use_sw_bkof_counter =0, hw generates this value = (lfsr_data & sw_mtu_cw_reg)

0x0A081450 WCSS_A_MTU_BKOF_CNT_3**Type:** RW**Reset State:** 0x00000001

This register contains the bkof_cnt of bkof 3. In one of the modes, sw can control this counter and run the back offs with this value. But this is normal way of operating. In this mode, sw needs to take care of the random number generation also

WCSS_A_MTU_BKOF_CNT_3

Bits	Name	Description
15:0	BKOF_CNT_3	This register contains the bkof_counter value for the back-off 4. In the sw controlled mode, (sw_mtu_use_sw_bkof_counter = 1), sw will have to write this register before starting the back-off. if sw_mtu_use_sw_bkof_counter = 0, hw generates this value = (lfsr_data & sw_mtu_cw_reg)

0x0A081454 WCSS_A_MTU_BKOF_CNT_4**Type:** RW**Reset State:** 0x00000002

This register contains the bkof_cnt of bkof 4. In one of the modes, sw can control this counter and run the back offs with this value. But this is normal way of operating. In this mode, sw needs to take care of the random number generation also

WCSS_A_MTU_BKOF_CNT_4

Bits	Name	Description
15:0	BKOF_CNT_4	This register contains the bkof_counter value for the back-off 4. In the sw controlled mode, (sw_mtu_use_sw_bkof_counter = 1), sw will have to write this register before starting the back-off. if sw_mtu_use_sw_bkof_counter = 0, hw generates this value = (lfsr_data & sw_mtu_cw_reg)

0x0A081458 WCSS_A_MTU_BKOF_CNT_5**Type:** RW**Reset State:** 0x00000005

This register contains the bkof_cnt of bkof 5. In one of the modes, sw can control this counter and run the back offs with this value. But this is normal way of operating. In this mode, sw needs to take care of the random number generation also

WCSS_A_MTU_BKOF_CNT_5

Bits	Name	Description
15:0	BKOF_CNT_5	This register contains the bkof_counter value for the back-off 5. In the sw controlled mode, (sw_mtu_use_sw_bkof_counter = 1), sw will have to write this register before starting the back-off. if sw_mtu_use_sw_bkof_counter = 0, hw generates this value = (lfsr_data & sw_mtu_cw_reg)

0x0A08145C WCSS_A_MTU_BKOF_CNT_6**Type:** RW**Reset State:** 0x00000006

This register contains the bkof_cnt of bkof 6. In one of the modes, sw can control this counter and run the back offs with this value. But this is normal way of operating. In this mode, sw needs to take care of the random number generation also.

WCSS_A_MTU_BKOF_CNT_6

Bits	Name	Description
15:0	BKOF_CNT_6	This register contains the bkof_counter value for the back-off 6. In the sw controlled mode, (sw_mtu_use_sw_bkof_counter = 1), sw will have to write this register before starting the back-off. if sw_mtu_use_sw_bkof_counter = 0, hw generates this value = (lfsr_data & sw_mtu_cw_reg)

0x0A081460 WCSS_A_MTU_BKOF_CNT_7**Type:** RW**Reset State:** 0x00000000

This register contains the bkof_cnt of bkof 7. In one of the modes, sw can control this counter and run the back offs with this value. But this is normal way of operating. In this mode, sw needs to take care of the random number generation also.

WCSS_A_MTU_BKOF_CNT_7

Bits	Name	Description
15:0	BKOF_CNT_7	This register contains the bkof_counter value for the back-off 7. In the sw controlled mode, (sw_mtu_use_sw_bkof_counter = 1), sw will have to write this register before starting the back-off. if sw_mtu_use_sw_bkof_counter = 0, hw generates this value = (lfsr_data & sw_mtu_cw_reg)

0x0A081464 WCSS_A_MTU_BKOF_CONTROL**Type:** RW**Reset State:** 0xFF000000

This register contains all the controls for the back-off engines.

WCSS_A_MTU_BKOF_CONTROL

Bits	Name	Description
31:24	SW_MTU_SUPPRESS_COLLISION_INT	1 bit for each back off is provided. bit 0 for back 0, bit 1 for back 1... this bit is set to 1, the back off will not send interrupt to s/w for collision case (int status = 2'b11) warm up also will not be generated
23:16	SW_MTU_QUEUE_DATA_AVAIL	1 bit for each back is provided. bit 0 for back 0, bit 1 for back 1... data avail bits from the s/w. Unless this bit is set, s/w won't get interrupt (irrespective of the status of back-off).
15:8	SW_MTU_STALL_BKOF	1 bit for each back is provided. bit 0 for back 0, bit 1 for back 1... if this bit is set, the corresponding back-off will be stalled. Basically the back-off counter -- will be stopped. This is used to stall all the back-offs while beacon is supposed to be sent or for any other reasons.
7:0	SW_MTU_USE_SW_BKOF_COUNTER	1 bit for each back is provided. bit 0 for back 0, bit 1 for back 1... if this bit is set, the back_off counter is set by sw. Otherwise, HW will use the cw_reg & lfsdr

0x0A081468 WCSS_A_MTU_CW_REG_CONTROL_FOR_BACKOFF_0**Type:** RW**Reset State:** 0x00000007

This register contains the cw_reg value that needs to be used by hw. For backoff engine_0 This register also has the control bit that initiates the back-off after the hw issued an interrupt to s/w for its attention.

WCSS_A_MTU_CW_REG_CONTROL_FOR_BACKOFF_0

Bits	Name	Description
19	SW_MTU_CW_UPDATE_VALID	this is write only register The sw_mtu_cw_reg will be updated only when this bit is set. Otherwise the hw update will take place
18:17	SW_MTU_CW_ACTION_0	This field indicates the action that bkoff-engine need to do to the cw_reg. 00-> no change, 01-> reset cw (cw_reg=cw_min), 10-> ++cw, hw does the boundary checks and resets the cw if it reaches the cw_max automatically
16	SW_MTU_WAIT_DONE_P_0	This bit is write only. The HW will be waiting for this signal to start the next back-off after issuing the interrupt by the back-off engine

WCSS_A_MTU_CW_REG_CONTROL_FOR_BACKOFF_0 (cont.)

Bits	Name	Description
15:0	SW_MTU_CW_REG_0	This register contains the cw_reg value for the back-off 0

0x0A08146C WCSS_A_MTU_CW_REG_CONTROL_FOR_BACKOFF_1**Type:** RW**Reset State:** 0x00000007

This register contains the cw_reg value that needs to be used by hw . For backoff engine_1 This register also has the control bit that initiates the back-off after the hw issued an interrupt to s/w for its attention.

WCSS_A_MTU_CW_REG_CONTROL_FOR_BACKOFF_1

Bits	Name	Description
19	SW_MTU_CW_UPDATE_VA LID	this is write only register The sw_mtu_cw_reg will be updated only when this bit is set. Otherwise the hw update will take place
18:17	SW_MTU_CW_ACTION_1	This field indicates the action that bkoff-engine need to do to the cw_reg. 00-> no change, 01-> reset cw (cw_reg=cw_min), 10-> ++cw, hw does the boundary checks and resets the cw if it reaches the cw_max automatically
16	SW_MTU_WAIT_DONE_P_ 1	This bit is write only. The HW will be waiting for this signal to start the next back-off after issuing the interrupt by the back-off engine
15:0	SW_MTU_CW_REG_1	This register contains the cw_reg value for the back-off 1

0x0A081470 WCSS_A_MTU_CW_REG_CONTROL_FOR_BACKOFF_2**Type:** RW**Reset State:** 0x00000007

This register contains the cw_reg value that needs to be used by hw. For backoff engine_2 This register also has the control bit that initiates the back-off after the hw issued an interrupt to s/w for its attention.

WCSS_A_MTU_CW_REG_CONTROL_FOR_BACKOFF_2

Bits	Name	Description
19	SW_MTU_CW_UPDATE_VA LID	this is write only register The sw_mtu_cw_reg will be updated only when this bit is set. Otherwise the hw update will take place
18:17	SW_MTU_CW_ACTION_2	This field indicates the action that bkoff-engine need to do to the cw_reg. 00-> no change, 01-> reset cw (cw_reg=cw_min), 10-> ++cw, hw does the boundary checks and resets the cw if it reaches the cw_max automatically

WCSS_A_MTU_CW_REG_CONTROL_FOR_BACKOFF_2 (cont.)

Bits	Name	Description
16	SW_MTU_WAIT_DONE_P_2	This bit is write only. The HW will be waiting for this signal to start the next back-off after issuing the interrupt by the back-off engine
15:0	SW_MTU_CW_REG_2	This register contains the cw_reg value for the back-off 2

0x0A081474 WCSS_A_MTU_CW_REG_CONTROL_FOR_BACKOFF_3**Type:** RW**Reset State:** 0x00000007

This register contains the cw_reg value that needs to be used by hw . For backoff engine_3 This register also has the control bit that initiates the back-off after the hw issued an interrupt to s/w for its attention.

WCSS_A_MTU_CW_REG_CONTROL_FOR_BACKOFF_3

Bits	Name	Description
19	SW_MTU_CW_UPDATE_VA LID	this is write only register The sw_mtu_cw_reg will be updated only when this bit is set. Otherwise the hw update will take place
18:17	SW_MTU_CW_ACTION_3	This field indicates the action that bkoff-engine need to do to the cw_reg. 00-> no change, 01-> reset cw (cw_reg=cw_min), 10-> ++cw, hw does the boundary checks and resets the cw if it reaches the cw_max automatically
16	SW_MTU_WAIT_DONE_P_3	This bit is write only. The HW will be waiting for this signal to start the next back-off after issuing the interrupt by the back-off engine
15:0	SW_MTU_CW_REG_3	This register contains the cw_reg value for the back-off 3

0x0A081478 WCSS_A_MTU_CW_REG_CONTROL_FOR_BACKOFF_4**Type:** RW**Reset State:** 0x00000007

This register contains the cw_reg value that needs to be used by hw . For backoff engine_4 This register also has the control bit that initiates the back-off after the hw issued an interrupt to s/w for its attention.

WCSS_A_MTU_CW_REG_CONTROL_FOR_BACKOFF_4

Bits	Name	Description
19	SW_MTU_CW_UPDATE_VA LID	this is write only register The sw_mtu_cw_reg will be updated only when this bit is set. Otherwise the hw update will take place

WCSS_A_MTU_CW_REG_CONTROL_FOR_BACKOFF_4 (cont.)

Bits	Name	Description
18:17	SW_MTU_CW_ACTION_4	This field indicates the action that bkoff-engine need to do to the cw_reg. 00-> no change, 01-> reset cw (cw_reg=cw_min), 10-> ++cw, hw does the boundary checks and resets the cw if it reaches the cw_max automatically
16	SW_MTU_WAIT_DONE_P_4	This bit is write only. The HW will be waiting for this signal to start the next back-off after issuing the interrupt by the back-off engine
15:0	SW_MTU_CW_REG_4	This register contains the cw_reg value for the back-off 4

0x0A08147C WCSS_A_MTU_CW_REG_CONTROL_FOR_BACKOFF_5**Type:** RW**Reset State:** 0x00000007

This register contains the cw_reg value that needs to be used by hw . For backoff engine_5 This register also has the control bit that initiates the back-off after the hw issued an interrupt to s/w for its attention.

WCSS_A_MTU_CW_REG_CONTROL_FOR_BACKOFF_5

Bits	Name	Description
19	SW_MTU_CW_UPDATE_VA LID	this is write only register The sw_mtu_cw_reg will be updated only when this bit is set. Otherwise the hw update will take place
18:17	SW_MTU_CW_ACTION_5	This field indicates the action that bkoff-engine need to do to the cw_reg. 00-> no change, 01-> reset cw (cw_reg=cw_min), 10-> ++cw, hw does the boundary checks and resets the cw if it reaches the cw_max automatically
16	SW_MTU_WAIT_DONE_P_5	This bit is write only. The HW will be waiting for this signal to start the next back-off after issuing the interrupt by the back-off engine
15:0	SW_MTU_CW_REG_5	This register contains the cw_reg value for the back-off 5

0x0A081480 WCSS_A_MTU_CW_REG_CONTROL_FOR_BACKOFF_6**Type:** RW**Reset State:** 0x00000007

This register contains the cw_reg value that needs to be used by hw . For backoff engine_6 This register also has the control bit that initiates the back-off after the hw issued an interrupt to s/w for its attention.

WCSS_A_MTU_CW_REG_CONTROL_FOR_BACKOFF_6

Bits	Name	Description
19	SW_MTU_CW_UPDATE_VA LID	this is write only register The sw_mtu_cw_reg will be updated only when this bit is set. Otherwise the hw update will take place
18:17	SW_MTU_CW_ACTION_6	This field indicates the action that bkoff-engine need to do to the cw_reg. 00-> no change, 01-> reset cw (cw_reg=cw_min), 10-> ++cw, hw does the boundary checks and resets the cw if it reaches the cw_max automatically
16	SW_MTU_WAIT_DONE_P_6	This bit is write only. The HW will be waiting for this signal to start the next back-off after issuing the interrupt by the back-off engine
15:0	SW_MTU_CW_REG_6	This register contains the cw_reg value for the back-off 6

0x0A081484 WCSS_A_MTU_CW_REG_CONTROL_FOR_BACKOFF_7**Type:** RW**Reset State:** 0x00000007

This register contains the cw_reg value that needs to be used by hw . For backoff engine_7 This register also has the control bit that initiates the back-off after the hw issued an interrupt to s/w for its attention.

WCSS_A_MTU_CW_REG_CONTROL_FOR_BACKOFF_7

Bits	Name	Description
19	SW_MTU_CW_UPDATE_VA LID	this is write only register The sw_mtu_cw_reg will be updated only when this bit is set. Otherwise the hw update will take place
18:17	SW_MTU_CW_ACTION_7	This field indicates the action that bkoff-engine need to do to the cw_reg. 00-> no change, 01-> reset cw (cw_reg=cw_min), 10-> ++cw, hw does the boundary checks and resets the cw if it reaches the cw_max automatically
16	SW_MTU_WAIT_DONE_P_7	This bit is write only. The HW will be waiting for this signal to start the next back-off after issuing the interrupt by the back-off engine
15:0	SW_MTU_CW_REG_7	This register contains the cw_reg value for the back-off 7

0x0A081488 WCSS_A_MTU_MTU_INTERRUPT_STATUS**Type:** RW**Reset State:** 0x00000000

This register contains the status of all the interrupts that are generated by mtu Basically enevry interrupt has a corresponding register bit which will be set on sending interrupt pulse to the interrupt module and this bit is reset by writing '1' to it. sw has higher priority over hw

WCSS_A_MTU_MTU_INTERRUPT_STATUS

Bits	Name	Type	Description
31	SECONDARY_CCA_TO	R	the secondary_cca_to is sampled and s/w can read its status through this register
30	SIFS_TO	R	the sifs_to is sampled and s/w can read its status through this register
29	PIFS_TO	R	the pifs_to is sampled and s/w can read its status through this register
28	MTU_MCU_WATCH_DOG_ENABLE_DISABLE_ERROR	RW	if s/w tries to write wrong combinations to the watch dog enable disable bits this interrupt will be generated
27	MTU_MCU_WATCH_DOG_PROTECTION_ERROR	RW	if s/w tries to write to the watch dog register and if the protection vector matches, this interrupt will be generated.
26:24	MTU_MCU_WATCH_DOG_TIMER_FLAG	RW	watch dog interrupt flags. 24 is for thr0, bit 25 is for thr1, bit 26 is fr thr3
23:16	MTU_MCU_TSF_INT_FLAG	RW	8 registers for 8 tsf based match register interrupts. bit 0 for match 0, 1 for match 1 ...
15:8	MTU_MCU_BKOF_INT_FLAG	RW	8 registers for 8 back off interrupts. bit 0 for timer 0, 1 for timer 1 ...
7:0	MTU_MCU_TIMER_INT_FLAG	RW	8 registers for 8 timer interrupts. bit 0 for timer 0, 1 for timer 1 ...

0x0A08148C WCSS_A_MTU_SW_MATCH_REGISTER_0**Type:** RW**Reset State:** 0x00000000

This register contains the match value for tsf based interrupts. Basically whenever the tsf is equal to this register, corresponding interrupt will be generated. This register is also used the timer 0 in up mode. Basically when the timer is operated in up mode, when the Timer 0 reaches this value, an interrupt is generated

WCSS_A_MTU_SW_MATCH_REGISTER_0

Bits	Name	Description
31:30	SW_TSF_SEL_0	Lower bits to select the tsf will trigger the interrupt when the timer value reaches 3'b00: BSS TSF; 3'b01: IBSS TSF; 3'b10: P2P group owner TSF; 3'b11: P2P client TSF 3'b100: BSS2 TSF
29	SW_TSF_SEL_0_HI	Upper bit to select the tsf will trigger the interrupt when the timer value reaches
23:0	SW_MATCH_REGISTER_0	This register contains the match value for tsf based counter interrupt 0, timer 0 (in up mode)

0x0A081490 WCSS_A_MTU_SW_MATCH_REGISTER_1**Type:** RW**Reset State:** 0x00000EFF

This register contains the match value for tsf based interrupts. Basically whenever the tsf is equal to this register, corresponding interrupt will be generated. This register is also used the timer 1 in up mode. Basically when the timer is operated in up mode, when the Timer 1 reaches this value, an interrupt is generated

WCSS_A_MTU_SW_MATCH_REGISTER_1

Bits	Name	Description
31:30	SW_TSF_SEL_1	Lower bits to select the tsf will trigger the interrupt when the timer value reaches 3'b00: BSS TSF; 3'b01: IBSS TSF; 3'b10: P2P group owner TSF; 3'b11: P2P client TSF 3'b100: BSS2 TSF
29	SW_TSF_SEL_1_HI	Upper bit to select the tsf will trigger the interrupt when the timer value reaches
23:0	SW_MATCH_REGISTER_1	This register contains the match value for tsf based counter interrupt 1, timer 1 (in up mode)

0x0A081494 WCSS_A_MTU_SW_MATCH_REGISTER_2**Type:** RW**Reset State:** 0x00000EFF

This register contains the match value for tsf based interrupts. Basically whenever the tsf is equal to this register, corresponding interrupt will be generated. This register is also used the timer 2 in up mode. Basically when the timer is operated in up mode, when the Timer 2 reaches this value, an interrupt is generated

WCSS_A_MTU_SW_MATCH_REGISTER_2

Bits	Name	Description
31:30	SW_TSF_SEL_2	Lower bits to select the tsf will trigger the interrupt when the timer value reaches 3'b00: BSS TSF; 3'b01: IBSS TSF; 3'b10: P2P group owner TSF; 3'b11: P2P client TSF 3'b100: BSS2 TSF
29	SW_TSF_SEL_2_HI	Upper bit to select the tsf will trigger the interrupt when the timer value reaches
23:0	SW_MATCH_REGISTER_2	This register contains the match value for tsf based counter interrupt 2, timer 2 (in up mode)

0x0A081498 WCSS_A_MTU_SW_MATCH_REGISTER_3**Type:** RW**Reset State:** 0x00000000

This register contains the match value for tsf based interrupts. Basically whenever the tsf is equal to this register, corresponding interrupt will be generated. This register is also used the timer 3 in up mode. Basically when the timer is operated in up mode, when the Timer 3 reaches this value, an interrupt is generated

WCSS_A_MTU_SW_MATCH_REGISTER_3

Bits	Name	Description
31:30	SW_TSF_SEL_3	Lower bits to select the tsf will trigger the interrupt when the timer value reaches 3'b00: BSS TSF; 3'b01: IBSS TSF; 3'b10: P2P group owner TSF; 3'b11: P2P client TSF 3'b100: BSS2 TSF
29	SW_TSF_SEL_3_HI	Upper bit to select the tsf will trigger the interrupt when the timer value reaches
23:0	SW_MATCH_REGISTER_3	This register contains the match value for tsf based counter interrupt 3, timer 3 (in up mode)

0x0A08149C WCSS_A_MTU_SW_MATCH_REGISTER_4**Type:** RW**Reset State:** 0x00000000

This register contains the match value for tsf based interrupts. Basically whenever the tsf is equal to this register, corresponding interrupt will be generated. This register is also used the timer 4 in up mode. Basically when the timer is operated in up mode, when the Timer 4 reaches this value, an interrupt is generated

WCSS_A_MTU_SW_MATCH_REGISTER_4

Bits	Name	Description
31:30	SW_TSF_SEL_4	Lower bits to select the tsf will trigger the interrupt when the timer value reaches 3'b00: BSS TSF; 3'b01: IBSS TSF; 3'b10: P2P group owner TSF; 3'b11: P2P client TSF 3'b100: BSS2 TSF
29	SW_TSF_SEL_4_HI	Upper bit to select the tsf will trigger the interrupt when the timer value reaches
23:0	SW_MATCH_REGISTER_4	This register contains the match value for tsf based counter interrupt 4, timer 4 (in up mode)

0x0A0814A0 WCSS_A_MTU_SW_MATCH_REGISTER_5**Type:** RW**Reset State:** 0x00000EFF

This register contains the match value for tsf based interrupts. Basically whenever the tsf is equal to this register, corresponding interrupt will be generated. This register is also used the timer 5 in up mode. Basically when the timer is operated in up mode, when the Timer 5 reaches this value, an interrupt is generated

WCSS_A_MTU_SW_MATCH_REGISTER_5

Bits	Name	Description
31:30	SW_TSF_SEL_5	Lower bits to select the tsf will trigger the interrupt when the timer value reaches 3'b00: BSS TSF; 3'b01: IBSS TSF; 3'b10: P2P group owner TSF; 3'b11: P2P client TSF 3'b100: BSS2 TSF
29	SW_TSF_SEL_5_HI	Upper bit to select the tsf will trigger the interrupt when the timer value reaches
23:0	SW_MATCH_REGISTER_5	This register contains the match value for tsf based counter interrupt 5, timer 5 (in up mode)

0x0A0814A4 WCSS_A_MTU_SW_MATCH_REGISTER_6**Type:** RW**Reset State:** 0x00000EFF

This register contains the match value for tsf based interrupts. Basically whenever the tsf is equal to this register, corresponding interrupt will be generated. This register is also used the timer 6 in up mode. Basically when the timer is operated in up mode, when the Timer 6 reaches this value, an interrupt is generated

WCSS_A_MTU_SW_MATCH_REGISTER_6

Bits	Name	Description
31:30	SW_TSF_SEL_6	Lower bits to select the tsf will trigger the interrupt when the timer value reaches 3'b00: BSS TSF; 3'b01: IBSS TSF; 3'b10: P2P group owner TSF; 3'b11: P2P client TSF 3'b100: BSS2 TSF
29	SW_TSF_SEL_6_HI	Upper bit to select the tsf will trigger the interrupt when the timer value reaches
23:0	SW_MATCH_REGISTER_6	This register contains the match value for tsf based counter interrupt 6, timer 6 (in up mode)

0x0A0814A8 WCSS_A_MTU_SW_MATCH_REGISTER_7**Type:** RW**Reset State:** 0x00000EFF

This register contains the match value for tsf based interrupts. Basically whenever the tsf is equal to this register, corresponding interrupt will be generated. This register is also used the timer 7 in up mode. Basically when the timer is operated in up mode, when the Timer 7 reaches this value, an interrupt is generated

WCSS_A_MTU_SW_MATCH_REGISTER_7

Bits	Name	Description
31:30	SW_TSF_SEL_7	Lower bits to select the tsf will trigger the interrupt when the timer value reaches 3'b00: BSS TSF; 3'b01: IBSS TSF; 3'b10: P2P group owner TSF; 3'b11: P2P client TSF 3'b100: BSS2 TSF
29	SW_TSF_SEL_7_HI	Upper bit to select the tsf will trigger the interrupt when the timer value reaches
23:0	SW_MATCH_REGISTER_7	This register contains the match value for tsf based counter interrupt 7, timer 7 (in up mode)

0x0A0814AC WCSS_A_MTU_SW_CW_MIN_CW_MAX_0**Type:** RW**Reset State:** 0x03FF000F

Contains the cw_min and cw_max for back-off 0

WCSS_A_MTU_SW_CW_MIN_CW_MAX_0

Bits	Name	Description
31:16	SW_CW_MAX_0	This register contains the cw_max for back-off 0. The default is 7ff. Any suggested value is welcome
15:0	SW_CW_MIN_0	This register contains the cw_min for back-off 0. The default is 3f. Any suggested value is welcome

0x0A0814B0 WCSS_A_MTU_SW_CW_MIN_CW_MAX_1**Type:** RW**Reset State:** 0x03FF000F

Contains the cw_min and cw_max for back-off 1

WCSS_A_MTU_SW_CW_MIN_CW_MAX_1

Bits	Name	Description
31:16	SW_CW_MAX_1	This register contains the cw_max for back-off 1. The default is 7ff. Any suggested value is welcome
15:0	SW_CW_MIN_1	This register contains the cw_min for back-off 1. The default is 3f. Any suggested value is welcome

0x0A0814B4 WCSS_A_MTU_SW_CW_MIN_CW_MAX_2**Type:** RW**Reset State:** 0x00070003

Contains the cw_min and cw_max for back-off 2

WCSS_A_MTU_SW_CW_MIN_CW_MAX_2

Bits	Name	Description
31:16	SW_CW_MAX_2	This register contains the cw_max for back-off 2. The default is 7ff. Any suggested value is welcome
15:0	SW_CW_MIN_2	This register contains the cw_min for back-off 2. The default is 3f. Any suggested value is welcome

0x0A0814B8 WCSS_A_MTU_SW_CW_MIN_CW_MAX_3**Type:** RW**Reset State:** 0x00070003

Contains the cw_min and cw_max for back-off 3

WCSS_A_MTU_SW_CW_MIN_CW_MAX_3

Bits	Name	Description
31:16	SW_CW_MAX_3	This register contains the cw_max for back-off 3. The default is 7ff. Any suggested value is welcome
15:0	SW_CW_MIN_3	This register contains the cw_min for back-off 3. The default is 3f. Any suggested value is welcome

0x0A0814BC WCSS_A_MTU_SW_CW_MIN_CW_MAX_4**Type:** RW**Reset State:** 0x00070003

Contains the cw_min and cw_max for back-off 4

WCSS_A_MTU_SW_CW_MIN_CW_MAX_4

Bits	Name	Description
31:16	SW_CW_MAX_4	This register contains the cw_max for back-off 4. The default is 7ff. Any suggested value is welcome
15:0	SW_CW_MIN_4	This register contains the cw_min for back-off 4. The default is 3f. Any suggested value is welcome

0x0A0814C0 WCSS_A_MTU_SW_CW_MIN_CW_MAX_5

Type: RW

Reset State: 0x000F0007

Contains the cw_min and cw_max for back-off 5

WCSS_A_MTU_SW_CW_MIN_CW_MAX_5

Bits	Name	Description
31:16	SW_CW_MAX_5	This register contains the cw_max for back-off 5. The default is 7ff. Any suggested value is welcome
15:0	SW_CW_MIN_5	This register contains the cw_min for back-off 5. The default is 3f. Any suggested value is welcome

0x0A0814C4 WCSS_A_MTU_SW_CW_MIN_CW_MAX_6

Type: RW

Reset State: 0x03FF000F

Contains the cw_min and cw_max for back-off 6

WCSS_A_MTU_SW_CW_MIN_CW_MAX_6

Bits	Name	Description
31:16	SW_CW_MAX_6	This register contains the cw_max for back-off 6. The default is 7ff. Any suggested value is welcome
15:0	SW_CW_MIN_6	This register contains the cw_min for back-off 6. The default is 3f. Any suggested value is welcome

0x0A0814C8 WCSS_A_MTU_SW_CW_MIN_CW_MAX_7**Type:** RW**Reset State:** 0x03FF000F

Contains the cw_min and cw_max for back-off 7

WCSS_A_MTU_SW_CW_MIN_CW_MAX_7

Bits	Name	Description
31:16	SW_CW_MAX_7	This register contains the cw_max for back-off 7. The default is 7ff. Any suggested value is welcome
15:0	SW_CW_MIN_7	This register contains the cw_min for back-off 7. The default is 3f. Any suggested value is welcome

0x0A0814CC WCSS_A_MTU_CCA_COUNTER0**Type:** RW**Reset State:** 0x00000000

this register contains the primary cca count. the primary cca is sampled at every us and incremented if the cca is LOW

WCSS_A_MTU_CCA_COUNTER0

Bits	Name	Description
31:0	CCA_COUNTER0	contains the cca primary counter. s/w write has higher periority over h/w update

0x0A0814D0 WCSS_A_MTU_CCA_COUNTER1**Type:** RW**Reset State:** 0x00000000

this register contains the cca count. the selection to the cca used by this counter is programmable -
--

WCSS_A_MTU_CCA_COUNTER1

Bits	Name	Description
31:0	CCA_COUNTER1	depending on the cca_count1_sel-- a programmable bit, the enable to this counter is selected as shown below s/w write has higher periority over h/w update cca_term0 = ~rxp_mtu_secondary_cca; cca_term1 = ~rxp_mtu_secondary_cca & ~rxp_mtu_primary_cca; cca_term2 = rxp_mtu_secondary_cca & ~rxp_mtu_primary_cca; cca_term3 = ~rxp_mtu_secondary_cca cca_term4 = rxp_mtu_secondary_cca cca_term5 = ~cca_term1; cca_term6 = ~cca_term3; cca_term7 = ~cca_term4; cca_count0_en_temp = ~rxp_mtu_primary_cca; 3'b000: cca_count1_en_temp = cca_term0; 3'b001: cca_count1_en_temp = cca_term1; 3'b010: cca_count1_en_temp = cca_term2; 3'b011: cca_count1_en_temp = cca_term3; 3'b100: cca_count1_en_temp = cca_term4; 3'b101: cca_count1_en_temp = cca_term5; 3'b110: cca_count1_en_temp = cca_term6; 3'b111: cca_count1_en_temp = cca_term7; cca_count1_en_final = consider_rx_cca_only ? (cca_count1_en_temp & ~txp_mtu_phytx_idle) : cca_count1_en_temp;

0x0A0814D4 WCSS_A_MTU_CCA_COUNTER2**Type:** RW**Reset State:** 0x00000000

this register contains the cca count. the selection to the cca used by this counter is programmable -
--

WCSS_A_MTU_CCA_COUNTER2

Bits	Name	Description
31:0	CCA_COUNTER2	depending on the cca_count2_sel-- a programmable bit, the enable to this counter is selected as shown below s/w write has higher periority over h/w update 3'b000: cca_count2_en_temp = ~rxp_mtu_primary_cca & ~rxp_mtu_secondary_cca & ~rxp_mtu_tertiary_cca & ~rxp_mtu_quaternary_cca; 3'b001: cca_count2_en_temp = ~rxp_mtu_tertiary_cca; 3'b010: cca_count2_en_temp = ~rxp_mtu_quaternary_cca; 3'b011: cca_count2_en_temp = ~rxp_mtu_tertiary_cca & ~rxp_mtu_quaternary_cca; 3'b100: cca_count2_en_temp = ~rxp_mtu_tertiary_cca 3'b101: cca_count2_en_temp = rxp_mtu_tertiary_cca & ~rxp_mtu_quaternary_cca; 3'b110: cca_count2_en_temp = rxp_mtu_tertiary_cca 3'b111: cca_count2_en_temp = 1'b0; cca_count2_en_final = consider_rx_cca_only ? (cca_count2_en_temp & ~txp_mtu_phytx_idle) : cca_count2_en_temp;

0x0A0814D8 WCSS_A_MTU_WATCH_DOG_TIMER**Type:** RW**Reset State:** 0x00000000

This is watch dog timer register. Basically, when the watch dog timer is enabled, this timer keeps ++ at every us. Once it reaches watch dog thr0, an interrupt is generated when it reaches thr1 another interrupt is generated (higher priority). On reaching thr3, another higher priority interrupt is generated. Typically this is used to reset the chip

WCSS_A_MTU_WATCH_DOG_TIMER

Bits	Name	Description
31:30	WATCH_DOG_TIMER_ENABLE_DISABLE	Watch dog timer and the corresponding interrupts are enabled only when these bits are 10. 01 is disable. Any other combination will not be accepted and if sw writes 00 or 11, the status is maintained. We may have an interrupt to this condition as well as for any writes to the watch dog register without protection vector match
29:24	WATCH_DOG_TIMER_PROTECTION_VECTOR	This is write only register. when it is read ZEROs are returned. Writes to the watch dog register can happen (including the enable/disable) only when this field matches the watch_dog_thr0[25:24],thr1[25:24],thr2[25:24]
23:0	WATCH_DOG_TIMER	if it is enabled, it keeps ++ at every us. S/W can write to it only when the protection vector matches the concat of the 2 bit protection bits from the watch dog threshold register. The protection vector can't be 00 or 3f

0x0A0814DC WCSS_A_MTU_WATCH_DOG_THR0**Type:** RW**Reset State:** 0x00000000

this register contains thr0 for watch dog

WCSS_A_MTU_WATCH_DOG_THR0

Bits	Name	Description
27:26	THR0_PROTECTION_FIELD	This register is writable only when this field is 11
25:24	WATCH_DOG_PROTECTION_VECTOR0	2 bits of the protection vector of watch_dog_timer
23:0	WATCH_DOG_THR0	contains the watch dog thr0. It is writable by s/w only when its protection field =11

0x0A0814E0 WCSS_A_MTU_WATCH_DOG_THR1**Type:** RW**Reset State:** 0x00000000

this register contains thr1 for wathch dog

WCSS_A_MTU_WATCH_DOG_THR1

Bits	Name	Description
27:26	THR1_PROTECTION_FIELD	This register is writable only when this field is 10
25:24	WATCH_DOG_PROTECTIO N_VECTOR1	2 bits of the protection vector of watch_dog_timer
23:0	WATCH_DOG_THR1	contains the watch dog thr1. It is writable by s/w only when its protection field =10

0x0A0814E4 WCSS_A_MTU_WATCH_DOG_THR2**Type:** RW**Reset State:** 0x00000000

this register contains thr2 for wathch dog

WCSS_A_MTU_WATCH_DOG_THR2

Bits	Name	Description
27:26	THR2_PROTECTION_FIELD	This register is writable only when this field is 01
25:24	WATCH_DOG_PROTECTIO N_VECTOR2	2 bits of the protection vector of watch_dog_timer
23:0	WATCH_DOG_THR2	contains the watch dog thr2. It is writable by s/w only when its protection field =01

0x0A0814E8 WCSS_A_MTU_BKOF_SW_INT_TYPE**Type:** RW**Reset State:** 0x00000000

this register contains the status of the interrupt type for the back off engines

WCSS_A_MTU_BKOF_SW_INT_TYPE

Bits	Name	Description
15:0	BKOF_SW_INT_TYPE	contains the type of the interrupt from the back off engines. 1:0 is bkoff 0, 3:2 for bkoff 1 and so on. Value 2'b00: No interrupt pending, or cleared by SW Value 2'b01: Backoff engine expired successfully (no collision with a higher priority backoff engine) Value 2'b10: There was an internal collision between this and a higher priority backoff engine. SW needs to re-initialize the backoff engine. Value 2'b11: Dataavailable got programmed after the backoff engine had already run and reached the BKOFF_DONE state. At the moment of programming Dataavailable, the CCA was low. SW needs to re-program the backoff engine. This is read only register. But any write 1 to this register will clear the status for the corresponding back-off. For ex. write of a 1 to bit 0 will clear 1:0, write of a 1 to bit 1 clears the 3:2 and so on. Only 11:0 are enabled for write operation, but will also automatically clear on a write. Corresponds to reset_bkof_sw_int_type_p in RTL

0x0A0814EC WCSS_A_MTU_CCA_CONTROL_REG**Type:** RW**Reset State:** 0x00000000

this register contains the controls to select the cca that needs to be used for a particular back_off engine

WCSS_A_MTU_CCA_CONTROL_REG

Bits	Name	Description
31	SW_MTU_BKOF_GOTO_ID LE	If the back-offs are stuck for some reason, this bit will take the back-offs to idle We see a hang in the lab on a netlist which has no test buses. But could not be reproduced. To be on safe side this bit is added
30	SW_MTU_PIFS_RAW_CCA _SEL_NEW	Work with bit[25:24] to select the cca to be used for pifs.
29	SW_MTU_CCA_FLAG	this is software controlled cca. if the sw_mtu_sel is 11, the corresponding back off engine uses this register bit as its cca
28:26	SW_MTU_PIFS_FINAL_CC A_SEL	We want follow the same strategy we followed for back-off cca since the last_slot_time doesn't make sense for pifs, we kept the last_slot_time corresponds to back-off 0 000 -- raw CCA 001 -- (raw CCA && early_pkt_det_n) 010 -- (raw CCA && pkt_det_n) 011 - - early_pkt_det_n 100 -- pkt_det_n 101 -- (raw CCA && (early_pkt_det_n ~last_slot_time)) 110 -- (raw CCA && pkt_det_n && (early_pkt_det_n ~last_slot_time))

WCSS_A_MTU_CCA_CONTROL_REG (cont.)

Bits	Name	Description
25:24	SW_MTU_PIFS_RAW_CCA_SEL	It was not clear which cca to be used for pifs. So a programmability is provided for this also. if bit[30] = 0, 00 selects primary cca, 01 selects secondary cca, 10 selects combined cca(basically and of both primary and secondary) 11 select the sw_mtu_cca_flag which is sw config registe bit. if bit[30] = 1, 00 selects tertiary cca, 01 selects quaternary cca, 10 selects all cca(basically and of primary, secondary, tertiary and quaternary) Finall cca selection is done based on the next 3 bits-- sw_mtu_pifs_final_cca_sel
23:16	SW_MTU_EIFS_CCA_SEL	1 bit for each back off. It was not clear which cca to use for eifs. So a programmability is provided such that either the same cca that is used for difs is used for eifs also or just primary cca is selected. If this bit is 0, the primary cca is used for eifs (for the corresponding back off). Otherwise the cca that is used for the back off is used for eifs also
15:0	SW_MTU_RAW_CCA_SEL	2 bits for each back off. 1:0 for back off 0, 3:2 back off 1,... and so on if the corresponding mtu_cca_control.sw_mtu_raw_cca_sel=0, 00 selects primary cca, 01 selects secondary cca, 10 selects combined cca(basically and of both primary and secondary) 11 select the sw_mtu_cca_flag which is sw config registe bit if the corresponding mtu_cca_control.sw_mtu_raw_cca_sel=1, 00 selects tertiary cca, 01 selects quaternary cca, 10 selects all cca(basically and of primary, secondary, tertiary and quaternary)

0x0A0814F0 WCSS_A_MTU_TIMER_8**Type:** RW**Reset State:** 0x00FFFFFF

This register contains the timer 8 value. This timer is rd/writable,up/down Based on the configuration, this can be used either for up or down counter and generates an interrupt on reaching Zero in down mode. in up mode interrupt is generated when the timer reaches the sw_match_register_8.

WCSS_A_MTU_TIMER_8

Bits	Name	Description
23:0	TIMER_8	This register contains the timer 8 value. This timer is rd/writable,up/down

0x0A0814F4 WCSS_A_MTU_TIMER_9**Type:** RW**Reset State:** 0x00FFFFFF

This register contains the timer 9 value. This timer is rd/writable,up/down Based on the configuration, this can be used either for up or down counter and generates an interrupt on reaching

Zero in down mode. in up mode interrupt is generated when the timer reaches the sw_match_register_9.

WCSS_A_MTU_TIMER_9

Bits	Name	Description
23:0	TIMER_9	This register contains the timer 9 value. This timer is rd/writable,up/down

0x0A0814F8 WCSS_A_MTU_TIMER_10

Type: RW

Reset State: 0x00FFFFFF

This register contains the timer c value. This timer is rd/writable,up/down Based on the configuration, this can be used either for up or down counter and generates an interrupt on reaching Zero in down mode. in up mode interrupt is generated when the timer reaches the sw_match_register_10.

WCSS_A_MTU_TIMER_10

Bits	Name	Description
23:0	TIMER_10	This register contains the timer 10 value. This timer is rd/writable,up/down

0x0A0814FC WCSS_A_MTU_TIMER_11

Type: RW

Reset State: 0x00FFFFFF

This register contains the timer 11 value. This timer is rd/writable,up/down Based on the configuration, this can be used either for up or down counter and generates an interrupt on reaching Zero in down mode. in up mode interrupt is generated when the timer reaches the sw_match_register_11.

WCSS_A_MTU_TIMER_11

Bits	Name	Description
23:0	TIMER_11	This register contains the timer 11 value. This timer is rd/writable,up/down

0x0A081510 WCSS_A_MTU_SW_MATCH_REGISTER_8**Type:** RW**Reset State:** 0x00000EFF

This register contains the match value for tsf based interrupts. Basically whenever the tsf is equal to this register, corresponding interrupt will be generated. This register is also used the timer 8 in up mode. Basically when the timer is operated in up mode, when the Timer 8 reaches this value, an interrupt is generated

WCSS_A_MTU_SW_MATCH_REGISTER_8

Bits	Name	Description
31:30	SW_TSF_SEL_8	Lower bits to select the tsf will trigger the interrupt when the timer value reaches 3'b00: BSS TSF; 3'b01: IBSS TSF; 3'b10: P2P group owner TSF; 3'b11: P2P client TSF 3'b100: BSS2 TSF
29	SW_TSF_SEL_8_HI	Upper bit to select the tsf will trigger the interrupt when the timer value reaches
23:0	SW_MATCH_REGISTER_8	This register contains the match value for tsf based counter interrupt 8, timer 8 (in up mode)

0x0A081514 WCSS_A_MTU_SW_MATCH_REGISTER_9**Type:** RW**Reset State:** 0x00000EFF

This register contains the match value for tsf based interrupts. Basically whenever the tsf is equal to this register, corresponding interrupt will be generated. This register is also used the timer 9 in up mode. Basically when the timer is operated in up mode, when the Timer 9 reaches this value, an interrupt is generated

WCSS_A_MTU_SW_MATCH_REGISTER_9

Bits	Name	Description
31:30	SW_TSF_SEL_9	Lower bits to select the tsf will trigger the interrupt when the timer value reaches 3'b00: BSS TSF; 3'b01: IBSS TSF; 3'b10: P2P group owner TSF; 3'b11: P2P client TSF 3'b100: BSS2 TSF
29	SW_TSF_SEL_9_HI	Upper bit to select the tsf will trigger the interrupt when the timer value reaches
23:0	SW_MATCH_REGISTER_9	This register contains the match value for tsf based counter interrupt 9, timer 9 (in up mode)

0x0A081518 WCSS_A_MTU_SW_MATCH_REGISTER_10**Type:** RW**Reset State:** 0x00000EFF

This register contains the match value for tsf based interrupts. Basically whenever the tsf is equal to this register, corresponding interrupt will be generated. This register is also used the timer 10 in up mode. Basically when the timer is operated in up mode, when the Timer 10 reaches this value, an interrupt is generated

WCSS_A_MTU_SW_MATCH_REGISTER_10

Bits	Name	Description
31:30	SW_TSF_SEL_10	Lower bits to select the tsf will trigger the interrupt when the timer value reaches 3'b00: BSS TSF; 3'b01: IBSS TSF; 3'b10: P2P group owner TSF; 3'b11: P2P client TSF 3'b100: BSS2 TSF
29	SW_TSF_SEL_10_HI	Upper bit to select the tsf will trigger the interrupt when the timer value reaches
23:0	SW_MATCH_REGISTER_10	This register contains the match value for tsf based counter interrupt 10, timer 10 (in up mode)

0x0A08151C WCSS_A_MTU_SW_MATCH_REGISTER_11**Type:** RW**Reset State:** 0x00000EFF

This register contains the match value for tsf based interrupts. Basically whenever the tsf is equal to this register, corresponding interrupt will be generated. This register is also used the timer 11 in up mode. Basically when the timer is operated in up mode, when the Timer 11 reaches this value, an interrupt is generated

WCSS_A_MTU_SW_MATCH_REGISTER_11

Bits	Name	Description
31:30	SW_TSF_SEL_11	Lower bits to select the tsf will trigger the interrupt when the timer value reaches 3'b00: BSS TSF; 3'b01: IBSS TSF; 3'b10: P2P group owner TSF; 3'b11: P2P client TSF 3'b100: BSS2 TSF
29	SW_TSF_SEL_11_HI	Upper bit to select the tsf will trigger the interrupt when the timer value reaches
23:0	SW_MATCH_REGISTER_11	This register contains the match value for tsf based counter interrupt 11, timer 11 (in up mode)

0x0A081534 WCSS_A_MTU_BKOF_CONTROL2**Type:** RW**Reset State:** 0x000E0000

This register contains all the controls for the back-off engines.

WCSS_A_MTU_BKOF_CONTROL2

Bits	Name	Description
31	SW_MTU_CONSIDER_CSI_BKOF_EN	When set, csi bkof enable signal from MTU for HW controlled sending CSI frame in calibration and delayed explicit beamform will be enabled
30	SW_MTU_CONSIDER_QOS_NULL_BKOF_EN	When set, qosnull bkof enable signal from MTU for HW controlled sending qosnull frame in implicit beamform will be enabled
19:16	SW_MTU_PKT_DET_TO_CNT_THR	pkt_det_to_cnt is compared with this register and generates a signal. This is used by tpe to start requesting BA. Basically allow tpe to get the bitmap updated.

0x0A081538 WCSS_A_MTU_CCA_CONTROL_REG2**Type:** RW**Reset State:** 0x00000000

this register contains the controls to select the cca that needs to be used for a particular back_off engine

WCSS_A_MTU_CCA_CONTROL_REG2

Bits	Name	Description
19	CONSIDER_RX_CCA_ONLY	since the cca will be asserted even for the tx, we may want to ave count of only rx busy if this bit is set, the cca counters will be ++ only if txp_mtu_phytx_idle txp_mtu_phytx_idle =1
18:16	CCA_COUNT2_SEL	these bits select the count enable to the cca_count2 3'b000: cca_count2_en_temp = cca_term0; 3'b001: cca_count2_en_temp = cca_term1; 3'b010: cca_count2_en_temp = cca_term2; 3'b011: cca_count2_en_temp = cca_term3; 3'b100: cca_count2_en_temp = cca_term4; 3'b101: cca_count2_en_temp = cca_term5; 3'b110: cca_count2_en_temp = cca_term6; 3'b111: cca_count2_en_temp = cca_term7; cca_count2_en_final = consider_rx_cca_only ? (cca_count2_en_temp & ~txp_mtu_phytx_idle) : cca_count2_en_temp;

WCSS_A_MTU_CCA_CONTROL_REG2 (cont.)

Bits	Name	Description
14:12	CCA_COUNT1_SEL	these bits select the count enable to the cca_count1 3'b000: cca_count1_en_temp = cca_term0; 3'b001: cca_count1_en_temp = cca_term1; 3'b010: cca_count1_en_temp = cca_term2; 3'b011: cca_count1_en_temp = cca_term3; 3'b100: cca_count1_en_temp = cca_term4; 3'b101: cca_count1_en_temp = cca_term5; 3'b110: cca_count1_en_temp = cca_term6; 3'b111: cca_count1_en_temp = cca_term7; cca_count1_en_final = consider_rx_cca_only ? (cca_count1_en_temp & ~txp_mtu_phytx_idle) : cca_count1_en_temp;
7	CONSIDER_RX_CCA_ONLY_345	since the cca will be asserted even for the tx, we may want to ave count of only rx busy if this bit is set, the cca counters 3,4,5 will be ++ only if txp_mtu_phytx_idle txp_mtu_phytx_idle = 1
6:4	CCA_COUNT5_SEL	these bits select the count enable to the cca_count5 3'b000: cca_count5_en_temp = cca_term0; 3'b001: cca_count5_en_temp = cca_term1; 3'b010: cca_count5_en_temp = cca_term2; 3'b011: cca_count5_en_temp = cca_term3; 3'b100: cca_count5_en_temp = cca_term4; 3'b101: cca_count5_en_temp = cca_term5; 3'b110: cca_count5_en_temp = cca_term6; 3'b111: cca_count5_en_temp = cca_term7; cca_count5_en_final = consider_rx_cca_only_345 ? (cca_count5_en_temp & ~txp_mtu_phytx_idle) : cca_count5_en_temp;
2:0	CCA_COUNT4_SEL	these bits select the count enable to the cca_count4 3'b000: cca_count4_en_temp = cca_term0; 3'b001: cca_count4_en_temp = cca_term1; 3'b010: cca_count4_en_temp = cca_term2; 3'b011: cca_count4_en_temp = cca_term3; 3'b100: cca_count4_en_temp = cca_term4; 3'b101: cca_count4_en_temp = cca_term5; 3'b110: cca_count4_en_temp = cca_term6; 3'b111: cca_count4_en_temp = cca_term7; cca_count4_en_final = consider_rx_cca_only_345 ? (cca_count4_en_temp & ~txp_mtu_phytx_idle) : cca_count4_en_temp;

0x0A08154C WCSS_A_MTU_TIMER_CONTROL11TO8**Type:** RW**Reset State:** 0x00000000

This register contains all the controls for the timers 11 to 8.

WCSS_A_MTU_TIMER_CONTROL11TO8

Bits	Name	Description
19:8	SW_MTU_CONTINUOUS_V ALID	One bit for timer. if this bit is set, the timer will be loaded either with zero or the corresponding match register(depending on up/down). Basically if this bit is set the timer works as divided by N

WCSS_A_MTU_TIMER_CONTROL11TO8 (cont.)

Bits	Name	Description
7:0	SW_MTU_BASIC_UNIT_SELECT11TO8	2 bit for each timer is provided. bit [1:0] for timer 0, bit [3:2] for timer 1... To make these timers more flexible, 3 basic units are provided. if the control =0, the timer will be stopped (this is the only way to disable the counters if the control =01, timer is ++ at every clk if the control =10, timer is ++ at every one us if the control =11, timer is ++ at every programmable pulse. Basically in this mode, the period of the pulse is also controlled by s/w. Note that this programmable pulse is only one for all the timers

0x0A081550 WCSS_A_MTU_MTU_GLOBAL_CONTROL**Type:** RW**Reset State:** 0x19000000

Contains the global controls for mtu

WCSS_A_MTU_MTU_GLOBAL_CONTROL

Bits	Name	Description
31:24	SW_MTU_SECONDARY_CCA_LIMIT	Whenever the secondary cca is high for this limit (in us) time, mtu_mcu_secondary_cca_to will be generated. This functionality is exactly similar to pifs to except that the cca used is always secondary
7:0	SW_MTU_SUPPRESS_RF_WARMUP	If this bit is set, mtu will not send rf_warmup to txp Any suggested value is welcome bitfield [19:12] sw_mtu_suppress_rf_warmup_for_collision configuration 8'b0 bitfieldComment If this bit is set, mtu will not send rf_warmup to txp for the collision case Any suggested value is welcome

0x0A081554 WCSS_A_MTU_MTU_INTERRUPT_STATUS11TO8**Type:** RW**Reset State:** 0x00000000

This register contains the status of all the interrupts that are generated by mtu Basically every interrupt has a corresponding register bit which will be set on sending interrupt pulse to the interrupt module and this bit is reset by writing '1' to it. sw has higher priority over hw

WCSS_A_MTU_MTU_INTERRUPT_STATUS11TO8

Bits	Name	Type	Description
31:17	TXP_SIFS_DELAY_CNT_FOR_INNAV	R	The internal count is exposed. Basically this counter keeps ++ after the rising edge of pkt_det. The resolution is one clock (it is 25ns in 40Mhz mode, 12.5 if mac is selected of 80Mhz)

WCSS_A_MTU_MTU_INTERRUPT_STATUS11TO8 (cont.)

Bits	Name	Type	Description
11:8	MTU_MCU_TSF_INT_FLAG11TO8	RW	4 registers for upper 4 tsf based match register interrupts. bit 0 for match 0, 1 for match 1 ...
7:4	MTU_MCU_BKOF_INT_FLAG11TO8	RW	4 registers for upper 4 back off interrupts. bit 0 for timer 0, 1 for timer 1 ...
3:0	MTU_MCU_TIMER_INT_FLAG11TO8	RW	4 registers for upper 4 timer interrupts. bit 0 for timer 0, 1 for timer 1 ...

0x0A081558 WCSS_A_MTU_MTU_GLOBAL_CONTROL2**Type:** RW**Reset State:** 0x000000EE

Contains the global controls for mtu

WCSS_A_MTU_MTU_GLOBAL_CONTROL2

Bits	Name	Description
15:12	SW_MTU_RTS_NAV_TO_LIMIT_U	The upper bits of sw_mtu_rts_nav_to_limit
11	SW_MTU_IGNORE_NAV_FOR_EIFS_FCS	Set to 1 to ignore the nav_greater-than-64 check. Whenever rxp indicates that fcs error, the eifs selection will be set regardless of the nav count.
10	SW_MTU_CONSIDER_SKIP_DIFS	if this bit is set, the skip_difs from the rxp is used.. to take care of any bug in other module. Since this is not a main line function, better to have a protection
9	SW_MTU_SELECT_LATE_PKT_DET_FOR_ACK_TO	if this bit is set, the late_pkt_det is used for ack_to. By default the early_pkt-det should be used. This bit should never be set in the system. The configuration is provided for unknown flexibility.. Should be removed in the next spin atleast
8	SW_MTU_NAV_BASED_ON_RTS_VALID	The shadow_nav_cnt logic is enabled only when this bit is set. The nav update for RTS is done irrespective of this bit but the rts_nav_to logic is not enabled if this bit is set
7:0	SW_MTU_RTS_NAV_TO_LIMIT	This register represents the timeout count in the following On receiving an RTS based NAV, a shadow NAV counter will keep running the previous NAV setting. The RTS NAV will be used to set the NAV. A (Programmable) timeout counter is started. If no pkdet_n is detected before the timeout count expires, the shadow NAV will be used again to set the NAV. If a pkdet_n is seen before the timeout expires, the RTS based NAV will continue. The 'shadow' NAV can now be discarded. The default is a randomly chosen. Any suggested value is welcome

0x0A08155C WCSS_A_MTU_SW_MTU_SHADOW_NAV_CNT**Type:** RW**Reset State:** 0x000000FF

This register contains the nav value in shadow register NAV is updated by hw as well as by the s/w. s/w has higher priority over the h/w updates

WCSS_A_MTU_SW_MTU_SHADOW_NAV_CNT

Bits	Name	Description
31:0	SW_MTU_SHADOW_NAV_CNT	This register contains the nav value. There has to some default. So I kept ff The value is in micro seconds. This register just follows the nav_cnt but it will differ during the rts_nav update. When there is rts NAV update, the current NAV is copied to this register and keeps updated. If there is rts_nav_to, this value is copied to the nav_cnt.

0x0A081560 WCSS_A_MTU_PRIMARY_CCA_HISTOGRAM_LO**Type:** R**Reset State:** 0xFFFFFFFF

This register contains the lower 32 bits of primary_cca_histogram

WCSS_A_MTU_PRIMARY_CCA_HISTOGRAM_LO

Bits	Name	Description
31:0	PRIMARY_CCA_HISTOGRAM31TO0	the primary cca is sampled at every slot time and shifted through a shift register s/w can read the shift register value through this address

0x0A081564 WCSS_A_MTU_PRIMARY_CCA_HISTOGRAM_HI**Type:** R**Reset State:** 0xFFFFFFFF

This register contains the higher 32 bits of primary_cca_histogram

WCSS_A_MTU_PRIMARY_CCA_HISTOGRAM_HI

Bits	Name	Description
31:0	PRIMARY_CCA_HISTOGRAM63TO32	the primary cca is sampled at every slot time and shifted through a shift register s/w can read the shift register value through this address

0x0A081568 WCSS_A_MTU_SECONDARY_CCA_HISTOGRAM_LO**Type:** R**Reset State:** 0xFFFFFFFF

This register contains the lower 32 bits of secondary_cca_histogram

WCSS_A_MTU_SECONDARY_CCA_HISTOGRAM_LO

Bits	Name	Description
31:0	SECONDARY_CCA_HISTOGRAM31TO0	the secondary cca is sampled at every slot time and shifted through a shift register s/w can read the shift register value through this address

0x0A08156C WCSS_A_MTU_SECONDARY_CCA_HISTOGRAM_HI**Type:** R**Reset State:** 0xFFFFFFFF

This register contains the higher 32 bits of secondary_cca_histogram

WCSS_A_MTU_SECONDARY_CCA_HISTOGRAM_HI

Bits	Name	Description
31:0	SECONDARY_CCA_HISTOGRAM63TO32	the secondary cca is sampled at every slot time and shifted through a shift register s/w can read the shift register value through this address

0x0A081570 WCSS_A_MTU_TIMEOUT_LIMIT**Type:** RW**Reset State:** 0x000640F5

This register contains the Timeouts for rsp_to and ..

WCSS_A_MTU_TIMEOUT_LIMIT

Bits	Name	Description
24:10	SW_MTU_SIFS_LIMIT_FOR_WARMUP	This is the time within SIFS up to which we will mask out warmup (in clks) The default is 10 usec. This needs to be changed. Right now it is made same as what we have in Libra (10 usec) for 40Mhz clock ($25\text{ns} \times 4000 = 10,000\text{ns} = 10\text{usec}$). If the clock is 80Mhz, this should be $2 \times 400 = 800$
9	SW_MTU_NOT_A_BUG_FIX	if the interrupts are back to back, though it should not be, there is a corner case the system could hang. So configurable fix..

WCSS_A_MTU_TIMEOUT_LIMIT (cont.)

Bits	Name	Description
8	SW_MTU_FORGET_PKT_DET_TO	Since this logic is to take care of fsm lock or misbehavior of early pkt det, we don't want this to be enabled by default
7:0	SW_MTU_PKT_DET_TO_LIMIT	To make sure the fsm don't get in a loop while waiting for response time out, we use more reliable pkt_det from PHY and run a counter when the fsm is not in idle and the pkt_det is HIGH. When the counter reaches this limit, the fsm comes out and gives a time out interrupt to s/w. The default is f5. Basically a big value The limit is in micro seconds

0x0A081574 WCSS_A_MTU_CCA_FINAL_CONTROL_REG**Type:** RW**Reset State:** 0x00000000

this register contains the controls to select the final cca that needs to be used for a particular back_off engine

WCSS_A_MTU_CCA_FINAL_CONTROL_REG

Bits	Name	Description
23:0	SW_MTU_FINAL_CCA_SEL	3 bits for each back off. 2:0 for back off 0, 5:3 back off 1,... and so on 000 -- CCA 001 -- (CCA && early_pkt_det_n) 010 -- (CCA && pkt_det_n) 011 -- early_pkt_det_n 100 -- pkt_det_n 101 -- (CCA && (early_pkt_det_n ~last_slot_time)) 110 -- (CCA && pkt_det_n && (early_pkt_det_n ~last_slot_time))

0x0A08157C WCSS_A_MTU_EARLY_INTERRUPT_LIMITS**Type:** RW**Reset State:** 0x00050104

this register contains the time limits for early interrupt to s/w, txp back_off engine. The early interrupt to s/w is sw_mtu_early_sw_int_limit_us+sw_mtu_early_sw_int_limit_us_clks

WCSS_A_MTU_EARLY_INTERRUPT_LIMITS

Bits	Name	Description
21:16	SW_MTU_TXP_SIFS_WIDTH_LIMIT	sw_mtu_txp_sifs_flag will be set as per the limit. After this limit the flag will just follow the pkt_det
15:8	SW_MTU_EARLY_SW_INT_LIMIT_CLKS	This gives the delay in clks -- used by pifs and the back off interrupts See the comments in above field for mor info

WCSS_A_MTU_EARLY_INTERRUPT_LIMITS (cont.)

Bits	Name	Description
7:0	SW_MTU_EARLY_SW_INT_LIMIT_US	This gives the delay in microseconds -- used by back off interrupts basically an early slot pulse is generated whenever the the slot count == this limit and one_usec count == sw_mtu_early_sw_int_limit_clks. The sw_mtu_early_sw_int_limit_clks is used to provide a resolution of one clk. Since the one_usec counter is a down counter the programming should be in terms of delay. Basicall i we want .25 us early, we should program $1-.25 = .75 = 90$ clks(approx)

0x0A081580 WCSS_A_MTU_BTC_CONFIG_REGISTERS**Type:** RW**Reset State:** 0x00000000

this register contains the Controls for btc

WCSS_A_MTU_BTC_CONFIG_REGISTERS

Bits	Name	Description
25	SW_MTU_TX_GRANT_SEL	'0': select bt_active, '1': select lte_tx_grant signal for sw_mtu_btc_cca_control
24	SW_MTU_CONSIDER_QUATERNARY_CCA	If this is set rxp_mtu_quaternary_cca is considered in generation of mtu_btc_tx_rx_busy
23	SW_MTU_CONSIDER_TERTIARY_CCA	If this is set rxp_mtu_tertiary_cca is considered in generation of mtu_btc_tx_rx_busy
22	SW_MTU_BTC_PIFS_CONTROL	If this is set the final btc signals are considered in pifs. To consider both high and low prio sw_mtu_consider_low_pri_btc_active also need to be set
21:6	SW_MTU_BTC_CCA_CONTROL	This controls the ANDING of BTC to CCA of the 8 back-offs 2 bits for each back-off (Only 8 nack-offs are considered) 21:6 00 --> don't consider btc from mcu/btc 01 --> Just simply aND with CCA 10 --> Don't consider till the slot count = 1
5	SW_MTU_CONSIDER_SECONDARYCCA	If this is set rxp_mtu_secondary_cca is considered in generation of mtu_btc_tx_rx_busy
4	SW_MTU_CONSIDER_PRIMARYCCA	If this is set rxp_mtu_primary_cca is considered in generation of mtu_btc_tx_rx_busy
3	SW_MTU_CONSIDER_EARLYPKTDETN	If this is set rxp_mtu_early_pktdet_n is considered in generation of mtu_btc_tx_rx_busy
2	SW_MTU_CONSIDER_PKTDETN	If this is set rxp_mtu_pktdet_n is considered in generation of mtu_btc_tx_rx_busy
1	SW_MTU_CONSIDER_PHYTXIDLE	If this is set txp_mtu_phytx_idle is considered in generation of mtu_btc_tx_rx_busy

WCSS_A_MTU_BTC_CONFIG_REGISTERS (cont.)

Bits	Name	Description
0	SW_MTU_CONSIDER_NAV_ACTIVE	If this is set nav_active is considered in generation of mtu_btc_tx_rx_busy

0x0A08158C WCSS_A_MTU_MTU_FOR_HMAC_CONTROLS**Type:** RW**Reset State:** 0xFF181003

This register contains the new registers that are added for hmac -- gen5/6

WCSS_A_MTU_MTU_FOR_HMAC_CONTROLS

Bits	Name	Description
31:24	SW_MTU_HW_BACKOFF_VALID	if this bit is set the hw back-offs will be enabled. By default this bit is enabled One per backoff.
20	SW_MTU_ALLOW_PMU_DISABLE	When cleared, the tsf timer will be always running When set, the tsf timer will be run when pmu_mtu_tsf_timer_en is asserted.
19	SW_MTU_ALLOW_PMU_UPDATE	For UAPSD, we allowed PMU to be able to restore the tsf in MTU. This is made programmable to avoid any issues with this
17	SW_MTU_BTAMP_MODE	When set, indicates it's in bt-amp mode.
16	SW_MTU_SLR_LIMITS_ARE_VALID	SLR limits are valid only if this bit is set and the data rate is SLR
15:8	SW_MTU_SLR_EARLY_PKT_DET_MISS_LIMIT	this is used in respons_to(ack_to) logic for SLR if the system is expecting a response and if the early pkt_det is HIGH for this limit, an interrupt is generated (old name is cca_miss_limit) Units are 4 usec
7	SW_MTU_CONSIDER_LSIG_CCA	only when this bit is set, the lsig from phy is considered in MTU
6	SW_MTU_CONSIDER_LOW_PRI_BTC_ACTIVE	only when this bit is set, the mcu_mtu_bt_low_pri_active is considered in generation of final cca
5	SW_MTU_IBSS_VALID	If this bit is set the back-off 0 is assigned for beacon...
4	SW_MTU_DTIM_CNT_EN	Enables dtim counter. This bit needs to be set to enable dtim logic to be enabled
2:0	SW_RSP_TO_CONSIDER_VECTOR	This register contains valid vector for ack_to. Basically bit 0 -- corresponds to mtu_mcu_timeout_missing_early_pkt_det_p bit 1 -- corresponds to mtu_mcu_timeout_missing_push_p bit 2 -- corresponds to mtu_mcu_timeout_pkt_det_p

0x0A081590 WCSS_A_MTU_MTU_DTIM_CNT_AND_PERIOD**Type:** RW**Reset State:** 0x00000001

This register contains dtim_cnt dtim_period

WCSS_A_MTU_MTU_DTIM_CNT_AND_PERIOD

Bits	Name	Description
31:16	DTIM_CNT	Basically this is the internal dtim counter. basically whenever this counter reaches, dtim_period -1 is loaded. this keeps ++ at every tbtt
15:0	DTIM_PERIOD	This register contains the dtim period. The period is defined in number of beacons if this is 1, every beacon is DTIM. 2 means alternate beacons are dtim

0x0A081594 WCSS_A_MTU_DTIM_THRSH_CNT_AND_LIMIT**Type:** RW**Reset State:** 0x00000000

This register contains dtim_thrsh_limit and dtim_thrsh_limit

WCSS_A_MTU_DTIM_THRSH_CNT_AND_LIMIT

Bits	Name	Description
31:16	DTIM_THRSH_CNT	Internal counter ++ at every usec to keep track of the dtim_thrsh_limit. This counter is effective only when sw_mtu_dtim_cnt_en is 1
15:0	DTIM_THRSH_LIMIT	This register contains the dtim_thrsh_limit. This number indicates the time in usec. Basically this number indicates upto what time during the DTIM beacon, MC and BC can be ransmited

0x0A081598 WCSS_A_MTU_TBTT_L**Type:** RW**Reset State:** 0x7FFFFFFF

Lower 32 bits of tbtt register

WCSS_A_MTU_TBTT_L

Bits	Name	Description
31:0	TBTT_L	Lower 32 bits of TBTT register.

0x0A08159C WCSS_A_MTU_TBTT_H**Type:** RW**Reset State:** 0x00000000

Upper 32 bits of tbtt resgiter

WCSS_A_MTU_TBTT_H

Bits	Name	Description
31:0	TBTT_H	Upper 32 bits of TBTT register.

0x0A0815A0 WCSS_A_MTU_M_TBTT_L**Type:** RW**Reset State:** 0x7FFFFFFF

Lower 32 bits of multiple bssid tbtt register

WCSS_A_MTU_M_TBTT_L

Bits	Name	Description
31:0	M_TBTT_L	Lower 32 bits of TBTT register.

0x0A0815A4 WCSS_A_MTU_M_TBTT_H**Type:** RW**Reset State:** 0x00000000

Upper 32 bits of multiple bssid tbtt resgiter

WCSS_A_MTU_M_TBTT_H

Bits	Name	Description
31:0	M_TBTT_H	Upper 32 bits of TBTT register.

0x0A0815A8 WCSS_A_MTU_BCN_BSSID_INTV**Type:** RW**Reset State:** 0x00100064

Beacon Interval Register is programmed by s/w and it determined the period between two consecutive TBTT. This also contains the mbssid interval in usec

WCSS_A_MTU_BCN_BSSID_INTV

Bits	Name	Description
29	SW_MTU_MBSSID_ENABLE	Multiple bssid are enabled only when this bit is set
28	SW_MTU_TBTT_ENABLE	Beacons are enabled only when this bit is set
27:24	SW_MTU_MAX_BSSIDS	Beacons are enabled only when this bit is set
23:16	SW_MTU_BSSID_INTV	This bits determine the period between two consecutive beacons of different bssids. This value is in n 256 micro seconds. Each count is 256 usec
15:0	SW_MTU_BEACON_INTV	This bits determine the period between two consecutive beacon. This value is in ms message.

0x0A0815AC WCSS_A_MTU_VALID_BSSID_BITMAP**Type:** RW**Reset State:** 0x00070001

This register contains the valid bssids in the system

WCSS_A_MTU_VALID_BSSID_BITMAP

Bits	Name	Description
31:16	IBSS_BCN_CW_LIMIT	This register contains the cw limit for ibss
15:0	SW_MTU_VALID_BSSID_BITMAP	A 16 bit register indicates which bssid is enabled Bit 0 should be enabled for single bssid By default only one bssid is enabled message.

0x0A0815B0 WCSS_A_MTU_IBSS_BKOF_CNT**Type:** RW**Reset State:** 0x00000000

This register contains the bkof_cnt of ibss . The units are slot number.

WCSS_A_MTU_IBSS_BKOF_CNT

Bits	Name	Description
15:0	IBSS_BKOF_CNT	This register contains the bkof_counter value for the ibss_bkof_cnt this value = (lfsr_data & lfsr_value)

0x0A0815B4 WCSS_A_MTU_LFSR_DATA_1_0**Type:** R**Reset State:** 0x00000000

This register contains the lfsr_data of bkof 1 and 0.

WCSS_A_MTU_LFSR_DATA_1_0

Bits	Name	Description
31:16	LFSR_DATA_1	This register contains the lfsr_data value for the bkof_1
15:0	LFSR_DATA_0	This register contains the lfsr_data value for the bkof_0

0x0A0815B8 WCSS_A_MTU_LFSR_DATA_3_2**Type:** R**Reset State:** 0x00000000

This register contains the lfsr_data of bkof 3 and 2.

WCSS_A_MTU_LFSR_DATA_3_2

Bits	Name	Description
31:16	LFSR_DATA_3	This register contains the lfsr_data value for the bkof_3
15:0	LFSR_DATA_2	This register contains the lfsr_data value for the bkof_2

0x0A0815BC WCSS_A_MTU_LFSR_DATA_5_4**Type:** R**Reset State:** 0x00000000

This register contains the lfsr_data of bkof 5 and 4.

WCSS_A_MTU_LFSR_DATA_5_4

Bits	Name	Description
31:16	LFSR_DATA_5	This register contains the lfsr_data value for the bkof_5
15:0	LFSR_DATA_4	This register contains the lfsr_data value for the bkof_4

0x0A0815C0 WCSS_A_MTU_LFSR_DATA_7_6**Type:** R**Reset State:** 0x00000000

This register contains the lfsr_data of bkof 7 and 6.

WCSS_A_MTU_LFSR_DATA_7_6

Bits	Name	Description
31:16	LFSR_DATA_7	This register contains the lfsr_data value for the bkof_7
15:0	LFSR_DATA_6	This register contains the lfsr_data value for the bkof_6

0x0A0815C4 WCSS_A_MTU_LONG_SHORT_XMIT_LIMIT_BKOF_3_TO_0**Type:** R**Reset State:** 0x47474747

This register contains the sort and long retry limts for backoff cw adjustment

WCSS_A_MTU_LONG_SHORT_XMIT_LIMIT_BKOF_3_TO_0

Bits	Name	Description
31:0	SW_LONG_SHORT_XMIT_LIMIT_BKOF_3_TO_0	This register contains the short and long retrylimts for bkof-3 to 0. 3:0 is short, 7:4 is long for back off 0. 11:8 is short, 15:12 is long for back off 1. 19:16 is short, 23:20 is long for back off 2. 27:24 is short, 31:28 is long for back off 3.

0x0A0815C8 WCSS_A_MTU_LONG_SHORT_XMIT_LIMIT_BKOF_7_TO_4**Type:** R**Reset State:** 0x47474747

This register contains the sort and long retry limts for backoff cw adjustment

WCSS_A_MTU_LONG_SHORT_XMIT_LIMIT_BKOF_7_TO_4

Bits	Name	Description
31:0	SW_LONG_SHORT_XMIT_LIMIT_BKOF_7_TO_4	This register contains the short and long retrylimts for bkof-7 to 4. 3:0 is short, 7:4 is long for back off 4. 11:8 is short, 15:12 is long for back off 5. 19:16 is short, 23:20 is long for back off 6. 27:24 is short, 31:28 is long for back off 7.

0x0A0815CC WCSS_A_MTU_WARMUP_LIMIT**Type:** RW**Reset State:** 0x00000025

This register contains the threshold number for txp_mpi_rf_warmup.

WCSS_A_MTU_WARMUP_LIMIT

Bits	Name	Type	Description
11:8	TXP_MPI_RF_WARMUP_COUNTER	R	counter number of txp_mpi_rf_warmup high duration.
7	TXP_MPI_RF_WARMUP_LOW_FLAG	R	When assert, indicates the duration of txp_mpi_rf_warmup is less than sw_mtu_warmup_thrd_low.
6	TXP_MPI_RF_WARMUP_HIGH_FLAG	R	When assert, indicates the duration of txp_mpi_rf_warmup is longer than sw_mtu_warmup_thrd_high.
5:4	SW_MTU_WARMUP_THRD_LOW	RW	This register contains the lower limit value (in unit of usec) for txp_mpi_rf_warmup.
3:0	SW_MTU_WARMUP_THRD_HIGH	RW	This register contains the upper limit value (in unit of usec) for txp_mpi_rf_warmup.

0x0A0815D0 WCSS_A_MTU_ONE_SEC_LIMIT**Type:** RW**Reset State:** 0x000F423F

This register contains the threshold number for one_sec pulse

WCSS_A_MTU_ONE_SEC_LIMIT

Bits	Name	Description
19:0	SW_MTU_ONE_SEC_LIMIT	limit value to be programmed to get one sec pulse. Basically this is number of usec. this value is loaded into a counter and keep getting decremented at every one_usec_p.

0x0A0815D4 WCSS_A_MTU_MTU_GLOBAL_TIMER**Type:** RW**Reset State:** 0x00000000

This register contains a global timer to be used by all modules including s/w if it wants

WCSS_A_MTU_MTU_GLOBAL_TIMER

Bits	Name	Description
31:0	MTU_GLOBAL_TIMER	This is a global timer increments at every one usec. This is different from tsf timer

0x0A0815D8 WCSS_A_MTU_TSF_TIMER_FOR_SIM_BSS_LO**Type:** RW**Reset State:** 0x00000000

This register contains the lower 32 bits of the tsf timer used only during simultaneous bss and ibss mode This corresponds to BSS tsf during simultaneous mode. In none Simultaneous mode, this is not valid

WCSS_A_MTU_TSF_TIMER_FOR_SIM_BSS_LO

Bits	Name	Description
31:0	TSF_TIMER_FOR_SIM_BSS_LO	This register contains the lower 32 bits of the tsf timer used only during simultaneous bss and ibss mode This corresponds to BSS tsf during simultaneous mode. In none Simultaneous mode, this is not valid

0x0A0815DC WCSS_A_MTU_TSF_TIMER_FOR_SIM_BSS_HI**Type:** RW**Reset State:** 0x00000000

This register contains the uppder 32 bits of the tsf timer used only during simultaneous bss and ibss mode This corresponds to BSS tsf during simultaneous mode. In none Simultaneous mode, this is not valid

WCSS_A_MTU_TSF_TIMER_FOR_SIM_BSS_HI

Bits	Name	Description
31:0	TSF_TIMER_FOR_SIM_BSS_HI	This register contains the uppder 32 bits of the tsf timer used only during simultaneous bss and ibss mode This corresponds to BSS tsf during simultaneous mode. In none Simultaneous mode, this is not valid

0x0A0815E0 WCSS_A_MTU_SW_MTU_SIFS_LIMIT_FORINNAV**Type:** RW**Reset State:** 0x00000190

This is register that contains the SIFS Limit specified in number of clocks rather than in usec

WCSS_A_MTU_SW_MTU_SIFS_LIMIT_FORINNAV

Bits	Name	Description
14:0	SW_MTU_SIFS_LIMIT_FORINNAV	This is register that contains the SIFS Limit specified in number of clocks rather than in usec The default is 15 usec. This needs to be changed. Right now it is made same as what we have in Libra (14 usec) for 40Mhz clock ($25\text{ns} \times 560 = 14,000\text{ns} = 14\text{usec}$). If the clock is 80Mhz, this should be $2 \times 560 = 1130$

0x0A0815E4 WCSS_A_MTU_SW_MTU_P2P_GO_TSF_TIMER_LO

Type: RW

Reset State: 0x00000000

This is register that contains the lower 32 bits of the p2p group owner tsf timer

WCSS_A_MTU_SW_MTU_P2P_GO_TSF_TIMER_LO

Bits	Name	Description
31:0	P2P_GO_TSF_TIMER_LO	The lower 32 bits of the p2p group owner tsf timer In unit of us.

0x0A0815E8 WCSS_A_MTU_SW_MTU_P2P_GO_TSF_TIMER_HI

Type: RW

Reset State: 0x00000000

This is register that contains the upper 32 bits of the p2p group owner tsf timer

WCSS_A_MTU_SW_MTU_P2P_GO_TSF_TIMER_HI

Bits	Name	Description
31:0	P2P_GO_TSF_TIMER_HI	The upper 32 bits of the p2p group owner tsf timer In unit of us.

0x0A0815EC WCSS_A_MTU_SW_MTU_P2P_CLIENT_TSF_TIMER_LO

Type: RW

Reset State: 0x00000000

This is register that contains the lower 32 bits of the p2p client owner tsf timer

WCSS_A_MTU_SW_MTU_P2P_CLIENT_TSF_TIMER_LO

Bits	Name	Description
31:0	P2P_CLIENT_TSF_TIMER_LO	The lower 32 bits of the p2p client owner tsf timer In unit of us.

0x0A0815F0 WCSS_A_MTU_SW_MTU_P2P_CLIENT_TSF_TIMER_HI**Type:** RW**Reset State:** 0x00000000

This is register that contains the upper 32 bits of the p2p client owner tsf timer

WCSS_A_MTU_SW_MTU_P2P_CLIENT_TSF_TIMER_HI

Bits	Name	Description
31:0	P2P_CLIENT_TSF_TIMER_HI	The upper 32 bits of the p2p client owner tsf timer In unit of us.

0x0A0815F4 WCSS_A_MTU_SW_MTU_P2P_CTRL**Type:** RW**Reset State:** 0x00000040

This is register that contains the control bits for p2p feature

WCSS_A_MTU_SW_MTU_P2P_CTRL

Bits	Name	Description
27	SW_MTU_BSS2_CLIENT_TSF_TIMER_EN	Set 1 to enable the bss2 client tsf timer
26	P2P_TSF_INSERTION_SEL_HI	upper bit to select the tsf value to be inserted in beacons in TPE
25	LTE_TSF_INSERTION_SEL_HI	upper bit to select the tsf value to be provided to lte csm
24	TSF_FOR_TBTT_SEL_HI	upper bit to select the tsf to be used for tbtt
23:21	TSF_FOR_MBSSID_TBTT_SEL	To select the tsf to be used for mbssid tbtt 3'b00: BSS TSF; 3'b01: IBSS TSF; 3'b10: P2P group owner TSF; 3'b11: P2P client TSF 3'b100 : BSS2 TSF
20:19	TSF_FOR_TBTT_SEL	Lower bits to select the tsf to be used for tbtt 3'b00: BSS TSF; 3'b01: IBSS TSF; 3'b10: P2P group owner TSF; 3'b11: P2P client TSF 3'b100 : BSS2 TSF

WCSS_A_MTU_SW_MTU_P2P_CTRL (cont.)

Bits	Name	Description
18:17	LTE_TSF_INSERTION_SEL	Lower bits to select the TSF value to be provided to lte csm 3'b00: BSS TSF; 3'b01: IBSS TSF; 3'b10: P2P group owner TSF; 3'b11: P2P client TSF 3'b100 : BSS2 TSF
16	SW_BEACON_TRANSMIT_EN	Writing a '1' to trigger a beacon transmission. Always read zero
15:8	P2P_TX_BOUNDARY_THR ESHOLD	The bcoff will be disabled after the transmit_boundary_timer reaches this value
6:4	GLOBAL_TIMER_SRC_SEL	To select the source for the global timer 3'b000: BSS TSF; 3'b001: IBSS TSF; 3'b010: P2P group owner TSF; 3'b011: P2P client TSF 3'b100: internal timer; 3'b101: BSS2 TSF
3:2	P2P_TSF_INSERTION_SEL	Lower bits to select the TSF value to be inserted in beacons in TPE 3'b00: BSS TSF; 3'b01: IBSS TSF; 3'b10: P2P group owner TSF; 3'b11: P2P client TSF 3'b100 : BSS2 TSF
1	P2P_CLIENT_TSF_TIMER_EN	Set to 1 to enable the p2p client timer
0	P2P_GO_TSF_TIMER_EN	Set to 1 to enable the p2p group owner timer

0x0A0815F8 WCSS_A_MTU_SW_MTU_P2P_TRANSMIT_BOUNDARY**Type:** RW**Reset State:** 0x00000000

This is register that contains the config bits of the transmit boundary timer

WCSS_A_MTU_SW_MTU_P2P_TRANSMIT_BOUNDARY

Bits	Name	Description
31	P2P_TRANSMIT_BOUNDARY_EN	Set 1 to enable the transmit boundary feature.
27:20	P2P_INT_BOUNDARY	When the counter reaches this value, an interrupt will be generated. In unit of us.
18:0	P2P_TRANSMIT_BOUNDARY	A down counter that SW can program so that no frame exchanges will go over this boundary. When value 0 is reached, no more transmissions happen, until a new none zero value is programmed or this feature is disabled. In unit of us.

0x0A0815FC WCSS_A_MTU_SW_MTU_P2P_WAIT_FOR_BEACON**Type:** RW**Reset State:** 0x00020000

This is register that contains the config bits of the wait for beacon timer

WCSS_A_MTU_SW_MTU_P2P_WAIT_FOR_BEACON

Bits	Name	Description
31	P2P_WAIT_FOR_BEACON_EN	Set 1 to enable the wait for beacon feature.
17:16	P2P_WAIT_FOR_BEACON_SEL	To select which beacon to wait for. 2'b00: IBSS beacon, 2'b01: BSS beacon, 2'b10: p2p client beacon, 2'b11: bss2 client beacon.
15:0	P2P_MAX_WAIT_DURATION	max duration of waiting for the p2p beacon. Start at p2p tbtt. In unit of us.

0x0A081600 WCSS_A_MTU_TERTIARY_CCA_HISTOGRAM_LO**Type:** R**Reset State:** 0xFFFFFFFF

This register contains the lower 32 bits of tertiary_cca_histogram

WCSS_A_MTU_TERTIARY_CCA_HISTOGRAM_LO

Bits	Name	Description
31:0	TERTIARY_CCA_HISTOGRAM31TO0	the tertiary cca is sampled at every slot time and shifted through a shift register s/w can read the shift register value through this address

0x0A081604 WCSS_A_MTU_TERTIARY_CCA_HISTOGRAM_HI**Type:** R**Reset State:** 0xFFFFFFFF

This register contains the higher 32 bits of tertiary_cca_histogram

WCSS_A_MTU_TERTIARY_CCA_HISTOGRAM_HI

Bits	Name	Description
31:0	TERTIARY_CCA_HISTOGRAM63TO32	the tertiary cca is sampled at every slot time and shifted through a shift register s/w can read the shift register value through this address

0x0A081608 WCSS_A_MTU_QUATERNARY_CCA_HISTOGRAM_LO**Type:** R**Reset State:** 0xFFFFFFFF

This register contains the lower 32 bits of quaternary_cca_histogram

WCSS_A_MTU_QUATERNARY_CCA_HISTOGRAM_LO

Bits	Name	Description
31:0	QUATERNARY_CCA_HISTOGRAM31TO0	the quaternary cca is sampled at every slot time and shifted through a shift register s/w can read the shift register value through this address

0x0A08160C WCSS_A_MTU_QUATERNARY_CCA_HISTOGRAM_HI**Type:** R**Reset State:** 0xFFFFFFFF

This register contains the higher 32 bits of quaternary_cca_histogram

WCSS_A_MTU_QUATERNARY_CCA_HISTOGRAM_HI

Bits	Name	Description
31:0	QUATERNARY_CCA_HISTOGRAM63TO32	the quaternary cca is sampled at every slot time and shifted through a shift register s/w can read the shift register value through this address

0x0A081610 WCSS_A_MTU_MTU_CCA_CONTROL**Type:** RW**Reset State:** 0x00001919

Contains the 3rd and 4th cca controls for mtu

WCSS_A_MTU_MTU_CCA_CONTROL

Bits	Name	Description
23:16	SW_MTU_RAW_CCA_SEL_NEW	Work with cca_control_reg.sw_mtu_raw_cca_sel to select the cca for backoffs. bit16 for bkof0, bit17 for bkof1 ...
15:8	SW_MTU_QUATERNARY_CCA_LIMIT	Whenever the quaternary cca is high for this limit (in us) time, mtu_mcu_quaternary_cca_to will be generated.
7:0	SW_MTU_TERTIARY_CCA_LIMIT	Whenever the tertiary cca is high for this limit (in us) time, mtu_mcu_tertiary_cca_to will be generated.

0x0A081614 WCSS_A_MTU_MTU_CCA_STATUS**Type:** R**Reset State:** 0x00000000

Contains the 3rd and 4th cca status for mtu

WCSS_A_MTU_MTU_CCA_STATUS

Bits	Name	Description
1	QUATERNARY_CCA_TO	the quaternary_cca_to is sampled and s/w can read its status through this register
0	TERTIARY_CCA_TO	the tertiary_cca_to is sampled and s/w can read its status through this register

0x0A081618 WCSS_A_MTU_ADD_TSF_TIMER_LO**Type:** RW**Reset State:** 0x00000000

This register contains the lower 32 bits of the value to be added to the tsf timer

WCSS_A_MTU_ADD_TSF_TIMER_LO

Bits	Name	Description
31:0	ADD_TSF_TIMER_LO	This register contains the lower 32 bits of the value to be added to the tsf timer

0x0A08161C WCSS_A_MTU_ADD_TSF_TIMER_HI**Type:** RW**Reset State:** 0x00000000

This register contains the upper 32 bits of the value to be added to the tsf timer

WCSS_A_MTU_ADD_TSF_TIMER_HI

Bits	Name	Description
31:0	ADD_TSF_TIMER_HI	This register contains the upper 32 bits of the value to be added to the tsf timer

0x0A081620 WCSS_A_MTU_TSF_TIMER_CTRL**Type:** R**Reset State:** 0x00000000

This register control the add function to the tsf timer

WCSS_A_MTU_TSF_TIMER_CTRL

Bits	Name	Description
0	ADD_TSF_TIMER	Write 1 to this bit so the sum of the add_tsf_timer_[hi lo] value and the current tsf timer will be loaded to the tsf timer.

0x0A081624 WCSS_A_MTU_MTU_LTE_CXM_TSF_LOW**Type:** R**Reset State:** 0x00000000

This register contains the lower 32-bits tsf value captured by the coex_frame_sync signal

WCSS_A_MTU_MTU_LTE_CXM_TSF_LOW

Bits	Name	Description
31:0	MTU_LTE_CXM_TSF_LOW	The lower 32-bits tsf value captured by the capture_clk signal

0x0A081628 WCSS_A_MTU_MTU_LTE_CXM_TSF_HIGH**Type:** R**Reset State:** 0x00000000

This register contains the upper 32-bits tsf value captured by the coex_frame_sync signal

WCSS_A_MTU_MTU_LTE_CXM_TSF_HIGH

Bits	Name	Description
31:0	MTU_LTE_CXM_TSF_HIGH	The upper 32-bits tsf value captured by the capture_clk signal

0x0A08162C WCSS_A_MTU_SW_SECONDARY_CCA_HISTOGRAM_MASK**Type:** RW**Reset State:** 0x0007F800

This register contains the mask bits of the secondary cca histogram for the bw signaling frame exchange

WCSS_A_MTU_SW_SECONDARY_CCA_HISTOGRAM_MASK

Bits	Name	Description
31:0	SW_SECONDARY_CCA_HISTOGRAM_MASK	The mask bits for the secondary cca histogram for 6mbps rts data (sampling rate is from sw_cca_histogram_mask_misc.cca_histogram_sampling_rate)

0x0A081630 WCSS_A_MTU_SW_TERTIARY_CCA_HISTOGRAM_MASK**Type:** RW**Reset State:** 0x0007F800

This register contains the mask bits of the tertiary cca histogram for the bw signaling frame exchange

WCSS_A_MTU_SW_TERTIARY_CCA_HISTOGRAM_MASK

Bits	Name	Description
31:0	SW_TERTIARY_CCA_HISTOGRAM_MASK	The mask bits for the tertiary cca histogram for 6mbps rts data (sampling rate is from sw_cca_histogram_mask_misc.cca_histogram_sampling_rate)

0x0A081634 WCSS_A_MTU_SW_QUATERNARY_CCA_HISTOGRAM_MASK**Type:** RW**Reset State:** 0x0007F800

This register contains the mask bits of the quaternary cca histogram for the bw signaling frame exchange

WCSS_A_MTU_SW_QUATERNARY_CCA_HISTOGRAM_MASK

Bits	Name	Description
31:0	SW_QUATERNARY_CCA_HISTOGRAM_MASK	The mask bits for the quaternary cca histogram for 6mbps rts data (sampling rate is from sw_cca_histogram_mask_misc.cca_histogram_sampling_rate)

0x0A081638 WCSS_A_MTU_SW_CCA_HISTOGRAM_MASK_MISC**Type:** RW**Reset State:** 0x00000302

This register contains the shift value of the cca histogram for the bw signaling frame exchange

WCSS_A_MTU_SW_CCA_HISTOGRAM_MASK_MISC

Bits	Name	Type	Description
13:8	SW_CCA_HISTOGRAM_SAMPLING_RATE	RW	cca histogram sampling rate in unit of usec
6	MTU_TPE_80MHZ_CCA_IS_IDLE_NON6MBPS	R	The 80mhz cca status (masked and shift)
5	MTU_TPE_40MHZ_CCA_IS_IDLE_NON6MBPS	R	The 40mhz cca status (masked and shift)
4	MTU_TPE_80MHZ_CCA_IS_IDLE_6MBPS	R	The 80mhz cca status (masked without shift)
3	MTU_TPE_40MHZ_CCA_IS_IDLE_6MBPS	R	The 40mhz cca status (masked without shift)
2:0	SW_CCA_HISTOGRAM_MASK_SHIFT	RW	Right shift the mask bits of the cca histogram for non-6mbps rts data

0x0A08163C WCSS_A_MTU_CCA_SECONDARY_HISTOGRAM**Type:** R**Reset State:** 0xFFFFFFFF

This register contains the secondary cca histogram

WCSS_A_MTU_CCA_SECONDARY_HISTOGRAM

Bits	Name	Description
31:0	CCA_SECONDARY_HISTOGRAM	The secondary cca is sampled at every 2us time and shifted through a shift register

0x0A081640 WCSS_A_MTU_CCA_TERTIARY_HISTOGRAM**Type:** R**Reset State:** 0xFFFFFFFF

This register contains the tertiary cca histogram

WCSS_A_MTU_CCA_TERTIARY_HISTOGRAM

Bits	Name	Description
31:0	CCA_TERTIARY_HISTOGRAM	The tertiary cca is sampled at every 3us time and shifted through a shift register

0x0A081644 WCSS_A_MTU_CCA_QUATERNARY_HISTOGRAM**Type:** R**Reset State:** 0xFFFFFFFF

This register contains the quaternary cca histogram

WCSS_A_MTU_CCA_QUATERNARY_HISTOGRAM

Bits	Name	Description
31:0	CCA_QUATERNARY_HISTOGRAM	The quaternary cca is sampled at every 3us time and shifted through a shift register

0x0A081648 WCSS_A_MTU_MTU_MISC_FOR_TVWS_1**Type:** RW**Reset State:** 0x00000000

This register contains the upper bits of the sifs/slot time related register bits for 11j/11af

WCSS_A_MTU_MTU_MISC_FOR_TVWS_1

Bits	Name	Description
31:28	SW_MTU_PUSH_MISS_LIMIT	The upper bits of sw_mtu_misc_limits.sw_mtu_push_miss_limit
27:24	SW_MTU_PKT_DET_TO_LIMIT	The upper bits of timeout_limit.sw_mtu_pkt_det_to_limit
23:20	SW_MTU_EARLY_PKT_DET_MISS_LIMIT	The upper bits of sw_mtu_bcn_slot_usec_sifs_limit.sw_mtu_early_pkt_det_miss_limit
19	SW_MTU_QUATERNARY_CCA_LIMIT	The upper bits of secpmdu cca limit for mtu_cca_control.sw_mtu_quaternary_cca_limit
18	SW_MTU_TERTIARY_CCA_LIMIT	The upper bits of secpmdu cca limit for mtu_cca_control.sw_mtu_tertiary_cca_limit
17	SW_MTU_SECONDARY_CCA_LIMIT	The upper bits of secpmdu cca limit for mtu_global_control.sw_mtu_secondary_cca_limit
16	SW_MTU_PIFS_LIMIT	The upper bits of pifs limit for eifs_pifs_slot_limit.sw_mtu_pifs_limit
15:0	DIFS_LIMIT_0TO7	The upper bits of difs limit for backoff0-7 [1:0] for bkof0, [3:2] for bkof1, [15:14] for bkof7

0x0A08164C WCSS_A_MTU_CCA_COUNTER3**Type:** RW**Reset State:** 0x00000000

this register contains the primary cca count. the primary cca is sampled at every us and incremented if the cca is LOW

WCSS_A_MTU_CCA_COUNTER3

Bits	Name	Description
31:0	CCA_COUNTER3	contains the cca primary counter. s/w write has higher periority over h/w update

0x0A081650 WCSS_A_MTU_CCA_COUNTER4**Type:** RW**Reset State:** 0x00000000

this register contains the cca count. the selection to the cca used by this counter is programmable - --

WCSS_A_MTU_CCA_COUNTER4

Bits	Name	Description
31:0	CCA_COUNTER4	depending on the cca_count4_sel-- a programmable bit, the enable to this counter is selected as shown below s/w write has higher periority over h/w update cca_term0 = ~rxp_mtu_secondary_cca; cca_term1 = ~rxp_mtu_secondary_cca & ~rxp_mtu_primary_cca; cca_term2 = rxp_mtu_secondary_cca & ~rxp_mtu_primary_cca; cca_term3 = ~rxp_mtu_secondary_cca cca_term4 = rxp_mtu_secondary_cca cca_term5 = ~cca_term1; cca_term6 = ~cca_term3; cca_term7 = ~cca_term4; cca_count3_en_temp = ~rxp_mtu_primary_cca; 3'b000: cca_count4_en_temp = cca_term0; 3'b001: cca_count4_en_temp = cca_term1; 3'b010: cca_count4_en_temp = cca_term2; 3'b011: cca_count4_en_temp = cca_term3; 3'b100: cca_count4_en_temp = cca_term4; 3'b101: cca_count4_en_temp = cca_term5; 3'b110: cca_count4_en_temp = cca_term6; 3'b111: cca_count4_en_temp = cca_term7; cca_count4_en_final = consider_rx_cca_only_345 ? (cca_count4_en_temp & ~txp_mtu_phytx_idle) : cca_count4_en_temp;

0x0A081654 WCSS_A_MTU_CCA_COUNTER5**Type:** RW**Reset State:** 0x00000000

this register contains the cca count. the selection to the cca used by this counter is programmable -
--

WCSS_A_MTU_CCA_COUNTER5

Bits	Name	Description
31:0	CCA_COUNTER5	depending on the cca_count5_sel-- a programmable bit, the enable to this counter is selected as shown below s/w write has higher periority over h/w update 3'b000: cca_count5_en_temp = ~rxp_mtu_primary_cca & ~rxp_mtu_secondary_cca & ~rxp_mtu_tertiary_cca & ~rxp_mtu_quaternary_cca; 3'b001: cca_count5_en_temp = ~rxp_mtu_tertiary_cca; 3'b010: cca_count5_en_temp = ~rxp_mtu_quaternary_cca; 3'b011: cca_count5_en_temp = ~rxp_mtu_tertiary_cca & ~rxp_mtu_quaternary_cca; 3'b100: cca_count5_en_temp = ~rxp_mtu_tertiary_cca 3'b101: cca_count5_en_temp = rxp_mtu_tertiary_cca & ~rxp_mtu_quaternary_cca; 3'b110: cca_count5_en_temp = rxp_mtu_tertiary_cca 3'b111: cca_count5_en_temp = 1'b0; cca_count5_en_final = consider_rx_cca_only_345 ? (cca_count5_en_temp & ~txp_mtu_phytx_idle) : cca_count5_en_temp;

0x0A081658 WCSS_A_MTU_BSS2_CLIENT_TSF_TIMER_LO

Type: RW

Reset State: 0x00000000

This register contains the lower 32 bits of the bss2 client tsf timer

WCSS_A_MTU_BSS2_CLIENT_TSF_TIMER_LO

Bits	Name	Description
31:0	BSS2_CLIENT_TSF_TIMER_LO	This register contains the lower 32 bits of bss2 client tsf timer

0x0A08165C WCSS_A_MTU_BSS2_CLIENT_TSF_TIMER_HI

Type: RW

Reset State: 0x00000000

This register contains the upper 32 bits of the bss2 client tsf timer

WCSS_A_MTU_BSS2_CLIENT_TSF_TIMER_HI

Bits	Name	Description
31:0	BSS2_CLIENT_TSF_TIMER_HI	This register contains the upper 32 bits of bss2 client tsf timer

0x0A081660 WCSS_A_MTU_QBOOST_CONTROL**Type:** RW**Reset State:** 0x00000000

This register contains the config bits for qboost

WCSS_A_MTU_QBOOST_CONTROL

Bits	Name	Description
15	QBOOST_EN	Set to 1 to enable the qboost feature
13:0	NAV_INCREASE_VALUE	In qboost, when the bmu_mtu_qboost_p signal is received, and the NAV counter value at that moment is smaller than this value, the NAV counter value gets set to the nav_increase_value

10.44 WCSS_A_PHYINT_PHYINT**0x0A015C00 WCSS_A_PHYINT_STATUS****Type:** RW**Reset State:** 0x0000

This register consolidates all the interrupt requests on Phy. All interrupt requests are active high. Interrupt bitfields ending in "_P" can be cleared by writing a "1" to the corresponding bit in this register. Interrupt bitfields not ending in "_P" must be cleared at source.

WCSS_A_PHYINT_STATUS

Bits	Name	Description
14	AGC_WATCHDOG_EXP	AGC FSM watchdog timer expired
13	PHYDBG_CAPTURE_DONE_P	Pronto2: When Phydbg completes a capture, it sets this interrupt. Useful for RTT3 channel-capture indication. Was: Txa header processor found a bad rate field from mpi data *** This interrupt does not exist in Gen5/6 ***
12	PMI_ALARM_P	The alarm condition in PMI
11	RCTL_ALARM_P	The alarm condition in RCTL
10	RBHPC_SOFEOTATN_P	The sof/eof interrupt from the RXB
9	TXCTL_ALARM_P	The interrupt from overall Tx Controller indicating error condition inside Phy TX
8	TACTL_ALARM_P	The interrupt from TACTL indicating error condition inside Phy Txa
7	MPI_MAXREQATN_P	The interrupt from MAC-Phy Interface (MPI) block indicating too many requests from txa or txb controller engines The threshold is configurable in mpi.reg

WCSS_A_PHYINT_STATUS (cont.)

Bits	Name	Description
6	MPI_WATCHDOG_P	The interrupt from MAC-Phy Interface (MPI) block indicating expiry of watchdog timer
5	MPI_TXPABORT_P	The interrupt from MAC-Phy Interface (MPI) block indicating Txp abort
4	TPC_ATN_P	The interrupt from Transmit Power Control (TPC) block signifying error in tpc loop convergence.
3	CAL_ATN_P	The interrupt from the Calibration (CAL) block signifying calibration failure.
2	AGC_RADARDET_P	Set whenever the AGC detects radar. *** This interrupt does not exist in Libra ***
1	AGC_CLRCHAN_P	Set whenever the channel is clear, i.e. CCA is high.
0	RFIF_CALIB_DONE_P	Interrupt from Iris on calibration completion. ** NOTE ** this bit is used for agc_pktdet_n interrupt in Virgo ** ** NOTE ** this bit was used for indicating vswr overload in Libra and Volans **

0x0A015C04 WCSS_A_PHYINT_FAST_MASK**Type:** RW**Reset State:** 0x0000

This register masks the Status register before the bits are combined to generate the fast interrupt to the embedded CPU. Setting a bit to "1" enables the interrupt, "0" masks the interrupt.

WCSS_A_PHYINT_FAST_MASK

Bits	Name	Description
14	AGC_WATCHDOG_EXP	Fast mask bit for bit[14] in STATUS register.
13	PHYDBG_CAPTURE_DONE_P	Fast mask bit for bit[13] in STATUS register.
12	PMI_ALARM_P	Fast mask bit for bit[12] in STATUS register.
11	RACTL_ALARM_P	Fast mask bit for bit[11] in STATUS register.
10	RBHPC_SOFEOFATN_P	Fast mask bit for bit[10] in STATUS register.
9	TXCTL_ALARM_P	Fast mask bit for bit[9] in STATUS register.
8	TACTL_ALARM_P	Fast mask bit for bit[8] in STATUS register.
7	MPI_MAXREQATN_P	Fast mask bit for bit[7] in STATUS register.
6	MPI_WATCHDOG_P	Fast mask bit for bit[6] in STATUS register.
5	MPI_TXPABORT_P	Fast mask bit for bit[5] in STATUS register.
4	TPC_ATN_P	Fast mask bit for bit[4] in STATUS register.

WCSS_A_PHYINT_FAST_MASK (cont.)

Bits	Name	Description
3	CAL_ATN_P	Fast mask bit for bit[3] in STATUS register.
2	AGC_RADARDET_P	Fast mask bit for bit[2] in STATUS register.
1	AGC_CLRCHAN_P	Fast mask bit for bit[1] in STATUS register.
0	RFIF_CALIB_DONE_P	Fast mask bit for bit[0] in STATUS register.

0x0A015C08 WCSS_A_PHYINT_SLOW_MASK**Type:** RW**Reset State:** 0x0000

This register masks the Status register before the bits are combined to generate the slow interrupt to the embedded CPU. Setting a bit to "1" enables the interrupt, "0" masks the interrupt.

WCSS_A_PHYINT_SLOW_MASK

Bits	Name	Description
14	AGC_WATCHDOG_EXP	Slow mask bit for bit[14] in STATUS register.
13	PHYDBG_CAPTURE_DONE_P	Slow mask bit for bit[13] in STATUS register.
12	PMI_ALARM_P	Slow mask bit for bit[12] in STATUS register.
11	RACTL_ALARM_P	Slow mask bit for bit[11] in STATUS register.
10	RBHPC_SOFEOFATN_P	Slow mask bit for bit[10] in STATUS register.
9	TXCTL_ALARM_P	Slow mask bit for bit[9] in STATUS register.
8	TACTL_ALARM_P	Slow mask bit for bit[8] in STATUS register.
7	MPI_MAXREQATN_P	Slow mask bit for bit[7] in STATUS register.
6	MPI_WATCHDOG_P	Slow mask bit for bit[6] in STATUS register.
5	MPI_TXPABORT_P	Slow mask bit for bit[5] in STATUS register.
4	TPC_ATN_P	Slow mask bit for bit[4] in STATUS register.
3	CAL_ATN_P	Slow mask bit for bit[3] in STATUS register.
2	AGC_RADARDET_P	Slow mask bit for bit[2] in STATUS register.
1	AGC_CLRCHAN_P	Slow mask bit for bit[1] in STATUS register.
0	RFIF_CALIB_DONE_P	Slow mask bit for bit[0] in STATUS register.

0x0A015C0C WCSS_A_PHYINT_HOST_MASK**Type:** RW**Reset State:** 0x0000

This register masks the Status register before the bits are combined to generate the PCI/USB interrupt to the Host processor. Setting a bit to "1" enables the interrupt, "0" masks the interrupt.

WCSS_A_PHYINT_HOST_MASK

Bits	Name	Description
14	AGC_WATCHDOG_EXP	Host mask bit for bit[14] in STATUS register.
13	PHYDBG_CAPTURE_DONE_P	Host mask bit for bit[13] in STATUS register.
12	PMI_ALARM_P	Host mask bit for bit[12] in STATUS register.
11	RACTL_ALARM_P	Host mask bit for bit[11] in STATUS register.
10	RBHPC_SOFEOFATN_P	Host interrupt mask bit for bit[10] in STATUS register.
9	TXCTL_ALARM_P	Host interrupt mask bit for bit[9] in STATUS register.
8	TACTL_ALARM_P	Host interrupt mask bit for bit[8] in STATUS register.
7	MPI_MAXREQATN_P	Host interrupt mask bit for bit[7] in STATUS register.
6	MPI_WATCHDOG_P	Host interrupt mask bit for bit[6] in STATUS register.
5	MPI_TXPABORT_P	Host interrupt mask bit for bit[5] in STATUS register.
4	TPC_ATN_P	Host interrupt mask bit for bit[4] in STATUS register.
3	CAL_ATN_P	Host interrupt mask bit for bit[3] in STATUS register.
2	AGC_RADARDET_P	Host interrupt mask bit for bit[2] in STATUS register.
1	AGC_CLRCHAN_P	Host interrupt mask bit for bit[1] in STATUS register.
0	RFIF_CALIB_DONE_P	Host interrupt mask bit for bit[0] in STATUS register.

10.45 WCSS_A_PMI_PMI**0x0A013800 WCSS_A_PMI_PMI_STATE****Type:** R**Reset State:** 0x00000000

Reflects the current value of the state machine.

WCSS_A_PMI_PMI_STATE

Bits	Name	Description
2:0	VALUE	PMI state.

0x0A013804 WCSS_A_PMI_RXA_PKTS**Type:** W**Reset State:** 0x00000000

Counts number of single packets only.

WCSS_A_PMI_RXA_PKTS

Bits	Name	Description
15:0	COUNT	RXA single packet event counter.

0x0A01380C WCSS_A_PMI_RXB_PACKETS**Type:** W**Reset State:** 0x00000000

Counts number of RXB packets.

WCSS_A_PMI_RXB_PACKETS

Bits	Name	Description
15:0	COUNT	RXB packet start event counter.

0x0A013810 WCSS_A_PMI_RXA_DATA_VALID**Type:** W**Reset State:** 0x00000000

Counts number of RXA data valid pulses.

WCSS_A_PMI_RXA_DATA_VALID

Bits	Name	Description
15:0	COUNT	RXA data valid pulse counter.

0x0A013814 WCSS_A_PMI_RXB_DATA_VALID**Type:** W**Reset State:** 0x00000000

Counts number of RXB data valid pulses.

WCSS_A_PMI_RXB_DATA_VALID

Bits	Name	Description
15:0	COUNT	RXB data valid pulse counter.

0x0A013818 WCSS_A_PMI_RXP_SHUTOFF**Type:** W**Reset State:** 0x00000000

Counts number of RXP shutoff pulses.

WCSS_A_PMI_RXP_SHUTOFF

Bits	Name	Description
15:0	COUNT	RXP shutoff pulse counter.

0x0A01381C WCSS_A_PMI_RXP_MU_EOP**Type:** W**Reset State:** 0x00000000

Counts number of RXP MU end-of-packet pulses.

WCSS_A_PMI_RXP_MU_EOP

Bits	Name	Description
15:0	COUNT	RXP MU eop pulse counter.

0x0A013820 WCSS_A_PMI_PHY_STARTS**Type:** W**Reset State:** 0x00000000

Counts number of RXP start pulses issued.

WCSS_A_PMI_PHY_STARTS

Bits	Name	Description
15:0	COUNT	RXP start pulses counter.

0x0A013824 WCSS_A_PMI_PHY_DATA_VAL**Type:** W**Reset State:** 0x00000000

Counts number of RXP data valid pulses.

WCSS_A_PMI_PHY_DATA_VAL

Bits	Name	Description
15:0	COUNT	RXP data valid pulses counter.

0x0A013828 WCSS_A_PMI_PHY_ABORT**Type:** W**Reset State:** 0x00000000

Counts number of RXP abort pulses.

WCSS_A_PMI_PHY_ABORT

Bits	Name	Description
15:0	COUNT	RXP abort pulses counter.

0x0A01382C WCSS_A_PMI_PHY_INT**Type:** W**Reset State:** 0x00000000

Counts number of interrupt pulses.

WCSS_A_PMI_PHY_INT

Bits	Name	Description
15:0	COUNT	PHYINT interrupt pulses counter.

0x0A013830 WCSS_A_PMI_WATCHDOG_TIMEOUT_GENERAL**Type:** RW**Reset State:** 0x0000

Max duration of non-idle time for PMI state machine before abort generated. Duration defined as $(x \times 128 + 128)$ clocks, where x is the value in this register.

WCSS_A_PMI_WATCHDOG_TIMEOUT_GENERAL

Bits	Name	Description
15:0	VALUE	PMI state machine general timeout value (disabled by default).

0x0A013834 WCSS_A_PMI_WATCHDOG_TIMEOUT_11A**Type:** RW**Reset State:** 0x0015

Max duration of time between 11a data valids before abort generated. Duration defined as $(x \times 128 + 128)$ clocks, where x is the value in this register.

WCSS_A_PMI_WATCHDOG_TIMEOUT_11A

Bits	Name	Description
15:0	VALUE	PMI state machine 11a-dval timeout value (default value 21).

0x0A013838 WCSS_A_PMI_WATCHDOG_TIMEOUT_11B**Type:** RW**Reset State:** 0x002A

Max duration of time between 11b data valids before abort generated. Duration defined as $(x \times 128 + 128)$ clocks, where x is the value in this register.

WCSS_A_PMI_WATCHDOG_TIMEOUT_11B

Bits	Name	Description
15:0	VALUE	PMI state machine 11b-dval timeout value (default value 42).

0x0A01383C WCSS_A_PMI_DISABLE_FLAGS**Type:** RW**Reset State:** 0x0000

Various functionality enables.

WCSS_A_PMI_DISABLE_FLAGS

Bits	Name	Description
3	CRC_FORCE_NON_AMPD_U_FLAG	When asserted, this switch forces the crc checker to assume pactes are not AMPDUs. Used for testing 11ac packets generated by phydbg.
2	STATS_UPDATE_DISABLE_FLAG	When asserted, this switch disables updating of the stats APB registers in the PMI.
1	CRC_DISABLE_FLAG	When asserted, this switch disables CRC checking in the PMI.
0	RXP_DISABLE_FLAG	When asserted, this switch disables all output to the RXP.

0x0A013840 WCSS_A_PMI_CRC_FSM_RESET

Type: W

Reset State: 0x0000

Accessing this register will generate a pulse that resets the CRC FSM

WCSS_A_PMI_CRC_FSM_RESET

Bits	Name	Description
0	PULSE	

0x0A013844 WCSS_A_PMI_RXPKT_FILTER_LSW

Type: RW

Reset State: 0x0000

LS 2 bytes of the RX packet filter pattern for CRC check. All-zeros pattern disables filtering.

WCSS_A_PMI_RXPKT_FILTER_LSW

Bits	Name	Description
15:0	DATA	

0x0A013848 WCSS_A_PMI_RXPKT_FILTER_MSW

Type: RW

Reset State: 0x0000

MS 2 bytes of the RX packet filter pattern for CRC check. All-zeros pattern disables filtering.

WCSS_A_PMI_RXPKT_FILTER_MSW

Bits	Name	Description
15:0	DATA	

0x0A01384C WCSS_A_PMI_CRC_DONE**Type:** W**Reset State:** 0x00000000

Number of packets that completed the CRC check.

WCSS_A_PMI_CRC_DONE

Bits	Name	Description
15:0	COUNT	

0x0A013850 WCSS_A_PMI_CRC_PASS**Type:** W**Reset State:** 0x00000000

Number of packets that passed the CRC check.

WCSS_A_PMI_CRC_PASS

Bits	Name	Description
15:0	COUNT	

0x0A013854 WCSS_A_PMI_AMPDU_BAD**Type:** W**Reset State:** 0x00000000

Number of packets that failed the AMPDU CRC check.

WCSS_A_PMI_AMPDU_BAD

Bits	Name	Description
15:0	COUNT	

0x0A013858 WCSS_A_PMI_AMPDU_GOOD**Type:** W**Reset State:** 0x00000000

Number of packets that passed the AMPDU CRC check.

WCSS_A_PMI_AMPDU_GOOD

Bits	Name	Description
15:0	COUNT	

0x0A01385C WCSS_A_PMI_AMPDU_DONE**Type:** W**Reset State:** 0x00000000

Number of packets that completed the AMPDU CRC check.

WCSS_A_PMI_AMPDU_DONE

Bits	Name	Description
15:0	COUNT	

0x0A013860 WCSS_A_PMI_LAST_STATS0**Type:** R**Reset State:** 0x00000000

Last statistics data transferred to RXP, bits[15:0].

WCSS_A_PMI_LAST_STATS0

Bits	Name	Description
15:0	DATA	

0x0A013864 WCSS_A_PMI_LAST_STATS1**Type:** R**Reset State:** 0x00000000

Last statistics data transferred to RXP, bits[31:16].

WCSS_A_PMI_LAST_STATS1

Bits	Name	Description
15:0	DATA	

0x0A013868 WCSS_A_PMI_LAST_STATS2**Type:** R**Reset State:** 0x00000000

Last statistics data transferred to RXP, bits[47:32].

WCSS_A_PMI_LAST_STATS2

Bits	Name	Description
15:0	DATA	

0x0A01386C WCSS_A_PMI_LAST_STATS3**Type:** R**Reset State:** 0x00000000

Last statistics data transferred to RXP, bits[63:48].

WCSS_A_PMI_LAST_STATS3

Bits	Name	Description
15:0	DATA	

10.46 WCSS_A_QTA_QTA

0x0A080FC0 WCSS_A_QTA_INTR**Type:** RW**Reset State:** 0x00000000

Interrupt status register.

WCSS_A_QTA_INTR

Bits	Name	Description
26	RX_FIFO_UFLOW_INTR	Receive FIFO Underflow interrupt status bit. Software must write a 1 to this bit to clear it.
25	RX_FIFO_OFLOW_INTR	Receive FIFO Overflow interrupt status bit. Software must write a 1 to this bit to clear it.
24	RX_FIFO_RDY_INTR	Receive FIFO Ready interrupt status bit: Indicates when the Receive FIFO is ready to be drained of software characters. This interrupt is set every clock cycle while the FIFO fill level is greater than the Receive FIFO Trigger Level (rx_trig_level). Software must write a 1 to this bit to clear it.
17	TX_FIFO_OFLOW_INTR	Transmit FIFO Overflow interrupt status bit. Software must write a 1 to this bit to clear it.
16	TX_FIFO_RDY_INTR	Transmit FIFO Ready interrupt status bit: Indicates when the Transmit FIFO is ready to accept more software characters for transmission. This interrupt is set every clock cycle while the FIFO fill level is less than or equal to the Transmit FIFO Trigger Level (tx_trig_level). Software must write a 1 to this bit to clear it.
8	RX_HW_CHAR_INTR	Received Hardware Character interrupt status bit. Software must write a 1 to this bit to clear it.
1	PARITY_ERR_INTR	UART Parity Error interrupt status bit. Software must write a 1 to this bit to clear it.
0	FRAME_ERR_INTR	UART Frame Error interrupt status bit. Software must write a 1 to this bit to clear it.

0x0A080FC4 WCSS_A_QTA_INTR_EN**Type:** RW**Reset State:** 0x00000000

Interrupt enable register.

WCSS_A_QTA_INTR_EN

Bits	Name	Description
26	RX_FIFO_UFLOW_INTR_EN	Interrupt enable for the Receive FIFO Underflow interrupt.
25	RX_FIFO_OFLOW_INTR_EN	Interrupt enable for the Receive FIFO Overflow interrupt.
24	RX_FIFO_RDY_INTR_EN	Interrupt enable for the Receive FIFO Ready interrupt.
17	TX_FIFO_OFLOW_INTR_EN	Interrupt enable for the Transmit FIFO Overflow interrupt.
16	TX_FIFO_RDY_INTR_EN	Interrupt enable for the Transmit FIFO Ready interrupt.

WCSS_A_QTA_INTR_EN (cont.)

Bits	Name	Description
8	RX_HW_CHAR_INTR_EN	Interrupt enable for the Received Hardware Character interrupt.
1	PARITY_ERR_INTR_EN	Interrupt enable for the Parity Error interrupt.
0	FRAME_ERR_INTR_EN	Interrupt enable for the Frame Error interrupt.

0x0A080FC8 WCSS_A_QTA_UART_CTRL**Type:** RW**Reset State:** 0x03280060

UART control register.

WCSS_A_QTA_UART_CTRL

Bits	Name	Description
31:24	BAUD_RATE_NUMERATOR	Sets the numerator for the ratio of the system clock frequency to the baud rate.
23:16	BAUD_RATE_DENOMINATOR	Sets the denominator for the ratio of the system clock frequency to the baud rate.
6	TX_INV_EN	When set, enables the UART's serial TX output inverter.
5	RX_INV_EN	When set, enables the UART's serial RX input inverter.
4	POWER_DOWN	When set, puts the UART into power-down mode. The UART transmitter will complete its current transmission and then disable all further transmissions. The UART receiver will discard the current character being received and then ignore all further receptions.
3	PARITY_EN	Enables parity bit generation in the UART transmitter, and parity checking in the UART receiver.
2	PARITY_ODD	When parity is enabled, this bit specifies whether the parity is odd or even. When 1, the parity is odd; when 0, it is even.
1:0	NUM_STOPS	Specifies the number of stop bits (1-4) at the end of each character. The number of stop bits per character = num_stops + 1.

0x0A080FCC WCSS_A_QTA_HW_CTRL**Type:** RW**Reset State:** 0x00000014

Hardware layer control register.

WCSS_A_QTA_HW_CTRL

Bits	Name	Type	Description
31:24	LAST_RX_HW_CHAR	R	This field captures the last received hardware character that was passed on to the Signaling layer.
15	TX_BUFFER_STATE	R	State of the 1-character-deep Hardware TX Buffer. When 1, the buffer is full; when 0, the buffer is empty.
14:8	TX_BUFFER_CONTENTS	R	This field captures the hardware character presently stored in the TX Buffer. Resend bit excluded as only non-Resend-Requests are stored in this buffer.
7	RESEND_REQ_FLAG	R	Indicates the present state of the Resend Request flag. When set, it indicates that a Resend Request is pending transmission.
4	EXCLUSIVE_RESEND_EN	RW	Exclusive Resend Enable: When set, the QTA module will treat Signal State bits within Resend Request characters as invalid. When clear, the Signal State bits are valid regardless of whether the character is a Resend Request.
3	RESEND_REQ_PASSTHRU_EN	RW	Resend Request Pass-Through Enable: When set, the QTA module will pass received Resend Request characters through to the Signaling layer for processing. NB: This feature is unsupported for the WLAN version of the QTA module, because the Signaling interface for the WLAN version does not support Resend signaling.
2	RESEND_RESP_EN	RW	Resend Response Enable: When this bit is set, the QTA module will automatically respond to received Resend Requests by resending its last (non-Resend-Request) hardware character. Otherwise, the QTA module will not respond to Resend Requests received from the peer device.
1	RETRANSMIT	W	Retransmit TX Buffer contents: Writing a 1 to this bit causes the last (non-Resend-Request) hardware character to be re-transmitted. This bit is self-clearing.
0	RESEND_REQUEST	W	Send a Resend Request: Writing a 1 to this bit causes a Resend Request character to be queued for transmission. This bit is self-clearing.

0x0A080FD0 WCSS_A_QTA_HW_TX_BUFFER_MODIFY**Type:** W**Reset State:** 0x00000000

TX Buffer modify register.

WCSS_A_QTA_HW_TX_BUFFER_MODIFY

Bits	Name	Description
6:0	DATA	Data to be written to the TX Buffer. Writing to this register causes the contents of the TX Buffer to be modified to the value of the data field, and causes the TX buffer state to be set to full.

0x0A080FD4 WCSS_A_QTA_SW_CTRL**Type:** RW**Reset State:** 0x00000000

Software layer control register.

WCSS_A_QTA_SW_CTRL

Bits	Name	Type	Description
29:24	RX_FILL	R	Indicates the number of software characters presently queued in the Receive FIFO.
21	RX_FLUSH	W	Receive FIFO Flush: Writing a 1 to this bit flushes the contents of the Receive FIFO. This bit is self-clearing.
20:16	RX_TRIG_LEVEL	RW	Receive FIFO Trigger Level. See the rx_fifo_rdy_int interrupt status register description.
13:8	TX_FILL	R	Indicates the number of software characters presently queued in the Transmit FIFO.
5	TX_FLUSH	W	Transmit FIFO Flush: Writing a 1 to this bit flushes the contents of the Transmit FIFO. This bit is self-clearing.
4:0	TX_TRIG_LEVEL	RW	Transmit FIFO Trigger Level. See the tx_fifo_rdy_int interrupt status register description.

0x0A080FD8 WCSS_A_QTA_SW_TX_FIFO_IN**Type:** W**Reset State:** 0x00000000

Software Transmit FIFO write data register.

WCSS_A_QTA_SW_TX_FIFO_IN

Bits	Name	Description
7:0	DATA	Software character to be transmitted.

0x0A080FDC WCSS_A_QTA_SW_RX_FIFO_OUT**Type:** R**Reset State:** 0x00000000

Software Receive FIFO read data register.

WCSS_A_QTA_SW_RX_FIFO_OUT

Bits	Name	Description
7:0	DATA	Software character received.

10.47 WCSS_A_RACTL_RACTL**0x0A00F000 WCSS_A_RACTL_PTC_LOOP_EN****Type:** RW**Reset State:** 0x003F

This registers provides the enable signals for the amplitude, frequency and channel tracking functions within the RXA PTC module. When each bit is asserted, the corresponding tracking loop is enabled.

WCSS_A_RACTL_PTC_LOOP_EN

Bits	Name	Description
9	PTCPHASETRK_DSBL	Disables VHTLTF phase-tracking functionality
5	TIME_ADJ_EN	TDC timing adjustment enable.
4	PH_ADJ_EN	TDC phase adjustment enable.
3	CHAN_EN	Channel tracking enable. If asserted, PTC channel tracking loop will detect channel phase changes and modify the channel estimate samples received from the channel estimator module.
2	FREQ_EN	Frequency tracking enable. If asserted, PTC frequency tracking loop will track reference oscillator frequency drift and modify the phase of the incoming samples received from the phase imbalance corrector module.
1	AMP_EN_MIMO	Amplitude tracking enable for mimo symbols. If asserted, PTC amplitude tracking loop will track and modify the amplitude of the incoming samples received from the phase imbalance corrector module (mimo symbols only)
0	AMP_EN_SIMO	Amplitude tracking enable for simo symbols. If asserted, PTC amplitude tracking loop will track and modify the amplitude of the incoming samples received from the phase imbalance corrector module (simo symbols only)

0x0A00F004 WCSS_A_RACTL_PTC_LOOP_GAIN**Type:** RW**Reset State:** 0x002A

This registers provides the gain factors for the ampiltude, frequency and channel tracking functions within the RXA PTC module.

WCSS_A_RACTL_PTC_LOOP_GAIN

Bits	Name	Description
7:4	CHAN_GAIN	Channel tracking loop gain. 0x0: CGAIN_2 (gain = 2) 0x1: CGAIN_3 (gain = 3) 0x2: CGAIN_4 (gain = 4) 0x3: CGAIN_4TOO (gain = 4)
3:2	FREQ_GAIN	Frequency tracking loop gain. 0x0: FGNO_AVG (No averaging) 0x1: FGAIN_2 (gain = 2) 0x2: FGAIN_3 (gain = 3) 0x3: FGAIN_4 (gain = 4)
1:0	AMP_GAIN	Amplitude tracking loop gain. 0x0: AGNO_AVG (No averaging) 0x1: AGAIN_3 (gain = 3) 0x2: AGAIN_4 (gain = 4) 0x3: AGAIN_5 (gain = 5)

0x0A00F008 WCSS_A_RACTL_PTC_CHAN_FREQ**Type:** RW**Reset State:** 0x087E

This value represents the ratio of sampling frequency to RF channel frequency using 10-bit quantisation

WCSS_A_RACTL_PTC_CHAN_FREQ

Bits	Name	Description
11:0	PTC_CHAN_FREQ	Default setting = 12'd2174

0x0A00F00C WCSS_A_RACTL_RACTL_CFG**Type:** RW**Reset State:** 0x7D35

This register contains all the controls to enable various packet conditioning/decoding parameters

WCSS_A_RACTL_RACTL_CFG

Bits	Name	Description
14	UNSUPP_CHK_11AC_PAID_GIDMISM_EN	*** Added in Pronto2 *** When enabled and when unsupported conditions in 11ac happens either from gid and/or paid mismatch, MAC will get a new level signal indicating Ractl has entered deferral mode.
13	UNSUPP_CHK_11N11AC_GENERAL_EN	*** Added in Pronto2 *** When enabled and when general unsupported conditions in 11n and 11ac happens (-except- gid and/or paid mismatch), MAC will get a new level signal indicating Ractl has entered deferral mode.
12	NDP_EN_11N	This bit enables the checking of NDP(ZLF) packets in 11n
11	AC_DEMOD_EN	Enables reception of AC packets
10	SERVFLD_OLVD_EN	This bit enables service-field-overloading for bw-capability exchange in legacy rts/cts frames. Default enabled, pmi byte5[6:4] gets data from de-scrambler input.
9	BEHAV_11N_EN	When enabled, the signal-field check of 11ac will be similar to 11n, i.e. the crc failure will be treated as a packet-abort condition. Any other condition will qualify for unsupported packet and deferral will be based on Isig. If this bit is 0, irrespective of unsupported or crc-failed vhtsiga, deferral will be done based on Isig.
8	AC_QAM256_SUPP_EN	This bit enables declaration of 256 QAM as supported rate in 11AC
7	BW_MSB_CHECK_EN	Asserted to indicate the vhtsiga bits [2:0] will be checked against agc bw [2:0] settings. 0 => vhtsiga bits [1:0] will be checked against agc bw [1:0] settings. Note default changed in Pronto assuming the msb has been changed in the latest 11ac spec to be reserved and so should not be checked.
6	SU_AID_CHECK_EN	Asserted high to indicate that when SU VHTSIGA is received, the pAID bits will be checked against the PROG_P_AID register contents to decide whether the packet was meant for this SU station Default 0 => vhtsiga bits [21:13] 9 bits will NOT be checked against PROG_P_AID reg settings.
5	AC_VHTB_CHK_EN_MU	This bit enables the 11ac VHTSIGB checking in MU mode Note that the contents of VHTSIGB are used by RACTL/RAHDR in MU mode so checking CRC is preferable. Unlike SU where contents of VHTSIGB are ignored and thus an invalid-CRC VHTSIGB is not indicative of a faulty packet, A CRC-failed VHTSIGB in MU-case can be indicative of a faulty packet.
4	AC_MU_RXPSHUTOFF_EN	This bit enables the 11ac MU packets to be terminated by packet-completion-indication from RXP. By default, the min of effective length computed from Isig and vhtsigb is used to signal completion within Phy itself.
3	AC_VHTB_CHK_EN	This bit enables the 11ac VHTSIGB checking in SU mode
2	SERVICE_EN	This bit enables the SERVICE field content check in non-11ac packets

WCSS_A_RACTL_RACTL_CFG (cont.)

Bits	Name	Description
1	SOUNDING_BIT_CHK_EN	This bit enables checking of sounding bit in 11n htsg for unsupported htsg purpose (applies only to non-ndp packets)
0	LSIG_BIT3_CHK_EN	This bit enables checking the LSIG[3] bit which must be 1 for legacy packets, 11n mm and 11ac packets

0x0A00F010 WCSS_A_RACTL_RXA_ABORT**Type:** W**Reset State:** 0x0000

Writing any value to this register instructs the RX-A controller to abort the current packet and return the RXA module to idle mode.

WCSS_A_RACTL_RXA_ABORT

Bits	Name	Description
0	RXA_ABORT	Writing any value to this bit will trigger an abort of the RX-A module. Reading this register will return a 0 value.

0x0A00F014 WCSS_A_RACTL_RACTL_TOFFSET**Type:** RW**Reset State:** 0x0000

This register contains a timing offset value that is added to the timing correction value provided by the BTCF module. This addition takes place in the TDC module.

WCSS_A_RACTL_RACTL_TOFFSET

Bits	Name	Description
7:0	RACTL_TOFFSET	Timing offset value fed to the TDC module. Default setting = 8'h0.

0x0A00F018 WCSS_A_RACTL_RACTL_DELTAT_IN**Type:** RW**Reset State:** 0x0000

This register contains a value that is used in PTC as a timing offset with the existing timing offset inside PTC

WCSS_A_RACTL_RACTL_DELTAT_IN

Bits	Name	Description
15	APB_DT_OVERRIDE	If asserted, the RACTL module passes the APB timing offset value to the PTC module. ** This is not used in Pronto **
10:0	APB_DT	Default setting = 11'h0. **In pronto, this has been reused from existing dt_override register (meant for overriding BTCF deltat earlier)

0x0A00F01C WCSS_A_RACTL_RACTL_DELTAT_OUT**Type:** R**Reset State:** 0x00000000

This register allows the host to observe timing value that is fed to the TDC module

WCSS_A_RACTL_RACTL_DELTAT_OUT

Bits	Name	Description
8:0	TDC_DT	Current timing correction value being used by TDC module provided by PTC Note the current timing adjust value is in PTC_DELTAAMP_TIMINGOFFSET_OUT register

0x0A00F020 WCSS_A_RACTL_RACTL_DELTAF_IN**Type:** RW**Reset State:** 0x0000

Test frequency correction value that overrides the value provided by the BTCF module.

WCSS_A_RACTL_RACTL_DELTAF_IN

Bits	Name	Description
15	APB_DFUPDT_OVERRIDE	Use this to indicate PTC to use the initially received value rather than updating it.
14	APB_DFINIT_OVERRIDE	If asserted, the RACTL module passes the APB frequency error value to the TDC module instead of that received from BTCF.
9:0	APB_DF	Test frequency correction value that overrides the internally-calculated value. Default setting = 10'h0.

0x0A00F024 WCSS_A_RACTL_RACTL_DELTA_F_OUT**Type:** R**Reset State:** 0x00000000

Current frequency correction value being used by TDC module

WCSS_A_RACTL_RACTL_DELTA_F_OUT

Bits	Name	Description
11:0	PTC_DF	Current frequency correction value being used by TDC module

0x0A00F028 WCSS_A_RACTL_PTC_DELTAAMP_TIMINGOFFSET_OUT**Type:** R**Reset State:** 0x00000000

Current frequency correction value being used by PTC module

WCSS_A_RACTL_PTC_DELTAAMP_TIMINGOFFSET_OUT

Bits	Name	Description
9:8	PTC_DT	Current timing offset correction value being provided PTC to TDC
7:0	PTC_DA	Current amplitude correction value being used by PTC module

0x0A00F034 RESERVED**0x0A00F038 WCSS_A_RACTL_RACTL_PKT_COUNT****Type:** W**Reset State:** 0x00000000

Number of packet burst start signals received by the RACTL module

WCSS_A_RACTL_RACTL_PKT_COUNT

Bits	Name	Description
11:0	RACTL_SPKTCOUNT	Number of packet burst start signals received by the RACTL module from AGC

0x0A00F03C WCSS_A_RACTL_RACTL_SYMCOUNT**Type:** W**Reset State:** 0x00000000

Number of OFDM symbol start signals generated by the RACTL module

WCSS_A_RACTL_RACTL_SYMCOUNT

Bits	Name	Description
15:0	RACTL_SYMCOUNT	Number of OFDM symbol start signals generated by the RACTL module

0x0A00F040 WCSS_A_RACTL_RACTL_PKTCOUNT_AC20**Type:** W**Reset State:** 0x00000000

Number of AC20 packets detected by RACTL

WCSS_A_RACTL_RACTL_PKTCOUNT_AC20

Bits	Name	Description
7:0	AC20_SPKTCOUNT	

0x0A00F044 WCSS_A_RACTL_RACTL_PKTCOUNT_AC40**Type:** W**Reset State:** 0x00000000

Number of AC40 packets detected by RACTL

WCSS_A_RACTL_RACTL_PKTCOUNT_AC40

Bits	Name	Description
7:0	AC40_SPKTCOUNT	

0x0A00F048 WCSS_A_RACTL_RACTL_PKTCOUNT_AC80**Type:** W**Reset State:** 0x00000000

Number of AC80 packets detected by RACTL

WCSS_A_RACTL_RACTL_PKTCOUNT_AC80

Bits	Name	Description
7:0	AC80_SPKTCOUNT	

0x0A00F04C WCSS_A_RACTL_RACTL_PKTCOUNT_AC_SU**Type:** W**Reset State:** 0x00000000

Number of AC SU packets detected by RACTL

WCSS_A_RACTL_RACTL_PKTCOUNT_AC_SU

Bits	Name	Description
7:0	AC_SU_SPKTCOUNT	

0x0A00F050 WCSS_A_RACTL_RACTL_PKTCOUNT_AC_MU**Type:** W**Reset State:** 0x00000000

Number of AC MU packets detected by RACTL

WCSS_A_RACTL_RACTL_PKTCOUNT_AC_MU

Bits	Name	Description
7:0	AC_MU_SPKTCOUNT	

0x0A00F054 WCSS_A_RACTL_CHE_PKTCOUNT**Type:** W**Reset State:** 0x00000000

Number of packet burst start signals received by the CHE module

WCSS_A_RACTL_CHE_PKTCOUNT

Bits	Name	Description
7:0	CHE_SPKTCOUNT	Number of packet burst start signals received by the CHE module

0x0A00F058 WCSS_A_RACTL_PTC_PKTCount**Type:** W**Reset State:** 0x00000000

Number of packet burst start signals received by the PTC module

WCSS_A_RACTL_PTC_PKTCount

Bits	Name	Description
11:0	PTC_SPKTCOUNT	Number of packet burst start signals received by the PTC module (all packet types)

0x0A00F05C WCSS_A_RACTL_PTC_SYMCount**Type:** W**Reset State:** 0x00000000

Number of OFDM symbol start signals received by the PTC module

WCSS_A_RACTL_PTC_SYMCount

Bits	Name	Description
15:0	PTC_SYMCOUNT	Number of OFDM symbol start signals received by the PTC module

0x0A00F060 WCSS_A_RACTL_PTC_PKTCount_20M**Type:** W**Reset State:** 0x00000000

Number of packet burst start signals received by the PTC module in 20MHz modes

WCSS_A_RACTL_PTC_PKTCount_20M

Bits	Name	Description
11:0	PTC_SPKTCOUNT_20M	Number of packet burst start signals received by the PTC module in 20MHz modes (all packet types) This includes the 20M 11n packets falsely detected as 40M packets

0x0A00F064 WCSS_A_RACTL_PTC_PKTCount_40M**Type:** W**Reset State:** 0x00000000

Number of packet burst start signals received by the PTC module in 40MHz modes

WCSS_A_RACTL_PTC_PKTCount_40M

Bits	Name	Description
11:0	PTC_SPKTCOUNT_40M	Number of packet burst start signals received by the PTC module in 40MHz and 80MHz modes (all packet types) This includes the 40M 11n packets falsely detected as 80M packets

0x0A00F068 WCSS_A_RACTL_PTC_PKTCount_SIMO

Type: W

Reset State: 0x00000000

Number of packet burst start signals received by the PTC module in simo mode

WCSS_A_RACTL_PTC_PKTCount_SIMO

Bits	Name	Description
11:0	PTC_SPKTCOUNT_SIMO	Number of packet burst start signals received by the PTC module in simo modes. This includes the 20M and 40M 11n and legacy mode packets but does not include stbc mode packets

0x0A00F06C WCSS_A_RACTL_PTC_PKTCount_STBC

Type: W

Reset State: 0x00000000

Number of packet burst start signals received by the PTC module in stbc modes

WCSS_A_RACTL_PTC_PKTCount_STBC

Bits	Name	Description
11:0	PTC_SPKTCOUNT_STBC	Number of packet burst start signals received by the PTC module in stbc modes. This includes the 20M and 40M 11n mode packets.

0x0A00F070 WCSS_A_RACTL_PTC_PKTCount_GF

Type: W

Reset State: 0x00000000

Number of packet burst start signals received by the PTC module in all Greenfield modes

WCSS_A_RACTL_PTC_PKT_COUNT_GF

Bits	Name	Description
11:0	PTC_SPKTCOUNT_GF	Number of packet burst start signals received by the PTC module in all Greenfield modes.

0x0A00F074 WCSS_A_RACTL_PTC_PKT_COUNT_MM**Type:** W**Reset State:** 0x00000000

Number of packet burst start signals received by the PTC module in all Mixed-mode packets

WCSS_A_RACTL_PTC_PKT_COUNT_MM

Bits	Name	Description
11:0	PTC_SPKTCOUNT_MM	Number of packet burst start signals received by the PTC module in all Mixed-mode modes.

0x0A00F078 WCSS_A_RACTL_FFT_OPSCALE**Type:** RW**Reset State:** 0x0000

Amount of output attenuation to be applied by the FFT module

WCSS_A_RACTL_FFT_OPSCALE

Bits	Name	Description
1:0	FFT_OPSCALE	Amount of output attenuation to be applied by the FFT module 0x0: GAIN_1 (gain = 1) 0x1: GAIN_1_2 (gain = 1/2) 0x2: GAIN_1_4 (gain = 1/4) 0x3: GAIN_1_8 (gain = 1/8)

0x0A00F07C WCSS_A_RACTL_RACTL_STATE**Type:** R**Reset State:** 0x00000000

Current state of RACTL state machine

WCSS_A_RACTL_RACTL_STATE

Bits	Name	Description
14:10	PREV1_STATE	Previous-but-1 state of RACTL state machine
9:5	PREV0_STATE	Previous state of RACTL state machine
4:0	CURRENT_STATE	Current state of RACTL state machine

0x0A00F080 WCSS_A_RACTL_AGC_ABORT_COUNT**Type:** W**Reset State:** 0x00000000

Number of abort signals received from the AGC module

WCSS_A_RACTL_AGC_ABORT_COUNT

Bits	Name	Description
15:0	AGC_ABORTCOUNT	Number of abort signals received from the AGC module

0x0A00F084 WCSS_A_RACTL_SIGNAL_INVALID_COUNT_LSIG**Type:** W**Reset State:** 0x00000000

Number of packets received with invalid LEGACY SIGNAL field values

WCSS_A_RACTL_SIGNAL_INVALID_COUNT_LSIG

Bits	Name	Description
15:0	INVALID_SFPCOUNT	Number of packets received with invalid SIGNAL field values

0x0A00F088 WCSS_A_RACTL_SERV_INVALID_COUNT**Type:** W**Reset State:** 0x00000000

Number of packets received with invalid SERVICE field values

WCSS_A_RACTL_SERV_INVALID_COUNT

Bits	Name	Description
15:0	INVALID_SERVCOUNT	Number of packets received with invalid SERVICE field values

0x0A00F08C WCSS_A_RACTL_INTERRUPT_COUNT**Type:** W**Reset State:** 0x00000000

Combined number of interrupts received by RAHDR and RACTL

WCSS_A_RACTL_INTERRUPT_COUNT

Bits	Name	Description
15:0	INTERRUPT_COUNT	Total number of interrupts. For Volans: Does not include the NDP packets that qualify for CSI transfer. Does not include the "Unsupported configuration" (Note that for Volans, 1rx 4ss configuration is allowed.)

0x0A00F090 WCSS_A_RACTL_LAST_LS_SIGNAL**Type:** R**Reset State:** 0x0000

This register holds the LS 16 bits of the last decoded standard SIGNAL field value. On reset, the register is set to 0

WCSS_A_RACTL_LAST_LS_SIGNAL

Bits	Name	Description
15:0	VALUE	LS 16 bits of the last decoded standard SIGNAL field

0x0A00F094 WCSS_A_RACTL_LAST_MS_SIGNAL**Type:** R**Reset State:** 0x0000

This register holds the MS 16 bits of the last decoded standard SIGNAL field value. On reset, the register is set to 0

WCSS_A_RACTL_LAST_MS_SIGNAL

Bits	Name	Description
7:0	VALUE	MS 8 bits of the last decoded standard SIGNAL field

0x0A00F098 WCSS_A_RACTL_LAST_MCS**Type:** R**Reset State:** 0x00000000

The last MCS received by the controller

WCSS_A_RACTL_LAST_MCS

Bits	Name	Description
6:0	VALUE	

0x0A00F09C WCSS_A_RACTL_STROBE_UNSUPPHTSIG**Type:** W**Reset State:** 0x00000000

Initiates write of the 1st occurrence of unsupported htsig

WCSS_A_RACTL_STROBE_UNSUPPHTSIG

Bits	Name	Description
0	INITIATE	Write to this register to do only one write to registers 9C to A0 above.

0x0A00F0A0 WCSS_A_RACTL_LAST_UNSUPHTSIGNAL0**Type:** R**Reset State:** 0x0000

This register holds the LS 16 bits of the last decoded 'unsupported' HTSIG SIGNAL field value. On reset, the register is set to 0

WCSS_A_RACTL_LAST_UNSUPHTSIGNAL0

Bits	Name	Description
15:0	VALUE	LS 16 bits of the last decoded 'unsupported' HTSIG SIGNAL field

0x0A00F0A4 WCSS_A_RACTL_LAST_UNSUPHTSIGNAL1**Type:** R**Reset State:** 0x0000

This register holds bits[31:16] of the last decoded 'unsupported' HTSIG SIGNAL field value. On reset, the register is set to 0

WCSS_A_RACTL_LAST_UNSUPHTSIGNAL1

Bits	Name	Description
15:0	VALUE	bits [31:16] of the last decoded 'unsupported' HTSIG SIGNAL field

0x0A00F0A8 WCSS_A_RACTL_LAST_UNSUPHTSIGNAL2

Type: R

Reset State: 0x0000

This register holds bits [47:32] of the last decoded 'unsupported' HTSIG SIGNAL field value. On reset, the register is set to 0

WCSS_A_RACTL_LAST_UNSUPHTSIGNAL2

Bits	Name	Description
15:0	VALUE	MS 16 bits of the last decoded 'unsupported' HTSIG SIGNAL field

0x0A00F0AC WCSS_A_RACTL_LAST_MXED_SIG0

Type: R

Reset State: 0x0000

This register will be used to primarily diagnose 11n MM and 11AC packet demodulation issues.

WCSS_A_RACTL_LAST_MXED_SIG0

Bits	Name	Description
15:0	VALUE	

0x0A00F0B0 WCSS_A_RACTL_LAST_MXED_SIG1

Type: R

Reset State: 0x0000

This register will be used to primarily diagnose 11n MM and 11AC packet demodulation issues.

WCSS_A_RACTL_LAST_MXED_SIG1

Bits	Name	Description
15:0	VALUE	

0x0A00F0B4 WCSS_A_RACTL_LAST_MXED_SIG2**Type:** R**Reset State:** 0x0000

This register will be used to primarily diagnose 11n MM and 11AC packet demodulation issues.

WCSS_A_RACTL_LAST_MXED_SIG2

Bits	Name	Description
15:0	VALUE	

0x0A00F0B8 WCSS_A_RACTL_STATISTICS_SELECT**Type:** RW**Reset State:** 0x0000

This register selects the format of the 8-byte stats field appended to the end of a received packet

WCSS_A_RACTL_STATISTICS_SELECT

Bits	Name	Description
2:0	VALUE	Stats format selector 0x0: AGC_CONSOLIDATED1 ({bpcf_statword, agc_statword[31:0]}) 0x1: AGC_CONSOLIDATED2 ({bpcf_statword, agc_statword[63:32]}) 0x2: TDC_CONSOLIDATED ({bpcf_statword, tdc_statword}) 0x3: CAL_ONLY ({cal3_statdata, cal2_statdata, cal1_statdata, cal0_statdata} for Virgo, {cal1_statdata, cal0_statdata} for Spica) 0x4: KNOWN_PATTERN (64 hex) 0x5: FSCALE_CONSOLIDATED ({fscale_bpcf_gain3, fscale_bpcf_gain2, fscale_bpcf_gain1, fscale_bpcf_gain0, bpcf_statword, agc_statword[59:32]} for Virgo, {fscale_bpcf_gain1, fscale_bpcf_gain0, bpcf_statword, agc_statword[59:32]} for Spica)

0x0A00F0BC WCSS_A_RACTL_DAGC_TGTBO**Type:** RW**Reset State:** 0x001E

Fscale Target Backoff for Digital Gain computation. The gain is set this amount of dBs below the gain that would fully saturate the receiver. The backoff is specified in 0.5 dB steps.

WCSS_A_RACTL_DAGC_TGTBO

Bits	Name	Description
5:0	BACK_OFF	Backoff value. Default backoff is 30.

0x0A00F0C0 WCSS_A_RACTL_CMDWDFILL**Type:** RW**Reset State:** 0x0003

Sets the options for software to get sig,htsig,service field in the command word

WCSS_A_RACTL_CMDWDFILL

Bits	Name	Description
1	SERVICE_EN	Asserted to enable transmission of sig,htsig from RAHDR to PMI. Default is enable.
0	HDR_EN	Asserted to enable transmission of sig,htsig rom RAHDR to PMI. Default is enable.

0x0A00F0C4 WCSS_A_RACTL_RACTL_STATUS**Type:** RW**Reset State:** 0x0000**WCSS_A_RACTL_RACTL_STATUS**

Bits	Name	Type	Description
13	PKTABORTSCOPE_CLR	RW	Clear for bits [9:1]
12	PKT_PENDING_CLR	RW	Clear for bit 0.

WCSS_A_RACTL_RACTL_STATUS (cont.)

Bits	Name	Type	Description
9:1	PKTABORTSCOPE	R	When sampled, shows the reason for ractl packet abort 0 - ssf_invalid 1 - htsig_invalid 2 - not_avalid 3 - serv_invalid 4 - state machine watchdog expired 5 - unsupported mode with given antenna configuration (NOTE that for Volans, 1rx all streams are supported for CSI transfer. However, the definition of this bit has been retained) 6 - unsupported htsig (NOTE that for Volans, NDP packets are supported for CSI transfer. However, the definition of this bit has been retained) 7 - spare 8 - spare
0	PKT_PENDING	R	When sampled, shows if a packet is pending for processing by RACTL

0x0A00F0C8 WCSS_A_RACTL_NES2_MCSBOUND0**Type:** RW**Reset State:** 0x1715

Programmable lower and upper bounds of mcs fields to detect Nes of 2

WCSS_A_RACTL_NES2_MCSBOUND0

Bits	Name	Description
14:8	MCSUB0	Upper bound0. This value will be used with 11n40 flag to declare Nes of 2
6:0	MCSLB0	Lower bound0. This value will be used with 11n40 flag to declare Nes of 2

0x0A00F0CC WCSS_A_RACTL_NES2_MCSBOUND1**Type:** RW**Reset State:** 0x1F1C

Programmable lower and upper bounds of mcs fields to detect Nes of 2

WCSS_A_RACTL_NES2_MCSBOUND1

Bits	Name	Description
14:8	MCSUB1	Upper bound1. This value will be used with 11n40 flag to declare Nes of 2
6:0	MCSLB1	Lower bound1. This value will be used with 11n40 flag to declare Nes of 2

0x0A00F0D0 WCSS_A_RACTL_NES2_MCSBOUND2**Type:** RW**Reset State:** 0x3434

Programmable lower and upper bounds of mcs fields to detect Nes of 2

WCSS_A_RACTL_NES2_MCSBOUND2

Bits	Name	Description
14:8	MCSUB2	Upper bound2. This value will be used with 11n40 flag to declare Nes of 2
6:0	MCSLB2	Lower bound2. This value will be used with 11n40 flag to declare Nes of 2

0x0A00F0D4 WCSS_A_RACTL_NES2_MCSBOUND3**Type:** RW**Reset State:** 0x4C46

Programmable lower and upper bounds of mcs fields to detect Nes of 2

WCSS_A_RACTL_NES2_MCSBOUND3

Bits	Name	Description
14:8	MCSUB3	Upper bound3. This value will be used with 11n40 flag to declare Nes of 2
6:0	MCSLB3	Lower bound3. This value will be used with 11n40 flag to declare Nes of 2

0x0A00F0D8 WCSS_A_RACTL_LSIGLEN_LOWERBOUND**Type:** RW**Reset State:** 0x0149

Programmable lower bound of length field in LSIG to detect htsig in a MM packet

WCSS_A_RACTL_LSIGLEN_LOWERBOUND

Bits	Name	Description
9:5	LSIGLENLB_AC	The length field in the lsig is required to be equal to or more than this value to look for an vhtsiga in the next symbol
4:0	LSIGLENLB_N	The length field in the lsig is required to be equal to or more than this value to look for an htsig in the next symbol

0x0A00F0DC WCSS_A_RACTL_RACTL_INTR_EN**Type:** RW**Reset State:** 0x0060

The enables to report the interrupts to Phyint. Bit 0 is for enabling interrupt from rahdr and the other ones are in sequence corresponding to register RACTL_STATUS. By default, only watchdog-expiry and unsupported configuration-related interrupts are enabled

WCSS_A_RACTL_RACTL_INTR_EN

Bits	Name	Description
7	UNSUPHTSIG_INTRMASKEN	
6	UNSUPCFG_INTRMASKEN	
5	WATCHDOGEXP_INTRMASKEN	
4	SERVINVALID_INTRMASKEN	
3	NOTAVALID_INTRMASKEN	
2	HTSIG_INVALID_INTRMASKEN	
1	SSF_INVALID_INTRMASKEN	
0	RAHDR_INTRMASKEN	

0x0A00F0E0 WCSS_A_RACTL_RACTL_SUPP_BWMISMCOUNT**Type:** R**Reset State:** 0x00000000

RACTL counter to count the number of times rxa demodulated a packet at a lower bandwidth compared to that specified by agc. Note that counter is in hardware and the slave is only a readonly.

WCSS_A_RACTL_RACTL_SUPP_BWMISMCOUNT

Bits	Name	Description
14:10	BW80FOUND20MISM	The dynamic bandwidth setting is 80Mhz but 20M packet detected
9:5	BW80FOUND40MISM	The dynamic bandwidth setting is 80Mhz but 40M packet detected
4:0	BW40FOUND20MISM	The dynamic bandwidth setting is 40Mhz but 20M packet detected

0x0A00F0E4 WCSS_A_RACTL_PTC_TH**Type:** RW**Reset State:** 0x0088**WCSS_A_RACTL_PTC_TH**

Bits	Name	Description
7:4	IBPSK	
3:0	QBPSK	

0x0A00F0E8 WCSS_A_RACTL_BWMISMCNTRST**Type:** RW**Reset State:** 0x0000**WCSS_A_RACTL_BWMISMCNTRST**

Bits	Name	Description
1	RST_UNSUPP	Resets the counters which count the UNsupported bandwidth mismatches
0	RST_SUPP	Resets the counters which count the supported bandwidth mismatches

0x0A00F0F0 WCSS_A_RACTL_LMAP_OPSCALE1_SISO**Type:** RW**Reset State:** 0x28EC

This registers provides the rate-qam-size dependent output scaling to LMAP simo distance outputs

WCSS_A_RACTL_LMAP_OPSCALE1_SISO

Bits	Name	Description
14:12	SISO_MCS4_16QA3BY4	
11:9	SISO_MCS3_16QA1BY2	
8:6	SISO_MCS2_QPSK3BY4	
5:3	SISO_MCS1_QPSK1BY2	
2:0	SISO_MCS0_BPSK1BY2	

0x0A00F0F4 WCSS_A_RACTL_LMAP_OPSCALE2_SISO**Type:** RW**Reset State:** 0x0053

This registers provides the rate-qam-size dependent output scaling to LMAP simo distance outputs

WCSS_A_RACTL_LMAP_OPSCALE2_SISO

Bits	Name	Description
14:12	SISO_MCS9_256QA5BY6	
11:9	SISO_MCS8_256QA3BY4	
8:6	SISO_MCS7_64QA5BY6	
5:3	SISO_MCS6_64QA3BY4	
2:0	SISO_MCS5_64QA2BY3	

0x0A00F0F8 WCSS_A_RACTL_LMAP_OPSCALE1_STBC**Type:** RW**Reset State:** 0x26E4

This registers provides the rate-qam-size dependent output scaling to LMAP stbc distance outputs

WCSS_A_RACTL_LMAP_OPSCALE1_STBC

Bits	Name	Description
14:12	STBC_MCS4_16QA3BY4	
11:9	STBC_MCS3_16QA1BY2	
8:6	STBC_MCS2_QPSK3BY4	
5:3	STBC_MCS1_QPSK1BY2	
2:0	STBC_MCS0_BPSK1BY2	

0x0A00F0FC WCSS_A_RACTL_LMAP_OPSCALE2_STBC**Type:** RW**Reset State:** 0x0052

This registers provides the rate-qam-size dependent output scaling to LMAP stbc distance outputs

WCSS_A_RACTL_LMAP_OPSCALE2_STBC

Bits	Name	Description
14:12	STBC_MCS9_256QA5BY6	
11:9	STBC_MCS8_256QA3BY4	
8:6	STBC_MCS7_64QA5BY6	
5:3	STBC_MCS6_64QA3BY4	
2:0	STBC_MCS5_64QA2BY3	

0x0A00F100 WCSS_A_RACTL_LMAP_OPSCALE3**Type:** RW**Reset State:** 0x0109

This registers provides the rate-qam-size dependent output scaling to LMAP distance outputs

WCSS_A_RACTL_LMAP_OPSCALE3

Bits	Name	Description
8:6	LGCY_BPSK3BY4	
5:3	STBC_MCS10_256QA7BY8	
2:0	SISO_MCS10_256QA7BY8	

0x0A00F104 WCSS_A_RACTL_LMAP_TRK_CTRL**Type:** RW**Reset State:** 0x0005

This registers provides the channel tracking controls

WCSS_A_RACTL_LMAP_TRK_CTRL

Bits	Name	Description
2:1	LMAP_TRACK_GAIN	Programmable tracking loop gain. Default setting = 2'h2. 0x0: LOOPGAIN_8 (Loop gain = 1/8) 0x1: LOOPGAIN_16 (Loop gain = 1/16) 0x2: LOOPGAIN_32 (Loop gain = 1/32) 0x3: INVALID (Undefined)
0	LMAP_TRACK_EN	Channel tracking enable. If de-asserted, the channel tracking loop will not modify channel estimate samples received from the channel estimator module. Default setting = 1'b1.

0x0A00F108 WCSS_A_RACTL_LMAP_DUPCOMB**Type:** RW**Reset State:** 0x0000

This register holds 2 bits - one (bit 1) is applicable till the header is demodulated and the other (bit 0) for the data symbols in general. The value in bit 0 of this bit register specifies whether duplicate tones need to be combined in the RXA before simo slicer function in the data portions of the symbol. Note that the contents of this register is modified by ractl's mode detection logic such that HT-duplicate packets should always do MRC over duplicate tones irrespective of this setting. Stated otherwise, this register only affects legacy duplicate mode packets.

WCSS_A_RACTL_LMAP_DUPCOMB

Bits	Name	Description
1	DUPTONE_COMB_EN_SF	Off by default. When high, duplicate tones combining will take place only for signal fields (legacy or htsig) duplicate symbols. Note that due to hardware processing latencies, the settings of this bit will affect the duplicate combining of the service-data symbol of the legacy duplicate packet.
0	DUPTONE_COMB_EN	Off by default. When high, duplicate tones combining will take place only for legacy duplicate symbols. Note that due to hardware processing latencies, the settings of this bit will <u>not</u> affect the duplicate combining of the service-data symbol of the legacy duplicate packet.

0x0A00F10C WCSS_A_RACTL_LMAP_SNR_SEL**Type:** RW**Reset State:** 0x0001**WCSS_A_RACTL_LMAP_SNR_SEL**

Bits	Name	Description
0	ENABLE_SNRCOMP	When asserted, selects the LMAP SNR computation function. This is used in CSI report as well as per-symbol snr metric

0x0A00F110 WCSS_A_RACTL_RACTL_SPL**Type:** RW**Reset State:** 0x0006

This register holds some configuration bits to handle ractl-agc signalling of txop signal (only for MM packets and when LSIG and HTSIG both pass) and ractl-rahdr signalling to steer transfer of data to pmi.

WCSS_A_RACTL_RACTL_SPL

Bits	Name	Description
15	RESET_TXOP	Off by default. When high, the txop signal from ractl will be cleared. This clearing mechanism is devised since the txop signal is not cleared with regular clearing mechanism in ractl like agc_start and agc_abort
2	NDP_PMI_XFER	ON by default. When high, the command-word and statistics bytes go to pmi. When low, nothing goes.
1	TXOP_SIG_NDP	On by default. When low, specifies that txop signal need not be toggled at all for ndp (zlf) packets
0	TXOP_SIG_SUPPHTSIG	Off by default. When high, specifies that txop calculation to be done for supported htsig Note that txop calculation is always done on unsupported htsig.

0x0A00F114 WCSS_A_RACTL_WATCHDOG1**Type:** RW**Reset State:** 0x0000

Watchdog timer max values in header detection states Watchdog timer maxvalues in microsec

WCSS_A_RACTL_WATCHDOG1

Bits	Name	Description
15:12	WD_HTSIG	Watchdog timer maxvalue in HTSIG state; Minimum value 5. Value of 0 disables this timer.
11:8	WD_SF3	Watchdog timer maxvalue in SF3 state; Minimum value 5. Value of 0 disables this timer.
7:4	WD_SF2	Watchdog timer maxvalue in SF2 state; Minimum value 5. Value of 0 disables this timer.
3:0	WD_SF1	Watchdog timer maxvalue in SF1 state; Minimum value 5. Value of 0 disables this timer.

0x0A00F118 WCSS_A_RACTL_WATCHDOG2**Type:** RW**Reset State:** 0x0000

Watchdog timer max values in extra short and long training, servdata and recovery states
Watchdog timer maxvalues in microsec

WCSS_A_RACTL_WATCHDOG2

Bits	Name	Description
15:12	WD_WF	Watchdog timer maxvalue in WAITFINISH state; Minimum value 11. Value of 0 disables this timer.
11:8	WD_SERV	Watchdog timer maxvalue in SERV state; Minimum value 5. Value of 0 disables this timer.
7:4	WD_EXTRALTF	Watchdog timer maxvalue in extra LTF states; Minimum value 5. Value of 0 disables this timer.
3:0	WD_STS	Watchdog timer maxvalue in STS state; Minimum value 7. Value of 0 disables this timer.

0x0A00F11C WCSS_A_RACTL_WATCHDOG3**Type:** RW**Reset State:** 0x0000

Watchdog timer max values in data state and eopcount state Watchdog timer maxvalues in microsec

WCSS_A_RACTL_WATCHDOG3

Bits	Name	Description
15	WDEXP_ABORT_EN	Enable for generating internal ractl abort based on expiry of watchdog timers listed in the registers WATCHDOG1, WATCHDOG2 and WATCHDOG3
12:0	WD_DAT	Watchdog timer maxvalue in DATA and EOPCNT state. Value of 0 disables this timer.

0x0A00F120 WCSS_A_RACTL_RACTL_UNSUPP_BWMISMCOUNT**Type:** R**Reset State:** 0x00000000

RACTL counter to count the number of times rxa encountered packet at a higher bandwidth compared to that specified by agc. Note that counter is in hardware and the slave is only a readonly.

WCSS_A_RACTL_RACTL_UNSUPP_BWMISMCOUNT

Bits	Name	Description
14:10	BW20FOUND80MISM	The dynamic bandwidth setting is 20Mhz but 80M packet detected
9:5	BW40FOUND80MISM	The dynamic bandwidth setting is 40Mhz but 80M packet detected

WCSS_A_RACTL_RACTL_UNSUPP_BWMISM COUNT (cont.)

Bits	Name	Description
4:0	BW20FOUND40MISM	The dynamic bandwidth setting is 20Mhz but 40M packet detected

0x0A00F124 WCSS_A_RACTL_RACTL_LAST_BW**Type:** R**Reset State:** 0x00000000**WCSS_A_RACTL_RACTL_LAST_BW**

Bits	Name	Description
15:8	DYN	Most recent rightmost {prev4_dyn, prev3_dyn, prev2_dyn, prev1_dyn}
7:0	STATIC	Most recent rightmost {prev4_static, prev3_static, prev2_static, prev1_static}

0x0A00F130 WCSS_A_RACTL_RACTL_MINSNRSTATS**Type:** R**Reset State:** 0x00000000**WCSS_A_RACTL_RACTL_MINSNRSTATS**

Bits	Name	Description
7:0	STATS	Min-SNR provided by LMAP.

0x0A00F140 WCSS_A_RACTL_SIGNAL_INVALID_COUNT_HTSIG**Type:** W**Reset State:** 0x00000000

Number of packets received with invalid HT SIGNAL field values

WCSS_A_RACTL_SIGNAL_INVALID_COUNT_HTSIG

Bits	Name	Description
15:0	INVALID_SFCOUNT	Number of packets received with invalid SIGNAL field values

0x0A00F144 WCSS_A_RACTL_RXFIR_SCALING_COEFF**Type:** RW**Reset State:** 0x0001**WCSS_A_RACTL_RXFIR_SCALING_COEFF**

Bits	Name	Description
0	DC_NOTCH_EN	When high, selects rxfir scaling coefficients (to be used by FFT) appropriate with rxfir operation WITH dc notch. Note to use this register bit with appropriate AGC bit settings

0x0A00F148 WCSS_A_RACTL_CSI_CODEBOOK_SNR_TH**Type:** RW**Reset State:** 0x0000**WCSS_A_RACTL_CSI_CODEBOOK_SNR_TH**

Bits	Name	Description
7:0	VALUE	Threshold used to compare against the lmap min SNR for adaptive codebook determination. Program $\text{snr_th} = (\text{snr_db} - 22) * 4$, where snr_db is the signed SNR threshold in dB. snr_db range is -10 dB to 53.75 dB, corresponding to snr_th range is -128 to 127. Use with CSI_CONFIG reg below and $\text{adaptive_codebook_en} = 1$

0x0A00F14C WCSS_A_RACTL_CSI_CONFIG**Type:** RW**Reset State:** 0x61C3**WCSS_A_RACTL_CSI_CONFIG**

Bits	Name	Description
15	CSI_ABORT_CLR	When set to 1, CSI abort counter is cleared
14	CSI_XFER_EN	When set to 1, CSI-transfer is enabled. This influences CHE and PHYCSXI functionality for CSI transfer
13	CSI_WATCHDOG_EN	When set to 1, the watchdog timer in PHYCSXI is on and expires after 10 usec (max possible CSI xfer time) Note - has power impact. Switch this off if power-saving is desired.
12	NOTSOUNDINGPPDU_GEN ERATE_CSI	When set to 1, generate CSI report for nonsounding valid PPDU (MCS < 8 length > 0, not_sounding is 1)

WCSS_A_RACTL_CSI_CONFIG (cont.)

Bits	Name	Description
9	ADAPTIVE_CODEBOOK_EN	When set to 1, the SNR being used as a part of feedback report is used against a SNR threshold (see above reg CSI_CODEBOOK_SNR_TH) If threshold crossed, use codebook value 0, otherwise use codebook value 1. Note that when adaptive_codebook selection is used, the supported codebook values for 11n are 1 and 3 only. By default the codebook values from this register bits above are used.
8	CODEBOOK_11AC	SU: 0 => 2 (psi) and 4 (phi) bits, 1 => 4 (psi) and 6 (phi) bits MU: 0 => 5 (psi) and 7 (phi) bits, 1 => 7 (psi) and 9 (phi) bits
7:6	CODEBOOK_11N	0 => 1 (psi) and 3 (phi) bits, 1 => 2 (psi) and 4 (phi) bits, 2 => 3 (psi) and 5 (phi) bits, 3 => 4 (psi) and 6 (phi) bits
5:4	NG_11AC	0x0: NG_52 (Ng=1 => No grouping (52, 108, 234 sc) -- 30, 58, 122 sc for dsnr) 0x1: NG_30 (Ng=2 => 30, 58, 122 sc -- 16, 30, 62 for dsnr) 0x2: NG_16 (Ng=4 => 16, 30, 62 sc -- 10, 16, 32 for dsnr) 0x3: NG_RESVD
3:2	NG_11N	0x0: NG_56 (Ng=1 => No grouping (56 sc)) 0x1: NG_30 (Ng=2 => 30 sc) 0x2: NG_16 (Ng=4 => 16 sc) 0x3: NG_RESVD
1:0	NB_11N	0x0: NB_4 (Nb=4) 0x1: NB_5 (Nb=5) 0x2: NB_6 (Nb=6) 0x3: NB_8 (Nb=8)

0x0A00F150 WCSS_A_RACTL_BF_GAIN_DENOM**Type:** RW**Reset State:** 0x00D3**WCSS_A_RACTL_BF_GAIN_DENOM**

Bits	Name	Description
8:0	VALUE	The numerator comes from CHE. The denominator will be subtracted from numerator to provide bf_gain value The beamforming gain denominator should be programmed $(10 \cdot \log_{10}(2^{22}) - \text{tdc.target_dagc_bo}) \cdot 4$. Units: 0.25 dB.

0x0A00F154 WCSS_A_RACTL_CSI_STATS1**Type:** W**Reset State:** 0x00000000

Number of CSI transfers initiated by RACTL in various cases

WCSS_A_RACTL_CSI_STATS1

Bits	Name	Description
11:0	MACREQ_NDP_SOUNDING	Counts the number of MAC-requested CSI transfers for NDP sounding packets. These packets will have MCS between 8 and 31 and non-STBC packets. The HTSIG is 'unsupported' type [Case2 - (mcs_ge8_n_lt32 & length_0 & not_sounding_0 & stbc_0 & ndp_sounding_cfg_modf)] It also covers the 11ac ndp packet counters too.

0x0A00F158 WCSS_A_RACTL_CSI_STATS2

Type: R

Reset State: 0x00000000

Number of CSI aborts initiated by RACTL in various cases

WCSS_A_RACTL_CSI_STATS2

Bits	Name	Description
7:0	ABORTCNT	

0x0A00F15C WCSS_A_RACTL_CHE_SMOOTHING

Type: RW

Reset State: 0x003D

WCSS_A_RACTL_CHE_SMOOTHING

Bits	Name	Description
15	DSBL_PPDU	When dsbl is high, disables smoothing of channel-estimate for packets with payload.
14	DSBL_NDP	When dsbl is high, disables smoothing of channel-estimate for packets without payload (i.e. for 11n and 11ac ndp packets)
7:0	NOISE_OFFSET	When dsbl (above) is low, the rms error between smoothed and unsmoothed channel estimates is compared with the difference of noise stimate and agc rssi and decision made to use smoothing or not. Stated otherwise - smoothing is applied on channel estimate when the $N < TH$, where N is the noise estimate and TH is this register expressed in 0.25 dB steps.

0x0A00F160 WCSS_A_RACTL_PKTCNT_SMOOTHED**Type:** RW**Reset State:** 0x0000

Number of packets with smoothing done

WCSS_A_RACTL_PKTCNT_SMOOTHED

Bits	Name	Type	Description
15	CLR_CNT	RW	Clears the above counters
14:8	SMOOTHED	R	Packets where smoothed channel-estimated generated
6:0	SMOOTH_CANDIDATE	R	Packets candidate for smoothing

0x0A00F164 WCSS_A_RACTL_RACTL_CRCCHKCFG**Type:** RW**Reset State:** 0x0004**WCSS_A_RACTL_RACTL_CRCCHKCFG**

Bits	Name	Description
2:1	CRCCHK_EN_VHTSIGB	Do CRC-field check in VHTSIGB over VHTSIGB only (==2'd0) Do CRC-field check in VHTSIGB over VHTSIGA and VHTSIGB (==2'd1) Do CRC-field check in VHTSIGB over LSIG and VHTSIGA and VHTSIGB (==2'd2 or 2'd3)
0	CRCCHK_EN_VHTSIGA	Do CRC-field check of VHTSIGA over VHTSIGA only (==0) or check over LSIG and VHTSIGA (==1)

0x0A00F16C WCSS_A_RACTL_FORCE_DBGPKTTYPE**Type:** RW**Reset State:** 0x0000

Forces the packet-type of received packet based on forced Ibpsk or Qbpskchecks in RACTL main FSM Note that the following packet-types will be inferred from the combination of symtype checks: {sym0_Ibpskforce, sym1_Ibpskforce, sym2_Ibpskforce} implies Legacy 6 mbps {sym0_Ibpskforce, sym1_Ibpskforce, sym2_Qbpskforce} implies 11AC {sym0_Ibpskforce, sym1_Qbpskforce, Not applicable } implies 11n_MM {sym0_Qbpskforce, Not applicable , Not applicable } implies 11n_GF

WCSS_A_RACTL_FORCE_DBGPKTTYPE

Bits	Name	Description
5	SYM2_QBPSKFORCE	
4	SYM2_IBPSKFORCE	
3	SYM1_QBPSKFORCE	
2	SYM1_IBPSKFORCE	
1	SYM0_QBPSKFORCE	
0	SYM0_IBPSKFORCE	

0x0A00F170 WCSS_A_RACTL_RACTL_MU**Type:** R**Reset State:** 0x00000000**WCSS_A_RACTL_RACTL_MU**

Bits	Name	Description
8:7	SELF_NUMSTS	Registers the number of streams for the current client in the MU scenario
6:5	SELF_STS_START	Registers the starting stream index for the current client in the MU scenario
4	MU_FLAG	Determined flag to indicate that packet was not meant for the current client
3	PKTNOTFORME	Decoded mu-flag
2:0	DECODED_GID_ENTRY	From the gid field of vhtsiga and the programmed LUT

0x0A00F174 WCSS_A_RACTL_VIT_LSIG**Type:** RW**Reset State:** 0x00E6

This registers provides threshold against which Lsig Vit metric to be compared to assess Lsig reliability

WCSS_A_RACTL_VIT_LSIG

Bits	Name	Description
15	DSBL_RELIABILITY_CHK	
7:0	THRESHOLD	Threshold==0 will always indicate an error. Increase threshold to reduce LSIG error detection.

0x0A00F180 WCSS_A_RACTL_MCSFBSNRTH1**Type:** RW**Reset State:** 0x00C8

The estimated SNR obtained from combination of LMAP min-SNR [and optionally beamforming gain] are compared against this threshold to determine the MCS value that is suitable for reception by this client receiver. The thresholds provided by Allert are in are in dBs. And they have an offset of 22 dBs. The values here are (Allert's values -22) * 4. Multiplication by 4 is to get 0.25dB resolution in comparison. Also note that the values are actually signed 2's complement numbers represented as unsigned numbers, i.e. a value of 255 is -1.

WCSS_A_RACTL_MCSFBSNRTH1

Bits	Name	Description
7:0	VALUE	

0x0A00F184 WCSS_A_RACTL_MCSFBSNRTH2**Type:** RW**Reset State:** 0x00DC

The estimated SNR obtained from combination of LMAP min-SNR [and optionally beamforming gain] are compared against this threshold to determine the MCS value that is suitable for reception by this client receiver. The thresholds provided by Allert are in are in dBs. And they have an offset of 22 dBs. The values here are (Allert's values -22) * 4. Multiplication by 4 is to get 0.25dB resolution in comparison. Also note that the values are actually signed 2's complement numbers represented as unsigned numbers, i.e. a value of 255 is -1.

WCSS_A_RACTL_MCSFBSNRTH2

Bits	Name	Description
7:0	VALUE	

0x0A00F188 WCSS_A_RACTL_MCSFBSNRTH3**Type:** RW**Reset State:** 0x00E0

The estimated SNR obtained from combination of LMAP min-SNR [and optionally beamforming gain] are compared against this threshold to determine the MCS value that is suitable for reception by this client receiver. The thresholds provided by Allert are in are in dBs. And they have an offset of 22 dBs. The values here are (Allert's values -22) * 4. Multiplication by 4 is to get 0.25dB resolution in comparison. Also note that the values are actually signed 2's complement numbers represented as unsigned numbers, i.e. a value of 255 is -1.

WCSS_A_RACTL_MCSFBSNRTH3

Bits	Name	Description
7:0	VALUE	

0x0A00F18C WCSS_A_RACTL_MCSFBSNRTH4**Type:** RW**Reset State:** 0x00F4

The estimated SNR obtained from combination of LMAP min-SNR [and optionally beamforming gain] are compared against this threshold to determine the MCS value that is suitable for reception by this client receiver. The thresholds provided by Allert are in are in dBs. And they have an offset of 22 dBs. The values here are (Allert's values -22) * 4. Multiplication by 4 is to get 0.25dB resolution in comparison. Also note that the values are actually signed 2's complement numbers represented as unsigned numbers, i.e. a value of 255 is -1.

WCSS_A_RACTL_MCSFBSNRTH4

Bits	Name	Description
7:0	VALUE	

0x0A00F190 WCSS_A_RACTL_MCSFBSNRTH5**Type:** RW**Reset State:** 0x00FF

The estimated SNR obtained from combination of LMAP min-SNR [and optionally beamforming gain] are compared against this threshold to determine the MCS value that is suitable for reception by this client receiver. The thresholds provided by Allert are in are in dBs. And they have an offset of 22 dBs. The values here are (Allert's values -22) * 4. Multiplication by 4 is to get 0.25dB resolution in comparison. Also note that the values are actually signed 2's complement numbers represented as unsigned numbers, i.e. a value of 255 is -1.

WCSS_A_RACTL_MCSFBSNRTH5

Bits	Name	Description
7:0	VALUE	

0x0A00F194 WCSS_A_RACTL_MCSFBSNRTH6**Type:** RW**Reset State:** 0x000A

The estimated SNR obtained from combination of LMAP min-SNR [and optionally beamforming gain] are compared against this threshold to determine the MCS value that is suitable for reception by this client receiver. The thresholds provided by Allert are in are in dBs. And they have an offset of 22 dBs. The values here are (Allert's values -22) * 4. Multiplication by 4 is to get 0.25dB resolution in comparison. Also note that the values are actually signed 2's complement numbers represented as unsigned numbers, i.e. a value of 255 is -1.

WCSS_A_RACTL_MCSFBSNRTH6

Bits	Name	Description
7:0	VALUE	

0x0A00F198 WCSS_A_RACTL_MCSFBSNRTH7

Type: RW

Reset State: 0x0013

The estimated SNR obtained from combination of LMAP min-SNR [and optionally beamforming gain] are compared against this threshold to determine the MCS value that is suitable for reception by this client receiver. The thresholds provided by Allert are in are in dBs. And they have an offset of 22 dBs. The values here are (Allert's values -22) * 4. Multiplication by 4 is to get 0.25dB resolution in comparison. Also note that the values are actually signed 2's complement numbers represented as unsigned numbers, i.e. a value of 255 is -1.

WCSS_A_RACTL_MCSFBSNRTH7

Bits	Name	Description
7:0	VALUE	

0x0A00F19C WCSS_A_RACTL_MCSFBSNRTH8

Type: RW

Reset State: 0x001B

The estimated SNR obtained from combination of LMAP min-SNR [and optionally beamforming gain] are compared against this threshold to determine the MCS value that is suitable for reception by this client receiver. The thresholds provided by Allert are in are in dBs. And they have an offset of 22 dBs. The values here are (Allert's values -22) * 4. Multiplication by 4 is to get 0.25dB resolution in comparison. Also note that the values are actually signed 2's complement numbers represented as unsigned numbers, i.e. a value of 255 is -1.

WCSS_A_RACTL_MCSFBSNRTH8

Bits	Name	Description
7:0	VALUE	

0x0A00F1A0 WCSS_A_RACTL_MCSFBSNRTH9**Type:** RW**Reset State:** 0x0022

The estimated SNR obtained from combination of LMAP min-SNR [and optionally beamforming gain] are compared against this threshold to determine the MCS value that is suitable for reception by this client receiver. The thresholds provided by Allert are in dBs. And they have an offset of 22 dBs. The values here are (Allert's values -22) * 4. Multiplication by 4 is to get 0.25dB resolution in comparison. Also note that the values are actually signed 2's complement numbers represented as unsigned numbers, i.e. a value of 255 is -1.

WCSS_A_RACTL_MCSFBSNRTH9

Bits	Name	Description
7:0	VALUE	

0x0A00F1A4 WCSS_A_RACTL_MSA_REG0**Type:** RW**Reset State:** 0x0000

Membership-status Array (MSA) lookup

WCSS_A_RACTL_MSA_REG0

Bits	Name	Description
15	MSA_ENTRY_15	MU membership status.
14	MSA_ENTRY_14	
13	MSA_ENTRY_13	
12	MSA_ENTRY_12	
11	MSA_ENTRY_11	
10	MSA_ENTRY_10	
9	MSA_ENTRY_9	
8	MSA_ENTRY_8	
7	MSA_ENTRY_7	
6	MSA_ENTRY_6	
5	MSA_ENTRY_5	
4	MSA_ENTRY_4	
3	MSA_ENTRY_3	

WCSS_A_RACTL_MSA_REG0 (cont.)

Bits	Name	Description
2	MSA_ENTRY_2	
1	MSA_ENTRY_1	
0	MSA_ENTRY_0	

0x0A00F1A8 WCSS_A_RACTL_MSA_REG1**Type:** RW**Reset State:** 0x0000

Membership-status Array (MSA) lookup

WCSS_A_RACTL_MSA_REG1

Bits	Name	Description
15	MSA_ENTRY_31	MU membership status.
14	MSA_ENTRY_30	
13	MSA_ENTRY_29	
12	MSA_ENTRY_28	
11	MSA_ENTRY_27	
10	MSA_ENTRY_26	
9	MSA_ENTRY_25	
8	MSA_ENTRY_24	
7	MSA_ENTRY_23	
6	MSA_ENTRY_22	
5	MSA_ENTRY_21	
4	MSA_ENTRY_20	
3	MSA_ENTRY_19	
2	MSA_ENTRY_18	
1	MSA_ENTRY_17	
0	MSA_ENTRY_16	

0x0A00F1AC WCSS_A_RACTL_MSA_REG2**Type:** RW**Reset State:** 0x0000

Membership-status Array (MSA) lookup

WCSS_A_RACTL_MSA_REG2

Bits	Name	Description
15	MSA_ENTRY_47	MU membership status.
14	MSA_ENTRY_46	
13	MSA_ENTRY_45	
12	MSA_ENTRY_44	
11	MSA_ENTRY_43	
10	MSA_ENTRY_42	
9	MSA_ENTRY_41	
8	MSA_ENTRY_40	
7	MSA_ENTRY_39	
6	MSA_ENTRY_38	
5	MSA_ENTRY_37	
4	MSA_ENTRY_36	
3	MSA_ENTRY_35	
2	MSA_ENTRY_34	
1	MSA_ENTRY_33	
0	MSA_ENTRY_32	

0x0A00F1B0 WCSS_A_RACTL_MSA_REG3**Type:** RW**Reset State:** 0x0000

Membership-status Array (MSA) lookup

WCSS_A_RACTL_MSA_REG3

Bits	Name	Description
15	MSA_ENTRY_63	MU membership status.

WCSS_A_RACTL_MSA_REG3 (cont.)

Bits	Name	Description
14	MSA_ENTRY_62	
13	MSA_ENTRY_61	
12	MSA_ENTRY_60	
11	MSA_ENTRY_59	
10	MSA_ENTRY_58	
9	MSA_ENTRY_57	
8	MSA_ENTRY_56	
7	MSA_ENTRY_55	
6	MSA_ENTRY_54	
5	MSA_ENTRY_53	
4	MSA_ENTRY_52	
3	MSA_ENTRY_51	
2	MSA_ENTRY_50	
1	MSA_ENTRY_49	
0	MSA_ENTRY_48	

0x0A00F1B4 WCSS_A_RACTL_UPA_REG0**Type:** RW**Reset State:** 0x0000

User-Position Array (UPA) lookup

WCSS_A_RACTL_UPA_REG0

Bits	Name	Description
15:14	UPA_ENTRY_7	MU user position in the group.
13:12	UPA_ENTRY_6	
11:10	UPA_ENTRY_5	
9:8	UPA_ENTRY_4	
7:6	UPA_ENTRY_3	
5:4	UPA_ENTRY_2	
3:2	UPA_ENTRY_1	
1:0	UPA_ENTRY_0	

0x0A00F1B8 WCSS_A_RACTL_UPA_REG1**Type:** RW**Reset State:** 0x0000

User-Position Array (UPA) lookup

WCSS_A_RACTL_UPA_REG1

Bits	Name	Description
15:14	UPA_ENTRY_15	MU user position in the group.
13:12	UPA_ENTRY_14	
11:10	UPA_ENTRY_13	
9:8	UPA_ENTRY_12	
7:6	UPA_ENTRY_11	
5:4	UPA_ENTRY_10	
3:2	UPA_ENTRY_9	
1:0	UPA_ENTRY_8	

0x0A00F1BC WCSS_A_RACTL_UPA_REG2**Type:** RW**Reset State:** 0x0000

User-Position Array (UPA) lookup

WCSS_A_RACTL_UPA_REG2

Bits	Name	Description
15:14	UPA_ENTRY_23	MU user position in the group.
13:12	UPA_ENTRY_22	
11:10	UPA_ENTRY_21	
9:8	UPA_ENTRY_20	
7:6	UPA_ENTRY_19	
5:4	UPA_ENTRY_18	
3:2	UPA_ENTRY_17	
1:0	UPA_ENTRY_16	

0x0A00F1C0 WCSS_A_RACTL_UPA_REG3**Type:** RW**Reset State:** 0x0000

User-Position Array (UPA) lookup

WCSS_A_RACTL_UPA_REG3

Bits	Name	Description
15:14	UPA_ENTRY_31	MU user position in the group.
13:12	UPA_ENTRY_30	
11:10	UPA_ENTRY_29	
9:8	UPA_ENTRY_28	
7:6	UPA_ENTRY_27	
5:4	UPA_ENTRY_26	
3:2	UPA_ENTRY_25	
1:0	UPA_ENTRY_24	

0x0A00F1C4 WCSS_A_RACTL_UPA_REG4**Type:** RW**Reset State:** 0x0000

User-Position Array (UPA) lookup

WCSS_A_RACTL_UPA_REG4

Bits	Name	Description
15:14	UPA_ENTRY_39	MU user position in the group.
13:12	UPA_ENTRY_38	
11:10	UPA_ENTRY_37	
9:8	UPA_ENTRY_36	
7:6	UPA_ENTRY_35	
5:4	UPA_ENTRY_34	
3:2	UPA_ENTRY_33	
1:0	UPA_ENTRY_32	

0x0A00F1C8 WCSS_A_RACTL_UPA_REG5**Type:** RW**Reset State:** 0x0000

User-Position Array (UPA) lookup

WCSS_A_RACTL_UPA_REG5

Bits	Name	Description
15:14	UPA_ENTRY_47	MU user position in the group.
13:12	UPA_ENTRY_46	
11:10	UPA_ENTRY_45	
9:8	UPA_ENTRY_44	
7:6	UPA_ENTRY_43	
5:4	UPA_ENTRY_42	
3:2	UPA_ENTRY_41	
1:0	UPA_ENTRY_40	

0x0A00F1CC WCSS_A_RACTL_UPA_REG6**Type:** RW**Reset State:** 0x0000

User-Position Array (UPA) lookup

WCSS_A_RACTL_UPA_REG6

Bits	Name	Description
15:14	UPA_ENTRY_55	MU user position in the group.
13:12	UPA_ENTRY_54	
11:10	UPA_ENTRY_53	
9:8	UPA_ENTRY_52	
7:6	UPA_ENTRY_51	
5:4	UPA_ENTRY_50	
3:2	UPA_ENTRY_49	
1:0	UPA_ENTRY_48	

0x0A00F1D0 WCSS_A_RACTL_UPA_REG7**Type:** RW**Reset State:** 0x0000

User-Position Array (UPA) lookup

WCSS_A_RACTL_UPA_REG7

Bits	Name	Description
15:14	UPA_ENTRY_63	MU user position in the group.
13:12	UPA_ENTRY_62	
11:10	UPA_ENTRY_61	
9:8	UPA_ENTRY_60	
7:6	UPA_ENTRY_59	
5:4	UPA_ENTRY_58	
3:2	UPA_ENTRY_57	
1:0	UPA_ENTRY_56	

10.48 WCSS_A_RBAPB_RBAPB**0x0A014400 WCSS_A_RBAPB_CONTROL_RXB_DSCR****Type:** RW**Reset State:** 0x00000000

Controls RXB Descrambler.

WCSS_A_RBAPB_CONTROL_RXB_DSCR

Bits	Name	Description
1	SFD_DETECT	Disable descrambled short SFD detector. 0x0: ENABLE (enable descrambled short SFD detector) 0x1: DISABLE (disable descrambled short SFD detector)
0	DESCRAMBLER	Disable RXB Descrambler. 0x0: ENABLE (enable descrambler) 0x1: DISABLE (disable descrambler)

0x0A014404 WCSS_A_RBAPB_COUNT_SSFD**Type:** W**Reset State:** 0x00000000

Counts the number of short Start-of-Frame delimiters detected.

WCSS_A_RBAPB_COUNT_SSFD

Bits	Name	Description
15:0	COUNT	Number of short Start-of-Frame delimiters detected

0x0A014408 WCSS_A_RBAPB_COUNT_LSFD**Type:** W**Reset State:** 0x00000000

Counts the number of long Start-of-Frame delimiters detected.

WCSS_A_RBAPB_COUNT_LSFD

Bits	Name	Description
15:0	COUNT	Number of long Start-of-Frame delimiters detected

0x0A01440C WCSS_A_RBAPB_IGN_ILL_PLCP**Type:** RW**Reset State:** 0x00FB

This register is used to enable checking of the following fields of the PLCP header: CRC16, Rate field and Service Field. These correctness check are all described in standard IEEE802.11b. The CRC16 should check, the service field should have its reserved bits set to zero and the rate field should contain one of the supported rates (in multiples of 100 kbit/s): 8'd10, 8'd20, 8'd55 or 8'd110.

WCSS_A_RBAPB_IGN_ILL_PLCP

Bits	Name	Description
10	IGN_SIG	Used to ignore check for valid signal fields and pass packets with signal field errors Valid rate fields contents are 10, 20, 55 and 110 (decimal), denoting 1,2, 5.5 and 11 Mbit/s. 0x0: ENABLE (Enable signal field check) 0x1: IGNORE (Ignore signal field check)

WCSS_A_RBAPB_IGN_ILL_PLCP (cont.)

Bits	Name	Description
9	IGN_SVC	Used to ignore check for valid service fields and pass packets with signal field errors. A valid service field has all bits equal to zero for 1, 2 and 5.5 Mbit/s mode. The most significant bit may be non-zero in 11 Mbit/s mode. 0x0: ENABLE (Enable service field check) 0x1: IGNORE (Ignore service field check)
8	IGN_CRC	Used to ignore CRC check and pass packets with PLCP CRC errors. 0x0: ENABLE (Enable CRC16 check) 0x1: IGNORE (Ignore CRC16 check)
7:0	SVC_MASK	Service Field Mask. The field is applied using a logical 'and' to the service field before the illegal service field check is applied. By default the 'coupled clocks' bit is masked out, in order not to reject packets transmitted by products with uncoupled clocks or products which couple clocks only for higher bit rates.

0x0A014410 WCSS_A_RBAPB_COUNT_PLCP_CRC_ERRORS**Type:** W**Reset State:** 0x00000000

Counter number of CRC errors.

WCSS_A_RBAPB_COUNT_PLCP_CRC_ERRORS

Bits	Name	Description
15:0	COUNT	CRC error counter.

0x0A014414 WCSS_A_RBAPB_COUNT_PLCP_SIG_ERRORS**Type:** W**Reset State:** 0x00000000

Counter number of signal field errors.

WCSS_A_RBAPB_COUNT_PLCP_SIG_ERRORS

Bits	Name	Description
15:0	COUNT	Illegal signal field counter.

0x0A014418 WCSS_A_RBAPB_COUNT_PLCP_SVC_ERRORS**Type:** W**Reset State:** 0x00000000

Counter number of service field errors.

WCSS_A_RBAPB_COUNT_PLCP_SVC_ERRORS

Bits	Name	Description
15:0	COUNT	Illegals service field counter.

0x0A01441C WCSS_A_RBAPB_COUNT_RX_BYTES**Type:** W**Reset State:** 0x00000000

Counts number of payload bytes received by the RXB receiver (output by the header processor).

WCSS_A_RBAPB_COUNT_RX_BYTES

Bits	Name	Description
15:0	COUNT	Byte counter.

0x0A014420 WCSS_A_RBAPB_COUNT_MPI_SOF**Type:** W**Reset State:** 0x00000000

Counts number of Start-Of_Frame signals generated by header processor and sent to MPI.

WCSS_A_RBAPB_COUNT_MPI_SOF

Bits	Name	Description
15:0	COUNT	rbhdr_mpi_sof event counter.

0x0A014424 WCSS_A_RBAPB_COUNT_MPI_EOF**Type:** W**Reset State:** 0x00000000

Counts number of End-Of_Frame signals generated by header processor and sent to MPI.

WCSS_A_RBAPB_COUNT_MPI_EOF

Bits	Name	Description
15:0	COUNT	rbhdr_mpi_eof event counter.

0x0A014428 WCSS_A_RBAPB_SFD_DET_TMO**Type:** RW**Reset State:** 0x007D

SFD Detetction threshold is specified in microseconds. Power-on reset value is 125 decimal. Includes demodulation and descrambling delay. This register is used to lead a timer that aborts the SFD detection process. It should be as set as low as possible to reduce the receiver dead time to a minimum.

WCSS_A_RBAPB_SFD_DET_TMO

Bits	Name	Description
7:0	TIME	SFD detection timeout. 0x7D: POR (power-on reset value)

0x0A01442C WCSS_A_RBAPB_SLR_SFD_DET_TMO**Type:** RW**Reset State:** 0x01F4

SFD Detetction threshold is specified in microseconds. Power-on reset value is 500 decimal. Includes demodulation and descrambling delay. This register is used to set a timer that aborts the SFD detection process for SLR packets. It should be as set as low as possible to reduce the receiver dead time to a minimum.

WCSS_A_RBAPB_SLR_SFD_DET_TMO

Bits	Name	Description
8:0	TIME	SLR SFD detection timeout. 0x1F4: POR (power-on reset value)

0x0A014430 WCSS_A_RBAPB_COUNT_RXB_START**Type:** W**Reset State:** 0x00000000

Counts number of RXB_START events received by the RXB controler.

WCSS_A_RBAPB_COUNT_RXB_START

Bits	Name	Description
15:0	COUNT	RXB_START Counter

0x0A014438 WCSS_A_RBAPB_COUNT_RXB_ABORT**Type:** W**Reset State:** 0x00000000

Counts number of RXB_ABORT events received by the RXB controller.

WCSS_A_RBAPB_COUNT_RXB_ABORT

Bits	Name	Description
15:0	COUNT	RXB_ABORT event counter.

0x0A01443C WCSS_A_RBAPB_RXB_RESET**Type:** W**Reset State:** 0x00000000

Resets RXB controller state machine.

WCSS_A_RBAPB_RXB_RESET

Bits	Name	Description
0	RST	Reset RXB.

0x0A014440 WCSS_A_RBAPB_CONTROL_BARKER_TIMING_TRACKING**Type:** RW**Reset State:** 0x0006

Control Barker timing tracking.

WCSS_A_RBAPB_CONTROL_BARKER_TIMING_TRACKING

Bits	Name	Description
2	FEEDBACK	Enable Barker timing tracking. 0x0: DISABLE (do not use timing tracking feedback) 0x1: ENABLE (use timing tracking feedback)

WCSS_A_RBAPB_CONTROL_BARKER_TIMING_TRACKING (cont.)

Bits	Name	Description
1:0	THRESHOLD	Timing error threshold.

0x0A014448 WCSS_A_RBAPB_CMF_PLL_COEFF_A**Type:** RW**Reset State:** 0x02DE

CMF PLL Loop Coefficient A

WCSS_A_RBAPB_CMF_PLL_COEFF_A

Bits	Name	Description
9:0	COEFF_A	default value -290

0x0A01444C WCSS_A_RBAPB_CMF_PLL_COEFF_B**Type:** RW**Reset State:** 0x0127

CMF PLL Loop Coefficient B

WCSS_A_RBAPB_CMF_PLL_COEFF_B

Bits	Name	Description
9:0	COEFF_B	default value 295

0x0A014450 WCSS_A_RBAPB_CMF_PLL_COEFF_A_SLR_TRACK**Type:** RW**Reset State:** 0x0378

CMF PLL Loop Coefficient A used for SLR packets after SFD has been detected

WCSS_A_RBAPB_CMF_PLL_COEFF_A_SLR_TRACK

Bits	Name	Description
9:0	COEFF_A_TRACK	default value -136

0x0A014454 WCSS_A_RBAPB_CMF_PLL_COEFF_B_SLR_TRACK**Type:** RW**Reset State:** 0x008A

CMF PLL Loop Coefficient B used for SLR packets after SFD has been detected

WCSS_A_RBAPB_CMF_PLL_COEFF_B_SLR_TRACK

Bits	Name	Description
9:0	COEFF_B_TRACK	default value 138

0x0A014458 WCSS_A_RBAPB_ANTENNA_DIVERSITY_MODE**Type:** RW**Reset State:** 0x0000

This register defines the behaviour of the antenna diversity logic.

WCSS_A_RBAPB_ANTENNA_DIVERSITY_MODE

Bits	Name	Description
1:0	VALUE	Antenna diversity selector mode field 0x0: DISABLE (no diversity switching) 0x1: ALWAYS (always switch to diversity mode) 0x2: SNR_SWITCH (switch to diversity mode if SNR exceeds threshold) 0x3: RESV (reserved)

0x0A01445C WCSS_A_RBAPB_SNR_THRESHOLD**Type:** RW**Reset State:** 0x0019

This threshold is used to decide whether the SNR of the current packet is sufficient to support diversity

WCSS_A_RBAPB_SNR_THRESHOLD

Bits	Name	Description
5:0	VALUE	SNR threshold value

0x0A014460 WCSS_A_RBAPB_SNR_AVERAGER_GAIN**Type:** RW**Reset State:** 0x0028

Barker and CCK demod SNR averager loop gain

WCSS_A_RBAPB_SNR_AVERAGER_GAIN

Bits	Name	Description
5:4	RBCKK_HANDOFF	CCK demod SNR averager loop gain - default value = 2^6
3:2	RBBRK_HANDOFF	Barker demod SNR averager loop gain after antenna diversity selection - default value = 2^6
1:0	RBBRK_ANTSEL	Barker demod SNR averager loop gain during antenna diversity selection - default value = 2^4

0x0A014464 WCSS_A_RBAPB_LMS_AVERAGER_SHIFT**Type:** RW**Reset State:** 0x0014

Channel equaliser averager loop gain

WCSS_A_RBAPB_LMS_AVERAGER_SHIFT

Bits	Name	Description
5:3	SLR	LMS averager loop gain - range $2^{(0-7)}$, default value = 2^2
2:0	NON_SLR	LMS averager loop gain - range $2^{(0-7)}$, default value = 2^4

0x0A014468 WCSS_A_RBAPB_COUNT_DYNSEL_ANT0**Type:** W**Reset State:** 0x00000000

Counts the number of times antenna0 is selected when dynamic antenna selection is enabled.

WCSS_A_RBAPB_COUNT_DYNSEL_ANT0

Bits	Name	Description
15:0	COUNT	antenna0 event counter.

0x0A01446C WCSS_A_RBAPB_COUNT_DYNSEL_ANT1**Type:** W**Reset State:** 0x00000000

Counts the number of times antenna1 is selected when dynamic antenna selection is enabled.

WCSS_A_RBAPB_COUNT_DYNSEL_ANT1

Bits	Name	Description
15:0	COUNT	antenna1 event counter.

0x0A014470 WCSS_A_RBAPB_COUNT_DYNSEL_2ANT**Type:** W**Reset State:** 0x00000000

Counts the number of times both antennae are selected when dynamic antenna selection is enabled.

WCSS_A_RBAPB_COUNT_DYNSEL_2ANT

Bits	Name	Description
15:0	COUNT	both antennae event counter.

10.49 WCSS_A_RFIF_RFIF**0x0A015000 WCSS_A_RFIF_RF_BUS_CONTROL****Type:** RW**Reset State:** 0x0209

This set of registers are used to adjust timing parameters used, and to issue special commands, over the RF bus interface.

WCSS_A_RFIF_RF_BUS_CONTROL

Bits	Name	Description
11:8	IRIS_ASYNC_CAP_DLY	Iris begins driving asynchronous signals to Riva at the completion of an RF command. This field is used to delay the sampling of this data after a command has completed. The default value represents a delay of approx. 25 ns ((2+1) x 120M clocks - spec requirement) when bit 3 of this register is 0. The delay changes to be in units of 60M clocks after completion of a register read command when bit 3 in this register is 1.

WCSS_A_RFIF_RF_BUS_CONTROL (cont.)

Bits	Name	Description
7	ENABLE_80M_IRIS_CLK_OVERRIDE	This bit is used to override the control of the 80M clock used for RF calibration mode. When disabled, the clock enable for the 80M Iris clock is automatically disabled when we observe a calibration done signal from Iris (software must still enable the clock via bit 6 below). If there are any issues with this automatic clock disabling feature, this override bit can be enabled to re-activate manual control of disabling the 80M clock via bit 6 below. 0x0: OVERRIDE_DISABLED (RF 80M calibration clock override is disabled) 0x1: OVERRIDE_ENABLED (RF 80M calibration clock override is enabled)
6	ENABLE_80M_IRIS_CLK	This bit is used to enable/disable the 80M clock used for RF calibration mode. During RF calibration mode, an 80 MHz clock must be fed out on the cmd_clk signal, and it is this bit that is used to enable/disable that clock. Bit 2 in this register is used to select this clock as the source to Iris. Bit 2 must be set first, followed by this bit. And after RF calibration, this bit must be disabled first, followed by bit 2. 0x0: DISABLED (RF 80M calibration clock is disabled) 0x1: ENABLED (RF 80M calibration clock is enabled)
5:4	SET_RF_BUS_RD_DELAY	Controls the delay value used by the RF bus interface state machine for register reads. By default (2'b00), the RF bus interface state machine inserts 3 cmd_clk cycles before it samples the read data (on the 4th cmd_clk cycle). Each value set in this register increases the delay inserted by the RF bus interface state machine before it samples the read data, by 1/2 cmd_clk cycle - i.e. 2'b01 causes the insertion of 3.5 cmd_clk cycles, and 2'b10 causes the insertion of 4 cmd_clk cycles. 0x0: DEFAULT (Default 3 cmd_clk RF bus interface read delay) 0x1: HALF_CYCLE (Extra half cmd_clk cycle RF bus interface read delay) 0x2: ONE_CYCLE (Extra one cmd_clk cycle RF bus interface read delay) 0x3: INVALID (Invalid setting - defaults to 3 cmd_clk cycle read delay mode)
3	SET_RF_BUS_FREQ	Controls the frequency setting of the RF bus interface cmd_clk signal. For Riva, two RF bus frequency values are supported - 60 MHz (default) and 30 MHz. When set to 1, this bit sets the RF bus frequency to be 30 MHz. When 0, the default mode of 60 MHz is selected. 0x0: V_60M_MODE (RF bus interface frequency set to 60 MHz) 0x1: V_30M_MODE (RF bus interface frequency set to 30 MHz)

WCSS_A_RFIF_RF_BUS_CONTROL (cont.)

Bits	Name	Description
2	ENABLE_RF_CALIBRATION	This bit is used to enable/disable the RF calibration mode. During RF calibration mode, an 80 MHz clock must be fed out on the cmd_clk signal, and it is this bit that is used to enable the selection of the 80 MHz clock onto cmd_clk. The entire RF calibration mode is under software control - software needs to set this bit when the 80 MHz clock is required, and disable this bit when calibration is complete. 0x0: DISABLED (RF calibration mode disabled) 0x1: ENABLED (RF calibration mode enabled)
1	RESERVED	reserved
0	ENABLE_RF_BUS	This bit enables/disables access to the RF bus. When this bit is 0 (disabled), the RF bus pins are not driven by internal RFIF logic, but may be used to drive debug data over the RF bus. The ENABLE_DEBUG_MODE bit must be set for the debug mode to be in effect. To avoid any pin conflict, debug data is only driven on the RF bus pins, when both ENABLE_DEBUG_MODE is set and ENABLE_RF_BUS is disabled. 0x0: DISABLED (RF bus interface access disabled) 0x1: ENABLED (RF bus interface access enabled)

0x0A015004 WCSS_A_RFIF_RF_BUS_STATUS**Type:** R**Reset State:** 0x0000

These registers capture the status of the RF bus interface state machine and controlling logic.

WCSS_A_RFIF_RF_BUS_STATUS

Bits	Name	Description
4	CALIB_DONE	Holds the current state of the Iris calibration mode done signal. When calibration mode is selected, this signal will reset to 0. On completion of calibration, this signal will capture the done signal returned from Iris.
3:0	RF_BUS_SM_STATE	Current state of the RF bus interface state machine.

0x0A015008 WCSS_A_RFIF_ARB_CONTROL**Type:** RW**Reset State:** 0x000F

This set of registers are used by the arbitration module to enable/disable data path access to the RF bus interface.

WCSS_A_RFIF_ARB_CONTROL

Bits	Name	Description
3	ENABLE_TXGAIN_COMMA NDS	Enables/disables the Tx gain commands coming from the TXFIR. 0x0: DISABLED (TX Gain commands issued to RF bus interface are disabled) 0x1: ENABLED (TX Gain commands issued to RF bus interface are enabled)
2	ENABLE_RXGAIN_COMMA NDS	Enables/disables the Rx gain commands coming from the AGC. 0x0: DISABLED (RX Gain commands issued to RF bus interface are disabled) 0x1: ENABLED (RX Gain commands issued to RF bus interface are enabled)
1	ENABLE_PHYHW_COMMA NDS	This bit enables/disables accesses by PHY H/W over the RF bus interface. 0x0: DISABLED (PHY H/W sourced accesses to/from RF bus interface are disabled) 0x1: ENABLED (PHY H/W sourced accesses to/from RF bus interface are enabled)
0	ENABLE_RFAPB_COMMAN DS	This bit enables/disables accesses by the RFAPB interface over the RF bus interface. 0x0: DISABLED (RFAPB sourced accesses to/from RF bus interface are disabled) 0x1: ENABLED (RFAPB sourced accesses to/from RF bus interface are enabled)

0x0A01500C WCSS_A_RFIF_ARB_COMMAND**Type:** RW**Reset State:** 0x000F

This set of registers are used by the arbitration module to issue RF commands requested by software over the RF interface bus.

WCSS_A_RFIF_ARB_COMMAND

Bits	Name	Description
6:4	RF_CMD_DATA_WRD1	This field holds the first data word of the software requested RF command. Currently the only commands with data supported are the RF clock enable command, and the RF reset command, of which both require only a single data word. The encoding definitions provided are for the clock enable command only. The RF reset command data word should be set to all zero. 0x0: DISABLE_19P2 (Disables Iris clock and selects 19.2M input from PMIC) 0x1: ENABLE_19P2 (Enables Iris clock and selects 19.2M input from PMIC) 0x2: DISABLE_38P4 (Disables Iris clock and selects 38.4M input from PMIC) 0x3: ENABLE_38P4 (Enables Iris clock and selects 38.4M input from PMIC) 0x4: RESERVED_WRD4 (Reserved) 0x5: RESERVED_WRD5 (Reserved) 0x6: DISABLE_48 (Disables Iris clock and selects 48M XO input) 0x7: ENABLE_48 (Enables Iris clock and selects 48M XO input)
3:1	RF_CMD	This field holds the RF_CMD to be issued. Default command is set to RF_RESET command. 0x0: CLOCK_ENABLE_CMD (Iris clock enable command. Used to enable/disable clocks and set frequency) 0x1: MODE_CMD (Iris mode command. Used to control TX and RX enable.) 0x2: RESERVED_CMD2 (Command not supported.) 0x3: RESERVED_CMD3 (Command not supported.) 0x4: RESERVED_CMD4 (Command not supported.) 0x5: RESERVED_CMD5 (Command not supported.) 0x6: RESERVED_CMD6 (Command not supported.) 0x7: RESET_CMD (Iris reset command. Resets Iris.)
0	ISSUE_RF_CMD	This bit is used to initiate an RF command over the RF bus interface. Writing a 1 to this bit requests the arbiter to issue the command loaded in the RF_CMD field. Note that only RF commands with 1 data word are currently supported by this means. When issuing a command (by writing a 1 to this bit), the value in this bit will clear to zero to indicate that the current command is in progress. When the command has completed, the value of this bit will return to 1. Reading this bit returns the status described in the encoding section. 0x0: CMD_IN_PROGRESS (RF command has been requested and the command is in progress.) 0x1: IDLE (RF command is complete, or waiting for new RF command.)

0x0A015010 WCSS_A_RFIF_ARB_STATUS**Type:** R**Reset State:** 0x0000

These registers hold the current status of the arbitration module, and are used for debug and status checking.

WCSS_A_RFIF_ARB_STATUS

Bits	Name	Description
15:12	RXGAIN_ABORTED_CMDS	Holds the number of RXGAIN commands aborted (by request).
11:8	RFAPB_DROPPED_CMDS	Holds the number of commands dropped because RFAPB accesses are disabled.
7:4	BLOCKED_CMDS	Holds the total number of RF commands blocked because the RF Bus has not been enabled.
3:0	ARB_SM_STATE	Captures the current state of the Arbitration state machine module. Used for debugging purposes.

0x0A015014 WCSS_A_RFIF_BT_CTRL_CFG**Type:** RW**Reset State:** 0x0003

This register configures the behavior of the BT input signals with regards to the PHY.

WCSS_A_RFIF_BT_CTRL_CFG

Bits	Name	Description
2	BT_RX_SHARING_EN	Enables simultaneous RX sharing of antenna with the BT device.
1	BT_MEDIUM_SHUTDOWN	Controls whether or not the bt_medium input is used to shutdown the PHY. 1 means it will, 0 means it won't.
0	BT_TX_SHUTDOWN	Controls whether or not the bt_tx input is used to shutdown the PHY. 1 means it will, 0 means it won't.

0x0A015018 WCSS_A_RFIF_TX_EN_CFG**Type:** RW**Reset State:** 0x0000

This register controls the behaviour of the TX_EN signal.

WCSS_A_RFIF_TX_EN_CFG

Bits	Name	Description
1	TX_OVERRIDE_VAL	If the Tx enable override is enabled, then this is the overriding value for the Tx.
0	OVERRIDE_EN	Setting this bit to 1 enables the override of the Tx enable.

0x0A01501C WCSS_A_RFIF_RESET_CTRL**Type:** RW**Reset State:** 0x0000

Reset control register.

WCSS_A_RFIF_RESET_CTRL

Bits	Name	Description
4	STATS_RESET	Setting this to 1 resets all of the current debug statistics counters.
3	PHYHW_SM_RESET	Setting this to 1 resets all of the PHY H/W interface state machines.
2	RF_BUS_IF_SM_RESET	Setting this to 1 resets the RF bus interface state machine. This should only be used if hang has somehow occurred in this state machine.
1	ARB_SM_RESET	Setting this to 1 resets the arbitration state machine. This should only be used if hang has somehow occurred in this state machine.
0	RFAPB_SM_RESET	Setting this to 1 resets the RFAPB state machine. This should only be used if hang has somehow occurred in this state machine.

0x0A015020 WCSS_A_RFIF_PHY_HW_STATUS**Type:** R**Reset State:** 0x0000

These registers hold the current status of the PHY HW module, and are used for debug and status checking.

WCSS_A_RFIF_PHY_HW_STATUS

Bits	Name	Description
15:10	PHYHW_DROPPED_CMDS	Holds the number of commands dropped because PHY H/W and/or TX & RX gain commands are disabled.

WCSS_A_RFIF_PHY_HW_STATUS (cont.)

Bits	Name	Description
9:8	MODE_SM_STATE	Captures the current state of the MODE state machine module. Used for debugging purposes.
7:5	REG_RDWR_SM_STATE	Captures the current state of the Register read/write state machine module. Used for debugging purposes.
4:3	TXGAIN_SM_STATE	Captures the current state of the TXGAIN state machine module. Used for debugging purposes.
2:0	RXGAIN_SM_STATE	Captures the current state of the RXGAIN state machine module. Used for debugging purposes.

0x0A015024 WCSS_A_RFIF_RFAPB_STATUS**Type:** R**Reset State:** 0x0000

Captures current status of the RFAPB module.

WCSS_A_RFIF_RFAPB_STATUS

Bits	Name	Description
2:0	RFAPB_SM_STATE	Captures the current state of the RFAPB state machine module. Used for debugging purposes.

0x0A015028 WCSS_A_RFIF_ERROR_CONDS**Type:** RW**Reset State:** 0x0000

Captures error conditions if they occur.

WCSS_A_RFIF_ERROR_CONDS

Bits	Name	Description
0	WR_FIFO_RD_EMPTY_ER R	This condition indicates that the write data fifo (data to be written to IRIS) became empty. This should never occur. A write of 1 will clear this condition.

0x0A01502C WCSS_A_RFIF_HKADC_RD_ADDR**Type:** RW**Reset State:** 0x0535

Specifies the Iris read address for HKADC operations.

WCSS_A_RFIF_HKADC_RD_ADDR

Bits	Name	Description
10:0	RD_ADDR	This field contains the read address for HKADC operations.

0x0A015030 WCSS_A_RFIF_HKADC_WR_ADDR

Type: RW

Reset State: 0x0534

Specifies the Iris write address for HKADC operations.

WCSS_A_RFIF_HKADC_WR_ADDR

Bits	Name	Description
10:0	WR_ADDR	This field contains the write address for HKADC operations.

0x0A015034 WCSS_A_RFIF_HKADC_CONV_TIMER

Type: RW

Reset State: 0x007F

Specifies the time in units of 12.5ns (80MHz) that the RFIF will wait for the HKADC conversion to occur, before issuing a read to read the data from Iris.

WCSS_A_RFIF_HKADC_CONV_TIMER

Bits	Name	Description
6:0	WAIT_CNT	Total time to wait for the HKADC conversion to complete (in units of 12.5ns).

0x0A015400+ WCSS_A_RFIF_RX_GAINn_0, n=[0..47] 0x10*n

Type: RW

Reset State: 0x0000

Read write interface to the RX Gain look up Table

WCSS_A_RFIF_RX_GAINn_0

Bits	Name	Description
15:0	RXGAIN_VALUE_0	

0x0A015404+ WCSS_A_RFIF_RX_GAINn_1, n=[0..47]
0x10*n
Type: RW**Reset State:** 0x0000

Read write interface to the RX Gain look up Table

WCSS_A_RFIF_RX_GAINn_1

Bits	Name	Description
15:0	RXGAIN_VALUE_1	

0x0A015408+ WCSS_A_RFIF_RX_GAINn_2, n=[0..47]
0x10*n
Type: RW**Reset State:** 0x0000

Read write interface to the RX Gain look up Table

WCSS_A_RFIF_RX_GAINn_2

Bits	Name	Description
6:0	RXGAIN_VALUE_2	

0x0A015800+ WCSS_A_RFIF_TX_GAINn_0, n=[0..31]
0x20*n
Type: RW**Reset State:** 0x0000

Read write interface to the TX Gain look up Table

WCSS_A_RFIF_TX_GAINn_0

Bits	Name	Description
15:0	TXGAIN_VALUE_0	

**0x0A015804+ WCSS_A_RFIF_TX_GAINn_1, n=[0..31]
0x20*n****Type:** RW**Reset State:** 0x0000

Read write interface to the TX Gain look up Table

WCSS_A_RFIF_TX_GAINn_1

Bits	Name	Description
15:0	TXGAIN_VALUE_1	

**0x0A015808+ WCSS_A_RFIF_TX_GAINn_2, n=[0..31]
0x20*n****Type:** RW**Reset State:** 0x0000

Read write interface to the TX Gain look up Table

WCSS_A_RFIF_TX_GAINn_2

Bits	Name	Description
15:0	TXGAIN_VALUE_2	

**0x0A01580C+ WCSS_A_RFIF_TX_GAINn_3, n=[0..31]
0x20*n****Type:** RW**Reset State:** 0x0000

Read write interface to the TX Gain look up Table

WCSS_A_RFIF_TX_GAINn_3

Bits	Name	Description
15:0	TXGAIN_VALUE_3	

**0x0A015810+ WCSS_A_RFIF_TX_GAINn_4, n=[0..31]
0x20*n****Type:** RW**Reset State:** 0x0000

Read write interface to the TX Gain look up Table

WCSS_A_RFIF_TX_GAINn_4

Bits	Name	Description
15:0	TXGAIN_VALUE_4	

0x0A015814+ WCSS_A_RFIF_TX_GAINn_5, n=[0..31]
0x20*n

Type: RW

Reset State: 0x0000

Read write interface to the TX Gain look up Table

WCSS_A_RFIF_TX_GAINn_5

Bits	Name	Description
10:0	TXGAIN_VALUE_5	

10.50 WCSS_A_RPE_RPE

0x0A082400 WCSS_A_RPE_MEM_BASE_ADDR

Type: RW

Reset State: 0x00000000

It gives the base address of the memory where all bitmaps and related flags are stored per station and per queue basis

WCSS_A_RPE_MEM_BASE_ADDR

Bits	Name	Description
31:0	CFG_BASEADDR	Base address for the RPE data descriptor memory

0x0A082404 WCSS_A_RPE_FULLSTATE_STATIONS_QUEUES

Type: RW

Reset State: 0x00080006

This register indicates the max number of stations we are associated with and the max number of Queues per station

WCSS_A_RPE_FULLSTATE_STATIONS_QUEUES

Bits	Name	Description
19:16	CFG_FULLSTATE_QUEUES	Max number of Queues per station default value 8 indicates 8 queues qids 0-7
4:0	CFG_FULLSTATE_STATIONS	The number here indicates the number of stations default value 6 indicates station ids 0- 5

0x0A082408 WCSS_A_RPE_SM_STUCK_CNT_VALUE**Type:** RW**Reset State:** 0x00000600

The register has the time out value used by RPE when any statemachine hangs

WCSS_A_RPE_SM_STUCK_CNT_VALUE

Bits	Name	Description
11:0	SM_STUCK_CNT_VALUE	This value indicates the timeout value in terms of number of clock cycles for pronto it is 40Mhz

0x0A08240C WCSS_A_RPE_RPE_CONFIG**Type:** RW**Reset State:** 0x00000080

gives some configurations for RPE

WCSS_A_RPE_RPE_CONFIG

Bits	Name	Description
27	RPE_CLEAR_COUNTERS	writing 1 will clear all counters, HW will make it zero
19:16	RESERVED	reserved
14:8	RSRC_ENTRY_DIS	This field controls the number of resorces used for fragmentation. one resource is always used. (for only 2 resources are used only bit 0 is useful here) These seven bits can be used to turn of any resource table entry for fragment gathering. Each bit indicates the resource table entry number. bit 8 is for resource table 1, resource table 0 is always on. Resource entries should be on/off sequentially only, not randomly. for example 7'b0001111 is okay 7'b0101011 is not okay

WCSS_A_RPE_RPE_CONFIG (cont.)

Bits	Name	Description
7	OUOFORD_PKT_FWD	Whenever a window moves due to one of internal flushing events (internal events described in bitfield 1) then the BA bitmap does not change but reorder bitmap changes if any packets that arrive in reorder for the holes then those packets need to be forwarded when this bit is set else they are dropped 1: Drop out of order packets arriving in reorder 0: Forward out of order packets packets arriving in reorder
5	NONFRAG_FLUSH_DIS	1'b1: When set resource entries for frag pkts are not flushed when non-frag packets are received for a staid/qid which has valid entry 1'b0: When non-frag pkts are received for a staid/qid which has valid resource entry then that resource is flushed
4	HOST_BAR_DROP_DIS	1'b1: when set then the BAR packets which have same sequence number as next expected packet will be dropped by RPE and not host 1'b0: The BAR packet which has same sequence number as next expected packet will be sent to host with opcode DROP_CUR and host will drop the packet
3	BDPDU_NOTAVAIL_DROP_DIS	This is mostly relevant fragmented packets since we create frag BD. set to 1'b1: the packets will not be dropped when bmu indicates no bdpdu is avail, then we will wait for it to beavailable and will not drop the packet. 1'b0: default we will drop the packets when bdpdu is notavailable
2	REORDER_DIS	when set the h/w buffer reordering support is disabled(Feature that was in Libra also)
0	LRU_OR_RR	This register should always be set to Zero Initiall it was reserved for LRU which was not implemented. No one should touch this register. If it is set to 1-> RPE will not work

0x0A082410 WCSS_A_RPE_ROUTING_FLAG**Type:** RW**Reset State:** 0x00FF0003

This register will allow you to program routing table for RPE

WCSS_A_RPE_ROUTING_FLAG

Bits	Name	Description
23:16	ROUTING_DROP_PACKET	All the BAD (error) packets will be pushed to this WQ default value gives the DROP wq == decimal 255 -- duplicate det packets during bit map updated -- duplicate or flushing fragments -- out of order etc -- ...
7:0	ROUTING_GOOD_PACKET	All good packets will be forwarded to WQ programmed in this feild Defalut WQ is DPU

0x0A082414 WCSS_A_RPE_SW_FLUSH_BITMAP_CACHE**Type:** RW**Reset State:** 0x00000000

This register allows s/w to flush the caches for fullstate or partialstate bitmaps

WCSS_A_RPE_SW_FLUSH_BITMAP_CACHE

Bits	Name	Description
4	CFG_FLUSH_PARTIALSTATE_CACHE	Writing to this will flush the bitmap maintained in the cache set to 1'b1: fullstate bitmap in the cache is written to memory after the flush H/W will invalidate the cache
0	CFG_FLUSH_FULLSTATE_CACHE	Writing to this will flush the fullstate bitmap maintained in the cache set to 1'b1: fullstate bitmap in the cache is written to memory after the flush H/W will invalidate the cache

0x0A082418 WCSS_A_RPE_FULLSTATE_BITMAP_CACHE_DW1**Type:** R**Reset State:** 0x00000000

when read it gives the lower 32 bits of fullstate bitmap that is in the cache. The staid/qid of this cache is in register stationid_qid_4_fullstate_bitmap

WCSS_A_RPE_FULLSTATE_BITMAP_CACHE_DW1

Bits	Name	Description
31:0	BITMAP_CACHE_DW1	Indicates following fields ENTRY_VAL 0 RSVD 6:1 REORDR_BUF 7 RETRY_CHK 8 FLUSH_FRAG 9 ORDR_CHK 10 FRAG_CHK 11 IEEE_2K_CHK 12 WIN_SIZE 18:13 SSN_VAL 19 SSN 31:20

0x0A08241C WCSS_A_RPE_FULLSTATE_BITMAP_CACHE_DW2**Type:** R**Reset State:** 0x00000000

when read it gives the middle 32 bits 63:32 of fullstate bitmap that is in the cache. The staid/qid of this cache is in register stationid_qid_4_fullstate_bitmap

WCSS_A_RPE_FULLSTATE_BITMAP_CACHE_DW2

Bits	Name	Description
31:0	BITMAP_CACHE_DW2	This is the actual bitmap which shows lower 32 bits 31:0

0x0A082420 WCSS_A_RPE_FULLSTATE_BITMAP_CACHE_DW3**Type:** R**Reset State:** 0x00000000

when read it gives the upper 32 bits 95:64 of fullstate bitmap that is in the cache. The staid/qid of this cache is in register stationid_qid_4_fullstate_bitmap

WCSS_A_RPE_FULLSTATE_BITMAP_CACHE_DW3

Bits	Name	Description
31:0	BITMAP_CACHE_DW3	This is bitmap which shows upper 32 bits

0x0A082424 WCSS_A_RPE_STATIONID_QID_4_FULLSTATE_BITMAP**Type:** R**Reset State:** 0x00000000

when read it gives you which station/qid the above bitmap belongs to

WCSS_A_RPE_STATIONID_QID_4_FULLSTATE_BITMAP

Bits	Name	Description
19:16	QID_4_FULLSTATE_BITMAP	Qid for the fullstatebitmap in cache
4:0	STATIONID_4_FULLSTATE_BITMAP	Station id for the fullstatebitmap in cache

0x0A082428 WCSS_A_RPE_PARTIALSTATE_BITMAP_CACHE_DW1**Type:** R**Reset State:** 0x00000000

when read it gives the lower 32 bits 31:0 of partialstate bitmap that is in the cache. The staid/qid of this cache is in register stationid_qid_4_partialstate_bitmap

WCSS_A_RPE_PARTIALSTATE_BITMAP_CACHE_DW1

Bits	Name	Description
31:0	BITMAP_CACHE_DW1	Indicates following fields ENTRY_VAL 0 RSVD 6:1 REORDR_BUF 7 RETRY_CHK 8 FLUSH_FRAG 9 ORDR_CHK 10 FRAG_CHK 11 IEEE_2K_CHK 12 WIN_SIZE 18:13 SSN_VAL 19 SSN 31:20

0x0A08242C WCSS_A_RPE_PARTIALSTATE_BITMAP_CACHE_DW2**Type:** R**Reset State:** 0x00000000

when read it gives the middle 32 bits 63:32 of partialstate bitmap that is in the cache. The staid/qid of this cache is in register stationid_qid_4_partialstate_bitmap

WCSS_A_RPE_PARTIALSTATE_BITMAP_CACHE_DW2

Bits	Name	Description
31:0	BITMAP_CACHE_DW2	This is the actual bitmap which shows lower 32 bits

0x0A082430 WCSS_A_RPE_PARTIALSTATE_BITMAP_CACHE_DW3**Type:** R**Reset State:** 0x00000000

when read it gives you upper 32 bits of partialstate bitmap

WCSS_A_RPE_PARTIALSTATE_BITMAP_CACHE_DW3

Bits	Name	Description
31:0	BITMAP_CACHE_DW3	This is bitmap which shows upper 32 bits

0x0A082434 WCSS_A_RPE_STATIONID_QID_4_PARTIALSTATE_BITMAP**Type:** R**Reset State:** 0x00000000

when read it gives you which station/qid the above bitmap belongs to

WCSS_A_RPE_STATIONID_QID_4_PARTIALSTATE_BITMAP

Bits	Name	Description
19:16	QID_4_PARTIALSTATE_BITMAP	Qid for the partialstatebitmap in cache
4:0	STATIONID_4_PARTIALSTATE_BITMAP	Station id for the partialstatebitmap in cache

0x0A082438 WCSS_A_RPE_FRAGMENT_RESRC_ENTRY1_DW1**Type:** R**Reset State:** 0x00000000

when read it gives you lower 32 bits of 1st resource of the 8 MPDUs maintained in RPE

WCSS_A_RPE_FRAGMENT_RESRC_ENTRY1_DW1

Bits	Name	Description
31:0	RESRC1_DW1	Indicates following fields ENTRY_VAL 0 STAID 8:1 QID 13:9 SSN 25:14 FRAG_NUM 29:26 LRU_ORDR 30 RSRC_ENA 31

0x0A08243C WCSS_A_RPE_FRAGMENT_RESRC_ENTRY1_DW2**Type:** R**Reset State:** 0x00000000

when read it gives you upper 32 bits of 1st resource of the 8 MPDUs maintained in RPE

WCSS_A_RPE_FRAGMENT_RESRC_ENTRY1_DW2

Bits	Name	Description
31:0	RESRC1_DW2	Indicates following fields BD_INDX 19:0 RSVD 31:20

0x0A082440 WCSS_A_RPE_FRAGMENT_RESRC_ENTRY2_DW1**Type:** R**Reset State:** 0x00000000

when read it gives you lower 32 bits of 2nd resource of the 8 MPDUs maintained in RPE

WCSS_A_RPE_FRAGMENT_RESRC_ENTRY2_DW1

Bits	Name	Description
31:0	RESRC2_DW1	Indicates following fields ENTRY_VAL 0 STAID 8:1 QID 13:9 SSN 25:14 FRAG_NUM 29:26 LRU_ORDR 30 RSRC_ENA 31

0x0A082444 WCSS_A_RPE_FRAGMENT_RESRC_ENTRY2_DW2**Type:** R**Reset State:** 0x00000000

when read it gives you upper 32 bits of 2nd resource of the 8 MPDUs maintained in RPE

WCSS_A_RPE_FRAGMENT_RESRC_ENTRY2_DW2

Bits	Name	Description
31:0	RESRC2_DW2	Indicates following fields BD_INDX 19:0 RSVD 31:20

0x0A082478 WCSS_A_RPE_ERR_INT_ENABLE**Type:** RW**Reset State:** 0x00000000

This register allows you to enable any error interrupt

WCSS_A_RPE_ERR_INT_ENABLE

Bits	Name	Description
8	B_2K_JUMP_SN_IN_BASES SION_INT_EN	When a 1 is written to this bit will enable interrupt whenever AMPDU packet is received whose SN has 2k jump
7	B_2K_JUMP_SSN_IN_BAR_ INT_EN	When a 1 is written to this bit will enable interrupt whenever BAR frame whose SSN has 2k jump
6	BAR_IN_NON_BASESSION _INT_EN	When a 1 is written to this bit will enable interrupt whenever BAR frames are received in BA session
5	FRAGPKT_IN_BASESSION _INT_EN	When a 1 is written to this bit will enable interrupt whenever fragmented packets are received in BA session
4	BU_SM_STUCK_ERR_INT_ EN	When a 1 is written to this bit will enable interrupt when bitmap update state machine is hung
3	DD_SM_STUCK_ERR_INT_ EN	When a 1 is written to this bit will enable interrupt when duplicate detect state machine is hung
2	BMU_BD_AVAIL_ERR_INT_ ENABLE	When a 1 is written to this bit will enable interrupt due to noavailable Bd/PDU in BMU
1	GBI_ERR_INT_EN	When a 1 is written to this bit will enable interrupt due to gbi error
0	GAM_ERR_INT_EN	When a 1 is written to this bit will enable interrupt due to gam error

0x0A08247C WCSS_A_RPE_ERR_INT_STATUS**Type:** RW**Reset State:** 0x00000000

This register status of error interrupt

WCSS_A_RPE_ERR_INT_STATUS

Bits	Name	Type	Description
31:28	REASON_CODE	R	Gives the reason for which the above 4 BA session related interrupts have occurred 4'b0000 : No error 4'b0001 : Fragmented packets received in BA session 4'b0010 : BAR frame received in NON BA session 4'b0100 : BAR frame received with a 2k jump in SSN 4'b1000 : AMPDU frame received with 2k jump in Seq Number
24:21	INT_QID	RW	gives QID of for which above interrupt occurred, if two interrupts happen one after another later one will overwrite the old one. Reason code will indicate which one of the interrupt was caused by this qid
20:16	INT_STAID	RW	gives STAID of for which above interrupt occurred, if two interrupts happen one after another later one will overwrite the old one. Reason code will indicate which interrupt was caused by this STAID
8	B_2K_JUMP_SN_IN_BASE SSION_INT_STATUS	RW	When this bit is set it indicates that a AMPDU packet is received whose SN has 2k jump Writing a 1 will clear this interrupt and the reason code
7	B_2K_JUMP_SSN_IN_BAR _INT_STATUS	RW	When this bit is set it indicates that a BAR frame is received whose SSN has 2k jump Writing a 1 will clear this interrupt and the reason code
6	BAR_IN_NON_BASESESSION _INT_STATUS	RW	When this bit is set it indicates that a BAR frame is received in a non BA session. Writing a 1 will clear this interrupt and the reason code
5	FRAGPKT_IN_BASESESSION _INT_STATUS	RW	When this bit is 1 it will indicate that fragmented packets are received in BA session. Writing a 1 will clear this interrupt and the reason code
4	BU_SM_STUCK_ERR_INT_ STATUS	RW	When this bit is 1 indicates bitmap update state machine is hung writing a 1 to this bit will clear the interrupt
3	DD_SM_STUCK_ERR_INT_ STATUS	RW	When this bit is 1 indicates duplicate detect state machine is hung writing a 1 to this bit will clear the interrupt
2	BMU_BD_AVAIL_ERR_INT_ STATUS	RW	When this bit is 1 indicates BMU has ran out of BDs writing a 1 to this bit will clear the interrupt
1	GBI_ERR_INT_STATUS	RW	When this bit is 1 indicates gbi error interrupt has occurred writing a 1 to this bit will clear the interrupt
0	GAM_ERR_INT_STATUS	RW	When this bit is 1 indicates gam error interrupt has occurred writing a 1 to this bit will clear the interrupt

0x0A082480 WCSS_A_RPE_CUR_STATE_BU_DD**Type:** R**Reset State:** 0x00010001

gives current state of state machine bitmap update and duplicate detect This register can be used to check the state if bit 5 or 6 in err_int_status is set

WCSS_A_RPE_CUR_STATE_BU_DD

Bits	Name	Description
28:16	CUR_STATE_BU	current status of bitmap update state machine
7:0	CUR_STATE_DD	current status of duplicate detect state machine

0x0A082484 WCSS_A_RPE_CUR_STATE_RA_RR

Type: R

Reset State: 0x00010001

gives current state of state machine reassembly and reorder This register can be used to check the state if bit 3 or 4 in err_int_status is set

WCSS_A_RPE_CUR_STATE_RA_RR

Bits	Name	Description
27:16	CUR_STATE_RA	current status of reassembly state machine
3:0	CUR_STATE_RR	current status of reorder state machine

0x0A082488 WCSS_A_RPE_CUR_STATE_DMA

Type: R

Reset State: 0x00000001

gives current state of state machine DMA control This register can be used to check the state if bit 7 in err_int_status is set

WCSS_A_RPE_CUR_STATE_DMA

Bits	Name	Description
15:0	CUR_STATE_DMA	current status of DMA control state machine

0x0A08248C WCSS_A_RPE_SEND_PKT_STATS

Type: RW

Reset State: 0x00000000

This registers gives count of all packets sent it rolls over after reaching max count

WCSS_A_RPE_SEND_PKT_STATS

Bits	Name	Description
31:0	SEND_PKT_STATS	gives stats for sent packets

0x0A082490 WCSS_A_RPE_DROP_PKT_STATS**Type:** RW**Reset State:** 0x00000000

This register gives count of all packets dropped by RPE it rolls over after reaching max count

WCSS_A_RPE_DROP_PKT_STATS

Bits	Name	Description
31:0	DROP_PKT_STATS	gives stats for drop packets

0x0A082494 WCSS_A_RPE_GET_BMU_FAIL_STATS**Type:** R**Reset State:** 0x00000000

This registers gives statistics for how many times get bmu failed it rolls over after reaching max count

WCSS_A_RPE_GET_BMU_FAIL_STATS

Bits	Name	Description
15:0	GET_BMU_FAIL_STATS	Gives stats for get bmu failures

0x0A082500 WCSS_A_RPE_RPE_BITMAP_DUPLICATE_CNTR**Type:** R**Reset State:** 0x00000000

Gives the number of bitmap updates and duplicate detect received from RXP

WCSS_A_RPE_RPE_BITMAP_DUPLICATE_CNTR

Bits	Name	Description
28:16	BITMAP_UPDATE_CNTR	Counter value for bitmap update and it rolls over after reaching max cnt
11:0	DUPLICATE_DETECT_CNTR	Counter value for duplicate detect and it rolls over after reaching max cnt

0x0A082504 WCSS_A_RPE_RPE_SW_FLUSH_RSRC_ENTRY**Type:** RW**Reset State:** 0x00000000

This register allows sw flush any fragment resources SW can program staid and qid for which if an resource entry for reassembly exists then it needs to be flushed

WCSS_A_RPE_RPE_SW_FLUSH_RSRC_ENTRY

Bits	Name	Type	Description
19:16	SW_FLUSH_RSRC_ENTRY_QUEUE	RW	This register gives queue id along with above station for which entry needs to be flushed
12:8	SW_FLUSH_RSRC_STAID	RW	This register indicates the station for which sw wants to flush any resource held for this staion
1	SW_FLUSH_RSRC_ENTRY_STATUS	R	This gives staus of flush if high that means it is still oongoing when it is zero it indicates we are ready for another flush
0	SW_FLUSH_RSRC_ENTRY	RW	writing 1 to this bit and then writing staid and qid in this same register will aloow H/w to flush any resource held for this staion and queue this bit is automatically cleared when flush is done

0x0A082508 WCSS_A_RPE_RPE_SW_BLOCK_PKTS**Type:** RW**Reset State:** 0x00000000

This register gives staid and qid for which all packets needs to be blocked. blocking means all packets to that staid/qid will be dropped If there are any packets in reorder buffer before block s/w needs to forward forward them using fwd dpu feature by sw

WCSS_A_RPE_RPE_SW_BLOCK_PKTS

Bits	Name	Description
19:16	SW_BLOCK_QID	This register gives queue id along with above station for which packets needs to be blocked
12:8	SW_BLOCK_STAID	This register indicates the station for which sw wants to block any packets coming in to this staion and the qid given below
0	SW_BLOCK_REQ	writing 1 to this bit and then writing staid and qid in this same register will make RPE drop any packets coming to this particular staid and qid

0x0A08250C WCSS_A_RPE_REORDER_BITMAP_CACHE_DW1**Type:** R**Reset State:** 0x00000000

when read it gives the lower 32 bits of reorder bitmap

WCSS_A_RPE_REORDER_BITMAP_CACHE_DW1

Bits	Name	Description
31:0	REORDER_CACHE_DW1	This is the actual reorder bitmap which shows lower 32 bits

0x0A082510 WCSS_A_RPE_REORDER_BITMAP_CACHE_DW2**Type:** R**Reset State:** 0x00000000

when read it gives the middle 32 bits of reorder bitmap

WCSS_A_RPE_REORDER_BITMAP_CACHE_DW2

Bits	Name	Description
31:0	REORDER_CACHE_DW2	This is the actual reorder bitmap which shows upper 32 bits

0x0A082514 WCSS_A_RPE_REORDER_BITMAP_CACHE_DW3**Type:** R**Reset State:** 0x00000000

when read it gives the upper 32 bits of reorder bitmap

WCSS_A_RPE_REORDER_BITMAP_CACHE_DW3

Bits	Name	Description
31:0	BITMAP_CACHE_DW3	[11:0] reorder SSN [12] reorder SSN val [18:13]reorder window size [31:19] reserved

10.51 WCSS_A_RXACLKCTRL_RXACLKCTRL**0x0A016000 WCSS_A_RXACLKCTRL_ROOT_CLK_EN****Type:** RW**Reset State:** 0x0000**WCSS_A_RXACLKCTRL_ROOT_CLK_EN**

Bits	Name	Description
2	CLK80_40D	
1	CLK160	
0	CLK80	

0x0A016004 WCSS_A_RXACLKCTRL_APB_BLOCK_CLK_EN**Type:** RW**Reset State:** 0xFFFF**WCSS_A_RXACLKCTRL_APB_BLOCK_CLK_EN**

Bits	Name	Description
15	PHYCSXI	
14	FSCALE	
13	RXA	
12	RAHDR	
11	RADSCR	

WCSS_A_RXCLKCTRL_APB_BLOCK_CLK_EN (cont.)

Bits	Name	Description
10	VIT	
9	DINT	
8	LMAP	
7	PTC	
6	RACTL_APB	
5	CHE	
4	INF	
3	FFT	
2	TDC	
1	BTCF	
0	RACTL	

0x0A016008 WCSS_A_RXCLKCTRL_APB_BLOCK_DYN_CLKG_DISABLE**Type:** RW**Reset State:** 0x0040**WCSS_A_RXCLKCTRL_APB_BLOCK_DYN_CLKG_DISABLE**

Bits	Name	Description
15	PHYCSXI	
14	FSCALE	
13	RXA	
12	RAHDR	
11	RADSCR	
10	VIT	
9	DINT	
8	LMAP	
7	PTC	
6	RACTL_APB	
5	CHE	
4	INF	
3	FFT	
2	TDC	

WCSS_A_RXCLKCTRL_APB_BLOCK_DYN_CLKG_DISABLE (cont.)

Bits	Name	Description
1	BTCF	
0	RACTL	

0x0A01600C WCSS_A_RXCLKCTRL_DYN_ANTSW_CLK_CTRL_DISABLE**Type:** RW**Reset State:** 0x0000**WCSS_A_RXCLKCTRL_DYN_ANTSW_CLK_CTRL_DISABLE**

Bits	Name	Description
0	ANTSW_CTRL	

0x0A016010 WCSS_A_RXCLKCTRL_MAIN1_FAST_CLK_CTRL**Type:** RW**Reset State:** 0x0152

Control for the auxiliary fast clock coming to Phy from external clock control logic.

WCSS_A_RXCLKCTRL_MAIN1_FAST_CLK_CTRL

Bits	Name	Description
12	DBG_FORCE_FAST_ALWAYS	This is a debug-only mode configuration. Should always be left at 0 for normal purposes. When high, forces the fast_clk to be the source of clock for ALL 11AC packet types.
11	COMPLIANT_BEHAV	This bit when set to 1 enforces phy-mac interface signalling (related to 256qam processing) to comply with other modes (11a, 11n and 11ac) at the cost of increased latency in 20M and 40M modes depending on tx_bw setting. The following applies to how the clkdivratio for 20M packets is decided in compliant_behav case: *** The design forces this value to 1 when tx_bw_config is 1 or more. *** The value is forced to 0 when tx_bw_config is 0. The following applies to how the clkdivratio for 40M packets is decided- *** The design forces the value to 1 if tx_bw_config is 1. Otherwise a value of 1 or 2 can be used. When this bit is set to 0, all 256QAM STBC 20M packets will get 160MHz fastclk; all 256QAM STBC 40M packets will get either 160 or 320 MHz clock depending on extnclkdivmode40 above.

WCSS_A_RXCLKCTRL_MAIN1_FAST_CLK_CTRL (cont.)

Bits	Name	Description
10:5	CLKCHGDLY	Specifies the delay (in number of 80MHz clocks) between assertion of fast_clk_en in ractl (in main2 domain) to when the processing of MCS 8 and 9 STBC data-symbols is going to start. This delay is required so as to allow multi-cycle operation of the enable. This delay should include CDC delays inside PHY as well as clock-generation delay. Tune this to the minimum required delay.
4:3	EXTNCLKDIVMODE40	This divmode determines the frequency of the clock being used by functional logic to demodulate 40MHz MCS8 and 9 STBC pkts. A value of 2 implies the clk used has same freq as that of the input fast clk. A value of 1 implies the clk used has one-half freq as that of the input fast clk. A value of 0 implies the clk used has one-quarter freq as that of the input fast clk. Compare this with the clock-division mechanism based on channel bandwidth configuration. Note that setting the value to < 2 will increase the latency of 40M packets to > 11 usec.
2:1	EXTNCLKDIVMODE20	This divmode determines the frequency of the clock being used by functional logic to demodulate 20MHz MCS8 and 9 STBC pkts. A value of 2 implies the clk used has same freq as that of the input fast clk. A value of 1 implies the clk used has one-half freq as that of the input fast clk. A value of 0 implies the clk used has one-quarter freq as that of the input fast clk. Compare this with the clock-division mechanism based on channel bandwidth configuration. Note that setting the value to 0 will increase the latency of 20M packets to > 12 usec so a value of 0 should never be used. A value of 2 is usable at the expense of power consumption penalty. *** NOT actually used ***
0	FORCE_CLK_DISABLE	When high, forcefully disables the clock to save power. When low, the functional logic decides the duration of the enable. Note that functional logic will ensure this clock is enabled only when STBC data symbols in 11AC MCS8 and MCS9 packets mode are being demodulated by RX.

10.52 WCSS_A_RXCLKCTRL_RXCLKCTRL**0x0A016400 WCSS_A_RXCLKCTRL_ROOT_CLK_EN****Type:** RW**Reset State:** 0x0000

WCSS_A_RXCLKCTRL_ROOT_CLK_EN

Bits	Name	Description
9	CLK3BY4_S	
8	CLK1BY4_S	
7	CLK160_D	
6	CLK80_S	
5	CLK40	
4	CLK80_40D	
3	CLK80_40S	
2	CLK160	
1	CLK_MIF_PHYDBG_INTF	
0	CLK80	0x0: DISABLE (root clock is turned off) 0x0: ENABLE (root clock is turned on)

0x0A016404 WCSS_A_RXCLKCTRL_APB_BLOCK_CLK_EN**Type:** RW**Reset State:** 0x1531**WCSS_A_RXCLKCTRL_APB_BLOCK_CLK_EN**

Bits	Name	Description
14	AGC_RTT	
13	RXFIR_DWNS	
12	AGC_APB	
11	AGC	
10	XBAR	
9	RFIF	
8	PHYDBG_APB	
7	PHYDBG	
6	CAL_EST	
5	CAL	
4	RXFIR	
3	PMI	
2	PHYINT	

WCSS_A_RXCLKCTRL_APB_BLOCK_CLK_EN (cont.)

Bits	Name	Description
1	TMUX	
0	RXB	0x0: DISABLE (per block static gate is turned off) 0x1: ENABLE (per block static gate is turned on)

0x0A016408 WCSS_A_RXCLKCTRL_APB_BLOCK_DYN_CLKG_DISABLE**Type:** RW**Reset State:** 0x0010**WCSS_A_RXCLKCTRL_APB_BLOCK_DYN_CLKG_DISABLE**

Bits	Name	Description
8	PMIRDET	
6	PMICCA	
5	XBAR	
4	CAL_APB	
3	CAL	
2	RXFIR	
1	PMI	
0	RXB	0x0: ENABLE (per block dynamic gate is turned off and allow only dynamic control of clock gate through state machines) 0x1: DISABLE (per block dynamic clock gating is disabled)

0x0A01640C WCSS_A_RXCLKCTRL_APB_BLOCK_DYN_CLKG_DISABLE_AGC**Type:** RW**Reset State:** 0x0008

Since AGC has a bunch of sub blocks they need to have their own dynamic clock gate disable

WCSS_A_RXCLKCTRL_APB_BLOCK_DYN_CLKG_DISABLE_AGC

Bits	Name	Description
8	CCADET	
7	DET11AC80	
6	DET11N40	

WCSS_A_RXCLKCTRL_APB_BLOCK_DYN_CLKG_DISABLE_AGC (cont.)

Bits	Name	Description
5	DET11A	
4	DET11B	
3	APB	
2	RDET	
1	PWR	
0	MAIN	

0x0A016410 WCSS_A_RXCLKCTRL_APB_SYNC_RESET**Type:** W**Reset State:** 0x0000**WCSS_A_RXCLKCTRL_APB_SYNC_RESET**

Bits	Name	Description
0	VALUE	

0x0A016414 WCSS_A_RXCLKCTRL_CLK_PHY_MAIN2_3BY4_DIV_BYPASS**Type:** RW**Reset State:** 0x0000**WCSS_A_RXCLKCTRL_CLK_PHY_MAIN2_3BY4_DIV_BYPASS**

Bits	Name	Description
0	BYP	

0x0A016418 WCSS_A_RXCLKCTRL_BW_CONFIG**Type:** R**Reset State:** 0x00000000**WCSS_A_RXCLKCTRL_BW_CONFIG**

Bits	Name	Description
3:2	TX_VAL	
1:0	RX_VAL	

0x0A016424 WCSS_A_RXCLKCTRL_FAST_CH_SW_PAGE_SEL**Type:** RW**Reset State:** 0x0000**WCSS_A_RXCLKCTRL_FAST_CH_SW_PAGE_SEL**

Bits	Name	Description
0	PAGE_SEL	Page select 0 selects lower half of the memories. 1 selects upper half. The affected memories are RX I/Q corr, RX Gain LUT, TX Gain LUT, TX I/Q corr, DPD and TPC.

0x0A016428 WCSS_A_RXCLKCTRL_ADC_FIFO_SYNC_RST**Type:** RW**Reset State:** 0x0000**WCSS_A_RXCLKCTRL_ADC_FIFO_SYNC_RST**

Bits	Name	Description
0	RST	This has been mainly added to reset the FIFO to a known state after the clocks on wr and rd side have been running. Mainly for external ADC-clock modes. This must be resetted manually before setting again.

10.53 WCSS_A_RXP_RXP**0x0A080800 WCSS_A_RXP_CONFIG****Type:** RW**Reset State:** 0x09000100

It contains RXP Configuration registers. These configurations (with the exception of soft abort and mask_subsq_fcs_err_within_ampdu) can be changed at any time by SW but HW will only take effect during the gap of packets so that SW will not corrupt packets during configuration.

WCSS_A_RXP_CONFIG

Bits	Name	Description
31:24	PHY_CMD_MIN_SIZE	RXP requires atleast minimum of 9 cmd bytes for processing packet if it is not 9 then packets will be dropped, but it can be configured to whatever value

WCSS_A_RXP_CONFIG (cont.)

Bits	Name	Description
23:19	RESERVED	reserved for future use
18	CFG_BDPDU_CLEAN_DIS	When gam error occurs it is catastrophic so we will reset the system But just to be safe RXP will clean up all BD/PDUs, this needs some extra state machine check. when this bit is set it will disable the BD/PDU cleanup and SW can go ahead and clean it after reset
17	CFG_MPR_IE_LENCHK_DISS	For the newly added IE parsing of channel switch and quiet element there is a lenchk done to make sure Channel switch is indeed 3bytes and quiet element is indeed 6 bytes setting this bit to 1 this check can be disabled
16	CFG_FLT_UNPROT_ACTION_RMF_DIS	All action frames are supposed to be RMF frames(encrypted management frames) But some action frames can be unprotected, in those cases RXP sets RMF bit in BD to 0 so DPU can bypass the decryption and pass it to host SW by writing 1 to this bit this feature can be disabled, so that all action frames will have RMF set to 1 and DPU can drop the unprotected RMF frames and firmware can process these packets from Error WQ.
15	CFG_11AC_AMPDU_EOP_INDICATION_EN	If set, RXP will generate 11ac ampdu end of packet signal to PHY. By default, it will NOT generate this eop signal. If 11ac_eop is generated to PHY, PHY will stop receiving the current packet. This is used to save power in case of MU-MIMO packets
14	CFG_MPR_DISABLE_WAIT_FOR_FILTER_IDLE	When set, this will disable the 'wait for filter to finish, when eop is received' function. This is a bug fix. This configuration bit will allow the bug fix to be disabled and the rxp logic to be reverted to the original implementation. This allows for backwards compatibility.
13	RRI_CLEAR_ON_FCS_FAIL	If set, this will enable RRI regeneration when the first RRI got followed by and FCS err.
12	MASK_SUBSQ_FCS_ERR_WITHIN_AMPDU	If set, this will mask out fcs error notification to the MTU after an MPDU within an AMPDU passes FCS check. This is to prevent doing EIFS in this case.
11	SOFT_PMI_ABORT_EN	Soft PMI abort enable.
10	CFG_HAS_FCS_EN	If set, RXP will enable writing 4-byte fcs into BD/PDU and RXP flag has_fcs flag will be set to 1. By default, the RXP DMA will NOT write fcs into BD/PDU and MPDU length does not include the 4-byte fcs.

WCSS_A_RXP_CONFIG (cont.)

Bits	Name	Description
9	CFG_RSV_BMU_EN	If set, RXP will enable sending reservation command to BMU. By default, the RXP DMA will NOT send reservation command to BMU at start of packet.
8	CFG_HAS_PHY_CMD	If set, RXP will write PHY Command into BD. By default, the PHY Command will not be put into BD so that SW can adjust the "mpdu_header_start_offset" to save the BD space for MPDU data. Note that the TimeStamp will be in BD in any case.
7	CFG_PHY_PROTOCOL_CHK_DIS	When set it disables RXP from doing any PHY protocol check Like checking PHY cmd length and stats length endcomment bitfield [6] cfg_disable_dlm_proc configuration 1'b0 bitfieldComment If set to disable, the Delimiter Processing is disabled for A-MPDU packets and the whole A-MPDU packet will be sent to mCPU without MPDU Processing. The byp_dlm_proc and byp_mpdu_proc will be asserted for A-MPDU packets. By default, Delimiter Processing is enabled. Note: This cfg_disable_dlm_proc will affect only A-MPDU packets, but not MPDU packets.
5	CFG_DISABLE_MPDU_PROC	If set to disable, the MPDU Processing is disabled for MPDU packets and the MPDU packets will be sent to mCPU without MPDU processing. This means NO FCS check, NO address filtering, and NO binary search. The packets are assumed passing MPDU processing. By default, MPDU Processing is enabled. Note: This cfg_disable_mpdu_proc will affect only MPDU packets, but not A-MPDU packets. This means MPDU within A-MPDU will be processed normally (without bypass) if cfg_disable_dlm_proc is not asserted for A-MPDU packets.
4	CFG_DISABLE_FCS_CHECK	If set to disable, the FCS Check is disabled and all packets will be treated as passing FCS Check. By default, it is enabled and RXP will check FCS on MPDU packets. The counter associated with this config is fcs_err_cnt. If not disabled, this counter will increment if a packet has FCS error.
3	CFG_MAX_PKTLEN_CHECK_EN	The max packet length check enable bit. By default, the max_pktlen check is disabled. If set, it will check MPDU packet length against max_pktlen register and check A-MPDU packet length against max_len_ampdu register. But it will not affect max_len_mpdu_in_ampdu which is checked in delimiter processing and can be disabled by asserting cfg_disable_dlm_proc.
2	CFG_ADDR_FILTER_EN	If set, the Address Filter will be enabled. By default, it is disabled and all packets will be treated as passing Address Filtering.

WCSS_A_RXP_CONFIG (cont.)

Bits	Name	Description
1	CFG_GEN_SHUTOFF_EN	If set, RXP will generate shutoff signal to PHY. By default, it will NOT generate shutoff signal. If shutoff is generated to PHY, PHY will stop receiving the current packet. This is used to save power.
0	CFG_RXP_EN	If enabled, RXP will receive packets from PHY. By default, the RXP is disabled and will not receive any packet from PHY. This bit can be changed at any time, but will only take effect when there is start pulse from PHY.

0x0A080808 WCSS_A_RXP_MAX_PKTLEN**Type:** RW**Reset State:** 0x00000040

The Max MPDU Packet Length in terms of the number bytes. It includes mac_header+payload+fcs, but does not include PHY data.

WCSS_A_RXP_MAX_PKTLEN

Bits	Name	Description
6:0	MAX_PKTLEN	The Max MPDU Packet Length in terms of the number 128 bytes.

0x0A08080C WCSS_A_RXP_SW_BD_REG**Type:** RW**Reset State:** 0x00000000

Register used by SW to write a DWord to BD

WCSS_A_RXP_SW_BD_REG

Bits	Name	Description
31:0	SW_BD_REG	SW mainly uses this register to write to location 38 in DWORD

0x0A080810 WCSS_A_RXP_MPDU_HEADER_START_OFFSET**Type:** RW**Reset State:** 0x00000040

The MPDU header start offset in BD which is in the unit of data word (4-byte).

WCSS_A_RXP_MPDU_HEADER_START_OFFSET

Bits	Name	Description
7:2	MPDU_HEADER_START_OFFSET	MPDU header start offset in BD. Note that the programmable bits are [7:2] and the lower [1:0] bits are always 2'b00 to guarantee the offset address is 4-byte aligned. If bit[7] is 0, the MPDU header starts at BD, but if bit[7] is 1, the MPDU header starts at PDU. Note that It is not valid to have the value of 6'h3f for bit[7:2] (or 8'hfc for bit[7:0]). The default value is 6'ha which means the offset address of 8'h28. In this case, the PHY Cmd bytes will not be written in BD so that the MPDU header can start right after TimeStamp field. However, for diagnostic purpose, the PHY Cmd can be written into BD by setting cfg_has_phy_cmd bit in config register so that this offset value will be adjusted to 6'h10 which means the offset address of 8'h40.

0x0A08082C WCSS_A_RXP_SEARCH_ADDR1_PTR

Type: RW

Reset State: 0x00000000

SW can set the addr1 pointers of Binary Search.

WCSS_A_RXP_SEARCH_ADDR1_PTR

Bits	Name	Description
16	ADDR1_POINTERS_VALID	When set, the pointers are valid.
13:8	ADDR1_SEARCH_HI_PTR	Indicate the addr1 pointer of Binary Search upper bound.
5:0	ADDR1_SEARCH_LO_PTR	Indicate the addr1 pointer of Binary Search lower bound.

0x0A080830 WCSS_A_RXP_SEARCH_ADDR2_PTR

Type: RW

Reset State: 0x00000000

SW can set the addr2 pointers of Binary Search.

WCSS_A_RXP_SEARCH_ADDR2_PTR

Bits	Name	Description
16	ADDR2_POINTERS_VALID	When set, the pointers are valid.
13:8	ADDR2_SEARCH_HI_PTR	Indicate the addr2 pointer of Binary Search upper bound.
5:0	ADDR2_SEARCH_LO_PTR	Indicate the addr2 pointer of Binary Search lower bound.

0x0A080834 WCSS_A_RXP_SEARCH_ADDR3_PTR**Type:** RW**Reset State:** 0x00000000

SW can set the addr3 pointers of Binary Search.

WCSS_A_RXP_SEARCH_ADDR3_PTR

Bits	Name	Description
16	ADDR3_POINTERS_VALID	When set, the pointers are valid.
13:8	ADDR3_SEARCH_HI_PTR	Indicate the addr3 pointer of Binary Search upper bound.
5:0	ADDR3_SEARCH_LO_PTR	Indicate the addr3 pointer of Binary Search lower bound.

0x0A080838 WCSS_A_RXP_SEARCH_TABLE_CMD**Type:** RW**Reset State:** 0x00000000

This is action register for indirect access to Binary Search Table. When SW writes to this action register, it will trigger the interaction between Search_Table_Data[0-1] registers and Binary Search Table. For writing data into memory table, it is important for SW to write into Search_Table_Data[0-1] registers before Search_Table_Cmd. For reading data from memory table, SW only needs to write into Search_Table_Cmd and the read data will be available in Search_Table_Data[0-1] registers.

WCSS_A_RXP_SEARCH_TABLE_CMD

Bits	Name	Type	Description
9	STATUS	R	read only: If set to 1 a binary search table read or write is ongoing. SW should read and make sure this bit is 0 before issuing another cmd write into this register.
8	WRITE	RW	Indicate write/read. If 1, write. If 0, read.
4:0	TABLE_PTR	RW	Indicate the action should take effect on this entry number in Binary Search Table.

0x0A08083C WCSS_A_RXP_SEARCH_TABLE_DATA0**Type:** RW**Reset State:** 0x00000000

This is data register 0 for indirect access to Binary Search Table. There are totally 3 Data registers for Binary Search address Table.

WCSS_A_RXP_SEARCH_TABLE_DATA0

Bits	Name	Description
31:0	MAC_ADDR	MAC_Addr[31:0] portion of MAC_Addr[47:0]. Note: The order is Little Endian: MAC_Addr[31:0] = {Byte_3, Byte_2, Byte_1, Byte_0}

0x0A080840 WCSS_A_RXP_SEARCH_TABLE_DATA1**Type:** RW**Reset State:** 0x00000000

This is data register 1 for indirect access to Binary Search Table. There are totally 3 Data registers for Binary Search address Table.

WCSS_A_RXP_SEARCH_TABLE_DATA1

Bits	Name	Description
31:29	DPU_MAN_MC_BC_SIG	This feild is used to configure the DPU signature for Management mc/bc packets
28	RSVD	reserved for future use

WCSS_A_RXP_SEARCH_TABLE_DATA1 (cont.)

Bits	Name	Description
27	VALID	Set when the sta address is valid
26	RMF	Set when robust managemnt frmases require BIP or decryption
25	FRAME_TRANSLATION	This bit is used to decide if a per station frame translation is required. When this bit is set and the bit in type/subtype is set then those packets for that station will require frame translation, so FT bit in BD will be set
24	DROP	When set, on a address hit, the frame will be dropped. The setting of this bit is only valid when 'valid configuration' is set
23:16	ADDR_ID	The addr_id[7:0].
15:0	MAC_ADDR	MAC_Addr[47:32] portion of MAC_Addr[47:0]. Note: The order is Little Endian: MAC_Addr[47:32] = {Byte_5, Byte_4}

0x0A080844 WCSS_A_RXP_MAX_LEN_AMPDU**Type:** RW**Reset State:** 0x0000FFFF

The Max Length for A-MPDU packet.

WCSS_A_RXP_MAX_LEN_AMPDU

Bits	Name	Description
16:0	MAX_LEN	The Max Length for A-MPDU packet.

0x0A080848 WCSS_A_RXP_MAX_LEN_MPDU_IN_AMPDU**Type:** RW**Reset State:** 0x00000FFF

The Max Length for MPDU in A-MPDU packet. This is used for delimiter processing to determine whether it finds a good delimiter.

WCSS_A_RXP_MAX_LEN_MPDU_IN_AMPDU

Bits	Name	Description
13:0	MAX_LEN	The Max Length for MPDU in A-MPDU packet.

0x0A08084C WCSS_A_RXP_DMA_MAX_RSV_PDU**Type:** RW**Reset State:** 0x00000008

The Max number of PDUs that RXP DMA can reserve at a time. The is to avoid the scenario that RXP drop big packet just because it fails to reserve a huge number of PDUs at the start of the packet.

WCSS_A_RXP_DMA_MAX_RSV_PDU

Bits	Name	Description
3:0	MAX_RSV_PDU	The Max number of PDUs that RXP DMA can reserve at a time.

0x0A080850 WCSS_A_RXP_DMA_BURST_THRESHOLD**Type:** RW**Reset State:** 0x00000008

The data word threshold that RXP DMA control can start burst MPDU data to MIF. The data unit is 1 word (or 4 bytes). By default, the threshold is 8 words (or 32 bytes). RXP MPDU data is accumulated in DMA FIFO. The DMA control will not send data to MIF if the number of data word in DMA FIFO does not reach the threshold. This is to used to control the burstness of RXP data on AHB.

WCSS_A_RXP_DMA_BURST_THRESHOLD

Bits	Name	Description
5:0	BURST_THRESHOLD	The data word threshold that DMA can start burst MPDU data to MIF.

0x0A080854 WCSS_A_RXP_CLEAR_STATS**Type:** RW**Reset State:** 0x00000000

This register is used to clear all stats.

WCSS_A_RXP_CLEAR_STATS

Bits	Name	Description
0	CLEAR_STATS	When set, all statistics counters will be cleared When set, the HW will automatically clear this bit.

0x0A080858 WCSS_A_RXP_PHY_MPDU_CNT

Type: R

Reset State: 0x00000000

The number of MPDU packets are received from PHY.

WCSS_A_RXP_PHY_MPDU_CNT

Bits	Name	Description
31:0	PHY_MPDU_CNT	PHY MPDU Cnt.

0x0A08085C WCSS_A_RXP_PHY_AMPDU_CNT

Type: R

Reset State: 0x00000000

The number of A-MPDU packets are received from PHY.

WCSS_A_RXP_PHY_AMPDU_CNT

Bits	Name	Description
31:0	PHY_AMPDU_CNT	PHY A-MPDU Cnt.

0x0A080864 WCSS_A_RXP_PHY_ABORT_CNT

Type: R

Reset State: 0x00000000

The number of aborted packets from PHY.

WCSS_A_RXP_PHY_ABORT_CNT

Bits	Name	Description
7:0	PHY_ABORT_CNT	PHY Abort Packet Cnt.

0x0A080868 WCSS_A_RXP_PHY_SHUTOFF_CNT**Type:** R**Reset State:** 0x00000000

The number of shutoff pulses are generated by RXP to PHY.

WCSS_A_RXP_PHY_SHUTOFF_CNT

Bits	Name	Description
15:8	PHY_11AC_EOP_CNT	This is also logically a shutoff count, but this is RXP shutting of due of EOF bit set in the 11ac AMPDU indicating end of useful AMPDU data. PHY will stop giving anymore data.
7:0	PHY_SHUTOFF_CNT	PHY Shutoff Cnt. This is number of packets that RXP terminated due to packet error or early beacon termination

0x0A08086C WCSS_A_RXP_DLM_FIFO_FULL_CNT**Type:** R**Reset State:** 0x00000000

The number of packets have error because DLM FIFO is full.

WCSS_A_RXP_DLM_FIFO_FULL_CNT

Bits	Name	Description
7:0	DLM_FIFO_FULL_CNT	DLM FIFO Full Cnt.

0x0A080870 WCSS_A_RXP_DLM_ERR_CNT**Type:** R**Reset State:** 0x00000000

The number of packets have error because Delimiter is bad. When a error is detected in the A-MPDU delimiter this count is incremented. But when RXP starts searching for the next A-MPDU

delimiter, this count is NOT incremented. The implementation uses 'seen_dlm_err' flag. This flag is cleared for the first delimiter or after a good delimiter is found. It is set after a bad delimiter is found. If this flag is set, then don't increment the counter. This will increment: - Whenever the next delimiter (using the previous good delimiter information) is not found in the expected location, or is bad.

WCSS_A_RXP_DLM_ERR_CNT

Bits	Name	Description
7:0	DLM_ERR_CNT	DLM Error Cnt.

0x0A080874 WCSS_A_RXP_FAIL_FILTER_CNT

Type: R

Reset State: 0x00000000

The number of packets have failed Address Filtering.

WCSS_A_RXP_FAIL_FILTER_CNT

Bits	Name	Description
15:0	FAIL_FILTER_CNT	Fail Address Filter Cnt.

0x0A080878 WCSS_A_RXP_FAIL_MAX_PKTLEN_CNT

Type: R

Reset State: 0x00000000

The number of packets have failed Max MPDU or AMPDU packet length check.

WCSS_A_RXP_FAIL_MAX_PKTLEN_CNT

Bits	Name	Description
7:0	FAIL_MAX_PKTLEN_CNT	Fail Max MPDU or AMPDU Packet Length Check Cnt.

0x0A08087C WCSS_A_RXP_FCS_ERR_CNT

Type: R

Reset State: 0x00000000

The number of packets have FCS error.

WCSS_A_RXP_FCS_ERR_CNT

Bits	Name	Description
31:0	FCS_ERR_CNT	FCS Error Cnt.

0x0A080880 WCSS_A_RXP_DMA_SEND_CNT**Type:** R**Reset State:** 0x00000000

The number of packets are sent to CPU by RXP Write DMA.

WCSS_A_RXP_DMA_SEND_CNT

Bits	Name	Description
31:0	DMA_SEND_CNT	The number of packets are sent to CPU by RXP Write DMA. All packets in current system are all pushed by RXP refer to rpe dma send cnt for that. This counter increments only if RXP pushes packets directly to DXE

0x0A080884 WCSS_A_RXP_DMA_DROP_CNT**Type:** R**Reset State:** 0x00000000

The number of packets are dropped by RXP Write DMA. This could increment due to: Failed filter PHY Abort AFTER the 9th command word BD/PDU errors If packet with command phase shorter or longer than actual command length specified is sent, the RXP will increment dma_drop_cnt instead of phy_err_drop_cnt (unless actual command phase < 9 words, in which case phy_err_drop_cnt will be incremented) Will NOT increment if end of psdu (as defined in psdu length by PHY command) occurs during search for next delimiter in AMPDU; this is because a new packet is not found, so there should not be a drop)

WCSS_A_RXP_DMA_DROP_CNT

Bits	Name	Description
15:0	DMA_DROP_CNT	The number of packets are dropped by RXP Write DMA. Drop due to filter error, packet len error or Get bmu error any packet dropped at DMA due to any of above errors, these can be drop at DMA marked packets which may have filter fail errors

0x0A080888 WCSS_A_RXP_DMA_GET_BMU_FAIL_CNT**Type:** R**Reset State:** 0x00000000

The number of packets are dropped due to insufficient BD/PDUs BMU BD or PDU.

WCSS_A_RXP_DMA_GET_BMU_FAIL_CNT

Bits	Name	Description
15:0	DMA_GET_BMU_FAIL_CNT	DMA No BMU BD/PDU Cnt.

0x0A080890 WCSS_A_RXP_TYPE_SUBTYPE_FILTER_CNT**Type:** R**Reset State:** 0x00000000

The number of packets dropped because of their type/subtype

WCSS_A_RXP_TYPE_SUBTYPE_FILTER_CNT

Bits	Name	Description
7:0	DROPPED_COUNT	number of frames dropped

0x0A080894 WCSS_A_RXP_INCORRECT_LENGTH_FILTER_CNT**Type:** R**Reset State:** 0x00000000

The length of the frame was not sufficient to do all the address filter checks. This may increment if dlm_fifo_full occurs.

WCSS_A_RXP_INCORRECT_LENGTH_FILTER_CNT

Bits	Name	Description
7:0	DROPPED_COUNT	number of frames dropped

0x0A080898 WCSS_A_RXP_ADDR1_BLOCK_FILTER_CNT**Type:** R**Reset State:** 0x00000000

Number of frames blocked based on the address one

WCSS_A_RXP_ADDR1_BLOCK_FILTER_CNT

Bits	Name	Description
7:0	DROPPED_COUNT	number of frames dropped

0x0A08089C WCSS_A_RXP_ADDR1_HIT_NO_PASS_CNT**Type:** R**Reset State:** 0x00000000

Number of frames that got a hit in the binary search table but were instructed to be dropped because of the drop setting in the binary search table

WCSS_A_RXP_ADDR1_HIT_NO_PASS_CNT

Bits	Name	Description
7:0	DROPPED_COUNT	number of frames dropped

0x0A0808A0 WCSS_A_RXP_ADDR1_DROP_CNT**Type:** R**Reset State:** 0x00000000

Number of frames that got dropped because there was no hit in the binary search table

WCSS_A_RXP_ADDR1_DROP_CNT

Bits	Name	Description
7:0	DROPPED_COUNT	number of frames dropped

0x0A0808A4 WCSS_A_RXP_ADDR2_HIT_NO_PASS_CNT**Type:** R**Reset State:** 0x00000000

Number of frames that got a hit in the binary search table but were instructed to be dropped because of the drop setting in the binary search table

WCSS_A_RXP_ADDR2_HIT_NO_PASS_CNT

Bits	Name	Description
7:0	DROPPED_COUNT	number of frames dropped

0x0A0808A8 WCSS_A_RXP_ADDR2_DROP_CNT**Type:** R**Reset State:** 0x00000000

Number of frames that got dropped because there was no hit in the binary search table

WCSS_A_RXP_ADDR2_DROP_CNT

Bits	Name	Description
7:0	DROPPED_COUNT	number of frames dropped

0x0A0808AC WCSS_A_RXP_ADDR3_HIT_NO_PASS_CNT**Type:** R**Reset State:** 0x00000000

Number of frames that got a hit in the binary search table but were instructed to be dropped because of the drop setting in the binary search table

WCSS_A_RXP_ADDR3_HIT_NO_PASS_CNT

Bits	Name	Description
7:0	DROPPED_COUNT	number of frames dropped

0x0A0808B0 WCSS_A_RXP_ADDR3_DROP_CNT**Type:** R**Reset State:** 0x00000000

Number of frames that got dropped because there was no hit in the binary search table

WCSS_A_RXP_ADDR3_DROP_CNT

Bits	Name	Description
7:0	DROPPED_COUNT	number of frames dropped

0x0A0808B4 WCSS_A_RXP_MPDU_IN_AMPDU_CNT**Type:** R**Reset State:** 0x00000000

The number of MPDU packets are split from A-MPDU packets.

WCSS_A_RXP_MPDU_IN_AMPDU_CNT

Bits	Name	Description
31:0	MPDU_IN_AMPDU_CNT	The number of MPDU packets are split from A-MPDU packets.

0x0A0808B8 WCSS_A_RXP_PHY_ERR_DROP_CNT**Type:** R**Reset State:** 0x00000000

The number of packets are dropped by RXP PHY Processing because of some apparent PHY protocol errors, such as: PHY Cmd Len is smaller than 9 bytes. PHY Cmd Len is larger than 28 bytes. PHY Stats Len is not equal to 8 bytes. Exception - if packet with command phase shorter longer than actual command length is sent, the RXP will increment dma_drop_cnt instead (however, if actual command length is < 9, this will be counted as a phy_err_drop) Note: The packets are dropped before reaching RXP Write DMA.

WCSS_A_RXP_PHY_ERR_DROP_CNT

Bits	Name	Description
7:0	PHY_ERR_DROP_CNT	The number of packets are dropped by RXP PHY Processing.

0x0A0808BC WCSS_A_RXP_RADAR_DETECT_STATUS**Type:** RW**Reset State:** 0x00000000

Radar detect status register. Pointers in radar detect circular buffer. Equal pointers means that the buffer is empty. There can be 0-63 entries in this circular buffer.

WCSS_A_RXP_RADAR_DETECT_STATUS

Bits	Name	Type	Description
24	RDR_DETECT_INT	R	status of the radar detect interrupt signal
21:16	NR_ENTRIES	R	Number of entries filled in radar detect circular buffer. (0-63)
13:8	TAIL_PTR	RW	tail pointer of radar detect circular buffer
5:0	HEAD_PTR	RW	head pointer of radar detect circular buffer

0x0A0808C0 WCSS_A_RXP_RADAR_DETECT_CONTROL**Type:** RW**Reset State:** 0x00000000

Radar detect control register.

WCSS_A_RXP_RADAR_DETECT_CONTROL

Bits	Name	Description
8	RADAR_DETECT_ENABLE	Enable/disable storing of radar detect logs
5:0	MIN_NR_LOGS_GEN_INT	Minimum number of stored logs to generate radar detect interrupt. 0 is a special case, which means to disable the interrupt.

0x0A0808C4 WCSS_A_RXP_RADAR_DETECT_LLOGS**Type:** RW**Reset State:** 0x00000000

Radar detect lower word register.

WCSS_A_RXP_RADAR_DETECT_LLOGS

Bits	Name	Description
31:0	RADAR_DETECT_LDATAL	On read, the lower 4 bytes from the radar detect log as pointed by the tail pointer is returned. On write, the lower 4 bytes are written into the location pointed by the head pointer.

0x0A0808C8 WCSS_A_RXP_RADAR_DETECT_ULOGS**Type:** RW**Reset State:** 0x00000000

Radar detect upper word register.

WCSS_A_RXP_RADAR_DETECT_ULOGS

Bits	Name	Description
31:0	RADAR_DETECT_UDATAL	On read, the lower 4 bytes from the radar detect log as pointed by the tail pointer is returned. After the read, the tail pointer is increased by one, except if that would cause it to become greater than the head pointer. On write, the lower 4 bytes are written into the location pointed by the head pointer. After the write, the head pointer is increased by one. If this would cause the head pointer to become equal to the tail pointer, the tail pointer would also be incremented.

0x0A0808CC WCSS_A_RXP_RADAR_DETECT_LOGS**Type:** RW**Reset State:** 0x00000000

Radar detect logs counter register

WCSS_A_RXP_RADAR_DETECT_LOGS

Bits	Name	Type	Description
31	RADAR_DETECT_LOGS_CNT_CLEAR	RW	When set, clears count of PHY written radar detect logs. This bit will automatically be cleared by HW.
15:0	RADAR_DETECT_LOGS_CNT	R	Number of radar detect logs received from PHY.

0x0A0808D0 WCSS_A_RXP_SOFT_PMI_ABORT**Type:** RW**Reset State:** 0x00000000

Soft PMI Abort register

WCSS_A_RXP_SOFT_PMI_ABORT

Bits	Name	Description
0	SOFT_PMI_ABORT	Write only bit; when 1 is written to this bit, and soft_pmi_abort_en is 1, then a soft PHY abort will be generated. Error counters will be incremented as if an actual abort had occurred.

0x0A0808D4 WCSS_A_RXP_START_ERR_CNT**Type:** R**Reset State:** 0x00000000

The number of start errors on the PMI interface (pmi_rxp_starts received while still waiting for previous packet to complete).

WCSS_A_RXP_START_ERR_CNT

Bits	Name	Description
7:0	PHY_START_ERR_CNT	PHY Start Error Cnt.

0x0A0808D8 WCSS_A_RXP_CCA_AND_EXT_TIMEOUT**Type:** RW**Reset State:** 0x3C064090

Configuration bits for cca extensions and timeout functionality.

WCSS_A_RXP_CCA_AND_EXT_TIMEOUT

Bits	Name	Description
31:21	CCA_PKTDET_6US_EXT_VALUE	This is extension of packet detect, which is done only for 11b packets for 80Mhz clock : 11'd480 for 160Mhz clock : 11'd960 for 320Mhz clock : 11'd1920

WCSS_A_RXP_CCA_AND_EXT_TIMEOUT (cont.)

Bits	Name	Description
20:10	CCA_PKTDET_EXT_VALUE	This is an additional extension of packet detect, if this is not needed it should be programmed to 1, otherwise HW will use the default value of 11'd400 80Mhz clocks 80Hmz clock : 11'd400 160Mhz clock: 11'd800 320Mhz clock: 11'd1600
9	CCA_PKDET_EXTENTION_ALLOWED	when set it allows for 6us extn on ofdm packets
8:4	FLT_TIMEOUT_TIME	This value represent that maximum time (in 256 OR 4096 120MHZ clk cycle units) that the rxp filter will wait for two sequential MPDU header bytes to be received. If this time is not met, the rxp filter will drop the frame by indicating that the filter is done, but the frame does not pass the filter. The 'timeout_error_cnt' register contents will increase by 1.
2	CCA_PKDET_EXTENTION_OLDLOGIC_DIS	when packet det gets deasserted 5us extension is done, and at the end of 5us we look for commandbyte 6 for packet type whether it is ofdm and then do 6us additional extension. But PHY may not always give this info within 5us hence ofdm signal is used for this (which is the new logic). This logic is only enabled when you write 1 to this bit
1	FLT_TIMEOUT_COUNT_UNIT	When clear, the timeout will count in 256 clk cycle units When set, the timeout counter will count in 4096 clk cycle units
0	FLT_TIMEOUT_CHECK_DISABLED	When set, the rxp filter timeout check is disabled

0x0A0808DC WCSS_A_RXP_TIMEOUT_ERROR_CNT**Type:** R**Reset State:** 0x00000000

Number of times the rxp filter encounters a timeout As a result of this, the frame will be dropped

WCSS_A_RXP_TIMEOUT_ERROR_CNT

Bits	Name	Description
7:0	TIMEOUT_COUNT	Number of times the rxp filter encounters a timeout

0x0A0808E0 WCSS_A_RXP_STALL_TIMEOUT_CNT**Type:** R**Reset State:** 0x00000000

Number of times stall timeout occurs. As a result of this, the frame will be dropped

WCSS_A_RXP_STALL_TIMEOUT_CNT

Bits	Name	Description
7:0	STALL_TIMEOUT_COUNT	Number of times the PMI interface encounters a stall timeout

0x0A0808EC WCSS_A_RXP_CFG_FLT_NAV_CONTROL_RESPONSE_REQ**Type:** RW**Reset State:** 0x00000000

NAV control register and response requested register. The NAV PHY delay compensation can be programmed here. Also the control of how the NAV needs to be updated, or not at all, when the CCA was not asserted when a frame got received (and dropped) is configured here.

WCSS_A_RXP_CFG_FLT_NAV_CONTROL_RESPONSE_REQ

Bits	Name	Description
25	CFG_FLT_RESPONSE_REQ_WITH_HT_CONTROL	When set and the Reponse interrupt is generated, and the incoming frame has an HT control field (this is: incoming frame is QoS data frame and has the 'order' bit in the frame control field set), the response interrupt is only generated after the HT control field has been fully received and will be located in the reponse_header_info5 field. Setting this bit will also enable the generation of the 'HTC present' reporting in the enhanced_sequence_number_message_type set 1: Will enable RXP to process frames with HTC feild set 0: RXP will ignore HTC feild hence whenever this bit is zero block_frames_with_ht_control_field(extra frame filter register) should be 1
24:21	CFG_FLT_NAV_RESERVED	reserved bits
20	CFG_FLT_NAV_DELAY_COMPENSATION_VALUE_EN	When set, enables the NAV delay compensation functionality. This is, the NAV update value given to the MTU is compensated for the PHY delay. The compensation value is given in the bits: cfg_ft_nav_delay_compensation_value
19:16	CFG_FLT_NAV_DELAY_COMPENSATION_VALUE	NAV compensation value in usec.

WCSS_A_RXP_CFG_FLT_NAV_CONTROL_RESPONSE_REQ (cont.)

Bits	Name	Description
10	CFG_FLT_REGISTER_CCA_AT_START_OF_OBSERVE_TIME	When NOT set: no action When set: The RXP filter samples the setting of the primary cca signal at the start of the cca observe time. This sampled CCA is later used to determine if an NAV update is needed or NOT. The cca observe time is configured with other bits in this register.
9	CFG_FLT_REGISTER_CCA_AT_END_OF_OBSERVE_TIME	When NOT set: no action When set: The RXP filter samples the setting of the primary cca signal at the end of the cca observe time. This sampled CCA is later used to determine if an NAV update is needed or NOT. The cca observe time is configured with other bits in this register.
8	CFG_FLT_REGISTER_CCA_DURING_OBSERVE_TIME	When NOT set: no action When set: The RXP filter samples the setting of the primary cca signal during the entire duration of the cca observe time. If the cca goes low anywhere during this time, this is recorded and fixed. In that case, even if the CCA went high a little later, the sampled CCA value that is later used to determine if an NAV update is needed or NOT, will have a low value. The cca observe time is configured with other bits in this register.
7	CFG_FLT_REGISTER_CCA_LEVEL_DURING_DATA_TIME	When NOT set: no action When set: The 'data reception time' as indicated by pktDET_n being low, is part of the cca observe time.
6	CFG_FLT_REGISTER_CCA_LEVEL_DURING_PREAMBLE_TIME	When NOT set: no action When set: The 'data reception time' as determined by early_pktDET_n being low, is part of the cca observe time.
5	CFG_FLT_SKIP_DIFS_WHEN_NAV_UPDATE_MASKED	When NOT set: no action When set: When the RXP filter drops a frame and the NAV update is masked (due to enabling the NAV update masking features in this register), the 'skip_difs' signal to the MTU will be set.
4	CFG_FLT_NO_NAV_UPDATE_ON_DROP_AND_BELOW_CCA	When NOT set: no action When set: When the RXP filter drops a frame due to any reason, AND the cca at the sample point was high, the RXP will NOT generate a NAV update based on the dropped frame.
3	CFG_FLT_NO_NAV_UPDATE_ON_EXPLICIT_DROP_AND_BELOW_CCA	When NOT set: no action When set: When the RXP filter drops a frame due to an explicit drop configuration in the address table, AND the cca at the sample point was high, the RXP will NOT generate a NAV update based on the dropped frame.
2	CFG_FLT_NO_NAV_UPDATE_ON_EXPLICIT_DROP	When NOT set: no action When set: When the RXP filter drops a frame due to an explicit drop configuration in the address table, the RXP will not generate a NAV update based on the dropped frame. The setting of the CCA is NOT taken into account

WCSS_A_RXP_CFG_FLT_NAV_CONTROL_RESPONSE_REQ (cont.)

Bits	Name	Description
1	CFG_FLT_CLEAR_SKIP_NEXT_DIFS_ON_EARLY_PKT_DET	When NOT set: no action When set: The RXP filter can generate a signal to the MTU to skip DIFS. See other register bits to enable this. This 'skip' signal will be cleared when early_pkt_det_n gets asserted.
0	CFG_FLT_CLEAR_SKIP_NEXT_DIFS_ON_CCA	When NOT set: no action When set: The RXP filter can generate a signal to the MTU to skip DIFS. See other register bits to enable this. This 'skip' signal will be cleared when primary cca gets asserted.

0x0A0808F0 WCSS_A_RXP_CFG_FLT_TYPE_SUBTYPE_RX_DISABLE0**Type:** RW**Reset State:** 0x00000000

Control overwrite functionality for what was programmed in type/subtype config table (upper 32 bits)

WCSS_A_RXP_CFG_FLT_TYPE_SUBTYPE_RX_DISABLE0

Bits	Name	Description
31:0	CFG_FLT_TYPE_SUBTYPE_RX_DISABLE0	Control some overwrite functionality for whatever was programmed in the Type/subtype configuration table. When set, a frame of the corresponding type should be dropped.

0x0A0808F4 WCSS_A_RXP_CFG_FLT_TYPE_SUBTYPE_RX_DISABLE1**Type:** RW**Reset State:** 0x00000000

Control overwrite functionality for what was programmed in type/subtype config table (lower 32 bits)

WCSS_A_RXP_CFG_FLT_TYPE_SUBTYPE_RX_DISABLE1

Bits	Name	Description
31:0	CFG_FLT_TYPE_SUBTYPE_RX_DISABLE1	Control some overwrite functionality for whatever was programmed in the Type/subtype configuration table. When set, a frame of the corresponding type should be dropped.

0x0A080900 WCSS_A_RXP_FRAME_FILTER_CONFIG**Type:** RW**Reset State:** 0x00000000

This is a register set of 64 sequential word addresses: Each of these 64 words represent the filter configuration for a incoming frame type/subtype The register programming address per type/subtype is created as follows: ("This registers address" + {Type [b3, b2], Subtype [b7,b6,b5,b4], 2'b00}) where 'Type [b3,b2]' and 'Subtype [b7,b6,b5,b4]' are the IEEE standard encodings for the frames. A summary of the bit funtions is as follows:

WCSS_A_RXP_FRAME_FILTER_CONFIG

Bits	Name	Description
21	G5_DROP_AT_DMA	When set, the RXP will drop the incoming frame (even when the CRC and filter function passes) at the RXP DMA engine. This feature is usefull to prevent that control frames like the TRS, ACK etc, get unnecessarily written into memory
20	NONHT_BW_INDICATION_ENABLE	when set then the this particular type/subtype will need to be checked for bw information. If bit 0 of TA is checked to see if BW information is valid.
19	ROUTING_FLAG_SEL	when set : rxp will use dpu routing flag that is programmed in config3 bits 29:24 when not set : rxp will use dpu routing flag that is programmed in config3 bits 21:16
18	FRAME_TRANSLATION_REQ_EN	When set Frame translation by UMA is enabled
17	PM_BIT_EVAL_REQUIRED_ENABLE	when this bit is set the pm bit is evaluated for this type/subtype packet
16	RESERVED3	It is reserved for future use, it is read writable
15	ADDR3_ACCEPT_REMAINING_ENABLE	when enabled, and no hit was found for the Addr3 binary search, the incoming frame for this type/subtype will pass the Addr3 filter. The ADDR3 index code returned is 255 for multicast/broadcast and 254 for unicast.
14	ADDR3_BINARY_SEARCH_FILTER_ENABLE	When enabled, Addr3 is submitted to the RXP Addr3 binary search filter. When not enabled, Addr3 filtering is NOT done, The returned ADDR3 index code is 253 and the 'Addr3_index_invalid' bit in the RXP status flags in the BD will be set

WCSS_A_RXP_FRAME_FILTER_CONFIG (cont.)

Bits	Name	Description
13	ADDR2_ACCEPT_REMAINING_ENABLE	when enabled, and no hit was found for the Addr2 binary search, the incoming frame for this type/subtype will pass the Addr2 filter. The ADDR2 index code returned is 255 for multicast/broadcast and 254 for unicast.
12	ADDR2_BINARY_SEARCH_FILTER_ENABLE	When enabled, Addr2 is submitted to the RXP Addr2 binary search filter. When not enabled, Addr2 filtering is NOT done, The returned ADDR2 index code is 253 and the 'Addr2_index_invalid' bit in the RXP status flags in the BD will be set
11	ADDR1_ACCEPT_REMAINING_UNICAST_ENABLE	when enabled, and no hit was found for the unicast address in the Addr1 binary search, the incoming unicast frame for this type/subtype will pass the Addr1 filter. The index code returned is: 254
10	ADDR1_ACCEPT_REMAINING_MULTICAST_ENABLE	when enabled, and no hit was found for the multicast/broadcast address in the Addr1 binary search, the incoming multicast/broadcast frame for this type/subtype will pass the Addr1 filter. The index code returned is: 255.
9	ADDR1_BINARY_SEARCH_FILTER_ENABLE	When enabled, Addr1 is submitted to the RXP Addr1 binary search filter. When not enabled, Addr1 filtering is NOT done, The returned ADDR1 index code is 253 and the 'Addr1_index_invalid' bit in the RXP status flags in the BD will be set
8	ADDR1_BLOCK_UNICAST_ENABLE	when enabled, a unicast frame for this type/subtype will not pass the filter.
7	ADDR1_BLOCK_MULTICAST_ENABLE	when enabled, a multicast frame for this type/subtype will not pass the filter.
6	ADDR1_BLOCK_BROADCAST_ENABLE	when enabled, a broadcast frame for this type/subtype will not pass the filter.
5	RESERVED4	It is reserved for future use, it is read writable
4	RESERVED5	It is reserved for future use, it is read writable
3	RESERVED6	It is reserved for future use, it is read writable
2	RESERVED7	It is reserved for future use, it is read writable

WCSS_A_RXP_FRAME_FILTER_CONFIG (cont.)

Bits	Name	Description
1	BLOCK_RECEPTION_ENABLE	When enabled, all frames of this type/subtype will be blocked and will NOT pass the type/subtype/address filter check.
0	RESERVED8	It is reserved for future use, it is read writable

0x0A080A04 WCSS_A_RXP_STALL_TIMEOUT**Type:** RW**Reset State:** 0x00012000

Stall timeout functionality - allows RXP to return to idle if PMI interface stalls.

WCSS_A_RXP_STALL_TIMEOUT

Bits	Name	Description
16	STALL_TIMEOUT_EN	Enable stall timeout functionality on the MPI.
15:0	STALL_TIMEOUT_TIME	Number of cycles of no activity during PMI transaction before stall timeout occurs.

0x0A080A50 WCSS_A_RXP_CONFIG2**Type:** RW**Reset State:** 0x010381DF

It contains RXP Configuration registers, related to gen5_6 functionality added

WCSS_A_RXP_CONFIG2

Bits	Name	Description
31	CFG_DLM_MPDULEN_MSBITS_DIS	for 11ac the the Delimiter now holds mpdu length more than 4k. enabling this bit will allow rxp to apture those bits in MSB of mpdu length for 11ac only
30	CFG_MPR_NEED_MRQ_STATS	By default all the values in stat bytes will go in the BD in PHY stats location When this bit is set then the dynamic stats bus(link adaptation related info) will get over written on bytes 6 and 7 of PHY stats
29	CFG_FLT_VERSION_CHECKING_DIS	when set it will disable version checking for all packets. By default version checking is always done

WCSS_A_RXP_CONFIG2 (cont.)

Bits	Name	Description
28	NAV_UPDATE_DOES_NOT_EXCLUDE_SELF	When set, the NAV will also be set for frames destined to this STA. When NOT set, the NAV is only set for unicast frames received that are destined for other STAs. For both settings the NAV will be set for all BC/MC frames.
27	MASK_OUT_AMSDU_FRAMES_IN_AMPDU_ENABLE	When set, and amsdu frames embedded in AMPDU frames are received, the amsdu frames will be dropped.
26	ADDR2_AUTH_EXTRACT_ENABLE	When set, the RXP will extract the 'authentication' setting from the addr 2 binary search entry (bit 2 of the 3 bit wide DPU tag entry), and insert it in the BE. In this mode, the DPU tag when based on the ADDR 2 DPU TAG entry will only be 2 bits wide instead of the regular 3 bits.
25	PASS_LSIG_PROT_INFO_TO_TPE_ENABLE	When set, the RXP will pass the setting of the LSIG protection signal to the TPE together with the other response required info.
24	DEFAULT_PUSH_WQ_OVERRWRITE_ENABLE	When set, the RXP will by default push frames to WQ indicated in "default_push_wq". When NOT set, the RXP will by default push frames to WQ 2.
23:16	DEFAULT_PUSH_WQ	Field is only used when
15	BD_CLEAR_ENABLE_ENABLE	When set, the RXP will clear (set to zero) the bytes located in the BD between the PMI command bytes and the MPDU header bytes.
14	FILTER_RPE_TRACE_ENABLE	When set, the RXP filter interaction result with the RPE will be passed on to the BMU. That can result in the BMU pushing this info into the BMU HW tracing list.
13	DMA_RPE_TRACE_ENABLE	When set, the RXP DMA interaction result with the RPE will be passed on to the BMU. That can result in the BMU pushing this info into the BMU HW tracing list.
12	CFG_FLT_DEFAULT_ROUTING_DIS	By default rxp uses the <code>cfgflt_dpu_man_rtg_flg</code> , <code>cfgflt_dpu_data_rtg_flag</code> to fill in dpu routing flags. If this bit is set then RXP will use the per type/subtype routing selection.
11	CFG_FLT_PHY_ABORT_DISABLE	By default phy abort is enabled when this bit is set it is disabled. By default PHY RX will be aborted (shutdown) when the frame does not pass the type/subtype/address filter. This will save extra power.
10	CFG_FLT_NAV_SET_ENABLE_ON_ABORT	When enabled and the incoming frame does not pass the type/subtype/address filter and is aborted before the FCS is calculated, update the NAV.
9	CFG_FLT_NAV_CLEAR_ENABLE_ON_ABORT	When enabled, and the incoming frame does not pass the type/subtype/address filter and is aborted before the FCS is calculated, the NAV will be cleared.

WCSS_A_RXP_CONFIG2 (cont.)

Bits	Name	Description
8	CFG_FLT_HW_SW_BASED_RESPONSE_SELECTION_ENABLE	when set, the response generation path: SW based or HW based is controlled based on programming in the type/subtype configuration table, done through register "frame_filter_config". Note that if this control is required, the 'cfgflt_generate_hw_response_enabled' should be cleared, otherwise this programming is overruled. When 'cfgflt_hw_sw_based_response_selection_enable' is set, 'cfgflt_generate_hw_response_enabled' is clear and 'G5_generate_HW_Response' for an incoming frame is set, the RXP will generate a pulse to the TPE to generate a response frame if also the following conditions are met: addr1 is a unicast address, caused a hit in the addr. lookup table AND incoming frame is a: Management frame OR data frame OR QoS data frame with ack policy set to normal OR RTS frame OR PSPOLL frame OR BAR_frame When 'cfgflt_hw_sw_based_response_selection_enable' is set, 'cfgflt_generate_hw_response_enabled' is clear and 'G5_generate_SW_RRI' for an incoming frame is set, the RXP will generate the RRI interrupt ,so that SW can start generating the response frame, if also the following conditions are met: addr1 is a unicast address, caused a hit in the addr. lookup table AND incoming frame is a: Management frame OR data frame OR QoS data frame with ack policy set to normal OR RTS frame OR PSPOLL frame OR BAR_frame
7	CFG_FLT_FILTER_GO_TO_IDLE_ON_CLEAR_ENABLED	Gen 4 workaround: When set, and the MPDU PROC statemachine signals 'clear' to the filter, the filter will return to its idle state, forcing it to break of all interaction with other modules
6	CFG_MPR_BMU_BA_UPDATE_ENABLED	When set, enables the rxp interface with the BMU for Block Ack update
5	CFG_MPDU_PROC_WAIT_FOR_IDLE_FILTER_ENABLED	When set, enables bug fix from gen 4. This forces the MPDU PROC statemachine to wait for the rxp filter to get into an idle state before aborting a frame.
4	CFG_MPR_TPE_INTERFACE_ENABLED	When set to 1 the interface to the TPE is enabled
3	CFG_FLT_GENERATE_HW_RESPONSE_ENABLED	When set to 1 the global hw response generation logic within the RXP filter is enabled for the following conditions: addr1 is a unicast address, caused a hit in the addr. lookup table AND incoming frame is a: Management frame OR data frame OR QoS data frame with ack policy set to normal OR RTS frame OR PSPOLL frame OR BAR_frame
2	BMU_PWR_UPDATE_ENABLED	When set, enables the rxp interface with the BMU for power management updates related to PWR Mgt bit in frame control field, PS POLL reception and APSD trigger frame reception
1	CFG_BD_DPU_SOFTMACFIELDS_UPDATE_ENABLED	when you write a 1 the DPU related fields that softmac updates is enabled
0	CFG_RPE_INTERFACE_ENABLED	when you write a 1 rpe interface is enabled That means duplicate detect interface function from the rxp filter is enabled as well as the bitmap update function from teh rxp dma controller

0x0A080A54 WCSS_A_RXP_CONFIG3**Type:** RW**Reset State:** 0x07180200

It contains RXP Configuration registers, related to gen5_6 functionality added

WCSS_A_RXP_CONFIG3

Bits	Name	Description
29:24	CFG_FLT_DPU_MAN_RTG_FLG	This is the routing flag to which DPU will push management packets to, this is the value that is in the dpu_routing feild feild in RXBD for management packets
21:16	CFG_FLT_DPU_DATA_RTG_FLG	This is the routing flag to which DPU will push data packets to, this is the value that is in the dpu_routing feild feild in RXBD for data packets
14	CFG_RSSI_MODE_SEL	This bit selects the two modes of rssi data collection when this bit is 1'b0 : select Mode 0, the byte2(RxFIR0) and Byte3(RxFIR1) of PHY STATS are selected to be stored when this bit is 1'b1 : select Mode 1, the greater of Byte2 and Byte3 and Byte4(SNR) of PHY stats are selected to be stored
11:0	CFG_FLT_RPE_HANDSHAKE_TIMEOUT_VALUE	The RXP filter interacts with the RPE. At the start of this interaction a timeout counter is initialized. This register provides the timeout value for this interaction in number of clk cycles

0x0A080A58 WCSS_A_RXP_SEARCH_TABLE_DATA2**Type:** RW**Reset State:** 0x00000000

This is data register 2 for indirect access to Binary Search Table. It contains all the DPU related configurations that the RXP will insert into the BD There are totally 3 Data registers for Binary Search address Table.

WCSS_A_RXP_SEARCH_TABLE_DATA2

Bits	Name	Description
31:29	DPU_MC_BC_TAG	ThE DPU routing flag that will be inserted in the BD
28:21	DPU_MAN_MC_BC_DESCR_IPTOR_INDEX	ThE DPU descriptor index that will be inserted in the BD for mc/bc management packets
20	DPU_NE	When set, the RXP will set the 'ne' field in the bd to 1'b1
19	WEP_KEY_ID_EXTRACT_ENABLE	When set, the RXP filter will extract the Key id from encrypted frames (assuming that the encryption is wep)
18:16	DPU_TAG	ThE DPU routing flag that will be inserted in the BD

WCSS_A_RXP_SEARCH_TABLE_DATA2 (cont.)

Bits	Name	Description
15:8	DPU_MC_BC_DPU_DESC	This feild contains dpu descriptor index for BC/MC QoS/NonQoS frames RXP will insert this field in the DPU descriptor field in the BD.
7:0	DPU_DESCRIPTOR_INDEX	ThE DPU descriptor index that will be inserted in the BD for unicast data or unicast robust management frames(RMF)

0x0A080A5C WCSS_A_RXP_BEACON_TSF_TIM_EXTRACT_CTRL**Type:** RW**Reset State:** 0x00000003

This is controls what information needs to be extracted from the beacon and under what condition

WCSS_A_RXP_BEACON_TSF_TIM_EXTRACT_CTRL

Bits	Name	Description
31	BSS2_BEACON_PARSE_EN	when this bit is set it indicates that we can receive a client2 bss beacons so the beacon parsing checks whether received beacon is for BSS,P2P or BSS2(BSS CLient2) mode and accordingly TSF comparison and update is done 1'b0: no BSS beacons for client2 allowed 1'b1: BSS beacons for client2 allowed
30	EARLY_TIM_PROCESS	when this bit is set then the tim not set interrupt is generated immediatly after TIM bit is received and will not wait for the FCS complete
29	P2P_BEACON_PARS_EN	when this bit is set then the system is indicating that we can receive p2p beacons so the beacon parsing checks whether received beacon is for P2P or BSS mode and accordingly TSF comparison and update is done 1'b0: no P2P beacons allowed, only BSS beacons allowed 1'b1: P2P/BSS beacons allowed
28	SIM_IBSS_BSS_BEACON_PARS_EN	when this bit is set then the system is indicating that there is IBSS/BSS co-existance so the beacon parsing checks whether received beacon is for IBSS or BSS mode and accordingly TSF comparison and update is done 1'b1: IBSS/BSS beacons allowed 1'b0: no IBSS beacons allowed only BSS beacons allowed NOTE: when bit[29:28] = 2'b11 P2P/IBSS/BSS beacons allowed
27	BEACON_CFP_AWARENESS_ENABLE	When set, and beacon processing is enabled, the beacon will also be evaluated for the presence of the CF parameter set. If present, the CFP remaining duration period is extracted and the NAV is set to that value.
26:16	ASSOCIATION_ID	The association id of the station. This field is used to know which tim bit from the TIM element is related to this station

WCSS_A_RXP_BEACON_TSF_TIM_EXTRACT_CTRL (cont.)

Bits	Name	Description
15:8	CFG_MPR_SSID_LENGTH	When 'enable_beacon_ssid_check' is set and 'enable_beacon_partial_ssid_check_ok' is NOT set, RXP will check the SSID length field of the received beacon. That length field will need to have the same value as this register, otherwise the beacon filter will indicate a mismatch. When 'enable_beacon_ssid_check' is set and 'enable_beacon_partial_ssid_check_ok' is set as well, this length field indicates how many sequential bytes in a row of the beacon SSID field, starting from the first byte of the SSID field received, will have to match the programmed SSID bytes, in order for the beacon to pass the filter. Note that in this case, the SSID length field itself does not necessarily need to match, except that the length field in the beacon needs to be larger than the value programmed here.
5	ENABLE_BEACON_SSID_CHECK	When set, the tsf update (based on IBSS or BSS rules) or tim field extraction will only happen if the incoming beacon has an SSID 3 field which matches the contents of the 'SSID_field' register, and 'SSID_length' field in this register
2	DISABLE_BEACON_DTIM_PARSING	When set, RXP will not generate wakeup pulse if broadcast packets are pending
1	ENABLE_BEACON_TIM_PARSING	When set, beacon tim field extraction and TIM field interrupt pulse generation is enabled
0	ENABLE_BEACON_TSF_PARSING	When set, beacon timestamp extraction and TSF update are enabled

0x0A080A60 WCSS_A_RXP_TSF_COMPENSATION1_VALUE**Type:** RW**Reset State:** 0x00000000

This register contains offset values to add the received TSF value to compensate for the delay in the PHY receive chain

WCSS_A_RXP_TSF_COMPENSATION1_VALUE

Bits	Name	Description
31:24	TSF_11B_11MBPS_COMP_VALUE	The TSF compensation value when the beacon is received at a 11b rate of 11Mbps
23:16	TSF_11B_5_5MBPS_COMP_VALUE	The TSF compensation value when the beacon is received at a 11b rate of 5.5Mbps
15:8	TSF_11B_2MBPS_COMP_VALUE	The TSF compensation value when the beacon is received at a 11b rate of 2Mbps
7:0	TSF_11B_1MBPS_COMP_VALUE	The TSF compensation value when the beacon is received at a 11b rate of 1Mbps

0x0A080A64 WCSS_A_RXP_TSF_COMPENSATION2_VALUE**Type:** RW**Reset State:** 0x00000000

This register contains offset values to add the received TSF value to compensate for the delay in the PHY receive chain

WCSS_A_RXP_TSF_COMPENSATION2_VALUE

Bits	Name	Description
23:16	TSF_11AG_OTHER_RATE_COMP_VALUE	The TSF compensation value when the beacon is received at any other 11a or 11g rate
15:8	TSF_11AG_54MBPS_COMP_VALUE	The TSF compensation value when the beacon is received at an 11a or 11g rate of 54Mbps
7:0	TSF_11AG_6MBPS_COMP_VALUE	The TSF compensation value when the beacon is received at an 11a or 11g rate of 6Mbps

0x0A080A68 WCSS_A_RXP_SSID_VALUES_B0_B3**Type:** RW**Reset State:** 0x00000000

This register contains SSID bytes 0 - 3

WCSS_A_RXP_SSID_VALUES_B0_B3

Bits	Name	Description
31:0	SSID_VALUES_BYTES_B0_B3	The SSID bytes, in LE format

0x0A080A6C WCSS_A_RXP_SSID_VALUES_B4_B7**Type:** RW**Reset State:** 0x00000000

This register contains SSID bytes 4 - 7

WCSS_A_RXP_SSID_VALUES_B4_B7

Bits	Name	Description
31:0	SSID_VALUES_BYTES_B4_B7	The SSID bytes, in LE format

0x0A080A70 WCSS_A_RXP_SSID_VALUES_B8_B11**Type:** RW**Reset State:** 0x00000000

This register contains SSID bytes 8 - 11

WCSS_A_RXP_SSID_VALUES_B8_B11

Bits	Name	Description
31:0	SSID_VALUES_BYTES_B8_11	The SSID bytes, in LE format

0x0A080A74 WCSS_A_RXP_SSID_VALUES_B12_B15**Type:** RW**Reset State:** 0x00000000

This register contains SSID bytes 12 - 15

WCSS_A_RXP_SSID_VALUES_B12_B15

Bits	Name	Description
31:0	SSID_VALUES_BYTES_B12_15	The SSID bytes, in LE format

0x0A080A78 WCSS_A_RXP_SSID_VALUES_B16_B19**Type:** RW**Reset State:** 0x00000000

This register contains SSID bytes 16 - 19

WCSS_A_RXP_SSID_VALUES_B16_B19

Bits	Name	Description
31:0	SSID_VALUES_BYTES_B16_19	The SSID bytes, in LE format

0x0A080A7C WCSS_A_RXP_SSID_VALUES_B20_B23**Type:** RW**Reset State:** 0x00000000

This register contains SSID bytes 20 - 23

WCSS_A_RXP_SSID_VALUES_B20_B23

Bits	Name	Description
31:0	SSID_VALUES_BYTES_B20_23	The SSID bytes, in LE format

0x0A080A80 WCSS_A_RXP_SSID_VALUES_B24_B27

Type: RW

Reset State: 0x00000000

This register contains SSID bytes 24 - 27

WCSS_A_RXP_SSID_VALUES_B24_B27

Bits	Name	Description
31:0	SSID_VALUES_BYTES_B24_27	The SSID bytes, in LE format

0x0A080A84 WCSS_A_RXP_SSID_VALUES_B28_B31

Type: RW

Reset State: 0x00000000

This register contains SSID bytes 28 - 31

WCSS_A_RXP_SSID_VALUES_B28_B31

Bits	Name	Description
31:0	SSID_VALUES_BYTES_B28_31	The SSID bytes, in LE format

0x0A080A88 WCSS_A_RXP_BEACON_ADDR2_LOW_FIELD

Type: RW

Reset State: 0x00000000

This register contains the lowest 4 bytes of the address 2 field of the beacon, in case a match function is enabled

WCSS_A_RXP_BEACON_ADDR2_LOW_FIELD

Bits	Name	Description
31:0	BEACON_ADDR2_LSB_BYTES	The 4 LS Bytes of the addr2, LE format

0x0A080A8C WCSS_A_RXP_BEACON_ADDR2_HIGH_FIELD**Type:** RW**Reset State:** 0x00000000

This register contains the highest 2 bytes of the address 2 field of the beacon, in case a match function is enabled

WCSS_A_RXP_BEACON_ADDR2_HIGH_FIELD

Bits	Name	Description
15:0	BEACON_ADDR2_MSB_BYTES	The 2 MS Bytes of the addr2, LE format

0x0A080A90 WCSS_A_RXP_BEACON_ADDR3_LOW_FIELD**Type:** RW**Reset State:** 0x00000000

This register contains the lowest 4 bytes of the address 3 field of the beacon, in case a match function is enabled

WCSS_A_RXP_BEACON_ADDR3_LOW_FIELD

Bits	Name	Description
31:0	BEACON_ADDR3_LSB_BYTES	The 4 LS Bytes of the addr3, LE format

0x0A080A94 WCSS_A_RXP_BEACON_ADDR3_HIGH_FIELD**Type:** RW**Reset State:** 0x00000000

This register contains the highest 2 bytes of the address 3 field of the beacon, in case a match function is enabled

WCSS_A_RXP_BEACON_ADDR3_HIGH_FIELD

Bits	Name	Description
15:0	BEACON_ADDR3_MSB_BYTES	The 2 MS Bytes of the addr3, LE format

0x0A080A98 WCSS_A_RXP_FLT_DUPL_CNT**Type:** R**Reset State:** 0x00000000

Number of frames that got dropped at the rxp filter because the RPE indicated to the RXP filter that the frame was a duplicate

WCSS_A_RXP_FLT_DUPL_CNT

Bits	Name	Description
7:0	FLT_DUPL_DROPPED_COUNT	number of frames dropped

0x0A080A9C WCSS_A_RXP_FLT_RPE_HANDSHAKE_TIMEOUT_CNT**Type:** R**Reset State:** 0x00000000

Number of times that the handshake between the rxp filter and the RPE timedout

WCSS_A_RXP_FLT_RPE_HANDSHAKE_TIMEOUT_CNT

Bits	Name	Description
19:16	DMA_RPE_TIMEOUT_COUNT	number of timeouts that occurred on the RXP DMA <-> RPE interface
3:0	FLT_RPE_TIMEOUT_COUNT	number of timeouts that occurred on the RXP Filter <-> RPE interface

0x0A080B00 WCSS_A_RXP_QID_MAPPING0**Type:** RW**Reset State:** 0x00000008

Controls how incoming data flows get mapped to a qid

WCSS_A_RXP_QID_MAPPING0

Bits	Name	Description
4:0	MAN_AND_DATA_QID	When the incoming frame is a management frame or plain (NON QoS) data frame, the contents of this register represents the qid assigned to this type of frames

0x0A080B04 WCSS_A_RXP_QID_MAPPING1**Type:** RW**Reset State:** 0x03020100

Controls how incoming data flows get mapped to a qid

WCSS_A_RXP_QID_MAPPING1

Bits	Name	Description
28:24	QOS_TID3_QID	When the incoming frame is a QoS data frame with TID set to 3, the contents of this register represents the qid assigned to this type of frames
20:16	QOS_TID2_QID	When the incoming frame is a QoS data frame with TID set to 2, the contents of this register represents the qid assigned to this type of frames
12:8	QOS_TID1_QID	When the incoming frame is a QoS data frame with TID set to 1, the contents of this register represents the qid assigned to this type of frames
4:0	QOS_TID0_QID	When the incoming frame is a QoS data frame with TID set to 0, the contents of this register represents the qid assigned to this type of frames

0x0A080B08 WCSS_A_RXP_QID_MAPPING2**Type:** RW**Reset State:** 0x07060504

Controls how incoming data flows get mapped to a qid

WCSS_A_RXP_QID_MAPPING2

Bits	Name	Description
28:24	QOS_TID7_QID	When the incoming frame is a QoS data frame with TID set to 7, the contents of this register represents the qid assigned to this type of frames
20:16	QOS_TID6_QID	When the incoming frame is a QoS data frame with TID set to 6, the contents of this register represents the qid assigned to this type of frames
12:8	QOS_TID5_QID	When the incoming frame is a QoS data frame with TID set to 5, the contents of this register represents the qid assigned to this type of frames
4:0	QOS_TID4_QID	When the incoming frame is a QoS data frame with TID set to 4, the contents of this register represents the qid assigned to this type of frames

0x0A080B0C WCSS_A_RXP_QID_MAPPING3**Type:** RW**Reset State:** 0x00000000

Controls how incoming data flows get mapped to a qid

WCSS_A_RXP_QID_MAPPING3

Bits	Name	Description
28:24	QOS_TID11_QID	When the incoming frame is a QoS data frame with TID set to 11, the contents of this register represents the qid assigned to this type of frames
20:16	QOS_TID10_QID	When the incoming frame is a QoS data frame with TID set to 10, the contents of this register represents the qid assigned to this type of frames
12:8	QOS_TID9_QID	When the incoming frame is a QoS data frame with TID set to 9, the contents of this register represents the qid assigned to this type of frames
4:0	QOS_TID8_QID	When the incoming frame is a QoS data frame with TID set to 8, the contents of this register represents the qid assigned to this type of frames

0x0A080B10 WCSS_A_RXP_QID_MAPPING4**Type:** RW**Reset State:** 0x00000000

Controls how incoming data flows get mapped to a qid

WCSS_A_RXP_QID_MAPPING4

Bits	Name	Description
28:24	QOS_TID15_QID	When the incoming frame is a QoS data frame with TID set to 15, the contents of this register represents the qid assigned to this type of frames
20:16	QOS_TID14_QID	When the incoming frame is a QoS data frame with TID set to 14, the contents of this register represents the qid assigned to this type of frames
12:8	QOS_TID13_QID	When the incoming frame is a QoS data frame with TID set to 13, the contents of this register represents the qid assigned to this type of frames
4:0	QOS_TID12_QID	When the incoming frame is a QoS data frame with TID set to 12, the contents of this register represents the qid assigned to this type of frames

0x0A080B14 WCSS_A_RXP_PUSH_WQ_CTRL**Type:** RW**Reset State:** 0x02020202

Controls how frames that fall in the unicast other of multicast 'other categories' get routed through the system

WCSS_A_RXP_PUSH_WQ_CTRL

Bits	Name	Description
31:24	ADDR2_255_CATEGORY_PUSH_WQ_INDEX	When the incoming frame will end up with staid 255 for addr 2, the frame will be routed to this WQ.
23:16	ADDR2_254_CATEGORY_PUSH_WQ_INDEX	When the incoming frame will end up with staid 254 for addr 2, the frame will be routed to this WQ.
15:8	ADDR1_255_CATEGORY_PUSH_WQ_INDEX	When the incoming frame will end up with staid 255 for addr 1 (multicast other), the frame will be routed to this WQ.
7:0	ADDR1_254_CATEGORY_PUSH_WQ_INDEX	When the incoming frame will end up with staid 254 for addr 1 (unicast other), the frame will be routed to this WQ.

0x0A080B18 WCSS_A_RXP_PUSH_WQ_CTRL2**Type:** RW**Reset State:** 0x00020202

Controls how frames that fall in the unicast other of multicast 'other categories' get routed through the system

WCSS_A_RXP_PUSH_WQ_CTRL2

Bits	Name	Description
23:16	UNFILTERED_FRAME_PUSH_WQ_INDEX	When the incoming frame will does not get processed in any address filtering, all three staid indexes are invalid and set to 253, the frame will be routed to this WQ
15:8	ADDR3_255_CATEGORY_PUSH_WQ_INDEX	When the incoming frame will end up with staid 255 for addr 3, the frame will be routed to this WQ.
7:0	ADDR3_254_CATEGORY_PUSH_WQ_INDEX	When the incoming frame will end up with staid 254 for addr 3, the frame will be routed to this WQ.

0x0A080B1C WCSS_A_RXP_EXTRA_FRAME_FILTER_CONTROL**Type:** RW**Reset State:** 0x000001DD

additional control for how frames are filtered

WCSS_A_RXP_EXTRA_FRAME_FILTER_CONTROL

Bits	Name	Description
15:9	RESERVED	Reserved for future filter functions
8	BLOCK_FRAMES_WITH_HT_CONTROL_FIELD	When set, frames that have a HT-control field present as specified in 11n are blocked from being received
7	BLOCK_MULTI_TID_BLOCK_ACK_REQ	When set, the multi TID block ACK Request variant as specified in 11n is blocked from being received
6	BLOCK_RESERVED_BLOCK_ACK_REQ	When set, the reserved block ACK Request variant as specified in 11n is blocked from being received
5	BLOCK_COMPRESSED_BLOCK_ACK_REQ	When set, the compressed block ACK Request variant as specified in 11n is blocked from being received
4	BLOCK_BASIC_BLOCK_ACK_REQ	When set, the basic block ACK Request variant as specified in 11n is blocked from being received
3	BLOCK_MULTI_TID_BLOCK_ACK	When set, the multi TID block ACK variant as specified in 11n is blocked from being received
2	BLOCK_RESERVED_BLOCK_ACK	When set, the reserved block ACK variant as specified in 11n is blocked from being received
1	BLOCK_COMPRESSED_BLOCK_ACK	When set, the compressed block ACK variant as specified in 11n is blocked from being received
0	BLOCK_BASIC_BLOCK_ACK	When set, the basic block ACK variant as specified in 11n is blocked from being received

0x0A080B20 WCSS_A_RXP_RXP_GROUPED_INTERRUPT_ENABLE**Type:** RW**Reset State:** 0x00000000

interrupt Enables for RXP errors

WCSS_A_RXP_RXP_GROUPED_INTERRUPT_ENABLE

Bits	Name	Description
27	DMA_GAM_ERROR	Write 1 to enable this interrupt, so that whenever there is error on the AHB bus (gam error) an interrupt will be generated to firmware.
26	RXP_SYSTEM_OUT_OF_MEMORY_FIRMWARE	Write 1 to enable this interrupt, so that whenever system runs out of memory this interrupt will be set. this is duplicate the one in bit 13 goes to host and this interrupt goes to firware only
25	P2P_BEACON_CTWING_DETECT_INT	when set enables this interrupt The NOA CTWin value is compared with the value programmed in p2p_parsing_control2 register if there is any change then this interrupt is generated Writing 1 to this bit will clear this interrupt this interrupt goes to firmware only
24	P2P_BEACON_OPPPS_CHNG_DETECT_INT_EN	when set enables this interrupt The NOA OppPS value is compared with the value programmed in p2p_parsing_control2 register if there is any change then this interrupt is generated this interrupt goes to firmware only --NOTE---- This interrupt is overloaded with NDPA received interrupt for ECO as partial fix for addr3 issue in VHTCB, to enable NDPA interrupt set bit 17 in spare_and_clkgate_dis register.
23	P2P_TIM_BIT_CLEAR_INT	When set, p2p beacon has tim bit clear this interrupt goes to firmware only
22	P2P_TIM_BIT_SET_INT	When set, tim bit set in p2p beacon this interrupt goes to firmware only
21	P2P_TIM_BEACON_CRC_ERROR_INT	When set, p2p beacon has CRC error this interrupt goes to firmware only
20	P2P_TIM_BASED_BROADCAST_FRAMES_PENDING_INTERRUPT	When set, indicates tim bit indicates broadcast frames are pending from a p2p beacon this interrupt goes to firmware only
19	P2P_BEACON_NOA_ABSENT_INT_EN	When set, P2P beacon which has no NOA element interrupt will be enabled this interrupt goes to firmware only
18	P2P_BEACON_NOA_INDEX_EQUAL_ONE_DETECT_INT_EN	when set, P2P beacon NOA index value equal to one interrupt will be enabled this interrupt goes to firmware only
17	P2P_BEACON_NOA_INDEX_CHANGE_DETECT_INT_EN	The NOA, index feild is compared with the index value programmed in p2p_parsing_control register if there is any change the this interrupt is generated When set, P2p Beacon NOA index value change detect interrupt is enabled this interrupt goes to firmware only
16	RXP_RADAR_INTERRUPT	When set, radar detect interrupt will be generated

WCSS_A_RXP_RXP_GROUPED_INTERRUPT_ENABLE (cont.)

Bits	Name	Description
15	RXP_MORE_DATA_SET_INTERRUPT	When set, and a frame is received that passes the filter, is a management, data or QoS data frame and the more data bit in the frame is SET an interrupt pulse will be generated. That status info of the corresponding frame is captured in the other fields more_data_int_control register. this interrupt goes to host only
14	RXP_RMF_FRAME_LEN_ERROR	Write 1 to enable this interrupt, whenever we receive a RMF BC/MC frame with frame less than 48bytes(which is minimum) then this interrupt is generated. this interrupt goes to host only
13	RXP_SYSTEM_OUT_OF_MEMORY	Write 1 to enable this interrupt, so that whenever system runs out of memory this interrupt will be set. this interrupt goes to host only
12	RXP_MORE_DATA_CLEAR_INTERRUPT	When set, and a frame is received that passes the filter, is a management, data or QoS data frame and the more data bit in the frame is NOT set an interrupt pulse will be generated. That status info of the corresponding frame is captured in the other fields more_data_int_control register. this interrupt goes to host only
11	RXP_ADDR2_MISS_INTERRUPT	When set, and a frame is received that has an addr2 miss (and addr 1 is a hit), an 'addr2 miss interrupt will be generated. this interrupt goes to host only
10	RXP_MISSED_EOSP_INT	When set, interrupt is generated when at least one other frame with EOSP got received while the first EOSP frame reception got reported this interrupt goes to firmware only
9	RXP_EOSP_INT	When set, the EOSP interrupt will get generated if eossp frame is received this interrupt goes to firmware only
8	TIM_BIT_CLEAR_INT	When set, beacon has tim bit clear this interrupt goes to firmware only
7	TIM_BIT_SET_INT	When set, tim bit set in beacon this interrupt goes to firmware only
6	TIM_BEACON_CRC_ERROR_INTERRUPT	When set, beacon has CRC error this interrupt goes to firmware only
5	TIM_BASED_BROADCAST_FRAMES_PENDING_INT	When set, indicates tim bit indicates broadcast frames are pending this interrupt goes to host only
4	ASYNC_FIFO_FULL_ERROR_INTERRUPT_ENABLE	When set, and the async fifo gets full, an interrupt will be generated this interrupt goes to host only
3	GET_BMU_ERR_INTERRUPT_ENABLE	When set, and the interaction with the bmu causes an error, an interrupt will be generated this interrupt goes to host only
2	INVALID_RATE_INDEX_INTERRUPT_ENABLE	When set, and an invalid rate index is generated based on information from the PHY, an interrupt will be generated this interrupt goes to host only
1	RPE_DUPL_INTERFACE_TIMEOUT_INTERRUPT_ENABLE	When set, the rpe duplicate detect interface handshake timeout error interrupt generation is enabled this interrupt goes to host only
0	RPE_BITMAP_INTERFACE_TIMEOUT_INTERRUPT_ENABLE	When set, the rpe bitmap interface handshake timeout error interrupt generation is enabled this interrupt goes to host only

0x0A080B24 WCSS_A_RXP_RXP_GROUPED_INTERRUPT_STATUS**Type:** RW**Reset State:** 0x00000000

RXP combined interrupt status, writing 1 to any bit will clear the corresponding interrupt

WCSS_A_RXP_RXP_GROUPED_INTERRUPT_STATUS

Bits	Name	Description
31	DROP_RPE_FIFO_FULL	whenever a frame is dropped due to system out of memory this bit will indicate if there was a rpe fifo full that caused this packet drop 1: if syste_out_of memory is set and this bit is one packet dropped due RPE fifo full 0: if syste_out_of memory is set and this bit is zero then there may BMU BD/PDU issue Writing 1 to this bit will clear this status
30	PACKET_PRIORITY	whenever a frame is dropped due to system out of memory this bit will indicate what was the priority of the frame that was dropped 1: high priority frame dropped 0: Low priority frame dropped Writing 1 to this bit will clear this status
27	DMA_GAM_ERROR	when set indicates there is error on the AHB bus (gam error) writing 1 to this bit will clear the interrupt
26	RXP_SYSTEM_OUT_OF_MEMORY_FIRMWARE_INT	when set it indicates system is running out of memory, which allows software to indicate to AP to back off on sending frames by sending frames with PM bit set Writing 1 to this bit will clear this interrupt this interrupt goes to firmware only
25	P2P_BEACON_CTWING_DETECT_INT	The NOA CTWin value is compared with the value programmed in p2p_parsing_control2 register if there is any change then this interrupt is generated Writing 1 to this bit will clear this interrupt this interrupt goes to firmware only
24	P2P_BEACON_OPPPS_CHNG_DETECT_INT	The NOA OppPS value is compared with the value programmed in p2p_parsing_control2 register if there is any change then this interrupt is generated this interrupt goes to firmware only
23	P2P_TIM_BIT_CLEAR_INT	When set, p2p beacon has tim bit clear Writing 1 to this bit will clear this interrupt this interrupt goes to firmware only
22	P2P_TIM_BIT_SET_INT	When set, tim bit set in p2p beacon Writing 1 to this bit will clear this interrupt this interrupt goes to firmware only
21	P2P_TIM_BEACON_CRC_ERROR_INT	When set, p2p beacon has CRC error Writing 1 to this bit will clear this interrupt this interrupt goes to firmware only
20	P2P_TIM_BASED_BROADCAST_FRAMES_PENDING_INTERRUPT	When set, indicates tim bit in p2p beacon indicates broadcast frames are pending Writing 1 to this bit will clear this interrupt this interrupt goes to firmware only
19	P2P_BEACON_NOA_ABSENT_INT	When set, P2P beacon which has no NOA element interrupt is generated Writing 1 to this bit will clear this interrupt this interrupt goes to firmware only
18	P2P_BEACON_NOA_INDEX_EQUAL_ONE_DETECT_INT	when set, indicates P2P beacon NOA index value equal to one interrupt is generated Writing 1 to this bit will clear this interrupt this interrupt goes to firmware only

WCSS_A_RXP_RXP_GROUPED_INTERRUPT_STATUS (cont.)

Bits	Name	Description
17	P2P_BEACON_NOA_IDX_CHNG_DETECT_INT	The NOA, index feild is compared with the index value programmed in p2p_parsing_control register if there is any change the this interrupt is generated Writing 1 to this bit will clear this interrupt this interrupt goes to firmware only
16	RXP_RADAR_INTERRUPT	When set, radar detect interrupt will be generated Writing 1 to this bit will clear this interrupt
15	RXP_MORE_DATA_SET_INTERRUPT	When set, and a frame is received that passes the filter, is a management, data or QoS data frame and the more data bit in the frame is SET an interrupt pulse will be generated. That status info of the corresponding frame is captured in the other fields more_data_int_control register. Writing 1 to this bit will clear this interrupt this interrupt goes to host only
14	RXP_RMF_FRAME_LEN_ERR_INT	When set it indicates that we have received a RMF BC/MC frame with frame less than 48bytes(which is minimum) then this interrupt is generated. Writing 1 to this bit will clear this interrupt this interrupt goes to host only
13	RXP_SYSTEM_OUT_OF_MEMORY_INT	when set it indicates system is running out of memory, which allows software to indicate to AP to back off on sending frmaes by sending frames with PM bit set Writing 1 to this bit will clear this interrupt this interrupt goes to host only
12	RXP_MORE_DATA_CLEAR_INTERRUPT	When set, and a frame is received that passes the filter, is a management, data or QoS data frame and the more data bit in the frame is NOT set an interrupt pulse will be generated. That status info of the corresponding frame is captured in the other fields more_data_int_control register. Writing 1 to this bit will clear this interrupt this interrupt goes to host only
11	RXP_ADDR2_MISS_INTERRUPT	When set, and a frame is received that has an addr2 miss (and addr 1 is a hit), an 'addr2 miss interrupt will be generated. Writing 1 to this bit will clear this interrupt this interrupt goes to host only
10	RXP_MISSED_EOSP_INT	When set, interrupt is generated when at least one other frame with EOSP got received while the first EOSP frame reception got reported Writing 1 to this bit will clear this interrupt this interrupt goes to firmware only
9	RXP_EOSP_INT	When set, the EOSP interrupt will get generated if eosps frame is received Writing 1 to this bit will clear this interrupt this interrupt goes to firmware only
8	TIM_BIT_CLEAR_INT	When set, beacon has tim bit clear Writing 1 to this bit will clear this interrupt this interrupt goes to firmware only
7	TIM_BIT_SET_INT	When set, tim bit set in beacon Writing 1 to this bit will clear this interrupt this interrupt goes to firmware only
6	TIM_BEACON_CRC_ERROR_INT	When set, beacon has CRC error Writing 1 to this bit will clear this interrupt this interrupt goes to firmware only
5	TIM_BASED_BROADCAST_FRAMES_PENDING_INT	When set, indicates tim bit indicates broadcast frames are pending Writing 1 to this bit will clear this interrupt this interrupt goes to firmware only

WCSS_A_RXP_RXP_GROUPED_INTERRUPT_STATUS (cont.)

Bits	Name	Description
4	ASYNC_FIFO_FULL_ERR_INT	When set, and the async fifo got full Writing 1 to this bit will clear this interrupt this interrupt goes to host only
3	GET_BMU_ERR_INT	When set, the interaction with the bmu caused an error Writing 1 to this bit will clear this interrupt this interrupt goes to host only
2	INVALID_RATE_INDEX_INT	When set, an invalid rate index is generated based on information from the PHY Writing 1 to this bit will clear this interrupt this interrupt goes to host only
1	RPE_DUPL_INTERFACE_TIMEOUT_INT	When set, the rpe duplicate detect interface handshake timeout error has been detected Writing 1 to this bit will clear this interrupt this interrupt goes to host only
0	RPE_BITMAP_INTERFACE_TIMEOUT_INT	When set, the rpe bitmap interface handshake timeout error has been detected Writing 1 to this bit will clear this interrupt this interrupt goes to host only

0x0A080B28 WCSS_A_RXP_DEFAULT_HW_INSERT_DPU_PARAM**Type:** RW**Reset State:** 0x00000000

DPU related parameters to be put in the BD when the "bd_dpu_softmacfields_update_enable" is set and the frame that is received does not cause a hit in the filter, but falls in the unicast/multicast others category, and is still allowed to come in.

WCSS_A_RXP_DEFAULT_HW_INSERT_DPU_PARAM

Bits	Name	Description
14:7	DPU_ROUTING_FLAG	The setting for the DPU_routing_flag bits in the BD
6:4	DPU_TAG	The setting for the DPU TAG bits in the BD
3:1	RX_KEY_ID_BITS	The setting for the RX KEY ID bits in the BD
0	NE_BIT	The setting for the NE bit in the BD

0x0A080B44 WCSS_A_RXP_PSPOLL_NAV_VALUE**Type:** RW**Reset State:** 0x00000000

indicates the NAV value that will be passed on to the MTU when a PSPOLL frame is received

WCSS_A_RXP_PSPOLL_NAV_VALUE

Bits	Name	Description
7:0	PSPOLL_NAV_VALUE	The NAV value that will be passed on to the MTU when a PSPOLL frame is received

0x0A080B48 WCSS_A_RXP_PMU_INTERFACE_CONTROL**Type:** RW**Reset State:** 0x00000000

Controls the PMU interfacing for Beacon Mode Power Save.

WCSS_A_RXP_PMU_INTERFACE_CONTROL

Bits	Name	Description
10	PMU_SLEEP_PULSE_GENERATED	When set, a sleep pulse has been generated to the PMU Write 0 to clear this bit
9	WAKEUP_ON_BC_TIM_SET_PULSE_GENERATED	When set, a wakeup pulse has been generated to the PMU indicating a beacon is received with the broadcast tim bit set Write 0 to clear this bit
8	WAKEUP_ON_STA_TIM_SET_PULSE_GENERATED	When set, a wakeup pulse has been generated to the PMU indicating a beacon is received with the STA tim bit set Write 0 to clear this bit
3	SLEEP_ON_BC_TIM_CLEAR	When set, a sleep pulse will be generated to the PMU when a beacon is received with the broadcast tim bit clear NOTE: when both 'pmu_sleep_on_sta_tim_clear' and 'sleep_on_bc_tim_clear' are set, both requirements need to be met in order to generate the sleep pulse
2	PMU_SLEEP_ON_STA_TIM_CLEAR	When set, a sleep pulse will be generated to the PMU when a beacon is received with the sta tim bit clear
1	WAKEUP_ON_SBC_TIM_SET	When set, a wakeup pulse will be generated to the PMU when a beacon is received with the broadcast tim bit set
0	WAKEUP_ON_STA_TIM_SET	When set, a wakeup pulse will be generated to the PMU when a beacon is received with the STA tim bit set

0x0A080B50 WCSS_A_RXP_RATE_INDEX_HISTORY**Type:** R**Reset State:** 0x00000000

Gives an overview of the rate indexes of the last 2 frames received

WCSS_A_RXP_RATE_INDEX_HISTORY

Bits	Name	Description
24:16	PREVIOUS_RECEIVED_RATE_INDEX	The rate index of the previous frame received
8:0	LAST_RECEIVED_RATE_INDEX	The rate index of the last frame received

0x0A080B58 WCSS_A_RXP_ADDR2_MISS_INFO**Type:** R**Reset State:** 0x00000000

The lowest 32 bits of the addr 2 received that caused an addr 2 miss interrupt

WCSS_A_RXP_ADDR2_MISS_INFO

Bits	Name	Description
31:0	ADDR2_BIT31_0	The lowest 32 bits of the addr 2 received

0x0A080B5C WCSS_A_RXP_ADDR2_MISS_CTRL**Type:** R**Reset State:** 0x00000000

Addr 2 miss interrupt control and status

WCSS_A_RXP_ADDR2_MISS_CTRL

Bits	Name	Description
23:16	ADDR1_INDEX	The corresponding addr 1 index of the frame received
15:0	ADDR2_BIT47_32	The upper 16 bits of the addr 2 received

0x0A080B60 WCSS_A_RXP_EOSP_INT_CONTROL**Type:** RW**Reset State:** 0x00000000

EOSP interrupt control and status

WCSS_A_RXP_EOSP_INT_CONTROL

Bits	Name	Description
31	EOSP_INT_ENABLE	When set, and a frame is received that passes the filter and the EOSP bit in the QoS control field is set, an interrupt pulse will be generated. That status info of the corresponding frame is captured in the other fields in this register.
30	EOSP_INT_STATUS	When set, the EOSP interrupt got generated
29	EOSP_INT_MISSED	When set, at least one other frame with EOSP set got received while the first EOSP frame reception got reported
27:24	TID	The tid of the qos control field
23:16	ADDR3_INDEX	The Addr 3 index of the frame received
15:8	ADDR2_INDEX	The Addr 2 index of the frame received
7:0	ADDR1_INDEX	The Addr 1 index of the frame received

0x0A080B64 WCSS_A_RXP_MORE_DATA_CLEAR_INT_CONTROL**Type:** RW**Reset State:** 0x00000000

More data bit clear interrupt control and status

WCSS_A_RXP_MORE_DATA_CLEAR_INT_CONTROL

Bits	Name	Description
31	MORE_DATA_CLEAR_INT_ENABLE	When set, and a frame is received that passes the filter, is a management, data or QoS data frame and the more data bit in the frame is NOT set an interrupt pulse will be generated. That status info of the corresponding frame is captured in the other fields in this register.
30	MORE_DATA_CLEAR_INT_STATUS	When set, the more data bit clear interrupt got generated
29	MORE_DATA_INT_MISSED	When set, at least one other frame with more data bit clear got received before SW got to process the interrupt. while the first EOSP frame reception got reported

WCSS_A_RXP_MORE_DATA_CLEAR_INT_CONTROL (cont.)

Bits	Name	Description
28	QOS_FRAME	when set indicates it is a qos frame
27:24	TID	tid of qos control feild
23:16	ADDR3_INDEX	The Addr 3 index of the frame received
15:8	ADDR2_INDEX	The Addr 2 index of the frame received
7:0	ADDR1_INDEX	The Addr 1 index of the frame received

0x0A080B6C WCSS_A_RXP_CONFIG4**Type:** RW**Reset State:** 0x00000000

It contains RXP Configuration registers.

WCSS_A_RXP_CONFIG4

Bits	Name	Description
28	WLAN_RX_GRANT	This bit gives status of wlan_rx_grant. Write 1 to this bit to clear it 0: No rx grant generated 1: rx grant is generated
27	WLAN_RX_GRANT_ENAB LE	This bit is used to enable the wlan_rx grant feature 0: disable the WLAN_RX_Grants control of WLAN RX 1: enable the WLAN_RX_Grants control of WLAN RX
26:11	TID_PRIORITY	This bit gives priority for frames on a tid basis this is used in support for Better Qos By default TID 6 and 7 have high priority 1: High priority 0: Low priority
10	MANAGEMENT_PRIORITY	This bit gives priority for management frames this is used in support for Better Qos 1: High priority 0: Low priority
9	NONQOS_PRIORITY	This bit gives priority for non qos frames this is used in support for Better Qos 1: High priority 0: Low priority
8:1	UNKNOWN_MC_STAID	when bit 0 is set then that means we are continuing address2 look up for an unknown multicast so the staid assigned for these types of packet should be set here

WCSS_A_RXP_CONFIG4 (cont.)

Bits	Name	Description
0	CFG_FLT_ADDR1_BC_ENTRY_PRESENT	When this bit is set it indicates that a broadcast entry is present in address 1 when it is set it allows RXP to accept unknown multicast packets and continue address2 lookup

0x0A080B74 WCSS_A_RXP_BEACON_P2P_CLIENT_ADDR_LOW_FIELD**Type:** RW**Reset State:** 0x00000000

This register contains the lowest 4 bytes of the address 2 field of the p2p beacon, in case a match function is enabled

WCSS_A_RXP_BEACON_P2P_CLIENT_ADDR_LOW_FIELD

Bits	Name	Description
31:0	BEACON_P2P_CLIENT_ADDR_LSB_BYTES	The 4 LS Bytes of the addr2, LE format

0x0A080B78 WCSS_A_RXP_BEACON_P2P_CLIENT_ADDR_HIGH_FIELD**Type:** RW**Reset State:** 0x00000000

This register contains the highest 2 bytes of the address 3 field of the beacon, in case a match function is enabled

WCSS_A_RXP_BEACON_P2P_CLIENT_ADDR_HIGH_FIELD

Bits	Name	Description
15:0	BEACON_P2P_CLIENT_ADDR_MSB_BYTES	The 2 MS Bytes of the addr2, LE format

0x0A080B7C WCSS_A_RXP_P2P_BEACON_PARSE_CONTROL**Type:** RW**Reset State:** 0x09000000

This register contains all the configurations necessary for handling p2p beacon

WCSS_A_RXP_P2P_BEACON_PARSE_CONTROL

Bits	Name	Description
31:24	OUI_TYPE_VALUE	This is the OUI type value in P2P IE in the beacon. By default it is 09 indicating it is P2P version 1.0
23:16	NOA_INDEX_VALUE	The Index field value of NOA subelement in P2P element in beacon RXP will parse the beacon for this index field and if there is any change in value an interrupt is generated.
15:5	P2P_ASSOCIATION_ID	The association id of the station for p2p beacons. This field is used to know which tim bit from the TIM element is related to this station
2	P2P_BEACON_NOA_PARSING_DIS	as long as pronto is configured to receive p2p beacons noa parsing will be done. NOA parsing can be disabled by writing 1 to this bit

0x0A080B80 WCSS_A_RXP_MORE_DATA_SET_INT_CONTROL**Type:** RW**Reset State:** 0x00000000

More data bit set interrupt related status

WCSS_A_RXP_MORE_DATA_SET_INT_CONTROL

Bits	Name	Description
29	MORE_DATA_INT_MISSED	When set, at least one other frame with more data bit set got received before SW got to process the interrupt.
28	QOS_FRAME	when set indicates it is a qos frame
27:24	TID	tid of qos control field
23:16	ADDR3_INDEX	The Addr 3 index of the frame received
15:8	ADDR2_INDEX	The Addr 2 index of the frame received
7:0	ADDR1_INDEX	The Addr 1 index of the frame received

0x0A080B84 WCSS_A_RXP_RULE1_CNTR_CTRL**Type:** RW**Reset State:** 0x00000000

Has all controls to generate rule1 counter

WCSS_A_RXP_RULE1_CNTR_CTRL

Bits	Name	Description
31:28	RULE1_QID_MASK	Only the bit that are set here are considered for counting when mask is set for any bit the corresponding bit in rule1_qid_en is valid
27:24	RULE1_QID_EN	Only the bit that are set here are considered for counting Program all the qid bit values that needs to be considered for counting The setting of the bits whose corresponding bits in the rule1_qid_mask are set to 0 are ignored.
23:20	RULE1_STAID_MASK	Only the bit that are set here are considered for counting when mask is set for any bit the corresponding bit in rule1_staid_en is valid
19:16	RULE1_STAID_EN	Program all the staid bit values that needs to be considered for counting The setting of the bits whose corresponding bits in the rule1_staid_mask are set to 0 are ignored
15:10	RULE1_TYPESUBTYPE_MASK	Only the bit that are set here are considered for counting when mask is set for any bit the corresponding bit in rule1_typesubtype_en is valid bits[15:14] = FC type bits[13:10] = FC subtype
9:4	RULE1_TYPESUBTYPE_EN	Program all the type/subtype bit values that needs to be considered for counting. The setting of the bits whose corresponding bits in the rule1_typesubtype_mask are set to 0 are ignored. bits[9:8] = FC type bits[7:4] = FC subtype
3	RULE1_HT_PRESENT_MASK	When set, ht present/or not present check is enabled
2	RULE1_HT_PRESENT_EN	program the requirement of HT being present or not. This field is ignored if bit rule1_ht_present_mask is not set.
0	RULE1_CNTR_EN	when set enables the counter

0x0A080B88 WCSS_A_RXP_RULE2_CNTR_CTRL**Type:** RW**Reset State:** 0x00000000

Has all controls to generate rule1 counter

WCSS_A_RXP_RULE2_CNTR_CTRL

Bits	Name	Description
31:28	RULE2_QID_MASK	Only the bit that are set here are considered for counting when mask is set for any bit the corresponding bit in rule2_qid_en is valid
27:24	RULE2_QID_EN	Only the bit that are set here are considered for counting Program all the qid bit values that needs to be considered for counting The setting of the bits whose corresponding bits in the rule2_qid_mask are set to 0 are ignored.
23:20	RULE2_STAID_MASK	Only the bit that are set here are considered for counting when mask is set for any bit the corresponding bit in rule2_staid_en is valid
19:16	RULE2_STAID_EN	Program all the staid bit values that needs to be considered for counting The setting of the bits whose corresponding bits in the rule2_staid_mask are set to 0 are ignored
15:10	RULE2_TYPESUBTYPE_MASK	Only the bit that are set here are considered for counting when mask is set for any bit the corresponding bit in rule2_typesubtype_en is valid bits[15:14] = FC type bits[13:10] = FC subtype
9:4	RULE2_TYPESUBTYPE_EN	Program all the type/subtype bit values that needs to be considered for counting. The setting of the bits whose corresponding bits in the rule2_typesubtype_mask are set to 0 are ignored. bits[9:8] = FC type bits[7:4] = FC subtype
3	RULE2_HT_PRESENT_MASK	When set, ht present/or not present check is enabled
2	RULE2_HT_PRESENT_EN	program the requirement of HT being present or not. This field is ignored if bit rule2_ht_present_mask is not set.
0	RULE2_CNTR_EN	when set enables the counter

0x0A080B8C WCSS_A_RXP_RULE1_CNTR**Type:** R**Reset State:** 0x00000000

Rule 1 counter, if all conditions in rule1_ctrl pass then this counter gets incremented

WCSS_A_RXP_RULE1_CNTR

Bits	Name	Description
7:0	RULE1_CNTR	This counter indicates number of packets received which passes rule 1 control conditions

0x0A080B90 WCSS_A_RXP_RULE2_CNTR**Type:** R**Reset State:** 0x00000000

Rule 2 counter, if all conditions in rule2_ctrl pass then this counter gets incremented

WCSS_A_RXP_RULE2_CNTR

Bits	Name	Description
7:0	RULE2_CNTR	This counter indicates number of packets received which passes rule 2 control conditions

0x0A080B94 WCSS_A_RXP_DROP_DUE_TO_DROP_AT_DMA_CNT**Type:** R**Reset State:** 0x00000000

The number of packets are dropped by RXP due to drop at DMA bit set in type/subtype filter(these are mostly control packets)

WCSS_A_RXP_DROP_DUE_TO_DROP_AT_DMA_CNT

Bits	Name	Description
15:0	DROP_DUE_TO_DROP_DMA_CMD_CNT	The number of packets are dropped by RXP due to drop at DMA bit being set these are usually all control packets.

0x0A080B98 WCSS_A_RXP_RX_TIME_CNTR**Type:** R**Reset State:** 0x00000000

The number of early pkt detect at every usec

WCSS_A_RXP_RX_TIME_CNTR

Bits	Name	Description
23:0	RX_TIME_CNT	The number of usec that have early pkt detect high

0x0A080BA0 WCSS_A_RXP_NDP_PKT_CNTR**Type:** R**Reset State:** 0x00000000

The number of NDP pkts received

WCSS_A_RXP_NDP_PKT_CNTR

Bits	Name	Description
15:0	NDP_PKT_CNT	Number of NDP packets received

0x0A080BA4 WCSS_A_RXP_P2P_BEACON_PARSE_CONTROL2**Type:** RW**Reset State:** 0x00000000

This register contains some more configurations necessary for handling p2p beacon

WCSS_A_RXP_P2P_BEACON_PARSE_CONTROL2

Bits	Name	Description
7	NOA_OPPPS_VALUE	This value is set by SW, set 1 indicates group owner is using opportunistic power save, set to 0 indicates that it is disabled This valuw is compared to that in the beacon any change in this will generate a interrupt

WCSS_A_RXP_P2P_BEACON_PARSE_CONTROL2 (cont.)

Bits	Name	Description
6:0	NOA_CTWIN_VALUE	Client Traffic window of value 0-6 is programmed in this register RXP will parse the beacon for this n CTWin of NOA element and if there is any change in value an interrupt is generated. This value is a period of time in TU after TBTT during which P2P group owner is present

0x0A080BA8 WCSS_A_RXP_BSS2_BEACON_PARSE_CONTROL**Type:** RW**Reset State:** 0x28250000

This register contains all the configurations necessary for handling bss2 beacon

WCSS_A_RXP_BSS2_BEACON_PARSE_CONTROL

Bits	Name	Description
31:24	IE_EID1	These bits can be programmed to any IE(except zero length IEs) in the beacon that needs to be parsed By default it is "Quiet" IE. Also the programmed in this register can not be same as IE_eid0
23:16	IE_EID0	These bits can be programmed to any IE(except zero length IE) in the beacon that needs to be parsed By default it is "Channel Switch" IE. Also the programmed in this register can not be same as IE_eid1
13	IE_EID1_PARSE_ENABLE	SW need to set this bit whenever SW wants to enable the parsing of the IE programmed in bits [31:24]
12	IE_EID0_PARSE_ENABLE	SW need to set this bit whenever SW wants to enable the parsing of the IE programmed in bits [23:16]
10:0	BSS2_ASSOCIATION_ID	The association id of the station for bss2 beacons. This field is used to know which tim bit from the TIM element is related to this station

0x0A080BAC WCSS_A_RXP_BEACON_BSS2_ADDR_LOW_FIELD**Type:** RW**Reset State:** 0x00000000

This register contains the lowest 4 bytes of the address 2 field of the bss2 beacon, in case a match function is enabled

WCSS_A_RXP_BEACON_BSS2_ADDR_LOW_FIELD

Bits	Name	Description
31:0	BEACON_BSS2_CLIENT_ADDR_LSB_BYTES	The 4 LS Bytes of the addr2, LE format

0x0A080BB0 WCSS_A_RXP_BEACON_BSS2_CLIENT_ADDR_HIGH_FIELD**Type:** RW**Reset State:** 0x00000000

This register contains the highest 2 bytes of the address 3 field of the beacon, in case a match function is enabled

WCSS_A_RXP_BEACON_BSS2_CLIENT_ADDR_HIGH_FIELD

Bits	Name	Description
15:0	BEACON_BSS2_CLIENT_ADDR_MSB_BYTES	The 2 MS Bytes of the addr2, LE format

0x0A080BB4 WCSS_A_RXP_RXP_GROUPED2_INTERRUPT_ENABLE**Type:** RW**Reset State:** 0x00000000

interrupt Enables for RXP errors

WCSS_A_RXP_RXP_GROUPED2_INTERRUPT_ENABLE

Bits	Name	Description
15	P2P_EID1_ABSENT_INTERRUPT	When set, enables interrupt when received beacon is p2p beacon and has reached end of beacon and has not found IE with EID1 programmed in bss2_beacon_extract_control register feild [31:24] this interrupt goes to firmware only
14	P2P_EID0_ABSENT_INTERRUPT	When set, enables interrupt when received beacon is p2p beacon and has reached end of the beacon and has not found IE with EID0 programmed bss2_beacon_extract_control register feild [23:15] this interrupt goes to firmware only
13	P2P_EID1_PRESENT_INTERRUPT	When set, enables interrupt when received beacon is p2p beacon and has IE with EID1 programmed in bss2_beacon_extract_control register feild [31:24] this interrupt goes to firmware only

WCSS_A_RXP_RXP_GROUPED2_INTERRUPT_ENABLE (cont.)

Bits	Name	Description
12	P2P_EID0_PRESENT_INT_EN	When set, enables interrupt when received beacon is p2p beacon and has found IE with EID0 programmed bss2_beacon_extract_control register feild [23:15] this interrupt goes to firmware only
11	BSS_EID1_ABSENT_INT_EN	When set, received beacon is Bss beacon and has reached end of beacon and has not found IE with EID1 programmed in bss2_beacon_extract_control register feild [31:24] this interrupt goes to firmware only
10	BSS_EID0_ABSENT_INT_EN	When set, enables interrupt when received beacon is Bss beacon and has reached end of the beacon and has not found IE with EID0 programmed bss2_beacon_extract_control register feild [23:15] this interrupt goes to firmware only
9	BSS_EID1_PRESENT_INT_EN	When set, enables interrupt when received beacon is Bss beacon and has found IE with EID1 programmed in bss2_beacon_extract_control register feild [31:24] this interrupt goes to firmware only
8	BSS_EID0_PRESENT_INT_EN	When set, enables interrupt when received beacon is Bss beacon and has found IE with EID0 programmed bss2_beacon_extract_control register feild [23:15] this interrupt goes to firmware only
7	BSS2_EID1_ABSENT_INT_EN	When set, enables interrupt when received beacon is Bss beacon for client 2 and has reached end of beacon and has not found IE with EID1 programmed in bss2_beacon_extract_control register feild [31:24] this interrupt goes to firmware only
6	BSS2_EID0_ABSENT_INT_EN	When set, enables interrupt when received beacon is Bss beacon for client2 and has reached end of the beacon and has not found IE with EID0 programmed bss2_beacon_extract_control register feild [23:15] this interrupt goes to firmware only
5	BSS2_EID1_PRESENT_INT_EN	When set, enables interrupt when received beacon is Bss beacon for client2 and has IE with EID1 programmed in bss2_beacon_extract_control register feild [31:24] this interrupt goes to firmware only
4	BSS2_EID0_PRESENT_INT_EN	When set, enables interrupt when received beacon is Bss beacon for client2 and has IE with EID0 programmed bss2_beacon_extract_control register feild [23:15] this interrupt goes to firmware only

WCSS_A_RXP_RXP_GROUPED2_INTERRUPT_ENABLE (cont.)

Bits	Name	Description
3	BSS2_TIM_BIT_CLEAR_INT_EN	When set, it enables the beacon has tim bit clear interrupt this interrupt goes to firmware only
2	BSS2_TIM_BIT_SET_INT_EN	When set, it enables tim bit set in beacon this interrupt goes to firmware only
1	BSS2_TIM_BEACON_CRC_ERR_INT_EN	When set, it enables the beacon has CRC error int this interrupt goes to firmware only
0	BSS2_TIM_BASED_BROADCAST_FRAMES_PENDING_INT_EN	When set, enables interrupt for detecting tim bit which indicates broadcast frames are pending this interrupt goes to firmware only

0x0A080BB8 WCSS_A_RXP_RXP_GROUPED2_INTERRUPT_STATUS**Type:** RW**Reset State:** 0x00000000

RXP combined interrupt status, writing 1 to any bit will clear the corresponding interrupt

WCSS_A_RXP_RXP_GROUPED2_INTERRUPT_STATUS

Bits	Name	Type	Description
31:27	DMA_STATE_ON_GAME_ERROR	R	gives status of dma statemachine when gam error occurred
15	P2P_EID1_ABSENT_INT	RW	When set, received beacon is p2p beacon and has reached end of beacon and has not found IE with EID1 programmed in bss2_beacon_extract_control register feild [31:24] Writing 1 to this bit will clear this interrupt this interrupt goes to firmware only
14	P2P_EID0_ABSENT_INT	RW	When set, received beacon is p2p beacon and has reached end of the beacon and has not found IE with EID0 programmed bss2_beacon_extract_control register feild [23:15] Writing 1 to this bit will clear this interrupt this interrupt goes to firmware only
13	P2P_EID1_PRESENT_INT	RW	When set, received beacon is p2p beacon and has IE with EID1 programmed in bss2_beacon_extract_control register feild [31:24] Writing 1 to this bit will clear this interrupt this interrupt goes to firmware only

WCSS_A_RXP_RXP_GROUPED2_INTERRUPT_STATUS (cont.)

Bits	Name	Type	Description
12	P2P_EID0_PRESENT_INT	RW	When set, received beacon is p2p beacon and has IE with EID0 programmed bss2_beacon_extract_control register feild [23:15] Writing 1 to this bit will clear this interrupt this interrupt goes to firmware only
11	BSS_EID1_ABSENT_INT	RW	When set, received beacon is Bss beacon and has reached end of beacon and has not found IE with EID1 programmed in bss2_beacon_extract_control register feild [31:24] Writing 1 to this bit will clear this interrupt this interrupt goes to firmware only
10	BSS_EID0_ABSENT_INT	RW	When set, received beacon is Bss beacon and has reached end of the beacon and has not found IE with EID0 programmed bss2_beacon_extract_control register feild [23:15] Writing 1 to this bit will clear this interrupt this interrupt goes to firmware only
9	BSS_EID1_PRESENT_INT	RW	When set, received beacon is Bss beacon and has IE with EID1 programmed in bss2_beacon_extract_control register feild [31:24] Writing 1 to this bit will clear this interrupt this interrupt goes to firmware only
8	BSS_EID0_PRESENT_INT	RW	When set, received beacon is Bss beacon and has IE with EID0 programmed bss2_beacon_extract_control register feild [23:15] Writing 1 to this bit will clear this interrupt this interrupt goes to firmware only
7	BSS2_EID1_ABSENT_INT	RW	When set, received beacon is Bss beacon for client 2 and has reached end of beacon and has not found IE with EID1 programmed in bss2_beacon_extract_control register feild [31:24] Writing 1 to this bit will clear this interrupt this interrupt goes to firmware only
6	BSS2_EID0_ABSENT_INT	RW	When set, received beacon is Bss beacon for client2 and has reached end of the beacon and has not found IE with EID0 programmed bss2_beacon_extract_control register feild [23:15] Writing 1 to this bit will clear this interrupt this interrupt goes to firmware only
5	RXP_BSS2_EID1_PRESEN T_INT	RW	When set, received beacon is Bss beacon for client2 and has found IE with EID1 programmed in bss2_beacon_extract_control register feild [31:24] Writing 1 to this bit will clear this interrupt this interrupt goes to firmware only

WCSS_A_RXP_RXP_GROUPED2_INTERRUPT_STATUS (cont.)

Bits	Name	Type	Description
4	RXP_BSS2_EID0_PRESENT_INT	RW	When set, received beacon is Bss beacon for client2 and has found IE with EID0 programmed bss2_beacon_extract_control register feild [23:15] Writing 1 to this bit will clear this interrupt this interrupt goes to firmware only
3	BSS2_TIM_BIT_CLEAR_INT	RW	When set, beacon has tim bit clear Writing 1 to this bit will clear this interrupt this interrupt goes to firmware only
2	BSS2_TIM_BIT_SET_INT	RW	When set, tim bit set in beacon Writing 1 to this bit will clear this interrupt this interrupt goes to firmware only
1	BSS2_TIM_BEACON_CRC_ERR_INT	RW	When set, beacon has CRC error Writing 1 to this bit will clear this interrupt this interrupt goes to firmware only
0	BSS2_TIM_BASED_BROADCAST_FRAMES_PENDING_INT	RW	When set, indicates tim bit indicates broadcast frames are pending Writing 1 to this bit will clear this interrupt this interrupt goes to firmware only

0x0A080BBC WCSS_A_RXP_RXP_QBOOST_CONTROL**Type:** RW**Reset State:** 0x00000000

Qboost control register

WCSS_A_RXP_RXP_QBOOST_CONTROL

Bits	Name	Description
15:8	QBOOST_STAID	When Qboost is enabled the STAID programmed here is staid of the address2 we are associated with
4	QBOOST_QOS_ACK_POLICY_NORMAL	when this bit is set and the incoming frame is a QoS frame, the ACK policy shall be set to Normal (or implicit BA request) in order for the frame to be considered as a Qboost candidate trigger frame
3	QBOOST_PWR_STATE_CHANGE_EN	when this bit is set, the Power State bit in the incoming shall NOT be set, in order for the frame to be considered as a Qboost candidate trigger frame
2	QBOOST_QOSNULL_PKT_EN	when this bit is set qos NULL packets will be subject to Qboost

WCSS_A_RXP_RXP_QBOOST_CONTROL (cont.)

Bits	Name	Description
1	QBOOST_BAR_PKT_EN	when this bit is set bar packets will be subject to Qboost
0	QBOOST_QOSDATA_PKT_EN	when this bit is set qos data packets will be subject to Qboost

0x0A080BC0 WCSS_A_RXP_RXP_P2P_CHANNELSWITCH_BODY**Type:** R**Reset State:** 0x00000000

This register has contents of Channel switch announcement IE for a p2p beacon

WCSS_A_RXP_RXP_P2P_CHANNELSWITCH_BODY

Bits	Name	Description
23:0	RXP_P2P_CHANNELSWITCH_IE_BODY	This register has contents of channel switch IE Bits [7:0] : Has Channel switch count field of CS IE Bits [15:8] : Has New Channel number field of CS IE Bits [23:16] : Has Channel switch Mode field of CS IE

0x0A080BC4 WCSS_A_RXP_RXP_BSS_CHANNELSWITCH_BODY**Type:** R**Reset State:** 0x00000000

This register has contents of Channel switch announcement IE for a bss beacon

WCSS_A_RXP_RXP_BSS_CHANNELSWITCH_BODY

Bits	Name	Description
23:0	RXP_BSS_CHANNELSWITCH_IE_BODY	This register has contents of channel switch IE Bits [7:0] : Has Channel switch count field of CS IE Bits [15:8] : Has New Channel number field of CS IE Bits [23:16] : Has Channel switch Mode field of CS IE

0x0A080BC8 WCSS_A_RXP_RXP_BSS2_CHANNELSWITCH_BODY**Type:** R**Reset State:** 0x00000000

This register has contents of Channel switch announcement IE for a bss2 beacon

WCSS_A_RXP_RXP_BSS2_CHANNELSWITCH_BODY

Bits	Name	Description
23:0	RXP_BSS2_CHANNELSWITCH_IE_BODY	This register has contents of channel switch IE Bits [7:0] : Has Channel switch count field of CS IE Bits [15:8] : Has New Channel number field of CS IE Bits [23:16] : Has Channel switch Mode field of CS IE

0x0A080BCC WCSS_A_RXP_RXP_P2P_QUIETIE_LSBYTES

Type: R

Reset State: 0x00000000

This register has contents of quiet element IE for a p2p beacon

WCSS_A_RXP_RXP_P2P_QUIETIE_LSBYTES

Bits	Name	Description
31:0	RXP_P2P_QUIETIE_LSB	This register has lower 4 bytes of contents of Quiet Element IE refer to following byte descriptions for correct byte positions Bits [15:0] : Has field Quiet offset Bits [31:16] : Has field Quiet Duration

0x0A080BD0 WCSS_A_RXP_RXP_BSS_QUIETIE_LSBYTES

Type: R

Reset State: 0x00000000

This register has contents of quiet element IE for a bss beacon

WCSS_A_RXP_RXP_BSS_QUIETIE_LSBYTES

Bits	Name	Description
31:0	RXP_BSS_QUIETIE_LSB	This register has lower 4 bytes of contents of Quiet Element IE refer to following byte descriptions for correct byte positions Bits [15:0] : Has field Quiet offset Bits [31:16] : Has field Quiet Duration

0x0A080BD4 WCSS_A_RXP_RXP_BSS2_QUIETIE_LSBYTES**Type:** R**Reset State:** 0x00000000

This register has contents of quiet element IE for a bss2 beacon

WCSS_A_RXP_RXP_BSS2_QUIETIE_LSBYTES

Bits	Name	Description
31:0	RXP_BSS2_QUIETIE_LSB	This register has lower 4 bytes of contents of Quiet Element IE refer to following byte descriptions for correct byte positions Bits [15:0] : Has feild Quiet offset Bits [31:16] : Has field Quiet Duration

0x0A080BD8 WCSS_A_RXP_RXP_P2P_QUIETIE_MSBYTES**Type:** R**Reset State:** 0x00000000

This register has contents of quiet element IE for a p2p beacon

WCSS_A_RXP_RXP_P2P_QUIETIE_MSBYTES

Bits	Name	Description
15:0	RXP_P2P_QUIETIE_MSB	This register has upper 2 bytes of contents of Quiet Element IE refer to following byte descriptions for correct byte positions Bits [7:0] : Has field Quiet period Bits [15:8] : Has field Quiet count

0x0A080BDC WCSS_A_RXP_RXP_BSS_QUIETIE_MSBYTES**Type:** R**Reset State:** 0x00000000

This register has contents of quiet element IE for a bss beacon

WCSS_A_RXP_RXP_BSS_QUIETIE_MSBYTES

Bits	Name	Description
15:0	RXP_BSS_QUIETIE_MSB	This register has upper 2 bytes of contents of Quiet Element IE refer to following byte descriptions for correct byte positions Bits [7:0] : Has field Quiet period Bits [15:8] : Has field Quiet count

0x0A080BE0 WCSS_A_RXP_RXP_BSS2_QUIETIE_MSBYTES**Type:** R**Reset State:** 0x00000000

This register has contents of quiet element IE for a bss2 beacon

WCSS_A_RXP_RXP_BSS2_QUIETIE_MSBYTES

Bits	Name	Description
15:0	RXP_BSS2_QUIETIE_MSB	This register has upper 2 bytes of contents of Quiet Element IE refer to following byte descriptions for correct byte positions Bits [7:0] : Has field Quiet period Bits [15:8] : Has field Quiet count

10.54 WCSS_A_RXP_RXP**0x0A080800 WCSS_A_RXP_CONFIG****Type:** RW**Reset State:** 0x09000100

It contains RXP Configuration registers. These configurations (with the exception of soft abort and mask_subsq_fcs_err_within_ampdu) can be changed at any time by SW but HW will only take effect during the gap of packets so that SW will not corrupt packets during configuration.

WCSS_A_RXP_CONFIG

Bits	Name	Description
31:24	PHY_CMD_MIN_SIZE	RXP requires atleast minimum of 9 cmd bytes for processing packet if it is not 9 then packets will be dropped, but it can be configured to whatever value
23:19	RESERVED	reserved for future use
18	CFG_BDPDU_CLEAN_DIS	When gam error occurs it is catastrophic so we will reset the system But just to be safe RXP will clean up all BD/PDUs, this needs some extra state machine check. when this bit is set it will disable the BD/PDU cleanup and SW can go ahead and clean it after reset
17	CFG_MPR_IE_LENCHK_DIS	For the newly added IE parsing of channel switch and quiet element there is a lenchk done to make sure Channel switch is indeed 3bytes and quiet element is indeed 6 bytes setting this bit to 1 this check can be disabled

WCSS_A_RXP_CONFIG (cont.)

Bits	Name	Description
16	CFG_FLT_UNPROT_ACTION_RMF_DIS	All action frames are supposed to be RMF frames(encrypted management frames) But some action frames can be unprotected, in those cases RXP sets RMF bit in BD to 0 so DPU can bypass the decryption and pass it to host SW by writing 1 to this bit this feature can be disabled, so that all action frames will have RMF set to 1 and DPU can drop the unprotected RMF frames and firmware can process these packets from Error WQ.
15	CFG_11AC_AMPDU_EOP_INDICATION_EN	If set, RXP will generate 11ac ampdu end of packet signal to PHY. By default, it will NOT generate this eop signal. If 11ac_eop is generated to PHY, PHY will stop receiving the current packet. This is used to save power in case of MU-MIMO packets
14	CFG_MPR_DISABLE_WAIT_FOR_FILTER_IDLE	When set, this will disable the 'wait for filter to finish, when eop is received' function. This is a bug fix. This configuration bit will allow the bug fix to be disabled and the rxp logic to be reverted to the original implementation. This allows for backwards compatibility.
13	RRI_CLEAR_ON_FCS_FAIL	If set, this will enable RRI regeneration when the first RRI got followed by and FCS err.
12	MASK_SUBSQ_FCS_ERR_WITHIN_AMPDU	If set, this will mask out fcs error notification to the MTU after an MPDU within an AMPDU passes FCS check. This is to prevent doing EIFS in this case.
11	SOFT_PMI_ABORT_EN	Soft PMI abort enable.
10	CFG_HAS_FCS_EN	If set, RXP will enable writing 4-byte fcs into BD/PDU and RXP flag has_fcs flag will be set to 1. By default, the RXP DMA will NOT write fcs into BD/PDU and MPDU length does not include the 4-byte fcs.
9	CFG_RSV_BMU_EN	If set, RXP will enable sending reservation command to BMU. By default, the RXP DMA will NOT send reservation command to BMU at start of packet.
8	CFG_HAS_PHY_CMD	If set, RXP will write PHY Command into BD. By default, the PHY Command will not be put into BD so that SW can adjust the "mpdu_header_start_offset" to save the BD space for MPDU data. Note that the TimeStamp will be in BD in any case.

WCSS_A_RXP_CONFIG (cont.)

Bits	Name	Description
7	CFG_PHY_PROTOCOL_CHK_DIS	When set it disables RXP from doing any PHY protocol check Like checking PHY cmd length and stats length endcomment bitfield [6] cfg_disable_dlm_proc configuration 1'b0 bitfieldComment If set to disable, the Delimiter Processing is disabled for A-MPDU packets and the whole A-MPDU packet will be sent to mCPU without MPDU Processing. The byp_dlm_proc and byp_mpdu_proc will be asserted for A-MPDU packets. By default, Delimiter Processing is enabled. Note: This cfg_disable_dlm_proc will affect only A-MPDU packets, but not MPDU packets.
5	CFG_DISABLE_MPDU_PROC	If set to disable, the MPDU Processing is disabled for MPDU packets and the MPDU packets will be sent to mCPU without MPDU processing. This means NO FCS check, NO address filtering, and NO binary search. The packets are assumed passing MPDU processing. By default, MPDU Processing is enabled. Note: This cfg_disable_mpdu_proc will affect only MPDU packets, but not A-MPDU packets. This means MPDU within A-MPDU will be processed normally (without bypass) if cfg_disable_dlm_proc is not asserted for A-MPDU packets.
4	CFG_DISABLE_FCS_CHECK	If set to disable, the FCS Check is disabled and all packets will be treated as passing FCS Check. By default, it is enabled and RXP will check FCS on MPDU packets. The counter associated with this config is fcs_err_cnt. If not disabled, this counter will increment if a packet has FCS error.
3	CFG_MAX_PKTLEN_CHECK_EN	The max packet length check enable bit. By default, the max_pktlen check is disabled. If set, it will check MPDU packet length against max_pktlen register and check A-MPDU packet length against max_len_ampdu register. But it will not affect max_len_mpdu_in_ampdu which is checked in delimiter processing and can be disabled by asserting cfg_disable_dlm_proc.
2	CFG_ADDR_FILTER_EN	If set, the Address Filter will be enabled. By default, it is disabled and all packets will be treated as passing Address Filtering.
1	CFG_GEN_SHUTOFF_EN	If set, RXP will generate shutoff signal to PHY. By default, it will NOT generate shutoff signal. If shutoff is generated to PHY, PHY will stop receiving the current packet. This is used to save power.
0	CFG_RXP_EN	If enabled, RXP will receive packets from PHY. By default, the RXP is disabled and will not receive any packet from PHY. This bit can be changed at any time, but will only take effect when there is start pulse from PHY.

0x0A080808 WCSS_A_RXP_MAX_PKTLEN**Type:** RW**Reset State:** 0x00000040

The Max MPDU Packet Length in terms of the number bytes. It includes mac_header+payload+fcs, but does not include PHY data.

WCSS_A_RXP_MAX_PKTLEN

Bits	Name	Description
6:0	MAX_PKTLEN	The Max MPDU Packet Length in terms of the number 128 bytes.

0x0A08080C WCSS_A_RXP_SW_BD_REG**Type:** RW**Reset State:** 0x00000000

Register used by SW to write a DWord to BD

WCSS_A_RXP_SW_BD_REG

Bits	Name	Description
31:0	SW_BD_REG	SW mainly uses this register to write to location 38 in DWORD

0x0A080810 WCSS_A_RXP_MPDU_HEADER_START_OFFSET**Type:** RW**Reset State:** 0x00000040

The MPDU header start offset in BD which is in the unit of data word (4-byte).

WCSS_A_RXP_MPDU_HEADER_START_OFFSET

Bits	Name	Description
7:2	MPDU_HEADER_START_OFFSET	MPDU header start offset in BD. Note that the programmable bits are [7:2] and the lower [1:0] bits are always 2'b00 to guarantee the offset address is 4-byte aligned. If bit[7] is 0, the MPDU header starts at BD, but if bit[7] is 1, the MPDU header starts at PDU. Note that It is not valid to have the value of 6'h3f for bit[7:2] (or 8'hfc for bit[7:0]). The default value is 6'ha which means the offset address of 8'h28. In this case, the PHY Cmd bytes will not be written in BD so that the MPDU header can start right after TimeStamp field. However, for diagnostic purpose, the PHY Cmd can be written into BD by setting cfg_has_phy_cmd bit in config register so that this offset value will be adjusted to 6'h10 which means the offset address of 8'h40.

0x0A08082C WCSS_A_RXP_SEARCH_ADDR1_PTR**Type:** RW**Reset State:** 0x00000000

SW can set the addr1 pointers of Binary Search.

WCSS_A_RXP_SEARCH_ADDR1_PTR

Bits	Name	Description
16	ADDR1_POINTERS_VALID	When set, the pointers are valid.
13:8	ADDR1_SEARCH_HI_PTR	Indicate the addr1 pointer of Binary Search upper bound.
5:0	ADDR1_SEARCH_LO_PTR	Indicate the addr1 pointer of Binary Search lower bound.

0x0A080830 WCSS_A_RXP_SEARCH_ADDR2_PTR**Type:** RW**Reset State:** 0x00000000

SW can set the addr2 pointers of Binary Search.

WCSS_A_RXP_SEARCH_ADDR2_PTR

Bits	Name	Description
16	ADDR2_POINTERS_VALID	When set, the pointers are valid.
13:8	ADDR2_SEARCH_HI_PTR	Indicate the addr2 pointer of Binary Search upper bound.
5:0	ADDR2_SEARCH_LO_PTR	Indicate the addr2 pointer of Binary Search lower bound.

0x0A080834 WCSS_A_RXP_SEARCH_ADDR3_PTR**Type:** RW**Reset State:** 0x00000000

SW can set the addr3 pointers of Binary Search.

WCSS_A_RXP_SEARCH_ADDR3_PTR

Bits	Name	Description
16	ADDR3_POINTERS_VALID	When set, the pointers are valid.
13:8	ADDR3_SEARCH_HI_PTR	Indicate the addr3 pointer of Binary Search upper bound.
5:0	ADDR3_SEARCH_LO_PTR	Indicate the addr3 pointer of Binary Search lower bound.

0x0A080838 WCSS_A_RXP_SEARCH_TABLE_CMD**Type:** RW**Reset State:** 0x00000000

This is action register for indirect access to Binary Search Table. When SW writes to this action register, it will trigger the interaction between Search_Table_Data[0-1] registers and Binary Search Table. For writing data into memory table, it is important for SW to write into Search_Table_Data[0-1] registers before Search_Table_Cmd. For reading data from memory table, SW only needs to write into Search_Table_Cmd and the read data will be available in Search_Table_Data[0-1] registers.

WCSS_A_RXP_SEARCH_TABLE_CMD

Bits	Name	Type	Description
9	STATUS	R	read only: If set to 1 a binary search table read or write is ongoing. SW should read and make sure this bit is 0 before issuing another cmd write into this register.
8	WRITE	RW	Indicate write/read. If 1, write. If 0, read.
4:0	TABLE_PTR	RW	Indicate the action should take effect on this entry number in Binary Search Table.

0x0A08083C WCSS_A_RXP_SEARCH_TABLE_DATA0**Type:** RW**Reset State:** 0x00000000

This is data register 0 for indirect access to Binary Search Table. There are totally 3 Data registers for Binary Search address Table.

WCSS_A_RXP_SEARCH_TABLE_DATA0

Bits	Name	Description
31:0	MAC_ADDR	MAC_Addr[31:0] portion of MAC_Addr[47:0]. Note: The order is Little Endian: MAC_Addr[31:0] = {Byte_3, Byte_2, Byte_1, Byte_0}

0x0A080840 WCSS_A_RXP_SEARCH_TABLE_DATA1**Type:** RW**Reset State:** 0x00000000

This is data register 1 for indirect access to Binary Search Table. There are totally 3 Data registers for Binary Search address Table.

WCSS_A_RXP_SEARCH_TABLE_DATA1

Bits	Name	Description
31:29	DPU_MAN_MC_BC_SIG	This feild is used to configure the DPU signature for Management mc/bc packets
28	RSVD	reserved for future use

WCSS_A_RXP_SEARCH_TABLE_DATA1 (cont.)

Bits	Name	Description
27	VALID	Set when the sta address is valid
26	RMF	Set when robust managemnt frmases require BIP or decryption
25	FRAME_TRANSLATION	This bit is used to decide if a per station frame translation is required. When this bit is set and the bit in type/subtype is set then those packets for that station will require frame translation, so FT bit in BD will be set
24	DROP	When set, on a address hit, the frame will be dropped. The setting of this bit is only valid when 'valid configuration' is set
23:16	ADDR_ID	The addr_id[7:0].
15:0	MAC_ADDR	MAC_Addr[47:32] portion of MAC_Addr[47:0]. Note: The order is Little Endian: MAC_Addr[47:32] = {Byte_5, Byte_4}

0x0A080844 WCSS_A_RXP_MAX_LEN_AMPDU**Type:** RW**Reset State:** 0x0000FFFF

The Max Length for A-MPDU packet.

WCSS_A_RXP_MAX_LEN_AMPDU

Bits	Name	Description
16:0	MAX_LEN	The Max Length for A-MPDU packet.

0x0A080848 WCSS_A_RXP_MAX_LEN_MPDU_IN_AMPDU**Type:** RW**Reset State:** 0x00000FFF

The Max Length for MPDU in A-MPDU packet. This is used for delimiter processing to determine whether it finds a good delimiter.

WCSS_A_RXP_MAX_LEN_MPDU_IN_AMPDU

Bits	Name	Description
13:0	MAX_LEN	The Max Length for MPDU in A-MPDU packet.

0x0A08084C WCSS_A_RXP_DMA_MAX_RSV_PDU**Type:** RW**Reset State:** 0x00000008

The Max number of PDUs that RXP DMA can reserve at a time. The is to avoid the scenario that RXP drop big packet just because it fails to reserve a huge number of PDUs at the start of the packet.

WCSS_A_RXP_DMA_MAX_RSV_PDU

Bits	Name	Description
3:0	MAX_RSV_PDU	The Max number of PDUs that RXP DMA can reserve at a time.

0x0A080850 WCSS_A_RXP_DMA_BURST_THRESHOLD**Type:** RW**Reset State:** 0x00000008

The data word threshold that RXP DMA control can start burst MPDU data to MIF. The data unit is 1 word (or 4 bytes). By default, the threshold is 8 words (or 32 bytes). RXP MPDU data is accumulated in DMA FIFO. The DMA control will not send data to MIF if the number of data word in DMA FIFO does not reach the threshold. This is to used to control the burstness of RXP data on AHB.

WCSS_A_RXP_DMA_BURST_THRESHOLD

Bits	Name	Description
5:0	BURST_THRESHOLD	The data word threshold that DMA can start burst MPDU data to MIF.

0x0A080854 WCSS_A_RXP_CLEAR_STATS**Type:** RW**Reset State:** 0x00000000

This register is used to clear all stats.

WCSS_A_RXP_CLEAR_STATS

Bits	Name	Description
0	CLEAR_STATS	When set, all statistics counters will be cleared When set, the HW will automatically clear this bit.

0x0A080858 WCSS_A_RXP_PHY_MPDU_CNT

Type: R

Reset State: 0x00000000

The number of MPDU packets are received from PHY.

WCSS_A_RXP_PHY_MPDU_CNT

Bits	Name	Description
31:0	PHY_MPDU_CNT	PHY MPDU Cnt.

0x0A08085C WCSS_A_RXP_PHY_AMPDU_CNT

Type: R

Reset State: 0x00000000

The number of A-MPDU packets are received from PHY.

WCSS_A_RXP_PHY_AMPDU_CNT

Bits	Name	Description
31:0	PHY_AMPDU_CNT	PHY A-MPDU Cnt.

0x0A080864 WCSS_A_RXP_PHY_ABORT_CNT

Type: R

Reset State: 0x00000000

The number of aborted packets from PHY.

WCSS_A_RXP_PHY_ABORT_CNT

Bits	Name	Description
7:0	PHY_ABORT_CNT	PHY Abort Packet Cnt.

0x0A080868 WCSS_A_RXP_PHY_SHUTOFF_CNT**Type:** R**Reset State:** 0x00000000

The number of shutoff pulses are generated by RXP to PHY.

WCSS_A_RXP_PHY_SHUTOFF_CNT

Bits	Name	Description
15:8	PHY_11AC_EOP_CNT	This is also logically a shutoff count, but this is RXP shutting of due of EOF bit set in the 11ac AMPDU indicating end of useful AMPDU data. PHY will stop giving anymore data.
7:0	PHY_SHUTOFF_CNT	PHY Shutoff Cnt. This is number of packets that RXP terminated due to packet error or early beacon termination

0x0A08086C WCSS_A_RXP_DLM_FIFO_FULL_CNT**Type:** R**Reset State:** 0x00000000

The number of packets have error because DLM FIFO is full.

WCSS_A_RXP_DLM_FIFO_FULL_CNT

Bits	Name	Description
7:0	DLM_FIFO_FULL_CNT	DLM FIFO Full Cnt.

0x0A080870 WCSS_A_RXP_DLM_ERR_CNT**Type:** R**Reset State:** 0x00000000

The number of packets have error because Delimiter is bad. When a error is detected in the A-MPDU delimiter this count is incremented. But when RXP starts searching for the next A-MPDU

delimiter, this count is NOT incremented. The implementation uses 'seen_dlm_err' flag. This flag is cleared for the first delimiter or after a good delimiter is found. It is set after a bad delimiter is found. If this flag is set, then don't increment the counter. This will increment: - Whenever the next delimiter (using the previous good delimiter information) is not found in the expected location, or is bad.

WCSS_A_RXP_DLM_ERR_CNT

Bits	Name	Description
7:0	DLM_ERR_CNT	DLM Error Cnt.

0x0A080874 WCSS_A_RXP_FAIL_FILTER_CNT

Type: R

Reset State: 0x00000000

The number of packets have failed Address Filtering.

WCSS_A_RXP_FAIL_FILTER_CNT

Bits	Name	Description
15:0	FAIL_FILTER_CNT	Fail Address Filter Cnt.

0x0A080878 WCSS_A_RXP_FAIL_MAX_PKTLEN_CNT

Type: R

Reset State: 0x00000000

The number of packets have failed Max MPDU or AMPDU packet length check.

WCSS_A_RXP_FAIL_MAX_PKTLEN_CNT

Bits	Name	Description
7:0	FAIL_MAX_PKTLEN_CNT	Fail Max MPDU or AMPDU Packet Length Check Cnt.

0x0A08087C WCSS_A_RXP_FCS_ERR_CNT

Type: R

Reset State: 0x00000000

The number of packets have FCS error.

WCSS_A_RXP_FCS_ERR_CNT

Bits	Name	Description
31:0	FCS_ERR_CNT	FCS Error Cnt.

0x0A080880 WCSS_A_RXP_DMA_SEND_CNT**Type:** R**Reset State:** 0x00000000

The number of packets are sent to CPU by RXP Write DMA.

WCSS_A_RXP_DMA_SEND_CNT

Bits	Name	Description
31:0	DMA_SEND_CNT	The number of packets are sent to CPU by RXP Write DMA. All packets in current system are all pushed by RXP refer to rpe dma send cnt for that. This counter increments only if RXP pushes packets directly to DXE

0x0A080884 WCSS_A_RXP_DMA_DROP_CNT**Type:** R**Reset State:** 0x00000000

The number of packets are dropped by RXP Write DMA. This could increment due to: Failed filter PHY Abort AFTER the 9th command word BD/PDU errors If packet with command phase shorter or longer than actual command length specified is sent, the RXP will increment dma_drop_cnt instead of phy_err_drop_cnt (unless actual command phase < 9 words, in which case phy_err_drop_cnt will be incremented) Will NOT increment if end of psdu (as defined in psdu length by PHY command) occurs during search for next delimiter in AMPDU; this is because a new packet is not found, so there should not be a drop)

WCSS_A_RXP_DMA_DROP_CNT

Bits	Name	Description
15:0	DMA_DROP_CNT	The number of packets are dropped by RXP Write DMA. Drop due to filter error, packet len error or Get bmu error any packet dropped at DMA due to any of above errors, these can be drop at DMA marked packets which may have filter fail errors

0x0A080888 WCSS_A_RXP_DMA_GET_BMU_FAIL_CNT**Type:** R**Reset State:** 0x00000000

The number of packets are dropped due to insufficient BD/PDUs BMU BD or PDU.

WCSS_A_RXP_DMA_GET_BMU_FAIL_CNT

Bits	Name	Description
15:0	DMA_GET_BMU_FAIL_CNT	DMA No BMU BD/PDU Cnt.

0x0A080890 WCSS_A_RXP_TYPE_SUBTYPE_FILTER_CNT**Type:** R**Reset State:** 0x00000000

The number of packets dropped because of their type/subtype

WCSS_A_RXP_TYPE_SUBTYPE_FILTER_CNT

Bits	Name	Description
7:0	DROPPED_COUNT	number of frames dropped

0x0A080894 WCSS_A_RXP_INCORRECT_LENGTH_FILTER_CNT**Type:** R**Reset State:** 0x00000000

The length of the frame was not sufficient to do all the address filter checks. This may increment if dlm_fifo_full occurs.

WCSS_A_RXP_INCORRECT_LENGTH_FILTER_CNT

Bits	Name	Description
7:0	DROPPED_COUNT	number of frames dropped

0x0A080898 WCSS_A_RXP_ADDR1_BLOCK_FILTER_CNT**Type:** R**Reset State:** 0x00000000

Number of frames blocked based on the address one

WCSS_A_RXP_ADDR1_BLOCK_FILTER_CNT

Bits	Name	Description
7:0	DROPPED_COUNT	number of frames dropped

0x0A08089C WCSS_A_RXP_ADDR1_HIT_NO_PASS_CNT**Type:** R**Reset State:** 0x00000000

Number of frames that got a hit in the binary search table but were instructed to be dropped because of the drop setting in the binary search table

WCSS_A_RXP_ADDR1_HIT_NO_PASS_CNT

Bits	Name	Description
7:0	DROPPED_COUNT	number of frames dropped

0x0A0808A0 WCSS_A_RXP_ADDR1_DROP_CNT**Type:** R**Reset State:** 0x00000000

Number of frames that got dropped because there was no hit in the binary search table

WCSS_A_RXP_ADDR1_DROP_CNT

Bits	Name	Description
7:0	DROPPED_COUNT	number of frames dropped

0x0A0808A4 WCSS_A_RXP_ADDR2_HIT_NO_PASS_CNT**Type:** R**Reset State:** 0x00000000

Number of frames that got a hit in the binary search table but were instructed to be dropped because of the drop setting in the binary search table

WCSS_A_RXP_ADDR2_HIT_NO_PASS_CNT

Bits	Name	Description
7:0	DROPPED_COUNT	number of frames dropped

0x0A0808A8 WCSS_A_RXP_ADDR2_DROP_CNT**Type:** R**Reset State:** 0x00000000

Number of frames that got dropped because there was no hit in the binary search table

WCSS_A_RXP_ADDR2_DROP_CNT

Bits	Name	Description
7:0	DROPPED_COUNT	number of frames dropped

0x0A0808AC WCSS_A_RXP_ADDR3_HIT_NO_PASS_CNT**Type:** R**Reset State:** 0x00000000

Number of frames that got a hit in the binary search table but were instructed to be dropped because of the drop setting in the binary search table

WCSS_A_RXP_ADDR3_HIT_NO_PASS_CNT

Bits	Name	Description
7:0	DROPPED_COUNT	number of frames dropped

0x0A0808B0 WCSS_A_RXP_ADDR3_DROP_CNT**Type:** R**Reset State:** 0x00000000

Number of frames that got dropped because there was no hit in the binary search table

WCSS_A_RXP_ADDR3_DROP_CNT

Bits	Name	Description
7:0	DROPPED_COUNT	number of frames dropped

0x0A0808B4 WCSS_A_RXP_MPDU_IN_AMPDU_CNT**Type:** R**Reset State:** 0x00000000

The number of MPDU packets are split from A-MPDU packets.

WCSS_A_RXP_MPDU_IN_AMPDU_CNT

Bits	Name	Description
31:0	MPDU_IN_AMPDU_CNT	The number of MPDU packets are split from A-MPDU packets.

0x0A0808B8 WCSS_A_RXP_PHY_ERR_DROP_CNT**Type:** R**Reset State:** 0x00000000

The number of packets are dropped by RXP PHY Processing because of some apparent PHY protocol errors, such as: PHY Cmd Len is smaller than 9 bytes. PHY Cmd Len is larger than 28 bytes. PHY Stats Len is not equal to 8 bytes. Exception - if packet with command phase shorter longer than actual command length is sent, the RXP will increment dma_drop_cnt instead (however, if actual command length is < 9, this will be counted as a phy_err_drop) Note: The packets are dropped before reaching RXP Write DMA.

WCSS_A_RXP_PHY_ERR_DROP_CNT

Bits	Name	Description
7:0	PHY_ERR_DROP_CNT	The number of packets are dropped by RXP PHY Processing.

0x0A0808BC WCSS_A_RXP_RADAR_DETECT_STATUS**Type:** RW**Reset State:** 0x00000000

Radar detect status register. Pointers in radar detect circular buffer. Equal pointers means that the buffer is empty. There can be 0-63 entries in this circular buffer.

WCSS_A_RXP_RADAR_DETECT_STATUS

Bits	Name	Type	Description
24	RDR_DETECT_INT	R	status of the radar detect interrupt signal
21:16	NR_ENTRIES	R	Number of entries filled in radar detect circular buffer. (0-63)
13:8	TAIL_PTR	RW	tail pointer of radar detect circular buffer
5:0	HEAD_PTR	RW	head pointer of radar detect circular buffer

0x0A0808C0 WCSS_A_RXP_RADAR_DETECT_CONTROL**Type:** RW**Reset State:** 0x00000000

Radar detect control register.

WCSS_A_RXP_RADAR_DETECT_CONTROL

Bits	Name	Description
8	RADAR_DETECT_ENABLE	Enable/disable storing of radar detect logs
5:0	MIN_NR_LOGS_GEN_INT	Minimum number of stored logs to generate radar detect interrupt. 0 is a special case, which means to disable the interrupt.

0x0A0808C4 WCSS_A_RXP_RADAR_DETECT_LLOGS**Type:** RW**Reset State:** 0x00000000

Radar detect lower word register.

WCSS_A_RXP_RADAR_DETECT_LLOGS

Bits	Name	Description
31:0	RADAR_DETECT_LDATA	On read, the lower 4 bytes from the radar detect log as pointed by the tail pointer is returned. On write, the lower 4 bytes are written into the location pointed by the head pointer.

0x0A0808C8 WCSS_A_RXP_RADAR_DETECT_ULOGS**Type:** RW**Reset State:** 0x00000000

Radar detect upper word register.

WCSS_A_RXP_RADAR_DETECT_ULOGS

Bits	Name	Description
31:0	RADAR_DETECT_UDATA	On read, the lower 4 bytes from the radar detect log as pointed by the tail pointer is returned. After the read, the tail pointer is increased by one, except if that would cause it to become greater than the head pointer. On write, the lower 4 bytes are written into the location pointed by the head pointer. After the write, the head pointer is increased by one. If this would cause the head pointer to become equal to the tail pointer, the tail pointer would also be incremented.

0x0A0808CC WCSS_A_RXP_RADAR_DETECT_LOGS**Type:** RW**Reset State:** 0x00000000

Radar detect logs counter register

WCSS_A_RXP_RADAR_DETECT_LOGS

Bits	Name	Type	Description
31	RADAR_DETECT_LOGS_CNT_CLEAR	RW	When set, clears count of PHY written radar detect logs. This bit will automatically be cleared by HW.
15:0	RADAR_DETECT_LOGS_CNT	R	Number of radar detect logs received from PHY.

0x0A0808D0 WCSS_A_RXP_SOFT_PMI_ABORT**Type:** RW**Reset State:** 0x00000000

Soft PMI Abort register

WCSS_A_RXP_SOFT_PMI_ABORT

Bits	Name	Description
0	SOFT_PMI_ABORT	Write only bit; when 1 is written to this bit, and soft_pmi_abort_en is 1, then a soft PHY abort will be generated. Error counters will be incremented as if an actual abort had occurred.

0x0A0808D4 WCSS_A_RXP_START_ERR_CNT**Type:** R**Reset State:** 0x00000000

The number of start errors on the PMI interface (pmi_rxp_starts received while still waiting for previous packet to complete).

WCSS_A_RXP_START_ERR_CNT

Bits	Name	Description
7:0	PHY_START_ERR_CNT	PHY Start Error Cnt.

0x0A0808D8 WCSS_A_RXP_CCA_AND_EXT_TIMEOUT**Type:** RW**Reset State:** 0x3C064090

Configuration bits for cca extensions and timeout functionality.

WCSS_A_RXP_CCA_AND_EXT_TIMEOUT

Bits	Name	Description
31:21	CCA_PKTDET_6US_EXT_VALUE	This is extension of packet detect, which is done only for 11b packets for 80Mhz clock : 11'd480 for 160Mhz clock : 11'd960 for 320Mhz clock : 11'd1920

WCSS_A_RXP_CCA_AND_EXT_TIMEOUT (cont.)

Bits	Name	Description
20:10	CCA_PKTDET_EXT_VALUE	This is an additional extension of packet detect, if this is not needed it should be programmed to 1, otherwise HW will use the default value of 11'd400 80Mhz clocks 80Hmz clock : 11'd400 160Mhz clock: 11'd800 320Mhz clock: 11'd1600
9	CCA_PKDET_EXTENTION_ALLOWED	when set it allows for 6us extn on ofdm packets
8:4	FLT_TIMEOUT_TIME	This value represent that maximum time (in 256 OR 4096 120MHZ clk cycle units) that the rxp filter will wait for two sequential MPDU header bytes to be received. If this time is not met, the rxp filter will drop the frame by indicating that the filter is done, but the frame does not pass the filter. The 'timeout_error_cnt' register contents will increase by 1.
2	CCA_PKDET_EXTENTION_OLDLOGIC_DIS	when packet det gets deasserted 5us extension is done, and at the end of 5us we look for commandbyte 6 for packet type whether it is ofdm and then do 6us additional extension. But PHY may not always give this info within 5us hence ofdm signal is used for this (which is the new logic). This logic is only enabled when you write 1 to this bit
1	FLT_TIMEOUT_COUNT_UNIT	When clear, the timeout will count in 256 clk cycle units When set, the timeout counter will count in 4096 clk cycle units
0	FLT_TIMEOUT_CHECK_DISABLED	When set, the rxp filter timeout check is disabled

0x0A0808DC WCSS_A_RXP_TIMEOUT_ERROR_CNT**Type:** R**Reset State:** 0x00000000

Number of times the rxp filter encounters a timeout As a result of this, the frame will be dropped

WCSS_A_RXP_TIMEOUT_ERROR_CNT

Bits	Name	Description
7:0	TIMEOUT_COUNT	Number of times the rxp filter encounters a timeout

0x0A0808E0 WCSS_A_RXP_STALL_TIMEOUT_CNT**Type:** R**Reset State:** 0x00000000

Number of times stall timeout occurs. As a result of this, the frame will be dropped

WCSS_A_RXP_STALL_TIMEOUT_CNT

Bits	Name	Description
7:0	STALL_TIMEOUT_COUNT	Number of times the PMI interface encounters a stall timeout

0x0A0808EC WCSS_A_RXP_CFG_FLT_NAV_CONTROL_RESPONSE_REQ**Type:** RW**Reset State:** 0x00000000

NAV control register and response requested register. The NAV PHY delay compensation can be programmed here. Also the control of how the NAV needs to be updated, or not at all, when the CCA was not asserted when a frame got received (and dropped) is configured here.

WCSS_A_RXP_CFG_FLT_NAV_CONTROL_RESPONSE_REQ

Bits	Name	Description
25	CFG_FLT_RESPONSE_REQ_WITH_HT_CONTROL	When set and the Reponse interrupt is generated, and the incoming frame has an HT control field (this is: incoming frame is QoS data frame and has the 'order' bit in the frame control field set), the response interrupt is only generated after the HT control field has been fully received and will be located in the reponse_header_info5 field. Setting this bit will also enable the generation of the 'HTC present' reporting in the enhanced_sequence_number_message_type set 1: Will enable RXP to process frames with HTC feild set 0: RXP will ignore HTC feild hence whenever this bit is zero block_frames_with_ht_control_field(extra frame filter register) should be 1
24:21	CFG_FLT_NAV_RESERVED	reserved bits
20	CFG_FLT_NAV_DELAY_COMPENSATION_VALUE_EN	When set, enables the NAV delay compensation functionality. This is, the NAV update value given to the MTU is compensated for the PHY delay. The compensation value is given in the bits: cfg_ft_nav_delay_compensation_value
19:16	CFG_FLT_NAV_DELAY_COMPENSATION_VALUE	NAV compensation value in usec.

WCSS_A_RXP_CFG_FLT_NAV_CONTROL_RESPONSE_REQ (cont.)

Bits	Name	Description
10	CFG_FLT_REGISTER_CCA_AT_START_OF_OBSERVE_TIME	When NOT set: no action When set: The RXP filter samples the setting of the primary cca signal at the start of the cca observe time. This sampled CCA is later used to determine if an NAV update is needed or NOT. The cca observe time is configured with other bits in this register.
9	CFG_FLT_REGISTER_CCA_AT_END_OF_OBSERVE_TIME	When NOT set: no action When set: The RXP filter samples the setting of the primary cca signal at the end of the cca observe time. This sampled CCA is later used to determine if an NAV update is needed or NOT. The cca observe time is configured with other bits in this register.
8	CFG_FLT_REGISTER_CCA_DURING_OBSERVE_TIME	When NOT set: no action When set: The RXP filter samples the setting of the primary cca signal during the entire duration of the cca observe time. If the cca goes low anywhere during this time, this is recorded and fixed. In that case, even if the CCA went high a little later, the sampled CCA value that is later used to determine if an NAV update is needed or NOT, will have a low value. The cca observe time is configured with other bits in this register.
7	CFG_FLT_REGISTER_CCA_LEVEL_DURING_DATA_TIME	When NOT set: no action When set: The 'data reception time' as indicated by pktDET_n being low, is part of the cca observe time.
6	CFG_FLT_REGISTER_CCA_LEVEL_DURING_PREAMBLE_TIME	When NOT set: no action When set: The 'data reception time' as determined by early_pktDET_n being low, is part of the cca observe time.
5	CFG_FLT_SKIP_DIFS_WHEN_NAV_UPDATE_MASKED	When NOT set: no action When set: When the RXP filter drops a frame and the NAV update is masked (due to enabling the NAV update masking features in this register), the 'skip_difs' signal to the MTU will be set.
4	CFG_FLT_NO_NAV_UPDATE_ON_DROP_AND_BELOW_CCA	When NOT set: no action When set: When the RXP filter drops a frame due to any reason, AND the cca at the sample point was high, the RXP will NOT generate a NAV update based on the dropped frame.
3	CFG_FLT_NO_NAV_UPDATE_ON_EXPLICIT_DROP_AND_BELOW_CCA	When NOT set: no action When set: When the RXP filter drops a frame due to an explicit drop configuration in the address table, AND the cca at the sample point was high, the RXP will NOT generate a NAV update based on the dropped frame.
2	CFG_FLT_NO_NAV_UPDATE_ON_EXPLICIT_DROP	When NOT set: no action When set: When the RXP filter drops a frame due to an explicit drop configuration in the address table, the RXP will not generate a NAV update based on the dropped frame. The setting of the CCA is NOT taken into account

WCSS_A_RXP_CFG_FLT_NAV_CONTROL_RESPONSE_REQ (cont.)

Bits	Name	Description
1	CFG_FLT_CLEAR_SKIP_NEXT_DIFS_ON_EARLY_PKT_DET	When NOT set: no action When set: The RXP filter can generate a signal to the MTU to skip DIFS. See other register bits to enable this. This 'skip' signal will be cleared when early_pkt_det_n gets asserted.
0	CFG_FLT_CLEAR_SKIP_NEXT_DIFS_ON_CCA	When NOT set: no action When set: The RXP filter can generate a signal to the MTU to skip DIFS. See other register bits to enable this. This 'skip' signal will be cleared when primary cca gets asserted.

0x0A0808F0 WCSS_A_RXP_CFG_FLT_TYPE_SUBTYPE_RX_DISABLE0**Type:** RW**Reset State:** 0x00000000

Control overwrite functionality for what was programmed in type/subtype config table (upper 32 bits)

WCSS_A_RXP_CFG_FLT_TYPE_SUBTYPE_RX_DISABLE0

Bits	Name	Description
31:0	CFG_FLT_TYPE_SUBTYPE_RX_DISABLE0	Control some overwrite functionality for whatever was programmed in the Type/subtype configuration table. When set, a frame of the corresponding type should be dropped.

0x0A0808F4 WCSS_A_RXP_CFG_FLT_TYPE_SUBTYPE_RX_DISABLE1**Type:** RW**Reset State:** 0x00000000

Control overwrite functionality for what was programmed in type/subtype config table (lower 32 bits)

WCSS_A_RXP_CFG_FLT_TYPE_SUBTYPE_RX_DISABLE1

Bits	Name	Description
31:0	CFG_FLT_TYPE_SUBTYPE_RX_DISABLE1	Control some overwrite functionality for whatever was programmed in the Type/subtype configuration table. When set, a frame of the corresponding type should be dropped.

0x0A080900 WCSS_A_RXP_FRAME_FILTER_CONFIG**Type:** RW**Reset State:** 0x00000000

This is a register set of 64 sequential word addresses: Each of these 64 words represent the filter configuration for a incoming frame type/subtype. The register programming address per type/subtype is created as follows: ("This registers address" + {Type [b3, b2], Subtype [b7,b6,b5,b4], 2'b00}) where 'Type [b3,b2]' and 'Subtype [b7,b6,b5,b4]' are the IEEE standard encodings for the frames. A detailed description of the RXP filter specification can be found in the NOVA MAC architecture document. A summary of the bit functions is as follows:

WCSS_A_RXP_FRAME_FILTER_CONFIG

Bits	Name	Description
21	G5_DROP_AT_DMA	When set, the RXP will drop the incoming frame (even when the CRC and filter function passes) at the RXP DMA engine. This feature is useful to prevent that control frames like the TRS, ACK etc, get unnecessarily written into memory
20	NONHT_BW_INDICATION_ENABLE	when set then this particular type/subtype will need to be checked for bw information. If bit 0 of TA is checked to see if BW information is valid.
19	ROUTING_FLAG_SEL	when set : rxp will use dpv routing flag that is programmed in config3 bits 29:24 when not set : rxp will use dpv routing flag that is programmed in config3 bits 21:16
18	FRAME_TRANSLATION_REQ_EN	When set Frame translation by UMA is enabled
17	PM_BIT_EVAL_REQUIRED_ENABLE	when this bit is set the pm bit is evaluated for this type/subtype packet
16	RESERVED3	It is reserved for future use, it is read writable
15	ADDR3_ACCEPT_REMAINING_ENABLE	when enabled, and no hit was found for the Addr3 binary search, the incoming frame for this type/subtype will pass the Addr3 filter. The ADDR3 index code returned is 255 for multicast/broadcast and 254 for unicast.
14	ADDR3_BINARY_SEARCH_FILTER_ENABLE	When enabled, Addr3 is submitted to the RXP Addr3 binary search filter. When not enabled, Addr3 filtering is NOT done, The returned ADDR3 index code is 253 and the 'Addr3_index_invalid' bit in the RXP status flags in the BD will be set

WCSS_A_RXP_FRAME_FILTER_CONFIG (cont.)

Bits	Name	Description
13	ADDR2_ACCEPT_REMAINING_ENABLE	when enabled, and no hit was found for the Addr2 binary search, the incoming frame for this type/subtype will pass the Addr2 filter. The ADDR2 index code returned is 255 for multicast/broadcast and 254 for unicast.
12	ADDR2_BINARY_SEARCH_FILTER_ENABLE	When enabled, Addr2 is submitted to the RXP Addr2 binary search filter. When not enabled, Addr2 filtering is NOT done, The returned ADDR2 index code is 253 and the 'Addr2_index_invalid' bit in the RXP status flags in the BD will be set
11	ADDR1_ACCEPT_REMAINING_UNICAST_ENABLE	when enabled, and no hit was found for the unicast address in the Addr1 binary search, the incoming unicast frame for this type/subtype will pass the Addr1 filter. The index code returned is: 254
10	ADDR1_ACCEPT_REMAINING_MULTICAST_ENABLE	when enabled, and no hit was found for the multicast/broadcast address in the Addr1 binary search, the incoming multicast/broadcast frame for this type/subtype will pass the Addr1 filter. The index code returned is: 255.
9	ADDR1_BINARY_SEARCH_FILTER_ENABLE	When enabled, Addr1 is submitted to the RXP Addr1 binary search filter. When not enabled, Addr1 filtering is NOT done, The returned ADDR1 index code is 253 and the 'Addr1_index_invalid' bit in the RXP status flags in the BD will be set
8	ADDR1_BLOCK_UNICAST_ENABLE	when enabled, a unicast frame for this type/subtype will not pass the filter.
7	ADDR1_BLOCK_MULTICAST_ENABLE	when enabled, a multicast frame for this type/subtype will not pass the filter.
6	ADDR1_BLOCK_BROADCAST_ENABLE	when enabled, a broadcast frame for this type/subtype will not pass the filter.
5	RESERVED4	It is reserved for future use, it is read writable
4	RESERVED5	It is reserved for future use, it is read writable
3	RESERVED6	It is reserved for future use, it is read writable
2	RESERVED7	It is reserved for future use, it is read writable

WCSS_A_RXP_FRAME_FILTER_CONFIG (cont.)

Bits	Name	Description
1	BLOCK_RECEPTION_ENABLE	When enabled, all frames of this type/subtype will be blocked and will NOT pass the type/subtype/address filter check.
0	RESERVED8	It is reserved for future use, it is read writable

0x0A080A04 WCSS_A_RXP_STALL_TIMEOUT**Type:** RW**Reset State:** 0x00012000

Stall timeout functionality - allows RXP to return to idle if PMI interface stalls.

WCSS_A_RXP_STALL_TIMEOUT

Bits	Name	Description
16	STALL_TIMEOUT_EN	Enable stall timeout functionality on the MPI.
15:0	STALL_TIMEOUT_TIME	Number of cycles of no activity during PMI transaction before stall timeout occurs.

0x0A080A50 WCSS_A_RXP_CONFIG2**Type:** RW**Reset State:** 0x010381DF

It contains RXP Configuration registers, related to gen5_6 functionality added

WCSS_A_RXP_CONFIG2

Bits	Name	Description
31	CFG_DLM_MPDULEN_MSB ITS_DIS	for 11ac the the Delimiter now holds mpdu length more than 4k. enabling this bit will allow rxp to apture those bits in MSB of mpdu length for 11ac only
30	CFG_MPR_NEED_MRQ_ST ATS	By default all the values in stat bytes will go in the BD in PHY stats location When this bit is set then the dynamic stats bus(link adaptation related info) will get over written on bytes 6 and 7 of PHY stats

WCSS_A_RXP_CONFIG2 (cont.)

Bits	Name	Description
29	CFG_FLT_VERSION_CHEC K_DIS	when set it will disable version checking for all packets. By default version checking is always done
28	NAV_UPDATE_DOES_NOT _EXCLUDE_SELF	When set, the NAV will also be set for frames destined to this STA. When NOT set, the NAV is only set for unicast frames received that are destined for other STAs. For both settings the NAV will be set for all BC/MC frames.
27	MASK_OUT_AMSDU_FRAM ES_IN_AMPDU_ENABLE	When set, and amsdu frames embedded in AMPDU frames are received, the amsdu frames will be dropped
26	ADDR2_AUTH_EXTRACT_E NABLE	When set, the RXP will extract the 'authentication setting from the addr 2 binary search entry (bit 2 of the 3 bit wide DPU tag entry), and insert in the BE. In this mode, the DPU tag when based on the ADDR 2 DPU TAG entry will only be 2 bits wide instead of the regular 3 bits.
25	PASS_LSIG_PROT_INFO_T O_TPE_ENABLE	When set, the RXP will pass the setting of the LSIG protection signal to the TPE together with the other response required info.
24	DEFAULT_PUSH_WQ_OVE RWRITE_ENABLE	When set, the RXP will by default push frames to WQ indicated in "default_push_wq". When NOT set, the RXP will by default push frames to WQ 2.
23:16	DEFAULT_PUSH_WQ	Field is only used when
15	BD_CLEAR_ENABLE_ENAB LE	When set, the RXP will clear (set to zero) the bytes located in the BD between the PMI command bytes and the MPDU header bytes.
14	FILTER_RPE_TRACE_ENA BLE	When set, the RXP filter interaction result with the RPE will be passed on to the BMU. That can result in the BMU pushing this info into the BMU HW tracing list
13	DMA_RPE_TRACE_ENABL E	When set, the RXP DMA interaction result with the RPE will be passed on to the BMU. That can result in the BMU pushing this info into the BMU HW tracing list
12	CFG_FLT_DEFAULT_ROUTI NG_DIS	By default rxp uses the cfgflt_dpu_man_rtg_flg, cfgflt_dpu_data_rtg_flg to fill in dpu routing flags. If this bit is set then RXP will use the per type/subtype routing selection

WCSS_A_RXP_CONFIG2 (cont.)

Bits	Name	Description
11	CFG_FLT_PHY_ABORT_DS	by default phy abort is enabled when this bit is set it is disabled By default PHY RX will be aborted (shutdown) when the frame does not pass the type/subtype/address filter. This will save extra power
10	CFG_FLT_NAV_SET_ENABLE_ON_ABORT	When enabled and the incoming frame does not pass the type/subtype/address filter and is aborted before the FCS is calculated, update the NAV,
9	CFG_FLT_NAV_CLEAR_ENABLE_ON_ABORT	When enabled, and the incoming frame does not pass the type/subtype/address filter and is aborted before the FCS is calculated, the NAV will be cleared.
8	CFG_FLT_HW_SW_BASED_RESPONSE_SELECTION_ENABLE	when set, the response generation path: SW based or HW based is controlled based on programming in the type/subtype configuration table, done through register "frame_filter_config". Note that if this control is required, the 'cfgflt_generate_hw_response_enabled' should be cleared, otherwise this programming is overruled. When 'cfgflt_hw_sw_based_response_selection_enable' is set, 'cfgflt_generate_hw_response_enabled' is clear and 'G5_generate_HW_Response' for an incoming frame is set, the RXP will generate a pulse to the TPE to generate a response frame if also the following conditions are met: addr1 is a unicast address, caused a hit in the addr. lookup table AND incoming frame is a: Management frame OR data frame OR QoS data frame with ack policy set to normal OR RTS frame OR PSPOLL frame OR BAR_frame When 'cfgflt_hw_sw_based_response_selection_enable' is set, 'cfgflt_generate_hw_response_enabled' is clear and 'G5_generate_SW_RRI' for an incoming frame is set, the RXP will generate the RRI interrupt ,so that SW can start generating the response frame, if also the following conditions are met: addr1 is a unicast address, caused a hit in the addr. lookup table AND incoming frame is a: Management frame OR data frame OR QoS data frame with ack policy set to normal OR RTS frame OR PSPOLL frame OR BAR_frame
7	CFG_FLT_FILTER_GO_TO_IDLE_ON_CLEAR_ENABLED	Gen 4 workaround: When set, and the MPDU PROC statemachine signals 'clear' to the filter, the filter will return to its idle state, forcing it to break of all interaction with other modules
6	CFG_MPR_BMU_BA_UPDATE_ENABLED	When set, enables the rxp interface with the BMU for Block Ack update
5	CFG_MPDU_PROC_WAIT_FOR_IDLE_FILTER_ENABLED	When set, enables bug fix from gen 4. This forces the MPDU PROC statemachine to wait for the rxp filter to get into an idle state before aborting a frame.

WCSS_A_RXP_CONFIG2 (cont.)

Bits	Name	Description
4	CFG_MPR_TPE_INTERFAC E_ENABLED	When set to 1 the interface to the TPE is enabled
3	CFG_FLT_GENERATE_HW _RESPONSE_ENABLED	When set to 1 the global hw response generation logic within the RXP filter is enabled for the following conditions: addr1 is a unicast address, caused a hit in the addr. lookup table AND incoming frame is a: Management frame OR data frame OR QoS data frame with ack policy set to normal OR RTS frame OR PSPOLL frame OR BAR_frame
2	BMU_PWR_UPDATE_ENAB LED	When set, enables the rxp interface with the BMU for power management updates related to PWR Mgt bit in frame control field, PS POLL reception and APSD trigger frame reception
1	CFG_BD_DPU_SOFTMACFI ELDS_UPDATE_ENABLE	when you write a 1 the DPU related fields that softmac updates is enabled
0	CFG_RPE_INTERFACE_EN ABLE	when you write a 1 rpe interface is enabled That means duplicate detect interface function from the rxp filter is enabled as well as the bitmap update function from teh rxp dma controller

0x0A080A54 WCSS_A_RXP_CONFIG3**Type:** RW**Reset State:** 0x07180200

It contains RXP Configuration registers, related to gen5_6 functionality added

WCSS_A_RXP_CONFIG3

Bits	Name	Description
29:24	CFG_FLT_DPU_MAN_RTG_ FLG	This is the routing flag to which DPU will push management packets to, this is the value that is in the dpu_routing feild feild in RXBD for management packets
21:16	CFG_FLT_DPU_DATA_RTG_ FLG	This is the routing flag to which DPU will push data packets to, this is the value that is in the dpu_routing feild feild in RXBD for data packets
14	CFG_RSSI_MODE_SEL	This bit selects the two modes of rssi data collection when this bit is 1'b0 : select Mode 0, the byte2(RxFIR0) and Byte3(RxFIR1) of PHY STATS are selected to be stored when this bit is 1'b1 : select Mode 1, the greater of Byte2 and Byte3 and Byte4(SNR) of PHY stats are selected to be stored

WCSS_A_RXP_CONFIG3 (cont.)

Bits	Name	Description
11:0	CFG_FLT_RPE_HANDSHAKE_TIMEOUT_VALUE	The RXP filter interacts with the RPE. At the start of this interaction a timeout counter is initialized. This register provides the timeout value for this interaction in number of clk cycles

0x0A080A58 WCSS_A_RXP_SEARCH_TABLE_DATA2**Type:** RW**Reset State:** 0x00000000

This is data register 2 for indirect access to Binary Search Table. It contains all the DPU related configurations that the RXP will insert into the BD. There are totally 3 Data registers for Binary Search address Table.

WCSS_A_RXP_SEARCH_TABLE_DATA2

Bits	Name	Description
31:29	DPU_MC_BC_TAG	The DPU routing flag that will be inserted in the BD
28:21	DPU_MAN_MC_BC_DESCRIPTOR_INDEX	The DPU descriptor index that will be inserted in the BD for mc/bc management packets
20	DPU_NE	When set, the RXP will set the 'ne' field in the bd to 1'b1
19	WEP_KEY_ID_EXTRACT_ENABLE	When set, the RXP filter will extract the Key id from encrypted frames (assuming that the encryption is wep)
18:16	DPU_TAG	The DPU routing flag that will be inserted in the BD
15:8	DPU_MC_BC_DPU_DESC	This field contains dpu descriptor index for BC/MC QoS/NonQoS frames RXP will insert this field in the DPU descriptor field in the BD.
7:0	DPU_DESCRIPTOR_INDEX	The DPU descriptor index that will be inserted in the BD for unicast data or unicast robust management frames(RMF)

0x0A080A5C WCSS_A_RXP_BEACON_TSF_TIM_EXTRACT_CTRL**Type:** RW**Reset State:** 0x00000003

This controls what information needs to be extracted from the beacon and under what condition

WCSS_A_RXP_BEACON_TSF_TIM_EXTRACT_CTRL

Bits	Name	Description
31	BSS2_BEACON_PARSE_EN	when this bit is set it indicates that we can receive a client2 bss beacons so the beacon parsing checks whether received beacon is for BSS,P2P or BSS2(BSS CLient2) mode and accordingly TSF comparison and update is done 1'b0: no BSS beacons for client2 allowed 1'b1: BSS beacons for client2 allowed
30	EARLY_TIM_PROCESS	when this bit is set then the tim not set interrupt is generated immediatly after TIM bit is received and will not wait for the FCS complete
29	P2P_BEACON_PARS_EN	when this bit is set then the system is indicating that we can receive p2p beacons so the beacon parsing checks whether received beacon is for P2P or BSS mode and accordingly TSF comparison and update is done 1'b0: no P2P beacons allowed, only BSS beacons allowed 1'b1: P2P/BSS beacons allowed
28	SIM_IBSS_BSS_BEACON_PARS_EN	when this bit is set then the system is indicating that there is IBSS/BSS co-existence so the beacon parsing checks whether received beacon is for IBSS or BSS mode and accordingly TSF comparison and update is done 1'b1: IBSS/BSS beacons allowed 1'b0: no IBSS beacons allowed only BSS beacons allowed NOTE: when bit[29:28] = 2'b11 P2P/IBSS/BSS beacons allowed
27	BEACON_CFP_AWARENESS_ENABLE	When set, and beacon processing is enabled, the beacon will also be evaluated for the presence of the CF parameter set. If present, the CFP remaining duration period is extracted and the NAV is set to that value.
26:16	ASSOCIATION_ID	The association id of the station. This field is used to know which tim bit from the TIM element is related to this station
15:8	CFG_MPR_SSID_LENGTH	When 'enable_beacon_ssid_check' is set and 'enable_beacon_partial_ssid_check_ok' is NOT set, RXP will check the SSID length field of the received beacon. That length field will need to have the same value as this register, otherwise the the beacon filter will indicate a mismatch. When 'enable_beacon_ssid_check' is set and 'enable_beacon_partial_ssid_check_ok' is set as well, this length field indicates how many sequential bytes in a row of the beacon SSID field, starting from the first byte of the SSID field recieved, will have to match the programmed SSID bytes, in order for the beacon to pass the filter. Note that in this case, the SSID length field itself does not necessarily need to match, except that the length field in the beacon needs to be larger then the value programmed here.

WCSS_A_RXP_BEACON_TSF_TIM_EXTRACT_CTRL (cont.)

Bits	Name	Description
5	ENABLE_BEACON_SSID_CHECK	When set, the tsf update (based on IBSS or BSS rules) or tim field extraction will only happen if the incoming beacon has an SSID 3 field which matches the contents of the 'SSID_field' register, and 'SSID_length' field in this register
2	DISABLE_BEACON_DTIM_PARSING	When set, RXP will not generate wakeup pulse if broadcast packets are pending
1	ENABLE_BEACON_TIM_PARSING	When set, beacon tim field extraction and TIM field interrupt pulse generation is enabled
0	ENABLE_BEACON_TSF_PARSING	When set, beacon timestamp extraction and TSF update are enabled

0x0A080A60 WCSS_A_RXP_TSF_COMPENSATION1_VALUE**Type:** RW**Reset State:** 0x00000000

This register contains offset values to add the received TSF value to compensate for the delay in the PHY receive chain

WCSS_A_RXP_TSF_COMPENSATION1_VALUE

Bits	Name	Description
31:24	TSF_11B_11MBPS_COMP_VALUE	The TSF compensation value when the beacon is received at a 11b rate of 11Mbps
23:16	TSF_11B_5_5MBPS_COMP_VALUE	The TSF compensation value when the beacon is received at a 11b rate of 5.5Mbps
15:8	TSF_11B_2MBPS_COMP_VALUE	The TSF compensation value when the beacon is received at a 11b rate of 2Mbps
7:0	TSF_11B_1MBPS_COMP_VALUE	The TSF compensation value when the beacon is received at a 11b rate of 1Mbps

0x0A080A64 WCSS_A_RXP_TSF_COMPENSATION2_VALUE**Type:** RW**Reset State:** 0x00000000

This register contains offset values to add the received TSF value to compensate for the delay in the PHY receive chain

WCSS_A_RXP_TSF_COMPENSATION2_VALUE

Bits	Name	Description
23:16	TSF_11AG_OTHER_RATE_COMP_VALUE	The TSF compensation value when the beacon is received at any other 11a or 11g rate
15:8	TSF_11AG_54MBPS_COMP_VALUE	The TSF compensation value when the beacon is received at an 11a or 11g rate of 54Mbps
7:0	TSF_11AG_6MBPS_COMP_VALUE	The TSF compensation value when the beacon is received at an 11a or 11g rate of 6Mbps

0x0A080A68 WCSS_A_RXP_SSID_VALUES_B0_B3**Type:** RW**Reset State:** 0x00000000

This register contains SSID bytes 0 - 3

WCSS_A_RXP_SSID_VALUES_B0_B3

Bits	Name	Description
31:0	SSID_VALUES_BYTES_B0_B3	The SSID bytes, in LE format

0x0A080A6C WCSS_A_RXP_SSID_VALUES_B4_B7**Type:** RW**Reset State:** 0x00000000

This register contains SSID bytes 4 - 7

WCSS_A_RXP_SSID_VALUES_B4_B7

Bits	Name	Description
31:0	SSID_VALUES_BYTES_B4_B7	The SSID bytes, in LE format

0x0A080A70 WCSS_A_RXP_SSID_VALUES_B8_B11**Type:** RW**Reset State:** 0x00000000

This register contains SSID bytes 8 - 11

WCSS_A_RXP_SSID_VALUES_B8_B11

Bits	Name	Description
31:0	SSID_VALUES_BYTES_B8_11	The SSID bytes, in LE format

0x0A080A74 WCSS_A_RXP_SSID_VALUES_B12_B15**Type:** RW**Reset State:** 0x00000000

This register contains SSID bytes 12 - 15

WCSS_A_RXP_SSID_VALUES_B12_B15

Bits	Name	Description
31:0	SSID_VALUES_BYTES_B12_15	The SSID bytes, in LE format

0x0A080A78 WCSS_A_RXP_SSID_VALUES_B16_B19**Type:** RW**Reset State:** 0x00000000

This register contains SSID bytes 16 - 19

WCSS_A_RXP_SSID_VALUES_B16_B19

Bits	Name	Description
31:0	SSID_VALUES_BYTES_B16_19	The SSID bytes, in LE format

0x0A080A7C WCSS_A_RXP_SSID_VALUES_B20_B23**Type:** RW**Reset State:** 0x00000000

This register contains SSID bytes 20 - 23

WCSS_A_RXP_SSID_VALUES_B20_B23

Bits	Name	Description
31:0	SSID_VALUES_BYTES_B20_23	The SSID bytes, in LE format

0x0A080A80 WCSS_A_RXP_SSID_VALUES_B24_B27**Type:** RW**Reset State:** 0x00000000

This register contains SSID bytes 24 - 27

WCSS_A_RXP_SSID_VALUES_B24_B27

Bits	Name	Description
31:0	SSID_VALUES_BYTES_B24_27	The SSID bytes, in LE format

0x0A080A84 WCSS_A_RXP_SSID_VALUES_B28_B31**Type:** RW**Reset State:** 0x00000000

This register contains SSID bytes 28 - 31

WCSS_A_RXP_SSID_VALUES_B28_B31

Bits	Name	Description
31:0	SSID_VALUES_BYTES_B28_31	The SSID bytes, in LE format

0x0A080A88 WCSS_A_RXP_BEACON_ADDR2_LOW_FIELD**Type:** RW**Reset State:** 0x00000000

This register contains the lowest 4 bytes of the address 2 field of the beacon, in case a match function is enabled

WCSS_A_RXP_BEACON_ADDR2_LOW_FIELD

Bits	Name	Description
31:0	BEACON_ADDR2_LSB_BYTES	The 4 LS Bytes of the addr2, LE format

0x0A080A8C WCSS_A_RXP_BEACON_ADDR2_HIGH_FIELD**Type:** RW**Reset State:** 0x00000000

This register contains the highest 2 bytes of the address 2 field of the beacon, in case a match function is enabled

WCSS_A_RXP_BEACON_ADDR2_HIGH_FIELD

Bits	Name	Description
15:0	BEACON_ADDR2_MSB_BYTES	The 2 MS Bytes of the addr2, LE format

0x0A080A90 WCSS_A_RXP_BEACON_ADDR3_LOW_FIELD**Type:** RW**Reset State:** 0x00000000

This register contains the lowest 4 bytes of the address 3 field of the beacon, in case a match function is enabled

WCSS_A_RXP_BEACON_ADDR3_LOW_FIELD

Bits	Name	Description
31:0	BEACON_ADDR3_LSB_BYTES	The 4 LS Bytes of the addr3, LE format

0x0A080A94 WCSS_A_RXP_BEACON_ADDR3_HIGH_FIELD**Type:** RW**Reset State:** 0x00000000

This register contains the highest 2 bytes of the address 3 field of the beacon, in case a match function is enabled

WCSS_A_RXP_BEACON_ADDR3_HIGH_FIELD

Bits	Name	Description
15:0	BEACON_ADDR3_MSB_BYTES	The 2 MS Bytes of the addr3, LE format

0x0A080A98 WCSS_A_RXP_FLT_DUPL_CNT**Type:** R**Reset State:** 0x00000000

Number of frames that got dropped at the rxp filter because the RPE indicated to the RXP filter that the frame was a duplicate

WCSS_A_RXP_FLT_DUPL_CNT

Bits	Name	Description
7:0	FLT_DUPL_DROPPED_COUNT	number of frames dropped

0x0A080A9C WCSS_A_RXP_FLT_RPE_HANDSHAKE_TIMEOUT_CNT**Type:** R**Reset State:** 0x00000000

Number of times that the handshake between the rxp filter and the RPE timedout

WCSS_A_RXP_FLT_RPE_HANDSHAKE_TIMEOUT_CNT

Bits	Name	Description
19:16	DMA_RPE_TIMEOUT_COUNT	number of timeouts that occurred on the RXP DMA <-> RPE interface
3:0	FLT_RPE_TIMEOUT_COUNT	number of timeouts that occurred on the RXP Filter <-> RPE interface

0x0A080B00 WCSS_A_RXP_QID_MAPPING0**Type:** RW**Reset State:** 0x00000008

Controls how incoming data flows get mapped to a qid

WCSS_A_RXP_QID_MAPPING0

Bits	Name	Description
4:0	MAN_AND_DATA_QID	When the incoming frame is a management frame or plain (NON QoS) data frame, the contents of this register represents the qid assigned to this type of frames

0x0A080B04 WCSS_A_RXP_QID_MAPPING1**Type:** RW**Reset State:** 0x03020100

Controls how incoming data flows get mapped to a qid

WCSS_A_RXP_QID_MAPPING1

Bits	Name	Description
28:24	QOS_TID3_QID	When the incoming frame is a QoS data frame with TID set to 3, the contents of this register represents the qid assigned to this type of frames
20:16	QOS_TID2_QID	When the incoming frame is a QoS data frame with TID set to 2, the contents of this register represents the qid assigned to this type of frames

WCSS_A_RXP_QID_MAPPING1 (cont.)

Bits	Name	Description
12:8	QOS_TID1_QID	When the incoming frame is a QoS data frame with TID set to 1, the contents of this register represents the qid assigned to this type of frames
4:0	QOS_TID0_QID	When the incoming frame is a QoS data frame with TID set to 0, the contents of this register represents the qid assigned to this type of frames

0x0A080B08 WCSS_A_RXP_QID_MAPPING2**Type:** RW**Reset State:** 0x07060504

Controls how incoming data flows get mapped to a qid

WCSS_A_RXP_QID_MAPPING2

Bits	Name	Description
28:24	QOS_TID7_QID	When the incoming frame is a QoS data frame with TID set to 7, the contents of this register represents the qid assigned to this type of frames
20:16	QOS_TID6_QID	When the incoming frame is a QoS data frame with TID set to 6, the contents of this register represents the qid assigned to this type of frames
12:8	QOS_TID5_QID	When the incoming frame is a QoS data frame with TID set to 5, the contents of this register represents the qid assigned to this type of frames
4:0	QOS_TID4_QID	When the incoming frame is a QoS data frame with TID set to 4, the contents of this register represents the qid assigned to this type of frames

0x0A080B0C WCSS_A_RXP_QID_MAPPING3**Type:** RW**Reset State:** 0x00000000

Controls how incoming data flows get mapped to a qid

WCSS_A_RXP_QID_MAPPING3

Bits	Name	Description
28:24	QOS_TID11_QID	When the incoming frame is a QoS data frame with TID set to 11, the contents of this register represents the qid assigned to this type of frames
20:16	QOS_TID10_QID	When the incoming frame is a QoS data frame with TID set to 10, the contents of this register represents the qid assigned to this type of frames
12:8	QOS_TID9_QID	When the incoming frame is a QoS data frame with TID set to 9, the contents of this register represents the qid assigned to this type of frames
4:0	QOS_TID8_QID	When the incoming frame is a QoS data frame with TID set to 8, the contents of this register represents the qid assigned to this type of frames

0x0A080B10 WCSS_A_RXP_QID_MAPPING4**Type:** RW**Reset State:** 0x00000000

Controls how incoming data flows get mapped to a qid

WCSS_A_RXP_QID_MAPPING4

Bits	Name	Description
28:24	QOS_TID15_QID	When the incoming frame is a QoS data frame with TID set to 15, the contents of this register represents the qid assigned to this type of frames
20:16	QOS_TID14_QID	When the incoming frame is a QoS data frame with TID set to 14, the contents of this register represents the qid assigned to this type of frames
12:8	QOS_TID13_QID	When the incoming frame is a QoS data frame with TID set to 13, the contents of this register represents the qid assigned to this type of frames
4:0	QOS_TID12_QID	When the incoming frame is a QoS data frame with TID set to 12, the contents of this register represents the qid assigned to this type of frames

0x0A080B14 WCSS_A_RXP_PUSH_WQ_CTRL**Type:** RW**Reset State:** 0x02020202

Controls how frames that fall in the unicast other of multicast 'other categories' get routed through the system

WCSS_A_RXP_PUSH_WQ_CTRL

Bits	Name	Description
31:24	ADDR2_255_CATEGORY_PUSH_WQ_INDEX	When the incoming frame will end up with staid 255 for addr 2, the frame will be routed to this WQ.
23:16	ADDR2_254_CATEGORY_PUSH_WQ_INDEX	When the incoming frame will end up with staid 254 for addr 2, the frame will be routed to this WQ.
15:8	ADDR1_255_CATEGORY_PUSH_WQ_INDEX	When the incoming frame will end up with staid 255 for addr 1 (multicast other), the frame will be routed to this WQ.
7:0	ADDR1_254_CATEGORY_PUSH_WQ_INDEX	When the incoming frame will end up with staid 254 for addr 1(unicast other), the frame will be routed to this WQ.

0x0A080B18 WCSS_A_RXP_PUSH_WQ_CTRL2**Type:** RW**Reset State:** 0x00020202

Controls how frames that fall in the unicast other of multicast 'other categories' get routed through the system

WCSS_A_RXP_PUSH_WQ_CTRL2

Bits	Name	Description
23:16	UNFILTERED_FRAME_PUSH_WQ_INDEX	When the incoming frame will does not get processed in any address filtering, all three staid indexes are invalid and set to 253, the frame will be routed to this WQ
15:8	ADDR3_255_CATEGORY_PUSH_WQ_INDEX	When the incoming frame will end up with staid 255 for addr 3, the frame will be routed to this WQ.
7:0	ADDR3_254_CATEGORY_PUSH_WQ_INDEX	When the incoming frame will end up with staid 254 for addr 3, the frame will be routed to this WQ.

0x0A080B1C WCSS_A_RXP_EXTRA_FRAME_FILTER_CONTROL**Type:** RW**Reset State:** 0x000001DD

additional control for how frames are filtered

WCSS_A_RXP_EXTRA_FRAME_FILTER_CONTROL

Bits	Name	Description
15:9	RESERVED	Reserved for future filter functions
8	BLOCK_FRAMES_WITH_HT_CONTROL_FIELD	When set, frames that have a HT-control field present as specified in 11n are blocked from being received
7	BLOCK_MULTI_TID_BLOCK_ACK_REQ	When set, the multi TID block ACK Request variant as specified in 11n is blocked from being received
6	BLOCK_RESERVED_BLOCK_ACK_REQ	When set, the reserved block ACK Request variant as specified in 11n is blocked from being received
5	BLOCK_COMPRESSED_BLOCK_ACK_REQ	When set, the compressed block ACK Request variant as specified in 11n is blocked from being received
4	BLOCK_BASIC_BLOCK_ACK_REQ	When set, the basic block ACK Request variant as specified in 11n is blocked from being received
3	BLOCK_MULTI_TID_BLOCK_ACK	When set, the multi TID block ACK variant as specified in 11n is blocked from being received
2	BLOCK_RESERVED_BLOCK_ACK	When set, the reserved block ACK variant as specified in 11n is blocked from being received
1	BLOCK_COMPRESSED_BLOCK_ACK	When set, the compressed block ACK variant as specified in 11n is blocked from being received
0	BLOCK_BASIC_BLOCK_ACK	When set, the basic block ACK variant as specified in 11n is blocked from being received

0x0A080B20 WCSS_A_RXP_RXP_GROUPED_INTERRUPT_ENABLE**Type:** RW**Reset State:** 0x00000000

interrupt Enables for RXP errors

WCSS_A_RXP_RXP_GROUPED_INTERRUPT_ENABLE

Bits	Name	Description
27	DMA_GAM_ERROR	Write 1 to enable this interrupt, so that whenever there is error on the AHB bus (gam error) an interrupt will be generated to firmware.
26	RXP_SYSTEM_OUT_OF_MEMORY_FIRMWARE	Write 1 to enable this interrupt, so that whenever system runs out of memory this interrupt will be set. this is duplicate the one in bit 13 goes to host and this interrupt goes to firware only
25	P2P_BEACON_CTWING_DETECT_INT	when set enables this interrupt The NOA CTWin value is compared with the value programmed in p2p_parsing_control2 register if there is any change then this interrupt is generated Writing 1 to this bit will clear this interrupt this interrupt goes to firmware only
24	P2P_BEACON_OPPPS_CHNG_DETECT_INT_EN	when set enables this interrupt The NOA OppPS value is compared with the value programmed in p2p_parsing_control2 register if there is any change then this interrupt is generated this interrupt goes to firmware only --NOTE---- This interrupt is overloaded with NDPA received interrupt for ECO as partial fix for addr3 issue in VHTCB, to enable NDPA interrupt set bit 17 in spare_and_clkgate_dis register.
23	P2P_TIM_BIT_CLEAR_INT	When set, p2p beacon has tim bit clear this interrupt goes to firmware only
22	P2P_TIM_BIT_SET_INT	When set, tim bit set in p2p beacon this interrupt goes to firmware only
21	P2P_TIM_BEACON_CRC_ERROR_INT	When set, p2p beacon has CRC error this interrupt goes to firmware only
20	P2P_TIM_BASED_BROADCAST_FRAMES_PENDING_INTERRUPT	When set, indicates tim bit indicates broadcast frames are pending from a p2p beacon this interrupt goes to firmware only
19	P2P_BEACON_NOA_ABSENT_INT_EN	When set, P2P beacon which has no NOA element interrupt will be enabled this interrupt goes to firmware only
18	P2P_BEACON_NOA_INDEX_EQUAL_ONE_DETECT_INT_EN	when set, P2P beacon NOA index value equal to one interrupt will be enabled this interrupt goes to firmware only

WCSS_A_RXP_RXP_GROUPED_INTERRUPT_ENABLE (cont.)

Bits	Name	Description
17	P2P_BEACON_NOA_INDX_CHNG_DETECT_INT_EN	The NOA, index feild is compared with the index value programmed in p2p_parsing_control register if there is any change the this interrupt is generated When set, P2p Beacon NOA index value change detect interrupt is enabled this interrupt goes to firmware only
16	RXP_RADAR_INTERRUPT	When set, radar detect interrupt will be generated
15	RXP_MORE_DATA_SET_INTERRUPT	When set, and a frame is received that passes the filter, is a management, data or QoS data frame and the more data bit in the frame is SET an interrupt pulse will be generated. That status info of the corresponding frame is captured in the other fields more_data_int_control register. this interrupt goes to host only
14	RXP_RMF_FRAME_LEN_ERR	Write 1 to enable this interrupt, whenever we receive a RMF BC/MC frame with frame less than 48bytes(which is minimum) then this interrupt is generated. this interrupt goes to host only
13	RXP_SYSTEM_OUT_OF_MEMORY	Write 1 to enable this interrupt, so that whenever system runs out of memory this interrupt will be set. this interrupt goes to host only
12	RXP_MORE_DATA_CLEAR_INTERRUPT	When set, and a frame is received that passes the filter, is a management, data or QoS data frame and the more data bit in the frame is NOT set an interrupt pulse will be generated. That status info of the corresponding frame is captured in the other fields more_data_int_control register. this interrupt goes to host only
11	RXP_ADDR2_MISS_INTERRUPT	When set, and a frame is received that has an addr2 miss (and addr 1 is a hit), an 'addr2 miss interrupt will be generated. this interrupt goes to host only
10	RXP_MISSED_EOSP_INT	When set, interrupt is generated when at least one other frame with EOSP got received while the first EOSP frame reception got reported this interrupt goes to firmware only
9	RXP_EOSP_INT	When set, the EOSP interrupt will get generated if eossp frame is receieved this interrupt goes to firmware only
8	TIM_BIT_CLEAR_INT	When set, beacon has tim bit clear this interrupt goes to firmware only
7	TIM_BIT_SET_INT	When set, tim bit set in beacon this interrupt goes to firmware only

WCSS_A_RXP_RXP_GROUPED_INTERRUPT_ENABLE (cont.)

Bits	Name	Description
6	TIM_BEACON_CRC_ERR_INT	When set, beacon has CRC error this interrupt goes to firmware only
5	TIM_BASED_BROADCAST_FRAMES_PENDING_INT	When set, indicates tim bit indicates broadcast frames are pending this interrupt goes to host only
4	ASYNC_FIFO_FULL_ERR_INT_ENABLE	When set, and the async fifo gets full, an interrupt will be generated this interrupt goes to host only
3	GET_BMU_ERR_INT_ENABLE	When set, and the interaction with the bmu causes an error, an interrupt will be generated this interrupt goes to host only
2	INVALID_RATE_INDEX_INT_ENABLE	When set, and an invalid rate index is generated based on information from the PHY, an interrupt will be generated this interrupt goes to host only
1	RPE_DUPL_INTERFACE_TIMEOUT_INT_ENABLE	When set, the rpe duplicate detect interface handshake timeout error interrupt generation is enabled this interrupt goes to host only
0	RPE_BITMAP_INTERFACE_TIMEOUT_INT_ENABLE	When set, the rpe bitmap interface handshake timeout error interrupt generation is enabled this interrupt goes to host only

0x0A080B24 WCSS_A_RXP_RXP_GROUPED_INTERRUPT_STATUS**Type:** RW**Reset State:** 0x00000000

RXP combined interrupt status, writing 1 to any bit will clear the corresponding interrupt

WCSS_A_RXP_RXP_GROUPED_INTERRUPT_STATUS

Bits	Name	Description
31	DROP_RPE_FIFO_FULL	whenever a frame is dropped due to system out of memory this bit will indicate if there was a rpe fifo full that caused this packet drop 1: if syste_out_of memory is set and this bit is one packet dropped due RPE fifo full 0: if syste_out_of memory is set and this bit is zero then there may BMU BD/PDU issue Writing 1 to this bit will clear this status

WCSS_A_RXP_RXP_GROUPED_INTERRUPT_STATUS (cont.)

Bits	Name	Description
30	PACKET_PRIORITY	whenever a frame is dropped due to system out of memory this bit will indicate what was the priority of the frame that was dropped 1: high priority frame dropped 0: Low priority frame dropped Writing 1 to this bit will clear this status
27	DMA_GAM_ERROR	when set indicates there is error on the AHB bus (gam error) writing 1 to this bit will clear the interrupt
26	RXP_SYSTEM_OUT_OF_MEMORY_FIRMWARE_INT	when set it indicates system is running out of memory, which allows software to indicate to AP to back off on sending frames by sending frames with PM bit set Writing 1 to this bit will clear this interrupt this interrupt goes to firmware only
25	P2P_BEACON_CTWING_DETECT_INT	The NOA CTWin value is compared with the value programmed in p2p_parsing_control2 register if there is any change then this interrupt is generated Writing 1 to this bit will clear this interrupt this interrupt goes to firmware only
24	P2P_BEACON_OPPPS_CHNG_DETECT_INT	The NOA OppPS value is compared with the value programmed in p2p_parsing_control2 register if there is any change then this interrupt is generated this interrupt goes to firmware only
23	P2P_TIM_BIT_CLEAR_INT	When set, p2p beacon has tim bit clear Writing 1 to this bit will clear this interrupt this interrupt goes to firmware only
22	P2P_TIM_BIT_SET_INT	When set, tim bit set in p2p beacon Writing 1 to this bit will clear this interrupt this interrupt goes to firmware only
21	P2P_TIM_BEACON_CRC_ERROR_INT	When set, p2p beacon has CRC error Writing 1 to this bit will clear this interrupt this interrupt goes to firmware only
20	P2P_TIM_BASED_BROADCAST_FRAMES_PENDING_INT	When set, indicates tim bit in p2p beacon indicates broadcast frames are pending Writing 1 to this bit will clear this interrupt this interrupt goes to firmware only
19	P2P_BEACON_NOA_ABSENT_INT	When set, P2P beacon which has no NOA element interrupt is generated Writing 1 to this bit will clear this interrupt this interrupt goes to firmware only
18	P2P_BEACON_NOA_INDEX_EQ_ONE_DETECT_INT	when set, indicates P2P beacon NOA index value equal to one interrupt is generated Writing 1 to this bit will clear this interrupt this interrupt goes to firmware only

WCSS_A_RXP_RXP_GROUPED_INTERRUPT_STATUS (cont.)

Bits	Name	Description
17	P2P_BEACON_NOA_INDX_CHNG_DETECT_INT	The NOA, index feild is compared with the index value programmed in p2p_parsing_control register if there is any change the this interrupt is generated Writing 1 to this bit will clear this interrupt this interrupt goes to firmware only
16	RXP_RADAR_INTERRUPT	When set, radar detect interrupt will be generated Writing 1 to this bit will clear this interrupt
15	RXP_MORE_DATA_SET_INTERRUPT	When set, and a frame is received that passes the filter, is a management, data or QoS data frame and the more data bit in the frame is SET an interrupt pulse will be generated. That status info of the corresponding frame is captured in the other fields more_data_int_control register. Writing 1 to this bit will clear this interrupt this interrupt goes to host only
14	RXP_RMF_FRAME_LEN_ERR_INT	When set it indicates that we have received a RMF BC/MC frame with frame less than 48bytes(which is minimum) then this interrupt is generated. Writing 1 to this bit will clear this interrupt this interrupt goes to host only
13	RXP_SYSTEM_OUT_OF_MEMORY_INT	when set it indicates system is running out of memory, which allows software to indicate to AP to back off on sending frmaes by sending frames with PM bit set Writing 1 to this bit will clear this interrupt this interrupt goes to host only
12	RXP_MORE_DATA_CLEAR_INTERRUPT	When set, and a frame is received that passes the filter, is a management, data or QoS data frame and the more data bit in the frame is NOT set an interrupt pulse will be generated. That status info of the corresponding frame is captured in the other fields more_data_int_control register. Writing 1 to this bit will clear this interrupt this interrupt goes to host only
11	RXP_ADDR2_MISS_INTERRUPT	When set, and a frame is received that has an addr2 miss (and addr 1 is a hit), an 'addr2 miss interrupt will be generated. Writing 1 to this bit will clear this interrupt this interrupt goes to host only
10	RXP_MISSED_EOSP_INT	When set, interrupt is generated when at least one other frame with EOSP got received while the first EOSP frame reception got reported Writing 1 to this bit will clear this interrupt this interrupt goes to firmware only
9	RXP_EOSP_INT	When set, the EOSP interrupt will get generated if eosp frame is receieved Writing 1 to this bit will clear this interrupt this interrupt goes to firmware only

WCSS_A_RXP_RXP_GROUPED_INTERRUPT_STATUS (cont.)

Bits	Name	Description
8	TIM_BIT_CLEAR_INT	When set, beacon has tim bit clear Writing 1 to this bit will clear this interrupt this interrupt goes to firmware only
7	TIM_BIT_SET_INT	When set, tim bit set in beacon Writing 1 to this bit will clear this interrupt this interrupt goes to firmware only
6	TIM_BEACON_CRC_ERR_INT	When set, beacon has CRC error Writing 1 to this bit will clear this interrupt this interrupt goes to firmware only
5	TIM_BASED_BROADCAST_FRAMES_PENDING_INT	When set, indicates tim bit indicates broadcast frames are pending Writing 1 to this bit will clear this interrupt this interrupt goes to firmware only
4	ASYNC_FIFO_FULL_ERR_INT	When set, and the async fifo got full Writing 1 to this bit will clear this interrupt this interrupt goes to host only
3	GET_BMU_ERR_INT	When set, the interaction with the bmu caused an error Writing 1 to this bit will clear this interrupt this interrupt goes to host only
2	INVALID_RATE_INDEX_INT	When set, an invalid rate index is generated based on information from the PHY Writing 1 to this bit will clear this interrupt this interrupt goes to host only
1	RPE_DUPL_INTERFACE_TIMEOUT_INT	When set, the rpe duplicate detect interface handshake timeout error has been detected Writing 1 to this bit will clear this interrupt this interrupt goes to host only
0	RPE_BITMAP_INTERFACE_TIMEOUT_INT	When set, the rpe bitmap interface handshake timeout error has been detected Writing 1 to this bit will clear this interrupt this interrupt goes to host only

0x0A080B28 WCSS_A_RXP_DEFAULT_HW_INSERT_DPU_PARAM**Type:** RW**Reset State:** 0x00000000

DPU related parameters to be put in the BD when the "bd_dpu_softmacfields_update_enable" is set and the frame that is received does not cause a hit in the filter, but falls in the unicast/multicast others category, and is still allowed to come in.

WCSS_A_RXP_DEFAULT_HW_INSERT_DPU_PARAM

Bits	Name	Description
14:7	DPU_ROUTING_FLAG	The setting for the DPU_routing_flag bits in the BD
6:4	DPU_TAG	The setting for the DPU TAG bits in the BD
3:1	RX_KEY_ID_BITS	The setting for the RX KEY ID bits in the BD
0	NE_BIT	The setting for the NE bit in the BD

0x0A080B44 WCSS_A_RXP_PSPOLL_NAV_VALUE**Type:** RW**Reset State:** 0x00000000

indicates the NAV value that will be passed on to the MTU when a PSPOLL frame is received

WCSS_A_RXP_PSPOLL_NAV_VALUE

Bits	Name	Description
7:0	PSPOLL_NAV_VALUE	The NAV value that will be passed on to the MTU when a PSPOLL frame is received

0x0A080B48 WCSS_A_RXP_PMU_INTERFACE_CONTROL**Type:** RW**Reset State:** 0x00000000

Controls the PMU interfacing for Beacon Mode Power Save.

WCSS_A_RXP_PMU_INTERFACE_CONTROL

Bits	Name	Description
10	PMU_SLEEP_PULSE_GENERATED	When set, a sleep pulse has been generated to the PMU Write 0 to clear this bit
9	WAKEUP_ON_BC_TIM_SET_PULSE_GENERATED	When set, a wakeup pulse has been generated to the PMU indicating a beacon is received with the broadcast tim bit set Write 0 to clear this bit

WCSS_A_RXP_PMU_INTERFACE_CONTROL (cont.)

Bits	Name	Description
8	WAKEUP_ON_STA_TIM_SET_PULSE_GENERATED	When set, a wakeup pulse has been generated to the PMU indicating a beacon is received with the STA tim bit set Write 0 to clear this bit
3	SLEEP_ON_BC_TIM_CLEAR	When set, a sleep pulse will be generated to the PMU when a beacon is received with the broadcast tim bit clear NOTE: when both 'pmu_sleep_on_sta_tim_clear' and 'sleep_on_bc_tim_clear' are set, both requirements need to be met in order to generate the sleep pulse
2	PMU_SLEEP_ON_STA_TIM_CLEAR	When set, a sleep pulse will be generated to the PMU when a beacon is received with the sta tim bit clear
1	WAKEUP_ON_SBC_TIM_SET	When set, a wakeup pulse will be generated to the PMU when a beacon is received with the broadcast tim bit set
0	WAKEUP_ON_STA_TIM_SET	When set, a wakeup pulse will be generated to the PMU when a beacon is received with the STA tim bit set

0x0A080B50 WCSS_A_RXP_RATE_INDEX_HISTORY**Type:** R**Reset State:** 0x00000000

Gives an overview of the rate indexes of the last 2 frames received

WCSS_A_RXP_RATE_INDEX_HISTORY

Bits	Name	Description
24:16	PREVIOUS_RECEIVED_RATE_INDEX	The rate index of the previous frame received
8:0	LAST_RECEIVED_RATE_INDEX	The rate index of the last frame received

0x0A080B58 WCSS_A_RXP_ADDR2_MISS_INFO**Type:** R**Reset State:** 0x00000000

The lowest 32 bits of the addr 2 received that caused an addr 2 miss interrupt

WCSS_A_RXP_ADDR2_MISS_INFO

Bits	Name	Description
31:0	ADDR2_BIT31_0	The lowest 32 bits of the addr 2 received

0x0A080B5C WCSS_A_RXP_ADDR2_MISS_CTRL**Type:** R**Reset State:** 0x00000000

Addr 2 miss interrupt control and status

WCSS_A_RXP_ADDR2_MISS_CTRL

Bits	Name	Description
23:16	ADDR1_INDEX	The corresponding addr 1 index of the frame received
15:0	ADDR2_BIT47_32	The upper 16 bits of the addr 2 received

0x0A080B60 WCSS_A_RXP_EOSP_INT_CONTROL**Type:** RW**Reset State:** 0x00000000

EOSP interrupt control and status

WCSS_A_RXP_EOSP_INT_CONTROL

Bits	Name	Description
31	EOSP_INT_ENABLE	When set, and a frame is received that passes the filter and the EOSP bit in the QoS control field is set, an interrupt pulse will be generated. That status info of the corresponding frame is captured in the other fields in this register.
30	EOSP_INT_STATUS	When set, the EOSP interrupt got generated
29	EOSP_INT_MISSED	When set, at least one other frame with EOSP set got received while the first EOSP frame reception got reported
27:24	TID	The tid of the qos control field

WCSS_A_RXP_EOSP_INT_CONTROL (cont.)

Bits	Name	Description
23:16	ADDR3_INDEX	The Addr 3 index of the frame received
15:8	ADDR2_INDEX	The Addr 2 index of the frame received
7:0	ADDR1_INDEX	The Addr 1 index of the frame received

0x0A080B64 WCSS_A_RXP_MORE_DATA_CLEAR_INT_CONTROL**Type:** RW**Reset State:** 0x00000000

More data bit clear interrupt control and status

WCSS_A_RXP_MORE_DATA_CLEAR_INT_CONTROL

Bits	Name	Description
31	MORE_DATA_CLEAR_INT_ENABLE	When set, and a frame is received that passes the filter, is a management, data or QoS data frame and the more data bit in the frame is NOT set an interrupt pulse will be generated. That status info of the corresponding frame is captured in the other fields in this register.
30	MORE_DATA_CLEAR_INT_STATUS	When set, the more data bit clear interrupt got generated
29	MORE_DATA_INT_MISSED	When set, at least one other frame with more data bit clear got received before SW got to process the interrupt. while the first EOSP frame reception got reported
28	QOS_FRAME	when set indicates it is a qos frame
27:24	TID	tid of qos control feild
23:16	ADDR3_INDEX	The Addr 3 index of the frame received
15:8	ADDR2_INDEX	The Addr 2 index of the frame received
7:0	ADDR1_INDEX	The Addr 1 index of the frame received

0x0A080B6C WCSS_A_RXP_CONFIG4**Type:** RW**Reset State:** 0x00000000

It contains RXP Configuration registers.

WCSS_A_RXP_CONFIG4

Bits	Name	Description
28	WLAN_RX_GRANT	This bit gives status of wlan_rx_grant. Write 1 to this bit to clear it 0: No rx grant generated 1: rx grant is generated
27	WLAN_RX_GRANT_ENABLE	This bit is used to enable the wlan_rx grant feature 0: disable the WLAN_RX_Grants control of WLAN RX 1: enable the WLAN_RX_Grants control of WLAN RX
26:11	TID_PRIORITY	This bit gives priority for frames on a tid basis this is used in support for Better Qos By default TID 6 and 7 have high priority 1: High priority 0: Low priority
10	MANAGEMENT_PRIORITY	This bit gives priority for management frames this is used in support for Better Qos 1: High priority 0: Low priority
9	NONQOS_PRIORITY	This bit gives priority for non qos frames this is used in support for Better Qos 1: High priority 0: Low priority
8:1	UNKNOWN_MC_STAID	when bit 0 is set then that means we are continuing address2 look up for an unknown multicast so the staid assigned for these types of packet should be set here
0	CFG_FLT_ADDR1_BC_ENTRY_PRESENT	When this bit is set it indicates that a broadcast entry is present in address 1 when it is set it allows RXP to accept unknown multicast packets and continue address2 lookup

0x0A080B74 WCSS_A_RXP_BEACON_P2P_CLIENT_ADDR_LOW_FIELD**Type:** RW**Reset State:** 0x00000000

This register contains the lowest 4 bytes of the address 2 field of the p2p beacon, in case a match function is enabled

WCSS_A_RXP_BEACON_P2P_CLIENT_ADDR_LOW_FIELD

Bits	Name	Description
31:0	BEACON_P2P_CLIENT_ADDR_LSB_BYTES	The 4 LS Bytes of the addr2, LE format

0x0A080B78 WCSS_A_RXP_BEACON_P2P_CLIENT_ADDR_HIGH_FIELD**Type:** RW**Reset State:** 0x00000000

This register contains the highest 2 bytes of the address 3 field of the beacon, in case a match function is enabled

WCSS_A_RXP_BEACON_P2P_CLIENT_ADDR_HIGH_FIELD

Bits	Name	Description
15:0	BEACON_P2P_CLIENT_ADDR_MSB_BYTES	The 2 MS Bytes of the addr2, LE format

0x0A080B7C WCSS_A_RXP_P2P_BEACON_PARSE_CONTROL**Type:** RW**Reset State:** 0x09000000

This register contains all the configurations necessary for handling p2p beacon

WCSS_A_RXP_P2P_BEACON_PARSE_CONTROL

Bits	Name	Description
31:24	OUI_TYPE_VALUE	This is the OUI type value in P2P IE in the beacon. By default it is 09 indicating it is P2P version 1.0
23:16	NOA_INDEX_VALUE	The Index field value of NOA subelement in P2P element in beacon RXP will parse the beacon for this index field and if there is any change in value an interrupt is generated.
15:5	P2P_ASSOCIATION_ID	The association id of the station for p2p beacons. This field is used to know which tim bit from the TIM element is related to this station
2	P2P_BEACON_NOA_PARSING_DIS	as long as pronto is configured to receive p2p beacons noa parsing will be done. NOA parsing can be disabled by writing 1 to this bit

0x0A080B80 WCSS_A_RXP_MORE_DATA_SET_INT_CONTROL**Type:** RW**Reset State:** 0x00000000

More data bit set interrupt related status

WCSS_A_RXP_MORE_DATA_SET_INT_CONTROL

Bits	Name	Description
29	MORE_DATA_INT_MISSED	When set, at least one other frame with more data bit set got received before SW got to process the interrupt.
28	QOS_FRAME	when set indicates it is a qos frame
27:24	TID	tid of qos control feild
23:16	ADDR3_INDEX	The Addr 3 index of the frame received
15:8	ADDR2_INDEX	The Addr 2 index of the frame received
7:0	ADDR1_INDEX	The Addr 1 index of the frame received

0x0A080B84 WCSS_A_RXP_RULE1_CNTR_CTRL**Type:** RW**Reset State:** 0x00000000

Has all controls to generate rule1 counter

WCSS_A_RXP_RULE1_CNTR_CTRL

Bits	Name	Description
31:28	RULE1_QID_MASK	Only the bit that are set here are considered for counting when mask is set for any bit the corresponding bit in rule1_qid_en is valid
27:24	RULE1_QID_EN	Only the bit that are set here are considered for counting Program all the qid bit values that needs to be considered for counting The setting of the bits whose corresponding bits in the rule1_qid_mask are set to 0 are ignored.

WCSS_A_RXP_RULE1_CNTR_CTRL (cont.)

Bits	Name	Description
23:20	RULE1_STAID_MASK	Only the bit that are set here are considered for counting when mask is set for any bit the corresponding bit in rule1_staid_en is valid
19:16	RULE1_STAID_EN	Program all the staid bit values that needs to be considered for counting The setting of the bits whose corresponding bits in the rule1_staid_mask are set to 0 are ignored
15:10	RULE1_TYPESUBTYPE_MASK	Only the bit that are set here are considered for counting when mask is set for any bit the corresponding bit in rule1_typesubtype_en is valid bits[15:14] = FC type bits[13:10] = FC subtype
9:4	RULE1_TYPESUBTYPE_EN	Program all the type/subtype bit values that needs to be considered for counting. The setting of the bits whose corresponding bits in the rule1_typesubtype_mask are set to 0 are ignored. bits[9:8] = FC type bits[7:4] = FC subtype
3	RULE1_HT_PRESENT_MASK	When set, ht present/or not present check is enabled
2	RULE1_HT_PRESENT_EN	program the requirement of HT being present or not. This field is ignored if bit rule1_ht_present_mask is not set.
0	RULE1_CNTR_EN	when set enables the counter

0x0A080B88 WCSS_A_RXP_RULE2_CNTR_CTRL**Type:** RW**Reset State:** 0x00000000

Has all controls to generate rule1 counter

WCSS_A_RXP_RULE2_CNTR_CTRL

Bits	Name	Description
31:28	RULE2_QID_MASK	Only the bit that are set here are considered for counting when mask is set for any bit the corresponding bit in rule2_qid_en is valid

WCSS_A_RXP_RULE2_CNTR_CTRL (cont.)

Bits	Name	Description
27:24	RULE2_QID_EN	Only the bit that are set here are considered for counting Program all the qid bit values that needs to be considered for counting The setting of the bits whose corresponding bits in the rule2_qid_mask are set to 0 are ignored.
23:20	RULE2_STAID_MASK	Only the bit that are set here are considered for counting when mask is set for any bit the corresponding bit in rule2_staid_en is valid
19:16	RULE2_STAID_EN	Program all the staid bit values that needs to be considered for counting The setting of the bits whose corresponding bits in the rule2_staid_mask are set to 0 are ignored
15:10	RULE2_TYPESUBTYPE_MASK	Only the bit that are set here are considered for counting when mask is set for any bit the corresponding bit in rule2_typesubtype_en is valid bits[15:14] = FC type bits[13:10] = FC subtype
9:4	RULE2_TYPESUBTYPE_EN	Program all the type/subtype bit values that needs to be considered for counting. The setting of the bits whose corresponding bits in the rule2_typesubtype_mask are set to 0 are ignored. bits[9:8] = FC type bits[7:4] = FC subtype
3	RULE2_HT_PRESENT_MASK	When set, ht present/or not present check is enabled
2	RULE2_HT_PRESENT_EN	program the requirement of HT being present or not. This field is ignored if bit rule2_ht_present_mask is not set.
0	RULE2_CNTR_EN	when set enables the counter

0x0A080B8C WCSS_A_RXP_RULE1_CNTR**Type:** R**Reset State:** 0x00000000

Rule 1 counter, if all conditions in rule1_ctrl pass then this counter gets incremented

WCSS_A_RXP_RULE1_CNTR

Bits	Name	Description
7:0	RULE1_CNTR	This counter indicates number of packets received which passes rule 1 control conditions

0x0A080B90 WCSS_A_RXP_RULE2_CNTR**Type:** R**Reset State:** 0x00000000

Rule 2 counter, if all conditions in rule2_ctrl pass then this counter gets incremented

WCSS_A_RXP_RULE2_CNTR

Bits	Name	Description
7:0	RULE2_CNTR	This counter indicates number of packets received which passes rule 2 control conditions

0x0A080B94 WCSS_A_RXP_DROP_DUE_TO_DROP_AT_DMA_CNT**Type:** R**Reset State:** 0x00000000

The number of packets are dropped by RXP due to drop at DMA bit set in type/subtype filter(these are mostly control packets)

WCSS_A_RXP_DROP_DUE_TO_DROP_AT_DMA_CNT

Bits	Name	Description
15:0	DROP_DUE_TO_DROP_DMA_CMD_CNT	The number of packets are dropped by RXP due to drop at DMA bit being set these are usually all control packets.

0x0A080B98 WCSS_A_RXP_RX_TIME_CNTR**Type:** R**Reset State:** 0x00000000

The number of early pkt detect at every usec

WCSS_A_RXP_RX_TIME_CNTR

Bits	Name	Description
23:0	RX_TIME_CNT	The number of usec that have early pkt detect high

0x0A080BA0 WCSS_A_RXP_NDP_PKT_CNTR**Type:** R**Reset State:** 0x00000000

The number of NDP pkts received

WCSS_A_RXP_NDP_PKT_CNTR

Bits	Name	Description
15:0	NDP_PKT_CNT	Number of NDP packets received

0x0A080BA4 WCSS_A_RXP_P2P_BEACON_PARSE_CONTROL2**Type:** RW**Reset State:** 0x00000000

This register contains some more configurations necessary for handling p2p beacon

WCSS_A_RXP_P2P_BEACON_PARSE_CONTROL2

Bits	Name	Description
7	NOA_OPPPS_VALUE	This value is set by SW, set 1 indicates group owner is using opportunistic power save, set to 0 indicates that it is disabled This valuw is compared to that in the beacon any change in this will generate a interrupt
6:0	NOA_CTWIN_VALUE	Client Traffic window of value 0-6 is programmed in this register RXP will parse the beacon for this n CTWin of NOA element and if there is any change in value an interrupt is generated. This value is a period of time in TU after TBTT during which P2P group owner is present

0x0A080BA8 WCSS_A_RXP_BSS2_BEACON_PARSE_CONTROL**Type:** RW**Reset State:** 0x28250000

This register contains all the configurations necessary for handling bss2 beacon

WCSS_A_RXP_BSS2_BEACON_PARSE_CONTROL

Bits	Name	Description
31:24	IE_EID1	These bits can be programmed to any IE(except zero length IEs) in the beacon that needs to be parsed By default it is "Quiet" IE. Also the programmed in this register can not be same as IE_eid0
23:16	IE_EID0	These bits can be programmed to any IE(except zero length IE) in the beacon that needs to be parsed By default it is "Channel Switch" IE.Also the programmed in this register can not be same as IE_eid1
13	IE_EID1_PARSE_ENABLE	SW need to set this bit whenever SW wants to enable the parsing of the IE programmed in bits [31:24]
12	IE_EID0_PARSE_ENABLE	SW need to set this bit whenever SW wants to enable the parsing of the IE programmed in bits [23:16]
10:0	BSS2_ASSOCIATION_ID	The association id of the station for bss2 beacons. This field is used to know which tim bit from the TIM element is related to this station

0x0A080BAC WCSS_A_RXP_BEACON_BSS2_ADDR_LOW_FIELD**Type:** RW**Reset State:** 0x00000000

This register contains the lowest 4 bytes of the address 2 field of the bss2 beacon, in case a match function is enabled

WCSS_A_RXP_BEACON_BSS2_ADDR_LOW_FIELD

Bits	Name	Description
31:0	BEACON_BSS2_CLIENT_ADDR_LSB_BYTES	The 4 LS Bytes of the addr2, LE format

0x0A080BB0 WCSS_A_RXP_BEACON_BSS2_CLIENT_ADDR_HIGH_FIELD**Type:** RW**Reset State:** 0x00000000

This register contains the highest 2 bytes of the address 3 field of the beacon, in case a match function is enabled

WCSS_A_RXP_BEACON_BSS2_CLIENT_ADDR_HIGH_FIELD

Bits	Name	Description
15:0	BEACON_BSS2_CLIENT_ADDR_MSB_BYTES	The 2 MS Bytes of the addr2, LE format

0x0A080BB4 WCSS_A_RXP_RXP_GROUPED2_INTERRUPT_ENABLE**Type:** RW**Reset State:** 0x00000000

interrupt Enables for RXP errors

WCSS_A_RXP_RXP_GROUPED2_INTERRUPT_ENABLE

Bits	Name	Description
15	P2P_EID1_ABSENT_INT_EN	When set, enables interrupt when received beacon is p2p beacon and has reached end of beacon and has not found IE with EID1 programmed in bss2_beacon_extract_control register feild [31:24] this interrupt goes to firmware only
14	P2P_EID0_ABSENT_INT_EN	When set, enables interrupt when received beacon is p2p beacon and has reached end of the beacon and has not found IE with EID0 programmed bss2_beacon_extract_control register feild [23:15] this interrupt goes to firmware only
13	P2P_EID1_PRESENT_INT_EN	When set, enables interrupt when received beacon is p2p beacon and has IE with EID1 programmed in bss2_beacon_extract_control register feild [31:24] this interrupt goes to firmware only
12	P2P_EID0_PRESENT_INT_EN	When set, enables interrupt when received beacon is p2p beacon and has found IE with EID0 programmed bss2_beacon_extract_control register feild [23:15] this interrupt goes to firmware only
11	BSS_EID1_ABSENT_INT_EN	When set, received beacon is Bss beacon and has reached end of beacon and has not found IE with EID1 programmed in bss2_beacon_extract_control register feild [31:24] this interrupt goes to firmware only
10	BSS_EID0_ABSENT_INT_EN	When set, enables interrupt when received beacon is Bss beacon and has reached end of the beacon and has not found IE with EID0 programmed bss2_beacon_extract_control register feild [23:15] this interrupt goes to firmware only

WCSS_A_RXP_RXP_GROUPE2_INTERRUPT_ENABLE (cont.)

Bits	Name	Description
9	BSS_EID1_PRESENT_INT_EN	When set, enables interrupt when received beacon is Bss beacon and has found IE with EID1 programmed in bss2_beacon_extract_control register feild [31:24] this interrupt goes to firmware only
8	BSS_EID0_PRESENT_INT_EN	When set, enables interrupt when received beacon is Bss beacon and has found IE with EID0 programmed bss2_beacon_extract_control register feild [23:15] this interrupt goes to firmware only
7	BSS2_EID1_ABSENT_INT_EN	When set, enables interrupt when received beacon is Bss beacon for client 2 and has reached end of beacon and has not found IE with EID1 programmed in bss2_beacon_extract_control register feild [31:24] this interrupt goes to firmware only
6	BSS2_EID0_ABSENT_INT_EN	When set, enables interrupt when received beacon is Bss beacon for client2 and has reached end of the beacon and has not found IE with EID0 programmed bss2_beacon_extract_control register feild [23:15] this interrupt goes to firmware only
5	BSS2_EID1_PRESENT_INT_EN	When set, enables interrupt when received beacon is Bss beacon for client2 and has IE with EID1 programmed in bss2_beacon_extract_control register feild [31:24] this interrupt goes to firmware only
4	BSS2_EID0_PRESENT_INT_EN	When set, enables interrupt when received beacon is Bss beacon for client2 and has IE with EID0 programmed bss2_beacon_extract_control register feild [23:15] this interrupt goes to firmware only
3	BSS2_TIM_BIT_CLEAR_INT_EN	When set, it enables the beacon has tim bit clear interrupt this interrupt goes to firmware only
2	BSS2_TIM_BIT_SET_INT_EN	When set, it enables tim bit set in beacon this interrupt goes to firmware only
1	BSS2_TIM_BEACON_CRC_ERR_INT_EN	When set, it enables the beacon has CRC error int this interrupt goes to firmware only
0	BSS2_TIM_BASED_BROADCAST_FRAMES_PENDING_INT_EN	When set, enables interrupt for detecting tim bit which indicates broadcast frames are pending this interrupt goes to firmware only

0x0A080BB8 WCSS_A_RXP_RXP_GROUPED2_INTERRUPT_STATUS**Type:** RW**Reset State:** 0x00000000

RXP combined interrupt status, writing 1 to any bit will clear the corresponding interrupt

WCSS_A_RXP_RXP_GROUPED2_INTERRUPT_STATUS

Bits	Name	Type	Description
31:27	DMA_STATE_ON_GAMERR OR	R	gives status of dma statemachine when gam error occured
15	P2P_EID1_ABSENT_INT	RW	When set, received beacon is p2p beacon and has reached end of beacon and has not found IE with EID1 programmed in bss2_beacon_extract_control register feild [31:24] Writing 1 to this bit will clear this interrupt this interrupt goes to firmware only
14	P2P_EID0_ABSENT_INT	RW	When set, received beacon is p2p beacon and has reached end of the beacon and has not found IE with EID0 programmed bss2_beacon_extract_control register feild [23:15] Writing 1 to this bit will clear this interrupt this interrupt goes to firmware only
13	P2P_EID1_PRESENT_INT	RW	When set, received beacon is p2p beacon and has IE with EID1 programmed in bss2_beacon_extract_control register feild [31:24] Writing 1 to this bit will clear this interrupt this interrupt goes to firmware only
12	P2P_EID0_PRESENT_INT	RW	When set, received beacon is p2p beacon and has IE with EID0 programmed bss2_beacon_extract_control register feild [23:15] Writing 1 to this bit will clear this interrupt this interrupt goes to firmware only
11	BSS_EID1_ABSENT_INT	RW	When set, received beacon is Bss beacon and has reached end of beacon and has not found IE with EID1 programmed in bss2_beacon_extract_control register feild [31:24] Writing 1 to this bit will clear this interrupt this interrupt goes to firmware only
10	BSS_EID0_ABSENT_INT	RW	When set, received beacon is Bss beacon and has reached end of the beacon and has not found IE with EID0 programmed bss2_beacon_extract_control register feild [23:15] Writing 1 to this bit will clear this interrupt this interrupt goes to firmware only

WCSS_A_RXP_RXP_GROUPED2_INTERRUPT_STATUS (cont.)

Bits	Name	Type	Description
9	BSS_EID1_PRESENT_INT	RW	When set, received beacon is Bss beacon and has IE with EID1 programmed in bss2_beacon_extract_control register feild [31:24] Writing 1 to this bit will clear this interrupt this interrupt goes to firmware only
8	BSS_EID0_PRESENT_INT	RW	When set, received beacon is Bss beacon and has IE with EID0 programmed bss2_beacon_extract_control register feild [23:15] Writing 1 to this bit will clear this interrupt this interrupt goes to firmware only
7	BSS2_EID1_ABSENT_INT	RW	When set, received beacon is Bss beacon for client 2 and has reached end of beacon and has not found IE with EID1 programmed in bss2_beacon_extract_control register feild [31:24] Writing 1 to this bit will clear this interrupt this interrupt goes to firmware only
6	BSS2_EID0_ABSENT_INT	RW	When set, received beacon is Bss beacon for client2 and has reached end of the beacon and has not found IE with EID0 programmed bss2_beacon_extract_control register feild [23:15] Writing 1 to this bit will clear this interrupt this interrupt goes to firmware only
5	RXP_BSS2_EID1_PRESEN T_INT	RW	When set, received beacon is Bss beacon for client2 and has found IE with EID1 programmed in bss2_beacon_extract_control register feild [31:24] Writing 1 to this bit will clear this interrupt this interrupt goes to firmware only
4	RXP_BSS2_EID0_PRESEN T_INT	RW	When set, received beacon is Bss beacon for client2 and has found IE with EID0 programmed bss2_beacon_extract_control register feild [23:15] Writing 1 to this bit will clear this interrupt this interrupt goes to firmware only
3	BSS2_TIM_BIT_CLEAR_IN T	RW	When set, beacon has tim bit clear Writing 1 to this bit will clear this interrupt this interrupt goes to firmware only
2	BSS2_TIM_BIT_SET_INT	RW	When set, tim bit set in beacon Writing 1 to this bit will clear this interrupt this interrupt goes to firmware only
1	BSS2_TIM_BEACON_CRC_ ERR_INT	RW	When set, beacon has CRC error Writing 1 to this bit will clear this interrupt this interrupt goes to firmware only

WCSS_A_RXP_RXP_GROUPED2_INTERRUPT_STATUS (cont.)

Bits	Name	Type	Description
0	BSS2_TIM_BASED_BROADCAST_FRAMES_PENDING_INT	RW	When set, indicates tim bit indicates broadcast frames are pending Writing 1 to this bit will clear this interrupt this interrupt goes to firmware only

0x0A080BBC WCSS_A_RXP_RXP_QBOOST_CONTROL**Type:** RW**Reset State:** 0x00000000

Qboost control register

WCSS_A_RXP_RXP_QBOOST_CONTROL

Bits	Name	Description
15:8	QBOOST_STAID	When Qboost is enabled the STAID programmed here is staid of the address2 we are associated with
4	QBOOST_QOS_ACK_POL_NORMAL	when this bit is set and the incoming frame is a QoS frame, the ACK policy shall be set to Normal (or implicit BA request) in order for the frame to be considered as a Qboost candidate trigger frame
3	QBOOST_PWR_STATE_CHK_EN	when this bit is set, the Power State bit in the incoming shall NOT be set, in order for the frame to be considered as a Qboost candidate trigger frame
2	QBOOST_QOSNULL_PKT_EN	when this bit is set qos NULL packets will be subject to Qboost
1	QBOOST_BAR_PKT_EN	when this bit is set bar packets will be subject to Qboost
0	QBOOST_QOSDATA_PKT_EN	when this bit is set qos data packets will be subject to Qboost

0x0A080BC0 WCSS_A_RXP_RXP_P2P_CHANNELSWITCH_BODY**Type:** R**Reset State:** 0x00000000

This register has contents of Channel switch announcement IE for a p2p beacon

WCSS_A_RXP_RXP_P2P_CHANNELSWITCH_BODY

Bits	Name	Description
23:0	RXP_P2P_CHANNELSWITCH_IE_BODY	This register has contents of channel switch IE Bits [7:0] : Has Channel switch count field of CS IE Bits [15:8] : Has New Channel number field of CS IE Bits [23:16] : Has Channel switch Mode field of CS IE

0x0A080BC4 WCSS_A_RXP_RXP_BSS_CHANNELSWITCH_BODY**Type:** R**Reset State:** 0x00000000

This register has contents of Channel switch announcement IE for a bss beacon

WCSS_A_RXP_RXP_BSS_CHANNELSWITCH_BODY

Bits	Name	Description
23:0	RXP_BSS_CHANNELSWITCH_IE_BODY	This register has contents of channel switch IE Bits [7:0] : Has Channel switch count field of CS IE Bits [15:8] : Has New Channel number field of CS IE Bits [23:16] : Has Channel switch Mode field of CS IE

0x0A080BC8 WCSS_A_RXP_RXP_BSS2_CHANNELSWITCH_BODY**Type:** R**Reset State:** 0x00000000

This register has contents of Channel switch announcement IE for a bss2 beacon

WCSS_A_RXP_RXP_BSS2_CHANNELSWITCH_BODY

Bits	Name	Description
23:0	RXP_BSS2_CHANNELSWITCH_IE_BODY	This register has contents of channel switch IE Bits [7:0] : Has Channel switch count field of CS IE Bits [15:8] : Has New Channel number field of CS IE Bits [23:16] : Has Channel switch Mode field of CS IE

0x0A080BCC WCSS_A_RXP_RXP_P2P_QUIETIE_LSBYTES**Type:** R**Reset State:** 0x00000000

This register has contents of quiet element IE for a p2p beacon

WCSS_A_RXP_RXP_P2P_QUIETIE_LSBYTES

Bits	Name	Description
31:0	RXP_P2P_QUIETIE_LSB	This register has lower 4 bytes of contents of Quiet Element IE refer to following byte descriptions for correct byte positions Bits [15:0] : Has feild Quiet offset Bits [31:16] : Has field Quiet Duration

0x0A080BD0 WCSS_A_RXP_RXP_BSS_QUIETIE_LSBYTES**Type:** R**Reset State:** 0x00000000

This register has contents of quiet element IE for a bss beacon

WCSS_A_RXP_RXP_BSS_QUIETIE_LSBYTES

Bits	Name	Description
31:0	RXP_BSS_QUIETIE_LSB	This register has lower 4 bytes of contents of Quiet Element IE refer to following byte descriptions for correct byte positions Bits [15:0] : Has feild Quiet offset Bits [31:16] : Has field Quiet Duration

0x0A080BD4 WCSS_A_RXP_RXP_BSS2_QUIETIE_LSBYTES**Type:** R**Reset State:** 0x00000000

This register has contents of quiet element IE for a bss2 beacon

WCSS_A_RXP_RXP_BSS2_QUIETIE_LSBYTES

Bits	Name	Description
31:0	RXP_BSS2_QUIETIE_LSB	This register has lower 4 bytes of contents of Quiet Element IE refer to following byte descriptions for correct byte positions Bits [15:0] : Has feild Quiet offset Bits [31:16] : Has field Quiet Duration

0x0A080BD8 WCSS_A_RXP_RXP_P2P_QUIETIE_MSBYTES**Type:** R**Reset State:** 0x00000000

This register has contents of quiet element IE for a p2p beacon

WCSS_A_RXP_RXP_P2P_QUIETIE_MSBYTES

Bits	Name	Description
15:0	RXP_P2P_QUIETIE_MSB	This register has upper 2 bytes of contents of Quiet Element IE refer to following byte descriptions for correct byte positions Bits [7:0] : Has field Quiet period Bits [15:8] : Has field Quiet count

0x0A080BDC WCSS_A_RXP_RXP_BSS_QUIETIE_MSBYTES**Type:** R**Reset State:** 0x00000000

This register has contents of quiet element IE for a bss beacon

WCSS_A_RXP_RXP_BSS_QUIETIE_MSBYTES

Bits	Name	Description
15:0	RXP_BSS_QUIETIE_MSB	This register has upper 2 bytes of contents of Quiet Element IE refer to following byte descriptions for correct byte positions Bits [7:0] : Has field Quiet period Bits [15:8] : Has field Quiet count

0x0A080BE0 WCSS_A_RXP_RXP_BSS2_QUIETIE_MSBYTES**Type:** R**Reset State:** 0x00000000

This register has contents of quiet element IE for a bss2 beacon

WCSS_A_RXP_RXP_BSS2_QUIETIE_MSBYTES

Bits	Name	Description
15:0	RXP_BSS2_QUIETIE_MSB	This register has upper 2 bytes of contents of Quiet Element IE refer to following byte descriptions for correct byte positions Bits [7:0] : Has field Quiet period Bits [15:8] : Has field Quiet count

10.55 WCSS_A_TAQAM_TAQAM

0x0A012C00 WCSS_A_TAQAM_SCALE_BPSK

Type: RW

Reset State: 0x007F

Amplitude scale factor for BPSK modulation

WCSS_A_TAQAM_SCALE_BPSK

Bits	Name	Description
8:0	REG	

0x0A012C08 WCSS_A_TAQAM_SCALE_QPSK

Type: RW

Reset State: 0x005A

Amplitude scale factor for QPSK modulation

WCSS_A_TAQAM_SCALE_QPSK

Bits	Name	Description
8:0	REG	

0x0A012C0C WCSS_A_TAQAM_SCALE_QAM16

Type: RW

Reset State: 0x0050

Amplitude scale factor for 16QAM modulation

WCSS_A_TAQAM_SCALE_QAM16

Bits	Name	Description
8:0	REG	

0x0A012C10 WCSS_A_TAQAM_SCALE_QAM64

Type: RW

Reset State: 0x004E

Amplitude scale factor for 64QAM modulation

WCSS_A_TAQAM_SCALE_QAM64

Bits	Name	Description
8:0	REG	

0x0A012C14 WCSS_A_TAQAM_SCALE_QAM256

Type: RW

Reset State: 0x004E

Amplitude scale factor for 256QAM modulation

WCSS_A_TAQAM_SCALE_QAM256

Bits	Name	Description
8:0	REG	

0x0A012C18 WCSS_A_TAQAM_SCALE_LEGACY20

Type: RW

Reset State: 0x0089

Amplitude scale factor for legacy packets and 11n/ac legacy symbols

WCSS_A_TAQAM_SCALE_LEGACY20

Bits	Name	Description
8:0	REG	

0x0A012C1C WCSS_A_TAQAM_SCALE_LEGACY40

Type: RW

Reset State: 0x0060

Amplitude scale factor for legacy 40MHz duplicate packets and 11n/ac 40MHz legacy symbols

WCSS_A_TAQAM_SCALE_LEGACY40

Bits	Name	Description
8:0	REG	

0x0A012C20 WCSS_A_TAQAM_SCALE_LEGACY80**Type:** RW**Reset State:** 0x0085

Amplitude scale factor for legacy 80MHz duplicate packets and 11n/ac 80MHz legacy symbols

WCSS_A_TAQAM_SCALE_LEGACY80

Bits	Name	Description
8:0	REG	

0x0A012C24 WCSS_A_TAQAM_SCALE_HT20**Type:** RW**Reset State:** 0x0085

Amplitude scale factor for (V)HT-LTF, HT-SIG in GF mode, VHT-SIG-B and Data 11n/ac 20MHz symbols

WCSS_A_TAQAM_SCALE_HT20

Bits	Name	Description
8:0	REG	

0x0A012C28 WCSS_A_TAQAM_SCALE_HT40**Type:** RW**Reset State:** 0x005B

Amplitude scale factor for (V)HT-LTF, HT-SIG in GF mode, VHT-SIG-B and Data 11n/ac 40MHz symbols

WCSS_A_TAQAM_SCALE_HT40

Bits	Name	Description
8:0	REG	

0x0A012C2C WCSS_A_TAQAM_SCALE_HT80**Type:** RW**Reset State:** 0x007B

Amplitude scale factor for VHT-LTF, VHT-SIG-B and Data 11ac 80MHz symbols

WCSS_A_TAQAM_SCALE_HT80

Bits	Name	Description
8:0	REG	

0x0A012C30 WCSS_A_TAQAM_AC**Type:** RW**Reset State:** 0x03E7

Programmable features to mitigate risk of spec change

WCSS_A_TAQAM_AC

Bits	Name	Description
9	SB_ROT_EN	
8	CYC_ROT_EN	
7:0	PILOT80	

10.56 WCSS_A_TBAPB_TBAPB**0x0A012800 WCSS_A_TBAPB_CRC16_CTRL****Type:** RW**Reset State:** 0x0000

Controls CRC16 generator operation that calculates IEEE802.11b PLCP header CRC. For debugging purposes.

WCSS_A_TBAPB_CRC16_CTRL

Bits	Name	Description
0	DIS	When set disables CRC16 calculation. Should be 0 (enabled) for normal operation. 0x0: ENABLE (CRC16 calculation enabled) 0x1: DISABLE (CRC16 calculation disabled)

0x0A012804 WCSS_A_TBAPB_SCRAMBLE_CTRL**Type:** RW**Reset State:** 0x0000

Controls data scrambler operation that scrambled IEEE802.11b frame. For debugging purposes.

WCSS_A_TBAPB_SCRAMBLE_CTRL

Bits	Name	Description
0	DIS	When set disables data scrambling by bypassing scrambler. Should be 0 (enabled) for normal operation. 0x0: ENABLE (scrambler enabled) 0x1: DISABLE (scrambler bypassed)

0x0A01280C WCSS_A_TBAPB_LINEAR_DELAY_ANT1

Type: RW

Reset State: 0x0002

Stores the select for the delay diversity value to be applied to txb chain2 samples in barker and cck modulated data

WCSS_A_TBAPB_LINEAR_DELAY_ANT1

Bits	Name	Description
7:6	CCK11DIVDLY	0 implies no delay to chain2 cck11 modulated data 1 implies 1 delay (1 20MHz sample) to chain2 cck11 modulated data 2 implies 2 delays 3 implies 3 delays
5:3	CCK5DIVDLY	0 implies no delay to chain2 cck5 modulated data 1 implies 1 delay (1 20MHz sample) to chain2 cck5 modulated data 2 implies 2 delays 3 implies 3 delays 4 implies 4 delays 5 implies 5 delays 6 implies 6 delays 7 implies 7 delays
2:0	BRKDIVDLY	0 implies no delay to chain2 barker modulated data 1 implies 1 delay (1 20MHz sample) to chain2 barker modulated data 2 implies 2 delays 3 implies 3 delays 4 implies 4 delays 5 implies 5 delays 6 implies 6 delays 7 implies 7 delays

0x0A012810 WCSS_A_TBAPB_LINEAR_DELAY_ANT2

Type: RW

Reset State: 0x0004

Stores the select for the delay diversity value to be applied to txb chain3 samples in barker and cck modulated data

WCSS_A_TBAPB_LINEAR_DELAY_ANT2

Bits	Name	Description
7:6	CCK11DIVDLY	0 implies no delay to chain3 cck11 modulated data 1 implies 1 delay (1 20MHz sample) to chain3 cck11 modulated data 2 implies 2 delays 3 implies 3 delays
5:3	CCK5DIVDLY	0 implies no delay to chain3 cck5 modulated data 1 implies 1 delay (1 20MHz sample) to chain3 cck5 modulated data 2 implies 2 delays 3 implies 3 delays 4 implies 4 delays 5 implies 5 delays 6 implies 6 delays 7 implies 7 delays
2:0	BRKDIVDLY	0 implies no delay to chain3 barker modulated data 1 implies 1 delay (1 20MHz sample) to chain3 barker modulated data 2 implies 2 delays 3 implies 3 delays 4 implies 4 delays 5 implies 5 delays 6 implies 6 delays 7 implies 7 delays

0x0A012814 WCSS_A_TBAPB_LINEAR_DELAY_ANT3**Type:** RW**Reset State:** 0x0006

Stores the select for the delay diversity value to be applied to txb chain4 samples in barker and cck modulated data

WCSS_A_TBAPB_LINEAR_DELAY_ANT3

Bits	Name	Description
7:6	CCK11DIVDLY	0 implies no delay to chain4 cck11 modulated data 1 implies 1 delay (1 20MHz sample) to chain4 cck11 modulated data 2 implies 2 delays 3 implies 3 delays
5:3	CCK5DIVDLY	0 implies no delay to chain4 cck5 modulated data 1 implies 1 delay (1 20MHz sample) to chain4 cck5 modulated data 2 implies 2 delays 3 implies 3 delays 4 implies 4 delays 5 implies 5 delays 6 implies 6 delays 7 implies 7 delays
2:0	BRKDIVDLY	0 implies no delay to chain4 barker modulated data 1 implies 1 delay (1 20MHz sample) to chain4 barker modulated data 2 implies 2 delays 3 implies 3 delays 4 implies 4 delays 5 implies 5 delays 6 implies 6 delays 7 implies 7 delays

10.57 WCSS_A_TDC_TDC

0x0A014C00 WCSS_A_TDC_TDCSYM_IN

Type: W

Reset State: 0x00000000

Number of multiples of 80 baseband samples received by the TDC module (from the AGC module)

WCSS_A_TDC_TDCSYM_IN

Bits	Name	Description
15:0	TDCSYM_IN	Number of multiples of 80 baseband samples received by the TDC module (from the AGC module)

0x0A014C04 WCSS_A_TDC_TDCSYM_OUT

Type: W

Reset State: 0x00000000

Number of symbol start pulses received by the TDC module (from the RACTL module)

WCSS_A_TDC_TDCSYM_OUT

Bits	Name	Description
15:0	TDCSYM_OUT	Number of symbol start pulses received by the TDC module (from the RACTL module)

0x0A014C08 WCSS_A_TDC_TDCPKT_IN

Type: W

Reset State: 0x00000000

Number of packet start signals received by the TDC module

WCSS_A_TDC_TDCPKT_IN

Bits	Name	Description
15:0	TDCPKT_IN	Number of packet start signals received by the TDC module

0x0A014C0C WCSS_A_TDC_CWDET_COUNT**Type:** W**Reset State:** 0x00000000

Number of times that the CW detector was triggered

WCSS_A_TDC_CWDET_COUNT

Bits	Name	Description
15:0	CW_COUNT	Number of times that the CW detector was triggered

0x0A014C10 WCSS_A_TDC_CWDET_STATUS**Type:** R**Reset State:** 0x00000000

Indicates whether CW has been detected by CWDET module, and if so at what location.

WCSS_A_TDC_CWDET_STATUS

Bits	Name	Description
6	CWDET_EN	Indicates that CW notch is currently enabled.
5:0	CWDET_LOC	Location of last CW notch detected.

0x0A014C14 WCSS_A_TDC_CWDET_OVERRIDE**Type:** RW**Reset State:** 0x0000

Location of APB-programmable fixed CW notch

WCSS_A_TDC_CWDET_OVERRIDE

Bits	Name	Description
6	APB_MODE	APB control of CW notch_location signal. If this bit is set, the CW detector will override the calculated value of this signal with the APB-programmed value in the CWDET_NOTCH_LOC register. Default value = 1'b0;

WCSS_A_TDC_CWDET_OVERRIDE (cont.)

Bits	Name	Description
5:0	NOTCH_LOC	Location of APB-programmable fixed CW notch Default value = 6'h0

0x0A014C18 WCSS_A_TDC_CWDET_TRIGGER**Type:** W**Reset State:** 0x0000

This register manually triggers the update of the CW notch enable and location in the AGC

WCSS_A_TDC_CWDET_TRIGGER

Bits	Name	Description
0	TRIGGER	Writing any value to this register will pulse the cw_detected signal to the AGC if the APB_MODE bit has been set in the CWDET_NOTCH_LOC register

0x0A014C1C WCSS_A_TDC_APB_GAIN_THRESH_LLP64_1RX**Type:** RW**Reset State:** 0x0016

This register contains the desired value of the TDC power calculator for 1 active chain for an ideal target backoff when receiving non-11n or 20MHz 11n packets.

WCSS_A_TDC_APB_GAIN_THRESH_LLP64_1RX

Bits	Name	Description
4:0	THRESH	Default value = 5'd23 Adjusting this value will result in a different peak-to-mean ratio for the input data to the RXA engine. The default value, 5'd23, will result in a backoff value of 12dB. To set a different backoff, set to: $\text{round}(16 + 2 * \log_2(175) - T * \log_2(10)/5)$, where T is the target backoff in dB

0x0A014C20 WCSS_A_TDC_APB_GAIN_THRESH_LLP128_1RX**Type:** RW**Reset State:** 0x0014

This register contains the desired value of the TDC power calculator for 1 active chain for an ideal target backoff when receiving 40MHz 11n packets.

WCSS_A_TDC_APB_GAIN_THRESH_LL128_1RX

Bits	Name	Description
4:0	THRESH	Default value = 5'd21 Adjusting this value will result in a different peak-to-mean ratio for the input data to the RXA engine. The default value, 5'd21, will result in a backoff value of 12dB. To set a different backoff, set to: $\text{round}(14 + 2 * \log_2(175) - T * \log_2(10)/5)$, where T is the target backoff in dB

0x0A014C24 WCSS_A_TDC_APB_GAIN_THRESH_HTLTF64_1RX

Type: RW

Reset State: 0x0016

This register contains the desired value of the TDC power calculator for 1 active chain for an ideal target backoff when receiving 20MHz 11n mixed-mode packets with 1 active chain.

WCSS_A_TDC_APB_GAIN_THRESH_HTLTF64_1RX

Bits	Name	Description
4:0	THRESH	Default value = 5'd23 Adjusting this value will result in a different peak-to-mean ratio for the input data to the RXA engine. The default value, 5'd23, will result in a backoff value of 12dB. To set a different backoff, set to: $\text{round}(18 + 2 * \log_2(79) - T * \log_2(10)/5)$, where T is the target backoff in dB

0x0A014C28 WCSS_A_TDC_APB_GAIN_THRESH_HTLTF128_1RX

Type: RW

Reset State: 0x0014

This register contains the desired value of the TDC power calculator for 1 active chain for an ideal target backoff when receiving 40MHz 11n mixed-mode packets with 1 active chain.

WCSS_A_TDC_APB_GAIN_THRESH_HTLTF128_1RX

Bits	Name	Description
4:0	THRESH	Default value = 5'd21 Adjusting this value will result in a different peak-to-mean ratio for the input data to the RXA engine. The default value, 5'd21, will result in a backoff value of 12dB. To set a different backoff, set to: $\text{round}(16+2*\log_2(79)-T*\log_2(10)/5)$, where T is the target backoff in dB

0x0A014C2C WCSS_A_TDC_APB_GAIN_THRESH_LLP256_1RX**Type:** RW**Reset State:** 0x0016

This register contains the desired value of the TDC power calculator for 1 active chain for an ideal target backoff when receiving 40MHz 11n packets.

WCSS_A_TDC_APB_GAIN_THRESH_LLP256_1RX

Bits	Name	Description
4:0	THRESH	Default value = 5'd21 Adjusting this value will result in a different peak-to-mean ratio for the input data to the RXA engine. The default value, 5'd21, will result in a backoff value of 12dB. To set a different backoff, set to: $\text{round}(14+2*\log_2(175)-T*\log_2(10)/5)$, where T is the target backoff in dB

0x0A014C30 WCSS_A_TDC_APB_GAIN_THRESH_VHTLTF256_1RX**Type:** RW**Reset State:** 0x0016

This register contains the desired value of the TDC power calculator for 1 active chain for an ideal target backoff when receiving 40MHz 11n mixed-mode packets with 1 active chain.

WCSS_A_TDC_APB_GAIN_THRESH_VHTLTF256_1RX

Bits	Name	Description
4:0	THRESH	Default value = 5'd21 Adjusting this value will result in a different peak-to-mean ratio for the input data to the RXA engine. The default value, 5'd21, will result in a backoff value of 12dB. To set a different backoff, set to: $\text{round}(16+2*\log_2(79)-T*\log_2(10)/5)$, where T is the target backoff in dB

0x0A014C34 WCSS_A_TDC_RAM_OVERFLOW**Type:** R**Reset State:** 0x00000000

Bit[0] of this register indicates the value of the TDC RAM overflow flag

WCSS_A_TDC_RAM_OVERFLOW

Bits	Name	Description
0	RAM_OVERFLOW	Bit[0] of this register indicates the value of the TDC RAM overflow flag

10.58 WCSS_A_TPC_TPC**0x0A010000 WCSS_A_TPC_TXPWR_OVERRIDE_11A0****Type:** RW**Reset State:** 0x0027

Register allows to override the automatic power control algorithm on transmit chains.

WCSS_A_TPC_TXPWR_OVERRIDE_11A0

Bits	Name	Description
10:6	RF_POWER	Override power control loop and write directly to the RF chip
5:0	FINE_POWER	Override power control loop and write directly to the gain multipliers

0x0A010004 WCSS_A_TPC_TXPWR_OVERRIDE_11B0**Type:** RW**Reset State:** 0x0027

Register allows to override the automatic power control algorithm on transmit chains.

WCSS_A_TPC_TXPWR_OVERRIDE_11B0

Bits	Name	Description
10:6	RF_POWER	Override power control loop and write directly to the RF chip

WCSS_A_TPC_TXPWR_OVERRIDE_11B0 (cont.)

Bits	Name	Description
5:0	FINE_POWER	Override power control loop and write directly to the gain multipliers

0x0A010008 WCSS_A_TPC_TXPWR_ENABLE**Type:** RW**Reset State:** 0x0003

controls the power gain mode.

WCSS_A_TPC_TXPWR_ENABLE

Bits	Name	Description
2	FIXED_RF_GAIN	fixed_rf_gain and en (clpc_en) are used to set the tpc mode. When fixed_rf_gain is set, a single rf gain value is used for the tx power control. CLPC Modes 1 and 2 require fixed_rf_gain be set. In Mode 1, rf gain will be static, in Mode 2, the gain value will be adjusted based on feedback from the PDADC.
1	OVERRIDE	When set the rf and digital gain values from the TXPWR_OVERRIDE register are used for all transmit power values.
0	EN	en (clpc_en) and fixed_rf_gain are used to set the tpc mode. When en is set, the gain LUTs will be updated based on the feedback from the PDADC. CLPC Modes 2 and 3 require en be set.

0x0A01000C WCSS_A_TPC_ADC_CTRL_GET_ADC**Type:** W**Reset State:** 0x00000000

ADC control port. Starts conversion.

WCSS_A_TPC_ADC_CTRL_GET_ADC

Bits	Name	Description
0	GET_ADC	Write to this register to get ADC readout (start the conversion). This register was used in Titan to perform temperature measurement. Not used in Taurus.

0x0A010010 WCSS_A_TPC_ADC_STATUS**Type:** R**Reset State:** 0x00000000

ADC status port, shows when ADC is ready (conversion finished).

WCSS_A_TPC_ADC_STATUS

Bits	Name	Description
2	FAILED	0x0: SUCCESS (Busy went low following successful adc conversion) 0x1: FAILURE (Busy went low following a power-adc response timeout)
1	BUSY_P	When 0, either failure happened or result of power measurement is available in SENSED_PWR register. 0x0: IDLE (Idle) 0x1: BUSY (ADC Busy in Power Measurement)
0	BUSY_T	Busy with a APB-initiated conversion. To be polled until 0. When 0, either failure happened or result of conversion is available in ADC_DATA register. 0x0: IDLE (Idle) 0x1: Reserved

0x0A010014 WCSS_A_TPC_MAN_TXPWR**Type:** W**Reset State:** 0x00000000

Manual TX gain command strobe.

WCSS_A_TPC_MAN_TXPWR

Bits	Name	Description
0	STRB	Write to this register to manually send out a TX gain command when in override mode.

0x0A010018 WCSS_A_TPC_MAX_RF_GAIN**Type:** RW**Reset State:** 0x001F

Contains the maximum RF gain to which the power control loop operates. This is different for the 2.4 vs. 5GHz bands.

WCSS_A_TPC_MAX_RF_GAIN

Bits	Name	Description
4:0	VALUE	Default 15, i.e. 20dBm (2.4GHz value) Should be set to 9 (16dBm) for 5GHz.

0x0A01001C WCSS_A_TPC_MIN_RF_GAIN

Type: RW

Reset State: 0x0000

Contains the minimum RF gain to which the power control loop operates. This is different for the 2.4 vs. 5GHz bands.

WCSS_A_TPC_MIN_RF_GAIN

Bits	Name	Description
4:0	VALUE	Default 15, i.e. 20dBm (2.4GHz value) Should be set to 9 (16dBm) for 5GHz.

0x0A010020 WCSS_A_TPC_MAX_DIGITAL_GAIN

Type: RW

Reset State: 0x000F

Max value of tx digital gain range.

WCSS_A_TPC_MAX_DIGITAL_GAIN

Bits	Name	Description
3:0	VALUE	Default value = 15

0x0A010024 WCSS_A_TPC_MID_DIGITAL_GAIN

Type: RW

Reset State: 0x0008

Mid value of tx digital gain range. Must be set to 1 or more!

WCSS_A_TPC_MID_DIGITAL_GAIN

Bits	Name	Description
3:0	VALUE	Default value = 8

0x0A010028 WCSS_A_TPC_MIN_DIGITAL_GAIN**Type:** RW**Reset State:** 0x0000

Min value of tx digital gain range.

WCSS_A_TPC_MIN_DIGITAL_GAIN

Bits	Name	Description
3:0	VALUE	Default value = 0

0x0A010030 WCSS_A_TPC_RC_DELAY**Type:** RW**Reset State:** 0x04B0

Contains the delay between the start of TX and the start of power measurement.

WCSS_A_TPC_RC_DELAY

Bits	Name	Description
12:0	DELAY	Default 1200, ie 15uS @ 80MHz clock. Value automatically adjusted for static bandwidth so that duration for given APB value is constant for all bandwidths

0x0A010034 WCSS_A_TPC_SENSED_PWR0**Type:** R**Reset State:** 0x00000000

Contains data from the Pwr ADCs which sensed the cygnus power during transmit operation

WCSS_A_TPC_SENSED_PWR0

Bits	Name	Description
10:0	VALUE	PWRADC0 readback during transmit.

0x0A010038 WCSS_A_TPC_ADCTIMEOUT**Type:** RW**Reset State:** 0x0900

Contains Pwr ADC response timeout value

WCSS_A_TPC_ADCTIMEOUT

Bits	Name	Description
15:0	VALUE	Specifies the number of operating clocks that power-adc state machine can be in non-idle state and beyond that point, the power-adc response failure will be declared and state machine will be brought to idle. Value automatically adjusted for static bandwidth so that duration for given APB value is constant for all bandwidths.

0x0A01003C WCSS_A_TPC_APBACCESS**Type:** RW**Reset State:** 0x0000

Selects APB access control

WCSS_A_TPC_APBACCESS

Bits	Name	Description
0	SELECT	Set this bit to high to enable APB access of TPC memory

0x0A010040 WCSS_A_TPC_PTADC_FSM_RST**Type:** W**Reset State:** 0x00000000

This register does the resetting of the state machine which interacts with power-adc analog interface

WCSS_A_TPC_PTADC_FSM_RST

Bits	Name	Description
0	RST	Write to this register to reset

0x0A010048 WCSS_A_TPC_PDADC_RANGE**Type:** RW**Reset State:** 0x0001

This register defines which portion of the 11-bit input PDADC sample is fed to the 8-bit LUT

WCSS_A_TPC_PDADC_RANGE

Bits	Name	Description
1:0	SELECT	2'h0 : selects bits [10:3] of input PDADC sample 2'h1 : selects bits [9:2] of input PDADC sample 2'h2 : selects bits [8:1] of input PDADC sample 2'h3 : selects bits [7:0] of input PDADC sample

0x0A01004C WCSS_A_TPC_PDADC_OFFSET**Type:** RW**Reset State:** 0x0000

Offset value to be subtracted from raw PDADC value

WCSS_A_TPC_PDADC_OFFSET

Bits	Name	Description
10:0	VALUE	

0x0A010050 WCSS_A_TPC_GAIN_OFFSET1**Type:** RW**Reset State:** 0x0000

Open Loop gain values for converting digital to an absolute digital gain value for the TXFIR fine gain block.

WCSS_A_TPC_GAIN_OFFSET1

Bits	Name	Description
11:6	OL_11B	

WCSS_A_TPC_GAIN_OFFSET1 (cont.)

Bits	Name	Description
5:0	OL_OFDM	

0x0A010054 WCSS_A_TPC_GAIN_OFFSET2**Type:** RW**Reset State:** 0x030C

Closed Loop gain values for converting digital to an absolute digital gain value for the TXFIR fine gain block.

WCSS_A_TPC_GAIN_OFFSET2

Bits	Name	Description
11:6	CL_11B	
5:0	CL_OFDM	

0x0A010058 WCSS_A_TPC_DEMANDED_PWR_RANGE**Type:** RW**Reset State:** 0x03E0

Selects a subrange of demanded power values over which CLPC is active. For demanded power levels outside this range, the TPC will operate in open-loop mode.

WCSS_A_TPC_DEMANDED_PWR_RANGE

Bits	Name	Description
9:5	MAX	Set max end of demanded power range
4:0	MIN	Set min end of demanded power range

0x0A01005C WCSS_A_TPC_LAST_DEMANDED_PWR**Type:** R**Reset State:** 0x00000000

Stored value of demanded power for the last transmitted packet

WCSS_A_TPC_LAST_DEMANDED_PWR

Bits	Name	Description
5	LOW_PWR	Low power setting for last transmitted packet
4:0	PWR_VALUE	Demanded power value for last transmitted packet

0x0A010800+ WCSS_A_TPC_POWERDET0_RAMn, n=[0..255]**0x4*n****Type:** RW**Reset State:** 0x00000000

Calibration data for the power detector

WCSS_A_TPC_POWERDET0_RAMn

Bits	Name	Description
5:0	POWER	

0x0A010C00+ WCSS_A_TPC_RF_GAIN_LUT0n, n=[0..31]**0x4*n****Type:** RW**Reset State:** 0x00000000

RF Course gain LUT

WCSS_A_TPC_RF_GAIN_LUT0n

Bits	Name	Description
4:0	RF_GAIN	

0x0A010C80+ WCSS_A_TPC_DIGITAL_GAIN_LUT0n, n=[0..31]**0x4*n****Type:** RW**Reset State:** 0x00000000

Digital fine gain LUT

WCSS_A_TPC_DIGITAL_GAIN_LUT0n

Bits	Name	Description
3:0	DIG_GAIN	

10.59 WCSS_A_TPE_TPE

0x0A082000 WCSS_A_TPE_EDCF_TXOP_0_1

Type: RW

Reset State: 0x06000600

edcf txop corresponds to back-offs 0 and 1

WCSS_A_TPE_EDCF_TXOP_0_1

Bits	Name	Description
31:16	EDCF_TXOP_1	TXOP that is used while transmitting mpdus from the q corresponds to back-off 1 This valus is in micro seconds.
15:0	EDCF_TXOP_0	TXOP that is used while transmitting mpdus from the q corresponds to back-off 0 This valus is in micro seconds.

0x0A082004 WCSS_A_TPE_EDCF_TXOP_2_3

Type: RW

Reset State: 0x06000600

edcf txop corresponds to back-offs 1 and 2

WCSS_A_TPE_EDCF_TXOP_2_3

Bits	Name	Description
31:16	EDCF_TXOP_3	TXOP that is used while transmitting mpdus from the q corresponds to back-off 3 This valus is in micro seconds.
15:0	EDCF_TXOP_2	TXOP that is used while transmitting mpdus from the q corresponds to back-off 2 This valus is in micro seconds.

0x0A082008 WCSS_A_TPE_EDCF_TXOP_4_5

Type: RW

Reset State: 0x06000600

edcf txop corresponds to back-offs 4 and 5

WCSS_A_TPE_EDCF_TXOP_4_5

Bits	Name	Description
31:16	EDCF_TXOP_5	TXOP that is used while transmitting mpdus from the q corresponds to back-off 5 This value is in micro seconds.
15:0	EDCF_TXOP_4	TXOP that is used while transmitting mpdus from the q corresponds to back-off 4 This value is in micro seconds.

0x0A08200C WCSS_A_TPE_EDCF_TXOP_6_7**Type:** RW**Reset State:** 0x06000600

edcf txop corresponds to back-offs 6 and 7

WCSS_A_TPE_EDCF_TXOP_6_7

Bits	Name	Description
31:16	EDCF_TXOP_7	TXOP that is used while transmitting mpdus from the q corresponds to back-off 7 This value is in micro seconds.
15:0	EDCF_TXOP_6	TXOP that is used while transmitting mpdus from the q corresponds to back-off 6 This value is in micro seconds.

0x0A082010 WCSS_A_TPE_SW_TEMPLATE_BASE_ADDR**Type:** RW**Reset State:** 0x00005000

This register contains the implicit beamform bkof qosnull template address.

WCSS_A_TPE_SW_TEMPLATE_BASE_ADDR

Bits	Name	Description
31:0	SW_BKOF_QOSNULL_TEMPLATE_ADDR	This address is used to specify the implicit beamform bkof qosnull template address.

0x0A082014 WCSS_A_TPE_SW_TPE_HW_CONTROL_REG**Type:** RW**Reset State:** 0x01840A01

This register contains all the h/w control config registers. s/w needs to set them as per h/w requirements. These may be either config for bug fixes or for future feature expansion or ...

WCSS_A_TPE_SW_TPE_HW_CONTROL_REG

Bits	Name	Description
31	IGNORE_TXP_ERROR_INT_ECO	Set '1' to ignore the txp_mim_internal_error_int_p, so tpe won't issue soft reset because of the assertion of the interrupt
30:25	RESERVED	Reserved spare register bit
24:16	SW_11G_DUPLICATE_CTRL_INDEX_OFFSET	This is same as sw_11g_ctrl_index_offset but is used only for (40Mhz) pkts If 11b stations are in, in 11g protection This sw_11g_duplicate_ctrl_index_offset + 11G Rate index of the ctrl pkt from rate table is used to derive the actual ctrl rate index -- used for RTS,CTS,BAR there is no separate valid bit for duplicate rate. sw_11g_ctrl_index_offset_valid is used
11	SW_DTIM_TERMINATE_VALID	when set, dtim will be terminated when p2p/bt/lte termination happened. When cleared, same behavior as in Volans.
10	SW_P2P_TX_BOUNDARY_TERMINATE_VALID	when set, TPE will terminate the current transmission when p2p tx boundary txop reaches 0.
9	SW_GEN6_3WIRE_VALID	when set, the gen6 bt functionality is enabled to generate wlan_Active
8	SW_11G_OFDM_VALID	If this bit is set, the compensation reg is added to duration Also the sifs value in duration calculation is made 10
7:4	SW_11G_OFDM_COMPENSATION	In the case of 11G OFDM, we add 6usec to duration. The value is given in Register for flexibility
1:0	SW_COMP_DIV_DELAY_REG	The div function delay is specified in this register. Actual MCP is = sw_comp_div_delay_reg+2. By default 1+2=3 cycles

0x0A082018 WCSS_A_TPE_SW_SOFTWARE_TX_CONTROL_REG1**Type:** RW**Reset State:** 0x00300000

This register contains the controls needed for s/w triggered transmission

WCSS_A_TPE_SW_SOFTWARE_TX_CONTROL_REG1

Bits	Name	Type	Description
29:24	SW_WAIT_COMMAND2	RW	This register contains the wait command2. This command is used by TXP while sending pkt in the case of TXOP burst, it is used only first time and later onwards follows the station parameters
21:16	SW_WAIT_COMMAND1	RW	This register contains the wait command1. The tpe main waits as per this wait command before the request is considered for arbitration. The definition is as per TXP wait command. Any value beyond 21Hex is treated as no wait. by default it is no wait
10	SW_TX_GAM_SELECT	RW	When set, TPE will read the sw template from FDAHB. This is only used to send the beamform report from CFG memory in 11n sw assist mode When clear, TPE will read the sw template from CAHB.
9:7	SW_TX_SESSION_CONTROL	RW	This register contains the type of transmission s/w wants 000 --> nothing 001 --> data and use back-off index provided by s/w 010 --> data and use the staid, Qid given by s/w 011 --> template transmission, addr given by template addr
6	SW_TX_RAW_PKT_VALID_NEED_11AC_DELIMITER	RW	if this bit is set the transmission is done in raw pkt mode, but TPE will add delimiter/ofdm paddings if the rate is 11ac
5	SW_TX_RAW_PKT_VALID	RW	if this bit is set the transmission is done in raw pkt mode
4	SW_TX_TERMINATE_VALID	RW	during s/w controlled tx, if this bit is set, the on going transmission will be terminated as if the txop is not available. The current transmission is ended gracefully. Do we send cf_end?? We won't
3:1	TPE_TX_TIMING_FEEDBACK	R	the feedback bits from h/w. They will be made zero on start_valid 000 --> Nothing is going on 001 --> waiting in command1 010 --> command1 is done and command2 is going on 011 --> tx done 111 --> ack to 100 --> aging

WCSS_A_TPE_SW_SOFTWARE_TX_CONTROL_REG1 (cont.)

Bits	Name	Type	Description
0	SW_TX_START_VALID	RW	by writing 1 to this bit s/w indicates to h/w that the s/w tx controls are valid and h/w should start working on the s/w request sw needs to toggle this by writing 0 and then 1

0x0A08201C WCSS_A_TPE_SW_SOFTWARE_TX_ADDRESS**Type:** RW**Reset State:** 0x00005000

This register contains the template address.

WCSS_A_TPE_SW_SOFTWARE_TX_ADDRESS

Bits	Name	Description
31:0	SW_SOFTWARE_TX_ADDRESS	S/W programs this register during the s/w initiated transmission in the case of template transmission, this is templated address session control 001 --> [3:0] = backoff index, other bits are not used [31:16] = txop for this transmission 010 --> [3:0] = QID of the stream [11:4] = STAID [31:16] = txop for this transmission 011 --> [31:0] = template memory address --> template transmission is always txop=0

0x0A082020 WCSS_A_TPE_SW_MAX_BYTES_AND_TX_TIME_IN_AMPDU**Type:** RW**Reset State:** 0x02000200

This register contains the max_tx (in micro sec) time that needs to be considered to terminate the AMPDU

WCSS_A_TPE_SW_MAX_BYTES_AND_TX_TIME_IN_AMPDU

Bits	Name	Description
28:16	SW_MAX_AMPDU_TX_TIME	This is the max tx_time that an AMPDU can take. This is another AMPDU termination condition
12:0	SW_MAX_AMPDU_TX_TIME_FOR_MIXED_MODE	During the mixed mode, we should not exceed this limit. This is considered in mixed mode only

0x0A082024 WCSS_A_TPE_SW_MAX_MPDUS_IN_AMPDU_AND_MISC**Type:** RW**Reset State:** 0x7F002010

This register contains the max no. of mpdus that can be packed in one AMPDU

WCSS_A_TPE_SW_MAX_MPDUS_IN_AMPDU_AND_MISC

Bits	Name	Description
31:16	SW_PROTECTION_THR	The threshold that needs to be used for protection... RTS/CTS to self etc
15	SW_TX_AMPDU_TERMINATE_AT_RATE_CHANGE	if this bit is set, Whenever there is a rate change, ampdu will be terminated
14	SW_TX_AMPDU_WINDOW_CHECK_VALID	the AMPDU window shcek is done in TX only when this bit is set Once the sata structure init is done in lab also, we can take out this bi
13	SW_AMPDU_TERMINATE_FROM_BD_VALID	when set, TPE will terminate the ampdu based on the bd bit from bmu.
9	SW_CONSIDER_80_MHZ	80MHZ rates are considered only when this bit is set
8	SW_GLOBAL_SB_MODE	indicates the side band mode. MAC simply copies this bit in the phy commands This bit is meaningfull only when the CB is valid.
7	SW_GLOBAL_CB_VALID	40MHZ rates are considered only when this bit is set
6:0	SW_MAX_MPDUS_IN_AMPDU	This field indicates the number of MPDUs that can be packed in one AMPDU

0x0A082028 WCSS_A_TPE_SW_STA_DESCR_ADDR**Type:** RW**Reset State:** 0x00005000

This register contains the sta descriptor start address

WCSS_A_TPE_SW_STA_DESCR_ADDR

Bits	Name	Description
31:0	SW_STA_DESCR_ADDR	the actual start address of descr is sw_sta_descr_addr + staid * (sw_global_no_of_qids * 16*4) Basically 64 bytes are reserved for each sta

0x0A082034 WCSS_A_TPE_SW_PM**Type:** RW**Reset State:** 0xF096280A

S/W-configurable PM bit

WCSS_A_TPE_SW_PM

Bits	Name	Description
31:24	SW_TX_SIFS_A_MODE_CYCLES	The lower bits of SIFS cycles that needs to be programmed in tXP commands for A and G (non B) mode pkts This number is multiplied by 8 and used internally. F0 corresponds to hex 780 corresponds 1920(decimal) which is 16usec. The upper bits are in sw_pm_2.sw_tx_sifs_a_mode_cycles
23:16	SW_TX_SIFS_B_MODE_CYCLES	The SIFS cycles that needs to be programmed in tXP commands for B mode pkts This number is multiplied by 8 and used internally. 96 corresponds to hex x4b0 corresponds 1200(decimal) which is 10usec.
15	SW_PROMOTE_BT_LOW_PRIORITY	if this bit is set, the lower priority BT is treated as Higher priority This should not be set in normal conditions. I don't think we ever set this bit Provided so that if we get in to any system issues, this can be turned on
14	SW_CONSIDER_TXP_ERROR_IN_TPE_ERROR	If this bit is set the tpe_mcu_error includes the TXP Error too
13	SW_CONSIDER_RESPONSE_WHILE_WAITING	if this bit is set, response pkt will be sent even tpe is busy
12	SW_IGNORE_DMA_INDEX	The dma index from rxp is ignored
11	SW_IGNORE_DMA_CONFIRM	The dma_confirm from RXP is ignored -- to eliminate any potential Hangs in the rxp-tpe interface
10	SW_LSIG_PROTECTION_TYPE	If this bit is set the entire TXOP is used for LSIG protection. Otherwise what ever duration is seen in pkt is used for lsig
9	SW_LSIG_PROTECTION_VALID	If this bit is set the lsig protection is considered. Entire TXOP or only the next_pkt is done based on lsig_protection_type
8	SW_CONSIDER_SECOND_PKT_READ	If this bit is set the next_pkt read and the duration computation for non zero TXOP and frags is done as per spec. Otherwise we don't include next pkt for the computation. This will be removed once the verification is done

WCSS_A_TPE_SW_PM (cont.)

Bits	Name	Description
7	SW_CONSIDER_TX_BOUNDARY_TXOP	if this bit is set, the tx_boundary_txop is considered in TPE
6	SW_BT_TERMINATE_VALID	if this bit is set, mproc will terminate if the btc_is_active (after validating it with configs) btc_active_final & sw_bt_terminate_valid = terminate mproc
5	SW_CONSIDER_BT_AVAIL_TIME_CNT	if this bit is set, the btc_tpe_avail_time_cnt is considered in tPE
4	SW_CONSIDER_BT_LOW_PRI	if this bit is set, the btc low priority request is considered
3	SW_CONSIDER_BT_HIGH_PRI	if this bit is set, the btc high priority request is considered
2	SW_CONSIDER_WARMUP_DELAY_FOR_SIFS_CMD	If this bit is set, all the sifs_ommand instructions will wait warm_upenable signal from mtu. This is to reduce the warmup pulse width for the cases in which PHY has less delay. Setting this bit to 0 is Taurus compatible
1	SW_BCN_TERMINATE_VALID	if this bit is set, the current transmission will be terminated if it is TBTT time
0	SW_PM	Reserved. The pm bit for data is in sta descriptor now.

0x0A08203C WCSS_A_TPE_SW_11G_CTRL_INDEX_OFFSET**Type:** RW**Reset State:** 0x0003FC00

offset for 11g protection rates

WCSS_A_TPE_SW_11G_CTRL_INDEX_OFFSET

Bits	Name	Description
31:24	SW_AMPDU_EXTRA_DURATION	This value is added to AMPDU duration if the current one is not last one in the TXOP. Since we don't know the exact duration of the next AMPDU, we just add a constant value. By default it is ZERO
18:10	SW_CF_END_TXOP_THR	CF-END Will be sent only if the remaining TXOP is more than this value Default is 255 usec
9	SW_11G_CTRL_INDEX_OFFSET_VALID	if this bit is set, sw_11g_ctrl_index_offset is used basically this is nothing but use_11b_protection_mode Do we need to validate with sta_mode??
8:0	SW_11G_CTRL_INDEX_OFFSET	If 11b stations are in, in 11g protection This sw_11g_index_offset + 11G Rate index of the ctrl pkt from rate table is used to derive the actual ctrl rate index -- used for RTS,CTS,BAR

0x0A082040 WCSS_A_TPE_RATE_TABLE_SRAM_CONTROL**Type:** RW**Reset State:** 0x80000000

controls sw operations on TPE rate table. To start a read transaction, check first if bit 31 is 1. If 1, then proceed, writing 0 to bit 31, address to [29:21], 0 to bit 1, and 1 to bit 0. When bit 31 turns 1, then read the data bits. To start a write operation, check first if bit 31 is 1. If 1, then proceed, write 0 to bit 31, address to [29:21], data to [20:4], 1 to bit 1, and 1 to bit 0. When bit 31 turns 1, then write is done.

WCSS_A_TPE_RATE_TABLE_SRAM_CONTROL

Bits	Name	Description
31	RATE_SRAM_XACT_DONE	1 indicates transaction complete - write to 0 at start of operation
29:21	RATE_SRAM_XACT_ADDR	address for SW rate table sram read/write
20:3	RATE_SRAM_XACT_DATA	rate table sram data [3] - ctrl resp lpe bit ([26] in rate table) [7:4] - ctrl rate index ([17:14] in rate table) [12:8] - cntrl rep tx power ([31:27] in rate table) [16:13] - resp rate index ([35:32] in rate table) [20:17] - 11g cntrl rate index minus 11b base offset ([61:58] in rate table)
1	RATE_SRAM_XACT_WRITE_READZ	1 indicates write operation, 0 indicates read operation
0	RATE_SRAM_XACT_START	Write only bit that initiates SW rate sram transaction

0x0A082044 WCSS_A_TPE_RATE_TABLE_SRAM_CONTROL2**Type:** RW**Reset State:** 0x00000000

controls sw operations on TPE rate table.

WCSS_A_TPE_RATE_TABLE_SRAM_CONTROL2

Bits	Name	Description
9:0	RATE_SRAM_XACT_DATA	rate table sram data [4:0] - upper bits of ctrl rate index ([22:18] in rate table) [9:5] - upper bits of resp rate index ([40:36] in rate table)

0x0A082050 WCSS_A_TPE_SW_BEACON_BASE_ADDR**Type:** RW**Reset State:** 0x00005000

This register contains the beacon base address. THIS REGISTER CAN BE REMOVED --
SIMPLY USE THE OFFSET FROM THE TEMPLATE

WCSS_A_TPE_SW_BEACON_BASE_ADDR

Bits	Name	Description
31:0	SW_BEACON_BASE_ADDR	S/w needs to program this register based on the chip configuration h/w uses this base address and depending on the sw_beacon_max_size, it derives the beacon addresses for all the bssids accordingly

0x0A082054 WCSS_A_TPE_SW_BEACON_MAX_SIZE_MISC**Type:** RW**Reset State:** 0x00000800

This register contains the max size of the beacon. Make sure the size is in bytes but always word boundary i.e. the lsb 2 bits are always expected to be Zeros

WCSS_A_TPE_SW_BEACON_MAX_SIZE_MISC

Bits	Name	Description
22	SW_BCN_UN_LOCK_BY_SW	s/w unlocks the beacon ownership by writing 1 to this register. Typically after updating the beacon, s/w writes to this register It is write only register
21:18	SW_UPDATE_BCN_INDEX	the beacon index that s/w wants to update
17	MAIN_SW_UPDATE_BCN_GNT_BY_HW	h/w grant to the s/w request to update beacon. Basically after requesting to update a beacon s/w will poll this register. Once this bit is set s/w can go ahead and update the beacon template. Once the s/w changes the beacon it needs to unlock by writing 1 to sw_bcn_un_lock_by_sw
16	SW_UPDATE_BCN_REQ	by writing this bit s/w requests h/w to grant to update beacon. when it is read, main_sw_bcn_update_pending_valid is reflected this bit will be set when s/w is written but reset by writing to sw_bcn_un_lock_by_sw
15:0	SW_BEACON_MAX_SIZE	This register is used to derive the start address of the beacons. This is used in multiple bssid base_address of beacon[index] = sw_beacon_base_addr + index * sw_beacon_max_size

0x0A082058 WCSS_A_TPE_SW_GLOBAL_CF_END_DATA_POWER_TEMPLATE**Type:** RW**Reset State:** 0x00000000**WCSS_A_TPE_SW_GLOBAL_CF_END_DATA_POWER_TEMPLATE**

Bits	Name	Description
30	FEC_MODE	

WCSS_A_TPE_SW_GLOBAL_CF_END_DATA_POWER_TEMPLATE (cont.)

Bits	Name	Description
29	LPE_BIT	
27:26	SW_CONSIDER_WLAN_RX_FOR_BT_ECO_VECTOR	basically this is config register to enable the ECO in Libra 2.0. While giving channel to BT Low priority request, the RX path (the packet det from PHY) is used. Two bits are provide for pkt_det and rxp filter pass Bit 0 of the vector i.e. bit 26 of this register enables consideration of rxp_tpeflt_pass in the logic, Bit 1 of the vector i.e. bit 27 of this register enables consideration of rxp_mtu_pktdet_n in the logic,
25	AMPDU_VALID	
24:20	TX_POWER	
18:16	TX_ANTENNA_EN	
15:14	STBC_VALID	
13	SGI_VALID	
12:11	NSS_11B_MODE	
8:3	STA_RATE	
2:0	STA_MODE	

0x0A082080 WCSS_A_TPE_TPE_BMU_INTERFACE_CNTS**Type:** R**Reset State:** 0x00000000

TPE<->BMU interface counters

WCSS_A_TPE_TPE_BMU_INTERFACE_CNTS

Bits	Name	Description
31:24	TPE_BMU_FEEDBACK_CNT	Increments when tpe_bmu_feedback_p is sent.
23:16	BMU_TPE_FEEDBACK_DONE_CNT	Increments when bmu_tpe_feedback_done_p is sent.
15:8	TPE_BMU_GET_INFO_CNT	Increments when tpe_bmu_get_info_p is sent.
7:0	BMU_TPE_GET_INFO_VALID_CNT	Increments when bmu_tpe_get_info_valid_p is sent.

0x0A082084 WCSS_A_TPE_RXP_TPE_INTERFACE_CNTS**Type:** R**Reset State:** 0x00000000

RXP<->TPE interface counters

WCSS_A_TPE_RXP_TPE_INTERFACE_CNTS

Bits	Name	Description
31:24	RXP_TPE_FRAME_INFO_CNT	Increments when rxp_tpe_frame_info_p is sent.
23:16	RXP_TPE_AMPDU_END_CNT	Increments when rxp_tpe_ampdu_end_p is sent.
15:8	RXP_TPE_DMA_CONFIRM_CNT	Increments when rxp_tpe_dma_confirm_p is sent.
7:0	RXP_TPE_BAR_INFO_CNT	Increments when rxp_tpe_bar_info_p is sent.

0x0A082088 WCSS_A_TPE_MTU_TPE_INTERFACE_CNTS

Type: R

Reset State: 0x00000000

MTU<->TPE interface counters

WCSS_A_TPE_MTU_TPE_INTERFACE_CNTS

Bits	Name	Description
23:16	MTU_TPE_BCN_REQ_VALID_CNT	Increments when mtu_tpe_bcn_req_valid_p is sent.
15:8	MTU_TPE_REQ_VALID_CNT	Increments when mtu_tpe_req_valid_p is sent.
7:0	TPE_MTU_FEEDBACK_VALID_ID_CNT	Increments when tpe_mtu_feedback_valid_p is sent.

0x0A08208C WCSS_A_TPE_RPE_TPE_INTERFACE_CNTS

Type: R

Reset State: 0x00000000

RPE<->TPE interface counters

WCSS_A_TPE_RPE_TPE_INTERFACE_CNTS

Bits	Name	Description
15:8	TPE_RPE_BITMAP_REQ_CNT	Increments when tpe_rpe_bitmap_req_p is sent.
7:0	RPE_TPE_BITMAP_VLD_CNT	Increments when rpe_tpe_bitmap_vld_p is sent.

0x0A082090 WCSS_A_TPE_ERROR_INT_CNTS**Type:** R**Reset State:** 0x00000000

Error counters

WCSS_A_TPE_ERROR_INT_CNTS

Bits	Name	Description
23:16	MPROC_ERROR2_CNT	
15:8	MPROC_ERROR1_CNT	
7:0	TXP_ERROR_INT_CNT	Increments when txp signals error interrupt

0x0A0820C0 WCSS_A_TPE_UNICAST_BYTES_LOWER**Type:** R**Reset State:** 0x00000000

lower unicast bytes

WCSS_A_TPE_UNICAST_BYTES_LOWER

Bits	Name	Description
31:0	TX_UNICAST_BYTES_LOWER	

0x0A0820C4 WCSS_A_TPE_UNICAST_BYTES_UPPER**Type:** R**Reset State:** 0x00000000

upper unicast bytes

WCSS_A_TPE_UNICAST_BYTES_UPPER

Bits	Name	Description
31:0	TX_UNICAST_BYTES_UPPER	

0x0A0820C8 WCSS_A_TPE_MULTICAST_BYTES_LOWER**Type:** R**Reset State:** 0x00000000

lower multicast bytes

WCSS_A_TPE_MULTICAST_BYTES_LOWER

Bits	Name	Description
31:0	TX_MULTICAST_BYTES_LOWER	

0x0A0820CC WCSS_A_TPE_MULTICAST_BYTES_UPPER**Type:** R**Reset State:** 0x00000000

upper multicast bytes

WCSS_A_TPE_MULTICAST_BYTES_UPPER

Bits	Name	Description
31:0	TX_MULTICAST_BYTES_UPPER	

0x0A0820D0 WCSS_A_TPE_BROADCAST_BYTES_LOWER**Type:** R**Reset State:** 0x00000000

lower broadcast bytes

WCSS_A_TPE_BROADCAST_BYTES_LOWER

Bits	Name	Description
31:0	TX_BROADCAST_BYTES_LOWER	

0x0A0820D4 WCSS_A_TPE_BROADCAST_BYTES_UPPER**Type:** R**Reset State:** 0x00000000

upper broadcast bytes

WCSS_A_TPE_BROADCAST_BYTES_UPPER

Bits	Name	Description
31:0	TX_BROADCAST_BYTES_UPPER	

0x0A082100 WCSS_A_TPE_STATISTIC_COUNTERS_CNTL

Type: RW

Reset State: 0x00000004

statistic counters control

WCSS_A_TPE_STATISTIC_COUNTERS_CNTL

Bits	Name	Description
2	COPY_VISTA_COUNTERS	copy 64 vista counters over into stats registers
1	CLEAR_VISTA_COUNTERS	clear vista counters
0	CLEAR_GLOBAL_COUNTERS	clear global counters

0x0A082104 WCSS_A_TPE_TPE_STATUS_REG_LOWER

Type: R

Reset State: 0x00000000

tpe status - should be all 0 when idle and da are all 0

WCSS_A_TPE_TPE_STATUS_REG_LOWER

Bits	Name	Description
31:0	TPE_STATUS	uspu_testbus[4:0], // 32:28 uspm_testbus[5:0], // 27:22 wpg_testbus[4:0], // 21:17 twr_testbus[5:0], // 16:11 tpe_mproc_test[5:0], // 10:5 tpe_main_test[22:18] // 4:0

0x0A082108 WCSS_A_TPE_TPE_STATUS_REG_UPPER

Type: R

Reset State: 0x00000000

tpe status upper - should be all 0 when idle and da are all 0

WCSS_A_TPE_TPE_STATUS_REG_UPPER

Bits	Name	Description
12:0	TPE_STATUS	mtu_tpe_response_to, // 44 bmu_tpe_bo_data_avail[5:0], // 43:38 usps_testbus[4:0], // 37:33

0x0A082110 WCSS_A_TPE_GOT_ACK_FOR_BACK_INFO_REG**Type:** R**Reset State:** 0x00000000

staidd, qid for got_ack_for_back_info interrupt

WCSS_A_TPE_GOT_ACK_FOR_BACK_INFO_REG

Bits	Name	Description
12:5	GOT_ACK_FOR_BACK_STAID	got ack for back staidd
4:0	GOT_ACK_FOR_BACK_QID	got ack for back qid

0x0A082118 WCSS_A_TPE_SW_DELAY_WAIT_CMD_CNT_REG**Type:** RW**Reset State:** 0xFF1E801E

This is used to delay the wait command (so that the warmup pulse can be sent ahead in time)

WCSS_A_TPE_SW_DELAY_WAIT_CMD_CNT_REG

Bits	Name	Description
31:24	SW_AGING_CNT_THRSHLD	(in terms of 16 us). The sw_template request if not sent after this there should time will be dropped
23:16	SW_WAIT_CYCLES_THRESHOLD	(in terms of 8 cycles). this value is num of clocks before we send warmup. we will be delaying the wait command, sending warmup, and then sending the wait command with sw_wait_cycles_threshold to the txp. the default value is 240 cycles, which is 2 us
15	SW_SEND_WARMUP_LATER_EN	send warmup later when we send wait_cycles (e.g. after cts-to-self) instead of right at the beginning of sifs. this "later" is determined by sw_wait_cycles_threshold
13:8	SW_DELAY_CYCLE_FOR_NO_WAIT_TX	For beacons, Template tx, the transmission will happen immediately This is a problem for Warm up. We can't use pifs because the trigger to TPE already after pifs_to.

WCSS_A_TPE_SW_DELAY_WAIT_CMD_CNT_REG (cont.)

Bits	Name	Description
5:0	SW_DELAY_WAIT_CMD_CNT	This is in terms of 8 clk cycles, so a value of 'd30 would mean 240 clk cycles, or 2us

0x0A08211C WCSS_A_TPE_SW_MEDIUM_TIME_USED_AC0**Type:** RW**Reset State:** 0x00000000

This register contains the medium time used for AC0 This is internal to h/w. s/w can write to these register at its risk. Basically the corresponding AC should not be enabled for Admission control while updating this register(update should be only in debug mode.) To make sure that there are no side effects, s/w needs to disable the h/w control over the backoffs by writing appropriate value to sw_mtu_bckof_disable_valid_vector

WCSS_A_TPE_SW_MEDIUM_TIME_USED_AC0

Bits	Name	Description
15:0	SW_MEDIUM_TIME_USED_AC0	Medium time used for AC0. Each count corresponds to 32usec

0x0A082120 WCSS_A_TPE_SW_MEDIUM_TIME_USED_AC1**Type:** RW**Reset State:** 0x00000000

This register contains the medium time used for AC1 This is internal to h/w. s/w can write to these register at its risk. Basically the corresponding AC should not be enabled for Admission control while updating this register(update should be only in debug mode.) To make sure that there are no side effects, s/w needs to disable the h/w control over the backoffs by writing appropriate value to sw_mtu_bckof_disable_valid_vector

WCSS_A_TPE_SW_MEDIUM_TIME_USED_AC1

Bits	Name	Description
15:0	SW_MEDIUM_TIME_USED_AC1	Medium time used for AC1. Each count corresponds to 32usec

0x0A082124 WCSS_A_TPE_SW_MEDIUM_TIME_USED_AC2**Type:** RW**Reset State:** 0x00000000

This register contains the medium time used for AC2 This is internal to h/w. s/w can write to these register at its risk. Basically the corresponding AC should not be enabled for Admission control while updating this register(update should be only in debug mode.) To make sure that there are no side effects, s/w needs to disable the h/w control over the backoffs by writing appropriate value to sw_mtu_bckof_disable_valid_vector

WCSS_A_TPE_SW_MEDIUM_TIME_USED_AC2

Bits	Name	Description
15:0	SW_MEDIUM_TIME_USED_AC2	Medium time used for AC2. Each count corresponds to 32usec

0x0A082128 WCSS_A_TPE_SW_MEDIUM_TIME_USED_AC3

Type: RW

Reset State: 0x00000000

This register contains the medium time used for AC3 This is internal to h/w. s/w can write to these register at its risk. Basically the corresponding AC should not be enabled for Admission control while updating this register(update should be only in debug mode.) To make sure that there are no side effects, s/w needs to disable the h/w control over the backoffs by writing appropriate value to sw_mtu_bckof_disable_valid_vector

WCSS_A_TPE_SW_MEDIUM_TIME_USED_AC3

Bits	Name	Description
15:0	SW_MEDIUM_TIME_USED_AC3	Medium time used for AC3. Each count corresponds to 32usec

0x0A08212C WCSS_A_TPE_SW_MEDIUM_TIME_THR_AC0_AC1

Type: RW

Reset State: 0xEFFFFFFF

This register contains the medium time threshold used for AC0 and AC1 Though the register will be written immediately, the internal logic will take this value at next one second pulse or when a pkt has been transmitted from the corresponding AC

WCSS_A_TPE_SW_MEDIUM_TIME_THR_AC0_AC1

Bits	Name	Description
31:16	SW_MEDIUM_TIME_THR_AC1	Medium time threshold for AC1. Each count corresponds to 32usec
15:0	SW_MEDIUM_TIME_THR_AC0	Medium time threshold for AC0. Each count corresponds to 32usec

0x0A082130 WCSS_A_TPE_SW_MEDIUM_TIME_THR_AC2_AC3**Type:** RW**Reset State:** 0xEFFFFFFF

This register contains the medium time threshold used for AC2 and AC3. Though the register will be written immediately, the internal logic will take this value at next one second pulse or when a pkt has been transmitted from the corresponding AC

WCSS_A_TPE_SW_MEDIUM_TIME_THR_AC2_AC3

Bits	Name	Description
31:16	SW_MEDIUM_TIME_THR_AC3	Medium time threshold for AC3. Each count corresponds to 32usec
15:0	SW_MEDIUM_TIME_THR_AC2	Medium time threshold for AC2. Each count corresponds to 32usec

0x0A082134 WCSS_A_TPE_SW_ADMISSION_CONTROL_CONTROL_REG**Type:** RW**Reset State:** 0x0000E4E4

This register contains the control for Admission Control feature. Basically it contains the back-off to AC mapping and enables to Admission control

WCSS_A_TPE_SW_ADMISSION_CONTROL_CONTROL_REG

Bits	Name	Description
31:24	SW_MTU_BCKOF_DISABLE_VALID_VECTOR	Enable bits for backoff disable control. [24] --> bkof0, [25]-> bkof 1, [26] -> bkof 27 This is override to disable back-off. Bug protection + also used if s/w wants to update the sw_medium_time_used_acs.
23:16	SW_AC_VALID_FOR_BKOF_VECTOR	Enable bits for Admission control. [16] --> bkof0, [17]-> bkof 1, [23] -> bkof 7 Only the traffic that corresponds to the back-off for which the enable is set is considered for admission control
15:0	SW_AC_TO_BKOF_LOOKUP_VECTOR	The AC that needs to be used for back-off 0, 1, 2, ...7 [1:0] -> bkof0, [3:2] -> bkof1,..[15:14]-> bkof7

0x0A08213C WCSS_A_TPE_SW_LIFE_TIME_DROP_CNT**Type:** RW**Reset State:** 0x00000000

Counter that counts the dropped pkts due to life time enforcement

WCSS_A_TPE_SW_LIFE_TIME_DROP_CNT

Bits	Name	Description
31:0	SW_LIFE_TIME_DROP_CNT	Whenever a pkt is dropped due to life time enforcement, this count will be incremented. Writing to this register clears the counter. s/w is given higher priority over h/w

0x0A082140 WCSS_A_TPE_TPE_MCU_BD_BASED_TX_INT_FEEDBACK**Type:** R**Reset State:** 0x00000000

Contains the feedback info for the bd based tx interrupt

WCSS_A_TPE_TPE_MCU_BD_BASED_TX_INT_FEEDBACK

Bits	Name	Description
31:0	TPE_MCU_BD_BASED_TX_INT_FEEDBACK	[0] is ack_to_valid [3:1] feedback that is sent to bmu -- Definition is as follows. S/w Doesn't need this. Kept this for any debug purpose and if s/w comes up with new requirement in future, these can be used TPE_BMU_FB_RESET 3'b000 TPE_BMU_FB_DROP 3'b001 TPE_BMU_FB_DROP_AND_REARB 3'b010 TPE_BMU_FB_RESET_AND_REARB 3'b011 TPE_BMU_FB_RESET_AND_CHK_RETRY 3'b100 TPE_BMU_FB_RESET_CHK_RTRY_AND_REARB 3'b101 TPE_BMU_FB_RESET_AND_REARB_UPDATE_RETRY 3'b110 [8:4] QID [15:9] STAID [31:16] bd index

0x0A082144 WCSS_A_TPE_TPE_MCU_BD_BASED_TX_INT_FEEDBACK_1**Type:** R**Reset State:** 0x00000000

Contains the feedback info for the bd based tx interrupt

WCSS_A_TPE_TPE_MCU_BD_BASED_TX_INT_FEEDBACK_1

Bits	Name	Description
31:0	TPE_MCU_BD_BASED_TX_INT_FEEDBACK_1	[0] is ack_to_valid [3:1] feedback that is sent to bmu -- Definition is as follows. S/w Doesn't need this. Kept this for any debug purpose and if s/w comes up with new requirement in future, these can be used TPE_BMU_FB_RESET 3'b000 TPE_BMU_FB_DROP 3'b001 TPE_BMU_FB_DROP_AND_REARB 3'b010 TPE_BMU_FB_RESET_AND_REARB 3'b011 TPE_BMU_FB_RESET_AND_CHK_RETRY 3'b100 TPE_BMU_FB_RESET_CHK_RTRY_AND_REARB 3'b101 TPE_BMU_FB_RESET_AND_REARB_UPDATE_RETRY 3'b110 [8:4] QID [15:9] STAID [31:16] bd index

0x0A082148 WCSS_A_TPE_WRONG_BA_SSN_TID_FEEDBACK**Type:** R**Reset State:** 0x00000000

Contains the feedback info for the wrong BA interrupt. For the BAR sent, it a BA is received with different SSN or TID, an interrupt is set This register contains the Staid and tid of the corresponding flow

WCSS_A_TPE_WRONG_BA_SSN_TID_FEEDBACK

Bits	Name	Description
11:8	WRONG_BA_TID_FEEDBACK	This contains the tid of the flow for which a wrong BA is received
7:0	WRONG_BA_STAID_FEEDBACK	This contains the station id of the flow for which a wrong BA is received

0x0A08214C WCSS_A_TPE_SW_CAPPED_POWER_CONTROL**Type:** RW**Reset State:** 0x00000000

This register contains the capped power controls -- bit vector and the capped power

WCSS_A_TPE_SW_CAPPED_POWER_CONTROL

Bits	Name	Description
26:24	SW_PKT_BASED_POWER_BITMAP	The capped power is used only if the corresponding bit is set Bit 0 is for All response pkts -- ack, ba, cts in response to RTS Bit 1 is for RTS,CTS to self Bit 2 is for the rest of types
20:16	SW_CAPPED_POWER_VALUE	The capped power. The power that needs to be used for data and ack. Note the minimum of this and the one specified in R2P is used
15:0	SW_CAPPED_POWER_BIT_MAP	Station bit vector for capped power. If the corresponding bit is set, only then the capped power is used for the data and ack

0x0A082150 WCSS_A_TPE_SW_ACK_TO_DUE_TO_BTC_COUNT**Type:** RW**Reset State:** 0x00000000

A counter which increments every time when there is ack_to due to BTC

WCSS_A_TPE_SW_ACK_TO_DUE_TO_BT_COUNT

Bits	Name	Description
15:0	SW_ACK_TO_DUE_TO_BT_C_COUNT	Whenever there is ack_to due to BTC, this counter is ++

0x0A082154 WCSS_A_TPE_SW_LOW_PRIORITY_BT_REJECTED_COUNT**Type:** RW**Reset State:** 0x00000000

A counter which ++ whenever the low priority BT is rejected due to the wlan is busy

WCSS_A_TPE_SW_LOW_PRIORITY_BT_REJECTED_COUNT

Bits	Name	Description
15:0	SW_LOW_PRIORITY_BT_REJECTED_COUNT	Whenever there is no grant to BT due to the wlan is busy (for low priority BT) this counter increments

0x0A082158 WCSS_A_TPE_SW_TERMINATE_CURRENT_TX_VALID**Type:** RW**Reset State:** 0x00000000

To support terminate at TBTT in STA mode

WCSS_A_TPE_SW_TERMINATE_CURRENT_TX_VALID

Bits	Name	Description
0	SW_TERMINATE_CURRENT_TX_VALID	If this bit is set, tpe will terminate the current transmission. Basically this needs to be used by s/w when the chip is in station mode. Though we lived with this in earlier chips, it is realized that we might occupy the channel such that the AP might delay the beacons which is not good for the power save of other stations. Basically s/w simply have tsf based timers to trigger TBTT and enable this bit. This is not a sticky bit. This bit is reset if the tpe is in idle

0x0A08215C WCSS_A_TPE_RTT_REGISTER**Type:** RW**Reset State:** 0x00000000

RTT registers for InNav Positioning

WCSS_A_TPE_RTT_REGISTER

Bits	Name	Type	Description
31	RTT_INFO_VALID	RW	When set to 1, this RTT register contains new valid information. The SW can clear this bit by writing '0'.
30:24	FIR0_RSSI	R	The measured RSSI value from FIR-0
23:16	SNR	R	The measured SBR value
15:11	TX_STAID	R	The TX STAID corresponding to this measurement
10:0	RTT	R	The measured RTT value in 80MHz clk cycles

0x0A082160 WCSS_A_TPE_BEAMFORM_CTRL**Type:** RW**Reset State:** 0x00000000

Contains the control signals for calibration and beamform

WCSS_A_TPE_BEAMFORM_CTRL

Bits	Name	Description
31:26	RESERVED	Move to ndp_beamform_ctrl.sounding_frame_response_rate_index due to bit-width increased because of new rate.
25	CLEAR_WAITING_FOR_SE NDING_QOSNULL	In SW controlled implicit beamform, SW has to write this bit so new qosnull request will be accepted SW can also use this to clear the waiting if anything goes wrong
24	ENABLE_MPI_CSI_EN	The mcfg_phyicsxi_xfercsi_nonndp will be always on (as a backup if anything goes wrong).
23	CLEAR_WAITING_FOR_SE NDING_CSI	In SW controlled calibration/explicit beamform, SW has to write this bit so new csi request will be accepted SW can also use this to clear the waiting if anything goes wrong
22	EXPLICIT_IMMEDIATE_CTS _RSP_EN	When set, the 'cts response' explicit beamform will be acked with immediate cts and later ack+csi feedback, regardless the setting in bit19. This is not currently implemented. When cleared, delayed feedback will be used if bit19 is set.
21	EXPLICIT_IMMEDIATE_ACK _RSP_EN	When set, the ack 'response' explicit beamform will be acked with immediate ack/ba+csi feedback, regardless the setting in bit19. When cleared, delayed feedback will be used if bit19 is set.
20	EXPLICIT_IMMEDIATE_NO_ RSP_EN	When set, the 'no response' explicit beamform will be acked with immediate csi feedback, regardless the setting in bit19. When cleared, delayed feedback will be used if bit19 is set.
19	HW_EXPLICIT_BEAMFORM _DELAY_FEEDBACK	Set to '1' to enable the hw control csi delayed feedback for explicit beamform (when the support is enabled, and not ruled by the immediate feedback bits).

WCSS_A_TPE_BEAMFORM_CTRL (cont.)

Bits	Name	Description
18	EXPLICIT_BEAMFORM_EN	Set to '1' to enable the CSI explicit beamform support.
17:14	SENDING_BKOF_QOSNULL_RETRY_THRSHLD	To setup the retry threshold when sending the bkof qosnull in hw-controlled implicit beamform mode.
13:10	SENDING_BKOF_CSI_RETRY_THRSHLD	To setup the retry threshold when sending the bkof csi in hw-controlled calibration and explicit beamform mode.
7	SW_NONCOMPRESSED_EXPLICIT_BEAMFORM_EN	When set, sw control noncompressed explicit beamform is enabled. Note no hw support for this mode.
6	SW_COMPRESSED_EXPLICIT_BEAMFORM_EN	When set, sw control compressed explicit beamform is enabled. Note no hw support for this mode.
5	HW_CONTROLLED_IMPLICIT_BEAMFORM_EN	When set, implicit is under hw control. If a qosnull is waiting to be sent and bakoff 6 expires. the qosnull will be sent.
4	HW_CONTROLLED_CALIBRATION_EN	When set, calibration is under hw control. If a csi is waiting to be sent and bakoff 7 expires. the csi will be sent.
3	IGNORE_CSI_FIELD_FOR_CAL_START_NDP	Set to '1' to ignore the csi field for ndp based calibration cal start frame
2	SW_SEND_NDP2_VALID	Set to '1' to force the HW to send the ndp frame to responded to the ndp frame in the ndp based calibration. The first response for this frame exchange will be a normal ack/cts. Clear to '0' so no ndp is sent. The first response for the ndp based calibration will be a sounding frame.
1	IMPLICIT_BEAMFORM_EN	Set to '1' to enable implicit beamform support
0	SW_CALIBRATION_FRAME_EXCHANGE_ENABLED	Set to '1' to enable calibration support

0x0A082164 WCSS_A_TPE_CFG_CSI_ADDR**Type:** RW**Reset State:** 0x00000000

Contains the address of the CSI template

WCSS_A_TPE_CFG_CSI_ADDR

Bits	Name	Description
31:0	CFG_CSI_ADDR	Contains the address of the CSI frame template

0x0A082168 WCSS_A_TPE_INTERRUPT_ENABLE**Type:** RW**Reset State:** 0x00000000

Contains the enable bits for which interrupt has to be generated

WCSS_A_TPE_INTERRUPT_ENABLE

Bits	Name	Description
12	RTT3_T2_CAPTURE_INT_EN	Set to '1' to enable the rtt3 interrupt when ack is send in respond to the received TMF so SW can stop the channel capture
11	RTT3_T4_CAPTURE_INT_EN	Set to '1' to enable the rtt3 interrupt when (t4-t1) and absolute t1 value is ready in the register
10	SW_TEMPLATE_WRONG_CMD_INT_EN	Set to '1' to enable the invalid sw command interrupt
9	EXPLICITBF_QOSNULL_INT_EN	Set to '1' to enable the explicit beamform qosnull+ndp interrupt when the ndp is received followed by the qosnull request frame.
8	SW_TEMPLATE_SENT_INT_EN	Set to '1' to enable interrupt after sw template is sent
7	GOT_ACK_FOR_BACK_INT_EN	Set to '1' to enable got_ack_for_back interrupt when expecting ba but received ack
6	ERROR_VALID_INT_EN	Set to '1' to enable error interrupt. It can be either the mproc enters error state (pkt length is '0' for non qosnull/datanull pkt), or when enabled by sw, the txp timeout or internal error interrupts
5	WRONG_SSN_OR_TID_ERROR_INT_EN	Set to '1' to enable wrong_ssn_or_tid interrupt when bar is sent and the tid/ssn of the received ba are not matched
4	NONCOMPRESSED_EXPLICIT_BEAMFORM_INT_EN	Set to '1' to enable noncompressed explicit beamform interrupt for sw-controlled non compressed explicit beamform transmission
3	COMPRESSED_EXPLICIT_BEAMFORM_INT_EN	Set to '1' to enable compressed explicit beamform interrupt for sw-controlled compressed explicit beamform transmission
2	CSI_EXPLICIT_BEAMFORM_INT_EN	Set to '1' to enable csi explicit beamform interrupt for sw controlled delayed csi explicit beamform transmission
1	IMPLICIT_BEAMFORM_INT_EN	Set to '1' to enable implicit beamform interrupt for sw controlled non-SIFS sounding frame transmission
0	CALIBRATION_INT_EN	Set to '1' to enable calibration interrupt for sw controlled calibration CSI frame transmission

0x0A08216C WCSS_A_TPE_INTERRUPT_STATUS**Type:** RW**Reset State:** 0x00000000

Contains the interrupt status, writing 1 to any bit will clear the corresponding interrupt

WCSS_A_TPE_INTERRUPT_STATUS

Bits	Name	Description
12	RTT3_T2_CAPTURE_INT_STATUS	When set, indicates that ack is send in responded to the received TMF so SW can stop the channel capture Write '1' to clear the interrupt.
11	RTT3_T4_CAPTURE_INT_STATUS	When set, indicates that (t4-t1) and absolute t1 value is ready in the register Write '1' to clear the interrupt.
10	SW_TEMPLATE_WRONG_CMD_INT_STATUS	When set, indicates that there's an invalid sw template command Write '1' to clear the interrupt.
9	EXPLICITBF_QOSNULL_INT_STATUS	When set, indicates that explicit bf qosdata+ndp frame exchange is received Write '1' to clear the interrupt.
8	SW_TEMPLATE_SENT_INT_STATUS	When set, indicates that sw template transmit is completed Write '1' to clear the interrupt.
7	GOT_ACK_FOR_BACK_INT_STATUS	When set, indicates that when expecting ba but received ack Write '1' to clear the interrupt.
6	ERROR_VALID_INT_STATUS	When set, indicates the mproc error state, or when enabled by sw, the txp timeout or internal error interrupts Write '1' to clear the interrupt.
5	WRONG_SSN_OR_TID_ERROR_INT_STATUS	When set, indicates that bar is sent and the tid/ssn of the received ba are not matched Write '1' to clear the interrupt.
4	NONCOMPRESSED_EXPLICIT_BEAMFORM_INT_STATUS	When set, indicates the noncompressed explicit beamform frame request is received. SW has to control the non compressed explicit beamform transmission Write '1' to clear the interrupt.
3	COMPRESSED_EXPLICIT_BEAMFORM_INT_STATUS	When set, indicate the compressed explicit beamform frame req is received. SW has to control the compressed explicit beamform transmission Write '1' to clear the interrupt.
2	CSI_EXPLICIT_BEAMFORM_INT_STATUS	When set, indicates the explicit beamform frame is received. SW has to control the delayed csi explicit beamform transmission Write '1' to clear the interrupt.
1	IMPLICIT_BEAMFORM_INT_STATUS	When set, indicates the implicit beamform frame is received. SW has to send the non-SIFS sounding frame transmission at bkof Write '1' to clear the interrupt.
0	CALIBRATION_INT_STATUS	When set, indicates the calibration frame exchnage step1 is done. SW has to control the CSI frame transmission Write '1' to clear the interrupt.

0x0A082170 WCSS_A_TPE_BEAMFORM_STATUS**Type:** R**Reset State:** 0x00000000

Contains the status for calibration and beamform.

WCSS_A_TPE_BEAMFORM_STATUS

Bits	Name	Description
25	WAITING_FOR_SEND_QOSNULL	A qosnull is waiting to be sent. HW won't accept any new qosnull request until the signal is cleared by 1. qosnull is sent. 2. dropped due to retry limit 3. sw write to the clear register bit
24	WAITING_FOR_SEND_CSI	A CSI is waiting to be sent. HW won't accept any new csi request until the signal is cleared by 1. csi is sent. 2. csi is dropped due to retry limit 3. sw write to the clear register bit
23:20	SENDING_QOSNULL_CSI_RETRY_CNT	Retry counter for the bkof qosnull
19:16	SENDING_BKOF_CSI_RETRY_CNT	Retry counter for the bkof csi
15:8	ADDR2_STAID	The addr2_staaid for SW to send the bkof qosnull in sw controlled implicit beamform
7:0	ADDR1_STAID	The addr1_staaid for SW to send the bkof qosnull in sw controlled implicit beamform

0x0A082174 WCSS_A_TPE_CALIBRATION_CNT**Type:** R**Reset State:** 0x00000000

Contains the debug counter for calibration

WCSS_A_TPE_CALIBRATION_CNT

Bits	Name	Description
23:16	CALIBRATION_SENT_CNT	Calibration sent count, increase when the bkof csi is sent or dropped.
15:8	CALIBRATION_ABORT_CNT	Calibration abort count, increase when calibration complete frame is not received
7:0	CALIBRATION_REQ_CNT	Calibration request count

0x0A082178 WCSS_A_TPE_BEAMFORM_CNT**Type:** R**Reset State:** 0x00000000

Contains the debug counter for implicit/explicit beamform`

WCSS_A_TPE_BEAMFORM_CNT

Bits	Name	Description
31:24	EXPLICIT_BEAMFORM_CSI_SENT_CNT	Explicit Beamform sent count, increase when the bkof csi is sent or dropped
23:16	EXPLICIT_BEAMFORM_REQ_CNT	Explicit beamform request count
15:8	IMPLICIT_BEAMFORM_QOSNULL_SENT_CNT	Implicit Beamform sent count, increase when the bkof qosnull is sent or dropped
7:0	IMPLICIT_BEAMFORM_REQ_CNT	Implicit Beamform request count

0x0A08217C WCSS_A_TPE_SW_PM_2**Type:** RW**Reset State:** 0x00000000

sw controlled pm bit

WCSS_A_TPE_SW_PM_2

Bits	Name	Description
31:28	SW_TX_SIFS_A_MODE_CYCLES	The upper bits of SIFS cycles that needs to be programmed in tXP commands for A and G (non B) mode pkts (The lower bits are in sw_pm.sw_tx_sifs_a_mode_cycles)
9:0	SW_PM	SW configurable PM bit for control/response frame which were sent by HW template before. bit0 - ACK. bit1 - RTS. bit2 - CTS. bit3 - BAR. bit4 - BA. bit5 - CF_END bit6 - ACK CTRL WRAPPER. bit7 - CTS CTRL WRAPPER. bit8 - Null feedback in 11ac bit9 - BA CTRL WRAPPER

0x0A082198 WCSS_A_TPE_SW_11AC_PADDING**Type:** RW**Reset State:** 0x0100794E

The SMDA version of A-MPDU delimiters to fill in the last ofdm symbol for 11ac

WCSS_A_TPE_SW_11AC_PADDING

Bits	Name	Description
31:0	SW_11AC_PADDING_CONTENTS	The SMDA version of A-MPDU delimiters to fill in the last ofdm symbol for 11ac

0x0A0821B0 WCSS_A_TPE_NDP_BEAMFORM_CTRL**Type:** RW**Reset State:** 0x00000000

Contains the control signals for beamform qosnull + ndp

WCSS_A_TPE_NDP_BEAMFORM_CTRL

Bits	Name	Description
24	SW_EXPLICITBF_NDP_NO_NCOMPRESSED_EN	When set, ndp based noncompressed explicit beamform is enabled.
23	SW_EXPLICITBF_NDP_COMPRESSED_EN	When set, ndp based compressed explicit beamform is enabled.
22	SW_EXPLICITBF_NDP_CSI_EN	When set, ndp based csi explicit beamform is enabled.
21:13	SOUNDING_FRAME_RESPONSE_RATE_INDEX_40MHZ	To be used in a sounding frame response at 40MHz when the response rate from the rate table is 11b or 11g rate.
12:4	SOUNDING_FRAME_RESPONSE_RATE_INDEX	To be used in a sounding frame response at 20MHz when the response rate from the rate table is 11b or 11g rate.
3	ENABLE_MPI_NDP_EN	The mcfg_phyicsxi_xfercsi_ndp will be always on (as a backup if anything goes wrong).
2	SW_EXPLICITBF_QOSNULL_HW_CTRL_EN	Set to '1' to enable the hw controlled delay feedback of beamform qosnull+ndp
1	SW_EXPLICITBF_QOSNULL_IMMEDIATE_EN	Set to '1' to enable the immediate feedback of beamform qosnull+ndp
0	SW_EXPLICITBF_QOSNULL_NDP_EN	Set to '1' to enable the explicit beamform qosnull + ndp

0x0A0821B4 WCSS_A_TPE_BEAMFORM_QOSNULL_NDP_CNT**Type:** R**Reset State:** 0x00000000

Contains the debug counter for beamform qosnull+ndp

WCSS_A_TPE_BEAMFORM_QOSNULL_NDP_CNT

Bits	Name	Description
23:16	EXPLICITBF_QOSNULL_SENT_CNT	explicit beamform feedback sent count
15:8	EXPLICITBF_QOSNULL_NDP_RECV_CNT	explicit beamform ndp receive count

WCSS_A_TPE_BEAMFORM_QOSNULL_NDP_CNT (cont.)

Bits	Name	Description
7:0	EXPLICITBF_QOSNULL_RE Q_CNT	explicit beamform qosnull request count

0x0A0821B8 WCSS_A_TPE_CXM_CTRL**Type:** RW**Reset State:** 0x00000080

Contains the control and status of cxm logic

WCSS_A_TPE_CXM_CTRL

Bits	Name	Type	Description
7	TPE_MTU_GEN6_3WIRE_WLAN_ACTIVE	R	tpe_mtu_gen6_3wire_wlan_active status
6	WMAC_RFIF_BT_TX_120	R	btc_active_final status
5	WMAC_RFIF_BT_MEDIUM_120	R	btc_active_final status
4	BTC_ACTIVE_FINAL	R	btc_active_final status
3	LTE_CXM_WLAN_TX_GRANT	R	lte_cxm_wlan_tx_grant status
2	WLAN_TX_GRANT	R	wlan_tx_grant status
1	SW_WLAN_TX_GRANT_SE L	RW	When enabled(bit[0]), 0: btc_active from pta controls tx, 1:lte_cxm_wlan_tx_grant from timer controls tx
0	SW_WLAN_TX_GRANT_CT RL_EN	RW	Set 1 to enable the wlan_tx_grant control logic. Clear 0 to always allow transmission

0x0A0821BC WCSS_A_TPE_SW_11AC_CTRL**Type:** RW**Reset State:** 0x00000200

Contains the control signals 11ac related

WCSS_A_TPE_SW_11AC_CTRL

Bits	Name	Description
28	SW_DONT_SEND_NULL_F EEDBACK	Set to 1 to disable sending null feedback. The FSM will go back to idle and do nothing if the VHTCB is not ready

WCSS_A_TPE_SW_11AC_CTRL (cont.)

Bits	Name	Description
27:16	SW_EXTRA_PADDING_FOR_11AC	The extra padding (in number of words) to be added to mimic mu packet generation. Debug purpose only.
15:0	SW_MAX_AMPDU_TX_TIME_11AC	The maximum ampdu tx time in 11ac mode

0x0A0821E4 WCSS_A_TPE_SW_NONHT_BW_CTRL**Type:** RW**Reset State:** 0x00000000

Contains the BW indication configuration

WCSS_A_TPE_SW_NONHT_BW_CTRL

Bits	Name	Description
6	SW_NAV_CHECK_FOR_RTS_EN	When set, always check if nav is zero before sending cts. Otherwise it will only be checked for BW signaling.
5:0	SW_RTS_BW_RETRY_THRESHOLD	In static mode, when rts retry counters is larger than the threshold, data will be sent out at 20Mhz rate.

0x0A0821E8 WCSS_A_TPE_SW_GLOBAL_CF_END_DATA_POWER_TEMPLATE_40MHZ**Type:** RW**Reset State:** 0x00000000**WCSS_A_TPE_SW_GLOBAL_CF_END_DATA_POWER_TEMPLATE_40MHZ**

Bits	Name	Description
30	FEC_MODE	
29	LPE_BIT	
25	AMPDU_VALID	
24:20	TX_POWER	
19	RESERVED	
18:16	TX_ANTENNA_EN	
15:14	STBC_VALID	
13	SGI_VALID	
12:11	NSS_11B_MODE	
8:3	STA_RATE	

WCSS_A_TPE_SW_GLOBAL_CF_END_DATA_POWER_TEMPLATE_40MHZ (cont.)

Bits	Name	Description
2:0	STA_MODE	

0x0A0821EC WCSS_A_TPE_SW_GLOBAL_CF_END_DATA_POWER_TEMPLATE_80MHZ**Type:** RW**Reset State:** 0x00000000**WCSS_A_TPE_SW_GLOBAL_CF_END_DATA_POWER_TEMPLATE_80MHZ**

Bits	Name	Description
30	FEC_MODE	
29	LPE_BIT	
25	AMPDU_VALID	
24:20	TX_POWER	
19	RESERVED	
18:16	TX_ANTENNA_EN	
15:14	STBC_VALID	
13	SGI_VALID	
12:11	NSS_11B_MODE	
8:3	STA_RATE	
2:0	STA_MODE	

0x0A0821F0 WCSS_A_TPE_SW_WARMUP_DELAY**Type:** RW**Reset State:** 0x00000000

This register contains the control signals for warmup

WCSS_A_TPE_SW_WARMUP_DELAY

Bits	Name	Description
23:20	SW_DELAY_WAIT_CMD_CNT_UPPER	The upper bits of sw_delay_wait_cmd_cnt_reg.sw_delay_wait_cmd_cnt
19:16	SW_WAIT_CYCLES_THRESHOLD_UPPER	(in terms of 8 cycles). The upper bits of sw_delay_wait_cmd_cnt_reg.sw_wait_cycles_threshold
15:12	SW_WARMUP_TIME_UPPER	The upper bits of wait time to be programmed in txp commands. The lower bits are [10:0]

WCSS_A_TPE_SW_WARMUP_DELAY (cont.)

Bits	Name	Description
11	SW_FIX_WARMUP_DELAY_EN	Set this bit to 1 to enable the fix warmup delay
10:0	SW_WARMUP_TIME	The wait time to be programmed in txp commands. This is the time between tpe asserts the warmup and the time when txp sends the first phy cmd.

0x0A0821F4 WCSS_A_TPE_MAIN_MPROC_REQ_CNT**Type:** R**Reset State:** 0x00000000

This register contains the main_mproc request counter per session

WCSS_A_TPE_MAIN_MPROC_REQ_CNT

Bits	Name	Description
27:24	SESSION_IS_BACKOFF_CSI_QOSNULL_CNT	
23:20	SESSION_IS_BACKOFF_CNT	
19:16	SESSION_IS_BEACON_CN	
15:12	SESSION_IS_SW_TEMPLATE_CNT	
11:8	SESSION_IS_SW_STA_QID_CNT	
7:4	SESSION_IS_SW_BACKOFF_INDEX_CNT	
3:0	SESSION_IS_RXP_CNT	

0x0A0821F8 WCSS_A_TPE_SW_11AC_BEAMFORM_CTRL**Type:** RW**Reset State:** 0x00000000

This register contains the control signals for 11ac beamform

WCSS_A_TPE_SW_11AC_BEAMFORM_CTRL

Bits	Name	Description
9	SW_11N_LINK_ADAPTATION_EN	Set to 1 to enable the 11n link adaptation feature

WCSS_A_TPE_SW_11AC_BEAMFORM_CTRL (cont.)

Bits	Name	Description
8	SW_11AC_LINK_ADAPTATION_EN	Set to 1 to enable the 11ac link adaptation feature
7:3	SW_CHECK_CFG_READY_TIME	The time (in micro second) when TPE should check the cfg_ready signal to see if the report is ready or not.
0	SW_11AC_BEAMFORM_EN	Set to 1 to enable the 11ac beamform feature

0x0A0821FC WCSS_A_TPE_SW_11AC_CNT1**Type:** R**Reset State:** 0x00000000

Contains the debug counters for 11ac beamform

WCSS_A_TPE_SW_11AC_CNT1

Bits	Name	Description
31:24	NULLFEEDBACK_SENT_COUNT	11ac null feedback sent count
23:16	VHTC_SENT_CNT	11ac vhtc sent count
15:8	NDP_RCVD_CNT	11ac ndp (following ndpa) receive count
7:0	NDPA_RCVD_CNT	11ac ndpa received count

0x0A0821AC WCSS_A_TPE_SW_11AC_CNT2**Type:** R**Reset State:** 0x00000000

Contains the debug counters for 11ac beamform / link adaptation

WCSS_A_TPE_SW_11AC_CNT2

Bits	Name	Description
23:16	NDPA_MRQ_REQ_CNT	11ac ndpa mrq request count
15:8	PPDU_MRQ_REQ_CNT	11ac ppdu mrq request receive count
7:0	BRPOLL_RCVD_CNT	11ac BR Poll received count

0x0A0821A8 WCSS_A_TPE_SW_11AC_VHTCB_RATE**Type:** RW**Reset State:** 0x00000000

Contains the rate for sending 11ac vhtcb/null feedback SW has to program this register dynamically following RA. The primary or secondary rate should be programmed.

WCSS_A_TPE_SW_11AC_VHTCB_RATE

Bits	Name	Description
26:18	VHTCB_RATE_80MHZ	the rate index to be used to send vhtcb/null feedback at 80Mhz
17:9	VHTCB_RATE_40MHZ	the rate index to be used to send vhtcb/null feedback at 40Mhz
8:0	VHTCB_RATE_20MHZ	the rate index to be used to send vhtcb/null feedback at 20Mhz

0x0A0821A0 WCSS_A_TPE_SW_VHTCB_ADDR2_0_LO**Type:** RW**Reset State:** 0x00000000

Contains the lower 32 bits of addr2 to send vhtcb/null feedback

WCSS_A_TPE_SW_VHTCB_ADDR2_0_LO

Bits	Name	Description
31:0	VHTCB_ADDR2_0_LO	the lower 32 bits of addr2 to send vhtcb/null feedback

0x0A0821A4 WCSS_A_TPE_SW_VHTCB_ADDR2_0_HI**Type:** RW**Reset State:** 0x00000000

Contains the upper 16 bits of addr2 to send vhtcb/null feedback

WCSS_A_TPE_SW_VHTCB_ADDR2_0_HI

Bits	Name	Description
15:0	VHTCB_ADDR2_0_HI	the upper 16 bits of addr2 to send vhtcb/null feedback

0x0A0821C0 WCSS_A_TPE_SW_VHTCB_ADDR2_1_LO**Type:** RW**Reset State:** 0x00000000

Contains the lower 32 bits of addr2 to send vhtcb/null feedback

WCSS_A_TPE_SW_VHTCB_ADDR2_1_LO

Bits	Name	Description
31:0	VHTCB_ADDR2_1_LO	the lower 32 bits of addr2 to send vhtcb/null feedback

0x0A0821C4 WCSS_A_TPE_SW_VHTCB_ADDR2_1_HI**Type:** RW**Reset State:** 0x00000000

Contains the upper 16 bits of addr2 to send vhtcb/null feedback

WCSS_A_TPE_SW_VHTCB_ADDR2_1_HI

Bits	Name	Description
15:0	VHTCB_ADDR2_1_HI	the upper 16 bits of addr2 to send vhtcb/null feedback

0x0A0821C8 WCSS_A_TPE_SW_VHTCB_ADDR2_LOOKUP**Type:** RW**Reset State:** 0x00000000

Contains the lookup table to select the sw_vhtcb_addr2 registers

WCSS_A_TPE_SW_VHTCB_ADDR2_LOOKUP

Bits	Name	Description
31:0	VHTCB_ADDR2_LOOKUP	2 bits for each staid to select which vhtcb_addr2 registers for sending vhtcb/nullfeedback in response to NPD 00: sw_vhtcb_addr2_0, 01: sw_vhtcb_addr2_1, 10, 11: reserved

0x0A0821CC WCSS_A_TPE_SW_RTT3_ABSOLUTE**Type:** R**Reset State:** 0x00000000

Contains the absolute value of t1 for rtt3

WCSS_A_TPE_SW_RTT3_ABSOLUTE

Bits	Name	Description
31:0	RTT_REF	The absolute value of r1

0x0A0821D0 WCSS_A_TPE_SW_RTT3_DELTA**Type:** R**Reset State:** 0x00010000

Contains the value of (t4-t1) for rtt3

WCSS_A_TPE_SW_RTT3_DELTA

Bits	Name	Description
16	RESP_TO_TMF_STATUS	The bit will be set when SW writes 1 to sw_rtt3_ctrl.rtt3_resp_to_tmf_en (bit4), and will be cleared after ack is sent in response to tmf.
15	DUPLICATE_PKT	1 indicates this is a duplicate/retried TMF that has been dropped by rxp
14	RTT_TYPE	Taken from pmi stats directly (through rxp). 1 indicates it's a valid t4-t1 value
13:0	RTT_DELTA	The (t4-t1) value

0x0A0821D4 WCSS_A_TPE_SW_RTT3_CTRL**Type:** RW**Reset State:** 0x00000000

Contains the rtt3 configuration

WCSS_A_TPE_SW_RTT3_CTRL

Bits	Name	Type	Description
31:16	RTT3_VALIDATE_TA_HI	RW	The upper bits of the TA of TMF to be validate
4	RTT3_RESP_TO_TMF_EN	W	When bit3 is set, SW has to write 1 to this bit field so HW will send ack for TMF. Once an ack is sent for a TMF, no more ack will be sent when tmf is received until this bit has been set to 1. Always read 0.

WCSS_A_TPE_SW_RTT3_CTRL (cont.)

Bits	Name	Type	Description
3	RTT3_ENABLE_NO_ACK_T O_TMF	RW	when clear to 0, the ack will always be sent for TMF regardless bit4 setting when set to 1, enable the feature of not sending ack to TMF. SW has to write 1 to bit4 so ack will be sent in response to TMF. No further ack will be sent to TMF afterwards and SW has to re-enable bit4 so the ack will be sent.
2	RTT3_NONHT_RESP_RAT E_EN	RW	Set to 1 to enable ht/vht response rate to TMF/TMRF
1	RTT3_RESPONDER_EN	RW	Set to 1 to enable rtt3 feature as responder
0	RTT3_INITIATOR_EN	RW	Set to 1 to enable rtt3 feature as initiator

0x0A0821D8 WCSS_A_TPE_SW_RTT3_CTRL2**Type:** RW**Reset State:** 0x00000000

Contains the rtt3 configuration

WCSS_A_TPE_SW_RTT3_CTRL2

Bits	Name	Description
31:0	RTT3_VALIDATE_TA_LO	The lower bits of the TA of TMF to be validate

0x0A0821DC WCSS_A_TPE_SW_11AF_CTRL**Type:** RW**Reset State:** 0x00000000

Contains the 11/11af configuration

WCSS_A_TPE_SW_11AF_CTRL

Bits	Name	Description
15:8	SW_11AF_SIFS	The SIFS time for duration calculation in 11j/11af mode (when bit5 is enabled)
5	SW_11AF_EN	To enable the scale factor for tx time and SIFS in 11j/11af
4:0	SW_11AF_SCALE_FACTOR	The scale factor for tx time in 11j/11af mode (when bit5 is enabled) Theses are the allowed numbers to be programmed: 2,4,5,8,10,16

0x0A0821E0 WCSS_A_TPE_SW_QBOOST_CTRL**Type:** RW**Reset State:** 0x00000000

Contains the 11/11af configuration

WCSS_A_TPE_SW_QBOOST_CTRL

Bits	Name	Type	Description
31:16	SW_QBOOST_TXOP	RW	To program the txop when sending data in qboost when bit2 is set to 1
3	MAIN_MPROC_QBOOST_VALID	R	main_mproc_qboost_valid status
2	SW_QBOOST_TXOP_SEL	RW	Set to 1 to select the max txop from the register value in bit[31:16] Clear to 0 then the existing logic will be used
1	SW_QBOOST_PROTECTION_SEL	RW	Set to 1 to disable sending protection frame before sending the data Clear to 0 then the existing logic will be used
0	SW_QBOOST_EN	RW	Set to 1 to enable the qboost feature

10.60 WCSS_A_TXCLKCTRL_TXCLKCTRL**0x0A013000 WCSS_A_TXCLKCTRL_ROOT_CLK_EN****Type:** RW**Reset State:** 0x0000**WCSS_A_TXCLKCTRL_ROOT_CLK_EN**

Bits	Name	Description
3	CLK60	
2	CLK80_3BY4	
1	CLK160	
0	CLK80	

0x0A013004 WCSS_A_TXCLKCTRL_APB_BLOCK_CLK_EN**Type:** RW**Reset State:** 0x00FF

WCSS_A_TXCLKCTRL_APB_BLOCK_CLK_EN

Bits	Name	Description
8	TATMUX	
7	TPC	
6	TXFIR_APB	
5	TXFIR	
4	TXB	
3	TXA_APB	
2	TXA	
1	TXCTL	
0	MPI	

0x0A013008 WCSS_A_TXCLKCTRL_APB_BLOCK_DYN_CLKG_DISABLE**Type:** RW**Reset State:** 0x0048**WCSS_A_TXCLKCTRL_APB_BLOCK_DYN_CLKG_DISABLE**

Bits	Name	Description
7	TPC	
6	TXFIR_APB	
5	TXFIR	
4	TXB	
3	TXA_APB	
2	TXA	
1	TXCTL	
0	MPI	

10.61 WCSS_A_TXCTL_TXCTL**0x0A012000 WCSS_A_TXCTL_FIR_MODE****Type:** RW**Reset State:** 0x00C4

This register is used to switch in a test waveform feed to the TX FIR instead of the normal packet data stream and to manually enable the DACs and RF chains during the test mode. !! NOTE : the user must also appropriately program the DAC_SHUTDOWN register to ensure that the DAC standby and shutdown registers are configured correctly

WCSS_A_TXCTL_FIR_MODE

Bits	Name	Description
9:8	RESERVED	reserved
7	DIS_11MBPS	When set disables secondary antennae (i.e. all antennae except ant0) for 11 Mbps mode. Should be 1 (antennae disabled) for normal operation. 0x0: ENABLE (secondary antennae enabled for 11 Mbps mode) 0x1: DISABLE (secondary antennae disabled for 11 Mbps mode)
6	DIS_5MBPS	When set disables secondary antennae (i.e. all antennae except ant0) for 5.5 Mbps mode. Should be 1 (antennae disabled) for normal operation. 0x0: ENABLE (secondary antennae enabled for 5.5 Mbps mode) 0x1: DISABLE (secondary antennae disabled for 5.5 Mbps mode)
2	ANT_EN	4'bxxx0 -> none (No DAC/RF chains enabled) 4'bxxx1 -> zero (DAC/RF chain 0 enabled) This field defines which TX CHAINS (DAC and RF) are enabled in the waveform (test) mode of operation and also for packets that request the default TX enable configuration. Default 1'b1, ie Chain 0 enabled.
1	RESERVED	reserved
0	SELECT_FIRMODE	This bit selects the input to the TX FIR, 0=Normal, 1=WFM generator. Default 0 0x0: NORMAL (Normal operation, packet data fed to TX FIR) 0x1: GENERATOR (Waveform operation, waveform generator output fed to TX FIR)

0x0A012004 WCSS_A_TXCTL_PA_GUARD

Type: RW

Reset State: 0x0000

Specifies guard interval duration before enabling PAs. Delay between assertion of T/R and assertion of PA enable to protect PAs from transients.

WCSS_A_TXCTL_PA_GUARD

Bits	Name	Description
9:0	TIMER	Specifies the time that the TX takes between enabling T/R switch and enabling PAs. Measured in increments of 6.25 ns for Spica (12.5 ns for Libra) from 0 to 6.4 us (12.8us for Libra). Default 20, ie 0.125us in Gen5, 0.25uS in Libra 0x0: MIN (0 ns) 0x1: STEP (6.25 ns for Spica, 12.5 ns for Libra) 0x3FF: MAX (6.4 us for Spica, 12.8 us for Libra)

0x0A012008 WCSS_A_TXCTL_RAMP_UP_11A**Type:** RW**Reset State:** 0x004C

Specifies ramp up time for OFDM packets. Delay between RF enable and time that IQ waveform is presented on IQ outputs.

WCSS_A_TXCTL_RAMP_UP_11A

Bits	Name	Description
9:0	TIMER	Specifies the time that the TX takes to ramp up before the base band data can be shifted into it. Measured in increments of 6.25 ns for Spica (12.5 ns for Libra) from 0 to 6.4 us (12.8us for Libra). Default 40, ie 0.25us in Gen5, 0.5uS in Libra 0x0: MIN (0 ns) 0x1: STEP (6.25 ns for Spica, 12.5 ns for Libra) 0x3FF: MAX (6.4 us for Spica, 12.8 us for Libra)

0x0A01200C WCSS_A_TXCTL_RAMP_UP_11B**Type:** RW**Reset State:** 0x0024

Specifies ramp up time for 802.11b packets. Delay between RF enable and time that IQ waveform is presented on IQ outputs.

WCSS_A_TXCTL_RAMP_UP_11B

Bits	Name	Description
9:0	TIMER	Specifies the time that the TX takes to ramp up before the base band data can be shifted into it. Measured in increments of 6.25 ns for Spica (12.5 ns for Libra) from 0 to 6.4 us (12.8us for Libra). Default 40, ie 0.25us in Gen5, 0.5uS in Libra 0x0: MIN (0 ns) 0x1: STEP (6.25 ns for Spica, 12.5 ns for Libra) 0x3FF: MAX (6.4 us for Spica, 12.8 us for Libra)

0x0A012010 WCSS_A_TXCTL_RAMP_DOWN**Type:** RW**Reset State:** 0x00A0

Specifies ramp down time. Delay between end of IQ waveform and de-assertion of the RF enable to accomodate the delay through the DACs.

WCSS_A_TXCTL_RAMP_DOWN

Bits	Name	Description
9:0	TIMER	Specifies the time that the TX takes to ramp up before the base band data can be shifted into it. Measured in increments of 6.25 ns for Spica (12.5 ns for Libra) from 0 to 6.4 us (12.8us for Libra). Default 160, ie 1.0us in Gen5, 2.0uS in Libra 0x0: MIN (0 ns) 0x1: STEP (6.25 ns for Spica, 12.5 ns for Libra) 0x3FF: MAX (6.4 us for Spica, 12.8 us for Libra)

0x0A012014 RESERVED**0x0A012018 WCSS_A_TXCTL_DAC_CONTROL****Type:** RW**Reset State:** 0x0001

This register controls DAC and RF enable signals such as standby and TX enables. Also, can control the DAC STANDBY and RF TX enable signals

WCSS_A_TXCTL_DAC_CONTROL

Bits	Name	Description
9	TXEN_OVERRIDE_EN	0x0: NORMAL (RF TX enable signals in normal mode) 0x1: OVERRIDE (RF TX enable signals in override mode)

WCSS_A_TXCTL_DAC_CONTROL (cont.)

Bits	Name	Description
8	DAC_OVERRIDE_EN	0x0: NORMAL (DAC Standby signals in normal mode) 0x1: OVERRIDE (DAC Standby signals in override mode)
4	TXEN0_OVERRIDE_VAL	0x0: TXEN_OFF (Chain 0 RF off (txen de-asserted)) 0x1: TXEN_ON (Chain 0 RF on (txen asserted))
0	CH0STDBY_OVERRIDE_VAL	0x0: DAC_ON (Chain 0 DAC on (standby de-asserted)) 0x1: DAC_OFF (Chain 0 DAC off (standby asserted))

0x0A01201C WCSS_A_TXCTL_DAC_STDBY**Type:** RW**Reset State:** 0x0181

This register controls the timing of DAC STANDBY signals

WCSS_A_TXCTL_DAC_STDBY

Bits	Name	Description
8:0	TIMING	Each bit corresponds to one of the possible state of the TXCTL state machine. The states are as follows: Bit 0 = TXCTL_IDLE Bit 1 = TXCTL_RFWARM Bit 2 = TXCTL_START Bit 3 = reserved (unused) Bit 4 = TXCTL_RAMPUP Bit 5 = TXCTL_TX Bit 6 = TXCTL_SHUTOFF Bit 7 = TXCTL_TPC Bit 8 = TXCTL_OVERRIDE

0x0A012020 WCSS_A_TXCTL_PLCP_OVERRIDE**Type:** RW**Reset State:** 0x0000

This register contains an override mask and override values for HT-SIG fields.

WCSS_A_TXCTL_PLCP_OVERRIDE

Bits	Name	Description
8:7	NESS	Override value for Ness field
6	RSVR	Override value for Reserved field
5	SOUND	Override value for Not-sounding field
4	SMOOTH	Override value for Smoothing field
3:0	MASK	Override mask for {Ness, Reserved, Not_sounding, Smoothing} fields

0x0A012024 WCSS_A_TXCTL_FSHIFT**Type:** RW**Reset State:** 0x0017

When enabled, this register defines the value of the frequency shift applied to quarter and half-bandwidth packets. For instance, a 20MHz packet in an 80MHz channel is considered as a quarter-bandwidth packet and its frequency shift would be defined by the bw14 field. Similarly, the frequency shift for a 20MHz packet in a 40MHz channel or 40MHz packet in an 80MHz channel would be defined by the bw12 field

WCSS_A_TXCTL_FSHIFT

Bits	Name	Description
4:2	BW14	0x0: ZERO (No shift) 0x1: P10 (+10MHz shift) 0x2: P20 (+20MHz shift) 0x3: P30 (+30MHz shift) 0x4: RESERVED (No shift) 0x5: M30 (-30MHz shift) 0x6: M20 (-20MHz shift) 0x7: M10 (-10MHz shift)
1:0	BW12	0x0: ZERO (No shift) 0x1: P20 (+static_BW/4 shift) 0x2: RESERVED (No shift) 0x3: M20 (-static_BW/4 shift)

0x0A012028 WCSS_A_TXCTL_SOFTRESET**Type:** W**Reset State:** 0x0000

Resets the PHY transmitter

WCSS_A_TXCTL_SOFTRESET

Bits	Name	Description
0	RESET	When this register is written to with any value, a active high pulse is generated which will reset the whole of Phy Tx.

0x0A01202C WCSS_A_TXCTL_WATCHDOG_START**Type:** RW**Reset State:** 0x0000

Allowed time for state machine to remain in START state

WCSS_A_TXCTL_WATCHDOG_START

Bits	Name	Description
15:0	CYCLES	Allowed period of time in this state = (Register value * 128 * 12.5)ns. The state machine automatically adjusts for 20/40/80MHz bandwidth settings. Setting a value of 0 disables the watchdog.

0x0A012030 WCSS_A_TXCTL_WATCHDOG_RFWARM

Type: RW

Reset State: 0x0009

Allowed time for state machine to remain in RFWARM state

WCSS_A_TXCTL_WATCHDOG_RFWARM

Bits	Name	Description
15:0	CYCLES	Allowed period of time in this state = (Register value * 128 * 12.5)ns. The state machine automatically adjusts for 20/40/80MHz bandwidth settings. Setting a value of 0 disables the watchdog.

0x0A012034 WCSS_A_TXCTL_WATCHDOG_PAGUARD

Type: RW

Reset State: 0x0000

Allowed time for state machine to remain in PAGUARD state

WCSS_A_TXCTL_WATCHDOG_PAGUARD

Bits	Name	Description
15:0	CYCLES	Allowed period of time in this state = (Register value * 128 * 12.5)ns. The state machine automatically adjusts for 20/40/80MHz bandwidth settings. Setting a value of 0 disables the watchdog.

0x0A012038 WCSS_A_TXCTL_WATCHDOG_RAMPUP

Type: RW

Reset State: 0x0000

Allowed time for state machine to remain in RAMPUP state

WCSS_A_TXCTL_WATCHDOG_RAMPUP

Bits	Name	Description
15:0	CYCLES	Allowed period of time in this state = (Register value * 128 * 12.5)ns. The state machine automatically adjusts for 20/40/80MHz bandwidth settings. Setting a value of 0 disables the watchdog.

0x0A01203C WCSS_A_TXCTL_WATCHDOG_TX**Type:** RW**Reset State:** 0x0000

Allowed time for state machine to remain in TX state

WCSS_A_TXCTL_WATCHDOG_TX

Bits	Name	Description
15:0	CYCLES	Allowed period of time in this state = (Register value * 128 * 12.5)ns. The state machine automatically adjusts for 20/40/80MHz bandwidth settings. Setting a value of 0 disables the watchdog.

0x0A012040 WCSS_A_TXCTL_WATCHDOG_SHUTOFF**Type:** RW**Reset State:** 0x0000

Allowed time for state machine to remain in SHUTOFF state

WCSS_A_TXCTL_WATCHDOG_SHUTOFF

Bits	Name	Description
15:0	CYCLES	Allowed period of time in this state = (Register value * 128 * 12.5)ns. The state machine automatically adjusts for 20/40/80MHz bandwidth settings. Setting a value of 0 disables the watchdog.

0x0A012044 WCSS_A_TXCTL_WATCHDOG_TPC**Type:** RW**Reset State:** 0x0000

Allowed time for state machine to remain in TPC state

WCSS_A_TXCTL_WATCHDOG_TPC

Bits	Name	Description
15:0	CYCLES	Allowed period of time in this state = (Register value * 128 * 12.5)ns. The state machine automatically adjusts for 20/40/80MHz bandwidth settings. Setting a value of 0 disables the watchdog.

0x0A012048 WCSS_A_TXCTL_MAX_TXBUSY**Type:** RW**Reset State:** 0x0000

Holds the max active duration of the txctl_mpi_txbusy signal (measured in 160MHz clock cycles)
Default value of 0 disables the timeout event

WCSS_A_TXCTL_MAX_TXBUSY

Bits	Name	Description
15:0	VALUE	

0x0A01204C WCSS_A_TXCTL_MAX_TXENB**Type:** RW**Reset State:** 0x0000

Holds the max active duration of the RF TX enables (measured in 160MHz clock cycles) Default value of 0 disables the timeout event

WCSS_A_TXCTL_MAX_TXENB

Bits	Name	Description
15:0	VALUE	

0x0A012050 WCSS_A_TXCTL_TXBUSY_ALARM**Type:** W**Reset State:** 0x00000000

Counts number of TXBUSY timeout events

WCSS_A_TXCTL_TXBUSY_ALARM

Bits	Name	Description
15:0	COUNT	

0x0A012054 WCSS_A_TXCTL_TXEN_ALARM**Type:** W**Reset State:** 0x00000000

Counts number of TXEN timeout events

WCSS_A_TXCTL_TXEN_ALARM

Bits	Name	Description
15:0	COUNT	

0x0A012058 WCSS_A_TXCTL_TXCTL_STATE**Type:** R**Reset State:** 0x00000000

Stores last four states of the TXCTL state machine.

WCSS_A_TXCTL_TXCTL_STATE

Bits	Name	Description
15:0	VALUE	Bits[15:12] = Previous (N-3) state Bits[11:8] = Previous (N-2) state Bits[7:4] = Previous (N-1) state Bits[3:0] = Current state

0x0A01205C WCSS_A_TXCTL_LAST_PKT_PARAMS1**Type:** R**Reset State:** 0x00000000

Stores the packet type, RATE field, Nss, BW mode and subband params for the last command word received.

WCSS_A_TXCTL_LAST_PKT_PARAMS1

Bits	Name	Description
12:6	RATE	MCS Rate parameter

WCSS_A_TXCTL_LAST_PKT_PARAMS1 (cont.)

Bits	Name	Description
5:4	BANDWIDTH	Bandwidth mode parameter 00 = 20MHz packet 01 = 40MHz packet 10 = 80MHz packet 11 = Reserved
3	DUPLICATE	Subband duplicate parameter
2:0	PKT_TYPE	Packet type parameter

0x0A012060 WCSS_A_TXCTL_LAST_PKT_PACKET_LENGTH**Type:** R**Reset State:** 0x00000000

Stores the PACKET length parameter for the last command word received.

WCSS_A_TXCTL_LAST_PKT_PACKET_LENGTH

Bits	Name	Description
15:0	VALUE	PACKET length parameter

0x0A012064 WCSS_A_TXCTL_LAST_PKT_LSIG_LENGTH**Type:** R**Reset State:** 0x00000000

Stores the LSIG length parameter for the last command word received.

WCSS_A_TXCTL_LAST_PKT_LSIG_LENGTH

Bits	Name	Description
11:0	VALUE	LSIG length parameter

0x0A012068 WCSS_A_TXCTL_LAST_TX_POWER**Type:** R**Reset State:** 0x00000000

Stores the TX power parameter for the last command word received.

WCSS_A_TXCTL_LAST_TX_POWER

Bits	Name	Description
4:0	VALUE	Packet TX power parameter

0x0A01206C WCSS_A_TXCTL_WATCHDOG_ALARM**Type:** W**Reset State:** 0x00000000

Counts number of WATCHDOG_ALARM abort pulses generated by the TXCTL module.

WCSS_A_TXCTL_WATCHDOG_ALARM

Bits	Name	Description
15:0	COUNT	

0x0A012070 WCSS_A_TXCTL_PKT_INVALID_ALARM**Type:** W**Reset State:** 0x00000000

Counts number of PKT_INVALID_ALARM abort pulses generated by the TXCTL module.

WCSS_A_TXCTL_PKT_INVALID_ALARM

Bits	Name	Description
15:0	COUNT	

0x0A012074 WCSS_A_TXCTL_TXA_ABORT_ALARM**Type:** W**Reset State:** 0x00000000

Counts number of abort pulses generated by the TXA module.

WCSS_A_TXCTL_TXA_ABORT_ALARM

Bits	Name	Description
15:0	COUNT	

0x0A012078 WCSS_A_TXCTL_LEGACY_PKTS**Type:** W**Reset State:** 0x00000000

Counts number of IEEE 802.11a packets transmitted.

WCSS_A_TXCTL_LEGACY_PKTS

Bits	Name	Description
15:0	COUNT	Legacy packet counter.

0x0A01207C WCSS_A_TXCTL_GF_SIMO_20_PKTS**Type:** W**Reset State:** 0x00000000

Counts number of GF SIMO 20MHz packets transmitted.

WCSS_A_TXCTL_GF_SIMO_20_PKTS

Bits	Name	Description
15:0	COUNT	GF SIMO 20MHz packet counter.

0x0A012080 WCSS_A_TXCTL_GF_SIMO_40_PKTS**Type:** W**Reset State:** 0x00000000

Counts number of GF SIMO 40MHz packets transmitted.

WCSS_A_TXCTL_GF_SIMO_40_PKTS

Bits	Name	Description
15:0	COUNT	GF SIMO 40MHz packet counter.

0x0A012084 WCSS_A_TXCTL_MM_SIMO_20_PKTS**Type:** W**Reset State:** 0x00000000

Counts number of mixed-mode SIMO 20MHz packets transmitted.

WCSS_A_TXCTL_MM_SIMO_20_PKTS

Bits	Name	Description
15:0	COUNT	mixed-mode SIMO 20MHz packet counter.

0x0A012088 WCSS_A_TXCTL_MM_SIMO_40_PKTS**Type:** W**Reset State:** 0x00000000

Counts number of mixed-mode SIMO 40MHz packets transmitted.

WCSS_A_TXCTL_MM_SIMO_40_PKTS

Bits	Name	Description
15:0	COUNT	mixed-mode SIMO 40MHz packet counter.

0x0A01208C WCSS_A_TXCTL_AC_SIMO_20_PKTS**Type:** W**Reset State:** 0x00000000

Counts number of 11ac SIMO 20MHz packets transmitted.

WCSS_A_TXCTL_AC_SIMO_20_PKTS

Bits	Name	Description
15:0	COUNT	11ac SIMO 20MHz packet counter.

0x0A012090 WCSS_A_TXCTL_AC_SIMO_40_PKTS**Type:** W**Reset State:** 0x00000000

Counts number of 11ac SIMO 40MHz packets transmitted.

WCSS_A_TXCTL_AC_SIMO_40_PKTS

Bits	Name	Description
15:0	COUNT	11ac SIMO 40MHz packet counter.

0x0A012094 WCSS_A_TXCTL_AC_SIMO_80_PKTS**Type:** W**Reset State:** 0x00000000

Counts number of 11ac SIMO 80MHz packets transmitted.

WCSS_A_TXCTL_AC_SIMO_80_PKTS

Bits	Name	Description
15:0	COUNT	11ac SIMO 80MHz packet counter.

0x0A012098 WCSS_A_TXCTL_TXB_SHORT_PKTS**Type:** W**Reset State:** 0x00000000

Counts number of TXB packets with short preamble transmitted.

WCSS_A_TXCTL_TXB_SHORT_PKTS

Bits	Name	Description
15:0	COUNT	TXB short packet counter.

0x0A01209C WCSS_A_TXCTL_TXB_LONG_PKTS**Type:** W**Reset State:** 0x00000000

Counts number of TXB packets with long preamble transmitted.

WCSS_A_TXCTL_TXB_LONG_PKTS

Bits	Name	Description
15:0	COUNT	TXB long packet counter.

0x0A0120A0 WCSS_A_TXCTL_TXFIR_DONE**Type:** W**Reset State:** 0x00000000

Counts the number of packets completed at output of TXFIR.

WCSS_A_TXCTL_TXFIR_DONE

Bits	Name	Description
15:0	COUNT	TXFIR done event counter

0x0A0120A4 WCSS_A_TXCTL_MISSING_WARMUP**Type:** W**Reset State:** 0x00000000

Counts the number of packets that are initiated by TXP start pulse rather than the assertion of the TXP RF warmup flag.

WCSS_A_TXCTL_MISSING_WARMUP

Bits	Name	Description
15:0	COUNT	Missing RF warmup flag event counter

0x0A0120A8 WCSS_A_TXCTL_ALARMS**Type:** R**Reset State:** 0x0000

Monitor of all contributors to TXA abort

WCSS_A_TXCTL_ALARMS

Bits	Name	Description
6:0	VALUE	Bit[0] = TXCTL watchdog timed out Bit[1] = Incorrect bandwidth field in command word Bit[2] = Invalid 11b packet parameters error Bit[3] = TX enable not asserted during active transmission Bit[4] = RF_WARMUP signal not asserted for sufficient length of time Bit[5] = Abort received from TACTL Bit[6] = Abort received from MPI

0x0A0120AC WCSS_A_TXCTL_RESET_ALARMS**Type:** W**Reset State:** 0x0000

Resets the alarm flags

WCSS_A_TXCTL_RESET_ALARMS

Bits	Name	Description
0	RESET_ALARMS	When this register is written to with any value, a active high pulse is generated which will reset all alarm flags in the TACTL alarms register.

10.62 WCSS_A_TXFIR_TXFIR

0x0A018000 WCSS_A_TXFIR_CFG

Type: RW

Reset State: 0x003E

Sets the various modes for the Tx-FIR

WCSS_A_TXFIR_CFG

Bits	Name	Description
6	TVWS_EN	When this field is set, new filtering logic to support TV white space channels is enabled
5	PEAK_BYPASS_OVERRIDE_EN	When this field is set, dynamic bypass of the smart clipping logic will be disabled (low power) and the peak_bypass field will be used to either statically bypass or not bypass this logic
4	PEAK_BYPASS	When this field is set, the smart clipping logic is bypassed (as long as the peak_bypass_override_en field is also set)
3	DPD_BYPASS	When this field is set, the digital pre-distorter logic is bypassed
2	LOLEAKAGE_BYPASS	When this field is set, the LO leakage correction logic is bypassed
1	IQIMB_BYPASS	When this field is set, the IQ imbalance correction logic is bypassed
0	CHAIN0FIRBYPASS	When set bypasses the upsampling and filtering for chain 0 The output of the actual filter is 12-bits and hence drops the LSB of the inputs to present 9-bits at the output

0x0A018004 WCSS_A_TXFIR_LOLEAKAGE_SHIFT

Type: RW

Reset State: 0x0000

This register defines the number of left shifts applied to the LO leakage value derived from the gain-based look-up table in the TXFIR

WCSS_A_TXFIR_LOLEAKAGE_SHIFT

Bits	Name	Description
1:0	CH0	

0x0A01800C WCSS_A_TXFIR_COEFF_QUARTER_0**Type:** RW**Reset State:** 0x0099

2's complement 10-bit OFDM quarter-band FIR coefficient 0

WCSS_A_TXFIR_COEFF_QUARTER_0

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A018010 WCSS_A_TXFIR_COEFF_QUARTER_1**Type:** RW**Reset State:** 0x006D

2's complement 10-bit OFDM quarter-band FIR coefficient 1

WCSS_A_TXFIR_COEFF_QUARTER_1

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A018014 WCSS_A_TXFIR_COEFF_QUARTER_2**Type:** RW**Reset State:** 0x0099

2's complement 10-bit OFDM quarter-band FIR coefficient 2

WCSS_A_TXFIR_COEFF_QUARTER_2

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A018018 WCSS_A_TXFIR_COEFF_QUARTER_3**Type:** RW**Reset State:** 0x03A9

2's complement 10-bit OFDM quarter-band FIR coefficient 3

WCSS_A_TXFIR_COEFF_QUARTER_3

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A01801C WCSS_A_TXFIR_COEFF_QUARTER_4**Type:** RW**Reset State:** 0x0035

2's complement 10-bit OFDM quarter-band FIR coefficient 4

WCSS_A_TXFIR_COEFF_QUARTER_4

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A018020 WCSS_A_TXFIR_COEFF_QUARTER_5**Type:** RW**Reset State:** 0x002C

2's complement 10-bit OFDM quarter-band FIR coefficient 5

WCSS_A_TXFIR_COEFF_QUARTER_5

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A018024 WCSS_A_TXFIR_COEFF_QUARTER_6**Type:** RW**Reset State:** 0x007D

2's complement 10-bit OFDM quarter-band FIR coefficient 6

WCSS_A_TXFIR_COEFF_QUARTER_6

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A018028 WCSS_A_TXFIR_COEFF_QUARTER_7**Type:** RW**Reset State:** 0x0050

2's complement 10-bit OFDM quarter-band FIR coefficient 7

WCSS_A_TXFIR_COEFF_QUARTER_7

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A01802C WCSS_A_TXFIR_COEFF_QUARTER_8**Type:** RW**Reset State:** 0x0047

2's complement 10-bit OFDM quarter-band FIR coefficient 8

WCSS_A_TXFIR_COEFF_QUARTER_8

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A018030 WCSS_A_TXFIR_COEFF_QUARTER_9**Type:** RW**Reset State:** 0x006F

2's complement 10-bit OFDM quarter-band FIR coefficient 9

WCSS_A_TXFIR_COEFF_QUARTER_9

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A018034 WCSS_A_TXFIR_COEFF_QUARTER_10**Type:** RW**Reset State:** 0x0051

2's complement 10-bit OFDM quarter-band FIR coefficient 10

WCSS_A_TXFIR_COEFF_QUARTER_10

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A018038 WCSS_A_TXFIR_COEFF_QUARTER_11**Type:** RW**Reset State:** 0x004C

2's complement 10-bit OFDM quarter-band FIR coefficient 11

WCSS_A_TXFIR_COEFF_QUARTER_11

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A01803C WCSS_A_TXFIR_COEFF_QUARTER_12**Type:** RW**Reset State:** 0x0057

2's complement 10-bit OFDM quarter-band FIR coefficient 12

WCSS_A_TXFIR_COEFF_QUARTER_12

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A018040 WCSS_A_TXFIR_COEFF_QUARTER_13**Type:** RW**Reset State:** 0x004A

2's complement 10-bit OFDM quarter-band FIR coefficient 13

WCSS_A_TXFIR_COEFF_QUARTER_13

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A018044 WCSS_A_TXFIR_COEFF_QUARTER_14**Type:** RW**Reset State:** 0x0047

2's complement 10-bit OFDM quarter-band FIR coefficient 14

WCSS_A_TXFIR_COEFF_QUARTER_14

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A018048 WCSS_A_TXFIR_COEFF_QUARTER_15**Type:** RW**Reset State:** 0x0041

2's complement 10-bit OFDM quarter-band FIR coefficient 15

WCSS_A_TXFIR_COEFF_QUARTER_15

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A01804C WCSS_A_TXFIR_COEFF_QUARTER_16**Type:** RW**Reset State:** 0x003E

2's complement 10-bit OFDM quarter-band FIR coefficient 16

WCSS_A_TXFIR_COEFF_QUARTER_16

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A018050 WCSS_A_TXFIR_COEFF_QUARTER_17**Type:** RW**Reset State:** 0x003C

2's complement 10-bit OFDM quarter-band FIR coefficient 17

WCSS_A_TXFIR_COEFF_QUARTER_17

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A018054 WCSS_A_TXFIR_COEFF_QUARTER_18**Type:** RW**Reset State:** 0x002D

2's complement 10-bit OFDM quarter-band FIR coefficient 18

WCSS_A_TXFIR_COEFF_QUARTER_18

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A018058 WCSS_A_TXFIR_COEFF_QUARTER_19**Type:** RW**Reset State:** 0x002D

2's complement 10-bit OFDM quarter-band FIR coefficient 19

WCSS_A_TXFIR_COEFF_QUARTER_19

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A01805C WCSS_A_TXFIR_COEFF_QUARTER_20**Type:** RW**Reset State:** 0x0027

2's complement 10-bit OFDM quarter-band FIR coefficient 20

WCSS_A_TXFIR_COEFF_QUARTER_20

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A018060 WCSS_A_TXFIR_COEFF_QUARTER_21**Type:** RW**Reset State:** 0x0017

2's complement 10-bit OFDM quarter-band FIR coefficient 21

WCSS_A_TXFIR_COEFF_QUARTER_21

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A018064 WCSS_A_TXFIR_COEFF_QUARTER_22**Type:** RW**Reset State:** 0x0014

2's complement 10-bit OFDM quarter-band FIR coefficient 22

WCSS_A_TXFIR_COEFF_QUARTER_22

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A018068 WCSS_A_TXFIR_COEFF_QUARTER_23**Type:** RW**Reset State:** 0x000C

2's complement 10-bit OFDM quarter-band FIR coefficient 23

WCSS_A_TXFIR_COEFF_QUARTER_23

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A01806C WCSS_A_TXFIR_COEFF_QUARTER_24**Type:** RW**Reset State:** 0x0000

2's complement 10-bit OFDM quarter-band FIR coefficient 24

WCSS_A_TXFIR_COEFF_QUARTER_24

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A018070 WCSS_A_TXFIR_COEFF_QUARTER_25**Type:** RW**Reset State:** 0x03FC

2's complement 10-bit OFDM quarter-band FIR coefficient 25

WCSS_A_TXFIR_COEFF_QUARTER_25

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A018074 WCSS_A_TXFIR_COEFF_QUARTER_26**Type:** RW**Reset State:** 0x03F9

2's complement 10-bit OFDM quarter-band FIR coefficient 26

WCSS_A_TXFIR_COEFF_QUARTER_26

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A018078 WCSS_A_TXFIR_COEFF_QUARTER_27**Type:** RW**Reset State:** 0x03F8

2's complement 10-bit OFDM quarter-band FIR coefficient 27

WCSS_A_TXFIR_COEFF_QUARTER_27

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A01807C WCSS_A_TXFIR_COEFF_QUARTER_28**Type:** RW**Reset State:** 0x03F7

2's complement 10-bit OFDM quarter-band FIR coefficient 28

WCSS_A_TXFIR_COEFF_QUARTER_28

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A018080 WCSS_A_TXFIR_COEFF_QUARTER_29**Type:** RW**Reset State:** 0x03F8

2's complement 10-bit OFDM quarter-band FIR coefficient 29

WCSS_A_TXFIR_COEFF_QUARTER_29

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A018084 WCSS_A_TXFIR_COEFF_QUARTER_30**Type:** RW**Reset State:** 0x0000

2's complement 10-bit OFDM quarter-band FIR coefficient 30

WCSS_A_TXFIR_COEFF_QUARTER_30

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A018088 WCSS_A_TXFIR_COEFF_QUARTER_31**Type:** RW**Reset State:** 0x0000

2's complement 10-bit OFDM quarter-band FIR coefficient 31

WCSS_A_TXFIR_COEFF_QUARTER_31

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A01808C WCSS_A_TXFIR_COEFF_QUARTER_32**Type:** RW**Reset State:** 0x0000

2's complement 10-bit OFDM quarter-band FIR coefficient 32

WCSS_A_TXFIR_COEFF_QUARTER_32

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A018090 WCSS_A_TXFIR_COEFF_QUARTER_33**Type:** RW**Reset State:** 0x0000

2's complement 10-bit OFDM quarter-band FIR coefficient 33

WCSS_A_TXFIR_COEFF_QUARTER_33

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A018094 WCSS_A_TXFIR_COEFF_QUARTER_34**Type:** RW**Reset State:** 0x0000

2's complement 10-bit OFDM quarter-band FIR coefficient 34

WCSS_A_TXFIR_COEFF_QUARTER_34

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A018098 WCSS_A_TXFIR_COEFF_QUARTER_35**Type:** RW**Reset State:** 0x0000

2's complement 10-bit OFDM quarter-band FIR coefficient 35

WCSS_A_TXFIR_COEFF_QUARTER_35

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A01809C WCSS_A_TXFIR_COEFF_QUARTER_36**Type:** RW**Reset State:** 0x0000

2's complement 10-bit OFDM quarter-band FIR coefficient 36

WCSS_A_TXFIR_COEFF_QUARTER_36

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A0180A0 WCSS_A_TXFIR_COEFF_QUARTER_37**Type:** RW**Reset State:** 0x0000

2's complement 10-bit OFDM quarter-band FIR coefficient 37

WCSS_A_TXFIR_COEFF_QUARTER_37

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A0180A4 WCSS_A_TXFIR_COEFF_QUARTER_38**Type:** RW**Reset State:** 0x0000

2's complement 10-bit OFDM quarter-band FIR coefficient 38

WCSS_A_TXFIR_COEFF_QUARTER_38

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A0180A8 WCSS_A_TXFIR_COEFF_QUARTER_39**Type:** RW**Reset State:** 0x0000

2's complement 10-bit OFDM quarter-band FIR coefficient 39

WCSS_A_TXFIR_COEFF_QUARTER_39

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A0180AC WCSS_A_TXFIR_COEFF_QUARTER_40**Type:** RW**Reset State:** 0x0000

2's complement 10-bit OFDM quarter-band FIR coefficient 40

WCSS_A_TXFIR_COEFF_QUARTER_40

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A0180B0 WCSS_A_TXFIR_COEFF_QUARTER_41**Type:** RW**Reset State:** 0x0000

2's complement 10-bit OFDM quarter-band FIR coefficient 41

WCSS_A_TXFIR_COEFF_QUARTER_41

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A0180B4 WCSS_A_TXFIR_COEFF_HALF_0**Type:** RW**Reset State:** 0x003D

2's complement 10-bit OFDM half-band FIR coefficient 0

WCSS_A_TXFIR_COEFF_HALF_0

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A0180B8 WCSS_A_TXFIR_COEFF_HALF_1**Type:** RW**Reset State:** 0x0052

2's complement 10-bit OFDM half-band FIR coefficient 1

WCSS_A_TXFIR_COEFF_HALF_1

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A0180BC WCSS_A_TXFIR_COEFF_HALF_2**Type:** RW**Reset State:** 0x009B

2's complement 10-bit OFDM half-band FIR coefficient 2

WCSS_A_TXFIR_COEFF_HALF_2

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A0180C0 WCSS_A_TXFIR_COEFF_HALF_3**Type:** RW**Reset State:** 0x005A

2's complement 10-bit OFDM half-band FIR coefficient 3

WCSS_A_TXFIR_COEFF_HALF_3

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A0180C4 WCSS_A_TXFIR_COEFF_HALF_4**Type:** RW**Reset State:** 0x0095

2's complement 10-bit OFDM half-band FIR coefficient 4

WCSS_A_TXFIR_COEFF_HALF_4

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A0180C8 WCSS_A_TXFIR_COEFF_HALF_5**Type:** RW**Reset State:** 0x006B

2's complement 10-bit OFDM half-band FIR coefficient 5

WCSS_A_TXFIR_COEFF_HALF_5

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A0180CC WCSS_A_TXFIR_COEFF_HALF_6**Type:** RW**Reset State:** 0x00BF

2's complement 10-bit OFDM half-band FIR coefficient 6

WCSS_A_TXFIR_COEFF_HALF_6

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A0180D0 WCSS_A_TXFIR_COEFF_HALF_7**Type:** RW**Reset State:** 0x0094

2's complement 10-bit OFDM half-band FIR coefficient 7

WCSS_A_TXFIR_COEFF_HALF_7

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A0180D4 WCSS_A_TXFIR_COEFF_HALF_8**Type:** RW**Reset State:** 0x009B

2's complement 10-bit OFDM half-band FIR coefficient 8

WCSS_A_TXFIR_COEFF_HALF_8

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A0180D8 WCSS_A_TXFIR_COEFF_HALF_9**Type:** RW**Reset State:** 0x0085

2's complement 10-bit OFDM half-band FIR coefficient 9

WCSS_A_TXFIR_COEFF_HALF_9

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A0180DC WCSS_A_TXFIR_COEFF_HALF_10**Type:** RW**Reset State:** 0x007A

2's complement 10-bit OFDM half-band FIR coefficient 10

WCSS_A_TXFIR_COEFF_HALF_10

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A0180E0 WCSS_A_TXFIR_COEFF_HALF_11**Type:** RW**Reset State:** 0x0071

2's complement 10-bit OFDM half-band FIR coefficient 11

WCSS_A_TXFIR_COEFF_HALF_11

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A0180E4 WCSS_A_TXFIR_COEFF_HALF_12**Type:** RW**Reset State:** 0x0038

2's complement 10-bit OFDM half-band FIR coefficient 12

WCSS_A_TXFIR_COEFF_HALF_12

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A0180E8 WCSS_A_TXFIR_COEFF_HALF_13**Type:** RW**Reset State:** 0x0027

2's complement 10-bit OFDM half-band FIR coefficient 13

WCSS_A_TXFIR_COEFF_HALF_13

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A0180EC WCSS_A_TXFIR_COEFF_HALF_14**Type:** RW**Reset State:** 0x0025

2's complement 10-bit OFDM half-band FIR coefficient 14

WCSS_A_TXFIR_COEFF_HALF_14

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A0180F0 WCSS_A_TXFIR_COEFF_HALF_15**Type:** RW**Reset State:** 0x001F

2's complement 10-bit OFDM half-band FIR coefficient 15

WCSS_A_TXFIR_COEFF_HALF_15

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A0180F4 WCSS_A_TXFIR_COEFF_HALF_16**Type:** RW**Reset State:** 0x000B

2's complement 10-bit OFDM half-band FIR coefficient 16

WCSS_A_TXFIR_COEFF_HALF_16

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A0180F8 WCSS_A_TXFIR_COEFF_HALF_17**Type:** RW**Reset State:** 0x03C9

2's complement 10-bit OFDM half-band FIR coefficient 17

WCSS_A_TXFIR_COEFF_HALF_17

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A0180FC WCSS_A_TXFIR_COEFF_HALF_18**Type:** RW**Reset State:** 0x03D0

2's complement 10-bit OFDM half-band FIR coefficient 18

WCSS_A_TXFIR_COEFF_HALF_18

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A018100 WCSS_A_TXFIR_COEFF_HALF_19**Type:** RW**Reset State:** 0x03D8

2's complement 10-bit OFDM half-band FIR coefficient 19

WCSS_A_TXFIR_COEFF_HALF_19

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A018104 WCSS_A_TXFIR_COEFF_HALF_20**Type:** RW**Reset State:** 0x0000

2's complement 10-bit OFDM half-band FIR coefficient 20

WCSS_A_TXFIR_COEFF_HALF_20

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A018108 WCSS_A_TXFIR_COEFF_HALF_21**Type:** RW**Reset State:** 0x0000

2's complement 10-bit OFDM half-band FIR coefficient 21

WCSS_A_TXFIR_COEFF_HALF_21

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A01810C WCSS_A_TXFIR_COEFF_HALF_22**Type:** RW**Reset State:** 0x0000

2's complement 10-bit OFDM half-band FIR coefficient 22

WCSS_A_TXFIR_COEFF_HALF_22

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A018110 WCSS_A_TXFIR_COEFF_HALF_23**Type:** RW**Reset State:** 0x0000

2's complement 10-bit OFDM half-band FIR coefficient 23

WCSS_A_TXFIR_COEFF_HALF_23

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A018114 WCSS_A_TXFIR_COEFF_HALF_24**Type:** RW**Reset State:** 0x0000

2's complement 10-bit OFDM half-band FIR coefficient 24

WCSS_A_TXFIR_COEFF_HALF_24

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A018118 WCSS_A_TXFIR_COEFF_HALF_25**Type:** RW**Reset State:** 0x0000

2's complement 10-bit OFDM half-band FIR coefficient 25

WCSS_A_TXFIR_COEFF_HALF_25

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A01811C WCSS_A_TXFIR_COEFF_HALF_26**Type:** RW**Reset State:** 0x0000

2's complement 10-bit OFDM half-band FIR coefficient 26

WCSS_A_TXFIR_COEFF_HALF_26

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A018120 WCSS_A_TXFIR_COEFF_HALF_27**Type:** RW**Reset State:** 0x0000

2's complement 10-bit OFDM half-band FIR coefficient 27

WCSS_A_TXFIR_COEFF_HALF_27

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A018124 WCSS_A_TXFIR_COEFF_HALF_28**Type:** RW**Reset State:** 0x0000

2's complement 10-bit OFDM half-band FIR coefficient 28

WCSS_A_TXFIR_COEFF_HALF_28

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A018128 WCSS_A_TXFIR_COEFF_HALF_29**Type:** RW**Reset State:** 0x0000

2's complement 10-bit OFDM half-band FIR coefficient 29

WCSS_A_TXFIR_COEFF_HALF_29

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A01812C WCSS_A_TXFIR_COEFF_HALF_30**Type:** RW**Reset State:** 0x0000

2's complement 10-bit OFDM half-band FIR coefficient 30

WCSS_A_TXFIR_COEFF_HALF_30

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A018130 WCSS_A_TXFIR_COEFF_HALF_31**Type:** RW**Reset State:** 0x0000

2's complement 10-bit OFDM half-band FIR coefficient 31

WCSS_A_TXFIR_COEFF_HALF_31

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A018134 WCSS_A_TXFIR_COEFF_HALF_32**Type:** RW**Reset State:** 0x0000

2's complement 10-bit OFDM half-band FIR coefficient 32

WCSS_A_TXFIR_COEFF_HALF_32

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A018138 WCSS_A_TXFIR_COEFF_HALF_33**Type:** RW**Reset State:** 0x0000

2's complement 10-bit OFDM half-band FIR coefficient 33

WCSS_A_TXFIR_COEFF_HALF_33

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A01813C WCSS_A_TXFIR_COEFF_HALF_34**Type:** RW**Reset State:** 0x0000

2's complement 10-bit OFDM half-band FIR coefficient 34

WCSS_A_TXFIR_COEFF_HALF_34

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A018140 WCSS_A_TXFIR_COEFF_HALF_35**Type:** RW**Reset State:** 0x0000

2's complement 10-bit OFDM half-band FIR coefficient 35

WCSS_A_TXFIR_COEFF_HALF_35

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A018144 WCSS_A_TXFIR_COEFF_HALF_36**Type:** RW**Reset State:** 0x0000

2's complement 10-bit OFDM half-band FIR coefficient 36

WCSS_A_TXFIR_COEFF_HALF_36

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A018148 WCSS_A_TXFIR_COEFF_HALF_37**Type:** RW**Reset State:** 0x0000

2's complement 10-bit OFDM half-band FIR coefficient 37

WCSS_A_TXFIR_COEFF_HALF_37

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A01814C WCSS_A_TXFIR_COEFF_HALF_38**Type:** RW**Reset State:** 0x0000

2's complement 10-bit OFDM half-band FIR coefficient 38

WCSS_A_TXFIR_COEFF_HALF_38

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A018150 WCSS_A_TXFIR_COEFF_HALF_39**Type:** RW**Reset State:** 0x0000

2's complement 10-bit OFDM half-band FIR coefficient 39

WCSS_A_TXFIR_COEFF_HALF_39

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A018154 WCSS_A_TXFIR_COEFF_HALF_40**Type:** RW**Reset State:** 0x0000

2's complement 10-bit OFDM half-band FIR coefficient 40

WCSS_A_TXFIR_COEFF_HALF_40

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A018158 WCSS_A_TXFIR_COEFF_HALF_41**Type:** RW**Reset State:** 0x0000

2's complement 10-bit OFDM half-band FIR coefficient 41

WCSS_A_TXFIR_COEFF_HALF_41

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A01815C WCSS_A_TXFIR_COEFF_FULL_0**Type:** RW**Reset State:** 0x0030

2's complement 10-bit OFDM full-band FIR coefficient 0

WCSS_A_TXFIR_COEFF_FULL_0

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A018160 WCSS_A_TXFIR_COEFF_FULL_1**Type:** RW**Reset State:** 0x0072

2's complement 10-bit OFDM full-band FIR coefficient 1

WCSS_A_TXFIR_COEFF_FULL_1

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A018164 WCSS_A_TXFIR_COEFF_FULL_2**Type:** RW**Reset State:** 0x00E2

2's complement 10-bit OFDM full-band FIR coefficient 2

WCSS_A_TXFIR_COEFF_FULL_2

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A018168 WCSS_A_TXFIR_COEFF_FULL_3**Type:** RW**Reset State:** 0x0104

2's complement 10-bit OFDM full-band FIR coefficient 3

WCSS_A_TXFIR_COEFF_FULL_3

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A01816C WCSS_A_TXFIR_COEFF_FULL_4**Type:** RW**Reset State:** 0x0126

2's complement 10-bit OFDM full-band FIR coefficient 4

WCSS_A_TXFIR_COEFF_FULL_4

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A018170 WCSS_A_TXFIR_COEFF_FULL_5**Type:** RW**Reset State:** 0x00EB

2's complement 10-bit OFDM full-band FIR coefficient 5

WCSS_A_TXFIR_COEFF_FULL_5

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A018174 WCSS_A_TXFIR_COEFF_FULL_6**Type:** RW**Reset State:** 0x00F4

2's complement 10-bit OFDM full-band FIR coefficient 6

WCSS_A_TXFIR_COEFF_FULL_6

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A018178 WCSS_A_TXFIR_COEFF_FULL_7**Type:** RW**Reset State:** 0x00A8

2's complement 10-bit OFDM full-band FIR coefficient 7

WCSS_A_TXFIR_COEFF_FULL_7

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A01817C WCSS_A_TXFIR_COEFF_FULL_8**Type:** RW**Reset State:** 0x0086

2's complement 10-bit OFDM full-band FIR coefficient 8

WCSS_A_TXFIR_COEFF_FULL_8

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A018180 WCSS_A_TXFIR_COEFF_FULL_9**Type:** RW**Reset State:** 0x0004

2's complement 10-bit OFDM full-band FIR coefficient 9

WCSS_A_TXFIR_COEFF_FULL_9

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A018184 WCSS_A_TXFIR_COEFF_FULL_10**Type:** RW**Reset State:** 0x03D3

2's complement 10-bit OFDM full-band FIR coefficient 10

WCSS_A_TXFIR_COEFF_FULL_10

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A018188 WCSS_A_TXFIR_COEFF_FULL_11**Type:** RW**Reset State:** 0x039A

2's complement 10-bit OFDM full-band FIR coefficient 11

WCSS_A_TXFIR_COEFF_FULL_11

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A01818C WCSS_A_TXFIR_COEFF_FULL_12**Type:** RW**Reset State:** 0x03B8

2's complement 10-bit OFDM full-band FIR coefficient 12

WCSS_A_TXFIR_COEFF_FULL_12

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A018190 WCSS_A_TXFIR_COEFF_FULL_13**Type:** RW**Reset State:** 0x03C0

2's complement 10-bit OFDM full-band FIR coefficient 13

WCSS_A_TXFIR_COEFF_FULL_13

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A018194 WCSS_A_TXFIR_COEFF_FULL_14**Type:** RW**Reset State:** 0x03E6

2's complement 10-bit OFDM full-band FIR coefficient 14

WCSS_A_TXFIR_COEFF_FULL_14

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A018198 WCSS_A_TXFIR_COEFF_FULL_15**Type:** RW**Reset State:** 0x0002

2's complement 10-bit OFDM full-band FIR coefficient 15

WCSS_A_TXFIR_COEFF_FULL_15

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A01819C WCSS_A_TXFIR_COEFF_FULL_16**Type:** RW**Reset State:** 0x0020

2's complement 10-bit OFDM full-band FIR coefficient 16

WCSS_A_TXFIR_COEFF_FULL_16

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A0181A0 WCSS_A_TXFIR_COEFF_FULL_17**Type:** RW**Reset State:** 0x002B

2's complement 10-bit OFDM full-band FIR coefficient 17

WCSS_A_TXFIR_COEFF_FULL_17

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A0181A4 WCSS_A_TXFIR_COEFF_FULL_18**Type:** RW**Reset State:** 0x001C

2's complement 10-bit OFDM full-band FIR coefficient 18

WCSS_A_TXFIR_COEFF_FULL_18

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A0181A8 WCSS_A_TXFIR_COEFF_FULL_19**Type:** RW**Reset State:** 0x000B

2's complement 10-bit OFDM full-band FIR coefficient 19

WCSS_A_TXFIR_COEFF_FULL_19

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A0181AC WCSS_A_TXFIR_COEFF_FULL_20**Type:** RW**Reset State:** 0x0000

2's complement 10-bit OFDM full-band FIR coefficient 20

WCSS_A_TXFIR_COEFF_FULL_20

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A0181B0 WCSS_A_TXFIR_COEFF_FULL_21**Type:** RW**Reset State:** 0x0000

2's complement 10-bit OFDM full-band FIR coefficient 21

WCSS_A_TXFIR_COEFF_FULL_21

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A0181B4 WCSS_A_TXFIR_COEFF_FULL_22**Type:** RW**Reset State:** 0x0000

2's complement 10-bit OFDM full-band FIR coefficient 22

WCSS_A_TXFIR_COEFF_FULL_22

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A0181B8 WCSS_A_TXFIR_COEFF_FULL_23**Type:** RW**Reset State:** 0x0000

2's complement 10-bit OFDM full-band FIR coefficient 23

WCSS_A_TXFIR_COEFF_FULL_23

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A0181BC WCSS_A_TXFIR_COEFF_FULL_24**Type:** RW**Reset State:** 0x0000

2's complement 10-bit OFDM full-band FIR coefficient 24

WCSS_A_TXFIR_COEFF_FULL_24

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A0181C0 WCSS_A_TXFIR_COEFF_FULL_25**Type:** RW**Reset State:** 0x0000

2's complement 10-bit OFDM full-band FIR coefficient 25

WCSS_A_TXFIR_COEFF_FULL_25

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A0181C4 WCSS_A_TXFIR_COEFF_FULL_26**Type:** RW**Reset State:** 0x0000

2's complement 10-bit OFDM full-band FIR coefficient 26

WCSS_A_TXFIR_COEFF_FULL_26

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A0181C8 WCSS_A_TXFIR_COEFF_FULL_27**Type:** RW**Reset State:** 0x0000

2's complement 10-bit OFDM full-band FIR coefficient 27

WCSS_A_TXFIR_COEFF_FULL_27

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A0181CC WCSS_A_TXFIR_COEFF_FULL_28**Type:** RW**Reset State:** 0x0000

2's complement 10-bit OFDM full-band FIR coefficient 28

WCSS_A_TXFIR_COEFF_FULL_28

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A0181D0 WCSS_A_TXFIR_COEFF_FULL_29**Type:** RW**Reset State:** 0x0000

2's complement 10-bit OFDM full-band FIR coefficient 29

WCSS_A_TXFIR_COEFF_FULL_29

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A0181D4 WCSS_A_TXFIR_COEFF_FULL_30**Type:** RW**Reset State:** 0x0000

2's complement 10-bit OFDM full-band FIR coefficient 30

WCSS_A_TXFIR_COEFF_FULL_30

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A0181D8 WCSS_A_TXFIR_COEFF_FULL_31**Type:** RW**Reset State:** 0x0000

2's complement 10-bit OFDM full-band FIR coefficient 31

WCSS_A_TXFIR_COEFF_FULL_31

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A0181DC WCSS_A_TXFIR_COEFF_FULL_32**Type:** RW**Reset State:** 0x0000

2's complement 10-bit OFDM full-band FIR coefficient 32

WCSS_A_TXFIR_COEFF_FULL_32

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A0181E0 WCSS_A_TXFIR_COEFF_FULL_33**Type:** RW**Reset State:** 0x0000

2's complement 10-bit OFDM full-band FIR coefficient 33

WCSS_A_TXFIR_COEFF_FULL_33

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A0181E4 WCSS_A_TXFIR_COEFF_FULL_34**Type:** RW**Reset State:** 0x0000

2's complement 10-bit OFDM full-band FIR coefficient 34

WCSS_A_TXFIR_COEFF_FULL_34

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A0181E8 WCSS_A_TXFIR_COEFF_FULL_35**Type:** RW**Reset State:** 0x0000

2's complement 10-bit OFDM full-band FIR coefficient 35

WCSS_A_TXFIR_COEFF_FULL_35

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A0181EC WCSS_A_TXFIR_COEFF_FULL_36**Type:** RW**Reset State:** 0x0000

2's complement 10-bit OFDM full-band FIR coefficient 36

WCSS_A_TXFIR_COEFF_FULL_36

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A0181F0 WCSS_A_TXFIR_COEFF_FULL_37**Type:** RW**Reset State:** 0x0000

2's complement 10-bit OFDM full-band FIR coefficient 37

WCSS_A_TXFIR_COEFF_FULL_37

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A0181F4 WCSS_A_TXFIR_COEFF_FULL_38**Type:** RW**Reset State:** 0x0000

2's complement 10-bit OFDM full-band FIR coefficient 38

WCSS_A_TXFIR_COEFF_FULL_38

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A0181F8 WCSS_A_TXFIR_COEFF_FULL_39**Type:** RW**Reset State:** 0x0000

2's complement 10-bit OFDM full-band FIR coefficient 39

WCSS_A_TXFIR_COEFF_FULL_39

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A0181FC WCSS_A_TXFIR_COEFF_FULL_40**Type:** RW**Reset State:** 0x0000

2's complement 10-bit OFDM full-band FIR coefficient 40

WCSS_A_TXFIR_COEFF_FULL_40

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A018200 WCSS_A_TXFIR_COEFF_FULL_41**Type:** RW**Reset State:** 0x0000

2's complement 10-bit OFDM full-band FIR coefficient 41

WCSS_A_TXFIR_COEFF_FULL_41

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A018204 WCSS_A_TXFIR_COEFF_TXB_20_0**Type:** RW**Reset State:** 0x0014

2's complement 10-bit Tx-B FIR20 coefficient 0

WCSS_A_TXFIR_COEFF_TXB_20_0

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A018208 WCSS_A_TXFIR_COEFF_TXB_20_1**Type:** RW**Reset State:** 0x003D

2's complement 10-bit Tx-B FIR20 coefficient 1

WCSS_A_TXFIR_COEFF_TXB_20_1

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A01820C WCSS_A_TXFIR_COEFF_TXB_20_2**Type:** RW**Reset State:** 0x007A

2's complement 10-bit Tx-B FIR20 coefficient 2

WCSS_A_TXFIR_COEFF_TXB_20_2

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A018210 WCSS_A_TXFIR_COEFF_TXB_20_3**Type:** RW**Reset State:** 0x00C2

2's complement 10-bit Tx-B FIR20 coefficient 3

WCSS_A_TXFIR_COEFF_TXB_20_3

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A018214 WCSS_A_TXFIR_COEFF_TXB_20_4**Type:** RW**Reset State:** 0x00FD

2's complement 10-bit Tx-B FIR20 coefficient 4

WCSS_A_TXFIR_COEFF_TXB_20_4

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A018218 WCSS_A_TXFIR_COEFF_TXB_20_5**Type:** RW**Reset State:** 0x0114

2's complement 10-bit Tx-B FIR20 coefficient 5

WCSS_A_TXFIR_COEFF_TXB_20_5

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A01821C WCSS_A_TXFIR_COEFF_TXB_20_6**Type:** RW**Reset State:** 0x00FD

2's complement 10-bit Tx-B FIR20 coefficient 6

WCSS_A_TXFIR_COEFF_TXB_20_6

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A018220 WCSS_A_TXFIR_COEFF_TXB_20_7**Type:** RW**Reset State:** 0x00BF

2's complement 10-bit Tx-B FIR20 coefficient 7

WCSS_A_TXFIR_COEFF_TXB_20_7

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A018224 WCSS_A_TXFIR_COEFF_TXB_20_8**Type:** RW**Reset State:** 0x0072

2's complement 10-bit Tx-B FIR20 coefficient 8

WCSS_A_TXFIR_COEFF_TXB_20_8

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A018228 WCSS_A_TXFIR_COEFF_TXB_20_9**Type:** RW**Reset State:** 0x0030

2's complement 10-bit Tx-B FIR20 coefficient 9

WCSS_A_TXFIR_COEFF_TXB_20_9

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A01822C WCSS_A_TXFIR_COEFF_TXB_20_10**Type:** RW**Reset State:** 0x000A

2's complement 10-bit Tx-B FIR20 coefficient 10

WCSS_A_TXFIR_COEFF_TXB_20_10

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A018230 WCSS_A_TXFIR_COEFF_TXB_20_11**Type:** RW**Reset State:** 0x03FA

2's complement 10-bit Tx-B FIR20 coefficient 11

WCSS_A_TXFIR_COEFF_TXB_20_11

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A018234 WCSS_A_TXFIR_COEFF_TXB_20_12**Type:** RW**Reset State:** 0x0000

2's complement 10-bit Tx-B FIR20 coefficient 12

WCSS_A_TXFIR_COEFF_TXB_20_12

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A018238 WCSS_A_TXFIR_COEFF_TXB_20_13**Type:** RW**Reset State:** 0x0000

2's complement 10-bit Tx-B FIR20 coefficient 13

WCSS_A_TXFIR_COEFF_TXB_20_13

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A01823C WCSS_A_TXFIR_COEFF_TXB_20_14**Type:** RW**Reset State:** 0x0000

2's complement 10-bit Tx-B FIR20 coefficient 14

WCSS_A_TXFIR_COEFF_TXB_20_14

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A018240 WCSS_A_TXFIR_COEFF_TXB_20_15**Type:** RW**Reset State:** 0x0000

2's complement 10-bit Tx-B FIR20 coefficient 15

WCSS_A_TXFIR_COEFF_TXB_20_15

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A018244 WCSS_A_TXFIR_COEFF_TXB_40_0**Type:** RW**Reset State:** 0x0008

2's complement 10-bit Tx-B FIR40 coefficient 0

WCSS_A_TXFIR_COEFF_TXB_40_0

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A018248 WCSS_A_TXFIR_COEFF_TXB_40_1**Type:** RW**Reset State:** 0x000E

2's complement 10-bit Tx-B FIR40 coefficient 1

WCSS_A_TXFIR_COEFF_TXB_40_1

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A01824C WCSS_A_TXFIR_COEFF_TXB_40_2**Type:** RW**Reset State:** 0x0018

2's complement 10-bit Tx-B FIR40 coefficient 2

WCSS_A_TXFIR_COEFF_TXB_40_2

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A018250 WCSS_A_TXFIR_COEFF_TXB_40_3**Type:** RW**Reset State:** 0x0025

2's complement 10-bit Tx-B FIR40 coefficient 3

WCSS_A_TXFIR_COEFF_TXB_40_3

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A018254 WCSS_A_TXFIR_COEFF_TXB_40_4**Type:** RW**Reset State:** 0x0035

2's complement 10-bit Tx-B FIR40 coefficient 4

WCSS_A_TXFIR_COEFF_TXB_40_4

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A018258 WCSS_A_TXFIR_COEFF_TXB_40_5**Type:** RW**Reset State:** 0x0046

2's complement 10-bit Tx-B FIR40 coefficient 5

WCSS_A_TXFIR_COEFF_TXB_40_5

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A01825C WCSS_A_TXFIR_COEFF_TXB_40_6**Type:** RW**Reset State:** 0x0058

2's complement 10-bit Tx-B FIR40 coefficient 6

WCSS_A_TXFIR_COEFF_TXB_40_6

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A018260 WCSS_A_TXFIR_COEFF_TXB_40_7**Type:** RW**Reset State:** 0x0069

2's complement 10-bit Tx-B FIR40 coefficient 7

WCSS_A_TXFIR_COEFF_TXB_40_7

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A018264 WCSS_A_TXFIR_COEFF_TXB_40_8**Type:** RW**Reset State:** 0x0078

2's complement 10-bit Tx-B FIR40 coefficient 8

WCSS_A_TXFIR_COEFF_TXB_40_8

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A018268 WCSS_A_TXFIR_COEFF_TXB_40_9**Type:** RW**Reset State:** 0x0083

2's complement 10-bit Tx-B FIR40 coefficient 9

WCSS_A_TXFIR_COEFF_TXB_40_9

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A01826C WCSS_A_TXFIR_COEFF_TXB_40_10**Type:** RW**Reset State:** 0x0089

2's complement 10-bit Tx-B FIR40 coefficient 10

WCSS_A_TXFIR_COEFF_TXB_40_10

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A018270 WCSS_A_TXFIR_COEFF_TXB_40_11**Type:** RW**Reset State:** 0x0089

2's complement 10-bit Tx-B FIR40 coefficient 11

WCSS_A_TXFIR_COEFF_TXB_40_11

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A018274 WCSS_A_TXFIR_COEFF_TXB_40_12**Type:** RW**Reset State:** 0x0083

2's complement 10-bit Tx-B FIR40 coefficient 12

WCSS_A_TXFIR_COEFF_TXB_40_12

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A018278 WCSS_A_TXFIR_COEFF_TXB_40_13**Type:** RW**Reset State:** 0x0078

2's complement 10-bit Tx-B FIR40 coefficient 13

WCSS_A_TXFIR_COEFF_TXB_40_13

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A01827C WCSS_A_TXFIR_COEFF_TXB_40_14**Type:** RW**Reset State:** 0x0069

2's complement 10-bit Tx-B FIR40 coefficient 14

WCSS_A_TXFIR_COEFF_TXB_40_14

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A018280 WCSS_A_TXFIR_COEFF_TXB_40_15**Type:** RW**Reset State:** 0x0056

2's complement 10-bit Tx-B FIR40 coefficient 15

WCSS_A_TXFIR_COEFF_TXB_40_15

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A018284 WCSS_A_TXFIR_COEFF_TXB_40_16**Type:** RW**Reset State:** 0x0043

2's complement 10-bit Tx-B FIR40 coefficient 16

WCSS_A_TXFIR_COEFF_TXB_40_16

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A018288 WCSS_A_TXFIR_COEFF_TXB_40_17**Type:** RW**Reset State:** 0x0030

2's complement 10-bit Tx-B FIR40 coefficient 17

WCSS_A_TXFIR_COEFF_TXB_40_17

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A01828C WCSS_A_TXFIR_COEFF_TXB_40_18**Type:** RW**Reset State:** 0x001F

2's complement 10-bit Tx-B FIR40 coefficient 18

WCSS_A_TXFIR_COEFF_TXB_40_18

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A018290 WCSS_A_TXFIR_COEFF_TXB_40_19**Type:** RW**Reset State:** 0x0012

2's complement 10-bit Tx-B FIR40 coefficient 19

WCSS_A_TXFIR_COEFF_TXB_40_19

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A018294 WCSS_A_TXFIR_COEFF_TXB_40_20**Type:** RW**Reset State:** 0x0008

2's complement 10-bit Tx-B FIR40 coefficient 20

WCSS_A_TXFIR_COEFF_TXB_40_20

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A018298 WCSS_A_TXFIR_COEFF_TXB_40_21**Type:** RW**Reset State:** 0x0002

2's complement 10-bit Tx-B FIR40 coefficient 21

WCSS_A_TXFIR_COEFF_TXB_40_21

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A01829C WCSS_A_TXFIR_COEFF_TXB_40_22**Type:** RW**Reset State:** 0x03FF

2's complement 10-bit Tx-B FIR40 coefficient 22

WCSS_A_TXFIR_COEFF_TXB_40_22

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A0182A0 WCSS_A_TXFIR_COEFF_TXB_40_23**Type:** RW**Reset State:** 0x03FB

2's complement 10-bit Tx-B FIR40 coefficient 23

WCSS_A_TXFIR_COEFF_TXB_40_23

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A0182A4 WCSS_A_TXFIR_COEFF_TXB_40_24**Type:** RW**Reset State:** 0x0000

2's complement 10-bit Tx-B FIR40 coefficient 24

WCSS_A_TXFIR_COEFF_TXB_40_24

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A0182A8 WCSS_A_TXFIR_COEFF_TXB_40_25**Type:** RW**Reset State:** 0x0000

2's complement 10-bit Tx-B FIR40 coefficient 25

WCSS_A_TXFIR_COEFF_TXB_40_25

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A0182AC WCSS_A_TXFIR_COEFF_TXB_40_26**Type:** RW**Reset State:** 0x0000

2's complement 10-bit Tx-B FIR40 coefficient 26

WCSS_A_TXFIR_COEFF_TXB_40_26

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A0182B0 WCSS_A_TXFIR_COEFF_TXB_40_27**Type:** RW**Reset State:** 0x0000

2's complement 10-bit Tx-B FIR40 coefficient 27

WCSS_A_TXFIR_COEFF_TXB_40_27

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A0182B4 WCSS_A_TXFIR_COEFF_TXB_40_28**Type:** RW**Reset State:** 0x0000

2's complement 10-bit Tx-B FIR40 coefficient 28

WCSS_A_TXFIR_COEFF_TXB_40_28

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A0182B8 WCSS_A_TXFIR_COEFF_TXB_40_29**Type:** RW**Reset State:** 0x0000

2's complement 10-bit Tx-B FIR40 coefficient 29

WCSS_A_TXFIR_COEFF_TXB_40_29

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A0182BC WCSS_A_TXFIR_COEFF_TXB_40_30**Type:** RW**Reset State:** 0x0000

2's complement 10-bit Tx-B FIR40 coefficient 30

WCSS_A_TXFIR_COEFF_TXB_40_30

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A0182C0 WCSS_A_TXFIR_COEFF_TXB_40_31**Type:** RW**Reset State:** 0x0000

2's complement 10-bit Tx-B FIR40 coefficient 31

WCSS_A_TXFIR_COEFF_TXB_40_31

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A0182C4 WCSS_A_TXFIR_COEFF_TXB_80_0**Type:** RW**Reset State:** 0x0005

2's complement 10-bit Tx-B FIR80 coefficient 0

WCSS_A_TXFIR_COEFF_TXB_80_0

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A0182C8 WCSS_A_TXFIR_COEFF_TXB_80_1**Type:** RW**Reset State:** 0x0005

2's complement 10-bit Tx-B FIR80 coefficient 1

WCSS_A_TXFIR_COEFF_TXB_80_1

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A0182CC WCSS_A_TXFIR_COEFF_TXB_80_2**Type:** RW**Reset State:** 0x0006

2's complement 10-bit Tx-B FIR80 coefficient 2

WCSS_A_TXFIR_COEFF_TXB_80_2

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A0182D0 WCSS_A_TXFIR_COEFF_TXB_80_3**Type:** RW**Reset State:** 0x0009

2's complement 10-bit Tx-B FIR80 coefficient 3

WCSS_A_TXFIR_COEFF_TXB_80_3

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A0182D4 WCSS_A_TXFIR_COEFF_TXB_80_4**Type:** RW**Reset State:** 0x000B

2's complement 10-bit Tx-B FIR80 coefficient 4

WCSS_A_TXFIR_COEFF_TXB_80_4

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A0182D8 WCSS_A_TXFIR_COEFF_TXB_80_5**Type:** RW**Reset State:** 0x000E

2's complement 10-bit Tx-B FIR80 coefficient 5

WCSS_A_TXFIR_COEFF_TXB_80_5

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A0182DC WCSS_A_TXFIR_COEFF_TXB_80_6**Type:** RW**Reset State:** 0x0012

2's complement 10-bit Tx-B FIR80 coefficient 6

WCSS_A_TXFIR_COEFF_TXB_80_6

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A0182E0 WCSS_A_TXFIR_COEFF_TXB_80_7**Type:** RW**Reset State:** 0x0016

2's complement 10-bit Tx-B FIR80 coefficient 7

WCSS_A_TXFIR_COEFF_TXB_80_7

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A0182E4 WCSS_A_TXFIR_COEFF_TXB_80_8**Type:** RW**Reset State:** 0x0019

2's complement 10-bit Tx-B FIR80 coefficient 8

WCSS_A_TXFIR_COEFF_TXB_80_8

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A0182E8 WCSS_A_TXFIR_COEFF_TXB_80_9**Type:** RW**Reset State:** 0x001E

2's complement 10-bit Tx-B FIR80 coefficient 9

WCSS_A_TXFIR_COEFF_TXB_80_9

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A0182EC WCSS_A_TXFIR_COEFF_TXB_80_10**Type:** RW**Reset State:** 0x0022

2's complement 10-bit Tx-B FIR80 coefficient 10

WCSS_A_TXFIR_COEFF_TXB_80_10

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A0182F0 WCSS_A_TXFIR_COEFF_TXB_80_11**Type:** RW**Reset State:** 0x0027

2's complement 10-bit Tx-B FIR80 coefficient 11

WCSS_A_TXFIR_COEFF_TXB_80_11

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A0182F4 WCSS_A_TXFIR_COEFF_TXB_80_12**Type:** RW**Reset State:** 0x002B

2's complement 10-bit Tx-B FIR80 coefficient 12

WCSS_A_TXFIR_COEFF_TXB_80_12

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A0182F8 WCSS_A_TXFIR_COEFF_TXB_80_13**Type:** RW**Reset State:** 0x002F

2's complement 10-bit Tx-B FIR80 coefficient 13

WCSS_A_TXFIR_COEFF_TXB_80_13

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A0182FC WCSS_A_TXFIR_COEFF_TXB_80_14**Type:** RW**Reset State:** 0x0034

2's complement 10-bit Tx-B FIR80 coefficient 14

WCSS_A_TXFIR_COEFF_TXB_80_14

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A018300 WCSS_A_TXFIR_COEFF_TXB_80_15**Type:** RW**Reset State:** 0x0038

2's complement 10-bit Tx-B FIR80 coefficient 15

WCSS_A_TXFIR_COEFF_TXB_80_15

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A018304 WCSS_A_TXFIR_COEFF_TXB_80_16**Type:** RW**Reset State:** 0x003B

2's complement 10-bit Tx-B FIR80 coefficient 16

WCSS_A_TXFIR_COEFF_TXB_80_16

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A018308 WCSS_A_TXFIR_COEFF_TXB_80_17**Type:** RW**Reset State:** 0x003E

2's complement 10-bit Tx-B FIR80 coefficient 17

WCSS_A_TXFIR_COEFF_TXB_80_17

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A01830C WCSS_A_TXFIR_COEFF_TXB_80_18**Type:** RW**Reset State:** 0x0041

2's complement 10-bit Tx-B FIR80 coefficient 18

WCSS_A_TXFIR_COEFF_TXB_80_18

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A018310 WCSS_A_TXFIR_COEFF_TXB_80_19**Type:** RW**Reset State:** 0x0043

2's complement 10-bit Tx-B FIR80 coefficient 19

WCSS_A_TXFIR_COEFF_TXB_80_19

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A018314 WCSS_A_TXFIR_COEFF_TXB_80_20**Type:** RW**Reset State:** 0x0044

2's complement 10-bit Tx-B FIR80 coefficient 20

WCSS_A_TXFIR_COEFF_TXB_80_20

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A018318 WCSS_A_TXFIR_COEFF_TXB_80_21**Type:** RW**Reset State:** 0x0045

2's complement 10-bit Tx-B FIR80 coefficient 21

WCSS_A_TXFIR_COEFF_TXB_80_21

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A01831C WCSS_A_TXFIR_COEFF_TXB_80_22**Type:** RW**Reset State:** 0x0045

2's complement 10-bit Tx-B FIR80 coefficient 22

WCSS_A_TXFIR_COEFF_TXB_80_22

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A018320 WCSS_A_TXFIR_COEFF_TXB_80_23**Type:** RW**Reset State:** 0x0044

2's complement 10-bit Tx-B FIR80 coefficient 23

WCSS_A_TXFIR_COEFF_TXB_80_23

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A018324 WCSS_A_TXFIR_COEFF_TXB_80_24**Type:** RW**Reset State:** 0x0042

2's complement 10-bit Tx-B FIR80 coefficient 24

WCSS_A_TXFIR_COEFF_TXB_80_24

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A018328 WCSS_A_TXFIR_COEFF_TXB_80_25**Type:** RW**Reset State:** 0x0040

2's complement 10-bit Tx-B FIR80 coefficient 25

WCSS_A_TXFIR_COEFF_TXB_80_25

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A01832C WCSS_A_TXFIR_COEFF_TXB_80_26**Type:** RW**Reset State:** 0x003D

2's complement 10-bit Tx-B FIR80 coefficient 26

WCSS_A_TXFIR_COEFF_TXB_80_26

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A018330 WCSS_A_TXFIR_COEFF_TXB_80_27**Type:** RW**Reset State:** 0x0039

2's complement 10-bit Tx-B FIR80 coefficient 27

WCSS_A_TXFIR_COEFF_TXB_80_27

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A018334 WCSS_A_TXFIR_COEFF_TXB_80_28**Type:** RW**Reset State:** 0x0035

2's complement 10-bit Tx-B FIR80 coefficient 28

WCSS_A_TXFIR_COEFF_TXB_80_28

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A018338 WCSS_A_TXFIR_COEFF_TXB_80_29**Type:** RW**Reset State:** 0x0031

2's complement 10-bit Tx-B FIR80 coefficient 29

WCSS_A_TXFIR_COEFF_TXB_80_29

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A01833C WCSS_A_TXFIR_COEFF_TXB_80_30**Type:** RW**Reset State:** 0x002C

2's complement 10-bit Tx-B FIR80 coefficient 30

WCSS_A_TXFIR_COEFF_TXB_80_30

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A018340 WCSS_A_TXFIR_COEFF_TXB_80_31**Type:** RW**Reset State:** 0x0027

2's complement 10-bit Tx-B FIR80 coefficient 31

WCSS_A_TXFIR_COEFF_TXB_80_31

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A018344 WCSS_A_TXFIR_COEFF_TXB_80_32**Type:** RW**Reset State:** 0x0023

2's complement 10-bit Tx-B FIR80 coefficient 32

WCSS_A_TXFIR_COEFF_TXB_80_32

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A018348 WCSS_A_TXFIR_COEFF_TXB_80_33**Type:** RW**Reset State:** 0x001E

2's complement 10-bit Tx-B FIR80 coefficient 33

WCSS_A_TXFIR_COEFF_TXB_80_33

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A01834C WCSS_A_TXFIR_COEFF_TXB_80_34**Type:** RW**Reset State:** 0x0019

2's complement 10-bit Tx-B FIR80 coefficient 34

WCSS_A_TXFIR_COEFF_TXB_80_34

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A018350 WCSS_A_TXFIR_COEFF_TXB_80_35**Type:** RW**Reset State:** 0x0015

2's complement 10-bit Tx-B FIR80 coefficient 35

WCSS_A_TXFIR_COEFF_TXB_80_35

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A018354 WCSS_A_TXFIR_COEFF_TXB_80_36**Type:** RW**Reset State:** 0x0011

2's complement 10-bit Tx-B FIR80 coefficient 36

WCSS_A_TXFIR_COEFF_TXB_80_36

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A018358 WCSS_A_TXFIR_COEFF_TXB_80_37**Type:** RW**Reset State:** 0x000D

2's complement 10-bit Tx-B FIR80 coefficient 37

WCSS_A_TXFIR_COEFF_TXB_80_37

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A01835C WCSS_A_TXFIR_COEFF_TXB_80_38**Type:** RW**Reset State:** 0x000A

2's complement 10-bit Tx-B FIR80 coefficient 38

WCSS_A_TXFIR_COEFF_TXB_80_38

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A018360 WCSS_A_TXFIR_COEFF_TXB_80_39**Type:** RW**Reset State:** 0x0007

2's complement 10-bit Tx-B FIR80 coefficient 39

WCSS_A_TXFIR_COEFF_TXB_80_39

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A018364 WCSS_A_TXFIR_COEFF_TXB_80_40**Type:** RW**Reset State:** 0x0005

2's complement 10-bit Tx-B FIR80 coefficient 40

WCSS_A_TXFIR_COEFF_TXB_80_40

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A018368 WCSS_A_TXFIR_COEFF_TXB_80_41**Type:** RW**Reset State:** 0x0003

2's complement 10-bit Tx-B FIR80 coefficient 41

WCSS_A_TXFIR_COEFF_TXB_80_41

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A01836C WCSS_A_TXFIR_COEFF_TXB_80_42**Type:** RW**Reset State:** 0x0001

2's complement 10-bit Tx-B FIR80 coefficient 42

WCSS_A_TXFIR_COEFF_TXB_80_42

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A018370 WCSS_A_TXFIR_COEFF_TXB_80_43**Type:** RW**Reset State:** 0x0000

2's complement 10-bit Tx-B FIR80 coefficient 43

WCSS_A_TXFIR_COEFF_TXB_80_43

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A018374 WCSS_A_TXFIR_COEFF_TXB_80_44**Type:** RW**Reset State:** 0x03FF

2's complement 10-bit Tx-B FIR80 coefficient 44

WCSS_A_TXFIR_COEFF_TXB_80_44

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A018378 WCSS_A_TXFIR_COEFF_TXB_80_45**Type:** RW**Reset State:** 0x03FF

2's complement 10-bit Tx-B FIR80 coefficient 45

WCSS_A_TXFIR_COEFF_TXB_80_45

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A01837C WCSS_A_TXFIR_COEFF_TXB_80_46**Type:** RW**Reset State:** 0x03FF

2's complement 10-bit Tx-B FIR80 coefficient 46

WCSS_A_TXFIR_COEFF_TXB_80_46

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A018380 WCSS_A_TXFIR_COEFF_TXB_80_47**Type:** RW**Reset State:** 0x03FC

2's complement 10-bit Tx-B FIR80 coefficient 47

WCSS_A_TXFIR_COEFF_TXB_80_47

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A018384 WCSS_A_TXFIR_COEFF_TXB_80_48**Type:** RW**Reset State:** 0x0000

2's complement 10-bit Tx-B FIR80 coefficient 48

WCSS_A_TXFIR_COEFF_TXB_80_48

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A018388 WCSS_A_TXFIR_COEFF_TXB_80_49**Type:** RW**Reset State:** 0x0000

2's complement 10-bit Tx-B FIR80 coefficient 49

WCSS_A_TXFIR_COEFF_TXB_80_49

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A01838C WCSS_A_TXFIR_COEFF_TXB_80_50**Type:** RW**Reset State:** 0x0000

2's complement 10-bit Tx-B FIR80 coefficient 50

WCSS_A_TXFIR_COEFF_TXB_80_50

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A018390 WCSS_A_TXFIR_COEFF_TXB_80_51**Type:** RW**Reset State:** 0x0000

2's complement 10-bit Tx-B FIR80 coefficient 51

WCSS_A_TXFIR_COEFF_TXB_80_51

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A018394 WCSS_A_TXFIR_COEFF_TXB_80_52**Type:** RW**Reset State:** 0x0000

2's complement 10-bit Tx-B FIR80 coefficient 52

WCSS_A_TXFIR_COEFF_TXB_80_52

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A018398 WCSS_A_TXFIR_COEFF_TXB_80_53**Type:** RW**Reset State:** 0x0000

2's complement 10-bit Tx-B FIR80 coefficient 53

WCSS_A_TXFIR_COEFF_TXB_80_53

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A01839C WCSS_A_TXFIR_COEFF_TXB_80_54**Type:** RW**Reset State:** 0x0000

2's complement 10-bit Tx-B FIR80 coefficient 54

WCSS_A_TXFIR_COEFF_TXB_80_54

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A0183A0 WCSS_A_TXFIR_COEFF_TXB_80_55**Type:** RW**Reset State:** 0x0000

2's complement 10-bit Tx-B FIR80 coefficient 55

WCSS_A_TXFIR_COEFF_TXB_80_55

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A0183A4 WCSS_A_TXFIR_COEFF_TXB_80_56**Type:** RW**Reset State:** 0x0000

2's complement 10-bit Tx-B FIR80 coefficient 56

WCSS_A_TXFIR_COEFF_TXB_80_56

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A0183A8 WCSS_A_TXFIR_COEFF_TXB_80_57**Type:** RW**Reset State:** 0x0000

2's complement 10-bit Tx-B FIR80 coefficient 57

WCSS_A_TXFIR_COEFF_TXB_80_57

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A0183AC WCSS_A_TXFIR_COEFF_TXB_80_58**Type:** RW**Reset State:** 0x0000

2's complement 10-bit Tx-B FIR80 coefficient 58

WCSS_A_TXFIR_COEFF_TXB_80_58

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A0183B0 WCSS_A_TXFIR_COEFF_TXB_80_59**Type:** RW**Reset State:** 0x0000

2's complement 10-bit Tx-B FIR80 coefficient 59

WCSS_A_TXFIR_COEFF_TXB_80_59

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A0183B4 WCSS_A_TXFIR_COEFF_TXB_80_60**Type:** RW**Reset State:** 0x0000

2's complement 10-bit Tx-B FIR80 coefficient 60

WCSS_A_TXFIR_COEFF_TXB_80_60

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A0183B8 WCSS_A_TXFIR_COEFF_TXB_80_61**Type:** RW**Reset State:** 0x0000

2's complement 10-bit Tx-B FIR80 coefficient 61

WCSS_A_TXFIR_COEFF_TXB_80_61

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A0183BC WCSS_A_TXFIR_COEFF_TXB_80_62**Type:** RW**Reset State:** 0x0000

2's complement 10-bit Tx-B FIR80 coefficient 62

WCSS_A_TXFIR_COEFF_TXB_80_62

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A0183C0 WCSS_A_TXFIR_COEFF_TXB_80_63**Type:** RW**Reset State:** 0x0000

2's complement 10-bit Tx-B FIR80 coefficient 63

WCSS_A_TXFIR_COEFF_TXB_80_63

Bits	Name	Description
9:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A0183C4 WCSS_A_TXFIR_COEFF_PEAKW_0**Type:** RW**Reset State:** 0x01FE

2's complement 9-bit peak windowing FIR coefficient 0

WCSS_A_TXFIR_COEFF_PEAKW_0

Bits	Name	Description
8:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A0183C8 WCSS_A_TXFIR_COEFF_PEAKW_1**Type:** RW**Reset State:** 0x01FF

2's complement 9-bit peak windowing FIR coefficient 1

WCSS_A_TXFIR_COEFF_PEAKW_1

Bits	Name	Description
8:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A0183CC WCSS_A_TXFIR_COEFF_PEAKW_2**Type:** RW**Reset State:** 0x0000

2's complement 9-bit peak windowing FIR coefficient 2

WCSS_A_TXFIR_COEFF_PEAKW_2

Bits	Name	Description
8:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A0183D0 WCSS_A_TXFIR_COEFF_PEAKW_3**Type:** RW**Reset State:** 0x0003

2's complement 9-bit peak windowing FIR coefficient 3

WCSS_A_TXFIR_COEFF_PEAKW_3

Bits	Name	Description
8:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A0183D4 WCSS_A_TXFIR_COEFF_PEAKW_4**Type:** RW**Reset State:** 0x0006

2's complement 9-bit peak windowing FIR coefficient 4

WCSS_A_TXFIR_COEFF_PEAKW_4

Bits	Name	Description
8:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A0183D8 WCSS_A_TXFIR_COEFF_PEAKW_5**Type:** RW**Reset State:** 0x000B

2's complement 9-bit peak windowing FIR coefficient 5

WCSS_A_TXFIR_COEFF_PEAKW_5

Bits	Name	Description
8:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A0183DC WCSS_A_TXFIR_COEFF_PEAKW_6**Type:** RW**Reset State:** 0x0010

2's complement 9-bit peak windowing FIR coefficient 6

WCSS_A_TXFIR_COEFF_PEAKW_6

Bits	Name	Description
8:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A0183E0 WCSS_A_TXFIR_COEFF_PEAKW_7**Type:** RW**Reset State:** 0x0016

2's complement 9-bit peak windowing FIR coefficient 7

WCSS_A_TXFIR_COEFF_PEAKW_7

Bits	Name	Description
8:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A0183E4 WCSS_A_TXFIR_COEFF_PEAKW_8**Type:** RW**Reset State:** 0x001A

2's complement 9-bit peak windowing FIR coefficient 8

WCSS_A_TXFIR_COEFF_PEAKW_8

Bits	Name	Description
8:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A0183E8 WCSS_A_TXFIR_COEFF_PEAKW_9**Type:** RW**Reset State:** 0x001D

2's complement 9-bit peak windowing FIR coefficient 9

WCSS_A_TXFIR_COEFF_PEAKW_9

Bits	Name	Description
8:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A0183EC WCSS_A_TXFIR_COEFF_PEAKW_10**Type:** RW**Reset State:** 0x001E

2's complement 9-bit peak windowing FIR coefficient 10

WCSS_A_TXFIR_COEFF_PEAKW_10

Bits	Name	Description
8:0	COEFFVAL	2's complement scaled, quantized coefficient

0x0A0183F0 WCSS_A_TXFIR_PEAKW_CLIP_THRESHOLD_0**Type:** RW**Reset State:** 0x0032

8-bit peak windowing clip threshold for power index 0

WCSS_A_TXFIR_PEAKW_CLIP_THRESHOLD_0

Bits	Name	Description
7:0	COEFFVAL	Clip threshold for MPI power index 0

0x0A0183F4 WCSS_A_TXFIR_PEAKW_CLIP_THRESHOLD_1**Type:** RW**Reset State:** 0x0032

8-bit peak windowing clip threshold for power index 1

WCSS_A_TXFIR_PEAKW_CLIP_THRESHOLD_1

Bits	Name	Description
7:0	COEFFVAL	Clip threshold for MPI power index 1

0x0A0183F8 WCSS_A_TXFIR_PEAKW_CLIP_THRESHOLD_2**Type:** RW**Reset State:** 0x0032

8-bit peak windowing clip threshold for power index 2

WCSS_A_TXFIR_PEAKW_CLIP_THRESHOLD_2

Bits	Name	Description
7:0	COEFFVAL	Clip threshold for MPI power index 2

0x0A0183FC WCSS_A_TXFIR_PEAKW_CLIP_THRESHOLD_3**Type:** RW**Reset State:** 0x0032

8-bit peak windowing clip threshold for power index 3

WCSS_A_TXFIR_PEAKW_CLIP_THRESHOLD_3

Bits	Name	Description
7:0	COEFFVAL	Clip threshold for MPI power index 3

0x0A018400 WCSS_A_TXFIR_PEAKW_CLIP_THRESHOLD_4**Type:** RW**Reset State:** 0x0032

8-bit peak windowing clip threshold for power index 4

WCSS_A_TXFIR_PEAKW_CLIP_THRESHOLD_4

Bits	Name	Description
7:0	COEFFVAL	Clip threshold for MPI power index 4

0x0A018404 WCSS_A_TXFIR_PEAKW_CLIP_THRESHOLD_5**Type:** RW**Reset State:** 0x0032

8-bit peak windowing clip threshold for power index 5

WCSS_A_TXFIR_PEAKW_CLIP_THRESHOLD_5

Bits	Name	Description
7:0	COEFFVAL	Clip threshold for MPI power index 5

0x0A018408 WCSS_A_TXFIR_PEAKW_CLIP_THRESHOLD_6**Type:** RW**Reset State:** 0x0032

8-bit peak windowing clip threshold for power index 6

WCSS_A_TXFIR_PEAKW_CLIP_THRESHOLD_6

Bits	Name	Description
7:0	COEFFVAL	Clip threshold for MPI power index 6

0x0A01840C WCSS_A_TXFIR_PEAKW_CLIP_THRESHOLD_7**Type:** RW**Reset State:** 0x0032

8-bit peak windowing clip threshold for power index 7

WCSS_A_TXFIR_PEAKW_CLIP_THRESHOLD_7

Bits	Name	Description
7:0	COEFFVAL	Clip threshold for MPI power index 7

0x0A018410 WCSS_A_TXFIR_PEAKW_CLIP_THRESHOLD_8**Type:** RW**Reset State:** 0x0032

8-bit peak windowing clip threshold for power index 8

WCSS_A_TXFIR_PEAKW_CLIP_THRESHOLD_8

Bits	Name	Description
7:0	COEFFVAL	Clip threshold for MPI power index 8

0x0A018414 WCSS_A_TXFIR_PEAKW_CLIP_THRESHOLD_9**Type:** RW**Reset State:** 0x0032

8-bit peak windowing clip threshold for power index 9

WCSS_A_TXFIR_PEAKW_CLIP_THRESHOLD_9

Bits	Name	Description
7:0	COEFFVAL	Clip threshold for MPI power index 9

0x0A018418 WCSS_A_TXFIR_PEAKW_CLIP_THRESHOLD_10**Type:** RW**Reset State:** 0x0032

8-bit peak windowing clip threshold for power index 10

WCSS_A_TXFIR_PEAKW_CLIP_THRESHOLD_10

Bits	Name	Description
7:0	COEFFVAL	Clip threshold for MPI power index 10

0x0A01841C WCSS_A_TXFIR_PEAKW_CLIP_THRESHOLD_11**Type:** RW**Reset State:** 0x0032

8-bit peak windowing clip threshold for power index 11

WCSS_A_TXFIR_PEAKW_CLIP_THRESHOLD_11

Bits	Name	Description
7:0	COEFFVAL	Clip threshold for MPI power index 11

0x0A018420 WCSS_A_TXFIR_PEAKW_CLIP_THRESHOLD_12**Type:** RW**Reset State:** 0x0032

8-bit peak windowing clip threshold for power index 12

WCSS_A_TXFIR_PEAKW_CLIP_THRESHOLD_12

Bits	Name	Description
7:0	COEFFVAL	Clip threshold for MPI power index 12

0x0A018424 WCSS_A_TXFIR_PEAKW_CLIP_THRESHOLD_13**Type:** RW**Reset State:** 0x0032

8-bit peak windowing clip threshold for power index 13

WCSS_A_TXFIR_PEAKW_CLIP_THRESHOLD_13

Bits	Name	Description
7:0	COEFFVAL	Clip threshold for MPI power index 13

0x0A018428 WCSS_A_TXFIR_PEAKW_CLIP_THRESHOLD_14**Type:** RW**Reset State:** 0x0032

8-bit peak windowing clip threshold for power index 14

WCSS_A_TXFIR_PEAKW_CLIP_THRESHOLD_14

Bits	Name	Description
7:0	COEFFVAL	Clip threshold for MPI power index 14

0x0A01842C WCSS_A_TXFIR_PEAKW_CLIP_THRESHOLD_15**Type:** RW**Reset State:** 0x0032

8-bit peak windowing clip threshold for power index 15

WCSS_A_TXFIR_PEAKW_CLIP_THRESHOLD_15

Bits	Name	Description
7:0	COEFFVAL	Clip threshold for MPI power index 15

0x0A018430 WCSS_A_TXFIR_PEAKW_CLIP_THRESHOLD_16**Type:** RW**Reset State:** 0x0032

8-bit peak windowing clip threshold for power index 16

WCSS_A_TXFIR_PEAKW_CLIP_THRESHOLD_16

Bits	Name	Description
7:0	COEFFVAL	Clip threshold for MPI power index 16

0x0A018434 WCSS_A_TXFIR_PEAKW_CLIP_THRESHOLD_17**Type:** RW**Reset State:** 0x0032

8-bit peak windowing clip threshold for power index 17

WCSS_A_TXFIR_PEAKW_CLIP_THRESHOLD_17

Bits	Name	Description
7:0	COEFFVAL	Clip threshold for MPI power index 17

0x0A018438 WCSS_A_TXFIR_PEAKW_CLIP_THRESHOLD_18**Type:** RW**Reset State:** 0x0032

8-bit peak windowing clip threshold for power index 18

WCSS_A_TXFIR_PEAKW_CLIP_THRESHOLD_18

Bits	Name	Description
7:0	COEFFVAL	Clip threshold for MPI power index 18

0x0A01843C WCSS_A_TXFIR_PEAKW_CLIP_THRESHOLD_19**Type:** RW**Reset State:** 0x0032

8-bit peak windowing clip threshold for power index 19

WCSS_A_TXFIR_PEAKW_CLIP_THRESHOLD_19

Bits	Name	Description
7:0	COEFFVAL	Clip threshold for MPI power index 19

0x0A018440 WCSS_A_TXFIR_PEAKW_CLIP_THRESHOLD_20**Type:** RW**Reset State:** 0x0032

8-bit peak windowing clip threshold for power index 20

WCSS_A_TXFIR_PEAKW_CLIP_THRESHOLD_20

Bits	Name	Description
7:0	COEFFVAL	Clip threshold for MPI power index 20

0x0A018444 WCSS_A_TXFIR_PEAKW_CLIP_THRESHOLD_21**Type:** RW**Reset State:** 0x0032

8-bit peak windowing clip threshold for power index 21

WCSS_A_TXFIR_PEAKW_CLIP_THRESHOLD_21

Bits	Name	Description
7:0	COEFFVAL	Clip threshold for MPI power index 21

0x0A018448 WCSS_A_TXFIR_PEAKW_CLIP_THRESHOLD_22**Type:** RW**Reset State:** 0x0032

8-bit peak windowing clip threshold for power index 22

WCSS_A_TXFIR_PEAKW_CLIP_THRESHOLD_22

Bits	Name	Description
7:0	COEFFVAL	Clip threshold for MPI power index 22

0x0A01844C WCSS_A_TXFIR_PEAKW_CLIP_THRESHOLD_23**Type:** RW**Reset State:** 0x0032

8-bit peak windowing clip threshold for power index 23

WCSS_A_TXFIR_PEAKW_CLIP_THRESHOLD_23

Bits	Name	Description
7:0	COEFFVAL	Clip threshold for MPI power index 23

0x0A018450 WCSS_A_TXFIR_PEAKW_CLIP_THRESHOLD_24**Type:** RW**Reset State:** 0x0032

8-bit peak windowing clip threshold for power index 24

WCSS_A_TXFIR_PEAKW_CLIP_THRESHOLD_24

Bits	Name	Description
7:0	COEFFVAL	Clip threshold for MPI power index 24

0x0A018454 WCSS_A_TXFIR_PEAKW_CLIP_THRESHOLD_25**Type:** RW**Reset State:** 0x0032

8-bit peak windowing clip threshold for power index 25

WCSS_A_TXFIR_PEAKW_CLIP_THRESHOLD_25

Bits	Name	Description
7:0	COEFFVAL	Clip threshold for MPI power index 25

0x0A018458 WCSS_A_TXFIR_PEAKW_CLIP_THRESHOLD_26**Type:** RW**Reset State:** 0x0032

8-bit peak windowing clip threshold for power index 26

WCSS_A_TXFIR_PEAkW_CLIP_THRESHOLD_26

Bits	Name	Description
7:0	COEFFVAL	Clip threshold for MPI power index 26

0x0A01845C WCSS_A_TXFIR_PEAkW_CLIP_THRESHOLD_27**Type:** RW**Reset State:** 0x0032

8-bit peak windowing clip threshold for power index 27

WCSS_A_TXFIR_PEAkW_CLIP_THRESHOLD_27

Bits	Name	Description
7:0	COEFFVAL	Clip threshold for MPI power index 27

0x0A018460 WCSS_A_TXFIR_PEAkW_CLIP_THRESHOLD_28**Type:** RW**Reset State:** 0x0032

8-bit peak windowing clip threshold for power index 28

WCSS_A_TXFIR_PEAkW_CLIP_THRESHOLD_28

Bits	Name	Description
7:0	COEFFVAL	Clip threshold for MPI power index 28

0x0A018464 WCSS_A_TXFIR_PEAkW_CLIP_THRESHOLD_29**Type:** RW**Reset State:** 0x0032

8-bit peak windowing clip threshold for power index 29

WCSS_A_TXFIR_PEAKW_CLIP_THRESHOLD_29

Bits	Name	Description
7:0	COEFFVAL	Clip threshold for MPI power index 29

0x0A018468 WCSS_A_TXFIR_PEAKW_CLIP_THRESHOLD_30**Type:** RW**Reset State:** 0x0032

8-bit peak windowing clip threshold for power index 30

WCSS_A_TXFIR_PEAKW_CLIP_THRESHOLD_30

Bits	Name	Description
7:0	COEFFVAL	Clip threshold for MPI power index 30

0x0A01846C WCSS_A_TXFIR_PEAKW_CLIP_THRESHOLD_31**Type:** RW**Reset State:** 0x0032

8-bit peak windowing clip threshold for power index 31

WCSS_A_TXFIR_PEAKW_CLIP_THRESHOLD_31

Bits	Name	Description
7:0	COEFFVAL	Clip threshold for MPI power index 31

0x0A018470 WCSS_A_TXFIR_Q_INVERT**Type:** RW**Reset State:** 0x0000

Register to enable sign inversion of the Q component of each chain

WCSS_A_TXFIR_Q_INVERT

Bits	Name	Description
0	ENABLE	Per-chain enable for Q-component sign inversion of each chain

0x0A018474 WCSS_A_TXFIR_INTERP_X2**Type:** RW**Reset State:** 0x0000

Register to enable interpolation by 2 instead of 1.5. This is used for P2P emulation

WCSS_A_TXFIR_INTERP_X2

Bits	Name	Description
0	ENABLE	Interpolation-by-2 enable

0x0A018478 WCSS_A_TXFIR_IQCORR_CENTRE_TAP**Type:** RW**Reset State:** 0x0002

This register defines the centre tap value in the IQ corrector complex filter

WCSS_A_TXFIR_IQCORR_CENTRE_TAP

Bits	Name	Description
2:0	VALUE	IQ corrector centre tap location

0x0A01847C WCSS_A_TXFIR_GAIN_OFFSET_OFDM**Type:** RW**Reset State:** 0x0000

This register stores the gain offset (dB) applied to the demanded power for OFDM packets. The adjusted value is used to adjust the scaling profile applied in the DPD module.

WCSS_A_TXFIR_GAIN_OFFSET_OFDM

Bits	Name	Description
5:0	VALUE	

0x0A018480 WCSS_A_TXFIR_FINEGAIN_OFFSET**Type:** RW**Reset State:** 0x0000

This register stores the gain offset applied to the fine gain value provided by the TPC module.
Valid range = 0 to 24, i.e. +0dB to +6dB inclusive

WCSS_A_TXFIR_FINEGAIN_OFFSET

Bits	Name	Description
5:0	VALUE	

0x0A018484 WCSS_A_TXFIR_GAIN_OFFSET_11B**Type:** RW**Reset State:** 0x0000

This register stores the gain offset (dB) applied to the demanded power for 11b packets. The adjusted value is used to adjust the scaling profile applied in the DPD module.

WCSS_A_TXFIR_GAIN_OFFSET_11B

Bits	Name	Description
5:0	VALUE	

0x0A018488 WCSS_A_TXFIR_TXFIR_FIFO_RESET**Type:** W**Reset State:** 0x0000

Writing any value to this register will generate an active-high reset pulse for the TXFIR output FIFO.

WCSS_A_TXFIR_TXFIR_FIFO_RESET

Bits	Name	Description
0	TXFIR_FIFO_RESET	

**0x0A018800+ WCSS_A_TXFIR_TXCAL_MEM0n_0, n=[0..23]
0x20*n**
Type: RW**Reset State:** 0x0000

Contains TX IQ imbalance correction and TX carrier suppression values for each RF gain setting

WCSS_A_TXFIR_TXCAL_MEM0n_0

Bits	Name	Description
15:9	IQPHASEIMB_COEFF_Q0_0	2's complement IQ phase imbalance correction value, coefficient q0 for chain 0
8:0	IQPHASEIMB_COEFF_I0	2's complement IQ phase imbalance correction value, coefficient i0 for chain 0

**0x0A018804+ WCSS_A_TXFIR_TXCAL_MEM0n_1, n=[0..23]
0x20*n**
Type: RW**Reset State:** 0x0000

Contains TX IQ imbalance correction and TX carrier suppression values for each RF gain setting

WCSS_A_TXFIR_TXCAL_MEM0n_1

Bits	Name	Description
15:11	IQPHASEIMB_COEFF_Q1_0	2's complement IQ phase imbalance correction value, coefficient q1 for chain 0
10:2	IQPHASEIMB_COEFF_I1	2's complement IQ phase imbalance correction value, coefficient i1 for chain 0
1:0	IQPHASEIMB_COEFF_Q0_1	2's complement IQ phase imbalance correction value, coefficient q0 for chain 0

**0x0A018808+ WCSS_A_TXFIR_TXCAL_MEM0n_2, n=[0..23]
0x20*n**
Type: RW**Reset State:** 0x0000

Contains TX IQ imbalance correction and TX carrier suppression values for each RF gain setting

WCSS_A_TXFIR_TXCAL_MEM0n_2

Bits	Name	Description
15:13	IQPHASEIMB_COEFF_Q2_0	2's complement IQ phase imbalance correction value, coefficient q2 for chain 0
12:4	IQPHASEIMB_COEFF_I2	2's complement IQ phase imbalance correction value, coefficient i2 for chain 0
3:0	IQPHASEIMB_COEFF_Q1_1	2's complement IQ phase imbalance correction value, coefficient q1 for chain 0

0x0A01880C+WCSS_A_TXFIR_TXCAL_MEM0n_3, n=[0..23]
0x20*n

Type: RW

Reset State: 0x0000

Contains TX IQ imbalance correction and TX carrier suppression values for each RF gain setting

WCSS_A_TXFIR_TXCAL_MEM0n_3

Bits	Name	Description
15	IQPHASEIMB_COEFF_Q3_0	2's complement IQ phase imbalance correction value, coefficient q3 for chain 0
14:6	IQPHASEIMB_COEFF_I3	2's complement IQ phase imbalance correction value, coefficient i3 for chain 0
5:0	IQPHASEIMB_COEFF_Q2_1	2's complement IQ phase imbalance correction value, coefficient q2 for chain 0

0x0A018810+ WCSS_A_TXFIR_TXCAL_MEM0n_4, n=[0..23]
0x20*n

Type: RW

Reset State: 0x0000

Contains TX IQ imbalance correction and TX carrier suppression values for each RF gain setting

WCSS_A_TXFIR_TXCAL_MEM0n_4

Bits	Name	Description
15:8	IQPHASEIMB_COEFF_I4_0	2's complement IQ phase imbalance correction value, coefficient i4 for chain 0
7:0	IQPHASEIMB_COEFF_Q3_1	2's complement IQ phase imbalance correction value, coefficient q3 for chain 0

0x0A018814+ WCSS_A_TXFIR_TXCAL_MEM0n_5, n=[0..23]
0x20*n
Type: RW**Reset State:** 0x0000

Contains TX IQ imbalance correction and TX carrier suppression values for each RF gain setting

WCSS_A_TXFIR_TXCAL_MEM0n_5

Bits	Name	Description
15:10	IQ_AMPIMB_COEFF_0	2's complement IQ amplitude imbalance correction factor.
9:1	IQPHASEIMB_COEFF_Q4	2's complement IQ phase imbalance correction value, coefficient q4 for chain 0
0	IQPHASEIMB_COEFF_I4_1	2's complement IQ phase imbalance correction value, coefficient i4 for chain 0

0x0A018818+ WCSS_A_TXFIR_TXCAL_MEM0n_6, n=[0..23]
0x20*n
Type: RW**Reset State:** 0x0000

Contains TX IQ imbalance correction and TX carrier suppression values for each RF gain setting

WCSS_A_TXFIR_TXCAL_MEM0n_6

Bits	Name	Description
14:9	TXLOLEAKAGE_Q	2's complement Transmit chain local oscillator leakage for chain 0, Q rail

WCSS_A_TXFIR_TXCAL_MEM0n_6 (cont.)

Bits	Name	Description
8:3	TXLOLEAKAGE_I	2's complement Transmit chain local oscillator leakage for chain 0, I rail
2:0	IQ_AMPIMB_COEFF_1	2's complement IQ amplitude imbalance correction factor.

0x0A019000+ WCSS_A_TXFIR_DPD_MEM0n_0, n=[0..127]
0x10*n
Type: RW**Reset State:** 0x0000

Contains TX pre-distorter parameter values for each RF gain setting

WCSS_A_TXFIR_DPD_MEM0n_0

Bits	Name	Description
15:12	DPD_AOFFSET_0	DPD amplitude gain offset for each section on chain 0
11:0	DPD_THRESHOLD	DPD amplitude threshold for each section on chain 0

0x0A019004+ WCSS_A_TXFIR_DPD_MEM0n_1, n=[0..127]
0x10*n
Type: RW**Reset State:** 0x0000

Contains TX pre-distorter parameter values for each RF gain setting

WCSS_A_TXFIR_DPD_MEM0n_1

Bits	Name	Description
15:8	DPD_AGAIN_0	DPD amplitude gain change for each section on chain 0
7:0	DPD_AOFFSET_1	DPD amplitude gain offset for each section on chain 0

0x0A019008+ WCSS_A_TXFIR_DPD_MEM0n_2, n=[0..127]
0x10*n
Type: RW**Reset State:** 0x0000

Contains TX pre-distorter parameter values for each RF gain setting

WCSS_A_TXFIR_DPD_MEM0n_2

Bits	Name	Description
15:4	DPD_POFFSET	DPD phase gain offset for each section on chain 0
3:0	DPD_AGAIN_1	DPD amplitude gain change for each section on chain 0

0x0A01900C+WCSS_A_TXFIR_DPD_MEM0n_3, n=[0..127] 0x10*n

Type: RW

Reset State: 0x0000

Contains TX pre-distorter parameter values for each RF gain setting

WCSS_A_TXFIR_DPD_MEM0n_3

Bits	Name	Description
11:0	DPD_PGAIN	DPD phase gain change for each section on chain 0

10.63 WCSS_A_TXP_TXP

0x0A080400 WCSS_A_TXP_TXP_CMDF_DIN

Type: W

Reset State: 0x00000000

Command write port of TXP FIFO.

WCSS_A_TXP_TXP_CMDF_DIN

Bits	Name	Description
31:0	TXP_CMDF_WDATA	Write-only, 32-bit bitfield for commands from the software to the TXP.

0x0A080404 WCSS_A_TXP_TXP_CMDF_CONTROL**Type:** RW**Reset State:** 0x00010024

TXP command FIFO control register.

WCSS_A_TXP_TXP_CMDF_CONTROL

Bits	Name	Type	Description
16	TPE_CAHBZ	RW	This bit controls where the TXP commands will come from. 1 for TPE, 0 for CAHB (i.e. from SW).
7:0	FREE_ENTRIES	R	Read-only field that shows free entries remaining in TXP command FIFO

0x0A080408 WCSS_A_TXP_TXP_TIMEOUT_TIMER**Type:** RW**Reset State:** 0x00000000

TXP Timeout Timer - will interrupt cpu if TXP state machines do not return to idle within prespecified time.

WCSS_A_TXP_TXP_TIMEOUT_TIMER

Bits	Name	Description
31	TIMER_EN	Timeout timer enable bit.
23:0	TIMEOUT_TIMER_VAL	Timeout timer value - in microseconds

0x0A08040C WCSS_A_TXP_TXP_STATUS**Type:** R**Reset State:** 0x00000000

Shows status of TXP.

WCSS_A_TXP_TXP_STATUS

Bits	Name	Description
18:17	GAS_STATUS	Status of GAS in TXP.
16	GAM_BUSY	Shows whether GAM in TXP is busy.
15:8	ERROR_CODE	Read-only field that shows TXP error code (0 if no error). Command FIFO - Over flow: 8'b0001_0000; GAM control - AHB error returned from GAM: 8'b0010_0000; Invalid BD header: 8'b0010_0001; tx_size with 0 bytes: 8'b0010_0010; MPDU Header longer than 1st pdu: 8'b0010_0011; Data starts outside 1st pdu: 8'b0010_0100; MPDU header longer than MPDU: 8'b0010_0101; MPDU data_offset < MPDU_hdr_off: 8'b0010_0110; PINF - PHY sends abort/BTC Active: 8'b0011_0000; - FIFO underrun: 8'b0011_0001; - New transmission with no PHY clkstart: 8'b0011_0010; - New transmission, but PHY not idle: 8'b0011_0011; - PHY goes idle prematurely: 8'b0011_0100; FCS - Two consecutive FCS Starts received (last was FCS32): 8'b0100_0001; - FCS8 Start followed by FCS8 Start: 8'b0100_0010; - Two consecutive FCS Stops received (last was FCS8): 8'b0100_0011; - FCS32 Start followed by FCS8 Start: 8'b0100_0100; - FCS32 Start followed by FCS8 Stop: 8'b0100_0101; - FCS8 Start followed by FCS32 Start: 8'b0100_0110; - FCS8 Start followed by FCS32 Stop: 8'b0100_0111; Main FSM - Invalid command type: 8'b0101_0000; - PHY command with 0 bytes: 8'b0101_0001; - tx_sw with 0 bytes: 8'b0101_0010;
7:0	LAST_CMD_NR	Read-only field that shows last command number successfully completed by TXP. TXP extracts the last 8 bits of the command and blindly copies it to this register This is the CMD index nr field supplied by s/w.

0x0A080410 WCSS_A_TXP_TXP_BTC_ACTIVE_ABORT_ENABLE**Type:** RW**Reset State:** 0x00000000

Controls if BTC active will cause TXP to be abort. SW would then have to soft reset the TXP.

WCSS_A_TXP_TXP_BTC_ACTIVE_ABORT_ENABLE

Bits	Name	Description
0	BTC_ACTIVE	Allows BTC active to cause TXP abort.

0x0A080418 WCSS_A_TXP_TXP_WARMUP_CNTL**Type:** RW**Reset State:** 0x00000530

TXP warmup control register - added for Virgo

WCSS_A_TXP_TXP_WARMUP_CNTL

Bits	Name	Description
13:8	DELAY_PHY_START_COUNT	Number of clock cycles to delay the phy start
5	TPE_TXP_RF_WARMUP_END_P_EN	allow tpe_txp_rf_warmup_end_p to affect txp_mpi_warmup
4	TPE_TXP_RF_WARMUP_P_EN	allow tpe_txp_rf_warmup_p to affect txp_mpi_warmup

0x0A080484 WCSS_A_TXP_TXP_NR_FRAMES_XMIT**Type:** RW**Reset State:** 0x00000000

Number of frames (PPDU) transmitted

WCSS_A_TXP_TXP_NR_FRAMES_XMIT

Bits	Name	Description
15:0	NR_FRAMES_XMIT	Number of frames transmitted; stuck at 0xffff

0x0A080488 WCSS_A_TXP_TXP_PHY_ABORTS**Type:** RW**Reset State:** 0x00000000

Number of times that txp_mpi_abort or mpi_txp_abort is asserted (counter is located before the loopback mux). The counters increment once for every low to high transition in the signal.

WCSS_A_TXP_TXP_PHY_ABORTS

Bits	Name	Description
31:16	NR_PHY_ABORT_RCVD	Number of times mpi_txp_abort (or fphy_txp_abort) is asserted.

WCSS_A_TXP_TXP_PHY_ABORTS (cont.)

Bits	Name	Description
15:0	NR_PHY_ABORT_GEN	Number of times txp_mpi_abort (or txp_fphy_abort) is asserted

0x0A08048C WCSS_A_TXP_TXP_NR_END_INTERRUPTS**Type:** RW**Reset State:** 0x00000000

Number of end interrupts sent by TXP.

WCSS_A_TXP_TXP_NR_END_INTERRUPTS

Bits	Name	Description
15:0	TXP_NR_END_INTERRUPT S	Number of end interrupts sent to s/w. Once it is reached max(0xFFFF), hw stops updating SW needs to reset to restart this counter again

0x0A080490 WCSS_A_TXP_TXP_NR_CMDS**Type:** RW**Reset State:** 0x00000000

Number of TXP commands processed.

WCSS_A_TXP_TXP_NR_CMDS

Bits	Name	Description
15:0	TXP_NR_CMDS	Number of commands processed

0x0A080494 WCSS_A_TXP_TXP_FSM_STATUS**Type:** R**Reset State:** 0x00000000

TXP internal FSM status

WCSS_A_TXP_TXP_FSM_STATUS

Bits	Name	Description
23:0	TXP_FSM_STATUS	txp_mcu_status, // [23] 1 is busy enc_txp_pinf_phy_cs_s[3:0], // [22:19] phy domain, own't be accurate when it's active enc_txp_fcs_cs_s[3:0], // [18:15] enc_txp_gamc_cs_s[4:0], // [14:10] enc_txp_main_cs_s[4:0], // [9:5] enc_txp_pinf_mac_cs_s[1:0], // [4:3] enc_txp_wait_cs_s[2:0]

0x0A08049C WCSS_A_TXP_TXP_NR_MPI**Type:** R**Reset State:** 0x00000000

Number of MPI request and data valid Note this register is only used for debug purpose. The counters are in phy clock domain, and is not sync to the mac domain. It's only valid when we are debugging txp is hang

WCSS_A_TXP_TXP_NR_MPI

Bits	Name	Description
31:16	TXP_NR_MPI_DATA_VALID	Number of mpi_data_valid (before loopback mux)
15:0	TXP_NR_MPI_REQ	Number of mpi request (after loopback mux)

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11.1 SPMI_CFG_TOP

11.1.1 SPMI_GENI_CFG

0x0200A000 SPMI_GENI_CLK_CTRL

Type: RW
Clock: AHB_CLK
Reset State: 0x00000000

GENI clock control register.

SPMI_GENI_CLK_CTRL

Bits	Name	Description
0	SER_CLK_SEL	Serial clock select. When clear(0), GENI PROGRAM RAM, and other serial clock domain logic is connected to AHB clock, this is required for the configuration phase. When set(1), GENI PROGRAM RAM, and other serial clock domain logic is connected to Serial clock, this is required for the command execution phase.

0x0200A008 SPMI_GENI_FW_REVISION

Type: RW
Clock: AHB_CLK
Reset State: 0x0000FFFF

GENI FW revision register.

In the full-duplex mode, the TX and RX protocols might be not the same.

For example, the same Geni instance might support RC-5 protocol for TX and RC-6 for RX. Therefore there are 2 FW Revision registers, one for TX and one for RX (if needed).

For half-duplex protocols (SPMI, RFFE, SSBI2, ...) both FW revision registers will contains the same value.

SPMI_GENI_FW_REVISION

Bits	Name	Description
15:6	PROTOCOL	TX protocol implemented.
5:0	VERSION	TX FW version.

0x0200A00C SPMI_GENI_S_FW_REVISION

Type: RW

Clock: AHB_CLK

Reset State: 0x0000FFFF

GENI secondary FW revision register.

In the full-duplex mode, the TX and RX protocols might be not the same.

For example, the same Geni instance might support RC-5 protocol for TX and

RC-6 for RX. Therefore there are 2 FW Revision registers, one for TX and one for RX (if needed).

For half-duplex protocols (SPMI, RFFE, SSBI2, ...) both FW revision registers will contains the same value.

SPMI_GENI_S_FW_REVISION

Bits	Name	Description
15:6	PROTOCOL	RX protocol implemented.
5:0	VERSION	RX FW version.

0x0200A010 SPMI_GENI_FORCE_DEFAULT_REG

Type: W

Clock: AHB_CLK

Reset State: 0x00000000

GENI force default register. Writing '1' to this register forces default values on GENI's outputs to pads. The default values are part of the GENI image (i.e. GENI_CFG_REG)

SPMI_GENI_FORCE_DEFAULT_REG

Bits	Name	Description
0	FORCE_DEFAULT	force default bit.

0x0200A014 SPMI_GENI_OUTPUT_CTRL**Type:** RW**Clock:** AHB_CLK**Reset State:** 0x00000000

GENI output control register.

SPMI_GENI_OUTPUT_CTRL

Bits	Name	Description
2	SOE2_EN	forces geni_soe02 to be 0 when set, i.e. geni_sout2 is high-Z.
1	SOE1_EN	forces geni_soe1 to be 0 when set, i.e. geni_sout1 is high-Z.
0	SOE0_EN	forces geni_soe0 to be 0 when set, i.e. geni_sout0 is high-Z.

0x0200A018 SPMI_GENI_CGC_CTRL**Type:** RW**Clock:** AHB_CLK**Reset State:** 0x00000078

GENI clock gating control register.

SPMI_GENI_CGC_CTRL

Bits	Name	Description
9	PROG_RAM_SCLK_OFF	When set, SCLK to program RAM is turned off. Should be used only when SCLK_CGC_ON bit is set, otherwise the clock is turned off automatically by HW when not required.
8	PROG_RAM_HCLK_OFF	When set, HCLK to program RAM is turned off
6	EXT_CLK_CGC_ON	When set, HW controls the CGC. When cleared, the CGC is always enabled.
5	RX_CLK_CGC_ON	When set, HW controls the CGC. When cleared, the CGC is enabled taking into consideration clock division
4	TX_CLK_CGC_ON	When set, HW controls the CGC. When cleared, the CGC is enabled taking into consideration clock division

SPMI_GENI_CGC_CTRL (cont.)

Bits	Name	Description
3	SCLK_CGC_ON	When set, HW controls the CGC. When cleared, the CGC is always enabled. This bit might be cleared only in half-duplex mode and when the SCLK isn't divided internally; use RX_CLK_ON and TX_CLK_ON to control CGCs in that cases.
2	DATA_AHB_CLK_CGC_ON	When set, HW controls the CGC. When cleared, the CGC is always enabled.
1	CFG_AHB_WR_CLK_CGC_ON	When set, HW controls the CGC. When cleared, the CGC is always enabled.
0	CFG_AHB_CLK_CGC_ON	When set, HW controls the CGC. When cleared, the CGC is always enabled.

0x0200A028 SPMI_GENI_CTRL**Type:** RW**Clock:** AHB_CLK**Reset State:** 0x00000000

Geni control register.

Command which is in progress during GENI disabling will complete its process and the next command will pend.

SPMI_GENI_CTRL

Bits	Name	Description
1	S_GENI_DISABLE	when 1, secondary sequencer is disabled, when 0 it is enabled
0	M_GENI_DISABLE	when 1, main sequencer is disabled, when 0 it is enabled

0x0200A02C SPMI_GENI_STATUS**Type:** R**Clock:** AHB_CLK**Reset State:** 0x00000000

Geni status register

SPMI_GENI_STATUS

Bits	Name	Description
20:16	S_GENI_CMD_FSM_STATE	secondary command interface state

SPMI_GENI_STATUS (cont.)

Bits	Name	Description
12	S_GENI_CMD_ACTIVE	when 0, secondary sequencer is idle, when 1, secondary sequencer is executing a command
8:4	M_GENI_CMD_FSM_STATE	main command interface state
0	M_GENI_CMD_ACTIVE	when 0, main sequencer is idle, when 1, main sequencer is executing a command

0x0200A034 SPMI_GENI_SER_CLK_CFG**Type:** RW**Clock:** AHB_CLK**Reset State:** 0x00004002

Geni serial clock configuration register

SPMI_GENI_SER_CLK_CFG

Bits	Name	Description
25:14	TX_CLK_DIV_VALUE	Main sequencer (TX) clock division value. The TX clock's frequency is $SER_CLK / (TX_CLK_DIV_VALUE * 2^{RX_SER_CLK_EN})$
13:2	RX_CLK_DIV_VALUE	Secondary sequencer (RX) clock division value. The RX clock's frequency is $SER_CLK / (RX_CLK_DIV_VALUE * 2^{TX_SER_CLK_EN})$
1	TX_SER_CLK_EN	Main sequencer (TX) clock enable
0	RX_SER_CLK_EN	Secondary sequencer (RX) clock enable

0x0200A038 SPMI_GENI_TX_WATERMARK_REG**Type:** RW**Clock:** AHB_CLK**Reset State:** 0x00000000

Geni TX FIFO Watermark register

SPMI_GENI_TX_WATERMARK_REG

Bits	Name	Description
3:0	TX_WATERMARK	Sets the condition for assertion of TX_WATERMARK interrupt bit. When 0 - watermark = 1, when 1 - watermark = 2, when 2 - watermark = 3, else - reserved

0x0200A03C SPMI_GENI_RX_WATERMARK_REG**Type:** RW**Clock:** AHB_CLK**Reset State:** 0x00000000

Geni RX FIFO Watermark register

SPMI_GENI_RX_WATERMARK_REG

Bits	Name	Description
3:0	RX_WATERMARK	Sets the condition for assertion of RX_WATERMARK interrupt bit. When 0 - watermark = 1, when 1 - watermark = 2, when 2 - watermark = 3, else - reserved

0x0200A040 SPMI_GENI_STOP_REG**Type:** W**Clock:** AHB_CLK**Reset State:** 0x00000000

Geni stop register

SPMI_GENI_STOP_REG

Bits	Name	Description
1	S_GENI_STOP	Writing one forces immediate termination of the secondary sequencer
0	M_GENI_STOP	Writing one forces immediate termination of the main sequencer

0x0200A044 SPMI_GENI_SW_COMP_REG**Type:** RW**Clock:** AHB_CLK**Reset State:** 0x00000000

Geni SW comparator input register

SPMI_GENI_SW_COMP_REG

Bits	Name	Description
2:0	SW_COMP	SW input to Geni's comparator unit

0x0200A048 SPMI_GENI_PROG_ROM_CTRL_REG**Type:** RW**Clock:** AHB_CLK**Reset State:** 0x00000001

Geni Program ROM control register

SPMI_GENI_PROG_ROM_CTRL_REG

Bits	Name	Description
0	PROG_ROM_EN	Program ROM may be generated in parallel to Program RAM (depends on a generic). HW accesses the Program ROM if this register is set to high.

0x0200A100 SPMI_GENI_CFG_REG0**Type:** RW**Clock:** AHB_CLK**Reset State:** 0x0202187F

Geni Configuration Register 0.

SPMI_GENI_CFG_REG0

Bits	Name	Description
29	GENI_TX_MODE	
28:21	COND_COMP_IN_0_SEL_RST_VAL	
20:13	COND_COMP_IN_VEC	

SPMI_GENI_CFG_REG0 (cont.)

Bits	Name	Description
12	M_DATA_CNT_INIT_VALUE_INCR	
11	DATA_CNT_MODE	
8:3	M_DATA_CNT_EN_BITS	
2	EXT_SECURITY_EN	
1	EXT_ARB_EN	
0	M_PROG_RAM_SEC_WORD_EN	

0x0200A104 SPMI_GENI_CFG_REG1**Type:** RW**Clock:** AHB_CLK**Reset State:** 0x00027F7B

Geni Configuration Register 1.

SPMI_GENI_CFG_REG1

Bits	Name	Description
17:9	TX_PARAM_TABLE_VEC_1	
8:0	TX_PARAM_TABLE_VEC_0	

0x0200A108 SPMI_GENI_CFG_REG2**Type:** RW**Clock:** AHB_CLK**Reset State:** 0x000076BF

Geni Configuration Register 2.

SPMI_GENI_CFG_REG2

Bits	Name	Description
17:9	TX_PARAM_TABLE_VEC_3	
8:0	TX_PARAM_TABLE_VEC_2	

0x0200A10C SPMI_GENI_CFG_REG3**Type:** RW**Clock:** AHB_CLK**Reset State:** 0x0007F8FE

Geni Configuration Register 3.

SPMI_GENI_CFG_REG3

Bits	Name	Description
19:10	TX_DATA_TABLE_VEC_1	
9:0	TX_DATA_TABLE_VEC_0	

0x0200A110 SPMI_GENI_CFG_REG4**Type:** RW**Clock:** AHB_CLK**Reset State:** 0x000FFEFE

Geni Configuration Register 4.

SPMI_GENI_CFG_REG4

Bits	Name	Description
19:10	TX_DATA_TABLE_VEC_3	
9:0	TX_DATA_TABLE_VEC_2	

0x0200A114 SPMI_GENI_CFG_REG5**Type:** RW**Clock:** AHB_CLK**Reset State:** 0x208802AB

Geni Configuration Register 5.

SPMI_GENI_CFG_REG5

Bits	Name	Description
29	DRIVE_DEFAULT_ON_STA RT_EN	
28	TX_BP_MODE	
27	TX_SOE_EDGE_SEL	

SPMI_GENI_CFG_REG5 (cont.)

Bits	Name	Description
26	TX_SOUT_EDGE_SEL	
25	TX_DEFAULT_PRIM_SOE_VALUE	
24	TX_DEFAULT_PRIM_SOUT_VALUE	
23	TX_DEFAULT_SOE_VALUE	
22	TX_DEFAULT_SOUT_VALU E	
20:18	TX_CONST1_EFF_SIZE	
9:2	TX_CONST1_REG	
1	TX_PAR_MODE	
0	TX_PAR_CALC_EN	

0x0200A118 SPMI_GENI_CFG_REG6**Type:** RW**Clock:** AHB_CLK**Reset State:** 0x00005512

Geni Configuration Register 6.

SPMI_GENI_CFG_REG6

Bits	Name	Description
31:30	M_TIME_CNT_FR_DIV	
29	M_TIME_CNT_EN_FR_SEL	
25:21	M_GP_CNT_INIT_VAL	
20	M_GP_CNT_DIRECTION	
19	TX_SOUT_MODUL_START_ VAL	
17:16	TX_SOUT_MODUL_STOP_ DUR	
15	TX_SOUT_MODUL_EN	
14	TX_LOAD_PAR_EN	
13	TX_PARAM_DUMMY_PAR_ EN	
12	TX_PARAM_PAR_EN	

SPMI_GENI_CFG_REG6 (cont.)

Bits	Name	Description
11	TX_FIFO_DUMMY_PAR_EN	
10	TX_FIFO_PAR_EN	
8	IO2_CONST_EFF_SIZE	
5:4	IO2_CONST_REG	
3	IO2_DEFAULT_PRIM_SOE_VALUE	
2	IO2_DEFAULT_PRIM_SOUT_VALUE	
1	IO2_DEFAULT_SOE_VALUE	
0	IO2_DEFAULT_SOUT_VALU E	

0x0200A11C SPMI_GENI_CFG_REG7**Type:** RW**Clock:** AHB_CLK**Reset State:** 0x03880001

Geni Configuration Register 7.

SPMI_GENI_CFG_REG7

Bits	Name	Description
26	GENI_RX_MODE	
25	RX_PAR_MODE	
24	RX_PAR_CALC_EN	
23:21	RX_IO_SIN_SEL	
19:18	RX_SIN_SEL	
17	RX_DATA_SRC	
16:9	S_COND_COMP_IN_0_SEL _RST_VAL	
8:1	S_COND_COMP_IN_VEC	
0	S_DATA_CNT_MODE	

0x0200A120 SPMI_GENI_CFG_REG8**Type:** RW**Clock:** AHB_CLK**Reset State:** 0x0000021F

Geni Configuration Register 8.

SPMI_GENI_CFG_REG8

Bits	Name	Description
15:11	S_GP_CNT_INIT_VAL	
10	S_GP_CNT_DIRECTION	
9	S_DATA_CNT_INIT_VALUE_INCR	
6:1	S_DATA_CNT_EN_BITS	
0	S_PROG_RAM_SEC_WORD_EN	

0x0200A124 SPMI_GENI_CFG_REG9**Type:** RW**Clock:** AHB_CLK**Reset State:** 0x0007F8FE

Geni Configuration Register 9.

SPMI_GENI_CFG_REG9

Bits	Name	Description
19:10	RX_DATA_TABLE_VEC_1	
9:0	RX_DATA_TABLE_VEC_0	

0x0200A128 SPMI_GENI_CFG_REG10**Type:** RW**Clock:** AHB_CLK**Reset State:** 0x000FFEFE

Geni Configuration Register 10.

SPMI_GENI_CFG_REG10

Bits	Name	Description
19:10	RX_DATA_TABLE_VEC_3	
9:0	RX_DATA_TABLE_VEC_2	

0x0200A12C SPMI_GENI_CFG_REG11**Type:** RW**Clock:** AHB_CLK**Reset State:** 0x00011000

Geni Configuration Register 11.

SPMI_GENI_CFG_REG11

Bits	Name	Description
23:22	PRIM_11_TX_DURATION	
21:20	PRIM_10_TX_DURATION	
19:18	PRIM_9_TX_DURATION	
17:16	PRIM_8_TX_DURATION	
15:14	PRIM_7_TX_DURATION	
13:12	PRIM_6_TX_DURATION	
11:10	PRIM_5_TX_DURATION	
9:8	PRIM_4_TX_DURATION	
7:6	PRIM_3_TX_DURATION	
5:4	PRIM_2_TX_DURATION	
3:2	PRIM_1_TX_DURATION	
1:0	PRIM_0_TX_DURATION	

0x0200A130 SPMI_GENI_CFG_REG12**Type:** RW**Clock:** AHB_CLK**Reset State:** 0x00000000

Geni Configuration Register 12.

SPMI_GENI_CFG_REG12

Bits	Name	Description
23:22	PRIM_23_TX_DURATION	
21:20	PRIM_22_TX_DURATION	
19:18	PRIM_21_TX_DURATION	
17:16	PRIM_20_TX_DURATION	
15:14	PRIM_19_TX_DURATION	
13:12	PRIM_18_TX_DURATION	
11:10	PRIM_17_TX_DURATION	
9:8	PRIM_16_TX_DURATION	
7:6	PRIM_15_TX_DURATION	
5:4	PRIM_14_TX_DURATION	
3:2	PRIM_13_TX_DURATION	
1:0	PRIM_12_TX_DURATION	

0x0200A134 SPMI_GENI_CFG_REG13**Type:** RW**Clock:** AHB_CLK**Reset State:** 0x00011000

Geni Configuration Register 13.

SPMI_GENI_CFG_REG13

Bits	Name	Description
23:22	PRIM_11_RX_DURATION	
21:20	PRIM_10_RX_DURATION	
19:18	PRIM_9_RX_DURATION	
17:16	PRIM_8_RX_DURATION	
15:14	PRIM_7_RX_DURATION	
13:12	PRIM_6_RX_DURATION	
11:10	PRIM_5_RX_DURATION	
9:8	PRIM_4_RX_DURATION	
7:6	PRIM_3_RX_DURATION	
5:4	PRIM_2_RX_DURATION	

SPMI_GENI_CFG_REG13 (cont.)

Bits	Name	Description
3:2	PRIM_1_RX_DURATION	
1:0	PRIM_0_RX_DURATION	

0x0200A138 SPMI_GENI_CFG_REG14**Type:** RW**Clock:** AHB_CLK**Reset State:** 0x00000000

Geni Configuration Register 14.

SPMI_GENI_CFG_REG14

Bits	Name	Description
23:22	PRIM_23_RX_DURATION	
21:20	PRIM_22_RX_DURATION	
19:18	PRIM_21_RX_DURATION	
17:16	PRIM_20_RX_DURATION	
15:14	PRIM_19_RX_DURATION	
13:12	PRIM_18_RX_DURATION	
11:10	PRIM_17_RX_DURATION	
9:8	PRIM_16_RX_DURATION	
7:6	PRIM_15_RX_DURATION	
5:4	PRIM_14_RX_DURATION	
3:2	PRIM_13_RX_DURATION	
1:0	PRIM_12_RX_DURATION	

0x0200A13C SPMI_GENI_CFG_REG15**Type:** RW**Clock:** AHB_CLK**Reset State:** 0x002DC896

Geni Configuration Register 15.

SPMI_GENI_CFG_REG15

Bits	Name	Description
23:21	PRIM_7_TX_FUNC_SEL	
20:18	PRIM_6_TX_FUNC_SEL	
17:15	PRIM_5_TX_FUNC_SEL	
14:12	PRIM_4_TX_FUNC_SEL	
11:9	PRIM_3_TX_FUNC_SEL	
8:6	PRIM_2_TX_FUNC_SEL	
5:3	PRIM_1_TX_FUNC_SEL	
2:0	PRIM_0_TX_FUNC_SEL	

0x0200A140 SPMI_GENI_CFG_REG16**Type:** RW**Clock:** AHB_CLK**Reset State:** 0x00A00001

Geni Configuration Register 16.

SPMI_GENI_CFG_REG16

Bits	Name	Description
23:21	PRIM_15_TX_FUNC_SEL	
20:18	PRIM_14_TX_FUNC_SEL	
17:15	PRIM_13_TX_FUNC_SEL	
14:12	PRIM_12_TX_FUNC_SEL	
11:9	PRIM_11_TX_FUNC_SEL	
8:6	PRIM_10_TX_FUNC_SEL	
5:3	PRIM_9_TX_FUNC_SEL	
2:0	PRIM_8_TX_FUNC_SEL	

0x0200A144 SPMI_GENI_CFG_REG17**Type:** RW**Clock:** AHB_CLK**Reset State:** 0x00000007

Geni Configuration Register 17.

SPMI_GENI_CFG_REG17

Bits	Name	Description
23:21	PRIM_23_TX_FUNC_SEL	
20:18	PRIM_22_TX_FUNC_SEL	
17:15	PRIM_21_TX_FUNC_SEL	
14:12	PRIM_20_TX_FUNC_SEL	
11:9	PRIM_19_TX_FUNC_SEL	
8:6	PRIM_18_TX_FUNC_SEL	
5:3	PRIM_17_TX_FUNC_SEL	
2:0	PRIM_16_TX_FUNC_SEL	

0x0200A148 SPMI_GENI_CFG_REG18**Type:** RW**Clock:** AHB_CLK**Reset State:** 0x00000082

Geni Configuration Register 18.

SPMI_GENI_CFG_REG18

Bits	Name	Description
23	PRIM_23_TX_PAR_FIRST	
22	PRIM_22_TX_PAR_FIRST	
21	PRIM_21_TX_PAR_FIRST	
20	PRIM_20_TX_PAR_FIRST	
19	PRIM_19_TX_PAR_FIRST	
18	PRIM_18_TX_PAR_FIRST	
17	PRIM_17_TX_PAR_FIRST	
16	PRIM_16_TX_PAR_FIRST	
15	PRIM_15_TX_PAR_FIRST	
14	PRIM_14_TX_PAR_FIRST	
13	PRIM_13_TX_PAR_FIRST	
12	PRIM_12_TX_PAR_FIRST	
11	PRIM_11_TX_PAR_FIRST	
10	PRIM_10_TX_PAR_FIRST	

SPMI_GENI_CFG_REG18 (cont.)

Bits	Name	Description
9	PRIM_9_TX_PAR_FIRST	
8	PRIM_8_TX_PAR_FIRST	
7	PRIM_7_TX_PAR_FIRST	
6	PRIM_6_TX_PAR_FIRST	
5	PRIM_5_TX_PAR_FIRST	
4	PRIM_4_TX_PAR_FIRST	
3	PRIM_3_TX_PAR_FIRST	
2	PRIM_2_TX_PAR_FIRST	
1	PRIM_1_TX_PAR_FIRST	
0	PRIM_0_TX_PAR_FIRST	

0x0200A14C SPMI_GENI_CFG_REG19**Type:** RW**Clock:** AHB_CLK**Reset State:** 0x00000000

Geni Configuration Register 19.

SPMI_GENI_CFG_REG19

Bits	Name	Description
23	PRIM_23_TX_DUMMY_DATA_EN	
22	PRIM_22_TX_DUMMY_DATA_EN	
21	PRIM_21_TX_DUMMY_DATA_EN	
20	PRIM_20_TX_DUMMY_DATA_EN	
19	PRIM_19_TX_DUMMY_DATA_EN	
18	PRIM_18_TX_DUMMY_DATA_EN	
17	PRIM_17_TX_DUMMY_DATA_EN	
16	PRIM_16_TX_DUMMY_DATA_EN	

SPMI_GENI_CFG_REG19 (cont.)

Bits	Name	Description
15	PRIM_15_TX_DUMMY_DATA_EN	
14	PRIM_14_TX_DUMMY_DATA_EN	
13	PRIM_13_TX_DUMMY_DATA_EN	
12	PRIM_12_TX_DUMMY_DATA_EN	
11	PRIM_11_TX_DUMMY_DATA_EN	
10	PRIM_10_TX_DUMMY_DATA_EN	
9	PRIM_9_TX_DUMMY_DATA_EN	
8	PRIM_8_TX_DUMMY_DATA_EN	
7	PRIM_7_TX_DUMMY_DATA_EN	
6	PRIM_6_TX_DUMMY_DATA_EN	
5	PRIM_5_TX_DUMMY_DATA_EN	
4	PRIM_4_TX_DUMMY_DATA_EN	
3	PRIM_3_TX_DUMMY_DATA_EN	
2	PRIM_2_TX_DUMMY_DATA_EN	
1	PRIM_1_TX_DUMMY_DATA_EN	
0	PRIM_0_TX_DUMMY_DATA_EN	

0x0200A150 SPMI_GENI_CFG_REG20**Type:** RW**Clock:** AHB_CLK**Reset State:** 0x00000008

Geni Configuration Register 20.

SPMI_GENI_CFG_REG20

Bits	Name	Description
23	PRIM_23_TX_DUMMY_PAR AM_EN	
22	PRIM_22_TX_DUMMY_PAR AM_EN	
21	PRIM_21_TX_DUMMY_PAR AM_EN	
20	PRIM_20_TX_DUMMY_PAR AM_EN	
19	PRIM_19_TX_DUMMY_PAR AM_EN	
18	PRIM_18_TX_DUMMY_PAR AM_EN	
17	PRIM_17_TX_DUMMY_PAR AM_EN	
16	PRIM_16_TX_DUMMY_PAR AM_EN	
15	PRIM_15_TX_DUMMY_PAR AM_EN	
14	PRIM_14_TX_DUMMY_PAR AM_EN	
13	PRIM_13_TX_DUMMY_PAR AM_EN	
12	PRIM_12_TX_DUMMY_PAR AM_EN	
11	PRIM_11_TX_DUMMY_PAR AM_EN	
10	PRIM_10_TX_DUMMY_PAR AM_EN	
9	PRIM_9_TX_DUMMY_PARA M_EN	
8	PRIM_8_TX_DUMMY_PARA M_EN	
7	PRIM_7_TX_DUMMY_PARA M_EN	
6	PRIM_6_TX_DUMMY_PARA M_EN	
5	PRIM_5_TX_DUMMY_PARA M_EN	
4	PRIM_4_TX_DUMMY_PARA M_EN	

SPMI_GENI_CFG_REG20 (cont.)

Bits	Name	Description
3	PRIM_3_TX_DUMMY_PARAM_EN	
2	PRIM_2_TX_DUMMY_PARAM_EN	
1	PRIM_1_TX_DUMMY_PARAM_EN	
0	PRIM_0_TX_DUMMY_PARAM_EN	

0x0200A154 SPMI_GENI_CFG_REG21**Type:** RW**Clock:** AHB_CLK**Reset State:** 0x00004400

Geni Configuration Register 21.

SPMI_GENI_CFG_REG21

Bits	Name	Description
23	PRIM_23_RX_SI_EN	
22	PRIM_22_RX_SI_EN	
21	PRIM_21_RX_SI_EN	
20	PRIM_20_RX_SI_EN	
19	PRIM_19_RX_SI_EN	
18	PRIM_18_RX_SI_EN	
17	PRIM_17_RX_SI_EN	
16	PRIM_16_RX_SI_EN	
15	PRIM_15_RX_SI_EN	
14	PRIM_14_RX_SI_EN	
13	PRIM_13_RX_SI_EN	
12	PRIM_12_RX_SI_EN	
11	PRIM_11_RX_SI_EN	
10	PRIM_10_RX_SI_EN	
9	PRIM_9_RX_SI_EN	
8	PRIM_8_RX_SI_EN	
7	PRIM_7_RX_SI_EN	

SPMI_GENI_CFG_REG21 (cont.)

Bits	Name	Description
6	PRIM_6_RX_SI_EN	
5	PRIM_5_RX_SI_EN	
4	PRIM_4_RX_SI_EN	
3	PRIM_3_RX_SI_EN	
2	PRIM_2_RX_SI_EN	
1	PRIM_1_RX_SI_EN	
0	PRIM_0_RX_SI_EN	

0x0200A158 SPMI_GENI_CFG_REG22**Type:** RW**Clock:** AHB_CLK**Reset State:** 0x00005C00

Geni Configuration Register 22.

SPMI_GENI_CFG_REG22

Bits	Name	Description
23	PRIM_23_RX_PAR_EN	
22	PRIM_22_RX_PAR_EN	
21	PRIM_21_RX_PAR_EN	
20	PRIM_20_RX_PAR_EN	
19	PRIM_19_RX_PAR_EN	
18	PRIM_18_RX_PAR_EN	
17	PRIM_17_RX_PAR_EN	
16	PRIM_16_RX_PAR_EN	
15	PRIM_15_RX_PAR_EN	
14	PRIM_14_RX_PAR_EN	
13	PRIM_13_RX_PAR_EN	
12	PRIM_12_RX_PAR_EN	
11	PRIM_11_RX_PAR_EN	
10	PRIM_10_RX_PAR_EN	
9	PRIM_9_RX_PAR_EN	
8	PRIM_8_RX_PAR_EN	

SPMI_GENI_CFG_REG22 (cont.)

Bits	Name	Description
7	PRIM_7_RX_PAR_EN	
6	PRIM_6_RX_PAR_EN	
5	PRIM_5_RX_PAR_EN	
4	PRIM_4_RX_PAR_EN	
3	PRIM_3_RX_PAR_EN	
2	PRIM_2_RX_PAR_EN	
1	PRIM_1_RX_PAR_EN	
0	PRIM_0_RX_PAR_EN	

0x0200A15C SPMI_GENI_CFG_REG23**Type:** RW**Clock:** AHB_CLK**Reset State:** 0x00000400

Geni Configuration Register 23.

SPMI_GENI_CFG_REG23

Bits	Name	Description
23	PRIM_23_RX_PAR_FIRST	
22	PRIM_22_RX_PAR_FIRST	
21	PRIM_21_RX_PAR_FIRST	
20	PRIM_20_RX_PAR_FIRST	
19	PRIM_19_RX_PAR_FIRST	
18	PRIM_18_RX_PAR_FIRST	
17	PRIM_17_RX_PAR_FIRST	
16	PRIM_16_RX_PAR_FIRST	
15	PRIM_15_RX_PAR_FIRST	
14	PRIM_14_RX_PAR_FIRST	
13	PRIM_13_RX_PAR_FIRST	
12	PRIM_12_RX_PAR_FIRST	
11	PRIM_11_RX_PAR_FIRST	
10	PRIM_10_RX_PAR_FIRST	
9	PRIM_9_RX_PAR_FIRST	

SPMI_GENI_CFG_REG23 (cont.)

Bits	Name	Description
8	PRIM_8_RX_PAR_FIRST	
7	PRIM_7_RX_PAR_FIRST	
6	PRIM_6_RX_PAR_FIRST	
5	PRIM_5_RX_PAR_FIRST	
4	PRIM_4_RX_PAR_FIRST	
3	PRIM_3_RX_PAR_FIRST	
2	PRIM_2_RX_PAR_FIRST	
1	PRIM_1_RX_PAR_FIRST	
0	PRIM_0_RX_PAR_FIRST	

0x0200A160 SPMI_GENI_CFG_REG24**Type:** RW**Clock:** AHB_CLK**Reset State:** 0x00001800

Geni Configuration Register 24.

SPMI_GENI_CFG_REG24

Bits	Name	Description
23	PRIM_23_ACC_PAR_EN	
22	PRIM_22_ACC_PAR_EN	
21	PRIM_21_ACC_PAR_EN	
20	PRIM_20_ACC_PAR_EN	
19	PRIM_19_ACC_PAR_EN	
18	PRIM_18_ACC_PAR_EN	
17	PRIM_17_ACC_PAR_EN	
16	PRIM_16_ACC_PAR_EN	
15	PRIM_15_ACC_PAR_EN	
14	PRIM_14_ACC_PAR_EN	
13	PRIM_13_ACC_PAR_EN	
12	PRIM_12_ACC_PAR_EN	
11	PRIM_11_ACC_PAR_EN	
10	PRIM_10_ACC_PAR_EN	

SPMI_GENI_CFG_REG24 (cont.)

Bits	Name	Description
9	PRIM_9_ACC_PAR_EN	
8	PRIM_8_ACC_PAR_EN	
7	PRIM_7_ACC_PAR_EN	
6	PRIM_6_ACC_PAR_EN	
5	PRIM_5_ACC_PAR_EN	
4	PRIM_4_ACC_PAR_EN	
3	PRIM_3_ACC_PAR_EN	
2	PRIM_2_ACC_PAR_EN	
1	PRIM_1_ACC_PAR_EN	
0	PRIM_0_ACC_PAR_EN	

0x0200A164 SPMI_GENI_CFG_REG25**Type:** RW**Clock:** AHB_CLK**Reset State:** 0x00555555

Geni Configuration Register 25.

SPMI_GENI_CFG_REG25

Bits	Name	Description
23:22	PRIM_11_IO2_FUNC_SEL	
21:20	PRIM_10_IO2_FUNC_SEL	
19:18	PRIM_9_IO2_FUNC_SEL	
17:16	PRIM_8_IO2_FUNC_SEL	
15:14	PRIM_7_IO2_FUNC_SEL	
13:12	PRIM_6_IO2_FUNC_SEL	
11:10	PRIM_5_IO2_FUNC_SEL	
9:8	PRIM_4_IO2_FUNC_SEL	
7:6	PRIM_3_IO2_FUNC_SEL	
5:4	PRIM_2_IO2_FUNC_SEL	
3:2	PRIM_1_IO2_FUNC_SEL	
1:0	PRIM_0_IO2_FUNC_SEL	

0x0200A168 SPMI_GENI_CFG_REG26**Type:** RW**Clock:** AHB_CLK**Reset State:** 0x00000290

Geni Configuration Register 26.

SPMI_GENI_CFG_REG26

Bits	Name	Description
23:22	PRIM_23_IO2_FUNC_SEL	
21:20	PRIM_22_IO2_FUNC_SEL	
19:18	PRIM_21_IO2_FUNC_SEL	
17:16	PRIM_20_IO2_FUNC_SEL	
15:14	PRIM_19_IO2_FUNC_SEL	
13:12	PRIM_18_IO2_FUNC_SEL	
11:10	PRIM_17_IO2_FUNC_SEL	
9:8	PRIM_16_IO2_FUNC_SEL	
7:6	PRIM_15_IO2_FUNC_SEL	
5:4	PRIM_14_IO2_FUNC_SEL	
3:2	PRIM_13_IO2_FUNC_SEL	
1:0	PRIM_12_IO2_FUNC_SEL	

0x0200A16C SPMI_GENI_CFG_REG27**Type:** RW**Clock:** AHB_CLK**Reset State:** 0x00000000

Geni Configuration Register 27.

SPMI_GENI_CFG_REG27

Bits	Name	Description
23	PRIM_23_TX_FORCE_DATA_VALUE	
22	PRIM_22_TX_FORCE_DATA_VALUE	
21	PRIM_21_TX_FORCE_DATA_VALUE	

SPMI_GENI_CFG_REG27 (cont.)

Bits	Name	Description
20	PRIM_20_TX_FORCE_DATA_VALUE	
19	PRIM_19_TX_FORCE_DATA_VALUE	
18	PRIM_18_TX_FORCE_DATA_VALUE	
17	PRIM_17_TX_FORCE_DATA_VALUE	
16	PRIM_16_TX_FORCE_DATA_VALUE	
15	PRIM_15_TX_FORCE_DATA_VALUE	
14	PRIM_14_TX_FORCE_DATA_VALUE	
13	PRIM_13_TX_FORCE_DATA_VALUE	
12	PRIM_12_TX_FORCE_DATA_VALUE	
11	PRIM_11_TX_FORCE_DATA_VALUE	
10	PRIM_10_TX_FORCE_DATA_VALUE	
9	PRIM_9_TX_FORCE_DATA_VALUE	
8	PRIM_8_TX_FORCE_DATA_VALUE	
7	PRIM_7_TX_FORCE_DATA_VALUE	
6	PRIM_6_TX_FORCE_DATA_VALUE	
5	PRIM_5_TX_FORCE_DATA_VALUE	
4	PRIM_4_TX_FORCE_DATA_VALUE	
3	PRIM_3_TX_FORCE_DATA_VALUE	
2	PRIM_2_TX_FORCE_DATA_VALUE	
1	PRIM_1_TX_FORCE_DATA_VALUE	

SPMI_GENI_CFG_REG27 (cont.)

Bits	Name	Description
0	PRIM_0_TX_FORCE_DATA_VALUE	

0x0200A170 SPMI_GENI_CFG_REG28**Type:** RW**Clock:** AHB_CLK**Reset State:** 0x00000000

Geni Configuration Register 28.

SPMI_GENI_CFG_REG28

Bits	Name	Description
23	PRIM_23_RX_FORCE_DATA_VALUE	
22	PRIM_22_RX_FORCE_DATA_VALUE	
21	PRIM_21_RX_FORCE_DATA_VALUE	
20	PRIM_20_RX_FORCE_DATA_VALUE	
19	PRIM_19_RX_FORCE_DATA_VALUE	
18	PRIM_18_RX_FORCE_DATA_VALUE	
17	PRIM_17_RX_FORCE_DATA_VALUE	
16	PRIM_16_RX_FORCE_DATA_VALUE	
15	PRIM_15_RX_FORCE_DATA_VALUE	
14	PRIM_14_RX_FORCE_DATA_VALUE	
13	PRIM_13_RX_FORCE_DATA_VALUE	
12	PRIM_12_RX_FORCE_DATA_VALUE	
11	PRIM_11_RX_FORCE_DATA_VALUE	

SPMI_GENI_CFG_REG28 (cont.)

Bits	Name	Description
10	PRIM_10_RX_FORCE_DATA_VALUE	
9	PRIM_9_RX_FORCE_DATA_VALUE	
8	PRIM_8_RX_FORCE_DATA_VALUE	
7	PRIM_7_RX_FORCE_DATA_VALUE	
6	PRIM_6_RX_FORCE_DATA_VALUE	
5	PRIM_5_RX_FORCE_DATA_VALUE	
4	PRIM_4_RX_FORCE_DATA_VALUE	
3	PRIM_3_RX_FORCE_DATA_VALUE	
2	PRIM_2_RX_FORCE_DATA_VALUE	
1	PRIM_1_RX_FORCE_DATA_VALUE	
0	PRIM_0_RX_FORCE_DATA_VALUE	

0x0200A174 SPMI_GENI_CFG_REG29**Type:** RW**Clock:** AHB_CLK**Reset State:** 0x00000000

Geni Configuration Register 29.

SPMI_GENI_CFG_REG29

Bits	Name	Description
23	PRIM_23_TX_TIME_COUNTER_EN	
22	PRIM_22_TX_TIME_COUNTER_EN	
21	PRIM_21_TX_TIME_COUNTER_EN	

SPMI_GENI_CFG_REG29 (cont.)

Bits	Name	Description
20	PRIM_20_TX_TIME_COUNT ER_EN	
19	PRIM_19_TX_TIME_COUNT ER_EN	
18	PRIM_18_TX_TIME_COUNT ER_EN	
17	PRIM_17_TX_TIME_COUNT ER_EN	
16	PRIM_16_TX_TIME_COUNT ER_EN	
15	PRIM_15_TX_TIME_COUNT ER_EN	
14	PRIM_14_TX_TIME_COUNT ER_EN	
13	PRIM_13_TX_TIME_COUNT ER_EN	
12	PRIM_12_TX_TIME_COUNT ER_EN	
11	PRIM_11_TX_TIME_COUNT ER_EN	
10	PRIM_10_TX_TIME_COUNT ER_EN	
9	PRIM_9_TX_TIME_COUNT ER_EN	
8	PRIM_8_TX_TIME_COUNT ER_EN	
7	PRIM_7_TX_TIME_COUNT ER_EN	
6	PRIM_6_TX_TIME_COUNT ER_EN	
5	PRIM_5_TX_TIME_COUNT ER_EN	
4	PRIM_4_TX_TIME_COUNT ER_EN	
3	PRIM_3_TX_TIME_COUNT ER_EN	
2	PRIM_2_TX_TIME_COUNT ER_EN	
1	PRIM_1_TX_TIME_COUNT ER_EN	

SPMI_GENI_CFG_REG29 (cont.)

Bits	Name	Description
0	PRIM_0_TX_TIME_COUNT ER_EN	

0x0200A178 SPMI_GENI_CFG_REG30**Type:** RW**Clock:** AHB_CLK**Reset State:** 0x00000000

Geni Configuration Register 30.

SPMI_GENI_CFG_REG30

Bits	Name	Description
23	PRIM_23_RX_TIME_COUNTER_EN	
22	PRIM_22_RX_TIME_COUNTER_EN	
21	PRIM_21_RX_TIME_COUNTER_EN	
20	PRIM_20_RX_TIME_COUNTER_EN	
19	PRIM_19_RX_TIME_COUNTER_EN	
18	PRIM_18_RX_TIME_COUNTER_EN	
17	PRIM_17_RX_TIME_COUNTER_EN	
16	PRIM_16_RX_TIME_COUNTER_EN	
15	PRIM_15_RX_TIME_COUNTER_EN	
14	PRIM_14_RX_TIME_COUNTER_EN	
13	PRIM_13_RX_TIME_COUNTER_EN	
12	PRIM_12_RX_TIME_COUNTER_EN	
11	PRIM_11_RX_TIME_COUNTER_EN	

SPMI_GENI_CFG_REG30 (cont.)

Bits	Name	Description
10	PRIM_10_RX_TIME_COUNTER_EN	
9	PRIM_9_RX_TIME_COUNTER_EN	
8	PRIM_8_RX_TIME_COUNTER_EN	
7	PRIM_7_RX_TIME_COUNTER_EN	
6	PRIM_6_RX_TIME_COUNTER_EN	
5	PRIM_5_RX_TIME_COUNTER_EN	
4	PRIM_4_RX_TIME_COUNTER_EN	
3	PRIM_3_RX_TIME_COUNTER_EN	
2	PRIM_2_RX_TIME_COUNTER_EN	
1	PRIM_1_RX_TIME_COUNTER_EN	
0	PRIM_0_RX_TIME_COUNTER_EN	

**0x0200A200+ SPMI_GENI_CFG_RAMn, n=[0..255]
0x4*n**

Type: RW

Clock: AHB_CLK

Reset State: 0x00000000

GENI Program RAM set of registers.

SPMI_GENI_CFG_RAMn

Bits	Name	Description
20:0	TBD	

11.1.2 SPMI_CFG

SPMI Configuration

11.1.2.1 SPMI_CFG

SPMI Configuration

**0x0200A700+ SPMI_PERIPHm_2OWNER_TABLE_REG, m=[0..255]
0x4*m**

Type: RW

Clock: AHB_CLK

Reset State: 0x00000000

Peripheral 2 owner Table; Each Peripheral has 4 bits indicating its associated owner. The APID2PPID is used verify that each mapping table request is given the correct APID

SPMI_PERIPHm_2OWNER_TABLE_REG

Bits	Name	Type	Description
27:16	APID2PPID	R	APID to PPID mapping
2:0	PERIPH2OWNER	RW	Owner index for peripheral m.

**0x0200AB00+ SPMI_MAPPING_TABLE_REGk, k=[0..254]
0x4*k**

Type: RW

Clock: AHB_CLK

Reset State: 0x00000000

Mapping table register; this table is used for conversion of the 12-bit physical peripheral ID (4-bit SID + 8 high address bits, PPID) to the 8-bit application peripheral ID (APID).

SPMI_MAPPING_TABLE_REGk

Bits	Name	Description
21:18	BIT_INDEX	Mapping logic table contents.
17	BRANCH_RESULT_0_FLAG	Mapping logic table contents.
16:9	BRANCH_RESULT_0	Mapping logic table contents.
8	BRANCH_RESULT_1_FLAG	Mapping logic table contents.
7:0	BRANCH_RESULT_1	Mapping logic table contents.

0x0200AF00 SPMI_MID_REG**Type:** RW**Clock:** AHB_CLK**Reset State:** 0x00000000

This register defines the MID (master ID) of the SPMI master on the bus.

SPMI_MID_REG

Bits	Name	Description
1:0	MID	This register contains the SPMI master MID.

0x0200AF04 SPMI_CFG_REG**Type:** RW**Clock:** AHB_CLK**Reset State:** 0x00000020

This register holds different configuration of the SPMI-controller.

SPMI_CFG_REG

Bits	Name	Description
16:12	ARBITER_CTRL	Reserved bits for arbiter control need.
11	ASYNC_SSC_DET_INT_DIS	when '0'(default) mean the spmi_m async_ssc_detect use the internal indication for SSC,this can be used also for multi master (unlike ssc_detect).
10	SSC_Q1_DELAY_DIS	when set,disable the delay that placed between Q1 to Q2 regs in the SSC_DETECT
9	ARB_UNKNOWN_EN	when '0'(default) mean that if ssc_window_en='1',ssc will be detect only when master_win slave_win (suitable for single master) if it set to '1'ssc will be detected when master_win slave_win arb_unknown(suitable for multi master).if ssc_window_en='0' arb_unknown_en is dont_care.
8	SSC_WINDOW_EN	when set ,SSC is detected only in specific window when it is expected in order to reduce glitch risk
7	SSC_DET_INT_DIS	when '0'(default) mean the spmi_m use the internal indication for SSC (suitable for single master ,when set use the external bus for ssc detection suitable for multi master).
6	FORCE_MPM_CLK_REQ_MM	When set request clock for cofiguration, to enable transferring the values to the ser_clock domain
5	FORCE_ARB_AFTER_MASTER_TO	When set, BOM starts an arbitration after false master arbitration (i.e. SSC hasn't been asserted within 64 us after an arbitration win.

SPMI_CFG_REG (cont.)

Bits	Name	Description
4	BUS_IDLE_CONN_MODE	When asserted, sdata is deasserted on a rising edge when connecting on bus idle. When deasserted, sdata is deasserted on the falling edge.
3	FORCE_MASTER_WRITE_ON_ERROR	When set, the arbiter will treat a slave command as master write in case of a parity error in a command frame.
2	FORCE_MPM_CLK_REQ	When set, the mpm_clk_req is high.
1	ARBITER_BYPASS	Set to enable bypass of SPMI bus arbitration. Used in case where SPMI-controller is a single master and there are no RCS slaves on the bus. Clear to disable the bypass.
0	ARBITER_ENABLE	Set to enable arbitration process (MID must be valid at time of assertion). All other bits in SPMI_CFG_REG are allowed to change value only while ARBITER_ENABLE bit remains low. Deassertion of this bit must be followed by arst to avoid errors

0x0200AF08 SPMI_SEC_DISABLE_REG**Type:** RW**Clock:** AHB_CLK**Reset State:** 0x00000000

This register disables the use of all **security** features. Used for boot initialization.

SPMI_SEC_DISABLE_REG

Bits	Name	Description
0	DISABLE_ SECURITY	Set to '1' to disable and bypass all security features. Set to '0' to enable them.

0x0200AF10 SPMI_CGC_CTRL**Type:** RW**Clock:** AHB_CLK**Reset State:** 0x00000000

SPMI clock gating control register.

SPMI_CGC_CTRL

Bits	Name	Description
6	MAP_LOGIC_CLK_CGC_ON	When set, HW CGC control enabled. When clear, the CGC is always open
5	RPU_CLK_CGC_ON	When set, HW CGC control enabled. When clear, the CGC is always open
4	MWB_CLK_CGC_ON	When set, HW CGC control enabled. When clear, the CGC is always open
3	PIC_CLK_CGC_ON	When set, HW CGC control enabled. When clear, the CGC is always open
2	PAC_CLK_CGC_ON	When set, HW CGC control enabled. When clear, the CGC is always open
1	CFG_AHB_BRIDGE_WRC_LK_CGC_ON	When set, HW CGC control enabled. When clear, the CGC is always open
0	CFG_AHB_BRIDGE_CLK_CGC_ON	When set, HW CGC control enabled. When clear, the CGC is always open

0x0200AF14 SPMI_MWB_ENABLE_REG**Type:** RW**Clock:** AHB_CLK**Reset State:** 0x00000000

This register enables master write buffer. It is used in order to prevent receiving an PMIC interrupt until the PIC database is initialized..

SPMI_MWB_ENABLE_REG

Bits	Name	Description
0	MWB_ENABLE	Set to '1' to enable master write buffer (i.e. allow PMIC interrupts to enter the PIC. Set to '0' to disable.

0x0200AF54 SPMI_CMPR_EN_REG**Type:** RW**Clock:** AHB_CLK**Reset State:** 0x00000000**SPMI_CMPR_EN_REG**

Bits	Name	Description
0	CMPR_ENABLE	

0x0200B000 SPMI_PROTOCOL_IRQ_STATUS**Type:** R**Clock:** AHB_CLK**Reset State:** 0x00000000

SPMI-protocol interrupt status register. The bits in this register are sticky. In order to clear some bit in this register, corresponding bit should be set in the SPMI_PROTOCOL_IRQ_CLEAR register.

SPMI_PROTOCOL_IRQ_STATUS

Bits	Name	Description
11	ARBITER_DISCONNECTED	Disconnection was detected.
10	ARBITER_CONNECTED	Connection was detected.
9	PERIH_IRQ_LOST	A Peripheral interrupt was lost.
8	UNEXPECTED_SSC	Unexpected SSC was detected.
7	NO_RESPONSE_DATA_FRAME_DETECTED	No response data-frame was detected.
6	NO_RESPONSE_CMD_FRAME_DETECTED	No response frame was detected.
5	FALSE_MASTER_ARBITRATION_WIN	Master has won the arbitration but didn't initiate any command sequence within TBT.
4	FALSE_BUS_REQUEST	Bus arbitration request was asserted but no master/slave won.
3	UNSUPPORTED_COMMAND	Unsupported SPMI command was detected.
2	DATA_ADDR_FRAME_PARITY_ERROR	A parity error was detected in a data or address frame driven by an external master/slave.
1	SLAVE_CMD_FRAME_PARITY_ERROR	A parity error was detected in a command-frame driven by a slave.
0	MASTER_CMD_FRAME_PARITY_ERROR	A parity error was detected in a command-frame driven by an external master.

0x0200B004 SPMI_PROTOCOL_IRQ_ENABLE**Type:** RW**Clock:** AHB_CLK**Reset State:** 0x00000000

SPMI protocol interrupt enable register. This register is used to enable the corresponding functions in the SPMI_PROTOCOL_IRQ_STATUS register. Setting (1) a bit in the SPMI_PROTOCOL_IRQ_ENABLE register causes an interrupt to be generated, if the corresponding bit in the SPMI_PROTOCOL_IRQ_STATUS register is set. Clearing (0) a bit in

the SPMI_PROTOCOL_IRQ_ENABLE register causes the setting of the corresponding bit in the SPMI_PROTOCOL_IRQ_STATUS register to have no effect on the interrupt.

SPMI_PROTOCOL_IRQ_ENABLE

Bits	Name	Description
11	ARBITER_DISCONNECTED	Enable bit of ARBITER_DISCONNECTED interrupt.
10	ARBITER_CONNECTED	Enable bit of ARBITER_CONNECTED interrupt.
9	PERIH_IRQ_LOST	Enable bit of PERIH_IRQ_LOST interrupt.
8	UNEXPECTED_SSC	
7	NO_RESPONSE_DATA_FRAME_DETECTED	Enable bit of NO_RESPONSE_DATA_FRAME_DETECTED interrupt.
6	NO_RESPONSE_CMD_FRAME_DETECTED	Enable bit of NO_RESPONSE_CMD_FRAME_DETECTED interrupt.
5	FALSE_MASTER_ARBITRATION_WIN	Enable bit of FALSE_MASTER_ARBITRATION_WIN interrupt.
4	FALSE_BUS_REQUEST	Enable bit of FALSE_BUS_REQUEST interrupt.
3	UNSUPPORTED_COMMAND	Enable bit of UNSUPPORTED_COMMAND interrupt.
2	DATA_ADDR_FRAME_PARITY_ERROR	Enable bit of DATA_ADDR_FRAME_PARITY_ERROR interrupt.
1	SLAVE_CMD_FRAME_PARITY_ERROR	Enable bit of SLAVE_CMD_FRAME_PARITY_ERROR interrupt.
0	MASTER_CMD_FRAME_PARITY_ERROR	Enable bit of MASTER_CMD_FRAME_PARITY_ERROR interrupt.

0x0200B008 SPMI_PROTOCOL_IRQ_CLEAR

Type: W

Clock: AHB_CLK

Reset State: 0x00000000

SPMI protocol interrupt clear register. Setting one of this register bits generates a pulse which clears corresponding bit in SPMI_PROTOCOL_IRQ_STATUS register.

SPMI_PROTOCOL_IRQ_CLEAR

Bits	Name	Description
11	ARBITER_DISCONNECTED	Clear bit of ARBITER_DISCONNECTED interrupt.
10	ARBITER_CONNECTED	Clear bit of ARBITER_CONNECTED interrupt.
9	PERIH_IRQ_LOST	Clear bit of PERIH_IRQ_LOST interrupt.
8	UNEXPECTED_SSC	Clear bit of UNEXPECTED_SSC interrupt.

SPMI_PROTOCOL_IRQ_CLEAR (cont.)

Bits	Name	Description
7	NO_RESPONSE_DATA_FRAME_DETECTED	Clear bit of NO_RESPONSE_DATA_FRAME_DETECTED interrupt.
6	NO_RESPONSE_CMD_FRAME_DETECTED	Clear bit of NO_RESPONSE_CMD_FRAME_DETECTED interrupt.
5	FALSE_MASTER_ARBITRATION_WIN	Clear bit of FALSE_MASTER_ARBITRATION_WIN interrupt.
4	FALSE_BUS_REQUEST	Clear bit of FALSE_BUS_REQUEST interrupt.
3	UNSUPPORTED_COMMAND	Clear bit of UNSUPPORTED_COMMAND interrupt.
2	DATA_ADDR_FRAME_PARITY_ERROR	Clear bit of DATA_ADDR_FRAME_PARITY_ERROR interrupt.
1	SLAVE_CMD_FRAME_PARITY_ERROR	Clear bit of SLAVE_CMD_FRAME_PARITY_ERROR interrupt.
0	MASTER_CMD_FRAME_PARITY_ERROR	Clear bit of MASTER_CMD_FRAME_PARITY_ERROR interrupt.

0x0200B00C SPMI_PROTOCOL_IRQ_EN_SET**Type:** W**Clock:** AHB_CLK**Reset State:** 0x00000000

SPMI protocol interrupt enable set register. This register is used to set the corresponding bits in the SPMI_PROTOCOL_IRQ_ENABLE register. Setting one of this register bits generates a pulse which sets corresponding bit in SPMI_PROTOCOL_IRQ_ENABLE register.

SPMI_PROTOCOL_IRQ_EN_SET

Bits	Name	Description
11	ARBITER_DISCONNECTED	Set-Enable bit of ARBITER_DISCONNECTED interrupt.
10	ARBITER_CONNECTED	Set-Enable bit of ARBITER_CONNECTED interrupt.
9	PERIH_IRQ_LOST	Set-Enable bit of PERIH_IRQ_LOST interrupt.
8	UNEXPECTED_SSC	Set-Enable bit of UNEXPECTED_SSC interrupt.
7	NO_RESPONSE_DATA_FRAME_DETECTED	Set-Enable bit of NO_RESPONSE_DATA_FRAME_DETECTED interrupt.
6	NO_RESPONSE_CMD_FRAME_DETECTED	Set-Enable bit of NO_RESPONSE_CMD_FRAME_DETECTED interrupt.
5	FALSE_MASTER_ARBITRATION_WIN	Set-Enable bit of FALSE_MASTER_ARBITRATION_WIN interrupt.
4	FALSE_BUS_REQUEST	Set-Enable bit of FALSE_BUS_REQUEST interrupt.

SPMI_PROTOCOL_IRQ_EN_SET (cont.)

Bits	Name	Description
3	UNSUPPORTED_COMMAND	Set-Enable bit of UNSUPPORTED_COMMAND interrupt.
2	DATA_ADDR_FRAME_PARITY_ERROR	Set-Enable bit of DATA_ADDR_FRAME_PARITY_ERROR interrupt.
1	SLAVE_CMD_FRAME_PARITY_ERROR	Set-Enable bit of SLAVE_CMD_FRAME_PARITY_ERROR interrupt.
0	MASTER_CMD_FRAME_PARITY_ERROR	Set-Enable bit of MASTER_CMD_FRAME_PARITY_ERROR interrupt.

0x0200B010 SPMI_PROTOCOL_IRQ_EN_CLEAR**Type:** W**Clock:** AHB_CLK**Reset State:** 0x00000000

SPMI protocol interrupt enable clear register. This register is used to clear the corresponding bits in the SPMI_PROTOCOL_IRQ_ENABLE register. Setting one of this register bits generates a pulse which clears corresponding bit in SPMI_PROTOCOL_IRQ_ENABLE register.

SPMI_PROTOCOL_IRQ_EN_CLEAR

Bits	Name	Description
11	ARBITER_DISCONNECTED	Clear-Enable bit of ARBITER_DISCONNECTED interrupt.
10	ARBITER_CONNECTED	Clear-Enable bit of ARBITER_CONNECTED interrupt.
9	PERIH_IRQ_LOST	Clear-Enable bit of PERIH_IRQ_LOST interrupt.
8	UNEXPECTED_SSC	Clear-Enable bit of UNEXPECTED_SSC interrupt.
7	NO_RESPONSE_DATA_FRAME_DETECTED	Clear-Enable bit of NO_RESPONSE_DATA_FRAME_DETECTED interrupt.
6	NO_RESPONSE_CMD_FRAME_DETECTED	Clear-Enable bit of NO_RESPONSE_CMD_FRAME_DETECTED interrupt.
5	FALSE_MASTER_ARBITRATION_WIN	Clear-Enable bit of FALSE_MASTER_ARBITRATION_WIN interrupt.
4	FALSE_BUS_REQUEST	Clear-Enable bit of FALSE_BUS_REQUEST interrupt.
3	UNSUPPORTED_COMMAND	Clear-Enable bit of UNSUPPORTED_COMMAND interrupt.
2	DATA_ADDR_FRAME_PARITY_ERROR	Clear-Enable bit of DATA_ADDR_FRAME_PARITY_ERROR interrupt.
1	SLAVE_CMD_FRAME_PARITY_ERROR	Clear-Enable bit of SLAVE_CMD_FRAME_PARITY_ERROR interrupt.

SPMI_PROTOCOL_IRQ_EN_CLEAR (cont.)

Bits	Name	Description
0	MASTER_CMD_FRAME_PARITY_ERROR	Clear-Enable bit of MASTER_CMD_FRAME_PARITY_ERROR interrupt.

0x0200C004 SPMI_HW_SW_EVENTS_CTRL**Type:** RW**Clock:** AHB_CLK**Reset State:** 0x00000000

SPMI HW/SW Events selector

SPMI_HW_SW_EVENTS_CTRL

Bits	Name	Description
2:0	HW_SW_EVENTS_SEL	when 3'd1 spmi_hw_sw_events is spmi hw events, when 3'd2 spmi_hw_sw_events is spmi sw events, when 3'd3 spmi_hw_sw_events is geni hw events, when 3'd4 spmi_hw_sw_events is geni sw events, else, spmi_hw_sw_events is 32'd0

**0x0200C008+ SPMI_HW_SW_EVENTS_BITWISE_REGj, j=[0..7]
0x4*j****Type:** RW**Clock:** AHB_CLK**Reset State:** 0x00000000

SPMI HW/SW Bitwise Events selector - to be used instead of SPMI_HW_SW_EVENTS_CTRL.
for each bit in spmi_hw_sw_events there are 2 control bits, a selector and an enable.

SPMI_HW_SW_EVENTS_BITWISE_REGj

Bits	Name	Description
31	HW_EVENTS_EN_D	when 0, bit is disabled, when 1, bit is enabled
24	HW_EVENTS_SEL_D	when 0, selects SPMI_HW_EVENTS, when 1,selects GENI_HW_EVENTS
23	HW_EVENTS_EN_C	when 0, bit is disabled, when 1, bit is enabled
16	HW_EVENTS_SEL_C	when 0, selects SPMI_HW_EVENTS, when 1,selects GENI_HW_EVENTS
15	HW_EVENTS_EN_B	when 0, bit is disabled, when 1, bit is enabled

SPMI_HW_SW_EVENTS_BITWISE_REGj (cont.)

Bits	Name	Description
8	HW_EVENTS_SEL_B	when 0, selects SPMI_HW_EVENTS, when 1,selects GENI_HW_EVENTS
7	HW_EVENTS_EN_A	when 0, bit is disabled, when 1, bit is enabled
0	HW_EVENTS_SEL_A	when 0, selects SPMI_HW_EVENTS, when 1,selects GENI_HW_EVENTS

11.2 SPMI_PIC

**0x03900000+ SPMI_PIC_OWNERm_ACC_STATUSn, m=[0..5], n=[0..3]
4096*m+4*n**

Type: R

Clock: AHB_CLK

Reset State: 0x00000000

Accumulated status register.

SPMI_PIC_OWNERm_ACC_STATUSn

Bits	Name	Description
31:0	INT_ACC_STATUS	256 accumulated interrupt bits; one bit per peripheral. The n-th register contains accumulated interrupts of peripherals $[32*(n+1)-1:32*n]$. m defines the owner

**0x03800000+ SPMI_PIC_ACC_ENABLEn, n=[0..127]
4096*n**

Type: RW

Clock: AHB_CLK

Reset State: 0x00000000

Accumulated interrupt enable register.

SPMI_PIC_ACC_ENABLEn

Bits	Name	Description
0	INT_ACC_ENABLE	256 interrupt enable bits. Each bit enables a peripheral's accumulated interrupt.

**0x03800004+ SPMI_PIC_IRQ_STATUSn, n=[0..127]
4096*n**

Type: R
Clock: AHB_CLK
Reset State: 0x00000000

Peripheral interrupt status register.

SPMI_PIC_IRQ_STATUSn

Bits	Name	Description
7:0	INT_STATUS	256 bytes of interrupts. Byte per peripheral to indicate individual interrupts.

**0x03800008+ SPMI_PIC_IRQ_CLEARn, n=[0..127]
4096*n**

Type: W
Clock: AHB_CLK
Reset State: 0x00000000

Interrupt clear register.

SPMI_PIC_IRQ_CLEARn

Bits	Name	Description
7:0	INT_CLEAR	256 bytes of interrupt clear commands. Byte per peripheral to clear individual interrupts.

11.3 PMIC_ARB_CORE

The PMIC arbiter core are used to configure the PMIC arbiter itself and for software environments to issue commands over the SPMI bus.

11.3.1 PMIC_ARB_CORE_SW

The PMIC arbiter core are used to configure the PMIC arbiter itself and for software environments to issue commands over the SPMI bus.

0x0200F004 PMIC_ARB_MISC_CONFIG

Type: RW
Clock: HCLK
Reset State: 0x00000002

This register provides a master switch to globally enable/disable transaction done interrupts.

PMIC_ARB_MISC_CONFIG

Bits	Name	Description
1	CLK_GATE_EN	Enable/disable dynamic hardware clock gating for the PMIC arbiter. 1: Enable 0: Disable
0	DONE_IRQ_EN	Globally enable/disable all transaction done interrupts. 1: Enable 0: Disable

0x0200F008 PMIC_ARB_PVC_INTF_CTL

Type: RW

Clock: HCLK

Reset State: 0x00000000

This register is used to configure aspects of the PMIC arbiter's PVC interfaces. Global and per-port enable/disable control over PVC ports is provided.

PMIC_ARB_PVC_INTF_CTL

Bits	Name	Description
0	PVC_INTF_EN	Global enable/disable for all PVC ports.

0x0200F00C PMIC_ARB_PVC_INTF_STATUS

Type: R

Clock: HCLK

Reset State: 0x00000000

This register is used to provide status of the PMIC arbiter's PVC interface as a whole. Provides status as to whether any PVC port has a request that has been received and is not yet complete.

PMIC_ARB_PVC_INTF_STATUS

Bits	Name	Description
0	PVC_INTF_BUSY	Status bit indicating whether or not there is a currently outstanding PMIC command on a PVC interface.

**0x0200F100+ PMIC_ARB_PRIORITIESn, n=[0..5]
0x4*n****Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

This register specifies which arbiter port has priority n. Priority 0 is the highest priority, priority 1 is the next highest, and so on.

Arbiter ports are numbered as follows: software is port 0, PVC ports are numbered from 1 to the total number of PVC ports.

PMIC_ARB_PRIORITIESn

Bits	Name	Description
2:0	PORT	Number of the port with priority n.

**0x0200F200+ PMIC_ARB_PVC_PORTn_CTL, n=[0..4]
0x4*n****Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

This register controls the settings for individual PVC ports. The SPMI priority of commands from this port and whether or not the port is enabled are controlled by this register.

PMIC_ARB_PVC_PORTn_CTL

Bits	Name	Description
1	SPMI_PRIORITY	The priority the SPMI master should use when executing SPMI commands that are requested by this port. Note that this is not priority for the PMIC arbiter, but priority on the SPMI bus in the event that other (off-chip) devices are also trying to use the SPMI bus. 1: High priority 0: Low priority
0	PVC_PORT_EN	Enables/disables this particular PVC port. 1: Enable 0: Disable

**0x0200F280+ PMIC_ARB_PVC_PORTn_STATUS, n=[0..4]
0x4*n****Type:** R**Clock:** HCLK**Reset State:** 0x00000000

This register provides per-port status on the PMIC arbiter's PVC interface.

PMIC_ARB_PVC_PORTn_STATUS

Bits	Name	Description
1	PVC_PORT_FAILURE	Status bit indicating whether or not a PMIC command has failed due to an SPMI protocol error. This bit is cleared on a read of this register.
0	PVC_PORT_BUSY	Status bit indicating a PMIC command is in progress.

0x0200F400+ PMIC_ARB_PVCn_ADDRm, n=[0..4], m=[0..3]
0x20*n+0x4*
m

Type: RW

Clock: HCLK

Reset State: 0x00000000

PVC table entry for PVC port n, address index m.

PMIC_ARB_PVCn_ADDRm

Bits	Name	Description
19:16	SID	SPMI slave ID
15:0	ADDRESS	PMIC register address

0x0200F800+ PMIC_ARB_REG_CHNLn, n=[0..127]
0x4*n

Type: RW

Clock: HCLK

Reset State: 0x00000000

SW register address translation table.

PMIC_ARB_REG_CHNLn

Bits	Name	Description
19:16	SID	SPMI slave ID
15:8	ADDRESS	8 MSB bits of the registers present in the PMIC

11.4 PMIC_ARB_CORE_REGISTERS

The PMIC arbiter are for software environments to issue commands over the SPMI bus.

11.4.1 PMIC_ARB_CORE_SW_DEC_CHANNELS

The PMIC arbiter are for software environments to issue commands over the SPMI bus.

0x02400000+ PMIC_ARBq_CHNLn_CMD, q=[0..5], n=[0..127]
0x01000*q+0
x08000*n

Type: RW

Clock: HCLK

Reset State: 0x00000000

A write to this register triggers an SPMI command. This register is responsible for specifying all of the information needed to define the command to be performed (except for write data). The SPMI opcode, slave ID, address, and byte count are all written to this register and need to be set with a single write.

NOTE: A write to this register will queue up and SPMI command and until that command finished, further writes to this register will be ignored.

NOTE: The command register is write-only for PMIC arbiter 1.0.0 (will always read as zero).

PMIC_ARBq_CHNLn_CMD

Bits	Name	Type	Description
31:27	OPCODE	RW	SPMI opcode, as defined by the SPMI master. See the SPMI master documents for further details. 0x0: EXTENDED_REG_WRITE_LONG 0x1: EXTENDED_REG_READ_LONG 0x2: EXTENDED_REG_WRITE 0x3: RESET 0x4: SLEEP 0x5: SHUTDOWN 0x6: WAKEUP 0x7: AUTHENTICATE 0x8: MASTER_READ 0x9: MASTER_WRITE 0xA: TRANSFER_BUS_OWNERSHIP 0xB: DEVICE_DESC_BLOCK_MASTER 0xC: DEVICE_DESC_BLOCK_SLAVE 0xD: EXTENDED_REGISTER_READ 0xE: REGISTER_WRITE 0xF: REGISTER_READ 0x10: REGISTER_0_WRITE
26	PRIORITY	W	Controls whether the SPMI command should be sent with high or low priority. 1: High priority 0: Low priority

PMIC_ARBq_CHNLn_CMD (cont.)

Bits	Name	Type	Description
11:4	ADDRESS	W	8 LSB bits of address for the register the SPMI command is to access.
2:0	BYTE_CNT	W	Indicates the number of bytes to read or write for operations that span multiple bytes. For single-byte operations, as well as operations other than read/write operations, this should be set to zero. 111: Eight bytes 110: Seven bytes 101: Six bytes 100: Five bytes 011: Four bytes 010: Three bytes 001: Two bytes 000: One byte

0x02400004+ PMIC_ARBq_CHNLn_CONFIG, q=[0..5], n=[0..127]**0x01000*q+0****x08000*n****Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

Configuration register for software channels: provides the ability to enable/disable interrupts indicating transactions are done on a per-channel basis.

PMIC_ARBq_CHNLn_CONFIG

Bits	Name	Description
0	IRQ_EN	Enable/disable transaction done interrupt for channel n. 1: Enable 0: Disable

0x02400008+ PMIC_ARBq_CHNLn_STATUS, q=[0..5], n=[0..127]**0x01000*q+0****x08000*n****Type:** R**Clock:** HCLK**Reset State:** 0x00000001

Provides status of the most recent SPMI command issued to channel n.

PMIC_ARBq_CHNLn_STATUS

Bits	Name	Description
3	DROPPED	Resets to 0 upon the issue of the new command. 1: Indicates that a write to this channel illegally occurred during handling of the last command and was subsequently dropped. 0: Indicates that no writes to the channel occurred while the most recent command was being handled.
2	DENIED	Resets to 0 upon the issue of a new command. This bit should only be considered when DONE=1. 1: Indicates that the most recent command was denied by the SPMI master. 0: Indicates that the most recent command was not denied.
1	FAILURE	Resets to 0 upon the issue of a new command. This bit should only be considered when DONE=1. 1: Indicates that the most recent command has failed. 0: Indicates that the most recent command has completed successfully.
0	DONE	Resets to 0 upon the issue of a new command. This bit indicates completion, but not necessarily success, and should be observed in conjunction with DENIED and FAILED to determine the status of an operation. 1: Indicates that the most recent command is finished. 0: Indicates that the most recent command is still pending.

0x02400010+ PMIC_ARBq_CHNLn_WDATA0, q=[0..5], n=[0..127]
0x01000*q+0
x08000*n

Type: RW

Clock: HCLK

Reset State: 0x00000000

Stores the lower 32-bits of data for an SPMI write. Starting with least significant bytes first, PMIC arbiter will grab only the bytes it needs for the write operation as specified in the BYTE_CNT of the command register.

PMIC_ARBq_CHNLn_WDATA0

Bits	Name	Description
31:0	DATA	Data to write to PMIC.

0x02400014+ PMIC_ARBq_CHNLn_WDATA1, q=[0..5], n=[0..127]
0x01000*q+0
x08000*n

Type: RW

Clock: HCLK

Reset State: 0x00000000

Stores the upper 32-bits of data for an SPMI write. Starting with least significant bytes first, PMIC arbiter will grab only the bytes it needs for the write operation as specified in the BYTE_CNT of the command register. Thus, this register is only needed for write operations larger than four bytes.

PMIC_ARBq_CHNLn_WDATA1

Bits	Name	Description
31:0	DATA	Data to write to PMIC.

0x02400018+ PMIC_ARBq_CHNLn_RDATA0, q=[0..5], n=[0..127]

**0x01000*q+0
x08000*n**

Type: R

Clock: HCLK

Reset State: 0x00000000

Stores the lower 32-bits of data from an SPMI read. Starting with the least significant bytes first, PMIC arbiter will fill only the bytes it needs for the read operation, as specified in the BYTE_CNT of the command register.

PMIC_ARBq_CHNLn_RDATA0

Bits	Name	Description
31:0	DATA	Data read from PMIC.

0x0240001C+ PMIC_ARBq_CHNLn_RDATA1, q=[0..5], n=[0..127]

**0x01000*q+0
x08000*n**

Type: R

Clock: HCLK

Reset State: 0x00000000

Stores the upper 32-bits of data from an SPMI read. Starting with the least significant bytes first, PMIC arbiter will fill only the bytes it needs for the read operation, as specified in the BYTE_CNT of the command register. Thus, this register is only needed for read operations larger than four bytes.

PMIC_ARBq_CHNLn_RDATA1

Bits	Name	Description
31:0	DATA	Data read from PMIC.

0x02400020+ PMIC_ARBq_CHNLn_RDATA2, q=[0..5], n=[0..127]**0x1000*q+0x08000*n****Type:** R**Clock:** HCLK**Reset State:** 0x00000000

Stores the additional two bytes needed for a device descriptor read command, which returns ten bytes of data.

PMIC_ARBq_CHNLn_RDATA2

Bits	Name	Description
15:0	DATA	Data read from PMIC.

11.5 PMIC_ARB_CORE_REGISTERS_OBS

The PMIC arbiter observer are for software environments to issue read only commands over the SPMI bus.

11.5.1 PMIC_ARB_CORE_SW_DEC_CHANNELS_OBS

The PMIC arbiter observer are for software environments to issue read only commands over the SPMI bus.

0x02C00000+ PMIC_ARB_OBSq_CHNLn_CMD, q=[0..5], n=[0..127]**0x01000*q+0x08000*n****Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

A write to this register triggers an SPMI command. This register is responsible for specifying all of the information needed to define the command to be performed (except for write data). The SPMI opcode, slave ID, address, and byte count are all written to this register and need to be set with a single write.

NOTE: A write to this register will queue up an SPMI command and until that command finished, further writes to this register will be ignored.

NOTE: The command register is write-only for PMIC arbiter 1.0.0 (will always read as zero).

PMIC_ARB_OBSq_CHNLn_CMD

Bits	Name	Type	Description
31:27	OPCODE	RW	SPMI opcode, as defined by the SPMI master. See the SPMI master documents for further details. 0x1: EXTENDED_REG_READ_LONG

PMIC_ARB_OBSq_CHNLn_CMD (cont.)

Bits	Name	Type	Description
26	PRIORITY	W	Controls whether the SPMI command should be sent with high or low priority. 1: High priority 0: Low priority
11:4	ADDRESS	W	Contains 8 LSB bits of address for the register the SPMI command is to access.
2:0	BYTE_CNT	W	Indicates the number of bytes to read or write for operations that span multiple bytes. For single-byte operations, as well as operations other than read/write operations, this should be set to zero. 111: Eight bytes 110: Seven bytes 101: Six bytes 100: Five bytes 011: Four bytes 010: Three bytes 001: Two bytes 000: One byte

0x02C00004+ PMIC_ARB_OBSq_CHNLn_CONFIG, q=[0..5], n=[0..127]

0x01000*q+0
x08000*n

Type: RW**Clock:** HCLK**Reset State:** 0x00000000

Configuration register for software channels: provides the ability to enable/disable interrupts indicating transactions are done on a per-channel basis.

PMIC_ARB_OBSq_CHNLn_CONFIG

Bits	Name	Description
0	IRQ_EN	Enable/disable transaction done interrupt for channel n. 1: Enable 0: Disable

0x02C00008+ PMIC_ARB_OBSq_CHNLn_STATUS, q=[0..5], n=[0..127]

0x01000*q+0
x08000*n

Type: R**Clock:** HCLK**Reset State:** 0x00000001

Provides status of the most recent SPMI command issued to channel n.

PMIC_ARB_OBSq_CHNLn_STATUS

Bits	Name	Description
3	DROPPED	Resets to 0 upon the issue of the new command. 1: Indicates that a write to this channel illegally occurred during handling of the last command and was subsequently dropped. 0: Indicates that no writes to the channel occurred while the most recent command was being handled.
2	DENIED	Resets to 0 upon the issue of a new command. This bit should only be considered when DONE=1. 1: Indicates that the most recent command was denied by the SPMI master. 0: Indicates that the most recent command was not denied.
1	FAILURE	Resets to 0 upon the issue of a new command. This bit should only be considered when DONE=1. 1: Indicates that the most recent command has failed. 0: Indicates that the most recent command has completed successfully.
0	DONE	Resets to 0 upon the issue of a new command. This bit indicates completion, but not necessarily success, and should be observed in conjunction with DENIED and FAILED to determine the status of an operation. 1: Indicates that the most recent command is finished. 0: Indicates that the most recent command is still pending.

0x02C00018+ PMIC_ARB_OBSq_CHNLn_RDATA0, q=[0..5], n=[0..127]

0x01000*q+0
x08000*n

Type: R**Clock:** HCLK**Reset State:** 0x00000000

Stores the lower 32-bits of data from an SPMI read. Starting with the least significant bytes first, PMIC arbiter will fill only the bytes it needs for the read operation, as specified in the BYTE_CNT of the command register.

PMIC_ARB_OBSq_CHNLn_RDATA0

Bits	Name	Description
31:0	DATA	Data read from PMIC.

0x02C0001C+ PMIC_ARB_OBSq_CHNLn_RDATA1, q=[0..5], n=[0..127]

0x01000*q+0
x08000*n

Type: R**Clock:** HCLK**Reset State:** 0x00000000

Stores the upper 32-bits of data from an SPMI read. Starting with the least significant bytes first, PMIC arbiter will fill only the bytes it needs for the read operation, as specified in the BYTE_CNT of the command register. Thus, this register is only needed for read operations larger than four bytes.

PMIC_ARB_OBSq_CHNLn_RDATA1

Bits	Name	Description
31:0	DATA	Data read from PMIC.

12 Qualcomm Debug Subsystem

12.1 QDSS_APB

12.1.1 QDSS_DAPROM

0x00800000 QDSS_CORESIGHT_CSR_ROM_WORD

Type: R
Clock: PCLKDBG
Reset State: 0x00001003

QDSS_CORESIGHT_CSR_ROM_WORD

Bits	Name	Description
31:12	ADDR_OFFSET	
11:2	RFU	
1	FORMAT	
0	PRESENT	

0x00800004 QDSS_STM_ROM_WORD

Type: R
Clock: PCLKDBG
Reset State: 0x00002003

QDSS STM ROM Word Register

QDSS_STM_ROM_WORD

Bits	Name	Description
31:12	ADDR_OFFSET	Address offset of entry from the base of the ROM table.
11:2	RFU	Reserved for Future Use.

QDSS_STM_ROM_WORD (cont.)

Bits	Name	Description
1	FORMAT	Selects 8-bit or 32-bit format for ROM table entries. Reads as 0x1 meaning 32-bit format.
0	PRESENT	Entry present. Reads as 0x1 meaning ROM table entry present.

0x00800008 QDSS_CTI0_ROM_WORD**Type:** R**Clock:** PCLKDBG**Reset State:** 0x00010003

QDSS CTI0 ROM Word Register

QDSS_CTI0_ROM_WORD

Bits	Name	Description
31:12	ADDR_OFFSET	Address offset of entry from the base of the ROM table.
11:2	RFU	Reserved for Future Use.
1	FORMAT	Selects 8-bit or 32-bit format for ROM table entries. Reads as 0x1 meaning 32-bit format.
0	PRESENT	Entry present. Reads as 0x1 meaning ROM table entry present.

0x0080000C QDSS_CTI1_ROM_WORD**Type:** R**Clock:** PCLKDBG**Reset State:** 0x00011003

QDSS CTI1 ROM Word Register

QDSS_CTI1_ROM_WORD

Bits	Name	Description
31:12	ADDR_OFFSET	Address offset of entry from the base of the ROM table.
11:2	RFU	Reserved for Future Use.
1	FORMAT	Selects 8-bit or 32-bit format for ROM table entries. Reads as 0x1 meaning 32-bit format.
0	PRESENT	Entry present. Reads as 0x1 meaning ROM table entry present.

0x00800010 QDSS_CT12_ROM_WORD**Type:** R**Clock:** PCLKDBG**Reset State:** 0x00012003

QDSS CT12 ROM Word Register

QDSS_CT12_ROM_WORD

Bits	Name	Description
31:12	ADDR_OFFSET	Address offset of entry from the base of the ROM table.
11:2	RFU	Reserved for Future Use.
1	FORMAT	Selects 8-bit or 32-bit format for ROM table entries. Reads as 0x1 meaning 32-bit format.
0	PRESENT	Entry present. Reads as 0x1 meaning ROM table entry present.

0x00800014 QDSS_CT13_ROM_WORD**Type:** R**Clock:** PCLKDBG**Reset State:** 0x00013003

QDSS CT13 ROM Word Register

QDSS_CT13_ROM_WORD

Bits	Name	Description
31:12	ADDR_OFFSET	Address offset of entry from the base of the ROM table.
11:2	RFU	Reserved for Future Use.
1	FORMAT	Selects 8-bit or 32-bit format for ROM table entries. Reads as 0x1 meaning 32-bit format.
0	PRESENT	Entry present. Reads as 0x1 meaning ROM table entry present.

0x00800018 QDSS_CT14_ROM_WORD**Type:** R**Clock:** PCLKDBG**Reset State:** 0x00014003

QDSS CT14 ROM Word Register

QDSS_CTI4_ROM_WORD

Bits	Name	Description
31:12	ADDR_OFFSET	Address offset of entry from the base of the ROM table.
11:2	RFU	Reserved for Future Use.
1	FORMAT	Selects 8-bit or 32-bit format for ROM table entries. Reads as 0x1 meaning 32-bit format.
0	PRESENT	Entry present. Reads as 0x1 meaning ROM table entry present.

0x0080001C QDSS_CTI5_ROM_WORD**Type:** R**Clock:** PCLKDBG**Reset State:** 0x00015003

QDSS CTI5 ROM Word Register

QDSS_CTI5_ROM_WORD

Bits	Name	Description
31:12	ADDR_OFFSET	Address offset of entry from the base of the ROM table.
11:2	RFU	Reserved for Future Use.
1	FORMAT	Selects 8-bit or 32-bit format for ROM table entries. Reads as 0x1 meaning 32-bit format.
0	PRESENT	Entry present. Reads as 0x1 meaning ROM table entry present.

0x00800020 QDSS_CTI6_ROM_WORD**Type:** R**Clock:** PCLKDBG**Reset State:** 0x00016003

QDSS CTI6 ROM Word Register

QDSS_CTI6_ROM_WORD

Bits	Name	Description
31:12	ADDR_OFFSET	Address offset of entry from the base of the ROM table.
11:2	RFU	Reserved for Future Use.
1	FORMAT	Selects 8-bit or 32-bit format for ROM table entries. Reads as 0x1 meaning 32-bit format.

QDSS_CTI6_ROM_WORD (cont.)

Bits	Name	Description
0	PRESENT	Entry present. Reads as 0x1 meaning ROM table entry present.

0x00800024 QDSS_CTI7_ROM_WORD**Type:** R**Clock:** PCLKDBG**Reset State:** 0x00017003

QDSS CTI7 ROM Word Register

QDSS_CTI7_ROM_WORD

Bits	Name	Description
31:12	ADDR_OFFSET	Address offset of entry from the base of the ROM table.
11:2	RFU	Reserved for Future Use.
1	FORMAT	Selects 8-bit or 32-bit format for ROM table entries. Reads as 0x1 meaning 32-bit format.
0	PRESENT	Entry present. Reads as 0x1 meaning ROM table entry present.

0x00800028 QDSS_CTI8_ROM_WORD**Type:** R**Clock:** PCLKDBG**Reset State:** 0x00018003

QDSS CTI8 ROM Word Register

QDSS_CTI8_ROM_WORD

Bits	Name	Description
31:12	ADDR_OFFSET	Address offset of entry from the base of the ROM table.
11:2	RFU	Reserved for Future Use.
1	FORMAT	Selects 8-bit or 32-bit format for ROM table entries. Reads as 0x1 meaning 32-bit format.
0	PRESENT	Entry present. Reads as 0x1 meaning ROM table entry present.

0x0080002C QDSS_TPIU_ROM_WORD**Type:** R**Clock:** PCLKDBG**Reset State:** 0x00020003

QDSS TPIU ROM Word Register

QDSS_TPIU_ROM_WORD

Bits	Name	Description
31:12	ADDR_OFFSET	Address offset of entry from the base of the ROM table.
11:2	RFU	Reserved for Future Use.
1	FORMAT	Selects 8-bit or 32-bit format for ROM table entries. Reads as 0x1 meaning 32-bit format.
0	PRESENT	Entry present. Reads as 0x1 meaning ROM table entry present.

0x00800030 QDSS_TFUN0_ROM_WORD**Type:** R**Clock:** PCLKDBG**Reset State:** 0x00021003

QDSS Trace Funnel 0 ROM Word Register

QDSS_TFUN0_ROM_WORD

Bits	Name	Description
31:12	ADDR_OFFSET	Address offset of entry from the base of the ROM table.
11:2	RFU	Reserved for Future Use.
1	FORMAT	Selects 8-bit or 32-bit format for ROM table entries. Reads as 0x1 meaning 32-bit format.
0	PRESENT	Entry present. Reads as 0x1 meaning ROM table entry present.

0x00800034 QDSS_TREPL_ROM_WORD**Type:** R**Clock:** PCLKDBG**Reset State:** 0x00024003

QDSS Trace Replicator ROM Word Register

QDSS_TREPL_ROM_WORD

Bits	Name	Description
31:12	ADDR_OFFSET	Address offset of entry from the base of the ROM table.
11:2	RFU	Reserved for Future Use.
1	FORMAT	Selects 8-bit or 32-bit format for ROM table entries. Reads as 0x1 meaning 32-bit format.
0	PRESENT	Entry present. Reads as 0x1 meaning ROM table entry present.

0x00800038 QDSS_ETBETF_ROM_WORD**Type:** R**Clock:** PCLKDBG**Reset State:** 0x00025003

QDSS ETB/ETF ROM Word Register

QDSS_ETBETF_ROM_WORD

Bits	Name	Description
31:12	ADDR_OFFSET	Address offset of entry from the base of the ROM table.
11:2	RFU	Reserved for Future Use.
1	FORMAT	Selects 8-bit or 32-bit format for ROM table entries. Reads as 0x1 meaning 32-bit format.
0	PRESENT	Entry present. Reads as 0x1 meaning ROM table entry present.

0x0080003C QDSS_ETR_ROM_WORD**Type:** R**Clock:** PCLKDBG**Reset State:** 0x00026003

QDSS ETR ROM Word Register

QDSS_ETR_ROM_WORD

Bits	Name	Description
31:12	ADDR_OFFSET	Address offset of entry from the base of the ROM table.
11:2	RFU	Reserved for Future Use.
1	FORMAT	Selects 8-bit or 32-bit format for ROM table entries. Reads as 0x1 meaning 32-bit format.

QDSS_ETR_ROM_WORD (cont.)

Bits	Name	Description
0	PRESENT	Entry present. Reads as 0x1 meaning ROM table entry present.

0x00800040 QDSS_VENUS_CTI_ROM_WORD**Type:** R**Clock:** PCLKDBG**Reset State:** 0x00030003

VENUS CTI ROM Word Register

QDSS_VENUS_CTI_ROM_WORD

Bits	Name	Description
31:12	ADDR_OFFSET	Address offset of entry from the base of the ROM table.
11:2	RFU	Reserved for Future Use.
1	FORMAT	Selects 8-bit or 32-bit format for ROM table entries. Reads as 0x1 meaning 32-bit format.
0	PRESENT	Entry present. Reads as 0x1 meaning ROM table entry present.

0x00800044 QDSS_PRONTO_A9_ETM_ROM_WORD**Type:** R**Clock:** PCLKDBG**Reset State:** 0x00034003

Pronto A9 ETM ROM Word Register

QDSS_PRONTO_A9_ETM_ROM_WORD

Bits	Name	Description
31:12	ADDR_OFFSET	Address offset of entry from the base of the ROM table.
11:2	RFU	Reserved for Future Use.
1	FORMAT	Selects 8-bit or 32-bit format for ROM table entries. Reads as 0x1 meaning 32-bit format.
0	PRESENT	Entry present. Reads as 0x1 meaning ROM table entry present.

0x00800048 QDSS_PRONTO_A9_CTI_ROM_WORD

Type: R
Clock: PCLKDBG
Reset State: 0x00035003

Pronto A9 CTI ROM Word Register

QDSS_PRONTO_A9_CTI_ROM_WORD

Bits	Name	Description
31:12	ADDR_OFFSET	Address offset of entry from the base of the ROM table.
11:2	RFU	Reserved for Future Use.
1	FORMAT	Selects 8-bit or 32-bit format for ROM table entries. Reads as 0x1 meaning 32-bit format.
0	PRESENT	Entry present. Reads as 0x1 meaning ROM table entry present.

0x0080004C QDSS_MSS_Q6_CTI_ROM_WORD

Type: R
Clock: PCLKDBG
Reset State: 0x00038003

MSS Q6 CTI ROM Word Register

QDSS_MSS_Q6_CTI_ROM_WORD

Bits	Name	Description
31:12	ADDR_OFFSET	Address offset of entry from the base of the ROM table.
11:2	RFU	Reserved for Future Use.
1	FORMAT	Selects 8-bit or 32-bit format for ROM table entries. Reads as 0x1 meaning 32-bit format.
0	PRESENT	Entry present. Reads as 0x1 meaning ROM table entry present.

0x00800050 QDSS_RPM_CM3_CTI_ROM_WORD

Type: R
Clock: PCLKDBG
Reset State: 0x0003C003

RPM CM3 CTI ROM Word Register

QDSS_RPM_CM3_CTI_ROM_WORD

Bits	Name	Description
31:12	ADDR_OFFSET	Address offset of entry from the base of the ROM table.
11:2	RFU	Reserved for Future Use.
1	FORMAT	Selects 8-bit or 32-bit format for ROM table entries. Reads as 0x1 meaning 32-bit format.
0	PRESENT	Entry present. Reads as 0x1 meaning ROM table entry present.

0x00800054 QDSS_A53_APB_ROM_TABLE_ROM_WORD**Type:** R**Clock:** PCLKDBG**Reset State:** 0x00040003

A53 APB ROM table ROM Word Register

QDSS_A53_APB_ROM_TABLE_ROM_WORD

Bits	Name	Description
31:12	ADDR_OFFSET	Address offset of entry from the base of the ROM table.
11:2	RFU	Reserved for Future Use.
1	FORMAT	Selects 8-bit or 32-bit format for ROM table entries. Reads as 0x1 meaning 32-bit format.
0	PRESENT	Entry present. Reads as 0x1 meaning ROM table entry present.

0x00800058 QDSS_A53_TFUNNEL_ROM_WORD**Type:** R**Clock:** PCLKDBG**Reset State:** 0x00041003

A53 TFUNNEL ROM Word Register

QDSS_A53_TFUNNEL_ROM_WORD

Bits	Name	Description
31:12	ADDR_OFFSET	Address offset of entry from the base of the ROM table.
11:2	RFU	Reserved for Future Use.
1	FORMAT	Selects 8-bit or 32-bit format for ROM table entries. Reads as 0x1 meaning 32-bit format.

QDSS_A53_TFUNNEL_ROM_WORD (cont.)

Bits	Name	Description
0	PRESENT	Entry present. Reads as 0x1 meaning ROM table entry present.

0x00800060 QDSS_A53_PC0_PM_ROM_WORD**Type:** R**Clock:** PCLKDBG**Reset State:** 0x00051003

A53 PC0 PM ROM Word Register

QDSS_A53_PC0_PM_ROM_WORD

Bits	Name	Description
31:12	ADDR_OFFSET	Address offset of entry from the base of the ROM table.
11:2	RFU	Reserved for Future Use.
1	FORMAT	Selects 8-bit or 32-bit format for ROM table entries. Reads as 0x1 meaning 32-bit format.
0	PRESENT	Entry present. Reads as 0x1 meaning ROM table entry present.

0x00800068 QDSS_A53_PC1_PM_ROM_WORD**Type:** R**Clock:** PCLKDBG**Reset State:** 0x00053003

A53 PC1 PM ROM Word Register

QDSS_A53_PC1_PM_ROM_WORD

Bits	Name	Description
31:12	ADDR_OFFSET	Address offset of entry from the base of the ROM table.
11:2	RFU	Reserved for Future Use.
1	FORMAT	Selects 8-bit or 32-bit format for ROM table entries. Reads as 0x1 meaning 32-bit format.
0	PRESENT	Entry present. Reads as 0x1 meaning ROM table entry present.

0x00800070 QDSS_A53_PC2_PM_ROM_WORD

Type: R
Clock: PCLKDBG
Reset State: 0x00055003

A53 PC2 PM ROM Word Register

QDSS_A53_PC2_PM_ROM_WORD

Bits	Name	Description
31:12	ADDR_OFFSET	Address offset of entry from the base of the ROM table.
11:2	RFU	Reserved for Future Use.
1	FORMAT	Selects 8-bit or 32-bit format for ROM table entries. Reads as 0x1 meaning 32-bit format.
0	PRESENT	Entry present. Reads as 0x1 meaning ROM table entry present.

0x00800078 QDSS_A53_PC3_PM_ROM_WORD

Type: R
Clock: PCLKDBG
Reset State: 0x00057003

A53 PC3 PM ROM Word Register

QDSS_A53_PC3_PM_ROM_WORD

Bits	Name	Description
31:12	ADDR_OFFSET	Address offset of entry from the base of the ROM table.
11:2	RFU	Reserved for Future Use.
1	FORMAT	Selects 8-bit or 32-bit format for ROM table entries. Reads as 0x1 meaning 32-bit format.
0	PRESENT	Entry present. Reads as 0x1 meaning ROM table entry present.

0x0080007C QDSS_A53_PC0_CTI_ROM_WORD

Type: R
Clock: PCLKDBG
Reset State: 0x00058003

A53 PC0 CTI ROM Word Register

QDSS_A53_PC0_CTI_ROM_WORD

Bits	Name	Description
31:12	ADDR_OFFSET	Address offset of entry from the base of the ROM table.
11:2	RFU	Reserved for Future Use.
1	FORMAT	Selects 8-bit or 32-bit format for ROM table entries. Reads as 0x1 meaning 32-bit format.
0	PRESENT	Entry present. Reads as 0x1 meaning ROM table entry present.

0x00800080 QDSS_A53_PC1_CTI_ROM_WORD**Type:** R**Clock:** PCLKDBG**Reset State:** 0x00059003

A53 PC1 CTI ROM Word Register

QDSS_A53_PC1_CTI_ROM_WORD

Bits	Name	Description
31:12	ADDR_OFFSET	Address offset of entry from the base of the ROM table.
11:2	RFU	Reserved for Future Use.
1	FORMAT	Selects 8-bit or 32-bit format for ROM table entries. Reads as 0x1 meaning 32-bit format.
0	PRESENT	Entry present. Reads as 0x1 meaning ROM table entry present.

0x00800084 QDSS_A53_PC2_CTI_ROM_WORD**Type:** R**Clock:** PCLKDBG**Reset State:** 0x0005A003

A53 PC2 CTI ROM Word Register

QDSS_A53_PC2_CTI_ROM_WORD

Bits	Name	Description
31:12	ADDR_OFFSET	Address offset of entry from the base of the ROM table.
11:2	RFU	Reserved for Future Use.
1	FORMAT	Selects 8-bit or 32-bit format for ROM table entries. Reads as 0x1 meaning 32-bit format.

QDSS_A53_PC2_CTI_ROM_WORD (cont.)

Bits	Name	Description
0	PRESENT	Entry present. Reads as 0x1 meaning ROM table entry present.

0x00800088 QDSS_A53_PC3_CTI_ROM_WORD**Type:** R**Clock:** PCLKDBG**Reset State:** 0x0005B003

A53 PC3 CTI ROM Word Register

QDSS_A53_PC3_CTI_ROM_WORD

Bits	Name	Description
31:12	ADDR_OFFSET	Address offset of entry from the base of the ROM table.
11:2	RFU	Reserved for Future Use.
1	FORMAT	Selects 8-bit or 32-bit format for ROM table entries. Reads as 0x1 meaning 32-bit format.
0	PRESENT	Entry present. Reads as 0x1 meaning ROM table entry present.

0x0080008C QDSS_A53_PC0_PFT_ROM_WORD**Type:** R**Clock:** PCLKDBG**Reset State:** 0x0005C003

A53 PC0 PFT ROM Word Register

QDSS_A53_PC0_PFT_ROM_WORD

Bits	Name	Description
31:12	ADDR_OFFSET	Address offset of entry from the base of the ROM table.
11:2	RFU	Reserved for Future Use.
1	FORMAT	Selects 8-bit or 32-bit format for ROM table entries. Reads as 0x1 meaning 32-bit format.
0	PRESENT	Entry present. Reads as 0x1 meaning ROM table entry present.

0x00800090 QDSS_A53_PC1_PFT_ROM_WORD

Type: R
Clock: PCLKDBG
Reset State: 0x0005D003

A53 PC1 PFT ROM Word Register

QDSS_A53_PC1_PFT_ROM_WORD

Bits	Name	Description
31:12	ADDR_OFFSET	Address offset of entry from the base of the ROM table.
11:2	RFU	Reserved for Future Use.
1	FORMAT	Selects 8-bit or 32-bit format for ROM table entries. Reads as 0x1 meaning 32-bit format.
0	PRESENT	Entry present. Reads as 0x1 meaning ROM table entry present.

0x00800094 QDSS_A53_PC2_PFT_ROM_WORD

Type: R
Clock: PCLKDBG
Reset State: 0x0005E003

A53 PC2 PFT ROM Word Register

QDSS_A53_PC2_PFT_ROM_WORD

Bits	Name	Description
31:12	ADDR_OFFSET	Address offset of entry from the base of the ROM table.
11:2	RFU	Reserved for Future Use.
1	FORMAT	Selects 8-bit or 32-bit format for ROM table entries. Reads as 0x1 meaning 32-bit format.
0	PRESENT	Entry present. Reads as 0x1 meaning ROM table entry present.

0x00800098 QDSS_A53_PC3_PFT_ROM_WORD

Type: R
Clock: PCLKDBG
Reset State: 0x0005F003

A53 PC3 PFT ROM Word Register

QDSS_A53_PC3_PFT_ROM_WORD

Bits	Name	Description
31:12	ADDR_OFFSET	Address offset of entry from the base of the ROM table.
11:2	RFU	Reserved for Future Use.
1	FORMAT	Selects 8-bit or 32-bit format for ROM table entries. Reads as 0x1 meaning 32-bit format.
0	PRESENT	Entry present. Reads as 0x1 meaning ROM table entry present.

0x0080009C QDSS_QDSS_WRAPPER_2X1_FUNNEL_ROM_WORD**Type:** R**Clock:** PCLKDBG**Reset State:** 0x00068003**QDSS_QDSS_WRAPPER_2X1_FUNNEL_ROM_WORD**

Bits	Name	Description
31:12	ADDR_OFFSET	
11:2	RFU	
1	FORMAT	
0	PRESENT	

0x008000A0 QDSS_QDSS_WRAPPER_8X1_FUNNEL_ROM_WORD**Type:** R**Clock:** PCLKDBG**Reset State:** 0x00069003**QDSS_QDSS_WRAPPER_8X1_FUNNEL_ROM_WORD**

Bits	Name	Description
31:12	ADDR_OFFSET	
11:2	RFU	
1	FORMAT	
0	PRESENT	

0x008000A4 QDSS_HWE_EXPD_MUX_CFG_ROM_WORD

Type: R
Clock: PCLKDBG
Reset State: 0x0006C003

HWE_EXPD_MUX_CFG ROM Word Register

QDSS_HWE_EXPD_MUX_CFG_ROM_WORD

Bits	Name	Description
31:12	ADDR_OFFSET	Address offset of entry from the base of the ROM table.
11:2	RFU	Reserved for Future Use.
1	FORMAT	Selects 8-bit or 32-bit format for ROM table entries. Reads as 0x1 meaning 32-bit format.
0	PRESENT	Entry present. Reads as 0x1 meaning ROM table entry present.

0x008000A8 QDSS_DBGUI_CFG_ROM_WORD

Type: R
Clock: PCLKDBG
Reset State: 0x0006D003

DBGUI_CFG ROM Word Register

QDSS_DBGUI_CFG_ROM_WORD

Bits	Name	Description
31:12	ADDR_OFFSET	Address offset of entry from the base of the ROM table.
11:2	RFU	Reserved for Future Use.
1	FORMAT	Selects 8-bit or 32-bit format for ROM table entries. Reads as 0x1 meaning 32-bit format.
0	PRESENT	Entry present. Reads as 0x1 meaning ROM table entry present.

12.1.2 QDSS_QDSSCSR**0x00801008 QDSS_CS_QDSSCSR_SWSPADREG0**

Type: RW
Clock: PDBGCLK
Reset State: 0x00000000

Software Scratchpad Register 0

QDSS_CS_QDSSCSR_SWSPADREG0

Bits	Name	Description
31:0	SWSPADRW	General read-write scratchpad register bits.

0x0080100C QDSS_CS_QDSSCSR_SWSPADREG1**Type:** RW**Clock:** PDBGCLK**Reset State:** 0x00000000

Software Scratchpad Register 1

QDSS_CS_QDSSCSR_SWSPADREG1

Bits	Name	Description
31:0	SWSPADRW	General read-write scratchpad register bits.

0x00801010 QDSS_CS_QDSSCSR_STMTRANSCTRL**Type:** RW**Clock:** PDBGCLK**Reset State:** 0x00000000

STM Burst Translator Control Register

QDSS_CS_QDSSCSR_STMTRANSCTRL

Bits	Name	Type	Description
31:4	RFU	R	Reserved for Future Use.
3	STMTRANENAB3	RW	Enables STM Burst Translator on STMAWID3[7:0]. A one means STM Burst Translator on STMAWID3[7:0] is enabled.
2	STMTRANENAB2	RW	Enables STM Burst Translator on STMAWID2[7:0]. A one means STM Burst Translator on STMAWID2[7:0] is enabled.
1	STMTRANENAB1	RW	Enables STM Burst Translator on STMAWID1[7:0]. A one means STM Burst Translator on STMAWID1[7:0] is enabled.
0	STMTRANENAB0	RW	Enables STM Burst Translator on STMAWID0[7:0]. A one means STM Burst Translator on STMAWID0[7:0] is enabled.

0x00801014 QDSS_CS_QDSSCSR_STMAWIDCTRL**Type:** RW**Clock:** PDBGCLK**Reset State:** 0x00000000

STM Burst Translator AWID Match Control Register

QDSS_CS_QDSSCSR_STMAWIDCTRL

Bits	Name	Description
31:24	STMAWID3	Burst Translator 3 AXI AWID filter. If STMTRANENAB3 is set and an incoming AXI transaction AWID matches this field, then burst translation is performed on that transaction.
23:16	STMAWID2	Burst Translator 2 AXI AWID filter. If STMTRANENAB2 is set and an incoming AXI transaction AWID matches this field, then burst translation is performed on that transaction.
15:8	STMAWID1	Burst Translator 1 AXI AWID filter. If STMTRANENAB1 is set and an incoming AXI transaction AWID matches this field, then burst translation is performed on that transaction.
7:0	STMAWID0	Burst Translator 0 AXI AWID filter. If STMTRANENAB0 is set and an incoming AXI transaction AWID matches this field, then burst translation is performed on that transaction.

0x00801018 QDSS_CS_QDSSCSR_STMCHNOFST0**Type:** RW**Clock:** PDBGCLK**Reset State:** 0x00000000

STM Burst Translator Channel Offset Register 0

QDSS_CS_QDSSCSR_STMCHNOFST0

Bits	Name	Type	Description
31:27	RFU	R	Reserved for Future Use.
26:18	STMCHNOFST2	RW	STPv2 channel offset for Burst Translator 2. If STMTRANENAB2 is set and an incoming AXI transaction AWID matches the AWID filter, then burst translation uses this STPv2 channel offset.
17:9	STMCHNOFST1	RW	STPv2 channel offset for Burst Translator 1. If STMTRANENAB1 is set and an incoming AXI transaction AWID matches the AWID filter, then burst translation uses this STPv2 channel offset.

QDSS_CS_QDSSCSR_STMCHNOFST0 (cont.)

Bits	Name	Type	Description
8:0	STMCHNOFST0	RW	STPv2 channel offset for Burst Translator 0. If STMTRANENAB0 is set and an incoming AXI transaction AWID matches the AWID filter, then burst translation uses this STPv2 channel offset.

0x0080101C QDSS_CS_QDSSCSR_STMCHNOFST1**Type:** RW**Clock:** PDBGCLK**Reset State:** 0x00000000

STM Burst Translator Channel Offset Register 1

QDSS_CS_QDSSCSR_STMCHNOFST1

Bits	Name	Type	Description
31:9	RFU	R	Reserved for Future Use.
8:0	STMCHNOFST3	RW	STPv2 channel offset for Burst Translator 3. If STMTRANENAB3 is set and an incoming AXI transaction AWID matches the AWID filter, then burst translation uses this STPv2 channel offset.

0x00801020 QDSS_CS_QDSSCSR_STMEXTHWCTRL0**Type:** RW**Clock:** PDBGCLK**Reset State:** 0x00000000

STM Extended Hardware Event Interface Control Register 0

QDSS_CS_QDSSCSR_STMEXTHWCTRL0

Bits	Name	Description
31:28	HEBS7	For bit 7 of the STM hardware event interface, this bit determines which bit of the incoming 512 bit hardware event bus is provided as the input.
27:24	HEBS6	For bit 6 of the STM hardware event interface, this bit determines which bit of the incoming 512 bit hardware event bus is provided as the input.
23:20	HEBS5	For bit 5 of the STM hardware event interface, this bit determines which bit of the incoming 512 bit hardware event bus is provided as the input.

QDSS_CS_QDSSCSR_STMEXTHWCTRL0 (cont.)

Bits	Name	Description
19:16	HEBS4	For bit 4 of the STM hardware event interface, this bit determines which bit of the incoming 512 bit hardware event bus is provided as the input.
15:12	HEBS3	For bit 3 of the STM hardware event interface, this bit determines which bit of the incoming 512 bit hardware event bus is provided as the input.
11:8	HEBS2	For bit 2 of the STM hardware event interface, this bit determines which bit of the incoming 512 bit hardware event bus is provided as the input.
7:4	HEBS1	For bit 1 of the STM hardware event interface, this bit determines which bit of the incoming 512 bit hardware event bus is provided as the input.
3:0	HEBS0	For bit 0 of the STM hardware event interface, this bit determines which bit of the incoming 512 bit hardware event bus is provided as the input.

0x00801024 QDSS_CS_QDSSCSR_STMEXTHWCTRL1**Type:** RW**Clock:** PDBGCLK**Reset State:** 0x00000000

STM Extended Hardware Event Interface Control Register 1

QDSS_CS_QDSSCSR_STMEXTHWCTRL1

Bits	Name	Description
31:28	HEBS15	For bit 15 of the STM hardware event interface, this bit determines which bit of the incoming 512 bit hardware event bus is provided as the input.
27:24	HEBS14	For bit 14 of the STM hardware event interface, this bit determines which bit of the incoming 512 bit hardware event bus is provided as the input.
23:20	HEBS13	For bit 13 of the STM hardware event interface, this bit determines which bit of the incoming 512 bit hardware event bus is provided as the input.
19:16	HEBS12	For bit 12 of the STM hardware event interface, this bit determines which bit of the incoming 512 bit hardware event bus is provided as the input.
15:12	HEBS11	For bit 11 of the STM hardware event interface, this bit determines which bit of the incoming 512 bit hardware event bus is provided as the input.

QDSS_CS_QDSSCSR_STMEXTHWCTRL1 (cont.)

Bits	Name	Description
11:8	HEBS10	For bit 10 of the STM hardware event interface, this bit determines which bit of the incoming 512 bit hardware event bus is provided as the input.
7:4	HEBS9	For bit 9 of the STM hardware event interface, this bit determines which bit of the incoming 512 bit hardware event bus is provided as the input.
3:0	HEBS8	For bit 8 of the STM hardware event interface, this bit determines which bit of the incoming 512 bit hardware event bus is provided as the input.

0x00801028 QDSS_CS_QDSSCSR_STMEXTHWCTRL2**Type:** RW**Clock:** PDBGCLK**Reset State:** 0x00000000

STM Extended Hardware Event Interface Control Register 2

QDSS_CS_QDSSCSR_STMEXTHWCTRL2

Bits	Name	Description
31:28	HEBS23	For bit 23 of the STM hardware event interface, this bit determines which bit of the incoming 512 bit hardware event bus is provided as the input.
27:24	HEBS22	For bit 22 of the STM hardware event interface, this bit determines which bit of the incoming 512 bit hardware event bus is provided as the input.
23:20	HEBS21	For bit 21 of the STM hardware event interface, this bit determines which bit of the incoming 512 bit hardware event bus is provided as the input.
19:16	HEBS20	For bit 20 of the STM hardware event interface, this bit determines which bit of the incoming 512 bit hardware event bus is provided as the input.
15:12	HEBS19	For bit 19 of the STM hardware event interface, this bit determines which bit of the incoming 512 bit hardware event bus is provided as the input.
11:8	HEBS18	For bit 18 of the STM hardware event interface, this bit determines which bit of the incoming 512 bit hardware event bus is provided as the input.
7:4	HEBS17	For bit 17 of the STM hardware event interface, this bit determines which bit of the incoming 512 bit hardware event bus is provided as the input.

QDSS_CS_QDSSCSR_STMEXTHWCTRL2 (cont.)

Bits	Name	Description
3:0	HEBS16	For bit 16 of the STM hardware event interface, this bit determines which bit of the incoming 512 bit hardware event bus is provided as the input.

0x0080102C QDSS_CS_QDSSCSR_STMEXTHWCTRL3**Type:** RW**Clock:** PDBGCLK**Reset State:** 0x00000000

STM Extended Hardware Event Interface Control Register 3

QDSS_CS_QDSSCSR_STMEXTHWCTRL3

Bits	Name	Description
31:28	HEBS31	For bit 31 of the STM hardware event interface, this bit determines which bit of the incoming 512 bit hardware event bus is provided as the input.
27:24	HEBS30	For bit 30 of the STM hardware event interface, this bit determines which bit of the incoming 512 bit hardware event bus is provided as the input.
23:20	HEBS29	For bit 29 of the STM hardware event interface, this bit determines which bit of the incoming 512 bit hardware event bus is provided as the input.
19:16	HEBS28	For bit 28 of the STM hardware event interface, this bit determines which bit of the incoming 512 bit hardware event bus is provided as the input.
15:12	HEBS27	For bit 27 of the STM hardware event interface, this bit determines which bit of the incoming 512 bit hardware event bus is provided as the input.
11:8	HEBS26	For bit 26 of the STM hardware event interface, this bit determines which bit of the incoming 512 bit hardware event bus is provided as the input.
7:4	HEBS25	For bit 25 of the STM hardware event interface, this bit determines which bit of the incoming 512 bit hardware event bus is provided as the input.
3:0	HEBS24	For bit 24 of the STM hardware event interface, this bit determines which bit of the incoming 512 bit hardware event bus is provided as the input.

0x00801030 QDSS_CS_QDSSCSR_USBBAMCTRL**Type:** RW**Clock:** PDBGCLK**Reset State:** 0x00000000

USB NDP BAM Control Register

QDSS_CS_QDSSCSR_USBBAMCTRL

Bits	Name	Type	Description
31:3	RFU	R	Reserved for Future Use.
2	USBENAB	RW	USB trace sink enable. BLKSIZE and USBENAB must be configured before ETR and NDP BAM are configured. A one means USB trace sinking enabled once ETR and NDP BAM blocks have been configured.
1:0	BLKSIZE	RW	Configures the number of bytes that will be allowed to be written by the ETR between successive EOB indications. This is an average value over time so the EOB descriptors could deviate from this size by plus or minus 15*8 bytes. 0x3: ENUM_3 (2048 bytes) 0x2: ENUM_2 (1024 bytes) 0x1: ENUM_1 (512 bytes) 0x0: ENUM_0 (256 bytes)

0x00801034 QDSS_CS_QDSSCSR_USBFLSHCTRL**Type:** RW**Clock:** PDBGCLK**Reset State:** 0x00000000

USB Trace Sink Flush Control Register

QDSS_CS_QDSSCSR_USBFLSHCTRL

Bits	Name	Type	Description
31:20	RFU	R	Reserved for Future Use.
19	TRIGEOT	RW	NDP BAM EOT control upon ECT trigger reception by the NDP BAM interface. A one means EOT descriptor creation upon ECT trigger reception enabled.
18	PERFLSHEOT	RW	NDP BAM EOT control upon periodic flush request by the NDP BAM interface. A one means EOT descriptor creation upon periodic flush creation enabled.
17:2	PERFLSHTHRS	RW	NDP BAM interface periodic flush periodicity in NDP BAM clock cycles control. When PERFLSH is set, determines flush periodicity in NDP BAM clock cycles.

QDSS_CS_QDSSCSR_USBFLSHCTRL (cont.)

Bits	Name	Type	Description
1	PERFLSH	RW	NDP BAM periodic flush control. A one means periodic flushing enabled and occurs every PERFLSHTHRS NDP BAM clock cycles.
0	EOT	RW	Manual NDP BAM EOT descriptor creation control. A LOW to HIGH transition causes manual creation of an EOT descriptor.

0x00801038 QDSS_CS_QDSSCSR_TIMESTAMPCTRL**Type:** RW**Clock:** PDBGCLK**Reset State:** 0x00000000

Timestamp Control Register

QDSS_CS_QDSSCSR_TIMESTAMPCTRL

Bits	Name	Type	Description
31:2	RFU	R	Reserved for Future Use.
1	TIMESTAMPDISABLE	RW	Enable for the master timestamp generator. This register should be set high (generator disabled) prior to writing CAPTURE. A one means master timestamp generator disabled.
0	CAPTURE	RW	Capture enable for AOTIMEVAL0,1 and QDSSTIMEVAL0,1. A LOW to HIGH transition triggers capturing of values into AOTIMEVAL0,1 and QDSSTIMEVAL0,1.

0x0080103C QDSS_CS_QDSSCSR_AOTIMEVAL0**Type:** R**Clock:** PDBGCLK**Reset State:** 0x00000000

MPM AO Counter Capture Register 0

QDSS_CS_QDSSCSR_AOTIMEVAL0

Bits	Name	Description
31:0	TIMEVAL	This register contains the captured value of the lower 32 bits of the 56 bit MPM AO system counter reference. Capture occurs on cc_qdss_pclkdbg_clk so when the timestamp clock is disabled in the system clock controller, cc_qdss_pclkdbg_clk must be active for this value to be captured.

0x00801040 QDSS_CS_QDSSCSR_AOTIMEVAL1**Type:** R**Clock:** PDBGCLK**Reset State:** 0x00000000

MPM AO Counter Capture Register 1

QDSS_CS_QDSSCSR_AOTIMEVAL1

Bits	Name	Description
31:24	RFU	Reserved for Future Use.
23:0	TIMEVAL	This register contains the captured value of the upper 24 bits of the 56 bit MPM AO system counter reference. Capture occurs on cc_qdss_pclkdbg_clk so when the timestamp clock is disabled in the system clock controller, cc_qdss_pclkdbg_clk must be active for this value to be captured.

0x00801044 QDSS_CS_QDSSCSR_QDSSTIMEVAL0**Type:** R**Clock:** PDBGCLK**Reset State:** 0x00000000

QDSS Timestamp Counter Capture Value Register 0

QDSS_CS_QDSSCSR_QDSSTIMEVAL0

Bits	Name	Description
31:0	TIMEVAL	This register contains the captured value of the lower 32 bits of the 64 bit QDSS timestamp counter. Capture occurs on cc_qdss_pclkdbg_clk so when the timestamp clock is disabled in the system clock controller, cc_qdss_pclkdbg_clk must be active for this value to be captured.

0x00801048 QDSS_CS_QDSSCSR_QDSSTIMEVAL1

Type: R
Clock: PDBGCLK
Reset State: 0x00000000

QDSS Timestamp Counter Capture Value Register 1

QDSS_CS_QDSSCSR_QDSSTIMEVAL1

Bits	Name	Description
31:0	TIMEVAL	This register contains the captured value of the upper 32 bits of the 64 bit QDSS timestamp counter. Capture occurs on cc_qdss_pclkdbg_clk so when the timestamp clock is disabled in the system clock controller, cc_qdss_pclkdbg_clk must be active for this value to be captured.

0x0080104C QDSS_CS_QDSSCSR_QDSSTIMELOAD0

Type: RW
Clock: PDBGCLK
Reset State: 0x00000000

QDSS Timestamp Counter Load Value Register 0

QDSS_CS_QDSSCSR_QDSSTIMELOAD0

Bits	Name	Description
31:0	TIMELOAD	Lower 32 bits of the value to be loaded into the QDSS timestamp reference by the RPM post VDDCX minimization. Loading occurs once both QDSSTIMELOAD0 and 1 have been written irrespective of the order in which they are written. It should be ensured that the timestamp counter clock is inactive whilst writing the TIMELOAD values.

0x00801050 QDSS_CS_QDSSCSR_QDSSTIMELOAD1

Type: RW
Clock: PDBGCLK
Reset State: 0x00000000

QDSS Timestamp Counter Load Value Register 1

QDSS_CS_QDSSCSR_QDSSTIMELOAD1

Bits	Name	Description
31:0	TIMELOAD	Upper 32 bits of the value to be loaded into the QDSS timestamp reference by the RPM post VDDCX minimization. Loading occurs once both QDSSTIMELOAD0 and 1 have been written irrespective of the order in which they are written. It should be ensured that the timestamp counter clock is inactive whilst writing the TIMELOAD values.

0x00801054 QDSS_CS_QDSSCSR_DAPMSAVAL**Type:** RW**Clock:** PDBGCLK**Reset State:** 0x00000000

QDSS DAP AHB-AP MSA Value Register

QDSS_CS_QDSSCSR_DAPMSAVAL

Bits	Name	Type	Description
31:1	RFU	R	Reserved for Future Use.
0	MSAVAL	RW	Sets the value of the Modem Self Authentication (MSA) bit for outgoing DAP AHB-AP transactions. Setting this bit HIGH is masked by the QDSS modem_IDEN fuse.

0x0080105C QDSS_CS_QDSSCSR_QDSSCLKIPI**Type:** RW**Clock:** PDBGCLK**Reset State:** 0x00000000

QDSS Clock Frequency Vote IPI Register

QDSS_CS_QDSSCSR_QDSSCLKIPI

Bits	Name	Type	Description
31:1	RFU	R	Reserved for Future Use.
0	CLKIPI	RW	Sets the value of the QDSS clock vote IPI to RPM. A one activates IRQ to RPM CPU.

0x00801060 QDSS_CS_QDSSCSR_QDSSPWRREQIGNORE**Type:** RW**Clock:** PDBGCLK**Reset State:** 0x00000000

QDSS CDBGPWRUPREQ Ignore Register

QDSS_CS_QDSSCSR_QDSSPWRREQIGNORE

Bits	Name	Description
31:0	PWRREQIGNORE	Enables up to 32 receivers of QDSS CDBGPWRUPREQ to ignore all internal hardware implications of CDBGPWRUPREQ. For example, the RPM Cortex M3 will use this to undo CDBGPWRUPREQ to their local HCLK sleep FSM. When RPM's ignore bit is removed, the GCC must be polled to determine when the RPM's HCLK has been re-activated.

0x00801064 QDSS_CS_QDSSCSR_QDSSSPARE**Type:** RW**Clock:** PDBGCLK**Reset State:** 0x00000000

QDSS Spare Register

QDSS_CS_QDSSCSR_QDSSSPARE

Bits	Name	Description
31:0	SPARE	32 spare RW bits.

0x00801068 QDSS_CS_QDSSCSR_IPCAT**Type:** R**Clock:** PDBGCLK**Reset State:** 0x30000000

QDSS IPCAT Register

QDSS_CS_QDSSCSR_IPCAT

Bits	Name	Description
31:28	MAJ	Indicates different interface version. Major version changes are not backwards compatible.
27:16	MIN	Indicates expanded functionality. Minor version adds functionality while being backward compatible for existing features.

QDSS_CS_QDSSCSR_IPCAT (cont.)

Bits	Name	Description
15:0	STEP	Indicates a change in the hardware which is not intended to impact software compatibility.

0x0080106C QDSS_CS_QDSSCSR_ETRIRQCTRL**Type:** RW**Clock:** PDBGCLK**Reset State:** 0x00000000

QDSS ETR Byte Counter Interrupt Control Register

QDSS_CS_QDSSCSR_ETRIRQCTRL

Bits	Name	Description
31:0	IRQBYTECNTVAL	This 32 bit value defines the number of bytes that, when moved by the ETR AXI interface, causes an interrupt that may be used by an EE to know how much data is present in memory requiring copy to some other location. A zero setting disables the interrupt. A one setting means 8 bytes, two 16 bytes, etc. In other words, the value in this register is the interrupt threshold times 8 bytes.

0x00801070 QDSS_CS_QDSSCSR_PRESERVEETF**Type:** R**Clock:** PDBGCLK**Reset State:** 0x00000000

QDSS ETB/ETF WDOG RESET Preservation Register

QDSS_CS_QDSSCSR_PRESERVEETF

Bits	Name	Description
31:23	RFU	Reserved for Future Use.
22	STS_MEMERR	Used to determine if an error on the AXI master interface caused the TMCReady to be set instead of a clean transition due to flush and stop on a crash reset
21	AFVALIDS	Used to determine if the flush got stuck for some reason as part of the flush and stop that occurs on a crash reset
20	STS_EMPTY	Used to determine valid trace being present in the trace memory on a crash reset
19	ACQCOMP	Similar usage as TMCReady bit on a crash reset

QDSS_CS_QDSSCSR_PRESERVEETF (cont.)

Bits	Name	Description
18	STS_TMCREADY	Used to determine we properly transitioned into the stopped state with flush on stop on a crash reset
17	STS_TRIGGERED	Indicates a trigger event being detected while trace capture was enabled when the crash reset occurred
16	FULL	Full bit to help determine if the buffer wrapped or got full at least one. This is used to decide whether the data should be dumped from location 0 or from the value of write pointer
15:0	RWP_PRESERVE	Write pointer value to keep track of where to start dumping trace from i.e. figuring the oldest and newest data in the circular buffer

0x00801074 QDSS_CS_QDSSCSR_PRESERVEETR0**Type:** R**Clock:** PDBGCLK**Reset State:** 0x00000000

QDSS ETR WDOG RESET Preservation Register0

QDSS_CS_QDSSCSR_PRESERVEETR0

Bits	Name	Description
31:0	RWP_PRESERVE	Write pointer value to keep track of where to start dumping trace from i.e. figuring the oldest and newest data in the circular buffer

0x00801078 QDSS_CS_QDSSCSR_PRESERVEETR1**Type:** R**Clock:** PDBGCLK**Reset State:** 0x00000000

QDSS ETR WDOG RESET Preservation Register1

QDSS_CS_QDSSCSR_PRESERVEETR1

Bits	Name	Description
31:23	RFU	Reserved for Future Use.
22	STS_MEMERR	Used to determine if an error on the AXI master interface caused the TMCReady to be set instead of a clean transition due to flush and stop on a crash reset
21	AFVALIDS	Used to determine if the flush got stuck for some reason as part of the flush and stop that occurs on a crash reset

QDSS_CS_QDSSCSR_PRESERVEETR1 (cont.)

Bits	Name	Description
20	STS_EMPTY	Used to determine valid trace being present in the trace memory on a crash reset
19	ACQCOMP	Similar usage as TMCReady bit on a crash reset
18	STS_TMCREADY	Used to determine we properly transitioned into the stopped state with flush on stop on a crash reset
17	STS_TRIGGERED	Indicates a trigger event being detected while trace capture was enabled when the crash reset occurred
16	FULL	Full bit to help determine if the buffer wrapped or got full at least one. This is used to decide whether the data should be dumped from location 0 or from the value of write pointer
15:8	RFU15_8	Reserved for Future Use.
7:0	RWP_PRESERVE	Write pointer value to keep track of where to start dumping trace from i.e. figuring the oldest and newest data in the circular buffer

0x00801F00 QDSS_CS_QDSSCSR_ITCTL**Type:** R**Clock:** PDBGCLK**Reset State:** 0x00000000

QDSS Integration Control (Coresight Programmer's Model) Register

QDSS_CS_QDSSCSR_ITCTL

Bits	Name	Description
31:0	RFU	Reads as zero since no integration functionality is implemented.

0x00801FA0 QDSS_CS_QDSSCSR_CLAIMSET**Type:** R**Clock:** PDBGCLK**Reset State:** 0x00000000

QDSS Claim Tag Set (Coresight Programmer's Model) Register

QDSS_CS_QDSSCSR_CLAIMSET

Bits	Name	Description
31:0	RFU	Reads as zero since no integration functionality is implemented.

0x00801FA4 QDSS_CS_QDSSCSR_CLAIMCLR

Type: R
Clock: PDBGCLK
Reset State: 0x00000000

QDSS Claim Tag Clear (Coresight Programmer's Model) Register

QDSS_CS_QDSSCSR_CLAIMCLR

Bits	Name	Description
31:0	RFU	Reads as zero since no integration functionality is implemented.

0x00801FB0 QDSS_CS_QDSSCSR_LOCKACCESS

Type: W
Clock: PDBGCLK
Reset State: Undefined

QDSS Lock Access (Coresight Programmer's Model) Register

QDSS_CS_QDSSCSR_LOCKACCESS

Bits	Name	Description
31:0	LOCKACCESS	Write access code. A write of 0xC5ACCE55 enables further writes to the QDSS Coresight CSR Slave from software.

0x00801FB4 QDSS_CS_QDSSCSR_LOCKSTATUS

Type: R
Clock: PDBGCLK
Reset State: 0x00000003

QDSS Lock Status (Coresight Programmer's Model) Register

QDSS_CS_QDSSCSR_LOCKSTATUS

Bits	Name	Description
31:3	RFU	Reserved for Future Use.
2	FIELD_8_BIT_LOCK	Reads as zero since this component does not implement an 8 bit lock access register.

QDSS_CS_QDSSCSR_LOCKSTATUS (cont.)

Bits	Name	Description
1	ACCESS	Access permission. This bit reflects the inversion of the current state of the UNLOCK bit within this component. 0x1: HIGH (Write access blocked. Reads permitted.) 0x0: LOW (Access permitted.)
0	LOCK_CONTROL	Reads as one since lock exists for this component.

0x00801FB8 QDSS_CS_QDSSCSR_AUTHSTATUS

Type: R
Clock: PDBGCLK
Reset State: 0x00000000

QDSS Authentication Status (Coresight Programmer's Model) Register

QDSS_CS_QDSSCSR_AUTHSTATUS

Bits	Name	Description
31:0	RFU	Reads as zero since no integration functionality is implemented.

0x00801FC8 QDSS_CS_QDSSCSR_DEVICEID

Type: R
Clock: PDBGCLK
Reset State: 0x00000000

QDSS Device Identification (Coresight Programmer's Model) Register

QDSS_CS_QDSSCSR_DEVICEID

Bits	Name	Description
31:8	RFU	Reserved for Future Use.
7:0	DEVID	This register is implementation defined for each Part Number and Designer. Unused for this component so reads as zeros.

0x00801FCC QDSS_CS_QDSSCSR_DEVICETYPE

Type: R
Clock: PDBGCLK
Reset State: 0x00000004

QDSS Device Type (Coresight Programmer's Model) Register

QDSS_CS_QDSSCSR_DEVICETYPE

Bits	Name	Description
31:8	RFU	Reserved for Future Use.
7:4	SUBTYPE	Reads as zeros meaning Other sub type.
3:0	MAJTYPE	Reads as 0x4 meaning Debug Control major type.

0x00801FD0 QDSS_CS_QDSSCSR_PERIPHID4**Type:** R**Clock:** PDBGCLK**Reset State:** 0x0000000X

QDSS Peripheral ID4 (Coresight Programmer's Model) Register

QDSS_CS_QDSSCSR_PERIPHID4

Bits	Name	Description
31:8	RFU	Reserved for Future Use.
7:4	FIELD_4KB_COUNT	Number of 4kB blocks. Reads as zero or 4kB block size.
3:0	JEP106_CONTINUATION	JEP106 continuation code. Reads as TARGETID[11:8] from MSM JTAG ID fuses [11:8]. This field should be set to bits [10:7] of Qualcomm's JEDEC assigned JEP106 code.

0x00801FD4 QDSS_CS_QDSSCSR_PERIPHID5**Type:** R**Clock:** PDBGCLK**Reset State:** 0x00000000

QDSS Peripheral ID5 (Coresight Programmer's Model) Register

QDSS_CS_QDSSCSR_PERIPHID5

Bits	Name	Description
31:0	RFU	Reads as zero per Coresight Architecture.

0x00801FD8 QDSS_CS_QDSSCSR_PERIPHID6

Type: R
Clock: PDBGCLK
Reset State: 0x00000000

QDSS Peripheral ID6 (Coresight Programmer's Model) Register

QDSS_CS_QDSSCSR_PERIPHID6

Bits	Name	Description
31:0	RFU	Reads as zero per Coresight Architecture.

0x00801FDC QDSS_CS_QDSSCSR_PERIPHID7

Type: R
Clock: PDBGCLK
Reset State: 0x00000000

QDSS Peripheral ID7 (Coresight Programmer's Model) Register

QDSS_CS_QDSSCSR_PERIPHID7

Bits	Name	Description
31:0	RFU	Reads as zero per Coresight Architecture.

0x00801FE0 QDSS_CS_QDSSCSR_PERIPHID0

Type: R
Clock: PDBGCLK
Reset State: 0x00000040

QDSS Peripheral ID0 (Coresight Programmer's Model) Register

QDSS_CS_QDSSCSR_PERIPHID0

Bits	Name	Description
31:8	RFU	Reserved for Future Use.
7:0	PARTNUM	PartNumber[7:0]. [7:6] is Subsystem/core family ID (e.g. denote QDSS family or generation). [5:4] is Subsystem/core configuration options (e.g. denote cache options, etc.). [3:2] is Subsystem/core fuse options (e.g. indicate modem fuse options, etc.). [1:0] is Subsystem/core future use field. Reads as 0x40.

0x00801FE4 QDSS_CS_QDSSCSR_PERIPHID1

Type: R
Clock: PDBGCLK
Reset State: 0x000000X4

QDSS Peripheral ID1 (Coresight Programmer's Model) Register

QDSS_CS_QDSSCSR_PERIPHID1

Bits	Name	Description
31:8	RFU	Reserved for Future Use.
7:4	JEP106_IDENTITY_3_0	JEF106 Identity Code[3:0]. Reads as TARGETID[4:1] from MSM JTAG ID fuses [4:1] which will always be 0x0 for Qualcomm.
3:0	PARTNUM	PartNumber[11:8]. [11:8] is 0x4 meaning Qualcomm designed Coresight component in QDSS. Reads as 0x4.

0x00801FE8 QDSS_CS_QDSSCSR_PERIPHID2

Type: R
Clock: PDBGCLK
Reset State: 0b0000000000000000000000000000000011XXX

QDSS Peripheral ID2 (Coresight Programmer's Model) Register

QDSS_CS_QDSSCSR_PERIPHID2

Bits	Name	Description
31:8	RFU	Reserved for Future Use.
7:4	MAJREV	Major revision number of this peripheral. Reads as 0x0 for this version of QDSS.
3	JEDEC	Reads as one which indicates that a JEDEC assigned value is used.
2:0	JEP106_IDENTITY_6_4	JEF106 Identity Code[6:4]. Reads as TARGETID[7:5] from MSM JTAG ID fuses [7:5] which will always be 3'b111 for Qualcomm.

0x00801FEC QDSS_CS_QDSSCSR_PERIPHID3

Type: R
Clock: PDBGCLK
Reset State: 0x00000000

QDSS Peripheral ID3 (Coresight Programmer's Model) Register

QDSS_CS_QDSSCSR_PERIPHID3

Bits	Name	Description
31:8	RFU	Reserved for Future Use.
7:4	REV_AND	This field indicates minor errata fixes specific to this design. Reads as zeros for this version of QDSS.
3:0	CUSTOMER_MODIFIED	Implementation defined. Always reads as zeros for QDSS.

0x00801FF0 QDSS_CS_QDSSCSR_COMPID0**Type:** R**Clock:** PDBGCLK**Reset State:** 0x0000000D

QDSS Component ID0 (Coresight Programmer's Model) Register

QDSS_CS_QDSSCSR_COMPID0

Bits	Name	Description
31:8	RFU	Reserved for Future Use.
7:0	PREAMBLE_7_0	Coresight mandates this read as 0x0D.

0x00801FF4 QDSS_CS_QDSSCSR_COMPID1**Type:** R**Clock:** PDBGCLK**Reset State:** 0x00000090

QDSS Component ID1 (Coresight Programmer's Model) Register

QDSS_CS_QDSSCSR_COMPID1

Bits	Name	Description
31:8	RFU	Reserved for Future Use.
7:4	PREAMBLE_15_12	Coresight mandates this read as 0x9 or Coresight class component.
3:0	PREAMBLE_11_8	Coresight mandates this read as 0x0.

0x00801FF8 QDSS_CS_QDSSCSR_COMPID2

Type: R
Clock: PDBGCLK
Reset State: 0x00000005

QDSS Component ID2 (Coresight Programmer's Model) Register

QDSS_CS_QDSSCSR_COMPID2

Bits	Name	Description
31:8	RFU	Reserved for Future Use.
7:0	PREAMBLE_23_16	Coresight mandates this read as 0x05.

0x00801FFC QDSS_CS_QDSSCSR_COMPID3

Type: R
Clock: PDBGCLK
Reset State: 0x000000B1

QDSS Component ID3 (Coresight Programmer's Model) Register

QDSS_CS_QDSSCSR_COMPID3

Bits	Name	Description
31:8	RFU	Reserved for Future Use.
7:0	PREAMBLE_31_24	Coresight mandates this read as 0xB1.

12.1.3 CXSTM_2_32_32_TRUE

The CoreSight CXSTM is instrumentation trace macrocell with software and hardware stimulus ports that generates trace according to MIPI STPv2 protocol.

12.1.3.1 QDSS_CXSTM_2_32_32_TRUE

The CoreSight CXSTM is instrumentation trace macrocell with software and hardware stimulus ports that generates trace according to MIPI STPv2 protocol.

0x00802C04 QDSS_STMDMASTARTR

Type: W
Reset State: Undefined

QDSS_STMDMASTARTR

Bits	Name	Description
0	START	Start a DMA transfer 0x1: START (Start a DMA transfer)

0x00802C08 QDSS_STMDMASTOPR**Type:** W**Reset State:** Undefined**QDSS_STMDMASTOPR**

Bits	Name	Description
0	STOP	Stop an active DMA transfer 0x1: STOP (Stop an active DMA transfer)

0x00802C0C QDSS_STMDMASTATR**Type:** R**Reset State:** 0x00000000**QDSS_STMDMASTATR**

Bits	Name	Description
0	STATUS	Stop an active DMA transfer 0x0: IDLE (interface is idle) 0x1: ACTIVE (interface is active)

0x00802C10 QDSS_STMDMACTLR**Type:** RW**Reset State:** 0x00000000**QDSS_STMDMACTLR**

Bits	Name	Description
3:2	SENS	Determines the sensitivity of the DMA request to the current buffer level in the STM: 0x0: EMPTY (FIFO is less than 25% full) 0x1: ENUM_25 (FIFO is less than 50% full) 0x2: ENUM_50 (FIFO is less than 75% full) 0x3: ENUM_75 (FIFO is less than 100% full)

0x00802CFC QDSS_STMDMAIDR**Type:** R**Reset State:** 0x00000002**QDSS_STMDMAIDR**

Bits	Name	Description
11:8	VENDSPEC	The VENDSPEC field identifies any vendor specific modifications or mappings. 0x0: VENDSPEC (Vendor specific information.)
7:4	CLASSREV	The CLASSREV field identifies the revision of the programmer's model. 0x0: REVISION_CONTROL (Revision.)
3:0	CLASS	The CLASS field identifies the programmer's model. 0x2: CONTROL (DMA Control.)

0x00802D00 QDSS_STMHEER**Type:** RW**Reset State:** Undefined**QDSS_STMHEER**

Bits	Name	Description
31:0	HEE	

0x00802D20 QDSS_STMHETER**Type:** RW**Reset State:** Undefined**QDSS_STMHETER**

Bits	Name	Description
31:0	HETE	

0x00802D64 QDSS_STMHEMCR**Type:** RW**Reset State:** 0b000000000000000000000000XX00XX0

QDSS_STMHEMCR

Bits	Name	Type	Description
7	RESERVED	RW	Reserved
6	TRIGCLEAR	W	0x0: NO_EFFECT (No effect) 0x1: CLEAR (Clears TRIGSTATUS if TRIGSTATUS is b1)
5	TRIGSTATUS	R	0x0: NOT_OCCURRED (Trigger has not occurred) 0x1: OCCURRED (Trigger has occurred)
4	TRIGCTL	RW	0x0: MULTI_SHOT (Triggers are multi-shot) 0x1: SINGLE_SHOT (Triggers are single-shot)
2	ERRDETECT	R	Enable error detection on the hardware event tracing 0x0: DISABLE (Error detection disabled) 0x1: ENABLE (Error detection enabled.)
1	COMPEN	RW	Enable leading zero suppression of hardware event data values in the trace stream 0x0: DISABLE (Compression disabled) 0x1: ENABLE (Compression enabled.)
0	EN	RW	Enable Hardware Event tracing 0x0: DISABLE (Disabled) 0x1: ENABLE (Enabled)

0x00802DF4 QDSS_STMHEMASTR**Type:** R**Reset State:** 0x00000080**QDSS_STMHEMASTR**

Bits	Name	Description
15:0	MASTER	The STPv2 master number for hardware event trace: 0x80: MASTER_1 (Hardware events are associated with master 0x80.)

0x00802DF8 QDSS_STMHEFEAT1R**Type:** R**Reset State:** 0x00100035**QDSS_STMHEFEAT1R**

Bits	Name	Description
23:15	NUMHE	The number of hardware events supported by the STM: 0x20: EVENTS (32 hardware events.)

QDSS_STMHEFEAT1R (cont.)

Bits	Name	Description
5:4	HECOMP	0x3: RO (Data compression support is programmable. STMHEMCR.COMPEN is implemented)
3	HEMASTR	0x0: RO (STMHEMASTR is read-only.)
2	HEERR	0x1: IMPLEMENTED (Hardware event error detection implemented. STMHEMCR.ERRDETECT is implemented.)
0	HETER	0x1: IMPLEMENTED (STMHETER implemented.)

0x00802DFC QDSS_STMHEIDR**Type:** R**Reset State:** 0x00000001**QDSS_STMHEIDR**

Bits	Name	Description
11:8	VENDSPEC	The VENDSPEC field identifies any vendor specific modifications or mappings: 0x0: VENDSPEC (Vendor specific information.)
7:4	CLASSREV	The CLASSREV field identifies the revision of the programmers model: 0x0: REVISION_CONTROL (Revision.)
3:0	CLASS	The CLASS field identifies the programmers model: 0x1: CONTROL (Hardware Event Control.)

0x00802E00 QDSS_STMSPER**Type:** RW**Reset State:** 0x00000000**QDSS_STMSPER**

Bits	Name	Description
31:0	SPE	

0x00802E20 QDSS_STMSPTER**Type:** RW**Reset State:** 0x00000000

QDSS_STMSPTER

Bits	Name	Description
31:0	SPTTE	

0x00802E60 QDSS_STMSPSCR**Type:** RW**Reset State:** 0xXXX00000**QDSS_STMSPSCR**

Bits	Name	Description
31:20	PORTSEL	This field defines which stimulus ports the STMSPTER and/or STMSPER apply to
1:0	PORTCTL	0x0: NOT_USED (Port selection not used) 0x1: STMSPTER_ONLY (Port selection applies only to the STMSPTER) 0x2: RESERVED (Reserved) 0x3: STMSPER_AND_STMSPTER (Port selection applies to both the STMSPER and STMSPTER)

0x00802E68 QDSS_STMSPOVERRIDER**Type:** RW**Reset State:** 0bXXXXXXXXXXXXXXXXX0000000000000000**QDSS_STMSPOVERRIDER**

Bits	Name	Description
31:15	PORTSEL	
2	OVERTS	0x0: DISABLED (Override not enabled) 0x1: ENABLED (Override enabled)
1:0	OVERCTL	0x0: DISABLED (Override controls disabled) 0x1: GUARANTEED (Ports selected by PORTSEL always behave as Guaranteed transactions) 0x2: INVARIANT_TIMING (Ports selected by PORTSEL always behave as Invariant Timing transactions) 0x3: RESERVED (Reserved)

0x00802E6C QDSS_STMSPMOVERRIDER**Type:** RW**Reset State:** 0b000000000XXXXXXXXX0000000000000000**QDSS_STMSPMOVERRIDER**

Bits	Name	Description
22:15	MASTSEL	
0	MASTCTL	0x0: DISABLED (Override controls disabled. STMSPMOVERRIDER applies equally to all masters) 0x1: ENABLED (Master selection enabled. STMSPMOVERRIDER applies to the masters selected by MASTSEL)

0x00802E70 QDSS_STMSPTRIGCSR**Type:** RW**Reset State:** 0b000000000000000000000000000000XX0**QDSS_STMSPTRIGCSR**

Bits	Name	Type	Description
4	RESERVED	RW	Reserved
3	RESERVED	RW	Reserved
2	TRIGCLEAR	W	0x0: NO_EFFECT (No effect) 0x1: CLEAR (Clears TRIGSTATUS if TRIGSTATUS is b1)
1	TRIGSTATUS	R	0x0: NOT_OCCURRED (Trigger has not occurred) 0x1: OCCURRED (Trigger has occurred)
0	TRIGCTL	RW	0x0: MULTI_SHOT (Triggers are multi-shot) 0x1: SINGLE_SHOT (Triggers are single-shot)

0x00802E80 QDSS_STMTCSR**Type:** RW**Reset State:** 0b000000000XXXXXXXXX0000000000000100**QDSS_STMTCSR**

Bits	Name	Type	Description
23	BUSY	R	STM is busy, for example the STM trace FIFO is not empty: 0x0: IDLE (STM is not busy.) 0x1: BUSY (STM is busy.)
22:16	RESERVED	RW	Reserved

QDSS_STMTCSR (cont.)

Bits	Name	Type	Description
5	COMPEN	RW	0x0: DISABLED (Compression disabled, data transfers are transmitted at the size of the transaction.) 0x1: ENABLED (Compression enabled, data transfers are compressed to save bandwidth.)
2	SYNCEN	R	STMSYNCR is implemented so this value is RAO. 0x1: RAO (STMSYNCR implemented.)
1	TSEN	RW	This bit controls if timestamp requests are ignored or not: 0x0: DISABLE (Timestamping disabled. Requests for timestamp generation are ignored, and stimulus port writes selecting timestamping are treated as if it were not selected.) 0x1: ENABLE (Timestamping enabled. If stimulus writes select timestamping, a timestamp is output according to STPv2.)
0	EN	RW	Global STM enable: 0x0: DISABLE (STM disabled) 0x1: ENABLE (STM enabled)

0x00802E84 QDSS_STMTSSTIMR**Type:** W**Reset State:** Undefined**QDSS_STMTSSTIMR**

Bits	Name	Description
0	FORCETS	0x1: FORCETS (Force Timestamp Stimulus)

0x00802E8C QDSS_STMTSFREQR**Type:** RW**Reset State:** 0x00000000**QDSS_STMTSFREQR**

Bits	Name	Description
31:0	FREQ	The timestamp frequency in Hz

0x00802E90 QDSS_STMSYNCR**Type:** RW**Reset State:** 0x00000000**QDSS_STMSYNCR**

Bits	Name	Description
12	MODE	0x0: N_BYTES (COUNT[11:0] defines a value N. Synchronization period is N bytes) 0x1: ENUM_2_N_BYTES (COUNT[11:7] defines a value N. Synchronization period is 2^N bytes. N must be in the range of 12 to 27 inclusive and other values are Unpredictable)
11:2	COUNT	Counter value for the number of bytes between synchronization packets. Reads return the value of this register.

0x00802E94 QDSS_STMAUXCR**Type:** RW**Reset State:** 0x00000000**QDSS_STMAUXCR**

Bits	Name	Description
4	AFREADYHIGH	0x0: NO_OVERRIDE (No override, AFREADY is controlled by the state of the STM.) 0x1: OVERRIDE (Override, AFREADY is driven HIGH.)
3	CLKON	0x0: NO_OVERRIDE (No override, clock gate is controlled by the state of STM.) 0x1: OVERRIDE (Override, clock is enabled.)
2	PRIORINVDIS	0x0: INVERSION_ENABLED (Priority inversion, when AXI flush is finished, HW gets priority until HW flush is done) 0x1: INVERSION_DISABLED (Priority inversion disabled, AXI always has priority over HW.)
1	ASYNCPPE	0x0: DISABLE (ASYNCP priority is always lower than trace.) 0x1: ENABLE (ASYNCP priority escalates on second synchronization request.)
0	FIFOAF	0x0: DISABLED (Auto-flush disabled.) 0x1: ENABLED (Auto-flush enabled.)

0x00802EA0 QDSS_STMSPFEAT1R**Type:** R**Reset State:** 0x006587D1

QDSS_STMSPFEAT1R

Bits	Name	Description
23:22	SWOEN	0x1: NOT_IMPLEMENTED (STMTCSR.SWOEN not implemented.)
21:20	SYNCEN	0x2: RAO (STMTCSR.SYNCEN implemented but always reads as b1.)
19:18	HWTEN	0x1: NOT_IMPLEMENTED (STMTCSR.HWTEN not implemented.)
17:16	TSPRESCALE	0x1: NOT_IMPLEMENTED (Timestamp prescale not implemented.)
15:14	TRIGCTL	0x2: MULTI_SHOT_AND_SINGLE_SHOT (Multi-shot and single-shot triggers supported. STMTRIGCSR implemented.)
13:10	RESERVED	Reserved
9:8	SYNC	0x3: MODE (STMSYNCR implemented with MODE control.)
7	FORCETS	0x1: IMPLEMENTED (STMTSSTIMR bit [0] implemented.)
6	TSFREQ	0x1: RW (STMTSFREQR is read-write.)
5:4	TS	0x1: ABSOLUT (Absolute timestamps implemented.)
3:0	PROT	0x1: STPV2 (STPV2 protocol.)

0x00802EA4 QDSS_STMSPFEAT2R**Type:** R**Reset State:** 0x000104F2**QDSS_STMSPFEAT2R**

Bits	Name	Description
17:16	SPTYPE	0x1: EXTENDED_ONLY (Only Extended Stimulus Ports implemented.)
15:12	DSIZE	0x0: ENUM_32_BIT (32-bit data.)
10:9	SPTRTYPE	0x2: INVARIANT_TIMING_AND_GUARANTEED (Both invariant timing and guaranteed transactions are supported)
8:7	PRIVMASK	0x1: NOT_IMPLEMENTED (STMPRIVMASKR not implemented.)
6	SPOVERRIDE	0x1: IMPLEMENTED (STMSPOVERRIDER and STMSPMOVERRIDER implemented.)
5:4	SPCOMP	0x3: PROGRAMMABLE (Data compression support is programmable. STMTCSR.COMPEN is implemented.)
2	SPER	0x0: IMPLEMENTED (STMSPER is implemented.)
1:0	SPTER	0x2: IMPLEMENTED (STMSPTER is implemented.)

0x00802EA8 QDSS_STMSPFEAT3R**Type:** R**Reset State:** 0x0000007F

Indicates the features of the STM.

QDSS_STMSPFEAT3R

Bits	Name	Description
6:0	NUMMAST	0x7F: NUMMAST (128 masters are implemented.)

0x00802EE8 QDSS_STMITTRIGGER**Type:** W**Reset State:** Undefined

Integration Test for Cross-Trigger Outputs Register

QDSS_STMITTRIGGER

Bits	Name	Description
3	ASYNCOOUT_W	Sets the value of the ASYNCOOUT output in integration mode: 0x1: HIGH (Drive logic 1 on ASYNCOOUT output.) 0x0: LOW (Drive logic 0 on ASYNCOOUT output.)
2	TRIGOUTHETE_W	Sets the value of the TRIGOUTHETE output in integration mode: 0x1: HIGH (Drive logic 1 on TRIGOUTHETE output.) 0x0: LOW (Drive logic 0 on TRIGOUTHETE output.)
1	TRIGOUTSW_W	Sets the value of the TRIGOUTSW output in integration mode: 0x1: HIGH (Drive logic 1 on TRIGOUTSW output.) 0x0: LOW (Drive logic 0 on TRIGOUTSW output.)
0	TRIGOUTSPTE_W	Sets the value of the TRIGOUTSPTE output in integration mode: 0x1: HIGH (Drive logic 1 on TRIGOUTSPTE output.) 0x0: LOW (Drive logic 0 on TRIGOUTSPTE output.)

0x00802EEC QDSS_STMITATBDATA0**Type:** W**Reset State:** Undefined

Controls the value of the ATDATAM output in integration mode:

QDSS_STMITATBDATA0

Bits	Name	Description
4	ATDATAM31_W	Set the value of the ATDATAM[31] output: 0x1: HIGH (Drive logic 1 on ATDATAM[31] output.) 0x0: LOW (Drive logic 0 on ATDATAM[31] output.)
3	ATDATAM23_W	Set the value of the ATDATAM[23] output: 0x1: HIGH (Drive logic 1 on ATDATAM[23] output.) 0x0: LOW (Drive logic 0 on ATDATAM[23] output.)
2	ATDATAM15_W	Set the value of the ATDATAM[15] output: 0x1: HIGH (Drive logic 1 on ATDATAM[15] output.) 0x0: LOW (Drive logic 0 on ATDATAM[15] output.)
1	ATDATAM7_W	Set the value of the ATDATAM[7] output: 0x1: HIGH (Drive logic 1 on ATDATAM[7] output.) 0x0: LOW (Drive logic 0 on ATDATAM[7] output.)
0	ATDATAM0_W	Set the value of the ATDATAM[0] output: 0x1: HIGH (Drive logic 1 on ATDATAM[0] output.) 0x0: LOW (Drive logic 0 on ATDATAM[0] output.)

0x00802EF0 QDSS_STMITATBCTR2**Type:** R**Reset State:** Undefined

Returns the value of the ATREADYM and AFVALIDM inputs in integration mode.

QDSS_STMITATBCTR2

Bits	Name	Description
1	AFVALIDM_R	Reads the value of the AFVALIDM input: 0x1: HIGH (Pin is at logic 1.) 0x0: LOW (Pin is at logic 0.)
0	ATREADYM_R	Reads the value of the ATREADYM input: 0x1: HIGH (Pin is at logic 1.) 0x0: LOW (Pin is at logic 0.)

0x00802EF4 QDSS_STMITATBID**Type:** W**Reset State:** Undefined

Controls the value of the ATIDM output in integration mode.

QDSS_STMITATBID

Bits	Name	Description
6:0	ATIDM_W	Sets the value of the ATIDM output.

0x00802EF8 QDSS_STMITATBCTR0**Type:** W**Reset State:** Undefined

Controls the value of the ATVALIDM, AFREADYM, and ATBYTESM outputs in integration mode.

QDSS_STMITATBCTR0

Bits	Name	Description
9:8	ATBYTESM_W	Sets the value of the ATBYTESM output: 0x3: ENUM_11 (Drive logic 2'b11 on the ATBYTESM output.) 0x2: ENUM_10 (Drive logic 2'b10 on the ATBYTESM output.) 0x1: ENUM_01 (Drive logic 2'b01 on the ATBYTESM output.) 0x0: ENUM_00 (Drive logic 2'b00 on the ATBYTESM output.)
1	AFREADYM_W	Sets the value of the AFREADYM output: 0x1: HIGH (Drive logic 1 on the AFREADYM output.) 0x0: LOW (Drive logic 0 on the AFREADYM output.)
0	ATVALIDM_W	Sets the value of the ATVALIDM output: 0x1: HIGH (Drive logic 1 on the ATVALIDM output.) 0x0: LOW (Drive logic 0 on the ATVALIDM output.)

0x00802F00 QDSS_STMITCTRL**Type:** RW**Reset State:** 0x00000000**QDSS_STMITCTRL**

Bits	Name	Description
0	INTEGRATION_MODE	Enables the component to switch from functional mode to integration mode and back. If no integration functionality is implemented, this register must read as zero. 0x1: INTEGRATION_MODE (Enable integration mode.) 0x0: FUNCTIONAL_MODE (Disable integration mode.)

0x00802FA0 QDSS_STMCLAIMSET**Type:** RW**Reset State:** 0x0000000F**QDSS_STMCLAIMSET**

Bits	Name	Type	Description
3:0	CLAIMSET_W	W	Each bit is considered separately: 0 = no effect, 1 = set this bit in the claim tag.
3:0	CLAIMSET_R	R	This claim tag bit is implemented 0xF: CLAIM_TAG_IMPLEMENTED_BITS (These bits are present within the claim tag field)

0x00802FA4 QDSS_STMCLAIMCLR**Type:** RW**Reset State:** 0x00000000**QDSS_STMCLAIMCLR**

Bits	Name	Type	Description
3:0	CLAIMCLR_W	W	Each bit is considered separately: 0 = no effect, 1 = clear this bit in the claim tag.
3:0	CLAIMCLR_R	R	The value present reflects the current setting of the Claim Tag.

0x00802FB0 QDSS_STMLAR**Type:** W**Reset State:** Undefined**QDSS_STMLAR**

Bits	Name	Description
31:0	ACCESS_W	A write of 0xC5ACCE55 enables further write access to this device. An invalid write has the affect of removing write access. 0xC5ACCE55: UNLOCK (unlock the protection register to enable write access)

0x00802FB4 QDSS_STMLSR**Type:** R**Reset State:** 0x00000003

QDSS_STMLSR

Bits	Name	Description
2	LOCKTYPE	Indicates if the Lock Access Register, 0xFB0, is implemented as 8-bit or 32-bit. 0x0: ENUM_32_BIT (This component implements a 32-bit Lock Access Register.)
1	LOCKGRANT	Returns the current status of the Lock. 0x0: ACCESS_PERMITTED (Write access is allowed to this device.) 0x1: DEVICE_LOCKED (Write access to the component is blocked. All writes to control registers are ignored. Reads are permitted.)
0	LOCKEXIST	Indicates that a lock control mechanism exists for this device. 0x0: LOCK_NOT_PRESENT (No lock control mechanism exists, writes to the Lock Access Register, 0xFB0, are ignored.) 0x1: LOCK_PRESENT (Lock control mechanism is present.)

0x00802FC8 QDSS_STMDEVID**Type:** R**Reset State:** 0x00010000**QDSS_STMDEVID**

Bits	Name	Description
16:0	NUMSP	This value indicates the number of stimulus ports implemented. 0x10000: NUMBER_OF_PORTS (65536 stimulus ports implemented.)

0x00802FCC QDSS_STMDEVTYPE**Type:** R**Reset State:** 0x00000063**QDSS_STMDEVTYPE**

Bits	Name	Description
7:4	SUB_TYPE	Sub-classification within the major category: 0x6: CORESIGHT_STM (This component generates trace based on software and hardware stimulus.)
3:0	RESERVED	Reserved

0x00802FE0 QDSS_STMPIDR0**Type:** R**Reset State:** 0x00000062**QDSS_STMPIDR0**

Bits	Name	Description
7:0	PART_NUMBER_BITS7TO0	Bits [7:0] of the component part number. This is selected by the designer of the component. 0x62: CORESIGHT_STM_PART_NUMBER_7_0 (Lowest 8 bits of the Part Number, 0x962.)

0x00802FE4 QDSS_STMPIDR1**Type:** R**Reset State:** 0x000000B9**QDSS_STMPIDR1**

Bits	Name	Description
7:4	JEP106_BITS3TO0	Bits [3:0] of the JEDEC identity code indicating the designer of the component, together with the continuation code. 0xB: ARM_JEP106_IDENTITY_CODE_7_0 (Lowest 4 bits of the JEP106 Identity Code.)
3:0	PART_NUMBER_BITS11TO8	Bits [11:8] of the component part number. This is selected by the designer of the component. 0x9: CORESIGHT_STM_PART_NUMBER_11_8 (Upper 4 bits of the Part Number, 0x907.)

0x00802FE8 QDSS_STMPIDR2**Type:** R**Reset State:** 0x0000001B**QDSS_STMPIDR2**

Bits	Name	Description
7:4	REVISION	The Revision field is an incremental value starting at 0x0 for the first design of this component. This only increases by 1 for both major and minor revisions and is used as a look-up to establish the exact major and minor revision. 0x1: R0P1 (This device is at r0p1.)
3	JEDEC	Always set. Indicates that a JEDEC assigned value is used. 0x1: JEDEC_IDENTITY (The designer ID is specified by JEDEC (http://www.jedec.org).)

QDSS_STMPIDR2 (cont.)

Bits	Name	Description
2:0	JEP106_BITS6TO4	Bits [6:4] of the JEDEC identity code indicating the designer of the component, together with the continuation code. 0x3: ARM_JEP106_IDENTITY_CODE_6_4 (Upper 3 bits of the JEP106 Identity Code.)

0x00802FEC QDSS_STMPIDR3**Type:** R**Reset State:** 0x00000000**QDSS_STMPIDR3**

Bits	Name	Description
7:4	REVAND	This field indicates minor errata fixes specific to this design, for example metal fixes after implementation. In most cases this field is zero. It is recommended that component designers ensure this field can be changed by a metal fix if required, for example by driving it from registers that reset to zero. 0x0: NO_MODS (Indicates that there have been no metal fixes to this component.)
3:0	CUSTOMER_MODIFIED	Where the component is reusable IP, this value indicates if the customer has modified the behavior of the component. In most cases this field is zero. 0x0: NO_MODS (Indicates that there have been no modifications made.)

0x00802FD0 QDSS_STMPIDR4**Type:** R**Reset State:** 0x00000004**QDSS_STMPIDR4**

Bits	Name	Description
7:4	FOURKB_COUNT	This is a 4-bit value that indicates the total contiguous size of the memory window used by this component in powers of 2 from the standard 4KB. If a component only requires the standard 4KB, this must read as 0x0, 4KB only. For 8KB set to 0x1, for 16KB set to 0x2, for 32KB set to 0x3, and so on. 0x0: ENUM_4KB (Indicates that the device only occupies 4KB of memory.)
3:0	JEP106_CONT	JEDEC continuation code indicating the designer of the component, together with the identity code. 0x4: ARM_JEP106_CONTINUATION_CODE (Indicates that ARM's JEDEC identity code is on the 5th bank.)

0x00802FD4 QDSS_STMPIDR5**Type:** RW**Reset State:** Undefined**QDSS_STMPIDR5**

Bits	Name	Description
31:0	STMPIDR5	

0x00802FD8 QDSS_STMPIDR6**Type:** RW**Reset State:** Undefined**QDSS_STMPIDR6**

Bits	Name	Description
31:0	STMPIDR6	

0x00802FDC QDSS_STMPIDR7**Type:** RW**Reset State:** Undefined**QDSS_STMPIDR7**

Bits	Name	Description
31:0	STMPIDR7	

0x00802FF0 QDSS_STMCIDR0**Type:** R**Reset State:** 0x0000000D**QDSS_STMCIDR0**

Bits	Name	Description
7:0	PREAMBLE	Contains bits [24:31] of the component identification. 0xD: ENUM_0D (Identification value.)

0x00802FF4 QDSS_STMCIDR1**Type:** R**Reset State:** 0x00000090**QDSS_STMCIDR1**

Bits	Name	Description
7:4	CLASS	Class of the component, for example, ROM table or CoreSight component. 0x9: CORESIGHT_COMPONENT (Indicates the component is a CoreSight component.)
3:0	PREAMBLE	Contains bits [19:16] of the component identification. 0x0: ENUM_0 (Identification value.)

0x00802FF8 QDSS_STMCIDR2**Type:** R**Reset State:** 0x00000005**QDSS_STMCIDR2**

Bits	Name	Description
7:0	PREAMBLE	Contains bits [15:8] of the component identification. 0x5: ENUM_05 (Identification value.)

0x00802FFC QDSS_STMCIDR3**Type:** R**Reset State:** 0x000000B1**QDSS_STMCIDR3**

Bits	Name	Description
7:0	PREAMBLE	Contains bits [7:0] of the component identification. 0xB1: B1 (Identification value.)

12.1.4 CTIO_CSCTI

The CTI allows mapping a trigger on the Trigger interface to a signal on the Channel interface and vice versa. The mapping is user-programmable. When the CTI receives a channel event it maps this onto a trigger output. This enables subsystems to cross trigger with each other. The receiving and transmitting of triggers is performed through the trigger interface.

12.1.4.1 QDSS_CTIO_CSCTI

The CTI allows mapping a trigger on the Trigger interface to a signal on the Channel interface and vice versa. The mapping is user-programmable. When the CTI receives a channel event it maps this onto a trigger output. This enables subsystems to cross trigger with each other. The receiving and transmitting of triggers is performed through the trigger interface.

0x00810000 QDSS_CTIO_CTICONTROL

Type: RW

Reset State: 0x00000000

QDSS_CTIO_CTICONTROL

Bits	Name	Description
0	GLBEN	0x1: ENABLED (When this bit is 1, cross triggering mapping logic functionality is enabled.) 0x0: DISABLED (When this bit is 0, all cross triggering mapping logic functionality is disabled.)

0x00810010 QDSS_CTIO_CTIINTACK

Type: W

Reset State: 0x00000000

QDSS_CTIO_CTIINTACK

Bits	Name	Description
7:0	INTACK	

0x00810014 QDSS_CTIO_CTIAPPSET

Type: W

Reset State: 0x00000000

QDSS_CTIO_CTIAPPSET

Bits	Name	Description
3:0	APPSET	

0x00810018 QDSS_CTIO_CTIAPPCLEAR**Type:** W**Reset State:** 0x00000000**QDSS_CTIO_CTIAPPCLEAR**

Bits	Name	Description
3:0	APPCLEAR	

0x0081001C QDSS_CTIO_CTIAPPPULSE**Type:** W**Reset State:** 0x00000000**QDSS_CTIO_CTIAPPPULSE**

Bits	Name	Description
3:0	APPULSE	

0x00810020 QDSS_CTIO_CTIINEN0**Type:** RW**Reset State:** 0x00000000**QDSS_CTIO_CTIINEN0**

Bits	Name	Description
3:0	TRIGINEN	

0x00810024 QDSS_CTIO_CTIINEN1**Type:** RW**Reset State:** 0x00000000**QDSS_CTIO_CTIINEN1**

Bits	Name	Description
3:0	TRIGINEN	

0x00810028 QDSS_CTIO_CTIINEN2**Type:** RW**Reset State:** 0x00000000**QDSS_CTIO_CTIINEN2**

Bits	Name	Description
3:0	TRIGINEN	

0x0081002C QDSS_CTIO_CTIINEN3**Type:** RW**Reset State:** 0x00000000**QDSS_CTIO_CTIINEN3**

Bits	Name	Description
3:0	TRIGINEN	

0x00810030 QDSS_CTIO_CTIINEN4**Type:** RW**Reset State:** 0x00000000**QDSS_CTIO_CTIINEN4**

Bits	Name	Description
3:0	TRIGINEN	

0x00810034 QDSS_CTIO_CTIINEN5**Type:** RW**Reset State:** 0x00000000**QDSS_CTIO_CTIINEN5**

Bits	Name	Description
3:0	TRIGINEN	

0x00810038 QDSS_CTIO_CTIINEN6**Type:** RW**Reset State:** 0x00000000**QDSS_CTIO_CTIINEN6**

Bits	Name	Description
3:0	TRIGINEN	

0x0081003C QDSS_CTIO_CTIINEN7**Type:** RW**Reset State:** 0x00000000**QDSS_CTIO_CTIINEN7**

Bits	Name	Description
3:0	TRIGINEN	

0x008100A0 QDSS_CTIO_CTIOUTEN0**Type:** RW**Reset State:** 0x00000000**QDSS_CTIO_CTIOUTEN0**

Bits	Name	Description
3:0	TRIGOUTEN	

0x008100A4 QDSS_CTIO_CTIOUTEN1**Type:** RW**Reset State:** 0x00000000**QDSS_CTIO_CTIOUTEN1**

Bits	Name	Description
3:0	TRIGOUTEN	

0x008100A8 QDSS_CTIO_CTIOUTEN2**Type:** RW**Reset State:** 0x00000000**QDSS_CTIO_CTIOUTEN2**

Bits	Name	Description
3:0	TRIGOUTEN	

0x008100AC QDSS_CTIO_CTIOUTEN3**Type:** RW**Reset State:** 0x00000000**QDSS_CTIO_CTIOUTEN3**

Bits	Name	Description
3:0	TRIGOUTEN	

0x008100B0 QDSS_CTIO_CTIOUTEN4**Type:** RW**Reset State:** 0x00000000**QDSS_CTIO_CTIOUTEN4**

Bits	Name	Description
3:0	TRIGOUTEN	

0x008100B4 QDSS_CTIO_CTIOUTEN5**Type:** RW**Reset State:** 0x00000000**QDSS_CTIO_CTIOUTEN5**

Bits	Name	Description
3:0	TRIGOUTEN	

0x008100B8 QDSS_CTIO_CTIOUTEN6**Type:** RW**Reset State:** 0x00000000**QDSS_CTIO_CTIOUTEN6**

Bits	Name	Description
3:0	TRIGOUTEN	

0x008100BC QDSS_CTIO_CTIOUTEN7**Type:** RW**Reset State:** 0x00000000**QDSS_CTIO_CTIOUTEN7**

Bits	Name	Description
3:0	TRIGOUTEN	

0x00810130 QDSS_CTIO_CTITRIGINSTATUS**Type:** R**Reset State:** Undefined**QDSS_CTIO_CTITRIGINSTATUS**

Bits	Name	Description
7:0	TRIGINSTATUS	

0x00810134 QDSS_CTIO_CTITRIGOUTSTATUS**Type:** R**Reset State:** 0x00000000**QDSS_CTIO_CTITRIGOUTSTATUS**

Bits	Name	Description
7:0	TRIGOUTSTATUS	

0x00810138 QDSS_CTIO_CTICHINSTATUS**Type:** R**Reset State:** Undefined**QDSS_CTIO_CTICHINSTATUS**

Bits	Name	Description
3:0	CTICHINSTATUS	

0x0081013C QDSS_CTIO_CTICHOUTSTATUS**Type:** R**Reset State:** 0x00000000**QDSS_CTIO_CTICHOUTSTATUS**

Bits	Name	Description
3:0	CTICHOUTSTATUS	

0x00810140 QDSS_CTIO_CTIGATE**Type:** RW**Reset State:** 0x0000000F**QDSS_CTIO_CTIGATE**

Bits	Name	Description
3	CTIGATEEN3	
2	CTIGATEEN2	
1	CTIGATEEN1	
0	CTIGATEEN0	

0x00810144 QDSS_CTIO_ASICCTL**Type:** RW**Reset State:** 0x00000000**QDSS_CTIO_ASICCTL**

Bits	Name	Description
7:0	ASICCTL	

0x00810EDC QDSS_CTIO_ITCHINACK**Type:** W**Reset State:** 0x00000000**QDSS_CTIO_ITCHINACK**

Bits	Name	Description
3:0	CTCHINACK	

0x00810EE0 QDSS_CTIO_ITTRIGINACK**Type:** W**Reset State:** 0x00000000**QDSS_CTIO_ITTRIGINACK**

Bits	Name	Description
7:0	CTTRIGINACK	

0x00810EE4 QDSS_CTIO_ITCHOUT**Type:** W**Reset State:** 0x00000000**QDSS_CTIO_ITCHOUT**

Bits	Name	Description
3:0	CTCHOUT	

0x00810EE8 QDSS_CTIO_ITTRIGOUT**Type:** W**Reset State:** 0x00000000**QDSS_CTIO_ITTRIGOUT**

Bits	Name	Description
7:0	CTTRIGOUT	

0x00810EEC QDSS_CTIO_ITCHOUTACK**Type:** R**Reset State:** 0x00000000**QDSS_CTIO_ITCHOUTACK**

Bits	Name	Description
3:0	CTCHOUTACK	

0x00810EF0 QDSS_CTIO_ITTRIGOUTACK**Type:** R**Reset State:** 0x00000000**QDSS_CTIO_ITTRIGOUTACK**

Bits	Name	Description
7:0	CTTRIGOUTACK	

0x00810EF4 QDSS_CTIO_ITCHIN**Type:** R**Reset State:** 0x00000000**QDSS_CTIO_ITCHIN**

Bits	Name	Description
3:0	CTCHIN	

0x00810EF8 QDSS_CTIO_ITTRIGIN**Type:** R**Reset State:** 0x00000000**QDSS_CTIO_ITTRIGIN**

Bits	Name	Description
7:0	CTTRIGIN	

0x00810F00 QDSS_CTIO_ITCTRL**Type:** RW**Reset State:** 0x00000000**QDSS_CTIO_ITCTRL**

Bits	Name	Description
0	INTEGRATION_MODE	Allows the component to switch from functional mode to integration mode or back. 0x1: INTEGRATION_MODE (Enable integration mode) 0x0: FUNCTIONAL_MODE (Disable integration mode)

0x00810FA0 QDSS_CTIO_CLAIMSET**Type:** RW**Reset State:** 0x0000000F

This is used in conjunction with Claim Tag Clear Register, CLAIMCLR. This register forms one half of the Claim Tag value. This location allows individual bits to be set, write, and returns the number of bits that can be set, read.

QDSS_CTIO_CLAIMSET

Bits	Name	Type	Description
3:0	CLAIMSET_W	W	Each bit is considered separately: 0 = no effect, 1 = set this bit in the claim tag.
3:0	CLAIMSET_R	R	This claim tag bit is implemented 0xF: CLAIM_TAG_IMPLEMENTED_BITS (These bits are present within the claim tag field)

0x00810FA4 QDSS_CTIO_CLAIMCLR**Type:** RW**Reset State:** 0x00000000

This register is used in conjunction with Claim Tag Set Register, CLAIMSET. This register forms one half of the Claim Tag value. This location enables individual bits to be cleared, write, and returns the current Claim Tag value, read.

QDSS_CTIO_CLAIMCLR

Bits	Name	Type	Description
3:0	CLAIMCLR_W	W	Each bit is considered separately: 0 = no effect, 1 = clear this bit in the claim tag.

QDSS_CTIO_CLAIMCLR (cont.)

Bits	Name	Type	Description
3:0	CLAIMCLR_R	R	The value present reflects the current setting of the Claim Tag.

0x00810FB0 QDSS_CTIO_LAR**Type:** W**Reset State:** Undefined**QDSS_CTIO_LAR**

Bits	Name	Description
31:0	ACCESS_W	A write of 0xC5ACCE55 enables further write access to this device. A write of any value other than 0xC5ACCE55 will have the affect of removing write access. 0xC5ACCE55: UNLOCK (Unlock the protection register to enable write access)

0x00810FB4 QDSS_CTIO_LSR**Type:** R**Reset State:** 0x00000003**QDSS_CTIO_LSR**

Bits	Name	Description
2	LOCKTYPE	Indicates if the Lock Access Register (0xFB0) is implemented as 8-bit or 32-bit 0x0: ENUM_32_BIT (This component implements a 32-bit Lock Access Register)
1	LOCKGRANT	Returns the current status of the Lock. This bit reads as 0 when read from an external debugger (PADDRDBG31 = 1) since external debugger accesses are not subject to Lock Registers. 0x0: ACCESS_PERMITTED (Write access is permitted to this device) 0x1: DEVICE_LOCKED (Write access to the component is blocked. All writes to control registers are ignored. Reads are permitted)
0	LOCKEXIST	Indicates that a lock control mechanism exists for this device. This bit reads as 0 when read from an external debugger (PADDRDBG31 = 1) since external debugger accesses are not subject to Lock Registers. 0x0: LOCK_NOT_PRESENT (No lock control mechanism exists, writes to the Lock Access register (0xFB0) will be ignored) 0x1: LOCK_PRESENT (Lock control mechanism is present)

0x00810FB8 QDSS_CTIO_AUTHSTATUS**Type:** R**Reset State:** 0xFFFFFFFF0X

Reports what functionality is currently permitted by the authentication interface.

QDSS_CTIO_AUTHSTATUS

Bits	Name	Description
7:6	SNID	Indicates the security level for secure non-invasive debug 0x0: ENUM_2_B00 (Functionality not implemented.)
5:4	SID	Indicates the security level for secure invasive debug 0x0: ENUM_2_B00 (Functionality not implemented.)
3:2	NSNID	Indicates the security level for non-secure non-invasive debug 0x2: ENUM_2_B10 (Non-secure Non-Invasive debug functionality currently disabled.) 0x3: ENUM_2_B11 (Non-secure Non-Invasive debug functionality currently enabled.)
1:0	NSID	Indicates the security level for non-secure invasive debug 0x2: ENUM_2_B10 (Non-secure Invasive debug functionality currently disabled.) 0x3: ENUM_2_B11 (Non-secure Invasive debug functionality currently enabled.)

0x00810FC8 QDSS_CTIO_DEVID**Type:** R**Reset State:** 0x00040800

This register indicates the capabilities of the CTI.

QDSS_CTIO_DEVID

Bits	Name	Description
19:16	NUMCH	Number of ECT channels available.
15:8	NUMTRIG	Number of ECT triggers available.
4:0	EXTMUXNUM	Indicates the number of multiplexing available on Trigger Inputs and Trigger Outputs using ASICCTL. Default value of 5'b00000 indicating no multiplexing present. Reflects the value of the Verilog `define EXTMUXNUM that the user must alter accordingly.

0x00810FE0 QDSS_CTIO_PERIPHID0**Type:** R**Reset State:** 0x00000006

Part of the set of Peripheral Identification registers. Contains part of the designer specific part number.

QDSS_CTIO_PERIPHID0

Bits	Name	Description
7:0	PART_0	Bits [7:0] of the component's part number. This is selected by the designer of the component. 0x6: CTI_PART_NUMBER_7_0 (Lowest 8 bits of the Part Number (0x906))

0x00810FE4 QDSS_CTIO_PERIPHID1**Type:** R**Reset State:** 0x000000B9

Part of the set of Peripheral Identification registers. Contains part of the designer specific part number and part of the designer identity.

QDSS_CTIO_PERIPHID1

Bits	Name	Description
7:4	DES_0	Bits 3:0 of the JEDEC identity code indicating the designer of the component (along with the continuation code) 0xB: ARM_JEP106_IDENTITY_CODE_3_0 (Lowest 4 bits of the JEP106 Identity Code)
3:0	PART_1	Bits [11:8] of the component's part number. This is selected by the designer of the component. 0x9: CTI_PART_NUMBER_11_8 (Upper 4 bits of the Part Number (0x961))

0x00810FE8 QDSS_CTIO_PERIPHID2**Type:** R**Reset State:** 0x0000004B

Part of the set of Peripheral Identification registers. Contains part of the designer identity and the product revision.

QDSS_CTIO_PERIPHID2

Bits	Name	Description
7:4	REVISION	The Revision field is an incremental value starting at 0x0 for the first design of this component. This only increases by 1 for both major and minor revisions and is simply used as a look-up to establish the exact major/minor revision. 0x4: R0P4 (This device is at r0p4)
3	JEDEC	Always set. Indicates that a JEDEC assigned value is used 0x1: JEDEC_IDENTITY (The designer ID is specified by JEDEC (http://www.jedec.org))
2:0	DES_1	Bits 6:4 of the JEDEC identity code indicating the designer of the component (along with the continuation code) 0x3: ARM_JEP106_IDENTITY_CODE_6_4 (Upper 3 bits of the JEP106 Identity Code)

0x00810FEC QDSS_CTIO_PERIPHID3**Type:** R**Reset State:** 0x00000000

Part of the set of Peripheral Identification registers. Contains the RevAnd and Customer Modified fields.

QDSS_CTIO_PERIPHID3

Bits	Name	Description
7:4	REVAND	This field indicates minor errata fixes specific to this design, for example metal fixes after implementation. In most cases this field is zero. It is recommended that component designers ensure this field can be changed by a metal fix if required, for example by driving it from registers that reset to zero. 0x0: NO_MODIFICATION (Indicates that there have been no metal fixes to this component)
3:0	CMOD	Where the component is reusable IP, this value indicates if the customer has modified the behavior of the component. In most cases this field is zero. 0x0: NO_MODIFICATION (Indicates that there have been no modifications made)

0x00810FD0 QDSS_CTIO_PERIPHID4**Type:** R**Reset State:** 0x00000004

Part of the set of Peripheral Identification registers. Contains part of the designer identity and the memory footprint indicator.

QDSS_CTIO_PERIPHID4

Bits	Name	Description
7:4	SIZE	This is a 4-bit value that indicates the total contiguous size of the memory window used by this component in powers of 2 from the standard 4KB. If a component only requires the standard 4KB then this should read as 0x0, 4KB only, for 8KB set to 0x1, 16KB == 0x2, 32KB == 0x3, and so on. 0x0: ENUM_4KB (Indicates that the device only occupies 4kB of memory)
3:0	DES_2	JEDEC continuation code indicating the designer of the component (along with the identity code) 0x4: ARM_JEP106_CONTINUATION_CODE (Indicates that ARM's JEDEC identity code is on the 5th bank)

0x00810FD4 QDSS_CTIO_PERIPHID5

Type: RW

Reset State: Undefined

Reserved

QDSS_CTIO_PERIPHID5

Bits	Name	Description
31:0	PERIPHID5	

0x00810FD8 QDSS_CTIO_PERIPHID6

Type: RW

Reset State: Undefined

Reserved

QDSS_CTIO_PERIPHID6

Bits	Name	Description
31:0	PERIPHID6	

0x00810FDC QDSS_CTIO_PERIPHID7**Type:** RW**Reset State:** Undefined

Reserved

QDSS_CTIO_PERIPHID7

Bits	Name	Description
31:0	PERIPHID7	

0x00810FF0 QDSS_CTIO_COMPID0**Type:** R**Reset State:** 0x0000000D

A component identification register, that indicates that the identification registers are present.

QDSS_CTIO_COMPID0

Bits	Name	Description
7:0	PRMBL_0	Contains bits [7:0] of the component identification 0xD: ENUM_0D (Identification value)

0x00810FF4 QDSS_CTIO_COMPID1**Type:** R**Reset State:** 0x00000090

A component identification register, that indicates that the identification registers are present. This register also indicates the component class.

QDSS_CTIO_COMPID1

Bits	Name	Description
7:4	CLASS	Class of the component. E.g. ROM table, CoreSight component etc. Constitutes bits [15:12] of the component identification. 0x9: CORESIGHT_COMPONENT (Indicates the component is a CoreSight component)
3:0	PRMBL_1	Contains bits [11:8] of the component identification 0x0: ENUM_0 (Identification value)

0x00810FF8 QDSS_CTIO_COMPID2**Type:** R**Reset State:** 0x00000005

A component identification register, that indicates that the identification registers are present.

QDSS_CTIO_COMPID2

Bits	Name	Description
7:0	PRMBL_2	Contains bits [23:16] of the component identification 0x5: ENUM_05 (Identification value)

0x00810FFC QDSS_CTIO_COMPID3**Type:** R**Reset State:** 0x000000B1

A component identification register, that indicates that the identification registers are present.

QDSS_CTIO_COMPID3

Bits	Name	Description
7:0	PRMBL_3	Contains bits [31:24] of the component identification 0xB1: B1 (Identification value)

12.1.5 CTI1_CSCTI

The CTI allows mapping a trigger on the Trigger interface to a signal on the Channel interface and vice versa. The mapping is user-programmable. When the CTI receives a channel event it maps this onto a trigger output. This enables subsystems to cross trigger with each other. The receiving and transmitting of triggers is performed through the trigger interface.

12.1.5.1 QDSS_CTII_CSCTI

The CTI allows mapping a trigger on the Trigger interface to a signal on the Channel interface and vice versa. The mapping is user-programmable. When the CTI receives a channel event it maps this onto a trigger output. This enables subsystems to cross trigger with each other. The receiving and transmitting of triggers is performed through the trigger interface.

0x00811000 QDSS_CTI1_CTICONTROL**Type:** RW**Reset State:** 0x00000000**QDSS_CTI1_CTICONTROL**

Bits	Name	Description
0	GLBEN	0x1: ENABLED (When this bit is 1, cross triggering mapping logic functionality is enabled.) 0x0: DISABLED (When this bit is 0, all cross triggering mapping logic functionality is disabled.)

0x00811010 QDSS_CTI1_CTIINTACK**Type:** W**Reset State:** 0x00000000**QDSS_CTI1_CTIINTACK**

Bits	Name	Description
7:0	INTACK	

0x00811014 QDSS_CTI1_CTIAPPSET**Type:** W**Reset State:** 0x00000000**QDSS_CTI1_CTIAPPSET**

Bits	Name	Description
3:0	APPSET	

0x00811018 QDSS_CTI1_CTIAPPCLEAR**Type:** W**Reset State:** 0x00000000**QDSS_CTI1_CTIAPPCLEAR**

Bits	Name	Description
3:0	APPCLEAR	

0x0081101C QDSS_CTI1_CTIAPPPULSE**Type:** W**Reset State:** 0x00000000**QDSS_CTI1_CTIAPPPULSE**

Bits	Name	Description
3:0	APPULSE	

0x00811020 QDSS_CTI1_CTIINEN0**Type:** RW**Reset State:** 0x00000000**QDSS_CTI1_CTIINEN0**

Bits	Name	Description
3:0	TRIGINEN	

0x00811024 QDSS_CTI1_CTIINEN1**Type:** RW**Reset State:** 0x00000000**QDSS_CTI1_CTIINEN1**

Bits	Name	Description
3:0	TRIGINEN	

0x00811028 QDSS_CTI1_CTIINEN2**Type:** RW**Reset State:** 0x00000000**QDSS_CTI1_CTIINEN2**

Bits	Name	Description
3:0	TRIGINEN	

0x0081102C QDSS_CTI1_CTIINEN3**Type:** RW**Reset State:** 0x00000000**QDSS_CTI1_CTIINEN3**

Bits	Name	Description
3:0	TRIGINEN	

0x00811030 QDSS_CTI1_CTIINEN4**Type:** RW**Reset State:** 0x00000000**QDSS_CTI1_CTIINEN4**

Bits	Name	Description
3:0	TRIGINEN	

0x00811034 QDSS_CTI1_CTIINEN5**Type:** RW**Reset State:** 0x00000000**QDSS_CTI1_CTIINEN5**

Bits	Name	Description
3:0	TRIGINEN	

0x00811038 QDSS_CTI1_CTIINEN6**Type:** RW**Reset State:** 0x00000000**QDSS_CTI1_CTIINEN6**

Bits	Name	Description
3:0	TRIGINEN	

0x0081103C QDSS_CTI1_CTIINEN7**Type:** RW**Reset State:** 0x00000000**QDSS_CTI1_CTIINEN7**

Bits	Name	Description
3:0	TRIGINEN	

0x008110A0 QDSS_CTI1_CTIOUTEN0**Type:** RW**Reset State:** 0x00000000**QDSS_CTI1_CTIOUTEN0**

Bits	Name	Description
3:0	TRIGOUTEN	

0x008110A4 QDSS_CTI1_CTIOUTEN1**Type:** RW**Reset State:** 0x00000000**QDSS_CTI1_CTIOUTEN1**

Bits	Name	Description
3:0	TRIGOUTEN	

0x008110A8 QDSS_CTI1_CTIOUTEN2**Type:** RW**Reset State:** 0x00000000**QDSS_CTI1_CTIOUTEN2**

Bits	Name	Description
3:0	TRIGOUTEN	

0x008110AC QDSS_CTI1_CTIOUTEN3**Type:** RW**Reset State:** 0x00000000**QDSS_CTI1_CTIOUTEN3**

Bits	Name	Description
3:0	TRIGOUTEN	

0x008110B0 QDSS_CTI1_CTIOUTEN4**Type:** RW**Reset State:** 0x00000000**QDSS_CTI1_CTIOUTEN4**

Bits	Name	Description
3:0	TRIGOUTEN	

0x008110B4 QDSS_CTI1_CTIOUTEN5**Type:** RW**Reset State:** 0x00000000**QDSS_CTI1_CTIOUTEN5**

Bits	Name	Description
3:0	TRIGOUTEN	

0x008110B8 QDSS_CTI1_CTIOUTEN6**Type:** RW**Reset State:** 0x00000000**QDSS_CTI1_CTIOUTEN6**

Bits	Name	Description
3:0	TRIGOUTEN	

0x008110BC QDSS_CTI1_CTIOUTEN7**Type:** RW**Reset State:** 0x00000000**QDSS_CTI1_CTIOUTEN7**

Bits	Name	Description
3:0	TRIGOUTEN	

0x00811130 QDSS_CTI1_CTITRIGINSTATUS**Type:** R**Reset State:** Undefined**QDSS_CTI1_CTITRIGINSTATUS**

Bits	Name	Description
7:0	TRIGINSTATUS	

0x00811134 QDSS_CTI1_CTITRIGOUTSTATUS**Type:** R**Reset State:** 0x00000000**QDSS_CTI1_CTITRIGOUTSTATUS**

Bits	Name	Description
7:0	TRIGOUTSTATUS	

0x00811138 QDSS_CTI1_CTICHINSTAUS**Type:** R**Reset State:** Undefined**QDSS_CTI1_CTICHINSTAUS**

Bits	Name	Description
3:0	CTICHINSTAUS	

0x0081113C QDSS_CTI1_CTICHOUTSTATUS**Type:** R**Reset State:** 0x00000000**QDSS_CTI1_CTICHOUTSTATUS**

Bits	Name	Description
3:0	CTICHOUTSTATUS	

0x00811140 QDSS_CTI1_CTIGATE**Type:** RW**Reset State:** 0x0000000F**QDSS_CTI1_CTIGATE**

Bits	Name	Description
3	CTIGATEEN3	
2	CTIGATEEN2	
1	CTIGATEEN1	
0	CTIGATEEN0	

0x00811144 QDSS_CTI1_ASICCTL**Type:** RW**Reset State:** 0x00000000**QDSS_CTI1_ASICCTL**

Bits	Name	Description
7:0	ASICCTL	

0x00811EDC QDSS_CTI1_ITCHINACK**Type:** W**Reset State:** 0x00000000**QDSS_CTI1_ITCHINACK**

Bits	Name	Description
3:0	CTCHINACK	

0x00811EE0 QDSS_CTI1_ITTRIGINACK**Type:** W**Reset State:** 0x00000000**QDSS_CTI1_ITTRIGINACK**

Bits	Name	Description
7:0	CTTRIGINACK	

0x00811EE4 QDSS_CTI1_ITCHOUT**Type:** W**Reset State:** 0x00000000**QDSS_CTI1_ITCHOUT**

Bits	Name	Description
3:0	CTCHOUT	

0x00811EE8 QDSS_CTI1_ITTRIGOUT**Type:** W**Reset State:** 0x00000000**QDSS_CTI1_ITTRIGOUT**

Bits	Name	Description
7:0	CTTRIGOUT	

0x00811EEC QDSS_CTI1_ITCHOUTACK**Type:** R**Reset State:** 0x00000000**QDSS_CTI1_ITCHOUTACK**

Bits	Name	Description
3:0	CTCHOUTACK	

0x00811EF0 QDSS_CTI1_ITTRIGOUTACK**Type:** R**Reset State:** 0x00000000**QDSS_CTI1_ITTRIGOUTACK**

Bits	Name	Description
7:0	CTTRIGOUTACK	

0x00811EF4 QDSS_CTI1_ITCHIN**Type:** R**Reset State:** 0x00000000**QDSS_CTI1_ITCHIN**

Bits	Name	Description
3:0	CTCHIN	

0x00811EF8 QDSS_CTI1_ITTRIGIN**Type:** R**Reset State:** 0x00000000**QDSS_CTI1_ITTRIGIN**

Bits	Name	Description
7:0	CTTRIGIN	

0x00811F00 QDSS_CTI1_ITCTRL**Type:** RW**Reset State:** 0x00000000**QDSS_CTI1_ITCTRL**

Bits	Name	Description
0	INTEGRATION_MODE	Allows the component to switch from functional mode to integration mode or back. 0x1: INTEGRATION_MODE (Enable integration mode) 0x0: FUNCTIONAL_MODE (Disable integration mode)

0x00811FA0 QDSS_CTI1_CLAIMSET**Type:** RW**Reset State:** 0x0000000F

This is used in conjunction with Claim Tag Clear Register, CLAIMCLR. This register forms one half of the Claim Tag value. This location allows individual bits to be set, write, and returns the number of bits that can be set, read.

QDSS_CTI1_CLAIMSET

Bits	Name	Type	Description
3:0	CLAIMSET_W	W	Each bit is considered separately: 0 = no effect, 1 = set this bit in the claim tag.
3:0	CLAIMSET_R	R	This claim tag bit is implemented 0xF: CLAIM_TAG_IMPLEMENTED_BITS (These bits are present within the claim tag field)

0x00811FA4 QDSS_CTI1_CLAIMCLR**Type:** RW**Reset State:** 0x00000000

This register is used in conjunction with Claim Tag Set Register, CLAIMSET. This register forms one half of the Claim Tag value. This location enables individual bits to be cleared, write, and returns the current Claim Tag value, read.

QDSS_CTI1_CLAIMCLR

Bits	Name	Type	Description
3:0	CLAIMCLR_W	W	Each bit is considered separately: 0 = no effect, 1 = clear this bit in the claim tag.
3:0	CLAIMCLR_R	R	The value present reflects the current setting of the Claim Tag.

0x00811FB0 QDSS_CTI1_LAR**Type:** W**Reset State:** Undefined

QDSS_CTI1_LAR

Bits	Name	Description
31:0	ACCESS_W	A write of 0xC5ACCE55 enables further write access to this device. A write of any value other than 0xC5ACCE55 will have the affect of removing write access. 0xC5ACCE55: UNLOCK (Unlock the protection register to enable write access)

0x00811FB4 QDSS_CTI1_LSR**Type:** R**Reset State:** 0x00000003**QDSS_CTI1_LSR**

Bits	Name	Description
2	LOCKTYPE	Indicates if the Lock Access Register (0xFB0) is implemented as 8-bit or 32-bit 0x0: ENUM_32_BIT (This component implements a 32-bit Lock Access Register)
1	LOCKGRANT	Returns the current status of the Lock. This bit reads as 0 when read from an external debugger (PADDRDBG31 = 1) since external debugger accesses are not subject to Lock Registers. 0x0: ACCESS_PERMITTED (Write access is permitted to this device) 0x1: DEVICE_LOCKED (Write access to the component is blocked. All writes to control registers are ignored. Reads are permitted)
0	LOCKEXIST	Indicates that a lock control mechanism exists for this device. This bit reads as 0 when read from an external debugger (PADDRDBG31 = 1) since external debugger accesses are not subject to Lock Registers. 0x0: LOCK_NOT_PRESENT (No lock control mechanism exists, writes to the Lock Access register (0xFB0) will be ignored) 0x1: LOCK_PRESENT (Lock control mechanism is present)

0x00811FB8 QDSS_CTI1_AUTHSTATUS**Type:** R**Reset State:** 0xFFFFFFFF0X

Reports what functionality is currently permitted by the authentication interface.

QDSS_CTL1_AUTHSTATUS

Bits	Name	Description
7:6	SNID	Indicates the security level for secure non-invasive debug 0x0: ENUM_2_B00 (Functionality not implemented.)
5:4	SID	Indicates the security level for secure invasive debug 0x0: ENUM_2_B00 (Functionality not implemented.)
3:2	NSNID	Indicates the security level for non-secure non-invasive debug 0x2: ENUM_2_B10 (Non-secure Non-Invasive debug functionality currently disabled.) 0x3: ENUM_2_B11 (Non-secure Non-Invasive debug functionality currently enabled.)
1:0	NSID	Indicates the security level for non-secure invasive debug 0x2: ENUM_2_B10 (Non-secure Invasive debug functionality currently disabled.) 0x3: ENUM_2_B11 (Non-secure Invasive debug functionality currently enabled.)

0x00811FC8 QDSS_CTL1_DEVID**Type:** R**Reset State:** 0x00040800

This register indicates the capabilities of the CTI.

QDSS_CTL1_DEVID

Bits	Name	Description
19:16	NUMCH	Number of ECT channels available.
15:8	NUMTRIG	Number of ECT triggers available.
4:0	EXTMUXNUM	Indicates the number of multiplexing available on Trigger Inputs and Trigger Outputs using ASICCTL. Default value of 5'b00000 indicating no multiplexing present. Reflects the value of the Verilog `define EXTMUXNUM that the user must alter accordingly.

0x00811FE0 QDSS_CTL1_PERIPHID0**Type:** R**Reset State:** 0x00000006

Part of the set of Peripheral Identification registers. Contains part of the designer specific part number.

QDSS_CTI1_PERIPHID0

Bits	Name	Description
7:0	PART_0	Bits [7:0] of the component's part number. This is selected by the designer of the component. 0x6: CTI_PART_NUMBER_7_0 (Lowest 8 bits of the Part Number (0x906))

0x00811FE4 QDSS_CTI1_PERIPHID1**Type:** R**Reset State:** 0x000000B9

Part of the set of Peripheral Identification registers. Contains part of the designer specific part number and part of the designer identity.

QDSS_CTI1_PERIPHID1

Bits	Name	Description
7:4	DES_0	Bits 3:0 of the JEDEC identity code indicating the designer of the component (along with the continuation code) 0xB: ARM_JEP106_IDENTITY_CODE_3_0 (Lowest 4 bits of the JEP106 Identity Code)
3:0	PART_1	Bits [11:8] of the component's part number. This is selected by the designer of the component. 0x9: CTI_PART_NUMBER_11_8 (Upper 4 bits of the Part Number (0x961))

0x00811FE8 QDSS_CTI1_PERIPHID2**Type:** R**Reset State:** 0x0000004B

Part of the set of Peripheral Identification registers. Contains part of the designer identity and the product revision.

QDSS_CTI1_PERIPHID2

Bits	Name	Description
7:4	REVISION	The Revision field is an incremental value starting at 0x0 for the first design of this component. This only increases by 1 for both major and minor revisions and is simply used as a look-up to establish the exact major/minor revision. 0x4: R0P4 (This device is at r0p4)

QDSS_CTI1_PERIPHID2 (cont.)

Bits	Name	Description
3	JEDEC	Always set. Indicates that a JEDEC assigned value is used 0x1: JEDEC_IDENTITY (The designer ID is specified by JEDEC (http://www.jedec.org))
2:0	DES_1	Bits 6:4 of the JEDEC identity code indicating the designer of the component (along with the continuation code) 0x3: ARM_JEP106_IDENTITY_CODE_6_4 (Upper 3 bits of the JEP106 Identity Code)

0x00811FEC QDSS_CTI1_PERIPHID3**Type:** R**Reset State:** 0x00000000

Part of the set of Peripheral Identification registers. Contains the RevAnd and Customer Modified fields.

QDSS_CTI1_PERIPHID3

Bits	Name	Description
7:4	REVAND	This field indicates minor errata fixes specific to this design, for example metal fixes after implementation. In most cases this field is zero. It is recommended that component designers ensure this field can be changed by a metal fix if required, for example by driving it from registers that reset to zero. 0x0: NO_MODIFICATION (Indicates that there have been no metal fixes to this component)
3:0	CMOD	Where the component is reusable IP, this value indicates if the customer has modified the behavior of the component. In most cases this field is zero. 0x0: NO_MODIFICATION (Indicates that there have been no modifications made)

0x00811FD0 QDSS_CTI1_PERIPHID4**Type:** R**Reset State:** 0x00000004

Part of the set of Peripheral Identification registers. Contains part of the designer identity and the memory footprint indicator.

QDSS_CTI1_PERIPHID4

Bits	Name	Description
7:4	SIZE	This is a 4-bit value that indicates the total contiguous size of the memory window used by this component in powers of 2 from the standard 4KB. If a component only requires the standard 4KB then this should read as 0x0, 4KB only, for 8KB set to 0x1, 16KB == 0x2, 32KB == 0x3, and so on. 0x0: ENUM_4KB (Indicates that the device only occupies 4kB of memory)
3:0	DES_2	JEDEC continuation code indicating the designer of the component (along with the identity code) 0x4: ARM_JEP106_CONTINUATION_CODE (Indicates that ARM's JEDEC identity code is on the 5th bank)

0x00811FD4 QDSS_CTI1_PERIPHID5**Type:** RW**Reset State:** Undefined

Reserved

QDSS_CTI1_PERIPHID5

Bits	Name	Description
31:0	PERIPHID5	

0x00811FD8 QDSS_CTI1_PERIPHID6**Type:** RW**Reset State:** Undefined

Reserved

QDSS_CTI1_PERIPHID6

Bits	Name	Description
31:0	PERIPHID6	

0x00811FDC QDSS_CTI1_PERIPHID7**Type:** RW**Reset State:** Undefined

Reserved

QDSS_CTI1_PERIPHID7

Bits	Name	Description
31:0	PERIPHID7	

0x00811FF0 QDSS_CTI1_COMPID0

Type: R

Reset State: 0x0000000D

A component identification register, that indicates that the identification registers are present.

QDSS_CTI1_COMPID0

Bits	Name	Description
7:0	PRMBL_0	Contains bits [7:0] of the component identification 0xD: ENUM_0D (Identification value)

0x00811FF4 QDSS_CTI1_COMPID1

Type: R

Reset State: 0x00000090

A component identification register, that indicates that the identification registers are present. This register also indicates the component class.

QDSS_CTI1_COMPID1

Bits	Name	Description
7:4	CLASS	Class of the component. E.g. ROM table, CoreSight component etc. Constitutes bits [15:12] of the component identification. 0x9: CORESIGHT_COMPONENT (Indicates the component is a CoreSight component)
3:0	PRMBL_1	Contains bits [11:8] of the component identification 0x0: ENUM_0 (Identification value)

0x00811FF8 QDSS_CTI1_COMPID2

Type: R

Reset State: 0x00000005

A component identification register, that indicates that the identification registers are present.

QDSS_CTI1_COMPID2

Bits	Name	Description
7:0	PRMBL_2	Contains bits [23:16] of the component identification 0x5: ENUM_05 (Identification value)

0x00811FFC QDSS_CTI1_COMPID3**Type:** R**Reset State:** 0x000000B1

A component identification register, that indicates that the identification registers are present.

QDSS_CTI1_COMPID3

Bits	Name	Description
7:0	PRMBL_3	Contains bits [31:24] of the component identification 0xB1: B1 (Identification value)

12.1.6 CTI2_CSCTI

The CTI allows mapping a trigger on the Trigger interface to a signal on the Channel interface and vice versa. The mapping is user-programmable. When the CTI receives a channel event it maps this onto a trigger output. This enables subsystems to cross trigger with each other. The receiving and transmitting of triggers is performed through the trigger interface.

12.1.6.1 QDSS_CTI2_CSCTI

The CTI allows mapping a trigger on the Trigger interface to a signal on the Channel interface and vice versa. The mapping is user-programmable. When the CTI receives a channel event it maps this onto a trigger output. This enables subsystems to cross trigger with each other. The receiving and transmitting of triggers is performed through the trigger interface.

0x00812000 QDSS_CTI2_CTICONTROL**Type:** RW**Reset State:** 0x00000000

QDSS_CT12_CTICONTROL

Bits	Name	Description
0	GLBEN	0x1: ENABLED (When this bit is 1, cross triggering mapping logic functionality is enabled.) 0x0: DISABLED (When this bit is 0, all cross triggering mapping logic functionality is disabled.)

0x00812010 QDSS_CT12_CTIINTACK**Type:** W**Reset State:** 0x00000000**QDSS_CT12_CTIINTACK**

Bits	Name	Description
7:0	INTACK	

0x00812014 QDSS_CT12_CTIAPPSET**Type:** W**Reset State:** 0x00000000**QDSS_CT12_CTIAPPSET**

Bits	Name	Description
3:0	APPSET	

0x00812018 QDSS_CT12_CTIAPPCLEAR**Type:** W**Reset State:** 0x00000000**QDSS_CT12_CTIAPPCLEAR**

Bits	Name	Description
3:0	APPCLEAR	

0x0081201C QDSS_CT12_CTIAPPPULSE**Type:** W**Reset State:** 0x00000000

QDSS_CT12_CTIAPPPULSE

Bits	Name	Description
3:0	APPULSE	

0x00812020 QDSS_CT12_CTIINEN0**Type:** RW**Reset State:** 0x00000000**QDSS_CT12_CTIINEN0**

Bits	Name	Description
3:0	TRIGINEN	

0x00812024 QDSS_CT12_CTIINEN1**Type:** RW**Reset State:** 0x00000000**QDSS_CT12_CTIINEN1**

Bits	Name	Description
3:0	TRIGINEN	

0x00812028 QDSS_CT12_CTIINEN2**Type:** RW**Reset State:** 0x00000000**QDSS_CT12_CTIINEN2**

Bits	Name	Description
3:0	TRIGINEN	

0x0081202C QDSS_CT12_CTIINEN3**Type:** RW**Reset State:** 0x00000000

QDSS_CT12_CTIINEN3

Bits	Name	Description
3:0	TRIGINEN	

0x00812030 QDSS_CT12_CTIINEN4**Type:** RW**Reset State:** 0x00000000**QDSS_CT12_CTIINEN4**

Bits	Name	Description
3:0	TRIGINEN	

0x00812034 QDSS_CT12_CTIINEN5**Type:** RW**Reset State:** 0x00000000**QDSS_CT12_CTIINEN5**

Bits	Name	Description
3:0	TRIGINEN	

0x00812038 QDSS_CT12_CTIINEN6**Type:** RW**Reset State:** 0x00000000**QDSS_CT12_CTIINEN6**

Bits	Name	Description
3:0	TRIGINEN	

0x0081203C QDSS_CT12_CTIINEN7**Type:** RW**Reset State:** 0x00000000

QDSS_CT12_CTIINEN7

Bits	Name	Description
3:0	TRIGINEN	

0x008120A0 QDSS_CT12_CTIOUTEN0**Type:** RW**Reset State:** 0x00000000**QDSS_CT12_CTIOUTEN0**

Bits	Name	Description
3:0	TRIGOUTEN	

0x008120A4 QDSS_CT12_CTIOUTEN1**Type:** RW**Reset State:** 0x00000000**QDSS_CT12_CTIOUTEN1**

Bits	Name	Description
3:0	TRIGOUTEN	

0x008120A8 QDSS_CT12_CTIOUTEN2**Type:** RW**Reset State:** 0x00000000**QDSS_CT12_CTIOUTEN2**

Bits	Name	Description
3:0	TRIGOUTEN	

0x008120AC QDSS_CT12_CTIOUTEN3**Type:** RW**Reset State:** 0x00000000

QDSS_CT12_CTIOUTEN3

Bits	Name	Description
3:0	TRIGOUTEN	

0x008120B0 QDSS_CT12_CTIOUTEN4**Type:** RW**Reset State:** 0x00000000**QDSS_CT12_CTIOUTEN4**

Bits	Name	Description
3:0	TRIGOUTEN	

0x008120B4 QDSS_CT12_CTIOUTEN5**Type:** RW**Reset State:** 0x00000000**QDSS_CT12_CTIOUTEN5**

Bits	Name	Description
3:0	TRIGOUTEN	

0x008120B8 QDSS_CT12_CTIOUTEN6**Type:** RW**Reset State:** 0x00000000**QDSS_CT12_CTIOUTEN6**

Bits	Name	Description
3:0	TRIGOUTEN	

0x008120BC QDSS_CT12_CTIOUTEN7**Type:** RW**Reset State:** 0x00000000

QDSS_CT12_CTIOUTEN7

Bits	Name	Description
3:0	TRIGOUTEN	

0x00812130 QDSS_CT12_CTITRIGINSTATUS**Type:** R**Reset State:** Undefined**QDSS_CT12_CTITRIGINSTATUS**

Bits	Name	Description
7:0	TRIGINSTATUS	

0x00812134 QDSS_CT12_CTITRIGOUTSTATUS**Type:** R**Reset State:** 0x00000000**QDSS_CT12_CTITRIGOUTSTATUS**

Bits	Name	Description
7:0	TRIGOUTSTATUS	

0x00812138 QDSS_CT12_CTICHINSTSTATUS**Type:** R**Reset State:** Undefined**QDSS_CT12_CTICHINSTSTATUS**

Bits	Name	Description
3:0	CTICHINSTSTATUS	

0x0081213C QDSS_CT12_CTICHOUTSTATUS**Type:** R**Reset State:** 0x00000000

QDSS_CT12_CTICHOUTSTATUS

Bits	Name	Description
3:0	CTICHOUTSTATUS	

0x00812140 QDSS_CT12_CTIGATE**Type:** RW**Reset State:** 0x0000000F**QDSS_CT12_CTIGATE**

Bits	Name	Description
3	CTIGATEEN3	
2	CTIGATEEN2	
1	CTIGATEEN1	
0	CTIGATEEN0	

0x00812144 QDSS_CT12_ASICCTL**Type:** RW**Reset State:** 0x00000000**QDSS_CT12_ASICCTL**

Bits	Name	Description
7:0	ASICCTL	

0x00812EDC QDSS_CT12_ITCHINACK**Type:** W**Reset State:** 0x00000000**QDSS_CT12_ITCHINACK**

Bits	Name	Description
3:0	CTCHINACK	

0x00812EE0 QDSS_CT12_ITTRIGINACK**Type:** W**Reset State:** 0x00000000**QDSS_CT12_ITTRIGINACK**

Bits	Name	Description
7:0	CTTRIGINACK	

0x00812EE4 QDSS_CT12_ITCHOUT**Type:** W**Reset State:** 0x00000000**QDSS_CT12_ITCHOUT**

Bits	Name	Description
3:0	CTCHOUT	

0x00812EE8 QDSS_CT12_ITTRIGOUT**Type:** W**Reset State:** 0x00000000**QDSS_CT12_ITTRIGOUT**

Bits	Name	Description
7:0	CTTRIGOUT	

0x00812EEC QDSS_CT12_ITCHOUTACK**Type:** R**Reset State:** 0x00000000**QDSS_CT12_ITCHOUTACK**

Bits	Name	Description
3:0	CTCHOUTACK	

0x00812EF0 QDSS_CT12_ITTRIGOUTACK**Type:** R**Reset State:** 0x00000000**QDSS_CT12_ITTRIGOUTACK**

Bits	Name	Description
7:0	CTTRIGOUTACK	

0x00812EF4 QDSS_CT12_ITCHIN**Type:** R**Reset State:** 0x00000000**QDSS_CT12_ITCHIN**

Bits	Name	Description
3:0	CTCHIN	

0x00812EF8 QDSS_CT12_ITTRIGIN**Type:** R**Reset State:** 0x00000000**QDSS_CT12_ITTRIGIN**

Bits	Name	Description
7:0	CTTRIGIN	

0x00812F00 QDSS_CT12_ITCTRL**Type:** RW**Reset State:** 0x00000000**QDSS_CT12_ITCTRL**

Bits	Name	Description
0	INTEGRATION_MODE	Allows the component to switch from functional mode to integration mode or back. 0x1: INTEGRATION_MODE (Enable integration mode) 0x0: FUNCTIONAL_MODE (Disable integration mode)

0x00812FA0 QDSS_CT12_CLAIMSET**Type:** RW**Reset State:** 0x0000000F

This is used in conjunction with Claim Tag Clear Register, CLAIMCLR. This register forms one half of the Claim Tag value. This location allows individual bits to be set, write, and returns the number of bits that can be set, read.

QDSS_CT12_CLAIMSET

Bits	Name	Type	Description
3:0	CLAIMSET_W	W	Each bit is considered separately: 0 = no effect, 1 = set this bit in the claim tag.
3:0	CLAIMSET_R	R	This claim tag bit is implemented 0xF: CLAIM_TAG_IMPLEMENTED_BITS (These bits are present within the claim tag field)

0x00812FA4 QDSS_CT12_CLAIMCLR**Type:** RW**Reset State:** 0x00000000

This register is used in conjunction with Claim Tag Set Register, CLAIMSET. This register forms one half of the Claim Tag value. This location enables individual bits to be cleared, write, and returns the current Claim Tag value, read.

QDSS_CT12_CLAIMCLR

Bits	Name	Type	Description
3:0	CLAIMCLR_W	W	Each bit is considered separately: 0 = no effect, 1 = clear this bit in the claim tag.
3:0	CLAIMCLR_R	R	The value present reflects the current setting of the Claim Tag.

0x00812FB0 QDSS_CT12_LAR**Type:** W**Reset State:** Undefined

QDSS_CTI2_LAR

Bits	Name	Description
31:0	ACCESS_W	A write of 0xC5ACCE55 enables further write access to this device. A write of any value other than 0xC5ACCE55 will have the affect of removing write access. 0xC5ACCE55: UNLOCK (Unlock the protection register to enable write access)

0x00812FB4 QDSS_CTI2_LSR**Type:** R**Reset State:** 0x00000003**QDSS_CTI2_LSR**

Bits	Name	Description
2	LOCKTYPE	Indicates if the Lock Access Register (0xFB0) is implemented as 8-bit or 32-bit 0x0: ENUM_32_BIT (This component implements a 32-bit Lock Access Register)
1	LOCKGRANT	Returns the current status of the Lock. This bit reads as 0 when read from an external debugger (PADDRDBG31 = 1) since external debugger accesses are not subject to Lock Registers. 0x0: ACCESS_PERMITTED (Write access is permitted to this device) 0x1: DEVICE_LOCKED (Write access to the component is blocked. All writes to control registers are ignored. Reads are permitted)
0	LOCKEXIST	Indicates that a lock control mechanism exists for this device. This bit reads as 0 when read from an external debugger (PADDRDBG31 = 1) since external debugger accesses are not subject to Lock Registers. 0x0: LOCK_NOT_PRESENT (No lock control mechanism exists, writes to the Lock Access register (0xFB0) will be ignored) 0x1: LOCK_PRESENT (Lock control mechanism is present)

0x00812FB8 QDSS_CTI2_AUTHSTATUS**Type:** R**Reset State:** 0xFFFFFFFF0X

Reports what functionality is currently permitted by the authentication interface.

QDSS_CTL2_AUTHSTATUS

Bits	Name	Description
7:6	SNID	Indicates the security level for secure non-invasive debug 0x0: ENUM_2_B00 (Functionality not implemented.)
5:4	SID	Indicates the security level for secure invasive debug 0x0: ENUM_2_B00 (Functionality not implemented.)
3:2	NSNID	Indicates the security level for non-secure non-invasive debug 0x2: ENUM_2_B10 (Non-secure Non-Invasive debug functionality currently disabled.) 0x3: ENUM_2_B11 (Non-secure Non-Invasive debug functionality currently enabled.)
1:0	NSID	Indicates the security level for non-secure invasive debug 0x2: ENUM_2_B10 (Non-secure Invasive debug functionality currently disabled.) 0x3: ENUM_2_B11 (Non-secure Invasive debug functionality currently enabled.)

0x00812FC8 QDSS_CTL2_DEVID**Type:** R**Reset State:** 0x00040800

This register indicates the capabilities of the CTI.

QDSS_CTL2_DEVID

Bits	Name	Description
19:16	NUMCH	Number of ECT channels available.
15:8	NUMTRIG	Number of ECT triggers available.
4:0	EXTMUXNUM	Indicates the number of multiplexing available on Trigger Inputs and Trigger Outputs using ASICCTL. Default value of 5'b00000 indicating no multiplexing present. Reflects the value of the Verilog `define EXTMUXNUM that the user must alter accordingly.

0x00812FE0 QDSS_CTL2_PERIPHID0**Type:** R**Reset State:** 0x00000006

Part of the set of Peripheral Identification registers. Contains part of the designer specific part number.

QDSS_CTID2_PERIPHID0

Bits	Name	Description
7:0	PART_0	Bits [7:0] of the component's part number. This is selected by the designer of the component. 0x6: CTI_PART_NUMBER_7_0 (Lowest 8 bits of the Part Number (0x906))

0x00812FE4 QDSS_CTID2_PERIPHID1**Type:** R**Reset State:** 0x000000B9

Part of the set of Peripheral Identification registers. Contains part of the designer specific part number and part of the designer identity.

QDSS_CTID2_PERIPHID1

Bits	Name	Description
7:4	DES_0	Bits 3:0 of the JEDEC identity code indicating the designer of the component (along with the continuation code) 0xB: ARM_JEP106_IDENTITY_CODE_3_0 (Lowest 4 bits of the JEP106 Identity Code)
3:0	PART_1	Bits [11:8] of the component's part number. This is selected by the designer of the component. 0x9: CTI_PART_NUMBER_11_8 (Upper 4 bits of the Part Number (0x961))

0x00812FE8 QDSS_CTID2_PERIPHID2**Type:** R**Reset State:** 0x0000004B

Part of the set of Peripheral Identification registers. Contains part of the designer identity and the product revision.

QDSS_CTID2_PERIPHID2

Bits	Name	Description
7:4	REVISION	The Revision field is an incremental value starting at 0x0 for the first design of this component. This only increases by 1 for both major and minor revisions and is simply used as a look-up to establish the exact major/minor revision. 0x4: R0P4 (This device is at r0p4)

QDSS_CTID2_PERIPHID2 (cont.)

Bits	Name	Description
3	JEDEC	Always set. Indicates that a JEDEC assigned value is used 0x1: JEDEC_IDENTITY (The designer ID is specified by JEDEC (http://www.jedec.org))
2:0	DES_1	Bits 6:4 of the JEDEC identity code indicating the designer of the component (along with the continuation code) 0x3: ARM_JEP106_IDENTITY_CODE_6_4 (Upper 3 bits of the JEP106 Identity Code)

0x00812FEC QDSS_CTID2_PERIPHID3**Type:** R**Reset State:** 0x00000000

Part of the set of Peripheral Identification registers. Contains the RevAnd and Customer Modified fields.

QDSS_CTID2_PERIPHID3

Bits	Name	Description
7:4	REVAND	This field indicates minor errata fixes specific to this design, for example metal fixes after implementation. In most cases this field is zero. It is recommended that component designers ensure this field can be changed by a metal fix if required, for example by driving it from registers that reset to zero. 0x0: NO_MODIFICATION (Indicates that there have been no metal fixes to this component)
3:0	CMOD	Where the component is reusable IP, this value indicates if the customer has modified the behavior of the component. In most cases this field is zero. 0x0: NO_MODIFICATION (Indicates that there have been no modifications made)

0x00812FD0 QDSS_CTID2_PERIPHID4**Type:** R**Reset State:** 0x00000004

Part of the set of Peripheral Identification registers. Contains part of the designer identity and the memory footprint indicator.

QDSS_CTI2_PERIPHID4

Bits	Name	Description
7:4	SIZE	This is a 4-bit value that indicates the total contiguous size of the memory window used by this component in powers of 2 from the standard 4KB. If a component only requires the standard 4KB then this should read as 0x0, 4KB only, for 8KB set to 0x1, 16KB == 0x2, 32KB == 0x3, and so on. 0x0: ENUM_4KB (Indicates that the device only occupies 4kB of memory)
3:0	DES_2	JEDEC continuation code indicating the designer of the component (along with the identity code) 0x4: ARM_JEP106_CONTINUATION_CODE (Indicates that ARM's JEDEC identity code is on the 5th bank)

0x00812FD4 QDSS_CTI2_PERIPHID5**Type:** RW**Reset State:** Undefined

Reserved

QDSS_CTI2_PERIPHID5

Bits	Name	Description
31:0	PERIPHID5	

0x00812FD8 QDSS_CTI2_PERIPHID6**Type:** RW**Reset State:** Undefined

Reserved

QDSS_CTI2_PERIPHID6

Bits	Name	Description
31:0	PERIPHID6	

0x00812FDC QDSS_CTI2_PERIPHID7**Type:** RW**Reset State:** Undefined

Reserved

QDSS_CTID2_PERIPHID7

Bits	Name	Description
31:0	PERIPHID7	

0x00812FF0 QDSS_CTID2_COMPID0

Type: R

Reset State: 0x0000000D

A component identification register, that indicates that the identification registers are present.

QDSS_CTID2_COMPID0

Bits	Name	Description
7:0	PRMBL_0	Contains bits [7:0] of the component identification 0xD: ENUM_0D (Identification value)

0x00812FF4 QDSS_CTID2_COMPID1

Type: R

Reset State: 0x00000090

A component identification register, that indicates that the identification registers are present. This register also indicates the component class.

QDSS_CTID2_COMPID1

Bits	Name	Description
7:4	CLASS	Class of the component. E.g. ROM table, CoreSight component etc. Constitutes bits [15:12] of the component identification. 0x9: CORESIGHT_COMPONENT (Indicates the component is a CoreSight component)
3:0	PRMBL_1	Contains bits [11:8] of the component identification 0x0: ENUM_0 (Identification value)

0x00812FF8 QDSS_CTID2_COMPID2

Type: R

Reset State: 0x00000005

A component identification register, that indicates that the identification registers are present.

QDSS_CTID2_COMPID2

Bits	Name	Description
7:0	PRMBL_2	Contains bits [23:16] of the component identification 0x5: ENUM_05 (Identification value)

0x00812FFC QDSS_CTID2_COMPID3**Type:** R**Reset State:** 0x000000B1

A component identification register, that indicates that the identification registers are present.

QDSS_CTID2_COMPID3

Bits	Name	Description
7:0	PRMBL_3	Contains bits [31:24] of the component identification 0xB1: B1 (Identification value)

12.1.7 CTI3_CSCTI

The CTI allows mapping a trigger on the Trigger interface to a signal on the Channel interface and vice versa. The mapping is user-programmable. When the CTI receives a channel event it maps this onto a trigger output. This enables subsystems to cross trigger with each other. The receiving and transmitting of triggers is performed through the trigger interface.

12.1.7.1 QDSS_CTID3_CSCTI

The CTI allows mapping a trigger on the Trigger interface to a signal on the Channel interface and vice versa. The mapping is user-programmable. When the CTI receives a channel event it maps this onto a trigger output. This enables subsystems to cross trigger with each other. The receiving and transmitting of triggers is performed through the trigger interface.

0x00813000 QDSS_CTID3_CTICONTROL**Type:** RW**Reset State:** 0x00000000

QDSS_CT13_CTICONTROL

Bits	Name	Description
0	GLBEN	0x1: ENABLED (When this bit is 1, cross triggering mapping logic functionality is enabled.) 0x0: DISABLED (When this bit is 0, all cross triggering mapping logic functionality is disabled.)

0x00813010 QDSS_CT13_CTIINTACK**Type:** W**Reset State:** 0x00000000**QDSS_CT13_CTIINTACK**

Bits	Name	Description
7:0	INTACK	

0x00813014 QDSS_CT13_CTIAPPSET**Type:** W**Reset State:** 0x00000000**QDSS_CT13_CTIAPPSET**

Bits	Name	Description
3:0	APPSET	

0x00813018 QDSS_CT13_CTIAPPCLEAR**Type:** W**Reset State:** 0x00000000**QDSS_CT13_CTIAPPCLEAR**

Bits	Name	Description
3:0	APPCLEAR	

0x0081301C QDSS_CT13_CTIAPPPULSE**Type:** W**Reset State:** 0x00000000

QDSS_CT13_CTIAPPPULSE

Bits	Name	Description
3:0	APPULSE	

0x00813020 QDSS_CT13_CTIINEN0**Type:** RW**Reset State:** 0x00000000**QDSS_CT13_CTIINEN0**

Bits	Name	Description
3:0	TRIGINEN	

0x00813024 QDSS_CT13_CTIINEN1**Type:** RW**Reset State:** 0x00000000**QDSS_CT13_CTIINEN1**

Bits	Name	Description
3:0	TRIGINEN	

0x00813028 QDSS_CT13_CTIINEN2**Type:** RW**Reset State:** 0x00000000**QDSS_CT13_CTIINEN2**

Bits	Name	Description
3:0	TRIGINEN	

0x0081302C QDSS_CT13_CTIINEN3**Type:** RW**Reset State:** 0x00000000

QDSS_CT13_CTIINEN3

Bits	Name	Description
3:0	TRIGINEN	

0x00813030 QDSS_CT13_CTIINEN4**Type:** RW**Reset State:** 0x00000000**QDSS_CT13_CTIINEN4**

Bits	Name	Description
3:0	TRIGINEN	

0x00813034 QDSS_CT13_CTIINEN5**Type:** RW**Reset State:** 0x00000000**QDSS_CT13_CTIINEN5**

Bits	Name	Description
3:0	TRIGINEN	

0x00813038 QDSS_CT13_CTIINEN6**Type:** RW**Reset State:** 0x00000000**QDSS_CT13_CTIINEN6**

Bits	Name	Description
3:0	TRIGINEN	

0x0081303C QDSS_CT13_CTIINEN7**Type:** RW**Reset State:** 0x00000000

QDSS_CT13_CTIINEN7

Bits	Name	Description
3:0	TRIGINEN	

0x008130A0 QDSS_CT13_CTIOUTEN0**Type:** RW**Reset State:** 0x00000000**QDSS_CT13_CTIOUTEN0**

Bits	Name	Description
3:0	TRIGOUTEN	

0x008130A4 QDSS_CT13_CTIOUTEN1**Type:** RW**Reset State:** 0x00000000**QDSS_CT13_CTIOUTEN1**

Bits	Name	Description
3:0	TRIGOUTEN	

0x008130A8 QDSS_CT13_CTIOUTEN2**Type:** RW**Reset State:** 0x00000000**QDSS_CT13_CTIOUTEN2**

Bits	Name	Description
3:0	TRIGOUTEN	

0x008130AC QDSS_CT13_CTIOUTEN3**Type:** RW**Reset State:** 0x00000000

QDSS_CT13_CTIOUTEN3

Bits	Name	Description
3:0	TRIGOUTEN	

0x008130B0 QDSS_CT13_CTIOUTEN4**Type:** RW**Reset State:** 0x00000000**QDSS_CT13_CTIOUTEN4**

Bits	Name	Description
3:0	TRIGOUTEN	

0x008130B4 QDSS_CT13_CTIOUTEN5**Type:** RW**Reset State:** 0x00000000**QDSS_CT13_CTIOUTEN5**

Bits	Name	Description
3:0	TRIGOUTEN	

0x008130B8 QDSS_CT13_CTIOUTEN6**Type:** RW**Reset State:** 0x00000000**QDSS_CT13_CTIOUTEN6**

Bits	Name	Description
3:0	TRIGOUTEN	

0x008130BC QDSS_CT13_CTIOUTEN7**Type:** RW**Reset State:** 0x00000000

QDSS_CT13_CTIOUTEN7

Bits	Name	Description
3:0	TRIGOUTEN	

0x00813130 QDSS_CT13_CTITRIGINSTATUS**Type:** R**Reset State:** Undefined**QDSS_CT13_CTITRIGINSTATUS**

Bits	Name	Description
7:0	TRIGINSTATUS	

0x00813134 QDSS_CT13_CTITRIGOUTSTATUS**Type:** R**Reset State:** 0x00000000**QDSS_CT13_CTITRIGOUTSTATUS**

Bits	Name	Description
7:0	TRIGOUTSTATUS	

0x00813138 QDSS_CT13_CTICHINSTSTATUS**Type:** R**Reset State:** Undefined**QDSS_CT13_CTICHINSTSTATUS**

Bits	Name	Description
3:0	CTICHINSTSTATUS	

0x0081313C QDSS_CT13_CTICHOUTSTATUS**Type:** R**Reset State:** 0x00000000

QDSS_CT13_CTICHOUTSTATUS

Bits	Name	Description
3:0	CTICHOUTSTATUS	

0x00813140 QDSS_CT13_CTIGATE**Type:** RW**Reset State:** 0x0000000F**QDSS_CT13_CTIGATE**

Bits	Name	Description
3	CTIGATEEN3	
2	CTIGATEEN2	
1	CTIGATEEN1	
0	CTIGATEEN0	

0x00813144 QDSS_CT13_ASICCTL**Type:** RW**Reset State:** 0x00000000**QDSS_CT13_ASICCTL**

Bits	Name	Description
7:0	ASICCTL	

0x00813EDC QDSS_CT13_ITCHINACK**Type:** W**Reset State:** 0x00000000**QDSS_CT13_ITCHINACK**

Bits	Name	Description
3:0	CTCHINACK	

0x00813EE0 QDSS_CT13_ITTRIGINACK**Type:** W**Reset State:** 0x00000000**QDSS_CT13_ITTRIGINACK**

Bits	Name	Description
7:0	CTTRIGINACK	

0x00813EE4 QDSS_CT13_ITCHOUT**Type:** W**Reset State:** 0x00000000**QDSS_CT13_ITCHOUT**

Bits	Name	Description
3:0	CTCHOUT	

0x00813EE8 QDSS_CT13_ITTRIGOUT**Type:** W**Reset State:** 0x00000000**QDSS_CT13_ITTRIGOUT**

Bits	Name	Description
7:0	CTTRIGOUT	

0x00813EEC QDSS_CT13_ITCHOUTACK**Type:** R**Reset State:** 0x00000000**QDSS_CT13_ITCHOUTACK**

Bits	Name	Description
3:0	CTCHOUTACK	

0x00813EF0 QDSS_CT13_ITTRIGOUTACK**Type:** R**Reset State:** 0x00000000**QDSS_CT13_ITTRIGOUTACK**

Bits	Name	Description
7:0	CTTRIGOUTACK	

0x00813EF4 QDSS_CT13_ITCHIN**Type:** R**Reset State:** 0x00000000**QDSS_CT13_ITCHIN**

Bits	Name	Description
3:0	CTCHIN	

0x00813EF8 QDSS_CT13_ITTRIGIN**Type:** R**Reset State:** 0x00000000**QDSS_CT13_ITTRIGIN**

Bits	Name	Description
7:0	CTTRIGIN	

0x00813F00 QDSS_CT13_ITCTRL**Type:** RW**Reset State:** 0x00000000**QDSS_CT13_ITCTRL**

Bits	Name	Description
0	INTEGRATION_MODE	Allows the component to switch from functional mode to integration mode or back. 0x1: INTEGRATION_MODE (Enable integration mode) 0x0: FUNCTIONAL_MODE (Disable integration mode)

0x00813FA0 QDSS_CT13_CLAIMSET**Type:** RW**Reset State:** 0x0000000F

This is used in conjunction with Claim Tag Clear Register, CLAIMCLR. This register forms one half of the Claim Tag value. This location allows individual bits to be set, write, and returns the number of bits that can be set, read.

QDSS_CT13_CLAIMSET

Bits	Name	Type	Description
3:0	CLAIMSET_W	W	Each bit is considered separately: 0 = no effect, 1 = set this bit in the claim tag.
3:0	CLAIMSET_R	R	This claim tag bit is implemented 0xF: CLAIM_TAG_IMPLEMENTED_BITS (These bits are present within the claim tag field)

0x00813FA4 QDSS_CT13_CLAIMCLR**Type:** RW**Reset State:** 0x00000000

This register is used in conjunction with Claim Tag Set Register, CLAIMSET. This register forms one half of the Claim Tag value. This location enables individual bits to be cleared, write, and returns the current Claim Tag value, read.

QDSS_CT13_CLAIMCLR

Bits	Name	Type	Description
3:0	CLAIMCLR_W	W	Each bit is considered separately: 0 = no effect, 1 = clear this bit in the claim tag.
3:0	CLAIMCLR_R	R	The value present reflects the current setting of the Claim Tag.

0x00813FB0 QDSS_CT13_LAR**Type:** W**Reset State:** Undefined

QDSS_CT13_LAR

Bits	Name	Description
31:0	ACCESS_W	A write of 0xC5ACCE55 enables further write access to this device. A write of any value other than 0xC5ACCE55 will have the affect of removing write access. 0xC5ACCE55: UNLOCK (Unlock the protection register to enable write access)

0x00813FB4 QDSS_CT13_LSR**Type:** R**Reset State:** 0x00000003**QDSS_CT13_LSR**

Bits	Name	Description
2	LOCKTYPE	Indicates if the Lock Access Register (0xFB0) is implemented as 8-bit or 32-bit 0x0: ENUM_32_BIT (This component implements a 32-bit Lock Access Register)
1	LOCKGRANT	Returns the current status of the Lock. This bit reads as 0 when read from an external debugger (PADDRDBG31 = 1) since external debugger accesses are not subject to Lock Registers. 0x0: ACCESS_PERMITTED (Write access is permitted to this device) 0x1: DEVICE_LOCKED (Write access to the component is blocked. All writes to control registers are ignored. Reads are permitted)
0	LOCKEXIST	Indicates that a lock control mechanism exists for this device. This bit reads as 0 when read from an external debugger (PADDRDBG31 = 1) since external debugger accesses are not subject to Lock Registers. 0x0: LOCK_NOT_PRESENT (No lock control mechanism exists, writes to the Lock Access register (0xFB0) will be ignored) 0x1: LOCK_PRESENT (Lock control mechanism is present)

0x00813FB8 QDSS_CT13_AUTHSTATUS**Type:** R**Reset State:** 0xFFFFFFFF0X

Reports what functionality is currently permitted by the authentication interface.

QDSS_CTI3_AUTHSTATUS

Bits	Name	Description
7:6	SNID	Indicates the security level for secure non-invasive debug 0x0: ENUM_2_B00 (Functionality not implemented.)
5:4	SID	Indicates the security level for secure invasive debug 0x0: ENUM_2_B00 (Functionality not implemented.)
3:2	NSNID	Indicates the security level for non-secure non-invasive debug 0x2: ENUM_2_B10 (Non-secure Non-Invasive debug functionality currently disabled.) 0x3: ENUM_2_B11 (Non-secure Non-Invasive debug functionality currently enabled.)
1:0	NSID	Indicates the security level for non-secure invasive debug 0x2: ENUM_2_B10 (Non-secure Invasive debug functionality currently disabled.) 0x3: ENUM_2_B11 (Non-secure Invasive debug functionality currently enabled.)

0x00813FC8 QDSS_CTI3_DEVID**Type:** R**Reset State:** 0x00040800

This register indicates the capabilities of the CTI.

QDSS_CTI3_DEVID

Bits	Name	Description
19:16	NUMCH	Number of ECT channels available.
15:8	NUMTRIG	Number of ECT triggers available.
4:0	EXTMUXNUM	Indicates the number of multiplexing available on Trigger Inputs and Trigger Outputs using ASICCTL. Default value of 5'b00000 indicating no multiplexing present. Reflects the value of the Verilog `define EXTMUXNUM that the user must alter accordingly.

0x00813FE0 QDSS_CTI3_PERIPHID0**Type:** R**Reset State:** 0x00000006

Part of the set of Peripheral Identification registers. Contains part of the designer specific part number.

QDSS_CT13_PERIPHID0

Bits	Name	Description
7:0	PART_0	Bits [7:0] of the component's part number. This is selected by the designer of the component. 0x6: CTI_PART_NUMBER_7_0 (Lowest 8 bits of the Part Number (0x906))

0x00813FE4 QDSS_CT13_PERIPHID1**Type:** R**Reset State:** 0x000000B9

Part of the set of Peripheral Identification registers. Contains part of the designer specific part number and part of the designer identity.

QDSS_CT13_PERIPHID1

Bits	Name	Description
7:4	DES_0	Bits 3:0 of the JEDEC identity code indicating the designer of the component (along with the continuation code) 0xB: ARM_JEP106_IDENTITY_CODE_3_0 (Lowest 4 bits of the JEP106 Identity Code)
3:0	PART_1	Bits [11:8] of the component's part number. This is selected by the designer of the component. 0x9: CTI_PART_NUMBER_11_8 (Upper 4 bits of the Part Number (0x961))

0x00813FE8 QDSS_CT13_PERIPHID2**Type:** R**Reset State:** 0x0000004B

Part of the set of Peripheral Identification registers. Contains part of the designer identity and the product revision.

QDSS_CT13_PERIPHID2

Bits	Name	Description
7:4	REVISION	The Revision field is an incremental value starting at 0x0 for the first design of this component. This only increases by 1 for both major and minor revisions and is simply used as a look-up to establish the exact major/minor revision. 0x4: R0P4 (This device is at r0p4)

QDSS_CT13_PERIPHID2 (cont.)

Bits	Name	Description
3	JEDEC	Always set. Indicates that a JEDEC assigned value is used 0x1: JEDEC_IDENTITY (The designer ID is specified by JEDEC (http://www.jedec.org))
2:0	DES_1	Bits 6:4 of the JEDEC identity code indicating the designer of the component (along with the continuation code) 0x3: ARM_JEP106_IDENTITY_CODE_6_4 (Upper 3 bits of the JEP106 Identity Code)

0x00813FEC QDSS_CT13_PERIPHID3**Type:** R**Reset State:** 0x00000000

Part of the set of Peripheral Identification registers. Contains the RevAnd and Customer Modified fields.

QDSS_CT13_PERIPHID3

Bits	Name	Description
7:4	REVAND	This field indicates minor errata fixes specific to this design, for example metal fixes after implementation. In most cases this field is zero. It is recommended that component designers ensure this field can be changed by a metal fix if required, for example by driving it from registers that reset to zero. 0x0: NO_MODIFICATION (Indicates that there have been no metal fixes to this component)
3:0	CMOD	Where the component is reusable IP, this value indicates if the customer has modified the behavior of the component. In most cases this field is zero. 0x0: NO_MODIFICATION (Indicates that there have been no modifications made)

0x00813FD0 QDSS_CT13_PERIPHID4**Type:** R**Reset State:** 0x00000004

Part of the set of Peripheral Identification registers. Contains part of the designer identity and the memory footprint indicator.

QDSS_CTI3_PERIPHID4

Bits	Name	Description
7:4	SIZE	This is a 4-bit value that indicates the total contiguous size of the memory window used by this component in powers of 2 from the standard 4KB. If a component only requires the standard 4KB then this should read as 0x0, 4KB only, for 8KB set to 0x1, 16KB == 0x2, 32KB == 0x3, and so on. 0x0: ENUM_4KB (Indicates that the device only occupies 4kB of memory)
3:0	DES_2	JEDEC continuation code indicating the designer of the component (along with the identity code) 0x4: ARM_JEP106_CONTINUATION_CODE (Indicates that ARM's JEDEC identity code is on the 5th bank)

0x00813FD4 QDSS_CTI3_PERIPHID5**Type:** RW**Reset State:** Undefined

Reserved

QDSS_CTI3_PERIPHID5

Bits	Name	Description
31:0	PERIPHID5	

0x00813FD8 QDSS_CTI3_PERIPHID6**Type:** RW**Reset State:** Undefined

Reserved

QDSS_CTI3_PERIPHID6

Bits	Name	Description
31:0	PERIPHID6	

0x00813FDC QDSS_CTI3_PERIPHID7**Type:** RW**Reset State:** Undefined

Reserved

QDSS_CT13_PERIPID7

Bits	Name	Description
31:0	PERIPID7	

0x00813FF0 QDSS_CT13_COMPID0

Type: R

Reset State: 0x0000000D

A component identification register, that indicates that the identification registers are present.

QDSS_CT13_COMPID0

Bits	Name	Description
7:0	PRMBL_0	Contains bits [7:0] of the component identification 0xD: ENUM_0D (Identification value)

0x00813FF4 QDSS_CT13_COMPID1

Type: R

Reset State: 0x00000090

A component identification register, that indicates that the identification registers are present. This register also indicates the component class.

QDSS_CT13_COMPID1

Bits	Name	Description
7:4	CLASS	Class of the component. E.g. ROM table, CoreSight component etc. Constitutes bits [15:12] of the component identification. 0x9: CORESIGHT_COMPONENT (Indicates the component is a CoreSight component)
3:0	PRMBL_1	Contains bits [11:8] of the component identification 0x0: ENUM_0 (Identification value)

0x00813FF8 QDSS_CT13_COMPID2

Type: R

Reset State: 0x00000005

A component identification register, that indicates that the identification registers are present.

QDSS_CT13_COMPID2

Bits	Name	Description
7:0	PRMBL_2	Contains bits [23:16] of the component identification 0x5: ENUM_05 (Identification value)

0x00813FFC QDSS_CT13_COMPID3**Type:** R**Reset State:** 0x000000B1

A component identification register, that indicates that the identification registers are present.

QDSS_CT13_COMPID3

Bits	Name	Description
7:0	PRMBL_3	Contains bits [31:24] of the component identification 0xB1: B1 (Identification value)

12.1.8 CTI4_CSCTI

The CTI allows mapping a trigger on the Trigger interface to a signal on the Channel interface and vice versa. The mapping is user-programmable. When the CTI receives a channel event it maps this onto a trigger output. This enables subsystems to cross trigger with each other. The receiving and transmitting of triggers is performed through the trigger interface.

12.1.8.1 QDSS_CT14_CSCTI

The CTI allows mapping a trigger on the Trigger interface to a signal on the Channel interface and vice versa. The mapping is user-programmable. When the CTI receives a channel event it maps this onto a trigger output. This enables subsystems to cross trigger with each other. The receiving and transmitting of triggers is performed through the trigger interface.

0x00814000 QDSS_CT14_CTICONTROL**Type:** RW**Reset State:** 0x00000000

QDSS_CT14_CTICONTROL

Bits	Name	Description
0	GLBEN	0x1: ENABLED (When this bit is 1, cross triggering mapping logic functionality is enabled.) 0x0: DISABLED (When this bit is 0, all cross triggering mapping logic functionality is disabled.)

0x00814010 QDSS_CT14_CTIINTACK**Type:** W**Reset State:** 0x00000000**QDSS_CT14_CTIINTACK**

Bits	Name	Description
7:0	INTACK	

0x00814014 QDSS_CT14_CTIAPPSET**Type:** W**Reset State:** 0x00000000**QDSS_CT14_CTIAPPSET**

Bits	Name	Description
3:0	APPSET	

0x00814018 QDSS_CT14_CTIAPPCLEAR**Type:** W**Reset State:** 0x00000000**QDSS_CT14_CTIAPPCLEAR**

Bits	Name	Description
3:0	APPCLEAR	

0x0081401C QDSS_CT14_CTIAPPPULSE**Type:** W**Reset State:** 0x00000000

QDSS_CTIA_CTIAPPPULSE

Bits	Name	Description
3:0	APPULSE	

0x00814020 QDSS_CTIA_CTIINEN0**Type:** RW**Reset State:** 0x00000000**QDSS_CTIA_CTIINEN0**

Bits	Name	Description
3:0	TRIGINEN	

0x00814024 QDSS_CTIA_CTIINEN1**Type:** RW**Reset State:** 0x00000000**QDSS_CTIA_CTIINEN1**

Bits	Name	Description
3:0	TRIGINEN	

0x00814028 QDSS_CTIA_CTIINEN2**Type:** RW**Reset State:** 0x00000000**QDSS_CTIA_CTIINEN2**

Bits	Name	Description
3:0	TRIGINEN	

0x0081402C QDSS_CTIA_CTIINEN3**Type:** RW**Reset State:** 0x00000000

QDSS_CT14_CTIINEN3

Bits	Name	Description
3:0	TRIGINEN	

0x00814030 QDSS_CT14_CTIINEN4**Type:** RW**Reset State:** 0x00000000**QDSS_CT14_CTIINEN4**

Bits	Name	Description
3:0	TRIGINEN	

0x00814034 QDSS_CT14_CTIINEN5**Type:** RW**Reset State:** 0x00000000**QDSS_CT14_CTIINEN5**

Bits	Name	Description
3:0	TRIGINEN	

0x00814038 QDSS_CT14_CTIINEN6**Type:** RW**Reset State:** 0x00000000**QDSS_CT14_CTIINEN6**

Bits	Name	Description
3:0	TRIGINEN	

0x0081403C QDSS_CT14_CTIINEN7**Type:** RW**Reset State:** 0x00000000

QDSS_CT14_CTIINEN7

Bits	Name	Description
3:0	TRIGINEN	

0x008140A0 QDSS_CT14_CTIOUTEN0**Type:** RW**Reset State:** 0x00000000**QDSS_CT14_CTIOUTEN0**

Bits	Name	Description
3:0	TRIGOUTEN	

0x008140A4 QDSS_CT14_CTIOUTEN1**Type:** RW**Reset State:** 0x00000000**QDSS_CT14_CTIOUTEN1**

Bits	Name	Description
3:0	TRIGOUTEN	

0x008140A8 QDSS_CT14_CTIOUTEN2**Type:** RW**Reset State:** 0x00000000**QDSS_CT14_CTIOUTEN2**

Bits	Name	Description
3:0	TRIGOUTEN	

0x008140AC QDSS_CT14_CTIOUTEN3**Type:** RW**Reset State:** 0x00000000

QDSS_CT14_CTIOUTEN3

Bits	Name	Description
3:0	TRIGOUTEN	

0x008140B0 QDSS_CT14_CTIOUTEN4**Type:** RW**Reset State:** 0x00000000**QDSS_CT14_CTIOUTEN4**

Bits	Name	Description
3:0	TRIGOUTEN	

0x008140B4 QDSS_CT14_CTIOUTEN5**Type:** RW**Reset State:** 0x00000000**QDSS_CT14_CTIOUTEN5**

Bits	Name	Description
3:0	TRIGOUTEN	

0x008140B8 QDSS_CT14_CTIOUTEN6**Type:** RW**Reset State:** 0x00000000**QDSS_CT14_CTIOUTEN6**

Bits	Name	Description
3:0	TRIGOUTEN	

0x008140BC QDSS_CT14_CTIOUTEN7**Type:** RW**Reset State:** 0x00000000

QDSS_CTIA_CTIOUTEN7

Bits	Name	Description
3:0	TRIGOUTEN	

0x00814130 QDSS_CTIA_CTITRIGINSTATUS**Type:** R**Reset State:** Undefined**QDSS_CTIA_CTITRIGINSTATUS**

Bits	Name	Description
7:0	TRIGINSTATUS	

0x00814134 QDSS_CTIA_CTITRIGOUTSTATUS**Type:** R**Reset State:** 0x00000000**QDSS_CTIA_CTITRIGOUTSTATUS**

Bits	Name	Description
7:0	TRIGOUTSTATUS	

0x00814138 QDSS_CTIA_CTICHINSTAUS**Type:** R**Reset State:** Undefined**QDSS_CTIA_CTICHINSTAUS**

Bits	Name	Description
3:0	CTICHINSTAUS	

0x0081413C QDSS_CTIA_CTICHOUTSTATUS**Type:** R**Reset State:** 0x00000000

QDSS_CT14_CTICHOUTSTATUS

Bits	Name	Description
3:0	CTICHOUTSTATUS	

0x00814140 QDSS_CT14_CTIGATE**Type:** RW**Reset State:** 0x0000000F**QDSS_CT14_CTIGATE**

Bits	Name	Description
3	CTIGATEEN3	
2	CTIGATEEN2	
1	CTIGATEEN1	
0	CTIGATEEN0	

0x00814144 QDSS_CT14_ASICCTL**Type:** RW**Reset State:** 0x00000000**QDSS_CT14_ASICCTL**

Bits	Name	Description
7:0	ASICCTL	

0x00814EDC QDSS_CT14_ITCHINACK**Type:** W**Reset State:** 0x00000000**QDSS_CT14_ITCHINACK**

Bits	Name	Description
3:0	CTCHINACK	

0x00814EE0 QDSS_CT14_ITTRIGINACK**Type:** W**Reset State:** 0x00000000**QDSS_CT14_ITTRIGINACK**

Bits	Name	Description
7:0	CTTRIGINACK	

0x00814EE4 QDSS_CT14_ITCHOUT**Type:** W**Reset State:** 0x00000000**QDSS_CT14_ITCHOUT**

Bits	Name	Description
3:0	CTCHOUT	

0x00814EE8 QDSS_CT14_ITTRIGOUT**Type:** W**Reset State:** 0x00000000**QDSS_CT14_ITTRIGOUT**

Bits	Name	Description
7:0	CTTRIGOUT	

0x00814EEC QDSS_CT14_ITCHOUTACK**Type:** R**Reset State:** 0x00000000**QDSS_CT14_ITCHOUTACK**

Bits	Name	Description
3:0	CTCHOUTACK	

0x00814EF0 QDSS_CT14_ITTRIGOUTACK**Type:** R**Reset State:** 0x00000000**QDSS_CT14_ITTRIGOUTACK**

Bits	Name	Description
7:0	CTTRIGOUTACK	

0x00814EF4 QDSS_CT14_ITCHIN**Type:** R**Reset State:** 0x00000000**QDSS_CT14_ITCHIN**

Bits	Name	Description
3:0	CTCHIN	

0x00814EF8 QDSS_CT14_ITTRIGIN**Type:** R**Reset State:** 0x00000000**QDSS_CT14_ITTRIGIN**

Bits	Name	Description
7:0	CTTRIGIN	

0x00814F00 QDSS_CT14_ITCTRL**Type:** RW**Reset State:** 0x00000000**QDSS_CT14_ITCTRL**

Bits	Name	Description
0	INTEGRATION_MODE	Allows the component to switch from functional mode to integration mode or back. 0x1: INTEGRATION_MODE (Enable integration mode) 0x0: FUNCTIONAL_MODE (Disable integration mode)

0x00814FA0 QDSS_CT14_CLAIMSET**Type:** RW**Reset State:** 0x0000000F

This is used in conjunction with Claim Tag Clear Register, CLAIMCLR. This register forms one half of the Claim Tag value. This location allows individual bits to be set, write, and returns the number of bits that can be set, read.

QDSS_CT14_CLAIMSET

Bits	Name	Type	Description
3:0	CLAIMSET_W	W	Each bit is considered separately: 0 = no effect, 1 = set this bit in the claim tag.
3:0	CLAIMSET_R	R	This claim tag bit is implemented 0xF: CLAIM_TAG_IMPLEMENTED_BITS (These bits are present within the claim tag field)

0x00814FA4 QDSS_CT14_CLAIMCLR**Type:** RW**Reset State:** 0x00000000

This register is used in conjunction with Claim Tag Set Register, CLAIMSET. This register forms one half of the Claim Tag value. This location enables individual bits to be cleared, write, and returns the current Claim Tag value, read.

QDSS_CT14_CLAIMCLR

Bits	Name	Type	Description
3:0	CLAIMCLR_W	W	Each bit is considered separately: 0 = no effect, 1 = clear this bit in the claim tag.
3:0	CLAIMCLR_R	R	The value present reflects the current setting of the Claim Tag.

0x00814FB0 QDSS_CT14_LAR**Type:** W**Reset State:** Undefined

QDSS_CTl4_LAR

Bits	Name	Description
31:0	ACCESS_W	A write of 0xC5ACCE55 enables further write access to this device. A write of any value other than 0xC5ACCE55 will have the affect of removing write access. 0xC5ACCE55: UNLOCK (Unlock the protection register to enable write access)

0x00814FB4 QDSS_CTl4_LSR**Type:** R**Reset State:** 0x00000003**QDSS_CTl4_LSR**

Bits	Name	Description
2	LOCKTYPE	Indicates if the Lock Access Register (0xFB0) is implemented as 8-bit or 32-bit 0x0: ENUM_32_BIT (This component implements a 32-bit Lock Access Register)
1	LOCKGRANT	Returns the current status of the Lock. This bit reads as 0 when read from an external debugger (PADDRDBG31 = 1) since external debugger accesses are not subject to Lock Registers. 0x0: ACCESS_PERMITTED (Write access is permitted to this device) 0x1: DEVICE_LOCKED (Write access to the component is blocked. All writes to control registers are ignored. Reads are permitted)
0	LOCKEXIST	Indicates that a lock control mechanism exists for this device. This bit reads as 0 when read from an external debugger (PADDRDBG31 = 1) since external debugger accesses are not subject to Lock Registers. 0x0: LOCK_NOT_PRESENT (No lock control mechanism exists, writes to the Lock Access register (0xFB0) will be ignored) 0x1: LOCK_PRESENT (Lock control mechanism is present)

0x00814FB8 QDSS_CTl4_AUTHSTATUS**Type:** R**Reset State:** 0xFFFFFFFF0X

Reports what functionality is currently permitted by the authentication interface.

QDSS_CTL4_AUTHSTATUS

Bits	Name	Description
7:6	SNID	Indicates the security level for secure non-invasive debug 0x0: ENUM_2_B00 (Functionality not implemented.)
5:4	SID	Indicates the security level for secure invasive debug 0x0: ENUM_2_B00 (Functionality not implemented.)
3:2	NSNID	Indicates the security level for non-secure non-invasive debug 0x2: ENUM_2_B10 (Non-secure Non-Invasive debug functionality currently disabled.) 0x3: ENUM_2_B11 (Non-secure Non-Invasive debug functionality currently enabled.)
1:0	NSID	Indicates the security level for non-secure invasive debug 0x2: ENUM_2_B10 (Non-secure Invasive debug functionality currently disabled.) 0x3: ENUM_2_B11 (Non-secure Invasive debug functionality currently enabled.)

0x00814FC8 QDSS_CTL4_DEVID**Type:** R**Reset State:** 0x00040800

This register indicates the capabilities of the CTI.

QDSS_CTL4_DEVID

Bits	Name	Description
19:16	NUMCH	Number of ECT channels available.
15:8	NUMTRIG	Number of ECT triggers available.
4:0	EXTMUXNUM	Indicates the number of multiplexing available on Trigger Inputs and Trigger Outputs using ASICCTL. Default value of 5'b00000 indicating no multiplexing present. Reflects the value of the Verilog `define EXTMUXNUM that the user must alter accordingly.

0x00814FE0 QDSS_CTL4_PERIPHID0**Type:** R**Reset State:** 0x00000006

Part of the set of Peripheral Identification registers. Contains part of the designer specific part number.

QDSS_CTI4_PERIPHID0

Bits	Name	Description
7:0	PART_0	Bits [7:0] of the component's part number. This is selected by the designer of the component. 0x6: CTI_PART_NUMBER_7_0 (Lowest 8 bits of the Part Number (0x906))

0x00814FE4 QDSS_CTI4_PERIPHID1**Type:** R**Reset State:** 0x000000B9

Part of the set of Peripheral Identification registers. Contains part of the designer specific part number and part of the designer identity.

QDSS_CTI4_PERIPHID1

Bits	Name	Description
7:4	DES_0	Bits 3:0 of the JEDEC identity code indicating the designer of the component (along with the continuation code) 0xB: ARM_JEP106_IDENTITY_CODE_3_0 (Lowest 4 bits of the JEP106 Identity Code)
3:0	PART_1	Bits [11:8] of the component's part number. This is selected by the designer of the component. 0x9: CTI_PART_NUMBER_11_8 (Upper 4 bits of the Part Number (0x961))

0x00814FE8 QDSS_CTI4_PERIPHID2**Type:** R**Reset State:** 0x0000004B

Part of the set of Peripheral Identification registers. Contains part of the designer identity and the product revision.

QDSS_CTI4_PERIPHID2

Bits	Name	Description
7:4	REVISION	The Revision field is an incremental value starting at 0x0 for the first design of this component. This only increases by 1 for both major and minor revisions and is simply used as a look-up to establish the exact major/minor revision. 0x4: R0P4 (This device is at r0p4)

QDSS_CT14_PERIPHID2 (cont.)

Bits	Name	Description
3	JEDEC	Always set. Indicates that a JEDEC assigned value is used 0x1: JEDEC_IDENTITY (The designer ID is specified by JEDEC (http://www.jedec.org))
2:0	DES_1	Bits 6:4 of the JEDEC identity code indicating the designer of the component (along with the continuation code) 0x3: ARM_JEP106_IDENTITY_CODE_6_4 (Upper 3 bits of the JEP106 Identity Code)

0x00814FEC QDSS_CT14_PERIPHID3**Type:** R**Reset State:** 0x00000000

Part of the set of Peripheral Identification registers. Contains the RevAnd and Customer Modified fields.

QDSS_CT14_PERIPHID3

Bits	Name	Description
7:4	REVAND	This field indicates minor errata fixes specific to this design, for example metal fixes after implementation. In most cases this field is zero. It is recommended that component designers ensure this field can be changed by a metal fix if required, for example by driving it from registers that reset to zero. 0x0: NO_MODIFICATION (Indicates that there have been no metal fixes to this component)
3:0	CMOD	Where the component is reusable IP, this value indicates if the customer has modified the behavior of the component. In most cases this field is zero. 0x0: NO_MODIFICATION (Indicates that there have been no modifications made)

0x00814FD0 QDSS_CT14_PERIPHID4**Type:** R**Reset State:** 0x00000004

Part of the set of Peripheral Identification registers. Contains part of the designer identity and the memory footprint indicator.

QDSS_CT14_PERIPHID4

Bits	Name	Description
7:4	SIZE	This is a 4-bit value that indicates the total contiguous size of the memory window used by this component in powers of 2 from the standard 4KB. If a component only requires the standard 4KB then this should read as 0x0, 4KB only, for 8KB set to 0x1, 16KB == 0x2, 32KB == 0x3, and so on. 0x0: ENUM_4KB (Indicates that the device only occupies 4kB of memory)
3:0	DES_2	JEDEC continuation code indicating the designer of the component (along with the identity code) 0x4: ARM_JEP106_CONTINUATION_CODE (Indicates that ARM's JEDEC identity code is on the 5th bank)

0x00814FD4 QDSS_CT14_PERIPHID5**Type:** RW**Reset State:** Undefined

Reserved

QDSS_CT14_PERIPHID5

Bits	Name	Description
31:0	PERIPHID5	

0x00814FD8 QDSS_CT14_PERIPHID6**Type:** RW**Reset State:** Undefined

Reserved

QDSS_CT14_PERIPHID6

Bits	Name	Description
31:0	PERIPHID6	

0x00814FDC QDSS_CT14_PERIPHID7**Type:** RW**Reset State:** Undefined

Reserved

QDSS_CTIA_PERIPHID7

Bits	Name	Description
31:0	PERIPHID7	

0x00814FF0 QDSS_CTIA_COMPID0

Type: R

Reset State: 0x0000000D

A component identification register, that indicates that the identification registers are present.

QDSS_CTIA_COMPID0

Bits	Name	Description
7:0	PRMBL_0	Contains bits [7:0] of the component identification 0xD: ENUM_0D (Identification value)

0x00814FF4 QDSS_CTIA_COMPID1

Type: R

Reset State: 0x00000090

A component identification register, that indicates that the identification registers are present. This register also indicates the component class.

QDSS_CTIA_COMPID1

Bits	Name	Description
7:4	CLASS	Class of the component. E.g. ROM table, CoreSight component etc. Constitutes bits [15:12] of the component identification. 0x9: CORESIGHT_COMPONENT (Indicates the component is a CoreSight component)
3:0	PRMBL_1	Contains bits [11:8] of the component identification 0x0: ENUM_0 (Identification value)

0x00814FF8 QDSS_CTIA_COMPID2

Type: R

Reset State: 0x00000005

A component identification register, that indicates that the identification registers are present.

QDSS_CTID_COMPID2

Bits	Name	Description
7:0	PRMBL_2	Contains bits [23:16] of the component identification 0x5: ENUM_05 (Identification value)

0x00814FFC QDSS_CTID_COMPID3**Type:** R**Reset State:** 0x000000B1

A component identification register, that indicates that the identification registers are present.

QDSS_CTID_COMPID3

Bits	Name	Description
7:0	PRMBL_3	Contains bits [31:24] of the component identification 0xB1: B1 (Identification value)

12.1.9 CTI5_CSCTI

The CTI allows mapping a trigger on the Trigger interface to a signal on the Channel interface and vice versa. The mapping is user-programmable. When the CTI receives a channel event it maps this onto a trigger output. This enables subsystems to cross trigger with each other. The receiving and transmitting of triggers is performed through the trigger interface.

12.1.9.1 QDSS_CTID_CSCTI

The CTI allows mapping a trigger on the Trigger interface to a signal on the Channel interface and vice versa. The mapping is user-programmable. When the CTI receives a channel event it maps this onto a trigger output. This enables subsystems to cross trigger with each other. The receiving and transmitting of triggers is performed through the trigger interface.

0x00815000 QDSS_CTID_CTICONTROL**Type:** RW**Reset State:** 0x00000000

QDSS_CT15_CTICONTROL

Bits	Name	Description
0	GLBEN	0x1: ENABLED (When this bit is 1, cross triggering mapping logic functionality is enabled.) 0x0: DISABLED (When this bit is 0, all cross triggering mapping logic functionality is disabled.)

0x00815010 QDSS_CT15_CTIINTACK**Type:** W**Reset State:** 0x00000000**QDSS_CT15_CTIINTACK**

Bits	Name	Description
7:0	INTACK	

0x00815014 QDSS_CT15_CTIAPPSET**Type:** W**Reset State:** 0x00000000**QDSS_CT15_CTIAPPSET**

Bits	Name	Description
3:0	APPSET	

0x00815018 QDSS_CT15_CTIAPPCLEAR**Type:** W**Reset State:** 0x00000000**QDSS_CT15_CTIAPPCLEAR**

Bits	Name	Description
3:0	APPCLEAR	

0x0081501C QDSS_CT15_CTIAPPPULSE**Type:** W**Reset State:** 0x00000000

QDSS_CT15_CTIAPPPULSE

Bits	Name	Description
3:0	APPULSE	

0x00815020 QDSS_CT15_CTIINEN0**Type:** RW**Reset State:** 0x00000000**QDSS_CT15_CTIINEN0**

Bits	Name	Description
3:0	TRIGINEN	

0x00815024 QDSS_CT15_CTIINEN1**Type:** RW**Reset State:** 0x00000000**QDSS_CT15_CTIINEN1**

Bits	Name	Description
3:0	TRIGINEN	

0x00815028 QDSS_CT15_CTIINEN2**Type:** RW**Reset State:** 0x00000000**QDSS_CT15_CTIINEN2**

Bits	Name	Description
3:0	TRIGINEN	

0x0081502C QDSS_CT15_CTIINEN3**Type:** RW**Reset State:** 0x00000000

QDSS_CT15_CTIINEN3

Bits	Name	Description
3:0	TRIGINEN	

0x00815030 QDSS_CT15_CTIINEN4**Type:** RW**Reset State:** 0x00000000**QDSS_CT15_CTIINEN4**

Bits	Name	Description
3:0	TRIGINEN	

0x00815034 QDSS_CT15_CTIINEN5**Type:** RW**Reset State:** 0x00000000**QDSS_CT15_CTIINEN5**

Bits	Name	Description
3:0	TRIGINEN	

0x00815038 QDSS_CT15_CTIINEN6**Type:** RW**Reset State:** 0x00000000**QDSS_CT15_CTIINEN6**

Bits	Name	Description
3:0	TRIGINEN	

0x0081503C QDSS_CT15_CTIINEN7**Type:** RW**Reset State:** 0x00000000

QDSS_CT15_CTIINEN7

Bits	Name	Description
3:0	TRIGINEN	

0x008150A0 QDSS_CT15_CTIOUTEN0**Type:** RW**Reset State:** 0x00000000**QDSS_CT15_CTIOUTEN0**

Bits	Name	Description
3:0	TRIGOUTEN	

0x008150A4 QDSS_CT15_CTIOUTEN1**Type:** RW**Reset State:** 0x00000000**QDSS_CT15_CTIOUTEN1**

Bits	Name	Description
3:0	TRIGOUTEN	

0x008150A8 QDSS_CT15_CTIOUTEN2**Type:** RW**Reset State:** 0x00000000**QDSS_CT15_CTIOUTEN2**

Bits	Name	Description
3:0	TRIGOUTEN	

0x008150AC QDSS_CT15_CTIOUTEN3**Type:** RW**Reset State:** 0x00000000

QDSS_CT15_CTIOUTEN3

Bits	Name	Description
3:0	TRIGOUTEN	

0x008150B0 QDSS_CT15_CTIOUTEN4**Type:** RW**Reset State:** 0x00000000**QDSS_CT15_CTIOUTEN4**

Bits	Name	Description
3:0	TRIGOUTEN	

0x008150B4 QDSS_CT15_CTIOUTEN5**Type:** RW**Reset State:** 0x00000000**QDSS_CT15_CTIOUTEN5**

Bits	Name	Description
3:0	TRIGOUTEN	

0x008150B8 QDSS_CT15_CTIOUTEN6**Type:** RW**Reset State:** 0x00000000**QDSS_CT15_CTIOUTEN6**

Bits	Name	Description
3:0	TRIGOUTEN	

0x008150BC QDSS_CT15_CTIOUTEN7**Type:** RW**Reset State:** 0x00000000

QDSS_CT15_CTIOUTEN7

Bits	Name	Description
3:0	TRIGOUTEN	

0x00815130 QDSS_CT15_CTITRIGINSTATUS**Type:** R**Reset State:** Undefined**QDSS_CT15_CTITRIGINSTATUS**

Bits	Name	Description
7:0	TRIGINSTATUS	

0x00815134 QDSS_CT15_CTITRIGOUTSTATUS**Type:** R**Reset State:** 0x00000000**QDSS_CT15_CTITRIGOUTSTATUS**

Bits	Name	Description
7:0	TRIGOUTSTATUS	

0x00815138 QDSS_CT15_CTICHINSTAUS**Type:** R**Reset State:** Undefined**QDSS_CT15_CTICHINSTAUS**

Bits	Name	Description
3:0	CTICHINSTAUS	

0x0081513C QDSS_CT15_CTICHOUTSTATUS**Type:** R**Reset State:** 0x00000000

QDSS_CT15_CTICHOUTSTATUS

Bits	Name	Description
3:0	CTICHOUTSTATUS	

0x00815140 QDSS_CT15_CTIGATE**Type:** RW**Reset State:** 0x0000000F**QDSS_CT15_CTIGATE**

Bits	Name	Description
3	CTIGATEEN3	
2	CTIGATEEN2	
1	CTIGATEEN1	
0	CTIGATEEN0	

0x00815144 QDSS_CT15_ASICCTL**Type:** RW**Reset State:** 0x00000000**QDSS_CT15_ASICCTL**

Bits	Name	Description
7:0	ASICCTL	

0x00815EDC QDSS_CT15_ITCHINACK**Type:** W**Reset State:** 0x00000000**QDSS_CT15_ITCHINACK**

Bits	Name	Description
3:0	CTCHINACK	

0x00815EE0 QDSS_CT15_ITTRIGINACK**Type:** W**Reset State:** 0x00000000**QDSS_CT15_ITTRIGINACK**

Bits	Name	Description
7:0	CTTRIGINACK	

0x00815EE4 QDSS_CT15_ITCHOUT**Type:** W**Reset State:** 0x00000000**QDSS_CT15_ITCHOUT**

Bits	Name	Description
3:0	CTCHOUT	

0x00815EE8 QDSS_CT15_ITTRIGOUT**Type:** W**Reset State:** 0x00000000**QDSS_CT15_ITTRIGOUT**

Bits	Name	Description
7:0	CTTRIGOUT	

0x00815EEC QDSS_CT15_ITCHOUTACK**Type:** R**Reset State:** 0x00000000**QDSS_CT15_ITCHOUTACK**

Bits	Name	Description
3:0	CTCHOUTACK	

0x00815EF0 QDSS_CT15_ITTRIGOUTACK**Type:** R**Reset State:** 0x00000000**QDSS_CT15_ITTRIGOUTACK**

Bits	Name	Description
7:0	CTTRIGOUTACK	

0x00815EF4 QDSS_CT15_ITCHIN**Type:** R**Reset State:** 0x00000000**QDSS_CT15_ITCHIN**

Bits	Name	Description
3:0	CTCHIN	

0x00815EF8 QDSS_CT15_ITTRIGIN**Type:** R**Reset State:** 0x00000000**QDSS_CT15_ITTRIGIN**

Bits	Name	Description
7:0	CTTRIGIN	

0x00815F00 QDSS_CT15_ITCTRL**Type:** RW**Reset State:** 0x00000000**QDSS_CT15_ITCTRL**

Bits	Name	Description
0	INTEGRATION_MODE	Allows the component to switch from functional mode to integration mode or back. 0x1: INTEGRATION_MODE (Enable integration mode) 0x0: FUNCTIONAL_MODE (Disable integration mode)

0x00815FA0 QDSS_CT15_CLAIMSET**Type:** RW**Reset State:** 0x0000000F

This is used in conjunction with Claim Tag Clear Register, CLAIMCLR. This register forms one half of the Claim Tag value. This location allows individual bits to be set, write, and returns the number of bits that can be set, read.

QDSS_CT15_CLAIMSET

Bits	Name	Type	Description
3:0	CLAIMSET_W	W	Each bit is considered separately: 0 = no effect, 1 = set this bit in the claim tag.
3:0	CLAIMSET_R	R	This claim tag bit is implemented 0xF: CLAIM_TAG_IMPLEMENTED_BITS (These bits are present within the claim tag field)

0x00815FA4 QDSS_CT15_CLAIMCLR**Type:** RW**Reset State:** 0x00000000

This register is used in conjunction with Claim Tag Set Register, CLAIMSET. This register forms one half of the Claim Tag value. This location enables individual bits to be cleared, write, and returns the current Claim Tag value, read.

QDSS_CT15_CLAIMCLR

Bits	Name	Type	Description
3:0	CLAIMCLR_W	W	Each bit is considered separately: 0 = no effect, 1 = clear this bit in the claim tag.
3:0	CLAIMCLR_R	R	The value present reflects the current setting of the Claim Tag.

0x00815FB0 QDSS_CT15_LAR**Type:** W**Reset State:** Undefined

QDSS_CTI5_LAR

Bits	Name	Description
31:0	ACCESS_W	A write of 0xC5ACCE55 enables further write access to this device. A write of any value other than 0xC5ACCE55 will have the affect of removing write access. 0xC5ACCE55: UNLOCK (Unlock the protection register to enable write access)

0x00815FB4 QDSS_CTI5_LSR**Type:** R**Reset State:** 0x00000003**QDSS_CTI5_LSR**

Bits	Name	Description
2	LOCKTYPE	Indicates if the Lock Access Register (0xFB0) is implemented as 8-bit or 32-bit 0x0: ENUM_32_BIT (This component implements a 32-bit Lock Access Register)
1	LOCKGRANT	Returns the current status of the Lock. This bit reads as 0 when read from an external debugger (PADDRDBG31 = 1) since external debugger accesses are not subject to Lock Registers. 0x0: ACCESS_PERMITTED (Write access is permitted to this device) 0x1: DEVICE_LOCKED (Write access to the component is blocked. All writes to control registers are ignored. Reads are permitted)
0	LOCKEXIST	Indicates that a lock control mechanism exists for this device. This bit reads as 0 when read from an external debugger (PADDRDBG31 = 1) since external debugger accesses are not subject to Lock Registers. 0x0: LOCK_NOT_PRESENT (No lock control mechanism exists, writes to the Lock Access register (0xFB0) will be ignored) 0x1: LOCK_PRESENT (Lock control mechanism is present)

0x00815FB8 QDSS_CTI5_AUTHSTATUS**Type:** R**Reset State:** 0xFFFFFFFF0X

Reports what functionality is currently permitted by the authentication interface.

QDSS_CTL5_AUTHSTATUS

Bits	Name	Description
7:6	SNID	Indicates the security level for secure non-invasive debug 0x0: ENUM_2_B00 (Functionality not implemented.)
5:4	SID	Indicates the security level for secure invasive debug 0x0: ENUM_2_B00 (Functionality not implemented.)
3:2	NSNID	Indicates the security level for non-secure non-invasive debug 0x2: ENUM_2_B10 (Non-secure Non-Invasive debug functionality currently disabled.) 0x3: ENUM_2_B11 (Non-secure Non-Invasive debug functionality currently enabled.)
1:0	NSID	Indicates the security level for non-secure invasive debug 0x2: ENUM_2_B10 (Non-secure Invasive debug functionality currently disabled.) 0x3: ENUM_2_B11 (Non-secure Invasive debug functionality currently enabled.)

0x00815FC8 QDSS_CTL5_DEVID**Type:** R**Reset State:** 0x00040800

This register indicates the capabilities of the CTI.

QDSS_CTL5_DEVID

Bits	Name	Description
19:16	NUMCH	Number of ECT channels available.
15:8	NUMTRIG	Number of ECT triggers available.
4:0	EXTMUXNUM	Indicates the number of multiplexing available on Trigger Inputs and Trigger Outputs using ASICCTL. Default value of 5'b00000 indicating no multiplexing present. Reflects the value of the Verilog `define EXTMUXNUM that the user must alter accordingly.

0x00815FE0 QDSS_CTL5_PERIPHID0**Type:** R**Reset State:** 0x00000006

Part of the set of Peripheral Identification registers. Contains part of the designer specific part number.

QDSS_CT15_PERIPHID0

Bits	Name	Description
7:0	PART_0	Bits [7:0] of the component's part number. This is selected by the designer of the component. 0x6: CTI_PART_NUMBER_7_0 (Lowest 8 bits of the Part Number (0x906))

0x00815FE4 QDSS_CT15_PERIPHID1**Type:** R**Reset State:** 0x000000B9

Part of the set of Peripheral Identification registers. Contains part of the designer specific part number and part of the designer identity.

QDSS_CT15_PERIPHID1

Bits	Name	Description
7:4	DES_0	Bits 3:0 of the JEDEC identity code indicating the designer of the component (along with the continuation code) 0xB: ARM_JEP106_IDENTITY_CODE_3_0 (Lowest 4 bits of the JEP106 Identity Code)
3:0	PART_1	Bits [11:8] of the component's part number. This is selected by the designer of the component. 0x9: CTI_PART_NUMBER_11_8 (Upper 4 bits of the Part Number (0x961))

0x00815FE8 QDSS_CT15_PERIPHID2**Type:** R**Reset State:** 0x0000004B

Part of the set of Peripheral Identification registers. Contains part of the designer identity and the product revision.

QDSS_CT15_PERIPHID2

Bits	Name	Description
7:4	REVISION	The Revision field is an incremental value starting at 0x0 for the first design of this component. This only increases by 1 for both major and minor revisions and is simply used as a look-up to establish the exact major/minor revision. 0x4: R0P4 (This device is at r0p4)

QDSS_CT15_PERIPHID2 (cont.)

Bits	Name	Description
3	JEDEC	Always set. Indicates that a JEDEC assigned value is used 0x1: JEDEC_IDENTITY (The designer ID is specified by JEDEC (http://www.jedec.org))
2:0	DES_1	Bits 6:4 of the JEDEC identity code indicating the designer of the component (along with the continuation code) 0x3: ARM_JEP106_IDENTITY_CODE_6_4 (Upper 3 bits of the JEP106 Identity Code)

0x00815FEC QDSS_CT15_PERIPHID3**Type:** R**Reset State:** 0x00000000

Part of the set of Peripheral Identification registers. Contains the RevAnd and Customer Modified fields.

QDSS_CT15_PERIPHID3

Bits	Name	Description
7:4	REVAND	This field indicates minor errata fixes specific to this design, for example metal fixes after implementation. In most cases this field is zero. It is recommended that component designers ensure this field can be changed by a metal fix if required, for example by driving it from registers that reset to zero. 0x0: NO_MODIFICATION (Indicates that there have been no metal fixes to this component)
3:0	CMOD	Where the component is reusable IP, this value indicates if the customer has modified the behavior of the component. In most cases this field is zero. 0x0: NO_MODIFICATION (Indicates that there have been no modifications made)

0x00815FD0 QDSS_CT15_PERIPHID4**Type:** R**Reset State:** 0x00000004

Part of the set of Peripheral Identification registers. Contains part of the designer identity and the memory footprint indicator.

QDSS_CTI5_PERIPHID4

Bits	Name	Description
7:4	SIZE	This is a 4-bit value that indicates the total contiguous size of the memory window used by this component in powers of 2 from the standard 4KB. If a component only requires the standard 4KB then this should read as 0x0, 4KB only, for 8KB set to 0x1, 16KB == 0x2, 32KB == 0x3, and so on. 0x0: ENUM_4KB (Indicates that the device only occupies 4kB of memory)
3:0	DES_2	JEDEC continuation code indicating the designer of the component (along with the identity code) 0x4: ARM_JEP106_CONTINUATION_CODE (Indicates that ARM's JEDEC identity code is on the 5th bank)

0x00815FD4 QDSS_CTI5_PERIPHID5**Type:** RW**Reset State:** Undefined

Reserved

QDSS_CTI5_PERIPHID5

Bits	Name	Description
31:0	PERIPHID5	

0x00815FD8 QDSS_CTI5_PERIPHID6**Type:** RW**Reset State:** Undefined

Reserved

QDSS_CTI5_PERIPHID6

Bits	Name	Description
31:0	PERIPHID6	

0x00815FDC QDSS_CTI5_PERIPHID7**Type:** RW**Reset State:** Undefined

Reserved

QDSS_CT15_PERIPHID7

Bits	Name	Description
31:0	PERIPHID7	

0x00815FF0 QDSS_CT15_COMPID0

Type: R

Reset State: 0x0000000D

A component identification register, that indicates that the identification registers are present.

QDSS_CT15_COMPID0

Bits	Name	Description
7:0	PRMBL_0	Contains bits [7:0] of the component identification 0xD: ENUM_0D (Identification value)

0x00815FF4 QDSS_CT15_COMPID1

Type: R

Reset State: 0x00000090

A component identification register, that indicates that the identification registers are present. This register also indicates the component class.

QDSS_CT15_COMPID1

Bits	Name	Description
7:4	CLASS	Class of the component. E.g. ROM table, CoreSight component etc. Constitutes bits [15:12] of the component identification. 0x9: CORESIGHT_COMPONENT (Indicates the component is a CoreSight component)
3:0	PRMBL_1	Contains bits [11:8] of the component identification 0x0: ENUM_0 (Identification value)

0x00815FF8 QDSS_CT15_COMPID2

Type: R

Reset State: 0x00000005

A component identification register, that indicates that the identification registers are present.

QDSS_CTI5_COMPID2

Bits	Name	Description
7:0	PRMBL_2	Contains bits [23:16] of the component identification 0x5: ENUM_05 (Identification value)

0x00815FFC QDSS_CTI5_COMPID3**Type:** R**Reset State:** 0x000000B1

A component identification register, that indicates that the identification registers are present.

QDSS_CTI5_COMPID3

Bits	Name	Description
7:0	PRMBL_3	Contains bits [31:24] of the component identification 0xB1: B1 (Identification value)

12.1.10 CTI6_CSCTI

The CTI allows mapping a trigger on the Trigger interface to a signal on the Channel interface and vice versa. The mapping is user-programmable. When the CTI receives a channel event it maps this onto a trigger output. This enables subsystems to cross trigger with each other. The receiving and transmitting of triggers is performed through the trigger interface.

12.1.10.1 QDSS_CTI6_CSCTI

The CTI allows mapping a trigger on the Trigger interface to a signal on the Channel interface and vice versa. The mapping is user-programmable. When the CTI receives a channel event it maps this onto a trigger output. This enables subsystems to cross trigger with each other. The receiving and transmitting of triggers is performed through the trigger interface.

0x00816000 QDSS_CTI6_CTICONTROL**Type:** RW**Reset State:** 0x00000000

QDSS_CT16_CTICONTROL

Bits	Name	Description
0	GLBEN	0x1: ENABLED (When this bit is 1, cross triggering mapping logic functionality is enabled.) 0x0: DISABLED (When this bit is 0, all cross triggering mapping logic functionality is disabled.)

0x00816010 QDSS_CT16_CTIINTACK**Type:** W**Reset State:** 0x00000000**QDSS_CT16_CTIINTACK**

Bits	Name	Description
7:0	INTACK	

0x00816014 QDSS_CT16_CTIAPPSET**Type:** W**Reset State:** 0x00000000**QDSS_CT16_CTIAPPSET**

Bits	Name	Description
3:0	APPSET	

0x00816018 QDSS_CT16_CTIAPPCLEAR**Type:** W**Reset State:** 0x00000000**QDSS_CT16_CTIAPPCLEAR**

Bits	Name	Description
3:0	APPCLEAR	

0x0081601C QDSS_CT16_CTIAPPPULSE**Type:** W**Reset State:** 0x00000000

QDSS_CT16_CTIAPPPULSE

Bits	Name	Description
3:0	APPULSE	

0x00816020 QDSS_CT16_CTIINEN0**Type:** RW**Reset State:** 0x00000000**QDSS_CT16_CTIINEN0**

Bits	Name	Description
3:0	TRIGINEN	

0x00816024 QDSS_CT16_CTIINEN1**Type:** RW**Reset State:** 0x00000000**QDSS_CT16_CTIINEN1**

Bits	Name	Description
3:0	TRIGINEN	

0x00816028 QDSS_CT16_CTIINEN2**Type:** RW**Reset State:** 0x00000000**QDSS_CT16_CTIINEN2**

Bits	Name	Description
3:0	TRIGINEN	

0x0081602C QDSS_CT16_CTIINEN3**Type:** RW**Reset State:** 0x00000000

QDSS_CT16_CTIINEN3

Bits	Name	Description
3:0	TRIGINEN	

0x00816030 QDSS_CT16_CTIINEN4**Type:** RW**Reset State:** 0x00000000**QDSS_CT16_CTIINEN4**

Bits	Name	Description
3:0	TRIGINEN	

0x00816034 QDSS_CT16_CTIINEN5**Type:** RW**Reset State:** 0x00000000**QDSS_CT16_CTIINEN5**

Bits	Name	Description
3:0	TRIGINEN	

0x00816038 QDSS_CT16_CTIINEN6**Type:** RW**Reset State:** 0x00000000**QDSS_CT16_CTIINEN6**

Bits	Name	Description
3:0	TRIGINEN	

0x0081603C QDSS_CT16_CTIINEN7**Type:** RW**Reset State:** 0x00000000

QDSS_CT16_CTINEN7

Bits	Name	Description
3:0	TRIGINEN	

0x008160A0 QDSS_CT16_CTIOUTEN0**Type:** RW**Reset State:** 0x00000000**QDSS_CT16_CTIOUTEN0**

Bits	Name	Description
3:0	TRIGOUTEN	

0x008160A4 QDSS_CT16_CTIOUTEN1**Type:** RW**Reset State:** 0x00000000**QDSS_CT16_CTIOUTEN1**

Bits	Name	Description
3:0	TRIGOUTEN	

0x008160A8 QDSS_CT16_CTIOUTEN2**Type:** RW**Reset State:** 0x00000000**QDSS_CT16_CTIOUTEN2**

Bits	Name	Description
3:0	TRIGOUTEN	

0x008160AC QDSS_CT16_CTIOUTEN3**Type:** RW**Reset State:** 0x00000000

QDSS_CT16_CTIOUTEN3

Bits	Name	Description
3:0	TRIGOUTEN	

0x008160B0 QDSS_CT16_CTIOUTEN4**Type:** RW**Reset State:** 0x00000000**QDSS_CT16_CTIOUTEN4**

Bits	Name	Description
3:0	TRIGOUTEN	

0x008160B4 QDSS_CT16_CTIOUTEN5**Type:** RW**Reset State:** 0x00000000**QDSS_CT16_CTIOUTEN5**

Bits	Name	Description
3:0	TRIGOUTEN	

0x008160B8 QDSS_CT16_CTIOUTEN6**Type:** RW**Reset State:** 0x00000000**QDSS_CT16_CTIOUTEN6**

Bits	Name	Description
3:0	TRIGOUTEN	

0x008160BC QDSS_CT16_CTIOUTEN7**Type:** RW**Reset State:** 0x00000000

QDSS_CT16_CTIOUTEN7

Bits	Name	Description
3:0	TRIGOUTEN	

0x00816130 QDSS_CT16_CTITRIGINSTATUS**Type:** R**Reset State:** Undefined**QDSS_CT16_CTITRIGINSTATUS**

Bits	Name	Description
7:0	TRIGINSTATUS	

0x00816134 QDSS_CT16_CTITRIGOUTSTATUS**Type:** R**Reset State:** 0x00000000**QDSS_CT16_CTITRIGOUTSTATUS**

Bits	Name	Description
7:0	TRIGOUTSTATUS	

0x00816138 QDSS_CT16_CTICHINSTSTATUS**Type:** R**Reset State:** Undefined**QDSS_CT16_CTICHINSTSTATUS**

Bits	Name	Description
3:0	CTICHINSTSTATUS	

0x0081613C QDSS_CT16_CTICHOUTSTATUS**Type:** R**Reset State:** 0x00000000

QDSS_CTI6_CTICHOUTSTATUS

Bits	Name	Description
3:0	CTICHOUTSTATUS	

0x00816140 QDSS_CTI6_CTIGATE**Type:** RW**Reset State:** 0x0000000F**QDSS_CTI6_CTIGATE**

Bits	Name	Description
3	CTIGATEEN3	
2	CTIGATEEN2	
1	CTIGATEEN1	
0	CTIGATEEN0	

0x00816144 QDSS_CTI6_ASICCTL**Type:** RW**Reset State:** 0x00000000**QDSS_CTI6_ASICCTL**

Bits	Name	Description
7:0	ASICCTL	

0x00816EDC QDSS_CTI6_ITCHINACK**Type:** W**Reset State:** 0x00000000**QDSS_CTI6_ITCHINACK**

Bits	Name	Description
3:0	CTCHINACK	

0x00816EE0 QDSS_CTI6_ITTRIGINACK**Type:** W**Reset State:** 0x00000000**QDSS_CTI6_ITTRIGINACK**

Bits	Name	Description
7:0	CTTRIGINACK	

0x00816EE4 QDSS_CTI6_ITCHOUT**Type:** W**Reset State:** 0x00000000**QDSS_CTI6_ITCHOUT**

Bits	Name	Description
3:0	CTCHOUT	

0x00816EE8 QDSS_CTI6_ITTRIGOUT**Type:** W**Reset State:** 0x00000000**QDSS_CTI6_ITTRIGOUT**

Bits	Name	Description
7:0	CTTRIGOUT	

0x00816EEC QDSS_CTI6_ITCHOUTACK**Type:** R**Reset State:** 0x00000000**QDSS_CTI6_ITCHOUTACK**

Bits	Name	Description
3:0	CTCHOUTACK	

0x00816EF0 QDSS_CT16_ITTRIGOUTACK**Type:** R**Reset State:** 0x00000000**QDSS_CT16_ITTRIGOUTACK**

Bits	Name	Description
7:0	CTTRIGOUTACK	

0x00816EF4 QDSS_CT16_ITCHIN**Type:** R**Reset State:** 0x00000000**QDSS_CT16_ITCHIN**

Bits	Name	Description
3:0	CTCHIN	

0x00816EF8 QDSS_CT16_ITTRIGIN**Type:** R**Reset State:** 0x00000000**QDSS_CT16_ITTRIGIN**

Bits	Name	Description
7:0	CTTRIGIN	

0x00816F00 QDSS_CT16_ITCTRL**Type:** RW**Reset State:** 0x00000000**QDSS_CT16_ITCTRL**

Bits	Name	Description
0	INTEGRATION_MODE	Allows the component to switch from functional mode to integration mode or back. 0x1: INTEGRATION_MODE (Enable integration mode) 0x0: FUNCTIONAL_MODE (Disable integration mode)

0x00816FA0 QDSS_CTI6_CLAIMSET**Type:** RW**Reset State:** 0x0000000F

This is used in conjunction with Claim Tag Clear Register, CLAIMCLR. This register forms one half of the Claim Tag value. This location allows individual bits to be set, write, and returns the number of bits that can be set, read.

QDSS_CTI6_CLAIMSET

Bits	Name	Type	Description
3:0	CLAIMSET_W	W	Each bit is considered separately: 0 = no effect, 1 = set this bit in the claim tag.
3:0	CLAIMSET_R	R	This claim tag bit is implemented 0xF: CLAIM_TAG_IMPLEMENTED_BITS (These bits are present within the claim tag field)

0x00816FA4 QDSS_CTI6_CLAIMCLR**Type:** RW**Reset State:** 0x00000000

This register is used in conjunction with Claim Tag Set Register, CLAIMSET. This register forms one half of the Claim Tag value. This location enables individual bits to be cleared, write, and returns the current Claim Tag value, read.

QDSS_CTI6_CLAIMCLR

Bits	Name	Type	Description
3:0	CLAIMCLR_W	W	Each bit is considered separately: 0 = no effect, 1 = clear this bit in the claim tag.
3:0	CLAIMCLR_R	R	The value present reflects the current setting of the Claim Tag.

0x00816FB0 QDSS_CTI6_LAR**Type:** W**Reset State:** Undefined

QDSS_CTI6_LAR

Bits	Name	Description
31:0	ACCESS_W	A write of 0xC5ACCE55 enables further write access to this device. A write of any value other than 0xC5ACCE55 will have the affect of removing write access. 0xC5ACCE55: UNLOCK (Unlock the protection register to enable write access)

0x00816FB4 QDSS_CTI6_LSR**Type:** R**Reset State:** 0x00000003**QDSS_CTI6_LSR**

Bits	Name	Description
2	LOCKTYPE	Indicates if the Lock Access Register (0xFB0) is implemented as 8-bit or 32-bit 0x0: ENUM_32_BIT (This component implements a 32-bit Lock Access Register)
1	LOCKGRANT	Returns the current status of the Lock. This bit reads as 0 when read from an external debugger (PADDRDBG31 = 1) since external debugger accesses are not subject to Lock Registers. 0x0: ACCESS_PERMITTED (Write access is permitted to this device) 0x1: DEVICE_LOCKED (Write access to the component is blocked. All writes to control registers are ignored. Reads are permitted)
0	LOCKEXIST	Indicates that a lock control mechanism exists for this device. This bit reads as 0 when read from an external debugger (PADDRDBG31 = 1) since external debugger accesses are not subject to Lock Registers. 0x0: LOCK_NOT_PRESENT (No lock control mechanism exists, writes to the Lock Access register (0xFB0) will be ignored) 0x1: LOCK_PRESENT (Lock control mechanism is present)

0x00816FB8 QDSS_CTI6_AUTHSTATUS**Type:** R**Reset State:** 0xFFFFFFFF0X

Reports what functionality is currently permitted by the authentication interface.

QDSS_CTI6_AUTHSTATUS

Bits	Name	Description
7:6	SNID	Indicates the security level for secure non-invasive debug 0x0: ENUM_2_B00 (Functionality not implemented.)
5:4	SID	Indicates the security level for secure invasive debug 0x0: ENUM_2_B00 (Functionality not implemented.)
3:2	NSNID	Indicates the security level for non-secure non-invasive debug 0x2: ENUM_2_B10 (Non-secure Non-Invasive debug functionality currently disabled.) 0x3: ENUM_2_B11 (Non-secure Non-Invasive debug functionality currently enabled.)
1:0	NSID	Indicates the security level for non-secure invasive debug 0x2: ENUM_2_B10 (Non-secure Invasive debug functionality currently disabled.) 0x3: ENUM_2_B11 (Non-secure Invasive debug functionality currently enabled.)

0x00816FC8 QDSS_CTI6_DEVID**Type:** R**Reset State:** 0x00040800

This register indicates the capabilities of the CTI.

QDSS_CTI6_DEVID

Bits	Name	Description
19:16	NUMCH	Number of ECT channels available.
15:8	NUMTRIG	Number of ECT triggers available.
4:0	EXTMUXNUM	Indicates the number of multiplexing available on Trigger Inputs and Trigger Outputs using ASICCTL. Default value of 5'b00000 indicating no multiplexing present. Reflects the value of the Verilog `define EXTMUXNUM that the user must alter accordingly.

0x00816FE0 QDSS_CTI6_PERIPHID0**Type:** R**Reset State:** 0x00000006

Part of the set of Peripheral Identification registers. Contains part of the designer specific part number.

QDSS_CTI6_PERIPHID0

Bits	Name	Description
7:0	PART_0	Bits [7:0] of the component's part number. This is selected by the designer of the component. 0x6: CTI_PART_NUMBER_7_0 (Lowest 8 bits of the Part Number (0x906))

0x00816FE4 QDSS_CTI6_PERIPHID1**Type:** R**Reset State:** 0x000000B9

Part of the set of Peripheral Identification registers. Contains part of the designer specific part number and part of the designer identity.

QDSS_CTI6_PERIPHID1

Bits	Name	Description
7:4	DES_0	Bits 3:0 of the JEDEC identity code indicating the designer of the component (along with the continuation code) 0xB: ARM_JEP106_IDENTITY_CODE_3_0 (Lowest 4 bits of the JEP106 Identity Code)
3:0	PART_1	Bits [11:8] of the component's part number. This is selected by the designer of the component. 0x9: CTI_PART_NUMBER_11_8 (Upper 4 bits of the Part Number (0x961))

0x00816FE8 QDSS_CTI6_PERIPHID2**Type:** R**Reset State:** 0x0000004B

Part of the set of Peripheral Identification registers. Contains part of the designer identity and the product revision.

QDSS_CTI6_PERIPHID2

Bits	Name	Description
7:4	REVISION	The Revision field is an incremental value starting at 0x0 for the first design of this component. This only increases by 1 for both major and minor revisions and is simply used as a look-up to establish the exact major/minor revision. 0x4: R0P4 (This device is at r0p4)

QDSS_CTL6_PERIPHID2 (cont.)

Bits	Name	Description
3	JEDEC	Always set. Indicates that a JEDEC assigned value is used 0x1: JEDEC_IDENTITY (The designer ID is specified by JEDEC (http://www.jedec.org))
2:0	DES_1	Bits 6:4 of the JEDEC identity code indicating the designer of the component (along with the continuation code) 0x3: ARM_JEP106_IDENTITY_CODE_6_4 (Upper 3 bits of the JEP106 Identity Code)

0x00816FEC QDSS_CTL6_PERIPHID3**Type:** R**Reset State:** 0x00000000

Part of the set of Peripheral Identification registers. Contains the RevAnd and Customer Modified fields.

QDSS_CTL6_PERIPHID3

Bits	Name	Description
7:4	REVAND	This field indicates minor errata fixes specific to this design, for example metal fixes after implementation. In most cases this field is zero. It is recommended that component designers ensure this field can be changed by a metal fix if required, for example by driving it from registers that reset to zero. 0x0: NO_MODIFICATION (Indicates that there have been no metal fixes to this component)
3:0	CMOD	Where the component is reusable IP, this value indicates if the customer has modified the behavior of the component. In most cases this field is zero. 0x0: NO_MODIFICATION (Indicates that there have been no modifications made)

0x00816FD0 QDSS_CTL6_PERIPHID4**Type:** R**Reset State:** 0x00000004

Part of the set of Peripheral Identification registers. Contains part of the designer identity and the memory footprint indicator.

QDSS_CTI6_PERIPHID4

Bits	Name	Description
7:4	SIZE	This is a 4-bit value that indicates the total contiguous size of the memory window used by this component in powers of 2 from the standard 4KB. If a component only requires the standard 4KB then this should read as 0x0, 4KB only, for 8KB set to 0x1, 16KB == 0x2, 32KB == 0x3, and so on. 0x0: ENUM_4KB (Indicates that the device only occupies 4kB of memory)
3:0	DES_2	JEDEC continuation code indicating the designer of the component (along with the identity code) 0x4: ARM_JEP106_CONTINUATION_CODE (Indicates that ARM's JEDEC identity code is on the 5th bank)

0x00816FD4 QDSS_CTI6_PERIPHID5**Type:** RW**Reset State:** Undefined

Reserved

QDSS_CTI6_PERIPHID5

Bits	Name	Description
31:0	PERIPHID5	

0x00816FD8 QDSS_CTI6_PERIPHID6**Type:** RW**Reset State:** Undefined

Reserved

QDSS_CTI6_PERIPHID6

Bits	Name	Description
31:0	PERIPHID6	

0x00816FDC QDSS_CTI6_PERIPHID7**Type:** RW**Reset State:** Undefined

Reserved

QDSS_CTI6_PERIPHID7

Bits	Name	Description
31:0	PERIPHID7	

0x00816FF0 QDSS_CTI6_COMPID0

Type: R

Reset State: 0x0000000D

A component identification register, that indicates that the identification registers are present.

QDSS_CTI6_COMPID0

Bits	Name	Description
7:0	PRMBL_0	Contains bits [7:0] of the component identification 0xD: ENUM_0D (Identification value)

0x00816FF4 QDSS_CTI6_COMPID1

Type: R

Reset State: 0x00000090

A component identification register, that indicates that the identification registers are present. This register also indicates the component class.

QDSS_CTI6_COMPID1

Bits	Name	Description
7:4	CLASS	Class of the component. E.g. ROM table, CoreSight component etc. Constitutes bits [15:12] of the component identification. 0x9: CORESIGHT_COMPONENT (Indicates the component is a CoreSight component)
3:0	PRMBL_1	Contains bits [11:8] of the component identification 0x0: ENUM_0 (Identification value)

0x00816FF8 QDSS_CTI6_COMPID2

Type: R

Reset State: 0x00000005

A component identification register, that indicates that the identification registers are present.

QDSS_CTI6_COMPID2

Bits	Name	Description
7:0	PRMBL_2	Contains bits [23:16] of the component identification 0x5: ENUM_05 (Identification value)

0x00816FFC QDSS_CTI6_COMPID3**Type:** R**Reset State:** 0x000000B1

A component identification register, that indicates that the identification registers are present.

QDSS_CTI6_COMPID3

Bits	Name	Description
7:0	PRMBL_3	Contains bits [31:24] of the component identification 0xB1: B1 (Identification value)

12.1.11 CTI7_CSCTI

The CTI allows mapping a trigger on the Trigger interface to a signal on the Channel interface and vice versa. The mapping is user-programmable. When the CTI receives a channel event it maps this onto a trigger output. This enables subsystems to cross trigger with each other. The receiving and transmitting of triggers is performed through the trigger interface.

12.1.11.1 QDSS_CTI7_CSCTI

The CTI allows mapping a trigger on the Trigger interface to a signal on the Channel interface and vice versa. The mapping is user-programmable. When the CTI receives a channel event it maps this onto a trigger output. This enables subsystems to cross trigger with each other. The receiving and transmitting of triggers is performed through the trigger interface.

0x00817000 QDSS_CTI7_CTICONTROL**Type:** RW**Reset State:** 0x00000000

QDSS_CTI7_CTICONTROL

Bits	Name	Description
0	GLBEN	0x1: ENABLED (When this bit is 1, cross triggering mapping logic functionality is enabled.) 0x0: DISABLED (When this bit is 0, all cross triggering mapping logic functionality is disabled.)

0x00817010 QDSS_CTI7_CTIINTACK**Type:** W**Reset State:** 0x00000000**QDSS_CTI7_CTIINTACK**

Bits	Name	Description
7:0	INTACK	

0x00817014 QDSS_CTI7_CTIAPPSET**Type:** W**Reset State:** 0x00000000**QDSS_CTI7_CTIAPPSET**

Bits	Name	Description
3:0	APPSET	

0x00817018 QDSS_CTI7_CTIAPPCLEAR**Type:** W**Reset State:** 0x00000000**QDSS_CTI7_CTIAPPCLEAR**

Bits	Name	Description
3:0	APPCLEAR	

0x0081701C QDSS_CTI7_CTIAPPPULSE**Type:** W**Reset State:** 0x00000000

QDSS_CTI7_CTIAPPPULSE

Bits	Name	Description
3:0	APPULSE	

0x00817020 QDSS_CTI7_CTIINEN0**Type:** RW**Reset State:** 0x00000000**QDSS_CTI7_CTIINEN0**

Bits	Name	Description
3:0	TRIGINEN	

0x00817024 QDSS_CTI7_CTIINEN1**Type:** RW**Reset State:** 0x00000000**QDSS_CTI7_CTIINEN1**

Bits	Name	Description
3:0	TRIGINEN	

0x00817028 QDSS_CTI7_CTIINEN2**Type:** RW**Reset State:** 0x00000000**QDSS_CTI7_CTIINEN2**

Bits	Name	Description
3:0	TRIGINEN	

0x0081702C QDSS_CTI7_CTIINEN3**Type:** RW**Reset State:** 0x00000000

QDSS_CT17_CTIINEN3

Bits	Name	Description
3:0	TRIGINEN	

0x00817030 QDSS_CT17_CTIINEN4**Type:** RW**Reset State:** 0x00000000**QDSS_CT17_CTIINEN4**

Bits	Name	Description
3:0	TRIGINEN	

0x00817034 QDSS_CT17_CTIINEN5**Type:** RW**Reset State:** 0x00000000**QDSS_CT17_CTIINEN5**

Bits	Name	Description
3:0	TRIGINEN	

0x00817038 QDSS_CT17_CTIINEN6**Type:** RW**Reset State:** 0x00000000**QDSS_CT17_CTIINEN6**

Bits	Name	Description
3:0	TRIGINEN	

0x0081703C QDSS_CT17_CTIINEN7**Type:** RW**Reset State:** 0x00000000

QDSS_CTI7_CTIINEN7

Bits	Name	Description
3:0	TRIGINEN	

0x008170A0 QDSS_CTI7_CTIOUTEN0**Type:** RW**Reset State:** 0x00000000**QDSS_CTI7_CTIOUTEN0**

Bits	Name	Description
3:0	TRIGOUTEN	

0x008170A4 QDSS_CTI7_CTIOUTEN1**Type:** RW**Reset State:** 0x00000000**QDSS_CTI7_CTIOUTEN1**

Bits	Name	Description
3:0	TRIGOUTEN	

0x008170A8 QDSS_CTI7_CTIOUTEN2**Type:** RW**Reset State:** 0x00000000**QDSS_CTI7_CTIOUTEN2**

Bits	Name	Description
3:0	TRIGOUTEN	

0x008170AC QDSS_CTI7_CTIOUTEN3**Type:** RW**Reset State:** 0x00000000

QDSS_CT17_CTIOUTEN3

Bits	Name	Description
3:0	TRIGOUTEN	

0x008170B0 QDSS_CT17_CTIOUTEN4**Type:** RW**Reset State:** 0x00000000**QDSS_CT17_CTIOUTEN4**

Bits	Name	Description
3:0	TRIGOUTEN	

0x008170B4 QDSS_CT17_CTIOUTEN5**Type:** RW**Reset State:** 0x00000000**QDSS_CT17_CTIOUTEN5**

Bits	Name	Description
3:0	TRIGOUTEN	

0x008170B8 QDSS_CT17_CTIOUTEN6**Type:** RW**Reset State:** 0x00000000**QDSS_CT17_CTIOUTEN6**

Bits	Name	Description
3:0	TRIGOUTEN	

0x008170BC QDSS_CT17_CTIOUTEN7**Type:** RW**Reset State:** 0x00000000

QDSS_CTI7_CTIOUTEN7

Bits	Name	Description
3:0	TRIGOUTEN	

0x00817130 QDSS_CTI7_CTITRIGINSTATUS**Type:** R**Reset State:** Undefined**QDSS_CTI7_CTITRIGINSTATUS**

Bits	Name	Description
7:0	TRIGINSTATUS	

0x00817134 QDSS_CTI7_CTITRIGOUTSTATUS**Type:** R**Reset State:** 0x00000000**QDSS_CTI7_CTITRIGOUTSTATUS**

Bits	Name	Description
7:0	TRIGOUTSTATUS	

0x00817138 QDSS_CTI7_CTICHINSTSTATUS**Type:** R**Reset State:** Undefined**QDSS_CTI7_CTICHINSTSTATUS**

Bits	Name	Description
3:0	CTICHINSTSTATUS	

0x0081713C QDSS_CTI7_CTICHOUTSTATUS**Type:** R**Reset State:** 0x00000000

QDSS_CTI7_CTICHOUTSTATUS

Bits	Name	Description
3:0	CTICHOUTSTATUS	

0x00817140 QDSS_CTI7_CTIGATE**Type:** RW**Reset State:** 0x0000000F**QDSS_CTI7_CTIGATE**

Bits	Name	Description
3	CTIGATEEN3	
2	CTIGATEEN2	
1	CTIGATEEN1	
0	CTIGATEEN0	

0x00817144 QDSS_CTI7_ASICCTL**Type:** RW**Reset State:** 0x00000000**QDSS_CTI7_ASICCTL**

Bits	Name	Description
7:0	ASICCTL	

0x00817EDC QDSS_CTI7_ITCHINACK**Type:** W**Reset State:** 0x00000000**QDSS_CTI7_ITCHINACK**

Bits	Name	Description
3:0	CTCHINACK	

0x00817EE0 QDSS_CTI7_ITTRIGINACK**Type:** W**Reset State:** 0x00000000**QDSS_CTI7_ITTRIGINACK**

Bits	Name	Description
7:0	CTTRIGINACK	

0x00817EE4 QDSS_CTI7_ITCHOUT**Type:** W**Reset State:** 0x00000000**QDSS_CTI7_ITCHOUT**

Bits	Name	Description
3:0	CTCHOUT	

0x00817EE8 QDSS_CTI7_ITTRIGOUT**Type:** W**Reset State:** 0x00000000**QDSS_CTI7_ITTRIGOUT**

Bits	Name	Description
7:0	CTTRIGOUT	

0x00817EEC QDSS_CTI7_ITCHOUTACK**Type:** R**Reset State:** 0x00000000**QDSS_CTI7_ITCHOUTACK**

Bits	Name	Description
3:0	CTCHOUTACK	

0x00817EF0 QDSS_CTI7_ITTRIGOUTACK**Type:** R**Reset State:** 0x00000000**QDSS_CTI7_ITTRIGOUTACK**

Bits	Name	Description
7:0	CTTRIGOUTACK	

0x00817EF4 QDSS_CTI7_ITCHIN**Type:** R**Reset State:** 0x00000000**QDSS_CTI7_ITCHIN**

Bits	Name	Description
3:0	CTCHIN	

0x00817EF8 QDSS_CTI7_ITTRIGIN**Type:** R**Reset State:** 0x00000000**QDSS_CTI7_ITTRIGIN**

Bits	Name	Description
7:0	CTTRIGIN	

0x00817F00 QDSS_CTI7_ITCTRL**Type:** RW**Reset State:** 0x00000000**QDSS_CTI7_ITCTRL**

Bits	Name	Description
0	INTEGRATION_MODE	Allows the component to switch from functional mode to integration mode or back. 0x1: INTEGRATION_MODE (Enable integration mode) 0x0: FUNCTIONAL_MODE (Disable integration mode)

0x00817FA0 QDSS_CTI7_CLAIMSET**Type:** RW**Reset State:** 0x0000000F

This is used in conjunction with Claim Tag Clear Register, CLAIMCLR. This register forms one half of the Claim Tag value. This location allows individual bits to be set, write, and returns the number of bits that can be set, read.

QDSS_CTI7_CLAIMSET

Bits	Name	Type	Description
3:0	CLAIMSET_W	W	Each bit is considered separately: 0 = no effect, 1 = set this bit in the claim tag.
3:0	CLAIMSET_R	R	This claim tag bit is implemented 0xF: CLAIM_TAG_IMPLEMENTED_BITS (These bits are present within the claim tag field)

0x00817FA4 QDSS_CTI7_CLAIMCLR**Type:** RW**Reset State:** 0x00000000

This register is used in conjunction with Claim Tag Set Register, CLAIMSET. This register forms one half of the Claim Tag value. This location enables individual bits to be cleared, write, and returns the current Claim Tag value, read.

QDSS_CTI7_CLAIMCLR

Bits	Name	Type	Description
3:0	CLAIMCLR_W	W	Each bit is considered separately: 0 = no effect, 1 = clear this bit in the claim tag.
3:0	CLAIMCLR_R	R	The value present reflects the current setting of the Claim Tag.

0x00817FB0 QDSS_CTI7_LAR**Type:** W**Reset State:** Undefined

QDSS_CTI7_LAR

Bits	Name	Description
31:0	ACCESS_W	A write of 0xC5ACCE55 enables further write access to this device. A write of any value other than 0xC5ACCE55 will have the affect of removing write access. 0xC5ACCE55: UNLOCK (Unlock the protection register to enable write access)

0x00817FB4 QDSS_CTI7_LSR**Type:** R**Reset State:** 0x00000003**QDSS_CTI7_LSR**

Bits	Name	Description
2	LOCKTYPE	Indicates if the Lock Access Register (0xFB0) is implemented as 8-bit or 32-bit 0x0: ENUM_32_BIT (This component implements a 32-bit Lock Access Register)
1	LOCKGRANT	Returns the current status of the Lock. This bit reads as 0 when read from an external debugger (PADDRDBG31 = 1) since external debugger accesses are not subject to Lock Registers. 0x0: ACCESS_PERMITTED (Write access is permitted to this device) 0x1: DEVICE_LOCKED (Write access to the component is blocked. All writes to control registers are ignored. Reads are permitted)
0	LOCKEXIST	Indicates that a lock control mechanism exists for this device. This bit reads as 0 when read from an external debugger (PADDRDBG31 = 1) since external debugger accesses are not subject to Lock Registers. 0x0: LOCK_NOT_PRESENT (No lock control mechanism exists, writes to the Lock Access register (0xFB0) will be ignored) 0x1: LOCK_PRESENT (Lock control mechanism is present)

0x00817FB8 QDSS_CTI7_AUTHSTATUS**Type:** R**Reset State:** 0xFFFFFFFF0X

Reports what functionality is currently permitted by the authentication interface.

QDSS_CTI7_AUTHSTATUS

Bits	Name	Description
7:6	SNID	Indicates the security level for secure non-invasive debug 0x0: ENUM_2_B00 (Functionality not implemented.)
5:4	SID	Indicates the security level for secure invasive debug 0x0: ENUM_2_B00 (Functionality not implemented.)
3:2	NSNID	Indicates the security level for non-secure non-invasive debug 0x2: ENUM_2_B10 (Non-secure Non-Invasive debug functionality currently disabled.) 0x3: ENUM_2_B11 (Non-secure Non-Invasive debug functionality currently enabled.)
1:0	NSID	Indicates the security level for non-secure invasive debug 0x2: ENUM_2_B10 (Non-secure Invasive debug functionality currently disabled.) 0x3: ENUM_2_B11 (Non-secure Invasive debug functionality currently enabled.)

0x00817FC8 QDSS_CTI7_DEVID**Type:** R**Reset State:** 0x00040800

This register indicates the capabilities of the CTI.

QDSS_CTI7_DEVID

Bits	Name	Description
19:16	NUMCH	Number of ECT channels available.
15:8	NUMTRIG	Number of ECT triggers available.
4:0	EXTMUXNUM	Indicates the number of multiplexing available on Trigger Inputs and Trigger Outputs using ASICCTL. Default value of 5'b00000 indicating no multiplexing present. Reflects the value of the Verilog `define EXTMUXNUM that the user must alter accordingly.

0x00817FE0 QDSS_CTI7_PERIPHID0**Type:** R**Reset State:** 0x00000006

Part of the set of Peripheral Identification registers. Contains part of the designer specific part number.

QDSS_CTI7_PERIPHID0

Bits	Name	Description
7:0	PART_0	Bits [7:0] of the component's part number. This is selected by the designer of the component. 0x6: CTI_PART_NUMBER_7_0 (Lowest 8 bits of the Part Number (0x906))

0x00817FE4 QDSS_CTI7_PERIPHID1**Type:** R**Reset State:** 0x000000B9

Part of the set of Peripheral Identification registers. Contains part of the designer specific part number and part of the designer identity.

QDSS_CTI7_PERIPHID1

Bits	Name	Description
7:4	DES_0	Bits 3:0 of the JEDEC identity code indicating the designer of the component (along with the continuation code) 0xB: ARM_JEP106_IDENTITY_CODE_3_0 (Lowest 4 bits of the JEP106 Identity Code)
3:0	PART_1	Bits [11:8] of the component's part number. This is selected by the designer of the component. 0x9: CTI_PART_NUMBER_11_8 (Upper 4 bits of the Part Number (0x961))

0x00817FE8 QDSS_CTI7_PERIPHID2**Type:** R**Reset State:** 0x0000004B

Part of the set of Peripheral Identification registers. Contains part of the designer identity and the product revision.

QDSS_CTI7_PERIPHID2

Bits	Name	Description
7:4	REVISION	The Revision field is an incremental value starting at 0x0 for the first design of this component. This only increases by 1 for both major and minor revisions and is simply used as a look-up to establish the exact major/minor revision. 0x4: R0P4 (This device is at r0p4)

QDSS_CTI7_PERIPHID2 (cont.)

Bits	Name	Description
3	JEDEC	Always set. Indicates that a JEDEC assigned value is used 0x1: JEDEC_IDENTITY (The designer ID is specified by JEDEC (http://www.jedec.org))
2:0	DES_1	Bits 6:4 of the JEDEC identity code indicating the designer of the component (along with the continuation code) 0x3: ARM_JEP106_IDENTITY_CODE_6_4 (Upper 3 bits of the JEP106 Identity Code)

0x00817FEC QDSS_CTI7_PERIPHID3**Type:** R**Reset State:** 0x00000000

Part of the set of Peripheral Identification registers. Contains the RevAnd and Customer Modified fields.

QDSS_CTI7_PERIPHID3

Bits	Name	Description
7:4	REVAND	This field indicates minor errata fixes specific to this design, for example metal fixes after implementation. In most cases this field is zero. It is recommended that component designers ensure this field can be changed by a metal fix if required, for example by driving it from registers that reset to zero. 0x0: NO_MODIFICATION (Indicates that there have been no metal fixes to this component)
3:0	CMOD	Where the component is reusable IP, this value indicates if the customer has modified the behavior of the component. In most cases this field is zero. 0x0: NO_MODIFICATION (Indicates that there have been no modifications made)

0x00817FD0 QDSS_CTI7_PERIPHID4**Type:** R**Reset State:** 0x00000004

Part of the set of Peripheral Identification registers. Contains part of the designer identity and the memory footprint indicator.

QDSS_CTI7_PERIPHID4

Bits	Name	Description
7:4	SIZE	This is a 4-bit value that indicates the total contiguous size of the memory window used by this component in powers of 2 from the standard 4KB. If a component only requires the standard 4KB then this should read as 0x0, 4KB only, for 8KB set to 0x1, 16KB == 0x2, 32KB == 0x3, and so on. 0x0: ENUM_4KB (Indicates that the device only occupies 4kB of memory)
3:0	DES_2	JEDEC continuation code indicating the designer of the component (along with the identity code) 0x4: ARM_JEP106_CONTINUATION_CODE (Indicates that ARM's JEDEC identity code is on the 5th bank)

0x00817FD4 QDSS_CTI7_PERIPHID5**Type:** RW**Reset State:** Undefined

Reserved

QDSS_CTI7_PERIPHID5

Bits	Name	Description
31:0	PERIPHID5	

0x00817FD8 QDSS_CTI7_PERIPHID6**Type:** RW**Reset State:** Undefined

Reserved

QDSS_CTI7_PERIPHID6

Bits	Name	Description
31:0	PERIPHID6	

0x00817FDC QDSS_CTI7_PERIPHID7**Type:** RW**Reset State:** Undefined

Reserved

QDSS_CTI7_PERIPHID7

Bits	Name	Description
31:0	PERIPHID7	

0x00817FF0 QDSS_CTI7_COMPID0

Type: R

Reset State: 0x0000000D

A component identification register, that indicates that the identification registers are present.

QDSS_CTI7_COMPID0

Bits	Name	Description
7:0	PRMBL_0	Contains bits [7:0] of the component identification 0xD: ENUM_0D (Identification value)

0x00817FF4 QDSS_CTI7_COMPID1

Type: R

Reset State: 0x00000090

A component identification register, that indicates that the identification registers are present. This register also indicates the component class.

QDSS_CTI7_COMPID1

Bits	Name	Description
7:4	CLASS	Class of the component. E.g. ROM table, CoreSight component etc. Constitutes bits [15:12] of the component identification. 0x9: CORESIGHT_COMPONENT (Indicates the component is a CoreSight component)
3:0	PRMBL_1	Contains bits [11:8] of the component identification 0x0: ENUM_0 (Identification value)

0x00817FF8 QDSS_CTI7_COMPID2

Type: R

Reset State: 0x00000005

A component identification register, that indicates that the identification registers are present.

QDSS_CTI7_COMPID2

Bits	Name	Description
7:0	PRMBL_2	Contains bits [23:16] of the component identification 0x5: ENUM_05 (Identification value)

0x00817FFC QDSS_CTI7_COMPID3**Type:** R**Reset State:** 0x000000B1

A component identification register, that indicates that the identification registers are present.

QDSS_CTI7_COMPID3

Bits	Name	Description
7:0	PRMBL_3	Contains bits [31:24] of the component identification 0xB1: B1 (Identification value)

12.1.12 CTI8_CSCTI

The CTI allows mapping a trigger on the Trigger interface to a signal on the Channel interface and vice versa. The mapping is user-programmable. When the CTI receives a channel event it maps this onto a trigger output. This enables subsystems to cross trigger with each other. The receiving and transmitting of triggers is performed through the trigger interface.

12.1.12.1 QDSS_CTI8_CSCTI

The CTI allows mapping a trigger on the Trigger interface to a signal on the Channel interface and vice versa. The mapping is user-programmable. When the CTI receives a channel event it maps this onto a trigger output. This enables subsystems to cross trigger with each other. The receiving and transmitting of triggers is performed through the trigger interface.

0x00818000 QDSS_CTI8_CTICONTROL**Type:** RW**Reset State:** 0x00000000

QDSS_CT18_CTICONTROL

Bits	Name	Description
0	GLBEN	0x1: ENABLED (When this bit is 1, cross triggering mapping logic functionality is enabled.) 0x0: DISABLED (When this bit is 0, all cross triggering mapping logic functionality is disabled.)

0x00818010 QDSS_CT18_CTIINTACK**Type:** W**Reset State:** 0x00000000**QDSS_CT18_CTIINTACK**

Bits	Name	Description
7:0	INTACK	

0x00818014 QDSS_CT18_CTIAPPSET**Type:** W**Reset State:** 0x00000000**QDSS_CT18_CTIAPPSET**

Bits	Name	Description
3:0	APPSET	

0x00818018 QDSS_CT18_CTIAPPCLEAR**Type:** W**Reset State:** 0x00000000**QDSS_CT18_CTIAPPCLEAR**

Bits	Name	Description
3:0	APPCLEAR	

0x0081801C QDSS_CT18_CTIAPPPULSE**Type:** W**Reset State:** 0x00000000

QDSS_CT18_CTIAPPPULSE

Bits	Name	Description
3:0	APPULSE	

0x00818020 QDSS_CT18_CTIINEN0**Type:** RW**Reset State:** 0x00000000**QDSS_CT18_CTIINEN0**

Bits	Name	Description
3:0	TRIGINEN	

0x00818024 QDSS_CT18_CTIINEN1**Type:** RW**Reset State:** 0x00000000**QDSS_CT18_CTIINEN1**

Bits	Name	Description
3:0	TRIGINEN	

0x00818028 QDSS_CT18_CTIINEN2**Type:** RW**Reset State:** 0x00000000**QDSS_CT18_CTIINEN2**

Bits	Name	Description
3:0	TRIGINEN	

0x0081802C QDSS_CT18_CTIINEN3**Type:** RW**Reset State:** 0x00000000

QDSS_CT18_CTIINEN3

Bits	Name	Description
3:0	TRIGINEN	

0x00818030 QDSS_CT18_CTIINEN4**Type:** RW**Reset State:** 0x00000000**QDSS_CT18_CTIINEN4**

Bits	Name	Description
3:0	TRIGINEN	

0x00818034 QDSS_CT18_CTIINEN5**Type:** RW**Reset State:** 0x00000000**QDSS_CT18_CTIINEN5**

Bits	Name	Description
3:0	TRIGINEN	

0x00818038 QDSS_CT18_CTIINEN6**Type:** RW**Reset State:** 0x00000000**QDSS_CT18_CTIINEN6**

Bits	Name	Description
3:0	TRIGINEN	

0x0081803C QDSS_CT18_CTIINEN7**Type:** RW**Reset State:** 0x00000000

QDSS_CT18_CTIINEN7

Bits	Name	Description
3:0	TRIGINEN	

0x008180A0 QDSS_CT18_CTIOUTEN0**Type:** RW**Reset State:** 0x00000000**QDSS_CT18_CTIOUTEN0**

Bits	Name	Description
3:0	TRIGOUTEN	

0x008180A4 QDSS_CT18_CTIOUTEN1**Type:** RW**Reset State:** 0x00000000**QDSS_CT18_CTIOUTEN1**

Bits	Name	Description
3:0	TRIGOUTEN	

0x008180A8 QDSS_CT18_CTIOUTEN2**Type:** RW**Reset State:** 0x00000000**QDSS_CT18_CTIOUTEN2**

Bits	Name	Description
3:0	TRIGOUTEN	

0x008180AC QDSS_CT18_CTIOUTEN3**Type:** RW**Reset State:** 0x00000000

QDSS_CT18_CTIOUTEN3

Bits	Name	Description
3:0	TRIGOUTEN	

0x008180B0 QDSS_CT18_CTIOUTEN4**Type:** RW**Reset State:** 0x00000000**QDSS_CT18_CTIOUTEN4**

Bits	Name	Description
3:0	TRIGOUTEN	

0x008180B4 QDSS_CT18_CTIOUTEN5**Type:** RW**Reset State:** 0x00000000**QDSS_CT18_CTIOUTEN5**

Bits	Name	Description
3:0	TRIGOUTEN	

0x008180B8 QDSS_CT18_CTIOUTEN6**Type:** RW**Reset State:** 0x00000000**QDSS_CT18_CTIOUTEN6**

Bits	Name	Description
3:0	TRIGOUTEN	

0x008180BC QDSS_CT18_CTIOUTEN7**Type:** RW**Reset State:** 0x00000000

QDSS_CT18_CTIOUTEN7

Bits	Name	Description
3:0	TRIGOUTEN	

0x00818130 QDSS_CT18_CTITRIGINSTATUS**Type:** R**Reset State:** Undefined**QDSS_CT18_CTITRIGINSTATUS**

Bits	Name	Description
7:0	TRIGINSTATUS	

0x00818134 QDSS_CT18_CTITRIGOUTSTATUS**Type:** R**Reset State:** 0x00000000**QDSS_CT18_CTITRIGOUTSTATUS**

Bits	Name	Description
7:0	TRIGOUTSTATUS	

0x00818138 QDSS_CT18_CTICHINSTSTATUS**Type:** R**Reset State:** Undefined**QDSS_CT18_CTICHINSTSTATUS**

Bits	Name	Description
3:0	CTICHINSTSTATUS	

0x0081813C QDSS_CT18_CTICHOUTSTATUS**Type:** R**Reset State:** 0x00000000

QDSS_CT18_CTICHOUTSTATUS

Bits	Name	Description
3:0	CTICHOUTSTATUS	

0x00818140 QDSS_CT18_CTIGATE**Type:** RW**Reset State:** 0x0000000F**QDSS_CT18_CTIGATE**

Bits	Name	Description
3	CTIGATEEN3	
2	CTIGATEEN2	
1	CTIGATEEN1	
0	CTIGATEEN0	

0x00818144 QDSS_CT18_ASICCTL**Type:** RW**Reset State:** 0x00000000**QDSS_CT18_ASICCTL**

Bits	Name	Description
7:0	ASICCTL	

0x00818EDC QDSS_CT18_ITCHINACK**Type:** W**Reset State:** 0x00000000**QDSS_CT18_ITCHINACK**

Bits	Name	Description
3:0	CTCHINACK	

0x00818EE0 QDSS_CT18_ITTRIGINACK**Type:** W**Reset State:** 0x00000000**QDSS_CT18_ITTRIGINACK**

Bits	Name	Description
7:0	CTTRIGINACK	

0x00818EE4 QDSS_CT18_ITCHOUT**Type:** W**Reset State:** 0x00000000**QDSS_CT18_ITCHOUT**

Bits	Name	Description
3:0	CTCHOUT	

0x00818EE8 QDSS_CT18_ITTRIGOUT**Type:** W**Reset State:** 0x00000000**QDSS_CT18_ITTRIGOUT**

Bits	Name	Description
7:0	CTTRIGOUT	

0x00818EEC QDSS_CT18_ITCHOUTACK**Type:** R**Reset State:** 0x00000000**QDSS_CT18_ITCHOUTACK**

Bits	Name	Description
3:0	CTCHOUTACK	

0x00818EF0 QDSS_CT18_ITTRIGOUTACK**Type:** R**Reset State:** 0x00000000**QDSS_CT18_ITTRIGOUTACK**

Bits	Name	Description
7:0	CTTRIGOUTACK	

0x00818EF4 QDSS_CT18_ITCHIN**Type:** R**Reset State:** 0x00000000**QDSS_CT18_ITCHIN**

Bits	Name	Description
3:0	CTCHIN	

0x00818EF8 QDSS_CT18_ITTRIGIN**Type:** R**Reset State:** 0x00000000**QDSS_CT18_ITTRIGIN**

Bits	Name	Description
7:0	CTTRIGIN	

0x00818F00 QDSS_CT18_ITCTRL**Type:** RW**Reset State:** 0x00000000**QDSS_CT18_ITCTRL**

Bits	Name	Description
0	INTEGRATION_MODE	Allows the component to switch from functional mode to integration mode or back. 0x1: INTEGRATION_MODE (Enable integration mode) 0x0: FUNCTIONAL_MODE (Disable integration mode)

0x00818FA0 QDSS_CT18_CLAIMSET**Type:** RW**Reset State:** 0x0000000F

This is used in conjunction with Claim Tag Clear Register, CLAIMCLR. This register forms one half of the Claim Tag value. This location allows individual bits to be set, write, and returns the number of bits that can be set, read.

QDSS_CT18_CLAIMSET

Bits	Name	Type	Description
3:0	CLAIMSET_W	W	Each bit is considered separately: 0 = no effect, 1 = set this bit in the claim tag.
3:0	CLAIMSET_R	R	This claim tag bit is implemented 0xF: CLAIM_TAG_IMPLEMENTED_BITS (These bits are present within the claim tag field)

0x00818FA4 QDSS_CT18_CLAIMCLR**Type:** RW**Reset State:** 0x00000000

This register is used in conjunction with Claim Tag Set Register, CLAIMSET. This register forms one half of the Claim Tag value. This location enables individual bits to be cleared, write, and returns the current Claim Tag value, read.

QDSS_CT18_CLAIMCLR

Bits	Name	Type	Description
3:0	CLAIMCLR_W	W	Each bit is considered separately: 0 = no effect, 1 = clear this bit in the claim tag.
3:0	CLAIMCLR_R	R	The value present reflects the current setting of the Claim Tag.

0x00818FB0 QDSS_CT18_LAR**Type:** W**Reset State:** Undefined

QDSS_CTl8_LAR

Bits	Name	Description
31:0	ACCESS_W	A write of 0xC5ACCE55 enables further write access to this device. A write of any value other than 0xC5ACCE55 will have the affect of removing write access. 0xC5ACCE55: UNLOCK (Unlock the protection register to enable write access)

0x00818FB4 QDSS_CTl8_LSR**Type:** R**Reset State:** 0x00000003**QDSS_CTl8_LSR**

Bits	Name	Description
2	LOCKTYPE	Indicates if the Lock Access Register (0xFB0) is implemented as 8-bit or 32-bit 0x0: ENUM_32_BIT (This component implements a 32-bit Lock Access Register)
1	LOCKGRANT	Returns the current status of the Lock. This bit reads as 0 when read from an external debugger (PADDRDBG31 = 1) since external debugger accesses are not subject to Lock Registers. 0x0: ACCESS_PERMITTED (Write access is permitted to this device) 0x1: DEVICE_LOCKED (Write access to the component is blocked. All writes to control registers are ignored. Reads are permitted)
0	LOCKEXIST	Indicates that a lock control mechanism exists for this device. This bit reads as 0 when read from an external debugger (PADDRDBG31 = 1) since external debugger accesses are not subject to Lock Registers. 0x0: LOCK_NOT_PRESENT (No lock control mechanism exists, writes to the Lock Access register (0xFB0) will be ignored) 0x1: LOCK_PRESENT (Lock control mechanism is present)

0x00818FB8 QDSS_CTl8_AUTHSTATUS**Type:** R**Reset State:** 0xFFFFFFFF0X

Reports what functionality is currently permitted by the authentication interface.

QDSS_CT18_AUTHSTATUS

Bits	Name	Description
7:6	SNID	Indicates the security level for secure non-invasive debug 0x0: ENUM_2_B00 (Functionality not implemented.)
5:4	SID	Indicates the security level for secure invasive debug 0x0: ENUM_2_B00 (Functionality not implemented.)
3:2	NSNID	Indicates the security level for non-secure non-invasive debug 0x2: ENUM_2_B10 (Non-secure Non-Invasive debug functionality currently disabled.) 0x3: ENUM_2_B11 (Non-secure Non-Invasive debug functionality currently enabled.)
1:0	NSID	Indicates the security level for non-secure invasive debug 0x2: ENUM_2_B10 (Non-secure Invasive debug functionality currently disabled.) 0x3: ENUM_2_B11 (Non-secure Invasive debug functionality currently enabled.)

0x00818FC8 QDSS_CT18_DEVID**Type:** R**Reset State:** 0x00040800

This register indicates the capabilities of the CTI.

QDSS_CT18_DEVID

Bits	Name	Description
19:16	NUMCH	Number of ECT channels available.
15:8	NUMTRIG	Number of ECT triggers available.
4:0	EXTMUXNUM	Indicates the number of multiplexing available on Trigger Inputs and Trigger Outputs using ASICCTL. Default value of 5'b00000 indicating no multiplexing present. Reflects the value of the Verilog `define EXTMUXNUM that the user must alter accordingly.

0x00818FE0 QDSS_CT18_PERIPHID0**Type:** R**Reset State:** 0x00000006

Part of the set of Peripheral Identification registers. Contains part of the designer specific part number.

QDSS_CT18_PERIPHID0

Bits	Name	Description
7:0	PART_0	Bits [7:0] of the component's part number. This is selected by the designer of the component. 0x6: CTI_PART_NUMBER_7_0 (Lowest 8 bits of the Part Number (0x906))

0x00818FE4 QDSS_CT18_PERIPHID1**Type:** R**Reset State:** 0x000000B9

Part of the set of Peripheral Identification registers. Contains part of the designer specific part number and part of the designer identity.

QDSS_CT18_PERIPHID1

Bits	Name	Description
7:4	DES_0	Bits 3:0 of the JEDEC identity code indicating the designer of the component (along with the continuation code) 0xB: ARM_JEP106_IDENTITY_CODE_3_0 (Lowest 4 bits of the JEP106 Identity Code)
3:0	PART_1	Bits [11:8] of the component's part number. This is selected by the designer of the component. 0x9: CTI_PART_NUMBER_11_8 (Upper 4 bits of the Part Number (0x961))

0x00818FE8 QDSS_CT18_PERIPHID2**Type:** R**Reset State:** 0x0000004B

Part of the set of Peripheral Identification registers. Contains part of the designer identity and the product revision.

QDSS_CT18_PERIPHID2

Bits	Name	Description
7:4	REVISION	The Revision field is an incremental value starting at 0x0 for the first design of this component. This only increases by 1 for both major and minor revisions and is simply used as a look-up to establish the exact major/minor revision. 0x4: R0P4 (This device is at r0p4)

QDSS_CT18_PERIPHID2 (cont.)

Bits	Name	Description
3	JEDEC	Always set. Indicates that a JEDEC assigned value is used 0x1: JEDEC_IDENTITY (The designer ID is specified by JEDEC (http://www.jedec.org))
2:0	DES_1	Bits 6:4 of the JEDEC identity code indicating the designer of the component (along with the continuation code) 0x3: ARM_JEP106_IDENTITY_CODE_6_4 (Upper 3 bits of the JEP106 Identity Code)

0x00818FEC QDSS_CT18_PERIPHID3**Type:** R**Reset State:** 0x00000000

Part of the set of Peripheral Identification registers. Contains the RevAnd and Customer Modified fields.

QDSS_CT18_PERIPHID3

Bits	Name	Description
7:4	REVAND	This field indicates minor errata fixes specific to this design, for example metal fixes after implementation. In most cases this field is zero. It is recommended that component designers ensure this field can be changed by a metal fix if required, for example by driving it from registers that reset to zero. 0x0: NO_MODIFICATION (Indicates that there have been no metal fixes to this component)
3:0	CMOD	Where the component is reusable IP, this value indicates if the customer has modified the behavior of the component. In most cases this field is zero. 0x0: NO_MODIFICATION (Indicates that there have been no modifications made)

0x00818FD0 QDSS_CT18_PERIPHID4**Type:** R**Reset State:** 0x00000004

Part of the set of Peripheral Identification registers. Contains part of the designer identity and the memory footprint indicator.

QDSS_CT18_PERIPHID4

Bits	Name	Description
7:4	SIZE	This is a 4-bit value that indicates the total contiguous size of the memory window used by this component in powers of 2 from the standard 4KB. If a component only requires the standard 4KB then this should read as 0x0, 4KB only, for 8KB set to 0x1, 16KB == 0x2, 32KB == 0x3, and so on. 0x0: ENUM_4KB (Indicates that the device only occupies 4kB of memory)
3:0	DES_2	JEDEC continuation code indicating the designer of the component (along with the identity code) 0x4: ARM_JEP106_CONTINUATION_CODE (Indicates that ARM's JEDEC identity code is on the 5th bank)

0x00818FD4 QDSS_CT18_PERIPHID5**Type:** RW**Reset State:** Undefined

Reserved

QDSS_CT18_PERIPHID5

Bits	Name	Description
31:0	PERIPHID5	

0x00818FD8 QDSS_CT18_PERIPHID6**Type:** RW**Reset State:** Undefined

Reserved

QDSS_CT18_PERIPHID6

Bits	Name	Description
31:0	PERIPHID6	

0x00818FDC QDSS_CT18_PERIPHID7**Type:** RW**Reset State:** Undefined

Reserved

QDSS_CT18_PERIPHID7

Bits	Name	Description
31:0	PERIPHID7	

0x00818FF0 QDSS_CT18_COMPID0

Type: R

Reset State: 0x0000000D

A component identification register, that indicates that the identification registers are present.

QDSS_CT18_COMPID0

Bits	Name	Description
7:0	PRMBL_0	Contains bits [7:0] of the component identification 0xD: ENUM_0D (Identification value)

0x00818FF4 QDSS_CT18_COMPID1

Type: R

Reset State: 0x00000090

A component identification register, that indicates that the identification registers are present. This register also indicates the component class.

QDSS_CT18_COMPID1

Bits	Name	Description
7:4	CLASS	Class of the component. E.g. ROM table, CoreSight component etc. Constitutes bits [15:12] of the component identification. 0x9: CORESIGHT_COMPONENT (Indicates the component is a CoreSight component)
3:0	PRMBL_1	Contains bits [11:8] of the component identification 0x0: ENUM_0 (Identification value)

0x00818FF8 QDSS_CT18_COMPID2

Type: R

Reset State: 0x00000005

A component identification register, that indicates that the identification registers are present.

QDSS_CT18_COMPID2

Bits	Name	Description
7:0	PRMBL_2	Contains bits [23:16] of the component identification 0x5: ENUM_05 (Identification value)

0x00818FFC QDSS_CT18_COMPID3**Type:** R**Reset State:** 0x000000B1

A component identification register, that indicates that the identification registers are present.

QDSS_CT18_COMPID3

Bits	Name	Description
7:0	PRMBL_3	Contains bits [31:24] of the component identification 0xB1: B1 (Identification value)

12.1.13 CSTPIU_CSTPIU

The TPIU acts as a bridge between the on-chip trace data, with separate IDs, to a data stream, encapsulating IDs where required, that is then captured by a Trace Port Analyzer (TPA). The internal formatter inserts source ID signals into the data packet stream so that trace data can be re-associated with its trace source. It contains an Asynchronous FIFO that enables trace data to be driven out at a speed that is not dependent on the on-chip bus clock. The internal trace-out block serializes formatted data before it goes off-chip. The TPIU includes a pattern generator unit that provides a simple set of defined bit sequences or patterns that can be output over the Trace Port and be detected by the TPA or other associated Trace Capture Device (TCD). The TCD can use these patterns to indicate if it is possible to increase or to decrease the trace port clock speed.

12.1.13.1 QDSS_CSTPIU_CSTPIU**0x00820000 QDSS_CSTPIU_SUPPORTED_PORT_SIZES****Type:** R**Reset State:** 0bXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX1**QDSS_CSTPIU_SUPPORTED_PORT_SIZES**

Bits	Name	Description
31	PORT_SIZE_32	0x1: SUPPORTED (Port size is supported.) 0x0: NOT_SUPPORTED (Port size is not supported.)

QDSS_CSTPIU_SUPPORTED_PORT_SIZES (cont.)

Bits	Name	Description
30	PORT_SIZE_31	0x1: SUPPORTED (Port size is supported.) 0x0: NOT_SUPPORTED (Port size is not supported.)
29	PORT_SIZE_30	0x1: SUPPORTED (Port size is supported.) 0x0: NOT_SUPPORTED (Port size is not supported.)
28	PORT_SIZE_29	0x1: SUPPORTED (Port size is supported.) 0x0: NOT_SUPPORTED (Port size is not supported.)
27	PORT_SIZE_28	0x1: SUPPORTED (Port size is supported.) 0x0: NOT_SUPPORTED (Port size is not supported.)
26	PORT_SIZE_27	0x1: SUPPORTED (Port size is supported.) 0x0: NOT_SUPPORTED (Port size is not supported.)
25	PORT_SIZE_26	0x1: SUPPORTED (Port size is supported.) 0x0: NOT_SUPPORTED (Port size is not supported.)
24	PORT_SIZE_25	0x1: SUPPORTED (Port size is supported.) 0x0: NOT_SUPPORTED (Port size is not supported.)
23	PORT_SIZE_24	0x1: SUPPORTED (Port size is supported.) 0x0: NOT_SUPPORTED (Port size is not supported.)
22	PORT_SIZE_23	0x1: SUPPORTED (Port size is supported.) 0x0: NOT_SUPPORTED (Port size is not supported.)
21	PORT_SIZE_22	0x1: SUPPORTED (Port size is supported.) 0x0: NOT_SUPPORTED (Port size is not supported.)
20	PORT_SIZE_21	0x1: SUPPORTED (Port size is supported.) 0x0: NOT_SUPPORTED (Port size is not supported.)
19	PORT_SIZE_20	0x1: SUPPORTED (Port size is supported.) 0x0: NOT_SUPPORTED (Port size is not supported.)
18	PORT_SIZE_19	0x1: SUPPORTED (Port size is supported.) 0x0: NOT_SUPPORTED (Port size is not supported.)
17	PORT_SIZE_18	0x1: SUPPORTED (Port size is supported.) 0x0: NOT_SUPPORTED (Port size is not supported.)
16	PORT_SIZE_17	0x1: SUPPORTED (Port size is supported.) 0x0: NOT_SUPPORTED (Port size is not supported.)
15	PORT_SIZE_16	0x1: SUPPORTED (Port size is supported.) 0x0: NOT_SUPPORTED (Port size is not supported.)
14	PORT_SIZE_15	0x1: SUPPORTED (Port size is supported.) 0x0: NOT_SUPPORTED (Port size is not supported.)
13	PORT_SIZE_14	0x1: SUPPORTED (Port size is supported.) 0x0: NOT_SUPPORTED (Port size is not supported.)
12	PORT_SIZE_13	0x1: SUPPORTED (Port size is supported.) 0x0: NOT_SUPPORTED (Port size is not supported.)

QDSS_CSTPIU_SUPPORTED_PORT_SIZES (cont.)

Bits	Name	Description
11	PORT_SIZE_12	0x1: SUPPORTED (Port size is supported.) 0x0: NOT_SUPPORTED (Port size is not supported.)
10	PORT_SIZE_11	0x1: SUPPORTED (Port size is supported.) 0x0: NOT_SUPPORTED (Port size is not supported.)
9	PORT_SIZE_10	0x1: SUPPORTED (Port size is supported.) 0x0: NOT_SUPPORTED (Port size is not supported.)
8	PORT_SIZE_9	0x1: SUPPORTED (Port size is supported.) 0x0: NOT_SUPPORTED (Port size is not supported.)
7	PORT_SIZE_8	0x1: SUPPORTED (Port size is supported.) 0x0: NOT_SUPPORTED (Port size is not supported.)
6	PORT_SIZE_7	0x1: SUPPORTED (Port size is supported.) 0x0: NOT_SUPPORTED (Port size is not supported.)
5	PORT_SIZE_6	0x1: SUPPORTED (Port size is supported.) 0x0: NOT_SUPPORTED (Port size is not supported.)
4	PORT_SIZE_5	0x1: SUPPORTED (Port size is supported.) 0x0: NOT_SUPPORTED (Port size is not supported.)
3	PORT_SIZE_4	0x1: SUPPORTED (Port size is supported.) 0x0: NOT_SUPPORTED (Port size is not supported.)
2	PORT_SIZE_3	0x1: SUPPORTED (Port size is supported.) 0x0: NOT_SUPPORTED (Port size is not supported.)
1	PORT_SIZE_2	0x1: SUPPORTED (Port size is supported.) 0x0: NOT_SUPPORTED (Port size is not supported.)
0	PORT_SIZE_1	0x1: SUPPORTED (Port size is supported.) 0x0: NOT_SUPPORTED (Port size is not supported.)

0x00820004 QDSS_CSTPIU_CURRENT_PORT_SIZE**Type:** RW**Reset State:** 0x00000001**QDSS_CSTPIU_CURRENT_PORT_SIZE**

Bits	Name	Description
31	PORT_SIZE_32	0x1: VALID (Current Port size is 32.) 0x0: INVALID (Current Port size is not 32.)
30	PORT_SIZE_31	0x1: VALID (Current Port size is 31.) 0x0: INVALID (Current Port size is not 31.)
29	PORT_SIZE_30	0x1: VALID (Current Port size is 30.) 0x0: INVALID (Current Port size is not 30.)

QDSS_CSTPIU_CURRENT_PORT_SIZE (cont.)

Bits	Name	Description
28	PORT_SIZE_29	0x1: VALID (Current Port size is 29.) 0x0: INVALID (Current Port size is not 29.)
27	PORT_SIZE_28	0x1: VALID (Current Port size is 28.) 0x0: INVALID (Current Port size is not 28.)
26	PORT_SIZE_27	0x1: VALID (Current Port size is 27.) 0x0: INVALID (Current Port size is not 27.)
25	PORT_SIZE_26	0x1: VALID (Current Port size is 26.) 0x0: INVALID (Current Port size is not 26.)
24	PORT_SIZE_25	0x1: VALID (Current Port size is 25.) 0x0: INVALID (Current Port size is not 25.)
23	PORT_SIZE_24	0x1: VALID (Current Port size is 24.) 0x0: INVALID (Current Port size is not 24.)
22	PORT_SIZE_23	0x1: VALID (Current Port size is 23.) 0x0: INVALID (Current Port size is not 23.)
21	PORT_SIZE_22	0x1: VALID (Current Port size is 22.) 0x0: INVALID (Current Port size is not 22.)
20	PORT_SIZE_21	0x1: VALID (Current Port size is 21.) 0x0: INVALID (Current Port size is not 21.)
19	PORT_SIZE_20	0x1: VALID (Current Port size is 20.) 0x0: INVALID (Current Port size is not 20.)
18	PORT_SIZE_19	0x1: VALID (Current Port size is 19.) 0x0: INVALID (Current Port size is not 19.)
17	PORT_SIZE_18	0x1: VALID (Current Port size is 18.) 0x0: INVALID (Current Port size is not 18.)
16	PORT_SIZE_17	0x1: VALID (Current Port size is 17.) 0x0: INVALID (Current Port size is not 17.)
15	PORT_SIZE_16	0x1: VALID (Current Port size is 16.) 0x0: INVALID (Current Port size is not 16.)
14	PORT_SIZE_15	0x1: VALID (Current Port size is 15.) 0x0: INVALID (Current Port size is not 15.)
13	PORT_SIZE_14	0x1: VALID (Current Port size is 14.) 0x0: INVALID (Current Port size is not 14.)
12	PORT_SIZE_13	0x1: VALID (Current Port size is 13.) 0x0: INVALID (Current Port size is not 13.)
11	PORT_SIZE_12	0x1: VALID (Current Port size is 12.) 0x0: INVALID (Current Port size is not 12.)
10	PORT_SIZE_11	0x1: VALID (Current Port size is 11.) 0x0: INVALID (Current Port size is not 11.)

QDSS_CSTPIU_CURRENT_PORT_SIZE (cont.)

Bits	Name	Description
9	PORT_SIZE_10	0x1: VALID (Current Port size is 10.) 0x0: INVALID (Current Port size is not 10.)
8	PORT_SIZE_9	0x1: VALID (Current Port size is 9.) 0x0: INVALID (Current Port size is not 9.)
7	PORT_SIZE_8	0x1: VALID (Current Port size is 8.) 0x0: INVALID (Current Port size is not 8.)
6	PORT_SIZE_7	0x1: VALID (Current Port size is 7.) 0x0: INVALID (Current Port size is not 7.)
5	PORT_SIZE_6	0x1: VALID (Current Port size is 6.) 0x0: INVALID (Current Port size is not 6.)
4	PORT_SIZE_5	0x1: VALID (Current Port size is 5.) 0x0: INVALID (Current Port size is not 5.)
3	PORT_SIZE_4	0x1: VALID (Current Port size is 4.) 0x0: INVALID (Current Port size is not 4.)
2	PORT_SIZE_3	0x1: VALID (Current Port size is 3.) 0x0: INVALID (Current Port size is not 3.)
1	PORT_SIZE_2	0x1: VALID (Current Port size is 2.) 0x0: INVALID (Current Port size is not 2.)
0	PORT_SIZE_1	0x1: VALID (Current Port size is 1.) 0x0: INVALID (Current Port size is not 1.)

0x00820100 QDSS_CSTPIU_SUPPORTED_TRIGGER_MODES**Type:** R**Reset State:** 0x0000011F**QDSS_CSTPIU_SUPPORTED_TRIGGER_MODES**

Bits	Name	Description
17	TRGRUN	0x1: TRGRUNNING (A trigger has occurred but the counter is not at zero.) 0x0: TRGNOTRUNNING (Either a trigger has not occurred or the counter is at zero.)
16	TRIGGERED	0x1: TRIGGER_OCCURRED (Trigger occurred.) 0x0: TRIGGER_NOT_OCCURRED (Trigger not occurred.)
8	TCOUNT8	0x1: IMPLEMENTED (8-bit wide counter register implemented.) 0x0: NOT_IMPLEMENTED (8-bit wide counter register not implemented.)

QDSS_CSTPIU_SUPPORTED_TRIGGER_MODES (cont.)

Bits	Name	Description
4	MULT64K	0x1: SUPPORTED (Multiply the Trigger Counter by 65536 supported.) 0x0: NOT_SUPPORTED (Multiply the Trigger Counter by 65536 not supported.)
3	MULT256	0x1: SUPPORTED (Multiply the Trigger Counter by 256 supported.) 0x0: NOT_SUPPORTED (Multiply the Trigger Counter by 256 not supported.)
2	MULT16	0x1: SUPPORTED (Multiply the Trigger Counter by 16 supported.) 0x0: NOT_SUPPORTED (Multiply the Trigger Counter by 16 not supported.)
1	MULT4	0x1: SUPPORTED (Multiply the Trigger Counter by 4 supported.) 0x0: NOT_SUPPORTED (Multiply the Trigger Counter by 4 not supported.)
0	MULT2	0x1: SUPPORTED (Multiply the Trigger Counter by 2 supported.) 0x0: NOT_SUPPORTED (Multiply the Trigger Counter by 2 not supported.)

0x00820104 QDSS_CSTPIU_TRIGGER_COUNTER_VALUE**Type:** RW**Reset State:** 0x00000000**QDSS_CSTPIU_TRIGGER_COUNTER_VALUE**

Bits	Name	Description
7:0	TRIGCOUNT	

0x00820108 QDSS_CSTPIU_TRIGGER_MULTIPLIER**Type:** RW**Reset State:** 0x00000000**QDSS_CSTPIU_TRIGGER_MULTIPLIER**

Bits	Name	Description
4	MULT64K	0x1: ENABLED (Multiplier enabled.) 0x0: DISABLED (Multiplier disabled.)
3	MULT256	0x1: ENABLED (Multiplier enabled.) 0x0: DISABLED (Multiplier disabled.)
2	MULT16	0x1: ENABLED (Multiplier enabled.) 0x0: DISABLED (Multiplier disabled.)

QDSS_CSTPIU_TRIGGER_MULTIPLIER (cont.)

Bits	Name	Description
1	MULT4	0x1: ENABLED (Multiplier enabled.) 0x0: DISABLED (Multiplier disabled.)
0	MULT2	0x1: ENABLED (Multiplier enabled.) 0x0: DISABLED (Multiplier disabled.)

0x00820208 QDSS_CSTPIU_TPRCR**Type:** RW**Reset State:** 0x00000000**QDSS_CSTPIU_TPRCR**

Bits	Name	Description
7:0	PATTCOUNT	

0x00820300 QDSS_CSTPIU_FFSR**Type:** R**Reset State:** 0b0000000000000000000000000000XX0**QDSS_CSTPIU_FFSR**

Bits	Name	Description
2	TCPRESENT	0x1: HIGH (TRACECTL pin present.) 0x0: LOW (TRACECTL pin not present.)
1	FTSTOPPED	0x1: HIGH (Formatter has stopped.) 0x0: LOW (Formatter has not stopped.)
0	FLINPROG	0x1: HIGH (AFVALIDS is HIGH.) 0x0: LOW (AFVALIDS is LOW.)

0x00820304 QDSS_CSTPIU_FFCR**Type:** RW**Reset State:** 0x00000000

QDSS_CSTPIU_FFCR

Bits	Name	Type	Description
13	STOPTRIG	RW	0x1: ENABLED (Enable stopping the formatter after a Trigger Event is observed.) 0x0: DISABLED (Disable stopping the formatter after a Trigger Event is observed.)
12	STOPFL	RW	0x1: ENABLED (Enable stopping the formatter on return of AFREADYYS.) 0x0: DISABLED (Disable stopping the formatter on return of AFREADYYS.)
10	TRIGFL	RW	0x1: ENABLED (Enable trigger indication on AFREADYYS being returned.) 0x0: DISABLED (Disable trigger indication on AFREADYYS being returned.)
9	TRIG EVT	RW	0x1: ENABLED (Enable trigger indication on a Trigger Event.) 0x0: DISABLED (Disable trigger indication on a Trigger Event.)
8	TRIGIN	RW	0x1: ENABLED (Enable trigger indication on TRIGIN being asserted.) 0x0: DISABLED (Disable trigger indication on TRIGIN being asserted.)
6	FONMAN_R	R	0x0: LOW (Manual Flush is not initiated) 0x1: HIGH (Manual Flush is initiated)
6	FONMAN_W	W	0x0: LOW (Manual Flush is not initiated) 0x1: HIGH (Manual Flush is initiated)
5	FONTRIG	RW	0x1: ENABLED (Enable generation of flush when a Trigger Event occurs.) 0x0: DISABLED (Disable generation of flush when a Trigger Event occurs.)
4	FONFLIN	RW	0x1: ENABLED (Enable generation of flush using the FLUSHIN interface.) 0x0: DISABLED (Disable generation of flush using the FLUSHIN interface.)
1	ENFCONT	RW	0x1: ENABLED (Continuous Formatting Enabled.) 0x0: DISABLED (Continuous Formatting Disabled.)
0	ENFTC	RW	0x1: ENABLED (Formatting enabled.) 0x0: DISABLED (Formatting disabled.)

0x00820308 QDSS_CSTPIU_FSCR**Type:** RW**Reset State:** 0x00000040

QDSS_CSTPIU_FSCR

Bits	Name	Description
11:0	CYCCOUNT	

0x00820400 QDSS_CSTPIU_EXTCTL_IN_PORT**Type:** R**Reset State:** Undefined**QDSS_CSTPIU_EXTCTL_IN_PORT**

Bits	Name	Description
7:0	EXTCTLIN	

0x00820404 QDSS_CSTPIU_EXTCTL_OUT_PORT**Type:** RW**Reset State:** 0x00000000**QDSS_CSTPIU_EXTCTL_OUT_PORT**

Bits	Name	Description
7:0	EXTCTLOUT	

0x00820EE4 QDSS_CSTPIU_ITTRFLINACK**Type:** W**Reset State:** 0x00000000**QDSS_CSTPIU_ITTRFLINACK**

Bits	Name	Description
1	FLUSHINACK	0x1: HIGH (Set the value of FLUSHINACK to 1.) 0x0: LOW (Set the value of FLUSHINACK to 0.)
0	TRIGINACK	0x1: HIGH (Set the value of TRIGINACK to 1.) 0x0: LOW (Set the value of TRIGINACK to 0.)

0x00820EE8 QDSS_CSTPIU_ITTRFLIN**Type:** R**Reset State:** Undefined

QDSS_CSTPIU_ITTRFLIN

Bits	Name	Description
1	FLUSHIN	0x1: HIGH (FLUSHIN is HIGH.) 0x0: LOW (FLUSHIN is LOW.)
0	TRIGIN	0x1: HIGH (TRIGIN is HIGH.) 0x0: LOW (TRIGIN is LOW.)

0x00820EEC QDSS_CSTPIU_ITATBDATA0**Type:** R**Reset State:** Undefined**QDSS_CSTPIU_ITATBDATA0**

Bits	Name	Description
4	ATDATA_31	0x1: HIGH (ATDATAS[31] is 1.) 0x0: LOW (ATDATAS[31] is 0.)
3	ATDATA_23	0x1: HIGH (ATDATAS[23] is 1.) 0x0: LOW (ATDATAS[23] is 0.)
2	ATDATA_15	0x1: HIGH (ATDATAS[15] is 1.) 0x0: LOW (ATDATAS[15] is 0.)
1	ATDATA_7	0x1: HIGH (ATDATAS[7] is 1.) 0x0: LOW (ATDATAS[7] is 0.)
0	ATDATA_0	0x1: HIGH (ATDATAS[0] is 1.) 0x0: LOW (ATDATAS[0] is 0.)

0x00820EF0 QDSS_CSTPIU_ITATBCTR2**Type:** W**Reset State:** 0x00000000**QDSS_CSTPIU_ITATBCTR2**

Bits	Name	Description
1	AFVALID	0x1: HIGH (Set the value of AFVALID to 1.) 0x0: LOW (Set the value of AFVALID to 0.)
0	ATREADY	0x1: HIGH (Set the value of ATREADY to 1.) 0x0: LOW (Set the value of ATREADY to 0.)

0x00820EF4 QDSS_CSTPIU_ITATBCTR1**Type:** R**Reset State:** Undefined**QDSS_CSTPIU_ITATBCTR1**

Bits	Name	Description
6:0	ATID	

0x00820EF8 QDSS_CSTPIU_ITATBCTR0**Type:** R**Reset State:** Undefined**QDSS_CSTPIU_ITATBCTR0**

Bits	Name	Description
9:8	ATBYTES	
1	AFREADY	0x1: HIGH (AFREADY is 1.) 0x0: LOW (AFREADY is 0.)
0	ATVALID	0x1: HIGH (ATVALID is 1.) 0x0: LOW (ATVALID is 0.)

0x00820F00 QDSS_CSTPIU_ITCTRL**Type:** RW**Reset State:** 0x00000000**QDSS_CSTPIU_ITCTRL**

Bits	Name	Description
0	INTEGRATION_MODE	Allows the component to switch from functional mode to integration mode or back. 0x1: INTEGRATION_MODE (Enable integration mode) 0x0: FUNCTIONAL_MODE (Disable integration mode)

0x00820FA0 QDSS_CSTPIU_CLAIMSET**Type:** RW**Reset State:** 0x0000000F

This is used in conjunction with Claim Tag Clear Register, CLAIMCLR. This register forms one half of the Claim Tag value. This location allows individual bits to be set, write, and returns the number of bits that can be set, read.

QDSS_CSTPIU_CLAIMSET

Bits	Name	Type	Description
3:0	CLAIMSET_W	W	Each bit is considered separately: 0 = no effect, 1 = set this bit in the claim tag.
3:0	CLAIMSET_R	R	This claim tag bit is implemented 0xF: CLAIM_TAG_IMPLEMENTED_BITS (These bits are present within the claim tag field)

0x00820FA4 QDSS_CSTPIU_CLAIMCLR

Type: RW

Reset State: 0x00000000

This register is used in conjunction with Claim Tag Set Register, CLAIMSET. This register forms one half of the Claim Tag value. This location enables individual bits to be cleared, write, and returns the current Claim Tag value, read.

QDSS_CSTPIU_CLAIMCLR

Bits	Name	Type	Description
3:0	CLAIMCLR_W	W	Each bit is considered separately: 0 = no effect, 1 = clear this bit in the claim tag.
3:0	CLAIMCLR_R	R	The value present reflects the current setting of the Claim Tag.

0x00820FB0 QDSS_CSTPIU_LAR

Type: W

Reset State: Undefined

QDSS_CSTPIU_LAR

Bits	Name	Description
31:0	ACCESS_W	A write of 0xC5ACCE55 enables further write access to this device. A write of any value other than 0xC5ACCE55 will have the affect of removing write access. 0xC5ACCE55: UNLOCK (Unlock the protection register to enable write access)

0x00820FB4 QDSS_CSTPIU_LSR**Type:** R**Reset State:** 0x00000003**QDSS_CSTPIU_LSR**

Bits	Name	Description
2	LOCKTYPE	Indicates if the Lock Access Register (0xFB0) is implemented as 8-bit or 32-bit 0x0: ENUM_32_BIT (This component implements a 32-bit Lock Access Register)
1	LOCKGRANT	Returns the current status of the Lock. This bit reads as 0 when read from an external debugger (PADDRDBG31 = 1) since external debugger accesses are not subject to Lock Registers. 0x0: ACCESS_PERMITTED (Write access is permitted to this device) 0x1: DEVICE_LOCKED (Write access to the component is blocked. All writes to control registers are ignored. Reads are permitted)
0	LOCKEXIST	Indicates that a lock control mechanism exists for this device. This bit reads as 0 when read from an external debugger (PADDRDBG31 = 1) since external debugger accesses are not subject to Lock Registers. 0x0: LOCK_NOT_PRESENT (No lock control mechanism exists, writes to the Lock Access register (0xFB0) will be ignored) 0x1: LOCK_PRESENT (Lock control mechanism is present)

0x00820FC8 QDSS_CSTPIU_DEVID**Type:** R**Reset State:** 0x000000A0

This register indicates the capabilities of the TPIU.

QDSS_CSTPIU_DEVID

Bits	Name	Description
11	SWOUARTNRZ	Indicates whether Serial Wire Output (UART/NRZ) is supported. 0x0: ENUM_0 (Serial Wire Output (UART/NRZ) not supported.)
10	SWOMAN	Indicates whether Serial Wire Output (Manchester) is supported. 0x0: ENUM_0 (Serial Wire Output (Manchester) not supported.)
9	TCLKDATA	Indicates whether trace clock + data is supported. 0x0: ENUM_0 (Trace clock + data is supported.)
8:6	FIFOSIZE	FIFO size in powers of 2. 0x2: ENUM_2 (FIFO size of 4 entries (16 bytes).)

QDSS_CSTPIU_DEVID (cont.)

Bits	Name	Description
5	CLKRELAT	Indicates the relationship between ATCLK and TRACECLKIN. 0x1: ASYNC (ATCLK and TRACECLKIN are asynchronous.)
4:0	MUXNUM	

0x00820FE0 QDSS_CSTPIU_PERIPHID0**Type:** R**Reset State:** 0x00000012

Part of the set of Peripheral Identification registers. Contains part of the designer specific part number.

QDSS_CSTPIU_PERIPHID0

Bits	Name	Description
7:0	PART_0	Bits [7:0] of the component's part number. This is selected by the designer of the component. 0x12: TPIU_PART_NUMBER_7_0 (Lowest 8 bits of the Part Number (0x912))

0x00820FE4 QDSS_CSTPIU_PERIPHID1**Type:** R**Reset State:** 0x000000B9

Part of the set of Peripheral Identification registers. Contains part of the designer specific part number and part of the designer identity.

QDSS_CSTPIU_PERIPHID1

Bits	Name	Description
7:4	DES_0	Bits 3:0 of the JEDEC identity code indicating the designer of the component (along with the continuation code) 0xB: ARM_JEP106_IDENTITY_CODE_3_0 (Lowest 4 bits of the JEP106 Identity Code)
3:0	PART_1	Bits [11:8] of the component's part number. This is selected by the designer of the component. 0x9: TPIU_PART_NUMBER_11_8 (Upper 4 bits of the Part Number (0x912))

0x00820FE8 QDSS_CSTPIU_PERIPHID2**Type:** R**Reset State:** 0x0000004B

Part of the set of Peripheral Identification registers. Contains part of the designer identity and the product revision.

QDSS_CSTPIU_PERIPHID2

Bits	Name	Description
7:4	REVISION	The Revision field is an incremental value starting at 0x0 for the first design of this component. This only increases by 1 for both major and minor revisions and is simply used as a look-up to establish the exact major/minor revision. 0x4: R0P4 (This device is at r0p4)
3	JEDEC	Always set. Indicates that a JEDEC assigned value is used 0x1: JEDEC_IDENTITY (The designer ID is specified by JEDEC (http://www.jedec.org))
2:0	DES_1	Bits 6:4 of the JEDEC identity code indicating the designer of the component (along with the continuation code) 0x3: ARM_JEP106_IDENTITY_CODE_6_4 (Upper 3 bits of the JEP106 Identity Code)

0x00820FEC QDSS_CSTPIU_PERIPHID3**Type:** R**Reset State:** 0x00000000

Part of the set of Peripheral Identification registers. Contains the RevAnd and Customer Modified fields.

QDSS_CSTPIU_PERIPHID3

Bits	Name	Description
7:4	REVAND	This field indicates minor errata fixes specific to this design, for example metal fixes after implementation. In most cases this field is zero. It is recommended that component designers ensure this field can be changed by a metal fix if required, for example by driving it from registers that reset to zero. 0x0: NO_MODIFICATION (Indicates that there have been no metal fixes to this component)
3:0	CMOD	Where the component is reusable IP, this value indicates if the customer has modified the behavior of the component. In most cases this field is zero. 0x0: NO_MODIFICATION (Indicates that there have been no modifications made)

0x00820FD0 QDSS_CSTPIU_PERIPHID4**Type:** R**Reset State:** 0x00000004

Part of the set of Peripheral Identification registers. Contains part of the designer identity and the memory footprint indicator.

QDSS_CSTPIU_PERIPHID4

Bits	Name	Description
7:4	SIZE	This is a 4-bit value that indicates the total contiguous size of the memory window used by this component in powers of 2 from the standard 4KB. If a component only requires the standard 4KB then this should read as 0x0, 4KB only, for 8KB set to 0x1, 16KB == 0x2, 32KB == 0x3, and so on. 0x0: ENUM_4KB (Indicates that the device only occupies 4kB of memory)
3:0	DES_2	JEDEC continuation code indicating the designer of the component (along with the identity code) 0x4: ARM_JEP106_CONTINUATION_CODE (Indicates that ARM's JEDEC identity code is on the 5th bank)

0x00820FD4 QDSS_CSTPIU_PERIPHID5**Type:** RW**Reset State:** Undefined

Reserved

QDSS_CSTPIU_PERIPHID5

Bits	Name	Description
31:0	PERIPHID5	

0x00820FD8 QDSS_CSTPIU_PERIPHID6**Type:** RW**Reset State:** Undefined

Reserved

QDSS_CSTPIU_PERIPHID6

Bits	Name	Description
31:0	PERIPHID6	

0x00820FDC QDSS_CSTPIU_PERIPHID7**Type:** RW**Reset State:** Undefined

Reserved

QDSS_CSTPIU_PERIPHID7

Bits	Name	Description
31:0	PERIPHID7	

0x00820FF0 QDSS_CSTPIU_COMPID0**Type:** R**Reset State:** 0x0000000D

A component identification register, that indicates that the identification registers are present.

QDSS_CSTPIU_COMPID0

Bits	Name	Description
7:0	PRMBL_0	Contains bits [7:0] of the component identification 0xD: ENUM_0D (Identification value)

0x00820FF4 QDSS_CSTPIU_COMPID1**Type:** R**Reset State:** 0x00000090

A component identification register, that indicates that the identification registers are present. This register also indicates the component class.

QDSS_CSTPIU_COMPID1

Bits	Name	Description
7:4	CLASS	Class of the component. E.g. ROM table, CoreSight component etc. Constitutes bits [15:12] of the component identification. 0x9: CORESIGHT_COMPONENT (Indicates the component is a CoreSight component)
3:0	PRMBL_1	Contains bits [11:8] of the component identification 0x0: ENUM_0 (Identification value)

0x00820FF8 QDSS_CSTPIU_COMPID2**Type:** R**Reset State:** 0x00000005

A component identification register, that indicates that the identification registers are present.

QDSS_CSTPIU_COMPID2

Bits	Name	Description
7:0	PRMBL_2	Contains bits [23:16] of the component identification 0x5: ENUM_05 (Identification value)

0x00820FFC QDSS_CSTPIU_COMPID3**Type:** R**Reset State:** 0x000000B1

A component identification register, that indicates that the identification registers are present.

QDSS_CSTPIU_COMPID3

Bits	Name	Description
7:0	PRMBL_3	Contains bits [31:24] of the component identification 0xB1: B1 (Identification value)

12.1.14 IN_FUN0_CXATBFUNNEL_128W8SP

The Funnel is used when there is more than one trace source.

12.1.14.1 QDSS_IN_FUN0_CXATBFUNNEL_128W8SP

The Funnel is used when there is more than one trace source.

0x00821000 QDSS_IN_FUN0_CTRL_REG**Type:** RW**Reset State:** 0x00000300

The Funnel Control Register enables the slave ports and defines the hold time of the slave ports. Hold time refers to the number of transactions that are output on the funnel master port from the same slave while that slave port atvalidsx is HIGH. Hold time does not refer to clock cycles in this context

QDSS_IN_FUN0_CTRL_REG

Bits	Name	Description
11:8	HT	The formatting scheme can easily become inefficient if fast switching occurs, so, where possible, this must be minimized. If a source has nothing to transmit, then another source is selected irrespective of the minimum number of transactions. The maximum value that can be entered is 0xE and this equates to 15 cycles. 0xF is reserved. 0x0: HOLD_TIME_1 (1 transactions hold time) 0x1: HOLD_TIME_2 (2 transactions hold time) 0x2: HOLD_TIME_3 (3 transactions hold time) 0x3: HOLD_TIME_4 (4 transactions hold time) 0x4: HOLD_TIME_5 (5 transactions hold time) 0x5: HOLD_TIME_6 (6 transactions hold time) 0x6: HOLD_TIME_7 (7 transactions hold time) 0x7: HOLD_TIME_8 (8 transactions hold time) 0x8: HOLD_TIME_9 (9 transactions hold time) 0x9: HOLD_TIME_10 (10 transactions hold time) 0xA: HOLD_TIME_11 (11 transactions hold time) 0xB: HOLD_TIME_12 (12 transactions hold time) 0xC: HOLD_TIME_13 (13 transactions hold time) 0xD: HOLD_TIME_14 (14 transactions hold time) 0xE: HOLD_TIME_15 (15 transactions hold time)
7	ENS7	Setting this bit enables this input, or slave, port. If the bit is not set then this has the effect of excluding the port from the priority selection scheme. The reset value is all clear, that is, all ports disabled. 0x1: ENABLE (Slave port enabled) 0x0: DISABLE (Slave port disabled)
6	ENS6	Setting this bit enables this input, or slave, port. If the bit is not set then this has the effect of excluding the port from the priority selection scheme. The reset value is all clear, that is, all ports disabled. 0x1: ENABLE (Slave port enabled) 0x0: DISABLE (Slave port disabled)
5	ENS5	Setting this bit enables this input, or slave, port. If the bit is not set then this has the effect of excluding the port from the priority selection scheme. The reset value is all clear, that is, all ports disabled. 0x1: ENABLE (Slave port enabled) 0x0: DISABLE (Slave port disabled)
4	ENS4	Setting this bit enables this input, or slave, port. If the bit is not set then this has the effect of excluding the port from the priority selection scheme. The reset value is all clear, that is, all ports disabled. 0x1: ENABLE (Slave port enabled) 0x0: DISABLE (Slave port disabled)

QDSS_IN_FUN0_CTRL_REG (cont.)

Bits	Name	Description
3	ENS3	Setting this bit enables this input, or slave, port. If the bit is not set then this has the effect of excluding the port from the priority selection scheme. The reset value is all clear, that is, all ports disabled. 0x1: ENABLE (Slave port enabled) 0x0: DISABLE (Slave port disabled)
2	ENS2	Setting this bit enables this input, or slave, port. If the bit is not set then this has the effect of excluding the port from the priority selection scheme. The reset value is all clear, that is, all ports disabled. 0x1: ENABLE (Slave port enabled) 0x0: DISABLE (Slave port disabled)
1	ENS1	Setting this bit enables this input, or slave, port. If the bit is not set then this has the effect of excluding the port from the priority selection scheme. The reset value is all clear, that is, all ports disabled. 0x1: ENABLE (Slave port enabled) 0x0: DISABLE (Slave port disabled)
0	ENS0	Setting this bit enables this input, or slave, port. If the bit is not set then this has the effect of excluding the port from the priority selection scheme. The reset value is all clear, that is, all ports disabled. 0x1: ENABLE (Slave port enabled) 0x0: DISABLE (Slave port disabled)

0x00821004 QDSS_IN_FUN0_PRIORITY_CTRL_REG**Type:** RW**Reset State:** 0x00000000

The Priority Control Register defines the order in which inputs are selected. Each 3-bit field represents a priority for each particular slave interface. Location 0 has the priority value for the first slave port. Location 1 is the priority value for the second slave port, Location 2 is the third, down to location 7, which has the priority value of the eighth slave port. Values represent the priority value for each port number. If you want to give highest priority to a particular slave port, the corresponding port must be programmed with the lowest value. Typically this is likely to be a port that has more important data or that has a small FIFO and is therefore likely to overflow.

If you want to give lowest priority to a particular slave port, the corresponding slave port must be programmed with the highest value. Typically this is likely to be a device that has a large FIFO that is less likely to overflow or a source that has information that is of lower importance.

QDSS_IN_FUN0_PRIORITY_CTRL_REG

Bits	Name	Description
23:21	PRIPORT7	Priority value of the eighth port. The value written into this location is the value that you want to assign the eighth slave port. If a priority value has been entered for multiple different slave ports then the arbitration logic selects the lowest port number of them. 0x0: PRORITY_LEVEL_0 (This value represents the highest priority.) 0x1: PRORITY_LEVEL_1 (2nd highest priority) 0x2: PRORITY_LEVEL_2 (3rd highest priority) 0x3: PRORITY_LEVEL_3 (4th highest priority) 0x4: PRORITY_LEVEL_4 (5th highest priority) 0x5: PRORITY_LEVEL_5 (6th highest priority) 0x6: PRORITY_LEVEL_6 (7th highest priority) 0x7: PRORITY_LEVEL_7 (This value represents the lowest priority.)
20:18	PRIPORT6	Priority value of the 7th port. 0x0: PRORITY_LEVEL_0 (This value represents the highest priority.) 0x1: PRORITY_LEVEL_1 (2nd highest priority) 0x2: PRORITY_LEVEL_2 (3rd highest priority) 0x3: PRORITY_LEVEL_3 (4th highest priority) 0x4: PRORITY_LEVEL_4 (5th highest priority) 0x5: PRORITY_LEVEL_5 (6th highest priority) 0x6: PRORITY_LEVEL_6 (7th highest priority) 0x7: PRORITY_LEVEL_7 (This value represents the lowest priority.)
17:15	PRIPORT5	Priority value of the 6th port. 0x0: PRORITY_LEVEL_0 (This value represents the highest priority.) 0x1: PRORITY_LEVEL_1 (2nd highest priority) 0x2: PRORITY_LEVEL_2 (3rd highest priority) 0x3: PRORITY_LEVEL_3 (4th highest priority) 0x4: PRORITY_LEVEL_4 (5th highest priority) 0x5: PRORITY_LEVEL_5 (6th highest priority) 0x6: PRORITY_LEVEL_6 (7th highest priority) 0x7: PRORITY_LEVEL_7 (This value represents the lowest priority.)

QDSS_IN_FUN0_PRIORITY_CTRL_REG (cont.)

Bits	Name	Description
14:12	PRIPORT4	Priority value of the 5th port. 0x0: PRORITY_LEVEL_0 (This value represents the highest priority.) 0x1: PRORITY_LEVEL_1 (2nd highest priority) 0x2: PRORITY_LEVEL_2 (3rd highest priority) 0x3: PRORITY_LEVEL_3 (4th highest priority) 0x4: PRORITY_LEVEL_4 (5th highest priority) 0x5: PRORITY_LEVEL_5 (6th highest priority) 0x6: PRORITY_LEVEL_6 (7th highest priority) 0x7: PRORITY_LEVEL_7 (This value represents the lowest priority.)
11:9	PRIPORT3	Priority value of the 4th port. 0x0: PRORITY_LEVEL_0 (This value represents the highest priority.) 0x1: PRORITY_LEVEL_1 (2nd highest priority) 0x2: PRORITY_LEVEL_2 (3rd highest priority) 0x3: PRORITY_LEVEL_3 (4th highest priority) 0x4: PRORITY_LEVEL_4 (5th highest priority) 0x5: PRORITY_LEVEL_5 (6th highest priority) 0x6: PRORITY_LEVEL_6 (7th highest priority) 0x7: PRORITY_LEVEL_7 (This value represents the lowest priority.)
8:6	PRIPORT2	Priority value of the 3rd port. 0x0: PRORITY_LEVEL_0 (This value represents the highest priority.) 0x1: PRORITY_LEVEL_1 (2nd highest priority) 0x2: PRORITY_LEVEL_2 (3rd highest priority) 0x3: PRORITY_LEVEL_3 (4th highest priority) 0x4: PRORITY_LEVEL_4 (5th highest priority) 0x5: PRORITY_LEVEL_5 (6th highest priority) 0x6: PRORITY_LEVEL_6 (7th highest priority) 0x7: PRORITY_LEVEL_7 (This value represents the lowest priority.)
5:3	PRIPORT1	Priority value of the 2nd port. 0x0: PRORITY_LEVEL_0 (This value represents the highest priority.) 0x1: PRORITY_LEVEL_1 (2nd highest priority) 0x2: PRORITY_LEVEL_2 (3rd highest priority) 0x3: PRORITY_LEVEL_3 (4th highest priority) 0x4: PRORITY_LEVEL_4 (5th highest priority) 0x5: PRORITY_LEVEL_5 (6th highest priority) 0x6: PRORITY_LEVEL_6 (7th highest priority) 0x7: PRORITY_LEVEL_7 (This value represents the lowest priority.)

QDSS_IN_FUN0_PRIORITY_CTRL_REG (cont.)

Bits	Name	Description
2:0	PRIPORT0	Priority value of the first slave port. The value written into this location is the value that you want to assign the first slave port. 0x0: PRORITY_LEVEL_0 (This value represents the highest priority.) 0x1: PRORITY_LEVEL_1 (2nd highest priority) 0x2: PRORITY_LEVEL_2 (3rd highest priority) 0x3: PRORITY_LEVEL_3 (4th highest priority) 0x4: PRORITY_LEVEL_4 (5th highest priority) 0x5: PRORITY_LEVEL_5 (6th highest priority) 0x6: PRORITY_LEVEL_6 (7th highest priority) 0x7: PRORITY_LEVEL_7 (This value represents the lowest priority.)

0x00821F00 QDSS_IN_FUN0_ITCTRL**Type:** RW**Reset State:** 0x00000000

This register is used to enable topology detection. This register enables the component to switch from a functional mode, the default behavior, to integration mode where the inputs and outputs of the component can be directly controlled for the purpose of integration testing and topology solving

Note

When a device has been in integration mode, it might not function with the original behavior. After performing integration or topology detection, you must reset the system to ensure correct behavior of CoreSight and other connected system components that are affected by the integration or topology detection.

QDSS_IN_FUN0_ITCTRL

Bits	Name	Description
0	INTEGRATION_MODE	Allows the component to switch from functional mode to integration mode or back. If no integration functionality is implemented, this register must read as zero. 0x1: INTEGRATION_MODE (Enable integration mode) 0x0: FUNCTIONAL_MODE (Disable integration mode)

0x00821FA0 QDSS_IN_FUN0_CLAIMSET**Type:** RW**Reset State:** 0x0000000F

This is used in conjunction with Claim Tag Clear Register, CLAIMCLR. This register forms one half of the Claim Tag value. This location allows individual bits to be set, write, and returns the number of bits that can be set, read.

QDSS_IN_FUN0_CLAIMSET

Bits	Name	Type	Description
3:0	CLAIMSET_W	W	Each bit is considered separately: 0 = no effect, 1 = set this bit in the claim tag.
3:0	CLAIMSET_R	R	This claim tag bit is implemented 0xF: CLAIM_TAG_IMPLEMENTED_BITS (These bits are present within the claim tag field)

0x00821FA4 QDSS_IN_FUN0_CLAIMCLR

Type: RW

Reset State: 0x00000000

This register is used in conjunction with Claim Tag Set Register, CLAIMSET. This register forms one half of the Claim Tag value. This location enables individual bits to be cleared, write, and returns the current Claim Tag value, read.

QDSS_IN_FUN0_CLAIMCLR

Bits	Name	Type	Description
3:0	CLAIMCLR_W	W	Each bit is considered separately: 0 = no effect, 1 = clear this bit in the claim tag.
3:0	CLAIMCLR_R	R	The value present reflects the current setting of the Claim Tag.

0x00821FB0 QDSS_IN_FUN0_LOCKACCESS

Type: W

Reset State: Undefined

This is used to enable write access to device registers.

QDSS_IN_FUN0_LOCKACCESS

Bits	Name	Description
31:0	ACCESS_W	A write of 0xC5ACCE55 enables further write access to this device. An invalid write will have the affect of removing write access. 0xC5ACCE55: UNLOCK (Unlock the protection register to enable write access) 0xDEADBEEF: LOCK (invalid write value to enable the protection register and block write access)

0x00821FC8 QDSS_IN_FUN0_DEVID**Type:** R**Reset State:** 0x00000038

This indicates the capabilities of the CoreSight Funnel.

QDSS_IN_FUN0_DEVID

Bits	Name	Description
7:4	SCHEME	This value indicates the priority scheme implemented within the Funnel 0x3: STATIC_PRIORITY (Slave ports may be programmed to have higher or lower priority with respect to each other)
3:0	RESERVED	Reserved

0x00821FE0 QDSS_IN_FUN0_PERIPHID0**Type:** R**Reset State:** 0x00000008

Part of the set of Peripheral Identification registers. Contains part of the designer specific part number.

QDSS_IN_FUN0_PERIPHID0

Bits	Name	Description
7:0	PART_NUMBER_BITS7TO0	Bits [7:0] of the component's part number. This is selected by the designer of the component. 0x8: CORESIGHT_FUNNEL_PART_NUMBER_7_0 (Lowest 8 bits of the Part Number (0x100))

0x00821FE4 QDSS_IN_FUN0_PERIPHID1**Type:** R**Reset State:** 0x000000B9

Part of the set of Peripheral Identification registers. Contains part of the designer specific part number and part of the designer identity.

QDSS_IN_FUN0_PERIPHID1

Bits	Name	Description
7:4	JEP106_BITS3TO0	Bits 3:0 of the JEDEC identity code indicating the designer of the component (along with the continuation code) 0xB: ARM_JEP106_IDENTITY_CODE_7_0 (Lowest 4 bits of the JEP106 Identity Code)
3:0	PART_NUMBER_BITS11TO8	Bits [11:8] of the component's part number. This is selected by the designer of the component. 0x9: CORESIGHT_FUNNEL_PART_NUMBER_11_8 (Upper 4 bits of the Part Number (0x100))

0x00821FE8 QDSS_IN_FUN0_PERIPHID2**Type:** R**Reset State:** 0x0000002B

Part of the set of Peripheral Identification registers. Contains part of the designer identity and the product revision.

QDSS_IN_FUN0_PERIPHID2

Bits	Name	Description
7:4	REVISION	The Revision field is an incremental value starting at 0x0 for the first design of this component. This only increases by 1 for both major and minor revisions and is simply used as a look-up to establish the exact major/minor revision. 0x2: R2P0 (This device is at r2p0)
3	JEDEC	Always set. Indicates that a JEDEC assigned value is used 0x1: JEDEC_IDENTITY (The designer ID is specified by JEDEC (http://www.jedec.org))
2:0	JEP106_BITS6TO4	Bits 6:4 of the JEDEC identity code indicating the designer of the component (along with the continuation code) 0x3: ARM_JEP106_IDENTITY_CODE_6_4 (Upper 3 bits of the JEP106 Identity Code)

0x00821FEC QDSS_IN_FUN0_PERIPHID3**Type:** R**Reset State:** 0x00000000

Part of the set of Peripheral Identification registers. Contains the RevAnd and Customer Modified fields.

QDSS_IN_FUN0_PERIPHID3

Bits	Name	Description
7:4	REVAND	This field indicates minor errata fixes specific to this design, for example metal fixes after implementation. In most cases this field is zero. It is recommended that component designers ensure this field can be changed by a metal fix if required, for example by driving it from registers that reset to zero. 0x0: NO_MODS (Indicates that there have been no metal fixes to this component)
3:0	CUSTOMER_MODIFIED	Where the component is reusable IP, this value indicates if the customer has modified the behavior of the component. In most cases this field is zero. 0x0: NO_MODS (Indicates that there have been no modifications made)

0x00821FD0 QDSS_IN_FUN0_PERIPHID4**Type:** R**Reset State:** 0x00000004

Part of the set of Peripheral Identification registers. Contains part of the designer identity and the memory footprint indicator.

QDSS_IN_FUN0_PERIPHID4

Bits	Name	Description
7:4	FOURKB_COUNT	This is a 4-bit value that indicates the total contiguous size of the memory window used by this component in powers of 2 from the standard 4KB. If a component only requires the standard 4KB then this should read as 0x0, 4KB only, for 8KB set to 0x1, 16KB == 0x2, 32KB == 0x3, and so on. 0x0: ENUM_4KB (Indicates that the device only occupies 4kB of memory)
3:0	JEP106_CONT	JEDEC continuation code indicating the designer of the component (along with the identity code) 0x4: ARM_JEP106_CONTINUATION_CODE (Indicates that ARM's JEDEC identity code is on the 5th bank)

0x00821FD4 QDSS_IN_FUN0_PERIPHID5**Type:** RW**Reset State:** Undefined

Reserved

QDSS_IN_FUN0_PERIPHID5

Bits	Name	Description
31:0	PERIPHID5	

0x00821FD8 QDSS_IN_FUN0_PERIPHID6**Type:** RW**Reset State:** Undefined

Reserved

QDSS_IN_FUN0_PERIPHID6

Bits	Name	Description
31:0	PERIPHID6	

0x00821FDC QDSS_IN_FUN0_PERIPHID7**Type:** RW**Reset State:** Undefined

Reserved

QDSS_IN_FUN0_PERIPHID7

Bits	Name	Description
31:0	PERIPHID7	

0x00821FF0 QDSS_IN_FUN0_COMPID0**Type:** R**Reset State:** 0x0000000D

A component identification register, that indicates that the identification registers are present.

QDSS_IN_FUN0_COMPID0

Bits	Name	Description
7:0	PREAMBLE	Contains bits [24:31] of the component identification 0xD: ENUM_0D (Identification value)

0x00821FF4 QDSS_IN_FUN0_COMPID1**Type:** R**Reset State:** 0x00000090

A component identification register, that indicates that the identification registers are present. This register also indicates the component class.

QDSS_IN_FUN0_COMPID1

Bits	Name	Description
7:4	CLASS	Class of the component. E.g. ROM table, CoreSight component etc. 0x9: CORESIGHT_COMPONENT (Indicates the component is a CoreSight component)
3:0	PREAMBLE	Contains bits [19:16] of the component identification 0x0: ENUM_0 (Identification value)

0x00821FF8 QDSS_IN_FUN0_COMPID2**Type:** R**Reset State:** 0x00000005

A component identification register, that indicates that the identification registers are present.

QDSS_IN_FUN0_COMPID2

Bits	Name	Description
7:0	PREAMBLE	Contains bits [15:8] of the component identification 0x5: ENUM_05 (Identification value)

0x00821FFC QDSS_IN_FUN0_COMPID3**Type:** R**Reset State:** 0x000000B1

A component identification register, that indicates that the identification registers are present.

QDSS_IN_FUN0_COMPID3

Bits	Name	Description
7:0	PREAMBLE	Contains bits [7:0] of the component identification 0xB1: B1 (Identification value)

12.1.15 REPL64_CXATBREPLICATOR_64WP

Replicator component forks ATB data stream into multiple streams

12.1.15.1 QDSS_REPL64_CXATBREPLICATOR_64WP

Replicator component forks ATB data stream into multiple streams

0x00824000 QDSS_REPL64_IDFILTER0

Type: RW

Reset State: 0x00000000

Allows the setting of ID filter for Master 0.

QDSS_REPL64_IDFILTER0

Bits	Name	Description
7	ID0_70_7F	Enable or disable ID filtering for id's 0x70 to 0x7F 0x1: HIGH (Enable filtering.) 0x0: LOW (Disable filtering.)
6	ID0_60_6F	Enable or disable ID filtering for id's 0x60 to 0x6F 0x1: HIGH (Enable filtering.) 0x0: LOW (Disable filtering.)
5	ID0_50_5F	Enable or disable ID filtering for id's 0x50 to 0x5F 0x1: HIGH (Enable filtering.) 0x0: LOW (Disable filtering.)
4	ID0_40_4F	Enable or disable ID filtering for id's 0x40 to 0x4F 0x1: HIGH (Enable filtering.) 0x0: LOW (Disable filtering.)
3	ID0_30_3F	Enable or disable ID filtering for id's 0x30 to 0x3F 0x1: HIGH (Enable filtering.) 0x0: LOW (Disable filtering.)
2	ID0_20_2F	Enable or disable ID filtering for id's 0x20 to 0x2F 0x1: HIGH (Enable filtering.) 0x0: LOW (Disable filtering.)

QDSS_REPL64_IDFILTER0 (cont.)

Bits	Name	Description
1	ID0_10_1F	Enable or disable ID filtering for id's 0x10 to 0x1F 0x1: HIGH (Enable filtering.) 0x0: LOW (Disable filtering.)
0	ID0_0_F	Enable or disable ID filtering for id's 0x0 to 0xF 0x1: HIGH (Enable filtering.) 0x0: LOW (Disable filtering.)

0x00824004 QDSS_REPL64_IDFILTER1**Type:** RW**Reset State:** 0x00000000

Allows the setting of ID filter for Master 1.

QDSS_REPL64_IDFILTER1

Bits	Name	Description
7	ID1_70_7F	Enable or disable ID filtering for id's 0x70 to 0x7F 0x1: HIGH (Enable filtering.) 0x0: LOW (Disable filtering.)
6	ID1_60_6F	Enable or disable ID filtering for id's 0x60 to 0x6F 0x1: HIGH (Enable filtering.) 0x0: LOW (Disable filtering.)
5	ID1_50_5F	Enable or disable ID filtering for id's 0x50 to 0x5F 0x1: HIGH (Enable filtering.) 0x0: LOW (Disable filtering.)
4	ID1_40_4F	Enable or disable ID filtering for id's 0x40 to 0x4F 0x1: HIGH (Enable filtering.) 0x0: LOW (Disable filtering.)
3	ID1_30_3F	Enable or disable ID filtering for id's 0x30 to 0x3F 0x1: HIGH (Enable filtering.) 0x0: LOW (Disable filtering.)
2	ID1_20_2F	Enable or disable ID filtering for id's 0x20 to 0x2F 0x1: HIGH (Enable filtering.) 0x0: LOW (Disable filtering.)
1	ID1_10_1F	Enable or disable ID filtering for id's 0x10 to 0x1F 0x1: HIGH (Enable filtering.) 0x0: LOW (Disable filtering.)
0	ID1_0_F	Enable or disable ID filtering for id's 0x0 to 0xF 0x1: HIGH (Enable filtering.) 0x0: LOW (Disable filtering.)

0x00824EFC QDSS_REPL64_ITATBCTR0**Type:** W**Reset State:** Undefined

Controls the value of the ATVALIDM0, ATVALIDM1 and ATREADYDYS outputs in integration mode.

QDSS_REPL64_ITATBCTR0

Bits	Name	Description
4	ATREADYDYS_W	Sets the value of the ATREADYDYS output: 0x1: HIGH (Drive logic 1 on the ATREADYDYS output.) 0x0: LOW (Drive logic 0 on the ATREADYDYS output.)
2	ATVALIDM1_W	Sets the value of the ATVALIDM1 output: 0x1: HIGH (Drive logic 1 on the ATVALIDM1 output.) 0x0: LOW (Drive logic 0 on the ATVALIDM1 output.)
0	ATVALIDM0_W	Sets the value of the ATVALIDM0 output: 0x1: HIGH (Drive logic 1 on the ATVALIDM0 output.) 0x0: LOW (Drive logic 0 on the ATVALIDM0 output.)

0x00824EF8 QDSS_REPL64_ITATBCTR1**Type:** R**Reset State:** Undefined

Returns the value of the ATREADYM0, ATREADYM1 and ATVALIDDS inputs in integration mode.

QDSS_REPL64_ITATBCTR1

Bits	Name	Description
3	ATVALIDDS_R	Reads the value of the ATVALIDDS input: 0x1: HIGH (Pin is at logic 1.) 0x0: LOW (Pin is at logic 0.)
1	ATREADYM1_R	Reads the value of the ATREADYM1 input: 0x1: HIGH (Pin is at logic 1.) 0x0: LOW (Pin is at logic 0.)
0	ATREADYM0_R	Reads the value of the ATREADYM0 input: 0x1: HIGH (Pin is at logic 1.) 0x0: LOW (Pin is at logic 0.)

0x00824F00 QDSS_REPL64_ITCTRL**Type:** RW**Reset State:** 0x00000000**QDSS_REPL64_ITCTRL**

Bits	Name	Description
0	INTEGRATION_MODE	Enables the component to switch from functional mode to integration mode and back. If no integration functionality is implemented, this register must read as zero. 0x1: INTEGRATION_MODE (Enable integration mode.) 0x0: FUNCTIONAL_MODE (Disable integration mode.)

0x00824FA0 QDSS_REPL64_CLAIMSET**Type:** RW**Reset State:** 0x0000000F**QDSS_REPL64_CLAIMSET**

Bits	Name	Type	Description
3:0	CLAIMSET_W	W	Each bit is considered separately: 0 = no effect, 1 = set this bit in the claim tag.
3:0	CLAIMSET_R	R	This claim tag bit is implemented 0xF: CLAIM_TAG_IMPLEMENTED_BITS (These bits are present within the claim tag field)

0x00824FA4 QDSS_REPL64_CLAIMCLR**Type:** RW**Reset State:** 0x00000000**QDSS_REPL64_CLAIMCLR**

Bits	Name	Type	Description
3:0	CLAIMCLR_W	W	Each bit is considered separately: 0 = no effect, 1 = clear this bit in the claim tag.
3:0	CLAIMCLR_R	R	The value present reflects the current setting of the Claim Tag.

0x00824FB0 QDSS_REPL64_LAR**Type:** W**Reset State:** Undefined**QDSS_REPL64_LAR**

Bits	Name	Description
31:0	ACCESS_W	A write of 0xC5ACCE55 enables further write access to this device. An invalid write has the affect of removing write access. 0xC5ACCE55: UNLOCK (unlock the protection register to enable write access)

0x00824FB4 QDSS_REPL64_LSR**Type:** R**Reset State:** 0x00000003**QDSS_REPL64_LSR**

Bits	Name	Description
2	LOCKTYPE	Indicates if the Lock Access Register, 0xFB0, is implemented as 8-bit or 32-bit. 0x0: ENUM_32_BIT (This component implements a 32-bit Lock Access Register.)
1	LOCKGRANT	Returns the current status of the Lock. 0x0: ACCESS_PERMITTED (Write access is allowed to this device.) 0x1: DEVICE_LOCKED (Write access to the component is blocked. All writes to control registers are ignored. Reads are permitted.)
0	LOCKEXIST	Indicates that a lock control mechanism exists for this device. 0x0: LOCK_NOT_PRESENT (No lock control mechanism exists, writes to the Lock Access Register, 0xFB0, are ignored.) 0x1: LOCK_PRESENT (Lock control mechanism is present.)

0x00824FB8 QDSS_REPL64_AUTHSTATUS**Type:** R**Reset State:** 0x00000000

Reports the required **security** level and current status of those enables. Where functionality changes on a given **security** level then this change in status must be reported in this register

QDSS_REPL64_AUTHSTATUS

Bits	Name	Description
7:6	SNID	Indicates the security level for secure non-invasive debug 0x0: ENUM_2_B00 (Functionality not implemented or controlled elsewhere)
5:4	SID	Indicates the security level for secure invasive debug 0x0: ENUM_2_B00 (Functionality not implemented or controlled elsewhere)
3:2	NSNID	Indicates the security level for non-secure non-invasive debug 0x0: ENUM_2_B00 (Functionality not implemented or controlled elsewhere)
1:0	NSID	Indicates the security level for non-secure invasive debug 0x0: ENUM_2_B00 (Functionality not implemented or controlled elsewhere)

0x00824FC8 QDSS_REPL64_DEVID**Type:** R**Reset State:** 0x00000002**QDSS_REPL64_DEVID**

Bits	Name	Description
3:0	PORTNUM	This value indicates the number of master ports implemented. 0x2: NUMBER_OF_PORTS (2 master ports implemented.)

0x00824FCC QDSS_REPL64_DEVTYPE**Type:** R**Reset State:** 0x00000022**QDSS_REPL64_DEVTYPE**

Bits	Name	Description
7:4	SUB_TYPE	Sub-classification within the major category: 0x2: CORESIGHT_REPLICATOR (This component replicates traces from single source to multiple targets.)
3:0	RESERVED	Reserved

0x00824FE0 QDSS_REPL64_PIDR0**Type:** R**Reset State:** 0x00000009**QDSS_REPL64_PIDR0**

Bits	Name	Description
7:0	PART_NUMBER_BITS7TO0	Bits [7:0] of the component part number. This is selected by the designer of the component. 0x9: CORESIGHT_REPLICATOR_PART_NUMBER_7_0 (Lowest 8 bits of the Part Number, 0x909.)

0x00824FE4 QDSS_REPL64_PIDR1**Type:** R**Reset State:** 0x000000B9**QDSS_REPL64_PIDR1**

Bits	Name	Description
7:4	JEP106_BITS3TO0	Bits [3:0] of the JEDEC identity code indicating the designer of the component, together with the continuation code. 0xB: ARM_JEP106_IDENTITY_CODE_7_0 (Lowest 4 bits of the JEP106 Identity Code.)
3:0	PART_NUMBER_BITS11TO8	Bits [11:8] of the component part number. This is selected by the designer of the component. 0x9: CORESIGHT_REPLICATOR_PART_NUMBER_11_8 (Upper 4 bits of the Part Number, 0x907.)

0x00824FE8 QDSS_REPL64_PIDR2**Type:** R**Reset State:** 0x0000001B**QDSS_REPL64_PIDR2**

Bits	Name	Description
7:4	REVISION	The Revision field is an incremental value starting at 0x0 for the first design of this component. This only increases by 1 for both major and minor revisions and is used as a look-up to establish the exact major and minor revision. 0x1: R1P0 (This device is at r1p0.)
3	JEDEC	Always set. Indicates that a JEDEC assigned value is used. 0x1: JEDEC_IDENTITY (The designer ID is specified by JEDEC (http://www.jedec.org).)

QDSS_REPL64_PIDR2 (cont.)

Bits	Name	Description
2:0	JEP106_BITS6TO4	Bits [6:4] of the JEDEC identity code indicating the designer of the component, together with the continuation code. 0x3: ARM_JEP106_IDENTITY_CODE_6_4 (Upper 3 bits of the JEP106 Identity Code.)

0x00824FEC QDSS_REPL64_PIDR3**Type:** R**Reset State:** 0x00000000**QDSS_REPL64_PIDR3**

Bits	Name	Description
7:4	REVAND	This field indicates minor errata fixes specific to this design, for example metal fixes after implementation. In most cases this field is zero. It is recommended that component designers ensure this field can be changed by a metal fix if required, for example by driving it from registers that reset to zero. 0x0: NO_MODS (Indicates that there have been no metal fixes to this component.)
3:0	CUSTOMER_MODIFIED	Where the component is reusable IP, this value indicates if the customer has modified the behavior of the component. In most cases this field is zero. 0x0: NO_MODS (Indicates that there have been no modifications made.)

0x00824FD0 QDSS_REPL64_PIDR4**Type:** R**Reset State:** 0x00000004**QDSS_REPL64_PIDR4**

Bits	Name	Description
7:4	FOURKB_COUNT	This is a 4-bit value that indicates the total contiguous size of the memory window used by this component in powers of 2 from the standard 4KB. If a component only requires the standard 4KB, this must read as 0x0, 4KB only. For 8KB set to 0x1, for 16KB set to 0x2, for 32KB set to 0x3, and so on. 0x0: ENUM_4KB (Indicates that the device only occupies 4KB of memory.)
3:0	JEP106_CONT	JEDEC continuation code indicating the designer of the component, together with the identity code. 0x4: ARM_JEP106_CONTINUATION_CODE (Indicates that ARM's JEDEC identity code is on the 5th bank.)

0x00824FD4 QDSS_REPL64_PIDR5**Type:** RW**Reset State:** Undefined**QDSS_REPL64_PIDR5**

Bits	Name	Description
31:0	PIDR5	

0x00824FD8 QDSS_REPL64_PIDR6**Type:** RW**Reset State:** Undefined**QDSS_REPL64_PIDR6**

Bits	Name	Description
31:0	PIDR6	

0x00824FDC QDSS_REPL64_PIDR7**Type:** RW**Reset State:** Undefined**QDSS_REPL64_PIDR7**

Bits	Name	Description
31:0	PIDR7	

0x00824FF0 QDSS_REPL64_CIDR0**Type:** R**Reset State:** 0x0000000D**QDSS_REPL64_CIDR0**

Bits	Name	Description
7:0	PREAMBLE	Contains bits [24:31] of the component identification. 0xD: ENUM_0D (Identification value.)

0x00824FF4 QDSS_REPL64_CIDR1**Type:** R**Reset State:** 0x00000090**QDSS_REPL64_CIDR1**

Bits	Name	Description
7:4	CLASS	Class of the component, for example, ROM table or CoreSight component. 0x9: CORESIGHT_COMPONENT (Indicates the component is a CoreSight component.)
3:0	PREAMBLE	Contains bits [19:16] of the component identification. 0x0: ENUM_0 (Identification value.)

0x00824FF8 QDSS_REPL64_CIDR2**Type:** R**Reset State:** 0x00000005**QDSS_REPL64_CIDR2**

Bits	Name	Description
7:0	PREAMBLE	Contains bits [15:8] of the component identification. 0x5: ENUM_05 (Identification value.)

0x00824FFC QDSS_REPL64_CIDR3**Type:** R**Reset State:** 0x000000B1**QDSS_REPL64_CIDR3**

Bits	Name	Description
7:0	PREAMBLE	Contains bits [7:0] of the component identification. 0xB1: B1 (Identification value.)

12.1.16 ETFETB_CXTMC_F128W8K

The CoreSight TMC enables use of local memory as a trace buffer.

12.1.16.1 QDSS ETFETB_CXTMC_F128W8K

The CoreSight TMC enables use of local memory as a trace buffer.

0x00825004 QDSS ETFETB_RSZ**Type:** R**Reset State:** 0x00000800**QDSS ETFETB_RSZ**

Bits	Name	Description
30:0	RSZ	

0x0082500C QDSS ETFETB_STS**Type:** R**Reset State:** 0x0000000C

TMC Status register

QDSS ETFETB_STS

Bits	Name	Description
4	EMPTY	0x1: HIGH (Trace buffer is empty) 0x0: LOW (Trace buffer is not empty)
3	FEMPTY	0x1: HIGH (Trace capture has been completed and all captured trace data has been written to the trace memory) 0x0: LOW (Trace capture has not yet completed)
2	TMCREADY	0x1: HIGH (Trace capture has been completed and all captured trace data has been written to the trace memory) 0x0: LOW (Trace capture has not yet completed)
1	TRIGGERED	0x1: HIGH (Trigger event has occurred) 0x0: LOW (Trigger event has not occurred)
0	FULL	0x1: HIGH 0x0: LOW

0x00825010 QDSS ETFETB_RRD**Type:** R**Reset State:** Undefined**QDSS ETFETB_RRD**

Bits	Name	Description
31:0	RRD	Reads return data from Trace RAM

0x00825014 QDSS ETFETB_RRP**Type:** RW**Reset State:** 0x00000000**QDSS ETFETB_RRP**

Bits	Name	Description
12:0	RRP	

0x00825018 QDSS ETFETB_RWP**Type:** RW**Reset State:** 0x00000000**QDSS ETFETB_RWP**

Bits	Name	Description
12:0	RWP	

0x0082501C QDSS ETFETB_TRG**Type:** RW**Reset State:** 0x00000000**QDSS ETFETB_TRG**

Bits	Name	Description
10:0	TRG	

0x00825020 QDSS ETFETB_CTL**Type:** RW**Reset State:** 0x00000000**QDSS ETFETB_CTL**

Bits	Name	Description
0	TRACECAPTEN	0x0: LOW (Trace capture is disabled.) 0x1: HIGH (Trace capture is enabled.)

0x00825024 QDSS ETFETB_RWD**Type:** W**Reset State:** Undefined**QDSS ETFETB_RWD**

Bits	Name	Description
31:0	RWD	

0x00825028 QDSS ETFETB_MODE**Type:** RW**Reset State:** 0x00000000**QDSS ETFETB_MODE**

Bits	Name	Description
1:0	MODE	Selects the operating mode. 0x3: RESERVED 0x2: HW_FIFO 0x1: SW_FIFO 0x0: CIRCULAR_BUFFER

0x0082502C QDSS ETFETB_LBUFLEVEL**Type:** R**Reset State:** 0x00000000**QDSS ETFETB_LBUFLEVEL**

Bits	Name	Description
11:0	LBUFLEVEL	Indicates the maximum fill level of the trace memory in 32-bit words since this register was last read.

0x00825030 QDSS ETFETB_CBUFLEVEL**Type:** R**Reset State:** 0x00000000**QDSS ETFETB_CBUFLEVEL**

Bits	Name	Description
11:0	CBUFLEVEL	Indicates the current fill level of the trace memory in 32-bit words.

0x00825034 QDSS ETFETB_BUFWM**Type:** RW**Reset State:** 0x00000000**QDSS ETFETB_BUFWM**

Bits	Name	Description
10:0	BUFWM	Indicates the desired threshold vacancy level in 32-bit words in the trace memory.

0x00825300 QDSS ETFETB_FFSR**Type:** R**Reset State:** 0x00000002

This register indicates the status of the Formatter and the status of Flush request.

QDSS ETFETB_FFSR

Bits	Name	Description
1	FTSTOPPED	0x0: LOW (Trace capture has not yet completed.) 0x1: HIGH (Trace capture has been completed and all captured trace data has been written to the trace memory.)
0	RESERVED	Reserved

0x00825304 QDSS ETFETB_FFCR**Type:** RW**Reset State:** 0x00000000**QDSS ETFETB_FFCR**

Bits	Name	Type	Description
14	RESERVED	W	Reserved
13	STOPONTRIG EVT	RW	0x0: LOW (Formatting is not stopped when trigger event is observed.) 0x1: HIGH (Formatting is stopped when trigger event is observed.)
12	STOPONFL	RW	0x0: LOW (Formatting is not stopped when FLUSH is completed.) 0x1: HIGH (Formatting is stopped when FLUSH is completed. Any part-completed packets are drained off into the trace memory with a post-amble.)

QDSS_ETFETB_FFCR (cont.)

Bits	Name	Type	Description
10	TRIGONFL	RW	0x0: LOW (A trigger is not indicated on the trace stream when AFREADY is returned HIGH.) 0x1: HIGH (A trigger is indicated on the trace stream when AFREADY is returned HIGH.)
9	TRIGONTRIGEV	RW	0x0: LOW (A trigger is not indicated on the trace stream when a Trigger Event occurs.) 0x1: HIGH (A trigger is indicated on the trace stream when a Trigger event occurs.)
8	TRIGONTRIGIN	RW	0x0: LOW (A trigger is not indicated on the trace stream when a rising edge is detected on the TRIGIN input.) 0x1: HIGH (A trigger is indicated on the trace stream when a rising edge is detected on the TRIGIN input.)
6	FLUSHMAN_R	R	Manually generate a flush of the system. Setting this bit causes a flush to be generated. If TraceCaptEn bit in CTL register is 0, then writes to this bit are ignored. In Circular buffer mode and Hardware-FIFO mode, this bit is cleared automatically when AFREADY is sampled HIGH. In Software-FIFO mode, this bit is cleared when the flush data is written to trace memory. This bit is clear on reset. 0x0: LOW (Manual Flush is not initiated) 0x1: HIGH (Manual Flush is initiated)
6	FLUSHMAN_W	W	Manually generate a flush of the system. Setting this bit causes a flush to be generated. If TraceCaptEn bit in CTL register is 0, then writes to this bit are ignored. In Circular buffer mode and Hardware-FIFO mode, this bit is cleared automatically when AFREADY is sampled HIGH. In Software-FIFO mode, this bit is cleared when the flush data is written to trace memory. This bit is clear on reset. 0x0: LOW (Manual Flush is not initiated) 0x1: HIGH (Manual Flush is initiated)
5	FONTRIGEV	RW	0x0: LOW (Flush-on-trigger-event disabled) 0x1: HIGH (Flush-on-trigger-event enabled)
4	FONFLIN	RW	Setting this bit enables the detection of transitions on the FLUSHIN input by the TMC. If this bit is set and the Formatter has not already stopped, a rising edge on FLUSHIN initiates a flush request. This bit is clear on reset. 0x0: LOW (Flush-on-FLUSHIN feature is disabled) 0x1: HIGH (Flush-on-FLUSHIN feature is enabled)

QDSS ETFETB_FFCR (cont.)

Bits	Name	Type	Description
1	ENTI	RW	Setting this bit enables the insertion of triggers in the formatted trace stream. A trigger is indicated by inserting one byte of data 8'h00 with ATID 7'h7D in the trace stream. Trigger indication on the trace stream is additionally controlled by the register bits TrigOnFI, TrigOnTrigEvt and TrigOnTrigIn in the FFCR register, 0x304. This bit can only be changed when TMCReady=1 and TraceCaptEn=0. This bit takes effect only when the EnFt register bit in this register is set. If EnTI bit is set to 1 when EnFt is 0, it results in formatting being enabled. This bit is clear on reset. 0x1: TRIGGER_INSERTION_ENABLED (Embed triggers into formatted trace stream) 0x0: TRIGGER_INSERTION_DISABLED (Triggers are not embedded into the formatted strace stream)
0	ENFT	RW	0x1: HIGH (Formatting enabled) 0x0: LOW (Formatting disabled.)

0x00825308 QDSS ETFETB_PSCR**Type:** RW**Reset State:** 0x00000000**QDSS ETFETB_PSCR**

Bits	Name	Description
4:0	PSCOUNT	

0x00825ED0 QDSS ETFETB_ITATBMDATA0**Type:** W**Reset State:** 0x00000000**QDSS ETFETB_ITATBMDATA0**

Bits	Name	Description
16	ATDATAMBIT127	Control the value of ATDATAM[127] output of TMC 0x1: HIGH (Drive logic 1 on this output) 0x0: LOW (Drive logic 0 on this output)
15	ATDATAMBIT119	Control the value of ATDATAM[119] output of TMC 0x1: HIGH (Drive logic 1 on this output) 0x0: LOW (Drive logic 0 on this output)

QDSS ETFETB_ITATBMDATA0 (cont.)

Bits	Name	Description
14	ATDATAMBIT111	Control the value of ATDATAM[111] output of TMC 0x1: HIGH (Drive logic 1 on this output) 0x0: LOW (Drive logic 0 on this output)
13	ATDATAMBIT103	Control the value of ATDATAM[103] output of TMC 0x1: HIGH (Drive logic 1 on this output) 0x0: LOW (Drive logic 0 on this output)
12	ATDATAMBIT95	Control the value of ATDATAM[95] output of TMC 0x1: HIGH (Drive logic 1 on this output) 0x0: LOW (Drive logic 0 on this output)
11	ATDATAMBIT87	Control the value of ATDATAM[87] output of TMC 0x1: HIGH (Drive logic 1 on this output) 0x0: LOW (Drive logic 0 on this output)
10	ATDATAMBIT79	Control the value of ATDATAM[79] output of TMC 0x1: HIGH (Drive logic 1 on this output) 0x0: LOW (Drive logic 0 on this output)
9	ATDATAMBIT71	Control the value of ATDATAM[71] output of TMC 0x1: HIGH (Drive logic 1 on this output) 0x0: LOW (Drive logic 0 on this output)
8	ATDATAMBIT63	Control the value of ATDATAM[63] output of TMC 0x1: HIGH (Drive logic 1 on this output) 0x0: LOW (Drive logic 0 on this output)
7	ATDATAMBIT55	Control the value of ATDATAM[55] output of TMC 0x1: HIGH (Drive logic 1 on this output) 0x0: LOW (Drive logic 0 on this output)
6	ATDATAMBIT47	Control the value of ATDATAM[47] output of TMC 0x1: HIGH (Drive logic 1 on this output) 0x0: LOW (Drive logic 0 on this output)
5	ATDATAMBIT39	Control the value of ATDATAM[39] output of TMC 0x1: HIGH (Drive logic 1 on this output) 0x0: LOW (Drive logic 0 on this output)
4	ATDATAMBIT31	Control the value of ATDATAM[31] output of TMC 0x1: HIGH (Drive logic 1 on this output) 0x0: LOW (Drive logic 0 on this output)
3	ATDATAMBIT23	Control the value of ATDATAM[23] output of TMC 0x1: HIGH (Drive logic 1 on this output) 0x0: LOW (Drive logic 0 on this output)
2	ATDATAMBIT15	Control the value of ATDATAM[15] output of TMC 0x1: HIGH (Drive logic 1 on this output) 0x0: LOW (Drive logic 0 on this output)

QDSS ETFETB_ITATBMDATA0 (cont.)

Bits	Name	Description
1	ATDATAMBIT7	Control the value of ATDATAM[7] output of TMC 0x1: HIGH (Drive logic 1 on this output) 0x0: LOW (Drive logic 0 on this output)
0	ATDATAMBIT0	Control the value of ATDATAM[0] output of TMC 0x1: HIGH (Drive logic 1 on this output) 0x0: LOW (Drive logic 0 on this output)

0x00825ED8 QDSS ETFETB_ITATBMCTR1**Type:** W**Reset State:** 0x00000000**QDSS ETFETB_ITATBMCTR1**

Bits	Name	Description
6:0	ATIDM	Control the value of ATIDM output from TMC. The value written to this field is driven on the ATIDM output of the TMC.

0x00825EDC QDSS ETFETB_ITATBMCTR0**Type:** W**Reset State:** 0x00000000**QDSS ETFETB_ITATBMCTR0**

Bits	Name	Description
11:8	ATBYTESM	Control the value of ATBYTESM output from TMC. The value written to this field is driven on the ATBYTESM output of the TMC.
1	AFREADYM	Set the value of AFREADYM output 0x1: HIGH (Drive logic 1 on AFREADYM output) 0x0: LOW (Drive logic 0 on AFREADYM output)
0	ATVALIDM	Set the value of ATVALIDM output 0x1: HIGH (Drive logic 1 on ATVALIDM output) 0x0: LOW (Drive logic 0 on ATVALIDM output)

0x00825EE0 QDSS ETFETB_ITMISCOP0**Type:** W**Reset State:** 0x00000000

QDSS ETFETB_ITMISCOP0

Bits	Name	Description
1	FULL	Set the value of the FULL output. 0x1: HIGH (Drive logic 1 on FULL output) 0x0: LOW (Drive logic 0 on FULL output)
0	ACQCOMP	Set the value of the ACQCOMP output. 0x1: HIGH (Drive logic 1 on ACQCOMP output) 0x0: LOW (Drive logic 0 on ACQCOMP output)

0x00825EE8 QDSS ETFETB_ITTRFLIN**Type:** R**Reset State:** 0x00000000**QDSS ETFETB_ITTRFLIN**

Bits	Name	Description
1	FLUSHIN	Read the value of the FLUSHIN input. 0x1: HIGH (Pin is at logic 1) 0x0: LOW (Pin is at logic 0)
0	TRIGIN	Read the value of the TRIGIN input. 0x1: HIGH (Pin is at logic 1) 0x0: LOW (Pin is at logic 0)

0x00825EEC QDSS ETFETB_ITATBDATA0**Type:** R**Reset State:** 0x00000000**QDSS ETFETB_ITATBDATA0**

Bits	Name	Description
16	ATDATASBIT127	Read the value of ATDATAS[127] input to TMC 0x1: HIGH (Pin is at logic 1) 0x0: LOW (Pin is at logic 0)
15	ATDATASBIT119	Read the value of ATDATAS[119] input to TMC 0x1: HIGH (Pin is at logic 1) 0x0: LOW (Pin is at logic 0)
14	ATDATASBIT111	Read the value of ATDATAS[111] input to TMC 0x1: HIGH (Pin is at logic 1) 0x0: LOW (Pin is at logic 0)

QDSS ETFETB_ITATBDATA0 (cont.)

Bits	Name	Description
13	ATDATASBIT103	Read the value of ATDATAS[103] input to TMC 0x1: HIGH (Pin is at logic 1) 0x0: LOW (Pin is at logic 0)
12	ATDATASBIT95	Read the value of ATDATAS[95] input to TMC 0x1: HIGH (Pin is at logic 1) 0x0: LOW (Pin is at logic 0)
11	ATDATASBIT87	Read the value of ATDATAS[87] input to TMC 0x1: HIGH (Pin is at logic 1) 0x0: LOW (Pin is at logic 0)
10	ATDATASBIT79	Read the value of ATDATAS[79] input to TMC 0x1: HIGH (Pin is at logic 1) 0x0: LOW (Pin is at logic 0)
9	ATDATASBIT71	Read the value of ATDATAS[71] input to TMC 0x1: HIGH (Pin is at logic 1) 0x0: LOW (Pin is at logic 0)
8	ATDATASBIT63	Read the value of ATDATAS[63] input to TMC 0x1: HIGH (Pin is at logic 1) 0x0: LOW (Pin is at logic 0)
7	ATDATASBIT55	Read the value of ATDATAS[55] input to TMC 0x1: HIGH (Pin is at logic 1) 0x0: LOW (Pin is at logic 0)
6	ATDATASBIT47	Read the value of ATDATAS[47] input to TMC 0x1: HIGH (Pin is at logic 1) 0x0: LOW (Pin is at logic 0)
5	ATDATASBIT39	Read the value of ATDATAS[39] input to TMC 0x1: HIGH (Pin is at logic 1) 0x0: LOW (Pin is at logic 0)
4	ATDATASBIT31	Read the value of ATDATAS[31] input to TMC 0x1: HIGH (Pin is at logic 1) 0x0: LOW (Pin is at logic 0)
3	ATDATASBIT23	Read the value of ATDATAS[23] input to TMC 0x1: HIGH (Pin is at logic 1) 0x0: LOW (Pin is at logic 0)
2	ATDATASBIT15	Read the value of ATDATAS[15] input to TMC 0x1: HIGH (Pin is at logic 1) 0x0: LOW (Pin is at logic 0)
1	ATDATASBIT7	Read the value of ATDATAS[7] input to TMC 0x1: HIGH (Pin is at logic 1) 0x0: LOW (Pin is at logic 0)

QDSS ETFETB_ITATBDATA0 (cont.)

Bits	Name	Description
0	ATDATASBIT0	Read the value of ATDATAS[0] input to TMC 0x1: HIGH (Pin is at logic 1) 0x0: LOW (Pin is at logic 0)

0x00825EF0 QDSS ETFETB_ITATBCTR2**Type:** W**Reset State:** 0x00000000**QDSS ETFETB_ITATBCTR2**

Bits	Name	Description
2	SYNCREQS	Set the value of SYNCREQS output 0x1: HIGH (Drive logic 1 on SYNCREQS output) 0x0: LOW (Drive logic 0 on SYNCREQS output)
1	AFVALIDS	Set the value of AFVALIDS output 0x1: HIGH (Drive logic 1 on AFVALIDS output) 0x0: LOW (Drive logic 0 on AFVALIDS output)
0	ATREADYDYS	Set the value of ATREADYDYS output 0x1: HIGH (Drive logic 1 on ATREADYDYS output) 0x0: LOW (Drive logic 0 on ATREADYDYS output)

0x00825F00 QDSS ETFETB_ITCTRL**Type:** RW**Reset State:** 0x00000000**QDSS ETFETB_ITCTRL**

Bits	Name	Description
0	INTEGRATION_MODE	Allows the component to switch from functional mode to integration mode or back. 0x1: INTEGRATION_MODE (Enable integration mode) 0x0: FUNCTIONAL_MODE (Disable integration mode)

0x00825FA0 QDSS ETFETB_CLAIMSET**Type:** RW**Reset State:** 0x0000000F

This is used in conjunction with Claim Tag Clear Register, CLAIMCLR. This register forms one half of the Claim Tag value. This location allows individual bits to be set, write, and returns the number of bits that can be set, read.

QDSS ETFETB CLAIMSET

Bits	Name	Type	Description
3:0	CLAIMSET_W	W	Each bit is considered separately: 0 = no effect, 1 = set this bit in the claim tag.
3:0	CLAIMSET_R	R	This claim tag bit is implemented 0xF: CLAIM_TAG_IMPLEMENTED_BITS (These bits are present within the claim tag field)

0x00825FA4 QDSS ETFETB CLAIMCLR

Type: RW

Reset State: 0x00000000

This register is used in conjunction with Claim Tag Set Register, CLAIMSET. This register forms one half of the Claim Tag value. This location enables individual bits to be cleared, write, and returns the current Claim Tag value, read.

QDSS ETFETB CLAIMCLR

Bits	Name	Type	Description
3:0	CLAIMCLR_W	W	Each bit is considered separately: 0 = no effect, 1 = clear this bit in the claim tag.
3:0	CLAIMCLR_R	R	The value present reflects the current setting of the Claim Tag.

0x00825FB0 QDSS ETFETB_LAR

Type: W

Reset State: Undefined

QDSS ETFETB_LAR

Bits	Name	Description
31:0	ACCESS_W	A write of 0xC5ACCE55 enables further write access to this device. A write of any value other than 0xC5ACCE55 will have the affect of removing write access. 0xC5ACCE55: UNLOCK (Unlock the protection register to enable write access)

0x00825FB4 QDSS ETFETB_LSR**Type:** R**Reset State:** 0x00000003**QDSS ETFETB_LSR**

Bits	Name	Description
2	LOCKTYPE	Indicates if the Lock Access Register (0xFB0) is implemented as 8-bit or 32-bit 0x0: ENUM_32_BIT (This component implements a 32-bit Lock Access Register)
1	LOCKGRANT	Returns the current status of the Lock. This bit reads as 0 when read from an external debugger (PADDRDBG31 = 1) since external debugger accesses are not subject to Lock Registers. 0x0: ACCESS_PERMITTED (Write access is permitted to this device) 0x1: DEVICE_LOCKED (Write access to the component is blocked. All writes to control registers are ignored. Reads are permitted)
0	LOCKEXIST	Indicates that a lock control mechanism exists for this device. This bit reads as 0 when read from an external debugger (PADDRDBG31 = 1) since external debugger accesses are not subject to Lock Registers. 0x0: LOCK_NOT_PRESENT (No lock control mechanism exists, writes to the Lock Access register (0xFB0) will be ignored) 0x1: LOCK_PRESENT (Lock control mechanism is present)

0x00825FC8 QDSS ETFETB_DEVID**Type:** R**Reset State:** 0x00000580

This register indicates the capabilities of the CoreSight TMC.

QDSS ETFETB_DEVID

Bits	Name	Description
10:8	MEMWIDTH	0x2: ENUM_32_BITS (Memory interface databus is 32-bits wide.) 0x3: ENUM_64_BITS (Memory interface databus is 64-bits wide.) 0x4: ENUM_128_BITS (Memory interface databus is 128-bits wide.) 0x5: ENUM_256_BITS (Memory interface databus is 256-bits wide.)

QDSS ETFETB_DEVID (cont.)

Bits	Name	Description
7:6	CONFIGTYPE	This value indicates TMC configuration type - ETB, ETF or ETR configuration. 0x0: TMC_CONFIGURED_AS_ETB_EMBEDDED_TRACE_BUFFER (When configured as an ETB, trace data is stored in local SRAM through the memory interface. Trace data can only be retrieved through APB register reads.) 0x1: TMC_CONFIGURED_AS_ETR_EMBEDDED_TRACE_ROUTER (When configured as an ETR, the TMC stores trace data in the system memory through the AXI interface. Trace data can only be retrieved through APB register reads.) 0x2: Reserved
5	CLKSCHEME	This value indicates the TMC RAM clocking scheme used i.e.. whether the TMC RAM operates synchronously or asynchronously to CLK. 0x0: TMC_RAM_SYNCHRONOUS_TO_CLK (The TMC RAM operates synchronously to CLK.)
4:0	RESERVED	Reserved

0x00825FE0 QDSS ETFETB_PERIPHID0**Type:** R**Reset State:** 0x00000061

Part of the set of Peripheral Identification registers. Contains part of the designer specific part number.

QDSS ETFETB_PERIPHID0

Bits	Name	Description
7:0	PART_NUMBER_BITS7TO0	Bits [7:0] of the component's part number. This is selected by the designer of the component. 0x61: CORESIGHT_TMC_PART_NUMBER_7_0 (Lowest 8 bits of the Part Number (0x961))

0x00825FE4 QDSS ETFETB_PERIPHID1**Type:** R**Reset State:** 0x000000B9

Part of the set of Peripheral Identification registers. Contains part of the designer specific part number and part of the designer identity.

QDSS ETFETB_PERIPHID1

Bits	Name	Description
7:4	JEP106_BITS3TO0	Bits 3:0 of the JEDEC identity code indicating the designer of the component (along with the continuation code) 0xB: ARM_JEP106_IDENTITY_CODE_7_0 (Lowest 4 bits of the JEP106 Identity Code)
3:0	PART_NUMBER_BITS11TO8	Bits [11:8] of the component's part number. This is selected by the designer of the component. 0x9: CORESIGHT_TMC_PART_NUMBER_11_8 (Upper 4 bits of the Part Number (0x961))

0x00825FE8 QDSS ETFETB_PERIPHID2**Type:** R**Reset State:** 0x0000001B

Part of the set of Peripheral Identification registers. Contains part of the designer identity and the product revision.

QDSS ETFETB_PERIPHID2

Bits	Name	Description
7:4	REVISION	The Revision field is an incremental value starting at 0x0 for the first design of this component. This only increases by 1 for both major and minor revisions and is simply used as a look-up to establish the exact major/minor revision. 0x1: R0P1 (This device is at r0p1)
3	JEDEC	Always set. Indicates that a JEDEC assigned value is used 0x1: JEDEC_IDENTITY (The designer ID is specified by JEDEC (http://www.jedec.org))
2:0	JEP106_BITS6TO4	Bits 6:4 of the JEDEC identity code indicating the designer of the component (along with the continuation code) 0x3: ARM_JEP106_IDENTITY_CODE_6_4 (Upper 3 bits of the JEP106 Identity Code)

0x00825FEC QDSS ETFETB_PERIPHID3**Type:** R**Reset State:** 0x00000000

Part of the set of Peripheral Identification registers. Contains the RevAnd and Customer Modified fields.

QDSS ETFETB_PERIPHID3

Bits	Name	Description
7:4	REVAND	This field indicates minor errata fixes specific to this design, for example metal fixes after implementation. In most cases this field is zero. It is recommended that component designers ensure this field can be changed by a metal fix if required, for example by driving it from registers that reset to zero. 0x0: NO_MODS (Indicates that there have been no metal fixes to this component)
3:0	CUSTOMER_MODIFIED	Where the component is reusable IP, this value indicates if the customer has modified the behavior of the component. In most cases this field is zero. 0x0: NO_MODS (Indicates that there have been no modifications made)

0x00825FD0 QDSS ETFETB_PERIPHID4**Type:** R**Reset State:** 0x00000004

Part of the set of Peripheral Identification registers. Contains part of the designer identity and the memory footprint indicator.

QDSS ETFETB_PERIPHID4

Bits	Name	Description
7:4	FOURKB_COUNT	This is a 4-bit value that indicates the total contiguous size of the memory window used by this component in powers of 2 from the standard 4KB. If a component only requires the standard 4KB then this should read as 0x0, 4KB only, for 8KB set to 0x1, 16KB == 0x2, 32KB == 0x3, and so on. 0x0: ENUM_4KB (Indicates that the device only occupies 4kB of memory)
3:0	JEP106_CONT	JEDEC continuation code indicating the designer of the component (along with the identity code) 0x4: ARM_JEP106_CONTINUATION_CODE (Indicates that ARM's JEDEC identity code is on the 5th bank)

0x00825FD4 QDSS ETFETB_PERIPHID5**Type:** RW**Reset State:** Undefined

Reserved

QDSS ETFETB_PERIPHID5

Bits	Name	Description
31:0	PERIPHID5	

0x00825FD8 QDSS ETFETB_PERIPHID6**Type:** RW**Reset State:** Undefined

Reserved

QDSS ETFETB_PERIPHID6

Bits	Name	Description
31:0	PERIPHID6	

0x00825FDC QDSS ETFETB_PERIPHID7**Type:** RW**Reset State:** Undefined

Reserved

QDSS ETFETB_PERIPHID7

Bits	Name	Description
31:0	PERIPHID7	

0x00825FF0 QDSS ETFETB_COMPID0**Type:** R**Reset State:** 0x0000000D

A component identification register, that indicates that the identification registers are present.

QDSS ETFETB_COMPID0

Bits	Name	Description
7:0	PREAMBLE	Contains bits [7:0] of the component identification 0xD: ENUM_OD (Identification value)

0x00825FF4 QDSS ETFETB_COMPID1**Type:** R**Reset State:** 0x00000090

A component identification register, that indicates that the identification registers are present. This register also indicates the component class.

QDSS ETFETB_COMPID1

Bits	Name	Description
7:4	CLASS	Class of the component. E.g. ROM table, CoreSight component etc. Constitutes bits [15:12] of the component identification. 0x9: CORESIGHT_COMPONENT (Indicates the component is a CoreSight component)
3:0	PREAMBLE	Contains bits [11:8] of the component identification 0x0: ENUM_0 (Identification value)

0x00825FF8 QDSS ETFETB_COMPID2**Type:** R**Reset State:** 0x00000005

A component identification register, that indicates that the identification registers are present.

QDSS ETFETB_COMPID2

Bits	Name	Description
7:0	PREAMBLE	Contains bits [23:16] of the component identification 0x5: ENUM_05 (Identification value)

0x00825FFC QDSS ETFETB_COMPID3**Type:** R**Reset State:** 0x000000B1

A component identification register, that indicates that the identification registers are present.

QDSS ETFETB_COMPID3

Bits	Name	Description
7:0	PREAMBLE	Contains bits [31:24] of the component identification 0xB1: B1 (Identification value)

12.1.17 ETR_CXTMC_R64W32D

The CoreSight TMC enables use of local memory as a trace buffer.

12.1.17.1 QDSS_ETR_CXTMC_R64W32D

The CoreSight TMC enables use of local memory as a trace buffer.

0x00826004 QDSS_ETR_RSZ

Type: RW

Reset State: 0x00000400

QDSS_ETR_RSZ

Bits	Name	Description
30:0	RSZ_ETR	

0x0082600C QDSS_ETR_STS

Type: RW

Reset State: 0x0000000C

TMC Status register

QDSS_ETR_STS

Bits	Name	Type	Description
5	MEMERR_W	W	0x1: HIGH (Clear the Memory Error bit.) 0x0: LOW (No effect)
5	MEMERR_R	R	0x1: HIGH (Memory Error has occurred.) 0x0: LOW (No Memory error has occurred)
4	EMPTY	R	0x1: HIGH (Trace buffer is empty) 0x0: LOW (Trace buffer is not empty)
3	FEMPTY	R	0x1: HIGH (Trace capture has been completed and all captured trace data has been written to the trace memory) 0x0: LOW (Trace capture has not yet completed)
2	TMCREADY	R	0x1: HIGH (Trace capture has been completed and all captured trace data has been written to the trace memory) 0x0: LOW (Trace capture has not yet completed)
1	TRIGGERED	R	0x1: HIGH (Trigger event has occurred) 0x0: LOW (Trigger event has not occurred)

QDSS_ETR_STS (cont.)

Bits	Name	Type	Description
0	FULL	R	0x1: HIGH 0x0: LOW

0x00826010 QDSS_ETR_RRD**Type:** R**Reset State:** Undefined**QDSS_ETR_RRD**

Bits	Name	Description
31:0	RRD	Reads return data from Trace RAM

0x00826014 QDSS_ETR_RRP**Type:** RW**Reset State:** 0x00000000**QDSS_ETR_RRP**

Bits	Name	Description
31:0	RRP	

0x00826018 QDSS_ETR_RWP**Type:** RW**Reset State:** 0x00000000**QDSS_ETR_RWP**

Bits	Name	Description
31:0	RWP	

0x0082601C QDSS_ETR_TRG**Type:** RW**Reset State:** 0x00000000

QDSS_ETR_TRG

Bits	Name	Description
31:0	TRG	

0x00826020 QDSS_ETR_CTL**Type:** RW**Reset State:** 0x00000000**QDSS_ETR_CTL**

Bits	Name	Description
0	TRACECAPTEN	0x0: LOW (Trace capture is disabled.) 0x1: HIGH (Trace capture is enabled.)

0x00826024 QDSS_ETR_RWD**Type:** W**Reset State:** Undefined**QDSS_ETR_RWD**

Bits	Name	Description
31:0	RWD	

0x00826028 QDSS_ETR_MODE**Type:** RW**Reset State:** 0x00000000**QDSS_ETR_MODE**

Bits	Name	Description
1:0	MODE	Selects the operating mode. 0x3: RESERVED 0x2: HW_FIFO 0x1: SW_FIFO 0x0: CIRCULAR_BUFFER

0x0082602C QDSS_ETR_LBUFLEVEL**Type:** R**Reset State:** 0x00000000**QDSS_ETR_LBUFLEVEL**

Bits	Name	Description
30:0	LBUFLEVEL	Indicates the maximum fill level of the trace memory in 32-bit words since this register was last read.

0x00826030 QDSS_ETR_CBUFLEVEL**Type:** R**Reset State:** 0x00000000**QDSS_ETR_CBUFLEVEL**

Bits	Name	Description
30:0	CBUFLEVEL	Indicates the current fill level of the trace memory in 32-bit words.

0x00826034 QDSS_ETR_BUFWM**Type:** RW**Reset State:** 0x00000000**QDSS_ETR_BUFWM**

Bits	Name	Description
29:0	BUFFWM	Indicates the desired threshold vacancy level in 32-bit words in the trace memory.

0x00826038 QDSS_ETR_RRPFI**Type:** RW**Reset State:** 0x00000000**QDSS_ETR_RRPFI**

Bits	Name	Description
7:0	RRPFI	

0x0082603C QDSS_ETR_RWPHI**Type:** RW**Reset State:** 0x00000000**QDSS_ETR_RWPHI**

Bits	Name	Description
7:0	RWPHI	

0x00826110 QDSS_ETR_AXICTL**Type:** RW**Reset State:** 0x00000000**QDSS_ETR_AXICTL**

Bits	Name	Description
11:8	WRBURSTLEN	
7	SCATTERGATHERMODE	This bit indicates whether trace memory is accessed as a single buffer in system memory or as a linked-list based scatter-gather memory. This is ignored when the TMC is in Disabled state (TraceCaptEn=0 and TMCReady=1). 0x0: SINGLE_BUFFER (Trace memory is a single contiguous block of system memory) 0x1: SCATTER_GATHER (Trace memory is spread over multiple blocks of system memory based on a linked-list mechanism.)
5	CACHECTRLBIT3	This bit controls the value driven on the ARCCACHEM[3]/AWCCACHEM[3] signal on the AXI interface when performing AXI transfers. If CacheCtrlBit1 is 0, then this bit must also be 0 to comply with AXI protocol. Setting this bit to 1 when the CacheCtrlBit1 is 0 will result in Unpredictable behavior. 0x0: DO_NOT_CACHE_ALLOCATE_ON_WRITES (Do not cache allocate on writes) 0x1: CACHE_ALLOCATE_ON_WRITES (If the transfer is a write and it misses in the cache then it should be allocated.)
4	CACHECTRLBIT2	This bit controls the value driven on the ARCCACHEM[2]/AWCCACHEM[2] signal on the AXI interface when performing AXI transfers. If CacheCtrlBit1 is 0, then this bit must also be 0 to comply with AXI protocol. Setting this bit to 1 when the CacheCtrlBit1 is 0 will result in Unpredictable behavior. 0x0: DO_NOT_CACHE_ALLOCATE_ON_READS (Do not cache allocate on reads) 0x1: CACHE_ALLOCATE_ON_READS (If the transfer is a read and it misses in the cache then it should be allocated.)

QDSS_ETR_AXICTL (cont.)

Bits	Name	Description
3	CACHECTRLBIT1	This bit controls the value driven on the ARCACHEM[1]/AWCACHEM[1] signal on the AXI interface when performing AXI transfers. 0x0: NON_CACHEABLE (Non-cacheable) 0x1: CACHEABLE (Cacheable. The transaction at the final destination does not have to match the characteristics of the original transaction. For reads this means that a location can be pre-fetched or can be fetched just once for multiple read transactions. For writes this means that a number of different writes can be merged together. To determine if a transaction should be cached this bit should be used in conjunction with the Read Allocate (RA) and Write Allocate (WA) bits.)
2	CACHECTRLBIT0	This bit controls the value driven on the ARCACHEM[0]/AWCACHEM[0] signal on the AXI interface when performing AXI transfers. 0x0: NON_BUFFERABLE (Non-bufferable) 0x1: BUFFERABLE (Bufferable)
1	PROTCTRLBIT1	0x0: SECURE_ACCESS (Secure access) 0x1: NON_SECURE_ACCESS (Non-secure access)
0	PROTCTRLBIT0	0x0: NORMAL_ACCESS (Normal access) 0x1: PRIVILEGED_ACCESS (Privileged access. A privileged processing mode might have a greater level of access within a system.)

0x00826118 QDSS_ETR_DBALO**Type:** RW**Reset State:** 0x00000000**QDSS_ETR_DBALO**

Bits	Name	Description
31:0	BUFADDRLO	Holds the lower 32 bits of the 40-bit address used to locate the trace buffer in system memory

0x0082611C QDSS_ETR_DBAHI**Type:** RW**Reset State:** 0x00000000

QDSS_ETR_DBAHI

Bits	Name	Description
7:0	BUFADDRHI	Holds the upper 8 bits of the 40-bit address used to locate the trace buffer in system memory

0x00826300 QDSS_ETR_FFSR**Type:** R**Reset State:** 0x00000002

This register indicates the status of the Formatter and the status of Flush request.

QDSS_ETR_FFSR

Bits	Name	Description
1	FTSTOPPED	0x0: LOW (Trace capture has not yet completed.) 0x1: HIGH (Trace capture has been completed and all captured trace data has been written to the trace memory.)
0	RESERVED	Reserved

0x00826304 QDSS_ETR_FFCR**Type:** RW**Reset State:** 0x00000000**QDSS_ETR_FFCR**

Bits	Name	Type	Description
13	STOPONTRIG EVT	RW	0x0: LOW (Formatting is not stopped when trigger event is observed.) 0x1: HIGH (Formatting is stopped when trigger event is observed.)
12	STOPONFL	RW	0x0: LOW (Formatting is not stopped when FLUSH is completed.) 0x1: HIGH (Formatting is stopped when FLUSH is completed. Any part-completed packets are drained off into the trace memory with a post-amble.)
10	TRIGONFL	RW	0x0: LOW (A trigger is not indicated on the trace stream when AFREADY is returned HIGH.) 0x1: HIGH (A trigger is indicated on the trace stream when AFREADY is returned HIGH.)

QDSS_ETR_FFCR (cont.)

Bits	Name	Type	Description
9	TRIGONTRIGEV	RW	0x0: LOW (A trigger is not indicated on the trace stream when a Trigger Event occurs.) 0x1: HIGH (A trigger is indicated on the trace stream when a Trigger event occurs.)
8	TRIGONTRIGIN	RW	0x0: LOW (A trigger is not indicated on the trace stream when a rising edge is detected on the TRIGIN input.) 0x1: HIGH (A trigger is indicated on the trace stream when a rising edge is detected on the TRIGIN input.)
6	FLUSHMAN_R	R	Manually generate a flush of the system. Setting this bit causes a flush to be generated. If TraceCaptEn bit in CTL register is 0, then writes to this bit are ignored. In Circular buffer mode and Hardware-FIFO mode, this bit is cleared automatically when AFREADY is sampled HIGH. In Software-FIFO mode, this bit is cleared when the flush data is written to trace memory. This bit is clear on reset. 0x0: LOW (Manual Flush is not initiated) 0x1: HIGH (Manual Flush is initiated)
6	FLUSHMAN_W	W	Manually generate a flush of the system. Setting this bit causes a flush to be generated. If TraceCaptEn bit in CTL register is 0, then writes to this bit are ignored. In Circular buffer mode and Hardware-FIFO mode, this bit is cleared automatically when AFREADY is sampled HIGH. In Software-FIFO mode, this bit is cleared when the flush data is written to trace memory. This bit is clear on reset. 0x0: LOW (Manual Flush is not initiated) 0x1: HIGH (Manual Flush is initiated)
5	FONTRIGEV	RW	0x0: LOW (Flush-on-trigger-event disabled) 0x1: HIGH (Flush-on-trigger-event enabled)
4	FONFLIN	RW	Setting this bit enables the detection of transitions on the FLUSHIN input by the TMC. If this bit is set and the Formatter has not already stopped, a rising edge on FLUSHIN initiates a flush request. This bit is clear on reset. 0x0: LOW (Flush-on-FLUSHIN feature is disabled) 0x1: HIGH (Flush-on-FLUSHIN feature is enabled)

QDSS_ETR_FFCR (cont.)

Bits	Name	Type	Description
1	ENTI	RW	Setting this bit enables the insertion of triggers in the formatted trace stream. A trigger is indicated by inserting one byte of data 8'h00 with ATID 7'h7D in the trace stream. Trigger indication on the trace stream is additionally controlled by the register bits TrigOnFI, TrigOnTrigEvt and TrigOnTrigIn in the FFCR register, 0x304. This bit can only be changed when TMCReady=1 and TraceCaptEn=0. This bit takes effect only when the EnFt register bit in this register is set. If EnTI bit is set to 1 when EnFt is 0, it results in formatting being enabled. This bit is clear on reset. 0x1: TRIGGER_INSERTION_ENABLED (Embed triggers into formatted trace stream) 0x0: TRIGGER_INSERTION_DISABLED (Triggers are not embedded into the formatted strace stream)
0	ENFT	RW	0x1: HIGH (Formatting enabled) 0x0: LOW (Formatting disabled.)

0x00826308 QDSS_ETR_PSCR**Type:** RW**Reset State:** 0x00000000**QDSS_ETR_PSCR**

Bits	Name	Description
4:0	PSCOUNT	

0x00826EE0 QDSS_ETR_ITMISCOP0**Type:** W**Reset State:** 0x00000000**QDSS_ETR_ITMISCOP0**

Bits	Name	Description
1	FULL	Set the value of the FULL output. 0x1: HIGH (Drive logic 1 on FULL output) 0x0: LOW (Drive logic 0 on FULL output)
0	ACQCOMP	Set the value of the ACQCOMP output. 0x1: HIGH (Drive logic 1 on ACQCOMP output) 0x0: LOW (Drive logic 0 on ACQCOMP output)

0x00826EE8 QDSS_ETR_ITTRFLIN**Type:** R**Reset State:** 0x00000000**QDSS_ETR_ITTRFLIN**

Bits	Name	Description
1	FLUSHIN	Read the value of the FLUSHIN input. 0x1: HIGH (Pin is at logic 1) 0x0: LOW (Pin is at logic 0)
0	TRIGIN	Read the value of the TRIGIN input. 0x1: HIGH (Pin is at logic 1) 0x0: LOW (Pin is at logic 0)

0x00826EEC QDSS_ETR_ITATBDATA0**Type:** R**Reset State:** 0x00000000**QDSS_ETR_ITATBDATA0**

Bits	Name	Description
8	ATDATASBIT63	Read the value of ATDATAS[63] input to TMC 0x1: HIGH (Pin is at logic 1) 0x0: LOW (Pin is at logic 0)
7	ATDATASBIT55	Read the value of ATDATAS[55] input to TMC 0x1: HIGH (Pin is at logic 1) 0x0: LOW (Pin is at logic 0)
6	ATDATASBIT47	Read the value of ATDATAS[47] input to TMC 0x1: HIGH (Pin is at logic 1) 0x0: LOW (Pin is at logic 0)
5	ATDATASBIT39	Read the value of ATDATAS[39] input to TMC 0x1: HIGH (Pin is at logic 1) 0x0: LOW (Pin is at logic 0)
4	ATDATASBIT31	Read the value of ATDATAS[31] input to TMC 0x1: HIGH (Pin is at logic 1) 0x0: LOW (Pin is at logic 0)
3	ATDATASBIT23	Read the value of ATDATAS[23] input to TMC 0x1: HIGH (Pin is at logic 1) 0x0: LOW (Pin is at logic 0)
2	ATDATASBIT15	Read the value of ATDATAS[15] input to TMC 0x1: HIGH (Pin is at logic 1) 0x0: LOW (Pin is at logic 0)

QDSS_ETR_ITATBDATA0 (cont.)

Bits	Name	Description
1	ATDATASBIT7	Read the value of ATDATAS[7] input to TMC 0x1: HIGH (Pin is at logic 1) 0x0: LOW (Pin is at logic 0)
0	ATDATASBIT0	Read the value of ATDATAS[0] input to TMC 0x1: HIGH (Pin is at logic 1) 0x0: LOW (Pin is at logic 0)

0x00826EF0 QDSS_ETR_ITATBCTR2**Type:** W**Reset State:** 0x00000000**QDSS_ETR_ITATBCTR2**

Bits	Name	Description
2	SYNCREQS	Set the value of SYNCREQS output 0x1: HIGH (Drive logic 1 on SYNCREQS output) 0x0: LOW (Drive logic 0 on SYNCREQS output)
1	AFVALIDS	Set the value of AFVALIDS output 0x1: HIGH (Drive logic 1 on AFVALIDS output) 0x0: LOW (Drive logic 0 on AFVALIDS output)
0	ATREADYDYS	Set the value of ATREADYDYS output 0x1: HIGH (Drive logic 1 on ATREADYDYS output) 0x0: LOW (Drive logic 0 on ATREADYDYS output)

0x00826F00 QDSS_ETR_ITCTRL**Type:** RW**Reset State:** 0x00000000**QDSS_ETR_ITCTRL**

Bits	Name	Description
0	INTEGRATION_MODE	Allows the component to switch from functional mode to integration mode or back. 0x1: INTEGRATION_MODE (Enable integration mode) 0x0: FUNCTIONAL_MODE (Disable integration mode)

0x00826FA0 QDSS_ETR_CLAIMSET**Type:** RW**Reset State:** 0x0000000F

This is used in conjunction with Claim Tag Clear Register, CLAIMCLR. This register forms one half of the Claim Tag value. This location allows individual bits to be set, write, and returns the number of bits that can be set, read.

QDSS_ETR_CLAIMSET

Bits	Name	Type	Description
3:0	CLAIMSET_W	W	Each bit is considered separately: 0 = no effect, 1 = set this bit in the claim tag.
3:0	CLAIMSET_R	R	This claim tag bit is implemented 0xF: CLAIM_TAG_IMPLEMENTED_BITS (These bits are present within the claim tag field)

0x00826FA4 QDSS_ETR_CLAIMCLR**Type:** RW**Reset State:** 0x00000000

This register is used in conjunction with Claim Tag Set Register, CLAIMSET. This register forms one half of the Claim Tag value. This location enables individual bits to be cleared, write, and returns the current Claim Tag value, read.

QDSS_ETR_CLAIMCLR

Bits	Name	Type	Description
3:0	CLAIMCLR_W	W	Each bit is considered separately: 0 = no effect, 1 = clear this bit in the claim tag.
3:0	CLAIMCLR_R	R	The value present reflects the current setting of the Claim Tag.

0x00826FB0 QDSS_ETR_LAR**Type:** W**Reset State:** Undefined

QDSS_ETR_LAR

Bits	Name	Description
31:0	ACCESS_W	A write of 0xC5ACCE55 enables further write access to this device. A write of any value other than 0xC5ACCE55 will have the affect of removing write access. 0xC5ACCE55: UNLOCK (Unlock the protection register to enable write access)

0x00826FB4 QDSS_ETR_LSR**Type:** R**Reset State:** 0x00000003**QDSS_ETR_LSR**

Bits	Name	Description
2	LOCKTYPE	Indicates if the Lock Access Register (0xFB0) is implemented as 8-bit or 32-bit 0x0: ENUM_32_BIT (This component implements a 32-bit Lock Access Register)
1	LOCKGRANT	Returns the current status of the Lock. This bit reads as 0 when read from an external debugger (PADDRDBG31 = 1) since external debugger accesses are not subject to Lock Registers. 0x0: ACCESS_PERMITTED (Write access is permitted to this device) 0x1: DEVICE_LOCKED (Write access to the component is blocked. All writes to control registers are ignored. Reads are permitted)
0	LOCKEXIST	Indicates that a lock control mechanism exists for this device. This bit reads as 0 when read from an external debugger (PADDRDBG31 = 1) since external debugger accesses are not subject to Lock Registers. 0x0: LOCK_NOT_PRESENT (No lock control mechanism exists, writes to the Lock Access register (0xFB0) will be ignored) 0x1: LOCK_PRESENT (Lock control mechanism is present)

0x00826FC8 QDSS_ETR_DEVID**Type:** R**Reset State:** 0x00002B40

This register indicates the capabilities of the CoreSight TMC.

QDSS_ETR_DEVID

Bits	Name	Description
13:11	WBUF_DEPTH	0x2: ENUM_4_ENTRIES (Depth of the Write buffer is 4 entries.) 0x3: ENUM_8_ENTRIES (Depth of the Write buffer is 8 entries.) 0x4: ENUM_16_ENTRIES (Depth of the Write buffer is 16 entries.) 0x5: ENUM_32_ENTRIES (Depth of the Write buffer is 32 entries.)
10:8	MEMWIDTH	0x2: ENUM_32_BITS (Memory interface databus is 32-bits wide.) 0x3: ENUM_64_BITS (Memory interface databus is 64-bits wide.) 0x4: ENUM_128_BITS (Memory interface databus is 128-bits wide.) 0x5: ENUM_256_BITS (Memory interface databus is 256-bits wide.)
7:6	CONFIGTYPE	This value indicates TMC configuration type - ETB, ETF or ETR configuration. 0x0: TMC_CONFIGURED_AS_ETB_EMBEDDED_TRACE_BUFFER (When configured as an ETB, trace data is stored in local SRAM through the memory interface. Trace data can only be retrieved through APB register reads.) 0x1: TMC_CONFIGURED_AS_ETR_EMBEDDED_TRACE_ROUTER (When configured as an ETR, the TMC stores trace data in the system memory through the AXI interface. Trace data can only be retrieved through APB register reads.) 0x2: Reserved
5	CLKSCHEME	This value indicates the TMC RAM clocking scheme used i.e., whether the TMC RAM operates synchronously or asynchronously to CLK. 0x0: TMC_RAM_SYNCHRONOUS_TO_CLK (The TMC RAM operates synchronously to CLK.)
4:0	RESERVED	Reserved

0x00826FE0 QDSS_ETR_PERIPHID0**Type:** R**Reset State:** 0x00000061

Part of the set of Peripheral Identification registers. Contains part of the designer specific part number.

QDSS_ETR_PERIPHID0

Bits	Name	Description
7:0	PART_NUMBER_BITS7TO0	Bits [7:0] of the component's part number. This is selected by the designer of the component. 0x61: CORESIGHT_TMC_PART_NUMBER_7_0 (Lowest 8 bits of the Part Number (0x961))

0x00826FE4 QDSS_ETR_PERIPHID1**Type:** R**Reset State:** 0x000000B9

Part of the set of Peripheral Identification registers. Contains part of the designer specific part number and part of the designer identity.

QDSS_ETR_PERIPHID1

Bits	Name	Description
7:4	JEP106_BITS3TO0	Bits 3:0 of the JEDEC identity code indicating the designer of the component (along with the continuation code) 0xB: ARM_JEP106_IDENTITY_CODE_7_0 (Lowest 4 bits of the JEP106 Identity Code)
3:0	PART_NUMBER_BITS11TO8	Bits [11:8] of the component's part number. This is selected by the designer of the component. 0x9: CORESIGHT_TMC_PART_NUMBER_11_8 (Upper 4 bits of the Part Number (0x961))

0x00826FE8 QDSS_ETR_PERIPHID2**Type:** R**Reset State:** 0x0000001B

Part of the set of Peripheral Identification registers. Contains part of the designer identity and the product revision.

QDSS_ETR_PERIPHID2

Bits	Name	Description
7:4	REVISION	The Revision field is an incremental value starting at 0x0 for the first design of this component. This only increases by 1 for both major and minor revisions and is simply used as a look-up to establish the exact major/minor revision. 0x1: R0P1 (This device is at r0p1)

QDSS_ETR_PERIPHID2 (cont.)

Bits	Name	Description
3	JEDEC	Always set. Indicates that a JEDEC assigned value is used 0x1: JEDEC_IDENTITY (The designer ID is specified by JEDEC (http://www.jedec.org))
2:0	JEP106_BITS6TO4	Bits 6:4 of the JEDEC identity code indicating the designer of the component (along with the continuation code) 0x3: ARM_JEP106_IDENTITY_CODE_6_4 (Upper 3 bits of the JEP106 Identity Code)

0x00826FEC QDSS_ETR_PERIPHID3**Type:** R**Reset State:** 0x00000000

Part of the set of Peripheral Identification registers. Contains the RevAnd and Customer Modified fields.

QDSS_ETR_PERIPHID3

Bits	Name	Description
7:4	REVAND	This field indicates minor errata fixes specific to this design, for example metal fixes after implementation. In most cases this field is zero. It is recommended that component designers ensure this field can be changed by a metal fix if required, for example by driving it from registers that reset to zero. 0x0: NO_MODS (Indicates that there have been no metal fixes to this component)
3:0	CUSTOMER_MODIFIED	Where the component is reusable IP, this value indicates if the customer has modified the behavior of the component. In most cases this field is zero. 0x0: NO_MODS (Indicates that there have been no modifications made)

0x00826FD0 QDSS_ETR_PERIPHID4**Type:** R**Reset State:** 0x00000004

Part of the set of Peripheral Identification registers. Contains part of the designer identity and the memory footprint indicator.

QDSS_ETR_PERIPHID4

Bits	Name	Description
7:4	FOURKB_COUNT	This is a 4-bit value that indicates the total contiguous size of the memory window used by this component in powers of 2 from the standard 4KB. If a component only requires the standard 4KB then this should read as 0x0, 4KB only, for 8KB set to 0x1, 16KB == 0x2, 32KB == 0x3, and so on. 0x0: ENUM_4KB (Indicates that the device only occupies 4kB of memory)
3:0	JEP106_CONT	JEDEC continuation code indicating the designer of the component (along with the identity code) 0x4: ARM_JEP106_CONTINUATION_CODE (Indicates that ARM's JEDEC identity code is on the 5th bank)

0x00826FD4 QDSS_ETR_PERIPHID5**Type:** RW**Reset State:** Undefined

Reserved

QDSS_ETR_PERIPHID5

Bits	Name	Description
31:0	PERIPHID5	

0x00826FD8 QDSS_ETR_PERIPHID6**Type:** RW**Reset State:** Undefined

Reserved

QDSS_ETR_PERIPHID6

Bits	Name	Description
31:0	PERIPHID6	

0x00826FDC QDSS_ETR_PERIPHID7**Type:** RW**Reset State:** Undefined

Reserved

QDSS_ETR_PERIPHID7

Bits	Name	Description
31:0	PERIPHID7	

0x00826FF0 QDSS_ETR_COMPID0

Type: R

Reset State: 0x0000000D

A component identification register, that indicates that the identification registers are present.

QDSS_ETR_COMPID0

Bits	Name	Description
7:0	PREAMBLE	Contains bits [7:0] of the component identification 0xD: ENUM_OD (Identification value)

0x00826FF4 QDSS_ETR_COMPID1

Type: R

Reset State: 0x00000090

A component identification register, that indicates that the identification registers are present. This register also indicates the component class.

QDSS_ETR_COMPID1

Bits	Name	Description
7:4	CLASS	Class of the component. E.g. ROM table, CoreSight component etc. Constitutes bits [15:12] of the component identification. 0x9: CORESIGHT_COMPONENT (Indicates the component is a CoreSight component)
3:0	PREAMBLE	Contains bits [11:8] of the component identification 0x0: ENUM_0 (Identification value)

0x00826FF8 QDSS_ETR_COMPID2

Type: R

Reset State: 0x00000005

A component identification register, that indicates that the identification registers are present.

QDSS_ETR_COMPID2

Bits	Name	Description
7:0	PREAMBLE	Contains bits [23:16] of the component identification 0x5: ENUM_05 (Identification value)

0x00826FFC QDSS_ETR_COMPID3**Type:** R**Reset State:** 0x000000B1

A component identification register, that indicates that the identification registers are present.

QDSS_ETR_COMPID3

Bits	Name	Description
7:0	PREAMBLE	Contains bits [31:24] of the component identification 0xB1: B1 (Identification value)

12.2 QDSS_AHB**12.2.1 NDPBAM_BAM_TOP****12.2.1.1 QDSS_NDPBAM_BAM_BAM****0x00885100 QDSS_NDPBAM_BAM_BAM_AHB_MASTER_ERR_ADDR_LSB****Type:** R**Clock:** bam_clk**Reset State:** 0x00000000

This Register holds the information about the data error in the AHB Buffer during bus Access.

The value stored is the value of the first error occurred.

Clear the BAM_HRESP_ERR_IRQ interrupt to clear the recorded error.

Available in BAM only

Relevant only on 36 bit address BAM.

QDSS_NDPBAM_BAM_BAM_AHB_MASTER_ERR_ADDR_LSB

Bits	Name	Description
31:0	BAM_ERR_ADDR	32 LSB of HADDR

0x00885104 QDSS_NDPBAM_BAM_BAM_AHB_MASTER_ERR_ADDR_MSB**Type:** R**Clock:** bam_clk**Reset State:** 0x00000000

This Register holds the information about the data error in the AHB Buffer during bus Access.

The value stored is the value of the first error occurred.

Clear the BAM_HRESP_ERR_IRQ interrupt to clear the recorded error.

Available in BAM only

Relevant only on 36 bit address BAM.

QDSS_NDPBAM_BAM_BAM_AHB_MASTER_ERR_ADDR_MSB

Bits	Name	Description
3:0	BAM_ERR_ADDR	4 MSB of Address (bits 35 to 32) of HADDR

**0x00886020+ QDSS_NDPBAM_BAM_BAM_P_TRUST_REGn, n=[0..30]
0x4*n****Type:** RW**Clock:** bam_clk**Reset State:** 0x00000000

This Register holds the BAM Trusted configurations.

This register cannot be reset by any SW reset. Only HW reset applies.

QDSS_NDPBAM_BAM_BAM_P_TRUST_REGn

Bits	Name	Description
12:8	RESERVED	Reserved
7:3	BAM_P_SUP_GROUP	Super-Group is the upper division of the pipes compared to the pipe-group division. When any pipe locks or unlocks the BAM, all the pipes which are in a different super-group will NOT be affected. Only the pipes within the same super-group of the locker are affected. Usually pipes which belongs to different peripherals which share the same BAM, would be in a different super-groups

QDSS_NDPBAM_BAM_BAM_P_TRUST_REGn (cont.)

Bits	Name	Description
2:0	BAM_P_EE	This Field Indicates the EE # (0-7) to which the Pipe Interrupt belongs to. The BAM interrupt will fire to the EE # set at BAM_P_TRUST_REGn registers. The number of supported EEs is reflect in the NUM_EES field in the BAM_REVISION register. 3'b000 - EE0 3'b001 - EE1 3'b010 - EE2 3'b011 - EE3 3'b100 - EE4 3'b101 - EE5 3'b110 - EE6 3'b111 - EE7

**0x00887000+ QDSS_NDPBAM_BAM_BAM_IRQ_SRCS_EEn, n=[0..7]
4096*n****Type:** R**Clock:** bam_clk**Reset State:** 0x00000000

This Interrupt Status Register provides a single interrupt bit per each pipe in the BAM and a single bit for the BAM (not pipe related) interrupt. This register is read only and provides fast indication for the software to decide which pipe triggered the interrupt. After checking this register, SW reads the BAM_P_IRQ_STTSn register for the pipe interrupt reason and BAM_IRQ_STTS for the BAM interrupt reason.

This register has an alias - BAM_IRQ_SRCS register.

Registers with n=[1..7] exist only when the BAM supports multiple EEs (NUM_EES field in the BAM_REVISION register).

QDSS_NDPBAM_BAM_BAM_IRQ_SRCS_EEn

Bits	Name	Description
31	BAM_IRQ	BAM Global interrupt
30:0	P_IRQ	An interrupt bit per each pipe in the BAM.

**0x00887004+ QDSS_NDPBAM_BAM_BAM_IRQ_SRCS_MSK_EEn, n=[0..7]
4096*n****Type:** RW**Clock:** bam_clk**Reset State:** 0x00000000

Interrupt Sources Mask register allows to mask the interrupts by their source, whether BAM or a specific pipe.

This register has an alias - BAM_IRQ_SRCS_MSK register.

Registers with n=[1..7] exist only when the BAM supports multiple EEs (NUM_EES field in the BAM_REVISION register).

QDSS_NDPBAM_BAM_BAM_IRQ_SRCS_MSK_EEn

Bits	Name	Description
31	BAM_IRQ_MSK	BAM Global Interrupt Mask 0x1: ENABLE_BAM_INTERRUPT (Enable BAM interrupt) 0x0: DISABLE_BAM_INTERRUPT (Disable BAM interrupt)
30:0	P_IRQ_MSK	An Interrupt Mask bit per each pipe in the BAM. 0x1: ENABLE_PIPE_INTERRUPT (Enable Pipe interrupt) 0x0: DISABLE_PIPE_INTERRUPT (Disable Pipe interrupt)

0x00887008+ QDSS_NDPBAM_BAM_BAM_IRQ_SRCS_UNMASKED_EEn, n=[0..7] 4096*n

Type: R

Clock: bam_clk

Reset State: 0x00000000

The register shows the interrupts sources like (BAM_IRQ_SRCS) although the interrupts from the BAM and the pipes are not masked on the lower level.

This register has an alias - BAM_IRQ_SRCS_UNMASKED register.

Registers with n=[1..7] exist only when the BAM supports multiple EEs (NUM_EES field in the BAM_REVISION register).

QDSS_NDPBAM_BAM_BAM_IRQ_SRCS_UNMASKED_EEn

Bits	Name	Description
31	BAM_IRQ_UNMASKED	BAM Global Interrupt after Mask
30:0	P_IRQ_UNMASKED	An Interrupt bit per each pipe in the BAM, after Mask

0x0088700C+ QDSS_NDPBAM_BAM_BAM_PIPE_ATTR_EEn, n=[0..7] 4096*n

Type: R

Clock: bam_clk

Reset State: 0x00000000

The register shows the pipes attributed to a specific EE, and an indication if BAM is enabled.

The reset value written above is true only if EE>0. For BAM_PIPE_ATTR_EE0 the default value is: Bits [MAX_PIPES-1:0] = 1, Bits [31: MAX_PIPES] = 0.

QDSS_NDPBAM_BAM_BAM_PIPE_ATTR_EEn

Bits	Name	Description
31	BAM_ENABLED	1'b1 - BAM is Enabled. 1'b0 - BAM Disabled.
30:0	P_ATTR	If bit 'i' == 1'b1 - Pipe 'i' is attributed to this EE.

0x00897000+ QDSS_NDPBAM_BAM_BAM_P_CTRLn, n=[0..30] 4096*n

Type: RW

Clock: bam_clk

Reset State: 0x00000000

Pipe Control register provides various controls for the pipe.

QDSS_NDPBAM_BAM_BAM_P_CTRLn

Bits	Name	Description
20:16	P_LOCK_GROUP	Pipe's lock group. Upon a lock request on this pipe, all pipes related to different pipe group and different EE (see PIPE_TRUST_REG) will be locked.
11	P_WRITE_NWD	BAM-Lite feature. Applicable to BAM2BAM producer pipes only. When this bit is set for producer B2B pipe, NWD bit (bit number 28 in the second word of descriptor) will be written with EOT into the generated descriptor for the consumer usage. This bit is not applicable in pipes belong to a Peripheral which uses messaging_only (e.g. USB2).
10:9	P_PREFETCH_LIMIT	Bam Lite feature Selects the number of bytes to be prefetched by the Bam Lite, once a peripheral requests to read data from a pipe. This feature is relevant for Consumer modes only. 2'b00 - 32 bytes at most (INCR8) 2'b01 - 16 bytes at most (INCR4) 2'b10 - 4 bytes at most (SINGLE) Available in BAM-Lite only

QDSS_NDPBAM_BAM_BAM_P_CTRLn (cont.)

Bits	Name	Description
8:7	P_AUTO_EOB_SEL	Bam Lite feature. Selects the number of bytes to be processed before creating End Of Block indication internally by the BAM. See P_AUTO_EOB. Descriptors generated by this feature (bam to bam producer mode) will always be of the chosen size, unless an End Of Transfer, causes a smaller descriptor to be created. 2'b00 - 512 Bytes 2'b01 - 256 Bytes 2'b10 - 128 Bytes 2'b11 - 64 Bytes Available in BAM-Lite only
6	P_AUTO_EOB	BAM Lite feature. When Enabled, the BAM Lite will ignore any incoming End Of Block indications from the peripheral (if any). Instead, the BAM Lite will generate it's own End Of Block indications internally, every P_AUTO_EOB_SEL bytes. This feature is relevant for Producer Bam 2 Bam modes only. It is a must for peripherals which are not able to produce End Of Block indications. 1'b1 - Enable 1'b0 - Disable Available in BAM-Lite only
5	P_SYS_MODE	This bit sets the pipe to work in System Mode, where the Software is the second side of the pipe. 1'b1 - System mode. 1'b0 - BAM to BAM mode.
4	P_SYS_STRM	Streaming Enable. Relevant to System Producer BAM mode only. When enabled, BAM will not close descriptors with End Of Transfer notifications received from peripheral, but continue writing the data to the same descriptor. 1'b1 - Enable Streaming Mode 1'b0 - None-Streaming Mode. This field is NOT supported in BAM-NDP (i.e. no streaming mode in NDP).
3	P_DIRECTION	This bit denotes pipe direction. 1'b1 - The pipe can be written (producer mode) 1'b0 - The pipe can be read (consumer mode)
1	P_EN	1'b1 - Enable Pipe 1'b0 - Disable Pipe

**0x00897004+ QDSS_NDPBAM_BAM_BAM_P_RSTn, n=[0..30]
4096*n****Type:** W**Clock:** bam_clk**Reset State:** 0x00000000

Pipe Reset register provides Pipe Reset ability. Software needs to write 1 to this register to make the Pipe Reset. Writing 0 to this register will finish the Pipe Reset. Writing zero is needed prior to Peripheral reconfiguration for the same pipe.

Note: Software can invoke pipe reset only after it make sure that the pipe is 'quiet'.

('quiet' - No data transaction or command-descriptor is active at the time.

BAM_P_RSTn, n=[0..30]

QDSS_NDPBAM_BAM_BAM_P_RSTn

Bits	Name	Description
0	P_SW_RST	This resets the pipe and it's registers, (Both Flip-Flops and RAM). Software can invoke pipe reset only after it make sure that the pipe is 'quiet'. 1'b1 - Reset 1'b0 - Do Nothing

**0x00897008+ QDSS_NDPBAM_BAM_BAM_P_HALTn, n=[0..30]
4096*n****Type:** RW**Clock:** bam_clk**Reset State:** 0x00000000

Pipe Halt register Enables/Disables the Halt Sequence. Refer to the Software Procedures chapter for the Halt Sequence Procedure.

This is a self-modifying register.

QDSS_NDPBAM_BAM_BAM_P_HALTn

Bits	Name	Description
1	P_PROD_HALTED	This bit is used to inform the Consumer about the Producer being in Halt mode now. This is a part of the Halt procedure. See the Software Procedures chapter. 1'b1 - Sets this bit to 1 1'b0 - Does Nothing This bit will be cleared by the HW.

QDSS_NDPBAM_BAM_BAM_P_HALTn (cont.)

Bits	Name	Description
0	P_HALT	When Enabled, the Pipe will enter Halt Mode. 1'b1 - Enable Halt 1'b0 - Disable Halt For B2B Producer pipes, this bit is self modifying. It doesn't need the SW to clear it. See the Software Procedure chapter for the detailed flow.

**0x00897010+ QDSS_NDPBAM_BAM_BAM_P_IRQ_STTSn, n=[0..30]
4096*n****Type:** R**Clock:** bam_clk**Reset State:** 0x00000000

Pipe Interrupt status register describing the reason for the ongoing interrupt. This is an informative only register and therefore read only. It also has informative bits which are not interrupts.

Clearing the interrupt bits is done by writing to P_IRQ_CLR register.

QDSS_NDPBAM_BAM_BAM_P_IRQ_STTSn

Bits	Name	Description
7	P_HRESP_ERR_IRQ	This interrupt Indicates that an Erroneous HResponse has been received by the AHB Master. Additional Information about the error may be read at BAM_AHB_MASTER_ERR_* Registers. Clearing this interrupt also clears the BAM_AHB_MASTER_ERR_* Registers. Available in BAM only
6	P_PIPE_RST_ERROR_IRQ	Unsuccessful Pipe Reset operation. Pipe reset timer expired.
5	P_TRNSFR_END_IRQ	For producer this interrupt happens whenever the last AHB burst of a transfer happens For Consumer when ack_on_success closes a transfer
4	P_ERR_IRQ	Pipe Error interrupt indicates that an error has happened. TBD: Error reason is not yet defined. Interrupt not in use.
3	P_OUT_OF_DESC_IRQ	Out of descriptors interrupt has several meanings depending on the operation mode: Producer - no free space in the descriptors fifo for adding a descriptor Consumer - no descriptors in the descriptors fifo for processing IO Vectors Producer - No descriptors to read (and therefore no space for data to write to the pipe) IO Vectors Consumer - No descriptors to read (and therefore no data to read from pipe)

QDSS_NDPBAM_BAM_BAM_P_IRQ_STTSn (cont.)

Bits	Name	Description
2	P_WAKE_IRQ	Wake peripheral interrupt signals the driver to wake up the peripheral (USB for example) for transmission via the current pipe. This interrupt is issued whenever an Event comes to the specific pipe.
1	P_TIMER_IRQ	Inactivity timer has reached its threshold value Interrupt
0	P_PRCSD_DESC_IRQ	This Interrupt rises when BAM finishes processing an IO Vector which has INT bit selected

**0x00897014+ QDSS_NDPBAM_BAM_BAM_P_IRQ_CLRn, n=[0..30]
4096*n****Type:** W**Clock:** bam_clk**Reset State:** 0x00000000

Writing to this register causes the interrupt to clear.

BAM_P_IRQ_CLRn, n=[0..30]

QDSS_NDPBAM_BAM_BAM_P_IRQ_CLRn

Bits	Name	Description
7	P_HRESP_ERR_CLR	1'b1 - Clear 1'b0 - Unchanged
6	P_PIPE_RST_ERROR_CLR	1'b1 - Clear 1'b0 - Unchanged
5	P_TRNSFR_END_CLR	1'b1 - Clear 1'b0 - Unchanged
4	P_ERR_CLR	1'b1 - Clear 1'b0 - Unchanged
3	P_OUT_OF_DESC_CLR	1'b1 - Clear 1'b0 - Unchanged
2	P_WAKE_CLR	Clear Wake peripheral interrupt by writing this bit 1'b1 - Clear 1'b0 - Unchanged
1	P_TIMER_CLR	Clear Inactivity timer interrupt by writing this bit 1'b1 - Clear 1'b0 - Unchanged
0	P_PRCSD_DESC_CLR	Clear Descriptor Processed interrupt by writing this bit 1'b1 - Clear 1'b0 - Unchanged

**0x00897018+ QDSS_NDPBAM_BAM_BAM_P_IRQ_ENn, n=[0..30]
4096*n****Type:** RW**Clock:** bam_clk**Reset State:** 0x00000000

Enable bits for the Interrupts in the Pipe Interrupt Status register.

QDSS_NDPBAM_BAM_BAM_P_IRQ_ENn

Bits	Name	Description
7	P_HRESP_ERR_EN	1'b1 - Enable 1'b0 - Disable
6	P_PIPE_RST_ERROR_EN	1'b1 - Enable 1'b0 - Disable
5	P_TRNSFR_END_EN	1'b1 - Enable 1'b0 - Disable
4	P_ERR_EN	1'b1 - Enable 1'b0 - Disable
3	P_OUT_OF_DESC_EN	1'b1 - Enable 1'b0 - Disable
2	P_WAKE_EN	Enable Wake peripheral interrupt 1'b1 - Enable 1'b0 - Disable
1	P_TIMER_EN	Enable Inactivity timer Interrupt 1'b1 - Enable 1'b0 - Disable
0	P_PRCSD_DESC_EN	Enable Descriptor Processed Interrupt 1'b1 - Enable 1'b0 - Disable

**0x0089701C+ QDSS_NDPBAM_BAM_BAM_P_TIMERn, n=[0..30]
4096*n****Type:** R**Clock:** bam_clk**Reset State:** 0x00000000

This register counts the idle time of the pipe.

BAM_P_TIMERn, n=[0..30]

QDSS_NDPBAM_BAM_BAM_P_TIMERn

Bits	Name	Description
15:0	P_TIMER	Inactivity timer counter. 16 bits = 64k*0.1ms ~ 6 seconds

**0x00897020+ QDSS_NDPBAM_BAM_BAM_P_TIMER_CTRLn, n=[0..30]
4096*n****Type:** RW**Clock:** bam_clk**Reset State:** 0x00000000

This register sets various controls and the threshold of the inactivity for interrupt triggering after long idle time.

The resolution of the pipe inactivity timers are in units set by a common counter. This common counter is controlled by a PARAMETER value of the BAM, and uses a separate clock, the inactivity_timers_clk. This clock can be slower than the bam_clk. The intent of the design is to use the sleep_clk, which is an always on 32kHz clock. This allows the bam_clk to be turned-off, while the base counter keeps counting. The width of this counter, as well of the rollover value are set using BAM PARAMETER. These values, taking the inactivity_timers_clk frequency define the pulse frequency that the base counter gives, and the threshold value can be calculated and set to interrupt after a specific time, regardless of the bam_clk frequency, and independent of clock power save features of the bam_clk. The timers continue counting even if the clock is turned off due to BAM dynamic hardware clock request voting.

The timer configuration is based on the inactivity_timers_clk period and the INACTIVITY_TIMER_WIDTH parameter constant. These parameters should be taken into account when setting this register. For example: for inactivity_timer_clk period of 1us and parameter is 3 and P_TIMER_THRSHLD is 10 will indicate the $2^3 * 1\mu s * 10$ which is 80us of inactivity in a pipe before sending an interrupt.

The general formula is: $2^{\text{INACTIVITY_TIMER_WIDTH}} * \text{clock_period} * \text{P_TIMER_TRSHLD}$.

QDSS_NDPBAM_BAM_BAM_P_TIMER_CTRLn

Bits	Name	Description
31	P_TIMER_RST	To begin using the inactivity timer the SW should write the threshold and write 1 to this bit. writing the value 0 will hold the timer reset until value 1 is written again. Then the timer will start down counting from the P_TIMER_THRESHOLD value 1'b1 - Active 1'b0 - Reset
30	P_TIMER_RUN	Will be used along with P_TIMER_MODE Not implemented at this time. Set to zero(0)
29	P_TIMER_MODE	Not implemented at this time. Set to zero(0)

QDSS_NDPBAM_BAM_BAM_P_TIMER_CTRLn (cont.)

Bits	Name	Description
15:0	P_TIMER_TRSHLD	Inactivity timer threshold for interrupt generation.

**0x00897024+ QDSS_NDPBAM_BAM_BAM_P_PRDCR_SDBNDn, n=[0..30]
4096*n****Type:** R**Clock:** bam_clk**Reset State:** 0x00000000

This Register reflects the current status of the Producer Sideband interface as the peripheral sees it and some additional internal bits related to its management.

QDSS_NDPBAM_BAM_BAM_P_PRDCR_SDBNDn

Bits	Name	Description
24	BAM_P_SB_UPDATED	sideband_inform_bytes_avail_updated_producer. relevant for system mode only. The sideband-Inform block which responsible for reading descriptor has updated the SB block (which responsible for publishing to peripheral) with all of its read descriptors.
20	BAM_P_TOGGLE	Pipe Bytes Free Toggle value. The toggle polarity of the publishment (each publishment changes its polarity).
19:16	BAM_P_CTRL	Pipe Bytes Free Ctrl value. The possible values are only: 0000 or 0001 (means desc-fifo-full).
15:0	BAM_P_BYTES_FREE	Pipe Bytes Free value. The accumulated bytes_free which are available for peripheral to use

**0x00897028+ QDSS_NDPBAM_BAM_BAM_P_CNSMR_SDBNDn, n=[0..30]
4096*n****Type:** R**Clock:** bam_clk**Reset State:** 0x00000000

This Register reflects the current status of the Consumer Sideband interface as the peripheral sees it and some additional internal bits related to its management.

QDSS_NDPBAM_BAM_BAM_P_CNSMR_SDBNDn

Bits	Name	Description
24	BAM_P_SB_UPDATED	sideband_inform_bytes_avail_updated_consumer. The sideband-Inform block which responsible for reading descriptor has updated the SB block (which responsible for publishing to peripheral) with all of its read descriptors.

QDSS_NDPBAM_BAM_BAM_P_CNSTR_SDBNDn (cont.)

Bits	Name	Description
23	BAM_P_WAIT_4_ACK	waiting_for_ack_bytes_avail_r. BAM is waiting for Peripheral to acknowledge the former publish.
22	BAM_P_ACK_TOGGLE	ack_bytes_avail_toggle. The Peripheral's acknowledgement for the publish. This is a toggling signal (each ack will be inverted to its former).
21	BAM_P_ACK_TOGGLE_R	ack_bytes_avail_toggle_r. The Peripheral's sampled acknowledgement for the publish. This is a toggling signal (each ack will be inverted to its former).
20	BAM_P_TOGGLE	Pipe Bytes Avail Toggle value. The toggle polarity of the publishment (each publishment changes its polarity).
19:16	BAM_P_CTRL	Pipe Bytes Avail Ctrl value. The possible values of BAM_P_CTRL field are: 0100 - Means immediate command (could be only in bam-ndp). 0001 - Means End-Of-Transaction (EOT). 0011 - means End-Of-Transaction with Notify-When-Done (NWD). 0000 - Neither Immediate-command nor EOT nor NWD.
15:0	BAM_P_BYTES_AVAIL	Pipe Bytes AVAIL value. The size of the current publish.

**0x00897800+ QDSS_NDPBAM_BAM_BAM_P_SW_OFSTSn, n=[0..30]
4096*n****Type:** RW**Clock:** bam_clk**Reset State:** 0x00000000

This register denotes the pointer Offset of the first Not Acknowledged Descriptor. Meaning all Descriptors prior to this one have been processed by the BAM. This register is only used by the Software. After receiving an interrupt, software reads this register in order to know what descriptors has been processed.

When the BAM works as Producer, Software will usually read this value after Descriptor Processed Interrupt or after the End of Transfer Interrupt.

When the BAM works as Consumer, Software will usually read this value after Descriptor Processed Interrupt.

NOTE This is non relevant in BAM to BAM modes.

NOTE Although being Writable, Software should never write to this register.

NOTE When using 2 pipes BAM, REVISION 0x03, this registers is Read Only.

BAM_P_SW_OFSTSn, n=[0..30]

QDSS_NDPBAM_BAM_BAM_P_SW_OFSTSn

Bits	Name	Description
31:16	SW_OFST_IN_DESC	Offset inside the Descriptor. This value is used by the Software only when a BAM is working as System Producer. Then the SW can read this value to know how many Bytes were written to the current descriptor if it was not closed down. This is useful for Streaming mode. This Field is NOT available in BAM-NDP
15:0	SW_DESC_OFST	Descriptor FIFO offset.

**0x00897818+ QDSS_NDPBAM_BAM_BAM_P_EVNT_REGn, n=[0..30]
4096*n****Type:** RW**Clock:** bam_clk**Reset State:** 0x00000000

This Register is where Read/Write events are sent to. If current BAM works as Producer, it will receive Read events into this register from the Consumer. If the Consumer is another BAM, it will write read events to this register. If the Consumer is the Software, it will prepare descriptors in the descriptor FIFO and update the DESC_FIFO_PEER_OFST value in this register.

If current BAM works as Consumer, opposite behavior is assumed.

BAM_P_EVNT_REGn, n=[0..30]

QDSS_NDPBAM_BAM_BAM_P_EVNT_REGn

Bits	Name	Description
31:16	P_BYTES_CONSUMED	This field is used only in the B2B mode. It indicates the number of bits consumed by the Consumer. Consumer sends this value to the Producer as a part of a read event, for the Producer to know how many bytes were consumed. When working in System mode, this value should be set to zero. BAM to BAM Producer writes zero in this value.
15:0	P_DESC_FIFO_PEER_OFST	This field indicates the Descriptor Fifo Offset Pointer of the Peer BAM (The BAM or the Software which creates descriptors). For example when current BAM works as consumer for the current pipe, Producer BAM (or the software) updates this value to let the current pipe know about the new descriptors added in the Descriptor Fifo.

**0x0089781C+ QDSS_NDPBAM_BAM_BAM_P_DESC_FIFO_ADDRn, n=[0..30]
4096*n****Type:** RW**Clock:** bam_clk**Reset State:** 0x00000000

This register stores the Address of the Descriptor Fifo beginning.

NOTE This register is used by all modes.

BAM_P_DESC_FIFO_ADDRn, n=[0..30]

QDSS_NDPBAM_BAM_BAM_P_DESC_FIFO_ADDRn

Bits	Name	Description
31:0	P_DESC_FIFO_ADDR	Address of the Descriptors Fifo. This address must be 8 bytes aligned.

**0x00897820+ QDSS_NDPBAM_BAM_BAM_P_FIFO_SIZESn, n=[0..30]
4096*n****Type:** RW**Clock:** bam_clk**Reset State:** 0x00000000

The least significant field is the Descriptor Fifo Size.

The most significant field is the Data Fifo Size.

NOTE This Descriptor Fifo Size is used by all modes. The Data Fifo Size is non relevant in System Modes.

BAM_P_FIFO_SIZESn, n=[0..30]

QDSS_NDPBAM_BAM_BAM_P_FIFO_SIZESn

Bits	Name	Description
31:16	P_DATA_FIFO_SIZE	Size of the Data Address Space. ALWAYS write 16'b0 to this field of P_DATA_FIFO_SIZE unless you work in BAM2BAM producer mode (also known as peer mode). NOTE: Data fifo size should not be bigger than 32KB.
15:0	P_DESC_FIFO_SIZE	Size of the Descriptors Fifo. Must be 8 byte aligned. Each descriptor is 8 bytes. Size of the descriptor fifo must contain an integer number of Descriptors. Size of descriptor fifo must be 16 bytes and above.

**0x00897824+ QDSS_NDPBAM_BAM_BAM_P_DATA_FIFO_ADDRn, n=[0..30]
4096*n****Type:** RW**Clock:** bam_clk**Reset State:** 0x00000000

This register stores the Beginning of the Data Address space.

This register is non relevant in System Modes.

BAM_P_DATA_FIFO_ADDRn, n=[0..30]

QDSS_NDPBAM_BAM_BAM_P_DATA_FIFO_ADDRn

Bits	Name	Description
31:0	P_DATA_FIFO_ADDR	Data Address Space beginning.

**0x00897828+ QDSS_NDPBAM_BAM_BAM_P_EVNT_GEN_TRSHLDn, n=[0..30]
4096*n****Type:** RW**Clock:** bam_clk**Reset State:** 0x00000000

This register configures the threshold value for Read/Write event generation by the BAM towards another BAM. When aBAM pushes/pops this number of bytes to/from the pipe an event to the other BAM will be generated. A counter summing up the bytes pushed/popped is reset when event is issued, thus restarting count for the threshold.

Events are also issued after End Of Transfer received from the peripheral (Producer case), non regarding whether the pushed bytes count has reached the threshold or not.

This register is non relevant in System Modes.

BAM_P_EVNT_GEN_TRSHLDn, n=[0..30]

QDSS_NDPBAM_BAM_BAM_P_EVNT_GEN_TRSHLDn

Bits	Name	Description
15:0	P_TRSHLD	Threshold value. The maximum allowed value is 32kByte.

**0x0089782C+ QDSS_NDPBAM_BAM_BAM_P_EVNT_DEST_ADDRn, n=[0..30]
4096*n****Type:** RW**Clock:** bam_clk**Reset State:** 0x00000000

This register stores the Event Destination Address which is the address of BAM_P_EVNT_REG in the BAM on the other side of the Pipe.

Producer BAM will send Write Events to this address.

Consumer BAM will send Read Events to this address.

This is relevant only for BAM to BAM modes. In System modes, BAM doesn't issue Write/Read events.

BAM_P_EVNT_DEST_ADDRn, n=[0..30]

QDSS_NDPBAM_BAM_BAM_P_EVNT_DEST_ADDRn

Bits	Name	Description
31:0	P_EVNT_DEST_ADDR	Address of the Event Register in the peer BAM (second end of the pipe). Driver will configure this when initializing the pipe.

0x00897900+ 4096*n QDSS_NDPBAM_BAM_BAM_P_PSM_CNTXT_3_LSBn, n=[0..30]

Type: RW

Clock: bam_clk

Reset State: 0x00000000

Internal Context of Pipe & Writeback state machines.

Relevant only on 36 bit address BAM.

NOTE When using 2 pipes BAM, REVISION 0x03, this registers is Read Only.

BAM_P_PSM_CNTXT_3_n, n=[0..30]

QDSS_NDPBAM_BAM_BAM_P_PSM_CNTXT_3_LSBn

Bits	Name	Description
31:0	PSM_DESC_ADDR	32 LSB of The Data Address provided in the current descriptor from the Writeback state machine.

0x00897904+ 4096*n QDSS_NDPBAM_BAM_BAM_P_PSM_CNTXT_3_MSBn, n=[0..30]

Type: RW

Clock: bam_clk

Reset State: 0x00000000

Internal Context of Pipe & Writeback state machines.

Relevant only on 36 bit address BAM.

NOTE When using 2 pipes BAM, REVISION 0x03, this registers is Read Only.

BAM_P_PSM_CNTXT_3_n, n=[0..30]

QDSS_NDPBAM_BAM_BAM_P_PSM_CNTXT_3_MSBn

Bits	Name	Description
3:0	PSM_DESC_ADDR	4 MSB of address (bits 35 to 32) of the Data Address provided in the current descriptor from the Writeback state machine.

0x00897910+ 4096*n QDSS_NDPBAM_BAM_BAM_P_DESC_FIFO_ADDR_LSBn, n=[0..30]

Type: RW

Clock: bam_clk

Reset State: 0x00000000

This register stores the LSB of Address of the Descriptor Fifo beginning.

Relevant only on 36 bit address BAM.

NOTE This register is used by all modes.

BAM_P_DESC_FIFO_ADDRn, n=[0..30]

QDSS_NDPBAM_BAM_BAM_P_DESC_FIFO_ADDR_LSBn

Bits	Name	Description
31:0	P_DESC_FIFO_ADDR	32 LSB of address of the Descriptors Fifo. This address must be 8 bytes aligned.

0x00897914+ 4096*n QDSS_NDPBAM_BAM_BAM_P_DESC_FIFO_ADDR_MSBn, n=[0..30]

Type: RW

Clock: bam_clk

Reset State: 0x00000000

This register stores the MSB of Address of the Descriptor Fifo beginning.

Relevant only on 36 bit address BAM.

NOTE This register is used by all modes.

BAM_P_DESC_FIFO_ADDRn, n=[0..30]

QDSS_NDPBAM_BAM_BAM_P_DESC_FIFO_ADDR_MSBn

Bits	Name	Description
3:0	P_DESC_FIFO_ADDR	4 MSB of address (bits 35 to 32) of Address of the Descriptors Fifo. This address must be 8 bytes aligned.

**0x00897920+ QDSS_NDPBAM_BAM_BAM_P_DATA_FIFO_ADDR_LSBn, n=[0..30]
4096*n****Type:** RW**Clock:** bam_clk**Reset State:** 0x00000000

This register stores the Beginning of the Data Address space.

This register is non relevant in System Modes.

Relevant only on 36 bit address BAM.

BAM_P_DATA_FIFO_ADDRn, n=[0..30]

QDSS_NDPBAM_BAM_BAM_P_DATA_FIFO_ADDR_LSBn

Bits	Name	Description
31:0	P_DATA_FIFO_ADDR	32 LSB of Data Address Space beginning. Relevant only on 36 bit address BAM.

**0x00897924+ QDSS_NDPBAM_BAM_BAM_P_DATA_FIFO_ADDR_MSBn, n=[0..30]
4096*n****Type:** RW**Clock:** bam_clk**Reset State:** 0x00000000

This register stores the Beginning of the Data Address space.

This register is non relevant in System Modes.

Relevant only on 36 bit address BAM.

BAM_P_DATA_FIFO_ADDRn, n=[0..30]

QDSS_NDPBAM_BAM_BAM_P_DATA_FIFO_ADDR_MSBn

Bits	Name	Description
3:0	P_DATA_FIFO_ADDR	4 MSB of Address (bits 35 to 32) of Data Address Space beginning. Relevant only on 36 bit address BAM.

**0x00897930+ QDSS_NDPBAM_BAM_BAM_P_EVNT_DEST_ADDR_LSBn, n=[0..30]
4096*n****Type:** RW**Clock:** bam_clk**Reset State:** 0x00000000

This register stores the Event Destination Address which is the address of BAM_P_EVNT_REG in the BAM on the other side of the Pipe.

Producer BAM will send Write Events to this address.

Consumer BAM will send Read Events to this address.

This is relevant only for BAM to BAM modes. In System modes, BAM doesn't issue Write/Read events.

Relevant only on 36 bit address BAM.

BAM_P_EVNT_DEST_ADDRn, n=[0..30]

QDSS_NDPBAM_BAM_BAM_P_EVNT_DEST_ADDR_LSBn

Bits	Name	Description
31:0	P_EVNT_DEST_ADDR	32 LSB of Address of the Event Register in the peer BAM (second end of the pipe). Driver will configure this when initializing the pipe. Relevant only on 36 bit address BAM.

**0x00897934+ QDSS_NDPBAM_BAM_BAM_P_EVNT_DEST_ADDR_MSBn, n=[0..30]
4096*n****Type:** RW**Clock:** bam_clk**Reset State:** 0x00000000

This register stores the Event Destination Address which is the address of BAM_P_EVNT_REG in the BAM on the other side of the Pipe.

Producer BAM will send Write Events to this address.

Consumer BAM will send Read Events to this address.

This is relevant only for BAM to BAM modes. In System modes, BAM doesn't issue Write/Read events.

Relevant only on 36 bit address BAM.

BAM_P_EVNT_DEST_ADDRn, n=[0..30]

QDSS_NDPBAM_BAM_BAM_P_EVNT_DEST_ADDR_MSBn

Bits	Name	Description
3:0	P_EVNT_DEST_ADDR	4 MSB of Address (bits 35 to 32) of the Event Register in the peer BAM (second end of the pipe). Driver will configure this when initializing the pipe. Relevant only on 36 bit address BAM.

13 SPDM Wrapper

13.1 SPDM_SPDM_OLEM

0x00041020+ SPDM_OLEM_CYCLE_Cn_OUT, n=[0..15]
0x4*n

Type: R
Clock: HCLK
Reset State: 0x00000000

The OLEM_CYCLE_Cn_OUT register configures the following:

SPDM_OLEM_CYCLE_Cn_OUT

Bits	Name	Description
31:0	RESULT	This value is the number of clock cycles that were detected during the interval period.

0x000410A0+ SPDM_OLEM_EVENT_Cn_OUT, n=[0..31]
0x4*n

Type: R
Clock: HCLK
Reset State: 0x00000000

The OLEM_EVENT_Cn_OUT register configures the following:

SPDM_OLEM_EVENT_Cn_OUT

Bits	Name	Description
31:0	RESULT	This value is the number of event cycles that were detected during the interval period.

0x00041120+ SPDM_OLEM_MULTIBIT_Cn_OUT, n=[0..7]
0x4*n

Type: R
Clock: HCLK
Reset State: 0x00000000

The OLEM_MULTIBIT_Cn_OUT register configures the following:

SPDM_OLEM_MULTIBIT_Cn_OUT

Bits	Name	Description
31:0	RESULT	This value is the accumulation of the multi-bit values coming from the realtime monitors during the interval period.

14 System NoC

0x00580008

SNOC_OBS_FAULTEN

Type: RW

Reset State: 0x00000000

SNOC_OBS_FAULTEN

Bits	Name	Description
0	FAULTEN	Enable Fault output

0x0058000C SNOC_OBS_ERRVLD

Type: R

Reset State: Undefined

SNOC_OBS_ERRVLD

Bits	Name	Description
0	ERRVLD	Error logged Valid

0x00580010 SNOC_OBS_ERRCLR

Type: RW

Reset State: 0x00000000

SNOC_OBS_ERRCLR

Bits	Name	Description
0	ERRCLR	Clr ErrVld status

0x00580014 SNOC_OBS_ERRLOG0**Type:** R**Reset State:** 0b1XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX

Header: Lock, Opcode, Len1, ErrCode values

SNOC_OBS_ERRLOG0

Bits	Name	Description
31	FORMAT	Format of ErrLog0 register
25:16	LEN1	Header: Len1 value
10:8	ERRCODE	Header: Error Code value
4:1	OPC	Header: Opcode value
0	LOCK	Header: Lock bit value

0x00580018 SNOC_OBS_ERRLOG1**Type:** R**Reset State:** Undefined**SNOC_OBS_ERRLOG1**

Bits	Name	Description
28:0	ERRLOG1	Header: Routed Isb value

0x00580020 SNOC_OBS_ERRLOG3**Type:** R**Reset State:** Undefined**SNOC_OBS_ERRLOG3**

Bits	Name	Description
31:0	ERRLOG3	Header: Addr Isb value

0x00580024 SNOC_OBS_ERRLOG4**Type:** R**Reset State:** Undefined

SNOC_OBS_ERRLOG4

Bits	Name	Description
0	ERRLOG4	Header: Addr msb value

0x00580028 SNOC_OBS_ERRLOG5**Type:** R**Reset State:** Undefined**SNOC_OBS_ERRLOG5**

Bits	Name	Description
15:0	ERRLOG5	Header: User lsb value

0x00581024 SNOC_MSS_TFILT_USERBASE**Type:** RW**Reset State:** 0x00000000**SNOC_MSS_TFILT_USERBASE**

Bits	Name	Description
15:0	USERBASE	This register contains the User base used to filter requests.

0x00581028 SNOC_MSS_TFILT_USERMASK**Type:** RW**Reset State:** 0x00000000**SNOC_MSS_TFILT_USERMASK**

Bits	Name	Description
15:0	USERMASK	This register contains the User mask used to filter requests.

0x005810A4 SNOC_MSS_NAV_TFILT_USERBASE**Type:** RW**Reset State:** 0x00000000

SNOC_MSS_NAV_TFILT_USERBASE

Bits	Name	Description
15:0	USERBASE	This register contains the User base used to filter requests.

0x005810A8 SNOC_MSS_NAV_TFILT_USERMASK**Type:** RW**Reset State:** 0x00000000**SNOC_MSS_NAV_TFILT_USERMASK**

Bits	Name	Description
15:0	USERMASK	This register contains the User mask used to filter requests.

0x00581100 SNOC_VENUS0_TFILT_ID_COREID**Type:** R**Reset State:** 0x785A6C09**SNOC_VENUS0_TFILT_ID_COREID**

Bits	Name	Description
31:8	CORECHECKSUM	Field containing a checksum of the parameters of the IP.
7:0	CORETYPEID	Field identifying the type of IP.

0x00581104 SNOC_VENUS0_TFILT_ID_REVISIONID**Type:** R**Reset State:** 0x01222B00**SNOC_VENUS0_TFILT_ID_REVISIONID**

Bits	Name	Description
31:8	FLEXNOCID	Field containing the build revision of the software used to generate the IP HDL code.
7:0	USERID	Field containing a user defined value, not used anywhere inside the IP itself.

0x00581108 SNOC_VENUS0_TFILT_MODE**Type:** RW**Reset State:** 0x00000000**SNOC_VENUS0_TFILT_MODE**

Bits	Name	Description
0	MODE	Register Mode is a 1-bit register that sets the filtering mode as follows: handshake Mode = 0 or latency Mode = 1.

0x0058110C SNOC_VENUS0_TFILT_ADDRBASE_LOW**Type:** RW**Reset State:** 0x00000000**SNOC_VENUS0_TFILT_ADDRBASE_LOW**

Bits	Name	Description
31:0	ADDRBASE_LOW	Address base LSB register.

0x00581110 SNOC_VENUS0_TFILT_ADDRBASE_HIGH**Type:** RW**Reset State:** 0x00000000**SNOC_VENUS0_TFILT_ADDRBASE_HIGH**

Bits	Name	Description
0	ADDRBASE_HIGH	Address base MSB register.

0x00581114 SNOC_VENUS0_TFILT_ADDRWINDOWSIZE**Type:** RW**Reset State:** 0x00000000**SNOC_VENUS0_TFILT_ADDRWINDOWSIZE**

Bits	Name	Description
5:0	ADDRWINDOWSIZE	Register AddrWindowSize contains the encoded address mask used to filter packets: the effective Mask value is equal to $\sim(2^{**}\text{AddrWindowSize} - 1)$. A packet is a candidate when $\text{ReqInfo.Addr} \& \text{AddrMask} = \text{AddrBase} \& \text{AddrMask}$.

0x00581118 SNOC_VENUS0_TFILT_SRCIDBASE**Type:** RW**Reset State:** 0x00000000**SNOC_VENUS0_TFILT_SRCIDBASE**

Bits	Name	Description
14:0	SRCIDBASE	Reserved, null-width register. Register SrcIdBase is only available for specific designs, and contains the source ID base used to filter packets. The number of bits is set with parameter wSrcId.

0x0058111C SNOC_VENUS0_TFILT_SRCIDMASK**Type:** RW**Reset State:** 0x00000000**SNOC_VENUS0_TFILT_SRCIDMASK**

Bits	Name	Description
14:0	SRCIDMASK	Reserved, null-width register. Register SrcIdMask is only available for specific designs, and contains the source ID mask used to filter packets. The number of bits is set with parameter wSrcId. A packet is a candidate when $\text{ReqInfo.SrcId} + \text{SrcIdMask} = \text{SrcIdBase} + \text{SrcIdMask}$.

0x00581120 SNOC_VENUS0_TFILT_OPCODE**Type:** RW**Reset State:** 0x00000000**SNOC_VENUS0_TFILT_OPCODE**

Bits	Name	Description
1	WREN	When set to 1, selects WR requests.
0	RDEN	When set to 1, selects RD requests.

0x00581124 SNOC_VENUS0_TFILT_USERBASE**Type:** RW**Reset State:** 0x00000000

SNOC_VENUS0_TFILT_USERBASE

Bits	Name	Description
15:0	USERBASE	This register contains the User base used to filter requests.

0x00581128 SNOC_VENUS0_TFILT_USERMASK**Type:** RW**Reset State:** 0x00000000**SNOC_VENUS0_TFILT_USERMASK**

Bits	Name	Description
15:0	USERMASK	This register contains the User mask used to filter requests.

0x00581180 SNOC_VFE_TFILT_ID_COREID**Type:** R**Reset State:** 0x80ED5E09**SNOC_VFE_TFILT_ID_COREID**

Bits	Name	Description
31:8	CORECHECKSUM	Field containing a checksum of the parameters of the IP.
7:0	CORETYPEID	Field identifying the type of IP.

0x00581184 SNOC_VFE_TFILT_ID_REVISIONID**Type:** R**Reset State:** 0x01222B00**SNOC_VFE_TFILT_ID_REVISIONID**

Bits	Name	Description
31:8	FLEXNOCID	Field containing the build revision of the software used to generate the IP HDL code.
7:0	USERID	Field containing a user defined value, not used anywhere inside the IP itself.

0x00581188 SNOC_VFE_TFILT_MODE**Type:** RW**Reset State:** 0x00000000**SNOC_VFE_TFILT_MODE**

Bits	Name	Description
0	MODE	Register Mode is a 1-bit register that sets the filtering mode as follows: handshake Mode = 0 or latency Mode = 1.

0x0058118C SNOC_VFE_TFILT_ADDRBASE_LOW**Type:** RW**Reset State:** 0x00000000**SNOC_VFE_TFILT_ADDRBASE_LOW**

Bits	Name	Description
31:0	ADDRBASE_LOW	Address base LSB register.

0x00581190 SNOC_VFE_TFILT_ADDRBASE_HIGH**Type:** RW**Reset State:** 0x00000000**SNOC_VFE_TFILT_ADDRBASE_HIGH**

Bits	Name	Description
0	ADDRBASE_HIGH	Address base MSB register.

0x00581194 SNOC_VFE_TFILT_ADDRWINDOWSIZE**Type:** RW**Reset State:** 0x00000000**SNOC_VFE_TFILT_ADDRWINDOWSIZE**

Bits	Name	Description
5:0	ADDRWINDOWSIZE	Register AddrWindowSize contains the encoded address mask used to filter packets: the effective Mask value is equal to $\sim(2^{**}\text{AddrWindowSize} - 1)$. A packet is a candidate when $\text{ReqInfo.Addr} \& \text{AddrMask} = \text{AddrBase} \& \text{AddrMask}$.

0x00581198 SNOC_VFE_TFILT_SRCIDBASE**Type:** RW**Reset State:** 0x00000000**SNOC_VFE_TFILT_SRCIDBASE**

Bits	Name	Description
14:0	SRCIDBASE	Reserved, null-width register. Register SrcIdBase is only available for specific designs, and contains the source ID base used to filter packets. The number of bits is set with parameter wSrcId.

0x0058119C SNOC_VFE_TFILT_SRCIDMASK**Type:** RW**Reset State:** 0x00000000**SNOC_VFE_TFILT_SRCIDMASK**

Bits	Name	Description
14:0	SRCIDMASK	Reserved, null-width register. Register SrcIdMask is only available for specific designs, and contains the source ID mask used to filter packets. The number of bits is set with parameter wSrcId. A packet is a candidate when $\text{ReqInfo.SrcId} + \text{SrcIdMask} = \text{SrcIdBase} + \text{SrcIdMask}$.

0x005811A0 SNOC_VFE_TFILT_OPCODE**Type:** RW**Reset State:** 0x00000000**SNOC_VFE_TFILT_OPCODE**

Bits	Name	Description
1	WREN	When set to 1, selects WR requests.
0	RDEN	When set to 1, selects RD requests.

0x005811A4 SNOC_VFE_TFILT_USERBASE**Type:** RW**Reset State:** 0x00000000

SNOC_VFE_TFILT_USERBASE

Bits	Name	Description
15:0	USERBASE	This register contains the User base used to filter requests.

0x005811A8 SNOC_VFE_TFILT_USERMASK**Type:** RW**Reset State:** 0x00000000**SNOC_VFE_TFILT_USERMASK**

Bits	Name	Description
15:0	USERMASK	This register contains the User mask used to filter requests.

0x00581224 SNOC_QDSS_BAM_TFILT_USERBASE**Type:** RW**Reset State:** 0x00000000**SNOC_QDSS_BAM_TFILT_USERBASE**

Bits	Name	Description
15:0	USERBASE	This register contains the User base used to filter requests.

0x00581228 SNOC_QDSS_BAM_TFILT_USERMASK**Type:** RW**Reset State:** 0x00000000**SNOC_QDSS_BAM_TFILT_USERMASK**

Bits	Name	Description
15:0	USERMASK	This register contains the User mask used to filter requests.

0x00581280 SNOC_PCNOC_TFILT_ID_COREID**Type:** R**Reset State:** 0x5E04A609

SNOC_PCNOC_TFILT_ID_COREID

Bits	Name	Description
31:8	CORECHECKSUM	Field containing a checksum of the parameters of the IP.
7:0	CORETYPEID	Field identifying the type of IP.

0x00581284 SNOC_PCNOC_TFILT_ID_REVISIONID**Type:** R**Reset State:** 0x01222B00**SNOC_PCNOC_TFILT_ID_REVISIONID**

Bits	Name	Description
31:8	FLEXNOCID	Field containing the build revision of the software used to generate the IP HDL code.
7:0	USERID	Field containing a user defined value, not used anywhere inside the IP itself.

0x00581288 SNOC_PCNOC_TFILT_MODE**Type:** RW**Reset State:** 0x00000000**SNOC_PCNOC_TFILT_MODE**

Bits	Name	Description
0	MODE	Register Mode is a 1-bit register that sets the filtering mode as follows: handshake Mode = 0 or latency Mode = 1.

0x0058128C SNOC_PCNOC_TFILT_ADDRBASE_LOW**Type:** RW**Reset State:** 0x00000000**SNOC_PCNOC_TFILT_ADDRBASE_LOW**

Bits	Name	Description
31:0	ADDRBASE_LOW	Address base LSB register.

0x00581290 SNOC_PCNOC_TFILT_ADDRBASE_HIGH**Type:** RW**Reset State:** 0x00000000**SNOC_PCNOC_TFILT_ADDRBASE_HIGH**

Bits	Name	Description
0	ADDRBASE_HIGH	Address base MSB register.

0x00581294 SNOC_PCNOC_TFILT_ADDRWINDOWSIZE**Type:** RW**Reset State:** 0x00000000**SNOC_PCNOC_TFILT_ADDRWINDOWSIZE**

Bits	Name	Description
5:0	ADDRWINDOWSIZE	Register AddrWindowSize contains the encoded address mask used to filter packets: the effective Mask value is equal to $\sim(2^{**}\text{AddrWindowSize} - 1)$. A packet is a candidate when $\text{ReqInfo.Addr} \& \text{AddrMask} = \text{AddrBase} \& \text{AddrMask}$.

0x00581298 SNOC_PCNOC_TFILT_SRCIDBASE**Type:** RW**Reset State:** 0x00000000**SNOC_PCNOC_TFILT_SRCIDBASE**

Bits	Name	Description
14:0	SRCIDBASE	Reserved, null-width register. Register SrcIdBase is only available for specific designs, and contains the source ID base used to filter packets. The number of bits is set with parameter wSrcId.

0x0058129C SNOC_PCNOC_TFILT_SRCIDMASK**Type:** RW**Reset State:** 0x00000000

SNOC_PCNOC_TFILT_SRCIDMASK

Bits	Name	Description
14:0	SRCIDMASK	Reserved, null-width register. Register SrcIdMask is only available for specific designs, and contains the source ID mask used to filter packets. The number of bits is set with parameter wSrcId. A packet is a candidate when $\text{ReqInfo.SrcId} + \text{SrcIdMask} = \text{SrcIdBase} + \text{SrcIdMask}$.

0x005812A0 SNOC_PCNOC_TFILT_OPCODE**Type:** RW**Reset State:** 0x00000000**SNOC_PCNOC_TFILT_OPCODE**

Bits	Name	Description
1	WREN	When set to 1, selects WR requests.
0	RDEN	When set to 1, selects RD requests.

0x005812A4 SNOC_PCNOC_TFILT_USERBASE**Type:** RW**Reset State:** 0x00000000**SNOC_PCNOC_TFILT_USERBASE**

Bits	Name	Description
15:0	USERBASE	This register contains the User base used to filter requests.

0x005812A8 SNOC_PCNOC_TFILT_USERMASK**Type:** RW**Reset State:** 0x00000000**SNOC_PCNOC_TFILT_USERMASK**

Bits	Name	Description
15:0	USERMASK	This register contains the User mask used to filter requests.

0x00581300 SNOC_BIMC_TFILT_ID_COREID**Type:** R**Reset State:** 0xE7F8D409**SNOC_BIMC_TFILT_ID_COREID**

Bits	Name	Description
31:8	CORECHECKSUM	Field containing a checksum of the parameters of the IP.
7:0	CORETYPEID	Field identifying the type of IP.

0x00581304 SNOC_BIMC_TFILT_ID_REVISIONID**Type:** R**Reset State:** 0x01222B00**SNOC_BIMC_TFILT_ID_REVISIONID**

Bits	Name	Description
31:8	FLEXNOCID	Field containing the build revision of the software used to generate the IP HDL code.
7:0	USERID	Field containing a user defined value, not used anywhere inside the IP itself.

0x00581308 SNOC_BIMC_TFILT_MODE**Type:** RW**Reset State:** 0x00000000**SNOC_BIMC_TFILT_MODE**

Bits	Name	Description
0	MODE	Register Mode is a 1-bit register that sets the filtering mode as follows: handshake Mode = 0 or latency Mode = 1.

0x0058130C SNOC_BIMC_TFILT_ADDRBASE_LOW**Type:** RW**Reset State:** 0x00000000**SNOC_BIMC_TFILT_ADDRBASE_LOW**

Bits	Name	Description
31:0	ADDRBASE_LOW	Address base LSB register.

0x00581310 SNOC_BIMC_TFILT_ADDRBASE_HIGH**Type:** RW**Reset State:** 0x00000000**SNOC_BIMC_TFILT_ADDRBASE_HIGH**

Bits	Name	Description
0	ADDRBASE_HIGH	Address base MSB register.

0x00581314 SNOC_BIMC_TFILT_ADDRWINDOWSIZE**Type:** RW**Reset State:** 0x00000000**SNOC_BIMC_TFILT_ADDRWINDOWSIZE**

Bits	Name	Description
5:0	ADDRWINDOWSIZE	Register AddrWindowSize contains the encoded address mask used to filter packets: the effective Mask value is equal to $\sim(2^{**}\text{AddrWindowSize} - 1)$. A packet is a candidate when $\text{ReqInfo.Addr} \& \text{AddrMask} = \text{AddrBase} \& \text{AddrMask}$.

0x00581318 SNOC_BIMC_TFILT_SRCIDBASE**Type:** RW**Reset State:** 0x00000000**SNOC_BIMC_TFILT_SRCIDBASE**

Bits	Name	Description
14:0	SRCIDBASE	Reserved, null-width register. Register SrcIdBase is only available for specific designs, and contains the source ID base used to filter packets. The number of bits is set with parameter wSrcId.

0x0058131C SNOC_BIMC_TFILT_SRCIDMASK**Type:** RW**Reset State:** 0x00000000

SNOC_BIMC_TFILT_SRCIDMASK

Bits	Name	Description
14:0	SRCIDMASK	Reserved, null-width register. Register SrcIdMask is only available for specific designs, and contains the source ID mask used to filter packets. The number of bits is set with parameter wSrcId. A packet is a candidate when $\text{ReqInfo.SrcId} + \text{SrcIdMask} = \text{SrcIdBase} + \text{SrcIdMask}$.

0x00581320 SNOC_BIMC_TFILT_OPCODE**Type:** RW**Reset State:** 0x00000000**SNOC_BIMC_TFILT_OPCODE**

Bits	Name	Description
1	WREN	When set to 1, selects WR requests.
0	RDEN	When set to 1, selects RD requests.

0x00581324 SNOC_BIMC_TFILT_USERBASE**Type:** RW**Reset State:** 0x00000000**SNOC_BIMC_TFILT_USERBASE**

Bits	Name	Description
15:0	USERBASE	This register contains the User base used to filter requests.

0x00581328 SNOC_BIMC_TFILT_USERMASK**Type:** RW**Reset State:** 0x00000000**SNOC_BIMC_TFILT_USERMASK**

Bits	Name	Description
15:0	USERMASK	This register contains the User mask used to filter requests.

0x00581380 SNOC_WCSS_TFILT_ID_COREID**Type:** R**Reset State:** 0x30706C09**SNOC_WCSS_TFILT_ID_COREID**

Bits	Name	Description
31:8	CORECHECKSUM	Field containing a checksum of the parameters of the IP.
7:0	CORETYPEID	Field identifying the type of IP.

0x00581384 SNOC_WCSS_TFILT_ID_REVISIONID**Type:** R**Reset State:** 0x01222B00**SNOC_WCSS_TFILT_ID_REVISIONID**

Bits	Name	Description
31:8	FLEXNOCID	Field containing the build revision of the software used to generate the IP HDL code.
7:0	USERID	Field containing a user defined value, not used anywhere inside the IP itself.

0x00581388 SNOC_WCSS_TFILT_MODE**Type:** RW**Reset State:** 0x00000000**SNOC_WCSS_TFILT_MODE**

Bits	Name	Description
0	MODE	Register Mode is a 1-bit register that sets the filtering mode as follows: handshake Mode = 0 or latency Mode = 1.

0x0058138C SNOC_WCSS_TFILT_ADDRBASE_LOW**Type:** RW**Reset State:** 0x00000000**SNOC_WCSS_TFILT_ADDRBASE_LOW**

Bits	Name	Description
31:0	ADDRBASE_LOW	Address base LSB register.

0x00581390 SNOC_WCSS_TFILT_ADDRBASE_HIGH**Type:** RW**Reset State:** 0x00000000**SNOC_WCSS_TFILT_ADDRBASE_HIGH**

Bits	Name	Description
0	ADDRBASE_HIGH	Address base MSB register.

0x00581394 SNOC_WCSS_TFILT_ADDRWINDOWSIZE**Type:** RW**Reset State:** 0x00000000**SNOC_WCSS_TFILT_ADDRWINDOWSIZE**

Bits	Name	Description
5:0	ADDRWINDOWSIZE	Register AddrWindowSize contains the encoded address mask used to filter packets: the effective Mask value is equal to $\sim(2^{**}\text{AddrWindowSize} - 1)$. A packet is a candidate when $\text{ReqInfo.Addr} \& \text{AddrMask} = \text{AddrBase} \& \text{AddrMask}$.

0x00581398 SNOC_WCSS_TFILT_SRCIDBASE**Type:** RW**Reset State:** 0x00000000**SNOC_WCSS_TFILT_SRCIDBASE**

Bits	Name	Description
14:0	SRCIDBASE	Reserved, null-width register. Register SrcIdBase is only available for specific designs, and contains the source ID base used to filter packets. The number of bits is set with parameter wSrcId.

0x0058139C SNOC_WCSS_TFILT_SRCIDMASK**Type:** RW**Reset State:** 0x00000000

SNOC_WCSS_TFILT_SRCIDMASK

Bits	Name	Description
14:0	SRCIDMASK	Reserved, null-width register. Register SrcIdMask is only available for specific designs, and contains the source ID mask used to filter packets. The number of bits is set with parameter wSrcId. A packet is a candidate when $\text{ReqInfo.SrcId} + \text{SrcIdMask} = \text{SrcIdBase} + \text{SrcIdMask}$.

0x005813A0 SNOC_WCSS_TFILT_OPCODE**Type:** RW**Reset State:** 0x00000000**SNOC_WCSS_TFILT_OPCODE**

Bits	Name	Description
1	WREN	When set to 1, selects WR requests.
0	RDEN	When set to 1, selects RD requests.

0x005813A4 SNOC_WCSS_TFILT_USERBASE**Type:** RW**Reset State:** 0x00000000**SNOC_WCSS_TFILT_USERBASE**

Bits	Name	Description
15:0	USERBASE	This register contains the User base used to filter requests.

0x005813A8 SNOC_WCSS_TFILT_USERMASK**Type:** RW**Reset State:** 0x00000000**SNOC_WCSS_TFILT_USERMASK**

Bits	Name	Description
15:0	USERMASK	This register contains the User mask used to filter requests.

0x00581424 SNOC_QDSS_ETR_TFILT_USERBASE**Type:** RW**Reset State:** 0x00000000**SNOC_QDSS_ETR_TFILT_USERBASE**

Bits	Name	Description
15:0	USERBASE	This register contains the User base used to filter requests.

0x00581428 SNOC_QDSS_ETR_TFILT_USERMASK**Type:** RW**Reset State:** 0x00000000**SNOC_QDSS_ETR_TFILT_USERMASK**

Bits	Name	Description
15:0	USERMASK	This register contains the User mask used to filter requests.

0x00581480 SNOC_JPEG_TFILT_ID_COREID**Type:** R**Reset State:** 0x441A1609**SNOC_JPEG_TFILT_ID_COREID**

Bits	Name	Description
31:8	CORECHECKSUM	Field containing a checksum of the parameters of the IP.
7:0	CORETYPEID	Field identifying the type of IP.

0x00581484 SNOC_JPEG_TFILT_ID_REVISIONID**Type:** R**Reset State:** 0x01222B00**SNOC_JPEG_TFILT_ID_REVISIONID**

Bits	Name	Description
31:8	FLEXNOCID	Field containing the build revision of the software used to generate the IP HDL code.
7:0	USERID	Field containing a user defined value, not used anywhere inside the IP itself.

0x00581488 SNOC_JPEG_TFILT_MODE**Type:** RW**Reset State:** 0x00000000**SNOC_JPEG_TFILT_MODE**

Bits	Name	Description
0	MODE	Register Mode is a 1-bit register that sets the filtering mode as follows: handshake Mode = 0 or latency Mode = 1.

0x0058148C SNOC_JPEG_TFILT_ADDRBASE_LOW**Type:** RW**Reset State:** 0x00000000**SNOC_JPEG_TFILT_ADDRBASE_LOW**

Bits	Name	Description
31:0	ADDRBASE_LOW	Address base LSB register.

0x00581490 SNOC_JPEG_TFILT_ADDRBASE_HIGH**Type:** RW**Reset State:** 0x00000000**SNOC_JPEG_TFILT_ADDRBASE_HIGH**

Bits	Name	Description
0	ADDRBASE_HIGH	Address base MSB register.

0x00581494 SNOC_JPEG_TFILT_ADDRWINDOWSIZE**Type:** RW**Reset State:** 0x00000000**SNOC_JPEG_TFILT_ADDRWINDOWSIZE**

Bits	Name	Description
5:0	ADDRWINDOWSIZE	Register AddrWindowSize contains the encoded address mask used to filter packets: the effective Mask value is equal to $\sim(2^{**}\text{AddrWindowSize} - 1)$. A packet is a candidate when $\text{ReqInfo.Addr} \& \text{AddrMask} = \text{AddrBase} \& \text{AddrMask}$.

0x00581498 SNOC_JPEG_TFILT_SRCIDBASE**Type:** RW**Reset State:** 0x00000000**SNOC_JPEG_TFILT_SRCIDBASE**

Bits	Name	Description
14:0	SRCIDBASE	Reserved, null-width register. Register SrcIdBase is only available for specific designs, and contains the source ID base used to filter packets. The number of bits is set with parameter wSrcId.

0x0058149C SNOC_JPEG_TFILT_SRCIDMASK**Type:** RW**Reset State:** 0x00000000**SNOC_JPEG_TFILT_SRCIDMASK**

Bits	Name	Description
14:0	SRCIDMASK	Reserved, null-width register. Register SrcIdMask is only available for specific designs, and contains the source ID mask used to filter packets. The number of bits is set with parameter wSrcId. A packet is a candidate when $\text{ReqInfo.SrcId} + \text{SrcIdMask} = \text{SrcIdBase} + \text{SrcIdMask}$.

0x005814A0 SNOC_JPEG_TFILT_OPCODE**Type:** RW**Reset State:** 0x00000000**SNOC_JPEG_TFILT_OPCODE**

Bits	Name	Description
1	WREN	When set to 1, selects WR requests.
0	RDEN	When set to 1, selects RD requests.

0x005814A4 SNOC_JPEG_TFILT_USERBASE**Type:** RW**Reset State:** 0x00000000

SNOC_JPEG_TFILT_USERBASE

Bits	Name	Description
15:0	USERBASE	This register contains the User base used to filter requests.

0x005814A8 SNOC_JPEG_TFILT_USERMASK**Type:** RW**Reset State:** 0x00000000**SNOC_JPEG_TFILT_USERMASK**

Bits	Name	Description
15:0	USERMASK	This register contains the User mask used to filter requests.

0x00581500 SNOC_MDP0_TFILT_ID_COREID**Type:** R**Reset State:** 0xF70C5D09**SNOC_MDP0_TFILT_ID_COREID**

Bits	Name	Description
31:8	CORECHECKSUM	Field containing a checksum of the parameters of the IP.
7:0	CORETYPEID	Field identifying the type of IP.

0x00581504 SNOC_MDP0_TFILT_ID_REVISIONID**Type:** R**Reset State:** 0x01222B00**SNOC_MDP0_TFILT_ID_REVISIONID**

Bits	Name	Description
31:8	FLEXNOCID	Field containing the build revision of the software used to generate the IP HDL code.
7:0	USERID	Field containing a user defined value, not used anywhere inside the IP itself.

0x00581508 SNOC_MDP0_TFILT_MODE**Type:** RW**Reset State:** 0x00000000**SNOC_MDP0_TFILT_MODE**

Bits	Name	Description
0	MODE	Register Mode is a 1-bit register that sets the filtering mode as follows: handshake Mode = 0 or latency Mode = 1.

0x0058150C SNOC_MDP0_TFILT_ADDRBASE_LOW**Type:** RW**Reset State:** 0x00000000**SNOC_MDP0_TFILT_ADDRBASE_LOW**

Bits	Name	Description
31:0	ADDRBASE_LOW	Address base LSB register.

0x00581510 SNOC_MDP0_TFILT_ADDRBASE_HIGH**Type:** RW**Reset State:** 0x00000000**SNOC_MDP0_TFILT_ADDRBASE_HIGH**

Bits	Name	Description
0	ADDRBASE_HIGH	Address base MSB register.

0x00581514 SNOC_MDP0_TFILT_ADDRWINDOWSIZE**Type:** RW**Reset State:** 0x00000000**SNOC_MDP0_TFILT_ADDRWINDOWSIZE**

Bits	Name	Description
5:0	ADDRWINDOWSIZE	Register AddrWindowSize contains the encoded address mask used to filter packets: the effective Mask value is equal to $\sim(2^{**}\text{AddrWindowSize} - 1)$. A packet is a candidate when $\text{ReqInfo.Addr} \& \text{AddrMask} = \text{AddrBase} \& \text{AddrMask}$.

0x00581518 SNOC_MDP0_TFILT_SRCIDBASE**Type:** RW**Reset State:** 0x00000000**SNOC_MDP0_TFILT_SRCIDBASE**

Bits	Name	Description
14:0	SRCIDBASE	Reserved, null-width register. Register SrcIdBase is only available for specific designs, and contains the source ID base used to filter packets. The number of bits is set with parameter wSrcId.

0x0058151C SNOC_MDP0_TFILT_SRCIDMASK**Type:** RW**Reset State:** 0x00000000**SNOC_MDP0_TFILT_SRCIDMASK**

Bits	Name	Description
14:0	SRCIDMASK	Reserved, null-width register. Register SrcIdMask is only available for specific designs, and contains the source ID mask used to filter packets. The number of bits is set with parameter wSrcId. A packet is a candidate when $\text{ReqInfo.SrcId} + \text{SrcIdMask} = \text{SrcIdBase} + \text{SrcIdMask}$.

0x00581520 SNOC_MDP0_TFILT_OPCODE**Type:** RW**Reset State:** 0x00000000**SNOC_MDP0_TFILT_OPCODE**

Bits	Name	Description
1	WREN	When set to 1, selects WR requests.
0	RDEN	When set to 1, selects RD requests.

0x00581524 SNOC_MDP0_TFILT_USERBASE**Type:** RW**Reset State:** 0x00000000

SNOC_MDP0_TFILT_USERBASE

Bits	Name	Description
15:0	USERBASE	This register contains the User base used to filter requests.

0x00581528 SNOC_MDP0_TFILT_USERMASK**Type:** RW**Reset State:** 0x00000000**SNOC_MDP0_TFILT_USERMASK**

Bits	Name	Description
15:0	USERMASK	This register contains the User mask used to filter requests.

0x00582188 SNOC_TPRB3_TPROF_EN**Type:** RW**Reset State:** 0x00000000**SNOC_TPRB3_TPROF_EN**

Bits	Name	Description
0	EN	Register En is a 1-bit register that enables the transaction probe counter unit.

0x0058218C SNOC_TPRB3_TPROF_MODE**Type:** RW**Reset State:** 0x00000000**SNOC_TPRB3_TPROF_MODE**

Bits	Name	Description
1:0	MODE	Register Mode sets the counting mode per observed port. Each bit per observation port defines the incrementing mode. (Mode = 0 for Delay, Mode = 1 for Pending)

0x00582190 SNOC_TPRB3_TPROF_OBSERVEDSEL_0**Type:** RW**Reset State:** 0x00000000

SNOC_TPRB3_TPROF_OBSERVEDSEL_0

Bits	Name	Description
1:0	OBSERVEDSEL_0	Register ObservedSel_i selects the port to be observed: 0.

0x00582194 SNOC_TPRB3_TPROF_OBSERVEDSEL_1**Type:** RW**Reset State:** 0x00000000**SNOC_TPRB3_TPROF_OBSERVEDSEL_1**

Bits	Name	Description
1:0	OBSERVEDSEL_1	Register ObservedSel_i selects the port to be observed: 1.

0x005821A0 SNOC_TPRB3_TPROF_NTENURELINES_0**Type:** RW**Reset State:** 0x00000000**SNOC_TPRB3_TPROF_NTENURELINES_0**

Bits	Name	Description
3:0	NTENURELINES_0	Register nTenureLine_i indicates the number of transaction probe lines allocated to each observed port: 0.

0x005821AC SNOC_TPRB3_TPROF_THRESHOLDS_0_0**Type:** RW**Reset State:** 0x00000000**SNOC_TPRB3_TPROF_THRESHOLDS_0_0**

Bits	Name	Description
3:0	THRESHOLDS_0_0	Register Threshold_i_j contains the threshold index '0' that allows computation of threshold values.

0x005821B0 SNOC_TPRB3_TPROF_THRESHOLDS_0_1**Type:** RW**Reset State:** 0x00000000

SNOC_TPRB3_TPROF_THRESHOLDS_0_1

Bits	Name	Description
3:0	THRESHOLDS_0_1	Register Threshold_i_j contains the threshold index '1' that allows computation of threshold values.

0x005821B4 SNOC_TPRB3_TPROF_THRESHOLDS_0_2**Type:** RW**Reset State:** 0x00000000**SNOC_TPRB3_TPROF_THRESHOLDS_0_2**

Bits	Name	Description
3:0	THRESHOLDS_0_2	Register Threshold_i_j contains the threshold index '2' that allows computation of threshold values.

0x005821BC SNOC_TPRB3_TPROF_THRESHOLDS_1_0**Type:** RW**Reset State:** 0x00000000**SNOC_TPRB3_TPROF_THRESHOLDS_1_0**

Bits	Name	Description
3:0	THRESHOLDS_1_0	Register Threshold_i_j contains the threshold index '0' that allows computation of threshold values.

0x005821C0 SNOC_TPRB3_TPROF_THRESHOLDS_1_1**Type:** RW**Reset State:** 0x00000000**SNOC_TPRB3_TPROF_THRESHOLDS_1_1**

Bits	Name	Description
3:0	THRESHOLDS_1_1	Register Threshold_i_j contains the threshold index '1' that allows computation of threshold values.

0x005821C4 SNOC_TPRB3_TPROF_THRESHOLDS_1_2**Type:** RW**Reset State:** 0x00000000

SNOC_TPRB3_TPROF_THRESHOLDS_1_2

Bits	Name	Description
3:0	THRESHOLDS_1_2	Register Threshold_i_j contains the threshold index '2' that allows computation of threshold values.

0x005821EC SNOC_TPRB3_TPROF_OVERFLOWSTATUS**Type:** R**Reset State:** Undefined**SNOC_TPRB3_TPROF_OVERFLOWSTATUS**

Bits	Name	Description
1:0	OVERFLOWSTATUS	Register OverflowStatus is a status register that handles the overflow status on each observed port. OverflowStatus = nObservable. Bit 0 gives the status of observed port 0.

0x005821F0 SNOC_TPRB3_TPROF_OVERFLOWRESET**Type:** RW**Reset State:** 0x00000000**SNOC_TPRB3_TPROF_OVERFLOWRESET**

Bits	Name	Description
1:0	OVERFLOWRESET	Register OverflowReset is a pulse register that clears overflow status bits per observed port on each write access. OverflowReset = nObservable. Writing 0x2 clears the overflow status of observed port 1.

0x005821F4 SNOC_TPRB3_TPROF_PENDINGEVENTMODE**Type:** RW**Reset State:** 0x00000000**SNOC_TPRB3_TPROF_PENDINGEVENTMODE**

Bits	Name	Description
0	PENDINGEVENTMODE	Register pendingEventMode is a 1-bit register that configures the pending event mode. When set to 0 (CYCLE), and when register mode is set to PENDING, the pending event is generated on each cycle when the counter is greater than zero. When set to 1 (STOP) the pending event is generated on each stop event.

0x005821F8 SNOC_TPRB3_TPROF_PRESCALER**Type:** RW**Reset State:** 0x00000000**SNOC_TPRB3_TPROF_PRESCALER**

Bits	Name	Description
7:0	PRESCALER	8Register Prescaler is an-bit pre-scaling register that accepts any pre-scaling value between 1 (default) and 256. If set to 0, pre-scaling is disabled. If set to any other supported value 'n', the threshold counter value is divided by (n + 1).

0x00584C08 SNOC_TPRB3_SPROF_MAINCTL**Type:** RW**Reset State:** 0x00000000

Register MainCtl contains probe global control bits. The register has seven bit fields:

SNOC_TPRB3_SPROF_MAINCTL

Bits	Name	Type	Description
7	FILTBYTEALWAYSCHAINABLEEN	RW	When set to 0, Filter Byte / ByteEn Cnt will be connected to all statistics counters where odd Stat count are not chainable. When set to 1, Filter Byte / ByteEn Cnt will be connected only to chainable stat counter (even stat counter).
6	INTRUSIVEMODE	R	When set to 1, register field IntrusiveMode enables trace operation in Intrusive flow-control mode. When set to 0, the register enables trace operation in Overflow flow-control mode
5	STATCONDDUMP	RW	When set, register field StatCondDump enables the dump of a statistics frame to the range of counter values set for registers StatAlarmMin, StatAlarmMax, and AlarmMode. This field also renders register StatAlarmStatus inoperative. When parameter statisticsCounterAlarm is set to False, the StatCondDump register bit is reserved.
4	ALARMEN	RW	When set, register field AlarmEn enables the probe to collect alarm-related information. When the register field bit is null, both TraceAlarm and StatAlarm outputs are driven to 0.
3	STATEN	RW	When set to 1, register field StatEn enables statistics profiling. The probe sendS statistics results to the output for signal ObsTx. All statistics counters are cleared when the StatEn bit goes from 0 to 1. When set to 0, counters are disabled.

SNOC_TPRB3_SPROF_MAINCTL (cont.)

Bits	Name	Type	Description
2	PAYLOADEN	RW	Register field PayloadEn, when set to 1, enables traces to contain headers and payload. When set to 0, only headers are reported.
1	TRACEEN	RW	Register field TraceEn enables the probe to send filtered packets (Trace) on the ObsTx observation output.
0	ERREN	RW	Register field ErrEn enables the probe to send on the ObsTx output any packet with Error status, independently of filtering mechanisms, thus constituting a simple supplementary global filter.

0x00584C0C SNOC_TPRB3_SPROF_CFGCTL**Type:** RW**Reset State:** 0b0000000000000000000000000000X0**SNOC_TPRB3_SPROF_CFGCTL**

Bits	Name	Type	Description
1	ACTIVE	R	
0	GLOBALEN	RW	

0x00584C24 SNOC_TPRB3_SPROF_STATPERIOD**Type:** RW**Reset State:** 0x00000000**SNOC_TPRB3_SPROF_STATPERIOD**

Bits	Name	Description
4:0	STATPERIOD	Register StatPeriod is a 5-bit register that sets a period, within a range of 2-8 megacycles, during which statistics are collected before being dumped automatically. Setting the register effectively enables automatic mode operation for statistics collection. Note: The configured period is also used for statistics alarm generation. The period is calculated with the formula: $N_Cycle = 2^{min(23, StatPeriod)}$ When register StatPeriod is set to its default value 0, automatic dump mode is disabled, and register StatGo is activated for manual mode operation. Note: When parameter nStatisticsCounter is set to 0, statistics collection is disabled, with the consequence that all statistics collection registers, including StatPeriod, are reserved.

0x00584C28 SNOC_TPRB3_SPROF_STATGO**Type:** RW**Reset State:** 0x00000000**SNOC_TPRB3_SPROF_STATGO**

Bits	Name	Description
0	STATGO	Writing to 1-bit pulse register StatGo generates a statistics dump. The register is active when statistics collection operates in its manual mode, that is, when register StatPeriod is set to 0.

0x00584C2C SNOC_TPRB3_SPROF_STATALARMMIN**Type:** RW**Reset State:** 0x00000000**SNOC_TPRB3_SPROF_STATALARMMIN**

Bits	Name	Description
31:0	STATALARMMIN	Register StatAlarmMin contains the minimum count value used in statistics alarm comparisons. The number of bits is equal to twice the value set for parameter wStatisticsCounter. When parameter statisticsCounterAlarm is set to False, StatAlarmMin is reserved.

0x00584C30 SNOC_TPRB3_SPROF_STATALARMMAX**Type:** RW**Reset State:** 0x00000000**SNOC_TPRB3_SPROF_STATALARMMAX**

Bits	Name	Description
31:0	STATALARMMAX	Register StatAlarmMax contains the maximum count value used in statistics alarm comparisons. The number of bits is equal to twice the value set for parameter wStatisticsCounter. When parameter statisticsCounterAlarm is set to False, StatAlarmMax is reserved.

0x00584C34 SNOC_TPRB3_SPROF_STATALARMSSTATUS**Type:** R**Reset State:** Undefined

SNOC_TPRB3_SPROF_STATALARMSSTATUS

Bits	Name	Description
0	STATALARMSSTATUS	Register StatAlarmStatus is a read-only 1-bit register indicating that at least one statistics counter has exceeded the programmed values for registers StatAlarmMin or StatAlarmMax. Output signal StatAlarm is equal to the values stored in register MainCtl fields StatAlarmStatus and AlarmEn. When parameter statisticsCounterAlarm is set to False, StatAlarmStatus is reserved.

0x00584C38 SNOC_TPRB3_SPROF_STATALARMCCLR**Type:** RW**Reset State:** 0x00000000**SNOC_TPRB3_SPROF_STATALARMCCLR**

Bits	Name	Description
0	STATALARMCCLR	Register StatAlarmClr is a 1-bit write-only register. A write operation to this register clears the register StatAlarmStatus bit. When parameter statisticsCounterAlarm is set to False, StatAlarmClr is reserved.

0x00584D38 SNOC_TPRB3_SPROF_COUNTERS_0_SRC**Type:** RW**Reset State:** 0x00000000**SNOC_TPRB3_SPROF_COUNTERS_0_SRC**

Bits	Name	Description
5	EXTEVENT	When set to 1 counts the cycles where ExtEvent[IntEvent] = 1. It exists when nExtEvent > 0.
4:0	INTEVENT	Internal packet event

0x00584D3C SNOC_TPRB3_SPROF_COUNTERS_0_ALARMMODE**Type:** RW**Reset State:** 0x00000000

SNOC_TPRB3_SPROF_COUNTERS_0_ALARMMODE

Bits	Name	Description
1:0	COUNTERS_0_ALARMMODE	Register AlarmMode is a 2-bit register that is present when parameter statisticsCounterAlarm is set to True. The register defines the statistics-alarm behavior of the counter.

0x00584D40 SNOC_TPRB3_SPROF_COUNTERS_0_VAL**Type:** R**Reset State:** Undefined**SNOC_TPRB3_SPROF_COUNTERS_0_VAL**

Bits	Name	Description
15:0	COUNTERS_0_VAL	Register Val is a read-only register that is always present. The register contains the statistics counter value either pending StatAlarm output, or when statistics collection is suspended subsequent to triggers or signal statSuspend.

0x00584D4C SNOC_TPRB3_SPROF_COUNTERS_1_SRC**Type:** RW**Reset State:** 0x00000000**SNOC_TPRB3_SPROF_COUNTERS_1_SRC**

Bits	Name	Description
5	EXTEVENT	When set to 1 counts the cycles where ExtEvent[IntEvent] = 1. It exists when nExtEvent > 0.
4:0	INTEVENT	Internal packet event

0x00584D50 SNOC_TPRB3_SPROF_COUNTERS_1_ALARMMODE**Type:** RW**Reset State:** 0x00000000**SNOC_TPRB3_SPROF_COUNTERS_1_ALARMMODE**

Bits	Name	Description
1:0	COUNTERS_1_ALARMMODE	Register AlarmMode is a 2-bit register that is present when parameter statisticsCounterAlarm is set to True. The register defines the statistics-alarm behavior of the counter.

0x00584D54 SNOC_TPRB3_SPROF_COUNTERS_1_VAL**Type:** R**Reset State:** Undefined**SNOC_TPRB3_SPROF_COUNTERS_1_VAL**

Bits	Name	Description
15:0	COUNTERS_1_VAL	Register Val is a read-only register that is always present. The register contains the statistics counter value either pending StatAlarm output, or when statistics collection is suspended subsequent to triggers or signal statSuspend.

0x00584D60 SNOC_TPRB3_SPROF_COUNTERS_2_SRC**Type:** RW**Reset State:** 0x00000000**SNOC_TPRB3_SPROF_COUNTERS_2_SRC**

Bits	Name	Description
5	EXTEVENT	When set to 1 counts the cycles where ExtEvent[IntEvent] = 1. It exists when nExtEvent > 0.
4:0	INTEVENT	Internal packet event

0x00584D64 SNOC_TPRB3_SPROF_COUNTERS_2_ALARMMODE**Type:** RW**Reset State:** 0x00000000**SNOC_TPRB3_SPROF_COUNTERS_2_ALARMMODE**

Bits	Name	Description
1:0	COUNTERS_2_ALARMMODE	Register AlarmMode is a 2-bit register that is present when parameter statisticsCounterAlarm is set to True. The register defines the statistics-alarm behavior of the counter.

0x00584D68 SNOC_TPRB3_SPROF_COUNTERS_2_VAL**Type:** R**Reset State:** Undefined

SNOC_TPRB3_SPROF_COUNTERS_2_VAL

Bits	Name	Description
15:0	COUNTERS_2_VAL	Register Val is a read-only register that is always present. The register contains the statistics counter value either pending StatAlarm output, or when statistics collection is suspended subsequent to triggers or signal statSuspend.

0x00584D74 SNOC_TPRB3_SPROF_COUNTERS_3_SRC**Type:** RW**Reset State:** 0x00000000**SNOC_TPRB3_SPROF_COUNTERS_3_SRC**

Bits	Name	Description
5	EXTEVENT	When set to 1 counts the cycles where ExtEvent[IntEvent] = 1. It exists when nExtEvent > 0.
4:0	INTEVENT	Internal packet event

0x00584D78 SNOC_TPRB3_SPROF_COUNTERS_3_ALARMMODE**Type:** RW**Reset State:** 0x00000000**SNOC_TPRB3_SPROF_COUNTERS_3_ALARMMODE**

Bits	Name	Description
1:0	COUNTERS_3_ALARMMODE	Register AlarmMode is a 2-bit register that is present when parameter statisticsCounterAlarm is set to True. The register defines the statistics-alarm behavior of the counter.

0x00584D7C SNOC_TPRB3_SPROF_COUNTERS_3_VAL**Type:** R**Reset State:** Undefined**SNOC_TPRB3_SPROF_COUNTERS_3_VAL**

Bits	Name	Description
15:0	COUNTERS_3_VAL	Register Val is a read-only register that is always present. The register contains the statistics counter value either pending StatAlarm output, or when statistics collection is suspended subsequent to triggers or signal statSuspend.

0x00584D88 SNOC_TPRB3_SPROF_COUNTERS_4_SRC**Type:** RW**Reset State:** 0x00000000**SNOC_TPRB3_SPROF_COUNTERS_4_SRC**

Bits	Name	Description
5	EXTEVENT	When set to 1 counts the cycles where ExtEvent[IntEvent] = 1. It exists when nExtEvent > 0.
4:0	INTEVENT	Internal packet event

0x00584D8C SNOC_TPRB3_SPROF_COUNTERS_4_ALARMMODE**Type:** RW**Reset State:** 0x00000000**SNOC_TPRB3_SPROF_COUNTERS_4_ALARMMODE**

Bits	Name	Description
1:0	COUNTERS_4_ALARMMODE	Register AlarmMode is a 2-bit register that is present when parameter statisticsCounterAlarm is set to True. The register defines the statistics-alarm behavior of the counter.

0x00584D90 SNOC_TPRB3_SPROF_COUNTERS_4_VAL**Type:** R**Reset State:** Undefined**SNOC_TPRB3_SPROF_COUNTERS_4_VAL**

Bits	Name	Description
15:0	COUNTERS_4_VAL	Register Val is a read-only register that is always present. The register contains the statistics counter value either pending StatAlarm output, or when statistics collection is suspended subsequent to triggers or signal statSuspend.

0x00584D9C SNOC_TPRB3_SPROF_COUNTERS_5_SRC**Type:** RW**Reset State:** 0x00000000

SNOC_TPRB3_SPROF_COUNTERS_5_SRC

Bits	Name	Description
5	EXTEVENT	When set to 1 counts the cycles where ExtEvent[IntEvent] = 1. It exists when nExtEvent > 0.
4:0	INTEVENT	Internal packet event

0x00584DA0 SNOC_TPRB3_SPROF_COUNTERS_5_ALARMMODE**Type:** RW**Reset State:** 0x00000000**SNOC_TPRB3_SPROF_COUNTERS_5_ALARMMODE**

Bits	Name	Description
1:0	COUNTERS_5_ALARMMODE	Register AlarmMode is a 2-bit register that is present when parameter statisticsCounterAlarm is set to True. The register defines the statistics-alarm behavior of the counter.

0x00584DA4 SNOC_TPRB3_SPROF_COUNTERS_5_VAL**Type:** R**Reset State:** Undefined**SNOC_TPRB3_SPROF_COUNTERS_5_VAL**

Bits	Name	Description
15:0	COUNTERS_5_VAL	Register Val is a read-only register that is always present. The register contains the statistics counter value either pending StatAlarm output, or when statistics collection is suspended subsequent to triggers or signal statSuspend.

0x00584DB0 SNOC_TPRB3_SPROF_COUNTERS_6_SRC**Type:** RW**Reset State:** 0x00000000**SNOC_TPRB3_SPROF_COUNTERS_6_SRC**

Bits	Name	Description
5	EXTEVENT	When set to 1 counts the cycles where ExtEvent[IntEvent] = 1. It exists when nExtEvent > 0.
4:0	INTEVENT	Internal packet event

0x00584DB4 SNOC_TPRB3_SPROF_COUNTERS_6_ALARMMODE**Type:** RW**Reset State:** 0x00000000**SNOC_TPRB3_SPROF_COUNTERS_6_ALARMMODE**

Bits	Name	Description
1:0	COUNTERS_6_ALARMMODE	Register AlarmMode is a 2-bit register that is present when parameter statisticsCounterAlarm is set to True. The register defines the statistics-alarm behavior of the counter.

0x00584DB8 SNOC_TPRB3_SPROF_COUNTERS_6_VAL**Type:** R**Reset State:** Undefined**SNOC_TPRB3_SPROF_COUNTERS_6_VAL**

Bits	Name	Description
15:0	COUNTERS_6_VAL	Register Val is a read-only register that is always present. The register contains the statistics counter value either pending StatAlarm output, or when statistics collection is suspended subsequent to triggers or signal statSuspend.

0x00584DC4 SNOC_TPRB3_SPROF_COUNTERS_7_SRC**Type:** RW**Reset State:** 0x00000000**SNOC_TPRB3_SPROF_COUNTERS_7_SRC**

Bits	Name	Description
5	EXTEVENT	When set to 1 counts the cycles where ExtEvent[IntEvent] = 1. It exists when nExtEvent > 0.
4:0	INTEVENT	Internal packet event

0x00584DC8 SNOC_TPRB3_SPROF_COUNTERS_7_ALARMMODE**Type:** RW**Reset State:** 0x00000000

SNOC_TPRB3_SPROF_COUNTERS_7_ALARMMODE

Bits	Name	Description
1:0	COUNTERS_7_ALARMMODE	Register AlarmMode is a 2-bit register that is present when parameter statisticsCounterAlarm is set to True. The register defines the statistics-alarm behavior of the counter.

0x00584DCC SNOC_TPRB3_SPROF_COUNTERS_7_VAL**Type:** R**Reset State:** Undefined**SNOC_TPRB3_SPROF_COUNTERS_7_VAL**

Bits	Name	Description
15:0	COUNTERS_7_VAL	Register Val is a read-only register that is always present. The register contains the statistics counter value either pending StatAlarm output, or when statistics collection is suspended subsequent to triggers or signal statSuspend.

0x00584DD8 SNOC_TPRB3_SPROF_COUNTERS_8_SRC**Type:** RW**Reset State:** 0x00000000**SNOC_TPRB3_SPROF_COUNTERS_8_SRC**

Bits	Name	Description
5	EXTEVENT	When set to 1 counts the cycles where ExtEvent[IntEvent] = 1. It exists when nExtEvent > 0.
4:0	INTEVENT	Internal packet event

0x00584DDC SNOC_TPRB3_SPROF_COUNTERS_8_ALARMMODE**Type:** RW**Reset State:** 0x00000000**SNOC_TPRB3_SPROF_COUNTERS_8_ALARMMODE**

Bits	Name	Description
1:0	COUNTERS_8_ALARMMODE	Register AlarmMode is a 2-bit register that is present when parameter statisticsCounterAlarm is set to True. The register defines the statistics-alarm behavior of the counter.

0x00584DE0 SNOC_TPRB3_SPROF_COUNTERS_8_VAL**Type:** R**Reset State:** Undefined**SNOC_TPRB3_SPROF_COUNTERS_8_VAL**

Bits	Name	Description
15:0	COUNTERS_8_VAL	Register Val is a read-only register that is always present. The register contains the statistics counter value either pending StatAlarm output, or when statistics collection is suspended subsequent to triggers or signal statSuspend.

0x00584DEC SNOC_TPRB3_SPROF_COUNTERS_9_SRC**Type:** RW**Reset State:** 0x00000000**SNOC_TPRB3_SPROF_COUNTERS_9_SRC**

Bits	Name	Description
5	EXTEVENT	When set to 1 counts the cycles where ExtEvent[IntEvent] = 1. It exists when nExtEvent > 0.
4:0	INTEVENT	Internal packet event

0x00584DF0 SNOC_TPRB3_SPROF_COUNTERS_9_ALARMMODE**Type:** RW**Reset State:** 0x00000000**SNOC_TPRB3_SPROF_COUNTERS_9_ALARMMODE**

Bits	Name	Description
1:0	COUNTERS_9_ALARMMODE	Register AlarmMode is a 2-bit register that is present when parameter statisticsCounterAlarm is set to True. The register defines the statistics-alarm behavior of the counter.

0x00584DF4 SNOC_TPRB3_SPROF_COUNTERS_9_VAL**Type:** R**Reset State:** Undefined

SNOC_TPRB3_SPROF_COUNTERS_9_VAL

Bits	Name	Description
15:0	COUNTERS_9_VAL	Register Val is a read-only register that is always present. The register contains the statistics counter value either pending StatAlarm output, or when statistics collection is suspended subsequent to triggers or signal statSuspend.

0x00584E00 SNOC_TPRB3_SPROF_COUNTERS_10_SRC**Type:** RW**Reset State:** 0x00000000**SNOC_TPRB3_SPROF_COUNTERS_10_SRC**

Bits	Name	Description
5	EXTEVENT	When set to 1 counts the cycles where ExtEvent[IntEvent] = 1. It exists when nExtEvent > 0.
4:0	INTEVENT	Internal packet event

0x00584E04 SNOC_TPRB3_SPROF_COUNTERS_10_ALARMMODE**Type:** RW**Reset State:** 0x00000000**SNOC_TPRB3_SPROF_COUNTERS_10_ALARMMODE**

Bits	Name	Description
1:0	COUNTERS_10_ALARMMODE	Register AlarmMode is a 2-bit register that is present when parameter statisticsCounterAlarm is set to True. The register defines the statistics-alarm behavior of the counter.

0x00584E08 SNOC_TPRB3_SPROF_COUNTERS_10_VAL**Type:** R**Reset State:** Undefined**SNOC_TPRB3_SPROF_COUNTERS_10_VAL**

Bits	Name	Description
15:0	COUNTERS_10_VAL	Register Val is a read-only register that is always present. The register contains the statistics counter value either pending StatAlarm output, or when statistics collection is suspended subsequent to triggers or signal statSuspend.

0x00584E14 SNOC_TPRB3_SPROF_COUNTERS_11_SRC**Type:** RW**Reset State:** 0x00000000**SNOC_TPRB3_SPROF_COUNTERS_11_SRC**

Bits	Name	Description
5	EXTEVENT	When set to 1 counts the cycles where ExtEvent[IntEvent] = 1. It exists when nExtEvent > 0.
4:0	INTEVENT	Internal packet event

0x00584E18 SNOC_TPRB3_SPROF_COUNTERS_11_ALARMMODE**Type:** RW**Reset State:** 0x00000000**SNOC_TPRB3_SPROF_COUNTERS_11_ALARMMODE**

Bits	Name	Description
1:0	COUNTERS_11_ALARMMODE	Register AlarmMode is a 2-bit register that is present when parameter statisticsCounterAlarm is set to True. The register defines the statistics-alarm behavior of the counter.

0x00584E1C SNOC_TPRB3_SPROF_COUNTERS_11_VAL**Type:** R**Reset State:** Undefined**SNOC_TPRB3_SPROF_COUNTERS_11_VAL**

Bits	Name	Description
15:0	COUNTERS_11_VAL	Register Val is a read-only register that is always present. The register contains the statistics counter value either pending StatAlarm output, or when statistics collection is suspended subsequent to triggers or signal statSuspend.

0x00584E28 SNOC_TPRB3_SPROF_COUNTERS_12_SRC**Type:** RW**Reset State:** 0x00000000

SNOC_TPRB3_SPROF_COUNTERS_12_SRC

Bits	Name	Description
5	EXTEVENT	When set to 1 counts the cycles where ExtEvent[IntEvent] = 1. It exists when nExtEvent > 0.
4:0	INTEVENT	Internal packet event

0x00584E2C SNOC_TPRB3_SPROF_COUNTERS_12_ALARMMODE**Type:** RW**Reset State:** 0x00000000**SNOC_TPRB3_SPROF_COUNTERS_12_ALARMMODE**

Bits	Name	Description
1:0	COUNTERS_12_ALARMMODE	Register AlarmMode is a 2-bit register that is present when parameter statisticsCounterAlarm is set to True. The register defines the statistics-alarm behavior of the counter.

0x00584E30 SNOC_TPRB3_SPROF_COUNTERS_12_VAL**Type:** R**Reset State:** Undefined**SNOC_TPRB3_SPROF_COUNTERS_12_VAL**

Bits	Name	Description
15:0	COUNTERS_12_VAL	Register Val is a read-only register that is always present. The register contains the statistics counter value either pending StatAlarm output, or when statistics collection is suspended subsequent to triggers or signal statSuspend.

0x00584E3C SNOC_TPRB3_SPROF_COUNTERS_13_SRC**Type:** RW**Reset State:** 0x00000000**SNOC_TPRB3_SPROF_COUNTERS_13_SRC**

Bits	Name	Description
5	EXTEVENT	When set to 1 counts the cycles where ExtEvent[IntEvent] = 1. It exists when nExtEvent > 0.
4:0	INTEVENT	Internal packet event

0x00584E40 SNOC_TPRB3_SPROF_COUNTERS_13_ALARMMODE**Type:** RW**Reset State:** 0x00000000**SNOC_TPRB3_SPROF_COUNTERS_13_ALARMMODE**

Bits	Name	Description
1:0	COUNTERS_13_ALARMMODE	Register AlarmMode is a 2-bit register that is present when parameter statisticsCounterAlarm is set to True. The register defines the statistics-alarm behavior of the counter.

0x00584E44 SNOC_TPRB3_SPROF_COUNTERS_13_VAL**Type:** R**Reset State:** Undefined**SNOC_TPRB3_SPROF_COUNTERS_13_VAL**

Bits	Name	Description
15:0	COUNTERS_13_VAL	Register Val is a read-only register that is always present. The register contains the statistics counter value either pending StatAlarm output, or when statistics collection is suspended subsequent to triggers or signal statSuspend.

0x00584E50 SNOC_TPRB3_SPROF_COUNTERS_14_SRC**Type:** RW**Reset State:** 0x00000000**SNOC_TPRB3_SPROF_COUNTERS_14_SRC**

Bits	Name	Description
5	EXTEVENT	When set to 1 counts the cycles where ExtEvent[IntEvent] = 1. It exists when nExtEvent > 0.
4:0	INTEVENT	Internal packet event

0x00584E54 SNOC_TPRB3_SPROF_COUNTERS_14_ALARMMODE**Type:** RW**Reset State:** 0x00000000

SNOC_TPRB3_SPROF_COUNTERS_14_ALARMMODE

Bits	Name	Description
1:0	COUNTERS_14_ALARMMODE	Register AlarmMode is a 2-bit register that is present when parameter statisticsCounterAlarm is set to True. The register defines the statistics-alarm behavior of the counter.

0x00584E58 SNOC_TPRB3_SPROF_COUNTERS_14_VAL**Type:** R**Reset State:** Undefined**SNOC_TPRB3_SPROF_COUNTERS_14_VAL**

Bits	Name	Description
15:0	COUNTERS_14_VAL	Register Val is a read-only register that is always present. The register contains the statistics counter value either pending StatAlarm output, or when statistics collection is suspended subsequent to triggers or signal statSuspend.

0x00584E64 SNOC_TPRB3_SPROF_COUNTERS_15_SRC**Type:** RW**Reset State:** 0x00000000**SNOC_TPRB3_SPROF_COUNTERS_15_SRC**

Bits	Name	Description
5	EXTEVENT	When set to 1 counts the cycles where ExtEvent[IntEvent] = 1. It exists when nExtEvent > 0.
4:0	INTEVENT	Internal packet event

0x00584E68 SNOC_TPRB3_SPROF_COUNTERS_15_ALARMMODE**Type:** RW**Reset State:** 0x00000000**SNOC_TPRB3_SPROF_COUNTERS_15_ALARMMODE**

Bits	Name	Description
1:0	COUNTERS_15_ALARMMODE	Register AlarmMode is a 2-bit register that is present when parameter statisticsCounterAlarm is set to True. The register defines the statistics-alarm behavior of the counter.

0x00584E6C SNOC_TPRB3_SPROF_COUNTERS_15_VAL**Type:** R**Reset State:** Undefined**SNOC_TPRB3_SPROF_COUNTERS_15_VAL**

Bits	Name	Description
15:0	COUNTERS_15_VAL	Register Val is a read-only register that is always present. The register contains the statistics counter value either pending StatAlarm output, or when statistics collection is suspended subsequent to triggers or signal statSuspend.

0x00587008 SNOC_MSS_QOS_PRIORITY**Type:** RW**Reset State:** 0x00000000

Priority register.

SNOC_MSS_QOS_PRIORITY

Bits	Name	Description
3:2	P1	In Regulator mode, defines the HIGH hurry level. In Fixed/Limiter mode, defines the Urgency level for READ transactions.
1:0	P0	In Regulator mode, defines the LOW hurry level. In Fixed/Limiter mode, defines the Urgency level for WRITE transactions.

0x0058700C SNOC_MSS_QOS_MODE**Type:** RW**Reset State:** 0x00000000**SNOC_MSS_QOS_MODE**

Bits	Name	Description
1:0	MODE	Functional Mode: 0=Fixed, 1=Limiter, 2=Bypass, 3=Regulator.

0x00587010 SNOC_MSS_QOS_BANDWIDTH**Type:** RW**Reset State:** 0x00000000

SNOC_MSS_QOS_BANDWIDTH

Bits	Name	Description
15:0	BANDWIDTH	Defines the bandwidth threshold in 1/256th-byte-per-cycle units. In other words, the desired rate in MBps is divided by frequency in MHz of the NIU, and then multiplied by 256.

0x00587014 SNOC_MSS_QOS_SATURATION**Type:** RW**Reset State:** 0x00000000**SNOC_MSS_QOS_SATURATION**

Bits	Name	Description
9:0	SATURATION	Defines the size of the bandwidth counter, that is, the measurement window, in 16-byte units. In other words, the desired number of bytes divided by 16.

0x00587018 SNOC_MSS_QOS_EXTCONTROL**Type:** RW**Reset State:** 0x00000000

External inputs control.

SNOC_MSS_QOS_EXTCONTROL

Bits	Name	Description
2	INTCLKEN	Replace the External reference by the local clock.
1	EXTTHREN	ExtThr input controls Low/High priority instead of bandwidth threshold.
0	SOCKETQOSEN	Combines the Socket QoS with Regulator QoS.

0x00588008 SNOC_MSS_NAV_QOS_PRIORITY**Type:** RW**Reset State:** 0x00000000

Priority register.

SNOC_MSS_NAV_QOS_PRIORITY

Bits	Name	Description
3:2	P1	In Regulator mode, defines the HIGH hurry level. In Fixed/Limiter mode, defines the Urgency level for READ transactions.
1:0	P0	In Regulator mode, defines the LOW hurry level. In Fixed/Limiter mode, defines the Urgency level for WRITE transactions.

0x0058800C SNOC_MSS_NAV_QOS_MODE**Type:** RW**Reset State:** 0x00000000**SNOC_MSS_NAV_QOS_MODE**

Bits	Name	Description
1:0	MODE	Functional Mode: 0=Fixed, 1=Limiter, 2=Bypass, 3=Regulator.

0x00588018 SNOC_MSS_NAV_QOS_EXTCONTROL**Type:** RW**Reset State:** 0x00000000

External inputs control.

SNOC_MSS_NAV_QOS_EXTCONTROL

Bits	Name	Description
2	INTCLKEN	Replace the External reference by the local clock.
1	EXTTHREN	ExtThr input controls Low/High priority instead of bandwidth threshold.
0	SOCKETQOSEN	Combines the Socket QoS with Regulator QoS.

0x0058B000 SNOC_WCSS_QOS_ID_COREID**Type:** R**Reset State:** 0x49117304**SNOC_WCSS_QOS_ID_COREID**

Bits	Name	Description
31:8	CORECHECKSUM	Field containing a checksum of the parameters of the IP.

SNOC_WCSS_QOS_ID_COREID (cont.)

Bits	Name	Description
7:0	CORETYPEID	Field identifying the type of IP.

0x0058B004 SNOC_WCSS_QOS_ID_REVISIONID**Type:** R**Reset State:** 0x01222B00**SNOC_WCSS_QOS_ID_REVISIONID**

Bits	Name	Description
31:8	FLEXNOCID	Field containing the build revision of the software used to generate the IP HDL code.
7:0	USERID	Field containing a user defined value, not used anywhere inside the IP itself.

0x0058B008 SNOC_WCSS_QOS_PRIORITY**Type:** RW**Reset State:** 0x00000000

Priority register.

SNOC_WCSS_QOS_PRIORITY

Bits	Name	Description
3:2	P1	In Regulator mode, defines the HIGH hurry level. In Fixed/Limiter mode, defines the Urgency level for READ transactions.
1:0	P0	In Regulator mode, defines the LOW hurry level. In Fixed/Limiter mode, defines the Urgency level for WRITE transactions.

0x0058B00C SNOC_WCSS_QOS_MODE**Type:** RW**Reset State:** 0x00000000**SNOC_WCSS_QOS_MODE**

Bits	Name	Description
1:0	MODE	Functional Mode: 0=Fixed, 1=Limiter, 2=Bypass, 3=Regulator.

0x0058B010 SNOC_WCSS_QOS_BANDWIDTH**Type:** RW**Reset State:** 0x00000000**SNOC_WCSS_QOS_BANDWIDTH**

Bits	Name	Description
15:0	BANDWIDTH	Defines the bandwidth threshold in 1/256th-byte-per-cycle units. In other words, the desired rate in MBps is divided by frequency in MHz of the NIU, and then multiplied by 256.

0x0058B014 SNOC_WCSS_QOS_SATURATION**Type:** RW**Reset State:** 0x00000000**SNOC_WCSS_QOS_SATURATION**

Bits	Name	Description
9:0	SATURATION	Defines the size of the bandwidth counter, that is, the measurement window, in 16-byte units. In other words, the desired number of bytes divided by 16.

0x0058B018 SNOC_WCSS_QOS_EXTCONTROL**Type:** RW**Reset State:** 0x00000000

External inputs control.

SNOC_WCSS_QOS_EXTCONTROL

Bits	Name	Description
2	INTCLKEN	Replace the External reference by the local clock.
1	EXTTHREN	ExtThr input controls Low/High priority instead of bandwidth threshold.
0	SOCKETQOSEN	Combines the Socket QoS with Regulator QoS.

0x0058C000 SNOC_PCNOQ_QOS_ID_COREID**Type:** R**Reset State:** 0x70A2B704

SNOC_PCNOC_QOS_ID_COREID

Bits	Name	Description
31:8	CORECHECKSUM	Field containing a checksum of the parameters of the IP.
7:0	CORETYPEID	Field identifying the type of IP.

0x0058C004 SNOC_PCNOC_QOS_ID_REVISIONID**Type:** R**Reset State:** 0x01222B00**SNOC_PCNOC_QOS_ID_REVISIONID**

Bits	Name	Description
31:8	FLEXNOCID	Field containing the build revision of the software used to generate the IP HDL code.
7:0	USERID	Field containing a user defined value, not used anywhere inside the IP itself.

0x0058C008 SNOC_PCNOC_QOS_PRIORITY**Type:** RW**Reset State:** 0x00000000

Priority register.

SNOC_PCNOC_QOS_PRIORITY

Bits	Name	Description
3:2	P1	In Regulator mode, defines the HIGH hurry level. In Fixed/Limiter mode, defines the Urgency level for READ transactions.
1:0	P0	In Regulator mode, defines the LOW hurry level. In Fixed/Limiter mode, defines the Urgency level for WRITE transactions.

0x0058C00C SNOC_PCNOC_QOS_MODE**Type:** RW**Reset State:** 0x00000000**SNOC_PCNOC_QOS_MODE**

Bits	Name	Description
1:0	MODE	Functional Mode: 0=Fixed, 1=Limiter, 2=Bypass, 3=Regulator.

0x0058C010 SNOC_PCNOC_QOS_BANDWIDTH**Type:** RW**Reset State:** 0x00000000**SNOC_PCNOC_QOS_BANDWIDTH**

Bits	Name	Description
15:0	BANDWIDTH	Defines the bandwidth threshold in 1/256th-byte-per-cycle units. In other words, the desired rate in MBps is divided by frequency in MHz of the NIU, and then multiplied by 256.

0x0058C014 SNOC_PCNOC_QOS_SATURATION**Type:** RW**Reset State:** 0x00000000**SNOC_PCNOC_QOS_SATURATION**

Bits	Name	Description
9:0	SATURATION	Defines the size of the bandwidth counter, that is, the measurement window, in 16-byte units. In other words, the desired number of bytes divided by 16.

0x0058C018 SNOC_PCNOC_QOS_EXTCONTROL**Type:** RW**Reset State:** 0x00000000

External inputs control.

SNOC_PCNOC_QOS_EXTCONTROL

Bits	Name	Description
2	INTCLKEN	Replace the External reference by the local clock.
1	EXTTHREN	ExtThr input controls Low/High priority instead of bandwidth threshold.
0	SOCKETQOSEN	Combines the Socket QoS with Regulator QoS.

0x0058D000 SNOC_JPEG_QOS_ID_COREID**Type:** R**Reset State:** 0x86FB7004

SNOC_JPEG_QOS_ID_COREID

Bits	Name	Description
31:8	CORECHECKSUM	Field containing a checksum of the parameters of the IP.
7:0	CORETYPEID	Field identifying the type of IP.

0x0058D004 SNOC_JPEG_QOS_ID_REVISIONID**Type:** R**Reset State:** 0x01222B00**SNOC_JPEG_QOS_ID_REVISIONID**

Bits	Name	Description
31:8	FLEXNOCID	Field containing the build revision of the software used to generate the IP HDL code.
7:0	USERID	Field containing a user defined value, not used anywhere inside the IP itself.

0x0058D008 SNOC_JPEG_QOS_PRIORITY**Type:** RW**Reset State:** 0x00000000

Priority register.

SNOC_JPEG_QOS_PRIORITY

Bits	Name	Description
3:2	P1	In Regulator mode, defines the HIGH hurry level. In Fixed/Limiter mode, defines the Urgency level for READ transactions.
1:0	P0	In Regulator mode, defines the LOW hurry level. In Fixed/Limiter mode, defines the Urgency level for WRITE transactions.

0x0058D00C SNOC_JPEG_QOS_MODE**Type:** RW**Reset State:** 0x00000000**SNOC_JPEG_QOS_MODE**

Bits	Name	Description
1:0	MODE	Functional Mode: 0=Fixed, 1=Limiter, 2=Bypass, 3=Regulator.

0x0058D010 SNOC_JPEG_QOS_BANDWIDTH**Type:** RW**Reset State:** 0x00000000**SNOC_JPEG_QOS_BANDWIDTH**

Bits	Name	Description
16:0	BANDWIDTH	Defines the bandwidth threshold in 1/256th-byte-per-cycle units. In other words, the desired rate in MBps is divided by frequency in MHz of the NIU, and then multiplied by 256.

0x0058D014 SNOC_JPEG_QOS_SATURATION**Type:** RW**Reset State:** 0x00000000**SNOC_JPEG_QOS_SATURATION**

Bits	Name	Description
9:0	SATURATION	Defines the size of the bandwidth counter, that is, the measurement window, in 16-byte units. In other words, the desired number of bytes divided by 16.

0x0058D018 SNOC_JPEG_QOS_EXTCONTROL**Type:** RW**Reset State:** 0x00000000

External inputs control.

SNOC_JPEG_QOS_EXTCONTROL

Bits	Name	Description
2	INTCLKEN	Replace the External reference by the local clock.
1	EXTTHREN	ExtThr input controls Low/High priority instead of bandwidth threshold.
0	SOCKETQOSEN	Combines the Socket QoS with Regulator QoS.

0x0058E000 SNOC_MDP0_QOS_ID_COREID**Type:** R**Reset State:** 0xD1CE1F04

SNOC_MDP0_QOS_ID_COREID

Bits	Name	Description
31:8	CORECHECKSUM	Field containing a checksum of the parameters of the IP.
7:0	CORETYPEID	Field identifying the type of IP.

0x0058E004 SNOC_MDP0_QOS_ID_REVISIONID**Type:** R**Reset State:** 0x01222B00**SNOC_MDP0_QOS_ID_REVISIONID**

Bits	Name	Description
31:8	FLEXNOCID	Field containing the build revision of the software used to generate the IP HDL code.
7:0	USERID	Field containing a user defined value, not used anywhere inside the IP itself.

0x0058E008 SNOC_MDP0_QOS_PRIORITY**Type:** RW**Reset State:** 0x00000000

Priority register.

SNOC_MDP0_QOS_PRIORITY

Bits	Name	Description
3:2	P1	In Regulator mode, defines the HIGH hurry level. In Fixed/Limiter mode, defines the Urgency level for READ transactions.
1:0	P0	In Regulator mode, defines the LOW hurry level. In Fixed/Limiter mode, defines the Urgency level for WRITE transactions.

0x0058E00C SNOC_MDP0_QOS_MODE**Type:** RW**Reset State:** 0x00000000**SNOC_MDP0_QOS_MODE**

Bits	Name	Description
1:0	MODE	Functional Mode: 0=Fixed, 1=Limiter, 2=Bypass, 3=Regulator.

0x0058E010 SNOC_MDP0_QOS_BANDWIDTH**Type:** RW**Reset State:** 0x00000000**SNOC_MDP0_QOS_BANDWIDTH**

Bits	Name	Description
16:0	BANDWIDTH	Defines the bandwidth threshold in 1/256th-byte-per-cycle units. In other words, the desired rate in MBps is divided by frequency in MHz of the NIU, and then multiplied by 256.

0x0058E014 SNOC_MDP0_QOS_SATURATION**Type:** RW**Reset State:** 0x00000000**SNOC_MDP0_QOS_SATURATION**

Bits	Name	Description
9:0	SATURATION	Defines the size of the bandwidth counter, that is, the measurement window, in 16-byte units. In other words, the desired number of bytes divided by 16.

0x0058E018 SNOC_MDP0_QOS_EXTCONTROL**Type:** RW**Reset State:** 0x00000000

External inputs control.

SNOC_MDP0_QOS_EXTCONTROL

Bits	Name	Description
2	INTCLKEN	Replace the External reference by the local clock.
1	EXTTHREN	ExtThr input controls Low/High priority instead of bandwidth threshold.
0	SOCKETQOSEN	Combines the Socket QoS with Regulator QoS.

0x0058F000 SNOC_VENUS0_QOS_ID_COREID**Type:** R**Reset State:** 0x5D808504

SNOC_VENUS0_QOS_ID_COREID

Bits	Name	Description
31:8	CORECHECKSUM	Field containing a checksum of the parameters of the IP.
7:0	CORETYPEID	Field identifying the type of IP.

0x0058F004 SNOC_VENUS0_QOS_ID_REVISIONID**Type:** R**Reset State:** 0x01222B00**SNOC_VENUS0_QOS_ID_REVISIONID**

Bits	Name	Description
31:8	FLEXNOCID	Field containing the build revision of the software used to generate the IP HDL code.
7:0	USERID	Field containing a user defined value, not used anywhere inside the IP itself.

0x0058F008 SNOC_VENUS0_QOS_PRIORITY**Type:** RW**Reset State:** 0x00000000

Priority register.

SNOC_VENUS0_QOS_PRIORITY

Bits	Name	Description
3:2	P1	In Regulator mode, defines the HIGH hurry level. In Fixed/Limiter mode, defines the Urgency level for READ transactions.
1:0	P0	In Regulator mode, defines the LOW hurry level. In Fixed/Limiter mode, defines the Urgency level for WRITE transactions.

0x0058F00C SNOC_VENUS0_QOS_MODE**Type:** RW**Reset State:** 0x00000000**SNOC_VENUS0_QOS_MODE**

Bits	Name	Description
1:0	MODE	Functional Mode: 0=Fixed, 1=Limiter, 2=Bypass, 3=Regulator.

0x0058F010 SNOC_VENUS0_QOS_BANDWIDTH**Type:** RW**Reset State:** 0x00000000**SNOC_VENUS0_QOS_BANDWIDTH**

Bits	Name	Description
16:0	BANDWIDTH	Defines the bandwidth threshold in 1/256th-byte-per-cycle units. In other words, the desired rate in MBps is divided by frequency in MHz of the NIU, and then multiplied by 256.

0x0058F014 SNOC_VENUS0_QOS_SATURATION**Type:** RW**Reset State:** 0x00000000**SNOC_VENUS0_QOS_SATURATION**

Bits	Name	Description
9:0	SATURATION	Defines the size of the bandwidth counter, that is, the measurement window, in 16-byte units. In other words, the desired number of bytes divided by 16.

0x0058F018 SNOC_VENUS0_QOS_EXTCONTROL**Type:** RW**Reset State:** 0x00000000

External inputs control.

SNOC_VENUS0_QOS_EXTCONTROL

Bits	Name	Description
2	INTCLKEN	Replace the External reference by the local clock.
1	EXTTHREN	ExtThr input controls Low/High priority instead of bandwidth threshold.
0	SOCKETQOSEN	Combines the Socket QoS with Regulator QoS.

0x00590000 SNOC_VFE_QOS_ID_COREID**Type:** R**Reset State:** 0x2C7FDF04

SNOC_VFE_QOS_ID_COREID

Bits	Name	Description
31:8	CORECHECKSUM	Field containing a checksum of the parameters of the IP.
7:0	CORETYPEID	Field identifying the type of IP.

0x00590004 SNOC_VFE_QOS_ID_REVISIONID**Type:** R**Reset State:** 0x01222B00**SNOC_VFE_QOS_ID_REVISIONID**

Bits	Name	Description
31:8	FLEXNOCID	Field containing the build revision of the software used to generate the IP HDL code.
7:0	USERID	Field containing a user defined value, not used anywhere inside the IP itself.

0x00590008 SNOC_VFE_QOS_PRIORITY**Type:** RW**Reset State:** 0x00000000

Priority register.

SNOC_VFE_QOS_PRIORITY

Bits	Name	Description
3:2	P1	In Regulator mode, defines the HIGH hurry level. In Fixed/Limiter mode, defines the Urgency level for READ transactions.
1:0	P0	In Regulator mode, defines the LOW hurry level. In Fixed/Limiter mode, defines the Urgency level for WRITE transactions.

0x0059000C SNOC_VFE_QOS_MODE**Type:** RW**Reset State:** 0x00000000**SNOC_VFE_QOS_MODE**

Bits	Name	Description
1:0	MODE	Functional Mode: 0=Fixed, 1=Limiter, 2=Bypass, 3=Regulator.

0x00590010 SNOC_VFE_QOS_BANDWIDTH**Type:** RW**Reset State:** 0x00000000**SNOC_VFE_QOS_BANDWIDTH**

Bits	Name	Description
16:0	BANDWIDTH	Defines the bandwidth threshold in 1/256th-byte-per-cycle units. In other words, the desired rate in MBps is divided by frequency in MHz of the NIU, and then multiplied by 256.

0x00590014 SNOC_VFE_QOS_SATURATION**Type:** RW**Reset State:** 0x00000000**SNOC_VFE_QOS_SATURATION**

Bits	Name	Description
9:0	SATURATION	Defines the size of the bandwidth counter, that is, the measurement window, in 16-byte units. In other words, the desired number of bytes divided by 16.

0x00590018 SNOC_VFE_QOS_EXTCONTROL**Type:** RW**Reset State:** 0x00000000

External inputs control.

SNOC_VFE_QOS_EXTCONTROL

Bits	Name	Description
2	INTCLKEN	Replace the External reference by the local clock.
1	EXTTHREN	ExtThr input controls Low/High priority instead of bandwidth threshold.
0	SOCKETQOSEN	Combines the Socket QoS with Regulator QoS.

0x00591008 SNOC_QDSS_ETR_QOS_PRIORITY**Type:** RW**Reset State:** 0x00000000

Priority register.

SNOC_QDSS_ETR_QOS_PRIORITY

Bits	Name	Description
3:2	P1	In Regulator mode, defines the HIGH hurry level. In Fixed/Limiter mode, defines the Urgency level for READ transactions.
1:0	P0	In Regulator mode, defines the LOW hurry level. In Fixed/Limiter mode, defines the Urgency level for WRITE transactions.

0x0059100C SNOC_QDSS_ETR_QOS_MODE

Type: RW

Reset State: 0x00000000

SNOC_QDSS_ETR_QOS_MODE

Bits	Name	Description
1:0	MODE	Functional Mode: 0=Fixed, 1=Limiter, 2=Bypass, 3=Regulator.

0x00591018 SNOC_QDSS_ETR_QOS_EXTCONTROL

Type: RW

Reset State: 0x00000000

External inputs control.

SNOC_QDSS_ETR_QOS_EXTCONTROL

Bits	Name	Description
2	INTCLKEN	Replace the External reference by the local clock.
1	EXTTHREN	ExtThr input controls Low/High priority instead of bandwidth threshold.
0	SOCKETQOSEN	Combines the Socket QoS with Regulator QoS.

0x00592008 SNOC_QDSS_BAM_QOS_PRIORITY

Type: RW

Reset State: 0x00000000

Priority register.

SNOC_QDSS_BAM_QOS_PRIORITY

Bits	Name	Description
3:2	P1	In Regulator mode, defines the HIGH hurry level. In Fixed/Limiter mode, defines the Urgency level for READ transactions.
1:0	P0	In Regulator mode, defines the LOW hurry level. In Fixed/Limiter mode, defines the Urgency level for WRITE transactions.

0x0059200C SNOC_QDSS_BAM_QOS_MODE**Type:** RW**Reset State:** 0x00000000**SNOC_QDSS_BAM_QOS_MODE**

Bits	Name	Description
1:0	MODE	Functional Mode: 0=Fixed, 1=Limiter, 2=Bypass, 3=Regulator.

0x00592018 SNOC_QDSS_BAM_QOS_EXTCONTROL**Type:** RW**Reset State:** 0x00000000

External inputs control.

SNOC_QDSS_BAM_QOS_EXTCONTROL

Bits	Name	Description
2	INTCLKEN	Replace the External reference by the local clock.
1	EXTTHREN	ExtThr input controls Low/High priority instead of bandwidth threshold.
0	SOCKETQOSEN	Combines the Socket QoS with Regulator QoS.

15 Top Level Mode Multiplexing

15.1 TLMM_MPU1132_16_M22L12_AHB

15.2 TLMM_CSR

TLMM control and status

15.2.1 TLMM_CSR

TLMM control and status

0x01000000+ TLMM_GPIO_CFGn, n=[0..121]
0x1000*n

Type: RW

Clock: GCC_TLMM_AHB_CLK

Reset State: 0x00000001

The GPIO_CFG[n] controls the GPIO Output Enable, Drive strength, Pull and Alternate Function for GPIO[n]. The default PULL can vary for different GPIOs and can be found in the pinout spreadsheet. The default drive setting for all GPIOs is '000' (2 mA).

TLMM_GPIO_CFGn

Bits	Name	Description
10	GPIO_HIHYS_EN	Controls the hihys_en for GPIO[n]
9	GPIO_OE	Controls the OE for GPIO[n] when in GPIO mode.

TLMM_GPIO_CFGn (cont.)

Bits	Name	Description
8:6	DRV_STRENGTH	Controls the GPIO pad drive strength. This applies regardless of the FUNC_SEL field selection. 0x0: DRV_2_MA (Sets the drive strength to 2mA. When in RFFE mode, sets output cap to 5-26 pF) 0x1: DRV_4_MA (Sets the drive strength to 4mA. When in RFFE mode, sets output cap to 27-45 pF) 0x2: DRV_6_MA (Sets the drive strength to 6mA. When in RFFE mode, sets output cap to 46-60 pF) 0x3: DRV_8_MA (Sets the drive strength to 8mA. When in RFFE mode, sets output cap to 61-79 pF) 0x4: DRV_10_MA (Sets the drive strength to 10mA. When in RFFE mode, it is a don't care) 0x5: DRV_12_MA (Sets the drive strength to 12mA. When in RFFE mode, it is a don't care) 0x6: DRV_14_MA (Sets the drive strength to 14mA. When in RFFE mode, it is a don't care) 0x7: DRV_16_MA (Sets the drive strength to 16mA. When in RFFE mode, it is a don't care)
5:2	FUNC_SEL	Many of the GPIO pads have one or more functional hardware interfaces behind them. This field controls how the pad is used. Set this to the appropriate value for the function desired.
1:0	GPIO_PULL	The pad can be configured to employ an internal weak pull up, pull down, or keeper function. This applies regardless of the FUNC_SEL field selection. 0x0: NO_PULL (Disables all pull) 0x1: PULL_DOWN (Weak Pull-down) 0x2: KEEPER (Weak Keeper) 0x3: PULL_UP (Weak Pull-Up)

**0x01000004+ TLMM_GPIO_IN_OUTn, n=[0..121]
0x1000*n**

Type: RW

Clock: GCC_TLMM_AHB_CLK

Reset State: 0x00000000

The GPIO_IN_OUT[n] controls the Out and In on the GPIO[n]. The GPIO_IN bit is read-only.

TLMM_GPIO_IN_OUTn

Bits	Name	Type	Description
1	GPIO_OUT	RW	Controls the value of the GPIO Output
0	GPIO_IN	R	Allows you to read the Input value of the GPIO

**0x01000008+ TLMM_GPIO_INTR_CFGn, n=[0..121]
0x1000*n**
Type: RW**Clock:** GCC_TLMM_AHB_CLK**Reset State:** 0x000000E2

The GPIO_INTR_CFGn controls the summary and direct interrupt settings for GPIO[n]. Additional settings for direct connect interrupts can be found in the DIR_CONN_INTR_CFGn_* registers

TLMM_GPIO_INTR_CFGn

Bits	Name	Description
8	DIR_CONN_EN	This bit tells the TLMM that GPIO[n] is being used as a Direct Connect Interrupt. 0x1: ENABLE (Enable the GPIO as direct connect interrupt) 0x0: DISABLE (Disable the GPIO as direct connect interrupt)
7:5	TARGET_PROC	This field determines which processor a summary interrupt from GPIO[n] should get routed to. 0x0: WCSS (Route the GPIO[n] signal to WCSS summary interrupt) 0x1: SENSORS (Route the GPIO[n] signal to SENSORS summary interrupt) 0x2: Reserved 0x3: RPM (Route the GPIO[n] signal to RPM summary interrupt) 0x4: APSS (Route the GPIO[n] signal to APSS summary interrupt) 0x5: MSS (Route the GPIO[n] signal to MSS summary interrupt) 0x6: TZ (Route the GPIO[n] signal to TZ summary interrupt) 0x7: NONE (Don't route to any processor subsystem. Default)
4	INTR_RAW_STATUS_EN	This enables the RAW status for the summary interrupt on this GPIO. This is a power saving mechanism. Leave this disabled unless it is needed. 0x1: ENABLE 0x0: DISABLE
3:2	INTR_DECT_CTL	Controls the Edge or Level Detection of the Interrupt Controller 0x0: LEVEL (Level sensitive. INTR_POL_CTL should be set to '1' for active-high, '0' for active-low) 0x1: POS_EDGE (Positive-edge sensitive. INTR_POL_CTL should be '1') 0x2: NEG_EDGE (Negative-edge sensitive INTR_POL_CTL should be '1') 0x3: DUAL_EDGE (Sensitive to both edges. INTR_POL_CTL is a don't care for this case)
1	INTR_POL_CTL	Controls the Polarity Detection of the Interrupt Controller. Polarity 1 corresponds to active high and Polarity 0 corresponds to active low. 0x1: POLARITY_1 (Active-high) 0x0: POLARITY_0 (Active-low)

TLMM_GPIO_INTR_CFGn (cont.)

Bits	Name	Description
0	INTR_ENABLE	Controls if this GPIO generates a summary interrupt. 0x1: ENABLE 0x0: DISABLE

**0x0100000C+ TLMM_GPIO_INTR_STATUSn, n=[0..121]
0x1000*n**
Type: RW**Clock:** GCC_TLMM_AHB_CLK**Reset State:** 0x00000000

The GPIO_INTR_STATUS[n] indicates the status of the summary interrupt for GPIO[n].

TLMM_GPIO_INTR_STATUSn

Bits	Name	Description
0	INTR_STATUS	When read it returns the status for the interrupt on GPIO[n]. When written with a 1 in this position, it sets the interrupt for GPIO[n]. When it is written with a 0, it clears the interrupt

0x01100000 TLMM_CLK_GATE_EN**Type:** RW**Clock:** GCC_TLMM_AHB_CLK**Reset State:** 0x00000000

This register can enable the clock gating in the TLMM. By default, the clock gating is disabled.

TLMM_CLK_GATE_EN

Bits	Name	Description
2	AHB_HCLK_EN	Enables the Clock Gating on the AHB Clock. 0x1: ENABLE 0x0: DISABLE
1	SUMMARY_INTR_EN	Enables the Clock Gating on the Summary Interrupt Demets. 0x1: ENABLE 0x0: DISABLE
0	CRIF_READ_EN	Enables the Clock Gating on the Read Data Register on the CRIF interface. 0x1: ENABLE 0x0: DISABLE

0x01100004 TLMM_IE_CTRL_DISABLE**Type:** RW**Clock:** GCC_TLMM_AHB_CLK**Reset State:** 0x00000001

This register can enable/disable control of the pad input enable within the TLMM. By default the IE control is disabled. When disabled, the IE for all GPIOs is always high and the input path is always enabled. When IE_CTRL is enabled, the input path can enabled/disabled based on the direction of the function that is configured for the GPIO. This register only affects GPIOs. Some loopback tests require the IE to always be high. As such, this bit is required functionality for certain test modes.

TLMM_IE_CTRL_DISABLE

Bits	Name	Description
0	IE_CTRL_DISABLE	Enables control of the pad IE within TLMM. When disabled, the IE is tied high. 0x1: DISABLE 0x0: ENABLE

0x01100008 TLMM_MPM_WAKEUP_INT_EN_0**Type:** RW**Clock:** GCC_TLMM_AHB_CLK**Reset State:** 0x00000000

This register controls the input enable for the HV input path from the pad. Interrupt enables are present within the MPM. This register is specifically used to gate the HV path from the pad to the MPM for power savings.

TLMM_MPM_WAKEUP_INT_EN_0

Bits	Name	Description
31	GPIO_115	This is used to enable the HV interrupt path to the MPM: 0x0: DISABLE 0x1: ENABLE
30	GPIO_114	This is used to enable the HV interrupt path to the MPM: 0x0: DISABLE 0x1: ENABLE
29	GPIO_113	This is used to enable the HV interrupt path to the MPM: 0x0: DISABLE 0x1: ENABLE
28	GPIO_112	This is used to enable the HV interrupt path to the MPM: 0x0: DISABLE 0x1: ENABLE

TLMM_MPM_WAKEUP_INT_EN_0 (cont.)

Bits	Name	Description
27	GPIO_111	This is used to enable the HV interrupt path to the MPM: 0x0: DISABLE 0x1: ENABLE
26	GPIO_110	This is used to enable the HV interrupt path to the MPM: 0x0: DISABLE 0x1: ENABLE
25	GPIO_109	This is used to enable the HV interrupt path to the MPM: 0x0: DISABLE 0x1: ENABLE
24	GPIO_108	This is used to enable the HV interrupt path to the MPM: 0x0: DISABLE 0x1: ENABLE
23	GPIO_107	This is used to enable the HV interrupt path to the MPM: 0x0: DISABLE 0x1: ENABLE
22	GPIO_98	This is used to enable the HV interrupt path to the MPM: 0x0: DISABLE 0x1: ENABLE
21	GPIO_97	This is used to enable the HV interrupt path to the MPM: 0x0: DISABLE 0x1: ENABLE
20	GPIO_69	This is used to enable the HV interrupt path to the MPM: 0x0: DISABLE 0x1: ENABLE
19	GPIO_62	This is used to enable the HV interrupt path to the MPM: 0x0: DISABLE 0x1: ENABLE
18	GPIO_54	This is used to enable the HV interrupt path to the MPM: 0x0: DISABLE 0x1: ENABLE
17	GPIO_52	This is used to enable the HV interrupt path to the MPM: 0x0: DISABLE 0x1: ENABLE
16	GPIO_51	This is used to enable the HV interrupt path to the MPM: 0x0: DISABLE 0x1: ENABLE
15	GPIO_50	This is used to enable the HV interrupt path to the MPM: 0x0: DISABLE 0x1: ENABLE

TLMM_MPM_WAKEUP_INT_EN_0 (cont.)

Bits	Name	Description
14	GPIO_49	This is used to enable the HV interrupt path to the MPM: 0x0: DISABLE 0x1: ENABLE
13	GPIO_38	This is used to enable the HV interrupt path to the MPM: 0x0: DISABLE 0x1: ENABLE
12	GPIO_37	This is used to enable the HV interrupt path to the MPM: 0x0: DISABLE 0x1: ENABLE
11	GPIO_36	This is used to enable the HV interrupt path to the MPM: 0x0: DISABLE 0x1: ENABLE
10	GPIO_35	This is used to enable the HV interrupt path to the MPM: 0x0: DISABLE 0x1: ENABLE
9	GPIO_34	This is used to enable the HV interrupt path to the MPM: 0x0: DISABLE 0x1: ENABLE
8	GPIO_31	This is used to enable the HV interrupt path to the MPM: 0x0: DISABLE 0x1: ENABLE
7	GPIO_28	This is used to enable the HV interrupt path to the MPM: 0x0: DISABLE 0x1: ENABLE
6	GPIO_25	This is used to enable the HV interrupt path to the MPM: 0x0: DISABLE 0x1: ENABLE
5	GPIO_21	This is used to enable the HV interrupt path to the MPM: 0x0: DISABLE 0x1: ENABLE
4	GPIO_20	This is used to enable the HV interrupt path to the MPM: 0x0: DISABLE 0x1: ENABLE
3	GPIO_13	This is used to enable the HV interrupt path to the MPM: 0x0: DISABLE 0x1: ENABLE
2	GPIO_12	This is used to enable the HV interrupt path to the MPM: 0x0: DISABLE 0x1: ENABLE

TLMM_MPM_WAKEUP_INT_EN_0 (cont.)

Bits	Name	Description
1	GPIO_5	This is used to enable the HV interrupt path to the MPM: 0x0: DISABLE 0x1: ENABLE
0	GPIO_1	This is used to enable the HV interrupt path to the MPM: 0x0: DISABLE 0x1: ENABLE

0x0110000C TLMM_MPM_WAKEUP_INT_EN_1**Type:** RW**Clock:** GCC_TLMM_AHB_CLK**Reset State:** 0x00000000

This register controls the input enable for the HV input path from the pad. Interrupt enables are present within the MPM. This register is specifically used to gate the HV path from the pad to the MPM for power savings.

TLMM_MPM_WAKEUP_INT_EN_1

Bits	Name	Description
12	GPIO_68	This is used to enable the HV interrupt path to the MPM: 0x0: DISABLE 0x1: ENABLE
11	GPIO_66	This is used to enable the HV interrupt path to the MPM: 0x0: DISABLE 0x1: ENABLE
10	SDC2_DATA_3	This is used to enable the HV interrupt path to the MPM: 0x0: DISABLE 0x1: ENABLE
9	SDC2_DATA_1	This is used to enable the HV interrupt path to the MPM: 0x0: DISABLE 0x1: ENABLE
8	SDC1_DATA_3	This is used to enable the HV interrupt path to the MPM: 0x0: DISABLE 0x1: ENABLE
7	SDC1_DATA_1	This is used to enable the HV interrupt path to the MPM: 0x0: DISABLE 0x1: ENABLE
6	SRST_N	This is used to enable the HV interrupt path to the MPM: 0x0: DISABLE 0x1: ENABLE

TLMM_MPM_WAKEUP_INT_EN_1 (cont.)

Bits	Name	Description
5	GPIO_121	This is used to enable the HV interrupt path to the MPM: 0x0: DISABLE 0x1: ENABLE
4	GPIO_120	This is used to enable the HV interrupt path to the MPM: 0x0: DISABLE 0x1: ENABLE
3	GPIO_9	This is used to enable the HV interrupt path to the MPM: 0x0: DISABLE 0x1: ENABLE
2	GPIO_118	This is used to enable the HV interrupt path to the MPM: 0x0: DISABLE 0x1: ENABLE
1	GPIO_117	This is used to enable the HV interrupt path to the MPM: 0x0: DISABLE 0x1: ENABLE
0	GPIO_11	This is used to enable the HV interrupt path to the MPM: 0x0: DISABLE 0x1: ENABLE

0x01110000 TLMM_INT_JTAG_CTL**Type:** RW**Clock:** GCC_TLMM_AHB_CLK**Reset State:** 0x00000007

The TLMM_INT_JTAG_CTL register controls which processor TAP(s) are enabled for the primary JTAG port.

TLMM_INT_JTAG_CTL

Bits	Name	Description
3	APSS_TAP_ENA	Enables APSS JTAG
2	QDSS_TAP_ENA	Enables QDSS JTAG (Default set to 1)
1	ACC_TAP_ENA	Enables ACC JTAG (Default set to 1)
0	MSM_TAP_ENA	Enables MSM JTAG (Default set to 1)

0x01110004 TLMM_ETM_MODE**Type:** RW**Clock:** CC_TLMM_CLK**Reset State:** 0x00000000

The TLMM_ETM_MODE register is spare register.

TLMM_ETM_MODE

Bits	Name	Description
1:0	TRACE_OVER_SDC2	Spare bits 0x0: MODE0 0x1: MODE1 0x2: MODE2 0x3: MODE3

0x0111001C SPARE1**Type:** RW**Clock:** GCC_TLMM_AHB_CLK**Reset State:** 0x00000000

The SPARE1 register is a miscellaneous register for spare register bits and is reserved

SPARE1

Bits	Name	Description
31:0	MISC1	These bits are reserved.

0x01110020 SPARE2**Type:** RW**Clock:** GCC_TLMM_AHB_CLK**Reset State:** 0x00000000

The SPARE2 register is a miscellaneous register for spare register bits and is reserved

SPARE2

Bits	Name	Description
31:0	MISC2	These bits are reserved.

0x01110010 TLMM_HW_REVISION_NUMBER**Type:** R**Clock:** GCC_TLMM_AHB_CLK**Reset State:** Undefined

The HW_REVISION_NUMBER register contains the hardware revision number for the **MSM**. This information is identical to that stored in JTAG ID, which contains chip revision chip ID (including 'variant'), Qualcomm manufacturing ID, and a start bit.

TLMM_HW_REVISION_NUMBER

Bits	Name	Description
31:28	VERSION_ID	Version ID 0x0: FIRST_TAPE_OUT (will be 0 for the first tapeout)
27:12	PARTNUM	Future variants can be created through efuse blowing.
11:1	QUALCOMM_MFG_ID	00001110000
0	START_BIT	Always set '1'

0x01108000 TLMM_RFFE_CTL**Type:** RW**Clock:** GCC_TLMM_AHB_CLK**Reset State:** 0x00000000

This register controls the RFFE sr_ctl_en/sen_en TLMM outputs. Its purpose is to control the clock and data pads for RFFE on GPIO. It is used to enable/disable slew rate control for the RFFE pads. When used for RFFE, slew rate control must be enabled by setting the SR_CTL_EN bits must be set to '01'. When used for other purposes, slew rate control should be disabled. When disabled, the pad behaves in a similar manner to a regular GPIO pad. The default value is to keep the slew rate control disabled.

TLMM_RFFE_CTL

Bits	Name	Description
29	RFFE5_DATA_SEN_EN	Reserved
28:27	RFFE5_DATA_SR_CTL_EN	Select whether slew rate control is enabled/disabled. Set to '01' for RFFE mode
26	RFFE5_CLK_SEN_EN	Reserved
25:24	RFFE5_CLK_SR_CTL_EN	Select whether slew rate control is enabled/disabled. Set to '01' for RFFE mode
23	RFFE4_DATA_SEN_EN	Reserved

TLMM_RFFE_CTL (cont.)

Bits	Name	Description
22:21	RFFE4_DATA_SR_CTL_EN	Select whether slew rate control is enabled/disabled. Set to '01' for RFFE mode
20	RFFE4_CLK_SEN_EN	Reserved
19:18	RFFE4_CLK_SR_CTL_EN	Select whether slew rate control is enabled/disabled. Set to '01' for RFFE mode
17	RFFE3_DATA_SEN_EN	Reserved
16:15	RFFE3_DATA_SR_CTL_EN	Select whether slew rate control is enabled/disabled. Set to '01' for RFFE mode
14	RFFE3_CLK_SEN_EN	Reserved
13:12	RFFE3_CLK_SR_CTL_EN	Select whether slew rate control is enabled/disabled. Set to '01' for RFFE mode
11	RFFE2_DATA_SEN_EN	Reserved
10:9	RFFE2_DATA_SR_CTL_EN	Select whether slew rate control is enabled/disabled. Set to '01' for RFFE mode
8	RFFE2_CLK_SEN_EN	Reserved
7:6	RFFE2_CLK_SR_CTL_EN	Select whether slew rate control is enabled/disabled. Set to '01' for RFFE mode
5	RFFE1_DATA_SEN_EN	Reserved
4:3	RFFE1_DATA_SR_CTL_EN	Select whether slew rate control is enabled/disabled. Set to '01' for RFFE mode
2	RFFE1_CLK_SEN_EN	Reserved
1:0	RFFE1_CLK_SR_CTL_EN	Select whether slew rate control is enabled/disabled. Set to '01' for RFFE mode

0x0110D000 TLMM_RESOUT_HDRV_CTL**Type:** RW**Clock:** GCC_TLMM_AHB_CLK**Reset State:** 0x00000001

This registers controls the HDrive value for the resout pin.

TLMM_RESOUT_HDRV_CTL

Bits	Name	Description
2:0	RESOUT_N_HDRV	RESOUT_N HDrive Value

0x0110C000 TLMM_JTAG_HDRV_CTL**Type:** RW**Clock:** GCC_TLMM_AHB_CLK**Reset State:** 0x000003DB

This register controls the HDrive and Pull value for the JTAG pins.

TLMM_JTAG_HDRV_CTL

Bits	Name	Description
9:8	TDI_PULL	TDI Pull Value
7:6	TMS_PULL	TMS Pull Value
5:3	TMS_HDRV	TMS HDrive Value
2:0	TDO_HDRV	TDO HDrive Value

0x0110B000 TLMM_PMIC_HDRV_PULL_CTL**Type:** RW**Clock:** GCC_TLMM_AHB_CLK**Reset State:** 0x003C5800

This register controls the HDrive and Pull values for the PMIC interfaces pins.

TLMM_PMIC_HDRV_PULL_CTL

Bits	Name	Description
21:20	PMIC_SPMI_DATA_SR_CTL_EN	Select whether slew rate control is enabled/disabled. Set to '11' for SPMI mode.
19:18	PMIC_SPMI_CLK_SR_CTL_EN	Select whether slew rate control is enabled/disabled. Set to '11' for SPMI mode.
17	PMIC_SPMI_DATA_SEN_EN	Reserved
16	PMIC_SPMI_CLK_SEN_EN	Reserved
15:14	PMIC_SPMI_DATA_PULL	SPMI PMIC Data Pull Value
13:12	PMIC_SPMI_CLK_PULL	SPMI PMIC Clock Pull Value
11:9	PSHOLD_HDRV	PSHOLD HDrive Value
8:6	CXO_EN_HDRV	CXO_ENABLE HDrive Value
5:3	PMIC_SPMI_DATA_HDRV	SPMI PMIC Data HDrive Value
2:0	PMIC_SPMI_CLK_HDRV	SPMI PMIC Clock HDrive Value

0x0110A000 TLMM_SDC1_HDRV_PULL_CTL**Type:** RW**Clock:** GCC_TLMM_AHB_CLK**Reset State:** 0x00000BDB

This register controls the HDrive and Pull values for the SDC1 interface pins.

TLMM_SDC1_HDRV_PULL_CTL

Bits	Name	Description
14:13	SDC1_CLK_PULL	SDC1 CLK Pull Value
12:11	SDC1_CMD_PULL	SDC1 CMD Pull Value
10:9	SDC1_DATA_PULL	SDC1 DATA Pull Value
8:6	SDC1_CLK_HDRV	SDC1 CLOCK HDrive Value
5:3	SDC1_CMD_HDRV	SDC1 CMD HDrive Value
2:0	SDC1_DATA_HDRV	SDC1 DATA HDrive Value

0x01109000 TLMM_SDC2_HDRV_PULL_CTL**Type:** RW**Clock:** GCC_TLMM_AHB_CLK**Reset State:** 0x00000BDB

This register controls the HDrive and Pull values for the SDC2 interface pins.

TLMM_SDC2_HDRV_PULL_CTL

Bits	Name	Description
15:14	SDC2_CLK_PULL	SDC2 CLK Pull Value
13	SDC2_HYS_CTL	Enable/Disable for additional pad hysteresis. 0x0: DISABLE 0x1: ENABLE
12:11	SDC2_CMD_PULL	SDC2 CMD Pull Value
10:9	SDC2_DATA_PULL	SDC2 DATA Pull Value
8:6	SDC2_CLK_HDRV	SDC2 CLOCK HDrive Value
5:3	SDC2_CMD_HDRV	SDC2 CMD HDrive Value
2:0	SDC2_DATA_HDRV	SDC2 DATA HDrive Value

0x01107000 TLMM_MODE_PULL_CTL**Type:** RW**Clock:** GCC_TLMM_AHB_CLK**Reset State:** 0x00000005

This register controls the Pull values for the Mode Interface pins.

TLMM_MODE_PULL_CTL

Bits	Name	Description
3:2	MODE_1_PULL	Pull Value for mode_1 pin
1:0	MODE_0_PULL	Pull Value for mode_0 pin

0x0119D000 TLMM_QDSD_BOOT_CTL**Type:** RW**Clock:** GCC_TLMM_AHB_CLK**Reset State:** 0x00000000

This register controls the configuration for the QDSD and this register being written during PBL.

TLMM_QDSD_BOOT_CTL

Bits	Name	Description
15:0	QDSD_BOOT_REG	BOOT FROM SD CARD OR QDSD DISABLE FEATURES IS WRITTEN BY SW

0x0119E000 TLMM_QDSD_CONFIG_CTL**Type:** RW**Clock:** GCC_TLMM_AHB_CLK**Reset State:** 0x00000000

This register controls the configuration of QDSD modes,IO pad power and debounce value.

TLMM_QDSD_CONFIG_CTL

Bits	Name	Description
31:0	QDSD_CONFIG_REG	QDSD CONFIGURATION REGISTER

0x0119F000 TLMM_QDSD_STATUS_CTL**Type:** R**Clock:** GCC_TLMM_AHB_CLK**Reset State:** 0x13000081

This register contains the status of current mode and card detect GPIO used in QDSD.

TLMM_QDSD_STATUS_CTL

Bits	Name	Description
31:0	QDSD_STATUS_REG	QDSD STATUS REGISTER.If fuse for NIDnT disable is not blown, POR value will be 13000081. If fuse is blown, POR value will be 13000002.

0x011A1000 TLMM_QDSD_GPIO_CTL**Type:** RW**Clock:** GCC_TLMM_AHB_CLK**Reset State:** 0x00000000

This QDSD register configures the card detect GPIO

TLMM_QDSD_GPIO_CTL

Bits	Name	Description
15:0	QDSD_GPIO_REG	GPIO, POLARITY TO BE USED FOR CARD_DETECT IS PROGRAMMED BY SW

0x011A2000 TLMM_QDSD_INTR_ENABLE_CTL**Type:** RW**Clock:** GCC_TLMM_AHB_CLK**Reset State:** 0x00000000

This is a QDSD register.

TLMM_QDSD_INTR_ENABLE_CTL

Bits	Name	Description
15:0	QDSD_INTR_ENABLE_REG	QDSD INTERRUPT ENABLE REGISTER

0x011A3000 TLMM_QDSD_INTR_CLEAR_CTL**Type:** RW**Clock:** GCC_TLMM_AHB_CLK**Reset State:** 0x00000000

This is a QDSD register.

TLMM_QDSD_INTR_CLEAR_CTL

Bits	Name	Description
15:0	QDSD_INTR_CLEAR_REG	QDSD INTERRUPT CLEAR REGISTER

0x011A5000 TLMM_QDSD_SPARE1_CTL**Type:** RW**Clock:** GCC_TLMM_AHB_CLK**Reset State:** 0x00000000

This is a QDSD spare register.

TLMM_QDSD_SPARE1_CTL

Bits	Name	Description
31:0	QDSD_SPARE1_REG	QDSD SPARE REGISTER

0x011A6000 TLMM_QDSD_SPARE2_CTL**Type:** RW**Clock:** GCC_TLMM_AHB_CLK**Reset State:** 0x00000000

This is a QDSD spare register.

TLMM_QDSD_SPARE2_CTL

Bits	Name	Description
31:0	QDSD_SPARE2_REG	QDSD SPARE REGISTER

0x011A4000 TLMM_QDSD_TIMEOUT_VALUE_CTL**Type:** RW**Clock:** GCC_TLMM_AHB_CLK**Reset State:** 0x00000000

This is a QDSD register.

TLMM_QDSD_TIMEOUT_VALUE_CTL

Bits	Name	Description
31:0	QDSD_TIMEOUT_VALUE_REG	SW programs the timeout value of the QDSD timer to this register

0x0110F000 RESERVED**0x01106000+ TLMM_DIR_CONN_INTRn_CFG_WCSS, n=[0..1]
0x4*n****Type:** RW**Clock:** GCC_TLMM_AHB_CLK**Reset State:** 0x00000100

This register is used to select the GPIO to be routed to this direct connect interrupt and to control the polarity for each interrupt. The default polarity is active-high.

TLMM_DIR_CONN_INTRn_CFG_WCSS

Bits	Name	Description
8	POLARITY	Controls the Polarity Detection for the Direct Connect Interrupt. Polarity 1 corresponds to active high and Polarity 0 corresponds to active low. 0x1: POLARITY_1 (Active-high) 0x0: POLARITY_0 (Active-low)
6:0	GPIO_SEL	This field selects which GPIO is to be routed to direct connect interrupt 'n'.

**0x01105000+ TLMM_DIR_CONN_INTRn_CFG_SENSORS, n=[0..9]
0x4*n****Type:** RW**Clock:** GCC_TLMM_AHB_CLK**Reset State:** 0x00000100

This register is used to select the GPIO to be routed to this direct connect interrupt and to control the polarity for each interrupt. The default polarity is active-high.

TLMM_DIR_CONN_INTRn_CFG_SENSORS

Bits	Name	Description
8	POLARITY	Controls the Polarity Detection for the Direct Connect Interrupt. Polarity 1 corresponds to active high and Polarity 0 corresponds to active low. 0x1: POLARITY_1 (Active-high) 0x0: POLARITY_0 (Active-low)
6:0	GPIO_SEL	This field selects which GPIO is to be routed to direct connect interrupt 'n'.

0x01104000+ RESERVED
0x4*n

0x01103000+ TLMM_DIR_CONN_INTRn_CFG_RPM, n=[0..0]
0x4*n

Type: RW

Clock: GCC_TLMM_AHB_CLK

Reset State: 0x00000100

This register is used to select the GPIO to be routed to this direct connect interrupt and to control the polarity for each interrupt. The default polarity is active-high.

TLMM_DIR_CONN_INTRn_CFG_RPM

Bits	Name	Description
8	POLARITY	Controls the Polarity Detection for the Direct Connect Interrupt. Polarity 1 corresponds to active high and Polarity 0 corresponds to active low. 0x1: POLARITY_1 (Active-high) 0x0: POLARITY_0 (Active-low)
6:0	GPIO_SEL	This field selects which GPIO is to be routed to direct connect interrupt 'n'.

0x01102000+ TLMM_DIR_CONN_INTRn_CFG_APSS, n=[0..7]
0x4*n

Type: RW

Clock: GCC_TLMM_AHB_CLK

Reset State: 0x00000100

This register is used to select the GPIO to be routed to this direct connect interrupt and to control the polarity for each interrupt. The default polarity is active-high.

TLMM_DIR_CONN_INTRn_CFG_APSS

Bits	Name	Description
8	POLARITY	Controls the Polarity Detection for the Direct Connect Interrupt. Polarity 1 corresponds to active high and Polarity 0 corresponds to active low. 0x1: POLARITY_1 (Active-high) 0x0: POLARITY_0 (Active-low)
6:0	GPIO_SEL	This field selects which GPIO is to be routed to direct connect interrupt 'n'.

**0x01101000+ TLMM_DIR_CONN_INTRn_CFG_MSS, n=[0..1]
0x4*n****Type:** RW**Clock:** GCC_TLMM_AHB_CLK**Reset State:** 0x00000100

This register is used to select the GPIO to be routed to this direct connect interrupt and to control the polarity for each interrupt. The default polarity is active-high.

TLMM_DIR_CONN_INTRn_CFG_MSS

Bits	Name	Description
8	POLARITY	Controls the Polarity Detection for the Direct Connect Interrupt. Polarity 1 corresponds to active high and Polarity 0 corresponds to active low. 0x1: POLARITY_1 (Active-high) 0x0: POLARITY_0 (Active-low)
6:0	GPIO_SEL	This field selects which GPIO is to be routed to direct connect interrupt 'n'.

0x01200000 TLMM_GPIO_OUT_0**Type:** RW**Clock:** GCC_TLMM_AHB_CLK**Reset State:** 0x00000000

This register controls the output values of GPIO[31:0]. This register is designed for bit-bang or fast configuration of multiple output bits.

TLMM_GPIO_OUT_0

Bits	Name	Description
31:0	GPIO_OUT	Output values for GPIO[31:0].

0x01200004 TLMM_GPIO_OUT_1**Type:** RW**Clock:** GCC_TLMM_AHB_CLK**Reset State:** 0x00000000

This register controls the output values of GPIO[63:32]. This register is designed for bit-bang or fast configuration of multiple output bits.

TLMM_GPIO_OUT_1

Bits	Name	Description
31:0	GPIO_OUT	Output values for GPIO[63:32].

0x01200008 TLMM_GPIO_OUT_2**Type:** RW**Clock:** GCC_TLMM_AHB_CLK**Reset State:** 0x00000000

This register controls the output values of GPIO[95:64]. This register is designed for bit-bang or fast configuration of multiple output bits.

TLMM_GPIO_OUT_2

Bits	Name	Description
31:0	GPIO_OUT	Output values for GPIO[95:64].

0x0120000C TLMM_GPIO_OUT_3**Type:** RW**Clock:** GCC_TLMM_AHB_CLK**Reset State:** 0x00000000

This register controls the output values of GPIO[115:96]. This register is designed for bit-bang or fast configuration of multiple output bits.

TLMM_GPIO_OUT_3

Bits	Name	Description
25:0	GPIO_OUT	Output values for GPIO[121:96].

0x01200020 TLMM_GPIO_OUT_CLR_0**Type:** W**Clock:** GCC_TLMM_AHB_CLK**Reset State:** 0x00000000

This register can be used to clear the output values of GPIO[31:0]. Once the CLR bit is written to '1', the output of the corresponding GPIO is cleared to '0'. The CLR bit then automatically returns to '0' so that it can be used again. This register is designed for bit-bang or fast configuration of multiple output bits and alleviates the need for a read-->modify-->write operation.

TLMM_GPIO_OUT_CLR_0

Bits	Name	Description
31:0	GPIO_OUT_CLR	Clear bits for GPIO[31:0] output values.

0x01200024 TLMM_GPIO_OUT_CLR_1**Type:** W**Clock:** GCC_TLMM_AHB_CLK**Reset State:** 0x00000000

This register can be used to clear the output values of GPIO[63:32]. Once the CLR bit is written to '1', the output of the corresponding GPIO is cleared to '0'. The CLR bit then automatically returns to '0' so that it can be used again. This register is designed for bit-bang or fast configuration of multiple output bits and alleviates the need for a read-->modify-->write operation.

TLMM_GPIO_OUT_CLR_1

Bits	Name	Description
31:0	GPIO_OUT_CLR	Clear bits for GPIO[63:32] output values.

0x01200028 TLMM_GPIO_OUT_CLR_2**Type:** W**Clock:** GCC_TLMM_AHB_CLK**Reset State:** 0x00000000

This register can be used to clear the output values of GPIO[95:64]. Once the CLR bit is written to '1', the output of the corresponding GPIO is cleared to '0'. The CLR bit then automatically returns to '0' so that it can be used again. This register is designed for bit-bang or fast configuration of multiple output bits and alleviates the need for a read-->modify-->write operation.

TLMM_GPIO_OUT_CLR_2

Bits	Name	Description
31:0	GPIO_OUT_CLR	Clear bits for GPIO[95:64] output values.

0x0120002C TLMM_GPIO_OUT_CLR_3**Type:** W**Clock:** GCC_TLMM_AHB_CLK**Reset State:** 0x00000000

This register can be used to clear the output values of GPIO[115:96]. Once the CLR bit is written to '1', the output of the corresponding GPIO is cleared to '0'. The CLR bit then automatically returns to '0' so that it can be used again. This register is designed for bit-bang or fast configuration of multiple output bits and alleviates the need for a read-->modify-->write operation.

TLMM_GPIO_OUT_CLR_3

Bits	Name	Description
25:0	GPIO_OUT_CLR	Clear bits for GPIO[121:96] output values.

0x01200040 TLMM_GPIO_OUT_SET_0**Type:** W**Clock:** GCC_TLMM_AHB_CLK**Reset State:** 0x00000000

This register can be used to set the output values of GPIO[31:0]. Once the SET bit is written to '1', the output of the corresponding GPIO is set to '1'. The SET bit then automatically returns to '0' so that it can be used again. This register is designed for bit-bang or fast configuration of multiple output bits and alleviates the need for a read-->modify-->write operation.

TLMM_GPIO_OUT_SET_0

Bits	Name	Description
31:0	GPIO_OUT_SET	Set bits for GPIO[31:0] output values.

0x01200044 TLMM_GPIO_OUT_SET_1**Type:** W**Clock:** GCC_TLMM_AHB_CLK**Reset State:** 0x00000000

This register can be used to set the output values of GPIO[63:32]. Once the SET bit is written to '1', the output of the corresponding GPIO is set to '1'. The SET bit then automatically returns to '0' so that it can be used again. This register is designed for bit-bang or fast configuration of multiple output bits and alleviates the need for a read-->modify-->write operation.

TLMM_GPIO_OUT_SET_1

Bits	Name	Description
31:0	GPIO_OUT_SET	Set bits for GPIO[63:32] output values.

0x01200048 TLMM_GPIO_OUT_SET_2

Type: W

Clock: GCC_TLMM_AHB_CLK

Reset State: 0x00000000

This register can be used to set the output values of GPIO[95:64]. Once the SET bit is written to '1', the output of the corresponding GPIO is set to '1'. The SET bit then automatically returns to '0' so that it can be used again. This register is designed for bit-bang or fast configuration of multiple output bits and alleviates the need for a read-->modify-->write operation.

TLMM_GPIO_OUT_SET_2

Bits	Name	Description
31:0	GPIO_OUT_SET	Set bits for GPIO[95:64] output values.

0x0120004C TLMM_GPIO_OUT_SET_3

Type: W

Clock: GCC_TLMM_AHB_CLK

Reset State: 0x00000000

This register can be used to set the output values of GPIO[115:96]. Once the SET bit is written to '1', the output of the corresponding GPIO is set to '1'. The SET bit then automatically returns to '0' so that it can be used again. This register is designed for bit-bang or fast configuration of multiple output bits and alleviates the need for a read-->modify-->write operation.

TLMM_GPIO_OUT_SET_3

Bits	Name	Description
25:0	GPIO_OUT_SET	Set bits for GPIO[121:96] output values.

0x01200060 TLMM_GPIO_IN_0**Type:** R**Clock:** GCC_TLMM_AHB_CLK**Reset State:** 0x00000000

This register allows the input values for GPIO[31:0] to be read. This register is designed to read multiple input bits.

TLMM_GPIO_IN_0

Bits	Name	Description
31:0	GPIO_IN	GPIO[31:0] input values.

0x01200064 TLMM_GPIO_IN_1**Type:** R**Clock:** GCC_TLMM_AHB_CLK**Reset State:** 0x00000000

This register allows the input values for GPIO[63:32] to be read. This register is designed to read multiple input bits.

TLMM_GPIO_IN_1

Bits	Name	Description
31:0	GPIO_IN	GPIO[63:32] input values.

0x01200068 TLMM_GPIO_IN_2**Type:** R**Clock:** GCC_TLMM_AHB_CLK**Reset State:** 0x00000000

This register allows the input values for GPIO[95:64] to be read. This register is designed to read multiple input bits.

TLMM_GPIO_IN_2

Bits	Name	Description
31:0	GPIO_IN	GPIO[95:64] input values.

0x0120006C TLMM_GPIO_IN_3**Type:** R**Clock:** GCC_TLMM_AHB_CLK**Reset State:** 0x00000000

This register allows the input values for GPIO[115:96] to be read. This register is designed to read multiple input bits.

TLMM_GPIO_IN_3

Bits	Name	Description
25:0	GPIO_IN	GPIO[121:96] input values.

0x01200080 TLMM_GPIO_OE_0**Type:** RW**Clock:** GCC_TLMM_AHB_CLK**Reset State:** 0x00000000

This register controls the output enables of GPIO[31:0]. This register is designed for bit-bang or fast configuration of multiple output bits.

TLMM_GPIO_OE_0

Bits	Name	Description
31:0	GPIO_OE	GPIO[31:0] output enables.

0x01200084 TLMM_GPIO_OE_1**Type:** RW**Clock:** GCC_TLMM_AHB_CLK**Reset State:** 0x00000000

This register controls the output enables of GPIO[63:32]. This register is designed for bit-bang or fast configuration of multiple output bits.

TLMM_GPIO_OE_1

Bits	Name	Description
31:0	GPIO_OE	GPIO[63:32] output enables.

0x01200088 TLMM_GPIO_OE_2**Type:** RW**Clock:** GCC_TLMM_AHB_CLK**Reset State:** 0x00000000

This register controls the output enables of GPIO[95:64]. This register is designed for bit-bang or fast configuration of multiple output bits.

TLMM_GPIO_OE_2

Bits	Name	Description
31:0	GPIO_OE	GPIO[95:64] output enables.

0x0120008C TLMM_GPIO_OE_3**Type:** RW**Clock:** GCC_TLMM_AHB_CLK**Reset State:** 0x00000000

This register controls the output enables of GPIO[115:96]. This register is designed for bit-bang or fast configuration of multiple output bits.

TLMM_GPIO_OE_3

Bits	Name	Description
25:0	GPIO_OE	GPIO[121:96] output enables.

0x012000A0 TLMM_GPIO_OE_CLR_0**Type:** W**Clock:** GCC_TLMM_AHB_CLK**Reset State:** 0x00000000

The GPIO_OE_CLR_0 register bits can be used to clear the output enables of GPIO[31:0]. Once the CLR bit is written to '1', the OE pin of the corresponding GPIO is cleared to '0'. The CLR bit then automatically returns to '0' so that it can be used again. This register is designed for bit-bang or fast configuration of multiple OE bits and alleviates the need for a read-->modify-->write operation.

TLMM_GPIO_OE_CLR_0

Bits	Name	Description
31:0	GPIO_OE_CLR	Clear bits for GPIO[31:0] output enables.

0x012000A4 TLMM_GPIO_OE_CLR_1**Type:** W**Clock:** GCC_TLMM_AHB_CLK**Reset State:** 0x00000000

The GPIO_OE_CLR_1 register bits can be used to clear the output enables of GPIO[63:32]. Once the CLR bit is written to '1', the OE pin of the corresponding GPIO is cleared to '0'. The CLR bit then automatically returns to '0' so that it can be used again. This register is designed for bit-bang or fast configuration of multiple OE bits and alleviates the need for a read-->modify-->write operation.

TLMM_GPIO_OE_CLR_1

Bits	Name	Description
31:0	GPIO_OE_CLR	Clear bits for GPIO[63:32] output enables.

0x012000A8 TLMM_GPIO_OE_CLR_2**Type:** W**Clock:** GCC_TLMM_AHB_CLK**Reset State:** 0x00000000

The GPIO_OE_CLR_2 register bits can be used to clear the output enables of GPIO[95:64]. Once the CLR bit is written to '1', the OE pin of the corresponding GPIO is cleared to '0'. The CLR bit then automatically returns to '0' so that it can be used again. This register is designed for bit-bang or fast configuration of multiple OE bits and alleviates the need for a read-->modify-->write operation.

TLMM_GPIO_OE_CLR_2

Bits	Name	Description
31:0	GPIO_OE_CLR	Clear bits for GPIO[95:64] output enables.

0x012000AC TLMM_GPIO_OE_CLR_3**Type:** W**Clock:** GCC_TLMM_AHB_CLK**Reset State:** 0x00000000

The GPIO_OE_CLR_3 register bits can be used to clear the output enables of GPIO[115:96]. Once the CLR bit is written to '1', the OE pin of the corresponding GPIO is cleared to '0'. The CLR bit then automatically returns to '0' so that it can be used again. This register is designed for bit-bang or fast configuration of multiple OE bits and alleviates the need for a read-->modify-->write operation.

TLMM_GPIO_OE_CLR_3

Bits	Name	Description
25:0	GPIO_OE_CLR	Clear bits for GPIO[121:96] output enables.

0x012000C0 TLMM_GPIO_OE_SET_0**Type:** W**Clock:** GCC_TLMM_AHB_CLK**Reset State:** 0x00000000

This register can be used to set the output enables of GPIO[31:0]. Once the SET bit is written to '1', the OE pin of the corresponding GPIO is set to '1'. The SET bit then automatically returns to '0' so that it can be used again. This register is designed for bit-bang or fast configuration of multiple OE bits and alleviates the need for a read-->modify-->write operation.

TLMM_GPIO_OE_SET_0

Bits	Name	Description
31:0	GPIO_OE_SET	Set bits for GPIO[31:0] output enables.

0x012000C4 TLMM_GPIO_OE_SET_1**Type:** W**Clock:** GCC_TLMM_AHB_CLK**Reset State:** 0x00000000

This register can be used to set the output enables of GPIO[63:32]. Once the SET bit is written to '1', the OE pin of the corresponding GPIO is set to '1'. The SET bit then automatically returns to '0' so that it can be used again. This register is designed for bit-bang or fast configuration of multiple OE bits and alleviates the need for a read-->modify-->write operation.

TLMM_GPIO_OE_SET_1

Bits	Name	Description
31:0	GPIO_OE_SET	Set bits for GPIO[63:32] output enables.

0x012000C8 TLMM_GPIO_OE_SET_2**Type:** W**Clock:** GCC_TLMM_AHB_CLK**Reset State:** 0x00000000

This register can be used to set the output enables of GPIO[95:64]. Once the SET bit is written to '1', the OE pin of the corresponding GPIO is set to '1'. The SET bit then automatically returns to '0' so that it can be used again. This register is designed for bit-bang or fast configuration of multiple OE bits and alleviates the need for a read-->modify-->write operation.

TLMM_GPIO_OE_SET_2

Bits	Name	Description
31:0	GPIO_OE_SET	Set bits for GPIO[95:64] output enables.

0x012000CC TLMM_GPIO_OE_SET_3

Type: W

Clock: GCC_TLMM_AHB_CLK

Reset State: 0x00000000

This register can be used to set the output enables of GPIO[115:96]. Once the SET bit is written to '1', the OE pin of the corresponding GPIO is set to '1'. The SET bit then automatically returns to '0' so that it can be used again. This register is designed for bit-bang or fast configuration of multiple OE bits and alleviates the need for a read-->modify-->write operation.

TLMM_GPIO_OE_SET_3

Bits	Name	Description
25:0	GPIO_OE_SET	Set bits for GPIO[121:96] output enables.

Table 15-1 GPIO table

GPIO #	Function	Output Enable (gnd_tie=input) (vdd_tie=output)	GPIO_CFG[5:2] (FUNC_SEL)
gpio[0]			
	GPIO_IN_OUT(0)	GPIO_CFG(0)	0
	blsp_spi_mosi(1)	blsp_spi_mosi_en(1)	1
	blsp_uart_tx(1)	vdd_tie	2
	blsp_uim_data(1)	blsp_uim_data_en(1)	3
	dmic0_clk	vdd_tie	4
gpio[1]			
	GPIO_IN_OUT(1)	GPIO_CFG(1)	0
	blsp_spi_miso(1)	blsp_spi_miso_en(1)	1
	blsp_uart_rx(1)	gnd_tie	2
	blsp_uim_clk(1)	vdd_tie	3
	dmic0_data	gnd_tie	4
gpio[2]			
	GPIO_IN_OUT(2)	GPIO_CFG(2)	0
	blsp_spi_cs_n(1)	blsp_spi_cs_n_en(1)	1
	blsp_uart_cts_n(1)	gnd_tie	2
	blsp_i2c_sda(1)	blsp_i2c_sda_en(1)	3
gpio[3]			
	GPIO_IN_OUT(3)	GPIO_CFG(3)	0
	blsp_spi_clk(1)	blsp_spi_clk_en(1)	1
	blsp_uart_rfr_n(1)	vdd_tie	2
	blsp_i2c_scl(1)	blsp_i2c_scl_en(1)	3
gpio[4]			
	GPIO_IN_OUT(4)	GPIO_CFG(4)	0
	blsp_spi_mosi(2)	blsp_spi_mosi_en(2)	1
	blsp_uart_tx(2)	vdd_tie	2

Table 15-1 GPIO table

	blsp_uim_data(2)	blsp_uim_data_en(2)	3
	blsp1_spi_cs3_n	blsp1_spi_cs3_n_en	4
gpio[5]			
	GPIO_IN_OUT(5)	GPIO_CFG(5)	0
	blsp_spi_miso(2)	blsp_spi_miso_en(2)	1
	blsp_uart_rx(2)	gnd_tie	2
	blsp_uim_clk(2)	vdd_tie	3
	blsp2_spi_cs3_n	blsp2_spi_cs3_n_en	4
gpio[6]			
	GPIO_IN_OUT(6)	GPIO_CFG(6)	0
	blsp_spi_cs_n(2)	blsp_spi_cs_n_en(2)	1
	blsp_uart_cts_n(2)	gnd_tie	2
	blsp_i2c_sda(2)	blsp_i2c_sda_en(2)	3
gpio[7]			
	GPIO_IN_OUT(7)	GPIO_CFG(7)	0
	blsp_spi_clk(2)	blsp_spi_clk_en(2)	1
	blsp_uart_rfr_n(2)	vdd_tie	2
	blsp_i2c_scl(2)	blsp_i2c_scl_en(2)	3
gpio[8]			
	GPIO_IN_OUT(8)	GPIO_CFG(8)	0
	blsp_spi_mosi(3)	blsp_spi_mosi_en(3)	1
gpio[9]			
	GPIO_IN_OUT(9)	GPIO_CFG(9)	0
	blsp_spi_miso(3)	blsp_spi_miso_en(3)	1
gpio[10]			
	GPIO_IN_OUT(10)	GPIO_CFG(10)	0
	blsp_spi_cs_n(3)	blsp_spi_cs_n_en(3)	1
	blsp_i2c_sda(3)	blsp_i2c_sda_en(3)	2

Table 15-1 GPIO table

gpio[11]			
	GPIO_IN_OUT(11)	GPIO_CFG(11)	0
	blsp_spi_clk(3)	blsp_spi_clk_en(3)	1
	blsp_i2c_scl(3)	blsp_i2c_scl_en(3)	2
gpio[12]			
	GPIO_IN_OUT(12)	GPIO_CFG(12)	0
	blsp_spi_mosi(4)	blsp_spi_mosi_en(4)	1
	gcc_gp2_clk_b	vdd_tie	2
gpio[13]			
	GPIO_IN_OUT(13)	GPIO_CFG(13)	0
	blsp_spi_miso(4)	blsp_spi_miso_en(4)	1
	gcc_gp3_clk_b	vdd_tie	2
gpio[14]			
	GPIO_IN_OUT(14)	GPIO_CFG(14)	0
	blsp_spi_cs_n(4)	blsp_spi_cs_n_en(4)	1
	blsp_i2c_sda(4)	blsp_i2c_sda_en(4)	2
gpio[15]			
	GPIO_IN_OUT(15)	GPIO_CFG(15)	0
	blsp_spi_clk(4)	blsp_spi_clk_en(4)	1
	blsp_i2c_scl(4)	blsp_i2c_scl_en(4)	2
gpio[16]			
	GPIO_IN_OUT(16)	GPIO_CFG(16)	0
	blsp_spi_mosi(5)	blsp_spi_mosi_en(5)	1
	blsp1_spi_cs2_n	blsp1_spi_cs2_n_en	2
gpio[17]			
	GPIO_IN_OUT(17)	GPIO_CFG(17)	0
	blsp_spi_miso(5)	blsp_spi_miso_en(5)	1
	blsp2_spi_cs2_n	blsp2_spi_cs2_n_en	2

Table 15-1 GPIO table

gpio[18]			
	GPIO_IN_OUT(18)	GPIO_CFG(18)	0
	blsp_spi_cs_n(5)	blsp_spi_cs_n_en(5)	1
	blsp_i2c_sda(5)	blsp_i2c_sda_en(5)	2
gpio[19]			
	GPIO_IN_OUT(19)	GPIO_CFG(19)	0
	blsp_spi_clk(5)	blsp_spi_clk_en(5)	1
	blsp_i2c_scl(5)	blsp_i2c_scl_en(5)	2
gpio[20]			
	GPIO_IN_OUT(20)	GPIO_CFG(20)	0
	blsp_spi_mosi(6)	blsp_spi_mosi_en(6)	1
	gp_pdm_0a	gp_pdm_0a_en	2
gpio[21]			
	GPIO_IN_OUT(21)	GPIO_CFG(21)	0
	blsp_spi_miso(6)	blsp_spi_miso_en(6)	1
	gp_pdm_1b	gp_pdm_1b_en	2
gpio[22]			
	GPIO_IN_OUT(22)	GPIO_CFG(22)	0
	blsp_spi_cs_n(6)	blsp_spi_cs_n_en(6)	1
	blsp_i2c_sda(6)	blsp_i2c_sda_en(6)	2
gpio[23]			
	GPIO_IN_OUT(23)	GPIO_CFG(23)	0
	blsp_spi_clk(6)	blsp_spi_clk_en(6)	1
	blsp_i2c_scl(6)	blsp_i2c_scl_en(6)	2
gpio[24]			
	GPIO_IN_OUT(24)	GPIO_CFG(24)	0
	mdp_vsync_p	gnd_tie	1
gpio[25]			

Table 15-1 GPIO table

	GPIO_IN_OUT(25)	GPIO_CFG(25)	0
	mdp_vsync_s	gnd_tie	1
	gp_pdm_0b	gp_pdm_0b_en	2
gpio[26]			
	GPIO_IN_OUT(26)	GPIO_CFG(26)	0
	cam_mclk0	vdd_tie	1
gpio[27]			
	GPIO_IN_OUT(27)	GPIO_CFG(27)	0
	cam_mclk1	vdd_tie	1
gpio[28]			
	GPIO_IN_OUT(28)	GPIO_CFG(28)	0
	pwr_modem_enabled_a	vdd_tie	1
gpio[29]			
	GPIO_IN_OUT(29)	GPIO_CFG(29)	0
	cci_i2c_sda0	cci_i2c_sda0_en	1
gpio[30]			
	GPIO_IN_OUT(30)	GPIO_CFG(30)	0
	cci_i2c_scl0	cci_i2c_scl0_en	1
gpio[31]			
	GPIO_IN_OUT(31)	GPIO_CFG(31)	0
	cci_timer0	vdd_tie	1
	gp0_clk	vdd_tie	2
gpio[32]			
	GPIO_IN_OUT(32)	GPIO_CFG(32)	0
	cci_timer1	vdd_tie	1
	gp1_clk	vdd_tie	2
gpio[33]			
	GPIO_IN_OUT(33)	GPIO_CFG(33)	0

Table 15-1 GPIO table

	cci_async_in0	gnd_tie	1
gpio[34]			
	GPIO_IN_OUT(34)	GPIO_CFG(34)	0
	pwr_nav_enabled_a	vdd_tie	1
gpio[35]			
	GPIO_IN_OUT(35)	GPIO_CFG(35)	0
	pwr_crypto_enabled_a	vdd_tie	1
gpio[36]			
	GPIO_IN_OUT(36)	GPIO_CFG(36)	0
gpio[37]			
	GPIO_IN_OUT(37)	GPIO_CFG(37)	0
	blsp3_spi_cs2_n	blsp3_spi_cs2_n_en	1
	forced_usb_boot	gnd_tie	
gpio[38]			
	GPIO_IN_OUT(38)	GPIO_CFG(38)	0
	cci_timer2	vdd_tie	1
gpio[39]			
	GPIO_IN_OUT(39)	GPIO_CFG(39)	0
	wcss_bt_ssbi	wcss_bt_ssbi_en	1
gpio[40]			
	GPIO_IN_OUT(40)	GPIO_CFG(40)	0
	wcss_wlan_data(2)	wcss_wlan_data_en(2)	1
gpio[41]			
	GPIO_IN_OUT(41)	GPIO_CFG(41)	0
	wcss_wlan_data(1)	wcss_wlan_data_en(1)	1
gpio[42]			
	GPIO_IN_OUT(42)	GPIO_CFG(42)	0
	wcss_wlan_data(0)	wcss_wlan_data_en(0)	1

Table 15-1 GPIO table

gpio[43]			
	GPIO_IN_OUT(43)	GPIO_CFG(43)	0
	wcss_wlan_set	wcss_wlan_set_en	1
gpio[44]			
	GPIO_IN_OUT(44)	GPIO_CFG(44)	0
	wcss_wlan_clk	wcss_wlan_clk_en	1
gpio[45]			
	GPIO_IN_OUT(45)	GPIO_CFG(45)	0
	wcss_fm_ssbi	wcss_fm_ssbi_en	1
gpio[46]			
	GPIO_IN_OUT(46)	GPIO_CFG(46)	0
	wcss_fm_sdi	wcss_fm_sdi_en	1
gpio[47]			
	GPIO_IN_OUT(47)	GPIO_CFG(47)	0
	wcss_bt_ctl	vdd_tie	1
gpio[48]			
	GPIO_IN_OUT(48)	GPIO_CFG(48)	0
	wcss_bt_dat_stb	wcss_bt_dat_stb_en	1
gpio[49]			
	GPIO_IN_OUT(49)	GPIO_CFG(49)	0
	uim3_data	uim3_data_en	1
	gcc_gp1_clk_a	vdd_tie	2
gpio[50]			
	GPIO_IN_OUT(50)	GPIO_CFG(50)	0
	uim3_clk	vdd_tie	1
	gcc_gp2_clk_a	vdd_tie	2
gpio[51]			
	GPIO_IN_OUT(51)	GPIO_CFG(51)	0

Table 15-1 GPIO table

	uim3_reset	vdd_tie	1
	gcc_gp3_clk_a	vdd_tie	2
gpio[52]			
	GPIO_IN_OUT(52)	GPIO_CFG(52)	0
	uim3_present	gnd_tie	1
	gp_pdm_1a	gp_pdm_1a_en	2
gpio[53]			
	GPIO_IN_OUT(53)	GPIO_CFG(53)	0
	uim2_data	uim2_data_en	1
gpio[54]			
	GPIO_IN_OUT(54)	GPIO_CFG(54)	0
	uim2_clk	vdd_tie	1
gpio[55]			
	GPIO_IN_OUT(55)	GPIO_CFG(55)	0
	uim2_reset	vdd_tie	1
gpio[56]			
	GPIO_IN_OUT(56)	GPIO_CFG(56)	0
	uim2_present	gnd_tie	1
gpio[57]			
	GPIO_IN_OUT(57)	GPIO_CFG(57)	0
	uim1_data	uim1_data_en	1
gpio[58]			
	GPIO_IN_OUT(58)	GPIO_CFG(58)	0
	uim1_clk	vdd_tie	1
gpio[59]			
	GPIO_IN_OUT(59)	GPIO_CFG(59)	0
	uim1_reset	vdd_tie	1
gpio[60]			
	GPIO_IN_OUT(60)	GPIO_CFG(60)	0
	uim1_present	gnd_tie	1
gpio[61]			

Table 15-1 GPIO table

	GPIO_IN_OUT(61)	GPIO_CFG(61)	0
	uim_batt_alarm	gnd_tie	1
gpio[62]			
	GPIO_IN_OUT(62)	GPIO_CFG(62)	0
gpio[63]			
	GPIO_IN_OUT(63)	GPIO_CFG(63)	0
	cdc_pdm0_clk	cdc_pdm0_clk_en	1
gpio[64]			
	GPIO_IN_OUT(64)	GPIO_CFG(64)	0
	cdc_pdm0_sync	vdd_tie	1
gpio[65]			
	GPIO_IN_OUT(65)	GPIO_CFG(65)	0
	cdc_pdm0_tx0	gnd_tie	1
gpio[66]			
	GPIO_IN_OUT(66)	GPIO_CFG(66)	0
	cdc_pdm0_rx0	vdd_tie	1
gpio[67]			
	GPIO_IN_OUT(67)	GPIO_CFG(67)	0
	cdc_pdm0_rx1	vdd_tie	1
gpio[68]			
	GPIO_IN_OUT(68)	GPIO_CFG(68)	0
	cdc_pdm0_rx2	vdd_tie	1
gpio[69]			
	GPIO_IN_OUT(69)	GPIO_CFG(69)	0
	blsp3_spi_cs3_n	blsp3_spi_cs3_n_en	1
gpio[70]			
	GPIO_IN_OUT(70)	GPIO_CFG(70)	0
	rffe1_clk	vdd_tie	1
gpio[71]			
	GPIO_IN_OUT(71)	GPIO_CFG(71)	0
	rffe1_data	rffe1_data_en	1
gpio[72]			
	GPIO_IN_OUT(72)	GPIO_CFG(72)	0

Table 15-1 GPIO table

	rffe2_clk	vdd_tie	1
gpio[73]			
	GPIO_IN_OUT(73)	GPIO_CFG(73)	0
	rffe2_data	rffe2_data_en	1
gpio[74]			
	GPIO_IN_OUT(74)	GPIO_CFG(74)	0
	grfc(0)	vdd_tie	1
	rffe3_clk	vdd_tie	2
gpio[75]			
	GPIO_IN_OUT(75)	GPIO_CFG(75)	0
	grfc(1)	vdd_tie	1
gpio[76]			
	GPIO_IN_OUT(76)	GPIO_CFG(76)	0
	grfc(2)	vdd_tie	1
	rffe3_data	rffe3_data_en	2
gpio[77]			
	GPIO_IN_OUT(77)	GPIO_CFG(77)	0
	grfc(3)	vdd_tie	1
gpio[78]			
	GPIO_IN_OUT(78)	GPIO_CFG(78)	0
	grfc(4)	vdd_tie	1
gpio[79]			
	GPIO_IN_OUT(79)	GPIO_CFG(79)	0
	grfc(5)	vdd_tie	1
gpio[80]			
	GPIO_IN_OUT(80)	GPIO_CFG(80)	0
	grfc(6)	vdd_tie	1
	boot_config(0)	gnd_tie	
gpio[81]			
	GPIO_IN_OUT(81)	GPIO_CFG(81)	0
	grfc(7)	vdd_tie	1
	boot_config(1)	gnd_tie	
gpio[82]			

Table 15-1 GPIO table

	GPIO_IN_OUT(82)	GPIO_CFG(82)	0
	grfc(8)	vdd_tie	1
	boot_config(2)	gnd_tie	
gpio[83]			
	GPIO_IN_OUT(83)	GPIO_CFG(83)	0
	grfc(9)	vdd_tie	1
	boot_config(3)	gnd_tie	
gpio[84]			
	GPIO_IN_OUT(84)	GPIO_CFG(84)	0
	grfc(10)	vdd_tie	1
	boot_config(4)	gnd_tie	
gpio[85]			
	GPIO_IN_OUT(85)	GPIO_CFG(85)	0
	grfc(11)	vdd_tie	1
	sd_c_emmc_mode1p2_en	gnd_tie	
gpio[86]			
	GPIO_IN_OUT(86)	GPIO_CFG(86)	0
	grfc(12)	vdd_tie	1
	pa_indicator_1_or_2	vdd_tie	2
	boot_config(5)	gnd_tie	
gpio[87]			
	GPIO_IN_OUT(87)	GPIO_CFG(87)	0
	grfc(13)	vdd_tie	1
	boot_config(6)	gnd_tie	
gpio[88]			
	GPIO_IN_OUT(88)	GPIO_CFG(88)	0
	grfc(14)	vdd_tie	1
	boot_config(7)	gnd_tie	
gpio[89]			
	GPIO_IN_OUT(89)	GPIO_CFG(89)	0
	grfc(15)	vdd_tie	1
	boot_config(8)	gnd_tie	
gpio[90]			

Table 15-1 GPIO table

	GPIO_IN_OUT(90)	GPIO_CFG(90)	0
	grfc(16)	vdd_tie	1
	boot_config(9)	gnd_tie	
gpio[91]			
	GPIO_IN_OUT(91)	GPIO_CFG(91)	0
	grfc(17)	vdd_tie	1
	gps_tx_aggressor	gnd_tie	2
	boot_config(10)	gnd_tie	
gpio[92]			
	GPIO_IN_OUT(92)	GPIO_CFG(92)	0
	grfc(18)	vdd_tie	1
	boot_config(11)	gnd_tie	
gpio[93]			
	GPIO_IN_OUT(93)	GPIO_CFG(93)	0
	grfc(19)	vdd_tie	1
	boot_config(12)	gnd_tie	
gpio[94]			
	GPIO_IN_OUT(94)	GPIO_CFG(94)	0
	grfc(20)	vdd_tie	1
	boot_config(13)	gnd_tie	
gpio[95]			
	GPIO_IN_OUT(95)	GPIO_CFG(95)	0
	grfc(21)	vdd_tie	1
gpio[96]			
	GPIO_IN_OUT(96)	GPIO_CFG(96)	0
	grfc(24)	vdd_tie	1
gpio[97]			
	GPIO_IN_OUT(97)	GPIO_CFG(97)	0
	gcc_gp1_clk_b	vdd_tie	1
	boot_config(14)	gnd_tie	
gpio[98]			
	GPIO_IN_OUT(98)	GPIO_CFG(98)	0
	gp_pdm_2a	gp_pdm_2a_en	1

Table 15-1 GPIO table

gpio[99]			
	GPIO_IN_OUT(99)	GPIO_CFG(99)	0
	gsm0_tx_phase_d(0)	gsm0_tx_phase_d_en(0)	1
gpio[100]			
	GPIO_IN_OUT(100)	GPIO_CFG(100)	0
	gsm0_tx_phase_d(1)	gsm0_tx_phase_d_en(1)	1
gpio[101]			
	GPIO_IN_OUT(101)	GPIO_CFG(101)	0
	gsm1_tx_phase_d(0)	gsm1_tx_phase_d_en(0)	1
gpio[102]			
	GPIO_IN_OUT(102)	GPIO_CFG(102)	0
	gsm1_tx_phase_d(1)	gsm1_tx_phase_d_en(1)	1
gpio[103]			
	GPIO_IN_OUT(103)	GPIO_CFG(103)	0
	ssbi_wtr0_rx	ssbi_wtr0_rx_en	1
	rffe5_clk	vdd_tie	2
gpio[104]			
	GPIO_IN_OUT(104)	GPIO_CFG(104)	0
	ssbi_wtr0_tx	ssbi_wtr0_tx_en	1
	rffe5_data	rffe5_data_en	2
gpio[105]			
	GPIO_IN_OUT(105)	GPIO_CFG(105)	0
	ssbi_wtr1_rx	ssbi_wtr1_rx_en	1
	rffe4_clk	vdd_tie	2
gpio[106]			
	GPIO_IN_OUT(106)	GPIO_CFG(106)	0
	ssbi_wtr1_tx	ssbi_wtr1_tx_en	1
	rffe4_data	rffe4_data_en	2

Table 15-1 GPIO table

gpio[107]			
	GPIO_IN_OUT(107)	GPIO_CFG(107)	0
gpio[108]			
	GPIO_IN_OUT(108)	GPIO_CFG(108)	0
gpio[109]			
	GPIO_IN_OUT(109)	GPIO_CFG(109)	0
gpio[110]			
	GPIO_IN_OUT(110)	GPIO_CFG(110)	0
	blsp1_spi_cs1_n	blsp1_spi_cs1_n_en	1
	pri_mi2s_ws	pri_mi2s_ws_en	2
	gp_mn	gp_mn_en	3
gpio[111]			
	GPIO_IN_OUT(111)	GPIO_CFG(111)	0
gpio[112]			
	GPIO_IN_OUT(112)	GPIO_CFG(112)	0
	sec_mi2s_data1	sec_mi2s_data1_en	1
gpio[113]			
	GPIO_IN_OUT(113)	GPIO_CFG(113)	0
	pri_mi2s_sck	pri_mi2s_sck_en	1
	gp_pdm_2b	gp_pdm_2b_en	2
gpio[114]			
	GPIO_IN_OUT(114)	GPIO_CFG(114)	0
	pri_mi2s_data0	pri_mi2s_data0_en	1
gpio[115]			
	GPIO_IN_OUT(115)	GPIO_CFG(115)	0
	pri_mi2s_data1	pri_mi2s_data1_en	1
gpio[116]			
	GPIO_IN_OUT(116)	GPIO_CFG(116)	0
	pri_mi2s_mclk	pri_mi2s_mclk_en	1
gpio[117]			
	GPIO_IN_OUT(117)	GPIO_CFG(117)	0

Table 15-1 GPIO table

	sec_mi2s_ws	sec_mi2s_ws_en	1
gpio[118]			
	GPIO_IN_OUT(118)	GPIO_CFG(118)	0
	sec_mi2s_sck	sec_mi2s_sck_en	1
gpio[119]			
	GPIO_IN_OUT(119)	GPIO_CFG(119)	0
	sec_mi2s_data0	sec_mi2s_data0_en	1
gpio[120]			
	GPIO_IN_OUT(120)	GPIO_CFG(120)	0
	blsp3_spi_cs1_n	blsp3_spi_cs1_n_en	1
gpio[121]			
	GPIO_IN_OUT(121)	GPIO_CFG(121)	0
	sd_write_protect	gnd_tie	1
	blsp2_spi_cs1_n	blsp2_spi_cs1_n_en	2

16 Low Power Audio Subsystem

16.1 LPASS_LPA_IF

0x07708004 LPASS_LPAIF_HW_CONFIG

Type: R

Clock: CC_LPAIF_HCLK

Reset State: 0x06AC2902

LPAIF HW configuration for DMIC/SIF/I2S/PCM

LPASS_LPAIF_HW_CONFIG

Bits	Name	Description
30	SIF_EN	If SIF module is present in LPAIF
29	SEC_PCM_EN	Secondary PCM present
28	PRI_PCM_EN	Primary PCM present
27:26	QUAD_I2S_LINES	Number of lanes for Quaternary interface -1
25:24	QUAD_I2S_DIR	Quaternary interface direction: 0 - SPKR setting 1 - MIC setting 2- Bidirectional
23	QUAD_I2S_EN	Quaternary interface present
22:21	TER_I2S_LINES	Number of lanes for Tertiary interface -1
20:19	TER_I2S_DIR	Tertiary interface direction: 0 - SPKR setting 1 - MIC setting 2- Bidirectional
18	TER_I2S_EN	Tertiary interface present
17:16	SEC_I2S_LINES	Number of lanes for Secondary interface -1
15:14	SEC_I2S_DIR	Secondary interface direction: 0 - SPKR setting 1 - MIC setting 2- Bidirectional
13	SEC_I2S_EN	Secondary interface present

LPASS_LPAIF_HW_CONFIG (cont.)

Bits	Name	Description
12:11	PRI_I2S_LINES	Number of lanes for Primary interface -1
10:9	PRI_I2S_DIR	Primary interface direction: 0 - SPKR setting 1 - MIC setting 2- Bidirectional
8	PRI_I2S_EN	Primary interface present
7:6	SPKR_I2S_LINES	Number of lanes for Speaker interface -1, has to be less than Primary interface
5:4	SPKR_I2S_DIR	Speaker interface direction: 0 - SPKR setting 1 - MIC setting 2- Bidirectional
3	SPKR_I2S_EN	Speaker interface present
2:1	NUM_DMIC	Number of DMIC modules -1, meaningful if DMIC_EN=1
0	DMIC_EN	DMIC present

0x07708008 LPASS_LPAIF_HW_CONFIG2**Type:** R**Clock:** CC_LPAIF_HCLK**Reset State:** 0x00000009

LPAIF HW configuration for Read DMA/Write DMA/ Rate auto detect circuit

LPASS_LPAIF_HW_CONFIG2

Bits	Name	Description
7	SEC_RATE_DET_EN	Secondary Rate Auto-detect module present
6	PRI_RATE_DET_EN	Primary Rate Auto-detect module present
5:3	NUM_RDDMA	Number of Read DMAs -1
2:0	NUM_WRDMA	Number of Write DMAs -1

0x07709000+ LPASS_LPAIF_I2S_CTLa, a=[0..3]**0x1000*a****Type:** RW**Clock:** CC_LPAIF_HCLK**Reset State:** 0x000F0004

LPASS_LPAIF_I2S_CTLa

Bits	Name	Description
31	RESET	0x0: DISABLE 0x1: ENABLE
22	EN_LONG_RATE	Allow for frame size to be longer than I2S sample bit width, BIT_WIDTH 0x0: DISABLE 0x1: ENABLE
21:16	LONG_RATE	Used as WS rate when EN_LONG_RATE=1
15	LOOPBACK	SW: RW, HW: R Loopback control bit. 0x0: DISABLE (DISABLE) 0x1: ENABLE (ENABLE (only applicable to a=1=codec_mic, a=2=mi2s, & a=4=sec mic. If ENABLE is set for a=1, codec speaker is looped back to both wires of codec mic. For a=2, mi2s is looped back on itself. For a=4 sec speaker is looped backed to sec mic))
14	SPKR_EN	SW: RW, HW: R Enable for the speaker direction operation (out of the MSM). I2S interface will start transmitting serial data when enabled and trying to access a read DMA channel. Set the enable after the other control bits (speaker related, bit_width, ws_src) are configured. Disable before changing the other bits. 0x0: DISABLE (DISABLE) 0x1: ENABLE (ENABLE)
13:10	SPKR_MODE	SW: RW, HW: R Controls which I2S lines are used (if supported by HW) and multichannel operation 0x0: NONE (NONE) 0x1: SD0 (SD0 (data sent to SD0 only- stereo or mono)) 0x2: SD1 (SD1 (data sent to SD1 only- stereo or mono)) 0x3: SD2 (SD2 (data sent to SD2 only- stereo or mono)) 0x4: SD3 (SD3 (data sent to SD3 only- stereo or mono)) 0x5: QUAD01 (QUAD01 (SD0 & SD1 = quad channel mode)) 0x6: QUAD23 (QUAD23 (SD2 & SD3 = quad channel mode)) 0x7: ENUM_6CH (6CH (SD0, SD1 & SD2 = 6 channel mode)) 0x8: ENUM_8CH (8CH (SD0, SD1, SD2 & SD3 = 8 channel mode))
9	SPKR_MONO	SW: RW, HW: R In mono mode, the speaker path of I2S (out of MSM) will take each mono sample from memory and transmit it during both the right and left portions of the word select. Only applicable when a single I2S line is used in SPCR_MODE. 0x0: STEREO (STEREO) 0x1: MONO (MONO)

LPASS_LPAIF_I2S_CTLa (cont.)

Bits	Name	Description
8	MIC_EN	SW: RW, HW: R Enable for the microphone direction operation (into the MSM). I2S interface will start capturing data when enabled and trying to access a write DMA channel. Set the enable after the other control bits (mic related, bit_width, ws_src) are configured. Disable before changing the other bits. 0x0: DISABLE (DISABLE) 0x1: ENABLE (ENABLE)
7:4	MIC_MODE	SW: RW, HW: R Controls which I2S lines are used (if supported by HW) and multichannel operation 0x0: NONE (NONE) 0x1: SD0 (SD0 (data taken from SD0 only- stereo or mono)) 0x2: SD1 (SD1 (data taken from SD1 only- stereo or mono)) 0x3: SD2 (SD2 (data taken from SD2 only- stereo or mono)) 0x4: SD3 (SD3 (data taken from SD3 only- stereo or mono)) 0x5: QUAD01 (QUAD01 (SD0 & SD1 = quad channel mode)) 0x6: QUAD23 (QUAD23 (SD2 & SD3 = quad channel mode)) 0x7: ENUM_6CH (6CH (SD0, SD1 & SD2 = 6 channel mode)) 0x8: ENUM_8CH (8CH (SD0, SD1, SD2 & SD3 = 8 channel mode))
3	MIC_MONO	SW: RW, HW: R Controls whether only the left word is stored in memory (mono) or the left and right words are stored (stereo). In mono mode, the data in the right portion of WS is ignored. Only applicable when a single I2S line is used in MIC_MODE. 0x0: STEREO (STEREO) 0x1: MONO (MONO)
2	WS_SRC	SW: RW, HW: R Controls the source of the word select, internally generated or taken from outside. Does not affect the clock. Master/Slave bit clock mode is controlled in LPASS clk_ctl. 0x0: INT (INT (master mode)) 0x1: EXT (EXT (slave mode))
1:0	BIT_WIDTH	SW: RW, HW: R Controls the bit width of the I2S samples 0x0: ENUM_16 (16) 0x1: ENUM_24 (24) 0x2: ENUM_32 (32) 0x3: ENUM_25 (25)

16.2 LPASS_CODEC_CORE

0x0771C000 LPASS_CDC_CLK_RX_RESET_CTL

Type: RW

Clock: csr_wr_clk

Reset State: 0x00000000

RX Interpolation Path Reset Control Register. This register enables or disables RX1 ->RX3 filter register states.

LPASS_CDC_CLK_RX_RESET_CTL

Bits	Name	Description
2	RX3_RESET	Specifies RX Interpolation Path Reset State 0x0: DISABLE 0x1: ENABLE
1	RX2_RESET	Specifies RX Interpolation Path Reset State 0x0: DISABLE 0x1: ENABLE
0	RX1_RESET	Specifies RX Interpolation Path Reset State 0x0: DISABLE 0x1: ENABLE

0x0771C004 LPASS_CDC_CLK_TX_RESET_B1_CTL

Type: RW

Clock: csr_wr_clk

Reset State: 0x00000000

TX Decimation Path Reset Control Register. This register enables or disables TX1 ->TX4 filter register states.

LPASS_CDC_CLK_TX_RESET_B1_CTL

Bits	Name	Description
1	TX2_RESET	Specifies TX Interpolation Path Reset State 0x0: DISABLE 0x1: ENABLE
0	TX1_RESET	Specifies TX Interpolation Path Reset State 0x0: DISABLE 0x1: ENABLE

0x0771C008 LPASS_CDC_CLK_DMIC_B1_CTL**Type:** RW**Clock:** csr_wr_clk**Reset State:** 0x00000000**LPASS_CDC_CLK_DMIC_B1_CTL**

Bits	Name	Description
3:1	DMIC0_CLK_SEL	Specifies the clock frequency of the input DMIC data samples in the TX path. Div6 is only valid for mclk=12.288 0x0: DIV2 0x1: DIV3 0x2: DIV4 0x3: DIV6 0x4: DIV16
0	DMIC0_CLK_EN	DMIC0 Clock Enable 0x0: DISABLE 0x1: ENABLE

0x0771C00C LPASS_CDC_CLK_RX_I2S_CTL**Type:** RW**Clock:** csr_wr_clk**Reset State:** 0x00000013

RX I2S Control Register. This register is required for I2S operation. The I2S master mode select register bit is not present in **APQ8016**. Instead the I2S master mode selection is done via a hardware port which is controlled by LPASS software settings.

LPASS_CDC_CLK_RX_I2S_CTL

Bits	Name	Description
5	RX_I2S_MODE	0x0: MODE_32 0x1: MODE_16
4	RX_I2S_CLK_EN	0x0: RX_I2S_CLK_DIS 0x1: RX_I2S_CLK_EN
2:0	RX_I2S_FS_RATE	0x0: F_8_KHZ 0x1: F_16_KHZ 0x2: F_32_KHZ 0x3: F_48_KHZ 0x4: F_96_KHZ 0x5: F_192_KHZ

0x0771C010 LPASS_CDC_CLK_TX_I2S_CTL**Type:** RW**Clock:** csr_wr_clk**Reset State:** 0x00000013

TX I2S Control Register. This register is required for I2S operation. The I2S master mode select register bit is not present in **APQ8016**. Instead the I2S master mode selection is done via a hardware port which is controlled by LPASS software settings.

LPASS_CDC_CLK_TX_I2S_CTL

Bits	Name	Description
5	TX_I2S_MODE	0x0: MODE_32 0x1: MODE_16
4	TX_I2S_CLK_EN	0x0: TX_I2S_CLK_DIS 0x1: TX_I2S_CLK_EN
2:0	TX_I2S_FS_RATE	0x0: F_8_KHZ 0x1: F_16_KHZ 0x2: F_32_KHZ 0x3: F_48_KHZ 0x4: F_96_KHZ 0x5: F_192_KHZ

0x0771C014 LPASS_CDC_CLK_OTHR_RESET_B1_CTL**Type:** RW**Clock:** csr_wr_clk**Reset State:** 0x00000000

Connectivity Block Control Register.

LPASS_CDC_CLK_OTHR_RESET_B1_CTL

Bits	Name	Description
1	CONN_RESET	This reset is applied to both the connectivity block and the pdm (serializer/deserializer) blocks 0x0: DISABLE 0x1: ENABLE

0x0771C018 LPASS_CDC_CLK_TX_CLK_EN_B1_CTL**Type:** RW**Clock:** csr_wr_clk**Reset State:** 0x00000000

TX Decimation Path Clock Control Register. This register enables or disables MCLK to TX1->TX2

LPASS_CDC_CLK_TX_CLK_EN_B1_CTL

Bits	Name	Description
1	TX2_CLK_EN	TX Clock Enable 0x0: DISABLE 0x1: ENABLE
0	TX1_CLK_EN	TX Clock Enable 0x0: DISABLE 0x1: ENABLE

0x0771C01C LPASS_CDC_CLK_OTHR_CTL

Type: RW

Clock: csr_wr_clk

Reset State: 0x00000004

Connectivity Clock Control Register. This register enables or disables MCLK to the connectivity block.

LPASS_CDC_CLK_OTHR_CTL

Bits	Name	Description
2	CONN_CLK_EN	Connectivity Clock Enable 0x0: DISABLE 0x1: ENABLE

0x0771C020 LPASS_CDC_CLK_RX_B1_CTL

Type: RW

Clock: csr_wr_clk

Reset State: 0x00000000

RX Interpolation Path Clock Control Register. This register enables or disables RX1->RX3

LPASS_CDC_CLK_RX_B1_CTL

Bits	Name	Description
2	RX3_CLK_EN	RX Clock Enable 0x0: DISABLE 0x1: ENABLE

LPASS_CDC_CLK_RX_B1_CTL (cont.)

Bits	Name	Description
1	RX2_CLK_EN	RX Clock Enable 0x0: DISABLE 0x1: ENABLE
0	RX1_CLK_EN	RX Clock Enable 0x0: DISABLE 0x1: ENABLE

0x0771C024 LPASS_CDC_CLK_MCLK_CTL**Type:** RW**Clock:** csr_wr_clk**Reset State:** 0x00000000

Master Clock Control Register. This register enables or disables the MCLK input to Codec Core.

LPASS_CDC_CLK_MCLK_CTL

Bits	Name	Description
0	MCLK_EN	Master Clock Enable 0x0: DISABLE 0x1: ENABLE

0x0771C028 LPASS_CDC_CLK_PDM_CTL**Type:** RW**Clock:** csr_wr_clk**Reset State:** 0x00000000

PDM Clock Control Register. This register enables or disables PDM Clock Output and PDM FB Clock selection. Switching of the PDM FB Clock can only occur and when PDM Clock is DISABLE

LPASS_CDC_CLK_PDM_CTL

Bits	Name	Description
1	PDM_CLK_SEL	PDM Clock Selection 0x0: PDM_CLK 0x1: PDM_CLK_FB
0	PDM_EN	PDM Clock Enable 0x0: DISABLE 0x1: ENABLE

0x0771C02C LPASS_CDC_CLK_SD_CTL**Type:** RW**Clock:** csr_wr_clk**Reset State:** 0x00000000

SideTone Reset and Clock Control Register. This register enables or disables reset/clocks to the SideTone Path.

LPASS_CDC_CLK_SD_CTL

Bits	Name	Description
5	IIR2_RESET	IIR Reset 0x0: DISABLE 0x1: ENABLE
4	IIR1_RESET	IIR Reset 0x0: DISABLE 0x1: ENABLE
1	IIR2_CLK_EN	IIR Clock Enable 0x0: DISABLE 0x1: ENABLE
0	IIR1_CLK_EN	IIR Clock Enable 0x0: DISABLE 0x1: ENABLE

0x0771C02C+LPASS_CDC_RXn_B4_CTL, n=[1..3]**0x20*n****Type:** RW**Clock:** csr_wr_clk**Reset State:** 0x00000000**LPASS_CDC_RXn_B4_CTL**

Bits	Name	Description
2	RESERVED	Reserved
1:0	CUT_OFF_FREQ	Specifies the Cutoff frequency for the DC blocker inside the RX chain. 0x0: CF_NEG_3DB_4HZ 0x1: CF_NEG_3DB_75HZ 0x2: CF_NEG_3DB_150HZ

0x0771C030+ LPASS_CDC_RXn_B5_CTL, n=[1..3]**0x20*n****Type:** RW**Clock:** csr_wr_clk**Reset State:** 0x00000068

LPASS_CDC_RXn_B5_CTL

Bits	Name	Description
7:5	FS_RATE	Specifies the sampling rate of the incoming PCM samples in the RX chain. 0x0: F_8_KHZ 0x1: F_16_KHZ 0x2: F_32_KHZ 0x3: F_48_KHZ
3	RESERVED	Reserved
2	HPF_BP_SEL	Specifies if the DC blocker needs to be bypassed. 0x0: NO_BYPASS 0x1: BYPASS
1:0	RESERVED	Reserved

0x0771C034+ LPASS_CDC_RXn_B6_CTL, n=[1..3]
0x20*n
Type: RW**Clock:** csr_wr_clk**Reset State:** 0x00000000

Specifies the enable signals for the Gain multiplier (Volume controller). This register should be updated only once per 5 mclk cycles for well-defined operation.

LPASS_CDC_RXn_B6_CTL

Bits	Name	Type	Description
3	GAIN_UPD_STATUS	R	Specifies if the gain update has been completed. Automatically resets to IN_PROCESS when gain is updated. When the gain has been applied, changes state to COMPLETE. Read only bit. 0x0: IN_PROCESS 0x1: COMPLETE
2	MODE	RW	Specifies the mode in which gain is updated. 0x0: AUTOMATIC 0x1: MANUAL
1	HALF_DB	RW	Specifies the addition of a half db to the original gain. When set(1) and If the actual gain is 1, a gain of 1.5dB is applied. When set(1) and If the actual gain is -1, a gain of -0.5dB is applied. 0x0: DISABLE 0x1: ENABLE
0	MUTE_EN	RW	Specifies if the output of the volume controller needs to be muted. 0x0: DISABLE 0x1: ENABLE

0x0771C038+ LPASS_CDC_RXn_VOL_CTL_B1_CTL, n=[1..3] 0x20*n

Type: RW

Clock: csr_wr_clk

Reset State: 0x00000000

RX Timeout value for gain multipliers.

At 48KHz sampling rate (Fs), the maximum timeout is 1.37s $((256 * \text{TIMER_VAL} + 255) / \text{Fs})$

11111111 => $(256 * 255 + 255) / 48k = 1.365s$; $(256 * 255 + 255) / 8k = 8.19s$

10000000 => $(256 * 128 + 255) / 48k = 0.69s$; $(256 * 128 + 255) / 8k = 4.13s$

01000000 => $(256 * 64 + 255) / 48k = 0.347s$; $(256 * 64 + 255) / 8k = 2.08s$

00000000 => Immediate gain update without timer expiration and zero crossing detection. SW control mode.

LPASS_CDC_RXn_VOL_CTL_B1_CTL

Bits	Name	Description
7:0	TIMER	Specifies the Volume Controller timeout period in Fs cycles.

0x0771C03C+ LPASS_CDC_RXn_VOL_CTL_B2_CTL, n=[1..3] 0x20*n

Type: RW

Clock: csr_wr_clk

Reset State: 0x00000000

RX Channel Gain. The gain is approximate to +/- 0.3dB.

LPASS_CDC_RXn_VOL_CTL_B2_CTL

Bits	Name	Description
7:0	GAIN	Specifies the Volume Controller gain. The value is a twos complement number.

0x0771C0A0 LPASS_CDC_TOP_GAIN_UPDATE

Type: RW

Clock: csr_wr_clk

Reset State: 0x00000000

This register should be updated only once per 5 mclk cycles for well-defined operation.

LPASS_CDC_TOP_GAIN_UPDATE

Bits	Name	Description
0	UPDATE	When Manual mode is used to update Volume Controllers. When this bit is set(1) to high, starts the update process across all volume controllers configured in manual mode. 0x0: DISABLE 0x1: UPDATE

0x0771C0A4 LPASS_CDC_TOP_CTL**Type:** RW**Clock:** csr_wr_clk**Reset State:** 0x00000001**LPASS_CDC_TOP_CTL**

Bits	Name	Description
0	DIG_MCLK_FREQ	Indicates the source or frequency of the MCLK 0x0: F_12_288MHZ 0x1: F_9_6MHZ (default)

**0x0771C0C0+LPASS_CDC_IIRn_GAIN_B1_CTL, n=[1..2]
0x40*n****Type:** RW**Clock:** csr_wr_clk**Reset State:** 0x00000000**LPASS_CDC_IIRn_GAIN_B1_CTL**

Bits	Name	Description
7:0	GAIN1	Specifies the gain for the first Volume Controller. The value is a twos complement number.

**0x0771C0C4+LPASS_CDC_IIRn_GAIN_B2_CTL, n=[1..2]
0x40*n****Type:** RW**Clock:** csr_wr_clk**Reset State:** 0x00000000**LPASS_CDC_IIRn_GAIN_B2_CTL**

Bits	Name	Description
7:0	GAIN2	Specifies the gain for the second Volume Controller. The value is a twos complement number.

0x0771C0C8+LPASS_CDC_IIRn_GAIN_B3_CTL, n=[1..2]
0x40*n
Type: RW**Clock:** csr_wr_clk**Reset State:** 0x00000000**LPASS_CDC_IIRn_GAIN_B3_CTL**

Bits	Name	Description
7:0	GAIN3	Specifies the gain for the third Volume Controller. The value is a twos complement number.

0x0771C0CC LPASS_CDC_IIRn_GAIN_B4_CTL, n=[1..2]
+0x40*n
Type: RW**Clock:** csr_wr_clk**Reset State:** 0x00000000**LPASS_CDC_IIRn_GAIN_B4_CTL**

Bits	Name	Description
7:0	GAIN4	Specifies the gain for the fourth Volume Controller. The value is a twos complement number.

0x0771C0D0+LPASS_CDC_IIRn_GAIN_B5_CTL, n=[1..2]
0x40*n
Type: RW**Clock:** csr_wr_clk**Reset State:** 0x00000000

This register should be updated only once per 5 mclk cycles for well-defined operation.

LPASS_CDC_IIRn_GAIN_B5_CTL

Bits	Name	Type	Description
3	GAIN1_UPD_STATUS	R	Specifies if the gain update has been completed. Automatically resets to IN_PROCESS when gain is updated. When the gain has been applied, changes state to COMPLETE. Read only bit. 0x0: IN_PROCESS 0x1: COMPLETE
2	MODE1	RW	Specifies the mode in which gain is updated for the first Volume Controller. 0x0: AUTOMATIC 0x1: MANUAL

LPASS_CDC_IIRn_GAIN_B5_CTL (cont.)

Bits	Name	Type	Description
1	HALF_DB1_EN	RW	Specifies the addition of a half db to the original gain. When set(1) and If the actual gain is 1, a gain of 1.5dB is applied. When set(1) and If the actual gain is -1, a gain of -0.5dB is applied. 0x0: DISABLE 0x1: ENABLE
0	MUTE1_EN	RW	Specifies if the output of the first volume controller needs to be muted. 0x0: DISABLE 0x1: ENABLE

0x0771C0D4+LPASS_CDC_IIRn_GAIN_B6_CTL, n=[1..2]
0x40*n
Type: RW**Clock:** csr_wr_clk**Reset State:** 0x00000000**LPASS_CDC_IIRn_GAIN_B6_CTL**

Bits	Name	Type	Description
3	GAIN2_UPD_STATUS	R	Specifies if the gain update has been completed. Automatically resets to IN_PROCESS when gain is updated. When the gain has been applied, changes state to COMPLETE. Read only bit. 0x0: IN_PROCESS 0x1: COMPLETE
2	MODE2	RW	Specifies the mode in which gain is updated for the second Volume Controller. 0x0: AUTOMATIC 0x1: MANUAL
1	HALF_DB2_EN	RW	Specifies the addition of a half db to the original gain. When set(1) and If the actual gain is 1, a gain of 1.5dB is applied. When set(1) and If the actual gain is -1, a gain of -0.5dB is applied. 0x0: DISABLE 0x1: ENABLE
0	MUTE2_EN	RW	Specifies if the output of the second volume controller needs to be muted. 0x0: DISABLE 0x1: ENABLE

**0x0771C0D8+LPASS_CDC_IIRn_GAIN_B7_CTL, n=[1..2]
0x40*n**
Type: RW**Clock:** csr_wr_clk**Reset State:** 0x00000000**LPASS_CDC_IIRn_GAIN_B7_CTL**

Bits	Name	Type	Description
3	GAIN3_UPD_STATUS	R	Specifies if the gain update has been completed. Automatically resets to IN_PROCESS when gain is updated. When the gain has been applied, changes state to COMPLETE. Read only bit. 0x0: IN_PROCESS 0x1: COMPLETE
2	MODE3	RW	Specifies the mode in which gain is updated for the third Volume Controller. 0x0: AUTOMATIC 0x1: MANUAL
1	HALF_DB3_EN	RW	Specifies the addition of a half db to the original gain. When set(1) and If the actual gain is 1, a gain of 1.5dB is applied. When set(1) and If the actual gain is -1, a gain of -0.5dB is applied. 0x0: DISABLE 0x1: ENABLE
0	MUTE3_EN	RW	Specifies if the output of the third volume controller needs to be muted. 0x0: DISABLE 0x1: ENABLE

**0x0771C0DC LPASS_CDC_IIRn_GAIN_B8_CTL, n=[1..2]
+0x40*n**
Type: RW**Clock:** csr_wr_clk**Reset State:** 0x00000000**LPASS_CDC_IIRn_GAIN_B8_CTL**

Bits	Name	Type	Description
3	GAIN4_UPD_STATUS	R	Specifies if the gain update has been completed. Automatically resets to IN_PROCESS when gain is updated. When the gain has been applied, changes state to COMPLETE. Read only bit. 0x0: IN_PROCESS 0x1: COMPLETE
2	MODE4	RW	Specifies the mode in which gain is updated for the fourth Volume Controller. 0x0: AUTOMATIC 0x1: MANUAL

LPASS_CDC_IIRn_GAIN_B8_CTL (cont.)

Bits	Name	Type	Description
1	HALF_DB4_EN	RW	Specifies the addition of a half db to the original gain. When set(1) and If the actual gain is 1, a gain of 1.5dB is applied. When set(1) and If the actual gain is -1, a gain of -0.5dB is applied. 0x0: DISABLE 0x1: ENABLE
0	MUTE4_EN	RW	Specifies if the output of the fourth volume controller needs to be muted. 0x0: DISABLE 0x1: ENABLE

**0x0771C0E0+LPASS_CDC_IIRn_CTL, n=[1..2]
0x40*n**

Type: RW

Clock: csr_wr_clk

Reset State: 0x00000040

LPASS_CDC_IIRn_CTL

Bits	Name	Description
7:5	POSTGAIN	Set post IIR gain/attenuation 0x0: POS_12P0_DB 0x1: POS_6P0_DB 0x2: POS_0P0_DB 0x3: NEG_6P0_DB 0x4: NEG_12P0_DB 0x5: NEG_18P1_DB 0x6: NEG_24P1_DB 0x7: NEG_30P1_DB
4	FLT_STG5_EN	Enables the fifth biquad stage of the sidetone IIR filter. 0x0: DISABLE 0x1: ENABLE
3	FLT_STG4_EN	Enables the fourth biquad stage of the sidetone IIR filter. 0x0: DISABLE 0x1: ENABLE
2	FLT_STG3_EN	Enables the third biquad stage of the sidetone IIR filter. 0x0: DISABLE 0x1: ENABLE
1	FLT_STG2_EN	Enables the second biquad stage of the sidetone IIR filter. 0x0: DISABLE 0x1: ENABLE

LPASS_CDC_IIRn_CTL (cont.)

Bits	Name	Description
0	FLT_STG1_EN	Enables the first biquad stage of the sidetone IIR filter. 0x0: DISABLE 0x1: ENABLE

0x0771C0E4+LPASS_CDC_IIRn_GAIN_TIMER_CTL, n=[1..2]**0x40*n****Type:** RW**Clock:** csr_wr_clk**Reset State:** 0x00000000

At 48KHz sampling rate (Fs), the maximum timeout is 1.37s $((256 * \text{TIMER_VAL} + 255) / \text{Fs})$

11111111 => $(256 * 255 + 255) / 48k = 1.365s$; $(256 * 255 + 255) / 8k = 8.19s$

10000000 => $(256 * 128 + 255) / 48k = 0.69s$; $(256 * 128 + 255) / 8k = 4.13s$

01000000 => $(256 * 64 + 255) / 48k = 0.347s$; $(256 * 64 + 255) / 8k = 2.08s$

00000000 => Immediate gain update without timer expiration and zero crossing detection. SW control mode.

LPASS_CDC_IIRn_GAIN_TIMER_CTL

Bits	Name	Description
7:0	TIMER	Specifies the timeout period in Fs cycles for all Volume Controllers.

0x0771C0E8+LPASS_CDC_IIRn_COEF_B1_CTL, n=[1..2]**0x40*n****Type:** RW**Clock:** csr_wr_clk**Reset State:** 0x00000000**LPASS_CDC_IIRn_COEF_B1_CTL**

Bits	Name	Description
7	BYPASS	Set (1) to disable automatic pointer incrementing 0x0: AUTOINC 0x1: BYPASS

LPASS_CDC_IIRn_COEF_B1_CTL (cont.)

Bits	Name	Description
6:0	ADDR	Pointer to internal register bank to store the Sidetone IIR filter coefficients. This register auto increments every time CDC_IIRn_COEF_B2_CTL is written. The maximum allowed value is 0x18. Any value above this will corrupt the coefficient bank. Data is LSB first. IIR co-efficients are 30-bits long, the co-efficients are written in a bank of registers 1-byte at a time. Since only 30 bits are needed, the top 2 MSB bits of every 4th byte is dropped in the design. 0x0: BIQ_STG0_COEFF_B0 0x4: BIQ_STG0_COEFF_B1 0x8: BIQ_STG0_COEFF_B2 0xC: BIQ_STG0_COEFF_A1 0x10: BIQ_STG0_COEFF_A2 0x14: BIQ_STG1_COEFF_B0 0x18: BIQ_STG1_COEFF_B1 0x1C: BIQ_STG1_COEFF_B2 0x20: BIQ_STG1_COEFF_A1 0x24: BIQ_STG1_COEFF_A2 0x28: BIQ_STG2_COEFF_B0 0x2C: BIQ_STG2_COEFF_B1 0x30: BIQ_STG2_COEFF_B2 0x34: BIQ_STG2_COEFF_A1 0x38: BIQ_STG2_COEFF_A2 0x3C: BIQ_STG3_COEFF_B0 0x40: BIQ_STG3_COEFF_B1 0x44: BIQ_STG3_COEFF_B2 0x48: BIQ_STG3_COEFF_A1 0x4C: BIQ_STG3_COEFF_A2 0x50: BIQ_STG4_COEFF_B0 0x54: BIQ_STG4_COEFF_B1 0x58: BIQ_STG4_COEFF_B2 0x5C: BIQ_STG4_COEFF_A1 0x60: BIQ_STG4_COEFF_A2

**0x0771C0EC LPASS_CDC_IIRn_COEF_B2_CTL, n=[1..2]
+0x40*n**

Type: RW

Clock: csr_wr_clk

Reset State: 0x00000000

LPASS_CDC_IIRn_COEF_B2_CTL

Bits	Name	Description
7:0	VAL	Address is auto incremented at this stage.

0x0771C180 LPASS_CDC_CONN_RX1_B1_CTL**Type:** RW**Clock:** csr_wr_clk**Reset State:** 0x00000000**LPASS_CDC_CONN_RX1_B1_CTL**

Bits	Name	Description
5:3	MIX1_INP2_SEL	Configures connectivity mux, to choose the second input to mixer1 in the RX1 path. 0x0: ZERO 0x1: IIR1 0x2: IIR2 0x3: RX1 0x4: RX2 0x5: RX3
2:0	MIX1_INP1_SEL	Configures connectivity mux, to choose the first input to mixer1 in the RX1 path. 0x0: ZERO 0x1: IIR1 0x2: IIR2 0x3: RX1 0x4: RX2 0x5: RX3

0x0771C184 LPASS_CDC_CONN_RX1_B2_CTL**Type:** RW**Clock:** csr_wr_clk**Reset State:** 0x00000000**LPASS_CDC_CONN_RX1_B2_CTL**

Bits	Name	Description
2:0	MIX1_INP3_SEL	Configures connectivity mux, to choose the third input to mixer1 in the RX1 path. 0x0: ZERO 0x1: IIR1 0x2: IIR2 0x3: RX1 0x4: RX2 0x5: RX3

0x0771C188 LPASS_CDC_CONN_RX1_B3_CTL**Type:** RW**Clock:** csr_wr_clk**Reset State:** 0x00000000**LPASS_CDC_CONN_RX1_B3_CTL**

Bits	Name	Description
1:0	MIX2_INP1_SEL	Configures connectivity mux, to choose the input to mixer2 in the RX1 path. 0x0: ZERO 0x1: IIR1 0x2: IIR2

0x0771C18C LPASS_CDC_CONN_RX2_B1_CTL**Type:** RW**Clock:** csr_wr_clk**Reset State:** 0x00000000**LPASS_CDC_CONN_RX2_B1_CTL**

Bits	Name	Description
5:3	MIX1_INP2_SEL	Configures connectivity mux, to choose the second input to mixer1 in the RX2 path. 0x0: ZERO 0x1: IIR1 0x2: IIR2 0x3: RX1 0x4: RX2 0x5: RX3
2:0	MIX1_INP1_SEL	Configures connectivity mux, to choose the first input to mixer1 in the RX2 path. 0x0: ZERO 0x1: IIR1 0x2: IIR2 0x3: RX1 0x4: RX2 0x5: RX3

0x0771C190 LPASS_CDC_CONN_RX2_B2_CTL**Type:** RW**Clock:** csr_wr_clk**Reset State:** 0x00000000

LPASS_CDC_CONN_RX2_B2_CTL

Bits	Name	Description
2:0	MIX1_INP3_SEL	Configures connectivity mux, to choose the third input to mixer1 in the RX2 path. 0x0: ZERO 0x1: IIR1 0x2: IIR2 0x3: RX1 0x4: RX2 0x5: RX3

0x0771C194 LPASS_CDC_CONN_RX2_B3_CTL**Type:** RW**Clock:** csr_wr_clk**Reset State:** 0x00000000**LPASS_CDC_CONN_RX2_B3_CTL**

Bits	Name	Description
1:0	MIX2_INP1_SEL	Configures connectivity mux, to choose the input to mixer2 in the RX2 path. 0x0: ZERO 0x1: IIR1 0x2: IIR2

0x0771C198 LPASS_CDC_CONN_RX3_B1_CTL**Type:** RW**Clock:** csr_wr_clk**Reset State:** 0x00000000**LPASS_CDC_CONN_RX3_B1_CTL**

Bits	Name	Description
5:3	MIX1_INP2_SEL	Configures connectivity mux, to choose the second input to mixer1 in the RX3 path. 0x0: ZERO 0x1: IIR1 0x2: IIR2 0x3: RX1 0x4: RX2 0x5: RX3

LPASS_CDC_CONN_RX3_B1_CTL (cont.)

Bits	Name	Description
2:0	MIX1_INP1_SEL	Configures connectivity mux, to choose the first input to mixer1 in the RX3 path. 0x0: ZERO 0x1: IIR1 0x2: IIR2 0x3: RX1 0x4: RX2 0x5: RX3

0x0771C19C LPASS_CDC_CONN_RX3_B2_CTL**Type:** RW**Clock:** csr_wr_clk**Reset State:** 0x00000000**LPASS_CDC_CONN_RX3_B2_CTL**

Bits	Name	Description
2:0	MIX1_INP3_SEL	Configures connectivity mux, to choose the third input to mixer1 in the RX3 path. 0x0: ZERO 0x1: IIR1 0x2: IIR2 0x3: RX1 0x4: RX2 0x5: RX3

0x0771C1A0 LPASS_CDC_CONN_TX_B1_CTL**Type:** RW**Clock:** csr_wr_clk**Reset State:** 0x00000000**LPASS_CDC_CONN_TX_B1_CTL**

Bits	Name	Description
5:3	TX2_INP_SEL	Configures connectivity mux, to choose the input to TX2 (DEC2) path. 0x0: ZERO 0x1: ADC1 0x2: ADC2 0x3: ADC3 0x4: DMIC1 0x5: DMIC2

LPASS_CDC_CONN_TX_B1_CTL (cont.)

Bits	Name	Description
2:0	TX1_INP_SEL	Configures connectivity mux, to choose the input to TX1 (DEC1) path. 0x0: ZERO 0x1: ADC1 0x2: ADC2 0x3: ADC3 0x4: DMIC1 0x5: DMIC2

0x0771C1A8 LPASS_CDC_CONN_EQ1_B1_CTL**Type:** RW**Clock:** csr_wr_clk**Reset State:** 0x00000000**LPASS_CDC_CONN_EQ1_B1_CTL**

Bits	Name	Description
2:0	MIX_INP1_SEL	Configures connectivity mux, to choose the first input to the IIR1 path. 0x0: ZERO 0x1: DEC1 0x2: DEC2 0x3: RX1 0x4: RX2 0x5: RX3

0x0771C1AC LPASS_CDC_CONN_EQ1_B2_CTL**Type:** RW**Clock:** csr_wr_clk**Reset State:** 0x00000000**LPASS_CDC_CONN_EQ1_B2_CTL**

Bits	Name	Description
2:0	MIX_INP2_SEL	Configures connectivity mux, to choose the second input to the IIR1 path. 0x0: ZERO 0x1: DEC1 0x2: DEC2 0x3: RX1 0x4: RX2 0x5: RX3

0x0771C1B0 LPASS_CDC_CONN_EQ1_B3_CTL**Type:** RW**Clock:** csr_wr_clk**Reset State:** 0x00000000**LPASS_CDC_CONN_EQ1_B3_CTL**

Bits	Name	Description
2:0	MIX_INP3_SEL	Configures connectivity mux, to choose the third input to the IIR1 path. 0x0: ZERO 0x1: DEC1 0x2: DEC2 0x3: RX1 0x4: RX2 0x5: RX3

0x0771C1B4 LPASS_CDC_CONN_EQ1_B4_CTL**Type:** RW**Clock:** csr_wr_clk**Reset State:** 0x00000000**LPASS_CDC_CONN_EQ1_B4_CTL**

Bits	Name	Description
2:0	MIX_INP4_SEL	Configures connectivity mux, to choose the fourth input to the IIR1 path. 0x0: ZERO 0x1: DEC1 0x2: DEC2 0x3: RX1 0x4: RX2 0x5: RX3

0x0771C1B8 LPASS_CDC_CONN_EQ2_B1_CTL**Type:** RW**Clock:** csr_wr_clk**Reset State:** 0x00000000

LPASS_CDC_CONN_EQ2_B1_CTL

Bits	Name	Description
2:0	MIX_INP1_SEL	Configures connectivity mux, to choose the first input to the IIR2 path. 0x0: ZERO 0x1: DEC1 0x2: DEC2 0x3: RX1 0x4: RX2 0x5: RX3

0x0771C1BC LPASS_CDC_CONN_EQ2_B2_CTL**Type:** RW**Clock:** csr_wr_clk**Reset State:** 0x00000000**LPASS_CDC_CONN_EQ2_B2_CTL**

Bits	Name	Description
2:0	MIX_INP2_SEL	Configures connectivity mux, to choose the second input to the IIR2 path. 0x0: ZERO 0x1: DEC1 0x2: DEC2 0x3: RX1 0x4: RX2 0x5: RX3

0x0771C1C0 LPASS_CDC_CONN_EQ2_B3_CTL**Type:** RW**Clock:** csr_wr_clk**Reset State:** 0x00000000**LPASS_CDC_CONN_EQ2_B3_CTL**

Bits	Name	Description
2:0	MIX_INP3_SEL	Configures connectivity mux, to choose the third input to the IIR2 path. 0x0: ZERO 0x1: DEC1 0x2: DEC2 0x3: RX1 0x4: RX2 0x5: RX3

0x0771C1C4 LPASS_CDC_CONN_EQ2_B4_CTL**Type:** RW**Clock:** csr_wr_clk**Reset State:** 0x00000000**LPASS_CDC_CONN_EQ2_B4_CTL**

Bits	Name	Description
2:0	MIX_INP4_SEL	Configures connectivity mux, to choose the fourth input to the IIR2 path. 0x0: ZERO 0x1: DEC1 0x2: DEC2 0x3: RX1 0x4: RX2 0x5: RX3

0x0771C1C8 LPASS_CDC_CONN_TX_I2S_SD1_CTL**Type:** RW**Clock:** csr_wr_clk**Reset State:** 0x00000000**LPASS_CDC_CONN_TX_I2S_SD1_CTL**

Bits	Name	Description
0	I2S_TX1_R_CH_INP_SEL	Configures connectivity to TX_I2S_SD1 right channel output mux 0x0: RX_MIX2 0x1: RX_MIX3

**0x0771C260+ LPASS_CDC_TXn_VOL_CTL_TIMER, n=[1..2]
0x20*n****Type:** RW**Clock:** csr_wr_clk**Reset State:** 0x00000000

TX Timeout value for gain multipliers.

At 48KHz sampling rate (Fs), the maximum timeout is 1.37s $((256 * \text{TIMER_VAL} + 255) / F_s)$ 11111111 => $(256 * 255 + 255) / 48k = 1.365s$; $(256 * 255 + 255) / 8k = 8.19s$ 10000000 => $(256 * 128 + 255) / 48k = 0.69s$; $(256 * 128 + 255) / 8k = 4.13s$ 01000000 => $(256 * 64 + 255) / 48k = 0.347s$; $(256 * 64 + 255) / 8k = 2.08s$

00000000 => Immediate gain update without timer expiration and zero crossing detection. SW control mode.

LPASS_CDC_TXn_VOL_CTL_TIMER

Bits	Name	Description
7:0	PGA_TIMER	Specifies the Volume Controller timeout period in Fs cycles.

0x0771C264+ LPASS_CDC_TXn_VOL_CTL_GAIN, n=[1..2]
0x20*n
Type: RW**Clock:** csr_wr_clk**Reset State:** 0x00000000

TX Channel Gain. The gain is approximate to +/- 0.3dB.

LPASS_CDC_TXn_VOL_CTL_GAIN

Bits	Name	Description
7:0	PGA_GAIN	Specifies the Volume Controller gain. The value is a twos complement number.

0x0771C268+ LPASS_CDC_TXn_VOL_CTL_CFG, n=[1..2]
0x20*n
Type: RW**Clock:** csr_wr_clk**Reset State:** 0x00000000

Specifies the enable signals for the Gain multiplier (Volume controller). This register should be updated only once per 5 mclk cycles for well-defined operation.

LPASS_CDC_TXn_VOL_CTL_CFG

Bits	Name	Type	Description
3	GAIN_UPD_STATUS	R	Specifies if the gain update has been completed. Automatically resets to IN_PROCESS when gain is updated. When the gain has been applied, changes state to COMPLETE. Read only bit. 0x0: IN_PROCESS 0x1: COMPLETE
2	MODE	RW	Specifies the mode in which gain is updated. 0x0: AUTOMATIC 0x1: MANUAL
1	HALF_DB	RW	Specifies the addition of a half db to the original gain. When set(1) and If the actual gain is 1, a gain of 1.5dB is applied. When set(1) and If the actual gain is -1, a gain of -0.5dB is applied. 0x0: DISABLE 0x1: ENABLE

LPASS_CDC_TXn_VOL_CTL_CFG (cont.)

Bits	Name	Type	Description
0	MUTE_EN	RW	Specifies if the output of the volume controller needs to be muted. 0x0: DISABLE 0x1: ENABLE

**0x0771C26C+LPASS_CDC_TXn_MUX_CTL, n=[1..2]
0x20*n****Type:** RW**Clock:** csr_wr_clk**Reset State:** 0x00000000

Specifies the control signals for Multiplexer inputs in TX path.

LPASS_CDC_TXn_MUX_CTL

Bits	Name	Description
5:4	CUT_OFF_FREQ	Specifies the Cutoff frequency to be used by the High pass filter block. 0x0: CF_NEG_3DB_4HZ 0x1: CF_NEG_3DB_75HZ 0x2: CF_NEG_3DB_150HZ
3	HPF_BP_SEL	Specifies if the high pass controller needs to be bypassed or not. 0x0: NO_BYPASS 0x1: BYPASS
2:1	RESERVED	Reserved
0	ADC_DMIC_SEL	Specifies ADCMIC source selection path in CIC filter. 0x0: ADC 0x1: DMIC

**0x0771C270+ LPASS_CDC_TXn_CLK_FS_CTL, n=[1..2]
0x20*n****Type:** RW**Clock:** csr_wr_clk**Reset State:** 0x00000003

Specifies the sampling rate of the PCM signal.

LPASS_CDC_TXn_CLK_FS_CTL

Bits	Name	Description
2:0	FS_RATE	Specifies the sampling rate of the PCM signal. 0x0: F_8_KHZ 0x1: F_16_KHZ 0x2: F_32_KHZ 0x3: F_48_KHZ

0x0771C274+ LPASS_CDC_TXn_DMIC_CTL, n=[1..2]**0x20*n****Type:** RW**Clock:** csr_wr_clk**Reset State:** 0x00000000**LPASS_CDC_TXn_DMIC_CTL**

Bits	Name	Description
2:0	CLK_SEL	Specifies the clock frequency of the input DMIC data samples in the TX/ANC path. Div6 is only valid for mclk=12.288 0x0: DIV2 0x1: DIV3 0x2: DIV4 0x3: DIV6 0x4: DIV16

17 Mobile Display Subsystem

17.1 MDSS_MDSS_HW

MDSS HW register description

17.1.1 MDSS_MDSS_HW_MDSS_HW

MDSS HW

MDSS HW register description

0x01A00010 MDSS_MDSS_HW_INTR_STATUS

Type: R

Reset State: 0x00000000

MDSS Toplevel Interrupt Collection Status Register

Since there are a number of display IP related interrupt sources, this register indicates which display IP core has an interrupt line asserted to help out the display driver

MDSS_MDSS_HW_INTR_STATUS

Bits	Name	Description
4	DSI0	0x0: DSI0 Interrupt line is NOT asserted 0x1: DSI0 Interrupt line is asserted
0	MDP	0x0: MDP Interrupt line is NOT asserted 0x1: MDP Interrupt line is asserted

0x01A00014 MDSS_MDSS_HW_MDSS_SCRATCH_REGISTER_0

Type: RW

Reset State: 0x00000000

MDSS scratch register

MDSS_MDSS_HW_MDSS_SCRATCH_REGISTER_0

Bits	Name	Description
31:0	MDSS_SCRATCH_0	32 bit scratch register

0x01A00018 MDSS_MDSS_HW_MDSS_SCRATCH_REGISTER_1**Type:** RW**Reset State:** 0x00000000

MDSS scratch register

MDSS_MDSS_HW_MDSS_SCRATCH_REGISTER_1

Bits	Name	Description
31:0	MDSS_SCRATCH_1	32 bit scratch register

0x01A0001C MDSS_MDSS_HW_MDSS_SCRATCH_REGISTER_2**Type:** RW**Reset State:** 0x00000000

MDSS scratch register

MDSS_MDSS_HW_MDSS_SCRATCH_REGISTER_2

Bits	Name	Description
31:0	MDSS_SCRATCH_2	32 bit scratch register

0x01A00020 MDSS_MDSS_HW_MDSS_SCRATCH_REGISTER_3**Type:** RW**Reset State:** 0x00000000

MDSS scratch register

MDSS_MDSS_HW_MDSS_SCRATCH_REGISTER_3

Bits	Name	Description
31:0	MDSS_SCRATCH_3	32 bit scratch register

0x01A00024 MDSS_MDSS_HW_MDSS_SCRATCH_REGISTER_4**Type:** RW**Reset State:** 0x00000000

MDSS scratch register

MDSS_MDSS_HW_MDSS_SCRATCH_REGISTER_4

Bits	Name	Description
31:0	MDSS_SCRATCH_4	32 bit scratch register

0x01A00028 MDSS_MDSS_HW_MDSS_SCRATCH_REGISTER_5**Type:** RW**Reset State:** 0x00000000

MDSS scratch register

MDSS_MDSS_HW_MDSS_SCRATCH_REGISTER_5

Bits	Name	Description
31:0	MDSS_SCRATCH_5	32 bit scratch register

0x01A0002C MDSS_MDSS_HW_MDSS_SCRATCH_REGISTER_6**Type:** RW**Reset State:** 0x00000000

MDSS scratch register

MDSS_MDSS_HW_MDSS_SCRATCH_REGISTER_6

Bits	Name	Description
31:0	MDSS_SCRATCH_6	32 bit scratch register

0x01A00030 MDSS_MDSS_HW_MDSS_SCRATCH_REGISTER_7**Type:** RW**Reset State:** 0x00000000

MDSS scratch register

MDSS_MDSS_HW_MDSS_SCRATCH_REGISTER_7

Bits	Name	Description
31:0	MDSS_SCRATCH_7	32 bit scratch register

0x01A00034 MDSS_MDSS_HW_MDSS_SCRATCH_REGISTER_8**Type:** RW**Reset State:** 0x00000000

MDSS scratch register

MDSS_MDSS_HW_MDSS_SCRATCH_REGISTER_8

Bits	Name	Description
31:0	MDSS_SCRATCH_8	32 bit scratch register

0x01A00038 MDSS_MDSS_HW_MDSS_SCRATCH_REGISTER_9**Type:** RW**Reset State:** 0x00000000

MDSS scratch register

MDSS_MDSS_HW_MDSS_SCRATCH_REGISTER_9

Bits	Name	Description
31:0	MDSS_SCRATCH_9	32 bit scratch register

0x01A0003C MDSS_MDSS_HW_MDSS_SCRATCH_REGISTER_10**Type:** RW**Reset State:** 0x00000000

MDSS scratch register

MDSS_MDSS_HW_MDSS_SCRATCH_REGISTER_10

Bits	Name	Description
31:0	MDSS_SCRATCH_10	32 bit scratch register

0x01A00040 MDSS_MDSS_HW_MDSS_SCRATCH_REGISTER_11**Type:** RW**Reset State:** 0x00000000

MDSS scratch register

MDSS_MDSS_HW_MDSS_SCRATCH_REGISTER_11

Bits	Name	Description
31:0	MDSS_SCRATCH_11	32 bit scratch register

0x01A00044 MDSS_MDSS_HW_MDSS_SCRATCH_REGISTER_12**Type:** RW**Reset State:** 0x00000000

MDSS scratch register

MDSS_MDSS_HW_MDSS_SCRATCH_REGISTER_12

Bits	Name	Description
31:0	MDSS_SCRATCH_12	32 bit scratch register

0x01A00048 MDSS_MDSS_HW_MDSS_SCRATCH_REGISTER_13**Type:** RW**Reset State:** 0x00000000

MDSS scratch register

MDSS_MDSS_HW_MDSS_SCRATCH_REGISTER_13

Bits	Name	Description
31:0	MDSS_SCRATCH_13	32 bit scratch register

0x01A0004C MDSS_MDSS_HW_MDSS_SCRATCH_REGISTER_14**Type:** RW**Reset State:** 0x00000000

MDSS scratch register

MDSS_MDSS_HW_MDSS_SCRATCH_REGISTER_14

Bits	Name	Description
31:0	MDSS_SCRATCH_14	32 bit scratch register

0x01A00050 MDSS_MDSS_HW_MDSS_SCRATCH_REGISTER_15**Type:** RW**Reset State:** 0x00000000

MDSS scratch register

MDSS_MDSS_HW_MDSS_SCRATCH_REGISTER_15

Bits	Name	Description
31:0	MDSS_SCRATCH_15	32 bit scratch register

17.2 MDSS_MDP

MDP register description

17.2.1 MDSS_MDP_MDP

MDP

0x01A01004 MDSS_MDP_DISP_INTF_SEL**Type:** RW**Reset State:** 0x00000000

Display Interface Select

This should be set up BEFORE enabling the timing engines

double buffered: NO

MDSS_MDP_DISP_INTF_SEL

Bits	Name	Description
15:8	INTF_1	Interface 1 Connection 0x00: Disabled 0x01: DSI 0x05: LCD 0x0A: DSI Generic Testing Capture Model (used for testing ONLY)

0x01A01010 MDSS_MDP_INTR_EN**Type:** RW**Reset State:** 0x00000000

MDP5 Interrupt Enable Control registers

MDSS_MDP_INTR_EN

Bits	Name	Description
27	INTF_1_VSYNC	VSYSNC Interrupt 0x0: Interrupt Disabled 0x1: Interrupt Enabled
26	INTF_1_UNDERRUN	Underrun 0x0: Interrupt Disabled 0x1: Interrupt Enabled
20	PING_PONG_0_AUTOREFR ESH_DONE	Autorefresh Done/Start Status associated with Ping Pong FIFO 2 0x0: Interrupt Disabled 0x1: Interrupt Enabled
16	PING_PONG_0_WR_PTR	Tear Check Write pointer interrupt associated with Ping Pong FIFO 0 0x0: Interrupt Disabled 0x1: Interrupt Enabled
12	PING_PONG_0_RD_PTR	Tear Check Read pointer interrupt associated with Ping Pong FIFO 0 0x0: Interrupt Disabled 0x1: Interrupt Enabled
8	PING_PONG_0_DONE	Ping Pong FIFO 0 done 0x0: Interrupt Disabled 0x1: Interrupt Enabled
4	WB_2_DONE	WriteBack 2 Done 0x0: Interrupt Disabled 0x1: Interrupt Enabled
3	WD_TIMER_1_DONE	
2	WD_TIMER_0_DONE	
0	WB_0_DONE	WriteBack 0 Done 0x0: Interrupt Disabled 0x1: Interrupt Enabled

0x01A01014 MDSS_MDP_INTR_STATUS**Type:** R**Reset State:** 0x00000000

MDP5 Interrupt Status Control registers

MDSS_MDP_INTR_STATUS

Bits	Name	Description
27	INTF_1_VSYNC	VSYNC Interrupt 0x0: Interrupt Clear 0x1: Interrupt Detectd
26	INTF_1_UNDERRUN	Underrun 0x0: Interrupt Clear 0x1: Interrupt Detectd
20	PING_PONG_0_AUTOREFR ESH_DONE	Autorefresh Done/Start Status associated with Ping Pong FIFO 0 0x0: Interrupt Clear 0x1: Interrupt Detected
16	PING_PONG_0_WR_PTR	Tear Check Write pointer interrupt associated with Ping Pong FIFO 0 0x0: Interrupt Clear 0x1: Interrupt Detected
12	PING_PONG_0_RD_PTR	Tear Check Read pointer interrupt associated with Ping Pong FIFO 0 0x0: Interrupt Clear 0x1: Interrupt Detected
8	PING_PONG_0_DONE	Ping Pong FIFO 0 done 0x0: Interrupt Clear 0x1: Interrupt Detectd
4	WB_2_DONE	WriteBack 2 Done 0x0: Interrupt Disabled 0x1: Interrupt Enabled
3	WD_TIMER_1_DONE	
2	WD_TIMER_0_DONE	
0	WB_0_DONE	WriteBack 0 Done 0x0: Interrupt Disabled 0x1: Interrupt Enabled

0x01A01018 MDSS_MDP_INTR_CLEAR**Type:** W**Reset State:** 0x00000000

MDP5 Interrupt Clear Control registers

MDSS_MDP_INTR_CLEAR

Bits	Name	Description
27	INTF_1_VSYNC	Primary Panel VSYNC Interrupt Write 1 to clear
26	INTF_1_UNDERRUN	Primary Panel Underrun Write 1 to clear
20	PING_PONG_0_AUTOREFR ESH_DONE	Autorefresh Done/Start Status associated with Ping Pong FIFO 2 Write 1 to clear
16	PING_PONG_0_WR_PTR	Tear Check Write pointer interrupt associated with Ping Pong FIFO 2 Write 1 to clear
12	PING_PONG_0_RD_PTR	Tear Check Read pointer interrupt associated with Ping Pong FIFO 2 Write 1 to clear
8	PING_PONG_0_DONE	Ping Pong FIFO 0 done Write 1 to clear
4	WB_2_DONE	WriteBack 2 Done 0x0: Interrupt Disabled 0x1: Interrupt Enabled
3	WD_TIMER_1_DONE	
2	WD_TIMER_0_DONE	
0	WB_0_DONE	WriteBack 0 Done 0x0: Interrupt Disabled 0x1: Interrupt Enabled

0x01A0101C MDSS_MDP_HIST_INTR_EN**Type:** RW**Reset State:** 0x00000000

MDP5 Histogram Interrupt Enable Control registers

MDSS_MDP_HIST_INTR_EN

Bits	Name	Description
13	DSPP_0_RESET_SEQ_DONE	DSPP0 Histogram Reset Sequence Done Interrupt for virtual pipeline 0
12	DSPP_0_DONE	DSPP0 Histogram Done Interrupt for virtual pipeline 0
1	VIG_0_RESET_SEQ_DONE	ViG0 Histogram Reset Sequence Done Interrupt for virtual pipeline 0
0	VIG_0_DONE	ViG0 Histogram Done Interrupt for virtual pipeline 0

0x01A01020 MDSS_MDP_HIST_INTR_STATUS**Type:** R**Reset State:** 0x00000000

MDP5 Histogram Interrupt Status Control registers

MDSS_MDP_HIST_INTR_STATUS

Bits	Name	Description
13	DSPP_0_RESET_SEQ_DONE	DSPP0 Histogram Reset Sequence Done Interrupt for virtual pipeline 0
12	DSPP_0_DONE	DSPP0 Histogram Done Interrupt for virtual pipeline 0
1	VIG_0_RESET_SEQ_DONE	ViG0 Histogram Reset Sequence Done Interrupt for virtual pipeline 0
0	VIG_0_DONE	ViG0 Histogram Done Interrupt for virtual pipeline 0

0x01A01024 MDSS_MDP_HIST_INTR_CLEAR**Type:** W**Reset State:** 0x00000000

MDP5 Histogram Interrupt Clear Control registers

MDSS_MDP_HIST_INTR_CLEAR

Bits	Name	Description
13	DSPP_0_RESET_SEQ_DONE	DSPP0 Histogram Reset Sequence Done Interrupt for virtual pipeline 0
12	DSPP_0_DONE	DSPP0 Histogram Done Interrupt for virtual pipeline 0
1	VIG_0_RESET_SEQ_DONE	ViG0 Histogram Reset Sequence Done Interrupt for virtual pipeline 0
0	VIG_0_DONE	ViG0 Histogram Done Interrupt for virtual pipeline 0

0x01A01030 MDSS_MDP_TRAFFIC_SHAPER_RD_CLIENT_0**Type:** RW**Reset State:** 0x00000000

Traffic Shaper Configuration for read client0

This is mapped to ViG0 read client

double buffered: YES

MDSS_MDP_TRAFFIC_SHAPER_RD_CLIENT_0

Bits	Name	Description
31	EN	Traffic Shaper enable
7:0	BYTES_PER_CLOCK	Determines how many bytes each mdp clock is equivalent to. This is a 4.4 value which allows for up to 1/16 granularity A value of 0x01 implies 0.0625 (1/16) which is the smallest programmable value A value of 0xFF implies 15.9375 which is the largest programmable value

0x01A01034 MDSS_MDP_TRAFFIC_SHAPER_RD_CLIENT_1**Type:** RW**Reset State:** 0x00000000

Traffic Shaper Configuration for read client1

This is mapped to RGB0 read client

double buffered: YES

MDSS_MDP_TRAFFIC_SHAPER_RD_CLIENT_1

Bits	Name	Description
31	EN	Traffic Shaper enable
7:0	BYTES_PER_CLOCK	Determines how many bytes each mdp clock is equivalent to. This is a 4.4 value which allows for up to 1/16 granularity A value of 0x01 implies 0.0625 (1/16) which is the smallest programmable value A value of 0xFF implies 15.9375 which is the largest programmable value

0x01A0103C MDSS_MDP_TRAFFIC_SHAPER_RD_CLIENT_3**Type:** RW**Reset State:** 0x00000000

Traffic Shaper Configuration for read client3

This is mapped to RGB1 read client

double buffered: YES

MDSS_MDP_TRAFFIC_SHAPER_RD_CLIENT_3

Bits	Name	Description
31	EN	Traffic Shaper enable
7:0	BYTES_PER_CLOCK	Determines how many bytes each mdp clock is equivalent to. This is a 4.4 value which allows for up to 1/16 granularity A value of 0x01 implies 0.0625 (1/16) which is the smallest programmable value A value of 0xFF implies 15.9375 which is the largest programmable value

0x01A01048 MDSS_MDP_TRAFFIC_SHAPER_RD_CLIENT_6**Type:** RW**Reset State:** 0x00000000

Traffic Shaper Configuration for read client6

This is mapped to DMA0 read client

double buffered: YES

MDSS_MDP_TRAFFIC_SHAPER_RD_CLIENT_6

Bits	Name	Description
31	EN	Traffic Shaper enable
7:0	BYTES_PER_CLOCK	Determines how many bytes each mdp clock is equivalent to. This is a 4.4 value which allows for up to 1/16 granularity A value of 0x01 implies 0.0625 (1/16) which is the smallest programmable value A value of 0xFF implies 15.9375 which is the largest programmable value

0x01A01060 MDSS_MDP_TRAFFIC_SHAPER_WR_CLIENT_0**Type:** RW**Reset State:** 0x00000000

Traffic Shaper Configuration for write client0

This is mapped to WB0 write client

double buffered: YES

MDSS_MDP_TRAFFIC_SHAPER_WR_CLIENT_0

Bits	Name	Description
31	EN	Traffic Shaper enable
7:0	BYTES_PER_CLOCK	Determines how many bytes each mdp clock is equivalent to. This is a 4.4 value which allows for up to 1/16 granularity A value of 0x01 implies 0.0625 (1/16) which is the smallest programmable value A value of 0xFF implies 15.9375 which is the largest programmable value

0x01A01068 MDSS_MDP_TRAFFIC_SHAPER_WR_CLIENT_2**Type:** RW**Reset State:** 0x00000000

Traffic Shaper Configuration for write client2

This is mapped to WB2 write client

double buffered: YES

MDSS_MDP_TRAFFIC_SHAPER_WR_CLIENT_2

Bits	Name	Description
31	EN	Traffic Shaper enable
7:0	BYTES_PER_CLOCK	Determines how many bytes each mdp clock is equivalent to. This is a 4.4 value which allows for up to 1/16 granularity A value of 0x01 implies 0.0625 (1/16) which is the smallest programmable value A value of 0xFF implies 15.9375 which is the largest programmable value

0x01A01080 MDSS_MDP_SMP_ALLOC_W_0**Type:** RW**Reset State:** 0x00000000

Shared Memory Pool allocation configuration 0

double buffered: NO

MDSS_MDP_SMP_ALLOC_W_0

Bits	Name	Description
23:22	MMB_2_CTX_W	Macro-block 2 write context relation - always set to 0x0 when virtual pipelines are not supported
21:16	MMB_2_MUX_W	Macro-block 2 write selector relation
15:14	MMB_1_CTX_W	Macro-block 1 write context relation - always set to 0x0 when virtual pipelines are not supported
13:8	MMB_1_MUX_W	Macro-block 1 write selector relation
7:6	MMB_0_CTX_W	Macro-block 0 write context relation - always set to 0x0 when virtual pipelines are not supported
5:0	MMB_0_MUX_W	Macro-block 0 write selector relation

0x01A01084 MDSS_MDP_SMP_ALLOC_W_1**Type:** RW**Reset State:** 0x00000000

Shared Memory Pool allocation configuration 1

double buffered: NO

MDSS_MDP_SMP_ALLOC_W_1

Bits	Name	Description
23:22	MMB_5_CTX_W	Macro-block 5 write context relation - always set to 0x0 when virtual pipelines are not supported
21:16	MMB_5_MUX_W	Macro-block 5 write selector relation
15:14	MMB_4_CTX_W	Macro-block 4 write context relation - always set to 0x0 when virtual pipelines are not supported
13:8	MMB_4_MUX_W	Macro-block 4 write selector relation
7:6	MMB_3_CTX_W	Macro-block 3 write context relation - always set to 0x0 when virtual pipelines are not supported
5:0	MMB_3_MUX_W	Macro-block 3 write selector relation

0x01A01088 MDSS_MDP_SMP_ALLOC_W_2**Type:** RW**Reset State:** 0x00000000

Shared Memory Pool allocation configuration 2

double buffered: NO

MDSS_MDP_SMP_ALLOC_W_2

Bits	Name	Description
23:22	MMB_8_CTX_W	Macro-block 8 write context relation - always set to 0x0 when virtual pipelines are not supported
21:16	MMB_8_MUX_W	Macro-block 8 write selector relation
15:14	MMB_7_CTX_W	Macro-block 7 write context relation - always set to 0x0 when virtual pipelines are not supported
13:8	MMB_7_MUX_W	Macro-block 7 write selector relation
7:6	MMB_6_CTX_W	Macro-block 6 write context relation - always set to 0x0 when virtual pipelines are not supported
5:0	MMB_6_MUX_W	Macro-block 6 write selector relation

0x01A0108C MDSS_MDP_SMP_ALLOC_W_3

Type: RW

Reset State: 0x00000000

Shared Memory Pool allocation configuration 3

double buffered: NO

MDSS_MDP_SMP_ALLOC_W_3

Bits	Name	Description
23:22	MMB_11_CTX_W	Macro-block 11 write context relation - always set to 0x0 when virtual pipelines are not supported
21:16	MMB_11_MUX_W	Macro-block 11 write selector relation
15:14	MMB_10_CTX_W	Macro-block 10 write context relation - always set to 0x0 when virtual pipelines are not supported
13:8	MMB_10_MUX_W	Macro-block 10 write selector relation
7:6	MMB_9_CTX_W	Macro-block 9 write context relation - always set to 0x0 when virtual pipelines are not supported
5:0	MMB_9_MUX_W	Macro-block 9 write selector relation

0x01A01090 MDSS_MDP_SMP_ALLOC_W_4

Type: RW

Reset State: 0x00000000

Shared Memory Pool allocation configuration 4

double buffered: NO

MDSS_MDP_SMP_ALLOC_W_4

Bits	Name	Description
23:22	MMB_14_CTX_W	Macro-block 14 write context relation - always set to 0x0 when virtual pipelines are not supported
21:16	MMB_14_MUX_W	Macro-block 14 write selector relation
15:14	MMB_13_CTX_W	Macro-block 13 write context relation - always set to 0x0 when virtual pipelines are not supported
13:8	MMB_13_MUX_W	Macro-block 13 write selector relation
7:6	MMB_12_CTX_W	Macro-block 12 write context relation - always set to 0x0 when virtual pipelines are not supported
5:0	MMB_12_MUX_W	Macro-block 12 write selector relation

0x01A01094 MDSS_MDP_SMP_ALLOC_W_5**Type:** RW**Reset State:** 0x00000000

Shared Memory Pool allocation configuration 5

double buffered: NO

MDSS_MDP_SMP_ALLOC_W_5

Bits	Name	Description
23:22	MMB_17_CTX_W	Macro-block 17 write context relation - always set to 0x0 when virtual pipelines are not supported
21:16	MMB_17_MUX_W	Macro-block 17 write selector relation
15:14	MMB_16_CTX_W	Macro-block 16 write context relation - always set to 0x0 when virtual pipelines are not supported
13:8	MMB_16_MUX_W	Macro-block 16 write selector relation
7:6	MMB_15_CTX_W	Macro-block 15 write context relation - always set to 0x0 when virtual pipelines are not supported
5:0	MMB_15_MUX_W	Macro-block 15 write selector relation

0x01A01098 MDSS_MDP_SMP_ALLOC_W_6**Type:** RW**Reset State:** 0x00000000

Shared Memory Pool allocation configuration 6

double buffered: NO

MDSS_MDP_SMP_ALLOC_W_6

Bits	Name	Description
23:22	MMB_20_CTX_W	Macro-block 20 write context relation - always set to 0x0 when virtual pipelines are not supported
21:16	MMB_20_MUX_W	Macro-block 20 write selector relation
15:14	MMB_19_CTX_W	Macro-block 19 write context relation - always set to 0x0 when virtual pipelines are not supported
13:8	MMB_19_MUX_W	Macro-block 19 write selector relation
7:6	MMB_18_CTX_W	Macro-block 18 write context relation - always set to 0x0 when virtual pipelines are not supported
5:0	MMB_18_MUX_W	Macro-block 18 write selector relation

0x01A0109C MDSS_MDP_SMP_ALLOC_W_7**Type:** RW**Reset State:** 0x00000000

Shared Memory Pool allocation configuration 7

double buffered: NO

MDSS_MDP_SMP_ALLOC_W_7

Bits	Name	Description
23:22	MMB_23_CTX_W	Macro-block 23 write context relation - always set to 0x0 when virtual pipelines are not supported
21:16	MMB_23_MUX_W	Macro-block 23 write selector relation
15:14	MMB_22_CTX_W	Macro-block 22 write context relation - always set to 0x0 when virtual pipelines are not supported
13:8	MMB_22_MUX_W	Macro-block 22 write selector relation
7:6	MMB_21_CTX_W	Macro-block 21 write context relation - always set to 0x0 when virtual pipelines are not supported
5:0	MMB_21_MUX_W	Macro-block 21 write selector relation

0x01A010A0 MDSS_MDP_SMP_ALLOC_W_8**Type:** RW**Reset State:** 0x00000000

Shared Memory Pool allocation configuration 8

double buffered: NO

MDSS_MDP_SMP_ALLOC_W_8

Bits	Name	Description
23:22	MMB_26_CTX_W	Macro-block 26 write context relation - always set to 0x0 when virtual pipelines are not supported
21:16	MMB_26_MUX_W	Macro-block 26 write selector relation
15:14	MMB_25_CTX_W	Macro-block 25 write context relation - always set to 0x0 when virtual pipelines are not supported
13:8	MMB_25_MUX_W	Macro-block 25 write selector relation
7:6	MMB_24_CTX_W	Macro-block 24 write context relation - always set to 0x0 when virtual pipelines are not supported
5:0	MMB_24_MUX_W	Macro-block 24 write selector relation

0x01A010A4 MDSS_MDP_SMP_ALLOC_W_9**Type:** RW**Reset State:** 0x00000000

Shared Memory Pool allocation configuration 9

double buffered: NO

MDSS_MDP_SMP_ALLOC_W_9

Bits	Name	Description
23:22	MMB_29_CTX_W	Macro-block 29 write context relation - always set to 0x0 when virtual pipelines are not supported
21:16	MMB_29_MUX_W	Macro-block 29 write selector relation
15:14	MMB_28_CTX_W	Macro-block 28 write context relation - always set to 0x0 when virtual pipelines are not supported
13:8	MMB_28_MUX_W	Macro-block 28 write selector relation
7:6	MMB_27_CTX_W	Macro-block 27 write context relation - always set to 0x0 when virtual pipelines are not supported
5:0	MMB_27_MUX_W	Macro-block 27 write selector relation

0x01A010A8 MDSS_MDP_SMP_ALLOC_W_10**Type:** RW**Reset State:** 0x00000000

Shared Memory Pool allocation configuration 10

double buffered: NO

MDSS_MDP_SMP_ALLOC_W_10

Bits	Name	Description
23:22	MMB_32_CTX_W	Macro-block 32 write context relation - always set to 0x0 when virtual pipelines are not supported
21:16	MMB_32_MUX_W	Macro-block 32 write selector relation
15:14	MMB_31_CTX_W	Macro-block 31 write context relation - always set to 0x0 when virtual pipelines are not supported
13:8	MMB_31_MUX_W	Macro-block 31 write selector relation
7:6	MMB_30_CTX_W	Macro-block 30 write context relation - always set to 0x0 when virtual pipelines are not supported
5:0	MMB_30_MUX_W	Macro-block 30 write selector relation

0x01A01130 MDSS_MDP_SMP_ALLOC_R_0**Type:** RW**Reset State:** 0x00000000

Shared Memory Pool allocation configuration 0

double buffered: NO

MDSS_MDP_SMP_ALLOC_R_0

Bits	Name	Description
23:22	MMB_2_CTX_R	Macro-block 2 read context relation - always set to 0x0 when virtual pipelines are not supported
21:16	MMB_2_MUX_R	Macro-block 2 read selector relation
15:14	MMB_1_CTX_R	Macro-block 1 read context relation - always set to 0x0 when virtual pipelines are not supported
13:8	MMB_1_MUX_R	Macro-block 1 read selector relation
7:6	MMB_0_CTX_R	Macro-block 0 read context relation - always set to 0x0 when virtual pipelines are not supported
5:0	MMB_0_MUX_R	Macro-block 0 read selector relation

0x01A01134 MDSS_MDP_SMP_ALLOC_R_1**Type:** RW**Reset State:** 0x00000000

Shared Memory Pool allocation configuration 1

double buffered: NO

MDSS_MDP_SMP_ALLOC_R_1

Bits	Name	Description
23:22	MMB_5_CTX_R	Macro-block 5 read context relation - always set to 0x0 when virtual pipelines are not supported
21:16	MMB_5_MUX_R	Macro-block 5 read selector relation
15:14	MMB_4_CTX_R	Macro-block 4 read context relation - always set to 0x0 when virtual pipelines are not supported
13:8	MMB_4_MUX_R	Macro-block 4 read selector relation
7:6	MMB_3_CTX_R	Macro-block 3 read context relation - always set to 0x0 when virtual pipelines are not supported
5:0	MMB_3_MUX_R	Macro-block 3 read selector relation

0x01A01138 MDSS_MDP_SMP_ALLOC_R_2**Type:** RW**Reset State:** 0x00000000

Shared Memory Pool allocation configuration 2

double buffered: NO

MDSS_MDP_SMP_ALLOC_R_2

Bits	Name	Description
23:22	MMB_8_CTX_R	Macro-block 8 read context relation - always set to 0x0 when virtual pipelines are not supported
21:16	MMB_8_MUX_R	Macro-block 8 read selector relation
15:14	MMB_7_CTX_R	Macro-block 7 read context relation - always set to 0x0 when virtual pipelines are not supported
13:8	MMB_7_MUX_R	Macro-block 7 read selector relation
7:6	MMB_6_CTX_R	Macro-block 6 read context relation - always set to 0x0 when virtual pipelines are not supported
5:0	MMB_6_MUX_R	Macro-block 6 read selector relation

0x01A0113C MDSS_MDP_SMP_ALLOC_R_3**Type:** RW**Reset State:** 0x00000000

Shared Memory Pool allocation configuration 3

double buffered: NO

MDSS_MDP_SMP_ALLOC_R_3

Bits	Name	Description
23:22	MMB_11_CTX_R	Macro-block 11 read context relation - always set to 0x0 when virtual pipelines are not supported
21:16	MMB_11_MUX_R	Macro-block 11 read selector relation
15:14	MMB_10_CTX_R	Macro-block 10 read context relation - always set to 0x0 when virtual pipelines are not supported
13:8	MMB_10_MUX_R	Macro-block 10 read selector relation
7:6	MMB_9_CTX_R	Macro-block 9 read context relation - always set to 0x0 when virtual pipelines are not supported
5:0	MMB_9_MUX_R	Macro-block 9 read selector relation

0x01A01140 MDSS_MDP_SMP_ALLOC_R_4**Type:** RW**Reset State:** 0x00000000

Shared Memory Pool allocation configuration 4

double buffered: NO

MDSS_MDP_SMP_ALLOC_R_4

Bits	Name	Description
23:22	MMB_14_CTX_R	Macro-block 14 read context relation - always set to 0x0 when virtual pipelines are not supported
21:16	MMB_14_MUX_R	Macro-block 14 read selector relation
15:14	MMB_13_CTX_R	Macro-block 13 read context relation - always set to 0x0 when virtual pipelines are not supported
13:8	MMB_13_MUX_R	Macro-block 13 read selector relation
7:6	MMB_12_CTX_R	Macro-block 12 read context relation - always set to 0x0 when virtual pipelines are not supported
5:0	MMB_12_MUX_R	Macro-block 12 read selector relation

0x01A01144 MDSS_MDP_SMP_ALLOC_R_5**Type:** RW**Reset State:** 0x00000000

Shared Memory Pool allocation configuration 5

double buffered: NO

MDSS_MDP_SMP_ALLOC_R_5

Bits	Name	Description
23:22	MMB_17_CTX_R	Macro-block 17 read context relation - always set to 0x0 when virtual pipelines are not supported
21:16	MMB_17_MUX_R	Macro-block 17 read selector relation
15:14	MMB_16_CTX_R	Macro-block 16 read context relation - always set to 0x0 when virtual pipelines are not supported
13:8	MMB_16_MUX_R	Macro-block 16 read selector relation
7:6	MMB_15_CTX_R	Macro-block 15 read context relation - always set to 0x0 when virtual pipelines are not supported
5:0	MMB_15_MUX_R	Macro-block 15 read selector relation

0x01A01148 MDSS_MDP_SMP_ALLOC_R_6**Type:** RW**Reset State:** 0x00000000

Shared Memory Pool allocation configuration 6

double buffered: NO

MDSS_MDP_SMP_ALLOC_R_6

Bits	Name	Description
23:22	MMB_20_CTX_R	Macro-block 20 read context relation - always set to 0x0 when virtual pipelines are not supported
21:16	MMB_20_MUX_R	Macro-block 20 read selector relation
15:14	MMB_19_CTX_R	Macro-block 19 read context relation - always set to 0x0 when virtual pipelines are not supported
13:8	MMB_19_MUX_R	Macro-block 19 read selector relation
7:6	MMB_18_CTX_R	Macro-block 18 read context relation - always set to 0x0 when virtual pipelines are not supported
5:0	MMB_18_MUX_R	Macro-block 18 read selector relation

0x01A0114C MDSS_MDP_SMP_ALLOC_R_7**Type:** RW**Reset State:** 0x00000000

Shared Memory Pool allocation configuration 7

double buffered: NO

MDSS_MDP_SMP_ALLOC_R_7

Bits	Name	Description
23:22	MMB_23_CTX_R	Macro-block 23 read context relation - always set to 0x0 when virtual pipelines are not supported
21:16	MMB_23_MUX_R	Macro-block 23 read selector relation
15:14	MMB_22_CTX_R	Macro-block 22 read context relation - always set to 0x0 when virtual pipelines are not supported
13:8	MMB_22_MUX_R	Macro-block 22 read selector relation
7:6	MMB_21_CTX_R	Macro-block 21 read context relation - always set to 0x0 when virtual pipelines are not supported
5:0	MMB_21_MUX_R	Macro-block 21 read selector relation

0x01A01150 MDSS_MDP_SMP_ALLOC_R_8**Type:** RW**Reset State:** 0x00000000

Shared Memory Pool allocation configuration 8

double buffered: NO

MDSS_MDP_SMP_ALLOC_R_8

Bits	Name	Description
23:22	MMB_26_CTX_R	Macro-block 26 read context relation - always set to 0x0 when virtual pipelines are not supported
21:16	MMB_26_MUX_R	Macro-block 26 read selector relation
15:14	MMB_25_CTX_R	Macro-block 25 read context relation - always set to 0x0 when virtual pipelines are not supported
13:8	MMB_25_MUX_R	Macro-block 25 read selector relation
7:6	MMB_24_CTX_R	Macro-block 24 read context relation - always set to 0x0 when virtual pipelines are not supported
5:0	MMB_24_MUX_R	Macro-block 24 read selector relation

0x01A01154 MDSS_MDP_SMP_ALLOC_R_9**Type:** RW**Reset State:** 0x00000000

Shared Memory Pool allocation configuration 9

double buffered: NO

MDSS_MDP_SMP_ALLOC_R_9

Bits	Name	Description
23:22	MMB_29_CTX_R	Macro-block 29 read context relation - always set to 0x0 when virtual pipelines are not supported
21:16	MMB_29_MUX_R	Macro-block 29 read selector relation
15:14	MMB_28_CTX_R	Macro-block 28 read context relation - always set to 0x0 when virtual pipelines are not supported
13:8	MMB_28_MUX_R	Macro-block 28 read selector relation
7:6	MMB_27_CTX_R	Macro-block 27 read context relation - always set to 0x0 when virtual pipelines are not supported
5:0	MMB_27_MUX_R	Macro-block 27 read selector relation

0x01A01158 MDSS_MDP_SMP_ALLOC_R_10**Type:** RW**Reset State:** 0x00000000

Shared Memory Pool allocation configuration 10

double buffered: NO

MDSS_MDP_SMP_ALLOC_R_10

Bits	Name	Description
23:22	MMB_32_CTX_R	Macro-block 32 read context relation - always set to 0x0 when virtual pipelines are not supported
21:16	MMB_32_MUX_R	Macro-block 32 read selector relation
15:14	MMB_31_CTX_R	Macro-block 31 read context relation - always set to 0x0 when virtual pipelines are not supported
13:8	MMB_31_MUX_R	Macro-block 31 read selector relation
7:6	MMB_30_CTX_R	Macro-block 30 read context relation - always set to 0x0 when virtual pipelines are not supported
5:0	MMB_30_MUX_R	Macro-block 30 read selector relation

0x01A011E0 MDSS_MDP_QOS_REMAPPER_CLASS_0**Type:** RW**Reset State:** 0x000000E4

QoS remapper to translate health level to priority levels on the bus

Class 0 is used by the real time clients in the system (i.e. ViG/RGB/DMA pipes in line mode)

Class 1 is used by the non-real time clients in the system (i.e. DMA pipes in block mode)

WB modules are completely SW configurable and dont rely on watermarks.

Therefore they do NOT go through the remapper

double buffered: NO

MDSS_MDP_QOS_REMAPPER_CLASS_0

Bits	Name	Description
7:6	REQPRIORITY_3	Remapping ID for priority 3 onto the client port
5:4	REQPRIORITY_2	Remapping ID for priority 0 onto the client port
3:2	REQPRIORITY_1	Remapping ID for priority 1 onto the client port
1:0	REQPRIORITY_0	Remapping ID for priority 0 onto the client port

0x01A011E4 MDSS_MDP_QOS_REMAPPER_CLASS_1**Type:** RW**Reset State:** 0x000000E4

QoS remapper to translate health level to priority levels on the bus

Class 0 is used by the real time clients in the system (i.e. ViG/RGB/DMA pipes in line mode)

Class 1 is used by the non-real time clients in the system (i.e. DMA pipes in block mode)

WB modules are completely SW configurable and dont rely on watermarks.

Therefore they do NOT go through the remapper

double buffered: NO

MDSS_MDP_QOS_REMAPPER_CLASS_1

Bits	Name	Description
7:6	REQPRIORITY_3	Remapping ID for priority 3 onto the client port
5:4	REQPRIORITY_2	Remapping ID for priority 2 onto the client port
3:2	REQPRIORITY_1	Remapping ID for priority 1 onto the client port

MDSS_MDP_QOS_REMAPPER_CLASS_1 (cont.)

Bits	Name	Description
1:0	REQPRIORITY_0	Remapping ID for priority 0 onto the client port

0x01A011E8 MDSS_MDP_SW_RESET_CTRL**Type:** RW**Reset State:** 0x00000038

The SW_RESET_CTRL register is used for the controls required for SW reset

MDSS_MDP_SW_RESET_CTRL

Bits	Name	Description
7:0	HYSTERISIS_CTRL	

0x01A011EC MDSS_MDP_UPPER_NEWROI_PRIOR_TO_START**Type:** RW**Reset State:** 0x00000001

The newroi prior to start is used to flush out incorrect states and such between frames. The newroi is sent prior to the start signals from the PMU's. This field is for the PMU's in the upper pipeline only - there exists a separate register for the lower pipeline (LOWER_NEWROI_PRIOR_TO_START)

MDSS_MDP_UPPER_NEWROI_PRIOR_TO_START

Bits	Name	Description
0	ENABLE	Enable the mode to send out newroi prior to start

0x01A01200 MDSS_MDP_VIG_IGC_COLOR0_RAM_LUTN**Type:** RW**Reset State:** 0x00000000**MDSS_MDP_VIG_IGC_COLOR0_RAM_LUTN**

Bits	Name	Description
30:28	VIG_MASK	
25	INDEX_UPDATE	
24	VALUE_UPDATEN	

MDSS_MDP_VIG_IGC_COLOR0_RAM_LUTN (cont.)

Bits	Name	Description
23:16	INDEX	
11:0	VALUE	

0x01A01204 MDSS_MDP_VIG_IGC_COLOR1_RAM_LUTN**Type:** RW**Reset State:** 0x00000000**MDSS_MDP_VIG_IGC_COLOR1_RAM_LUTN**

Bits	Name	Description
30:28	VIG_MASK	
25	INDEX_UPDATE	
24	VALUE_UPDATEN	
23:16	INDEX	
11:0	VALUE	

0x01A01208 MDSS_MDP_VIG_IGC_COLOR2_RAM_LUTN**Type:** RW**Reset State:** 0x00000000**MDSS_MDP_VIG_IGC_COLOR2_RAM_LUTN**

Bits	Name	Description
30:28	VIG_MASK	
25	INDEX_UPDATE	
24	VALUE_UPDATEN	
23:16	INDEX	
11:0	VALUE	

0x01A01210 MDSS_MDP_RGB_IGC_COLOR0_RAM_LUTN**Type:** RW**Reset State:** 0x00000000

MDSS_MDP_RGB_IGC_COLOR0_RAM_LUTN

Bits	Name	Description
30:28	RGB_MASK	
25	INDEX_UPDATE	
24	VALUE_UPDATEN	
23:16	INDEX	
11:0	VALUE	

0x01A01214 MDSS_MDP_RGB_IGC_COLOR1_RAM_LUTN**Type:** RW**Reset State:** 0x00000000**MDSS_MDP_RGB_IGC_COLOR1_RAM_LUTN**

Bits	Name	Description
30:28	RGB_MASK	
25	INDEX_UPDATE	
24	VALUE_UPDATEN	
23:16	INDEX	
11:0	VALUE	

0x01A01218 MDSS_MDP_RGB_IGC_COLOR2_RAM_LUTN**Type:** RW**Reset State:** 0x00000000**MDSS_MDP_RGB_IGC_COLOR2_RAM_LUTN**

Bits	Name	Description
30:28	RGB_MASK	
25	INDEX_UPDATE	
24	VALUE_UPDATEN	
23:16	INDEX	
11:0	VALUE	

0x01A01220 MDSS_MDP_DMA_IGC_COLOR0_RAM_LUTN**Type:** RW**Reset State:** 0x00000000**MDSS_MDP_DMA_IGC_COLOR0_RAM_LUTN**

Bits	Name	Description
29:28	DMA_MASK	
25	INDEX_UPDATE	
24	VALUE_UPDATEN	
23:16	INDEX	
11:0	VALUE	

0x01A01224 MDSS_MDP_DMA_IGC_COLOR1_RAM_LUTN**Type:** RW**Reset State:** 0x00000000**MDSS_MDP_DMA_IGC_COLOR1_RAM_LUTN**

Bits	Name	Description
29:28	DMA_MASK	
25	INDEX_UPDATE	
24	VALUE_UPDATEN	
23:16	INDEX	
11:0	VALUE	

0x01A01228 MDSS_MDP_DMA_IGC_COLOR2_RAM_LUTN**Type:** RW**Reset State:** 0x00000000**MDSS_MDP_DMA_IGC_COLOR2_RAM_LUTN**

Bits	Name	Description
29:28	DMA_MASK	
25	INDEX_UPDATE	
24	VALUE_UPDATEN	
23:16	INDEX	
11:0	VALUE	

0x01A012A0 MDSS_MDP_UP_MISR_SEL**Type:** RW**Reset State:** 0x00000000

select Upper pipe MISR

MDSS_MDP_UP_MISR_SEL

Bits	Name	Description
7:0	BLOCK_ID	0x00:VIG0_unpack_0 0x01:VIG0_unpack_1 0x02:VIG0_unpack_2 0x03:VIG0_unpack_3 0x04:VIG0_QSEED 0x05:VIG0_CSC 0x06:VIG0_SPA 0x07:VIG0_IGC 0x08:RGB0_unpack_0 0x09:RGB0_unpack_1 0x0a:RGB0_unpack_2 0x0b:RGB0_unpack_3 0x0c:RGB0_scale 0x0d:RGB0_IGC 0x0e:unused 0x0f:unused 0x10:VIG1_unpack_0 0x11:VIG1_unpack_1 0x12:VIG1_unpack_2 0x13:VIG1_unpack_3 0x14:VIG1_QSEED 0x15:VIG1_CSC 0x16:VIG1_SPA 0x17:VIG1_IGC 0x18:RGB1_unpack_0 0x19:RGB1_unpack_1 0x1a:RGB1_unpack_2 0x1b:RGB1_unpack_3 0x1c:RGB1_scale 0x1d:RGB1_IGC 0x1e:unused 0x1f:unused 0x20:VIG2_unpack_0 0x21:VIG2_unpack_1 0x22:VIG2_unpack_2 0x23:VIG2_unpack_3 0x24:VIG2_QSEED 0x25:VIG2_CSC 0x26:unused 0x27:VIG2_IGC 0x28:RGB2_unpack_0 0x29:RGB2_unpack_1 0x2a:RGB2_unpack_2 0x2b:RGB2_unpack_3 0x2c:RGB2_scale 0x2d:RGB2_IGC

MDSS_MDP_UP_MISR_SEL (cont.)

Bits	Name	Description
		0x2e through 0x2f: unused 0x30:DMA0_unpack_0 0x31:DMA0_unpack_1 0x32:DMA0_unpack_2 0x33:DMA0_unpack_3 0x33:DMA0_IGC 0x38:DMA1_unpack_0 0x39:DMA1_unpack_1 0x3a:DMA1_unpack_2 0x3b:DMA1_unpack_3

0x01A012A4 MDSS_MDP_UP_MISR_CTRL**Type:** RW**Reset State:** 0x00000000

Misr control

MDSS_MDP_UP_MISR_CTRL

Bits	Name	Type	Description
10	MISR_STATUS_CLEAR	W	misr status is cleared by writing 1 to this field
9	MISR_STATUS	R	set to 1 when misr signature is updated
8	MISR_ENABLE	RW	1: enable the misr on next vsync 0: disable the misr. 1 to 0 transition will reset the misr logic
7:0	MISR_FRAME_COUNT	RW	number of frames of data to be input to misr before capturing the signature

0x01A012A8 MDSS_MDP_UP_MISR_SIGN_MDP**Type:** R**Reset State:** 0x00000000

MISR Signature for blocks in MDP domain

MDSS_MDP_UP_MISR_SIGN_MDP

Bits	Name	Description
31:0	MISR_CAPTURED	32 bit CRC value from MISR

0x01A012AC MDSS_MDP_CLK_CTRL0**Type:** RW**Reset State:** 0x00000000

Clock Gating Control

MDSS_MDP_CLK_CTRL0

Bits	Name	Description
31:30	SMP_CLK_HYSTERISIS_CTRL	0x0 : Clock will run for extra 4 cycles before is gated OFF 0x1 : Clock will run for extra 8 cycles before is gated OFF 0x2 : Clock will run for extra 16 cycles before is gated OFF 0x3 : Clock will run for extra 32 cycles before is gated OFF
29	SMP_CLK_FORCE_OFF	0x0 : Clock gating is active if CLK_FORCE_ON=0 and CLK_FORCE_OFF=0 0x1 : Force clock branch OFF if CLK_FORCE_ON=0, otherwise setting is ignored
28	SMP_CLK_FORCE_ON	0x0 : Clock gating is active if CLK_FORCE_ON=0 and CLK_FORCE_OFF=0 0x1 : Force clock branch ON
11:10	DMA0_CLK_HYSTERISIS_CTRL	0x0 : Clock will run for extra 4 cycles before is gated OFF 0x1 : Clock will run for extra 8 cycles before is gated OFF 0x2 : Clock will run for extra 16 cycles before is gated OFF 0x3 : Clock will run for extra 32 cycles before is gated OFF
9	DMA0_CLK_FORCE_OFF	0x0 : Clock gating is active if CLK_FORCE_ON=0 and CLK_FORCE_OFF=0 0x1 : Force clock branch OFF if CLK_FORCE_ON=0, otherwise setting is ignored
8	DMA0_CLK_FORCE_ON	0x0 : Clock gating is active if CLK_FORCE_ON=0 and CLK_FORCE_OFF=0 0x1 : Force clock branch ON
7:6	RGB0_CLK_HYSTERISIS_CTRL	0x0 : Clock will run for extra 4 cycles before is gated OFF 0x1 : Clock will run for extra 8 cycles before is gated OFF 0x2 : Clock will run for extra 16 cycles before is gated OFF 0x3 : Clock will run for extra 32 cycles before is gated OFF
5	RGB0_CLK_FORCE_OFF	0x0 : Clock gating is active if CLK_FORCE_ON=0 and CLK_FORCE_OFF=0 0x1 : Force clock branch OFF if CLK_FORCE_ON=0, otherwise setting is ignored
4	RGB0_CLK_FORCE_ON	0x0 : Clock gating is active if CLK_FORCE_ON=0 and CLK_FORCE_OFF=0 0x1 : Force clock branch ON

MDSS_MDP_CLK_CTRL0 (cont.)

Bits	Name	Description
3:2	VIG0_CLK_HYSTERISIS_CTRL	0x0 : Clock will run for extra 4 cycles before is gated OFF 0x1 : Clock will run for extra 8 cycles before is gated OFF 0x2 : Clock will run for extra 16 cycles before is gated OFF 0x3 : Clock will run for extra 32 cycles before is gated OFF
1	VIG0_CLK_FORCE_OFF	0x0 : Clock gating is active if CLK_FORCE_ON=0 and CLK_FORCE_OFF=0 0x1 : Force clock branch OFF if CLK_FORCE_ON=0, otherwise setting is ignored
0	VIG0_CLK_FORCE_ON	0x0 : Clock gating is active if CLK_FORCE_ON=0 and CLK_FORCE_OFF=0 0x1 : Force clock branch ON

0x01A012B0 MDSS_MDP_CLK_STATUS0**Type:** R**Reset State:** 0x00000000

Status of gated clock branches

MDSS_MDP_CLK_STATUS0

Bits	Name	Description
14	DMA0_IGC_LUT_CLK_ACTIVE	
13	DMA0_ROT_N_CLK_ACTIVE	
12	DMA0_CLK_ACTIVE	
11	RGB0_IGC_LUT_CLK_ACTIVE	
10	RGB0_SCL_LUT_CLK_ACTIVE	
9	RGB0_SCL_CLK_ACTIVE	
8	RGB0_CLK_ACTIVE	
6	SMP_CLK_ACTIVE	
5	VIG0_IGC_LUT_CLK_ACTIVE	
4	VIG0_PA_LUTV_CLK_ACTIVE	
3	VIG0_HIST_LUT_CLK_ACTIVE	

MDSS_MDP_CLK_STATUS0 (cont.)

Bits	Name	Description
2	VIG0_SCL_LUT_CLK_ACTIVE	
1	VIG0_SCL_CLK_ACTIVE	
0	VIG0_CLK_ACTIVE	

0x01A012B4 MDSS_MDP_CLK_CTRL1**Type:** RW**Reset State:** 0x00000000

Clock Gating Control

MDSS_MDP_CLK_CTRL1

Bits	Name	Description
31:30	AHB_CLK_HYSTERISIS_CTRL	0x0 : For AHB in upper pipe, clock will run for extra 4 cycles before is gated OFF 0x1 : Clock will run for extra 8 cycles before is gated OFF 0x2 : Clock will run for extra 16 cycles before is gated OFF 0x3 : Clock will run for extra 32 cycles before is gated OFF
29	AHB_CLK_FORCE_OFF	0x0 : For AHB in upper pipe, clock gating is active if CLK_FORCE_ON=0 and CLK_FORCE_OFF=0 0x1 : Force clock branch OFF if CLK_FORCE_ON=0, otherwise setting is ignored
28	AHB_CLK_FORCE_ON	0x0 : For AHB in upper pipe, clock gating is active if CLK_FORCE_ON=0 and CLK_FORCE_OFF=0 0x1 : Force clock branch ON
11:10	DMA1_CLK_HYSTERISIS_CTRL	0x0 : Clock will run for extra 4 cycles before is gated OFF 0x1 : Clock will run for extra 8 cycles before is gated OFF 0x2 : Clock will run for extra 16 cycles before is gated OFF 0x3 : Clock will run for extra 32 cycles before is gated OFF
9	DMA1_CLK_FORCE_OFF	0x0 : Clock gating is active if CLK_FORCE_ON=0 and CLK_FORCE_OFF=0 0x1 : Force clock branch OFF if CLK_FORCE_ON=0, otherwise setting is ignored
8	DMA1_CLK_FORCE_ON	0x0 : Clock gating is active if CLK_FORCE_ON=0 and CLK_FORCE_OFF=0 0x1 : Force clock branch ON
7:6	RGB1_CLK_HYSTERISIS_CTRL	0x0 : Clock will run for extra 4 cycles before is gated OFF 0x1 : Clock will run for extra 8 cycles before is gated OFF 0x2 : Clock will run for extra 16 cycles before is gated OFF 0x3 : Clock will run for extra 32 cycles before is gated OFF

MDSS_MDP_CLK_CTRL1 (cont.)

Bits	Name	Description
5	RGB1_CLK_FORCE_OFF	0x0 : Clock gating is active if CLK_FORCE_ON=0 and CLK_FORCE_OFF=0 0x1 : Force clock branch OFF if CLK_FORCE_ON=0, otherwise setting is ignored
4	RGB1_CLK_FORCE_ON	0x0 : Clock gating is active if CLK_FORCE_ON=0 and CLK_FORCE_OFF=0 0x1 : Force clock branch ON
3:2	VIG1_CLK_HYSTERISIS_CTRL	0x0 : Clock will run for extra 4 cycles before is gated OFF 0x1 : Clock will run for extra 8 cycles before is gated OFF 0x2 : Clock will run for extra 16 cycles before is gated OFF 0x3 : Clock will run for extra 32 cycles before is gated OFF
1	VIG1_CLK_FORCE_OFF	0x0 : Clock gating is active if CLK_FORCE_ON=0 and CLK_FORCE_OFF=0 0x1 : Force clock branch OFF if CLK_FORCE_ON=0, otherwise setting is ignored
0	VIG1_CLK_FORCE_ON	0x0 : Clock gating is active if CLK_FORCE_ON=0 and CLK_FORCE_OFF=0 0x1 : Force clock branch ON

0x01A012B8 MDSS_MDP_CLK_STATUS1**Type:** R**Reset State:** 0x00000000

Status of gated clock branches

MDSS_MDP_CLK_STATUS1

Bits	Name	Description
14	DMA1_IGC_LUT_CLK_ACTIVE	
13	DMA1_ROT_N_CLK_ACTIVE	
12	DMA1_CLK_ACTIVE	
11	RGB1_IGC_LUT_CLK_ACTIVE	
10	RGB1_SCL_LUT_CLK_ACTIVE	
9	RGB1_SCL_CLK_ACTIVE	
8	RGB1_CLK_ACTIVE	
6	AHB_CLK_ACTIVE	0x0: AHB clock for upper pipe is inactive 0x1: AHB clock for upper pipe is active

MDSS_MDP_CLK_STATUS1 (cont.)

Bits	Name	Description
5	VIG1_IGC_LUT_CLK_ACTIVE	
4	VIG1_PA_LUTV_CLK_ACTIVE	
3	VIG1_HIST_LUT_CLK_ACTIVE	
2	VIG1_SCL_LUT_CLK_ACTIVE	
1	VIG1_SCL_CLK_ACTIVE	
0	VIG1_CLK_ACTIVE	

0x01A012BC MDSS_MDP_CLK_CTRL2**Type:** RW**Reset State:** 0x00000000

Clock Gating Control

MDSS_MDP_CLK_CTRL2

Bits	Name	Description
23:22	VBIF_CLI_ARB_CLK_HYSTERISIS_CTRL	0x0 : Clock will run for extra 4 cycles before is gated OFF 0x1 : Clock will run for extra 8 cycles before is gated OFF 0x2 : Clock will run for extra 16 cycles before is gated OFF 0x3 : Clock will run for extra 32 cycles before is gated OFF
21	VBIF_CLI_ARB_CLK_FORCE_OFF	0x0 : Clock gating is active if CLK_FORCE_ON=0 and CLK_FORCE_OFF=0 0x1 : Force clock branch OFF if CLK_FORCE_ON=0, otherwise setting is ignored
19:18	VBIF_CLI_WB2_CLK_HYSTERISIS_CTRL	0x0 : Clock will run for extra 4 cycles before is gated OFF 0x1 : Clock will run for extra 8 cycles before is gated OFF 0x2 : Clock will run for extra 16 cycles before is gated OFF 0x3 : Clock will run for extra 32 cycles before is gated OFF
17	VBIF_CLI_WB2_CLK_FORCE_OFF	0x0 : Clock gating is active if CLK_FORCE_ON=0 and CLK_FORCE_OFF=0 0x1 : Force clock branch OFF if CLK_FORCE_ON=0, otherwise setting is ignored
16	VBIF_CLI_WB2_CLK_FORCE_ON	0x0 : Clock gating is active if CLK_FORCE_ON=0 and CLK_FORCE_OFF=0 0x1 : Force clock branch ON

MDSS_MDP_CLK_CTRL2 (cont.)

Bits	Name	Description
15:14	VBIF_CLI_ROT1_CLK_HYS TERISIS_CTRL	0x0 : Clock will run for extra 4 cycles before is gated OFF 0x1 : Clock will run for extra 8 cycles before is gated OFF 0x2 : Clock will run for extra 16 cycles before is gated OFF 0x3 : Clock will run for extra 32 cycles before is gated OFF
13	VBIF_CLI_ROT1_CLK_FOR CE_OFF	0x0 : Clock gating is active if CLK_FORCE_ON=0 and CLK_FORCE_OFF=0 0x1 : Force clock branch OFF if CLK_FORCE_ON=0, otherwise setting is ignored
12	VBIF_CLI_ROT1_CLK_FOR CE_ON	0x0 : Clock gating is active if CLK_FORCE_ON=0 and CLK_FORCE_OFF=0 0x1 : Force clock branch ON
11:10	VBIF_CLI_ROT0_CLK_HYS TERISIS_CTRL	0x0 : Clock will run for extra 4 cycles before is gated OFF 0x1 : Clock will run for extra 8 cycles before is gated OFF 0x2 : Clock will run for extra 16 cycles before is gated OFF 0x3 : Clock will run for extra 32 cycles before is gated OFF
9	VBIF_CLI_ROT0_CLK_FOR CE_OFF	0x0 : Clock gating is active if CLK_FORCE_ON=0 and CLK_FORCE_OFF=0 0x1 : Force clock branch OFF if CLK_FORCE_ON=0, otherwise setting is ignored
8	VBIF_CLI_ROT0_CLK_FOR CE_ON	0x0 : Clock gating is active if CLK_FORCE_ON=0 and CLK_FORCE_OFF=0 0x1 : Force clock branch ON

0x01A012C0 MDSS_MDP_CLK_STATUS2**Type:** R**Reset State:** 0x00000000

Status of gated clock branches

MDSS_MDP_CLK_STATUS2

Bits	Name	Description
15	VBIF_CLI_ARB_CLK_ACTIV E	
14	VBIF_CLI_WB2_CLK_ACTIV E	
13	VBIF_CLI_ROT1_CLK_ACTI VE	
12	VBIF_CLI_ROT0_CLK_ACTI VE	

0x01A012E0 MDSS_MDP_VIDEO_INTF_UNDERFLOW_CTL**Type:** RW**Reset State:** 0x00001111

VIDEO_INTF_UNDERFLOW_CTRL register controls the auto recovery mechanism on underflow case. Register reprogramming is not allowed when each video interface is running.

double buffered: NO

MDSS_MDP_VIDEO_INTF_UNDERFLOW_CTL

Bits	Name	Description
4	INTF_1_ERR_RECOVERY_EN	Enable auto recovery during underflow case in interface 1 if NEWROI_PRIOR_TO_START is disabled. 0x0 : Disalbe. SW is required to apply SW reset when underflow interrupt occurs 0x1 : Enable. No SW reset is required. Underflow interrupt is generated after auto recovery is finished

0x01A012F4 MDSS_MDP_SPLIT_DISPLAY_EN**Type:** RW**Reset State:** 0x00000000

Split display on/off control register. This register bit is not double buffered.

MDSS_MDP_SPLIT_DISPLAY_EN

Bits	Name	Description
0	ENABLE	Turning on/off split display configuration 0x0 : Enable 0x1 : Disable

0x01A012F8 MDSS_MDP_SPLIT_DISPLAY_UPPER_PIPE_CTRL**Type:** RW**Reset State:** 0x00000000

Main configuration of split display in mdp upper pipeline. This register are not double buffered.

MDSS_MDP_SPLIT_DISPLAY_UPPER_PIPE_CTRL

Bits	Name	Description
4	INTF_1_SW_TRG_MUX	Control SW trigger mux of interface 1 0x0 : No SW trigger change 0x1 : Use SW trigger of interface 2 for interface 1
2	SMART_PANEL_FREE_RUN	Select split display mode for smart panel. This register value should be identical to lower pipe's same name bit before enabling split display. 0x1 : Free running mode between panels 0x0 : Synchronous mode between panels
1	SPLIT_DISPLAY_TYPE	Specify panel type. This register value should be identical to lower pipe's same name bit before enabling split display. 0x1 : Smart Panel 0x0 : Dumb Panel

0x01A01300 MDSS_MDP_DSPP_IGC_COLOR0_RAM_LUTN**Type:** RW**Reset State:** 0x00000000**MDSS_MDP_DSPP_IGC_COLOR0_RAM_LUTN**

Bits	Name	Description
30:28	DSPP_MASK	
25	INDEX_UPDATE	
24	VALUE_UPDATEN	
23:16	INDEX	
11:0	VALUE	

0x01A01304 MDSS_MDP_DSPP_IGC_COLOR1_RAM_LUTN**Type:** RW**Reset State:** 0x00000000**MDSS_MDP_DSPP_IGC_COLOR1_RAM_LUTN**

Bits	Name	Description
30:28	DSPP_MASK	
25	INDEX_UPDATE	
24	VALUE_UPDATEN	
23:16	INDEX	

MDSS_MDP_DSPP_IGC_COLOR1_RAM_LUTN (cont.)

Bits	Name	Description
11:0	VALUE	

0x01A01308 MDSS_MDP_DSPP_IGC_COLOR2_RAM_LUTN**Type:** RW**Reset State:** 0x00000000**MDSS_MDP_DSPP_IGC_COLOR2_RAM_LUTN**

Bits	Name	Description
30:28	DSPP_MASK	
25	INDEX_UPDATE	
24	VALUE_UPDATEN	
23:16	INDEX	
11:0	VALUE	

0x01A01330 MDSS_MDP_MDP_VSYNC_SEL**Type:** RW**Reset State:** 0x00000000

MDP_VSYNC_SEL register is used to select either the vsyncs from GPIOs or DSI blocks to be used internally as primary, secondary and external vsyncs

double buffered: NO

MDSS_MDP_MDP_VSYNC_SEL

Bits	Name	Description
18	VSYNC_INVERT_0	0=Vsync for stream of Ping-pong 0 is active high; 1=Vsync for stream of Ping-pong 0 is active low
17	VSYNC_INVERT_1	0=Vsync for stream of Ping-pong 1 is active high; 1=Vsync for stream of Ping-pong 0 is active low
16	VSYNC_INVERT_2	0=Vsync for stream of Ping-pong 2 is active high; 1=Vsync for stream of Ping-pong 0 is active low

MDSS_MDP_MDP_VSYNC_SEL (cont.)

Bits	Name	Description
15:12	PING_PONG_0	0x0: Vsync0 from GPIO 0x1: Vsync1 from GPIO 0x2: Vsync2 from GPIO 0x3: Vsync(TE) from INTF_0 panel 0x4: Vsync(TE) from INTF_1 panel 0x5: Vsync(TE) from INTF_2 panel 0x6: Vsync(TE) from INTF_3 panel

0x01A01350 MDSS_MDP_LP_MISR_SEL**Type:** RW**Reset State:** 0x00000000

select Upper pipe MISR

MDSS_MDP_LP_MISR_SEL

Bits	Name	Description
7:0	BLOCK_ID	MISR block select for the MDP clock domain MISR The following are the block select encodings 0x00:cursor_unpack_0 0x01:cursor_unpack_1 0x02:cursor_unpack_2 0x03_0x07:unused 0x08:LMIX0_blend 0x09:LMIX0_GC 0x0a:LMIX1_blend 0x0b:LMIX1_GC 0x0c:LMIX2_blend 0x0d:LMIX2_GC 0x0e:LMIX3_blend 0x0f:LMIX3_GC 0x10:LMIX4_blend 0x11:LMIX4_GC 0x12:DSPP0_PA 0x13:DSPP0_IGC 0x14:DSPP0_GamutM 0x15:DSPP0_PCC 0x16:DSPP0_panelC_LUT 0x17:DSPP0_dither 0x18:DSPP1_PA 0x19:DSPP1_IGC 0x1a:DSPP1_GamutM 0x1b:DSPP1_PCC 0x1c:DSPP1_panelC_LUT 0x1d:DSPP1_dither 0x1e:DSPP2_PA 0x1f:DSPP2_IGC 0x20:DSPP2_GamutM 0x21:DSPP2_PCC 0x22:DSPP2_panelC_LUT 0x23:DSPP2_dither 0x24:ppb0_0 0x25:ppb0_1 0x26:ppb1_0 0x27:ppb1_1 0x28:ppb2_0 0x29:ppb2_1 0x2a:rot0_csc 0x2b:rot0_chromaD 0x2c:rot0_dither 0x2d:rot0_packer(remove) 0x30:rot0_rotate_0 0x31:rot0_rotate_1 0x32:rot0_rotate_2

MDSS_MDP_LP_MISR_SEL (cont.)

Bits	Name	Description
		0x33:rot0_rotate_3 0x34:rot0_bwc_0 0x35:rot0_bwc_1 0x36:rot0_bwc_2 0x37:rot0_bwc_3 0x38:rot1_csc 0x39:rot1_chromaD 0x3a:rot1_dither 0x3b:rot1_packer(remove) 0x3c:rot1_rotate_0 0x3d:rot1_rotate_1 0x3e:rot1_rotate_2 0x3f:rot1_rotate_3 0x40:rot1_bwc_0 0x41:rot1_bwc_1 0x42:rot1_bwc_2 0x43:rot1_bwc_3 0x44: 3D mux_0 0x45: 3Dmux_1 0x46: 3Dmux_2

0x01A01354 MDSS_MDP_LP_MISR_CTRL**Type:** RW**Reset State:** 0x00000000

Misr control MDP

MDSS_MDP_LP_MISR_CTRL

Bits	Name	Type	Description
10	MISR_STATUS_CLEAR	W	misr status is cleared by writing 1 to this field
9	MISR_STATUS	R	set to 1 when misr signature is updated
8	MISR_ENABLE	RW	1: enable the misr on next vsync 0: disable the misr. 1 to 0 transition will reset the misr logic
7:0	MISR_FRAME_COUNT	RW	number of frames of data to be input to misr before capturing the signature

0x01A01358 MDSS_MDP_LP_MISR_CTRL_HDMI**Type:** RW**Reset State:** 0x00000000

Misr control hdmi

MDSS_MDP_LP_MISR_CTRL_HDMI

Bits	Name	Type	Description
10	MISR_STATUS_CLEAR	W	misr status is cleared by writing 1 to this field
9	MISR_STATUS	R	set to 1 when misr signature is updated
8	MISR_ENABLE	RW	1: enable the misr on next vsync 0: disable the misr. 1 to 0 transition will reset the misr logic
7:0	MISR_FRAME_COUNT	RW	number of frames of data to be input to misr before capturing the signature

0x01A0135C MDSS_MDP_LP_MISR_CTRL_EDP**Type:** RW**Reset State:** 0x00000000

Misr control edp

MDSS_MDP_LP_MISR_CTRL_EDP

Bits	Name	Type	Description
10	MISR_STATUS_CLEAR	W	misr status is cleared by writing 1 to this field
9	MISR_STATUS	R	set to 1 when misr signature is updated
8	MISR_ENABLE	RW	1: enable the misr on next vsync 0: disable the misr. 1 to 0 transition will reset the misr logic
7:0	MISR_FRAME_COUNT	RW	number of frames of data to be input to misr before capturing the signature

0x01A01360 MDSS_MDP_LP_MISR_CTRL_DSI0**Type:** RW**Reset State:** 0x00000000

Misr control DSI0

MDSS_MDP_LP_MISR_CTRL_DSI0

Bits	Name	Type	Description
10	MISR_STATUS_CLEAR	W	misr status is cleared by writing 1 to this field
9	MISR_STATUS	R	set to 1 when misr signature is updated
8	MISR_ENABLE	RW	1: enable the misr on next vsync 0: disable the misr. 1 to 0 transition will reset the misr logic
7:0	MISR_FRAME_COUNT	RW	number of frames of data to be input to misr before capturing the signature

0x01A01364 MDSS_MDP_LP_MISR_CTRL_DSI1**Type:** RW**Reset State:** 0x00000000

Misr control DSI1

MDSS_MDP_LP_MISR_CTRL_DSI1

Bits	Name	Type	Description
10	MISR_STATUS_CLEAR	W	misr status is cleared by writing 1 to this field
9	MISR_STATUS	R	set to 1 when misr signature is updated
8	MISR_ENABLE	RW	1: enable the misr on next vsync 0: disable the misr. 1 to 0 transition will reset the misr logic
7:0	MISR_FRAME_COUNT	RW	number of frames of data to be input to misr before capturing the signature

0x01A01368 MDSS_MDP_LP_MISR_SIGN_MDP**Type:** R**Reset State:** 0x00000000

MISR Signature for blocks in MDP domain

MDSS_MDP_LP_MISR_SIGN_MDP

Bits	Name	Description
31:0	MISR_CAPTURED	32 bit CRC value from MISR

0x01A0136C MDSS_MDP_LP_MISR_SIGN_EDP**Type:** R**Reset State:** 0x00000000

MISR Signature for blocks in EDP domain

MDSS_MDP_LP_MISR_SIGN_EDP

Bits	Name	Description
31:0	MISR_CAPTURED	32 bit CRC value from MISR

0x01A01370 MDSS_MDP_LP_MISR_SIGN_HDMI**Type:** R**Reset State:** 0x00000000

MISR Signature for blocks in HDMI P clock domain

MDSS_MDP_LP_MISR_SIGN_HDMI

Bits	Name	Description
31:0	MISR_CAPTURED	32 bit CRC value from MISR

0x01A01374 MDSS_MDP_LP_MISR_SIGN_DSI0**Type:** R**Reset State:** 0x00000000

MISR Signature for blocks in DSI0 domain

MDSS_MDP_LP_MISR_SIGN_DSI0

Bits	Name	Description
31:0	MISR_CAPTURED	32 bit CRC value from MISR

0x01A01378 MDSS_MDP_LP_MISR_SIGN_DSI1**Type:** R**Reset State:** 0x00000000

MISR Signature for blocks in DSI1 domain

MDSS_MDP_LP_MISR_SIGN_DSI1

Bits	Name	Description
31:0	MISR_CAPTURED	32 bit CRC value from MISR

0x01A01380 MDSS_MDP_WD_TIMER_0_CTL**Type:** W**Reset State:** 0x00000000

This register is used to control Watchdog Timer 0 in mdp

MDSS_MDP_WD_TIMER_0_CTL

Bits	Name	Description
0	WD_TIMER_0_CLEAR	Clear the timer

0x01A01384 MDSS_MDP_WD_TIMER_0_CTL2**Type:** RW**Reset State:** 0x00000002

This register is used to control Watchdog Timer 0 in mdp

MDSS_MDP_WD_TIMER_0_CTL2

Bits	Name	Description
4:1	WD_TIMER_0_GRANULARITY	Configurable # of clock ticks granularity per main counter increment 0x0: 8 clock ticks 0x1: 16 clock ticks (default) 0x2: 24 clock ticks 0x3: 32 clock ticks 0x4: 40 clock ticks 0x5: 48 clock ticks 0x6: 56 clock ticks 0x7: 64 clock ticks 0x8: 72 clock ticks 0x9: 80 clock ticks 0xa: 88 clock ticks 0xb: 96 clock ticks 0xc: 104 clock ticks 0xd: 112 clock ticks 0xe: 120 clock ticks 0xf: 128 clock ticks

MDSS_MDP_WD_TIMER_0_CTL2 (cont.)

Bits	Name	Description
0	WD_TIMER_0_ENABLE	Enable the timer

0x01A01388 MDSS_MDP_WD_TIMER_0_LOAD_VALUE**Type:** RW**Reset State:** 0x00000000

The timer count up to value that would trigger WD interrupt

MDSS_MDP_WD_TIMER_0_LOAD_VALUE

Bits	Name	Description
31:0	WD_TIMER_0_LOAD_VALUE	Timer count value 0x0 : Free running mode, timer counter wraps Others: Timer count up to value in granularity of 16 clock tick periods eg. 0x2 : Translates into 2*16=32 clock tick periods

0x01A0138C MDSS_MDP_WD_TIMER_0_STATUS_VALUE**Type:** R**Reset State:** 0x00000000

The current timer count value

MDSS_MDP_WD_TIMER_0_STATUS_VALUE

Bits	Name	Description
31:0	WD_TIMER_0_STATUS_VALUE	

0x01A01390 MDSS_MDP_WD_TIMER_1_CTL**Type:** W**Reset State:** 0x00000000

This register is used to control Watchdog Timer 1 in mdp

MDSS_MDP_WD_TIMER_1_CTL

Bits	Name	Description
0	WD_TIMER_1_CLEAR	Clear the timer

0x01A01394 MDSS_MDP_WD_TIMER_1_CTL2**Type:** RW**Reset State:** 0x00000002

This register is used to control Watchdog Timer 1 in mdp

MDSS_MDP_WD_TIMER_1_CTL2

Bits	Name	Description
4:1	WD_TIMER_1_GRANULARITY	Configurable # of clock ticks granularity per main counter increment 0x0: 8 clock ticks 0x1: 16 clock ticks (default) 0x2: 24 clock ticks 0x3: 32 clock ticks 0x4: 40 clock ticks 0x5: 48 clock ticks 0x6: 56 clock ticks 0x7: 64 clock ticks 0x8: 72 clock ticks 0x9: 80 clock ticks 0xa: 88 clock ticks 0xb: 96 clock ticks 0xc: 104 clock ticks 0xd: 112 clock ticks 0xe: 120 clock ticks 0xf: 128 clock ticks
0	WD_TIMER_1_ENABLE	Enable the timer

0x01A01398 MDSS_MDP_WD_TIMER_1_LOAD_VALUE**Type:** RW**Reset State:** 0x00000000

The timer count up to value that would trigger WD interrupt

MDSS_MDP_WD_TIMER_1_LOAD_VALUE

Bits	Name	Description
31:0	WD_TIMER_1_LOAD_VALUE	Timer count value 0x0 : Free running mode, timer counter wraps Others: Timer count up to value in granularity of 16 clock tick periods eg. 0x2 : Translates into 2*16=32 clock tick periods

0x01A0139C MDSS_MDP_WD_TIMER_1_STATUS_VALUE**Type:** R**Reset State:** 0x00000000

The current timer count value

MDSS_MDP_WD_TIMER_1_STATUS_VALUE

Bits	Name	Description
31:0	WD_TIMER_1_STATUS_VALUE	

0x01A013A8 MDSS_MDP_CLK_CTRL3**Type:** RW**Reset State:** 0x00000000

Clock Gating Control

MDSS_MDP_CLK_CTRL3

Bits	Name	Description
27:26	ROT0_CLK_HYSTERISIS_CTRL	0x0 : Clock will run for extra 4 cycles before is gated OFF 0x1 : Clock will run for extra 8 cycles before is gated OFF 0x2 : Clock will run for extra 16 cycles before is gated OFF 0x3 : Clock will run for extra 32 cycles before is gated OFF
25	ROT0_CLK_FORCE_OFF	0x0 : Clock gating is active if CLK_FORCE_ON=0 and CLK_FORCE_OFF=0 0x1 : Force clock branch OFF if CLK_FORCE_ON=0, otherwise setting is ignored
24	ROT0_CLK_FORCE_ON	0x0 : Clock gating is active if CLK_FORCE_ON=0 and CLK_FORCE_OFF=0 0x1 : Force clock branch ON

MDSS_MDP_CLK_CTRL3 (cont.)

Bits	Name	Description
23:22	PP0_CLK_HYSTERISIS_CTRL	0x0 : Clock will run for extra 4 cycles before is gated OFF 0x1 : Clock will run for extra 8 cycles before is gated OFF 0x2 : Clock will run for extra 16 cycles before is gated OFF 0x3 : Clock will run for extra 32 cycles before is gated OFF
21	PP0_CLK_FORCE_OFF	0x0 : Clock gating is active if CLK_FORCE_ON=0 and CLK_FORCE_OFF=0 0x1 : Force clock branch OFF if CLK_FORCE_ON=0, otherwise setting is ignored
20	PP0_CLK_FORCE_ON	0x0 : Clock gating is active if CLK_FORCE_ON=0 and CLK_FORCE_OFF=0 0x1 : Force clock branch ON
19:18	CURSOR0_CLK_HYSTERISIS_CTRL	0x0 : Clock will run for extra 4 cycles before is gated OFF 0x1 : Clock will run for extra 8 cycles before is gated OFF 0x2 : Clock will run for extra 16 cycles before is gated OFF 0x3 : Clock will run for extra 32 cycles before is gated OFF
17	CURSOR0_CLK_FORCE_OFF	0x0 : Clock gating is active if CLK_FORCE_ON=0 and CLK_FORCE_OFF=0 0x1 : Force clock branch OFF if CLK_FORCE_ON=0, otherwise setting is ignored
16	CURSOR0_CLK_FORCE_ON	0x0 : Clock gating is active if CLK_FORCE_ON=0 and CLK_FORCE_OFF=0 0x1 : Force clock branch ON
15:14	LM_DSPP0_CLK_HYSTERISIS_CTRL	0x0 : Clock will run for extra 4 cycles before is gated OFF 0x1 : Clock will run for extra 8 cycles before is gated OFF 0x2 : Clock will run for extra 16 cycles before is gated OFF 0x3 : Clock will run for extra 32 cycles before is gated OFF
13	LM_DSPP0_CLK_FORCE_OFF	0x0 : Clock gating is active if CLK_FORCE_ON=0 and CLK_FORCE_OFF=0 0x1 : Force clock branch OFF if CLK_FORCE_ON=0, otherwise setting is ignored
12	LM_DSPP0_CLK_FORCE_ON	0x0 : Clock gating is active if CLK_FORCE_ON=0 and CLK_FORCE_OFF=0 0x1 : Force clock branch ON

0x01A013AC MDSS_MDP_CLK_STATUS3**Type:** R**Reset State:** 0x00000000

Status of gated clock branches

MDSS_MDP_CLK_STATUS3

Bits	Name	Description
22	ROT0_BLK_CLK_ACTIVE	
21	ROT0_LINE_CLK_ACTIVE	
20	ROT0_CLK_ACTIVE	
19	LM_DSPP0_IGC_LUT_CLK_ACTIVE	
18	LM_DSPP0_PA_LUTV_CLK_ACTIVE	
17	LM_DSPP0_HIST_LUT_CLK_ACTIVE	
16	LM_DSPP0_CLK_ACTIVE	
15	CURSOR0_CLK_ACTIVE	
14	LM_DSPP0_GAMUT_LUT_CLK_ACTIVE	
7	PP0_CLK_ACTIVE	

0x01A013B0 MDSS_MDP_CLK_CTRL4**Type:** RW**Reset State:** 0x00000000

Clock Gating Control

MDSS_MDP_CLK_CTRL4

Bits	Name	Description
31:30	AHB_CLK_HYSTERISIS_CTRL	0x0 : Clock will run for extra 4 cycles before is gated OFF 0x1 : Clock will run for extra 8 cycles before is gated OFF 0x2 : Clock will run for extra 16 cycles before is gated OFF 0x3 : Clock will run for extra 32 cycles before is gated OFF
29	AHB_CLK_FORCE_OFF	0x0 : For AHB in lower pipe, clock gating is active if CLK_FORCE_ON=0 and CLK_FORCE_OFF=0 0x1 : Force clock branch OFF if CLK_FORCE_ON=0, otherwise setting is ignored
28	AHB_CLK_FORCE_ON	0x0 : For AHB in lower pipe, clock gating is active if CLK_FORCE_ON=0 and CLK_FORCE_OFF=0 0x1 : Force clock branch ON

0x01A013B4 MDSS_MDP_CLK_STATUS4**Type:** R**Reset State:** 0x00000000

Status of gated clock branches

MDSS_MDP_CLK_STATUS4

Bits	Name	Description
6	AHB_CLK_ACTIVE	0x0: AHB clock for lower pipe is inactive 0x1: AHB clock for lower pipe is active

0x01A013B8 MDSS_MDP_CLK_CTRL5**Type:** RW**Reset State:** 0x00000004

Clock Gating Control

MDSS_MDP_CLK_CTRL5

Bits	Name	Description
31:30	VBIF_ARB_CLK_HYSTERISIS_CTRL	0x0 : Clock will run for extra 4 cycles before is gated OFF 0x1 : Clock will run for extra 8 cycles before is gated OFF 0x2 : Clock will run for extra 16 cycles before is gated OFF 0x3 : Clock will run for extra 32 cycles before is gated OFF
29	VBIF_ARB_CLK_FORCE_OFF	0x0 : Clock gating is active if CLK_FORCE_ON=0 and CLK_FORCE_OFF=0 0x1 : Force clock branch OFF if CLK_FORCE_ON=0, otherwise setting is ignored
27:26	WB2_CLK_HYSTERISIS_CTRL	0x0 : Clock will run for extra 4 cycles before is gated OFF 0x1 : Clock will run for extra 8 cycles before is gated OFF 0x2 : Clock will run for extra 16 cycles before is gated OFF 0x3 : Clock will run for extra 32 cycles before is gated OFF
25	WB2_CLK_FORCE_OFF	0x0 : Clock gating is active if CLK_FORCE_ON=0 and CLK_FORCE_OFF=0 0x1 : Force clock branch OFF if CLK_FORCE_ON=0, otherwise setting is ignored
24	WB2_CLK_FORCE_ON	0x0 : Clock gating is active if CLK_FORCE_ON=0 and CLK_FORCE_OFF=0 0x1 : Force clock branch ON
3:2	VBIF_CLI_CORE_CLK_HYSTERISIS_CTRL	

MDSS_MDP_CLK_CTRL5 (cont.)

Bits	Name	Description
1	VBIF_CLI_CORE_CLK_FOR CE_OFF	

0x01A013BC MDSS_MDP_CLK_STATUS5**Type:** R**Reset State:** 0x00000000

Status of gated clock branches

MDSS_MDP_CLK_STATUS5

Bits	Name	Description
24	VBIF_CLI_CORE_CLK_ACTIVE	
23	VBIF_ARB_CLK_ACTIVE	
22	WB2_BLK_CLK_ACTIVE	
21	WB2_LINE_CLK_ACTIVE	
20	WB2_CLK_ACTIVE	

0x01A013E0 MDSS_MDP_QOS_REMAPPER_LOWER_CLASS_0**Type:** RW**Reset State:** 0x000000E4

QoS remapper to translate health level to priority levels on the bus

Class 0 is used by the real time clients in the system (i.e. ViG/RGB/DMA pipes in line mode)

Class 1 is used by the non-real time clients in the system (i.e. DMA pipes in block mode)

WB modules are completely SW configurable and dont rely on watermarks.

Therefore they do NOT go through the remapper

double buffered: NO

MDSS_MDP_QOS_REMAPPER_LOWER_CLASS_0

Bits	Name	Description
7:6	REQPRIORITY_3	Remapping ID for priority 3 onto the client port
5:4	REQPRIORITY_2	Remapping ID for priority 0 onto the client port

MDSS_MDP_QOS_REMAPPER_LOWER_CLASS_0 (cont.)

Bits	Name	Description
3:2	REQPRIORITY_1	Remapping ID for priority 1 onto the client port
1:0	REQPRIORITY_0	Remapping ID for priority 0 onto the client port

0x01A013E4 MDSS_MDP_QOS_REMAPPER_LOWER_CLASS_1**Type:** RW**Reset State:** 0x000000E4

QoS remapper to translate health level to priority levels on the bus

Class 0 is used by the real time clients in the system (i.e. ViG/RGB/DMA pipes in line mode)

Class 1 is used by the non-real time clients in the system (i.e. DMA pipes in block mode)

WB modules are completely SW configurable and dont rely on watermarks.

Therefore they do NOT go through the remapper

double buffered: NO

MDSS_MDP_QOS_REMAPPER_LOWER_CLASS_1

Bits	Name	Description
7:6	REQPRIORITY_3	Remapping ID for priority 3 onto the client port
5:4	REQPRIORITY_2	Remapping ID for priority 2 onto the client port
3:2	REQPRIORITY_1	Remapping ID for priority 1 onto the client port
1:0	REQPRIORITY_0	Remapping ID for priority 0 onto the client port

0x01A013F0 MDSS_MDP_SPLIT_DISPLAY_LOWER_PIPE_CTRL**Type:** RW**Reset State:** 0x00000000

Main configuration of split display in mdp lower pipeline. This register are not double buffered.

MDSS_MDP_SPLIT_DISPLAY_LOWER_PIPE_CTRL

Bits	Name	Description
31:16	TE_LINE_INTERVAL_WATERMARK	TE interval interrupt is generated if smart panels for split display are not synchronized as many lines as this watermark value.

MDSS_MDP_SPLIT_DISPLAY_LOWER_PIPE_CTRL (cont.)

Bits	Name	Description
8	INTF_2_TIM_GEN_SYNC_MODE	Control synchronization mode for timing generator in interface 2 0x0 : No sync mode 0x1 : Use INTF_1's enable signal for interface 2
4	INTF_1_TIM_GEN_SYNC_MODE	Control synchronization mode for timing generator in interface 1 0x0 : No SW trigger change 0x1 : Use INTF_2's enable signal for interface 1
2	SMART_PANEL_FREE_RUNNING	Select split display mode for smart panel. This register value should be identical to upper pipe's same name bit before enabling split display. 0x1 : Free running mode between panels 0x0 : Synchronous mode between panels
1	SPLIT_DISPLAY_TYPE	Specify panel type. This register value should be identical to upper pipe's same name bit before enabling split display. 0x1 : Smart Panel 0x0 : Dumb Panel

0x01A013F4 MDSS_MDP_SPLIT_DISPLAY_TE_LINE_INTERVAL**Type:** R**Reset State:** 0x00000000

This indicates how many lines smart panels are not synced for split display

MDSS_MDP_SPLIT_DISPLAY_TE_LINE_INTERVAL

Bits	Name	Description
15:0	VALUE	The number of line interval for smart panel in split display mode

0x01A013F8 MDSS_MDP_LOWER_NEWROI_PRIOR_TO_START**Type:** RW**Reset State:** 0x00000001

The newroi prior to start is used to flush out incorrect states and such between frames. The newroi is sent prior to the start signals from the PMU's. This field is for the PMU's in the lower pipeline only - there exists a separate register for the upper pipeline (UPPER_NEWROI_PRIOR_TO_START)

MDSS_MDP_LOWER_NEWROI_PRIOR_TO_START

Bits	Name	Description
0	ENABLE	Enable the mode to send out newroi prior to start

0x01A013FC MDSS_MDP_INTF_SW_RESET_MASK**Type:** RW**Reset State:** 0x00000000

This register is used to mask the video/cmd mode SW_RESET within the disp_intf.

MDSS_MDP_INTF_SW_RESET_MASK

Bits	Name	Description
1:0	INTF_1_SW_RESET_MASK	Control mask for intf_1 (DSIO) 0x0 : Resets for both video and command modes 0x1 : Reset for video mode only 0x2 : Reset for command mode only 0x3 : No resets to both video and command

17.2.2 MDSS_MDP_CTL_0

MDP Control block

17.2.2.1 MDSS_MDP_CTL_0_CTL

MDP

MDP Control block

0x01A02000 MDSS_MDP_CTL_0_LAYER_0**Type:** RW**Reset State:** 0x00000000

Operation Mode Configuration for Virtual Pipe for the Upper Portion of the Pipeline Connected to LayerMixer0

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_CTL_0_LAYER_0

Bits	Name	Description
25	CURSOR_OUT	Cursor Connection for this virtual pipe The cursor layer will ALWAYS be the top most layer Note that cursor is ONLY applicable in line mode operation and should NOT be used in block mode processing 0x0: Cursor is NOT used 0x1: Cursor is enabled
24	BORDER_OUT	Border Color Connection for this virtual pipe The border color will ALWAYS be the base layer 0x0: Border color NOT used 0x1: Border color enabled
23:21	DMA_1_OUT	DMA1 Connection for this virtual pipe 0x0: DMA1 NOT used at all 0x1: Base Layer for layermixer 0 0x2: Stage0 FG for layermixer 0 0x3: Stage1 FG for layermixer 0 0x4: Stage2 FG for layermixer 0 0x5: Stage3 FG for layermixer 0
20:18	DMA_0_OUT	DMA0 Connection for this virtual pipe 0x0: DMA0 NOT used at all 0x1: Base Layer for layermixer 0 0x2: Stage0 FG for layermixer 0 0x3: Stage1 FG for layermixer 0 0x4: Stage2 FG for layermixer 0 0x5: Stage3 FG for layermixer 0
17:15	RGB_2_OUT	RGB2 Connection for this virtual pipe 0x0: RGB2 NOT used at all 0x1: Base Layer for layermixer 0 0x2: Stage0 FG for layermixer 0 0x3: Stage1 FG for layermixer 0 0x4: Stage2 FG for layermixer 0 0x5: Stage3 FG for layermixer 0
14:12	RGB_1_OUT	RGB1 Connection for this virtual pipe 0x0: RGB1 NOT used at all 0x1: Base Layer for layermixer 0 0x2: Stage0 FG for layermixer 0 0x3: Stage1 FG for layermixer 0 0x4: Stage2 FG for layermixer 0 0x5: Stage3 FG for layermixer 0

MDSS_MDP_CTL_0_LAYER_0 (cont.)

Bits	Name	Description
11:9	RGB_0_OUT	RGB0 Connection for this virtual pipe 0x0: RGB0 NOT used at all 0x1: Base Layer for layermixer 0 0x2: Stage0 FG for layermixer 0 0x3: Stage1 FG for layermixer 0 0x4: Stage2 FG for layermixer 0 0x5: Stage3 FG for layermixer 0
8:6	VIG_2_OUT	VIG2 Connection for this virtual pipe 0x0: VIG2 NOT used at all 0x1: Base Layer for layermixer 0 0x2: Stage0 FG for layermixer 0 0x3: Stage1 FG for layermixer 0 0x4: Stage2 FG for layermixer 0 0x5: Stage3 FG for layermixer 0
5:3	VIG_1_OUT	VIG1 Connection for this virtual pipe 0x0: VIG1 NOT used at all 0x1: Base Layer for layermixer 0 0x2: Stage0 FG for layermixer 0 0x3: Stage1 FG for layermixer 0 0x4: Stage2 FG for layermixer 0 0x5: Stage3 FG for layermixer 0
2:0	VIG_0_OUT	VIG0 Connection for this virtual pipe 0x0: VIG0 NOT used at all 0x1: Base Layer for layermixer 0 0x2: Stage0 FG for layermixer 0 0x3: Stage1 FG for layermixer 0 0x4: Stage2 FG for layermixer 0 0x5: Stage3 FG for layermixer 0

0x01A02004 MDSS_MDP_CTL_0_LAYER_1**Type:** RW**Reset State:** 0x00000000

Operation Mode Configuration for Virtual Pipe for the Upper Portion of the Pipeline Connected to Layer Mixer1

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_CTL_0_LAYER_1

Bits	Name	Description
25	CURSOR_OUT	Cursor Connection for this virtual pipe The cursor layer will ALWAYS be the top most layer Note that cursor is ONLY applicable in line mode operation and should NOT be used in block mode processing 0x0: Cursor is NOT used 0x1: Cursor is enabled
24	BORDER_OUT	Border Color Connection for this virtual pipe The border color will ALWAYS be the base layer 0x0: Border color NOT used 0x1: Border color enabled
23:21	DMA_1_OUT	DMA1 Connection for this virtual pipe 0x0: DMA1 NOT used at all 0x1: Base Layer for layermixer 1 0x2: Stage0 FG for layermixer 1 0x3: Stage1 FG for layermixer 1 0x4: Stage2 FG for layermixer 1 0x5: Stage3 FG for layermixer 1
20:18	DMA_0_OUT	DMA0 Connection for this virtual pipe 0x0: DMA0 NOT used at all 0x1: Base Layer for layermixer 1 0x2: Stage0 FG for layermixer 1 0x3: Stage1 FG for layermixer 1 0x4: Stage2 FG for layermixer 1 0x5: Stage3 FG for layermixer 1
17:15	RGB_2_OUT	RGB2 Connection for this virtual pipe 0x0: RGB2 NOT used at all 0x1: Base Layer for layermixer 1 0x2: Stage0 FG for layermixer 1 0x3: Stage1 FG for layermixer 1 0x4: Stage2 FG for layermixer 1 0x5: Stage3 FG for layermixer 1
14:12	RGB_1_OUT	RGB1 Connection for this virtual pipe 0x0: RGB1 NOT used at all 0x1: Base Layer for layermixer 1 0x2: Stage0 FG for layermixer 1 0x3: Stage1 FG for layermixer 1 0x4: Stage2 FG for layermixer 1 0x5: Stage3 FG for layermixer 1

MDSS_MDP_CTL_0_LAYER_1 (cont.)

Bits	Name	Description
11:9	RGB_0_OUT	RGB0 Connection for this virtual pipe 0x0: RGB0 NOT used at all 0x1: Base Layer for layermixer 1 0x2: Stage0 FG for layermixer 1 0x3: Stage1 FG for layermixer 1 0x4: Stage2 FG for layermixer 1 0x5: Stage3 FG for layermixer 1
8:6	VIG_2_OUT	VIG2 Connection for this virtual pipe 0x0: VIG2 NOT used at all 0x1: Base Layer for layermixer 1 0x2: Stage0 FG for layermixer 1 0x3: Stage1 FG for layermixer 1 0x4: Stage2 FG for layermixer 1 0x5: Stage3 FG for layermixer 1
5:3	VIG_1_OUT	VIG1 Connection for this virtual pipe 0x0: VIG1 NOT used at all 0x1: Base Layer for layermixer 1 0x2: Stage0 FG for layermixer 1 0x3: Stage1 FG for layermixer 1 0x4: Stage2 FG for layermixer 1 0x5: Stage3 FG for layermixer 1
2:0	VIG_0_OUT	VIG0 Connection for this virtual pipe 0x0: VIG0 NOT used at all 0x1: Base Layer for layermixer 1 0x2: Stage0 FG for layermixer 1 0x3: Stage1 FG for layermixer 1 0x4: Stage2 FG for layermixer 1 0x5: Stage3 FG for layermixer 1

0x01A02008 MDSS_MDP_CTL_0_LAYER_2**Type:** RW**Reset State:** 0x00000000

Operation Mode Configuration for Virtual Pipe for the Upper Portion of the Pipeline Connected to Layer Mixer2

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_CTL_0_LAYER_2

Bits	Name	Description
25	CURSOR_OUT	Cursor Connection for this virtual pipe The cursor layer will ALWAYS be the top most layer Note that cursor is ONLY applicable in line mode operation and should NOT be used in block mode processing 0x0: Cursor is NOT used 0x1: Cursor is enabled
24	BORDER_OUT	Border Color Connection for this virtual pipe The border color will ALWAYS be the base layer 0x0: Border color NOT used 0x1: Border color enabled
23:21	DMA_1_OUT	DMA1 Connection for this virtual pipe 0x0: DMA1 NOT used at all 0x1: Base Layer for layermixer 2 0x2: Stage0 FG for layermixer 2 0x3: Stage1 FG for layermixer 2 0x4: Stage2 FG for layermixer 2 0x5: Stage3 FG for layermixer 2
20:18	DMA_0_OUT	DMA0 Connection for this virtual pipe 0x0: DMA0 NOT used at all 0x1: Base Layer for layermixer 2 0x2: Stage0 FG for layermixer 2 0x3: Stage1 FG for layermixer 2 0x4: Stage2 FG for layermixer 2 0x5: Stage3 FG for layermixer 2
17:15	RGB_2_OUT	RGB2 Connection for this virtual pipe 0x0: RGB2 NOT used at all 0x1: Base Layer for layermixer 2 0x2: Stage0 FG for layermixer 2 0x3: Stage1 FG for layermixer 2 0x4: Stage2 FG for layermixer 2 0x5: Stage3 FG for layermixer 2
14:12	RGB_1_OUT	RGB1 Connection for this virtual pipe 0x0: RGB1 NOT used at all 0x1: Base Layer for layermixer 2 0x2: Stage0 FG for layermixer 2 0x3: Stage1 FG for layermixer 2 0x4: Stage2 FG for layermixer 2 0x5: Stage3 FG for layermixer 2

MDSS_MDP_CTL_0_LAYER_2 (cont.)

Bits	Name	Description
11:9	RGB_0_OUT	RGB0 Connection for this virtual pipe 0x0: RGB0 NOT used at all 0x1: Base Layer for layermixer 2 0x2: Stage0 FG for layermixer 2 0x3: Stage1 FG for layermixer 2 0x4: Stage2 FG for layermixer 2 0x5: Stage3 FG for layermixer 2
8:6	VIG_2_OUT	VIG2 Connection for this virtual pipe 0x0: VIG2 NOT used at all 0x1: Base Layer for layermixer 2 0x2: Stage0 FG for layermixer 2 0x3: Stage1 FG for layermixer 2 0x4: Stage2 FG for layermixer 2 0x5: Stage3 FG for layermixer 2
5:3	VIG_1_OUT	VIG1 Connection for this virtual pipe 0x0: VIG1 NOT used at all 0x1: Base Layer for layermixer 2 0x2: Stage0 FG for layermixer 2 0x3: Stage1 FG for layermixer 2 0x4: Stage2 FG for layermixer 2 0x5: Stage3 FG for layermixer 2
2:0	VIG_0_OUT	VIG0 Connection for this virtual pipe 0x0: VIG0 NOT used at all 0x1: Base Layer for layermixer 2 0x2: Stage0 FG for layermixer 2 0x3: Stage1 FG for layermixer 2 0x4: Stage2 FG for layermixer 2 0x5: Stage3 FG for layermixer 2

0x01A0200C MDSS_MDP_CTL_0_LAYER_3**Type:** RW**Reset State:** 0x00000000

Operation Mode Configuration for Virtual Pipe for the Upper Portion of the Pipeline Connected to Layer Mixer3

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_CTL_0_LAYER_3

Bits	Name	Description
25	CURSOR_OUT	Cursor Connection for this virtual pipe The cursor layer will ALWAYS be the top most layer Note that cursor is ONLY applicable in line mode operation and should NOT be used in block mode processing 0x0: Cursor is NOT used 0x1: Cursor is enabled
24	BORDER_OUT	Border Color Connection for this virtual pipe The border color will ALWAYS be the base layer 0x0: Border color NOT used 0x1: Border color enabled
23:21	DMA_1_OUT	DMA1 Connection for this virtual pipe 0x0: DMA1 NOT used at all 0x1: Base Layer for layermixer 3 0x2: Stage0 FG for layermixer 3 0x3: Stage1 FG for layermixer 3 0x4: Stage2 FG for layermixer 3 0x5: Stage3 FG for layermixer 3
20:18	DMA_0_OUT	DMA0 Connection for this virtual pipe 0x0: DMA0 NOT used at all 0x1: Base Layer for layermixer 3 0x2: Stage0 FG for layermixer 3 0x3: Stage1 FG for layermixer 3 0x4: Stage2 FG for layermixer 3 0x5: Stage3 FG for layermixer 3
17:15	RGB_2_OUT	RGB2 Connection for this virtual pipe 0x0: RGB2 NOT used at all 0x1: Base Layer for layermixer 3 0x2: Stage0 FG for layermixer 3 0x3: Stage1 FG for layermixer 3 0x4: Stage2 FG for layermixer 3 0x5: Stage3 FG for layermixer 3
14:12	RGB_1_OUT	RGB1 Connection for this virtual pipe 0x0: RGB1 NOT used at all 0x1: Base Layer for layermixer 3 0x2: Stage0 FG for layermixer 3 0x3: Stage1 FG for layermixer 3 0x4: Stage2 FG for layermixer 3 0x5: Stage3 FG for layermixer 3

MDSS_MDP_CTL_0_LAYER_3 (cont.)

Bits	Name	Description
11:9	RGB_0_OUT	RGB0 Connection for this virtual pipe 0x0: RGB0 NOT used at all 0x1: Base Layer for layermixer 3 0x2: Stage0 FG for layermixer 3 0x3: Stage1 FG for layermixer 3 0x4: Stage2 FG for layermixer 3 0x5: Stage3 FG for layermixer 3
8:6	VIG_2_OUT	VIG2 Connection for this virtual pipe 0x0: VIG2 NOT used at all 0x1: Base Layer for layermixer 3 0x2: Stage0 FG for layermixer 3 0x3: Stage1 FG for layermixer 3 0x4: Stage2 FG for layermixer 3 0x5: Stage3 FG for layermixer 3
5:3	VIG_1_OUT	VIG1 Connection for this virtual pipe 0x0: VIG1 NOT used at all 0x1: Base Layer for layermixer 3 0x2: Stage0 FG for layermixer 3 0x3: Stage1 FG for layermixer 3 0x4: Stage2 FG for layermixer 3 0x5: Stage3 FG for layermixer 3
2:0	VIG_0_OUT	VIG0 Connection for this virtual pipe 0x0: VIG0 NOT used at all 0x1: Base Layer for layermixer 3 0x2: Stage0 FG for layermixer 3 0x3: Stage1 FG for layermixer 3 0x4: Stage2 FG for layermixer 3 0x5: Stage3 FG for layermixer 3

0x01A02010 MDSS_MDP_CTL_0_LAYER_4**Type:** RW**Reset State:** 0x00000000

Operation Mode Configuration for Virtual Pipe for the Upper Portion of the Pipeline Connected to Layer Mixer4

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_CTL_0_LAYER_4

Bits	Name	Description
25	CURSOR_OUT	Cursor Connection for this virtual pipe The cursor layer will ALWAYS be the top most layer Note that cursor is ONLY applicable in line mode operation and should NOT be used in block mode processing 0x0: Cursor is NOT used 0x1: Cursor is enabled
24	BORDER_OUT	Border Color Connection for this virtual pipe The border color will ALWAYS be the base layer 0x0: Border color NOT used 0x1: Border color enabled
23:21	DMA_1_OUT	DMA1 Connection for this virtual pipe 0x0: DMA1 NOT used at all 0x1: Base Layer for layermixer 4 0x2: Stage0 FG for layermixer 4 0x3: Stage1 FG for layermixer 4 0x4: Stage2 FG for layermixer 4 0x5: Stage3 FG for layermixer 4
20:18	DMA_0_OUT	DMA0 Connection for this virtual pipe 0x0: DMA0 NOT used at all 0x1: Base Layer for layermixer 4 0x2: Stage0 FG for layermixer 4 0x3: Stage1 FG for layermixer 4 0x4: Stage2 FG for layermixer 4 0x5: Stage3 FG for layermixer 4
17:15	RGB_2_OUT	RGB2 Connection for this virtual pipe 0x0: RGB2 NOT used at all 0x1: Base Layer for layermixer 4 0x2: Stage0 FG for layermixer 4 0x3: Stage1 FG for layermixer 4 0x4: Stage2 FG for layermixer 4 0x5: Stage3 FG for layermixer 4
14:12	RGB_1_OUT	RGB1 Connection for this virtual pipe 0x0: RGB1 NOT used at all 0x1: Base Layer for layermixer 4 0x2: Stage0 FG for layermixer 4 0x3: Stage1 FG for layermixer 4 0x4: Stage2 FG for layermixer 4 0x5: Stage3 FG for layermixer 4

MDSS_MDP_CTL_0_LAYER_4 (cont.)

Bits	Name	Description
11:9	RGB_0_OUT	RGB0 Connection for this virtual pipe 0x0: RGB0 NOT used at all 0x1: Base Layer for layermixer 4 0x2: Stage0 FG for layermixer 4 0x3: Stage1 FG for layermixer 4 0x4: Stage2 FG for layermixer 4 0x5: Stage3 FG for layermixer 4
8:6	VIG_2_OUT	VIG2 Connection for this virtual pipe 0x0: VIG2 NOT used at all 0x1: Base Layer for layermixer 4 0x2: Stage0 FG for layermixer 4 0x3: Stage1 FG for layermixer 4 0x4: Stage2 FG for layermixer 4 0x5: Stage3 FG for layermixer 4
5:3	VIG_1_OUT	VIG1 Connection for this virtual pipe 0x0: VIG1 NOT used at all 0x1: Base Layer for layermixer 4 0x2: Stage0 FG for layermixer 4 0x3: Stage1 FG for layermixer 4 0x4: Stage2 FG for layermixer 4 0x5: Stage3 FG for layermixer 4
2:0	VIG_0_OUT	VIG0 Connection for this virtual pipe 0x0: VIG0 NOT used at all 0x1: Base Layer for layermixer 4 0x2: Stage0 FG for layermixer 4 0x3: Stage1 FG for layermixer 4 0x4: Stage2 FG for layermixer 4 0x5: Stage3 FG for layermixer 4

0x01A02014 MDSS_MDP_CTL_0_TOP**Type:** RW**Reset State:** 0x00001F00**Operation Mode Configuration for the Pipeline**

The purpose of the register is to define the complete data path of this operation and that includes all the way to the interfaces such as DSI/HDMI/eDP

Changing of the data path dynamically is NOT supported and requires stopping that data path, reconfiguring the connections, and restarting the datapath

MDSS_MDP_CTL_0_TOP

Bits	Name	Description
17	MODE_SEL	Interface Mode of the destination interface Needed to distinguish interfaces such as DSI that can be either mode DSI can also interleave streams of different types (i.e. command and video) to a single core 0x0: Video Mode 0x1: Command Mode1
16:15	STREAM_SEL	Stream Identification Used For Interleaving of Command Streams This field is a dont care for video mode streams 0x0: Stream 0 0x1: Stream 1 0x2: Stream 2
7:4	OUT_SEL	Output Ping-Pong Interface Connection to external display interface 0x0: No connection from Ping-Pong buffers to output interfaces 0x1: Interface 0 0x2: Interface 1 0x3: Interface 2 0x4: Interface 3
3:0	MEM_SEL	Memory Output Connection 0x0: NO memory output selected 0x1: Memory BLOCK mode output selected for rotation 0 0x2: Memory BLOCK mode output selected for rotation 1 0x3: Memory LINE mode output selected for rotation 0 0x4: Memory LINE mode output selected for rotation 1 0x5: Memory LINE mode output selected for linear memory writeback

0x01A02018 MDSS_MDP_CTL_0_FLUSH**Type:** RW**Reset State:** 0x00000000

Register Flush register

The flush register is used to indicate a pipe/layermixers registers are all programmed, and are safe to update to the back copy of the double buffered registers.

Each pipes/layermixers flush mask bit will be used only when the associated pipe/layermixer is running at active refresh pull mode, e.g., direct out going to LCDC/HDMI/DSI Video.

In such operation mode, the associated pipe/layermixers double buffered registers back copies are updated with the new values at the vsync boundary only when the flush mask bit has already been written a 1.

Each flush mask bit would also returned a 0 once the back copies are updated.

Before the back copies are updated, the associate mask bit would returned a 1 if it has been previously written a 1.

MDSS_MDP_CTL_0_FLUSH

Bits	Name	Description
30	TIMING_1	Timing Interface 1 are ready for flush
17	CTL	Control Pipe register is ready for flush This applies to upper pipe ONLY to change the adding/removal of layers
16	WB	Write Back registers are ready for flush
15	DSPP_2	DSPP2 registers are ready for flush
14	DSPP_1	DSPP1 registers are ready for flush
13	DSPP_0	DSPP0 registers are ready for flush
12	DMA_1	DMA_1 registers are ready for flush
11	DMA_0	DMA_0 registers are ready for flush
10	LAYER_MIXER_4	Layer Mixer4 registers are ready for flush This includes the cursor registers stored in the layer mixer register space
9	LAYER_MIXER_3	Layer Mixer3 registers are ready for flush This includes the cursor registers stored in the layer mixer register space
8	LAYER_MIXER_2	Layer Mixer2 registers are ready for flush This includes the cursor registers stored in the layer mixer register space
7	LAYER_MIXER_1	Layer Mixer1 registers are ready for flush This includes the cursor registers stored in the layer mixer register space
6	LAYER_MIXER_0	Layer Mixer0 registers are ready for flush This includes the cursor registers stored in the layer mixer register space
5	RGB_2	RGB2 registers are ready for flush
4	RGB_1	RGB1 registers are ready for flush
3	RGB_0	RGB0 registers are ready for flush

MDSS_MDP_CTL_0_FLUSH (cont.)

Bits	Name	Description
2	VIG_2	VIG2 registers are ready for flush
1	VIG_1	VIG1 registers are ready for flush
0	VIG_0	VIG0 registers are ready for flush

0x01A0201C MDSS_MDP_CTL_0_CTL_START**Type:** W**Reset State:** 0x00000000

Overlay Processor Start Signal for the given Control Operation

MDSS_MDP_CTL_0_CTL_START

Bits	Name	Description
0	START	A write to this register will kick off operation for the given operation control and activate all layers

0x01A02030 MDSS_MDP_CTL_0_SW_RESET**Type:** RW**Reset State:** 0x00000000

Software Reset of the given control operation

MDSS_MDP_CTL_0_SW_RESET

Bits	Name	Description
0	RESET	Writing to this bit will trigger a soft reset for the described data path This register is self-clearing and will return 0x1 if soft reset sequence is not complete Upon completion, this will return 0x0 Note that this will soft reset the peripheral display interfaces that are described in this path i.e. if driving intf_0 it will reset the pixel clock domain of intf_0 as well

0x01A02034 MDSS_MDP_CTL_0_SW_RESET_OVERRIDE**Type:** RW**Reset State:** 0x00000000

Software Reset Override of the given control operation

MDSS_MDP_CTL_0_SW_RESET_OVERRIDE

Bits	Name	Description
0	RESET	Writing to this bit will assert reset for the described data path This value contained in this register will be forced onto the datapath It will reset the state machines within the scheduler as well for the given datapath To use this register sequence, SW shall write a value of 0x1 to this register to assert reset Once SW wants to de-assert reset, SW shall write a value of 0x0 SW will need to ensure the duration of the value 0x1 is long enough to ensure the path is clean and all memory reads are complete The datapath will not be triggered if this bit is asserted

17.2.3 MDSS_MDP_CTL_1

MDP Control block

17.2.3.1 MDSS_MDP_CTL_1_CTL

MDP

MDP Control block

0x01A02200 MDSS_MDP_CTL_1_LAYER_0

Type: RW

Reset State: 0x00000000

Operation Mode Configuration for Virtual Pipe for the Upper Portion of the Pipeline Connected to LayerMixer0

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_CTL_1_LAYER_0

Bits	Name	Description
25	CURSOR_OUT	Cursor Connection for this virtual pipe The cursor layer will ALWAYS be the top most layer Note that cursor is ONLY applicable in line mode operation and should NOT be used in block mode processing 0x0: Cursor is NOT used 0x1: Cursor is enabled
24	BORDER_OUT	Border Color Connection for this virtual pipe The border color will ALWAYS be the base layer 0x0: Border color NOT used 0x1: Border color enabled
23:21	DMA_1_OUT	DMA1 Connection for this virtual pipe 0x0: DMA1 NOT used at all 0x1: Base Layer for layermixer 0 0x2: Stage0 FG for layermixer 0 0x3: Stage1 FG for layermixer 0 0x4: Stage2 FG for layermixer 0 0x5: Stage3 FG for layermixer 0
20:18	DMA_0_OUT	DMA0 Connection for this virtual pipe 0x0: DMA0 NOT used at all 0x1: Base Layer for layermixer 0 0x2: Stage0 FG for layermixer 0 0x3: Stage1 FG for layermixer 0 0x4: Stage2 FG for layermixer 0 0x5: Stage3 FG for layermixer 0
17:15	RGB_2_OUT	RGB2 Connection for this virtual pipe 0x0: RGB2 NOT used at all 0x1: Base Layer for layermixer 0 0x2: Stage0 FG for layermixer 0 0x3: Stage1 FG for layermixer 0 0x4: Stage2 FG for layermixer 0 0x5: Stage3 FG for layermixer 0
14:12	RGB_1_OUT	RGB1 Connection for this virtual pipe 0x0: RGB1 NOT used at all 0x1: Base Layer for layermixer 0 0x2: Stage0 FG for layermixer 0 0x3: Stage1 FG for layermixer 0 0x4: Stage2 FG for layermixer 0 0x5: Stage3 FG for layermixer 0

MDSS_MDP_CTL_1_LAYER_0 (cont.)

Bits	Name	Description
11:9	RGB_0_OUT	RGB0 Connection for this virtual pipe 0x0: RGB0 NOT used at all 0x1: Base Layer for layermixer 0 0x2: Stage0 FG for layermixer 0 0x3: Stage1 FG for layermixer 0 0x4: Stage2 FG for layermixer 0 0x5: Stage3 FG for layermixer 0
8:6	VIG_2_OUT	VIG2 Connection for this virtual pipe 0x0: VIG2 NOT used at all 0x1: Base Layer for layermixer 0 0x2: Stage0 FG for layermixer 0 0x3: Stage1 FG for layermixer 0 0x4: Stage2 FG for layermixer 0 0x5: Stage3 FG for layermixer 0
5:3	VIG_1_OUT	VIG1 Connection for this virtual pipe 0x0: VIG1 NOT used at all 0x1: Base Layer for layermixer 0 0x2: Stage0 FG for layermixer 0 0x3: Stage1 FG for layermixer 0 0x4: Stage2 FG for layermixer 0 0x5: Stage3 FG for layermixer 0
2:0	VIG_0_OUT	VIG0 Connection for this virtual pipe 0x0: VIG0 NOT used at all 0x1: Base Layer for layermixer 0 0x2: Stage0 FG for layermixer 0 0x3: Stage1 FG for layermixer 0 0x4: Stage2 FG for layermixer 0 0x5: Stage3 FG for layermixer 0

0x01A02204 MDSS_MDP_CTL_1_LAYER_1**Type:** RW**Reset State:** 0x00000000

Operation Mode Configuration for Virtual Pipe for the Upper Portion of the Pipeline Connected to Layer Mixer1

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_CTL_1_LAYER_1

Bits	Name	Description
25	CURSOR_OUT	Cursor Connection for this virtual pipe The cursor layer will ALWAYS be the top most layer Note that cursor is ONLY applicable in line mode operation and should NOT be used in block mode processing 0x0: Cursor is NOT used 0x1: Cursor is enabled
24	BORDER_OUT	Border Color Connection for this virtual pipe The border color will ALWAYS be the base layer 0x0: Border color NOT used 0x1: Border color enabled
23:21	DMA_1_OUT	DMA1 Connection for this virtual pipe 0x0: DMA1 NOT used at all 0x1: Base Layer for layermixer 1 0x2: Stage0 FG for layermixer 1 0x3: Stage1 FG for layermixer 1 0x4: Stage2 FG for layermixer 1 0x5: Stage3 FG for layermixer 1
20:18	DMA_0_OUT	DMA0 Connection for this virtual pipe 0x0: DMA0 NOT used at all 0x1: Base Layer for layermixer 1 0x2: Stage0 FG for layermixer 1 0x3: Stage1 FG for layermixer 1 0x4: Stage2 FG for layermixer 1 0x5: Stage3 FG for layermixer 1
17:15	RGB_2_OUT	RGB2 Connection for this virtual pipe 0x0: RGB2 NOT used at all 0x1: Base Layer for layermixer 1 0x2: Stage0 FG for layermixer 1 0x3: Stage1 FG for layermixer 1 0x4: Stage2 FG for layermixer 1 0x5: Stage3 FG for layermixer 1
14:12	RGB_1_OUT	RGB1 Connection for this virtual pipe 0x0: RGB1 NOT used at all 0x1: Base Layer for layermixer 1 0x2: Stage0 FG for layermixer 1 0x3: Stage1 FG for layermixer 1 0x4: Stage2 FG for layermixer 1 0x5: Stage3 FG for layermixer 1

MDSS_MDP_CTL_1_LAYER_1 (cont.)

Bits	Name	Description
11:9	RGB_0_OUT	RGB0 Connection for this virtual pipe 0x0: RGB0 NOT used at all 0x1: Base Layer for layermixer 1 0x2: Stage0 FG for layermixer 1 0x3: Stage1 FG for layermixer 1 0x4: Stage2 FG for layermixer 1 0x5: Stage3 FG for layermixer 1
8:6	VIG_2_OUT	VIG2 Connection for this virtual pipe 0x0: VIG2 NOT used at all 0x1: Base Layer for layermixer 1 0x2: Stage0 FG for layermixer 1 0x3: Stage1 FG for layermixer 1 0x4: Stage2 FG for layermixer 1 0x5: Stage3 FG for layermixer 1
5:3	VIG_1_OUT	VIG1 Connection for this virtual pipe 0x0: VIG1 NOT used at all 0x1: Base Layer for layermixer 1 0x2: Stage0 FG for layermixer 1 0x3: Stage1 FG for layermixer 1 0x4: Stage2 FG for layermixer 1 0x5: Stage3 FG for layermixer 1
2:0	VIG_0_OUT	VIG0 Connection for this virtual pipe 0x0: VIG0 NOT used at all 0x1: Base Layer for layermixer 1 0x2: Stage0 FG for layermixer 1 0x3: Stage1 FG for layermixer 1 0x4: Stage2 FG for layermixer 1 0x5: Stage3 FG for layermixer 1

0x01A02208 MDSS_MDP_CTL_1_LAYER_2**Type:** RW**Reset State:** 0x00000000

Operation Mode Configuration for Virtual Pipe for the Upper Portion of the Pipeline Connected to Layer Mixer2

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_CTL_1_LAYER_2

Bits	Name	Description
25	CURSOR_OUT	Cursor Connection for this virtual pipe The cursor layer will ALWAYS be the top most layer Note that cursor is ONLY applicable in line mode operation and should NOT be used in block mode processing 0x0: Cursor is NOT used 0x1: Cursor is enabled
24	BORDER_OUT	Border Color Connection for this virtual pipe The border color will ALWAYS be the base layer 0x0: Border color NOT used 0x1: Border color enabled
23:21	DMA_1_OUT	DMA1 Connection for this virtual pipe 0x0: DMA1 NOT used at all 0x1: Base Layer for layermixer 2 0x2: Stage0 FG for layermixer 2 0x3: Stage1 FG for layermixer 2 0x4: Stage2 FG for layermixer 2 0x5: Stage3 FG for layermixer 2
20:18	DMA_0_OUT	DMA0 Connection for this virtual pipe 0x0: DMA0 NOT used at all 0x1: Base Layer for layermixer 2 0x2: Stage0 FG for layermixer 2 0x3: Stage1 FG for layermixer 2 0x4: Stage2 FG for layermixer 2 0x5: Stage3 FG for layermixer 2
17:15	RGB_2_OUT	RGB2 Connection for this virtual pipe 0x0: RGB2 NOT used at all 0x1: Base Layer for layermixer 2 0x2: Stage0 FG for layermixer 2 0x3: Stage1 FG for layermixer 2 0x4: Stage2 FG for layermixer 2 0x5: Stage3 FG for layermixer 2
14:12	RGB_1_OUT	RGB1 Connection for this virtual pipe 0x0: RGB1 NOT used at all 0x1: Base Layer for layermixer 2 0x2: Stage0 FG for layermixer 2 0x3: Stage1 FG for layermixer 2 0x4: Stage2 FG for layermixer 2 0x5: Stage3 FG for layermixer 2

MDSS_MDP_CTL_1_LAYER_2 (cont.)

Bits	Name	Description
11:9	RGB_0_OUT	RGB0 Connection for this virtual pipe 0x0: RGB0 NOT used at all 0x1: Base Layer for layermixer 2 0x2: Stage0 FG for layermixer 2 0x3: Stage1 FG for layermixer 2 0x4: Stage2 FG for layermixer 2 0x5: Stage3 FG for layermixer 2
8:6	VIG_2_OUT	VIG2 Connection for this virtual pipe 0x0: VIG2 NOT used at all 0x1: Base Layer for layermixer 2 0x2: Stage0 FG for layermixer 2 0x3: Stage1 FG for layermixer 2 0x4: Stage2 FG for layermixer 2 0x5: Stage3 FG for layermixer 2
5:3	VIG_1_OUT	VIG1 Connection for this virtual pipe 0x0: VIG1 NOT used at all 0x1: Base Layer for layermixer 2 0x2: Stage0 FG for layermixer 2 0x3: Stage1 FG for layermixer 2 0x4: Stage2 FG for layermixer 2 0x5: Stage3 FG for layermixer 2
2:0	VIG_0_OUT	VIG0 Connection for this virtual pipe 0x0: VIG0 NOT used at all 0x1: Base Layer for layermixer 2 0x2: Stage0 FG for layermixer 2 0x3: Stage1 FG for layermixer 2 0x4: Stage2 FG for layermixer 2 0x5: Stage3 FG for layermixer 2

0x01A0220C MDSS_MDP_CTL_1_LAYER_3**Type:** RW**Reset State:** 0x00000000

Operation Mode Configuration for Virtual Pipe for the Upper Portion of the Pipeline Connected to Layer Mixer3

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_CTL_1_LAYER_3

Bits	Name	Description
25	CURSOR_OUT	Cursor Connection for this virtual pipe The cursor layer will ALWAYS be the top most layer Note that cursor is ONLY applicable in line mode operation and should NOT be used in block mode processing 0x0: Cursor is NOT used 0x1: Cursor is enabled
24	BORDER_OUT	Border Color Connection for this virtual pipe The border color will ALWAYS be the base layer 0x0: Border color NOT used 0x1: Border color enabled
23:21	DMA_1_OUT	DMA1 Connection for this virtual pipe 0x0: DMA1 NOT used at all 0x1: Base Layer for layermixer 3 0x2: Stage0 FG for layermixer 3 0x3: Stage1 FG for layermixer 3 0x4: Stage2 FG for layermixer 3 0x5: Stage3 FG for layermixer 3
20:18	DMA_0_OUT	DMA0 Connection for this virtual pipe 0x0: DMA0 NOT used at all 0x1: Base Layer for layermixer 3 0x2: Stage0 FG for layermixer 3 0x3: Stage1 FG for layermixer 3 0x4: Stage2 FG for layermixer 3 0x5: Stage3 FG for layermixer 3
17:15	RGB_2_OUT	RGB2 Connection for this virtual pipe 0x0: RGB2 NOT used at all 0x1: Base Layer for layermixer 3 0x2: Stage0 FG for layermixer 3 0x3: Stage1 FG for layermixer 3 0x4: Stage2 FG for layermixer 3 0x5: Stage3 FG for layermixer 3
14:12	RGB_1_OUT	RGB1 Connection for this virtual pipe 0x0: RGB1 NOT used at all 0x1: Base Layer for layermixer 3 0x2: Stage0 FG for layermixer 3 0x3: Stage1 FG for layermixer 3 0x4: Stage2 FG for layermixer 3 0x5: Stage3 FG for layermixer 3

MDSS_MDP_CTL_1_LAYER_3 (cont.)

Bits	Name	Description
11:9	RGB_0_OUT	RGB0 Connection for this virtual pipe 0x0: RGB0 NOT used at all 0x1: Base Layer for layermixer 3 0x2: Stage0 FG for layermixer 3 0x3: Stage1 FG for layermixer 3 0x4: Stage2 FG for layermixer 3 0x5: Stage3 FG for layermixer 3
8:6	VIG_2_OUT	VIG2 Connection for this virtual pipe 0x0: VIG2 NOT used at all 0x1: Base Layer for layermixer 3 0x2: Stage0 FG for layermixer 3 0x3: Stage1 FG for layermixer 3 0x4: Stage2 FG for layermixer 3 0x5: Stage3 FG for layermixer 3
5:3	VIG_1_OUT	VIG1 Connection for this virtual pipe 0x0: VIG1 NOT used at all 0x1: Base Layer for layermixer 3 0x2: Stage0 FG for layermixer 3 0x3: Stage1 FG for layermixer 3 0x4: Stage2 FG for layermixer 3 0x5: Stage3 FG for layermixer 3
2:0	VIG_0_OUT	VIG0 Connection for this virtual pipe 0x0: VIG0 NOT used at all 0x1: Base Layer for layermixer 3 0x2: Stage0 FG for layermixer 3 0x3: Stage1 FG for layermixer 3 0x4: Stage2 FG for layermixer 3 0x5: Stage3 FG for layermixer 3

0x01A02210 MDSS_MDP_CTL_1_LAYER_4**Type:** RW**Reset State:** 0x00000000

Operation Mode Configuration for Virtual Pipe for the Upper Portion of the Pipeline Connected to Layer Mixer4

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_CTL_1_LAYER_4

Bits	Name	Description
25	CURSOR_OUT	Cursor Connection for this virtual pipe The cursor layer will ALWAYS be the top most layer Note that cursor is ONLY applicable in line mode operation and should NOT be used in block mode processing 0x0: Cursor is NOT used 0x1: Cursor is enabled
24	BORDER_OUT	Border Color Connection for this virtual pipe The border color will ALWAYS be the base layer 0x0: Border color NOT used 0x1: Border color enabled
23:21	DMA_1_OUT	DMA1 Connection for this virtual pipe 0x0: DMA1 NOT used at all 0x1: Base Layer for layermixer 4 0x2: Stage0 FG for layermixer 4 0x3: Stage1 FG for layermixer 4 0x4: Stage2 FG for layermixer 4 0x5: Stage3 FG for layermixer 4
20:18	DMA_0_OUT	DMA0 Connection for this virtual pipe 0x0: DMA0 NOT used at all 0x1: Base Layer for layermixer 4 0x2: Stage0 FG for layermixer 4 0x3: Stage1 FG for layermixer 4 0x4: Stage2 FG for layermixer 4 0x5: Stage3 FG for layermixer 4
17:15	RGB_2_OUT	RGB2 Connection for this virtual pipe 0x0: RGB2 NOT used at all 0x1: Base Layer for layermixer 4 0x2: Stage0 FG for layermixer 4 0x3: Stage1 FG for layermixer 4 0x4: Stage2 FG for layermixer 4 0x5: Stage3 FG for layermixer 4
14:12	RGB_1_OUT	RGB1 Connection for this virtual pipe 0x0: RGB1 NOT used at all 0x1: Base Layer for layermixer 4 0x2: Stage0 FG for layermixer 4 0x3: Stage1 FG for layermixer 4 0x4: Stage2 FG for layermixer 4 0x5: Stage3 FG for layermixer 4

MDSS_MDP_CTL_1_LAYER_4 (cont.)

Bits	Name	Description
11:9	RGB_0_OUT	RGB0 Connection for this virtual pipe 0x0: RGB0 NOT used at all 0x1: Base Layer for layermixer 4 0x2: Stage0 FG for layermixer 4 0x3: Stage1 FG for layermixer 4 0x4: Stage2 FG for layermixer 4 0x5: Stage3 FG for layermixer 4
8:6	VIG_2_OUT	VIG2 Connection for this virtual pipe 0x0: VIG2 NOT used at all 0x1: Base Layer for layermixer 4 0x2: Stage0 FG for layermixer 4 0x3: Stage1 FG for layermixer 4 0x4: Stage2 FG for layermixer 4 0x5: Stage3 FG for layermixer 4
5:3	VIG_1_OUT	VIG1 Connection for this virtual pipe 0x0: VIG1 NOT used at all 0x1: Base Layer for layermixer 4 0x2: Stage0 FG for layermixer 4 0x3: Stage1 FG for layermixer 4 0x4: Stage2 FG for layermixer 4 0x5: Stage3 FG for layermixer 4
2:0	VIG_0_OUT	VIG0 Connection for this virtual pipe 0x0: VIG0 NOT used at all 0x1: Base Layer for layermixer 4 0x2: Stage0 FG for layermixer 4 0x3: Stage1 FG for layermixer 4 0x4: Stage2 FG for layermixer 4 0x5: Stage3 FG for layermixer 4

0x01A02214 MDSS_MDP_CTL_1_TOP**Type:** RW**Reset State:** 0x00001F00**Operation Mode Configuration for the Pipeline**

The purpose of the register is to define the complete data path of this operation and that includes all the way to the interfaces such as DSI/HDMI/eDP

Changing of the data path dynamically is NOT supported and requires stopping that data path, reconfiguring the connections, and restarting the datapath

MDSS_MDP_CTL_1_TOP

Bits	Name	Description
17	MODE_SEL	Interface Mode of the destination interface Needed to distinguish interfaces such as DSI that can be either mode DSI can also interleave streams of different types (i.e. command and video) to a single core 0x0: Video Mode 0x1: Command Mode1
16:15	STREAM_SEL	Stream Identification Used For Interleaving of Command Streams This field is a dont care for video mode streams 0x0: Stream 0 0x1: Stream 1 0x2: Stream 2
7:4	OUT_SEL	Output Ping-Pong Interface Connection to external display interface 0x0: No connection from Ping-Pong buffers to output interfaces 0x1: Interface 0 0x2: Interface 1 0x3: Interface 2 0x4: Interface 3
3:0	MEM_SEL	Memory Output Connection 0x0: NO memory output selected 0x1: Memory BLOCK mode output selected for rotation 0 0x2: Memory BLOCK mode output selected for rotation 1 0x3: Memory LINE mode output selected for rotation 0 0x4: Memory LINE mode output selected for rotation 1 0x5: Memory LINE mode output selected for linear memory writeback

0x01A02218 MDSS_MDP_CTL_1_FLUSH**Type:** RW**Reset State:** 0x00000000

Register Flush register

The flush register is used to indicate a pipe/layermixers registers are all programmed, and are safe to update to the back copy of the double buffered registers.

Each pipes/layermixers flush mask bit will be used only when the associated pipe/layermixer is running at active refresh pull mode, e.g., direct out going to LCDC/HDMI/DSI Video.

In such operation mode, the associated pipe/layermixers double buffered registers back copies are updated with the new values at the vsync boundary only when the flush mask bit has already been written a 1.

Each flush mask bit would also returned a 0 once the back copies are updated.

Before the back copies are updated, the associate mask bit would returned a 1 if it has been previously written a 1.

MDSS_MDP_CTL_1_FLUSH

Bits	Name	Description
30	TIMING_1	Timing Interface 1 are ready for flush
17	CTL	Control Pipe register is ready for flush This applies to upper pipe ONLY to change the adding/removal of layers
16	WB	Write Back registers are ready for flush
15	DSPP_2	DSPP2 registers are ready for flush
14	DSPP_1	DSPP1 registers are ready for flush
13	DSPP_0	DSPP0 registers are ready for flush
12	DMA_1	DMA_1 registers are ready for flush
11	DMA_0	DMA_0 registers are ready for flush
10	LAYER_MIXER_4	Layer Mixer4 registers are ready for flush This includes the cursor registers stored in the layer mixer register space
9	LAYER_MIXER_3	Layer Mixer3 registers are ready for flush This includes the cursor registers stored in the layer mixer register space
8	LAYER_MIXER_2	Layer Mixer2 registers are ready for flush This includes the cursor registers stored in the layer mixer register space
7	LAYER_MIXER_1	Layer Mixer1 registers are ready for flush This includes the cursor registers stored in the layer mixer register space
6	LAYER_MIXER_0	Layer Mixer0 registers are ready for flush This includes the cursor registers stored in the layer mixer register space
5	RGB_2	RGB2 registers are ready for flush
4	RGB_1	RGB1 registers are ready for flush
3	RGB_0	RGB0 registers are ready for flush

MDSS_MDP_CTL_1_FLUSH (cont.)

Bits	Name	Description
2	VIG_2	VIG2 registers are ready for flush
1	VIG_1	VIG1 registers are ready for flush
0	VIG_0	VIG0 registers are ready for flush

0x01A0221C MDSS_MDP_CTL_1_CTL_START**Type:** W**Reset State:** 0x00000000

Overlay Processor Start Signal for the given Control Operation

MDSS_MDP_CTL_1_CTL_START

Bits	Name	Description
0	START	A write to this register will kick off operation for the given operation control and activate all layers

0x01A02230 MDSS_MDP_CTL_1_SW_RESET**Type:** RW**Reset State:** 0x00000000

Software Reset of the given control operation

MDSS_MDP_CTL_1_SW_RESET

Bits	Name	Description
0	RESET	Writing to this bit will trigger a soft reset for the described data path This register is self-clearing and will return 0x1 if soft reset sequence is not complete Upon completion, this will return 0x0 Note that this will soft reset the peripheral display interfaces that are described in this path i.e. if driving intf_0 it will reset the pixel clock domain of intf_0 as well

0x01A02234 MDSS_MDP_CTL_1_SW_RESET_OVERRIDE**Type:** RW**Reset State:** 0x00000000

Software Reset Override of the given control operation

MDSS_MDP_CTL_1_SW_RESET_OVERRIDE

Bits	Name	Description
0	RESET	Writing to this bit will assert reset for the described data path This value contained in this register will be forced onto the datapath It will reset the state machines within the scheduler as well for the given datapath To use this register sequence, SW shall write a value of 0x1 to this register to assert reset Once SW wants to de-assert reset, SW shall write a value of 0x0 SW will need to ensure the duration of the value 0x1 is long enough to ensure the path is clean and all memory reads are complete The datapath will not be triggered if this bit is asserted

17.2.4 MDSS_MDP_CTL_2

MDP Control block

17.2.4.1 MDSS_MDP_CTL_2_CTL

MDP

MDP Control block

0x01A02400 MDSS_MDP_CTL_2_LAYER_0

Type: RW

Reset State: 0x00000000

Operation Mode Configuration for Virtual Pipe for the Upper Portion of the Pipeline Connected to LayerMixer0

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_CTL_2_LAYER_0

Bits	Name	Description
25	CURSOR_OUT	Cursor Connection for this virtual pipe The cursor layer will ALWAYS be the top most layer Note that cursor is ONLY applicable in line mode operation and should NOT be used in block mode processing 0x0: Cursor is NOT used 0x1: Cursor is enabled
24	BORDER_OUT	Border Color Connection for this virtual pipe The border color will ALWAYS be the base layer 0x0: Border color NOT used 0x1: Border color enabled
23:21	DMA_1_OUT	DMA1 Connection for this virtual pipe 0x0: DMA1 NOT used at all 0x1: Base Layer for layermixer 0 0x2: Stage0 FG for layermixer 0 0x3: Stage1 FG for layermixer 0 0x4: Stage2 FG for layermixer 0 0x5: Stage3 FG for layermixer 0
20:18	DMA_0_OUT	DMA0 Connection for this virtual pipe 0x0: DMA0 NOT used at all 0x1: Base Layer for layermixer 0 0x2: Stage0 FG for layermixer 0 0x3: Stage1 FG for layermixer 0 0x4: Stage2 FG for layermixer 0 0x5: Stage3 FG for layermixer 0
17:15	RGB_2_OUT	RGB2 Connection for this virtual pipe 0x0: RGB2 NOT used at all 0x1: Base Layer for layermixer 0 0x2: Stage0 FG for layermixer 0 0x3: Stage1 FG for layermixer 0 0x4: Stage2 FG for layermixer 0 0x5: Stage3 FG for layermixer 0
14:12	RGB_1_OUT	RGB1 Connection for this virtual pipe 0x0: RGB1 NOT used at all 0x1: Base Layer for layermixer 0 0x2: Stage0 FG for layermixer 0 0x3: Stage1 FG for layermixer 0 0x4: Stage2 FG for layermixer 0 0x5: Stage3 FG for layermixer 0

MDSS_MDP_CTL_2_LAYER_0 (cont.)

Bits	Name	Description
11:9	RGB_0_OUT	RGB0 Connection for this virtual pipe 0x0: RGB0 NOT used at all 0x1: Base Layer for layermixer 0 0x2: Stage0 FG for layermixer 0 0x3: Stage1 FG for layermixer 0 0x4: Stage2 FG for layermixer 0 0x5: Stage3 FG for layermixer 0
8:6	VIG_2_OUT	VIG2 Connection for this virtual pipe 0x0: VIG2 NOT used at all 0x1: Base Layer for layermixer 0 0x2: Stage0 FG for layermixer 0 0x3: Stage1 FG for layermixer 0 0x4: Stage2 FG for layermixer 0 0x5: Stage3 FG for layermixer 0
5:3	VIG_1_OUT	VIG1 Connection for this virtual pipe 0x0: VIG1 NOT used at all 0x1: Base Layer for layermixer 0 0x2: Stage0 FG for layermixer 0 0x3: Stage1 FG for layermixer 0 0x4: Stage2 FG for layermixer 0 0x5: Stage3 FG for layermixer 0
2:0	VIG_0_OUT	VIG0 Connection for this virtual pipe 0x0: VIG0 NOT used at all 0x1: Base Layer for layermixer 0 0x2: Stage0 FG for layermixer 0 0x3: Stage1 FG for layermixer 0 0x4: Stage2 FG for layermixer 0 0x5: Stage3 FG for layermixer 0

0x01A02404 MDSS_MDP_CTL_2_LAYER_1**Type:** RW**Reset State:** 0x00000000

Operation Mode Configuration for Virtual Pipe for the Upper Portion of the Pipeline Connected to Layer Mixer1

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_CTL_2_LAYER_1

Bits	Name	Description
25	CURSOR_OUT	Cursor Connection for this virtual pipe The cursor layer will ALWAYS be the top most layer Note that cursor is ONLY applicable in line mode operation and should NOT be used in block mode processing 0x0: Cursor is NOT used 0x1: Cursor is enabled
24	BORDER_OUT	Border Color Connection for this virtual pipe The border color will ALWAYS be the base layer 0x0: Border color NOT used 0x1: Border color enabled
23:21	DMA_1_OUT	DMA1 Connection for this virtual pipe 0x0: DMA1 NOT used at all 0x1: Base Layer for layermixer 1 0x2: Stage0 FG for layermixer 1 0x3: Stage1 FG for layermixer 1 0x4: Stage2 FG for layermixer 1 0x5: Stage3 FG for layermixer 1
20:18	DMA_0_OUT	DMA0 Connection for this virtual pipe 0x0: DMA0 NOT used at all 0x1: Base Layer for layermixer 1 0x2: Stage0 FG for layermixer 1 0x3: Stage1 FG for layermixer 1 0x4: Stage2 FG for layermixer 1 0x5: Stage3 FG for layermixer 1
17:15	RGB_2_OUT	RGB2 Connection for this virtual pipe 0x0: RGB2 NOT used at all 0x1: Base Layer for layermixer 1 0x2: Stage0 FG for layermixer 1 0x3: Stage1 FG for layermixer 1 0x4: Stage2 FG for layermixer 1 0x5: Stage3 FG for layermixer 1
14:12	RGB_1_OUT	RGB1 Connection for this virtual pipe 0x0: RGB1 NOT used at all 0x1: Base Layer for layermixer 1 0x2: Stage0 FG for layermixer 1 0x3: Stage1 FG for layermixer 1 0x4: Stage2 FG for layermixer 1 0x5: Stage3 FG for layermixer 1

MDSS_MDP_CTL_2_LAYER_1 (cont.)

Bits	Name	Description
11:9	RGB_0_OUT	RGB0 Connection for this virtual pipe 0x0: RGB0 NOT used at all 0x1: Base Layer for layermixer 1 0x2: Stage0 FG for layermixer 1 0x3: Stage1 FG for layermixer 1 0x4: Stage2 FG for layermixer 1 0x5: Stage3 FG for layermixer 1
8:6	VIG_2_OUT	VIG2 Connection for this virtual pipe 0x0: VIG2 NOT used at all 0x1: Base Layer for layermixer 1 0x2: Stage0 FG for layermixer 1 0x3: Stage1 FG for layermixer 1 0x4: Stage2 FG for layermixer 1 0x5: Stage3 FG for layermixer 1
5:3	VIG_1_OUT	VIG1 Connection for this virtual pipe 0x0: VIG1 NOT used at all 0x1: Base Layer for layermixer 1 0x2: Stage0 FG for layermixer 1 0x3: Stage1 FG for layermixer 1 0x4: Stage2 FG for layermixer 1 0x5: Stage3 FG for layermixer 1
2:0	VIG_0_OUT	VIG0 Connection for this virtual pipe 0x0: VIG0 NOT used at all 0x1: Base Layer for layermixer 1 0x2: Stage0 FG for layermixer 1 0x3: Stage1 FG for layermixer 1 0x4: Stage2 FG for layermixer 1 0x5: Stage3 FG for layermixer 1

0x01A02408 MDSS_MDP_CTL_2_LAYER_2**Type:** RW**Reset State:** 0x00000000

Operation Mode Configuration for Virtual Pipe for the Upper Portion of the Pipeline Connected to Layer Mixer2

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_CTL_2_LAYER_2

Bits	Name	Description
25	CURSOR_OUT	Cursor Connection for this virtual pipe The cursor layer will ALWAYS be the top most layer Note that cursor is ONLY applicable in line mode operation and should NOT be used in block mode processing 0x0: Cursor is NOT used 0x1: Cursor is enabled
24	BORDER_OUT	Border Color Connection for this virtual pipe The border color will ALWAYS be the base layer 0x0: Border color NOT used 0x1: Border color enabled
23:21	DMA_1_OUT	DMA1 Connection for this virtual pipe 0x0: DMA1 NOT used at all 0x1: Base Layer for layermixer 2 0x2: Stage0 FG for layermixer 2 0x3: Stage1 FG for layermixer 2 0x4: Stage2 FG for layermixer 2 0x5: Stage3 FG for layermixer 2
20:18	DMA_0_OUT	DMA0 Connection for this virtual pipe 0x0: DMA0 NOT used at all 0x1: Base Layer for layermixer 2 0x2: Stage0 FG for layermixer 2 0x3: Stage1 FG for layermixer 2 0x4: Stage2 FG for layermixer 2 0x5: Stage3 FG for layermixer 2
17:15	RGB_2_OUT	RGB2 Connection for this virtual pipe 0x0: RGB2 NOT used at all 0x1: Base Layer for layermixer 2 0x2: Stage0 FG for layermixer 2 0x3: Stage1 FG for layermixer 2 0x4: Stage2 FG for layermixer 2 0x5: Stage3 FG for layermixer 2
14:12	RGB_1_OUT	RGB1 Connection for this virtual pipe 0x0: RGB1 NOT used at all 0x1: Base Layer for layermixer 2 0x2: Stage0 FG for layermixer 2 0x3: Stage1 FG for layermixer 2 0x4: Stage2 FG for layermixer 2 0x5: Stage3 FG for layermixer 2

MDSS_MDP_CTL_2_LAYER_2 (cont.)

Bits	Name	Description
11:9	RGB_0_OUT	RGB0 Connection for this virtual pipe 0x0: RGB0 NOT used at all 0x1: Base Layer for layermixer 2 0x2: Stage0 FG for layermixer 2 0x3: Stage1 FG for layermixer 2 0x4: Stage2 FG for layermixer 2 0x5: Stage3 FG for layermixer 2
8:6	VIG_2_OUT	VIG2 Connection for this virtual pipe 0x0: VIG2 NOT used at all 0x1: Base Layer for layermixer 2 0x2: Stage0 FG for layermixer 2 0x3: Stage1 FG for layermixer 2 0x4: Stage2 FG for layermixer 2 0x5: Stage3 FG for layermixer 2
5:3	VIG_1_OUT	VIG1 Connection for this virtual pipe 0x0: VIG1 NOT used at all 0x1: Base Layer for layermixer 2 0x2: Stage0 FG for layermixer 2 0x3: Stage1 FG for layermixer 2 0x4: Stage2 FG for layermixer 2 0x5: Stage3 FG for layermixer 2
2:0	VIG_0_OUT	VIG0 Connection for this virtual pipe 0x0: VIG0 NOT used at all 0x1: Base Layer for layermixer 2 0x2: Stage0 FG for layermixer 2 0x3: Stage1 FG for layermixer 2 0x4: Stage2 FG for layermixer 2 0x5: Stage3 FG for layermixer 2

0x01A0240C MDSS_MDP_CTL_2_LAYER_3**Type:** RW**Reset State:** 0x00000000

Operation Mode Configuration for Virtual Pipe for the Upper Portion of the Pipeline Connected to Layer Mixer3

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_CTL_2_LAYER_3

Bits	Name	Description
25	CURSOR_OUT	Cursor Connection for this virtual pipe The cursor layer will ALWAYS be the top most layer Note that cursor is ONLY applicable in line mode operation and should NOT be used in block mode processing 0x0: Cursor is NOT used 0x1: Cursor is enabled
24	BORDER_OUT	Border Color Connection for this virtual pipe The border color will ALWAYS be the base layer 0x0: Border color NOT used 0x1: Border color enabled
23:21	DMA_1_OUT	DMA1 Connection for this virtual pipe 0x0: DMA1 NOT used at all 0x1: Base Layer for layermixer 3 0x2: Stage0 FG for layermixer 3 0x3: Stage1 FG for layermixer 3 0x4: Stage2 FG for layermixer 3 0x5: Stage3 FG for layermixer 3
20:18	DMA_0_OUT	DMA0 Connection for this virtual pipe 0x0: DMA0 NOT used at all 0x1: Base Layer for layermixer 3 0x2: Stage0 FG for layermixer 3 0x3: Stage1 FG for layermixer 3 0x4: Stage2 FG for layermixer 3 0x5: Stage3 FG for layermixer 3
17:15	RGB_2_OUT	RGB2 Connection for this virtual pipe 0x0: RGB2 NOT used at all 0x1: Base Layer for layermixer 3 0x2: Stage0 FG for layermixer 3 0x3: Stage1 FG for layermixer 3 0x4: Stage2 FG for layermixer 3 0x5: Stage3 FG for layermixer 3
14:12	RGB_1_OUT	RGB1 Connection for this virtual pipe 0x0: RGB1 NOT used at all 0x1: Base Layer for layermixer 3 0x2: Stage0 FG for layermixer 3 0x3: Stage1 FG for layermixer 3 0x4: Stage2 FG for layermixer 3 0x5: Stage3 FG for layermixer 3

MDSS_MDP_CTL_2_LAYER_3 (cont.)

Bits	Name	Description
11:9	RGB_0_OUT	RGB0 Connection for this virtual pipe 0x0: RGB0 NOT used at all 0x1: Base Layer for layermixer 3 0x2: Stage0 FG for layermixer 3 0x3: Stage1 FG for layermixer 3 0x4: Stage2 FG for layermixer 3 0x5: Stage3 FG for layermixer 3
8:6	VIG_2_OUT	VIG2 Connection for this virtual pipe 0x0: VIG2 NOT used at all 0x1: Base Layer for layermixer 3 0x2: Stage0 FG for layermixer 3 0x3: Stage1 FG for layermixer 3 0x4: Stage2 FG for layermixer 3 0x5: Stage3 FG for layermixer 3
5:3	VIG_1_OUT	VIG1 Connection for this virtual pipe 0x0: VIG1 NOT used at all 0x1: Base Layer for layermixer 3 0x2: Stage0 FG for layermixer 3 0x3: Stage1 FG for layermixer 3 0x4: Stage2 FG for layermixer 3 0x5: Stage3 FG for layermixer 3
2:0	VIG_0_OUT	VIG0 Connection for this virtual pipe 0x0: VIG0 NOT used at all 0x1: Base Layer for layermixer 3 0x2: Stage0 FG for layermixer 3 0x3: Stage1 FG for layermixer 3 0x4: Stage2 FG for layermixer 3 0x5: Stage3 FG for layermixer 3

0x01A02410 MDSS_MDP_CTL_2_LAYER_4**Type:** RW**Reset State:** 0x00000000

Operation Mode Configuration for Virtual Pipe for the Upper Portion of the Pipeline Connected to Layer Mixer4

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_CTL_2_LAYER_4

Bits	Name	Description
25	CURSOR_OUT	Cursor Connection for this virtual pipe The cursor layer will ALWAYS be the top most layer Note that cursor is ONLY applicable in line mode operation and should NOT be used in block mode processing 0x0: Cursor is NOT used 0x1: Cursor is enabled
24	BORDER_OUT	Border Color Connection for this virtual pipe The border color will ALWAYS be the base layer 0x0: Border color NOT used 0x1: Border color enabled
23:21	DMA_1_OUT	DMA1 Connection for this virtual pipe 0x0: DMA1 NOT used at all 0x1: Base Layer for layermixer 4 0x2: Stage0 FG for layermixer 4 0x3: Stage1 FG for layermixer 4 0x4: Stage2 FG for layermixer 4 0x5: Stage3 FG for layermixer 4
20:18	DMA_0_OUT	DMA0 Connection for this virtual pipe 0x0: DMA0 NOT used at all 0x1: Base Layer for layermixer 4 0x2: Stage0 FG for layermixer 4 0x3: Stage1 FG for layermixer 4 0x4: Stage2 FG for layermixer 4 0x5: Stage3 FG for layermixer 4
17:15	RGB_2_OUT	RGB2 Connection for this virtual pipe 0x0: RGB2 NOT used at all 0x1: Base Layer for layermixer 4 0x2: Stage0 FG for layermixer 4 0x3: Stage1 FG for layermixer 4 0x4: Stage2 FG for layermixer 4 0x5: Stage3 FG for layermixer 4
14:12	RGB_1_OUT	RGB1 Connection for this virtual pipe 0x0: RGB1 NOT used at all 0x1: Base Layer for layermixer 4 0x2: Stage0 FG for layermixer 4 0x3: Stage1 FG for layermixer 4 0x4: Stage2 FG for layermixer 4 0x5: Stage3 FG for layermixer 4

MDSS_MDP_CTL_2_LAYER_4 (cont.)

Bits	Name	Description
11:9	RGB_0_OUT	RGB0 Connection for this virtual pipe 0x0: RGB0 NOT used at all 0x1: Base Layer for layermixer 4 0x2: Stage0 FG for layermixer 4 0x3: Stage1 FG for layermixer 4 0x4: Stage2 FG for layermixer 4 0x5: Stage3 FG for layermixer 4
8:6	VIG_2_OUT	VIG2 Connection for this virtual pipe 0x0: VIG2 NOT used at all 0x1: Base Layer for layermixer 4 0x2: Stage0 FG for layermixer 4 0x3: Stage1 FG for layermixer 4 0x4: Stage2 FG for layermixer 4 0x5: Stage3 FG for layermixer 4
5:3	VIG_1_OUT	VIG1 Connection for this virtual pipe 0x0: VIG1 NOT used at all 0x1: Base Layer for layermixer 4 0x2: Stage0 FG for layermixer 4 0x3: Stage1 FG for layermixer 4 0x4: Stage2 FG for layermixer 4 0x5: Stage3 FG for layermixer 4
2:0	VIG_0_OUT	VIG0 Connection for this virtual pipe 0x0: VIG0 NOT used at all 0x1: Base Layer for layermixer 4 0x2: Stage0 FG for layermixer 4 0x3: Stage1 FG for layermixer 4 0x4: Stage2 FG for layermixer 4 0x5: Stage3 FG for layermixer 4

0x01A02414 MDSS_MDP_CTL_2_TOP**Type:** RW**Reset State:** 0x00001F00**Operation Mode Configuration for the Pipeline**

The purpose of the register is to define the complete data path of this operation and that includes all the way to the interfaces such as DSI/HDMI/eDP

Changing of the data path dynamically is NOT supported and requires stopping that data path, reconfiguring the connections, and restarting the datapath

MDSS_MDP_CTL_2_TOP

Bits	Name	Description
17	MODE_SEL	Interface Mode of the destination interface Needed to distinguish interfaces such as DSI that can be either mode DSI can also interleave streams of different types (i.e. command and video) to a single core 0x0: Video Mode 0x1: Command Mode1
16:15	STREAM_SEL	Stream Identification Used For Interleaving of Command Streams This field is a dont care for video mode streams 0x0: Stream 0 0x1: Stream 1 0x2: Stream 2
7:4	OUT_SEL	Output Ping-Pong Interface Connection to external display interface 0x0: No connection from Ping-Pong buffers to output interfaces 0x1: Interface 0 0x2: Interface 1 0x3: Interface 2 0x4: Interface 3
3:0	MEM_SEL	Memory Output Connection 0x0: NO memory output selected 0x1: Memory BLOCK mode output selected for rotation 0 0x2: Memory BLOCK mode output selected for rotation 1 0x3: Memory LINE mode output selected for rotation 0 0x4: Memory LINE mode output selected for rotation 1 0x5: Memory LINE mode output selected for linear memory writeback

0x01A02418 MDSS_MDP_CTL_2_FLUSH**Type:** RW**Reset State:** 0x00000000

Register Flush register

The flush register is used to indicate a pipe/layermixers registers are all programmed, and are safe to update to the back copy of the double buffered registers.

Each pipes/layermixers flush mask bit will be used only when the associated pipe/layermixer is running at active refresh pull mode, e.g., direct out going to LCDC/HDMI/DSI Video.

In such operation mode, the associated pipe/layermixers double buffered registers back copies are updated with the new values at the vsync boundary only when the flush mask bit has already been written a 1.

Each flush mask bit would also returned a 0 once the back copies are updated.

Before the back copies are updated, the associate mask bit would returned a 1 if it has been previously written a 1.

MDSS_MDP_CTL_2_FLUSH

Bits	Name	Description
30	TIMING_1	Timing Interface 1 are ready for flush
17	CTL	Control Pipe register is ready for flush This applies to upper pipe ONLY to change the adding/removal of layers
16	WB	Write Back registers are ready for flush
15	DSPP_2	DSPP2 registers are ready for flush
14	DSPP_1	DSPP1 registers are ready for flush
13	DSPP_0	DSPP0 registers are ready for flush
12	DMA_1	DMA_1 registers are ready for flush
11	DMA_0	DMA_0 registers are ready for flush
10	LAYER_MIXER_4	Layer Mixer4 registers are ready for flush This includes the cursor registers stored in the layer mixer register space
9	LAYER_MIXER_3	Layer Mixer3 registers are ready for flush This includes the cursor registers stored in the layer mixer register space
8	LAYER_MIXER_2	Layer Mixer2 registers are ready for flush This includes the cursor registers stored in the layer mixer register space
7	LAYER_MIXER_1	Layer Mixer1 registers are ready for flush This includes the cursor registers stored in the layer mixer register space
6	LAYER_MIXER_0	Layer Mixer0 registers are ready for flush This includes the cursor registers stored in the layer mixer register space
5	RGB_2	RGB2 registers are ready for flush
4	RGB_1	RGB1 registers are ready for flush
3	RGB_0	RGB0 registers are ready for flush

MDSS_MDP_CTL_2_FLUSH (cont.)

Bits	Name	Description
2	VIG_2	VIG2 registers are ready for flush
1	VIG_1	VIG1 registers are ready for flush
0	VIG_0	VIG0 registers are ready for flush

0x01A0241C MDSS_MDP_CTL_2_CTL_START**Type:** W**Reset State:** 0x00000000

Overlay Processor Start Signal for the given Control Operation

MDSS_MDP_CTL_2_CTL_START

Bits	Name	Description
0	START	A write to this register will kick off operation for the given operation control and activate all layers

0x01A02430 MDSS_MDP_CTL_2_SW_RESET**Type:** RW**Reset State:** 0x00000000

Software Reset of the given control operation

MDSS_MDP_CTL_2_SW_RESET

Bits	Name	Description
0	RESET	Writing to this bit will trigger a soft reset for the described data path This register is self-clearing and will return 0x1 if soft reset sequence is not complete Upon completion, this will return 0x0 Note that this will soft reset the peripheral display interfaces that are described in this path i.e. if driving intf_0 it will reset the pixel clock domain of intf_0 as well

0x01A02434 MDSS_MDP_CTL_2_SW_RESET_OVERRIDE**Type:** RW**Reset State:** 0x00000000

Software Reset Override of the given control operation

MDSS_MDP_CTL_2_SW_RESET_OVERRIDE

Bits	Name	Description
0	RESET	Writing to this bit will assert reset for the described data path This value contained in this register will be forced onto the datapath It will reset the state machines within the scheduler as well for the given datapath To use this register sequence, SW shall write a value of 0x1 to this register to assert reset Once SW wants to de-assert reset, SW shall write a value of 0x0 SW will need to ensure the duration of the value 0x1 is long enough to ensure the path is clean and all memory reads are complete The datapath will not be triggered if this bit is asserted

17.2.5 MDSS_MDP_CTL_3

MDP Control block

17.2.5.1 MDSS_MDP_CTL_3_CTL

MDP

MDP Control block

0x01A02600 MDSS_MDP_CTL_3_LAYER_0

Type: RW

Reset State: 0x00000000

Operation Mode Configuration for Virtual Pipe for the Upper Portion of the Pipeline Connected to LayerMixer0

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_CTL_3_LAYER_0

Bits	Name	Description
25	CURSOR_OUT	Cursor Connection for this virtual pipe The cursor layer will ALWAYS be the top most layer Note that cursor is ONLY applicable in line mode operation and should NOT be used in block mode processing 0x0: Cursor is NOT used 0x1: Cursor is enabled
24	BORDER_OUT	Border Color Connection for this virtual pipe The border color will ALWAYS be the base layer 0x0: Border color NOT used 0x1: Border color enabled
23:21	DMA_1_OUT	DMA1 Connection for this virtual pipe 0x0: DMA1 NOT used at all 0x1: Base Layer for layermixer 0 0x2: Stage0 FG for layermixer 0 0x3: Stage1 FG for layermixer 0 0x4: Stage2 FG for layermixer 0 0x5: Stage3 FG for layermixer 0
20:18	DMA_0_OUT	DMA0 Connection for this virtual pipe 0x0: DMA0 NOT used at all 0x1: Base Layer for layermixer 0 0x2: Stage0 FG for layermixer 0 0x3: Stage1 FG for layermixer 0 0x4: Stage2 FG for layermixer 0 0x5: Stage3 FG for layermixer 0
17:15	RGB_2_OUT	RGB2 Connection for this virtual pipe 0x0: RGB2 NOT used at all 0x1: Base Layer for layermixer 0 0x2: Stage0 FG for layermixer 0 0x3: Stage1 FG for layermixer 0 0x4: Stage2 FG for layermixer 0 0x5: Stage3 FG for layermixer 0
14:12	RGB_1_OUT	RGB1 Connection for this virtual pipe 0x0: RGB1 NOT used at all 0x1: Base Layer for layermixer 0 0x2: Stage0 FG for layermixer 0 0x3: Stage1 FG for layermixer 0 0x4: Stage2 FG for layermixer 0 0x5: Stage3 FG for layermixer 0

MDSS_MDP_CTL_3_LAYER_0 (cont.)

Bits	Name	Description
11:9	RGB_0_OUT	RGB0 Connection for this virtual pipe 0x0: RGB0 NOT used at all 0x1: Base Layer for layermixer 0 0x2: Stage0 FG for layermixer 0 0x3: Stage1 FG for layermixer 0 0x4: Stage2 FG for layermixer 0 0x5: Stage3 FG for layermixer 0
8:6	VIG_2_OUT	VIG2 Connection for this virtual pipe 0x0: VIG2 NOT used at all 0x1: Base Layer for layermixer 0 0x2: Stage0 FG for layermixer 0 0x3: Stage1 FG for layermixer 0 0x4: Stage2 FG for layermixer 0 0x5: Stage3 FG for layermixer 0
5:3	VIG_1_OUT	VIG1 Connection for this virtual pipe 0x0: VIG1 NOT used at all 0x1: Base Layer for layermixer 0 0x2: Stage0 FG for layermixer 0 0x3: Stage1 FG for layermixer 0 0x4: Stage2 FG for layermixer 0 0x5: Stage3 FG for layermixer 0
2:0	VIG_0_OUT	VIG0 Connection for this virtual pipe 0x0: VIG0 NOT used at all 0x1: Base Layer for layermixer 0 0x2: Stage0 FG for layermixer 0 0x3: Stage1 FG for layermixer 0 0x4: Stage2 FG for layermixer 0 0x5: Stage3 FG for layermixer 0

0x01A02604 MDSS_MDP_CTL_3_LAYER_1**Type:** RW**Reset State:** 0x00000000

Operation Mode Configuration for Virtual Pipe for the Upper Portion of the Pipeline Connected to Layer Mixer1

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_CTL_3_LAYER_1

Bits	Name	Description
25	CURSOR_OUT	Cursor Connection for this virtual pipe The cursor layer will ALWAYS be the top most layer Note that cursor is ONLY applicable in line mode operation and should NOT be used in block mode processing 0x0: Cursor is NOT used 0x1: Cursor is enabled
24	BORDER_OUT	Border Color Connection for this virtual pipe The border color will ALWAYS be the base layer 0x0: Border color NOT used 0x1: Border color enabled
23:21	DMA_1_OUT	DMA1 Connection for this virtual pipe 0x0: DMA1 NOT used at all 0x1: Base Layer for layermixer 1 0x2: Stage0 FG for layermixer 1 0x3: Stage1 FG for layermixer 1 0x4: Stage2 FG for layermixer 1 0x5: Stage3 FG for layermixer 1
20:18	DMA_0_OUT	DMA0 Connection for this virtual pipe 0x0: DMA0 NOT used at all 0x1: Base Layer for layermixer 1 0x2: Stage0 FG for layermixer 1 0x3: Stage1 FG for layermixer 1 0x4: Stage2 FG for layermixer 1 0x5: Stage3 FG for layermixer 1
17:15	RGB_2_OUT	RGB2 Connection for this virtual pipe 0x0: RGB2 NOT used at all 0x1: Base Layer for layermixer 1 0x2: Stage0 FG for layermixer 1 0x3: Stage1 FG for layermixer 1 0x4: Stage2 FG for layermixer 1 0x5: Stage3 FG for layermixer 1
14:12	RGB_1_OUT	RGB1 Connection for this virtual pipe 0x0: RGB1 NOT used at all 0x1: Base Layer for layermixer 1 0x2: Stage0 FG for layermixer 1 0x3: Stage1 FG for layermixer 1 0x4: Stage2 FG for layermixer 1 0x5: Stage3 FG for layermixer 1

MDSS_MDP_CTL_3_LAYER_1 (cont.)

Bits	Name	Description
11:9	RGB_0_OUT	RGB0 Connection for this virtual pipe 0x0: RGB0 NOT used at all 0x1: Base Layer for layermixer 1 0x2: Stage0 FG for layermixer 1 0x3: Stage1 FG for layermixer 1 0x4: Stage2 FG for layermixer 1 0x5: Stage3 FG for layermixer 1
8:6	VIG_2_OUT	VIG2 Connection for this virtual pipe 0x0: VIG2 NOT used at all 0x1: Base Layer for layermixer 1 0x2: Stage0 FG for layermixer 1 0x3: Stage1 FG for layermixer 1 0x4: Stage2 FG for layermixer 1 0x5: Stage3 FG for layermixer 1
5:3	VIG_1_OUT	VIG1 Connection for this virtual pipe 0x0: VIG1 NOT used at all 0x1: Base Layer for layermixer 1 0x2: Stage0 FG for layermixer 1 0x3: Stage1 FG for layermixer 1 0x4: Stage2 FG for layermixer 1 0x5: Stage3 FG for layermixer 1
2:0	VIG_0_OUT	VIG0 Connection for this virtual pipe 0x0: VIG0 NOT used at all 0x1: Base Layer for layermixer 1 0x2: Stage0 FG for layermixer 1 0x3: Stage1 FG for layermixer 1 0x4: Stage2 FG for layermixer 1 0x5: Stage3 FG for layermixer 1

0x01A02608 MDSS_MDP_CTL_3_LAYER_2**Type:** RW**Reset State:** 0x00000000

Operation Mode Configuration for Virtual Pipe for the Upper Portion of the Pipeline Connected to Layer Mixer2

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_CTL_3_LAYER_2

Bits	Name	Description
25	CURSOR_OUT	Cursor Connection for this virtual pipe The cursor layer will ALWAYS be the top most layer Note that cursor is ONLY applicable in line mode operation and should NOT be used in block mode processing 0x0: Cursor is NOT used 0x1: Cursor is enabled
24	BORDER_OUT	Border Color Connection for this virtual pipe The border color will ALWAYS be the base layer 0x0: Border color NOT used 0x1: Border color enabled
23:21	DMA_1_OUT	DMA1 Connection for this virtual pipe 0x0: DMA1 NOT used at all 0x1: Base Layer for layermixer 2 0x2: Stage0 FG for layermixer 2 0x3: Stage1 FG for layermixer 2 0x4: Stage2 FG for layermixer 2 0x5: Stage3 FG for layermixer 2
20:18	DMA_0_OUT	DMA0 Connection for this virtual pipe 0x0: DMA0 NOT used at all 0x1: Base Layer for layermixer 2 0x2: Stage0 FG for layermixer 2 0x3: Stage1 FG for layermixer 2 0x4: Stage2 FG for layermixer 2 0x5: Stage3 FG for layermixer 2
17:15	RGB_2_OUT	RGB2 Connection for this virtual pipe 0x0: RGB2 NOT used at all 0x1: Base Layer for layermixer 2 0x2: Stage0 FG for layermixer 2 0x3: Stage1 FG for layermixer 2 0x4: Stage2 FG for layermixer 2 0x5: Stage3 FG for layermixer 2
14:12	RGB_1_OUT	RGB1 Connection for this virtual pipe 0x0: RGB1 NOT used at all 0x1: Base Layer for layermixer 2 0x2: Stage0 FG for layermixer 2 0x3: Stage1 FG for layermixer 2 0x4: Stage2 FG for layermixer 2 0x5: Stage3 FG for layermixer 2

MDSS_MDP_CTL_3_LAYER_2 (cont.)

Bits	Name	Description
11:9	RGB_0_OUT	RGB0 Connection for this virtual pipe 0x0: RGB0 NOT used at all 0x1: Base Layer for layermixer 2 0x2: Stage0 FG for layermixer 2 0x3: Stage1 FG for layermixer 2 0x4: Stage2 FG for layermixer 2 0x5: Stage3 FG for layermixer 2
8:6	VIG_2_OUT	VIG2 Connection for this virtual pipe 0x0: VIG2 NOT used at all 0x1: Base Layer for layermixer 2 0x2: Stage0 FG for layermixer 2 0x3: Stage1 FG for layermixer 2 0x4: Stage2 FG for layermixer 2 0x5: Stage3 FG for layermixer 2
5:3	VIG_1_OUT	VIG1 Connection for this virtual pipe 0x0: VIG1 NOT used at all 0x1: Base Layer for layermixer 2 0x2: Stage0 FG for layermixer 2 0x3: Stage1 FG for layermixer 2 0x4: Stage2 FG for layermixer 2 0x5: Stage3 FG for layermixer 2
2:0	VIG_0_OUT	VIG0 Connection for this virtual pipe 0x0: VIG0 NOT used at all 0x1: Base Layer for layermixer 2 0x2: Stage0 FG for layermixer 2 0x3: Stage1 FG for layermixer 2 0x4: Stage2 FG for layermixer 2 0x5: Stage3 FG for layermixer 2

0x01A0260C MDSS_MDP_CTL_3_LAYER_3**Type:** RW**Reset State:** 0x00000000

Operation Mode Configuration for Virtual Pipe for the Upper Portion of the Pipeline Connected to Layer Mixer3

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_CTL_3_LAYER_3

Bits	Name	Description
25	CURSOR_OUT	Cursor Connection for this virtual pipe The cursor layer will ALWAYS be the top most layer Note that cursor is ONLY applicable in line mode operation and should NOT be used in block mode processing 0x0: Cursor is NOT used 0x1: Cursor is enabled
24	BORDER_OUT	Border Color Connection for this virtual pipe The border color will ALWAYS be the base layer 0x0: Border color NOT used 0x1: Border color enabled
23:21	DMA_1_OUT	DMA1 Connection for this virtual pipe 0x0: DMA1 NOT used at all 0x1: Base Layer for layermixer 3 0x2: Stage0 FG for layermixer 3 0x3: Stage1 FG for layermixer 3 0x4: Stage2 FG for layermixer 3 0x5: Stage3 FG for layermixer 3
20:18	DMA_0_OUT	DMA0 Connection for this virtual pipe 0x0: DMA0 NOT used at all 0x1: Base Layer for layermixer 3 0x2: Stage0 FG for layermixer 3 0x3: Stage1 FG for layermixer 3 0x4: Stage2 FG for layermixer 3 0x5: Stage3 FG for layermixer 3
17:15	RGB_2_OUT	RGB2 Connection for this virtual pipe 0x0: RGB2 NOT used at all 0x1: Base Layer for layermixer 3 0x2: Stage0 FG for layermixer 3 0x3: Stage1 FG for layermixer 3 0x4: Stage2 FG for layermixer 3 0x5: Stage3 FG for layermixer 3
14:12	RGB_1_OUT	RGB1 Connection for this virtual pipe 0x0: RGB1 NOT used at all 0x1: Base Layer for layermixer 3 0x2: Stage0 FG for layermixer 3 0x3: Stage1 FG for layermixer 3 0x4: Stage2 FG for layermixer 3 0x5: Stage3 FG for layermixer 3

MDSS_MDP_CTL_3_LAYER_3 (cont.)

Bits	Name	Description
11:9	RGB_0_OUT	RGB0 Connection for this virtual pipe 0x0: RGB0 NOT used at all 0x1: Base Layer for layermixer 3 0x2: Stage0 FG for layermixer 3 0x3: Stage1 FG for layermixer 3 0x4: Stage2 FG for layermixer 3 0x5: Stage3 FG for layermixer 3
8:6	VIG_2_OUT	VIG2 Connection for this virtual pipe 0x0: VIG2 NOT used at all 0x1: Base Layer for layermixer 3 0x2: Stage0 FG for layermixer 3 0x3: Stage1 FG for layermixer 3 0x4: Stage2 FG for layermixer 3 0x5: Stage3 FG for layermixer 3
5:3	VIG_1_OUT	VIG1 Connection for this virtual pipe 0x0: VIG1 NOT used at all 0x1: Base Layer for layermixer 3 0x2: Stage0 FG for layermixer 3 0x3: Stage1 FG for layermixer 3 0x4: Stage2 FG for layermixer 3 0x5: Stage3 FG for layermixer 3
2:0	VIG_0_OUT	VIG0 Connection for this virtual pipe 0x0: VIG0 NOT used at all 0x1: Base Layer for layermixer 3 0x2: Stage0 FG for layermixer 3 0x3: Stage1 FG for layermixer 3 0x4: Stage2 FG for layermixer 3 0x5: Stage3 FG for layermixer 3

0x01A02610 MDSS_MDP_CTL_3_LAYER_4**Type:** RW**Reset State:** 0x00000000

Operation Mode Configuration for Virtual Pipe for the Upper Portion of the Pipeline Connected to Layer Mixer4

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_CTL_3_LAYER_4

Bits	Name	Description
25	CURSOR_OUT	Cursor Connection for this virtual pipe The cursor layer will ALWAYS be the top most layer Note that cursor is ONLY applicable in line mode operation and should NOT be used in block mode processing 0x0: Cursor is NOT used 0x1: Cursor is enabled
24	BORDER_OUT	Border Color Connection for this virtual pipe The border color will ALWAYS be the base layer 0x0: Border color NOT used 0x1: Border color enabled
23:21	DMA_1_OUT	DMA1 Connection for this virtual pipe 0x0: DMA1 NOT used at all 0x1: Base Layer for layermixer 4 0x2: Stage0 FG for layermixer 4 0x3: Stage1 FG for layermixer 4 0x4: Stage2 FG for layermixer 4 0x5: Stage3 FG for layermixer 4
20:18	DMA_0_OUT	DMA0 Connection for this virtual pipe 0x0: DMA0 NOT used at all 0x1: Base Layer for layermixer 4 0x2: Stage0 FG for layermixer 4 0x3: Stage1 FG for layermixer 4 0x4: Stage2 FG for layermixer 4 0x5: Stage3 FG for layermixer 4
17:15	RGB_2_OUT	RGB2 Connection for this virtual pipe 0x0: RGB2 NOT used at all 0x1: Base Layer for layermixer 4 0x2: Stage0 FG for layermixer 4 0x3: Stage1 FG for layermixer 4 0x4: Stage2 FG for layermixer 4 0x5: Stage3 FG for layermixer 4
14:12	RGB_1_OUT	RGB1 Connection for this virtual pipe 0x0: RGB1 NOT used at all 0x1: Base Layer for layermixer 4 0x2: Stage0 FG for layermixer 4 0x3: Stage1 FG for layermixer 4 0x4: Stage2 FG for layermixer 4 0x5: Stage3 FG for layermixer 4

MDSS_MDP_CTL_3_LAYER_4 (cont.)

Bits	Name	Description
11:9	RGB_0_OUT	RGB0 Connection for this virtual pipe 0x0: RGB0 NOT used at all 0x1: Base Layer for layermixer 4 0x2: Stage0 FG for layermixer 4 0x3: Stage1 FG for layermixer 4 0x4: Stage2 FG for layermixer 4 0x5: Stage3 FG for layermixer 4
8:6	VIG_2_OUT	VIG2 Connection for this virtual pipe 0x0: VIG2 NOT used at all 0x1: Base Layer for layermixer 4 0x2: Stage0 FG for layermixer 4 0x3: Stage1 FG for layermixer 4 0x4: Stage2 FG for layermixer 4 0x5: Stage3 FG for layermixer 4
5:3	VIG_1_OUT	VIG1 Connection for this virtual pipe 0x0: VIG1 NOT used at all 0x1: Base Layer for layermixer 4 0x2: Stage0 FG for layermixer 4 0x3: Stage1 FG for layermixer 4 0x4: Stage2 FG for layermixer 4 0x5: Stage3 FG for layermixer 4
2:0	VIG_0_OUT	VIG0 Connection for this virtual pipe 0x0: VIG0 NOT used at all 0x1: Base Layer for layermixer 4 0x2: Stage0 FG for layermixer 4 0x3: Stage1 FG for layermixer 4 0x4: Stage2 FG for layermixer 4 0x5: Stage3 FG for layermixer 4

0x01A02614 MDSS_MDP_CTL_3_TOP**Type:** RW**Reset State:** 0x00001F00**Operation Mode Configuration for the Pipeline**

The purpose of the register is to define the complete data path of this operation and that includes all the way to the interfaces such as DSI/HDMI/eDP

Changing of the data path dynamically is NOT supported and requires stopping that data path, reconfiguring the connections, and restarting the datapath

MDSS_MDP_CTL_3_TOP

Bits	Name	Description
17	MODE_SEL	Interface Mode of the destination interface Needed to distinguish interfaces such as DSI that can be either mode DSI can also interleave streams of different types (i.e. command and video) to a single core 0x0: Video Mode 0x1: Command Mode1
16:15	STREAM_SEL	Stream Identification Used For Interleaving of Command Streams This field is a dont care for video mode streams 0x0: Stream 0 0x1: Stream 1 0x2: Stream 2
7:4	OUT_SEL	Output Ping-Pong Interface Connection to external display interface 0x0: No connection from Ping-Pong buffers to output interfaces 0x1: Interface 0 0x2: Interface 1 0x3: Interface 2 0x4: Interface 3
3:0	MEM_SEL	Memory Output Connection 0x0: NO memory output selected 0x1: Memory BLOCK mode output selected for rotation 0 0x2: Memory BLOCK mode output selected for rotation 1 0x3: Memory LINE mode output selected for rotation 0 0x4: Memory LINE mode output selected for rotation 1 0x5: Memory LINE mode output selected for linear memory writeback

0x01A02618 MDSS_MDP_CTL_3_FLUSH**Type:** RW**Reset State:** 0x00000000

Register Flush register

The flush register is used to indicate a pipe/layermixers registers are all programmed, and are safe to update to the back copy of the double buffered registers.

Each pipes/layermixers flush mask bit will be used only when the associated pipe/layermixer is running at active refresh pull mode, e.g., direct out going to LCDC/HDMI/DSI Video.

In such operation mode, the associated pipe/layermixers double buffered registers back copies are updated with the new values at the vsync boundary only when the flush mask bit has already been written a 1.

Each flush mask bit would also returned a 0 once the back copies are updated.

Before the back copies are updated, the associate mask bit would returned a 1 if it has been previously written a 1.

MDSS_MDP_CTL_3_FLUSH

Bits	Name	Description
30	TIMING_1	Timing Interface 1 are ready for flush
17	CTL	Control Pipe register is ready for flush This applies to upper pipe ONLY to change the adding/removal of layers
16	WB	Write Back registers are ready for flush
15	DSPP_2	DSPP2 registers are ready for flush
14	DSPP_1	DSPP1 registers are ready for flush
13	DSPP_0	DSPP0 registers are ready for flush
12	DMA_1	DMA_1 registers are ready for flush
11	DMA_0	DMA_0 registers are ready for flush
10	LAYER_MIXER_4	Layer Mixer4 registers are ready for flush This includes the cursor registers stored in the layer mixer register space
9	LAYER_MIXER_3	Layer Mixer3 registers are ready for flush This includes the cursor registers stored in the layer mixer register space
8	LAYER_MIXER_2	Layer Mixer2 registers are ready for flush This includes the cursor registers stored in the layer mixer register space
7	LAYER_MIXER_1	Layer Mixer1 registers are ready for flush This includes the cursor registers stored in the layer mixer register space
6	LAYER_MIXER_0	Layer Mixer0 registers are ready for flush This includes the cursor registers stored in the layer mixer register space
5	RGB_2	RGB2 registers are ready for flush
4	RGB_1	RGB1 registers are ready for flush
3	RGB_0	RGB0 registers are ready for flush

MDSS_MDP_CTL_3_FLUSH (cont.)

Bits	Name	Description
2	VIG_2	VIG2 registers are ready for flush
1	VIG_1	VIG1 registers are ready for flush
0	VIG_0	VIG0 registers are ready for flush

0x01A0261C MDSS_MDP_CTL_3_CTL_START**Type:** W**Reset State:** 0x00000000

Overlay Processor Start Signal for the given Control Operation

MDSS_MDP_CTL_3_CTL_START

Bits	Name	Description
0	START	A write to this register will kick off operation for the given operation control and activate all layers

0x01A02630 MDSS_MDP_CTL_3_SW_RESET**Type:** RW**Reset State:** 0x00000000

Software Reset of the given control operation

MDSS_MDP_CTL_3_SW_RESET

Bits	Name	Description
0	RESET	Writing to this bit will trigger a soft reset for the described data path This register is self-clearing and will return 0x1 if soft reset sequence is not complete Upon completion, this will return 0x0 Note that this will soft reset the peripheral display interfaces that are described in this path i.e. if driving intf_0 it will reset the pixel clock domain of intf_0 as well

0x01A02634 MDSS_MDP_CTL_3_SW_RESET_OVERRIDE**Type:** RW**Reset State:** 0x00000000

Software Reset Override of the given control operation

MDSS_MDP_CTL_3_SW_RESET_OVERRIDE

Bits	Name	Description
0	RESET	Writing to this bit will assert reset for the described data path This value contained in this register will be forced onto the datapath It will reset the state machines within the scheduler as well for the given datapath To use this register sequence, SW shall write a value of 0x1 to this register to assert reset Once SW wants to de-assert reset, SW shall write a value of 0x0 SW will need to ensure the duration of the value 0x1 is long enough to ensure the path is clean and all memory reads are complete The datapath will not be triggered if this bit is asserted

17.2.6 MDSS_MDP_CTL_4

MDP Control block

17.2.6.1 MDSS_MDP_CTL_4_CTL

MDP

MDP Control block

0x01A02800 MDSS_MDP_CTL_4_LAYER_0

Type: RW

Reset State: 0x00000000

Operation Mode Configuration for Virtual Pipe for the Upper Portion of the Pipeline Connected to LayerMixer0

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_CTL_4_LAYER_0

Bits	Name	Description
25	CURSOR_OUT	Cursor Connection for this virtual pipe The cursor layer will ALWAYS be the top most layer Note that cursor is ONLY applicable in line mode operation and should NOT be used in block mode processing 0x0: Cursor is NOT used 0x1: Cursor is enabled
24	BORDER_OUT	Border Color Connection for this virtual pipe The border color will ALWAYS be the base layer 0x0: Border color NOT used 0x1: Border color enabled
23:21	DMA_1_OUT	DMA1 Connection for this virtual pipe 0x0: DMA1 NOT used at all 0x1: Base Layer for layermixer 0 0x2: Stage0 FG for layermixer 0 0x3: Stage1 FG for layermixer 0 0x4: Stage2 FG for layermixer 0 0x5: Stage3 FG for layermixer 0
20:18	DMA_0_OUT	DMA0 Connection for this virtual pipe 0x0: DMA0 NOT used at all 0x1: Base Layer for layermixer 0 0x2: Stage0 FG for layermixer 0 0x3: Stage1 FG for layermixer 0 0x4: Stage2 FG for layermixer 0 0x5: Stage3 FG for layermixer 0
17:15	RGB_2_OUT	RGB2 Connection for this virtual pipe 0x0: RGB2 NOT used at all 0x1: Base Layer for layermixer 0 0x2: Stage0 FG for layermixer 0 0x3: Stage1 FG for layermixer 0 0x4: Stage2 FG for layermixer 0 0x5: Stage3 FG for layermixer 0
14:12	RGB_1_OUT	RGB1 Connection for this virtual pipe 0x0: RGB1 NOT used at all 0x1: Base Layer for layermixer 0 0x2: Stage0 FG for layermixer 0 0x3: Stage1 FG for layermixer 0 0x4: Stage2 FG for layermixer 0 0x5: Stage3 FG for layermixer 0

MDSS_MDP_CTL_4_LAYER_0 (cont.)

Bits	Name	Description
11:9	RGB_0_OUT	RGB0 Connection for this virtual pipe 0x0: RGB0 NOT used at all 0x1: Base Layer for layermixer 0 0x2: Stage0 FG for layermixer 0 0x3: Stage1 FG for layermixer 0 0x4: Stage2 FG for layermixer 0 0x5: Stage3 FG for layermixer 0
8:6	VIG_2_OUT	VIG2 Connection for this virtual pipe 0x0: VIG2 NOT used at all 0x1: Base Layer for layermixer 0 0x2: Stage0 FG for layermixer 0 0x3: Stage1 FG for layermixer 0 0x4: Stage2 FG for layermixer 0 0x5: Stage3 FG for layermixer 0
5:3	VIG_1_OUT	VIG1 Connection for this virtual pipe 0x0: VIG1 NOT used at all 0x1: Base Layer for layermixer 0 0x2: Stage0 FG for layermixer 0 0x3: Stage1 FG for layermixer 0 0x4: Stage2 FG for layermixer 0 0x5: Stage3 FG for layermixer 0
2:0	VIG_0_OUT	VIG0 Connection for this virtual pipe 0x0: VIG0 NOT used at all 0x1: Base Layer for layermixer 0 0x2: Stage0 FG for layermixer 0 0x3: Stage1 FG for layermixer 0 0x4: Stage2 FG for layermixer 0 0x5: Stage3 FG for layermixer 0

0x01A02804 MDSS_MDP_CTL_4_LAYER_1**Type:** RW**Reset State:** 0x00000000

Operation Mode Configuration for Virtual Pipe for the Upper Portion of the Pipeline Connected to Layer Mixer1

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_CTL_4_LAYER_1

Bits	Name	Description
25	CURSOR_OUT	Cursor Connection for this virtual pipe The cursor layer will ALWAYS be the top most layer Note that cursor is ONLY applicable in line mode operation and should NOT be used in block mode processing 0x0: Cursor is NOT used 0x1: Cursor is enabled
24	BORDER_OUT	Border Color Connection for this virtual pipe The border color will ALWAYS be the base layer 0x0: Border color NOT used 0x1: Border color enabled
23:21	DMA_1_OUT	DMA1 Connection for this virtual pipe 0x0: DMA1 NOT used at all 0x1: Base Layer for layermixer 1 0x2: Stage0 FG for layermixer 1 0x3: Stage1 FG for layermixer 1 0x4: Stage2 FG for layermixer 1 0x5: Stage3 FG for layermixer 1
20:18	DMA_0_OUT	DMA0 Connection for this virtual pipe 0x0: DMA0 NOT used at all 0x1: Base Layer for layermixer 1 0x2: Stage0 FG for layermixer 1 0x3: Stage1 FG for layermixer 1 0x4: Stage2 FG for layermixer 1 0x5: Stage3 FG for layermixer 1
17:15	RGB_2_OUT	RGB2 Connection for this virtual pipe 0x0: RGB2 NOT used at all 0x1: Base Layer for layermixer 1 0x2: Stage0 FG for layermixer 1 0x3: Stage1 FG for layermixer 1 0x4: Stage2 FG for layermixer 1 0x5: Stage3 FG for layermixer 1
14:12	RGB_1_OUT	RGB1 Connection for this virtual pipe 0x0: RGB1 NOT used at all 0x1: Base Layer for layermixer 1 0x2: Stage0 FG for layermixer 1 0x3: Stage1 FG for layermixer 1 0x4: Stage2 FG for layermixer 1 0x5: Stage3 FG for layermixer 1

MDSS_MDP_CTL_4_LAYER_1 (cont.)

Bits	Name	Description
11:9	RGB_0_OUT	RGB0 Connection for this virtual pipe 0x0: RGB0 NOT used at all 0x1: Base Layer for layermixer 1 0x2: Stage0 FG for layermixer 1 0x3: Stage1 FG for layermixer 1 0x4: Stage2 FG for layermixer 1 0x5: Stage3 FG for layermixer 1
8:6	VIG_2_OUT	VIG2 Connection for this virtual pipe 0x0: VIG2 NOT used at all 0x1: Base Layer for layermixer 1 0x2: Stage0 FG for layermixer 1 0x3: Stage1 FG for layermixer 1 0x4: Stage2 FG for layermixer 1 0x5: Stage3 FG for layermixer 1
5:3	VIG_1_OUT	VIG1 Connection for this virtual pipe 0x0: VIG1 NOT used at all 0x1: Base Layer for layermixer 1 0x2: Stage0 FG for layermixer 1 0x3: Stage1 FG for layermixer 1 0x4: Stage2 FG for layermixer 1 0x5: Stage3 FG for layermixer 1
2:0	VIG_0_OUT	VIG0 Connection for this virtual pipe 0x0: VIG0 NOT used at all 0x1: Base Layer for layermixer 1 0x2: Stage0 FG for layermixer 1 0x3: Stage1 FG for layermixer 1 0x4: Stage2 FG for layermixer 1 0x5: Stage3 FG for layermixer 1

0x01A02808 MDSS_MDP_CTL_4_LAYER_2**Type:** RW**Reset State:** 0x00000000

Operation Mode Configuration for Virtual Pipe for the Upper Portion of the Pipeline Connected to Layer Mixer2

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_CTL_4_LAYER_2

Bits	Name	Description
25	CURSOR_OUT	Cursor Connection for this virtual pipe The cursor layer will ALWAYS be the top most layer Note that cursor is ONLY applicable in line mode operation and should NOT be used in block mode processing 0x0: Cursor is NOT used 0x1: Cursor is enabled
24	BORDER_OUT	Border Color Connection for this virtual pipe The border color will ALWAYS be the base layer 0x0: Border color NOT used 0x1: Border color enabled
23:21	DMA_1_OUT	DMA1 Connection for this virtual pipe 0x0: DMA1 NOT used at all 0x1: Base Layer for layermixer 2 0x2: Stage0 FG for layermixer 2 0x3: Stage1 FG for layermixer 2 0x4: Stage2 FG for layermixer 2 0x5: Stage3 FG for layermixer 2
20:18	DMA_0_OUT	DMA0 Connection for this virtual pipe 0x0: DMA0 NOT used at all 0x1: Base Layer for layermixer 2 0x2: Stage0 FG for layermixer 2 0x3: Stage1 FG for layermixer 2 0x4: Stage2 FG for layermixer 2 0x5: Stage3 FG for layermixer 2
17:15	RGB_2_OUT	RGB2 Connection for this virtual pipe 0x0: RGB2 NOT used at all 0x1: Base Layer for layermixer 2 0x2: Stage0 FG for layermixer 2 0x3: Stage1 FG for layermixer 2 0x4: Stage2 FG for layermixer 2 0x5: Stage3 FG for layermixer 2
14:12	RGB_1_OUT	RGB1 Connection for this virtual pipe 0x0: RGB1 NOT used at all 0x1: Base Layer for layermixer 2 0x2: Stage0 FG for layermixer 2 0x3: Stage1 FG for layermixer 2 0x4: Stage2 FG for layermixer 2 0x5: Stage3 FG for layermixer 2

MDSS_MDP_CTL_4_LAYER_2 (cont.)

Bits	Name	Description
11:9	RGB_0_OUT	RGB0 Connection for this virtual pipe 0x0: RGB0 NOT used at all 0x1: Base Layer for layermixer 2 0x2: Stage0 FG for layermixer 2 0x3: Stage1 FG for layermixer 2 0x4: Stage2 FG for layermixer 2 0x5: Stage3 FG for layermixer 2
8:6	VIG_2_OUT	VIG2 Connection for this virtual pipe 0x0: VIG2 NOT used at all 0x1: Base Layer for layermixer 2 0x2: Stage0 FG for layermixer 2 0x3: Stage1 FG for layermixer 2 0x4: Stage2 FG for layermixer 2 0x5: Stage3 FG for layermixer 2
5:3	VIG_1_OUT	VIG1 Connection for this virtual pipe 0x0: VIG1 NOT used at all 0x1: Base Layer for layermixer 2 0x2: Stage0 FG for layermixer 2 0x3: Stage1 FG for layermixer 2 0x4: Stage2 FG for layermixer 2 0x5: Stage3 FG for layermixer 2
2:0	VIG_0_OUT	VIG0 Connection for this virtual pipe 0x0: VIG0 NOT used at all 0x1: Base Layer for layermixer 2 0x2: Stage0 FG for layermixer 2 0x3: Stage1 FG for layermixer 2 0x4: Stage2 FG for layermixer 2 0x5: Stage3 FG for layermixer 2

0x01A0280C MDSS_MDP_CTL_4_LAYER_3**Type:** RW**Reset State:** 0x00000000

Operation Mode Configuration for Virtual Pipe for the Upper Portion of the Pipeline Connected to Layer Mixer3

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_CTL_4_LAYER_3

Bits	Name	Description
25	CURSOR_OUT	Cursor Connection for this virtual pipe The cursor layer will ALWAYS be the top most layer Note that cursor is ONLY applicable in line mode operation and should NOT be used in block mode processing 0x0: Cursor is NOT used 0x1: Cursor is enabled
24	BORDER_OUT	Border Color Connection for this virtual pipe The border color will ALWAYS be the base layer 0x0: Border color NOT used 0x1: Border color enabled
23:21	DMA_1_OUT	DMA1 Connection for this virtual pipe 0x0: DMA1 NOT used at all 0x1: Base Layer for layermixer 3 0x2: Stage0 FG for layermixer 3 0x3: Stage1 FG for layermixer 3 0x4: Stage2 FG for layermixer 3 0x5: Stage3 FG for layermixer 3
20:18	DMA_0_OUT	DMA0 Connection for this virtual pipe 0x0: DMA0 NOT used at all 0x1: Base Layer for layermixer 3 0x2: Stage0 FG for layermixer 3 0x3: Stage1 FG for layermixer 3 0x4: Stage2 FG for layermixer 3 0x5: Stage3 FG for layermixer 3
17:15	RGB_2_OUT	RGB2 Connection for this virtual pipe 0x0: RGB2 NOT used at all 0x1: Base Layer for layermixer 3 0x2: Stage0 FG for layermixer 3 0x3: Stage1 FG for layermixer 3 0x4: Stage2 FG for layermixer 3 0x5: Stage3 FG for layermixer 3
14:12	RGB_1_OUT	RGB1 Connection for this virtual pipe 0x0: RGB1 NOT used at all 0x1: Base Layer for layermixer 3 0x2: Stage0 FG for layermixer 3 0x3: Stage1 FG for layermixer 3 0x4: Stage2 FG for layermixer 3 0x5: Stage3 FG for layermixer 3

MDSS_MDP_CTL_4_LAYER_3 (cont.)

Bits	Name	Description
11:9	RGB_0_OUT	RGB0 Connection for this virtual pipe 0x0: RGB0 NOT used at all 0x1: Base Layer for layermixer 3 0x2: Stage0 FG for layermixer 3 0x3: Stage1 FG for layermixer 3 0x4: Stage2 FG for layermixer 3 0x5: Stage3 FG for layermixer 3
8:6	VIG_2_OUT	VIG2 Connection for this virtual pipe 0x0: VIG2 NOT used at all 0x1: Base Layer for layermixer 3 0x2: Stage0 FG for layermixer 3 0x3: Stage1 FG for layermixer 3 0x4: Stage2 FG for layermixer 3 0x5: Stage3 FG for layermixer 3
5:3	VIG_1_OUT	VIG1 Connection for this virtual pipe 0x0: VIG1 NOT used at all 0x1: Base Layer for layermixer 3 0x2: Stage0 FG for layermixer 3 0x3: Stage1 FG for layermixer 3 0x4: Stage2 FG for layermixer 3 0x5: Stage3 FG for layermixer 3
2:0	VIG_0_OUT	VIG0 Connection for this virtual pipe 0x0: VIG0 NOT used at all 0x1: Base Layer for layermixer 3 0x2: Stage0 FG for layermixer 3 0x3: Stage1 FG for layermixer 3 0x4: Stage2 FG for layermixer 3 0x5: Stage3 FG for layermixer 3

0x01A02810 MDSS_MDP_CTL_4_LAYER_4**Type:** RW**Reset State:** 0x00000000

Operation Mode Configuration for Virtual Pipe for the Upper Portion of the Pipeline Connected to Layer Mixer4

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_CTL_4_LAYER_4

Bits	Name	Description
25	CURSOR_OUT	Cursor Connection for this virtual pipe The cursor layer will ALWAYS be the top most layer Note that cursor is ONLY applicable in line mode operation and should NOT be used in block mode processing 0x0: Cursor is NOT used 0x1: Cursor is enabled
24	BORDER_OUT	Border Color Connection for this virtual pipe The border color will ALWAYS be the base layer 0x0: Border color NOT used 0x1: Border color enabled
23:21	DMA_1_OUT	DMA1 Connection for this virtual pipe 0x0: DMA1 NOT used at all 0x1: Base Layer for layermixer 4 0x2: Stage0 FG for layermixer 4 0x3: Stage1 FG for layermixer 4 0x4: Stage2 FG for layermixer 4 0x5: Stage3 FG for layermixer 4
20:18	DMA_0_OUT	DMA0 Connection for this virtual pipe 0x0: DMA0 NOT used at all 0x1: Base Layer for layermixer 4 0x2: Stage0 FG for layermixer 4 0x3: Stage1 FG for layermixer 4 0x4: Stage2 FG for layermixer 4 0x5: Stage3 FG for layermixer 4
17:15	RGB_2_OUT	RGB2 Connection for this virtual pipe 0x0: RGB2 NOT used at all 0x1: Base Layer for layermixer 4 0x2: Stage0 FG for layermixer 4 0x3: Stage1 FG for layermixer 4 0x4: Stage2 FG for layermixer 4 0x5: Stage3 FG for layermixer 4
14:12	RGB_1_OUT	RGB1 Connection for this virtual pipe 0x0: RGB1 NOT used at all 0x1: Base Layer for layermixer 4 0x2: Stage0 FG for layermixer 4 0x3: Stage1 FG for layermixer 4 0x4: Stage2 FG for layermixer 4 0x5: Stage3 FG for layermixer 4

MDSS_MDP_CTL_4_LAYER_4 (cont.)

Bits	Name	Description
11:9	RGB_0_OUT	RGB0 Connection for this virtual pipe 0x0: RGB0 NOT used at all 0x1: Base Layer for layermixer 4 0x2: Stage0 FG for layermixer 4 0x3: Stage1 FG for layermixer 4 0x4: Stage2 FG for layermixer 4 0x5: Stage3 FG for layermixer 4
8:6	VIG_2_OUT	VIG2 Connection for this virtual pipe 0x0: VIG2 NOT used at all 0x1: Base Layer for layermixer 4 0x2: Stage0 FG for layermixer 4 0x3: Stage1 FG for layermixer 4 0x4: Stage2 FG for layermixer 4 0x5: Stage3 FG for layermixer 4
5:3	VIG_1_OUT	VIG1 Connection for this virtual pipe 0x0: VIG1 NOT used at all 0x1: Base Layer for layermixer 4 0x2: Stage0 FG for layermixer 4 0x3: Stage1 FG for layermixer 4 0x4: Stage2 FG for layermixer 4 0x5: Stage3 FG for layermixer 4
2:0	VIG_0_OUT	VIG0 Connection for this virtual pipe 0x0: VIG0 NOT used at all 0x1: Base Layer for layermixer 4 0x2: Stage0 FG for layermixer 4 0x3: Stage1 FG for layermixer 4 0x4: Stage2 FG for layermixer 4 0x5: Stage3 FG for layermixer 4

0x01A02814 MDSS_MDP_CTL_4_TOP**Type:** RW**Reset State:** 0x00001F00**Operation Mode Configuration for the Pipeline**

The purpose of the register is to define the complete data path of this operation and that includes all the way to the interfaces such as DSI/HDMI/eDP

Changing of the data path dynamically is NOT supported and requires stopping that data path, reconfiguring the connections, and restarting the datapath

MDSS_MDP_CTL_4_TOP

Bits	Name	Description
17	MODE_SEL	Interface Mode of the destination interface Needed to distinguish interfaces such as DSI that can be either mode DSI can also interleave streams of different types (i.e. command and video) to a single core 0x0: Video Mode 0x1: Command Mode1
16:15	STREAM_SEL	Stream Identification Used For Interleaving of Command Streams This field is a dont care for video mode streams 0x0: Stream 0 0x1: Stream 1 0x2: Stream 2
7:4	OUT_SEL	Output Ping-Pong Interface Connection to external display interface 0x0: No connection from Ping-Pong buffers to output interfaces 0x1: Interface 0 0x2: Interface 1 0x3: Interface 2 0x4: Interface 3
3:0	MEM_SEL	Memory Output Connection 0x0: NO memory output selected 0x1: Memory BLOCK mode output selected for rotation 0 0x2: Memory BLOCK mode output selected for rotation 1 0x3: Memory LINE mode output selected for rotation 0 0x4: Memory LINE mode output selected for rotation 1 0x5: Memory LINE mode output selected for linear memory writeback

0x01A02818 MDSS_MDP_CTL_4_FLUSH**Type:** RW**Reset State:** 0x00000000

Register Flush register

The flush register is used to indicate a pipe/layermixers registers are all programmed, and are safe to update to the back copy of the double buffered registers.

Each pipes/layermixers flush mask bit will be used only when the associated pipe/layermixer is running at active refresh pull mode, e.g., direct out going to LCDC/HDMI/DSI Video.

In such operation mode, the associated pipe/layermixers double buffered registers back copies are updated with the new values at the vsync boundary only when the flush mask bit has already been written a 1.

Each flush mask bit would also returned a 0 once the back copies are updated.

Before the back copies are updated, the associate mask bit would returned a 1 if it has been previously written a 1.

MDSS_MDP_CTL_4_FLUSH

Bits	Name	Description
30	TIMING_1	Timing Interface 1 are ready for flush
17	CTL	Control Pipe register is ready for flush This applies to upper pipe ONLY to change the adding/removal of layers
16	WB	Write Back registers are ready for flush
15	DSPP_2	DSPP2 registers are ready for flush
14	DSPP_1	DSPP1 registers are ready for flush
13	DSPP_0	DSPP0 registers are ready for flush
12	DMA_1	DMA_1 registers are ready for flush
11	DMA_0	DMA_0 registers are ready for flush
10	LAYER_MIXER_4	Layer Mixer4 registers are ready for flush This includes the cursor registers stored in the layer mixer register space
9	LAYER_MIXER_3	Layer Mixer3 registers are ready for flush This includes the cursor registers stored in the layer mixer register space
8	LAYER_MIXER_2	Layer Mixer2 registers are ready for flush This includes the cursor registers stored in the layer mixer register space
7	LAYER_MIXER_1	Layer Mixer1 registers are ready for flush This includes the cursor registers stored in the layer mixer register space
6	LAYER_MIXER_0	Layer Mixer0 registers are ready for flush This includes the cursor registers stored in the layer mixer register space
5	RGB_2	RGB2 registers are ready for flush
4	RGB_1	RGB1 registers are ready for flush
3	RGB_0	RGB0 registers are ready for flush

MDSS_MDP_CTL_4_FLUSH (cont.)

Bits	Name	Description
2	VIG_2	VIG2 registers are ready for flush
1	VIG_1	VIG1 registers are ready for flush
0	VIG_0	VIG0 registers are ready for flush

0x01A0281C MDSS_MDP_CTL_4_CTL_START**Type:** W**Reset State:** 0x00000000

Overlay Processor Start Signal for the given Control Operation

MDSS_MDP_CTL_4_CTL_START

Bits	Name	Description
0	START	A write to this register will kick off operation for the given operation control and activate all layers

0x01A02830 MDSS_MDP_CTL_4_SW_RESET**Type:** RW**Reset State:** 0x00000000

Software Reset of the given control operation

MDSS_MDP_CTL_4_SW_RESET

Bits	Name	Description
0	RESET	Writing to this bit will trigger a soft reset for the described data path This register is self-clearing and will return 0x1 if soft reset sequence is not complete Upon completion, this will return 0x0 Note that this will soft reset the peripheral display interfaces that are described in this path i.e. if driving intf_0 it will reset the pixel clock domain of intf_0 as well

0x01A02834 MDSS_MDP_CTL_4_SW_RESET_OVERRIDE**Type:** RW**Reset State:** 0x00000000

Software Reset Override of the given control operation

MDSS_MDP_CTL_4_SW_RESET_OVERRIDE

Bits	Name	Description
0	RESET	Writing to this bit will assert reset for the described data path This value contained in this register will be forced onto the datapath It will reset the state machines within the scheduler as well for the given datapath To use this register sequence, SW shall write a value of 0x1 to this register to assert reset Once SW wants to de-assert reset, SW shall write a value of 0x0 SW will need to ensure the duration of the value 0x1 is long enough to ensure the path is clean and all memory reads are complete The datapath will not be triggered if this bit is asserted

17.2.7 MDSS_MDP_VP_0

MDP VP hardware register description

VP

17.2.7.1 MDSS_MDP_VP_0_VIG_0

VIG hardware register description

17.2.7.1.1 MDSS_MDP_VP_0_VIG_0_VIG

VIG

0x01A05200 MDSS_MDP_VP_0_VIG_0_OP_MODE

Type: RW

Reset State: 0x00040000

Virtual Pipeline operation mode

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_VIG_0_OP_MODE

Bits	Name	Description
28	PA_CONT_MASK	0x0: Disable HSIC adj on CONTRAST 0x1: Enable HSIC adj on CONTRAST

MDSS_MDP_VP_0_VIG_0_OP_MODE (cont.)

Bits	Name	Description
27	PA_VAL_MASK	0x0: Disable HSIC adj on VALUE 0x1: Enable HSIC adj on VALUE
26	PA_SAT_MASK	0x0: Disable HSIC adj on SATURATION 0x1: Enable HSIC adj on SATURATION
25	PA_HUE_MASK	0x0: Disable HSIC adj on HUE 0x1: Enable HSIC adj on HUE
19	CSC_DST_DATA_FORMAT	Destination Data Format 0x0: RGB 0x1: YCbCr
18	CSC_SRC_DATA_FORMAT	Source Data Format 0x0: RGB 0x1: YCbCr
17	CSC_1_EN	Enables the color matrix conversion 1 0x0: Bypass 0x1: Enabled
10	PA_LUTV_EN	Histogram Enhancement Lookup Table enable
8	HIST_EN	Histogram Enable 0x0: Disabled 0x1: Enabled
7	PA_MEM_COL_SKY_MASK	0x0: Disable Sky adjustment 0x1: Enable Sky adjustment
6	PA_MEM_COL_FOL_MASK	0x0: Disable Foliage adjustment 0x1: Enable Foliage adjustment
5	PA_MEM_COL_SKIN_MASK	0x0: Disable Skin adjustment 0x1: Enable Skin adjustment
4	PA_EN	Enable Picture Adjustment 0x0: Disabled 0x1: Enabled
3	PA_MEM_PROTECT_ENABLE	0: global hsv adjustment is independent of memor color 1: global hsv adjustment is active if the pixel is not already processed by memory color contrast adjustment in global adjustment is always independ of memory color regardless
2	PA_SAT_ZERO_EXP_ENABLE	0: if PA_SAT_ADJ < -32, clamp PA_SAT_ADJ to 0 1: if PA_SAT_ADJ < -64, clamp PA_SAT_ADJ to 0

0x01A05204 MDSS_MDP_VP_0_VIG_0_QSEED2_CONFIG**Type:** RW**Reset State:** 0x00000000

QSEED configuration for each of the Virtual Pipelines

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_VIG_0_QSEED2_CONFIG

Bits	Name	Description
19:18	COMP3_VERT_FILTER_MODE	Component 3 (A) vertical filtering mode 0x0: Nearest Neighbor 0x1: Bilinear 0x2: PCMN
17:16	COMP3_HORZ_FILTER_MODE	Component 3 (A) horizontal filtering mode 0x0: Nearest Neighbor 0x1: Bilinear 0x2: PCMN
15:14	COMP1_2_VERT_FILTER_MODE	Component 1/2 (CrCb or RB) vertical filtering mode 0x0: Nearest Neighbor 0x1: Bilinear 0x2: PCMN
13:12	COMP1_2_HORZ_FILTER_MODE	Component 1/2 (CrCb or RB) horizontal filtering mode 0x0: Nearest Neighbor 0x1: Bilinear 0x2: PCMN
11:10	COMP0_VERT_FILTER_MODE	Component 0 (Y or G) vertical filtering mode 0x0: Nearest Neighbor 0x1: Bilinear 0x2: PCMN 0x3: Content Adaptive
9:8	COMP0_HORZ_FILTER_MODE	Component 0 (Y or G) horizontal filtering mode 0x0: Nearest Neighbor 0x1: Bilinear 0x2: PCMN 0x3: Content Adaptive
1	SCALEY_EN	Scaling Enable in the Y-direction 0x0: Disabled 0x1: Enabled
0	SCALEX_EN	Scaling Enable in the X-direction 0x0: Disabled 0x1: Enabled

0x01A05210 MDSS_MDP_VP_0_VIG_0_QSEED2_COMP0_3_PHASE_STEP_X**Type:** RW**Reset State:** 0x00000000

QSEED component 0 and 3 phase step for each of the Virtual Pipelines

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_VIG_0_QSEED2_COMP0_3_PHASE_STEP_X

Bits	Name	Description
25:0	PHASE_STEP_X	Unsigned Phase Step in the X-direction

0x01A05214 MDSS_MDP_VP_0_VIG_0_QSEED2_COMP0_3_PHASE_STEP_Y**Type:** RW**Reset State:** 0x00000000

QSEED component 0 and 3 phase step for each of the Virtual Pipelines

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_VIG_0_QSEED2_COMP0_3_PHASE_STEP_Y

Bits	Name	Description
25:0	PHASE_STEP_Y	Unsigned Phase Step in the Y-direction

0x01A05218 MDSS_MDP_VP_0_VIG_0_QSEED2_COMP1_2_PHASE_STEP_X**Type:** RW**Reset State:** 0x00000000

QSEED component 1 and 2 phase step for each of the Virtual Pipelines

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_VIG_0_QSEED2_COMP1_2_PHASE_STEP_X

Bits	Name	Description
25:0	PHASE_STEP_X	Unsigned Phase Step in the X-direction

0x01A0521C MDSS_MDP_VP_0_VIG_0_QSEED2_COMP1_2_PHASE_STEP_Y**Type:** RW**Reset State:** 0x00000000

QSEED component 1 and 2 phase step for each of the Virtual Pipelines

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_VIG_0_QSEED2_COMP1_2_PHASE_STEP_Y

Bits	Name	Description
25:0	PHASE_STEP_Y	Unsigned Phase Step in the Y-direction

0x01A05220 MDSS_MDP_VP_0_VIG_0_QSEED2_COMP0_3_INIT_PHASE_X**Type:** RW**Reset State:** 0x00000000

QSEED component 0 and 3 init phase for each of the Virtual Pipelines

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_VIG_0_QSEED2_COMP0_3_INIT_PHASE_X

Bits	Name	Description
25:0	INIT_PHASE_X	Signed INIT Phase in the X-direction

0x01A05224 MDSS_MDP_VP_0_VIG_0_QSEED2_COMP0_3_INIT_PHASE_Y**Type:** RW**Reset State:** 0x00000000

QSEED component 0 and 3 init phase for each of the Virtual Pipelines

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_VIG_0_QSEED2_COMP0_3_INIT_PHASE_Y

Bits	Name	Description
25:0	INIT_PHASE_Y	Signed INIT Phase in the Y-direction

0x01A05228 MDSS_MDP_VP_0_VIG_0_QSEED2_COMP1_2_INIT_PHASE_X**Type:** RW**Reset State:** 0x00000000

QSEED component 1 and 2 init phase for each of the Virtual Pipelines

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_VIG_0_QSEED2_COMP1_2_INIT_PHASE_X

Bits	Name	Description
25:0	INIT_PHASE_X	Signed INIT Phase in the X-direction

0x01A0522C MDSS_MDP_VP_0_VIG_0_QSEED2_COMP1_2_INIT_PHASE_Y**Type:** RW**Reset State:** 0x00000000

QSEED component 1 and 2 init phase for each of the Virtual Pipelines

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_VIG_0_QSEED2_COMP1_2_INIT_PHASE_Y

Bits	Name	Description
25:0	INIT_PHASE_Y	Signed INIT Phase in the Y-direction

0x01A05230 MDSS_MDP_VP_0_VIG_0_QSEED2_SHARP_SMOOTH_STRENGTH**Type:** RW**Reset State:** 0x00000000

QSEED configuration for each of the Virtual Pipelines

double buffered: YES

MDSS_MDP_VP_0_VIG_0_QSEED2_SHARP_SMOOTH_STRENGTH

Bits	Name	Description
8:0	STRENGTH	Sharpening/Smooth Strength used for content Adaptive Filtering S9 notation with a range from -256 to 255 0x0: No sharpening and No smoothing +ve: sharpening -ve: smoothing

0x01A05234 MDSS_MDP_VP_0_VIG_0_QSEED2_SHARP_THRESHOLD_EDGE**Type:** RW**Reset State:** 0x00000000

QSEED configuration for each of the Virtual Pipelines

double buffered: YES

MDSS_MDP_VP_0_VIG_0_QSEED2_SHARP_THRESHOLD_EDGE

Bits	Name	Description
9:0	THRESHOLD	Unsigned Threshold to select Edge/Normal Filter for Content Adaptive Filter

0x01A05238 MDSS_MDP_VP_0_VIG_0_QSEED2_SHARP_THRESHOLD_SMOOTH**Type:** RW**Reset State:** 0x00000000

QSEED configuration for each of the Virtual Pipelines

double buffered: YES

MDSS_MDP_VP_0_VIG_0_QSEED2_SHARP_THRESHOLD_SMOOTH

Bits	Name	Description
9:0	THRESHOLD	Unsigned Threshold to select Normal/Smooth Filter for Content Adaptive Filter

0x01A0523C MDSS_MDP_VP_0_VIG_0_QSEED2_SHARP_THRESHOLD_NOISE**Type:** RW**Reset State:** 0x00000000

QSEED configuration for each of the Virtual Pipelines

double buffered: YES

MDSS_MDP_VP_0_VIG_0_QSEED2_SHARP_THRESHOLD_NOISE

Bits	Name	Description
7:0	THRESHOLD	Unsigned Threshold to select Normal/Smooth Filter for Content Adaptive Filter

0x01A05250 MDSS_MDP_VP_0_VIG_0_QSEED2_NORMAL_FILTER_COEFF_N

Type: RW

Reset State: 0x00000000

MDSS_MDP_VP_0_VIG_0_QSEED2_NORMAL_FILTER_COEFF_N

Bits	Name	Description
11:0	VALUE	

0x01A05260 MDSS_MDP_VP_0_VIG_0_QSEED2_SMOOTH_FILTER_COEFF_N

Type: RW

Reset State: 0x00000000

MDSS_MDP_VP_0_VIG_0_QSEED2_SMOOTH_FILTER_COEFF_N

Bits	Name	Description
11:0	VALUE	

0x01A05270 MDSS_MDP_VP_0_VIG_0_QSEED2_EDGE_FILTER_COEFF_START

Type: W

Reset State: 0x00000000

QSEED SW trigger to calculate the edge filter coefficients

double buffered: NO

MDSS_MDP_VP_0_VIG_0_QSEED2_EDGE_FILTER_COEFF_START

Bits	Name	Description
0	START	SW trigger to initiate the calculate of edge coefficients in hardware The pipeline should be idle as smooth/normal filter coefficients are required This implies that the QSEED2 is not active during the trigger

0x01A05288 MDSS_MDP_VP_0_VIG_0_SKIN_ADJUST_PWL0**Type:** RW**Reset State:** 0x00000000

Skin Piece-wise Linear Adjustment Curve

double buffered: Yes

MDSS_MDP_VP_0_VIG_0_SKIN_ADJUST_PWL0

Bits	Name	Description
31:16	HUE_GAIN	S16, is a symmetrical trapezoidal curve, this is the slope of the 2 sides
10:0	HUE_MID	U11, this is the offset from the min and max hue of skin region which marks the top 2 corners of the trapezoid

0x01A0528C MDSS_MDP_VP_0_VIG_0_SKIN_ADJUST_PWL1**Type:** RW**Reset State:** 0x00000000

Skin Piece-wise Linear Adjustment Curve

double buffered: Yes

MDSS_MDP_VP_0_VIG_0_SKIN_ADJUST_PWL1

Bits	Name	Description
31:16	VAL_GAIN	S16, is a symmetrical trapezoidal curve, this is the slope of the 2 sides
15:0	SAT_GAIN	S16, is a symmetrical trapezoidal curve, this is the slope of the 2 sides

0x01A05290 MDSS_MDP_VP_0_VIG_0_SKIN_REGION_HUE**Type:** RW**Reset State:** 0x00000000

Skin Adjustment Hue Qualifier

double buffered: Yes

MDSS_MDP_VP_0_VIG_0_SKIN_REGION_HUE

Bits	Name	Description
26:16	MAX	if MAX > x17ff (i.e. 360 degree), the behaviour will be same as MAX = x1800
10:0	MIN	U11, Hue has to be within the inclusive range of MIN to MAX to qualify for skin tone adjustment on the pixel if MIN > x17ff (i.e. 360 degree), the behaviour will be same as MIN = x1800

0x01A05294 MDSS_MDP_VP_0_VIG_0_SKIN_REGION_SAT**Type:** RW**Reset State:** 0x00000000

Skin Adjustment Saturation Qualifier

double buffered: Yes

MDSS_MDP_VP_0_VIG_0_SKIN_REGION_SAT

Bits	Name	Description
15:8	MAX	Saturation has to be below or equal to this to qualify for skin tone adjustment on the pixel
7:0	MIN	U8, Saturation has to be above or equal to this to qualify for skin tone adjustment on the pixel

0x01A05298 MDSS_MDP_VP_0_VIG_0_SKIN_REGION_VAL**Type:** RW**Reset State:** 0x00000000

Skin Adjustment Value Qualifier

double buffered: Yes

MDSS_MDP_VP_0_VIG_0_SKIN_REGION_VAL

Bits	Name	Description
15:8	MAX	Value has to be below or equal to this to qualify for skin tone adjustment on the pixel
7:0	MIN	U8, Value has to be above or equal to this to qualify for skin tone adjustment on the pixel

0x01A0529C MDSS_MDP_VP_0_VIG_0_SKY_ADJUST_PWL0**Type:** RW**Reset State:** 0x00000000

Sky Piece-wise Linear Adjustment Curve

double buffered: Yes

MDSS_MDP_VP_0_VIG_0_SKY_ADJUST_PWL0

Bits	Name	Description
31:16	HUE_GAIN	S16, is a symmetrical trapezoidal curve, this is the slope of the 2 sides
10:0	HUE_MID	U11, this is the offset from the min and max hue of skin region which marks the top 2 corners of the trapezoid

0x01A052A0 MDSS_MDP_VP_0_VIG_0_SKY_ADJUST_PWL1**Type:** RW**Reset State:** 0x00000000

Sky Piece-wise Linear Adjustment Curve

double buffered: Yes

MDSS_MDP_VP_0_VIG_0_SKY_ADJUST_PWL1

Bits	Name	Description
31:16	VAL_GAIN	S16, is a symmetrical trapezoidal curve, this is the slope of the 2 sides
15:0	SAT_GAIN	S16, is a symmetrical trapezoidal curve, this is the slope of the 2 sides

0x01A052A4 MDSS_MDP_VP_0_VIG_0_SKY_REGION_HUE**Type:** RW**Reset State:** 0x00000000

Sky Adjustment Hue Qualifier

double buffered: Yes

MDSS_MDP_VP_0_VIG_0_SKY_REGION_HUE

Bits	Name	Description
26:16	MAX	if MAX > x17ff (i.e. 360 degree), the behaviour will be same as MAX = x1800
10:0	MIN	U11, Hue has to be within the inclusive range of MIN to MAX (U13) to qualify for sky tone adjustment on the pixel if MIN > x17ff (i.e. 360 degree), the behaviour will be same as MIN = x1800

0x01A052A8 MDSS_MDP_VP_0_VIG_0_SKY_REGION_SAT**Type:** RW**Reset State:** 0x00000000

Sky Adjustment Saturation Qualifier

double buffered: Yes

MDSS_MDP_VP_0_VIG_0_SKY_REGION_SAT

Bits	Name	Description
15:8	MAX	Saturation has to be below or equal to this to qualify for sky adjustment on the pixel
7:0	MIN	U8, Saturation has to be above or equal to this to qualify for sky adjustment on the pixel

0x01A052AC MDSS_MDP_VP_0_VIG_0_SKY_REGION_VAL**Type:** RW**Reset State:** 0x00000000

Sky Adjustment Value Qualifier

double buffered: Yes

MDSS_MDP_VP_0_VIG_0_SKY_REGION_VAL

Bits	Name	Description
15:8	MAX	Value has to be below or equal to this to qualify for sky adjustment on the pixel
7:0	MIN	U8, Value has to be above or equal to this to qualify for sky adjustment on the pixel

0x01A052B0 MDSS_MDP_VP_0_VIG_0_FOLIAGE_ADJUST_PWL0**Type:** RW**Reset State:** 0x00000000**MDSS_MDP_VP_0_VIG_0_FOLIAGE_ADJUST_PWL0**

Bits	Name	Description
31:16	HUE_GAIN	
10:0	HUE_MID	

0x01A052B4 MDSS_MDP_VP_0_VIG_0_FOLIAGE_ADJUST_PWL1**Type:** RW**Reset State:** 0x00000000**MDSS_MDP_VP_0_VIG_0_FOLIAGE_ADJUST_PWL1**

Bits	Name	Description
31:16	VAL_GAIN	
15:0	SAT_GAIN	

0x01A052B8 MDSS_MDP_VP_0_VIG_0_FOLIAGE_REGION_HUE**Type:** RW**Reset State:** 0x00000000

Foliage Adjustment Hue Qualifier

double buffered: Yes

MDSS_MDP_VP_0_VIG_0_FOLIAGE_REGION_HUE

Bits	Name	Description
26:16	MAX	if MAX > x17ff (i.e. 360 degree), the behaviour will be same as MAX = x1800
10:0	MIN	U11, Hue has to be within the inclusive range of MIN to MAX to qualify for foliage tone adjustment on the pixel if MIN > x17ff (i.e. 360 degree), the behaviour will be same as MIN = x1800

0x01A052BC MDSS_MDP_VP_0_VIG_0_FOLIAGE_REGION_SAT**Type:** RW**Reset State:** 0x00000000

Foilage Adjustment Saturation Qualifier

double buffered: Yes

MDSS_MDP_VP_0_VIG_0_FOLIAGE_REGION_SAT

Bits	Name	Description
15:8	MAX	Saturation has to be below or equal to this to qualify for foilage adjustment on the pixel
7:0	MIN	U8, Saturation has to be above or equal to this to qualify for foilage adjustment on the pixel

0x01A052C0 MDSS_MDP_VP_0_VIG_0_FOLIAGE_REGION_VAL**Type:** RW**Reset State:** 0x00000000

Foilage Adjustment Value Qualifier

double buffered: Yes

MDSS_MDP_VP_0_VIG_0_FOLIAGE_REGION_VAL

Bits	Name	Description
15:8	MAX	Value has to be below or equal to this to qualify for foilage adjustment on the pixel
7:0	MIN	U8, Value has to be above or equal to this to qualify for foilage adjustment on the pixel

0x01A052C4 MDSS_MDP_VP_0_VIG_0_HIST_COLLECT_CTL**Type:** W**Reset State:** 0x00000000

Histogram Collection Control

double buffered: YES

except for LOCK

MDSS_MDP_VP_0_VIG_0_HIST_COLLECT_CTL

Bits	Name	Description
0	LOCK	0: SW and HW swap their access on the 2 histogram tables; HW collects on a new table on the new frame, while SW accesses the histogram of the last frame on the table just swapped out by HW 1: current table will be overwritten for new collection on the next frame, the other table remains untouched for SW to continue to access

0x01A052C8 MDSS_MDP_VP_0_VIG_0_HIST_MIN_MAX**Type:** R**Reset State:** 0x00000000

MIN and MAX V value over the entire frame

double buffered: NO

MDSS_MDP_VP_0_VIG_0_HIST_MIN_MAX

Bits	Name	Description
23:16	MAX	Maximum V value of the last collected frame
7:0	MIN	Minimum V value of the last collected frame

0x01A052CC MDSS_MDP_VP_0_VIG_0_HIST_FRAME_COLLECTED**Type:** R**Reset State:** 0x00000000

Histogram Frame Number since the last rising START

double buffered: NO

MDSS_MDP_VP_0_VIG_0_HIST_FRAME_COLLECTED

Bits	Name	Description
31:0	FRAME_NUM	The frame number that the current histogram accessible by SW was collected from

0x01A052D0 MDSS_MDP_VP_0_VIG_0_HIST_SEL**Type:** R**Reset State:** 0x00000000

PA LUTV and Histogram Table Selection

double buffered: NO

MDSS_MDP_VP_0_VIG_0_HIST_SEL

Bits	Name	Description
4	HIST_SEL	0x0: Table 0 is available for SW and Table 1 is available for HW 0x1: Table 1 is available for SW and Table 0 is available for HW
0	PA_LUTV_SEL	0x0: LUT0 is available for SW and LUT1 is available for HW 0x1: LUT1 is available for SW and LUT0 is available for HW

0x01A052D4 MDSS_MDP_VP_0_VIG_0_HIST_SUM**Type:** R**Reset State:** 0x00000000

Accumulated V Value of the frame

double buffered: NO

MDSS_MDP_VP_0_VIG_0_HIST_SUM

Bits	Name	Description
31:0	HIST_SUM	This sum matches the histogram currently accessible by SW

0x01A052E0 MDSS_MDP_VP_0_VIG_0_HIST_V_DATAN**Type:** RW**Reset State:** 0x00000000**MDSS_MDP_VP_0_VIG_0_HIST_V_DATAN**

Bits	Name	Type	Description
31:24	INDEX	RW	
23:0	VALUE	R	

0x01A052F0 MDSS_MDP_VP_0_VIG_0_PA_LUTV**Type:** RW**Reset State:** 0x00000000

PA LUTV data

Auto-incremented by hardware or by software

256 entries are available for reading

double buffered: NO. Hardware maintains 2 copies

MDSS_MDP_VP_0_VIG_0_PA_LUTV

Bits	Name	Description
17	VALUE_UPDATEN	Software issue an write to the LUT 0: VALUE will be written to RAM at INDEX if INDEX_UPDATE=1 or at INDEX+1 if INDEX_UPDATE=0 1: VALUE will not be written to RAM, this setting is appropriate for example when INDEX needs to be updated for read
16	INDEX_UPDATE	Software issue an index update 0: No index update by software 1: Update LUT index to value specified by INDEX
15:8	INDEX	VALUE is stored in this INDEX of the HIST_LUTn, next write will be stored to the next higher INDEX
7:0	VALUE	Value read depends on the LUT select and which one is available for reading by software The read sequence should always begin with WR INDEX_UPDATE=1, INDEX=<rptr start location> subsequent RD will start from the start location and every read will auto-increment the rptr

0x01A05300 MDSS_MDP_VP_0_VIG_0_PA_LUTV_SWAP**Type:** W**Reset State:** 0x00000000

Histogram SWAP signal to indicate that the HIST_LUTn contents should be swapped.

Software accesses one LUT while HW accesses the other LUT

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_VIG_0_PA_LUTV_SWAP

Bits	Name	Description
0	SWAP	SWAP signal to flip buffers accessible by SW and made available to HW Achieves double buffering of the LUT values

0x01A05310 MDSS_MDP_VP_0_VIG_0_SPA_HUE_ADJUST**Type:** RW**Reset State:** 0x00000000

Source Picture HUE Adjustment

double buffered: YES

MDSS_MDP_VP_0_VIG_0_SPA_HUE_ADJUST

Bits	Name	Description
11:0	VALUE	Value in S12, recommend to be within exclusive range xA00 to x600 (i.e. -360 to +360 degree) x600 is equivalent to a shift of 360 degree if VALUE > x5FF in either negative or positive direction, it is circularly clipped (i.e. VALUE + x600 or VALUE - x600)

0x01A05314 MDSS_MDP_VP_0_VIG_0_SPA_SAT_ADJUST**Type:** RW**Reset State:** 0x00001A00

Source Picture Saturation Adjustment

double buffered: YES

MDSS_MDP_VP_0_VIG_0_SPA_SAT_ADJUST

Bits	Name	Description
15:8	THRSHLD	Below this threshold of saturation, no adjustment is applied to saturation. This field is in U8
7:0	VALUE	Value in S8

0x01A05318 MDSS_MDP_VP_0_VIG_0_SPA_VAL_ADJUST**Type:** RW**Reset State:** 0x00000000

Source Picture Adjustment

double buffered: YES

MDSS_MDP_VP_0_VIG_0_SPA_VAL_ADJUST

Bits	Name	Description
7:0	VALUE	Value in S8

0x01A0531C MDSS_MDP_VP_0_VIG_0_SPA_CONT_ADJUST**Type:** RW**Reset State:** 0x00000000

Source Picture Contrast Adjustment

double buffered: YES

MDSS_MDP_VP_0_VIG_0_SPA_CONT_ADJUST

Bits	Name	Description
7:0	VALUE	Value in S8

0x01A05320 MDSS_MDP_VP_0_VIG_0_CSC_1_MATRIX_COEFF_0**Type:** RW**Reset State:** 0x00000200

Color Space conversion coefficients for ALL of the Virtual Pipelines

double buffered: NO

MDSS_MDP_VP_0_VIG_0_CSC_1_MATRIX_COEFF_0

Bits	Name	Description
28:16	COEFF_12	Coefficient value in S4.9 format Row 1 Column 2
12:0	COEFF_11	Coefficient value in S4.9 format Row 1 Column 1

0x01A05324 MDSS_MDP_VP_0_VIG_0_CSC_1_MATRIX_COEFF_1**Type:** RW**Reset State:** 0x00000000

Color Space conversion coefficients for ALL of the Virtual Pipelines

double buffered: NO

MDSS_MDP_VP_0_VIG_0_CSC_1_MATRIX_COEFF_1

Bits	Name	Description
28:16	COEFF_21	Coefficient value in S4.9 format Row 2 Column 1
12:0	COEFF_13	Coefficient value in S4.9 format Row 1 Column 3

0x01A05328 MDSS_MDP_VP_0_VIG_0_CSC_1_MATRIX_COEFF_2**Type:** RW**Reset State:** 0x00000200

Color Space conversion coefficients for ALL of the Virtual Pipelines

double buffered: NO

MDSS_MDP_VP_0_VIG_0_CSC_1_MATRIX_COEFF_2

Bits	Name	Description
28:16	COEFF_23	Coefficient value in S4.9 format Row 2 Column 3
12:0	COEFF_22	Coefficient value in S4.9 format Row 2 Column 2

0x01A0532C MDSS_MDP_VP_0_VIG_0_CSC_1_MATRIX_COEFF_3**Type:** RW**Reset State:** 0x00000000

Color Space conversion coefficients for ALL of the Virtual Pipelines

double buffered: NO

MDSS_MDP_VP_0_VIG_0_CSC_1_MATRIX_COEFF_3

Bits	Name	Description
28:16	COEFF_32	Coefficient value in S4.9 format Row 3 Column 2
12:0	COEFF_31	Coefficient value in S4.9 format Row 3 Column 1

0x01A05330 MDSS_MDP_VP_0_VIG_0_CSC_1_MATRIX_COEFF_4**Type:** RW**Reset State:** 0x00000200

Color Space conversion coefficients for ALL of the Virtual Pipelines

double buffered: NO

MDSS_MDP_VP_0_VIG_0_CSC_1_MATRIX_COEFF_4

Bits	Name	Description
12:0	COEFF_33	Coefficient value in S4.9 format Row 3 Column 3

0x01A05334 MDSS_MDP_VP_0_VIG_0_CSC_1_COMP_0_PRE_CLAMP**Type:** RW**Reset State:** 0x000000FF

Component 0 pre-clamp value

double buffered: NO

MDSS_MDP_VP_0_VIG_0_CSC_1_COMP_0_PRE_CLAMP

Bits	Name	Description
15:8	LOW	Unsigned 8 bit pre-clamp value
7:0	HIGH	Unsigned 8 bit pre-clamp value

0x01A05338 MDSS_MDP_VP_0_VIG_0_CSC_1_COMP_1_PRE_CLAMP**Type:** RW**Reset State:** 0x000000FF

Component 1 pre-clamp value

double buffered: NO

MDSS_MDP_VP_0_VIG_0_CSC_1_COMP_1_PRE_CLAMP

Bits	Name	Description
15:8	LOW	Unsigned 8 bit pre-clamp value
7:0	HIGH	Unsigned 8 bit pre-clamp value

0x01A0533C MDSS_MDP_VP_0_VIG_0_CSC_1_COMP_2_PRE_CLAMP

Type: RW

Reset State: 0x000000FF

Component 2 pre-clamp value

double buffered: NO

MDSS_MDP_VP_0_VIG_0_CSC_1_COMP_2_PRE_CLAMP

Bits	Name	Description
15:8	LOW	Unsigned 8 bit pre-clamp value
7:0	HIGH	Unsigned 8 bit pre-clamp value

0x01A05340 MDSS_MDP_VP_0_VIG_0_CSC_1_COMP_0_POST_CLAMP

Type: RW

Reset State: 0x000000FF

Component 0 post-clamp value

double buffered: NO

MDSS_MDP_VP_0_VIG_0_CSC_1_COMP_0_POST_CLAMP

Bits	Name	Description
15:8	LOW	Unsigned 8 bit pre-clamp value
7:0	HIGH	Unsigned 8 bit pre-clamp value

0x01A05344 MDSS_MDP_VP_0_VIG_0_CSC_1_COMP_1_POST_CLAMP**Type:** RW**Reset State:** 0x000000FF

Component 1 post-clamp value

double buffered: NO

MDSS_MDP_VP_0_VIG_0_CSC_1_COMP_1_POST_CLAMP

Bits	Name	Description
15:8	LOW	Unsigned 8 bit pre-clamp value
7:0	HIGH	Unsigned 8 bit pre-clamp value

0x01A05348 MDSS_MDP_VP_0_VIG_0_CSC_1_COMP_2_POST_CLAMP**Type:** RW**Reset State:** 0x000000FF

Component 2 post-clamp value

double buffered: NO

MDSS_MDP_VP_0_VIG_0_CSC_1_COMP_2_POST_CLAMP

Bits	Name	Description
15:8	LOW	Unsigned 8 bit pre-clamp value
7:0	HIGH	Unsigned 8 bit pre-clamp value

0x01A0534C MDSS_MDP_VP_0_VIG_0_CSC_1_COMP_0_PRE_BIAS**Type:** RW**Reset State:** 0x00000000

Component 0 pre-bias value

double buffered: NO

MDSS_MDP_VP_0_VIG_0_CSC_1_COMP_0_PRE_BIAS

Bits	Name	Description
8:0	VALUE	Signed 9 bit pre-bias value

0x01A05350 MDSS_MDP_VP_0_VIG_0_CSC_1_COMP_1_PRE_BIAS**Type:** RW**Reset State:** 0x00000000

Component 1 pre-bias value

double buffered: NO

MDSS_MDP_VP_0_VIG_0_CSC_1_COMP_1_PRE_BIAS

Bits	Name	Description
8:0	VALUE	Signed 9 bit pre-bias value

0x01A05354 MDSS_MDP_VP_0_VIG_0_CSC_1_COMP_2_PRE_BIAS**Type:** RW**Reset State:** 0x00000000

Component 2 pre-bias value

double buffered: NO

MDSS_MDP_VP_0_VIG_0_CSC_1_COMP_2_PRE_BIAS

Bits	Name	Description
8:0	VALUE	Signed 9 bit pre-bias value

0x01A05358 MDSS_MDP_VP_0_VIG_0_CSC_1_COMP_0_POST_BIAS**Type:** RW**Reset State:** 0x00000000

Component 0 post-bias value

N double buffered: NO

MDSS_MDP_VP_0_VIG_0_CSC_1_COMP_0_POST_BIAS

Bits	Name	Description
8:0	VALUE	Signed 9 bit pre-bias value

0x01A0535C MDSS_MDP_VP_0_VIG_0_CSC_1_COMP_1_POST_BIAS**Type:** RW**Reset State:** 0x00000000

Component 1 post-bias value

double buffered: NO

MDSS_MDP_VP_0_VIG_0_CSC_1_COMP_1_POST_BIAS

Bits	Name	Description
8:0	VALUE	Signed 9 bit pre-bias value

0x01A05360 MDSS_MDP_VP_0_VIG_0_CSC_1_COMP_2_POST_BIAS**Type:** RW**Reset State:** 0x00000000

Component 2 post-bias value

double buffered: NO

MDSS_MDP_VP_0_VIG_0_CSC_1_COMP_2_POST_BIAS

Bits	Name	Description
8:0	VALUE	Signed 9 bit pre-bias value

0x01A05370 MDSS_MDP_VP_0_VIG_0_QSEED2_MISR_CTRL**Type:** RW**Reset State:** 0x00000000**MDSS_MDP_VP_0_VIG_0_QSEED2_MISR_CTRL**

Bits	Name	Type	Description
31	RUN_MODE	RW	
27:24	INPUT_SEL	RW	
10	STATUS_CLEAR	W	
9	STATUS	R	
8	ENABLE	RW	
7:0	FRAME_COUNT	RW	

0x01A05374 MDSS_MDP_VP_0_VIG_0_QSEED2_MISR_SIGNATURE**Type:** R**Reset State:** 0x00000000**MDSS_MDP_VP_0_VIG_0_QSEED2_MISR_SIGNATURE**

Bits	Name	Description
31:0	CAPTURED	

**0x01A06000+ MDSS_MDP_VP_0_VIG_0_HIST_V_DATAn_BIN_i, i=[0..255]
4*i****Type:** R**Reset State:** 0x00000000

Direct Register Mapped Histogram data collection

256 bins

double buffered: NO

MDSS_MDP_VP_0_VIG_0_HIST_V_DATAn_BIN_i

Bits	Name	Description
23:0	VALUE	Value

**0x01A06400+ MDSS_MDP_VP_0_VIG_0_PA_LUTV_IDX_i, i=[0..127]
4*i****Type:** RW**Reset State:** 0x00000000

Direct Register Mapped PA LUTV data

256 entries are available for writing and reading

double buffered: NO. Hardware maintains 2 copies

MDSS_MDP_VP_0_VIG_0_PA_LUTV_IDX_i

Bits	Name	Description
23:16	VALUE_UPPER	LUT content at location 2*IDX+1 to be written/read to/from the LUT accessible by software at the moment
7:0	VALUE_LOWER	LUT content at location 2*IDX to be written/read to/from the LUT accessible by software at the moment

17.2.7.1.2 MDSS_MDP_VP_0_VIG_0_SSPP

SSPP hardware register description

37.17.7.1.2.1 MDSS_MDP_VP_0_VIG_0_SSPP_SSPP

SSPP

SSPP hardware register description

0x01A05000 MDSS_MDP_VP_0_VIG_0_SSPP_SRC_SIZE

Type: RW

Reset State: 0x00000000

Source Size for the region of interest source image

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_VIG_0_SSPP_SRC_SIZE

Bits	Name	Description
31:16	ROI_H	Source Height of the Region Of Interest (ROI)
15:0	ROI_W	Source Width of the Region Of Interest (ROI) The exact supported size depends on the width of line buffers allocated for MDP.

0x01A05004 MDSS_MDP_VP_0_VIG_0_SSPP_SRC_IMG_SIZE

Type: RW

Reset State: 0x00000000

Source Size for the image in which the ROI resides

This value along with SRC_SIZE and SRC_XY will be used by hardware to calculate whether pixel repeat or overfetch is required along the ROI border.

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_VIG_0_SSPP_SRC_IMG_SIZE

Bits	Name	Description
31:16	SRC_H	Source Height. Note that this is NOT the ROI height

MDSS_MDP_VP_0_VIG_0_SSPP_SRC_IMG_SIZE (cont.)

Bits	Name	Description
15:0	SRC_W	Source Width. Note that this is NOT the ROI width

0x01A05008 MDSS_MDP_VP_0_VIG_0_SSPP_SRC_XY**Type:** RW**Reset State:** 0x00000000

Source Offset from the top-left corner of the source image for each of the region of interest dimensions

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_VIG_0_SSPP_SRC_XY

Bits	Name	Description
31:16	SRC_Y	Y-offset of source ROI in the source image
15:0	SRC_X	X-offset of source ROI in the source image

0x01A0500C MDSS_MDP_VP_0_VIG_0_SSPP_OUT_SIZE**Type:** RW**Reset State:** 0x00000000

Output Size of the layer.

Should be same as source size if scaling is NOT enabled.

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_VIG_0_SSPP_OUT_SIZE

Bits	Name	Description
31:16	DST_H	Height of Output ROI
15:0	DST_W	Width of Output ROI

0x01A05010 MDSS_MDP_VP_0_VIG_0_SSPP_OUT_XY**Type:** RW**Reset State:** 0x00000000

Output Offset from the top-left corner of the output image

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_VIG_0_SSPP_OUT_XY

Bits	Name	Description
31:16	DST_Y	Height of Output ROI
15:0	DST_X	Width of Output ROI

0x01A05014 MDSS_MDP_VP_0_VIG_0_SSPP_SRC0_ADDR**Type:** RW**Reset State:** 0x00000000

Plane 0 base address

For fetching BWC data, this plane refers to the compressed RAU data

double buffered: YES

when: On frame boundaries (vsync or SW trigger) when ADDR_SYNC_MODE is in frame mode and on line boundary if ADDR_SYNC_MODE in in line mode

MDSS_MDP_VP_0_VIG_0_SSPP_SRC0_ADDR

Bits	Name	Description
31:0	ADDR	Base byte address of the image single interleave color plane, Luma plane in pseudo-planar format, or color0 in planar format

0x01A05018 MDSS_MDP_VP_0_VIG_0_SSPP_SRC1_ADDR**Type:** RW**Reset State:** 0x00000000

Plane 1 base address

For fetching BWC data, this plane refers to the META data to fetch the lengths of the compressed RAU data stream

double buffered: YES

when: On frame boundaries (vsync or SW trigger) when ADDR_SYNC_MODE is in frame mode and on line boundary if ADDR_SYNC_MODE in in line mode

MDSS_MDP_VP_0_VIG_0_SSPP_SRC1_ADDR

Bits	Name	Description
31:0	ADDR	Base byte address of the chroma plane in pseudo-planar format, or color1 in planar format

0x01A0501C MDSS_MDP_VP_0_VIG_0_SSPP_SRC2_ADDR

Type: RW

Reset State: 0x00000000

Plane 2 base address

double buffered: YES

when: On frame boundaries (vsync or SW trigger) when ADDR_SYNC_MODE is in frame mode and on line boundary if ADDR_SYNC_MODE in in line mode

MDSS_MDP_VP_0_VIG_0_SSPP_SRC2_ADDR

Bits	Name	Description
31:0	ADDR	Base byte address of color2 in planar format

0x01A05020 MDSS_MDP_VP_0_VIG_0_SSPP_SRC3_ADDR

Type: RW

Reset State: 0x00000000

Plane 3 base address

double buffered: YES

when: On frame boundaries (vsync or SW trigger) when ADDR_SYNC_MODE is in frame mode and on line boundary if ADDR_SYNC_MODE in in line mode

MDSS_MDP_VP_0_VIG_0_SSPP_SRC3_ADDR

Bits	Name	Description
31:0	ADDR	Base byte address of color3 in planar format, or pseudo planar + alpha format

0x01A05024 MDSS_MDP_VP_0_VIG_0_SSPP_SRC_YSTRIDE0**Type:** RW**Reset State:** 0x00000000

Stride Values

Double-buffered

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_VIG_0_SSPP_SRC_YSTRIDE0

Bits	Name	Description
31:16	SRCP1_YSTRIDE	Plane 1 Y stride in bytes
15:0	SRCP0_YSTRIDE	Plane 0 Y stride in bytes

0x01A05028 MDSS_MDP_VP_0_VIG_0_SSPP_SRC_YSTRIDE1**Type:** RW**Reset State:** 0x00000000

Stride Values

Double-buffered

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_VIG_0_SSPP_SRC_YSTRIDE1

Bits	Name	Description
31:16	SRCP3_YSTRIDE	Plane 3 Y stride in bytes
15:0	SRCP2_YSTRIDE	Plane 2 Y stride in bytes

0x01A05030 MDSS_MDP_VP_0_VIG_0_SSPP_SRC_FORMAT**Type:** RW**Reset State:** 0x00000000

Source Format Configuration for each virtual pipeline.

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_VIG_0_SSPP_SRC_FORMAT

Bits	Name	Description
31:30	FRAME_FORMAT	Frame format In the current revision, there is no support for tiled formats 0x0: Linear, no tile 0x1: Tiled, 4x4
25:23	SRC_CHROMA_SAMP	Source Chroma Sampling 0x0: RGB/YUV 4:4:4 0x1: 4:2:2 - H2V1 0x2: 4:2:2 - H1V2 0x3: 4:2:0 - N2V1 0x4: 4:2:0 - N1V2
22	SOLID_FILL	Fetch feeds the lower pipeline with the CONSTANT_COLOR register as its output value. NOTHING is fetched Note that in order to use this feature then Interleaved FETCH_PLANES must be set as well and no chroma sub-sampling enabled
21	VC1_REDUCE	VC1 range reduction enable for YUV sources The reduce range is set by separate register VC1_RANGE
20:19	FETCH_PLANES	Determines the number of planes to fetch 0x0: Interleaved 0x1: Planar 0x2: Pseudo-planar
18	UNPACK_ALIGN	0x0: TO LSB 0x1: TO MSB
17	UNPACK_TIGHT	0x0: Loose 0x1: Tight
13:12	UNPACK_COUNT	Valid unpacking pattern count 0x0 = 1 component 0x1 = 2 components ... 0x3 = 4 components Unpacking applies to interleaved plane (chroma plane in pseudo-planar)
11	ROT90	Source image is to be rotated 90 degrees
10:9	SRC_BPP	Effective source byte per pixel 0x0: 1 byte 0x1: 2 bytes 0x2: 3 bytes 0x3: 4 bytes This is used ONLY when unpacking the interleaved plane

MDSS_MDP_VP_0_VIG_0_SSPP_SRC_FORMAT (cont.)

Bits	Name	Description
8	SRCC3_EN	Alpha Enable 0x0: Source has NO alpha 0x1: Source has alpha
7:6	SRCC3	Number of bits for component 3 (Alpha) 0x0: 1 bits 0x1: 4 bits 0x2: 6 bits 0x3: 8 bits
5:4	SRCC2	Number of bits for component 2 (R/Cr) 0x0: 4 bits 0x1: 5 bits 0x2: 6 bits 0x3: 8 bits
3:2	SRCC1	Number of bits for component 1 (B/Cb) 0x0: 4 bits 0x1: 5 bits 0x2: 6 bits 0x3: 8 bits
1:0	SRCC0	Number of bits for component 0 (G/Luma) 0x0: 4 bits 0x1: 5 bits 0x2: 6 bits 0x3: 8 bits

0x01A05034 MDSS_MDP_VP_0_VIG_0_SSPP_SRC_UNPACK_PATTERN**Type:** RW**Reset State:** 0x00000000

Unpacking pattern, maximum of 4 pattern elements starting from LSB to MSB (0 to 4).

Default is ARGB or C3-C2-C0-C1.

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_VIG_0_SSPP_SRC_UNPACK_PATTERN

Bits	Name	Description
25:24	ELEMENT3	0x0: Color0 0x1: Color1 0x2: Color2 0x3: Color3

MDSS_MDP_VP_0_VIG_0_SSPP_SRC_UNPACK_PATTERN (cont.)

Bits	Name	Description
17:16	ELEMENT2	0x0: Color0 0x1: Color1 0x2: Color2 0x3: Color3
9:8	ELEMENT1	0x0: Color0 0x1: Color1 0x2: Color2 0x3: Color3
1:0	ELEMENT0	0x0: Color0 0x1: Color1 0x2: Color2 0x3: Color3

0x01A05038 MDSS_MDP_VP_0_VIG_0_SSPP_SRC_OP_MODE**Type:** RW**Reset State:** 0x00000000

Operation Mode for each SSPP pipeline

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_VIG_0_SSPP_SRC_OP_MODE

Bits	Name	Description
31	SW_PIX_EXT_OVERRIDE	SW override for pixel extension values. 0x0: Use HW calculated pixel extensions 0x1: Use SW programmed pixel extensions
16	IGC_LUT_EN	Inverse Gamma Correction LUT Enable 0x0: Disabled - mapped Linearly from 8-bit to 12-bit 0x1: Enabled
14:13	FLIP_MODE	Flip Operation Lower Bit: Left/Right flip Upper Bit: Up/Down flip

0x01A0503C MDSS_MDP_VP_0_VIG_0_SSPP_SRC_CONSTANT_COLOR**Type:** RW**Reset State:** 0x00000000

Constant Color Register during solid fill mode

double buffered: NO

MDSS_MDP_VP_0_VIG_0_SSPP_SRC_CONSTANT_COLOR

Bits	Name	Description
31:24	COLOR3	Constant color driven for color3 during solid fill mode.
23:16	COLOR2	Constant color driven for color2 during solid fill mode.
15:8	COLOR1	Constant color driven for color1 during solid fill mode.
7:0	COLOR0	Constant color driven for color0 during solid fill mode.

0x01A05048 MDSS_MDP_VP_0_VIG_0_SSPP_FETCH_CONFIG

Type: RW

Reset State: 0x00000087

SSPP Client Fetch Configuration per Virtual Pipeline double buffered: NO

This register must NOT be altered during operation and MDP must be idle when changing values

MDSS_MDP_VP_0_VIG_0_SSPP_FETCH_CONFIG

Bits	Name	Description
22	PRI_ARB_DIS	
21	PMARK_DIS	
17:16	TILE_SIZE	Tile Size for processing of BLOCK formats Note that this only applies to blocks that operate in block mode which is limited to DMA source pipe. This setting has no effect for all other source pipes (ViG/RGB) 0x0: 128x128 Others: ILLEGAL / Not Supported
12	ADDR_SYNC_MODE	Boundary on which fetch address synchronizes. Setting this field to line forces the address updates on the next line which performs the software requested feature of updating on hsync. Note that updating on line boundaries will likely result in tearing on the panel since hardware is fetching from new memory buffer. 0x0: frame (default) 0x1: line

MDSS_MDP_VP_0_VIG_0_SSPP_FETCH_CONFIG (cont.)

Bits	Name	Description
9:8	MAL	Describes the Minimum Access Length for the source pipeline. This value should align with the DRAM's expected transaction alignment unit. This setting forces the source fetch pipe to be aligned to DRAM and implement re-alignment for maximum efficiency 0x0: 128b alignment (default) 0x1: 256b alignment 0x2: 512b alignment
7:4	MAX_BURST_SIZE	Maximum burst beat size. Hardware fetch engines operate on 128-bit bus Possible values: 0x8: burst of 8 0x4: burst of 4 0x2: burst of 2
3:0	REQ_DEPTH_LIMIT	Pending request limit, default max value of 0x7 means maximum pending requests allowed in the design Anything less limits the bus requests to that number + 1.

0x01A0504C MDSS_MDP_VP_0_VIG_0_SSPP_VC1_RANGE**Type:** RW**Reset State:** 0x00000707

VC1 Range Map for each virtual pipeline

ONLY valid for pipelines that suport YUV formats

double buffered: NO

MDSS_MDP_VP_0_VIG_0_SSPP_VC1_RANGE

Bits	Name	Description
10:8	RANGE_MAPUV	Chroma range map for VC1 range reduction, default of 7 is equivalent to setting it for main profile.
2:0	RANGE_MAPY	Luma range map for VC1 range reduction, default of 7 is equivalent to setting it for main profile.

0x01A05050 MDSS_MDP_VP_0_VIG_0_SSPP_REQPRIORITY_FIFO_WATERMARK_0**Type:** RW**Reset State:** 0x00000040

FIFO watermark to aid in generation of mdpREQPRIORITY

The watermark here refers to the # of 128-entries available in the Shared Memory Pool

The value programmed here should be aware of the amount programmed in the Shared Memory Pool.

So a value of 0 indicates completely FULL (i.e. there are no more entries available in the SMP)

A value of 100 indicates 100 entries in the Shared Memory Pool (each entry being 128-bit entries)

This is the lowestmost FIFO depth and should not be programmed higher than REQPRIORITY_FIFO_WATERMARK_1 or REQPRIORITY_FIFO_WATERMARK_2

An example programming would set the following:

REQPRIORITY_FIFO_WATERMARK_0 : 50 (meaning 50 entries free to be written into SMP - least room available)

REQPRIORITY_FIFO_WATERMARK_1 : 100 (meaning 100 entries free to be written into SMP - middle ground room available)

REQPRIORITY_FIFO_WATERMARK_2 : 150 (meaning 150 entries free to be written into SMP - most room available)

mdpREQPRIORITY = 11 (highest priority request) when above all 3 watermark levels (i.e. lots of space available in SMP)

mdpREQPRIORITY = 10 above FIFO_0 and FIFO_1 but below FIFO_2

mdpREQPRIORITY = 01 above FIFO_0 but below FIFO_1 and FIFO_2

mdpREQPRIORITY = 00 (lowest priority request) when below all 3 watermark levels (i.e. not a lot of space available in SMP so its good)

double buffered: YES

MDSS_MDP_VP_0_VIG_0_SSPP_REQPRIORITY_FIFO_WATERMARK_0

Bits	Name	Description
15:0	WATERMARK	Watermark Level for FIFO_0

0x01A05054 MDSS_MDP_VP_0_VIG_0_SSPP_REQPRIORITY_FIFO_WATERMARK_1

Type: RW

Reset State: 0x00000080

FIFO watermark to aid in generation of mdpREQPRIORITY

The watermark here refers to the # of 128-entries available in the Shared Memory Pool

The value programmed here should be aware of the amount programmed in the Shared Memory Pool.

So a value of 0 indicates completely FULL (i.e. there are no more entries available in the SMP)

A value of 100 indicates 100 entries in the Shared Memory Pool (each entry being 128-bit entries)

This is the middle FIFO depth and should not be programmed higher than
REQPRIORITY_FIFO_WATERMARK_2 or lower than
REQPRIORITY_FIFO_WATERMARK_0

An example programming would set the following:

REQPRIORITY_FIFO_WATERMARK_0 : 50 (meaning 50 entries free to be written into SMP - least room available)

REQPRIORITY_FIFO_WATERMARK_1 : 100 (meaning 100 entries free to be written into SMP - middle ground room available)

REQPRIORITY_FIFO_WATERMARK_2 : 150 (meaning 150 entries free to be written into SMP - most room available)

mdpREQPRIORITY = 11 (highest priority request) when above all 3 watermark levels (i.e. lots of space available in SMP)

mdpREQPRIORITY = 10 above FIFO_0 and FIFO_1 but below FIFO_2

mdpREQPRIORITY = 01 above FIFO_0 but below FIFO_1 and FIFO_2

mdpREQPRIORITY = 00 (lowest priority request) when below all 3 watermark levels (i.e. not a lot of space available in SMP so its good)

double buffered: YES

MDSS_MDP_VP_0_VIG_0_SSPP_REQPRIORITY_FIFO_WATERMARK_1

Bits	Name	Description
15:0	WATERMARK	Watermark Level for FIFO_1

0x01A05058 MDSS_MDP_VP_0_VIG_0_SSPP_REQPRIORITY_FIFO_WATERMARK_2

Type: RW

Reset State: 0x000000C0

FIFO watermark to aid in generation of mdpREQPRIORITY

The watermark here refers to the # of 128-entries available in the Shared Memory Pool

The value programmed here should be aware of the amount programmed in the Shared Memory Pool.

So a value of 0 indicates completely FULL (i.e. there are no more entries available in the SMP)

A value of 100 indicates 100 entries in the Shared Memory Pool (each entry being 128-bit entries)

This is the highest FIFO depth and should not be programmed lower than
REQPRIORITY_FIFO_WATERMARK_0 or REQPRIORITY_FIFO_WATERMARK_1

An example programming would set the following:

REQPRIORITY_FIFO_WATERMARK_0 : 50 (meaning 50 entries free to be written into SMP - least room available)

REQPRIORITY_FIFO_WATERMARK_1 : 100 (meaning 100 entries free to be written into SMP - middle ground room available)

REQPRIORITY_FIFO_WATERMARK_2 : 150 (meaning 150 entries free to be written into SMP - most room available)

mdpREQPRIORITY = 11 (highest priority request) when above all 3 watermark levels (i.e. lots of space available in SMP)

mdpREQPRIORITY = 10 above FIFO_0 and FIFO_1 but below FIFO_2

mdpREQPRIORITY = 01 above FIFO_0 but below FIFO_1 and FIFO_2

mdpREQPRIORITY = 00 (lowest priority request) when below all 3 watermark levels (i.e. not a lot of space available in SMP so its good)

double buffered: YES

MDSS_MDP_VP_0_VIG_0_SSPP_REQPRIORITY_FIFO_WATERMARK_2

Bits	Name	Description
15:0	WATERMARK	Watermark Level for FIFO_2

0x01A05070 MDSS_MDP_VP_0_VIG_0_SSPP_SRC_ADDR_SW_STATUS

Type: RW

Reset State: 0x00000000

Security Status of the source buffer plane

This is programmed by SW to indicate if the buffer is NON-SECURE or SECURE

This information will be presented to VBIF per transaction and reflected as part of the memory transaction

double buffered: YES

MDSS_MDP_VP_0_VIG_0_SSPP_SRC_ADDR_SW_STATUS

Bits	Name	Description
3	SRC3_ADDR_CP	This is the secure status bit for source plane 3 (if used) If the format doesnt fetch from that plane then this value is ignored Memory transactions will not complete if this status doesnt corespond to system programming in the rest of the path to memory SW should program SRC0_ADDR_CP, SRC1_ADDR_CP, SRC2_ADDR_CP, SRC3_ADDR_CP to the same value 0x0: Non Content Protected Buffer State 0x1: Content Protected Buffer State
2	SRC2_ADDR_CP	This is the secure status bit for source plane 2 (if used) If the format doesnt fetch from that plane then this value is ignored Memory transactions will not complete if this status doesnt corespond to system programming in the rest of the path to memory SW should program SRC0_ADDR_CP, SRC1_ADDR_CP, SRC2_ADDR_CP, SRC3_ADDR_CP to the same value 0x0: Non Content Protected Buffer State 0x1: Content Protected Buffer State
1	SRC1_ADDR_CP	This is the secure status bit for source plane 1 (if used) If the format doesnt fetch from that plane then this value is ignored Memory transactions will not complete if this status doesnt corespond to system programming in the rest of the path to memory SW should program SRC0_ADDR_CP, SRC1_ADDR_CP, SRC2_ADDR_CP, SRC3_ADDR_CP to the same value 0x0: Non Content Protected Buffer State 0x1: Content Protected Buffer State
0	SRC0_ADDR_CP	This is the secure status bit for source plane 0 (if used) If the format doesnt fetch from that plane then this value is ignored Memory transactions will not complete if this status doesnt corespond to system programming in the rest of the path to memory SW should program SRC0_ADDR_CP, SRC1_ADDR_CP, SRC2_ADDR_CP, SRC3_ADDR_CP to the same value 0x0: Non Content Protected Buffer State 0x1: Content Protected Buffer State

0x01A050A4 MDSS_MDP_VP_0_VIG_0_SSPP_CURRENT_SRC0_ADDR**Type:** R**Reset State:** 0x00000000

Current Plane 0 base address that Hardware is using

For fetching BWC data, this plane refers to the compressed RAU data

double buffered: N/A

MDSS_MDP_VP_0_VIG_0_SSPP_CURRENT_SRC0_ADDR

Bits	Name	Description
31:0	ADDR	Base byte address of the image single interleave color plane, Luma plane in pseudo-planar format, or color0 in planar format

0x01A050A8 MDSS_MDP_VP_0_VIG_0_SSPP_CURRENT_SRC1_ADDR

Type: R

Reset State: 0x00000000

Plane 1 base address that Hardware is using

For fetching BWC data, this plane refers to the META data to fetch the lengths of the compressed RAU data stream

double buffered: N/A

MDSS_MDP_VP_0_VIG_0_SSPP_CURRENT_SRC1_ADDR

Bits	Name	Description
31:0	ADDR	Base byte address of the chroma plane in pseudo-planar format, or color1 in planar format

0x01A050AC MDSS_MDP_VP_0_VIG_0_SSPP_CURRENT_SRC2_ADDR

Type: R

Reset State: 0x00000000

Plane 2 base address that Hardware is using

double buffered: N/A

MDSS_MDP_VP_0_VIG_0_SSPP_CURRENT_SRC2_ADDR

Bits	Name	Description
31:0	ADDR	Base byte address of color2 in planar format

0x01A050B0 MDSS_MDP_VP_0_VIG_0_SSPP_CURRENT_SRC3_ADDR

Type: R

Reset State: 0x00000000

Plane 3 base address that Hardware is using

double buffered: N/A

MDSS_MDP_VP_0_VIG_0_SSPP_CURRENT_SRC3_ADDR

Bits	Name	Description
31:0	ADDR	Base byte address of color3 in planar format, or pseudo planar + alpha format

0x01A050B4 MDSS_MDP_VP_0_VIG_0_SSPP_DECIMATION_CONFIG

Type: RW

Reset State: 0x00000000

Decimation Configuration

Note that there is no decimation on DMA pipe which means no decimation in line mode on DMA or block mode

double buffered: YES

MDSS_MDP_VP_0_VIG_0_SSPP_DECIMATION_CONFIG

Bits	Name	Description
11:8	HORZ_S	Horizontal Configuration for decimation; required to support downscaling greater than 1/4 S refers to the number of pixels skipped. The algorithm is always the following: read 1 pixel skip S pixels The algorithm is applied to ALL components. Decimation only supports the following values. All other values will result in undefined behaviour The decimation block will provide coarse scaling and should be used in conjunction with qseed2 to support all scaling ratios. 0x0: No decimation is performed 0x1: Read 1 pixel drop 1 pixel. Used to decimate by 2 horizontally 0x3: Read 1 pixel drop 3 pixels. Used to decimate by 4 horizontally 0x7: Read 1 pixel drop 7 pixels. Used to decimate by 8 horizontally 0xF: Read 1 pixel drop 15 pixels. Used to decimate by 16 horizontally

MDSS_MDP_VP_0_VIG_0_SSPP_DECIMATION_CONFIG (cont.)

Bits	Name	Description
3:0	VERT_S	Vertical Configuration for decimation; required to support downscaling greater than 1/4 S refers to the number of lines skipped. The algorithm is always the following: read 1 line skip S lines The algorithm is applied to ALL components. Decimation only supports the following values. All other values will result in undefined behaviour The decimation block will provide coarse scaling and should be used in conjunction with qseed2 to support all scaling ratios. 0x0: No decimation is performed 0x1: Read 1 line drop 1 line. Used to decimate by 2 horizontally 0x3: Read 1 line drop 3 lines. Used to decimate by 4 horizontally 0x7: Read 1 line drop 7 lines. Used to decimate by 8 horizontally 0xF: Read 1 line drop 15 lines. Used to decimate by 16 horizontally

0x01A050D0 MDSS_MDP_VP_0_VIG_0_SSPP_FETCH_SMP_WR_PLANE0**Type:** R**Reset State:** 0x00000000

Fetch to SMP on plane 0 write counter

double buffered: N/A

MDSS_MDP_VP_0_VIG_0_SSPP_FETCH_SMP_WR_PLANE0

Bits	Name	Description
15:0	COUNT	This counter is reset on every frame and is incremented for each write fetch makes to the SMP plane0 port This port is used for interleaved formats or the Y component for multi-plane source formats This should correlate to value SMP_UNPACK_RD_PLANE0 More useful for debug but can be used for profiling purposes as well.

0x01A050D4 MDSS_MDP_VP_0_VIG_0_SSPP_FETCH_SMP_WR_PLANE1**Type:** R**Reset State:** 0x00000000

Fetch to SMP on plane 1 write counter

double buffered: N/A

MDSS_MDP_VP_0_VIG_0_SSPP_FETCH_SMP_WR_PLANE1

Bits	Name	Description
15:0	COUNT	This counter is reset on every frame and is incremented for each write fetch makes to the SMP plane1 port This port is NOT used for interleaved formats but is used for U (planar) or UV (pseudo-planar) formats This should correlate to value SMP_UNPACK_RD_PLANE1 More useful for debug but can be used for profiling purposes as well.

0x01A050D8 MDSS_MDP_VP_0_VIG_0_SSPP_FETCH_SMP_WR_PLANE2**Type:** R**Reset State:** 0x00000000

Fetch to SMP on plane 2 write counter

double buffered: N/A

MDSS_MDP_VP_0_VIG_0_SSPP_FETCH_SMP_WR_PLANE2

Bits	Name	Description
15:0	COUNT	This counter is reset on every frame and is incremented for each write fetch makes to the SMP plane2 port This port is NOT used for interleaved formats but is used for V (planar) or A (pseudo-planar plus alpha) formats This should correlate to value SMP_UNPACK_RD_PLANE2 More useful for debug but can be used for profiling purposes as well.

0x01A050E0 MDSS_MDP_VP_0_VIG_0_SSPP_SMP_UNPACK_RD_PLANE0**Type:** R**Reset State:** 0x00000000

SMP to unpack on plane 0 read counter

double buffered: N/A

MDSS_MDP_VP_0_VIG_0_SSPP_SMP_UNPACK_RD_PLANE0

Bits	Name	Description
15:0	COUNT	This counter is reset on every frame and is incremented for each line read from SMP plane0 port This port is used for interleaved formats or the Y component for multi-plane source formats This should correlate to FETCH_SMP_WR_PLANE0 value More useful for debug but can be used for profiling purposes as well.

0x01A050E4 MDSS_MDP_VP_0_VIG_0_SSPP_SMP_UNPACK_RD_PLANE1**Type:** R**Reset State:** 0x00000000

SMP to unpack on plane 1 read counter

double buffered: N/A

MDSS_MDP_VP_0_VIG_0_SSPP_SMP_UNPACK_RD_PLANE1

Bits	Name	Description
15:0	COUNT	This counter is reset on every frame and is incremented for each line read from SMP plane1 port This port is NOT used for interleaved formats but is used for U (planar) or UV (pseudo-planar) formats This should correlate to FETCH_SMP_WR_PLANE1 value More useful for debug but can be used for profiling purposes as well.

0x01A050E8 MDSS_MDP_VP_0_VIG_0_SSPP_SMP_UNPACK_RD_PLANE2**Type:** R**Reset State:** 0x00000000

SMP to unpack on plane 2 read counter

double buffered: N/A

MDSS_MDP_VP_0_VIG_0_SSPP_SMP_UNPACK_RD_PLANE2

Bits	Name	Description
15:0	COUNT	This counter is reset on every frame and is incremented for each line read from SMP plane1 port This port is NOT used for interleaved formats but is used for V (planar) or A (pseudo-planar plus alpha) formats This should correlate to FETCH_SMP_WR_PLANE1 value More useful for debug but can be used for profiling purposes as well.

0x01A05100 MDSS_MDP_VP_0_VIG_0_SSPP_SW_PIX_EXT_C0_LR**Type:** RW**Reset State:** 0x00000000

SW Pixel Extension Override for Left/Right of Color 0

These value only take effect if SW_PIX_EXT_OVERRIDE is set in SRC_OP_MODE register

double buffered: YES

MDSS_MDP_VP_0_VIG_0_SSPP_SW_PIX_EXT_C0_LR

Bits	Name	Description
31:24	RIGHT_OVF	Signed Value for Right Overfetch value. +ve value indicates overfetch -ve value indicates pixel drop
23:16	RIGHT_RPT	Unsigned Value for Right Pixel Repeat value.
15:8	LEFT_OVF	Signed Value for Left Overfetch value. +ve value indicates overfetch -ve value indicates pixel drop
7:0	LEFT_RPT	Unsigned Value for Left Pixel Repeat value.

0x01A05104 MDSS_MDP_VP_0_VIG_0_SSPP_SW_PIX_EXT_C0_TB**Type:** RW**Reset State:** 0x00000000

SW Pixel Extension Override for Left/Right of Color 0

These value only take effect if SW_PIX_EXT_OVERRIDE is set in SRC_OP_MODE register

double buffered: YES

MDSS_MDP_VP_0_VIG_0_SSPP_SW_PIX_EXT_C0_TB

Bits	Name	Description
31:24	BOTTOM_OVF	Signed Value for Bottom Overfetch value. +ve value indicates overfetch -ve value indicates line drop
23:16	BOTTOM_RPT	
15:8	TOP_OVF	Signed Value for Top Overfetch value. +ve value indicates overfetch -ve value indicates line drop
7:0	TOP_RPT	Unsigned Value for TOP Line Repeat value.

0x01A05108 MDSS_MDP_VP_0_VIG_0_SSPP_SW_PIX_EXT_C0_REQ_PIXELS**Type:** RW**Reset State:** 0x00000000

SW Pixel Extension Override for pixel height/width of Color 0

These value only take effect if SW_PIX_EXT_OVERRIDE is set in SRC_OP_MODE register

double buffered: YES

MDSS_MDP_VP_0_VIG_0_SSPP_SW_PIX_EXT_C0_REQ_PIXELS

Bits	Name	Description
31:16	TOP_BOTTOM	Unsigned Value for pixel height post pixel extension.
15:0	LEFT_RIGHT	Unsigned Value for pixel width post pixel extension.

0x01A05110 MDSS_MDP_VP_0_VIG_0_SSPP_SW_PIX_EXT_C1C2_LR**Type:** RW**Reset State:** 0x00000000

SW Pixel Extension Override for Left/Right of Color 0

These value only take effect if SW_PIX_EXT_OVERRIDE is set in SRC_OP_MODE register

double buffered: YES

MDSS_MDP_VP_0_VIG_0_SSPP_SW_PIX_EXT_C1C2_LR

Bits	Name	Description
31:24	RIGHT_OVF	Signed Value for Right Overfetch value. +ve value indicates overfetch -ve value indicates pixel drop
23:16	RIGHT_RPT	Unsigned Value for Right Pixel Repeat value.
15:8	LEFT_OVF	Signed Value for Left Overfetch value. +ve value indicates overfetch -ve value indicates pixel drop
7:0	LEFT_RPT	Unsigned Value for Left Pixel Repeat value.

0x01A05114 MDSS_MDP_VP_0_VIG_0_SSPP_SW_PIX_EXT_C1C2_TB**Type:** RW**Reset State:** 0x00000000

SW Pixel Extension Override for Left/Right of Color 0

These value only take effect if SW_PIX_EXT_OVERRIDE is set in SRC_OP_MODE register
double buffered: YES

MDSS_MDP_VP_0_VIG_0_SSPP_SW_PIX_EXT_C1C2_TB

Bits	Name	Description
31:24	BOTTOM_OVF	Signed Value for Bottom Overfetch value. +ve value indicates overfetch -ve value indicates line drop
23:16	BOTTOM_RPT	
15:8	TOP_OVF	Signed Value for Top Overfetch value. +ve value indicates overfetch -ve value indicates line drop
7:0	TOP_RPT	Unsigned Value for TOP Line Repeat value.

0x01A05118 MDSS_MDP_VP_0_VIG_0_SSPP_SW_PIX_EXT_C1C2_REQ_PIXELS**Type:** RW**Reset State:** 0x00000000

SW Pixel Extension Override for pixel height/width of Color 0

These value only take effect if SW_PIX_EXT_OVERRIDE is set in SRC_OP_MODE register

double buffered: YES

MDSS_MDP_VP_0_VIG_0_SSPP_SW_PIX_EXT_C1C2_REQ_PIXELS

Bits	Name	Description
31:16	TOP_BOTTOM	Unsigned Value for pixel height post pixel extension.
15:0	LEFT_RIGHT	Unsigned Value for pixel width post pixel extension.

0x01A05120 MDSS_MDP_VP_0_VIG_0_SSPP_SW_PIX_EXT_C3_LR

Type: RW

Reset State: 0x00000000

SW Pixel Extension Override for Left/Right of Color 0

These value only take effect if SW_PIX_EXT_OVERRIDE is set in SRC_OP_MODE register

double buffered: YES

MDSS_MDP_VP_0_VIG_0_SSPP_SW_PIX_EXT_C3_LR

Bits	Name	Description
31:24	RIGHT_OVF	Signed Value for Right Overfetch value. +ve value indicates overfetch -ve value indicates pixel drop
23:16	RIGHT_RPT	Unsigned Value for Right Pixel Repeat value.
15:8	LEFT_OVF	Signed Value for Left Overfetch value. +ve value indicates overfetch -ve value indicates pixel drop
7:0	LEFT_RPT	Unsigned Value for Left Pixel Repeat value.

0x01A05124 MDSS_MDP_VP_0_VIG_0_SSPP_SW_PIX_EXT_C3_TB

Type: RW

Reset State: 0x00000000

SW Pixel Extension Override for Left/Right of Color 0

These value only take effect if SW_PIX_EXT_OVERRIDE is set in SRC_OP_MODE register

double buffered: YES

MDSS_MDP_VP_0_VIG_0_SSPP_SW_PIX_EXT_C3_TB

Bits	Name	Description
31:24	BOTTOM_OVF	Signed Value for Bottom Overfetch value. +ve value indicates overfetch -ve value indicates line drop
23:16	BOTTOM_RPT	
15:8	TOP_OVF	Signed Value for Top Overfetch value. +ve value indicates overfetch -ve value indicates line drop
7:0	TOP_RPT	Unsigned Value for TOP Line Repeat value.

0x01A05128 MDSS_MDP_VP_0_VIG_0_SSPP_SW_PIX_EXT_C3_REQ_PIXELS**Type:** RW**Reset State:** 0x00000000

SW Pixel Extension Override for pixel height/width of Color 0

These value only take effect if SW_PIX_EXT_OVERRIDE is set in SRC_OP_MODE register

double buffered: YES

MDSS_MDP_VP_0_VIG_0_SSPP_SW_PIX_EXT_C3_REQ_PIXELS

Bits	Name	Description
31:16	TOP_BOTTOM	Unsigned Value for pixel height post pixel extension.
15:0	LEFT_RIGHT	Unsigned Value for pixel width post pixel extension.

17.17.7.2 MDSS_MDP_VP_0_VIG_1

VIG hardware register description

17.17.7.2.1 MDSS_MDP_VP_0_VIG_1_VIG

VIG

0x01A07200 MDSS_MDP_VP_0_VIG_1_OP_MODE**Type:** RW**Reset State:** 0x00040000

Virtual Pipeline operation mode

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_VIG_1_OP_MODE

Bits	Name	Description
28	PA_CONT_MASK	0x0: Disable HSIC adj on CONTRAST 0x1: Enable HSIC adj on CONTRAST
27	PA_VAL_MASK	0x0: Disable HSIC adj on VALUE 0x1: Enable HSIC adj on VALUE
26	PA_SAT_MASK	0x0: Disable HSIC adj on SATURATION 0x1: Enable HSIC adj on SATURATION
25	PA_HUE_MASK	0x0: Disable HSIC adj on HUE 0x1: Enable HSIC adj on HUE
19	CSC_DST_DATA_FORMAT	Destination Data Format 0x0: RGB 0x1: YCbCr
18	CSC_SRC_DATA_FORMAT	Source Data Format 0x0: RGB 0x1: YCbCr
17	CSC_1_EN	Enables the color matrix conversion 1 0x0: Bypass 0x1: Enabled
10	PA_LUTV_EN	Histogram Enhancement Lookup Table enable
8	HIST_EN	Histogram Enable 0x0: Disabled 0x1: Enabled
7	PA_MEM_COL_SKY_MASK	0x0: Disable Sky adjustment 0x1: Enable Sky adjustment
6	PA_MEM_COL_FOL_MASK	0x0: Disable Foliage adjustment 0x1: Enable Foliage adjustment
5	PA_MEM_COL_SKIN_MASK	0x0: Disable Skin adjustment 0x1: Enable Skin adjustment
4	PA_EN	Enable Picture Adjustment 0x0: Disabled 0x1: Enabled
3	PA_MEM_PROTECT_ENABLE	0: global hsv adjustment is independent of memor color 1: global hsv adjustment is active if the pixel is not already processed by memory color contrast adjustment in global adjustment is always independ of memory color regardless

MDSS_MDP_VP_0_VIG_1_OP_MODE (cont.)

Bits	Name	Description
2	PA_SAT_ZERO_EXP_ENAB LE	0: if PA_SAT_ADJ < -32, clamp PA_SAT_ADJ to 0 1: if PA_SAT_ADJ < -64, clamp PA_SAT_ADJ to 0

0x01A07204 MDSS_MDP_VP_0_VIG_1_QSEED2_CONFIG**Type:** RW**Reset State:** 0x00000000

QSEED configuration for each of the Virtual Pipelines

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_VIG_1_QSEED2_CONFIG

Bits	Name	Description
19:18	COMP3_VERT_FILTER_MODE	Component 3 (A) vertical filtering mode 0x0: Nearest Neighbor 0x1: Bilinear 0x2: PCMN
17:16	COMP3_HORZ_FILTER_MODE	Component 3 (A) horizontal filtering mode 0x0: Nearest Neighbor 0x1: Bilinear 0x2: PCMN
15:14	COMP1_2_VERT_FILTER_MODE	Component 1/2 (CrCb or RB) vertical filtering mode 0x0: Nearest Neighbor 0x1: Bilinear 0x2: PCMN
13:12	COMP1_2_HORZ_FILTER_MODE	Component 1/2 (CrCb or RB) horizontal filtering mode 0x0: Nearest Neighbor 0x1: Bilinear 0x2: PCMN
11:10	COMP0_VERT_FILTER_MODE	Component 0 (Y or G) vertical filtering mode 0x0: Nearest Neighbor 0x1: Bilinear 0x2: PCMN 0x3: Content Adaptive
9:8	COMP0_HORZ_FILTER_MODE	Component 0 (Y or G) horizontal filtering mode 0x0: Nearest Neighbor 0x1: Bilinear 0x2: PCMN 0x3: Content Adaptive

MDSS_MDP_VP_0_VIG_1_QSEED2_CONFIG (cont.)

Bits	Name	Description
1	SCALEY_EN	Scaling Enable in the Y-direction 0x0: Disabled 0x1: Enabled
0	SCALEX_EN	Scaling Enable in the X-direction 0x0: Disabled 0x1: Enabled

0x01A07210 MDSS_MDP_VP_0_VIG_1_QSEED2_COMP0_3_PHASE_STEP_X**Type:** RW**Reset State:** 0x00000000

QSEED component 0 and 3 phase step for each of the Virtual Pipelines

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_VIG_1_QSEED2_COMP0_3_PHASE_STEP_X

Bits	Name	Description
25:0	PHASE_STEP_X	Unsigned Phase Step in the X-direction

0x01A07214 MDSS_MDP_VP_0_VIG_1_QSEED2_COMP0_3_PHASE_STEP_Y**Type:** RW**Reset State:** 0x00000000

QSEED component 0 and 3 phase step for each of the Virtual Pipelines

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_VIG_1_QSEED2_COMP0_3_PHASE_STEP_Y

Bits	Name	Description
25:0	PHASE_STEP_Y	Unsigned Phase Step in the Y-direction

0x01A07218 MDSS_MDP_VP_0_VIG_1_QSEED2_COMP1_2_PHASE_STEP_X**Type:** RW**Reset State:** 0x00000000

QSEED component 1 and 2 phase step for each of the Virtual Pipelines

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_VIG_1_QSEED2_COMP1_2_PHASE_STEP_X

Bits	Name	Description
25:0	PHASE_STEP_X	Unsigned Phase Step in the X-direction

0x01A0721C MDSS_MDP_VP_0_VIG_1_QSEED2_COMP1_2_PHASE_STEP_Y**Type:** RW**Reset State:** 0x00000000

QSEED component 1 and 2 phase step for each of the Virtual Pipelines

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_VIG_1_QSEED2_COMP1_2_PHASE_STEP_Y

Bits	Name	Description
25:0	PHASE_STEP_Y	Unsigned Phase Step in the Y-direction

0x01A07220 MDSS_MDP_VP_0_VIG_1_QSEED2_COMP0_3_INIT_PHASE_X**Type:** RW**Reset State:** 0x00000000

QSEED component 0 and 3 init phase for each of the Virtual Pipelines

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_VIG_1_QSEED2_COMP0_3_INIT_PHASE_X

Bits	Name	Description
25:0	INIT_PHASE_X	Signed INIT Phase in the X-direction

0x01A07224 MDSS_MDP_VP_0_VIG_1_QSEED2_COMP0_3_INIT_PHASE_Y**Type:** RW**Reset State:** 0x00000000

QSEED component 0 and 3 init phase for each of the Virtual Pipelines

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_VIG_1_QSEED2_COMP0_3_INIT_PHASE_Y

Bits	Name	Description
25:0	INIT_PHASE_Y	Signed INIT Phase in the Y-direction

0x01A07228 MDSS_MDP_VP_0_VIG_1_QSEED2_COMP1_2_INIT_PHASE_X**Type:** RW**Reset State:** 0x00000000

QSEED component 1 and 2 init phase for each of the Virtual Pipelines

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_VIG_1_QSEED2_COMP1_2_INIT_PHASE_X

Bits	Name	Description
25:0	INIT_PHASE_X	Signed INIT Phase in the X-direction

0x01A0722C MDSS_MDP_VP_0_VIG_1_QSEED2_COMP1_2_INIT_PHASE_Y**Type:** RW**Reset State:** 0x00000000

QSEED component 1 and 2 init phase for each of the Virtual Pipelines

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_VIG_1_QSEED2_COMP1_2_INIT_PHASE_Y

Bits	Name	Description
25:0	INIT_PHASE_Y	Signed INIT Phase in the Y-direction

0x01A07230 MDSS_MDP_VP_0_VIG_1_QSEED2_SHARP_SMOOTH_STRENGTH**Type:** RW**Reset State:** 0x00000000

QSEED configuration for each of the Virtual Pipelines

double buffered: YES

MDSS_MDP_VP_0_VIG_1_QSEED2_SHARP_SMOOTH_STRENGTH

Bits	Name	Description
8:0	STRENGTH	Sharpening/Smooth Strength used for content Adaptive Filtering S9 notation with a range from -256 to 255 0x0: No sharpening and No smoothing +ve: sharpening -ve: smoothing

0x01A07234 MDSS_MDP_VP_0_VIG_1_QSEED2_SHARP_THRESHOLD_EDGE**Type:** RW**Reset State:** 0x00000000

QSEED configuration for each of the Virtual Pipelines

double buffered: YES

MDSS_MDP_VP_0_VIG_1_QSEED2_SHARP_THRESHOLD_EDGE

Bits	Name	Description
9:0	THRESHOLD	Unsigned Threshold to select Edge/Normal Filter for Content Adaptive Filter

0x01A07238 MDSS_MDP_VP_0_VIG_1_QSEED2_SHARP_THRESHOLD_SMOOTH**Type:** RW**Reset State:** 0x00000000

QSEED configuration for each of the Virtual Pipelines

double buffered: YES

MDSS_MDP_VP_0_VIG_1_QSEED2_SHARP_THRESHOLD_SMOOTH

Bits	Name	Description
9:0	THRESHOLD	Unsigned Threshold to select Normal/Smooth Filter for Content Adaptive Filter

0x01A0723C MDSS_MDP_VP_0_VIG_1_QSEED2_SHARP_THRESHOLD_NOISE**Type:** RW**Reset State:** 0x00000000

QSEED configuration for each of the Virtual Pipelines

double buffered: YES

MDSS_MDP_VP_0_VIG_1_QSEED2_SHARP_THRESHOLD_NOISE

Bits	Name	Description
7:0	THRESHOLD	Unsigned Threshold to select Normal/Smooth Filter for Content Adaptive Filter

0x01A07250 MDSS_MDP_VP_0_VIG_1_QSEED2_NORMAL_FILTER_COEFF_N**Type:** RW**Reset State:** 0x00000000**MDSS_MDP_VP_0_VIG_1_QSEED2_NORMAL_FILTER_COEFF_N**

Bits	Name	Description
11:0	VALUE	

0x01A07260 MDSS_MDP_VP_0_VIG_1_QSEED2_SMOOTH_FILTER_COEFF_N**Type:** RW**Reset State:** 0x00000000**MDSS_MDP_VP_0_VIG_1_QSEED2_SMOOTH_FILTER_COEFF_N**

Bits	Name	Description
11:0	VALUE	

0x01A07270 MDSS_MDP_VP_0_VIG_1_QSEED2_EDGE_FILTER_COEFF_START**Type:** W**Reset State:** 0x00000000

QSEED SW trigger to calculate the edge filter coefficients

double buffered: NO

MDSS_MDP_VP_0_VIG_1_QSEED2_EDGE_FILTER_COEFF_START

Bits	Name	Description
0	START	SW trigger to initiate the calculate of edge coefficients in hardware The pipeline should be idle as smooth/normal filter coefficients are required This implies that the QSEED2 is not active during the trigger

0x01A07288 MDSS_MDP_VP_0_VIG_1_SKIN_ADJUST_PWL0**Type:** RW**Reset State:** 0x00000000

Skin Piece-wise Linear Adjustment Curve

double buffered: Yes

MDSS_MDP_VP_0_VIG_1_SKIN_ADJUST_PWL0

Bits	Name	Description
31:16	HUE_GAIN	S16, is a symmetrical trapezoidal curve, this is the slope of the 2 sides
10:0	HUE_MID	U11, this is the offset from the min and max hue of skin region which marks the top 2 corners of the trapezoid

0x01A0728C MDSS_MDP_VP_0_VIG_1_SKIN_ADJUST_PWL1**Type:** RW**Reset State:** 0x00000000

Skin Piece-wise Linear Adjustment Curve

double buffered: Yes

MDSS_MDP_VP_0_VIG_1_SKIN_ADJUST_PWL1

Bits	Name	Description
31:16	VAL_GAIN	S16, is a symmetrical trapezoidal curve, this is the slope of the 2 sides
15:0	SAT_GAIN	S16, is a symmetrical trapezoidal curve, this is the slope of the 2 sides

0x01A07290 MDSS_MDP_VP_0_VIG_1_SKIN_REGION_HUE**Type:** RW**Reset State:** 0x00000000

Skin Adjustment Hue Qualifier

double buffered: Yes

MDSS_MDP_VP_0_VIG_1_SKIN_REGION_HUE

Bits	Name	Description
26:16	MAX	if MAX > x17ff (i.e. 360 degree), the behaviour will be same as MAX = x1800
10:0	MIN	U11, Hue has to be within the inclusive range of MIN to MAX to qualify for skin tone adjustment on the pixel if MIN > x17ff (i.e. 360 degree), the behaviour will be same as MIN = x1800

0x01A07294 MDSS_MDP_VP_0_VIG_1_SKIN_REGION_SAT**Type:** RW**Reset State:** 0x00000000

Skin Adjustment Saturation Qualifier

double buffered: Yes

MDSS_MDP_VP_0_VIG_1_SKIN_REGION_SAT

Bits	Name	Description
15:8	MAX	Saturation has to be below or equal to this to qualify for skin tone adjustment on the pixel
7:0	MIN	U8, Saturation has to be above or equal to this to qualify for skin tone adjustment on the pixel

0x01A07298 MDSS_MDP_VP_0_VIG_1_SKIN_REGION_VAL**Type:** RW**Reset State:** 0x00000000

Skin Adjustment Value Qualifier

double buffered: Yes

MDSS_MDP_VP_0_VIG_1_SKIN_REGION_VAL

Bits	Name	Description
15:8	MAX	Value has to be below or equal to this to qualify for skin tone adjustment on the pixel
7:0	MIN	U8, Value has to be above or equal to this to qualify for skin tone adjustment on the pixel

0x01A0729C MDSS_MDP_VP_0_VIG_1_SKY_ADJUST_PWL0**Type:** RW**Reset State:** 0x00000000

Sky Piece-wise Linear Adjustment Curve

double buffered: Yes

MDSS_MDP_VP_0_VIG_1_SKY_ADJUST_PWL0

Bits	Name	Description
31:16	HUE_GAIN	S16, is a symmetrical trapezoidal curve, this is the slope of the 2 sides
10:0	HUE_MID	U11, this is the offset from the min and max hue of skin region which marks the top 2 corners of the trapezoid

0x01A072A0 MDSS_MDP_VP_0_VIG_1_SKY_ADJUST_PWL1**Type:** RW**Reset State:** 0x00000000

Sky Piece-wise Linear Adjustment Curve

double buffered: Yes

MDSS_MDP_VP_0_VIG_1_SKY_ADJUST_PWL1

Bits	Name	Description
31:16	VAL_GAIN	S16, is a symmetrical trapezoidal curve, this is the slope of the 2 sides
15:0	SAT_GAIN	S16, is a symmetrical trapezoidal curve, this is the slope of the 2 sides

0x01A072A4 MDSS_MDP_VP_0_VIG_1_SKY_REGION_HUE**Type:** RW**Reset State:** 0x00000000

Sky Adjustment Hue Qualifier

double buffered: Yes

MDSS_MDP_VP_0_VIG_1_SKY_REGION_HUE

Bits	Name	Description
26:16	MAX	if MAX > x17ff (i.e. 360 degree), the behaviour will be same as MAX = x1800
10:0	MIN	U11, Hue has to be within the inclusive range of MIN to MAX (U13) to qualify for sky tone adjustment on the pixel if MIN > x17ff (i.e. 360 degree), the behaviour will be same as MIN = x1800

0x01A072A8 MDSS_MDP_VP_0_VIG_1_SKY_REGION_SAT**Type:** RW**Reset State:** 0x00000000

Sky Adjustment Saturation Qualifier

double buffered: Yes

MDSS_MDP_VP_0_VIG_1_SKY_REGION_SAT

Bits	Name	Description
15:8	MAX	Saturation has to be below or equal to this to qualify for sky adjustment on the pixel
7:0	MIN	U8, Saturation has to be above or equal to this to qualify for sky adjustment on the pixel

0x01A072AC MDSS_MDP_VP_0_VIG_1_SKY_REGION_VAL**Type:** RW**Reset State:** 0x00000000

Sky Adjustment Value Qualifier

double buffered: Yes

MDSS_MDP_VP_0_VIG_1_SKY_REGION_VAL

Bits	Name	Description
15:8	MAX	Value has to be below or equal to this to qualify for sky adjustment on the pixel
7:0	MIN	U8, Value has to be above or equal to this to qualify for sky adjustment on the pixel

0x01A072B0 MDSS_MDP_VP_0_VIG_1_FOLIAGE_ADJUST_PWL0**Type:** RW**Reset State:** 0x00000000**MDSS_MDP_VP_0_VIG_1_FOLIAGE_ADJUST_PWL0**

Bits	Name	Description
31:16	HUE_GAIN	
10:0	HUE_MID	

0x01A072B4 MDSS_MDP_VP_0_VIG_1_FOLIAGE_ADJUST_PWL1**Type:** RW**Reset State:** 0x00000000**MDSS_MDP_VP_0_VIG_1_FOLIAGE_ADJUST_PWL1**

Bits	Name	Description
31:16	VAL_GAIN	
15:0	SAT_GAIN	

0x01A072B8 MDSS_MDP_VP_0_VIG_1_FOLIAGE_REGION_HUE**Type:** RW**Reset State:** 0x00000000

Foliage Adjustment Hue Qualifier

double buffered: Yes

MDSS_MDP_VP_0_VIG_1_FOLIAGE_REGION_HUE

Bits	Name	Description
26:16	MAX	if MAX > x17ff (i.e. 360 degree), the behaviour will be same as MAX = x1800
10:0	MIN	U11, Hue has to be within the inclusive range of MIN to MAX to qualify for foliage tone adjustment on the pixel if MIN > x17ff (i.e. 360 degree), the behaviour will be same as MIN = x1800

0x01A072BC MDSS_MDP_VP_0_VIG_1_FOLIAGE_REGION_SAT**Type:** RW**Reset State:** 0x00000000

Foliage Adjustment Saturation Qualifier

double buffered: Yes

MDSS_MDP_VP_0_VIG_1_FOLIAGE_REGION_SAT

Bits	Name	Description
15:8	MAX	Saturation has to be below or equal to this to qualify for foliage adjustment on the pixel
7:0	MIN	U8, Saturation has to be above or equal to this to qualify for foliage adjustment on the pixel

0x01A072C0 MDSS_MDP_VP_0_VIG_1_FOLIAGE_REGION_VAL**Type:** RW**Reset State:** 0x00000000

Foliage Adjustment Value Qualifier

double buffered: Yes

MDSS_MDP_VP_0_VIG_1_FOLIAGE_REGION_VAL

Bits	Name	Description
15:8	MAX	Value has to be below or equal to this to qualify for foliage adjustment on the pixel

MDSS_MDP_VP_0_VIG_1_FOLIAGE_REGION_VAL (cont.)

Bits	Name	Description
7:0	MIN	U8, Value has to be above or equal to this to qualify for foilage adjustment on the pixel

0x01A072C4 MDSS_MDP_VP_0_VIG_1_HIST_COLLECT_CTL**Type:** W**Reset State:** 0x00000000

Histogram Collection Control

double buffered: YES

except for LOCK

MDSS_MDP_VP_0_VIG_1_HIST_COLLECT_CTL

Bits	Name	Description
0	LOCK	0: SW and HW swap their access on the 2 histogram tables; HW collects on a new table on the new frame, while SW accesses the histogram of the last frame on the table just swapped out by HW 1: current table will be overwritten for new collection on the next frame, the other table remains untouched for SW to continue to access

0x01A072C8 MDSS_MDP_VP_0_VIG_1_HIST_MIN_MAX**Type:** R**Reset State:** 0x00000000

MIN and MAX V value over the entire frame

double buffered: NO

MDSS_MDP_VP_0_VIG_1_HIST_MIN_MAX

Bits	Name	Description
23:16	MAX	Maximum V value of the last collected frame
7:0	MIN	Minimum V value of the last collected frame

0x01A072CC MDSS_MDP_VP_0_VIG_1_HIST_FRAME_COLLECTED**Type:** R**Reset State:** 0x00000000

Histogram Frame Number since the last rising START

double buffered: NO

MDSS_MDP_VP_0_VIG_1_HIST_FRAME_COLLECTED

Bits	Name	Description
31:0	FRAME_NUM	The frame number that the current histogram accessible by SW was collected from

0x01A072D0 MDSS_MDP_VP_0_VIG_1_HIST_SEL**Type:** R**Reset State:** 0x00000000

PA LUTV and Histogram Table Selection

double buffered: NO

MDSS_MDP_VP_0_VIG_1_HIST_SEL

Bits	Name	Description
4	HIST_SEL	0x0: Table 0 is available for SW and Table 1 is available for HW 0x1: Table 1 is available for SW and Table 0 is available for HW
0	PA_LUTV_SEL	0x0: LUT0 is available for SW and LUT1 is available for HW 0x1: LUT1 is available for SW and LUT0 is available for HW

0x01A072D4 MDSS_MDP_VP_0_VIG_1_HIST_SUM**Type:** R**Reset State:** 0x00000000

Accumulated V Value of the frame

double buffered: NO

MDSS_MDP_VP_0_VIG_1_HIST_SUM

Bits	Name	Description
31:0	HIST_SUM	This sum matches the histogram currently accessible by SW

0x01A072E0 MDSS_MDP_VP_0_VIG_1_HIST_V_DATAN**Type:** RW**Reset State:** 0x00000000**MDSS_MDP_VP_0_VIG_1_HIST_V_DATAN**

Bits	Name	Type	Description
31:24	INDEX	RW	
23:0	VALUE	R	

0x01A072F0 MDSS_MDP_VP_0_VIG_1_PA_LUTV**Type:** RW**Reset State:** 0x00000000

PA LUTV data

Auto-incremented by hardware or by software

256 entries are available for reading

double buffered: NO. Hardware maintains 2 copies

MDSS_MDP_VP_0_VIG_1_PA_LUTV

Bits	Name	Description
17	VALUE_UPDATEN	Software issue an write to the LUT 0: VALUE will be written to RAM at INDEX if INDEX_UPDATE=1 or at INDEX+1 if INDEX_UPDATE=0 1: VALUE will not be written to RAM, this setting is appropriate for example when INDEX needs to be updated for read
16	INDEX_UPDATE	Software issue an index update 0: No index update by software 1: Update LUT index to value specified by INDEX
15:8	INDEX	VALUE is stored in this INDEX of the HIST_LUTn, next write will be stored to the next higher INDEX
7:0	VALUE	Value read depends on the LUT select and which one is available for reading by software The read sequence should always begin with WR INDEX_UPDATE=1, INDEX=<rprr start location> subsequent RD will start from the start location and every read will auto-increment the rprr

0x01A07300 MDSS_MDP_VP_0_VIG_1_PA_LUTV_SWAP**Type:** W**Reset State:** 0x00000000

Histogram SWAP signal to indicate that the HIST_LUTn contents should be swapped.

Software accesses one LUT while HW accesses the other LUT

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_VIG_1_PA_LUTV_SWAP

Bits	Name	Description
0	SWAP	SWAP signal to flip buffers accessible by SW and made available to HW Achieves double buffering of the LUT values

0x01A07310 MDSS_MDP_VP_0_VIG_1_SPA_HUE_ADJUST**Type:** RW**Reset State:** 0x00000000

Source Picture HUE Adjustment

double buffered: YES

MDSS_MDP_VP_0_VIG_1_SPA_HUE_ADJUST

Bits	Name	Description
11:0	VALUE	Value in S12, recommend to be within exclusive range xA00 to x600 (i.e. -360 to +360 degree) x600 is equivalent to a shift of 360 degree if VALUE > x5FF in either negative or positive direction, it is circularly clipped (i.e. VALUE + x600 or VALUE - x600)

0x01A07314 MDSS_MDP_VP_0_VIG_1_SPA_SAT_ADJUST**Type:** RW**Reset State:** 0x00001A00

Source Picture Saturation Adjustment

double buffered: YES

MDSS_MDP_VP_0_VIG_1_SPA_SAT_ADJUST

Bits	Name	Description
15:8	THRSHLD	Below this threshold of saturation, no adjustment is applied to saturation. This field is in U8
7:0	VALUE	Value in S8

0x01A07318 MDSS_MDP_VP_0_VIG_1_SPA_VAL_ADJUST**Type:** RW**Reset State:** 0x00000000

Source Picture Adjustment

double buffered: YES

MDSS_MDP_VP_0_VIG_1_SPA_VAL_ADJUST

Bits	Name	Description
7:0	VALUE	Value in S8

0x01A0731C MDSS_MDP_VP_0_VIG_1_SPA_CONT_ADJUST**Type:** RW**Reset State:** 0x00000000

Source Picture Contrast Adjustment

double buffered: YES

MDSS_MDP_VP_0_VIG_1_SPA_CONT_ADJUST

Bits	Name	Description
7:0	VALUE	Value in S8

0x01A07320 MDSS_MDP_VP_0_VIG_1_CSC_1_MATRIX_COEFF_0**Type:** RW**Reset State:** 0x00000200

Color Space conversion coefficients for ALL of the Virtual Pipelines

double buffered: NO

MDSS_MDP_VP_0_VIG_1_CSC_1_MATRIX_COEFF_0

Bits	Name	Description
28:16	COEFF_12	Coefficient value in S4.9 format Row 1 Column 2
12:0	COEFF_11	Coefficient value in S4.9 format Row 1 Column 1

0x01A07324 MDSS_MDP_VP_0_VIG_1_CSC_1_MATRIX_COEFF_1**Type:** RW**Reset State:** 0x00000000

Color Space conversion coefficients for ALL of the Virtual Pipelines

double buffered: NO

MDSS_MDP_VP_0_VIG_1_CSC_1_MATRIX_COEFF_1

Bits	Name	Description
28:16	COEFF_21	Coefficient value in S4.9 format Row 2 Column 1
12:0	COEFF_13	Coefficient value in S4.9 format Row 1 Column 3

0x01A07328 MDSS_MDP_VP_0_VIG_1_CSC_1_MATRIX_COEFF_2**Type:** RW**Reset State:** 0x00000200

Color Space conversion coefficients for ALL of the Virtual Pipelines

double buffered: NO

MDSS_MDP_VP_0_VIG_1_CSC_1_MATRIX_COEFF_2

Bits	Name	Description
28:16	COEFF_23	Coefficient value in S4.9 format Row 2 Column 3
12:0	COEFF_22	Coefficient value in S4.9 format Row 2 Column 2

0x01A0732C MDSS_MDP_VP_0_VIG_1_CSC_1_MATRIX_COEFF_3**Type:** RW**Reset State:** 0x00000000

Color Space conversion coefficients for ALL of the Virtual Pipelines

double buffered: NO

MDSS_MDP_VP_0_VIG_1_CSC_1_MATRIX_COEFF_3

Bits	Name	Description
28:16	COEFF_32	Coefficient value in S4.9 format Row 3 Column 2
12:0	COEFF_31	Coefficient value in S4.9 format Row 3 Column 1

0x01A07330 MDSS_MDP_VP_0_VIG_1_CSC_1_MATRIX_COEFF_4**Type:** RW**Reset State:** 0x00000200

Color Space conversion coefficients for ALL of the Virtual Pipelines

double buffered: NO

MDSS_MDP_VP_0_VIG_1_CSC_1_MATRIX_COEFF_4

Bits	Name	Description
12:0	COEFF_33	Coefficient value in S4.9 format Row 3 Column 3

0x01A07334 MDSS_MDP_VP_0_VIG_1_CSC_1_COMP_0_PRE_CLAMP**Type:** RW**Reset State:** 0x000000FF

Component 0 pre-clamp value

double buffered: NO

MDSS_MDP_VP_0_VIG_1_CSC_1_COMP_0_PRE_CLAMP

Bits	Name	Description
15:8	LOW	Unsigned 8 bit pre-clamp value

MDSS_MDP_VP_0_VIG_1_CSC_1_COMP_0_PRE_CLAMP (cont.)

Bits	Name	Description
7:0	HIGH	Unsigned 8 bit pre-clamp value

0x01A07338 MDSS_MDP_VP_0_VIG_1_CSC_1_COMP_1_PRE_CLAMP**Type:** RW**Reset State:** 0x000000FF

Component 1 pre-clamp value

double buffered: NO

MDSS_MDP_VP_0_VIG_1_CSC_1_COMP_1_PRE_CLAMP

Bits	Name	Description
15:8	LOW	Unsigned 8 bit pre-clamp value
7:0	HIGH	Unsigned 8 bit pre-clamp value

0x01A0733C MDSS_MDP_VP_0_VIG_1_CSC_1_COMP_2_PRE_CLAMP**Type:** RW**Reset State:** 0x000000FF

Component 2 pre-clamp value

double buffered: NO

MDSS_MDP_VP_0_VIG_1_CSC_1_COMP_2_PRE_CLAMP

Bits	Name	Description
15:8	LOW	Unsigned 8 bit pre-clamp value
7:0	HIGH	Unsigned 8 bit pre-clamp value

0x01A07340 MDSS_MDP_VP_0_VIG_1_CSC_1_COMP_0_POST_CLAMP**Type:** RW**Reset State:** 0x000000FF

Component 0 post-clamp value

double buffered: NO

MDSS_MDP_VP_0_VIG_1_CSC_1_COMP_0_POST_CLAMP

Bits	Name	Description
15:8	LOW	Unsigned 8 bit pre-clamp value
7:0	HIGH	Unsigned 8 bit pre-clamp value

0x01A07344 MDSS_MDP_VP_0_VIG_1_CSC_1_COMP_1_POST_CLAMP**Type:** RW**Reset State:** 0x000000FF

Component 1 post-clamp value

double buffered: NO

MDSS_MDP_VP_0_VIG_1_CSC_1_COMP_1_POST_CLAMP

Bits	Name	Description
15:8	LOW	Unsigned 8 bit pre-clamp value
7:0	HIGH	Unsigned 8 bit pre-clamp value

0x01A07348 MDSS_MDP_VP_0_VIG_1_CSC_1_COMP_2_POST_CLAMP**Type:** RW**Reset State:** 0x000000FF

Component 2 post-clamp value

double buffered: NO

MDSS_MDP_VP_0_VIG_1_CSC_1_COMP_2_POST_CLAMP

Bits	Name	Description
15:8	LOW	Unsigned 8 bit pre-clamp value
7:0	HIGH	Unsigned 8 bit pre-clamp value

0x01A0734C MDSS_MDP_VP_0_VIG_1_CSC_1_COMP_0_PRE_BIAS**Type:** RW**Reset State:** 0x00000000

Component 0 pre-bias value

double buffered: NO

MDSS_MDP_VP_0_VIG_1_CSC_1_COMP_0_PRE_BIAS

Bits	Name	Description
8:0	VALUE	Signed 9 bit pre-bias value

0x01A07350 MDSS_MDP_VP_0_VIG_1_CSC_1_COMP_1_PRE_BIAS

Type: RW

Reset State: 0x00000000

Component 1 pre-bias value

double buffered: NO

MDSS_MDP_VP_0_VIG_1_CSC_1_COMP_1_PRE_BIAS

Bits	Name	Description
8:0	VALUE	Signed 9 bit pre-bias value

0x01A07354 MDSS_MDP_VP_0_VIG_1_CSC_1_COMP_2_PRE_BIAS

Type: RW

Reset State: 0x00000000

Component 2 pre-bias value

double buffered: NO

MDSS_MDP_VP_0_VIG_1_CSC_1_COMP_2_PRE_BIAS

Bits	Name	Description
8:0	VALUE	Signed 9 bit pre-bias value

0x01A07358 MDSS_MDP_VP_0_VIG_1_CSC_1_COMP_0_POST_BIAS

Type: RW

Reset State: 0x00000000

Component 0 post-bias value

N double buffered: NO

MDSS_MDP_VP_0_VIG_1_CSC_1_COMP_0_POST_BIAS

Bits	Name	Description
8:0	VALUE	Signed 9 bit pre-bias value

0x01A0735C MDSS_MDP_VP_0_VIG_1_CSC_1_COMP_1_POST_BIAS**Type:** RW**Reset State:** 0x00000000

Component 1 post-bias value

double buffered: NO

MDSS_MDP_VP_0_VIG_1_CSC_1_COMP_1_POST_BIAS

Bits	Name	Description
8:0	VALUE	Signed 9 bit pre-bias value

0x01A07360 MDSS_MDP_VP_0_VIG_1_CSC_1_COMP_2_POST_BIAS**Type:** RW**Reset State:** 0x00000000

Component 2 post-bias value

double buffered: NO

MDSS_MDP_VP_0_VIG_1_CSC_1_COMP_2_POST_BIAS

Bits	Name	Description
8:0	VALUE	Signed 9 bit pre-bias value

0x01A07370 MDSS_MDP_VP_0_VIG_1_QSEED2_MISR_CTRL**Type:** RW**Reset State:** 0x00000000**MDSS_MDP_VP_0_VIG_1_QSEED2_MISR_CTRL**

Bits	Name	Type	Description
31	RUN_MODE	RW	
27:24	INPUT_SEL	RW	

MDSS_MDP_VP_0_VIG_1_QSEED2_MISR_CTRL (cont.)

Bits	Name	Type	Description
10	STATUS_CLEAR	W	
9	STATUS	R	
8	ENABLE	RW	
7:0	FRAME_COUNT	RW	

0x01A07374 MDSS_MDP_VP_0_VIG_1_QSEED2_MISR_SIGNATURE**Type:** R**Reset State:** 0x00000000**MDSS_MDP_VP_0_VIG_1_QSEED2_MISR_SIGNATURE**

Bits	Name	Description
31:0	CAPTURED	

**0x01A08000+ MDSS_MDP_VP_0_VIG_1_HIST_V_DATAn_BIN_i, i=[0..255]
4*i****Type:** R**Reset State:** 0x00000000

Direct Register Mapped Histogram data collection

256 bins

double buffered: NO

MDSS_MDP_VP_0_VIG_1_HIST_V_DATAn_BIN_i

Bits	Name	Description
23:0	VALUE	Value

**0x01A08400+ MDSS_MDP_VP_0_VIG_1_PA_LUTV_IDX_i, i=[0..127]
4*i****Type:** RW**Reset State:** 0x00000000

Direct Register Mapped PA LUTV data

256 entries are available for writing and reading

double buffered: NO. Hardware maintains 2 copies

MDSS_MDP_VP_0_VIG_1_PA_LUTV_IDX_i

Bits	Name	Description
23:16	VALUE_UPPER	LUT content at location 2*IDX+1 to be written/read to/from the LUT accessible by software at the moment
7:0	VALUE_LOWER	LUT content at location 2*IDX to be written/read to/from the LUT accessible by software at the moment

17.17.7.2.2 MDSS_MDP_VP_0_VIG_1_SSPP

SSPP hardware register description

37.17.7.2.2.1 MDSS_MDP_VP_0_VIG_1_SSPP_SSPP

SSPP

SSPP hardware register description

0x01A07000 MDSS_MDP_VP_0_VIG_1_SSPP_SRC_SIZE

Type: RW

Reset State: 0x00000000

Source Size for the region of interest source image

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_VIG_1_SSPP_SRC_SIZE

Bits	Name	Description
31:16	ROI_H	Source Height of the Region Of Interest (ROI)
15:0	ROI_W	Source Width of the Region Of Interest (ROI) The exact supported size depends on the width of line buffers allocated for MDP.

0x01A07004 MDSS_MDP_VP_0_VIG_1_SSPP_SRC_IMG_SIZE

Type: RW

Reset State: 0x00000000

Source Size for the image in which the ROI resides

This value along with SRC_SIZE and SRC_XY will be used by hardware to calculate whether pixel repeat or overfetch is required along the ROI border.

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_VIG_1_SSPP_SRC_IMG_SIZE

Bits	Name	Description
31:16	SRC_H	Source Height. Note that this is NOT the ROI height
15:0	SRC_W	Source Width. Note that this is NOT the ROI width

0x01A07008 MDSS_MDP_VP_0_VIG_1_SSPP_SRC_XY

Type: RW

Reset State: 0x00000000

Source Offset from the top-left corner of the source image for each of the region of interest dimensions

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_VIG_1_SSPP_SRC_XY

Bits	Name	Description
31:16	SRC_Y	Y-offset of source ROI in the source image
15:0	SRC_X	X-offset of source ROI in the source image

0x01A0700C MDSS_MDP_VP_0_VIG_1_SSPP_OUT_SIZE

Type: RW

Reset State: 0x00000000

Output Size of the layer.

Should be same as source size if scaling is NOT enabled.

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_VIG_1_SSPP_OUT_SIZE

Bits	Name	Description
31:16	DST_H	Height of Output ROI
15:0	DST_W	Width of Output ROI

0x01A07010 MDSS_MDP_VP_0_VIG_1_SSPP_OUT_XY**Type:** RW**Reset State:** 0x00000000

Output Offset from the top-left corner of the output image

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_VIG_1_SSPP_OUT_XY

Bits	Name	Description
31:16	DST_Y	Height of Output ROI
15:0	DST_X	Width of Output ROI

0x01A07014 MDSS_MDP_VP_0_VIG_1_SSPP_SRC0_ADDR**Type:** RW**Reset State:** 0x00000000

Plane 0 base address

For fetching BWC data, this plane refers to the compressed RAU data

double buffered: YES

when: On frame boundaries (vsync or SW trigger) when ADDR_SYNC_MODE is in frame mode and on line boundary if ADDR_SYNC_MODE in in line mode

MDSS_MDP_VP_0_VIG_1_SSPP_SRC0_ADDR

Bits	Name	Description
31:0	ADDR	Base byte address of the image single interleave color plane, Luma plane in pseudo-planar format, or color0 in planar format

0x01A07018 MDSS_MDP_VP_0_VIG_1_SSPP_SRC1_ADDR**Type:** RW**Reset State:** 0x00000000

Plane 1 base address

For fetching BWC data, this plane refers to the META data to fetch the lengths of the compressed RAU data stream

double buffered: YES

when: On frame boundaries (vsync or SW trigger) when ADDR_SYNC_MODE is in frame mode and on line boundary if ADDR_SYNC_MODE in in line mode

MDSS_MDP_VP_0_VIG_1_SSPP_SRC1_ADDR

Bits	Name	Description
31:0	ADDR	Base byte address of the chroma plane in pseudo-planar format, or color1 in planar format

0x01A0701C MDSS_MDP_VP_0_VIG_1_SSPP_SRC2_ADDR**Type:** RW**Reset State:** 0x00000000

Plane 2 base address

double buffered: YES

when: On frame boundaries (vsync or SW trigger) when ADDR_SYNC_MODE is in frame mode and on line boundary if ADDR_SYNC_MODE in in line mode

MDSS_MDP_VP_0_VIG_1_SSPP_SRC2_ADDR

Bits	Name	Description
31:0	ADDR	Base byte address of color2 in planar format

0x01A07020 MDSS_MDP_VP_0_VIG_1_SSPP_SRC3_ADDR**Type:** RW**Reset State:** 0x00000000

Plane 3 base address

double buffered: YES

when: On frame boundaries (vsync or SW trigger) when ADDR_SYNC_MODE is in frame mode and on line boundary if ADDR_SYNC_MODE in in line mode

MDSS_MDP_VP_0_VIG_1_SSPP_SRC3_ADDR

Bits	Name	Description
31:0	ADDR	Base byte address of color3 in planar format, or pseudo planar + alpha format

0x01A07024 MDSS_MDP_VP_0_VIG_1_SSPP_SRC_YSTRIDE0

Type: RW

Reset State: 0x00000000

Stride Values

Double-buffered

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_VIG_1_SSPP_SRC_YSTRIDE0

Bits	Name	Description
31:16	SRCP1_YSTRIDE	Plane 1 Y stride in bytes
15:0	SRCP0_YSTRIDE	Plane 0 Y stride in bytes

0x01A07028 MDSS_MDP_VP_0_VIG_1_SSPP_SRC_YSTRIDE1

Type: RW

Reset State: 0x00000000

Stride Values

Double-buffered

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_VIG_1_SSPP_SRC_YSTRIDE1

Bits	Name	Description
31:16	SRCP3_YSTRIDE	Plane 3 Y stride in bytes
15:0	SRCP2_YSTRIDE	Plane 2 Y stride in bytes

0x01A07030 MDSS_MDP_VP_0_VIG_1_SSPP_SRC_FORMAT**Type:** RW**Reset State:** 0x00000000

Source Format Configuration for each virtual pipeline.

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_VIG_1_SSPP_SRC_FORMAT

Bits	Name	Description
31:30	FRAME_FORMAT	Frame format In the current revision, there is no support for tiled formats 0x0: Linear, no tile 0x1: Tiled, 4x4
25:23	SRC_CHROMA_SAMP	Source Chroma Sampling 0x0: RGB/YUV 4:4:4 0x1: 4:2:2 - H2V1 0x2: 4:2:2 - H1V2 0x3: 4:2:0 - N2V1 0x4: 4:2:0 - N1V2
22	SOLID_FILL	Fetch feeds the lower pipeline with the CONSTANT_COLOR register as its output value. NOTHING is fetched Note that in order to use this feature then Interleaved FETCH_PLANES must be set as well and no chroma sub-sampling enabled
21	VC1_REDUCE	VC1 range reduction enable for YUV sources The reduce range is set by separate register VC1_RANGE
20:19	FETCH_PLANES	Determines the number of planes to fetch 0x0: Interleaved 0x1: Planar 0x2: Pseudo-planar
18	UNPACK_ALIGN	0x0: TO LSB 0x1: TO MSB
17	UNPACK_TIGHT	0x0: Loose 0x1: Tight

MDSS_MDP_VP_0_VIG_1_SSPP_SRC_FORMAT (cont.)

Bits	Name	Description
13:12	UNPACK_COUNT	Valid unpacking pattern count 0x0 = 1 component 0x1 = 2 components ... 0x3 = 4 components Unpacking applies to interleaved plane (chroma plane in pseudo-planar)
11	ROT90	Source image is to be rotated 90 degrees
10:9	SRC_BPP	Effective source byte per pixel 0x0: 1 byte 0x1: 2 bytes 0x2: 3 bytes 0x3:4 bytes This is used ONLY when unpacking the interleaved plane
8	SRCC3_EN	Alpha Enable 0x0: Source has NO alpha 0x1: Source has alpha
7:6	SRCC3	Number of bits for component 3 (Alpha) 0x0: 1 bits 0x1: 4 bits 0x2: 6 bits 0x3: 8 bits
5:4	SRCC2	Number of bits for component 2 (R/Cr) 0x0: 4 bits 0x1: 5 bits 0x2: 6 bits 0x3: 8 bits
3:2	SRCC1	Number of bits for component 1 (B/Cb) 0x0: 4 bits 0x1: 5 bits 0x2: 6 bits 0x3: 8 bits
1:0	SRCC0	Number of bits for component 0 (G/Luma) 0x0: 4 bits 0x1: 5 bits 0x2: 6 bits 0x3: 8 bits

0x01A07034 MDSS_MDP_VP_0_VIG_1_SSPP_SRC_UNPACK_PATTERN**Type:** RW**Reset State:** 0x00000000

Unpacking pattern, maximum of 4 pattern elements starting from LSB to MSB (0 to 4).

Default is ARGB or C3-C2-C0-C1.

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_VIG_1_SSPP_SRC_UNPACK_PATTERN

Bits	Name	Description
25:24	ELEMENT3	0x0: Color0 0x1: Color1 0x2: Color2 0x3: Color3
17:16	ELEMENT2	0x0: Color0 0x1: Color1 0x2: Color2 0x3: Color3
9:8	ELEMENT1	0x0: Color0 0x1: Color1 0x2: Color2 0x3: Color3
1:0	ELEMENT0	0x0: Color0 0x1: Color1 0x2: Color2 0x3: Color3

0x01A07038 MDSS_MDP_VP_0_VIG_1_SSPP_SRC_OP_MODE

Type: RW

Reset State: 0x00000000

Operation Mode for each SSPP pipeline

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_VIG_1_SSPP_SRC_OP_MODE

Bits	Name	Description
31	SW_PIX_EXT_OVERRIDE	SW override for pixel extension values. 0x0: Use HW calculated pixel extensions 0x1: Use SW programmed pixel extensions

MDSS_MDP_VP_0_VIG_1_SSPP_SRC_OP_MODE (cont.)

Bits	Name	Description
16	IGC_LUT_EN	Inverse Gamma Correction LUT Enable 0x0: Disabled - mapped Linearly from 8-bit to 12-bit 0x1: Enabled
14:13	FLIP_MODE	Flip Operation Lower Bit: Left/Right flip Upper Bit: Up/Down flip

0x01A0703C MDSS_MDP_VP_0_VIG_1_SSPP_SRC_CONSTANT_COLOR**Type:** RW**Reset State:** 0x00000000

Constant Color Register during solid fill mode

double buffered: NO

MDSS_MDP_VP_0_VIG_1_SSPP_SRC_CONSTANT_COLOR

Bits	Name	Description
31:24	COLOR3	Constant color driven for color3 during solid fill mode.
23:16	COLOR2	Constant color driven for color2 during solid fill mode.
15:8	COLOR1	Constant color driven for color1 during solid fill mode.
7:0	COLOR0	Constant color driven for color0 during solid fill mode.

0x01A07048 MDSS_MDP_VP_0_VIG_1_SSPP_FETCH_CONFIG**Type:** RW**Reset State:** 0x00000087

SSPP Client Fetch Configuration per Virtual Pipeline double buffered: NO

This register must NOT be altered during operation and MDP must be idle when changing values

MDSS_MDP_VP_0_VIG_1_SSPP_FETCH_CONFIG

Bits	Name	Description
22	PRI_ARB_DIS	
21	PMARK_DIS	

MDSS_MDP_VP_0_VIG_1_SSPP_FETCH_CONFIG (cont.)

Bits	Name	Description
17:16	TILE_SIZE	Tile Size for processing of BLOCK formats Note that this only applies to blocks that operate in block mode which is limited to DMA source pipe. This setting has no effect for all other source pipes (ViG/RGB) 0x0: 128x128 Others: ILLEGAL / Not Supported
12	ADDR_SYNC_MODE	Boundary on which fetch address synchronizes. Setting this field to line forces the address updates on the next line which performs the software requested feature of updating on hsync. Note that updating on line boundaries will likely result in tearing on the panel since hardware is fetching from new memory buffer. 0x0: frame (default) 0x1: line
9:8	MAL	Describes the Minimum Access Length for the source pipeline. This value should align with the DRAM's expected transaction alignment unit. This setting forces the source fetch pipe to be aligned to DRAM and implement re-alignment for maximum efficiency 0x0: 128b alignment (default) 0x1: 256b alignment 0x2: 512b alignment
7:4	MAX_BURST_SIZE	Maximum burst beat size. Hardware fetch engines operate on 128-bit bus Possible values: 0x8: burst of 8 0x4: burst of 4 0x2: burst of 2
3:0	REQ_DEPTH_LIMIT	Pending request limit, default max value of 0x7 means maximum pending requests allowed in the design Anything less limits the bus requests to that number + 1.

0x01A0704C MDSS_MDP_VP_0_VIG_1_SSPP_VC1_RANGE**Type:** RW**Reset State:** 0x00000707

VC1 Range Map for each virtual pipeline

ONLY valid for pipelines that suport YUV formats

double buffered: NO

MDSS_MDP_VP_0_VIG_1_SSPP_VC1_RANGE

Bits	Name	Description
10:8	RANGE_MAPUV	Chroma range map for VC1 range reduction, default of 7 is equivalent to setting it for main profile.
2:0	RANGE_MAPY	Luma range map for VC1 range reduction, default of 7 is equivalent to setting it for main profile.

0x01A07050 MDSS_MDP_VP_0_VIG_1_SSPP_REQPRIORITY_FIFO_WATERMARK_0**Type:** RW**Reset State:** 0x00000040

FIFO watermark to aid in generation of mdpREQPRIORITY

The watermark here refers to the # of 128-entries available in the Shared Memory Pool

The value programmed here should be aware of the amount programmed in the Shared Memory Pool.

So a value of 0 indicates completely FULL (i.e. there are no more entries available in the SMP)

A value of 100 indicates 100 entries in the Shared Memory Pool (each entry being 128-bit entries)

This is the lowestmost FIFO depth and should not be programmed higher than REQPRIORITY_FIFO_WATERMARK_1 or REQPRIORITY_FIFO_WATERMARK_2

An example programming would set the following:

REQPRIORITY_FIFO_WATERMARK_0 : 50 (meaning 50 entries free to be written into SMP - least room available)

REQPRIORITY_FIFO_WATERMARK_1 : 100 (meaning 100 entries free to be written into SMP - middle ground room available)

REQPRIORITY_FIFO_WATERMARK_2 : 150 (meaning 150 entries free to be written into SMP - most room available)

mdpREQPRIORITY = 11 (highest priority request) when above all 3 watermark levels (i.e. lots of space available in SMP)

mdpREQPRIORITY = 10 above FIFO_0 and FIFO_1 but below FIFO_2

mdpREQPRIORITY = 01 above FIFO_0 but below FIFO_1 and FIFO_2

mdpREQPRIORITY = 00 (lowest priority request) when below all 3 watermark levels (i.e. not a lot of space available in SMP so its good)

double buffered: YES

MDSS_MDP_VP_0_VIG_1_SSPP_REQPRIORITY_FIFO_WATERMARK_0

Bits	Name	Description
15:0	WATERMARK	Watermark Level for FIFO_0

0x01A07054 MDSS_MDP_VP_0_VIG_1_SSPP_REQPRIORITY_FIFO_WATERMARK_1**Type:** RW**Reset State:** 0x00000080

FIFO watermark to aid in generation of mdpREQPRIORITY

The watermark here refers to the # of 128-entries available in the Shared Memory Pool

The value programmed here should be aware of the amount programmed in the Shared Memory Pool.

So a value of 0 indicates completely FULL (i.e. there are no more entries available in the SMP)

A value of 100 indicates 100 entries in the Shared Memory Pool (each entry being 128-bit entries)

This is the middle FIFO depth and should not be programmed higher than
REQPRIORITY_FIFO_WATERMARK_2 or lower than
REQPRIORITY_FIFO_WATERMARK_0

An example programming would set the following:

REQPRIORITY_FIFO_WATERMARK_0 : 50 (meaning 50 entries free to be written into SMP - least room available)

REQPRIORITY_FIFO_WATERMARK_1 : 100 (meaning 100 entries free to be written into SMP - middle ground room available)

REQPRIORITY_FIFO_WATERMARK_2 : 150 (meaning 150 entries free to be written into SMP - most room available)

mdpREQPRIORITY = 11 (highest priority request) when above all 3 watermark levels (i.e. lots of space available in SMP)

mdpREQPRIORITY = 10 above FIFO_0 and FIFO_1 but below FIFO_2

mdpREQPRIORITY = 01 above FIFO_0 but below FIFO_1 and FIFO_2

mdpREQPRIORITY = 00 (lowest priority request) when below all 3 watermark levels (i.e. not a lot of space available in SMP so its good)

double buffered: YES

MDSS_MDP_VP_0_VIG_1_SSPP_REQPRIORITY_FIFO_WATERMARK_1

Bits	Name	Description
15:0	WATERMARK	Watermark Level for FIFO_1

0x01A07058 MDSS_MDP_VP_0_VIG_1_SSPP_REQPRIORITY_FIFO_WATERMARK_2**Type:** RW**Reset State:** 0x000000C0

FIFO watermark to aid in generation of mdpREQPRIORITY

The watermark here refers to the # of 128-entries available in the Shared Memory Pool

The value programmed here should be aware of the amount programmed in the Shared Memory Pool.

So a value of 0 indicates completely FULL (i.e. there are no more entries available in the SMP)

A value of 100 indicates 100 entries in the Shared Memory Pool (each entry being 128-bit entries)

This is the highest FIFO depth and should not be programmed lower than REQPRIORITY_FIFO_WATERMARK_0 or REQPRIORITY_FIFO_WATERMARK_1

An example programming would set the following:

REQPRIORITY_FIFO_WATERMARK_0 : 50 (meaning 50 entries free to be written into SMP - least room available)

REQPRIORITY_FIFO_WATERMARK_1 : 100 (meaning 100 entries free to be written into SMP - middle ground room available)

REQPRIORITY_FIFO_WATERMARK_2 : 150 (meaning 150 entries free to be written into SMP - most room available)

mdpREQPRIORITY = 11 (highest priority request) when above all 3 watermark levels (i.e. lots of space available in SMP)

mdpREQPRIORITY = 10 above FIFO_0 and FIFO_1 but below FIFO_2

mdpREQPRIORITY = 01 above FIFO_0 but below FIFO_1 and FIFO_2

mdpREQPRIORITY = 00 (lowest priority request) when below all 3 watermark levels (i.e. not a lot of space available in SMP so its good)

double buffered: YES

MDSS_MDP_VP_0_VIG_1_SSPP_REQPRIORITY_FIFO_WATERMARK_2

Bits	Name	Description
15:0	WATERMARK	Watermark Level for FIFO_2

0x01A07070 MDSS_MDP_VP_0_VIG_1_SSPP_SRC_ADDR_SW_STATUS**Type:** RW**Reset State:** 0x00000000

Security Status of the source buffer plane

This is programmed by SW to indicate if the buffer is NON-SECURE or SECURE

This information will be presented to VBIF per transaction and reflected as part of the memory transaction

double buffered: YES

MDSS_MDP_VP_0_VIG_1_SSPP_SRC_ADDR_SW_STATUS

Bits	Name	Description
3	SRC3_ADDR_CP	This is the secure status bit for source plane 3 (if used) If the format doesn't fetch from that plane then this value is ignored Memory transactions will not complete if this status doesn't correspond to system programming in the rest of the path to memory SW should program SRC0_ADDR_CP, SRC1_ADDR_CP, SRC2_ADDR_CP, SRC3_ADDR_CP to the same value 0x0: Non Content Protected Buffer State 0x1: Content Protected Buffer State
2	SRC2_ADDR_CP	This is the secure status bit for source plane 2 (if used) If the format doesn't fetch from that plane then this value is ignored Memory transactions will not complete if this status doesn't correspond to system programming in the rest of the path to memory SW should program SRC0_ADDR_CP, SRC1_ADDR_CP, SRC2_ADDR_CP, SRC3_ADDR_CP to the same value 0x0: Non Content Protected Buffer State 0x1: Content Protected Buffer State
1	SRC1_ADDR_CP	This is the secure status bit for source plane 1 (if used) If the format doesn't fetch from that plane then this value is ignored Memory transactions will not complete if this status doesn't correspond to system programming in the rest of the path to memory SW should program SRC0_ADDR_CP, SRC1_ADDR_CP, SRC2_ADDR_CP, SRC3_ADDR_CP to the same value 0x0: Non Content Protected Buffer State 0x1: Content Protected Buffer State

MDSS_MDP_VP_0_VIG_1_SSPP_SRC_ADDR_SW_STATUS (cont.)

Bits	Name	Description
0	SRC0_ADDR_CP	This is the secure status bit for source plane 0 (if used) If the format doesn't fetch from that plane then this value is ignored Memory transactions will not complete if this status doesn't correspond to system programming in the rest of the path to memory SW should program SRC0_ADDR_CP, SRC1_ADDR_CP, SRC2_ADDR_CP, SRC3_ADDR_CP to the same value 0x0: Non Content Protected Buffer State 0x1: Content Protected Buffer State

0x01A070A4 MDSS_MDP_VP_0_VIG_1_SSPP_CURRENT_SRC0_ADDR**Type:** R**Reset State:** 0x00000000

Current Plane 0 base address that Hardware is using

For fetching BWC data, this plane refers to the compressed RAU data

double buffered: N/A

MDSS_MDP_VP_0_VIG_1_SSPP_CURRENT_SRC0_ADDR

Bits	Name	Description
31:0	ADDR	Base byte address of the image single interleave color plane, Luma plane in pseudo-planar format, or color0 in planar format

0x01A070A8 MDSS_MDP_VP_0_VIG_1_SSPP_CURRENT_SRC1_ADDR**Type:** R**Reset State:** 0x00000000

Plane 1 base address that Hardware is using

For fetching BWC data, this plane refers to the META data to fetch the lengths of the compressed RAU data stream

double buffered: N/A

MDSS_MDP_VP_0_VIG_1_SSPP_CURRENT_SRC1_ADDR

Bits	Name	Description
31:0	ADDR	Base byte address of the chroma plane in pseudo-planar format, or color1 in planar format

0x01A070AC MDSS_MDP_VP_0_VIG_1_SSPP_CURRENT_SRC2_ADDR**Type:** R**Reset State:** 0x00000000

Plane 2 base address that Hardware is using

double buffered: N/A

MDSS_MDP_VP_0_VIG_1_SSPP_CURRENT_SRC2_ADDR

Bits	Name	Description
31:0	ADDR	Base byte address of color2 in planar format

0x01A070B0 MDSS_MDP_VP_0_VIG_1_SSPP_CURRENT_SRC3_ADDR**Type:** R**Reset State:** 0x00000000

Plane 3 base address that Hardware is using

double buffered: N/A

MDSS_MDP_VP_0_VIG_1_SSPP_CURRENT_SRC3_ADDR

Bits	Name	Description
31:0	ADDR	Base byte address of color3 in planar format, or pseudo planar + alpha format

0x01A070B4 MDSS_MDP_VP_0_VIG_1_SSPP_DECIMATION_CONFIG**Type:** RW**Reset State:** 0x00000000

Decimation Configuration

Note that there is no decimation on DMA pipe which means no decimation in line mode on DMA or block mode

double buffered: YES

MDSS_MDP_VP_0_VIG_1_SSPP_DECIMATION_CONFIG

Bits	Name	Description
11:8	HORZ_S	Horizontal Configuration for decimation; required to support downscaling greater than 1/4 S refers to the number of pixels skipped. The algorithm is always the following: read 1 pixel skip S pixels The algorithm is applied to ALL components. Decimation only supports the following values. All other values will result in undefined behaviour The decimation block will provide coarse scaling and should be used in conjunction with qseed2 to support all scaling ratios. 0x0: No decimation is performed 0x1: Read 1 pixel drop 1 pixel. Used to decimate by 2 horizontally 0x3: Read 1 pixel drop 3 pixels. Used to decimate by 4 horizontally 0x7: Read 1 pixel drop 7 pixels. Used to decimate by 8 horizontally 0xF: Read 1 pixel drop 15 pixels. Used to decimate by 16 horizontally
3:0	VERT_S	Vertical Configuration for decimation; required to support downscaling greater than 1/4 S refers to the number of lines skipped. The algorithm is always the following: read 1 line skip S lines The algorithm is applied to ALL components. Decimation only supports the following values. All other values will result in undefined behaviour The decimation block will provide coarse scaling and should be used in conjunction with qseed2 to support all scaling ratios. 0x0: No decimation is performed 0x1: Read 1 line drop 1 line. Used to decimate by 2 horizontally 0x3: Read 1 line drop 3 lines. Used to decimate by 4 horizontally 0x7: Read 1 line drop 7 lines. Used to decimate by 8 horizontally 0xF: Read 1 line drop 15 lines. Used to decimate by 16 horizontally

0x01A070D0 MDSS_MDP_VP_0_VIG_1_SSPP_FETCH_SMP_WR_PLANE0**Type:** R**Reset State:** 0x00000000

Fetch to SMP on plane 0 write counter

double buffered: N/A

MDSS_MDP_VP_0_VIG_1_SSPP_FETCH_SMP_WR_PLANE0

Bits	Name	Description
15:0	COUNT	This counter is reset on every frame and is incremented for each write fetch makes to the SMP plane0 port This port is used for interleaved formats or the Y component for multi-plane source formats This should correlate to value SMP_UNPACK_RD_PLANE0 More useful for debug but can be used for profiling purposes as well.

0x01A070D4 MDSS_MDP_VP_0_VIG_1_SSPP_FETCH_SMP_WR_PLANE1**Type:** R**Reset State:** 0x00000000

Fetch to SMP on plane 1 write counter

double buffered: N/A

MDSS_MDP_VP_0_VIG_1_SSPP_FETCH_SMP_WR_PLANE1

Bits	Name	Description
15:0	COUNT	This counter is reset on every frame and is incremented for each write fetch makes to the SMP plane1 port This port is NOT used for interleaved formats but is used for U (planar) or UV (pseudo-planar) formats This should correlate to value SMP_UNPACK_RD_PLANE1 More useful for debug but can be used for profiling purposes as well.

0x01A070D8 MDSS_MDP_VP_0_VIG_1_SSPP_FETCH_SMP_WR_PLANE2**Type:** R**Reset State:** 0x00000000

Fetch to SMP on plane 2 write counter

double buffered: N/A

MDSS_MDP_VP_0_VIG_1_SSPP_FETCH_SMP_WR_PLANE2

Bits	Name	Description
15:0	COUNT	This counter is reset on every frame and is incremented for each write fetch makes to the SMP plane2 port This port is NOT used for interleaved formats but is used for V (planar) or A (pseudo-planar plus alpha) formats This should correlate to value SMP_UNPACK_RD_PLANE2 More useful for debug but can be used for profiling purposes as well.

0x01A070E0 MDSS_MDP_VP_0_VIG_1_SSPP_SMP_UNPACK_RD_PLANE0**Type:** R**Reset State:** 0x00000000

SMP to unpack on plane 0 read counter

double buffered: N/A

MDSS_MDP_VP_0_VIG_1_SSPP_SMP_UNPACK_RD_PLANE0

Bits	Name	Description
15:0	COUNT	This counter is reset on every frame and is incremented for each line read from SMP plane0 port This port is used for interleaved formats or the Y component for multi-plane source formats This should correlate to FETCH_SMP_WR_PLANE0 value More useful for debug but can be used for profiling purposes as well.

0x01A070E4 MDSS_MDP_VP_0_VIG_1_SSPP_SMP_UNPACK_RD_PLANE1**Type:** R**Reset State:** 0x00000000

SMP to unpack on plane 1 read counter

double buffered: N/A

MDSS_MDP_VP_0_VIG_1_SSPP_SMP_UNPACK_RD_PLANE1

Bits	Name	Description
15:0	COUNT	This counter is reset on every frame and is incremented for each line read from SMP plane1 port This port is NOT used for interleaved formats but is used for U (planar) or UV (pseudo-planar) formats This should correlate to FETCH_SMP_WR_PLANE1 value More useful for debug but can be used for profiling purposes as well.

0x01A070E8 MDSS_MDP_VP_0_VIG_1_SSPP_SMP_UNPACK_RD_PLANE2**Type:** R**Reset State:** 0x00000000

SMP to unpack on plane 2 read counter

double buffered: N/A

MDSS_MDP_VP_0_VIG_1_SSPP_SMP_UNPACK_RD_PLANE2

Bits	Name	Description
15:0	COUNT	This counter is reset on every frame and is incremented for each line read from SMP plane1 port This port is NOT used for interleaved formats but is used for V (planar) or A (pseudo-planar plus alpha) formats This should correlate to FETCH_SMP_WR_PLANE1 value More useful for debug but can be used for profiling purposes as well.

0x01A07100 MDSS_MDP_VP_0_VIG_1_SSPP_SW_PIX_EXT_C0_LR**Type:** RW**Reset State:** 0x00000000

SW Pixel Extension Override for Left/Right of Color 0

These value only take effect if SW_PIX_EXT_OVERRIDE is set in SRC_OP_MODE register

double buffered: YES

MDSS_MDP_VP_0_VIG_1_SSPP_SW_PIX_EXT_C0_LR

Bits	Name	Description
31:24	RIGHT_OVF	Signed Value for Right Overfetch value. +ve value indicates overfetch -ve value indicates pixel drop
23:16	RIGHT_RPT	Unsigned Value for Right Pixel Repeat value.
15:8	LEFT_OVF	Signed Value for Left Overfetch value. +ve value indicates overfetch -ve value indicates pixel drop
7:0	LEFT_RPT	Unsigned Value for Left Pixel Repeat value.

0x01A07104 MDSS_MDP_VP_0_VIG_1_SSPP_SW_PIX_EXT_C0_TB**Type:** RW**Reset State:** 0x00000000

SW Pixel Extension Override for Left/Right of Color 0

These value only take effect if SW_PIX_EXT_OVERRIDE is set in SRC_OP_MODE register
double buffered: YES

MDSS_MDP_VP_0_VIG_1_SSPP_SW_PIX_EXT_C0_TB

Bits	Name	Description
31:24	BOTTOM_OVF	Signed Value for Bottom Overfetch value. +ve value indicates overfetch -ve value indicates line drop
23:16	BOTTOM_RPT	
15:8	TOP_OVF	Signed Value for Top Overfetch value. +ve value indicates overfetch -ve value indicates line drop
7:0	TOP_RPT	Unsigned Value for TOP Line Repeat value.

0x01A07108 MDSS_MDP_VP_0_VIG_1_SSPP_SW_PIX_EXT_C0_REQ_PIXELS**Type:** RW**Reset State:** 0x00000000

SW Pixel Extension Override for pixel height/width of Color 0

These value only take effect if SW_PIX_EXT_OVERRIDE is set in SRC_OP_MODE register

double buffered: YES

MDSS_MDP_VP_0_VIG_1_SSPP_SW_PIX_EXT_C0_REQ_PIXELS

Bits	Name	Description
31:16	TOP_BOTTOM	Unsigned Value for pixel height post pixel extension.
15:0	LEFT_RIGHT	Unsigned Value for pixel width post pixel extension.

0x01A07110 MDSS_MDP_VP_0_VIG_1_SSPP_SW_PIX_EXT_C1C2_LR

Type: RW

Reset State: 0x00000000

SW Pixel Extension Override for Left/Right of Color 0

These value only take effect if SW_PIX_EXT_OVERRIDE is set in SRC_OP_MODE register

double buffered: YES

MDSS_MDP_VP_0_VIG_1_SSPP_SW_PIX_EXT_C1C2_LR

Bits	Name	Description
31:24	RIGHT_OVF	Signed Value for Right Overfetch value. +ve value indicates overfetch -ve value indicates pixel drop
23:16	RIGHT_RPT	Unsigned Value for Right Pixel Repeat value.
15:8	LEFT_OVF	Signed Value for Left Overfetch value. +ve value indicates overfetch -ve value indicates pixel drop
7:0	LEFT_RPT	Unsigned Value for Left Pixel Repeat value.

0x01A07114 MDSS_MDP_VP_0_VIG_1_SSPP_SW_PIX_EXT_C1C2_TB

Type: RW

Reset State: 0x00000000

SW Pixel Extension Override for Left/Right of Color 0

These value only take effect if SW_PIX_EXT_OVERRIDE is set in SRC_OP_MODE register

double buffered: YES

MDSS_MDP_VP_0_VIG_1_SSPP_SW_PIX_EXT_C1C2_TB

Bits	Name	Description
31:24	BOTTOM_OVF	Signed Value for Bottom Overfetch value. +ve value indicates overfetch -ve value indicates line drop
23:16	BOTTOM_RPT	
15:8	TOP_OVF	Signed Value for Top Overfetch value. +ve value indicates overfetch -ve value indicates line drop
7:0	TOP_RPT	Unsigned Value for TOP Line Repeat value.

0x01A07118 MDSS_MDP_VP_0_VIG_1_SSPP_SW_PIX_EXT_C1C2_REQ_PIXELS**Type:** RW**Reset State:** 0x00000000

SW Pixel Extension Override for pixel height/width of Color 0

These value only take effect if SW_PIX_EXT_OVERRIDE is set in SRC_OP_MODE register

double buffered: YES

MDSS_MDP_VP_0_VIG_1_SSPP_SW_PIX_EXT_C1C2_REQ_PIXELS

Bits	Name	Description
31:16	TOP_BOTTOM	Unsigned Value for pixel height post pixel extension.
15:0	LEFT_RIGHT	Unsigned Value for pixel width post pixel extension.

0x01A07120 MDSS_MDP_VP_0_VIG_1_SSPP_SW_PIX_EXT_C3_LR**Type:** RW**Reset State:** 0x00000000

SW Pixel Extension Override for Left/Right of Color 0

These value only take effect if SW_PIX_EXT_OVERRIDE is set in SRC_OP_MODE register

double buffered: YES

MDSS_MDP_VP_0_VIG_1_SSPP_SW_PIX_EXT_C3_LR

Bits	Name	Description
31:24	RIGHT_OVF	Signed Value for Right Overfetch value. +ve value indicates overfetch -ve value indicates pixel drop
23:16	RIGHT_RPT	Unsigned Value for Right Pixel Repeat value.
15:8	LEFT_OVF	Signed Value for Left Overfetch value. +ve value indicates overfetch -ve value indicates pixel drop
7:0	LEFT_RPT	Unsigned Value for Left Pixel Repeat value.

0x01A07124 MDSS_MDP_VP_0_VIG_1_SSPP_SW_PIX_EXT_C3_TB**Type:** RW**Reset State:** 0x00000000

SW Pixel Extension Override for Left/Right of Color 0

These value only take effect if SW_PIX_EXT_OVERRIDE is set in SRC_OP_MODE register
double buffered: YES

MDSS_MDP_VP_0_VIG_1_SSPP_SW_PIX_EXT_C3_TB

Bits	Name	Description
31:24	BOTTOM_OVF	Signed Value for Bottom Overfetch value. +ve value indicates overfetch -ve value indicates line drop
23:16	BOTTOM_RPT	
15:8	TOP_OVF	Signed Value for Top Overfetch value. +ve value indicates overfetch -ve value indicates line drop
7:0	TOP_RPT	Unsigned Value for TOP Line Repeat value.

0x01A07128 MDSS_MDP_VP_0_VIG_1_SSPP_SW_PIX_EXT_C3_REQ_PIXELS**Type:** RW**Reset State:** 0x00000000

SW Pixel Extension Override for pixel height/width of Color 0

These value only take effect if SW_PIX_EXT_OVERRIDE is set in SRC_OP_MODE register

double buffered: YES

MDSS_MDP_VP_0_VIG_1_SSPP_SW_PIX_EXT_C3_REQ_PIXELS

Bits	Name	Description
31:16	TOP_BOTTOM	Unsigned Value for pixel height post pixel extension.
15:0	LEFT_RIGHT	Unsigned Value for pixel width post pixel extension.

17.17.7.3 MDSS_MDP_VP_0_VIG_2

VIG hardware register description

17.17.7.3.1 MDSS_MDP_VP_0_VIG_2_VIG

VIG

0x01A09200 MDSS_MDP_VP_0_VIG_2_OP_MODE

Type: RW

Reset State: 0x00040000

Virtual Pipeline operation mode

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_VIG_2_OP_MODE

Bits	Name	Description
28	PA_CONT_MASK	0x0: Disable HSIC adj on CONTRAST 0x1: Enable HSIC adj on CONTRAST
27	PA_VAL_MASK	0x0: Disable HSIC adj on VALUE 0x1: Enable HSIC adj on VALUE
26	PA_SAT_MASK	0x0: Disable HSIC adj on SATURATION 0x1: Enable HSIC adj on SATURATION
25	PA_HUE_MASK	0x0: Disable HSIC adj on HUE 0x1: Enable HSIC adj on HUE
19	CSC_DST_DATA_FORMAT	Destination Data Format 0x0: RGB 0x1: YCbCr
18	CSC_SRC_DATA_FORMAT	Source Data Format 0x0: RGB 0x1: YCbCr

MDSS_MDP_VP_0_VIG_2_OP_MODE (cont.)

Bits	Name	Description
17	CSC_1_EN	Enables the color matrix conversion 1 0x0: Bypass 0x1: Enabled
10	PA_LUTV_EN	Histogram Enhancement Lookup Table enable
8	HIST_EN	Histogram Enable 0x0: Disabled 0x1: Enabled
7	PA_MEM_COL_SKY_MASK	0x0: Disable Sky adjustment 0x1: Enable Sky adjustment
6	PA_MEM_COL_FOL_MASK	0x0: Disable Foliage adjustment 0x1: Enable Foliage adjustment
5	PA_MEM_COL_SKIN_MASK	0x0: Disable Skin adjustment 0x1: Enable Skin adjustment
4	PA_EN	Enable Picture Adjustment 0x0: Disabled 0x1: Enabled
3	PA_MEM_PROTECT_ENABLE	0: global hsv adjustment is independent of memor color 1: global hsv adjustment is active if the pixel is not already processed by memory color contrast adjustment in global adjustment is always independ of memory color regardless
2	PA_SAT_ZERO_EXP_ENABLE	0: if PA_SAT_ADJ < -32, clamp PA_SAT_ADJ to 0 1: if PA_SAT_ADJ < -64, clamp PA_SAT_ADJ to 0

0x01A09204 MDSS_MDP_VP_0_VIG_2_QSEED2_CONFIG**Type:** RW**Reset State:** 0x00000000

QSEED configuration for each of the Virtual Pipelines

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_VIG_2_QSEED2_CONFIG

Bits	Name	Description
19:18	COMP3_VERT_FILTER_MODE	Component 3 (A) vertical filtering mode 0x0: Nearest Neighbor 0x1: Bilinear 0x2: PCMN

MDSS_MDP_VP_0_VIG_2_QSEED2_CONFIG (cont.)

Bits	Name	Description
17:16	COMP3_HORZ_FILTER_MODE	Component 3 (A) horizontal filtering mode 0x0: Nearest Neighbor 0x1: Bilinear 0x2: PCMN
15:14	COMP1_2_VERT_FILTER_MODE	Component 1/2 (CrCb or RB) vertical filtering mode 0x0: Nearest Neighbor 0x1: Bilinear 0x2: PCMN
13:12	COMP1_2_HORZ_FILTER_MODE	Component 1/2 (CrCb or RB) horizontal filtering mode 0x0: Nearest Neighbor 0x1: Bilinear 0x2: PCMN
11:10	COMP0_VERT_FILTER_MODE	Component 0 (Y or G) vertical filtering mode 0x0: Nearest Neighbor 0x1: Bilinear 0x2: PCMN 0x3: Content Adaptive
9:8	COMP0_HORZ_FILTER_MODE	Component 0 (Y or G) horizontal filtering mode 0x0: Nearest Neighbor 0x1: Bilinear 0x2: PCMN 0x3: Content Adaptive
1	SCALEY_EN	Scaling Enable in the Y-direction 0x0: Disabled 0x1: Enabled
0	SCALEX_EN	Scaling Enable in the X-direction 0x0: Disabled 0x1: Enabled

0x01A09210 MDSS_MDP_VP_0_VIG_2_QSEED2_COMP0_3_PHASE_STEP_X**Type:** RW**Reset State:** 0x00000000

QSEED component 0 and 3 phase step for each of the Virtual Pipelines

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_VIG_2_QSEED2_COMP0_3_PHASE_STEP_X

Bits	Name	Description
25:0	PHASE_STEP_X	Unsigned Phase Step in the X-direction

0x01A09214 MDSS_MDP_VP_0_VIG_2_QSEED2_COMP0_3_PHASE_STEP_Y**Type:** RW**Reset State:** 0x00000000

QSEED component 0 and 3 phase step for each of the Virtual Pipelines

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_VIG_2_QSEED2_COMP0_3_PHASE_STEP_Y

Bits	Name	Description
25:0	PHASE_STEP_Y	Unsigned Phase Step in the Y-direction

0x01A09218 MDSS_MDP_VP_0_VIG_2_QSEED2_COMP1_2_PHASE_STEP_X**Type:** RW**Reset State:** 0x00000000

QSEED component 1 and 2 phase step for each of the Virtual Pipelines

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_VIG_2_QSEED2_COMP1_2_PHASE_STEP_X

Bits	Name	Description
25:0	PHASE_STEP_X	Unsigned Phase Step in the X-direction

0x01A0921C MDSS_MDP_VP_0_VIG_2_QSEED2_COMP1_2_PHASE_STEP_Y**Type:** RW**Reset State:** 0x00000000

QSEED component 1 and 2 phase step for each of the Virtual Pipelines

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_VIG_2_QSEED2_COMP1_2_PHASE_STEP_Y

Bits	Name	Description
25:0	PHASE_STEP_Y	Unsigned Phase Step in the Y-direction

0x01A09220 MDSS_MDP_VP_0_VIG_2_QSEED2_COMP0_3_INIT_PHASE_X

Type: RW

Reset State: 0x00000000

QSEED component 0 and 3 init phase for each of the Virtual Pipelines

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_VIG_2_QSEED2_COMP0_3_INIT_PHASE_X

Bits	Name	Description
25:0	INIT_PHASE_X	Signed INIT Phase in the X-direction

0x01A09224 MDSS_MDP_VP_0_VIG_2_QSEED2_COMP0_3_INIT_PHASE_Y

Type: RW

Reset State: 0x00000000

QSEED component 0 and 3 init phase for each of the Virtual Pipelines

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_VIG_2_QSEED2_COMP0_3_INIT_PHASE_Y

Bits	Name	Description
25:0	INIT_PHASE_Y	Signed INIT Phase in the Y-direction

0x01A09228 MDSS_MDP_VP_0_VIG_2_QSEED2_COMP1_2_INIT_PHASE_X

Type: RW

Reset State: 0x00000000

QSEED component 1 and 2 init phase for each of the Virtual Pipelines

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_VIG_2_QSEED2_COMP1_2_INIT_PHASE_X

Bits	Name	Description
25:0	INIT_PHASE_X	Signed INIT Phase in the X-direction

0x01A0922C MDSS_MDP_VP_0_VIG_2_QSEED2_COMP1_2_INIT_PHASE_Y

Type: RW

Reset State: 0x00000000

QSEED component 1 and 2 init phase for each of the Virtual Pipelines

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_VIG_2_QSEED2_COMP1_2_INIT_PHASE_Y

Bits	Name	Description
25:0	INIT_PHASE_Y	Signed INIT Phase in the Y-direction

0x01A09230 MDSS_MDP_VP_0_VIG_2_QSEED2_SHARP_SMOOTH_STRENGTH

Type: RW

Reset State: 0x00000000

QSEED configuration for each of the Virtual Pipelines

double buffered: YES

MDSS_MDP_VP_0_VIG_2_QSEED2_SHARP_SMOOTH_STRENGTH

Bits	Name	Description
8:0	STRENGTH	Sharpening/Smooth Strength used for content Adaptive Filtering S9 notation with a range from -256 to 255 0x0: No sharpening and No smoothing +ve: sharpening -ve: smoothing

0x01A09234 MDSS_MDP_VP_0_VIG_2_QSEED2_SHARP_THRESHOLD_EDGE**Type:** RW**Reset State:** 0x00000000

QSEED configuration for each of the Virtual Pipelines

double buffered: YES

MDSS_MDP_VP_0_VIG_2_QSEED2_SHARP_THRESHOLD_EDGE

Bits	Name	Description
9:0	THRESHOLD	Unsigned Threshold to select Edge/Normal Filter for Content Adaptive Filter

0x01A09238 MDSS_MDP_VP_0_VIG_2_QSEED2_SHARP_THRESHOLD_SMOOTH**Type:** RW**Reset State:** 0x00000000

QSEED configuration for each of the Virtual Pipelines

double buffered: YES

MDSS_MDP_VP_0_VIG_2_QSEED2_SHARP_THRESHOLD_SMOOTH

Bits	Name	Description
9:0	THRESHOLD	Unsigned Threshold to select Normal/Smooth Filter for Content Adaptive Filter

0x01A0923C MDSS_MDP_VP_0_VIG_2_QSEED2_SHARP_THRESHOLD_NOISE**Type:** RW**Reset State:** 0x00000000

QSEED configuration for each of the Virtual Pipelines

double buffered: YES

MDSS_MDP_VP_0_VIG_2_QSEED2_SHARP_THRESHOLD_NOISE

Bits	Name	Description
7:0	THRESHOLD	Unsigned Threshold to select Normal/Smooth Filter for Content Adaptive Filter

0x01A09250 MDSS_MDP_VP_0_VIG_2_QSEED2_NORMAL_FILTER_COEFF_N**Type:** RW**Reset State:** 0x00000000**MDSS_MDP_VP_0_VIG_2_QSEED2_NORMAL_FILTER_COEFF_N**

Bits	Name	Description
11:0	VALUE	

0x01A09260 MDSS_MDP_VP_0_VIG_2_QSEED2_SMOOTH_FILTER_COEFF_N**Type:** RW**Reset State:** 0x00000000**MDSS_MDP_VP_0_VIG_2_QSEED2_SMOOTH_FILTER_COEFF_N**

Bits	Name	Description
11:0	VALUE	

0x01A09270 MDSS_MDP_VP_0_VIG_2_QSEED2_EDGE_FILTER_COEFF_START**Type:** W**Reset State:** 0x00000000

QSEED SW trigger to calculate the edge filter coefficients

double buffered: NO

MDSS_MDP_VP_0_VIG_2_QSEED2_EDGE_FILTER_COEFF_START

Bits	Name	Description
0	START	SW trigger to initiate the calculate of edge coefficients in hardware The pipeline should be idle as smooth/normal filter coefficients are required This implies that the QSEED2 is not active during the trigger

0x01A09288 MDSS_MDP_VP_0_VIG_2_SKIN_ADJUST_PWL0**Type:** RW**Reset State:** 0x00000000

Skin Piece-wise Linear Adjustment Curve

double buffered: Yes

MDSS_MDP_VP_0_VIG_2_SKIN_ADJUST_PWL0

Bits	Name	Description
31:16	HUE_GAIN	S16, is a symmetrical trapezoidal curve, this is the slope of the 2 sides
10:0	HUE_MID	U11, this is the offset from the min and max hue of skin region which marks the top 2 corners of the trapezoid

0x01A0928C MDSS_MDP_VP_0_VIG_2_SKIN_ADJUST_PWL1**Type:** RW**Reset State:** 0x00000000

Skin Piece-wise Linear Adjustment Curve

double buffered: Yes

MDSS_MDP_VP_0_VIG_2_SKIN_ADJUST_PWL1

Bits	Name	Description
31:16	VAL_GAIN	S16, is a symmetrical trapezoidal curve, this is the slope of the 2 sides
15:0	SAT_GAIN	S16, is a symmetrical trapezoidal curve, this is the slope of the 2 sides

0x01A09290 MDSS_MDP_VP_0_VIG_2_SKIN_REGION_HUE**Type:** RW**Reset State:** 0x00000000

Skin Adjustment Hue Qualifier

double buffered: Yes

MDSS_MDP_VP_0_VIG_2_SKIN_REGION_HUE

Bits	Name	Description
26:16	MAX	if MAX > x17ff (i.e. 360 degree), the behaviour will be same as MAX = x1800
10:0	MIN	U11, Hue has to be within the inclusive range of MIN to MAX to qualify for skin tone adjustment on the pixel if MIN > x17ff (i.e. 360 degree), the behaviour will be same as MIN = x1800

0x01A09294 MDSS_MDP_VP_0_VIG_2_SKIN_REGION_SAT**Type:** RW**Reset State:** 0x00000000

Skin Adjustment Saturation Qualifier

double buffered: Yes

MDSS_MDP_VP_0_VIG_2_SKIN_REGION_SAT

Bits	Name	Description
15:8	MAX	Saturation has to be below or equal to this to qualify for skin tone adjustment on the pixel
7:0	MIN	U8, Saturation has to be above or equal to this to qualify for skin tone adjustment on the pixel

0x01A09298 MDSS_MDP_VP_0_VIG_2_SKIN_REGION_VAL**Type:** RW**Reset State:** 0x00000000

Skin Adjustment Value Qualifier

double buffered: Yes

MDSS_MDP_VP_0_VIG_2_SKIN_REGION_VAL

Bits	Name	Description
15:8	MAX	Value has to be below or equal to this to qualify for skin tone adjustment on the pixel
7:0	MIN	U8, Value has to be above or equal to this to qualify for skin tone adjustment on the pixel

0x01A0929C MDSS_MDP_VP_0_VIG_2_SKY_ADJUST_PWL0**Type:** RW**Reset State:** 0x00000000

Sky Piece-wise Linear Adjustment Curve

double buffered: Yes

MDSS_MDP_VP_0_VIG_2_SKY_ADJUST_PWL0

Bits	Name	Description
31:16	HUE_GAIN	S16, is a symmetrical trapezoidal curve, this is the slope of the 2 sides
10:0	HUE_MID	U11, this is the offset from the min and max hue of skin region which marks the top 2 corners of the trapezoid

0x01A092A0 MDSS_MDP_VP_0_VIG_2_SKY_ADJUST_PWL1**Type:** RW**Reset State:** 0x00000000

Sky Piece-wise Linear Adjustment Curve

double buffered: Yes

MDSS_MDP_VP_0_VIG_2_SKY_ADJUST_PWL1

Bits	Name	Description
31:16	VAL_GAIN	S16, is a symmetrical trapezoidal curve, this is the slope of the 2 sides
15:0	SAT_GAIN	S16, is a symmetrical trapezoidal curve, this is the slope of the 2 sides

0x01A092A4 MDSS_MDP_VP_0_VIG_2_SKY_REGION_HUE**Type:** RW**Reset State:** 0x00000000

Sky Adjustment Hue Qualifier

double buffered: Yes

MDSS_MDP_VP_0_VIG_2_SKY_REGION_HUE

Bits	Name	Description
26:16	MAX	if MAX > x17ff (i.e. 360 degree), the behaviour will be same as MAX = x1800
10:0	MIN	U11, Hue has to be within the inclusive range of MIN to MAX (U13) to qualify for sky tone adjustment on the pixel if MIN > x17ff (i.e. 360 degree), the behaviour will be same as MIN = x1800

0x01A092A8 MDSS_MDP_VP_0_VIG_2_SKY_REGION_SAT**Type:** RW**Reset State:** 0x00000000

Sky Adjustment Saturation Qualifier

double buffered: Yes

MDSS_MDP_VP_0_VIG_2_SKY_REGION_SAT

Bits	Name	Description
15:8	MAX	Saturation has to be below or equal to this to qualify for sky adjustment on the pixel
7:0	MIN	U8, Saturation has to be above or equal to this to qualify for sky adjustment on the pixel

0x01A092AC MDSS_MDP_VP_0_VIG_2_SKY_REGION_VAL**Type:** RW**Reset State:** 0x00000000

Sky Adjustment Value Qualifier

double buffered: Yes

MDSS_MDP_VP_0_VIG_2_SKY_REGION_VAL

Bits	Name	Description
15:8	MAX	Value has to be below or equal to this to qualify for sky adjustment on the pixel
7:0	MIN	U8, Value has to be above or equal to this to qualify for sky adjustment on the pixel

0x01A092B0 MDSS_MDP_VP_0_VIG_2_FOLIAGE_ADJUST_PWL0**Type:** RW**Reset State:** 0x00000000**MDSS_MDP_VP_0_VIG_2_FOLIAGE_ADJUST_PWL0**

Bits	Name	Description
31:16	HUE_GAIN	
10:0	HUE_MID	

0x01A092B4 MDSS_MDP_VP_0_VIG_2_FOLIAGE_ADJUST_PWL1**Type:** RW**Reset State:** 0x00000000**MDSS_MDP_VP_0_VIG_2_FOLIAGE_ADJUST_PWL1**

Bits	Name	Description
31:16	VAL_GAIN	
15:0	SAT_GAIN	

0x01A092B8 MDSS_MDP_VP_0_VIG_2_FOLIAGE_REGION_HUE**Type:** RW**Reset State:** 0x00000000

Foilage Adjustment Hue Qualifier

double buffered: Yes

MDSS_MDP_VP_0_VIG_2_FOLIAGE_REGION_HUE

Bits	Name	Description
26:16	MAX	if MAX > x17ff (i.e. 360 degree), the behaviour will be same as MAX = x1800
10:0	MIN	U11, Hue has to be within the inclusive range of MIN to MAX to qualify for foilage tone adjustment on the pixel if MIN > x17ff (i.e. 360 degree), the behaviour will be same as MIN = x1800

0x01A092BC MDSS_MDP_VP_0_VIG_2_FOLIAGE_REGION_SAT**Type:** RW**Reset State:** 0x00000000

Foilage Adjustment Saturation Qualifier

double buffered: Yes

MDSS_MDP_VP_0_VIG_2_FOLIAGE_REGION_SAT

Bits	Name	Description
15:8	MAX	Saturation has to be below or equal to this to qualify for foilage adjustment on the pixel

MDSS_MDP_VP_0_VIG_2_FOLIAGE_REGION_SAT (cont.)

Bits	Name	Description
7:0	MIN	U8, Saturation has to be above or equal to this to qualify for foilage adjustment on the pixel

0x01A092C0 MDSS_MDP_VP_0_VIG_2_FOLIAGE_REGION_VAL**Type:** RW**Reset State:** 0x00000000

Foilage Adjustment Value Qualifier

double buffered: Yes

MDSS_MDP_VP_0_VIG_2_FOLIAGE_REGION_VAL

Bits	Name	Description
15:8	MAX	Value has to be below or equal to this to qualify for foilage adjustment on the pixel
7:0	MIN	U8, Value has to be above or equal to this to qualify for foilage adjustment on the pixel

0x01A092C4 MDSS_MDP_VP_0_VIG_2_HIST_COLLECT_CTL**Type:** W**Reset State:** 0x00000000

Histogram Collection Control

double buffered: YES

except for LOCK

MDSS_MDP_VP_0_VIG_2_HIST_COLLECT_CTL

Bits	Name	Description
0	LOCK	0: SW and HW swap their access on the 2 histogram tables; HW collects on a new table on the new frame, while SW accesses the histogram of the last frame on the table just swapped out by HW 1: current table will be overwritten for new collection on the next frame, the other table remains untouched for SW to continue to access

0x01A092C8 MDSS_MDP_VP_0_VIG_2_HIST_MIN_MAX**Type:** R**Reset State:** 0x00000000

MIN and MAX V value over the entire frame

double buffered: NO

MDSS_MDP_VP_0_VIG_2_HIST_MIN_MAX

Bits	Name	Description
23:16	MAX	Maximum V value of the last collected frame
7:0	MIN	Minimum V value of the last collected frame

0x01A092CC MDSS_MDP_VP_0_VIG_2_HIST_FRAME_COLLECTED**Type:** R**Reset State:** 0x00000000

Histogram Frame Number since the last rising START

double buffered: NO

MDSS_MDP_VP_0_VIG_2_HIST_FRAME_COLLECTED

Bits	Name	Description
31:0	FRAME_NUM	The frame number that the current histogram accessible by SW was collected from

0x01A092D0 MDSS_MDP_VP_0_VIG_2_HIST_SEL**Type:** R**Reset State:** 0x00000000

PA LUTV and Histogram Table Selection

double buffered: NO

MDSS_MDP_VP_0_VIG_2_HIST_SEL

Bits	Name	Description
4	HIST_SEL	0x0: Table 0 is available for SW and Table 1 is available for HW 0x1: Table 1 is available for SW and Table 0 is available for HW

MDSS_MDP_VP_0_VIG_2_HIST_SEL (cont.)

Bits	Name	Description
0	PA_LUTV_SEL	0x0: LUT0 is available for SW and LUT1 is available for HW 0x1: LUT1 is available for SW and LUT0 is available for HW

0x01A092D4 MDSS_MDP_VP_0_VIG_2_HIST_SUM**Type:** R**Reset State:** 0x00000000

Accumulated V Value of the frame

double buffered: NO

MDSS_MDP_VP_0_VIG_2_HIST_SUM

Bits	Name	Description
31:0	HIST_SUM	This sum matches the histogram currently accessible by SW

0x01A092E0 MDSS_MDP_VP_0_VIG_2_HIST_V_DATAN**Type:** RW**Reset State:** 0x00000000**MDSS_MDP_VP_0_VIG_2_HIST_V_DATAN**

Bits	Name	Type	Description
31:24	INDEX	RW	
23:0	VALUE	R	

0x01A092F0 MDSS_MDP_VP_0_VIG_2_PA_LUTV**Type:** RW**Reset State:** 0x00000000

PA LUTV data

Auto-incremented by hardware or by software

256 entries are available for reading

double buffered: NO. Hardware maintains 2 copies

MDSS_MDP_VP_0_VIG_2_PA_LUTV

Bits	Name	Description
17	VALUE_UPDATEN	Software issue an write to the LUT 0: VALUE will be written to RAM at INDEX if INDEX_UPDATE=1 or at INDEX+1 if INDEX_UPDATE=0 1: VALUE will not be written to RAM, this setting is appropriate for example when INDEX needs to be updated for read
16	INDEX_UPDATE	Software issue an index update 0: No index update by software 1: Update LUT index to value specified by INDEX
15:8	INDEX	VALUE is stored in this INDEX of the HIST_LUTn, next write will be stored to the next higher INDEX
7:0	VALUE	Value read depends on the LUT select and which one is available for reading by software The read sequence should always begin with WR INDEX_UPDATE=1, INDEX=<rptr start location> subsequent RD will start from the start location and every read will auto-increment the rptr

0x01A09300 MDSS_MDP_VP_0_VIG_2_PA_LUTV_SWAP**Type:** W**Reset State:** 0x00000000

Histogram SWAP signal to indicate that the HIST_LUTn contents should be swapped.

Software accesses one LUT while HW accesses the other LUT

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_VIG_2_PA_LUTV_SWAP

Bits	Name	Description
0	SWAP	SWAP signal to flip buffers accessible by SW and made available to HW Achieves double buffering of the LUT values

0x01A09310 MDSS_MDP_VP_0_VIG_2_SPA_HUE_ADJUST**Type:** RW**Reset State:** 0x00000000

Source Picture HUE Adjustment

double buffered: YES

MDSS_MDP_VP_0_VIG_2_SPA_HUE_ADJUST

Bits	Name	Description
11:0	VALUE	Value in S12, recommend to be within exclusive range xA00 to x600 (i.e. -360 to +360 degree) x600 is equivalent to a shift of 360 degree if VALUE > x5FF in either negative or positive direction, it is circularly clipped (i.e. VALUE + x600 or VALUE - x600)

0x01A09314 MDSS_MDP_VP_0_VIG_2_SPA_SAT_ADJUST

Type: RW

Reset State: 0x00001A00

Source Picture Saturation Adjustment

double buffered: YES

MDSS_MDP_VP_0_VIG_2_SPA_SAT_ADJUST

Bits	Name	Description
15:8	THRSHLD	Below this threshold of saturation, no adjustment is applied to saturation. This field is in U8
7:0	VALUE	Value in S8

0x01A09318 MDSS_MDP_VP_0_VIG_2_SPA_VAL_ADJUST

Type: RW

Reset State: 0x00000000

Source Picture Adjustment

double buffered: YES

MDSS_MDP_VP_0_VIG_2_SPA_VAL_ADJUST

Bits	Name	Description
7:0	VALUE	Value in S8

0x01A0931C MDSS_MDP_VP_0_VIG_2_SPA_CONT_ADJUST**Type:** RW**Reset State:** 0x00000000

Source Picture Contrast Adjustment

double buffered: YES

MDSS_MDP_VP_0_VIG_2_SPA_CONT_ADJUST

Bits	Name	Description
7:0	VALUE	Value in S8

0x01A09320 MDSS_MDP_VP_0_VIG_2_CSC_1_MATRIX_COEFF_0**Type:** RW**Reset State:** 0x00000200

Color Space conversion coefficients for ALL of the Virtual Pipelines

double buffered: NO

MDSS_MDP_VP_0_VIG_2_CSC_1_MATRIX_COEFF_0

Bits	Name	Description
28:16	COEFF_12	Coefficient value in S4.9 format Row 1 Column 2
12:0	COEFF_11	Coefficient value in S4.9 format Row 1 Column 1

0x01A09324 MDSS_MDP_VP_0_VIG_2_CSC_1_MATRIX_COEFF_1**Type:** RW**Reset State:** 0x00000000

Color Space conversion coefficients for ALL of the Virtual Pipelines

double buffered: NO

MDSS_MDP_VP_0_VIG_2_CSC_1_MATRIX_COEFF_1

Bits	Name	Description
28:16	COEFF_21	Coefficient value in S4.9 format Row 2 Column 1

MDSS_MDP_VP_0_VIG_2_CSC_1_MATRIX_COEFF_1 (cont.)

Bits	Name	Description
12:0	COEFF_13	Coefficient value in S4.9 format Row 1 Column 3

0x01A09328 MDSS_MDP_VP_0_VIG_2_CSC_1_MATRIX_COEFF_2**Type:** RW**Reset State:** 0x00000200

Color Space conversion coefficients for ALL of the Virtual Pipelines

double buffered: NO

MDSS_MDP_VP_0_VIG_2_CSC_1_MATRIX_COEFF_2

Bits	Name	Description
28:16	COEFF_23	Coefficient value in S4.9 format Row 2 Column 3
12:0	COEFF_22	Coefficient value in S4.9 format Row 2 Column 2

0x01A0932C MDSS_MDP_VP_0_VIG_2_CSC_1_MATRIX_COEFF_3**Type:** RW**Reset State:** 0x00000000

Color Space conversion coefficients for ALL of the Virtual Pipelines

double buffered: NO

MDSS_MDP_VP_0_VIG_2_CSC_1_MATRIX_COEFF_3

Bits	Name	Description
28:16	COEFF_32	Coefficient value in S4.9 format Row 3 Column 2
12:0	COEFF_31	Coefficient value in S4.9 format Row 3 Column 1

0x01A09330 MDSS_MDP_VP_0_VIG_2_CSC_1_MATRIX_COEFF_4**Type:** RW**Reset State:** 0x00000200

Color Space conversion coefficients for ALL of the Virtual Pipelines

double buffered: NO

MDSS_MDP_VP_0_VIG_2_CSC_1_MATRIX_COEFF_4

Bits	Name	Description
12:0	COEFF_33	Coefficient value in S4.9 format Row 3 Column 3

0x01A09334 MDSS_MDP_VP_0_VIG_2_CSC_1_COMP_0_PRE_CLAMP

Type: RW

Reset State: 0x000000FF

Component 0 pre-clamp value

double buffered: NO

MDSS_MDP_VP_0_VIG_2_CSC_1_COMP_0_PRE_CLAMP

Bits	Name	Description
15:8	LOW	Unsigned 8 bit pre-clamp value
7:0	HIGH	Unsigned 8 bit pre-clamp value

0x01A09338 MDSS_MDP_VP_0_VIG_2_CSC_1_COMP_1_PRE_CLAMP

Type: RW

Reset State: 0x000000FF

Component 1 pre-clamp value

double buffered: NO

MDSS_MDP_VP_0_VIG_2_CSC_1_COMP_1_PRE_CLAMP

Bits	Name	Description
15:8	LOW	Unsigned 8 bit pre-clamp value
7:0	HIGH	Unsigned 8 bit pre-clamp value

0x01A0933C MDSS_MDP_VP_0_VIG_2_CSC_1_COMP_2_PRE_CLAMP**Type:** RW**Reset State:** 0x000000FF

Component 2 pre-clamp value

double buffered: NO

MDSS_MDP_VP_0_VIG_2_CSC_1_COMP_2_PRE_CLAMP

Bits	Name	Description
15:8	LOW	Unsigned 8 bit pre-clamp value
7:0	HIGH	Unsigned 8 bit pre-clamp value

0x01A09340 MDSS_MDP_VP_0_VIG_2_CSC_1_COMP_0_POST_CLAMP**Type:** RW**Reset State:** 0x000000FF

Component 0 post-clamp value

double buffered: NO

MDSS_MDP_VP_0_VIG_2_CSC_1_COMP_0_POST_CLAMP

Bits	Name	Description
15:8	LOW	Unsigned 8 bit pre-clamp value
7:0	HIGH	Unsigned 8 bit pre-clamp value

0x01A09344 MDSS_MDP_VP_0_VIG_2_CSC_1_COMP_1_POST_CLAMP**Type:** RW**Reset State:** 0x000000FF

Component 1 post-clamp value

double buffered: NO

MDSS_MDP_VP_0_VIG_2_CSC_1_COMP_1_POST_CLAMP

Bits	Name	Description
15:8	LOW	Unsigned 8 bit pre-clamp value
7:0	HIGH	Unsigned 8 bit pre-clamp value

0x01A09348 MDSS_MDP_VP_0_VIG_2_CSC_1_COMP_2_POST_CLAMP**Type:** RW**Reset State:** 0x000000FF

Component 2 post-clamp value

double buffered: NO

MDSS_MDP_VP_0_VIG_2_CSC_1_COMP_2_POST_CLAMP

Bits	Name	Description
15:8	LOW	Unsigned 8 bit pre-clamp value
7:0	HIGH	Unsigned 8 bit pre-clamp value

0x01A0934C MDSS_MDP_VP_0_VIG_2_CSC_1_COMP_0_PRE_BIAS**Type:** RW**Reset State:** 0x00000000

Component 0 pre-bias value

double buffered: NO

MDSS_MDP_VP_0_VIG_2_CSC_1_COMP_0_PRE_BIAS

Bits	Name	Description
8:0	VALUE	Signed 9 bit pre-bias value

0x01A09350 MDSS_MDP_VP_0_VIG_2_CSC_1_COMP_1_PRE_BIAS**Type:** RW**Reset State:** 0x00000000

Component 1 pre-bias value

double buffered: NO

MDSS_MDP_VP_0_VIG_2_CSC_1_COMP_1_PRE_BIAS

Bits	Name	Description
8:0	VALUE	Signed 9 bit pre-bias value

0x01A09354 MDSS_MDP_VP_0_VIG_2_CSC_1_COMP_2_PRE_BIAS**Type:** RW**Reset State:** 0x00000000

Component 2 pre-bias value

double buffered: NO

MDSS_MDP_VP_0_VIG_2_CSC_1_COMP_2_PRE_BIAS

Bits	Name	Description
8:0	VALUE	Signed 9 bit pre-bias value

0x01A09358 MDSS_MDP_VP_0_VIG_2_CSC_1_COMP_0_POST_BIAS**Type:** RW**Reset State:** 0x00000000

Component 0 post-bias value

N double buffered: NO

MDSS_MDP_VP_0_VIG_2_CSC_1_COMP_0_POST_BIAS

Bits	Name	Description
8:0	VALUE	Signed 9 bit pre-bias value

0x01A0935C MDSS_MDP_VP_0_VIG_2_CSC_1_COMP_1_POST_BIAS**Type:** RW**Reset State:** 0x00000000

Component 1 post-bias value

double buffered: NO

MDSS_MDP_VP_0_VIG_2_CSC_1_COMP_1_POST_BIAS

Bits	Name	Description
8:0	VALUE	Signed 9 bit pre-bias value

0x01A09360 MDSS_MDP_VP_0_VIG_2_CSC_1_COMP_2_POST_BIAS**Type:** RW**Reset State:** 0x00000000

Component 2 post-bias value

double buffered: NO

MDSS_MDP_VP_0_VIG_2_CSC_1_COMP_2_POST_BIAS

Bits	Name	Description
8:0	VALUE	Signed 9 bit pre-bias value

0x01A09370 MDSS_MDP_VP_0_VIG_2_QSEED2_MISR_CTRL**Type:** RW**Reset State:** 0x00000000**MDSS_MDP_VP_0_VIG_2_QSEED2_MISR_CTRL**

Bits	Name	Type	Description
31	RUN_MODE	RW	
27:24	INPUT_SEL	RW	
10	STATUS_CLEAR	W	
9	STATUS	R	
8	ENABLE	RW	
7:0	FRAME_COUNT	RW	

0x01A09374 MDSS_MDP_VP_0_VIG_2_QSEED2_MISR_SIGNATURE**Type:** R**Reset State:** 0x00000000**MDSS_MDP_VP_0_VIG_2_QSEED2_MISR_SIGNATURE**

Bits	Name	Description
31:0	CAPTURED	

**0x01A0A000+MDSS_MDP_VP_0_VIG_2_HIST_V_DATAn_BIN_i, i=[0..255]
4*i****Type:** R**Reset State:** 0x00000000

Direct Register Mapped Histogram data collection

256 bins

double buffered: NO

MDSS_MDP_VP_0_VIG_2_HIST_V_DATAn_BIN_i

Bits	Name	Description
23:0	VALUE	Value

**0x01A0A400+MDSS_MDP_VP_0_VIG_2_PA_LUTV_IDX_i, i=[0..127]
4*i****Type:** RW**Reset State:** 0x00000000

Direct Register Mapped PA LUTV data

256 entries are available for writing and reading

double buffered: NO. Hardware maintains 2 copies

MDSS_MDP_VP_0_VIG_2_PA_LUTV_IDX_i

Bits	Name	Description
23:16	VALUE_UPPER	LUT content at location 2*IDX+1 to be written/read to/from the LUT accessible by software at the moment
7:0	VALUE_LOWER	LUT content at location 2*IDX to be written/read to/from the LUT accessible by software at the moment

17.17.7.3.2 MDSS_MDP_VP_0_VIG_2_SSPP

SSPP hardware register description

37.17.7.3.2.1 MDSS_MDP_VP_0_VIG_2_SSPP_SSPP

SSPP

SSPP hardware register description

0x01A09000 MDSS_MDP_VP_0_VIG_2_SSPP_SRC_SIZE**Type:** RW**Reset State:** 0x00000000

Source Size for the region of interest source image

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_VIG_2_SSPP_SRC_SIZE

Bits	Name	Description
31:16	ROI_H	Source Height of the Region Of Interest (ROI)
15:0	ROI_W	Source Width of the Region Of Interest (ROI) The exact supported size depends on the width of line buffers allocated for MDP.

0x01A09004 MDSS_MDP_VP_0_VIG_2_SSPP_SRC_IMG_SIZE**Type:** RW**Reset State:** 0x00000000

Source Size for the image in which the ROI resides

This value along with SRC_SIZE and SRC_XY will be used by hardware to calculate whether pixel repeat or overfetch is required along the ROI border.

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_VIG_2_SSPP_SRC_IMG_SIZE

Bits	Name	Description
31:16	SRC_H	Source Height. Note that this is NOT the ROI height
15:0	SRC_W	Source Width. Note that this is NOT the ROI width

0x01A09008 MDSS_MDP_VP_0_VIG_2_SSPP_SRC_XY**Type:** RW**Reset State:** 0x00000000

Source Offset from the top-left corner of the source image for each of the region of interest dimensions

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_VIG_2_SSPP_SRC_XY

Bits	Name	Description
31:16	SRC_Y	Y-offset of source ROI in the source image
15:0	SRC_X	X-offset of source ROI in the source image

0x01A0900C MDSS_MDP_VP_0_VIG_2_SSPP_OUT_SIZE

Type: RW

Reset State: 0x00000000

Output Size of the layer.

Should be same as source size if scaling is NOT enabled.

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_VIG_2_SSPP_OUT_SIZE

Bits	Name	Description
31:16	DST_H	Height of Output ROI
15:0	DST_W	Width of Output ROI

0x01A09010 MDSS_MDP_VP_0_VIG_2_SSPP_OUT_XY

Type: RW

Reset State: 0x00000000

Output Offset from the top-left corner of the output image

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_VIG_2_SSPP_OUT_XY

Bits	Name	Description
31:16	DST_Y	Height of Output ROI
15:0	DST_X	Width of Output ROI

0x01A09014 MDSS_MDP_VP_0_VIG_2_SSPP_SRC0_ADDR**Type:** RW**Reset State:** 0x00000000

Plane 0 base address

For fetching BWC data, this plane refers to the compressed RAU data

double buffered: YES

when: On frame boundaries (vsync or SW trigger) when ADDR_SYNC_MODE is in frame mode and on line boundary if ADDR_SYNC_MODE in in line mode

MDSS_MDP_VP_0_VIG_2_SSPP_SRC0_ADDR

Bits	Name	Description
31:0	ADDR	Base byte address of the image single interleave color plane, Luma plane in pseudo-planar format, or color0 in planar format

0x01A09018 MDSS_MDP_VP_0_VIG_2_SSPP_SRC1_ADDR**Type:** RW**Reset State:** 0x00000000

Plane 1 base address

For fetching BWC data, this plane refers to the META data to fetch the lengths of the compressed RAU data stream

double buffered: YES

when: On frame boundaries (vsync or SW trigger) when ADDR_SYNC_MODE is in frame mode and on line boundary if ADDR_SYNC_MODE in in line mode

MDSS_MDP_VP_0_VIG_2_SSPP_SRC1_ADDR

Bits	Name	Description
31:0	ADDR	Base byte address of the chroma plane in pseudo-planar format, or color1 in planar format

0x01A0901C MDSS_MDP_VP_0_VIG_2_SSPP_SRC2_ADDR**Type:** RW**Reset State:** 0x00000000

Plane 2 base address

double buffered: YES

when: On frame boundaries (vsync or SW trigger) when ADDR_SYNC_MODE is in frame mode and on line boundary if ADDR_SYNC_MODE in in line mode

MDSS_MDP_VP_0_VIG_2_SSPP_SRC2_ADDR

Bits	Name	Description
31:0	ADDR	Base byte address of color2 in planar format

0x01A09020 MDSS_MDP_VP_0_VIG_2_SSPP_SRC3_ADDR

Type: RW

Reset State: 0x00000000

Plane 3 base address

double buffered: YES

when: On frame boundaries (vsync or SW trigger) when ADDR_SYNC_MODE is in frame mode and on line boundary if ADDR_SYNC_MODE in in line mode

MDSS_MDP_VP_0_VIG_2_SSPP_SRC3_ADDR

Bits	Name	Description
31:0	ADDR	Base byte address of color3 in planar format, or pseudo planar + alpha format

0x01A09024 MDSS_MDP_VP_0_VIG_2_SSPP_SRC_YSTRIDE0

Type: RW

Reset State: 0x00000000

Stride Values

Double-buffered

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_VIG_2_SSPP_SRC_YSTRIDE0

Bits	Name	Description
31:16	SRCP1_YSTRIDE	Plane 1 Y stride in bytes
15:0	SRCP0_YSTRIDE	Plane 0 Y stride in bytes

0x01A09028 MDSS_MDP_VP_0_VIG_2_SSPP_SRC_YSTRIDE1**Type:** RW**Reset State:** 0x00000000

Stride Values

Double-buffered

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_VIG_2_SSPP_SRC_YSTRIDE1

Bits	Name	Description
31:16	SRCP3_YSTRIDE	Plane 3 Y stride in bytes
15:0	SRCP2_YSTRIDE	Plane 2 Y stride in bytes

0x01A09030 MDSS_MDP_VP_0_VIG_2_SSPP_SRC_FORMAT**Type:** RW**Reset State:** 0x00000000

Source Format Configuration for each virtual pipeline.

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_VIG_2_SSPP_SRC_FORMAT

Bits	Name	Description
31:30	FRAME_FORMAT	Frame format In the current revision, there is no support for tiled formats 0x0: Linear, no tile 0x1: Tiled, 4x4
25:23	SRC_CHROMA_SAMP	Source Chroma Sampling 0x0: RGB/YUV 4:4:4 0x1: 4:2:2 - H2V1 0x2: 4:2:2 - H1V2 0x3: 4:2:0 - N2V1 0x4: 4:2:0 - N1V2

MDSS_MDP_VP_0_VIG_2_SSPP_SRC_FORMAT (cont.)

Bits	Name	Description
22	SOLID_FILL	Fetch feeds the lower pipeline with the CONSTANT_COLOR register as its output value. NOTHING is fetched Note that in order to use this feature then Interleaved FETCH_PLANES must be set as well and no chroma sub-sampling enabled
21	VC1_REDUCE	VC1 range reduction enable for YUV sources The reduce range is set by separate register VC1_RANGE
20:19	FETCH_PLANES	Determines the number of planes to fetch 0x0: Interleaved 0x1: Planar 0x2: Pseudo-planar
18	UNPACK_ALIGN	0x0: TO LSB 0x1: TO MSB
17	UNPACK_TIGHT	0x0: Loose 0x1: Tight
13:12	UNPACK_COUNT	Valid unpacking pattern count 0x0 = 1 component 0x1 = 2 components ... 0x3 = 4 components Unpacking applies to interleaved plane (chroma plane in pseudo-planar)
11	ROT90	Source image is to be rotated 90 degrees
10:9	SRC_BPP	Effective source byte per pixel 0x0: 1 byte 0x1: 2 bytes 0x2: 3 bytes 0x3: 4 bytes This is used ONLY when unpacking the interleaved plane
8	SRCC3_EN	Alpha Enable 0x0: Source has NO alpha 0x1: Source has alpha
7:6	SRCC3	Number of bits for component 3 (Alpha) 0x0: 1 bits 0x1: 4 bits 0x2: 6 bits 0x3: 8 bits

MDSS_MDP_VP_0_VIG_2_SSPP_SRC_FORMAT (cont.)

Bits	Name	Description
5:4	SRCC2	Number of bits for component 2 (R/Cr) 0x0: 4 bits 0x1: 5 bits 0x2: 6 bits 0x3: 8 bits
3:2	SRCC1	Number of bits for component 1 (B/Cb) 0x0: 4 bits 0x1: 5 bits 0x2: 6 bits 0x3: 8 bits
1:0	SRCC0	Number of bits for component 0 (G/Luma) 0x0: 4 bits 0x1: 5 bits 0x2: 6 bits 0x3: 8 bits

0x01A09034 MDSS_MDP_VP_0_VIG_2_SSPP_SRC_UNPACK_PATTERN**Type:** RW**Reset State:** 0x00000000

Unpacking pattern, maximum of 4 pattern elements starting from LSB to MSB (0 to 4).

Default is ARGB or C3-C2-C0-C1.

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_VIG_2_SSPP_SRC_UNPACK_PATTERN

Bits	Name	Description
25:24	ELEMENT3	0x0: Color0 0x1: Color1 0x2: Color2 0x3: Color3
17:16	ELEMENT2	0x0: Color0 0x1: Color1 0x2: Color2 0x3: Color3
9:8	ELEMENT1	0x0: Color0 0x1: Color1 0x2: Color2 0x3: Color3

MDSS_MDP_VP_0_VIG_2_SSPP_SRC_UNPACK_PATTERN (cont.)

Bits	Name	Description
1:0	ELEMENT0	0x0: Color0 0x1: Color1 0x2: Color2 0x3: Color3

0x01A09038 MDSS_MDP_VP_0_VIG_2_SSPP_SRC_OP_MODE**Type:** RW**Reset State:** 0x00000000

Operation Mode for each SSPP pipeline

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_VIG_2_SSPP_SRC_OP_MODE

Bits	Name	Description
31	SW_PIX_EXT_OVERRIDE	SW override for pixel extension values. 0x0: Use HW calculated pixel extensions 0x1: Use SW programmed pixel extensions
16	IGC_LUT_EN	Inverse Gamma Correction LUT Enable 0x0: Disabled - mapped Linearly from 8-bit to 12-bit 0x1: Enabled
14:13	FLIP_MODE	Flip Operation Lower Bit: Left/Right flip Upper Bit: Up/Down flip

0x01A0903C MDSS_MDP_VP_0_VIG_2_SSPP_SRC_CONSTANT_COLOR**Type:** RW**Reset State:** 0x00000000

Constant Color Register during solid fill mode

double buffered: NO

MDSS_MDP_VP_0_VIG_2_SSPP_SRC_CONSTANT_COLOR

Bits	Name	Description
31:24	COLOR3	Constant color driven for color3 during solid fill mode.

MDSS_MDP_VP_0_VIG_2_SSPP_SRC_CONSTANT_COLOR (cont.)

Bits	Name	Description
23:16	COLOR2	Constant color driven for color2 during solid fill mode.
15:8	COLOR1	Constant color driven for color1 during solid fill mode.
7:0	COLOR0	Constant color driven for color0 during solid fill mode.

0x01A09048 MDSS_MDP_VP_0_VIG_2_SSPP_FETCH_CONFIG**Type:** RW**Reset State:** 0x00000087

SSPP Client Fetch Configuration per Virtual Pipeline double buffered: NO

This register must NOT be altered during operation and MDP must be idle when changing values

MDSS_MDP_VP_0_VIG_2_SSPP_FETCH_CONFIG

Bits	Name	Description
22	PRI_ARB_DIS	
21	PMARK_DIS	
17:16	TILE_SIZE	Tile Size for processing of BLOCK formats Note that this only applies to blocks that operate in block mode which is limited to DMA source pipe. This setting has no effect for all other source pipes (ViG/RGB) 0x0: 128x128 Others: ILLEGAL / Not Supported
12	ADDR_SYNC_MODE	Boundary on which fetch address synchronizes. Setting this field to line forces the address updates on the next line which performs the software requested feature of updating on hsync. Note that updating on line boundaries will likely result in tearing on the panel since hardware is fetching from new memory buffer. 0x0: frame (default) 0x1: line
9:8	MAL	Describes the Minimum Access Length for the source pipeline. This value should align with the DRAM's expected transaction alignment unit. This setting forces the source fetch pipe to be aligned to DRAM and implement re-alignment for maximum efficiency 0x0: 128b alignment (default) 0x1: 256b alignment 0x2: 512b alignment

MDSS_MDP_VP_0_VIG_2_SSPP_FETCH_CONFIG (cont.)

Bits	Name	Description
7:4	MAX_BURST_SIZE	Maximum burst beat size. Hardware fetch engines operate on 128-bit bus Possible values: 0x8: burst of 8 0x4: burst of 4 0x2: burst of 2
3:0	REQ_DEPTH_LIMIT	Pending request limit, default max value of 0x7 means maximum pending requests allowed in the design Anything less limits the bus requests to that number + 1.

0x01A0904C MDSS_MDP_VP_0_VIG_2_SSPP_VC1_RANGE**Type:** RW**Reset State:** 0x00000707

VC1 Range Map for each virtual pipeline

ONLY valid for pipelines that suport YUV formats

double buffered: NO

MDSS_MDP_VP_0_VIG_2_SSPP_VC1_RANGE

Bits	Name	Description
10:8	RANGE_MAPUV	Chroma range map for VC1 range reduction, default of 7 is equivalent to setting it for main profile.
2:0	RANGE_MAPY	Luma range map for VC1 range reduction, default of 7 is equivalent to setting it for main profile.

0x01A09050 MDSS_MDP_VP_0_VIG_2_SSPP_REQPRIORITY_FIFO_WATERMARK_0**Type:** RW**Reset State:** 0x00000040

FIFO watermark to aid in generation of mdpREQPRIORITY

The watermark here refers to the # of 128-entries available in the Shared Memory Pool

The value programmed here should be aware of the amount programmed in the Shared Memory Pool.

So a value of 0 indicates completely FULL (i.e. there are no more entries available in the SMP)

A value of 100 indicates 100 entries in the Shared Memory Pool (each entry being 128-bit entries)

This is the lowestmost FIFO depth and should not be programmed higher than REQPRIORITY_FIFO_WATERMARK_1 or REQPRIORITY_FIFO_WATERMARK_2

An example programming would set the following:

REQPRIORITY_FIFO_WATERMARK_0 : 50 (meaning 50 entries free to be written into SMP - least room available)

REQPRIORITY_FIFO_WATERMARK_1 : 100 (meaning 100 entries free to be written into SMP - middle ground room available)

REQPRIORITY_FIFO_WATERMARK_2 : 150 (meaning 150 entries free to be written into SMP - most room available)

mdpREQPRIORITY = 11 (highest priority request) when above all 3 watermark levels (i.e. lots of space available in SMP)

mdpREQPRIORITY = 10 above FIFO_0 and FIFO_1 but below FIFO_2

mdpREQPRIORITY = 01 above FIFO_0 but below FIFO_1 and FIFO_2

mdpREQPRIORITY = 00 (lowest priority request) when below all 3 watermark levels (i.e. not a lot of space available in SMP so its good)

double buffered: YES

MDSS_MDP_VP_0_VIG_2_SSPP_REQPRIORITY_FIFO_WATERMARK_0

Bits	Name	Description
15:0	WATERMARK	Watermark Level for FIFO_0

0x01A09054 MDSS_MDP_VP_0_VIG_2_SSPP_REQPRIORITY_FIFO_WATERMARK_1

Type: RW

Reset State: 0x00000080

FIFO watermark to aid in generation of mdpREQPRIORITY

The watermark here refers to the # of 128-entries available in the Shared Memory Pool

The value programmed here should be aware of the amount programmed in the Shared Memory Pool.

So a value of 0 indicates completely FULL (i.e. there are no more entries available in the SMP)

A value of 100 indicates 100 entries in the Shared Memory Pool (each entry being 128-bit entries)

This is the middle FIFO depth and should not be programmed higher than REQPRIORITY_FIFO_WATERMARK_2 or lower than REQPRIORITY_FIFO_WATERMARK_0

An example programming would set the following:

REQPRIORITY_FIFO_WATERMARK_0 : 50 (meaning 50 entries free to be written into SMP - least room available)

REQPRIORITY_FIFO_WATERMARK_1 : 100 (meaning 100 entries free to be written into SMP - middle ground room available)

REQPRIORITY_FIFO_WATERMARK_2 : 150 (meaning 150 entries free to be written into SMP - most room available)

mdpREQPRIORITY = 11 (highest priority request) when above all 3 watermark levels (i.e. lots of space available in SMP)

mdpREQPRIORITY = 10 above FIFO_0 and FIFO_1 but below FIFO_2

mdpREQPRIORITY = 01 above FIFO_0 but below FIFO_1 and FIFO_2

mdpREQPRIORITY = 00 (lowest priority request) when below all 3 watermark levels (i.e. not a lot of space available in SMP so its good)

double buffered: YES

MDSS_MDP_VP_0_VIG_2_SSPP_REQPRIORITY_FIFO_WATERMARK_1

Bits	Name	Description
15:0	WATERMARK	Watermark Level for FIFO_1

0x01A09058 MDSS_MDP_VP_0_VIG_2_SSPP_REQPRIORITY_FIFO_WATERMARK_2

Type: RW

Reset State: 0x000000C0

FIFO watermark to aid in generation of mdpREQPRIORITY

The watermark here refers to the # of 128-entries available in the Shared Memory Pool

The value programmed here should be aware of the amount programmed in the Shared Memory Pool.

So a value of 0 indicates completely FULL (i.e. there are no more entries available in the SMP)

A value of 100 indicates 100 entries in the Shared Memory Pool (each entry being 128-bit entries)

This is the highest FIFO depth and should not be programmed lower than REQPRIORITY_FIFO_WATERMARK_0 or REQPRIORITY_FIFO_WATERMARK_1

An example programming would set the following:

REQPRIORITY_FIFO_WATERMARK_0 : 50 (meaning 50 entries free to be written into SMP - least room available)

REQPRIORITY_FIFO_WATERMARK_1 : 100 (meaning 100 entries free to be written into SMP - middle ground room available)

REQPRIORITY_FIFO_WATERMARK_2 : 150 (meaning 150 entries free to be written into SMP - most room available)

mdpREQPRIORITY = 11 (highest priority request) when above all 3 watermark levels (i.e. lots of space available in SMP)

mdpREQPRIORITY = 10 above FIFO_0 and FIFO_1 but below FIFO_2

mdpREQPRIORITY = 01 above FIFO_0 but below FIFO_1 and FIFO_2

mdpREQPRIORITY = 00 (lowest priority request) when below all 3 watermark levels (i.e. not a lot of space available in SMP so its good)

double buffered: YES

MDSS_MDP_VP_0_VIG_2_SSPP_REQPRIORITY_FIFO_WATERMARK_2

Bits	Name	Description
15:0	WATERMARK	Watermark Level for FIFO_2

0x01A09070 MDSS_MDP_VP_0_VIG_2_SSPP_SRC_ADDR_SW_STATUS

Type: RW

Reset State: 0x00000000

Security Status of the source buffer plane

This is programmed by SW to indicate if the buffer is NON-SECURE or SECURE

This information will be presented to VBIF per transaction and reflected as part of the memory transaction

double buffered: YES

MDSS_MDP_VP_0_VIG_2_SSPP_SRC_ADDR_SW_STATUS

Bits	Name	Description
3	SRC3_ADDR_CP	This is the secure status bit for source plane 3 (if used) If the format doesnt fetch from that plane then this value is ignored Memory transactions will not complete if this status doesnt corespond to system programming in the rest of the path to memory SW should program SRC0_ADDR_CP, SRC1_ADDR_CP, SRC2_ADDR_CP, SRC3_ADDR_CP to the same value 0x0: Non Content Protected Buffer State 0x1: Content Protected Buffer State

MDSS_MDP_VP_0_VIG_2_SSPP_SRC_ADDR_SW_STATUS (cont.)

Bits	Name	Description
2	SRC2_ADDR_CP	This is the secure status bit for source plane 2 (if used) If the format doesn't fetch from that plane then this value is ignored Memory transactions will not complete if this status doesn't correspond to system programming in the rest of the path to memory SW should program SRC0_ADDR_CP, SRC1_ADDR_CP, SRC2_ADDR_CP, SRC3_ADDR_CP to the same value 0x0: Non Content Protected Buffer State 0x1: Content Protected Buffer State
1	SRC1_ADDR_CP	This is the secure status bit for source plane 1 (if used) If the format doesn't fetch from that plane then this value is ignored Memory transactions will not complete if this status doesn't correspond to system programming in the rest of the path to memory SW should program SRC0_ADDR_CP, SRC1_ADDR_CP, SRC2_ADDR_CP, SRC3_ADDR_CP to the same value 0x0: Non Content Protected Buffer State 0x1: Content Protected Buffer State
0	SRC0_ADDR_CP	This is the secure status bit for source plane 0 (if used) If the format doesn't fetch from that plane then this value is ignored Memory transactions will not complete if this status doesn't correspond to system programming in the rest of the path to memory SW should program SRC0_ADDR_CP, SRC1_ADDR_CP, SRC2_ADDR_CP, SRC3_ADDR_CP to the same value 0x0: Non Content Protected Buffer State 0x1: Content Protected Buffer State

0x01A090A4 MDSS_MDP_VP_0_VIG_2_SSPP_CURRENT_SRC0_ADDR**Type:** R**Reset State:** 0x00000000

Current Plane 0 base address that Hardware is using

For fetching BWC data, this plane refers to the compressed RAU data

double buffered: N/A

MDSS_MDP_VP_0_VIG_2_SSPP_CURRENT_SRC0_ADDR

Bits	Name	Description
31:0	ADDR	Base byte address of the image single interleave color plane, Luma plane in pseudo-planar format, or color0 in planar format

0x01A090A8 MDSS_MDP_VP_0_VIG_2_SSPP_CURRENT_SRC1_ADDR**Type:** R**Reset State:** 0x00000000

Plane 1 base address that Hardware is using

For fetching BWC data, this plane refers to the META data to fetch the lengths of the compressed RAU data stream

double buffered: N/A

MDSS_MDP_VP_0_VIG_2_SSPP_CURRENT_SRC1_ADDR

Bits	Name	Description
31:0	ADDR	Base byte address of the chroma plane in pseudo-planar format, or color1 in planar format

0x01A090AC MDSS_MDP_VP_0_VIG_2_SSPP_CURRENT_SRC2_ADDR**Type:** R**Reset State:** 0x00000000

Plane 2 base address that Hardware is using

double buffered: N/A

MDSS_MDP_VP_0_VIG_2_SSPP_CURRENT_SRC2_ADDR

Bits	Name	Description
31:0	ADDR	Base byte address of color2 in planar format

0x01A090B0 MDSS_MDP_VP_0_VIG_2_SSPP_CURRENT_SRC3_ADDR**Type:** R**Reset State:** 0x00000000

Plane 3 base address that Hardware is using

double buffered: N/A

MDSS_MDP_VP_0_VIG_2_SSPP_CURRENT_SRC3_ADDR

Bits	Name	Description
31:0	ADDR	Base byte address of color3 in planar format, or pseudo planar + alpha format

0x01A090B4 MDSS_MDP_VP_0_VIG_2_SSPP_DECIMATION_CONFIG**Type:** RW**Reset State:** 0x00000000

Decimation Configuration

Note that there is no decimation on DMA pipe which means no decimation in line mode on DMA or block mode

double buffered: YES

MDSS_MDP_VP_0_VIG_2_SSPP_DECIMATION_CONFIG

Bits	Name	Description
11:8	HORZ_S	Horizontal Configuration for decimation; required to support downscaling greater than 1/4 S refers to the number of pixels skipped. The algorithm is always the following: read 1 pixel skip S pixels The algorithm is applied to ALL components. Decimation only supports the following values. All other values will result in undefined behaviour The decimation block will provide coarse scaling and should be used in conjunction with qseed2 to support all scaling ratios. 0x0: No decimation is performed 0x1: Read 1 pixel drop 1 pixel. Used to decimate by 2 horizontally 0x3: Read 1 pixel drop 3 pixels. Used to decimate by 4 horizontally 0x7: Read 1 pixel drop 7 pixels. Used to decimate by 8 horizontally 0xF: Read 1 pixel drop 15 pixels. Used to decimate by 16 horizontally
3:0	VERT_S	Vertical Configuration for decimation; required to support downscaling greater than 1/4 S refers to the number of lines skipped. The algorithm is always the following: read 1 line skip S lines The algorithm is applied to ALL components. Decimation only supports the following values. All other values will result in undefined behaviour The decimation block will provide coarse scaling and should be used in conjunction with qseed2 to support all scaling ratios. 0x0: No decimation is performed 0x1: Read 1 line drop 1 line. Used to decimate by 2 horizontally 0x3: Read 1 line drop 3 lines. Used to decimate by 4 horizontally 0x7: Read 1 line drop 7 lines. Used to decimate by 8 horizontally 0xF: Read 1 line drop 15 lines. Used to decimate by 16 horizontally

0x01A090D0 MDSS_MDP_VP_0_VIG_2_SSPP_FETCH_SMP_WR_PLANE0**Type:** R**Reset State:** 0x00000000

Fetch to SMP on plane 0 write counter

double buffered: N/A

MDSS_MDP_VP_0_VIG_2_SSPP_FETCH_SMP_WR_PLANE0

Bits	Name	Description
15:0	COUNT	This counter is reset on every frame and is incremented for each write fetch makes to the SMP plane0 port This port is used for interleaved formats or the Y component for multi-plane source formats This should correlate to value SMP_UNPACK_RD_PLANE0 More useful for debug but can be used for profiling purposes as well.

0x01A090D4 MDSS_MDP_VP_0_VIG_2_SSPP_FETCH_SMP_WR_PLANE1**Type:** R**Reset State:** 0x00000000

Fetch to SMP on plane 1 write counter

double buffered: N/A

MDSS_MDP_VP_0_VIG_2_SSPP_FETCH_SMP_WR_PLANE1

Bits	Name	Description
15:0	COUNT	This counter is reset on every frame and is incremented for each write fetch makes to the SMP plane1 port This port is NOT used for interleaved formats but is used for U (planar) or UV (pseudo-planar) formats This should correlate to value SMP_UNPACK_RD_PLANE1 More useful for debug but can be used for profiling purposes as well.

0x01A090D8 MDSS_MDP_VP_0_VIG_2_SSPP_FETCH_SMP_WR_PLANE2**Type:** R**Reset State:** 0x00000000

Fetch to SMP on plane 2 write counter

double buffered: N/A

MDSS_MDP_VP_0_VIG_2_SSPP_FETCH_SMP_WR_PLANE2

Bits	Name	Description
15:0	COUNT	This counter is reset on every frame and is incremented for each write fetch makes to the SMP plane2 port This port is NOT used for interleaved formats but is used for V (planar) or A (pseudo-planar plus alpha) formats This should correlate to value SMP_UNPACK_RD_PLANE2 More useful for debug but can be used for profiling purposes as well.

0x01A090E0 MDSS_MDP_VP_0_VIG_2_SSPP_SMP_UNPACK_RD_PLANE0**Type:** R**Reset State:** 0x00000000

SMP to unpack on plane 0 read counter

double buffered: N/A

MDSS_MDP_VP_0_VIG_2_SSPP_SMP_UNPACK_RD_PLANE0

Bits	Name	Description
15:0	COUNT	This counter is reset on every frame and is incremented for each line read from SMP plane0 port This port is used for interleaved formats or the Y component for multi-plane source formats This should correlate to FETCH_SMP_WR_PLANE0 value More useful for debug but can be used for profiling purposes as well.

0x01A090E4 MDSS_MDP_VP_0_VIG_2_SSPP_SMP_UNPACK_RD_PLANE1**Type:** R**Reset State:** 0x00000000

SMP to unpack on plane 1 read counter

double buffered: N/A

MDSS_MDP_VP_0_VIG_2_SSPP_SMP_UNPACK_RD_PLANE1

Bits	Name	Description
15:0	COUNT	This counter is reset on every frame and is incremented for each line read from SMP plane1 port This port is NOT used for interleaved formats but is used for U (planar) or UV (pseudo-planar) formats This should correlate to FETCH_SMP_WR_PLANE1 value More useful for debug but can be used for profiling purposes as well.

0x01A090E8 MDSS_MDP_VP_0_VIG_2_SSPP_SMP_UNPACK_RD_PLANE2**Type:** R**Reset State:** 0x00000000

SMP to unpack on plane 2 read counter

double buffered: N/A

MDSS_MDP_VP_0_VIG_2_SSPP_SMP_UNPACK_RD_PLANE2

Bits	Name	Description
15:0	COUNT	This counter is reset on every frame and is incremented for each line read from SMP plane1 port This port is NOT used for interleaved formats but is used for V (planar) or A (pseudo-planar plus alpha) formats This should correlate to FETCH_SMP_WR_PLANE1 value More useful for debug but can be used for profiling purposes as well.

0x01A09100 MDSS_MDP_VP_0_VIG_2_SSPP_SW_PIX_EXT_C0_LR**Type:** RW**Reset State:** 0x00000000

SW Pixel Extension Override for Left/Right of Color 0

These value only take effect if SW_PIX_EXT_OVERRIDE is set in SRC_OP_MODE register

double buffered: YES

MDSS_MDP_VP_0_VIG_2_SSPP_SW_PIX_EXT_C0_LR

Bits	Name	Description
31:24	RIGHT_OVF	Signed Value for Right Overfetch value. +ve value indicates overfetch -ve value indicates pixel drop
23:16	RIGHT_RPT	Unsigned Value for Right Pixel Repeat value.
15:8	LEFT_OVF	Signed Value for Left Overfetch value. +ve value indicates overfetch -ve value indicates pixel drop
7:0	LEFT_RPT	Unsigned Value for Left Pixel Repeat value.

0x01A09104 MDSS_MDP_VP_0_VIG_2_SSPP_SW_PIX_EXT_C0_TB**Type:** RW**Reset State:** 0x00000000

SW Pixel Extension Override for Left/Right of Color 0

These value only take effect if SW_PIX_EXT_OVERRIDE is set in SRC_OP_MODE register
double buffered: YES

MDSS_MDP_VP_0_VIG_2_SSPP_SW_PIX_EXT_C0_TB

Bits	Name	Description
31:24	BOTTOM_OVF	Signed Value for Bottom Overfetch value. +ve value indicates overfetch -ve value indicates line drop
23:16	BOTTOM_RPT	
15:8	TOP_OVF	Signed Value for Top Overfetch value. +ve value indicates overfetch -ve value indicates line drop
7:0	TOP_RPT	Unsigned Value for TOP Line Repeat value.

0x01A09108 MDSS_MDP_VP_0_VIG_2_SSPP_SW_PIX_EXT_C0_REQ_PIXELS**Type:** RW**Reset State:** 0x00000000

SW Pixel Extension Override for pixel height/width of Color 0

These value only take effect if SW_PIX_EXT_OVERRIDE is set in SRC_OP_MODE register

double buffered: YES

MDSS_MDP_VP_0_VIG_2_SSPP_SW_PIX_EXT_C0_REQ_PIXELS

Bits	Name	Description
31:16	TOP_BOTTOM	Unsigned Value for pixel height post pixel extension.
15:0	LEFT_RIGHT	Unsigned Value for pixel width post pixel extension.

0x01A09110 MDSS_MDP_VP_0_VIG_2_SSPP_SW_PIX_EXT_C1C2_LR

Type: RW

Reset State: 0x00000000

SW Pixel Extension Override for Left/Right of Color 0

These value only take effect if SW_PIX_EXT_OVERRIDE is set in SRC_OP_MODE register

double buffered: YES

MDSS_MDP_VP_0_VIG_2_SSPP_SW_PIX_EXT_C1C2_LR

Bits	Name	Description
31:24	RIGHT_OVF	Signed Value for Right Overfetch value. +ve value indicates overfetch -ve value indicates pixel drop
23:16	RIGHT_RPT	Unsigned Value for Right Pixel Repeat value.
15:8	LEFT_OVF	Signed Value for Left Overfetch value. +ve value indicates overfetch -ve value indicates pixel drop
7:0	LEFT_RPT	Unsigned Value for Left Pixel Repeat value.

0x01A09114 MDSS_MDP_VP_0_VIG_2_SSPP_SW_PIX_EXT_C1C2_TB

Type: RW

Reset State: 0x00000000

SW Pixel Extension Override for Left/Right of Color 0

These value only take effect if SW_PIX_EXT_OVERRIDE is set in SRC_OP_MODE register

double buffered: YES

MDSS_MDP_VP_0_VIG_2_SSPP_SW_PIX_EXT_C1C2_TB

Bits	Name	Description
31:24	BOTTOM_OVF	Signed Value for Bottom Overfetch value. +ve value indicates overfetch -ve value indicates line drop
23:16	BOTTOM_RPT	
15:8	TOP_OVF	Signed Value for Top Overfetch value. +ve value indicates overfetch -ve value indicates line drop
7:0	TOP_RPT	Unsigned Value for TOP Line Repeat value.

0x01A09118 MDSS_MDP_VP_0_VIG_2_SSPP_SW_PIX_EXT_C1C2_REQ_PIXELS**Type:** RW**Reset State:** 0x00000000

SW Pixel Extension Override for pixel height/width of Color 0

These value only take effect if SW_PIX_EXT_OVERRIDE is set in SRC_OP_MODE register

double buffered: YES

MDSS_MDP_VP_0_VIG_2_SSPP_SW_PIX_EXT_C1C2_REQ_PIXELS

Bits	Name	Description
31:16	TOP_BOTTOM	Unsigned Value for pixel height post pixel extension.
15:0	LEFT_RIGHT	Unsigned Value for pixel width post pixel extension.

0x01A09120 MDSS_MDP_VP_0_VIG_2_SSPP_SW_PIX_EXT_C3_LR**Type:** RW**Reset State:** 0x00000000

SW Pixel Extension Override for Left/Right of Color 0

These value only take effect if SW_PIX_EXT_OVERRIDE is set in SRC_OP_MODE register

double buffered: YES

MDSS_MDP_VP_0_VIG_2_SSPP_SW_PIX_EXT_C3_LR

Bits	Name	Description
31:24	RIGHT_OVF	Signed Value for Right Overfetch value. +ve value indicates overfetch -ve value indicates pixel drop
23:16	RIGHT_RPT	Unsigned Value for Right Pixel Repeat value.
15:8	LEFT_OVF	Signed Value for Left Overfetch value. +ve value indicates overfetch -ve value indicates pixel drop
7:0	LEFT_RPT	Unsigned Value for Left Pixel Repeat value.

0x01A09124 MDSS_MDP_VP_0_VIG_2_SSPP_SW_PIX_EXT_C3_TB**Type:** RW**Reset State:** 0x00000000

SW Pixel Extension Override for Left/Right of Color 0

These value only take effect if SW_PIX_EXT_OVERRIDE is set in SRC_OP_MODE register
double buffered: YES

MDSS_MDP_VP_0_VIG_2_SSPP_SW_PIX_EXT_C3_TB

Bits	Name	Description
31:24	BOTTOM_OVF	Signed Value for Bottom Overfetch value. +ve value indicates overfetch -ve value indicates line drop
23:16	BOTTOM_RPT	
15:8	TOP_OVF	Signed Value for Top Overfetch value. +ve value indicates overfetch -ve value indicates line drop
7:0	TOP_RPT	Unsigned Value for TOP Line Repeat value.

0x01A09128 MDSS_MDP_VP_0_VIG_2_SSPP_SW_PIX_EXT_C3_REQ_PIXELS**Type:** RW**Reset State:** 0x00000000

SW Pixel Extension Override for pixel height/width of Color 0

These value only take effect if SW_PIX_EXT_OVERRIDE is set in SRC_OP_MODE register

double buffered: YES

MDSS_MDP_VP_0_VIG_2_SSPP_SW_PIX_EXT_C3_REQ_PIXELS

Bits	Name	Description
31:16	TOP_BOTTOM	Unsigned Value for pixel height post pixel extension.
15:0	LEFT_RIGHT	Unsigned Value for pixel width post pixel extension.

17.17.7.4 MDSS_MDP_VP_0_RGB_0

RGB hardware register description

17.17.7.4.1 MDSS_MDP_VP_0_RGB_0_RGB

RGB

0x01A15300 MDSS_MDP_VP_0_RGB_0_MULTI2ONE_MISR_CTRL

Type: RW

Reset State: 0x00000000

MDSS_MDP_VP_0_RGB_0_MULTI2ONE_MISR_CTRL

Bits	Name	Type	Description
31	RUN_MODE	RW	
27:24	INPUT_SEL	RW	
10	STATUS_CLEAR	W	
9	STATUS	R	
8	ENABLE	RW	
7:0	FRAME_COUNT	RW	

0x01A15304 MDSS_MDP_VP_0_RGB_0_MULTI2ONE_MISR_SIGNATURE

Type: R

Reset State: 0x00000000

MDSS_MDP_VP_0_RGB_0_MULTI2ONE_MISR_SIGNATURE

Bits	Name	Description
31:0	CAPTURED	

17.17.7.4.2 MDSS_MDP_VP_0_RGB_0_SSPP

SSPP hardware register description

37.17.7.4.2.1 MDSS_MDP_VP_0_RGB_0_SSPP_SSPP

SSPP

SSPP hardware register description

0x01A15000 MDSS_MDP_VP_0_RGB_0_SSPP_SRC_SIZE

Type: RW

Reset State: 0x00000000

Source Size for the region of interest source image

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_RGB_0_SSPP_SRC_SIZE

Bits	Name	Description
31:16	ROI_H	Source Height of the Region Of Interest (ROI)
15:0	ROI_W	Source Width of the Region Of Interest (ROI) The exact supported size depends on the width of line buffers allocated for MDP.

0x01A15004 MDSS_MDP_VP_0_RGB_0_SSPP_SRC_IMG_SIZE

Type: RW

Reset State: 0x00000000

Source Size for the image in which the ROI resides

This value along with SRC_SIZE and SRC_XY will be used by hardware to calculate whether pixel repeat or overfetch is required along the ROI border.

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_RGB_0_SSPP_SRC_IMG_SIZE

Bits	Name	Description
31:16	SRC_H	Source Height. Note that this is NOT the ROI height

MDSS_MDP_VP_0_RGB_0_SSPP_SRC_IMG_SIZE (cont.)

Bits	Name	Description
15:0	SRC_W	Source Width. Note that this is NOT the ROI width

0x01A15008 MDSS_MDP_VP_0_RGB_0_SSPP_SRC_XY**Type:** RW**Reset State:** 0x00000000

Source Offset from the top-left corner of the source image for each of the region of interest dimensions

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_RGB_0_SSPP_SRC_XY

Bits	Name	Description
31:16	SRC_Y	Y-offset of source ROI in the source image
15:0	SRC_X	X-offset of source ROI in the source image

0x01A1500C MDSS_MDP_VP_0_RGB_0_SSPP_OUT_SIZE**Type:** RW**Reset State:** 0x00000000

Output Size of the layer.

Should be same as source size if scaling is NOT enabled.

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_RGB_0_SSPP_OUT_SIZE

Bits	Name	Description
31:16	DST_H	Height of Output ROI
15:0	DST_W	Width of Output ROI

0x01A15010 MDSS_MDP_VP_0_RGB_0_SSPP_OUT_XY**Type:** RW**Reset State:** 0x00000000

Output Offset from the top-left corner of the output image

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_RGB_0_SSPP_OUT_XY

Bits	Name	Description
31:16	DST_Y	Height of Output ROI
15:0	DST_X	Width of Output ROI

0x01A15014 MDSS_MDP_VP_0_RGB_0_SSPP_SRC0_ADDR**Type:** RW**Reset State:** 0x00000000

Plane 0 base address

For fetching BWC data, this plane refers to the compressed RAU data

double buffered: YES

when: On frame boundaries (vsync or SW trigger) when ADDR_SYNC_MODE is in frame mode and on line boundary if ADDR_SYNC_MODE in in line mode

MDSS_MDP_VP_0_RGB_0_SSPP_SRC0_ADDR

Bits	Name	Description
31:0	ADDR	Base byte address of the image single interleave color plane, Luma plane in pseudo-planar format, or color0 in planar format

0x01A15018 MDSS_MDP_VP_0_RGB_0_SSPP_SRC1_ADDR**Type:** RW**Reset State:** 0x00000000

Plane 1 base address

For fetching BWC data, this plane refers to the META data to fetch the lengths of the compressed RAU data stream

double buffered: YES

when: On frame boundaries (vsync or SW trigger) when ADDR_SYNC_MODE is in frame mode and on line boundary if ADDR_SYNC_MODE in in line mode

MDSS_MDP_VP_0_RGB_0_SSPP_SRC1_ADDR

Bits	Name	Description
31:0	ADDR	Base byte address of the chroma plane in pseudo-planar format, or color1 in planar format

0x01A1501C MDSS_MDP_VP_0_RGB_0_SSPP_SRC2_ADDR

Type: RW

Reset State: 0x00000000

Plane 2 base address

double buffered: YES

when: On frame boundaries (vsync or SW trigger) when ADDR_SYNC_MODE is in frame mode and on line boundary if ADDR_SYNC_MODE in in line mode

MDSS_MDP_VP_0_RGB_0_SSPP_SRC2_ADDR

Bits	Name	Description
31:0	ADDR	Base byte address of color2 in planar format

0x01A15020 MDSS_MDP_VP_0_RGB_0_SSPP_SRC3_ADDR

Type: RW

Reset State: 0x00000000

Plane 3 base address

double buffered: YES

when: On frame boundaries (vsync or SW trigger) when ADDR_SYNC_MODE is in frame mode and on line boundary if ADDR_SYNC_MODE in in line mode

MDSS_MDP_VP_0_RGB_0_SSPP_SRC3_ADDR

Bits	Name	Description
31:0	ADDR	Base byte address of color3 in planar format, or pseudo planar + alpha format

0x01A15024 MDSS_MDP_VP_0_RGB_0_SSPP_SRC_YSTRIDE0**Type:** RW**Reset State:** 0x00000000

Stride Values

Double-buffered

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_RGB_0_SSPP_SRC_YSTRIDE0

Bits	Name	Description
31:16	SRCP1_YSTRIDE	Plane 1 Y stride in bytes
15:0	SRCP0_YSTRIDE	Plane 0 Y stride in bytes

0x01A15028 MDSS_MDP_VP_0_RGB_0_SSPP_SRC_YSTRIDE1**Type:** RW**Reset State:** 0x00000000

Stride Values

Double-buffered

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_RGB_0_SSPP_SRC_YSTRIDE1

Bits	Name	Description
31:16	SRCP3_YSTRIDE	Plane 3 Y stride in bytes
15:0	SRCP2_YSTRIDE	Plane 2 Y stride in bytes

0x01A15030 MDSS_MDP_VP_0_RGB_0_SSPP_SRC_FORMAT**Type:** RW**Reset State:** 0x00000000

Source Format Configuration for each virtual pipeline.

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_RGB_0_SSPP_SRC_FORMAT

Bits	Name	Description
31:30	FRAME_FORMAT	Frame format In the current revision, there is no support for tiled formats 0x0: Linear, no tile 0x1: Tiled, 4x4
25:23	SRC_CHROMA_SAMP	Source Chroma Sampling 0x0: RGB/YUV 4:4:4 0x1: 4:2:2 - H2V1 0x2: 4:2:2 - H1V2 0x3: 4:2:0 - N2V1 0x4: 4:2:0 - N1V2
22	SOLID_FILL	Fetch feeds the lower pipeline with the CONSTANT_COLOR register as its output value. NOTHING is fetched Note that in order to use this feature then Interleaved FETCH_PLANES must be set as well and no chroma sub-sampling enabled
21	VC1_REDUCE	VC1 range reduction enable for YUV sources The reduce range is set by separate register VC1_RANGE
20:19	FETCH_PLANES	Determines the number of planes to fetch 0x0: Interleaved 0x1: Planar 0x2: Pseudo-planar
18	UNPACK_ALIGN	0x0: TO LSB 0x1: TO MSB
17	UNPACK_TIGHT	0x0: Loose 0x1: Tight
13:12	UNPACK_COUNT	Valid unpacking pattern count 0x0 = 1 component 0x1 = 2 components ... 0x3 = 4 components Unpacking applies to interleaved plane (chroma plane in pseudo-planar)
11	ROT90	Source image is to be rotated 90 degrees
10:9	SRC_BPP	Effective source byte per pixel 0x0: 1 byte 0x1: 2 bytes 0x2: 3 bytes 0x3: 4 bytes This is used ONLY when unpacking the interleaved plane

MDSS_MDP_VP_0_RGB_0_SSPP_SRC_FORMAT (cont.)

Bits	Name	Description
8	SRCC3_EN	Alpha Enable 0x0: Source has NO alpha 0x1: Source has alpha
7:6	SRCC3	Number of bits for component 3 (Alpha) 0x0: 1 bits 0x1: 4 bits 0x2: 6 bits 0x3: 8 bits
5:4	SRCC2	Number of bits for component 2 (R/Cr) 0x0: 4 bits 0x1: 5 bits 0x2: 6 bits 0x3: 8 bits
3:2	SRCC1	Number of bits for component 1 (B/Cb) 0x0: 4 bits 0x1: 5 bits 0x2: 6 bits 0x3: 8 bits
1:0	SRCC0	Number of bits for component 0 (G/Luma) 0x0: 4 bits 0x1: 5 bits 0x2: 6 bits 0x3: 8 bits

0x01A15034 MDSS_MDP_VP_0_RGB_0_SSPP_SRC_UNPACK_PATTERN**Type:** RW**Reset State:** 0x00000000

Unpacking pattern, maximum of 4 pattern elements starting from LSB to MSB (0 to 4).

Default is ARGB or C3-C2-C0-C1.

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_RGB_0_SSPP_SRC_UNPACK_PATTERN

Bits	Name	Description
25:24	ELEMENT3	0x0: Color0 0x1: Color1 0x2: Color2 0x3: Color3

MDSS_MDP_VP_0_RGB_0_SSPP_SRC_UNPACK_PATTERN (cont.)

Bits	Name	Description
17:16	ELEMENT2	0x0: Color0 0x1: Color1 0x2: Color2 0x3: Color3
9:8	ELEMENT1	0x0: Color0 0x1: Color1 0x2: Color2 0x3: Color3
1:0	ELEMENT0	0x0: Color0 0x1: Color1 0x2: Color2 0x3: Color3

0x01A15038 MDSS_MDP_VP_0_RGB_0_SSPP_SRC_OP_MODE**Type:** RW**Reset State:** 0x00000000

Operation Mode for each SSPP pipeline

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_RGB_0_SSPP_SRC_OP_MODE

Bits	Name	Description
31	SW_PIX_EXT_OVERRIDE	SW override for pixel extension values. 0x0: Use HW calculated pixel extensions 0x1: Use SW programmed pixel extensions
16	IGC_LUT_EN	Inverse Gamma Correction LUT Enable 0x0: Disabled - mapped Linearly from 8-bit to 12-bit 0x1: Enabled
14:13	FLIP_MODE	Flip Operation Lower Bit: Left/Right flip Upper Bit: Up/Down flip

0x01A1503C MDSS_MDP_VP_0_RGB_0_SSPP_SRC_CONSTANT_COLOR**Type:** RW**Reset State:** 0x00000000

Constant Color Register during solid fill mode

double buffered: NO

MDSS_MDP_VP_0_RGB_0_SSPP_SRC_CONSTANT_COLOR

Bits	Name	Description
31:24	COLOR3	Constant color driven for color3 during solid fill mode.
23:16	COLOR2	Constant color driven for color2 during solid fill mode.
15:8	COLOR1	Constant color driven for color1 during solid fill mode.
7:0	COLOR0	Constant color driven for color0 during solid fill mode.

0x01A15048 MDSS_MDP_VP_0_RGB_0_SSPP_FETCH_CONFIG

Type: RW

Reset State: 0x00000087

SSPP Client Fetch Configuration per Virtual Pipeline double buffered: NO

This register must NOT be altered during operation and MDP must be idle when changing values

MDSS_MDP_VP_0_RGB_0_SSPP_FETCH_CONFIG

Bits	Name	Description
22	PRI_ARB_DIS	
21	PMARK_DIS	
17:16	TILE_SIZE	Tile Size for processing of BLOCK formats Note that this only applies to blocks that operate in block mode which is limited to DMA source pipe. This setting has no effect for all other source pipes (ViG/RGB) 0x0: 128x128 Others: ILLEGAL / Not Supported
12	ADDR_SYNC_MODE	Boundary on which fetch address synchronizes. Setting this field to line forces the address updates on the next line which performs the software requested feature of updating on hsync. Note that updating on line boundaries will likely result in tearing on the panel since hardware is fetching from new memory buffer. 0x0: frame (default) 0x1: line

MDSS_MDP_VP_0_RGB_0_SSPP_FETCH_CONFIG (cont.)

Bits	Name	Description
9:8	MAL	Describes the Minimum Access Length for the source pipeline. This value should align with the DRAM's expected transaction alignment unit. This setting forces the source fetch pipe to be aligned to DRAM and implement re-alignment for maximum efficiency 0x0: 128b alignment (default) 0x1: 256b alignment 0x2: 512b alignment
7:4	MAX_BURST_SIZE	Maximum burst beat size. Hardware fetch engines operate on 128-bit bus Possible values: 0x8: burst of 8 0x4: burst of 4 0x2: burst of 2
3:0	REQ_DEPTH_LIMIT	Pending request limit, default max value of 0x7 means maximum pending requests allowed in the design Anything less limits the bus requests to that number + 1.

0x01A1504C MDSS_MDP_VP_0_RGB_0_SSPP_VC1_RANGE**Type:** RW**Reset State:** 0x00000707

VC1 Range Map for each virtual pipeline

ONLY valid for pipelines that suport YUV formats

double buffered: NO

MDSS_MDP_VP_0_RGB_0_SSPP_VC1_RANGE

Bits	Name	Description
10:8	RANGE_MAPUV	Chroma range map for VC1 range reduction, default of 7 is equivalent to setting it for main profile.
2:0	RANGE_MAPY	Luma range map for VC1 range reduction, default of 7 is equivalent to setting it for main profile.

0x01A15050 MDSS_MDP_VP_0_RGB_0_SSPP_REQPRIORITY_FIFO_WATERMARK_0**Type:** RW**Reset State:** 0x00000040

FIFO watermark to aid in generation of mdpREQPRIORITY

The watermark here refers to the # of 128-entries available in the Shared Memory Pool

The value programmed here should be aware of the amount programmed in the Shared Memory Pool.

So a value of 0 indicates completely FULL (i.e. there are no more entries available in the SMP)

A value of 100 indicates 100 entries in the Shared Memory Pool (each entry being 128-bit entries)

This is the lowestmost FIFO depth and should not be programmed higher than REQPRIORITY_FIFO_WATERMARK_1 or REQPRIORITY_FIFO_WATERMARK_2

An example programming would set the following:

REQPRIORITY_FIFO_WATERMARK_0 : 50 (meaning 50 entries free to be written into SMP - least room available)

REQPRIORITY_FIFO_WATERMARK_1 : 100 (meaning 100 entries free to be written into SMP - middle ground room available)

REQPRIORITY_FIFO_WATERMARK_2 : 150 (meaning 150 entries free to be written into SMP - most room available)

mdpREQPRIORITY = 11 (highest priority request) when above all 3 watermark levels (i.e. lots of space available in SMP)

mdpREQPRIORITY = 10 above FIFO_0 and FIFO_1 but below FIFO_2

mdpREQPRIORITY = 01 above FIFO_0 but below FIFO_1 and FIFO_2

mdpREQPRIORITY = 00 (lowest priority request) when below all 3 watermark levels (i.e. not a lot of space available in SMP so its good)

double buffered: YES

MDSS_MDP_VP_0_RGB_0_SSPP_REQPRIORITY_FIFO_WATERMARK_0

Bits	Name	Description
15:0	WATERMARK	Watermark Level for FIFO_0

0x01A15054 MDSS_MDP_VP_0_RGB_0_SSPP_REQPRIORITY_FIFO_WATERMARK_1

Type: RW

Reset State: 0x00000080

FIFO watermark to aid in generation of mdpREQPRIORITY

The watermark here refers to the # of 128-entries available in the Shared Memory Pool

The value programmed here should be aware of the amount programmed in the Shared Memory Pool.

So a value of 0 indicates completely FULL (i.e. there are no more entries available in the SMP)

A value of 100 indicates 100 entries in the Shared Memory Pool (each entry being 128-bit entries)

This is the middle FIFO depth and should not be programmed higher than
REQPRIORITY_FIFO_WATERMARK_2 or lower than
REQPRIORITY_FIFO_WATERMARK_0

An example programming would set the following:

REQPRIORITY_FIFO_WATERMARK_0 : 50 (meaning 50 entries free to be written into SMP - least room available)

REQPRIORITY_FIFO_WATERMARK_1 : 100 (meaning 100 entries free to be written into SMP - middle ground room available)

REQPRIORITY_FIFO_WATERMARK_2 : 150 (meaning 150 entries free to be written into SMP - most room available)

mdpREQPRIORITY = 11 (highest priority request) when above all 3 watermark levels (i.e. lots of space available in SMP)

mdpREQPRIORITY = 10 above FIFO_0 and FIFO_1 but below FIFO_2

mdpREQPRIORITY = 01 above FIFO_0 but below FIFO_1 and FIFO_2

mdpREQPRIORITY = 00 (lowest priority request) when below all 3 watermark levels (i.e. not a lot of space available in SMP so its good)

double buffered: YES

MDSS_MDP_VP_0_RGB_0_SSPP_REQPRIORITY_FIFO_WATERMARK_1

Bits	Name	Description
15:0	WATERMARK	Watermark Level for FIFO_1

0x01A15058 MDSS_MDP_VP_0_RGB_0_SSPP_REQPRIORITY_FIFO_WATERMARK_2

Type: RW

Reset State: 0x000000C0

FIFO watermark to aid in generation of mdpREQPRIORITY

The watermark here refers to the # of 128-entries available in the Shared Memory Pool

The value programmed here should be aware of the amount programmed in the Shared Memory Pool.

So a value of 0 indicates completely FULL (i.e. there are no more entries available in the SMP)

A value of 100 indicates 100 entries in the Shared Memory Pool (each entry being 128-bit entries)

This is the highest FIFO depth and should not be programmed lower than
REQPRIORITY_FIFO_WATERMARK_0 or REQPRIORITY_FIFO_WATERMARK_1

An example programming would set the following:

REQPRIORITY_FIFO_WATERMARK_0 : 50 (meaning 50 entries free to be written into SMP - least room available)

REQPRIORITY_FIFO_WATERMARK_1 : 100 (meaning 100 entries free to be written into SMP - middle ground room available)

REQPRIORITY_FIFO_WATERMARK_2 : 150 (meaning 150 entries free to be written into SMP - most room available)

mdpREQPRIORITY = 11 (highest priority request) when above all 3 watermark levels (i.e. lots of space available in SMP)

mdpREQPRIORITY = 10 above FIFO_0 and FIFO_1 but below FIFO_2

mdpREQPRIORITY = 01 above FIFO_0 but below FIFO_1 and FIFO_2

mdpREQPRIORITY = 00 (lowest priority request) when below all 3 watermark levels (i.e. not a lot of space available in SMP so its good)

double buffered: YES

MDSS_MDP_VP_0_RGB_0_SSPP_REQPRIORITY_FIFO_WATERMARK_2

Bits	Name	Description
15:0	WATERMARK	Watermark Level for FIFO_2

0x01A15070 MDSS_MDP_VP_0_RGB_0_SSPP_SRC_ADDR_SW_STATUS

Type: RW

Reset State: 0x00000000

Security Status of the source buffer plane

This is programmed by SW to indicate if the buffer is NON-SECURE or SECURE

This information will be presented to VBIF per transaction and reflected as part of the memory transaction

double buffered: YES

MDSS_MDP_VP_0_RGB_0_SSPP_SRC_ADDR_SW_STATUS

Bits	Name	Description
3	SRC3_ADDR_CP	This is the secure status bit for source plane 3 (if used) If the format doesnt fetch from that plane then this value is ignored Memory transactions will not complete if this status doesnt corespond to system programming in the rest of the path to memory SW should program SRC0_ADDR_CP, SRC1_ADDR_CP, SRC2_ADDR_CP, SRC3_ADDR_CP to the same value 0x0: Non Content Protected Buffer State 0x1: Content Protected Buffer State
2	SRC2_ADDR_CP	This is the secure status bit for source plane 2 (if used) If the format doesnt fetch from that plane then this value is ignored Memory transactions will not complete if this status doesnt corespond to system programming in the rest of the path to memory SW should program SRC0_ADDR_CP, SRC1_ADDR_CP, SRC2_ADDR_CP, SRC3_ADDR_CP to the same value 0x0: Non Content Protected Buffer State 0x1: Content Protected Buffer State
1	SRC1_ADDR_CP	This is the secure status bit for source plane 1 (if used) If the format doesnt fetch from that plane then this value is ignored Memory transactions will not complete if this status doesnt corespond to system programming in the rest of the path to memory SW should program SRC0_ADDR_CP, SRC1_ADDR_CP, SRC2_ADDR_CP, SRC3_ADDR_CP to the same value 0x0: Non Content Protected Buffer State 0x1: Content Protected Buffer State
0	SRC0_ADDR_CP	This is the secure status bit for source plane 0 (if used) If the format doesnt fetch from that plane then this value is ignored Memory transactions will not complete if this status doesnt corespond to system programming in the rest of the path to memory SW should program SRC0_ADDR_CP, SRC1_ADDR_CP, SRC2_ADDR_CP, SRC3_ADDR_CP to the same value 0x0: Non Content Protected Buffer State 0x1: Content Protected Buffer State

0x01A150A4 MDSS_MDP_VP_0_RGB_0_SSPP_CURRENT_SRC0_ADDR**Type:** R**Reset State:** 0x00000000

Current Plane 0 base address that Hardware is using

For fetching BWC data, this plane refers to the compressed RAU data

double buffered: N/A

MDSS_MDP_VP_0_RGB_0_SSPP_CURRENT_SRC0_ADDR

Bits	Name	Description
31:0	ADDR	Base byte address of the image single interleave color plane, Luma plane in pseudo-planar format, or color0 in planar format

0x01A150A8 MDSS_MDP_VP_0_RGB_0_SSPP_CURRENT_SRC1_ADDR

Type: R

Reset State: 0x00000000

Plane 1 base address that Hardware is using

For fetching BWC data, this plane refers to the META data to fetch the lengths of the compressed RAU data stream

double buffered: N/A

MDSS_MDP_VP_0_RGB_0_SSPP_CURRENT_SRC1_ADDR

Bits	Name	Description
31:0	ADDR	Base byte address of the chroma plane in pseudo-planar format, or color1 in planar format

0x01A150AC MDSS_MDP_VP_0_RGB_0_SSPP_CURRENT_SRC2_ADDR

Type: R

Reset State: 0x00000000

Plane 2 base address that Hardware is using

double buffered: N/A

MDSS_MDP_VP_0_RGB_0_SSPP_CURRENT_SRC2_ADDR

Bits	Name	Description
31:0	ADDR	Base byte address of color2 in planar format

0x01A150B0 MDSS_MDP_VP_0_RGB_0_SSPP_CURRENT_SRC3_ADDR

Type: R

Reset State: 0x00000000

Plane 3 base address that Hardware is using

double buffered: N/A

MDSS_MDP_VP_0_RGB_0_SSPP_CURRENT_SRC3_ADDR

Bits	Name	Description
31:0	ADDR	Base byte address of color3 in planar format, or pseudo planar + alpha format

0x01A150B4 MDSS_MDP_VP_0_RGB_0_SSPP_DECIMATION_CONFIG

Type: RW

Reset State: 0x00000000

Decimation Configuration

Note that there is no decimation on DMA pipe which means no decimation in line mode on DMA or block mode

double buffered: YES

MDSS_MDP_VP_0_RGB_0_SSPP_DECIMATION_CONFIG

Bits	Name	Description
11:8	HORZ_S	Horizontal Configuration for decimation; required to support downscaling greater than 1/4 S refers to the number of pixels skipped. The algorithm is always the following: read 1 pixel skip S pixels The algorithm is applied to ALL components. Decimation only supports the following values. All other values will result in undefined behaviour The decimation block will provide coarse scaling and should be used in conjunction with qseed2 to support all scaling ratios. 0x0: No decimation is performed 0x1: Read 1 pixel drop 1 pixel. Used to decimate by 2 horizontally 0x3: Read 1 pixel drop 3 pixels. Used to decimate by 4 horizontally 0x7: Read 1 pixel drop 7 pixels. Used to decimate by 8 horizontally 0xF: Read 1 pixel drop 15 pixels. Used to decimate by 16 horizontally

MDSS_MDP_VP_0_RGB_0_SSPP_DECIMATION_CONFIG (cont.)

Bits	Name	Description
3:0	VERT_S	Vertical Configuration for decimation; required to support downscaling greater than 1/4 S refers to the number of lines skipped. The algorithm is always the following: read 1 line skip S lines The algorithm is applied to ALL components. Decimation only supports the following values. All other values will result in undefined behaviour The decimation block will provide coarse scaling and should be used in conjunction with qseed2 to support all scaling ratios. 0x0: No decimation is performed 0x1: Read 1 line drop 1 line. Used to decimate by 2 horizontally 0x3: Read 1 line drop 3 lines. Used to decimate by 4 horizontally 0x7: Read 1 line drop 7 lines. Used to decimate by 8 horizontally 0xF: Read 1 line drop 15 lines. Used to decimate by 16 horizontally

0x01A150D0 MDSS_MDP_VP_0_RGB_0_SSPP_FETCH_SMP_WR_PLANE0**Type:** R**Reset State:** 0x00000000

Fetch to SMP on plane 0 write counter

double buffered: N/A

MDSS_MDP_VP_0_RGB_0_SSPP_FETCH_SMP_WR_PLANE0

Bits	Name	Description
15:0	COUNT	This counter is reset on every frame and is incremented for each write fetch makes to the SMP plane0 port This port is used for interleaved formats or the Y component for multi-plane source formats This should correlate to value SMP_UNPACK_RD_PLANE0 More useful for debug but can be used for profiling purposes as well.

0x01A150D4 MDSS_MDP_VP_0_RGB_0_SSPP_FETCH_SMP_WR_PLANE1**Type:** R**Reset State:** 0x00000000

Fetch to SMP on plane 1 write counter

double buffered: N/A

MDSS_MDP_VP_0_RGB_0_SSPP_FETCH_SMP_WR_PLANE1

Bits	Name	Description
15:0	COUNT	This counter is reset on every frame and is incremented for each write fetch makes to the SMP plane1 port This port is NOT used for interleaved formats but is used for U (planar) or UV (pseudo-planar) formats This should correlate to value SMP_UNPACK_RD_PLANE1 More useful for debug but can be used for profiling purposes as well.

0x01A150D8 MDSS_MDP_VP_0_RGB_0_SSPP_FETCH_SMP_WR_PLANE2**Type:** R**Reset State:** 0x00000000

Fetch to SMP on plane 2 write counter

double buffered: N/A

MDSS_MDP_VP_0_RGB_0_SSPP_FETCH_SMP_WR_PLANE2

Bits	Name	Description
15:0	COUNT	This counter is reset on every frame and is incremented for each write fetch makes to the SMP plane2 port This port is NOT used for interleaved formats but is used for V (planar) or A (pseudo-planar plus alpha) formats This should correlate to value SMP_UNPACK_RD_PLANE2 More useful for debug but can be used for profiling purposes as well.

0x01A150E0 MDSS_MDP_VP_0_RGB_0_SSPP_SMP_UNPACK_RD_PLANE0**Type:** R**Reset State:** 0x00000000

SMP to unpack on plane 0 read counter

double buffered: N/A

MDSS_MDP_VP_0_RGB_0_SSPP_SMP_UNPACK_RD_PLANE0

Bits	Name	Description
15:0	COUNT	This counter is reset on every frame and is incremented for each line read from SMP plane0 port This port is used for interleaved formats or the Y component for multi-plane source formats This should correlate to FETCH_SMP_WR_PLANE0 value More useful for debug but can be used for profiling purposes as well.

0x01A150E4 MDSS_MDP_VP_0_RGB_0_SSPP_SMP_UNPACK_RD_PLANE1**Type:** R**Reset State:** 0x00000000

SMP to unpack on plane 1 read counter

double buffered: N/A

MDSS_MDP_VP_0_RGB_0_SSPP_SMP_UNPACK_RD_PLANE1

Bits	Name	Description
15:0	COUNT	This counter is reset on every frame and is incremented for each line read from SMP plane1 port This port is NOT used for interleaved formats but is used for U (planar) or UV (pseudo-planar) formats This should correlate to FETCH_SMP_WR_PLANE1 value More useful for debug but can be used for profiling purposes as well.

0x01A150E8 MDSS_MDP_VP_0_RGB_0_SSPP_SMP_UNPACK_RD_PLANE2**Type:** R**Reset State:** 0x00000000

SMP to unpack on plane 2 read counter

double buffered: N/A

MDSS_MDP_VP_0_RGB_0_SSPP_SMP_UNPACK_RD_PLANE2

Bits	Name	Description
15:0	COUNT	This counter is reset on every frame and is incremented for each line read from SMP plane1 port This port is NOT used for interleaved formats but is used for V (planar) or A (pseudo-planar plus alpha) formats This should correlate to FETCH_SMP_WR_PLANE1 value More useful for debug but can be used for profiling purposes as well.

0x01A15100 MDSS_MDP_VP_0_RGB_0_SSPP_SW_PIX_EXT_C0_LR**Type:** RW**Reset State:** 0x00000000

SW Pixel Extension Override for Left/Right of Color 0

These value only take effect if SW_PIX_EXT_OVERRIDE is set in SRC_OP_MODE register

double buffered: YES

MDSS_MDP_VP_0_RGB_0_SSPP_SW_PIX_EXT_C0_LR

Bits	Name	Description
31:24	RIGHT_OVF	Signed Value for Right Overfetch value. +ve value indicates overfetch -ve value indicates pixel drop
23:16	RIGHT_RPT	Unsigned Value for Right Pixel Repeat value.
15:8	LEFT_OVF	Signed Value for Left Overfetch value. +ve value indicates overfetch -ve value indicates pixel drop
7:0	LEFT_RPT	Unsigned Value for Left Pixel Repeat value.

0x01A15104 MDSS_MDP_VP_0_RGB_0_SSPP_SW_PIX_EXT_C0_TB**Type:** RW**Reset State:** 0x00000000

SW Pixel Extension Override for Left/Right of Color 0

These value only take effect if SW_PIX_EXT_OVERRIDE is set in SRC_OP_MODE register

double buffered: YES

MDSS_MDP_VP_0_RGB_0_SSPP_SW_PIX_EXT_C0_TB

Bits	Name	Description
31:24	BOTTOM_OVF	Signed Value for Bottom Overfetch value. +ve value indicates overfetch -ve value indicates line drop
23:16	BOTTOM_RPT	
15:8	TOP_OVF	Signed Value for Top Overfetch value. +ve value indicates overfetch -ve value indicates line drop
7:0	TOP_RPT	Unsigned Value for TOP Line Repeat value.

0x01A15108 MDSS_MDP_VP_0_RGB_0_SSPP_SW_PIX_EXT_C0_REQ_PIXELS**Type:** RW**Reset State:** 0x00000000

SW Pixel Extension Override for pixel height/width of Color 0

These value only take effect if SW_PIX_EXT_OVERRIDE is set in SRC_OP_MODE register

double buffered: YES

MDSS_MDP_VP_0_RGB_0_SSPP_SW_PIX_EXT_C0_REQ_PIXELS

Bits	Name	Description
31:16	TOP_BOTTOM	Unsigned Value for pixel height post pixel extension.
15:0	LEFT_RIGHT	Unsigned Value for pixel width post pixel extension.

0x01A15110 MDSS_MDP_VP_0_RGB_0_SSPP_SW_PIX_EXT_C1C2_LR**Type:** RW**Reset State:** 0x00000000

SW Pixel Extension Override for Left/Right of Color 0

These value only take effect if SW_PIX_EXT_OVERRIDE is set in SRC_OP_MODE register

double buffered: YES

MDSS_MDP_VP_0_RGB_0_SSPP_SW_PIX_EXT_C1C2_LR

Bits	Name	Description
31:24	RIGHT_OVF	Signed Value for Right Overfetch value. +ve value indicates overfetch -ve value indicates pixel drop
23:16	RIGHT_RPT	Unsigned Value for Right Pixel Repeat value.
15:8	LEFT_OVF	Signed Value for Left Overfetch value. +ve value indicates overfetch -ve value indicates pixel drop
7:0	LEFT_RPT	Unsigned Value for Left Pixel Repeat value.

0x01A15114 MDSS_MDP_VP_0_RGB_0_SSPP_SW_PIX_EXT_C1C2_TB**Type:** RW**Reset State:** 0x00000000

SW Pixel Extension Override for Left/Right of Color 0

These value only take effect if SW_PIX_EXT_OVERRIDE is set in SRC_OP_MODE register
double buffered: YES

MDSS_MDP_VP_0_RGB_0_SSPP_SW_PIX_EXT_C1C2_TB

Bits	Name	Description
31:24	BOTTOM_OVF	Signed Value for Bottom Overfetch value. +ve value indicates overfetch -ve value indicates line drop
23:16	BOTTOM_RPT	
15:8	TOP_OVF	Signed Value for Top Overfetch value. +ve value indicates overfetch -ve value indicates line drop
7:0	TOP_RPT	Unsigned Value for TOP Line Repeat value.

0x01A15118 MDSS_MDP_VP_0_RGB_0_SSPP_SW_PIX_EXT_C1C2_REQ_PIXELS**Type:** RW**Reset State:** 0x00000000

SW Pixel Extension Override for pixel height/width of Color 0

These value only take effect if SW_PIX_EXT_OVERRIDE is set in SRC_OP_MODE register

double buffered: YES

MDSS_MDP_VP_0_RGB_0_SSPP_SW_PIX_EXT_C1C2_REQ_PIXELS

Bits	Name	Description
31:16	TOP_BOTTOM	Unsigned Value for pixel height post pixel extension.
15:0	LEFT_RIGHT	Unsigned Value for pixel width post pixel extension.

0x01A15120 MDSS_MDP_VP_0_RGB_0_SSPP_SW_PIX_EXT_C3_LR

Type: RW

Reset State: 0x00000000

SW Pixel Extension Override for Left/Right of Color 0

These value only take effect if SW_PIX_EXT_OVERRIDE is set in SRC_OP_MODE register

double buffered: YES

MDSS_MDP_VP_0_RGB_0_SSPP_SW_PIX_EXT_C3_LR

Bits	Name	Description
31:24	RIGHT_OVF	Signed Value for Right Overfetch value. +ve value indicates overfetch -ve value indicates pixel drop
23:16	RIGHT_RPT	Unsigned Value for Right Pixel Repeat value.
15:8	LEFT_OVF	Signed Value for Left Overfetch value. +ve value indicates overfetch -ve value indicates pixel drop
7:0	LEFT_RPT	Unsigned Value for Left Pixel Repeat value.

0x01A15124 MDSS_MDP_VP_0_RGB_0_SSPP_SW_PIX_EXT_C3_TB

Type: RW

Reset State: 0x00000000

SW Pixel Extension Override for Left/Right of Color 0

These value only take effect if SW_PIX_EXT_OVERRIDE is set in SRC_OP_MODE register

double buffered: YES

MDSS_MDP_VP_0_RGB_0_SSPP_SW_PIX_EXT_C3_TB

Bits	Name	Description
31:24	BOTTOM_OVF	Signed Value for Bottom Overfetch value. +ve value indicates overfetch -ve value indicates line drop
23:16	BOTTOM_RPT	
15:8	TOP_OVF	Signed Value for Top Overfetch value. +ve value indicates overfetch -ve value indicates line drop
7:0	TOP_RPT	Unsigned Value for TOP Line Repeat value.

0x01A15128 MDSS_MDP_VP_0_RGB_0_SSPP_SW_PIX_EXT_C3_REQ_PIXELS**Type:** RW**Reset State:** 0x00000000

SW Pixel Extension Override for pixel height/width of Color 0

These value only take effect if SW_PIX_EXT_OVERRIDE is set in SRC_OP_MODE register

double buffered: YES

MDSS_MDP_VP_0_RGB_0_SSPP_SW_PIX_EXT_C3_REQ_PIXELS

Bits	Name	Description
31:16	TOP_BOTTOM	Unsigned Value for pixel height post pixel extension.
15:0	LEFT_RIGHT	Unsigned Value for pixel width post pixel extension.

17.17.7.5 MDSS_MDP_VP_0_RGB_1

RGB hardware register description

17.17.7.5.1 MDSS_MDP_VP_0_RGB_1_RGB

RGB

0x01A17300 MDSS_MDP_VP_0_RGB_1_MULTI2ONE_MISR_CTRL**Type:** RW**Reset State:** 0x00000000

MDSS_MDP_VP_0_RGB_1_MULTI2ONE_MISR_CTRL

Bits	Name	Type	Description
31	RUN_MODE	RW	
27:24	INPUT_SEL	RW	
10	STATUS_CLEAR	W	
9	STATUS	R	
8	ENABLE	RW	
7:0	FRAME_COUNT	RW	

0x01A17304 MDSS_MDP_VP_0_RGB_1_MULTI2ONE_MISR_SIGNATURE**Type:** R**Reset State:** 0x00000000**MDSS_MDP_VP_0_RGB_1_MULTI2ONE_MISR_SIGNATURE**

Bits	Name	Description
31:0	CAPTURED	

17.17.7.5.2 MDSS_MDP_VP_0_RGB_1_SSPP

SSPP hardware register description

37.17.7.5.2.1 MDSS_MDP_VP_0_RGB_1_SSPP_SSPP

SSPP

SSPP hardware register description

0x01A17000 MDSS_MDP_VP_0_RGB_1_SSPP_SRC_SIZE**Type:** RW**Reset State:** 0x00000000

Source Size for the region of interest source image

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_RGB_1_SSPP_SRC_SIZE

Bits	Name	Description
31:16	ROI_H	Source Height of the Region Of Interest (ROI)
15:0	ROI_W	Source Width of the Region Of Interest (ROI) The exact supported size depends on the width of line buffers allocated for MDP.

0x01A17004 MDSS_MDP_VP_0_RGB_1_SSPP_SRC_IMG_SIZE**Type:** RW**Reset State:** 0x00000000

Source Size for the image in which the ROI resides

This value along with SRC_SIZE and SRC_XY will be used by hardware to calculate whether pixel repeat or overfetch is required along the ROI border.

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_RGB_1_SSPP_SRC_IMG_SIZE

Bits	Name	Description
31:16	SRC_H	Source Height. Note that this is NOT the ROI height
15:0	SRC_W	Source Width. Note that this is NOT the ROI width

0x01A17008 MDSS_MDP_VP_0_RGB_1_SSPP_SRC_XY**Type:** RW**Reset State:** 0x00000000

Source Offset from the top-left corner of the source image for each of the region of interest dimensions

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_RGB_1_SSPP_SRC_XY

Bits	Name	Description
31:16	SRC_Y	Y-offset of source ROI in the source image
15:0	SRC_X	X-offset of source ROI in the source image

0x01A1700C MDSS_MDP_VP_0_RGB_1_SSPP_OUT_SIZE**Type:** RW**Reset State:** 0x00000000

Output Size of the layer.

Should be same as source size if scaling is NOT enabled.

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_RGB_1_SSPP_OUT_SIZE

Bits	Name	Description
31:16	DST_H	Height of Output ROI
15:0	DST_W	Width of Output ROI

0x01A17010 MDSS_MDP_VP_0_RGB_1_SSPP_OUT_XY**Type:** RW**Reset State:** 0x00000000

Output Offset from the top-left corner of the output image

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_RGB_1_SSPP_OUT_XY

Bits	Name	Description
31:16	DST_Y	Height of Output ROI
15:0	DST_X	Width of Output ROI

0x01A17014 MDSS_MDP_VP_0_RGB_1_SSPP_SRC0_ADDR**Type:** RW**Reset State:** 0x00000000

Plane 0 base address

For fetching BWC data, this plane refers to the compressed RAU data

double buffered: YES

when: On frame boundaries (vsync or SW trigger) when ADDR_SYNC_MODE is in frame mode and on line boundary if ADDR_SYNC_MODE in in line mode

MDSS_MDP_VP_0_RGB_1_SSPP_SRC0_ADDR

Bits	Name	Description
31:0	ADDR	Base byte address of the image single interleave color plane, Luma plane in pseudo-planar format, or color0 in planar format

0x01A17018 MDSS_MDP_VP_0_RGB_1_SSPP_SRC1_ADDR

Type: RW

Reset State: 0x00000000

Plane 1 base address

For fetching BWC data, this plane refers to the META data to fetch the lengths of the compressed RAU data stream

double buffered: YES

when: On frame boundaries (vsync or SW trigger) when ADDR_SYNC_MODE is in frame mode and on line boundary if ADDR_SYNC_MODE in in line mode

MDSS_MDP_VP_0_RGB_1_SSPP_SRC1_ADDR

Bits	Name	Description
31:0	ADDR	Base byte address of the chroma plane in pseudo-planar format, or color1 in planar format

0x01A1701C MDSS_MDP_VP_0_RGB_1_SSPP_SRC2_ADDR

Type: RW

Reset State: 0x00000000

Plane 2 base address

double buffered: YES

when: On frame boundaries (vsync or SW trigger) when ADDR_SYNC_MODE is in frame mode and on line boundary if ADDR_SYNC_MODE in in line mode

MDSS_MDP_VP_0_RGB_1_SSPP_SRC2_ADDR

Bits	Name	Description
31:0	ADDR	Base byte address of color2 in planar format

0x01A17020 MDSS_MDP_VP_0_RGB_1_SSPP_SRC3_ADDR**Type:** RW**Reset State:** 0x00000000

Plane 3 base address

double buffered: YES

when: On frame boundaries (vsync or SW trigger) when ADDR_SYNC_MODE is in frame mode and on line boundary if ADDR_SYNC_MODE in in line mode

MDSS_MDP_VP_0_RGB_1_SSPP_SRC3_ADDR

Bits	Name	Description
31:0	ADDR	Base byte address of color3 in planar format, or pseudo planar + alpha format

0x01A17024 MDSS_MDP_VP_0_RGB_1_SSPP_SRC_YSTRIDE0**Type:** RW**Reset State:** 0x00000000

Stride Values

Double-buffered

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_RGB_1_SSPP_SRC_YSTRIDE0

Bits	Name	Description
31:16	SRCP1_YSTRIDE	Plane 1 Y stride in bytes
15:0	SRCP0_YSTRIDE	Plane 0 Y stride in bytes

0x01A17028 MDSS_MDP_VP_0_RGB_1_SSPP_SRC_YSTRIDE1**Type:** RW**Reset State:** 0x00000000

Stride Values

Double-buffered

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_RGB_1_SSPP_SRC_YSTRIDE1

Bits	Name	Description
31:16	SRCP3_YSTRIDE	Plane 3 Y stride in bytes
15:0	SRCP2_YSTRIDE	Plane 2 Y stride in bytes

0x01A17030 MDSS_MDP_VP_0_RGB_1_SSPP_SRC_FORMAT

Type: RW

Reset State: 0x00000000

Source Format Configuration for each virtual pipeline.

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_RGB_1_SSPP_SRC_FORMAT

Bits	Name	Description
31:30	FRAME_FORMAT	Frame format In the current revision, there is no support for tiled formats 0x0: Linear, no tile 0x1: Tiled, 4x4
25:23	SRC_CHROMA_SAMP	Source Chroma Sampling 0x0: RGB/YUV 4:4:4 0x1: 4:2:2 - H2V1 0x2: 4:2:2 - H1V2 0x3: 4:2:0 - N2V1 0x4: 4:2:0 - N1V2
22	SOLID_FILL	Fetch feeds the lower pipeline with the CONSTANT_COLOR register as its output value. NOTHING is fetched Note that in order to use this feature then Interleaved FETCH_PLANES must be set as well and no chroma sub-sampling enabled
21	VC1_REDUCE	VC1 range reduction enable for YUV sources The reduce range is set by separate register VC1_RANGE
20:19	FETCH_PLANES	Determines the number of planes to fetch 0x0: Interleaved 0x1: Planar 0x2: Pseudo-planar

MDSS_MDP_VP_0_RGB_1_SSPP_SRC_FORMAT (cont.)

Bits	Name	Description
18	UNPACK_ALIGN	0x0: TO LSB 0x1: TO MSB
17	UNPACK_TIGHT	0x0: Loose 0x1: Tight
13:12	UNPACK_COUNT	Valid unpacking pattern count 0x0 = 1 component 0x1 = 2 components ... 0x3 = 4 components Unpacking applies to interleaved plane (chroma plane in pseudo-planar)
11	ROT90	Source image is to be rotated 90 degrees
10:9	SRC_BPP	Effective source byte per pixel 0x0: 1 byte 0x1: 2 bytes 0x2: 3 bytes 0x3: 4 bytes This is used ONLY when unpacking the interleaved plane
8	SRCC3_EN	Alpha Enable 0x0: Source has NO alpha 0x1: Source has alpha
7:6	SRCC3	Number of bits for component 3 (Alpha) 0x0: 1 bits 0x1: 4 bits 0x2: 6 bits 0x3: 8 bits
5:4	SRCC2	Number of bits for component 2 (R/Cr) 0x0: 4 bits 0x1: 5 bits 0x2: 6 bits 0x3: 8 bits
3:2	SRCC1	Number of bits for component 1 (B/Cb) 0x0: 4 bits 0x1: 5 bits 0x2: 6 bits 0x3: 8 bits
1:0	SRCC0	Number of bits for component 0 (G/Luma) 0x0: 4 bits 0x1: 5 bits 0x2: 6 bits 0x3: 8 bits

0x01A17034 MDSS_MDP_VP_0_RGB_1_SSPP_SRC_UNPACK_PATTERN**Type:** RW**Reset State:** 0x00000000

Unpacking pattern, maximum of 4 pattern elements starting from LSB to MSB (0 to 4).

Default is ARGB or C3-C2-C0-C1.

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_RGB_1_SSPP_SRC_UNPACK_PATTERN

Bits	Name	Description
25:24	ELEMENT3	0x0: Color0 0x1: Color1 0x2: Color2 0x3: Color3
17:16	ELEMENT2	0x0: Color0 0x1: Color1 0x2: Color2 0x3: Color3
9:8	ELEMENT1	0x0: Color0 0x1: Color1 0x2: Color2 0x3: Color3
1:0	ELEMENT0	0x0: Color0 0x1: Color1 0x2: Color2 0x3: Color3

0x01A17038 MDSS_MDP_VP_0_RGB_1_SSPP_SRC_OP_MODE**Type:** RW**Reset State:** 0x00000000

Operation Mode for each SSPP pipeline

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_RGB_1_SSPP_SRC_OP_MODE

Bits	Name	Description
31	SW_PIX_EXT_OVERRIDE	SW override for pixel extension values. 0x0: Use HW calculated pixel extensions 0x1: Use SW programmed pixel extensions
16	IGC_LUT_EN	Inverse Gamma Correction LUT Enable 0x0: Disabled - mapped Linearly from 8-bit to 12-bit 0x1: Enabled
14:13	FLIP_MODE	Flip Operation Lower Bit: Left/Right flip Upper Bit: Up/Down flip

0x01A1703C MDSS_MDP_VP_0_RGB_1_SSPP_SRC_CONSTANT_COLOR**Type:** RW**Reset State:** 0x00000000

Constant Color Register during solid fill mode

double buffered: NO

MDSS_MDP_VP_0_RGB_1_SSPP_SRC_CONSTANT_COLOR

Bits	Name	Description
31:24	COLOR3	Constant color driven for color3 during solid fill mode.
23:16	COLOR2	Constant color driven for color2 during solid fill mode.
15:8	COLOR1	Constant color driven for color1 during solid fill mode.
7:0	COLOR0	Constant color driven for color0 during solid fill mode.

0x01A17048 MDSS_MDP_VP_0_RGB_1_SSPP_FETCH_CONFIG**Type:** RW**Reset State:** 0x00000087

SSPP Client Fetch Configuration per Virtual Pipeline double buffered: NO

This register must NOT be altered during operation and MDP must be idle when changing values

MDSS_MDP_VP_0_RGB_1_SSPP_FETCH_CONFIG

Bits	Name	Description
22	PRI_ARB_DIS	

MDSS_MDP_VP_0_RGB_1_SSPP_FETCH_CONFIG (cont.)

Bits	Name	Description
21	PMARK_DIS	
17:16	TILE_SIZE	Tile Size for processing of BLOCK formats Note that this only applies to blocks that operate in block mode which is limited to DMA source pipe. This setting has no effect for all other source pipes (ViG/RGB) 0x0: 128x128 Others: ILLEGAL / Not Supported
12	ADDR_SYNC_MODE	Boundary on which fetch address synchronizes. Setting this field to line forces the address updates on the next line which performs the software requested feature of updating on hsync. Note that updating on line boundaries will likely result in tearing on the panel since hardware is fetching from new memory buffer. 0x0: frame (default) 0x1: line
9:8	MAL	Describes the Minimum Access Length for the source pipeline. This value should align with the DRAM's expected transaction alignment unit. This setting forces the source fetch pipe to be aligned to DRAM and implement re-alignment for maximum efficiency 0x0: 128b alignment (default) 0x1: 256b alignment 0x2: 512b alignment
7:4	MAX_BURST_SIZE	Maximum burst beat size. Hardware fetch engines operate on 128-bit bus Possible values: 0x8: burst of 8 0x4: burst of 4 0x2: burst of 2
3:0	REQ_DEPTH_LIMIT	Pending request limit, default max value of 0x7 means maximum pending requests allowed in the design Anything less limits the bus requests to that number + 1.

0x01A1704C MDSS_MDP_VP_0_RGB_1_SSPP_VC1_RANGE**Type:** RW**Reset State:** 0x00000707

VC1 Range Map for each virtual pipeline

ONLY valid for pipelines that support YUV formats

double buffered: NO

MDSS_MDP_VP_0_RGB_1_SSPP_VC1_RANGE

Bits	Name	Description
10:8	RANGE_MAPUV	Chroma range map for VC1 range reduction, default of 7 is equivalent to setting it for main profile.
2:0	RANGE_MAPY	Luma range map for VC1 range reduction, default of 7 is equivalent to setting it for main profile.

0x01A17050 MDSS_MDP_VP_0_RGB_1_SSPP_REQPRIORITY_FIFO_WATERMARK_0**Type:** RW**Reset State:** 0x00000040

FIFO watermark to aid in generation of mdpREQPRIORITY

The watermark here refers to the # of 128-entries available in the Shared Memory Pool

The value programmed here should be aware of the amount programmed in the Shared Memory Pool.

So a value of 0 indicates completely FULL (i.e. there are no more entries available in the SMP)

A value of 100 indicates 100 entries in the Shared Memory Pool (each entry being 128-bit entries)

This is the lowestmost FIFO depth and should not be programmed higher than REQPRIORITY_FIFO_WATERMARK_1 or REQPRIORITY_FIFO_WATERMARK_2

An example programming would set the following:

REQPRIORITY_FIFO_WATERMARK_0 : 50 (meaning 50 entries free to be written into SMP - least room available)

REQPRIORITY_FIFO_WATERMARK_1 : 100 (meaning 100 entries free to be written into SMP - middle ground room available)

REQPRIORITY_FIFO_WATERMARK_2 : 150 (meaning 150 entries free to be written into SMP - most room available)

mdpREQPRIORITY = 11 (highest priority request) when above all 3 watermark levels (i.e. lots of space available in SMP)

mdpREQPRIORITY = 10 above FIFO_0 and FIFO_1 but below FIFO_2

mdpREQPRIORITY = 01 above FIFO_0 but below FIFO_1 and FIFO_2

mdpREQPRIORITY = 00 (lowest priority request) when below all 3 watermark levels (i.e. not a lot of space available in SMP so its good)

double buffered: YES

MDSS_MDP_VP_0_RGB_1_SSPP_REQPRIORITY_FIFO_WATERMARK_0

Bits	Name	Description
15:0	WATERMARK	Watermark Level for FIFO_0

0x01A17054 MDSS_MDP_VP_0_RGB_1_SSPP_REQPRIORITY_FIFO_WATERMARK_1**Type:** RW**Reset State:** 0x00000080

FIFO watermark to aid in generation of mdpREQPRIORITY

The watermark here refers to the # of 128-entries available in the Shared Memory Pool

The value programmed here should be aware of the amount programmed in the Shared Memory Pool.

So a value of 0 indicates completely FULL (i.e. there are no more entries available in the SMP)

A value of 100 indicates 100 entries in the Shared Memory Pool (each entry being 128-bit entries)

This is the middle FIFO depth and should not be programmed higher than REQPRIORITY_FIFO_WATERMARK_2 or lower than REQPRIORITY_FIFO_WATERMARK_0

An example programming would set the following:

REQPRIORITY_FIFO_WATERMARK_0 : 50 (meaning 50 entries free to be written into SMP - least room available)

REQPRIORITY_FIFO_WATERMARK_1 : 100 (meaning 100 entries free to be written into SMP - middle ground room available)

REQPRIORITY_FIFO_WATERMARK_2 : 150 (meaning 150 entries free to be written into SMP - most room available)

mdpREQPRIORITY = 11 (highest priority request) when above all 3 watermark levels (i.e. lots of space available in SMP)

mdpREQPRIORITY = 10 above FIFO_0 and FIFO_1 but below FIFO_2

mdpREQPRIORITY = 01 above FIFO_0 but below FIFO_1 and FIFO_2

mdpREQPRIORITY = 00 (lowest priority request) when below all 3 watermark levels (i.e. not a lot of space available in SMP so its good)

double buffered: YES

MDSS_MDP_VP_0_RGB_1_SSPP_REQPRIORITY_FIFO_WATERMARK_1

Bits	Name	Description
15:0	WATERMARK	Watermark Level for FIFO_1

0x01A17058 MDSS_MDP_VP_0_RGB_1_SSPP_REQPRIORITY_FIFO_WATERMARK_2**Type:** RW**Reset State:** 0x000000C0

FIFO watermark to aid in generation of mdpREQPRIORITY

The watermark here refers to the # of 128-entries available in the Shared Memory Pool

The value programmed here should be aware of the amount programmed in the Shared Memory Pool.

So a value of 0 indicates completely FULL (i.e. there are no more entries available in the SMP)

A value of 100 indicates 100 entries in the Shared Memory Pool (each entry being 128-bit entries)

This is the highest FIFO depth and should not be programmed lower than REQPRIORITY_FIFO_WATERMARK_0 or REQPRIORITY_FIFO_WATERMARK_1

An example programming would set the following:

REQPRIORITY_FIFO_WATERMARK_0 : 50 (meaning 50 entries free to be written into SMP - least room available)

REQPRIORITY_FIFO_WATERMARK_1 : 100 (meaning 100 entries free to be written into SMP - middle ground room available)

REQPRIORITY_FIFO_WATERMARK_2 : 150 (meaning 150 entries free to be written into SMP - most room available)

mdpREQPRIORITY = 11 (highest priority request) when above all 3 watermark levels (i.e. lots of space available in SMP)

mdpREQPRIORITY = 10 above FIFO_0 and FIFO_1 but below FIFO_2

mdpREQPRIORITY = 01 above FIFO_0 but below FIFO_1 and FIFO_2

mdpREQPRIORITY = 00 (lowest priority request) when below all 3 watermark levels (i.e. not a lot of space available in SMP so its good)

double buffered: YES

MDSS_MDP_VP_0_RGB_1_SSPP_REQPRIORITY_FIFO_WATERMARK_2

Bits	Name	Description
15:0	WATERMARK	Watermark Level for FIFO_2

0x01A17070 MDSS_MDP_VP_0_RGB_1_SSPP_SRC_ADDR_SW_STATUS**Type:** RW**Reset State:** 0x00000000

Security Status of the source buffer plane

This is programmed by SW to indicate if the buffer is NON-SECURE or SECURE

This information will be presented to VBIF per transaction and reflected as part of the memory transaction

double buffered: YES

MDSS_MDP_VP_0_RGB_1_SSPP_SRC_ADDR_SW_STATUS

Bits	Name	Description
3	SRC3_ADDR_CP	This is the secure status bit for source plane 3 (if used) If the format doesn't fetch from that plane then this value is ignored Memory transactions will not complete if this status doesn't correspond to system programming in the rest of the path to memory SW should program SRC0_ADDR_CP, SRC1_ADDR_CP, SRC2_ADDR_CP, SRC3_ADDR_CP to the same value 0x0: Non Content Protected Buffer State 0x1: Content Protected Buffer State
2	SRC2_ADDR_CP	This is the secure status bit for source plane 2 (if used) If the format doesn't fetch from that plane then this value is ignored Memory transactions will not complete if this status doesn't correspond to system programming in the rest of the path to memory SW should program SRC0_ADDR_CP, SRC1_ADDR_CP, SRC2_ADDR_CP, SRC3_ADDR_CP to the same value 0x0: Non Content Protected Buffer State 0x1: Content Protected Buffer State
1	SRC1_ADDR_CP	This is the secure status bit for source plane 1 (if used) If the format doesn't fetch from that plane then this value is ignored Memory transactions will not complete if this status doesn't correspond to system programming in the rest of the path to memory SW should program SRC0_ADDR_CP, SRC1_ADDR_CP, SRC2_ADDR_CP, SRC3_ADDR_CP to the same value 0x0: Non Content Protected Buffer State 0x1: Content Protected Buffer State

MDSS_MDP_VP_0_RGB_1_SSPP_SRC_ADDR_SW_STATUS (cont.)

Bits	Name	Description
0	SRC0_ADDR_CP	This is the secure status bit for source plane 0 (if used) If the format doesn't fetch from that plane then this value is ignored Memory transactions will not complete if this status doesn't correspond to system programming in the rest of the path to memory SW should program SRC0_ADDR_CP, SRC1_ADDR_CP, SRC2_ADDR_CP, SRC3_ADDR_CP to the same value 0x0: Non Content Protected Buffer State 0x1: Content Protected Buffer State

0x01A170A4 MDSS_MDP_VP_0_RGB_1_SSPP_CURRENT_SRC0_ADDR**Type:** R**Reset State:** 0x00000000

Current Plane 0 base address that Hardware is using

For fetching BWC data, this plane refers to the compressed RAU data

double buffered: N/A

MDSS_MDP_VP_0_RGB_1_SSPP_CURRENT_SRC0_ADDR

Bits	Name	Description
31:0	ADDR	Base byte address of the image single interleave color plane, Luma plane in pseudo-planar format, or color0 in planar format

0x01A170A8 MDSS_MDP_VP_0_RGB_1_SSPP_CURRENT_SRC1_ADDR**Type:** R**Reset State:** 0x00000000

Plane 1 base address that Hardware is using

For fetching BWC data, this plane refers to the META data to fetch the lengths of the compressed RAU data stream

double buffered: N/A

MDSS_MDP_VP_0_RGB_1_SSPP_CURRENT_SRC1_ADDR

Bits	Name	Description
31:0	ADDR	Base byte address of the chroma plane in pseudo-planar format, or color1 in planar format

0x01A170AC MDSS_MDP_VP_0_RGB_1_SSPP_CURRENT_SRC2_ADDR**Type:** R**Reset State:** 0x00000000

Plane 2 base address that Hardware is using

double buffered: N/A

MDSS_MDP_VP_0_RGB_1_SSPP_CURRENT_SRC2_ADDR

Bits	Name	Description
31:0	ADDR	Base byte address of color2 in planar format

0x01A170B0 MDSS_MDP_VP_0_RGB_1_SSPP_CURRENT_SRC3_ADDR**Type:** R**Reset State:** 0x00000000

Plane 3 base address that Hardware is using

double buffered: N/A

MDSS_MDP_VP_0_RGB_1_SSPP_CURRENT_SRC3_ADDR

Bits	Name	Description
31:0	ADDR	Base byte address of color3 in planar format, or pseudo planar + alpha format

0x01A170B4 MDSS_MDP_VP_0_RGB_1_SSPP_DECIMATION_CONFIG**Type:** RW**Reset State:** 0x00000000

Decimation Configuration

Note that there is no decimation on DMA pipe which means no decimation in line mode on DMA or block mode

double buffered: YES

MDSS_MDP_VP_0_RGB_1_SSPP_DECIMATION_CONFIG

Bits	Name	Description
11:8	HORZ_S	Horizontal Configuration for decimation; required to support downscaling greater than 1/4 S refers to the number of pixels skipped. The algorithm is always the following: read 1 pixel skip S pixels The algorithm is applied to ALL components. Decimation only supports the following values. All other values will result in undefined behaviour The decimation block will provide coarse scaling and should be used in conjunction with qseed2 to support all scaling ratios. 0x0: No decimation is performed 0x1: Read 1 pixel drop 1 pixel. Used to decimate by 2 horizontally 0x3: Read 1 pixel drop 3 pixels. Used to decimate by 4 horizontally 0x7: Read 1 pixel drop 7 pixels. Used to decimate by 8 horizontally 0xF: Read 1 pixel drop 15 pixels. Used to decimate by 16 horizontally
3:0	VERT_S	Vertical Configuration for decimation; required to support downscaling greater than 1/4 S refers to the number of lines skipped. The algorithm is always the following: read 1 line skip S lines The algorithm is applied to ALL components. Decimation only supports the following values. All other values will result in undefined behaviour The decimation block will provide coarse scaling and should be used in conjunction with qseed2 to support all scaling ratios. 0x0: No decimation is performed 0x1: Read 1 line drop 1 line. Used to decimate by 2 horizontally 0x3: Read 1 line drop 3 lines. Used to decimate by 4 horizontally 0x7: Read 1 line drop 7 lines. Used to decimate by 8 horizontally 0xF: Read 1 line drop 15 lines. Used to decimate by 16 horizontally

0x01A170D0 MDSS_MDP_VP_0_RGB_1_SSPP_FETCH_SMP_WR_PLANE0**Type:** R**Reset State:** 0x00000000

Fetch to SMP on plane 0 write counter

double buffered: N/A

MDSS_MDP_VP_0_RGB_1_SSPP_FETCH_SMP_WR_PLANE0

Bits	Name	Description
15:0	COUNT	This counter is reset on every frame and is incremented for each write fetch makes to the SMP plane0 port This port is used for interleaved formats or the Y component for multi-plane source formats This should correlate to value SMP_UNPACK_RD_PLANE0 More useful for debug but can be used for profiling purposes as well.

0x01A170D4 MDSS_MDP_VP_0_RGB_1_SSPP_FETCH_SMP_WR_PLANE1**Type:** R**Reset State:** 0x00000000

Fetch to SMP on plane 1 write counter

double buffered: N/A

MDSS_MDP_VP_0_RGB_1_SSPP_FETCH_SMP_WR_PLANE1

Bits	Name	Description
15:0	COUNT	This counter is reset on every frame and is incremented for each write fetch makes to the SMP plane1 port This port is NOT used for interleaved formats but is used for U (planar) or UV (pseudo-planar) formats This should correlate to value SMP_UNPACK_RD_PLANE1 More useful for debug but can be used for profiling purposes as well.

0x01A170D8 MDSS_MDP_VP_0_RGB_1_SSPP_FETCH_SMP_WR_PLANE2**Type:** R**Reset State:** 0x00000000

Fetch to SMP on plane 2 write counter

double buffered: N/A

MDSS_MDP_VP_0_RGB_1_SSPP_FETCH_SMP_WR_PLANE2

Bits	Name	Description
15:0	COUNT	This counter is reset on every frame and is incremented for each write fetch makes to the SMP plane2 port This port is NOT used for interleaved formats but is used for V (planar) or A (pseudo-planar plus alpha) formats This should correlate to value SMP_UNPACK_RD_PLANE2 More useful for debug but can be used for profiling purposes as well.

0x01A170E0 MDSS_MDP_VP_0_RGB_1_SSPP_SMP_UNPACK_RD_PLANE0**Type:** R**Reset State:** 0x00000000

SMP to unpack on plane 0 read counter

double buffered: N/A

MDSS_MDP_VP_0_RGB_1_SSPP_SMP_UNPACK_RD_PLANE0

Bits	Name	Description
15:0	COUNT	This counter is reset on every frame and is incremented for each line read from SMP plane0 port This port is used for interleaved formats or the Y component for multi-plane source formats This should correlate to FETCH_SMP_WR_PLANE0 value More useful for debug but can be used for profiling purposes as well.

0x01A170E4 MDSS_MDP_VP_0_RGB_1_SSPP_SMP_UNPACK_RD_PLANE1**Type:** R**Reset State:** 0x00000000

SMP to unpack on plane 1 read counter

double buffered: N/A

MDSS_MDP_VP_0_RGB_1_SSPP_SMP_UNPACK_RD_PLANE1

Bits	Name	Description
15:0	COUNT	This counter is reset on every frame and is incremented for each line read from SMP plane1 port This port is NOT used for interleaved formats but is used for U (planar) or UV (pseudo-planar) formats This should correlate to FETCH_SMP_WR_PLANE1 value More useful for debug but can be used for profiling purposes as well.

0x01A170E8 MDSS_MDP_VP_0_RGB_1_SSPP_SMP_UNPACK_RD_PLANE2**Type:** R**Reset State:** 0x00000000

SMP to unpack on plane 2 read counter

double buffered: N/A

MDSS_MDP_VP_0_RGB_1_SSPP_SMP_UNPACK_RD_PLANE2

Bits	Name	Description
15:0	COUNT	This counter is reset on every frame and is incremented for each line read from SMP plane1 port This port is NOT used for interleaved formats but is used for V (planar) or A (pseudo-planar plus alpha) formats This should correlate to FETCH_SMP_WR_PLANE1 value More useful for debug but can be used for profiling purposes as well.

0x01A17100 MDSS_MDP_VP_0_RGB_1_SSPP_SW_PIX_EXT_C0_LR**Type:** RW**Reset State:** 0x00000000

SW Pixel Extension Override for Left/Right of Color 0

These value only take effect if SW_PIX_EXT_OVERRIDE is set in SRC_OP_MODE register

double buffered: YES

MDSS_MDP_VP_0_RGB_1_SSPP_SW_PIX_EXT_C0_LR

Bits	Name	Description
31:24	RIGHT_OVF	Signed Value for Right Overfetch value. +ve value indicates overfetch -ve value indicates pixel drop
23:16	RIGHT_RPT	Unsigned Value for Right Pixel Repeat value.
15:8	LEFT_OVF	Signed Value for Left Overfetch value. +ve value indicates overfetch -ve value indicates pixel drop
7:0	LEFT_RPT	Unsigned Value for Left Pixel Repeat value.

0x01A17104 MDSS_MDP_VP_0_RGB_1_SSPP_SW_PIX_EXT_C0_TB**Type:** RW**Reset State:** 0x00000000

SW Pixel Extension Override for Left/Right of Color 0

These value only take effect if SW_PIX_EXT_OVERRIDE is set in SRC_OP_MODE register
double buffered: YES

MDSS_MDP_VP_0_RGB_1_SSPP_SW_PIX_EXT_C0_TB

Bits	Name	Description
31:24	BOTTOM_OVF	Signed Value for Bottom Overfetch value. +ve value indicates overfetch -ve value indicates line drop
23:16	BOTTOM_RPT	
15:8	TOP_OVF	Signed Value for Top Overfetch value. +ve value indicates overfetch -ve value indicates line drop
7:0	TOP_RPT	Unsigned Value for TOP Line Repeat value.

0x01A17108 MDSS_MDP_VP_0_RGB_1_SSPP_SW_PIX_EXT_C0_REQ_PIXELS**Type:** RW**Reset State:** 0x00000000

SW Pixel Extension Override for pixel height/width of Color 0

These value only take effect if SW_PIX_EXT_OVERRIDE is set in SRC_OP_MODE register

double buffered: YES

MDSS_MDP_VP_0_RGB_1_SSPP_SW_PIX_EXT_C0_REQ_PIXELS

Bits	Name	Description
31:16	TOP_BOTTOM	Unsigned Value for pixel height post pixel extension.
15:0	LEFT_RIGHT	Unsigned Value for pixel width post pixel extension.

0x01A17110 MDSS_MDP_VP_0_RGB_1_SSPP_SW_PIX_EXT_C1C2_LR

Type: RW

Reset State: 0x00000000

SW Pixel Extension Override for Left/Right of Color 0

These value only take effect if SW_PIX_EXT_OVERRIDE is set in SRC_OP_MODE register

double buffered: YES

MDSS_MDP_VP_0_RGB_1_SSPP_SW_PIX_EXT_C1C2_LR

Bits	Name	Description
31:24	RIGHT_OVF	Signed Value for Right Overfetch value. +ve value indicates overfetch -ve value indicates pixel drop
23:16	RIGHT_RPT	Unsigned Value for Right Pixel Repeat value.
15:8	LEFT_OVF	Signed Value for Left Overfetch value. +ve value indicates overfetch -ve value indicates pixel drop
7:0	LEFT_RPT	Unsigned Value for Left Pixel Repeat value.

0x01A17114 MDSS_MDP_VP_0_RGB_1_SSPP_SW_PIX_EXT_C1C2_TB

Type: RW

Reset State: 0x00000000

SW Pixel Extension Override for Left/Right of Color 0

These value only take effect if SW_PIX_EXT_OVERRIDE is set in SRC_OP_MODE register

double buffered: YES

MDSS_MDP_VP_0_RGB_1_SSPP_SW_PIX_EXT_C1C2_TB

Bits	Name	Description
31:24	BOTTOM_OVF	Signed Value for Bottom Overfetch value. +ve value indicates overfetch -ve value indicates line drop
23:16	BOTTOM_RPT	
15:8	TOP_OVF	Signed Value for Top Overfetch value. +ve value indicates overfetch -ve value indicates line drop
7:0	TOP_RPT	Unsigned Value for TOP Line Repeat value.

0x01A17118 MDSS_MDP_VP_0_RGB_1_SSPP_SW_PIX_EXT_C1C2_REQ_PIXELS**Type:** RW**Reset State:** 0x00000000

SW Pixel Extension Override for pixel height/width of Color 0

These value only take effect if SW_PIX_EXT_OVERRIDE is set in SRC_OP_MODE register

double buffered: YES

MDSS_MDP_VP_0_RGB_1_SSPP_SW_PIX_EXT_C1C2_REQ_PIXELS

Bits	Name	Description
31:16	TOP_BOTTOM	Unsigned Value for pixel height post pixel extension.
15:0	LEFT_RIGHT	Unsigned Value for pixel width post pixel extension.

0x01A17120 MDSS_MDP_VP_0_RGB_1_SSPP_SW_PIX_EXT_C3_LR**Type:** RW**Reset State:** 0x00000000

SW Pixel Extension Override for Left/Right of Color 0

These value only take effect if SW_PIX_EXT_OVERRIDE is set in SRC_OP_MODE register

double buffered: YES

MDSS_MDP_VP_0_RGB_1_SSPP_SW_PIX_EXT_C3_LR

Bits	Name	Description
31:24	RIGHT_OVF	Signed Value for Right Overfetch value. +ve value indicates overfetch -ve value indicates pixel drop
23:16	RIGHT_RPT	Unsigned Value for Right Pixel Repeat value.
15:8	LEFT_OVF	Signed Value for Left Overfetch value. +ve value indicates overfetch -ve value indicates pixel drop
7:0	LEFT_RPT	Unsigned Value for Left Pixel Repeat value.

0x01A17124 MDSS_MDP_VP_0_RGB_1_SSPP_SW_PIX_EXT_C3_TB**Type:** RW**Reset State:** 0x00000000

SW Pixel Extension Override for Left/Right of Color 0

These value only take effect if SW_PIX_EXT_OVERRIDE is set in SRC_OP_MODE register
double buffered: YES

MDSS_MDP_VP_0_RGB_1_SSPP_SW_PIX_EXT_C3_TB

Bits	Name	Description
31:24	BOTTOM_OVF	Signed Value for Bottom Overfetch value. +ve value indicates overfetch -ve value indicates line drop
23:16	BOTTOM_RPT	
15:8	TOP_OVF	Signed Value for Top Overfetch value. +ve value indicates overfetch -ve value indicates line drop
7:0	TOP_RPT	Unsigned Value for TOP Line Repeat value.

0x01A17128 MDSS_MDP_VP_0_RGB_1_SSPP_SW_PIX_EXT_C3_REQ_PIXELS**Type:** RW**Reset State:** 0x00000000

SW Pixel Extension Override for pixel height/width of Color 0

These value only take effect if SW_PIX_EXT_OVERRIDE is set in SRC_OP_MODE register

double buffered: YES

MDSS_MDP_VP_0_RGB_1_SSPP_SW_PIX_EXT_C3_REQ_PIXELS

Bits	Name	Description
31:16	TOP_BOTTOM	Unsigned Value for pixel height post pixel extension.
15:0	LEFT_RIGHT	Unsigned Value for pixel width post pixel extension.

17.17.7.6 MDSS_MDP_VP_0_RGB_2

RGB hardware register description

17.17.7.6.1 MDSS_MDP_VP_0_RGB_2_RGB

RGB

0x01A19300 MDSS_MDP_VP_0_RGB_2_MULTI2ONE_MISR_CTRL

Type: RW

Reset State: 0x00000000

MDSS_MDP_VP_0_RGB_2_MULTI2ONE_MISR_CTRL

Bits	Name	Type	Description
31	RUN_MODE	RW	
27:24	INPUT_SEL	RW	
10	STATUS_CLEAR	W	
9	STATUS	R	
8	ENABLE	RW	
7:0	FRAME_COUNT	RW	

0x01A19304 MDSS_MDP_VP_0_RGB_2_MULTI2ONE_MISR_SIGNATURE

Type: R

Reset State: 0x00000000

MDSS_MDP_VP_0_RGB_2_MULTI2ONE_MISR_SIGNATURE

Bits	Name	Description
31:0	CAPTURED	

17.17.7.6.2 MDSS_MDP_VP_0_RGB_2_SSPP

SSPP hardware register description

37.17.7.6.2.1 MDSS_MDP_VP_0_RGB_2_SSPP_SSPP

SSPP

SSPP hardware register description

0x01A19000 MDSS_MDP_VP_0_RGB_2_SSPP_SRC_SIZE

Type: RW

Reset State: 0x00000000

Source Size for the region of interest source image

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_RGB_2_SSPP_SRC_SIZE

Bits	Name	Description
31:16	ROI_H	Source Height of the Region Of Interest (ROI)
15:0	ROI_W	Source Width of the Region Of Interest (ROI) The exact supported size depends on the width of line buffers allocated for MDP.

0x01A19004 MDSS_MDP_VP_0_RGB_2_SSPP_SRC_IMG_SIZE

Type: RW

Reset State: 0x00000000

Source Size for the image in which the ROI resides

This value along with SRC_SIZE and SRC_XY will be used by hardware to calculate whether pixel repeat or overfetch is required along the ROI border.

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_RGB_2_SSPP_SRC_IMG_SIZE

Bits	Name	Description
31:16	SRC_H	Source Height. Note that this is NOT the ROI height

MDSS_MDP_VP_0_RGB_2_SSPP_SRC_IMG_SIZE (cont.)

Bits	Name	Description
15:0	SRC_W	Source Width. Note that this is NOT the ROI width

0x01A19008 MDSS_MDP_VP_0_RGB_2_SSPP_SRC_XY**Type:** RW**Reset State:** 0x00000000

Source Offset from the top-left corner of the source image for each of the region of interest dimensions

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_RGB_2_SSPP_SRC_XY

Bits	Name	Description
31:16	SRC_Y	Y-offset of source ROI in the source image
15:0	SRC_X	X-offset of source ROI in the source image

0x01A1900C MDSS_MDP_VP_0_RGB_2_SSPP_OUT_SIZE**Type:** RW**Reset State:** 0x00000000

Output Size of the layer.

Should be same as source size if scaling is NOT enabled.

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_RGB_2_SSPP_OUT_SIZE

Bits	Name	Description
31:16	DST_H	Height of Output ROI
15:0	DST_W	Width of Output ROI

0x01A19010 MDSS_MDP_VP_0_RGB_2_SSPP_OUT_XY**Type:** RW**Reset State:** 0x00000000

Output Offset from the top-left corner of the output image

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_RGB_2_SSPP_OUT_XY

Bits	Name	Description
31:16	DST_Y	Height of Output ROI
15:0	DST_X	Width of Output ROI

0x01A19014 MDSS_MDP_VP_0_RGB_2_SSPP_SRC0_ADDR**Type:** RW**Reset State:** 0x00000000

Plane 0 base address

For fetching BWC data, this plane refers to the compressed RAU data

double buffered: YES

when: On frame boundaries (vsync or SW trigger) when ADDR_SYNC_MODE is in frame mode and on line boundary if ADDR_SYNC_MODE in in line mode

MDSS_MDP_VP_0_RGB_2_SSPP_SRC0_ADDR

Bits	Name	Description
31:0	ADDR	Base byte address of the image single interleave color plane, Luma plane in pseudo-planar format, or color0 in planar format

0x01A19018 MDSS_MDP_VP_0_RGB_2_SSPP_SRC1_ADDR**Type:** RW**Reset State:** 0x00000000

Plane 1 base address

For fetching BWC data, this plane refers to the META data to fetch the lengths of the compressed RAU data stream

double buffered: YES

when: On frame boundaries (vsync or SW trigger) when ADDR_SYNC_MODE is in frame mode and on line boundary if ADDR_SYNC_MODE in in line mode

MDSS_MDP_VP_0_RGB_2_SSPP_SRC1_ADDR

Bits	Name	Description
31:0	ADDR	Base byte address of the chroma plane in pseudo-planar format, or color1 in planar format

0x01A1901C MDSS_MDP_VP_0_RGB_2_SSPP_SRC2_ADDR

Type: RW

Reset State: 0x00000000

Plane 2 base address

double buffered: YES

when: On frame boundaries (vsync or SW trigger) when ADDR_SYNC_MODE is in frame mode and on line boundary if ADDR_SYNC_MODE in in line mode

MDSS_MDP_VP_0_RGB_2_SSPP_SRC2_ADDR

Bits	Name	Description
31:0	ADDR	Base byte address of color2 in planar format

0x01A19020 MDSS_MDP_VP_0_RGB_2_SSPP_SRC3_ADDR

Type: RW

Reset State: 0x00000000

Plane 3 base address

double buffered: YES

when: On frame boundaries (vsync or SW trigger) when ADDR_SYNC_MODE is in frame mode and on line boundary if ADDR_SYNC_MODE in in line mode

MDSS_MDP_VP_0_RGB_2_SSPP_SRC3_ADDR

Bits	Name	Description
31:0	ADDR	Base byte address of color3 in planar format, or pseudo planar + alpha format

0x01A19024 MDSS_MDP_VP_0_RGB_2_SSPP_SRC_YSTRIDE0**Type:** RW**Reset State:** 0x00000000

Stride Values

Double-buffered

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_RGB_2_SSPP_SRC_YSTRIDE0

Bits	Name	Description
31:16	SRCP1_YSTRIDE	Plane 1 Y stride in bytes
15:0	SRCP0_YSTRIDE	Plane 0 Y stride in bytes

0x01A19028 MDSS_MDP_VP_0_RGB_2_SSPP_SRC_YSTRIDE1**Type:** RW**Reset State:** 0x00000000

Stride Values

Double-buffered

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_RGB_2_SSPP_SRC_YSTRIDE1

Bits	Name	Description
31:16	SRCP3_YSTRIDE	Plane 3 Y stride in bytes
15:0	SRCP2_YSTRIDE	Plane 2 Y stride in bytes

0x01A19030 MDSS_MDP_VP_0_RGB_2_SSPP_SRC_FORMAT**Type:** RW**Reset State:** 0x00000000

Source Format Configuration for each virtual pipeline.

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_RGB_2_SSPP_SRC_FORMAT

Bits	Name	Description
31:30	FRAME_FORMAT	Frame format In the current revision, there is no support for tiled formats 0x0: Linear, no tile 0x1: Tiled, 4x4
25:23	SRC_CHROMA_SAMP	Source Chroma Sampling 0x0: RGB/YUV 4:4:4 0x1: 4:2:2 - H2V1 0x2: 4:2:2 - H1V2 0x3: 4:2:0 - N2V1 0x4: 4:2:0 - N1V2
22	SOLID_FILL	Fetch feeds the lower pipeline with the CONSTANT_COLOR register as its output value. NOTHING is fetched Note that in order to use this feature then Interleaved FETCH_PLANES must be set as well and no chroma sub-sampling enabled
21	VC1_REDUCE	VC1 range reduction enable for YUV sources The reduce range is set by separate register VC1_RANGE
20:19	FETCH_PLANES	Determines the number of planes to fetch 0x0: Interleaved 0x1: Planar 0x2: Pseudo-planar
18	UNPACK_ALIGN	0x0: TO LSB 0x1: TO MSB
17	UNPACK_TIGHT	0x0: Loose 0x1: Tight
13:12	UNPACK_COUNT	Valid unpacking pattern count 0x0 = 1 component 0x1 = 2 components ... 0x3 = 4 components Unpacking applies to interleaved plane (chroma plane in pseudo-planar)
11	ROT90	Source image is to be rotated 90 degrees
10:9	SRC_BPP	Effective source byte per pixel 0x0: 1 byte 0x1: 2 bytes 0x2: 3 bytes 0x3: 4 bytes This is used ONLY when unpacking the interleaved plane

MDSS_MDP_VP_0_RGB_2_SSPP_SRC_FORMAT (cont.)

Bits	Name	Description
8	SRCC3_EN	Alpha Enable 0x0: Source has NO alpha 0x1: Source has alpha
7:6	SRCC3	Number of bits for component 3 (Alpha) 0x0: 1 bits 0x1: 4 bits 0x2: 6 bits 0x3: 8 bits
5:4	SRCC2	Number of bits for component 2 (R/Cr) 0x0: 4 bits 0x1: 5 bits 0x2: 6 bits 0x3: 8 bits
3:2	SRCC1	Number of bits for component 1 (B/Cb) 0x0: 4 bits 0x1: 5 bits 0x2: 6 bits 0x3: 8 bits
1:0	SRCC0	Number of bits for component 0 (G/Luma) 0x0: 4 bits 0x1: 5 bits 0x2: 6 bits 0x3: 8 bits

0x01A19034 MDSS_MDP_VP_0_RGB_2_SSPP_SRC_UNPACK_PATTERN**Type:** RW**Reset State:** 0x00000000

Unpacking pattern, maximum of 4 pattern elements starting from LSB to MSB (0 to 4).

Default is ARGB or C3-C2-C0-C1.

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_RGB_2_SSPP_SRC_UNPACK_PATTERN

Bits	Name	Description
25:24	ELEMENT3	0x0: Color0 0x1: Color1 0x2: Color2 0x3: Color3

MDSS_MDP_VP_0_RGB_2_SSPP_SRC_UNPACK_PATTERN (cont.)

Bits	Name	Description
17:16	ELEMENT2	0x0: Color0 0x1: Color1 0x2: Color2 0x3: Color3
9:8	ELEMENT1	0x0: Color0 0x1: Color1 0x2: Color2 0x3: Color3
1:0	ELEMENT0	0x0: Color0 0x1: Color1 0x2: Color2 0x3: Color3

0x01A19038 MDSS_MDP_VP_0_RGB_2_SSPP_SRC_OP_MODE**Type:** RW**Reset State:** 0x00000000

Operation Mode for each SSPP pipeline

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_RGB_2_SSPP_SRC_OP_MODE

Bits	Name	Description
31	SW_PIX_EXT_OVERRIDE	SW override for pixel extension values. 0x0: Use HW calculated pixel extensions 0x1: Use SW programmed pixel extensions
16	IGC_LUT_EN	Inverse Gamma Correction LUT Enable 0x0: Disabled - mapped Linearly from 8-bit to 12-bit 0x1: Enabled
14:13	FLIP_MODE	Flip Operation Lower Bit: Left/Right flip Upper Bit: Up/Down flip

0x01A1903C MDSS_MDP_VP_0_RGB_2_SSPP_SRC_CONSTANT_COLOR**Type:** RW**Reset State:** 0x00000000

Constant Color Register during solid fill mode

double buffered: NO

MDSS_MDP_VP_0_RGB_2_SSPP_SRC_CONSTANT_COLOR

Bits	Name	Description
31:24	COLOR3	Constant color driven for color3 during solid fill mode.
23:16	COLOR2	Constant color driven for color2 during solid fill mode.
15:8	COLOR1	Constant color driven for color1 during solid fill mode.
7:0	COLOR0	Constant color driven for color0 during solid fill mode.

0x01A19048 MDSS_MDP_VP_0_RGB_2_SSPP_FETCH_CONFIG

Type: RW

Reset State: 0x00000087

SSPP Client Fetch Configuration per Virtual Pipeline double buffered: NO

This register must NOT be altered during operation and MDP must be idle when changing values

MDSS_MDP_VP_0_RGB_2_SSPP_FETCH_CONFIG

Bits	Name	Description
22	PRI_ARB_DIS	
21	PMARK_DIS	
17:16	TILE_SIZE	Tile Size for processing of BLOCK formats Note that this only applies to blocks that operate in block mode which is limited to DMA source pipe. This setting has no effect for all other source pipes (ViG/RGB) 0x0: 128x128 Others: ILLEGAL / Not Supported
12	ADDR_SYNC_MODE	Boundary on which fetch address synchronizes. Setting this field to line forces the address updates on the next line which performs the software requested feature of updating on hsync. Note that updating on line boundaries will likely result in tearing on the panel since hardware is fetching from new memory buffer. 0x0: frame (default) 0x1: line

MDSS_MDP_VP_0_RGB_2_SSPP_FETCH_CONFIG (cont.)

Bits	Name	Description
9:8	MAL	Describes the Minimum Access Length for the source pipeline. This value should align with the DRAM's expected transaction alignment unit. This setting forces the source fetch pipe to be aligned to DRAM and implement re-alignment for maximum efficiency 0x0: 128b alignment (default) 0x1: 256b alignment 0x2: 512b alignment
7:4	MAX_BURST_SIZE	Maximum burst beat size. Hardware fetch engines operate on 128-bit bus Possible values: 0x8: burst of 8 0x4: burst of 4 0x2: burst of 2
3:0	REQ_DEPTH_LIMIT	Pending request limit, default max value of 0x7 means maximum pending requests allowed in the design Anything less limits the bus requests to that number + 1.

0x01A1904C MDSS_MDP_VP_0_RGB_2_SSPP_VC1_RANGE**Type:** RW**Reset State:** 0x00000707

VC1 Range Map for each virtual pipeline

ONLY valid for pipelines that suport YUV formats

double buffered: NO

MDSS_MDP_VP_0_RGB_2_SSPP_VC1_RANGE

Bits	Name	Description
10:8	RANGE_MAPUV	Chroma range map for VC1 range reduction, default of 7 is equivalent to setting it for main profile.
2:0	RANGE_MAPY	Luma range map for VC1 range reduction, default of 7 is equivalent to setting it for main profile.

0x01A19050 MDSS_MDP_VP_0_RGB_2_SSPP_REQPRIORITY_FIFO_WATERMARK_0**Type:** RW**Reset State:** 0x00000040

FIFO watermark to aid in generation of mdpREQPRIORITY

The watermark here refers to the # of 128-entries available in the Shared Memory Pool

The value programmed here should be aware of the amount programmed in the Shared Memory Pool.

So a value of 0 indicates completely FULL (i.e. there are no more entries available in the SMP)

A value of 100 indicates 100 entries in the Shared Memory Pool (each entry being 128-bit entries)

This is the lowestmost FIFO depth and should not be programmed higher than REQPRIORITY_FIFO_WATERMARK_1 or REQPRIORITY_FIFO_WATERMARK_2

An example programming would set the following:

REQPRIORITY_FIFO_WATERMARK_0 : 50 (meaning 50 entries free to be written into SMP - least room available)

REQPRIORITY_FIFO_WATERMARK_1 : 100 (meaning 100 entries free to be written into SMP - middle ground room available)

REQPRIORITY_FIFO_WATERMARK_2 : 150 (meaning 150 entries free to be written into SMP - most room available)

mdpREQPRIORITY = 11 (highest priority request) when above all 3 watermark levels (i.e. lots of space available in SMP)

mdpREQPRIORITY = 10 above FIFO_0 and FIFO_1 but below FIFO_2

mdpREQPRIORITY = 01 above FIFO_0 but below FIFO_1 and FIFO_2

mdpREQPRIORITY = 00 (lowest priority request) when below all 3 watermark levels (i.e. not a lot of space available in SMP so its good)

double buffered: YES

MDSS_MDP_VP_0_RGB_2_SSPP_REQPRIORITY_FIFO_WATERMARK_0

Bits	Name	Description
15:0	WATERMARK	Watermark Level for FIFO_0

0x01A19054 MDSS_MDP_VP_0_RGB_2_SSPP_REQPRIORITY_FIFO_WATERMARK_1

Type: RW

Reset State: 0x00000080

FIFO watermark to aid in generation of mdpREQPRIORITY

The watermark here refers to the # of 128-entries available in the Shared Memory Pool

The value programmed here should be aware of the amount programmed in the Shared Memory Pool.

So a value of 0 indicates completely FULL (i.e. there are no more entries available in the SMP)

A value of 100 indicates 100 entries in the Shared Memory Pool (each entry being 128-bit entries)

This is the middle FIFO depth and should not be programmed higher than
REQPRIORITY_FIFO_WATERMARK_2 or lower than
REQPRIORITY_FIFO_WATERMARK_0

An example programming would set the following:

REQPRIORITY_FIFO_WATERMARK_0 : 50 (meaning 50 entries free to be written into SMP - least room available)

REQPRIORITY_FIFO_WATERMARK_1 : 100 (meaning 100 entries free to be written into SMP - middle ground room available)

REQPRIORITY_FIFO_WATERMARK_2 : 150 (meaning 150 entries free to be written into SMP - most room available)

mdpREQPRIORITY = 11 (highest priority request) when above all 3 watermark levels (i.e. lots of space available in SMP)

mdpREQPRIORITY = 10 above FIFO_0 and FIFO_1 but below FIFO_2

mdpREQPRIORITY = 01 above FIFO_0 but below FIFO_1 and FIFO_2

mdpREQPRIORITY = 00 (lowest priority request) when below all 3 watermark levels (i.e. not a lot of space available in SMP so its good)

double buffered: YES

MDSS_MDP_VP_0_RGB_2_SSPP_REQPRIORITY_FIFO_WATERMARK_1

Bits	Name	Description
15:0	WATERMARK	Watermark Level for FIFO_1

0x01A19058 MDSS_MDP_VP_0_RGB_2_SSPP_REQPRIORITY_FIFO_WATERMARK_2

Type: RW

Reset State: 0x000000C0

FIFO watermark to aid in generation of mdpREQPRIORITY

The watermark here refers to the # of 128-entries available in the Shared Memory Pool

The value programmed here should be aware of the amount programmed in the Shared Memory Pool.

So a value of 0 indicates completely FULL (i.e. there are no more entries available in the SMP)

A value of 100 indicates 100 entries in the Shared Memory Pool (each entry being 128-bit entries)

This is the highest FIFO depth and should not be programmed lower than
REQPRIORITY_FIFO_WATERMARK_0 or REQPRIORITY_FIFO_WATERMARK_1

An example programming would set the following:

REQPRIORITY_FIFO_WATERMARK_0 : 50 (meaning 50 entries free to be written into SMP - least room available)

REQPRIORITY_FIFO_WATERMARK_1 : 100 (meaning 100 entries free to be written into SMP - middle ground room available)

REQPRIORITY_FIFO_WATERMARK_2 : 150 (meaning 150 entries free to be written into SMP - most room available)

mdpREQPRIORITY = 11 (highest priority request) when above all 3 watermark levels (i.e. lots of space available in SMP)

mdpREQPRIORITY = 10 above FIFO_0 and FIFO_1 but below FIFO_2

mdpREQPRIORITY = 01 above FIFO_0 but below FIFO_1 and FIFO_2

mdpREQPRIORITY = 00 (lowest priority request) when below all 3 watermark levels (i.e. not a lot of space available in SMP so its good)

double buffered: YES

MDSS_MDP_VP_0_RGB_2_SSPP_REQPRIORITY_FIFO_WATERMARK_2

Bits	Name	Description
15:0	WATERMARK	Watermark Level for FIFO_2

0x01A19070 MDSS_MDP_VP_0_RGB_2_SSPP_SRC_ADDR_SW_STATUS

Type: RW

Reset State: 0x00000000

Security Status of the source buffer plane

This is programmed by SW to indicate if the buffer is NON-SECURE or SECURE

This information will be presented to VBIF per transaction and reflected as part of the memory transaction

double buffered: YES

MDSS_MDP_VP_0_RGB_2_SSPP_SRC_ADDR_SW_STATUS

Bits	Name	Description
3	SRC3_ADDR_CP	This is the secure status bit for source plane 3 (if used) If the format doesnt fetch from that plane then this value is ignored Memory transactions will not complete if this status doesnt corespond to system programming in the rest of the path to memory SW should program SRC0_ADDR_CP, SRC1_ADDR_CP, SRC2_ADDR_CP, SRC3_ADDR_CP to the same value 0x0: Non Content Protected Buffer State 0x1: Content Protected Buffer State
2	SRC2_ADDR_CP	This is the secure status bit for source plane 2 (if used) If the format doesnt fetch from that plane then this value is ignored Memory transactions will not complete if this status doesnt corespond to system programming in the rest of the path to memory SW should program SRC0_ADDR_CP, SRC1_ADDR_CP, SRC2_ADDR_CP, SRC3_ADDR_CP to the same value 0x0: Non Content Protected Buffer State 0x1: Content Protected Buffer State
1	SRC1_ADDR_CP	This is the secure status bit for source plane 1 (if used) If the format doesnt fetch from that plane then this value is ignored Memory transactions will not complete if this status doesnt corespond to system programming in the rest of the path to memory SW should program SRC0_ADDR_CP, SRC1_ADDR_CP, SRC2_ADDR_CP, SRC3_ADDR_CP to the same value 0x0: Non Content Protected Buffer State 0x1: Content Protected Buffer State
0	SRC0_ADDR_CP	This is the secure status bit for source plane 0 (if used) If the format doesnt fetch from that plane then this value is ignored Memory transactions will not complete if this status doesnt corespond to system programming in the rest of the path to memory SW should program SRC0_ADDR_CP, SRC1_ADDR_CP, SRC2_ADDR_CP, SRC3_ADDR_CP to the same value 0x0: Non Content Protected Buffer State 0x1: Content Protected Buffer State

0x01A190A4 MDSS_MDP_VP_0_RGB_2_SSPP_CURRENT_SRC0_ADDR**Type:** R**Reset State:** 0x00000000

Current Plane 0 base address that Hardware is using

For fetching BWC data, this plane refers to the compressed RAU data

double buffered: N/A

MDSS_MDP_VP_0_RGB_2_SSPP_CURRENT_SRC0_ADDR

Bits	Name	Description
31:0	ADDR	Base byte address of the image single interleave color plane, Luma plane in pseudo-planar format, or color0 in planar format

0x01A190A8 MDSS_MDP_VP_0_RGB_2_SSPP_CURRENT_SRC1_ADDR

Type: R

Reset State: 0x00000000

Plane 1 base address that Hardware is using

For fetching BWC data, this plane refers to the META data to fetch the lengths of the compressed RAU data stream

double buffered: N/A

MDSS_MDP_VP_0_RGB_2_SSPP_CURRENT_SRC1_ADDR

Bits	Name	Description
31:0	ADDR	Base byte address of the chroma plane in pseudo-planar format, or color1 in planar format

0x01A190AC MDSS_MDP_VP_0_RGB_2_SSPP_CURRENT_SRC2_ADDR

Type: R

Reset State: 0x00000000

Plane 2 base address that Hardware is using

double buffered: N/A

MDSS_MDP_VP_0_RGB_2_SSPP_CURRENT_SRC2_ADDR

Bits	Name	Description
31:0	ADDR	Base byte address of color2 in planar format

0x01A190B0 MDSS_MDP_VP_0_RGB_2_SSPP_CURRENT_SRC3_ADDR

Type: R

Reset State: 0x00000000

Plane 3 base address that Hardware is using

double buffered: N/A

MDSS_MDP_VP_0_RGB_2_SSPP_CURRENT_SRC3_ADDR

Bits	Name	Description
31:0	ADDR	Base byte address of color3 in planar format, or pseudo planar + alpha format

0x01A190B4 MDSS_MDP_VP_0_RGB_2_SSPP_DECIMATION_CONFIG

Type: RW

Reset State: 0x00000000

Decimation Configuration

Note that there is no decimation on DMA pipe which means no decimation in line mode on DMA or block mode

double buffered: YES

MDSS_MDP_VP_0_RGB_2_SSPP_DECIMATION_CONFIG

Bits	Name	Description
11:8	HORZ_S	Horizontal Configuration for decimation; required to support downscaling greater than 1/4 S refers to the number of pixels skipped. The algorithm is always the following: read 1 pixel skip S pixels The algorithm is applied to ALL components. Decimation only supports the following values. All other values will result in undefined behaviour The decimation block will provide coarse scaling and should be used in conjunction with qseed2 to support all scaling ratios. 0x0: No decimation is performed 0x1: Read 1 pixel drop 1 pixel. Used to decimate by 2 horizontally 0x3: Read 1 pixel drop 3 pixels. Used to decimate by 4 horizontally 0x7: Read 1 pixel drop 7 pixels. Used to decimate by 8 horizontally 0xF: Read 1 pixel drop 15 pixels. Used to decimate by 16 horizontally

MDSS_MDP_VP_0_RGB_2_SSPP_DECIMATION_CONFIG (cont.)

Bits	Name	Description
3:0	VERT_S	Vertical Configuration for decimation; required to support downscaling greater than 1/4 S refers to the number of lines skipped. The algorithm is always the following: read 1 line skip S lines The algorithm is applied to ALL components. Decimation only supports the following values. All other values will result in undefined behaviour The decimation block will provide coarse scaling and should be used in conjunction with qseed2 to support all scaling ratios. 0x0: No decimation is performed 0x1: Read 1 line drop 1 line. Used to decimate by 2 horizontally 0x3: Read 1 line drop 3 lines. Used to decimate by 4 horizontally 0x7: Read 1 line drop 7 lines. Used to decimate by 8 horizontally 0xF: Read 1 line drop 15 lines. Used to decimate by 16 horizontally

0x01A190D0 MDSS_MDP_VP_0_RGB_2_SSPP_FETCH_SMP_WR_PLANE0**Type:** R**Reset State:** 0x00000000

Fetch to SMP on plane 0 write counter

double buffered: N/A

MDSS_MDP_VP_0_RGB_2_SSPP_FETCH_SMP_WR_PLANE0

Bits	Name	Description
15:0	COUNT	This counter is reset on every frame and is incremented for each write fetch makes to the SMP plane0 port This port is used for interleaved formats or the Y component for multi-plane source formats This should correlate to value SMP_UNPACK_RD_PLANE0 More useful for debug but can be used for profiling purposes as well.

0x01A190D4 MDSS_MDP_VP_0_RGB_2_SSPP_FETCH_SMP_WR_PLANE1**Type:** R**Reset State:** 0x00000000

Fetch to SMP on plane 1 write counter

double buffered: N/A

MDSS_MDP_VP_0_RGB_2_SSPP_FETCH_SMP_WR_PLANE1

Bits	Name	Description
15:0	COUNT	This counter is reset on every frame and is incremented for each write fetch makes to the SMP plane1 port This port is NOT used for interleaved formats but is used for U (planar) or UV (pseudo-planar) formats This should correlate to value SMP_UNPACK_RD_PLANE1 More useful for debug but can be used for profiling purposes as well.

0x01A190D8 MDSS_MDP_VP_0_RGB_2_SSPP_FETCH_SMP_WR_PLANE2**Type:** R**Reset State:** 0x00000000

Fetch to SMP on plane 2 write counter

double buffered: N/A

MDSS_MDP_VP_0_RGB_2_SSPP_FETCH_SMP_WR_PLANE2

Bits	Name	Description
15:0	COUNT	This counter is reset on every frame and is incremented for each write fetch makes to the SMP plane2 port This port is NOT used for interleaved formats but is used for V (planar) or A (pseudo-planar plus alpha) formats This should correlate to value SMP_UNPACK_RD_PLANE2 More useful for debug but can be used for profiling purposes as well.

0x01A190E0 MDSS_MDP_VP_0_RGB_2_SSPP_SMP_UNPACK_RD_PLANE0**Type:** R**Reset State:** 0x00000000

SMP to unpack on plane 0 read counter

double buffered: N/A

MDSS_MDP_VP_0_RGB_2_SSPP_SMP_UNPACK_RD_PLANE0

Bits	Name	Description
15:0	COUNT	This counter is reset on every frame and is incremented for each line read from SMP plane0 port This port is used for interleaved formats or the Y component for multi-plane source formats This should correlate to FETCH_SMP_WR_PLANE0 value More useful for debug but can be used for profiling purposes as well.

0x01A190E4 MDSS_MDP_VP_0_RGB_2_SSPP_SMP_UNPACK_RD_PLANE1**Type:** R**Reset State:** 0x00000000

SMP to unpack on plane 1 read counter

double buffered: N/A

MDSS_MDP_VP_0_RGB_2_SSPP_SMP_UNPACK_RD_PLANE1

Bits	Name	Description
15:0	COUNT	This counter is reset on every frame and is incremented for each line read from SMP plane1 port This port is NOT used for interleaved formats but is used for U (planar) or UV (pseudo-planar) formats This should correlate to FETCH_SMP_WR_PLANE1 value More useful for debug but can be used for profiling purposes as well.

0x01A190E8 MDSS_MDP_VP_0_RGB_2_SSPP_SMP_UNPACK_RD_PLANE2**Type:** R**Reset State:** 0x00000000

SMP to unpack on plane 2 read counter

double buffered: N/A

MDSS_MDP_VP_0_RGB_2_SSPP_SMP_UNPACK_RD_PLANE2

Bits	Name	Description
15:0	COUNT	This counter is reset on every frame and is incremented for each line read from SMP plane1 port This port is NOT used for interleaved formats but is used for V (planar) or A (pseudo-planar plus alpha) formats This should correlate to FETCH_SMP_WR_PLANE1 value More useful for debug but can be used for profiling purposes as well.

0x01A19100 MDSS_MDP_VP_0_RGB_2_SSPP_SW_PIX_EXT_C0_LR**Type:** RW**Reset State:** 0x00000000

SW Pixel Extension Override for Left/Right of Color 0

These value only take effect if SW_PIX_EXT_OVERRIDE is set in SRC_OP_MODE register

double buffered: YES

MDSS_MDP_VP_0_RGB_2_SSPP_SW_PIX_EXT_C0_LR

Bits	Name	Description
31:24	RIGHT_OVF	Signed Value for Right Overfetch value. +ve value indicates overfetch -ve value indicates pixel drop
23:16	RIGHT_RPT	Unsigned Value for Right Pixel Repeat value.
15:8	LEFT_OVF	Signed Value for Left Overfetch value. +ve value indicates overfetch -ve value indicates pixel drop
7:0	LEFT_RPT	Unsigned Value for Left Pixel Repeat value.

0x01A19104 MDSS_MDP_VP_0_RGB_2_SSPP_SW_PIX_EXT_C0_TB**Type:** RW**Reset State:** 0x00000000

SW Pixel Extension Override for Left/Right of Color 0

These value only take effect if SW_PIX_EXT_OVERRIDE is set in SRC_OP_MODE register

double buffered: YES

MDSS_MDP_VP_0_RGB_2_SSPP_SW_PIX_EXT_C0_TB

Bits	Name	Description
31:24	BOTTOM_OVF	Signed Value for Bottom Overfetch value. +ve value indicates overfetch -ve value indicates line drop
23:16	BOTTOM_RPT	
15:8	TOP_OVF	Signed Value for Top Overfetch value. +ve value indicates overfetch -ve value indicates line drop
7:0	TOP_RPT	Unsigned Value for TOP Line Repeat value.

0x01A19108 MDSS_MDP_VP_0_RGB_2_SSPP_SW_PIX_EXT_C0_REQ_PIXELS**Type:** RW**Reset State:** 0x00000000

SW Pixel Extension Override for pixel height/width of Color 0

These value only take effect if SW_PIX_EXT_OVERRIDE is set in SRC_OP_MODE register

double buffered: YES

MDSS_MDP_VP_0_RGB_2_SSPP_SW_PIX_EXT_C0_REQ_PIXELS

Bits	Name	Description
31:16	TOP_BOTTOM	Unsigned Value for pixel height post pixel extension.
15:0	LEFT_RIGHT	Unsigned Value for pixel width post pixel extension.

0x01A19110 MDSS_MDP_VP_0_RGB_2_SSPP_SW_PIX_EXT_C1C2_LR**Type:** RW**Reset State:** 0x00000000

SW Pixel Extension Override for Left/Right of Color 0

These value only take effect if SW_PIX_EXT_OVERRIDE is set in SRC_OP_MODE register

double buffered: YES

MDSS_MDP_VP_0_RGB_2_SSPP_SW_PIX_EXT_C1C2_LR

Bits	Name	Description
31:24	RIGHT_OVF	Signed Value for Right Overfetch value. +ve value indicates overfetch -ve value indicates pixel drop
23:16	RIGHT_RPT	Unsigned Value for Right Pixel Repeat value.
15:8	LEFT_OVF	Signed Value for Left Overfetch value. +ve value indicates overfetch -ve value indicates pixel drop
7:0	LEFT_RPT	Unsigned Value for Left Pixel Repeat value.

0x01A19114 MDSS_MDP_VP_0_RGB_2_SSPP_SW_PIX_EXT_C1C2_TB**Type:** RW**Reset State:** 0x00000000

SW Pixel Extension Override for Left/Right of Color 0

These value only take effect if SW_PIX_EXT_OVERRIDE is set in SRC_OP_MODE register
double buffered: YES

MDSS_MDP_VP_0_RGB_2_SSPP_SW_PIX_EXT_C1C2_TB

Bits	Name	Description
31:24	BOTTOM_OVF	Signed Value for Bottom Overfetch value. +ve value indicates overfetch -ve value indicates line drop
23:16	BOTTOM_RPT	
15:8	TOP_OVF	Signed Value for Top Overfetch value. +ve value indicates overfetch -ve value indicates line drop
7:0	TOP_RPT	Unsigned Value for TOP Line Repeat value.

0x01A19118 MDSS_MDP_VP_0_RGB_2_SSPP_SW_PIX_EXT_C1C2_REQ_PIXELS**Type:** RW**Reset State:** 0x00000000

SW Pixel Extension Override for pixel height/width of Color 0

These value only take effect if SW_PIX_EXT_OVERRIDE is set in SRC_OP_MODE register

double buffered: YES

MDSS_MDP_VP_0_RGB_2_SSPP_SW_PIX_EXT_C1C2_REQ_PIXELS

Bits	Name	Description
31:16	TOP_BOTTOM	Unsigned Value for pixel height post pixel extension.
15:0	LEFT_RIGHT	Unsigned Value for pixel width post pixel extension.

0x01A19120 MDSS_MDP_VP_0_RGB_2_SSPP_SW_PIX_EXT_C3_LR

Type: RW

Reset State: 0x00000000

SW Pixel Extension Override for Left/Right of Color 0

These value only take effect if SW_PIX_EXT_OVERRIDE is set in SRC_OP_MODE register

double buffered: YES

MDSS_MDP_VP_0_RGB_2_SSPP_SW_PIX_EXT_C3_LR

Bits	Name	Description
31:24	RIGHT_OVF	Signed Value for Right Overfetch value. +ve value indicates overfetch -ve value indicates pixel drop
23:16	RIGHT_RPT	Unsigned Value for Right Pixel Repeat value.
15:8	LEFT_OVF	Signed Value for Left Overfetch value. +ve value indicates overfetch -ve value indicates pixel drop
7:0	LEFT_RPT	Unsigned Value for Left Pixel Repeat value.

0x01A19124 MDSS_MDP_VP_0_RGB_2_SSPP_SW_PIX_EXT_C3_TB

Type: RW

Reset State: 0x00000000

SW Pixel Extension Override for Left/Right of Color 0

These value only take effect if SW_PIX_EXT_OVERRIDE is set in SRC_OP_MODE register

double buffered: YES

MDSS_MDP_VP_0_RGB_2_SSPP_SW_PIX_EXT_C3_TB

Bits	Name	Description
31:24	BOTTOM_OVF	Signed Value for Bottom Overfetch value. +ve value indicates overfetch -ve value indicates line drop
23:16	BOTTOM_RPT	
15:8	TOP_OVF	Signed Value for Top Overfetch value. +ve value indicates overfetch -ve value indicates line drop
7:0	TOP_RPT	Unsigned Value for TOP Line Repeat value.

0x01A19128 MDSS_MDP_VP_0_RGB_2_SSPP_SW_PIX_EXT_C3_REQ_PIXELS**Type:** RW**Reset State:** 0x00000000

SW Pixel Extension Override for pixel height/width of Color 0

These value only take effect if SW_PIX_EXT_OVERRIDE is set in SRC_OP_MODE register

double buffered: YES

MDSS_MDP_VP_0_RGB_2_SSPP_SW_PIX_EXT_C3_REQ_PIXELS

Bits	Name	Description
31:16	TOP_BOTTOM	Unsigned Value for pixel height post pixel extension.
15:0	LEFT_RIGHT	Unsigned Value for pixel width post pixel extension.

17.17.7.7 MDSS_MDP_VP_0_DMA_0

DMA hardware register description

DMA

17.17.7.7.1 MDSS_MDP_VP_0_DMA_0_SSPP

SSPP hardware register description

37.17.7.7.1.1 MDSS_MDP_VP_0_DMA_0_SSPP_SSPP

SSPP

SSPP hardware register description

17.17.7.8 MDSS_MDP_VP_0_DMA_1

DMA hardware register description

DMA

17.17.7.8.1 MDSS_MDP_VP_0_DMA_1_SSPP

SSPP hardware register description

37.17.7.8.1.1 MDSS_MDP_VP_0_DMA_1_SSPP_SSPP

SSPP

SSPP hardware register description

17.17.7.9 MDSS_MDP_VP_0_LAYER_0

Layer Mixer hardware register description

17.17.7.9.1 MDSS_MDP_VP_0_LAYER_0_LAYER

LAYER

Layer Mixer hardware register description

0x01A45000 MDSS_MDP_VP_0_LAYER_0_OP_MODE**Type:** RW**Reset State:** 0x00000000

LayerMixer Operation Control

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_LAYER_0_OP_MODE

Bits	Name	Description
4	BLEND3_COLOR3_OUT	Selects which source to drive on color3 component output for stage3 blend 0x0: BG Layer Color3 0x1: FG Layer Color3

MDSS_MDP_VP_0_LAYER_0_OP_MODE (cont.)

Bits	Name	Description
3	BLEND2_COLOR3_OUT	Selects which source to drive on color3 component output for stage2 blend 0x0: BG Layer Color3 0x1: FG Layer Color3
2	BLEND1_COLOR3_OUT	Selects which source to drive on color3 component output for stage1 blend 0x0: BG Layer Color3 0x1: FG Layer Color3
1	BLEND0_COLOR3_OUT	Selects which source to drive on color3 component output for stage0 blend 0x0: BG Layer Color3 0x1: FG Layer Color3
0	GC_LUT_EN	0x0: Disabled - mapped linearly from 12-bit to 8-bit 0x1: Enabled

0x01A45004 MDSS_MDP_VP_0_LAYER_0_OUT_SIZE**Type:** RW**Reset State:** 0x00000000

LayerMixer Output Size

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_LAYER_0_OUT_SIZE

Bits	Name	Description
31:16	DST_H	Destination Output Height
15:0	DST_W	Destination Output Width The exact supported size depends on the width of line buffers allocated for MDP.

0x01A45008 MDSS_MDP_VP_0_LAYER_0_BORDER_COLOR_0**Type:** RW**Reset State:** 0x00000000

LayerMixer Border Color Register

double buffered: YES

MDSS_MDP_VP_0_LAYER_0_BORDER_COLOR_0

Bits	Name	Description
27:16	COLOR1	Color1 in 12-bit format
11:0	COLOR0	Color0 in 12-bit format

0x01A45010 MDSS_MDP_VP_0_LAYER_0_BORDER_COLOR_1**Type:** RW**Reset State:** 0x00000000

LayerMixer Border Color Register

double buffered: YES

MDSS_MDP_VP_0_LAYER_0_BORDER_COLOR_1

Bits	Name	Description
27:16	COLOR3	Color3 in 12-bit format, LSB4 are ignored, should be programmed to 0
11:0	COLOR2	Color2 in 12-bit format

0x01A45020 MDSS_MDP_VP_0_LAYER_0_BLEND0_OP**Type:** RW**Reset State:** 0x00000000

LayerMixer Blend0 Operation Control

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_LAYER_0_BLEND0_OP

Bits	Name	Description
13	BG_TRANSP_EN	BG transparency enabled
12	BG_INV_MOD_ALPHA	Ability to select between alpha and inverse alpha right after the alpha modulation 0x0: Alpha 0x1: Inverse Alpha
11	BG_MOD_ALPHA	Modulate BG Alpha

MDSS_MDP_VP_0_LAYER_0_BLEND0_OP (cont.)

Bits	Name	Description
10	BG_INV_ALPHA_SEL	Ability to select between alpha and inverse alpha right after the alpha select 0x0: Alpha 0x1: Inverse Alpha
9:8	BG_ALPHA_SEL	Blending Alpha Select 0x0: FG constant alpha 0x1: BG constant alpha 0x2: FG per pixel alpha 0x3: BG per pixel alpha
5	FG TRANSP_EN	FG transparency enabled
4	FG_INV_MOD_ALPHA	Ability to select between alpha and inverse alpha right after the alpha modulation 0x0: Alpha 0x1: Inverse Alpha
3	FG_MOD_ALPHA	Modulate Alpha
2	FG_INV_ALPHA_SEL	Ability to select between alpha and inverse alpha right after the alpha select 0x0: Alpha 0x1: Inverse Alpha
1:0	FG_ALPHA_SEL	Blending Alpha Select 0x0: FG constant alpha 0x1: BG constant alpha 0x2: FG per pixel alpha 0x3: BG per pixel alpha

0x01A45024 MDSS_MDP_VP_0_LAYER_0_BLEND0_FG_ALPHA**Type:** RW**Reset State:** 0x00000000

Blend Layer0 FG alpha

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_LAYER_0_BLEND0_FG_ALPHA

Bits	Name	Description
7:0	CONSTANT_ALPHA	Constant Alpha

0x01A45028 MDSS_MDP_VP_0_LAYER_0_BLEND0_BG_ALPHA**Type:** RW**Reset State:** 0x00000000

Blend Layer0 FG alpha

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_LAYER_0_BLEND0_BG_ALPHA

Bits	Name	Description
7:0	CONSTANT_ALPHA	Constant Alpha

0x01A4502C MDSS_MDP_VP_0_LAYER_0_BLEND0_FG TRANSP_LOW0**Type:** RW**Reset State:** 0x00000000

Blend Layer0 Transparency Color Low Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_0_BLEND0_FG TRANSP_LOW0

Bits	Name	Description
27:16	TRANSPARENCY_C1	Transparency color check lower limit for Color1
11:0	TRANSPARENCY_C0	Transparency color check lower limit for Color0

0x01A45030 MDSS_MDP_VP_0_LAYER_0_BLEND0_FG TRANSP_LOW1**Type:** RW**Reset State:** 0x00000000

Blend Layer0 Transparency Color Low Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_0_BLEND0_FG TRANSP_LOW1

Bits	Name	Description
11:0	TRANSPARENCY_C2	Transparency color check lower limit for Color2

0x01A45034 MDSS_MDP_VP_0_LAYER_0_BLEND0_FG TRANSP_HIGH0**Type:** RW**Reset State:** 0x00000000

Blend Layer0 Transparency Color High Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_0_BLEND0_FG TRANSP_HIGH0

Bits	Name	Description
27:16	TRANSPARENCY_C1	Transparency color check higher limit for Color1
11:0	TRANSPARENCY_C0	Transparency color check higher limit for Color0

0x01A45038 MDSS_MDP_VP_0_LAYER_0_BLEND0_FG TRANSP_HIGH1**Type:** RW**Reset State:** 0x00000000

Blend Layer0 Transparency Color High Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_0_BLEND0_FG TRANSP_HIGH1

Bits	Name	Description
11:0	TRANSPARENCY_C2	Transparency color check higher limit for Color2

0x01A4503C MDSS_MDP_VP_0_LAYER_0_BLEND0_BG TRANSP_LOW0**Type:** RW**Reset State:** 0x00000000

Blend Layer0 Transparency Color Low Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_0_BLEND0_BG TRANSP_LOW0

Bits	Name	Description
27:16	TRANSPARENCY_C1	Transparency color check lower limit for Color1
11:0	TRANSPARENCY_C0	Transparency color check lower limit for Color0

0x01A45040 MDSS_MDP_VP_0_LAYER_0_BLEND0_BG TRANSP_LOW1**Type:** RW**Reset State:** 0x00000000

Blend Layer0 Transparency Color Low Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_0_BLEND0_BG TRANSP_LOW1

Bits	Name	Description
11:0	TRANSPARENCY_C2	Transparency color check lower limit for Color2

0x01A45044 MDSS_MDP_VP_0_LAYER_0_BLEND0_BG TRANSP_HIGH0**Type:** RW**Reset State:** 0x00000000

Blend Layer0 Transparency Color High Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_0_BLEND0_BG TRANSP_HIGH0

Bits	Name	Description
27:16	TRANSPARENCY_C1	Transparency color check higher limit for Color1
11:0	TRANSPARENCY_C0	Transparency color check higher limit for Color0

0x01A45048 MDSS_MDP_VP_0_LAYER_0_BLEND0_BG TRANSP_HIGH1**Type:** RW**Reset State:** 0x00000000

Blend Layer0 Transparency Color High Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_0_BLEND0_BG TRANSP_HIGH1

Bits	Name	Description
11:0	TRANSPARENCY_C2	Transparency color check higher limit for Color2

0x01A45050 MDSS_MDP_VP_0_LAYER_0_BLEND1_OP**Type:** RW**Reset State:** 0x00000000

LayerMixer Blend1 Operation Control

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_LAYER_0_BLEND1_OP

Bits	Name	Description
13	BG TRANSP_EN	BG transparency enabled
12	BG_INV_MOD_ALPHA	Ability to select between alpha and inverse alpha right after the alpha modulation 0x0: Alpha 0x1: Inverse Alpha
11	BG_MOD_ALPHA	Modulate BG Alpha
10	BG_INV_ALPHA_SEL	Ability to select between alpha and inverse alpha right after the alpha select 0x0: Alpha 0x1: Inverse Alpha
9:8	BG_ALPHA_SEL	Blending Alpha Select 0x0: FG constant alpha 0x1: BG constant alpha 0x2: FG per pixel alpha 0x3: BG per pixel alpha
5	FG TRANSP_EN	FG transparency enabled
4	FG_INV_MOD_ALPHA	Ability to select between alpha and inverse alpha right after the alpha modulation 0x0: Alpha 0x1: Inverse Alpha
3	FG_MOD_ALPHA	Modulate Alpha
2	FG_INV_ALPHA_SEL	Ability to select between alpha and inverse alpha right after the alpha select 0x0: Alpha 0x1: Inverse Alpha
1:0	FG_ALPHA_SEL	Blending Alpha Select 0x0: FG constant alpha 0x1: BG constant alpha 0x2: FG per pixel alpha 0x3: BG per pixel alpha

0x01A45054 MDSS_MDP_VP_0_LAYER_0_BLEND1_FG_ALPHA**Type:** RW**Reset State:** 0x00000000

Blend Layer1 FG alpha

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_LAYER_0_BLEND1_FG_ALPHA

Bits	Name	Description
7:0	CONSTANT_ALPHA	Constant Alpha

0x01A45058 MDSS_MDP_VP_0_LAYER_0_BLEND1_BG_ALPHA**Type:** RW**Reset State:** 0x00000000

Blend Layer1 FG alpha

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_LAYER_0_BLEND1_BG_ALPHA

Bits	Name	Description
7:0	CONSTANT_ALPHA	Constant Alpha

0x01A4505C MDSS_MDP_VP_0_LAYER_0_BLEND1_FG TRANSP_LOW0**Type:** RW**Reset State:** 0x00000000

Blend Layer1 Transparency Color Low Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_0_BLEND1_FG TRANSP_LOW0

Bits	Name	Description
27:16	TRANSPARENCY_C1	Transparency color check lower limit for Color1
11:0	TRANSPARENCY_C0	Transparency color check lower limit for Color0

0x01A45060 MDSS_MDP_VP_0_LAYER_0_BLEND1_FG TRANSP_LOW1**Type:** RW**Reset State:** 0x00000000

Blend Layer1 Transparency Color Low Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_0_BLEND1_FG TRANSP_LOW1

Bits	Name	Description
11:0	TRANSPARENCY_C2	Transparency color check lower limit for Color2

0x01A45064 MDSS_MDP_VP_0_LAYER_0_BLEND1_FG TRANSP_HIGH0**Type:** RW**Reset State:** 0x00000000

Blend Layer1 Transparency Color High Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_0_BLEND1_FG TRANSP_HIGH0

Bits	Name	Description
27:16	TRANSPARENCY_C1	Transparency color check higher limit for Color1
11:0	TRANSPARENCY_C0	Transparency color check higher limit for Color0

0x01A45068 MDSS_MDP_VP_0_LAYER_0_BLEND1_FG TRANSP_HIGH1**Type:** RW**Reset State:** 0x00000000

Blend Layer1 Transparency Color High Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_0_BLEND1_FG TRANSP_HIGH1

Bits	Name	Description
11:0	TRANSPARENCY_C2	Transparency color check higher limit for Color2

0x01A4506C MDSS_MDP_VP_0_LAYER_0_BLEND1_BG TRANSP_LOW0**Type:** RW**Reset State:** 0x00000000

Blend Layer1 Transparency Color Low Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_0_BLEND1_BG TRANSP_LOW0

Bits	Name	Description
27:16	TRANSPARENCY_C1	Transparency color check lower limit for Color1
11:0	TRANSPARENCY_C0	Transparency color check lower limit for Color0

0x01A45070 MDSS_MDP_VP_0_LAYER_0_BLEND1_BG TRANSP_LOW1**Type:** RW**Reset State:** 0x00000000

Blend Layer1 Transparency Color Low Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_0_BLEND1_BG TRANSP_LOW1

Bits	Name	Description
11:0	TRANSPARENCY_C2	Transparency color check lower limit for Color2

0x01A45074 MDSS_MDP_VP_0_LAYER_0_BLEND1_BG TRANSP_HIGH0**Type:** RW**Reset State:** 0x00000000

Blend Layer1 Transparency Color High Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_0_BLEND1_BG TRANSP_HIGH0

Bits	Name	Description
27:16	TRANSPARENCY_C1	Transparency color check higher limit for Color1
11:0	TRANSPARENCY_C0	Transparency color check higher limit for Color0

0x01A45078 MDSS_MDP_VP_0_LAYER_0_BLEND1_BG_TRANSP_HIGH1**Type:** RW**Reset State:** 0x00000000

Blend Layer1 Transparency Color High Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_0_BLEND1_BG_TRANSP_HIGH1

Bits	Name	Description
11:0	TRANSPARENCY_C2	Transparency color check higher limit for Color2

0x01A45080 MDSS_MDP_VP_0_LAYER_0_BLEND2_OP**Type:** RW**Reset State:** 0x00000000

LayerMixer Blend2 Operation Control

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_LAYER_0_BLEND2_OP

Bits	Name	Description
13	BG_TRANSP_EN	BG transparency enabled
12	BG_INV_MOD_ALPHA	Ability to select between alpha and inverse alpha right after the alpha modulation 0x0: Alpha 0x1: Inverse Alpha
11	BG_MOD_ALPHA	Modulate BG Alpha
10	BG_INV_ALPHA_SEL	Ability to select between alpha and inverse alpha right after the alpha select 0x0: Alpha 0x1: Inverse Alpha
9:8	BG_ALPHA_SEL	Blending Alpha Select 0x0: FG constant alpha 0x1: BG constant alpha 0x2: FG per pixel alpha 0x3: BG per pixel alpha
5	FG_TRANSP_EN	FG transparency enabled

MDSS_MDP_VP_0_LAYER_0_BLEND2_OP (cont.)

Bits	Name	Description
4	FG_INV_MOD_ALPHA	Ability to select between alpha and inverse alpha right after the alpha modulation 0x0: Alpha 0x1: Inverse Alpha
3	FG_MOD_ALPHA	Modulate Alpha
2	FG_INV_ALPHA_SEL	Ability to select between alpha and inverse alpha right after the alpha select 0x0: Alpha 0x1: Inverse Alpha
1:0	FG_ALPHA_SEL	Blending Alpha Select 0x0: FG constant alpha 0x1: BG constant alpha 0x2: FG per pixel alpha 0x3: BG per pixel alpha

0x01A45084 MDSS_MDP_VP_0_LAYER_0_BLEND2_FG_ALPHA**Type:** RW**Reset State:** 0x00000000

Blend Layer2 FG alpha

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_LAYER_0_BLEND2_FG_ALPHA

Bits	Name	Description
7:0	CONSTANT_ALPHA	Constant Alpha

0x01A45088 MDSS_MDP_VP_0_LAYER_0_BLEND2_BG_ALPHA**Type:** RW**Reset State:** 0x00000000

Blend Layer2 BG alpha

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_LAYER_0_BLEND2_BG_ALPHA

Bits	Name	Description
7:0	CONSTANT_ALPHA	Constant Alpha

0x01A4508C MDSS_MDP_VP_0_LAYER_0_BLEND2_FG TRANSP_LOW0**Type:** RW**Reset State:** 0x00000000

Blend Layer2 Transparency Color Low Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_0_BLEND2_FG TRANSP_LOW0

Bits	Name	Description
27:16	TRANSPARENCY_C1	Transparency color check lower limit for Color1
11:0	TRANSPARENCY_C0	Transparency color check lower limit for Color0

0x01A45090 MDSS_MDP_VP_0_LAYER_0_BLEND2_FG TRANSP_LOW1**Type:** RW**Reset State:** 0x00000000

Blend Layer2 Transparency Color Low Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_0_BLEND2_FG TRANSP_LOW1

Bits	Name	Description
11:0	TRANSPARENCY_C2	Transparency color check lower limit for Color2

0x01A45094 MDSS_MDP_VP_0_LAYER_0_BLEND2_FG TRANSP_HIGH0**Type:** RW**Reset State:** 0x00000000

Blend Layer2 Transparency Color High Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_0_BLEND2_FG TRANSP_HIGH0

Bits	Name	Description
27:16	TRANSPARENCY_C1	Transparency color check higher limit for Color1
11:0	TRANSPARENCY_C0	Transparency color check higher limit for Color0

0x01A45098 MDSS_MDP_VP_0_LAYER_0_BLEND2_FG TRANSP_HIGH1**Type:** RW**Reset State:** 0x00000000

Blend Layer2 Transparency Color High Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_0_BLEND2_FG TRANSP_HIGH1

Bits	Name	Description
11:0	TRANSPARENCY_C2	Transparency color check higher limit for Color2

0x01A4509C MDSS_MDP_VP_0_LAYER_0_BLEND2_BG TRANSP_LOW0**Type:** RW**Reset State:** 0x00000000

Blend Layer2 Transparency Color Low Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_0_BLEND2_BG TRANSP_LOW0

Bits	Name	Description
27:16	TRANSPARENCY_C1	Transparency color check lower limit for Color1
11:0	TRANSPARENCY_C0	Transparency color check lower limit for Color0

0x01A450A0 MDSS_MDP_VP_0_LAYER_0_BLEND2_BG TRANSP_LOW1**Type:** RW**Reset State:** 0x00000000

Blend Layer2 Transparency Color Low Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_0_BLEND2_BG TRANSP_LOW1

Bits	Name	Description
11:0	TRANSPARENCY_C2	Transparency color check lower limit for Color2

0x01A450A4 MDSS_MDP_VP_0_LAYER_0_BLEND2_BG TRANSP_HIGH0**Type:** RW**Reset State:** 0x00000000

Blend Layer2 Transparency Color High Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_0_BLEND2_BG TRANSP_HIGH0

Bits	Name	Description
27:16	TRANSPARENCY_C1	Transparency color check higher limit for Color1
11:0	TRANSPARENCY_C0	Transparency color check higher limit for Color0

0x01A450A8 MDSS_MDP_VP_0_LAYER_0_BLEND2_BG TRANSP_HIGH1**Type:** RW**Reset State:** 0x00000000

Blend Layer2 Transparency Color High Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_0_BLEND2_BG TRANSP_HIGH1

Bits	Name	Description
11:0	TRANSPARENCY_C2	Transparency color check higher limit for Color2

0x01A450B0 MDSS_MDP_VP_0_LAYER_0_BLEND3_OP**Type:** RW**Reset State:** 0x00000000

LayerMixer Blend3 Operation Control

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_LAYER_0_BLEND3_OP

Bits	Name	Description
13	BG TRANSP_EN	BG transparency enabled
12	BG_INV_MOD_ALPHA	Ability to select between alpha and inverse alpha right after the alpha modulation 0x0: Alpha 0x1: Inverse Alpha
11	BG_MOD_ALPHA	Modulate BG Alpha
10	BG_INV_ALPHA_SEL	Ability to select between alpha and inverse alpha right after the alpha select 0x0: Alpha 0x1: Inverse Alpha
9:8	BG_ALPHA_SEL	Blending Alpha Select 0x0: FG constant alpha 0x1: BG constant alpha 0x2: FG per pixel alpha 0x3: BG per pixel alpha
5	FG TRANSP_EN	FG transparency enabled
4	FG_INV_MOD_ALPHA	Ability to select between alpha and inverse alpha right after the alpha modulation 0x0: Alpha 0x1: Inverse Alpha
3	FG_MOD_ALPHA	Modulate Alpha
2	FG_INV_ALPHA_SEL	Ability to select between alpha and inverse alpha right after the alpha select 0x0: Alpha 0x1: Inverse Alpha
1:0	FG_ALPHA_SEL	Blending Alpha Select 0x0: FG constant alpha 0x1: BG constant alpha 0x2: FG per pixel alpha 0x3: BG per pixel alpha

0x01A450B4 MDSS_MDP_VP_0_LAYER_0_BLEND3_FG_ALPHA**Type:** RW**Reset State:** 0x00000000

Blend Layer3 FG alpha

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_LAYER_0_BLEND3_FG_ALPHA

Bits	Name	Description
7:0	CONSTANT_ALPHA	Constant Alpha

0x01A450B8 MDSS_MDP_VP_0_LAYER_0_BLEND3_BG_ALPHA**Type:** RW**Reset State:** 0x00000000

Blend Layer3 FG alpha

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_LAYER_0_BLEND3_BG_ALPHA

Bits	Name	Description
7:0	CONSTANT_ALPHA	Constant Alpha

0x01A450BC MDSS_MDP_VP_0_LAYER_0_BLEND3_FG TRANSP_LOW0**Type:** RW**Reset State:** 0x00000000

Blend Layer3 Transparency Color Low Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_0_BLEND3_FG TRANSP_LOW0

Bits	Name	Description
27:16	TRANSPARENCY_C1	Transparency color check lower limit for Color1
11:0	TRANSPARENCY_C0	Transparency color check lower limit for Color0

0x01A450C0 MDSS_MDP_VP_0_LAYER_0_BLEND3_FG TRANSP_LOW1**Type:** RW**Reset State:** 0x00000000

Blend Layer3 Transparency Color Low Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_0_BLEND3_FG TRANSP_LOW1

Bits	Name	Description
11:0	TRANSPARENCY_C2	Transparency color check lower limit for Color2

0x01A450C4 MDSS_MDP_VP_0_LAYER_0_BLEND3_FG TRANSP_HIGH0**Type:** RW**Reset State:** 0x00000000

Blend Layer3 Transparency Color High Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_0_BLEND3_FG TRANSP_HIGH0

Bits	Name	Description
27:16	TRANSPARENCY_C1	Transparency color check higher limit for Color1
11:0	TRANSPARENCY_C0	Transparency color check higher limit for Color0

0x01A450C8 MDSS_MDP_VP_0_LAYER_0_BLEND3_FG TRANSP_HIGH1**Type:** RW**Reset State:** 0x00000000

Blend Layer3 Transparency Color High Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_0_BLEND3_FG TRANSP_HIGH1

Bits	Name	Description
11:0	TRANSPARENCY_C2	Transparency color check higher limit for Color2

0x01A450CC MDSS_MDP_VP_0_LAYER_0_BLEND3_BG TRANSP_LOW0**Type:** RW**Reset State:** 0x00000000

Blend Layer3 Transparency Color Low Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_0_BLEND3_BG TRANSP_LOW0

Bits	Name	Description
27:16	TRANSPARENCY_C1	Transparency color check lower limit for Color1
11:0	TRANSPARENCY_C0	Transparency color check lower limit for Color0

0x01A450D0 MDSS_MDP_VP_0_LAYER_0_BLEND3_BG TRANSP_LOW1**Type:** RW**Reset State:** 0x00000000

Blend Layer3 Transparency Color Low Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_0_BLEND3_BG TRANSP_LOW1

Bits	Name	Description
11:0	TRANSPARENCY_C2	Transparency color check lower limit for Color2

0x01A450D4 MDSS_MDP_VP_0_LAYER_0_BLEND3_BG TRANSP_HIGH0**Type:** RW**Reset State:** 0x00000000

Blend Layer3 Transparency Color High Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_0_BLEND3_BG TRANSP_HIGH0

Bits	Name	Description
27:16	TRANSPARENCY_C1	Transparency color check higher limit for Color1
11:0	TRANSPARENCY_C0	Transparency color check higher limit for Color0

0x01A450D8 MDSS_MDP_VP_0_LAYER_0_BLEND3_BG TRANSP_HIGH1**Type:** RW**Reset State:** 0x00000000

Blend Layer3 Transparency Color High Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_0_BLEND3_BG_TRANSP_HIGH1

Bits	Name	Description
11:0	TRANSPARENCY_C2	Transparency color check higher limit for Color2

0x01A450DC MDSS_MDP_VP_0_LAYER_0_CURSOR_STRIDE**Type:** RW**Reset State:** 0x00000000

Cursor Stride

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_LAYER_0_CURSOR_STRIDE

Bits	Name	Description
15:0	STRIDE	Cursor Stride

0x01A450E0 MDSS_MDP_VP_0_LAYER_0_CURSOR_IMG_SIZE**Type:** RW**Reset State:** 0x00000000

Cursor Image Size

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_LAYER_0_CURSOR_IMG_SIZE

Bits	Name	Description
31:16	SRC_H	Total Cursor Source Height
15:0	SRC_W	Total Cursor Source Width

0x01A450E4 MDSS_MDP_VP_0_LAYER_0_CURSOR_SIZE**Type:** RW**Reset State:** 0x00000000

Cursor ROI Image Size

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_LAYER_0_CURSOR_SIZE

Bits	Name	Description
31:16	ROI_H	Cursor ROI Source Height
15:0	ROI_W	Cursor ROI Source Width

0x01A450E8 MDSS_MDP_VP_0_LAYER_0_CURSOR_XY

Type: RW

Reset State: 0x00000000

Cursor ROI Offset

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_LAYER_0_CURSOR_XY

Bits	Name	Description
31:16	SRC_Y	Y ROI coordinate
15:0	SRC_X	X ROI coordinate

0x01A450EC MDSS_MDP_VP_0_LAYER_0_CURSOR_FORMAT

Type: RW

Reset State: 0x00000000

Cursor Format

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_LAYER_0_CURSOR_FORMAT

Bits	Name	Description
2:0	FORMAT	Cursor Format 0x0: ARGB8888 0x2: ARGB1555 0x4: ARGB4444

0x01A450F0 MDSS_MDP_VP_0_LAYER_0_CURSOR_BASE_ADDR**Type:** RW**Reset State:** 0x00000000

Cursor Base Address

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_LAYER_0_CURSOR_BASE_ADDR

Bits	Name	Description
31:0	ADDR	Cursor Base-Address (byte-aligned)

0x01A450F4 MDSS_MDP_VP_0_LAYER_0_CURSOR_START_XY**Type:** RW**Reset State:** 0x00000000

Cursor Starting Coordinates

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_LAYER_0_CURSOR_START_XY

Bits	Name	Description
31:16	Y_START	Starting Y coordinate
15:0	X_START	Starting X coordinate

0x01A450F8 MDSS_MDP_VP_0_LAYER_0_CURSOR_BLEND_CONFIG**Type:** RW**Reset State:** 0x00000000

Cursor Blending Configuration

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_LAYER_0_CURSOR_BLEND_CONFIG

Bits	Name	Description
3	BLEND_TRANSP_EN	0x0: Disabled 0x1: Color keying is done on the cursor and blended result
2:1	BLEND_ALPHA_SEL	Blending Alpha Select 0x0: Constant Alpha 0x1: reserved 0x2: Cursor per pixel alpha 0x3: reserved
0	BLEND_EN	Cursor Blend Enable

0x01A450FC MDSS_MDP_VP_0_LAYER_0_CURSOR_BLEND_PARAM**Type:** RW**Reset State:** 0x00000000

Cursor Blending Parameter

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_LAYER_0_CURSOR_BLEND_PARAM

Bits	Name	Description
7:0	CONSTANT_ALPHA	Constant Alpha

0x01A45100 MDSS_MDP_VP_0_LAYER_0_CURSOR_BLEND_TRANSP_LOW0**Type:** RW**Reset State:** 0x00000000

Cursor Transparency Color Low Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_0_CURSOR_BLEND_TRANSP_LOW0

Bits	Name	Description
27:16	TRANSPARENCY_C1	Transparency color check lower limit for Color1
11:0	TRANSPARENCY_C0	Transparency color check lower limit for Color0

0x01A45104 MDSS_MDP_VP_0_LAYER_0_CURSOR_BLEND TRANSP_LOW1**Type:** RW**Reset State:** 0x00000000

Cursor Transparency Color Low Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_0_CURSOR_BLEND TRANSP_LOW1

Bits	Name	Description
11:0	TRANSPARENCY_C2	Transparency color check lower limit for Color2

0x01A45108 MDSS_MDP_VP_0_LAYER_0_CURSOR_BLEND TRANSP_HIGH0**Type:** RW**Reset State:** 0x00000000

Cursor Transparency Color High Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_0_CURSOR_BLEND TRANSP_HIGH0

Bits	Name	Description
27:16	TRANSPARENCY_C1	Transparency color check higher limit for Color1
11:0	TRANSPARENCY_C0	Transparency color check higher limit for Color0

0x01A4510C MDSS_MDP_VP_0_LAYER_0_CURSOR_BLEND TRANSP_HIGH1**Type:** RW**Reset State:** 0x00000000

Cursor Transparency Color High Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_0_CURSOR_BLEND TRANSP_HIGH1

Bits	Name	Description
11:0	TRANSPARENCY_C2	Transparency color check higher limit for Color2

0x01A45110 MDSS_MDP_VP_0_LAYER_0_GC_ARLUT_RED_STARTX_N**Type:** RW**Reset State:** 0x00000000**MDSS_MDP_VP_0_LAYER_0_GC_ARLUT_RED_STARTX_N**

Bits	Name	Description
21	VALUE_UPDATEN	
20	SEGMENT_UPDATE	
19:16	SEGMENT	
11:0	STARTX	

0x01A45114 MDSS_MDP_VP_0_LAYER_0_GC_ARLUT_RED_SLOPE_N**Type:** RW**Reset State:** 0x00000000**MDSS_MDP_VP_0_LAYER_0_GC_ARLUT_RED_SLOPE_N**

Bits	Name	Description
21	VALUE_UPDATEN	
20	SEGMENT_UPDATE	
19:16	SEGMENT	
14:0	SLOPE	

0x01A45118 MDSS_MDP_VP_0_LAYER_0_GC_ARLUT_RED_OFFSET_N**Type:** RW**Reset State:** 0x00000000**MDSS_MDP_VP_0_LAYER_0_GC_ARLUT_RED_OFFSET_N**

Bits	Name	Description
21	VALUE_UPDATEN	
20	SEGMENT_UPDATE	
19:16	SEGMENT	
14:0	OFFSET	

0x01A45120 MDSS_MDP_VP_0_LAYER_0_GC_ARLUT_GREEN_STARTX_N**Type:** RW**Reset State:** 0x00000000**MDSS_MDP_VP_0_LAYER_0_GC_ARLUT_GREEN_STARTX_N**

Bits	Name	Description
21	VALUE_UPDATEN	
20	SEGMENT_UPDATE	
19:16	SEGMENT	
11:0	STARTX	

0x01A45124 MDSS_MDP_VP_0_LAYER_0_GC_ARLUT_GREEN_SLOPE_N**Type:** RW**Reset State:** 0x00000000**MDSS_MDP_VP_0_LAYER_0_GC_ARLUT_GREEN_SLOPE_N**

Bits	Name	Description
21	VALUE_UPDATEN	
20	SEGMENT_UPDATE	
19:16	SEGMENT	
14:0	SLOPE	

0x01A45128 MDSS_MDP_VP_0_LAYER_0_GC_ARLUT_GREEN_OFFSET_N**Type:** RW**Reset State:** 0x00000000**MDSS_MDP_VP_0_LAYER_0_GC_ARLUT_GREEN_OFFSET_N**

Bits	Name	Description
21	VALUE_UPDATEN	
20	SEGMENT_UPDATE	
19:16	SEGMENT	
14:0	OFFSET	

0x01A45130 MDSS_MDP_VP_0_LAYER_0_GC_ARLUT_BLUE_STARTX_N**Type:** RW**Reset State:** 0x00000000**MDSS_MDP_VP_0_LAYER_0_GC_ARLUT_BLUE_STARTX_N**

Bits	Name	Description
21	VALUE_UPDATEN	
20	SEGMENT_UPDATE	
19:16	SEGMENT	
11:0	STARTX	

0x01A45134 MDSS_MDP_VP_0_LAYER_0_GC_ARLUT_BLUE_SLOPE_N**Type:** RW**Reset State:** 0x00000000**MDSS_MDP_VP_0_LAYER_0_GC_ARLUT_BLUE_SLOPE_N**

Bits	Name	Description
21	VALUE_UPDATEN	
20	SEGMENT_UPDATE	
19:16	SEGMENT	
14:0	SLOPE	

0x01A45138 MDSS_MDP_VP_0_LAYER_0_GC_ARLUT_BLUE_OFFSET_N**Type:** RW**Reset State:** 0x00000000**MDSS_MDP_VP_0_LAYER_0_GC_ARLUT_BLUE_OFFSET_N**

Bits	Name	Description
21	VALUE_UPDATEN	
20	SEGMENT_UPDATE	
19:16	SEGMENT	
14:0	OFFSET	

0x01A45200 MDSS_MDP_VP_0_LAYER_0_CURSOR_BASE_ADDR_SW_STATUS**Type:** RW**Reset State:** 0x00000000

Security Status of the source buffer plane

This is programmed by SW to indicate if the buffer is NON-SECURE or SECURE

This information will be presented to VBIF per transaction and reflected as part of the memory transaction

double buffered: YES

MDSS_MDP_VP_0_LAYER_0_CURSOR_BASE_ADDR_SW_STATUS

Bits	Name	Description
0	ADDR_CP	

0x01A45204 MDSS_MDP_VP_0_LAYER_0_CURSOR_FETCH_CONFIG**Type:** RW**Reset State:** 0x00000087

Cursor Client Fetch Configuration per Virtual Pipeline double buffered: NO

This register must NOT be altered during operation and MDP must be idle when changing values

MDSS_MDP_VP_0_LAYER_0_CURSOR_FETCH_CONFIG

Bits	Name	Description
17:16	TILE_SIZE	Tile Size for processing of BLOCK formats Note that this only applies to blocks that operate in block mode which is limited to DMA source pipe. This setting has no effect for all other source pipes (ViG/RGB) 0x0: 128x128 0x1: 64x64 0x2: 32x32
12	ADDR_SYNC_MODE	Boundary on which fetch address synchronizes. Setting this field to line forces the address updates on the next line which performs the software requested feature of updating on hsync. Note that updating on line boundaries will likely result in tearing on the panel since hardware is fetching from new memory buffer. 0x0: frame (default) 0x1: line

MDSS_MDP_VP_0_LAYER_0_CURSOR_FETCH_CONFIG (cont.)

Bits	Name	Description
9:8	MAL	Describes the Minimum Access Length for the source pipeline. This value should align with the DRAM's expected transaction alignment unit. This setting forces the source fetch pipe to be aligned to DRAM and implement re-alignment for maximum efficiency 0x0: 128b alignment (default) 0x1: 256b alignment 0x2: 512b alignment
7:4	MAX_BURST_SIZE	Maximum burst beat size. Hardware fetch engines operate on 128-bit bus Possible values: 0x8: burst of 8 0x4: burst of 4 0x2: burst of 2
3:0	REQ_DEPTH_LIMIT	Pending request limit, default max value of 0x7 means maximum pending requests allowed in the design Anything less limits the bus requests to that number + 1.

0x01A45210 MDSS_MDP_VP_0_LAYER_0_CURSOR_FETCH_SMP_WR_PLANE0**Type:** R**Reset State:** 0x00000000

Fetch to SMP on plane 0 write counter

double buffered: N/A

MDSS_MDP_VP_0_LAYER_0_CURSOR_FETCH_SMP_WR_PLANE0

Bits	Name	Description
15:0	COUNT	This counter is reset on every frame and is incremented for each write fetch makes to the SMP plane0 port This should correlate to value SMP_UNPACK_RD_PLANE0 More useful for debug but can be used for profiling purposes as well.

0x01A45214 MDSS_MDP_VP_0_LAYER_0_CURSOR_SMP_UNPACK_RD_PLANE0**Type:** R**Reset State:** 0x00000000

SMP to unpack on plane 0 read counter

double buffered: N/A

MDSS_MDP_VP_0_LAYER_0_CURSOR_SMP_UNPACK_RD_PLANE0

Bits	Name	Description
15:0	COUNT	This counter is reset on every frame and is incremented for each line read from SMP plane0 port This should correlate to FETCH_SMP_WR_PLANE0 value More useful for debug but can be used for profiling purposes as well.

0x01A45220 MDSS_MDP_VP_0_LAYER_0_CURSOR_REQPRIORITY_FIFO_WATERMARK_0**Type:** RW**Reset State:** 0x00000008

FIFO watermark to aid in generation of mdpREQPRIORITY

The watermark here refers to the dedicated cursor fetch memory.

The value programmed here should be aware of the amount of dedicated cursor fetch memory.

So a value of 0 indicates completely FULL (i.e. there are no more entries available in the dedicated memory)

A value of 100 indicates 100 entries in the Shared Memory Pool (each entry being 128-bit entries)

This is the lowestmost FIFO depth and should not be programmed higher than
REQPRIORITY_FIFO_WATERMARK_1 or REQPRIORITY_FIFO_WATERMARK_2

An example programming would set the following:

REQPRIORITY_FIFO_WATERMARK_0 : 0x08 (meaning 0x08 entries free to be written into dedicated cursor fetch memory - least room available)

REQPRIORITY_FIFO_WATERMARK_1 : 0x10 (meaning 0x10 entries free to be written into dedicated cursor fetch memory - middle ground room available)

REQPRIORITY_FIFO_WATERMARK_2 : 0x20 (meaning 0x20 entries free to be written into dedicated cursor fetch memory - most room available)

mdpREQPRIORITY = 11 (highest priority request) when above all 3 watermark levels (i.e. lots of space available in dedicated cursor fetch memory)

mdpREQPRIORITY = 10 above FIFO_0 and FIFO_1 but below FIFO_2

mdpREQPRIORITY = 01 above FIFO_0 but below FIFO_1 and FIFO_2

mdpREQPRIORITY = 00 (lowest priority request) when below all 3 watermark levels (i.e. not a lot of space available in dedicated cursor fetch memory so its good)

double buffered: YES

MDSS_MDP_VP_0_LAYER_0_CURSOR_REQPRIORITY_FIFO_WATERMARK_0

Bits	Name	Description
15:0	WATERMARK	Watermark Level for FIFO_0

0x01A45224 MDSS_MDP_VP_0_LAYER_0_CURSOR_REQPRIORITY_FIFO_WATERMARK_1**Type:** RW**Reset State:** 0x00000010

FIFO watermark to aid in generation of mdpREQPRIORITY

The watermark here refers to the dedicated cursor fetch memory.

The value programmed here should be aware of the amount of dedicated cursor fetch memory.

So a value of 0 indicates completely FULL (i.e. there are no more entries available in the dedicated memory)

A value of 100 indicates 100 entries in the Shared Memory Pool (each entry being 128-bit entries)

This is the lowestmost FIFO depth and should not be programmed higher than REQPRIORITY_FIFO_WATERMARK_1 or REQPRIORITY_FIFO_WATERMARK_2

An example programming would set the following:

REQPRIORITY_FIFO_WATERMARK_0 : 0x08 (meaning 0x08 entries free to be written into dedicated cursor fetch memory - least room available)

REQPRIORITY_FIFO_WATERMARK_1 : 0x10 (meaning 0x10 entries free to be written into dedicated cursor fetch memory - middle ground room available)

REQPRIORITY_FIFO_WATERMARK_2 : 0x20 (meaning 0x20 entries free to be written into dedicated cursor fetch memory - most room available)

mdpREQPRIORITY = 11 (highest priority request) when above all 3 watermark levels (i.e. lots of space available in dedicated cursor fetch memory)

mdpREQPRIORITY = 10 above FIFO_0 and FIFO_1 but below FIFO_2

mdpREQPRIORITY = 01 above FIFO_0 but below FIFO_1 and FIFO_2

mdpREQPRIORITY = 00 (lowest priority request) when below all 3 watermark levels (i.e. not a lot of space available in dedicated cursor fetch memory so its good)

double buffered: YES

MDSS_MDP_VP_0_LAYER_0_CURSOR_REQPRIORITY_FIFO_WATERMARK_1

Bits	Name	Description
15:0	WATERMARK	Watermark Level for FIFO_1

0x01A45228 MDSS_MDP_VP_0_LAYER_0_CURSOR_REQPRIORITY_FIFO_WATERMARK_2**Type:** RW**Reset State:** 0x00000020

FIFO watermark to aid in generation of mdpREQPRIORITY

The watermark here refers to the dedicated cursor fetch memory.

The value programmed here should be aware of the amount of dedicated cursor fetch memory.

So a value of 0 indicates completely FULL (i.e. there are no more entries available in the dedicated memory)

A value of 100 indicates 100 entries in the Shared Memory Pool (each entry being 128-bit entries)

This is the lowestmost FIFO depth and should not be programmed higher than REQPRIORITY_FIFO_WATERMARK_1 or REQPRIORITY_FIFO_WATERMARK_2

An example programming would set the following:

REQPRIORITY_FIFO_WATERMARK_0 : 0x08 (meaning 0x08 entries free to be written into dedicated cursor fetch memory - least room available)

REQPRIORITY_FIFO_WATERMARK_1 : 0x10 (meaning 0x10 entries free to be written into dedicated cursor fetch memory - middle ground room available)

REQPRIORITY_FIFO_WATERMARK_2 : 0x20 (meaning 0x20 entries free to be written into dedicated cursor fetch memory - most room available)

mdpREQPRIORITY = 11 (highest priority request) when above all 3 watermark levels (i.e. lots of space available in dedicated cursor fetch memory)

mdpREQPRIORITY = 10 above FIFO_0 and FIFO_1 but below FIFO_2

mdpREQPRIORITY = 01 above FIFO_0 but below FIFO_1 and FIFO_2

mdpREQPRIORITY = 00 (lowest priority request) when below all 3 watermark levels (i.e. not a lot of space available in dedicated cursor fetch memory so its good)

double buffered: YES

MDSS_MDP_VP_0_LAYER_0_CURSOR_REQPRIORITY_FIFO_WATERMARK_2

Bits	Name	Description
15:0	WATERMARK	Watermark Level for FIFO_2

0x01A45380 MDSS_MDP_VP_0_LAYER_0_GC_MISR_CTRL**Type:** RW**Reset State:** 0x00000000

MDSS_MDP_VP_0_LAYER_0_GC_MISR_CTRL

Bits	Name	Type	Description
31	RUN_MODE	RW	
27:24	INPUT_SEL	RW	
10	STATUS_CLEAR	W	
9	STATUS	R	
8	ENABLE	RW	
7:0	FRAME_COUNT	RW	

0x01A45384 MDSS_MDP_VP_0_LAYER_0_GC_MISR_SIGNATURE**Type:** R**Reset State:** 0x00000000**MDSS_MDP_VP_0_LAYER_0_GC_MISR_SIGNATURE**

Bits	Name	Description
31:0	CAPTURED	

17.17.7.10 MDSS_MDP_VP_0_LAYER_1

Layer Mixer hardware register description

17.17.7.10.1 MDSS_MDP_VP_0_LAYER_1_LAYER

LAYER

Layer Mixer hardware register description

0x01A46000 MDSS_MDP_VP_0_LAYER_1_OP_MODE**Type:** RW**Reset State:** 0x00000000

LayerMixer Operation Control

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_LAYER_1_OP_MODE

Bits	Name	Description
4	BLEND3_COLOR3_OUT	Selects which source to drive on color3 component output for stage3 blend 0x0: BG Layer Color3 0x1: FG Layer Color3
3	BLEND2_COLOR3_OUT	Selects which source to drive on color3 component output for stage2 blend 0x0: BG Layer Color3 0x1: FG Layer Color3
2	BLEND1_COLOR3_OUT	Selects which source to drive on color3 component output for stage1 blend 0x0: BG Layer Color3 0x1: FG Layer Color3
1	BLEND0_COLOR3_OUT	Selects which source to drive on color3 component output for stage0 blend 0x0: BG Layer Color3 0x1: FG Layer Color3
0	GC_LUT_EN	0x0: Disabled - mapped linearly from 12-bit to 8-bit 0x1: Enabled

0x01A46004 MDSS_MDP_VP_0_LAYER_1_OUT_SIZE**Type:** RW**Reset State:** 0x00000000

LayerMixer Output Size

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_LAYER_1_OUT_SIZE

Bits	Name	Description
31:16	DST_H	Destination Output Height
15:0	DST_W	Destination Output Width The exact supported size depends on the width of line buffers allocated for MDP.

0x01A46008 MDSS_MDP_VP_0_LAYER_1_BORDER_COLOR_0**Type:** RW**Reset State:** 0x00000000

LayerMixer Border Color Register

double buffered: YES

MDSS_MDP_VP_0_LAYER_1_BORDER_COLOR_0

Bits	Name	Description
27:16	COLOR1	Color1 in 12-bit format
11:0	COLOR0	Color0 in 12-bit format

0x01A46010 MDSS_MDP_VP_0_LAYER_1_BORDER_COLOR_1**Type:** RW**Reset State:** 0x00000000

LayerMixer Border Color Register

double buffered: YES

MDSS_MDP_VP_0_LAYER_1_BORDER_COLOR_1

Bits	Name	Description
27:16	COLOR3	Color3 in 12-bit format, LSB4 are ignored, should be programmed to 0
11:0	COLOR2	Color2 in 12-bit format

0x01A46020 MDSS_MDP_VP_0_LAYER_1_BLEND0_OP**Type:** RW**Reset State:** 0x00000000

LayerMixer Blend0 Operation Control

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_LAYER_1_BLEND0_OP

Bits	Name	Description
13	BG TRANSP_EN	BG transparency enabled
12	BG_INV_MOD_ALPHA	Ability to select between alpha and inverse alpha right after the alpha modulation 0x0: Alpha 0x1: Inverse Alpha
11	BG_MOD_ALPHA	Modulate BG Alpha
10	BG_INV_ALPHA_SEL	Ability to select between alpha and inverse alpha right after the alpha select 0x0: Alpha 0x1: Inverse Alpha
9:8	BG_ALPHA_SEL	Blending Alpha Select 0x0: FG constant alpha 0x1: BG constant alpha 0x2: FG per pixel alpha 0x3: BG per pixel alpha
5	FG TRANSP_EN	FG transparency enabled
4	FG_INV_MOD_ALPHA	Ability to select between alpha and inverse alpha right after the alpha modulation 0x0: Alpha 0x1: Inverse Alpha
3	FG_MOD_ALPHA	Modulate Alpha
2	FG_INV_ALPHA_SEL	Ability to select between alpha and inverse alpha right after the alpha select 0x0: Alpha 0x1: Inverse Alpha
1:0	FG_ALPHA_SEL	Blending Alpha Select 0x0: FG constant alpha 0x1: BG constant alpha 0x2: FG per pixel alpha 0x3: BG per pixel alpha

0x01A46024 MDSS_MDP_VP_0_LAYER_1_BLEND0_FG_ALPHA**Type:** RW**Reset State:** 0x00000000

Blend Layer0 FG alpha

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_LAYER_1_BLEND0_FG_ALPHA

Bits	Name	Description
7:0	CONSTANT_ALPHA	Constant Alpha

0x01A46028 MDSS_MDP_VP_0_LAYER_1_BLEND0_BG_ALPHA**Type:** RW**Reset State:** 0x00000000

Blend Layer0 FG alpha

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_LAYER_1_BLEND0_BG_ALPHA

Bits	Name	Description
7:0	CONSTANT_ALPHA	Constant Alpha

0x01A4602C MDSS_MDP_VP_0_LAYER_1_BLEND0_FG TRANSP_LOW0**Type:** RW**Reset State:** 0x00000000

Blend Layer0 Transparency Color Low Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_1_BLEND0_FG TRANSP_LOW0

Bits	Name	Description
27:16	TRANSPARENCY_C1	Transparency color check lower limit for Color1
11:0	TRANSPARENCY_C0	Transparency color check lower limit for Color0

0x01A46030 MDSS_MDP_VP_0_LAYER_1_BLEND0_FG TRANSP_LOW1**Type:** RW**Reset State:** 0x00000000

Blend Layer0 Transparency Color Low Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_1_BLEND0_FG TRANSP_LOW1

Bits	Name	Description
11:0	TRANSPARENCY_C2	Transparency color check lower limit for Color2

0x01A46034 MDSS_MDP_VP_0_LAYER_1_BLEND0_FG TRANSP_HIGH0**Type:** RW**Reset State:** 0x00000000

Blend Layer0 Transparency Color High Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_1_BLEND0_FG TRANSP_HIGH0

Bits	Name	Description
27:16	TRANSPARENCY_C1	Transparency color check higher limit for Color1
11:0	TRANSPARENCY_C0	Transparency color check higher limit for Color0

0x01A46038 MDSS_MDP_VP_0_LAYER_1_BLEND0_FG TRANSP_HIGH1**Type:** RW**Reset State:** 0x00000000

Blend Layer0 Transparency Color High Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_1_BLEND0_FG TRANSP_HIGH1

Bits	Name	Description
11:0	TRANSPARENCY_C2	Transparency color check higher limit for Color2

0x01A4603C MDSS_MDP_VP_0_LAYER_1_BLEND0_BG TRANSP_LOW0**Type:** RW**Reset State:** 0x00000000

Blend Layer0 Transparency Color Low Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_1_BLEND0_BG TRANSP_LOW0

Bits	Name	Description
27:16	TRANSPARENCY_C1	Transparency color check lower limit for Color1
11:0	TRANSPARENCY_C0	Transparency color check lower limit for Color0

0x01A46040 MDSS_MDP_VP_0_LAYER_1_BLEND0_BG TRANSP_LOW1**Type:** RW**Reset State:** 0x00000000

Blend Layer0 Transparency Color Low Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_1_BLEND0_BG TRANSP_LOW1

Bits	Name	Description
11:0	TRANSPARENCY_C2	Transparency color check lower limit for Color2

0x01A46044 MDSS_MDP_VP_0_LAYER_1_BLEND0_BG TRANSP_HIGH0**Type:** RW**Reset State:** 0x00000000

Blend Layer0 Transparency Color High Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_1_BLEND0_BG TRANSP_HIGH0

Bits	Name	Description
27:16	TRANSPARENCY_C1	Transparency color check higher limit for Color1
11:0	TRANSPARENCY_C0	Transparency color check higher limit for Color0

0x01A46048 MDSS_MDP_VP_0_LAYER_1_BLEND0_BG TRANSP_HIGH1**Type:** RW**Reset State:** 0x00000000

Blend Layer0 Transparency Color High Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_1_BLEND0_BG_TRANSP_HIGH1

Bits	Name	Description
11:0	TRANSPARENCY_C2	Transparency color check higher limit for Color2

0x01A46050 MDSS_MDP_VP_0_LAYER_1_BLEND1_OP**Type:** RW**Reset State:** 0x00000000

LayerMixer Blend1 Operation Control

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_LAYER_1_BLEND1_OP

Bits	Name	Description
13	BG_TRANSP_EN	BG transparency enabled
12	BG_INV_MOD_ALPHA	Ability to select between alpha and inverse alpha right after the alpha modulation 0x0: Alpha 0x1: Inverse Alpha
11	BG_MOD_ALPHA	Modulate BG Alpha
10	BG_INV_ALPHA_SEL	Ability to select between alpha and inverse alpha right after the alpha select 0x0: Alpha 0x1: Inverse Alpha
9:8	BG_ALPHA_SEL	Blending Alpha Select 0x0: FG constant alpha 0x1: BG constant alpha 0x2: FG per pixel alpha 0x3: BG per pixel alpha
5	FG_TRANSP_EN	FG transparency enabled
4	FG_INV_MOD_ALPHA	Ability to select between alpha and inverse alpha right after the alpha modulation 0x0: Alpha 0x1: Inverse Alpha
3	FG_MOD_ALPHA	Modulate Alpha
2	FG_INV_ALPHA_SEL	Ability to select between alpha and inverse alpha right after the alpha select 0x0: Alpha 0x1: Inverse Alpha

MDSS_MDP_VP_0_LAYER_1_BLEND1_OP (cont.)

Bits	Name	Description
1:0	FG_ALPHA_SEL	Blending Alpha Select 0x0: FG constant alpha 0x1: BG constant alpha 0x2: FG per pixel alpha 0x3: BG per pixel alpha

0x01A46054 MDSS_MDP_VP_0_LAYER_1_BLEND1_FG_ALPHA**Type:** RW**Reset State:** 0x00000000

Blend Layer1 FG alpha

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_LAYER_1_BLEND1_FG_ALPHA

Bits	Name	Description
7:0	CONSTANT_ALPHA	Constant Alpha

0x01A46058 MDSS_MDP_VP_0_LAYER_1_BLEND1_BG_ALPHA**Type:** RW**Reset State:** 0x00000000

Blend Layer1 FG alpha

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_LAYER_1_BLEND1_BG_ALPHA

Bits	Name	Description
7:0	CONSTANT_ALPHA	Constant Alpha

0x01A4605C MDSS_MDP_VP_0_LAYER_1_BLEND1_FG TRANSP_LOW0**Type:** RW**Reset State:** 0x00000000

Blend Layer1 Transparency Color Low Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_1_BLEND1_FG TRANSP_LOW0

Bits	Name	Description
27:16	TRANSPARENCY_C1	Transparency color check lower limit for Color1
11:0	TRANSPARENCY_C0	Transparency color check lower limit for Color0

0x01A46060 MDSS_MDP_VP_0_LAYER_1_BLEND1_FG TRANSP_LOW1

Type: RW

Reset State: 0x00000000

Blend Layer1 Transparency Color Low Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_1_BLEND1_FG TRANSP_LOW1

Bits	Name	Description
11:0	TRANSPARENCY_C2	Transparency color check lower limit for Color2

0x01A46064 MDSS_MDP_VP_0_LAYER_1_BLEND1_FG TRANSP_HIGH0

Type: RW

Reset State: 0x00000000

Blend Layer1 Transparency Color High Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_1_BLEND1_FG TRANSP_HIGH0

Bits	Name	Description
27:16	TRANSPARENCY_C1	Transparency color check higher limit for Color1
11:0	TRANSPARENCY_C0	Transparency color check higher limit for Color0

0x01A46068 MDSS_MDP_VP_0_LAYER_1_BLEND1_FG TRANSP_HIGH1

Type: RW

Reset State: 0x00000000

Blend Layer1 Transparency Color High Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_1_BLEND1_FG TRANSP_HIGH1

Bits	Name	Description
11:0	TRANSPARENCY_C2	Transparency color check higher limit for Color2

0x01A4606C MDSS_MDP_VP_0_LAYER_1_BLEND1_BG TRANSP_LOW0

Type: RW

Reset State: 0x00000000

Blend Layer1 Transparency Color Low Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_1_BLEND1_BG TRANSP_LOW0

Bits	Name	Description
27:16	TRANSPARENCY_C1	Transparency color check lower limit for Color1
11:0	TRANSPARENCY_C0	Transparency color check lower limit for Color0

0x01A46070 MDSS_MDP_VP_0_LAYER_1_BLEND1_BG TRANSP_LOW1

Type: RW

Reset State: 0x00000000

Blend Layer1 Transparency Color Low Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_1_BLEND1_BG TRANSP_LOW1

Bits	Name	Description
11:0	TRANSPARENCY_C2	Transparency color check lower limit for Color2

0x01A46074 MDSS_MDP_VP_0_LAYER_1_BLEND1_BG TRANSP_HIGH0

Type: RW

Reset State: 0x00000000

Blend Layer1 Transparency Color High Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_1_BLEND1_BG TRANSP_HIGH0

Bits	Name	Description
27:16	TRANSPARENCY_C1	Transparency color check higher limit for Color1
11:0	TRANSPARENCY_C0	Transparency color check higher limit for Color0

0x01A46078 MDSS_MDP_VP_0_LAYER_1_BLEND1_BG TRANSP_HIGH1

Type: RW

Reset State: 0x00000000

Blend Layer1 Transparency Color High Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_1_BLEND1_BG TRANSP_HIGH1

Bits	Name	Description
11:0	TRANSPARENCY_C2	Transparency color check higher limit for Color2

0x01A46080 MDSS_MDP_VP_0_LAYER_1_BLEND2_OP

Type: RW

Reset State: 0x00000000

LayerMixer Blend2 Operation Control

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_LAYER_1_BLEND2_OP

Bits	Name	Description
13	BG_TRANSP_EN	BG transparency enabled
12	BG_INV_MOD_ALPHA	Ability to select between alpha and inverse alpha right after the alpha modulation 0x0: Alpha 0x1: Inverse Alpha
11	BG_MOD_ALPHA	Modulate BG Alpha

MDSS_MDP_VP_0_LAYER_1_BLEND2_OP (cont.)

Bits	Name	Description
10	BG_INV_ALPHA_SEL	Ability to select between alpha and inverse alpha right after the alpha select 0x0: Alpha 0x1: Inverse Alpha
9:8	BG_ALPHA_SEL	Blending Alpha Select 0x0: FG constant alpha 0x1: BG constant alpha 0x2: FG per pixel alpha 0x3: BG per pixel alpha
5	FG TRANSP_EN	FG transparency enabled
4	FG_INV_MOD_ALPHA	Ability to select between alpha and inverse alpha right after the alpha modulation 0x0: Alpha 0x1: Inverse Alpha
3	FG_MOD_ALPHA	Modulate Alpha
2	FG_INV_ALPHA_SEL	Ability to select between alpha and inverse alpha right after the alpha select 0x0: Alpha 0x1: Inverse Alpha
1:0	FG_ALPHA_SEL	Blending Alpha Select 0x0: FG constant alpha 0x1: BG constant alpha 0x2: FG per pixel alpha 0x3: BG per pixel alpha

0x01A46084 MDSS_MDP_VP_0_LAYER_1_BLEND2_FG_ALPHA**Type:** RW**Reset State:** 0x00000000

Blend Layer2 FG alpha

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_LAYER_1_BLEND2_FG_ALPHA

Bits	Name	Description
7:0	CONSTANT_ALPHA	Constant Alpha

0x01A46088 MDSS_MDP_VP_0_LAYER_1_BLEND2_BG_ALPHA**Type:** RW**Reset State:** 0x00000000

Blend Layer2 FG alpha

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_LAYER_1_BLEND2_BG_ALPHA

Bits	Name	Description
7:0	CONSTANT_ALPHA	Constant Alpha

0x01A4608C MDSS_MDP_VP_0_LAYER_1_BLEND2_FG TRANSP_LOW0**Type:** RW**Reset State:** 0x00000000

Blend Layer2 Transparency Color Low Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_1_BLEND2_FG TRANSP_LOW0

Bits	Name	Description
27:16	TRANSPARENCY_C1	Transparency color check lower limit for Color1
11:0	TRANSPARENCY_C0	Transparency color check lower limit for Color0

0x01A46090 MDSS_MDP_VP_0_LAYER_1_BLEND2_FG TRANSP_LOW1**Type:** RW**Reset State:** 0x00000000

Blend Layer2 Transparency Color Low Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_1_BLEND2_FG TRANSP_LOW1

Bits	Name	Description
11:0	TRANSPARENCY_C2	Transparency color check lower limit for Color2

0x01A46094 MDSS_MDP_VP_0_LAYER_1_BLEND2_FG TRANSP_HIGH0**Type:** RW**Reset State:** 0x00000000

Blend Layer2 Transparency Color High Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_1_BLEND2_FG TRANSP_HIGH0

Bits	Name	Description
27:16	TRANSPARENCY_C1	Transparency color check higher limit for Color1
11:0	TRANSPARENCY_C0	Transparency color check higher limit for Color0

0x01A46098 MDSS_MDP_VP_0_LAYER_1_BLEND2_FG TRANSP_HIGH1**Type:** RW**Reset State:** 0x00000000

Blend Layer2 Transparency Color High Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_1_BLEND2_FG TRANSP_HIGH1

Bits	Name	Description
11:0	TRANSPARENCY_C2	Transparency color check higher limit for Color2

0x01A4609C MDSS_MDP_VP_0_LAYER_1_BLEND2_BG TRANSP_LOW0**Type:** RW**Reset State:** 0x00000000

Blend Layer2 Transparency Color Low Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_1_BLEND2_BG TRANSP_LOW0

Bits	Name	Description
27:16	TRANSPARENCY_C1	Transparency color check lower limit for Color1
11:0	TRANSPARENCY_C0	Transparency color check lower limit for Color0

0x01A460A0 MDSS_MDP_VP_0_LAYER_1_BLEND2_BG TRANSP_LOW1**Type:** RW**Reset State:** 0x00000000

Blend Layer2 Transparency Color Low Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_1_BLEND2_BG TRANSP_LOW1

Bits	Name	Description
11:0	TRANSPARENCY_C2	Transparency color check lower limit for Color2

0x01A460A4 MDSS_MDP_VP_0_LAYER_1_BLEND2_BG TRANSP_HIGH0**Type:** RW**Reset State:** 0x00000000

Blend Layer2 Transparency Color High Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_1_BLEND2_BG TRANSP_HIGH0

Bits	Name	Description
27:16	TRANSPARENCY_C1	Transparency color check higher limit for Color1
11:0	TRANSPARENCY_C0	Transparency color check higher limit for Color0

0x01A460A8 MDSS_MDP_VP_0_LAYER_1_BLEND2_BG TRANSP_HIGH1**Type:** RW**Reset State:** 0x00000000

Blend Layer2 Transparency Color High Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_1_BLEND2_BG TRANSP_HIGH1

Bits	Name	Description
11:0	TRANSPARENCY_C2	Transparency color check higher limit for Color2

0x01A460B0 MDSS_MDP_VP_0_LAYER_1_BLEND3_OP**Type:** RW**Reset State:** 0x00000000

LayerMixer Blend3 Operation Control

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_LAYER_1_BLEND3_OP

Bits	Name	Description
13	BG TRANSP_EN	BG transparency enabled
12	BG_INV_MOD_ALPHA	Ability to select between alpha and inverse alpha right after the alpha modulation 0x0: Alpha 0x1: Inverse Alpha
11	BG_MOD_ALPHA	Modulate BG Alpha
10	BG_INV_ALPHA_SEL	Ability to select between alpha and inverse alpha right after the alpha select 0x0: Alpha 0x1: Inverse Alpha
9:8	BG_ALPHA_SEL	Blending Alpha Select 0x0: FG constant alpha 0x1: BG constant alpha 0x2: FG per pixel alpha 0x3: BG per pixel alpha
5	FG TRANSP_EN	FG transparency enabled
4	FG_INV_MOD_ALPHA	Ability to select between alpha and inverse alpha right after the alpha modulation 0x0: Alpha 0x1: Inverse Alpha
3	FG_MOD_ALPHA	Modulate Alpha
2	FG_INV_ALPHA_SEL	Ability to select between alpha and inverse alpha right after the alpha select 0x0: Alpha 0x1: Inverse Alpha
1:0	FG_ALPHA_SEL	Blending Alpha Select 0x0: FG constant alpha 0x1: BG constant alpha 0x2: FG per pixel alpha 0x3: BG per pixel alpha

0x01A460B4 MDSS_MDP_VP_0_LAYER_1_BLEND3_FG_ALPHA**Type:** RW**Reset State:** 0x00000000

Blend Layer3 FG alpha

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_LAYER_1_BLEND3_FG_ALPHA

Bits	Name	Description
7:0	CONSTANT_ALPHA	Constant Alpha

0x01A460B8 MDSS_MDP_VP_0_LAYER_1_BLEND3_BG_ALPHA**Type:** RW**Reset State:** 0x00000000

Blend Layer3 FG alpha

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_LAYER_1_BLEND3_BG_ALPHA

Bits	Name	Description
7:0	CONSTANT_ALPHA	Constant Alpha

0x01A460BC MDSS_MDP_VP_0_LAYER_1_BLEND3_FG TRANSP_LOW0**Type:** RW**Reset State:** 0x00000000

Blend Layer3 Transparency Color Low Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_1_BLEND3_FG TRANSP_LOW0

Bits	Name	Description
27:16	TRANSPARENCY_C1	Transparency color check lower limit for Color1
11:0	TRANSPARENCY_C0	Transparency color check lower limit for Color0

0x01A460C0 MDSS_MDP_VP_0_LAYER_1_BLEND3_FG TRANSP_LOW1**Type:** RW**Reset State:** 0x00000000

Blend Layer3 Transparency Color Low Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_1_BLEND3_FG TRANSP_LOW1

Bits	Name	Description
11:0	TRANSPARENCY_C2	Transparency color check lower limit for Color2

0x01A460C4 MDSS_MDP_VP_0_LAYER_1_BLEND3_FG TRANSP_HIGH0**Type:** RW**Reset State:** 0x00000000

Blend Layer3 Transparency Color High Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_1_BLEND3_FG TRANSP_HIGH0

Bits	Name	Description
27:16	TRANSPARENCY_C1	Transparency color check higher limit for Color1
11:0	TRANSPARENCY_C0	Transparency color check higher limit for Color0

0x01A460C8 MDSS_MDP_VP_0_LAYER_1_BLEND3_FG TRANSP_HIGH1**Type:** RW**Reset State:** 0x00000000

Blend Layer3 Transparency Color High Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_1_BLEND3_FG TRANSP_HIGH1

Bits	Name	Description
11:0	TRANSPARENCY_C2	Transparency color check higher limit for Color2

0x01A460CC MDSS_MDP_VP_0_LAYER_1_BLEND3_BG TRANSP_LOW0**Type:** RW**Reset State:** 0x00000000

Blend Layer3 Transparency Color Low Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_1_BLEND3_BG TRANSP_LOW0

Bits	Name	Description
27:16	TRANSPARENCY_C1	Transparency color check lower limit for Color1
11:0	TRANSPARENCY_C0	Transparency color check lower limit for Color0

0x01A460D0 MDSS_MDP_VP_0_LAYER_1_BLEND3_BG TRANSP_LOW1**Type:** RW**Reset State:** 0x00000000

Blend Layer3 Transparency Color Low Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_1_BLEND3_BG TRANSP_LOW1

Bits	Name	Description
11:0	TRANSPARENCY_C2	Transparency color check lower limit for Color2

0x01A460D4 MDSS_MDP_VP_0_LAYER_1_BLEND3_BG TRANSP_HIGH0**Type:** RW**Reset State:** 0x00000000

Blend Layer3 Transparency Color High Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_1_BLEND3_BG TRANSP_HIGH0

Bits	Name	Description
27:16	TRANSPARENCY_C1	Transparency color check higher limit for Color1
11:0	TRANSPARENCY_C0	Transparency color check higher limit for Color0

0x01A460D8 MDSS_MDP_VP_0_LAYER_1_BLEND3_BG TRANSP_HIGH1**Type:** RW**Reset State:** 0x00000000

Blend Layer3 Transparency Color High Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_1_BLEND3_BG TRANSP_HIGH1

Bits	Name	Description
11:0	TRANSPARENCY_C2	Transparency color check higher limit for Color2

0x01A460DC MDSS_MDP_VP_0_LAYER_1_CURSOR_STRIDE**Type:** RW**Reset State:** 0x00000000

Cursor Stride

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_LAYER_1_CURSOR_STRIDE

Bits	Name	Description
15:0	STRIDE	Cursor Stride

0x01A460E0 MDSS_MDP_VP_0_LAYER_1_CURSOR_IMG_SIZE**Type:** RW**Reset State:** 0x00000000

Cursor Image Size

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_LAYER_1_CURSOR_IMG_SIZE

Bits	Name	Description
31:16	SRC_H	Total Cursor Source Height
15:0	SRC_W	Total Cursor Source Width

0x01A460E4 MDSS_MDP_VP_0_LAYER_1_CURSOR_SIZE**Type:** RW**Reset State:** 0x00000000

Cursor ROI Image Size

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_LAYER_1_CURSOR_SIZE

Bits	Name	Description
31:16	ROI_H	Cursor ROI Source Height
15:0	ROI_W	Cursor ROI Source Width

0x01A460E8 MDSS_MDP_VP_0_LAYER_1_CURSOR_XY**Type:** RW**Reset State:** 0x00000000

Cursor ROI Offset

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_LAYER_1_CURSOR_XY

Bits	Name	Description
31:16	SRC_Y	Y ROI coordinate
15:0	SRC_X	X ROI coordinate

0x01A460EC MDSS_MDP_VP_0_LAYER_1_CURSOR_FORMAT**Type:** RW**Reset State:** 0x00000000

Cursor Format

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_LAYER_1_CURSOR_FORMAT

Bits	Name	Description
2:0	FORMAT	Cursor Format 0x0: ARGB8888 0x2: ARGB1555 0x4: ARGB4444

0x01A460F0 MDSS_MDP_VP_0_LAYER_1_CURSOR_BASE_ADDR**Type:** RW**Reset State:** 0x00000000

Cursor Base Address

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_LAYER_1_CURSOR_BASE_ADDR

Bits	Name	Description
31:0	ADDR	Cursor Base-Address (byte-aligned)

0x01A460F4 MDSS_MDP_VP_0_LAYER_1_CURSOR_START_XY**Type:** RW**Reset State:** 0x00000000

Cursor Starting Coordinates

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_LAYER_1_CURSOR_START_XY

Bits	Name	Description
31:16	Y_START	Starting Y coordinate
15:0	X_START	Starting X coordinate

0x01A460F8 MDSS_MDP_VP_0_LAYER_1_CURSOR_BLEND_CONFIG**Type:** RW**Reset State:** 0x00000000

Cursor Blending Configuration

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_LAYER_1_CURSOR_BLEND_CONFIG

Bits	Name	Description
3	BLEND_TRANSP_EN	0x0: Disabled 0x1: Color keying is done on the cursor and blended result
2:1	BLEND_ALPHA_SEL	Blending Alpha Select 0x0: Constant Alpha 0x1: reserved 0x2: Cursor per pixel alpha 0x3: reserved
0	BLEND_EN	Cursor Blend Enable

0x01A460FC MDSS_MDP_VP_0_LAYER_1_CURSOR_BLEND_PARAM**Type:** RW**Reset State:** 0x00000000

Cursor Blending Parameter

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_LAYER_1_CURSOR_BLEND_PARAM

Bits	Name	Description
7:0	CONSTANT_ALPHA	Constant Alpha

0x01A46100 MDSS_MDP_VP_0_LAYER_1_CURSOR_BLEND_TRANSP_LOW0**Type:** RW**Reset State:** 0x00000000

Cursor Transparency Color Low Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_1_CURSOR_BLEND_TRANSP_LOW0

Bits	Name	Description
27:16	TRANSPARENCY_C1	Transparency color check lower limit for Color1
11:0	TRANSPARENCY_C0	Transparency color check lower limit for Color0

0x01A46104 MDSS_MDP_VP_0_LAYER_1_CURSOR_BLEND_TRANSP_LOW1

Type: RW

Reset State: 0x00000000

Cursor Transparency Color Low Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_1_CURSOR_BLEND_TRANSP_LOW1

Bits	Name	Description
11:0	TRANSPARENCY_C2	Transparency color check lower limit for Color2

0x01A46108 MDSS_MDP_VP_0_LAYER_1_CURSOR_BLEND_TRANSP_HIGH0

Type: RW

Reset State: 0x00000000

Cursor Transparency Color High Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_1_CURSOR_BLEND_TRANSP_HIGH0

Bits	Name	Description
27:16	TRANSPARENCY_C1	Transparency color check higher limit for Color1
11:0	TRANSPARENCY_C0	Transparency color check higher limit for Color0

0x01A4610C MDSS_MDP_VP_0_LAYER_1_CURSOR_BLEND_TRANSP_HIGH1

Type: RW

Reset State: 0x00000000

Cursor Transparency Color High Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_1_CURSOR_BLEND_TRANSP_HIGH1

Bits	Name	Description
11:0	TRANSPARENCY_C2	Transparency color check higher limit for Color2

0x01A46110 MDSS_MDP_VP_0_LAYER_1_GC_ARLUT_RED_STARTX_N

Type: RW

Reset State: 0x00000000

MDSS_MDP_VP_0_LAYER_1_GC_ARLUT_RED_STARTX_N

Bits	Name	Description
21	VALUE_UPDATEN	
20	SEGMENT_UPDATE	
19:16	SEGMENT	
11:0	STARTX	

0x01A46114 MDSS_MDP_VP_0_LAYER_1_GC_ARLUT_RED_SLOPE_N

Type: RW

Reset State: 0x00000000

MDSS_MDP_VP_0_LAYER_1_GC_ARLUT_RED_SLOPE_N

Bits	Name	Description
21	VALUE_UPDATEN	
20	SEGMENT_UPDATE	
19:16	SEGMENT	
14:0	SLOPE	

0x01A46118 MDSS_MDP_VP_0_LAYER_1_GC_ARLUT_RED_OFFSET_N

Type: RW

Reset State: 0x00000000

MDSS_MDP_VP_0_LAYER_1_GC_ARLUT_RED_OFFSET_N

Bits	Name	Description
21	VALUE_UPDATEN	

MDSS_MDP_VP_0_LAYER_1_GC_ARLUT_RED_OFFSET_N (cont.)

Bits	Name	Description
20	SEGMENT_UPDATE	
19:16	SEGMENT	
14:0	OFFSET	

0x01A46120 MDSS_MDP_VP_0_LAYER_1_GC_ARLUT_GREEN_STARTX_N**Type:** RW**Reset State:** 0x00000000**MDSS_MDP_VP_0_LAYER_1_GC_ARLUT_GREEN_STARTX_N**

Bits	Name	Description
21	VALUE_UPDATEN	
20	SEGMENT_UPDATE	
19:16	SEGMENT	
11:0	STARTX	

0x01A46124 MDSS_MDP_VP_0_LAYER_1_GC_ARLUT_GREEN_SLOPE_N**Type:** RW**Reset State:** 0x00000000**MDSS_MDP_VP_0_LAYER_1_GC_ARLUT_GREEN_SLOPE_N**

Bits	Name	Description
21	VALUE_UPDATEN	
20	SEGMENT_UPDATE	
19:16	SEGMENT	
14:0	SLOPE	

0x01A46128 MDSS_MDP_VP_0_LAYER_1_GC_ARLUT_GREEN_OFFSET_N**Type:** RW**Reset State:** 0x00000000

MDSS_MDP_VP_0_LAYER_1_GC_ARLUT_GREEN_OFFSET_N

Bits	Name	Description
21	VALUE_UPDATEN	
20	SEGMENT_UPDATE	
19:16	SEGMENT	
14:0	OFFSET	

0x01A46130 MDSS_MDP_VP_0_LAYER_1_GC_ARLUT_BLUE_STARTX_N**Type:** RW**Reset State:** 0x00000000**MDSS_MDP_VP_0_LAYER_1_GC_ARLUT_BLUE_STARTX_N**

Bits	Name	Description
21	VALUE_UPDATEN	
20	SEGMENT_UPDATE	
19:16	SEGMENT	
11:0	STARTX	

0x01A46134 MDSS_MDP_VP_0_LAYER_1_GC_ARLUT_BLUE_SLOPE_N**Type:** RW**Reset State:** 0x00000000**MDSS_MDP_VP_0_LAYER_1_GC_ARLUT_BLUE_SLOPE_N**

Bits	Name	Description
21	VALUE_UPDATEN	
20	SEGMENT_UPDATE	
19:16	SEGMENT	
14:0	SLOPE	

0x01A46138 MDSS_MDP_VP_0_LAYER_1_GC_ARLUT_BLUE_OFFSET_N**Type:** RW**Reset State:** 0x00000000

MDSS_MDP_VP_0_LAYER_1_GC_ARLUT_BLUE_OFFSET_N

Bits	Name	Description
21	VALUE_UPDATEN	
20	SEGMENT_UPDATE	
19:16	SEGMENT	
14:0	OFFSET	

0x01A46200 MDSS_MDP_VP_0_LAYER_1_CURSOR_BASE_ADDR_SW_STATUS**Type:** RW**Reset State:** 0x00000000

Security Status of the source buffer plane

This is programmed by SW to indicate if the buffer is NON-SECURE or SECURE

This information will be presented to VBIF per transaction and reflected as part of the memory transaction

double buffered: YES

MDSS_MDP_VP_0_LAYER_1_CURSOR_BASE_ADDR_SW_STATUS

Bits	Name	Description
0	ADDR_CP	

0x01A46204 MDSS_MDP_VP_0_LAYER_1_CURSOR_FETCH_CONFIG**Type:** RW**Reset State:** 0x00000087

Cursor Client Fetch Configuration per Virtual Pipeline double buffered: NO

This register must NOT be altered during operation and MDP must be idle when changing values

MDSS_MDP_VP_0_LAYER_1_CURSOR_FETCH_CONFIG

Bits	Name	Description
17:16	TILE_SIZE	Tile Size for processing of BLOCK formats Note that this only applies to blocks that operate in block mode which is limited to DMA source pipe. This setting has no effect for all other source pipes (ViG/RGB) 0x0: 128x128 0x1: 64x64 0x2: 32x32
12	ADDR_SYNC_MODE	Boundary on which fetch address synchronizes. Setting this field to line forces the address updates on the next line which performs the software requested feature of updating on hsync. Note that updating on line boundaries will likely result in tearing on the panel since hardware is fetching from new memory buffer. 0x0: frame (default) 0x1: line
9:8	MAL	Describes the Minimum Access Length for the source pipeline. This value should align with the DRAM's expected transaction alignment unit. This setting forces the source fetch pipe to be aligned to DRAM and implement re-alignment for maximum efficiency 0x0: 128b alignment (default) 0x1: 256b alignment 0x2: 512b alignment
7:4	MAX_BURST_SIZE	Maximum burst beat size. Hardware fetch engines operate on 128-bit bus Possible values: 0x8: burst of 8 0x4: burst of 4 0x2: burst of 2
3:0	REQ_DEPTH_LIMIT	Pending request limit, default max value of 0x7 means maximum pending requests allowed in the design Anything less limits the bus requests to that number + 1.

0x01A46210 MDSS_MDP_VP_0_LAYER_1_CURSOR_FETCH_SMP_WR_PLANE0**Type:** R**Reset State:** 0x00000000

Fetch to SMP on plane 0 write counter

double buffered: N/A

MDSS_MDP_VP_0_LAYER_1_CURSOR_FETCH_SMP_WR_PLANE0

Bits	Name	Description
15:0	COUNT	This counter is reset on every frame and is incremented for each write fetch makes to the SMP plane0 port This should correlate to value SMP_UNPACK_RD_PLANE0 More useful for debug but can be used for profiling purposes as well.

0x01A46214 MDSS_MDP_VP_0_LAYER_1_CURSOR_SMP_UNPACK_RD_PLANE0**Type:** R**Reset State:** 0x00000000

SMP to unpack on plane 0 read counter

double buffered: N/A

MDSS_MDP_VP_0_LAYER_1_CURSOR_SMP_UNPACK_RD_PLANE0

Bits	Name	Description
15:0	COUNT	This counter is reset on every frame and is incremented for each line read from SMP plane0 port This should correlate to FETCH_SMP_WR_PLANE0 value More useful for debug but can be used for profiling purposes as well.

0x01A46220 MDSS_MDP_VP_0_LAYER_1_CURSOR_REQPRIORITY_FIFO_WATERMARK_0**Type:** RW**Reset State:** 0x00000008

FIFO watermark to aid in generation of mdpREQPRIORITY

The watermark here refers to the dedicated cursor fetch memory.

The value programmed here should be aware of the amount of dedicated cursor fetch memory.

So a value of 0 indicates completely FULL (i.e. there are no more entries available in the dedicated memory)

A value of 100 indicates 100 entries in the Shared Memory Pool (each entry being 128-bit entries)

This is the lowestmost FIFO depth and should not be programmed higher than REQPRIORITY_FIFO_WATERMARK_1 or REQPRIORITY_FIFO_WATERMARK_2

An example programming would set the following:

REQPRIORITY_FIFO_WATERMARK_0 : 0x08 (meaning 0x08 entries free to be written into dedicated cursor fetch memory - least room available)

REQPRIORITY_FIFO_WATERMARK_1 : 0x10 (meaning 0x10 entries free to be written into dedicated cursor fetch memory - middle ground room available)

REQPRIORITY_FIFO_WATERMARK_2 : 0x20 (meaning 0x20 entries free to be written into dedicated cursor fetch memory - most room available)

mdpREQPRIORITY = 11 (highest priority request) when above all 3 watermark levels (i.e. lots of space available in dedicated cursor fetch memory)

mdpREQPRIORITY = 10 above FIFO_0 and FIFO_1 but below FIFO_2

mdpREQPRIORITY = 01 above FIFO_0 but below FIFO_1 and FIFO_2

mdpREQPRIORITY = 00 (lowest priority request) when below all 3 watermark levels (i.e. not a lot of space available in dedicated cursor fetch memory so its good)

double buffered: YES

MDSS_MDP_VP_0_LAYER_1_CURSOR_REQPRIORITY_FIFO_WATERMARK_0

Bits	Name	Description
15:0	WATERMARK	Watermark Level for FIFO_0

0x01A46224 MDSS_MDP_VP_0_LAYER_1_CURSOR_REQPRIORITY_FIFO_WATERMARK_1

Type: RW

Reset State: 0x00000010

FIFO watermark to aid in generation of mdpREQPRIORITY

The watermark here refers to the dedicated cursor fetch memory.

The value programmed here should be aware of the amount of dedicated cursor fetch memory.

So a value of 0 indicates completely FULL (i.e. there are no more entries available in the dedicated memory)

A value of 100 indicates 100 entries in the Shared Memory Pool (each entry being 128-bit entries)

This is the lowestmost FIFO depth and should not be programmed higher than REQPRIORITY_FIFO_WATERMARK_1 or REQPRIORITY_FIFO_WATERMARK_2

An example programming would set the following:

REQPRIORITY_FIFO_WATERMARK_0 : 0x08 (meaning 0x08 entries free to be written into dedicated cursor fetch memory - least room available)

REQPRIORITY_FIFO_WATERMARK_1 : 0x10 (meaning 0x10 entries free to be written into dedicated cursor fetch memory - middle ground room available)

REQPRIORITY_FIFO_WATERMARK_2 : 0x20 (meaning 0x20 entries free to be written into dedicated cursor fetch memory - most room available)

mdpREQPRIORITY = 11 (highest priority request) when above all 3 watermark levels (i.e. lots of space available in dedicated cursor fetch memory)

mdpREQPRIORITY = 10 above FIFO_0 and FIFO_1 but below FIFO_2

mdpREQPRIORITY = 01 above FIFO_0 but below FIFO_1 and FIFO_2

mdpREQPRIORITY = 00 (lowest priority request) when below all 3 watermark levels (i.e. not a lot of space available in dedicated cursor fetch memory so its good)

double buffered: YES

MDSS_MDP_VP_0_LAYER_1_CURSOR_REQPRIORITY_FIFO_WATERMARK_1

Bits	Name	Description
15:0	WATERMARK	Watermark Level for FIFO_1

0x01A46228 MDSS_MDP_VP_0_LAYER_1_CURSOR_REQPRIORITY_FIFO_WATERMARK_2

Type: RW

Reset State: 0x00000020

FIFO watermark to aid in generation of mdpREQPRIORITY

The watermark here refers to the dedicated cursor fetch memory.

The value programmed here should be aware of the amount of dedicated cursor fetch memory.

So a value of 0 indicates completely FULL (i.e. there are no more entries available in the dedicated memory)

A value of 100 indicates 100 entries in the Shared Memory Pool (each entry being 128-bit entries)

This is the lowestmost FIFO depth and should not be programmed higher than REQPRIORITY_FIFO_WATERMARK_1 or REQPRIORITY_FIFO_WATERMARK_2

An example programming would set the following:

REQPRIORITY_FIFO_WATERMARK_0 : 0x08 (meaning 0x08 entries free to be written into dedicated cursor fetch memory - least room available)

REQPRIORITY_FIFO_WATERMARK_1 : 0x10 (meaning 0x10 entries free to be written into dedicated cursor fetch memory - middle ground room available)

REQPRIORITY_FIFO_WATERMARK_2 : 0x20 (meaning 0x20 entries free to be written into dedicated cursor fetch memory - most room available)

mdpREQPRIORITY = 11 (highest priority request) when above all 3 watermark levels (i.e. lots of space available in dedicated cursor fetch memory)

mdpREQPRIORITY = 10 above FIFO_0 and FIFO_1 but below FIFO_2

mdpREQPRIORITY = 01 above FIFO_0 but below FIFO_1 and FIFO_2

mdpREQPRIORITY = 00 (lowest priority request) when below all 3 watermark levels (i.e. not a lot of space available in dedicated cursor fetch memory so its good)

double buffered: YES

MDSS_MDP_VP_0_LAYER_1_CURSOR_REQPRIORITY_FIFO_WATERMARK_2

Bits	Name	Description
15:0	WATERMARK	Watermark Level for FIFO_2

0x01A46380 MDSS_MDP_VP_0_LAYER_1_GC_MISR_CTRL

Type: RW

Reset State: 0x00000000

MDSS_MDP_VP_0_LAYER_1_GC_MISR_CTRL

Bits	Name	Type	Description
31	RUN_MODE	RW	
27:24	INPUT_SEL	RW	
10	STATUS_CLEAR	W	
9	STATUS	R	
8	ENABLE	RW	
7:0	FRAME_COUNT	RW	

0x01A46384 MDSS_MDP_VP_0_LAYER_1_GC_MISR_SIGNATURE

Type: R

Reset State: 0x00000000

MDSS_MDP_VP_0_LAYER_1_GC_MISR_SIGNATURE

Bits	Name	Description
31:0	CAPTURED	

17.17.7.11 MDSS_MDP_VP_0_LAYER_2

Layer Mixer hardware register description

17.17.7.11.1 MDSS_MDP_VP_0_LAYER_2_LAYER

LAYER

Layer Mixer hardware register description

0x01A47000 MDSS_MDP_VP_0_LAYER_2_OP_MODE**Type:** RW**Reset State:** 0x00000000

LayerMixer Operation Control

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_LAYER_2_OP_MODE

Bits	Name	Description
4	BLEND3_COLOR3_OUT	Selects which source to drive on color3 component output for stage3 blend 0x0: BG Layer Color3 0x1: FG Layer Color3
3	BLEND2_COLOR3_OUT	Selects which source to drive on color3 component output for stage2 blend 0x0: BG Layer Color3 0x1: FG Layer Color3
2	BLEND1_COLOR3_OUT	Selects which source to drive on color3 component output for stage1 blend 0x0: BG Layer Color3 0x1: FG Layer Color3
1	BLEND0_COLOR3_OUT	Selects which source to drive on color3 component output for stage0 blend 0x0: BG Layer Color3 0x1: FG Layer Color3
0	GC_LUT_EN	0x0: Disabled - mapped linearly from 12-bit to 8-bit 0x1: Enabled

0x01A47004 MDSS_MDP_VP_0_LAYER_2_OUT_SIZE**Type:** RW**Reset State:** 0x00000000

LayerMixer Output Size

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_LAYER_2_OUT_SIZE

Bits	Name	Description
31:16	DST_H	Destination Output Height
15:0	DST_W	Destination Output Width The exact supported size depends on the width of line buffers allocated for MDP.

0x01A47008 MDSS_MDP_VP_0_LAYER_2_BORDER_COLOR_0

Type: RW

Reset State: 0x00000000

LayerMixer Border Color Register

double buffered: YES

MDSS_MDP_VP_0_LAYER_2_BORDER_COLOR_0

Bits	Name	Description
27:16	COLOR1	Color1 in 12-bit format
11:0	COLOR0	Color0 in 12-bit format

0x01A47010 MDSS_MDP_VP_0_LAYER_2_BORDER_COLOR_1

Type: RW

Reset State: 0x00000000

LayerMixer Border Color Register

double buffered: YES

MDSS_MDP_VP_0_LAYER_2_BORDER_COLOR_1

Bits	Name	Description
27:16	COLOR3	Color3 in 12-bit format, LSB4 are ignored, should be programmed to 0
11:0	COLOR2	Color2 in 12-bit format

0x01A47020 MDSS_MDP_VP_0_LAYER_2_BLEND0_OP**Type:** RW**Reset State:** 0x00000000

LayerMixer Blend0 Operation Control

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_LAYER_2_BLEND0_OP

Bits	Name	Description
13	BG TRANSP_EN	BG transparency enabled
12	BG_INV_MOD_ALPHA	Ability to select between alpha and inverse alpha right after the alpha modulation 0x0: Alpha 0x1: Inverse Alpha
11	BG_MOD_ALPHA	Modulate BG Alpha
10	BG_INV_ALPHA_SEL	Ability to select between alpha and inverse alpha right after the alpha select 0x0: Alpha 0x1: Inverse Alpha
9:8	BG_ALPHA_SEL	Blending Alpha Select 0x0: FG constant alpha 0x1: BG constant alpha 0x2: FG per pixel alpha 0x3: BG per pixel alpha
5	FG TRANSP_EN	FG transparency enabled
4	FG_INV_MOD_ALPHA	Ability to select between alpha and inverse alpha right after the alpha modulation 0x0: Alpha 0x1: Inverse Alpha
3	FG_MOD_ALPHA	Modulate Alpha
2	FG_INV_ALPHA_SEL	Ability to select between alpha and inverse alpha right after the alpha select 0x0: Alpha 0x1: Inverse Alpha
1:0	FG_ALPHA_SEL	Blending Alpha Select 0x0: FG constant alpha 0x1: BG constant alpha 0x2: FG per pixel alpha 0x3: BG per pixel alpha

0x01A47024 MDSS_MDP_VP_0_LAYER_2_BLEND0_FG_ALPHA**Type:** RW**Reset State:** 0x00000000

Blend Layer0 FG alpha

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_LAYER_2_BLEND0_FG_ALPHA

Bits	Name	Description
7:0	CONSTANT_ALPHA	Constant Alpha

0x01A47028 MDSS_MDP_VP_0_LAYER_2_BLEND0_BG_ALPHA**Type:** RW**Reset State:** 0x00000000

Blend Layer0 FG alpha

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_LAYER_2_BLEND0_BG_ALPHA

Bits	Name	Description
7:0	CONSTANT_ALPHA	Constant Alpha

0x01A4702C MDSS_MDP_VP_0_LAYER_2_BLEND0_FG TRANSP_LOW0**Type:** RW**Reset State:** 0x00000000

Blend Layer0 Transparency Color Low Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_2_BLEND0_FG TRANSP_LOW0

Bits	Name	Description
27:16	TRANSPARENCY_C1	Transparency color check lower limit for Color1
11:0	TRANSPARENCY_C0	Transparency color check lower limit for Color0

0x01A47030 MDSS_MDP_VP_0_LAYER_2_BLEND0_FG TRANSP_LOW1**Type:** RW**Reset State:** 0x00000000

Blend Layer0 Transparency Color Low Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_2_BLEND0_FG TRANSP_LOW1

Bits	Name	Description
11:0	TRANSPARENCY_C2	Transparency color check lower limit for Color2

0x01A47034 MDSS_MDP_VP_0_LAYER_2_BLEND0_FG TRANSP_HIGH0**Type:** RW**Reset State:** 0x00000000

Blend Layer0 Transparency Color High Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_2_BLEND0_FG TRANSP_HIGH0

Bits	Name	Description
27:16	TRANSPARENCY_C1	Transparency color check higher limit for Color1
11:0	TRANSPARENCY_C0	Transparency color check higher limit for Color0

0x01A47038 MDSS_MDP_VP_0_LAYER_2_BLEND0_FG TRANSP_HIGH1**Type:** RW**Reset State:** 0x00000000

Blend Layer0 Transparency Color High Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_2_BLEND0_FG TRANSP_HIGH1

Bits	Name	Description
11:0	TRANSPARENCY_C2	Transparency color check higher limit for Color2

0x01A4703C MDSS_MDP_VP_0_LAYER_2_BLEND0_BG TRANSP_LOW0**Type:** RW**Reset State:** 0x00000000

Blend Layer0 Transparency Color Low Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_2_BLEND0_BG TRANSP_LOW0

Bits	Name	Description
27:16	TRANSPARENCY_C1	Transparency color check lower limit for Color1
11:0	TRANSPARENCY_C0	Transparency color check lower limit for Color0

0x01A47040 MDSS_MDP_VP_0_LAYER_2_BLEND0_BG TRANSP_LOW1**Type:** RW**Reset State:** 0x00000000

Blend Layer0 Transparency Color Low Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_2_BLEND0_BG TRANSP_LOW1

Bits	Name	Description
11:0	TRANSPARENCY_C2	Transparency color check lower limit for Color2

0x01A47044 MDSS_MDP_VP_0_LAYER_2_BLEND0_BG TRANSP_HIGH0**Type:** RW**Reset State:** 0x00000000

Blend Layer0 Transparency Color High Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_2_BLEND0_BG TRANSP_HIGH0

Bits	Name	Description
27:16	TRANSPARENCY_C1	Transparency color check higher limit for Color1
11:0	TRANSPARENCY_C0	Transparency color check higher limit for Color0

0x01A47048 MDSS_MDP_VP_0_LAYER_2_BLEND0_BG_TRANSP_HIGH1**Type:** RW**Reset State:** 0x00000000

Blend Layer0 Transparency Color High Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_2_BLEND0_BG_TRANSP_HIGH1

Bits	Name	Description
11:0	TRANSPARENCY_C2	Transparency color check higher limit for Color2

0x01A47050 MDSS_MDP_VP_0_LAYER_2_BLEND1_OP**Type:** RW**Reset State:** 0x00000000

LayerMixer Blend1 Operation Control

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_LAYER_2_BLEND1_OP

Bits	Name	Description
13	BG_TRANSP_EN	BG transparency enabled
12	BG_INV_MOD_ALPHA	Ability to select between alpha and inverse alpha right after the alpha modulation 0x0: Alpha 0x1: Inverse Alpha
11	BG_MOD_ALPHA	Modulate BG Alpha
10	BG_INV_ALPHA_SEL	Ability to select between alpha and inverse alpha right after the alpha select 0x0: Alpha 0x1: Inverse Alpha
9:8	BG_ALPHA_SEL	Blending Alpha Select 0x0: FG constant alpha 0x1: BG constant alpha 0x2: FG per pixel alpha 0x3: BG per pixel alpha
5	FG_TRANSP_EN	FG transparency enabled

MDSS_MDP_VP_0_LAYER_2_BLEND1_OP (cont.)

Bits	Name	Description
4	FG_INV_MOD_ALPHA	Ability to select between alpha and inverse alpha right after the alpha modulation 0x0: Alpha 0x1: Inverse Alpha
3	FG_MOD_ALPHA	Modulate Alpha
2	FG_INV_ALPHA_SEL	Ability to select between alpha and inverse alpha right after the alpha select 0x0: Alpha 0x1: Inverse Alpha
1:0	FG_ALPHA_SEL	Blending Alpha Select 0x0: FG constant alpha 0x1: BG constant alpha 0x2: FG per pixel alpha 0x3: BG per pixel alpha

0x01A47054 MDSS_MDP_VP_0_LAYER_2_BLEND1_FG_ALPHA**Type:** RW**Reset State:** 0x00000000

Blend Layer1 FG alpha

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_LAYER_2_BLEND1_FG_ALPHA

Bits	Name	Description
7:0	CONSTANT_ALPHA	Constant Alpha

0x01A47058 MDSS_MDP_VP_0_LAYER_2_BLEND1_BG_ALPHA**Type:** RW**Reset State:** 0x00000000

Blend Layer1 FG alpha

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_LAYER_2_BLEND1_BG_ALPHA

Bits	Name	Description
7:0	CONSTANT_ALPHA	Constant Alpha

0x01A4705C MDSS_MDP_VP_0_LAYER_2_BLEND1_FG TRANSP_LOW0**Type:** RW**Reset State:** 0x00000000

Blend Layer1 Transparency Color Low Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_2_BLEND1_FG TRANSP_LOW0

Bits	Name	Description
27:16	TRANSPARENCY_C1	Transparency color check lower limit for Color1
11:0	TRANSPARENCY_C0	Transparency color check lower limit for Color0

0x01A47060 MDSS_MDP_VP_0_LAYER_2_BLEND1_FG TRANSP_LOW1**Type:** RW**Reset State:** 0x00000000

Blend Layer1 Transparency Color Low Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_2_BLEND1_FG TRANSP_LOW1

Bits	Name	Description
11:0	TRANSPARENCY_C2	Transparency color check lower limit for Color2

0x01A47064 MDSS_MDP_VP_0_LAYER_2_BLEND1_FG TRANSP_HIGH0**Type:** RW**Reset State:** 0x00000000

Blend Layer1 Transparency Color High Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_2_BLEND1_FG TRANSP_HIGH0

Bits	Name	Description
27:16	TRANSPARENCY_C1	Transparency color check higher limit for Color1
11:0	TRANSPARENCY_C0	Transparency color check higher limit for Color0

0x01A47068 MDSS_MDP_VP_0_LAYER_2_BLEND1_FG TRANSP_HIGH1**Type:** RW**Reset State:** 0x00000000

Blend Layer1 Transparency Color High Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_2_BLEND1_FG TRANSP_HIGH1

Bits	Name	Description
11:0	TRANSPARENCY_C2	Transparency color check higher limit for Color2

0x01A4706C MDSS_MDP_VP_0_LAYER_2_BLEND1_BG TRANSP_LOW0**Type:** RW**Reset State:** 0x00000000

Blend Layer1 Transparency Color Low Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_2_BLEND1_BG TRANSP_LOW0

Bits	Name	Description
27:16	TRANSPARENCY_C1	Transparency color check lower limit for Color1
11:0	TRANSPARENCY_C0	Transparency color check lower limit for Color0

0x01A47070 MDSS_MDP_VP_0_LAYER_2_BLEND1_BG TRANSP_LOW1**Type:** RW**Reset State:** 0x00000000

Blend Layer1 Transparency Color Low Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_2_BLEND1_BG TRANSP_LOW1

Bits	Name	Description
11:0	TRANSPARENCY_C2	Transparency color check lower limit for Color2

0x01A47074 MDSS_MDP_VP_0_LAYER_2_BLEND1_BG TRANSP_HIGH0**Type:** RW**Reset State:** 0x00000000

Blend Layer1 Transparency Color High Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_2_BLEND1_BG TRANSP_HIGH0

Bits	Name	Description
27:16	TRANSPARENCY_C1	Transparency color check higher limit for Color1
11:0	TRANSPARENCY_C0	Transparency color check higher limit for Color0

0x01A47078 MDSS_MDP_VP_0_LAYER_2_BLEND1_BG TRANSP_HIGH1**Type:** RW**Reset State:** 0x00000000

Blend Layer1 Transparency Color High Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_2_BLEND1_BG TRANSP_HIGH1

Bits	Name	Description
11:0	TRANSPARENCY_C2	Transparency color check higher limit for Color2

0x01A47080 MDSS_MDP_VP_0_LAYER_2_BLEND2_OP**Type:** RW**Reset State:** 0x00000000

LayerMixer Blend2 Operation Control

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_LAYER_2_BLEND2_OP

Bits	Name	Description
13	BG TRANSP_EN	BG transparency enabled
12	BG_INV_MOD_ALPHA	Ability to select between alpha and inverse alpha right after the alpha modulation 0x0: Alpha 0x1: Inverse Alpha
11	BG_MOD_ALPHA	Modulate BG Alpha
10	BG_INV_ALPHA_SEL	Ability to select between alpha and inverse alpha right after the alpha select 0x0: Alpha 0x1: Inverse Alpha
9:8	BG_ALPHA_SEL	Blending Alpha Select 0x0: FG constant alpha 0x1: BG constant alpha 0x2: FG per pixel alpha 0x3: BG per pixel alpha
5	FG TRANSP_EN	FG transparency enabled
4	FG_INV_MOD_ALPHA	Ability to select between alpha and inverse alpha right after the alpha modulation 0x0: Alpha 0x1: Inverse Alpha
3	FG_MOD_ALPHA	Modulate Alpha
2	FG_INV_ALPHA_SEL	Ability to select between alpha and inverse alpha right after the alpha select 0x0: Alpha 0x1: Inverse Alpha
1:0	FG_ALPHA_SEL	Blending Alpha Select 0x0: FG constant alpha 0x1: BG constant alpha 0x2: FG per pixel alpha 0x3: BG per pixel alpha

0x01A47084 MDSS_MDP_VP_0_LAYER_2_BLEND2_FG_ALPHA**Type:** RW**Reset State:** 0x00000000

Blend Layer2 FG alpha

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_LAYER_2_BLEND2_FG_ALPHA

Bits	Name	Description
7:0	CONSTANT_ALPHA	Constant Alpha

0x01A47088 MDSS_MDP_VP_0_LAYER_2_BLEND2_BG_ALPHA**Type:** RW**Reset State:** 0x00000000

Blend Layer2 FG alpha

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_LAYER_2_BLEND2_BG_ALPHA

Bits	Name	Description
7:0	CONSTANT_ALPHA	Constant Alpha

0x01A4708C MDSS_MDP_VP_0_LAYER_2_BLEND2_FG TRANSP_LOW0**Type:** RW**Reset State:** 0x00000000

Blend Layer2 Transparency Color Low Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_2_BLEND2_FG TRANSP_LOW0

Bits	Name	Description
27:16	TRANSPARENCY_C1	Transparency color check lower limit for Color1
11:0	TRANSPARENCY_C0	Transparency color check lower limit for Color0

0x01A47090 MDSS_MDP_VP_0_LAYER_2_BLEND2_FG TRANSP_LOW1**Type:** RW**Reset State:** 0x00000000

Blend Layer2 Transparency Color Low Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_2_BLEND2_FG TRANSP_LOW1

Bits	Name	Description
11:0	TRANSPARENCY_C2	Transparency color check lower limit for Color2

0x01A47094 MDSS_MDP_VP_0_LAYER_2_BLEND2_FG TRANSP_HIGH0**Type:** RW**Reset State:** 0x00000000

Blend Layer2 Transparency Color High Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_2_BLEND2_FG TRANSP_HIGH0

Bits	Name	Description
27:16	TRANSPARENCY_C1	Transparency color check higher limit for Color1
11:0	TRANSPARENCY_C0	Transparency color check higher limit for Color0

0x01A47098 MDSS_MDP_VP_0_LAYER_2_BLEND2_FG TRANSP_HIGH1**Type:** RW**Reset State:** 0x00000000

Blend Layer2 Transparency Color High Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_2_BLEND2_FG TRANSP_HIGH1

Bits	Name	Description
11:0	TRANSPARENCY_C2	Transparency color check higher limit for Color2

0x01A4709C MDSS_MDP_VP_0_LAYER_2_BLEND2_BG TRANSP_LOW0**Type:** RW**Reset State:** 0x00000000

Blend Layer2 Transparency Color Low Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_2_BLEND2_BG TRANSP_LOW0

Bits	Name	Description
27:16	TRANSPARENCY_C1	Transparency color check lower limit for Color1
11:0	TRANSPARENCY_C0	Transparency color check lower limit for Color0

0x01A470A0 MDSS_MDP_VP_0_LAYER_2_BLEND2_BG TRANSP_LOW1**Type:** RW**Reset State:** 0x00000000

Blend Layer2 Transparency Color Low Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_2_BLEND2_BG TRANSP_LOW1

Bits	Name	Description
11:0	TRANSPARENCY_C2	Transparency color check lower limit for Color2

0x01A470A4 MDSS_MDP_VP_0_LAYER_2_BLEND2_BG TRANSP_HIGH0**Type:** RW**Reset State:** 0x00000000

Blend Layer2 Transparency Color High Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_2_BLEND2_BG TRANSP_HIGH0

Bits	Name	Description
27:16	TRANSPARENCY_C1	Transparency color check higher limit for Color1
11:0	TRANSPARENCY_C0	Transparency color check higher limit for Color0

0x01A470A8 MDSS_MDP_VP_0_LAYER_2_BLEND2_BG TRANSP_HIGH1**Type:** RW**Reset State:** 0x00000000

Blend Layer2 Transparency Color High Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_2_BLEND2_BG_TRANSP_HIGH1

Bits	Name	Description
11:0	TRANSPARENCY_C2	Transparency color check higher limit for Color2

0x01A470B0 MDSS_MDP_VP_0_LAYER_2_BLEND3_OP**Type:** RW**Reset State:** 0x00000000

LayerMixer Blend3 Operation Control

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_LAYER_2_BLEND3_OP

Bits	Name	Description
13	BG_TRANSP_EN	BG transparency enabled
12	BG_INV_MOD_ALPHA	Ability to select between alpha and inverse alpha right after the alpha modulation 0x0: Alpha 0x1: Inverse Alpha
11	BG_MOD_ALPHA	Modulate BG Alpha
10	BG_INV_ALPHA_SEL	Ability to select between alpha and inverse alpha right after the alpha select 0x0: Alpha 0x1: Inverse Alpha
9:8	BG_ALPHA_SEL	Blending Alpha Select 0x0: FG constant alpha 0x1: BG constant alpha 0x2: FG per pixel alpha 0x3: BG per pixel alpha
5	FG_TRANSP_EN	FG transparency enabled
4	FG_INV_MOD_ALPHA	Ability to select between alpha and inverse alpha right after the alpha modulation 0x0: Alpha 0x1: Inverse Alpha
3	FG_MOD_ALPHA	Modulate Alpha
2	FG_INV_ALPHA_SEL	Ability to select between alpha and inverse alpha right after the alpha select 0x0: Alpha 0x1: Inverse Alpha

MDSS_MDP_VP_0_LAYER_2_BLEND3_OP (cont.)

Bits	Name	Description
1:0	FG_ALPHA_SEL	Blending Alpha Select 0x0: FG constant alpha 0x1: BG constant alpha 0x2: FG per pixel alpha 0x3: BG per pixel alpha

0x01A470B4 MDSS_MDP_VP_0_LAYER_2_BLEND3_FG_ALPHA**Type:** RW**Reset State:** 0x00000000

Blend Layer3 FG alpha

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_LAYER_2_BLEND3_FG_ALPHA

Bits	Name	Description
7:0	CONSTANT_ALPHA	Constant Alpha

0x01A470B8 MDSS_MDP_VP_0_LAYER_2_BLEND3_BG_ALPHA**Type:** RW**Reset State:** 0x00000000

Blend Layer3 FG alpha

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_LAYER_2_BLEND3_BG_ALPHA

Bits	Name	Description
7:0	CONSTANT_ALPHA	Constant Alpha

0x01A470BC MDSS_MDP_VP_0_LAYER_2_BLEND3_FG TRANSP_LOW0**Type:** RW**Reset State:** 0x00000000

Blend Layer3 Transparency Color Low Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_2_BLEND3_FG TRANSP_LOW0

Bits	Name	Description
27:16	TRANSPARENCY_C1	Transparency color check lower limit for Color1
11:0	TRANSPARENCY_C0	Transparency color check lower limit for Color0

0x01A470C0 MDSS_MDP_VP_0_LAYER_2_BLEND3_FG TRANSP_LOW1

Type: RW

Reset State: 0x00000000

Blend Layer3 Transparency Color Low Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_2_BLEND3_FG TRANSP_LOW1

Bits	Name	Description
11:0	TRANSPARENCY_C2	Transparency color check lower limit for Color2

0x01A470C4 MDSS_MDP_VP_0_LAYER_2_BLEND3_FG TRANSP_HIGH0

Type: RW

Reset State: 0x00000000

Blend Layer3 Transparency Color High Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_2_BLEND3_FG TRANSP_HIGH0

Bits	Name	Description
27:16	TRANSPARENCY_C1	Transparency color check higher limit for Color1
11:0	TRANSPARENCY_C0	Transparency color check higher limit for Color0

0x01A470C8 MDSS_MDP_VP_0_LAYER_2_BLEND3_FG TRANSP_HIGH1

Type: RW

Reset State: 0x00000000

Blend Layer3 Transparency Color High Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_2_BLEND3_FG TRANSP_HIGH1

Bits	Name	Description
11:0	TRANSPARENCY_C2	Transparency color check higher limit for Color2

0x01A470CC MDSS_MDP_VP_0_LAYER_2_BLEND3_BG TRANSP_LOW0

Type: RW

Reset State: 0x00000000

Blend Layer3 Transparency Color Low Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_2_BLEND3_BG TRANSP_LOW0

Bits	Name	Description
27:16	TRANSPARENCY_C1	Transparency color check lower limit for Color1
11:0	TRANSPARENCY_C0	Transparency color check lower limit for Color0

0x01A470D0 MDSS_MDP_VP_0_LAYER_2_BLEND3_BG TRANSP_LOW1

Type: RW

Reset State: 0x00000000

Blend Layer3 Transparency Color Low Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_2_BLEND3_BG TRANSP_LOW1

Bits	Name	Description
11:0	TRANSPARENCY_C2	Transparency color check lower limit for Color2

0x01A470D4 MDSS_MDP_VP_0_LAYER_2_BLEND3_BG TRANSP_HIGH0

Type: RW

Reset State: 0x00000000

Blend Layer3 Transparency Color High Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_2_BLEND3_BG TRANSP_HIGH0

Bits	Name	Description
27:16	TRANSPARENCY_C1	Transparency color check higher limit for Color1
11:0	TRANSPARENCY_C0	Transparency color check higher limit for Color0

0x01A470D8 MDSS_MDP_VP_0_LAYER_2_BLEND3_BG TRANSP_HIGH1

Type: RW

Reset State: 0x00000000

Blend Layer3 Transparency Color High Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_2_BLEND3_BG TRANSP_HIGH1

Bits	Name	Description
11:0	TRANSPARENCY_C2	Transparency color check higher limit for Color2

0x01A470DC MDSS_MDP_VP_0_LAYER_2_CURSOR_STRIDE

Type: RW

Reset State: 0x00000000

Cursor Stride

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_LAYER_2_CURSOR_STRIDE

Bits	Name	Description
15:0	STRIDE	Cursor Stride

0x01A470E0 MDSS_MDP_VP_0_LAYER_2_CURSOR_IMG_SIZE

Type: RW

Reset State: 0x00000000

Cursor Image Size

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_LAYER_2_CURSOR_IMG_SIZE

Bits	Name	Description
31:16	SRC_H	Total Cursor Source Height
15:0	SRC_W	Total Cursor Source Width

0x01A470E4 MDSS_MDP_VP_0_LAYER_2_CURSOR_SIZE

Type: RW

Reset State: 0x00000000

Cursor ROI Image Size

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_LAYER_2_CURSOR_SIZE

Bits	Name	Description
31:16	ROI_H	Cursor ROI Source Height
15:0	ROI_W	Cursor ROI Source Width

0x01A470E8 MDSS_MDP_VP_0_LAYER_2_CURSOR_XY

Type: RW

Reset State: 0x00000000

Cursor ROI Offset

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_LAYER_2_CURSOR_XY

Bits	Name	Description
31:16	SRC_Y	Y ROI coordinate
15:0	SRC_X	X ROI coordinate

0x01A470EC MDSS_MDP_VP_0_LAYER_2_CURSOR_FORMAT**Type:** RW**Reset State:** 0x00000000

Cursor Format

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_LAYER_2_CURSOR_FORMAT

Bits	Name	Description
2:0	FORMAT	Cursor Format 0x0: ARGB8888 0x2: ARGB1555 0x4: ARGB4444

0x01A470F0 MDSS_MDP_VP_0_LAYER_2_CURSOR_BASE_ADDR**Type:** RW**Reset State:** 0x00000000

Cursor Base Address

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_LAYER_2_CURSOR_BASE_ADDR

Bits	Name	Description
31:0	ADDR	Cursor Base-Address (byte-aligned)

0x01A470F4 MDSS_MDP_VP_0_LAYER_2_CURSOR_START_XY**Type:** RW**Reset State:** 0x00000000

Cursor Starting Coordinates

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_LAYER_2_CURSOR_START_XY

Bits	Name	Description
31:16	Y_START	Starting Y coordinate
15:0	X_START	Starting X coordinate

0x01A470F8 MDSS_MDP_VP_0_LAYER_2_CURSOR_BLEND_CONFIG**Type:** RW**Reset State:** 0x00000000

Cursor Blending Configuration

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_LAYER_2_CURSOR_BLEND_CONFIG

Bits	Name	Description
3	BLEND_TRANSP_EN	0x0: Disabled 0x1: Color keying is done on the cursor and blended result
2:1	BLEND_ALPHA_SEL	Blending Alpha Select 0x0: Constant Alpha 0x1: reserved 0x2: Cursor per pixel alpha 0x3: reserved
0	BLEND_EN	Cursor Blend Enable

0x01A470FC MDSS_MDP_VP_0_LAYER_2_CURSOR_BLEND_PARAM**Type:** RW**Reset State:** 0x00000000

Cursor Blending Parameter

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_LAYER_2_CURSOR_BLEND_PARAM

Bits	Name	Description
7:0	CONSTANT_ALPHA	Constant Alpha

0x01A47100 MDSS_MDP_VP_0_LAYER_2_CURSOR_BLEND TRANSP_LOW0**Type:** RW**Reset State:** 0x00000000

Cursor Transparency Color Low Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_2_CURSOR_BLEND TRANSP_LOW0

Bits	Name	Description
27:16	TRANSPARENCY_C1	Transparency color check lower limit for Color1
11:0	TRANSPARENCY_C0	Transparency color check lower limit for Color0

0x01A47104 MDSS_MDP_VP_0_LAYER_2_CURSOR_BLEND TRANSP_LOW1**Type:** RW**Reset State:** 0x00000000

Cursor Transparency Color Low Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_2_CURSOR_BLEND TRANSP_LOW1

Bits	Name	Description
11:0	TRANSPARENCY_C2	Transparency color check lower limit for Color2

0x01A47108 MDSS_MDP_VP_0_LAYER_2_CURSOR_BLEND TRANSP_HIGH0**Type:** RW**Reset State:** 0x00000000

Cursor Transparency Color High Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_2_CURSOR_BLEND TRANSP_HIGH0

Bits	Name	Description
27:16	TRANSPARENCY_C1	Transparency color check higher limit for Color1
11:0	TRANSPARENCY_C0	Transparency color check higher limit for Color0

0x01A4710C MDSS_MDP_VP_0_LAYER_2_CURSOR_BLEND TRANSP_HIGH1**Type:** RW**Reset State:** 0x00000000

Cursor Transparency Color High Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_2_CURSOR_BLEND TRANSP_HIGH1

Bits	Name	Description
11:0	TRANSPARENCY_C2	Transparency color check higher limit for Color2

0x01A47110 MDSS_MDP_VP_0_LAYER_2_GC_ARLUT_RED_STARTX_N**Type:** RW**Reset State:** 0x00000000**MDSS_MDP_VP_0_LAYER_2_GC_ARLUT_RED_STARTX_N**

Bits	Name	Description
21	VALUE_UPDATEN	
20	SEGMENT_UPDATE	
19:16	SEGMENT	
11:0	STARTX	

0x01A47114 MDSS_MDP_VP_0_LAYER_2_GC_ARLUT_RED_SLOPE_N**Type:** RW**Reset State:** 0x00000000**MDSS_MDP_VP_0_LAYER_2_GC_ARLUT_RED_SLOPE_N**

Bits	Name	Description
21	VALUE_UPDATEN	
20	SEGMENT_UPDATE	
19:16	SEGMENT	
14:0	SLOPE	

0x01A47118 MDSS_MDP_VP_0_LAYER_2_GC_ARLUT_RED_OFFSET_N**Type:** RW**Reset State:** 0x00000000**MDSS_MDP_VP_0_LAYER_2_GC_ARLUT_RED_OFFSET_N**

Bits	Name	Description
21	VALUE_UPDATEN	
20	SEGMENT_UPDATE	
19:16	SEGMENT	
14:0	OFFSET	

0x01A47120 MDSS_MDP_VP_0_LAYER_2_GC_ARLUT_GREEN_STARTX_N**Type:** RW**Reset State:** 0x00000000**MDSS_MDP_VP_0_LAYER_2_GC_ARLUT_GREEN_STARTX_N**

Bits	Name	Description
21	VALUE_UPDATEN	
20	SEGMENT_UPDATE	
19:16	SEGMENT	
11:0	STARTX	

0x01A47124 MDSS_MDP_VP_0_LAYER_2_GC_ARLUT_GREEN_SLOPE_N**Type:** RW**Reset State:** 0x00000000**MDSS_MDP_VP_0_LAYER_2_GC_ARLUT_GREEN_SLOPE_N**

Bits	Name	Description
21	VALUE_UPDATEN	
20	SEGMENT_UPDATE	
19:16	SEGMENT	
14:0	SLOPE	

0x01A47128 MDSS_MDP_VP_0_LAYER_2_GC_ARLUT_GREEN_OFFSET_N**Type:** RW**Reset State:** 0x00000000**MDSS_MDP_VP_0_LAYER_2_GC_ARLUT_GREEN_OFFSET_N**

Bits	Name	Description
21	VALUE_UPDATEN	
20	SEGMENT_UPDATE	
19:16	SEGMENT	
14:0	OFFSET	

0x01A47130 MDSS_MDP_VP_0_LAYER_2_GC_ARLUT_BLUE_STARTX_N**Type:** RW**Reset State:** 0x00000000**MDSS_MDP_VP_0_LAYER_2_GC_ARLUT_BLUE_STARTX_N**

Bits	Name	Description
21	VALUE_UPDATEN	
20	SEGMENT_UPDATE	
19:16	SEGMENT	
11:0	STARTX	

0x01A47134 MDSS_MDP_VP_0_LAYER_2_GC_ARLUT_BLUE_SLOPE_N**Type:** RW**Reset State:** 0x00000000**MDSS_MDP_VP_0_LAYER_2_GC_ARLUT_BLUE_SLOPE_N**

Bits	Name	Description
21	VALUE_UPDATEN	
20	SEGMENT_UPDATE	
19:16	SEGMENT	
14:0	SLOPE	

0x01A47138 MDSS_MDP_VP_0_LAYER_2_GC_ARLUT_BLUE_OFFSET_N**Type:** RW**Reset State:** 0x00000000**MDSS_MDP_VP_0_LAYER_2_GC_ARLUT_BLUE_OFFSET_N**

Bits	Name	Description
21	VALUE_UPDATEN	
20	SEGMENT_UPDATE	
19:16	SEGMENT	
14:0	OFFSET	

0x01A47200 MDSS_MDP_VP_0_LAYER_2_CURSOR_BASE_ADDR_SW_STATUS**Type:** RW**Reset State:** 0x00000000

Security Status of the source buffer plane

This is programmed by SW to indicate if the buffer is NON-SECURE or SECURE

This information will be presented to VBIF per transaction and reflected as part of the memory transaction

double buffered: YES

MDSS_MDP_VP_0_LAYER_2_CURSOR_BASE_ADDR_SW_STATUS

Bits	Name	Description
0	ADDR_CP	

0x01A47204 MDSS_MDP_VP_0_LAYER_2_CURSOR_FETCH_CONFIG**Type:** RW**Reset State:** 0x00000087

Cursor Client Fetch Configuration per Virtual Pipeline double buffered: NO

This register must NOT be altered during operation and MDP must be idle when changing values

MDSS_MDP_VP_0_LAYER_2_CURSOR_FETCH_CONFIG

Bits	Name	Description
17:16	TILE_SIZE	Tile Size for processing of BLOCK formats Note that this only applies to blocks that operate in block mode which is limited to DMA source pipe. This setting has no effect for all other source pipes (ViG/RGB) 0x0: 128x128 0x1: 64x64 0x2: 32x32
12	ADDR_SYNC_MODE	Boundary on which fetch address synchronizes. Setting this field to line forces the address updates on the next line which performs the software requested feature of updating on hsync. Note that updating on line boundaries will likely result in tearing on the panel since hardware is fetching from new memory buffer. 0x0: frame (default) 0x1: line
9:8	MAL	Describes the Minimum Access Length for the source pipeline. This value should align with the DRAM's expected transaction alignment unit. This setting forces the source fetch pipe to be aligned to DRAM and implement re-alignment for maximum efficiency 0x0: 128b alignment (default) 0x1: 256b alignment 0x2: 512b alignment
7:4	MAX_BURST_SIZE	Maximum burst beat size. Hardware fetch engines operate on 128-bit bus Possible values: 0x8: burst of 8 0x4: burst of 4 0x2: burst of 2
3:0	REQ_DEPTH_LIMIT	Pending request limit, default max value of 0x7 means maximum pending requests allowed in the design Anything less limits the bus requests to that number + 1.

0x01A47210 MDSS_MDP_VP_0_LAYER_2_CURSOR_FETCH_SMP_WR_PLANE0**Type:** R**Reset State:** 0x00000000

Fetch to SMP on plane 0 write counter

double buffered: N/A

MDSS_MDP_VP_0_LAYER_2_CURSOR_FETCH_SMP_WR_PLANE0

Bits	Name	Description
15:0	COUNT	This counter is reset on every frame and is incremented for each write fetch makes to the SMP plane0 port This should correlate to value SMP_UNPACK_RD_PLANE0 More useful for debug but can be used for profiling purposes as well.

0x01A47214 MDSS_MDP_VP_0_LAYER_2_CURSOR_SMP_UNPACK_RD_PLANE0**Type:** R**Reset State:** 0x00000000

SMP to unpack on plane 0 read counter

double buffered: N/A

MDSS_MDP_VP_0_LAYER_2_CURSOR_SMP_UNPACK_RD_PLANE0

Bits	Name	Description
15:0	COUNT	This counter is reset on every frame and is incremented for each line read from SMP plane0 port This should correlate to FETCH_SMP_WR_PLANE0 value More useful for debug but can be used for profiling purposes as well.

0x01A47220 MDSS_MDP_VP_0_LAYER_2_CURSOR_REQPRIORITY_FIFO_WATERMARK_0**Type:** RW**Reset State:** 0x00000008

FIFO watermark to aid in generation of mdpREQPRIORITY

The watermark here refers to the dedicated cursor fetch memory.

The value programmed here should be aware of the amount of dedicated cursor fetch memory.

So a value of 0 indicates completely FULL (i.e. there are no more entries available in the dedicated memory)

A value of 100 indicates 100 entries in the Shared Memory Pool (each entry being 128-bit entries)

This is the lowestmost FIFO depth and should not be programmed higher than REQPRIORITY_FIFO_WATERMARK_1 or REQPRIORITY_FIFO_WATERMARK_2

An example programming would set the following:

REQPRIORITY_FIFO_WATERMARK_0 : 0x08 (meaning 0x08 entries free to be written into dedicated cursor fetch memory - least room available)

REQPRIORITY_FIFO_WATERMARK_1 : 0x10 (meaning 0x10 entries free to be written into dedicated cursor fetch memory - middle ground room available)

REQPRIORITY_FIFO_WATERMARK_2 : 0x20 (meaning 0x20 entries free to be written into dedicated cursor fetch memory - most room available)

mdpREQPRIORITY = 11 (highest priority request) when above all 3 watermark levels (i.e. lots of space available in dedicated cursor fetch memory)

mdpREQPRIORITY = 10 above FIFO_0 and FIFO_1 but below FIFO_2

mdpREQPRIORITY = 01 above FIFO_0 but below FIFO_1 and FIFO_2

mdpREQPRIORITY = 00 (lowest priority request) when below all 3 watermark levels (i.e. not a lot of space available in dedicated cursor fetch memory so its good)

double buffered: YES

MDSS_MDP_VP_0_LAYER_2_CURSOR_REQPRIORITY_FIFO_WATERMARK_0

Bits	Name	Description
15:0	WATERMARK	Watermark Level for FIFO_0

0x01A47224 MDSS_MDP_VP_0_LAYER_2_CURSOR_REQPRIORITY_FIFO_WATERMARK_1

Type: RW

Reset State: 0x00000010

FIFO watermark to aid in generation of mdpREQPRIORITY

The watermark here refers to the dedicated cursor fetch memory.

The value programmed here should be aware of the amount of dedicated cursor fetch memory.

So a value of 0 indicates completely FULL (i.e. there are no more entries available in the dedicated memory)

A value of 100 indicates 100 entries in the Shared Memory Pool (each entry being 128-bit entries)

This is the lowestmost FIFO depth and should not be programmed higher than REQPRIORITY_FIFO_WATERMARK_1 or REQPRIORITY_FIFO_WATERMARK_2

An example programming would set the following:

REQPRIORITY_FIFO_WATERMARK_0 : 0x08 (meaning 0x08 entries free to be written into dedicated cursor fetch memory - least room available)

REQPRIORITY_FIFO_WATERMARK_1 : 0x10 (meaning 0x10 entries free to be written into dedicated cursor fetch memory - middle ground room available)

REQPRIORITY_FIFO_WATERMARK_2 : 0x20 (meaning 0x20 entries free to be written into dedicated cursor fetch memory - most room available)

mdpREQPRIORITY = 11 (highest priority request) when above all 3 watermark levels (i.e. lots of space available in dedicated cursor fetch memory)

mdpREQPRIORITY = 10 above FIFO_0 and FIFO_1 but below FIFO_2

mdpREQPRIORITY = 01 above FIFO_0 but below FIFO_1 and FIFO_2

mdpREQPRIORITY = 00 (lowest priority request) when below all 3 watermark levels (i.e. not a lot of space available in dedicated cursor fetch memory so its good)

double buffered: YES

MDSS_MDP_VP_0_LAYER_2_CURSOR_REQPRIORITY_FIFO_WATERMARK_1

Bits	Name	Description
15:0	WATERMARK	Watermark Level for FIFO_1

0x01A47228 MDSS_MDP_VP_0_LAYER_2_CURSOR_REQPRIORITY_FIFO_WATERMARK_2

Type: RW

Reset State: 0x00000020

FIFO watermark to aid in generation of mdpREQPRIORITY

The watermark here refers to the dedicated cursor fetch memory.

The value programmed here should be aware of the amount of dedicated cursor fetch memory.

So a value of 0 indicates completely FULL (i.e. there are no more entries available in the dedicated memory)

A value of 100 indicates 100 entries in the Shared Memory Pool (each entry being 128-bit entries)

This is the lowestmost FIFO depth and should not be programmed higher than REQPRIORITY_FIFO_WATERMARK_1 or REQPRIORITY_FIFO_WATERMARK_2

An example programming would set the following:

REQPRIORITY_FIFO_WATERMARK_0 : 0x08 (meaning 0x08 entries free to be written into dedicated cursor fetch memory - least room available)

REQPRIORITY_FIFO_WATERMARK_1 : 0x10 (meaning 0x10 entries free to be written into dedicated cursor fetch memory - middle ground room available)

REQPRIORITY_FIFO_WATERMARK_2 : 0x20 (meaning 0x20 entries free to be written into dedicated cursor fetch memory - most room available)

mdpREQPRIORITY = 11 (highest priority request) when above all 3 watermark levels (i.e. lots of space available in dedicated cursor fetch memory)

mdpREQPRIORITY = 10 above FIFO_0 and FIFO_1 but below FIFO_2

mdpREQPRIORITY = 01 above FIFO_0 but below FIFO_1 and FIFO_2

mdpREQPRIORITY = 00 (lowest priority request) when below all 3 watermark levels (i.e. not a lot of space available in dedicated cursor fetch memory so its good)

double buffered: YES

MDSS_MDP_VP_0_LAYER_2_CURSOR_REQPRIORITY_FIFO_WATERMARK_2

Bits	Name	Description
15:0	WATERMARK	Watermark Level for FIFO_2

0x01A47380 MDSS_MDP_VP_0_LAYER_2_GC_MISR_CTRL

Type: RW

Reset State: 0x00000000

MDSS_MDP_VP_0_LAYER_2_GC_MISR_CTRL

Bits	Name	Type	Description
31	RUN_MODE	RW	
27:24	INPUT_SEL	RW	
10	STATUS_CLEAR	W	
9	STATUS	R	
8	ENABLE	RW	
7:0	FRAME_COUNT	RW	

0x01A47384 MDSS_MDP_VP_0_LAYER_2_GC_MISR_SIGNATURE

Type: R

Reset State: 0x00000000

MDSS_MDP_VP_0_LAYER_2_GC_MISR_SIGNATURE

Bits	Name	Description
31:0	CAPTURED	

17.17.7.12 MDSS_MDP_VP_0_LAYER_3

Layer Mixer hardware register description

17.17.7.12.1 MDSS_MDP_VP_0_LAYER_3_LAYER

LAYER

Layer Mixer hardware register description

0x01A48000 MDSS_MDP_VP_0_LAYER_3_OP_MODE**Type:** RW**Reset State:** 0x00000000

LayerMixer Operation Control

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_LAYER_3_OP_MODE

Bits	Name	Description
4	BLEND3_COLOR3_OUT	Selects which source to drive on color3 component output for stage3 blend 0x0: BG Layer Color3 0x1: FG Layer Color3
3	BLEND2_COLOR3_OUT	Selects which source to drive on color3 component output for stage2 blend 0x0: BG Layer Color3 0x1: FG Layer Color3
2	BLEND1_COLOR3_OUT	Selects which source to drive on color3 component output for stage1 blend 0x0: BG Layer Color3 0x1: FG Layer Color3
1	BLEND0_COLOR3_OUT	Selects which source to drive on color3 component output for stage0 blend 0x0: BG Layer Color3 0x1: FG Layer Color3
0	GC_LUT_EN	0x0: Disabled - mapped linearly from 12-bit to 8-bit 0x1: Enabled

0x01A48004 MDSS_MDP_VP_0_LAYER_3_OUT_SIZE**Type:** RW**Reset State:** 0x00000000

LayerMixer Output Size

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_LAYER_3_OUT_SIZE

Bits	Name	Description
31:16	DST_H	Destination Output Height
15:0	DST_W	Destination Output Width The exact supported size depends on the width of line buffers allocated for MDP.

0x01A48008 MDSS_MDP_VP_0_LAYER_3_BORDER_COLOR_0

Type: RW

Reset State: 0x00000000

LayerMixer Border Color Register

double buffered: YES

MDSS_MDP_VP_0_LAYER_3_BORDER_COLOR_0

Bits	Name	Description
27:16	COLOR1	Color1 in 12-bit format
11:0	COLOR0	Color0 in 12-bit format

0x01A48010 MDSS_MDP_VP_0_LAYER_3_BORDER_COLOR_1

Type: RW

Reset State: 0x00000000

LayerMixer Border Color Register

double buffered: YES

MDSS_MDP_VP_0_LAYER_3_BORDER_COLOR_1

Bits	Name	Description
27:16	COLOR3	Color3 in 12-bit format, LSB4 are ignored, should be programmed to 0
11:0	COLOR2	Color2 in 12-bit format

0x01A48020 MDSS_MDP_VP_0_LAYER_3_BLEND0_OP**Type:** RW**Reset State:** 0x00000000

LayerMixer Blend0 Operation Control

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_LAYER_3_BLEND0_OP

Bits	Name	Description
13	BG TRANSP_EN	BG transparency enabled
12	BG_INV_MOD_ALPHA	Ability to select between alpha and inverse alpha right after the alpha modulation 0x0: Alpha 0x1: Inverse Alpha
11	BG_MOD_ALPHA	Modulate BG Alpha
10	BG_INV_ALPHA_SEL	Ability to select between alpha and inverse alpha right after the alpha select 0x0: Alpha 0x1: Inverse Alpha
9:8	BG_ALPHA_SEL	Blending Alpha Select 0x0: FG constant alpha 0x1: BG constant alpha 0x2: FG per pixel alpha 0x3: BG per pixel alpha
5	FG TRANSP_EN	FG transparency enabled
4	FG_INV_MOD_ALPHA	Ability to select between alpha and inverse alpha right after the alpha modulation 0x0: Alpha 0x1: Inverse Alpha
3	FG_MOD_ALPHA	Modulate Alpha
2	FG_INV_ALPHA_SEL	Ability to select between alpha and inverse alpha right after the alpha select 0x0: Alpha 0x1: Inverse Alpha
1:0	FG_ALPHA_SEL	Blending Alpha Select 0x0: FG constant alpha 0x1: BG constant alpha 0x2: FG per pixel alpha 0x3: BG per pixel alpha

0x01A48024 MDSS_MDP_VP_0_LAYER_3_BLEND0_FG_ALPHA**Type:** RW**Reset State:** 0x00000000

Blend Layer0 FG alpha

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_LAYER_3_BLEND0_FG_ALPHA

Bits	Name	Description
7:0	CONSTANT_ALPHA	Constant Alpha

0x01A48028 MDSS_MDP_VP_0_LAYER_3_BLEND0_BG_ALPHA**Type:** RW**Reset State:** 0x00000000

Blend Layer0 FG alpha

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_LAYER_3_BLEND0_BG_ALPHA

Bits	Name	Description
7:0	CONSTANT_ALPHA	Constant Alpha

0x01A4802C MDSS_MDP_VP_0_LAYER_3_BLEND0_FG TRANSP_LOW0**Type:** RW**Reset State:** 0x00000000

Blend Layer0 Transparency Color Low Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_3_BLEND0_FG TRANSP_LOW0

Bits	Name	Description
27:16	TRANSPARENCY_C1	Transparency color check lower limit for Color1
11:0	TRANSPARENCY_C0	Transparency color check lower limit for Color0

0x01A48030 MDSS_MDP_VP_0_LAYER_3_BLEND0_FG TRANSP_LOW1**Type:** RW**Reset State:** 0x00000000

Blend Layer0 Transparency Color Low Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_3_BLEND0_FG TRANSP_LOW1

Bits	Name	Description
11:0	TRANSPARENCY_C2	Transparency color check lower limit for Color2

0x01A48034 MDSS_MDP_VP_0_LAYER_3_BLEND0_FG TRANSP_HIGH0**Type:** RW**Reset State:** 0x00000000

Blend Layer0 Transparency Color High Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_3_BLEND0_FG TRANSP_HIGH0

Bits	Name	Description
27:16	TRANSPARENCY_C1	Transparency color check higher limit for Color1
11:0	TRANSPARENCY_C0	Transparency color check higher limit for Color0

0x01A48038 MDSS_MDP_VP_0_LAYER_3_BLEND0_FG TRANSP_HIGH1**Type:** RW**Reset State:** 0x00000000

Blend Layer0 Transparency Color High Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_3_BLEND0_FG TRANSP_HIGH1

Bits	Name	Description
11:0	TRANSPARENCY_C2	Transparency color check higher limit for Color2

0x01A4803C MDSS_MDP_VP_0_LAYER_3_BLEND0_BG TRANSP_LOW0**Type:** RW**Reset State:** 0x00000000

Blend Layer0 Transparency Color Low Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_3_BLEND0_BG TRANSP_LOW0

Bits	Name	Description
27:16	TRANSPARENCY_C1	Transparency color check lower limit for Color1
11:0	TRANSPARENCY_C0	Transparency color check lower limit for Color0

0x01A48040 MDSS_MDP_VP_0_LAYER_3_BLEND0_BG TRANSP_LOW1**Type:** RW**Reset State:** 0x00000000

Blend Layer0 Transparency Color Low Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_3_BLEND0_BG TRANSP_LOW1

Bits	Name	Description
11:0	TRANSPARENCY_C2	Transparency color check lower limit for Color2

0x01A48044 MDSS_MDP_VP_0_LAYER_3_BLEND0_BG TRANSP_HIGH0**Type:** RW**Reset State:** 0x00000000

Blend Layer0 Transparency Color High Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_3_BLEND0_BG TRANSP_HIGH0

Bits	Name	Description
27:16	TRANSPARENCY_C1	Transparency color check higher limit for Color1
11:0	TRANSPARENCY_C0	Transparency color check higher limit for Color0

0x01A48048 MDSS_MDP_VP_0_LAYER_3_BLEND0_BG TRANSP_HIGH1**Type:** RW**Reset State:** 0x00000000

Blend Layer0 Transparency Color High Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_3_BLEND0_BG TRANSP_HIGH1

Bits	Name	Description
11:0	TRANSPARENCY_C2	Transparency color check higher limit for Color2

0x01A48050 MDSS_MDP_VP_0_LAYER_3_BLEND1_OP**Type:** RW**Reset State:** 0x00000000

LayerMixer Blend1 Operation Control

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_LAYER_3_BLEND1_OP

Bits	Name	Description
13	BG_TRANSP_EN	BG transparency enabled
12	BG_INV_MOD_ALPHA	Ability to select between alpha and inverse alpha right after the alpha modulation 0x0: Alpha 0x1: Inverse Alpha
11	BG_MOD_ALPHA	Modulate BG Alpha
10	BG_INV_ALPHA_SEL	Ability to select between alpha and inverse alpha right after the alpha select 0x0: Alpha 0x1: Inverse Alpha
9:8	BG_ALPHA_SEL	Blending Alpha Select 0x0: FG constant alpha 0x1: BG constant alpha 0x2: FG per pixel alpha 0x3: BG per pixel alpha
5	FG_TRANSP_EN	FG transparency enabled

MDSS_MDP_VP_0_LAYER_3_BLEND1_OP (cont.)

Bits	Name	Description
4	FG_INV_MOD_ALPHA	Ability to select between alpha and inverse alpha right after the alpha modulation 0x0: Alpha 0x1: Inverse Alpha
3	FG_MOD_ALPHA	Modulate Alpha
2	FG_INV_ALPHA_SEL	Ability to select between alpha and inverse alpha right after the alpha select 0x0: Alpha 0x1: Inverse Alpha
1:0	FG_ALPHA_SEL	Blending Alpha Select 0x0: FG constant alpha 0x1: BG constant alpha 0x2: FG per pixel alpha 0x3: BG per pixel alpha

0x01A48054 MDSS_MDP_VP_0_LAYER_3_BLEND1_FG_ALPHA**Type:** RW**Reset State:** 0x00000000

Blend Layer1 FG alpha

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_LAYER_3_BLEND1_FG_ALPHA

Bits	Name	Description
7:0	CONSTANT_ALPHA	Constant Alpha

0x01A48058 MDSS_MDP_VP_0_LAYER_3_BLEND1_BG_ALPHA**Type:** RW**Reset State:** 0x00000000

Blend Layer1 FG alpha

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_LAYER_3_BLEND1_BG_ALPHA

Bits	Name	Description
7:0	CONSTANT_ALPHA	Constant Alpha

0x01A4805C MDSS_MDP_VP_0_LAYER_3_BLEND1_FG TRANSP_LOW0**Type:** RW**Reset State:** 0x00000000

Blend Layer1 Transparency Color Low Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_3_BLEND1_FG TRANSP_LOW0

Bits	Name	Description
27:16	TRANSPARENCY_C1	Transparency color check lower limit for Color1
11:0	TRANSPARENCY_C0	Transparency color check lower limit for Color0

0x01A48060 MDSS_MDP_VP_0_LAYER_3_BLEND1_FG TRANSP_LOW1**Type:** RW**Reset State:** 0x00000000

Blend Layer1 Transparency Color Low Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_3_BLEND1_FG TRANSP_LOW1

Bits	Name	Description
11:0	TRANSPARENCY_C2	Transparency color check lower limit for Color2

0x01A48064 MDSS_MDP_VP_0_LAYER_3_BLEND1_FG TRANSP_HIGH0**Type:** RW**Reset State:** 0x00000000

Blend Layer1 Transparency Color High Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_3_BLEND1_FG TRANSP_HIGH0

Bits	Name	Description
27:16	TRANSPARENCY_C1	Transparency color check higher limit for Color1
11:0	TRANSPARENCY_C0	Transparency color check higher limit for Color0

0x01A48068 MDSS_MDP_VP_0_LAYER_3_BLEND1_FG TRANSP_HIGH1**Type:** RW**Reset State:** 0x00000000

Blend Layer1 Transparency Color High Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_3_BLEND1_FG TRANSP_HIGH1

Bits	Name	Description
11:0	TRANSPARENCY_C2	Transparency color check higher limit for Color2

0x01A4806C MDSS_MDP_VP_0_LAYER_3_BLEND1_BG TRANSP_LOW0**Type:** RW**Reset State:** 0x00000000

Blend Layer1 Transparency Color Low Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_3_BLEND1_BG TRANSP_LOW0

Bits	Name	Description
27:16	TRANSPARENCY_C1	Transparency color check lower limit for Color1
11:0	TRANSPARENCY_C0	Transparency color check lower limit for Color0

0x01A48070 MDSS_MDP_VP_0_LAYER_3_BLEND1_BG TRANSP_LOW1**Type:** RW**Reset State:** 0x00000000

Blend Layer1 Transparency Color Low Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_3_BLEND1_BG TRANSP_LOW1

Bits	Name	Description
11:0	TRANSPARENCY_C2	Transparency color check lower limit for Color2

0x01A48074 MDSS_MDP_VP_0_LAYER_3_BLEND1_BG TRANSP_HIGH0**Type:** RW**Reset State:** 0x00000000

Blend Layer1 Transparency Color High Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_3_BLEND1_BG TRANSP_HIGH0

Bits	Name	Description
27:16	TRANSPARENCY_C1	Transparency color check higher limit for Color1
11:0	TRANSPARENCY_C0	Transparency color check higher limit for Color0

0x01A48078 MDSS_MDP_VP_0_LAYER_3_BLEND1_BG TRANSP_HIGH1**Type:** RW**Reset State:** 0x00000000

Blend Layer1 Transparency Color High Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_3_BLEND1_BG TRANSP_HIGH1

Bits	Name	Description
11:0	TRANSPARENCY_C2	Transparency color check higher limit for Color2

0x01A48080 MDSS_MDP_VP_0_LAYER_3_BLEND2_OP**Type:** RW**Reset State:** 0x00000000

LayerMixer Blend2 Operation Control

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_LAYER_3_BLEND2_OP

Bits	Name	Description
13	BG TRANSP_EN	BG transparency enabled
12	BG_INV_MOD_ALPHA	Ability to select between alpha and inverse alpha right after the alpha modulation 0x0: Alpha 0x1: Inverse Alpha
11	BG_MOD_ALPHA	Modulate BG Alpha
10	BG_INV_ALPHA_SEL	Ability to select between alpha and inverse alpha right after the alpha select 0x0: Alpha 0x1: Inverse Alpha
9:8	BG_ALPHA_SEL	Blending Alpha Select 0x0: FG constant alpha 0x1: BG constant alpha 0x2: FG per pixel alpha 0x3: BG per pixel alpha
5	FG TRANSP_EN	FG transparency enabled
4	FG_INV_MOD_ALPHA	Ability to select between alpha and inverse alpha right after the alpha modulation 0x0: Alpha 0x1: Inverse Alpha
3	FG_MOD_ALPHA	Modulate Alpha
2	FG_INV_ALPHA_SEL	Ability to select between alpha and inverse alpha right after the alpha select 0x0: Alpha 0x1: Inverse Alpha
1:0	FG_ALPHA_SEL	Blending Alpha Select 0x0: FG constant alpha 0x1: BG constant alpha 0x2: FG per pixel alpha 0x3: BG per pixel alpha

0x01A48084 MDSS_MDP_VP_0_LAYER_3_BLEND2_FG_ALPHA**Type:** RW**Reset State:** 0x00000000

Blend Layer2 FG alpha

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_LAYER_3_BLEND2_FG_ALPHA

Bits	Name	Description
7:0	CONSTANT_ALPHA	Constant Alpha

0x01A48088 MDSS_MDP_VP_0_LAYER_3_BLEND2_BG_ALPHA**Type:** RW**Reset State:** 0x00000000

Blend Layer2 FG alpha

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_LAYER_3_BLEND2_BG_ALPHA

Bits	Name	Description
7:0	CONSTANT_ALPHA	Constant Alpha

0x01A4808C MDSS_MDP_VP_0_LAYER_3_BLEND2_FG TRANSP_LOW0**Type:** RW**Reset State:** 0x00000000

Blend Layer2 Transparency Color Low Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_3_BLEND2_FG TRANSP_LOW0

Bits	Name	Description
27:16	TRANSPARENCY_C1	Transparency color check lower limit for Color1
11:0	TRANSPARENCY_C0	Transparency color check lower limit for Color0

0x01A48090 MDSS_MDP_VP_0_LAYER_3_BLEND2_FG TRANSP_LOW1**Type:** RW**Reset State:** 0x00000000

Blend Layer2 Transparency Color Low Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_3_BLEND2_FG TRANSP_LOW1

Bits	Name	Description
11:0	TRANSPARENCY_C2	Transparency color check lower limit for Color2

0x01A48094 MDSS_MDP_VP_0_LAYER_3_BLEND2_FG TRANSP_HIGH0**Type:** RW**Reset State:** 0x00000000

Blend Layer2 Transparency Color High Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_3_BLEND2_FG TRANSP_HIGH0

Bits	Name	Description
27:16	TRANSPARENCY_C1	Transparency color check higher limit for Color1
11:0	TRANSPARENCY_C0	Transparency color check higher limit for Color0

0x01A48098 MDSS_MDP_VP_0_LAYER_3_BLEND2_FG TRANSP_HIGH1**Type:** RW**Reset State:** 0x00000000

Blend Layer2 Transparency Color High Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_3_BLEND2_FG TRANSP_HIGH1

Bits	Name	Description
11:0	TRANSPARENCY_C2	Transparency color check higher limit for Color2

0x01A4809C MDSS_MDP_VP_0_LAYER_3_BLEND2_BG TRANSP_LOW0**Type:** RW**Reset State:** 0x00000000

Blend Layer2 Transparency Color Low Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_3_BLEND2_BG TRANSP_LOW0

Bits	Name	Description
27:16	TRANSPARENCY_C1	Transparency color check lower limit for Color1
11:0	TRANSPARENCY_C0	Transparency color check lower limit for Color0

0x01A480A0 MDSS_MDP_VP_0_LAYER_3_BLEND2_BG TRANSP_LOW1**Type:** RW**Reset State:** 0x00000000

Blend Layer2 Transparency Color Low Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_3_BLEND2_BG TRANSP_LOW1

Bits	Name	Description
11:0	TRANSPARENCY_C2	Transparency color check lower limit for Color2

0x01A480A4 MDSS_MDP_VP_0_LAYER_3_BLEND2_BG TRANSP_HIGH0**Type:** RW**Reset State:** 0x00000000

Blend Layer2 Transparency Color High Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_3_BLEND2_BG TRANSP_HIGH0

Bits	Name	Description
27:16	TRANSPARENCY_C1	Transparency color check higher limit for Color1
11:0	TRANSPARENCY_C0	Transparency color check higher limit for Color0

0x01A480A8 MDSS_MDP_VP_0_LAYER_3_BLEND2_BG TRANSP_HIGH1**Type:** RW**Reset State:** 0x00000000

Blend Layer2 Transparency Color High Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_3_BLEND2_BG TRANSP_HIGH1

Bits	Name	Description
11:0	TRANSPARENCY_C2	Transparency color check higher limit for Color2

0x01A480B0 MDSS_MDP_VP_0_LAYER_3_BLEND3_OP**Type:** RW**Reset State:** 0x00000000

LayerMixer Blend3 Operation Control

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_LAYER_3_BLEND3_OP

Bits	Name	Description
13	BG TRANSP_EN	BG transparency enabled
12	BG_INV_MOD_ALPHA	Ability to select between alpha and inverse alpha right after the alpha modulation 0x0: Alpha 0x1: Inverse Alpha
11	BG_MOD_ALPHA	Modulate BG Alpha
10	BG_INV_ALPHA_SEL	Ability to select between alpha and inverse alpha right after the alpha select 0x0: Alpha 0x1: Inverse Alpha
9:8	BG_ALPHA_SEL	Blending Alpha Select 0x0: FG constant alpha 0x1: BG constant alpha 0x2: FG per pixel alpha 0x3: BG per pixel alpha
5	FG TRANSP_EN	FG transparency enabled
4	FG_INV_MOD_ALPHA	Ability to select between alpha and inverse alpha right after the alpha modulation 0x0: Alpha 0x1: Inverse Alpha
3	FG_MOD_ALPHA	Modulate Alpha
2	FG_INV_ALPHA_SEL	Ability to select between alpha and inverse alpha right after the alpha select 0x0: Alpha 0x1: Inverse Alpha

MDSS_MDP_VP_0_LAYER_3_BLEND3_OP (cont.)

Bits	Name	Description
1:0	FG_ALPHA_SEL	Blending Alpha Select 0x0: FG constant alpha 0x1: BG constant alpha 0x2: FG per pixel alpha 0x3: BG per pixel alpha

0x01A480B4 MDSS_MDP_VP_0_LAYER_3_BLEND3_FG_ALPHA**Type:** RW**Reset State:** 0x00000000

Blend Layer3 FG alpha

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_LAYER_3_BLEND3_FG_ALPHA

Bits	Name	Description
7:0	CONSTANT_ALPHA	Constant Alpha

0x01A480B8 MDSS_MDP_VP_0_LAYER_3_BLEND3_BG_ALPHA**Type:** RW**Reset State:** 0x00000000

Blend Layer3 FG alpha

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_LAYER_3_BLEND3_BG_ALPHA

Bits	Name	Description
7:0	CONSTANT_ALPHA	Constant Alpha

0x01A480BC MDSS_MDP_VP_0_LAYER_3_BLEND3_FG TRANSP_LOW0**Type:** RW**Reset State:** 0x00000000

Blend Layer3 Transparency Color Low Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_3_BLEND3_FG TRANSP_LOW0

Bits	Name	Description
27:16	TRANSPARENCY_C1	Transparency color check lower limit for Color1
11:0	TRANSPARENCY_C0	Transparency color check lower limit for Color0

0x01A480C0 MDSS_MDP_VP_0_LAYER_3_BLEND3_FG TRANSP_LOW1

Type: RW

Reset State: 0x00000000

Blend Layer3 Transparency Color Low Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_3_BLEND3_FG TRANSP_LOW1

Bits	Name	Description
11:0	TRANSPARENCY_C2	Transparency color check lower limit for Color2

0x01A480C4 MDSS_MDP_VP_0_LAYER_3_BLEND3_FG TRANSP_HIGH0

Type: RW

Reset State: 0x00000000

Blend Layer3 Transparency Color High Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_3_BLEND3_FG TRANSP_HIGH0

Bits	Name	Description
27:16	TRANSPARENCY_C1	Transparency color check higher limit for Color1
11:0	TRANSPARENCY_C0	Transparency color check higher limit for Color0

0x01A480C8 MDSS_MDP_VP_0_LAYER_3_BLEND3_FG TRANSP_HIGH1

Type: RW

Reset State: 0x00000000

Blend Layer3 Transparency Color High Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_3_BLEND3_FG TRANSP_HIGH1

Bits	Name	Description
11:0	TRANSPARENCY_C2	Transparency color check higher limit for Color2

0x01A480CC MDSS_MDP_VP_0_LAYER_3_BLEND3_BG TRANSP_LOW0

Type: RW

Reset State: 0x00000000

Blend Layer3 Transparency Color Low Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_3_BLEND3_BG TRANSP_LOW0

Bits	Name	Description
27:16	TRANSPARENCY_C1	Transparency color check lower limit for Color1
11:0	TRANSPARENCY_C0	Transparency color check lower limit for Color0

0x01A480D0 MDSS_MDP_VP_0_LAYER_3_BLEND3_BG TRANSP_LOW1

Type: RW

Reset State: 0x00000000

Blend Layer3 Transparency Color Low Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_3_BLEND3_BG TRANSP_LOW1

Bits	Name	Description
11:0	TRANSPARENCY_C2	Transparency color check lower limit for Color2

0x01A480D4 MDSS_MDP_VP_0_LAYER_3_BLEND3_BG TRANSP_HIGH0

Type: RW

Reset State: 0x00000000

Blend Layer3 Transparency Color High Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_3_BLEND3_BG TRANSP_HIGH0

Bits	Name	Description
27:16	TRANSPARENCY_C1	Transparency color check higher limit for Color1
11:0	TRANSPARENCY_C0	Transparency color check higher limit for Color0

0x01A480D8 MDSS_MDP_VP_0_LAYER_3_BLEND3_BG TRANSP_HIGH1

Type: RW

Reset State: 0x00000000

Blend Layer3 Transparency Color High Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_3_BLEND3_BG TRANSP_HIGH1

Bits	Name	Description
11:0	TRANSPARENCY_C2	Transparency color check higher limit for Color2

0x01A480DC MDSS_MDP_VP_0_LAYER_3_CURSOR_STRIDE

Type: RW

Reset State: 0x00000000

Cursor Stride

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_LAYER_3_CURSOR_STRIDE

Bits	Name	Description
15:0	STRIDE	Cursor Stride

0x01A480E0 MDSS_MDP_VP_0_LAYER_3_CURSOR_IMG_SIZE

Type: RW

Reset State: 0x00000000

Cursor Image Size

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_LAYER_3_CURSOR_IMG_SIZE

Bits	Name	Description
31:16	SRC_H	Total Cursor Source Height
15:0	SRC_W	Total Cursor Source Width

0x01A480E4 MDSS_MDP_VP_0_LAYER_3_CURSOR_SIZE

Type: RW

Reset State: 0x00000000

Cursor ROI Image Size

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_LAYER_3_CURSOR_SIZE

Bits	Name	Description
31:16	ROI_H	Cursor ROI Source Height
15:0	ROI_W	Cursor ROI Source Width

0x01A480E8 MDSS_MDP_VP_0_LAYER_3_CURSOR_XY

Type: RW

Reset State: 0x00000000

Cursor ROI Offset

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_LAYER_3_CURSOR_XY

Bits	Name	Description
31:16	SRC_Y	Y ROI coordinate
15:0	SRC_X	X ROI coordinate

0x01A480EC MDSS_MDP_VP_0_LAYER_3_CURSOR_FORMAT**Type:** RW**Reset State:** 0x00000000

Cursor Format

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_LAYER_3_CURSOR_FORMAT

Bits	Name	Description
2:0	FORMAT	Cursor Format 0x0: ARGB8888 0x2: ARGB1555 0x4: ARGB4444

0x01A480F0 MDSS_MDP_VP_0_LAYER_3_CURSOR_BASE_ADDR**Type:** RW**Reset State:** 0x00000000

Cursor Base Address

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_LAYER_3_CURSOR_BASE_ADDR

Bits	Name	Description
31:0	ADDR	Cursor Base-Address (byte-aligned)

0x01A480F4 MDSS_MDP_VP_0_LAYER_3_CURSOR_START_XY**Type:** RW**Reset State:** 0x00000000

Cursor Starting Coordinates

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_LAYER_3_CURSOR_START_XY

Bits	Name	Description
31:16	Y_START	Starting Y coordinate
15:0	X_START	Starting X coordinate

0x01A480F8 MDSS_MDP_VP_0_LAYER_3_CURSOR_BLEND_CONFIG**Type:** RW**Reset State:** 0x00000000

Cursor Blending Configuration

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_LAYER_3_CURSOR_BLEND_CONFIG

Bits	Name	Description
3	BLEND_TRANSP_EN	0x0: Disabled 0x1: Color keying is done on the cursor and blended result
2:1	BLEND_ALPHA_SEL	Blending Alpha Select 0x0: Constant Alpha 0x1: reserved 0x2: Cursor per pixel alpha 0x3: reserved
0	BLEND_EN	Cursor Blend Enable

0x01A480FC MDSS_MDP_VP_0_LAYER_3_CURSOR_BLEND_PARAM**Type:** RW**Reset State:** 0x00000000

Cursor Blending Parameter

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_LAYER_3_CURSOR_BLEND_PARAM

Bits	Name	Description
7:0	CONSTANT_ALPHA	Constant Alpha

0x01A48100 MDSS_MDP_VP_0_LAYER_3_CURSOR_BLEND TRANSP_LOW0**Type:** RW**Reset State:** 0x00000000

Cursor Transparency Color Low Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_3_CURSOR_BLEND TRANSP_LOW0

Bits	Name	Description
27:16	TRANSPARENCY_C1	Transparency color check lower limit for Color1
11:0	TRANSPARENCY_C0	Transparency color check lower limit for Color0

0x01A48104 MDSS_MDP_VP_0_LAYER_3_CURSOR_BLEND TRANSP_LOW1**Type:** RW**Reset State:** 0x00000000

Cursor Transparency Color Low Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_3_CURSOR_BLEND TRANSP_LOW1

Bits	Name	Description
11:0	TRANSPARENCY_C2	Transparency color check lower limit for Color2

0x01A48108 MDSS_MDP_VP_0_LAYER_3_CURSOR_BLEND TRANSP_HIGH0**Type:** RW**Reset State:** 0x00000000

Cursor Transparency Color High Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_3_CURSOR_BLEND TRANSP_HIGH0

Bits	Name	Description
27:16	TRANSPARENCY_C1	Transparency color check higher limit for Color1
11:0	TRANSPARENCY_C0	Transparency color check higher limit for Color0

0x01A4810C MDSS_MDP_VP_0_LAYER_3_CURSOR_BLEND TRANSP_HIGH1**Type:** RW**Reset State:** 0x00000000

Cursor Transparency Color High Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_3_CURSOR_BLEND TRANSP_HIGH1

Bits	Name	Description
11:0	TRANSPARENCY_C2	Transparency color check higher limit for Color2

0x01A48110 MDSS_MDP_VP_0_LAYER_3_GC_ARLUT_RED_STARTX_N**Type:** RW**Reset State:** 0x00000000**MDSS_MDP_VP_0_LAYER_3_GC_ARLUT_RED_STARTX_N**

Bits	Name	Description
21	VALUE_UPDATEN	
20	SEGMENT_UPDATE	
19:16	SEGMENT	
11:0	STARTX	

0x01A48114 MDSS_MDP_VP_0_LAYER_3_GC_ARLUT_RED_SLOPE_N**Type:** RW**Reset State:** 0x00000000**MDSS_MDP_VP_0_LAYER_3_GC_ARLUT_RED_SLOPE_N**

Bits	Name	Description
21	VALUE_UPDATEN	
20	SEGMENT_UPDATE	
19:16	SEGMENT	
14:0	SLOPE	

0x01A48118 MDSS_MDP_VP_0_LAYER_3_GC_ARLUT_RED_OFFSET_N**Type:** RW**Reset State:** 0x00000000**MDSS_MDP_VP_0_LAYER_3_GC_ARLUT_RED_OFFSET_N**

Bits	Name	Description
21	VALUE_UPDATEN	
20	SEGMENT_UPDATE	
19:16	SEGMENT	
14:0	OFFSET	

0x01A48120 MDSS_MDP_VP_0_LAYER_3_GC_ARLUT_GREEN_STARTX_N**Type:** RW**Reset State:** 0x00000000**MDSS_MDP_VP_0_LAYER_3_GC_ARLUT_GREEN_STARTX_N**

Bits	Name	Description
21	VALUE_UPDATEN	
20	SEGMENT_UPDATE	
19:16	SEGMENT	
11:0	STARTX	

0x01A48124 MDSS_MDP_VP_0_LAYER_3_GC_ARLUT_GREEN_SLOPE_N**Type:** RW**Reset State:** 0x00000000**MDSS_MDP_VP_0_LAYER_3_GC_ARLUT_GREEN_SLOPE_N**

Bits	Name	Description
21	VALUE_UPDATEN	
20	SEGMENT_UPDATE	
19:16	SEGMENT	
14:0	SLOPE	

0x01A48128 MDSS_MDP_VP_0_LAYER_3_GC_ARLUT_GREEN_OFFSET_N**Type:** RW**Reset State:** 0x00000000**MDSS_MDP_VP_0_LAYER_3_GC_ARLUT_GREEN_OFFSET_N**

Bits	Name	Description
21	VALUE_UPDATEN	
20	SEGMENT_UPDATE	
19:16	SEGMENT	
14:0	OFFSET	

0x01A48130 MDSS_MDP_VP_0_LAYER_3_GC_ARLUT_BLUE_STARTX_N**Type:** RW**Reset State:** 0x00000000**MDSS_MDP_VP_0_LAYER_3_GC_ARLUT_BLUE_STARTX_N**

Bits	Name	Description
21	VALUE_UPDATEN	
20	SEGMENT_UPDATE	
19:16	SEGMENT	
11:0	STARTX	

0x01A48134 MDSS_MDP_VP_0_LAYER_3_GC_ARLUT_BLUE_SLOPE_N**Type:** RW**Reset State:** 0x00000000**MDSS_MDP_VP_0_LAYER_3_GC_ARLUT_BLUE_SLOPE_N**

Bits	Name	Description
21	VALUE_UPDATEN	
20	SEGMENT_UPDATE	
19:16	SEGMENT	
14:0	SLOPE	

0x01A48138 MDSS_MDP_VP_0_LAYER_3_GC_ARLUT_BLUE_OFFSET_N**Type:** RW**Reset State:** 0x00000000**MDSS_MDP_VP_0_LAYER_3_GC_ARLUT_BLUE_OFFSET_N**

Bits	Name	Description
21	VALUE_UPDATEN	
20	SEGMENT_UPDATE	
19:16	SEGMENT	
14:0	OFFSET	

0x01A48200 MDSS_MDP_VP_0_LAYER_3_CURSOR_BASE_ADDR_SW_STATUS**Type:** RW**Reset State:** 0x00000000

Security Status of the source buffer plane

This is programmed by SW to indicate if the buffer is NON-SECURE or SECURE

This information will be presented to VBIF per transaction and reflected as part of the memory transaction

double buffered: YES

MDSS_MDP_VP_0_LAYER_3_CURSOR_BASE_ADDR_SW_STATUS

Bits	Name	Description
0	ADDR_CP	

0x01A48204 MDSS_MDP_VP_0_LAYER_3_CURSOR_FETCH_CONFIG**Type:** RW**Reset State:** 0x00000087

Cursor Client Fetch Configuration per Virtual Pipeline double buffered: NO

This register must NOT be altered during operation and MDP must be idle when changing values

MDSS_MDP_VP_0_LAYER_3_CURSOR_FETCH_CONFIG

Bits	Name	Description
17:16	TILE_SIZE	Tile Size for processing of BLOCK formats Note that this only applies to blocks that operate in block mode which is limited to DMA source pipe. This setting has no effect for all other source pipes (ViG/RGB) 0x0: 128x128 0x1: 64x64 0x2: 32x32
12	ADDR_SYNC_MODE	Boundary on which fetch address synchronizes. Setting this field to line forces the address updates on the next line which performs the software requested feature of updating on hsync. Note that updating on line boundaries will likely result in tearing on the panel since hardware is fetching from new memory buffer. 0x0: frame (default) 0x1: line
9:8	MAL	Describes the Minimum Access Length for the source pipeline. This value should align with the DRAM's expected transaction alignment unit. This setting forces the source fetch pipe to be aligned to DRAM and implement re-alignment for maximum efficiency 0x0: 128b alignment (default) 0x1: 256b alignment 0x2: 512b alignment
7:4	MAX_BURST_SIZE	Maximum burst beat size. Hardware fetch engines operate on 128-bit bus Possible values: 0x8: burst of 8 0x4: burst of 4 0x2: burst of 2
3:0	REQ_DEPTH_LIMIT	Pending request limit, default max value of 0x7 means maximum pending requests allowed in the design Anything less limits the bus requests to that number + 1.

0x01A48210 MDSS_MDP_VP_0_LAYER_3_CURSOR_FETCH_SMP_WR_PLANE0**Type:** R**Reset State:** 0x00000000

Fetch to SMP on plane 0 write counter

double buffered: N/A

MDSS_MDP_VP_0_LAYER_3_CURSOR_FETCH_SMP_WR_PLANE0

Bits	Name	Description
15:0	COUNT	This counter is reset on every frame and is incremented for each write fetch makes to the SMP plane0 port This should correlate to value SMP_UNPACK_RD_PLANE0 More useful for debug but can be used for profiling purposes as well.

0x01A48214 MDSS_MDP_VP_0_LAYER_3_CURSOR_SMP_UNPACK_RD_PLANE0**Type:** R**Reset State:** 0x00000000

SMP to unpack on plane 0 read counter

double buffered: N/A

MDSS_MDP_VP_0_LAYER_3_CURSOR_SMP_UNPACK_RD_PLANE0

Bits	Name	Description
15:0	COUNT	This counter is reset on every frame and is incremented for each line read from SMP plane0 port This should correlate to FETCH_SMP_WR_PLANE0 value More useful for debug but can be used for profiling purposes as well.

0x01A48220 MDSS_MDP_VP_0_LAYER_3_CURSOR_REQPRIORITY_FIFO_WATERMARK_0**Type:** RW**Reset State:** 0x00000008

FIFO watermark to aid in generation of mdpREQPRIORITY

The watermark here refers to the dedicated cursor fetch memory.

The value programmed here should be aware of the amount of dedicated cursor fetch memory.

So a value of 0 indicates completely FULL (i.e. there are no more entries available in the dedicated memory)

A value of 100 indicates 100 entries in the Shared Memory Pool (each entry being 128-bit entries)

This is the lowestmost FIFO depth and should not be programmed higher than REQPRIORITY_FIFO_WATERMARK_1 or REQPRIORITY_FIFO_WATERMARK_2

An example programming would set the following:

REQPRIORITY_FIFO_WATERMARK_0 : 0x08 (meaning 0x08 entries free to be written into dedicated cursor fetch memory - least room available)

REQPRIORITY_FIFO_WATERMARK_1 : 0x10 (meaning 0x10 entries free to be written into dedicated cursor fetch memory - middle ground room available)

REQPRIORITY_FIFO_WATERMARK_2 : 0x20 (meaning 0x20 entries free to be written into dedicated cursor fetch memory - most room available)

mdpREQPRIORITY = 11 (highest priority request) when above all 3 watermark levels (i.e. lots of space available in dedicated cursor fetch memory)

mdpREQPRIORITY = 10 above FIFO_0 and FIFO_1 but below FIFO_2

mdpREQPRIORITY = 01 above FIFO_0 but below FIFO_1 and FIFO_2

mdpREQPRIORITY = 00 (lowest priority request) when below all 3 watermark levels (i.e. not a lot of space available in dedicated cursor fetch memory so its good)

double buffered: YES

MDSS_MDP_VP_0_LAYER_3_CURSOR_REQPRIORITY_FIFO_WATERMARK_0

Bits	Name	Description
15:0	WATERMARK	Watermark Level for FIFO_0

0x01A48224 MDSS_MDP_VP_0_LAYER_3_CURSOR_REQPRIORITY_FIFO_WATERMARK_1

Type: RW

Reset State: 0x00000010

FIFO watermark to aid in generation of mdpREQPRIORITY

The watermark here refers to the dedicated cursor fetch memory.

The value programmed here should be aware of the amount of dedicated cursor fetch memory.

So a value of 0 indicates completely FULL (i.e. there are no more entries available in the dedicated memory)

A value of 100 indicates 100 entries in the Shared Memory Pool (each entry being 128-bit entries)

This is the lowestmost FIFO depth and should not be programmed higher than REQPRIORITY_FIFO_WATERMARK_1 or REQPRIORITY_FIFO_WATERMARK_2

An example programming would set the following:

REQPRIORITY_FIFO_WATERMARK_0 : 0x08 (meaning 0x08 entries free to be written into dedicated cursor fetch memory - least room available)

REQPRIORITY_FIFO_WATERMARK_1 : 0x10 (meaning 0x10 entries free to be written into dedicated cursor fetch memory - middle ground room available)

REQPRIORITY_FIFO_WATERMARK_2 : 0x20 (meaning 0x20 entries free to be written into dedicated cursor fetch memory - most room available)

mdpREQPRIORITY = 11 (highest priority request) when above all 3 watermark levels (i.e. lots of space available in dedicated cursor fetch memory)

mdpREQPRIORITY = 10 above FIFO_0 and FIFO_1 but below FIFO_2

mdpREQPRIORITY = 01 above FIFO_0 but below FIFO_1 and FIFO_2

mdpREQPRIORITY = 00 (lowest priority request) when below all 3 watermark levels (i.e. not a lot of space available in dedicated cursor fetch memory so its good)

double buffered: YES

MDSS_MDP_VP_0_LAYER_3_CURSOR_REQPRIORITY_FIFO_WATERMARK_1

Bits	Name	Description
15:0	WATERMARK	Watermark Level for FIFO_1

0x01A48228 MDSS_MDP_VP_0_LAYER_3_CURSOR_REQPRIORITY_FIFO_WATERMARK_2

Type: RW

Reset State: 0x00000020

FIFO watermark to aid in generation of mdpREQPRIORITY

The watermark here refers to the dedicated cursor fetch memory.

The value programmed here should be aware of the amount of dedicated cursor fetch memory.

So a value of 0 indicates completely FULL (i.e. there are no more entries available in the dedicated memory)

A value of 100 indicates 100 entries in the Shared Memory Pool (each entry being 128-bit entries)

This is the lowestmost FIFO depth and should not be programmed higher than REQPRIORITY_FIFO_WATERMARK_1 or REQPRIORITY_FIFO_WATERMARK_2

An example programming would set the following:

REQPRIORITY_FIFO_WATERMARK_0 : 0x08 (meaning 0x08 entries free to be written into dedicated cursor fetch memory - least room available)

REQPRIORITY_FIFO_WATERMARK_1 : 0x10 (meaning 0x10 entries free to be written into dedicated cursor fetch memory - middle ground room available)

REQPRIORITY_FIFO_WATERMARK_2 : 0x20 (meaning 0x20 entries free to be written into dedicated cursor fetch memory - most room available)

mdpREQPRIORITY = 11 (highest priority request) when above all 3 watermark levels (i.e. lots of space available in dedicated cursor fetch memory)

mdpREQPRIORITY = 10 above FIFO_0 and FIFO_1 but below FIFO_2

mdpREQPRIORITY = 01 above FIFO_0 but below FIFO_1 and FIFO_2

mdpREQPRIORITY = 00 (lowest priority request) when below all 3 watermark levels (i.e. not a lot of space available in dedicated cursor fetch memory so its good)

double buffered: YES

MDSS_MDP_VP_0_LAYER_3_CURSOR_REQPRIORITY_FIFO_WATERMARK_2

Bits	Name	Description
15:0	WATERMARK	Watermark Level for FIFO_2

0x01A48380 MDSS_MDP_VP_0_LAYER_3_GC_MISR_CTRL

Type: RW

Reset State: 0x00000000

MDSS_MDP_VP_0_LAYER_3_GC_MISR_CTRL

Bits	Name	Type	Description
31	RUN_MODE	RW	
27:24	INPUT_SEL	RW	
10	STATUS_CLEAR	W	
9	STATUS	R	
8	ENABLE	RW	
7:0	FRAME_COUNT	RW	

0x01A48384 MDSS_MDP_VP_0_LAYER_3_GC_MISR_SIGNATURE

Type: R

Reset State: 0x00000000

MDSS_MDP_VP_0_LAYER_3_GC_MISR_SIGNATURE

Bits	Name	Description
31:0	CAPTURED	

17.17.7.13 MDSS_MDP_VP_0_LAYER_4

Layer Mixer hardware register description

17.17.7.13.1 MDSS_MDP_VP_0_LAYER_4_LAYER

LAYER

Layer Mixer hardware register description

0x01A49000 MDSS_MDP_VP_0_LAYER_4_OP_MODE**Type:** RW**Reset State:** 0x00000000

LayerMixer Operation Control

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_LAYER_4_OP_MODE

Bits	Name	Description
4	BLEND3_COLOR3_OUT	Selects which source to drive on color3 component output for stage3 blend 0x0: BG Layer Color3 0x1: FG Layer Color3
3	BLEND2_COLOR3_OUT	Selects which source to drive on color3 component output for stage2 blend 0x0: BG Layer Color3 0x1: FG Layer Color3
2	BLEND1_COLOR3_OUT	Selects which source to drive on color3 component output for stage1 blend 0x0: BG Layer Color3 0x1: FG Layer Color3
1	BLEND0_COLOR3_OUT	Selects which source to drive on color3 component output for stage0 blend 0x0: BG Layer Color3 0x1: FG Layer Color3
0	GC_LUT_EN	0x0: Disabled - mapped linearly from 12-bit to 8-bit 0x1: Enabled

0x01A49004 MDSS_MDP_VP_0_LAYER_4_OUT_SIZE**Type:** RW**Reset State:** 0x00000000

LayerMixer Output Size

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_LAYER_4_OUT_SIZE

Bits	Name	Description
31:16	DST_H	Destination Output Height
15:0	DST_W	Destination Output Width The exact supported size depends on the width of line buffers allocated for MDP.

0x01A49008 MDSS_MDP_VP_0_LAYER_4_BORDER_COLOR_0

Type: RW

Reset State: 0x00000000

LayerMixer Border Color Register

double buffered: YES

MDSS_MDP_VP_0_LAYER_4_BORDER_COLOR_0

Bits	Name	Description
27:16	COLOR1	Color1 in 12-bit format
11:0	COLOR0	Color0 in 12-bit format

0x01A49010 MDSS_MDP_VP_0_LAYER_4_BORDER_COLOR_1

Type: RW

Reset State: 0x00000000

LayerMixer Border Color Register

double buffered: YES

MDSS_MDP_VP_0_LAYER_4_BORDER_COLOR_1

Bits	Name	Description
27:16	COLOR3	Color3 in 12-bit format, LSB4 are ignored, should be programmed to 0
11:0	COLOR2	Color2 in 12-bit format

0x01A49020 MDSS_MDP_VP_0_LAYER_4_BLEND0_OP**Type:** RW**Reset State:** 0x00000000

LayerMixer Blend0 Operation Control

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_LAYER_4_BLEND0_OP

Bits	Name	Description
13	BG TRANSP_EN	BG transparency enabled
12	BG_INV_MOD_ALPHA	Ability to select between alpha and inverse alpha right after the alpha modulation 0x0: Alpha 0x1: Inverse Alpha
11	BG_MOD_ALPHA	Modulate BG Alpha
10	BG_INV_ALPHA_SEL	Ability to select between alpha and inverse alpha right after the alpha select 0x0: Alpha 0x1: Inverse Alpha
9:8	BG_ALPHA_SEL	Blending Alpha Select 0x0: FG constant alpha 0x1: BG constant alpha 0x2: FG per pixel alpha 0x3: BG per pixel alpha
5	FG TRANSP_EN	FG transparency enabled
4	FG_INV_MOD_ALPHA	Ability to select between alpha and inverse alpha right after the alpha modulation 0x0: Alpha 0x1: Inverse Alpha
3	FG_MOD_ALPHA	Modulate Alpha
2	FG_INV_ALPHA_SEL	Ability to select between alpha and inverse alpha right after the alpha select 0x0: Alpha 0x1: Inverse Alpha
1:0	FG_ALPHA_SEL	Blending Alpha Select 0x0: FG constant alpha 0x1: BG constant alpha 0x2: FG per pixel alpha 0x3: BG per pixel alpha

0x01A49024 MDSS_MDP_VP_0_LAYER_4_BLEND0_FG_ALPHA**Type:** RW**Reset State:** 0x00000000

Blend Layer0 FG alpha

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_LAYER_4_BLEND0_FG_ALPHA

Bits	Name	Description
7:0	CONSTANT_ALPHA	Constant Alpha

0x01A49028 MDSS_MDP_VP_0_LAYER_4_BLEND0_BG_ALPHA**Type:** RW**Reset State:** 0x00000000

Blend Layer0 FG alpha

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_LAYER_4_BLEND0_BG_ALPHA

Bits	Name	Description
7:0	CONSTANT_ALPHA	Constant Alpha

0x01A4902C MDSS_MDP_VP_0_LAYER_4_BLEND0_FG TRANSP_LOW0**Type:** RW**Reset State:** 0x00000000

Blend Layer0 Transparency Color Low Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_4_BLEND0_FG TRANSP_LOW0

Bits	Name	Description
27:16	TRANSPARENCY_C1	Transparency color check lower limit for Color1
11:0	TRANSPARENCY_C0	Transparency color check lower limit for Color0

0x01A49030 MDSS_MDP_VP_0_LAYER_4_BLEND0_FG TRANSP_LOW1**Type:** RW**Reset State:** 0x00000000

Blend Layer0 Transparency Color Low Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_4_BLEND0_FG TRANSP_LOW1

Bits	Name	Description
11:0	TRANSPARENCY_C2	Transparency color check lower limit for Color2

0x01A49034 MDSS_MDP_VP_0_LAYER_4_BLEND0_FG TRANSP_HIGH0**Type:** RW**Reset State:** 0x00000000

Blend Layer0 Transparency Color High Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_4_BLEND0_FG TRANSP_HIGH0

Bits	Name	Description
27:16	TRANSPARENCY_C1	Transparency color check higher limit for Color1
11:0	TRANSPARENCY_C0	Transparency color check higher limit for Color0

0x01A49038 MDSS_MDP_VP_0_LAYER_4_BLEND0_FG TRANSP_HIGH1**Type:** RW**Reset State:** 0x00000000

Blend Layer0 Transparency Color High Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_4_BLEND0_FG TRANSP_HIGH1

Bits	Name	Description
11:0	TRANSPARENCY_C2	Transparency color check higher limit for Color2

0x01A4903C MDSS_MDP_VP_0_LAYER_4_BLEND0_BG TRANSP_LOW0**Type:** RW**Reset State:** 0x00000000

Blend Layer0 Transparency Color Low Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_4_BLEND0_BG TRANSP_LOW0

Bits	Name	Description
27:16	TRANSPARENCY_C1	Transparency color check lower limit for Color1
11:0	TRANSPARENCY_C0	Transparency color check lower limit for Color0

0x01A49040 MDSS_MDP_VP_0_LAYER_4_BLEND0_BG TRANSP_LOW1**Type:** RW**Reset State:** 0x00000000

Blend Layer0 Transparency Color Low Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_4_BLEND0_BG TRANSP_LOW1

Bits	Name	Description
11:0	TRANSPARENCY_C2	Transparency color check lower limit for Color2

0x01A49044 MDSS_MDP_VP_0_LAYER_4_BLEND0_BG TRANSP_HIGH0**Type:** RW**Reset State:** 0x00000000

Blend Layer0 Transparency Color High Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_4_BLEND0_BG TRANSP_HIGH0

Bits	Name	Description
27:16	TRANSPARENCY_C1	Transparency color check higher limit for Color1
11:0	TRANSPARENCY_C0	Transparency color check higher limit for Color0

0x01A49048 MDSS_MDP_VP_0_LAYER_4_BLEND0_BG_TRANSP_HIGH1**Type:** RW**Reset State:** 0x00000000

Blend Layer0 Transparency Color High Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_4_BLEND0_BG_TRANSP_HIGH1

Bits	Name	Description
11:0	TRANSPARENCY_C2	Transparency color check higher limit for Color2

0x01A49050 MDSS_MDP_VP_0_LAYER_4_BLEND1_OP**Type:** RW**Reset State:** 0x00000000

LayerMixer Blend1 Operation Control

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_LAYER_4_BLEND1_OP

Bits	Name	Description
13	BG_TRANSP_EN	BG transparency enabled
12	BG_INV_MOD_ALPHA	Ability to select between alpha and inverse alpha right after the alpha modulation 0x0: Alpha 0x1: Inverse Alpha
11	BG_MOD_ALPHA	Modulate BG Alpha
10	BG_INV_ALPHA_SEL	Ability to select between alpha and inverse alpha right after the alpha select 0x0: Alpha 0x1: Inverse Alpha
9:8	BG_ALPHA_SEL	Blending Alpha Select 0x0: FG constant alpha 0x1: BG constant alpha 0x2: FG per pixel alpha 0x3: BG per pixel alpha
5	FG_TRANSP_EN	FG transparency enabled

MDSS_MDP_VP_0_LAYER_4_BLEND1_OP (cont.)

Bits	Name	Description
4	FG_INV_MOD_ALPHA	Ability to select between alpha and inverse alpha right after the alpha modulation 0x0: Alpha 0x1: Inverse Alpha
3	FG_MOD_ALPHA	Modulate Alpha
2	FG_INV_ALPHA_SEL	Ability to select between alpha and inverse alpha right after the alpha select 0x0: Alpha 0x1: Inverse Alpha
1:0	FG_ALPHA_SEL	Blending Alpha Select 0x0: FG constant alpha 0x1: BG constant alpha 0x2: FG per pixel alpha 0x3: BG per pixel alpha

0x01A49054 MDSS_MDP_VP_0_LAYER_4_BLEND1_FG_ALPHA**Type:** RW**Reset State:** 0x00000000

Blend Layer1 FG alpha

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_LAYER_4_BLEND1_FG_ALPHA

Bits	Name	Description
7:0	CONSTANT_ALPHA	Constant Alpha

0x01A49058 MDSS_MDP_VP_0_LAYER_4_BLEND1_BG_ALPHA**Type:** RW**Reset State:** 0x00000000

Blend Layer1 FG alpha

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_LAYER_4_BLEND1_BG_ALPHA

Bits	Name	Description
7:0	CONSTANT_ALPHA	Constant Alpha

0x01A4905C MDSS_MDP_VP_0_LAYER_4_BLEND1_FG TRANSP_LOW0**Type:** RW**Reset State:** 0x00000000

Blend Layer1 Transparency Color Low Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_4_BLEND1_FG TRANSP_LOW0

Bits	Name	Description
27:16	TRANSPARENCY_C1	Transparency color check lower limit for Color1
11:0	TRANSPARENCY_C0	Transparency color check lower limit for Color0

0x01A49060 MDSS_MDP_VP_0_LAYER_4_BLEND1_FG TRANSP_LOW1**Type:** RW**Reset State:** 0x00000000

Blend Layer1 Transparency Color Low Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_4_BLEND1_FG TRANSP_LOW1

Bits	Name	Description
11:0	TRANSPARENCY_C2	Transparency color check lower limit for Color2

0x01A49064 MDSS_MDP_VP_0_LAYER_4_BLEND1_FG TRANSP_HIGH0**Type:** RW**Reset State:** 0x00000000

Blend Layer1 Transparency Color High Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_4_BLEND1_FG TRANSP_HIGH0

Bits	Name	Description
27:16	TRANSPARENCY_C1	Transparency color check higher limit for Color1
11:0	TRANSPARENCY_C0	Transparency color check higher limit for Color0

0x01A49068 MDSS_MDP_VP_0_LAYER_4_BLEND1_FG TRANSP_HIGH1**Type:** RW**Reset State:** 0x00000000

Blend Layer1 Transparency Color High Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_4_BLEND1_FG TRANSP_HIGH1

Bits	Name	Description
11:0	TRANSPARENCY_C2	Transparency color check higher limit for Color2

0x01A4906C MDSS_MDP_VP_0_LAYER_4_BLEND1_BG TRANSP_LOW0**Type:** RW**Reset State:** 0x00000000

Blend Layer1 Transparency Color Low Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_4_BLEND1_BG TRANSP_LOW0

Bits	Name	Description
27:16	TRANSPARENCY_C1	Transparency color check lower limit for Color1
11:0	TRANSPARENCY_C0	Transparency color check lower limit for Color0

0x01A49070 MDSS_MDP_VP_0_LAYER_4_BLEND1_BG TRANSP_LOW1**Type:** RW**Reset State:** 0x00000000

Blend Layer1 Transparency Color Low Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_4_BLEND1_BG TRANSP_LOW1

Bits	Name	Description
11:0	TRANSPARENCY_C2	Transparency color check lower limit for Color2

0x01A49074 MDSS_MDP_VP_0_LAYER_4_BLEND1_BG TRANSP_HIGH0**Type:** RW**Reset State:** 0x00000000

Blend Layer1 Transparency Color High Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_4_BLEND1_BG TRANSP_HIGH0

Bits	Name	Description
27:16	TRANSPARENCY_C1	Transparency color check higher limit for Color1
11:0	TRANSPARENCY_C0	Transparency color check higher limit for Color0

0x01A49078 MDSS_MDP_VP_0_LAYER_4_BLEND1_BG TRANSP_HIGH1**Type:** RW**Reset State:** 0x00000000

Blend Layer1 Transparency Color High Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_4_BLEND1_BG TRANSP_HIGH1

Bits	Name	Description
11:0	TRANSPARENCY_C2	Transparency color check higher limit for Color2

0x01A49080 MDSS_MDP_VP_0_LAYER_4_BLEND2_OP**Type:** RW**Reset State:** 0x00000000

LayerMixer Blend2 Operation Control

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_LAYER_4_BLEND2_OP

Bits	Name	Description
13	BG TRANSP_EN	BG transparency enabled
12	BG_INV_MOD_ALPHA	Ability to select between alpha and inverse alpha right after the alpha modulation 0x0: Alpha 0x1: Inverse Alpha
11	BG_MOD_ALPHA	Modulate BG Alpha
10	BG_INV_ALPHA_SEL	Ability to select between alpha and inverse alpha right after the alpha select 0x0: Alpha 0x1: Inverse Alpha
9:8	BG_ALPHA_SEL	Blending Alpha Select 0x0: FG constant alpha 0x1: BG constant alpha 0x2: FG per pixel alpha 0x3: BG per pixel alpha
5	FG TRANSP_EN	FG transparency enabled
4	FG_INV_MOD_ALPHA	Ability to select between alpha and inverse alpha right after the alpha modulation 0x0: Alpha 0x1: Inverse Alpha
3	FG_MOD_ALPHA	Modulate Alpha
2	FG_INV_ALPHA_SEL	Ability to select between alpha and inverse alpha right after the alpha select 0x0: Alpha 0x1: Inverse Alpha
1:0	FG_ALPHA_SEL	Blending Alpha Select 0x0: FG constant alpha 0x1: BG constant alpha 0x2: FG per pixel alpha 0x3: BG per pixel alpha

0x01A49084 MDSS_MDP_VP_0_LAYER_4_BLEND2_FG_ALPHA**Type:** RW**Reset State:** 0x00000000

Blend Layer2 FG alpha

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_LAYER_4_BLEND2_FG_ALPHA

Bits	Name	Description
7:0	CONSTANT_ALPHA	Constant Alpha

0x01A49088 MDSS_MDP_VP_0_LAYER_4_BLEND2_BG_ALPHA**Type:** RW**Reset State:** 0x00000000

Blend Layer2 FG alpha

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_LAYER_4_BLEND2_BG_ALPHA

Bits	Name	Description
7:0	CONSTANT_ALPHA	Constant Alpha

0x01A4908C MDSS_MDP_VP_0_LAYER_4_BLEND2_FG TRANSP_LOW0**Type:** RW**Reset State:** 0x00000000

Blend Layer2 Transparency Color Low Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_4_BLEND2_FG TRANSP_LOW0

Bits	Name	Description
27:16	TRANSPARENCY_C1	Transparency color check lower limit for Color1
11:0	TRANSPARENCY_C0	Transparency color check lower limit for Color0

0x01A49090 MDSS_MDP_VP_0_LAYER_4_BLEND2_FG TRANSP_LOW1**Type:** RW**Reset State:** 0x00000000

Blend Layer2 Transparency Color Low Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_4_BLEND2_FG TRANSP_LOW1

Bits	Name	Description
11:0	TRANSPARENCY_C2	Transparency color check lower limit for Color2

0x01A49094 MDSS_MDP_VP_0_LAYER_4_BLEND2_FG TRANSP_HIGH0**Type:** RW**Reset State:** 0x00000000

Blend Layer2 Transparency Color High Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_4_BLEND2_FG TRANSP_HIGH0

Bits	Name	Description
27:16	TRANSPARENCY_C1	Transparency color check higher limit for Color1
11:0	TRANSPARENCY_C0	Transparency color check higher limit for Color0

0x01A49098 MDSS_MDP_VP_0_LAYER_4_BLEND2_FG TRANSP_HIGH1**Type:** RW**Reset State:** 0x00000000

Blend Layer2 Transparency Color High Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_4_BLEND2_FG TRANSP_HIGH1

Bits	Name	Description
11:0	TRANSPARENCY_C2	Transparency color check higher limit for Color2

0x01A4909C MDSS_MDP_VP_0_LAYER_4_BLEND2_BG TRANSP_LOW0**Type:** RW**Reset State:** 0x00000000

Blend Layer2 Transparency Color Low Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_4_BLEND2_BG TRANSP_LOW0

Bits	Name	Description
27:16	TRANSPARENCY_C1	Transparency color check lower limit for Color1
11:0	TRANSPARENCY_C0	Transparency color check lower limit for Color0

0x01A490A0 MDSS_MDP_VP_0_LAYER_4_BLEND2_BG TRANSP_LOW1**Type:** RW**Reset State:** 0x00000000

Blend Layer2 Transparency Color Low Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_4_BLEND2_BG TRANSP_LOW1

Bits	Name	Description
11:0	TRANSPARENCY_C2	Transparency color check lower limit for Color2

0x01A490A4 MDSS_MDP_VP_0_LAYER_4_BLEND2_BG TRANSP_HIGH0**Type:** RW**Reset State:** 0x00000000

Blend Layer2 Transparency Color High Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_4_BLEND2_BG TRANSP_HIGH0

Bits	Name	Description
27:16	TRANSPARENCY_C1	Transparency color check higher limit for Color1
11:0	TRANSPARENCY_C0	Transparency color check higher limit for Color0

0x01A490A8 MDSS_MDP_VP_0_LAYER_4_BLEND2_BG TRANSP_HIGH1**Type:** RW**Reset State:** 0x00000000

Blend Layer2 Transparency Color High Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_4_BLEND2_BG_TRANSP_HIGH1

Bits	Name	Description
11:0	TRANSPARENCY_C2	Transparency color check higher limit for Color2

0x01A490B0 MDSS_MDP_VP_0_LAYER_4_BLEND3_OP**Type:** RW**Reset State:** 0x00000000

LayerMixer Blend3 Operation Control

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_LAYER_4_BLEND3_OP

Bits	Name	Description
13	BG_TRANSP_EN	BG transparency enabled
12	BG_INV_MOD_ALPHA	Ability to select between alpha and inverse alpha right after the alpha modulation 0x0: Alpha 0x1: Inverse Alpha
11	BG_MOD_ALPHA	Modulate BG Alpha
10	BG_INV_ALPHA_SEL	Ability to select between alpha and inverse alpha right after the alpha select 0x0: Alpha 0x1: Inverse Alpha
9:8	BG_ALPHA_SEL	Blending Alpha Select 0x0: FG constant alpha 0x1: BG constant alpha 0x2: FG per pixel alpha 0x3: BG per pixel alpha
5	FG_TRANSP_EN	FG transparency enabled
4	FG_INV_MOD_ALPHA	Ability to select between alpha and inverse alpha right after the alpha modulation 0x0: Alpha 0x1: Inverse Alpha
3	FG_MOD_ALPHA	Modulate Alpha
2	FG_INV_ALPHA_SEL	Ability to select between alpha and inverse alpha right after the alpha select 0x0: Alpha 0x1: Inverse Alpha

MDSS_MDP_VP_0_LAYER_4_BLEND3_OP (cont.)

Bits	Name	Description
1:0	FG_ALPHA_SEL	Blending Alpha Select 0x0: FG constant alpha 0x1: BG constant alpha 0x2: FG per pixel alpha 0x3: BG per pixel alpha

0x01A490B4 MDSS_MDP_VP_0_LAYER_4_BLEND3_FG_ALPHA**Type:** RW**Reset State:** 0x00000000

Blend Layer3 FG alpha

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_LAYER_4_BLEND3_FG_ALPHA

Bits	Name	Description
7:0	CONSTANT_ALPHA	Constant Alpha

0x01A490B8 MDSS_MDP_VP_0_LAYER_4_BLEND3_BG_ALPHA**Type:** RW**Reset State:** 0x00000000

Blend Layer3 FG alpha

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_LAYER_4_BLEND3_BG_ALPHA

Bits	Name	Description
7:0	CONSTANT_ALPHA	Constant Alpha

0x01A490BC MDSS_MDP_VP_0_LAYER_4_BLEND3_FG TRANSP_LOW0**Type:** RW**Reset State:** 0x00000000

Blend Layer3 Transparency Color Low Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_4_BLEND3_FG TRANSP_LOW0

Bits	Name	Description
27:16	TRANSPARENCY_C1	Transparency color check lower limit for Color1
11:0	TRANSPARENCY_C0	Transparency color check lower limit for Color0

0x01A490C0 MDSS_MDP_VP_0_LAYER_4_BLEND3_FG TRANSP_LOW1

Type: RW

Reset State: 0x00000000

Blend Layer3 Transparency Color Low Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_4_BLEND3_FG TRANSP_LOW1

Bits	Name	Description
11:0	TRANSPARENCY_C2	Transparency color check lower limit for Color2

0x01A490C4 MDSS_MDP_VP_0_LAYER_4_BLEND3_FG TRANSP_HIGH0

Type: RW

Reset State: 0x00000000

Blend Layer3 Transparency Color High Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_4_BLEND3_FG TRANSP_HIGH0

Bits	Name	Description
27:16	TRANSPARENCY_C1	Transparency color check higher limit for Color1
11:0	TRANSPARENCY_C0	Transparency color check higher limit for Color0

0x01A490C8 MDSS_MDP_VP_0_LAYER_4_BLEND3_FG TRANSP_HIGH1

Type: RW

Reset State: 0x00000000

Blend Layer3 Transparency Color High Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_4_BLEND3_FG TRANSP_HIGH1

Bits	Name	Description
11:0	TRANSPARENCY_C2	Transparency color check higher limit for Color2

0x01A490CC MDSS_MDP_VP_0_LAYER_4_BLEND3_BG TRANSP_LOW0

Type: RW

Reset State: 0x00000000

Blend Layer3 Transparency Color Low Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_4_BLEND3_BG TRANSP_LOW0

Bits	Name	Description
27:16	TRANSPARENCY_C1	Transparency color check lower limit for Color1
11:0	TRANSPARENCY_C0	Transparency color check lower limit for Color0

0x01A490D0 MDSS_MDP_VP_0_LAYER_4_BLEND3_BG TRANSP_LOW1

Type: RW

Reset State: 0x00000000

Blend Layer3 Transparency Color Low Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_4_BLEND3_BG TRANSP_LOW1

Bits	Name	Description
11:0	TRANSPARENCY_C2	Transparency color check lower limit for Color2

0x01A490D4 MDSS_MDP_VP_0_LAYER_4_BLEND3_BG TRANSP_HIGH0

Type: RW

Reset State: 0x00000000

Blend Layer3 Transparency Color High Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_4_BLEND3_BG TRANSP_HIGH0

Bits	Name	Description
27:16	TRANSPARENCY_C1	Transparency color check higher limit for Color1
11:0	TRANSPARENCY_C0	Transparency color check higher limit for Color0

0x01A490D8 MDSS_MDP_VP_0_LAYER_4_BLEND3_BG TRANSP_HIGH1

Type: RW

Reset State: 0x00000000

Blend Layer3 Transparency Color High Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_4_BLEND3_BG TRANSP_HIGH1

Bits	Name	Description
11:0	TRANSPARENCY_C2	Transparency color check higher limit for Color2

0x01A490DC MDSS_MDP_VP_0_LAYER_4_CURSOR_STRIDE

Type: RW

Reset State: 0x00000000

Cursor Stride

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_LAYER_4_CURSOR_STRIDE

Bits	Name	Description
15:0	STRIDE	Cursor Stride

0x01A490E0 MDSS_MDP_VP_0_LAYER_4_CURSOR_IMG_SIZE

Type: RW

Reset State: 0x00000000

Cursor Image Size

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_LAYER_4_CURSOR_IMG_SIZE

Bits	Name	Description
31:16	SRC_H	Total Cursor Source Height
15:0	SRC_W	Total Cursor Source Width

0x01A490E4 MDSS_MDP_VP_0_LAYER_4_CURSOR_SIZE

Type: RW

Reset State: 0x00000000

Cursor ROI Image Size

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_LAYER_4_CURSOR_SIZE

Bits	Name	Description
31:16	ROI_H	Cursor ROI Source Height
15:0	ROI_W	Cursor ROI Source Width

0x01A490E8 MDSS_MDP_VP_0_LAYER_4_CURSOR_XY

Type: RW

Reset State: 0x00000000

Cursor ROI Offset

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_LAYER_4_CURSOR_XY

Bits	Name	Description
31:16	SRC_Y	Y ROI coordinate
15:0	SRC_X	X ROI coordinate

0x01A490EC MDSS_MDP_VP_0_LAYER_4_CURSOR_FORMAT**Type:** RW**Reset State:** 0x00000000

Cursor Format

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_LAYER_4_CURSOR_FORMAT

Bits	Name	Description
2:0	FORMAT	Cursor Format 0x0: ARGB8888 0x2: ARGB1555 0x4: ARGB4444

0x01A490F0 MDSS_MDP_VP_0_LAYER_4_CURSOR_BASE_ADDR**Type:** RW**Reset State:** 0x00000000

Cursor Base Address

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_LAYER_4_CURSOR_BASE_ADDR

Bits	Name	Description
31:0	ADDR	Cursor Base-Address (byte-aligned)

0x01A490F4 MDSS_MDP_VP_0_LAYER_4_CURSOR_START_XY**Type:** RW**Reset State:** 0x00000000

Cursor Starting Coordinates

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_LAYER_4_CURSOR_START_XY

Bits	Name	Description
31:16	Y_START	Starting Y coordinate
15:0	X_START	Starting X coordinate

0x01A490F8 MDSS_MDP_VP_0_LAYER_4_CURSOR_BLEND_CONFIG**Type:** RW**Reset State:** 0x00000000

Cursor Blending Configuration

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_LAYER_4_CURSOR_BLEND_CONFIG

Bits	Name	Description
3	BLEND_TRANSP_EN	0x0: Disabled 0x1: Color keying is done on the cursor and blended result
2:1	BLEND_ALPHA_SEL	Blending Alpha Select 0x0: Constant Alpha 0x1: reserved 0x2: Cursor per pixel alpha 0x3: reserved
0	BLEND_EN	Cursor Blend Enable

0x01A490FC MDSS_MDP_VP_0_LAYER_4_CURSOR_BLEND_PARAM**Type:** RW**Reset State:** 0x00000000

Cursor Blending Parameter

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_LAYER_4_CURSOR_BLEND_PARAM

Bits	Name	Description
7:0	CONSTANT_ALPHA	Constant Alpha

0x01A49100 MDSS_MDP_VP_0_LAYER_4_CURSOR_BLEND TRANSP_LOW0**Type:** RW**Reset State:** 0x00000000

Cursor Transparency Color Low Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_4_CURSOR_BLEND TRANSP_LOW0

Bits	Name	Description
27:16	TRANSPARENCY_C1	Transparency color check lower limit for Color1
11:0	TRANSPARENCY_C0	Transparency color check lower limit for Color0

0x01A49104 MDSS_MDP_VP_0_LAYER_4_CURSOR_BLEND TRANSP_LOW1**Type:** RW**Reset State:** 0x00000000

Cursor Transparency Color Low Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_4_CURSOR_BLEND TRANSP_LOW1

Bits	Name	Description
11:0	TRANSPARENCY_C2	Transparency color check lower limit for Color2

0x01A49108 MDSS_MDP_VP_0_LAYER_4_CURSOR_BLEND TRANSP_HIGH0**Type:** RW**Reset State:** 0x00000000

Cursor Transparency Color High Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_4_CURSOR_BLEND TRANSP_HIGH0

Bits	Name	Description
27:16	TRANSPARENCY_C1	Transparency color check higher limit for Color1
11:0	TRANSPARENCY_C0	Transparency color check higher limit for Color0

0x01A4910C MDSS_MDP_VP_0_LAYER_4_CURSOR_BLEND TRANSP_HIGH1**Type:** RW**Reset State:** 0x00000000

Cursor Transparency Color High Value

double buffered: NO

MDSS_MDP_VP_0_LAYER_4_CURSOR_BLEND TRANSP_HIGH1

Bits	Name	Description
11:0	TRANSPARENCY_C2	Transparency color check higher limit for Color2

0x01A49110 MDSS_MDP_VP_0_LAYER_4_GC_ARLUT_RED_STARTX_N**Type:** RW**Reset State:** 0x00000000**MDSS_MDP_VP_0_LAYER_4_GC_ARLUT_RED_STARTX_N**

Bits	Name	Description
21	VALUE_UPDATEN	
20	SEGMENT_UPDATE	
19:16	SEGMENT	
11:0	STARTX	

0x01A49114 MDSS_MDP_VP_0_LAYER_4_GC_ARLUT_RED_SLOPE_N**Type:** RW**Reset State:** 0x00000000**MDSS_MDP_VP_0_LAYER_4_GC_ARLUT_RED_SLOPE_N**

Bits	Name	Description
21	VALUE_UPDATEN	
20	SEGMENT_UPDATE	
19:16	SEGMENT	
14:0	SLOPE	

0x01A49118 MDSS_MDP_VP_0_LAYER_4_GC_ARLUT_RED_OFFSET_N**Type:** RW**Reset State:** 0x00000000**MDSS_MDP_VP_0_LAYER_4_GC_ARLUT_RED_OFFSET_N**

Bits	Name	Description
21	VALUE_UPDATEN	
20	SEGMENT_UPDATE	
19:16	SEGMENT	
14:0	OFFSET	

0x01A49120 MDSS_MDP_VP_0_LAYER_4_GC_ARLUT_GREEN_STARTX_N**Type:** RW**Reset State:** 0x00000000**MDSS_MDP_VP_0_LAYER_4_GC_ARLUT_GREEN_STARTX_N**

Bits	Name	Description
21	VALUE_UPDATEN	
20	SEGMENT_UPDATE	
19:16	SEGMENT	
11:0	STARTX	

0x01A49124 MDSS_MDP_VP_0_LAYER_4_GC_ARLUT_GREEN_SLOPE_N**Type:** RW**Reset State:** 0x00000000**MDSS_MDP_VP_0_LAYER_4_GC_ARLUT_GREEN_SLOPE_N**

Bits	Name	Description
21	VALUE_UPDATEN	
20	SEGMENT_UPDATE	
19:16	SEGMENT	
14:0	SLOPE	

0x01A49128 MDSS_MDP_VP_0_LAYER_4_GC_ARLUT_GREEN_OFFSET_N**Type:** RW**Reset State:** 0x00000000**MDSS_MDP_VP_0_LAYER_4_GC_ARLUT_GREEN_OFFSET_N**

Bits	Name	Description
21	VALUE_UPDATEN	
20	SEGMENT_UPDATE	
19:16	SEGMENT	
14:0	OFFSET	

0x01A49130 MDSS_MDP_VP_0_LAYER_4_GC_ARLUT_BLUE_STARTX_N**Type:** RW**Reset State:** 0x00000000**MDSS_MDP_VP_0_LAYER_4_GC_ARLUT_BLUE_STARTX_N**

Bits	Name	Description
21	VALUE_UPDATEN	
20	SEGMENT_UPDATE	
19:16	SEGMENT	
11:0	STARTX	

0x01A49134 MDSS_MDP_VP_0_LAYER_4_GC_ARLUT_BLUE_SLOPE_N**Type:** RW**Reset State:** 0x00000000**MDSS_MDP_VP_0_LAYER_4_GC_ARLUT_BLUE_SLOPE_N**

Bits	Name	Description
21	VALUE_UPDATEN	
20	SEGMENT_UPDATE	
19:16	SEGMENT	
14:0	SLOPE	

0x01A49138 MDSS_MDP_VP_0_LAYER_4_GC_ARLUT_BLUE_OFFSET_N**Type:** RW**Reset State:** 0x00000000**MDSS_MDP_VP_0_LAYER_4_GC_ARLUT_BLUE_OFFSET_N**

Bits	Name	Description
21	VALUE_UPDATEN	
20	SEGMENT_UPDATE	
19:16	SEGMENT	
14:0	OFFSET	

0x01A49200 MDSS_MDP_VP_0_LAYER_4_CURSOR_BASE_ADDR_SW_STATUS**Type:** RW**Reset State:** 0x00000000

Security Status of the source buffer plane

This is programmed by SW to indicate if the buffer is NON-SECURE or SECURE

This information will be presented to VBIF per transaction and reflected as part of the memory transaction

double buffered: YES

MDSS_MDP_VP_0_LAYER_4_CURSOR_BASE_ADDR_SW_STATUS

Bits	Name	Description
0	ADDR_CP	

0x01A49204 MDSS_MDP_VP_0_LAYER_4_CURSOR_FETCH_CONFIG**Type:** RW**Reset State:** 0x00000087

Cursor Client Fetch Configuration per Virtual Pipeline double buffered: NO

This register must NOT be altered during operation and MDP must be idle when changing values

MDSS_MDP_VP_0_LAYER_4_CURSOR_FETCH_CONFIG

Bits	Name	Description
17:16	TILE_SIZE	Tile Size for processing of BLOCK formats Note that this only applies to blocks that operate in block mode which is limited to DMA source pipe. This setting has no effect for all other source pipes (ViG/RGB) 0x0: 128x128 0x1: 64x64 0x2: 32x32
12	ADDR_SYNC_MODE	Boundary on which fetch address synchronizes. Setting this field to line forces the address updates on the next line which performs the software requested feature of updating on hsync. Note that updating on line boundaries will likely result in tearing on the panel since hardware is fetching from new memory buffer. 0x0: frame (default) 0x1: line
9:8	MAL	Describes the Minimum Access Length for the source pipeline. This value should align with the DRAM's expected transaction alignment unit. This setting forces the source fetch pipe to be aligned to DRAM and implement re-alignment for maximum efficiency 0x0: 128b alignment (default) 0x1: 256b alignment 0x2: 512b alignment
7:4	MAX_BURST_SIZE	Maximum burst beat size. Hardware fetch engines operate on 128-bit bus Possible values: 0x8: burst of 8 0x4: burst of 4 0x2: burst of 2
3:0	REQ_DEPTH_LIMIT	Pending request limit, default max value of 0x7 means maximum pending requests allowed in the design Anything less limits the bus requests to that number + 1.

0x01A49210 MDSS_MDP_VP_0_LAYER_4_CURSOR_FETCH_SMP_WR_PLANE0**Type:** R**Reset State:** 0x00000000

Fetch to SMP on plane 0 write counter

double buffered: N/A

MDSS_MDP_VP_0_LAYER_4_CURSOR_FETCH_SMP_WR_PLANE0

Bits	Name	Description
15:0	COUNT	This counter is reset on every frame and is incremented for each write fetch makes to the SMP plane0 port This should correlate to value SMP_UNPACK_RD_PLANE0 More useful for debug but can be used for profiling purposes as well.

0x01A49214 MDSS_MDP_VP_0_LAYER_4_CURSOR_SMP_UNPACK_RD_PLANE0**Type:** R**Reset State:** 0x00000000

SMP to unpack on plane 0 read counter

double buffered: N/A

MDSS_MDP_VP_0_LAYER_4_CURSOR_SMP_UNPACK_RD_PLANE0

Bits	Name	Description
15:0	COUNT	This counter is reset on every frame and is incremented for each line read from SMP plane0 port This should correlate to FETCH_SMP_WR_PLANE0 value More useful for debug but can be used for profiling purposes as well.

0x01A49220 MDSS_MDP_VP_0_LAYER_4_CURSOR_REQPRIORITY_FIFO_WATERMARK_0**Type:** RW**Reset State:** 0x00000008

FIFO watermark to aid in generation of mdpREQPRIORITY

The watermark here refers to the dedicated cursor fetch memory.

The value programmed here should be aware of the amount of dedicated cursor fetch memory.

So a value of 0 indicates completely FULL (i.e. there are no more entries available in the dedicated memory)

A value of 100 indicates 100 entries in the Shared Memory Pool (each entry being 128-bit entries)

This is the lowestmost FIFO depth and should not be programmed higher than REQPRIORITY_FIFO_WATERMARK_1 or REQPRIORITY_FIFO_WATERMARK_2

An example programming would set the following:

REQPRIORITY_FIFO_WATERMARK_0 : 0x08 (meaning 0x08 entries free to be written into dedicated cursor fetch memory - least room available)

REQPRIORITY_FIFO_WATERMARK_1 : 0x10 (meaning 0x10 entries free to be written into dedicated cursor fetch memory - middle ground room available)

REQPRIORITY_FIFO_WATERMARK_2 : 0x20 (meaning 0x20 entries free to be written into dedicated cursor fetch memory - most room available)

mdpREQPRIORITY = 11 (highest priority request) when above all 3 watermark levels (i.e. lots of space available in dedicated cursor fetch memory)

mdpREQPRIORITY = 10 above FIFO_0 and FIFO_1 but below FIFO_2

mdpREQPRIORITY = 01 above FIFO_0 but below FIFO_1 and FIFO_2

mdpREQPRIORITY = 00 (lowest priority request) when below all 3 watermark levels (i.e. not a lot of space available in dedicated cursor fetch memory so its good)

double buffered: YES

MDSS_MDP_VP_0_LAYER_4_CURSOR_REQPRIORITY_FIFO_WATERMARK_0

Bits	Name	Description
15:0	WATERMARK	Watermark Level for FIFO_0

0x01A49224 MDSS_MDP_VP_0_LAYER_4_CURSOR_REQPRIORITY_FIFO_WATERMARK_1

Type: RW

Reset State: 0x00000010

FIFO watermark to aid in generation of mdpREQPRIORITY

The watermark here refers to the dedicated cursor fetch memory.

The value programmed here should be aware of the amount of dedicated cursor fetch memory.

So a value of 0 indicates completely FULL (i.e. there are no more entries available in the dedicated memory)

A value of 100 indicates 100 entries in the Shared Memory Pool (each entry being 128-bit entries)

This is the lowestmost FIFO depth and should not be programmed higher than REQPRIORITY_FIFO_WATERMARK_1 or REQPRIORITY_FIFO_WATERMARK_2

An example programming would set the following:

REQPRIORITY_FIFO_WATERMARK_0 : 0x08 (meaning 0x08 entries free to be written into dedicated cursor fetch memory - least room available)

REQPRIORITY_FIFO_WATERMARK_1 : 0x10 (meaning 0x10 entries free to be written into dedicated cursor fetch memory - middle ground room available)

REQPRIORITY_FIFO_WATERMARK_2 : 0x20 (meaning 0x20 entries free to be written into dedicated cursor fetch memory - most room available)

mdpREQPRIORITY = 11 (highest priority request) when above all 3 watermark levels (i.e. lots of space available in dedicated cursor fetch memory)

mdpREQPRIORITY = 10 above FIFO_0 and FIFO_1 but below FIFO_2

mdpREQPRIORITY = 01 above FIFO_0 but below FIFO_1 and FIFO_2

mdpREQPRIORITY = 00 (lowest priority request) when below all 3 watermark levels (i.e. not a lot of space available in dedicated cursor fetch memory so its good)

double buffered: YES

MDSS_MDP_VP_0_LAYER_4_CURSOR_REQPRIORITY_FIFO_WATERMARK_1

Bits	Name	Description
15:0	WATERMARK	Watermark Level for FIFO_1

0x01A49228 MDSS_MDP_VP_0_LAYER_4_CURSOR_REQPRIORITY_FIFO_WATERMARK_2

Type: RW

Reset State: 0x00000020

FIFO watermark to aid in generation of mdpREQPRIORITY

The watermark here refers to the dedicated cursor fetch memory.

The value programmed here should be aware of the amount of dedicated cursor fetch memory.

So a value of 0 indicates completely FULL (i.e. there are no more entries available in the dedicated memory)

A value of 100 indicates 100 entries in the Shared Memory Pool (each entry being 128-bit entries)

This is the lowestmost FIFO depth and should not be programmed higher than REQPRIORITY_FIFO_WATERMARK_1 or REQPRIORITY_FIFO_WATERMARK_2

An example programming would set the following:

REQPRIORITY_FIFO_WATERMARK_0 : 0x08 (meaning 0x08 entries free to be written into dedicated cursor fetch memory - least room available)

REQPRIORITY_FIFO_WATERMARK_1 : 0x10 (meaning 0x10 entries free to be written into dedicated cursor fetch memory - middle ground room available)

REQPRIORITY_FIFO_WATERMARK_2 : 0x20 (meaning 0x20 entries free to be written into dedicated cursor fetch memory - most room available)

mdpREQPRIORITY = 11 (highest priority request) when above all 3 watermark levels (i.e. lots of space available in dedicated cursor fetch memory)

mdpREQPRIORITY = 10 above FIFO_0 and FIFO_1 but below FIFO_2

mdpREQPRIORITY = 01 above FIFO_0 but below FIFO_1 and FIFO_2

mdpREQPRIORITY = 00 (lowest priority request) when below all 3 watermark levels (i.e. not a lot of space available in dedicated cursor fetch memory so its good)

double buffered: YES

MDSS_MDP_VP_0_LAYER_4_CURSOR_REQPRIORITY_FIFO_WATERMARK_2

Bits	Name	Description
15:0	WATERMARK	Watermark Level for FIFO_2

0x01A49380 MDSS_MDP_VP_0_LAYER_4_GC_MISR_CTRL

Type: RW

Reset State: 0x00000000

MDSS_MDP_VP_0_LAYER_4_GC_MISR_CTRL

Bits	Name	Type	Description
31	RUN_MODE	RW	
27:24	INPUT_SEL	RW	
10	STATUS_CLEAR	W	
9	STATUS	R	
8	ENABLE	RW	
7:0	FRAME_COUNT	RW	

0x01A49384 MDSS_MDP_VP_0_LAYER_4_GC_MISR_SIGNATURE

Type: R

Reset State: 0x00000000

MDSS_MDP_VP_0_LAYER_4_GC_MISR_SIGNATURE

Bits	Name	Description
31:0	CAPTURED	

17.17.7.14 MDSS_MDP_VP_0_DSPP_0

DSPP hardware register description

17.17.7.14.1 MDSS_MDP_VP_0_DSPP_0_DSPP

DSPP

DSPP hardware register description

0x01A55000 MDSS_MDP_VP_0_DSPP_0_OP_MODE**Type:** RW**Reset State:** 0x00000000

DSPP Operation Control

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_DSPP_0_OP_MODE

Bits	Name	Description
31	SIX_ZONE_VAL_MASK	0x0: Disable six zone adj on VAL 0x1: Enable six zone adj on VAL
30	SIX_ZONE_SAT_MASK	0x0: Disable six zone adj on SAT 0x1: Enable six zone adj on SAT
29	SIX_ZONE_HUE_MASK	0x0: Disable six zone adj on HUE 0x1: Enable six zone adj on HUE
28	PA_CONT_MASK	0x0: Disable HSIC adj on CONTRAST 0x1: Enable HSIC adj on CONTRAST
27	PA_VAL_MASK	0x0: Disable HSIC adj on VALUE 0x1: Enable HSIC adj on VALUE
26	PA_SAT_MASK	0x0: Disable HSIC adj on SATURATION 0x1: Enable HSIC adj on SATURATION
25	PA_HUE_MASK	0x0: Disable HSIC adj on HUE 0x1: Enable HSIC adj on HUE
22	GC_LUT_EN	0x0: Disabled - mapped linearly from 12-bit to 8-bit 0x1: Enabled
20	PA_EN	Picture Adjustment enable 0x0: DISABLED 0x1: ENABLED
19	PA_LUTV_EN	Histogram Enhancement Lookup Table enable
16	HIST_EN	Histogram Enable

MDSS_MDP_VP_0_DSPP_0_OP_MODE (cont.)

Bits	Name	Description
8	DST_DITHER_EN	Destination Dither enable 0x0: DISABLED 0x1: ENABLED
7	PA_MEM_COL_SKY_MASK	0x0: Disable Sky adjustment 0x1: Enable Sky adjustment
6	PA_MEM_COL_FOL_MASK	0x0: Disable Foliage adjustment 0x1: Enable Foliage adjustment
5	PA_MEM_COL_SKIN_MASK	0x0: Disable Skin adjustment 0x1: Enable Skin adjustment
4	PCC_EN	Panel Color Correction enable 0x0: DISABLED 0x1: ENABLED
2	PA_MEM_PROTECT_ENABLE	0: global hsv adjustment is independent of memor color 1: global hsv adjustment is active if the pixel is not already processed by memory color contrast adjustment in global adjustment is always independ of memory color regardless
1	PA_SAT_ZERO_EXP_ENABLE	0: if PA_SAT_ADJ < -32, clamp PA_SAT_ADJ to 0 1: if PA_SAT_ADJ < -64, clamp PA_SAT_ADJ to 0
0	IGC_LUT_EN	Inverse Gamma Correction LUT enable 0x0: DISABLED 0x1: ENABLED

**0x01A55030+ MDSS_MDP_VP_0_DSPP_0_PCC_PLANEC_CONSTANT_COEFF, c=[0..2]
4*c****Type:** RW**Reset State:** 0x00000000

Constant Coefficient for RED/GREEN/BLUE plane

double buffered: NO

MDSS_MDP_VP_0_DSPP_0_PCC_PLANEC_CONSTANT_COEFF

Bits	Name	Description
15:0	COEFF	Constant Coefficient for RED/GREEN/BLUE plane in S13.3 format

**0x01A55040+ MDSS_MDP_VP_0_DSPP_0_PCC_PLANEc_R_COEFF, c=[0..2]
4*c**

Type: RW

Reset State: 0x00000000

Red Coefficient for RED/GREEN/BLUE plane

double buffered: NO

MDSS_MDP_VP_0_DSPP_0_PCC_PLANEc_R_COEFF

Bits	Name	Description
17:0	COEFF	Red Coefficient for RED/GREEN/BLUE plane in S4.15 format

**0x01A55050+ MDSS_MDP_VP_0_DSPP_0_PCC_PLANEc_G_COEFF, c=[0..2]
4*c**

Type: RW

Reset State: 0x00000000

Red Coefficient for RED/GREEN/BLUE plane

double buffered: NO

MDSS_MDP_VP_0_DSPP_0_PCC_PLANEc_G_COEFF

Bits	Name	Description
17:0	COEFF	Green Coefficient for RED/GREEN/BLUE plane in S4.15 format

**0x01A55060+ MDSS_MDP_VP_0_DSPP_0_PCC_PLANEc_B_COEFF, c=[0..2]
4*c**

Type: RW

Reset State: 0x00000000

Red Coefficient for RED/GREEN/BLUE plane

double buffered: NO

MDSS_MDP_VP_0_DSPP_0_PCC_PLANEc_B_COEFF

Bits	Name	Description
17:0	COEFF	Blue Coefficient for RED/GREEN/BLUE plane in S4.15 format

**0x01A55070+ MDSS_MDP_VP_0_DSPP_0_PCC_PLANEc_RR_COEFF, c=[0..2]
4*c****Type:** RW**Reset State:** 0x00000000

Red Red Coefficient for RED/GREEN/BLUE plane

double buffered: NO

MDSS_MDP_VP_0_DSPP_0_PCC_PLANEc_RR_COEFF

Bits	Name	Description
27:0	COEFF	RED RED Coefficient for RED/GREEN/BLUE plane. Total format is S1.27

**0x01A55080+ MDSS_MDP_VP_0_DSPP_0_PCC_PLANEc_RG_COEFF, c=[0..2]
4*c****Type:** RW**Reset State:** 0x00000000

Red Green Coefficient for RED/GREEN/BLUE plane

double buffered: NO

MDSS_MDP_VP_0_DSPP_0_PCC_PLANEc_RG_COEFF

Bits	Name	Description
27:0	COEFF	RED GREEN Coefficient for RED/GREEN/BLUE plane. Total format is S1.27

**0x01A55090+ MDSS_MDP_VP_0_DSPP_0_PCC_PLANEc_RB_COEFF, c=[0..2]
4*c****Type:** RW**Reset State:** 0x00000000

Red Blue Coefficient for RED/GREEN/BLUE plane

double buffered: NO

MDSS_MDP_VP_0_DSPP_0_PCC_PLANEc_RB_COEFF

Bits	Name	Description
27:0	COEFF	RED BLUE Coefficient for RED/GREEN/BLUE plane. Total format is S1.27

**0x01A550A0+MDSS_MDP_VP_0_DSPP_0_PCC_PLANEc_GG_COEFF, c=[0..2]
4*c****Type:** RW**Reset State:** 0x00000000

Green Green Coefficient for RED/GREEN/BLUE plane

double buffered: NO

MDSS_MDP_VP_0_DSPP_0_PCC_PLANEc_GG_COEFF

Bits	Name	Description
27:0	COEFF	GREEN GREEN Coefficient for RED/GREEN/BLUE plane. Total format is S1.27

**0x01A550B0+MDSS_MDP_VP_0_DSPP_0_PCC_PLANEc_BG_COEFF, c=[0..2]
4*c****Type:** RW**Reset State:** 0x00000000

Blue Green Coefficient for RED/GREEN/BLUE plane

double buffered: NO

MDSS_MDP_VP_0_DSPP_0_PCC_PLANEc_BG_COEFF

Bits	Name	Description
27:0	COEFF	BLUE GREEN Coefficient for RED/GREEN/BLUE plane. Total format is S1.27

**0x01A550C0+MDSS_MDP_VP_0_DSPP_0_PCC_PLANEc_BB_COEFF, c=[0..2]
4*c****Type:** RW**Reset State:** 0x00000000

Blue Blue Coefficient for RED/GREEN/BLUE plane

double buffered: NO

MDSS_MDP_VP_0_DSPP_0_PCC_PLANEc_BB_COEFF

Bits	Name	Description
27:0	COEFF	BLUE BLUE Coefficient for RED/GREEN/BLUE plane. Total format is S1.27

**0x01A550D0+MDSS_MDP_VP_0_DSPP_0_PCC_PLANEc_RGB_COEFF_0, c=[0..2]
4*c**

Type: RW

Reset State: 0x00000000

Red Blue Green Coefficient for RED/GREEN/BLUE plane

double buffered: NO

MDSS_MDP_VP_0_DSPP_0_PCC_PLANEc_RGB_COEFF_0

Bits	Name	Description
7:0	COEFF	PCC coefficient data (sign + 7 fractional bits - S1.39) for RGB_0 and RGB_1 field

**0x01A550E0+MDSS_MDP_VP_0_DSPP_0_PCC_PLANEc_RGB_COEFF_1, c=[0..2]
4*c**

Type: RW

Reset State: 0x00000000

Red Blue Green for RED/GREEN/BLUE plane

double buffered: NO

MDSS_MDP_VP_0_DSPP_0_PCC_PLANEc_RGB_COEFF_1

Bits	Name	Description
31:0	COEFF	PCC coefficient data (remaining 32 fractional bits - S1.39) for RGB_0 and RGB_1 field

0x01A55150 MDSS_MDP_VP_0_DSPP_0_DITHER_BITDEPTH

Type: RW

Reset State: 0x00000000

Dither Strength

double buffered: NO

MDSS_MDP_VP_0_DSPP_0_DITHER_BITDEPTH

Bits	Name	Description
5:4	C2_BITS	Number of MSB bits on component2 to indicate the dithering boundary 0x0: 4bits 0x1: 5bits 0x2: 6bits 0x3: 8bits
3:2	C1_BITS	Number of MSB bits on component1 to indicate the dithering boundary 0x0: 4bits 0x1: 5bits 0x2: 6bits 0x3: 8bits
1:0	C0_BITS	Number of MSB bits on component0 to indicate the dithering boundary 0x0: 4bits 0x1: 5bits 0x2: 6bits 0x3: 8bits

0x01A55164 MDSS_MDP_VP_0_DSPP_0_DITHER_MATRIX_ROW0**Type:** RW**Reset State:** 0x00005D7F

Dither Strength Matrix Values Register for first row which covers entries [0-3]

double buffered: NO

MDSS_MDP_VP_0_DSPP_0_DITHER_MATRIX_ROW0

Bits	Name	Description
15:12	ENTRY_3	Entry 3 for the dither matrix
11:8	ENTRY_2	Entry 2 for the dither matrix
7:4	ENTRY_1	Entry 1 for the dither matrix
3:0	ENTRY_0	Entry 0 for the dither matrix

0x01A55168 MDSS_MDP_VP_0_DSPP_0_DITHER_MATRIX_ROW1**Type:** RW**Reset State:** 0x000091B3

Dither Strength Matrix Values Register for first row which covers entries [4-7]

double buffered: NO

MDSS_MDP_VP_0_DSPP_0_DITHER_MATRIX_ROW1

Bits	Name	Description
15:12	ENTRY_7	Entry 7 for the dither matrix
11:8	ENTRY_6	Entry 6 for the dither matrix
7:4	ENTRY_5	Entry 5 for the dither matrix
3:0	ENTRY_4	Entry 4 for the dither matrix

0x01A5516C MDSS_MDP_VP_0_DSPP_0_DITHER_MATRIX_ROW2

Type: RW

Reset State: 0x00006E4C

Dither Strength Matrix Values Register for first row which covers entries [8-11]

double buffered: NO

MDSS_MDP_VP_0_DSPP_0_DITHER_MATRIX_ROW2

Bits	Name	Description
15:12	ENTRY_11	Entry 11 for the dither matrix
11:8	ENTRY_10	Entry 10 for the dither matrix
7:4	ENTRY_9	Entry 9 for the dither matrix
3:0	ENTRY_8	Entry 8 for the dither matrix

0x01A55170 MDSS_MDP_VP_0_DSPP_0_DITHER_MATRIX_ROW3

Type: RW

Reset State: 0x0000A280

Dither Strength Matrix Values Register for first row which covers entries [12-15]

double buffered: NO

MDSS_MDP_VP_0_DSPP_0_DITHER_MATRIX_ROW3

Bits	Name	Description
15:12	ENTRY_15	Entry 15 for the dither matrix
11:8	ENTRY_14	Entry 14 for the dither matrix

MDSS_MDP_VP_0_DSPP_0_DITHER_MATRIX_ROW3 (cont.)

Bits	Name	Description
7:4	ENTRY_13	Entry 13 for the dither matrix
3:0	ENTRY_12	Entry 12 for the dither matrix

0x01A55210 MDSS_MDP_VP_0_DSPP_0_HIST_COLLECT_CTL**Type:** RW**Reset State:** 0x00000000

Histogram Collection Control

double buffered: YES

except for LOCK

MDSS_MDP_VP_0_DSPP_0_HIST_COLLECT_CTL

Bits	Name	Description
0	LOCK	0: SW and HW swap their access on the 2 histogram tables; HW collects on a new table on the new frame, while SW accesses the histogram of the last frame on the table just swapped out by HW 1: current table will be overwritten for new collection on the next frame, the other table remains untouched for SW to continue to access

0x01A55214 MDSS_MDP_VP_0_DSPP_0_HIST_MIN_MAX**Type:** R**Reset State:** 0x00000000

MIN and MAX V value over the entire frame

double buffered: NO

MDSS_MDP_VP_0_DSPP_0_HIST_MIN_MAX

Bits	Name	Description
23:16	MAX	Maximum V value of the last collected frame
7:0	MIN	Minimum V value of the last collected frame

0x01A55218 MDSS_MDP_VP_0_DSPP_0_HIST_FRAME_COLLECTED**Type:** R**Reset State:** 0x00000000

Histogram Frame Number since the last rising START

double buffered: NO

MDSS_MDP_VP_0_DSPP_0_HIST_FRAME_COLLECTED

Bits	Name	Description
31:0	FRAME_NUM	The frame number that the current histogram accessible by SW was collected from

0x01A5521C MDSS_MDP_VP_0_DSPP_0_HIST_SEL**Type:** R**Reset State:** 0x00000000

PA LUTV and Histogram Table Selection

double buffered: NO

MDSS_MDP_VP_0_DSPP_0_HIST_SEL

Bits	Name	Description
4	HIST_SEL	0x0: Table 0 is available for SW and Table 1 is available for HW 0x1: Table 1 is available for SW and Table 0 is available for HW
0	PA_LUTV_SEL	0x0: LUT0 is available for SW and LUT1 is available for HW 0x1: LUT1 is available for SW and LUT0 is available for HW

0x01A55220 MDSS_MDP_VP_0_DSPP_0_HIST_SUM**Type:** R**Reset State:** 0x00000000

Accumulated V Value of the frame

double buffered: NO

MDSS_MDP_VP_0_DSPP_0_HIST_SUM

Bits	Name	Description
31:0	HIST_SUM	This sum matches the histogram currently accessible by SW

0x01A5522C MDSS_MDP_VP_0_DSPP_0_HIST_V_DATAN**Type:** RW**Reset State:** 0x00000000**MDSS_MDP_VP_0_DSPP_0_HIST_V_DATAN**

Bits	Name	Type	Description
31:24	INDEX	RW	
23:0	VALUE	R	

0x01A55230 MDSS_MDP_VP_0_DSPP_0_PA_LUTV**Type:** RW**Reset State:** 0x00000000

PA LUTV data

Auto-incremented by hardware

256 entries are available for reading

double buffered: NO

MDSS_MDP_VP_0_DSPP_0_PA_LUTV

Bits	Name	Description
17	VALUE_UPDATEN	Software issue an write to the LUT 0: VALUE will be written to RAM at INDEX if INDEX_UPDATE=1 or at INDEX+1 if INDEX_UPDATE=0 1: VALUE will not be written to RAM, this setting is appropriate for example when INDEX needs to be updated for read
16	INDEX_UPDATE	Software issue an index update 0: No index update by software 1: Update LUT index to value specified by INDEX
15:8	INDEX	VALUE is stored in this INDEX of the PA_LUTV, next write will be stored to the next higher INDEX
7:0	VALUE	Value read depends on the LUT select and which one is available for reading by software The read sequence to the LUT should always begin with WR INDEX_UPDATE=1, INDEX=<rprr start location> subsequent RD will start from the start location and every read will auto-increment the rprr

0x01A55234 MDSS_MDP_VP_0_DSPP_0_PA_LUTV_SWAP**Type:** W**Reset State:** 0x00000000

Histogram SWAP signal to indicate that the PA_LUTV contents should be swapped

Software accesses one LUT while HW accesses the other LUT

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_DSPP_0_PA_LUTV_SWAP

Bits	Name	Description
0	SWAP	SWAP signal to flip buffers accessible by SW and made available to HW. Achieves double buffering of the LUT values

0x01A55238 MDSS_MDP_VP_0_DSPP_0_PA_HUE_ADJUST**Type:** RW**Reset State:** 0x00000000

Picture HUE Adjustment

double buffered: YES

MDSS_MDP_VP_0_DSPP_0_PA_HUE_ADJUST

Bits	Name	Description
11:0	VALUE	Value in S12, recommend to be within exclusive range xA00 to x600 (i.e. -360 to +360 degree) x600 is equivalent to a shift of 360 degree if VALUE > x5FF in either negative or positive direction, it is circularly clipped (i.e. VALUE + x600 or VALUE - x600)

0x01A5523C MDSS_MDP_VP_0_DSPP_0_PA_SAT_ADJUST**Type:** RW**Reset State:** 0x00001A00

Picture Saturation Adjustment

double buffered: YES

MDSS_MDP_VP_0_DSPP_0_PA_SAT_ADJUST

Bits	Name	Description
15:8	THRSHLD	Below this threshold of saturation, no adjustment is applied to saturation. This field is in U8
7:0	VALUE	Value in S8

0x01A55240 MDSS_MDP_VP_0_DSPP_0_PA_VAL_ADJUST**Type:** RW**Reset State:** 0x00000000

Picture Adjustment

double buffered: YES

MDSS_MDP_VP_0_DSPP_0_PA_VAL_ADJUST

Bits	Name	Description
7:0	VALUE	Value in S8

0x01A55244 MDSS_MDP_VP_0_DSPP_0_PA_CONT_ADJUST**Type:** RW**Reset State:** 0x00000000

Picture Contrast Adjustment

double buffered: YES

MDSS_MDP_VP_0_DSPP_0_PA_CONT_ADJUST

Bits	Name	Description
7:0	VALUE	Value in S8

0x01A55248 MDSS_MDP_VP_0_DSPP_0_SIX_ZONE_ADJ_CURVE_P0**Type:** RW**Reset State:** 0x00000000

Six Zone Adjust Curve Part 0

double buffered: No

MDSS_MDP_VP_0_DSPP_0_SIX_ZONE_ADJ_CURVE_P0

Bits	Name	Description
26	INDEX_UPDATE	Set this to 1 to point current LUT location to INDEX. The LUT location is autoincremented on every value write or read when there is no INDEX_UPDATE Before every series of read, there must be an INDEX_UPDATE
25	VALUE_UPDATEN	Set this to 0 to enable VALUE to be written to the current location of the LUT
24:16	INDEX	Index to one of the 384 location of the LUT
11:0	HUE	S12, recommend to be within exclusive range x2800 and x1800 (i.e. -360 to +360 exclusive) circularly clipped otherwise On write: Value to be updated to one of the 384 entry LUT On read : Value read out from LUT location pointed by INDEX

0x01A5524C MDSS_MDP_VP_0_DSPP_0_SIX_ZONE_ADJ_CURVE_P1**Type:** RW**Reset State:** 0x00000000

Six Zone Adjust Curve Part 1

double buffered: No

MDSS_MDP_VP_0_DSPP_0_SIX_ZONE_ADJ_CURVE_P1

Bits	Name	Description
27:16	SAT	S2.10, On write: Value to be updated to one of the 384 entry LUT On read : Value read out from LUT location pointed by INDEX
11:0	VALUE	S2.10, On write: Value to be updated to one of the 384 entry LUT On read : Value read out from LUT location pointed by INDEX

0x01A55250 MDSS_MDP_VP_0_DSPP_0_SIX_ZONE_THRESHOLDS**Type:** RW**Reset State:** 0x00000000

Six Zone Threshold Qualifier

double buffered: Yes

MDSS_MDP_VP_0_DSPP_0_SIX_ZONE_THRESHOLDS

Bits	Name	Description
23:16	VAL_MAX	U8, Value has to be within the inclusive range of MIN to MAX to qualify for six zone adjustment on the pixel
15:8	VAL_MIN	
7:0	SAT_MIN	U8, Saturation has to be above or equal to this threshold to qualify for six zone adjustment

0x01A55254 MDSS_MDP_VP_0_DSPP_0_SKIN_ADJUST_PWL0**Type:** RW**Reset State:** 0x00000000

Skin Piece-wise Linear Adjustment Curve

double buffered: Yes

MDSS_MDP_VP_0_DSPP_0_SKIN_ADJUST_PWL0

Bits	Name	Description
31:16	HUE_GAIN	S16, is a symmetrical trapezoidal curve, this is the slope of the 2 sides
10:0	HUE_MID	U11, this is the offset from the min and max hue of skin region which marks the top 2 corners of the trapezoid

0x01A55258 MDSS_MDP_VP_0_DSPP_0_SKIN_ADJUST_PWL1**Type:** RW**Reset State:** 0x00000000

Skin Piece-wise Linear Adjustment Curve

double buffered: Yes

MDSS_MDP_VP_0_DSPP_0_SKIN_ADJUST_PWL1

Bits	Name	Description
31:16	VAL_GAIN	S16, is a symmetrical trapezoidal curve, this is the slope of the 2 sides
15:0	SAT_GAIN	S16, is a symmetrical trapezoidal curve, this is the slope of the 2 sides

0x01A5525C MDSS_MDP_VP_0_DSPP_0_SKIN_REGION_HUE**Type:** RW**Reset State:** 0x00000000

Skin Adjustment Hue Qualifier

double buffered: Yes

MDSS_MDP_VP_0_DSPP_0_SKIN_REGION_HUE

Bits	Name	Description
26:16	MAX	if MAX > x5ff (i.e. 360 degree), the behaviour will be same as MAX = x600
10:0	MIN	U11, Hue has to be within the inclusive range of MIN to MAX to qualify for skin tone adjustment on the pixel if MIN > x5ff (i.e. 360 degree), the behaviour will be same as MIN = x600

0x01A55260 MDSS_MDP_VP_0_DSPP_0_SKIN_REGION_SAT**Type:** RW**Reset State:** 0x00000000

Skin Adjustment Saturation Qualifier

double buffered: Yes

MDSS_MDP_VP_0_DSPP_0_SKIN_REGION_SAT

Bits	Name	Description
15:8	MAX	Saturation has to be below or equal to this to qualify for skin tone adjustment on the pixel
7:0	MIN	U8, Saturation has to be above or equal to this to qualify for skin tone adjustment on the pixel

0x01A55264 MDSS_MDP_VP_0_DSPP_0_SKIN_REGION_VAL**Type:** RW**Reset State:** 0x00000000

Skin Adjustment Value Qualifier

double buffered: Yes

MDSS_MDP_VP_0_DSPP_0_SKIN_REGION_VAL

Bits	Name	Description
15:8	MAX	Value has to be below or equal to this to qualify for skin tone adjustment on the pixel
7:0	MIN	U8, Value has to be above or equal to this to qualify for skin tone adjustment on the pixel

0x01A55268 MDSS_MDP_VP_0_DSPP_0_SKY_ADJUST_PWL0**Type:** RW**Reset State:** 0x00000000

Sky Piece-wise Linear Adjustment Curve

double buffered: Yes

MDSS_MDP_VP_0_DSPP_0_SKY_ADJUST_PWL0

Bits	Name	Description
31:16	HUE_GAIN	S16, is a symmetrical trapezoidal curve, this is the slope of the 2 sides
10:0	HUE_MID	U11, this is the offset from the min and max hue of skin region which marks the top 2 corners of the trapezoid

0x01A5526C MDSS_MDP_VP_0_DSPP_0_SKY_ADJUST_PWL1**Type:** RW**Reset State:** 0x00000000

Sky Piece-wise Linear Adjustment Curve

double buffered: Yes

MDSS_MDP_VP_0_DSPP_0_SKY_ADJUST_PWL1

Bits	Name	Description
31:16	VAL_GAIN	S16, is a symmetrical trapezoidal curve, this is the slope of the 2 sides
15:0	SAT_GAIN	S16, is a symmetrical trapezoidal curve, this is the slope of the 2 sides

0x01A55270 MDSS_MDP_VP_0_DSPP_0_SKY_REGION_HUE**Type:** RW**Reset State:** 0x00000000

Sky Adjustment Hue Qualifier

double buffered: Yes

MDSS_MDP_VP_0_DSPP_0_SKY_REGION_HUE

Bits	Name	Description
26:16	MAX	if MAX > x5ff (i.e. 360 degree), the behaviour will be same as MAX = x600
10:0	MIN	U11, Hue has to be within the inclusive range of MIN to MAX to qualify for sky tone adjustment on the pixel if MIN > x5ff (i.e. 360 degree), the behaviour will be same as MIN = x600

0x01A55274 MDSS_MDP_VP_0_DSPP_0_SKY_REGION_SAT**Type:** RW**Reset State:** 0x00000000

Sky Adjustment Saturation Qualifier

double buffered: Yes

MDSS_MDP_VP_0_DSPP_0_SKY_REGION_SAT

Bits	Name	Description
15:8	MAX	Saturation has to be below or equal to this to qualify for sky adjustment on the pixel
7:0	MIN	U8, Saturation has to be above or equal to this to qualify for sky adjustment on the pixel

0x01A55278 MDSS_MDP_VP_0_DSPP_0_SKY_REGION_VAL**Type:** RW**Reset State:** 0x00000000

Sky Adjustment Value Qualifier

double buffered: Yes

MDSS_MDP_VP_0_DSPP_0_SKY_REGION_VAL

Bits	Name	Description
15:8	MAX	Value has to be below or equal to this to qualify for sky adjustment on the pixel
7:0	MIN	U8, Value has to be above or equal to this to qualify for sky adjustment on the pixel

0x01A5527C MDSS_MDP_VP_0_DSPP_0_FOLIAGE_ADJUST_PWL0**Type:** RW**Reset State:** 0x00000000**MDSS_MDP_VP_0_DSPP_0_FOLIAGE_ADJUST_PWL0**

Bits	Name	Description
31:16	HUE_GAIN	
10:0	HUE_MID	

0x01A55280 MDSS_MDP_VP_0_DSPP_0_FOLIAGE_ADJUST_PWL1**Type:** RW**Reset State:** 0x00000000**MDSS_MDP_VP_0_DSPP_0_FOLIAGE_ADJUST_PWL1**

Bits	Name	Description
31:16	VAL_GAIN	
15:0	SAT_GAIN	

0x01A55284 MDSS_MDP_VP_0_DSPP_0_FOLIAGE_REGION_HUE**Type:** RW**Reset State:** 0x00000000

Foilage Adjustment Hue Qualifier

double buffered: Yes

MDSS_MDP_VP_0_DSPP_0_FOLIAGE_REGION_HUE

Bits	Name	Description
26:16	MAX	if MAX > x5ff (i.e. 360 degree), the behaviour will be same as MAX = x600
10:0	MIN	U11, Hue has to be within the inclusive range of MIN to MAX to qualify for foliage tone adjustment on the pixel if MIN > x5ff (i.e. 360 degree), the behaviour will be same as MIN = x600

0x01A55288 MDSS_MDP_VP_0_DSPP_0_FOLIAGE_REGION_SAT**Type:** RW**Reset State:** 0x00000000

Foliage Adjustment Saturation Qualifier

double buffered: Yes

MDSS_MDP_VP_0_DSPP_0_FOLIAGE_REGION_SAT

Bits	Name	Description
15:8	MAX	Saturation has to be below or equal to this to qualify for foliage adjustment on the pixel
7:0	MIN	U8, Saturation has to be above or equal to this to qualify for foliage adjustment on the pixel

0x01A5528C MDSS_MDP_VP_0_DSPP_0_FOLIAGE_REGION_VAL**Type:** RW**Reset State:** 0x00000000

Foliage Adjustment Value Qualifier

double buffered: Yes

MDSS_MDP_VP_0_DSPP_0_FOLIAGE_REGION_VAL

Bits	Name	Description
15:8	MAX	Value has to be below or equal to this to qualify for foliage adjustment on the pixel
7:0	MIN	U8, Value has to be above or equal to this to qualify for foliage adjustment on the pixel

0x01A552B0 MDSS_MDP_VP_0_DSPP_0_GC_ARLUT_RED_STARTX_N**Type:** RW**Reset State:** 0x00000000**MDSS_MDP_VP_0_DSPP_0_GC_ARLUT_RED_STARTX_N**

Bits	Name	Description
21	VALUE_UPDATEN	
20	SEGMENT_UPDATE	
19:16	SEGMENT	
11:0	STARTX	

0x01A552B4 MDSS_MDP_VP_0_DSPP_0_GC_ARLUT_RED_SLOPE_N**Type:** RW**Reset State:** 0x00000000**MDSS_MDP_VP_0_DSPP_0_GC_ARLUT_RED_SLOPE_N**

Bits	Name	Description
21	VALUE_UPDATEN	
20	SEGMENT_UPDATE	
19:16	SEGMENT	
14:0	SLOPE	

0x01A552B8 MDSS_MDP_VP_0_DSPP_0_GC_ARLUT_RED_OFFSET_N**Type:** RW**Reset State:** 0x00000000**MDSS_MDP_VP_0_DSPP_0_GC_ARLUT_RED_OFFSET_N**

Bits	Name	Description
21	VALUE_UPDATEN	
20	SEGMENT_UPDATE	
19:16	SEGMENT	
14:0	OFFSET	

0x01A552C0 MDSS_MDP_VP_0_DSPP_0_GC_ARLUT_GREEN_STARTX_N**Type:** RW**Reset State:** 0x00000000**MDSS_MDP_VP_0_DSPP_0_GC_ARLUT_GREEN_STARTX_N**

Bits	Name	Description
21	VALUE_UPDATEN	
20	SEGMENT_UPDATE	
19:16	SEGMENT	
11:0	STARTX	

0x01A552C4 MDSS_MDP_VP_0_DSPP_0_GC_ARLUT_GREEN_SLOPE_N**Type:** RW**Reset State:** 0x00000000**MDSS_MDP_VP_0_DSPP_0_GC_ARLUT_GREEN_SLOPE_N**

Bits	Name	Description
21	VALUE_UPDATEN	
20	SEGMENT_UPDATE	
19:16	SEGMENT	
14:0	SLOPE	

0x01A552C8 MDSS_MDP_VP_0_DSPP_0_GC_ARLUT_GREEN_OFFSET_N**Type:** RW**Reset State:** 0x00000000**MDSS_MDP_VP_0_DSPP_0_GC_ARLUT_GREEN_OFFSET_N**

Bits	Name	Description
21	VALUE_UPDATEN	
20	SEGMENT_UPDATE	
19:16	SEGMENT	
14:0	OFFSET	

0x01A552D0 MDSS_MDP_VP_0_DSPP_0_GC_ARLUT_BLUE_STARTX_N**Type:** RW**Reset State:** 0x00000000**MDSS_MDP_VP_0_DSPP_0_GC_ARLUT_BLUE_STARTX_N**

Bits	Name	Description
21	VALUE_UPDATEN	
20	SEGMENT_UPDATE	
19:16	SEGMENT	
11:0	STARTX	

0x01A552D4 MDSS_MDP_VP_0_DSPP_0_GC_ARLUT_BLUE_SLOPE_N**Type:** RW**Reset State:** 0x00000000**MDSS_MDP_VP_0_DSPP_0_GC_ARLUT_BLUE_SLOPE_N**

Bits	Name	Description
21	VALUE_UPDATEN	
20	SEGMENT_UPDATE	
19:16	SEGMENT	
14:0	SLOPE	

0x01A552D8 MDSS_MDP_VP_0_DSPP_0_GC_ARLUT_BLUE_OFFSET_N**Type:** RW**Reset State:** 0x00000000**MDSS_MDP_VP_0_DSPP_0_GC_ARLUT_BLUE_OFFSET_N**

Bits	Name	Description
21	VALUE_UPDATEN	
20	SEGMENT_UPDATE	
19:16	SEGMENT	
14:0	OFFSET	

**0x01A56000+ MDSS_MDP_VP_0_DSPP_0_HIST_V_DATAn_BIN_i, i=[0..255]
4*i****Type:** R**Reset State:** 0x00000000

Direct Register Mapped Histogram data collection

256 bins

double buffered: NO.

MDSS_MDP_VP_0_DSPP_0_HIST_V_DATAn_BIN_i

Bits	Name	Description
23:0	VALUE	Value

**0x01A56400+ MDSS_MDP_VP_0_DSPP_0_PA_LUTV_IDX_i, i=[0..127]
4*i****Type:** RW**Reset State:** 0x00000000

Direct Register Mapped PA LUTV data

256 entries are available for writing and reading

double buffered: NO

MDSS_MDP_VP_0_DSPP_0_PA_LUTV_IDX_i

Bits	Name	Description
23:16	VALUE_UPPER	LUT content at location 2*IDX+1 to be written/read to/from the LUT accessible by software at the moment
7:0	VALUE_LOWER	LUT content at location 2*IDX to be written/read to/from the LUT accessible by software at the moment

17.17.7.15 MDSS_MDP_VP_0_DSPP_1

DSPP hardware register description

17.17.7.15.1 MDSS_MDP_VP_0_DSPP_1_DSPP

DSPP

DSPP hardware register description

0x01A57000 MDSS_MDP_VP_0_DSPP_1_OP_MODE**Type:** RW**Reset State:** 0x00000000

DSPP Operation Control

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_DSPP_1_OP_MODE

Bits	Name	Description
31	SIX_ZONE_VAL_MASK	0x0: Disable six zone adj on VAL 0x1: Enable six zone adj on VAL
30	SIX_ZONE_SAT_MASK	0x0: Disable six zone adj on SAT 0x1: Enable six zone adj on SAT
29	SIX_ZONE_HUE_MASK	0x0: Disable six zone adj on HUE 0x1: Enable six zone adj on HUE
28	PA_CONT_MASK	0x0: Disable HSIC adj on CONTRAST 0x1: Enable HSIC adj on CONTRAST
27	PA_VAL_MASK	0x0: Disable HSIC adj on VALUE 0x1: Enable HSIC adj on VALUE
26	PA_SAT_MASK	0x0: Disable HSIC adj on SATURATION 0x1: Enable HSIC adj on SATURATION
25	PA_HUE_MASK	0x0: Disable HSIC adj on HUE 0x1: Enable HSIC adj on HUE
22	GC_LUT_EN	0x0: Disabled - mapped linearly from 12-bit to 8-bit 0x1: Enabled
20	PA_EN	Picture Adjustment enable 0x0: DISABLED 0x1: ENABLED
19	PA_LUTV_EN	Histogram Enhancement Lookup Table enable
16	HIST_EN	Histogram Enable
8	DST_DITHER_EN	Destination Dither enable 0x0: DISABLED 0x1: ENABLED
7	PA_MEM_COL_SKY_MASK	0x0: Disable Sky adjustment 0x1: Enable Sky adjustment
6	PA_MEM_COL_FOL_MASK	0x0: Disable Foliage adjustment 0x1: Enable Foliage adjustment

MDSS_MDP_VP_0_DSPP_1_OP_MODE (cont.)

Bits	Name	Description
5	PA_MEM_COL_SKIN_MASK	0x0: Disable Skin adjustment 0x1: Enable Skin adjustment
4	PCC_EN	Panel Color Correction enable 0x0: DISABLED 0x1: ENABLED
2	PA_MEM_PROTECT_ENABLE	0: global hsv adjustment is independent of memor color 1: global hsv adjustment is active if the pixel is not already processed by memory color contrast adjustment in global adjustment is always independ of memory color regardless
1	PA_SAT_ZERO_EXP_ENABLE	0: if PA_SAT_ADJ < -32, clamp PA_SAT_ADJ to 0 1: if PA_SAT_ADJ < -64, clamp PA_SAT_ADJ to 0
0	IGC_LUT_EN	Inverse Gamma Correction LUT enable 0x0: DISABLED 0x1: ENABLED

**0x01A57030+ MDSS_MDP_VP_0_DSPP_1_PCC_PLANEc_CONSTANT_COEFF, c=[0..2]
4*c****Type:** RW**Reset State:** 0x00000000

Constant Coefficient for RED/GREEN/BLUE plane

double buffered: NO

MDSS_MDP_VP_0_DSPP_1_PCC_PLANEc_CONSTANT_COEFF

Bits	Name	Description
15:0	COEFF	Constant Coefficient for RED/GREEN/BLUE plane in S13.3 format

**0x01A57040+ MDSS_MDP_VP_0_DSPP_1_PCC_PLANEc_R_COEFF, c=[0..2]
4*c****Type:** RW**Reset State:** 0x00000000

Red Coefficient for RED/GREEN/BLUE plane

double buffered: NO

MDSS_MDP_VP_0_DSPP_1_PCC_PLANEc_R_COEFF

Bits	Name	Description
17:0	COEFF	Red Coefficient for RED/GREEN/BLUE plane in S4.15 format

0x01A57050+ MDSS_MDP_VP_0_DSPP_1_PCC_PLANEc_G_COEFF, c=[0..2]
4*c

Type: RW

Reset State: 0x00000000

Red Coefficient for RED/GREEN/BLUE plane

double buffered: NO

MDSS_MDP_VP_0_DSPP_1_PCC_PLANEc_G_COEFF

Bits	Name	Description
17:0	COEFF	Green Coefficient for RED/GREEN/BLUE plane in S4.15 format

0x01A57060+ MDSS_MDP_VP_0_DSPP_1_PCC_PLANEc_B_COEFF, c=[0..2]
4*c

Type: RW

Reset State: 0x00000000

Red Coefficient for RED/GREEN/BLUE plane

double buffered: NO

MDSS_MDP_VP_0_DSPP_1_PCC_PLANEc_B_COEFF

Bits	Name	Description
17:0	COEFF	Blue Coefficient for RED/GREEN/BLUE plane in S4.15 format

0x01A57070+ MDSS_MDP_VP_0_DSPP_1_PCC_PLANEc_RR_COEFF, c=[0..2]
4*c

Type: RW

Reset State: 0x00000000

Red Red Coefficient for RED/GREEN/BLUE plane

double buffered: NO

MDSS_MDP_VP_0_DSPP_1_PCC_PLANEc_RR_COEFF

Bits	Name	Description
27:0	COEFF	RED RED Coefficient for RED/GREEN/BLUE plane. Total format is S1.27

**0x01A57080+ MDSS_MDP_VP_0_DSPP_1_PCC_PLANEc_RG_COEFF, c=[0..2]
4*c****Type:** RW**Reset State:** 0x00000000

Red Green Coefficient for RED/GREEN/BLUE plane

double buffered: NO

MDSS_MDP_VP_0_DSPP_1_PCC_PLANEc_RG_COEFF

Bits	Name	Description
27:0	COEFF	RED GREEN Coefficient for RED/GREEN/BLUE plane. Total format is S1.27

**0x01A57090+ MDSS_MDP_VP_0_DSPP_1_PCC_PLANEc_RB_COEFF, c=[0..2]
4*c****Type:** RW**Reset State:** 0x00000000

Red Blue Coefficient for RED/GREEN/BLUE plane

double buffered: NO

MDSS_MDP_VP_0_DSPP_1_PCC_PLANEc_RB_COEFF

Bits	Name	Description
27:0	COEFF	RED BLUE Coefficient for RED/GREEN/BLUE plane. Total format is S1.27

**0x01A570A0+ MDSS_MDP_VP_0_DSPP_1_PCC_PLANEc_GG_COEFF, c=[0..2]
4*c****Type:** RW**Reset State:** 0x00000000

Green Green Coefficient for RED/GREEN/BLUE plane

double buffered: NO

MDSS_MDP_VP_0_DSPP_1_PCC_PLANEc_GG_COEFF

Bits	Name	Description
27:0	COEFF	GREEN GREEN Coefficient for RED/GREEN/BLUE plane. Total format is S1.27

0x01A570B0+MDSS_MDP_VP_0_DSPP_1_PCC_PLANEc_BG_COEFF, c=[0..2]
4*c

Type: RW

Reset State: 0x00000000

Blue Green Coefficient for RED/GREEN/BLUE plane

double buffered: NO

MDSS_MDP_VP_0_DSPP_1_PCC_PLANEc_BG_COEFF

Bits	Name	Description
27:0	COEFF	BLUE GREEN Coefficient for RED/GREEN/BLUE plane. Total format is S1.27

0x01A570C0+MDSS_MDP_VP_0_DSPP_1_PCC_PLANEc_BB_COEFF, c=[0..2]
4*c

Type: RW

Reset State: 0x00000000

Blue Blue Coefficient for RED/GREEN/BLUE plane

double buffered: NO

MDSS_MDP_VP_0_DSPP_1_PCC_PLANEc_BB_COEFF

Bits	Name	Description
27:0	COEFF	BLUE BLUE Coefficient for RED/GREEN/BLUE plane. Total format is S1.27

0x01A570D0+MDSS_MDP_VP_0_DSPP_1_PCC_PLANEc_RGB_COEFF_0, c=[0..2]
4*c

Type: RW

Reset State: 0x00000000

Red Blue Green Coefficient for RED/GREEN/BLUE plane

double buffered: NO

MDSS_MDP_VP_0_DSPP_1_PCC_PLANEc_RGB_COEFF_0

Bits	Name	Description
7:0	COEFF	PCC coefficient data (sign + 7 fractional bits - S1.39) for RGB_0 and RGB_1 field

**0x01A570E0+ MDSS_MDP_VP_0_DSPP_1_PCC_PLANEc_RGB_COEFF_1, c=[0..2]
4*c**

Type: RW

Reset State: 0x00000000

Red Blue Green for RED/GREEN/BLUE plane

double buffered: NO

MDSS_MDP_VP_0_DSPP_1_PCC_PLANEc_RGB_COEFF_1

Bits	Name	Description
31:0	COEFF	PCC coefficient data (remaining 32 fractional bits - S1.39) for RGB_0 and RGB_1 field

0x01A57150 MDSS_MDP_VP_0_DSPP_1_DITHER_BITDEPTH

Type: RW

Reset State: 0x00000000

Dither Strength

double buffered: NO

MDSS_MDP_VP_0_DSPP_1_DITHER_BITDEPTH

Bits	Name	Description
5:4	C2_BITS	Number of MSB bits on component2 to indicate the dithering boundary 0x0: 4bits 0x1: 5bits 0x2: 6bits 0x3: 8bits
3:2	C1_BITS	Number of MSB bits on component1 to indicate the dithering boundary 0x0: 4bits 0x1: 5bits 0x2: 6bits 0x3: 8bits

MDSS_MDP_VP_0_DSPP_1_DITHER_BITDEPTH (cont.)

Bits	Name	Description
1:0	C0_BITS	Number of MSB bits on component0 to indicate the dithering boundary 0x0: 4bits 0x1: 5bits 0x2: 6bits 0x3: 8bits

0x01A57164 MDSS_MDP_VP_0_DSPP_1_DITHER_MATRIX_ROW0**Type:** RW**Reset State:** 0x00005D7F

Dither Strength Matrix Values Register for first row which covers entries [0-3]

double buffered: NO

MDSS_MDP_VP_0_DSPP_1_DITHER_MATRIX_ROW0

Bits	Name	Description
15:12	ENTRY_3	Entry 3 for the dither matrix
11:8	ENTRY_2	Entry 2 for the dither matrix
7:4	ENTRY_1	Entry 1 for the dither matrix
3:0	ENTRY_0	Entry 0 for the dither matrix

0x01A57168 MDSS_MDP_VP_0_DSPP_1_DITHER_MATRIX_ROW1**Type:** RW**Reset State:** 0x000091B3

Dither Strength Matrix Values Register for first row which covers entries [4-7]

double buffered: NO

MDSS_MDP_VP_0_DSPP_1_DITHER_MATRIX_ROW1

Bits	Name	Description
15:12	ENTRY_7	Entry 7 for the dither matrix
11:8	ENTRY_6	Entry 6 for the dither matrix
7:4	ENTRY_5	Entry 5 for the dither matrix
3:0	ENTRY_4	Entry 4 for the dither matrix

0x01A5716C MDSS_MDP_VP_0_DSPP_1_DITHER_MATRIX_ROW2**Type:** RW**Reset State:** 0x00006E4C

Dither Strength Matrix Values Register for first row which covers entries [8-11]

double buffered: NO

MDSS_MDP_VP_0_DSPP_1_DITHER_MATRIX_ROW2

Bits	Name	Description
15:12	ENTRY_11	Entry 11 for the dither matrix
11:8	ENTRY_10	Entry 10 for the dither matrix
7:4	ENTRY_9	Entry 9 for the dither matrix
3:0	ENTRY_8	Entry 8 for the dither matrix

0x01A57170 MDSS_MDP_VP_0_DSPP_1_DITHER_MATRIX_ROW3**Type:** RW**Reset State:** 0x0000A280

Dither Strength Matrix Values Register for first row which covers entries [12-15]

double buffered: NO

MDSS_MDP_VP_0_DSPP_1_DITHER_MATRIX_ROW3

Bits	Name	Description
15:12	ENTRY_15	Entry 15 for the dither matrix
11:8	ENTRY_14	Entry 14 for the dither matrix
7:4	ENTRY_13	Entry 13 for the dither matrix
3:0	ENTRY_12	Entry 12 for the dither matrix

0x01A57210 MDSS_MDP_VP_0_DSPP_1_HIST_COLLECT_CTL**Type:** RW**Reset State:** 0x00000000

Histogram Collection Control

double buffered: YES

except for LOCK

MDSS_MDP_VP_0_DSPP_1_HIST_COLLECT_CTL

Bits	Name	Description
0	LOCK	0: SW and HW swap their access on the 2 histogram tables; HW collects on a new table on the new frame, while SW accesses the histogram of the last frame on the table just swapped out by HW 1: current table will be overwritten for new collection on the next frame, the other table remains untouched for SW to continue to access

0x01A57214 MDSS_MDP_VP_0_DSPP_1_HIST_MIN_MAX**Type:** R**Reset State:** 0x00000000

MIN and MAX V value over the entire frame

double buffered: NO

MDSS_MDP_VP_0_DSPP_1_HIST_MIN_MAX

Bits	Name	Description
23:16	MAX	Maximum V value of the last collected frame
7:0	MIN	Minimum V value of the last collected frame

0x01A57218 MDSS_MDP_VP_0_DSPP_1_HIST_FRAME_COLLECTED**Type:** R**Reset State:** 0x00000000

Histogram Frame Number since the last rising START

double buffered: NO

MDSS_MDP_VP_0_DSPP_1_HIST_FRAME_COLLECTED

Bits	Name	Description
31:0	FRAME_NUM	The frame number that the current histogram accessible by SW was collected from

0x01A5721C MDSS_MDP_VP_0_DSPP_1_HIST_SEL**Type:** R**Reset State:** 0x00000000

PA LUTV and Histogram Table Selection

double buffered: NO

MDSS_MDP_VP_0_DSPP_1_HIST_SEL

Bits	Name	Description
4	HIST_SEL	0x0: Table 0 is available for SW and Table 1 is available for HW 0x1: Table 1 is available for SW and Table 0 is available for HW
0	PA_LUTV_SEL	0x0: LUT0 is available for SW and LUT1 is available for HW 0x1: LUT1 is available for SW and LUT0 is available for HW

0x01A57220 MDSS_MDP_VP_0_DSPP_1_HIST_SUM**Type:** R**Reset State:** 0x00000000

Accumulated V Value of the frame

double buffered: NO

MDSS_MDP_VP_0_DSPP_1_HIST_SUM

Bits	Name	Description
31:0	HIST_SUM	This sum matches the histogram currently accessible by SW

0x01A5722C MDSS_MDP_VP_0_DSPP_1_HIST_V_DATAN**Type:** RW**Reset State:** 0x00000000**MDSS_MDP_VP_0_DSPP_1_HIST_V_DATAN**

Bits	Name	Type	Description
31:24	INDEX	RW	
23:0	VALUE	R	

0x01A57230 MDSS_MDP_VP_0_DSPP_1_PA_LUTV**Type:** RW**Reset State:** 0x00000000

PA LUTV data

Auto-incremented by hardware

256 entries are available for reading

double buffered: NO

MDSS_MDP_VP_0_DSPP_1_PA_LUTV

Bits	Name	Description
17	VALUE_UPDATEN	Software issue an write to the LUT 0: VALUE will be written to RAM at INDEX if INDEX_UPDATE=1 or at INDEX+1 if INDEX_UPDATE=0 1: VALUE will not be written to RAM, this setting is appropriate for example when INDEX needs to be updated for read
16	INDEX_UPDATE	Software issue an index update 0: No index update by software 1: Update LUT index to value specified by INDEX
15:8	INDEX	VALUE is stored in this INDEX of the PA_LUTV, next write will be stored to the next higher INDEX
7:0	VALUE	Value read depends on the LUT select and which one is available for reading by software The read sequence to the LUT should always begin with WR INDEX_UPDATE=1, INDEX=<rptr start location> subsequent RD will start from the start location and every read will auto-increment the rptr

0x01A57234 MDSS_MDP_VP_0_DSPP_1_PA_LUTV_SWAP**Type:** W**Reset State:** 0x00000000

Histogram SWAP signal to indicate that the PA_LUTV contents should be swapped

Software accesses one LUT while HW accesses the other LUT

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_DSPP_1_PA_LUTV_SWAP

Bits	Name	Description
0	SWAP	SWAP signal to flip buffers accessible by SW and made available to HW. Achieves double buffering of the LUT values

0x01A57238 MDSS_MDP_VP_0_DSPP_1_PA_HUE_ADJUST**Type:** RW**Reset State:** 0x00000000

Picture HUE Adjustment

double buffered: YES

MDSS_MDP_VP_0_DSPP_1_PA_HUE_ADJUST

Bits	Name	Description
11:0	VALUE	Value in S12, recommend to be within exclusive range xA00 to x600 (i.e. -360 to +360 degree) x600 is equivalent to a shift of 360 degree if VALUE > x5FF in either negative or positive direction, it is circularly clipped (i.e. VALUE + x600 or VALUE - x600)

0x01A5723C MDSS_MDP_VP_0_DSPP_1_PA_SAT_ADJUST**Type:** RW**Reset State:** 0x00001A00

Picture Saturation Adjustment

double buffered: YES

MDSS_MDP_VP_0_DSPP_1_PA_SAT_ADJUST

Bits	Name	Description
15:8	THRSHLD	Below this threshold of saturation, no adjustment is applied to saturation. This field is in U8
7:0	VALUE	Value in S8

0x01A57240 MDSS_MDP_VP_0_DSPP_1_PA_VAL_ADJUST**Type:** RW**Reset State:** 0x00000000

Picture Adjustment

double buffered: YES

MDSS_MDP_VP_0_DSPP_1_PA_VAL_ADJUST

Bits	Name	Description
7:0	VALUE	Value in S8

0x01A57244 MDSS_MDP_VP_0_DSPP_1_PA_CONT_ADJUST**Type:** RW**Reset State:** 0x00000000

Picture Contrast Adjustment

double buffered: YES

MDSS_MDP_VP_0_DSPP_1_PA_CONT_ADJUST

Bits	Name	Description
7:0	VALUE	Value in S8

0x01A57248 MDSS_MDP_VP_0_DSPP_1_SIX_ZONE_ADJ_CURVE_P0**Type:** RW**Reset State:** 0x00000000

Six Zone Adjust Curve Part 0

double buffered: No

MDSS_MDP_VP_0_DSPP_1_SIX_ZONE_ADJ_CURVE_P0

Bits	Name	Description
26	INDEX_UPDATE	Set this to 1 to point current LUT location to INDEX. The LUT location is autoincremented on every value write or read when there is no INDEX_UPDATE Before every series of read, there must be an INDEX_UPDATE
25	VALUE_UPDATEN	Set this to 0 to enable VALUE to be written to the current location of the LUT

MDSS_MDP_VP_0_DSPP_1_SIX_ZONE_ADJ_CURVE_P0 (cont.)

Bits	Name	Description
24:16	INDEX	Index to one of the 384 location of the LUT
11:0	HUE	S12, recommend to be within exclusive range x2800 and x1800 (i.e. -360 to +360 exclusive) circularly clipped otherwise On write: Value to be updated to one of the 384 entry LUT On read : Value read out from LUT location pointed by INDEX

0x01A5724C MDSS_MDP_VP_0_DSPP_1_SIX_ZONE_ADJ_CURVE_P1**Type:** RW**Reset State:** 0x00000000

Six Zone Adjust Curve Part 1

double buffered: No

MDSS_MDP_VP_0_DSPP_1_SIX_ZONE_ADJ_CURVE_P1

Bits	Name	Description
27:16	SAT	S2.10, On write: Value to be updated to one of the 384 entry LUT On read : Value read out from LUT location pointed by INDEX
11:0	VALUE	S2.10, On write: Value to be updated to one of the 384 entry LUT On read : Value read out from LUT location pointed by INDEX

0x01A57250 MDSS_MDP_VP_0_DSPP_1_SIX_ZONE_THRESHOLDS**Type:** RW**Reset State:** 0x00000000

Six Zone Threshold Qualifier

double buffered: Yes

MDSS_MDP_VP_0_DSPP_1_SIX_ZONE_THRESHOLDS

Bits	Name	Description
23:16	VAL_MAX	U8, Value has to be within the inclusive range of MIN to MAX to qualify for six zone adjustment on the pixel
15:8	VAL_MIN	
7:0	SAT_MIN	U8, Saturation has to be above or equal to this threshold to qualify for six zone adjustment

0x01A57254 MDSS_MDP_VP_0_DSPP_1_SKIN_ADJUST_PWL0**Type:** RW**Reset State:** 0x00000000

Skin Piece-wise Linear Adjustment Curve

double buffered: Yes

MDSS_MDP_VP_0_DSPP_1_SKIN_ADJUST_PWL0

Bits	Name	Description
31:16	HUE_GAIN	S16, is a symmetrical trapezoidal curve, this is the slope of the 2 sides
10:0	HUE_MID	U11, this is the offset from the min and max hue of skin region which marks the top 2 corners of the trapezoid

0x01A57258 MDSS_MDP_VP_0_DSPP_1_SKIN_ADJUST_PWL1**Type:** RW**Reset State:** 0x00000000

Skin Piece-wise Linear Adjustment Curve

double buffered: Yes

MDSS_MDP_VP_0_DSPP_1_SKIN_ADJUST_PWL1

Bits	Name	Description
31:16	VAL_GAIN	S16, is a symmetrical trapezoidal curve, this is the slope of the 2 sides
15:0	SAT_GAIN	S16, is a symmetrical trapezoidal curve, this is the slope of the 2 sides

0x01A5725C MDSS_MDP_VP_0_DSPP_1_SKIN_REGION_HUE**Type:** RW**Reset State:** 0x00000000

Skin Adjustment Hue Qualifier

double buffered: Yes

MDSS_MDP_VP_0_DSPP_1_SKIN_REGION_HUE

Bits	Name	Description
26:16	MAX	if MAX > x5ff (i.e. 360 degree), the behaviour will be same as MAX = x600
10:0	MIN	U11, Hue has to be within the inclusive range of MIN to MAX to qualify for skin tone adjustment on the pixel if MIN > x5ff (i.e. 360 degree), the behaviour will be same as MIN = x600

0x01A57260 MDSS_MDP_VP_0_DSPP_1_SKIN_REGION_SAT**Type:** RW**Reset State:** 0x00000000

Skin Adjustment Saturation Qualifier

double buffered: Yes

MDSS_MDP_VP_0_DSPP_1_SKIN_REGION_SAT

Bits	Name	Description
15:8	MAX	Saturation has to be below or equal to this to qualify for skin tone adjustment on the pixel
7:0	MIN	U8, Saturation has to be above or equal to this to qualify for skin tone adjustment on the pixel

0x01A57264 MDSS_MDP_VP_0_DSPP_1_SKIN_REGION_VAL**Type:** RW**Reset State:** 0x00000000

Skin Adjustment Value Qualifier

double buffered: Yes

MDSS_MDP_VP_0_DSPP_1_SKIN_REGION_VAL

Bits	Name	Description
15:8	MAX	Value has to be below or equal to this to qualify for skin tone adjustment on the pixel
7:0	MIN	U8, Value has to be above or equal to this to qualify for skin tone adjustment on the pixel

0x01A57268 MDSS_MDP_VP_0_DSPP_1_SKY_ADJUST_PWL0**Type:** RW**Reset State:** 0x00000000

Sky Piece-wise Linear Adjustment Curve

double buffered: Yes

MDSS_MDP_VP_0_DSPP_1_SKY_ADJUST_PWL0

Bits	Name	Description
31:16	HUE_GAIN	S16, is a symmetrical trapezoidal curve, this is the slope of the 2 sides
10:0	HUE_MID	U11, this is the offset from the min and max hue of skin region which marks the top 2 corners of the trapezoid

0x01A5726C MDSS_MDP_VP_0_DSPP_1_SKY_ADJUST_PWL1**Type:** RW**Reset State:** 0x00000000

Sky Piece-wise Linear Adjustment Curve

double buffered: Yes

MDSS_MDP_VP_0_DSPP_1_SKY_ADJUST_PWL1

Bits	Name	Description
31:16	VAL_GAIN	S16, is a symmetrical trapezoidal curve, this is the slope of the 2 sides
15:0	SAT_GAIN	S16, is a symmetrical trapezoidal curve, this is the slope of the 2 sides

0x01A57270 MDSS_MDP_VP_0_DSPP_1_SKY_REGION_HUE**Type:** RW**Reset State:** 0x00000000

Sky Adjustment Hue Qualifier

double buffered: Yes

MDSS_MDP_VP_0_DSPP_1_SKY_REGION_HUE

Bits	Name	Description
26:16	MAX	if MAX > x5ff (i.e. 360 degree), the behaviour will be same as MAX = x600
10:0	MIN	U11, Hue has to be within the inclusive range of MIN to MAX to qualify for sky tone adjustment on the pixel if MIN > x5ff (i.e. 360 degree), the behaviour will be same as MIN = x600

0x01A57274 MDSS_MDP_VP_0_DSPP_1_SKY_REGION_SAT**Type:** RW**Reset State:** 0x00000000

Sky Adjustment Saturation Qualifier

double buffered: Yes

MDSS_MDP_VP_0_DSPP_1_SKY_REGION_SAT

Bits	Name	Description
15:8	MAX	Saturation has to be below or equal to this to qualify for sky adjustment on the pixel
7:0	MIN	U8, Saturation has to be above or equal to this to qualify for sky adjustment on the pixel

0x01A57278 MDSS_MDP_VP_0_DSPP_1_SKY_REGION_VAL**Type:** RW**Reset State:** 0x00000000

Sky Adjustment Value Qualifier

double buffered: Yes

MDSS_MDP_VP_0_DSPP_1_SKY_REGION_VAL

Bits	Name	Description
15:8	MAX	Value has to be below or equal to this to qualify for sky adjustment on the pixel
7:0	MIN	U8, Value has to be above or equal to this to qualify for sky adjustment on the pixel

0x01A5727C MDSS_MDP_VP_0_DSPP_1_FOLIAGE_ADJUST_PWL0**Type:** RW**Reset State:** 0x00000000**MDSS_MDP_VP_0_DSPP_1_FOLIAGE_ADJUST_PWL0**

Bits	Name	Description
31:16	HUE_GAIN	
10:0	HUE_MID	

0x01A57280 MDSS_MDP_VP_0_DSPP_1_FOLIAGE_ADJUST_PWL1**Type:** RW**Reset State:** 0x00000000**MDSS_MDP_VP_0_DSPP_1_FOLIAGE_ADJUST_PWL1**

Bits	Name	Description
31:16	VAL_GAIN	
15:0	SAT_GAIN	

0x01A57284 MDSS_MDP_VP_0_DSPP_1_FOLIAGE_REGION_HUE**Type:** RW**Reset State:** 0x00000000

Foliage Adjustment Hue Qualifier

double buffered: Yes

MDSS_MDP_VP_0_DSPP_1_FOLIAGE_REGION_HUE

Bits	Name	Description
26:16	MAX	if MAX > x5ff (i.e. 360 degree), the behaviour will be same as MAX = x600
10:0	MIN	U11, Hue has to be within the inclusive range of MIN to MAX to qualify for foliage tone adjustment on the pixel if MIN > x5ff (i.e. 360 degree), the behaviour will be same as MIN = x600

0x01A57288 MDSS_MDP_VP_0_DSPP_1_FOLIAGE_REGION_SAT**Type:** RW**Reset State:** 0x00000000

Foilage Adjustment Saturation Qualifier

double buffered: Yes

MDSS_MDP_VP_0_DSPP_1_FOLIAGE_REGION_SAT

Bits	Name	Description
15:8	MAX	Saturation has to be below or equal to this to qualify for foilage adjustment on the pixel
7:0	MIN	U8, Saturation has to be above or equal to this to qualify for foilage adjustment on the pixel

0x01A5728C MDSS_MDP_VP_0_DSPP_1_FOLIAGE_REGION_VAL**Type:** RW**Reset State:** 0x00000000

Foilage Adjustment Value Qualifier

double buffered: Yes

MDSS_MDP_VP_0_DSPP_1_FOLIAGE_REGION_VAL

Bits	Name	Description
15:8	MAX	Value has to be below or equal to this to qualify for foilage adjustment on the pixel
7:0	MIN	U8, Value has to be above or equal to this to qualify for foilage adjustment on the pixel

0x01A572B0 MDSS_MDP_VP_0_DSPP_1_GC_ARLUT_RED_STARTX_N**Type:** RW**Reset State:** 0x00000000**MDSS_MDP_VP_0_DSPP_1_GC_ARLUT_RED_STARTX_N**

Bits	Name	Description
21	VALUE_UPDATEN	
20	SEGMENT_UPDATE	
19:16	SEGMENT	

MDSS_MDP_VP_0_DSPP_1_GC_ARLUT_RED_STARTX_N (cont.)

Bits	Name	Description
11:0	STARTX	

0x01A572B4 MDSS_MDP_VP_0_DSPP_1_GC_ARLUT_RED_SLOPE_N**Type:** RW**Reset State:** 0x00000000**MDSS_MDP_VP_0_DSPP_1_GC_ARLUT_RED_SLOPE_N**

Bits	Name	Description
21	VALUE_UPDATEN	
20	SEGMENT_UPDATE	
19:16	SEGMENT	
14:0	SLOPE	

0x01A572B8 MDSS_MDP_VP_0_DSPP_1_GC_ARLUT_RED_OFFSET_N**Type:** RW**Reset State:** 0x00000000**MDSS_MDP_VP_0_DSPP_1_GC_ARLUT_RED_OFFSET_N**

Bits	Name	Description
21	VALUE_UPDATEN	
20	SEGMENT_UPDATE	
19:16	SEGMENT	
14:0	OFFSET	

0x01A572C0 MDSS_MDP_VP_0_DSPP_1_GC_ARLUT_GREEN_STARTX_N**Type:** RW**Reset State:** 0x00000000**MDSS_MDP_VP_0_DSPP_1_GC_ARLUT_GREEN_STARTX_N**

Bits	Name	Description
21	VALUE_UPDATEN	
20	SEGMENT_UPDATE	

MDSS_MDP_VP_0_DSPP_1_GC_ARLUT_GREEN_STARTX_N (cont.)

Bits	Name	Description
19:16	SEGMENT	
11:0	STARTX	

0x01A572C4 MDSS_MDP_VP_0_DSPP_1_GC_ARLUT_GREEN_SLOPE_N**Type:** RW**Reset State:** 0x00000000**MDSS_MDP_VP_0_DSPP_1_GC_ARLUT_GREEN_SLOPE_N**

Bits	Name	Description
21	VALUE_UPDATEN	
20	SEGMENT_UPDATE	
19:16	SEGMENT	
14:0	SLOPE	

0x01A572C8 MDSS_MDP_VP_0_DSPP_1_GC_ARLUT_GREEN_OFFSET_N**Type:** RW**Reset State:** 0x00000000**MDSS_MDP_VP_0_DSPP_1_GC_ARLUT_GREEN_OFFSET_N**

Bits	Name	Description
21	VALUE_UPDATEN	
20	SEGMENT_UPDATE	
19:16	SEGMENT	
14:0	OFFSET	

0x01A572D0 MDSS_MDP_VP_0_DSPP_1_GC_ARLUT_BLUE_STARTX_N**Type:** RW**Reset State:** 0x00000000**MDSS_MDP_VP_0_DSPP_1_GC_ARLUT_BLUE_STARTX_N**

Bits	Name	Description
21	VALUE_UPDATEN	

MDSS_MDP_VP_0_DSPP_1_GC_ARLUT_BLUE_STARTX_N (cont.)

Bits	Name	Description
20	SEGMENT_UPDATE	
19:16	SEGMENT	
11:0	STARTX	

0x01A572D4 MDSS_MDP_VP_0_DSPP_1_GC_ARLUT_BLUE_SLOPE_N**Type:** RW**Reset State:** 0x00000000**MDSS_MDP_VP_0_DSPP_1_GC_ARLUT_BLUE_SLOPE_N**

Bits	Name	Description
21	VALUE_UPDATEN	
20	SEGMENT_UPDATE	
19:16	SEGMENT	
14:0	SLOPE	

0x01A572D8 MDSS_MDP_VP_0_DSPP_1_GC_ARLUT_BLUE_OFFSET_N**Type:** RW**Reset State:** 0x00000000**MDSS_MDP_VP_0_DSPP_1_GC_ARLUT_BLUE_OFFSET_N**

Bits	Name	Description
21	VALUE_UPDATEN	
20	SEGMENT_UPDATE	
19:16	SEGMENT	
14:0	OFFSET	

**0x01A58000+ MDSS_MDP_VP_0_DSPP_1_HIST_V_DATA_n_BIN_i, i=[0..255]
4*i*****Type:** R**Reset State:** 0x00000000

Direct Register Mapped Histogram data collection

256 bins

double buffered: NO.

MDSS_MDP_VP_0_DSPP_1_HIST_V_DATA_n_BIN_i

Bits	Name	Description
23:0	VALUE	Value

0x01A58400+ MDSS_MDP_VP_0_DSPP_1_PA_LUTV_IDX_i, i=[0..127] 4*i*

Type: RW

Reset State: 0x00000000

Direct Register Mapped PA LUTV data

256 entries are available for writing and reading

double buffered: NO

MDSS_MDP_VP_0_DSPP_1_PA_LUTV_IDX_i

Bits	Name	Description
23:16	VALUE_UPPER	LUT content at location 2*IDX+1 to be written/read to/from the LUT accessible by software at the moment
7:0	VALUE_LOWER	LUT content at location 2*IDX to be written/read to/from the LUT accessible by software at the moment

17.17.7.16 MDSS_MDP_VP_0_DSPP_2

DSPP hardware register description

17.17.7.16.1 MDSS_MDP_VP_0_DSPP_2_DSPP

DSPP

DSPP hardware register description

0x01A59000 MDSS_MDP_VP_0_DSPP_2_OP_MODE

Type: RW

Reset State: 0x00000000

DSPP Operation Control

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_DSPP_2_OP_MODE

Bits	Name	Description
31	SIX_ZONE_VAL_MASK	0x0: Disable six zone adj on VAL 0x1: Enable six zone adj on VAL
30	SIX_ZONE_SAT_MASK	0x0: Disable six zone adj on SAT 0x1: Enable six zone adj on SAT
29	SIX_ZONE_HUE_MASK	0x0: Disable six zone adj on HUE 0x1: Enable six zone adj on HUE
28	PA_CONT_MASK	0x0: Disable HSIC adj on CONTRAST 0x1: Enable HSIC adj on CONTRAST
27	PA_VAL_MASK	0x0: Disable HSIC adj on VALUE 0x1: Enable HSIC adj on VALUE
26	PA_SAT_MASK	0x0: Disable HSIC adj on SATURATION 0x1: Enable HSIC adj on SATURATION
25	PA_HUE_MASK	0x0: Disable HSIC adj on HUE 0x1: Enable HSIC adj on HUE
22	GC_LUT_EN	0x0: Disabled - mapped linearly from 12-bit to 8-bit 0x1: Enabled
20	PA_EN	Picture Adjustment enable 0x0: DISABLED 0x1: ENABLED
19	PA_LUTV_EN	Histogram Enhancement Lookup Table enable
16	HIST_EN	Histogram Enable
8	DST_DITHER_EN	Destination Dither enable 0x0: DISABLED 0x1: ENABLED
7	PA_MEM_COL_SKY_MASK	0x0: Disable Sky adjustment 0x1: Enable Sky adjustment
6	PA_MEM_COL_FOL_MASK	0x0: Disable Foliage adjustment 0x1: Enable Foliage adjustment
5	PA_MEM_COL_SKIN_MASK	0x0: Disable Skin adjustment 0x1: Enable Skin adjustment
4	PCC_EN	Panel Color Correction enable 0x0: DISABLED 0x1: ENABLED
2	PA_MEM_PROTECT_ENABLE	0: global hsv adjustment is independent of memor color 1: global hsv adjustment is active if the pixel is not already processed by memory color contrast adjustment in global adjustment is always independ of memory color regardless

MDSS_MDP_VP_0_DSPP_2_OP_MODE (cont.)

Bits	Name	Description
1	PA_SAT_ZERO_EXP_ENAB LE	0: if PA_SAT_ADJ < -32, clamp PA_SAT_ADJ to 0 1: if PA_SAT_ADJ < -64, clamp PA_SAT_ADJ to 0
0	IGC_LUT_EN	Inverse Gamma Correction LUT enable 0x0: DISABLED 0x1: ENABLED

**0x01A59030+ MDSS_MDP_VP_0_DSPP_2_PCC_PLANEc_CONSTANT_COEFF, c=[0..2]
4*c****Type:** RW**Reset State:** 0x00000000

Constant Coefficient for RED/GREEN/BLUE plane

double buffered: NO

MDSS_MDP_VP_0_DSPP_2_PCC_PLANEc_CONSTANT_COEFF

Bits	Name	Description
15:0	COEFF	Constant Coefficient for RED/GREEN/BLUE plane in S13.3 format

**0x01A59040+ MDSS_MDP_VP_0_DSPP_2_PCC_PLANEc_R_COEFF, c=[0..2]
4*c****Type:** RW**Reset State:** 0x00000000

Red Coefficient for RED/GREEN/BLUE plane

double buffered: NO

MDSS_MDP_VP_0_DSPP_2_PCC_PLANEc_R_COEFF

Bits	Name	Description
17:0	COEFF	Red Coefficient for RED/GREEN/BLUE plane in S4.15 format

**0x01A59050+ MDSS_MDP_VP_0_DSPP_2_PCC_PLANEc_G_COEFF, c=[0..2]
4*c****Type:** RW**Reset State:** 0x00000000

Red Coefficient for RED/GREEN/BLUE plane

double buffered: NO

MDSS_MDP_VP_0_DSPP_2_PCC_PLANEc_G_COEFF

Bits	Name	Description
17:0	COEFF	Green Coefficient for RED/GREEN/BLUE plane in S4.15 format

0x01A59060+ MDSS_MDP_VP_0_DSPP_2_PCC_PLANEc_B_COEFF, c=[0..2]
4*c

Type: RW

Reset State: 0x00000000

Red Coefficient for RED/GREEN/BLUE plane

double buffered: NO

MDSS_MDP_VP_0_DSPP_2_PCC_PLANEc_B_COEFF

Bits	Name	Description
17:0	COEFF	Blue Coefficient for RED/GREEN/BLUE plane in S4.15 format

0x01A59070+ MDSS_MDP_VP_0_DSPP_2_PCC_PLANEc_RR_COEFF, c=[0..2]
4*c

Type: RW

Reset State: 0x00000000

Red Red Coefficient for RED/GREEN/BLUE plane

double buffered: NO

MDSS_MDP_VP_0_DSPP_2_PCC_PLANEc_RR_COEFF

Bits	Name	Description
27:0	COEFF	RED RED Coefficient for RED/GREEN/BLUE plane. Total format is S1.27

0x01A59080+ MDSS_MDP_VP_0_DSPP_2_PCC_PLANEc_RG_COEFF, c=[0..2]
4*c

Type: RW

Reset State: 0x00000000

Red Green Coefficient for RED/GREEN/BLUE plane

double buffered: NO

MDSS_MDP_VP_0_DSPP_2_PCC_PLANEc_RG_COEFF

Bits	Name	Description
27:0	COEFF	RED GREEN Coefficient for RED/GREEN/BLUE plane. Total format is S1.27

**0x01A59090+ MDSS_MDP_VP_0_DSPP_2_PCC_PLANEc_RB_COEFF, c=[0..2]
4*c****Type:** RW**Reset State:** 0x00000000

Red Blue Coefficient for RED/GREEN/BLUE plane

double buffered: NO

MDSS_MDP_VP_0_DSPP_2_PCC_PLANEc_RB_COEFF

Bits	Name	Description
27:0	COEFF	RED BLUE Coefficient for RED/GREEN/BLUE plane. Total format is S1.27

**0x01A590A0+ MDSS_MDP_VP_0_DSPP_2_PCC_PLANEc_GG_COEFF, c=[0..2]
4*c****Type:** RW**Reset State:** 0x00000000

Green Green Coefficient for RED/GREEN/BLUE plane

double buffered: NO

MDSS_MDP_VP_0_DSPP_2_PCC_PLANEc_GG_COEFF

Bits	Name	Description
27:0	COEFF	GREEN GREEN Coefficient for RED/GREEN/BLUE plane. Total format is S1.27

**0x01A590B0+ MDSS_MDP_VP_0_DSPP_2_PCC_PLANEc_BG_COEFF, c=[0..2]
4*c****Type:** RW**Reset State:** 0x00000000

Blue Green Coefficient for RED/GREEN/BLUE plane

double buffered: NO

MDSS_MDP_VP_0_DSPP_2_PCC_PLANEc_BG_COEFF

Bits	Name	Description
27:0	COEFF	BLUE GREEN Coefficient for RED/GREEN/BLUE plane. Total format is S1.27

0x01A590C0+MDSS_MDP_VP_0_DSPP_2_PCC_PLANEc_BB_COEFF, c=[0..2]
4*c

Type: RW

Reset State: 0x00000000

Blue Blue Coefficient for RED/GREEN/BLUE plane

double buffered: NO

MDSS_MDP_VP_0_DSPP_2_PCC_PLANEc_BB_COEFF

Bits	Name	Description
27:0	COEFF	BLUE BLUE Coefficient for RED/GREEN/BLUE plane. Total format is S1.27

0x01A590D0+MDSS_MDP_VP_0_DSPP_2_PCC_PLANEc_RGB_COEFF_0, c=[0..2]
4*c

Type: RW

Reset State: 0x00000000

Red Blue Green Coefficient for RED/GREEN/BLUE plane

double buffered: NO

MDSS_MDP_VP_0_DSPP_2_PCC_PLANEc_RGB_COEFF_0

Bits	Name	Description
7:0	COEFF	PCC coefficient data (sign + 7 fractional bits - S1.39) for RGB_0 and RGB_1 field

0x01A590E0+MDSS_MDP_VP_0_DSPP_2_PCC_PLANEc_RGB_COEFF_1, c=[0..2]
4*c

Type: RW

Reset State: 0x00000000

Red Blue Green for RED/GREEN/BLUE plane

double buffered: NO

MDSS_MDP_VP_0_DSPP_2_PCC_PLANEc_RGB_COEFF_1

Bits	Name	Description
31:0	COEFF	PCC coefficient data (remaining 32 fractional bits - S1.39) for RGB_0 and RGB_1 field

0x01A59150 MDSS_MDP_VP_0_DSPP_2_DITHER_BITDEPTH**Type:** RW**Reset State:** 0x00000000

Dither Strength

double buffered: NO

MDSS_MDP_VP_0_DSPP_2_DITHER_BITDEPTH

Bits	Name	Description
5:4	C2_BITS	Number of MSB bits on component2 to indicate the dithering boundary 0x0: 4bits 0x1: 5bits 0x2: 6bits 0x3: 8bits
3:2	C1_BITS	Number of MSB bits on component1 to indicate the dithering boundary 0x0: 4bits 0x1: 5bits 0x2: 6bits 0x3: 8bits
1:0	C0_BITS	Number of MSB bits on component0 to indicate the dithering boundary 0x0: 4bits 0x1: 5bits 0x2: 6bits 0x3: 8bits

0x01A59164 MDSS_MDP_VP_0_DSPP_2_DITHER_MATRIX_ROW0**Type:** RW**Reset State:** 0x00005D7F

Dither Strength Matrix Values Register for first row which covers entries [0-3]

double buffered: NO

MDSS_MDP_VP_0_DSPP_2_DITHER_MATRIX_ROW0

Bits	Name	Description
15:12	ENTRY_3	Entry 3 for the dither matrix
11:8	ENTRY_2	Entry 2 for the dither matrix
7:4	ENTRY_1	Entry 1 for the dither matrix
3:0	ENTRY_0	Entry 0 for the dither matrix

0x01A59168 MDSS_MDP_VP_0_DSPP_2_DITHER_MATRIX_ROW1**Type:** RW**Reset State:** 0x000091B3

Dither Strength Matrix Values Register for first row which covers entries [4-7]

double buffered: NO

MDSS_MDP_VP_0_DSPP_2_DITHER_MATRIX_ROW1

Bits	Name	Description
15:12	ENTRY_7	Entry 7 for the dither matrix
11:8	ENTRY_6	Entry 6 for the dither matrix
7:4	ENTRY_5	Entry 5 for the dither matrix
3:0	ENTRY_4	Entry 4 for the dither matrix

0x01A5916C MDSS_MDP_VP_0_DSPP_2_DITHER_MATRIX_ROW2**Type:** RW**Reset State:** 0x00006E4C

Dither Strength Matrix Values Register for first row which covers entries [8-11]

double buffered: NO

MDSS_MDP_VP_0_DSPP_2_DITHER_MATRIX_ROW2

Bits	Name	Description
15:12	ENTRY_11	Entry 11 for the dither matrix
11:8	ENTRY_10	Entry 10 for the dither matrix
7:4	ENTRY_9	Entry 9 for the dither matrix
3:0	ENTRY_8	Entry 8 for the dither matrix

0x01A59170 MDSS_MDP_VP_0_DSPP_2_DITHER_MATRIX_ROW3**Type:** RW**Reset State:** 0x0000A280

Dither Strength Matrix Values Register for first row which covers entries [12-15]

double buffered: NO

MDSS_MDP_VP_0_DSPP_2_DITHER_MATRIX_ROW3

Bits	Name	Description
15:12	ENTRY_15	Entry 15 for the dither matrix
11:8	ENTRY_14	Entry 14 for the dither matrix
7:4	ENTRY_13	Entry 13 for the dither matrix
3:0	ENTRY_12	Entry 12 for the dither matrix

0x01A59210 MDSS_MDP_VP_0_DSPP_2_HIST_COLLECT_CTL**Type:** RW**Reset State:** 0x00000000

Histogram Collection Control

double buffered: YES

except for LOCK

MDSS_MDP_VP_0_DSPP_2_HIST_COLLECT_CTL

Bits	Name	Description
0	LOCK	0: SW and HW swap their access on the 2 histogram tables; HW collects on a new table on the new frame, while SW accesses the histogram of the last frame on the table just swapped out by HW 1: current table will be overwritten for new collection on the next frame, the other table remains untouched for SW to continue to access

0x01A59214 MDSS_MDP_VP_0_DSPP_2_HIST_MIN_MAX**Type:** R**Reset State:** 0x00000000

MIN and MAX V value over the entire frame

double buffered: NO

MDSS_MDP_VP_0_DSPP_2_HIST_MIN_MAX

Bits	Name	Description
23:16	MAX	Maximum V value of the last collected frame
7:0	MIN	Minimum V value of the last collected frame

0x01A59218 MDSS_MDP_VP_0_DSPP_2_HIST_FRAME_COLLECTED

Type: R

Reset State: 0x00000000

Histogram Frame Number since the last rising START

double buffered: NO

MDSS_MDP_VP_0_DSPP_2_HIST_FRAME_COLLECTED

Bits	Name	Description
31:0	FRAME_NUM	The frame number that the current histogram accessible by SW was collected from

0x01A5921C MDSS_MDP_VP_0_DSPP_2_HIST_SEL

Type: R

Reset State: 0x00000000

PA LUTV and Histogram Table Selection

double buffered: NO

MDSS_MDP_VP_0_DSPP_2_HIST_SEL

Bits	Name	Description
4	HIST_SEL	0x0: Table 0 is available for SW and Table 1 is available for HW 0x1: Table 1 is available for SW and Table 0 is available for HW
0	PA_LUTV_SEL	0x0: LUT0 is available for SW and LUT1 is available for HW 0x1: LUT1 is available for SW and LUT0 is available for HW

0x01A59220 MDSS_MDP_VP_0_DSPP_2_HIST_SUM**Type:** R**Reset State:** 0x00000000

Accumulated V Value of the frame

double buffered: NO

MDSS_MDP_VP_0_DSPP_2_HIST_SUM

Bits	Name	Description
31:0	HIST_SUM	This sum matches the histogram currently accessible by SW

0x01A5922C MDSS_MDP_VP_0_DSPP_2_HIST_V_DATAN**Type:** RW**Reset State:** 0x00000000**MDSS_MDP_VP_0_DSPP_2_HIST_V_DATAN**

Bits	Name	Type	Description
31:24	INDEX	RW	
23:0	VALUE	R	

0x01A59230 MDSS_MDP_VP_0_DSPP_2_PA_LUTV**Type:** RW**Reset State:** 0x00000000

PA LUTV data

Auto-incremented by hardware

256 entries are available for reading

double buffered: NO

MDSS_MDP_VP_0_DSPP_2_PA_LUTV

Bits	Name	Description
17	VALUE_UPDATEN	Software issue an write to the LUT 0: VALUE will be written to RAM at INDEX if INDEX_UPDATE=1 or at INDEX+1 if INDEX_UPDATE=0 1: VALUE will not be written to RAM, this setting is appropriate for example when INDEX needs to be updated for read

MDSS_MDP_VP_0_DSPP_2_PA_LUTV (cont.)

Bits	Name	Description
16	INDEX_UPDATE	Software issue an index update 0: No index update by software 1: Update LUT index to value specified by INDEX
15:8	INDEX	VALUE is stored in this INDEX of the PA_LUTV, next write will be stored to the next higher INDEX
7:0	VALUE	Value read depends on the LUT select and which one is available for reading by software The read sequence to the LUT should always begin with WR INDEX_UPDATE=1, INDEX=<rptr start location> subsequent RD will start from the start location and every read will auto-increment the rptr

0x01A59234 MDSS_MDP_VP_0_DSPP_2_PA_LUTV_SWAP**Type:** W**Reset State:** 0x00000000

Histogram SWAP signal to indicate that the PA_LUTV contents should be swapped

Software accesses one LUT while HW accesses the other LUT

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_VP_0_DSPP_2_PA_LUTV_SWAP

Bits	Name	Description
0	SWAP	SWAP signal to flip buffers accessible by SW and made available to HW. Achieves double buffering of the LUT values

0x01A59238 MDSS_MDP_VP_0_DSPP_2_PA_HUE_ADJUST**Type:** RW**Reset State:** 0x00000000

Picture HUE Adjustment

double buffered: YES

MDSS_MDP_VP_0_DSPP_2_PA_HUE_ADJUST

Bits	Name	Description
11:0	VALUE	Value in S12, recommend to be within exclusive range xA00 to x600 (i.e. -360 to +360 degree) x600 is equivalent to a shift of 360 degree if VALUE > x5FF in either negative or positive direction, it is circularly clipped (i.e. VALUE + x600 or VALUE - x600)

0x01A5923C MDSS_MDP_VP_0_DSPP_2_PA_SAT_ADJUST**Type:** RW**Reset State:** 0x00001A00

Picture Saturation Adjustment

double buffered: YES

MDSS_MDP_VP_0_DSPP_2_PA_SAT_ADJUST

Bits	Name	Description
15:8	THRSHLD	Below this threshold of saturation, no adjustment is applied to saturation. This field is in U8
7:0	VALUE	Value in S8

0x01A59240 MDSS_MDP_VP_0_DSPP_2_PA_VAL_ADJUST**Type:** RW**Reset State:** 0x00000000

Picture Adjustment

double buffered: YES

MDSS_MDP_VP_0_DSPP_2_PA_VAL_ADJUST

Bits	Name	Description
7:0	VALUE	Value in S8

0x01A59244 MDSS_MDP_VP_0_DSPP_2_PA_CONT_ADJUST**Type:** RW**Reset State:** 0x00000000

Picture Contrast Adjustment

double buffered: YES

MDSS_MDP_VP_0_DSPP_2_PA_CONT_ADJUST

Bits	Name	Description
7:0	VALUE	Value in S8

0x01A59248 MDSS_MDP_VP_0_DSPP_2_SIX_ZONE_ADJ_CURVE_P0

Type: RW

Reset State: 0x00000000

Six Zone Adjust Curve Part 0

double buffered: No

MDSS_MDP_VP_0_DSPP_2_SIX_ZONE_ADJ_CURVE_P0

Bits	Name	Description
26	INDEX_UPDATE	Set this to 1 to point current LUT location to INDEX. The LUT location is autoincremented on every value write or read when there is no INDEX_UPDATE Before every series of read, there must be an INDEX_UPDATE
25	VALUE_UPDATEN	Set this to 0 to enable VALUE to be written to the current location of the LUT
24:16	INDEX	Index to one of the 384 location of the LUT
11:0	HUE	S12, recommend to be within exclusive range x2800 and x1800 (i.e. -360 to +360 exclusive) circularly clipped otherwise On write: Value to be updated to one of the 384 entry LUT On read : Value read out from LUT location pointed by INDEX

0x01A5924C MDSS_MDP_VP_0_DSPP_2_SIX_ZONE_ADJ_CURVE_P1

Type: RW

Reset State: 0x00000000

Six Zone Adjust Curve Part 1

double buffered: No

MDSS_MDP_VP_0_DSPP_2_SIX_ZONE_ADJ_CURVE_P1

Bits	Name	Description
27:16	SAT	S2.10, On write: Value to be updated to one of the 384 entry LUT On read : Value read out from LUT location pointed by INDEX
11:0	VALUE	S2.10, On write: Value to be updated to one of the 384 entry LUT On read : Value read out from LUT location pointed by INDEX

0x01A59250 MDSS_MDP_VP_0_DSPP_2_SIX_ZONE_THRESHOLDS**Type:** RW**Reset State:** 0x00000000

Six Zone Threshold Qualifier

double buffered: Yes

MDSS_MDP_VP_0_DSPP_2_SIX_ZONE_THRESHOLDS

Bits	Name	Description
23:16	VAL_MAX	U8, Value has to be within the inclusive range of MIN to MAX to qualify for six zone adjustment on the pixel
15:8	VAL_MIN	
7:0	SAT_MIN	U8, Saturation has to be above or equal to this threshold to qualify for six zone adjustment

0x01A59254 MDSS_MDP_VP_0_DSPP_2_SKIN_ADJUST_PWL0**Type:** RW**Reset State:** 0x00000000

Skin Piece-wise Linear Adjustment Curve

double buffered: Yes

MDSS_MDP_VP_0_DSPP_2_SKIN_ADJUST_PWL0

Bits	Name	Description
31:16	HUE_GAIN	S16, is a symmetrical trapezoidal curve, this is the slope of the 2 sides
10:0	HUE_MID	U11, this is the offset from the min and max hue of skin region which marks the top 2 corners of the trapezoid

0x01A59258 MDSS_MDP_VP_0_DSPP_2_SKIN_ADJUST_PWL1**Type:** RW**Reset State:** 0x00000000

Skin Piece-wise Linear Adjustment Curve

double buffered: Yes

MDSS_MDP_VP_0_DSPP_2_SKIN_ADJUST_PWL1

Bits	Name	Description
31:16	VAL_GAIN	S16, is a symmetrical trapezoidal curve, this is the slope of the 2 sides
15:0	SAT_GAIN	S16, is a symmetrical trapezoidal curve, this is the slope of the 2 sides

0x01A5925C MDSS_MDP_VP_0_DSPP_2_SKIN_REGION_HUE**Type:** RW**Reset State:** 0x00000000

Skin Adjustment Hue Qualifier

double buffered: Yes

MDSS_MDP_VP_0_DSPP_2_SKIN_REGION_HUE

Bits	Name	Description
26:16	MAX	if MAX > x5ff (i.e. 360 degree), the behaviour will be same as MAX = x600
10:0	MIN	U11, Hue has to be within the inclusive range of MIN to MAX to qualify for skin tone adjustment on the pixel if MIN > x5ff (i.e. 360 degree), the behaviour will be same as MIN = x600

0x01A59260 MDSS_MDP_VP_0_DSPP_2_SKIN_REGION_SAT**Type:** RW**Reset State:** 0x00000000

Skin Adjustment Saturation Qualifier

double buffered: Yes

MDSS_MDP_VP_0_DSPP_2_SKIN_REGION_SAT

Bits	Name	Description
15:8	MAX	Saturation has to be below or equal to this to qualify for skin tone adjustment on the pixel
7:0	MIN	U8, Saturation has to be above or equal to this to qualify for skin tone adjustment on the pixel

0x01A59264 MDSS_MDP_VP_0_DSPP_2_SKIN_REGION_VAL**Type:** RW**Reset State:** 0x00000000

Skin Adjustment Value Qualifier

double buffered: Yes

MDSS_MDP_VP_0_DSPP_2_SKIN_REGION_VAL

Bits	Name	Description
15:8	MAX	Value has to be below or equal to this to qualify for skin tone adjustment on the pixel
7:0	MIN	U8, Value has to be above or equal to this to qualify for skin tone adjustment on the pixel

0x01A59268 MDSS_MDP_VP_0_DSPP_2_SKY_ADJUST_PWL0**Type:** RW**Reset State:** 0x00000000

Sky Piece-wise Linear Adjustment Curve

double buffered: Yes

MDSS_MDP_VP_0_DSPP_2_SKY_ADJUST_PWL0

Bits	Name	Description
31:16	HUE_GAIN	S16, is a symmetrical trapezoidal curve, this is the slope of the 2 sides
10:0	HUE_MID	U11, this is the offset from the min and max hue of skin region which marks the top 2 corners of the trapezoid

0x01A5926C MDSS_MDP_VP_0_DSPP_2_SKY_ADJUST_PWL1**Type:** RW**Reset State:** 0x00000000

Sky Piece-wise Linear Adjustment Curve

double buffered: Yes

MDSS_MDP_VP_0_DSPP_2_SKY_ADJUST_PWL1

Bits	Name	Description
31:16	VAL_GAIN	S16, is a symmetrical trapezoidal curve, this is the slope of the 2 sides
15:0	SAT_GAIN	S16, is a symmetrical trapezoidal curve, this is the slope of the 2 sides

0x01A59270 MDSS_MDP_VP_0_DSPP_2_SKY_REGION_HUE**Type:** RW**Reset State:** 0x00000000

Sky Adjustment Hue Qualifier

double buffered: Yes

MDSS_MDP_VP_0_DSPP_2_SKY_REGION_HUE

Bits	Name	Description
26:16	MAX	if MAX > x5ff (i.e. 360 degree), the behaviour will be same as MAX = x600
10:0	MIN	U11, Hue has to be within the inclusive range of MIN to MAX to qualify for sky tone adjustment on the pixel if MIN > x5ff (i.e. 360 degree), the behaviour will be same as MIN = x600

0x01A59274 MDSS_MDP_VP_0_DSPP_2_SKY_REGION_SAT**Type:** RW**Reset State:** 0x00000000

Sky Adjustment Saturation Qualifier

double buffered: Yes

MDSS_MDP_VP_0_DSPP_2_SKY_REGION_SAT

Bits	Name	Description
15:8	MAX	Saturation has to be below or equal to this to qualify for sky adjustment on the pixel
7:0	MIN	U8, Saturation has to be above or equal to this to qualify for sky adjustment on the pixel

0x01A59278 MDSS_MDP_VP_0_DSPP_2_SKY_REGION_VAL**Type:** RW**Reset State:** 0x00000000

Sky Adjustment Value Qualifier

double buffered: Yes

MDSS_MDP_VP_0_DSPP_2_SKY_REGION_VAL

Bits	Name	Description
15:8	MAX	Value has to be below or equal to this to qualify for sky adjustment on the pixel
7:0	MIN	U8, Value has to be above or equal to this to qualify for sky adjustment on the pixel

0x01A5927C MDSS_MDP_VP_0_DSPP_2_FOLIAGE_ADJUST_PWL0**Type:** RW**Reset State:** 0x00000000**MDSS_MDP_VP_0_DSPP_2_FOLIAGE_ADJUST_PWL0**

Bits	Name	Description
31:16	HUE_GAIN	
10:0	HUE_MID	

0x01A59280 MDSS_MDP_VP_0_DSPP_2_FOLIAGE_ADJUST_PWL1**Type:** RW**Reset State:** 0x00000000

MDSS_MDP_VP_0_DSPP_2_FOLIAGE_ADJUST_PWL1

Bits	Name	Description
31:16	VAL_GAIN	
15:0	SAT_GAIN	

0x01A59284 MDSS_MDP_VP_0_DSPP_2_FOLIAGE_REGION_HUE**Type:** RW**Reset State:** 0x00000000

Foilage Adjustment Hue Qualifier

double buffered: Yes

MDSS_MDP_VP_0_DSPP_2_FOLIAGE_REGION_HUE

Bits	Name	Description
26:16	MAX	if MAX > x5ff (i.e. 360 degree), the behaviour will be same as MAX = x600
10:0	MIN	U11, Hue has to be within the inclusive range of MIN to MAX to qualify for foliage tone adjustment on the pixel if MIN > x5ff (i.e. 360 degree), the behaviour will be same as MIN = x600

0x01A59288 MDSS_MDP_VP_0_DSPP_2_FOLIAGE_REGION_SAT**Type:** RW**Reset State:** 0x00000000

Foilage Adjustment Saturation Qualifier

double buffered: Yes

MDSS_MDP_VP_0_DSPP_2_FOLIAGE_REGION_SAT

Bits	Name	Description
15:8	MAX	Saturation has to be below or equal to this to qualify for foilage adjustment on the pixel
7:0	MIN	U8, Saturation has to be above or equal to this to qualify for foilage adjustment on the pixel

0x01A5928C MDSS_MDP_VP_0_DSPP_2_FOLIAGE_REGION_VAL**Type:** RW**Reset State:** 0x00000000

Foliage Adjustment Value Qualifier

double buffered: Yes

MDSS_MDP_VP_0_DSPP_2_FOLIAGE_REGION_VAL

Bits	Name	Description
15:8	MAX	Value has to be below or equal to this to qualify for foliage adjustment on the pixel
7:0	MIN	U8, Value has to be above or equal to this to qualify for foliage adjustment on the pixel

0x01A592B0 MDSS_MDP_VP_0_DSPP_2_GC_ARLUT_RED_STARTX_N**Type:** RW**Reset State:** 0x00000000**MDSS_MDP_VP_0_DSPP_2_GC_ARLUT_RED_STARTX_N**

Bits	Name	Description
21	VALUE_UPDATEN	
20	SEGMENT_UPDATE	
19:16	SEGMENT	
11:0	STARTX	

0x01A592B4 MDSS_MDP_VP_0_DSPP_2_GC_ARLUT_RED_SLOPE_N**Type:** RW**Reset State:** 0x00000000**MDSS_MDP_VP_0_DSPP_2_GC_ARLUT_RED_SLOPE_N**

Bits	Name	Description
21	VALUE_UPDATEN	
20	SEGMENT_UPDATE	
19:16	SEGMENT	
14:0	SLOPE	

0x01A592B8 MDSS_MDP_VP_0_DSPP_2_GC_ARLUT_RED_OFFSET_N**Type:** RW**Reset State:** 0x00000000**MDSS_MDP_VP_0_DSPP_2_GC_ARLUT_RED_OFFSET_N**

Bits	Name	Description
21	VALUE_UPDATEN	
20	SEGMENT_UPDATE	
19:16	SEGMENT	
14:0	OFFSET	

0x01A592C0 MDSS_MDP_VP_0_DSPP_2_GC_ARLUT_GREEN_STARTX_N**Type:** RW**Reset State:** 0x00000000**MDSS_MDP_VP_0_DSPP_2_GC_ARLUT_GREEN_STARTX_N**

Bits	Name	Description
21	VALUE_UPDATEN	
20	SEGMENT_UPDATE	
19:16	SEGMENT	
11:0	STARTX	

0x01A592C4 MDSS_MDP_VP_0_DSPP_2_GC_ARLUT_GREEN_SLOPE_N**Type:** RW**Reset State:** 0x00000000**MDSS_MDP_VP_0_DSPP_2_GC_ARLUT_GREEN_SLOPE_N**

Bits	Name	Description
21	VALUE_UPDATEN	
20	SEGMENT_UPDATE	
19:16	SEGMENT	
14:0	SLOPE	

0x01A592C8 MDSS_MDP_VP_0_DSPP_2_GC_ARLUT_GREEN_OFFSET_N**Type:** RW**Reset State:** 0x00000000**MDSS_MDP_VP_0_DSPP_2_GC_ARLUT_GREEN_OFFSET_N**

Bits	Name	Description
21	VALUE_UPDATEN	
20	SEGMENT_UPDATE	
19:16	SEGMENT	
14:0	OFFSET	

0x01A592D0 MDSS_MDP_VP_0_DSPP_2_GC_ARLUT_BLUE_STARTX_N**Type:** RW**Reset State:** 0x00000000**MDSS_MDP_VP_0_DSPP_2_GC_ARLUT_BLUE_STARTX_N**

Bits	Name	Description
21	VALUE_UPDATEN	
20	SEGMENT_UPDATE	
19:16	SEGMENT	
11:0	STARTX	

0x01A592D4 MDSS_MDP_VP_0_DSPP_2_GC_ARLUT_BLUE_SLOPE_N**Type:** RW**Reset State:** 0x00000000**MDSS_MDP_VP_0_DSPP_2_GC_ARLUT_BLUE_SLOPE_N**

Bits	Name	Description
21	VALUE_UPDATEN	
20	SEGMENT_UPDATE	
19:16	SEGMENT	
14:0	SLOPE	

0x01A592D8 MDSS_MDP_VP_0_DSPP_2_GC_ARLUT_BLUE_OFFSET_N**Type:** RW**Reset State:** 0x00000000**MDSS_MDP_VP_0_DSPP_2_GC_ARLUT_BLUE_OFFSET_N**

Bits	Name	Description
21	VALUE_UPDATEN	
20	SEGMENT_UPDATE	
19:16	SEGMENT	
14:0	OFFSET	

**0x01A5A000+MDSS_MDP_VP_0_DSPP_2_HIST_V_DATAn_BIN_i, i=[0..255]
4*i****Type:** R**Reset State:** 0x00000000

Direct Register Mapped Histogram data collection

256 bins

double buffered: NO.

MDSS_MDP_VP_0_DSPP_2_HIST_V_DATAn_BIN_i

Bits	Name	Description
23:0	VALUE	Value

**0x01A5A400+MDSS_MDP_VP_0_DSPP_2_PA_LUTV_IDX_i, i=[0..127]
4*i****Type:** RW**Reset State:** 0x00000000

Direct Register Mapped PA LUTV data

256 entries are available for writing and reading

double buffered: NO

MDSS_MDP_VP_0_DSPP_2_PA_LUTV_IDX_i

Bits	Name	Description
23:16	VALUE_UPPER	LUT content at location 2*IDX+1 to be written/read to/from the LUT accessible by software at the moment
7:0	VALUE_LOWER	LUT content at location 2*IDX to be written/read to/from the LUT accessible by software at the moment

17.17.8 MDSS_MDP_WB_0

MDP WB hardware register description

17.17.8.1 MDSS_MDP_WB_0_WB

WB

MDP WB hardware register description

0x01A65000 MDSS_MDP_WB_0_DST_FORMAT

Type: RW

Reset State: 0x00000000

Memory Write Back Output Format

double buffered: NO

MDSS_MDP_WB_0_DST_FORMAT

Bits	Name	Description
31:30	FRAME_FORMAT	FRAME_FORMAT In the current revision, there is no support for tiled formats 0x0: Linear, no tile
29:26	DST_CHROMA_SITE	Destination Chroma Site 0x0: A (cosite for YUV422) 0x1: B (offsite for YUV422) 0x2: C (N/A for YUV422) 0x3: D (N/A for YUV422) 0x4: E (N/A for YUV422) 0x5: F (N/A for YUV422) 0x6: G (N/A for YUV422) 0x7: H (N/A for YUV422) 0x8: I (N/A for YUV422)

MDSS_MDP_WB_0_DST_FORMAT (cont.)

Bits	Name	Description
25:23	DST_CHROMA_SAMP	Chroma Sampling 0x0: RGB/4:4:4 0x1: 4:2:2 - H2V1 0x2: 4:2:2 - H1V2 0x3: 4:2:0 - N2V1 0x4: 4:2:0 - N1V2
22	DST_DITHER_EN	Destination Dither Enable
20:19	WRITE_PLANES	Number of planes 0x0: Interleaved 0x1: Planar 0x2: Pseudo-planar
18	PACK_ALIGN	0x0: LSB 0x1: MSB
17	PACK_TIGHT	0x0: Loose 0x1: Tight
14	DST_ALPHA_X	0x0: Use alpha value from pipe 0x1: Use alpha value from register
13:12	PACK_COUNT	Valid packing elements count 0x0 = 1 element 0x1 = 2 elements 0x2 = 3 elements 0x3 = 4 elements
10:9	DST_BPP	Effective byte per pixel 0x0: 1 byte 0x1: 2 bytes 0x2: 3 bytes 0x3: 4 bytes
8	DSTC3_EN	0x1: Destination has alpha
7:6	DSTC3_OUT	Number of bits for component 3 output 0x0: 1 bit 0x1: 4 bits 0x2: 6 bits 0x3: 8 bits
5:4	DSTC2_OUT	Number of bits for component 2 output 0x0: 4 bits 0x1: 5 bits 0x2: 6 bits 0x3: 8 bits

MDSS_MDP_WB_0_DST_FORMAT (cont.)

Bits	Name	Description
3:2	DSTC1_OUT	Number of bits for component 1 output 0x0: 4 bits 0x1: 5 bits 0x2: 6 bits 0x3: 8 bits
1:0	DSTC0_OUT	Number of bits for component 0 output 0x0: 4 bits 0x1: 5 bits 0x2: 6 bits 0x3: 8 bits

0x01A65004 MDSS_MDP_WB_0_DST_OP_MODE**Type:** RW**Reset State:** 0x00000500

Destination Operation Mode

double buffered: NO

MDSS_MDP_WB_0_DST_OP_MODE

Bits	Name	Description
14	CHROMA_DWN_SAMPLE_V_MTHD	Vertical Chroma Down Sample method 0x0: non-pixel dropping 0x1: pixel dropping
13	CHROMA_DWN_SAMPLE_H_MTHD	Horizontal Chroma Down Sample method 0x0: non-pixel dropping 0x1: pixel dropping
12	CHROMA_DWN_SAMPLE_FORMAT	Chroma Down Sample format 0x0: YUV422 H2V1 0x1: YUV420
11	CHROMA_DWN_SAMPLE_EN	Chroma Down Sample enable for YUV444 NOTE: This is not valid for WB0. 0x0: Disabled 0x1: Enabled
10	CSC_DST_DATA_FORMAT	Destination Data Format 0x0: RGB 0x1: YCbCr
9	CSC_SRC_DATA_FORMAT	Source Data Format 0x0: RGB 0x1: YCbCr

MDSS_MDP_WB_0_DST_OP_MODE (cont.)

Bits	Name	Description
8	CSC_EN	Enables the color matrix conversion NOTE: This is not valid for WB0. 0x0: Bypass 0x1: Enabled
6	ROT_EN	0x0: Rotate Disabled 0x1: Rotate Enabled
5	ROT_MODE	0x0: Rotate 0 0x1: Rotate 90
4	BLOCK_SIZE	0x0: 64x64 Block Size 0x1: 128x128 Block Size

0x01A65008 MDSS_MDP_WB_0_DST_PACK_PATTERN**Type:** RW**Reset State:** 0x00000000

Write Back Pack Pattern

double buffered: NO

MDSS_MDP_WB_0_DST_PACK_PATTERN

Bits	Name	Description
25:24	ELEMENT3	0x0: Color0 0x1: Color1 0x2: Color2 0x3: Color3
17:16	ELEMENT2	0x0: Color0 0x1: Color1 0x2: Color2 0x3: Color3
9:8	ELEMENT1	0x0: Color0 0x1: Color1 0x2: Color2 0x3: Color3
1:0	ELEMENT0	0x0: Color0 0x1: Color1 0x2: Color2 0x3: Color3

0x01A6500C MDSS_MDP_WB_0_DST0_ADDR**Type:** RW**Reset State:** 0x00000000

Write Back Destination Address

For fetching BWC data, this plane refers to the compressed RAU data stream

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_WB_0_DST0_ADDR

Bits	Name	Description
31:0	ADDR	Destination Address (byte-aligned)

0x01A65010 MDSS_MDP_WB_0_DST1_ADDR**Type:** RW**Reset State:** 0x00000000

Write Back Destination Address

For fetching BWC data, this plane refers to the META data to fetch the lengths of the compressed RAU data stream

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_WB_0_DST1_ADDR

Bits	Name	Description
31:0	ADDR	Destination Address (byte-aligned)

0x01A65014 MDSS_MDP_WB_0_DST2_ADDR**Type:** RW**Reset State:** 0x00000000

Write Back Destination Address

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_WB_0_DST2_ADDR

Bits	Name	Description
31:0	ADDR	Destination Address (byte-aligned)

0x01A65018 MDSS_MDP_WB_0_DST3_ADDR**Type:** RW**Reset State:** 0x00000000

Write Back Destination Address

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_WB_0_DST3_ADDR

Bits	Name	Description
31:0	ADDR	Destination Address (byte-aligned)

0x01A6501C MDSS_MDP_WB_0_DST_YSTRIDE0**Type:** RW**Reset State:** 0x00000000

Stride Values

Double-buffered

double buffered: NO

MDSS_MDP_WB_0_DST_YSTRIDE0

Bits	Name	Description
31:16	DST1_YSTRIDE	Plane 1 Y stride in bytes
15:0	DST0_YSTRIDE	Plane 0 Y stride in bytes

0x01A65020 MDSS_MDP_WB_0_DST_YSTRIDE1**Type:** RW**Reset State:** 0x00000000

Stride Values

Double-buffered

double buffered: NO

MDSS_MDP_WB_0_DST_YSTRIDE1

Bits	Name	Description
31:16	DST3_YSTRIDE	Plane 3 Y stride in bytes
15:0	DST2_YSTRIDE	Plane 2 Y stride in bytes

0x01A65024 MDSS_MDP_WB_0_DST_DITHER_BITDEPTH

Type: RW

Reset State: 0x00000000

Write Dither Strength

double buffered: NO

MDSS_MDP_WB_0_DST_DITHER_BITDEPTH

Bits	Name	Description
5:4	C2_BITS	Number of MSB bits on component2 to indicate the output depth 0x0: 4bits 0x1: 5 bits 0x2: 6 bits 0x3: 8 bits
3:2	C1_BITS	Number of MSB bits on component1 to indicate the output depth 0x0: 4bits 0x1: 5 bits 0x2: 6 bits 0x3: 8 bits
1:0	C0_BITS	Number of MSB bits on component0 to indicate the output depth 0x0: 4bits 0x1: 5 bits 0x2: 6 bits 0x3: 8 bits

0x01A65030 MDSS_MDP_WB_0_DITHER_MATRIX_ROW0

Type: RW

Reset State: 0x00005D7F

Dither Strength Matrix Values Register for first row which covers entries [0-3]

double buffered: NO

MDSS_MDP_WB_0_DITHER_MATRIX_ROW0

Bits	Name	Description
15:12	ENTRY_3	Entry 3 for the dither matrix
11:8	ENTRY_2	Entry 2 for the dither matrix
7:4	ENTRY_1	Entry 1 for the dither matrix
3:0	ENTRY_0	Entry 0 for the dither matrix

0x01A65034 MDSS_MDP_WB_0_DITHER_MATRIX_ROW1

Type: RW

Reset State: 0x000091B3

Dither Strength Matrix Values Register for first row which covers entries [4-7]

double buffered: NO

MDSS_MDP_WB_0_DITHER_MATRIX_ROW1

Bits	Name	Description
15:12	ENTRY_7	Entry 7 for the dither matrix
11:8	ENTRY_6	Entry 6 for the dither matrix
7:4	ENTRY_5	Entry 5 for the dither matrix
3:0	ENTRY_4	Entry 4 for the dither matrix

0x01A65038 MDSS_MDP_WB_0_DITHER_MATRIX_ROW2

Type: RW

Reset State: 0x00006E4C

Dither Strength Matrix Values Register for first row which covers entries [8-11]

double buffered: NO

MDSS_MDP_WB_0_DITHER_MATRIX_ROW2

Bits	Name	Description
15:12	ENTRY_11	Entry 11 for the dither matrix
11:8	ENTRY_10	Entry 10 for the dither matrix
7:4	ENTRY_9	Entry 9 for the dither matrix

MDSS_MDP_WB_0_DITHER_MATRIX_ROW2 (cont.)

Bits	Name	Description
3:0	ENTRY_8	Entry 8 for the dither matrix

0x01A6503C MDSS_MDP_WB_0_DITHER_MATRIX_ROW3**Type:** RW**Reset State:** 0x0000A280

Dither Strength Matrix Values Register for first row which covers entries [12-15]

double buffered: NO

MDSS_MDP_WB_0_DITHER_MATRIX_ROW3

Bits	Name	Description
15:12	ENTRY_15	Entry 15 for the dither matrix
11:8	ENTRY_14	Entry 14 for the dither matrix
7:4	ENTRY_13	Entry 13 for the dither matrix
3:0	ENTRY_12	Entry 12 for the dither matrix

0x01A65048 MDSS_MDP_WB_0_DST_WRITE_CONFIG**Type:** RW**Reset State:** 0x00000008

Write-back Configuration double buffered: NO

This register must NOT be altered during operation and MDP must be idle when changing values

MDSS_MDP_WB_0_DST_WRITE_CONFIG

Bits	Name	Description
10	ADDR_GEN_MODE	
9:8	HIGHEST_BANK_BIT	
7:6	PRIORITY_LEVEL	Priority Level. Side-band priority reflecting the status of VBIF client. 0x3 - max priority 0x0 - min priority
5:4	REQ_PRIORITY	Request Priority. In-band priority level assigned when VBIF transaction is created. 0x3 - max priority 0x0 - min priority

MDSS_MDP_WB_0_DST_WRITE_CONFIG (cont.)

Bits	Name	Description
3:0	MAX_BURST_SIZE	Maximum burst beat size. Hardware fetch engines operate on 128-bit bus Possible values: 0x8: burst of 8 0x4: burst of 4 0x2: burst of 2 0x1: burst of 1

0x01A65054 MDSS_MDP_WB_0_ROTATOR_PIPE_DOWNSCALER**Type:** RW**Reset State:** 0x00000000**MDSS_MDP_WB_0_ROTATOR_PIPE_DOWNSCALER**

Bits	Name	Description
23:16	DNOSC_FACTOR_V	0x0: V_BYPASS 0x2: V_DNOSC_X2 0x4: V_DNOSC_X4 0x8: V_DNOSC_X8 0x10: V_DNOSC_X16 0x20: V_DNOSC_X32 0x40: V_DNOSC_X64
7:0	DNOSC_FACTOR_H	0x0: H_BYPASS 0x2: H_DNOSC_X2 0x4: H_DNOSC_X4 0x8: H_DNOSC_X8 0x10: H_DNOSC_X16 0x20: H_DNOSC_X32 0x40: H_DNOSC_X64

0x01A65074 MDSS_MDP_WB_0_OUT_SIZE**Type:** RW**Reset State:** 0x00000000

Output Size

Should be same as source is scaling is NOT enabled

double buffered: NO

MDSS_MDP_WB_0_OUT_SIZE

Bits	Name	Description
31:16	DST_H	Height of Output ROI
15:0	DST_W	Width of Output ROI

0x01A65078 MDSS_MDP_WB_0_DST_ALPHA_X_VALUE**Type:** RW**Reset State:** 0x00000000

Alpha value used for XRGB formats

double buffered: NO

MDSS_MDP_WB_0_DST_ALPHA_X_VALUE

Bits	Name	Description
7:0	VALUE	Alpha value used for XRGB formats

0x01A65260 MDSS_MDP_WB_0_CSC_MATRIX_COEFF_0**Type:** RW**Reset State:** 0x00000200

Color Space conversion coefficients

double buffered: NO

MDSS_MDP_WB_0_CSC_MATRIX_COEFF_0

Bits	Name	Description
28:16	COEFF_12	Coefficient value in S4.9 format Row 1 Column 2
12:0	COEFF_11	Coefficient value in S4.9 format Row 1 Column 1

0x01A65264 MDSS_MDP_WB_0_CSC_MATRIX_COEFF_1**Type:** RW**Reset State:** 0x00000000

Color Space conversion coefficients

double buffered: NO

MDSS_MDP_WB_0_CSC_MATRIX_COEFF_1

Bits	Name	Description
28:16	COEFF_21	Coefficient value in S4.9 format Row 2 Column 1
12:0	COEFF_13	Coefficient value in S4.9 format Row 1 Column 3

0x01A65268 MDSS_MDP_WB_0_CSC_MATRIX_COEFF_2

Type: RW

Reset State: 0x00000200

Color Space conversion coefficients

double buffered: NO

MDSS_MDP_WB_0_CSC_MATRIX_COEFF_2

Bits	Name	Description
28:16	COEFF_23	Coefficient value in S4.9 format Row 2 Column 3
12:0	COEFF_22	Coefficient value in S4.9 format Row 2 Column 2

0x01A6526C MDSS_MDP_WB_0_CSC_MATRIX_COEFF_3

Type: RW

Reset State: 0x00000000

Color Space conversion coefficients

double buffered: NO

MDSS_MDP_WB_0_CSC_MATRIX_COEFF_3

Bits	Name	Description
28:16	COEFF_32	Coefficient value in S4.9 format Row 3 Column 2
12:0	COEFF_31	Coefficient value in S4.9 format Row 3 Column 1

0x01A65270 MDSS_MDP_WB_0_CSC_MATRIX_COEFF_4**Type:** RW**Reset State:** 0x00000200

Color Space conversion coefficients

double buffered: NO

MDSS_MDP_WB_0_CSC_MATRIX_COEFF_4

Bits	Name	Description
12:0	COEFF_33	Coefficient value in S4.9 format Row 3 Column 3

0x01A65274 MDSS_MDP_WB_0_CSC_COMP0_PRECLAMP**Type:** RW**Reset State:** 0x000000FF

Component 0 pre-clamp value

double buffered: NO

MDSS_MDP_WB_0_CSC_COMP0_PRECLAMP

Bits	Name	Description
15:8	LOW	Unsigned 8 bit pre-clamp value
7:0	HIGH	Unsigned 8 bit pre-clamp value

0x01A65278 MDSS_MDP_WB_0_CSC_COMP1_PRECLAMP**Type:** RW**Reset State:** 0x000000FF

Component 1 pre-clamp value

double buffered: NO

MDSS_MDP_WB_0_CSC_COMP1_PRECLAMP

Bits	Name	Description
15:8	LOW	Unsigned 8 bit pre-clamp value
7:0	HIGH	Unsigned 8 bit pre-clamp value

0x01A6527C MDSS_MDP_WB_0_CSC_COMP2_PRECLAMP**Type:** RW**Reset State:** 0x000000FF

Component 2 pre-clamp value

double buffered: NO

MDSS_MDP_WB_0_CSC_COMP2_PRECLAMP

Bits	Name	Description
15:8	LOW	Unsigned 8 bit pre-clamp value
7:0	HIGH	Unsigned 8 bit pre-clamp value

0x01A65280 MDSS_MDP_WB_0_CSC_COMP0_POSTCLAMP**Type:** RW**Reset State:** 0x000000FF

Component 0 post-clamp value

double buffered: NO

MDSS_MDP_WB_0_CSC_COMP0_POSTCLAMP

Bits	Name	Description
15:8	LOW	Unsigned 8 bit pre-clamp value
7:0	HIGH	Unsigned 8 bit pre-clamp value

0x01A65284 MDSS_MDP_WB_0_CSC_COMP1_POSTCLAMP**Type:** RW**Reset State:** 0x000000FF

Component 1 post-clamp value

double buffered: NO

MDSS_MDP_WB_0_CSC_COMP1_POSTCLAMP

Bits	Name	Description
15:8	LOW	Unsigned 8 bit pre-clamp value
7:0	HIGH	Unsigned 8 bit pre-clamp value

0x01A65288 MDSS_MDP_WB_0_CSC_COMP2_POSTCLAMP**Type:** RW**Reset State:** 0x000000FF

Component 2 post-clamp value

double buffered: NO

MDSS_MDP_WB_0_CSC_COMP2_POSTCLAMP

Bits	Name	Description
15:8	LOW	Unsigned 8 bit pre-clamp value
7:0	HIGH	Unsigned 8 bit pre-clamp value

0x01A6528C MDSS_MDP_WB_0_CSC_COMP0_PREBIAS**Type:** RW**Reset State:** 0x00000000

Component 0 pre-bias value

double buffered: NO

MDSS_MDP_WB_0_CSC_COMP0_PREBIAS

Bits	Name	Description
8:0	VALUE	Signed 9 bit pre-bias value

0x01A65290 MDSS_MDP_WB_0_CSC_COMP1_PREBIAS**Type:** RW**Reset State:** 0x00000000

Component 1 pre-bias value

double buffered: NO

MDSS_MDP_WB_0_CSC_COMP1_PREBIAS

Bits	Name	Description
8:0	VALUE	Signed 9 bit pre-bias value

0x01A65294 MDSS_MDP_WB_0_CSC_COMP2_PREBIAS**Type:** RW**Reset State:** 0x00000000

Component 2 pre-bias value

double buffered: NO

MDSS_MDP_WB_0_CSC_COMP2_PREBIAS

Bits	Name	Description
8:0	VALUE	Signed 9 bit pre-bias value

0x01A65298 MDSS_MDP_WB_0_CSC_COMP0_POSTBIAS**Type:** RW**Reset State:** 0x00000000

Component 0 post-bias value

double buffered: NO

MDSS_MDP_WB_0_CSC_COMP0_POSTBIAS

Bits	Name	Description
8:0	VALUE	Signed 9 bit pre-bias value

0x01A6529C MDSS_MDP_WB_0_CSC_COMP1_POSTBIAS**Type:** RW**Reset State:** 0x00000000

Component 1 post-bias value

double buffered: NO

MDSS_MDP_WB_0_CSC_COMP1_POSTBIAS

Bits	Name	Description
8:0	VALUE	Signed 9 bit pre-bias value

0x01A652A0 MDSS_MDP_WB_0_CSC_COMP2_POSTBIAS**Type:** RW**Reset State:** 0x00000000

Component 2 post-bias value

double buffered: NO

MDSS_MDP_WB_0_CSC_COMP2_POSTBIAS

Bits	Name	Description
8:0	VALUE	Signed 9 bit pre-bias value

0x01A652B0 MDSS_MDP_WB_0_DST_ADDR_SW_STATUS**Type:** RW**Reset State:** 0x00000000

Security Status of the destination buffer plane

This provides a SW override to the hardware calculated content protection sent to VBIF

This value is logically or ed with the source content protection produced by hardware

This information will be presented to VBIF per transaction and reflected as part of the memory transaction

Note that the VBIF context banks need to be properly programmed for the output to be successful

double buffered: NO

MDSS_MDP_WB_0_DST_ADDR_SW_STATUS

Bits	Name	Description
0	DST_ADDR_CP	This is the secure status bit for destination plane 0 (if used) Memory transactions will not complete if this status doesnt correspond to system programming in the rest of the path to memory 0x0: Content Protected Buffer State 0x1: Non Content Protected Buffer State

17.17.9 MDSS_MDP_WB_1

MDP WB hardware register description

17.17.9.1 MDSS_MDP_WB_1_WB

WB

MDP WB hardware register description

0x01A65800 MDSS_MDP_WB_1_DST_FORMAT**Type:** RW**Reset State:** 0x00000000

Memory Write Back Output Format

double buffered: NO

MDSS_MDP_WB_1_DST_FORMAT

Bits	Name	Description
31:30	FRAME_FORMAT	FRAME_FORMAT In the current revision, there is no support for tiled formats 0x0: Linear, no tile
29:26	DST_CHROMA_SITE	Destination Chroma Site 0x0: A (cosite for YUV422) 0x1: B (offsite for YUV422) 0x2: C (N/A for YUV422) 0x3: D (N/A for YUV422) 0x4: E (N/A for YUV422) 0x5: F (N/A for YUV422) 0x6: G (N/A for YUV422) 0x7: H (N/A for YUV422) 0x8: I (N/A for YUV422)
25:23	DST_CHROMA_SAMP	Chroma Sampling 0x0: RGB/4:4:4 0x1: 4:2:2 - H2V1 0x2: 4:2:2 - H1V2 0x3: 4:2:0 - N2V1 0x4: 4:2:0 - N1V2
22	DST_DITHER_EN	Destination Dither Enable
20:19	WRITE_PLANES	Number of planes 0x0: Interleaved 0x1: Planar 0x2: Pseudo-planar
18	PACK_ALIGN	0x0: LSB 0x1: MSB

MDSS_MDP_WB_1_DST_FORMAT (cont.)

Bits	Name	Description
17	PACK_TIGHT	0x0: Loose 0x1: Tight
14	DST_ALPHA_X	0x0: Use alpha value from pipe 0x1: Use alpha value from register
13:12	PACK_COUNT	Valid packing elements count 0x0 = 1 element 0x1 = 2 elements 0x2 = 3 elements 0x3 = 4 elements
10:9	DST_BPP	Effective byte per pixel 0x0: 1 byte 0x1: 2 bytes 0x2: 3 bytes 0x3: 4 bytes
8	DSTC3_EN	0x1: Destination has alpha
7:6	DSTC3_OUT	Number of bits for component 3 output 0x0: 1 bit 0x1: 4 bits 0x2: 6 bits 0x3: 8 bits
5:4	DSTC2_OUT	Number of bits for component 2 output 0x0: 4 bits 0x1: 5 bits 0x2: 6 bits 0x3: 8 bits
3:2	DSTC1_OUT	Number of bits for component 1 output 0x0: 4 bits 0x1: 5 bits 0x2: 6 bits 0x3: 8 bits
1:0	DSTC0_OUT	Number of bits for component 0 output 0x0: 4 bits 0x1: 5 bits 0x2: 6 bits 0x3: 8 bits

0x01A65804 MDSS_MDP_WB_1_DST_OP_MODE**Type:** RW**Reset State:** 0x00000500

Destination Operation Mode

double buffered: NO

MDSS_MDP_WB_1_DST_OP_MODE

Bits	Name	Description
14	CHROMA_DWN_SAMPLE_V_MTHD	Vertical Chroma Down Sample method 0x0: non-pixel dropping 0x1: pixel dropping
13	CHROMA_DWN_SAMPLE_H_MTHD	Horizontal Chroma Down Sample method 0x0: non-pixel dropping 0x1: pixel dropping
12	CHROMA_DWN_SAMPLE_FORMAT	Chroma Down Sample format 0x0: YUV422 H2V1 0x1: YUV420
11	CHROMA_DWN_SAMPLE_EN	Chroma Down Sample enable for YUV444 NOTE: This is not valid for WB0. 0x0: Disabled 0x1: Enabled
10	CSC_DST_DATA_FORMAT	Destination Data Format 0x0: RGB 0x1: YCbCr
9	CSC_SRC_DATA_FORMAT	Source Data Format 0x0: RGB 0x1: YCbCr
8	CSC_EN	Enables the color matrix conversion NOTE: This is not valid for WB0. 0x0: Bypass 0x1: Enabled
6	ROT_EN	0x0: Rotate Disabled 0x1: Rotate Enabled
5	ROT_MODE	0x0: Rotate 0 0x1: Rotate 90
4	BLOCK_SIZE	0x0: 64x64 Block Size 0x1: 128x128 Block Size

0x01A65808 MDSS_MDP_WB_1_DST_PACK_PATTERN

Type: RW

Reset State: 0x00000000

Write Back Pack Pattern

double buffered: NO

MDSS_MDP_WB_1_DST_PACK_PATTERN

Bits	Name	Description
25:24	ELEMENT3	0x0: Color0 0x1: Color1 0x2: Color2 0x3: Color3
17:16	ELEMENT2	0x0: Color0 0x1: Color1 0x2: Color2 0x3: Color3
9:8	ELEMENT1	0x0: Color0 0x1: Color1 0x2: Color2 0x3: Color3
1:0	ELEMENT0	0x0: Color0 0x1: Color1 0x2: Color2 0x3: Color3

0x01A6580C MDSS_MDP_WB_1_DST0_ADDR**Type:** RW**Reset State:** 0x00000000

Write Back Destination Address

For fetching BWC data, this plane refers to the compressed RAU data stream

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_WB_1_DST0_ADDR

Bits	Name	Description
31:0	ADDR	Destination Address (byte-aligned)

0x01A65810 MDSS_MDP_WB_1_DST1_ADDR**Type:** RW**Reset State:** 0x00000000

Write Back Destination Address

For fetching BWC data, this plane refers to the META data to fetch the lengths of the compressed RAU data stream

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_WB_1_DST1_ADDR

Bits	Name	Description
31:0	ADDR	Destination Address (byte-aligned)

0x01A65814 MDSS_MDP_WB_1_DST2_ADDR

Type: RW

Reset State: 0x00000000

Write Back Destination Address

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_WB_1_DST2_ADDR

Bits	Name	Description
31:0	ADDR	Destination Address (byte-aligned)

0x01A65818 MDSS_MDP_WB_1_DST3_ADDR

Type: RW

Reset State: 0x00000000

Write Back Destination Address

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_WB_1_DST3_ADDR

Bits	Name	Description
31:0	ADDR	Destination Address (byte-aligned)

0x01A6581C MDSS_MDP_WB_1_DST_YSTRIDE0**Type:** RW**Reset State:** 0x00000000

Stride Values

Double-buffered

double buffered: NO

MDSS_MDP_WB_1_DST_YSTRIDE0

Bits	Name	Description
31:16	DST1_YSTRIDE	Plane 1 Y stride in bytes
15:0	DST0_YSTRIDE	Plane 0 Y stride in bytes

0x01A65820 MDSS_MDP_WB_1_DST_YSTRIDE1**Type:** RW**Reset State:** 0x00000000

Stride Values

Double-buffered

double buffered: NO

MDSS_MDP_WB_1_DST_YSTRIDE1

Bits	Name	Description
31:16	DST3_YSTRIDE	Plane 3 Y stride in bytes
15:0	DST2_YSTRIDE	Plane 2 Y stride in bytes

0x01A65824 MDSS_MDP_WB_1_DST_DITHER_BITDEPTH**Type:** RW**Reset State:** 0x00000000

Write Dither Strength

double buffered: NO

MDSS_MDP_WB_1_DST_DITHER_BITDEPTH

Bits	Name	Description
5:4	C2_BITS	Number of MSB bits on component2 to indicate the output depth 0x0: 4bits 0x1: 5 bits 0x2: 6 bits 0x3: 8 bits
3:2	C1_BITS	Number of MSB bits on component1 to indicate the output depth 0x0: 4bits 0x1: 5 bits 0x2: 6 bits 0x3: 8 bits
1:0	C0_BITS	Number of MSB bits on component0 to indicate the output depth 0x0: 4bits 0x1: 5 bits 0x2: 6 bits 0x3: 8 bits

0x01A65830 MDSS_MDP_WB_1_DITHER_MATRIX_ROW0**Type:** RW**Reset State:** 0x00005D7F

Dither Strength Matrix Values Register for first row which covers entries [0-3]

double buffered: NO

MDSS_MDP_WB_1_DITHER_MATRIX_ROW0

Bits	Name	Description
15:12	ENTRY_3	Entry 3 for the dither matrix
11:8	ENTRY_2	Entry 2 for the dither matrix
7:4	ENTRY_1	Entry 1 for the dither matrix
3:0	ENTRY_0	Entry 0 for the dither matrix

0x01A65834 MDSS_MDP_WB_1_DITHER_MATRIX_ROW1**Type:** RW**Reset State:** 0x000091B3

Dither Strength Matrix Values Register for first row which covers entries [4-7]

double buffered: NO

MDSS_MDP_WB_1_DITHER_MATRIX_ROW1

Bits	Name	Description
15:12	ENTRY_7	Entry 7 for the dither matrix
11:8	ENTRY_6	Entry 6 for the dither matrix
7:4	ENTRY_5	Entry 5 for the dither matrix
3:0	ENTRY_4	Entry 4 for the dither matrix

0x01A65838 MDSS_MDP_WB_1_DITHER_MATRIX_ROW2**Type:** RW**Reset State:** 0x00006E4C

Dither Strength Matrix Values Register for first row which covers entries [8-11]

double buffered: NO

MDSS_MDP_WB_1_DITHER_MATRIX_ROW2

Bits	Name	Description
15:12	ENTRY_11	Entry 11 for the dither matrix
11:8	ENTRY_10	Entry 10 for the dither matrix
7:4	ENTRY_9	Entry 9 for the dither matrix
3:0	ENTRY_8	Entry 8 for the dither matrix

0x01A6583C MDSS_MDP_WB_1_DITHER_MATRIX_ROW3**Type:** RW**Reset State:** 0x0000A280

Dither Strength Matrix Values Register for first row which covers entries [12-15]

double buffered: NO

MDSS_MDP_WB_1_DITHER_MATRIX_ROW3

Bits	Name	Description
15:12	ENTRY_15	Entry 15 for the dither matrix
11:8	ENTRY_14	Entry 14 for the dither matrix
7:4	ENTRY_13	Entry 13 for the dither matrix
3:0	ENTRY_12	Entry 12 for the dither matrix

0x01A65848 MDSS_MDP_WB_1_DST_WRITE_CONFIG**Type:** RW**Reset State:** 0x00000008

Write-back Configuration double buffered: NO

This register must NOT be altered during operation and MDP must be idle when changing values

MDSS_MDP_WB_1_DST_WRITE_CONFIG

Bits	Name	Description
10	ADDR_GEN_MODE	
9:8	HIGHEST_BANK_BIT	
7:6	PRIORITY_LEVEL	Priority Level. Side-band priority reflecting the status of VBIF client. 0x3 - max priority 0x0 - min priority
5:4	REQ_PRIORITY	Request Priority. In-band priority level assigned when VBIF transaction is created. 0x3 - max priority 0x0 - min priority
3:0	MAX_BURST_SIZE	Maximum burst beat size. Hardware fetch engines operate on 128-bit bus Possible values: 0x8: burst of 8 0x4: burst of 4 0x2: burst of 2 0x1: burst of 1

0x01A65854 MDSS_MDP_WB_1_ROTATOR_PIPE_DOWNSCALER**Type:** RW**Reset State:** 0x00000000**MDSS_MDP_WB_1_ROTATOR_PIPE_DOWNSCALER**

Bits	Name	Description
23:16	DNOSC_FACTOR_V	0x0: V_BYPASS 0x2: V_DNOSC_X2 0x4: V_DNOSC_X4 0x8: V_DNOSC_X8 0x10: V_DNOSC_X16 0x20: V_DNOSC_X32 0x40: V_DNOSC_X64

MDSS_MDP_WB_1_ROTATOR_PIPE_DOWNSCALER (cont.)

Bits	Name	Description
7:0	DNOSC_FACTOR_H	0x0: H_BYPASS 0x2: H_DNOSC_X2 0x4: H_DNOSC_X4 0x8: H_DNOSC_X8 0x10: H_DNOSC_X16 0x20: H_DNOSC_X32 0x40: H_DNOSC_X64

0x01A65874 MDSS_MDP_WB_1_OUT_SIZE**Type:** RW**Reset State:** 0x00000000

Output Size

Should be same as source is scaling is NOT enabled

double buffered: NO

MDSS_MDP_WB_1_OUT_SIZE

Bits	Name	Description
31:16	DST_H	Height of Output ROI
15:0	DST_W	Width of Output ROI

0x01A65878 MDSS_MDP_WB_1_DST_ALPHA_X_VALUE**Type:** RW**Reset State:** 0x00000000

Alpha value used for XRGB formats

double buffered: NO

MDSS_MDP_WB_1_DST_ALPHA_X_VALUE

Bits	Name	Description
7:0	VALUE	Alpha value used for XRGB formats

0x01A65A60 MDSS_MDP_WB_1_CSC_MATRIX_COEFF_0**Type:** RW**Reset State:** 0x00000200

Color Space conversion coefficients

double buffered: NO

MDSS_MDP_WB_1_CSC_MATRIX_COEFF_0

Bits	Name	Description
28:16	COEFF_12	Coefficient value in S4.9 format Row 1 Column 2
12:0	COEFF_11	Coefficient value in S4.9 format Row 1 Column 1

0x01A65A64 MDSS_MDP_WB_1_CSC_MATRIX_COEFF_1**Type:** RW**Reset State:** 0x00000000

Color Space conversion coefficients

double buffered: NO

MDSS_MDP_WB_1_CSC_MATRIX_COEFF_1

Bits	Name	Description
28:16	COEFF_21	Coefficient value in S4.9 format Row 2 Column 1
12:0	COEFF_13	Coefficient value in S4.9 format Row 1 Column 3

0x01A65A68 MDSS_MDP_WB_1_CSC_MATRIX_COEFF_2**Type:** RW**Reset State:** 0x00000200

Color Space conversion coefficients

double buffered: NO

MDSS_MDP_WB_1_CSC_MATRIX_COEFF_2

Bits	Name	Description
28:16	COEFF_23	Coefficient value in S4.9 format Row 2 Column 3
12:0	COEFF_22	Coefficient value in S4.9 format Row 2 Column 2

0x01A65A6C MDSS_MDP_WB_1_CSC_MATRIX_COEFF_3**Type:** RW**Reset State:** 0x00000000

Color Space conversion coefficients

double buffered: NO

MDSS_MDP_WB_1_CSC_MATRIX_COEFF_3

Bits	Name	Description
28:16	COEFF_32	Coefficient value in S4.9 format Row 3 Column 2
12:0	COEFF_31	Coefficient value in S4.9 format Row 3 Column 1

0x01A65A70 MDSS_MDP_WB_1_CSC_MATRIX_COEFF_4**Type:** RW**Reset State:** 0x00000200

Color Space conversion coefficients

double buffered: NO

MDSS_MDP_WB_1_CSC_MATRIX_COEFF_4

Bits	Name	Description
12:0	COEFF_33	Coefficient value in S4.9 format Row 3 Column 3

0x01A65A74 MDSS_MDP_WB_1_CSC_COMP0_PRECLAMP**Type:** RW**Reset State:** 0x000000FF

Component 0 pre-clamp value

double buffered: NO

MDSS_MDP_WB_1_CSC_COMP0_PRECLAMP

Bits	Name	Description
15:8	LOW	Unsigned 8 bit pre-clamp value
7:0	HIGH	Unsigned 8 bit pre-clamp value

0x01A65A78 MDSS_MDP_WB_1_CSC_COMP1_PRECLAMP**Type:** RW**Reset State:** 0x000000FF

Component 1 pre-clamp value

double buffered: NO

MDSS_MDP_WB_1_CSC_COMP1_PRECLAMP

Bits	Name	Description
15:8	LOW	Unsigned 8 bit pre-clamp value
7:0	HIGH	Unsigned 8 bit pre-clamp value

0x01A65A7C MDSS_MDP_WB_1_CSC_COMP2_PRECLAMP**Type:** RW**Reset State:** 0x000000FF

Component 2 pre-clamp value

double buffered: NO

MDSS_MDP_WB_1_CSC_COMP2_PRECLAMP

Bits	Name	Description
15:8	LOW	Unsigned 8 bit pre-clamp value
7:0	HIGH	Unsigned 8 bit pre-clamp value

0x01A65A80 MDSS_MDP_WB_1_CSC_COMP0_POSTCLAMP**Type:** RW**Reset State:** 0x000000FF

Component 0 post-clamp value

double buffered: NO

MDSS_MDP_WB_1_CSC_COMP0_POSTCLAMP

Bits	Name	Description
15:8	LOW	Unsigned 8 bit pre-clamp value
7:0	HIGH	Unsigned 8 bit pre-clamp value

0x01A65A84 MDSS_MDP_WB_1_CSC_COMP1_POSTCLAMP**Type:** RW**Reset State:** 0x000000FF

Component 1 post-clamp value

double buffered: NO

MDSS_MDP_WB_1_CSC_COMP1_POSTCLAMP

Bits	Name	Description
15:8	LOW	Unsigned 8 bit pre-clamp value
7:0	HIGH	Unsigned 8 bit pre-clamp value

0x01A65A88 MDSS_MDP_WB_1_CSC_COMP2_POSTCLAMP**Type:** RW**Reset State:** 0x000000FF

Component 2 post-clamp value

double buffered: NO

MDSS_MDP_WB_1_CSC_COMP2_POSTCLAMP

Bits	Name	Description
15:8	LOW	Unsigned 8 bit pre-clamp value
7:0	HIGH	Unsigned 8 bit pre-clamp value

0x01A65A8C MDSS_MDP_WB_1_CSC_COMP0_PREBIAS**Type:** RW**Reset State:** 0x00000000

Component 0 pre-bias value

double buffered: NO

MDSS_MDP_WB_1_CSC_COMP0_PREBIAS

Bits	Name	Description
8:0	VALUE	Signed 9 bit pre-bias value

0x01A65A90 MDSS_MDP_WB_1_CSC_COMP1_PREBIAS**Type:** RW**Reset State:** 0x00000000

Component 1 pre-bias value

double buffered: NO

MDSS_MDP_WB_1_CSC_COMP1_PREBIAS

Bits	Name	Description
8:0	VALUE	Signed 9 bit pre-bias value

0x01A65A94 MDSS_MDP_WB_1_CSC_COMP2_PREBIAS**Type:** RW**Reset State:** 0x00000000

Component 2 pre-bias value

double buffered: NO

MDSS_MDP_WB_1_CSC_COMP2_PREBIAS

Bits	Name	Description
8:0	VALUE	Signed 9 bit pre-bias value

0x01A65A98 MDSS_MDP_WB_1_CSC_COMP0_POSTBIAS**Type:** RW**Reset State:** 0x00000000

Component 0 post-bias value

double buffered: NO

MDSS_MDP_WB_1_CSC_COMP0_POSTBIAS

Bits	Name	Description
8:0	VALUE	Signed 9 bit pre-bias value

0x01A65A9C MDSS_MDP_WB_1_CSC_COMP1_POSTBIAS**Type:** RW**Reset State:** 0x00000000

Component 1 post-bias value

double buffered: NO

MDSS_MDP_WB_1_CSC_COMP1_POSTBIAS

Bits	Name	Description
8:0	VALUE	Signed 9 bit pre-bias value

0x01A65AA0 MDSS_MDP_WB_1_CSC_COMP2_POSTBIAS**Type:** RW**Reset State:** 0x00000000

Component 2 post-bias value

double buffered: NO

MDSS_MDP_WB_1_CSC_COMP2_POSTBIAS

Bits	Name	Description
8:0	VALUE	Signed 9 bit pre-bias value

0x01A65AB0 MDSS_MDP_WB_1_DST_ADDR_SW_STATUS**Type:** RW**Reset State:** 0x00000000

Security Status of the destination buffer plane

This provides a SW override to the hardware calculated content protection sent to VBIF

This value is logically or ed with the source content protection produced by hardware

This information will be presented to VBIF per transaction and reflected as part of the memory transaction

Note that the VBIF context banks need to be properly programmed for the output to be successful

double buffered: NO

MDSS_MDP_WB_1_DST_ADDR_SW_STATUS

Bits	Name	Description
0	DST_ADDR_CP	This is the secure status bit for destination plane 0 (if used) Memory transactions will not complete if this status doesnt correspond to system programming in the rest of the path to memory 0x0: Content Protected Buffer State 0x1: Non Content Protected Buffer State

17.17.10 MDSS_MDP_WB_2

MDP WB hardware register description

17.17.10.1 MDSS_MDP_WB_2_WB

WB

MDP WB hardware register description

0x01A66000 MDSS_MDP_WB_2_DST_FORMAT**Type:** RW**Reset State:** 0x00000000

Memory Write Back Output Format

double buffered: NO

MDSS_MDP_WB_2_DST_FORMAT

Bits	Name	Description
31:30	FRAME_FORMAT	FRAME_FORMAT In the current revision, there is no support for tiled formats 0x0: Linear, no tile
29:26	DST_CHROMA_SITE	Destination Chroma Site 0x0: A (cosite for YUV422) 0x1: B (offsite for YUV422) 0x2: C (N/A for YUV422) 0x3: D (N/A for YUV422) 0x4: E (N/A for YUV422) 0x5: F (N/A for YUV422) 0x6: G (N/A for YUV422) 0x7: H (N/A for YUV422) 0x8: I (N/A for YUV422)
25:23	DST_CHROMA_SAMP	Chroma Sampling 0x0: RGB/4:4:4 0x1: 4:2:2 - H2V1 0x2: 4:2:2 - H1V2 0x3: 4:2:0 - N2V1 0x4: 4:2:0 - N1V2
22	DST_DITHER_EN	Destination Dither Enable
20:19	WRITE_PLANES	Number of planes 0x0: Interleaved 0x1: Planar 0x2: Pseudo-planar
18	PACK_ALIGN	0x0: LSB 0x1: MSB
17	PACK_TIGHT	0x0: Loose 0x1: Tight
14	DST_ALPHA_X	0x0: Use alpha value from pipe 0x1: Use alpha value from register
13:12	PACK_COUNT	Valid packing elements count 0x0 = 1 element 0x1 = 2 elements 0x2 = 3 elements 0x3 = 4 elements
10:9	DST_BPP	Effective byte per pixel 0x0: 1 byte 0x1: 2 bytes 0x2: 3 bytes 0x3: 4 bytes
8	DSTC3_EN	0x1: Destination has alpha

MDSS_MDP_WB_2_DST_FORMAT (cont.)

Bits	Name	Description
7:6	DSTC3_OUT	Number of bits for component 3 output 0x0: 1 bit 0x1: 4 bits 0x2: 6 bits 0x3: 8 bits
5:4	DSTC2_OUT	Number of bits for component 2 output 0x0: 4 bits 0x1: 5 bits 0x2: 6 bits 0x3: 8 bits
3:2	DSTC1_OUT	Number of bits for component 1 output 0x0: 4 bits 0x1: 5 bits 0x2: 6 bits 0x3: 8 bits
1:0	DSTC0_OUT	Number of bits for component 0 output 0x0: 4 bits 0x1: 5 bits 0x2: 6 bits 0x3: 8 bits

0x01A66004 MDSS_MDP_WB_2_DST_OP_MODE**Type:** RW**Reset State:** 0x00000500

Destination Operation Mode

double buffered: NO

MDSS_MDP_WB_2_DST_OP_MODE

Bits	Name	Description
14	CHROMA_DWN_SAMPLE_V_MTHD	Vertical Chroma Down Sample method 0x0: non-pixel dropping 0x1: pixel dropping
13	CHROMA_DWN_SAMPLE_H_MTHD	Horizontal Chroma Down Sample method 0x0: non-pixel dropping 0x1: pixel dropping
12	CHROMA_DWN_SAMPLE_FORMAT	Chroma Down Sample format 0x0: YUV422 H2V1 0x1: YUV420

MDSS_MDP_WB_2_DST_OP_MODE (cont.)

Bits	Name	Description
11	CHROMA_DWN_SAMPLE_EN	Chroma Down Sample enable for YUV444 NOTE: This is not valid for WB0. 0x0: Disabled 0x1: Enabled
10	CSC_DST_DATA_FORMAT	Destination Data Format 0x0: RGB 0x1: YCbCr
9	CSC_SRC_DATA_FORMAT	Source Data Format 0x0: RGB 0x1: YCbCr
8	CSC_EN	Enables the color matrix conversion NOTE: This is not valid for WB0. 0x0: Bypass 0x1: Enabled
6	ROT_EN	0x0: Rotate Disabled 0x1: Rotate Enabled
5	ROT_MODE	0x0: Rotate 0 0x1: Rotate 90
4	BLOCK_SIZE	0x0: 64x64 Block Size 0x1: 128x128 Block Size

0x01A66008 MDSS_MDP_WB_2_DST_PACK_PATTERN**Type:** RW**Reset State:** 0x00000000

Write Back Pack Pattern

double buffered: NO

MDSS_MDP_WB_2_DST_PACK_PATTERN

Bits	Name	Description
25:24	ELEMENT3	0x0: Color0 0x1: Color1 0x2: Color2 0x3: Color3
17:16	ELEMENT2	0x0: Color0 0x1: Color1 0x2: Color2 0x3: Color3

MDSS_MDP_WB_2_DST_PACK_PATTERN (cont.)

Bits	Name	Description
9:8	ELEMENT1	0x0: Color0 0x1: Color1 0x2: Color2 0x3: Color3
1:0	ELEMENT0	0x0: Color0 0x1: Color1 0x2: Color2 0x3: Color3

0x01A6600C MDSS_MDP_WB_2_DST0_ADDR**Type:** RW**Reset State:** 0x00000000

Write Back Destination Address

For fetching BWC data, this plane refers to the compressed RAU data stream

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_WB_2_DST0_ADDR

Bits	Name	Description
31:0	ADDR	Destination Address (byte-aligned)

0x01A66010 MDSS_MDP_WB_2_DST1_ADDR**Type:** RW**Reset State:** 0x00000000

Write Back Destination Address

For fetching BWC data, this plane refers to the META data to fetch the lengths of the compressed RAU data stream

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_WB_2_DST1_ADDR

Bits	Name	Description
31:0	ADDR	Destination Address (byte-aligned)

0x01A66014 MDSS_MDP_WB_2_DST2_ADDR**Type:** RW**Reset State:** 0x00000000

Write Back Destination Address

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_WB_2_DST2_ADDR

Bits	Name	Description
31:0	ADDR	Destination Address (byte-aligned)

0x01A66018 MDSS_MDP_WB_2_DST3_ADDR**Type:** RW**Reset State:** 0x00000000

Write Back Destination Address

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_WB_2_DST3_ADDR

Bits	Name	Description
31:0	ADDR	Destination Address (byte-aligned)

0x01A6601C MDSS_MDP_WB_2_DST_YSTRIDE0**Type:** RW**Reset State:** 0x00000000

Stride Values

Double-buffered

double buffered: NO

MDSS_MDP_WB_2_DST_YSTRIDE0

Bits	Name	Description
31:16	DST1_YSTRIDE	Plane 1 Y stride in bytes
15:0	DST0_YSTRIDE	Plane 0 Y stride in bytes

0x01A66020 MDSS_MDP_WB_2_DST_YSTRIDE1

Type: RW

Reset State: 0x00000000

Stride Values

Double-buffered

double buffered: NO

MDSS_MDP_WB_2_DST_YSTRIDE1

Bits	Name	Description
31:16	DST3_YSTRIDE	Plane 3 Y stride in bytes
15:0	DST2_YSTRIDE	Plane 2 Y stride in bytes

0x01A66024 MDSS_MDP_WB_2_DST_DITHER_BITDEPTH

Type: RW

Reset State: 0x00000000

Write Dither Strength

double buffered: NO

MDSS_MDP_WB_2_DST_DITHER_BITDEPTH

Bits	Name	Description
5:4	C2_BITS	Number of MSB bits on component2 to indicate the output depth 0x0: 4bits 0x1: 5 bits 0x2: 6 bits 0x3: 8 bits

MDSS_MDP_WB_2_DST_DITHER_BITDEPTH (cont.)

Bits	Name	Description
3:2	C1_BITS	Number of MSB bits on component1 to indicate the output depth 0x0: 4bits 0x1: 5 bits 0x2: 6 bits 0x3: 8 bits
1:0	C0_BITS	Number of MSB bits on component0 to indicate the output depth 0x0: 4bits 0x1: 5 bits 0x2: 6 bits 0x3: 8 bits

0x01A66030 MDSS_MDP_WB_2_DITHER_MATRIX_ROW0**Type:** RW**Reset State:** 0x00005D7F

Dither Strength Matrix Values Register for first row which covers entries [0-3]

double buffered: NO

MDSS_MDP_WB_2_DITHER_MATRIX_ROW0

Bits	Name	Description
15:12	ENTRY_3	Entry 3 for the dither matrix
11:8	ENTRY_2	Entry 2 for the dither matrix
7:4	ENTRY_1	Entry 1 for the dither matrix
3:0	ENTRY_0	Entry 0 for the dither matrix

0x01A66034 MDSS_MDP_WB_2_DITHER_MATRIX_ROW1**Type:** RW**Reset State:** 0x000091B3

Dither Strength Matrix Values Register for first row which covers entries [4-7]

double buffered: NO

MDSS_MDP_WB_2_DITHER_MATRIX_ROW1

Bits	Name	Description
15:12	ENTRY_7	Entry 7 for the dither matrix

MDSS_MDP_WB_2_DITHER_MATRIX_ROW1 (cont.)

Bits	Name	Description
11:8	ENTRY_6	Entry 6 for the dither matrix
7:4	ENTRY_5	Entry 5 for the dither matrix
3:0	ENTRY_4	Entry 4 for the dither matrix

0x01A66038 MDSS_MDP_WB_2_DITHER_MATRIX_ROW2**Type:** RW**Reset State:** 0x00006E4C

Dither Strength Matrix Values Register for first row which covers entries [8-11]

double buffered: NO

MDSS_MDP_WB_2_DITHER_MATRIX_ROW2

Bits	Name	Description
15:12	ENTRY_11	Entry 11 for the dither matrix
11:8	ENTRY_10	Entry 10 for the dither matrix
7:4	ENTRY_9	Entry 9 for the dither matrix
3:0	ENTRY_8	Entry 8 for the dither matrix

0x01A6603C MDSS_MDP_WB_2_DITHER_MATRIX_ROW3**Type:** RW**Reset State:** 0x0000A280

Dither Strength Matrix Values Register for first row which covers entries [12-15]

double buffered: NO

MDSS_MDP_WB_2_DITHER_MATRIX_ROW3

Bits	Name	Description
15:12	ENTRY_15	Entry 15 for the dither matrix
11:8	ENTRY_14	Entry 14 for the dither matrix
7:4	ENTRY_13	Entry 13 for the dither matrix
3:0	ENTRY_12	Entry 12 for the dither matrix

0x01A66048 MDSS_MDP_WB_2_DST_WRITE_CONFIG**Type:** RW**Reset State:** 0x00000008

Write-back Configuration double buffered: NO

This register must NOT be altered during operation and MDP must be idle when changing values

MDSS_MDP_WB_2_DST_WRITE_CONFIG

Bits	Name	Description
10	ADDR_GEN_MODE	
9:8	HIGHEST_BANK_BIT	
7:6	PRIORITY_LEVEL	Priority Level. Side-band priority reflecting the status of VBIF client. 0x3 - max priority 0x0 - min priority
5:4	REQ_PRIORITY	Request Priority. In-band priority level assigned when VBIF transaction is created. 0x3 - max priority 0x0 - min priority
3:0	MAX_BURST_SIZE	Maximum burst beat size. Hardware fetch engines operate on 128-bit bus Possible values: 0x8: burst of 8 0x4: burst of 4 0x2: burst of 2 0x1: burst of 1

0x01A66054 MDSS_MDP_WB_2_ROTATOR_PIPE_DOWNSCALER**Type:** RW**Reset State:** 0x00000000**MDSS_MDP_WB_2_ROTATOR_PIPE_DOWNSCALER**

Bits	Name	Description
23:16	DNOSC_FACTOR_V	0x0: V_BYPASS 0x2: V_DNOSC_X2 0x4: V_DNOSC_X4 0x8: V_DNOSC_X8 0x10: V_DNOSC_X16 0x20: V_DNOSC_X32 0x40: V_DNOSC_X64

MDSS_MDP_WB_2_ROTATOR_PIPE_DOWNSCALER (cont.)

Bits	Name	Description
7:0	DNOSC_FACTOR_H	0x0: H_BYPASS 0x2: H_DNOSC_X2 0x4: H_DNOSC_X4 0x8: H_DNOSC_X8 0x10: H_DNOSC_X16 0x20: H_DNOSC_X32 0x40: H_DNOSC_X64

0x01A66074 MDSS_MDP_WB_2_OUT_SIZE**Type:** RW**Reset State:** 0x00000000

Output Size

Should be same as source is scaling is NOT enabled

double buffered: NO

MDSS_MDP_WB_2_OUT_SIZE

Bits	Name	Description
31:16	DST_H	Height of Output ROI
15:0	DST_W	Width of Output ROI

0x01A66078 MDSS_MDP_WB_2_DST_ALPHA_X_VALUE**Type:** RW**Reset State:** 0x00000000

Alpha value used for XRGB formats

double buffered: NO

MDSS_MDP_WB_2_DST_ALPHA_X_VALUE

Bits	Name	Description
7:0	VALUE	Alpha value used for XRGB formats

0x01A66260 MDSS_MDP_WB_2_CSC_MATRIX_COEFF_0**Type:** RW**Reset State:** 0x00000200

Color Space conversion coefficients

double buffered: NO

MDSS_MDP_WB_2_CSC_MATRIX_COEFF_0

Bits	Name	Description
28:16	COEFF_12	Coefficient value in S4.9 format Row 1 Column 2
12:0	COEFF_11	Coefficient value in S4.9 format Row 1 Column 1

0x01A66264 MDSS_MDP_WB_2_CSC_MATRIX_COEFF_1**Type:** RW**Reset State:** 0x00000000

Color Space conversion coefficients

double buffered: NO

MDSS_MDP_WB_2_CSC_MATRIX_COEFF_1

Bits	Name	Description
28:16	COEFF_21	Coefficient value in S4.9 format Row 2 Column 1
12:0	COEFF_13	Coefficient value in S4.9 format Row 1 Column 3

0x01A66268 MDSS_MDP_WB_2_CSC_MATRIX_COEFF_2**Type:** RW**Reset State:** 0x00000200

Color Space conversion coefficients

double buffered: NO

MDSS_MDP_WB_2_CSC_MATRIX_COEFF_2

Bits	Name	Description
28:16	COEFF_23	Coefficient value in S4.9 format Row 2 Column 3
12:0	COEFF_22	Coefficient value in S4.9 format Row 2 Column 2

0x01A6626C MDSS_MDP_WB_2_CSC_MATRIX_COEFF_3**Type:** RW**Reset State:** 0x00000000

Color Space conversion coefficients

double buffered: NO

MDSS_MDP_WB_2_CSC_MATRIX_COEFF_3

Bits	Name	Description
28:16	COEFF_32	Coefficient value in S4.9 format Row 3 Column 2
12:0	COEFF_31	Coefficient value in S4.9 format Row 3 Column 1

0x01A66270 MDSS_MDP_WB_2_CSC_MATRIX_COEFF_4**Type:** RW**Reset State:** 0x00000200

Color Space conversion coefficients

double buffered: NO

MDSS_MDP_WB_2_CSC_MATRIX_COEFF_4

Bits	Name	Description
12:0	COEFF_33	Coefficient value in S4.9 format Row 3 Column 3

0x01A66274 MDSS_MDP_WB_2_CSC_COMP0_PRECLAMP**Type:** RW**Reset State:** 0x000000FF

Component 0 pre-clamp value

double buffered: NO

MDSS_MDP_WB_2_CSC_COMP0_PRECLAMP

Bits	Name	Description
15:8	LOW	Unsigned 8 bit pre-clamp value
7:0	HIGH	Unsigned 8 bit pre-clamp value

0x01A66278 MDSS_MDP_WB_2_CSC_COMP1_PRECLAMP**Type:** RW**Reset State:** 0x000000FF

Component 1 pre-clamp value

double buffered: NO

MDSS_MDP_WB_2_CSC_COMP1_PRECLAMP

Bits	Name	Description
15:8	LOW	Unsigned 8 bit pre-clamp value
7:0	HIGH	Unsigned 8 bit pre-clamp value

0x01A6627C MDSS_MDP_WB_2_CSC_COMP2_PRECLAMP**Type:** RW**Reset State:** 0x000000FF

Component 2 pre-clamp value

double buffered: NO

MDSS_MDP_WB_2_CSC_COMP2_PRECLAMP

Bits	Name	Description
15:8	LOW	Unsigned 8 bit pre-clamp value
7:0	HIGH	Unsigned 8 bit pre-clamp value

0x01A66280 MDSS_MDP_WB_2_CSC_COMP0_POSTCLAMP**Type:** RW**Reset State:** 0x000000FF

Component 0 post-clamp value

double buffered: NO

MDSS_MDP_WB_2_CSC_COMP0_POSTCLAMP

Bits	Name	Description
15:8	LOW	Unsigned 8 bit pre-clamp value
7:0	HIGH	Unsigned 8 bit pre-clamp value

0x01A66284 MDSS_MDP_WB_2_CSC_COMP1_POSTCLAMP**Type:** RW**Reset State:** 0x000000FF

Component 1 post-clamp value

double buffered: NO

MDSS_MDP_WB_2_CSC_COMP1_POSTCLAMP

Bits	Name	Description
15:8	LOW	Unsigned 8 bit pre-clamp value
7:0	HIGH	Unsigned 8 bit pre-clamp value

0x01A66288 MDSS_MDP_WB_2_CSC_COMP2_POSTCLAMP**Type:** RW**Reset State:** 0x000000FF

Component 2 post-clamp value

double buffered: NO

MDSS_MDP_WB_2_CSC_COMP2_POSTCLAMP

Bits	Name	Description
15:8	LOW	Unsigned 8 bit pre-clamp value
7:0	HIGH	Unsigned 8 bit pre-clamp value

0x01A6628C MDSS_MDP_WB_2_CSC_COMP0_PREBIAS**Type:** RW**Reset State:** 0x00000000

Component 0 pre-bias value

double buffered: NO

MDSS_MDP_WB_2_CSC_COMP0_PREBIAS

Bits	Name	Description
8:0	VALUE	Signed 9 bit pre-bias value

0x01A66290 MDSS_MDP_WB_2_CSC_COMP1_PREBIAS**Type:** RW**Reset State:** 0x00000000

Component 1 pre-bias value

double buffered: NO

MDSS_MDP_WB_2_CSC_COMP1_PREBIAS

Bits	Name	Description
8:0	VALUE	Signed 9 bit pre-bias value

0x01A66294 MDSS_MDP_WB_2_CSC_COMP2_PREBIAS**Type:** RW**Reset State:** 0x00000000

Component 2 pre-bias value

double buffered: NO

MDSS_MDP_WB_2_CSC_COMP2_PREBIAS

Bits	Name	Description
8:0	VALUE	Signed 9 bit pre-bias value

0x01A66298 MDSS_MDP_WB_2_CSC_COMP0_POSTBIAS**Type:** RW**Reset State:** 0x00000000

Component 0 post-bias value

double buffered: NO

MDSS_MDP_WB_2_CSC_COMP0_POSTBIAS

Bits	Name	Description
8:0	VALUE	Signed 9 bit pre-bias value

0x01A6629C MDSS_MDP_WB_2_CSC_COMP1_POSTBIAS**Type:** RW**Reset State:** 0x00000000

Component 1 post-bias value

double buffered: NO

MDSS_MDP_WB_2_CSC_COMP1_POSTBIAS

Bits	Name	Description
8:0	VALUE	Signed 9 bit pre-bias value

0x01A662A0 MDSS_MDP_WB_2_CSC_COMP2_POSTBIAS**Type:** RW**Reset State:** 0x00000000

Component 2 post-bias value

double buffered: NO

MDSS_MDP_WB_2_CSC_COMP2_POSTBIAS

Bits	Name	Description
8:0	VALUE	Signed 9 bit pre-bias value

0x01A662B0 MDSS_MDP_WB_2_DST_ADDR_SW_STATUS**Type:** RW**Reset State:** 0x00000000

Security Status of the destination buffer plane

This provides a SW override to the hardware calculated content protection sent to VBIF

This value is logically or ed with the source content protection produced by hardware

This information will be presented to VBIF per transaction and reflected as part of the memory transaction

Note that the VBIF context banks need to be properly programmed for the output to be successful

double buffered: NO

MDSS_MDP_WB_2_DST_ADDR_SW_STATUS

Bits	Name	Description
0	DST_ADDR_CP	This is the secure status bit for destination plane 0 (if used) Memory transactions will not complete if this status doesnt corespond to system programming in the rest of the path to memory 0x0: Content Protected Buffer State 0x1: Non Content Protected Buffer State

17.17.11 MDSS_MDP_WB_3

MDP WB hardware register description

17.17.11.1 MDSS_MDP_WB_3_WB

WB

MDP WB hardware register description

0x01A66800 MDSS_MDP_WB_3_DST_FORMAT**Type:** RW**Reset State:** 0x00000000

Memory Write Back Output Format

double buffered: NO

MDSS_MDP_WB_3_DST_FORMAT

Bits	Name	Description
31:30	FRAME_FORMAT	FRAME_FORMAT In the current revision, there is no support for tiled formats 0x0: Linear, no tile
29:26	DST_CHROMA_SITE	Destination Chroma Site 0x0: A (cosite for YUV422) 0x1: B (offsite for YUV422) 0x2: C (N/A for YUV422) 0x3: D (N/A for YUV422) 0x4: E (N/A for YUV422) 0x5: F (N/A for YUV422) 0x6: G (N/A for YUV422) 0x7: H (N/A for YUV422) 0x8: I (N/A for YUV422)
25:23	DST_CHROMA_SAMP	Chroma Sampling 0x0: RGB/4:4:4 0x1: 4:2:2 - H2V1 0x2: 4:2:2 - H1V2 0x3: 4:2:0 - N2V1 0x4: 4:2:0 - N1V2
22	DST_DITHER_EN	Destination Dither Enable
20:19	WRITE_PLANES	Number of planes 0x0: Interleaved 0x1: Planar 0x2: Pseudo-planar
18	PACK_ALIGN	0x0: LSB 0x1: MSB
17	PACK_TIGHT	0x0: Loose 0x1: Tight
14	DST_ALPHA_X	0x0: Use alpha value from pipe 0x1: Use alpha value from register
13:12	PACK_COUNT	Valid packing elements count 0x0 = 1 element 0x1 = 2 elements 0x2 = 3 elements 0x3 = 4 elements
10:9	DST_BPP	Effective byte per pixel 0x0: 1 byte 0x1: 2 bytes 0x2: 3 bytes 0x3: 4 bytes
8	DSTC3_EN	0x1: Destination has alpha

MDSS_MDP_WB_3_DST_FORMAT (cont.)

Bits	Name	Description
7:6	DSTC3_OUT	Number of bits for component 3 output 0x0: 1 bit 0x1: 4 bits 0x2: 6 bits 0x3: 8 bits
5:4	DSTC2_OUT	Number of bits for component 2 output 0x0: 4 bits 0x1: 5 bits 0x2: 6 bits 0x3: 8 bits
3:2	DSTC1_OUT	Number of bits for component 1 output 0x0: 4 bits 0x1: 5 bits 0x2: 6 bits 0x3: 8 bits
1:0	DSTC0_OUT	Number of bits for component 0 output 0x0: 4 bits 0x1: 5 bits 0x2: 6 bits 0x3: 8 bits

0x01A66804 MDSS_MDP_WB_3_DST_OP_MODE**Type:** RW**Reset State:** 0x00000500

Destination Operation Mode

double buffered: NO

MDSS_MDP_WB_3_DST_OP_MODE

Bits	Name	Description
14	CHROMA_DWN_SAMPLE_V_MTHD	Vertical Chroma Down Sample method 0x0: non-pixel dropping 0x1: pixel dropping
13	CHROMA_DWN_SAMPLE_H_MTHD	Horizontal Chroma Down Sample method 0x0: non-pixel dropping 0x1: pixel dropping
12	CHROMA_DWN_SAMPLE_FORMAT	Chroma Down Sample format 0x0: YUV422 H2V1 0x1: YUV420

MDSS_MDP_WB_3_DST_OP_MODE (cont.)

Bits	Name	Description
11	CHROMA_DWN_SAMPLE_EN	Chroma Down Sample enable for YUV444 NOTE: This is not valid for WB0. 0x0: Disabled 0x1: Enabled
10	CSC_DST_DATA_FORMAT	Destination Data Format 0x0: RGB 0x1: YCbCr
9	CSC_SRC_DATA_FORMAT	Source Data Format 0x0: RGB 0x1: YCbCr
8	CSC_EN	Enables the color matrix conversion NOTE: This is not valid for WB0. 0x0: Bypass 0x1: Enabled
6	ROT_EN	0x0: Rotate Disabled 0x1: Rotate Enabled
5	ROT_MODE	0x0: Rotate 0 0x1: Rotate 90
4	BLOCK_SIZE	0x0: 64x64 Block Size 0x1: 128x128 Block Size

0x01A66808 MDSS_MDP_WB_3_DST_PACK_PATTERN**Type:** RW**Reset State:** 0x00000000

Write Back Pack Pattern

double buffered: NO

MDSS_MDP_WB_3_DST_PACK_PATTERN

Bits	Name	Description
25:24	ELEMENT3	0x0: Color0 0x1: Color1 0x2: Color2 0x3: Color3
17:16	ELEMENT2	0x0: Color0 0x1: Color1 0x2: Color2 0x3: Color3

MDSS_MDP_WB_3_DST_PACK_PATTERN (cont.)

Bits	Name	Description
9:8	ELEMENT1	0x0: Color0 0x1: Color1 0x2: Color2 0x3: Color3
1:0	ELEMENT0	0x0: Color0 0x1: Color1 0x2: Color2 0x3: Color3

0x01A6680C MDSS_MDP_WB_3_DST0_ADDR**Type:** RW**Reset State:** 0x00000000

Write Back Destination Address

For fetching BWC data, this plane refers to the compressed RAU data stream

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_WB_3_DST0_ADDR

Bits	Name	Description
31:0	ADDR	Destination Address (byte-aligned)

0x01A66810 MDSS_MDP_WB_3_DST1_ADDR**Type:** RW**Reset State:** 0x00000000

Write Back Destination Address

For fetching BWC data, this plane refers to the META data to fetch the lengths of the compressed RAU data stream

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_WB_3_DST1_ADDR

Bits	Name	Description
31:0	ADDR	Destination Address (byte-aligned)

0x01A66814 MDSS_MDP_WB_3_DST2_ADDR**Type:** RW**Reset State:** 0x00000000

Write Back Destination Address

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_WB_3_DST2_ADDR

Bits	Name	Description
31:0	ADDR	Destination Address (byte-aligned)

0x01A66818 MDSS_MDP_WB_3_DST3_ADDR**Type:** RW**Reset State:** 0x00000000

Write Back Destination Address

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_WB_3_DST3_ADDR

Bits	Name	Description
31:0	ADDR	Destination Address (byte-aligned)

0x01A6681C MDSS_MDP_WB_3_DST_YSTRIDE0**Type:** RW**Reset State:** 0x00000000

Stride Values

Double-buffered

double buffered: NO

MDSS_MDP_WB_3_DST_YSTRIDE0

Bits	Name	Description
31:16	DST1_YSTRIDE	Plane 1 Y stride in bytes
15:0	DST0_YSTRIDE	Plane 0 Y stride in bytes

0x01A66820 MDSS_MDP_WB_3_DST_YSTRIDE1

Type: RW

Reset State: 0x00000000

Stride Values

Double-buffered

double buffered: NO

MDSS_MDP_WB_3_DST_YSTRIDE1

Bits	Name	Description
31:16	DST3_YSTRIDE	Plane 3 Y stride in bytes
15:0	DST2_YSTRIDE	Plane 2 Y stride in bytes

0x01A66824 MDSS_MDP_WB_3_DST_DITHER_BITDEPTH

Type: RW

Reset State: 0x00000000

Write Dither Strength

double buffered: NO

MDSS_MDP_WB_3_DST_DITHER_BITDEPTH

Bits	Name	Description
5:4	C2_BITS	Number of MSB bits on component2 to indicate the output depth 0x0: 4bits 0x1: 5 bits 0x2: 6 bits 0x3: 8 bits

MDSS_MDP_WB_3_DST_DITHER_BITDEPTH (cont.)

Bits	Name	Description
3:2	C1_BITS	Number of MSB bits on component1 to indicate the output depth 0x0: 4bits 0x1: 5 bits 0x2: 6 bits 0x3: 8 bits
1:0	C0_BITS	Number of MSB bits on component0 to indicate the output depth 0x0: 4bits 0x1: 5 bits 0x2: 6 bits 0x3: 8 bits

0x01A66830 MDSS_MDP_WB_3_DITHER_MATRIX_ROW0**Type:** RW**Reset State:** 0x00005D7F

Dither Strength Matrix Values Register for first row which covers entries [0-3]

double buffered: NO

MDSS_MDP_WB_3_DITHER_MATRIX_ROW0

Bits	Name	Description
15:12	ENTRY_3	Entry 3 for the dither matrix
11:8	ENTRY_2	Entry 2 for the dither matrix
7:4	ENTRY_1	Entry 1 for the dither matrix
3:0	ENTRY_0	Entry 0 for the dither matrix

0x01A66834 MDSS_MDP_WB_3_DITHER_MATRIX_ROW1**Type:** RW**Reset State:** 0x000091B3

Dither Strength Matrix Values Register for first row which covers entries [4-7]

double buffered: NO

MDSS_MDP_WB_3_DITHER_MATRIX_ROW1

Bits	Name	Description
15:12	ENTRY_7	Entry 7 for the dither matrix

MDSS_MDP_WB_3_DITHER_MATRIX_ROW1 (cont.)

Bits	Name	Description
11:8	ENTRY_6	Entry 6 for the dither matrix
7:4	ENTRY_5	Entry 5 for the dither matrix
3:0	ENTRY_4	Entry 4 for the dither matrix

0x01A66838 MDSS_MDP_WB_3_DITHER_MATRIX_ROW2**Type:** RW**Reset State:** 0x00006E4C

Dither Strength Matrix Values Register for first row which covers entries [8-11]

double buffered: NO

MDSS_MDP_WB_3_DITHER_MATRIX_ROW2

Bits	Name	Description
15:12	ENTRY_11	Entry 11 for the dither matrix
11:8	ENTRY_10	Entry 10 for the dither matrix
7:4	ENTRY_9	Entry 9 for the dither matrix
3:0	ENTRY_8	Entry 8 for the dither matrix

0x01A6683C MDSS_MDP_WB_3_DITHER_MATRIX_ROW3**Type:** RW**Reset State:** 0x0000A280

Dither Strength Matrix Values Register for first row which covers entries [12-15]

double buffered: NO

MDSS_MDP_WB_3_DITHER_MATRIX_ROW3

Bits	Name	Description
15:12	ENTRY_15	Entry 15 for the dither matrix
11:8	ENTRY_14	Entry 14 for the dither matrix
7:4	ENTRY_13	Entry 13 for the dither matrix
3:0	ENTRY_12	Entry 12 for the dither matrix

0x01A66848 MDSS_MDP_WB_3_DST_WRITE_CONFIG**Type:** RW**Reset State:** 0x00000008

Write-back Configuration double buffered: NO

This register must NOT be altered during operation and MDP must be idle when changing values

MDSS_MDP_WB_3_DST_WRITE_CONFIG

Bits	Name	Description
10	ADDR_GEN_MODE	
9:8	HIGHEST_BANK_BIT	
7:6	PRIORITY_LEVEL	Priority Level. Side-band priority reflecting the status of VBIF client. 0x3 - max priority 0x0 - min priority
5:4	REQ_PRIORITY	Request Priority. In-band priority level assigned when VBIF transaction is created. 0x3 - max priority 0x0 - min priority
3:0	MAX_BURST_SIZE	Maximum burst beat size. Hardware fetch engines operate on 128-bit bus Possible values: 0x8: burst of 8 0x4: burst of 4 0x2: burst of 2 0x1: burst of 1

0x01A66854 MDSS_MDP_WB_3_ROTATOR_PIPE_DOWNSCALER**Type:** RW**Reset State:** 0x00000000**MDSS_MDP_WB_3_ROTATOR_PIPE_DOWNSCALER**

Bits	Name	Description
23:16	DNOSC_FACTOR_V	0x0: V_BYPASS 0x2: V_DNOSC_X2 0x4: V_DNOSC_X4 0x8: V_DNOSC_X8 0x10: V_DNOSC_X16 0x20: V_DNOSC_X32 0x40: V_DNOSC_X64

MDSS_MDP_WB_3_ROTATOR_PIPE_DOWNSCALER (cont.)

Bits	Name	Description
7:0	DNOSC_FACTOR_H	0x0: H_BYPASS 0x2: H_DNOSC_X2 0x4: H_DNOSC_X4 0x8: H_DNOSC_X8 0x10: H_DNOSC_X16 0x20: H_DNOSC_X32 0x40: H_DNOSC_X64

0x01A66874 MDSS_MDP_WB_3_OUT_SIZE**Type:** RW**Reset State:** 0x00000000

Output Size

Should be same as source is scaling is NOT enabled

double buffered: NO

MDSS_MDP_WB_3_OUT_SIZE

Bits	Name	Description
31:16	DST_H	Height of Output ROI
15:0	DST_W	Width of Output ROI

0x01A66878 MDSS_MDP_WB_3_DST_ALPHA_X_VALUE**Type:** RW**Reset State:** 0x00000000

Alpha value used for XRGB formats

double buffered: NO

MDSS_MDP_WB_3_DST_ALPHA_X_VALUE

Bits	Name	Description
7:0	VALUE	Alpha value used for XRGB formats

0x01A66A60 MDSS_MDP_WB_3_CSC_MATRIX_COEFF_0**Type:** RW**Reset State:** 0x00000200

Color Space conversion coefficients

double buffered: NO

MDSS_MDP_WB_3_CSC_MATRIX_COEFF_0

Bits	Name	Description
28:16	COEFF_12	Coefficient value in S4.9 format Row 1 Column 2
12:0	COEFF_11	Coefficient value in S4.9 format Row 1 Column 1

0x01A66A64 MDSS_MDP_WB_3_CSC_MATRIX_COEFF_1**Type:** RW**Reset State:** 0x00000000

Color Space conversion coefficients

double buffered: NO

MDSS_MDP_WB_3_CSC_MATRIX_COEFF_1

Bits	Name	Description
28:16	COEFF_21	Coefficient value in S4.9 format Row 2 Column 1
12:0	COEFF_13	Coefficient value in S4.9 format Row 1 Column 3

0x01A66A68 MDSS_MDP_WB_3_CSC_MATRIX_COEFF_2**Type:** RW**Reset State:** 0x00000200

Color Space conversion coefficients

double buffered: NO

MDSS_MDP_WB_3_CSC_MATRIX_COEFF_2

Bits	Name	Description
28:16	COEFF_23	Coefficient value in S4.9 format Row 2 Column 3
12:0	COEFF_22	Coefficient value in S4.9 format Row 2 Column 2

0x01A66A6C MDSS_MDP_WB_3_CSC_MATRIX_COEFF_3**Type:** RW**Reset State:** 0x00000000

Color Space conversion coefficients

double buffered: NO

MDSS_MDP_WB_3_CSC_MATRIX_COEFF_3

Bits	Name	Description
28:16	COEFF_32	Coefficient value in S4.9 format Row 3 Column 2
12:0	COEFF_31	Coefficient value in S4.9 format Row 3 Column 1

0x01A66A70 MDSS_MDP_WB_3_CSC_MATRIX_COEFF_4**Type:** RW**Reset State:** 0x00000200

Color Space conversion coefficients

double buffered: NO

MDSS_MDP_WB_3_CSC_MATRIX_COEFF_4

Bits	Name	Description
12:0	COEFF_33	Coefficient value in S4.9 format Row 3 Column 3

0x01A66A74 MDSS_MDP_WB_3_CSC_COMP0_PRECLAMP**Type:** RW**Reset State:** 0x000000FF

Component 0 pre-clamp value

double buffered: NO

MDSS_MDP_WB_3_CSC_COMP0_PRECLAMP

Bits	Name	Description
15:8	LOW	Unsigned 8 bit pre-clamp value
7:0	HIGH	Unsigned 8 bit pre-clamp value

0x01A66A78 MDSS_MDP_WB_3_CSC_COMP1_PRECLAMP**Type:** RW**Reset State:** 0x000000FF

Component 1 pre-clamp value

double buffered: NO

MDSS_MDP_WB_3_CSC_COMP1_PRECLAMP

Bits	Name	Description
15:8	LOW	Unsigned 8 bit pre-clamp value
7:0	HIGH	Unsigned 8 bit pre-clamp value

0x01A66A7C MDSS_MDP_WB_3_CSC_COMP2_PRECLAMP**Type:** RW**Reset State:** 0x000000FF

Component 2 pre-clamp value

double buffered: NO

MDSS_MDP_WB_3_CSC_COMP2_PRECLAMP

Bits	Name	Description
15:8	LOW	Unsigned 8 bit pre-clamp value
7:0	HIGH	Unsigned 8 bit pre-clamp value

0x01A66A80 MDSS_MDP_WB_3_CSC_COMP0_POSTCLAMP**Type:** RW**Reset State:** 0x000000FF

Component 0 post-clamp value

double buffered: NO

MDSS_MDP_WB_3_CSC_COMP0_POSTCLAMP

Bits	Name	Description
15:8	LOW	Unsigned 8 bit pre-clamp value
7:0	HIGH	Unsigned 8 bit pre-clamp value

0x01A66A84 MDSS_MDP_WB_3_CSC_COMP1_POSTCLAMP**Type:** RW**Reset State:** 0x000000FF

Component 1 post-clamp value

double buffered: NO

MDSS_MDP_WB_3_CSC_COMP1_POSTCLAMP

Bits	Name	Description
15:8	LOW	Unsigned 8 bit pre-clamp value
7:0	HIGH	Unsigned 8 bit pre-clamp value

0x01A66A88 MDSS_MDP_WB_3_CSC_COMP2_POSTCLAMP**Type:** RW**Reset State:** 0x000000FF

Component 2 post-clamp value

double buffered: NO

MDSS_MDP_WB_3_CSC_COMP2_POSTCLAMP

Bits	Name	Description
15:8	LOW	Unsigned 8 bit pre-clamp value
7:0	HIGH	Unsigned 8 bit pre-clamp value

0x01A66A8C MDSS_MDP_WB_3_CSC_COMP0_PREBIAS**Type:** RW**Reset State:** 0x00000000

Component 0 pre-bias value

double buffered: NO

MDSS_MDP_WB_3_CSC_COMP0_PREBIAS

Bits	Name	Description
8:0	VALUE	Signed 9 bit pre-bias value

0x01A66A90 MDSS_MDP_WB_3_CSC_COMP1_PREBIAS**Type:** RW**Reset State:** 0x00000000

Component 1 pre-bias value

double buffered: NO

MDSS_MDP_WB_3_CSC_COMP1_PREBIAS

Bits	Name	Description
8:0	VALUE	Signed 9 bit pre-bias value

0x01A66A94 MDSS_MDP_WB_3_CSC_COMP2_PREBIAS**Type:** RW**Reset State:** 0x00000000

Component 2 pre-bias value

double buffered: NO

MDSS_MDP_WB_3_CSC_COMP2_PREBIAS

Bits	Name	Description
8:0	VALUE	Signed 9 bit pre-bias value

0x01A66A98 MDSS_MDP_WB_3_CSC_COMP0_POSTBIAS**Type:** RW**Reset State:** 0x00000000

Component 0 post-bias value

double buffered: NO

MDSS_MDP_WB_3_CSC_COMP0_POSTBIAS

Bits	Name	Description
8:0	VALUE	Signed 9 bit pre-bias value

0x01A66A9C MDSS_MDP_WB_3_CSC_COMP1_POSTBIAS**Type:** RW**Reset State:** 0x00000000

Component 1 post-bias value

double buffered: NO

MDSS_MDP_WB_3_CSC_COMP1_POSTBIAS

Bits	Name	Description
8:0	VALUE	Signed 9 bit pre-bias value

0x01A66AA0 MDSS_MDP_WB_3_CSC_COMP2_POSTBIAS**Type:** RW**Reset State:** 0x00000000

Component 2 post-bias value

double buffered: NO

MDSS_MDP_WB_3_CSC_COMP2_POSTBIAS

Bits	Name	Description
8:0	VALUE	Signed 9 bit pre-bias value

0x01A66AB0 MDSS_MDP_WB_3_DST_ADDR_SW_STATUS**Type:** RW**Reset State:** 0x00000000

Security Status of the destination buffer plane

This provides a SW override to the hardware calculated content protection sent to VBIF

This value is logically or ed with the source content protection produced by hardware

This information will be presented to VBIF per transaction and reflected as part of the memory transaction

Note that the VBIF context banks need to be properly programmed for the output to be successful

double buffered: NO

MDSS_MDP_WB_3_DST_ADDR_SW_STATUS

Bits	Name	Description
0	DST_ADDR_CP	This is the secure status bit for destination plane 0 (if used) Memory transactions will not complete if this status doesnt correspond to system programming in the rest of the path to memory 0x0: Content Protected Buffer State 0x1: Non Content Protected Buffer State

17.17.12 MDSS_MDP_WB_4

MDP WB hardware register description

17.17.12.1 MDSS_MDP_WB_4_WB

WB

MDP WB hardware register description

0x01A67000 MDSS_MDP_WB_4_DST_FORMAT**Type:** RW**Reset State:** 0x00000000

Memory Write Back Output Format

double buffered: NO

MDSS_MDP_WB_4_DST_FORMAT

Bits	Name	Description
31:30	FRAME_FORMAT	FRAME_FORMAT In the current revision, there is no support for tiled formats 0x0: Linear, no tile
29:26	DST_CHROMA_SITE	Destination Chroma Site 0x0: A (cosite for YUV422) 0x1: B (offsite for YUV422) 0x2: C (N/A for YUV422) 0x3: D (N/A for YUV422) 0x4: E (N/A for YUV422) 0x5: F (N/A for YUV422) 0x6: G (N/A for YUV422) 0x7: H (N/A for YUV422) 0x8: I (N/A for YUV422)
25:23	DST_CHROMA_SAMP	Chroma Sampling 0x0: RGB/4:4:4 0x1: 4:2:2 - H2V1 0x2: 4:2:2 - H1V2 0x3: 4:2:0 - N2V1 0x4: 4:2:0 - N1V2
22	DST_DITHER_EN	Destination Dither Enable
20:19	WRITE_PLANES	Number of planes 0x0: Interleaved 0x1: Planar 0x2: Pseudo-planar
18	PACK_ALIGN	0x0: LSB 0x1: MSB
17	PACK_TIGHT	0x0: Loose 0x1: Tight
14	DST_ALPHA_X	0x0: Use alpha value from pipe 0x1: Use alpha value from register
13:12	PACK_COUNT	Valid packing elements count 0x0 = 1 element 0x1 = 2 elements 0x2 = 3 elements 0x3 = 4 elements
10:9	DST_BPP	Effective byte per pixel 0x0: 1 byte 0x1: 2 bytes 0x2: 3 bytes 0x3: 4 bytes
8	DSTC3_EN	0x1: Destination has alpha

MDSS_MDP_WB_4_DST_FORMAT (cont.)

Bits	Name	Description
7:6	DSTC3_OUT	Number of bits for component 3 output 0x0: 1 bit 0x1: 4 bits 0x2: 6 bits 0x3: 8 bits
5:4	DSTC2_OUT	Number of bits for component 2 output 0x0: 4 bits 0x1: 5 bits 0x2: 6 bits 0x3: 8 bits
3:2	DSTC1_OUT	Number of bits for component 1 output 0x0: 4 bits 0x1: 5 bits 0x2: 6 bits 0x3: 8 bits
1:0	DSTC0_OUT	Number of bits for component 0 output 0x0: 4 bits 0x1: 5 bits 0x2: 6 bits 0x3: 8 bits

0x01A67004 MDSS_MDP_WB_4_DST_OP_MODE**Type:** RW**Reset State:** 0x00000500

Destination Operation Mode

double buffered: NO

MDSS_MDP_WB_4_DST_OP_MODE

Bits	Name	Description
14	CHROMA_DWN_SAMPLE_V_MTHD	Vertical Chroma Down Sample method 0x0: non-pixel dropping 0x1: pixel dropping
13	CHROMA_DWN_SAMPLE_H_MTHD	Horizontal Chroma Down Sample method 0x0: non-pixel dropping 0x1: pixel dropping
12	CHROMA_DWN_SAMPLE_FORMAT	Chroma Down Sample format 0x0: YUV422 H2V1 0x1: YUV420

MDSS_MDP_WB_4_DST_OP_MODE (cont.)

Bits	Name	Description
11	CHROMA_DWN_SAMPLE_EN	Chroma Down Sample enable for YUV444 NOTE: This is not valid for WB0. 0x0: Disabled 0x1: Enabled
10	CSC_DST_DATA_FORMAT	Destination Data Format 0x0: RGB 0x1: YCbCr
9	CSC_SRC_DATA_FORMAT	Source Data Format 0x0: RGB 0x1: YCbCr
8	CSC_EN	Enables the color matrix conversion NOTE: This is not valid for WB0. 0x0: Bypass 0x1: Enabled
6	ROT_EN	0x0: Rotate Disabled 0x1: Rotate Enabled
5	ROT_MODE	0x0: Rotate 0 0x1: Rotate 90
4	BLOCK_SIZE	0x0: 64x64 Block Size 0x1: 128x128 Block Size

0x01A67008 MDSS_MDP_WB_4_DST_PACK_PATTERN**Type:** RW**Reset State:** 0x00000000

Write Back Pack Pattern

double buffered: NO

MDSS_MDP_WB_4_DST_PACK_PATTERN

Bits	Name	Description
25:24	ELEMENT3	0x0: Color0 0x1: Color1 0x2: Color2 0x3: Color3
17:16	ELEMENT2	0x0: Color0 0x1: Color1 0x2: Color2 0x3: Color3

MDSS_MDP_WB_4_DST_PACK_PATTERN (cont.)

Bits	Name	Description
9:8	ELEMENT1	0x0: Color0 0x1: Color1 0x2: Color2 0x3: Color3
1:0	ELEMENT0	0x0: Color0 0x1: Color1 0x2: Color2 0x3: Color3

0x01A6700C MDSS_MDP_WB_4_DST0_ADDR**Type:** RW**Reset State:** 0x00000000

Write Back Destination Address

For fetching BWC data, this plane refers to the compressed RAU data stream

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_WB_4_DST0_ADDR

Bits	Name	Description
31:0	ADDR	Destination Address (byte-aligned)

0x01A67010 MDSS_MDP_WB_4_DST1_ADDR**Type:** RW**Reset State:** 0x00000000

Write Back Destination Address

For fetching BWC data, this plane refers to the META data to fetch the lengths of the compressed RAU data stream

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_WB_4_DST1_ADDR

Bits	Name	Description
31:0	ADDR	Destination Address (byte-aligned)

0x01A67014 MDSS_MDP_WB_4_DST2_ADDR**Type:** RW**Reset State:** 0x00000000

Write Back Destination Address

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_WB_4_DST2_ADDR

Bits	Name	Description
31:0	ADDR	Destination Address (byte-aligned)

0x01A67018 MDSS_MDP_WB_4_DST3_ADDR**Type:** RW**Reset State:** 0x00000000

Write Back Destination Address

double buffered: YES

when: On frame boundaries (vsync or SW trigger)

MDSS_MDP_WB_4_DST3_ADDR

Bits	Name	Description
31:0	ADDR	Destination Address (byte-aligned)

0x01A6701C MDSS_MDP_WB_4_DST_YSTRIDE0**Type:** RW**Reset State:** 0x00000000

Stride Values

Double-buffered

double buffered: NO

MDSS_MDP_WB_4_DST_YSTRIDE0

Bits	Name	Description
31:16	DST1_YSTRIDE	Plane 1 Y stride in bytes
15:0	DST0_YSTRIDE	Plane 0 Y stride in bytes

0x01A67020 MDSS_MDP_WB_4_DST_YSTRIDE1

Type: RW

Reset State: 0x00000000

Stride Values

Double-buffered

double buffered: NO

MDSS_MDP_WB_4_DST_YSTRIDE1

Bits	Name	Description
31:16	DST3_YSTRIDE	Plane 3 Y stride in bytes
15:0	DST2_YSTRIDE	Plane 2 Y stride in bytes

0x01A67024 MDSS_MDP_WB_4_DST_DITHER_BITDEPTH

Type: RW

Reset State: 0x00000000

Write Dither Strength

double buffered: NO

MDSS_MDP_WB_4_DST_DITHER_BITDEPTH

Bits	Name	Description
5:4	C2_BITS	Number of MSB bits on component2 to indicate the output depth 0x0: 4bits 0x1: 5 bits 0x2: 6 bits 0x3: 8 bits

MDSS_MDP_WB_4_DST_DITHER_BITDEPTH (cont.)

Bits	Name	Description
3:2	C1_BITS	Number of MSB bits on component1 to indicate the output depth 0x0: 4bits 0x1: 5 bits 0x2: 6 bits 0x3: 8 bits
1:0	C0_BITS	Number of MSB bits on component0 to indicate the output depth 0x0: 4bits 0x1: 5 bits 0x2: 6 bits 0x3: 8 bits

0x01A67030 MDSS_MDP_WB_4_DITHER_MATRIX_ROW0**Type:** RW**Reset State:** 0x00005D7F

Dither Strength Matrix Values Register for first row which covers entries [0-3]

double buffered: NO

MDSS_MDP_WB_4_DITHER_MATRIX_ROW0

Bits	Name	Description
15:12	ENTRY_3	Entry 3 for the dither matrix
11:8	ENTRY_2	Entry 2 for the dither matrix
7:4	ENTRY_1	Entry 1 for the dither matrix
3:0	ENTRY_0	Entry 0 for the dither matrix

0x01A67034 MDSS_MDP_WB_4_DITHER_MATRIX_ROW1**Type:** RW**Reset State:** 0x000091B3

Dither Strength Matrix Values Register for first row which covers entries [4-7]

double buffered: NO

MDSS_MDP_WB_4_DITHER_MATRIX_ROW1

Bits	Name	Description
15:12	ENTRY_7	Entry 7 for the dither matrix

MDSS_MDP_WB_4_DITHER_MATRIX_ROW1 (cont.)

Bits	Name	Description
11:8	ENTRY_6	Entry 6 for the dither matrix
7:4	ENTRY_5	Entry 5 for the dither matrix
3:0	ENTRY_4	Entry 4 for the dither matrix

0x01A67038 MDSS_MDP_WB_4_DITHER_MATRIX_ROW2**Type:** RW**Reset State:** 0x00006E4C

Dither Strength Matrix Values Register for first row which covers entries [8-11]

double buffered: NO

MDSS_MDP_WB_4_DITHER_MATRIX_ROW2

Bits	Name	Description
15:12	ENTRY_11	Entry 11 for the dither matrix
11:8	ENTRY_10	Entry 10 for the dither matrix
7:4	ENTRY_9	Entry 9 for the dither matrix
3:0	ENTRY_8	Entry 8 for the dither matrix

0x01A6703C MDSS_MDP_WB_4_DITHER_MATRIX_ROW3**Type:** RW**Reset State:** 0x0000A280

Dither Strength Matrix Values Register for first row which covers entries [12-15]

double buffered: NO

MDSS_MDP_WB_4_DITHER_MATRIX_ROW3

Bits	Name	Description
15:12	ENTRY_15	Entry 15 for the dither matrix
11:8	ENTRY_14	Entry 14 for the dither matrix
7:4	ENTRY_13	Entry 13 for the dither matrix
3:0	ENTRY_12	Entry 12 for the dither matrix

0x01A67048 MDSS_MDP_WB_4_DST_WRITE_CONFIG**Type:** RW**Reset State:** 0x00000008

Write-back Configuration double buffered: NO

This register must NOT be altered during operation and MDP must be idle when changing values

MDSS_MDP_WB_4_DST_WRITE_CONFIG

Bits	Name	Description
10	ADDR_GEN_MODE	
9:8	HIGHEST_BANK_BIT	
7:6	PRIORITY_LEVEL	Priority Level. Side-band priority reflecting the status of VBIF client. 0x3 - max priority 0x0 - min priority
5:4	REQ_PRIORITY	Request Priority. In-band priority level assigned when VBIF transaction is created. 0x3 - max priority 0x0 - min priority
3:0	MAX_BURST_SIZE	Maximum burst beat size. Hardware fetch engines operate on 128-bit bus Possible values: 0x8: burst of 8 0x4: burst of 4 0x2: burst of 2 0x1: burst of 1

0x01A67054 MDSS_MDP_WB_4_ROTATOR_PIPE_DOWNSCALER**Type:** RW**Reset State:** 0x00000000**MDSS_MDP_WB_4_ROTATOR_PIPE_DOWNSCALER**

Bits	Name	Description
23:16	DNOSC_FACTOR_V	0x0: V_BYPASS 0x2: V_DNOSC_X2 0x4: V_DNOSC_X4 0x8: V_DNOSC_X8 0x10: V_DNOSC_X16 0x20: V_DNOSC_X32 0x40: V_DNOSC_X64

MDSS_MDP_WB_4_ROTATOR_PIPE_DOWNSCALER (cont.)

Bits	Name	Description
7:0	DNOSC_FACTOR_H	0x0: H_BYPASS 0x2: H_DNOSC_X2 0x4: H_DNOSC_X4 0x8: H_DNOSC_X8 0x10: H_DNOSC_X16 0x20: H_DNOSC_X32 0x40: H_DNOSC_X64

0x01A67074 MDSS_MDP_WB_4_OUT_SIZE**Type:** RW**Reset State:** 0x00000000

Output Size

Should be same as source is scaling is NOT enabled

double buffered: NO

MDSS_MDP_WB_4_OUT_SIZE

Bits	Name	Description
31:16	DST_H	Height of Output ROI
15:0	DST_W	Width of Output ROI

0x01A67078 MDSS_MDP_WB_4_DST_ALPHA_X_VALUE**Type:** RW**Reset State:** 0x00000000

Alpha value used for XRGB formats

double buffered: NO

MDSS_MDP_WB_4_DST_ALPHA_X_VALUE

Bits	Name	Description
7:0	VALUE	Alpha value used for XRGB formats

0x01A67260 MDSS_MDP_WB_4_CSC_MATRIX_COEFF_0**Type:** RW**Reset State:** 0x00000200

Color Space conversion coefficients

double buffered: NO

MDSS_MDP_WB_4_CSC_MATRIX_COEFF_0

Bits	Name	Description
28:16	COEFF_12	Coefficient value in S4.9 format Row 1 Column 2
12:0	COEFF_11	Coefficient value in S4.9 format Row 1 Column 1

0x01A67264 MDSS_MDP_WB_4_CSC_MATRIX_COEFF_1**Type:** RW**Reset State:** 0x00000000

Color Space conversion coefficients

double buffered: NO

MDSS_MDP_WB_4_CSC_MATRIX_COEFF_1

Bits	Name	Description
28:16	COEFF_21	Coefficient value in S4.9 format Row 2 Column 1
12:0	COEFF_13	Coefficient value in S4.9 format Row 1 Column 3

0x01A67268 MDSS_MDP_WB_4_CSC_MATRIX_COEFF_2**Type:** RW**Reset State:** 0x00000200

Color Space conversion coefficients

double buffered: NO

MDSS_MDP_WB_4_CSC_MATRIX_COEFF_2

Bits	Name	Description
28:16	COEFF_23	Coefficient value in S4.9 format Row 2 Column 3
12:0	COEFF_22	Coefficient value in S4.9 format Row 2 Column 2

0x01A6726C MDSS_MDP_WB_4_CSC_MATRIX_COEFF_3**Type:** RW**Reset State:** 0x00000000

Color Space conversion coefficients

double buffered: NO

MDSS_MDP_WB_4_CSC_MATRIX_COEFF_3

Bits	Name	Description
28:16	COEFF_32	Coefficient value in S4.9 format Row 3 Column 2
12:0	COEFF_31	Coefficient value in S4.9 format Row 3 Column 1

0x01A67270 MDSS_MDP_WB_4_CSC_MATRIX_COEFF_4**Type:** RW**Reset State:** 0x00000200

Color Space conversion coefficients

double buffered: NO

MDSS_MDP_WB_4_CSC_MATRIX_COEFF_4

Bits	Name	Description
12:0	COEFF_33	Coefficient value in S4.9 format Row 3 Column 3

0x01A67274 MDSS_MDP_WB_4_CSC_COMP0_PRECLAMP**Type:** RW**Reset State:** 0x000000FF

Component 0 pre-clamp value

double buffered: NO

MDSS_MDP_WB_4_CSC_COMP0_PRECLAMP

Bits	Name	Description
15:8	LOW	Unsigned 8 bit pre-clamp value
7:0	HIGH	Unsigned 8 bit pre-clamp value

0x01A67278 MDSS_MDP_WB_4_CSC_COMP1_PRECLAMP**Type:** RW**Reset State:** 0x000000FF

Component 1 pre-clamp value

double buffered: NO

MDSS_MDP_WB_4_CSC_COMP1_PRECLAMP

Bits	Name	Description
15:8	LOW	Unsigned 8 bit pre-clamp value
7:0	HIGH	Unsigned 8 bit pre-clamp value

0x01A6727C MDSS_MDP_WB_4_CSC_COMP2_PRECLAMP**Type:** RW**Reset State:** 0x000000FF

Component 2 pre-clamp value

double buffered: NO

MDSS_MDP_WB_4_CSC_COMP2_PRECLAMP

Bits	Name	Description
15:8	LOW	Unsigned 8 bit pre-clamp value
7:0	HIGH	Unsigned 8 bit pre-clamp value

0x01A67280 MDSS_MDP_WB_4_CSC_COMP0_POSTCLAMP**Type:** RW**Reset State:** 0x000000FF

Component 0 post-clamp value

double buffered: NO

MDSS_MDP_WB_4_CSC_COMP0_POSTCLAMP

Bits	Name	Description
15:8	LOW	Unsigned 8 bit pre-clamp value
7:0	HIGH	Unsigned 8 bit pre-clamp value

0x01A67284 MDSS_MDP_WB_4_CSC_COMP1_POSTCLAMP**Type:** RW**Reset State:** 0x000000FF

Component 1 post-clamp value

double buffered: NO

MDSS_MDP_WB_4_CSC_COMP1_POSTCLAMP

Bits	Name	Description
15:8	LOW	Unsigned 8 bit pre-clamp value
7:0	HIGH	Unsigned 8 bit pre-clamp value

0x01A67288 MDSS_MDP_WB_4_CSC_COMP2_POSTCLAMP**Type:** RW**Reset State:** 0x000000FF

Component 2 post-clamp value

double buffered: NO

MDSS_MDP_WB_4_CSC_COMP2_POSTCLAMP

Bits	Name	Description
15:8	LOW	Unsigned 8 bit pre-clamp value
7:0	HIGH	Unsigned 8 bit pre-clamp value

0x01A6728C MDSS_MDP_WB_4_CSC_COMP0_PREBIAS**Type:** RW**Reset State:** 0x00000000

Component 0 pre-bias value

double buffered: NO

MDSS_MDP_WB_4_CSC_COMP0_PREBIAS

Bits	Name	Description
8:0	VALUE	Signed 9 bit pre-bias value

0x01A67290 MDSS_MDP_WB_4_CSC_COMP1_PREBIAS**Type:** RW**Reset State:** 0x00000000

Component 1 pre-bias value

double buffered: NO

MDSS_MDP_WB_4_CSC_COMP1_PREBIAS

Bits	Name	Description
8:0	VALUE	Signed 9 bit pre-bias value

0x01A67294 MDSS_MDP_WB_4_CSC_COMP2_PREBIAS**Type:** RW**Reset State:** 0x00000000

Component 2 pre-bias value

double buffered: NO

MDSS_MDP_WB_4_CSC_COMP2_PREBIAS

Bits	Name	Description
8:0	VALUE	Signed 9 bit pre-bias value

0x01A67298 MDSS_MDP_WB_4_CSC_COMP0_POSTBIAS**Type:** RW**Reset State:** 0x00000000

Component 0 post-bias value

double buffered: NO

MDSS_MDP_WB_4_CSC_COMP0_POSTBIAS

Bits	Name	Description
8:0	VALUE	Signed 9 bit pre-bias value

0x01A6729C MDSS_MDP_WB_4_CSC_COMP1_POSTBIAS**Type:** RW**Reset State:** 0x00000000

Component 1 post-bias value

double buffered: NO

MDSS_MDP_WB_4_CSC_COMP1_POSTBIAS

Bits	Name	Description
8:0	VALUE	Signed 9 bit pre-bias value

0x01A672A0 MDSS_MDP_WB_4_CSC_COMP2_POSTBIAS**Type:** RW**Reset State:** 0x00000000

Component 2 post-bias value

double buffered: NO

MDSS_MDP_WB_4_CSC_COMP2_POSTBIAS

Bits	Name	Description
8:0	VALUE	Signed 9 bit pre-bias value

0x01A672B0 MDSS_MDP_WB_4_DST_ADDR_SW_STATUS**Type:** RW**Reset State:** 0x00000000

Security Status of the destination buffer plane

This provides a SW override to the hardware calculated content protection sent to VBIF

This value is logically or ed with the source content protection produced by hardware

This information will be presented to VBIF per transaction and reflected as part of the memory transaction

Note that the VBIF context banks need to be properly programmed for the output to be successful
double buffered: NO**MDSS_MDP_WB_4_DST_ADDR_SW_STATUS**

Bits	Name	Description
0	DST_ADDR_CP	This is the secure status bit for destination plane 0 (if used) Memory transactions will not complete if this status doesnt corespond to system programming in the rest of the path to memory 0x0: Content Protected Buffer State 0x1: Non Content Protected Buffer State

17.17.13 MDSS_MDP_INTF_0

Display Interface hardware register description

17.17.13.1 MDSS_MDP_INTF_0_INTF

INTF

Display Interface hardware register description

0x01A6B000 MDSS_MDP_INTF_0_TIMING_ENGINE_EN**Type:** RW**Reset State:** 0x00000000

Display Interface Timing Engine Enable register

double buffered: YES

when: On frame boundaries

MDSS_MDP_INTF_0_TIMING_ENGINE_EN

Bits	Name	Description
1	OSYNC_LOCK	
0	EN	Display Interface enable Should be the LAST register programmed for the interface Be sure to program ALL timing registers and setting up of the display pipeline BEFORE enabling this register Upon disabling of this bit, hardware will stop driving the interface at the NEXT vsync boundary A VSYNC interrupt WILL be generated but a VSYNC will NOT be generated on the interface

0x01A6B004 MDSS_MDP_INTF_0_INTF_CONFIG**Type:** RW**Reset State:** 0x00040000

Display Interface Configuration register.

Note that the configuration register ONLY applies to INTF_4 which transmits to MHL

MDSS_MDP_INTF_0_INTF_CONFIG

Bits	Name	Description
31	PROG_FETCH_START_EN	Programmable Fetch feature enable bit 0: programmable line counter disabled - default value to be backward compatible 1: programmable line counter enabled
30	ACTIVE_V_EN	Active V Enable
29	ACTIVE_H_EN	Active H Enable
28	VSYNC_INTERRUPT_MODE	vsync interrupt generation mode 0 = generate an interrupt every vsync 1 = generate an interrupt every 2 fields (applicable to interlace mode)
27:24	REPEAT_PIXEL_MODE	Pixel repeat mode 0x1 = repeat each pixel 2 times 0x2 = repeat each pixel 3 times ... 15 = repeat pixel 16 times

MDSS_MDP_INTF_0_INTF_CONFIG (cont.)

Bits	Name	Description
23	DSI_VIDEO_STOP_MODE	This bit controls how video mode stream is stopped when DSI dynamic refresh is triggered. It is not applicable for non-DSI interface. 0x0: disable timing engine when DSI sets the video stop signal; 0x1: halt timing engine counters when DSI sets the video stop signal
21	CHROMA_DWN_SAMPLE_H_MTHD	Horizontal Chroma Down Sample method 0x0: non-pixel dropping 0x1: pixel dropping
20	CHROMA_DWN_SAMPLE_EN	Chroma Down Sample enable for YUV444 0x0: Disabled 0x1: Enabled
18	CSC_DST_DATA_FORMAT	Destination Data Format 0x0: RGB 0x1: YCbCr
17	CSC_SRC_DATA_FORMAT	Source Data Format 0x0: RGB 0x1: YCbCr
16	CSC_EN	Enables the color matrix conversion 0x0: Bypass 0x1: Enabled
13:11	NIBBLE_SWAP_EN	Enable nibble swapping within each pixel byte NIBBLE_SWAP_EN[0]: 0 = no swapping for pixel_data [7:0]; 1 = enable swapping of pixel_data[3:0] with pixel_data[7:4] NIBBLE_SWAP_EN[1] = 0 = no swapping for pixel_data [15:8]; 1 = enable swapping of pixel_data[11:8] with pixel_data[15:12] NIBBLE_SWAP_EN[2] = 0 = no swapping for pixel_data [23:16]; 1 = enable swapping of pixel_data[19:16] with pixel_data[23:20]
10	PACK_ALIGN	0x0: LSB 0x1:MSB
9:8	PACK_422	0x0: Y2Cr1Y1Cb1 0x1: Cr1Y2Cb1Y1 0x2: Y2Cb1Y1Cr1 0x3: Cb1Y2Cr1Y1
6:4	OUT_MODE	0x0: YCbCr or RGB4444 seperate syncs 24-bit interface 0x1: RSVD 0x2: YCbCr 422 seperate syncs 16-bit interface
2	INTERLACE_INIT_SEL	0x0: start with even field 0x1: start with odd field
1	REPEAT_PIXEL	0x0: Do NOT repeat 0x1: Repeat

MDSS_MDP_INTF_0_INTF_CONFIG (cont.)

Bits	Name	Description
0	INTERLACE_MODE	double buffered: YES 0x0: Progressive Out 0x1: Interlace out

0x01A6B008 MDSS_MDP_INTF_0_HSYNC_CTL**Type:** RW**Reset State:** 0x00000000

Display Interface HSYNC Control

double buffered: YES

MDSS_MDP_INTF_0_HSYNC_CTL

Bits	Name	Description
31:16	HSYNC_PERIOD	HSYNC Period in pixel clocks. This is the start between start of HSYNC pulse and the start of the next HSYNC pulse. Note: hsync_period=width + h_porch where width=output image width h_porch=(h_back_porch+h_front_porch) h_back_porch starts at the beginning of hsync pulse
15:0	PULSE_WIDTH	HSYNC pulse width in pixel clocks

0x01A6B00C MDSS_MDP_INTF_0_VSYNC_PERIOD_F0**Type:** RW**Reset State:** 0x00000000

Display Interface VSYNC Control for field 0

double buffered: YES

MDSS_MDP_INTF_0_VSYNC_PERIOD_F0

Bits	Name	Description
31:0	VSYNC_PERIOD	VSYNC Period in pixel clocks. This is the time between start of vsync pulse and the start of the next vsync pulse. Note vsync_period=height+v_porch where height=output_image_height v_porch=(v_back_porch+v_front_porch) v_back_porch starts at the beginning of vsync pulse.

0x01A6B010 MDSS_MDP_INTF_0_VSYNC_PERIOD_F1**Type:** RW**Reset State:** 0x00000000

Display Interface VSYNC Control for field 1

double buffered: NO

MDSS_MDP_INTF_0_VSYNC_PERIOD_F1

Bits	Name	Description
31:0	VSYNC_PERIOD	VSYNC Period in pixel clocks. In progressive mode, this is the time between start of vsync pulse and the start of next vsync pulse Interlace mode this is the time between start of Field1 vsync pulse and the start of Field2 vsync pulse. Note: vsync_period = Height + v_porch where, Height = Output image height v_porch = (v_back_porch + v_front_porch) v_back_porch starts at the beginning of vsync pulse

0x01A6B014 MDSS_MDP_INTF_0_VSYNC_PULSE_WIDTH_F0**Type:** RW**Reset State:** 0x00000000

Display Interface VSYNC Control for field 0

double buffered: YES

MDSS_MDP_INTF_0_VSYNC_PULSE_WIDTH_F0

Bits	Name	Description
31:0	VSYNC_PERIOD	VSYNC Period width in pixel clocks

0x01A6B018 MDSS_MDP_INTF_0_VSYNC_PULSE_WIDTH_F1**Type:** RW**Reset State:** 0x00000000

Display Interface VSYNC Control for field 1

double buffered: NO

MDSS_MDP_INTF_0_VSYNC_PULSE_WIDTH_F1

Bits	Name	Description
31:0	VSYNC_PERIOD	VSYNC Period in pixel clocks for the 2nd field. NOT valid for progressive mode

0x01A6B01C MDSS_MDP_INTF_0_DISPLAY_V_START_F0**Type:** RW**Reset State:** 0x00000000

Display Interface Vertical Active Start Control for field 0

double buffered: YES

MDSS_MDP_INTF_0_DISPLAY_V_START_F0

Bits	Name	Description
31:0	DISPLAY_START_Y	Display Y Start Defines the time period between the field0 start of vsync pulse and the first displayed line position in pixel clocks $\text{display_start_y} = \text{v_back_porch} * \text{hcycle}$ where $\text{hcycle} = \text{horizontal_back_porch} + \text{horizontal_active} + \text{horizontal_front_porch}$ For eDP control the value programmed into this register differs slightly and should be programmed to the first active pixel and not the first pixel on that line! In particular, the programming for eDP should be: $\text{display_start_y} = (\text{v_back_porch} * \text{hcycle}) + \text{horizontal_back_porch}$

0x01A6B020 MDSS_MDP_INTF_0_DISPLAY_V_START_F1**Type:** RW**Reset State:** 0x00000000

Display Interface Vertical Active Start Control for field 1

double buffered: NO

MDSS_MDP_INTF_0_DISPLAY_V_START_F1

Bits	Name	Description
31:0	DISPLAY_START_Y	Display Y Start Defines the time period between the field1 start of vsync pulse and the first displayed line position in pixel clocks $\text{display_start_y} = \text{v_back_porch} * \text{hcycle}$ where $\text{hcycle} = \text{horizontal_back_porch} + \text{horizontal_active} + \text{horizontal_front_porch}$ For eDP control the value programmed into this register differs slightly and should be programmed to the first active pixel and not the first pixel on that line! In particular, the programming for eDP should be: $\text{display_start_y} = (\text{v_back_porch} * \text{hcycle}) + \text{horizontal_back_porch}$

0x01A6B024 MDSS_MDP_INTF_0_DISPLAY_V_END_F0**Type:** RW**Reset State:** 0x00000000

Display Interface Vertical Active End Control for field 0

double buffered: YES

MDSS_MDP_INTF_0_DISPLAY_V_END_F0

Bits	Name	Description
31:0	DISPLAY_END_Y	Display Y End Defines the time period between the field0 start of vsync pulse and the last displayed line position in pixel clocks $\text{display_end_y} = (\text{vertical_back_porch} + \text{vertical_active}) * \text{hcycle} - 1$ where hcycle = horizontal_back_porch + horizontal_active + horizontal_front_porch For eDP control the value programmed into this register differs slightly and should be programmed to the last active pixel and not the last pixel on that line! In particular, the programming for eDP should be: $\text{display_end_y} = (\text{vertical_back_porch} + \text{vertical_active}) * \text{hcycle} - 1 - \text{horizontal_front_porch}$

0x01A6B028 MDSS_MDP_INTF_0_DISPLAY_V_END_F1**Type:** RW**Reset State:** 0x00000000

Display Interface Vertical Active End Control for field 1

double buffered: NO

MDSS_MDP_INTF_0_DISPLAY_V_END_F1

Bits	Name	Description
31:0	DISPLAY_END_Y	Display Y End Defines the time period between the field0 start of vsync pulse and the last displayed line position in pixel clocks $\text{display_end_y} = (\text{vertical_back_porch} + \text{vertical_active}) * \text{hcycle} - 1$ where hcycle = horizontal_back_porch + horizontal_active + horizontal_front_porch For eDP control the value programmed into this register differs slightly and should be programmed to the last active pixel and not the last pixel on that line! In particular, the programming for eDP should be: $\text{display_end_y} = (\text{vertical_back_porch} + \text{vertical_active}) * \text{hcycle} - 1 - \text{horizontal_front_porch}$

0x01A6B02C MDSS_MDP_INTF_0_ACTIVE_V_START_F0**Type:** RW**Reset State:** 0x00000000

Display Interface Active Start Control for field 0

double buffered: YES

MDSS_MDP_INTF_0_ACTIVE_V_START_F0

Bits	Name	Description
31:0	ACTIVE_START_Y	Active Y Start

0x01A6B030 MDSS_MDP_INTF_0_ACTIVE_V_START_F1**Type:** RW**Reset State:** 0x00000000

Display Interface Active Start Control for field 1

double buffered: NO

MDSS_MDP_INTF_0_ACTIVE_V_START_F1

Bits	Name	Description
31:0	ACTIVE_START_Y	Active Y Start

0x01A6B034 MDSS_MDP_INTF_0_ACTIVE_V_END_F0**Type:** RW**Reset State:** 0x00000000

Display Interface Active End Control for field 0

double buffered: YES

MDSS_MDP_INTF_0_ACTIVE_V_END_F0

Bits	Name	Description
31:0	ACTIVE_END_Y	Active Y End

0x01A6B038 MDSS_MDP_INTF_0_ACTIVE_V_END_F1**Type:** RW**Reset State:** 0x00000000

Display Interface Active End Control for field 1

double buffered: NO

MDSS_MDP_INTF_0_ACTIVE_V_END_F1

Bits	Name	Description
31:0	ACTIVE_END_Y	Active Y End

0x01A6B03C MDSS_MDP_INTF_0_DISPLAY_HCTL**Type:** RW**Reset State:** 0x00000000

Display Interface HSYNC Control for field 1

double buffered: YES

MDSS_MDP_INTF_0_DISPLAY_HCTL

Bits	Name	Description
31:16	DISPLAY_END_X	Display End X
15:0	DISPLAY_START_X	Display Start X

0x01A6B040 MDSS_MDP_INTF_0_ACTIVE_HCTL**Type:** RW**Reset State:** 0x00000000

Display Interface HSYNC Control for field 1

double buffered: YES

MDSS_MDP_INTF_0_ACTIVE_HCTL

Bits	Name	Description
31:16	ACTIVE_END_X	Active End X
15:0	ACTIVE_START_X	Active Start X

0x01A6B044 MDSS_MDP_INTF_0_BORDER_COLOR**Type:** RW**Reset State:** 0x00000000

Display Interface Border Color

double buffered: NO

MDSS_MDP_INTF_0_BORDER_COLOR

Bits	Name	Description
23:0	BORDER_COLOR	Defines 24-bit Border Color of the inactive region of display

0x01A6B048 MDSS_MDP_INTF_0_UNDERFLOW_COLOR**Type:** RW**Reset State:** 0x00000000

Display Interface Underflow Color

double buffered: NO

MDSS_MDP_INTF_0_UNDERFLOW_COLOR

Bits	Name	Description
23:0	UNDERFLOW_COLOR	Defines 24-bit Underflow Color

0x01A6B04C MDSS_MDP_INTF_0_HSYNC_SKEW**Type:** RW**Reset State:** 0x00000000

Display Interface HSYNC Skew Control

double buffered: YES

MDSS_MDP_INTF_0_HSYNC_SKEW

Bits	Name	Description
15:0	HSYNC_SKEW	HSYNC Skew Defines the number of pixel clocks HSYNC active edge is delayed from VSYNC active edge

0x01A6B050 MDSS_MDP_INTF_0_POLARITY_CTL**Type:** RW**Reset State:** 0x00000000

Display Interface Polarity Control

double buffered: NO

MDSS_MDP_INTF_0_POLARITY_CTL

Bits	Name	Description
2	DEN	DEN Polarity 0x0: Active HIGH 0x1: Active LOW
1	VSYNC	VSYNC Polarity 0x0: Active HIGH 0x1: Active LOW
0	HSYNC	HSYNC Polarity 0x0: Active HIGH 0x1: Active LOW

0x01A6B058 MDSS_MDP_INTF_0_TP_COLOR0**Type:** RW**Reset State:** 0x00000000

Display Interface Test Pattern Color

double buffered: NO

MDSS_MDP_INTF_0_TP_COLOR0

Bits	Name	Description
23:0	COLOR0	Test Pattern Color

0x01A6B05C MDSS_MDP_INTF_0_TP_COLOR1**Type:** RW**Reset State:** 0x00000000

Display Interface Test Pattern Color

double buffered: NO

MDSS_MDP_INTF_0_TP_COLOR1

Bits	Name	Description
23:0	COLOR1	Test Pattern Color

0x01A6B084 MDSS_MDP_INTF_0_DSI_CMD_MODE_TRIGGER_EN**Type:** RW**Reset State:** 0x00000000

Enable for DSI trigger processing on DSI Command Mode interface

double buffered: NO

MDSS_MDP_INTF_0_DSI_CMD_MODE_TRIGGER_EN

Bits	Name	Description
5:4	DSI_CMD_TG_INTF_SEL	Select the stream to be triggered by DSI controller. Takes effect only if DSI_CMD_TG_EN is set to 1
0	DSI_CMD_TG_EN	DSI Command Trigger Enable. Set it to 1 to enable trigger processing 0: Disable trigger processing 1: Enable trigger processing

0x01A6B090 MDSS_MDP_INTF_0_PANEL_FORMAT**Type:** RW**Reset State:** 0x00002100

Panel Packer Configuration

double buffered: NO

MDSS_MDP_INTF_0_PANEL_FORMAT

Bits	Name	Description
16	COMP2_BIT_FLIP_EN	0x1: All bits of component 2 are flipped
15	COMP1_BIT_FLIP_EN	0x1: All bits of component 1 are flipped
14	COMP0_BIT_FLIP_EN	0x1: All bits of component 0 are flipped

MDSS_MDP_INTF_0_PANEL_FORMAT (cont.)

Bits	Name	Description
13:8	PACK_PATTERN	Packing Pattern used by the packer 0x21: RGB 0x24: RBG 0x12: BGR 0x18: BRG 0x06: GBR 0x09: GRB
7	PACK_ALIGN	Packing alignment (within a byte) for each colour component 0x0: LSB 0x1: MSB
5:4	COMP2_OUT	Number of bits for component 2 output 0x0: 4 bits 0x1: 5 bits 0x2: 6 bits 0x3: 8 bits
3:2	COMP1_OUT	Number of bits for component 1 output 0x0: 4 bits 0x1: 5 bits 0x2: 6 bits 0x3: 8 bits
1:0	COMP0_OUT	Number of bits for component 0 output 0x0: 4 bits 0x1: 5 bits 0x2: 6 bits 0x3: 8 bits

0x01A6B0A8 MDSS_MDP_INTF_0_FRAME_LINE_COUNT_EN**Type:** RW**Reset State:** 0x00000000

Frame and Line Count Enable

double buffered: NO

MDSS_MDP_INTF_0_FRAME_LINE_COUNT_EN

Bits	Name	Description
1	FRAME_COUNT_EN	Current Frame Count Enable This will provide SW information on the current frame being sent out on an active panel It will reset upon disabling of the timing engine and increment for every frame Note that there is possibility for rollover if enabled for long time. NO status is provided that rollover occurred
0	LINE_COUNT_EN	Current Line Count Enable This will provide SW information on the current line being sent out on an active panel This will include all blanking lines as well It will reset upon vsync and increment until the end of the frame.

0x01A6B0AC MDSS_MDP_INTF_0_FRAME_COUNT**Type:** R**Reset State:** 0x00000000

Current Frame Count since Timing Engine Enabled

double buffered: N/A

MDSS_MDP_INTF_0_FRAME_COUNT

Bits	Name	Description
31:0	VALUE	Current Frame Count - Base 1

0x01A6B0B0 MDSS_MDP_INTF_0_LINE_COUNT**Type:** R**Reset State:** 0x00000000

Current Line Count Including blanking

double buffered: N/A

MDSS_MDP_INTF_0_LINE_COUNT

Bits	Name	Description
15:0	VALUE	Current Line Count - Base 1

0x01A6B170 MDSS_MDP_INTF_0_PROG_FETCH_START**Type:** RW**Reset State:** 0x00000000

Programmable Fetch Start

double buffered: YES

MDSS_MDP_INTF_0_PROG_FETCH_START

Bits	Name	Description
31:0	VALUE	Programmable line value for the feature Note: The valid register setting shall be vsync counter value for the first pixel of VFP lines

0x01A6B180 MDSS_MDP_INTF_0_INTF_INTR_EN**Type:** RW**Reset State:** 0x00000000

Display Interface Interrupt Enable

MDSS_MDP_INTF_0_INTF_INTR_EN

Bits	Name	Description
0	PROG_LINE	Programmeble Line Interrupt 0x0: Interrupt Disabled 0x1: Interrupt Enabled

0x01A6B184 MDSS_MDP_INTF_0_INTF_INTR_STATUS**Type:** R**Reset State:** 0x00000000

Display Interface Interrupt Status

MDSS_MDP_INTF_0_INTF_INTR_STATUS

Bits	Name	Description
0	PROG_LINE	Programmeble Line Interrupt

0x01A6B188 MDSS_MDP_INTF_0_INTF_INTR_CLEAR**Type:** W**Reset State:** 0x00000000

Display Interface Interrupt Clear

MDSS_MDP_INTF_0_INTF_INTR_CLEAR

Bits	Name	Description
0	PROG_LINE	Programmeble Line Interrupt

0x01A6B18C MDSS_MDP_INTF_0_PROG_LINE_INTR_CONF**Type:** RW**Reset State:** 0xFFFFFFFF

Programmable Line Interrupt Config

double buffered: YES

MDSS_MDP_INTF_0_PROG_LINE_INTR_CONF

Bits	Name	Description
31:0	INTR_CORD	Pixel location in the panel video timing in terms of video raster structure including V/H blanks where the programmable line interrupt be generated When SW programe this register, it will be applied for the next frame after vsync

0x01A6B190 MDSS_MDP_INTF_0_VIDEO_MISR_CTRL**Type:** RW**Reset State:** 0x00000000

Video MISR Control

MDSS_MDP_INTF_0_VIDEO_MISR_CTRL

Bits	Name	Type	Description
31	RUN_MODE	RW	This register defines the operation mode for the MISR 0x0: Single run mode meaning it stops after the programmable number of frames. (i.e. legacy mode prior to MDP5 1.5.0) 0x1: Free running mode meaning it updates the misr signature every programmable number of frames and continues to run while MISR_ENABLE is set to 0x1
27:24	INPUT_SEL	RW	This field allows the selected input to propagate to the MISR in the event that multiple inputs want to share the same physical MISR For example, the video interface might want to MISR on active pixel data only or active+timing information as well This field would select which input is sent to the MISR
10	STATUS_CLEAR	W	This is a write only register that doesn't have any effect prior to MDP5 1.5.0 since there was no logic associated with it Write 0x1: Clears the MISR_STATUS register Write 0x0: Has no effect on the the MISR_STATUS register
9	STATUS	R	Read only register Set to 0x1 when misr signature has been updated
8	ENABLE	RW	MISR enable which starts/ends calculation of MISR on the next vsync/trigger event This essentially double buffers the enable so that the misr doesn't start/end immediately but rather at a defined time The MISR calculation would run until the end of programmed number of frames
7:0	FRAME_COUNT	RW	Number of frames to run before capturing

0x01A6B194 MDSS_MDP_INTF_0_VIDEO_MISR_SIGNATURE**Type:** R**Reset State:** 0x00000000

Video MISR signature

MDSS_MDP_INTF_0_VIDEO_MISR_SIGNATURE

Bits	Name	Description
31:0	CAPTURED	This register is the MISR value captured

0x01A6B198 MDSS_MDP_INTF_0_CMD_0_MISR_CTRL**Type:** RW**Reset State:** 0x00000000

DSI CMD0 MISR Control

MDSS_MDP_INTF_0_CMD_0_MISR_CTRL

Bits	Name	Type	Description
31	RUN_MODE	RW	This register defines the operation mode for the MISR 0x0: Single run mode meaning it stops after the programmable number of frames. (i.e. legacy mode prior to MDP5 1.5.0) 0x1: Free running mode meaning it updates the misr signature every programmable number of frames and continues to run while MISR_ENABLE is set to 0x1
27:24	INPUT_SEL	RW	This field allows the selected input to propagate to the MISR in the event that multiple inputs want to share the same physical MISR For example, the video interface might want to MISR on active pixel data only or active+timing information as well This field would select which input is sent to the MISR
10	STATUS_CLEAR	W	This is a write only register that doesn't have any effect prior to MDP5 1.5.0 since there was no logic associated with it Write 0x1: Clears the MISR_STATUS register Write 0x0: Has no effect on the the MISR_STATUS register
9	STATUS	R	Read only register Set to 0x1 when misr signature has been updated
8	ENABLE	RW	MISR enable which starts/ends calculation of MISR on the next vsync/trigger event This essentially double buffers the enable so that the misr doesn't start/end immediately but rather at a defined time The MISR calculation would run until the end of programmed number of frames
7:0	FRAME_COUNT	RW	Number of frames to run before capturing

0x01A6B19C MDSS_MDP_INTF_0_CMD_0_MISR_SIGNATURE**Type:** R**Reset State:** 0x00000000

DSI CMD0 MISR signature

MDSS_MDP_INTF_0_CMD_0_MISR_SIGNATURE

Bits	Name	Description
31:0	CAPTURED	This register is the MISR value captured

17.17.14 MDSS_MDP_INTF_1

Display Interface hardware register description

17.17.14.1 MDSS_MDP_INTF_1_INTF

INTF

Display Interface hardware register description

0x01A6B800 MDSS_MDP_INTF_1_TIMING_ENGINE_EN

Type: RW

Reset State: 0x00000000

Display Interface Timing Engine Enable register

double buffered: YES

when: On frame boundaries

MDSS_MDP_INTF_1_TIMING_ENGINE_EN

Bits	Name	Description
1	OSYNC_LOCK	
0	EN	Display Interface enable Should be the LAST register programmed for the interface Be sure to program ALL timing registers and setting up of the display pipeline BEFORE enabling this register Upon disabling of this bit, hardware will stop driving the interface at the NEXT vsync boundary A VSYNC interrupt WILL be generated but a VSYNC will NOT be generated on the interface

0x01A6B804 MDSS_MDP_INTF_1_INTF_CONFIG

Type: RW

Reset State: 0x00040000

Display Interface Configuration register.

Note that the configuration register ONLY applies to INTF_4 which transmits to MHL

MDSS_MDP_INTF_1_INTF_CONFIG

Bits	Name	Description
31	PROG_FETCH_START_EN	Programmable Fetch feature enable bit 0: programmable line counter disabled - default value to be backward compatible 1: programmable line counter enabled
30	ACTIVE_V_EN	Active V Enable
29	ACTIVE_H_EN	Active H Enable
28	VSYNC_INTERRUPT_MODE	vsync interrupt generation mode 0 = generate an interrupt every vsync 1 = generate an interrupt every 2 fields (applicable to interlace mode)
27:24	REPEAT_PIXEL_MODE	Pixel repeat mode 0x1 = repeat each pixel 2 times 0x2 = repeat each pixel 3 times ... 15 = repeat pixel 16 times
23	DSI_VIDEO_STOP_MODE	This bit controls how video mode stream is stopped when DSI dynamic refresh is triggered. It is not applicable for non-DSI interface. 0x0: disable timing engine when DSI sets the video stop signal; 0x1: halt timing engine counters when DSI sets the video stop signal
21	CHROMA_DWN_SAMPLE_H_MTHD	Horizontal Chroma Down Sample method 0x0: non-pixel dropping 0x1: pixel dropping
20	CHROMA_DWN_SAMPLE_EN	Chroma Down Sample enable for YUV444 0x0: Disabled 0x1: Enabled
18	CSC_DST_DATA_FORMAT	Destination Data Format 0x0: RGB 0x1: YCbCr
17	CSC_SRC_DATA_FORMAT	Source Data Format 0x0: RGB 0x1: YCbCr
16	CSC_EN	Enables the color matrix conversion 0x0: Bypass 0x1: Enabled

MDSS_MDP_INTF_1_INTF_CONFIG (cont.)

Bits	Name	Description
13:11	NIBBLE_SWAP_EN	Enable nibble swapping within each pixel byte NIBBLE_SWAP_EN[0]: 0 = no swapping for pixel_data [7:0]; 1 = enable swapping of pixel_data[3:0] with pixel_data[7:4] NIBBLE_SWAP_EN[1] = 0 = no swapping for pixel_data [15:8]; 1 = enable swapping of pixel_data[11:8] with pixel_data[15:12] NIBBLE_SWAP_EN[2] = 0 = no swapping for pixel_data [23:16]; 1 = enable swapping of pixel_data[19:16] with pixel_data[23:20]
10	PACK_ALIGN	0x0: LSB 0x1:MSB
9:8	PACK_422	0x0: Y2Cr1Y1Cb1 0x1: Cr1Y2Cb1Y1 0x2: Y2Cb1Y1Cr1 0x3: Cb1Y2Cr1Y1
6:4	OUT_MODE	0x0: YCbCr or RGB4444 seperate syncs 24-bit interface 0x1: RSVD 0x2: YCbCr 422 seperate syncs 16-bit interface
2	INTERLACE_INIT_SEL	0x0: start with even field 0x1: start with odd field
1	REPEAT_PIXEL	0x0: Do NOT repeat 0x1: Repeat
0	INTERLACE_MODE	double buffered: YES 0x0: Progressive Out 0x1: Interlace out

0x01A6B808 MDSS_MDP_INTF_1_HSYNC_CTL**Type:** RW**Reset State:** 0x00000000

Display Interface HSYNC Control

double buffered: YES

MDSS_MDP_INTF_1_HSYNC_CTL

Bits	Name	Description
31:16	HSYNC_PERIOD	HSYNC Period in pixel clocks. This is the start between start of HSYNC pulse and the start of the next HSYNC pulse. Note: hsync_period=width + h_porch where width=output image width h_porch=(h_back_porch+h_front_porch) h_back_porch starts at the beginning of hsync pulse
15:0	PULSE_WIDTH	HSYNC pulse width in pixel clocks

0x01A6B80C MDSS_MDP_INTF_1_VSYNC_PERIOD_F0**Type:** RW**Reset State:** 0x00000000

Display Interface VSYNC Control for field 0

double buffered: YES

MDSS_MDP_INTF_1_VSYNC_PERIOD_F0

Bits	Name	Description
31:0	VSYNC_PERIOD	VSYNC Period in pixel clocks. This is the time between start of vsync pulse and the start of the next vsync pulse. Note vsync_period=height+v_porch where height=output_image_height v_porch=(v_back_porch+v_front_porch) v_back_porch starts at the beginning of vsync pulse.

0x01A6B810 MDSS_MDP_INTF_1_VSYNC_PERIOD_F1**Type:** RW**Reset State:** 0x00000000

Display Interface VSYNC Control for field 1

double buffered: NO

MDSS_MDP_INTF_1_VSYNC_PERIOD_F1

Bits	Name	Description
31:0	VSYNC_PERIOD	VSYNC Period in pixel clocks. In progressive mode, this is the time between start of vsync pulse and the start of next vsync pulse Interlace mode this is the time between start of Field1 vsync pulse and the start of Field2 vsync pulse. Note: $\text{vsync_period} = \text{Height} + \text{v_porch}$ where, Height = Output image height $\text{v_porch} = (\text{v_back_porch} + \text{v_front_porch})$ v_back_porch starts at the beginning of vsync pulse

0x01A6B814 MDSS_MDP_INTF_1_VSYNC_PULSE_WIDTH_F0**Type:** RW**Reset State:** 0x00000000

Display Interface VSYNC Control for field 0

double buffered: YES

MDSS_MDP_INTF_1_VSYNC_PULSE_WIDTH_F0

Bits	Name	Description
31:0	VSYNC_PERIOD	VSYNC Period width in pixel clocks

0x01A6B818 MDSS_MDP_INTF_1_VSYNC_PULSE_WIDTH_F1**Type:** RW**Reset State:** 0x00000000

Display Interface VSYNC Control for field 1

double buffered: NO

MDSS_MDP_INTF_1_VSYNC_PULSE_WIDTH_F1

Bits	Name	Description
31:0	VSYNC_PERIOD	VSYNC Period in pixel clocks for the 2nd field. NOT valid for progressive mode

0x01A6B81C MDSS_MDP_INTF_1_DISPLAY_V_START_F0**Type:** RW**Reset State:** 0x00000000

Display Interface Vertical Active Start Control for field 0

double buffered: YES

MDSS_MDP_INTF_1_DISPLAY_V_START_F0

Bits	Name	Description
31:0	DISPLAY_START_Y	Display Y Start Defines the time period between the field0 start of vsync pulse and the first displayed line position in pixel clocks $\text{display_start_y} = \text{v_back_porch} * \text{hcycle}$ where $\text{hcycle} = \text{horizontal_back_porch} + \text{horizontal_active} + \text{horizontal_front_porch}$ For eDP control the value programmed into this register differs slightly and should be programmed to the first active pixel and not the first pixel on that line! In particular, the programming for eDP should be: $\text{display_start_y} = (\text{v_back_porch} * \text{hcycle}) + \text{horizontal_back_porch}$

0x01A6B820 MDSS_MDP_INTF_1_DISPLAY_V_START_F1**Type:** RW**Reset State:** 0x00000000

Display Interface Vertical Active Start Control for field 1

double buffered: NO

MDSS_MDP_INTF_1_DISPLAY_V_START_F1

Bits	Name	Description
31:0	DISPLAY_START_Y	Display Y Start Defines the time period between the field1 start of vsync pulse and the first displayed line position in pixel clocks $\text{display_start_y} = \text{v_back_porch} * \text{hcycle}$ where $\text{hcycle} = \text{horizontal_back_porch} + \text{horizontal_active} + \text{horizontal_front_porch}$ For eDP control the value programmed into this register differs slightly and should be programmed to the first active pixel and not the first pixel on that line! In particular, the programming for eDP should be: $\text{display_start_y} = (\text{v_back_porch} * \text{hcycle}) + \text{horizontal_back_porch}$

0x01A6B824 MDSS_MDP_INTF_1_DISPLAY_V_END_F0**Type:** RW**Reset State:** 0x00000000

Display Interface Vertical Active End Control for field 0

double buffered: YES

MDSS_MDP_INTF_1_DISPLAY_V_END_F0

Bits	Name	Description
31:0	DISPLAY_END_Y	Display Y End Defines the time period between the field0 start of vsync pulse and the last displayed line position in pixel clocks $\text{display_end_y} = (\text{vertical_back_porch} + \text{vertical_active}) * \text{hcycle} - 1$ where $\text{hcycle} = \text{horizontal_back_porch} + \text{horizontal_active} + \text{horizontal_front_porch}$ For eDP control the value programmed into this register differs slightly and should be programmed to the last active pixel and not the last pixel on that line! In particular, the programming for eDP should be: $\text{display_end_y} = (\text{vertical_back_porch} + \text{vertical_active}) * \text{hcycle} - 1 - \text{horizontal_front_porch}$

0x01A6B828 MDSS_MDP_INTF_1_DISPLAY_V_END_F1**Type:** RW**Reset State:** 0x00000000

Display Interface Vertical Active End Control for field 1

double buffered: NO

MDSS_MDP_INTF_1_DISPLAY_V_END_F1

Bits	Name	Description
31:0	DISPLAY_END_Y	Display Y End Defines the time period between the field0 start of vsync pulse and the last displayed line position in pixel clocks $\text{display_end_y} = (\text{vertical_back_porch} + \text{vertical_active}) * \text{hcycle} - 1$ where hcycle = horizontal_back_porch + horizontal_active + horizontal_front_porch For eDP control the value programmed into this register differs slightly and should be programmed to the last active pixel and not the last pixel on that line! In particular, the programming for eDP should be: $\text{display_end_y} = (\text{vertical_back_porch} + \text{vertical_active}) * \text{hcycle} - 1 - \text{horizontal_front_porch}$

0x01A6B82C MDSS_MDP_INTF_1_ACTIVE_V_START_F0**Type:** RW**Reset State:** 0x00000000

Display Interface Active Start Control for field 0

double buffered: YES

MDSS_MDP_INTF_1_ACTIVE_V_START_F0

Bits	Name	Description
31:0	ACTIVE_START_Y	Active Y Start

0x01A6B830 MDSS_MDP_INTF_1_ACTIVE_V_START_F1**Type:** RW**Reset State:** 0x00000000

Display Interface Active Start Control for field 1

double buffered: NO

MDSS_MDP_INTF_1_ACTIVE_V_START_F1

Bits	Name	Description
31:0	ACTIVE_START_Y	Active Y Start

0x01A6B834 MDSS_MDP_INTF_1_ACTIVE_V_END_F0

Type: RW

Reset State: 0x00000000

Display Interface Active End Control for field 0

double buffered: YES

MDSS_MDP_INTF_1_ACTIVE_V_END_F0

Bits	Name	Description
31:0	ACTIVE_END_Y	Active Y End

0x01A6B838 MDSS_MDP_INTF_1_ACTIVE_V_END_F1

Type: RW

Reset State: 0x00000000

Display Interface Active End Control for field 1

double buffered: NO

MDSS_MDP_INTF_1_ACTIVE_V_END_F1

Bits	Name	Description
31:0	ACTIVE_END_Y	Active Y End

0x01A6B83C MDSS_MDP_INTF_1_DISPLAY_HCTL

Type: RW

Reset State: 0x00000000

Display Interface HSYNC Control for field 1

double buffered: YES

MDSS_MDP_INTF_1_DISPLAY_HCTL

Bits	Name	Description
31:16	DISPLAY_END_X	Display End X
15:0	DISPLAY_START_X	Display Start X

0x01A6B840 MDSS_MDP_INTF_1_ACTIVE_HCTL**Type:** RW**Reset State:** 0x00000000

Display Interface HSYNC Control for field 1

double buffered: YES

MDSS_MDP_INTF_1_ACTIVE_HCTL

Bits	Name	Description
31:16	ACTIVE_END_X	Active End X
15:0	ACTIVE_START_X	Active Start X

0x01A6B844 MDSS_MDP_INTF_1_BORDER_COLOR**Type:** RW**Reset State:** 0x00000000

Display Interface Border Color

double buffered: NO

MDSS_MDP_INTF_1_BORDER_COLOR

Bits	Name	Description
23:0	BORDER_COLOR	Defines 24-bit Border Color of the inactive region of display

0x01A6B848 MDSS_MDP_INTF_1_UNDERFLOW_COLOR**Type:** RW**Reset State:** 0x00000000

Display Interface Underflow Color

double buffered: NO

MDSS_MDP_INTF_1_UNDERFLOW_COLOR

Bits	Name	Description
23:0	UNDERFLOW_COLOR	Defines 24-bit Underflow Color

0x01A6B84C MDSS_MDP_INTF_1_HSYNC_SKEW**Type:** RW**Reset State:** 0x00000000

Display Interface HSYNC Skew Control

double buffered: YES

MDSS_MDP_INTF_1_HSYNC_SKEW

Bits	Name	Description
15:0	HSYNC_SKEW	HSYNC Skew Defines the number of pixel clocks HSYNC active edge is delayed from VSYNC active edge

0x01A6B850 MDSS_MDP_INTF_1_POLARITY_CTL**Type:** RW**Reset State:** 0x00000000

Display Interface Polarity Control

double buffered: NO

MDSS_MDP_INTF_1_POLARITY_CTL

Bits	Name	Description
2	DEN	DEN Polarity 0x0: Active HIGH 0x1: Active LOW
1	VSYNC	VSYNC Polarity 0x0: Active HIGH 0x1: Active LOW
0	HSYNC	HSYNC Polarity 0x0: Active HIGH 0x1: Active LOW

0x01A6B858 MDSS_MDP_INTF_1_TP_COLOR0**Type:** RW**Reset State:** 0x00000000

Display Interface Test Pattern Color

double buffered: NO

MDSS_MDP_INTF_1_TP_COLOR0

Bits	Name	Description
23:0	COLOR0	Test Pattern Color

0x01A6B85C MDSS_MDP_INTF_1_TP_COLOR1**Type:** RW**Reset State:** 0x00000000

Display Interface Test Pattern Color

double buffered: NO

MDSS_MDP_INTF_1_TP_COLOR1

Bits	Name	Description
23:0	COLOR1	Test Pattern Color

0x01A6B884 MDSS_MDP_INTF_1_DSI_CMD_MODE_TRIGGER_EN**Type:** RW**Reset State:** 0x00000000

Enable for DSI trigger processing on DSI Command Mode interface

double buffered: NO

MDSS_MDP_INTF_1_DSI_CMD_MODE_TRIGGER_EN

Bits	Name	Description
5:4	DSI_CMD_TG_INTF_SEL	Select the stream to be triggered by DSI controller. Takes effect only if DSI_CMD_TG_EN is set to 1
0	DSI_CMD_TG_EN	DSI Command Trigger Enable. Set it to 1 to enable trigger processing 0: Disable trigger processing 1: Enable trigger processing

0x01A6B890 MDSS_MDP_INTF_1_PANEL_FORMAT**Type:** RW**Reset State:** 0x00002100

Panel Packer Configuration

double buffered: NO

MDSS_MDP_INTF_1_PANEL_FORMAT

Bits	Name	Description
16	COMP2_BIT_FLIP_EN	0x1: All bits of component 2 are flipped
15	COMP1_BIT_FLIP_EN	0x1: All bits of component 1 are flipped
14	COMP0_BIT_FLIP_EN	0x1: All bits of component 0 are flipped
13:8	PACK_PATTERN	Packing Pattern used by the packer 0x21: RGB 0x24: RBG 0x12: BGR 0x18: BRG 0x06: GBR 0x09: GRB
7	PACK_ALIGN	Packing alignment (within a byte) for each colour component 0x0: LSB 0x1: MSB
5:4	COMP2_OUT	Number of bits for component 2 output 0x0: 4 bits 0x1: 5 bits 0x2: 6 bits 0x3: 8 bits
3:2	COMP1_OUT	Number of bits for component 1 output 0x0: 4 bits 0x1: 5 bits 0x2: 6 bits 0x3: 8 bits
1:0	COMP0_OUT	Number of bits for component 0 output 0x0: 4 bits 0x1: 5 bits 0x2: 6 bits 0x3: 8 bits

0x01A6B8A8 MDSS_MDP_INTF_1_FRAME_LINE_COUNT_EN**Type:** RW**Reset State:** 0x00000000

Frame and Line Count Enable

double buffered: NO

MDSS_MDP_INTF_1_FRAME_LINE_COUNT_EN

Bits	Name	Description
1	FRAME_COUNT_EN	Current Frame Count Enable This will provide SW information on the current frame being sent out on an active panel It will reset upon disabling of the timing engine and increment for every frame Note that there is possibility for rollover if enabled for long time. NO status is provided that rollover occurred
0	LINE_COUNT_EN	Current Line Count Enable This will provide SW information on the current line being sent out on an active panel This will include all blanking lines as well It will reset upon vsync and increment until the end of the frame.

0x01A6B8AC MDSS_MDP_INTF_1_FRAME_COUNT**Type:** R**Reset State:** 0x00000000

Current Frame Count since Timing Engine Enabled

double buffered: N/A

MDSS_MDP_INTF_1_FRAME_COUNT

Bits	Name	Description
31:0	VALUE	Current Frame Count - Base 1

0x01A6B8B0 MDSS_MDP_INTF_1_LINE_COUNT**Type:** R**Reset State:** 0x00000000

Current Line Count Including blanking

double buffered: N/A

MDSS_MDP_INTF_1_LINE_COUNT

Bits	Name	Description
15:0	VALUE	Current Line Count - Base 1

0x01A6B970 MDSS_MDP_INTF_1_PROG_FETCH_START**Type:** RW**Reset State:** 0x00000000

Programmable Fetch Start

double buffered: YES

MDSS_MDP_INTF_1_PROG_FETCH_START

Bits	Name	Description
31:0	VALUE	Programmable line value for the feature Note: The valid register setting shall be vsync counter value for the first pixel of VFP lines

0x01A6B980 MDSS_MDP_INTF_1_INTF_INTR_EN**Type:** RW**Reset State:** 0x00000000

Display Interface Interrupt Enable

MDSS_MDP_INTF_1_INTF_INTR_EN

Bits	Name	Description
0	PROG_LINE	Programmeble Line Interrupt 0x0: Interrupt Disabled 0x1: Interrupt Enabled

0x01A6B984 MDSS_MDP_INTF_1_INTF_INTR_STATUS**Type:** R**Reset State:** 0x00000000

Display Interface Interrupt Status

MDSS_MDP_INTF_1_INTF_INTR_STATUS

Bits	Name	Description
0	PROG_LINE	Programmeble Line Interrupt

0x01A6B988 MDSS_MDP_INTF_1_INTF_INTR_CLEAR**Type:** W**Reset State:** 0x00000000

Display Interface Interrupt Clear

MDSS_MDP_INTF_1_INTF_INTR_CLEAR

Bits	Name	Description
0	PROG_LINE	Programmeble Line Interrupt

0x01A6B98C MDSS_MDP_INTF_1_PROG_LINE_INTR_CONF**Type:** RW**Reset State:** 0xFFFFFFFF

Programmable Line Interrupt Config

double buffered: YES

MDSS_MDP_INTF_1_PROG_LINE_INTR_CONF

Bits	Name	Description
31:0	INTR_CORD	Pixel location in the panel video timing in terms of video raster structure including V/H blanks where the programmable line interrupt be generated When SW programe this register, it will be applied for the next frame after vsync

0x01A6B990 MDSS_MDP_INTF_1_VIDEO_MISR_CTRL**Type:** RW**Reset State:** 0x00000000

Video MISR Control

MDSS_MDP_INTF_1_VIDEO_MISR_CTRL

Bits	Name	Type	Description
31	RUN_MODE	RW	This register defines the operation mode for the MISR 0x0: Single run mode meaning it stops after the programmable number of frames. (i.e. legacy mode prior to MDP5 1.5.0) 0x1: Free running mode meaning it updates the misr signature every programmable number of frames and continues to run while MISR_ENABLE is set to 0x1
27:24	INPUT_SEL	RW	This field allows the selected input to propagate to the MISR in the event that multiple inputs want to share the same physical MISR For example, the video interface might want to MISR on active pixel data only or active+timing information as well This field would select which input is sent to the MISR
10	STATUS_CLEAR	W	This is a write only register that doesn't have any effect prior to MDP5 1.5.0 since there was no logic associated with it Write 0x1: Clears the MISR_STATUS register Write 0x0: Has no effect on the the MISR_STATUS register
9	STATUS	R	Read only register Set to 0x1 when misr signature has been updated
8	ENABLE	RW	MISR enable which starts/ends calculation of MISR on the next vsync/trigger event This essentially double buffers the enable so that the misr doesn't start/end immediately but rather at a defined time The MISR calculation would run until the end of programmed number of frames
7:0	FRAME_COUNT	RW	Number of frames to run before capturing

0x01A6B994 MDSS_MDP_INTF_1_VIDEO_MISR_SIGNATURE**Type:** R**Reset State:** 0x00000000

Video MISR signature

MDSS_MDP_INTF_1_VIDEO_MISR_SIGNATURE

Bits	Name	Description
31:0	CAPTURED	This register is the MISR value captured

0x01A6B998 MDSS_MDP_INTF_1_CMD_0_MISR_CTRL**Type:** RW**Reset State:** 0x00000000

DSI CMD0 MISR Control

MDSS_MDP_INTF_1_CMD_0_MISR_CTRL

Bits	Name	Type	Description
31	RUN_MODE	RW	This register defines the operation mode for the MISR 0x0: Single run mode meaning it stops after the programmable number of frames. (i.e. legacy mode prior to MDP5 1.5.0) 0x1: Free running mode meaning it updates the misr signature every programmable number of frames and continues to run while MISR_ENABLE is set to 0x1
27:24	INPUT_SEL	RW	This field allows the selected input to propagate to the MISR in the event that multiple inputs want to share the same physical MISR For example, the video interface might want to MISR on active pixel data only or active+timing information as well This field would select which input is sent to the MISR
10	STATUS_CLEAR	W	This is a write only register that doesn't have any effect prior to MDP5 1.5.0 since there was no logic associated with it Write 0x1: Clears the MISR_STATUS register Write 0x0: Has no effect on the the MISR_STATUS register
9	STATUS	R	Read only register Set to 0x1 when misr signature has been updated
8	ENABLE	RW	MISR enable which starts/ends calculation of MISR on the next vsync/trigger event This essentially double buffers the enable so that the misr doesn't start/end immediately but rather at a defined time The MISR calculation would run until the end of programmed number of frames
7:0	FRAME_COUNT	RW	Number of frames to run before capturing

0x01A6B99C MDSS_MDP_INTF_1_CMD_0_MISR_SIGNATURE**Type:** R**Reset State:** 0x00000000

DSI CMD0 MISR signature

MDSS_MDP_INTF_1_CMD_0_MISR_SIGNATURE

Bits	Name	Description
31:0	CAPTURED	This register is the MISR value captured

17.17.15 MDSS_MDP_INTF_2

Display Interface hardware register description

17.17.15.1 MDSS_MDP_INTF_2_INTF

INTF

Display Interface hardware register description

0x01A6C000 MDSS_MDP_INTF_2_TIMING_ENGINE_EN

Type: RW

Reset State: 0x00000000

Display Interface Timing Engine Enable register

double buffered: YES

when: On frame boundaries

MDSS_MDP_INTF_2_TIMING_ENGINE_EN

Bits	Name	Description
1	OSYNC_LOCK	
0	EN	Display Interface enable Should be the LAST register programmed for the interface Be sure to program ALL timing registers and setting up of the display pipeline BEFORE enabling this register Upon disabling of this bit, hardware will stop driving the interface at the NEXT vsync boundary A VSYNC interrupt WILL be generated but a VSYNC will NOT be generated on the interface

0x01A6C004 MDSS_MDP_INTF_2_INTF_CONFIG

Type: RW

Reset State: 0x00040000

Display Interface Configuration register.

Note that the configuration register ONLY applies to INTF_4 which transmits to MHL

MDSS_MDP_INTF_2_INTF_CONFIG

Bits	Name	Description
31	PROG_FETCH_START_EN	Programmable Fetch feature enable bit 0: programmable line counter disabled - default value to be backward compatible 1: programmable line counter enabled
30	ACTIVE_V_EN	Active V Enable
29	ACTIVE_H_EN	Active H Enable
28	VSYNC_INTERRUPT_MODE	vsync interrupt generation mode 0 = generate an interrupt every vsync 1 = generate an interrupt every 2 fields (applicable to interlace mode)
27:24	REPEAT_PIXEL_MODE	Pixel repeat mode 0x1 = repeat each pixel 2 times 0x2 = repeat each pixel 3 times ... 15 = repeat pixel 16 times
23	DSI_VIDEO_STOP_MODE	This bit controls how video mode stream is stopped when DSI dynamic refresh is triggered. It is not applicable for non-DSI interface. 0x0: disable timing engine when DSI sets the video stop signal; 0x1: halt timing engine counters when DSI sets the video stop signal
21	CHROMA_DWN_SAMPLE_H_MTHD	Horizontal Chroma Down Sample method 0x0: non-pixel dropping 0x1: pixel dropping
20	CHROMA_DWN_SAMPLE_EN	Chroma Down Sample enable for YUV444 0x0: Disabled 0x1: Enabled
18	CSC_DST_DATA_FORMAT	Destination Data Format 0x0: RGB 0x1: YCbCr
17	CSC_SRC_DATA_FORMAT	Source Data Format 0x0: RGB 0x1: YCbCr
16	CSC_EN	Enables the color matrix conversion 0x0: Bypass 0x1: Enabled

MDSS_MDP_INTF_2_INTF_CONFIG (cont.)

Bits	Name	Description
13:11	NIBBLE_SWAP_EN	Enable nibble swapping within each pixel byte NIBBLE_SWAP_EN[0]: 0 = no swapping for pixel_data [7:0]; 1 = enable swapping of pixel_data[3:0] with pixel_data[7:4] NIBBLE_SWAP_EN[1] = 0 = no swapping for pixel_data [15:8]; 1 = enable swapping of pixel_data[11:8] with pixel_data[15:12] NIBBLE_SWAP_EN[2] = 0 = no swapping for pixel_data [23:16]; 1 = enable swapping of pixel_data[19:16] with pixel_data[23:20]
10	PACK_ALIGN	0x0: LSB 0x1:MSB
9:8	PACK_422	0x0: Y2Cr1Y1Cb1 0x1: Cr1Y2Cb1Y1 0x2: Y2Cb1Y1Cr1 0x3: Cb1Y2Cr1Y1
6:4	OUT_MODE	0x0: YCbCr or RGB4444 seperate syncs 24-bit interface 0x1: RSVD 0x2: YCbCr 422 seperate syncs 16-bit interface
2	INTERLACE_INIT_SEL	0x0: start with even field 0x1: start with odd field
1	REPEAT_PIXEL	0x0: Do NOT repeat 0x1: Repeat
0	INTERLACE_MODE	double buffered: YES 0x0: Progressive Out 0x1: Interlace out

0x01A6C008 MDSS_MDP_INTF_2_HSYNC_CTL**Type:** RW**Reset State:** 0x00000000

Display Interface HSYNC Control

double buffered: YES

MDSS_MDP_INTF_2_HSYNC_CTL

Bits	Name	Description
31:16	HSYNC_PERIOD	HSYNC Period in pixel clocks. This is the start between start of HSYNC pulse and the start of the next HSYNC pulse. Note: hsync_period=width + h_porch where width=output image width h_porch=(h_back_porch+h_front_porch) h_back_porch starts at the beginning of hsync pulse
15:0	PULSE_WIDTH	HSYNC pulse width in pixel clocks

0x01A6C00C MDSS_MDP_INTF_2_VSYNC_PERIOD_F0**Type:** RW**Reset State:** 0x00000000

Display Interface VSYNC Control for field 0

double buffered: YES

MDSS_MDP_INTF_2_VSYNC_PERIOD_F0

Bits	Name	Description
31:0	VSYNC_PERIOD	VSYNC Period in pixel clocks. This is the time between start of vsync pulse and the start of the next vsync pulse. Note vsync_period=height+v_porch where height=output_image_height v_porch=(v_back_porch+v_front_porch) v_back_porch starts at the beginning of vsync pulse.

0x01A6C010 MDSS_MDP_INTF_2_VSYNC_PERIOD_F1**Type:** RW**Reset State:** 0x00000000

Display Interface VSYNC Control for field 1

double buffered: NO

MDSS_MDP_INTF_2_VSYNC_PERIOD_F1

Bits	Name	Description
31:0	VSYNC_PERIOD	VSYNC Period in pixel clocks. In progressive mode, this is the time between start of vsync pulse and the start of next vsync pulse Interlace mode this is the time between start of Field1 vsync pulse and the start of Field2 vsync pulse. Note: $\text{vsync_period} = \text{Height} + \text{v_porch}$ where, Height = Output image height $\text{v_porch} = (\text{v_back_porch} + \text{v_front_porch})$ v_back_porch starts at the beginning of vsync pulse

0x01A6C014 MDSS_MDP_INTF_2_VSYNC_PULSE_WIDTH_F0**Type:** RW**Reset State:** 0x00000000

Display Interface VSYNC Control for field 0

double buffered: YES

MDSS_MDP_INTF_2_VSYNC_PULSE_WIDTH_F0

Bits	Name	Description
31:0	VSYNC_PERIOD	VSYNC Period width in pixel clocks

0x01A6C018 MDSS_MDP_INTF_2_VSYNC_PULSE_WIDTH_F1**Type:** RW**Reset State:** 0x00000000

Display Interface VSYNC Control for field 1

double buffered: NO

MDSS_MDP_INTF_2_VSYNC_PULSE_WIDTH_F1

Bits	Name	Description
31:0	VSYNC_PERIOD	VSYNC Period in pixel clocks for the 2nd field. NOT valid for progressive mode

0x01A6C01C MDSS_MDP_INTF_2_DISPLAY_V_START_F0**Type:** RW**Reset State:** 0x00000000

Display Interface Vertical Active Start Control for field 0

double buffered: YES

MDSS_MDP_INTF_2_DISPLAY_V_START_F0

Bits	Name	Description
31:0	DISPLAY_START_Y	Display Y Start Defines the time period between the field0 start of vsync pulse and the first displayed line position in pixel clocks $\text{display_start_y} = \text{v_back_porch} * \text{hcycle}$ where $\text{hcycle} = \text{horizontal_back_porch} + \text{horizontal_active} + \text{horizontal_front_porch}$ For eDP control the value programmed into this register differs slightly and should be programmed to the first active pixel and not the first pixel on that line! In particular, the programming for eDP should be: $\text{display_start_y} = (\text{v_back_porch} * \text{hcycle}) + \text{horizontal_back_porch}$

0x01A6C020 MDSS_MDP_INTF_2_DISPLAY_V_START_F1**Type:** RW**Reset State:** 0x00000000

Display Interface Vertical Active Start Control for field 1

double buffered: NO

MDSS_MDP_INTF_2_DISPLAY_V_START_F1

Bits	Name	Description
31:0	DISPLAY_START_Y	Display Y Start Defines the time period between the field1 start of vsync pulse and the first displayed line position in pixel clocks $\text{display_start_y} = \text{v_back_porch} * \text{hcycle}$ where $\text{hcycle} = \text{horizontal_back_porch} + \text{horizontal_active} + \text{horizontal_front_porch}$ For eDP control the value programmed into this register differs slightly and should be programmed to the first active pixel and not the first pixel on that line! In particular, the programming for eDP should be: $\text{display_start_y} = (\text{v_back_porch} * \text{hcycle}) + \text{horizontal_back_porch}$

0x01A6C024 MDSS_MDP_INTF_2_DISPLAY_V_END_F0**Type:** RW**Reset State:** 0x00000000

Display Interface Vertical Active End Control for field 0

double buffered: YES

MDSS_MDP_INTF_2_DISPLAY_V_END_F0

Bits	Name	Description
31:0	DISPLAY_END_Y	Display Y End Defines the time period between the field0 start of vsync pulse and the last displayed line position in pixel clocks $\text{display_end_y} = (\text{vertical_back_porch} + \text{vertical_active}) * \text{hcycle} - 1$ where $\text{hcycle} = \text{horizontal_back_porch} + \text{horizontal_active} + \text{horizontal_front_porch}$ For eDP control the value programmed into this register differs slightly and should be programmed to the last active pixel and not the last pixel on that line! In particular, the programming for eDP should be: $\text{display_end_y} = (\text{vertical_back_porch} + \text{vertical_active}) * \text{hcycle} - 1 - \text{horizontal_front_porch}$

0x01A6C028 MDSS_MDP_INTF_2_DISPLAY_V_END_F1**Type:** RW**Reset State:** 0x00000000

Display Interface Vertical Active End Control for field 1

double buffered: NO

MDSS_MDP_INTF_2_DISPLAY_V_END_F1

Bits	Name	Description
31:0	DISPLAY_END_Y	Display Y End Defines the time period between the field0 start of vsync pulse and the last displayed line position in pixel clocks $\text{display_end_y} = (\text{vertical_back_porch} + \text{vertical_active}) * \text{hcycle} - 1$ where $\text{hcycle} = \text{horizontal_back_porch} + \text{horizontal_active} + \text{horizontal_front_porch}$ For eDP control the value programmed into this register differs slightly and should be programmed to the last active pixel and not the last pixel on that line! In particular, the programming for eDP should be: $\text{display_end_y} = (\text{vertical_back_porch} + \text{vertical_active}) * \text{hcycle} - 1 - \text{horizontal_front_porch}$

0x01A6C02C MDSS_MDP_INTF_2_ACTIVE_V_START_F0**Type:** RW**Reset State:** 0x00000000

Display Interface Active Start Control for field 0

double buffered: YES

MDSS_MDP_INTF_2_ACTIVE_V_START_F0

Bits	Name	Description
31:0	ACTIVE_START_Y	Active Y Start

0x01A6C030 MDSS_MDP_INTF_2_ACTIVE_V_START_F1**Type:** RW**Reset State:** 0x00000000

Display Interface Active Start Control for field 1

double buffered: NO

MDSS_MDP_INTF_2_ACTIVE_V_START_F1

Bits	Name	Description
31:0	ACTIVE_START_Y	Active Y Start

0x01A6C034 MDSS_MDP_INTF_2_ACTIVE_V_END_F0

Type: RW

Reset State: 0x00000000

Display Interface Active End Control for field 0

double buffered: YES

MDSS_MDP_INTF_2_ACTIVE_V_END_F0

Bits	Name	Description
31:0	ACTIVE_END_Y	Active Y End

0x01A6C038 MDSS_MDP_INTF_2_ACTIVE_V_END_F1

Type: RW

Reset State: 0x00000000

Display Interface Active End Control for field 1

double buffered: NO

MDSS_MDP_INTF_2_ACTIVE_V_END_F1

Bits	Name	Description
31:0	ACTIVE_END_Y	Active Y End

0x01A6C03C MDSS_MDP_INTF_2_DISPLAY_HCTL

Type: RW

Reset State: 0x00000000

Display Interface HSYNC Control for field 1

double buffered: YES

MDSS_MDP_INTF_2_DISPLAY_HCTL

Bits	Name	Description
31:16	DISPLAY_END_X	Display End X
15:0	DISPLAY_START_X	Display Start X

0x01A6C040 MDSS_MDP_INTF_2_ACTIVE_HCTL**Type:** RW**Reset State:** 0x00000000

Display Interface HSYNC Control for field 1

double buffered: YES

MDSS_MDP_INTF_2_ACTIVE_HCTL

Bits	Name	Description
31:16	ACTIVE_END_X	Active End X
15:0	ACTIVE_START_X	Active Start X

0x01A6C044 MDSS_MDP_INTF_2_BORDER_COLOR**Type:** RW**Reset State:** 0x00000000

Display Interface Border Color

double buffered: NO

MDSS_MDP_INTF_2_BORDER_COLOR

Bits	Name	Description
23:0	BORDER_COLOR	Defines 24-bit Border Color of the inactive region of display

0x01A6C048 MDSS_MDP_INTF_2_UNDERFLOW_COLOR**Type:** RW**Reset State:** 0x00000000

Display Interface Underflow Color

double buffered: NO

MDSS_MDP_INTF_2_UNDERFLOW_COLOR

Bits	Name	Description
23:0	UNDERFLOW_COLOR	Defines 24-bit Underflow Color

0x01A6C04C MDSS_MDP_INTF_2_HSYNC_SKEW**Type:** RW**Reset State:** 0x00000000

Display Interface HSYNC Skew Control

double buffered: YES

MDSS_MDP_INTF_2_HSYNC_SKEW

Bits	Name	Description
15:0	HSYNC_SKEW	HSYNC Skew Defines the number of pixel clocks HSYNC active edge is delayed from VSYNC active edge

0x01A6C050 MDSS_MDP_INTF_2_POLARITY_CTL**Type:** RW**Reset State:** 0x00000000

Display Interface Polarity Control

double buffered: NO

MDSS_MDP_INTF_2_POLARITY_CTL

Bits	Name	Description
2	DEN	DEN Polarity 0x0: Active HIGH 0x1: Active LOW
1	VSYNC	VSYNC Polarity 0x0: Active HIGH 0x1: Active LOW
0	HSYNC	HSYNC Polarity 0x0: Active HIGH 0x1: Active LOW

0x01A6C058 MDSS_MDP_INTF_2_TP_COLOR0**Type:** RW**Reset State:** 0x00000000

Display Interface Test Pattern Color

double buffered: NO

MDSS_MDP_INTF_2_TP_COLOR0

Bits	Name	Description
23:0	COLOR0	Test Pattern Color

0x01A6C05C MDSS_MDP_INTF_2_TP_COLOR1**Type:** RW**Reset State:** 0x00000000

Display Interface Test Pattern Color

double buffered: NO

MDSS_MDP_INTF_2_TP_COLOR1

Bits	Name	Description
23:0	COLOR1	Test Pattern Color

0x01A6C084 MDSS_MDP_INTF_2_DSI_CMD_MODE_TRIGGER_EN**Type:** RW**Reset State:** 0x00000000

Enable for DSI trigger processing on DSI Command Mode interface

double buffered: NO

MDSS_MDP_INTF_2_DSI_CMD_MODE_TRIGGER_EN

Bits	Name	Description
5:4	DSI_CMD_TG_INTF_SEL	Select the stream to be triggered by DSI controller. Takes effect only if DSI_CMD_TG_EN is set to 1
0	DSI_CMD_TG_EN	DSI Command Trigger Enable. Set it to 1 to enable trigger processing 0: Disable trigger processing 1: Enable trigger processing

0x01A6C090 MDSS_MDP_INTF_2_PANEL_FORMAT**Type:** RW**Reset State:** 0x00002100

Panel Packer Configuration

double buffered: NO

MDSS_MDP_INTF_2_PANEL_FORMAT

Bits	Name	Description
16	COMP2_BIT_FLIP_EN	0x1: All bits of component 2 are flipped
15	COMP1_BIT_FLIP_EN	0x1: All bits of component 1 are flipped
14	COMP0_BIT_FLIP_EN	0x1: All bits of component 0 are flipped
13:8	PACK_PATTERN	Packing Pattern used by the packer 0x21: RGB 0x24: RBG 0x12: BGR 0x18: BRG 0x06: GBR 0x09: GRB
7	PACK_ALIGN	Packing alignment (within a byte) for each colour component 0x0: LSB 0x1: MSB
5:4	COMP2_OUT	Number of bits for component 2 output 0x0: 4 bits 0x1: 5 bits 0x2: 6 bits 0x3: 8 bits
3:2	COMP1_OUT	Number of bits for component 1 output 0x0: 4 bits 0x1: 5 bits 0x2: 6 bits 0x3: 8 bits
1:0	COMP0_OUT	Number of bits for component 0 output 0x0: 4 bits 0x1: 5 bits 0x2: 6 bits 0x3: 8 bits

0x01A6C0A8 MDSS_MDP_INTF_2_FRAME_LINE_COUNT_EN**Type:** RW**Reset State:** 0x00000000

Frame and Line Count Enable

double buffered: NO

MDSS_MDP_INTF_2_FRAME_LINE_COUNT_EN

Bits	Name	Description
1	FRAME_COUNT_EN	Current Frame Count Enable This will provide SW information on the current frame being sent out on an active panel It will reset upon disabling of the timing engine and increment for every frame Note that there is possibility for rollover if enabled for long time. NO status is provided that rollover occurred
0	LINE_COUNT_EN	Current Line Count Enable This will provide SW information on the current line being sent out on an active panel This will include all blanking lines as well It will reset upon vsync and increment until the end of the frame.

0x01A6C0AC MDSS_MDP_INTF_2_FRAME_COUNT**Type:** R**Reset State:** 0x00000000

Current Frame Count since Timing Engine Enabled

double buffered: N/A

MDSS_MDP_INTF_2_FRAME_COUNT

Bits	Name	Description
31:0	VALUE	Current Frame Count - Base 1

0x01A6C0B0 MDSS_MDP_INTF_2_LINE_COUNT**Type:** R**Reset State:** 0x00000000

Current Line Count Including blanking

double buffered: N/A

MDSS_MDP_INTF_2_LINE_COUNT

Bits	Name	Description
15:0	VALUE	Current Line Count - Base 1

0x01A6C170 MDSS_MDP_INTF_2_PROG_FETCH_START**Type:** RW**Reset State:** 0x00000000

Programmable Fetch Start

double buffered: YES

MDSS_MDP_INTF_2_PROG_FETCH_START

Bits	Name	Description
31:0	VALUE	Programmable line value for the feature Note: The valid register setting shall be vsync counter value for the first pixel of VFP lines

0x01A6C180 MDSS_MDP_INTF_2_INTF_INTR_EN**Type:** RW**Reset State:** 0x00000000

Display Interface Interrupt Enable

MDSS_MDP_INTF_2_INTF_INTR_EN

Bits	Name	Description
0	PROG_LINE	Programmeble Line Interrupt 0x0: Interrupt Disabled 0x1: Interrupt Enabled

0x01A6C184 MDSS_MDP_INTF_2_INTF_INTR_STATUS**Type:** R**Reset State:** 0x00000000

Display Interface Interrupt Status

MDSS_MDP_INTF_2_INTF_INTR_STATUS

Bits	Name	Description
0	PROG_LINE	Programmeble Line Interrupt

0x01A6C188 MDSS_MDP_INTF_2_INTF_INTR_CLEAR**Type:** W**Reset State:** 0x00000000

Display Interface Interrupt Clear

MDSS_MDP_INTF_2_INTF_INTR_CLEAR

Bits	Name	Description
0	PROG_LINE	Programmeble Line Interrupt

0x01A6C18C MDSS_MDP_INTF_2_PROG_LINE_INTR_CONF**Type:** RW**Reset State:** 0xFFFFFFFF

Programmable Line Interrupt Config

double buffered: YES

MDSS_MDP_INTF_2_PROG_LINE_INTR_CONF

Bits	Name	Description
31:0	INTR_CORD	Pixel location in the panel video timing in terms of video raster structure including V/H blanks where the programmable line interrupt be generated When SW programe this register, it will be applied for the next frame after vsync

0x01A6C190 MDSS_MDP_INTF_2_VIDEO_MISR_CTRL**Type:** RW**Reset State:** 0x00000000

Video MISR Control

MDSS_MDP_INTF_2_VIDEO_MISR_CTRL

Bits	Name	Type	Description
31	RUN_MODE	RW	This register defines the operation mode for the MISR 0x0: Single run mode meaning it stops after the programmable number of frames. (i.e. legacy mode prior to MDP5 1.5.0) 0x1: Free running mode meaning it updates the misr signature every programmable number of frames and continues to run while MISR_ENABLE is set to 0x1
27:24	INPUT_SEL	RW	This field allows the selected input to propagate to the MISR in the event that multiple inputs want to share the same physical MISR For example, the video interface might want to MISR on active pixel data only or active+timing information as well This field would select which input is sent to the MISR
10	STATUS_CLEAR	W	This is a write only register that doesn't have any effect prior to MDP5 1.5.0 since there was no logic associated with it Write 0x1: Clears the MISR_STATUS register Write 0x0: Has no effect on the the MISR_STATUS register
9	STATUS	R	Read only register Set to 0x1 when misr signature has been updated
8	ENABLE	RW	MISR enable which starts/ends calculation of MISR on the next vsync/trigger event This essentially double buffers the enable so that the misr doesn't start/end immediately but rather at a defined time The MISR calculation would run until the end of programmed number of frames
7:0	FRAME_COUNT	RW	Number of frames to run before capturing

0x01A6C194 MDSS_MDP_INTF_2_VIDEO_MISR_SIGNATURE**Type:** R**Reset State:** 0x00000000

Video MISR signature

MDSS_MDP_INTF_2_VIDEO_MISR_SIGNATURE

Bits	Name	Description
31:0	CAPTURED	This register is the MISR value captured

0x01A6C198 MDSS_MDP_INTF_2_CMD_0_MISR_CTRL**Type:** RW**Reset State:** 0x00000000

DSI CMD0 MISR Control

MDSS_MDP_INTF_2_CMD_0_MISR_CTRL

Bits	Name	Type	Description
31	RUN_MODE	RW	This register defines the operation mode for the MISR 0x0: Single run mode meaning it stops after the programmable number of frames. (i.e. legacy mode prior to MDP5 1.5.0) 0x1: Free running mode meaning it updates the misr signature every programmable number of frames and continues to run while MISR_ENABLE is set to 0x1
27:24	INPUT_SEL	RW	This field allows the selected input to propagate to the MISR in the event that multiple inputs want to share the same physical MISR For example, the video interface might want to MISR on active pixel data only or active+timing information as well This field would select which input is sent to the MISR
10	STATUS_CLEAR	W	This is a write only register that doesn't have any effect prior to MDP5 1.5.0 since there was no logic associated with it Write 0x1: Clears the MISR_STATUS register Write 0x0: Has no effect on the the MISR_STATUS register
9	STATUS	R	Read only register Set to 0x1 when misr signature has been updated
8	ENABLE	RW	MISR enable which starts/ends calculation of MISR on the next vsync/trigger event This essentially double buffers the enable so that the misr doesn't start/end immediately but rather at a defined time The MISR calculation would run until the end of programmed number of frames
7:0	FRAME_COUNT	RW	Number of frames to run before capturing

0x01A6C19C MDSS_MDP_INTF_2_CMD_0_MISR_SIGNATURE**Type:** R**Reset State:** 0x00000000

DSI CMD0 MISR signature

MDSS_MDP_INTF_2_CMD_0_MISR_SIGNATURE

Bits	Name	Description
31:0	CAPTURED	This register is the MISR value captured

17.17.16 MDSS_MDP_INTF_3

Display Interface hardware register description

17.17.16.1 MDSS_MDP_INTF_3_INTF

INTF

Display Interface hardware register description

0x01A6C800 MDSS_MDP_INTF_3_TIMING_ENGINE_EN

Type: RW

Reset State: 0x00000000

Display Interface Timing Engine Enable register

double buffered: YES

when: On frame boundaries

MDSS_MDP_INTF_3_TIMING_ENGINE_EN

Bits	Name	Description
1	OSYNC_LOCK	
0	EN	Display Interface enable Should be the LAST register programmed for the interface Be sure to program ALL timing registers and setting up of the display pipeline BEFORE enabling this register Upon disabling of this bit, hardware will stop driving the interface at the NEXT vsync boundary A VSYNC interrupt WILL be generated but a VSYNC will NOT be generated on the interface

0x01A6C804 MDSS_MDP_INTF_3_INTF_CONFIG

Type: RW

Reset State: 0x00040000

Display Interface Configuration register.

Note that the configuration register ONLY applies to INTF_4 which transmits to MHL

MDSS_MDP_INTF_3_INTF_CONFIG

Bits	Name	Description
31	PROG_FETCH_START_EN	Programmable Fetch feature enable bit 0: programmable line counter disabled - default value to be backward compatible 1: programmable line counter enabled
30	ACTIVE_V_EN	Active V Enable
29	ACTIVE_H_EN	Active H Enable
28	VSYNC_INTERRUPT_MODE	vsync interrupt generation mode 0 = generate an interrupt every vsync 1 = generate an interrupt every 2 fields (applicable to interlace mode)
27:24	REPEAT_PIXEL_MODE	Pixel repeat mode 0x1 = repeat each pixel 2 times 0x2 = repeat each pixel 3 times ... 15 = repeat pixel 16 times
23	DSI_VIDEO_STOP_MODE	This bit controls how video mode stream is stopped when DSI dynamic refresh is triggered. It is not applicable for non-DSI interface. 0x0: disable timing engine when DSI sets the video stop signal; 0x1: halt timing engine counters when DSI sets the video stop signal
21	CHROMA_DWN_SAMPLE_H_MTHD	Horizontal Chroma Down Sample method 0x0: non-pixel dropping 0x1: pixel dropping
20	CHROMA_DWN_SAMPLE_EN	Chroma Down Sample enable for YUV444 0x0: Disabled 0x1: Enabled
18	CSC_DST_DATA_FORMAT	Destination Data Format 0x0: RGB 0x1: YCbCr
17	CSC_SRC_DATA_FORMAT	Source Data Format 0x0: RGB 0x1: YCbCr
16	CSC_EN	Enables the color matrix conversion 0x0: Bypass 0x1: Enabled

MDSS_MDP_INTF_3_INTF_CONFIG (cont.)

Bits	Name	Description
13:11	NIBBLE_SWAP_EN	Enable nibble swapping within each pixel byte NIBBLE_SWAP_EN[0]: 0 = no swapping for pixel_data [7:0]; 1 = enable swapping of pixel_data[3:0] with pixel_data[7:4] NIBBLE_SWAP_EN[1] = 0 = no swapping for pixel_data [15:8]; 1 = enable swapping of pixel_data[11:8] with pixel_data[15:12] NIBBLE_SWAP_EN[2] = 0 = no swapping for pixel_data [23:16]; 1 = enable swapping of pixel_data[19:16] with pixel_data[23:20]
10	PACK_ALIGN	0x0: LSB 0x1:MSB
9:8	PACK_422	0x0: Y2Cr1Y1Cb1 0x1: Cr1Y2Cb1Y1 0x2: Y2Cb1Y1Cr1 0x3: Cb1Y2Cr1Y1
6:4	OUT_MODE	0x0: YCbCr or RGB4444 seperate syncs 24-bit interface 0x1: RSVD 0x2: YCbCr 422 seperate syncs 16-bit interface
2	INTERLACE_INIT_SEL	0x0: start with even field 0x1: start with odd field
1	REPEAT_PIXEL	0x0: Do NOT repeat 0x1: Repeat
0	INTERLACE_MODE	double buffered: YES 0x0: Progressive Out 0x1: Interlace out

0x01A6C808 MDSS_MDP_INTF_3_HSYNC_CTL**Type:** RW**Reset State:** 0x00000000

Display Interface HSYNC Control

double buffered: YES

MDSS_MDP_INTF_3_HSYNC_CTL

Bits	Name	Description
31:16	HSYNC_PERIOD	HSYNC Period in pixel clocks. This is the start between start of HSYNC pulse and the start of the next HSYNC pulse. Note: $hsync_period = width + h_porch$ where $width = \text{output image width}$ $h_porch = (h_back_porch + h_front_porch)$ h_back_porch starts at the beginning of hsync pulse
15:0	PULSE_WIDTH	HSYNC pulse width in pixel clocks

0x01A6C80C MDSS_MDP_INTF_3_VSYNC_PERIOD_F0**Type:** RW**Reset State:** 0x00000000

Display Interface VSYNC Control for field 0

double buffered: YES

MDSS_MDP_INTF_3_VSYNC_PERIOD_F0

Bits	Name	Description
31:0	VSYNC_PERIOD	VSYNC Period in pixel clocks. This is the time between start of vsync pulse and the start of the next vsync pulse. Note $vsync_period = height + v_porch$ where $height = \text{output_image_height}$ $v_porch = (v_back_porch + v_front_porch)$ v_back_porch starts at the beginning of vsync pulse.

0x01A6C810 MDSS_MDP_INTF_3_VSYNC_PERIOD_F1**Type:** RW**Reset State:** 0x00000000

Display Interface VSYNC Control for field 1

double buffered: NO

MDSS_MDP_INTF_3_VSYNC_PERIOD_F1

Bits	Name	Description
31:0	VSYNC_PERIOD	VSYNC Period in pixel clocks. In progressive mode, this is the time between start of vsync pulse and the start of next vsync pulse Interlace mode this is the time between start of Field1 vsync pulse and the start of Field2 vsync pulse. Note: $\text{vsync_period} = \text{Height} + \text{v_porch}$ where, Height = Output image height $\text{v_porch} = (\text{v_back_porch} + \text{v_front_porch})$ v_back_porch starts at the beginning of vsync pulse

0x01A6C814 MDSS_MDP_INTF_3_VSYNC_PULSE_WIDTH_F0**Type:** RW**Reset State:** 0x00000000

Display Interface VSYNC Control for field 0

double buffered: YES

MDSS_MDP_INTF_3_VSYNC_PULSE_WIDTH_F0

Bits	Name	Description
31:0	VSYNC_PERIOD	VSYNC Period width in pixel clocks

0x01A6C818 MDSS_MDP_INTF_3_VSYNC_PULSE_WIDTH_F1**Type:** RW**Reset State:** 0x00000000

Display Interface VSYNC Control for field 1

double buffered: NO

MDSS_MDP_INTF_3_VSYNC_PULSE_WIDTH_F1

Bits	Name	Description
31:0	VSYNC_PERIOD	VSYNC Period in pixel clocks for the 2nd field. NOT valid for progressive mode

0x01A6C81C MDSS_MDP_INTF_3_DISPLAY_V_START_F0**Type:** RW**Reset State:** 0x00000000

Display Interface Vertical Active Start Control for field 0

double buffered: YES

MDSS_MDP_INTF_3_DISPLAY_V_START_F0

Bits	Name	Description
31:0	DISPLAY_START_Y	Display Y Start Defines the time period between the field0 start of vsync pulse and the first displayed line position in pixel clocks $\text{display_start_y} = \text{v_back_porch} * \text{hcycle}$ where $\text{hcycle} = \text{horizontal_back_porch} + \text{horizontal_active} + \text{horizontal_front_porch}$ For eDP control the value programmed into this register differs slightly and should be programmed to the first active pixel and not the first pixel on that line! In particular, the programming for eDP should be: $\text{display_start_y} = (\text{v_back_porch} * \text{hcycle}) + \text{horizontal_back_porch}$

0x01A6C820 MDSS_MDP_INTF_3_DISPLAY_V_START_F1**Type:** RW**Reset State:** 0x00000000

Display Interface Vertical Active Start Control for field 1

double buffered: NO

MDSS_MDP_INTF_3_DISPLAY_V_START_F1

Bits	Name	Description
31:0	DISPLAY_START_Y	Display Y Start Defines the time period between the field1 start of vsync pulse and the first displayed line position in pixel clocks $\text{display_start_y} = \text{v_back_porch} * \text{hcycle}$ where $\text{hcycle} = \text{horizontal_back_porch} + \text{horizontal_active} + \text{horizontal_front_porch}$ For eDP control the value programmed into this register differs slightly and should be programmed to the first active pixel and not the first pixel on that line! In particular, the programming for eDP should be: $\text{display_start_y} = (\text{v_back_porch} * \text{hcycle}) + \text{horizontal_back_porch}$

0x01A6C824 MDSS_MDP_INTF_3_DISPLAY_V_END_F0**Type:** RW**Reset State:** 0x00000000

Display Interface Vertical Active End Control for field 0

double buffered: YES

MDSS_MDP_INTF_3_DISPLAY_V_END_F0

Bits	Name	Description
31:0	DISPLAY_END_Y	Display Y End Defines the time period between the field0 start of vsync pulse and the last displayed line position in pixel clocks $\text{display_end_y} = (\text{vertical_back_porch} + \text{vertical_active}) * \text{hcycle} - 1$ where $\text{hcycle} = \text{horizontal_back_porch} + \text{horizontal_active} + \text{horizontal_front_porch}$ For eDP control the value programmed into this register differs slightly and should be programmed to the last active pixel and not the last pixel on that line! In particular, the programming for eDP should be: $\text{display_end_y} = (\text{vertical_back_porch} + \text{vertical_active}) * \text{hcycle} - 1 - \text{horizontal_front_porch}$

0x01A6C828 MDSS_MDP_INTF_3_DISPLAY_V_END_F1**Type:** RW**Reset State:** 0x00000000

Display Interface Vertical Active End Control for field 1

double buffered: NO

MDSS_MDP_INTF_3_DISPLAY_V_END_F1

Bits	Name	Description
31:0	DISPLAY_END_Y	Display Y End Defines the time period between the field0 start of vsync pulse and the last displayed line position in pixel clocks $\text{display_end_y} = (\text{vertical_back_porch} + \text{vertical_active}) * \text{hcycle} - 1$ where hcycle = horizontal_back_porch + horizontal_active + horizontal_front_porch For eDP control the value programmed into this register differs slightly and should be programmed to the last active pixel and not the last pixel on that line! In particular, the programming for eDP should be: $\text{display_end_y} = (\text{vertical_back_porch} + \text{vertical_active}) * \text{hcycle} - 1 - \text{horizontal_front_porch}$

0x01A6C82C MDSS_MDP_INTF_3_ACTIVE_V_START_F0**Type:** RW**Reset State:** 0x00000000

Display Interface Active Start Control for field 0

double buffered: YES

MDSS_MDP_INTF_3_ACTIVE_V_START_F0

Bits	Name	Description
31:0	ACTIVE_START_Y	Active Y Start

0x01A6C830 MDSS_MDP_INTF_3_ACTIVE_V_START_F1**Type:** RW**Reset State:** 0x00000000

Display Interface Active Start Control for field 1

double buffered: NO

MDSS_MDP_INTF_3_ACTIVE_V_START_F1

Bits	Name	Description
31:0	ACTIVE_START_Y	Active Y Start

0x01A6C834 MDSS_MDP_INTF_3_ACTIVE_V_END_F0

Type: RW

Reset State: 0x00000000

Display Interface Active End Control for field 0

double buffered: YES

MDSS_MDP_INTF_3_ACTIVE_V_END_F0

Bits	Name	Description
31:0	ACTIVE_END_Y	Active Y End

0x01A6C838 MDSS_MDP_INTF_3_ACTIVE_V_END_F1

Type: RW

Reset State: 0x00000000

Display Interface Active End Control for field 1

double buffered: NO

MDSS_MDP_INTF_3_ACTIVE_V_END_F1

Bits	Name	Description
31:0	ACTIVE_END_Y	Active Y End

0x01A6C83C MDSS_MDP_INTF_3_DISPLAY_HCTL

Type: RW

Reset State: 0x00000000

Display Interface HSYNC Control for field 1

double buffered: YES

MDSS_MDP_INTF_3_DISPLAY_HCTL

Bits	Name	Description
31:16	DISPLAY_END_X	Display End X
15:0	DISPLAY_START_X	Display Start X

0x01A6C840 MDSS_MDP_INTF_3_ACTIVE_HCTL**Type:** RW**Reset State:** 0x00000000

Display Interface HSYNC Control for field 1

double buffered: YES

MDSS_MDP_INTF_3_ACTIVE_HCTL

Bits	Name	Description
31:16	ACTIVE_END_X	Active End X
15:0	ACTIVE_START_X	Active Start X

0x01A6C844 MDSS_MDP_INTF_3_BORDER_COLOR**Type:** RW**Reset State:** 0x00000000

Display Interface Border Color

double buffered: NO

MDSS_MDP_INTF_3_BORDER_COLOR

Bits	Name	Description
23:0	BORDER_COLOR	Defines 24-bit Border Color of the inactive region of display

0x01A6C848 MDSS_MDP_INTF_3_UNDERFLOW_COLOR**Type:** RW**Reset State:** 0x00000000

Display Interface Underflow Color

double buffered: NO

MDSS_MDP_INTF_3_UNDERFLOW_COLOR

Bits	Name	Description
23:0	UNDERFLOW_COLOR	Defines 24-bit Underflow Color

0x01A6C84C MDSS_MDP_INTF_3_HSYNC_SKEW**Type:** RW**Reset State:** 0x00000000

Display Interface HSYNC Skew Control

double buffered: YES

MDSS_MDP_INTF_3_HSYNC_SKEW

Bits	Name	Description
15:0	HSYNC_SKEW	HSYNC Skew Defines the number of pixel clocks HSYNC active edge is delayed from VSYNC active edge

0x01A6C850 MDSS_MDP_INTF_3_POLARITY_CTL**Type:** RW**Reset State:** 0x00000000

Display Interface Polarity Control

double buffered: NO

MDSS_MDP_INTF_3_POLARITY_CTL

Bits	Name	Description
2	DEN	DEN Polarity 0x0: Active HIGH 0x1: Active LOW
1	VSYNC	VSYNC Polarity 0x0: Active HIGH 0x1: Active LOW
0	HSYNC	HSYNC Polarity 0x0: Active HIGH 0x1: Active LOW

0x01A6C858 MDSS_MDP_INTF_3_TP_COLOR0**Type:** RW**Reset State:** 0x00000000

Display Interface Test Pattern Color

double buffered: NO

MDSS_MDP_INTF_3_TP_COLOR0

Bits	Name	Description
23:0	COLOR0	Test Pattern Color

0x01A6C85C MDSS_MDP_INTF_3_TP_COLOR1**Type:** RW**Reset State:** 0x00000000

Display Interface Test Pattern Color

double buffered: NO

MDSS_MDP_INTF_3_TP_COLOR1

Bits	Name	Description
23:0	COLOR1	Test Pattern Color

0x01A6C884 MDSS_MDP_INTF_3_DSI_CMD_MODE_TRIGGER_EN**Type:** RW**Reset State:** 0x00000000

Enable for DSI trigger processing on DSI Command Mode interface

double buffered: NO

MDSS_MDP_INTF_3_DSI_CMD_MODE_TRIGGER_EN

Bits	Name	Description
5:4	DSI_CMD_TG_INTF_SEL	Select the stream to be triggered by DSI controller. Takes effect only if DSI_CMD_TG_EN is set to 1
0	DSI_CMD_TG_EN	DSI Command Trigger Enable. Set it to 1 to enable trigger processing 0: Disable trigger processing 1: Enable trigger processing

0x01A6C890 MDSS_MDP_INTF_3_PANEL_FORMAT**Type:** RW**Reset State:** 0x00002100

Panel Packer Configuration

double buffered: NO

MDSS_MDP_INTF_3_PANEL_FORMAT

Bits	Name	Description
16	COMP2_BIT_FLIP_EN	0x1: All bits of component 2 are flipped
15	COMP1_BIT_FLIP_EN	0x1: All bits of component 1 are flipped
14	COMP0_BIT_FLIP_EN	0x1: All bits of component 0 are flipped
13:8	PACK_PATTERN	Packing Pattern used by the packer 0x21: RGB 0x24: RBG 0x12: BGR 0x18: BRG 0x06: GBR 0x09: GRB
7	PACK_ALIGN	Packing alignment (within a byte) for each colour component 0x0: LSB 0x1: MSB
5:4	COMP2_OUT	Number of bits for component 2 output 0x0: 4 bits 0x1: 5 bits 0x2: 6 bits 0x3: 8 bits
3:2	COMP1_OUT	Number of bits for component 1 output 0x0: 4 bits 0x1: 5 bits 0x2: 6 bits 0x3: 8 bits
1:0	COMP0_OUT	Number of bits for component 0 output 0x0: 4 bits 0x1: 5 bits 0x2: 6 bits 0x3: 8 bits

0x01A6C8A8 MDSS_MDP_INTF_3_FRAME_LINE_COUNT_EN**Type:** RW**Reset State:** 0x00000000

Frame and Line Count Enable

double buffered: NO

MDSS_MDP_INTF_3_FRAME_LINE_COUNT_EN

Bits	Name	Description
1	FRAME_COUNT_EN	Current Frame Count Enable This will provide SW information on the current frame being sent out on an active panel It will reset upon disabling of the timing engine and increment for every frame Note that there is possibility for rollover if enabled for long time. NO status is provided that rollover occurred
0	LINE_COUNT_EN	Current Line Count Enable This will provide SW information on the current line being sent out on an active panel This will include all blanking lines as well It will reset upon vsync and increment until the end of the frame.

0x01A6C8AC MDSS_MDP_INTF_3_FRAME_COUNT**Type:** R**Reset State:** 0x00000000

Current Frame Count since Timing Engine Enabled

double buffered: N/A

MDSS_MDP_INTF_3_FRAME_COUNT

Bits	Name	Description
31:0	VALUE	Current Frame Count - Base 1

0x01A6C8B0 MDSS_MDP_INTF_3_LINE_COUNT**Type:** R**Reset State:** 0x00000000

Current Line Count Including blanking

double buffered: N/A

MDSS_MDP_INTF_3_LINE_COUNT

Bits	Name	Description
15:0	VALUE	Current Line Count - Base 1

0x01A6C970 MDSS_MDP_INTF_3_PROG_FETCH_START**Type:** RW**Reset State:** 0x00000000

Programmable Fetch Start

double buffered: YES

MDSS_MDP_INTF_3_PROG_FETCH_START

Bits	Name	Description
31:0	VALUE	Programmable line value for the feature Note: The valid register setting shall be vsync counter value for the first pixel of VFP lines

0x01A6C980 MDSS_MDP_INTF_3_INTF_INTR_EN**Type:** RW**Reset State:** 0x00000000

Display Interface Interrupt Enable

MDSS_MDP_INTF_3_INTF_INTR_EN

Bits	Name	Description
0	PROG_LINE	Programmeble Line Interrupt 0x0: Interrupt Disabled 0x1: Interrupt Enabled

0x01A6C984 MDSS_MDP_INTF_3_INTF_INTR_STATUS**Type:** R**Reset State:** 0x00000000

Display Interface Interrupt Status

MDSS_MDP_INTF_3_INTF_INTR_STATUS

Bits	Name	Description
0	PROG_LINE	Programmeble Line Interrupt

0x01A6C988 MDSS_MDP_INTF_3_INTF_INTR_CLEAR**Type:** W**Reset State:** 0x00000000

Display Interface Interrupt Clear

MDSS_MDP_INTF_3_INTF_INTR_CLEAR

Bits	Name	Description
0	PROG_LINE	Programmeble Line Interrupt

0x01A6C98C MDSS_MDP_INTF_3_PROG_LINE_INTR_CONF**Type:** RW**Reset State:** 0xFFFFFFFF

Programmable Line Interrupt Config

double buffered: YES

MDSS_MDP_INTF_3_PROG_LINE_INTR_CONF

Bits	Name	Description
31:0	INTR_CORD	Pixel location in the panel video timing in terms of video raster structure including V/H blanks where the programmable line interrupt be generated When SW programe this register, it will be applied for the next frame after vsync

0x01A6C990 MDSS_MDP_INTF_3_VIDEO_MISR_CTRL**Type:** RW**Reset State:** 0x00000000

Video MISR Control

MDSS_MDP_INTF_3_VIDEO_MISR_CTRL

Bits	Name	Type	Description
31	RUN_MODE	RW	This register defines the operation mode for the MISR 0x0: Single run mode meaning it stops after the programmable number of frames. (i.e. legacy mode prior to MDP5 1.5.0) 0x1: Free running mode meaning it updates the misr signature every programmable number of frames and continues to run while MISR_ENABLE is set to 0x1
27:24	INPUT_SEL	RW	This field allows the selected input to propagate to the MISR in the event that multiple inputs want to share the same physical MISR For example, the video interface might want to MISR on active pixel data only or active+timing information as well This field would select which input is sent to the MISR
10	STATUS_CLEAR	W	This is a write only register that doesn't have any effect prior to MDP5 1.5.0 since there was no logic associated with it Write 0x1: Clears the MISR_STATUS register Write 0x0: Has no effect on the the MISR_STATUS register
9	STATUS	R	Read only register Set to 0x1 when misr signature has been updated
8	ENABLE	RW	MISR enable which starts/ends calculation of MISR on the next vsync/trigger event This essentially double buffers the enable so that the misr doesn't start/end immediately but rather at a defined time The MISR calculation would run until the end of programmed number of frames
7:0	FRAME_COUNT	RW	Number of frames to run before capturing

0x01A6C994 MDSS_MDP_INTF_3_VIDEO_MISR_SIGNATURE**Type:** R**Reset State:** 0x00000000

Video MISR signature

MDSS_MDP_INTF_3_VIDEO_MISR_SIGNATURE

Bits	Name	Description
31:0	CAPTURED	This register is the MISR value captured

0x01A6C998 MDSS_MDP_INTF_3_CMD_0_MISR_CTRL**Type:** RW**Reset State:** 0x00000000

DSI CMD0 MISR Control

MDSS_MDP_INTF_3_CMD_0_MISR_CTRL

Bits	Name	Type	Description
31	RUN_MODE	RW	This register defines the operation mode for the MISR 0x0: Single run mode meaning it stops after the programmable number of frames. (i.e. legacy mode prior to MDP5 1.5.0) 0x1: Free running mode meaning it updates the misr signature every programmable number of frames and continues to run while MISR_ENABLE is set to 0x1
27:24	INPUT_SEL	RW	This field allows the selected input to propagate to the MISR in the event that multiple inputs want to share the same physical MISR For example, the video interface might want to MISR on active pixel data only or active+timing information as well This field would select which input is sent to the MISR
10	STATUS_CLEAR	W	This is a write only register that doesn't have any effect prior to MDP5 1.5.0 since there was no logic associated with it Write 0x1: Clears the MISR_STATUS register Write 0x0: Has no effect on the the MISR_STATUS register
9	STATUS	R	Read only register Set to 0x1 when misr signature has been updated
8	ENABLE	RW	MISR enable which starts/ends calculation of MISR on the next vsync/trigger event This essentially double buffers the enable so that the misr doesn't start/end immediately but rather at a defined time The MISR calculation would run until the end of programmed number of frames
7:0	FRAME_COUNT	RW	Number of frames to run before capturing

0x01A6C99C MDSS_MDP_INTF_3_CMD_0_MISR_SIGNATURE**Type:** R**Reset State:** 0x00000000

DSI CMD0 MISR signature

MDSS_MDP_INTF_3_CMD_0_MISR_SIGNATURE

Bits	Name	Description
31:0	CAPTURED	This register is the MISR value captured

17.17.17 MDSS_MDP_PP_0

Ping Pong hardware register description

17.17.17.1 MDSS_MDP_PP_0_PP

PP

Ping Pong hardware register description

0x01A71000 MDSS_MDP_PP_0_TEAR_CHECK_EN

Type: RW

Reset State: 0x00000000

Register is used to enable tearing check on different displays.

double buffered: NO

MDSS_MDP_PP_0_TEAR_CHECK_EN

Bits	Name	Description
0	EN	Enable tear check 0x0: disabled 0x1: enabled

0x01A71004 MDSS_MDP_PP_0_SYNC_CONFIG_VSYNC

Type: RW

Reset State: 0x00000000

The SYNC_CONFIG_VSYNC register is a synchronization configuration register.

double buffered: NO

MDSS_MDP_PP_0_SYNC_CONFIG_VSYNC

Bits	Name	Description
20	VSYNC_IN_EN	Setting this bit will sync with the external frame sync input

MDSS_MDP_PP_0_SYNC_CONFIG_VSYNC (cont.)

Bits	Name	Description
19	VSYNC_COUNTER_EN	Setting this bit to enable the internal counter
18:0	VSYNC_COUNT	This scale value to be programmed is the ratio of the MDP VSYNC clock frequency to LCD panel frequency divided by the no. of rows (lines) in the LCD panel. For example, if the VSYNC clock is 133 MHz and the LCD is 176 lines (rows) and 220 pixels (columns) at 60 Hz, then the value to be programmed is:\n $133 * 10^6 / (60 * 176) = 12594$ That is, it takes 12594 MDP clocks to complete writing one line in the LCD panel.

0x01A71008 MDSS_MDP_PP_0_SYNC_CONFIG_HEIGHT**Type:** RW**Reset State:** 0x00000000

The SYNC_CONFIG_VSYNC register is a synchronization configuration register.

double buffered: NO

MDSS_MDP_PP_0_SYNC_CONFIG_HEIGHT

Bits	Name	Description
15:0	HEIGHT	Display height - 1. It is the vertical total number of lines

0x01A7100C MDSS_MDP_PP_0_SYNC_WRCOUNT**Type:** RW**Reset State:** 0x00000000

The MDP_SYNC_STATUS_P register is a primary synchronization status register. Writing to this register loads the line counter and is equivalent to providing a software Vsync.

double buffered: NO

MDSS_MDP_PP_0_SYNC_WRCOUNT

Bits	Name	Description
31:16	FRAME_COUNT	Writing to this register will load the counter according to the written value
15:0	LINE_COUNT	Writing to this register will load the counter according to the written value

0x01A71010 MDSS_MDP_PP_0_VSYNC_INIT_VAL**Type:** RW**Reset State:** 0x00000000

VSYNC_INIT_VAL defines the value with which the read pointer gets loaded at primary vsync edge.

double buffered: NO

MDSS_MDP_PP_0_VSYNC_INIT_VAL

Bits	Name	Description
15:0	INIT_VAL	Specify the init value to which the read pointer gets loaded at vsync edge

0x01A71014 MDSS_MDP_PP_0_INT_COUNT_VAL**Type:** R**Reset State:** 0x00000000

The INT_CNT_VAL register shows the internal line and frame counter for primary vsync.

double buffered: NO

MDSS_MDP_PP_0_INT_COUNT_VAL

Bits	Name	Description
31:16	FRAME_COUNT	Indication of the number of frames elapsed since enabling the Interface The counter increments once MDP has send the frame to the display peripheral The counter will reset upon enabling of the interface
15:0	LINE_COUNT	Indication of the current line count for the current frame This counter is applicable for both command and active panels. For active panels, the line count WILL include blanking information For command mode panels, there is no blanking information so it reflects the actual line count being displayed The counter increments at the end of each line The counter will reset at the start of the first line

0x01A71018 MDSS_MDP_PP_0_SYNC_THRESH**Type:** RW**Reset State:** 0x00000000

SYNC_THRESH register is a primary synchronization threshold register.

double buffered: NO

MDSS_MDP_PP_0_SYNC_THRESH

Bits	Name	Description
31:16	CONTINUE_THRESHOLD	The minimum number of lines the write pointer needs to be above the read pointer so that it is safe to write to the panel. (This check is not done for the first ROI line write of an update)
15:0	START_THRESHOLD	Allows the first ROI line write to an panel when read pointer is between the range of ROI start line and ROI start line plus this setting

0x01A7101C MDSS_MDP_PP_0_START_POS

Type: RW

Reset State: 0x00000000

START_POS defines the starting y position value for the vsync line counter at every frame

double buffered: NO

MDSS_MDP_PP_0_START_POS

Bits	Name	Description
15:0	POS	Specify the y position from which the start_threshold value is added and write is kicked off if the read pointer falls within that region

0x01A71020 MDSS_MDP_PP_0_RD_PTR_IRQ

Type: RW

Reset State: 0x00000000

RD_PTR_IRQ defines the read pointer of the display at which an interrupt has to be generated.

double buffered: NO

MDSS_MDP_PP_0_RD_PTR_IRQ

Bits	Name	Description
15:0	IRQ_LINE	Specify the read pointer value at which an interrupt has to be generated

0x01A71024 MDSS_MDP_PP_0_WR_PTR_IRQ**Type:** RW**Reset State:** 0x00000000

WR_PTR_IRQ defines the read pointer of the display at which an interrupt has to be generated.

double buffered: NO

MDSS_MDP_PP_0_WR_PTR_IRQ

Bits	Name	Description
15:0	IRQ_LINE	Specify the write pointer value at which an interrupt has to be generated

0x01A71028 MDSS_MDP_PP_0_OUT_LINE_COUNT**Type:** RW**Reset State:** 0x00000000

Output line counter as to when the update will occur

This is equivalent to the Y value of the panel update .

double buffered: NO

MDSS_MDP_PP_0_OUT_LINE_COUNT

Bits	Name	Description
15:0	OUT_Y	Starting Y value as to when the update will occur

0x01A7102C MDSS_MDP_PP_0_PP_LINE_COUNT**Type:** R**Reset State:** 0x00000000

Current write pointer line counter

double buffered: NO

MDSS_MDP_PP_0_PP_LINE_COUNT

Bits	Name	Description
15:0	COUNT	Contains the current line counter within the ping pong FIFO as used by tear check logic

0x01A71030 MDSS_MDP_PP_0_AUTOREFRESH_CONFIG**Type:** RW**Reset State:** 0x00000000**MDSS_MDP_PP_0_AUTOREFRESH_CONFIG**

Bits	Name	Description
31	ENABLE	
15:0	FRAME_COUNT	

17.17.18 MDSS_MDP_PP_1

Ping Pong hardware register description

17.17.18.1 MDSS_MDP_PP_1_PP

PP

Ping Pong hardware register description

0x01A71800 MDSS_MDP_PP_1_TEAR_CHECK_EN**Type:** RW**Reset State:** 0x00000000

Register is used to enable tearing check on different displays.

double buffered: NO

MDSS_MDP_PP_1_TEAR_CHECK_EN

Bits	Name	Description
0	EN	Enable tear check 0x0: disabled 0x1: enabled

0x01A71804 MDSS_MDP_PP_1_SYNC_CONFIG_VSYNC**Type:** RW**Reset State:** 0x00000000

The SYNC_CONFIG_VSYNC register is a synchronization configuration register.

double buffered: NO

MDSS_MDP_PP_1_SYNC_CONFIG_VSYNC

Bits	Name	Description
20	VSYNC_IN_EN	Setting this bit will sync with the external frame sync input
19	VSYNC_COUNTER_EN	Setting this bit to enable the internal counter
18:0	VSYNC_COUNT	This scale value to be programmed is the ratio of the MDP VSYNC clock frequency to LCD panel frequency divided by the no. of rows (lines) in the LCD panel. For example, if the VSYNC clock is 133 MHz and the LCD is 176 lines (rows) and 220 pixels (columns) at 60 Hz, then the value to be programmed is: $133 * 10^6 / (60 * 176) = 12594$ That is, it takes 12594 MDP clocks to complete writing one line in the LCD panel.

0x01A71808 MDSS_MDP_PP_1_SYNC_CONFIG_HEIGHT

Type: RW

Reset State: 0x00000000

The SYNC_CONFIG_VSYNC register is a synchronization configuration register.

double buffered: NO

MDSS_MDP_PP_1_SYNC_CONFIG_HEIGHT

Bits	Name	Description
15:0	HEIGHT	Display height - 1. It is the vertical total number of lines

0x01A7180C MDSS_MDP_PP_1_SYNC_WRCOUNT

Type: RW

Reset State: 0x00000000

The MDP_SYNC_STATUS_P register is a primary synchronization status register. Writing to this register loads the line counter and is equivalent to providing a software Vsync.

double buffered: NO

MDSS_MDP_PP_1_SYNC_WRCOUNT

Bits	Name	Description
31:16	FRAME_COUNT	Writing to this register will load the counter according to the written value

MDSS_MDP_PP_1_SYNC_WRCOUNT (cont.)

Bits	Name	Description
15:0	LINE_COUNT	Writing to this register will load the counter according to the written value

0x01A71810 MDSS_MDP_PP_1_VSYNC_INIT_VAL**Type:** RW**Reset State:** 0x00000000

VSYNC_INIT_VAL defines the value with which the read pointer gets loaded at primary vsync edge.

double buffered: NO

MDSS_MDP_PP_1_VSYNC_INIT_VAL

Bits	Name	Description
15:0	INIT_VAL	Specify the init value to which the read pointer gets loaded at vsync edge

0x01A71814 MDSS_MDP_PP_1_INT_COUNT_VAL**Type:** R**Reset State:** 0x00000000

The INT_CNT_VAL register shows the internal line and frame counter for primary vsync.

double buffered: NO

MDSS_MDP_PP_1_INT_COUNT_VAL

Bits	Name	Description
31:16	FRAME_COUNT	Indication of the number of frames elapsed since enabling the Interface The counter increments once MDP has send the frame to the display peripheral The counter will reset upon enabling of the interface
15:0	LINE_COUNT	Indication of the current line count for the current frame This counter is applicable for both command and active panels. For active panels, the line count WILL include blanking information For command mode panels, there is no blanking information so it reflects the actual line count being displayed The counter increments at the end of each line The counter will reset at the start of the first line

0x01A71818 MDSS_MDP_PP_1_SYNC_THRESH**Type:** RW**Reset State:** 0x00000000

SYNC_THRESH register is a primary synchronization threshold register.

double buffered: NO

MDSS_MDP_PP_1_SYNC_THRESH

Bits	Name	Description
31:16	CONTINUE_THRESHOLD	The minimum number of lines the write pointer needs to be above the read pointer so that it is safe to write to the panel. (This check is not done for the first ROI line write of an update)
15:0	START_THRESHOLD	Allows the first ROI line write to an panel when read pointer is between the range of ROI start line and ROI start line plus this setting

0x01A7181C MDSS_MDP_PP_1_START_POS**Type:** RW**Reset State:** 0x00000000

START_POS defines the starting y position value for the vsync line counter at every frame

double buffered: NO

MDSS_MDP_PP_1_START_POS

Bits	Name	Description
15:0	POS	Specify the y position from which the start_threshold value is added and write is kicked off if the read pointer falls within that region

0x01A71820 MDSS_MDP_PP_1_RD_PTR_IRQ**Type:** RW**Reset State:** 0x00000000

RD_PTR_IRQ defines the read pointer of the display at which an interrupt has to be generated.

double buffered: NO

MDSS_MDP_PP_1_RD_PTR_IRQ

Bits	Name	Description
15:0	IRQ_LINE	Specify the read pointer value at which an interrupt has to be generated

0x01A71824 MDSS_MDP_PP_1_WR_PTR_IRQ**Type:** RW**Reset State:** 0x00000000

WR_PTR_IRQ defines the read pointer of the display at which an interrupt has to be generated.

double buffered: NO

MDSS_MDP_PP_1_WR_PTR_IRQ

Bits	Name	Description
15:0	IRQ_LINE	Specify the write pointer value at which an interrupt has to be generated

0x01A71828 MDSS_MDP_PP_1_OUT_LINE_COUNT**Type:** RW**Reset State:** 0x00000000

Output line counter as to when the update will occur

This is equivalent to the Y value of the panel update .

double buffered: NO

MDSS_MDP_PP_1_OUT_LINE_COUNT

Bits	Name	Description
15:0	OUT_Y	Starting Y value as to when the update will occur

0x01A7182C MDSS_MDP_PP_1_PP_LINE_COUNT**Type:** R**Reset State:** 0x00000000

Current write pointer line counter

double buffered: NO

MDSS_MDP_PP_1_PP_LINE_COUNT

Bits	Name	Description
15:0	COUNT	Contains the current line counter within the ping pong FIFO as used by tear check logic

0x01A71830 MDSS_MDP_PP_1_AUTOREFRESH_CONFIG**Type:** RW**Reset State:** 0x00000000**MDSS_MDP_PP_1_AUTOREFRESH_CONFIG**

Bits	Name	Description
31	ENABLE	
15:0	FRAME_COUNT	

17.17.19 MDSS_MDP_PP_2

Ping Pong hardware register description

17.17.19.1 MDSS_MDP_PP_2_PP

PP

Ping Pong hardware register description

0x01A72000 MDSS_MDP_PP_2_TEAR_CHECK_EN**Type:** RW**Reset State:** 0x00000000

Register is used to enable tearing check on different displays.

double buffered: NO

MDSS_MDP_PP_2_TEAR_CHECK_EN

Bits	Name	Description
0	EN	Enable tear check 0x0: disabled 0x1: enabled

0x01A72004 MDSS_MDP_PP_2_SYNC_CONFIG_VSYNC**Type:** RW**Reset State:** 0x00000000

The SYNC_CONFIG_VSYNC register is a synchronization configuration register.

double buffered: NO

MDSS_MDP_PP_2_SYNC_CONFIG_VSYNC

Bits	Name	Description
20	VSYNC_IN_EN	Setting this bit will sync with the external frame sync input
19	VSYNC_COUNTER_EN	Setting this bit to enable the internal counter
18:0	VSYNC_COUNT	This scale value to be programmed is the ratio of the MDP VSYNC clock frequency to LCD panel frequency divided by the no. of rows (lines) in the LCD panel. For example, if the VSYNC clock is 133 MHz and the LCD is 176 lines (rows) and 220 pixels (columns) at 60 Hz, then the value to be programmed is: $\frac{133 \times 10^6}{(60 \times 176)} = 12594$ That is, it takes 12594 MDP clocks to complete writing one line in the LCD panel.

0x01A72008 MDSS_MDP_PP_2_SYNC_CONFIG_HEIGHT**Type:** RW**Reset State:** 0x00000000

The SYNC_CONFIG_VSYNC register is a synchronization configuration register.

double buffered: NO

MDSS_MDP_PP_2_SYNC_CONFIG_HEIGHT

Bits	Name	Description
15:0	HEIGHT	Display height - 1. It is the vertical total number of lines

0x01A7200C MDSS_MDP_PP_2_SYNC_WRCOUNT**Type:** RW**Reset State:** 0x00000000

The MDP_SYNC_STATUS_P register is a primary synchronization status register. Writing to this register loads the line counter and is equivalent to providing a software Vsync.

double buffered: NO

MDSS_MDP_PP_2_SYNC_WRCOUNT

Bits	Name	Description
31:16	FRAME_COUNT	Writing to this register will load the counter according to the written value
15:0	LINE_COUNT	Writing to this register will load the counter according to the written value

0x01A72010 MDSS_MDP_PP_2_VSYNC_INIT_VAL**Type:** RW**Reset State:** 0x00000000

VSYNC_INIT_VAL defines the value with which the read pointer gets loaded at primary vsync edge.

double buffered: NO

MDSS_MDP_PP_2_VSYNC_INIT_VAL

Bits	Name	Description
15:0	INIT_VAL	Specify the init value to which the read pointer gets loaded at vsync edge

0x01A72014 MDSS_MDP_PP_2_INT_COUNT_VAL**Type:** R**Reset State:** 0x00000000

The INT_CNT_VAL register shows the internal line and frame counter for primary vsync.

double buffered: NO

MDSS_MDP_PP_2_INT_COUNT_VAL

Bits	Name	Description
31:16	FRAME_COUNT	Indication of the number of frames elapsed since enabling the Interface The counter increments once MDP has send the frame to the display peripheral The counter will reset upon enabling of the interface

MDSS_MDP_PP_2_INT_COUNT_VAL (cont.)

Bits	Name	Description
15:0	LINE_COUNT	Indication of the current line count for the current frame This counter is applicable for both command and active panels. For active panels, the line count WILL include blanking information For command mode panels, there is no blanking information so it reflects the actual line count being displayed The counter increments at the end of each line The counter will reset at the start of the first line

0x01A72018 MDSS_MDP_PP_2_SYNC_THRESH**Type:** RW**Reset State:** 0x00000000

SYNC_THRESH register is a primary synchronization threshold register.

double buffered: NO

MDSS_MDP_PP_2_SYNC_THRESH

Bits	Name	Description
31:16	CONTINUE_THRESHOLD	The minimum number of lines the write pointer needs to be above the read pointer so that it is safe to write to the panel. (This check is not done for the first ROI line write of an update)
15:0	START_THRESHOLD	Allows the first ROI line write to an panel when read pointer is between the range of ROI start line and ROI start line plus this setting

0x01A7201C MDSS_MDP_PP_2_START_POS**Type:** RW**Reset State:** 0x00000000

START_POS defines the starting y position value for the vsync line counter at every frame

double buffered: NO

MDSS_MDP_PP_2_START_POS

Bits	Name	Description
15:0	POS	Specify the y position from which the start_threshold value is added and write is kicked off if the read pointer falls within that region

0x01A72020 MDSS_MDP_PP_2_RD_PTR_IRQ**Type:** RW**Reset State:** 0x00000000

RD_PTR_IRQ defines the read pointer of the display at which an interrupt has to be generated.

double buffered: NO

MDSS_MDP_PP_2_RD_PTR_IRQ

Bits	Name	Description
15:0	IRQ_LINE	Specify the read pointer value at which an interrupt has to be generated

0x01A72024 MDSS_MDP_PP_2_WR_PTR_IRQ**Type:** RW**Reset State:** 0x00000000

WR_PTR_IRQ defines the read pointer of the display at which an interrupt has to be generated.

double buffered: NO

MDSS_MDP_PP_2_WR_PTR_IRQ

Bits	Name	Description
15:0	IRQ_LINE	Specify the write pointer value at which an interrupt has to be generated

0x01A72028 MDSS_MDP_PP_2_OUT_LINE_COUNT**Type:** RW**Reset State:** 0x00000000

Output line counter as to when the update will occur

This is equivalent to the Y value of the panel update .

double buffered: NO

MDSS_MDP_PP_2_OUT_LINE_COUNT

Bits	Name	Description
15:0	OUT_Y	Starting Y value as to when the update will occur

0x01A7202C MDSS_MDP_PP_2_PP_LINE_COUNT**Type:** R**Reset State:** 0x00000000

Current write pointer line counter

double buffered: NO

MDSS_MDP_PP_2_PP_LINE_COUNT

Bits	Name	Description
15:0	COUNT	Contains the current line counter within the ping pong FIFO as used by tear check logic

0x01A72030 MDSS_MDP_PP_2_AUTOREFRESH_CONFIG**Type:** RW**Reset State:** 0x00000000**MDSS_MDP_PP_2_AUTOREFRESH_CONFIG**

Bits	Name	Description
31	ENABLE	
15:0	FRAME_COUNT	

17.17.20 MDSS_MDP_ENCR_NS_0_ENCR_NS**0x01A77004 MDSS_MDP_ENCR_NS_0_CRYPT0_SEG_SIZE****Type:** RW**Clock:** Core Clock**Reset State:** 0x00000000

This register specifies the total size of the segment. It must be set to the total number of bytes in line.

MDSS_MDP_ENCR_NS_0_CRYPT0_SEG_SIZE

Bits	Name	Description
31:0	SEG_SIZE	The number of bytes to process in total. Normally this register would be set to a value representing the end of encryption segment.

17.17.21 MDSS_MDP_ENCR_S_0_ENCR_S

0x01A77400 MDSS_MDP_ENCR_S_0_CRYPTO_CONFIG

Type: RW

Clock: Core Clock

Reset State: 0x00000000

This register enables/disables the crypto core operation. The register is double buffered and protected by the permissions logic.

MDSS_MDP_ENCR_S_0_CRYPTO_CONFIG

Bits	Name	Description
0	FRAME_ENCR_EN	Enables video frame encryption. The value is qualified at the start of each video frame.

0x01A77404+ MDSS_MDP_ENCR_S_0_CRYPTO_ENCR_KEYn, n=[0..3] 0x4*n

Type: W

Clock: Core Clock

Reset State: 0x00000000

These registers specify the keys for AES encryption algorithms. The registers are write only and can be accessed by Trustzonesoftware only.

MDSS_MDP_ENCR_S_0_CRYPTO_ENCR_KEYn

Bits	Name	Description
31:0	CRYPTO_ENCR_KEY	Specifies the key for AES encryption algorithms.

0x01A77414+ MDSS_MDP_ENCR_S_0_CRYPTO_ENCR_RIVn, n=[0..1] 0x4*n

Type: W

Clock: Core Clock

Reset State: 0x00000000

This register holds the Riv value used for Counter Initialization Vector computation according to HDCP2.1.

MDSS_MDP_ENCR_S_0_CRYPTO_ENCR_RIVn

Bits	Name	Description
31:0	CRYPTO_ENCR_RIV	Specifies the Riv value for AES encryption algorithms.

0x01A7741C MDSS_MDP_ENCR_S_0_CRYPTO_ENCR_STRCTR**Type:** W**Clock:** Core Clock**Reset State:** 0x00000000

This register holds the StreamCtr value used for Counter Initialization Vector computation according to HDCP2.1.

MDSS_MDP_ENCR_S_0_CRYPTO_ENCR_STRCTR

Bits	Name	Description
31:0	CRYPTO_ENCR_STRCTR	Specifies the StreamCtr value for AES encryption algorithms.

17.17.22 MDSS_MDP_AADC_0

Apical Assertive Dispay Core hardware register description

17.17.22.1 MDSS_MDP_AADC_0_AADC

AADC

Apical Assertive Dispay Core hardware register description

0x01A79000 MDSS_MDP_AADC_0_APICAL_CORE_BYPASS**Type:** RW**Reset State:** 0x00000101

Apical bypass and SW reset control

MDSS_MDP_AADC_0_APICAL_CORE_BYPASS

Bits	Name	Description
23:16	APICAL_CORE_RESERVED_23_16	RESERVED for future use.
8	APICAL_CORE_SW_RESET_EN	Enable SW reset control capability for Apical Iridix core - SHOULD always be set to 1'b1; (active high, default 0). This ensures that when ares and apical_core_sw_reset are ORed together, the in/out conversion blocks reset first to block propagation of any potential hazards out of the apical wrapper,, also clocks to the apical core are stopped preventing internal hazards.

MDSS_MDP_AADC_0_APICAL_CORE_BYPASS (cont.)

Bits	Name	Description
0	APICAL_CORE_BYPASS	0x0 : Pass pixel data through Apical core - Apical processing is enabled via APICAL_IRIDIX_CONTROL_0.IRIDIX_ENABLE 0x1 : Bypass Apical core entirely - this will also clock gate the core (default) Double-buffered: Yes

0x01A79004 MDSS_MDP_AADC_0_APICAL_CONTROL_REG0**Type:** RW**Reset State:** 0x00000003

The Data format port specifies the input data format so that the iridix core can process the different input data formats.

double buffered: Yes

MDSS_MDP_AADC_0_APICAL_CONTROL_REG0

Bits	Name	Description
7:2	CONTROL_REG0_RESERVED_7_2	Reserved - Should be zero, except for bit 3 which is hardcoded as 1b.
1:0	DATA_FORMAT	Data Format Select 0x0: YCbCr Color space 4:2:2 format, interleaved (Cb-Y-Cr-Y) 0x1: YCbCr Color space 4:2:2 format, parallel 0x2: YCbCr Color 4:4:4 format 0x3: RGB Color space

0x01A79008 MDSS_MDP_AADC_0_APICAL_CONTROL_REG1**Type:** RW**Reset State:** 0x00000000

Synchronization Format Register.

double buffered: Yes

MDSS_MDP_AADC_0_APICAL_CONTROL_REG1

Bits	Name	Description
7	AUTO_SIZE	Auto Size Register 0x0: Auto_Size off. The Iridix core will use Frame_Width, Frame_Height ports values 0x1: Auto_Size on. The Iridix core will calculate the frame geometry based on the active line and field signals. The size of current frame will be used for the estimation of the next frame. The Frame size will be updated when the Field signal toggles.
6:0	CONTROL_REG1_RESERVED_6_0	Reserved, should be set to 0x00.

0x01A7900C MDSS_MDP_AADC_0_APICAL_FRAME_SIZE**Type:** RW**Reset State:** 0x01000100

Frame Size Register.

double buffered: Yes

MDSS_MDP_AADC_0_APICAL_FRAME_SIZE

Bits	Name	Description
31:16	FRAME_WIDTH	Frame width in number of pixels in an active line. This port is used only in non-automatic mode(default)
15:0	FRAME_HEIGHT	Frame height in number of line in an active frame. This port is used only in non-automatic mode(default)

0x01A79010 MDSS_MDP_AADC_0_APICAL_IRIDIX_CONTROL_0**Type:** RW**Reset State:** 0x00000006

Iridix Control Register 0

double buffered: Yes

MDSS_MDP_AADC_0_APICAL_IRIDIX_CONTROL_0

Bits	Name	Description
7:5	IRIDIX_CONTROL_0_RESERVED_7_5	Reserved - Should be zero.

MDSS_MDP_AADC_0_APICAL_IRIDIX_CONTROL_0 (cont.)

Bits	Name	Description
4	IRIDIX_COLOR_CORRECTION	Color Correction/Control - Enabling this processing will result in more accurate colors processing. The color correction algorithm is required on gamma corrected sources. It reduces the saturation in dark areas when they are being amplified, and saturates bright parts when they get darker. 0x0: Disable color correction 0x1: Enable color correction
3	IRIDIX_LOCAL_CONTRAST_2D	Local contrast (2D) 0x0: Memory interfaces for 2D processing used 0x1: Memory interfaces for 2D processing bypassed, 2D processing disabled. This selection overrides LC0, LC1 settings. Note: If 2D Local contrast is not implemented then the bit[3] should be one.
2:1	IRIDIX_LOCAL_CONTRAST	Local Contrast Control - This is a supplementary module preserving local contrast in small areas. Depending on custom implementation LC module can be implemented differently and may have different settings. 1D Case : 0x0: disabled; 0x1: LC enabled, kernel 1x3 used 0x2: LC enabled, kernel 1x3 used 0x3: LC enabled, kernel 1x5 used 2D Case : 0x0: disabled; 0x1: LC enabled, kernel 1x3 used 0x2: LC enabled, kernel 3x3 used 0x3: LC enabled, kernel 3x5 used
0	IRIDIX_ENABLE	iridix Enable - This bit turns iridix processing ON and OFF. When iridix is OFF, the video data passes to the output without any changes. Disabling iridix® using this bit is equivalent to setting the Strength parameter to 0. Many internal modules run in idle mode when this bit is 0. The Latency Delays between the input and the output signals are the same in both ON and OFF modes. 0x0: Disable iridix processing 0x1: Enable iridix processing

0x01A79014 MDSS_MDP_AADC_0_APICAL_IRIDIX_CONTROL_1**Type:** RW**Reset State:** 0x00004600

This Port is reserved for debugging purposes by Apical. Do not modify.

MDSS_MDP_AADC_0_APICAL_IRIDIX_CONTROL_1

Bits	Name	Description
15:8	IRIDIX_CONTROL_1	Reserved for debugging purposes by Apical

0x01A79018 MDSS_MDP_AADC_0_APICAL_STRENGTH_MANUAL**Type:** RW**Reset State:** 0x00000000

Strength of iridix -Manual override value of iridix module's strength input (strength_ir) when option_select = 7 (dec).

MDSS_MDP_AADC_0_APICAL_STRENGTH_MANUAL

Bits	Name	Description
7:0	STRENGTH_MANUAL	Strength of iridix - This Port sets processing Strength. Minimum value is 0, maximum is 255. 0x00: Video data will not be processed at all, and will go to the output unchanged. 0xFF: Maximum strengthSTRENGTH_MANUAL

0x01A7901C MDSS_MDP_AADC_0_APICAL_VARIANCE**Type:** RW**Reset State:** 0x00000041

Affects the sensitivity of the transform to different areas of the image, and can be increased in order to emphasize small regions (e.g. faces).

If this parameter is set to zero, the sensitivity to the small areas is maximal. The transform becomes more local.

When the parameter is set to 0xFF the transform deemphasizes small local details, and transform becomes more global

e.g. the sky at the top of the frame can affect flowers at the bottom of the frame.

Default value is 36h but it depends on custom implementation.

Setting variance to zero will progressively increase the spatial sensitivity of the algorithm but it can lead to artificial-looking images in some cases.

double buffered: Yes

MDSS_MDP_AADC_0_APICAL_VARIANCE

Bits	Name	Description
7:4	VARIANCE_INTENSITY	Sets the degree of sensitivity in the luminance domain. Maximum Variance is 0xF, and minimum Variance is 0x0.
3:0	VARIANCE_SPACE	Sets the degree of spatial sensitivity of the algorithm. As this parameter is made smaller, the algorithm focuses on smaller regions within the image. Maximum Variance is 0xF, and minimum Variance is 0x0

0x01A79020 MDSS_MDP_AADC_0_APICAL_IRIDIX_DITHER**Type:** RW**Reset State:** 0x00000000

Iridix Dither Register: When the number of color gradations of a display device is small (for example 6 bit per color) pixel dithering can make gradients look smother.

There can be a situation when iridix output signal is 12 bits but the signal needs to be compressed into 8 bits.

The best way for this is to use dithering of least significant bits and then truncate these bits. The dithering can be performed by iridix Processor.

However, the truncation needs to be done by a hardware designer outside iridix.

double buffered: Yes

MDSS_MDP_AADC_0_APICAL_IRIDIX_DITHER

Bits	Name	Description
7:3	IRIDIX_DITHER_RESERVE_D_7_3	Reserved - Not used for now.
2:0	DITHER	Strength of Dithering. 0x0: No dithering 0x1: One least significant bit of the output signal is dithered 0x2: Two bits are dithered 0x3: Three bits are dithered 0x4: Four bits are dithered Others: Four bits are dithered

0x01A79024 MDSS_MDP_AADC_0_APICAL_DITEHR_CONTROL**Type:** RW**Reset State:** 0x00000000

Spatial Preceptual Model Dither

This module provides a 4-bit dither via a spatial perceptual model.

There is no temporal dependence of the dither pattern, therefore eliminating the possibility of flicker

note that temporal dither is available via the port iridix_dither.

double buffered: Yes

MDSS_MDP_AADC_0_APICAL_DITEHR_CONTROL

Bits	Name	Description
7:4	DITHER_CONTROL_RESE RVED_7_4	Reserved - Should be Zero.
3:2	DITHER_AMOUNT	Strength of Dithering. 0x0: Dither 1 bit 0x1: Dither 2 bit 0x2: Dither 3 bit 0x3: Dither 4 bit
1	DITHER_SHIFT_MODE	Shift Mode: 0x0: Output is MSB aligned 0x1: Output is LSB aligned
0	DITHER_ENABLE_DITHER	Enable Dither: Enables the dither module. 0: Disable dither module 1: Enable dither module

0x01A79028 MDSS_MDP_AADC_0_APICAL_LIMIT_AMPL

Type: RW

Reset State: 0x000000F0

Two parameters used to restrict the luminance space in which Iridix can adaptively generate tone curves for each pixel

double buffered: Yes

MDSS_MDP_AADC_0_APICAL_LIMIT_AMPL

Bits	Name	Description
7:4	BRIGHT_AMP_LIMIT	Bright amplification limit - The resultant tone curve cannot be lower than bright amplification limit line controlled by the bright amplification limit parameter. See Apical TRM Chapter 4 for more explanation. Maximum limit is 0xF, when the value is 0x0 there is no limit.

MDSS_MDP_AADC_0_APICAL_LIMIT_AMPL (cont.)

Bits	Name	Description
3:0	DARK_AMP_LIMIT	Dark amplification limit - The resultant tone curve cannot be higher than dark amplification limit line controlled by the dark amplification limit parameter. See Apical TRM Chapter 4 for more explanation. Maximum limit is 0xF, when the value is 0x0 there is no limit.

0x01A7902C MDSS_MDP_AADC_0_APICAL_SLOPE**Type:** RW**Reset State:** 0x00003C80

Slope Min and Max Limit used to restrict the slope of the tone-curve generated by Iridix.

double buffered: Yes

MDSS_MDP_AADC_0_APICAL_SLOPE

Bits	Name	Description
15:8	SLOPE_MAX	Slope Max Limit - Slope Max Limit is used to restrict the slope of the tone-curve generated by iridix. When Slope Max Limit parameter is set to 0xFF, the tone curve slope, generated by iridix is not limited (maximum slope 15). When this value is set to 00 iridix will not generate the tone-curve with slope greater than 16/15 (48°). Intermediate values of Slope Limit are linear interpolation between minimum and maximum values. Recommended Slope Max Limit range is 20h-60h. For noisy images less value should be used. See Chapter 4 for more explanation.
7:0	SLOPE_MIN	Slope Min Limit - Slope Min Limit is used to restrict the slope of the tone-curve generated by iridix. When Slope Min Limit parameter is set to 0x00, the tone curve slope, generated by iridix is not limited. When this value is set to FF iridix will not generate the tone-curve with slope less than 3/4. Intermediate values of Slope Limit are linear interpolation between minimum and maximum values. Recommended Slope Min Limit range is 00h-C0h. See Chapter 4 for more explanation.

0x01A79030 MDSS_MDP_AADC_0_APICAL_BW_LEVEL**Type:** RW**Reset State:** 0x03FF0000

Black and White Level of the Iridix IP.

The least significant bits [dw-1:0] are used; higher bits [15:dw] are unused. dw is the component data width. dw=8 for MDSS \n double buffered: Yes

MDSS_MDP_AADC_0_APICAL_BW_LEVEL

Bits	Name	Description
25:16	WHITE_LEVEL	White Level - The value stored in White Level Port will be used as white level for iridix processing in all unsigned data channels. Data above White level will not be processed and stay unchanged. Note: 1. For 12 bit variant minimum White Level is constrained as: White Level - Black Level > 1024. 2. For 10 bit variant minimum White Level is constrained as: White Level - Black Level > 256.
9:0	BLACK_LEVEL	Black Level - The value stored in Black Level Port will be used as zero level for iridix processing in all unsigned data channels. Data below Black level will not be processed and stay unchanged. Note: 1. For 12 bit variant maximum Black Level is constrained as: White Level - Black Level > 1024. 2. For 10 bit variant maximum Black Level is constrained as: White Level - Black Level > 256

0x01A79034 MDSS_MDP_AADC_0_APICAL_LOGO_POS

Type: RW

Reset State: 0x00000000

Apical Logo Position

double buffered: Yes

MDSS_MDP_AADC_0_APICAL_LOGO_POS

Bits	Name	Description
15:8	LOGO_LEFT	horizontal position of logo.
7:0	LOGO_TOP	vertical position of logo.

0x01A79038 MDSS_MDP_AADC_0_APICAL_LUT_FI1_0

Type: RW

Reset State: 0x00C70000

Asymmetry Function Lookup Table Entry 1-0

The Asymmetry Function Lookup Table port geometry is 33 words, each of which is 16 bits.

The Asymmetry function is used to balance the iridix effect between the dark and bright regions of the image.

double buffered: Yes

MDSS_MDP_AADC_0_APICAL_LUT_FI1_0

Bits	Name	Description
31:16	LUT_FI1	Asymmetry Function Lookup Table Entry 1.
15:0	LUT_FI0	Asymmetry Function Lookup Table Entry 0.

0x01A7903C MDSS_MDP_AADC_0_APICAL_LUT_FI3_2

Type: RW

Reset State: 0x023D0186

Asymmetry Function Lookup Table Entry 3-2

The Asymmetry Function Lookup Table port geometry is 33 words, each of which is 16 bits.

The Asymmetry function is used to balance the iridix effect between the dark and bright regions of the image.

double buffered: Yes

MDSS_MDP_AADC_0_APICAL_LUT_FI3_2

Bits	Name	Description
31:16	LUT_FI3	Asymmetry Function Lookup Table Entry 3.
15:0	LUT_FI2	Asymmetry Function Lookup Table Entry 2.

0x01A79040 MDSS_MDP_AADC_0_APICAL_LUT_FI5_4

Type: RW

Reset State: 0x039602ED

Asymmetry Function Lookup Table Entry 5-4

The Asymmetry Function Lookup Table port geometry is 33 words, each of which is 16 bits.

The Asymmetry function is used to balance the iridix effect between the dark and bright regions of the image.

double buffered: Yes

MDSS_MDP_AADC_0_APICAL_LUT_FI5_4

Bits	Name	Description
31:16	LUT_FI5	Asymmetry Function Lookup Table Entry 5.
15:0	LUT_FI4	Asymmetry Function Lookup Table Entry 4.

0x01A79044 MDSS_MDP_AADC_0_APICAL_LUT_FI7_6**Type:** RW**Reset State:** 0x04D60439

Asymmetry Function Lookup Table Entry 7-6

The Asymmetry Function Lookup Table port geometry is 33 words, each of which is 16 bits.

The Asymmetry function is used to balance the iridix effect between the dark and bright regions of the image.

double buffered: Yes

MDSS_MDP_AADC_0_APICAL_LUT_FI7_6

Bits	Name	Description
31:16	LUT_FI7	Asymmetry Function Lookup Table Entry 7.
15:0	LUT_FI6	Asymmetry Function Lookup Table Entry 6.

0x01A79048 MDSS_MDP_AADC_0_APICAL_LUT_FI9_8**Type:** RW**Reset State:** 0x05FF056D

Asymmetry Function Lookup Table Entry 9-8

The Asymmetry Function Lookup Table port geometry is 33 words, each of which is 16 bits.

The Asymmetry function is used to balance the iridix effect between the dark and bright regions of the image.

double buffered: Yes

MDSS_MDP_AADC_0_APICAL_LUT_FI9_8

Bits	Name	Description
31:16	LUT_FI9	Asymmetry Function Lookup Table Entry 9.
15:0	LUT_FI8	Asymmetry Function Lookup Table Entry 8.

0x01A7904C MDSS_MDP_AADC_0_APICAL_LUT_FI11_10**Type:** RW**Reset State:** 0x0714068C

Asymmetry Function Lookup Table Entry 11-10

The Asymmetry Function Lookup Table port geometry is 33 words, each of which is 16 bits.

The Asymmetry function is used to balance the iridix effect between the dark and bright regions of the image.

double buffered: Yes

MDSS_MDP_AADC_0_APICAL_LUT_FI11_10

Bits	Name	Description
31:16	LUT_FI11	Asymmetry Function Lookup Table Entry 11.
15:0	LUT_FI10	Asymmetry Function Lookup Table Entry 10.

0x01A79050 MDSS_MDP_AADC_0_APICAL_LUT_FI13_12**Type:** RW**Reset State:** 0x08190799

Asymmetry Function Lookup Table Entry 13-12

The Asymmetry Function Lookup Table port geometry is 33 words, each of which is 16 bits.

The Asymmetry function is used to balance the iridix effect between the dark and bright regions of the image.

double buffered: Yes

MDSS_MDP_AADC_0_APICAL_LUT_FI13_12

Bits	Name	Description
31:16	LUT_FI13	Asymmetry Function Lookup Table Entry 13.
15:0	LUT_FI12	Asymmetry Function Lookup Table Entry 12.

0x01A79054 MDSS_MDP_AADC_0_APICAL_LUT_FI15_14**Type:** RW**Reset State:** 0x090E0895

Asymmetry Function Lookup Table Entry 15-14

The Asymmetry Function Lookup Table port geometry is 33 words, each of which is 16 bits.

The Asymmetry function is used to balance the iridix effect between the dark and bright regions of the image.

double buffered: Yes

MDSS_MDP_AADC_0_APICAL_LUT_FI15_14

Bits	Name	Description
31:16	LUT_FI15	Asymmetry Function Lookup Table Entry 15.
15:0	LUT_FI14	Asymmetry Function Lookup Table Entry 14.

0x01A79058 MDSS_MDP_AADC_0_APICAL_LUT_FI17_16

Type: RW

Reset State: 0x09F80985

Asymmetry Function Lookup Table Entry 17-16

The Asymmetry Function Lookup Table port geometry is 33 words, each of which is 16 bits.

The Asymmetry function is used to balance the iridix effect between the dark and bright regions of the image.

double buffered: Yes

MDSS_MDP_AADC_0_APICAL_LUT_FI17_16

Bits	Name	Description
31:16	LUT_FI17	Asymmetry Function Lookup Table Entry 17.
15:0	LUT_FI16	Asymmetry Function Lookup Table Entry 16.

0x01A7905C MDSS_MDP_AADC_0_APICAL_LUT_FI19_18

Type: RW

Reset State: 0x0AD70A68

Asymmetry Function Lookup Table Entry 19-18

The Asymmetry Function Lookup Table port geometry is 33 words, each of which is 16 bits.

The Asymmetry function is used to balance the iridix effect between the dark and bright regions of the image.

double buffered: Yes

MDSS_MDP_AADC_0_APICAL_LUT_FI19_18

Bits	Name	Description
31:16	LUT_FI19	Asymmetry Function Lookup Table Entry 19.
15:0	LUT_FI18	Asymmetry Function Lookup Table Entry 18.

0x01A79060 MDSS_MDP_AADC_0_APICAL_LUT_FI21_20**Type:** RW**Reset State:** 0x0BAD0B43

Asymmetry Function Lookup Table Entry 21-20

The Asymmetry Function Lookup Table port geometry is 33 words, each of which is 16 bits.

The Asymmetry function is used to balance the iridix effect between the dark and bright regions of the image.

double buffered: Yes

MDSS_MDP_AADC_0_APICAL_LUT_FI21_20

Bits	Name	Description
31:16	LUT_FI21	Asymmetry Function Lookup Table Entry 21.
15:0	LUT_FI20	Asymmetry Function Lookup Table Entry 20.

0x01A79064 MDSS_MDP_AADC_0_APICAL_LUT_FI23_22**Type:** RW**Reset State:** 0x0C7D0C15

Asymmetry Function Lookup Table Entry 23-22

The Asymmetry Function Lookup Table port geometry is 33 words, each of which is 16 bits.

The Asymmetry function is used to balance the iridix effect between the dark and bright regions of the image.

double buffered: Yes

MDSS_MDP_AADC_0_APICAL_LUT_FI23_22

Bits	Name	Description
31:16	LUT_FI23	Asymmetry Function Lookup Table Entry 23.
15:0	LUT_FI22	Asymmetry Function Lookup Table Entry 22.

0x01A79068 MDSS_MDP_AADC_0_APICAL_LUT_FI25_24**Type:** RW**Reset State:** 0x0D480CE3

Asymmetry Function Lookup Table Entry 25-24

The Asymmetry Function Lookup Table port geometry is 33 words, each of which is 16 bits.

The Asymmetry function is used to balance the iridix effect between the dark and bright regions of the image.

double buffered: Yes

MDSS_MDP_AADC_0_APICAL_LUT_FI25_24

Bits	Name	Description
31:16	LUT_FI25	Asymmetry Function Lookup Table Entry 25.
15:0	LUT_FI24	Asymmetry Function Lookup Table Entry 24.

0x01A7906C MDSS_MDP_AADC_0_APICAL_LUT_FI27_26**Type:** RW**Reset State:** 0x0E0F0DAC

Asymmetry Function Lookup Table Entry 27-26

The Asymmetry Function Lookup Table port geometry is 33 words, each of which is 16 bits.

The Asymmetry function is used to balance the iridix effect between the dark and bright regions of the image.

double buffered: Yes

MDSS_MDP_AADC_0_APICAL_LUT_FI27_26

Bits	Name	Description
31:16	LUT_FI27	Asymmetry Function Lookup Table Entry 27.
15:0	LUT_FI26	Asymmetry Function Lookup Table Entry 26.

0x01A79070 MDSS_MDP_AADC_0_APICAL_LUT_FI29_28**Type:** RW**Reset State:** 0x0ED50E72

Asymmetry Function Lookup Table Entry 29-28

The Asymmetry Function Lookup Table port geometry is 33 words, each of which is 16 bits.

The Asymmetry function is used to balance the iridix effect between the dark and bright regions of the image.

double buffered: Yes

MDSS_MDP_AADC_0_APICAL_LUT_FI29_28

Bits	Name	Description
31:16	LUT_FI29	Asymmetry Function Lookup Table Entry 29.
15:0	LUT_FI28	Asymmetry Function Lookup Table Entry 28.

0x01A79074 MDSS_MDP_AADC_0_APICAL_LUT_FI31_30

Type: RW

Reset State: 0x0F9B0F38

Asymmetry Function Lookup Table Entry 31-30

The Asymmetry Function Lookup Table port geometry is 33 words, each of which is 16 bits.

The Asymmetry function is used to balance the iridix effect between the dark and bright regions of the image.

double buffered: Yes

MDSS_MDP_AADC_0_APICAL_LUT_FI31_30

Bits	Name	Description
31:16	LUT_FI31	Asymmetry Function Lookup Table Entry 31.
15:0	LUT_FI30	Asymmetry Function Lookup Table Entry 30.

0x01A79078 MDSS_MDP_AADC_0_APICAL_LUT_FI32

Type: RW

Reset State: 0x0FFF0000

Asymmetry Function Lookup Table Entry 32

The Asymmetry Function Lookup Table port geometry is 33 words, each of which is 16 bits.

The Asymmetry function is used to balance the iridix effect between the dark and bright regions of the image.

double buffered: Yes

MDSS_MDP_AADC_0_APICAL_LUT_FI32

Bits	Name	Description
31:16	LUT_FI32	Asymmetry Function Lookup Table Entry 32.

0x01A7907C MDSS_MDP_AADC_0_APICAL_LUT_CC1_0**Type:** RW**Reset State:** 0x011600FF

Color Correction Lookup Table Entry 1-0

The Color Correction Lookup Table port geometry is 33 words, each of which is 16 bits.

The Asymmetry function is used to balance the iridix effect between the dark and bright regions of the image.

double buffered: Yes

MDSS_MDP_AADC_0_APICAL_LUT_CC1_0

Bits	Name	Description
27:16	LUT_CC1	Color Correction Lookup Table Entry 1.
11:0	LUT_CC0	Color Correction Lookup Table Entry 0.

0x01A79080 MDSS_MDP_AADC_0_APICAL_LUT_CC3_2**Type:** RW**Reset State:** 0x0146012E

Color Correction Lookup Table Entry 3-2

The Color Correction Lookup Table port geometry is 33 words, each of which is 16 bits.

The Asymmetry function is used to balance the iridix effect between the dark and bright regions of the image.

double buffered: Yes

MDSS_MDP_AADC_0_APICAL_LUT_CC3_2

Bits	Name	Description
27:16	LUT_CC3	Color Correction Lookup Table Entry 3.
11:0	LUT_CC2	Color Correction Lookup Table Entry 2.

0x01A79084 MDSS_MDP_AADC_0_APICAL_LUT_CC5_4**Type:** RW**Reset State:** 0x0176015E

Color Correction Lookup Table Entry 5-4

The Color Correction Lookup Table port geometry is 33 words, each of which is 16 bits.

The Asymmetry function is used to balance the iridix effect between the dark and bright regions of the image.

double buffered: Yes

MDSS_MDP_AADC_0_APICAL_LUT_CC5_4

Bits	Name	Description
27:16	LUT_CC5	Color Correction Lookup Table Entry 5.
11:0	LUT_CC4	Color Correction Lookup Table Entry 4.

0x01A79088 MDSS_MDP_AADC_0_APICAL_LUT_CC7_6**Type:** RW**Reset State:** 0x01A6018E

Color Correction Lookup Table Entry 7-6

The Color Correction Lookup Table port geometry is 33 words, each of which is 16 bits.

The Asymmetry function is used to balance the iridix effect between the dark and bright regions of the image.

double buffered: Yes

MDSS_MDP_AADC_0_APICAL_LUT_CC7_6

Bits	Name	Description
27:16	LUT_CC7	Color Correction Lookup Table Entry 7.
11:0	LUT_CC6	Color Correction Lookup Table Entry 6.

0x01A7908C MDSS_MDP_AADC_0_APICAL_LUT_CC9_8**Type:** RW**Reset State:** 0x01D601BE

Color Correction Lookup Table Entry 9-8

The Color Correction Lookup Table port geometry is 33 words, each of which is 16 bits.

The Asymmetry function is used to balance the iridix effect between the dark and bright regions of the image.

double buffered: Yes

MDSS_MDP_AADC_0_APICAL_LUT_CC9_8

Bits	Name	Description
27:16	LUT_CC9	Color Correction Lookup Table Entry 9.
11:0	LUT_CC8	Color Correction Lookup Table Entry 8.

0x01A79090 MDSS_MDP_AADC_0_APICAL_LUT_CC11_10

Type: RW

Reset State: 0x020501EE

Color Correction Lookup Table Entry 11-10

The Color Correction Lookup Table port geometry is 33 words, each of which is 16 bits.

The Asymmetry function is used to balance the iridix effect between the dark and bright regions of the image.

double buffered: Yes

MDSS_MDP_AADC_0_APICAL_LUT_CC11_10

Bits	Name	Description
27:16	LUT_CC11	Color Correction Lookup Table Entry 11.
11:0	LUT_CC10	Color Correction Lookup Table Entry 10.

0x01A79094 MDSS_MDP_AADC_0_APICAL_LUT_CC13_12

Type: RW

Reset State: 0x0235021D

Color Correction Lookup Table Entry 13-12

The Color Correction Lookup Table port geometry is 33 words, each of which is 16 bits.

The Asymmetry function is used to balance the iridix effect between the dark and bright regions of the image.

double buffered: Yes

MDSS_MDP_AADC_0_APICAL_LUT_CC13_12

Bits	Name	Description
27:16	LUT_CC13	Color Correction Lookup Table Entry 13.
11:0	LUT_CC12	Color Correction Lookup Table Entry 12.

0x01A79098 MDSS_MDP_AADC_0_APICAL_LUT_CC15_14**Type:** RW**Reset State:** 0x0265024D

Color Correction Lookup Table Entry 15-14

The Color Correction Lookup Table port geometry is 33 words, each of which is 16 bits.

The Asymmetry function is used to balance the iridix effect between the dark and bright regions of the image.

double buffered: Yes

MDSS_MDP_AADC_0_APICAL_LUT_CC15_14

Bits	Name	Description
27:16	LUT_CC15	Color Correction Lookup Table Entry 15.
11:0	LUT_CC14	Color Correction Lookup Table Entry 14.

0x01A7909C MDSS_MDP_AADC_0_APICAL_LUT_CC17_16**Type:** RW**Reset State:** 0x0295027D

Color Correction Lookup Table Entry 17-16

The Color Correction Lookup Table port geometry is 33 words, each of which is 16 bits.

The Asymmetry function is used to balance the iridix effect between the dark and bright regions of the image.

double buffered: Yes

MDSS_MDP_AADC_0_APICAL_LUT_CC17_16

Bits	Name	Description
27:16	LUT_CC17	Color Correction Lookup Table Entry 17.
11:0	LUT_CC16	Color Correction Lookup Table Entry 16.

0x01A790A0 MDSS_MDP_AADC_0_APICAL_LUT_CC19_18**Type:** RW**Reset State:** 0x02C402AC

Color Correction Lookup Table Entry 19-18

The Color Correction Lookup Table port geometry is 33 words, each of which is 16 bits.

The Asymmetry function is used to balance the iridix effect between the dark and bright regions of the image.

double buffered: Yes

MDSS_MDP_AADC_0_APICAL_LUT_CC19_18

Bits	Name	Description
27:16	LUT_CC19	Color Correction Lookup Table Entry 19.
11:0	LUT_CC18	Color Correction Lookup Table Entry 18.

0x01A790A4 MDSS_MDP_AADC_0_APICAL_LUT_CC21_20**Type:** RW**Reset State:** 0x02F302DC

Color Correction Lookup Table Entry 21-20

The Color Correction Lookup Table port geometry is 33 words, each of which is 16 bits.

The Asymmetry function is used to balance the iridix effect between the dark and bright regions of the image.

double buffered: Yes

MDSS_MDP_AADC_0_APICAL_LUT_CC21_20

Bits	Name	Description
27:16	LUT_CC21	Color Correction Lookup Table Entry 21.
11:0	LUT_CC20	Color Correction Lookup Table Entry 20.

0x01A790A8 MDSS_MDP_AADC_0_APICAL_LUT_CC23_22**Type:** RW**Reset State:** 0x0323030B

Color Correction Lookup Table Entry 23-22

The Color Correction Lookup Table port geometry is 33 words, each of which is 16 bits.

The Asymmetry function is used to balance the iridix effect between the dark and bright regions of the image.

double buffered: Yes

MDSS_MDP_AADC_0_APICAL_LUT_CC23_22

Bits	Name	Description
27:16	LUT_CC23	Color Correction Lookup Table Entry 23.
11:0	LUT_CC22	Color Correction Lookup Table Entry 22.

0x01A790AC MDSS_MDP_AADC_0_APICAL_LUT_CC25_24

Type: RW

Reset State: 0x0352033A

Color Correction Lookup Table Entry 25-24

The Color Correction Lookup Table port geometry is 33 words, each of which is 16 bits.

The Asymmetry function is used to balance the iridix effect between the dark and bright regions of the image.

double buffered: Yes

MDSS_MDP_AADC_0_APICAL_LUT_CC25_24

Bits	Name	Description
27:16	LUT_CC25	Color Correction Lookup Table Entry 25.
11:0	LUT_CC24	Color Correction Lookup Table Entry 24.

0x01A790B0 MDSS_MDP_AADC_0_APICAL_LUT_CC27_26

Type: RW

Reset State: 0x0381036A

Color Correction Lookup Table Entry 27-26

The Color Correction Lookup Table port geometry is 33 words, each of which is 16 bits.

The Asymmetry function is used to balance the iridix effect between the dark and bright regions of the image.

double buffered: Yes

MDSS_MDP_AADC_0_APICAL_LUT_CC27_26

Bits	Name	Description
27:16	LUT_CC27	Color Correction Lookup Table Entry 27.
11:0	LUT_CC26	Color Correction Lookup Table Entry 26.

0x01A790B4 MDSS_MDP_AADC_0_APICAL_LUT_CC29_28**Type:** RW**Reset State:** 0x03B10399

Color Correction Lookup Table Entry 29-28

The Color Correction Lookup Table port geometry is 33 words, each of which is 16 bits.

The Asymmetry function is used to balance the iridix effect between the dark and bright regions of the image.

double buffered: Yes

MDSS_MDP_AADC_0_APICAL_LUT_CC29_28

Bits	Name	Description
27:16	LUT_CC29	Color Correction Lookup Table Entry 29.
11:0	LUT_CC28	Color Correction Lookup Table Entry 28.

0x01A790B8 MDSS_MDP_AADC_0_APICAL_LUT_CC31_30**Type:** RW**Reset State:** 0x03E003C8

Color Correction Lookup Table Entry 31-30

The Color Correction Lookup Table port geometry is 33 words, each of which is 16 bits.

The Asymmetry function is used to balance the iridix effect between the dark and bright regions of the image.

double buffered: Yes

MDSS_MDP_AADC_0_APICAL_LUT_CC31_30

Bits	Name	Description
27:16	LUT_CC31	Color Correction Lookup Table Entry 31.
11:0	LUT_CC30	Color Correction Lookup Table Entry 30.

0x01A790BC MDSS_MDP_AADC_0_APICAL_LUT_CC32**Type:** RW**Reset State:** 0x03F80000

Color Correction Lookup Table Entry 32

The Color Correction Lookup Table port geometry is 33 words, each of which is 16 bits.

The Asymmetry function is used to balance the iridix effect between the dark and bright regions of the image.

double buffered: Yes

MDSS_MDP_AADC_0_APICAL_LUT_CC32

Bits	Name	Description
27:16	LUT_CC32	Color Correction Lookup Table Entry 32.

0x01A790C8 MDSS_MDP_AADC_0_APICAL_STRENGTH_LIMIT**Type:** RW**Reset State:** 0x00000080

Imposes a cap on maximum dynamic range compression strength.

double buffered: No

MDSS_MDP_AADC_0_APICAL_STRENGTH_LIMIT

Bits	Name	Description
7:0	STRENGTH_LIMIT	Used to restrict the maximum value taken by strength_out: in all cases, strength_out <= strength_limit.

0x01A790CC MDSS_MDP_AADC_0_APICAL_CALIBRATION_AB**Type:** RW**Reset State:** 0x005F006E

Screen calibration parameter

four calibration parameters control the overall adaptation of displayed content to ambient light and backlight level.

They represent a combination of physical and perceptual factors which influence the apparent contrast of imagery as the user's eyes adapt to changing viewing conditions.

double buffered: No

MDSS_MDP_AADC_0_APICAL_CALIBRATION_AB

Bits	Name	Description
31:16	CALIBRATION_B	This parameter controls the adaptation of content to the screen brightness. As backlight is reduced in the presence of ambient light, shadow detail is progressively lost below the minimum grey level of the display. When calibrated correctly, this parameter ensures that image content remains viewable at all backlight levels.
15:0	CALIBRATION_A	This parameter is related to the reflectivity of the screen. The more reflective the screen, the greater is the loss of shadow detail as ambient light level increases. It is the objective of the Assertive Display core to maintain the same level of shadow detail which pertains in very low light, up to ambient light levels typical of outdoor viewing. When calibrated correctly, this parameter ensures that image content is neither lost nor over-enhanced leading to preservation of the viewing experience.

0x01A790D0 MDSS_MDP_AADC_0_APICAL_CALIBRATION_CD**Type:** RW**Reset State:** 0x00000013

Screen calibration parameter

four calibration parameters control the overall adaptation of displayed content to ambient light and backlight level.

They represent a combination of physical and perceptual factors which influence the apparent contrast of imagery as the user's eyes adapt to changing viewing conditions.

double buffered: No

MDSS_MDP_AADC_0_APICAL_CALIBRATION_CD

Bits	Name	Description
31:16	CALIBRATION_D	Typical movie, image and gaming content is 8-bit and encoded with a gamma curve which results in a dynamic range of around 1000:1. Most large displays have an effective contrast ratio larger than this under ideal viewing conditions (full backlight, low ambient light). However, many portable displays cannot reproduce such contrast ratios even under ideal conditions. When calibrated correctly, this parameter increases the effective contrast ratio of low contrast displays to match that of typical content.

MDSS_MDP_AADC_0_APICAL_CALIBRATION_CD (cont.)

Bits	Name	Description
15:0	CALIBRATION_C	This parameter controls the adaptation of content to overcome the limitations of the human eye sensitivity at very low light. Decreasing screen brightness in total darkness causes loss of image detail in shadow regions, especially on small displays, as the eyes are unable to adapt sufficiently. When calibrated correctly, this parameter enables restoration of such details.

0x01A790D4 MDSS_MDP_AADC_0_APICAL_OPTION_SELECT**Type:** RW**Reset State:** 0x00000000

Selects the overall operating mode of the core.- refer to Apical TRM Option_select Section for details

double buffered: ??

MDSS_MDP_AADC_0_APICAL_OPTION_SELECT

Bits	Name	Description
7:0	OPTION_SELECT	Selects the overall operating mode of the core. 0x0: Automatic BC, Automatic ACM 0x1: Manual BC, Automatic ACM 0x3/0x7: Manual BC, Manual ACM Others: Reserved - do not use

0x01A790D8 MDSS_MDP_AADC_0_APICAL_T_FILTER_CONTROL**Type:** RW**Reset State:** 0x00000001

Recursion depth of temporal filter

double buffered: No

MDSS_MDP_AADC_0_APICAL_T_FILTER_CONTROL

Bits	Name	Description
7:0	T_FILTER_CONTROL	The port t_filter_control sets the recursion depth of temporal filtering of photosensor measurements and is used to smooth changes in the strength of DRC in response to rapid changes in photosensor data. A larger value increases the recursion depth in the averaging of time-historic photosensor measurements and results in a slower response to changes in ambient light.

0x01A790DC MDSS_MDP_AADC_0_APICAL_BACKLIGHT_MINMAX**Type:** RW**Reset State:** 0xFFFF2800

Apical Backlight Min/Max Value

double buffered: No

MDSS_MDP_AADC_0_APICAL_BACKLIGHT_MINMAX

Bits	Name	Description
31:16	BACKLIGHT_MAX	Maximum output backlight value permitted.
15:0	BACKLIGHT_MIN	Minimum output backlight value permitted.

0x01A790E0 MDSS_MDP_AADC_0_APICAL_BACKLIGHT**Type:** RW**Reset State:** 0x00008000

Apical Screen brightness Value.

MDSS_MDP_AADC_0_APICAL_BACKLIGHT

Bits	Name	Description
15:0	BACKLIGHT	Screen brightness on a scale of 0 to backlight_scale.

0x01A790E8 MDSS_MDP_AADC_0_APICAL_BACKLIGHT_SCALE**Type:** RW**Reset State:** 0x0000FFFF

Apical Screen brightness Value.

double buffered: No

MDSS_MDP_AADC_0_APICAL_BACKLIGHT_SCALE

Bits	Name	Description
15:0	BACKLIGHT_SCALE	Digital value corresponding to the maximum possible backlight input value. The port backlight_scale should be set equal to the digital value corresponding to the maximum possible backlight brightness that the display can produce. For example, if the backlight level is 16-bit, then backlight_scale should be set to 0xFFFF, if it is an 8-bit range, then backlight_scale should be set to 0x00FF.

0x01A790EC MDSS_MDP_AADC_0_APICAL_AMBIENT_LIGHT

Type: RW

Reset State: 0x00000000

Ambient light level from photosensor

MDSS_MDP_AADC_0_APICAL_AMBIENT_LIGHT

Bits	Name	Description
15:0	AMBIENT_LIGHT	Real time control input from photosensor The core takes the input of the ambient light level in lux via the parameter ambient_light. If the value is available in linear lux units, then it should be entered directly via this parameter. For light sensors which provide a non-linear measure of ambient light, the LUT al_calib_lut should be programmed to convert this value into linear lux units.

0x01A790F0 MDSS_MDP_AADC_0_APICAL_AMBIENT_LIGHT_MIN

Type: RW

Reset State: 0x0000000E

Minimum Ambient light Register

double buffered: No

MDSS_MDP_AADC_0_APICAL_AMBIENT_LIGHT_MIN

Bits	Name	Description
15:0	AMBIENT_LIGHT_MIN	This value represents the incoming ambient light level which backlight_out is set to backlight_min.

0x01A790F4 MDSS_MDP_AADC_0_APICAL_FILTER**Type:** RW**Reset State:** 0x000606CA

Filter values for the input ambient_light

double buffered: No

MDSS_MDP_AADC_0_APICAL_FILTER

Bits	Name	Description
23:16	FILTER_B	This value sets the ambient light-independent component of the stability threshold, below which deviations in photosensor measurements do not lead to a response by the backlight. Larger values result in greater stability to changes in photosensor measurements. This component of the threshold becomes increasingly influential as average ambient light decreases.
15:0	FILTER_A	FilterA Settings [11:9]: Sets the recursion depth of the temporal filter used to smooth out rapid changes in detected ambient light. The larger this parameter, the greater the recursion depth, and the more slowly the backlight responds to changes in ambient light. [8:6] : Sets the ambient light-dependent component of the stability threshold, below which deviations in photosensor measurements do not lead to a response by the backlight. Larger values result in greater stability to changes in photosensor measurements. This component of the threshold becomes increasingly influential as average ambient light increases. [5:3] : Sets the rate at which backlight responds to changes in photosensor measurements, within the stability range established by the values in bits [8:6] and filter_b. [2:0] : Sets the asymmetry between the rate of decrease of backlight compared to the rate of increase of backlight. A value of 0 implies symmetric response, while increasing this value leads to a faster increase in backlight compared to decrease

0x01A790F8 MDSS_MDP_AADC_0_APICAL_CONFIG_BUFFER**Type:** RW**Reset State:** 0x00000003

Double buffering control for Video Pipeline Configuration Parameters

Together with config_buffer_global real time input for iridix core, which is driven by vp_db_update, these control the double buffering of the video pipeline configuration parameters inside Iridix core

double buffered: No

MDSS_MDP_AADC_0_APICAL_CONFIG_BUFFER

Bits	Name	Description
1:0	CONFIG_BUFFER_MODE	config_buffer_mode = 0 (dec) (Pass through): Allows the input configuration ports to pass through to the registers within the core. The internal configuration registers are updated as soon as the ports are changed. config_buffer_mode = 1 (dec) (Block Access): Stops the internal configuration registers from being updated even when the input port values change. The previous register values are held till the next update is allowed. config_buffer_mode = 2 (dec) (Update on VBI): Allows the internal configuration registers only to be updated during the vertical blanking interval (VBI). This ensures that the configuration is never updated during an active frame. config_buffer_mode = 3 (dec) (External): This method is based on the config_buffer_global signal and updates the internal configuration registers when the config_buffer_global signal is asserted. This signal can be generated externally by user logic.

0x01A79100 MDSS_MDP_AADC_0_APICAL_LUT_AL_CALIB1_0**Type:** RW**Reset State:** 0x08000000

Calibration Lookup Table Entry 1-0

The Calibration Lookup Table port geometry is 33 words, each of which is 16 bits.

This LUT is used to linearize the ambient light data

double buffered: No

MDSS_MDP_AADC_0_APICAL_LUT_AL_CALIB1_0

Bits	Name	Description
31:16	LUT_AL_CALIB1	Calibration Lookup Table Entry 1.
15:0	LUT_AL_CALIB0	Calibration Lookup Table Entry 0.

0x01A79104 MDSS_MDP_AADC_0_APICAL_LUT_AL_CALIB3_2**Type:** RW**Reset State:** 0x18001000

Calibration Lookup Table Entry 3-2

The Calibration Lookup Table port geometry is 33 words, each of which is 16 bits.

This LUT is used to linearize the ambient light data

double buffered: No

MDSS_MDP_AADC_0_APICAL_LUT_AL_CALIB3_2

Bits	Name	Description
31:16	LUT_AL_CALIB3	Calibration Lookup Table Entry 3.
15:0	LUT_AL_CALIB2	Calibration Lookup Table Entry 2.

0x01A79108 MDSS_MDP_AADC_0_APICAL_LUT_AL_CALIB5_4

Type: RW

Reset State: 0x28002000

Calibration Lookup Table Entry 5-4

The Calibration Lookup Table port geometry is 33 words, each of which is 16 bits.

This LUT is used to linearize the ambient light data

double buffered: No

MDSS_MDP_AADC_0_APICAL_LUT_AL_CALIB5_4

Bits	Name	Description
31:16	LUT_AL_CALIB5	Calibration Lookup Table Entry 5.
15:0	LUT_AL_CALIB4	Calibration Lookup Table Entry 4.

0x01A7910C MDSS_MDP_AADC_0_APICAL_LUT_AL_CALIB7_6

Type: RW

Reset State: 0x38003000

Calibration Lookup Table Entry 7-6

The Calibration Lookup Table port geometry is 33 words, each of which is 16 bits.

This LUT is used to linearize the ambient light data

double buffered: No

MDSS_MDP_AADC_0_APICAL_LUT_AL_CALIB7_6

Bits	Name	Description
31:16	LUT_AL_CALIB7	Calibration Lookup Table Entry 7.

MDSS_MDP_AADC_0_APICAL_LUT_AL_CALIB7_6 (cont.)

Bits	Name	Description
15:0	LUT_AL_CALIB6	Calibration Lookup Table Entry 6.

0x01A79110 MDSS_MDP_AADC_0_APICAL_LUT_AL_CALIB9_8**Type:** RW**Reset State:** 0x48004000

Calibration Lookup Table Entry 9-8

The Calibration Lookup Table port geometry is 33 words, each of which is 16 bits.

This LUT is used to linearize the ambient light data

double buffered: No

MDSS_MDP_AADC_0_APICAL_LUT_AL_CALIB9_8

Bits	Name	Description
31:16	LUT_AL_CALIB9	Calibration Lookup Table Entry 9.
15:0	LUT_AL_CALIB8	Calibration Lookup Table Entry 8.

0x01A79114 MDSS_MDP_AADC_0_APICAL_LUT_AL_CALIB11_10**Type:** RW**Reset State:** 0x58005000

Calibration Lookup Table Entry 11-10

The Calibration Lookup Table port geometry is 33 words, each of which is 16 bits.

This LUT is used to linearize the ambient light data

double buffered: No

MDSS_MDP_AADC_0_APICAL_LUT_AL_CALIB11_10

Bits	Name	Description
31:16	LUT_AL_CALIB11	Calibration Lookup Table Entry 11.
15:0	LUT_AL_CALIB10	Calibration Lookup Table Entry 10.

0x01A79118 MDSS_MDP_AADC_0_APICAL_LUT_AL_CALIB13_12**Type:** RW**Reset State:** 0x68006000

Calibration Lookup Table Entry 13-12

The Calibration Lookup Table port geometry is 33 words, each of which is 16 bits.

This LUT is used to linearize the ambient light data

double buffered: No

MDSS_MDP_AADC_0_APICAL_LUT_AL_CALIB13_12

Bits	Name	Description
31:16	LUT_AL_CALIB13	Calibration Lookup Table Entry 13.
15:0	LUT_AL_CALIB12	Calibration Lookup Table Entry 12.

0x01A7911C MDSS_MDP_AADC_0_APICAL_LUT_AL_CALIB15_14**Type:** RW**Reset State:** 0x78007000

Calibration Lookup Table Entry 15-14

The Calibration Lookup Table port geometry is 33 words, each of which is 16 bits.

This LUT is used to linearize the ambient light data

double buffered: No

MDSS_MDP_AADC_0_APICAL_LUT_AL_CALIB15_14

Bits	Name	Description
31:16	LUT_AL_CALIB15	Calibration Lookup Table Entry 15.
15:0	LUT_AL_CALIB14	Calibration Lookup Table Entry 14.

0x01A79120 MDSS_MDP_AADC_0_APICAL_LUT_AL_CALIB17_16**Type:** RW**Reset State:** 0x87FF8000

Calibration Lookup Table Entry 17-16

The Calibration Lookup Table port geometry is 33 words, each of which is 16 bits.

This LUT is used to linearize the ambient light data

double buffered: No

MDSS_MDP_AADC_0_APICAL_LUT_AL_CALIB17_16

Bits	Name	Description
31:16	LUT_AL_CALIB17	Calibration Lookup Table Entry 17.
15:0	LUT_AL_CALIB16	Calibration Lookup Table Entry 16.

0x01A79124 MDSS_MDP_AADC_0_APICAL_LUT_AL_CALIB19_18

Type: RW

Reset State: 0x97FF8FFF

Calibration Lookup Table Entry 19-18

The Calibration Lookup Table port geometry is 33 words, each of which is 16 bits.

This LUT is used to linearize the ambient light data

double buffered: No

MDSS_MDP_AADC_0_APICAL_LUT_AL_CALIB19_18

Bits	Name	Description
31:16	LUT_AL_CALIB19	Calibration Lookup Table Entry 19.
15:0	LUT_AL_CALIB18	Calibration Lookup Table Entry 18.

0x01A79128 MDSS_MDP_AADC_0_APICAL_LUT_AL_CALIB21_20

Type: RW

Reset State: 0xA7FF9FFF

Calibration Lookup Table Entry 21-20

The Calibration Lookup Table port geometry is 33 words, each of which is 16 bits.

This LUT is used to linearize the ambient light data

double buffered: No

MDSS_MDP_AADC_0_APICAL_LUT_AL_CALIB21_20

Bits	Name	Description
31:16	LUT_AL_CALIB21	Calibration Lookup Table Entry 21.

MDSS_MDP_AADC_0_APICAL_LUT_AL_CALIB21_20 (cont.)

Bits	Name	Description
15:0	LUT_AL_CALIB20	Calibration Lookup Table Entry 20.

0x01A7912C MDSS_MDP_AADC_0_APICAL_LUT_AL_CALIB23_22**Type:** RW**Reset State:** 0xB7FFAFFF

Calibration Lookup Table Entry 23-22

The Calibration Lookup Table port geometry is 33 words, each of which is 16 bits.

This LUT is used to linearize the ambient light data

double buffered: No

MDSS_MDP_AADC_0_APICAL_LUT_AL_CALIB23_22

Bits	Name	Description
31:16	LUT_AL_CALIB23	Calibration Lookup Table Entry 23.
15:0	LUT_AL_CALIB22	Calibration Lookup Table Entry 22.

0x01A79130 MDSS_MDP_AADC_0_APICAL_LUT_AL_CALIB25_24**Type:** RW**Reset State:** 0xC7FFBFFF

Calibration Lookup Table Entry 25-24

The Calibration Lookup Table port geometry is 33 words, each of which is 16 bits.

This LUT is used to linearize the ambient light data

double buffered: No

MDSS_MDP_AADC_0_APICAL_LUT_AL_CALIB25_24

Bits	Name	Description
31:16	LUT_AL_CALIB25	Calibration Lookup Table Entry 25.
15:0	LUT_AL_CALIB24	Calibration Lookup Table Entry 24.

0x01A79134 MDSS_MDP_AADC_0_APICAL_LUT_AL_CALIB27_26**Type:** RW**Reset State:** 0xD7FFCFFF

Calibration Lookup Table Entry 27-26

The Calibration Lookup Table port geometry is 33 words, each of which is 16 bits.

This LUT is used to linearize the ambient light data

double buffered: No

MDSS_MDP_AADC_0_APICAL_LUT_AL_CALIB27_26

Bits	Name	Description
31:16	LUT_AL_CALIB27	Calibration Lookup Table Entry 27.
15:0	LUT_AL_CALIB26	Calibration Lookup Table Entry 26.

0x01A79138 MDSS_MDP_AADC_0_APICAL_LUT_AL_CALIB29_28**Type:** RW**Reset State:** 0xE7FFDFFF

Calibration Lookup Table Entry 29-28

The Calibration Lookup Table port geometry is 33 words, each of which is 16 bits.

This LUT is used to linearize the ambient light data

double buffered: No

MDSS_MDP_AADC_0_APICAL_LUT_AL_CALIB29_28

Bits	Name	Description
31:16	LUT_AL_CALIB29	Calibration Lookup Table Entry 29.
15:0	LUT_AL_CALIB28	Calibration Lookup Table Entry 28.

0x01A7913C MDSS_MDP_AADC_0_APICAL_LUT_AL_CALIB31_30**Type:** RW**Reset State:** 0xF7FFEFFF

Calibration Lookup Table Entry 31-30

The Calibration Lookup Table port geometry is 33 words, each of which is 16 bits.

This LUT is used to linearize the ambient light data

double buffered: No

MDSS_MDP_AADC_0_APICAL_LUT_AL_CALIB31_30

Bits	Name	Description
31:16	LUT_AL_CALIB31	Calibration Lookup Table Entry 31.
15:0	LUT_AL_CALIB30	Calibration Lookup Table Entry 30.

0x01A79140 MDSS_MDP_AADC_0_APICAL_LUT_AL_CALIB32

Type: RW

Reset State: 0xFFFF0000

Calibration Lookup Table Entry 32

The Calibration Lookup Table port geometry is 33 words, each of which is 16 bits.

This LUT is used to linearize the ambient light data

double buffered: No

MDSS_MDP_AADC_0_APICAL_LUT_AL_CALIB32

Bits	Name	Description
31:16	LUT_AL_CALIB32	Calibration Lookup Table Entry 32.

0x01A79144 MDSS_MDP_AADC_0_APICAL_DRC_IN

Type: RW

Reset State: 0x00000000

Target strength of dynamic range compression.

MDSS_MDP_AADC_0_APICAL_DRC_IN

Bits	Name	Description
15:0	DRC_IN	Target strength of dynamic range compression.

0x01A79148 MDSS_MDP_AADC_0_APICAL_START_CALC

Type: RW

Reset State: 0x00000000

Start calculator pulse.

double buffered: NO

MDSS_MDP_AADC_0_APICAL_START_CALC

Bits	Name	Description
0	START_CALC	Start calculation register Start-Done Protocol of non-Video Pipeline Parameters: Double buffering is not required for the Assertive Display core parameters. Instead a simple start-done protocol is used. The external host will update the parameters and then transition the start_calc input from 0 to 1. After a rising edge is detected on this input, the Assertive Display core's calculator will use the input parameters to calculate the backlight and iridix module strength and the optimum backlight. When the results are available the calc_done status flag is asserted about 500 clock cycles after this edge. In a typical use case this calc_done flag can be used as an interrupt to start the next parameter update cycle. The calc_done flag is held high until the next start_calc rising edge arrives.

0x01A7914C MDSS_MDP_AADC_0_APICAL_STRENGTH_OUT

Type: R

Reset State: 0x00000000

Strength of iridix core output

MDSS_MDP_AADC_0_APICAL_STRENGTH_OUT

Bits	Name	Description
7:0	STRENGTH_OUT	The primary output of the calculator, passed to the internal iridix module's strength_ir signal. This is included as an output from the core for debug purposes only.

0x01A79150 MDSS_MDP_AADC_0_APICAL_DRC_OUT

Type: R

Reset State: 0x00000000

Calculated dynamic range compression output.

MDSS_MDP_AADC_0_APICAL_DRC_OUT

Bits	Name	Description
15:0	DRC_OUT	Calculated dynamic range compression output.

0x01A79154 MDSS_MDP_AADC_0_APICAL_BACKLIGHT_OUT**Type:** R**Reset State:** 0x00000000

Calculated backlight setting.

MDSS_MDP_AADC_0_APICAL_BACKLIGHT_OUT

Bits	Name	Description
15:0	BACKLIGHT_OUT	

0x01A79158 MDSS_MDP_AADC_0_APICAL_CALC_DONE**Type:** R**Reset State:** 0x00000000

Calculator operations finished flag.

MDSS_MDP_AADC_0_APICAL_CALC_DONE

Bits	Name	Description
0	CALC_DONE	Start calculation register Start-Done Protocol of non-Video Pipeline Parameters: Double buffering is not required for the Assertive Display core parameters. Instead a simple start-done protocol is used. The external host will update the parameters and then transition the start_calc input from 0 to 1. After a rising edge is detected on this input, the Assertive Display core's calculator will use the input parameters to calculate the backlight and iridix module strength and the optimum backlight. When the results are available the calc_done status flag is asserted about 500 clock cycles after this edge. In a typical use case this calc_done flag can be used as an interrupt to start the next parameter update cycle. The calc_done flag is held high until the next start_calc rising edge arrives.

17.17.23 MDSS_MDP_AADC_1

Apical Assertive Display Core hardware register description

17.17.23.1 MDSS_MDP_AADC_1_AADC

AADC

Apical Assertive Display Core hardware register description

0x01A79800 MDSS_MDP_AADC_1_APICAL_CORE_BYPASS

Type: RW

Reset State: 0x00000101

Apical bypass and SW reset control

MDSS_MDP_AADC_1_APICAL_CORE_BYPASS

Bits	Name	Description
23:16	APICAL_CORE_RESERVED_23_16	RESERVED for future use.
8	APICAL_CORE_SW_RESET_EN	Enable SW reset control capability for Apical Iridix core - SHOULD always be set to 1'b1; (active high, default 0). This ensures that when ares and apical_core_sw_reset are ORed together, the in/out conversion blocks reset first to block propagation of any potential hazards out of the apical wrapper,, also clocks to the apical core are stopped preventing internal hazards.
0	APICAL_CORE_BYPASS	0x0 : Pass pixel data through Apical core - Apical processing is enabled via APICAL_IRIDIX_CONTROL_0.IRIDIX_ENABLE 0x1 : Bypass Apical core entirely - this will also clock gate the core (default) Double-buffered: Yes

0x01A79804 MDSS_MDP_AADC_1_APICAL_CONTROL_REG0

Type: RW

Reset State: 0x00000003

The Data format port specifies the input data format so that the iridix core can process the different input data formats.

double buffered: Yes

MDSS_MDP_AADC_1_APICAL_CONTROL_REG0

Bits	Name	Description
7:2	CONTROL_REG0_RESERVED_7_2	Reserved - Should be zero, except for bit 3 which is hardcoded as 1b.
1:0	DATA_FORMAT	Data Format Select 0x0: YCbCr Color space 4:2:2 format, interleaved (Cb-Y-Cr-Y) 0x1: YCbCr Color space 4:2:2 format, parallel 0x2: YCbCr Color 4:4:4 format 0x3: RGB Color space

0x01A79808 MDSS_MDP_AADC_1_APICAL_CONTROL_REG1**Type:** RW**Reset State:** 0x00000000

Synchronization Format Register.

double buffered: Yes

MDSS_MDP_AADC_1_APICAL_CONTROL_REG1

Bits	Name	Description
7	AUTO_SIZE	Auto Size Register 0x0: Auto_Size off. The Iridix core will use Frame_Width, Frame_Height ports values 0x1: Auto_Size on. The Iridix core will calculate the frame geometry based on the active line and field signals. The size of current frame will be used for the estimation of the next frame. The Frame size will be updated when the Field signal toggles.
6:0	CONTROL_REG1_RESERVED_6_0	Reserved, should be set to 0x00.

0x01A7980C MDSS_MDP_AADC_1_APICAL_FRAME_SIZE**Type:** RW**Reset State:** 0x01000100

Frame Size Register.

double buffered: Yes

MDSS_MDP_AADC_1_APICAL_FRAME_SIZE

Bits	Name	Description
31:16	FRAME_WIDTH	Frame width in number of pixels in an active line. This port is used only in non-automatic mode(default)
15:0	FRAME_HEIGHT	Frame height in number of line in an active frame. This port is used only in non-automatic mode(default)

0x01A79810 MDSS_MDP_AADC_1_APICAL_IRIDIX_CONTROL_0**Type:** RW**Reset State:** 0x00000006

Iridix Control Register 0

double buffered: Yes

MDSS_MDP_AADC_1_APICAL_IRIDIX_CONTROL_0

Bits	Name	Description
7:5	IRIDIX_CONTROL_0_RESERVED_7_5	Reserved - Should be zero.
4	IRIDIX_COLOR_CORRECTION	Color Correction/Control - Enabling this processing will result in more accurate colors processing. The color correction algorithm is required on gamma corrected sources. It reduces the saturation in dark areas when they are being amplified, and saturates bright parts when they get darker. 0x0: Disable color correction 0x1: Enable color correction
3	IRIDIX_LOCAL_CONTRAST_2D	Local contrast (2D) 0x0: Memory interfaces for 2D processing used 0x1: Memory interfaces for 2D processing bypassed, 2D processing disabled. This selection overrides LC0, LC1 settings. Note: If 2D Local contrast is not implemented then the bit[3] should be one.

MDSS_MDP_AADC_1_APICAL_IRIDIX_CONTROL_0 (cont.)

Bits	Name	Description
2:1	IRIDIX_LOCAL_CONTRAST	Local Contrast Control - This is a supplementary module preserving local contrast in small areas. Depending on custom implementation LC module can be implemented differently and may have different settings. 1D Case : 0x0: disabled; 0x1: LC enabled, kernel 1x3 used 0x2: LC enabled, kernel 1x3 used 0x3: LC enabled, kernel 1x5 used 2D Case : 0x0: disabled; 0x1: LC enabled, kernel 1x3 used 0x2: LC enabled, kernel 3x3 used 0x3: LC enabled, kernel 3x5 used
0	IRIDIX_ENABLE	iridix Enable - This bit turns iridix processing ON and OFF. When iridix is OFF, the video data passes to the output without any changes. Disabling iridix®using this bit is equivalent to setting the Strength parameter to 0. Many internal modules run in idle mode when this bit is 0. The Latency Delays between the input and the output signals are the same in both ON and OFF modes. 0x0: Disable iridix processing 0x1: Enable iridix processing

0x01A79814 MDSS_MDP_AADC_1_APICAL_IRIDIX_CONTROL_1**Type:** RW**Reset State:** 0x00004600

This Port is reserved for debugging purposes by Apical. Do not modify.

MDSS_MDP_AADC_1_APICAL_IRIDIX_CONTROL_1

Bits	Name	Description
15:8	IRIDIX_CONTROL_1	Reserved for debugging purposes by Apical

0x01A79818 MDSS_MDP_AADC_1_APICAL_STRENGTH_MANUAL**Type:** RW**Reset State:** 0x00000000

Strength of iridix -Manual override value of iridix module's strength input (strength_ir) when option_select = 7 (dec).

MDSS_MDP_AADC_1_APICAL_STRENGTH_MANUAL

Bits	Name	Description
7:0	STRENGTH_MANUAL	Strength of iridix - This Port sets processing Strength. Minimum value is 0, maximum is 255. 0x00: Video data will not be processed at all, and will go to the output unchanged. 0xFF: Maximum strengthSTRENGTH_MANUAL

0x01A7981C MDSS_MDP_AADC_1_APICAL_VARIANCE

Type: RW

Reset State: 0x00000041

Affects the sensitivity of the transform to different areas of the image, and can be increased in order to emphasize small regions (e.g. faces).

If this parameter is set to zero, the sensitivity to the small areas is maximal. The transform becomes more local.

When the parameter is set to 0xFF the transform deemphasizes small local details, and transform becomes more global

e.g. the sky at the top of the frame can affect flowers at the bottom of the frame.

Default value is 36h but it depends on custom implementation.

Setting variance to zero will progressively increase the spatial sensitivity of the algorithm but it can lead to artificial-looking images in some cases.

double buffered: Yes

MDSS_MDP_AADC_1_APICAL_VARIANCE

Bits	Name	Description
7:4	VARIANCE_INTENSITY	Sets the degree of sensitivity in the luminance domain. Maximum Variance is 0xF, and minimum Variance is 0x0.
3:0	VARIANCE_SPACE	Sets the degree of spatial sensitivity of the algorithm. As this parameter is made smaller, the algorithm focuses on smaller regions within the image. Maximum Variance is 0xF, and minimum Variance is 0x0

0x01A79820 MDSS_MDP_AADC_1_APICAL_IRIDIX_DITHER**Type:** RW**Reset State:** 0x00000000

Iridix Dither Register: When the number of color gradations of a display device is small (for example 6 bit per color) pixel dithering can make gradients look smother.

There can be a situation when iridix output signal is 12 bits but the signal needs to be compressed into 8 bits.

The best way for this is to use dithering of least significant bits and then truncate these bits. The dithering can be performed by iridix Processor.

However, the truncation needs to be done by a hardware designer outside iridix.

double buffered: Yes

MDSS_MDP_AADC_1_APICAL_IRIDIX_DITHER

Bits	Name	Description
7:3	IRIDIX_DITHER_RESERVE D_7_3	Reserved - Not used for now.
2:0	DITHER	Strength of Dithering. 0x0: No dithering 0x1: One least significant bit of the output signal is dithered 0x2: Two bits are dithered 0x3: Three bits are dithered 0x4: Four bits are dithered Others: Four bits are dithered

0x01A79824 MDSS_MDP_AADC_1_APICAL_DITEHR_CONTROL**Type:** RW**Reset State:** 0x00000000

Spatial Preceptual Model Dither

This module provides a 4-bit dither via a spatial perceptual model.

There is no temporal dependence of the dither pattern, therefore eliminating the possibility of flicker

note that temporal dither is available via the port iridix_dither.

double buffered: Yes

MDSS_MDP_AADC_1_APICAL_DITEHR_CONTROL

Bits	Name	Description
7:4	DITHER_CONTROL_RESE RVED_7_4	Reserved - Should be Zero.
3:2	DITHER_AMOUNT	Strength of Dithering. 0x0: Dither 1 bit 0x1: Dither 2 bit 0x2: Dither 3 bit 0x3: Dither 4 bit
1	DITHER_SHIFT_MODE	Shift Mode: 0x0: Output is MSB aligned 0x1: Output is LSB aligned
0	DITHER_ENABLE_DITHER	Enable Dither: Enables the dither module. 0: Disable dither module 1: Enable dither module

0x01A79828 MDSS_MDP_AADC_1_APICAL_LIMIT_AMPL**Type:** RW**Reset State:** 0x000000F0

Two parameters used to restrict the luminance space in which Iridix can adaptively generate tone curves for each pixel

double buffered: Yes

MDSS_MDP_AADC_1_APICAL_LIMIT_AMPL

Bits	Name	Description
7:4	BRIGHT_AMP_LIMIT	Bright amplification limit - The resultant tone curve cannot be lower than bright amplification limit line controlled by the bright amplification limit parameter. See Apical TRM Chapter 4 for more explanation. Maximum limit is 0xF, when the value is 0x0 there is no limit.
3:0	DARK_AMP_LIMIT	Dark amplification limit - The resultant tone curve cannot be higher than dark amplification limit line controlled by the dark amplification limit parameter. See Apical TRM Chapter 4 for more explanation. Maximum limit is 0xF, when the value is 0x0 there is no limit.

0x01A7982C MDSS_MDP_AADC_1_APICAL_SLOPE**Type:** RW**Reset State:** 0x00003C80

Slope Min and Max Limit used to restrict the slope of the tone-curve generated by Iridix.

double buffered: Yes

MDSS_MDP_AADC_1_APICAL_SLOPE

Bits	Name	Description
15:8	SLOPE_MAX	Slope Max Limit - Slope Max Limit is used to restrict the slope of the tone-curve generated by iridix. When Slope Max Limit parameter is set to 0xFF, the tone curve slope, generated by iridix is not limited (maximum slope 15). When this value is set to 00 iridix will not generate the tone-curve with slope greater than 16/15 (48°). Intermediate values of Slope Limit are linear interpolation between minimum and maximum values. Recommended Slope Max Limit range is 20h-60h. For noisy images less value should be used. See Chapter 4 for more explanation.
7:0	SLOPE_MIN	Slope Min Limit - Slope Min Limit is used to restrict the slope of the tone-curve generated by iridix. When Slope Min Limit parameter is set to 0x00, the tone curve slope, generated by iridix is not limited. When this value is set to FF iridix will not generate the tone-curve with slope less than 3/4. Intermediate values of Slope Limit are linear interpolation between minimum and maximum values. Recommended Slope Min Limit range is 00h-C0h. See Chapter 4 for more explanation.

0x01A79830 MDSS_MDP_AADC_1_APICAL_BW_LEVEL**Type:** RW**Reset State:** 0x03FF0000

Black and White Level of the Iridix IP.

The least significant bits [dw-1:0] are used; higher bits [15:dw] are unused. dw is the component data width. dw=8 for MDSS \n double buffered: Yes

MDSS_MDP_AADC_1_APICAL_BW_LEVEL

Bits	Name	Description
25:16	WHITE_LEVEL	White Level - The value stored in White Level Port will be used as white level for iridix processing in all unsigned data channels. Data above White level will not be processed and stay unchanged. Note: 1. For 12 bit variant minimum White Level is constrained as: White Level - Black Level > 1024. 2. For 10 bit variant minimum White Level is constrained as: White Level - Black Level > 256.
9:0	BLACK_LEVEL	Black Level - The value stored in Black Level Port will be used as zero level for iridix processing in all unsigned data channels. Data below Black level will not be processed and stay unchanged. Note: 1. For 12 bit variant maximum Black Level is constrained as: White Level - Black Level > 1024. 2. For 10 bit variant maximum Black Level is constrained as: White Level - Black Level > 256

0x01A79834 MDSS_MDP_AADC_1_APICAL_LOGO_POS**Type:** RW**Reset State:** 0x00000000

Apical Logo Position

double buffered: Yes

MDSS_MDP_AADC_1_APICAL_LOGO_POS

Bits	Name	Description
15:8	LOGO_LEFT	horizontal position of logo.
7:0	LOGO_TOP	vertical position of logo.

0x01A79838 MDSS_MDP_AADC_1_APICAL_LUT_FI1_0**Type:** RW**Reset State:** 0x00C70000

Asymmetry Function Lookup Table Entry 1-0

The Asymmetry Function Lookup Table port geometry is 33 words, each of which is 16 bits.

The Asymmetry function is used to balance the iridix effect between the dark and bright regions of the image.

double buffered: Yes

MDSS_MDP_AADC_1_APICAL_LUT_FI1_0

Bits	Name	Description
31:16	LUT_FI1	Asymmetry Function Lookup Table Entry 1.
15:0	LUT_FI0	Asymmetry Function Lookup Table Entry 0.

0x01A7983C MDSS_MDP_AADC_1_APICAL_LUT_FI3_2**Type:** RW**Reset State:** 0x023D0186

Asymmetry Function Lookup Table Entry 3-2

The Asymmetry Function Lookup Table port geometry is 33 words, each of which is 16 bits.

The Asymmetry function is used to balance the iridix effect between the dark and bright regions of the image.

double buffered: Yes

MDSS_MDP_AADC_1_APICAL_LUT_FI3_2

Bits	Name	Description
31:16	LUT_FI3	Asymmetry Function Lookup Table Entry 3.
15:0	LUT_FI2	Asymmetry Function Lookup Table Entry 2.

0x01A79840 MDSS_MDP_AADC_1_APICAL_LUT_FI5_4**Type:** RW**Reset State:** 0x039602ED

Asymmetry Function Lookup Table Entry 5-4

The Asymmetry Function Lookup Table port geometry is 33 words, each of which is 16 bits.

The Asymmetry function is used to balance the iridix effect between the dark and bright regions of the image.

double buffered: Yes

MDSS_MDP_AADC_1_APICAL_LUT_FI5_4

Bits	Name	Description
31:16	LUT_FI5	Asymmetry Function Lookup Table Entry 5.
15:0	LUT_FI4	Asymmetry Function Lookup Table Entry 4.

0x01A79844 MDSS_MDP_AADC_1_APICAL_LUT_FI7_6**Type:** RW**Reset State:** 0x04D60439

Asymmetry Function Lookup Table Entry 7-6

The Asymmetry Function Lookup Table port geometry is 33 words, each of which is 16 bits.

The Asymmetry function is used to balance the iridix effect between the dark and bright regions of the image.

double buffered: Yes

MDSS_MDP_AADC_1_APICAL_LUT_FI7_6

Bits	Name	Description
31:16	LUT_FI7	Asymmetry Function Lookup Table Entry 7.
15:0	LUT_FI6	Asymmetry Function Lookup Table Entry 6.

0x01A79848 MDSS_MDP_AADC_1_APICAL_LUT_FI9_8**Type:** RW**Reset State:** 0x05FF056D

Asymmetry Function Lookup Table Entry 9-8

The Asymmetry Function Lookup Table port geometry is 33 words, each of which is 16 bits.

The Asymmetry function is used to balance the iridix effect between the dark and bright regions of the image.

double buffered: Yes

MDSS_MDP_AADC_1_APICAL_LUT_FI9_8

Bits	Name	Description
31:16	LUT_FI9	Asymmetry Function Lookup Table Entry 9.
15:0	LUT_FI8	Asymmetry Function Lookup Table Entry 8.

0x01A7984C MDSS_MDP_AADC_1_APICAL_LUT_FI11_10**Type:** RW**Reset State:** 0x0714068C

Asymmetry Function Lookup Table Entry 11-10

The Asymmetry Function Lookup Table port geometry is 33 words, each of which is 16 bits.

The Asymmetry function is used to balance the iridix effect between the dark and bright regions of the image.

double buffered: Yes

MDSS_MDP_AADC_1_APICAL_LUT_FI11_10

Bits	Name	Description
31:16	LUT_FI11	Asymmetry Function Lookup Table Entry 11.
15:0	LUT_FI10	Asymmetry Function Lookup Table Entry 10.

0x01A79850 MDSS_MDP_AADC_1_APICAL_LUT_FI13_12

Type: RW

Reset State: 0x08190799

Asymmetry Function Lookup Table Entry 13-12

The Asymmetry Function Lookup Table port geometry is 33 words, each of which is 16 bits.

The Asymmetry function is used to balance the iridix effect between the dark and bright regions of the image.

double buffered: Yes

MDSS_MDP_AADC_1_APICAL_LUT_FI13_12

Bits	Name	Description
31:16	LUT_FI13	Asymmetry Function Lookup Table Entry 13.
15:0	LUT_FI12	Asymmetry Function Lookup Table Entry 12.

0x01A79854 MDSS_MDP_AADC_1_APICAL_LUT_FI15_14

Type: RW

Reset State: 0x090E0895

Asymmetry Function Lookup Table Entry 15-14

The Asymmetry Function Lookup Table port geometry is 33 words, each of which is 16 bits.

The Asymmetry function is used to balance the iridix effect between the dark and bright regions of the image.

double buffered: Yes

MDSS_MDP_AADC_1_APICAL_LUT_FI15_14

Bits	Name	Description
31:16	LUT_FI15	Asymmetry Function Lookup Table Entry 15.
15:0	LUT_FI14	Asymmetry Function Lookup Table Entry 14.

0x01A79858 MDSS_MDP_AADC_1_APICAL_LUT_FI17_16**Type:** RW**Reset State:** 0x09F80985

Asymmetry Function Lookup Table Entry 17-16

The Asymmetry Function Lookup Table port geometry is 33 words, each of which is 16 bits.

The Asymmetry function is used to balance the iridix effect between the dark and bright regions of the image.

double buffered: Yes

MDSS_MDP_AADC_1_APICAL_LUT_FI17_16

Bits	Name	Description
31:16	LUT_FI17	Asymmetry Function Lookup Table Entry 17.
15:0	LUT_FI16	Asymmetry Function Lookup Table Entry 16.

0x01A7985C MDSS_MDP_AADC_1_APICAL_LUT_FI19_18**Type:** RW**Reset State:** 0x0AD70A68

Asymmetry Function Lookup Table Entry 19-18

The Asymmetry Function Lookup Table port geometry is 33 words, each of which is 16 bits.

The Asymmetry function is used to balance the iridix effect between the dark and bright regions of the image.

double buffered: Yes

MDSS_MDP_AADC_1_APICAL_LUT_FI19_18

Bits	Name	Description
31:16	LUT_FI19	Asymmetry Function Lookup Table Entry 19.
15:0	LUT_FI18	Asymmetry Function Lookup Table Entry 18.

0x01A79860 MDSS_MDP_AADC_1_APICAL_LUT_FI21_20**Type:** RW**Reset State:** 0x0BAD0B43

Asymmetry Function Lookup Table Entry 21-20

The Asymmetry Function Lookup Table port geometry is 33 words, each of which is 16 bits.

The Asymmetry function is used to balance the iridix effect between the dark and bright regions of the image.

double buffered: Yes

MDSS_MDP_AADC_1_APICAL_LUT_FI21_20

Bits	Name	Description
31:16	LUT_FI21	Asymmetry Function Lookup Table Entry 21.
15:0	LUT_FI20	Asymmetry Function Lookup Table Entry 20.

0x01A79864 MDSS_MDP_AADC_1_APICAL_LUT_FI23_22**Type:** RW**Reset State:** 0x0C7D0C15

Asymmetry Function Lookup Table Entry 23-22

The Asymmetry Function Lookup Table port geometry is 33 words, each of which is 16 bits.

The Asymmetry function is used to balance the iridix effect between the dark and bright regions of the image.

double buffered: Yes

MDSS_MDP_AADC_1_APICAL_LUT_FI23_22

Bits	Name	Description
31:16	LUT_FI23	Asymmetry Function Lookup Table Entry 23.
15:0	LUT_FI22	Asymmetry Function Lookup Table Entry 22.

0x01A79868 MDSS_MDP_AADC_1_APICAL_LUT_FI25_24**Type:** RW**Reset State:** 0x0D480CE3

Asymmetry Function Lookup Table Entry 25-24

The Asymmetry Function Lookup Table port geometry is 33 words, each of which is 16 bits.

The Asymmetry function is used to balance the iridix effect between the dark and bright regions of the image.

double buffered: Yes

MDSS_MDP_AADC_1_APICAL_LUT_FI25_24

Bits	Name	Description
31:16	LUT_FI25	Asymmetry Function Lookup Table Entry 25.
15:0	LUT_FI24	Asymmetry Function Lookup Table Entry 24.

0x01A7986C MDSS_MDP_AADC_1_APICAL_LUT_FI27_26

Type: RW

Reset State: 0x0E0F0DAC

Asymmetry Function Lookup Table Entry 27-26

The Asymmetry Function Lookup Table port geometry is 33 words, each of which is 16 bits.

The Asymmetry function is used to balance the iridix effect between the dark and bright regions of the image.

double buffered: Yes

MDSS_MDP_AADC_1_APICAL_LUT_FI27_26

Bits	Name	Description
31:16	LUT_FI27	Asymmetry Function Lookup Table Entry 27.
15:0	LUT_FI26	Asymmetry Function Lookup Table Entry 26.

0x01A79870 MDSS_MDP_AADC_1_APICAL_LUT_FI29_28

Type: RW

Reset State: 0x0ED50E72

Asymmetry Function Lookup Table Entry 29-28

The Asymmetry Function Lookup Table port geometry is 33 words, each of which is 16 bits.

The Asymmetry function is used to balance the iridix effect between the dark and bright regions of the image.

double buffered: Yes

MDSS_MDP_AADC_1_APICAL_LUT_FI29_28

Bits	Name	Description
31:16	LUT_FI29	Asymmetry Function Lookup Table Entry 29.
15:0	LUT_FI28	Asymmetry Function Lookup Table Entry 28.

0x01A79874 MDSS_MDP_AADC_1_APICAL_LUT_FI31_30**Type:** RW**Reset State:** 0x0F9B0F38

Asymmetry Function Lookup Table Entry 31-30

The Asymmetry Function Lookup Table port geometry is 33 words, each of which is 16 bits.

The Asymmetry function is used to balance the iridix effect between the dark and bright regions of the image.

double buffered: Yes

MDSS_MDP_AADC_1_APICAL_LUT_FI31_30

Bits	Name	Description
31:16	LUT_FI31	Asymmetry Function Lookup Table Entry 31.
15:0	LUT_FI30	Asymmetry Function Lookup Table Entry 30.

0x01A79878 MDSS_MDP_AADC_1_APICAL_LUT_FI32**Type:** RW**Reset State:** 0x0FFF0000

Asymmetry Function Lookup Table Entry 32

The Asymmetry Function Lookup Table port geometry is 33 words, each of which is 16 bits.

The Asymmetry function is used to balance the iridix effect between the dark and bright regions of the image.

double buffered: Yes

MDSS_MDP_AADC_1_APICAL_LUT_FI32

Bits	Name	Description
31:16	LUT_FI32	Asymmetry Function Lookup Table Entry 32.

0x01A7987C MDSS_MDP_AADC_1_APICAL_LUT_CC1_0**Type:** RW**Reset State:** 0x011600FF

Color Correction Lookup Table Entry 1-0

The Color Correction Lookup Table port geometry is 33 words, each of which is 16 bits.

The Asymmetry function is used to balance the iridix effect between the dark and bright regions of the image.

double buffered: Yes

MDSS_MDP_AADC_1_APICAL_LUT_CC1_0

Bits	Name	Description
27:16	LUT_CC1	Color Correction Lookup Table Entry 1.
11:0	LUT_CC0	Color Correction Lookup Table Entry 0.

0x01A79880 MDSS_MDP_AADC_1_APICAL_LUT_CC3_2**Type:** RW**Reset State:** 0x0146012E

Color Correction Lookup Table Entry 3-2

The Color Correction Lookup Table port geometry is 33 words, each of which is 16 bits.

The Asymmetry function is used to balance the iridix effect between the dark and bright regions of the image.

double buffered: Yes

MDSS_MDP_AADC_1_APICAL_LUT_CC3_2

Bits	Name	Description
27:16	LUT_CC3	Color Correction Lookup Table Entry 3.
11:0	LUT_CC2	Color Correction Lookup Table Entry 2.

0x01A79884 MDSS_MDP_AADC_1_APICAL_LUT_CC5_4**Type:** RW**Reset State:** 0x0176015E

Color Correction Lookup Table Entry 5-4

The Color Correction Lookup Table port geometry is 33 words, each of which is 16 bits.

The Asymmetry function is used to balance the iridix effect between the dark and bright regions of the image.

double buffered: Yes

MDSS_MDP_AADC_1_APICAL_LUT_CC5_4

Bits	Name	Description
27:16	LUT_CC5	Color Correction Lookup Table Entry 5.
11:0	LUT_CC4	Color Correction Lookup Table Entry 4.

0x01A79888 MDSS_MDP_AADC_1_APICAL_LUT_CC7_6

Type: RW

Reset State: 0x01A6018E

Color Correction Lookup Table Entry 7-6

The Color Correction Lookup Table port geometry is 33 words, each of which is 16 bits.

The Asymmetry function is used to balance the iridix effect between the dark and bright regions of the image.

double buffered: Yes

MDSS_MDP_AADC_1_APICAL_LUT_CC7_6

Bits	Name	Description
27:16	LUT_CC7	Color Correction Lookup Table Entry 7.
11:0	LUT_CC6	Color Correction Lookup Table Entry 6.

0x01A7988C MDSS_MDP_AADC_1_APICAL_LUT_CC9_8

Type: RW

Reset State: 0x01D601BE

Color Correction Lookup Table Entry 9-8

The Color Correction Lookup Table port geometry is 33 words, each of which is 16 bits.

The Asymmetry function is used to balance the iridix effect between the dark and bright regions of the image.

double buffered: Yes

MDSS_MDP_AADC_1_APICAL_LUT_CC9_8

Bits	Name	Description
27:16	LUT_CC9	Color Correction Lookup Table Entry 9.
11:0	LUT_CC8	Color Correction Lookup Table Entry 8.

0x01A79890 MDSS_MDP_AADC_1_APICAL_LUT_CC11_10**Type:** RW**Reset State:** 0x020501EE

Color Correction Lookup Table Entry 11-10

The Color Correction Lookup Table port geometry is 33 words, each of which is 16 bits.

The Asymmetry function is used to balance the iridix effect between the dark and bright regions of the image.

double buffered: Yes

MDSS_MDP_AADC_1_APICAL_LUT_CC11_10

Bits	Name	Description
27:16	LUT_CC11	Color Correction Lookup Table Entry 11.
11:0	LUT_CC10	Color Correction Lookup Table Entry 10.

0x01A79894 MDSS_MDP_AADC_1_APICAL_LUT_CC13_12**Type:** RW**Reset State:** 0x0235021D

Color Correction Lookup Table Entry 13-12

The Color Correction Lookup Table port geometry is 33 words, each of which is 16 bits.

The Asymmetry function is used to balance the iridix effect between the dark and bright regions of the image.

double buffered: Yes

MDSS_MDP_AADC_1_APICAL_LUT_CC13_12

Bits	Name	Description
27:16	LUT_CC13	Color Correction Lookup Table Entry 13.
11:0	LUT_CC12	Color Correction Lookup Table Entry 12.

0x01A79898 MDSS_MDP_AADC_1_APICAL_LUT_CC15_14**Type:** RW**Reset State:** 0x0265024D

Color Correction Lookup Table Entry 15-14

The Color Correction Lookup Table port geometry is 33 words, each of which is 16 bits.

The Asymmetry function is used to balance the iridix effect between the dark and bright regions of the image.

double buffered: Yes

MDSS_MDP_AADC_1_APICAL_LUT_CC15_14

Bits	Name	Description
27:16	LUT_CC15	Color Correction Lookup Table Entry 15.
11:0	LUT_CC14	Color Correction Lookup Table Entry 14.

0x01A7989C MDSS_MDP_AADC_1_APICAL_LUT_CC17_16**Type:** RW**Reset State:** 0x0295027D

Color Correction Lookup Table Entry 17-16

The Color Correction Lookup Table port geometry is 33 words, each of which is 16 bits.

The Asymmetry function is used to balance the iridix effect between the dark and bright regions of the image.

double buffered: Yes

MDSS_MDP_AADC_1_APICAL_LUT_CC17_16

Bits	Name	Description
27:16	LUT_CC17	Color Correction Lookup Table Entry 17.
11:0	LUT_CC16	Color Correction Lookup Table Entry 16.

0x01A798A0 MDSS_MDP_AADC_1_APICAL_LUT_CC19_18**Type:** RW**Reset State:** 0x02C402AC

Color Correction Lookup Table Entry 19-18

The Color Correction Lookup Table port geometry is 33 words, each of which is 16 bits.

The Asymmetry function is used to balance the iridix effect between the dark and bright regions of the image.

double buffered: Yes

MDSS_MDP_AADC_1_APICAL_LUT_CC19_18

Bits	Name	Description
27:16	LUT_CC19	Color Correction Lookup Table Entry 19.
11:0	LUT_CC18	Color Correction Lookup Table Entry 18.

0x01A798A4 MDSS_MDP_AADC_1_APICAL_LUT_CC21_20

Type: RW

Reset State: 0x02F302DC

Color Correction Lookup Table Entry 21-20

The Color Correction Lookup Table port geometry is 33 words, each of which is 16 bits.

The Asymmetry function is used to balance the iridix effect between the dark and bright regions of the image.

double buffered: Yes

MDSS_MDP_AADC_1_APICAL_LUT_CC21_20

Bits	Name	Description
27:16	LUT_CC21	Color Correction Lookup Table Entry 21.
11:0	LUT_CC20	Color Correction Lookup Table Entry 20.

0x01A798A8 MDSS_MDP_AADC_1_APICAL_LUT_CC23_22

Type: RW

Reset State: 0x0323030B

Color Correction Lookup Table Entry 23-22

The Color Correction Lookup Table port geometry is 33 words, each of which is 16 bits.

The Asymmetry function is used to balance the iridix effect between the dark and bright regions of the image.

double buffered: Yes

MDSS_MDP_AADC_1_APICAL_LUT_CC23_22

Bits	Name	Description
27:16	LUT_CC23	Color Correction Lookup Table Entry 23.
11:0	LUT_CC22	Color Correction Lookup Table Entry 22.

0x01A798AC MDSS_MDP_AADC_1_APICAL_LUT_CC25_24**Type:** RW**Reset State:** 0x0352033A

Color Correction Lookup Table Entry 25-24

The Color Correction Lookup Table port geometry is 33 words, each of which is 16 bits.

The Asymmetry function is used to balance the iridix effect between the dark and bright regions of the image.

double buffered: Yes

MDSS_MDP_AADC_1_APICAL_LUT_CC25_24

Bits	Name	Description
27:16	LUT_CC25	Color Correction Lookup Table Entry 25.
11:0	LUT_CC24	Color Correction Lookup Table Entry 24.

0x01A798B0 MDSS_MDP_AADC_1_APICAL_LUT_CC27_26**Type:** RW**Reset State:** 0x0381036A

Color Correction Lookup Table Entry 27-26

The Color Correction Lookup Table port geometry is 33 words, each of which is 16 bits.

The Asymmetry function is used to balance the iridix effect between the dark and bright regions of the image.

double buffered: Yes

MDSS_MDP_AADC_1_APICAL_LUT_CC27_26

Bits	Name	Description
27:16	LUT_CC27	Color Correction Lookup Table Entry 27.
11:0	LUT_CC26	Color Correction Lookup Table Entry 26.

0x01A798B4 MDSS_MDP_AADC_1_APICAL_LUT_CC29_28**Type:** RW**Reset State:** 0x03B10399

Color Correction Lookup Table Entry 29-28

The Color Correction Lookup Table port geometry is 33 words, each of which is 16 bits.

The Asymmetry function is used to balance the iridix effect between the dark and bright regions of the image.

double buffered: Yes

MDSS_MDP_AADC_1_APICAL_LUT_CC29_28

Bits	Name	Description
27:16	LUT_CC29	Color Correction Lookup Table Entry 29.
11:0	LUT_CC28	Color Correction Lookup Table Entry 28.

0x01A798B8 MDSS_MDP_AADC_1_APICAL_LUT_CC31_30**Type:** RW**Reset State:** 0x03E003C8

Color Correction Lookup Table Entry 31-30

The Color Correction Lookup Table port geometry is 33 words, each of which is 16 bits.

The Asymmetry function is used to balance the iridix effect between the dark and bright regions of the image.

double buffered: Yes

MDSS_MDP_AADC_1_APICAL_LUT_CC31_30

Bits	Name	Description
27:16	LUT_CC31	Color Correction Lookup Table Entry 31.
11:0	LUT_CC30	Color Correction Lookup Table Entry 30.

0x01A798BC MDSS_MDP_AADC_1_APICAL_LUT_CC32**Type:** RW**Reset State:** 0x03F80000

Color Correction Lookup Table Entry 32

The Color Correction Lookup Table port geometry is 33 words, each of which is 16 bits.

The Asymmetry function is used to balance the iridix effect between the dark and bright regions of the image.

double buffered: Yes

MDSS_MDP_AADC_1_APICAL_LUT_CC32

Bits	Name	Description
27:16	LUT_CC32	Color Correction Lookup Table Entry 32.

0x01A798C8 MDSS_MDP_AADC_1_APICAL_STRENGTH_LIMIT

Type: RW

Reset State: 0x00000080

Imposes a cap on maximum dynamic range compression strength.

double buffered: No

MDSS_MDP_AADC_1_APICAL_STRENGTH_LIMIT

Bits	Name	Description
7:0	STRENGTH_LIMIT	Used to restrict the maximum value taken by strength_out: in all cases, strength_out <= strength_limit.

0x01A798CC MDSS_MDP_AADC_1_APICAL_CALIBRATION_AB

Type: RW

Reset State: 0x005F006E

Screen calibration parameter

four calibration parameters control the overall adaptation of displayed content to ambient light and backlight level.

They represent a combination of physical and perceptual factors which influence the apparent contrast of imagery as the user's eyes adapt to changing viewing conditions.

double buffered: No

MDSS_MDP_AADC_1_APICAL_CALIBRATION_AB

Bits	Name	Description
31:16	CALIBRATION_B	This parameter controls the adaptation of content to the screen brightness. As backlight is reduced in the presence of ambient light, shadow detail is progressively lost below the minimum grey level of the display. When calibrated correctly, this parameter ensures that image content remains viewable at all backlight levels.
15:0	CALIBRATION_A	This parameter is related to the reflectivity of the screen. The more reflective the screen, the greater is the loss of shadow detail as ambient light level increases. It is the objective of the Assertive Display core to maintain the same level of shadow detail which pertains in very low light, up to ambient light levels typical of outdoor viewing. When calibrated correctly, this parameter ensures that image content is neither lost nor over-enhanced leading to preservation of the viewing experience.

0x01A798D0 MDSS_MDP_AADC_1_APICAL_CALIBRATION_CD**Type:** RW**Reset State:** 0x00000013

Screen calibration parameter

four calibration parameters control the overall adaptation of displayed content to ambient light and backlight level.

They represent a combination of physical and perceptual factors which influence the apparent contrast of imagery as the user's eyes adapt to changing viewing conditions.

double buffered: No

MDSS_MDP_AADC_1_APICAL_CALIBRATION_CD

Bits	Name	Description
31:16	CALIBRATION_D	Typical movie, image and gaming content is 8-bit and encoded with a gamma curve which results in a dynamic range of around 1000:1. Most large displays have an effective contrast ratio larger than this under ideal viewing conditions (full backlight, low ambient light). However, many portable displays cannot reproduce such contrast ratios even under ideal conditions. When calibrated correctly, this parameter increases the effective contrast ratio of low contrast displays to match that of typical content.

MDSS_MDP_AADC_1_APICAL_CALIBRATION_CD (cont.)

Bits	Name	Description
15:0	CALIBRATION_C	This parameter controls the adaptation of content to overcome the limitations of the human eye sensitivity at very low light. Decreasing screen brightness in total darkness causes loss of image detail in shadow regions, especially on small displays, as the eyes are unable to adapt sufficiently. When calibrated correctly, this parameter enables restoration of such details.

0x01A798D4 MDSS_MDP_AADC_1_APICAL_OPTION_SELECT**Type:** RW**Reset State:** 0x00000000

Selects the overall operating mode of the core.- refer to Apical TRM Option_select Section for details

double buffered:?

MDSS_MDP_AADC_1_APICAL_OPTION_SELECT

Bits	Name	Description
7:0	OPTION_SELECT	Selects the overall operating mode of the core. 0x0: Automatic BC, Automatic ACM 0x1: Manual BC, Automatic ACM 0x3/0x7: Manual BC, Manual ACM Others: Reserved - do not use

0x01A798D8 MDSS_MDP_AADC_1_APICAL_T_FILTER_CONTROL**Type:** RW**Reset State:** 0x00000001

Recursion depth of temporal filter

double buffered: No

MDSS_MDP_AADC_1_APICAL_T_FILTER_CONTROL

Bits	Name	Description
7:0	T_FILTER_CONTROL	The port t_filter_control sets the recursion depth of temporal filtering of photosensor measurements and is used to smooth changes in the strength of DRC in response to rapid changes in photosensor data. A larger value increases the recursion depth in the averaging of time-historic photosensor measurements and results in a slower response to changes in ambient light.

0x01A798DC MDSS_MDP_AADC_1_APICAL_BACKLIGHT_MINMAX**Type:** RW**Reset State:** 0xFFFF2800

Apical Backlight Min/Max Value

double buffered: No

MDSS_MDP_AADC_1_APICAL_BACKLIGHT_MINMAX

Bits	Name	Description
31:16	BACKLIGHT_MAX	Maximum output backlight value permitted.
15:0	BACKLIGHT_MIN	Minimum output backlight value permitted.

0x01A798E0 MDSS_MDP_AADC_1_APICAL_BACKLIGHT**Type:** RW**Reset State:** 0x00008000

Apical Screen brightness Value.

MDSS_MDP_AADC_1_APICAL_BACKLIGHT

Bits	Name	Description
15:0	BACKLIGHT	Screen brightness on a scale of 0 to backlight_scale.

0x01A798E8 MDSS_MDP_AADC_1_APICAL_BACKLIGHT_SCALE**Type:** RW**Reset State:** 0x0000FFFF

Apical Screen brightness Value.

double buffered: No

MDSS_MDP_AADC_1_APICAL_BACKLIGHT_SCALE

Bits	Name	Description
15:0	BACKLIGHT_SCALE	Digital value corresponding to the maximum possible backlight input value. The port backlight_scale should be set equal to the digital value corresponding to the maximum possible backlight brightness that the display can produce. For example, if the backlight level is 16-bit, then backlight_scale should be set to 0xFFFF, if it is an 8-bit range, then backlight_scale should be set to 0x00FF.

0x01A798EC MDSS_MDP_AADC_1_APICAL_AMBIENT_LIGHT

Type: RW

Reset State: 0x00000000

Ambient light level from photosensor

MDSS_MDP_AADC_1_APICAL_AMBIENT_LIGHT

Bits	Name	Description
15:0	AMBIENT_LIGHT	Real time control input from photosensor The core takes the input of the ambient light level in lux via the parameter ambient_light. If the value is available in linear lux units, then it should be entered directly via this parameter. For light sensors which provide a non-linear measure of ambient light, the LUT al_calib_lut should be programmed to convert this value into linear lux units.

0x01A798F0 MDSS_MDP_AADC_1_APICAL_AMBIENT_LIGHT_MIN

Type: RW

Reset State: 0x0000000E

Minimum Ambient light Register

double buffered: No

MDSS_MDP_AADC_1_APICAL_AMBIENT_LIGHT_MIN

Bits	Name	Description
15:0	AMBIENT_LIGHT_MIN	This value represents the incoming ambient light level which backlight_out is set to backlight_min.

0x01A798F4 MDSS_MDP_AADC_1_APICAL_FILTER**Type:** RW**Reset State:** 0x000606CA

Filter values for the input ambient_light

double buffered: No

MDSS_MDP_AADC_1_APICAL_FILTER

Bits	Name	Description
23:16	FILTER_B	This value sets the ambient light-independent component of the stability threshold, below which deviations in photosensor measurements do not lead to a response by the backlight. Larger values result in greater stability to changes in photosensor measurements. This component of the threshold becomes increasingly influential as average ambient light decreases.
15:0	FILTER_A	FilterA Settings [11:9]: Sets the recursion depth of the temporal filter used to smooth out rapid changes in detected ambient light. The larger this parameter, the greater the recursion depth, and the more slowly the backlight responds to changes in ambient light. [8:6] : Sets the ambient light-dependent component of the stability threshold, below which deviations in photosensor measurements do not lead to a response by the backlight. Larger values result in greater stability to changes in photosensor measurements. This component of the threshold becomes increasingly influential as average ambient light increases. [5:3] : Sets the rate at which backlight responds to changes in photosensor measurements, within the stability range established by the values in bits [8:6] and filter_b. [2:0] : Sets the asymmetry between the rate of decrease of backlight compared to the rate of increase of backlight. A value of 0 implies symmetric response, while increasing this value leads to a faster increase in backlight compared to decrease

0x01A798F8 MDSS_MDP_AADC_1_APICAL_CONFIG_BUFFER**Type:** RW**Reset State:** 0x00000003

Double buffering control for Video Pipeline Configuration Parameters

Together with config_buffer_global real time input for iridix core, which is driven by vp_db_update, these control the double buffering of the video pipeline configuration parameters inside Iridix core

double buffered: No

MDSS_MDP_AADC_1_APICAL_CONFIG_BUFFER

Bits	Name	Description
1:0	CONFIG_BUFFER_MODE	config_buffer_mode = 0 (dec) (Pass through): Allows the input configuration ports to pass through to the registers within the core. The internal configuration registers are updated as soon as the ports are changed. config_buffer_mode = 1 (dec) (Block Access): Stops the internal configuration registers from being updated even when the input port values change. The previous register values are held till the next update is allowed. config_buffer_mode = 2 (dec) (Update on VBI): Allows the internal configuration registers only to be updated during the vertical blanking interval (VBI). This ensures that the configuration is never updated during an active frame. config_buffer_mode = 3 (dec) (External): This method is based on the config_buffer_global signal and updates the internal configuration registers when the config_buffer_global signal is asserted. This signal can be generated externally by user logic.

0x01A79900 MDSS_MDP_AADC_1_APICAL_LUT_AL_CALIB1_0**Type:** RW**Reset State:** 0x08000000

Calibration Lookup Table Entry 1-0

The Calibration Lookup Table port geometry is 33 words, each of which is 16 bits.

This LUT is used to linearize the ambient light data

double buffered: No

MDSS_MDP_AADC_1_APICAL_LUT_AL_CALIB1_0

Bits	Name	Description
31:16	LUT_AL_CALIB1	Calibration Lookup Table Entry 1.
15:0	LUT_AL_CALIB0	Calibration Lookup Table Entry 0.

0x01A79904 MDSS_MDP_AADC_1_APICAL_LUT_AL_CALIB3_2**Type:** RW**Reset State:** 0x18001000

Calibration Lookup Table Entry 3-2

The Calibration Lookup Table port geometry is 33 words, each of which is 16 bits.

This LUT is used to linearize the ambient light data

double buffered: No

MDSS_MDP_AADC_1_APICAL_LUT_AL_CALIB3_2

Bits	Name	Description
31:16	LUT_AL_CALIB3	Calibration Lookup Table Entry 3.
15:0	LUT_AL_CALIB2	Calibration Lookup Table Entry 2.

0x01A79908 MDSS_MDP_AADC_1_APICAL_LUT_AL_CALIB5_4

Type: RW

Reset State: 0x28002000

Calibration Lookup Table Entry 5-4

The Calibration Lookup Table port geometry is 33 words, each of which is 16 bits.

This LUT is used to linearize the ambient light data

double buffered: No

MDSS_MDP_AADC_1_APICAL_LUT_AL_CALIB5_4

Bits	Name	Description
31:16	LUT_AL_CALIB5	Calibration Lookup Table Entry 5.
15:0	LUT_AL_CALIB4	Calibration Lookup Table Entry 4.

0x01A7990C MDSS_MDP_AADC_1_APICAL_LUT_AL_CALIB7_6

Type: RW

Reset State: 0x38003000

Calibration Lookup Table Entry 7-6

The Calibration Lookup Table port geometry is 33 words, each of which is 16 bits.

This LUT is used to linearize the ambient light data

double buffered: No

MDSS_MDP_AADC_1_APICAL_LUT_AL_CALIB7_6

Bits	Name	Description
31:16	LUT_AL_CALIB7	Calibration Lookup Table Entry 7.

MDSS_MDP_AADC_1_APICAL_LUT_AL_CALIB7_6 (cont.)

Bits	Name	Description
15:0	LUT_AL_CALIB6	Calibration Lookup Table Entry 6.

0x01A79910 MDSS_MDP_AADC_1_APICAL_LUT_AL_CALIB9_8**Type:** RW**Reset State:** 0x48004000

Calibration Lookup Table Entry 9-8

The Calibration Lookup Table port geometry is 33 words, each of which is 16 bits.

This LUT is used to linearize the ambient light data

double buffered: No

MDSS_MDP_AADC_1_APICAL_LUT_AL_CALIB9_8

Bits	Name	Description
31:16	LUT_AL_CALIB9	Calibration Lookup Table Entry 9.
15:0	LUT_AL_CALIB8	Calibration Lookup Table Entry 8.

0x01A79914 MDSS_MDP_AADC_1_APICAL_LUT_AL_CALIB11_10**Type:** RW**Reset State:** 0x58005000

Calibration Lookup Table Entry 11-10

The Calibration Lookup Table port geometry is 33 words, each of which is 16 bits.

This LUT is used to linearize the ambient light data

double buffered: No

MDSS_MDP_AADC_1_APICAL_LUT_AL_CALIB11_10

Bits	Name	Description
31:16	LUT_AL_CALIB11	Calibration Lookup Table Entry 11.
15:0	LUT_AL_CALIB10	Calibration Lookup Table Entry 10.

0x01A79918 MDSS_MDP_AADC_1_APICAL_LUT_AL_CALIB13_12**Type:** RW**Reset State:** 0x68006000

Calibration Lookup Table Entry 13-12

The Calibration Lookup Table port geometry is 33 words, each of which is 16 bits.

This LUT is used to linearize the ambient light data

double buffered: No

MDSS_MDP_AADC_1_APICAL_LUT_AL_CALIB13_12

Bits	Name	Description
31:16	LUT_AL_CALIB13	Calibration Lookup Table Entry 13.
15:0	LUT_AL_CALIB12	Calibration Lookup Table Entry 12.

0x01A7991C MDSS_MDP_AADC_1_APICAL_LUT_AL_CALIB15_14**Type:** RW**Reset State:** 0x78007000

Calibration Lookup Table Entry 15-14

The Calibration Lookup Table port geometry is 33 words, each of which is 16 bits.

This LUT is used to linearize the ambient light data

double buffered: No

MDSS_MDP_AADC_1_APICAL_LUT_AL_CALIB15_14

Bits	Name	Description
31:16	LUT_AL_CALIB15	Calibration Lookup Table Entry 15.
15:0	LUT_AL_CALIB14	Calibration Lookup Table Entry 14.

0x01A79920 MDSS_MDP_AADC_1_APICAL_LUT_AL_CALIB17_16**Type:** RW**Reset State:** 0x87FF8000

Calibration Lookup Table Entry 17-16

The Calibration Lookup Table port geometry is 33 words, each of which is 16 bits.

This LUT is used to linearize the ambient light data

double buffered: No

MDSS_MDP_AADC_1_APICAL_LUT_AL_CALIB17_16

Bits	Name	Description
31:16	LUT_AL_CALIB17	Calibration Lookup Table Entry 17.
15:0	LUT_AL_CALIB16	Calibration Lookup Table Entry 16.

0x01A79924 MDSS_MDP_AADC_1_APICAL_LUT_AL_CALIB19_18

Type: RW

Reset State: 0x97FF8FFF

Calibration Lookup Table Entry 19-18

The Calibration Lookup Table port geometry is 33 words, each of which is 16 bits.

This LUT is used to linearize the ambient light data

double buffered: No

MDSS_MDP_AADC_1_APICAL_LUT_AL_CALIB19_18

Bits	Name	Description
31:16	LUT_AL_CALIB19	Calibration Lookup Table Entry 19.
15:0	LUT_AL_CALIB18	Calibration Lookup Table Entry 18.

0x01A79928 MDSS_MDP_AADC_1_APICAL_LUT_AL_CALIB21_20

Type: RW

Reset State: 0xA7FF9FFF

Calibration Lookup Table Entry 21-20

The Calibration Lookup Table port geometry is 33 words, each of which is 16 bits.

This LUT is used to linearize the ambient light data

double buffered: No

MDSS_MDP_AADC_1_APICAL_LUT_AL_CALIB21_20

Bits	Name	Description
31:16	LUT_AL_CALIB21	Calibration Lookup Table Entry 21.

MDSS_MDP_AADC_1_APICAL_LUT_AL_CALIB21_20 (cont.)

Bits	Name	Description
15:0	LUT_AL_CALIB20	Calibration Lookup Table Entry 20.

0x01A7992C MDSS_MDP_AADC_1_APICAL_LUT_AL_CALIB23_22**Type:** RW**Reset State:** 0xB7FFAFFF

Calibration Lookup Table Entry 23-22

The Calibration Lookup Table port geometry is 33 words, each of which is 16 bits.

This LUT is used to linearize the ambient light data

double buffered: No

MDSS_MDP_AADC_1_APICAL_LUT_AL_CALIB23_22

Bits	Name	Description
31:16	LUT_AL_CALIB23	Calibration Lookup Table Entry 23.
15:0	LUT_AL_CALIB22	Calibration Lookup Table Entry 22.

0x01A79930 MDSS_MDP_AADC_1_APICAL_LUT_AL_CALIB25_24**Type:** RW**Reset State:** 0xC7FFBFFF

Calibration Lookup Table Entry 25-24

The Calibration Lookup Table port geometry is 33 words, each of which is 16 bits.

This LUT is used to linearize the ambient light data

double buffered: No

MDSS_MDP_AADC_1_APICAL_LUT_AL_CALIB25_24

Bits	Name	Description
31:16	LUT_AL_CALIB25	Calibration Lookup Table Entry 25.
15:0	LUT_AL_CALIB24	Calibration Lookup Table Entry 24.

0x01A79934 MDSS_MDP_AADC_1_APICAL_LUT_AL_CALIB27_26**Type:** RW**Reset State:** 0xD7FFCFFF

Calibration Lookup Table Entry 27-26

The Calibration Lookup Table port geometry is 33 words, each of which is 16 bits.

This LUT is used to linearize the ambient light data

double buffered: No

MDSS_MDP_AADC_1_APICAL_LUT_AL_CALIB27_26

Bits	Name	Description
31:16	LUT_AL_CALIB27	Calibration Lookup Table Entry 27.
15:0	LUT_AL_CALIB26	Calibration Lookup Table Entry 26.

0x01A79938 MDSS_MDP_AADC_1_APICAL_LUT_AL_CALIB29_28**Type:** RW**Reset State:** 0xE7FFDFFF

Calibration Lookup Table Entry 29-28

The Calibration Lookup Table port geometry is 33 words, each of which is 16 bits.

This LUT is used to linearize the ambient light data

double buffered: No

MDSS_MDP_AADC_1_APICAL_LUT_AL_CALIB29_28

Bits	Name	Description
31:16	LUT_AL_CALIB29	Calibration Lookup Table Entry 29.
15:0	LUT_AL_CALIB28	Calibration Lookup Table Entry 28.

0x01A7993C MDSS_MDP_AADC_1_APICAL_LUT_AL_CALIB31_30**Type:** RW**Reset State:** 0xF7FFEFFF

Calibration Lookup Table Entry 31-30

The Calibration Lookup Table port geometry is 33 words, each of which is 16 bits.

This LUT is used to linearize the ambient light data

double buffered: No

MDSS_MDP_AADC_1_APICAL_LUT_AL_CALIB31_30

Bits	Name	Description
31:16	LUT_AL_CALIB31	Calibration Lookup Table Entry 31.
15:0	LUT_AL_CALIB30	Calibration Lookup Table Entry 30.

0x01A79940 MDSS_MDP_AADC_1_APICAL_LUT_AL_CALIB32

Type: RW

Reset State: 0xFFFF0000

Calibration Lookup Table Entry 32

The Calibration Lookup Table port geometry is 33 words, each of which is 16 bits.

This LUT is used to linearize the ambient light data

double buffered: No

MDSS_MDP_AADC_1_APICAL_LUT_AL_CALIB32

Bits	Name	Description
31:16	LUT_AL_CALIB32	Calibration Lookup Table Entry 32.

0x01A79944 MDSS_MDP_AADC_1_APICAL_DRC_IN

Type: RW

Reset State: 0x00000000

Target strength of dynamic range compression.

MDSS_MDP_AADC_1_APICAL_DRC_IN

Bits	Name	Description
15:0	DRC_IN	Target strength of dynamic range compression.

0x01A79948 MDSS_MDP_AADC_1_APICAL_START_CALC

Type: RW

Reset State: 0x00000000

Start calculator pulse.

double buffered: NO

MDSS_MDP_AADC_1_APICAL_START_CALC

Bits	Name	Description
0	START_CALC	Start calculation register Start-Done Protocol of non-Video Pipeline Parameters: Double buffering is not required for the Assertive Display core parameters. Instead a simple start-done protocol is used. The external host will update the parameters and then transition the start_calc input from 0 to 1. After a rising edge is detected on this input, the Assertive Display core's calculator will use the input parameters to calculate the backlight and iridix module strength and the optimum backlight. When the results are available the calc_done status flag is asserted about 500 clock cycles after this edge. In a typical use case this calc_done flag can be used as an interrupt to start the next parameter update cycle. The calc_done flag is held high until the next start_calc rising edge arrives.

0x01A7994C MDSS_MDP_AADC_1_APICAL_STRENGTH_OUT

Type: R

Reset State: 0x00000000

Strength of iridix core output

MDSS_MDP_AADC_1_APICAL_STRENGTH_OUT

Bits	Name	Description
7:0	STRENGTH_OUT	The primary output of the calculator, passed to the internal iridix module's strength_ir signal. This is included as an output from the core for debug purposes only.

0x01A79950 MDSS_MDP_AADC_1_APICAL_DRC_OUT

Type: R

Reset State: 0x00000000

Calculated dynamic range compression output.

MDSS_MDP_AADC_1_APICAL_DRC_OUT

Bits	Name	Description
15:0	DRC_OUT	Calculated dynamic range compression output.

0x01A79954 MDSS_MDP_AADC_1_APICAL_BACKLIGHT_OUT**Type:** R**Reset State:** 0x00000000

Calculated backlight setting.

MDSS_MDP_AADC_1_APICAL_BACKLIGHT_OUT

Bits	Name	Description
15:0	BACKLIGHT_OUT	

0x01A79958 MDSS_MDP_AADC_1_APICAL_CALC_DONE**Type:** R**Reset State:** 0x00000000

Calculator operations finished flag.

MDSS_MDP_AADC_1_APICAL_CALC_DONE

Bits	Name	Description
0	CALC_DONE	Start calculation register Start-Done Protocol of non-Video Pipeline Parameters: Double buffering is not required for the Assertive Display core parameters. Instead a simple start-done protocol is used. The external host will update the parameters and then transition the start_calc input from 0 to 1. After a rising edge is detected on this input, the Assertive Display core's calculator will use the input parameters to calculate the backlight and iridix module strength and the optimum backlight. When the results are available the calc_done status flag is asserted about 500 clock cycles after this edge. In a typical use case this calc_done flag can be used as an interrupt to start the next parameter update cycle. The calc_done flag is held high until the next start_calc rising edge arrives.

17.18 MDSS_DSI_0

Basic description of the dsi controller block

17.18.1 MDSS_DSI_0_DSI

DSI

0x01A98004 MDSS_DSI_0_CTRL

Type: RW

Reset State: 0x00000000

DSI Control Register provides general control to the protocol, lane management and PHY layers

MDSS_DSI_0_CTRL

Bits	Name	Description
30	NEW_INTERLEAVE_MODE_EN	unused (Enable new interleave mode with H/W arbitration between long commands and BLLP periods.)
29:28	PACKET_BYTE_MSB_LSB_FLIP	Swap the bit ordering of each byte of a packet 0: no swap (bit 0 is lsb, bit 7 is msb) 1: swap each byte of payload (bit 0 is msb, bit 7 is lsb) 2: swap each byte of packet (bit 0 is msb, bit 7 is lsb)
26	RESERVED	Reserved
24	CRC_CHK_EN	Enable CRC verification of the received packets
20	ECC_CHK_EN	Enable ECC verification of the received packets
8	CLKLN_EN	Enable clock lane of the PHY
7	DLN3_EN	Enable data lane 3 of the PHY
6	DLN2_EN	Enable data lane 2 of the PHY
5	DLN1_EN	Enable data lane 1 of the PHY
4	DLN0_EN	Enable data lane 0 of the PHY
2	CMD_MODE_EN	Enable the Command Mode Engine.
1	VIDEO_MODE_EN	Enable the Video Mode Engine.
0	DSI_EN	Enable the DSI controller.

0x01A98008 MDSS_DSI_0_STATUS**Type:** RW**Reset State:** 0x00000000

DSI Status Register provides status of the DSI controller

MDSS_DSI_0_STATUS

Bits	Name	Type	Description
31	INTERLEAVE_OP_CONTENTION_CLR	W	writing a 1 to this field will clear this bit
31	INTERLEAVE_OP_CONTENTION	R	The controller generates command mode packets and video mode packets at the same time (i.e.: the command mode packets are longer than one BLLP period)
6	PHY_RESET_BUSY	R	DSI interface is busy sending the panel reset through the link
5	GENERIC_TRIGGER_BUSY	R	DSI interface is busy sending the generic trigger through the link
4	BTA_BUSY	R	DSI interface is busy waiting for read response, acknowledge or tearing effect signal from the peripheral
3	VIDEO_MODE_ENGINE_BUSY	R	Video Mode Engine is busy sending data from display engine
2	CMD_MODE_MDP_BUSY	R	Command Mode Engine is busy sending data from display engine
1	CMD_MODE_DMA_BUSY	R	Command Mode Engine is busy sending data from memory or reading data from the display panel
0	CMD_MODE_ENGINE_BUSY	R	Command Mode Engine Busy

0x01A9800C MDSS_DSI_0_FIFO_STATUS**Type:** RW**Reset State:** 0x11111000

Status of FIFOs in the DSI controller

MDSS_DSI_0_FIFO_STATUS

Bits	Name	Type	Description
31	DLN3_HS_FIFO_UNDERFLOW_CLR	W	writing a 1 to this field will clear this bit
31	DLN3_HS_FIFO_UNDERFLOW	R	The FIFO for HS data on Lane 3 underflows

MDSS_DSI_0_FIFO_STATUS (cont.)

Bits	Name	Type	Description
30	DLN3_HS_FIFO_OVERFLOW	R	The FIFO of Lane 3 overflows
30	DLN3_HS_FIFO_OVERFLOW_CLR	W	writing a 1 to this field will clear this bit
29	DLN3_HS_FIFO_FULL	R	The FIFO of Lane 3 is full
28	DLN3_HS_FIFO_EMPTY	R	The FIFO of Lane 3 is empty
27	DLN2_HS_FIFO_UNDERFLOW_CLR	W	writing a 1 to this field will clear this bit
27	DLN2_HS_FIFO_UNDERFLOW	R	The FIFO for HS data on Lane 2 underflows
26	DLN2_HS_FIFO_OVERFLOW_CLR	W	writing a 1 to this field will clear this bit
26	DLN2_HS_FIFO_OVERFLOW	R	The FIFO for HS data on Lane 2 overflows
25	DLN2_HS_FIFO_FULL	R	The FIFO for HS data on Lane 2 is full
24	DLN2_HS_FIFO_EMPTY	R	The FIFO for HS data on Lane 2 is empty
23	DLN1_HS_FIFO_UNDERFLOW	R	The FIFO for HS data on Lane 1 underflows
23	DLN1_HS_FIFO_UNDERFLOW_CLR	W	writing a 1 to this field will clear this bit
22	DLN1_HS_FIFO_OVERFLOW	R	The FIFO of Lane 1 overflows
22	DLN1_HS_FIFO_OVERFLOW_CLR	W	writing a 1 to this field will clear this bit
21	DLN1_HS_FIFO_FULL	R	The FIFO of Lane 1 is full
20	DLN1_HS_FIFO_EMPTY	R	The FIFO of Lane 1 is empty
19	DLN0_HS_FIFO_UNDERFLOW	R	The FIFO for HS data on Lane 0 underflows
19	DLN0_HS_FIFO_UNDERFLOW_CLR	W	writing a 1 to this field will clear this bit
18	DLN0_HS_FIFO_OVERFLOW_CLR	W	writing a 1 to this field will clear this bit
18	DLN0_HS_FIFO_OVERFLOW	R	The FIFO for HS data on Lane 0 overflows
17	DLN0_HS_FIFO_FULL	R	The FIFO for HS data on Lane 0 is full
16	DLN0_HS_FIFO_EMPTY	R	The FIFO for HS data on Lane 0 is empty
14	DLN0_LP_FIFO_OVERFLOW_CLR	W	unused

MDSS_DSI_0_FIFO_STATUS (cont.)

Bits	Name	Type	Description
14	DLN0_LP_FIFO_OVERFLOW	R	unused
13	DLN0_LP_FIFO_FULL	R	The FIFO for LP data on Lane 0 is full
12	DLN0_LP_FIFO_EMPTY	R	The FIFO for LP data on Lane 0 is empty
10	CMD_DMA_FIFO_UNDERFLOW	R	The Command mode engine dmafifo underflows
10	CMD_DMA_FIFO_UNDERFLOW_CLR	W	writing a 1 to this field will clear this bit
9	CMD_DMA_FIFO_WR_WATERMARK_REACH	R	The Command mode engine dmafifo write watermark is reached
9	CMD_DMA_FIFO_WR_WATERMARK_REACH_CLR	W	writing a 1 to this field will clear this bit
8	CMD_DMA_FIFO_RD_WATERMARK_REACH_CLR	W	writing a 1 to this field will clear this bit
8	CMD_DMA_FIFO_RD_WATERMARK_REACH	R	The Command mode engine dmafifo read watermark is reached
7	CMD_MDP_FIFO_UNDERFLOW_CLR	W	writing a 1 to this field will clear this bit
7	CMD_MDP_FIFO_UNDERFLOW	R	command mode pixel stream from MDP underflows
3	VIDEO_MDP_FIFO_UNDERFLOW	R	Unused
3	VIDEO_MDP_FIFO_UNDERFLOW_CLR	W	Unused
0	VIDEO_MDP_FIFO_OVERFLOW_CLR	W	Unused
0	VIDEO_MDP_FIFO_OVERFLOW	R	Unused

0x01A98010 MDSS_DSI_0_VIDEO_MODE_CTRL**Type:** RW**Reset State:** 0x00008000

DSI Video Mode Engine Control Register

MDSS_DSI_0_VIDEO_MODE_CTRL

Bits	Name	Description
31	LAST_LINE_INTERLEAVE_EN	Allow command mode operation interleaved on last line of video stream 0: disable 1: enable
28	PULSE_MODE_OPT	Select pulse mode option 0: no HSA and HE following VS/VE packet 1: send HSA and HE following VS/VE packet
24	HFP_PWR_MODE	Select power mode during the HFP period 0: send blanking packets in High Speed mode 1: low power stop mode (LP-11)
20	HBP_PWR_MODE	Select power mode during the HBP period 0: send blanking packets in High Speed mode 1: low power stop mode (LP-11)
16	HSA_PWR_MODE	Select power mode during the HSA period 0: send blanking packets in High Speed mode 1: low power stop mode (LP-11)
15	EOF_BLLP_PWR_MODE	Select power mode for the BLLP of the last line of a frame 0: send blanking packets during BLLP in high speed mode and block Command Mode packets 1: low power stop mode (LP-11) or let Command Mode Engine send packets in HS or LP mode
12	BLLP_PWR_MODE	DSI Power Mode for packets sent during BLLP period 0: send blanking packets during BLLP in high speed mode and block Command Mode packets 1: low power stop mode (LP-11) or let Command Mode Engine send packets in HS or LP mode
9:8	TRAFFIC_MODE	DSI video mode traffic sequence 0: non-burst mode with sync pulses 1: non-burst mode with sync start events 2: burst mode
5:4	DST_FORMAT	Pixel format. If there is interleaved Command Mode pixel packets (generated from MDP), the pixel depth of the Command Mode pixel packet should be greater or equal to the pixel depth of the Video mode pixel packet. 0: RGB565 1: RGB666 (packed) 2: RGB666 (loosely packed) 3: RGB888
1:0	VC	Virtual channel identifier

0x01A98014 MDSS_DSI_0_VIDEO_MODE_SYNC_DATATYPE**Type:** RW**Reset State:** 0x31211101

DSI Video Mode Sync Datatype Register

MDSS_DSI_0_VIDEO_MODE_SYNC_DATATYPE

Bits	Name	Description
29:24	HE	Specify the datatype of a hsync end packet
21:16	HS	Specify the datatype of a hsync start packet
13:8	VE	Specify the datatype of a vsync end packet
5:0	VS	Specify the datatype of a vsync start packet

0x01A98018 MDSS_DSI_0_VIDEO_MODE_PIXEL_DATATYPE**Type:** RW**Reset State:** 0x3E2E1E0E

DSI Video Mode Pixel Datatype Register

MDSS_DSI_0_VIDEO_MODE_PIXEL_DATATYPE

Bits	Name	Description
29:24	RGB888	Specify the datatype of an RGB888 pixel stream packet
21:16	RGB666	Specify the datatype of a loosely-packed RGB666 pixel stream packet
13:8	RGB666_PACKED	Specify the datatype of a packed RGB666 pixel stream packet
5:0	RGB565	Specify the datatype of an RGB565 pixel stream packet

0x01A9801C MDSS_DSI_0_VIDEO_MODE_BLANKING_DATATYPE**Type:** RW**Reset State:** 0x00001900

DSI Video Mode Blanking Packet Datatype Register

MDSS_DSI_0_VIDEO_MODE_BLANKING_DATATYPE

Bits	Name	Description
13:8	BLANK_PKT_DATATYPE	Specify the datatype of a blanking packet

MDSS_DSI_0_VIDEO_MODE_BLANKING_DATATYPE (cont.)

Bits	Name	Description
7:0	BLANK_PKT_DATA	This byte will be repeated in the payload portion of the blanking packet

0x01A98020 MDSS_DSI_0_VIDEO_MODE_DATA_CTRL**Type:** RW**Reset State:** 0x00000000

DSI Video Mode Data Control Register

MDSS_DSI_0_VIDEO_MODE_DATA_CTRL

Bits	Name	Description
14:12	RGB_SWAP	Swaps the R, G, and B channels of the dst pixels. This field should be used when different R,G, and B orderings are required. For example, if RGB_SWAP = 1 , RGB565 will become RBG556.
8	B_SEL	Bit Swaps the B Channel of the dst pixel. This field is used to reverse the ordering of the bits that is outputted for the color red. For example, for RGB888 color format, if the red component is 10000000 then if this field is enabled, 00000001 is output.
4	G_SEL	Bit Swaps the G Channel of the dst pixel. This field is used to reverse the ordering of the bits that is outputted for the color red. For example, for RGB888 color format, if the red component is 10000000 then if this field is enabled, 00000001 is output.
0	R_SEL	Bit Swaps the R Channel of the dst pixel. This field is used to reverse the ordering of the bits that is outputted for the color red. For example, for RGB888 color format, if the red component is 10000000 then if this field is enabled, 00000001 is output.

0x01A98024 MDSS_DSI_0_VIDEO_MODE_ACTIVE_H**Type:** RW**Reset State:** 0x00000000

Active Horizontal Display Region register.

The Active_H_Disp register defines the horizontal characteristics of the active window. (NOTE: video mode parameters are 0-based.)

double buffered: YES when DSI double-buffer is enabled by set DSI_TIMING_DB_MODE to 1.

MDSS_DSI_0_VIDEO_MODE_ACTIVE_H

Bits	Name	Description
31:16	ACTIVE_H_END	ACTIVE_H_END field tells the video mode engine where the horizontal viewing area ends (minus one) in pixel clock cycle relative to the horizontal sync signal.
15:0	ACTIVE_H_START	ACTIVE_H_START field tells the video mode engine where the horizontal viewing area starts (minus one) in pixel clock cycle relative to the horizontal sync signal.

0x01A98028 MDSS_DSI_0_VIDEO_MODE_ACTIVE_V**Type:** RW**Reset State:** 0x00000000

CRTC Active Vertical Display Region register.

The Active_V_Disp register defines the vertical characteristics of the active window. (NOTE: video mode parameters are 0-based.)

double buffered: YES when DSI double-buffer is enabled by set DSI_TIMING_DB_MODE to 1.

MDSS_DSI_0_VIDEO_MODE_ACTIVE_V

Bits	Name	Description
31:16	ACTIVE_V_END	ACTIVE_V_END field tells the video mode engine where the horizontal viewing area ends (minus one) in lines relative to the vertical sync signal
15:0	ACTIVE_V_START	ACTIVE_V_START field tells the video mode engine where the vertical viewing area starts (minus one) in lines relative to the vertical sync signal

0x01A9802C MDSS_DSI_0_VIDEO_MODE_TOTAL**Type:** RW**Reset State:** 0x00000000

Total Scan Area register.

TOTAL register defines the size of the total possible scanning area (NOTE: video mode parameters are 0-based.)

double buffered: YES when DSI double-buffer is enabled by set DSI_TIMING_DB_MODE to 1.

MDSS_DSI_0_VIDEO_MODE_TOTAL

Bits	Name	Description
31:16	VIDEO_V_TOTAL	Number of lines in one frame minus one. This value is equivalent to the period of the vertical sync
15:0	VIDEO_H_TOTAL	Number of pixels in one scan line (horizontal line) of the LCD controller minus one. This value is equivalent to the period of the horizontal sync signal (in pixel clock cycles).

0x01A98030 MDSS_DSI_0_VIDEO_MODE_HSYNC**Type:** RW**Reset State:** 0x00000000

HSync Control register which controls the horizontal screen position at which a horizontal sync packet is sent by the DSI controller to the panel. (NOTE: video mode parameters are 0-based.)

double buffered: YES when DSI double-buffer is enabled by set DSI_TIMING_DB_MODE to 1.

MDSS_DSI_0_VIDEO_MODE_HSYNC

Bits	Name	Description
31:16	HS_END	The position (minus one) where the horizontal sync ends in one line.
15:0	HS_START	The position (minus one) where the horizontal sync starts in one line.

0x01A98034 MDSS_DSI_0_VIDEO_MODE_VSYNC**Type:** RW**Reset State:** 0x00000000

VSynC Control register which controls the horizontal screen position at which a vertical sync packet is sent by the DSI controller to the panel. (NOTE: video mode parameters are 0-based.)

double buffered: YES when DSI double-buffer is enabled by set DSI_TIMING_DB_MODE to 1.

MDSS_DSI_0_VIDEO_MODE_VSYNC

Bits	Name	Description
31:16	VS_END	The position (minus one) where the vertical sync ends in one line.
15:0	VS_START	The position (minus one) where the vertical sync starts in one line.

0x01A98038 MDSS_DSI_0_VIDEO_MODE_VSYNC_VPOS**Type:** RW**Reset State:** 0x00000000

VSync Control register which controls the vertical screen position at which a vertical sync packet is sent by the DSI controller to the panel. (NOTE: video mode parameters are 0-based.)

double buffered: YES when DSI double-buffer is enabled by set DSI_TIMING_DB_MODE to 1.

MDSS_DSI_0_VIDEO_MODE_VSYNC_VPOS

Bits	Name	Description
31:16	VS_VPOS_END	The line (minus one) where the vertical sync ends in one frame.
15:0	VS_VPOS_START	The line (minus one) where the vertical sync starts in one frame.

0x01A9803C MDSS_DSI_0_COMMAND_MODE_DMA_CTRL**Type:** RW**Reset State:** 0x00000000

DSI Command Mode Control Register for DMA data control

MDSS_DSI_0_COMMAND_MODE_DMA_CTRL

Bits	Name	Description
31	BROADCAST_EN	Enable the DSI core to send packets from DMA at the same time with the other core (this bit has to be set for the other DSI core as well). If broadcast is enable, the core needs to wait until both cores have fetched entire command buffer before start sending dma packets. When this bit is set, the command buffer size (DSI_DMA_CMD_LENGTH) must be less than 0x200 bytes.
30	BROADCAST_MASTER	Indicate whether this core is the master or slave 0: this core is the slave 1: this core is the master
28	EMBEDDED_MODE	Select packet_type, BTA settings and packet header information 0: from the register 1: from the frame buffer
26	POWER_MODE	Select power mode for data transmission of both embedded and non-embedded commands
24	PACKET_TYPE	Select packet type for non-embedded commands
23:22	VC	Virtual channel identifier for pixel data or non-embedded commands
21:16	DT	Data type for pixel data or non-embedded commands

MDSS_DSI_0_COMMAND_MODE_DMA_CTRL (cont.)

Bits	Name	Description
15:0	WC	Exact number of bytes in one packet for pixel data or non-embedded commands

0x01A98040 MDSS_DSI_0_COMMAND_MODE_MDP_CTRL**Type:** RW**Reset State:** 0x06100006

DSI Command Mode Control Register for MDP data control

MDSS_DSI_0_COMMAND_MODE_MDP_CTRL

Bits	Name	Description
31	LINE_SPLIT	Break a line into small packets
30:28	RGB_SWAP1	Swaps the R, G, and B channels of the dst pixels for stream 1 (if STREAM1_DST_FORMAT_SEL = 1, otherwise rgb swap for stream1 is defined by RGB_SWAP). This field should be used when different R,G, and B orderings are required. For example, if RGB_SWAP = 1 , RGB565 will become RBG556.
27:24	DST_FORMAT1	Pixel format for stream 1 (if STREAM1_DST_FORMAT_SEL = 1, otherwise pixel format for stream1 is defined by DST_FORMAT) 0 = RGB111 1 = reserved 2 = reserved 3 = RGB332 (data_src_width must be in multiple of 2 pixels) 4 = RGB444 (data_src_width must be in multiple of 2 pixels) 5 = reserved 6 = RGB565 7 = RGB666 8 = RGB888 others = reserved
23:20	INTERLEAVE_MAX	Maximum number of Command Mode RGB packets to send within one horizontal blanking period of the Video Mode frame (sw must ensure that they can fit in one BLLP period)
18:16	RGB_SWAP	Swaps the R, G, and B channels of the dst pixels. This field should be used when different R,G, and B orderings are required. For example, if RGB_SWAP = 1 , RGB565 will become RBG556.

MDSS_DSI_0_COMMAND_MODE_MDP_CTRL (cont.)

Bits	Name	Description
15	STREAM1_DST_FORMAT_SEL	Pixel format selection for stream 1. If MDP sends two streams to DSI and these streams have different output formats, then the pixel clock must be set according to the smallest pixel depth between the two. Otherwise, underflow may occur. Ex.: If stream0 is in RGB888 and stream1 is in RGB565, then pixel clock must be greater than dsiclk/2. 0: DST_FORMAT (same as stream0) 1: DST_FORMAT1
13	B_SEL1	Bit Swaps the B Channel of the dst pixel for stream 1 (if STREAM1_DST_FORMAT_SEL = 1, otherwise bit swap for stream1 is defined by B_SEL). This field is used to reverse the ordering of the bits that is outputted for the color red. For example, for RGB888 color format, if the red component is 10000000 then if this field is enabled, 00000001 is output.
12	B_SEL	Bit Swaps the B Channel of the dst pixel. This field is used to reverse the ordering of the bits that is outputted for the color red. For example, for RGB888 color format, if the red component is 10000000 then if this field is enabled, 00000001 is output.
9	G_SEL1	Bit Swaps the G Channel of the dst pixel for stream 1 (if STREAM1_DST_FORMAT_SEL = 1, otherwise bit swap for stream1 is defined by G_SEL). This field is used to reverse the ordering of the bits that is outputted for the color red. For example, for RGB888 color format, if the red component is 10000000 then if this field is enabled, 00000001 is output.
8	G_SEL	Bit Swaps the G Channel of the dst pixel. This field is used to reverse the ordering of the bits that is outputted for the color red. For example, for RGB888 color format, if the red component is 10000000 then if this field is enabled, 00000001 is output.
7	BYTE_MSB_LSB_FLIP1	Swap the bit ordering of each byte of the dst pixel for stream 1 (if STREAM1_DST_FORMAT_SEL = 1, otherwise the byte endian is defined by BYTE_MSB_LSB_FLIP). 0: bit 0 is lsb, bit 7 is msb 1: bit 0 is msb, bit 0 is lsb
6	BYTE_MSB_LSB_FLIP	Swap the bit ordering of each byte of the dst pixel (this is done after the RGB swap and R/G/B_SEL). 0: bit 0 is lsb, bit 7 is msb 1: bit 0 is msb, bit 0 is lsb
5	R_SEL1	Bit Swaps the R Channel of the dst pixel for stream 1 (if STREAM1_DST_FORMAT_SEL = 1, otherwise bit swap for stream1 is defined by R_SEL). This field is used to reverse the ordering of the bits that is outputted for the color red. For example, for RGB888 color format, if the red component is 10000000 then if this field is enabled, 00000001 is output.
4	R_SEL	Bit Swaps the R Channel of the dst pixel. This field is used to reverse the ordering of the bits that is outputted for the color red. For example, for RGB888 color format, if the red component is 10000000 then if this field is enabled, 00000001 is output.

MDSS_DSI_0_COMMAND_MODE_MDP_CTRL (cont.)

Bits	Name	Description
3:0	DST_FORMAT	Destination pixel format. For video mode operation interleave with MDP pixel packets, this is required: command mode pixel depth >= video mode pixel depth (check video mode pixel format in DSI_VIDEO_MODE_PIXEL_DATATYPE) 0 = RGB111 1 = reserved 2 = reserved 3 = RGB332 (data_src_width must be in multiple of 2 pixels) 4 = RGB444 (data_src_width must be in multiple of 2 pixels) 5 = reserved 6 = RGB565 7 = RGB666 8 = RGB888 others = reserved

0x01A98044 MDSS_DSI_0_COMMAND_MODE_MDP_DCS_CMD_CTRL**Type:** RW**Reset State:** 0x00003C2C

This register controls DCS command insertion to MDP data

MDSS_DSI_0_COMMAND_MODE_MDP_DCS_CMD_CTRL

Bits	Name	Description
18	STREAM2_INSERT_DCS_COMMAND	Indicate whether a DCS command has to be inserted as the first byte of the payload of the pixel data packet of the stream 2 0: no command is inserted 1: value in the DCS_COMMAND field will be inserted
17	INSERT_DCS_COMMAND2	Indicate whether a DCS command has to be inserted as the first byte of the payload of the pixel data packet of the stream 1 if COMMAND_MODE_DMA_STREAM_SEL is set to 1 or 2, otherwise it is for stream 0 0: no command is inserted 1: value in the DCS_COMMAND field will be inserted
16	INSERT_DCS_COMMAND	Indicate whether a DCS command has to be inserted as the first byte of payload of the pixel data packet for stream 0 if COMMAND_MODE_DMA_STREAM_SEL is set to 0 or 2, otherwise it is for stream 1 0: no command is inserted 1: value in the DCS_COMMAND field will be inserted
15:8	WR_MEM_CONTINUE	DCS command for write_memory_continue

MDSS_DSI_0_COMMAND_MODE_MDP_DCS_CMD_CTRL (cont.)

Bits	Name	Description
7:0	WR_MEM_START	DCS command for write_memory_start

0x01A98048 MDSS_DSI_0_DMA_CMD_OFFSET**Type:** RW**Reset State:** 0x00000000

SECURE

DSI Command Offset Register

MDSS_DSI_0_DMA_CMD_OFFSET

Bits	Name	Description
31:0	CMD_OFFSET	ZERO 2 : 0 Memory offset (in bytes) for DSI command, qword aligned, i.e.: bit 2:0 are tied to 0. This value has to be multiple of 8.

0x01A9804C MDSS_DSI_0_DMA_CMD_LENGTH**Type:** RW**Reset State:** 0x00000000

SECURE

DSI Command Length Register

MDSS_DSI_0_DMA_CMD_LENGTH

Bits	Name	Description
23:0	CMD_LENGTH	Number of commands (in bytes). This value has to be multiple of 4.

0x01A98050 MDSS_DSI_0_DMA_FIFO_CTRL**Type:** RW**Reset State:** 0x00000000

DSI DMA FIFO Control Register

MDSS_DSI_0_DMA_FIFO_CTRL

Bits	Name	Description
5:4	READ_WATERMARK	Allow fifo reads for a packet when the number of unread entries is above this watermark or larger than the packet size (read_watermark has to be smaller than or equal to write_watermark) 0: FIFO is 1/2 full 1: FIFO is 3/4 full 2: FIFO is 7/8 full 3: FIFO is 15/16 full
1:0	WRITE_WATERMARK	Fills FIFO under the following conditions 0: FIFO is 1/2 empty 1: FIFO is 1/4 empty 2: FIFO is 1/8 empty 3: FIFO is 1/16 empty

0x01A98054 MDSS_DSI_0_DMA_NULL_PACKET_DATA**Type:** RW**Reset State:** 0x00000900

DSI DMA Null Packet Data Register

MDSS_DSI_0_DMA_NULL_PACKET_DATA

Bits	Name	Description
13:8	NULL_DATATYPE	Datatype of null packet
7:0	NULL_DATA	This byte will be repeated in the payload portion of the embedded null packet

0x01A98058 MDSS_DSI_0_COMMAND_MODE_MDP_STREAM0_CTRL**Type:** RW**Reset State:** 0x00000000

DSI MDP stream 0 control register

MDSS_DSI_0_COMMAND_MODE_MDP_STREAM0_CTRL

Bits	Name	Description
31:16	MDP0_WC	Exact number of Word (byte) count for one packet
9:8	MDP0_VC	Virtual channel

MDSS_DSI_0_COMMAND_MODE_MDP_STREAM0_CTRL (cont.)

Bits	Name	Description
5:0	MDP0_DT	Packet data type

0x01A9805C MDSS_DSI_0_COMMAND_MODE_MDP_STREAM0_TOTAL**Type:** RW**Reset State:** 0x00000000

DSI MDP stream 0 frame dimension (NOTE: unlike video mode parameters which are 0-based, these parameters in command mode are 1-based, i.e.: the actual number of pixels are set.)

MDSS_DSI_0_COMMAND_MODE_MDP_STREAM0_TOTAL

Bits	Name	Description
31:16	MDP0_V_TOTAL	Number of lines in 1 frame
15:0	MDP0_H_TOTAL	Number of pixels in 1 line

0x01A98060 MDSS_DSI_0_COMMAND_MODE_MDP_STREAM1_CTRL**Type:** RW**Reset State:** 0x00000000

DSI MDP stream 1 control register

MDSS_DSI_0_COMMAND_MODE_MDP_STREAM1_CTRL

Bits	Name	Description
31:16	MDP1_WC	Exact number of word (byte) count for one packet
9:8	MDP1_VC	Virtual channel
5:0	MDP1_DT	Packet data type

0x01A98064 MDSS_DSI_0_COMMAND_MODE_MDP_STREAM1_TOTAL**Type:** RW**Reset State:** 0x00000000

DSI MDP stream 1 frame dimension (NOTE: unlike video mode parameters which are 0-based, these parameters in command mode are 1-based, i.e.: the actual number of pixels are set.)

MDSS_DSI_0_COMMAND_MODE_MDP_STREAM1_TOTAL

Bits	Name	Description
31:16	MDP1_V_TOTAL	Exact number of lines in 1 frame
15:0	MDP1_H_TOTAL	Exact number of pixels in 1 line

0x01A98068 MDSS_DSI_0_ACK_ERR_STATUS**Type:** RW**Reset State:** 0x00000000

DSI Error Report Register stores errors and acknowledge from the DSI display module.

This register stores errors from the error report packet and error in the ECC and CRC of the returned packet

MDSS_DSI_0_ACK_ERR_STATUS

Bits	Name	Type	Description
28	ACK_CLR	W	clear ACK by writing a '1' and then a '0' to this bit
28	ACK	R	Transmission acknowledged by the DSI display module
24	ERROR	R	There is an error report returned from the DSI display module (or-ed error bits 0 - 15)
24	ERROR_CLR	W	clear ERROR by writing a '1' and then a '0' to this bit
23	RDBK_INCOMPLETE_PACKET_ERR	R	Received incomplete read packet
23	RDBK_INCOMPLETE_PACKET_ERR_CLR	W	clear RDBK_INCOMPLETE_PACKET_ERR by writing a '1' and then a '0' to this bit
20	RDBK_DATA_CRC_ERR_CLR	W	clear RDBK_DATA_CRC_ERR by writing a '1' and then a '0' to this bit
20	RDBK_DATA_CRC_ERR	R	Incorrect CRC detected in read packet
17	RDBK_DATA_MULTI_ECC_ERR	R	Multi-bit ECC detected in read packet and not corrected
17	RDBK_DATA_MULTI_ECC_ERR_CLR	W	clear RDBK_DATA_MULTI_ECC_ERR by writing a '1' and then a '0' to this bit
16	RDBK_DATA_ECC_ERR	R	Single bit ECC error detected in read packet and corrected by DSI controller
16	RDBK_DATA_ECC_ERR_CLR	W	clear RDBK_DATA_ECC_ERR by writing a '1' and then a '0' to this bit
15	PANEL_SPECIFIC_ERR_CLR	W	Clear PANEL_SPECIFIC_ERR by writing a '1' and then a '0' to this bit

MDSS_DSI_0_ACK_ERR_STATUS (cont.)

Bits	Name	Type	Description
15	PANEL_SPECIFIC_ERR	R	This bit correspond to bit 15 (DSI protocol violation) of the error report. Refer to the meaning by this description rather than by the name of the register field
13	PROTOCOL_VIOLATION_CLR	W	Clear Protocol violation by writing a '1' and then a '0' to this bit
13	PROTOCOL_VIOLATION	R	This bit correspond to bit 13 (invalid transmission length) of the error report. Refer to the meaning by this description rather than by the name of the register field.
12	VC_ERR	R	Invalid virtual channel ID detected by the DSI display module
12	VC_ERR_CLR	W	clear VC_ERR by writing a '1' and then a '0' to this bit
11	DT_ERR_CLR	W	clear DT_ERR by writing a '1' and then a '0' to this bit
11	DT_ERR	R	DSI data type not recognized by the DSI display module
10	CRC_ERR_CLR	W	clear CRC_ERR by writing a '1' and then a '0' to this bit
10	CRC_ERR	R	Checksum error for long packet detected by the DSI display module
9	MULTI_ECC_ERR_CLR	W	clear MULTI_ECC_ERR by writing a '1' and then a '0' to this bit
9	MULTI_ECC_ERR	R	Multi-bit ECC detected and not corrected by the DSI display module
8	ECC_ERR	R	Single bit ECC detected and corrected by the DSI display module
8	ECC_ERR_CLR	W	clear ECC_ERR by writing a '1' and then a '0' to this bit
7	CONTENTION	R	Contention Detected by Peripheral
7	CONTENTION_CLR	W	clear CONTENTION by writing a '1' and then a '0' to this bit
6	FALSE_CTRL_ERR_CLR	W	clear FALSE_CTRL_ERR by writing a '1' and then a '0' to this bit
6	FALSE_CTRL_ERR	R	False Control Error detected by the DSI display module
5	TIMEOUT	R	timeout error (Low Power forward transmission, bus turn around, High Speed reverse transmission) detected by the DSI display module
5	TIMEOUT_CLR	W	clear TIMEOUT by writing a '1' and then a '0' to this bit
4	LP_ERR	R	Low-power transmit sync error detected by the DSI display module
4	LP_ERR_CLR	W	clear LP_ERR by writing a '1' and then a '0' to this bit
3	ESC_ERR_CLR	W	clear ESC_ERR by writing a '1' and then a '0' to this bit
3	ESC_ERR	R	Escape Mode Entry Command Error detected by the DSI display module

MDSS_DSI_0_ACK_ERR_STATUS (cont.)

Bits	Name	Type	Description
2	EOT_ERR_CLR	W	clear EOT_ERR by writing a '1' and then a '0' to this bit
2	EOT_ERR	R	EoT Sync Error detected by the DSI display module
1	SOT_SYNC_ERR	R	SoT Sync Error detected by the DSI display module
1	SOT_SYNC_ERR_CLR	W	clear SOT_SYNC_ERR by writing a '1' and then a '0' to this bit
0	SOT_ERR_CLR	W	clear SOT_ERR by writing a '1' and then a '0' to this bit
0	SOT_ERR	R	SoT Error detected by the DSI display module

0x01A9806C MDSS_DSI_0_RDBK_DATA0**Type:** R**Reset State:** 0x00000000

DSI Read Data 0 Register

MDSS_DSI_0_RDBK_DATA0

Bits	Name	Description
31:0	RD_DATA0	Data returned from the DSI display module

0x01A98070 MDSS_DSI_0_RDBK_DATA1**Type:** R**Reset State:** 0x00000000

DSI Read Data 1 Register

MDSS_DSI_0_RDBK_DATA1

Bits	Name	Description
31:0	RD_DATA1	Data returned from the DSI display module

0x01A98074 MDSS_DSI_0_RDBK_DATA2**Type:** R**Reset State:** 0x00000000

DSI Read Data 2 Register

MDSS_DSI_0_RDBK_DATA2

Bits	Name	Description
31:0	RD_DATA2	Data returned from the DSI display module

0x01A98078 MDSS_DSI_0_RDBK_DATA3**Type:** R**Reset State:** 0x00000000

DSI Read Data 3 Register

MDSS_DSI_0_RDBK_DATA3

Bits	Name	Description
31:0	RD_DATA3	Data returned from the DSI display module

0x01A9807C MDSS_DSI_0_RDBK_DATATYPE0**Type:** RW**Reset State:** 0x22211211

DSI Read Response Datatype Register 0

MDSS_DSI_0_RDBK_DATATYPE0

Bits	Name	Description
29:24	DCS_SHORT_RD_2_BYTE	Specify the datatype of a DCS short read response with 2 byte returned
21:16	DCS_SHORT_RD_1_BYTE	Specify the datatype of a DCS short read response with 1 byte returned
13:8	GENERIC_SHORT_RD_2_BYTE	Specify the datatype of a generic short read response with 2 byte returned
5:0	GENERIC_SHORT_RD_1_BYTE	Specify the datatype of a generic short read response with 1 byte returned

0x01A98080 MDSS_DSI_0_RDBK_DATATYPE1**Type:** RW**Reset State:** 0x001C1A02

DSI Read Response Datatype Register 1

MDSS_DSI_0_RDBK_DATATYPE1

Bits	Name	Description
21:16	DCS_LONG_RD	Specify the datatype of a DCS long read response
13:8	GENERIC_LONG_RD	Specify the datatype of a generic long read response
5:0	ERROR_REPORT	Specify the datatype of an acknowledge with error report

0x01A98084 MDSS_DSI_0_TRIG_CTRL**Type:** RW**Reset State:** 0x00000000

DSI Trigger Control Register

When DSI is enabled and CRTC1 or 2 is used for other non-DSI panels, the display engine path trigger must be set to manual internal, otherwise the CRTC will be stalled

MDSS_DSI_0_TRIG_CTRL

Bits	Name	Description
31	TE_SEL	Select TE path 0: Tearing effect (external trigger) returns from data link 1: Tearing effect (external trigger) returns from a dedicated pin
28	COMMAND_MODE_DMA_MDP_ORDER	This bit field indicates whether DMA or MDP path will win the arbitration if both requests the link at the same time 0: DMA has higher priority than display processor 1: Display processor has higher priority than DMA
24	COMMAND_MODE_MDP_REQ_ORDER	If PHY TE is required for one of the MDP stream while the other stream is ready to transmit, this field determines which will win the arbitration 0: Command Mode MDP PHY TE request has higher priority than Command Mode MDP data transmission request 1: Command Mode MDP Data transmission request has higher priority than Command Mode MDP PHY TE request
12	BLOCK_DMA_WITHIN_FRAME	If Command Mode DMA operation is triggered in the middle of a frame of the stream selected by COMMAND_MODE_DMA_STREAM_SEL, delay the operation until end of frame, otherwise service the dma operation right away

MDSS_DSI_0_TRIG_CTRL (cont.)

Bits	Name	Description
9:8	COMMAND_MODE_DMA_STREAM_SEL	If COMMAND_MODE_DMA_TRIGGER_SEL = SOF/EOF or both COMMAND_MODE_DMA_TRIGGER_SEL and COMMAND_MODE_MDP_TRIGGER_SEL = TE or sw+TE, the Command Mode DMA path is triggered at frame boundary of the MDP stream selected by this field. 0 = command dma triggered by stream 0 1 = command dma triggered by stream 1 2 = command dma triggered by stream2 0: stream 0 1: stream 1 2: stream 2 3: reserved
6:4	COMMAND_MODE_MDP_TRIGGER_SEL	TRIGGER_SEL field is used to select which trigger is used for Command Mode display processor path. (The stream to be triggered is selected by COMMAND_MODE_DMA_STREAM_SEL) 0: none 1: reserved0 2: TE 3: reserved1 4: sw trigger 5: reserved2 6: sw trigger and TE 7: reserved3
2:0	COMMAND_MODE_DMA_TRIGGER_SEL	DMA_TRIGGER_SEL field is used to select which trigger is used for Command Mode DMA path. 0: none 1: Start/end of frame (determined by COMMAND_MODE_DMA_MDP_ORDER) 2: TE 3: reserved0 4: sw trigger 5: sw trigger and start/end of frame (COMMAND_MODE_DMA_MDP_ORDER) 6: sw trigger and TE 7: reserved1

0x01A98088 MDSS_DSI_0_EXT_MUX**Type:** RW**Reset State:** 0x00000000

DSI External Pin Mux Register - select IO pin for tearing effect and panel reset

MDSS_DSI_0_EXT_MUX

Bits	Name	Description
31:20	EXT_TE_HSYNC_TOTAL	Defines the total number of hsync per frame
19:8	EXT_TE_HSYNC_TRIG_CN T	Defines the number of hsync needed to trigger the engine if TE_MODE is set to 1
7	EXT_TE_POL	Polarity of external tearing effect signal 0: TE occurs at the rising edge 1: TE occurs at the falling edge
5:4	EXT_TE_MODE	Select TE mode 0: vsync (edge detection) 1: vsync (pulse width detection) 2: vsync and hsync combined in 1 signal (edge detection) 3: vsync and hsync combined in 1 signal (pulse width detection)
3:0	EXT_TE_MUX	Input port for external tearing effect (external trigger). Valid range: 0 to 1

0x01A9808C MDSS_DSI_0_EXT_TE_PULSE_DETECT_CTRL**Type:** RW**Reset State:** 0x00000000

DSI External Tearing Effect Pulse Width Detection Control Register

MDSS_DSI_0_EXT_TE_PULSE_DETECT_CTRL

Bits	Name	Description
31:16	TE_VSYNC_MIN_WIDTH	Defines the minimum width (in escclk) of the vsync pulse in the TE signal
15:0	TE_HSYNC_MAX_WIDTH	Defines the maximum width (in escclk) of the hsync pulse in the TE signal

0x01A98090 MDSS_DSI_0_CMD_MODE_DMA_SW_TRIGGER**Type:** RW**Reset State:** 0x00000000

DSI Command Mode DMA Software Trigger Register

MDSS_DSI_0_CMD_MODE_DMA_SW_TRIGGER

Bits	Name	Description
0	SW_TRIGGER	Software can start a DMA trigger for the Command Mode engine by writing a 1 to this bit

0x01A98094 MDSS_DSI_0_CMD_MODE_MDP_SW_TRIGGER**Type:** RW**Reset State:** 0x00000000

DSI Command Mode Display Processor Path Software Trigger Register

MDSS_DSI_0_CMD_MODE_MDP_SW_TRIGGER

Bits	Name	Description
0	SW_TRIGGER	Software can start an MPD trigger for the Command Mode engine by writing a 1 to this bit

0x01A98098 MDSS_DSI_0_CMD_MODE_BTA_SW_TRIGGER**Type:** RW**Reset State:** 0x00000000

DSI Command Mode Bus-Turn-Around Software Trigger Register

MDSS_DSI_0_CMD_MODE_BTA_SW_TRIGGER

Bits	Name	Description
0	SW_TRIGGER	Software can start a BTA trigger for the Command Mode engine by writing a 1 to this bit

0x01A9809C MDSS_DSI_0_RESET_SW_TRIGGER**Type:** RW**Reset State:** 0x00000000

DSI Reset Software Trigger Register

MDSS_DSI_0_RESET_SW_TRIGGER

Bits	Name	Description
0	SW_TRIGGER	Software can trigger reset through the data link by writing a 1 to this bit

0x01A980A0 MDSS_DSI_0_MISR_CMD_CTRL**Type:** RW**Reset State:** 0x00000000

DSI misr control for command mode operation - misr is collected for packets output from packetizer

MDSS_DSI_0_MISR_CMD_CTRL

Bits	Name	Type	Description
28	EN		enable Command Mode misr accumulation (all the misr's are reset to 0 when it is disable)
24	WIDTH_SEL		misr width of packets generated from dma, mdp stream 0 and mdp stream1 data 0: generate misr in 8 bits 1: generate misr in 16 bits
23:16	MDP_STREAM1_MISR	R	8-bit misr for packets generated from MDP stream 1
15:8	MDP_STREAM0_MISR	R	8-bit misr for packets generated from MDP stream 0
7:0	DMA_MISR	R	8-bit misr for packets generated from memory

0x01A980A4 MDSS_DSI_0_MISR_VIDEO_CTRL**Type:** RW**Reset State:** 0x00000000

DSI misr control for video mode operation - misr is collected for packets output from packetizer

MDSS_DSI_0_MISR_VIDEO_CTRL

Bits	Name	Type	Description
28	STOP_EN		select misr accumulation restart or stop after N frame (specified in MAX_FRAME_COUNT) 0: auto reset every N frame 1: stop after N frames

MDSS_DSI_0_MISR_VIDEO_CTRL (cont.)

Bits	Name	Type	Description
24	WIDTH_SEL		misr width of packets generated from dma, mdp stream 0 and mdp stream1 data 0: generate misr in 8 bits 1: generate misr in 16 bits
16	EN		enable Video Mode misr accumulation (the misr is reset to 0 when it is disable)
15:8	MAX_FRAME_COUNT		internal misr is reset when the number of frames generated by the video mode engine reaches this number and the MISR field in this register is updated before the reset
7:0	MISR	R	misr for video mode packets

0x01A980A8 MDSS_DSI_0_LANE_STATUS**Type:** R**Reset State:** 0x00000000

DSI Lane Status Register

MDSS_DSI_0_LANE_STATUS

Bits	Name	Description
16	DLN0_DIRECTION	Indicate transmission direction of Data Lane 0
12	CLKLN_ULPS_ACTIVE_NOT	Indicate whether Clock Lane is in Ultra-Low power state
11	DLN3_ULPS_ACTIVE_NOT	Indicate whether Data Lane 3 is in Ultra-Low power state
10	DLN2_ULPS_ACTIVE_NOT	Indicate whether Data Lane 2 is in Ultra-Low power state
9	DLN1_ULPS_ACTIVE_NOT	Indicate whether Data Lane 1 is in Ultra-Low power state
8	DLN0_ULPS_ACTIVE_NOT	Indicate whether Data Lane 0 is in Ultra-Low power state
4	CLKLN_STOPSTATE	Indicate whether Clock Lane is in Stop state
3	DLN3_STOPSTATE	Indicate whether Data Lane 3 is in Stop state
2	DLN2_STOPSTATE	Indicate whether Data Lane 2 is in Stop state
1	DLN1_STOPSTATE	Indicate whether Data Lane 1 is in Stop state
0	DLN0_STOPSTATE	Indicate whether Data Lane 0 is in Stop state

0x01A980AC MDSS_DSI_0_LANE_CTRL**Type:** RW**Reset State:** 0x00000000

DSI Lane Control Register

MDSS_DSI_0_LANE_CTRL

Bits	Name	Description
28	CLKLN_HS_FORCE_REQU EST	Force Clock Lane to enter HS mode always
20	CLKLN_FORCE_TX_STOP	Force Clock Lane to go into transmit mode and the lane module state machine is forced into the Stop state
19	DLN3_FORCE_TX_STOP	Force Data Lane 3 to go into transmit mode and the lane module state machine is forced into the Stop state
18	DLN2_FORCE_TX_STOP	Force Data Lane 2 to go into transmit mode and the lane module state machine is forced into the Stop state
17	DLN1_FORCE_TX_STOP	Force Data Lane 1 to go into transmit mode and the lane module state machine is forced into the Stop state
16	DLN0_FORCE_TX_STOP	Force Data Lane 0 to go into transmit mode and the lane module state machine is forced into the Stop state
12	CLKLN_ULPS_EXIT	Request Clock Lane to exit Ultra-Low power state
11	DLN3_ULPS_EXIT	Request Data Lane 3 to exit Ultra-Low power state
10	DLN2_ULPS_EXIT	Request Data Lane 2 to exit Ultra-Low power state
9	DLN1_ULPS_EXIT	Request Data Lane 1 to exit Ultra-Low power state
8	DLN0_ULPS_EXIT	Request Data Lane 0 to exit Ultra-Low power state
4	CLKLN_ULPS_REQUEST	Request Clock Lane to enter Ultra-Low power state
3	DLN3_ULPS_REQUEST	Request Data Lane 3 to enter Ultra-Low power state
2	DLN2_ULPS_REQUEST	Request Data Lane 2 to enter Ultra-Low power state
1	DLN1_ULPS_REQUEST	Request Data Lane 1 to enter Ultra-Low power state
0	DLN0_ULPS_REQUEST	Request Data Lane 0 to enter Ultra-Low power state

0x01A980B0 MDSS_DSI_0_LANE_SWAP_CTRL**Type:** RW**Reset State:** 0x00000000

DSI Lane Swapping Control Register

MDSS_DSI_0_LANE_SWAP_CTRL

Bits	Name	Description
2:0	DLN_SWAP_SEL	PHY lane mapping for data lane 0, 1, 2, 3 output from the DSI controller 000 = 0, 1, 2, 3 001 = 3, 0, 1, 2 010 = 2, 3, 0, 1 011 = 1, 2, 3, 0 100 = 0, 3, 2, 1 101 = 1, 0, 3, 2 110 = 2, 1, 0, 3 111 = 3, 2, 1, 0

0x01A980B4 MDSS_DSI_0_DLN0_PHY_ERR**Type:** RW**Reset State:** 0x00088888

DSI PHY Error Register - reports errors returned from the PHY. If the mask bit for a specific error is set to 1, turn request will be deasserted if the corresponding error occurs and controller will wait for the link to return to forward stopstate.

NOTE: turn request abort/interrupt will be generated if the mask field(s) in DSI_DLN0_PHY_ERR is set to 1. (Interrupt will be generated if the mask field(s) in DSI_INT_CTRL is set to 1; error status will be generated if the mask field(s) in DSI_ERR_INT_MASK0 is set to 0)

MDSS_DSI_0_DLN0_PHY_ERR

Bits	Name	Type	Description
19	DLN0_ERR_CONTENTION_LP1_MASK		Option for controller to ignore this error 0: no action 1: abort turn request if error occurs
16	DLN0_ERR_CONTENTION_LP1_CLR	W	Clear DLN0_ERR_CONTENTION_LP1 by writing a '1' to this field
16	DLN0_ERR_CONTENTION_LP1	R	Contention detected while lane 0 is driven high
15	DLN0_ERR_CONTENTION_LP0_MASK		Option for controller to ignore this error 0: no action 1: abort turn request if error occurs
12	DLN0_ERR_CONTENTION_LP0_CLR	W	Clear DLN0_ERR_CONTENTION_LP0 by writing a '1' to this field

MDSS_DSI_0_DLN0_PHY_ERR (cont.)

Bits	Name	Type	Description
12	DLN0_ERR_CONTENTION_LP0	R	Contention detected while lane 0 is driven low
11	DLN0_ERR_CONTROL_MASK		Option for controller to ignore this error 0: no action 1: abort turn request if error occurs
8	DLN0_ERR_CONTROL_CLR	W	Clear DLN0_ERR_CONTROL by writing a '1' to this field
8	DLN0_ERR_CONTROL	R	An incorrect line state sequence is detected
7	DLN0_ERR_SYNC_ESC_MASK		Option for controller to ignore this error 0: no action 1: abort turn request if error occurs
4	DLN0_ERR_SYNC_ESC	R	Number of bits received during a low power data transmission is not a multiple of eight when the transmission ends
4	DLN0_ERR_SYNC_ESC_CLR	W	Clear DLN0_ERR_SYNC_ESC by writing a '1' to this field
3	DLN0_ERR_ESC_MASK		Option for controller to ignore this error 0: no action 1: abort turn request if error occurs
0	DLN0_ERR_ESC_CLR	W	Clear DLN0_ERR_ESC by writing a '1' to this field
0	DLN0_ERR_ESC	R	Unrecognized escape entry command received

0x01A980B8 MDSS_DSI_0_LP_TIMER_CTRL**Type:** RW**Reset State:** 0xFFFFFFFF

DSI Timeout Control Register for Low Power Mode Transmission

MDSS_DSI_0_LP_TIMER_CTRL

Bits	Name	Description
31:16	BTA_TO	Bus turn around timeout (in escclk)
15:0	LP_RX_TO	Low power transmission reverse timeout (in Escclk)

0x01A980BC MDSS_DSI_0_HS_TIMER_CTRL**Type:** RW**Reset State:** 0x0000FFFF

DSI Timeout Control Register for High Speed Mode Transmission

MDSS_DSI_0_HS_TIMER_CTRL

Bits	Name	Description
28	HS_TX_TO_STOP_EN	Stop High Speed Transmission immediately at timeout
19:16	TIMER_RESOLUTION	Control all 3 timeout counter increment steps by power of 2 of this field. Ex.: 0 = increment every escclk; 1 = increment every 2 escclk; 3 = increment every 8 escclk;
15:0	HS_TX_TO	High speed transmission timeout (in escclk)

0x01A980C0 MDSS_DSI_0_TIMEOUT_STATUS**Type:** RW**Reset State:** 0x00000000

DSI Timeout Status Register

MDSS_DSI_0_TIMEOUT_STATUS

Bits	Name	Type	Description
8	BTA_TIMEOUT_CLR	W	writing a 1 will clear this bit
8	BTA_TIMEOUT	R	Indicates Bus Turnaround timeout
4	LP_RX_TIMEOUT_CLR	W	writing a 1 will clear this bit
4	LP_RX_TIMEOUT	R	Indicates Low Power reverse transmission timeout
0	HS_TX_TIMEOUT_CLR	W	writing a 1 will clear this bit
0	HS_TX_TIMEOUT	R	Indicates High Speed forward transmission timeout

0x01A980C4 MDSS_DSI_0_CLKOUT_TIMING_CTRL**Type:** RW**Reset State:** 0x00000000

DSI PHY Clock Lane Timing Control Register

MDSS_DSI_0_CLKOUT_TIMING_CTRL

Bits	Name	Description
13:8	T_CLK_POST	Number of byteclk cycles (minus one) that the transmitter shall continue sending HS clock after the last associated Data Lane has transitioned to Low Power mode
5:0	T_CLK_PRE	Number of byteclk cycles (minus one) that the high speed clock shall be driven prior to any associated Data Lane beginning the transition from Low Power to High Speed mode

0x01A980C8 MDSS_DSI_0_EOT_PACKET**Type:** RW**Reset State:** 0x010F0F08

DSI EOT Packet Register - packet to append at the end of a High Speed Data Burst

MDSS_DSI_0_EOT_PACKET

Bits	Name	Description
31:24	ECC	ECC of the EOT packet
23:8	WC	Word count of the EOT packet
7:0	DI	Data identifier (virtual channel and datatype) of the EOT packet

0x01A980CC MDSS_DSI_0_EOT_PACKET_CTRL**Type:** RW**Reset State:** 0x00000000

DSI EOT Packet Control Register

MDSS_DSI_0_EOT_PACKET_CTRL

Bits	Name	Description
4	RX_EOT_IGNORE	Specify whether the controller should ignore the EOT packet if it is received from reverse transmission
0	TX_EOT_APPEND	Specify whether the EOT packet has to be appended at the end of each forward High Speed data burst

0x01A980D0 MDSS_DSI_0_GENERIC_ESC_TX_TRIGGER**Type:** RW**Reset State:** 0x00000000

DSI Generic Escape Mode Trigger Register

MDSS_DSI_0_GENERIC_ESC_TX_TRIGGER

Bits	Name	Description
0	SW_TRIGGER	Software can trigger a generic escape mode command through DISP_DSI_PHY_DLN0_TX_TRIGGER_ESC[3] by writing a 1 to this bit

0x01A9810C MDSS_DSI_0_ERR_INT_MASK0**Type:** RW**Reset State:** 0x7FFF3BFF

DSI Error Interrupt Enable Register - enable any event to drive the error status (DSI_INT_CTRL.DSI_ERROR_STAT). If the DSI_INT_CTRL.DSI_ERROR_MASK is set to 1, interrupt will be generated.

NOTE: Interrupt will be generated if the mask field(s) in DSI_INT_CTRL is set to 1; error status will be generated if the mask field(s) in DSI_ERR_INT_MASK0 is set to 0.

MDSS_DSI_0_ERR_INT_MASK0

Bits	Name	Description
30	DLN3_HS_FIFO_UNDERFLOW_MASK	Mask Lane Layer High Speed FIFO for data lane 3 interrupt (DSI_STATUS reg, bit 31) 0: allow generation of error status 1: no error status generated
29	DLN2_HS_FIFO_UNDERFLOW_MASK	Mask Lane Layer High Speed FIFO for data lane 2 interrupt (DSI_STATUS reg, bit 27) 0: allow generation of error status 1: no error status generated
28	DPHY_DSIPLL_UNLOCK_MASK	Mask dsipll unlock (to avoid sending out unnecessary interrupts, set this bit to 1 before dsipll locks) 0: allow generation of error status 1: no error status generated
27	DLN1_HS_FIFO_UNDERFLOW_MASK	Mask Lane Layer High Speed FIFO for data lane 1 interrupt (DSI_FIFO_STATUS reg, bit 23) 0: allow generation of error status 1: no error status generated

MDSS_DSI_0_ERR_INT_MASK0 (cont.)

Bits	Name	Description
26	DLN0_HS_FIFO_UNDERFLOW_MASK	Mask Lane Layer High Speed FIFO for data lane 0 interrupt (DSI_FIFO_STATUS reg, bit 19) 0: allow generation of error status 1: no error status generated
25	DLN0_ERR_CONTENTION_LP1_MASK	Mask contention while data lane 0 is driven high interrupt (DSI_DLNO_PHY_ERR reg, bit 16) 0: allow generation of error status (this should be set when dsipll is ready or when dsipll bypass is set to 1) 1: no error status generated
24	DLN0_ERR_CONTENTION_LP0_MASK	Mask contention while data lane 0 is driven low interrupt (DSI_DLNO_PHY_ERR reg, bit 12) 0: allow generation of error status 1: no error status generated
23	DLN0_ERR_CONTROL_MASK	Mask incorrect LP RX state sequence interrupt (DSI_DLNO_PHY_ERR reg, bit 8) 0: allow generation of error status 1: no error status generated
22	DLN0_ERR_SYNC_ESCAPE_MASK	Mask LP RX data that is not byte-aligned interrupt (DSI_DLNO_PHY_ERR reg, bit 4) 0: allow generation of error status 1: no error status generated
21	DLN0_ERR_ESCAPE_MASK	Mask incorrect LP RX escape entry interrupt (DSI_DLNO_PHY_ERR reg, bit 0) 0: allow generation of error status 1: no error status generated
20	DLN0_LP_FIFO_OVERFLOW_MASK	unused (Mask Lane Layer Low Power FIFO for data lane 0 interrupt (DSI_FIFO_STATUS reg, bit 14)) 0: allow generation of error status 1: no error status generated
19	DLN3_HS_FIFO_OVERFLOW_MASK	Mask Lane Layer High Speed FIFO for data lane 3 interrupt (DSI_STATUS reg, bit 30) 0: allow generation of error status 1: no error status generated
18	DLN2_HS_FIFO_OVERFLOW_MASK	Mask Lane Layer High Speed FIFO for data lane 2 interrupt (DSI_STATUS reg, bit 26) 0: allow generation of error status 1: no error status generated
17	DLN1_HS_FIFO_OVERFLOW_MASK	Mask Lane Layer High Speed FIFO for data lane 1 interrupt (DSI_FIFO_STATUS reg, bit 22) 0: allow generation of error status 1: no error status generated

MDSS_DSI_0_ERR_INT_MASK0 (cont.)

Bits	Name	Description
16	DLN0_HS_FIFO_OVERFLOW_MASK	Mask Lane Layer High Speed FIFO for data lane 0 interrupt (DSI_FIFO_STATUS reg, bit 18) 0: allow generation of error status 1: no error status generated
13	VID_MDP_FIFO_UNDERFLOW_MASK	Unused
12	VID_MDP_FIFO_OVERFLOW_MASK	Unused
11	CMD_MDP_FIFO_UNDERFLOW_MASK	Mask Display Engine FIFO underflow interrupt (DSI_FIFO_STATUS reg, bit 7) 0: allow generation of error status 1: no error status generated
9	CMD_DMA_FIFO_UNDERFLOW_MASK	Mask Command mode dma FIFO underflow interrupt (DSI_FIFO_STATUS reg, bit 10) 0: allow generation of error status 1: no error status generated
8	INTERLEAVE_OP_CONTENTION_MASK	Mask interleave operation contention interrupt (DSI_STATUS reg, bit 31) 0: allow generation of error status 1: no error status generated
7	BTA_TO_MASK	Mask bus turnaround timeout interrupt (DSI_TIMEOUT_STATUS reg, bit 8) 0: allow generation of error status 1: no error status generated
6	HS_TX_TO_MASK	Mask High Speed forward transmission timeout interrupt (DSI_TIMEOUT_STATUS reg, bit 0) 0: allow generation of error status 1: no error status generated
5	LP_RX_TO_MASK	Mask Low Power reverse transmission timeout interrupt (DSI_TIMEOUT_STATUS reg, bit 4) 0: allow generation of error status 1: no error status generated
4	ERROR_PACKET_MASK	Mask error packet returned from peripheral interrupt (DSI_ACK_ERR_STATUS reg, bit 24) 0: allow generation of error status 1: no error status generated
3	RDBK_INCOMPLETE_PACKET_ERR_MASK	Mask incomplete read packet error interrupt (DSI_ACK_ERR_STATUS reg, bit 23) 0: allow generation of error status 1: no error status generated

MDSS_DSI_0_ERR_INT_MASK0 (cont.)

Bits	Name	Description
2	RDBK_DATA_CRC_ERR_MASK	Mask crc error in the read packet interrupt (DSI_ACK_ERR_STATUS reg, bit 20) 0: allow generation of error status 1: no error status generated
1	RDBK_DATA_MULTI_ECC_ERR_MASK	Mask multi bit ecc error in the read packet interrupt (DSI_ACK_ERR_STATUS reg, bit 17) 0: allow generation of error status 1: no error status generated
0	RDBK_DATA_ECC_ERR_MASK	Mask single bit ecc error in the read packet interrupt (DSI_ACK_ERR_STATUS reg, bit 16) 0: allow generation of error status 1: no error status generated

0x01A98110 MDSS_DSI_0_INT_CTRL**Type:** RW**Reset State:** 0x2220AA02

DSI Interrupt Control Register.

NOTE: Interrupt will be generated if the mask field(s) in DSI_INT_CTRL is set to 1; error status will be generated if the mask field(s) in DSI_ERR_INT_MASK0 is set to 0.

MDSS_DSI_0_INT_CTRL

Bits	Name	Type	Description
29	DSI_DYNAMIC_REFRESH_DONE_MASK		DSI Dynamic Refresh Mask. If set to 1, interrupt generation is enabled. 0: no interrupt generated 1: allow generation of interrupt
28	DSI_DYNAMIC_REFRESH_DONE_STAT	R	DSI Dynamic Refresh status. Set when the Dynamic Refresh has completed.
28	DSI_DYNAMIC_REFRESH_DONE_AK	W	DSI Dynamic Refresh Acknowledge. If set to 1, status and interrupt is cleared.
25	DSI_ERROR_MASK		DSI Error Mask. If set to 1, interrupt generation is enabled. 0: no interrupt generated 1: allow generation of interrupt
24	DSI_ERROR_AK	W	DSI Error Acknowledge. If set to 1, status and interrupt is cleared.
24	DSI_ERROR_STAT	R	DSI Error status. Set when the DSI Error has happened.

MDSS_DSI_0_INT_CTRL (cont.)

Bits	Name	Type	Description
21	DSI_BTA_DONE_MASK		DSI BTA Done Mask. If set to 1, interrupt generation is enabled. 0: no interrupt generated 1: allow generation of interrupt
20	DSI_BTA_DONE_AK	W	DSI BTA Done Acknowledge. If set to 1, status and interrupt is cleared.
20	DSI_BTA_DONE_STAT	R	DSI BTA Done Status (sw triggered BTA, embededd BTA or BTA generated for TE). Set when a bus turnaround has completed.
17	DSI_VIDEO_MODE_DONE_MASK		DSI Video Mode Engine path Done Mask. If set to 1, interrupt generation is enabled. 0: no interrupt generated 1: allow generation of interrupt
16	DSI_VIDEO_MODE_DONE_AK	W	DSI Video Mode Engine path Done Acknowledge. If set to 1, status and interrupt is cleared.
16	DSI_VIDEO_MODE_DONE_STAT	R	DSI Video Mode Engine status. Set when the DSI Video Mode Engine has finished transferring data to the panel.
15	DSI_CMD_MDP_STREAM2_DONE_MASK		DSI Command Mode Engine MDP stream 2 Done Mask. If set to 1, interrupt generation is enabled. 0: no interrupt generated 1: allow generation of interrupt
14	DSI_CMD_MDP_STREAM2_DONE_AK	W	DSI Command Mode Engine MDP stream 2 Done Acknowledge. If set to 1, status and interrupt is cleared.
14	DSI_CMD_MDP_STREAM2_DONE_STAT	R	DSI Command Mode Engine MDP stream 2 status. Set when the DSI Command Mode Engine has finished transferring stream 2 data to the panel.
13	DSI_CMD_MDP_STREAM1_DONE_MASK		DSI Command Mode Engine MDP stream 1 Done Mask. If set to 1, interrupt generation is enabled. 0: no interrupt generated 1: allow generation of interrupt
12	DSI_CMD_MDP_STREAM1_DONE_AK	W	DSI Command Mode Engine MDP stream 1 Done Acknowledge. If set to 1, status and interrupt is cleared.
12	DSI_CMD_MDP_STREAM1_DONE_STAT	R	DSI Command Mode Engine MDP stream 1 status. Set when the DSI Command Mode Engine has finished transferring stream 1 data to the panel.
11	DSI_CMD_MDP_STREAM0_DONE_MASK		DSI Command Mode Engine MDP stream 0 Done Mask. If set to 1, interrupt generation is enabled. 0: no interrupt generated 1: allow generation of interrupt
10	DSI_CMD_MDP_STREAM0_DONE_STAT	R	DSI Command Mode Engine MDP stream 0 status. Set when the DSI Command Mode Engine has finished transferring stream 0 data to the panel.

MDSS_DSI_0_INT_CTRL (cont.)

Bits	Name	Type	Description
10	DSI_CMD_MDP_STREAM0_DONE_AK	W	DSI Command Mode Engine MDP stream 0 Done Acknowledge. If set to 1, status and interrupt is cleared.
9	DSI_CMD_MODE_MDP_DONE_MASK		DSI Command Mode Engine MDP path Done Mask. If set to 1, interrupt generation is enabled. 0: no interrupt generated 1: allow generation of interrupt
8	DSI_CMD_MODE_MDP_DONE_STAT	R	DSI Command Mode Engine MDP path status. Set when the DSI Command Mode Engine has finished transferring MDP stream data selected by COMMAND_MODE_DMA_STREAM_SEL to the panel.
8	DSI_CMD_MODE_MDP_DONE_AK	W	DSI Command Mode Engine MDP path Done Acknowledge. If set to 1, status and interrupt is cleared.
1	DSI_CMD_MODE_DMA_DONE_MASK		DSI Command Mode Engine DMA path Done Mask. If set to 1, interrupt generation is enabled. 0: no interrupt generated 1: allow generation of interrupt
0	DSI_CMD_MODE_DMA_DONE_AK	W	DSI Command Mode Engine DMA path Done Acknowledge. If set to 1, status and interrupt is cleared.
0	DSI_CMD_MODE_DMA_DONE_STAT	R	DSI Command Mode Engine DMA status. Set when the DSI Command Mode Engine has finished transferring dma data to the panel.

0x01A98114 MDSS_DSI_0_IOBIST_CTRL**Type:** RW**Reset State:** 0x00000000

DSI IOBIST Control Register

MDSS_DSI_0_IOBIST_CTRL

Bits	Name	Description
31:0	DSI_IOBIST_CTRL	This register has 2 modes of operation: command and timing mode. All configuration fields should be written first, then enable (bit 0) should be set to 1 next and start (bit 5) is set to 1 last. COMMAND MODE bit 0 = DSI IOBIST ENABLE. Active high; bit 5 = IOBIST start. Active high; bit 6 = Select power mode for data transfer. 0 = high speed (HS), 1 = low power (LP) bit 7 = Data pattern mode. 0 = Use data pattern word, 1 = PRBS pattern. bit 10:8 = multiple of finite random run (N): if bit 12 is finite, the finite run is 256 x (N+1) runs for HS PRBS and user data, or 16 x (N+1) for LP PRBS bit 12 = Infinite run 0 for infinite; 1 for 256 x (N+1) runs of HS PRBS, 16 x (N+1) runs of LP PRBS, 256 x (N+1) runs of HS user data, or 8 runs of LP user data. bit 14:13 = DSIBIST_DSIPHY_LANE_SWAP, select which lane is used as lane0 for LPDT: 0 = DLN0 1 = DLN1 2 = DLN2 3 = DLN3 bit 15 = Select instruction type 0 = command mode, 1 = DSI timing control mode. bit 23:16 = data pattern byte (used when bit 7 is set to 0) other bits = reserved TIMING MODE bit 0 = DSI IOBIST ENABLE. Active high; bit 5 = IOBIST start. Active high; bit 15 = Select instruction type 0 = command mode, 1 = DSI timing control mode. (If you write one DSI timing data with the select instruction type == 1, you have to change the select instruction type from 1 to 0 for timing data latch control.) bit 23:16 = DSI timing data; bit 28:24 = DSI timing register selection: 0x10 = DSI_T_CLK_PRE (data width[5:0]) 0x11 = DSI_T_CLK_POST (data width[5:0])

0x01A98118 MDSS_DSI_0_SOFT_RESET**Type:** RW**Reset State:** 0x00000000

DSI soft reset Register

MDSS_DSI_0_SOFT_RESET

Bits	Name	Description
0	DSI_SOFT_RESET	Resets all internal logic in dsiclk, pclk, byteclk, escclk, ahbm_hclk domain. All enable bits in DSI_CTRL should be already set to 0 when exiting soft reset (DSI_SOFT_RESET changing from 1 to 0). The dynamic clocks should be forced on when DSI_SOFT_RESET is set to 1. (This reset does not apply to DSI registers which are in ahbs_hclk domain.)

0x01A9811C MDSS_DSI_0_CLK_CTRL**Type:** RW**Reset State:** 0x00000000

DSI Clock Control Register. The dynamic clocks should be forced on when DSI_SOFT_RESET is set to 1 or when DSI_EN is changed from 1 to 0

MDSS_DSI_0_CLK_CTRL

Bits	Name	Description
20	DSI_FORCE_ON_DYN_PCLK	DSI PCLK clock Branch Force On. Set to one to always force on all PCLK clock branches.
17:16	DSI_DSICLK_HYSTERISIS1_CTRL	DSI DSICLK Hysterisis Control
15:14	DSI_AHBM_HCLK_HYSTERISIS1_CTRL	DSI AHBM_HCLK Hysterisis Control
13:12	DSI_AHBS_HCLK_HYSTERISIS1_CTRL	unused
11	DSI_FORCE_ON_DYN_BYTECLK	DSI Dynamic BYTECLK Clock Branch Force On. Set to one to always force on this clock branch.
10	DSI_FORCE_ON_DYN_DSICLK	Unused.
9	DSI_FORCE_ON_DYN_AHBM_HCLK	DSI Dynamic AHBM_HCLK (MCLK) Clock Branch Force On. Set to one to always force on this clock branch.
8	DSI_FORCE_ON_DYN_AHBS_HCLK	DSI Dynamic AHBS_HCLK (REGCLK) Clock Branch Force On. Set to one to always force on this clock branch.
5	DSI_ESCCLK_ON	Turns on/off ESCCLK for the DSI. If set to 1, clock is ON.

MDSS_DSI_0_CLK_CTRL (cont.)

Bits	Name	Description
4	DSI_BYTECLK_ON	Turns on/off BYTECLK for the DSI. If set to 1, clock is ON.
3	DSI_DSICLK_ON	Unused.
2	DSI_PCLK_ON	Turns on/off PCLK for the DSI. If set to 1, clock is ON.
1	DSI_AHBM_SCLK_ON	Turns on/off AHBM_HCLK for the DSI. If set to 1, clock is ON.
0	DSI_AHBS_HCLK_ON	Turns on/off AHBS_HCLK for the DSI. If set to 1, clock is ON.

0x01A98120 MDSS_DSI_0_CLK_STATUS**Type:** RW**Reset State:** 0x00000000

DSI Clock Status Register

MDSS_DSI_0_CLK_STATUS

Bits	Name	Type	Description
16	DSIPLL_UNLOCKED_CLR	W	Write only. Clear dsipll_unlocked status by writing to this register field
16	DSIPLL_UNLOCKED	R	Read only. Dsi pll unlock status 0: dsipll not unlocked 1: dsipll unlocked
15	DSI_CAM_BIST_PCLK_ACTIVE	R	Active Status of PCLK branch of Camera (CSI) BIST block of Application Layer. Read-only field that shows the status of this clock branch. If set to one, clock is on. If set to zero, clock is off.
14	DSI_VID_PCLK_ACTIVE	R	Active Status of PCLK branch of Video Mode engine of Application Layer. Read-only field that shows the status of this clock branch. If set to one, clock is on. If set to zero, clock is off.
13	DSI_CMD_PCLK_ACTIVE	R	Active Status of PCLK branch of Command Mode engine of Application Layer. Read-only field that shows the status of this clock branch. If set to one, clock is on. If set to zero, clock is off.
12	DSI_DYN_CMD_PCLK_ACTIVE	R	Active Status of the Dynamic PCLK branch of Command Mode engine of Application Layer. Read-only field that shows the status of this clock branch. If set to one, clock is on. If set to zero, clock is off.
10	DSI_DYN_PCLK_ACTIVE	R	DSI Dynamic PCLK Active Status. Read-only field that shows the status of this clock branch. If set to one, clock is on. If set to zero, clock is off.

MDSS_DSI_0_CLK_STATUS (cont.)

Bits	Name	Type	Description
9	DSI_AON_PCLK_ACTIVE	R	DSI Always ON PCLK Active Status. Read-only field that shows the status of this clock branch. If set to one, clock is on. If set to zero, clock is off.
8	DSI_AON_ESCCLK_ACTIVE	R	DSI Always ON ESCCLK Active Status. Read-only field that shows the status of this clock branch. If set to one, clock is on. If set to zero, clock is off.
7	DSI_DYN_BYTECLK_ACTIVE	R	DSI Dynamic BYTECLK Active Status. Read-only field that shows the status of this clock branch. If set to one, clock is on. If set to zero, clock is off.
6	DSI_AON_BYTECLK_ACTIVE	R	DSI Always ON BYTECLK Active Status. Read-only field that shows the status of this clock branch. If set to one, clock is on. If set to zero, clock is off.
5	DSI_DYN_DSICLK_ACTIVE	R	Unused.
4	DSI_AON_DSICLK_ACTIVE	R	Unused.
3	DSI_DYN_AHBS_HCLK_ACTIVE	R	DSI Dynamic AHBS_HCLK Active Status. Read-only field that shows the status of this clock branch. If set to one, clock is on. If set to zero, clock is off.
2	DSI_AON_AHBS_HCLK_ACTIVE	R	DSI Always ON AHBS_HCLK Active Status. Read-only field that shows the status of this clock branch. If set to one, clock is on. If set to zero, clock is off.
1	DSI_DYN_AHBM_HCLK_ACTIVE	R	DSI Dynamic AHBM_HCLK Active Status. Read-only field that shows the status of this clock branch. If set to one, clock is on. If set to zero, clock is off.
0	DSI_AON_AHBM_HCLK_ACTIVE	R	DSI Always ON AHBM_HCLK Active Status. Read-only field that shows the status of this clock branch. If set to one, clock is on. If set to zero, clock is off.

0x01A9812C MDSS_DSI_0_PHY_SW_RESET**Type:** RW**Reset State:** 0x00000000

DSI PHY SW Reset Register

MDSS_DSI_0_PHY_SW_RESET

Bits	Name	Description
24	PHY_REGULATOR_HW_RESET	THIS BIT IS ONLY USED IN DSI CORE 1. PHY software reset for the regulator. A reset pulse is issued by writing 1 followed by a 0 to this bit.
16	PHY_LN_HW_RESET	PHY software reset for the LANES. A reset pulse is issued by writing 1 followed by a 0 to this bit.

MDSS_DSI_0_PHY_SW_RESET (cont.)

Bits	Name	Description
8	PHY_PLL_HW_RESET	PHY software reset for the PLL. A reset pulse is issued by writing 1 followed by a 0 to this bit.
4	PHY_SW_RESET_POL	polarity of all reset signals for PHY PLL, LANES and regulator. 0 = active high; 1 = active low
0	PHY_SW_RESET	PHY hardware reset for all PHY PLL, LANES and REGULATOR (for DSI CORE 1 only). A reset pulse is issued by writing 1 followed by a 0 to this bit.

0x01A98130 MDSS_DSI_0_AXI2AHB_CTRL**Type:** RW**Reset State:** 0x00000000

DSI AXI to AHB bridge control Register

MDSS_DSI_0_AXI2AHB_CTRL

Bits	Name	Description
0	MSM_ACCUMULATE_EN	Read data accumulate enable 1 = allows for all read data to be returned prior to feeding it out 0 = returns read data immediately

0x01A98134 MDSS_DSI_0_MISR_CMD_MDP0_32BIT**Type:** R**Reset State:** 0xFFFFFFFF

DSI Command MDP stream 0 32-bit misr

MDSS_DSI_0_MISR_CMD_MDP0_32BIT

Bits	Name	Description
31:0	MISR	32-bit misr for packets generated from MDP stream 0

0x01A98138 MDSS_DSI_0_MISR_CMD_MDP1_32BIT**Type:** R**Reset State:** 0xFFFFFFFF

DSI Command MDP stream 0 32-bit misr

MDSS_DSI_0_MISR_CMD_MDP1_32BIT

Bits	Name	Description
31:0	MISR	32-bit misr for packets generated from MDP stream 0

0x01A9813C MDSS_DSI_0_MISR_CMD_DMA_32BIT**Type:** R**Reset State:** 0xFFFFFFFF

DSI Command dma 32-bit misr

MDSS_DSI_0_MISR_CMD_DMA_32BIT

Bits	Name	Description
31:0	MISR	32-bit misr for packets generated from memory

0x01A98140 MDSS_DSI_0_MISR_VIDEO_32BIT**Type:** R**Reset State:** 0xFFFFFFFF

DSI Video Mode 32-bit misr

MDSS_DSI_0_MISR_VIDEO_32BIT

Bits	Name	Description
31:0	MISR	32-bit misr for packets generated from MDP's DSI Video Mode stream

0x01A98144 MDSS_DSI_0_LANE_MISR_CTRL**Type:** RW**Reset State:** 0x00000000

DSI Lane MISR control

MDSS_DSI_0_LANE_MISR_CTRL

Bits	Name	Description
0	EN	Enable MISR calculation of the lane data at the output of the lane distribution layer

0x01A98148 MDSS_DSI_0_LANE0_MISR**Type:** R**Reset State:** 0x0000FFFF

DSI Lane 0 MISR value

MDSS_DSI_0_LANE0_MISR

Bits	Name	Description
15:0	MISR	MISR value of the lane data at the output of the lane distribution layer

0x01A9814C MDSS_DSI_0_LANE1_MISR**Type:** R**Reset State:** 0x0000FFFF

DSI Lane 1 MISR value

MDSS_DSI_0_LANE1_MISR

Bits	Name	Description
15:0	MISR	MISR value of the lane data at the output of the lane distribution layer

0x01A98150 MDSS_DSI_0_LANE2_MISR**Type:** R**Reset State:** 0x0000FFFF

DSI Lane 2 MISR value

MDSS_DSI_0_LANE2_MISR

Bits	Name	Description
15:0	MISR	MISR value of the lane data at the output of the lane distribution layer

0x01A98154 MDSS_DSI_0_LANE3_MISR**Type:** R**Reset State:** 0x0000FFFF

DSI Lane 3 MISR value

MDSS_DSI_0_LANE3_MISR

Bits	Name	Description
15:0	MISR	MISR value of the lane data at the output of the lane distribution layer

0x01A98194 MDSS_DSI_0_COMMAND_MODE_MDP_IDLE_CTRL

Type: RW

Reset State: 0x00000000

Idle insertion between Command MDP packets

MDSS_DSI_0_COMMAND_MODE_MDP_IDLE_CTRL

Bits	Name	Description
14:12	EN	enable idle insertion between Command Mode MDP packets
9:0	LENGTH	Number of dsiclk cycles of idle time to insert between Command Mode MDP packets. This idle counter is added in the application layer. To ensure the DSI link will go into stopstate between Command Mode MDP packets, the LENGTH must be long enough to cover the time the link takes to switch between High Speed mode to Low Power mode (LP11). 0: disable 1: insert idle for stream0 2: insert idle for stream1 3: insertion idle for both streams

0x01A981B8 MDSS_DSI_0_COMMAND_MODE_MDP_CTRL2

Type: RW

Reset State: 0x00000006

Command Mode MDP Control2

MDSS_DSI_0_COMMAND_MODE_MDP_CTRL2

Bits	Name	Description
16	BURST_MODE	Enable frame transfer in one burst 0: a frame can be transferred in multiple High Speed data bursts 1: enable frame transfer in one High Speed data burst if there is no gaps between lines from MDP

MDSS_DSI_0_COMMAND_MODE_MDP_CTRL2 (cont.)

Bits	Name	Description
14:12	INPUT_RGB_SWAP	Swaps the R, G, and B channels of the input command mode pixels from MDP. This field should be used when different R,G, and B orderings are required.
10:8	RGB_SWAP2	Swaps the R, G, and B channels of the dst pixels. This field should be used when different R,G, and B orderings are required. For example, if RGB_SWAP = 1 , RGB565 will become RBG556.
7	BYTE_MSB_LSB_FLIP2	Swap the bit ordering of each byte of the dst pixel (this is done after the RGB swap and R/G/B_SEL). 0: bit 0 is lsb, bit 7 is msb 1: bit 0 is msb, bit 7 is lsb
6	B_SEL2	Bit Swaps the B Channel of the dst pixel. This field is used to reverse the ordering of the bits that is outputted for the color red. For example, for RGB888 color format, if the red component is 10000000 then if this field is enabled, 00000001 is output.
5	G_SEL2	Bit Swaps the G Channel of the dst pixel. This field is used to reverse the ordering of the bits that is outputted for the color red. For example, for RGB888 color format, if the red component is 10000000 then if this field is enabled, 00000001 is output.
4	R_SEL2	Bit Swaps the R Channel of the dst pixel. This field is used to reverse the ordering of the bits that is outputted for the color red. For example, for RGB888 color format, if the red component is 10000000 then if this field is enabled, 00000001 is output.
3:0	DST_FORMAT2	Pixel format for stream 2 0 = RGB111 1 = reserved 2 = reserved 3 = RGB332 (data_src_width must be in multiple of 2 pixels) 4 = RGB444 (data_src_width must be in multiple of 2 pixels) 5 = reserved 6 = RGB565 7 = RGB666 8 = RGB888 others = reserved

0x01A981BC MDSS_DSI_0_COMMAND_MODE_MDP_STREAM2_CTRL**Type:** RW**Reset State:** 0x00000000

Command Mode MDP Stream2 control

MDSS_DSI_0_COMMAND_MODE_MDP_STREAM2_CTRL

Bits	Name	Description
31:16	MDP2_WC	Stream 2 packet word count (number of bytes of the packet payload)
9:8	MDP2_VC	Stream 2 packet virtual channel
5:0	MDP2_DT	Stream 2 packet datatype

0x01A981C0 MDSS_DSI_0_COMMAND_MODE_MDP_STREAM2_TOTAL**Type:** RW**Reset State:** 0x00000000

Command Mode MDP Stream2 resolution

MDSS_DSI_0_COMMAND_MODE_MDP_STREAM2_TOTAL

Bits	Name	Description
31:16	MDP2_V_TOTAL	Stream 2 height (in lines)
15:0	MDP2_H_TOTAL	Stream 2 width (in pixels)

0x01A981C4 MDSS_DSI_0_MISR_CMD_MDP2_8BIT**Type:** R**Reset State:** 0x00000000

Command Mode MDP Stream2 8-bit MISR value

MDSS_DSI_0_MISR_CMD_MDP2_8BIT

Bits	Name	Description
7:0	MISR	The 8-bit MISR value

0x01A981C8 MDSS_DSI_0_MISR_CMD_MDP2_32BIT**Type:** R**Reset State:** 0xFFFFFFFF

Command Mode MDP Stream2 32-bit MISR value

MDSS_DSI_0_MISR_CMD_MDP2_32BIT

Bits	Name	Description
31:0	MISR	The 32-bit MISR value

0x01A981CC MDSS_DSI_0_VBIF_CTRL**Type:** RW**Reset State:** 0x00000000

VBIF interface control register

MDSS_DSI_0_VBIF_CTRL

Bits	Name	Description
1:0	PRIORITY	Value of VBIF request priority and priority level

0x01A981D0 MDSS_DSI_0_AES_CTRL**Type:** RW**Reset State:** 0x00290000

AES counter interface control register

MDSS_DSI_0_AES_CTRL

Bits	Name	Description
21:16	AES_DATATYPE	Data type of the AES counter packet
9:8	AES_VC	Virtual channel of the AES counter packet
4	AES_SEL	Send counter packet at 0 = video mode vsync 1 = command mode sof
0	AES_EN	Enable AES counter packet generation

0x01A981D4 MDSS_DSI_0_RDBK_DATA_CTRL**Type:** RW**Reset State:** 0x00000000

Read Data Control register

MDSS_DSI_0_RDBK_DATA_CTRL

Bits	Name	Type	Description
23:16	RDBK_DATA_COUNT	R	Number of bytes read
0	RDBK_DATA_CLR		Clear read back data and counter by writing a 1 followed by a 0 to this bit

0x01A981E4 MDSS_DSI_0_DSI_TIMING_FLUSH**Type:** RW**Reset State:** 0x00000000

DSI Timing Flush registers

MDSS_DSI_0_DSI_TIMING_FLUSH

Bits	Name	Description
0	DSI_TIMING_FLUSH	DSI Timing Flush register, SW set this bit and post-write trigger will flush the dsi timing register to pclk domain. This flush bit would also returned a 0 once the back copies are updated. Before the back copies are updated, this bit would returned a 1 if it has been previously written a 1.

0x01A981E8 MDSS_DSI_0_DSI_TIMING_DB_MODE**Type:** RW**Reset State:** 0x00000000

DSI Timing Double-buffer Mode

MDSS_DSI_0_DSI_TIMING_DB_MODE

Bits	Name	Description
0	DSI_TIMING_DB_MODE	Register bit to enable DSI timing DB 0: Disable DSI timing register double-buffering - fall back to legacy mode 1: Enable DSI timing register double-buffering

0x01A981F0 MDSS_DSI_0_SCRATCH_REGISTER_0**Type:** RW**Reset State:** 0x00000000

DSI Scratch register

MDSS_DSI_0_SCRATCH_REGISTER_0

Bits	Name	Description
31:0	DSI_SCRATCH_0	32 bit scratch register

0x01A981F8 MDSS_DSI_0_SCRATCH_REGISTER_1**Type:** RW**Reset State:** 0x00000000

DSI Scratch register

MDSS_DSI_0_SCRATCH_REGISTER_1

Bits	Name	Description
31:0	DSI_SCRATCH_1	32 bit scratch register

0x01A981FC MDSS_DSI_0_SCRATCH_REGISTER_2**Type:** RW**Reset State:** 0x00000000

DSI Scratch register

MDSS_DSI_0_SCRATCH_REGISTER_2

Bits	Name	Description
31:0	DSI_SCRATCH_2	32 bit scratch register

0x01A98200 MDSS_DSI_0_DYNAMIC_REFRESH_CTRL**Type:** RW**Reset State:** 0x00000000

DSI Dynamic Refresh Control Register

MDSS_DSI_0_DYNAMIC_REFRESH_CTRL

Bits	Name	Type	Description
31	READY	R	This register is ready for write when the read value of this bit is 1
12	HW_PLL_SWI_MODE		HW access mode 0 = write PLL_DATA0-31 before PLL delay and PLL_DATA32-35 after PLL delay; 1 = only write PLL_DATA0-31 before PLL delay
8	SW_TRIGGER		Trigger dynamic refresh operation by writing '1' once to this field
4	RESET		Reset dynamic refresh operation by writing a '1' followed by a '0' to this field
0	PLL_SWI_CTRL		Select PLL Register access control. This field should not be changed when dynamic refresh operation is busy. 0: SW control 1: HW control

0x01A98204 MDSS_DSI_0_DYNAMIC_REFRESH_PIPE_DELAY**Type:** RW**Reset State:** 0x00000000

DSI Dynamic Refresh Pipe Delay Control Register

MDSS_DSI_0_DYNAMIC_REFRESH_PIPE_DELAY

Bits	Name	Description
31:0	PIPE_DELAY	Delay (in escclk cycles) between the last video Hsync and the PLL register flush. The delay should be long enough for the High Speed data transmission for the Hsync to be completed.

0x01A98208 MDSS_DSI_0_DYNAMIC_REFRESH_PIPE_DELAY2**Type:** RW**Reset State:** 0x00000000

DSI Dynamic Refresh Pipe Delay Control Register 2

MDSS_DSI_0_DYNAMIC_REFRESH_PIPE_DELAY2

Bits	Name	Description
31:0	PIPE_DELAY2	Second delay (in escclk cycles) between the PLL register flush and the end of dynamic refresh operation. The delay should be used when LANE_CTRL.CLKLN_HS_FORCE_REQUEST register is set to 1.

0x01A9820C MDSS_DSI_0_DYNAMIC_REFRESH_PLL_DELAY**Type:** RW**Reset State:** 0x00000000

DSI Dynamic Refresh PLL Delay Control Register

MDSS_DSI_0_DYNAMIC_REFRESH_PLL_DELAY

Bits	Name	Description
31:0	PLL_DELAY	Delay (in escclk cycles) between the PLL register flush and the end of dynamic refresh operation. The delay should be long enough for PLL to lock.

0x01A98210 MDSS_DSI_0_DYNAMIC_REFRESH_STATUS**Type:** R**Reset State:** 0x00000000

DSI Dynamic Refresh Status Register

MDSS_DSI_0_DYNAMIC_REFRESH_STATUS

Bits	Name	Description
13	PLL_DELAY_DONE	DSI Dynamic Refresh pll delay completion status
12	PLL_DELAY_BUSY	DSI Dynamic Refresh pll delay status
9	PIPE_DELAY2_DONE	
8	PIPE_DELAY2_BUSY	
5	PIPE_DELAY_DONE	DSI Dynamic Refresh pipe delay completion status
4	PIPE_DELAY_BUSY	DSI Dynamic Refresh pipe delay busy status
0	BUSY	DSI Dynamic Refresh operation busy status

0x01A98214 MDSS_DSI_0_DYNAMIC_REFRESH_PLL_CTRL0**Type:** RW**Reset State:** 0x00000000

DSI Dynamic Refresh PLL Register Control

MDSS_DSI_0_DYNAMIC_REFRESH_PLL_CTRL0

Bits	Name	Description
31:24	PLL_DATA1	data of PLL register access 1
23:16	PLL_ADDR1	address of PLL register access 1
15:8	PLL_DATA0	data of PLL register access 0
7:0	PLL_ADDR0	address of PLL register access 0.

0x01A98218 MDSS_DSI_0_DYNAMIC_REFRESH_PLL_CTRL1**Type:** RW**Reset State:** 0x00000000

DSI Dynamic Refresh PLL Register Control

MDSS_DSI_0_DYNAMIC_REFRESH_PLL_CTRL1

Bits	Name	Description
31:24	PLL_DATA3	data of PLL register access 3
23:16	PLL_ADDR3	address of PLL register access 3
15:8	PLL_DATA2	data of PLL register access 2
7:0	PLL_ADDR2	address of PLL register access 2.

0x01A9821C MDSS_DSI_0_DYNAMIC_REFRESH_PLL_CTRL2**Type:** RW**Reset State:** 0x00000000

DSI Dynamic Refresh PLL Register Control

MDSS_DSI_0_DYNAMIC_REFRESH_PLL_CTRL2

Bits	Name	Description
31:24	PLL_DATA5	data of PLL register access 5
23:16	PLL_ADDR5	address of PLL register access 5

MDSS_DSI_0_DYNAMIC_REFRESH_PLL_CTRL2 (cont.)

Bits	Name	Description
15:8	PLL_DATA4	data of PLL register access 4
7:0	PLL_ADDR4	address of PLL register access 4.

0x01A98220 MDSS_DSI_0_DYNAMIC_REFRESH_PLL_CTRL3**Type:** RW**Reset State:** 0x00000000

DSI Dynamic Refresh PLL Register Control

MDSS_DSI_0_DYNAMIC_REFRESH_PLL_CTRL3

Bits	Name	Description
31:24	PLL_DATA7	data of PLL register access 7
23:16	PLL_ADDR7	address of PLL register access 7
15:8	PLL_DATA6	data of PLL register access 6
7:0	PLL_ADDR6	address of PLL register access 6.

0x01A98224 MDSS_DSI_0_DYNAMIC_REFRESH_PLL_CTRL4**Type:** RW**Reset State:** 0x00000000

DSI Dynamic Refresh PLL Register Control

MDSS_DSI_0_DYNAMIC_REFRESH_PLL_CTRL4

Bits	Name	Description
31:24	PLL_DATA9	data of PLL register access 9
23:16	PLL_ADDR9	address of PLL register access 9
15:8	PLL_DATA8	data of PLL register access 8
7:0	PLL_ADDR8	address of PLL register access 8.

0x01A98228 MDSS_DSI_0_DYNAMIC_REFRESH_PLL_CTRL5**Type:** RW**Reset State:** 0x00000000

DSI Dynamic Refresh PLL Register Control

MDSS_DSI_0_DYNAMIC_REFRESH_PLL_CTRL5

Bits	Name	Description
31:24	PLL_DATA11	data of PLL register access 11
23:16	PLL_ADDR11	address of PLL register access 11
15:8	PLL_DATA10	data of PLL register access 10
7:0	PLL_ADDR10	address of PLL register access 10.

0x01A9822C MDSS_DSI_0_DYNAMIC_REFRESH_PLL_CTRL6**Type:** RW**Reset State:** 0x00000000

DSI Dynamic Refresh PLL Register Control

MDSS_DSI_0_DYNAMIC_REFRESH_PLL_CTRL6

Bits	Name	Description
31:24	PLL_DATA13	data of PLL register access 13
23:16	PLL_ADDR13	address of PLL register access 13
15:8	PLL_DATA12	data of PLL register access 12
7:0	PLL_ADDR12	address of PLL register access 12.

0x01A98230 MDSS_DSI_0_DYNAMIC_REFRESH_PLL_CTRL7**Type:** RW**Reset State:** 0x00000000

DSI Dynamic Refresh PLL Register Control

MDSS_DSI_0_DYNAMIC_REFRESH_PLL_CTRL7

Bits	Name	Description
31:24	PLL_DATA15	data of PLL register access 15
23:16	PLL_ADDR15	address of PLL register access 15
15:8	PLL_DATA14	data of PLL register access 14
7:0	PLL_ADDR14	address of PLL register access 14.

0x01A98234 MDSS_DSI_0_DYNAMIC_REFRESH_PLL_CTRL8**Type:** RW**Reset State:** 0x00000000

DSI Dynamic Refresh PLL Register Control

MDSS_DSI_0_DYNAMIC_REFRESH_PLL_CTRL8

Bits	Name	Description
31:24	PLL_DATA17	data of PLL register access 17
23:16	PLL_ADDR17	address of PLL register access 17
15:8	PLL_DATA16	data of PLL register access 16
7:0	PLL_ADDR16	address of PLL register access 16.

0x01A98238 MDSS_DSI_0_DYNAMIC_REFRESH_PLL_CTRL9**Type:** RW**Reset State:** 0x00000000

DSI Dynamic Refresh PLL Register Control

MDSS_DSI_0_DYNAMIC_REFRESH_PLL_CTRL9

Bits	Name	Description
31:24	PLL_DATA19	data of PLL register access 19
23:16	PLL_ADDR19	address of PLL register access 19
15:8	PLL_DATA18	data of PLL register access 18
7:0	PLL_ADDR18	address of PLL register access 18.

0x01A9823C MDSS_DSI_0_DYNAMIC_REFRESH_PLL_CTRL10**Type:** RW**Reset State:** 0x00000000

DSI Dynamic Refresh PLL Register Control

MDSS_DSI_0_DYNAMIC_REFRESH_PLL_CTRL10

Bits	Name	Description
31:24	PLL_DATA21	data of PLL register access 21
23:16	PLL_ADDR21	address of PLL register access 21

MDSS_DSI_0_DYNAMIC_REFRESH_PLL_CTRL10 (cont.)

Bits	Name	Description
15:8	PLL_DATA20	data of PLL register access 20
7:0	PLL_ADDR20	address of PLL register access 20.

0x01A98240 MDSS_DSI_0_DYNAMIC_REFRESH_PLL_CTRL11**Type:** RW**Reset State:** 0x00000000

DSI Dynamic Refresh PLL Register Control

MDSS_DSI_0_DYNAMIC_REFRESH_PLL_CTRL11

Bits	Name	Description
31:24	PLL_DATA23	data of PLL register access 23
23:16	PLL_ADDR23	address of PLL register access 23
15:8	PLL_DATA22	data of PLL register access 22
7:0	PLL_ADDR22	address of PLL register access 22.

0x01A98244 MDSS_DSI_0_DYNAMIC_REFRESH_PLL_CTRL12**Type:** RW**Reset State:** 0x00000000

DSI Dynamic Refresh PLL Register Control

MDSS_DSI_0_DYNAMIC_REFRESH_PLL_CTRL12

Bits	Name	Description
31:24	PLL_DATA25	data of PLL register access 25
23:16	PLL_ADDR25	address of PLL register access 25
15:8	PLL_DATA24	data of PLL register access 24
7:0	PLL_ADDR24	address of PLL register access 24.

0x01A98248 MDSS_DSI_0_DYNAMIC_REFRESH_PLL_CTRL13**Type:** RW**Reset State:** 0x00000000

DSI Dynamic Refresh PLL Register Control

MDSS_DSI_0_DYNAMIC_REFRESH_PLL_CTRL13

Bits	Name	Description
31:24	PLL_DATA27	data of PLL register access 27
23:16	PLL_ADDR27	address of PLL register access 27
15:8	PLL_DATA26	data of PLL register access 26
7:0	PLL_ADDR26	address of PLL register access 26.

0x01A9824C MDSS_DSI_0_DYNAMIC_REFRESH_PLL_CTRL14**Type:** RW**Reset State:** 0x00000000

DSI Dynamic Refresh PLL Register Control

MDSS_DSI_0_DYNAMIC_REFRESH_PLL_CTRL14

Bits	Name	Description
31:24	PLL_DATA29	data of PLL register access 29
23:16	PLL_ADDR29	address of PLL register access 29
15:8	PLL_DATA28	data of PLL register access 28
7:0	PLL_ADDR28	address of PLL register access 28.

0x01A98250 MDSS_DSI_0_DYNAMIC_REFRESH_PLL_CTRL15**Type:** RW**Reset State:** 0x00000000

DSI Dynamic Refresh PLL Register Control

MDSS_DSI_0_DYNAMIC_REFRESH_PLL_CTRL15

Bits	Name	Description
31:24	PLL_DATA31	data of PLL register access 31
23:16	PLL_ADDR31	address of PLL register access 31
15:8	PLL_DATA30	data of PLL register access 30
7:0	PLL_ADDR30	address of PLL register access 30.

0x01A98254 MDSS_DSI_0_DYNAMIC_REFRESH_PLL_CTRL16**Type:** RW**Reset State:** 0x00000000

DSI Dynamic Refresh PLL Register Control

MDSS_DSI_0_DYNAMIC_REFRESH_PLL_CTRL16

Bits	Name	Description
31:24	PLL_DATA33	data of PLL register access 33
23:16	PLL_ADDR33	address of PLL register access 33
15:8	PLL_DATA32	data of PLL register access 32
7:0	PLL_ADDR32	address of PLL register access 32.

0x01A98258 MDSS_DSI_0_DYNAMIC_REFRESH_PLL_CTRL17**Type:** RW**Reset State:** 0x00000000

DSI Dynamic Refresh PLL Register Control

MDSS_DSI_0_DYNAMIC_REFRESH_PLL_CTRL17

Bits	Name	Description
31:24	PLL_DATA35	data of PLL register access 35
23:16	PLL_ADDR35	address of PLL register access 35
15:8	PLL_DATA34	data of PLL register access 34
7:0	PLL_ADDR34	address of PLL register access 34.

17.18.2 MDSS_DSI_0_PHY_PLL_QSERDES_PLL**0x01A98300 MDSS_DSI_0_PHY_PLL_UNIPHY_PLL_REFCLK_CFG****Type:** RW**Clock:** WCLK**Reset State:** 0x00000000

Reference clock divider and feedback pre-dividers

MDSS_DSI_0_PHY_PLL_UNIPHY_PLL_REFCLK_CFG

Bits	Name	Description
7	RSVD_7	
6	PLL_CLOCK_BYPASS_ANA	Bypass PLL and use reference clock as the PLL output clock Default value is 0
5:4	PLL_FBCLK_PREDIV	FB pre-divider x0: divide FB clock by 1 x1: divide FB clock by 2 Normal operational value : Refer to PLL Frequency programming table in Integration guideline Default value is 00
3:2	PLL_REFCLK_DIV	Reference clock divider ratio. 00 : div by 1 01 : div by 2 10 : div by 3 11 : div by 4 Normal operational value : Refer to PLL Frequency programming table in Integration guideline Default value is 0000
1	RSVD_1	
0	PLL_REFCLK_DBLR	Reference clock doubler ratio. 0: div by 1 1: div by 2 Normal operational value : Refer to PLL Frequency programming table in Integration guideline Default value is 0

0x01A98304 MDSS_DSI_0_PHY_PLL_UNIPHY_PLL_POSTDIV1_CFG**Type:** RW**Clock:** WCLK**Reset State:** 0x00000009

Divide ratio for post-divider 1 and SVS mode selection

MDSS_DSI_0_PHY_PLL_UNIPHY_PLL_POSTDIV1_CFG

Bits	Name	Description
7:5	RSVD_7_5	

MDSS_DSI_0_PHY_PLL_UNIPHY_PLL_POSTDIV1_CFG (cont.)

Bits	Name	Description
4	PLL_SVS_MODE	SVS Mode Enable 0: Normal operation mode 1: SVS operation mode In SVS mode, all clocks will be half the frequency of normal operation mode. Normal operational mode setting is 0. Default value is 0
3:0	PLL_POSTDIV1	Output Divider setting for oCLK1. 0000: div by1 0001: div by2 ... 1111: div by16 Normal operational value : Refer to PLL Frequency programming table in Integration guideline Default up value is 0000

0x01A98308 MDSS_DSI_0_PHY_PLL_UNIPHY_PLL_CHGPUMP_CFG**Type:** RW**Clock:** WCLK**Reset State:** 0x00000002

Charge pump Control

MDSS_DSI_0_PHY_PLL_UNIPHY_PLL_CHGPUMP_CFG

Bits	Name	Description
7	PLL_IDAC_FILTER_CAP_SEL	select filter cap of IDAC
6:5	RSVD_6_5	
4	PLL_IDAC_PTAT_STR_EN	Enable PTAT strength Normal operational mode setting is 0 Default value is 0
3:0	PLL_CP_IDAC	Charge charge pump current Normal operational mode setting is 0010 Default value is 0010

0x01A9830C MDSS_DSI_0_PHY_PLL_UNIPHY_PLL_VCOLPF_CFG**Type:** RW**Clock:** WCLK**Reset State:** 0x00000030

Loop filter and VCO control

MDSS_DSI_0_PHY_PLL_UNIPHY_PLL_VCOLPF_CFG

Bits	Name	Description
7:6	RSVD_7_6	
5	PD_HP_VCO	Power down high-performance VCO Default value is 1 Normal value: refer to frequency plan
4	PD_LP_VCO	Power down low-power VCO Default value is 1 Normal value: refer to frequency plan
3	PLL_VCO_SEL	Select VCO (LP or HP) Normal operational mode setting is 0 Default value is 0
2:1	PLL_C3_SEL	Select C3 in LPF Normal operational mode setting is 00 Default value is 00
0	PLL_BYPASS_P3	Bypass P3 in LPF Normal operational mode setting is 0 Default value is 0

0x01A98310 MDSS_DSI_0_PHY_PLL_UNIPHY_PLL_VREG_CFG**Type:** RW**Clock:** WCLK**Reset State:** 0x00000000

Regulator control

MDSS_DSI_0_PHY_PLL_UNIPHY_PLL_VREG_CFG

Bits	Name	Description
7:3	RSVD_7_3	
2	RESERVED	Reserved

MDSS_DSI_0_PHY_PLL_UNIPHY_PLL_VREG_CFG (cont.)

Bits	Name	Description
1	PLL_POSTDIV1_BYPASS_B	Bypass post-divider 1 0 : post-divider is bypassed (output of postdiv 1 is same as vco_clk) 1: post-divider is functional (output of postdiv 1 is a divided clock) Normal operational mode setting is 0 Default value is 0
0	PLL_BAND_SEL	Select PLL operating band Normal operational mode setting: refer to programming guideline Default value is 0

0x01A98314 MDSS_DSI_0_PHY_PLL_UNIPHY_PLL_PWRGEN_CFG**Type:** RW**Clock:** WCLK**Reset State:** 0x00000000

Powergen control

MDSS_DSI_0_PHY_PLL_UNIPHY_PLL_PWRGEN_CFG

Bits	Name	Description
7:4	RSVD_7_4	
3	PLL_IBIAS_CAL_EN	Bias calibration enable
2:0	PLL_PWRGEN_CTRL	PWRGEN block control (reserved) Normal operational mode setting is 000 Default value is 000

0x01A98318 MDSS_DSI_0_PHY_PLL_UNIPHY_PLL_DMUX_CFG**Type:** RW**Clock:** WCLK**Reset State:** 0x00000000

Digital mux control setting

MDSS_DSI_0_PHY_PLL_UNIPHY_PLL_DMUX_CFG

Bits	Name	Description
7:5	RSVD_7_5	
4:1	PLL_DMUX_SEL	PLL digital test mux select Default value is 0000

MDSS_DSI_0_PHY_PLL_UNIPHY_PLL_DMUX_CFG (cont.)

Bits	Name	Description
0	PLL_DMUX_EN	PLL digital test mux enable Default value is 0

0x01A9831C MDSS_DSI_0_PHY_PLL_UNIPHY_PLL_AMUX_CFG**Type:** RW**Clock:** WCLK**Reset State:** 0x00000000

Analog mux control setting

MDSS_DSI_0_PHY_PLL_UNIPHY_PLL_AMUX_CFG

Bits	Name	Description
7:5	RSVD_7_5	
4:1	PLL_AMUX_SEL	PLL analog test mux select Default value is 00000
0	PLL_AMUX_EN	PLL analog test mux enable Default value is 0

0x01A98320 MDSS_DSI_0_PHY_PLL_UNIPHY_PLL_GLB_CFG**Type:** RW**Clock:** WCLK**Reset State:** 0x00000000

Global configuration for PLL

MDSS_DSI_0_PHY_PLL_UNIPHY_PLL_GLB_CFG

Bits	Name	Description
7:4	PLL_GLB_CFG	Reserved Normal operational mode setting is 000000 Default value is 000000
3	PLL_ENABLE	Enable PLL, e.g., enable analg side of the PLL and start calibration process Normal operational mode setting is 1 Default value is 0
2	PLL_PWRGEN_PWRDN_B	power down the powergen block Normal operational mode setting is 1 Default value is 0

MDSS_DSI_0_PHY_PLL_UNIPHY_PLL_GLB_CFG (cont.)

Bits	Name	Description
1	PLL_LDO_PWRDN_B	LDO Power Down. 1---Normal operation 0---Digital block power down mode Normal operational mode setting is 1 Default value is 0
0	PLL_PWRDN_B	Global power-down signal for PLL; this is an active low signal, i.e., when PLL_PWRDN_B = 0, PLL is in power-down mode. CSR is still accessible Normal operational mode setting is 1 Default value is 0

0x01A98324 MDSS_DSI_0_PHY_PLL_UNIPHY_PLL_POSTDIV2_CFG**Type:** RW**Clock:** WCLK**Reset State:** 0x00000007

Divide ratio for post divider number 2

MDSS_DSI_0_PHY_PLL_UNIPHY_PLL_POSTDIV2_CFG

Bits	Name	Description
7:0	PLL_POSTDIV2	Output Divider setting for oCLK2 00000000: div by1 00000001: div by2 ... 11111111: div by256 Normal operational value : Refer to PLL Frequency programming table in Integration guideline Default value is 0000_0111

0x01A98328 MDSS_DSI_0_PHY_PLL_UNIPHY_PLL_POSTDIV3_CFG**Type:** RW**Clock:** WCLK**Reset State:** 0x00000007

Divide ratio for post divider number 3

MDSS_DSI_0_PHY_PLL_UNIPHY_PLL_POSTDIV3_CFG

Bits	Name	Description
7:0	PLL_POSTDIV3	Output Divider setting for oCLK3 00000000: div by1 00000001: div by2 ... 11111111: div by256 Normal operational value : Refer to PLL Frequency programming table in Integration guideline Defalut value is 0000_0111

0x01A9832C MDSS_DSI_0_PHY_PLL_UNIPHY_PLL_LPFR_CFG**Type:** RW**Clock:** WCLK**Reset State:** 0x00000000

Resistance setting of loop filter

MDSS_DSI_0_PHY_PLL_UNIPHY_PLL_LPFR_CFG

Bits	Name	Description
7:0	PLL_LPF_R	Loop filter resistance selection Normal operational mode setting is 00000000 Default value is 00000000

0x01A98330 MDSS_DSI_0_PHY_PLL_UNIPHY_PLL_LPFC1_CFG**Type:** RW**Clock:** WCLK**Reset State:** 0x00000003

Capacitance C1 setting of loop filter

MDSS_DSI_0_PHY_PLL_UNIPHY_PLL_LPFC1_CFG

Bits	Name	Description
7:0	PLL_LPF_C1	Loop filter C1 selection Normal operational mode setting is 00000011 Default value is 00000011

0x01A98334 MDSS_DSI_0_PHY_PLL_UNIPHY_PLL_LPFC2_CFG**Type:** RW**Clock:** WCLK**Reset State:** 0x00000003

Capacitance C2 setting of loop filter

MDSS_DSI_0_PHY_PLL_UNIPHY_PLL_LPFC2_CFG

Bits	Name	Description
7:0	PLL_LPF_C2	Loop filter C2 selection Normal operational mode setting is 00000011 Default value is 00000011

0x01A98338 MDSS_DSI_0_PHY_PLL_UNIPHY_PLL_SDM_CFG0**Type:** RW**Clock:** WCLK**Reset State:** 0x00000076

Sigma-delta modulator setting- cfg0

MDSS_DSI_0_PHY_PLL_UNIPHY_PLL_SDM_CFG0

Bits	Name	Description
7	RSVD_7	
6	PLL_SDM_BYP	Bypass sigma-delta modulator and run PLL in integer N mode Normal operational mode setting is 1 Default value is 1
5:0	PLL_SDM_BYP_DIV	Integer N mode feedback clock divider value Normal operational mode setting is 110110 Default value is 110110

0x01A9833C MDSS_DSI_0_PHY_PLL_UNIPHY_PLL_SDM_CFG1**Type:** RW**Clock:** WCLK**Reset State:** 0x0000004C

Sigma-delta modulator setting- cfg1

MDSS_DSI_0_PHY_PLL_UNIPHY_PLL_SDM_CFG1

Bits	Name	Description
7	RSVD_7	
6	PLL_SDM_DITHER_EN	Enable dithering Normal operational mode setting is 1 Default value is 1
5:0	PLL_SDM_DC_OFFSET	Integer portion of fractional N mode divider Normal operational mode setting is 001100 Default value is 001100

0x01A98340 MDSS_DSI_0_PHY_PLL_UNIPHY_PLL_SDM_CFG2**Type:** RW**Clock:** WCLK**Reset State:** 0x00000000

Sigma-delta modulator setting- cfg2

MDSS_DSI_0_PHY_PLL_UNIPHY_PLL_SDM_CFG2

Bits	Name	Description
7:0	PLL_SDM_FREQ_SEED_7_0	Fractional N mode divider fractional portion, bits 7:0 sdm_freq_seed[17:16] are always 00. fddiv value= (sdm_dc_+1)+(sdm_freq_seed<15:0>/2^16) ex:(6+1)+2^15/2^16=7+0.5=7.5 Normal operational mode setting is 00000000 Default value is 00000000

0x01A98344 MDSS_DSI_0_PHY_PLL_UNIPHY_PLL_SDM_CFG3**Type:** RW**Clock:** WCLK**Reset State:** 0x000000C0

Sigma-delta modulator setting- cfg3

MDSS_DSI_0_PHY_PLL_UNIPHY_PLL_SDM_CFG3

Bits	Name	Description
7:0	PLL_SDM_FREQ_SEED_15_8	Fractional N mode divider fractional portion, bits 15:8 Normal operational mode setting is 11000000 Default value is 11000000

0x01A98348 MDSS_DSI_0_PHY_PLL_UNIPHY_PLL_SDM_CFG4**Type:** RW**Clock:** WCLK**Reset State:** 0x00000000

Sigma-delta modulator setting- cfg4

MDSS_DSI_0_PHY_PLL_UNIPHY_PLL_SDM_CFG4

Bits	Name	Description
7:2	RSVD_7_2	
1:0	PLL_SDM_FREQ_SEED_17_16	Fractional N mode divider fractional portion, bits 17:16 Normal operational mode setting is 00 Default value is 00

0x01A9834C MDSS_DSI_0_PHY_PLL_UNIPHY_PLL_SSC_CFG0**Type:** RW**Clock:** WCLK**Reset State:** 0x0000009A

Spread-spectrum clocking setting- cfg0

MDSS_DSI_0_PHY_PLL_UNIPHY_PLL_SSC_CFG0

Bits	Name	Description
7	PLL_SSC_BYP	Bypass SSC circuit Normal operational mode setting is 1 Default value is 1
6:0	PLL_SSC_KDIV	Divider to generate 2MHz clock from refclk. This clock is used by the Spread spectrum portion of the sigma-delta block. It is mainly changing the fractional portion of feedback divider value, e.g. if refclk=100MHz, kdiv = 00110001 Normal operational mode setting is 0011010 Default value is 0011010

0x01A98350 MDSS_DSI_0_PHY_PLL_UNIPHY_PLL_SSC_CFG1**Type:** RW**Clock:** WCLK**Reset State:** 0x00000000

Spread-spectrum clocking setting- cfg1

MDSS_DSI_0_PHY_PLL_UNIPHY_PLL_SSC_CFG1

Bits	Name	Description
7:0	PLL_SSC_TRIANG_INC_7_0	SSC triangle incr, bits 7:0 Normal operational mode setting is 00000000 Default value is 00000000

0x01A98354 MDSS_DSI_0_PHY_PLL_UNIPHY_PLL_SSC_CFG2**Type:** RW**Clock:** WCLK**Reset State:** 0x00000000

Spread-spectrum clocking setting- cfg2

MDSS_DSI_0_PHY_PLL_UNIPHY_PLL_SSC_CFG2

Bits	Name	Description
7:0	PLL_SSC_TRIANG_INC_15_8	SSC triangle incr, bits 9:8 Normal operational mode setting is 00 Default value is 00

0x01A98358 MDSS_DSI_0_PHY_PLL_UNIPHY_PLL_SSC_CFG3**Type:** RW**Clock:** WCLK**Reset State:** 0x00000020

Spread-spectrum clocking setting- cfg2

MDSS_DSI_0_PHY_PLL_UNIPHY_PLL_SSC_CFG3

Bits	Name	Description
7:0	PLL_SSC_TRIANG_STEPS	SSC triangle steps Normal operational mode setting is 100000 Default value is 100000

0x01A9835C MDSS_DSI_0_PHY_PLL_UNIPHY_PLL_LKDET_CFG0**Type:** RW**Clock:** WCLK**Reset State:** 0x00000010

Lock detect setting - cfg0

MDSS_DSI_0_PHY_PLL_UNIPHY_PLL_LKDET_CFG0

Bits	Name	Description
7:0	PLL_LCKDET_MCNT	mpll lock detect maximum count value for the fref and fddiv counters Normal operational mode setting is 00010000 Default value is 00010000

0x01A98360 MDSS_DSI_0_PHY_PLL_UNIPHY_PLL_LKDET_CFG1**Type:** RW**Clock:** WCLK**Reset State:** 0x0000001A

Lock detect setting - cfg1

MDSS_DSI_0_PHY_PLL_UNIPHY_PLL_LKDET_CFG1

Bits	Name	Description
7:0	PLL_LOCKDET_WAIT	Number of cycles to wait (2 cycles incremental) Normal operational mode setting is 00011010 Default value is 00011010

0x01A98364 MDSS_DSI_0_PHY_PLL_UNIPHY_PLL_LKDET_CFG2**Type:** RW**Clock:** WCLK**Reset State:** 0x0000000D

Lock detect setting - cfg2

MDSS_DSI_0_PHY_PLL_UNIPHY_PLL_LKDET_CFG2

Bits	Name	Description
7	RSVD_7	
6	FORCE_PLL_LOCK	0: PLL_RDY and pll_lockdet from PLL Lock detect will be used 1: PLL_RDY and pll_lockdet from PLL Lock detect will be bypassed. In other words, it will force the PLL_RDY to go high and pll_unlockdet to go low. This bit is only used as a back up in case the PLL Lock Detect circuit does not operate correctly. Normal operational mode setting is 0 Default value is 0

MDSS_DSI_0_PHY_PLL_UNIPHY_PLL_LKDET_CFG2 (cont.)

Bits	Name	Description
5:2	PLL_LOCKDET_PPM	PPM setting for PLL lock detect circuitry Normal operational mode setting is 0011 Default value is 0011
1	PLL_LOCKDET_CTRL	control phy_tx_lockdet Normal operational mode setting is 0 Default value is 0
0	PLL_LOCKDET_EN	Enable lockdet Normal operational mode setting is 1 Default value is 1

0x01A9836C MDSS_DSI_0_PHY_PLL_UNIPHY_PLL_CAL_CFG0**Type:** RW**Clock:** WCLK**Reset State:** 0x0000000A

PLL calibration setting - cfg0

MDSS_DSI_0_PHY_PLL_UNIPHY_PLL_CAL_CFG0

Bits	Name	Description
7	RSVD_7	
6	PLL_LOAD_CAL_STR	Load saved PLL calibration string for offset, vco, and kvdd calibration Normal operational mode setting is 0 Default value is 0
5	PLL_OPEN_LOOP	Configure PLL in open loop mode Normal operational mode setting is 0 Default value is 0
4:3	PLL_CAL_WAITTIME	Idle time for VCO frequency adjustment during Calibration Normal operational mode setting is 01 Default value is 01
2:0	PLL_CAL_MODE	Calibration Mode 000: skip calibration 001: offset only 010: offset --> kvco --> offset 011: offset --> kvdd --> offset 100: offset --> kvdd --> offset --> kvco --> offset 101: offset --> kvco --> offset --> kvdd --> offset Normal operational mode setting is 010 Default value is 010

0x01A98370 MDSS_DSI_0_PHY_PLL_UNIPHY_PLL_CAL_CFG1**Type:** RW**Clock:** WCLK**Reset State:** 0x00000004

PLL calibration setting - cfg1

MDSS_DSI_0_PHY_PLL_UNIPHY_PLL_CAL_CFG1

Bits	Name	Description
7	PLL_POS_KVSLP	Slope polarity of KVCO Normal operational mode setting is 0 Default value is 0
6	RSVD_6	
5:4	PLL_STARTUP_TIME	Control PLL startup pulse time 00: pll_start pulse width is 5 cycles of refclk 01: pll_start pulse width is 10 cycles of refclk 10: pll_start pulse width is 15 cycles of refclk 11: pll_start pulse width is 20 cycles of refclk
3:0	PLL_VREF_CONF	KVCO Calibration Reference voltage setting Normal operational mode setting is 0100 Default value is 0100

0x01A98374 MDSS_DSI_0_PHY_PLL_UNIPHY_PLL_CAL_CFG2**Type:** RW**Clock:** WCLK**Reset State:** 0x00000004

PLL calibration setting - cfg2

MDSS_DSI_0_PHY_PLL_UNIPHY_PLL_CAL_CFG2

Bits	Name	Description
7:0	PLL_VREG_CONF	KVDD Calibration Reference voltage setting Normal operational mode setting is 11110000 Default value is 11110000

0x01A98378 MDSS_DSI_0_PHY_PLL_UNIPHY_PLL_CAL_CFG3**Type:** RW**Clock:** WCLK**Reset State:** 0x0000002B

PLL calibration setting - cfg3

MDSS_DSI_0_PHY_PLL_UNIPHY_PLL_CAL_CFG3

Bits	Name	Description
7:0	PLL_VCO_MAN_OFFSET	Manual selection of VCO frequency sub-banding (offset) Normal operational mode setting is 0101011 Default value is 0101011

0x01A9837C MDSS_DSI_0_PHY_PLL_UNIPHY_PLL_CAL_CFG4

Type: RW

Clock: WCLK

Reset State: 0x00000006

PLL calibration setting - cfg4

MDSS_DSI_0_PHY_PLL_UNIPHY_PLL_CAL_CFG4

Bits	Name	Description
7:5	RSVD_7_5	
4:0	PLL_VCO_MAN_SLOPE	Manual selection of Kvco slope Normal operational mode setting is 00110 Default value is 00110

0x01A98380 MDSS_DSI_0_PHY_PLL_UNIPHY_PLL_CAL_CFG5

Type: RW

Clock: WCLK

Reset State: 0x00000006

PLL calibration setting - cfg5

MDSS_DSI_0_PHY_PLL_UNIPHY_PLL_CAL_CFG5

Bits	Name	Description
7	RSVD_7	
6:0	PLL_VCO_MAN_KVDD	Manual selction of KVDD parameters Normal operational mode setting is 00110 Default value is 00110

0x01A98384 MDSS_DSI_0_PHY_PLL_UNIPHY_PLL_CAL_CFG6**Type:** RW**Clock:** WCLK**Reset State:** 0x0000002B

PLL calibration setting - cfg6

MDSS_DSI_0_PHY_PLL_UNIPHY_PLL_CAL_CFG6

Bits	Name	Description
7:0	PLL_CAL_KVCO_COUNT_7_0	Target Calibration KVCO Use This equation: KV_COUNT = kvco *vdd/6, bits 7:0 Normal operational mode setting is 00101011 Default value is 00101011

0x01A98388 MDSS_DSI_0_PHY_PLL_UNIPHY_PLL_CAL_CFG7**Type:** RW**Clock:** WCLK**Reset State:** 0x00000000

PLL calibration setting - cfg7

MDSS_DSI_0_PHY_PLL_UNIPHY_PLL_CAL_CFG7

Bits	Name	Description
7:2	RSVD_7_2	
1:0	PLL_CAL_KVCO_COUNT_9_8	Target Calibration KVCO Use This equation: KV_COUNT = kvco *vdd/6, bits 9:8 Normal operational mode setting is 00 Default value is 00

0x01A9838C MDSS_DSI_0_PHY_PLL_UNIPHY_PLL_CAL_CFG8**Type:** RW**Clock:** WCLK**Reset State:** 0x00000086

PLL calibration setting - cfg8

MDSS_DSI_0_PHY_PLL_UNIPHY_PLL_CAL_CFG8

Bits	Name	Description
7:0	PLL_CAL_REFCLK_DIV_7_0	Reference divider number to get to 5us measurement period, bits 7:0 Normal operational mode setting is 10000110 Default value is 10000110

0x01A98390 MDSS_DSI_0_PHY_PLL_UNIPHY_PLL_CAL_CFG9**Type:** RW**Clock:** WCLK**Reset State:** 0x00000000

PLL calibration setting - cfg9

MDSS_DSI_0_PHY_PLL_UNIPHY_PLL_CAL_CFG9

Bits	Name	Description
7:2	RSVD_7_2	
1:0	PLL_CAL_REFCLK_DIV_9_8	Reference divider number to get to 5us measurement period, bits 9:8 Normal operational mode setting is 00 Default value is 00

0x01A98394 MDSS_DSI_0_PHY_PLL_UNIPHY_PLL_CAL_CFG10**Type:** RW**Clock:** WCLK**Reset State:** 0x000000E6

PLL calibration setting - cfg10

MDSS_DSI_0_PHY_PLL_UNIPHY_PLL_CAL_CFG10

Bits	Name	Description
7:0	PLL_VCO_FREQ_7_0	Target Calibration VCO frequency, bits 7:0 Normal operational mode setting is 11100110 Default value is 11100110

0x01A98398 MDSS_DSI_0_PHY_PLL_UNIPHY_PLL_CAL_CFG11**Type:** RW**Clock:** WCLK**Reset State:** 0x00000002

PLL calibration setting - cfg11

MDSS_DSI_0_PHY_PLL_UNIPHY_PLL_CAL_CFG11

Bits	Name	Description
7:3	RSVD_7_3	
2:0	PLL_VCO_FREQ_10_8	Target Calibration VCO frequency, bits 10:8 Normal operational mode setting is 010 Default value is 010

0x01A9839C MDSS_DSI_0_PHY_PLL_UNIPHY_PLL_EFUSE_CFG**Type:** RW**Clock:** WCLK**Reset State:** 0x00000000

PLL efuse setting

MDSS_DSI_0_PHY_PLL_UNIPHY_PLL_EFUSE_CFG

Bits	Name	Description
7:6	RSVD_7_6	
5	PLL_EFUSE_OVRRISE_SEL	0: Use EFUSE value selected by PLLPHY_DRVSTR_EFUSE_SEL 1: Override with PLLPHY_EFUSE value. Default value is 0
4	PLL_DRVSTR_EFUSE_SEL	1: Select the efuse used for datalane impedance calibration 0: Select dedicated PLL calibration efuse Default value is 0
3:0	PLL_EFUSE	efuse override value Default value is 0000

0x01A983A4 MDSS_DSI_0_PHY_PLL_UNIPHY_PLL_CTRL_42**Type:** RW**Clock:** WCLK**Reset State:** 0x00000000**MDSS_DSI_0_PHY_PLL_UNIPHY_PLL_CTRL_42**

Bits	Name	Description
7:0	RSVD_7_0	

0x01A983A8 MDSS_DSI_0_PHY_PLL_UNIPHY_PLL_CTRL_43**Type:** RW**Clock:** WCLK**Reset State:** 0x00000000**MDSS_DSI_0_PHY_PLL_UNIPHY_PLL_CTRL_43**

Bits	Name	Description
7:0	RSVD_7_0	

0x01A983AC MDSS_DSI_0_PHY_PLL_UNIPHY_PLL_CTRL_44**Type:** RW**Clock:** WCLK**Reset State:** 0x00000000**MDSS_DSI_0_PHY_PLL_UNIPHY_PLL_CTRL_44**

Bits	Name	Description
7:0	RSVD_7_0	

0x01A983B0 MDSS_DSI_0_PHY_PLL_UNIPHY_PLL_CTRL_45**Type:** RW**Clock:** WCLK**Reset State:** 0x00000000**MDSS_DSI_0_PHY_PLL_UNIPHY_PLL_CTRL_45**

Bits	Name	Description
7:0	RSVD_7_0	

0x01A983B4 MDSS_DSI_0_PHY_PLL_UNIPHY_PLL_CTRL_46**Type:** RW**Clock:** WCLK**Reset State:** 0x00000000

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MDSS_DSI_0_PHY_PLL_UNIPHY_PLL_CTRL_46

Bits	Name	Description
7:0	RSVD_7_0	

0x01A983B8 MDSS_DSI_0_PHY_PLL_UNIPHY_PLL_CTRL_47**Type:** RW**Clock:** WCLK**Reset State:** 0x00000000

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MDSS_DSI_0_PHY_PLL_UNIPHY_PLL_CTRL_47

Bits	Name	Description
7:0	RSVD_7_0	

0x01A983BC MDSS_DSI_0_PHY_PLL_UNIPHY_PLL_CTRL_48**Type:** RW**Clock:** WCLK**Reset State:** 0x00000000

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MDSS_DSI_0_PHY_PLL_UNIPHY_PLL_CTRL_48

Bits	Name	Description
7:0	RSVD_7_0	

0x01A983C0 MDSS_DSI_0_PHY_PLL_UNIPHY_PLL_STATUS**Type:** R**Clock:** WCLK**Reset State:** 0x00000000

PLL status signals

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MDSS_DSI_0_PHY_PLL_UNIPHY_PLL_STATUS

Bits	Name	Description
7:1	RSVD_7_1	
0	PLL_RDY	Status of pll_lockdet signal Default value is 0

0x01A983D4 MDSS_DSI_0_PHY_PLL_UNIPHY_PLL_CTRL_54**Type:** R**Clock:** WCLK**Reset State:** 0x00000000**MDSS_DSI_0_PHY_PLL_UNIPHY_PLL_CTRL_54**

Bits	Name	Description
7:0	RSVD_7_0	

17.18.3 MDSS_DSI_0_PHY_CM_MIPI_DSI_LN**0x01A98500+ MDSS_DSI_0_PHY_DSIPHY_LNn_CFG0, n=[0..3]
0x40*n****Type:** RW**Clock:** WCLK**Reset State:** 0x00000000**MDSS_DSI_0_PHY_DSIPHY_LNn_CFG0**

Bits	Name	Description
7:4	DSIPHY_TX_HSEN_PGM_DLY	DSIPHY_TX_HSEN_PGM_DLY Normal operational mode setting is 0000. Power up value is 0000

MDSS_DSI_0_PHY_DSIPHY_LNn_CFG0 (cont.)

Bits	Name	Description
3:0	DSIPHY_HSTX_PGM_DLY	DSIPHY_HSTX_PGM_DLY 0000: 0ps delay 0001: 20ps delay 0010: 40ps delay ... 1111: 300ps delay Power up value is 0000

**0x01A98504+ MDSS_DSI_0_PHY_DSIPHY_LNn_CFG1, n=[0..3]
0x40*n****Type:** RW**Clock:** WCLK**Reset State:** 0x000000D0**MDSS_DSI_0_PHY_DSIPHY_LNn_CFG1**

Bits	Name	Description
7:6	DSIPHY_HSTX_SLEW	DSIPHY Slew Rate Control 00: HSTX Load>2pF 01: 1.33pF<HSTX load<2pF 10: 0.66pF<HSTX load<1.33pF 11: 0pF<HSTX load<0.66pF Normal operational mode setting is 11. Power up value is 11
5	DSIPHY_HSEN_PGM_RSV D	DSIPHY program reserved Power up value is 0
4	DSIPHY_HSEN_PGM_DLY_ BYPASS	DSIPHY HSTX program delay enable low active Power up value is 1
3:2	RSVD_3_2	
1:0	DSIPHY_PEMPH_DLY_SEL	DSIPHY_PEMPH_DLY_SEL 00: 196ps 01: 254ps 10: 344ps 11:644ps Power up value is 00

**0x01A98508+ MDSS_DSI_0_PHY_DSIPHY_LNn_CFG2, n=[0..3]
0x40*n**
Type: RW**Clock:** WCLK**Reset State:** 0x00000000**MDSS_DSI_0_PHY_DSIPHY_LNn_CFG2**

Bits	Name	Description
7:5	DSIPHY_PEMPH_STRBOT	Determines how many Pre-Emphasis branches for the bottom termination are enabled. 000: HSTX load<2pF 100: HSTX 2pF<load<4pF 111: HSTX 4pF<load<6pF Normal operational mode setting is 000. Power up value is 000.
4	RSVD_4	.
3:1	DSIPHY_PEMPH_STRTOP	Determines how many Pre-Emphasis branches for the top termination are enabled 000: HSTX load<2pF 100: HSTX 2pF<load<4pF 111: HSTX 4pF<load<6pF Normal operational mode setting is 000. Power up value is 000.
0	DSIPHY_PEMPH_EN	Enable Signal for Driver Pre-Emphasis. 0 : Pre-Emphasis is Disabled 1 : Pre-Emphasis is Enabled Normal operational mode setting is 0. Power up value is 0

**0x01A9850C+MDSS_DSI_0_PHY_DSIPHY_LNn_CFG3, n=[0..3]
0x40*n**
Type: RW**Clock:** WCLK**Reset State:** 0x00000000**MDSS_DSI_0_PHY_DSIPHY_LNn_CFG3**

Bits	Name	Description
7:6	RSVD_7_6	
5:4	RESERVED	Reserved

MDSS_DSI_0_PHY_DSIPHY_LNn_CFG3 (cont.)

Bits	Name	Description
3:0	DSIPHY_LPRX_DLY	3:2: Escape Clock pulse width adjustment 1:0: Escape Clock Delay element. {00, 01}: 0ns Clock pulse width adjustment and 1.5ns Escape Clock delay {00, 10}: 0ns Clock pulse width adjustment and 3ns Escape Clock delay ... {01, 00}: 1.5ns Clock pulse width adjustment and 0ns Escape Clock delay {01, 01}: 1.5ns Clock pulse width adjustment and 1.5ns Escape Clock delay ... {10, 01}: 3ns Clock pulse width adjustment and 1.5ns Escape Clock delay ... Increment of 1.5ns for both Clock pulse width adjustment and Clock Delay Power up value is 0000

0x01A98510+ MDSS_DSI_0_PHY_DSIPHY_LNn_CFG4, n=[0..3]
0x40*n
Type: RW**Clock:** WCLK**Reset State:** 0x00000000

Adjust T_HS_ZERO Period.

MDSS_DSI_0_PHY_DSIPHY_LNn_CFG4

Bits	Name	Description
7:5	RSVD_7_5	
4:0	LANE_T_HS_ZERO_ADJ	This Field is used to adjust T_HS_ZERO period. Assert early or late between lanes. [4]: 0-late, 1-early [3:0]: max 16 x 2-UI . Must less than T_HS_PREPARE when [4] is 1, less than T_HS_ZERO when [4] is 0 Power up value is 0_0000

**0x01A98530+ MDSS_DSI_0_PHY_DSIPHY_LNn_HSTX_STR_STATUS, n=[0..3]
0x40*n****Type:** R**Clock:** WCLK**Reset State:** 0x00000000**MDSS_DSI_0_PHY_DSIPHY_LNn_HSTX_STR_STATUS**

Bits	Name	Description
7:4	DSIPHY_HSTX_STR_HSTO P_STATUS	Power up value is 0000
3:0	DSIPHY_HSTX_STR_HSBO T_STATUS	Power up value is 0000

**0x01A98534+ MDSS_DSI_0_PHY_RSVD_1_PHY_LNn, n=[0..3]
0x40*n****Type:** R**Clock:** WCLK**Reset State:** 0x00000000**MDSS_DSI_0_PHY_RSVD_1_PHY_LNn**

Bits	Name	Description
7:0	RSVD_7_0	

0x01A98600 MDSS_DSI_0_PHY_DSIPHY_LNCK_CFG0**Type:** RW**Clock:** WCLK**Reset State:** 0x00000000**MDSS_DSI_0_PHY_DSIPHY_LNCK_CFG0**

Bits	Name	Description
7:4	DSIPHY_TX_HSEN_PGM_D LY	DSIPHY_TX_HSEN_PGM_DLY 0000: 0ps delay 0001: 20ps delay 0010: 40ps delay ... 1111: 300ps delay Normal operational mode setting is 0000. Power up value is 0000

MDSS_DSI_0_PHY_DSIPHY_LNCK_CFG0 (cont.)

Bits	Name	Description
3:0	DSIPHY_HSTX_PGM_DLY	DSIPHY_HSTX_PGM_DLY Power up value is 0000

0x01A98604 MDSS_DSI_0_PHY_DSIPHY_LNCK_CFG1**Type:** RW**Clock:** WCLK**Reset State:** 0x000000D0**MDSS_DSI_0_PHY_DSIPHY_LNCK_CFG1**

Bits	Name	Description
7:6	DSIPHY_HSTX_SLEW	DSIPHY Slew Rate Control Normal operational mode setting is 11. Power up value is 11
5	DSIPHY_HSEN_PGM_RSV D	DSIPHY program reserved Power up value is 0
4	DSIPHY_HSEN_PGM_DLY_ BYPASS	DSIPHY HSTX program delay enable low active Power up value is 1
3:2	RSVD_3_2	
1:0	DSIPHY_PEMPH_DLY_SEL	DSIPHY_PEMPH_DLY_SEL Power up value is 00

0x01A98608 MDSS_DSI_0_PHY_DSIPHY_LNCK_CFG2**Type:** RW**Clock:** WCLK**Reset State:** 0x00000000**MDSS_DSI_0_PHY_DSIPHY_LNCK_CFG2**

Bits	Name	Description
7:5	DSIPHY_PEMPH_STRBOT	Determines how many Pre-Emphasis branches for the bottom termination are enabled. Normal operational mode setting is 000. Power up value is 000.

MDSS_DSI_0_PHY_DSIPHY_LNCK_CFG2 (cont.)

Bits	Name	Description
4	RSVD_4	.
3:1	DSIPHY_PEMPH_STRTOP	Determines how many Pre-Emphasis branches for the top termination are enabled Normal operational mode setting is 000. Power up value is 000.
0	DSIPHY_PEMPH_EN	Enable Signal for Driver Pre-Emphasis. 0 : Pre-Emphasis is Disabled 1 : Pre-Emphasis is Enabled Normal operational mode setting is 0. Power up value is 0

0x01A9860C MDSS_DSI_0_PHY_DSIPHY_LNCK_CFG3**Type:** RW**Clock:** WCLK**Reset State:** 0x00000000**MDSS_DSI_0_PHY_DSIPHY_LNCK_CFG3**

Bits	Name	Description
7:6	RSVD_7_6	
5:4	RESERVED	Reserved
3:0	DSIPHY_LPRX_DLY	3:2 : Escape Clock pulse width adjustment 1:0 : Escape Clock Delay element.ps delay {00, 01}: 0ns Clock pulse width adjustment and 1.5ns Escape Clock delay {00, 10}: 0ns Clock pulse width adjustment and 3ns Escape Clock delay ... {01,00}: 1.5ns Clock pulse width adjustment and 0ns Escape Clock delay {01,01}: 1.5ns Clock pulse width adjustment and 1.5ns Escape Clock delay ... {10,01}: 3ns Clock pulse width adjustment and 1.5ns Escape Clock delay ... Increment of 1.5ns for both Clock pulse width adjustment and Clock Delay Power up value is 0000

0x01A98610 MDSS_DSI_0_PHY_DSIPHY_LNCK_CFG4**Type:** RW**Clock:** WCLK**Reset State:** 0x00000000

Adjust T_HS_ZERO Period.

MDSS_DSI_0_PHY_DSIPHY_LNCK_CFG4

Bits	Name	Description
7	CLKLANE_HSTX_INV	This field can be used to invert clock lane HS DATA after T_CLK_ZERO Power up value is 0
6:5	RSVD_6_5	
4:0	LANE_T_HS_ZERO_ADJ	This Field is used to adjust T_HS_ZERO period. Assert early or late between lanes. [4]: 0-late, 1-early [3:0]: max 16 x 2-UI . Must less than T_HS_PREPARE when [4] is 1, less than T_HS_ZERO when [4] is 0 Power up value is 0_0000

0x01A98630 MDSS_DSI_0_PHY_DSIPHY_LNCK_HSTX_STR_STATUS**Type:** R**Clock:** WCLK**Reset State:** 0x00000000**MDSS_DSI_0_PHY_DSIPHY_LNCK_HSTX_STR_STATUS**

Bits	Name	Description
7:4	DSIPHY_HSTX_STR_HSTOP_STATUS	Power up value is 0000
3:0	DSIPHY_HSTX_STR_HSBO T_STATUS	Power up value is 0000

0x01A98634 MDSS_DSI_0_PHY_RSVD_1_PHY_LNCK**Type:** R**Clock:** WCLK**Reset State:** 0x00000000

MDSS_DSI_0_PHY_RSVD_1_PHY_LNCK

Bits	Name	Description
7:0	RSVD_7_0	

0x01A98640 MDSS_DSI_0_PHY_DSIPHY_TIMING_CTRL_0**Type:** RW**Clock:** WCLK**Reset State:** 0x00000018

tHS_CLK_ZERO Timing Parameter Per UI based 9-bit Counter bit7_0.

MDSS_DSI_0_PHY_DSIPHY_TIMING_CTRL_0

Bits	Name	Description
7:0	DSIPHY_T_CLK_ZERO	DSIPHY tCLK-ZERO Timing Parameter UI Calculation: $((\text{DSIPHY_T_CLK_ZERO_8}, \text{DSIPHY_T_CLK_ZERO}[7:0]) \gg 1) * 2 + 2$ DSIPHY_TIMING_CTRL_3[0] contains DSIPHY_T_CLK_ZERO_8. DSIPHY_T_CLK_ZERO_8 only possibly need to be used for 1.5Gbps lane rate Normal operational mode setting is 0b0_0000_0111. Power up value is 0b0_0000_0111

0x01A98644 MDSS_DSI_0_PHY_DSIPHY_TIMING_CTRL_1**Type:** RW**Clock:** WCLK**Reset State:** 0x00000004

tHS_CLK-TRAIL Timing Parameter Per UI based 8-bit counter.

MDSS_DSI_0_PHY_DSIPHY_TIMING_CTRL_1

Bits	Name	Description
7:0	DSIPHY_T_CLK_TRAIL	DSIPHY tCLK-TRAIL Timing Parameter UI Calculation: $(\text{DSIPHY_T_CLK_TRAIL} \gg 1) * 2 + 2$ Normal operational mode setting is 0b0000_0100. Power up value is 0b0000_0100

0x01A98648 MDSS_DSI_0_PHY_DSIPHY_TIMING_CTRL_2**Type:** RW**Clock:** WCLK**Reset State:** 0x00000007

tHS_CLK_PREPARE Timing Parameter Per UI based 8-bit Counter.

MDSS_DSI_0_PHY_DSIPHY_TIMING_CTRL_2

Bits	Name	Description
7:0	DSIPHY_T_CLK_PREPARE	DSIPHY tCLK-PREPARE Timing Parameter UI Calculation: (DSIPHY_T_CLK_PREPARE >> 1) * 2 + 2 Normal operational mode setting is 0b0001_1000. Power up value is 0b0001_1000

0x01A9864C MDSS_DSI_0_PHY_DSIPHY_TIMING_CTRL_3**Type:** RW**Clock:** WCLK**Reset State:** 0x00000000

tHS_CLK_ZERO Timing Parameter Per UI based 9-bit Counter bit8.

MDSS_DSI_0_PHY_DSIPHY_TIMING_CTRL_3

Bits	Name	Description
7:1	RSVD_7_1	
0	DSIPHY_T_CLK_ZERO_8	DSIPHY tCLK-ZERO Timing Parameter bit8. UI Calculation: {(DSIPHY_T_CLK_ZERO_8, DSIPHY_T_CLK_ZERO[7:0]) >> 1} * 2 + 2 DSIPHY_T_CLK_ZERO_8 only possibly need to be used for 1.5Gbps lane rate. Normal operational mode setting is 0b0_0000_0111. Power up value is 0b0_0000_0111

0x01A98650 MDSS_DSI_0_PHY_DSIPHY_TIMING_CTRL_4**Type:** RW**Clock:** WCLK**Reset State:** 0x00000008

tHS_EXIT Timing Parameter Per UI based 8-bit Counter.

MDSS_DSI_0_PHY_DSIPHY_TIMING_CTRL_4

Bits	Name	Description
7:0	DSIPHY_T_HS_EXIT	DSIPHY tHS-EXIT Timing Parameter UI Calculation: (DSIPHY_T_HS_EXIT>>1) * 2 + 2 Normal operational mode setting is 0b0000_1000. Power up value is 0b0000_1000

0x01A98654 MDSS_DSI_0_PHY_DSIPHY_TIMING_CTRL_5**Type:** RW**Clock:** WCLK**Reset State:** 0x00000018

tHS_ZERO Timing Parameter Per UI based 8-bit Counter.

MDSS_DSI_0_PHY_DSIPHY_TIMING_CTRL_5

Bits	Name	Description
7:0	DSIPHY_T_HS_ZERO	DSIPHY tHS-ZERO Timing Parameter. Due to design implementation limitation, tHS-ZERO has to be set greater than 8 (>8). Set to 24 for typical operation and above 24 when HS settle time need to be expended. Any value set between 0 and 24 will cause un-expected high speed data transfer behavior such as data loss or miss sync pattern. UI Calculation: (DSIPHY_T_HS_ZERO>>1) * 2 + 2 Normal operational mode setting is 0b0001_1000. Power up value is 0b0001_1000

0x01A98658 MDSS_DSI_0_PHY_DSIPHY_TIMING_CTRL_6**Type:** RW**Clock:** WCLK**Reset State:** 0x00000009

tHS_PREPARE Timing Parameter Per UI based 8-bit Counter.

MDSS_DSI_0_PHY_DSIPHY_TIMING_CTRL_6

Bits	Name	Description
7:0	DSIPHY_T_HS_PREPARE	DSIPHY tHS-PREPARE Timing Parameter UI Calculation: $(\text{DSIPHY_T_HS_PREPARE} \gg 1) * 2 + 2$ Normal operational mode setting is 0b0000_1001. Power up value is 0b0000_1001

0x01A9865C MDSS_DSI_0_PHY_DSIPHY_TIMING_CTRL_7**Type:** RW**Clock:** WCLK**Reset State:** 0x00000008

tHS_TRAIL Timing Parameter Per UI based 8-bit Counter.

MDSS_DSI_0_PHY_DSIPHY_TIMING_CTRL_7

Bits	Name	Description
7:0	DSIPHY_T_HS_TRAIL	DSIPHY tHS-TRAIL Timing Parameter UI Calculation: $(\text{DSIPHY_T_HS_TRAIL} \gg 1) * 2 + 2$ Normal operational mode setting is 0b0000_1000. Power up value is 0b0000_1000

0x01A98660 MDSS_DSI_0_PHY_DSIPHY_TIMING_CTRL_8**Type:** RW**Clock:** WCLK**Reset State:** 0x00000001

tHS_RQST Timing Parameter Per UI based 8-bit Counter.

MDSS_DSI_0_PHY_DSIPHY_TIMING_CTRL_8

Bits	Name	Description
7:0	DSIPHY_T_HS_RQST	DSIPHY tHS-RQST Timing Parameter UI Calculation: $(\text{DSIPHY_T_HS_RQST} \gg 1) * 2 + 2$ Normal operational mode setting is 0b0000_0001. Power up value is 0b0000_0001

0x01A98664 MDSS_DSI_0_PHY_DSIPHY_TIMING_CTRL_9**Type:** RW**Clock:** WCLK**Reset State:** 0x00000013

tLP_TA_SURE and tLP_TA_GO Timing Parameter Per LP UI based 3-bit Counter.

MDSS_DSI_0_PHY_DSIPHY_TIMING_CTRL_9

Bits	Name	Description
7	RSVD_7	
6:4	DSIPHY_T_TA_SURE	DSIPHY tTA-SURE Timing Parameter. tTA-SURE be set to 2 for typical operation and above 2 if bus turn around time need to be expended. value may need to be set to 0 or 1 to meet minimum T_ta_sure requirement when rx_tlp _x is 2/3 of tx_tlp _x during bus direction turn around process. Normal operational mode setting is 010 Power up value is 001
3	RSVD_3	
2:0	DSIPHY_T_TA_GO	DSIPHY tTA-GO Timing Parameter Normal operational mode setting is 011. Power up value is 011

0x01A98668 MDSS_DSI_0_PHY_DSIPHY_TIMING_CTRL_10**Type:** RW**Clock:** WCLK**Reset State:** 0x00000004

tLP_TA_GET Timing Parameter Per LP UI based 3-bit Counter.

MDSS_DSI_0_PHY_DSIPHY_TIMING_CTRL_10

Bits	Name	Description
7:3	RSVD_7_3	
2:0	DSIPHY_T_TA_GET	DSIPHY tTA-GET Timing Parameter Normal operational mode setting is 0b100. Power up value is 0b100

0x01A9866C MDSS_DSI_0_PHY_DSIPHY_TIMING_CTRL_11**Type:** RW**Clock:** WCLK**Reset State:** 0x00000000

TX Escape Trigger[3] programmable command.

MDSS_DSI_0_PHY_DSIPHY_TIMING_CTRL_11

Bits	Name	Description
7:0	DSIPHY_TRIG3_CMD	DSIPHY TX Escape Trigger[3] programmable command sequence Normal operational mode setting is 0b0000_0000. Power up value is 0b0000_0000

0x01A98670 MDSS_DSI_0_PHY_DSIPHY_CTRL_0**Type:** RW**Clock:** WCLK**Reset State:** 0x00000000

Note : The preferred name for this register is DSIPHY_GLBL_PWR_CFG

MDSS_DSI_0_PHY_DSIPHY_CTRL_0

Bits	Name	Description
7	RSVD_7	
6	DSIPHY_DIGTOP_PWRDN_B	Digital Control block Power Down. 1---Normal operation 0---Digital block power down mode Power up value is 0
5	DSIPHY_CAL_PWRDN_B	Digital Control block Power Down. 1: Normal calibration operation 0: Digital Control block powerdown Normal operational mode setting is 1 Power up value is 0
4	DSIPHY_DLN3_SHUTDOWNB	DSIPHY Data Lane 3 Power Down. 0: Normal operation 1: Data LN3 NPL Block powerdown Note: Preferred name for *_SHUTDOWNB bits are *_PWRDN_B Normal operational mode setting is 0. Power up value is 1

MDSS_DSI_0_PHY_DSIPHY_CTRL_0 (cont.)

Bits	Name	Description
3	DSIPHY_DLN2_SHUTDOWNB	DSIPHY Data Lane 2 Power Down. 0: Normal operation 1: Data LN2 NPL Block powerdown Note: Preferred name for *_SHUTDOWNB bits are *_PWRDN_B Normal operational mode setting is 0. Power up value is 1
2	DSIPHY_CLK_SHUTDOWNB	DSIPHY Clock Lane Power Down. 0: Normal operation 1: NPL Block powerdown Note: Preferred name for *_SHUTDOWNB bits are *_PWRDN_B Normal operational mode setting is 0. Power up value is 1
1	DSIPHY_DLN1_SHUTDOWNB	DSIPHY Data Lane 1Power Down. 0: Normal operation 1: Data LN1 NPL Block powerdown Note: Preferred name for *_SHUTDOWNB bits are *_PWRDN_B Normal operational mode setting is 0. Power up value is 1
0	DSIPHY_DLN0_SHUTDOWNB	DSIPHY Data Lane 0 Power Down. 0: Normal operation 1: Data LN0 NPL Block powerdown Note: Preferred name for *_SHUTDOWNB bits are *_PWRDN_B Normal operational mode setting is 0. Power up value is 1

0x01A98674 MDSS_DSI_0_PHY_DSIPHY_CTRL_1**Type:** RW**Clock:** WCLK**Reset State:** 0x00000000

Note : Preferred name for this register is DSIPHY_GLBL_RESET_CFG

MDSS_DSI_0_PHY_DSIPHY_CTRL_1

Bits	Name	Description
7	DSIPHY_SW_RESET	RESET for Digital blocks except control registers . Active high signal. 1: Assertion 0: Deassertion Normal operational mode setting is 0. Power up value is 0

MDSS_DSI_0_PHY_DSIPHY_CTRL_1 (cont.)

Bits	Name	Description
6:0	RSVD_6_0	

0x01A98678 MDSS_DSI_0_PHY_DSIPHY_CTRL_2**Type:** RW**Clock:** WCLK**Reset State:** 0x00000000**MDSS_DSI_0_PHY_DSIPHY_CTRL_2**

Bits	Name	Description
7:6	DCC_COMP_WAIT_TIME	Duty Cycle Correction analog compare time. 00: TODO. 01: TODO 10: TODO. 11: TODO. Power up value is 00
5:4	DCC_WAKEUP_TIME	Duty Cycle Correction wake up time. 00: TODO. 01: TODO 10: TODO. 11: TODO. Power up value is 00
3	RSVD_3	
2	DCC_SPEEDUP_EN	Duty Cycle Correction speed up mode. 0: use compare timer. 1: bypass compare timer. Power up value is 0
1	DCC_SEL	Duty Cycle Correction Select. 0: bypass Duty Cycle Correction. 1: select Duty Cycle Correction. Power up value is 0
0	DCC_EN	Start Duty Cycle Correction enable start. 0: disable Duty Cycle Correction Select. 1: start Duty Cycle Correction Select. Power up value is 0

0x01A9867C MDSS_DSI_0_PHY_DSIPHY_CTRL_3**Type:** RW**Clock:** WCLK**Reset State:** 0x00000008**MDSS_DSI_0_PHY_DSIPHY_CTRL_3**

Bits	Name	Description
7	RSVD_7	
6:0	DCC_SEL_P_PGM	Duty Cycle Correction PMOS programe seeting. Power up value is 000_1000

0x01A98680 MDSS_DSI_0_PHY_DSIPHY_CTRL_4**Type:** RW**Clock:** WCLK**Reset State:** 0x00000008**MDSS_DSI_0_PHY_DSIPHY_CTRL_4**

Bits	Name	Description
7	RSVD_7	
6:0	DCC_SEL_N_PGM	Duty Cycle Correction NMOS programe setting. Power up value is 000_1000

0x01A98684 MDSS_DSI_0_PHY_DSIPHY_STRENGTH_CTRL_0**Type:** RW**Clock:** WCLK**Reset State:** 0x00000000

MDSS_DSI_0_PHY_DSIPHY_STRENGTH_CTRL_0

Bits	Name	Description
7:4	DSIPHY_STR_LP_N	0000: 2-9.5 pF load. 0010: 9.5-18 pF load 0100: 18-25 pF load. 0110: 25-35 pF load. 1100: 35-65 pF load. 1111: 65-70 pF load. Operational mode value is 1111 Power up value is 0000
3:0	DSIPHY_STR_LP_P	0000: 2-9.5 pF load. 0010: 9.5-18 pF load 0100: 18-25 pF load. 0110: 25-35 pF load. 1100: 35-65 pF load. 1111: 65-70 pF load. Operational mode value is 1111 Power up value is 0000

0x01A98688 MDSS_DSI_0_PHY_DSIPHY_STRENGTH_CTRL_1**Type:** RW**Clock:** WCLK**Reset State:** 0x00000000

Note: The preferred name of this register is DSIPHY_GLBL_LPRX_CFG

MDSS_DSI_0_PHY_DSIPHY_STRENGTH_CTRL_1

Bits	Name	Description
7:4	RSVD_7_4	
3	DSIPHY_LN_CTRL_3	HS Contention Disable. 1 : Disable HS Mode Contention Detection Power up value is 0
2	DSIPHY_LN_CTRL_2	Enable for Low Power Receiver. 1 : LP RX is enabled 0 : LP RX is disabled Note; The preferred name for this register bit should be DSIPHY_LPRX_EN Power up value is 0

MDSS_DSI_0_PHY_DSIPHY_STRENGTH_CTRL_1 (cont.)

Bits	Name	Description
1	DSIPHY_LN_CTRL_1	Enable for Contention Detection Receiver. 1 : CD RX is enabled. 0 : CD RX is disabled Note : The preferred name for this register bit should be DSIPHY_CDRX_EN Power up value is 0
0	RSVD_0	

0x01A9868C MDSS_DSI_0_PHY_DSIPHY_CAL_HW_TRIGGER**Type:** RW**Clock:** WCLK**Reset State:** 0x00000000**MDSS_DSI_0_PHY_DSIPHY_CAL_HW_TRIGGER**

Bits	Name	Description
7:1	RSVD_7_1	
0	TRIGGER	Trigger the hw calibration of the pull-up and pull-down strength for the High-speed transmitters. Normal operational mode setting is 0 Power up value is 0

0x01A98690 MDSS_DSI_0_PHY_DSIPHY_CAL_SW_CFG0**Type:** RW**Clock:** WCLK**Reset State:** 0x00000000**MDSS_DSI_0_PHY_DSIPHY_CAL_SW_CFG0**

Bits	Name	Description
7:5	RSVD_7_5	
4	SW_CAL_TX_BOT_EN	High speed transmitter pull-down strength calibration enable. Normal operational mode setting is 0 Power up value is 0
3:1	RSVD_3_1	

MDSS_DSI_0_PHY_DSIPHY_CAL_SW_CFG0 (cont.)

Bits	Name	Description
0	SW_CAL_TX_TOP_EN	High speed transmitter pull-up strength calibration enable. Normal operational mode setting is 0 Power up value is 0

0x01A98694 MDSS_DSI_0_PHY_DSIPHY_CAL_SW_CFG1**Type:** RW**Clock:** WCLK**Reset State:** 0x00000000**MDSS_DSI_0_PHY_DSIPHY_CAL_SW_CFG1**

Bits	Name	Description
7:4	SW_CAL_TX_BOT_STR	High speed transmitter pull-down strength setting. 1111=Strongest 0000=Weakest strength. Normal operational mode setting is 0000 Power up value is 0000
3:0	SW_CAL_TX_TOP_STR	High speed transmitter pull-up strength setting. 1111=Strongest 0000=Weakest strength. Normal operational mode setting is 0000 Power up value is 0000

0x01A98698 MDSS_DSI_0_PHY_DSIPHY_CAL_SW_CFG2**Type:** RW**Clock:** WCLK**Reset State:** 0x00000000**MDSS_DSI_0_PHY_DSIPHY_CAL_SW_CFG2**

Bits	Name	Description
7:1	RSVD_7_1	
0	CALIBRATION_SEL	Select calibration mode 0: hardware control 1: software control Normal operational mode setting is 0 Power up value is 0

0x01A9869C MDSS_DSI_0_PHY_DSIPHY_CAL_HW_CFG0**Type:** RW**Clock:** WCLK**Reset State:** 0x00000001**MDSS_DSI_0_PHY_DSIPHY_CAL_HW_CFG0**

Bits	Name	Description
7:5	RSVD_7_5	
4	CAL_POLYR_MODE	DSIPHY Poly Resistor Select. To enable the Poly Resistor to be used for impedance calibration. When asserted, the MIPI_DSI_LDO pin will be connected to the reference poly resistor. Note: DSIPHY_REGULATOR_CTRL_0[DSIPH_REG_EN] must be set to 0 before asserting CAL_POLYR_MODE Normal operational mode setting is 0 Power up value is 0
3:1	RSVD_3_1	
0	HW_CAL_EN	To enable the hardware calibration Normal operational mode setting is 1 Power up value is 1

0x01A986A0 MDSS_DSI_0_PHY_DSIPHY_CAL_HW_CFG1**Type:** RW**Clock:** WCLK**Reset State:** 0x0000005A**MDSS_DSI_0_PHY_DSIPHY_CAL_HW_CFG1**

Bits	Name	Description
7:4	LOWER_THRESHOLD	If number of 1s out of samples of CAL_WEAK(Comparator output signal) is below this value, strength will be incremented Normal operational mode setting is 0101 Power up value is 0101
3:0	UPPER_THRESHOLD	If number of 1s out of samples of CAL_WEAK(Comparator output signal) is above this value, strength will be incremented Normal operational mode setting is 1010 Power up value is 1010

0x01A986A4 MDSS_DSI_0_PHY_DSIPHY_CAL_HW_CFG2**Type:** RW**Clock:** WCLK**Reset State:** 0x00000010**MDSS_DSI_0_PHY_DSIPHY_CAL_HW_CFG2**

Bits	Name	Description
7:5	RSVD_7_5	
4:0	PC_STR_TOO_WEAK_COUNT	Number of samples of CAL_WEAK(Comparator output signal) to take for each loop. This is used for counting the samples of analog calibrator compactor output str_too_weak signal. The sample counter are based on dsi pll refclk cycle. This register value must be set greater than UPPER_THRESHOLD in order to have correct HW calibration sequence. Normal operational mode setting is 1_0000 Power up value is 1_0000

0x01A986A8 MDSS_DSI_0_PHY_DSIPHY_CAL_HW_CFG3**Type:** RW**Clock:** WCLK**Reset State:** 0x0000000A**MDSS_DSI_0_PHY_DSIPHY_CAL_HW_CFG3**

Bits	Name	Description
7:0	COMPARATOR_SETTLE_TIME_7_0	Number of ESCCLK cycles to wait before taking another set of 16 samples of CAL_WEAK(Comparator output signal) Normal operational mode setting is 0000_1010 Power up value is 0000_1010

0x01A986AC MDSS_DSI_0_PHY_DSIPHY_CAL_HW_CFG4**Type:** RW**Clock:** WCLK**Reset State:** 0x00000000**MDSS_DSI_0_PHY_DSIPHY_CAL_HW_CFG4**

Bits	Name	Description
7:2	RSVD_7_2	

MDSS_DSI_0_PHY_DSIPHY_CAL_HW_CFG4 (cont.)

Bits	Name	Description
1:0	COMPARATOR_SETTLE_TIME_9_8	Number of ESCCLK cycles to wait before taking another set of 16 samples of CAL_WEAK(Comparator output signal) Normal operational mode setting is 00 Power up value is 00

0x01A986CC MDSS_DSI_0_PHY_RSVD_1_PHY_DIGTOP**Type:** RW**Clock:** WCLK**Reset State:** 0x00000000**MDSS_DSI_0_PHY_RSVD_1_PHY_DIGTOP**

Bits	Name	Description
7:0	RSVD_7_0	

0x01A986D0 MDSS_DSI_0_PHY_DSIPHY_GLBL_MISR_CTRL**Type:** RW**Clock:** WCLK**Reset State:** 0x00000000**MDSS_DSI_0_PHY_DSIPHY_GLBL_MISR_CTRL**

Bits	Name	Description
7:4	RSVD_7_4	
3:0	DSIPHY_MISR_EN	DSIPHY_MISR_en<i> is MISR enable for each data lane <i>. Power up value is 0000

0x01A986DC MDSS_DSI_0_PHY_DSIPHY_LDO_CNTRL**Type:** RW**Clock:** WCLK**Reset State:** 0x00000000

MDSS_DSI_0_PHY_DSIPHY_LDO_CNTRL

Bits	Name	Description
7	RSVD_7	
6	LDO_MODE_1P2_EN	LDO Mode. DSIPHY Regulator 1.2V Enable. 1: DSIPHY Regulator is enabled. 0: DSIPHY Regulator is disabled. Normal operational mode setting is 0. DSIPHY_REGULATOR_CTRL_0[DSIPHY_REG_EN] must be set to 0 when LDO_MODE_EN is set to 1 Power up value is 0
5:3	LDO_VREF_SEL	DSIPHY Regulator Reference Select. Selects the value of reference voltage for the DSIPHY LDO Regulator. When LDO_VREF_DIV_EN = 1 000 : 0.400V 001 : 0.415V 010 : 0.385V 011 : 0.370V 100 : 0.430V 101 : 0.445V 110 : 0.460V 111 : 0.475V When LDO_VREF_CGM_EN = 1 000 : 0.400V 001 : 0.415V 010 : 0.385V 011 : 0.370V 100 : 0.430V 101 : 0.430V 110 : 0.430V 111 : 0.430V Note: LDO_VREF_CGM_EN and LDO_VREF_DIV_EN both cannot be 1. Setting both to 1 will disable both CGM_EN and DIV_EN Normal operational mode setting is 000 Power up value is 000
2	LDO_VREF_DIV_EN	To enable resistor divider as reference voltage Power up value is 0
1	LDO_VREF_CGM_EN	To enable Constant Gm as reference voltage Power up value is 0

MDSS_DSI_0_PHY_DSIPHY_LDO_CNTRL (cont.)

Bits	Name	Description
0	LDO_MODE_0P4_EN	LDO Mode. DSIPHY Regulator 0.4V Enable. 1: DSIPHY Regulator is enabled. 0: DSIPHY Regulator is disabled. Normal operational mode setting is 0. DSIPHY_REGULATOR_CTRL_0[DSIPHY_REG_EN] must be set to 0 when LDO_MODE_EN is set to 1 Power up value is 0

0x01A986E0 MDSS_DSI_0_PHY_DSIPHY_REGULATOR_CAL_STATUS0**Type:** R**Clock:** WCLK**Reset State:** 0x00000000**MDSS_DSI_0_PHY_DSIPHY_REGULATOR_CAL_STATUS0**

Bits	Name	Description
7:5	RSVD_7_5	
4	HW_CALIBRATOR_BUSY	To monitor hardware calibration status 1 : Calibration in progress 0 : Calibration complete
3:1	RSVD_3_1	
0	PC_STR_TOO_WEAK	To monitor CAL_WEAK(Comparator output signal)

0x01A986E4 MDSS_DSI_0_PHY_DSIPHY_REGULATOR_CAL_STATUS1**Type:** R**Clock:** WCLK**Reset State:** 0x00000000**MDSS_DSI_0_PHY_DSIPHY_REGULATOR_CAL_STATUS1**

Bits	Name	Description
7:4	HW_CAL_BOT_STR	Bottom calibrated impedance strength value. Stable after calibration is complete
3:0	HW_CAL_TOP_STR	Top calibrated impedance strength value. Stable after calibration is complete

0x01A986E8 MDSS_DSI_0_PHY_DSIPHY_GLBL_STATUS_DCC_SEL_P**Type:** R**Clock:** WCLK**Reset State:** 0x00000008**MDSS_DSI_0_PHY_DSIPHY_GLBL_STATUS_DCC_SEL_P**

Bits	Name	Description
7	DCC_DONE_SEL_P	Duty Cycle Correction Done SEL_P. Power up value is 0
6:0	DCC_SEL_P_STATUS	Duty Cycle Correction SEL_P status. Power up value is 000_1000

0x01A986EC MDSS_DSI_0_PHY_DSIPHY_GLBL_STATUS_DCC_SEL_N**Type:** R**Clock:** WCLK**Reset State:** 0x00000008**MDSS_DSI_0_PHY_DSIPHY_GLBL_STATUS_DCC_SEL_N**

Bits	Name	Description
7	DCC_DONE_SEL_N	Duty Cycle Correction Done SEL_N. Power up value is 0
6:0	DCC_SEL_N_STATUS	Duty Cycle Correction SEL_N status. Power up value is 000_1000

0x01A98700 MDSS_DSI_0_PHY_DSIPHY_REVISION_ID0**Type:** R**Clock:** WCLK**Reset State:** 0x00000000**MDSS_DSI_0_PHY_DSIPHY_REVISION_ID0**

Bits	Name	Description
7:0	STEP_7_0	Revision ID0

0x01A98704 MDSS_DSI_0_PHY_DSIPHY_REVISION_ID1

Type: R
Clock: WCLK
Reset State: 0x00000000

MDSS_DSI_0_PHY_DSIPHY_REVISION_ID1

Bits	Name	Description
7:0	STEP_15_8	Revision ID1

0x01A98708 MDSS_DSI_0_PHY_DSIPHY_REVISION_ID2

Type: R
Clock: WCLK
Reset State: 0x00000000

MDSS_DSI_0_PHY_DSIPHY_REVISION_ID2

Bits	Name	Description
7:0	MINOR_7_0	Revision ID2

0x01A9870C MDSS_DSI_0_PHY_DSIPHY_REVISION_ID3

Type: R
Clock: WCLK
Reset State: 0x00000010

MDSS_DSI_0_PHY_DSIPHY_REVISION_ID3

Bits	Name	Description
7:4	MAJOR	Revision ID3_major
3:0	MINOR_11_8	Revision ID3_minor

17.18.4 MDSS_DSI_0_PHY_REG_CM_MIPI_DSI_REG**0x01A98780 MDSS_DSI_0_PHY_REG_DSIPHY_REGULATOR_CTRL_0**

Type: RW
Clock: WCLK
Reset State: 0x00000000

MDSS_DSI_0_PHY_REG_DSIPHY_REGULATOR_CTRL_0

Bits	Name	Description
7:4	RSVD_7_4	
3:1	DSIPHY_REG_REF_SEL	DSIPHY Regulator Reference Select. Selects the value of reference voltage for the DSIPHY DCDC Regulator. When DSIPHY_REGULATOR_TEST[DIV_EN] = 1 000 : 0.400V 001 : 0.415V 010 : 0.385V 011 : 0.370V 100 : 0.430V 101 : 0.445V 110 : 0.460V 111 : 0.475V When DSIPHY_REGULATOR_TEST[NP_CGM_EN] = 1 000 : 0.400V 001 : 0.415V 010 : 0.385V 011 : 0.370V 100 : 0.430V 101 : 0.430V 110 : 0.430V 111 : 0.430V Note: NP_CGM_EN and DIV_EN both cannot be 1. Setting both to 1 will disable both CGM_EN and DIV_EN Normal operational mode setting is 000 Power up value is 000
0	DSIPHY_REG_EN	DC DC Mode. DSIPHY Regulator Enable. 1: DSIPHY Regulator is enabled. 0: DSIPHY Regulator is disabled. Normal operational mode setting is 1 and DSIPHY_LDO_CNTRL[LDO_MODE_EN] must be set to 0 Power up value is 0

0x01A98784 MDSS_DSI_0_PHY_REG_DSIPHY_REGULATOR_CTRL_1**Type:** RW**Clock:** WCLK**Reset State:** 0x00000006

MDSS_DSI_0_PHY_REG_DSIPHY_REGULATOR_CTRL_1

Bits	Name	Description
7:0	DSIPHY_REG_CONT_DISCHARGE_CYCLES	DSIPHY Continuous Mode Cycle Control. Used to control the number of cycles of continuous discharge mode. Normal operational mode setting is 0000_0110 Power up value is 0000_0110

0x01A98788 MDSS_DSI_0_PHY_REG_DSIPHY_REGULATOR_CTRL_2**Type:** RW**Clock:** WCLK**Reset State:** 0x00000004**MDSS_DSI_0_PHY_REG_DSIPHY_REGULATOR_CTRL_2**

Bits	Name	Description
7:0	DSIPHY_REG_CONT_CHARGE_CYCLES	DSIPHY Continuous Mode I Cycle Control. Used to control the number of cycles of continuous charge mode. Normal operational mode setting is 0000_0100 Power up value is 0000_0100

0x01A9878C MDSS_DSI_0_PHY_REG_DSIPHY_REGULATOR_CTRL_3**Type:** RW**Clock:** WCLK**Reset State:** 0x00000000**MDSS_DSI_0_PHY_REG_DSIPHY_REGULATOR_CTRL_3**

Bits	Name	Description
7:4	RSVD_7_4	
3:0	DSIPHY_REG_CLK_DIV_RATIO	DSIPHY REFCLK CLOCK DIVIDER Selection Bits. 0000 : div by1 0001 : div by2 0010 : div by3 . . . 1111--- div by16 Power up value is 0000

0x01A98790 MDSS_DSI_0_PHY_REG_DSIPHY_REGULATOR_CTRL_4**Type:** RW**Clock:** WCLK**Reset State:** 0x00000000**MDSS_DSI_0_PHY_REG_DSIPHY_REGULATOR_CTRL_4**

Bits	Name	Description
7:6	RSVD_7_6	
5	DIV_EN	To enable resistor divider as reference voltage Power up value is 0
4	NP_CGM_ENABLE	To enable constant Gm as reference voltage Power up value is 0
3:0	RSVD_3_0	

0x01A98794 MDSS_DSI_0_PHY_REG_DSIPHY_REGULATOR_CTRL_5**Type:** RW**Clock:** WCLK**Reset State:** 0x00000007**MDSS_DSI_0_PHY_REG_DSIPHY_REGULATOR_CTRL_5**

Bits	Name	Description
7:5	RSVD_7_5	
4	DSIPHY_REG_SWCTRL_DISCHARGE_EN	SWI switch discharge mode for NP_PSW_CTRL NP_NSW_CTRL NP_EMI_CTRL. Power up value is 0
3	DSIPHY_REG_SWI_FORCE_PWR_SWITCH	SWI control discharge enable for NP_PSW_CTRL NP_NSW_CTRL NP_EMI_CTRL. Power up value is 0
2	DSIPHY_REG_SWI_PSW_CTRL	SWI control bit for NP_PSW_CTRL. Power up value is 1
1	DSIPHY_REG_SWI_NSW_CTRL	SWI control bit for NP_NSW_CTRL. Power up value is 1

MDSS_DSI_0_PHY_REG_DSIPHY_REGULATOR_CTRL_5 (cont.)

Bits	Name	Description
0	DSIPHY_REG_SWI_EMI_CTRL	SWI control bit for NP_EMI_CTRL. Power up value is 1

0x01A98798 MDSS_DSI_0_PHY_REG_DSIPHY_REGULATOR_CAL_PWR_CFG**Type:** RW**Clock:** WCLK**Reset State:** 0x00000000**MDSS_DSI_0_PHY_REG_DSIPHY_REGULATOR_CAL_PWR_CFG**

Bits	Name	Description
7:5	RSVD_7_5	
4	SW_RESET	RESET for Digital blocks except control registers . Active high signal. 1: Assertion 0: Deassertion Normal operational mode setting is 0. Power up value is 0
3:1	RSVD_3_1	
0	DSIPHY_REG_PWRDN_B	DSIPHY Regulator NPL Power Down. 1 : Normal operation.Note: After the Regulator power down bit be set from '1' to '0' (powerup), regulator takes minumum 250uS worst case to charge up the external capacitor for MIPI DC/DC converter. 0 : Regulator NPL Block powerdown Normal operational mode setting is 1 Power up value is 0

0x01A9879C MDSS_DSI_0_PHY_REG_DSIPHY_REGULATOR_CTRL_6**Type:** RW**Clock:** WCLK**Reset State:** 0x00000015

MDSS_DSI_0_PHY_REG_DSIPHY_REGULATOR_CTRL_6

Bits	Name	Description
7:0	INIT_DISCHARGE_CYCLES	DSIPHY INIT_DISCHARGE delay Cycle Control. Used to control the number of cycles of continuous power_up_discharge mode. Normal operational mode setting is 0001_0101 Power up value is 0001_0101

0x01A987A0 MDSS_DSI_0_PHY_REG_DSIPHY_REGULATOR_RSVD_1**Type:** RW**Clock:** WCLK**Reset State:** 0x00000000**MDSS_DSI_0_PHY_REG_DSIPHY_REGULATOR_RSVD_1**

Bits	Name	Description
7:0	RSVD_7_0	

17.19 MDSS_VBIF

VBIF block

Decode space for top block.

17.19.1 VBIF_MDP

Description of the VBIF core

17.19.1.1 MDSS_VBIF_VBIFCORE

VBIF core Registers

Description of the VBIF core

0x01AC8004 MDSS_VBIF_VBIF_CLKON**Type:** RW**Reset State:** 0x00000000

Clock force ON for both DDR and OCMEM AXI clock.

MDSS_VBIF_VBIF_CLKON

Bits	Name	Description
15:8	HYSTERESIS_VALUE	Hysteresis counter value for all CGCs inside VBIF.
1	RESERVED	Reserved
0	FORCE_ON	0: dynamic clock gating on. 1: force on, always clock is on for all AXI clock domain both DDR and OCMEM domain without test bus logic.

0x01AC8008 MDSS_VBIF_VBIF_CLK_FORCE_CTRL0**Type:** RW**Reset State:** 0x00000000

individual force enables for each clock branch.

MDSS_VBIF_VBIF_CLK_FORCE_CTRL0

Bits	Name	Description
7	FORCE_ON_C7	force on for AXI clock of XIN7.
6	FORCE_ON_C6	force on for AXI clock of XIN6.
5	FORCE_ON_C5	force on for AXI clock of XIN5.
4	FORCE_ON_C4	force on for AXI clock of XIN4.
3	FORCE_ON_C3	force on for AXI clock of XIN3.
2	FORCE_ON_C2	force on for AXI clock of XIN2.
1	FORCE_ON_C1	force on for AXI clock of XIN1.
0	FORCE_ON_C0	0: dynamic clock gating on for AXI clock of XIN0. 1: force on, always clock is on for AXI clock of XIN0.

0x01AC800C MDSS_VBIF_VBIF_CLK_FORCE_CTRL1**Type:** RW**Reset State:** 0x00000000

individual force enables for each clock branch.

MDSS_VBIF_VBIF_CLK_FORCE_CTRL1

Bits	Name	Description
15	FORCE_ON_DCD	force on the DCD output for AXI DDR clock.

MDSS_VBIF_VBIF_CLK_FORCE_CTRL1 (cont.)

Bits	Name	Description
9	FORCE_ON_QOS	force on for all of QOS logics AXI clock.
8	FORCE_ON_PERF	force on for all of performance logics AXI,OCMEM clock.
7	RESERVED	Reserved
5	RESERVED	Reserved
4	RESERVED	Reserved
3	FORCE_ON_MM_MMU	force on for AXI clock of MM_MMU.
2	FORCE_ON_AHB2AHB	force on for AXI clock of ahb2ahb bridge(Master side).
0	FORCE_ON_ARB	0: dynamic clock gating on for all of ARBs(AXI Master block) AXI clock . 1: force on, always clock is on for all of ARBs AXI clock.

0x01AC8010 MDSS_VBIF_VBIF_QOS_OVERRIDE_EN**Type:** RW**Reset State:** 0x00000000

Enable VBIF to override the request priority and priority level signals provided by the client ports.

MDSS_VBIF_VBIF_QOS_OVERRIDE_EN

Bits	Name	Description
23	OVERRIDE_PRILVL_EN_C7	Enable VBIF to override the priority level for client 7.
22	OVERRIDE_PRILVL_EN_C6	Enable VBIF to override the priority level for client 6.
21	OVERRIDE_PRILVL_EN_C5	Enable VBIF to override the priority level for client 5.
20	OVERRIDE_PRILVL_EN_C4	Enable VBIF to override the priority level for client 4.
19	OVERRIDE_PRILVL_EN_C3	Enable VBIF to override the priority level for client 3.
18	OVERRIDE_PRILVL_EN_C2	Enable VBIF to override the priority level for client 2.
17	OVERRIDE_PRILVL_EN_C1	Enable VBIF to override the priority level for client 1.
16	OVERRIDE_PRILVL_EN_C0	Enable VBIF to override the priority level for client 0.
7	OVERRIDE_REQPRI_EN_C7	Enable VBIF to override the request priority for client 7.
6	OVERRIDE_REQPRI_EN_C6	Enable VBIF to override the request priority for client 6.
5	OVERRIDE_REQPRI_EN_C5	Enable VBIF to override the request priority for client 5.
4	OVERRIDE_REQPRI_EN_C4	Enable VBIF to override the request priority for client 4.

MDSS_VBIF_VBIF_QOS_OVERRIDE_EN (cont.)

Bits	Name	Description
3	OVERRIDE_REQPRI_EN_C3	Enable VBIF to override the request priority for client 3.
2	OVERRIDE_REQPRI_EN_C2	Enable VBIF to override the request priority for client 2.
1	OVERRIDE_REQPRI_EN_C1	Enable VBIF to override the request priority for client 1.
0	OVERRIDE_REQPRI_EN_C0	Enable VBIF to override the request priority for client 0.

0x01AC8014 MDSS_VBIF_VBIF_PRILVL_ACT_PROP_EN**Type:** RW**Reset State:** 0x00000000

Enable VBIF to only propagate priority level for active clients.

MDSS_VBIF_VBIF_PRILVL_ACT_PROP_EN

Bits	Name	Description
7	PRILVL_ACT_PROP_EN_C7	Enable VBIF to only propagate priority level for client 7 if it is active.
6	PRILVL_ACT_PROP_EN_C6	Enable VBIF to only propagate priority level for client 6 if it is active.
5	PRILVL_ACT_PROP_EN_C5	Enable VBIF to only propagate priority level for client 5 if it is active.
4	PRILVL_ACT_PROP_EN_C4	Enable VBIF to only propagate priority level for client 4 if it is active.
3	PRILVL_ACT_PROP_EN_C3	Enable VBIF to only propagate priority level for client 3 if it is active.
2	PRILVL_ACT_PROP_EN_C2	Enable VBIF to only propagate priority level for client 2 if it is active.
1	PRILVL_ACT_PROP_EN_C1	Enable VBIF to only propagate priority level for client 1 if it is active.
0	PRILVL_ACT_PROP_EN_C0	Enable VBIF to only propagate priority level for client 0 if it is active.

0x01AC8018 MDSS_VBIF_VBIF_QOS_OVERRIDE_REQPRI**Type:** RW**Reset State:** 0x00000000

Request priority override value

MDSS_VBIF_VBIF_QOS_OVERRIDE_REQPRI

Bits	Name	Description
15:14	OVERRIDE_REQPRI_C7	Request priority override value for client 7.
13:12	OVERRIDE_REQPRI_C6	Request priority override value for client 6.
11:10	OVERRIDE_REQPRI_C5	Request priority override value for client 5.
9:8	OVERRIDE_REQPRI_C4	Request priority override value for client 4.
7:6	OVERRIDE_REQPRI_C3	Request priority override value for client 3.
5:4	OVERRIDE_REQPRI_C2	Request priority override value for client 2.
3:2	OVERRIDE_REQPRI_C1	Request priority override value for client 1.
1:0	OVERRIDE_REQPRI_C0	Request priority override value for client 0.

0x01AC801C MDSS_VBIF_VBIF_QOS_OVERRIDE_PRILVL

Type: RW

Reset State: 0x00000000

Request priority override value

MDSS_VBIF_VBIF_QOS_OVERRIDE_PRILVL

Bits	Name	Description
15:14	OVERRIDE_PRILVL_C7	Priority level override value for client 7.
13:12	OVERRIDE_PRILVL_C6	Priority level override value for client 6.
11:10	OVERRIDE_PRILVL_C5	Priority level override value for client 5.
9:8	OVERRIDE_PRILVL_C4	Priority level override value for client 4.
7:6	OVERRIDE_PRILVL_C3	Priority level override value for client 3.
5:4	OVERRIDE_PRILVL_C2	Priority level override value for client 2.
3:2	OVERRIDE_PRILVL_C1	Priority level override value for client 1.
1:0	OVERRIDE_PRILVL_C0	Priority level override value for client 0.

0x01AC8020 MDSS_VBIF_VBIF_QOS_REMAP_00

Type: RW

Reset State: 0x00000000

Post-arbitration QOS remapping table entry 0

MDSS_VBIF_VBIF_QOS_REMAP_00

Bits	Name	Description
15:14	QOS_REMAP_00_C7	Post-arbitration QOS remapping table entry0 for client 7.
13:12	QOS_REMAP_00_C6	Post-arbitration QOS remapping table entry0 for client 6.
11:10	QOS_REMAP_00_C5	Post-arbitration QOS remapping table entry0 for client 5.
9:8	QOS_REMAP_00_C4	Post-arbitration QOS remapping table entry0 for client 4.
7:6	QOS_REMAP_00_C3	Post-arbitration QOS remapping table entry0 for client 3.
5:4	QOS_REMAP_00_C2	Post-arbitration QOS remapping table entry0 for client 2.
3:2	QOS_REMAP_00_C1	Post-arbitration QOS remapping table entry0 for client 1.
1:0	QOS_REMAP_00_C0	Post-arbitration QOS remapping table entry0 for client 0.

0x01AC8024 MDSS_VBIF_VBIF_QOS_REMAP_01**Type:** RW**Reset State:** 0x00005555

Post-arbitration QOS remapping table entry 1

MDSS_VBIF_VBIF_QOS_REMAP_01

Bits	Name	Description
15:14	QOS_REMAP_01_C7	Post-arbitration QOS remapping table entry1 for client 7.
13:12	QOS_REMAP_01_C6	Post-arbitration QOS remapping table entry1 for client 6.
11:10	QOS_REMAP_01_C5	Post-arbitration QOS remapping table entry1 for client 5.
9:8	QOS_REMAP_01_C4	Post-arbitration QOS remapping table entry1 for client 4.
7:6	QOS_REMAP_01_C3	Post-arbitration QOS remapping table entry1 for client 3.
5:4	QOS_REMAP_01_C2	Post-arbitration QOS remapping table entry1 for client 2.
3:2	QOS_REMAP_01_C1	Post-arbitration QOS remapping table entry1 for client 1.
1:0	QOS_REMAP_01_C0	Post-arbitration QOS remapping table entry1 for client 0.

0x01AC8028 MDSS_VBIF_VBIF_QOS_REMAP_10**Type:** RW**Reset State:** 0x0000AAAA

Post-arbitration QOS remapping table entry 2

MDSS_VBIF_VBIF_QOS_REMAP_10

Bits	Name	Description
15:14	QOS_REMAP_10_C7	Post-arbitration QOS remapping table entry2 for client 7.
13:12	QOS_REMAP_10_C6	Post-arbitration QOS remapping table entry2 for client 6.
11:10	QOS_REMAP_10_C5	Post-arbitration QOS remapping table entry2 for client 5.
9:8	QOS_REMAP_10_C4	Post-arbitration QOS remapping table entry2 for client 4.
7:6	QOS_REMAP_10_C3	Post-arbitration QOS remapping table entry2 for client 3.
5:4	QOS_REMAP_10_C2	Post-arbitration QOS remapping table entry2 for client 2.
3:2	QOS_REMAP_10_C1	Post-arbitration QOS remapping table entry2 for client 1.
1:0	QOS_REMAP_10_C0	Post-arbitration QOS remapping table entry2 for client 0.

0x01AC802C MDSS_VBIF_VBIF_QOS_REMAP_11**Type:** RW**Reset State:** 0x0000FFFF

Post-arbitration QOS remapping table entry 3

MDSS_VBIF_VBIF_QOS_REMAP_11

Bits	Name	Description
15:14	QOS_REMAP_11_C7	Post-arbitration QOS remapping table entry3 for client 7.
13:12	QOS_REMAP_11_C6	Post-arbitration QOS remapping table entry3 for client 6.
11:10	QOS_REMAP_11_C5	Post-arbitration QOS remapping table entry3 for client 5.
9:8	QOS_REMAP_11_C4	Post-arbitration QOS remapping table entry3 for client 4.
7:6	QOS_REMAP_11_C3	Post-arbitration QOS remapping table entry3 for client 3.
5:4	QOS_REMAP_11_C2	Post-arbitration QOS remapping table entry3 for client 2.
3:2	QOS_REMAP_11_C1	Post-arbitration QOS remapping table entry3 for client 1.
1:0	QOS_REMAP_11_C0	Post-arbitration QOS remapping table entry3 for client 0.

0x01AC80AC MDSS_VBIF_VBIF_WRITE_GATHER_EN**Type:** RW**Reset State:** 0x00000000

By default the XIN blocks forward address channel write requests to the arbiter block without consideration for whether they have received all the write data.

This can disadvantageous as it can result in the output write data bus being under utilized.

MDSS_VBIF_VBIF_WRITE_GATHER_EN

Bits	Name	Description
7	VBIF_WRITE_GATHER_EN_C7	Enable write gathering for Client 7
6	VBIF_WRITE_GATHER_EN_C6	Enable write gathering for Client 6
5	VBIF_WRITE_GATHER_EN_C5	Enable write gathering for Client 5
4	VBIF_WRITE_GATHER_EN_C4	Enable write gathering for Client 4
3	VBIF_WRITE_GATHER_EN_C3	Enable write gathering for Client 3
2	VBIF_WRITE_GATHER_EN_C2	Enable write gathering for Client 2
1	VBIF_WRITE_GATHER_EN_C1	Enable write gathering for Client 1
0	VBIF_WRITE_GATHER_EN_C0	Enable write gathering for Client 0

0x01AC80B0 MDSS_VBIF_VBIF_IN_RD_LIM_CONF0

Type: RW

Reset State: 0x00080808

The maximum number of pending reads from each Client is programmable.

MDSS_VBIF_VBIF_IN_RD_LIM_CONF0

Bits	Name	Description
31:24	IN_RD_LIM_C3	The maximum number of pending reads from Client 3. Must be <= the HW parameter that XBAR_IN_RD_LIM
23:16	IN_RD_LIM_C2	The maximum number of pending reads from Client 2. Must be <= the HW parameter that XBAR_IN_RD_LIM
15:8	IN_RD_LIM_C1	The maximum number of pending reads from Client 1. Must be <= the HW parameter that XBAR_IN_RD_LIM
7:0	IN_RD_LIM_C0	The maximum number of pending reads from Client 0. Must be <= the HW parameter that XBAR_IN_RD_LIM

0x01AC80B4 MDSS_VBIF_VBIF_IN_RD_LIM_CONF1**Type:** RW**Reset State:** 0x08000800

The maximum number of pending reads from each Client is programmable.

MDSS_VBIF_VBIF_IN_RD_LIM_CONF1

Bits	Name	Description
31:24	IN_RD_LIM_C7	The maximum number of pending reads from Client 7. Must be <= the HW parameter that XBAR_IN_RD_LIM
23:16	IN_RD_LIM_C6	The maximum number of pending reads from Client 6. Must be <= the HW parameter that XBAR_IN_RD_LIM
15:8	IN_RD_LIM_C5	The maximum number of pending reads from Client 5. Must be <= the HW parameter that XBAR_IN_RD_LIM
7:0	IN_RD_LIM_C4	The maximum number of pending reads from Client 4. Must be <= the HW parameter that XBAR_IN_RD_LIM

0x01AC80C0 MDSS_VBIF_VBIF_IN_WR_LIM_CONF0**Type:** RW**Reset State:** 0x10000000

The maximum number of pending writes from each Client is programmable.

MDSS_VBIF_VBIF_IN_WR_LIM_CONF0

Bits	Name	Description
31:24	IN_WR_LIM_C3	The maximum number of pending writes from Client 3. Must be <= the HW parameter that XBAR_IN_WR_LIM
23:16	IN_WR_LIM_C2	The maximum number of pending writes from Client 2. Must be <= the HW parameter that XBAR_IN_WR_LIM
15:8	IN_WR_LIM_C1	The maximum number of pending writes from Client 1. Must be <= the HW parameter that XBAR_IN_WR_LIM
7:0	IN_WR_LIM_C0	The maximum number of pending writes from Client 0. Must be <= the HW parameter that XBAR_IN_WR_LIM

0x01AC80C4 MDSS_VBIF_VBIF_IN_WR_LIM_CONF1**Type:** RW**Reset State:** 0x00100000

The maximum number of pending writes from each Client is programmable.

MDSS_VBIF_VBIF_IN_WR_LIM_CONF1

Bits	Name	Description
31:24	IN_WR_LIM_C7	The maximum number of pending writes from Client 7. Must be <= the HW parameter that XBAR_IN_WR_LIM
23:16	IN_WR_LIM_C6	The maximum number of pending writes from Client 6. Must be <= the HW parameter that XBAR_IN_WR_LIM
15:8	IN_WR_LIM_C5	The maximum number of pending writes from Client 5. Must be <= the HW parameter that XBAR_IN_WR_LIM
7:0	IN_WR_LIM_C4	The maximum number of pending writes from Client 4. Must be <= the HW parameter that XBAR_IN_WR_LIM

0x01AC80D0 MDSS_VBIF_VBIF_OUT_RD_LIM_CONF0

Type: RW

Reset State: 0x00000010

The maximum number of pending reads from each AXI is programmable.

MDSS_VBIF_VBIF_OUT_RD_LIM_CONF0

Bits	Name	Description
7:0	OUT_RD_LIM_A0	The maximum number of pending reads from AXI 0. Must be <= the HW parameter that XBAR_OUT_RD_LIM

0x01AC80D4 MDSS_VBIF_VBIF_OUT_WR_LIM_CONF0

Type: RW

Reset State: 0x00000010

The maximum number of pending reads from each AXI is programmable.

MDSS_VBIF_VBIF_OUT_WR_LIM_CONF0

Bits	Name	Description
7:0	OUT_WR_LIM_A0	The maximum number of pending writes from AXI 0. Must be <= the HW parameter that XBAR_OUT_WR_LIM

0x01AC80D8 MDSS_VBIF_VBIF_DDR_OUT_MAX_BURST**Type:** RW**Reset State:** 0x00000707

The maximum burst length from any DDR AXI is programmable.

The client bursts will be broken up according to these limits, although the client will not be aware of it.

MDSS_VBIF_VBIF_DDR_OUT_MAX_BURST

Bits	Name	Description
15:8	DDR_OUT_MAX_WR_BURST	The maximum number of write beats minus 1 allowed on the AXI bus (arbiter outputs). Must be < the HW parameter that XBAR_OUT_WR_BURST
7:0	DDR_OUT_MAX_RD_BURST	The maximum number of read beats minus 1 allowed on the AXI bus (arbiter outputs). Must be < the HW parameter that XBAR_OUT_RD_BURST

0x01AC80E0 MDSS_VBIF_VBIF_ISO_SYNC**Type:** RW**Reset State:** 0x00000000

Enables iso-sync operation between client and arbiter, 1-bit per client. Client & AXI clocks must be iso-sync and have balanced clock trees.

MDSS_VBIF_VBIF_ISO_SYNC

Bits	Name	Description
7	ISO_SYNC_C7	Enables for Client 7
6	ISO_SYNC_C6	Enables for Client 6
5	ISO_SYNC_C5	Enables for Client 5
4	ISO_SYNC_C4	Enables for Client 4
3	ISO_SYNC_C3	Enables for Client 3
2	ISO_SYNC_C2	Enables for Client 2
1	ISO_SYNC_C1	Enables for Client 1
0	ISO_SYNC_C0	Enables for Client 0

0x01AC80F0 MDSS_VBIF_VBIF_ARB_CTL**Type:** RW**Reset State:** 0x00000010

Each arbiter can be programmed to utilize fixed priority, slave-aware or Round-Robin arbitration

MDSS_VBIF_VBIF_ARB_CTL

Bits	Name	Description
4	ROUND_ROBIN_ARB_EN_A0	Enables Round-Robin arbitration for AXI port 0. If enabled, Slave-Aware and Longest-Time-Waiting arbitrations are ignored.
0	FIXED_ARB_EN_A0	Enables fixed priority arbitration for AXI port 0. If enabled, the other arbitrations are ignored

0x01AC80F4 MDSS_VBIF_VBIF_DDR_ARB_CONF0**Type:** RW**Reset State:** 0x00000000

Client limits for DDR AXI, 5 bits per client.

Each client limit is the maximum number of consecutive grants minus 1 to give that client before allowing requests from other clients.

This is used for slave-aware and longest-time-waiting arbitration.

MDSS_VBIF_VBIF_DDR_ARB_CONF0

Bits	Name	Description
29:25	DDR_CLIENT_LIM_C5	Client limits for Client 5
24:20	DDR_CLIENT_LIM_C4	Client limits for Client 4
19:15	DDR_CLIENT_LIM_C3	Client limits for Client 3
14:10	DDR_CLIENT_LIM_C2	Client limits for Client 2
9:5	DDR_CLIENT_LIM_C1	Client limits for Client 1
4:0	DDR_CLIENT_LIM_C0	Client limits for Client 0

0x01AC80F8 MDSS_VBIF_VBIF_DDR_ARB_CONF1**Type:** RW**Reset State:** 0x00000000

Client limits

MDSS_VBIF_VBIF_DDR_ARB_CONF1

Bits	Name	Description
9:5	DDR_CLIENT_LIM_C7	Client limits for Client 7
4:0	DDR_CLIENT_LIM_C6	Client limits for Client 6

0x01AC8100 MDSS_VBIF_VBIF_FIXED_ARB_CONF0**Type:** RW**Reset State:** 0x76543210

A list of Client number for Fixed arbitration

The highest priority client number on highest number of register name.

For example, If total unnumber of client is 12, and client2 has highest priority should be write 2 into FIXED_ARB_LIST_11. and ignore LIST_12 ~ 15.

MDSS_VBIF_VBIF_FIXED_ARB_CONF0

Bits	Name	Description
31:28	FIXED_ARB_LIST_7	A list of Client number for Fixed arbitration
27:24	FIXED_ARB_LIST_6	A list of Client number for Fixed arbitration
23:20	FIXED_ARB_LIST_5	A list of Client number for Fixed arbitration
19:16	FIXED_ARB_LIST_4	A list of Client number for Fixed arbitration
15:12	FIXED_ARB_LIST_3	A list of Client number for Fixed arbitration
11:8	FIXED_ARB_LIST_2	A list of Client number for Fixed arbitration
7:4	FIXED_ARB_LIST_1	A list of Client number for Fixed arbitration
3:0	FIXED_ARB_LIST_0	A list of Client number for Fixed arbitration

0x01AC8124 MDSS_VBIF_VBIF_ROUND_ROBIN_QOS_ARB**Type:** RW**Reset State:** 0x00000003

Round-Robin Arbiter with QoS signal that from MM core to VBIF via input signal.

MDSS_VBIF_VBIF_ROUND_ROBIN_QOS_ARB

Bits	Name	Description
1	MMU_RR_QOS_EN	Enable Round-Robin Arbiter with QoS for MMU Virtual Address Arbiter.
0	RR_QOS_EN	Enable Round-Robin Arbiter with QoS for AXI Master Arbiter. If Round-Robin Arbiter is disabled, ignore this register.

0x01AC8160 MDSS_VBIF_VBIF_OUT_AXI_AMEMTYPE_CONF0**Type:** RW**Reset State:** 0x22222222

When a request is finally granted, this AXI signal is controlled on a per-client basis.

This is driven to AXI port but does not directly impact VBIF functionality.

MDSS_VBIF_VBIF_OUT_AXI_AMEMTYPE_CONF0

Bits	Name	Description
30:28	AXI_AMEMTYPE_C7	A MEMTYPE signal for Client 7
26:24	AXI_AMEMTYPE_C6	A MEMTYPE signal for Client 6
22:20	AXI_AMEMTYPE_C5	A MEMTYPE signal for Client 5
18:16	AXI_AMEMTYPE_C4	A MEMTYPE signal for Client 4
14:12	AXI_AMEMTYPE_C3	A MEMTYPE signal for Client 3
10:8	AXI_AMEMTYPE_C2	A MEMTYPE signal for Client 2
6:4	AXI_AMEMTYPE_C1	A MEMTYPE signal for Client 1
2:0	AXI_AMEMTYPE_C0	A MEMTYPE signal for Client 0

0x01AC8168 MDSS_VBIF_VBIF_OUT_AXI_AINST_OVERRIDE_EN**Type:** RW**Reset State:** 0x00000000

Enable VBIF to override the AINST signals provided by the client ports or MM-MMU.

MDSS_VBIF_VBIF_OUT_AXI_AINST_OVERRIDE_EN

Bits	Name	Description
7	AXI_AINST_OVERRIDE_EN_C7	Enable VBIF to override the AINST signal for Client 7

MDSS_VBIF_VBIF_OUT_AXI_AINST_OVERRIDE_EN (cont.)

Bits	Name	Description
6	AXI_AINST_OVERRIDE_EN_C6	Enable VBIF to override the AINST signal for Client 6
5	AXI_AINST_OVERRIDE_EN_C5	Enable VBIF to override the AINST signal for Client 5
4	AXI_AINST_OVERRIDE_EN_C4	Enable VBIF to override the AINST signal for Client 4
3	AXI_AINST_OVERRIDE_EN_C3	Enable VBIF to override the AINST signal for Client 3
2	AXI_AINST_OVERRIDE_EN_C2	Enable VBIF to override the AINST signal for Client 2
1	AXI_AINST_OVERRIDE_EN_C1	Enable VBIF to override the AINST signal for Client 1
0	AXI_AINST_OVERRIDE_EN_C0	Enable VBIF to override the AINST signal for client 0.

0x01AC816C MDSS_VBIF_VBIF_OUT_AXI_AINST**Type:** RW**Reset State:** 0x00000000

When a request is finally granted, this AXI signal is controlled on a per-client basis.

This is driven to AXI port but does not directly impact VBIF functionality.

Only valid when VBIF_OUT_AXI_AINST_OVERRIDE_EN is enabled

MDSS_VBIF_VBIF_OUT_AXI_AINST

Bits	Name	Description
7	AXI_AINST_C7	AINST signal for Client 7
6	AXI_AINST_C6	AINST signal for Client 6
5	AXI_AINST_C5	AINST signal for Client 5
4	AXI_AINST_C4	AINST signal for Client 4
3	AXI_AINST_C3	AINST signal for Client 3
2	AXI_AINST_C2	AINST signal for Client 2
1	AXI_AINST_C1	AINST signal for Client 1
0	AXI_AINST_C0	AINST signal for Client 0

0x01AC8170 MDSS_VBIF_VBIF_OUT_AXI_ASHARED**Type:** RW**Reset State:** 0x000000FF

When a request is finally granted, this AXI signal is controlled on a per-client basis.

This is driven to AXI port but does not directly impact VBIF functionality.

MDSS_VBIF_VBIF_OUT_AXI_ASHARED

Bits	Name	Description
7	AXI_ASHARED_C7	ASHARED signal for Client 7
6	AXI_ASHARED_C6	ASHARED signal for Client 6
5	AXI_ASHARED_C5	ASHARED signal for Client 5
4	AXI_ASHARED_C4	ASHARED signal for Client 4
3	AXI_ASHARED_C3	ASHARED signal for Client 3
2	AXI_ASHARED_C2	ASHARED signal for Client 2
1	AXI_ASHARED_C1	ASHARED signal for Client 1
0	AXI_ASHARED_C0	ASHARED signal for Client 0

0x01AC8174 MDSS_VBIF_VBIF_OUT_AXI_AINNERSHARED**Type:** RW**Reset State:** 0x00000000

When a request is finally granted, this AXI signal is controlled on a per-client basis.

This is driven to AXI port but does not directly impact VBIF functionality.

MDSS_VBIF_VBIF_OUT_AXI_AINNERSHARED

Bits	Name	Description
7	AXI_AINNERSHARED_C7	AINNERSHARED signal for Client 7
6	AXI_AINNERSHARED_C6	AINNERSHARED signal for Client 6
5	AXI_AINNERSHARED_C5	AINNERSHARED signal for Client 5
4	AXI_AINNERSHARED_C4	AINNERSHARED signal for Client 4
3	AXI_AINNERSHARED_C3	AINNERSHARED signal for Client 3
2	AXI_AINNERSHARED_C2	AINNERSHARED signal for Client 2
1	AXI_AINNERSHARED_C1	AINNERSHARED signal for Client 1
0	AXI_AINNERSHARED_C0	AINNERSHARED signal for Client 0

0x01AC8178 MDSS_VBIF_VBIF_OUT_AXI_AOOO_EN**Type:** RW**Reset State:** 0x000000FF

Enable signal for controlling AOOOWR/AOOORD signals on AXI protocol, this signal is controlled on a per-client basis.

Actually, AOOOWR/RD signals are controlled by parent core(via client I/F port) dynamically.

But, it can be controlled by SW for performance test, debugging, etc

If enable this register, ignore cooo value of client port and SW value applies to AOOO signal.

MDSS_VBIF_VBIF_OUT_AXI_AOOO_EN

Bits	Name	Description
7	AXI_AOOO_EN_C7	SW AOOOAOOO_EN signal for Client 7
6	AXI_AOOO_EN_C6	SW AOOOAOOO_EN signal for Client 6
5	AXI_AOOO_EN_C5	SW AOOOAOOO_EN signal for Client 5
4	AXI_AOOO_EN_C4	SW AOOOAOOO_EN signal for Client 4
3	AXI_AOOO_EN_C3	SW AOOOAOOO_EN signal for Client 3
2	AXI_AOOO_EN_C2	SW AOOOAOOO_EN signal for Client 2
1	AXI_AOOO_EN_C1	SW AOOOAOOO_EN signal for Client 1
0	AXI_AOOO_EN_C0	SW AOOO enable signal for Client 0

0x01AC817C MDSS_VBIF_VBIF_OUT_AXI_AOOO**Type:** RW**Reset State:** 0x00FF00FF

When a request is finally granted and AXI_AOOO_EN is enabled, this AXI signal is controlled on a per-client basis.

If AXI_AOOO_EN is disable, ignore this value.

MDSS_VBIF_VBIF_OUT_AXI_AOOO

Bits	Name	Description
23	AXI_AOOORD_C7	AOOORD signal for Client 7 if AXI_AOOO_EN_C7
22	AXI_AOOORD_C6	AOOORD signal for Client 6 if AXI_AOOO_EN_C6
21	AXI_AOOORD_C5	AOOORD signal for Client 5 if AXI_AOOO_EN_C5
20	AXI_AOOORD_C4	AOOORD signal for Client 4 if AXI_AOOO_EN_C4

MDSS_VBIF_VBIF_OUT_AXI_AOOO (cont.)

Bits	Name	Description
19	AXI_AOOORD_C3	AOOORD signal for Client 3 if AXI_AOOO_EN_C3
18	AXI_AOOORD_C2	AOOORD signal for Client 2 if AXI_AOOO_EN_C2
17	AXI_AOOORD_C1	AOOORD signal for Client 1 if AXI_AOOO_EN_C1
16	AXI_AOOORD_C0	AOOORD signal for Client 0 if AXI_AOOO_EN_C0
7	AXI_AOOOWR_C7	AOOOWR signal for Client 7 if AXI_AOOO_EN_C7
6	AXI_AOOOWR_C6	AOOOWR signal for Client 6 if AXI_AOOO_EN_C6
5	AXI_AOOOWR_C5	AOOOWR signal for Client 5 if AXI_AOOO_EN_C5
4	AXI_AOOOWR_C4	AOOOWR signal for Client 4 if AXI_AOOO_EN_C4
3	AXI_AOOOWR_C3	AOOOWR signal for Client 3 if AXI_AOOO_EN_C3
2	AXI_AOOOWR_C2	AOOOWR signal for Client 2 if AXI_AOOO_EN_C2
1	AXI_AOOOWR_C1	AOOOWR signal for Client 1 if AXI_AOOO_EN_C1
0	AXI_AOOOWR_C0	AOOOWR signal for Client 0 if AXI_AOOO_EN_C0

0x01AC8180 MDSS_VBIF_VBIF_OUT_AXI_ANOALLOCATE**Type:** RW**Reset State:** 0x00000000

When a request is finally granted, this AXI signal is controlled on a per-client basis.

This is driven to AXI port but does not directly impact VBIF functionality.

MDSS_VBIF_VBIF_OUT_AXI_ANOALLOCATE

Bits	Name	Description
7	AXI_ANOALLOCATE_C7	ANOALLOCATE signal for Client 7
6	AXI_ANOALLOCATE_C6	ANOALLOCATE signal for Client 6
5	AXI_ANOALLOCATE_C5	ANOALLOCATE signal for Client 5
4	AXI_ANOALLOCATE_C4	ANOALLOCATE signal for Client 4
3	AXI_ANOALLOCATE_C3	ANOALLOCATE signal for Client 3
2	AXI_ANOALLOCATE_C2	ANOALLOCATE signal for Client 2
1	AXI_ANOALLOCATE_C1	ANOALLOCATE signal for Client 1
0	AXI_ANOALLOCATE_C0	ANOALLOCATE signal for Client 0

0x01AC8184 MDSS_VBIF_VBIF_WD_TIMEOUT**Type:** RW**Reset State:** 0x00000000

Write Data Timeout function. enable bit per Client.

Each Client block generate a interrupt when write data timeout with enabled.

MDSS_VBIF_VBIF_WD_TIMEOUT

Bits	Name	Description
19:16	WD_TIMEOUT_LOG2	This field (shared by all Clients) contains log2(write data timeout), i.e. the bit number of the timeout counter which, when 1, will trigger the timeout. For example, 15 here means the timeout is 32768 AXI clock cycles. This field is ignored when AXI_WD_TIMEOUT_EN=0
7	WD_TIMEOUT_EN_C7	Write Data Timeout enable for Client 7
6	WD_TIMEOUT_EN_C6	Write Data Timeout enable for Client 6
5	WD_TIMEOUT_EN_C5	Write Data Timeout enable for Client 5
4	WD_TIMEOUT_EN_C4	Write Data Timeout enable for Client 4
3	WD_TIMEOUT_EN_C3	Write Data Timeout enable for Client 3
2	WD_TIMEOUT_EN_C2	Write Data Timeout enable for Client 2
1	WD_TIMEOUT_EN_C1	Write Data Timeout enable for Client 1
0	WD_TIMEOUT_EN_C0	Write Data Timeout enable for Client 0

0x01AC8190 MDSS_VBIF_VBIF_PND_ERR**Type:** R**Reset State:** 0x00000000

Pending error status. per client

When error is happened and not masked, this register is set to 1.

But masked, not appear in this register.

MDSS_VBIF_VBIF_PND_ERR

Bits	Name	Description
7	CLIENT_PND_ERR_C7	Indicates than an error in Client 7 occured since the last reset or ERR_CLR was set.

MDSS_VBIF_VBIF_PND_ERR (cont.)

Bits	Name	Description
6	CLIENT_PND_ERR_C6	Indicates than an error in Client 6 occured since the last reset or ERR_CLR was set.
5	CLIENT_PND_ERR_C5	Indicates than an error in Client 5 occured since the last reset or ERR_CLR was set.
4	CLIENT_PND_ERR_C4	Indicates than an error in Client 4 occured since the last reset or ERR_CLR was set.
3	CLIENT_PND_ERR_C3	Indicates than an error in Client 3 occured since the last reset or ERR_CLR was set.
2	CLIENT_PND_ERR_C2	Indicates than an error in Client 2 occured since the last reset or ERR_CLR was set.
1	CLIENT_PND_ERR_C1	Indicates than an error in Client 1 occured since the last reset or ERR_CLR was set.
0	CLIENT_PND_ERR_C0	Indicates than an error in Client 0 occured since the last reset or ERR_CLR was set.

0x01AC8194 MDSS_VBIF_VBIF_SRC_ERR**Type:** R**Reset State:** 0x00000000

Source error status. per client

When error is happened, this register is set to 1 evenif masked.

MDSS_VBIF_VBIF_SRC_ERR

Bits	Name	Description
7	CLIENT_SRC_ERR_C7	Indicates than an error in Client 7 occured since the last reset or ERR_CLR was set.
6	CLIENT_SRC_ERR_C6	Indicates than an error in Client 6 occured since the last reset or ERR_CLR was set.
5	CLIENT_SRC_ERR_C5	Indicates than an error in Client 5 occured since the last reset or ERR_CLR was set.
4	CLIENT_SRC_ERR_C4	Indicates than an error in Client 4 occured since the last reset or ERR_CLR was set.
3	CLIENT_SRC_ERR_C3	Indicates than an error in Client 3 occured since the last reset or ERR_CLR was set.
2	CLIENT_SRC_ERR_C2	Indicates than an error in Client 2 occured since the last reset or ERR_CLR was set.
1	CLIENT_SRC_ERR_C1	Indicates than an error in Client 1 occured since the last reset or ERR_CLR was set.

MDSS_VBIF_VBIF_SRC_ERR (cont.)

Bits	Name	Description
0	CLIENT_SRC_ERR_C0	Indicates than an error in Client 0 occurred since the last reset or ERR_CLR was set.

0x01AC8198 MDSS_VBIF_VBIF_ERR_MASK**Type:** RW**Reset State:** 0x00000000

Error Mask. per client.

When this mask register is set to 1, error is not propagated to Interrupt signal port.

But, can check the error within SRC_ERR register and ERR_INFO register.

MDSS_VBIF_VBIF_ERR_MASK

Bits	Name	Description
7	CLIENT_ERR_MASK_C7	Error mask for Client 7
6	CLIENT_ERR_MASK_C6	Error mask for Client 6
5	CLIENT_ERR_MASK_C5	Error mask for Client 5
4	CLIENT_ERR_MASK_C4	Error mask for Client 4
3	CLIENT_ERR_MASK_C3	Error mask for Client 3
2	CLIENT_ERR_MASK_C2	Error mask for Client 2
1	CLIENT_ERR_MASK_C1	Error mask for Client 1
0	CLIENT_ERR_MASK_C0	Error mask for Client 0

0x01AC819C MDSS_VBIF_VBIF_ERR_CLR**Type:** RW**Reset State:** 0x00000000

Error clear(interrupt clear). per client

After Client error has been captured, this field should be written to 1 in order to clear the previous error and enable future errors to be captured.

NOT need to write to 0, This field will be changed to 0 automatically when error is cleared.

MDSS_VBIF_VBIF_ERR_CLR

Bits	Name	Description
7	CLIENT_ERR_CLR_C7	Error Clear for Client 7
6	CLIENT_ERR_CLR_C6	Error Clear for Client 6
5	CLIENT_ERR_CLR_C5	Error Clear for Client 5
4	CLIENT_ERR_CLR_C4	Error Clear for Client 4
3	CLIENT_ERR_CLR_C3	Error Clear for Client 3
2	CLIENT_ERR_CLR_C2	Error Clear for Client 2
1	CLIENT_ERR_CLR_C1	Error Clear for Client 1
0	CLIENT_ERR_CLR_C0	Error Clear for Client 0

0x01AC81A0 MDSS_VBIF_VBIF_ERR_INFO**Type:** RW**Reset State:** 0x00000000

Error informations. only the first error is captured. The error must be cleared to enable future errors to be captured.

MDSS_VBIF_VBIF_ERR_INFO

Bits	Name	Type	Description
15:8	CLIENT_MID	R	The value on Client MID when the last error occurred. Note that value of this CLIENT_MID register will be 0 when write responses are disabled(XBAR_INn_WR_RESP=0)
7:6	AXI_RESP	R	The value of AXI_RRSP or AXI_BRSP when the last error occurred.
5:4	ERR_TYPE	R	The type of the last error. Read only
3:0	CLIENT_SEL		select number of Client to can read follow error info. (type, resp, mid and tid) For example, if this field is 3, can read error information about Client 3 error.

0x01AC8200 MDSS_VBIF_VBIF_XIN_HALT_CTRL0**Type:** RW**Reset State:** 0x00000000

VBIF Client HALT REQ

MDSS_VBIF_VBIF_XIN_HALT_CTRL0

Bits	Name	Description
7	HALT_REQ_C7	HALT request and internal SW-reset for Client I/F 7 .
6	HALT_REQ_C6	HALT request and internal SW-reset for Client I/F 6 .
5	HALT_REQ_C5	HALT request and internal SW-reset for Client I/F 5 .
4	HALT_REQ_C4	HALT request and internal SW-reset for Client I/F 4 .
3	HALT_REQ_C3	HALT request and internal SW-reset for Client I/F 3 .
2	HALT_REQ_C2	HALT request and internal SW-reset for Client I/F 2 .
1	HALT_REQ_C1	HALT request and internal SW-reset for Client I/F 1 .
0	HALT_REQ_C0	HALT request and internal SW-reset for Client I/F 0. When asserted, that Client I/F block(XIN) does not accept new requests/write data from all XINs(Client I/F block) and does not return the RESP/read data to MM core client. And also does not send the request to AXI. And waiting until return all outstanding RESP/read data from AXI Slave(NoC). When all RESPs are returned, ACK/IDLE/writes/reads_done are asserted. Finally, when that XIN block is IDLE, assert the internal sw-reset automatically for that XIN. MUST set to '0' to this field again after ACK/IDLE goes to High.

0x01AC8204 MDSS_VBIF_VBIF_XIN_HALT_CTRL1**Type:** R**Reset State:** 0x00FF0000

VBIF Client HALT ACK/IDLE

MDSS_VBIF_VBIF_XIN_HALT_CTRL1

Bits	Name	Description
23	HALT_IDLE_C7	IDLE for Client 7 .
22	HALT_IDLE_C6	IDLE for Client 6 .
21	HALT_IDLE_C5	IDLE for Client 5 .
20	HALT_IDLE_C4	IDLE for Client 4 .
19	HALT_IDLE_C3	IDLE for Client 3 .
18	HALT_IDLE_C2	IDLE for Client 2 .
17	HALT_IDLE_C1	IDLE for Client 1 .

MDSS_VBIF_VBIF_XIN_HALT_CTRL1 (cont.)

Bits	Name	Description
16	HALT_IDLE_C0	IDLE for Client 0 .
7	HALT_ACK_C7	Acknowledge about the Halt Request of the Client 7 .
6	HALT_ACK_C6	Acknowledge about the Halt Request of the Client 6 .
5	HALT_ACK_C5	Acknowledge about the Halt Request of the Client 5 .
4	HALT_ACK_C4	Acknowledge about the Halt Request of the Client 4 .
3	HALT_ACK_C3	Acknowledge about the Halt Request of the Client 3 .
2	HALT_ACK_C2	Acknowledge about the Halt Request of the Client 2 .
1	HALT_ACK_C1	Acknowledge about the Halt Request of the Client 1 .
0	HALT_ACK_C0	Acknowledge about the Halt Request of the Client 0. Reset value is '0'. After Halt requests, this acknowledge field is set to '1' when Client is IDLE.

0x01AC8208 MDSS_VBIF_VBIF_AXI_HALT_CTRL0**Type:** RW**Reset State:** 0x00000000

VBIF AXI HALT REQ

MDSS_VBIF_VBIF_AXI_HALT_CTRL0

Bits	Name	Description
0	HALT_REQ	HALT request for all VBIF AXI I/F(both DDR and OCMEM) . When asserted, AXI Master block(ARB) does not accept new requests from all XINs(Client I/F block) and MM-MMU(for HTW). And waiting until return all outstanding RESP/read data from AXI Slave(NoC). When all RESPs are returned, ACK/IDLE is asserted. This HATL request requires HW reset for VBIF.

0x01AC820C MDSS_VBIF_VBIF_AXI_HALT_CTRL1**Type:** R**Reset State:** 0x00000010

VBIF AXI HALT ACK/IDLE

MDSS_VBIF_VBIF_AXI_HALT_CTRL1

Bits	Name	Description
4	HALT_IDLE_A0	IDLE for AXI I/F 0 .
0	HALT_ACK	Acknowledge about the Halt Request of the AXI I/F. Reset value is '0'. After Halt requests, this acknowledge field is set to '1' when all AXI I/F are IDLE.

0x01AC8260 MDSS_VBIF_VBIF_STM_CONF**Type:** RW**Reset State:** 0x00000000

configuration for STM HW Events.

MDSS_VBIF_VBIF_STM_CONF

Bits	Name	Description
4:0	STM_GROUP_SEL	Select the STM Group TBD

0x01AC8270 MDSS_VBIF_VBIF_WR_BUF_FOR_AP**Type:** RW**Reset State:** 0x00000000

Write data buffer for AXI performance.

MDSS_VBIF_VBIF_WR_BUF_FOR_AP

Bits	Name	Description
0	WR_BUF_FOR_AP_A0	Optional feature: Write data buffer for AXI performance. A.K.A. Hold back AXI write request til write data is ready. (Write is granted by ARB, then write data from XIN is buffered in write data FIFO or passed through to AXI, then write request is issued to AXI, then wait for that write request (not data) to be accepted, then resume arbitration. So while this feature may improve efficiency of write data channel on AXI, it may reduce performance of both reads and writes for this core.) for AXI Master I/F 0.

**0x01AC8300+MDSS_VBIF_VBIF_PERF_CNT_ENn, n=[0..3]
0x4*n****Type:** RW**Reset State:** 0x00000000

Performance counter enable.

MDSS_VBIF_VBIF_PERF_CNT_ENn

Bits	Name	Description
0	PERF_CNT_EN	Performance counter enable

**0x01AC8320+MDSS_VBIF_VBIF_PERF_CNT_CLRn, n=[0..3]
0x4*n****Type:** RW**Reset State:** 0x00000000

Performance counter clear.

MDSS_VBIF_VBIF_PERF_CNT_CLRn

Bits	Name	Description
0	PERF_CNT_CLR	Performance counter clear

**0x01AC8340+MDSS_VBIF_VBIF_PERF_CNT_SELn, n=[0..3]
0x4*n****Type:** RW**Reset State:** 0x00000000

Performance counter select.

MDSS_VBIF_VBIF_PERF_CNT_SELn

Bits	Name	Description
7:0	PERF_CNT_SEL	Performance counter select

**0x01AC8360+MDSS_VBIF_VBIF_PERF_CNT_Ln, n=[0..3]
0x4*n****Type:** R**Reset State:** 0x00000000

Performance counter signature - low.

MDSS_VBIF_VBIF_PERF_CNT_Ln

Bits	Name	Description
31:0	PERF_CNT_LOW	Performance counter signature - low

0x01AC8380+MDSS_VBIF_VBIF_PERF_CNT_Hn, n=[0..3]
0x4*n

Type: R

Reset State: 0x00000000

Performance counter signature - high.

MDSS_VBIF_VBIF_PERF_CNT_Hn

Bits	Name	Description
15:0	PERF_CNT_HIGH	Performance counter signature - high

0x01AC8400+MDSS_VBIF_VBIF_PERF_PWR_CNT_ENn, n=[0..2]
0x4*n

Type: RW

Reset State: 0x00000000

Power performance counter enable.

MDSS_VBIF_VBIF_PERF_PWR_CNT_ENn

Bits	Name	Description
0	PERF_PWR_CNT_EN	Power performance counter enable

0x01AC8420+MDSS_VBIF_VBIF_PERF_PWR_CNT_CLRn, n=[0..2]
0x4*n

Type: RW

Reset State: 0x00000000

Power performance counter clear.

MDSS_VBIF_VBIF_PERF_PWR_CNT_CLRn

Bits	Name	Description
0	PERF_PWR_CNT_CLR	Power performance counter clear

**0x01AC8440+MDSS_VBIF_VBIF_PERF_PWR_CNT_Ln, n=[0..2]
0x4*n**
Type: R**Reset State:** 0x00000000

Power performance counter signature - low.

MDSS_VBIF_VBIF_PERF_PWR_CNT_Ln

Bits	Name	Description
31:0	PERF_PWR_CNT_LOW	Power performance counter signature - high.

**0x01AC8460+MDSS_VBIF_VBIF_PERF_PWR_CNT_Hn, n=[0..2]
0x4*n**
Type: R**Reset State:** 0x00000000

Power performance counter signature - high.

MDSS_VBIF_VBIF_PERF_PWR_CNT_Hn

Bits	Name	Description
15:0	PERF_PWR_CNT_HIGH	Power performance counter signature - high

**0x01AC8480+MDSS_VBIF_VBIF_MISR_CFG_An, n=[0..0]
0x4*n**
Type: RW**Reset State:** 0x00000000

Arbiter MISR configuration register

MDSS_VBIF_VBIF_MISR_CFG_An

Bits	Name	Type	Description
31	MISR_ERR	R	Indicates an error occurred. Currently this can only occur if the MISR is set to generate signatures based on read channel transactions and a transaction is processed where ASIZE does not match the bus width.
11:4	MISR_MID		Transactions must match this MID to be considered in the MISR signature unless MISR_IGNORE_MID is set.
3	MISR_IGNORE_MID		When set to 1 MISR_MID is ignored. Useful only in a limited number of circumstances.
2:1	MISR_INPUT_SEL		MISR input select. 0 - Address channel transactions. 1 - Read channel transactions. 2 - Write channel transactions. 3 - Reserved.

MDSS_VBIF_VBIF_MISR_CFG_An (cont.)

Bits	Name	Type	Description
0	MISR_EN		MISR enable

0x01AC8490 MDSS_VBIF_VBIF_MISR_COUNTER**Type:** RW**Reset State:** 0x00000000

MISR counter

MDSS_VBIF_VBIF_MISR_COUNTER

Bits	Name	Description
19:0	MISR_COUNTER	Specifies the number of transactions to accumulate in the MISR signature. If set to 0 it specifies an unlimited number of transactions. Note that in the case of read/write data transactions, a transaction consists of all the data beats associated with an address channel request. In other words the internal counter does not increment until the last data beat is seen.

**0x01AC8494+ MDSS_VBIF_VBIF_MISR_CLR_An, n=[0..0]
0x4*n****Type:** RW**Reset State:** 0x00000000

Clear MISR signature.

MDSS_VBIF_VBIF_MISR_CLR_An

Bits	Name	Description
0	MISR_CLR	Clear signature. This register should be set to 1 and back to 0 between tests if a second set of MISR signatures is to be collected. Also clears any errors.

**0x01AC84A4 MDSS_VBIF_VBIF_MISR_A0_SIGn, n=[0..3]
+0x4*n****Type:** R**Reset State:** 0x00000000

Arbiter 0 MISR signature.

MDSS_VBIF_VBIF_MISR_A0_SIGn

Bits	Name	Description
31:0	MISR_SIG	MISR signature

0x01AC9040 MDSS_VBIF_VBIF_STM_EN**Type:** RW**Reset State:** 0x00000000

Enable STM HW Events.

MDSS_VBIF_VBIF_STM_EN

Bits	Name	Description
0	STM_OUT_EN	Enable to start VBIF HW Events via qdss_stm_event output port

0x01ACA000 MDSS_VBIF_VBIF_DDR_OUTPUT_RECOVERABLE_HALT_CTRL0**Type:** RW**Reset State:** 0x00000000

Pause VBIF output

MDSS_VBIF_VBIF_DDR_OUTPUT_RECOVERABLE_HALT_CTRL0

Bits	Name	Description
0	DDR_OUTPUT_RECOVERABLE_HALT_REQ	When set this cause VBIF to stop issuing transactions onto the system bus.

0x01ACA004 MDSS_VBIF_VBIF_DDR_OUTPUT_RECOVERABLE_HALT_CTRL1**Type:** R**Reset State:** 0x00000000

Acknowledge for VBIF pause

MDSS_VBIF_VBIF_DDR_OUTPUT_RECOVERABLE_HALT_CTRL1

Bits	Name	Description
0	DDR_OUTPUT_RECOVERABLE_HALT_ACK	When this bit is set, after SW has issued a pause request, it indicates that VBIF is no longer sending out transactions and all outstanding transactions in the arbiters have been completed.

18 SDC1

18.1 SDC1_SDCC_SDCC5

0x07824000 SDC1_SDCC_MCI_POWER

Type: RW

Clock: same rate as HCLK

Reset State: 0x00000000

(ARM name: MCIPower). This register must be updated to the MCLK domain so subsequent writes to this register will be ignored until 3 MCLK and 3 HCLK have passed. SW should wait until bit 0 (MCLK_REG_WR_ACTIVE) of MCI_STATUS2 register is 0, after writing to MCI_POWER register and before accessing this register again.

SDC1_SDCC_MCI_POWER

Bits	Name	Description
11	DML_SW_RST_WAIT_IDLE_DIS	If this bit is set, then SW_RST_REQ procedure before BAM_TRANS_RESET is not executed automatically.
10	SW_RST_WAIT_IDLE_DIS	If this bit is set, then SW_RST_REQ procedure is not executed automatically.
9	SW_RST_REQ	Before executing SW_RST or SW_RST_CONFIG, this bit should be set and SW needs to poll on it until it's being cleared. The bit is cleared after AXI master has finished processing the outstanding transactions and it's in IDLE state.
8	SW_RST_CONFIG	SW reset bit for the core without resetting configuration registers (MCI_POWER, MCI_CLK, MCI_INT_MASKn). In that case, SW will be able to reset the internal state machines, data path, etc. and it will be able to proceed to next transaction without reconfiguring the SDCC registers.
7	SW_RST	SW reset bit for the core. The impact of this reset bit will be similar to the impact of the reset from SW register in CLK_CTRL block. All SW registers and all internal logic is reset.

SDC1_SDCC_MCI_POWER (cont.)

Bits	Name	Description
6	OPEN_DRAIN	MCICMD Output Control. This bit is reserved for the use mentioned in the Description but it is not tested. The open drain mode is preserved such that when enabled, the command output is driven low when the logic is low and undriven (so that the pull up can pull high) when the logic is high. Whether this gets used in the system is up to SW/system but the HW does support this mode.
0	CONTROL	This register is used to control the operation of the memory controller. value 0: power-off value 1: power-on

0x07824004 SDC1_SDCC_MCI_CLK**Type:** RW**Clock:** same rate as HCLK**Reset State:** 0x01008000

(ARM name: MCIClock). This register must be updated to the MCLK domain so subsequent writes to this register will be ignored until 3 MCLK and 3 HCLK have passed. SW should wait until bit 0 (MCLK_REG_WR_ACTIVE) of MCI_STATUS2 register is 0, after writing to MCI_POWER register and before accessing this register again.

SDC1_SDCC_MCI_CLK

Bits	Name	Description
31	PWRSAVE_DLL	When this bit is set, DLL will get clock during transactions but not in idle. When this bit is cleared, DLL will get clock all the time so sdc4_dll_lock won't be de-asserted
30	SDIO_TRANS	When this bit is set in SDIO transaction, START_BIT_ERR status bit won't be raised. The reason is that interrupt period which is de-asserted only after the RX command can cause to wrong START_BIT_ERR detection. The START_BIT_ERR bit won't appear in MCI_STATUS when SDIO_TRANS is set
29	HCLK_IDLE_GATING	Additional CGC for HCLK was added which gates the HCLK for status bits. If this bit is set then HCLK is gated for these registers during IDLE
28	MCLK_IDLE_GATING	Additional CGC for MCLK was added, which gates the MCLK for status bits. If this bit is set then MCLK is gated for these registers during IDLE
27	INT_MCLK_ON	An internal CGC is located in MCLK domain. SW can enable the internal clock at IDLE time by writing '1' to this bit. If the bit is cleared, then the clock is gated at IDLE time

SDC1_SDCC_MCI_CLK (cont.)

Bits	Name	Description
26	SDCC_CLK_EXT_EN	Value 0 (default) - sdcc_clkout is driven via the clock pad and external clock can't be used for testing the cm_dll_sdc4. Value 1 - External clock can be used as a test clock.
25	RX_FLOW_TIMING	Configuration bit which selects the cycle which RxFlowControl will be asserted when UHS mode is used Value 0 (default) - RxFlowControl is asserted one clock before the end bit is received in DPSM (NextBlkTCnt in DPSM equals to 1). Value 1 - RxFlowControl is asserted two clocks before the end bit is received in DPSM (NextBlkTCnt in DPSM equals to 2).
24:23	SDC4_MCLK_SEL	Selects the sdc4_mclk - input clock of cm_dll_sdc4 Value 1 - feedback clock from CLK pad Value 2 (default) - free running MCLK (gated between the transactions) Value 3 - free running MCLK divided by 2
22	CLK_INV	When set(1), the input clock which is inserted into the input macros is inverted.
21	IO_PAD_PWR_SWITCH	Indication to Pad that power has to be switched from 3.3V to 1.8V.
20	CLK_FB_DLY_SEL	Selects the source of feedback clock used by the controller. Value 1 - output of cm_dll_sdc4 is used as feedback clock Value 0 - feedback clock from CLK pad is used
19:18	SD_DEV_SEL	Select the active device if more than one device are connected in shared bus mode. (range: '00' - '11').
17	HCLKON_SW_EN	SW enable of AHB clock request of SDCC5. value 0: HW clock request mechanism is used value 1: clock request signal is always high
16:14	SELECT_IN	Select to latch data and command coming in: value 000: on the falling edge of internal MCLK. value 001: on the rising edge of internal MCLK. value 010: using feedback clock (default). value 011: DDR mode. value 100: UHS mode - MCLK is used internally. value 101: UHS mode - divided frequency (MCLK/2) is used internally. value 110: e.MMC5.0 DDR200 mode.
13	INVERT_OUT	Clear (0) to change data and command going out on the falling edge. Set (1) to change data and command going out on the rising edge.
12	FLOW_ENA	Enable flow control: value 0: disable (default) value 1: enable

SDC1_SDCC_MCI_CLK (cont.)

Bits	Name	Description
11:10	WIDEBUS	Enable wide bus mode: value 00: 1 bit mode value 10: 4 bit mode value 01 or 11: 8 bit mode
9	PWRSAVE	Disable Prime-Cell MCI clock output when bus is idle to save power. value 0: Power save disabled - always enabled if bit 8 is set. value 1: Power save enabled - clock enabled only when bus is active and bit 8 is set.
8	ENABLE	Enable Prime-Cell MCI bus clock: value 0: clock disabled value 1: clock enabled

0x07824008 SDC1_SDCC_MCI_ARGUMENT**Type:** RW**Clock:** same rate as HCLK**Reset State:** 0x00000000**SDC1_SDCC_MCI_ARGUMENT**

Bits	Name	Description
31:0	CMD_ARG	Command argument.

0x0782400C SDC1_SDCC_MCI_CMD**Type:** RW**Clock:** same rate as HCLK**Reset State:** 0x00000000

(ARM name: MMCCCommand) CPSM: Command Path State Machine. This register must be updated to the MCLK domain so subsequent writes to this register will be ignored until 3 MCLK and 3 HCLK have passed. SW should wait until bit 0 (MCLK_REG_WR_ACTIVE) of MCI_STATUS2 register is 0, after writing to MCI_POWER register and before accessing this register again.

SDC1_SDCC_MCI_CMD

Bits	Name	Description
17	AUTO_CMD21	If set (1), CPSM sends CMD21 automatically before sending CMD17 or CMD18. This sequence is needed for CDR auto-calibration of cm_dll_sdc4. CMD21 is supported in eMMC4.5 Spec.

SDC1_SDCC_MCI_CMD (cont.)

Bits	Name	Description
16	AUTO_CMD19	If set (1), CPSM sends CMD19 automatically before sending CMD17 or CMD18. This sequence is needed for CDR auto-calibration of cm_dll_sdc4. CMD19 is supported in SD3.0 Spec.
15	CCS_DISABLE	If set (1), CPSM sends Command_Completion_Signal (CCS) disable sequence to the external CE-ATA device.
14	CCS_ENABLE	If set (1), CPSM waits for CCS from external card CE-ATA device.
13	MCIABORT	Signals the next command will be an abort (stop) command. This bit always read as (0) due to the hardware implementation. See a related note in next field
12	DAT_CMD	If set (1) indicates that this is a Command with Data. Note - MCIABORT and DAT_CMD bits don't have any effect on the CPSM. They are used to signal the SDIO Interrupt state machine that a Multi Block (examples: CMD18, CMD25, CMD53, etc. The rule is that each command with data must set this bit) data transaction is taking place. The DAT_CMD bit signals the start of the data transaction while the MCIABORT bit signals the end. For a pure SD Memory card, these bits are a don't care and should be set to zero by convention. However, some software drivers may chose to manipulate the bits in order to maintain compatibility with Combo cards and/or pure SDIO cards.
11	PROG_ENA	If set (1), PROG_DONE status bit will be asserted when busy is de-asserted. This bit is to be used with a stop or status command after a block write is performed. Does not effect CPSM.
10	ENABLE	The action of writing to this register (MCI_CMD) with this bit set (1), triggers CPSM to leave the Idle State.
9	PENDING	If set (1), CPSM waits for 'CmdPend' from the DPSM before it starts sending a command. This bit is designed to be used with SD CMD18 (READ_MULTIPLE_BLOCK) and SD CMD25 (WRITE_MULTIPLE_BLOCK) which must be terminated with a SD CMD12 (STOP_TRANSMISSION). After CMD18 or CMD25 is sent, the DPSM is enabled to send or receive data of amount MCI_DATA_LENGTH. After the DPSM finishes the transferring the data, it sends iCmdPend to the CPSM and goes idle. When the CPSM receives this signal, it sends the currently loaded command which software would normally establish as CMD12.
8	INTERRUPT	If set (1), CPSM disables command timer and waits for interrupt request (Card Response). The START, CMD_INDEX, ARGUMENT, and CRC are a total of 48 bits in length. If the INTERRUPT bit is zero, the CPSM will wait (63-47) 16 MCLK ticks for a response before going back to IDLE. If the INTERRUPT bit is set, the CPSM will wait indefinitely for a card response.

SDC1_SDCC_MCI_CMD (cont.)

Bits	Name	Description
7	LONGRSP	If set (1), receives a 136-bit long response.
6	RESPONSE	If set (1), CPSM waits for a response subject to INTERRUPT above.
5:0	CMD_INDEX	Command index.

0x07824010 SDC1_SDCC_MCI_RESP_CMD**Type:** R**Clock:** same rate as HCLK**Reset State:** 0x00000000

(ARM name: MCIRspCmd)

SDC1_SDCC_MCI_RESP_CMD

Bits	Name	Description
5:0	RESPCMD	Response command index.

**0x07824014+ SDC1_SDCC_MCI_RESPn, n=[0..3]
0x4*n****Type:** R**Clock:** same rate as HCLK**Reset State:** 0x00000000

(ARM name: MCIResponse0-3). Note that MCIResponse3 has 31 bits; the other three (*RESP0, *RESP1, *RESP2) have 32 bits.

MCI_RESPn, n=[0..3]

SDC1_SDCC_MCI_RESPn

Bits	Name	Description
31:0	STATUS	Card status.

0x07824024 SDC1_SDCC_MCI_DATA_TIMER**Type:** RW**Clock:** same rate as HCLK**Reset State:** 0x00000000

(ARM name: MCIDataTimer).

There is no way to disable this timer. The timer starts counting as soon as a transaction is initiated. The time counts in MCLK ticks.

SDC1_SDCC_MCI_DATA_TIMER

Bits	Name	Description
31:0	DATA_TIME	Data timeout period.

0x07824028 SDC1_SDCC_MCI_DATA_LENGTH

Type: RW

Clock: same rate as HCLK

Reset State: 0x00000000

(ARM name: MCIDataLength)

Total number of bytes in the transaction regardless of mode (stream or block). In block mode, must be an exact multiple of block size. During an infinite transfer the value of DATALENGTH should be programmed to 0.

SDC1_SDCC_MCI_DATA_LENGTH

Bits	Name	Description
24:0	DATALENGTH	Data length value.

0x0782402C SDC1_SDCC_MCI_DATA_CTL

Type: RW

Clock: same rate as HCLK

Reset State: 0x00100000

(ARM name: MCIDataCtrl). Data Path State Machine (DPSM). This register must be updated to the MCLK domain so subsequent writes to this register will be ignored until 3 MCLK and 3 HCLK have passed. SW should wait until bit 0 (MCLK_REG_WR_ACTIVE) of MCI_STATUS2 register is 0, after writing to MCI_POWER register and before accessing this register again.

SDC1_SDCC_MCI_DATA_CTL

Bits	Name	Description
21	SW_SDC4_CMD19	The input of cm_dll_sdc4 - sdc4_cmd19 can be driven by SW if AUTO_CMD19 (bit 16 in MCI_CMD) feature is not used. Value 1 - sdc4_cmd19 is driven by SW Value 0 (default) - sdc4_cmd19 is driven by HW

SDC1_SDCC_MCI_DATA_CTL (cont.)

Bits	Name	Description
20	RX_DATA_PEND	If set (1), timeout counter will start counting only after the RX command (with data) was sent instead of counting from initialization of DPSM. The feature is enabled by default.
19	AUTO_PROG_DONE	If set (1), automatic detection of PROG_DONE condition is executed without sending CMD12, CMD13, CMD52 or any other 'dummy' command. SW should set this bit in the following cases only: 1. When sending CMD53 for SDIO write transaction. 2. When sending CMD24. 3. When sending CMD25 and CMD23 was issued before to inform card about the exact number of blocks to be written (The rule is that this bit can be used for each command with data that doesn't need CMD12 for terminating the data or when it's prefixed by CMD23). 4. When sending CMD19 for testing bus procedure. CMD12 is not required in this case.
18	INFINITE_TRANSFER	If set (1), infinite transfer is enabled. MCI_DATA_LENGTH register should set to 0 during infinite transfer.
17	DATA_PEND	If set (1), DPSM waits for 'DataPend' from the CPSM before it enables the DPSM. This bit is designed to be used with SD CMD24 and CMD25 (WRITE_SINGLE_BLOCK and WRITE_MULTIPLE_BLOCK) to automatically start the DPSM after a normal (non-error) response is received. This register should be written with the enable bit and the pending bit asserted before MCI_CMD is enabled.
16:4	BLOCKSIZE	Data block length in bytes (1 to 4096).
3	DM_ENABLE	Enable DM interface: 0: DM disabled 1: DM enabled
2	MODE	Data transfer mode: 0: block data transfer 1: stream data transfer
1	DIRECTION	Data transfer direction: 0: controller to card 1: card to controller
0	ENABLE	Data transfer enabled. If zero, DPSM unconditionally reset.

0x07824030 SDC1_SDCC_MCI_DATA_COUNT**Type:** R**Clock:** same rate as HCLK**Reset State:** 0x00000000

(ARM name: MCIDataCnt)

NOTE There is no mechanism to ensure that a read of this counter will be accurate as it is not synchronized. The reason being the value read is async to the clock domain in which the reading is done. It is only useful as a debug tool for diagnostic purposes only.

SDC1_SDCC_MCI_DATA_COUNT

Bits	Name	Description
24:0	DATACOUNT	Value of data counter in MciDPSM block. Represents remaining data of transaction.

0x07824034 SDC1_SDCC_MCI_STATUS

Type: R

Clock: same rate as HCLK

Reset State: 0x000C0000

(ARM name: MCISStatus)

This register may need exclusion for QCSR POR Test for bits 25 and 22.

Static[30:26], [24:23] and [10:0]

These remain asserted until they are cleared by writing to the appropriate bit in the Clear Register (see MCI_CLEAR)

XCIS

The means used to send the SDIO external-card-interrupt-signal (XCIS) to the SDCC depends on the transfer mode: 1-bit, 4-bit, or 8-bit. Under 1 bit transfer mode, the XCIS has a dedicated physical connection to the SDCC. Hence once it is asserted, it remain at a high level. Under 4 and 8 bit transfe modes, the XCIS is time multiplexed with physical connection data bit 1. The XCIS is recognized as valid during specific time slots on data bit 1 by the SDCC. How this 'recognition' takes places shows up as different behaviors on SDIO_INTR_OPER, SDIO_INTR, and the actual interrupt sent to the ARM through MCI_INT_MASKn.

Special[25]

The SDIO_INTR_OPER indicator reflects the true state of XCIS as it would be observed inside the external SDIO card. As such, there is no bit 25 in MCI_CLEAR. The only way to clear SDIO_INTR_OPER is to clear the appropriate bit in the external card Common Card Control Registers (CCCR). Like 'static' above and 'dynamic' below, bit 25 in MCI_INT_MASKn controls actually sending this indicator value to the ARM. This indicator of XCIS is used for normal SDCC operation.

Special[22]

The SDIO_INTR indicator reflects that there has been a low-to-high transition on the connection as explained under XCIS above. This indicator is cleared via bit 22 of MCI_CLEAR but any low-to-high transition on the connection will set this indicator again. Further note that this indicator is not routed through the MCI_INT_MASKn to the ARM. For bit 22, the raw value of the connection is routed through the MCI_INT_MASKn to the ARM. As this raw connection is not aware of

specific time slots, the signal sent to the ARM generally toggles in an unpredictable fashion. This is generally not useful during normal SDCC operation. However, for wake-up with clocks off, the raw connection is desirable.

Dynamic[21:11]

These change state depending on the state of the underlying logic (for example, FIFO full and empty flags are asserted and de-asserted as data while written to the FIFO).

SDC1_SDCC_MCI_STATUS

Bits	Name	Description
31	STATUS2_INT	One of the error bits in MCI_STATUS2[4:1] register is set
30	AUTO_CMD19_TIMEOUT	Response or tuning pattern wasn't received after automatic CMD19's transmission. This status bit should be read only if AUTO_CMD19 (MCI_CMD(16)) was set.
29	BOOT_TIMEOUT	Data wasn't received within the valid time (according to MCI_CCS_TIMER) from the start of boot operation.
28	BOOT_ACK_ERR	Acknowledge pattern wasn't received correctly or not within the valid time (according to MCI_ACK_TIMER) from the start of boot operation.
27	BOOT_ACK_REC	Acknowledge pattern was received correctly.
26	CCS_TIMEOUT	CE-ATA Command Completion Signal timeout.
25	SDIO_INTR_OPER	SDIO interrupt indicator for normal operation.
24	ATA_CMD_COMPL	CE-ATA Command Completion Signal has been detected.
23	PROG_DONE	Programming done.
22	SDIO_INTR	SDIO interrupt indicator for wake-up.
21	RXDATA_AVLBL	Data available in receive FIFO. At least 1 word in the RX FIFO. SW can read 1 word only from the FIFO.
20	TXDATA_AVLBL	Data available in transmit FIFO. At least 1 word in the TX FIFO.
19	RXFIFO_EMPTY	Receive FIFO empty. SW can't read FIFO.
18	TXFIFO_EMPTY	Transmit FIFO empty. SW can write 8 words into the FIFO.
17	RXFIFO_FULL	Receive FIFO full. SW can read 8 words from the FIFO.
16	TXFIFO_FULL	Transmit FIFO full. TX FIFO contains 8 words. SW can't write to FIFO.
15	RXFIFO_HALF_FULL	Receive FIFO half full. SW can read 8 words from the FIFO.
14	TXFIFO_HALF_FULL	Transmit FIFO half full. SW can write 8 words into the FIFO.
13	RXACTIVE	Data receive in progress.
12	TXACTIVE	Data transmit in progress.
11	CMD_ACTIVE	Command transfer in progress.
10	DATA_BLK_END	Data block sent / received (CRC check passed).

SDC1_SDCC_MCI_STATUS (cont.)

Bits	Name	Description
9	START_BIT_ERR	Start Bit Error flag
8	DATAEND	Data end (data counter is zero).
7	CMD_SENT	Command sent (no response required).
6	CMD_RESPONSE_END	Command response received (CRC check passed).
5	RX_OVERRUN	Receive FIFO overrun error.
4	TX_UNDERRUN	Transmit FIFO underrun error.
3	DATA_TIMEOUT	Data timeout.
2	CMD_TIMEOUT	Command response timeout.
1	DATA_CRC_FAIL	Data block sent / received (CRC check failed).
0	CMD_CRC_FAIL	Command response received (CRC check failed).

0x07824038 SDC1_SDCC_MCI_CLEAR**Type:** W**Clock:** same rate as HCLK**Reset State:** 0x00000000

(ARM name: MCIClear)

SDC1_SDCC_MCI_CLEAR

Bits	Name	Description
30	AUTO_CMD19_TIMEOUT_CLR	Clears AUTO_CMD19_TIMEOUT flag.
29	BOOT_TIMEOUT_CLR	Clears BOOT_TIMEOUT flag.
28	BOOT_ACK_ERR_CLR	Clears BOOT_ACK_ERR flag.
27	BOOT_ACK_REC_CLR	Clears BOOT_ACK_REC flag.
26	CCS_TIMEOUT_CLR	Clears CCSTimeOut flag.
24	ATA_CMD_COMPL_CLR	Clears AtaCmdCompl flag.
23	PROG_DONE_CLR	Clears ProgDone flag.
22	SDIO_INTR_CLR	Clears SDIOInt flag.
10	DATA_BLK_END_CLR	Clears DataBlockEnd flag.
9	START_BIT_ERR_CLR	Clears StartBitErr flag.
8	DATA_END_CLR	Clears DataEnd flag.
7	CMD_SENT_CLR	Clears commandSent flag.

SDC1_SDCC_MCI_CLEAR (cont.)

Bits	Name	Description
6	CMD_RESP_END_CLT	Clears CmdRespEnd flag.
5	RX_OVERRUN_CLR	Clears RxOverrunClr flag.
4	TX_UNDERRUN_CLR	Clears TxUnderrun flag.
3	DATA_TIMEOUT_CLR	Clears DataTimeOut flag.
2	CMD_TIMEOUT_CLR	Clears CmdTimOutflag.
1	DATA_CRC_FAIL_CLR	Clears DataCrcFail flag.
0	CMD_CRC_FAIL_CLR	Clears CmdCrcFail flag.

**0x0782403C+ SDC1_SDCC_MCI_INT_MASKn, n=[0..1]
0x4*n****Type:** RW**Clock:** same rate as HCLK**Reset State:** 0x00000000

(ARM name: MCIMask0)

SDC1_SDCC_MCI_INT_MASKn

Bits	Name	Description
31	MASK31	MASK STATUS2_INT flag.
30	MASK30	MASK AUTO_CMD19_TIMEOUT flag.
29	MASK29	MASK BOOT_TIMEOUT flag.
28	MASK28	MASK BOOT_ACK_ERR flag.
27	MASK27	MASK BOOT_ACK_REC flag.
26	MASK26	MASK CCSTimeOut flag.
25	MASK25	MASK SDIOIntOper flag.
24	MASK24	MASK AtaCmdCompl flag.
23	MASK23	MASK ProgDone flag.
22	MASK22	MASK SDIOInt flag.
21	MASK21	MASK RxDataAvlbl flag.
20	MASK20	MASK TxDataAvlbl flag.
19	MASK19	MASK RxFifoEmpty flag.
18	MASK18	MASK TxFifoEmpty flag.
17	MASK17	MASK RxFifoFull flag.
16	MASK16	MASK TxFifoFull flag.

SDC1_SDCC_MCI_INT_MASKn (cont.)

Bits	Name	Description
15	MASK15	MASK RxFifoHalfFull flag.
14	MASK14	MASK TxFifoHalfFull flag.
13	MASK13	MASK RxActive flag.
12	MASK12	MASK TxActive flag.
11	MASK11	MASK CmdActive flag.
10	MASK10	MASK DataBlockEnd flag.
9	MASK9	MASK StartBitErr
8	MASK8	MASK DataEnd flag.
7	MASK7	MASK CmdSent flag.
6	MASK6	MASK CmdRespEnd flag.
5	MASK5	MASK RxOverrun flag.
4	MASK4	MASK TxOverrun flag.
3	MASK3	MASK DataTimeOut flag.
2	MASK2	MASK CmdTimeOut flag.
1	MASK1	MASK DataCmdCrcFail flag.
0	MASK0	MASK CmdCrcFail flag.

0x07824044 SDC1_SDCC_MCI_FIFO_COUNT**Type:** R**Clock:** same rate as HCLK**Reset State:** 0x00000000

The SDCC core operates in two clock domains MCLK and HCLK. The CPSM and DPSM operate in MCLK domain and MciFifoDmaCtl (FIFO/DMA Controller) operates in HCLK domain. The MCI_FIFO_COUNT is a counter of FIFO Controller that monitors how many words of data are still needed to be transferred through the FIFO. At the beginning of data transaction the MCI_FIFO_COUNT counter will be loaded with the (MCI_DATA_LENGTH / 4) value and then will count down during the data transaction until it reach zero. This register is only useful for debug purposes and should not be used for normal operation since it does not reflect data which may or may not be in the pipeline.

SDC1_SDCC_MCI_FIFO_COUNT

Bits	Name	Description
23:0	DATA_COUNT	Remaining data.

0x07824048 SDC1_SDCC_MCI_BOOT**Type:** RW**Clock:** same rate as HCLK**Reset State:** 0x00000000

Handling the boot operation.

SDC1_SDCC_MCI_BOOT

Bits	Name	Description
3	EARLY_ASSERT_CMD_LINE	If set to '1' then the CMD line is asserted by HW when all the data was received on the SD bus
2	BOOT_ACK_EN	If set to '1' then Host waits for acknowledge pattern after initiating the boot operation.
1	BOOT_EN	When this bit is asserted, the boot operation is initiated in both of the modes.
0	BOOT_MODE	If set to '1' then CMD line is low during the boot operation. If boot_mode = '0', then CMD0 with the argument 0xFFFFF0 is sent.

0x0782404C SDC1_SDCC_MCI_BOOT_ACK_TIMER**Type:** RW**Clock:** same rate as HCLK**Reset State:** 0x00000000

Timer for boot operation - the time until receiving the acknowledge pattern.

SDC1_SDCC_MCI_BOOT_ACK_TIMER

Bits	Name	Description
31:0	BOOT_ACK_TIMER	Timer for counting the cycles from initiating the boot operation until the acknowledge pattern is accepted.

0x07824054 SDC1_SDCC_MCI_EMULATION_DLY_LINE**Type:** RW**Clock:** same rate as HCLK**Reset State:** 0x00000000

The register is responsible for handling DCM which is used in emulation supports 256 values of shift intervals so 8 bits are required (0:255). Bits [31:30] are used for emulation purposes only.

The procedure for activating the DCM is:

1. Setting bit 28 - DCM_RESET.
2. Clearing bit 28 - DCM_RESET.
3. Waiting until bit 29 DCM_LOCKED is high.
4. Writing a new phase in SD_CLK_DLY_CTRL field and setting the DCM_START bit (bit 30).

The clock's placement was executed by the DCM when DCM_DONE is asserted.

SDC1_SDCC_MCI_EMULATION_DLY_LINE

Bits	Name	Description
31	DCM_DONE	The shift process was finished in DCM and CMD19 can be sent towards the card. This is a read-only bit.
30	DCM_START	A new value of SD_CLK_DLY_CTRL is ready. Clears the DCM_DONE bit.
29	DCM_LOCKED	Reading the status of the LOCKED DCM's output which indicates that the output clock is ready for use.
28	DCM_RESET	Driving reset to DCM.
7:0	SD_CLK_DLY_CTRL	Input to DCM block in FPGA (8 bits). Controlling the change of the delay interval. The value's range may be 0-255 in DCM block.

0x07824058 SDC1_SDCC_MCI_CCS_TIMER

Type: RW

Clock: same rate as HCLK

Reset State: 0x00000000

This register is used for operation with CE-ATA devices. The CCS timer starts to count at the end of SD transaction and it advances with every MCLK clock cycle. In boot operation this register is used for storing the number of cycles from initiating the boot operation until the first data is received.

SDC1_SDCC_MCI_CCS_TIMER

Bits	Name	Description
31:0	CCS_TIMER	CE-ATA Command Completion Signal timeout period.

0x0782405C SDC1_SDCC_MCI_RESPONSE_MASK

Type: RW

Clock: same rate as HCLK

Reset State: 0x00000000

SDC1_SDCC_MCI_RESPONSE_MASK

Bits	Name	Description
31:0	RESPONSE_MASK	Mask used to verify the 32 status error bits. Active only when the data pending bit is set in the MCI_DATA_CTL register. When bit of this mask is set the corresponding bit from response [39:8] is checked to be 0 i.e. shows there no error. If all bits from mask shows no error the DPSM will be activated without waiting for software write.

0x07824060 SDC1_SDCC_MCI_DLL_CONFIG**Type:** RW**Clock:** same rate as HCLK**Reset State:** 0x60006400

This register configures the cm_dll_sdc4 for UHS mode.

SDC1_SDCC_MCI_DLL_CONFIG

Bits	Name	Description
31	SDC4_DIS_DOUT	When this bit equals to '0', the data which comes out from the dll is valid. When it equals to '1' this data isn't valid. The inversion of this bit is connected to dll's input - dll_en_dout
30	DLL_RST	Setting this bit to 1 resets cm_dll_sdc4. cm_dll_sdc4 should be reset every time the MCLK frequency is changed.
29	PDN	Power Down Value 0 - analog blocks are enabled Value 1 - analog blocks are powered down (default)
28	CK_INTP_SEL	Selects interpolator output
27	CK_INTP_EN	Enable clock interpolation for finer resolution
26:24	MCLK_FREQ	Frequency of MCLK Value 000 -100 - 112 (MHz) Value 001 -112 - 125 Value 010 -125 - 137 Value 011 -137 - 150 Value 100 -150 - 162 Value 101 -162 - 175 Value 110 -175 - 187 Value 111 -187 - 200

SDC1_SDCC_MCI_DLL_CONFIG (cont.)

Bits	Name	Description
23:20	CDR_SELECT	CDR phase select (gray coded) Value 0000 - phase 0 Value 0001 - phase 1 Value 0011 - phase 2 Value 0010 - phase 3 Value 0110 - phase 4 Value 0111 - phase 5 Value 0101 - phase 6 Value 0100 - phase 7 Value 1100 - phase 8 Value 1101 - phase 9 Value 1111 - phase 10 Value 1110 - phase 11 Value 1010 - phase 12 Value 1011 - phase 13 Value 1001 - phase 14 Value 1000 - phase 15
19	CDR_EXT_EN	Enable external control of cdr phase select
18	CK_OUT_EN	Enable output clock (default value is '1')
17	CDR_EN	Enable CDR function
16	DLL_EN	Enable DLL function
15:14	CDR_UPD_RATE	CDR update rate, low pass filtering window of CDR Max update rate: Value 00 - 0.5 MCLK frequency Value 01 - 0.25 MCLK Frequency (default) Value 10 - 0.125 MCLK frequency Value 11 - 1/16 MCLK Frequency
13:12	DLL_UPD_RATE	DLL update rate Value 00 - sdc4_mclk/10 Value 01 - sdc4_mclk/20 Value 10 - sdc4_mclk/40 (default) Value 11 - sdc4_mclk/80
11:10	DLL_PHASE_DET	DLL phase detector low pass average window Value 00 - 2 cycles Value 01 - 4 cycles (default) Value 10 - 8 cycles Value 11 - 16 cycles
9	CDR_ALGORITHM_SEL	CDR algorithm select Value 0 - Edge (default) Value 1 - Level

SDC1_SDCC_MCI_DLL_CONFIG (cont.)

Bits	Name	Description
0	CMD_DAT_TRACK_SEL	Select DAT or CMD lines for CDR tracking Value 0 - track DAT inputs (default) Value 1 - track CMD input

0x07824068 SDC1_SDCC_MCI_DLL_STATUS**Type:** R**Clock:** same rate as HCLK**Reset State:** 0x00000000

This register is connected to sdc4_status output of cm_dll_sdc4.

SDC1_SDCC_MCI_DLL_STATUS

Bits	Name	Description
12	SDC4_DTEST_MUXSEL	Digital test mux select
11	DDR_DLL_LOCK_JDR	Reserved
10:9	SDC4_DLL_DTEST_OUT_A TPG	SDC4_DLL_DTEST_OUT output from DLL after a clock mux with sampled JDRs for ATPG mode. This bit helps collecting coverage on DLL's output
8	SDC4_DLL_LOCK_ATPG	SDC4_DLL_LOCK output from DLL after a clock mux with sampled JDRs for ATPG mode. This bit helps collecting coverage on DLL's output
7	DLL_LOCK	DLL lock status Value 0 - Not locked Value 1 - Locked

SDC1_SDCC_MCI_DLL_STATUS (cont.)

Bits	Name	Description
6:3	CDR_PHASE	CDR phase select (gray coded) Value 0000 - phase 0 Value 0001 - phase 1 Value 0011 - phase 2 Value 0010 - phase 3 Value 0110 - phase 4 Value 0111 - phase 5 Value 0101 - phase 6 Value 0100 - phase 7 Value 1100 - phase 8 Value 1101 - phase 9 Value 1111 - phase 10 Value 1110 - phase 11 Value 1010 - phase 12 Value 1011 - phase 13 Value 1001 - phase 14 Value 1000 - phase 15
2	DDLL_COARSE_CAL	Value 1 - done Value 0 - not done

0x0782406C SDC1_SDCC_MCI_STATUS2**Type:** R**Clock:** same rate as HCLK**Reset State:** 0x00000000

Additional status and error bits.

SDC1_SDCC_MCI_STATUS2

Bits	Name	Description
8	BAM_AXI_MASTER_ERR	Error occurred on AXI master for BAM transaction. It is enabled if the field BAM_ERR_EN is set in MCI_HC_MODE register
7	AUTO_CDC_SW_CALIB_ERR	calibration error is received (CDC_ERROR_CODE isn't 0) during auto SW CDC calibration.
6	AUTO_CDC_SW_CALIB_TIMEOUT	CALIBRATION_DONE isn't received within the number of cycles which configured in CDC_CAL_TIMEOUT_THRESHOLD field (from CDC_AUTO_SW_CAL_CFG register).
5	DATA_AXI_MASTER_ERR	Error occurred on AXI master for data transaction
4	DATA_END_BIT_ERROR	The bit is set when the end bit of the received block was not 1 on all the data lines
3	CMD_END_BIT_ERROR	The bit is set when the end bit of the received response was not 1 on the command line

SDC1_SDCC_MCI_STATUS2 (cont.)

Bits	Name	Description
2	FIFO_EMPTY_ERROR	FIFO read attempt when FIFO was empty
1	FIFO_FULL_ERROR	write attempt when FIFO was full
0	MCLK_REG_WR_ACTIVE	Value 0 - No active register write to MCLK domain Value 1 - Active register write to MCLK domain. The bit indicates if a write operation to one of the following registers is in process (synchronization between HCLK and MCLK): MCI_POWER MCI_CLOCK MCI_CMD MCI_DATA_CTL

0x07824070 SDC1_SDCC_MCI_GENERICS**Type:** R**Clock:** same rate as HCLK**Reset State:** 0x00000000

Generics list of each SDCC instance.

SDC1_SDCC_MCI_GENERICS

Bits	Name	Description
29	SWITCHABLE_SIGNALING_VOLTAGE	The core supports switching of IO signaling voltage.
28	BUS_18V_SUPPORT	The core supports 1.8v power supply to the bus.
27	BUS_30V_SUPPORT	The core supports 3.0v power supply to the bus.
26:23	SD_DATA_WIDTH	Number of DAT lines Value 4 - 4 DAT lines (SD, SDIO) Value 8 - 8 DAT lines (MMC, eMMC)
22:10	RAM_SIZE	Size of RAM size: Optional Values: 512, 1024, 2048 or 4096 bytes.
9	USE_SPS	Value 1 - BAM and DML are used and USB core is connected to SDCC5, Value 0 - BAM and DML are not used.
8:6	NUM_OF_DEV	Number of eSD or eSDIO devices supported on shared SD bus. Can be set from 1 to 4.
5:1	MAX_PIPES	Number of simultaneous parallel pipes supported by the BAM and attached peripheral pair. Can be set from 1 to 31.

SDC1_SDCC_MCI_GENERIC (cont.)

Bits	Name	Description
0	USE_DLL_SDC4	Enables the instantiation of cm_dll_sdc4. Value 0 - cm_dll_sdc4 isn't integrated with SDCC5 Value 1 - cm_dll_sdc4 is used with SDCC5.

0x07824074 SDC1_SDCC_MCI_FIFO_STATUS**Type:** R**Clock:** same rate as HCLK**Reset State:** 0x00007800

Status bit of FIFO's fill level. Can be used for long accesses to FIFO using the FIFO_ALT register

SDC1_SDCC_MCI_FIFO_STATUS

Bits	Name	Description
18	RX_FIFO_512	512 bytes are ready in FIFO during RX transaction
17	RX_FIFO_256	256 bytes are ready in FIFO during RX transaction
16	RX_FIFO_128	128 bytes are ready in FIFO during RX transaction
15	RX_FIFO_64	64 bytes are ready in FIFO during RX transaction
14	TX_FIFO_512	available space in FIFO for 512 bytes during TX transaction
13	TX_FIFO_256	available space in FIFO for 256 bytes during TX transaction
12	TX_FIFO_128	available space in FIFO for 128 bytes during TX transaction
11	TX_FIFO_64	available space in FIFO for 64 bytes during TX transaction
10:0	FIFO_FILL_LEVEL	reflects the actual fill level of FIFO DPRAM during TX or RX transaction. Supported values are 0, 4, 8, ..., 1020, 1024

0x07824078 SDC1_SDCC_MCI_HC_MODE**Type:** RW**Clock:** same rate as HCLK**Reset State:** 0x00000000**SDC1_SDCC_MCI_HC_MODE**

Bits	Name	Type	Description
21	PROGDONE_WO_CMD_RSP	RW	'0' (Default): we wait for cmd response end to detect progdone (legacy behavior). '1': we start looking for progdone without waiting for cmd response end but after 10 cycles from end bit of command

SDC1_SDCC_MCI_HC_MODE (cont.)

Bits	Name	Type	Description
20	AUTO_CMD12_CLR_CMDACTIVE	RW	'0' (Default): auto_cmd12 is not clearing cmdactive (bug fix). '1': auto_cmd12 is clearing cmdactive like previous behavior
19	WRAP_ERROR	RW	When the bit is set, the core responds with errors on WRAP transaction in configuration bus.
18:17	BUSY_CHECK_VALID_PERIOD	RW	Configuring the wait period between CRC token to BUSY period. Value 0 - waiting 7 cycles. The values of BUSY_CHECK_VALID_PERIOD is added to 7 cycles
16	BUSY_CHECK_VALID_ALWAYS	RW	In DDR200 mode, the end of busy should be checked after the CRC token was received. There is a counter of 4 cycles for this purpose. If this bit is set then the end of busy will be checked right after the crc token.
15	BAM_ERR_EN	RW	Enabling the assertion of BAM_AXI_MASTER_ERR field in MCISTATUS2
14	DIS_RST_AFTER_CRC_TO_KEN_DDR200	RW	Disabling the option to reset the RCLK's data registers in IO macro after each CRC token. The reset is needed if redundant RCLK is sent and BUSY is sampled into the FIFO as a data bit.
13	FF_CLK_SW_RST_DIS	RW	Disabling the SW reset in FF clock domain
12	IO_MACRO_SW_RST_PERIOD	R	When moving to DDR mode, reset period is generated for dividers. There is possibility to read from this bit if this period was finished
11	CLOCK_AFTER_EOB_DIS	RW	Disabling the 8 cycles after the CRC token when TxFlowControl is asserted before the end of the block
10	ADMA_INT_DATA	RW	If this bit is set, then the ADMA interrupt is set when the data was already transferred. If this bit is cleared, then the ADMA interrupt is set when the requests related to the descriptors were finished on address channel before the data itself is processed
9	BAM_CLK_EN_ACT	RW	If this bit is set, then bam_clk is gated when hc mode is enabled
8	ADMA_HPROT_DIS	R	If cleared then HPROT is changed in last burst of write transaction if block_count_en is 1, otherwise it will be for each end of descriptor. Not applicable for Gandalf - AXI Noc
7	ADMA_NON_WRD_ALIGN_DIS	R	If set the fix for non word aligned transfers in ADMA is disabled. Not applicable for Gandalf - AXI Noc
6	DEASSERT_HREADY_DATA_DIS	RW	if set, then hready won't be deasserted for access to Transfer Mode register in HC address. By default, the hready is deasserted for enabling the completion of synchronization period to MCLK before the next access
5	DEASSERT_HREADY_COMMAND_DIS	RW	if set ,then hready won't be deasserted for access to Command register in HC address. By default, the hready is deasserted for enabling the completion of synchronization period to MCLK before the next access

SDC1_SDCC_MCI_HC_MODE (cont.)

Bits	Name	Type	Description
4	IRQ_PCLK_DIS	RW	From tag sdcc5_p3q3r50, the interrupt and status bits are handled in PCLK domain only. If this bit is set, then the legacy architecture is used
3	SINGLE_NON32_ERROR	RW	if set , then a response error will be sent if single configuration access is executed in legacy mode with size of non 32 bits
2	WAIT_DLL_LOCK	RW	If this bit is set, the host controller isn writing to the async fifo unless DLL_LOCK of MCI_DLL_STATUS is equals to '1'
1	CYCLES_AFTER_EOB_DIS	RW	Enable bit for the fix that TxFlowControl isn't set until end of the block (CRC token reception) in the middle of TX transaction. The fix is required for releasing each BAM's descriptor separately without waiting to the next descriptor.
0	HC_MODE_EN	RW	Enabling the Host Controller Specification by setting this bit

**0x07824080+ SDC1_SDCC_MCI_FIFOn, n=[0..15]
0x4*n****Type:** RW**Clock:** same rate as HCLK**Reset State:** 0x00000000**SDC1_SDCC_MCI_FIFOn**

Bits	Name	Description
31:0	DATA	FIFO data. This register is aliased to 16 words, 0x080 - 0xBC. In SDCC3, an access to the FIFO was executed with 16 words so 16 addresses were defined. However, in SDCC5 only 8 words are used in each burst so only 8 addresses from the range should be used. When writing to the external SD_Card, writes to the data path FIFO. When reading from the external SD_Card, reads from the data path FIFO.

0x078240DC SDC1_SDCC_MCI_PWRCTL_STATUS_REG**Type:** R**Clock:** PCLK**Reset State:** 0x00000000

HC Spec Defines power and IO voltage switch bits. The information is sent by a dedicated interrupt to the QCT SW driver for handling the power and voltage tasks with PMIC. The different tasks are defined in this register

SDC1_SDCC_MCI_PWRCTL_STATUS_REG

Bits	Name	Description
3	IO_HIGH_V	SDCC driver orders that the I/O signaling level be switched to 3.0v. This interrupt is triggered when SDCC driver writes '0' to offset 03Eh in HC register (bit 3) or when SW reset for all is handled and last supported voltage was 1.8V. This logic is enabled when SWITCHABLE_SIGNALING_VOLTAGE=true
2	IO_LOW_V	SDCC driver orders that the I/O signaling level be switched to 1.8v. This interrupt is triggered when SDCC driver writes '1' to offset 03Eh in HC register (bit 3). This logic is enabled when SWITCHABLE_SIGNALING_VOLTAGE=true
1	BUS_ON	SDCC driver orders that the SD bus power be switched on. This interrupt is triggered when SDCC driver writes '1' to offset 029h in HC register (bit 0).
0	BUS_OFF	SDCC driver orders that the SD bus power be switched off. This interrupt is triggered when SDCC driver writes '0' to offset 029h in HC register (bit 0) or when SW reset for all is handled.

0x078240E0 SDC1_SDCC_MCI_PWRCTL_MASK_REG

Type: RW

Clock: PCLK

Reset State: 0x00000000

Mask register for MCI_PWRCTL_STATUS_REG

SDC1_SDCC_MCI_PWRCTL_MASK_REG

Bits	Name	Description
3	IO_HIGH_V	'1' - interrupt is enabled / '0' - interrupt is masked.
2	IO_LOW_V	'1' - interrupt is enabled / '0' - interrupt is masked.
1	BUS_ON	'1' - interrupt is enabled / '0' - interrupt is masked.
0	BUS_OFF	'1' - interrupt is enabled / '0' - interrupt is masked.

0x078240E4 SDC1_SDCC_MCI_PWRCTL_CLEAR_REG

Type: W

Clock: PCLK

Reset State: 0x00000000

Clear register for MCI_PWRCTL_STATUS_REG

SDC1_SDCC_MCI_PWRCTL_CLEAR_REG

Bits	Name	Description
3	IO_HIGH_V	Writing '1' clears bit 03 in PWRCTL_STATUS_REG
2	IO_LOW_V	Writing '1' clears bit 02 in PWRCTL_STATUS_REG
1	BUS_ON	Writing '1' clears bit 01 in PWRCTL_STATUS_REG
0	BUS_OFF	Writing '1' clears bit 00 in PWRCTL_STATUS_REG

0x078240E8 SDC1_SDCC_MCI_PWRCTL_CTL_REG

Type: RW

Clock: PCLK

Reset State: 0x00000000

HC Spec Defines power and IO voltage switch bits. The information is sent by a dedicated interrupt to the QCT SW driver for handling the power and voltage tasks with PMIC and then the QCT SW driver needs to report success or fail to this register.

SDC1_SDCC_MCI_PWRCTL_CTL_REG

Bits	Name	Description
9	HC_CARD_STATE_STABLE	This bit is driving CARD_STATE_STABLE bit in Present State Register
8	RST_NOT_WAIT_PWRCTL_REG	RST process doesn't need to wait for accessing to MCI_PWRCTL_CTL_REG register
7:6	HC_WRITE_PROTECT_PAD	Value 0 inversion of HC_WRITE_PROTECT is connected to bit 19 in Present State Register. Value 1 input from PAD is used. Value 2 inversion of input from PAD is used
5	CARD_DETECT_PIN_LEVEL	This bit reflects the inverse value of the SD_CD# pin. Debouncing is not performed on this bit. This bit may be valid when Card State Stable is set to 1, but it is not guaranteed because of propagation delay. Use of this bit is limited to testing since it must be debounced by software. value 1: card present value 0: no card present
4	HC_WRITE_PROTECT	This bit allows to disable the writing. value 0: write enabled value 1: write protected
3	IO_SIG_SWITCH_FAIL	Host controller clears this bit when it fires either I/O Low V interrupt or I/O High V interrupt. SDCC power control software writes '1' to indicate the signal switch has failed.

SDC1_SDCC_MCI_PWRCTL_CTL_REG (cont.)

Bits	Name	Description
2	IO_SIG_SWITCH_SUCCESS	Host controller clears this bit when it fires either I/O Low V interrupt or I/O High V interrupt. SDCC power control software writes '1' to indicate the signal switch has completed successfully.
1	BUS_ON_OFF_FAIL	Host controller clears this bit when it fires either Bus ON interrupt or Bus OFF interrupt. SDCC power control software writes '1' to indicate the requested operation has failed.
0	BUS_ON_OFF_SUCCESS	Host controller clears this bit when it fires either Bus ON interrupt or Bus OFF interrupt. SDCC power control software writes '1' to indicate the requested operation has completed successfully.

0x078240EC SDC1_SDCC_MCI_CLEAR2**Type:** W**Clock:** same rate as HCLK**Reset State:** 0x00000000

Clearing the error bits in MCI_STATUS2 register

SDC1_SDCC_MCI_CLEAR2

Bits	Name	Description
8	BAM_AXI_MASTER_ERR_CLR	BAM_AXI_MASTER_ERR clear bit
7	AUTO_CDC_SW_CALIB_ERR_CLR	AUTO_CDC_SW_CALIB_ERR clear bit.
6	AUTO_CDC_SW_CALIB_TOUT_CLR	AUTO_CDC_SW_CALIB_TOUT clear bit.
5	DATA_AXI_MASTER_ERR_CLR	DATA_AXI_MASTER_ERR clear bit
4	DATA_END_BIT_ERROR_CLR	DATA_END_BIT_ERROR clear bit
3	CMD_END_BIT_ERROR_CLR	CMD_END_BIT_ERROR clear bit
2	FIFO_EMPTY_ERROR_CLR	FIFO_EMPTY_ERROR clear bit
1	FIFO_FULL_ERROR_CLR	FIFO_FULL_ERROR clear bit

**0x078240F0+ SDC1_SDCC_MCI_INT_MASKINT2_n, n=[0..1]
0x4*n****Type:** RW**Clock:** same rate as HCLK**Reset State:** 0x00000000

Mask register for the MCI_STATUS2 register for each interrupt line

SDC1_SDCC_MCI_INT_MASKINT2_n

Bits	Name	Description
8	MASK8	BAM_AXI_MASTER_ERR mask bit
7	MASK7	AUTO_CDC_SW_CALIB_ERR mask bit.
6	MASK6	AUTO_CDC_SW_CALIB_TOUT mask bit.
5	MASK5	DATA_AXI_MASTER_ERR mask bit
4	MASK4	DATA_END_BIT_ERROR mask bit
3	MASK3	CMD_END_BIT_ERROR mask bit
2	MASK2	FIFO_EMPTY_ERROR mask bit
1	MASK1	FIFO_FULL_ERROR mask bit

0x07824100 SDC1_SDCC_CHAR_CMD

Type: RW

Clock: HCLK

Reset State: 0x00000000

Characterization config - 1 CMD line

SDC1_SDCC_CHAR_CMD

Bits	Name	Type	Description
15:8	CMDIN_ACTUAL	R	CMDIN storage for comparison. The incoming data is stored for debugging feature
7:0	CMDOUT_DATA_DIN_EXP	RW	CMDOUT for transmitting, CMDIN for compare. in TX testing the data is written in the register and should be transmitted outside from the core. In RX testing the incoming data is compared to the register's value

0x0782412C SDC1_SDCC_QSB_VALUES

Type: R

Clock: same rate as HCLK

Reset State: 0x00007991

configurable QSB values. Not applicabale for Gandalf - AXI Noc

SDC1_SDCC_QSB_VALUES

Bits	Name	Description
15	NON_BUF_BAM_BLOCK_END	Non bufferable/strongly ordered (hprot == 4'b0001/ memtype == 0) is driven in end of transaction by default. If setting this bit it will be driven for each block end
14	NOALLOCATE_VALUE	Driving noallocate value
13	RESERVED	Reserved
12	HPROT_VALUE_EN	Enabling the hprot values
11:8	HPROT_FOR_RDATA	hprot for read data. should be 9 by default so memtype is 2 (cachable)
7:4	HPROT_FOR_WDATA	hprot for write data during transaction or block. should be 9 by default so memtype is 2 (cachable)
3:0	HPROT_FOR_LAST_WDATA	hprot for last write data. should be 1 by default so memtype is 0 (strongly ordered)

0x07824130 SDC1_SDCC_QSB_AXI_VALUES**Type:** RW**Clock:** same rate as HCLK**Reset State:** 0x3E2218A8

configurable QSB AXI values

SDC1_SDCC_QSB_AXI_VALUES

Bits	Name	Type	Description
30	QSB_AXI_REDIRECT	RW	Driving QSB AXI output
29	QSB_AXI_ABURST	RW	Driving QSB AXI output
28:26	NUM_OUTSTANDING_DATA	RW	Configuring the number of outstanding transactions for the data. The descriptors accesses have more one outstanding request
25	PROCEED_AXI_AFTER_ERROR	RW	When error occurs on AXI master the traffic can continue or terminate. If this bit is set then the traffic continues and more requests are transmitted on the address channel
24	QSB_AXI_AFULL_CALC	RW	If this bit is set, then afull isn't '0' constantly but can be equal to '1' if all the beats have 8 bytes on the strobe lines
23	ONE_MID_SUPPORT	RW	If this bit is set, then only 1 MID is used for descriptors and data
22:20	RESERVED	RW	Reserved
19	SDCC5_HALT_REQ	R	Reading SDCC5_HALT_REQ input

SDC1_SDCC_QSB_AXI_VALUES (cont.)

Bits	Name	Type	Description
18	SDCC5_HALT_ACK	R	Reading SDCC5_HALT_ACK output
17	SDCC5_M_IDLE	R	Reading SDCC5_M_IDLE output
16	SDCC5_HALT_ACK_SW_EN	RW	Enabling the SW value for SDCC5_HALT_ACK
15	SDCC5_HALT_ACK_SW	RW	Driving SDCC5_HALT_ACK from SW and not by HW
14	SDCC5_M_IDLE_DIS	RW	If this bit is set, then sdcc5_m_idle will be tied low
13	QSB_AXI_INTERLEAVING_EN	RW	If the bit is set, then the controller can support interleaving between two bursts from the Noc. This option is disabled by default because Noc doesn't support interleaving but only re-ordering
12	RESERVED	RW	Reserved
11	RESERVED	RW	Reserved
10:9	RESERVED	RW	Reserved
8:6	RESERVED	RW	Reserved
5	RESERVED	RW	Reserved
4	RESERVED	RW	Reserved
3	RESERVED	RW	Reserved
2	QSB_AXI_OOOWR	RW	Driving QSB AXI output
1	QSB_AXI_OOORD	RW	Driving QSB AXI output
0	QSB_AXI_AFULL	RW	Driving QSB AXI output

0x07824134 SDC1_SDCC_AXI_DATA_ERR_DEBUG**Type:** R**Clock:** same rate as HCLK**Reset State:** 0x00000000

Debug details after error on AXI I/F

SDC1_SDCC_AXI_DATA_ERR_DEBUG

Bits	Name	Description
31	RESP_ERR_WR	If set, then the failed transaction was a write access
30:28	RESP_ERR_TID	The TID of the failed transaction
27:13	DESC_ADDR_35_21	Upper 15 bits of current address in address channel
12:0	DESC_ADDR_12_0	Lower 13 bits of current address in address channel

0x07824138 SDC1_SDCC_T4_DLY_CTRL**Type:** RW**Clock:** same rate as HCLK**Reset State:** 0x0000382B

Control register of T/4 Delay element which is used in e.MMC5.0 DDR mode.

SDC1_SDCC_T4_DLY_CTRL

Bits	Name	Description
15:14	T4_DLY_SPARE_CTL	
13	T4_DLY_POWERDOWN_EN	Powerdown enable. When asserted (default), it powers down the T/4 macro and shuts off all bias circuits.
12:11	T4_DLY_TESTOUT_SEL	Test output select. When toggled, it presents either the strb_in or strb_out on the analog test port for debug.
10	T4_DLY_BYPASS_EN	Bypass enable. When asserted it presents the strb_in on the strb_out port.
9	T4_DLY_CAL_START	Calibration start input. When asserted, it indicates the start of the calibration cycle.
8	T4_DLY_CAL_OVERRIDE	Calibration override input. When asserted, the internal delay line calibration is bypassed and the tby4_user_ctl<7:0> is used as the default code for the delay line.
7:0	T4_DLY_TARGET_CODE	T/4 delay target code. 0x2b gives delay of 1.25ns

0x0782413C SDC1_SDCC_T4_DLY_STAT**Type:** R**Clock:** same rate as HCLK**Reset State:** 0x00000000

Status register T/4 Delay element which is used in e.MMC5.0 DDR mode.

SDC1_SDCC_T4_DLY_STAT

Bits	Name	Description
15:14	T4_DLY_SPARE_OUT	Spare status bits.
13	T4_DLY_CAL_DLY_CODE_FAST_RANGE	Calibrated delay code, indicates the fast range selection bit status.
12:6	T4_DLY_CAL_DLY_CODE_ISEL	Calibrated delay code, indicates the isel programming value for the bias generator.
5:4	T4_DLY_CAL_DLY_CODE_RSEL	Calibrated delay code, indicates the rsel programming value for the bias generator.
3:0	T4_DLY_STATE_DESC	T/4 state descriptor. Indicates the calibration status and result. value 4'b0000: Indicates Reset mode, waiting for cal_start signal. value 4'b0111: Indicates the fast range decision completion. value 4'b0100: Indicates rsel adjustment decision. value 4'b1011: Indicates isel adjustment decision. value 4'b1110: Indicates Normal end-of-cal. value 4'b1111: Indicates Maximum Delay Reached

0x07824140 SDC1_SDCC_CSR_CDC_CTLR_CFG0**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

The CSR_CDC_CTLR_CFG0 register configures the following:

SDC1_SDCC_CSR_CDC_CTLR_CFG0

Bits	Name	Description
25	STAGGER_CAL_ENA	1'b1: Stagger calibration of write and read CDCs once after the other to reduce peak voltage drop. 1'b0: Both write and read CDCs calibrated simultaneously This bit is introduced on the x2 core, so not present on x0 and x1 versions of DIM/PHY.
24	TRACKING_CAL_ENA	1'b1: Relationship between subunits and oscillator count is learnt and used for tracking calibration. 1'b0: Periodic calibration triggers cause full recalibration

SDC1_SDCC_CSR_CDC_CTLR_CFG0 (cont.)

Bits	Name	Description
23:20	OSC_COUNT_ERR_TOLERANCE	Magnitude of the difference between oscillator count and target count when the algorithm assumes convergence and stops further search.
19	TRACK_CALIB_MODE	1'b0: Tracking calibration uses previous Target_count value. 1'b1 Accurate mode: Target_count is recomputed at the start of tracking calibration
18	FULL_DELAY	1'b0: unitstep and subunit caps are from calibration or SW settings 1'b1: All unit step and subunit caps are selected for the delay arrays.
17	HW_AUTOCAL_ENA	1'b1: Enables periodic auto calibration 1'b0: Periodic calibration is disabled
16	SW_TRIG_FULL_CALIB	A low to high transition on this register bit triggers the CDC calibration statemachine to initiate full calibration of the standby delay array. Also resets tracking mode for the delay arrays.
13:12	OSC_PRE_DIV	The factor by which the raw oscillator frequency is divided before being given to the counting logic. 2'b00 Divide by 1 2'b01 Divide by 2 2'b10 Divide by 4 2'b11 Divide by 8
10:0	TMUX_CHAR	Cchar is a characterized constant equal to the ratio of output mux offset delay to average unit step delay. This register is programmed by multiplying the constant with 1024 and rounding to the nearest integer. E.g. if Cchar = 0.67, $RO(0.67 * 1024) = 686$, hence program TMUX_CHAR = 11'h2AE

0x07824144 SDC1_SDCC_CSR_CDC_CTLR_CFG1**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

The CSR_CDC_CTLR_CFG1 register configures the following:

SDC1_SDCC_CSR_CDC_CTLR_CFG1

Bits	Name	Description
26:24	OSC_COUNT_DELAY	Number of oscillator cycles from the end of reference gate signal to assertion of oscillation counter done. Legal values are from 0 to 4.

SDC1_SDCC_CSR_CDC_CTLR_CFG1 (cont.)

Bits	Name	Description
23:20	STANDBY_DELAY	This is the number of reference clock cycles from when a newly calibrated delay array is made active to when the current active delay array is put into standby mode. 4'b0: Disables the counter. Hardware waits for explicit acknowledgement of delay array switch from CDC. Others: Explicit acknowledgement mode is disabled and the counter is used
18:16	DEL_MODE_DELAY	Number of reference clock cycles between putting a newly calibrated delay array in delay mode to selecting it as the active delay array. The minimum allowed value is 1
14:12	OSC_MODE_DELAY	Number of reference clock cycles between putting a delay array in oscillator mode to starting the oscillation counter. This delay is allowed for the oscillations to stabilize through the delay array. The minimum allowed value is 1
10:8	DECODER_DELAY	Number of reference clock cycles allowed for unitstep and subunit counts to be encoded into the unit and finecap select lines on the DDA. The minimum allowed value is 1.
6:4	DIVIDER_DELAY	Number of reference clock cycles allowed for arithmetic division operations. The minimum allowed value is 1
2:0	MULTIPLIER_DELAY	Number of reference clock cycles allowed for arithmetic multiplication operations. The minimum allowed value is 1

0x07824148 SDC1_SDCC_CSR_CDC_CAL_TIMER_CFG0**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

The CSR_CDC_CAL_TIMER_CFG0 register configures the following:

SDC1_SDCC_CSR_CDC_CAL_TIMER_CFG0

Bits	Name	Description
24	INVALID_TIMER_ENA	1'b1: Enable the timer counting number of missed calibration intervals after which the CDC is invalid. 1'b0: Timer disabled A low to high transition on this register bit loads the timer with programmed value below.
23:20	INVALID_TIMER_VAL	Number of missed calibration intervals after which CDC invalid state is asserted.

SDC1_SDCC_CSR_CDC_CAL_TIMER_CFG0 (cont.)

Bits	Name	Description
16	TIMER_ENA	1'b1: Enables 32 kHz sleep clock based periodic autocalibration timer. 1'b0: Timer disabled A low to high transition on this register bit loads the timer with programmed value below.
15:0	TIMER_VAL	Count value for periodic autocalibration interval based on sleep clock.

0x0782414C SDC1_SDCC_CSR_CDC_CAL_TIMER_CFG1**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

The CSR_CDC_CAL_TIMER_CFG1 register configures the following:

SDC1_SDCC_CSR_CDC_CAL_TIMER_CFG1

Bits	Name	Description
12	FF_TIMER_ENA	1'b1: Enables reference clock based periodic autocalibration timer. 1'b0: Timer disabled A low to high transition on this register bit loads the timer with programmed value below.
9:0	FF_TIMER_VAL	Count value for periodic autocalibration interval based on reference clock.

0x07824150 SDC1_SDCC_CSR_CDC_REFCOUNT_CFG**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

The CSR_CDC_REFCOUNT_CFG register configures the following:

SDC1_SDCC_CSR_CDC_REFCOUNT_CFG

Bits	Name	Description
31:16	TREF	Reference clock period programmed in pico seconds.
13:8	CCAL_REF_COUNT	Number of reference clock cycles used for oscillation counts during coarse calibration. Maximum supported value is 59.
5:0	FCAL_REF_COUNT	Number of reference clock cycles used for oscillation counts during fine calibration. Maximum supported value is 59

0x07824154 SDC1_SDCC_CSR_CDC_COARSE_CAL_CFG**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

The register CSR_CDC_COARSE_CAL_CFG configures the following:

SDC1_SDCC_CSR_CDC_COARSE_CAL_CFG

Bits	Name	Description
12:8	CCAL_SUBUNIT_CAPS	Subunit count needed for an effective unit step count of 32 during coarse calibration. This value is also used by software oscillator mode.
5:0	CCAL_UNITSTEPS	unitstep count needed for an effective unit step count of 32 during coarse calibration.

0x07824158 SDC1_SDCC_CSR_CDC_RSVD_CFG**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

The CSR_CDC_RSVD_CFG register configures the following:

SDC1_SDCC_CSR_CDC_RSVD_CFG

Bits	Name	Description
15:0	TEMP_FIELD	Reserved register

0x0782415C SDC1_SDCC_CSR_CDC_OFFSET_CFG**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

The CSR_CDC_OFFSET_CFG register configures the following:

SDC1_SDCC_CSR_CDC_OFFSET_CFG

Bits	Name	Description
20	SUBUNIT_OFFSET_MODE	1'b1: The programmed offset is directly loaded as subunit count 1'b0: Programmed offset is added/subtracted from the subunit count

SDC1_SDCC_CSR_CDC_OFFSET_CFG (cont.)

Bits	Name	Description
19	SUBUNIT_OFFSET_SIGN	1'b1: Offset is subtracted from the subunit count 1'b0: Offset is added to the subunit count
16:12	SUBUNIT_OFFSET	Software subunit offset
8	UNIT_OFFSET_MODE	1'b1: The programmed offset is directly loaded as unit step count 1'b0: Programmed offset is added/subtracted from the unit step count
7	UNIT_OFFSET_SIGN	1'b1: Offset is subtracted from the unitstep count 1'b0: Offset is added to the unitstep count
5:0	UNIT_OFFSET	Software unitstep offset

0x07824160 SDC1_SDCC_CSR_CDC_DELAY_CFG**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

The CSR_CDC_DELAY_CFG register configures the following:

SDC1_SDCC_CSR_CDC_DELAY_CFG

Bits	Name	Description
28	TARGET_COUNT_ENA	1'b1: Enable software specified Target_count for calibration instead of value calculated by hardware. 1'b0: Hardware computed Target_count used.
27:16	TARGET_COUNT	Software specified Target_count to which the statemachine calibrates
11:0	DELAY_VAL	Delay required from CDC programmed in pico seconds. This value has to be greater than the minimum delay supported by DDA Tmin.

0x07824164 SDC1_SDCC_CSR_CDC_SW_MODE_CFG**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

The CSR_CDC_SW_MODE_CFG register configures the following:

SDC1_SDCC_CSR_CDC_SW_MODE_CFG

Bits	Name	Description
3	SW_DEL_MODE	A low to high transition on this register bit puts the delay array selected by SW_DA_SEL into delay mode and connects it to the DDA output. This register has to remain set for the duration of SW controlled delay mode.
2	SW_OSC_MODE	Reserved register.
1	SW_DA_SEL	Selects the delay array inside DDA during software delay or oscillator mode.
0	SW_LOAD	1'b1: Enable loading of the software offsets from the offset configuration register. This register has to remain set for the duration of SW mode operation.

0x0782416C SDC1_SDCC_CSR_CDC_SW_OVRD_CFG**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

The CSR_CDC_SW_OVRD_CFG register configures the following:

SDC1_SDCC_CSR_CDC_SW_OVRD_CFG

Bits	Name	Description
16	SW_REF_GATE	Write 1'b1 to start the reference window. Write 1'b0 to end the reference window. Number of reference cycle in the window will be reported in SW_REF_GATE_COUNT in CDC_STATUS0
11	SW_OVRD_DA1_OSC_EN	In software override mode, 1'b1 written to this bit puts delay array 1 in oscillator-mode
10	SW_OVRD_DA1_IN_EN	In software override mode, 1'b1 written to this bit puts delay array 1 in delay-mode
9	SW_OVRD_DA0_OSC_EN	In software override mode, 1'b1 written to this bit puts delay array 0 in oscillator-mode
8	SW_OVRD_DA0_IN_EN	In software override mode, 1'b1 written to this bit puts delay array 0 in delay-mode
3	SW_OVRD_CDC_COUNTERR_RST	1'b1 written to this bit resets the oscillation counter. Follow this with 1'b0 before setting reference gate.
2	SW_OVRD_LOAD_DA_SEL	In software override mode, this bit select the delay array to which unit and subunit offset values from CDC_OFFSET_CFG register will be loaded.
1	SW_OVRD_ACTV_DA_SEL	In software override mode, this register selects the delay array connected to the CDC output

SDC1_SDCC_CSR_CDC_SW_OVRD_CFG (cont.)

Bits	Name	Description
0	SW_OVRD_ENA	1'b1: Written to this bit puts the CDC interface in software override mode where the below register bits control the CDC. 1'b0: CDC is under hardware control

0x07824170 SDC1_SDCC_CSR_CDC_SLAVE_DDA_CFG**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

The CSR_CDC_SLAVE_DDA_CFG register configures the following:

SDC1_SDCC_CSR_CDC_SLAVE_DDA_CFG

Bits	Name	Description
17	SLAV_DDA_OFFSET_MODE	1'b1: The programmed offset is directly loaded as unit-step count 1'b0: Programmed offset is added/subtracted from the unit-step count
16	SLAV_DDA_OFFSET_SIGN	1'b1: Offset is subtracted from the unit-step count. This subtraction feature is not supported on x0 and x1 PHY cores (bit RESERVED). Supported on the x2 core version. 1'b0: Offset is added to the unit-step count
15:12	SLAVE_DDA_OFFSET	unit-step offset for slave DDA.
10:0	SLAVE_DDA_DELAY	Delay required from slave DDA programmed in pico seconds.

0x07824174 SDC1_SDCC_CSR_CDC_STATUS0**Type:** R**Clock:** HCLK**Reset State:** 0x0000000C

The CSR_CDC_STATUS0 register configures the following:

SDC1_SDCC_CSR_CDC_STATUS0

Bits	Name	Description
26:24	CDC_ERROR_CODE	Error code for internal errors: 3'b000: No error 3'b001: Programmed delay less than Tmux 3'b010: Programmed delay less than Tmin 3'b011: Unit-step overflow: Overflow during fine calibration. Desired delay cannot be realized even when all unit-steps are selected. 3'b100: Unit-step underflow 3'b101: Unit-step overflow: Overflow during coarse calibration. Estimated unit steps needed to realize the desired delay are greater than the total number of unit-steps in design. Note: The CDC_ERROR_CODE is not a sticky register and is updated after each CDC calibration.
23:16	SW_REF_GATE_COUNT	Number of reference clock cycles in the software generated oscillation gate window during SW override mode
15:4	OSC_COUNT	Current value of oscillator count
3	CURR_SEL_DA	This bit indicates the current active delay array connected to the output. 1'b0: DA0 is selected 1'b1: DA1 is selected
2	CTLR_SM_IDLE	1'b1 Indicates that the calibration state machine is idle 1'b0 indicates calibration state machine is busy
1	OSC_DONE	This status register is set after oscillations count is finished by CDC. It gets reset when a new oscillator count is started.
0	CALIBRATION_DONE	1'b1 indicates that calibration is done and CURR_SUBUNIT_STEPS, CURR_UNIT_STEPS status registers are valid.

0x07824178 SDC1_SDCC_CSR_CDC_STATUS1**Type:** R**Clock:** HCLK**Reset State:** 0x00000000

The CSR_CDC_STATUS1 register configures the following:

SDC1_SDCC_CSR_CDC_STATUS1

Bits	Name	Description
27:16	CURR_DELAY_VALUE	HW Computed value of current delay after calibration by the state machine. This register is valid if CALIBRATION_DONE status bit is set.

SDC1_SDCC_CSR_CDC_STATUS1 (cont.)

Bits	Name	Description
11:0	CURR_TMUX_DELAY	Current value of output mux offset delay computed by HW.

0x0782417C SDC1_SDCC_CSR_CDC_STATUS2**Type:** R**Clock:** HCLK**Reset State:** 0x10331033

The CSR_CDC_STATUS2 register configures the following:

SDC1_SDCC_CSR_CDC_STATUS2

Bits	Name	Description
28:24	DA1_SUBUNITS	Current value of subunit cap settings applied to delay array 1
21:16	DA1_UNITSTEPS	Current value of unit steps applied to delay array 1
12:8	DA0_SUBUNITS	Current value of subunit cap settings applied to delay array 0
5:0	DA0_UNITSTEPS	Current value of unit steps applied to delay array 0

0x07824180 SDC1_SDCC_CSR_CDC_STATUS3**Type:** R**Clock:** HCLK**Reset State:** 0x00000000

The CSR_CDC_STATUS3 register configures the following:

SDC1_SDCC_CSR_CDC_STATUS3

Bits	Name	Description
31:24	NUM_OF_OSC_ITER	Number of oscillator iterations before convergence was achieved during tracking
23:12	COUNT_ERROR	Difference between target and actual oscillator counts at the start of tracking calibration
11:0	CURR_TARGET_COUNT	HW computed value of current Target_count used by the calibration state machine.

0x07824184 SDC1_SDCC_CSR_CDC_STATUS4**Type:** R**Clock:** HCLK**Reset State:** 0x000000FF

The CSR_CDC_STATUS4 register configures the following:

SDC1_SDCC_CSR_CDC_STATUS4

Bits	Name	Description
7:4	SLAVE_DDA_DA1_TAPS	Number of unit taps applied to delay array 1 of slave DDA
3:0	SLAVE_DDA_DA0_TAPS	Number of unit taps applied to delay array 0 of slave DDA

0x07824188 SDC1_SDCC_CSR_CDC_GEN_CFG**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000**SDC1_SDCC_CSR_CDC_GEN_CFG**

Bits	Name	Description
2	CDC_SDDA_EN	Enable CDC Slave DDA to generate 3/4T RCLK from cdc_out_2 output of CDC: 0 (default) - CDC Slave DDA is disabled and 3/4T clock is a falling edge of cdc_out_1 - CDC Slave DDA is enabled and 3/4T clock is a rising edge of cdc_out_2.
1	CDC_SWITCH_RC_EN	Enablement of CDC power RC (LPF) switch 1'b1: Enabled 1'b0: Disabled (default)
0	CDC_SWITCH_BYPASS_OFF	Enablement of CDC power bypass switch 1'b1: Disabled 1'b0: Enabled (default)

0x0782418C SDC1_SDCC_CDC_AUTO_SW_CAL_CFG**Type:** RW**Clock:** HCLK**Reset State:** 0x00F08000**SDC1_SDCC_CDC_AUTO_SW_CAL_CFG**

Bits	Name	Description
25	CDC_CAL_TIMEOUT_ENABLE	Enable for cdc calibration timeout counter.

SDC1_SDCC_CDC_AUTO_SW_CAL_CFG (cont.)

Bits	Name	Description
24	CDC_AUTO_SW_CAL_ENABLE	Enable for the Auto SW calibration Feature.
23:16	CDC_CAL_TIMEOUT_THRESHOLD	Sets the threshold for CDC calibration timeout during sw calibration triggered by the HW FSM. counts in 16xFF clock. Default is 0xF0 which is 255x16 FF clocks, which gives threshold of ~200 us
15:0	AUTO_TRIG_INTERVAL	Sets the load value of the counter that counts the interval between each sw calibration triggered by the HW FSM. this counter counts in sleep clock. Default is 0x8000 which is ~32,000 sleep clocks, which gives interval of ~1 sec

0x07824190 SDC1_SDCC_DDR200_STAT**Type:** R**Clock:** HCLK**Reset State:** 0x00000000**SDC1_SDCC_DDR200_STAT**

Bits	Name	Description
0	CDC_CAL_VALID	CDC Calibration status: 0 - CDC isn't calibrated, 1 - CDC is calibrated and valid

0x07824194 SDC1_SDCC_DDR200_CFG**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

Configuration bits for e.MMC5.0 DDR200 mode

SDC1_SDCC_DDR200_CFG

Bits	Name	Description
10	FF_CLK_DIS	The ff_clk to CDC/TBY4 is enabled by default. It can be disabled by setting this bit. The ff_clk is enabled to CDC and TBY4 according to CDC_T4_DLY_SEL field
9	VOLTAGE_MUX_SEL	select between 1.8v and 1.2v signals on sdcc5 IO interface with device. default is 0 to choose 1.8v

SDC1_SDCC_DDR200_CFG (cont.)

Bits	Name	Description
8:7	CDC_TRAFFIC_SEL	Selecting the method of cdc_traffic's assertions: Value 0 (default) - auto calibration with stand by feature. Value 1 - SW calibration before any transaction. Value 2 - automatic calibration with HW assertion of cdc_traffic.
6	START_CDC_TRAFFIC	This bit starts the cdc traffic assertion after calibration done is received.
5	CRC_TOKEN_SAMPL_FALL_EDGE	If set, the start bit of incoming CRC token in DDR200 mode is sampled in falling edge. If this bit is cleared (default) then the start bit is detected in rising edge
4	DATIN_SAMPL_FALL_EDGE	If set, the start bit of incoming data in DDR200 mode is sampled in falling edge. If this bit is cleared (default) then the start bit is detected in rising edge
3	CDC_T4_TEST_OUT_SEL	controls the mux between dll test_out and CDC test out. Default value is 0 (dll is chosen). When set - CDC test_out is chosen.
2	CMDIN_EDGE_SEL	Select sampling edge of CMD input when CMDIN_RCLK_EN is 1: 0 (default) - falling edge of RCLK, 1 - rising edge of RCLK
1	CMDIN_RCLK_EN	Enable CMD input sampling with RCLK: 0 (default) - CMD input is sampled by SD DLL output clock, 1 - CMD input is sampled by delayed RCLK
0	CDC_T4_DLY_SEL	Select block for RCLK delay: 0 (default) - use CM_DLL_SDC4

0x07824198 SDC1_SDCC_TIME_COUNT_CTRL**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

Control the performance counters

SDC1_SDCC_TIME_COUNT_CTRL

Bits	Name	Description
1	CMD_PERIOD	When this bit is set the commands periods are counted to the CMD_TIME counter
0	ENABLE	Enabling the performance counters

0x0782419C SDC1_SDCC_FLOW_CTRL_TIME_COUNT**Type:** R**Clock:** HCLK**Reset State:** 0x00000000

Counting the Flow Control period

SDC1_SDCC_FLOW_CTRL_TIME_COUNT

Bits	Name	Description
31:0	FLOW_CTRL_TIME	FlowControl periods are counted

0x078241A0 SDC1_SDCC_BUSY_TIME_COUNT

Type: R

Clock: HCLK

Reset State: 0x00000000

Counting the BUSY periods

SDC1_SDCC_BUSY_TIME_COUNT

Bits	Name	Description
31:0	BUSY_TIME	In Tx transaction, the BUSY periods are counted until the progdone is asserted. In Rx transaction, the periods between the blocks without flowcontrol is counted including the time until the first block is received after the command.

0x078241A4 SDC1_SDCC_DATA_TIME_COUNT

Type: R

Clock: HCLK

Reset State: 0x00000000

Counting the time of transaction

SDC1_SDCC_DATA_TIME_COUNT

Bits	Name	Description
31:0	DATA_TIME_CNT	In Tx transaction, the time is from ENabling the DPSM until progdone is asserted. In Rx transaction, the time is from the command transmission to de-assertion of RxActive.

0x078241A8 SDC1_SDCC_CMD_TIME_COUNT

Type: R

Clock: HCLK

Reset State: 0x00000000

Counting the time of transaction

SDC1_SDCC_CMD_TIME_COUNT

Bits	Name	Description
31:0	CMD_TIME	Counting the Command-Response periods of commands which relate to the transaction including auto commands. The counter is active when CMD_PERIOD is set.

0x078241AC SDC1_SDCC_MCI_DLL_CONFIG_2

Type: RW

Clock: same rate as HCLK

Reset State: 0x00000000

Second register for DLL's configuration

SDC1_SDCC_MCI_DLL_CONFIG_2

Bits	Name	Description
3:2	DDR_TRAFFIC_INIT_SEL	Selecting how DDR_TRAFFIC_INIT will be driven to DLL. Value 00 - HW drives the DDR_TRAFFIC_INIT when DPSM is active and when CPSM is active if RCLK is expected during response. Value 01 - The value is driven by SW by SW_DDR_TRAFFIC_INIT bit. Value 10 - DDR_TRAFFIC_INIT to DLL is driven when sdc4_mclk is not gated. off.
1	DDR_TRAFFIC_INIT_SW	The signal should be high 3-4 cycles before RCLK is toggling. It should be high when MCLK is toggling.
0	DDR_CAL_EN	The signal is set after the configuration of MCI_DDR_CONFIG is finished. When the signal is low the TCXO domain is being reset.

0x078241B0 SDC1_SDCC_MCI_DDR_CONFIG

Type: RW

Clock: same rate as HCLK

Reset State: 0x80040853

Configuration of ddr_config input of DLL

SDC1_SDCC_MCI_DDR_CONFIG

Bits	Name	Description
31	PRG_DLY_EN	Enables Programmable delay for RCLK_IN
30	EXT_PRG_RCLK_DLY_EN	Enables external control of Programmable RCLK delayline

SDC1_SDCC_MCI_DDR_CONFIG (cont.)

Bits	Name	Description
29:27	EXT_PRG_RCLK_DLY_COD E	Used only if EXT_PRG_RCLK_DLY_EN is set. Configuring Programmable RCLK Delayline process code number.
26:21	EXT_PRG_RCLK_DLY	Used only if EXT_PRG_RCLK_DLY_EN is set. Programmable RCLK Delay line fine delay control.
20:12	TCXO_CYCLES_DLY_LINE	Number of TCXO clock cycles between finish of one delay line calibration and beginning of next delay line calibration.
11:9	TCXO_CYCLES_CNT	Number of TCXO clock cycles for which down counter is enabled for before making each inc/dec decision.
8:0	PRG_RCLK_DLY	These bits are controlled the programmable delay line. This bus is binary coded. Decimal value = (TCXO period * Decimal value of TCXO_CYCLES_CNT)/(2*delay needed). For 1.25ns the formula is $(52n*4/2*1.25) = 83$ in decimal.

0x07824400+ SDC1_SDCC_MCI_FIFO_ALTn, n=[0..255]
0x4*n
Type: RW**Clock:** same rate as HCLK**Reset State:** 0x00000000

Alternative register to MCI_FIFO (0x80-0xBC). The SW can use this register range (0x400-0x7FC) for accessing all the RAM range (1024 bytes) in FIFO mode. The advantage is that any byte in the RAM can be accessed without repeating the same address

SDC1_SDCC_MCI_FIFO_ALTn

Bits	Name	Description
31:0	DATA	WORD is written or received to/from the RAM in FIFO mode

18.2 SDC1_SDCC_SDCC5_HC**0x07824900 SDC1_SDCC_HC_REG_0_2****Type:** RW**Clock:** same rate as HCLK**Reset State:** 0x00000000

Argument 2 Register in SD Host Controller Standard Specification.

SDC1_SDCC_HC_REG_0_2

Bits	Name	Description
31:0	ARG_2	Argument 2 field

0x07824904 SDC1_SDCC_HC_REG_4_6**Type:** RW**Clock:** same rate as HCLK**Reset State:** 0x00000000

Combination of two registers from the SD Host Controller Standard Specification:

-Block Size Register(15-0)

-Block Count Register(31-16)

SDC1_SDCC_HC_REG_4_6

Bits	Name	Description
31:16	BLK_CNT_FOR_CUR_TRANS	Blocks Count For Current Transfer field from Block Count Register
14:12	BLK_SIZE_HST_SDMA_BUFFER	Host SDMA Buffer Boundary field from Block Size Register
11:00	BLK_SIZE_TRANS	Transfer Block Size field from Block Size Register

0x07824908 SDC1_SDCC_HC_REG_8_A**Type:** RW**Clock:** same rate as HCLK**Reset State:** 0x00000000

Argument 1 Register in SD Host Controller Standard Specification.

SDC1_SDCC_HC_REG_8_A

Bits	Name	Description
31:00	CMD_ARG_1	Command Argument 1 field

0x0782490C SDC1_SDCC_HC_REG_C_E**Type:** RW**Clock:** same rate as HCLK**Reset State:** 0x00000000

Combination of two registers from the SD Host Controller Standard Specification:

-Transfer Mode Register(15-0)

-Command Register(31-16)

SDC1_SDCC_HC_REG_C_E

Bits	Name	Description
29:24	CMD_INDX	Command Index field from Command Register
23:22	CMD_TYPE	Command Type field from Command Register
21	CMD_DATA_PRESENT_SEL	Data Present Select field from Command Register
20	CMD_INDX_CHECK_EN	Command Index Check Enable field from Command Register
19	CMD_CRC_CHECK_EN	Command CRC Check Enable field from Command Register
17:16	CMD_RESP_TYPE_SEL	Response Type Select field from Command Register
5	TRANS_MODE_MULTI_SINGLE_BLK_SEL	Multi/Single Block Select from Transfer Mode Register
4	TRANS_MODE_DATA_DIRECTION_SEL	Data Transfer Direction Select from Transfer Mode Register
3:2	TRANS_MODE_AUTO_CMD_EN	Auto CMD Enable from Transfer Mode Register
1	TRANS_MODE_BLK_CNT_EN	Block Count Enable field from Transfer Mode Register
0	TRANS_MODE_DMA_EN	DMA Enable field from Transfer Mode Register

0x07824910 SDC1_SDCC_HC_REG_10_12**Type:** R**Clock:** same rate as HCLK**Reset State:** 0x00000000

Response Register in SD Host Controller Standard Specification.

SDC1_SDCC_HC_REG_10_12

Bits	Name	Description
31:0	CMD_RESP	Command Response field - Response bits 0-31

0x07824914 SDC1_SDCC_HC_REG_14_16**Type:** R**Clock:** same rate as HCLK**Reset State:** 0x00000000

Response Register in SD Host Controller Standard Specification.

SDC1_SDCC_HC_REG_14_16

Bits	Name	Description
31:0	CMD_RESP	Command Response field - Long Response bits 31-63

0x07824918 SDC1_SDCC_HC_REG_18_1A**Type:** R**Clock:** same rate as HCLK**Reset State:** 0x00000000

Response Register in SD Host Controller Standard Specification.

SDC1_SDCC_HC_REG_18_1A

Bits	Name	Description
31:0	CMD_RESP	Command Response field - Long Response bits 64-95

0x0782491C SDC1_SDCC_HC_REG_1C_1E**Type:** R**Clock:** same rate as HCLK**Reset State:** 0x00000000

Response Register in SD Host Controller Standard Specification.

SDC1_SDCC_HC_REG_1C_1E

Bits	Name	Description
31:0	CMD_RESP	Command Response field - Long Response bits 96-127

0x07824920 SDC1_SDCC_HC_REG_20_22**Type:** RW**Clock:** same rate as HCLK**Reset State:** 0x00000000

Buffer Data Port Register in SD Host Controller Standard Specification.

SDC1_SDCC_HC_REG_20_22

Bits	Name	Description
31:24	BUF_DATA_PORT_3	Byte 3 of Buffer Data
23:16	BUF_DATA_PORT_2	Byte 2 of Buffer Data
15:8	BUF_DATA_PORT_1	Byte 1 of Buffer Data
7:0	BUF_DATA_PORT_0	Byte 0 of Buffer Data

0x07824924 SDC1_SDCC_HC_REG_24_26**Type:** R**Clock:** same rate as HCLK**Reset State:** 0x00080000

Present State Register in SD Host Controller Standard Specification.

SDC1_SDCC_HC_REG_24_26

Bits	Name	Description
25	SIGNALING_18_SWITCHING_STS	Bit it is cleared, by hardware, when host driver writes any value to 1.8V Signaling Enable field, indicating that a signal switching operation has started. Bit is set, by hardware, when switching operation succeeds.
24	PRESENT_STATE_CMD_LINE_SIGNAL_LEVEL	CMD Linr Signal Level field
23:20	PRESENT_STATE_DAT_3_0_LINE_SIGNAL_LEVEL	DAT[3:0] Line Signal Level field
19	PRESENT_STATE_WRITE_PROTECT_SWITCH_PIN_LEVEL	Write Protect Switch Pin Level field
18	PRESENT_STATE_CARD_DETECT_PIN_LEVEL	Card Detect Pin Level field
17	PRESENT_STATE_CARD_STATE_STABLE	Card State Stable field
16	PRESENT_STATE_CARD_INSERTED	Card Inserted field

SDC1_SDCC_HC_REG_24_26 (cont.)

Bits	Name	Description
11	PRESENT_STATE_BUF_RD_EN	Buffer Read Enable field
10	PRESENT_STATE_BUF_WR_EN	Buffer Write Enable field
9	PRESENT_STATE_RD_TRANS_ACT	Read Transfer Active field
8	PRESENT_STATE_WR_TRANS_ACT	Write Transfer Active field
3	PRESENT_STATE_RETUNING_REQ	Re-Tuning Request field
2	PRESENT_STATE_DATA_LINE_ACT	DAT Linr Active field
1	PRESENT_STATE_CMD_INHIBIT_DAT	Command Inhibit (DAT) field
0	PRESENT_STATE_CMD_INHIBIT_CMD	Command Inhibit (CMD) field

0x07824928 SDC1_SDCC_HC_REG_28_2A**Type:** RW**Clock:** same rate as HCLK**Reset State:** 0x00000000

Combination of four registers from the SD Host Controller Standard Specification:

-Host Control 1 Register(7-0)

-Power Control Register(15-8)

-Block Gap Control Register(23-16)

-Wakeup Control Register(31-24)

SDC1_SDCC_HC_REG_28_2A

Bits	Name	Description
26	WKUP_EVENT_EN_ON_SD_CARD_REMOVAL	Wakeup Event Enable On SD Card Removal field from Wakeup Control Register
25	WKUP_EVENT_EN_ON_SD_CARD_INSERTION	Wakeup Event Enable On SD Card Insertion field from Wakeup Control Register
24	WKUP_EVENT_EN_ON_SD_CARD_INT	Wakeup Event Enable On Card Interrupt field from Wakeup Control Register
19	BLK_GAP_CTL_INT	Interrupt At Block Gap field from Block Gap Control Register

SDC1_SDCC_HC_REG_28_2A (cont.)

Bits	Name	Description
18	BLK_GAP_CTL_RD_WAIT	Read Wait Control field from Block Gap Control Register
17	BLK_GAP_CTL_CONTINUE_REQ	Continue Request field from Block Gap Control Register
16	BLK_GAP_CTL_STOP_GAP_REQ	Stop At Block Gap Request field from Block Gap Control Register
11:9	PWR_CTL_SD_BUS_VOLTAGE_SEL	SD Bus Voltage Select field from Power Control Register
8	PWR_CTL_SD_BUS_PWR	SD Bus Power field from Power Control Register
7	HST_CTL1_CARD_DETECT_SIGNAL_SEL	Card Detect Signal Selection field from Host Control 1 Register
6	HST_CTL1_CARD_DETECT_TEST_LEVEL	Card Detect Test Level field from Host Control 1 Register
5	HST_CTL1_EXTENDED_DATA_TRANS_WIDTH	Extended Data Transfer Width field from Host Control 1 Register
4:3	HST_CTL1_DMA_SEL	DMA Select field from Host Control 1 Register
2	HST_CTL1_HS_EN	High Speed Enable field from Host Control 1 Register
1	HST_CTL1_DATA_TRANS_WIDTH	Data Transfer Width field from Host Control 1 Register
0	HST_CTL1_LED_CTL	LED Control field from Host Control 1 Register

0x0782492C SDC1_SDCC_HC_REG_2C_2E**Type:** RW**Clock:** same rate as HCLK**Reset State:** 0x00000001

Combination of three registers from the SD Host Controller Standard Specification:

-Clock Control Register(15-0)

-Timeout Control Register(23-16)

-Software Reset Register(31-24)

SDC1_SDCC_HC_REG_2C_2E

Bits	Name	Type	Description
26	SW_RST_DAT_LINE	RW	Software Reset For DAT Line field from Software Reset Register
25	SW_RST_CMD_LINE	RW	Software Reset For CMD Line field from Software Reset Register

SDC1_SDCC_HC_REG_2C_2E (cont.)

Bits	Name	Type	Description
24	SW_RST_FOR_ALL	RW	Software Reset For All field from Software Reset Register
19:16	DATA_TIMEOUT_COUNTER	RW	Data Timeout Counter Value field from Timeout Control Register. The reserved value 0xF is translated to maximum number of cycles - $2^{32}-1$
15:8	CLK_CTL_SDCLK_FREQ_SEL	RW	SDCLK Frequency Select field from Clock Control Register
7:6	CLK_CTL_SDCLK_FREQ_SEL_MSB	RW	Upper Bits of SDCLK Frequency Select field from Clock Control Register
5	CLK_CTL_GEN_SEL	RW	Clock Generator Select field from Clock Control Register
2	CLK_CTL_SDCLK_EN	RW	SD Clock Enable field from Clock Control Register
1	CLK_CTL_INTERNAL_CLK_STABLE	R	Internal Clock Stable field from Clock Control Register
0	CLK_CTL_INTERNAL_CLK_EN	RW	Internal Clock Enable field from Clock Control Register

0x07824930 SDC1_SDCC_HC_REG_30_32**Type:** RW**Clock:** same rate as HCLK**Reset State:** 0x00000000

Combination of two registers from the SD Host Controller Standard Specification:

-Normal Interrupt Status Register(15-0)

-Error Interrupt Status Register(31-16)

SDC1_SDCC_HC_REG_30_32

Bits	Name	Type	Description
31:28	ERR_INT_STS_VENDOR_SPECIFIC	RW	Vendor Specific Error Status field from Error Interrupt Status Register
26	ERR_INT_STS_TUNING_ERROR	RW	Tuning Error field from Error Interrupt Status Register
25	ERR_INT_STS_ADMA_ERR	RW	ADMA Error field from Error Interrupt Status Register
24	ERR_INT_STS_AUTO_CMD_ERR	RW	Auto CMD Error field from Error Interrupt Status Register
23	ERR_INT_STS_CURRENT_LIMIT_ERR	RW	Current Limit Error field from Error Interrupt Status Register
22	ERR_INT_STS_DATA_END_BIT_ERR	RW	Data End Bit Error field from Error Interrupt Status Register

SDC1_SDCC_HC_REG_30_32 (cont.)

Bits	Name	Type	Description
21	ERR_INT_STS_DATA_CRC_ERR	RW	Data CRC Error field from Error Interrupt Status Register
20	ERR_INT_STS_DATA_TIMEOUT_ERR	RW	Data Timeout Error field from Error Interrupt Status Register
19	ERR_INT_STS_CMD_IDX_ERR	RW	Command Index Error field from Error Interrupt Status Register
18	ERR_INT_STS_CMD_END_BIT_ERR	RW	Command End Bit Error field from Error Interrupt Status Register
17	ERR_INT_STS_CMD_CRC_ERR	RW	Command CRC Error field from Error Interrupt Status Register
16	ERR_INT_STS_CMD_TIMEOUT_ERR	RW	Command Timeout Error field from Error Interrupt Status Register
15	NORMAL_INT_STS_ERR_INT	R	Error Interrupt field from Normal Interrupt Status Register
12	NORMAL_INT_STS_RETUNING_EVENT	R	Re-Tuning Event field from Normal Interrupt Status Register
11	NORMAL_INT_STS_INT_C	R	INT_C field from Normal Interrupt Status Register
10	NORMAL_INT_STS_INT_B	R	INT_B field from Normal Interrupt Status Register
9	NORMAL_INT_STS_INT_A	R	INT_A field from Normal Interrupt Status Register
8	NORMAL_INT_STS_CARD_INT	R	Card Interrupt field from Normal Interrupt Status Register
7	NORMAL_INT_STS_CARD_REMOVAL	RW	Card Removal field from Normal Interrupt Status Register
6	NORMAL_INT_STS_CARD_INSERTION	RW	Card Insertion field from Normal Interrupt Status Register
5	NORMAL_INT_STS_BUF_RD_READY	RW	Buffer Read Ready field from Normal Interrupt Status Register
4	NORMAL_INT_STS_BUF_WR_READY	RW	Buffer Write Ready field from Normal Interrupt Status Register
3	NORMAL_INT_STS_DMA_INT	RW	DMA Interrupt field from Normal Interrupt Status Register
2	NORMAL_INT_STS_BLK_GAP_EVENT	RW	Block Gap Event field from Normal Interrupt Status Register
1	NORMAL_INT_STS_TRANS_COMPLETE	RW	Transfer Complete field from Normal Interrupt Status Register
0	NORMAL_INT_STS_CMD_COMPLETE	RW	Command Complete field from Normal Interrupt Status Register

0x07824934 SDC1_SDCC_HC_REG_34_36**Type:** RW**Clock:** same rate as HCLK**Reset State:** 0x00000000

Combination of two registers from the SD Host Controller Standard Specification:

-Normal Interrupt Status Enable Register(15-0)

-Error Interrupt Status Enable Register(31-16)

SDC1_SDCC_HC_REG_34_36

Bits	Name	Description
31:28	ERR_INT_STS_EN_VENDOR_SPECIFIC_ERR	Vendor Specific Error Status Enable field from Error Interrupt Status Enable Register
26	ERR_INT_STS_EN_TUNING_ERR	Tuning Error Status Enable field from Error Interrupt Status Enable Register
25	ERR_INT_STS_EN_ADMA_ERR	ADMA Error Status Enable field from Error Interrupt Status Enable Register
24	ERR_INT_STS_EN_AUTO_CMD_ERR	Auto CMD Error Status Enable field from Error Interrupt Status Enable Register
23	ERR_INT_STS_EN_CURRENT_LIMIT_ERR	Current Limit Error Status Enable field from Error Interrupt Status Enable Register
22	ERR_INT_STS_EN_DATA_END_BIT_ERR	Data End Bit Error Status Enable field from Error Interrupt Status Enable Register
21	ERR_INT_STS_EN_DATA_CRC_ERR	Data CRC Error Status Enable field from Error Interrupt Status Enable Register
20	ERR_INT_STS_EN_DATA_TIMEOUT	Data Timeout Error Status Enable field from Error Interrupt Status Enable Register
19	ERR_INT_STS_EN_CMD_INDEX_ERR	Command Index Error Status Enable field from Error Interrupt Status Enable Register
18	ERR_INT_STS_EN_CMD_END_BIT_ERR	Command End Bit Error Status Enable field from Error Interrupt Status Enable Register
17	ERR_INT_STS_EN_CMD_CRC_ERR	Command CRC Error Status Enable field from Error Interrupt Status Enable Register
16	ERR_INT_STS_EN_CMD_TIMEOUT	Command Timeout Error Status Enable field from Error Interrupt Status Enable Register
12	NORMAL_INT_STS_EN_RE_TUNING_EVENT	Re-Tuning Event Status Enable field from Normal Interrupt Status Enable Register
11	NORMAL_INT_STS_EN_INT_C	INT_C Status Enable field from Normal Interrupt Status Enable Register
10	NORMAL_INT_STS_EN_INT_B	INT_B Status Enable field from Normal Interrupt Status Enable Register

SDC1_SDCC_HC_REG_34_36 (cont.)

Bits	Name	Description
9	NORMAL_INT_STS_EN_INT_A	INT_A Status Enable field from Normal Interrupt Status Enable Register
8	NORMAL_INT_STS_EN_CARD_INT	Card Interrupt Status Enable field from Normal Interrupt Status Enable Register
7	NORMAL_INT_STS_EN_CARD_REMOVAL	Card Removal Status Enable field from Normal Interrupt Status Enable Register
6	NORMAL_INT_STS_EN_CARD_INSERTION	Card Insertion Status Enable field from Normal Interrupt Status Enable Register
5	NORMAL_INT_STS_EN_BUFFER_RD_READY	Buffer Read Ready Status Enable field from Normal Interrupt Status Enable Register
4	NORMAL_INT_STS_EN_BUFFER_WR_READY	Buffer Write Ready Status Enable field from Normal Interrupt Status Enable Register
3	NORMAL_INT_STS_EN_DMA_A_INT	DMA Interrupt Status Enable field from Normal Interrupt Status Enable Register
2	NORMAL_INT_STS_EN_BLOCK_GAP_EVENT	Block Gap Event Status Enable field from Normal Interrupt Status Enable Register
1	NORMAL_INT_STS_EN_TRANSFER_COMPLETE	Transfer Complete Status Enable field from Normal Interrupt Status Enable Register
0	NORMAL_INT_STS_EN_COMMAND_COMPLETE	Command Complete Status Enable field from Normal Interrupt Status Enable Register

0x07824938 SDC1_SDCC_HC_REG_38_3A**Type:** RW**Clock:** same rate as HCLK**Reset State:** 0x00000000

Combination of two registers from the SD Host Controller Standard Specification:

-Normal Interrupt Signal Enable Register(15-0)

-Error Interrupt Signal Enable Register(31-16)

SDC1_SDCC_HC_REG_38_3A

Bits	Name	Type	Description
31:28	ERR_INT_SIGNAL_EN_VENDOR_SPECIFIC_ERR	RW	Vendor Specific Error Signal Enable field from Error Interrupt Signal Enable Register
26	ERR_INT_SIGNAL_EN_TUNING_ERR	RW	Tuning Error Signal Enable field from Error Interrupt Signal Enable Register
25	ERR_INT_SIGNAL_EN_ADMA_ERR	RW	ADMA Error Signal Enable field from Error Interrupt Signal Enable Register

SDC1_SDCC_HC_REG_38_3A (cont.)

Bits	Name	Type	Description
24	ERR_INT_SIGNAL_EN_AU TO_CMD_ERR	RW	Auto CMD Error Signal Enable field from Error Interrupt Signal Enable Register
23	ERR_INT_SIGNAL_EN_CU RRENT_LIMIT_ERR	RW	Current Limit Error Signal Enable field from Error Interrupt Signal Enable Register
22	ERR_INT_SIGNAL_EN_DAT A_END_BIT_ERR	RW	Data End Bit Error Signal Enable field from Error Interrupt Signal Enable Register
21	ERR_INT_SIGNAL_EN_DAT A_CRC_ERR	RW	Data CRC Error Signal Enable field from Error Interrupt Signal Enable Register
20	ERR_INT_SIGNAL_EN_DAT A_TIMEOUT	RW	Data Timeout Error Signal Enable field from Error Interrupt Signal Enable Register
19	ERR_INT_SIGNAL_EN_CM D_INDX_ERR	RW	Command Index Error Signal Enable field from Error Interrupt Signal Enable Register
18	ERR_INT_SIGNAL_EN_CM D_END_BIT_ERR	RW	Command End Bit Error Signal Enable field from Error Interrupt Signal Enable Register
17	ERR_INT_SIGNAL_EN_CM D_CRC_ERR	RW	Command CRC Error Signal Enable field from Error Interrupt Signal Enable Register
16	ERR_INT_SIGNAL_EN_CM D_TIMEOUT	RW	Command Timeout Error Signal Enable field from Error Interrupt Signal Enable Register
15	NORMAL_INT_SIGNAL_EN _ERR_INT	R	Error Interrupt Signal Enable field from Normal Interrupt Signal Enable Register
12	NORMAL_INT_SIGNAL_EN _RETUNING_EVENT	RW	Re-Tuning Event Signal Enable field from Normal Interrupt Signal Enable Register
11	NORMAL_INT_SIGNAL_EN _INT_C	RW	INT_C Signal Enable field from Normal Interrupt Signal Enable Register
10	NORMAL_INT_SIGNAL_EN _INT_B	RW	INT_B Signal Enable field from Normal Interrupt Signal Enable Register
9	NORMAL_INT_SIGNAL_EN _INT_A	RW	INT_A Signal Enable field from Normal Interrupt Signal Enable Register
8	NORMAL_INT_SIGNAL_EN _CARD_INT	RW	Card Interrupt Signal Enable field from Normal Interrupt Signal Enable Register
7	NORMAL_INT_SIGNAL_EN _CARD_REMOVAL	RW	Card Removal Signal Enable field from Normal Interrupt Signal Enable Register
6	NORMAL_INT_SIGNAL_EN _CARD_INSERTION	RW	Card Insertion Signal Enable field from Normal Interrupt Signal Enable Register
5	NORMAL_INT_SIGNAL_EN _BUF_RD_READY	RW	Buffer Read Ready Signal Enable field from Normal Interrupt Signal Enable Register
4	NORMAL_INT_SIGNAL_EN _BUF_WR_READY	RW	Buffer Write Ready Signal Enable field from Normal Interrupt Signal Enable Register
3	NORMAL_INT_SIGNAL_EN _DMA_INT	RW	DMA Interrupt Signal Enable field from Normal Interrupt Signal Enable Register

SDC1_SDCC_HC_REG_38_3A (cont.)

Bits	Name	Type	Description
2	NORMAL_INT_SIGNAL_EN_BLK_GAP_EVENT	RW	Block Gap Event Signal Enable field from Normal Interrupt Signal Enable Register
1	NORMAL_INT_SIGNAL_EN_TRANS_COMPLETE	RW	Transfer Complete Signal Enable field from Normal Interrupt Signal Enable Register
0	NORMAL_INT_SIGNAL_EN_CMD_COMPLETE	RW	Command Complete Signal Enable field from Normal Interrupt Signal Enable Register

0x0782493C SDC1_SDCC_HC_REG_3C_3E**Type:** RW**Clock:** same rate as HCLK**Reset State:** 0x00000000

Combination of two registers from the SD Host Controller Standard Specification:

-Auto CMD Error Status Register(15-0)

-Host Control 2 Register(31-16)

SDC1_SDCC_HC_REG_3C_3E

Bits	Name	Type	Description
31	HST_CTL2_PRESET_VALUE_EN	RW	Preset Value Enable field from Host Control2 Register
30	HST_CTL2_ASYNC_INTERRUPT_ENABLE	RW	Asynchronous Interrupt Enable field from Host Control2 Register
23	HST_CTL2_SAMPLING_CLOCK_SELECT	RW	Sampling Clock Select field from Host Control2 Register
22	HST_CTL2_EXECUTE_TUNING	RW	Execute Tuning field from Host Control2 Register
21:20	HST_CTL2_DRIVER_STRENGTH_SELECT	RW	Driver Strength Select field from Host Control2 Register
19	HST_CTL2_SIGNALING_1_8V_ENABLE	RW	1.8V Signaling Enable field from Host Control2 Register
18:16	HST_CTL2_UHS_MODE_SELECT	RW	UHS Mode Select field from Host Control2 Register
7	AUTO_CMD_NOT_ISSUED_BY_AUTO_CMD12	R	Command Not issued By Auto CMD12 Error field from Auto CMD Error Status Register
4	AUTO_CMD_INDEX_ERROR	R	Auto CMD Index Error field from Auto CMD Error Status Register
3	AUTO_CMD_END_BIT_ERROR	R	Auto CMD End Bit Error field from Auto CMD Error Status Register

SDC1_SDCC_HC_REG_3C_3E (cont.)

Bits	Name	Type	Description
2	AUTO_CMD_CRC_ERR	R	Auto CMD CRC Error field from Auto CMD Error Status Register
1	AUTO_CMD_TIMEOUT	R	Auto CMD Timeout Error field from Auto CMD Error Status Register
0	AUTO_CMD12_NOT_EXEC	R	Auto CMD12 Not Executed field from Auto CMD Error Status Register

0x07824940 SDC1_SDCC_HC_REG_40_42**Type:** R**Clock:** same rate as HCLK**Reset State:** 0x3029C8B2

The 32 LSBs of the Capabilities Register in SD Host Controller Standard Specification

SDC1_SDCC_HC_REG_40_42

Bits	Name	Description
31:30	CAPABILITIES_SLOT_TYPE	Slot Type field from Capabilities Register
29	CAPABILITIES_ASYNC_INT_SUPPORT	Asynchronous Interrupt Support field from Capabilities Register
28	CAPABILITIES_SYS_BUS_SUPPORT_64_BIT	64 bit System Bus Support field from Capabilities Register
26	CAPABILITIES_VOLTAGE_SUPPORT_1_8V	Voltage Support 1.8v field from Capabilities Register
25	CAPABILITIES_VOLTAGE_SUPPORT_3_0V	Voltage Support 3.0v field from Capabilities Register
24	CAPABILITIES_VOLTAGE_SUPPORT_3_3V	Voltage Support 3.3v field from Capabilities Register
23	CAPABILITIES_SUSPEND_RESUME_SUPPORT	Suspend Resume Support field from Capabilities Register
22	CAPABILITIES_SDMA_SUPPORT	SDMA Support field from Capabilities Register
21	CAPABILITIES_HS_SUPPORT	High Speed Support field from Capabilities Register
19	CAPABILITIES_ADMA2_SUPPORT	ADMA2 Support field from Capabilities Register
18	CAPABILITIES_SUPPORT_8_BIT	8 bit Supported For Embedded Product field from Capabilities Register
17:16	CAPABILITIES_MAX_BLK_LENGTH	Max Block Length field from Capabilities Register

SDC1_SDCC_HC_REG_40_42 (cont.)

Bits	Name	Description
15:8	CAPABILITIES_BASE_SDCLK_FREQ	Base Clock Frequency For SD Clock field from Capabilities Register
7	CAPABILITIES_TIMEOUT_CLK_UNIT	Timeout Clock Unit field from Capabilities Register
5:0	CAPABILITIES_TIMEOUT_CLK_FREQ	Timeout Clock Frequency field from Capabilities Register

0x07824944 SDC1_SDCC_HC_REG_44_46**Type:** R**Clock:** same rate as HCLK**Reset State:** 0x00008007

The 32 MSBs of the Capabilities Register in SD Host Controller Standard Specification

SDC1_SDCC_HC_REG_44_46

Bits	Name	Description
23:16	CAPABILITIES_CLK_MULTIPLIER	Clock Multiplier field from Capabilities Register
15:14	CAPABILITIES_RETUNING_MODE	Re-Tuning Modes field from Capabilities Register
13	CAPABILITIES_USE_TUNING_FOR_SDR50	Use Tuning for SDR50 field from Capabilities Register
11:8	CAPABILITIES_TIMER_CNT_FOR_RETUNING	Timer Count for Re-Tuning field from Capabilities Register
6	CAPABILITIES_DRIVER_TYPE_D_SUPPORT	Driver Type D Support field from Capabilities Register
5	CAPABILITIES_DRIVER_TYPE_C_SUPPORT	Driver Type C Support field from Capabilities Register
4	CAPABILITIES_DRIVER_TYPE_A_SUPPORT	Driver Type A Support field from Capabilities Register
2	CAPABILITIES_DDR_50_SUPPORT	DDR50 Support field from Capabilities Register
1	CAPABILITIES_SDR_104_SUPPORT	SDR104 Support field from Capabilities Register
0	CAPABILITIES_SDR_50_SUPPORT	SDR50 Support field from Capabilities Register

0x07824948 SDC1_SDCC_HC_REG_48_4A**Type:** R**Clock:** same rate as HCLK**Reset State:** 0x00000000

Maximum Current Register in SD Host Controller Standard Specification.

SDC1_SDCC_HC_REG_48_4A

Bits	Name	Description
23:16	MAX_CURRENT_1_8V	Maximum Current for 1.8v field from Maximum Current Capabilities Register
15:08	MAX_CURRENT_3_0V	Maximum Current for 3.0v field from Maximum Current Capabilities Register
07:00	MAX_CURRENT_3_3V	Maximum Current for 3.3v field from Maximum Current Capabilities Register

0x07824950 SDC1_SDCC_HC_REG_50_52**Type:** W**Clock:** same rate as HCLK**Reset State:** 0x00000000

Combination of two registers from the SD Host Controller Standard Specification:

-Force Event Register for Auto CMD Error Status(15-0)

-Force Event Register for Error Interrupt Status(31-16)

SDC1_SDCC_HC_REG_50_52

Bits	Name	Description
31:28	FORCE_EVENT_FOR_ERR_VENDOR_SPECIFIC_ERR_STS	Force Event for Vendor Specific Error Status field from Force Event Register for Error Interrupt Status
25	FORCE_EVENT_FOR_ERR_ADMA	Force Event for ADMA Error field from Force Event Register for Error Interrupt Status
24	FORCE_EVENT_FOR_ERR_AUTO_CMD	Force Event for Auto CMD Error field from Force Event Register for Error Interrupt Status
23	FORCE_EVENT_FOR_ERR_CURRENT_LIMIT	Force Event for Current Limit Error field from Force Event Register for Error Interrupt Status
22	FORCE_EVENT_FOR_ERR_DATA_END_BIT	Force Event for Data End Bit Error field from Force Event Register for Error Interrupt Status
21	FORCE_EVENT_FOR_ERR_DATA_CRC	Force Event for Data CRC Error field from Force Event Register for Error Interrupt Status

SDC1_SDCC_HC_REG_50_52 (cont.)

Bits	Name	Description
20	FORCE_EVENT_FOR_ERR_DATA_TIMEOUT	Force Event for Data Timeout Error field from Force Event Register for Error Interrupt Status
19	FORCE_EVENT_FOR_ERR_CMD_INDX	Force Event for Command Index Error field from Force Event Register for Error Interrupt Status
18	FORCE_EVENT_FOR_ERR_CMD_END_BIT	Force Event for Command End Bit Error field from Force Event Register for Error Interrupt Status
17	FORCE_EVENT_FOR_ERR_CMD_CRC	Force Event for Command CRC Error field from Force Event Register for Error Interrupt Status
16	FORCE_EVENT_FOR_ERR_CMD_TIMEOUT	Force Event for Command Timeout Error field from Force Event Register for Error Interrupt Status
7	FORCE_EVENT_FOR_CMD_NOT_ISSUED_BY_AUTO_CMD12	Force Event for Command Not Issued By Auto CMD12 Error field from Force Event Register for Auto CMD Error Status
4	FORCE_EVENT_FOR_AUTO_CMD_INDX_ERR	Force Event for Auto CMD Index Error field from Force Event Register for Auto CMD Error Status
3	FORCE_EVENT_FOR_AUTO_CMD_END_BIT_ERR	Force Event for Auto CMD End Bit Error field from Force Event Register for Auto CMD Error Status
2	FORCE_EVENT_FOR_AUTO_CMD_CRC_ERR	Force Event for Auto CMD CRC Error field from Force Event Register for Auto CMD Error Status
1	FORCE_EVENT_FOR_AUTO_CMD_TIMEOUT_ERR	Force Event for Auto CMD Timeout Error field from Force Event Register for Auto CMD Error Status
0	FORCE_EVENT_FOR_AUTO_CMD12_NOT_EXEC	Force Event for Auto CMD12 Not Executed field from Force Event Register for Auto CMD Error Status

0x07824954 SDC1_SDCC_HC_REG_54_56**Type:** R**Clock:** same rate as HCLK**Reset State:** 0x00000000

The 7 LSBs are the ADMA Error Status Register in SD Host Controller Standard Specification.
The Bits 31-8 are unused.

SDC1_SDCC_HC_REG_54_56

Bits	Name	Description
2	ADMA_LENGTH_MISMATCH_ERR	ADMA Length Mismatch Error field from ADMA Error Status Register
1:0	ADMA_ERR_STATE	ADMA Error State field from ADMA Error Status Register

0x07824958 SDC1_SDCC_HC_REG_58_5A**Type:** RW**Clock:** same rate as HCLK**Reset State:** 0x00000000

The 32 LSBs of the ADMA System Address Register in SD Host Controller Standard Specification

SDC1_SDCC_HC_REG_58_5A

Bits	Name	Description
31:00	ADMA_SYS_ADDRESS	ADMA System Address field from ADMA System Address Register

0x0782495C SDC1_SDCC_HC_REG_5C_5E**Type:** RW**Clock:** same rate as HCLK**Reset State:** 0x00000000

The 32 MSBs of the ADMA System Address Register in SD Host Controller Standard Specification

SDC1_SDCC_HC_REG_5C_5E

Bits	Name	Description
31:00	ADMA_SYS_ADDRESS_64	upper 32 bits of ADMA System Address field from ADMA System Address Register

0x07824960 SDC1_SDCC_HC_REG_60_62**Type:** R**Clock:** same rate as HCLK**Reset State:** 0x00040000

Combination of two registers from the SD Host Controller Standard Specification:

-Preset Value for Initialization Register(15-0)

-Preset Value for Default Speed Register(31-16)

SDC1_SDCC_HC_REG_60_62

Bits	Name	Description
31:30	DEFAULT_SPEED_DRIVER_STRENGTH_SEL	Driver Strength Select Value field from Preset Value for Default Speed

SDC1_SDCC_HC_REG_60_62 (cont.)

Bits	Name	Description
26	DEFAULT_SPEED_CLK_GEN_SEL	Clock Generator Select Value field from Preset Value for Default Speed
25:16	DEFAULT_SPEED_SDCLK_FREQ_SEL	SDCLK Frequency Select Value field from Preset Value for Default Speed
15:14	INIT_DRIVER_STRENGTH_SEL	Driver Strength Select Value field from Preset Value for Initialization
10	INIT_CLK_GEN_SEL	Clock Generator Select Value field from Preset Value for Initialization
9:0	INIT_SDCLK_FREQ_SEL	SDCLK Frequency Select Value field from Preset Value for Initialization

0x07824964 SDC1_SDCC_HC_REG_64_66**Type:** R**Clock:** same rate as HCLK**Reset State:** 0x00040002

Combination of two registers from the SD Host Controller Standard Specification:

-Preset Value for High Speed Register(15-0)

-Preset Value for SDR12 Register

SDC1_SDCC_HC_REG_64_66

Bits	Name	Description
31:30	SDR12_DRIVER_STRENGTH_SEL	Driver Strength Select Value field from Preset Value for SDR12
26	SDR12_CLK_GEN_SEL	Clock Generator Select Value field from Preset Value for SDR12
25:16	SDR12_SDCLK_FREQ_SEL	SDCLK Frequency Select Value field from Preset Value for SDR12
15:14	HS_DRIVER_STRENGTH_SEL	Driver Strength Select Value field from Preset Value for High Speed
10	HS_CLK_GEN_SEL	Clock Generator Select Value field from Preset Value for High Speed
9:0	HS_SDCLK_FREQ_SEL	SDCLK Frequency Select Value field from Preset Value for High Speed

0x07824968 SDC1_SDCC_HC_REG_68_6A**Type:** R**Clock:** same rate as HCLK**Reset State:** 0x00010002

Combination of two registers from the SD Host Controller Standard Specification:

-Preset Value for SDR25 Register(15-0)

-Preset Value for SDR50 Register

SDC1_SDCC_HC_REG_68_6A

Bits	Name	Description
31:30	SDR50_DRIVER_STRENGTH_SEL	Driver Strength Select Value field from Preset Value for SDR50
26	SDR50_CLK_GEN_SEL	Clock Generator Select Value field from Preset Value for SDR50
25:16	SDR50_SDCLK_FREQ_SEL	SDCLK Frequency Select Value field from Preset Value for SDR50
15:14	SDR25_DRIVER_STRENGTH_SEL	Driver Strength Select Value field from Preset Value for SDR25
10	SDR25_CLK_GEN_SEL	Clock Generator Select Value field from Preset Value for SDR25
9:0	SDR25_SDCLK_FREQ_SEL	SDCLK Frequency Select Value field from Preset Value for SDR25

0x0782496C SDC1_SDCC_HC_REG_6C_6E**Type:** R**Clock:** same rate as HCLK**Reset State:** 0x00020000

Combination of two registers from the SD Host Controller Standard Specification:

-Preset Value for SDR104 Register(15-0)

-Preset Value for DDR50 Register

SDC1_SDCC_HC_REG_6C_6E

Bits	Name	Description
31:30	DDR50_DRIVER_STRENGTH_SEL	Driver Strength Select Value field from Preset Value for DDR50
26	DDR50_CLK_GEN_SEL	Clock Generator Select Value field from Preset Value for DDR50
25:16	DDR50_SDCLK_FREQ_SEL	SDCLK Frequency Select Value field from Preset Value for DDR50
15:14	SDR104_DRIVER_STRENGTH_SEL	Driver Strength Select Value field from Preset Value for SDR104

SDC1_SDCC_HC_REG_6C_6E (cont.)

Bits	Name	Description
10	SDR104_CLK_GEN_SEL	Clock Generator Select Value field from Preset Value for SDR104
9:0	SDR104_SDCLK_FREQ_SEL	SDCLK Frequency Select Value field from Preset Value for SDR104

0x078249E0 SDC1_SDCC_HC_REG_E0_E2**Type:** RW**Clock:** same rate as HCLK**Reset State:** 0x00000000

Shared Bus Control Register in SD Host Controller Standard Specification.

SDC1_SDCC_HC_REG_E0_E2

Bits	Name	Type	Description
30:24	SHARED_BUS_CTL_BACK_END_PWR	RW	Back End Power Control field from Shared Bus Control Register
22:20	SHARED_BUS_INT_PIN_SEL	RW	
18:16	SHARED_BUS_CTL_CLK_PIN_SEL	RW	Clock Pin Select field from Shared Bus Control Register
14:8	SHARED_BUS_CTL_BUS_WIDTH_PRESET	R	Bus Width Preset field from Shared Bus Control Register
5:4	SHARED_BUS_CTL_NUM_INTERRUPT_INPUT_PINS	R	Number Of Interrupt Input Pins field from Shared Bus Control Register
2:0	SHARED_BUS_CTL_NUM_CLK_PINS	R	Number of Clock Pins field from Shared Bus Control Register

0x078249FC SDC1_SDCC_HC_REG_FC_FE**Type:** RW**Clock:** same rate as HCLK**Reset State:** 0x00020000

Combination of two registers from the SD Host Controller Standard Specification:

-Slot Interrupt Status Register(15-0)

-Host Controller Version Register

SDC1_SDCC_HC_REG_FC_FE

Bits	Name	Type	Description
31:24	HC_VENDOR_VERSION_NUM	R	Vendor Version Number field from Host Controller Version Register
23:16	HC_SPEC_VERSION_NUM	R	Specification Version Number field from Host Controller Version Register
7:0	INT_SIGNAL_FOR_EACH_SLOT	RW	Interrupt Signal For Each Slot field from Slot Interrupt Status Register

0x07824A00 SDC1_SDCC_HC_REG_DLL_CONFIG**Type:** RW**Clock:** same rate as HCLK**Reset State:** 0x60006400

This register configures the DLL's inputs.

SDC1_SDCC_HC_REG_DLL_CONFIG

Bits	Name	Description
31	SDC4_DIS_DOUT	When this bit equals to '0', the data which comes out from the dll is valid. When it equals to '1' this data isn't valid. The inversion of this bit is connected to dll's input - dll_en_dout
30	DLL_RST	Setting this bit to 1 resets cm_dll_sdc4. cm_dll_sdc4 should be reset every time the MCLK frequency is changed.
29	PDN	Power Down Value 0 - analog blocks are enabled Value 1 - analog blocks are powered down (default)
28	CK_INTP_SEL	Selects interpolator output
27	CK_INTP_EN	Enable clock interpolation for finer resolution
26:24	MCLK_FREQ	Frequency of MCLK Value 000 -100 - 112 (MHz) Value 001 -112 - 125 Value 010 -125 - 137 Value 011 -137 - 150 Value 100 -150 - 162 Value 101 -162 - 175 Value 110 -175 - 187 Value 111 -187 - 200

SDC1_SDCC_HC_REG_DLL_CONFIG (cont.)

Bits	Name	Description
23:20	CDR_SELECT	CDR phase select (gray coded) Value 0000 - phase 0 Value 0001 - phase 1 Value 0011 - phase 2 Value 0010 - phase 3 Value 0110 - phase 4 Value 0111 - phase 5 Value 0101 - phase 6 Value 0100 - phase 7 Value 1100 - phase 8 Value 1101 - phase 9 Value 1111 - phase 10 Value 1110 - phase 11 Value 1010 - phase 12 Value 1011 - phase 13 Value 1001 - phase 14 Value 1000 - phase 15
19	CDR_EXT_EN	Enable external control of cdr phase select
18	CK_OUT_EN	Enable output clock (default value is '1')
17	CDR_EN	Enable CDR function
16	DLL_EN	Enable DLL function
15:14	CDR_UPD_RATE	CDR update rate, low pass filtering window of CDR Max update rate: Value 00 - 0.5 MCLK frequency Value 01 - 0.25 MCLK Frequency (default) Value 10 - 0.125 MCLK frequency Value 11 - 1/16 MCLK Frequency
13:12	DLL_UPD_RATE	DLL update rate Value 00 - sdc4_mclk/10 Value 01 - sdc4_mclk/20 Value 10 - sdc4_mclk/40 (default) Value 11 - sdc4_mclk/80
11:10	DLL_PHASE_DET	DLL phase detector low pass average window Value 00 - 2 cycles Value 01 - 4 cycles (default) Value 10 - 8 cycles Value 11 - 16 cycles
9	CDR_ALGORITHM_SEL	CDR algorithm select Value 0 - Edge (default) Value 1 - Level

SDC1_SDCC_HC_REG_DLL_CONFIG (cont.)

Bits	Name	Description
0	CMD_DAT_TRACK_SEL	Select DAT or CMD lines for CDR tracking Value 0 - track DAT inputs (default) Value 1 - track CMD input

0x07824A08 SDC1_SDCC_HC_REG_DLL_STATUS**Type:** R**Clock:** same rate as HCLK**Reset State:** 0x00000000

This register is connected to cm_dll_sdc4's status output.

SDC1_SDCC_HC_REG_DLL_STATUS

Bits	Name	Description
12	SDC4_DTEST_MUXSEL	Digital test mux select
11	DDR_DLL_LOCK_JDR	Reserved
10:9	SDC4_DLL_DTEST_OUT_A TPG	SDC4_DLL_DTEST_OUT output from DLL after a clock mux with sampled JDRs for ATPG mode. This bit helps collecting coverage on DLL's output
8	SDC4_DLL_LOCK_ATPG	SDC4_DLL_LOCK output from DLL after a clock mux with sampled JDRs for ATPG mode. This bit helps collecting coverage on DLL's output
7	DLL_LOCK	DLL lock status Value 0 - Not locked Value 1 - Locked

SDC1_SDCC_HC_REG_DLL_STATUS (cont.)

Bits	Name	Description
6:3	CDR_PHASE	CDR phase select (gray coded) Value 0000 - phase 0 Value 0001 - phase 1 Value 0011 - phase 2 Value 0010 - phase 3 Value 0110 - phase 4 Value 0111 - phase 5 Value 0101 - phase 6 Value 0100 - phase 7 Value 1100 - phase 8 Value 1101 - phase 9 Value 1111 - phase 10 Value 1110 - phase 11 Value 1010 - phase 12 Value 1011 - phase 13 Value 1001 - phase 14 Value 1000 - phase 15
2	DDLL_COARSE_CAL	Value 1 - done Value 0 - not done

0x07824A0C SDC1_SDCC_HC_VENDOR_SPECIFIC_FUNC**Type:** RW**Clock:** same rate as HCLK**Reset State:** 0x00000A1C**SDC1_SDCC_HC_VENDOR_SPECIFIC_FUNC**

Bits	Name	Description
31:22	HC_AUTO_CMD_COUNTER	This counter will determine the time between two auto commands transmissions
21:19	HC_SELECT_IN	Selecting the SD controller's mode. Equivalent to SELECT_IN filed in MCI_CLK
18	HC_SELECT_IN_EN	Enabling the selection of mode via the vendor specific register
17	ADMA_DESC_PRIORITY	If set (1) then the ADMA's descriptors SM gets priority on the data accesses. Not Applicabale for Gandalf.
16	HC_IO_PAD_PWR_SWITCH	Indication to Pad that power has to be switched from 3.3V to 1.8V.
15	HC_IO_PAD_PWR_SWITCH_EN	This Bit enables the use of HC_IO_PAD_PWR_SWITCH
14	HC_IGNORE_CTOUT	If set (1), CPSM disables command timer and waits for interrupt request (Card Response). See note #4 below.

SDC1_SDCC_HC_VENDOR_SPECIFIC_FUNC (cont.)

Bits	Name	Description
13	HC_SW_SDC4_CMD19	The input of cm_dll_sdc4 - sdc4_cmd19 can be driven by SW if AUTO_CMD19 (bit 16 in MCI_CMD) feature is not used. Value 1 - sdc4_cmd19 is driven by SW Value 0 (default) - sdc4_cmd19 is driven by HW
12	HC_INVERT_OUT	value: 0 data and cmd are driven by falling edge value 1: data and cmd are driven by rising edge.
11	HC_INVERT_OUT_EN	value 1: HC_INVERT_OUT is driven by bit 12 of HC_VENDOR_SPECIFIC_FUNC (HC_INVERT_OUT) value 0: HC_INVERT_OUT is driven by HST_CTL1_HS_EN .
10	MODE	Data transfer mode: 0: block data transfer 1: stream data transfer. Can be used in bus testing procedure
9:8	SDC4_MCLK_SEL	Selects the sdc4_mclk - input clock of cm_dll_sdc4 Value 1 - feedback clock from CLK pad Value 2 (default) - free running MCLK (gated between the transactions) Value 3 - free running MCLK divided by 2
7	ADMA_ERR_SIZE_EN	enabling interrupt for mismatch error in ADMA2
6	HC_AUTO_CMD21_EN	Enabling bit for the auto command 21 option. value 0: disable value 1:enable
5	HC_AUTO_CMD19_EN	Enabling bit for the auto command 19 option. value 0: disable value 1:enable
4	HC_HCLKON_SW_EN	SW enable of AHB clock request of SDCC5. value 0: HW clock request mechanism is used value 1: clock request signal is always high
3	ADMA_BUSREQ_EN	enabling the fix in ADMA2 for busrequest
2	FLOW_EN	Enable flow control: value 0: disable value 1: enable (default)
1	PWRSAVE	Disable Prime-Cell MCI clock output when bus is idle to save power. value 0: Power save disabled - always enabled if bit 8 is set. value 1: Power save enabled - clock enabled only when bus is active and bit 8 is set.
0	CLK_INV	When set(1), the input clock which is inserted into the input macros is inverted.

0x07824A10 SDC1_SDCC_HC_VENDOR_SPECIFIC_FUNC2**Type:** RW**Clock:** same rate as HCLK**Reset State:** 0xF88218A8**SDC1_SDCC_HC_VENDOR_SPECIFIC_FUNC2**

Bits	Name	Type	Description
31	QSB_AXI_ABURST	RW	Driving QSB AXI output
30:28	NUM_OUTSTANDING_DATA	RW	Configuring the number of outstanding transactions for the data. The descriptors accesses have more one outstanding request
27	PROCEED_AXI_AFTER_ERROR	RW	When error occurs on AXI master the traffic can continue or terminate. If this bit is set then the traffic continues and more requests are transmitted on the address channel
26	QSB_AXI_AFULL_CALC	RW	If this bit is set, then afull isn't '0' constantly but can be equal to '1' if all the beats have 8 bytes on the strobe lines
25	ONE_MID_SUPPORT	RW	If this bit is set, then only 1 MID is used for descriptors and data
24:22	RESERVED	RW	Reserved
21	HC_SW_RST_REQ	RW	Before executing HC_SW_RST_FOR_ALL or HC_SW_RST_FOR_DAT this bit can be set and SW needs to polls on it until it's being cleared. The bit is cleared after AXI master (QMB) has finished processing the outstanding transactions and it's in IDLE state. This bit can be useful for HPI feature.
20	HC_SW_RST_WAIT_IDLE_DIS	RW	When this bit is cleared, the HC_SW_RST is executed only after the AXI master (QMB) is in IDLE state. This bit Can be used for HPI feature instead of polling on HC_SW_RST_REQ bit. This bit should be cleared for SW reset from unknown states.
19	SDCC5_HALT_REQ	R	Reading SDCC5_HALT_REQ input
18	SDCC5_HALT_ACK	R	Reading SDCC5_HALT_ACK output
17	SDCC5_M_IDLE	R	Reading SDCC5_M_IDLE output
16	SDCC5_HALT_ACK_SW_EN	RW	Enabling the SW value for SDCC5_HALT_ACK
15	SDCC5_HALT_ACK_SW	RW	Driving SDCC5_HALT_ACK from SW and not by HW
14	SDCC5_M_IDLE_DIS	RW	If this bit is set, then sdcc5_m_idle will be tied low
13	QSB_AXI_INTERLEAVING_EN	RW	If the bit is set, then the controller can support interleaving between two bursts from the Noc. This option is disabled by default because Noc doesn't support interleaving but only re-ordering
12	RESERVED	RW	Reserved

SDC1_SDCC_HC_VENDOR_SPECIFIC_FUNC2 (cont.)

Bits	Name	Type	Description
11	RESERVED	RW	Reserved
10:9	RESERVED	RW	Reserved
8:6	RESERVED	RW	Reserved
5	RESERVED	RW	Reserved
4	RESERVED	RW	Reserved
3	RESERVED	RW	Reserved
2	QSB_AXI_OOOWR	RW	Driving QSB AXI output
1	QSB_AXI_OOORD	RW	Driving QSB AXI output
0	QSB_AXI_AFULL	RW	Driving QSB AXI output

0x07824A14 SDC1_SDCC_HC_VENDOR_SPECIFIC_ADMA_ERR_ADDR0**Type:** R**Clock:** same rate as HCLK**Reset State:** 0x00000000

Last AHB address phase before response error was detected

SDC1_SDCC_HC_VENDOR_SPECIFIC_ADMA_ERR_ADDR0

Bits	Name	Description
31	RESP_ERR_WR	If set, then the failed transaction was a write access
30:28	RESP_ERR_TID	The TID of the failed transaction
27:13	DESC_ADDR_35_21	Upper 15 bits of current address in address channel
12:0	DESC_ADDR_12_0	Lower 13 bits of current address in address channel

0x07824A18 SDC1_SDCC_HC_VENDOR_SPECIFIC_ADMA_ERR_ADDR1**Type:** R**Clock:** same rate as HCLK**Reset State:** 0x00000000

Last AHB attributes before response error was detected

SDC1_SDCC_HC_VENDOR_SPECIFIC_ADMA_ERR_ADDR1

Bits	Name	Description
21:20	AHB_MON_HTRANS	Htrans control bits from AHB in the last address phase before getting the response error
19	AHB_MON_WO_REQ	indication that the current master got the response error w/o asserting hbusreq
18:16	AHB_MON_HBURST	Hburst control bits from AHB in the last address phase before getting the response error
15	AHB_MON_HWRITE	Hwrite control bits from AHB in the last address phase before getting the response error
14:12	AHB_MON_HSIZE	Hsize control bits from AHB in the last address phase before getting the response error
11:8	AHB_MON_HPROT	Hprot control bits from AHB in the last address phase before getting the response error
7:0	AHB_MON_CUR_MASTER	Indication on the last bus master which accessed to AHB bus

0x07824A1C SDC1_SDCC_HC_VENDOR_SPECIFIC_CAPABILITIES0**Type:** RW**Clock:** same rate as HCLK**Reset State:** 0x00000000

driving the HC_REG_40_42 register

SDC1_SDCC_HC_VENDOR_SPECIFIC_CAPABILITIES0

Bits	Name	Description
31:30	VS_CAPABILITIES_SLOT_T YPE	Slot Type field from Capabilities Register
29	VS_CAPABILITIES_ASYNC_ INT_SUPPORT	Asynchronous Interrupt Support field from Capabilities Register
28	VS_CAPABILITIES_SYS_BU S_SUPPORT_64_BIT	64 bit System Bus Support field from Capabilities Register
26	VS_CAPABILITIES_VOLTAG E_SUPPORT_1_8V	Voltage Support 1.8v field from Capabilities Register
25	VS_CAPABILITIES_VOLTAG E_SUPPORT_3_0V	Voltage Support 3.0v field from Capabilities Register
24	VS_CAPABILITIES_VOLTAG E_SUPPORT_3_3V	Voltage Support 3.3v field from Capabilities Register
23	VS_CAPABILITIES_SUSPE ND_RESUME_SUPPORT	Suspend Resume Support field from Capabilities Register

SDC1_SDCC_HC_VENDOR_SPECIFIC_CAPABILITIES0 (cont.)

Bits	Name	Description
22	VS_CAPABILITIES_SDMA_SUPPORT	SDMA Support field from Capabilities Register
21	VS_CAPABILITIES_HS_SUPPORT	High Speed Support field from Capabilities Register
19	VS_CAPABILITIES_ADMA2_SUPPORT	ADMA2 Support field from Capabilities Register
18	VS_CAPABILITIES_SUPPОРТ_8_BIT	8 bit Supported For Embedded Product field from Capabilities Register
17:16	VS_CAPABILITIES_MAX_BLOCK_LENGTH	Max Block Length field from Capabilities Register
15:8	VS_CAPABILITIES_BASE_SDCLK_FREQ	Base Clock Frequency For SD Clock field from Capabilities Register
7	VS_CAPABILITIES_TIMEOUT_CLK_UNIT	Timeout Clock Unit field from Capabilities Register
5:0	VS_CAPABILITIES_TIMEOUT_CLK_FREQ	Timeout Clock Frequency field from Capabilities Register

0x07824A20 SDC1_SDCC_HC_VENDOR_SPECIFIC_CAPABILITIES1**Type:** RW**Clock:** same rate as HCLK**Reset State:** 0x02008007

driving the HC_REG_44_46 register and spec version

SDC1_SDCC_HC_VENDOR_SPECIFIC_CAPABILITIES1

Bits	Name	Description
31:24	SPEC_VERSION	driving HC_SPEC_VERSION_NUM field in HC_REG_FC_FE register
23:16	VS_CAPABILITIES_CLK_MULTIPLIER	Clock Multiplier field from Capabilities Register
15:14	VS_CAPABILITIES_RETUNING_MODE	Re-Tuning Modes field from Capabilities Register
13	VS_CAPABILITIES_USE_TUNING_FOR_SDR50	Use Tuning for SDR50 field from Capabilities Register
11:8	VS_CAPABILITIES_TIMER_CNT_FOR_RETUNING	Timer Count for Re-Tuning field from Capabilities Register
6	VS_CAPABILITIES_DRIVER_TYPE_D_SUPPORT	Driver Type D Support field from Capabilities Register

SDC1_SDCC_HC_VENDOR_SPECIFIC_CAPABILITIES1 (cont.)

Bits	Name	Description
5	VS_CAPABILITIES_DRIVER_TYPE_C_SUPPORT	Driver Type C Support field from Capabilities Register
4	VS_CAPABILITIES_DRIVER_TYPE_A_SUPPORT	Driver Type A Support field from Capabilities Register
2	VS_CAPABILITIES_DDR_50_SUPPORT	DDR50 Support field from Capabilities Register
1	VS_CAPABILITIES_SDR_104_SUPPORT	SDR104 Support field from Capabilities Register
0	VS_CAPABILITIES_SDR_50_SUPPORT	SDR50 Support field from Capabilities Register

0x07824A24 SDC1_SDCC_HC_VENDOR_SPECIFIC_T4_DLY_CTRL**Type:** RW**Clock:** same rate as HCLK**Reset State:** 0x0000382B

Control register of T/4 Delay element which is used in e.MMC5.0 DDR mode.

SDC1_SDCC_HC_VENDOR_SPECIFIC_T4_DLY_CTRL

Bits	Name	Description
15:14	T4_DLY_SPARE_CTL	
13	T4_DLY_POWERDOWN_EN	Powerdown enable. When asserted (default), it powers down the T/4 macro and shuts off all bias circuits.
12:11	T4_DLY_TESTOUT_SEL	Test output select. When toggled, it presents either the strb_in or strb_out on the analog test port for debug.
10	T4_DLY_BYPASS_EN	Bypass enable. When asserted it presents the strb_in on the strb_out port.
9	T4_DLY_CAL_START	Calibration start input. When asserted, it indicates the start of the calibration cycle.
8	T4_DLY_CAL_OVERRIDE	Calibration override input. When asserted, the internal delay line calibration is bypassed and the tby4_user_ctl<7:0> is used as the default code for the delay line.
7:0	T4_DLY_TARGET_CODE	T/4 delay target code. 0x2b gives delay of 1.25ns

0x07824A28 SDC1_SDCC_HC_VENDOR_SPECIFIC_T4_DLY_STAT**Type:** R**Clock:** same rate as HCLK**Reset State:** 0x00000000

Status register T/4 Delay element which is used in e.MMC5.0 DDR mode.

SDC1_SDCC_HC_VENDOR_SPECIFIC_T4_DLY_STAT

Bits	Name	Description
15:14	T4_DLY_SPARE_OUT	Spare status bits.
13	T4_DLY_CAL_DLY_CODE_FAST_RANGE	Calibrated delay code, indicates the fast range selection bit status.
12:6	T4_DLY_CAL_DLY_CODE_ISEL	Calibrated delay code, indicates the isel programming value for the bias generator.
5:4	T4_DLY_CAL_DLY_CODE_RSEL	Calibrated delay code, indicates the rsel programming value for the bias generator.
3:0	T4_DLY_STATE_DESC	T/4 state descriptor. Indicates the calibration status and result. value 4'b0000: Indicates Reset mode, waiting for cal_start signal. value 4'b0111: Indicates the fast range decision completion. value 4'b0100: Indicates rsel adjustment decision. value 4'b1011: Indicates isel adjustment decision. value 4'b1110: Indicates Normal end-of-cal. value 4b'1111: Indicates Maximum Delay Reached

0x07824A30 SDC1_SDCC_HC_VENDOR_SPECIFIC_CSR_CDC_CTLR_CFG0**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

The HC_VENDOR_SPECIFIC_CSR_CDC_CTLR_CFG0 register configures the following:

SDC1_SDCC_HC_VENDOR_SPECIFIC_CSR_CDC_CTLR_CFG0

Bits	Name	Description
25	STAGGER_CAL_ENA	1'b1: Stagger calibration of write and read CDCs once after the other to reduce peak voltage drop. 1'b0: Both write and read CDCs calibrated simultaneously This bit is introduced on the x2 core, so not present on x0 and x1 versions of DIM/PHY.
24	TRACKING_CAL_ENA	1'b1: Relationship between subunits and oscillator count is learnt and used for tracking calibration. 1'b0: Periodic calibration triggers cause full recalibration

SDC1_SDCC_HC_VENDOR_SPECIFIC_CSR_CDC_CTLR_CFG0 (cont.)

Bits	Name	Description
23:20	OSC_COUNT_ERR_TOLERANCE	Magnitude of the difference between oscillator count and target count when the algorithm assumes convergence and stops further search.
19	TRACK_CALIB_MODE	1'b0: Tracking calibration uses previous Target_count value. 1'b1 Accurate mode: Target_count is recomputed at the start of tracking calibration
18	FULL_DELAY	1'b0: unitstep and subunit caps are from calibration or SW settings 1'b1: All unit step and subunit caps are selected for the delay arrays.
17	HW_AUTOCAL_ENA	1'b1: Enables periodic auto calibration 1'b0: Periodic calibration is disabled
16	SW_TRIG_FULL_CALIB	A low to high transition on this register bit triggers the CDC calibration statemachine to initiate full calibration of the standby delay array. Also resets tracking mode for the delay arrays.
13:12	OSC_PRE_DIV	The factor by which the raw oscillator frequency is divided before being given to the counting logic. 2'b00 Divide by 1 2'b01 Divide by 2 2'b10 Divide by 4 2'b11 Divide by 8
10:0	TMUX_CHAR	Cchar is a characterized constant equal to the ratio of output mux offset delay to average unit step delay. This register is programmed by multiplying the constant with 1024 and rounding to the nearest integer. E.g. if Cchar = 0.67, $RO(0.67 * 1024) = 686$, hence program TMUX_CHAR = 11'h2AE

0x07824A34 SDC1_SDCC_HC_VENDOR_SPECIFIC_CSR_CDC_CTLR_CFG1**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

The HC_VENDOR_SPECIFIC_CSR_CDC_CTLR_CFG1 register configures the following:

SDC1_SDCC_HC_VENDOR_SPECIFIC_CSR_CDC_CTLR_CFG1

Bits	Name	Description
26:24	OSC_COUNT_DELAY	Number of oscillator cycles from the end of reference gate signal to assertion of oscillation counter done. Legal values are from 0 to 4.

SDC1_SDCC_HC_VENDOR_SPECIFIC_CSR_CDC_CTLR_CFG1 (cont.)

Bits	Name	Description
23:20	STANDBY_DELAY	This is the number of reference clock cycles from when a newly calibrated delay array is made active to when the current active delay array is put into standby mode. 4'b0: Disables the counter. Hardware waits for explicit acknowledgement of delay array switch from CDC. Others: Explicit acknowledgement mode is disabled and the counter is used
18:16	DEL_MODE_DELAY	Number of reference clock cycles between putting a newly calibrated delay array in delay mode to selecting it as the active delay array. The minimum allowed value is 1
14:12	OSC_MODE_DELAY	Number of reference clock cycles between putting a delay array in oscillator mode to starting the oscillation counter. This delay is allowed for the oscillations to stabilize through the delay array. The minimum allowed value is 1
10:8	DECODER_DELAY	Number of reference clock cycles allowed for unitstep and subunit counts to be encoded into the unit and finecap select lines on the DDA. The minimum allowed value is 1.
6:4	DIVIDER_DELAY	Number of reference clock cycles allowed for arithmetic division operations. The minimum allowed value is 1
2:0	MULTIPLIER_DELAY	Number of reference clock cycles allowed for arithmetic multiplication operations. The minimum allowed value is 1

0x07824A38 SDC1_SDCC_HC_VENDOR_SPECIFIC_CSR_CDC_CAL_TIMER_CFG0**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

The HC_VENDOR_SPECIFIC_CSR_CDC_CAL_TIMER_CFG0 register configures the following:

SDC1_SDCC_HC_VENDOR_SPECIFIC_CSR_CDC_CAL_TIMER_CFG0

Bits	Name	Description
24	INVALID_TIMER_ENA	1'b1: Enable the timer counting number of missed calibration intervals after which the CDC is invalid. 1'b0: Timer disabled A low to high transition on this register bit loads the timer with programmed value below.
23:20	INVALID_TIMER_VAL	Number of missed calibration intervals after which CDC invalid state is asserted.

SDC1_SDCC_HC_VENDOR_SPECIFIC_CSR_CDC_CAL_TIMER_CFG0 (cont.)

Bits	Name	Description
16	TIMER_ENA	1'b1: Enables 32 kHz sleep clock based periodic autocalibration timer. 1'b0: Timer disabled A low to high transition on this register bit loads the timer with programmed value below.
15:0	TIMER_VAL	Count value for periodic autocalibration interval based on sleep clock.

0x07824A3C SDC1_SDCC_HC_VENDOR_SPECIFIC_CSR_CDC_CAL_TIMER_CFG1**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

The HC_VENDOR_SPECIFIC_CSR_CDC_CAL_TIMER_CFG1 register configures the following:

SDC1_SDCC_HC_VENDOR_SPECIFIC_CSR_CDC_CAL_TIMER_CFG1

Bits	Name	Description
12	FF_TIMER_ENA	1'b1: Enables reference clock based periodic autocalibration timer. 1'b0: Timer disabled A low to high transition on this register bit loads the timer with programmed value below.
9:0	FF_TIMER_VAL	Count value for periodic autocalibration interval based on reference clock.

0x07824A40 SDC1_SDCC_HC_VENDOR_SPECIFIC_CSR_CDC_REFCOUNT_CFG**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

The HC_VENDOR_SPECIFIC_CSR_CDC_REFCOUNT_CFG register configures the following:

SDC1_SDCC_HC_VENDOR_SPECIFIC_CSR_CDC_REFCOUNT_CFG

Bits	Name	Description
31:16	TREF	Reference clock period programmed in pico seconds.
13:8	CCAL_REF_COUNT	Number of reference clock cycles used for oscillation counts during coarse calibration. Maximum supported value is 59.

SDC1_SDCC_HC_VENDOR_SPECIFIC_CSR_CDC_REFCOUNT_CFG (cont.)

Bits	Name	Description
5:0	FCAL_REF_COUNT	Number of reference clock cycles used for oscillation counts during fine calibration. Maximum supported value is 59

0x07824A44 SDC1_SDCC_HC_VENDOR_SPECIFIC_CSR_CDC_COARSE_CAL_CFG**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

The register HC_VENDOR_SPECIFIC_CSR_CDC_COARSE_CAL_CFG configures the following:

SDC1_SDCC_HC_VENDOR_SPECIFIC_CSR_CDC_COARSE_CAL_CFG

Bits	Name	Description
12:8	CCAL_SUBUNIT_CAPS	Subunit count needed for an effective unit step count of 32 during coarse calibration. This value is also used by software oscillator mode.
5:0	CCAL_UNITSTEPS	unitstep count needed for an effective unit step count of 32 during coarse calibration.

0x07824A48 SDC1_SDCC_HC_VENDOR_SPECIFIC_CSR_CDC_RSVD_CFG**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

The HC_VENDOR_SPECIFIC_CSR_CDC_RSVD_CFG register configures the following:

SDC1_SDCC_HC_VENDOR_SPECIFIC_CSR_CDC_RSVD_CFG

Bits	Name	Description
15:0	TEMP_FIELD	Reserved register

0x07824A4C SDC1_SDCC_HC_VENDOR_SPECIFIC_CSR_CDC_OFFSET_CFG**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

The HC_VENDOR_SPECIFIC_CSR_CDC_OFFSET_CFG register configures the following:

SDC1_SDCC_HC_VENDOR_SPECIFIC_CSR_CDC_OFFSET_CFG

Bits	Name	Description
20	SUBUNIT_OFFSET_MODE	1'b1: The programmed offset is directly loaded as subunit count 1'b0: Programmed offset is added/subtracted from the subunit count
19	SUBUNIT_OFFSET_SIGN	1'b1: Offset is subtracted from the subunit count 1'b0: Offset is added to the subunit count
16:12	SUBUNIT_OFFSET	Software subunit offset
8	UNIT_OFFSET_MODE	1'b1: The programmed offset is directly loaded as unit step count 1'b0: Programmed offset is added/subtracted from the unit step count
7	UNIT_OFFSET_SIGN	1'b1: Offset is subtracted from the unitstep count 1'b0: Offset is added to the unitstep count
5:0	UNIT_OFFSET	Software unitstep offset

0x07824A50 SDC1_SDCC_HC_VENDOR_SPECIFIC_CSR_CDC_DELAY_CFG**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

The HC_VENDOR_SPECIFIC_CSR_CDC_DELAY_CFG register configures the following:

SDC1_SDCC_HC_VENDOR_SPECIFIC_CSR_CDC_DELAY_CFG

Bits	Name	Description
28	TARGET_COUNT_ENA	1'b1: Enable software specified Target_count for calibration instead of value calculated by hardware. 1'b0: Hardware computed Target_count used.
27:16	TARGET_COUNT	Software specified Target_count to which the statemachine calibrates
11:0	DELAY_VAL	Delay required from CDC programmed in pico seconds. This value has to be greater than the minimum delay supported by DDA Tmin.

0x07824A54 SDC1_SDCC_HC_VENDOR_SPECIFIC_CSR_CDC_SW_MODE_CFG**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

The HC_VENDOR_SPECIFIC_CSR_CDC_SW_MODE_CFG register configures the following:

SDC1_SDCC_HC_VENDOR_SPECIFIC_CSR_CDC_SW_MODE_CFG

Bits	Name	Description
3	SW_DEL_MODE	A low to high transition on this register bit puts the delay array selected by SW_DA_SEL into delay mode and connects it to the DDA output. This register has to remain set for the duration of SW controlled delay mode.
2	SW_OSC_MODE	Reserved register.
1	SW_DA_SEL	Selects the delay array inside DDA during software delay or oscillator mode.
0	SW_LOAD	1'b1: Enable loading of the software offsets from the offset configuration register. This register has to remain set for the duration of SW mode operation.

0x07824A5C SDC1_SDCC_HC_VENDOR_SPECIFIC_CSR_CDC_SW_OVRD_CFG**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

The HC_VENDOR_SPECIFIC_CSR_CDC_SW_OVRD_CFG register configures the following:

SDC1_SDCC_HC_VENDOR_SPECIFIC_CSR_CDC_SW_OVRD_CFG

Bits	Name	Description
16	SW_REF_GATE	Write 1'b1 to start the reference window. Write 1'b0 to end the reference window. Number of reference cycle in the window will be reported in SW_REF_GATE_COUNT in CDC_STATUS0
11	SW_OVRD_DA1_OSC_EN	In software override mode, 1'b1 written to this bit puts delay array 1 in oscillator-mode
10	SW_OVRD_DA1_IN_EN	In software override mode, 1'b1 written to this bit puts delay array 1 in delay-mode
9	SW_OVRD_DA0_OSC_EN	In software override mode, 1'b1 written to this bit puts delay array 0 in oscillator-mode
8	SW_OVRD_DA0_IN_EN	In software override mode, 1'b1 written to this bit puts delay array 0 in delay-mode
3	SW_OVRD_CDC_COUNTEN_R_RST	1'b1 written to this bit resets the oscillation counter. Follow this with 1'b0 before setting reference gate.
2	SW_OVRD_LOAD_DA_SEL	In software override mode, this bit select the delay array to which unit and subunit offset values from CDC_OFFSET_CFG register will be loaded.
1	SW_OVRD_ACTV_DA_SEL	In software override mode, this register selects the delay array connected to the CDC output

SDC1_SDCC_HC_VENDOR_SPECIFIC_CSR_CDC_SW_OVRD_CFG (cont.)

Bits	Name	Description
0	SW_OVRD_ENA	1'b1: written to this bit puts the CDC interface in software override mode where the below register bits control the CDC. 1'b0: CDC is under hardware control

0x07824A60 SDC1_SDCC_HC_VENDOR_SPECIFIC_CSR_CDC_SLAVE_DDA_CFG**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

The HC_VENDOR_SPECIFIC_CSR_CDC_SLAVE_DDA_CFG register configures the following:

SDC1_SDCC_HC_VENDOR_SPECIFIC_CSR_CDC_SLAVE_DDA_CFG

Bits	Name	Description
17	SLAV_DDA_OFFSET_MODE	1'b1: The programmed offset is directly loaded as unit-step count 1'b0: Programmed offset is added/subtracted from the unit-step count
16	SLAV_DDA_OFFSET_SIGN	1'b1: Offset is subtracted from the unit-step count. This subtraction feature is not supported on x0 and x1 PHY cores (bit RESERVED). Supported on the x2 core version. 1'b0: Offset is added to the unit-step count
15:12	SLAVE_DDA_OFFSET	unit-step offset for slave DDA.
10:0	SLAVE_DDA_DELAY	Delay required from slave DDA programmed in pico seconds.

0x07824A64 SDC1_SDCC_HC_VENDOR_SPECIFIC_CSR_CDC_STATUS0**Type:** R**Clock:** HCLK**Reset State:** 0x0000000C

The HC_VENDOR_SPECIFIC_CSR_CDC_STATUS0 register configures the following:

SDC1_SDCC_HC_VENDOR_SPECIFIC_CSR_CDC_STATUS0

Bits	Name	Description
26:24	CDC_ERROR_CODE	Error code for internal errors: 3'b000: No error 3'b001: Programmed delay less than Tmux 3'b010: Programmed delay less than Tmin 3'b011: Unit-step overflow: Overflow during fine calibration. Desired delay cannot be realized even when all unit-steps are selected. 3'b100: Unit-step underflow 3'b101: Unit-step overflow: Overflow during coarse calibration. Estimated unit steps needed to realize the desired delay are greater than the total number of unit-steps in design. Note: The CDC_ERROR_CODE is not a sticky register and is updated after each CDC calibration.
23:16	SW_REF_GATE_COUNT	Number of reference clock cycles in the software generated oscillation gate window during SW override mode
15:4	OSC_COUNT	Current value of oscillator count
3	CURR_SEL_DA	This bit indicates the current active delay array connected to the output. 1'b0: DA0 is selected 1'b1: DA1 is selected
2	CTLR_SM_IDLE	1'b1 Indicates that the calibration state machine is idle 1'b0 indicates calibration state machine is busy
1	OSC_DONE	This status register is set after oscillations count is finished by CDC. It gets reset when a new oscillator count is started.
0	CALIBRATION_DONE	1'b1 indicates that calibration is done and CURR_SUBUNIT_STEPS, CURR_UNIT_STEPS status registers are valid.

0x07824A68 SDC1_SDCC_HC_VENDOR_SPECIFIC_CSR_CDC_STATUS1**Type:** R**Clock:** HCLK**Reset State:** 0x00000000

The HC_VENDOR_SPECIFIC_CSR_CDC_STATUS1 register configures the following:

SDC1_SDCC_HC_VENDOR_SPECIFIC_CSR_CDC_STATUS1

Bits	Name	Description
27:16	CURR_DELAY_VALUE	HW Computed value of current delay after calibration by the state machine. This register is valid if CALIBRATION_DONE status bit is set.

SDC1_SDCC_HC_VENDOR_SPECIFIC_CSR_CDC_STATUS1 (cont.)

Bits	Name	Description
11:0	CURR_TMUX_DELAY	Current value of output mux offset delay computed by HW.

0x07824A6C SDC1_SDCC_HC_VENDOR_SPECIFIC_CSR_CDC_STATUS2**Type:** R**Clock:** HCLK**Reset State:** 0x10331033

The HC_VENDOR_SPECIFIC_CSR_CDC_STATUS2 register configures the following:

SDC1_SDCC_HC_VENDOR_SPECIFIC_CSR_CDC_STATUS2

Bits	Name	Description
28:24	DA1_SUBUNITS	Current value of subunit cap settings applied to delay array 1
21:16	DA1_UNITSTEPS	Current value of unit steps applied to delay array 1
12:8	DA0_SUBUNITS	Current value of subunit cap settings applied to delay array 0
5:0	DA0_UNITSTEPS	Current value of unit steps applied to delay array 0

0x07824A70 SDC1_SDCC_HC_VENDOR_SPECIFIC_CSR_CDC_STATUS3**Type:** R**Clock:** HCLK**Reset State:** 0x00000000

The HC_VENDOR_SPECIFIC_CSR_CDC_STATUS3 register configures the following:

SDC1_SDCC_HC_VENDOR_SPECIFIC_CSR_CDC_STATUS3

Bits	Name	Description
31:24	NUM_OF_OSC_ITER	Number of oscillator iterations before convergence was achieved during tracking
23:12	COUNT_ERROR	Difference between target and actual oscillator counts at the start of tracking calibration
11:0	CURR_TARGET_COUNT	HW computed value of current Target_count used by the calibration state machine.

0x07824A74 SDC1_SDCC_HC_VENDOR_SPECIFIC_CSR_CDC_STATUS4**Type:** R**Clock:** HCLK**Reset State:** 0x000000FF

The HC_VENDOR_SPECIFIC_CSR_CDC_STATUS4 register configures the following:

SDC1_SDCC_HC_VENDOR_SPECIFIC_CSR_CDC_STATUS4

Bits	Name	Description
7:4	SLAVE_DDA_DA1_TAPS	Number of unit taps applied to delay array 1 of slave DDA
3:0	SLAVE_DDA_DA0_TAPS	Number of unit taps applied to delay array 0 of slave DDA

0x07824A78 SDC1_SDCC_HC_VENDOR_SPECIFIC_CSR_CDC_GEN_CFG**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000**SDC1_SDCC_HC_VENDOR_SPECIFIC_CSR_CDC_GEN_CFG**

Bits	Name	Description
2	CDC_SDDA_EN	Enable CDC Slave DDA to generate 3/4T RCLK from cdc_out_2 output of CDC: 0 (default) - CDC Slave DDA is disabled and 3/4T clock is a falling edge of cdc_out_2, 1 - CDC Slave DDA is enabled and 3/4T clock is a rising edge of cdc_out_2.
1	CDC_SWITCH_RC_EN	Enablement of CDC power RC (LPF) switch 1'b1: Enabled 1'b0: Disabled (default)
0	CDC_SWITCH_BYPASS_OFF	Enablement of CDC power bypass switch 1'b1: Disabled 1'b0: Enabled (default)

0x07824A7C SDC1_SDCC_HC_VENDOR_SPECIFIC_CDC_AUTO_SW_CAL_CFG**Type:** RW**Clock:** HCLK**Reset State:** 0x00F08000**SDC1_SDCC_HC_VENDOR_SPECIFIC_CDC_AUTO_SW_CAL_CFG**

Bits	Name	Description
25	CDC_CAL_TIMEOUT_ENABLE	Enable for cdc calibration timeout counter.

SDC1_SDCC_HC_VENDOR_SPECIFIC_CDC_AUTO_SW_CAL_CFG (cont.)

Bits	Name	Description
24	CDC_AUTO_SW_CAL_ENABLE	Enable for the Auto SW calibration Feature.
23:16	CDC_CAL_TIMEOUT_THRESHOLD	Sets the threshold for CDC calibration timeout during sw calibration triggered by the HW FSM. counts in 16xFF clock. Default is 0xF0 which is 255x16 FF clocks, which gives threshold of ~200 us
15:0	AUTO_TRIG_INTERVAL	Sets the load value of the counter that counts the interval between each sw calibration triggered by the HW FSM. this counter counts in sleep clock. Default is 0x8000 which is ~32,000 sleep clocks, which gives interval of ~1 sec

0x07824A80 SDC1_SDCC_HC_VENDOR_SPECIFIC_DDR200_STAT**Type:** R**Clock:** HCLK**Reset State:** 0x00000000**SDC1_SDCC_HC_VENDOR_SPECIFIC_DDR200_STAT**

Bits	Name	Description
0	CDC_CAL_VALID	CDC Calibration status: 0 - CDC isn't calibrated, 1 - CDC is calibrated and valid

0x07824A84 SDC1_SDCC_HC_VENDOR_SPECIFIC_DDR200_CFG**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000**SDC1_SDCC_HC_VENDOR_SPECIFIC_DDR200_CFG**

Bits	Name	Description
10	FF_CLK_DIS	The ff_clk to CDC/TBY4 is enabled by default. It can be disabled by setting this bit. The ff_clk is enabled to CDC and TBY4 according to CDC_T4_DLY_SEL field
9	VOLTAGE_MUX_SEL	select between 1.8v and 1.2v signals on sdcc5 IO interface with device. default is 0 to choose 1.8v
8:7	CDC_TRAFFIC_SEL	Selecting the method of cdc_traffic's assertions: Value 0 (default) - auto calibration with stand by feature. Value 1 - SW calibration before any transaction. Value 2 - automatic calibration with HW assertion of cdc_traffic

SDC1_SDCC_HC_VENDOR_SPECIFIC_DDR200_CFG (cont.)

Bits	Name	Description
6	START_CDC_TRAFFIC	This bit starts the cdc traffic assertion after calibration done is received
5	CRC_TOKEN_SAMPL_FALL_EDGE	If set, the start bit of incoming CRC token in DDR200 mode is sampled in falling edge. If this bit is cleared (default) then the start bit is detected in rising edge
4	DATIN_SAMPL_FALL_EDGE	If set, the start bit of incoming data in DDR200 mode is sampled in falling edge. If this bit is cleared (default) then the start bit is detected in rising edge
3	CDC_T4_TEST_OUT_SEL	controls the mux between cm_dll_sdc4 testout and CDC test out. When set - CDC test_out is chosen
2	CMDIN_EDGE_SEL	Select sampling edge of CMD input when CMDIN_RCLK_EN is 1: 0 (default) - falling edge of RCLK, 1 - rising edge of RCLK
1	CMDIN_RCLK_EN	Enable CMD input sampling with RCLK: 0 (default) - CMD input is sampled by SD DLL output clock, 1 - CMD input is sampled by delayed RCLK
0	CDC_T4_DLY_SEL	Select block for RCLK delay: 0 (default) - CM_DLL_SDC4

0x07824A88 SDC1_SDCC_HC_VENDOR_SPECIFIC_TIME_COUNT_CTRL**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

Control the performance counters

SDC1_SDCC_HC_VENDOR_SPECIFIC_TIME_COUNT_CTRL

Bits	Name	Description
1	CMD_PERIOD	When this bit is set the commands periods are counted to the CMD_TIME counter
0	ENABLE	Enabling the performance counters

0x07824A8C SDC1_SDCC_HC_VENDOR_SPECIFIC_FLOW_CTRL_TIME_COUNT**Type:** R**Clock:** HCLK**Reset State:** 0x00000000

Counting the Flow Control period

SDC1_SDCC_HC_VENDOR_SPECIFIC_FLOW_CTRL_TIME_COUNT

Bits	Name	Description
31:0	FLOW_CTRL_TIME	FlowControl periods are counted

0x07824A90 SDC1_SDCC_HC_VENDOR_SPECIFIC_BUSY_TIME_COUNT**Type:** R**Clock:** HCLK**Reset State:** 0x00000000

Counting the BUSY periods

SDC1_SDCC_HC_VENDOR_SPECIFIC_BUSY_TIME_COUNT

Bits	Name	Description
31:0	BUSY_TIME	In Tx transaction, the BUSY periods are counted until the progdone is asserted. In Rx transaction, the periods between the blocks without flowcontrol is counted including the time until the first block is received after the command.

0x07824A94 SDC1_SDCC_HC_VENDOR_SPECIFIC_DATA_TIME_COUNT**Type:** R**Clock:** HCLK**Reset State:** 0x00000000

Counting the time of transaction

SDC1_SDCC_HC_VENDOR_SPECIFIC_DATA_TIME_COUNT

Bits	Name	Description
31:0	DATA_TIME	In Tx transaction, the time is from ENabling the DPSM until progdone is asserted. In Rx transaction, the time is from the command transmission to de-assertion of RxActive.

0x07824A98 SDC1_SDCC_HC_VENDOR_SPECIFIC_CMD_TIME_COUNT**Type:** R**Clock:** HCLK**Reset State:** 0x00000000

Counting the time of transaction

SDC1_SDCC_HC_VENDOR_SPECIFIC_CMD_TIME_COUNT

Bits	Name	Description
31:0	CMD_TIME	Counting the Command-Response periods of commands which relate to the transaction including auto commands. The counter is active when CMD_PERIOD is set.

0x07824A9C SDC1_SDCC_HC_VENDOR_SPECIFIC_ERR_INT_STS

Type: R

Clock: HCLK

Reset State: 0x00000000

Details of the vendor specific errors

SDC1_SDCC_HC_VENDOR_SPECIFIC_ERR_INT_STS

Bits	Name	Description
1	AUTO_CDC_SW_CALIB_ERR	calibration error is received (CDC_ERROR_CODE isn't 0) during auto SW CDC calibration
0	AUTO_CDC_SW_CALIB_TIMEOUT	CALIBRATION_DONE isn't received within the number of cycles which configured in CDC_CAL_TIMEOUT_THRESHOLD field (from CDC_AUTO_SW_CAL_CFG register)

0x07824AA0 SDC1_SDCC_HC_VENDOR_SPECIFIC_PWRCTL_STATUS_REG

Type: R

Clock: PCLK

Reset State: 0x00000000

HC Spec Defines power and IO voltage switch bits. The information is sent by a dedicated interrupt to the QCT SW driver for handling the power and voltage tasks with PMIC. The different tasks are defined in this register

SDC1_SDCC_HC_VENDOR_SPECIFIC_PWRCTL_STATUS_REG

Bits	Name	Description
3	IO_HIGH_V	SDCC driver orders that the I/O signaling level be switched to 3.0v. This interrupt is triggered when SDCC driver writes '0' to offset 03Eh in HC register (bit 3) or when SW reset for all is handled and last supported voltage was 1.8V. This logic is enabled when SWITCHABLE_SIGNALING_VOLTAGE=true

SDC1_SDCC_HC_VENDOR_SPECIFIC_PWRCTL_STATUS_REG (cont.)

Bits	Name	Description
2	IO_LOW_V	SDCC driver orders that the I/O signaling level be switched to 1.8v. This interrupt is triggered when SDCC driver writes '1' to offset 03Eh in HC register (bit 3). This logic is enabled when SWITCHABLE_SIGNALING_VOLTAGE=true
1	BUS_ON	SDCC driver orders that the SD bus power be switched on. This interrupt is triggered when SDCC driver writes '1' to offset 029h in HC register (bit 0).
0	BUS_OFF	SDCC driver orders that the SD bus power be switched off. This interrupt is triggered when SDCC driver writes '0' to offset 029h in HC register(bit 0) or when SW reset for all is handled.

0x07824AA4 SDC1_SDCC_HC_VENDOR_SPECIFIC_PWRCTL_MASK_REG**Type:** RW**Clock:** PCLK**Reset State:** 0x00000000

Mask register for MCI_PWRCTL_STATUS_REG

SDC1_SDCC_HC_VENDOR_SPECIFIC_PWRCTL_MASK_REG

Bits	Name	Description
3	IO_HIGH_V	'1' - interrupt is enabled / '0' - interrupt is masked.
2	IO_LOW_V	'1' - interrupt is enabled / '0' - interrupt is masked.
1	BUS_ON	'1' - interrupt is enabled / '0' - interrupt is masked.
0	BUS_OFF	'1' - interrupt is enabled / '0' - interrupt is masked.

0x07824AA8 SDC1_SDCC_HC_VENDOR_SPECIFIC_PWRCTL_CLEAR_REG**Type:** W**Clock:** PCLK**Reset State:** 0x00000000

Clear register for MCI_PWRCTL_STATUS_REG

SDC1_SDCC_HC_VENDOR_SPECIFIC_PWRCTL_CLEAR_REG

Bits	Name	Description
3	IO_HIGH_V	Writing '1' clears bit 03 in PWRCTL_STATUS_REG
2	IO_LOW_V	Writing '1' clears bit 02 in PWRCTL_STATUS_REG
1	BUS_ON	Writing '1' clears bit 01 in PWRCTL_STATUS_REG

SDC1_SDCC_HC_VENDOR_SPECIFIC_PWRCTL_CLEAR_REG (cont.)

Bits	Name	Description
0	BUS_OFF	Writing '1' clears bit 00 in PWRCTL_STATUS_REG

0x07824AAC SDC1_SDCC_HC_VENDOR_SPECIFIC_PWRCTL_CTL_REG**Type:** RW**Clock:** PCLK**Reset State:** 0x00000000

HC Spec Defines power and IO voltage switch bits. The information is sent by a dedicated interrupt to the QCT SW driver for handling the power and voltage tasks with PMIC and then the QCT SW driver needs to report success or fail to this register

SDC1_SDCC_HC_VENDOR_SPECIFIC_PWRCTL_CTL_REG

Bits	Name	Description
9	HC_CARD_STATE_STABLE	This bit is driving CARD_STATE_STABLE bit in Present State Register
8	RST_NOT_WAIT_PWRCTL_REG	RST process doesn't need to wait for accessing to MCI_PWRCTL_CTL_REG register
7:6	HC_WRITE_PROTECT_PAD	Value 0 inversion of HC_WRITE_PROTECT is connected to bit 19 in Present State Register. Value 1 input from PAD is used. Value 2 inversion of input from PAD is used
5	CARD_DETECT_PIN_LEVEL	This bit reflects the inverse value of the SDCD# pin. Debouncing is not performed on this bit. This bit may be valid when Card State Stable is set to 1, but it is not guaranteed because of propagation delay. Use of this bit is limited to testing since it must be debounced by software. value 1: card present value 0: no card present
4	HC_WRITE_PROTECT	This bit allows to disable the writing. value 0: write enabled value 1: write protected
3	IO_SIG_SWITCH_FAIL	Host controller clears this bit when it fires either I/O Low V interrupt or I/O High V interrupt. SDCC power control software writes '1' to indicate the signal switch has failed.
2	IO_SIG_SWITCH_SUCCESS	Host controller clears this bit when it fires either I/O Low V interrupt or I/O High V interrupt. SDCC power control software writes '1' to indicate the signal switch has completed successfully.
1	BUS_ON_OFF_FAIL	Host controller clears this bit when it fires either Bus ON interrupt or Bus OFF interrupt. SDCC power control software writes '1' to indicate the requested operation has failed.

SDC1_SDCC_HC_VENDOR_SPECIFIC_PWRCTL_CTL_REG (cont.)

Bits	Name	Description
0	BUS_ON_OFF_SUCCESS	Host controller clears this bit when it fires either Bus ON interrupt or Bus OFF interrupt. SDCC power control software writes '1' to indicate the requested operation has completed successfully.

0x07824AB0 SDC1_SDCC_HC_VENDOR_SPECIFIC_FUNC3**Type:** RW**Clock:** same rate as HCLK**Reset State:** 0x00000000

Vendor specific register #3

SDC1_SDCC_HC_VENDOR_SPECIFIC_FUNC3

Bits	Name	Description
4	QSB_AXI_REDIRECT	Driving QSB AXI output
3	PWRSAVE_DLL	When this bit is set, DLL will get clock during transactions but not in idle. When this bit is cleared, DLL will get clock all the time so sdc4_dll_lock won't be de-asserted
2	SDIO_TRANS	When this bit is set in SDIO transaction, START_BIT_ERR status bit won't be raised. The reason is that interrupt period which is de-asserted only after the RX command can cause to wrong START_BIT_ERR detection. The START_BIT_ERR bit won't appear in MCI_STATUS when SDIO_TRANS is set
1	HCLK_IDLE_GATING	Additional CGC for HCLK was added, which gates the HCLK for status bits. If this bit is set then HCLK is gated for these registers during IDLE
0	MCLK_IDLE_GATING	Additional CGC for MCLK was added, which gates the MCLK for status bits. If this bit is set then MCLK is gated for these registers during IDLE

0x07824AB4 SDC1_SDCC_HC_REG_DLL_CONFIG_2**Type:** RW**Clock:** same rate as HCLK**Reset State:** 0x00000000**SDC1_SDCC_HC_REG_DLL_CONFIG_2**

Bits	Name	Description
3:2	DDR_TRAFFIC_INIT_SEL	
1	DDR_TRAFFIC_INIT_SW	

SDC1_SDCC_HC_REG_DLL_CONFIG_2 (cont.)

Bits	Name	Description
0	DDR_CAL_EN	

0x07824AB8 SDC1_SDCC_HC_REG_DDR_CONFIG**Type:** RW**Clock:** same rate as HCLK**Reset State:** 0x80040853

Configuration of ddr_config input of DLL.

SDC1_SDCC_HC_REG_DDR_CONFIG

Bits	Name	Description
31	PRG_DLY_EN	Enables Programmable delay for RCLK_IN
30	EXT_PRG_RCLK_DLY_EN	Enables external control of Programmable RCLK delayline
29:27	EXT_PRG_RCLK_DLY_CODE	Used only if EXT_PRG_RCLK_DLY_EN is set. Configuring Programmable RCLK Delayline process code number.
26:21	EXT_PRG_RCLK_DLY	Used only if EXT_PRG_RCLK_DLY_EN is set. Programmable RCLK Delay line fine delay control.
20:12	TCXO_CYCLES_DLY_LINE	Number of TCXO clock cycles between finish of one delay line calibration and beginning of next delay line calibration.
11:9	TCXO_CYCLES_CNT	Number of TCXO clock cycles for which down counter is enabled for before making each inc/dec decision.
8:0	PRG_RCLK_DLY	These bits are controlled the programmable delay line. This bus is binary coded. Decimal value = (TCXO period * Decimal value of TCXO_CYCLES_CNT)/(2*delay needed). For 1.25ns the formula is $(52n^4/2*1.25) = 83$ in decimal.

19 SDC2

19.1 SDC2_SDCC_SDCC5

0x07864000 SDC2_SDCC_MCI_POWER

Type: RW

Clock: same rate as HCLK

Reset State: 0x00000000

(ARM name: MCIPower). This register must be updated to the MCLK domain so subsequent writes to this register will be ignored until 3 MCLK and 3 HCLK have passed. SW should wait until bit 0 (MCLK_REG_WR_ACTIVE) of MCI_STATUS2 register is 0, after writing to MCI_POWER register and before accessing this register again.

SDC2_SDCC_MCI_POWER

Bits	Name	Description
11	DML_SW_RST_WAIT_IDLE_DIS	If this bit is set, then SW_RST_REQ procedure before BAM_TRANS_RESET is not executed automatically.
10	SW_RST_WAIT_IDLE_DIS	If this bit is set, then SW_RST_REQ procedure is not executed automatically.
9	SW_RST_REQ	Before executing SW_RST or SW_RST_CONFIG, this bit should be set and SW needs to poll on it until it's being cleared. The bit is cleared after AXI master has finished processing the outstanding transactions and it's in IDLE state.
8	SW_RST_CONFIG	SW reset bit for the core without resetting configuration registers (MCI_POWER, MCI_CLK, MCI_INT_MASKn). In that case, SW will be able to reset the internal state machines, data path, etc. and it will be able to proceed to next transaction without reconfiguring the SDCC registers.
7	SW_RST	SW reset bit for the core. The impact of this reset bit will be similar to the impact of the reset from SW register in CLK_CTRL block. All SW registers and all internal logic is reset.

SDC2_SDCC_MCI_POWER (cont.)

Bits	Name	Description
6	OPEN_DRAIN	MCICMD Output Control. This bit is reserved for the use mentioned in the Description but it is not tested. The open drain mode is preserved such that when enabled, the command output is driven low when the logic is low and undriven (so that the pull up can pull high) when the logic is high. Whether this gets used in the system is up to SW/system but the HW does support this mode.
0	CONTROL	This register is used to control the operation of the memory controller. value 0: power-off value 1: power-on

0x07864004 SDC2_SDCC_MCI_CLK**Type:** RW**Clock:** same rate as HCLK**Reset State:** 0x01008000

(ARM name: MCIClock). This register must be updated to the MCLK domain so subsequent writes to this register will be ignored until 3 MCLK and 3 HCLK have passed. SW should wait until bit 0 (MCLK_REG_WR_ACTIVE) of MCI_STATUS2 register is 0, after writing to MCI_POWER register and before accessing this register again.

SDC2_SDCC_MCI_CLK

Bits	Name	Description
31	PWRSAVE_DLL	When this bit is set, DLL will get clock during transactions but not in idle. When this bit is cleared, DLL will get clock all the time so sdc4_dll_lock won't be de-asserted
30	SDIO_TRANS	When this bit is set in SDIO transaction, START_BIT_ERR status bit won't be raised. The reason is that interrupt period which is de-asserted only after the RX command can cause to wrong START_BIT_ERR detection. The START_BIT_ERR bit won't appear in MCI_STATUS when SDIO_TRANS is set
29	HCLK_IDLE_GATING	Additional CGC for HCLK was added, which gates the HCLK for status bits. If this bit is set then HCLK is gated for these registers during IDLE
28	MCLK_IDLE_GATING	Additional CGC for MCLK was added, which gates the MCLK for status bits. If this bit is set then MCLK is gated for these registers during IDLE
27	INT_MCLK_ON	An internal CGC is located in MCLK domain. SW can enable the internal clock at IDLE time by writing '1' to this bit. If the bit is cleared, then the clock is gated at IDLE time

SDC2_SDCC_MCI_CLK (cont.)

Bits	Name	Description
26	SDCC_CLK_EXT_EN	Value 0 (default) - sdcc_clkout is driven via the clock pad and external clock can't be used for testing the cm_dll_sdc4. Value 1 - External clock can be used as a test clock.
25	RX_FLOW_TIMING	Configuration bit which selects the cycle which RxFlowControl will be asserted when UHS mode is used Value 0 (default) - RxFlowControl is asserted one clock before the end bit is received in DPSM (NextBlkTCnt in DPSM equals to 1). Value 1 - RxFlowControl is asserted two clocks before the end bit is received in DPSM (NextBlkTCnt in DPSM equals to 2).
24:23	SDC4_MCLK_SEL	Selects the sdc4_mclk - input clock of cm_dll_sdc4 Value 1 - feedback clock from CLK pad Value 2 (default) - free running MCLK (gated between the transactions) Value 3 - free running MCLK divided by 2
22	CLK_INV	When set(1), the input clock which is inserted into the input macros is inverted.
21	IO_PAD_PWR_SWITCH	Indication to Pad that power has to be switched from 3.3V to 1.8V.
20	CLK_FB_DLY_SEL	Selects the source of feedback clock used by the controller. Value 1 - output of cm_dll_sdc4 is used as feedback clock Value 0 - feedback clock from CLK pad is used
19:18	SD_DEV_SEL	Select the active device if more than one device are connected in shared bus mode. (range: '00' - '11').
17	HCLKON_SW_EN	SW enable of AHB clock request of SDCC5. value 0: HW clock request mechanism is used value 1: clock request signal is always high
16:14	SELECT_IN	Select to latch data and command coming in: value 000: on the falling edge of internal MCLK. value 001: on the rising edge of internal MCLK. value 010: using feedback clock (default). value 011: DDR mode. value 100: UHS mode - MCLK is used internally. value 101: UHS mode - divided frequency (MCLK/2) is used internally. value 110: e.MMC5.0 DDR200 mode.
13	INVERT_OUT	Clear (0) to change data and command going out on the falling edge. Set (1) to change data and command going out on the rising edge.
12	FLOW_ENA	Enable flow control: value 0: disable (default) value 1: enable

SDC2_SDCC_MCI_CLK (cont.)

Bits	Name	Description
11:10	WIDEBUS	Enable wide bus mode: value 00: 1 bit mode value 10: 4 bit mode value 01 or 11: 8 bit mode
9	PWRSAVE	Disable Prime-Cell MCI clock output when bus is idle to save power. value 0: Power save disabled - always enabled if bit 8 is set. value 1: Power save enabled - clock enabled only when bus is active and bit 8 is set.
8	ENABLE	Enable Prime-Cell MCI bus clock: value 0: clock disabled value 1: clock enabled

0x07864008 SDC2_SDCC_MCI_ARGUMENT**Type:** RW**Clock:** same rate as HCLK**Reset State:** 0x00000000**SDC2_SDCC_MCI_ARGUMENT**

Bits	Name	Description
31:0	CMD_ARG	Command argument.

0x0786400C SDC2_SDCC_MCI_CMD**Type:** RW**Clock:** same rate as HCLK**Reset State:** 0x00000000

(ARM name: MMCCCommand) CPSM: Command Path State Machine. This register must be updated to the MCLK domain so subsequent writes to this register will be

ignored until 3 MCLK and 3 HCLK have passed. SW should wait until bit 0 (MCLK_REG_WR_ACTIVE) of

MCI_STATUS2 register is 0, after writing to MCI_POWER register and before accessing this register

again.

SDC2_SDCC_MCI_CMD

Bits	Name	Description
17	AUTO_CMD21	If set (1), CPSM sends CMD21 automatically before sending CMD17 or CMD18. This sequence is needed for CDR auto-calibration of cm_dll_sdc4. CMD21 is supported in eMMC4.5 Spec.
16	AUTO_CMD19	If set (1), CPSM sends CMD19 automatically before sending CMD17 or CMD18. This sequence is needed for CDR auto-calibration of cm_dll_sdc4. CMD19 is supported in SD3.0 Spec.
15	CCS_DISABLE	If set (1), CPSM sends Command_Completion_Signal (CCS) disable sequence to the external CE-ATA device.
14	CCS_ENABLE	If set (1), CPSM waits for CCS from external card CE-ATA device.
13	MCIABORT	Signals the next command will be an abort (stop) command. This bit always read as (0) due to the hardware implementation. See a related note in next field
12	DAT_CMD	If set (1) indicates that this is a Command with Data. Note - MCIABORT and DAT_CMD bits do not have any effect on the CPSM. They are used to signal the SDIO Interrupt state machine that a Multi Block (examples: CMD18, CMD25, CMD53, etc. The rule is that each command with data must set this bit) data transaction is taking place. The DAT_CMD bit signals the start of the data transaction while the MCIABORT bit signals the end. For a pure SD Memory card, these bits are a don't care and should be set to zero by convention. However, some software drivers may chose to manipulate the bits in order to maintain compatibility with Combo cards and/or pure SDIO cards.
11	PROG_ENA	If set (1), PROG_DONE status bit will be asserted when busy is de-asserted. This bit is to be used with a stop or status command after a block write is performed. Does not effect CPSM.
10	ENABLE	The action of writing to this register (MCI_CMD) with this bit set (1), triggers CPSM to leave the Idle State.
9	PENDING	If set (1), CPSM waits for 'CmdPend' from the DPSM before it starts sending a command. This bit is designed to be used with SD CMD18 (READ_MULTIPLE_BLOCK) and SD CMD25 (WRITE_MULTIPLE_BLOCK) which must be terminated with a SD CMD12 (STOP_TRANSMISSION). After CMD18 or CMD25 is sent, the DPSM is enabled to send or receive data of amount MCI_DATA_LENGTH. After the DPSM finishes the transferring the data, it sends ¡CmdPend! to the CPSM and goes idle. When the CPSM receives this signal, it sends the currently loaded command which software would normally establish as CMD12.

SDC2_SDCC_MCI_CMD (cont.)

Bits	Name	Description
8	INTERRUPT	If set (1), CPSM disables command timer and waits for interrupt request (Card Response). The START, CMD_INDEX, ARGUMENT, and CRC are a total of 48 bits in length. If the INTERRUPT bit is zero, the CPSM will wait (63-47) 16 MCLK ticks for a response before going back to IDLE. If the INTERRUPT bit is set, the CPSM will wait indefinitely for a card response.
7	LONGRSP	If set (1), receives a 136-bit long response.
6	RESPONSE	If set (1), CPSM waits for a response subject to INTERRUPT above.
5:0	CMD_INDEX	Command index.

0x07864010 SDC2_SDCC_MCI_RESP_CMD**Type:** R**Clock:** same rate as HCLK**Reset State:** 0x00000000

(ARM name: MCIRspCmd)

SDC2_SDCC_MCI_RESP_CMD

Bits	Name	Description
5:0	RESPCMD	Response command index.

**0x07864014+ SDC2_SDCC_MCI_RESPn, n=[0..3]
0x4*n****Type:** R**Clock:** same rate as HCLK**Reset State:** 0x00000000

(ARM name: MCIResponse0-3). Note that MCIResponse3 has 31 bits; the other three (*RESP0, *RESP1, *RESP2) have 32 bits.

MCI_RESPn, n=[0..3]

SDC2_SDCC_MCI_RESPn

Bits	Name	Description
31:0	STATUS	Card status.

0x07864024 SDC2_SDCC_MCI_DATA_TIMER**Type:** RW**Clock:** same rate as HCLK**Reset State:** 0x00000000

(ARM name: MCIDataTimer).

There is no way to disable this timer. The timer starts counting as soon as a transaction is initiated. The time counts in MCLK ticks.

SDC2_SDCC_MCI_DATA_TIMER

Bits	Name	Description
31:0	DATA_TIME	Data timeout period.

0x07864028 SDC2_SDCC_MCI_DATA_LENGTH**Type:** RW**Clock:** same rate as HCLK**Reset State:** 0x00000000

(ARM name: MCIDataLength)

Total number of bytes in the transaction regardless of mode (stream or block). In block mode, must be an exact multiple of block size. During an infinite transfer the value of DATALENGTH should be programmed to 0.

SDC2_SDCC_MCI_DATA_LENGTH

Bits	Name	Description
24:0	DATALLENGTH	Data length value.

0x0786402C SDC2_SDCC_MCI_DATA_CTL**Type:** RW**Clock:** same rate as HCLK**Reset State:** 0x00100000

(ARM name: MCIDataCtrl). Data Path State Machine (DPSM). This register must be updated to the MCLK domain so subsequent writes to this register will be

ignored until 3 MCLK and 3 HCLK have passed. SW should wait until bit 0 (MCLK_REG_WR_ACTIVE) of

MCI_STATUS2 register is 0, after writing to MCI_POWER register and before accessing this register

again.

SDC2_SDCC_MCI_DATA_CTL

Bits	Name	Description
21	SW_SDC4_CMD19	The input of cm_dll_sdc4 - sdc4_cmd19 can be driven by SW if AUTO_CMD19 (bit 16 in MCI_CMD) feature is not used. Value 1 - sdc4_cmd19 is driven by SW Value 0 (default) - sdc4_cmd19 is driven by HW
20	RX_DATA_PEND	If set (1), timeout counter will start counting only after the RX command (with data) was sent instead of counting from initialization of DPSM. The feature is enabled by default.
19	AUTO_PROG_DONE	If set (1), automatic detection of PROG_DONE condition is executed without sending CMD12, CMD13, CMD52 or any other 'dummy' command. SW should set this bit in the following cases only: 1. When sending CMD53 for SDIO write transaction. 2. When sending CMD24. 3. When sending CMD25 and CMD23 was issued before to inform card about the exact number of blocks to be written (The rule is that this bit can be used for each command with data that doesn't need CMD12 for terminating the data or when it's prefixed by CMD23). 4. When sending CMD19 for testing bus procedure. CMD12 is not required in this case.
18	INFINITE_TRANSFER	If set (1), infinite transfer is enabled. MCI_DATA_LENGTH register should set to 0 during infinite transfer.
17	DATA_PEND	If set (1), DPSM waits for 'DataPend' from the CPSM before it enables the DPSM. This bit is designed to be used with SD CMD24 and CMD25 (WRITE_SINGLE_BLOCK and WRITE_MULTIPLE_BLOCK) to automatically start the DPSM after a normal (non-error) response is received. This register should be written with the enable bit and the pending bit asserted before MCI_CMD is enabled.
16:4	BLOCKSIZE	Data block length in bytes (1 to 4096).
3	DM_ENABLE	Enable DM interface: 0: DM disabled 1: DM enabled
2	MODE	Data transfer mode: 0: block data transfer 1: stream data transfer
1	DIRECTION	Data transfer direction: 0: controller to card 1: card to controller
0	ENABLE	Data transfer enabled. If zero, DPSM unconditionally reset.

0x07864030 SDC2_SDCC_MCI_DATA_COUNT**Type:** R**Clock:** same rate as HCLK**Reset State:** 0x00000000

(ARM name: MCIDataCnt)

NOTE There is no mechanism to ensure that a read of this counter will be accurate as it is not synchronized. The reason being the value read is async to the clock domain in which the reading is done. It is only useful as a debug tool for diagnostic purposes only.

SDC2_SDCC_MCI_DATA_COUNT

Bits	Name	Description
24:0	DATA_COUNT	Value of data counter in MciDPSM block. Represents remaining data of transaction.

0x07864034 SDC2_SDCC_MCI_STATUS**Type:** R**Clock:** same rate as HCLK**Reset State:** 0x000C0000

(ARM name: MCIStatus)

This register may need exclusion for QCSR POR Test for bits 25 and 22.

Static[30:26], [24:23] and [10:0]

These remain asserted until they are cleared by writing to the appropriate bit in the Clear Register (see MCI_CLEAR)

XCIS

The means used to send the SDIO external-card-interrupt-signal (XCIS) to the SDCC depends on the transfer mode: 1-bit, 4-bit, or 8-bit. Under 1 bit transfer mode, the XCIS has a dedicated physical connection to the SDCC. Hence once it is asserted, it remain at a high level. Under 4 and 8 bit transfe modes, the XCIS is time multiplexed with physical connection data bit 1. The XCIS is recognized as valid during specific time slots on data bit 1 by the SDCC. How this 'recognition' takes places shows up as different behaviors on SDIO_INTR_OPER, SDIO_INTR, and the actual interrupt sent to the ARM through MCI_INT_MASKn.

Special[25]

The SDIO_INTR_OPER indicator reflects the true state of XCIS as it would be observed inside the external SDIO card. As such, there is no bit 25 in MCI_CLEAR. The only way to clear SDIO_INTR_OPER is to clear the appropriate bit in the external card Common Card Control Registers (CCCR). Like 'static' above and 'dynamic' below, bit 25 in MCI_INT_MASKn controls

actually sending this indicator value to the ARM. This indicator of XCIS is used for normal SDCC operation.

Special[22]

The SDIO_INTR indicator reflects that there has been a low-to-high transition on the connection as explained under XCIS above. This indicator is cleared via bit 22 of MCI_CLEAR but any low-to-high transition on the connection will set this indicator again. Further note that this indicator is not routed through the MCI_INT_MASKn to the ARM. For bit 22, the raw value of the connection is routed through the MCI_INT_MASKn to the ARM. As this raw connection is not aware of specific time slots, the signal sent to the ARM generally toggles in an unpredictable fashion. This is generally not useful during normal SDCC operation. However, for wake-up with clocks off, the raw connection is desirable.

Dynamic[21:11]

These change state depending on the state of the underlying logic (for example, FIFO full and empty flags are asserted and de-asserted as data while written to the FIFO).

SDC2_SDCC_MCI_STATUS

Bits	Name	Description
31	STATUS2_INT	One of the error bits in MCI_STATUS2[4:1] register is set
30	AUTO_CMD19_TIMEOUT	Response or tuning pattern wasn't received after automatic CMD19's transmission. This status bit should be read only if AUTO_CMD19 (MCI_CMD(16)) was set.
29	BOOT_TIMEOUT	Data wasn't received within the valid time (according to MCI_CCS_TIMER) from the start of boot operation.
28	BOOT_ACK_ERR	Acknowledge pattern wasn't received correctly or not within the valid time (according to MCI_ACK_TIMER) from the start of boot operation.
27	BOOT_ACK_REC	Acknowledge pattern was received correctly.
26	CCS_TIMEOUT	CE-ATA Command Completion Signal timeout.
25	SDIO_INTR_OPER	SDIO interrupt indicator for normal operation.
24	ATA_CMD_COMPL	CE-ATA Command Completion Signal has been detected.
23	PROG_DONE	Programming done.
22	SDIO_INTR	SDIO interrupt indicator for wake-up.
21	RXDATA_AVLBL	Data available in receive FIFO. At least 1 word in the RX FIFO. SW can read 1 word only from the FIFO.
20	TXDATA_AVLBL	Data available in transmit FIFO. At least 1 word in the TX FIFO.
19	RXFIFO_EMPTY	Receive FIFO empty. SW can't read FIFO.
18	TXFIFO_EMPTY	Transmit FIFO empty. SW can write 8 words into the FIFO.
17	RXFIFO_FULL	Receive FIFO full. SW can read 8 words from the FIFO.

SDC2_SDCC_MCI_STATUS (cont.)

Bits	Name	Description
16	TXFIFO_FULL	Transmit FIFO full. TX FIFO contains 8 words. SW can't write to FIFO.
15	RXFIFO_HALF_FULL	Receive FIFO half full. SW can read 8 words from the FIFO.
14	TXFIFO_HALF_FULL	Transmit FIFO half full. SW can write 8 words into the FIFO.
13	RXACTIVE	Data receive in progress.
12	TXACTIVE	Data transmit in progress.
11	CMD_ACTIVE	Command transfer in progress.
10	DATA_BLK_END	Data block sent / received (CRC check passed).
9	START_BIT_ERR	Start Bit Error flag
8	DATAEND	Data end (data counter is zero).
7	CMD_SENT	Command sent (no response required).
6	CMD_RESPONSE_END	Command response received (CRC check passed).
5	RX_OVERRUN	Receive FIFO overrun error.
4	TX_UNDERRUN	Transmit FIFO underrun error.
3	DATA_TIMEOUT	Data timeout.
2	CMD_TIMEOUT	Command response timeout.
1	DATA_CRC_FAIL	Data block sent / received (CRC check failed).
0	CMD_CRC_FAIL	Command response received (CRC check failed).

0x07864038 SDC2_SDCC_MCI_CLEAR**Type:** W**Clock:** same rate as HCLK**Reset State:** 0x00000000

(ARM name: MCIClear)

SDC2_SDCC_MCI_CLEAR

Bits	Name	Description
30	AUTO_CMD19_TIMEOUT_CLR	Clears AUTO_CMD19_TIMEOUT flag.
29	BOOT_TIMEOUT_CLR	Clears BOOT_TIMEOUT flag.
28	BOOT_ACK_ERR_CLR	Clears BOOT_ACK_ERR flag.
27	BOOT_ACK_REC_CLR	Clears BOOT_ACK_REC flag.
26	CCS_TIMEOUT_CLR	Clears CCSTimeOut flag.

SDC2_SDCC_MCI_CLEAR (cont.)

Bits	Name	Description
24	ATA_CMD_COMPL_CLR	Clears AtaCmdCompl flag.
23	PROG_DONE_CLR	Clears ProgDone flag.
22	SDIO_INTR_CLR	Clears SDIOInt flag.
10	DATA_BLK_END_CLR	Clears DataBlockEnd flag.
9	START_BIT_ERR_CLR	Clears StartBitErr flag.
8	DATA_END_CLR	Clears DataEnd flag.
7	CMD_SENT_CLR	Clears commandSent flag.
6	CMD_RESP_END_CLT	Clears CmdRespEnd flag.
5	RX_OVERRUN_CLR	Clears RxOverrunClr flag.
4	TX_UNDERRUN_CLR	Clears TxUnderrun flag.
3	DATA_TIMEOUT_CLR	Clears DataTimeOut flag.
2	CMD_TIMEOUT_CLR	Clears CmdTimOutflag.
1	DATA_CRC_FAIL_CLR	Clears DataCrcFail flag.
0	CMD_CRC_FAIL_CLR	Clears CmdCrcFail flag.

**0x0786403C+ SDC2_SDCC_MCI_INT_MASKn, n=[0..1]
0x4*n****Type:** RW**Clock:** same rate as HCLK**Reset State:** 0x00000000

(ARM name: MCIMask0)

SDC2_SDCC_MCI_INT_MASKn

Bits	Name	Description
31	MASK31	MASK STATUS2_INT flag.
30	MASK30	MASK AUTO_CMD19_TIMEOUT flag.
29	MASK29	MASK BOOT_TIMEOUT flag.
28	MASK28	MASK BOOT_ACK_ERR flag.
27	MASK27	MASK BOOT_ACK_REC flag.
26	MASK26	MASK CCSTimeOut flag.
25	MASK25	MASK SDIOIntOper flag.
24	MASK24	MASK AtaCmdCompl flag.
23	MASK23	MASK ProgDone flag.

SDC2_SDCC_MCI_INT_MASKn (cont.)

Bits	Name	Description
22	MASK22	MASK SDIOInt flag.
21	MASK21	MASK RxDataAvlbl flag.
20	MASK20	MASK TxDataAvlbl flag.
19	MASK19	MASK RxFifoEmpty flag.
18	MASK18	MASK TxFifoEmpty flag.
17	MASK17	MASK RxFifoFull flag.
16	MASK16	MASK TxFifoFull flag.
15	MASK15	MASK RxFifoHalfFull flag.
14	MASK14	MASK TxFifoHalfFull flag.
13	MASK13	MASK RxActive flag.
12	MASK12	MASK TxActive flag.
11	MASK11	MASK CmdActive flag.
10	MASK10	MASK DataBlockEnd flag.
9	MASK9	MASK StartBitErr
8	MASK8	MASK DataEnd flag.
7	MASK7	MASK CmdSent flag.
6	MASK6	MASK CmdRespEnd flag.
5	MASK5	MASK RxOverrun flag.
4	MASK4	MASK TxOverrun flag.
3	MASK3	MASK DataTimeOut flag.
2	MASK2	MASK CmdTimeOut flag.
1	MASK1	MASK DataCmdCrcFail flag.
0	MASK0	MASK CmdCrcFail flag.

0x07864044 SDC2_SDCC_MCI_FIFO_COUNT**Type:** R**Clock:** same rate as HCLK**Reset State:** 0x00000000

The SDCC core operates in two clock domains MCLK and HCLK. The CPSM and DPSM operate in MCLK domain and MciFifoDmaCtl (FIFO/DMA Controller) operates in HCLK domain. The MCI_FIFO_COUNT is a counter of FIFO Controller that monitors how many words of data are still needed to be transferred through the FIFO. At the beginning of data transaction the MCI_FIFO_COUNT counter will be loaded with the (MCI_DATA_LENGTH / 4) value and then

will count down during the data transaction until it reach zero. This register is only useful for debug purposes and should not be used for normal operation since it does not reflect data which may or may not be in the pipeline.

SDC2_SDCC_MCI_FIFO_COUNT

Bits	Name	Description
23:0	DATA_COUNT	Remaining data.

0x07864048 SDC2_SDCC_MCI_BOOT

Type: RW

Clock: same rate as HCLK

Reset State: 0x00000000

Handling the boot operation.

SDC2_SDCC_MCI_BOOT

Bits	Name	Description
3	EARLY_ASSERT_CMD_LINE	If set to '1' then the CMD line is asserted by HW when all the data was received on the SD bus
2	BOOT_ACK_EN	If set to '1' then Host waits for acknowledge pattern after initiating the boot operation.
1	BOOT_EN	When this bit is asserted, the boot operation is initiated in both of the modes.
0	BOOT_MODE	If set to '1' then CMD line is low during the boot operation. If boot_mode = '0', then CMD0 with the argument 0xFFFFFFFFFA is sent.

0x0786404C SDC2_SDCC_MCI_BOOT_ACK_TIMER

Type: RW

Clock: same rate as HCLK

Reset State: 0x00000000

Timer for boot operation - the time until receiving the acknowledge pattern.

SDC2_SDCC_MCI_BOOT_ACK_TIMER

Bits	Name	Description
31:0	BOOT_ACK_TIMER	Timer for counting the cycles from initiating the boot operation until the acknowledge pattern is accepted.

0x07864054 SDC2_SDCC_MCI_EMULATION_DLY_LINE**Type:** RW**Clock:** same rate as HCLK**Reset State:** 0x00000000

The register is responsible for handling DCM which is used in emulation supports 256 values of shift intervals so 8 bits are required (0:255). Bits [31:30] are used for emulation purposes only.

The procedure for activating the DCM is:

1. Setting bit 28 - DCM_RESET.
2. Clearing bit 28 - DCM_RESET.
3. Waiting until bit 29 DCM_LOCKED is high.
4. Writing a new phase in SD_CLK_DLY_CTRL field and setting the DCM_START bit (bit 30).

The clock's placement was executed by the DCM when DCM_DONE is asserted.

SDC2_SDCC_MCI_EMULATION_DLY_LINE

Bits	Name	Description
31	DCM_DONE	The shift process was finished in DCM and CMD19 can be sent towards the card. This is a read-only bit.
30	DCM_START	A new value of SD_CLK_DLY_CTRL is ready. Clears the DCM_DONE bit.
29	DCM_LOCKED	Reading the status of the LOCKED DCM's output which indicates that the output clock is ready for use.
28	DCM_RESET	Driving reset to DCM.
7:0	SD_CLK_DLY_CTRL	Input to DCM block in FPGA (8 bits). Controlling the change of the delay interval. The value's range may be 0-255 in DCM block.

0x07864058 SDC2_SDCC_MCI_CCS_TIMER**Type:** RW**Clock:** same rate as HCLK**Reset State:** 0x00000000

This register is used for operation with CE-ATA devices. The CCS timer starts to count at the end of SD transaction and it advances with every MCLK clock cycle. In boot operation this register is used for storing the number of cycles from initiating the boot operation until the first data is received.

SDC2_SDCC_MCI_CCS_TIMER

Bits	Name	Description
31:0	CCS_TIMER	CE-ATA Command Completion Signal timeout period.

0x0786405C SDC2_SDCC_MCI_RESPONSE_MASK**Type:** RW**Clock:** same rate as HCLK**Reset State:** 0x00000000**SDC2_SDCC_MCI_RESPONSE_MASK**

Bits	Name	Description
31:0	RESPONSE_MASK	Mask used to verify the 32 status error bits. Active only when the data pending bit is set in the MCI_DATA_CTL register. When bit of this mask is set the corresponding bit from response [39:8] is checked to be 0 i.e. shows there no error. If all bits from mask shows no error the DPSM will be activated without waiting for software write.

0x07864060 SDC2_SDCC_MCI_DLL_CONFIG**Type:** RW**Clock:** same rate as HCLK**Reset State:** 0x60006400

This register configures the cm_dll_sdc4 for UHS mode.

SDC2_SDCC_MCI_DLL_CONFIG

Bits	Name	Description
31	SDC4_DIS_DOUT	When this bit equals to '0', the data which comes out from the dll is valid. When it equals to '1' this data isn't valid. The inversion of this bit is connected to dll's input - dll_en_dout
30	DLL_RST	Setting this bit to 1 resets cm_dll_sdc4. cm_dll_sdc4 should be reset every time the MCLK frequency is changed.
29	PDN	Power Down Value 0 - analog blocks are enabled Value 1 - analog blocks are powered down (default)
28	CK_INTP_SEL	Selects interpolator output
27	CK_INTP_EN	Enable clock interpolation for finer resolution

SDC2_SDCC_MCI_DLL_CONFIG (cont.)

Bits	Name	Description
26:24	MCLK_FREQ	Frequency of MCLK Value 000 -100 - 112 (MHz) Value 001 -112 - 125 Value 010 -125 - 137 Value 011 -137 - 150 Value 100 -150 - 162 Value 101 -162 - 175 Value 110 -175 - 187 Value 111 -187 - 200
23:20	CDR_SELECT	CDR phase select (gray coded) Value 0000 - phase 0 Value 0001 - phase 1 Value 0011 - phase 2 Value 0010 - phase 3 Value 0110 - phase 4 Value 0111 - phase 5 Value 0101 - phase 6 Value 0100 - phase 7 Value 1100 - phase 8 Value 1101 - phase 9 Value 1111 - phase 10 Value 1110 - phase 11 Value 1010 - phase 12 Value 1011 - phase 13 Value 1001 - phase 14 Value 1000 - phase 15
19	CDR_EXT_EN	Enable external control of cdr phase select
18	CK_OUT_EN	Enable output clock (default value is '1')
17	CDR_EN	Enable CDR function
16	DLL_EN	Enable DLL function
15:14	CDR_UPD_RATE	CDR update rate, low pass filtering window of CDR Max update rate: Value 00 - 0.5 MCLK frequency Value 01 - 0.25 MCLK Frequency (default) Value 10 - 0.125 MCLK frequency Value 11 - 1/16 MCLK Frequency
13:12	DLL_UPD_RATE	DLL update rate Value 00 - sdc4_mclk/10 Value 01 - sdc4_mclk/20 Value 10 - sdc4_mclk/40 (default) Value 11 - sdc4_mclk/80

SDC2_SDCC_MCI_DLL_CONFIG (cont.)

Bits	Name	Description
11:10	DLL_PHASE_DET	DLL phase detector low pass average window Value 00 - 2 cycles Value 01 - 4 cycles (default) Value 10 - 8 cycles Value 11 - 16 cycles
9	CDR_ALGORITHM_SEL	CDR algorithm select Value 0 - Edge (default) Value 1 - Level
0	CMD_DAT_TRACK_SEL	Select DAT or CMD lines for CDR tracking Value 0 - track DAT inputs (default) Value 1 - track CMD input

0x07864068 SDC2_SDCC_MCI_DLL_STATUS**Type:** R**Clock:** same rate as HCLK**Reset State:** 0x00000000

This register is connected to sdc4_status output of cm_dll_sdc4.

SDC2_SDCC_MCI_DLL_STATUS

Bits	Name	Description
12	SDC4_DTEST_MUXSEL	Digital test mux select
11	DDR_DLL_LOCK_JDR	Reserved
10:9	SDC4_DLL_DTEST_OUT_A TPG	SDC4_DLL_DTEST_OUT output from DLL after a clock mux with sampled JDRs for ATPG mode. This bit helps collecting coverage on DLL's output
8	SDC4_DLL_LOCK_ATPG	SDC4_DLL_LOCK output from DLL after a clock mux with sampled JDRs for ATPG mode. This bit helps collecting coverage on DLL's output
7	DLL_LOCK	DLL lock status Value 0 - Not locked Value 1 - Locked

SDC2_SDCC_MCI_DLL_STATUS (cont.)

Bits	Name	Description
6:3	CDR_PHASE	CDR phase select (gray coded) Value 0000 - phase 0 Value 0001 - phase 1 Value 0011 - phase 2 Value 0010 - phase 3 Value 0110 - phase 4 Value 0111 - phase 5 Value 0101 - phase 6 Value 0100 - phase 7 Value 1100 - phase 8 Value 1101 - phase 9 Value 1111 - phase 10 Value 1110 - phase 11 Value 1010 - phase 12 Value 1011 - phase 13 Value 1001 - phase 14 Value 1000 - phase 15
2	DDLL_COARSE_CAL	Value 1 - done Value 0 - not done

0x0786406C SDC2_SDCC_MCI_STATUS2**Type:** R**Clock:** same rate as HCLK**Reset State:** 0x00000000

Additional status and error bits.

SDC2_SDCC_MCI_STATUS2

Bits	Name	Description
8	BAM_AXI_MASTER_ERR	Error occurred on AXI master for BAM transaction. It is enabled if the field BAM_ERR_EN is set in MCI_HC_MODE register
7	AUTO_CDC_SW_CALIB_ERR	calibration error is received (CDC_ERROR_CODE isn't 0) during auto SW CDC calibration.
6	AUTO_CDC_SW_CALIB_TIMEOUT	CALIBRATION_DONE isn't received within the number of cycles which configured in CDC_CAL_TIMEOUT_THRESHOLD field (from CDC_AUTO_SW_CAL_CFG register).
5	DATA_AXI_MASTER_ERR	Error occurred on AXI master for data transaction
4	DATA_END_BIT_ERROR	The bit is set when the end bit of the received block was not 1 on all the data lines
3	CMD_END_BIT_ERROR	The bit is set when the end bit of the received response was not 1 on the command line

SDC2_SDCC_MCI_STATUS2 (cont.)

Bits	Name	Description
2	FIFO_EMPTY_ERROR	FIFO read attempt when FIFO was empty
1	FIFO_FULL_ERROR	write attempt when FIFO was full
0	MCLK_REG_WR_ACTIVE	Value 0 - No active register write to MCLK domain Value 1 - Active register write to MCLK domain. The bit indicates if a write operation to one of the following registers is in process (synchronization between HCLK and MCLK): MCI_POWER MCI_CLOCK MCI_CMD MCI_DATA_CTL

0x07864070 SDC2_SDCC_MCI_GENERICS**Type:** R**Clock:** same rate as HCLK**Reset State:** 0x00000000

Generics list of each SDCC instance.

SDC2_SDCC_MCI_GENERICS

Bits	Name	Description
29	SWITCHABLE_SIGNALING_VOLTAGE	The core supports switching of IO signaling voltage.
28	BUS_18V_SUPPORT	The core supports 1.8v power supply to the bus.
27	BUS_30V_SUPPORT	The core supports 3.0v power supply to the bus.
26:23	SD_DATA_WIDTH	Number of DAT lines Value 4 - 4 DAT lines (SD, SDIO) Value 8 - 8 DAT lines (MMC, eMMC)
22:10	RAM_SIZE	Size of RAM size: Optional Values: 512, 1024, 2048 or 4096 bytes.
9	USE_SPS	Value 1 - BAM and DML are used and USB core is connected to SDCC5, Value 0 - BAM and DML are not used.
8:6	NUM_OF_DEV	Number of eSD or eSDIO devices supported on shared SD bus. Can be set from 1 to 4.
5:1	MAX_PIPES	Number of simultaneous parallel pipes supported by the BAM and attached peripheral pair. Can be set from 1 to 31.

SDC2_SDCC_MCI_GENERIC (cont.)

Bits	Name	Description
0	USE_DLL_SDC4	Enables the instantiation of cm_dll_sdc4. Value 0 - cm_dll_sdc4 isn't integrated with SDCC5 Value 1 - cm_dll_sdc4 is used with SDCC5.

0x07864074 SDC2_SDCC_MCI_FIFO_STATUS**Type:** R**Clock:** same rate as HCLK**Reset State:** 0x00007800

Status bit of FIFO's fill level. Can be used for long accesses to FIFO using the FIFO_ALT register

SDC2_SDCC_MCI_FIFO_STATUS

Bits	Name	Description
18	RX_FIFO_512	512 bytes are ready in FIFO during RX transaction
17	RX_FIFO_256	256 bytes are ready in FIFO during RX transaction
16	RX_FIFO_128	128 bytes are ready in FIFO during RX transaction
15	RX_FIFO_64	64 bytes are ready in FIFO during RX transaction
14	TX_FIFO_512	available space in FIFO for 512 bytes during TX transaction
13	TX_FIFO_256	available space in FIFO for 256 bytes during TX transaction
12	TX_FIFO_128	available space in FIFO for 128 bytes during TX transaction
11	TX_FIFO_64	available space in FIFO for 64 bytes during TX transaction
10:0	FIFO_FILL_LEVEL	reflects the actual fill level of FIFO DPRAM during TX or RX transaction. Supported values are 0, 4, 8, ..., 1020, 1024

0x07864078 SDC2_SDCC_MCI_HC_MODE**Type:** RW**Clock:** same rate as HCLK**Reset State:** 0x00000000

Enabling the Host Controller Specification

SDC2_SDCC_MCI_HC_MODE

Bits	Name	Type	Description
21	PROGDONE_WO_CMD_RESP	RW	'0' (Default): we wait for cmd response end to detect progdone (legacy behavior). '1': we start looking for progdone without waiting for cmd response end but after 10 cycles from end bit of command
20	AUTO_CMD12_CLR_CMDACTIVE	RW	'0' (Default): auto_cmd12 is not clearing cmdactive (bug fix). '1': auto_cmd12 is clearing cmdactive like previous behavior
19	WRAP_ERROR	RW	When the bit is set, the core responds with errors on WRAP transaction in configuration bus.
18:17	BUSY_CHECK_VALID_PERIOD	RW	Configuring the wait period between CRC token to BUSY period. Value 0 - waiting 7 cycles. The values of BUSY_CHECK_VALID_PERIOD is added to 7 cycles
16	BUSY_CHECK_VALID_ALWAYS	RW	In DDR200 mode, the end of busy should be checked after the CRC token was received. There is a counter of 4 cycles for this purpose. If this bit is set then the end of busy will be checked right after the crc token.
15	BAM_ERR_EN	RW	Enabling the assertion of BAM_AXI_MASTER_ERR field in MCISTATUS2
14	DIS_RST_AFTER_CRC_TO_KEN_DDR200	RW	Disabling the option to reset the RCLK's data registers in IO macro after each CRC token. The reset is needed if redundant RCLK is sent and BUSY is sampled into the FIFO as a data bit.
13	FF_CLK_SW_RST_DIS	RW	Disabling the SW reset in FF clock domain
12	IO_MACRO_SW_RST_PERIOD	R	When moving to DDR mode, reset period is generated for dividers. There is possibility to read from this bit if this period was finished
11	CLOCK_AFTER_EOB_DIS	RW	Disabling the 8 cycles after the CRC token when TxFlowControl is asserted before the end of the block
10	ADMA_INT_DATA	RW	If this bit is set, then the ADMA interrupt is set when the data was already transferred. If this bit is cleared, then the ADMA interrupt is set when the requests related to the descriptors were finished on address channel before the data itself is processed
9	BAM_CLK_EN_ACT	RW	If this bit is set, then bam_clk is gated when hc mode is enabled
8	ADMA_HPROT_DIS	R	If cleared then HPROT is changed in last burst of write transaction if block_count_en is 1, otherwise it will be for each end of descriptor. Not applicable for Gandalf - AXI Noc
7	ADMA_NON_WRD_ALIGN_DIS	R	If set the fix for non word aligned transfers in ADMA is disabled. Not applicable for Gandalf - AXI Noc

SDC2_SDCC_MCI_HC_MODE (cont.)

Bits	Name	Type	Description
6	DEASSERT_HREADY_DAT_DIS	RW	if set, then hready won't be deasserted for access to Transfer Mode register in HC address. By default, the hready is deasserted for enabling the completion of synchronization period to MCLK before the next access
5	DEASSERT_HREADY_CMD_DIS	RW	if set ,then hready won't be deasserted for access to Command register in HC address. By default, the hready is deasserted for enabling the completion of synchronization period to MCLK before the next access
4	IRQ_PCLK_DIS	RW	From tag sdcc5_p3q3r50, the interrupt and status bits are handled in PCLK domain only. If this bit is set, then the legacy architecture is used
3	SINGLE_NON32_ERROR	RW	if set , then a response error will be sent if single configuration access is executed in legacy mode with size of non 32 bits
2	WAIT_DLL_LOCK	RW	If this bit is set, the host controller isn writing to the async fifo unless DLL_LOCK of MCI_DLL_STATUS is equals to '1'
1	CYCLES_AFTER_EOB_DIS	RW	Enable bit for the fix that TxFlowControl isn't set until end of the block (CRC token reception) in the middle of TX transaction. The fix is required for releasing each BAM's descriptor seperately without waiting to the next descriptor.
0	HC_MODE_EN	RW	Enabling the Host Controller Specification by setting this bit

0x07864080+ SDC2_SDCC_MCI_FIFO_n, n=[0..15]
0x4*n

Type: RW

Clock: same rate as HCLK

Reset State: 0x00000000

SDC2_SDCC_MCI_FIFO_n

Bits	Name	Description
31:0	DATA	FIFO data. This register is aliased to 16 words, 0x080 - 0xBC. In SDCC3, an access to the FIFO was executed with 16 words so 16 addresses were defined. However, in SDCC5 only 8 words are used in each burst so only 8 addresses from the range should be used. When writing to the external SD_Card, writes to the data path FIFO. When reading from the external SD_Card, reads from the data path FIFO.

0x078640DC SDC2_SDCC_MCI_PWRCTL_STATUS_REG**Type:** R**Clock:** PCLK**Reset State:** 0x00000000

HC Spec Defines power and IO voltage switch bits. The information is sent by a dedicated interrupt to the QCT SW driver for handling the power and voltage tasks with PMIC. The different tasks are defined in this register

SDC2_SDCC_MCI_PWRCTL_STATUS_REG

Bits	Name	Description
3	IO_HIGH_V	SDCC driver orders that the I/O signaling level be switched to 3.0v. This interrupt is triggered when SDCC driver writes '0' to offset 03Eh in HC register (bit 3) or when SW reset for all is handled and last supported voltage was 1.8V. This logic is enabled when SWITCHABLE_SIGNALING_VOLTAGE=true
2	IO_LOW_V	SDCC driver orders that the I/O signaling level be switched to 1.8v. This interrupt is triggered when SDCC driver writes '1' to offset 03Eh in HC register (bit 3). This logic is enabled when SWITCHABLE_SIGNALING_VOLTAGE=true
1	BUS_ON	SDCC driver orders that the SD bus power be switched on. This interrupt is triggered when SDCC driver writes '1' to offset 029h in HC register (bit 0).
0	BUS_OFF	SDCC driver orders that the SD bus power be switched off. This interrupt is triggered when SDCC driver writes '0' to offset 029h in HC register(bit 0) or when SW reset for all is handled.

0x078640E0 SDC2_SDCC_MCI_PWRCTL_MASK_REG**Type:** RW**Clock:** PCLK**Reset State:** 0x00000000

Mask register for MCI_PWRCTL_STATUS_REG

SDC2_SDCC_MCI_PWRCTL_MASK_REG

Bits	Name	Description
3	IO_HIGH_V	'1' - interrupt is enabled / '0' - interrupt is masked.
2	IO_LOW_V	'1' - interrupt is enabled / '0' - interrupt is masked.
1	BUS_ON	'1' - interrupt is enabled / '0' - interrupt is masked.
0	BUS_OFF	'1' - interrupt is enabled / '0' - interrupt is masked.

0x078640E4 SDC2_SDCC_MCI_PWRCTL_CLEAR_REG**Type:** W**Clock:** PCLK**Reset State:** 0x00000000

Clear register for MCI_PWRCTL_STATUS_REG

SDC2_SDCC_MCI_PWRCTL_CLEAR_REG

Bits	Name	Description
3	IO_HIGH_V	Writing '1' clears bit 03 in PWRCTL_STATUS_REG
2	IO_LOW_V	Writing '1' clears bit 02 in PWRCTL_STATUS_REG
1	BUS_ON	Writing '1' clears bit 01 in PWRCTL_STATUS_REG
0	BUS_OFF	Writing '1' clears bit 00 in PWRCTL_STATUS_REG

0x078640E8 SDC2_SDCC_MCI_PWRCTL_CTL_REG**Type:** RW**Clock:** PCLK**Reset State:** 0x00000000

HC Spec Defines power and IO voltage switch bits. The information is sent by a dedicated interrupt to the QCT SW driver for handling the power and voltage tasks with PMIC and then the QCT SW driver needs to report success or fail to this register

SDC2_SDCC_MCI_PWRCTL_CTL_REG

Bits	Name	Description
9	HC_CARD_STATE_STABLE	This bit is driving CARD_STATE_STABLE bit in Present State Register
8	RST_NOT_WAIT_PWRCTL_REG	RST process doesn't need to wait for accessing to MCI_PWRCTL_CTL_REG register
7:6	HC_WRITE_PROTECT_PAD	Value 0 inversion of HC_WRITE_PROTECT is connected to bit 19 in Present State Register. Value 1 input from PAD is used. Value 2 inversion of input from PAD is used
5	CARD_DETECT_PIN_LEVEL	This bit reflects the inverse value of the SDCD# pin. Debouncing is not performed on this bit. This bit may be valid when Card State Stable is set to 1, but it is not guaranteed because of propagation delay. Use of this bit is limited to testing since it must be debounced by software. value 1: card present value 0: no card present

SDC2_SDCC_MCI_PWRCTL_CTL_REG (cont.)

Bits	Name	Description
4	HC_WRITE_PROTECT	This bit allows to disable the writing. value 0: write enabled value 1: write protected
3	IO_SIG_SWITCH_FAIL	Host controller clears this bit when it fires either I/O Low V interrupt or I/O High V interrupt. SDCC power control software writes '1' to indicate the signal switch has failed.
2	IO_SIG_SWITCH_SUCCESS	Host controller clears this bit when it fires either I/O Low V interrupt or I/O High V interrupt. SDCC power control software writes '1' to indicate the signal switch has completed successfully.
1	BUS_ON_OFF_FAIL	Host controller clears this bit when it fires either Bus ON interrupt or Bus OFF interrupt. SDCC power control software writes '1' to indicate the requested operation has failed.
0	BUS_ON_OFF_SUCCESS	Host controller clears this bit when it fires either Bus ON interrupt or Bus OFF interrupt. SDCC power control software writes '1' to indicate the requested operation has completed successfully.

0x078640EC SDC2_SDCC_MCI_CLEAR2**Type:** W**Clock:** same rate as HCLK**Reset State:** 0x00000000

Clearing the error bits in MCI_STATUS2 register

SDC2_SDCC_MCI_CLEAR2

Bits	Name	Description
8	BAM_AXI_MASTER_ERR_CLR	BAM_AXI_MASTER_ERR clear bit
7	AUTO_CDC_SW_CALIB_ERR_CLR	AUTO_CDC_SW_CALIB_ERR clear bit.
6	AUTO_CDC_SW_CALIB_TOUT_CLR	AUTO_CDC_SW_CALIB_TOUT clear bit.
5	DATA_AXI_MASTER_ERR_CLR	DATA_AXI_MASTER_ERR clear bit
4	DATA_END_BIT_ERROR_CLR	DATA_END_BIT_ERROR clear bit
3	CMD_END_BIT_ERROR_CLR	CMD_END_BIT_ERROR clear bit
2	FIFO_EMPTY_ERROR_CLR	FIFO_EMPTY_ERROR clear bit
1	FIFO_FULL_ERROR_CLR	FIFO_FULL_ERROR clear bit

**0x078640F0+ SDC2_SDCC_MCI_INT_MASKINT2_n, n=[0..1]
0x4*n**
Type: RW**Clock:** same rate as HCLK**Reset State:** 0x00000000

Mask register for the MCI_STATUS2 register for each interrupt line

SDC2_SDCC_MCI_INT_MASKINT2_n

Bits	Name	Description
8	MASK8	BAM_AXI_MASTER_ERR mask bit
7	MASK7	AUTO_CDC_SW_CALIB_ERR mask bit.
6	MASK6	AUTO_CDC_SW_CALIB_TOUT mask bit.
5	MASK5	DATA_AXI_MASTER_ERR mask bit
4	MASK4	DATA_END_BIT_ERROR mask bit
3	MASK3	CMD_END_BIT_ERROR mask bit
2	MASK2	FIFO_EMPTY_ERROR mask bit
1	MASK1	FIFO_FULL_ERROR mask bit

0x07864100 SDC2_SDCC_CHAR_CMD
Type: RW**Clock:** HCLK**Reset State:** 0x00000000

Characterization config - 1 CMD line

SDC2_SDCC_CHAR_CMD

Bits	Name	Type	Description
15:8	CMDIN_ACTUAL	R	CMDIN storage for comparison. The incoming data is stored for debugging feature
7:0	CMDOUT_DATA_DIN_EXP	RW	CMDOUT for transmitting, CMDIN for compare. in TX testing the data is written in the register and should be transmitted outside from the core. In RX testing the incoming data is compared to the register's value

0x0786412C SDC2_SDCC_QSB_VALUES**Type:** R**Clock:** same rate as HCLK**Reset State:** 0x00007991

configurable QSB values. Not applicabale for Gandalf - AXI Noc

SDC2_SDCC_QSB_VALUES

Bits	Name	Description
15	NON_BUF_BAM_BLOCK_END	Non bufferable/strongly ordered (hprot == 4'b0001/ memtype == 0) is driven in end of transaction by default. If setting this bit it will be driven for each block end
14	NOALLOCATE_VALUE	Driving noallocate value
13	RESERVED	Reserved
12	HPROT_VALUE_EN	Enabling the hprot values
11:8	HPROT_FOR_RDATA	hprot for read data. should be 9 by default so memtype is 2 (cachable)
7:4	HPROT_FOR_WDATA	hprot for write data during transaction or block. should be 9 by default so memtype is 2 (cachable)
3:0	HPROT_FOR_LAST_WDATA	hprot for last write data. should be 1 by default so memtype is 0 (strongly ordered)

0x07864130 SDC2_SDCC_QSB_AXI_VALUES**Type:** RW**Clock:** same rate as HCLK**Reset State:** 0x3E2218A8

configurable QSB AXI values

SDC2_SDCC_QSB_AXI_VALUES

Bits	Name	Type	Description
30	QSB_AXI_REDIRECT	RW	Driving QSB AXI output
29	QSB_AXI_ABURST	RW	Driving QSB AXI output
28:26	NUM_OUTSTANDING_DATA	RW	Configuring the number of outstanding transactions for the data. The descriptors accesses have more one outstanding request
25	PROCEED_AXI_AFTER_ERROR	RW	When error occurs on AXI master the traffic can continue or terminate. If this bit is set then the traffic continues and more requests are transmitted on the address channel

SDC2_SDCC_QSB_AXI_VALUES (cont.)

Bits	Name	Type	Description
24	QSB_AXI_AFULL_CALC	RW	If this bit is set, then afull isn't '0' constantly but can be equal to '1' if all the beats have 8 bytes on the strobe lines
23	ONE_MID_SUPPORT	RW	If this bit is set, then only 1 MID is used for descriptors and data
22:20	RESERVED	RW	Reserved
19	SDCC5_HALT_REQ	R	Reading SDCC5_HALT_REQ input
18	SDCC5_HALT_ACK	R	Reading SDCC5_HALT_ACK output
17	SDCC5_M_IDLE	R	Reading SDCC5_M_IDLE output
16	SDCC5_HALT_ACK_SW_EN	RW	Enabling the SW value for SDCC5_HALT_ACK
15	SDCC5_HALT_ACK_SW	RW	Driving SDCC5_HALT_ACK from SW and not by HW
14	SDCC5_M_IDLE_DIS	RW	If this bit is set, then sdcc5_m_idle will be tied low
13	QSB_AXI_INTERLEAVING_EN	RW	If the bit is set, then the controller can support interleaving between two bursts from the Noc. This option is disabled by default because Noc doesn't support interleaving but only re-ordering
12	RESERVED	RW	Reserved
11	RESERVED	RW	Reserved
10:9	RESERVED	RW	Reserved
8:6	RESERVED	RW	Reserved
5	RESERVED	RW	Reserved
4	RESERVED	RW	Reserved
3	RESERVED	RW	Reserved
2	QSB_AXI_OOOWR	RW	Driving QSB AXI output
1	QSB_AXI_OOORD	RW	Driving QSB AXI output
0	QSB_AXI_AFULL	RW	Driving QSB AXI output

0x07864134 SDC2_SDCC_AXI_DATA_ERR_DEBUG**Type:** R**Clock:** same rate as HCLK**Reset State:** 0x00000000

Debug details after error on AXI I/F

SDC2_SDCC_AXI_DATA_ERR_DEBUG

Bits	Name	Description
31	RESP_ERR_WR	If set, then the failed transaction was a write access
30:28	RESP_ERR_TID	The TID of the failed transaction
27:13	DESC_ADDR_35_21	Upper 15 bits of current address in address channel
12:0	DESC_ADDR_12_0	Lower 13 bits of current address in address channel

0x07864138 SDC2_SDCC_T4_DLY_CTRL**Type:** RW**Clock:** same rate as HCLK**Reset State:** 0x0000382B

Control register of T/4 Delay element which is used in e.MMC5.0 DDR mode.

SDC2_SDCC_T4_DLY_CTRL

Bits	Name	Description
15:14	T4_DLY_SPARE_CTL	
13	T4_DLY_POWERDOWN_EN	Powerdown enable. When asserted (default), it powers down the T/4 macro and shuts off all bias circuits.
12:11	T4_DLY_TESTOUT_SEL	Test output select. When toggled, it presents either the strb_in or strb_out on the analog test port for debug.
10	T4_DLY_BYPASS_EN	Bypass enable. When asserted it presents the strb_in on the strb_out port.
9	T4_DLY_CAL_START	Calibration start input. When asserted, it indicates the start of the calibration cycle.
8	T4_DLY_CAL_OVERRIDE	Calibration override input. When asserted, the internal delay line calibration is bypassed and the tby4_user_ctl<7:0> is used as the default code for the delay line.
7:0	T4_DLY_TARGET_CODE	T/4 delay target code. 0x2b gives delay of 1.25ns

0x0786413C SDC2_SDCC_T4_DLY_STAT**Type:** R**Clock:** same rate as HCLK**Reset State:** 0x00000000

Status register T/4 Delay element which is used in e.MMC5.0 DDR mode.

SDC2_SDCC_T4_DLY_STAT

Bits	Name	Description
15:14	T4_DLY_SPARE_OUT	Spare status bits.
13	T4_DLY_CAL_DLY_CODE_FAST_RANGE	Calibrated delay code, indicates the fast range selection bit status.
12:6	T4_DLY_CAL_DLY_CODE_ISEL	Calibrated delay code, indicates the isel programming value for the bias generator.
5:4	T4_DLY_CAL_DLY_CODE_RSEL	Calibrated delay code, indicates the rsel programming value for the bias generator.
3:0	T4_DLY_STATE_DESC	T/4 state descriptor. Indicates the calibration status and result. value 4'b0000: Indicates Reset mode, waiting for cal_start signal. value 4'b0111: Indicates the fast range decision completion. value 4'b0100: Indicates rsel adjustment decision. value 4'b1011: Indicates isel adjustment decision. value 4'b1110: Indicates Normal end-of-cal. value 4b'1111: Indicates Maximum Delay Reached

0x07864140 SDC2_SDCC_CSR_CDC_CTLR_CFG0**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

The CSR_CDC_CTLR_CFG0 register configures the following:

SDC2_SDCC_CSR_CDC_CTLR_CFG0

Bits	Name	Description
25	STAGGER_CAL_ENA	1'b1: Stagger calibration of write and read CDCs once after the other to reduce peak voltage drop. 1'b0: Both write and read CDCs calibrated simultaneously This bit is introduced on the x2 core, so not present on x0 and x1 versions of DIM/PHY.
24	TRACKING_CAL_ENA	1'b1: Relationship between subunits and oscillator count is learnt and used for tracking calibration. 1'b0: Periodic calibration triggers cause full recalibration
23:20	OSC_COUNT_ERR_TOLERANCE	Magnitude of the difference between oscillator count and target count when the algorithm assumes convergence and stops further search.
19	TRACK_CALIB_MODE	1'b0: Tracking calibration uses previous Target_count value. 1'b1 Accurate mode: Target_count is recomputed at the start of tracking calibration

SDC2_SDCC_CSR_CDC_CTLR_CFG0 (cont.)

Bits	Name	Description
18	FULL_DELAY	1'b0: unitstep and subunit caps are from calibration or SW settings 1'b1: All unit step and subunit caps are selected for the delay arrays.
17	HW_AUTOCAL_ENA	1'b1: Enables periodic auto calibration 1'b0: Periodic calibration is disabled
16	SW_TRIG_FULL_CALIB	A low to high transition on this register bit triggers the CDC calibration statemachine to initiate full calibration of the standby delay array. Also resets tracking mode for the delay arrays.
13:12	OSC_PRE_DIV	The factor by which the raw oscillator frequency is divided before being given to the counting logic. 2'b00 Divide by 1 2'b01 Divide by 2 2'b10 Divide by 4 2'b11 Divide by 8
10:0	TMUX_CHAR	Cchar is a characterized constant equal to the ratio of output mux offset delay to average unit step delay. This register is programmed by multiplying the constant with 1024 and rounding to the nearest integer. E.g. if Cchar = 0.67, $RO(0.67 * 1024) = 686$, hence program TMUX_CHAR = 11'h2AE

0x07864144 SDC2_SDCC_CSR_CDC_CTLR_CFG1**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

The CSR_CDC_CTLR_CFG1 register configures the following:

SDC2_SDCC_CSR_CDC_CTLR_CFG1

Bits	Name	Description
26:24	OSC_COUNT_DELAY	Number of oscillator cycles from the end of reference gate signal to assertion of oscillation counter done. Legal values are from 0 to 4.
23:20	STANDBY_DELAY	This is the number of reference clock cycles from when a newly calibrated delay array is made active to when the current active delay array is put into standby mode. 4'b0: Disables the counter. Hardware waits for explicit acknowledgement of delay array switch from CDC. Others: Explicit acknowledgement mode is disabled and the counter is used
18:16	DEL_MODE_DELAY	Number of reference clock cycles between putting a newly calibrated delay array in delay mode to selecting it as the active delay array. The minimum allowed value is 1

SDC2_SDCC_CSR_CDC_CTLR_CFG1 (cont.)

Bits	Name	Description
14:12	OSC_MODE_DELAY	Number of reference clock cycles between putting a delay array in oscillator mode to starting the oscillation counter. This delay is allowed for the oscillations to stabilize through the delay array. The minimum allowed value is 1
10:8	DECODER_DELAY	Number of reference clock cycles allowed for unitstep and subunit counts to be encoded into the unit and finecap select lines on the DDA. The minimum allowed value is 1.
6:4	DIVIDER_DELAY	Number of reference clock cycles allowed for arithmetic division operations. The minimum allowed value is 1
2:0	MULTIPLIER_DELAY	Number of reference clock cycles allowed for arithmetic multiplication operations. The minimum allowed value is 1

0x07864148 SDC2_SDCC_CSR_CDC_CAL_TIMER_CFG0**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

The CSR_CDC_CAL_TIMER_CFG0 register configures the following:

SDC2_SDCC_CSR_CDC_CAL_TIMER_CFG0

Bits	Name	Description
24	INVALID_TIMER_ENA	1'b1: Enable the timer counting number of missed calibration intervals after which the CDC is invalid. 1'b0: Timer disabled A low to high transition on this register bit loads the timer with programmed value below.
23:20	INVALID_TIMER_VAL	Number of missed calibration intervals after which CDC invalid state is asserted.
16	TIMER_ENA	1'b1: Enables 32 kHz sleep clock based periodic autocalibration timer. 1'b0: Timer disabled A low to high transition on this register bit loads the timer with programmed value below.
15:0	TIMER_VAL	Count value for periodic autocalibration interval based on sleep clock.

0x0786414C SDC2_SDCC_CSR_CDC_CAL_TIMER_CFG1**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

The CSR_CDC_CAL_TIMER_CFG1 register configures the following:

SDC2_SDCC_CSR_CDC_CAL_TIMER_CFG1

Bits	Name	Description
12	FF_TIMER_ENA	1'b1: Enables reference clock based periodic autocalibration timer. 1'b0: Timer disabled A low to high transition on this register bit loads the timer with programmed value below.
9:0	FF_TIMER_VAL	Count value for periodic autocalibration interval based on reference clock.

0x07864150 SDC2_SDCC_CSR_CDC_REFCOUNT_CFG**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

The CSR_CDC_REFCOUNT_CFG register configures the following:

SDC2_SDCC_CSR_CDC_REFCOUNT_CFG

Bits	Name	Description
31:16	TREF	Reference clock period programmed in pico seconds.
13:8	CCAL_REF_COUNT	Number of reference clock cycles used for oscillation counts during coarse calibration. Maximum supported value is 59.
5:0	FCAL_REF_COUNT	Number of reference clock cycles used for oscillation counts during fine calibration. Maximum supported value is 59

0x07864154 SDC2_SDCC_CSR_CDC_COARSE_CAL_CFG**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

The register CSR_CDC_COARSE_CAL_CFG configures the following:

SDC2_SDCC_CSR_CDC_COARSE_CAL_CFG

Bits	Name	Description
12:8	CCAL_SUBUNIT_CAPS	Subunit count needed for an effective unit step count of 32 during coarse calibration. This value is also used by software oscillator mode.
5:0	CCAL_UNITSTEPS	unitstep count needed for an effective unit step count of 32 during coarse calibration.

0x07864158 SDC2_SDCC_CSR_CDC_RSVD_CFG**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

The CSR_CDC_RSVD_CFG register configures the following:

SDC2_SDCC_CSR_CDC_RSVD_CFG

Bits	Name	Description
15:0	TEMP_FIELD	Reserved register

0x0786415C SDC2_SDCC_CSR_CDC_OFFSET_CFG**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

The CSR_CDC_OFFSET_CFG register configures the following:

SDC2_SDCC_CSR_CDC_OFFSET_CFG

Bits	Name	Description
20	SUBUNIT_OFFSET_MODE	1'b1: The programmed offset is directly loaded as subunit count 1'b0: Programmed offset is added/subtracted from the subunit count
19	SUBUNIT_OFFSET_SIGN	1'b1: Offset is subtracted from the subunit count 1'b0: Offset is added to the subunit count
16:12	SUBUNIT_OFFSET	Software subunit offset
8	UNIT_OFFSET_MODE	1'b1: The programmed offset is directly loaded as unit step count 1'b0: Programmed offset is added/subtracted from the unit step count

SDC2_SDCC_CSR_CDC_OFFSET_CFG (cont.)

Bits	Name	Description
7	UNIT_OFFSET_SIGN	1'b1: Offset is subtracted from the unitstep count 1'b0: Offset is added to the unitstep count
5:0	UNIT_OFFSET	Software unitstep offset

0x07864160 SDC2_SDCC_CSR_CDC_DELAY_CFG**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

The CSR_CDC_DELAY_CFG register configures the following:

SDC2_SDCC_CSR_CDC_DELAY_CFG

Bits	Name	Description
28	TARGET_COUNT_ENA	1'b1: Enable software specified Target_count for calibration instead of value calculated by hardware. 1'b0: Hardware computed Target_count used.
27:16	TARGET_COUNT	Software specified Target_count to which the statemachine calibrates
11:0	DELAY_VAL	Delay required from CDC programmed in pico seconds. This value has to be greater than the minimum delay supported by DDA Tmin.

0x07864164 SDC2_SDCC_CSR_CDC_SW_MODE_CFG**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

The CSR_CDC_SW_MODE_CFG register configures the following:

SDC2_SDCC_CSR_CDC_SW_MODE_CFG

Bits	Name	Description
3	SW_DEL_MODE	A low to high transition on this register bit puts the delay array selected by SW_DA_SEL into delay mode and connects it to the DDA output. This register has to remain set for the duration of SW controlled delay mode.
2	SW_OSC_MODE	Reserved register.
1	SW_DA_SEL	Selects the delay array inside DDA during software delay or oscillator mode.

SDC2_SDCC_CSR_CDC_SW_MODE_CFG (cont.)

Bits	Name	Description
0	SW_LOAD	1'b1: Enable loading of the software offsets from the offset configuration register. This register has to remain set for the duration of SW mode operation.

0x0786416C SDC2_SDCC_CSR_CDC_SW_OVRD_CFG**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

The CSR_CDC_SW_OVRD_CFG register configures the following:

SDC2_SDCC_CSR_CDC_SW_OVRD_CFG

Bits	Name	Description
16	SW_REF_GATE	Write 1'b1 to start the reference window. Write 1'b0 to end the reference window. Number of reference cycle in the window will be reported in SW_REF_GATE_COUNT in CDC_STATUS0
11	SW_OVRD_DA1_OSC_EN	In software override mode, 1'b1 written to this bit puts delay array 1 in oscillator-mode
10	SW_OVRD_DA1_IN_EN	In software override mode, 1'b1 written to this bit puts delay array 1 in delay-mode
9	SW_OVRD_DA0_OSC_EN	In software override mode, 1'b1 written to this bit puts delay array 0 in oscillator-mode
8	SW_OVRD_DA0_IN_EN	In software override mode, 1'b1 written to this bit puts delay array 0 in delay-mode
3	SW_OVRD_CDC_COUNTERR_RST	1'b1 written to this bit resets the oscillation counter. Follow this with 1'b0 before setting reference gate.
2	SW_OVRD_LOAD_DA_SEL	In software override mode, this bit select the delay array to which unit and subunit offset values from CDC_OFFSET_CFG register will be loaded.
1	SW_OVRD_ACTV_DA_SEL	In software override mode, this register selects the delay array connected to the CDC output
0	SW_OVRD_ENA	1'b1 written to this bit puts the CDC interface in software override mode where the below register bits control the CDC. 1'b0: CDC is under hardware control

0x07864170 SDC2_SDCC_CSR_CDC_SLAVE_DDA_CFG**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

The CSR_CDC_SLAVE_DDA_CFG register configures the following:

SDC2_SDCC_CSR_CDC_SLAVE_DDA_CFG

Bits	Name	Description
17	SLAV_DDA_OFFSET_MODE	1'b1: The programmed offset is directly loaded as unit-step count 1'b0: Programmed offset is added/subtracted from the unit-step count
16	SLAV_DDA_OFFSET_SIGN	1'b1: Offset is subtracted from the unit-step count. This subtraction feature is not supported on x0 and x1 PHY cores (bit RESERVED). Supported on the x2 core version. 1'b0: Offset is added to the unit-step count
15:12	SLAVE_DDA_OFFSET	unit-step offset for slave DDA.
10:0	SLAVE_DDA_DELAY	Delay required from slave DDA programmed in pico seconds.

0x07864174 SDC2_SDCC_CSR_CDC_STATUS0**Type:** R**Clock:** HCLK**Reset State:** 0x0000000C

The CSR_CDC_STATUS0 register configures the following:

SDC2_SDCC_CSR_CDC_STATUS0

Bits	Name	Description
26:24	CDC_ERROR_CODE	Error code for internal errors: 3'b000: No error 3'b001: Programmed delay less than Tmux 3'b010: Programmed delay less than Tmin 3'b011: Unit-step overflow: Overflow during fine calibration. Desired delay cannot be realized even when all unit-steps are selected. 3'b100: Unit-step underflow 3'b101: Unit-step overflow: Overflow during coarse calibration. Estimated unit steps needed to realize the desired delay are greater than the total number of unit-steps in design. Note: The CDC_ERROR_CODE is not a sticky register and is updated after each CDC calibration.

SDC2_SDCC_CSR_CDC_STATUS0 (cont.)

Bits	Name	Description
23:16	SW_REF_GATE_COUNT	Number of reference clock cycles in the software generated oscillation gate window during SW override mode
15:4	OSC_COUNT	Current value of oscillator count
3	CURR_SEL_DA	This bit indicates the current active delay array connected to the output. 1'b0: DA0 is selected 1'b1: DA1 is selected
2	CTLR_SM_IDLE	1'b1 Indicates that the calibration state machine is idle 1'b0 indicates calibration state machine is busy
1	OSC_DONE	This status register is set after oscillations count is finished by CDC. It gets reset when a new oscillator count is started.
0	CALIBRATION_DONE	1'b1 indicates that calibration is done and CURR_SUBUNIT_STEPS, CURR_UNIT_STEPS status registers are valid.

0x07864178 SDC2_SDCC_CSR_CDC_STATUS1**Type:** R**Clock:** HCLK**Reset State:** 0x00000000

The CSR_CDC_STATUS1 register configures the following:

SDC2_SDCC_CSR_CDC_STATUS1

Bits	Name	Description
27:16	CURR_DELAY_VALUE	HW Computed value of current delay after calibration by the state machine. This register is valid if CALIBRATION_DONE status bit is set.
11:0	CURR_TMUX_DELAY	Current value of output mux offset delay computed by HW.

0x0786417C SDC2_SDCC_CSR_CDC_STATUS2**Type:** R**Clock:** HCLK**Reset State:** 0x10331033

The CSR_CDC_STATUS2 register configures the following:

SDC2_SDCC_CSR_CDC_STATUS2

Bits	Name	Description
28:24	DA1_SUBUNITS	Current value of subunit cap settings applied to delay array 1
21:16	DA1_UNITSTEPS	Current value of unit steps applied to delay array 1
12:8	DA0_SUBUNITS	Current value of subunit cap settings applied to delay array 0
5:0	DA0_UNITSTEPS	Current value of unit steps applied to delay array 0

0x07864180 SDC2_SDCC_CSR_CDC_STATUS3**Type:** R**Clock:** HCLK**Reset State:** 0x00000000

The CSR_CDC_STATUS3 register configures the following:

SDC2_SDCC_CSR_CDC_STATUS3

Bits	Name	Description
31:24	NUM_OF_OSC_ITER	Number of oscillator iterations before convergence was achieved during tracking
23:12	COUNT_ERROR	Difference between target and actual oscillator counts at the start of tracking calibration
11:0	CURR_TARGET_COUNT	HW computed value of current Target_count used by the calibration state machine.

0x07864184 SDC2_SDCC_CSR_CDC_STATUS4**Type:** R**Clock:** HCLK**Reset State:** 0x000000FF

The CSR_CDC_STATUS4 register configures the following:

SDC2_SDCC_CSR_CDC_STATUS4

Bits	Name	Description
7:4	SLAVE_DDA_DA1_TAPS	Number if unit taps applied to delay array 1 of slave DDA
3:0	SLAVE_DDA_DA0_TAPS	Number if unit taps applied to delay array 0 of slave DDA

0x07864188 SDC2_SDCC_CSR_CDC_GEN_CFG**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000**SDC2_SDCC_CSR_CDC_GEN_CFG**

Bits	Name	Description
2	CDC_SDDA_EN	Enable CDC Slave DDA to generate 3/4T RCLK from cdc_out_2 output of CDC: 0 (default) - CDC Slave DDA is disabled and 3/4T clock is a falling edge of cdc_out, 1 - CDC Slave DDA is enabled and 3/4T clock is a rising edge of cdc_out_2.
1	CDC_SWITCH_RC_EN	Enablement of CDC power RC (LPF) switch 1'b1: Enabled 1'b0: Disabled (default)
0	CDC_SWITCH_BYPASS_OF F	Enablement of CDC power bypass switch 1'b1: Disabled 1'b0: Enabled (default)

0x0786418C SDC2_SDCC_CDC_AUTO_SW_CAL_CFG**Type:** RW**Clock:** HCLK**Reset State:** 0x00F08000

Auto SW calibration configuration bits.

SDC2_SDCC_CDC_AUTO_SW_CAL_CFG

Bits	Name	Description
25	CDC_CAL_TIMEOUT_ENABLE	Enable for cdc calibration timeout counter.
24	CDC_AUTO_SW_CAL_ENABLE	Enable for the Auto SW calibration Feature.
23:16	CDC_CAL_TIMEOUT_THRESHOLD	Sets the threshold for CDC calibration timeout during sw calibration triggered by the HW FSM. counts in 16xFF clock. Default is 0xF0 which is 255x16 FF clocks, which gives threshold of ~200 us
15:0	AUTO_TRIG_INTERVAL	Sets the load value of the counter that counts the interval between each sw calibration triggered by the HW FSM. this counter counts in sleep clock. Default is 0x8000 which is ~32,000 sleep clocks, which gives interval of ~1 sec

0x07864190 SDC2_SDCC_DDR200_STAT**Type:** R**Clock:** HCLK**Reset State:** 0x00000000

Status bits for e.MMC5.0 DDR200 mode.

SDC2_SDCC_DDR200_STAT

Bits	Name	Description
0	CDC_CAL_VALID	CDC Calibration status: 0 - CDC isn't calibrated, 1 - CDC is calibrated and valid

0x07864194 SDC2_SDCC_DDR200_CFG**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

Configuration bits for e.MMC5.0 DDR200 mode

SDC2_SDCC_DDR200_CFG

Bits	Name	Description
10	FF_CLK_DIS	The ff_clk to CDC/TBY4 is enabled by default. It can be disabled by setting this bit. The ff_clk is enabled to CDC and TBY4 according to CDC_T4_DLY_SEL field
9	VOLTAGE_MUX_SEL	select between 1.8v and 1.2v signals on sdcc5 IO interface with device. default is 0 to choose 1.8v
8:7	CDC_TRAFFIC_SEL	Selecting the method of cdc_traffic's assertions: Value 0 (default) - auto calibration with stand by feature. Value 1 - SW calibration before any transaction. Value 2 - automatic calibration with HW assertion of cdc_traffic.
6	START_CDC_TRAFFIC	This bit starts the cdc traffic assertion after calibration done is received.
5	CRC_TOKEN_SAMPL_FALL_EDGE	If set, the start bit of incoming CRC token in DDR200 mode is sampled in falling edge. If this bit is cleared (default) then the start bit is detected in rising edge
4	DATIN_SAMPL_FALL_EDGE	If set, the start bit of incoming data in DDR200 mode is sampled in falling edge. If this bit is cleared (default) then the start bit is detected in rising edge
3	CDC_T4_TEST_OUT_SEL	controls the mux between dll test_out and CDC test out. Default value is 0 (dll is chosen). When set - CDC test_out is chosen.
2	CMDIN_EDGE_SEL	Select sampling edge of CMD input when CMDIN_RCLK_EN is 1: 0 (default) - falling edge of RCLK, 1 - rising edge of RCLK

SDC2_SDCC_DDR200_CFG (cont.)

Bits	Name	Description
1	CMDIN_RCLK_EN	Enable CMD input sampling with RCLK: 0 (default) - CMD input is sampled by SD DLL output clock, 1 - CMD input is sampled by delayed RCLK
0	CDC_T4_DLY_SEL	Select block for RCLK delay: 0 (default) - use CM_DLL_SDC4

0x07864198 SDC2_SDCC_TIME_COUNT_CTRL**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

Control the performance counters

SDC2_SDCC_TIME_COUNT_CTRL

Bits	Name	Description
1	CMD_PERIOD	When this bit is set the commands periods are counted to the CMD_TIME counter
0	ENABLE	Enabling the performance counters

0x0786419C SDC2_SDCC_FLOW_CTRL_TIME_COUNT**Type:** R**Clock:** HCLK**Reset State:** 0x00000000

Counting the Flow Control period

SDC2_SDCC_FLOW_CTRL_TIME_COUNT

Bits	Name	Description
31:0	FLOW_CTRL_TIME	FlowControl periods are counted

0x078641A0 SDC2_SDCC_BUSY_TIME_COUNT**Type:** R**Clock:** HCLK**Reset State:** 0x00000000

Counting the BUSY periods

SDC2_SDCC_BUSY_TIME_COUNT

Bits	Name	Description
31:0	BUSY_TIME	In Tx transaction, the BUSY periods are counted until the progdone is asserted. In Rx transaction, the periods between the blocks without flowcontrol is counted including the time until the first block is received after the command.

0x078641A4 SDC2_SDCC_DATA_TIME_COUNT**Type:** R**Clock:** HCLK**Reset State:** 0x00000000

Counting the time of transaction

SDC2_SDCC_DATA_TIME_COUNT

Bits	Name	Description
31:0	DATA_TIME_CNT	In Tx transaction, the time is from ENabling the DPSM until progdone is asserted. In Rx transaction, the time is from the command transmission to de-assertion of RxActive.

0x078641A8 SDC2_SDCC_CMD_TIME_COUNT**Type:** R**Clock:** HCLK**Reset State:** 0x00000000

Counting the time of transaction

SDC2_SDCC_CMD_TIME_COUNT

Bits	Name	Description
31:0	CMD_TIME	Counting the Command-Response periods of commands which relate to the transaction including auto commands. The counter is active when CMD_PERIOD is set.

0x078641AC SDC2_SDCC_MCI_DLL_CONFIG_2**Type:** RW**Clock:** same rate as HCLK**Reset State:** 0x00000000

Second register for DLL's configuration

SDC2_SDCC_MCI_DLL_CONFIG_2

Bits	Name	Description
3:2	DDR_TRAFFIC_INIT_SEL	Selecting how DDR_TRAFFIC_INIT will be driven to DLL. Value 00 - HW drives the DDR_TRAFFIC_INIT when DPSM is active and when CPSM is active if RCLK is expected during response. Value 01 - The value is driven by SW by SW_DDR_TRAFFIC_INIT bit. Value 10 - DDR_TRAFFIC_INIT to DLL is driven when sdc4_mclk is not gated. off.
1	DDR_TRAFFIC_INIT_SW	The signal should be high 3-4 cycles before RCLK is toggling. It should be high when MCLK is toggling.
0	DDR_CAL_EN	The signal is set after the configuration of MCI_DDR_CONFIG is finished. When the signal is low the TCXO domain is being reset.

0x078641B0 SDC2_SDCC_MCI_DDR_CONFIG

Type: RW

Clock: same rate as HCLK

Reset State: 0x80040853

Configuration of ddr_config input of DLL

SDC2_SDCC_MCI_DDR_CONFIG

Bits	Name	Description
31	PRG_DLY_EN	Enables Programmable delay for RCLK_IN
30	EXT_PRG_RCLK_DLY_EN	Enables external control of Programmable RCLK delayline
29:27	EXT_PRG_RCLK_DLY_CODE	Used only if EXT_PRG_RCLK_DLY_EN is set. Configuring Programmable RCLK Delayline process code number.
26:21	EXT_PRG_RCLK_DLY	Used only if EXT_PRG_RCLK_DLY_EN is set. Programmable RCLK Delay line fine delay control.
20:12	TCXO_CYCLES_DLY_LINE	Number of TCXO clock cycles between finish of one delay line calibration and beginning of next delay line calibration.
11:9	TCXO_CYCLES_CNT	Number of TCXO clock cycles for which down counter is enabled for before making each inc/dec decision.
8:0	PRG_RCLK_DLY	These bits are controlled the programmable delay line. This bus is binary coded. Decimal value = (TCXO period * Decimal value of TCXO_CYCLES_CNT)/(2*delay needed). For 1.25ns the formula is $(52n \cdot 4/2 \cdot 1.25) = 83$ in decimal.

**0x07864400+ SDC2_SDCC_MCI_FIFO_ALTn, n=[0..255]
0x4*n****Type:** RW**Clock:** same rate as HCLK**Reset State:** 0x00000000

Alternative register to MCI_FIFO (0x80-0xBC). The SW can use this register range (0x400-0x7FC) for accessing all the RAM range (1024 bytes) in FIFO mode. The advantage is that any byte in the RAM can be accessed without repeating the same address

SDC2_SDCC_MCI_FIFO_ALTn

Bits	Name	Description
31:0	DATA	WORD is written or received to/from the RAM in FIFO mode

19.2 SDC2_SDCC_SDCC5_HC**0x07864900 SDC2_SDCC_HC_REG_0_2****Type:** RW**Clock:** same rate as HCLK**Reset State:** 0x00000000

Argument 2 Register in SD Host Controller Standard Specification.

SDC2_SDCC_HC_REG_0_2

Bits	Name	Description
31:0	ARG_2	Argument 2 field

0x07864904 SDC2_SDCC_HC_REG_4_6**Type:** RW**Clock:** same rate as HCLK**Reset State:** 0x00000000

Combination of two registers from the SD Host Controller Standard Specification:

-Block Size Register(15-0)

-Block Count Register(31-16)

SDC2_SDCC_HC_REG_4_6

Bits	Name	Description
31:16	BLK_CNT_FOR_CUR_TRANS	Blocks Count For Current Transfer field from Block Count Register
14:12	BLK_SIZE_HST_SDMA_BUFFER	Host SDMA Buffer Boundary field from Block Size Register
11:00	BLK_SIZE_TRANS	Transfer Block Size field from Block Size Register

0x07864908 SDC2_SDCC_HC_REG_8_A**Type:** RW**Clock:** same rate as HCLK**Reset State:** 0x00000000

Argument 1 Register in SD Host Controller Standard Specification.

SDC2_SDCC_HC_REG_8_A

Bits	Name	Description
31:00	CMD_ARG_1	Command Argument 1 field

0x0786490C SDC2_SDCC_HC_REG_C_E**Type:** RW**Clock:** same rate as HCLK**Reset State:** 0x00000000

Combination of two registers from the SD Host Controller Standard Specification:

-Transfer Mode Register(15-0)

-Command Register(31-16)

SDC2_SDCC_HC_REG_C_E

Bits	Name	Description
29:24	CMD_INDX	Command Index field from Command Register
23:22	CMD_TYPE	Command Type field from Command Register
21	CMD_DATA_PRESENT_SEL	Data Present Select field from Command Register
20	CMD_INDX_CHECK_EN	Command Index Check Enable field from Command Register
19	CMD_CRC_CHECK_EN	Command CRC Check Enable field from Command Register

SDC2_SDCC_HC_REG_C_E (cont.)

Bits	Name	Description
17:16	CMD_RESP_TYPE_SEL	Response Type Select field from Command Register
5	TRANS_MODE_MULTI_SINGLE_BLK_SEL	Multi/Single Block Select from Transfer Mode Register
4	TRANS_MODE_DATA_DIRECTION_SEL	Data Transfer Direction Select from Transfer Mode Register
3:2	TRANS_MODE_AUTO_CMD_EN	Auto CMD Enable from Transfer Mode Register
1	TRANS_MODE_BLK_CNT_EN	Block Count Enable field from Transfer Mode Register
0	TRANS_MODE_DMA_EN	DMA Enable field from Transfer Mode Register

0x07864910 SDC2_SDCC_HC_REG_10_12**Type:** R**Clock:** same rate as HCLK**Reset State:** 0x00000000

Response Register in SD Host Controller Standard Specification.

SDC2_SDCC_HC_REG_10_12

Bits	Name	Description
31:0	CMD_RESP	Command Response field - Response bits 0-31

0x07864914 SDC2_SDCC_HC_REG_14_16**Type:** R**Clock:** same rate as HCLK**Reset State:** 0x00000000

Response Register in SD Host Controller Standard Specification.

SDC2_SDCC_HC_REG_14_16

Bits	Name	Description
31:0	CMD_RESP	Command Response field - Long Response bits 31-63

0x07864918 SDC2_SDCC_HC_REG_18_1A**Type:** R**Clock:** same rate as HCLK**Reset State:** 0x00000000

Response Register in SD Host Controller Standard Specification.

SDC2_SDCC_HC_REG_18_1A

Bits	Name	Description
31:0	CMD_RESP	Command Response field - Long Response bits 64-95

0x0786491C SDC2_SDCC_HC_REG_1C_1E**Type:** R**Clock:** same rate as HCLK**Reset State:** 0x00000000

Response Register in SD Host Controller Standard Specification.

SDC2_SDCC_HC_REG_1C_1E

Bits	Name	Description
31:0	CMD_RESP	Command Response field - Long Response bits 96-127

0x07864920 SDC2_SDCC_HC_REG_20_22**Type:** RW**Clock:** same rate as HCLK**Reset State:** 0x00000000

Buffer Data Port Register in SD Host Controller Standard Specification.

SDC2_SDCC_HC_REG_20_22

Bits	Name	Description
31:24	BUF_DATA_PORT_3	Byte 3 of Buffer Data
23:16	BUF_DATA_PORT_2	Byte 2 of Buffer Data
15:8	BUF_DATA_PORT_1	Byte 1 of Buffer Data
7:0	BUF_DATA_PORT_0	Byte 0 of Buffer Data

0x07864924 SDC2_SDCC_HC_REG_24_26**Type:** R**Clock:** same rate as HCLK**Reset State:** 0x00080000

Present State Register in SD Host Controller Standard Specification.

SDC2_SDCC_HC_REG_24_26

Bits	Name	Description
25	SIGANLING_18_SWITCHING_STS	Bit it is cleared, by hardware, when host driver writes any value to 1.8V Signaling Enable field, indicating that a signal switching operation has started. Bit is set, by hardware, when switching operation succeeds.
24	PRESENT_STATE_CMD_LINE_SIGNAL_LEVEL	CMD Linr Signal Level field
23:20	PRESENT_STATE_DAT_3_0_LINE_SIGNAL_LEVEL	DAT[3:0] Line Signal Level field
19	PRESENT_STATE_WR_PROTECT_SWITCH_PIN_LEVEL	Write Protect Switch Pin Level field
18	PRESENT_STATE_CARD_DETECT_PIN_LEVEL	Card Detect Pin Level field
17	PRESENT_STATE_CARD_STATE_STABLE	Card State Stable field
16	PRESENT_STATE_CARD_INSERTED	Card Inserted field
11	PRESENT_STATE_BUF_READ_EN	Buffer Read Enable field
10	PRESENT_STATE_BUF_WRITE_EN	Buffer Write Enable field
9	PRESENT_STATE_READ_TRANSFER_ACTIVE	Read Transfer Active field
8	PRESENT_STATE_WRITE_TRANSFER_ACTIVE	Write Transfer Active field
3	PRESENT_STATE_RETUNING_REQ	Re-Tuning Request field
2	PRESENT_STATE_DAT_LINE_ACTIVE	DAT Linr Active field
1	PRESENT_STATE_CMD_INHIBIT_DAT	Command Inhibit (DAT) field
0	PRESENT_STATE_CMD_INHIBIT_CMD	Command Inhibit (CMD) field

0x07864928 SDC2_SDCC_HC_REG_28_2A**Type:** RW**Clock:** same rate as HCLK**Reset State:** 0x00000000

Combination of four registers from the SD Host Controller Standard Specification:

-Host Control 1 Register(7-0)

-Power Control Register(15-8)

-Block Gap Control Register(23-16)

-Wakeup Control Register(31-24)

SDC2_SDCC_HC_REG_28_2A

Bits	Name	Description
26	WKUP_EVENT_EN_ON_SD_CARD_REMOVAL	Wakeup Event Enable On SD Card Removal field from Wakeup Control Register
25	WKUP_EVENT_EN_ON_SD_CARD_INSERTION	Wakeup Event Enable On SD Card Insertion field from Wakeup Control Register
24	WKUP_EVENT_EN_ON_SD_CARD_INT	Wakeup Event Enable On Card Interrupt field from Wakeup Control Register
19	BLK_GAP_CTL_INT	Interrupt At Block Gap field from Block Gap Control Register
18	BLK_GAP_CTL_RD_WAIT	Read Wait Control field from Block Gap Control Register
17	BLK_GAP_CTL_CONTINUE_REQ	Continue Request field from Block Gap Control Register
16	BLK_GAP_CTL_STOP_GAP_REQ	Stop At Block Gap Request field from Block Gap Control Register
11:9	PWR_CTL_SD_BUS_VOLTAGE_SEL	SD Bus Voltage Select field from Power Control Register
8	PWR_CTL_SD_BUS_PWR	SD Bus Power field from Power Control Register
7	HST_CTL1_CARD_DETECT_SIGNAL_SEL	Card Detect Signal Selection field from Host Control 1 Register
6	HST_CTL1_CARD_DETECT_TEST_LEVEL	Card Detect Test Level field from Host Control 1 Register
5	HST_CTL1_EXTENDED_DATA_TRANS_WIDTH	Extended Data Transfer Width field from Host Control 1 Register
4:3	HST_CTL1_DMA_SEL	DMA Select field from Host Control 1 Register
2	HST_CTL1_HS_EN	High Speed Enable field from Host Control 1 Register
1	HST_CTL1_DATA_TRANS_WIDTH	Data Transfer Width field from Host Control 1 Register
0	HST_CTL1_LED_CTL	LED Control field from Host Control 1 Register

0x0786492C SDC2_SDCC_HC_REG_2C_2E**Type:** RW**Clock:** same rate as HCLK**Reset State:** 0x00000001

Combination of three registers from the SD Host Controller Standard Specification:

-Clock Control Register(15-0)

-Timeout Control Register(23-16)

-Software Reset Register(31-24)

SDC2_SDCC_HC_REG_2C_2E

Bits	Name	Type	Description
26	SW_RST_DAT_LINE	RW	Software Reset For DAT Line field from Software Reset Register
25	SW_RST_CMD_LINE	RW	Software Reset For CMD Line field from Software Reset Register
24	SW_RST_FOR_ALL	RW	Software Reset For All field from Software Reset Register
19:16	DATA_TIMEOUT_COUNTER	RW	Data Timeout Counter Value field from Timeout Control Register. The reserved value 0xF is translated to maximum number of cycles - $2^{32}-1$
15:8	CLK_CTL_SDCLK_FREQ_SEL	RW	SDCLK Frequency Select field from Clock Control Register
7:6	CLK_CTL_SDCLK_FREQ_SEL_MSB	RW	Upper Bits of SDCLK Frequency Select field from Clock Control Register
5	CLK_CTL_GEN_SEL	RW	Clock Generator Select field from Clock Control Register
2	CLK_CTL_SDCLK_EN	RW	SD Clock Enable field from Clock Control Register
1	CLK_CTL_INTERNAL_CLK_STABLE	R	Internal Clock Stable field from Clock Control Register
0	CLK_CTL_INTERNAL_CLK_EN	RW	Internal Clock Enable field from Clock Control Register

0x07864930 SDC2_SDCC_HC_REG_30_32**Type:** RW**Clock:** same rate as HCLK**Reset State:** 0x00000000

Combination of two registers from the SD Host Controller Standard Specification:

-Normal Interrupt Status Register(15-0)

-Error Interrupt Status Register(31-16)

SDC2_SDCC_HC_REG_30_32

Bits	Name	Type	Description
31:28	ERR_INT_STS_VENDOR_SPECIFIC	RW	Vendor Specific Error Status field from Error Interrupt Status Register
26	ERR_INT_STS_TUNING_ERROR	RW	Tuning Error field from Error Interrupt Status Register
25	ERR_INT_STS_ADMA_ERR	RW	ADMA Error field from Error Interrupt Status Register
24	ERR_INT_STS_AUTO_CMD_ERR	RW	Auto CMD Error field from Error Interrupt Status Register
23	ERR_INT_STS_CURRENT_LIMIT_ERR	RW	Current Limit Error field from Error Interrupt Status Register
22	ERR_INT_STS_DATA_END_BIT_ERR	RW	Data End Bit Error field from Error Interrupt Status Register
21	ERR_INT_STS_DATA_CRC_ERR	RW	Data CRC Error field from Error Interrupt Status Register
20	ERR_INT_STS_DATA_TIMEOUT_ERR	RW	Data Timeout Error field from Error Interrupt Status Register
19	ERR_INT_STS_CMD_INDEX_ERR	RW	Command Index Error field from Error Interrupt Status Register
18	ERR_INT_STS_CMD_END_BIT_ERR	RW	Command End Bit Error field from Error Interrupt Status Register
17	ERR_INT_STS_CMD_CRC_ERR	RW	Command CRC Error field from Error Interrupt Status Register
16	ERR_INT_STS_CMD_TIMEOUT_ERR	RW	Command Timeout Error field from Error Interrupt Status Register
15	NORMAL_INT_STS_ERROR_INTERRUPT	R	Error Interrupt field from Normal Interrupt Status Register
12	NORMAL_INT_STS_RETUNING_EVENT	R	Re-Tuning Event field from Normal Interrupt Status Register
11	NORMAL_INT_STS_INT_C	R	INT_C field from Normal Interrupt Status Register
10	NORMAL_INT_STS_INT_B	R	INT_B field from Normal Interrupt Status Register
9	NORMAL_INT_STS_INT_A	R	INT_A field from Normal Interrupt Status Register
8	NORMAL_INT_STS_CARD_INTERRUPT	R	Card Interrupt field from Normal Interrupt Status Register
7	NORMAL_INT_STS_CARD_REMOVAL	RW	Card Removal field from Normal Interrupt Status Register
6	NORMAL_INT_STS_CARD_INSERTION	RW	Card Insertion field from Normal Interrupt Status Register

SDC2_SDCC_HC_REG_30_32 (cont.)

Bits	Name	Type	Description
5	NORMAL_INT_STS_BUF_RD_READY	RW	Buffer Read Ready field from Normal Interrupt Status Register
4	NORMAL_INT_STS_BUF_WR_READY	RW	Buffer Write Ready field from Normal Interrupt Status Register
3	NORMAL_INT_STS_DMA_INTERRUPT	RW	DMA Interrupt field from Normal Interrupt Status Register
2	NORMAL_INT_STS_BLK_GAP_EVENT	RW	Block Gap Event field from Normal Interrupt Status Register
1	NORMAL_INT_STS_TRANSFER_COMPLETE	RW	Transfer Complete field from Normal Interrupt Status Register
0	NORMAL_INT_STS_CMD_COMPLETE	RW	Command Complete field from Normal Interrupt Status Register

0x07864934 SDC2_SDCC_HC_REG_34_36**Type:** RW**Clock:** same rate as HCLK**Reset State:** 0x00000000

Combination of two registers from the SD Host Controller Standard Specification:

-Normal Interrupt Status Enable Register(15-0)

-Error Interrupt Status Enable Register(31-16)

SDC2_SDCC_HC_REG_34_36

Bits	Name	Description
31:28	ERR_INT_STS_EN_VENDOR_SPECIFIC_ERR	Vendor Specific Error Status Enable field from Error Interrupt Status Enable Register
26	ERR_INT_STS_EN_TUNING_ERR	Tuning Error Status Enable field from Error Interrupt Status Enable Register
25	ERR_INT_STS_EN_ADMA_ERR	ADMA Error Status Enable field from Error Interrupt Status Enable Register
24	ERR_INT_STS_EN_AUTO_CMD_ERR	Auto CMD Error Status Enable field from Error Interrupt Status Enable Register
23	ERR_INT_STS_EN_CURRENT_LIMIT_ERR	Current Limit Error Status Enable field from Error Interrupt Status Enable Register
22	ERR_INT_STS_EN_DATA_END_BIT_ERR	Data End Bit Error Status Enable field from Error Interrupt Status Enable Register
21	ERR_INT_STS_EN_DATA_CRC_ERR	Data CRC Error Status Enable field from Error Interrupt Status Enable Register

SDC2_SDCC_HC_REG_34_36 (cont.)

Bits	Name	Description
20	ERR_INT_STS_EN_DATA_TIMEOUT	Data Timeout Error Status Enable field from Error Interrupt Status Enable Register
19	ERR_INT_STS_EN_CMD_IDX_ERR	Command Index Error Status Enable field from Error Interrupt Status Enable Register
18	ERR_INT_STS_EN_CMD_END_BIT_ERR	Command End Bit Error Status Enable field from Error Interrupt Status Enable Register
17	ERR_INT_STS_EN_CMD_CRC_ERR	Command CRC Error Status Enable field from Error Interrupt Status Enable Register
16	ERR_INT_STS_EN_CMD_TIMEOUT	Command Timeout Error Status Enable field from Error Interrupt Status Enable Register
12	NORMAL_INT_STS_EN_RE_TUNING_EVENT	Re-Tuning Event Status Enable field from Normal Interrupt Status Enable Register
11	NORMAL_INT_STS_EN_INT_C	INT_C Status Enable field from Normal Interrupt Status Enable Register
10	NORMAL_INT_STS_EN_INT_B	INT_B Status Enable field from Normal Interrupt Status Enable Register
9	NORMAL_INT_STS_EN_INT_A	INT_A Status Enable field from Normal Interrupt Status Enable Register
8	NORMAL_INT_STS_EN_CARD_INT	Card Interrupt Status Enable field from Normal Interrupt Status Enable Register
7	NORMAL_INT_STS_EN_CARD_REMOVAL	Card Removal Status Enable field from Normal Interrupt Status Enable Register
6	NORMAL_INT_STS_EN_CARD_INSERTION	Card Insertion Status Enable field from Normal Interrupt Status Enable Register
5	NORMAL_INT_STS_EN_BUFFER_RD_READY	Buffer Read Ready Status Enable field from Normal Interrupt Status Enable Register
4	NORMAL_INT_STS_EN_BUFFER_WR_READY	Buffer Write Ready Status Enable field from Normal Interrupt Status Enable Register
3	NORMAL_INT_STS_EN_DMA_A_INT	DMA Interrupt Status Enable field from Normal Interrupt Status Enable Register
2	NORMAL_INT_STS_EN_BLOCK_GAP_EVENT	Block Gap Event Status Enable field from Normal Interrupt Status Enable Register
1	NORMAL_INT_STS_EN_TRANSFER_COMPLETE	Transfer Complete Status Enable field from Normal Interrupt Status Enable Register
0	NORMAL_INT_STS_EN_COMMAND_COMPLETE	Command Complete Status Enable field from Normal Interrupt Status Enable Register

0x07864938 SDC2_SDCC_HC_REG_38_3A**Type:** RW**Clock:** same rate as HCLK**Reset State:** 0x00000000

Combination of two registers from the SD Host Controller Standard Specification:

-Normal Interrupt Signal Enable Register(15-0)

-Error Interrupt Signal Enable Register(31-16)

SDC2_SDCC_HC_REG_38_3A

Bits	Name	Type	Description
31:28	ERR_INT_SIGNAL_EN_VE NDOR_SPECIFIC_ERR	RW	Vendor Specific Error Signal Enable field from Error Interrupt Signal Enable Register
26	ERR_INT_SIGNAL_EN_TU NING_ERR	RW	Tuning Error Signal Enable field from Error Interrupt Signal Enable Register
25	ERR_INT_SIGNAL_EN_AD MA_ERR	RW	ADMA Error Signal Enable field from Error Interrupt Signal Enable Register
24	ERR_INT_SIGNAL_EN_AU TO_CMD_ERR	RW	Auto CMD Error Signal Enable field from Error Interrupt Signal Enable Register
23	ERR_INT_SIGNAL_EN_CU RRENT_LIMIT_ERR	RW	Current Limit Error Signal Enable field from Error Interrupt Signal Enable Register
22	ERR_INT_SIGNAL_EN_DAT A_END_BIT_ERR	RW	Data End Bit Error Signal Enable field from Error Interrupt Signal Enable Register
21	ERR_INT_SIGNAL_EN_DAT A_CRC_ERR	RW	Data CRC Error Signal Enable field from Error Interrupt Signal Enable Register
20	ERR_INT_SIGNAL_EN_DAT A_TIMEOUT	RW	Data Timeout Error Signal Enable field from Error Interrupt Signal Enable Register
19	ERR_INT_SIGNAL_EN_CM D_INDXX_ERR	RW	Command Index Error Signal Enable field from Error Interrupt Signal Enable Register
18	ERR_INT_SIGNAL_EN_CM D_END_BIT_ERR	RW	Command End Bit Error Signal Enable field from Error Interrupt Signal Enable Register
17	ERR_INT_SIGNAL_EN_CM D_CRC_ERR	RW	Command CRC Error Signal Enable field from Error Interrupt Signal Enable Register
16	ERR_INT_SIGNAL_EN_CM D_TIMEOUT	RW	Command Timeoutout Error Signal Enable field from Error Interrupt Signal Enable Register
15	NORMAL_INT_SIGNAL_EN _ERR_INT	R	Error Interrupt Signal Enable field from Normal Interrupt Signal Enable Register
12	NORMAL_INT_SIGNAL_EN _RETUNING_EVENT	RW	Re-Tuning Event Signal Enable field from Normal Interrupt Signal Enable Register
11	NORMAL_INT_SIGNAL_EN _INT_C	RW	INT_C Signal Enable field from Normal Interrupt Signal Enable Register

SDC2_SDCC_HC_REG_38_3A (cont.)

Bits	Name	Type	Description
10	NORMAL_INT_SIGNAL_EN_INT_B	RW	INT_B Signal Enable field from Normal Interrupt Signal Enable Register
9	NORMAL_INT_SIGNAL_EN_INT_A	RW	INT_A Signal Enable field from Normal Interrupt Signal Enable Register
8	NORMAL_INT_SIGNAL_EN_CARD_INT	RW	Card Interrupt Signal Enable field from Normal Interrupt Signal Enable Register
7	NORMAL_INT_SIGNAL_EN_CARD_REMOVAL	RW	Card Removal Signal Enable field from Normal Interrupt Signal Enable Register
6	NORMAL_INT_SIGNAL_EN_CARD_INSERTION	RW	Card Insertion Signal Enable field from Normal Interrupt Signal Enable Register
5	NORMAL_INT_SIGNAL_EN_BUF_RD_READY	RW	Buffer Read Ready Signal Enable field from Normal Interrupt Signal Enable Register
4	NORMAL_INT_SIGNAL_EN_BUF_WR_READY	RW	Buffer Write Ready Signal Enable field from Normal Interrupt Signal Enable Register
3	NORMAL_INT_SIGNAL_EN_DMA_INT	RW	DMA Interrupt Signal Enable field from Normal Interrupt Signal Enable Register
2	NORMAL_INT_SIGNAL_EN_BLK_GAP_EVENT	RW	Block Gap Event Signal Enable field from Normal Interrupt Signal Enable Register
1	NORMAL_INT_SIGNAL_EN_TRANS_COMPLETE	RW	Transfer Complete Signal Enable field from Normal Interrupt Signal Enable Register
0	NORMAL_INT_SIGNAL_EN_CMD_COMPLETE	RW	Command Complete Signal Enable field from Normal Interrupt Signal Enable Register

0x0786493C SDC2_SDCC_HC_REG_3C_3E**Type:** RW**Clock:** same rate as HCLK**Reset State:** 0x00000000

Combination of two registers from the SD Host Controller Standard Specification:

-Auto CMD Error Status Register(15-0)

-Host Control 2 Register(31-16)

SDC2_SDCC_HC_REG_3C_3E

Bits	Name	Type	Description
31	HST_CTL2_PRESET_VALUE_EN	RW	Preset Value Enable field from Host Control2 Register
30	HST_CTL2_ASYNC_INT_EN	RW	Asynchronous Interrupt Enable field from Host Control2 Register

SDC2_SDCC_HC_REG_3C_3E (cont.)

Bits	Name	Type	Description
23	HST_CTL2_SAMPL_CLK_SEL	RW	Sampling Clock Select field from Host Control2 Register
22	HST_CTL2_EXEC_TUNING	RW	Execute Tuning field from Host Control2 Register
21:20	HST_CTL2_DRIVER_STRENGTH_SEL	RW	Driver Strength Select field from Host Control2 Register
19	HST_CTL2_SIGNALING_1_8_EN	RW	1.8V Signaling Enable field from Host Control2 Register
18:16	HST_CTL2_UHS_MODE_SEL	RW	UHS Mode Select field from Host Control2 Register
7	AUTO_CMD_NOT_ISSUED_BY_AUTO_CMD12	R	Command Not issued By Auto CMD12 Error field from Auto CMD Error Status Register
4	AUTO_CMD_INDX_ERR	R	Auto CMD Index Error field from Auto CMD Error Status Register
3	AUTO_CMD_END_BIT_ERR	R	Auto CMD End Bit Error field from Auto CMD Error Status Register
2	AUTO_CMD_CRC_ERR	R	Auto CMD CRC Error field from Auto CMD Error Status Register
1	AUTO_CMD_TIMEOUT	R	Auto CMD Timeout Error field from Auto CMD Error Status Register
0	AUTO_CMD12_NOT_EXEC	R	Auto CMD12 Not Executed field from Auto CMD Error Status Register

0x07864940 SDC2_SDCC_HC_REG_40_42**Type:** R**Clock:** same rate as HCLK**Reset State:** 0x3029C8B2

The 32 LSBs of the Capabilities Register in SD Host Controller Standard Specification

SDC2_SDCC_HC_REG_40_42

Bits	Name	Description
31:30	CAPABILITIES_SLOT_TYPE	Slot Type field from Capabilities Register
29	CAPABILITIES_ASYNC_INT_SUPPORT	Asynchronous Interrupt Support field from Capabilities Register
28	CAPABILITIES_SYS_BUS_SUPPORT_64_BIT	64 bit System Bus Support field from Capabilities Register
26	CAPABILITIES_VOLTAGE_SUPPORT_1_8V	Voltage Support 1.8v field from Capabilities Register

SDC2_SDCC_HC_REG_40_42 (cont.)

Bits	Name	Description
25	CAPABILITIES_VOLTAGE_SUPPORT_3_0V	Voltage Support 3.0v field from Capabilities Register
24	CAPABILITIES_VOLTAGE_SUPPORT_3_3V	Voltage Support 3.3v field from Capabilities Register
23	CAPABILITIES_SUSPEND_RESUME_SUPPORT	Suspend Resume Support field from Capabilities Register
22	CAPABILITIES_SDMA_SUPPORT	SDMA Support field from Capabilities Register
21	CAPABILITIES_HS_SUPPORT	High Speed Support field from Capabilities Register
19	CAPABILITIES_ADMA2_SUPPORT	ADMA2 Support field from Capabilities Register
18	CAPABILITIES_SUPPORT_8_BIT	8 bit Supported For Embedded Product field from Capabilities Register
17:16	CAPABILITIES_MAX_BLK_LENGTH	Max Block Length field from Capabilities Register
15:8	CAPABILITIES_BASE_SDCLK_FREQ	Base Clock Frequency For SD Clock field from Capabilities Register
7	CAPABILITIES_TIMEOUT_CLOCK_UNIT	Timeout Clock Unit field from Capabilities Register
5:0	CAPABILITIES_TIMEOUT_CLOCK_FREQ	Timeout Clock Frequency field from Capabilities Register

0x07864944 SDC2_SDCC_HC_REG_44_46**Type:** R**Clock:** same rate as HCLK**Reset State:** 0x00008007

The 32 MSBs of the Capabilities Register in SD Host Controller Standard Specification

SDC2_SDCC_HC_REG_44_46

Bits	Name	Description
23:16	CAPABILITIES_CLK_MULTIPLIER	Clock Multiplier field from Capabilities Register
15:14	CAPABILITIES_RETUNING_MODE	Re-Tuning Modes field from Capabilities Register
13	CAPABILITIES_USE_TUNING_FOR_SDR50	Use Tuning for SDR50 field from Capabilities Register

SDC2_SDCC_HC_REG_44_46 (cont.)

Bits	Name	Description
11:8	CAPABILITIES_TIMER_CNT_FOR_RETUNING	Timer Count for Re-Tuning field from Capabilities Register
6	CAPABILITIES_DRIVER_TYPE_D_SUPPORT	Driver Type D Support field from Capabilities Register
5	CAPABILITIES_DRIVER_TYPE_C_SUPPORT	Driver Type C Support field from Capabilities Register
4	CAPABILITIES_DRIVER_TYPE_A_SUPPORT	Driver Type A Support field from Capabilities Register
2	CAPABILITIES_DDR_50_SUPPORT	DDR50 Support field from Capabilities Register
1	CAPABILITIES_SDR_104_SUPPORT	SDR104 Support field from Capabilities Register
0	CAPABILITIES_SDR_50_SUPPORT	SDR50 Support field from Capabilities Register

0x07864948 SDC2_SDCC_HC_REG_48_4A**Type:** R**Clock:** same rate as HCLK**Reset State:** 0x00000000

Maximum Current Register in SD Host Controller Standard Specification.

SDC2_SDCC_HC_REG_48_4A

Bits	Name	Description
23:16	MAX_CURRENT_1_8V	Maximum Current for 1.8v field from Maximum Current Capabilities Register
15:08	MAX_CURRENT_3_0V	Maximum Current for 3.0v field from Maximum Current Capabilities Register
07:00	MAX_CURRENT_3_3V	Maximum Current for 3.3v field from Maximum Current Capabilities Register

0x07864950 SDC2_SDCC_HC_REG_50_52**Type:** W**Clock:** same rate as HCLK**Reset State:** 0x00000000

Combination of two registers from the SD Host Controller Standard Specification:

-Force Event Register for Auto CMD Error Status(15-0)

-Force Event Register for Error Interrupt Status(31-16)

SDC2_SDCC_HC_REG_50_52

Bits	Name	Description
31:28	FORCE_EVENT_FOR_ERR_VENDOR_SPECIFIC_ERR_STS	Force Event for Vendor Specific Error Status field from Force Event Register for Error Interrupt Status
25	FORCE_EVENT_FOR_ERR_ADMA	Force Event for ADMA Error field from Force Event Register for Error Interrupt Status
24	FORCE_EVENT_FOR_ERR_AUTO_CMD	Force Event for Auto CMD Error field from Force Event Register for Error Interrupt Status
23	FORCE_EVENT_FOR_ERR_CURRENT_LIMIT	Force Event for Current Limit Error field from Force Event Register for Error Interrupt Status
22	FORCE_EVENT_FOR_ERR_DATA_END_BIT	Force Event for Data End Bit Error field from Force Event Register for Error Interrupt Status
21	FORCE_EVENT_FOR_ERR_DATA_CRC	Force Event for Data CRC Error field from Force Event Register for Error Interrupt Status
20	FORCE_EVENT_FOR_ERR_DATA_TIMEOUT	Force Event for Data Timeout Error field from Force Event Register for Error Interrupt Status
19	FORCE_EVENT_FOR_ERR_CMD_INDX	Force Event for Command Index Error field from Force Event Register for Error Interrupt Status
18	FORCE_EVENT_FOR_ERR_CMD_END_BIT	Force Event for Command End Bit Error field from Force Event Register for Error Interrupt Status
17	FORCE_EVENT_FOR_ERR_CMD_CRC	Force Event for Command CRC Error field from Force Event Register for Error Interrupt Status
16	FORCE_EVENT_FOR_ERR_CMD_TIMEOUT	Force Event for Command Timeout Error field from Force Event Register for Error Interrupt Status
7	FORCE_EVENT_FOR_CMD_NOT_ISSUED_BY_AUTO_CMD12	Force Event for Command Not Issued By Auto CMD12 Error field from Force Event Register for Auto CMD Error Status
4	FORCE_EVENT_FOR_AUTO_CMD_INDX_ERR	Force Event for Auto CMD Index Error field from Force Event Register for Auto CMD Error Status
3	FORCE_EVENT_FOR_AUTO_CMD_END_BIT_ERR	Force Event for Auto CMD End Bit Error field from Force Event Register for Auto CMD Error Status
2	FORCE_EVENT_FOR_AUTO_CMD_CRC_ERR	Force Event for Auto CMD CRC Error field from Force Event Register for Auto CMD Error Status
1	FORCE_EVENT_FOR_AUTO_CMD_TIMEOUT_ERR	Force Event for Auto CMD Timeout Error field from Force Event Register for Auto CMD Error Status
0	FORCE_EVENT_FOR_AUTO_CMD12_NOT_EXEC	Force Event for Auto CMD12 Not Executed field from Force Event Register for Auto CMD Error Status

0x07864954 SDC2_SDCC_HC_REG_54_56**Type:** R**Clock:** same rate as HCLK**Reset State:** 0x00000000

The 7 LSBs are the ADMA Error Status Register in SD Host Controller Standard Specification.
The Bits 31-8 are unused.

SDC2_SDCC_HC_REG_54_56

Bits	Name	Description
2	ADMA_LENGTH_MISMATCH_ERR	ADMA Length Mismatch Error field from ADMA Error Status Register
1:0	ADMA_ERR_STATE	ADMA Error State field from ADMA Error Status Register

0x07864958 SDC2_SDCC_HC_REG_58_5A**Type:** RW**Clock:** same rate as HCLK**Reset State:** 0x00000000

The 32 LSBs of the ADMA System Address Register in SD Host Controller Standard Specification

SDC2_SDCC_HC_REG_58_5A

Bits	Name	Description
31:00	ADMA_SYS_ADDRESS	ADMA System Address field from ADMA System Address Register

0x0786495C SDC2_SDCC_HC_REG_5C_5E**Type:** RW**Clock:** same rate as HCLK**Reset State:** 0x00000000

The 32 MSBs of the ADMA System Address Register in SD Host Controller Standard Specification

SDC2_SDCC_HC_REG_5C_5E

Bits	Name	Description
31:00	ADMA_SYS_ADDRESS_64	upper 32 bits of ADMA System Address field from ADMA System Address Register

0x07864960 SDC2_SDCC_HC_REG_60_62**Type:** R**Clock:** same rate as HCLK**Reset State:** 0x00040000

Combination of two registers from the SD Host Controller Standard Specification:

-Preset Value for Initialization Register(15-0)

-Preset Value for Default Speed Register(31-16)

SDC2_SDCC_HC_REG_60_62

Bits	Name	Description
31:30	DEFAULT_SPEED_DRIVER_STRENGTH_SEL	Driver Strength Select Value field from Preset Value for Default Speed
26	DEFAULT_SPEED_CLK_GEN_SEL	Clock Generator Select Value field from Preset Value for Default Speed
25:16	DEFAULT_SPEED_SDCLK_FREQ_SEL	SDCLK Frequency Select Value field from Preset Value for Default Speed
15:14	INIT_DRIVER_STRENGTH_SEL	Driver Strength Select Value field from Preset Value for Initialization
10	INIT_CLK_GEN_SEL	Clock Generator Select Value field from Preset Value for Initialization
9:0	INIT_SDCLK_FREQ_SEL	SDCLK Frequency Select Value field from Preset Value for Initialization

0x07864964 SDC2_SDCC_HC_REG_64_66**Type:** R**Clock:** same rate as HCLK**Reset State:** 0x00040002

Combination of two registers from the SD Host Controller Standard Specification:

-Preset Value for High Speed Register(15-0)

-Preset Value for SDR12 Register

SDC2_SDCC_HC_REG_64_66

Bits	Name	Description
31:30	SDR12_DRIVER_STRENGTH_SEL	Driver Strength Select Value field from Preset Value for SDR12
26	SDR12_CLK_GEN_SEL	Clock Generator Select Value field from Preset Value for SDR12

SDC2_SDCC_HC_REG_64_66 (cont.)

Bits	Name	Description
25:16	SDR12_SDCLK_FREQ_SEL	SDCLK Frequency Select Value field from Preset Value for SDR12
15:14	HS_DRIVER_STRENGTH_SEL	Driver Strength Select Value field from Preset Value for High Speed
10	HS_CLK_GEN_SEL	Clock Generator Select Value field from Preset Value for High Speed
9:0	HS_SDCLK_FREQ_SEL	SDCLK Frequency Select Value field from Preset Value for High Speed

0x07864968 SDC2_SDCC_HC_REG_68_6A**Type:** R**Clock:** same rate as HCLK**Reset State:** 0x00010002

Combination of two registers from the SD Host Controller Standard Specification:

-Preset Value for SDR25 Register(15-0)

-Preset Value for SDR50 Register

SDC2_SDCC_HC_REG_68_6A

Bits	Name	Description
31:30	SDR50_DRIVER_STRENGTH_SEL	Driver Strength Select Value field from Preset Value for SDR50
26	SDR50_CLK_GEN_SEL	Clock Generator Select Value field from Preset Value for SDR50
25:16	SDR50_SDCLK_FREQ_SEL	SDCLK Frequency Select Value field from Preset Value for SDR50
15:14	SDR25_DRIVER_STRENGTH_SEL	Driver Strength Select Value field from Preset Value for SDR25
10	SDR25_CLK_GEN_SEL	Clock Generator Select Value field from Preset Value for SDR25
9:0	SDR25_SDCLK_FREQ_SEL	SDCLK Frequency Select Value field from Preset Value for SDR25

0x0786496C SDC2_SDCC_HC_REG_6C_6E**Type:** R**Clock:** same rate as HCLK**Reset State:** 0x00020000

Combination of two registers from the SD Host Controller Standard Specification:

-Preset Value for SDR104 Register(15-0)

-Preset Value for DDR50 Register

SDC2_SDCC_HC_REG_6C_6E

Bits	Name	Description
31:30	DDR50_DRIVER_STRENGTH_SEL	Driver Strength Select Value field from Preset Value for DDR50
26	DDR50_CLK_GEN_SEL	Clock Generator Select Value field from Preset Value for DDR50
25:16	DDR50_SDCLK_FREQ_SEL	SDCLK Frequency Select Value field from Preset Value for DDR50
15:14	SDR104_DRIVER_STRENGTH_SEL	Driver Strength Select Value field from Preset Value for SDR104
10	SDR104_CLK_GEN_SEL	Clock Generator Select Value field from Preset Value for SDR104
9:0	SDR104_SDCLK_FREQ_SEL	SDCLK Frequency Select Value field from Preset Value for SDR104

0x078649E0 SDC2_SDCC_HC_REG_E0_E2

Type: RW

Clock: same rate as HCLK

Reset State: 0x00000000

Shared Bus Control Register in SD Host Controller Standard Specification.

SDC2_SDCC_HC_REG_E0_E2

Bits	Name	Type	Description
30:24	SHARED_BUS_CTL_BACK_END_PWR	RW	Back End Power Control field from Shared Bus Control Register
22:20	SHARED_BUS_INT_PIN_SEL	RW	
18:16	SHARED_BUS_CTL_CLK_PIN_SEL	RW	Clock Pin Select field from Shared Bus Control Register
14:8	SHARED_BUS_CTL_BUS_WIDTH_PRESET	R	Bus Width Preset field from Shared Bus Control Register
5:4	SHARED_BUS_CTL_NUM_INTERRUPT_INPUT_PINS	R	Number Of Interrupt Input Pins field from Shared Bus Control Register
2:0	SHARED_BUS_CTL_NUM_CLK_PINS	R	Number of Clock Pins field from Shared Bus Control Register

0x078649FC SDC2_SDCC_HC_REG_FC_FE**Type:** RW**Clock:** same rate as HCLK**Reset State:** 0x00020000

Combination of two registers from the SD Host Controller Standard Specification:

-Slot Interrupt Status Register(15-0)

-Host Controller Version Register

SDC2_SDCC_HC_REG_FC_FE

Bits	Name	Type	Description
31:24	HC_VENDOR_VERSION_NUM	R	Vendor Version Number field from Host Controller Version Register
23:16	HC_SPEC_VERSION_NUM	R	Specification Version Number field from Host Controller Version Register
7:0	INT_SIGNAL_FOR_EACH_SLOT	RW	Interrupt Signal For Each Slot field from Slot Interrupt Status Register

0x07864A00 SDC2_SDCC_HC_REG_DLL_CONFIG**Type:** RW**Clock:** same rate as HCLK**Reset State:** 0x60006400

This register configures the DLL's inputs.

SDC2_SDCC_HC_REG_DLL_CONFIG

Bits	Name	Description
31	SDC4_DIS_DOUT	When this bit equals to '0', the data which comes out from the dll is valid. When it equals to '1' this data isn't valid. The inversion of this bit is connected to dll's input - dll_en_dout
30	DLL_RST	Setting this bit to 1 resets cm_dll_sdc4. cm_dll_sdc4 should be reset every time the MCLK frequency is changed.
29	PDN	Power Down Value 0 - analog blocks are enabled Value 1 - analog blocks are powered down (default)
28	CK_INTP_SEL	Selects interpolator output
27	CK_INTP_EN	Enable clock interpolation for finer resolution

SDC2_SDCC_HC_REG_DLL_CONFIG (cont.)

Bits	Name	Description
26:24	MCLK_FREQ	Frequency of MCLK Value 000 -100 - 112 (MHz) Value 001 -112 - 125 Value 010 -125 - 137 Value 011 -137 - 150 Value 100 -150 - 162 Value 101 -162 - 175 Value 110 -175 - 187 Value 111 -187 - 200
23:20	CDR_SELECT	CDR phase select (gray coded) Value 0000 - phase 0 Value 0001 - phase 1 Value 0011 - phase 2 Value 0010 - phase 3 Value 0110 - phase 4 Value 0111 - phase 5 Value 0101 - phase 6 Value 0100 - phase 7 Value 1100 - phase 8 Value 1101 - phase 9 Value 1111 - phase 10 Value 1110 - phase 11 Value 1010 - phase 12 Value 1011 - phase 13 Value 1001 - phase 14 Value 1000 - phase 15
19	CDR_EXT_EN	Enable external control of cdr phase select
18	CK_OUT_EN	Enable output clock (default value is '1')
17	CDR_EN	Enable CDR function
16	DLL_EN	Enable DLL function
15:14	CDR_UPD_RATE	CDR update rate, low pass filtering window of CDR Max update rate: Value 00 - 0.5 MCLK frequency Value 01 - 0.25 MCLK Frequency (default) Value 10 - 0.125 MCLK frequency Value 11 - 1/16 MCLK Frequency
13:12	DLL_UPD_RATE	DLL update rate Value 00 - sdc4_mclk/10 Value 01 - sdc4_mclk/20 Value 10 - sdc4_mclk/40 (default) Value 11 - sdc4_mclk/80

SDC2_SDCC_HC_REG_DLL_CONFIG (cont.)

Bits	Name	Description
11:10	DLL_PHASE_DET	DLL phase detector low pass average window Value 00 - 2 cycles Value 01 - 4 cycles (default) Value 10 - 8 cycles Value 11 - 16 cycles
9	CDR_ALGORITHM_SEL	CDR algorithm select Value 0 - Edge (default) Value 1 - Level
0	CMD_DAT_TRACK_SEL	Select DAT or CMD lines for CDR tracking Value 0 - track DAT inputs (default) Value 1 - track CMD input

0x07864A08 SDC2_SDCC_HC_REG_DLL_STATUS**Type:** R**Clock:** same rate as HCLK**Reset State:** 0x00000000

This register is connected to cm_dll_sdc4's status output.

SDC2_SDCC_HC_REG_DLL_STATUS

Bits	Name	Description
12	SDC4_DTEST_MUXSEL	Digital test mux select
11	DDR_DLL_LOCK_JDR	Reserved
10:9	SDC4_DLL_DTEST_OUT_A TPG	SDC4_DLL_DTEST_OUT output from DLL after a clock mux with sampled JDRs for ATPG mode. This bit helps collecting coverage on DLL's output
8	SDC4_DLL_LOCK_ATPG	SDC4_DLL_LOCK output from DLL after a clock mux with sampled JDRs for ATPG mode. This bit helps collecting coverage on DLL's output
7	DLL_LOCK	DLL lock status Value 0 - Not locked Value 1 - Locked

SDC2_SDCC_HC_REG_DLL_STATUS (cont.)

Bits	Name	Description
6:3	CDR_PHASE	CDR phase select (gray coded) Value 0000 - phase 0 Value 0001 - phase 1 Value 0011 - phase 2 Value 0010 - phase 3 Value 0110 - phase 4 Value 0111 - phase 5 Value 0101 - phase 6 Value 0100 - phase 7 Value 1100 - phase 8 Value 1101 - phase 9 Value 1111 - phase 10 Value 1110 - phase 11 Value 1010 - phase 12 Value 1011 - phase 13 Value 1001 - phase 14 Value 1000 - phase 15
2	DDLL_COARSE_CAL	Value 1 - done Value 0 - not done

0x07864A0C SDC2_SDCC_HC_VENDOR_SPECIFIC_FUNC**Type:** RW**Clock:** same rate as HCLK**Reset State:** 0x00000A1C**SDC2_SDCC_HC_VENDOR_SPECIFIC_FUNC**

Bits	Name	Description
31:22	HC_AUTO_CMD_COUNTER	This counter will determine the time between two auto commands transmissions
21:19	HC_SELECT_IN	Selecting the SD controller's mode. Equivalent to SELECT_IN filed in MCI_CLK
18	HC_SELECT_IN_EN	Enabling the selection of mode via the vendor specific register
17	ADMA_DESC_PRIORITY	If set (1) then the ADMA's descriptors SM gets priority on the data accesses. Not Applicabale for Gandalf.
16	HC_IO_PAD_PWR_SWITCH	Indication to Pad that power has to be switched from 3.3V to 1.8V.
15	HC_IO_PAD_PWR_SWITCH_EN	This Bit enables the use of HC_IO_PAD_PWR_SWITCH
14	HC_IGNORE_CTOUT	If set (1), CPSM disables command timer and waits for interrupt request (Card Response). See note #4 below.

SDC2_SDC_C_HC_VENDOR_SPECIFIC_FUNC (cont.)

Bits	Name	Description
13	HC_SW_SDC4_CMD19	The input of cm_dll_sdc4 - sdc4_cmd19 can be driven by SW if AUTO_CMD19 (bit 16 in MCI_CMD) feature is not used. Value 1 - sdc4_cmd19 is driven by SW Value 0 (default) - sdc4_cmd19 is driven by HW
12	HC_INVERT_OUT	value: 0 data and cmd are driven by falling edge value 1: data and cmd are driven by rising edge.
11	HC_INVERT_OUT_EN	value 1: HC_INVERT_OUT is driven by bit 12 of HC_VENDOR_SPECIFIC_FUNC (HC_INVERT_OUT) value 0: HC_INVERT_OUT is driven by HST_CTL1_HS_EN .
10	MODE	Data transfer mode: 0: block data transfer 1: stream data transfer. Can be used in bus testing procedure
9:8	SDC4_MCLK_SEL	Selects the sdc4_mclk - input clock of cm_dll_sdc4 Value 1 - feedback clock from CLK pad Value 2 (default) - free running MCLK (gated between the transactions) Value 3 - free running MCLK divided by 2
7	ADMA_ERR_SIZE_EN	enabling interrupt for mismatch error in ADMA2
6	HC_AUTO_CMD21_EN	Enabling bit for the auto command 21 option. value 0: disable value 1:enable
5	HC_AUTO_CMD19_EN	Enabling bit for the auto command 19 option. value 0: disable value 1:enable
4	HC_HCLKON_SW_EN	SW enable of AHB clock request of SDCC5. value 0: HW clock request mechanism is used value 1: clock request signal is always high
3	ADMA_BUSREQ_EN	enabling the fix in ADMA2 for busrequest
2	FLOW_EN	Enable flow control: value 0: disable value 1: enable (default)
1	PWRSAVE	Disable Prime-Cell MCI clock output when bus is idle to save power. value 0: Power save disabled - always enabled if bit 8 is set. value 1: Power save enabled - clock enabled only when bus is active and bit 8 is set.
0	CLK_INV	When set(1), the input clock which is inserted into the input macros is inverted.

0x07864A10 SDC2_SDCC_HC_VENDOR_SPECIFIC_FUNC2**Type:** RW**Clock:** same rate as HCLK**Reset State:** 0xF88218A8**SDC2_SDCC_HC_VENDOR_SPECIFIC_FUNC2**

Bits	Name	Type	Description
31	QSB_AXI_ABURST	RW	Driving QSB AXI output
30:28	NUM_OUTSTANDING_DATA	RW	Configuring the number of outstanding transactions for the data. The descriptors accesses have more one outstanding request
27	PROCEED_AXI_AFTER_ERROR	RW	When error occurs on AXI master the traffic can continue or terminate. If this bit is set then the traffic continues and more requests are transmitted on the address channel
26	QSB_AXI_AFULL_CALC	RW	If this bit is set, then afull isn't '0' constantly but can be equal to '1' if all the beats have 8 bytes on the strobe lines
25	ONE_MID_SUPPORT	RW	If this bit is set, then only 1 MID is used for descriptors and data
24:22	RESERVED	RW	Reserved
21	HC_SW_RST_REQ	RW	Before executing HC_SW_RST_FOR_ALL or HC_SW_RST_FOR_DAT this bit can be set and SW needs to polls on it until it's being cleared. The bit is cleared after AXI master (QMB) has finished processing the outstanding transactions and it's in IDLE state. This bit can be useful for HPI feature.
20	HC_SW_RST_WAIT_IDLE_DIS	RW	When this bit is cleared, the HC_SW_RST is executed only after the AXI master (QMB) is in IDLE state. This bit Can be used for HPI feature instead of polling on HC_SW_RST_REQ bit. This bit should be cleared for SW reset from unknown states.
19	SDCC5_HALT_REQ	R	Reading SDCC5_HALT_REQ input
18	SDCC5_HALT_ACK	R	Reading SDCC5_HALT_ACK output
17	SDCC5_M_IDLE	R	Reading SDCC5_M_IDLE output
16	SDCC5_HALT_ACK_SW_EN	RW	Enabling the SW value for SDCC5_HALT_ACK
15	SDCC5_HALT_ACK_SW	RW	Driving SDCC5_HALT_ACK from SW and not by HW
14	SDCC5_M_IDLE_DIS	RW	If this bit is set, then sdcc5_m_idle will be tied low
13	QSB_AXI_INTERLEAVING_EN	RW	If the bit is set, then the controller can support interleaving between two bursts from the Noc. This option is disabled by default because Noc doesn't support interleaving but only re-ordering
12	RESERVED	RW	Reserved

SDC2_SDCC_HC_VENDOR_SPECIFIC_FUNC2 (cont.)

Bits	Name	Type	Description
11	RESERVED	RW	Reserved
10:9	RESERVED	RW	Reserved
8:6	RESERVED	RW	Reserved
5	RESERVED	RW	Reserved
4	RESERVED	RW	Reserved
3	RESERVED	RW	Reserved
2	QSB_AXI_OOOWR	RW	Driving QSB AXI output
1	QSB_AXI_OOORD	RW	Driving QSB AXI output
0	QSB_AXI_AFULL	RW	Driving QSB AXI output

0x07864A14 SDC2_SDCC_HC_VENDOR_SPECIFIC_ADMA_ERR_ADDR0**Type:** R**Clock:** same rate as HCLK**Reset State:** 0x00000000

Last AHB address phase before response error was detected

SDC2_SDCC_HC_VENDOR_SPECIFIC_ADMA_ERR_ADDR0

Bits	Name	Description
31	RESP_ERR_WR	If set, then the failed transaction was a write access
30:28	RESP_ERR_TID	The TID of the failed transaction
27:13	DESC_ADDR_35_21	Upper 15 bits of current address in address channel
12:0	DESC_ADDR_12_0	Lower 13 bits of current address in address channel

0x07864A18 SDC2_SDCC_HC_VENDOR_SPECIFIC_ADMA_ERR_ADDR1**Type:** R**Clock:** same rate as HCLK**Reset State:** 0x00000000

Last AHB attributes before response error was detected

SDC2_SDCC_HC_VENDOR_SPECIFIC_ADMA_ERR_ADDR1

Bits	Name	Description
21:20	AHB_MON_HTRANS	Htrans control bits from AHB in the last address phase before getting the response error
19	AHB_MON_WO_REQ	indication that the current master got the response error w/o asserting hbusreq
18:16	AHB_MON_HBURST	Hburst control bits from AHB in the last address phase before getting the response error
15	AHB_MON_HWRITE	Hwrite control bits from AHB in the last address phase before getting the response error
14:12	AHB_MON_HSIZE	Hsize control bits from AHB in the last address phase before getting the response error
11:8	AHB_MON_HPROT	Hprot control bits from AHB in the last address phase before getting the response error
7:0	AHB_MON_CUR_MASTER	Indication on the last bus master which accessed to AHB bus

0x07864A1C SDC2_SDCC_HC_VENDOR_SPECIFIC_CAPABILITIES0**Type:** RW**Clock:** same rate as HCLK**Reset State:** 0x00000000

driving the HC_REG_40_42 register

SDC2_SDCC_HC_VENDOR_SPECIFIC_CAPABILITIES0

Bits	Name	Description
31:30	VS_CAPABILITIES_SLOT_T YPE	Slot Type field from Capabilities Register
29	VS_CAPABILITIES_ASYNC_ INT_SUPPORT	Asynchronous Interrupt Support field from Capabilities Register
28	VS_CAPABILITIES_SYS_BU S_SUPPORT_64_BIT	64 bit System Bus Support field from Capabilities Register
26	VS_CAPABILITIES_VOLTAG E_SUPPORT_1_8V	Voltage Support 1.8v field from Capabilities Register
25	VS_CAPABILITIES_VOLTAG E_SUPPORT_3_0V	Voltage Support 3.0v field from Capabilities Register
24	VS_CAPABILITIES_VOLTAG E_SUPPORT_3_3V	Voltage Support 3.3v field from Capabilities Register
23	VS_CAPABILITIES_SUSPE ND_RESUME_SUPPORT	Suspend Resume Support field from Capabilities Register

SDC2_SDCC_HC_VENDOR_SPECIFIC_CAPABILITIES0 (cont.)

Bits	Name	Description
22	VS_CAPABILITIES_SDMA_SUPPORT	SDMA Support field from Capabilities Register
21	VS_CAPABILITIES_HS_SUPPORT	High Speed Support field from Capabilities Register
19	VS_CAPABILITIES_ADMA2_SUPPORT	ADMA2 Support field from Capabilities Register
18	VS_CAPABILITIES_SUPPORT_8_BIT	8 bit Supported For Embedded Product field from Capabilities Register
17:16	VS_CAPABILITIES_MAX_BLOCK_LENGTH	Max Block Length field from Capabilities Register
15:8	VS_CAPABILITIES_BASE_SDCLK_FREQ	Base Clock Frequency For SD Clock field from Capabilities Register
7	VS_CAPABILITIES_TIMEOUT_CLK_UNIT	Timeout Clock Unit field from Capabilities Register
5:0	VS_CAPABILITIES_TIMEOUT_CLK_FREQ	Timeout Clock Frequency field from Capabilities Register

0x07864A20 SDC2_SDCC_HC_VENDOR_SPECIFIC_CAPABILITIES1**Type:** RW**Clock:** same rate as HCLK**Reset State:** 0x02008007

driving the HC_REG_44_46 register and spec version

SDC2_SDCC_HC_VENDOR_SPECIFIC_CAPABILITIES1

Bits	Name	Description
31:24	SPEC_VERSION	driving HC_SPEC_VERSION_NUM field in HC_REG_FC_FE register
23:16	VS_CAPABILITIES_CLK_MULTIPLIER	Clock Multiplier field from Capabilities Register
15:14	VS_CAPABILITIES_RETUNING_MODE	Re-Tuning Modes field from Capabilities Register
13	VS_CAPABILITIES_USE_TUNING_FOR_SDR50	Use Tuning for SDR50 field from Capabilities Register
11:8	VS_CAPABILITIES_TIMER_COUNT_FOR_RETUNING	Timer Count for Re-Tuning field from Capabilities Register
6	VS_CAPABILITIES_DRIVER_TYPE_D_SUPPORT	Driver Type D Support field from Capabilities Register

SDC2_SDCC_HC_VENDOR_SPECIFIC_CAPABILITIES1 (cont.)

Bits	Name	Description
5	VS_CAPABILITIES_DRIVER_TYPE_C_SUPPORT	Driver Type C Support field from Capabilities Register
4	VS_CAPABILITIES_DRIVER_TYPE_A_SUPPORT	Driver Type A Support field from Capabilities Register
2	VS_CAPABILITIES_DDR_50_SUPPORT	DDR50 Support field from Capabilities Register
1	VS_CAPABILITIES_SDR_104_SUPPORT	SDR104 Support field from Capabilities Register
0	VS_CAPABILITIES_SDR_50_SUPPORT	SDR50 Support field from Capabilities Register

0x07864A24 SDC2_SDCC_HC_VENDOR_SPECIFIC_T4_DLY_CTRL**Type:** RW**Clock:** same rate as HCLK**Reset State:** 0x0000382B

Control register of T/4 Delay element which is used in e.MMC5.0 DDR mode.

SDC2_SDCC_HC_VENDOR_SPECIFIC_T4_DLY_CTRL

Bits	Name	Description
15:14	T4_DLY_SPARE_CTL	
13	T4_DLY_POWERDOWN_EN	Powerdown enable. When asserted (default), it powers down the T/4 macro and shuts off all bias circuits.
12:11	T4_DLY_TESTOUT_SEL	Test output select. When toggled, it presents either the strb_in or strb_out on the analog test port for debug.
10	T4_DLY_BYPASS_EN	Bypass enable. When asserted it presents the strb_in on the strb_out port.
9	T4_DLY_CAL_START	Calibration start input. When asserted, it indicates the start of the calibration cycle.
8	T4_DLY_CAL_OVERRIDE	Calibration override input. When asserted, the internal delay line calibration is bypassed and the tby4_user_ctl<7:0> is used as the default code for the delay line.
7:0	T4_DLY_TARGET_CODE	T/4 delay target code. 0x2b gives delay of 1.25ns

0x07864A28 SDC2_SDCC_HC_VENDOR_SPECIFIC_T4_DLY_STAT**Type:** R**Clock:** same rate as HCLK**Reset State:** 0x00000000

Status register T/4 Delay element which is used in e.MMC5.0 DDR mode.

SDC2_SDCC_HC_VENDOR_SPECIFIC_T4_DLY_STAT

Bits	Name	Description
15:14	T4_DLY_SPARE_OUT	Spare status bits.
13	T4_DLY_CAL_DLY_CODE_FAST_RANGE	Calibrated delay code, indicates the fast range selection bit status.
12:6	T4_DLY_CAL_DLY_CODE_ISEL	Calibrated delay code, indicates the isel programming value for the bias generator.
5:4	T4_DLY_CAL_DLY_CODE_RSEL	Calibrated delay code, indicates the rsel programming value for the bias generator.
3:0	T4_DLY_STATE_DESC	T/4 state descriptor. Indicates the calibration status and result. value 4'b0000: Indicates Reset mode, waiting for cal_start signal. value 4'b0111: Indicates the fast range decision completion. value 4'b0100: Indicates rsel adjustment decision. value 4'b1011: Indicates isel adjustment decision. value 4'b1110: Indicates Normal end-of-cal. value 4b'1111: Indicates Maximum Delay Reached

0x07864A30 SDC2_SDCC_HC_VENDOR_SPECIFIC_CSR_CDC_CTLR_CFG0**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

The HC_VENDOR_SPECIFIC_CSR_CDC_CTLR_CFG0 register configures the following:

SDC2_SDCC_HC_VENDOR_SPECIFIC_CSR_CDC_CTLR_CFG0

Bits	Name	Description
25	STAGGER_CAL_ENA	1'b1: Stagger calibration of write and read CDCs once after the other to reduce peak voltage drop. 1'b0: Both write and read CDCs calibrated simultaneously This bit is introduced on the x2 core, so not present on x0 and x1 versions of DIM/PHY.
24	TRACKING_CAL_ENA	1'b1: Relationship between subunits and oscillator count is learnt and used for tracking calibration. 1'b0: Periodic calibration triggers cause full recalibration

SDC2_SDCC_HC_VENDOR_SPECIFIC_CSR_CDC_CTLR_CFG0 (cont.)

Bits	Name	Description
23:20	OSC_COUNT_ERR_TOLERANCE	Magnitude of the difference between oscillator count and target count when the algorithm assumes convergence and stops further search.
19	TRACK_CALIB_MODE	1'b0: Tracking calibration uses previous Target_count value. 1'b1 Accurate mode: Target_count is recomputed at the start of tracking calibration
18	FULL_DELAY	1'b0: unitstep and subunit caps are from calibration or SW settings 1'b1: All unit step and subunit caps are selected for the delay arrays.
17	HW_AUTOCAL_ENA	1'b1: Enables periodic auto calibration 1'b0: Periodic calibration is disabled
16	SW_TRIG_FULL_CALIB	A low to high transition on this register bit triggers the CDC calibration statemachine to initiate full calibration of the standby delay array. Also resets tracking mode for the delay arrays.
13:12	OSC_PRE_DIV	The factor by which the raw oscillator frequency is divided before being given to the counting logic. 2'b00 Divide by 1 2'b01 Divide by 2 2'b10 Divide by 4 2'b11 Divide by 8
10:0	TMUX_CHAR	Cchar is a characterized constant equal to the ratio of output mux offset delay to average unit step delay. This register is programmed by multiplying the constant with 1024 and rounding to the nearest integer. E.g. if Cchar = 0.67, $RO(0.67 * 1024) = 686$, hence program TMUX_CHAR = 11'h2AE

0x07864A34 SDC2_SDCC_HC_VENDOR_SPECIFIC_CSR_CDC_CTLR_CFG1**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

The HC_VENDOR_SPECIFIC_CSR_CDC_CTLR_CFG1 register configures the following:

SDC2_SDCC_HC_VENDOR_SPECIFIC_CSR_CDC_CTLR_CFG1

Bits	Name	Description
26:24	OSC_COUNT_DELAY	Number of oscillator cycles from the end of reference gate signal to assertion of oscillation counter done. Legal values are from 0 to 4.

SDC2_SDCC_HC_VENDOR_SPECIFIC_CSR_CDC_CTLR_CFG1 (cont.)

Bits	Name	Description
23:20	STANDBY_DELAY	This is the number of reference clock cycles from when a newly calibrated delay array is made active to when the current active delay array is put into standby mode. 4'b0: Disables the counter. Hardware waits for explicit acknowledgement of delay array switch from CDC. Others: Explicit acknowledgement mode is disabled and the counter is used
18:16	DEL_MODE_DELAY	Number of reference clock cycles between putting a newly calibrated delay array in delay mode to selecting it as the active delay array. The minimum allowed value is 1
14:12	OSC_MODE_DELAY	Number of reference clock cycles between putting a delay array in oscillator mode to starting the oscillation counter. This delay is allowed for the oscillations to stabilize through the delay array. The minimum allowed value is 1
10:8	DECODER_DELAY	Number of reference clock cycles allowed for unitstep and subunit counts to be encoded into the unit and finecap select lines on the DDA. The minimum allowed value is 1.
6:4	DIVIDER_DELAY	Number of reference clock cycles allowed for arithmetic division operations. The minimum allowed value is 1
2:0	MULTIPLIER_DELAY	Number of reference clock cycles allowed for arithmetic multiplication operations. The minimum allowed value is 1

0x07864A38 SDC2_SDCC_HC_VENDOR_SPECIFIC_CSR_CDC_CAL_TIMER_CFG0**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

The HC_VENDOR_SPECIFIC_CSR_CDC_CAL_TIMER_CFG0 register configures the following:

SDC2_SDCC_HC_VENDOR_SPECIFIC_CSR_CDC_CAL_TIMER_CFG0

Bits	Name	Description
24	INVALID_TIMER_ENA	1'b1: Enable the timer counting number of missed calibration intervals after which the CDC is invalid. 1'b0: Timer disabled A low to high transition on this register bit loads the timer with programmed value below.
23:20	INVALID_TIMER_VAL	Number of missed calibration intervals after which CDC invalid state is asserted.

SDC2_SDCC_HC_VENDOR_SPECIFIC_CSR_CDC_CAL_TIMER_CFG0 (cont.)

Bits	Name	Description
16	TIMER_ENA	1'b1: Enables 32 kHz sleep clock based periodic autocalibration timer. 1'b0: Timer disabled A low to high transition on this register bit loads the timer with programmed value below.
15:0	TIMER_VAL	Count value for periodic autocalibration interval based on sleep clock.

0x07864A3C SDC2_SDCC_HC_VENDOR_SPECIFIC_CSR_CDC_CAL_TIMER_CFG1**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

The HC_VENDOR_SPECIFIC_CSR_CDC_CAL_TIMER_CFG1 register configures the following:

SDC2_SDCC_HC_VENDOR_SPECIFIC_CSR_CDC_CAL_TIMER_CFG1

Bits	Name	Description
12	FF_TIMER_ENA	1'b1: Enables reference clock based periodic autocalibration timer. 1'b0: Timer disabled A low to high transition on this register bit loads the timer with programmed value below.
9:0	FF_TIMER_VAL	Count value for periodic autocalibration interval based on reference clock.

0x07864A40 SDC2_SDCC_HC_VENDOR_SPECIFIC_CSR_CDC_REFCOUNT_CFG**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

The HC_VENDOR_SPECIFIC_CSR_CDC_REFCOUNT_CFG register configures the following:

SDC2_SDCC_HC_VENDOR_SPECIFIC_CSR_CDC_REFCOUNT_CFG

Bits	Name	Description
31:16	TREF	Reference clock period programmed in pico seconds.
13:8	CCAL_REF_COUNT	Number of reference clock cycles used for oscillation counts during coarse calibration. Maximum supported value is 59.

SDC2_SDCC_HC_VENDOR_SPECIFIC_CSR_CDC_REFCOUNT_CFG (cont.)

Bits	Name	Description
5:0	FCAL_REF_COUNT	Number of reference clock cycles used for oscillation counts during fine calibration. Maximum supported value is 59

0x07864A44 SDC2_SDCC_HC_VENDOR_SPECIFIC_CSR_CDC_COARSE_CAL_CFG**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

The register HC_VENDOR_SPECIFIC_CSR_CDC_COARSE_CAL_CFG configures the following:

SDC2_SDCC_HC_VENDOR_SPECIFIC_CSR_CDC_COARSE_CAL_CFG

Bits	Name	Description
12:8	CCAL_SUBUNIT_CAPS	Subunit count needed for an effective unit step count of 32 during coarse calibration. This value is also used by software oscillator mode.
5:0	CCAL_UNITSTEPS	unitstep count needed for an effective unit step count of 32 during coarse calibration.

0x07864A48 SDC2_SDCC_HC_VENDOR_SPECIFIC_CSR_CDC_RSVD_CFG**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

The HC_VENDOR_SPECIFIC_CSR_CDC_RSVD_CFG register configures the following:

SDC2_SDCC_HC_VENDOR_SPECIFIC_CSR_CDC_RSVD_CFG

Bits	Name	Description
15:0	TEMP_FIELD	Reserved register

0x07864A4C SDC2_SDCC_HC_VENDOR_SPECIFIC_CSR_CDC_OFFSET_CFG**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

The HC_VENDOR_SPECIFIC_CSR_CDC_OFFSET_CFG register configures the following:

SDC2_SDCC_HC_VENDOR_SPECIFIC_CSR_CDC_OFFSET_CFG

Bits	Name	Description
20	SUBUNIT_OFFSET_MODE	1'b1: The programmed offset is directly loaded as subunit count 1'b0: Programmed offset is added/subtracted from the subunit count
19	SUBUNIT_OFFSET_SIGN	1'b1: Offset is subtracted from the subunit count 1'b0: Offset is added to the subunit count
16:12	SUBUNIT_OFFSET	Software subunit offset
8	UNIT_OFFSET_MODE	1'b1: The programmed offset is directly loaded as unit step count 1'b0: Programmed offset is added/subtracted from the unit step count
7	UNIT_OFFSET_SIGN	1'b1: Offset is subtracted from the unitstep count 1'b0: Offset is added to the unitstep count
5:0	UNIT_OFFSET	Software unitstep offset

0x07864A50 SDC2_SDCC_HC_VENDOR_SPECIFIC_CSR_CDC_DELAY_CFG**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

The HC_VENDOR_SPECIFIC_CSR_CDC_DELAY_CFG register configures the following:

SDC2_SDCC_HC_VENDOR_SPECIFIC_CSR_CDC_DELAY_CFG

Bits	Name	Description
28	TARGET_COUNT_ENA	1'b1: Enable software specified Target_count for calibration instead of value calculated by hardware. 1'b0: Hardware computed Target_count used.
27:16	TARGET_COUNT	Software specified Target_count to which the statemachine calibrates
11:0	DELAY_VAL	Delay required from CDC programmed in pico seconds. This value has to be greater than the minimum delay supported by DDA Tmin.

0x07864A54 SDC2_SDCC_HC_VENDOR_SPECIFIC_CSR_CDC_SW_MODE_CFG**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

The HC_VENDOR_SPECIFIC_CSR_CDC_SW_MODE_CFG register configures the following:

SDC2_SDCC_HC_VENDOR_SPECIFIC_CSR_CDC_SW_MODE_CFG

Bits	Name	Description
3	SW_DEL_MODE	A low to high transition on this register bit puts the delay array selected by SW_DA_SEL into delay mode and connects it to the DDA output. This register has to remain set for the duration of SW controlled delay mode.
2	SW_OSC_MODE	Reserved register.
1	SW_DA_SEL	Selects the delay array inside DDA during software delay or oscillator mode.
0	SW_LOAD	1'b1: Enable loading of the software offsets from the offset configuration register. This register has to remain set for the duration of SW mode operation.

0x07864A5C SDC2_SDCC_HC_VENDOR_SPECIFIC_CSR_CDC_SW_OVRD_CFG**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

The HC_VENDOR_SPECIFIC_CSR_CDC_SW_OVRD_CFG register configures the following:

SDC2_SDCC_HC_VENDOR_SPECIFIC_CSR_CDC_SW_OVRD_CFG

Bits	Name	Description
16	SW_REF_GATE	Write 1'b1 to start the reference window. Write 1'b0 to end the reference window. Number of reference cycle in the window will be reported in SW_REF_GATE_COUNT in CDC_STATUS0
11	SW_OVRD_DA1_OSC_EN	In software override mode, 1'b1 written to this bit puts delay array 1 in oscillator-mode
10	SW_OVRD_DA1_IN_EN	In software override mode, 1'b1 written to this bit puts delay array 1 in delay-mode
9	SW_OVRD_DA0_OSC_EN	In software override mode, 1'b1 written to this bit puts delay array 0 in oscillator-mode
8	SW_OVRD_DA0_IN_EN	In software override mode, 1'b1 written to this bit puts delay array 0 in delay-mode
3	SW_OVRD_CDC_COUNTER_RST	1'b1 written to this bit resets the oscillation counter. Follow this with 1'b0 before setting reference gate.
2	SW_OVRD_LOAD_DA_SEL	In software override mode, this bit select the delay array to which unit and subunit offset values from CDC_OFFSET_CFG register will be loaded.
1	SW_OVRD_ACTV_DA_SEL	In software override mode, this register selects the delay array connected to the CDC output

SDC2_SDCC_HC_VENDOR_SPECIFIC_CSR_CDC_SW_OVRD_CFG (cont.)

Bits	Name	Description
0	SW_OVRD_ENA	1'b1: written to this bit puts the CDC interface in software override mode where the below register bits control the CDC. 1'b0: CDC is under hardware control

0x07864A60 SDC2_SDCC_HC_VENDOR_SPECIFIC_CSR_CDC_SLAVE_DDA_CFG**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

The HC_VENDOR_SPECIFIC_CSR_CDC_SLAVE_DDA_CFG register configures the following:

SDC2_SDCC_HC_VENDOR_SPECIFIC_CSR_CDC_SLAVE_DDA_CFG

Bits	Name	Description
17	SLAV_DDA_OFFSET_MODE	1'b1: The programmed offset is directly loaded as unit-step count 1'b0: Programmed offset is added/subtracted from the unit-step count
16	SLAV_DDA_OFFSET_SIGN	1'b1: Offset is subtracted from the unit-step count. This subtraction feature is not supported on x0 and x1 PHY cores (bit RESERVED). Supported on the x2 core version. 1'b0: Offset is added to the unit-step count
15:12	SLAVE_DDA_OFFSET	unit-step offset for slave DDA.
10:0	SLAVE_DDA_DELAY	Delay required from slave DDA programmed in pico seconds.

0x07864A64 SDC2_SDCC_HC_VENDOR_SPECIFIC_CSR_CDC_STATUS0**Type:** R**Clock:** HCLK**Reset State:** 0x0000000C

The HC_VENDOR_SPECIFIC_CSR_CDC_STATUS0 register configures the following:

SDC2_SDCC_HC_VENDOR_SPECIFIC_CSR_CDC_STATUS0

Bits	Name	Description
26:24	CDC_ERROR_CODE	Error code for internal errors: 3'b000: No error 3'b001: Programmed delay less than Tmux 3'b010: Programmed delay less than Tmin 3'b011: Unit-step overflow: Overflow during fine calibration. Desired delay cannot be realized even when all unit-steps are selected. 3'b100: Unit-step underflow 3'b101: Unit-step overflow: Overflow during coarse calibration. Estimated unit steps needed to realize the desired delay are greater than the total number of unit-steps in design. Note: The CDC_ERROR_CODE is not a sticky register and is updated after each CDC calibration.
23:16	SW_REF_GATE_COUNT	Number of reference clock cycles in the software generated oscillation gate window during SW override mode
15:4	OSC_COUNT	Current value of oscillator count
3	CURR_SEL_DA	This bit indicates the current active delay array connected to the output. 1'b0: DA0 is selected 1'b1: DA1 is selected
2	CTLR_SM_IDLE	1'b1 Indicates that the calibration state machine is idle 1'b0 indicates calibration state machine is busy
1	OSC_DONE	This status register is set after oscillations count is finished by CDC. It gets reset when a new oscillator count is started.
0	CALIBRATION_DONE	1'b1 indicates that calibration is done and CURR_SUBUNIT_STEPS, CURR_UNIT_STEPS status registers are valid.

0x07864A68 SDC2_SDCC_HC_VENDOR_SPECIFIC_CSR_CDC_STATUS1**Type:** R**Clock:** HCLK**Reset State:** 0x00000000

The HC_VENDOR_SPECIFIC_CSR_CDC_STATUS1 register configures the following:

SDC2_SDCC_HC_VENDOR_SPECIFIC_CSR_CDC_STATUS1

Bits	Name	Description
27:16	CURR_DELAY_VALUE	HW Computed value of current delay after calibration by the state machine. This register is valid if CALIBRATION_DONE status bit is set.

SDC2_SDCC_HC_VENDOR_SPECIFIC_CSR_CDC_STATUS1 (cont.)

Bits	Name	Description
11:0	CURR_TMUX_DELAY	Current value of output mux offset delay computed by HW.

0x07864A6C SDC2_SDCC_HC_VENDOR_SPECIFIC_CSR_CDC_STATUS2**Type:** R**Clock:** HCLK**Reset State:** 0x10331033

The HC_VENDOR_SPECIFIC_CSR_CDC_STATUS2 register configures the following:

SDC2_SDCC_HC_VENDOR_SPECIFIC_CSR_CDC_STATUS2

Bits	Name	Description
28:24	DA1_SUBUNITS	Current value of subunit cap settings applied to delay array 1
21:16	DA1_UNITSTEPS	Current value of unit steps applied to delay array 1
12:8	DA0_SUBUNITS	Current value of subunit cap settings applied to delay array 0
5:0	DA0_UNITSTEPS	Current value of unit steps applied to delay array 0

0x07864A70 SDC2_SDCC_HC_VENDOR_SPECIFIC_CSR_CDC_STATUS3**Type:** R**Clock:** HCLK**Reset State:** 0x00000000

The HC_VENDOR_SPECIFIC_CSR_CDC_STATUS3 register configures the following:

SDC2_SDCC_HC_VENDOR_SPECIFIC_CSR_CDC_STATUS3

Bits	Name	Description
31:24	NUM_OF_OSC_ITER	Number of oscillator iterations before convergence was achieved during tracking
23:12	COUNT_ERROR	Difference between target and actual oscillator counts at the start of tracking calibration
11:0	CURR_TARGET_COUNT	HW computed value of current Target_count used by the calibration state machine.

0x07864A74 SDC2_SDCC_HC_VENDOR_SPECIFIC_CSR_CDC_STATUS4**Type:** R**Clock:** HCLK**Reset State:** 0x000000FF

The HC_VENDOR_SPECIFIC_CSR_CDC_STATUS4 register configures the following:

SDC2_SDCC_HC_VENDOR_SPECIFIC_CSR_CDC_STATUS4

Bits	Name	Description
7:4	SLAVE_DDA_DA1_TAPS	Number of unit taps applied to delay array 1 of slave DDA
3:0	SLAVE_DDA_DA0_TAPS	Number of unit taps applied to delay array 0 of slave DDA

0x07864A78 SDC2_SDCC_HC_VENDOR_SPECIFIC_CSR_CDC_GEN_CFG**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000**SDC2_SDCC_HC_VENDOR_SPECIFIC_CSR_CDC_GEN_CFG**

Bits	Name	Description
2	CDC_SDDA_EN	Enable CDC Slave DDA to generate 3/4T RCLK from cdc_out_2 output of CDC: 0 (default) - CDC Slave DDA is disabled and 3/4T clock is a falling edge of cdc_out_1, 1 - CDC Slave DDA is enabled and 3/4T clock is a rising edge of cdc_out_2.
1	CDC_SWITCH_RC_EN	Enablement of CDC power RC (LPF) switch 1'b1: Enabled 1'b0: Disabled (default)
0	CDC_SWITCH_BYPASS_OFF	Enablement of CDC power bypass switch 1'b1: Disabled 1'b0: Enabled (default)

0x07864A7C SDC2_SDCC_HC_VENDOR_SPECIFIC_CDC_AUTO_SW_CAL_CFG**Type:** RW**Clock:** HCLK**Reset State:** 0x00F08000

Auto SW calibration configuration bits.

SDC2_SDCC_HC_VENDOR_SPECIFIC_CDC_AUTO_SW_CAL_CFG

Bits	Name	Description
25	CDC_CAL_TIMEOUT_ENABLE	Enable for cdc calibration timeout counter.
24	CDC_AUTO_SW_CAL_ENABLE	Enable for the Auto SW calibration Feature.
23:16	CDC_CAL_TIMEOUT_THRESHOLD	Sets the threshold for CDC calibration timeout during sw calibration triggered by the HW FSM. counts in 16xFF clock. Default is 0xF0 which is 255x16 FF clocks, which gives threshold of ~200 us
15:0	AUTO_TRIG_INTERVAL	Sets the load value of the counter that counts the interval between each sw calibration triggered by the HW FSM. this counter counts in sleep clock. Default is 0x8000 which is ~32,000 sleep clocks, which gives interval of ~1 sec

0x07864A80 SDC2_SDCC_HC_VENDOR_SPECIFIC_DDR200_STAT**Type:** R**Clock:** HCLK**Reset State:** 0x00000000

Status bits for e.MMC5.0 DDR200 mode.

SDC2_SDCC_HC_VENDOR_SPECIFIC_DDR200_STAT

Bits	Name	Description
0	CDC_CAL_VALID	CDC Calibration status: 0 - CDC isn't calibrated, 1 - CDC is calibrated and valid

0x07864A84 SDC2_SDCC_HC_VENDOR_SPECIFIC_DDR200_CFG**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

Configuration bits for e.MMC5.0 DDR200 mode.

SDC2_SDCC_HC_VENDOR_SPECIFIC_DDR200_CFG

Bits	Name	Description
10	FF_CLK_DIS	The ff_clk to CDC/TBY4 is enabled by default. It can be disabled by setting this bit. The ff_clk is enabled to CDC and TBY4 according to CDC_T4_DLY_SEL field

SDC2_SDCC_HC_VENDOR_SPECIFIC_DDR200_CFG (cont.)

Bits	Name	Description
9	VOLTAGE_MUX_SEL	select between 1.8v and 1.2v signals on sdcc5 IO interface with device. default is 0 to choose 1.8v
8:7	CDC_TRAFFIC_SEL	Selecting the method of cdc_traffic's assertions: Value 0 (default) - auto calibration with stand by feature. Value 1 - SW calibration before any transaction. Value 2 - automatic calibration with HW assertion of cdc_traffic
6	START_CDC_TRAFFIC	This bit starts the cdc traffic assertion after calibration done is received
5	CRC_TOKEN_SAMPL_FALL_EDGE	If set, the start bit of incoming CRC token in DDR200 mode is sampled in falling edge. If this bit is cleared (default) then the start bit is detected in rising edge
4	DATIN_SAMPL_FALL_EDGE	If set, the start bit of incoming data in DDR200 mode is sampled in falling edge. If this bit is cleared (default) then the start bit is detected in rising edge
3	CDC_T4_TEST_OUT_SEL	controls the mux between cm_dll_sdc4 testout and CDC test out. When set - CDC test_out is chosen
2	CMDIN_EDGE_SEL	Select sampling edge of CMD input when CMDIN_RCLK_EN is 1: 0 (default) - falling edge of RCLK, 1 - rising edge of RCLK
1	CMDIN_RCLK_EN	Enable CMD input sampling with RCLK: 0 (default) - CMD input is sampled by SD DLL output clock, 1 - CMD input is sampled by delayed RCLK
0	CDC_T4_DLY_SEL	Select block for RCLK delay: 0 (default) - CM_DLL_SDC4

0x07864A88 SDC2_SDCC_HC_VENDOR_SPECIFIC_TIME_COUNT_CTRL**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

Control the performance counters

SDC2_SDCC_HC_VENDOR_SPECIFIC_TIME_COUNT_CTRL

Bits	Name	Description
1	CMD_PERIOD	When this bit is set the commands periods are counted to the CMD_TIME counter
0	ENABLE	Enabling the performance counters

0x07864A8C SDC2_SDCC_HC_VENDOR_SPECIFIC_FLOW_CTRL_TIME_COUNT**Type:** R**Clock:** HCLK**Reset State:** 0x00000000

Counting the Flow Control period

SDC2_SDCC_HC_VENDOR_SPECIFIC_FLOW_CTRL_TIME_COUNT

Bits	Name	Description
31:0	FLOW_CTRL_TIME	FlowControl periods are counted

0x07864A90 SDC2_SDCC_HC_VENDOR_SPECIFIC_BUSY_TIME_COUNT**Type:** R**Clock:** HCLK**Reset State:** 0x00000000

Counting the BUSY periods

SDC2_SDCC_HC_VENDOR_SPECIFIC_BUSY_TIME_COUNT

Bits	Name	Description
31:0	BUSY_TIME	In Tx transaction, the BUSY periods are counted until the progdone is asserted. In Rx transaction, the periods between the blocks without flowcontrol is counted including the time until the first block is received after the command.

0x07864A94 SDC2_SDCC_HC_VENDOR_SPECIFIC_DATA_TIME_COUNT**Type:** R**Clock:** HCLK**Reset State:** 0x00000000

Counting the time of transaction

SDC2_SDCC_HC_VENDOR_SPECIFIC_DATA_TIME_COUNT

Bits	Name	Description
31:0	DATA_TIME	In Tx transaction, the time is from ENabling the DPSM until progdone is asserted. In Rx transaction, the time is from the command transmission to de-assertion of RxActive.

0x07864A98 SDC2_SDCC_HC_VENDOR_SPECIFIC_CMD_TIME_COUNT**Type:** R**Clock:** HCLK**Reset State:** 0x00000000

Counting the time of transaction

SDC2_SDCC_HC_VENDOR_SPECIFIC_CMD_TIME_COUNT

Bits	Name	Description
31:0	CMD_TIME	Counting the Command-Response periods of commands which relate to the transaction including auto commands. The counter is active when CMD_PERIOD is set.

0x07864A9C SDC2_SDCC_HC_VENDOR_SPECIFIC_ERR_INT_STS**Type:** R**Clock:** HCLK**Reset State:** 0x00000000

Details of the vendor specific errors

SDC2_SDCC_HC_VENDOR_SPECIFIC_ERR_INT_STS

Bits	Name	Description
1	AUTO_CDC_SW_CALIB_ERR	calibration error is received (CDC_ERROR_CODE isn't 0) during auto SW CDC calibration
0	AUTO_CDC_SW_CALIB_TIMEOUT	CALIBRATION_DONE isn't received within the number of cycles which configured in CDC_CAL_TIMEOUT_THRESHOLD field (from CDC_AUTO_SW_CAL_CFG register)

0x07864AA0 SDC2_SDCC_HC_VENDOR_SPECIFIC_PWRCTL_STATUS_REG**Type:** R**Clock:** PCLK**Reset State:** 0x00000000

HC Spec Defines power and IO voltage switch bits. The information is sent by a dedicated interrupt to the QCT SW driver for handling the power and voltage tasks with PMIC. The different tasks are defined in this register

SDC2_SDCC_HC_VENDOR_SPECIFIC_PWRCTL_STATUS_REG

Bits	Name	Description
3	IO_HIGH_V	SDCC driver orders that the I/O signaling level be switched to 3.0v. This interrupt is triggered when SDCC driver writes '0' to offset 03Eh in HC register (bit 3) or when SW reset for all is handled and last supported voltage was 1.8V. This logic is enabled when SWITCHABLE_SIGNALING_VOLTAGE=true
2	IO_LOW_V	SDCC driver orders that the I/O signaling level be switched to 1.8v. This interrupt is triggered when SDCC driver writes '1' to offset 03Eh in HC register (bit 3). This logic is enabled when SWITCHABLE_SIGNALING_VOLTAGE=true
1	BUS_ON	SDCC driver orders that the SD bus power be switched on. This interrupt is triggered when SDCC driver writes '1' to offset 029h in HC register (bit 0).
0	BUS_OFF	SDCC driver orders that the SD bus power be switched off. This interrupt is triggered when SDCC driver writes '0' to offset 029h in HC register (bit 0) or when SW reset for all is handled.

0x07864AA4 SDC2_SDCC_HC_VENDOR_SPECIFIC_PWRCTL_MASK_REG**Type:** RW**Clock:** PCLK**Reset State:** 0x00000000

Mask register for MCI_PWRCTL_STATUS_REG

SDC2_SDCC_HC_VENDOR_SPECIFIC_PWRCTL_MASK_REG

Bits	Name	Description
3	IO_HIGH_V	'1' - interrupt is enabled / '0' - interrupt is masked.
2	IO_LOW_V	'1' - interrupt is enabled / '0' - interrupt is masked.
1	BUS_ON	'1' - interrupt is enabled / '0' - interrupt is masked.
0	BUS_OFF	'1' - interrupt is enabled / '0' - interrupt is masked.

0x07864AA8 SDC2_SDCC_HC_VENDOR_SPECIFIC_PWRCTL_CLEAR_REG**Type:** W**Clock:** PCLK**Reset State:** 0x00000000

Clear register for MCI_PWRCTL_STATUS_REG

SDC2_SDCC_HC_VENDOR_SPECIFIC_PWRCTL_CLEAR_REG

Bits	Name	Description
3	IO_HIGH_V	Writing '1' clears bit 03 in PWRCTL_STATUS_REG
2	IO_LOW_V	Writing '1' clears bit 02 in PWRCTL_STATUS_REG
1	BUS_ON	Writing '1' clears bit 01 in PWRCTL_STATUS_REG
0	BUS_OFF	Writing '1' clears bit 00 in PWRCTL_STATUS_REG

0x07864AAC SDC2_SDCC_HC_VENDOR_SPECIFIC_PWRCTL_CTL_REG**Type:** RW**Clock:** PCLK**Reset State:** 0x00000000

HC Spec Defines power and IO voltage switch bits. The information is sent by a dedicated interrupt to the QCT SW driver for handling the power and voltage tasks with PMIC and then the QCT SW driver needs to report success or fail to this register.

SDC2_SDCC_HC_VENDOR_SPECIFIC_PWRCTL_CTL_REG

Bits	Name	Description
9	HC_CARD_STATE_STABLE	This bit is driving CARD_STATE_STABLE bit in Present State Register
8	RST_NOT_WAIT_PWRCTL_REG	RST process doesn't need to wait for accessing to MCI_PWRCTL_CTL_REG register
7:6	HC_WRITE_PROTECT_PAD	Value 0 inversion of HC_WRITE_PROTECT is connected to bit 19 in Present State Register. Value 1 input from PAD is used. Value 2 inversion of input from PAD is used
5	CARD_DETECT_PIN_LEVEL	This bit reflects the inverse value of the SDCCD# pin. Debouncing is not performed on this bit. This bit may be valid when Card State Stable is set to 1, but it is not guaranteed because of propagation delay. Use of this bit is limited to testing since it must be debounced by software. value 1: card present value 0: no card present
4	HC_WRITE_PROTECT	This bit allows to disable the writing. value 0: write enabled value 1: write protected
3	IO_SIG_SWITCH_FAIL	Host controller clears this bit when it fires either I/O Low V interrupt or I/O High V interrupt. SDCC power control software writes '1' to indicate the signal switch has failed.

SDC2_SDCC_HC_VENDOR_SPECIFIC_PWRCTL_CTL_REG (cont.)

Bits	Name	Description
2	IO_SIG_SWITCH_SUCCES S	Host controller clears this bit when it fires either I/O Low V interrupt or I/O High V interrupt. SDCC power control software writes '1' to indicate the signal switch has completed successfully.
1	BUS_ON_OFF_FAIL	Host controller clears this bit when it fires either Bus ON interrupt or Bus OFF interrupt. SDCC power control software writes '1' to indicate the requested operation has failed.
0	BUS_ON_OFF_SUCCESS	Host controller clears this bit when it fires either Bus ON interrupt or Bus OFF interrupt. SDCC power control software writes '1' to indicate the requested operation has completed successfully.

0x07864AB0 SDC2_SDCC_HC_VENDOR_SPECIFIC_FUNC3**Type:** RW**Clock:** same rate as HCLK**Reset State:** 0x00000000

Vendor specific register #3

SDC2_SDCC_HC_VENDOR_SPECIFIC_FUNC3

Bits	Name	Description
4	QSB_AXI_REDIRECT	Driving QSB AXI output
3	PWRSAVE_DLL	When this bit is set, DLL will get clock during transactions but not in idle. When this bit is cleared, DLL will get clock all the time so sdc4_dll_lock won't be de-asserted
2	SDIO_TRANS	When this bit is set in SDIO transaction, START_BIT_ERR status bit won't be raised. The reason is that interrupt period which is de-asserted only after the RX command can cause to wrong START_BIT_ERR detection. The START_BIT_ERR bit won't appear in MCI_STATUS when SDIO_TRANS is set
1	HCLK_IDLE_GATING	Additional CGC for HCLK was added, which gates the HCLK for status bits. If this bit is set then HCLK is gated for these registers during IDLE
0	MCLK_IDLE_GATING	Additional CGC for MCLK was added, which gates the MCLK for status bits. If this bit is set then MCLK is gated for these registers during IDLE

0x07864AB4 SDC2_SDCC_HC_REG_DLL_CONFIG_2**Type:** RW**Clock:** same rate as HCLK**Reset State:** 0x00000000

SDC2_SDCC_HC_REG_DLL_CONFIG_2

Bits	Name	Description
3:2	DDR_TRAFFIC_INIT_SEL	
1	DDR_TRAFFIC_INIT_SW	
0	DDR_CAL_EN	

0x07864AB8 SDC2_SDCC_HC_REG_DDR_CONFIG**Type:** RW**Clock:** same rate as HCLK**Reset State:** 0x80040853

Configuration of ddr_config input of DLL

SDC2_SDCC_HC_REG_DDR_CONFIG

Bits	Name	Description
31	PRG_DLY_EN	Enables Programmable delay for RCLK_IN
30	EXT_PRG_RCLK_DLY_EN	Enables external control of Programmable RCLK delayline
29:27	EXT_PRG_RCLK_DLY_CODE	Used only if EXT_PRG_RCLK_DLY_EN is set. Configuring Programmable RCLK Delayline process code number.
26:21	EXT_PRG_RCLK_DLY	Used only if EXT_PRG_RCLK_DLY_EN is set. Programmable RCLK Delay line fine delay control.
20:12	TCXO_CYCLES_DLY_LINE	Number of TCXO clock cycles between finish of one delay line calibration and beginning of next delay line calibration.
11:9	TCXO_CYCLES_CNT	Number of TCXO clock cycles for which down counter is enabled for before making each inc/dec decision.
8:0	PRG_RCLK_DLY	These bits are controlled the programmable delay line. This bus is binary coded. Decimal value = (TCXO period * Decimal value of TCXO_CYCLES_CNT)/(2*delay needed). For 1.25ns the formula is $(52n*4/2*1.25) = 83$ in decimal.

20 BLSP

20.1 BLSP1_BAM

20.1.1 BLSP1_BLSP_BAM_BAM_BAM

0x07884000 BLSP1_BLSP_BAM_BAM_BAM_CTRL

Type: RW

Clock: bam_clk

Reset State: 0x00020000

BAM Control register allows global controls for the BAM.

BLSP1_BLSP_BAM_BAM_BAM_CTRL

Bits	Name	Description
20	BAM_MESS_ONLY_CANCE L_WB	When active (1) - BAM will DROP former descriptor write-backing and Interrupt firing when former descriptor has been closed (accumulated=acknoledged) and ack_on_sucess with EOT ans size 0 arrived (messaging only). When legacy mode is used (0) - This said a case then the former descriptor has been closed (accumulated=acknowledged) and ack_on_sucess with EOT and size 0 arrived (messaging only). In this case the FIFO-pointer has been already advanced before the ack_on_sucess arives and therefore we do write-back to former descriptor and fire interrupt. 1'b1 - Drop WB and Interrupt 1'b0 - legacy.
19	CACHE_MISS_ERR_RESP_ EN	When set to '1', upon local ahb access results with cache miss, the bam_ndp will not stall the bus, and finish the access with error response. This bit is relevant for BAM_NDP only.

BLSP1_BLSP_BAM_BAM_BAM_CTRL (cont.)

Bits	Name	Description
18:17	LOCAL_CLK_GATING	These Bits enables power saving by using a local clock gating cell. Bit 17: 1'b1 - CGC is on, so that the clock is controled by the HW - this is the reset value. 1'b0 - CGC is off so that the clock is free runing Bit 18: Reserved bit for future improvements.
16	IBC_DISABLE	This Bit enables power saving by disabling the inactivity clock timer/counter. This is useful when: 1) The BAM is Enabled but idling for long periods. 2) The BAM is Enabled and running but the inactivity timers/counters are not required by the SW for operating the BAM. When the BAM is Disabled, it automatically shuts down those timers to save power. 1'b1 - Enable Power Saving Mode 1'b0 - Disable Power Saving Mode
15	BAM_CACHED_DESC_STORE	This Bit enables storage of a cached Descriptor into RAM when doing Context Switch and loading it back when returning to this pipe. This is relevant for System Modes Only. BAM to BAM Consumer mode doesn't have this option. Enabling this makes the Context Switch process slightly longer but reduces AHB Descriptor Fetch time later on. It also reduces the number of descriptor fetches done by the BAM if BAM is treating more than 2 pipes. 1'b1 - Enabled 1'b0 - Disabled Available in BAM only
14:13	BAM_DESC_CACHE_SEL	This Selector chooses which Descriptor Caching mode will be used. This is relevant for System Producer or Consumer modes only. 2'b00 - Cache when current descriptor has less than 64 bytes left. 2'b01 - Cache when current descriptor has less than 128 bytes left. 2'b10 - Immediate Auto cache - cache will happen whenever there are no descriptors cached. This might cause unnecessary descriptors reads when switching between more than 2 pipes. It is the most effective when each pipe processes more bytes than there are in an average descriptor (small descriptors, big block size), or when working with 2 or less pipes. 2'b11 - Cache when descriptor is needed only. This results in 1 descriptor present in cache at most. Available in BAM only
11:5	RESERVED	Reserved

BLSP1_BLSP_BAM_BAM_BAM_CTRL (cont.)

Bits	Name	Description
4	BAM_EN_ACCUM	When Enabled, BAM will accumulate data written by the peripheral in Direct Mode into INCR8 Bursts. This is an optimization feature for Producer Direct Mode cases. 1'b1 - Enabled 1'b0 - Disabled Available in BAM only
1	BAM_EN	After reset the BAM wakes up in Disabled Mode. While Disabled, BAM operates in Legacy mode (all the transfers from the peripheral go around the BAM as if it doesn't exist) Software Enables this bit to allow BAM operation. This bit doesn't affect the PIPE enable bit in BAM_P_CTRL register. The only possibility to Disable the BAM after it has been enabled is by Hardware or Software reset. 1'b1 - Enabled 1'b0 - Disabled
0	BAM_SW_RST	This will reset the BAM & all Pipes. Software may use this to reset the BAM. As long as the bit remains high the BAM remains in reset state. Software should clear this bit 1'b1 - Reset state 1'b0 - Normal state

0x07884040 BLSP1_BLSP_BAM_BAM_BAM_TIMER**Type:** R**Clock:** bam_clk**Reset State:** 0x00000000

This register counts the idle time of the BAM (all the pipes are in idle state).

BLSP1_BLSP_BAM_BAM_BAM_TIMER

Bits	Name	Description
15:0	TIMER	Inactivity timer counter. 16 bits = 64k*0.1ms ~ 6 seconds

0x07884044 BLSP1_BLSP_BAM_BAM_BAM_TIMER_CTRL**Type:** RW**Clock:** bam_clk**Reset State:** 0x00000000

This register sets various controls and the threshold of the inactivity for interrupt triggering after long idle time.

The register can be written and read only when INACTIVITY_TIMERS_SUPPORTED generic equals to 1.

BLSP1_BLSP_BAM_BAM_BAM_TIMER_CTRL

Bits	Name	Description
31	TIMER_RST	To begin using the inactivity timer the SW should write the threshold and write 1 to this bit. writing the value 0 will hold the timer reset until value 1 is written again. Then the timer will start down counting from the TIMER_THRESOLD value 1'b1 - Active 1'b0 - Reset
30	TIMER_RUN	Will be used along with TIMER_MODE Not implemented at this time. Set to zero(0)
29	TIMER_MODE	Not implemented at this time. Set to zero(0)
15:0	TIMER_TRSHLD	Inactivity timer threshold for interrupt generation.

0x07884008 BLSP1_BLSP_BAM_BAM_BAM_DESC_CNT_TRSHLD

Type: RW

Clock: bam_clk

Reset State: 0x00000001

This Register holds a threshold value for the counter summing the Size of the Descriptors Provided. This value is Global for all pipes but it is relevant for System Producer BAM mode and Both Consumer modes, where Descriptors summing takes place in order to provide the information to the peripheral via pipe bytes free (producer) or pipe bytes available interfaces.

BLSP1_BLSP_BAM_BAM_BAM_DESC_CNT_TRSHLD

Bits	Name	Description
15:0	CNT_TRSHLD	Threshold value. The maximum allowed value is 32kByte. The minimum allowed value is 1 (value 0 is not allowed). Available in BAM only

0x07884014 BLSP1_BLSP_BAM_BAM_BAM_IRQ_STTS

Type: R

Clock: bam_clk

Reset State: 0x00000000

Interrupt status register describing the reason for the ongoing interrupt. This is an informative only register and therefore read only.

Clearing the interrupt bits is done by writing to BAM_IRQ_CLR register.

BLSP1_BLSP_BAM_BAM_BAM_IRQ_STTS

Bits	Name	Description
4	BAM_TIMER_IRQ	Inactivity timer has reached its threshold value Interrupt
3	BAM_EMPTY_IRQ	This interrupt is for DEBUG purpose only. It should not be used in normal SW flows. The interrupt indicates the BAMs buffer is empty. This bit can become high only when BAM_DATA_ERASE or BAM_DATA_FLUSH is high in BAM_TEST_BUS_SEL register.
2	BAM_ERROR_IRQ	This Indicates a Fatal Error has happened in the BAM. Currently this might happen due to an UnSuccessful Pipe Reset operation.
1	BAM_HRESP_ERR_IRQ	This interrupt Indicates that an Erroneous HResponse has been received by the AHB Master. Additional Information about the error may be read at BAM_AHB_MASTER_ERR_* Registers. Clearing this interrupt also clears the BAM_AHB_MASTER_ERR_* Registers. Available in BAM only

0x07884018 BLSP1_BLSP_BAM_BAM_BAM_IRQ_CLR

Type: W

Clock: bam_clk

Reset State: 0x00000000

Writing to this register causes the interrupt to clear.

BLSP1_BLSP_BAM_BAM_BAM_IRQ_CLR

Bits	Name	Description
4	BAM_TIMER_CLR	Clear Inactivity timer interrupt by writing this bit 1'b1 - Clear 1'b0 - Unchanged
3	BAM_EMPTY_CLR	1'b1 - Clear 1'b0 - Unchanged Available in BAM-Lite only
2	BAM_ERROR_CLR	1'b1 - Clear 1'b0 - Unchanged

BLSP1_BLSP_BAM_BAM_BAM_IRQ_CLR (cont.)

Bits	Name	Description
1	BAM_HRESP_ERR_CLR	1'b1 - Clear 1'b0 - Unchanged Available in BAM only

0x0788401C BLSP1_BLSP_BAM_BAM_BAM_IRQ_EN**Type:** RW**Clock:** bam_clk**Reset State:** 0x00000000

Enable bits for the Interrupts in the BAM Interrupt Status register.

BLSP1_BLSP_BAM_BAM_BAM_IRQ_EN

Bits	Name	Description
4	BAM_TIMER_EN	Enables Inactivity timer interrupt by writing this bit 1'b1 - Enable 1'b0 - Disable
3	BAM_EMPTY_EN	1'b1 - Enable 1'b0 - Disable Available in BAM-Lite only
2	BAM_ERROR_EN	1'b1 - Enable 1'b0 - Disable
1	BAM_HRESP_ERR_EN	1'b1 - Enable 1'b0 - Disable Available in BAM only

0x0788407C BLSP1_BLSP_BAM_BAM_BAM_CNFG_BITS**Type:** RW**Clock:** bam_clk**Reset State:** 0x00000000

This Register holds the BAM configuration bits for bug fixes. It is highly recommended to follow the directions for each bit and set it accordingly.

BLSP1_BLSP_BAM_BAM_BAM_CNFG_BITS

Bits	Name	Description
31	AOS_OVERFLOW_PRVNT	This bit enables overflow prevention when Sideband block accumulates ack_on_success updates from peripherals to values higher than 0x7fff. '1' - enable ack-on_success overflow prevention. If current ack_on_success update causes overflow accept_ack_on_success_toggle will not approve the update until SB accumulator is cleared. '0' - disable ack-on_success overflow prevention. Available in BAM-NDP only. Default value is '0' (Legacy). Recommended value is '1'.
30	MULTIPLE_EVENTS_DESC_AVAIL_EN	This is an enable bit for supporting multiple events received by a Producer BAM2BAM and makes BAM-NDP to notify a correct desc_cache_avail bus. It fixed CR: QCTDD00209370.
29	MULTIPLE_EVENTS_SIZE_EN	This is an enable bit for supporting multiple events received by a Producer BAM2BAM and makes BAM-NDP delivering correct descriptor sizes on local_ahb bit. It fixed CR: QCTDD00209379.
28	BAM_ZLT_W_CD_SUPPORT	Recommended value: 1 Enables support of ZLT (i.e. a zero-sized descriptor) before a command-descriptor. Specifically it supports a case on a pipe without any data-descriptors but with ZLT descriptors and Command-descriptors only. This allows putting NWD bit in the ZLT descriptor (instead of in the CMD) which results in better performance than using the NWD in the command-descriptor. Available in BAM-Lite Only.
27	BAM_CD_ENABLE	Recommended value: 1 Enables the command descriptor feature.
26	BAM_AU_ACCUMED	Recommended value: 1 This bit fixes a bug in the Ack Update state machine, where an overflow happened while counting descriptors and reaching more than 64kB of calculated sizes. 1'b1 - Enable Fix 1'b0 - Disable Available in BAM/BAM-NDP only

BLSP1_BLSP_BAM_BAM_BAM_CNFG_BITS (cont.)

Bits	Name	Description
25	BAM_PSM_P_HD_DATA	Recommended value: 1 This bit allows pipe state machine to ignore retransmission requests if a pipe has just been initialized and process those as a regular fetch request. (consumer modes only). When this bit disabled, BAM could fetch descriptors for a pipe which was reset and no descriptors were added yet, if a retransmission request followed after the reset.. 1'b1 - Enable Fix 1'b0 - Disable Available in BAM only
24	BAM_REG_P_EN	Recommended value: 1 This bit fixes the pipe configuration signals mux for the current active pipe in 2 pipes BAM. When disabled, internal state machines might get into enabled states while the pipe is disabled. This would typically happen after pipe reset. 1'b1 - Enable 1'b0 - Disable Available in BAM/BAM-NDP only
23	BAM_WB_DSC_AVL_P_RST	Recommended value: 1 This bit allows the Writeback state machine to reset the vector indicating there're available descriptors when a pipe reset occurs. If disabled, BAM might fetch descriptors after resetting and reconfiguring a pipe, even though no Event (descriptors) was provided.. 1'b1 - Enable 1'b0 - Disable Available in BAM only
22	BAM_WB_RETR_SVPNT	Recommended value: 1 This bit fixes a bug where a pipe which was reset, still stored its retransmission savepoint, but into the illegal's pipe address space, thus hurting the last pipe of the BAM if the BAM had a total 4, 8 or 16 pipes. This is relevant for Producer to System modes only. (CR-0000151585) 1'b1 - Enabled 1'b0 - Disable Available in BAM/BAM-NDP only
21	BAM_WB_CSW_ACK_IDL	Recommended value: 1 This bit allows the Writeback state machine to switch into idle state prior to visiting disabled state. This is needed when context switching from mode X to another pipe of mode X is well. This is required to fix a bug in the 2 pipes BAM. 1'b1 - Enable 1'b0 - Disable Available in BAM/BAM-NDP only

BLSP1_BLSP_BAM_BAM_BAM_CNFG_BITS (cont.)

Bits	Name	Description
20	BAM_WB_BLK_CSW	Recommended value: 1 When Enabled, this bit does not allow context switch to happen in the Writeback state machine until it has created a descriptor. This is relevant when the descriptor fifo is becoming full and there's no space to create a descriptor, while another pipe is context switching. This might result in the descriptor not to be created ever, if it was the last one for that pipe. Relevant for Producer BAM-to-BAM mode only. 1'b1 - Enable 1'b0 - Disable Available in BAM only
19	BAM_WB_P_RES	Recommended value: 1 This bit allows to reset all internal registers of the Writeback state machine when performing pipe reset. 1'b1 - Enable 1'b0 - Disable Available in BAM only
18	BAM_SI_P_RES	Recommended value: 1 This bit allows to reset all internal registers of the Sideband Inform state machine when performing pipe reset. 1'b1 - Enable 1'b0 - Disable Available in BAM only
17	BAM_AU_P_RES	Recommended value: 1 This bit allows to reset all internal registers of the Ack Update state machine when performing pipe reset. 1'b1 - Enable 1'b0 - Disable Available in BAM only
16	BAM_PSM_P_RES	Recommended value: 1 This bit allows to reset all internal registers of the Pipe state machine when performing pipe reset.. 1'b1 - Enable 1'b0 - Disable Available in BAM only
15	BAM_PSM_CSW_REQ	Recommended value: 1 This bit forces the context switch request from pipe state machine to RAM controller not to last longer than the slave requested. (2 Pipes BAM bug fix). 1'b1 - Enable 1'b0 - Disable Available in BAM only

BLSP1_BLSP_BAM_BAM_BAM_CNFG_BITS (cont.)

Bits	Name	Description
14	BAM_SB_CLK_REQ	Recommended value: 1 This bit allows the clock request from the sideband block to propagate into the BAM's common clock request. 1'b1 - Propagate Sideband Clock Request 1'b0 - Disable Available in BAM only
13	BAM_IBC_DISABLE	Recommended value: 1 This bit helps to save power by allowing the BAM to keep the inactivity base counter in reset when BAM is disabled or when SW configures IBC_DISABLE bit high. Irrelevant field for BAM-NDP. 1'b1 - Enable Power Saving 1'b0 - Disable Power Saving
12	BAM_NO_EXT_P_RST	Recommended value: 1 This bit allows the BAM / BAM Lite to ignore the externally connected blocks (eDML) when doing pipe reset. The BAM, once instructed to pipe reset, first thing lets the externally connected block know a reset is needed. Then it waits for the externally connected block to Acknowledge it is ready for the pipe reset (meaning it doesn't push any data for the reseted pipe) and then the BAM Lite completes the pipe reset operation internally. When disabled, the BAM doesn't require any Acknowledge from the external block to perform pipe reset. 1'b1 - Enable external block pipe reset 1'b0 - Disable - ignore external block pipe reset
11	BAM_FULL_PIPE	Recommended value: 0 This enables the BAM support for a BAM to BAM Producer which insists to write to a full pipe. When 0, BAM might issue data overflow if producers write to a full pipe. When 1 BAM will not allow this and lower HReady when peripheral tries to do so. Once space is freed in the pipe, Hready will rise and the flow will continue. 1'b1 - Enable 1'b0 - Disable Available in BAM only
3	RESERVED	Reserved
2	BAM_PIPE_CNFG	Recommended value: 1 Pipe SM upgrade for writing EOT bit to the previous descriptor. It's invoked only when EOB arrives in the end of a descriptor. It is highly recommended to set this bit high. Leaving it low might cause incorrect Pipe Bytes Free value reported to peripheral in rare cases. 1'b1 - Enable 1'b0 - Disable Available in BAM only
1:0	RESERVED	Reserved

0x07885000 BLSP1_BLSP_BAM_BAM_BAM_REVISION**Type:** R**Clock:** bam_clk**Reset State:** Undefined

This Register holds a hard coded revision number of the BAM RTL. This corresponds to the BAM target number in the IMS. Also additional BAM settings are reflected here.

BLSP1_BLSP_BAM_BAM_BAM_REVISION

Bits	Name	Description
31:24	INACTIV_TMR_BASE	This indicates the width of the inactivity timers base counter.
23	CMD_DESC_EN	This indicates BAM has Command Descriptor feature enabled in HW.
22:21	DESC_CACHE_DEPTH	This field indicates the Per-Pipe-Descriptor-Cache. could be 1-4: 2'b00 - Descriptor Cache Depth of 1. 2'b01 - Descriptor Cache Depth of 2. 2'b10 - Descriptor Cache Depth of 3. 2'b11 - Descriptor Cache Depth of 4. Valid only for BAM_NDP.
20	NUM_INACTIV_TMRS	This bit is checked only if INACTIV_TMRS_EXST = 1. This bit indicates how many inactivity timers are implemented according to value of INACTIVITY_TIMERS_SUPPORTED generic: 1'b1 - One timer for global BAM activity (INACTIVITY_TIMERS_SUPPORTED = 1) 1'b0 - MAX_PIPES timers for each pipe activity (INACTIVITY_TIMERS_SUPPORTED = 2)
19	INACTIV_TMRS_EXST	This indicates BAM has inactivity timers. 1'b1 - Timers available (INACTIVITY_TIMERS_SUPPORTED = 1 or 2) 1'b0 - No timers available (INACTIVITY_TIMERS_SUPPORTED = 0)
18	HIGH_FREQUENCY_BAM	This indicates BAM has Double Sampling on the Ack On Success interface. 1'b1 - Ack On Success double sampled 1'b0 - No double sampling of Ack On Success
17	BAM_HAS_NO_BYPASS	This field indicates BAM has no bypass on the AHB Data bus, from the Peripheral AHB Master to the fabric. When Bypass is available, it is the default routing when BAM is disabled. 1'b1 - No Bypass 1'b0 - Bypass Exists
16	RESERVED	Reserved
15	RESERVED	Reserved

BLSP1_BLSP_BAM_BAM_BAM_REVISION (cont.)

Bits	Name	Description
14	AXI_ACTIVE	This field indicates BAM_NDP uses internal AXI bridge on the master port.
13:12	CE_BUFFER_SIZE	This field indicates the size (in Data words) of the buffer which stores the command elements. Each command element includes 4 words. 2'b00 - 4 Words (one command element). 2'b10 - 8 Words (2 command elements). 2'b11 - 16 Words (4 command elements). Valid Only if CMD_DESC_EN = 1'b1.
11:8	NUM_EES	This indicates the Number of Interrupt Lines (Execution Environments) supported by the BAM. When 1 EE is supported, only the BAM_IRQ_SRC*_EE0 registers exist. When 4 EEs are supported, all BAM_IRQ_SRC*_EE _n registers exist for n=[0..3].
7:0	RESERVED	Reserved

0x07885008 BLSP1_BLSP_BAM_BAM_BAM_NUM_PIPES**Type:** R**Clock:** bam_clk**Reset State:** Undefined

This Register holds a the hard coded number of pipes in the BAM. Since each BAM may have a different number of pipes instantiated in HW, this register allows the SW to know this parameter. The reset state value doesn't have any meaning in this register since it is a constant.

BLSP1_BLSP_BAM_BAM_BAM_NUM_PIPES

Bits	Name	Description
31:24	RESERVED	Reserved
23:16	RESERVED	Reserved
15:14	BAM_DATA_ADDR_BUS_WIDTH	This field indicates BAM data bus address width: 2'b00 - 32bit address width (legacy - all chips before Gandalf) 2'b01 - 36bit address width (Gandalf and later) 2'b1x - Reserved for future widening
7:0	BAM_NUM_PIPES	This field contains the number of pipes available in this BAM

0x07885024 BLSP1_BLSP_BAM_BAM_BAM_AHB_MASTER_ERR_CTRL**Type:** R**Clock:** bam_clk**Reset State:** 0x00000000

This Register holds the information about the data error in the AHB Buffer during bus Access.

The value stored is the value of the first error occurred.

Clear the BAM_HRESP_ERR_IRQ interrupt to clear the recorded error.

Available in BAM only

BLSP1_BLSP_BAM_BAM_BAM_AHB_MASTER_ERR_CTRL

Bits	Name	Description
22:18	BAM_ERR_HVMID	HVMID
17	BAM_ERR_DIRECT_MODE	DIRECT MODE
16:12	BAM_ERR_HCID	HCID
11:8	BAM_ERR_HPROT	HPROT
7:5	BAM_ERR_HBURST	HBURST
4:3	BAM_ERR_HSIZE	HSIZE
2	BAM_ERR_HWRITE	HWRITE
1:0	BAM_ERR_HTRANS	HTRANS

0x07885028 BLSP1_BLSP_BAM_BAM_BAM_AHB_MASTER_ERR_ADDR**Type:** R**Clock:** bam_clk**Reset State:** 0x00000000

This Register holds the information about the data error in the AHB Buffer during bus Access.

The value stored is the value of the first error occurred.

Clear the BAM_HRESP_ERR_IRQ interrupt to clear the recorded error.

Available in BAM only

BLSP1_BLSP_BAM_BAM_BAM_AHB_MASTER_ERR_ADDR

Bits	Name	Description
31:0	BAM_ERR_ADDR	HADDR

0x0788502C BLSP1_BLSP_BAM_BAM_BAM_AHB_MASTER_ERR_DATA**Type:** R**Clock:** bam_clk**Reset State:** 0x00000000

This Register holds the information about the data error in the AHB Buffer during bus Access.

The value stored is the value of the first error occurred.

Clear the BAM_HRESP_ERR_IRQ interrupt to clear the recorded error.

Available in BAM only

BLSP1_BLSP_BAM_BAM_BAM_AHB_MASTER_ERR_DATA

Bits	Name	Description
31:0	BAM_ERR_DATA	HDATA

0x07885100 BLSP1_BLSP_BAM_BAM_BAM_AHB_MASTER_ERR_ADDR_LSB**Type:** R**Clock:** bam_clk**Reset State:** 0x00000000

This Register holds the information about the data error in the AHB Buffer during bus Access.

The value stored is the value of the first error occurred.

Clear the BAM_HRESP_ERR_IRQ interrupt to clear the recorded error.

Available in BAM only

Relevant only on 36 bit address BAM.

BLSP1_BLSP_BAM_BAM_BAM_AHB_MASTER_ERR_ADDR_LSB

Bits	Name	Description
31:0	BAM_ERR_ADDR	32 LSB of HADDR

0x07885104 BLSP1_BLSP_BAM_BAM_BAM_AHB_MASTER_ERR_ADDR_MSB**Type:** R**Clock:** bam_clk**Reset State:** 0x00000000

This Register holds the information about the data error in the AHB Buffer during bus Access.

The value stored is the value of the first error occurred.

Clear the BAM_HRESP_ERR_IRQ interrupt to clear the recorded error.

Available in BAM only

Relevant only on 36 bit address BAM.

BLSP1_BLSP_BAM_BAM_BAM_AHB_MASTER_ERR_ADDR_MSB

Bits	Name	Description
3:0	BAM_ERR_ADDR	4 MSB of Address (bits 35 to 32) of HADDR

0x07886000 BLSP1_BLSP_BAM_BAM_BAM_TRUST_REG

Type: RW

Clock: bam_clk

Reset State: 0x00000000

This Register holds the BAM Trusted configurations.

This register cannot be reset by any SW reset. Only HW reset applies.

BLSP1_BLSP_BAM_BAM_BAM_TRUST_REG

Bits	Name	Description
13	LOCK_EE_CTRL	This bit controls if the EE setting defined in the TRUST registers will be taken into account for the pipe lock grouping decoding. 1'b0 - Only P_LOCK_GROUP is checked for pipe locking. 1'b1 - Both BAM_P_EE and P_LOCK_GROUP are checked for pipe locking.
12:8	RESERVED	Reserved
7	BAM_RST_BLOCK	When enabled, the BAM Global SW reset will not be available for usage. This is in order to deny Global SW reset requests by the Global security group. 1'b1 - Enable 1'b0 - Disable

BLSP1_BLSP_BAM_BAM_BAM_TRUST_REG (cont.)

Bits	Name	Description
2:0	BAM_EE	This Field Indicates the EE # (0-7) to which the BAM Interrupt belongs to. The pipe interrupts will fire on the EE # set at BAM_P_TRUST_REGn registers. The number of supported EEs is reflect in the NUM_EES field in the BAM_REVISION register. 3'b000 - EE0 3'b001 - EE1 3'b010 - EE2 3'b011 - EE3 3'b100 - EE4 3'b101 - EE5 3'b110 - EE6 3'b111 - EE7

0x07886020+ BLSP1_BLSP_BAM_BAM_BAM_P_TRUST_REGn, n=[0..30]
0x4*n

Type: RW

Clock: bam_clk

Reset State: 0x00000000

This Register holds the BAM Trusted configurations.

This register cannot be reset by any SW reset. Only HW reset applies. When using a non-secured BAM, it is reset by the BAM HW Reset.

BLSP1_BLSP_BAM_BAM_BAM_P_TRUST_REGn

Bits	Name	Description
12:8	RESERVED	Reserved
7:3	BAM_P_SUP_GROUP	Super-Group is the upper division of the pipes compared to the pipe-group division. When any pipe locks or unlocks the BAM, all the pipes which are in a different super-group will NOT be affected. Only the pipes within the same super-group of the locker are affected. Usually pipes which belongs to different peripherals which share the same BAM, would be in a different super-groups

BLSP1_BLSP_BAM_BAM_BAM_P_TRUST_REGn (cont.)

Bits	Name	Description
2:0	BAM_P_EE	This Field Indicates the EE # (0-7) to which the Pipe Interrupt belongs to. The BAM interrupt will fire to the EE # set at BAM_P_TRUST_REGn registers. The number of supported EEs is reflect in the NUM_EES field in the BAM_REVISION register. 3'b000 - EE0 3'b001 - EE1 3'b010 - EE2 3'b011 - EE3 3'b100 - EE4 3'b101 - EE5 3'b110 - EE6 3'b111 - EE7

**0x07887000+ BLSP1_BLSP_BAM_BAM_BAM_IRQ_SRCS_EEn, n=[0..7]
4096*n****Type:** R**Clock:** bam_clk**Reset State:** 0x00000000

This Interrupt Status Register provides a single interrupt bit per each pipe in the BAM and a single bit for the BAM (not pipe related) interrupt. This register is read only and provides fast indication for the software to decide which pipe triggered the interrupt. After checking this register, SW reads the BAM_P_IRQ_STTSn register for the pipe interrupt reason and BAM_IRQ_STTS for the BAM interrupt reason.

This register has an alias - BAM_IRQ_SRCS register.

Registers with n=[1..7] exist only when the BAM supports multiple EEs (NUM_EES field in the BAM_REVISION register).

BLSP1_BLSP_BAM_BAM_BAM_IRQ_SRCS_EEn

Bits	Name	Description
31	BAM_IRQ	BAM Global interrupt
30:0	P_IRQ	An interrupt bfit per each pipe in the BAM.

**0x07887004+ BLSP1_BLSP_BAM_BAM_BAM_IRQ_SRCS_MSK_EEn, n=[0..7]
4096*n****Type:** RW**Clock:** bam_clk**Reset State:** 0x00000000

Interrupt Sources Mask register allows to mask the interrupts by their source, whether BAM or a specific pipe.

This register has an alias - BAM_IRQ_SRCS_MSK register.

Registers with n=[1..7] exist only when the BAM supports multiple EEs (NUM_EES field in the BAM_REVISION register).

BLSP1_BLSP_BAM_BAM_BAM_IRQ_SRCS_MSK_EEn

Bits	Name	Description
31	BAM_IRQ_MSK	BAM Global Interrupt Mask 0x1: ENABLE_BAM_INTERRUPT (Enable BAM interrupt) 0x0: DISABLE_BAM_INTERRUPT (Disable BAM interrupt)
30:0	P_IRQ_MSK	An Interrupt Mask bit per each pipe in the BAM. 0x1: ENABLE_PIPE_INTERRUPT (Enable Pipe interrupt) 0x0: DISABLE_PIPE_INTERRUPT (Disable Pipe interrupt)

0x07887008+ BLSP1_BLSP_BAM_BAM_BAM_IRQ_SRCS_UNMASKED_EEn, n=[0..7] 4096*n

Type: R

Clock: bam_clk

Reset State: 0x00000000

The register shows the interrupts sources like (BAM_IRQ_SRCS) although the interrupts from the BAM and the pipes are not masked on the lower level.

This register has an alias - BAM_IRQ_SRCS_UNMASKED register.

Registers with n=[1..7] exist only when the BAM supports multiple EEs (NUM_EES field in the BAM_REVISION register).

BLSP1_BLSP_BAM_BAM_BAM_IRQ_SRCS_UNMASKED_EEn

Bits	Name	Description
31	BAM_IRQ_UNMASKED	BAM Global Interrupt after Mask
30:0	P_IRQ_UNMASKED	An Interrupt bit per each pipe in the BAM, after Mask

0x0788700C+ BLSP1_BLSP_BAM_BAM_BAM_PIPE_ATTR_EEn, n=[0..7] 4096*n

Type: R

Clock: bam_clk

Reset State: 0x00000000

The register shows the pipes attributed to a specific EE, and an indication if BAM is enabled.

The reset value written above is true only if EE>0. For BAM_PIPE_ATTR_EE0 the default value is: Bits [MAX_PIPES-1:0] = 1 , Bits [31: MAX_PIPES] = 0 .

BLSP1_BLSP_BAM_BAM_BAM_PIPE_ATTR_EEn

Bits	Name	Description
31	BAM_ENABLED	1'b1 - BAM is Enabled. 1'b0 - BAM Disabled.
30:0	P_ATTR	If bit 'i' == 1'b1 - Pipe 'i' is attributed to this EE.

0x07887010 BLSP1_BLSP_BAM_BAM_BAM_IRQ_SRCS

Type: R

Clock: bam_clk

Reset State: 0x00000000

This Interrupt Status Register provides a single interrupt bit per each pipe in the BAM and a single bit for the BAM (not pipe related) interrupt. This register is read only and provides fast indication for the software to decide which pipe triggered the interrupt. After checking out this register, SW reads the BAM_P_IRQ_STTSn register for the pipe interrupt reason and BAM_IRQ_STTS for the BAM interrupt reason.

This register points to the physical BAM_IRQ_SRCS_EE0 register.

BLSP1_BLSP_BAM_BAM_BAM_IRQ_SRCS

Bits	Name	Description
31	BAM_IRQ	BAM Global interrupt
30:0	P_IRQ	An interrupt bit per each pipe in the BAM.

0x07887014 BLSP1_BLSP_BAM_BAM_BAM_IRQ_SRCS_MSK

Type: RW

Clock: bam_clk

Reset State: 0x00000000

Interrupt Sources Mask register allows to mask the interrupts by their source, whether BAM or a specific pipe.

This register points to the physical BAM_IRQ_SRCS_MSK_EE0 register.

BLSP1_BLSP_BAM_BAM_BAM_IRQ_SRCS_MSK

Bits	Name	Description
31	BAM_IRQ_MSK	BAM Global Interrupt Mask 0x1: ENABLE_BAM_INTERRUPT (Enable BAM interrupt) 0x0: DISABLE_BAM_INTERRUPT (Disable BAM interrupt)
30:0	P_IRQ_MSK	An Interrupt Mask bit per each pipe in the BAM. 0x1: ENABLE_PIPE_INTERRUPT (Enable Pipe interrupt) 0x0: DISABLE_PIPE_INTERRUPT (Disable Pipe interrupt)

0x07887018 BLSP1_BLSP_BAM_BAM_BAM_IRQ_SRCS_UNMASKED**Type:** R**Clock:** bam_clk**Reset State:** 0x00000000

The register shows the interrupts sources like (BAM_IRQ_SRCS) although the interrupts from the BAM and the pipes are not masked on the lower level. This register points to the physical BAM_IRQ_SRCS_UNMASKED_EE0 register.

BLSP1_BLSP_BAM_BAM_BAM_IRQ_SRCS_UNMASKED

Bits	Name	Description
31	BAM_IRQ_UNMASKED	BAM Global Interrupt after Mask
30:0	P_IRQ_UNMASKED	An Interrupt bit per each pipe in the BAM, after Mask

0x07897000+ 4096*n BLSP1_BLSP_BAM_BAM_BAM_P_CTRLn, n=[0..30]**Type:** RW**Clock:** bam_clk**Reset State:** 0x00000000

Pipe Control register provides various controls for the pipe.

BLSP1_BLSP_BAM_BAM_BAM_P_CTRLn

Bits	Name	Description
20:16	P_LOCK_GROUP	Pipe's lock group. Upon a lock request on this pipe, all pipes related to different pipe group and different EE (see PIPE_TRUST_REG) will be locked.

BLSP1_BLSP_BAM_BAM_BAM_P_CTRLn (cont.)

Bits	Name	Description
11	P_WRITE_NWD	BAM-Lite feature. Applicable to BAM2BAM producer pipes only. When this bit is set for producer B2B pipe, NWD bit (bit number 28 in the second word of descriptor) will be written with EOT into the generated descriptor for the consumer usage. This bit is not applicable in pipes belong to a Peripheral which uses messaging_only (e.g. USB2).
10:9	P_PREFETCH_LIMIT	Bam Lite feature Selects the number of bytes to be prefetched by the Bam Lite, once a peripheral requests to read data from a pipe. This feature is relevant for Consumer modes only. 2'b00 - 32 bytes at most (INCR8) 2'b01 - 16 bytes at most (INCR4) 2'b10 - 4 bytes at most (SINGLE) Available in BAM-Lite only
8:7	P_AUTO_EOB_SEL	Bam Lite feature. Selects the number of bytes to be processed before creating End Of Block indication internally by the BAM. See P_AUTO_EOB. Descriptors generated by this feature (bam to bam producer mode) will always be of the chosen size, unless an End Of Transfer, causes a smaller descriptor to be created. 2'b00 - 512 Bytes 2'b01 - 256 Bytes 2'b10 - 128 Bytes 2'b11 - 64 Bytes Available in BAM-Lite only
6	P_AUTO_EOB	BAM Lite feature. When Enabled, the BAM Lite will ignore any incoming End Of Block indications from the peripheral (if any). Instead, the BAM Lite will generate it's own End Of Block indications internally, every P_AUTO_EOB_SEL bytes. This feature is relevant for Producer Bam 2 Bam modes only. It is a must for peripherals which are not able to produce End Of Block indications. 1'b1 - Enable 1'b0 - Disable Available in BAM-Lite only
5	P_SYS_MODE	This bit sets the pipe to work in System Mode, where the Software is the second side of the pipe. 1'b1 - System mode. 1'b0 - BAM to BAM mode.

BLSP1_BLSP_BAM_BAM_BAM_P_CTRLn (cont.)

Bits	Name	Description
4	P_SYS_STRM	Streaming Enable. Relevant to System Producer BAM mode only. When enabled, BAM will not close descriptors with End Of Transfer notifications received from peripheral, but continue writing the data to the same descriptor. 1'b1 - Enable Streaming Mode 1'b0 - None-Streaming Mode. This field is NOT supported in BAM-NDP (i.e. no streaming mode in NDP).
3	P_DIRECTION	This bit denotes pipe direction. 1'b1 - The pipe can be written (producer mode) 1'b0 - The pipe can be read (consumer mode)
1	P_EN	1'b1 - Enable Pipe 1'b0 - Disable Pipe

**0x07897004+ BLSP1_BLSP_BAM_BAM_BAM_P_RSTn, n=[0..30]
4096*n****Type:** W**Clock:** bam_clk**Reset State:** 0x00000000

Pipe Reset register provides Pipe Reset ability. Software needs to write 1 to this register to make the Pipe Reset. Writing 0 to this register will finish the Pipe Reset. Writing zero is needed prior to Peripheral reconfiguration for the same pipe.

Note: Software can invoke pipe reset only after it make sure that the pipe is 'quiet'.

('quiet' - No data transaction or command-descriptor is active at the time.

BAM_P_RSTn, n=[0..30]

BLSP1_BLSP_BAM_BAM_BAM_P_RSTn

Bits	Name	Description
0	P_SW_RST	This resets the pipe and it's registers, (Both Flip-Flops and RAM). Software can invoke pipe reset only after it make sure that the pipe is 'quiet'. 1'b1 - Reset 1'b0 - Do Nothing

**0x07897008+ BLSP1_BLSP_BAM_BAM_BAM_P_HALTn, n=[0..30]
4096*n****Type:** RW**Clock:** bam_clk**Reset State:** 0x00000000

Pipe Halt register Enables/Disables the Halt Sequence. Refer to the Software Procedures chapter for the Halt Sequence Procedure.

This is a self-modifying register.

BLSP1_BLSP_BAM_BAM_BAM_P_HALTn

Bits	Name	Description
1	P_PROD_HALTED	This bit is used to inform the Consumer about the Producer being in Halt mode now. This is a part of the Halt procedure. See the Software Procedures chapter. 1'b1 - Sets this bit to 1 1'b0 - Does Nothing This bit will be cleared by the HW.
0	P_HALT	When Enabled, the Pipe will enter Halt Mode. 1'b1 - Enable Halt 1'b0 - Disable Halt For B2B Producer pipes, this bit is self modifying. It doesn't need the SW to clear it. See the Software Procedure chapter for the detailed flow.

0x07897010+ BLSP1_BLSP_BAM_BAM_BAM_P_IRQ_STTSn, n=[0..30] 4096*n

Type: R

Clock: bam_clk

Reset State: 0x00000000

Pipe Interrupt status register describing the reason for the ongoing interrupt. This is an informative only register and therefore read only. It also has informative bits which are not interrupts.

Clearing the interrupt bits is done by writing to P_IRQ_CLR register.

BLSP1_BLSP_BAM_BAM_BAM_P_IRQ_STTSn

Bits	Name	Description
7	P_HRESP_ERR_IRQ	This interrupt Indicates that an Erroneous HResponse has been received by the AHB Master. Additional Information about the error may be read at BAM_AHB_MASTER_ERR_* Registers. Clearing this interrupt also clears the BAM_AHB_MASTER_ERR_* Registers. Available in BAM only
6	P_PIPE_RST_ERROR_IRQ	Unsuccessful Pipe Reset operation. Pipe reset timer expired.
5	P_TRNSFR_END_IRQ	For producer this interrupt happens whenever the last AHB burst of a transfer happens For Consumer when ack_on_success closes a transfer

BLSP1_BLSP_BAM_BAM_BAM_P_IRQ_STTSn (cont.)

Bits	Name	Description
4	P_ERR_IRQ	Pipe Error interrupt indicates that an error has happened. TBD: Error reason is not yet defined. Interrupt not in use.
3	P_OUT_OF_DESC_IRQ	Out of descriptors interrupt has several meanings depending on the operation mode: Producer - no free space in the descriptors fifo for adding a descriptor Consumer - no descriptors in the descriptors fifo for processing IO Vectors Producer - No descriptors to read (and therefore no space for data to write to the pipe) IO Vectors Consumer - No descriptors to read (and therefore no data to read from pipe)
2	P_WAKE_IRQ	Wake peripheral interrupt signals the driver to wake up the peripheral (USB for example) for transmission via the current pipe. This interrupt is issued whenever an Event comes to the specific pipe.
1	P_TIMER_IRQ	Inactivity timer has reached its threshold value Interrupt
0	P_PRCSN_DESC_IRQ	This Interrupt rises when BAM finishes processing an IO Vector which has INT bit selected

**0x07897014+ BLSP1_BLSP_BAM_BAM_BAM_P_IRQ_CLRN, n=[0..30]
4096*n****Type:** W**Clock:** bam_clk**Reset State:** 0x00000000

Writing to this register causes the interrupt to clear.

BAM_P_IRQ_CLRN, n=[0..30]

BLSP1_BLSP_BAM_BAM_BAM_P_IRQ_CLRN

Bits	Name	Description
7	P_HRESP_ERR_CLR	1'b1 - Clear 1'b0 - Unchanged
6	P_PIPE_RST_ERROR_CLR	1'b1 - Clear 1'b0 - Unchanged
5	P_TRNSFR_END_CLR	1'b1 - Clear 1'b0 - Unchanged
4	P_ERR_CLR	1'b1 - Clear 1'b0 - Unchanged
3	P_OUT_OF_DESC_CLR	1'b1 - Clear 1'b0 - Unchanged

BLSP1_BLSP_BAM_BAM_BAM_P_IRQ_CLRn (cont.)

Bits	Name	Description
2	P_WAKE_CLR	Clear Wake peripheral interrupt by writing this bit 1'b1 - Clear 1'b0 - Unchanged
1	P_TIMER_CLR	Clear Inactivity timer interrupt by writing this bit 1'b1 - Clear 1'b0 - Unchanged
0	P_PRCSD_DESC_CLR	Clear Descriptor Processed interrupt by writing this bit 1'b1 - Clear 1'b0 - Unchanged

**0x07897018+ BLSP1_BLSP_BAM_BAM_BAM_P_IRQ_ENn, n=[0..30]
4096*n****Type:** RW**Clock:** bam_clk**Reset State:** 0x00000000

Enable bits for the Interrupts in the Pipe Interrupt Status register.

BLSP1_BLSP_BAM_BAM_BAM_P_IRQ_ENn

Bits	Name	Description
7	P_HRESP_ERR_EN	1'b1 - Enable 1'b0 - Disable
6	P_PIPE_RST_ERROR_EN	1'b1 - Enable 1'b0 - Disable
5	P_TRNSFR_END_EN	1'b1 - Enable 1'b0 - Disable
4	P_ERR_EN	1'b1 - Enable 1'b0 - Disable
3	P_OUT_OF_DESC_EN	1'b1 - Enable 1'b0 - Disable
2	P_WAKE_EN	Enable Wake peripheral interrupt 1'b1 - Enable 1'b0 - Disable
1	P_TIMER_EN	Enable Inactivity timer Interrupt 1'b1 - Enable 1'b0 - Disable
0	P_PRCSD_DESC_EN	Enable Descriptor Processed Interrupt 1'b1 - Enable 1'b0 - Disable

**0x0789701C+ BLSP1_BLSP_BAM_BAM_BAM_P_TIMERn, n=[0..30]
4096*n****Type:** R**Clock:** bam_clk**Reset State:** 0x00000000

This register counts the idle time of the pipe.

BAM_P_TIMERn, n=[0..30]

BLSP1_BLSP_BAM_BAM_BAM_P_TIMERn

Bits	Name	Description
15:0	P_TIMER	Inactivity timer counter. 16 bits = 64k*0.1ms ~ 6 seconds

**0x07897020+ BLSP1_BLSP_BAM_BAM_BAM_P_TIMER_CTRLn, n=[0..30]
4096*n****Type:** RW**Clock:** bam_clk**Reset State:** 0x00000000

This register sets various controls and the threshold of the inactivity for interrupt triggering after long idle time.

The resolution of the pipe inactivity timers are in units set by a common counter. This common counter is controlled by a PARAMETER value of the BAM, and uses a separate clock, the inactivity_timers_clk. This clock can be slower than the bam_clk. The intent of the design is to use the sleep_clk, which is an always on 32kHz clock. This allows the bam_clk to be turned-off, while the base counter keeps counting. The width of this counter, as well of the rollover value are set using BAM PARAMETER. These values, taking the inactivity_timers_clk frequency define the pulse frequency that the base counter gives, and the threshold value can be calculated and set to interrupt after a specific time, regardless of the bam_clk frequency, and independent of clock power save features of the bam_clk. The timers continue counting even if the clock is turned off due to BAM dynamic hardware clock request voting.

The timer configuration is based on the inactivity_timers_clk period and the INACTIVITY_TIMER_WIDTH parameter constant. These parameters should be taken into account when setting this register. For example: for inactivity_timer_clk period of 1us and parameter is 3 and P_TIMER_THRSHLD is 10 will indicate the $2^3 * 1\mu s * 10$ which is 80us of inactivity in a pipe before sending an interrupt.

The general formula is: $2^{\text{INACTIVITY_TIMER_WIDTH}} * \text{clock_period} * \text{P_TIMER_THRSHLD}$.

BLSP1_BLSP_BAM_BAM_BAM_P_TIMER_CTRLn

Bits	Name	Description
31	P_TIMER_RST	To begin using the inactivity timer the SW should write the threshold and write 1 to this bit. writing the value 0 will hold the timer reset until value 1 is written again. Then the timer will start down counting from the P_TIMER_THRESHOLD value 1'b1 - Active 1'b0 - Reset
30	P_TIMER_RUN	Will be used along with P_TIMER_MODE Not implemented at this time. Set to zero(0)
29	P_TIMER_MODE	Not implemented at this time. Set to zero(0)
15:0	P_TIMER_TRSHLD	Inactivity timer threshold for interrupt generation.

**0x07897024+ BLSP1_BLSP_BAM_BAM_BAM_P_PRDCR_SDBNDn, n=[0..30]
4096*n****Type:** R**Clock:** bam_clk**Reset State:** 0x00000000

This Register reflects the current status of the Producer Sideband interface as the peripheral sees it and some additional internal bits related to its management.

BLSP1_BLSP_BAM_BAM_BAM_P_PRDCR_SDBNDn

Bits	Name	Description
24	BAM_P_SB_UPDATED	sideband_inform_bytes_avail_updated_producer. relevant for system mode only. The sideband-Inform block which responsible for reading descriptor has updated the SB block (which responsible for publishing to peripheral) with all of its read descriptors.
20	BAM_P_TOGGLE	Pipe Bytes Free Toggle value. The toggle polarity of the publishment (each publishment changes its polarity).
19:16	BAM_P_CTRL	Pipe Bytes Free Ctrl value. The possible values are only: 0000 or 0001 (means desc-fifo-full).
15:0	BAM_P_BYTES_FREE	Pipe Bytes Free value. The accumulated bytes_free which are available for peripheral to use

**0x07897028+ BLSP1_BLSP_BAM_BAM_BAM_P_CNSTRM_SDBNDn, n=[0..30]
4096*n****Type:** R**Clock:** bam_clk**Reset State:** 0x00000000

This Register reflects the current status of the Consumer Sideband interface as the peripheral sees it and some additional internal bits related to its management.

BLSP1_BLSP_BAM_BAM_BAM_P_CNSTRM_SDBNDn

Bits	Name	Description
24	BAM_P_SB_UPDATED	sideband_inform_bytes_avail_updated_consumer. The sideband-Inform block which responsible for reading descriptor has updated the SB block (which responsible for publishing to peripheral) with all of its read descriptors.
23	BAM_P_WAIT_4_ACK	waiting_for_ack_bytes_avail_r. BAM is waiting for Peripheral to acknowledge the former publish.
22	BAM_P_ACK_TOGGLE	ack_bytes_avail_toggle. The Peripheral's acknowledgement for the publish. This is a toggling signal (each ack will be inverted to its former).
21	BAM_P_ACK_TOGGLE_R	ack_bytes_avail_toggle_r. The Peripheral's sampled acknowledgement for the publish. This is a toggling signal (each ack will be inverted to its former).
20	BAM_P_TOGGLE	Pipe Bytes Avail Toggle value. The toggle polarity of the publishment (each publishment changes its polarity).
19:16	BAM_P_CTRL	Pipe Bytes Avail Ctrl value. The possible values of BAM_P_CTRL field are: 0100 - Means immediate command (could be only in bam-ndp). 0001 - Means End-Of-Transaction (EOT). 0011 - means End-Of-Transaction with Notify-When-Done (NWD). 0000 - Neither Immediate-command nor EOT nor NWD.
15:0	BAM_P_BYTES_AVAIL	Pipe Bytes AVAIL value. The size of the current publish.

**0x07897800+ BLSP1_BLSP_BAM_BAM_BAM_P_SW_OFSTSn, n=[0..30]
4096*n****Type:** RW**Clock:** bam_clk**Reset State:** 0x00000000

This register denotes the pointer Offset of the first Not Acknowledged Descriptor. Meaning all Descriptors prior to this one have been processed by the BAM. This register is only used by the Software. After receiving an interrupt, software reads this register in order to know what descriptors has been processed.

When the BAM works as Producer, Software will usually read this value after Descriptor Processed Interrupt or after the End of Transfer Interrupt.

When the BAM works as Consumer, Software will usually read this value after Descriptor Processed Interrupt.

NOTE This is non relevant in BAM to BAM modes.

NOTE Although being Writable, Software should never write to this register.

NOTE When using 2 pipes BAM, REVISION 0x03, this registers is Read Only.

BAM_P_SW_OFSTSn, n=[0..30]

BLSP1_BLSP_BAM_BAM_BAM_P_SW_OFSTSn

Bits	Name	Description
31:16	SW_OFST_IN_DESC	Offset inside the Descriptor. This value is used by the Software only when a BAM is working as System Producer. Then the SW can read this value to know how many Bytes were written to the current descriptor if it was not closed down. This is useful for Streaming mode. This Field is NOT available in BAM-NDP
15:0	SW_DESC_OFST	Descriptor FIFO offset.

0x07897804+ BLSP1_BLSP_BAM_BAM_BAM_P_AU_PSM_CNTXT_1_n, n=[0..30] 4096*n

Type: RW

Clock: bam_clk

Reset State: 0x00000000

Internal Context of Ack Update state machine.

NOTE When using 2 pipes BAM, REVISION 0x03, this registers is Read Only.

BAM_P_AU_PSM_CNTXT_1_n, n=[0..30]

BLSP1_BLSP_BAM_BAM_BAM_P_AU_PSM_CNTXT_1_n

Bits	Name	Description
31:16	AU_PSM_ACCUMED	PSM: The acknowledged (EOB/T) number of bytes since last Write Event. AU: The accumulation of the descriptor sizes for the present ack_on_success event. This is never bigger than the ACKED field. Once the ACCUMED values are bigger than Event Threshold, an Event will be issued, the ACKED value will decrease by the ACCUMED size and this value will be zeroed. This also serves as the Bytes Consumed value in the Event.
15:0	AU_ACKED	AU: The left over of the number of bytes that were received via the ack_on_success, that did not cause a descriptor closure. This value is down counting upon issuing Read Event. PSM:

**0x07897808+ BLSP1_BLSP_BAM_BAM_BAM_P_PSM_CNTXT_2_n, n=[0..30]
4096*n****Type:** RW**Clock:** bam_clk**Reset State:** 0x00000000

Internal Context of Pipe & Writeback state machines.

NOTE When using 2 pipes BAM, REVISION 0x03, this registers is Read Only.

BAM_P_PSM_CNTXT_2_n, n=[0..30]

BLSP1_BLSP_BAM_BAM_BAM_P_PSM_CNTXT_2_n

Bits	Name	Description
31	PSM_DESC_VALID	The Descriptor Valid bit provided by the Writeback state machine to Pipe state machine.
30	PSM_DESC_IRQ	The Descriptor Interrupt bit provided by the Writeback state machine to pipe state machine.
29	PSM_DESC_IRQ_DONE	The Descriptor Interrupt needed selection bit.
28:25	PSM_GENERAL_BITS	General Context Switch Bits
24:22	PSM_CONS_STATE	State of the Pipe Consumer state machine.
21:19	PSM_PROD_SYS_STATE	State of the Pipe Producer System state machine.
18:16	PSM_PROD_B2B_STATE	State of the Pipe Producer BAM to BAM state machine.
15:0	PSM_DESC_SIZE	The Descriptor Size provided by the Writeback state machine to pipe state machine.

**0x0789780C+ BLSP1_BLSP_BAM_BAM_BAM_P_PSM_CNTXT_3_n, n=[0..30]
4096*n****Type:** RW**Clock:** bam_clk**Reset State:** 0x00000000

Internal Context of Pipe & Writeback state machines.

NOTE When using 2 pipes BAM, REVISION 0x03, this registers is Read Only.

BAM_P_PSM_CNTXT_3_n, n=[0..30]

BLSP1_BLSP_BAM_BAM_BAM_P_PSM_CNTXT_3_n

Bits	Name	Description
31:0	PSM_DESC_ADDR	The Data Address provided in the current descriptor from the Writeback state machine.

**0x07897810+ BLSP1_BLSP_BAM_BAM_BAM_P_PSM_CNTXT_4_n, n=[0..30]
4096*n****Type:** RW**Clock:** bam_clk**Reset State:** 0x00000000

Internal Context of Pipe & Writeback state machines.

NOTE When using 2 pipes BAM, REVISION 0x03, this registers is Read Only.

BAM_P_PSM_CNTXT_4_n, n=[0..30]

BLSP1_BLSP_BAM_BAM_BAM_P_PSM_CNTXT_4_n

Bits	Name	Description
31:16	PSM_DESC_OFST	The running Descriptor Fifo Offset of the Writeback state machine
15:0	PSM_SAVED_ACCUMED_SIZE	The Saved Accumulated Size from the Pipe state machine.

**0x07897814+ BLSP1_BLSP_BAM_BAM_BAM_P_PSM_CNTXT_5_n, n=[0..30]
4096*n****Type:** RW**Clock:** bam_clk**Reset State:** 0x00000000

Internal Context of Pipe & Writeback state machines.

NOTE When using 2 pipes BAM, REVISION 0x03, this registers is Read Only.

BAM_P_PSM_CNTXT_5_n, n=[0..30]

BLSP1_BLSP_BAM_BAM_BAM_P_PSM_CNTXT_5_n

Bits	Name	Description
31:16	PSM_BLOCK_BYTE_CNT	The Byte count inside the current Block.
15:0	PSM_OFST_IN_DESC	The Offset inside a register. This is used by pipe state machine to compute offset inside a register after retransmission

**0x07897818+ BLSP1_BLSP_BAM_BAM_BAM_P_EVNT_REGn, n=[0..30]
4096*n****Type:** RW**Clock:** bam_clk**Reset State:** 0x00000000

This Register is where Read/Write events are sent to. If current BAM works as Producer, it will receive Read events into this register from the Consumer. If the Consumer is another BAM, it will

write read events to this register. If the Consumer is the Software, it will prepare descriptors in the descriptor FIFO and update the DESC_FIFO_PEER_OFST value in this register.

If current BAM works as Consumer, opposite behavior is assumed.

BAM_P_EVNT_REGn, n=[0..30]

BLSP1_BLSP_BAM_BAM_BAM_P_EVNT_REGn

Bits	Name	Description
31:16	P_BYTES_CONSUMED	This field is used only in the B2B mode. It indicates the number of bits consumed by the Consumer. Consumer sends this value to the Producer as a part of a read event, for the Producer to know how many bytes were consumed. When working in System mode, this value should be set to zero. BAM to BAM Producer writes zero in this value.
15:0	P_DESC_FIFO_PEER_OFST	This field indicates the Descriptor Fifo Offset Pointer of the Peer BAM (The BAM or the Software which creates descriptors). For example when current BAM works as consumer for the current pipe, Producer BAM (or the software) updates this value to let the current pipe know about the new descriptors added in the Descriptor Fifo.

0x0789781C+ BLSP1_BLSP_BAM_BAM_BAM_P_DESC_FIFO_ADDRn, n=[0..30] 4096*n

Type: RW

Clock: bam_clk

Reset State: 0x00000000

This register stores the Address of the Descriptor Fifo beginning.

NOTE This register is used by all modes.

BAM_P_DESC_FIFO_ADDRn, n=[0..30]

BLSP1_BLSP_BAM_BAM_BAM_P_DESC_FIFO_ADDRn

Bits	Name	Description
31:0	P_DESC_FIFO_ADDR	Address of the Descriptors Fifo. This address must be 8 bytes aligned.

0x07897820+ BLSP1_BLSP_BAM_BAM_BAM_P_FIFO_SIZESn, n=[0..30] 4096*n

Type: RW

Clock: bam_clk

Reset State: 0x00000000

The least significant field is the Descriptor Fifo Size.

The most significant field is the Data Fifo Size.

NOTE This Descriptor Fifo Size is used by all modes. The Data Fifo Size is non relevant in System Modes.

BAM_P_FIFO_SIZES_n, n=[0..30]

BLSP1_BLSP_BAM_BAM_BAM_P_FIFO_SIZES_n

Bits	Name	Description
31:16	P_DATA_FIFO_SIZE	Size of the Data Address Space. ALWAYS write 16'b0 to this field of P_DATA_FIFO_SIZE unless you work in BAM2BAM producer mode (also known as peer mode). NOTE: Data fifo size should not be bigger than 32KB.
15:0	P_DESC_FIFO_SIZE	Size of the Descriptors Fifo. Must be 8 byte aligned. Each descriptor is 8 bytes. Size of the descriptor fifo must contain an integer number of Descriptors. Size of descriptor fifo must be 16 bytes and above.

0x07897824+ BLSP1_BLSP_BAM_BAM_BAM_P_DATA_FIFO_ADDR_n, n=[0..30] 4096*n

Type: RW

Clock: bam_clk

Reset State: 0x00000000

This register stores the Beginning of the Data Address space.

This register is non relevant in System Modes.

BAM_P_DATA_FIFO_ADDR_n, n=[0..30]

BLSP1_BLSP_BAM_BAM_BAM_P_DATA_FIFO_ADDR_n

Bits	Name	Description
31:0	P_DATA_FIFO_ADDR	Data Address Space beginning.

0x07897828+ BLSP1_BLSP_BAM_BAM_BAM_P_EVNT_GEN_TRSHLD_n, n=[0..30] 4096*n

Type: RW

Clock: bam_clk

Reset State: 0x00000000

This register configures the threshold value for Read/Write event generation by the BAM towards another BAM. When aBAM pushes/pops this number of bytes to/from the pipe an event to the

other BAM will be generated. A counter summing up the bytes pushed/popped is reset when event is issued, thus restarting count for the threshold.

Events are also issued after End Of Transfer received from the peripheral (Producer case), non regarding whether the pushed bytes count has reached the threshold or not.

This register is non relevant in System Modes.

BAM_P_EVNT_GEN_TRSHLDn, n=[0..30]

BLSP1_BLSP_BAM_BAM_BAM_P_EVNT_GEN_TRSHLDn

Bits	Name	Description
15:0	P_TRSHLD	Threshold value. The maximum allowed value is 32kByte.

0x0789782C+ BLSP1_BLSP_BAM_BAM_BAM_P_EVNT_DEST_ADDRn, n=[0..30] 4096*n

Type: RW

Clock: bam_clk

Reset State: 0x00000000

This register stores the Event Destination Address which is the address of BAM_P_EVNT_REG in the BAM on the other side of the Pipe.

Producer BAM will send Write Events to this address.

Consumer BAM will send Read Events to this address.

This is relevant only for BAM to BAM modes. In System modes, BAM doesn't issue Write/Read events.

BAM_P_EVNT_DEST_ADDRn, n=[0..30]

BLSP1_BLSP_BAM_BAM_BAM_P_EVNT_DEST_ADDRn

Bits	Name	Description
31:0	P_EVNT_DEST_ADDR	Address of the Event Register in the peer BAM (second end of the pipe). Driver will configure this when initializing the pipe.

0x07897830+ BLSP1_BLSP_BAM_BAM_BAM_P_DF_CNTXT_n, n=[0..30] 4096*n

Type: RW

Clock: bam_clk

Reset State: 0x00000000

This register is used on in BAM-NDP core.

BAM_P_DF_CNTXT_n, n=[0..30]

BLSP1_BLSP_BAM_BAM_BAM_P_DF_CNTXT_n

Bits	Name	Description
31:16	WB_ACCUMULATED	Relevant in B2B Producer mode only. This field stores the current running accumulation of the total amount of data that has been written to the data fifo. When it goes bigger then event_threshold a write event occurs.
15:0	DF_DESC_OFST	Holds the descriptor offset of the Descriptor-Fetcher block which is responsible for fetching descriptors before publishing them to the peripheral.

0x07897834+ BLSP1_BLSP_BAM_BAM_BAM_P_RETR_CNTXT_n, n=[0..30] 4096*n

Type: RW

Clock: bam_clk

Reset State: 0x00000000

Internal Context stored for retransmission.

This is being stored by Ack Update state machine in consumer mode or by the Writeback state machine in producer mode.

This is being loaded by the Writeback state machine when retransmission happens.

NOTE When using 2 pipes BAM, REVISION 0x03, this registers is Read Only.

BAM_P_RETR_CNTXT_n, n=[0..30]

BLSP1_BLSP_BAM_BAM_BAM_P_RETR_CNTXT_n

Bits	Name	Description
31:16	RETR_DESC_OFST	Descriptor Fifo Offset for retransmission.
15:0	RETR_OFST_IN_DESC	The Offset inside the Descriptor for retransmission.

0x07897838+ BLSP1_BLSP_BAM_BAM_BAM_P_SI_CNTXT_n, n=[0..30] 4096*n

Type: RW

Clock: bam_clk

Reset State: 0x00000000

Internal Context of Sideband Inform state machine.

NOTE When using 2 pipes BAM, REVISION 0x03, this registers is Read Only.

BAM_P_SI_CNTXT_n, n=[0..30]

BLSP1_BLSP_BAM_BAM_BAM_P_SI_CNTXT_n

Bits	Name	Description
15:0	SI_DESC_OFST	The Descriptor Fifo Offset pointer stored by the Sideband Inform state machine.

**0x07897900+ BLSP1_BLSP_BAM_BAM_BAM_P_PSM_CNTXT_3_LSBn, n=[0..30]
4096*n****Type:** RW**Clock:** bam_clk**Reset State:** 0x00000000

Internal Context of Pipe & Writeback state machines.

Relevant only on 36 bit address BAM.

NOTE When using 2 pipes BAM, REVISION 0x03, this registers is Read Only.

BAM_P_PSM_CNTXT_3_n, n=[0..30]

BLSP1_BLSP_BAM_BAM_BAM_P_PSM_CNTXT_3_LSBn

Bits	Name	Description
31:0	PSM_DESC_ADDR	32 LSB of The Data Address provided in the current descriptor from the Writeback state machine.

**0x07897904+ BLSP1_BLSP_BAM_BAM_BAM_P_PSM_CNTXT_3_MSBn, n=[0..30]
4096*n****Type:** RW**Clock:** bam_clk**Reset State:** 0x00000000

Internal Context of Pipe & Writeback state machines.

Relevant only on 36 bit address BAM.

NOTE When using 2 pipes BAM, REVISION 0x03, this registers is Read Only.

BAM_P_PSM_CNTXT_3_n, n=[0..30]

BLSP1_BLSP_BAM_BAM_BAM_P_PSM_CNTXT_3_MSBn

Bits	Name	Description
3:0	PSM_DESC_ADDR	4 MSB of address (bits 35 to 32) of the Data Address provided in the current descriptor from the Writeback state machine.

**0x07897910+ BLSP1_BLSP_BAM_BAM_BAM_P_DESC_FIFO_ADDR_LSBn, n=[0..30]
4096*n****Type:** RW**Clock:** bam_clk**Reset State:** 0x00000000

This register stores the LSB of Address of the Descriptor Fifo beginning.

Relevant only on 36 bit address BAM.

NOTE This register is used by all modes.

BAM_P_DESC_FIFO_ADDRn, n=[0..30]

BLSP1_BLSP_BAM_BAM_BAM_P_DESC_FIFO_ADDR_LSBn

Bits	Name	Description
31:0	P_DESC_FIFO_ADDR	32 LSB of address of the Descriptors Fifo. This address must be 8 bytes aligned.

**0x07897914+ BLSP1_BLSP_BAM_BAM_BAM_P_DESC_FIFO_ADDR_MSBn, n=[0..30]
4096*n****Type:** RW**Clock:** bam_clk**Reset State:** 0x00000000

This register stores the MSB of Address of the Descriptor Fifo beginning.

Relevant only on 36 bit address BAM.

NOTE This register is used by all modes.

BAM_P_DESC_FIFO_ADDRn, n=[0..30]

BLSP1_BLSP_BAM_BAM_BAM_P_DESC_FIFO_ADDR_MSBn

Bits	Name	Description
3:0	P_DESC_FIFO_ADDR	4 MSB of address (bits 35 to 32) of Address of the Descriptors Fifo. This address must be 8 bytes aligned.

**0x07897920+ BLSP1_BLSP_BAM_BAM_BAM_P_DATA_FIFO_ADDR_LSBn, n=[0..30]
4096*n****Type:** RW**Clock:** bam_clk**Reset State:** 0x00000000

This register stores the Beginning of the Data Address space.

This register is non relevant in System Modes.

Relevant only on 36 bit address BAM.

BAM_P_DATA_FIFO_ADDRn, n=[0..30]

BLSP1_BLSP_BAM_BAM_BAM_P_DATA_FIFO_ADDR_LSBn

Bits	Name	Description
31:0	P_DATA_FIFO_ADDR	32 LSB of Data Address Space beginning. Relevant only on 36 bit address BAM.

0x07897924+ BLSP1_BLSP_BAM_BAM_BAM_P_DATA_FIFO_ADDR_MSBn, n=[0..30] 4096*n

Type: RW

Clock: bam_clk

Reset State: 0x00000000

This register stores the Beginning of the Data Address space.

This register is non relevant in System Modes.

Relevant only on 36 bit address BAM.

BAM_P_DATA_FIFO_ADDRn, n=[0..30]

BLSP1_BLSP_BAM_BAM_BAM_P_DATA_FIFO_ADDR_MSBn

Bits	Name	Description
3:0	P_DATA_FIFO_ADDR	4 MSB of Address (bits 35 to 32) of Data Address Space beginning. Relevant only on 36 bit address BAM.

0x07897930+ BLSP1_BLSP_BAM_BAM_BAM_P_EVNT_DEST_ADDR_LSBn, n=[0..30] 4096*n

Type: RW

Clock: bam_clk

Reset State: 0x00000000

This register stores the Event Destination Address which is the address of BAM_P_EVNT_REG in the BAM on the other side of the Pipe.

Producer BAM will send Write Events to this address.

Consumer BAM will send Read Events to this address.

This is relevant only for BAM to BAM modes. In System modes, BAM doesn't issue Write/Read events.

Relevant only on 36 bit address BAM.

BAM_P_EVNT_DEST_ADDRn, n=[0..30]

BLSP1_BLSP_BAM_BAM_BAM_P_EVNT_DEST_ADDR_LSBn

Bits	Name	Description
31:0	P_EVNT_DEST_ADDR	32 LSB of Address of the Event Register in the peer BAM (second end of the pipe). Driver will configure this when initializing the pipe. Relevant only on 36 bit address BAM.

0x07897934+ BLSP1_BLSP_BAM_BAM_BAM_P_EVNT_DEST_ADDR_MSBn, n=[0..30] 4096*n

Type: RW

Clock: bam_clk

Reset State: 0x00000000

This register stores the Event Destination Address which is the address of BAM_P_EVNT_REG in the BAM on the other side of the Pipe.

Producer BAM will send Write Events to this address.

Consumer BAM will send Read Events to this address.

This is relevant only for BAM to BAM modes. In System modes, BAM doesn't issue Write/Read events.

Relevant only on 36 bit address BAM.

BAM_P_EVNT_DEST_ADDRn, n=[0..30]

BLSP1_BLSP_BAM_BAM_BAM_P_EVNT_DEST_ADDR_MSBn

Bits	Name	Description
3:0	P_EVNT_DEST_ADDR	4 MSB of Address (bits 35 to 32) of the Event Register in the peer BAM (second end of the pipe). Driver will configure this when initializing the pipe. Relevant only on 36 bit address BAM.

20.2 BLSP1_BLSP_UART0_UART_DM

0x078AF000 BLSP1_BLSP_UART0_UART_DM_MR1

Type: RW

Clock: AHB_CLK

Reset State: 0x00000000

The UART_DM_MR1 register is the UART mode register #1. It is used, along with UART_DM_MR2, to configure the operational mode of the UART.

BLSP1_BLSP_UART0_UART_DM_MR1

Bits	Name	Description
31:8	AUTO_RFR_LEVEL1	These bits are used, along with field AUTO_RFR_LEVEL0 to program the level in the receive FIFO at which the Ready-For-Receive signal (RFR_N) toggles, if programmed to do so (see RX_RDY_CTL field of this register). When RX-FIFO level is greater than the value in fields AUTO_RFR_LEVEL1 and AUTO_RFR_LEVEL0, signal will indicate Not-Ready (RFR_N high). Otherwise signal will indicate Ready RFR_N low. Configured value should be higher/equal than RFRW value, otherwise the watermark will never be reached. Value of this register is in words and can be programmed from 1 to $2^{\text{RAM_ADDR_WIDTH}}$. - Only RAM_ADDR_WIDTH + 1:8 bits of this field are generated.
7	RX_RDY_CTL	Setting (1) this bit enables the automatic ready-for-receiving (RFR_N) control for the receiver. RFR_N is turned off, or set (1), when a valid start bit is received and the RX channel FIFO is at the level programmed in bits [31:8,5:0] of this register. When the FIFO level falls back to the programmed level, the RFR_N signal is turned on, or clear (0). When this feature is off, the RFR_N signal can be used by normal signal port bit manipulation (see UART_DM_CR register).
6	CTS_CTL	When this bit is set(1), the transmitter checks the CTS_N input to determine whether to begin transmission of a new character. If CTS_N is low, the character is sent. Otherwise the transmitter continues marking until CTS_N goes low, then the next character is transmitted. A change on CTS_N during the transmission of a character has no effect on that character. When this bit is clear(0), the CTS_N input for the channel has no effect on the transmitter.
5:0	AUTO_RFR_LEVEL0	See the description of field AUTO_RFR_LEVEL1.

0x078AF004 BLSP1_BLSP_UART0_UART_DM_MR2**Type:** RW**Clock:** AHB_CLK**Reset State:** 0x00000000

See description of UART_DM_MR1 register.

BLSP1_BLSP_UART0_UART_DM_MR2

Bits	Name	Description
10	RFR_CTS_LOOPBACK	Enables internal loopback between RFR_N of RX channel and CTS_N of TX channel if set. For this feature to be enabled, data loopback must also be enabled (LOOPBACK bit of this register).
9	RX_ERROR_CHAR_OFF	When this bit is set, characters received with parity or framing errors get discarded and not moved to the RX channel. Otherwise they enter RX channel as any valid character.
8	RX_BREAK_ZERO_CHAR_OFF	When this bit is set, the zero character received at rx_break event get discarded and not moved to the RX channel. Otherwise it enters RX-channel as any valid character.
7	LOOPBACK	When set - data output of transmitter is fed back to the data input of the receiver, i.e. TX-pin is connected to RX-pin. Might be used with flow-control loopback (RFR_CTS_LOOPBACK bit above).
6	ERROR_MODE	This bit controls the operation of the two status register bits for the RX channel (parity or framing error and received break, see UART_DM_SR register). - When clear (0), the UART operates in character mode and the status bits apply only to the last character received. - When set (1), the UART operates in block mode and both bits are sticky, i.e. they go high once an error/break-event is detected and go low only after reset error status command is issued (see UART_DM_CR register).
5:4	BITS_PER_CHAR	These bits determine how many data bits are transmitted or received per character, not including the start, stop, and parity bits. 0x0: ENUM_5_BITS (5 bits) 0x1: ENUM_6_BITS (6 bits) 0x2: ENUM_7_BITS (7 bits) 0x3: ENUM_8_BITS (8 bits)
3:2	STOP_BIT_LEN	This field programs the duration of the stop bit that is appended to each transmitted character. The stop bit duration can be 9/16, 1, 1+ 9/16, and 2 bit times. The receiver only samples the stop bit once, regardless of the programmed stop bit length. This configuration only applies to UART/IrDA interfaces, for SIM interface see SIM_STOP_BIT_LEN field of UART_DM_SIM_CFG register. 0x0: ENUM_0_563 (0.563 (9/16) bit times) 0x1: ENUM_1_000_BIT_TIME (1.000 bit time) 0x2: ENUM_1_563 (1.563 (1+9/16) bit times) 0x3: ENUM_2_000_BIT_TIMES (2.000 bit times)

BLSP1_BLSP_UART0_UART_DM_MR2 (cont.)

Bits	Name	Description
1:0	PARITY_MODE	These bits determine which parity mode is used. The user can select between odd, even, space, or no parity. 0x0: NO_PARITY (no parity - last data-bit is followed by a stop-bit) 0x1: ODD_PARITY (odd parity - parity-bit ensures number of ones in data-bits+parity-bit is odd) 0x2: EVEN_PARITY (even parity - parity-bit ensures number of ones in data-bits+parity-bit is even) 0x3: SPACE_PARITY (space parity - parity is always '0')

0x078AF008 BLSP1_BLSP_UART0_UART_DM_CSR_SR_DEPRECATED**Type:** RW**Clock:** AHB_CLK**Reset State:** Undefined

This register is deprecated and remains only for backward-compatibility. Use UART_DM_CSR or UART_DM_SR.

BLSP1_BLSP_UART0_UART_DM_CSR_SR_DEPRECATED

Bits	Name	Description
31:0	UART_DM_CSR_SR_DEPRECATED	

0x078AF010 BLSP1_BLSP_UART0_UART_DM_CR_MISR_DEPRECATED**Type:** RW**Clock:** AHB_CLK**Reset State:** Undefined

This register is deprecated and remains only for backward-compatibility. Use UART_DM_CR or UART_DM_MISR.

BLSP1_BLSP_UART0_UART_DM_CR_MISR_DEPRECATED

Bits	Name	Description
31:0	UART_DM_CR_MISR_DEPRECATED	

0x078AF014 BLSP1_BLSP_UART0_UART_DM_IMR_ISR_DEPRECATED**Type:** RW**Clock:** AHB_CLK**Reset State:** Undefined

This register is deprecated and remains only for backward-compatibility. Use UART_DM_IMR or UART_DM_ISR.

BLSP1_BLSP_UART0_UART_DM_IMR_ISR_DEPRECATED

Bits	Name	Description
31:0	UART_DM_IMR_ISR_DEPRECATED	

0x078AF018 BLSP1_BLSP_UART0_UART_DM_IPR**Type:** RW**Clock:** AHB_CLK**Reset State:** 0xFFFFFFFF9F

The UART_DM_IPR register is the UART interrupt programming register.

BLSP1_BLSP_UART0_UART_DM_IPR

Bits	Name	Description
31:7	STALE_TIMEOUT_MSB	The stale character time-out duration field contains a number from 1 to $2^{30} - 1$. This number determines how many character times must elapse before a stale event is generated. This character time is defined as 10 times the bit rate. It is reset whenever a data bit is received. Never set this fields to 0 while RX channel is operational. Do not re-configure while an RX-transfer is active. Note the discontinuity in the bit assignments.
6	SAMPLE_DATA	Currently not in use. Setting (1) to this bit enables the sample data mode. In this mode start bit is sampled along with the data of the UART character. Works only in 'Packet mode' (See the CR register, CHANNEL_COMMAND bit for more information).
5	RESERVED	
4:0	STALE_TIMEOUT_LSB	This bitfield is the LSbits of the STALE_TIMEOUT bitfield.

0x078AF01C BLSP1_BLSP_UART0_UART_DM_TFWR**Type:** RW**Clock:** AHB_CLK**Reset State:** 0x00000000

The UART_DM_TFWR register is the UART transmit FIFO watermark register.

BLSP1_BLSP_UART0_UART_DM_TFWR

Bits	Name	Description
31:0	TFW	These bits contain a number, between 0 and $2^{\text{RAM_ADDR_WIDTH}} - 1$, that determines the level of the transmit (TX) FIFO at which space_avail_req signal is asserted in BAM mode or TXLEV interrupt is asserted in non-BAM mode. The signal/interrupt is asserted when the FIFO level (number of TX words in the FIFO) is less than or equal to the value in TFWR. Notice that the value is in words. See UART_DM_ISR register for TXLEV interrupt description. - Only RAM_ADDR_WIDTH -1:0 bits are generated.

0x078AF020 BLSP1_BLSP_UART0_UART_DM_RFWR**Type:** RW**Clock:** AHB_CLK**Reset State:** Undefined

The UART_DM_RFWR register is the UART receive FIFO watermark register.

BLSP1_BLSP_UART0_UART_DM_RFWR

Bits	Name	Description
31:0	RFW	These bits contain a number, between 0 and $2^{\text{RAM_ADDR_WIDTH}} - 1$, that determines the level of the receive (RX) FIFO at which data_avail_req signal is asserted in BAM mode or RXLEV interrupt is asserted in non-BAM mode. The signal/interrupt is asserted when the FIFO level (number of words in the RX FIFO) is greater than the value in RFWR. Configured value should be lower/equal than AUTO_RFR_LEVEL value, otherwise the watermark will never be reached. Notice that the value is in words. See UART_DM_ISR register for TXLEV interrupt description - Only RAM_ADDR_WIDTH -1:0 bits are generated.

0x078AF024 BLSP1_BLSP_UART0_UART_DM_HCR

Type: RW
Clock: AHB_CLK
Reset State: 0x00000000

Configures Hunt-Character value.

BLSP1_BLSP_UART0_UART_DM_HCR

Bits	Name	Description
7:0	DATA	The UART_DM_HCR register is the UART hunt character register, which contains the character for which the receiver looks to assert a hunt character received interrupt.

0x078AF034 BLSP1_BLSP_UART0_UART_DM_DMRX

Type: RW
Clock: AHB_CLK
Reset State: 0x00000000

Initializes RX transfer and configures maximal length in bytes of the transfer.

BLSP1_BLSP_UART0_UART_DM_DMRX

Bits	Name	Description
24:0	RX_DM_CRCI_CHARS	A write to RX_DM_CRCI_CHARS initializes RX transfer in SW (FIFO) mode, with written value configuring the maximal length in bytes of the transfer. In BAM-mode, only the value held in this register impacts the data-flow and not the write-event. A write-value of 0 is not valid. Read of UART_DM_DMRX register is returns the number of characters that were received since the end of the last RX transfer. To read back the last value written to this register use UART_DM_DMRX_DBG register. Refer to UART_DM_BCR register description for additional constraints on the value to be written to this register.

0x078AF038 BLSP1_BLSP_UART0_UART_DM_IRDA_RX_TOTAL_SNAP_DEPRECATED

Type: RW
Clock: AHB_CLK
Reset State: Undefined

This register is deprecated and remains only for backward-compatibility. Use UART_DM_IRDA or UART_DM_RX_TOTAL_SNAP.

BLSP1_BLSP_UART0_UART_DM_IRDA_RX_TOTAL_SNAP_DEPRECATED

Bits	Name	Description
31:0	UART_DM_IRDA_RX_TOTAL_SNAP_DEPRECATED	

0x078AF03C BLSP1_BLSP_UART0_UART_DM_DMEN**Type:** RW**Clock:** AHB_CLK**Reset State:** 0x00000000

The UART_DM_DMEN register defines the DMA used by the core and the data packing mode for TX and RX channels.

BLSP1_BLSP_UART0_UART_DM_DMEN

Bits	Name	Description
5	RX_SC_ENABLE	- Set (1) for work in Single-Character mode for RX channel (every character received is zero-padded into a word). - Clearing this bit requires resetting the receiver.
4	TX_SC_ENABLE	- Set (1) for work in Single-Character mode for TX channel (Only LSB byte of each word get transmitted). - Clearing this bit requires resetting the transmitter.
3	RX_BAM_ENABLE	- Set (1) this bit to enable RX BAM interface (BAM-mode). - Clear (0) this bit to disable RX BAM interface (SW/FIFO mode). Clearing this bit requires resetting the receiver (see UART_DM_CR register).
2	TX_BAM_ENABLE	- Set (1) this bit to enable TX BAM interface (BAM-mode). - Clear (0) this bit to disable TX BAM interface (SW/FIFO mode). Clearing this bit requires resetting the transmitter(see UART_DM_CR register).
1	RESERVED_1	
0	RESERVED_2	

0x078AF040 BLSP1_BLSP_UART0_UART_DM_NO_CHARS_FOR_TX**Type:** RW**Clock:** AHB_CLK**Reset State:** 0x00000000

Initialization of TX transfer in SW/FIFO mode.

BLSP1_BLSP_UART0_UART_DM_NO_CHARS_FOR_TX

Bits	Name	Description
23:0	TX_TOTAL_TRANS_LEN	The total number of characters for transmission. Before writing a new value, the TX FIFO must be empty as indicated by TX_READY interrupt (see UART_DM_ISR register for description) or after a TX-channel reset. It is used by the TX-channel in SW/FIFO mode to calculate how many characters to transmit in the last word. Any additional writes to the TX FIFO beyond TX_TOTAL_TRANS_LEN characters will be discarded. A TX_READY interrupt is triggered after TX_TOTAL_TRANS_LEN number of characters were moved from the TX FIFO to the unpacking buffer (not all may have been transmitted at that point though).

0x078AF044 BLSP1_BLSP_UART0_UART_DM_BADR**Type:** RW**Clock:** AHB_CLK**Reset State:** Undefined

Configures FIFO division between RX-FIFO and TX-FIFO.

BLSP1_BLSP_UART0_UART_DM_BADR

Bits	Name	Description
31:2	RX_BASE_ADDR	RX FIFO base address. Both FIFOs use the same RAM ($2^{\text{RAM_ADDR_WIDTH}}$, 32-bit entries). This register controls the division of the memory between the RX and TX FIFOs. The division must be a multiple of 4 entries for legacy reasons, therefore bits 1:0 of this register are not used. The size of the TX FIFO equals RX_BASE_ADDR. The size of the RX FIFO is $2^{\text{RAM_ADDR_WIDTH}} - \text{RX_BASE_ADDR}$. * The default is $\text{RX_BASE_ADDR} = 2^{(\text{RAM_ADDR_WIDTH} - 1)}$ * Only $\text{RAM_ADDR_WIDTH} - 1:2$ bits are generated.

0x078AF04C BLSP1_BLSP_UART0_UART_DM_TXFS**Type:** R**Clock:** AHB_CLK**Reset State:** Undefined

TX channel status register.

BLSP1_BLSP_UART0_UART_DM_TXFS

Bits	Name	Description
31:14	TX_FIFO_STATE_MSB	The msb of TX_FIFO_STATE bitfield. - Only RAM_ADDR_WIDTH +4 +3 -1:14 bits are generated.
13:10	TX_ASYNC_FIFO_STATE	TX asynchronous fifo status - number of characters in the TX asynchronous FIFO.
9:7	TX_BUFFER_STATE	TX unpacking buffer status - number of bytes in the unpacking buffer
6:0	TX_FIFO_STATE_LSB	TX FIFO status - number of words in the TX FIFO

0x078AF050 BLSP1_BLSP_UART0_UART_DM_RXFS**Type:** R**Clock:** AHB_CLK**Reset State:** Undefined

RX channel status register.

BLSP1_BLSP_UART0_UART_DM_RXFS

Bits	Name	Description
31:14	RX_FIFO_STATE_MSB	The msb of TX_FIFO_STATE bitfield. Only RAM_ADDR_WIDTH +4 +3 -1:14 bits are generated.
13:10	RX_ASYNC_FIFO_STATE	RX asynchronous fifo status - number of characters in the RX asynchronous FIFO.
9:7	RX_BUFFER_STATE	RX packing buffer status - number of bytes in the packing buffer
6:0	RX_FIFO_STATE_LSB	The number of entries in the RX FIFO that have at least one valid word. It is incremented each time a word is moved from the packing register to the RX FIFO. It is decremented each time a word is read from the RX FIFO. Note: The Uart does not keep track of non-valid characters in each word.

0x078AF060 BLSP1_BLSP_UART0_UART_DM_MISR_MODE**Type:** RW**Clock:** WR_CLK**Reset State:** Undefined

BLSP1_BLSP_UART0_UART_DM_MISR_MODE

Bits	Name	Description
31:2	RESERVED	
1:0	MODE	0x0: DISABLED (Disabled) 0x1: ENABLED_TX_TEST (Enabled, TX test) 0x2: ENABLED_RX_TEST (Enabled, RX test)

0x078AF064 BLSP1_BLSP_UART0_UART_DM_MISR_RESET**Type:** W**Clock:** WR_CLK**Reset State:** Undefined**BLSP1_BLSP_UART0_UART_DM_MISR_RESET**

Bits	Name	Description
31:1	RESERVED	
0	RESET	

0x078AF068 BLSP1_BLSP_UART0_UART_DM_MISR_EXPORT**Type:** RW**Clock:** WR_CLK**Reset State:** Undefined**BLSP1_BLSP_UART0_UART_DM_MISR_EXPORT**

Bits	Name	Description
31:1	RESERVED	
0	EXPORT	When clear (0), testbus is driven by what is sent into the MISR. (i.e. what is the result of the muxing of all the input data streams with <BLOCK>_TEST_MODE) When set (1), testbus is driven by the current state of the MISR. (debug typically)

0x078AF06C BLSP1_BLSP_UART0_UART_DM_MISR_VAL**Type:** R**Clock:** WR_CLK**Reset State:** Undefined

BLSP1_BLSP_UART0_UART_DM_MISR_VAL

Bits	Name	Description
9:0	VAL	Current MISR state

0x078AF070 BLSP1_BLSP_UART0_UART_DM_TF_RF_DEPRECATED**Type:** RW**Clock:** AHB_CLK**Reset State:** Undefined

This register is deprecated and remains only for backward-compatibility. Use UART_DM_TF or UART_DM_RF.

BLSP1_BLSP_UART0_UART_DM_TF_RF_DEPRECATED

Bits	Name	Description
31:0	UART_DM_TF_RF_DEPRECATED	

0x078AF074 BLSP1_BLSP_UART0_UART_DM_TF_RF_2_DEPRECATED**Type:** RW**Clock:** AHB_CLK**Reset State:** Undefined

This register is deprecated and remains only for backward-compatibility. Use UART_DM_TF_2 or UART_DM_RF_2.

BLSP1_BLSP_UART0_UART_DM_TF_RF_2_DEPRECATED

Bits	Name	Description
31:0	UART_DM_TF_RF_2_DEPRECATED	

0x078AF078 BLSP1_BLSP_UART0_UART_DM_TF_RF_3_DEPRECATED**Type:** RW**Clock:** AHB_CLK**Reset State:** Undefined

This register is deprecated and remains only for backward-compatibility. Use UART_DM_TF_3 or UART_DM_RF_3.

BLSP1_BLSP_UART0_UART_DM_TF_RF_3_DEPRECATED

Bits	Name	Description
31:0	UART_DM_TF_RF_3_DEPRECATED	

0x078AF07C BLSP1_BLSP_UART0_UART_DM_TF_RF_4_DEPRECATED**Type:** RW**Clock:** AHB_CLK**Reset State:** Undefined

This register is deprecated and remains only for backward-compatibility. Use UART_DM_TF_4 or UART_DM_RF_4.

BLSP1_BLSP_UART0_UART_DM_TF_RF_4_DEPRECATED

Bits	Name	Description
31:0	UART_DM_TF_RF_4_DEPRECATED	

0x078AF080 BLSP1_BLSP_UART0_UART_DM_SIM_CFG**Type:** RW**Clock:** AHB_CLK**Reset State:** 0x00000000

The UART_DM_SIM_CFG register is used to configure the SIM interface for the UART.

* This register is generated only when SIM_GLUE_GEN generic equals 1.

BLSP1_BLSP_UART0_UART_DM_SIM_CFG

Bits	Name	Description
17	UIM_TX_MODE	When this bit is set (1), any re-transmission signal from the UIM card will be ignored (The transmission portion of the SIM interface operates in block mode T=1). When clear (0), re-transmission will be executed upon request (the transmission portion of the SIM interface operates in asynchronous transfer mode T=0). Note that the operation of this function is conditional upon SIM_SEL field being set (1) to designate the UIM_IF mode of operation.

BLSP1_BLSP_UART0_UART_DM_SIM_CFG (cont.)

Bits	Name	Description
16	UIM_RX_MODE	When this bit is set (1), error-signal indication will not be generated by the receiver (the receive portion of the SIM interface operates in block mode T=1). When clear (0), error signal will be generated when a character error is detected (the receive portion of the SIM interface operates in asynchronous transfer mode T=0). Note that the operation of this function is conditional upon SIM_SEL field being set (1) to designate the UIM_IF mode of operation.
15:8	SIM_STOP_BIT_LEN	In UIM_IF mode, this field programs the duration of the stop bit that is appended to each transmitted character. The stop bit duration can be 1 to 254 bit times. The receiver only samples the stop bit once, regardless of the programmed stop bit length. 0x1: ENUM_1_BIT_TIMES (1 bit times) 0x2: ENUM_2_BIT_TIMES (2 bit times) 11 111110: 254 bit times)
7	SIM_CLK_ON	When this bit is set(1), the UIM_CLK is turned on at either the TD8 or TD4 frequency, as indicated by bit6(UIM_CLK_TD8_SEL). When this bit is cleared, the UIM_CLK is stopped either LOW or HIGH, as indicated by bit 5(UIM_CLK_STOP_HIGH).
6	SIM_CLK_TD8_SEL	This bit selects the UIM_CLK frequency at which the UIM_CLK is turned on. 0x1: TD8 (UIM_CLK runs at the TD8 frequency (same frequency as SIM-SRC_CLK)) 0x0: TD4 (UIM_CLK runs at the TD4 frequency (1/2 the frequency of SIM-SRC_CLK))
5	SIM_CLK_STOP_HIGH	This bit indicates at what point - high or low - the UIM_CLK will stop. 0x1: HIGH (high) 0x0: LOW (low)
4	RESERVED	
3	MASK_RX	Setting (1) this bit masks off any 0 on the RX line, which prevents the receiver from seeing the START bit. Set (1) this bit if you do not want to receive the data you transmit.
2	SWAP_D	Setting (1) this bit causes the UIM_IF to swap the 8 data bits (making MSB into LSB, bit 6 into bit1, and so on) before it is read by the microprocessor.
1	INV_D	Set (1) this bit to cause the UIM_IF to invert the 8 data bit before it is read by the microprocessor.
0	SIM_SEL	When set (1), UART_DM will operate in SIM/UIM interface mode. When cleared (0), SIM/UIM interface is disabled.

0x078AF0A0 BLSP1_BLSP_UART0_UART_DM_CSR**Type:** RW**Clock:** AHB_CLK**Reset State:** 0x00000000

The UART_DM_CSR register is the UART clock selection register. This register is used to determine the bit rate for the transmitter and receiver. The transmitter and receiver can be clocked at different rates.

Data-rate of receiver/transmitter is UART-CLK frequency divided by the integer matching the configured value below:

0xF = 16

0xE = 32

0xD = 48

0xC = 64

0xB = 96

0xA = 128

0x9 = 192

0x8 = 256

0x7 = 384

0x6 = 512

0x5 = 768

0x4 = 1536

0x3 = 3072

0x2 = 6144

0x1 = 12288

0x0 = 24576

BLSP1_BLSP_UART0_UART_DM_CSR

Bits	Name	Description
7:4	UART_RX_CLK_SEL	Use the values above to select the receive bit rate.
3:0	UART_TX_CLK_SEL	Use the values above to select the transmit bit rate.

0x078AF0A4 BLSP1_BLSP_UART0_UART_DM_SR**Type:** R**Clock:** AHB_CLK**Reset State:** Undefined

The UART_DM_SR register is the UART status register. This register is used to obtain the current state of the UART subsystem. This register is updated asynchronously.

BLSP1_BLSP_UART0_UART_DM_SR

Bits	Name	Description
11:10	TRANS_END_TRIGGER	RX-transfer-end event type. Stores trigger for last transfer end: '00' for no trigger, '01' for DMRX-event, '10' for timeout stale, '11' for SW stale.
9	TRANS_ACTIVE	Transfer Active bit.
8	RX_BREAK_START_LAST	When break start is detected, this bit is set (1). When break end was detected, this bit is reset (0).
7	HUNT_CHAR	When set (1), this bit indicates that the programmed character in the UART_DM_HCR register was received. If programmed, this condition causes the RXHUNT bit to be set (1) in the UART_DM_ISR register. This bit is cleared by issuing a reset error status command.
6	RX_BREAK	When set (1), this bit indicates that an all-zero character of the programmed length was received without a stop bit. If a break begins in the middle of a character, it is detected if the condition remains until the end of the next character time. Only one all zero character is written into the FIFO upon receiving the break. After the RX input returns high for at least half a bit time, then more characters are accepted by the receiver. Whenever the break condition starts or stops as defined above, the RXBREAK bit is set (1) in the UART_DM_ISR register. This bit is appended to the character in the receive FIFO.
5	PAR_FRAME_ERR	A parity error occurs if parity is programmed to be checked and an incoming character is received with incorrect parity. A framing error occurs when the RX input is low while the stop bit is being sampled. This bit is sampled in the middle of the first stop bit position, regardless of the programmed length of the stop bit. This error bit is appended to the character in the receive FIFO.
4	UART_OVERRUN	An overrun error occurs if one or more incoming characters are lost. This happens when the receive FIFO is full, the packing buffer is full and another character has been received in the shift register. The character in the receive shift register is lost. Upon overrun, this status bit remains high until cleared by a reset error status command.
3	TXEMT	This bit is set (1) when the transmitter under-runs. It is set after the transmission of the stop bit at the end of a character, if there are no characters waiting to be sent. It is reset when a character is written to the transmit FIFO.

BLSP1_BLSP_UART0_UART_DM_SR (cont.)

Bits	Name	Description
2	TXRDY	This bit indicates that the transmit FIFO has space in it. It is reset when the FIFO becomes full and set when space becomes available again. Any characters that are written to the FIFO while this bit is low is lost.
1	RXFULL	This bit is set (1) when the receive FIFO becomes full. It is reset when the CPU reads a word.
0	RXRDY	This bit is set (1) when there is a word in the receive FIFO waiting to be read. It is reset when the last word currently stored in the FIFO is read.

0x078AF0A8 BLSP1_BLSP_UART0_UART_DM_CR**Type:** RW**Clock:** AHB_CLK**Reset State:** 0x00000000

The UART_DM_CR register is the UART command register. This register is used to issue specific commands to the UART subsystem. This register is updated asynchronously.

CAUTION Do not reset the transmitter and disable it at the same time. Do not reset the receiver and disable it at the same time.

Channel-commands:

0x00 - Null Command - Does nothing.

0x01 - Reset Receiver - Resets the receiver as if a hardware reset were issued. The receiver is disabled and the FIFO, packing buffer and shift registers are flushed.

0x02 - Reset Transmitter - Resets the transmitter as if a hardware reset were issued. The transmitter signal goes high (marking) and the FIFO, unpacking register and shift register are flushed.

0x03 - Reset error status - Clears the overrun error and hunt char received status bits in both the character and block error modes. In the block error mode, it clears the error status and received break.

0x04 - Reset break change interrupt - Clears the break change interrupt status bit.

0x05 - Start break - Forces the transmitter signal low. The transmitter must be enabled. If the transmitter is busy, the break is started when all characters in the transmit FIFO and the transmit shift register have been completely sent.

0x06 - Stop break - If executed while channel is breaking, this command causes the transmitter signal to go high. The signal remains high for at least one bit time before sending out a new character.

0x07 - Reset CTS_N - Clears ISR bit 5.

0x08 - Reset stale interrupt - Clears the stale interrupt.

0x09 - Packet mode - Turns on the sample data mode, which causes the receiver to sample the receive data stream at 16 times the programmed baud rate. The data is sampled with the start of the start bit, or the first data bit, and continued until the marking state. To exit this state, write 1100 in the command field.

0x0A - test_parity_on - Internal use only.

0x0B - test_frame_on - Internal use only.

0x0C - Mode reset - Turns off the sample data mode.

0x0D - Set RFR - Asserts the ready for receiving signal (actual flow-control port is active-low (rfr_n) and will go low following this command).

0x0E - Reset RFR - Deasserts the ready for receiving signal (actual flow-control port is active-low (rfr_n) and will go high following this command).

0x0F - uart_reset_int - Internal use only.

0x10 - Reset TX_ERROR - Clears TX_ERROR

0x11 - Clear TX_DONE - Clears the TX_DONE interrupt (ISR bit 9)

0x12 - Reset break start interrupt - Clears the break start interrupt status bit.

0x13 - Reset break end interrupt - Clears the break end interrupt status bit.

0x14 - Reset par_frame_err interrupt - Clears the par_frame_err interrupt status bit.

0x15 - Start RX BAM IFC - Initialize producer sidebands towards BAM after reset.

0x16 - Start TX BAM IFC - Initialize consumer sidebands towards BAM after reset.

0x17 - WWT counter enable - Enable WWT counter & IRQ.

0x18 - WWT counter disable - Disable WWT counter & IRQ.

0x19 - Clear TX_WR_ERROR IRQ - Clears TX write error interrupt status bit.

0x1A - Clear RX_RD_ERROR IRQ - Clears RX read error interrupt status bit.

0x1B - Clear TX_COMP IRQ - Clears TX complete interrupt status bit.

0x1C - Clear WWT IRQ - Clears WWT timeout interrupt status bit.

General-commands:

0x0 - Null Command - Does nothing.

0x1 - CR Protection Enable - Enables CR HW protection. When two consecutive writes to the CR are detected, the second write is delayed until the command of the first write is finished. The delay is done by de-asserting the AHB ready and this ensures that the first command completes and the second one will be executed right afterward.

0x2 - CR Protection Disable - Disables CR HW protection. SW is responsible for managing delay between writes to the CR register.

0x3 - Reset TX-Ready interrupt - Clears the TX_READY interrupt.

0x4 - SW Force Stale - Causes a stale event (even if stale events are disabled).

0x5 - Enable Stale Event - Enables the stale event mechanism

0x6 - Disable Stale Event - Enables the stale event mechanism

0x7 - RESERVED

BLSP1_BLSP_UART0_UART_DM_CR

Bits	Name	Description
11	CHANNEL_COMMAND_MSB	This is the msb of the CHANNEL_COMMAND bitfield.
10:8	GENERAL_COMMAND	Writing the appropriate value to these bits executes the commands that are listed in Table .
7:4	CHANNEL_COMMAND_LSB	Writing the appropriate value to these bits along with bit 11,executes the channel-command as listed in the register decription.
3	UART_TX_DISABLE	This command terminates the operation of the transmitter after any character in the transmit shift register is sent.
2	UART_TX_EN	This command enables the operation of the transmitter.
1	UART_RX_DISABLE	This command immediately terminates the operation of the channel receiver and any incoming characters are lost. None of the receiver status bits are affected by this command and characters that are already in the receive FIFO remain there.
0	UART_RX_EN	This command enables the channel receiver.

0x078AF0AC BLSP1_BLSP_UART0_UART_DM_MISR

Type: R

Clock: AHB_CLK

Reset State: 0x00000000

BLSP1_BLSP_UART0_UART_DM_MISR

Bits	Name	Description
16:0	UART_MISR	The UART_DM_MISR register is the masked interrupt status register. A read from this register returns the 'AND' of the UART_DM_ISR and the UART_DM_IMR registers. This register is updated asynchronously. This bit field is $\text{misr} \leq (\text{isr}(16 \text{ DOWNT} 7) \text{ AND } \text{imr}(16 \text{ DOWNT} 7)) \text{ \& '0' \& } (\text{isr}(5 \text{ DOWNT} 0) \text{ AND } \text{imr}(5 \text{ DOWNT} 0)).$

0x078AF0B0 BLSP1_BLSP_UART0_UART_DM_IMR**Type:** RW**Clock:** AHB_CLK**Reset State:** 0x00000000

The UART_DM_IMR register is the UART interrupt mask register. This register is used to enable the corresponding functions in the UART_DM_ISR register. Setting (1) a bit in the UART_DM_IMR register causes an interrupt to be generated, if the corresponding bit in the UART_DM_ISR register is set. Clearing (0) a bit in the UART_DM_IMR register causes the setting of the corresponding bit in the UART_DM_ISR register to have no effect on the interrupt signal. (See UART_DM_ISR register)

Bit 6 of the UART_DM_IMR register, CURRENT_CTS, is slightly different. If the current value of the CTS is one (1), no interrupt is generated. However, the user can use bit 6 in this register to mask the CTS value when reading the UART_DM_MISR register or as a general-purpose bit.

BLSP1_BLSP_UART0_UART_DM_IMR

Bits	Name	Description
16	WWT_IRQ	Masking bit of the matching field interrupt bit (See UART_DM_ISR for field descriptions).
15	TXCOMP_IRQ	Masking bit of the matching field interrupt bit (See UART_DM_ISR for field descriptions).
14	RX_RD_ERROR_IRQ	Masking bit of the matching field interrupt bit (See UART_DM_ISR for field descriptions).
13	TX_WR_ERROR_IRQ	Masking bit of the matching field interrupt bit (See UART_DM_ISR for field descriptions).
12	PAR_FRAME_ERR_IRQ	Masking bit of the matching field interrupt bit (See UART_DM_ISR for field descriptions).
11	RXBREAK_END	Masking bit of the matching field interrupt bit (See UART_DM_ISR for field descriptions).
10	RXBREAK_START	Masking bit of the matching field interrupt bit (See UART_DM_ISR for field descriptions).
9	TX_DONE	Masking bit of the matching field interrupt bit (See UART_DM_ISR for field descriptions).
8	TX_ERROR	Masking bit of the matching field interrupt bit (See UART_DM_ISR for field descriptions).
7	TX_READY	Masking bit of the matching field interrupt bit (See UART_DM_ISR for field descriptions).
6	CURRENT_CTS	Masking bit of the matching field interrupt bit (See UART_DM_ISR for field descriptions).
5	DELTA_CTS	Masking bit of the matching field interrupt bit (See UART_DM_ISR for field descriptions).

BLSP1_BLSP_UART0_UART_DM_IMR (cont.)

Bits	Name	Description
4	RXLEV	Masking bit of the matching field interrupt bit (See UART_DM_ISR for field descriptions).
3	RXSTALE	Masking bit of the matching field interrupt bit (See UART_DM_ISR for field descriptions).
2	RXBREAK_CHANGE	Masking bit of the matching field interrupt bit (See UART_DM_ISR for field descriptions).
1	RXHUNT	Masking bit of the matching field interrupt bit (See UART_DM_ISR for field descriptions).
0	TXLEV	Masking bit of the matching field interrupt bit (See UART_DM_ISR for field descriptions).

0x078AF0B4 BLSP1_BLSP_UART0_UART_DM_ISR**Type:** R**Clock:** AHB_CLK**Reset State:** Undefined

The UART_DM_ISR register is the UART interrupt status register. This register provides the current status of the possible interrupt conditions. If one of these bits is set (1), and the corresponding bit in the UART_DM_IMR register is set (1), an interrupt is generated (with the exception of bit 6, CURRENT_CTS). If the corresponding bit in the UART_DM_IMR register is clear (0), setting one of these bits has no effect on the UART interrupt request signal. This register is updated asynchronously.

BLSP1_BLSP_UART0_UART_DM_ISR

Bits	Name	Description
16	WWT_IRQ	WWT timeout reached interrupt. Fired after WWT timer was enabled and no data was received during the configured time that follows. Cleared by issuing CHANNEL_COMMAND = CLEAR_WWT_IRQ (UART_DM_CR register).
15	TXCOMP_IRQ	TX-complete interrupt. Fired when TX channel finished transmitting all of TX-transfer's data and no new transfer was initiated. Cleared by issuing CHANNEL_COMMAND = CLEAR_TX_COMP_IRQ (UART_DM_CR register).
14	RX_RD_ERROR_IRQ	RX-FIFO empty read error. Fired when AHB master is trying to read from an empty RX-FIFO. Cleared by issuing CHANNEL_COMMAND = CLEAR_RX_RD_ERROR_IRQ (UART_DM_CR register).
13	TX_WR_ERROR_IRQ	TX-FIFO full write error. Fired when AHB master is trying to write into a full TX-FIFO. Cleared by issuing CHANNEL_COMMAND = CLEAR_TX_WR_ERROR_IRQ (UART_DM_CR register).

BLSP1_BLSP_UART0_UART_DM_ISR (cont.)

Bits	Name	Description
12	PAR_FRAME_ERR_IRQ	This bit is asserted in the same condition at which PAR_FRAME_ERR status bit is asserted (See UART_DM_SR register). Cleared by issuing CHANNEL_COMMAND = RESET_PAR_FRAME_ERR_IRQ (UART_DM_CR register).
11	RXBREAK_END	This bit is set (1) if the end of a break condition is detected. Cleared by issuing CHANNEL_COMMAND = RESET_BREAK_END_IRQ (UART_DM_CR register).
10	RXBREAK_START	This bit is set (1) if the start of a break condition is detected. Cleared by issuing CHANNEL_COMMAND = RESET_BREAK_START_IRQ (UART_DM_CR register).
9	TX_DONE	This bit is used only in UIM/SIM_IF mode. Set (1) this bit when the last byte in the transmitter has been sent. Cleared by issuing CHANNEL_COMMAND = CLEAR_TX_DONE_IRQ (UART_DM_CR register). - This bit is generated only when SIM_GLUE_GEN generic equals 1.
8	TX_ERROR	Only used in UIM_IF mode. Set (1) this bit to indicate that there has been parity error during transmission. Cleared by issuing CHANNEL_COMMAND = CLEAR_TX_ERROR_IRQ (UART_DM_CR register). - This bit is generated only when SIM_GLUE_GEN generic equals 1.
7	TX_READY	This bit, when set(1), indicates that: 1. TX FIFO is empty. 2. The value written in NO_CHARS_FOR_TX register equals the number of bytes written to the TX FIFO since the register was updated. Note: There may be characters in the unpack buffer or in the shift register. Cleared by issuing GENERAL_COMMAND = RESET_TX_READY_IRQ (UART_DM_CR register)
6	CURRENT_CTS	This bit indicates the current state of the CTS input, high ('1') meaning peer is not ready for data reception, low ('0') - peer is ready. The toggling of this bit does not generate an interrupt.
5	DELTA_CTS	This bit, when set (1), indicates that the CTS input has changed states. Cleared by issuing CHANNEL_COMMAND = RESET_CTS_N (UART_DM_CR register).
4	RXLEV	This bit is set when a word is loaded into the receive FIFO that brings the total number of words in the FIFO above the programmed watermark level in the FIFO watermark register (UART_DM_RFWR). This bit is cleared after enough words have been read to bring the level equal to or below the programmed watermark level.
3	RXSTALE	This bit indicates that an RX-transfer has finished (See UART_DM HPG for more information). Cleared by issuing CHANNEL_COMMAND = RESET_STALE_IRQ (UART_DM_CR register).

BLSP1_BLSP_UART0_UART_DM_ISR (cont.)

Bits	Name	Description
2	RXBREAK_CHANGE	This bit is set (1) if the start or end of a break condition is detected. Cleared by issuing CHANNEL_COMMAND = RESET_BREAK_CHANGE_IRQ (UART_DM_CR register). A detected break is defined as an all-zeros character with an invalid stop bit AND LOW PARITY. The end of a break is defined as a mark condition (1) at least 1/2 bit width. A detected break occupies one position in the Rx FIFO.
1	RXHUNT	This bit is set (1) when an incoming character matches the value programmed into the UART_DM_HCR register. Cleared by issuing CHANNEL_COMMAND = RESET_ERROR_STATUS (UART_DM_CR register).
0	TXLEV	This bit is set (1) when a word which is transferred from the transmit FIFO to the transmit shift register brings the total number of words in the FIFO below or equal to the programmed watermark level in the UART_DM_TFWR register. This bit is cleared (0) after enough words have been written to the FIFO to bring the level above the programmed watermark level. Since this bit does not have a Clear command, during interrupt handling this bit must be masked (using the matching bit in UART_DM_IMR) until writing additional data for transmission is required. After data is written, this interrupt-bit should be unmasked.

0x078AF0B8 BLSP1_BLSP_UART0_UART_DM_IRDA**Type:** RW**Clock:** AHB_CLK**Reset State:** 0x00000000

The UART_DM_IRDA register enables the IRDA function, selects a loopback, and allows inversion of RX and TX data. This register also controls the IRDA transceiver that optionally interfaces between the UART and the DP_RX_DATA and DP_TX_DATA pins.

The UART proper is a standard RX/TX configuration that converts the serial pin input and output lines to 8-bit data for the microprocessor. The configuration contains receive and transmit blocks, a control block, a bit rate generator for RX and TX, and FIFO interfaces with the microprocessor data bus. The IRDA register provides the means of switching IRDA circuits into the serial I/O lines and the IRDA function is normally switched out. The IRDA modulator/demodulator may be enabled or disabled under microprocessor control. When enabled, the RX and TX data paths have individual inversion select bits, so external true or inverted drivers may be used in the signal path.

* The register is generated only when IRDA_IFC_GEN generic equals 1.

BLSP1_BLSP_UART0_UART_DM_IRDA

Bits	Name	Description
4	MEDIUM_RATE_EN	- Set (1) for 1/4 bit-time pulse length (Medium rate) - Clear (0) for 3/16 bit-time pulse length (Low rate)

BLSP1_BLSP_UART0_UART_DM_IRDA (cont.)

Bits	Name	Description
3	IRDA_LOOPBACK	- This bit loops the IRDA modulated TX line back into the IRDA RX line. The normal UART loopback mode must be off in order to see this effect.
2	INVERT_IRDA_TX	This bit inverts the polarity of the IRDA modulated signal on the DP_TX_DATA pin. - Set (1) this bit for an inverted polarity. - Clear (0) this bit for a non-inverted polarity.
1	INVERT_IRDA_RX	This bit inverts the polarity of the IRDA received signal on the DP_RX_DATA pin. - Set (1) this bit for inverted the polarity. - Clear (0) this bit for non-inverted polarity.
0	IRDA_EN	- Set (1) this bit to enable the IRDA transceiver. - Clear (0) this bit to bypass the IRDA transceiver and ignore the other bits of this register.

0x078AF0BC BLSP1_BLSP_UART0_UART_DM_RX_TOTAL_SNAP**Type:** R**Clock:** AHB_CLK**Reset State:** 0x00000000**BLSP1_BLSP_UART0_UART_DM_RX_TOTAL_SNAP**

Bits	Name	Description
23:0	RX_TOTAL_BYTES	?RX_TOTAL_SNAP register holds the number of characters received since the end of last Rx transfer. It is updated at the end of an Rx transfer (see UART_DM HPG for more information). Rx transfer ends when one of the conditions is met: - The number of characters which were received since the end of the previous transfer equals the value which was written to DMRX at ?Transfer initialization?. - A stale event occurred (flush operation already performed if was needed)

0x078AF0C0 BLSP1_BLSP_UART0_UART_DM_WWT_TIMEOUT**Type:** RW**Clock:** AHB_CLK**Reset State:** 0x00000000**BLSP1_BLSP_UART0_UART_DM_WWT_TIMEOUT**

Bits	Name	Description
24:0	WWT_TIMEOUT	timeout value for WWT mechanism.

0x078AF0C4 BLSP1_BLSP_UART0_UART_DM_CLK_CTRL**Type:** RW**Clock:** AHB_CLK**Reset State:** 0x00000000**BLSP1_BLSP_UART0_UART_DM_CLK_CTRL**

Bits	Name	Description
23	UART_IRDA_CLK_CGC_OPEN	Forces UART IRDA-clock (uart_tx_clk) CGC to open when set.
22	UART_SIM_CLK_CGC_OPEN	Forces UART SIM-clock (uart_tx_clk) CGC to open when set.
21	UART_RX_CLK_CGC_OPEN	Forces UART RX-clock (uart_tx_clk) CGC to open when set.
20	UART_TX_CLK_CGC_OPEN	Forces UART TX-clock (uart_tx_clk) CGC to open when set.
19	RESERVED_1	
18	RESERVED_2	
17	RESERVED_3	
16	RESERVED_4	
15	RESERVED_5	
14	AHB_RX_BAM_CLK_CGC_OPEN	Forces AHB RX-BAM-clock (hrx_bam_clk) CGC to open when set.
13	AHB_TX_BAM_CLK_CGC_OPEN	Forces AHB TX-BAM-clock (htx_bam_clk) CGC to open when set.
12	RESERVED_6	
11	RESERVED_7	
10	AHB_RX_CLK_CGC_OPEN	Forces AHB RX-clock (hrx_clk) CGC to open when set.
9	AHB_TX_CLK_CGC_OPEN	Forces AHB TX-clock (htx_clk) CGC to open when set.
8	AHB_WR_CLK_CGC_OPEN	Forces AHB write-clock (hwr_clk) CGC to open when set.
7	RESERVED_8	
6	RESERVED_9	
5	RX_ENABLE_CGC_OPT	CGCs will gate AHB & UART clocks due to RX-DISABLE command/RX-RESET command only when set.
4	TX_ENABLE_CGC_OPT	CGCs will gate AHB & UART clocks due to TX-DISABLE command/TX-RESET command only when set.
3	RESERVED_10	
2	RESERVED_11	
1	RESERVED_12	

BLSP1_BLSP_UART0_UART_DM_CLK_CTRL (cont.)

Bits	Name	Description
0	AHB_CLK_CGC_CLOSE	Setting this bit to '1' will completely gate AHB-clock to the core. Only allowed register acces during the duration this bit is set is the access to reset this bit. All other register accesses are not allowed and might result in errors. When set to '0', AHB clock is not gated at the root of the core and can be gated by the other gating features.

0x078AF0C8 BLSP1_BLSP_UART0_UART_DM_BCR**Type:** RW**Clock:** AHB_CLK**Reset State:** 0x00000000**BLSP1_BLSP_UART0_UART_DM_BCR**

Bits	Name	Description
5	RX_DMRX_1BYTE_RES_EN	Allows DMRX values other than a multiple of 16 if set.
4	RX_STALE_IRQ_DMRX_EQUAL	If set, Stale irq will be asserted when (DMRX==valid_char_count) and stale is not enabled.
3	RESERVED	
2	RX_DMRX_LOW_EN	DMRX lower than valid_char_cnt logic is enabled if bit is set
1	STALE_IRQ_EMPTY	Stale interrupt when RX-FIFO is empty will fire if bit is set.
0	TX_BREAK_DISABLE	TX-pin value when transmitter disabled while in break state is '0' if set, else '1'

0x078AF0CC BLSP1_BLSP_UART0_UART_DM_RX_TRANS_CTRL**Type:** RW**Clock:** AHB_CLK**Reset State:** 0x00000000**BLSP1_BLSP_UART0_UART_DM_RX_TRANS_CTRL**

Bits	Name	Description
2	RX_DMRX_CYCLIC_EN	If set - Auto-Re-Activated RX-transfer will be re-activated with a DMRX value from the DMRX register, making a DMRX-transfer-end-event possible. Otherwise, Auto-Re-Activated RX-transfer cannot end with a DMRX event unless its value is written to the DMRX register throughout the transfer.
1	RX_TRANS_AUTO_RE_ACTIVATE	RX-transfer will be automatically re-activated after last data of previous transfer was read - if set. Otherwise, transfers are only activated by writing to DMRX register.

BLSP1_BLSP_UART0_UART_DM_RX_TRANS_CTRL (cont.)

Bits	Name	Description
0	RX_STALE_AUTO_RE_EN	RX-Stale automatic re-enable - on if set.

0x078AF0D4 BLSP1_BLSP_UART0_UART_DM_FSM_STATUS**Type:** R**Clock:** AHB_CLK**Reset State:** 0x00000000**BLSP1_BLSP_UART0_UART_DM_FSM_STATUS**

Bits	Name	Description
29:28	TX_COMP_FSM	Value of TX-COMP FSM state: '00' for empty, '01' non-empty, '10' for empty-check, '11' for empty-check-fail.
27	RESERVED_1	
26:24	RX_PACK_FSM	Value of RX-packing FSM state: '000' for idle, '001' for fifo_write, '010' for wait_ack, '011' for flush_check, '100' for flushing, '101' for flush_done.
22:23	RESERVED_2	
21:20	RX_TRANS_FSM	Value of RX-transfer FSM state: '00' for inactive, '01' for active, '10' for active_dmr, '11' for active_flush.
19	RESERVED_3	
18:16	TX_TRANS_FSM	Value of TX-transfer FSM state: '000' for idle, '001' for bam_idle, '010' for active, '011' for inactive, '100' for last_trap, '101' for nwd.
15	RESERVED_4	
14:12	RX_PRO_TRANS_END_FSM	Value of RX_PRO_TRANS_END FSM state: '000' for disabled, '001' for idle, '010' for 2char_dphase, '011' for 1char_waiting, '100' for 1char dphase, '101' for mess_only_dphase.
11	RESERVED_5	
10:8	RX_PRO_ACTIVE_FSM	Value of RX_PRO_ACTIVE FSM state: '000' for disabled, '001' for isactive, '010' for wait_check_1, '011' for wait_check_1, '100' for pf_pending, '101' for wait_update_1, '110' for wait_update_2, '111' for inactive?
7	RESERVED_6	
6:4	TX_CON_TRANS_END_FSM	Value of TX_CON_TRANS_END FSM state: '000' for disabled, '001' for idle, '010' for 2char_dphase, '011' for 1char_waiting, '100' for 1char dphase, '101' for mess_only_dphase, '110' for normal_dphase.
3:1	RESERVED_7	
0	RX_TRANSFER_ACTIVE	1 is RX transfer is currently active, 0 if inactive.

0x078AF0DC BLSP1_BLSP_UART0_UART_DM_GENERICS

Type: R
Clock: AHB_CLK
Reset State: 0x00000000

BLSP1_BLSP_UART0_UART_DM_GENERICS

Bits	Name	Description
7	GENERIC_BAM_IFC	Value of 'BAM_IFC_GEN' generic.
6	GENERIC_DM_IFC	Value of 'DM_IFC_GEN' generic.
5	GENERIC_IRDA_IFC	Value of 'IRDA_IFC_GEN' generic
4	GENERIC_SIM_GLUE	Value of 'SIM_GLUE_GEN' generic
3:0	GENERIC_RAM_ADDR_WIDTH	Value of 'RAM_ADDR_WIDTH' generic.

0x078AF100 BLSP1_BLSP_UART0_UART_DM_TF

Type: W
Clock: AHB_CLK
Reset State: Undefined

In order to support bursts of 4, 0x70-0x7C address space is reserved for the UART_DM TX FIFO.

BLSP1_BLSP_UART0_UART_DM_TF

Bits	Name	Description
31:0	UART_TF	The UART_TF register is the UART transmit FIFO register. This register is used to access the channel transmit FIFO. If the FIFO is full at the time of writing, the characters are lost and an interrupt is generated (see UART_DM_ISR register).

0x078AF104 BLSP1_BLSP_UART0_UART_DM_TF_2

Type: W
Clock: AHB_CLK
Reset State: Undefined

BLSP1_BLSP_UART0_UART_DM_TF_2

Bits	Name	Description
31:0	UART_TF	See UART_DM_TF register.

0x078AF108 BLSP1_BLSP_UART0_UART_DM_TF_3**Type:** W**Clock:** AHB_CLK**Reset State:** Undefined**BLSP1_BLSP_UART0_UART_DM_TF_3**

Bits	Name	Description
31:0	UART_TF	See UART_DM_TF register.

0x078AF10C BLSP1_BLSP_UART0_UART_DM_TF_4**Type:** W**Clock:** AHB_CLK**Reset State:** Undefined**BLSP1_BLSP_UART0_UART_DM_TF_4**

Bits	Name	Description
31:0	UART_TF	See UART_DM_TF register.

0x078AF110 BLSP1_BLSP_UART0_UART_DM_TF_5**Type:** W**Clock:** AHB_CLK**Reset State:** Undefined**BLSP1_BLSP_UART0_UART_DM_TF_5**

Bits	Name	Description
31:0	UART_TF	See UART_DM_TF register.

0x078AF114 BLSP1_BLSP_UART0_UART_DM_TF_6**Type:** W**Clock:** AHB_CLK**Reset State:** Undefined**BLSP1_BLSP_UART0_UART_DM_TF_6**

Bits	Name	Description
31:0	UART_TF	See UART_DM_TF register.

0x078AF118 BLSP1_BLSP_UART0_UART_DM_TF_7**Type:** W**Clock:** AHB_CLK**Reset State:** Undefined**BLSP1_BLSP_UART0_UART_DM_TF_7**

Bits	Name	Description
31:0	UART_TF	See UART_DM_TF register.

0x078AF11C BLSP1_BLSP_UART0_UART_DM_TF_8**Type:** W**Clock:** AHB_CLK**Reset State:** Undefined**BLSP1_BLSP_UART0_UART_DM_TF_8**

Bits	Name	Description
31:0	UART_TF	See UART_DM_TF register.

0x078AF120 BLSP1_BLSP_UART0_UART_DM_TF_9**Type:** W**Clock:** AHB_CLK**Reset State:** Undefined**BLSP1_BLSP_UART0_UART_DM_TF_9**

Bits	Name	Description
31:0	UART_TF	See UART_DM_TF register.

0x078AF124 BLSP1_BLSP_UART0_UART_DM_TF_10**Type:** W**Clock:** AHB_CLK**Reset State:** Undefined**BLSP1_BLSP_UART0_UART_DM_TF_10**

Bits	Name	Description
31:0	UART_TF	See UART_DM_TF register.

0x078AF128 BLSP1_BLSP_UART0_UART_DM_TF_11

Type: W
Clock: AHB_CLK
Reset State: Undefined

BLSP1_BLSP_UART0_UART_DM_TF_11

Bits	Name	Description
31:0	UART_TF	See UART_DM_TF register.

0x078AF12C BLSP1_BLSP_UART0_UART_DM_TF_12

Type: W
Clock: AHB_CLK
Reset State: Undefined

BLSP1_BLSP_UART0_UART_DM_TF_12

Bits	Name	Description
31:0	UART_TF	See UART_DM_TF register.

0x078AF130 BLSP1_BLSP_UART0_UART_DM_TF_13

Type: W
Clock: AHB_CLK
Reset State: Undefined

BLSP1_BLSP_UART0_UART_DM_TF_13

Bits	Name	Description
31:0	UART_TF	See UART_DM_TF register.

0x078AF134 BLSP1_BLSP_UART0_UART_DM_TF_14

Type: W
Clock: AHB_CLK
Reset State: Undefined

BLSP1_BLSP_UART0_UART_DM_TF_14

Bits	Name	Description
31:0	UART_TF	See UART_DM_TF register.

0x078AF138 BLSP1_BLSP_UART0_UART_DM_TF_15**Type:** W**Clock:** AHB_CLK**Reset State:** Undefined**BLSP1_BLSP_UART0_UART_DM_TF_15**

Bits	Name	Description
31:0	UART_TF	See UART_DM_TF register.

0x078AF13C BLSP1_BLSP_UART0_UART_DM_TF_16**Type:** W**Clock:** AHB_CLK**Reset State:** Undefined**BLSP1_BLSP_UART0_UART_DM_TF_16**

Bits	Name	Description
31:0	UART_TF	See UART_DM_TF register.

0x078AF140 BLSP1_BLSP_UART0_UART_DM_RF**Type:** R**Clock:** AHB_CLK**Reset State:** Undefined

The UART_DM_RF register is the UART receive FIFO register, which is used to access the channel receive FIFO. In order to support bursts of 16, 0x140-0x17C address space is reserved for the UART_DM RX FIFO.

BLSP1_BLSP_UART0_UART_DM_RF

Bits	Name	Description
31:0	UART_RF	This register returns the next word in the receive FIFO. After the read is completed, the FIFO read pointer is updated to make the next word available.

0x078AF144 BLSP1_BLSP_UART0_UART_DM_RF_2**Type:** R**Clock:** AHB_CLK**Reset State:** Undefined

BLSP1_BLSP_UART0_UART_DM_RF_2

Bits	Name	Description
31:0	UART_RF	see UART_DM_RF register.

0x078AF148 BLSP1_BLSP_UART0_UART_DM_RF_3**Type:** R**Clock:** AHB_CLK**Reset State:** Undefined**BLSP1_BLSP_UART0_UART_DM_RF_3**

Bits	Name	Description
31:0	UART_RF	see UART_DM_RF register.

0x078AF14C BLSP1_BLSP_UART0_UART_DM_RF_4**Type:** R**Clock:** AHB_CLK**Reset State:** Undefined**BLSP1_BLSP_UART0_UART_DM_RF_4**

Bits	Name	Description
31:0	UART_RF	see UART_DM_RF register.

0x078AF150 BLSP1_BLSP_UART0_UART_DM_RF_5**Type:** R**Clock:** AHB_CLK**Reset State:** Undefined**BLSP1_BLSP_UART0_UART_DM_RF_5**

Bits	Name	Description
31:0	UART_RF	see UART_DM_RF register.

0x078AF154 BLSP1_BLSP_UART0_UART_DM_RF_6

Type: R
Clock: AHB_CLK
Reset State: Undefined

BLSP1_BLSP_UART0_UART_DM_RF_6

Bits	Name	Description
31:0	UART_RF	see UART_DM_RF register.

0x078AF158 BLSP1_BLSP_UART0_UART_DM_RF_7

Type: R
Clock: AHB_CLK
Reset State: Undefined

BLSP1_BLSP_UART0_UART_DM_RF_7

Bits	Name	Description
31:0	UART_RF	see UART_DM_RF register.

0x078AF15C BLSP1_BLSP_UART0_UART_DM_RF_8

Type: R
Clock: AHB_CLK
Reset State: Undefined

BLSP1_BLSP_UART0_UART_DM_RF_8

Bits	Name	Description
31:0	UART_RF	see UART_DM_RF register.

0x078AF160 BLSP1_BLSP_UART0_UART_DM_RF_9

Type: R
Clock: AHB_CLK
Reset State: Undefined

BLSP1_BLSP_UART0_UART_DM_RF_9

Bits	Name	Description
31:0	UART_RF	see UART_DM_RF register.

0x078AF164 BLSP1_BLSP_UART0_UART_DM_RF_10

Type: R
Clock: AHB_CLK
Reset State: Undefined

BLSP1_BLSP_UART0_UART_DM_RF_10

Bits	Name	Description
31:0	UART_RF	see UART_DM_RF register.

0x078AF168 BLSP1_BLSP_UART0_UART_DM_RF_11

Type: R
Clock: AHB_CLK
Reset State: Undefined

BLSP1_BLSP_UART0_UART_DM_RF_11

Bits	Name	Description
31:0	UART_RF	see UART_DM_RF register.

0x078AF16C BLSP1_BLSP_UART0_UART_DM_RF_12

Type: R
Clock: AHB_CLK
Reset State: Undefined

BLSP1_BLSP_UART0_UART_DM_RF_12

Bits	Name	Description
31:0	UART_RF	see UART_DM_RF register.

0x078AF170 BLSP1_BLSP_UART0_UART_DM_RF_13

Type: R
Clock: AHB_CLK
Reset State: Undefined

BLSP1_BLSP_UART0_UART_DM_RF_13

Bits	Name	Description
31:0	UART_RF	see UART_DM_RF register.

0x078AF174 BLSP1_BLSP_UART0_UART_DM_RF_14

Type: R
Clock: AHB_CLK
Reset State: Undefined

BLSP1_BLSP_UART0_UART_DM_RF_14

Bits	Name	Description
31:0	UART_RF	see UART_DM_RF register.

0x078AF178 BLSP1_BLSP_UART0_UART_DM_RF_15

Type: R
Clock: AHB_CLK
Reset State: Undefined

BLSP1_BLSP_UART0_UART_DM_RF_15

Bits	Name	Description
31:0	UART_RF	see UART_DM_RF register.

0x078AF17C BLSP1_BLSP_UART0_UART_DM_RF_16

Type: R
Clock: AHB_CLK
Reset State: Undefined

BLSP1_BLSP_UART0_UART_DM_RF_16

Bits	Name	Description
31:0	UART_RF	see UART_DM_RF register.

0x078AF180 BLSP1_BLSP_UART0_UART_DM_UIM_CFG

Type: RW
Clock: AHB_CLK
Reset State: 0x00001703

Configuration register for UIM-controller (linked with this UART_DM). Reset value relevant to UIM-controller HW-reset and not a UART_DM HW-reset. Writing to this register requires waiting until UIM-WRITE is done (indication by status-bit or interrupt) before attempting an additional write/read one of UIM_CFG/UIM_CMD registers.

BLSP1_BLSP_UART0_UART_DM_UIM_CFG

Bits	Name	Description
15	BATT_ALARM_QUICK_DROP_EN	When set, deactivation sequence resulted from battery-alarm will drop all UIM-lines immediately. Otherwise, lines will be dropped gradually.
14	TESTBUS_EN	When set, UIM-controller will output testbus signals on the testbus port. Otherwise, testbus port will be driven with zeros.
13	SW_RESET	When set, UIM-controller will be at reset state.
12	MODE18	Configures PAD's working reference voltage (set to '0' for 3V, '1' for 1.8V).
11	RESERVED_1	
10	PMIC_ALARM_EN	Enables messaging on the alarm-line to notify PMIC to drop card's voltage supply. Set to enable messaging, clear to disable.
9	BATT_ALARM_TRIGGER_EN	Enables triggering of deactivation sequence when battery-alarm detected. Set to enable triggering, clear to disable.
8	UIM_RMV_TRIGGER_EN	Enables triggering of deactivation sequence when card-removal detected. Set to enable triggering, clear to disable.
7	RESERVED_2	
6	UIM_CARD_EVENTS_ENABLE	When cleared, UIM-controller will not react to any card-event (insertion/removal). When set, UIM-controller reacts to these events and considered active. This bit should be set only after/during UIM_PRESENT_POLARITY bit is configured so UIM-controller will not interpret card-events incorrectly due to a wrong polarity configuration (the reset-value of the polarity might not match the actual one).
5	UIM_PRESENT_POLARITY	Determines value interpretation of uim_present signal. Setting this bit to '1' indicates that a uim-card present in the uim-slot will return the value '1' on uim_present line & '0' when a card is not present. Setting this bit to '0' indicates that a uim-card present in the uim-slot will return the value '0' on uim_present line & '1' when a card is not present.
4:0	EVENT_DEBOUNCE_TIME	Configures debounce-time of card-removal/insertion events (in sleep-clk cycles).

0x078AF184 BLSP1_BLSP_UART0_UART_DM_UIM_CMD**Type:** W**Clock:** AHB_CLK**Reset State:** Undefined

Command register for UIM-controller (linked with this UART_DM). Writing to this register requires waiting until UIM-WRITE is done (indication by status-bit or interrupt) before attempting an additional write/read one of UIM_CFG/UIM_CMD registers.

BLSP1_BLSP_UART0_UART_DM_UIM_CMD

Bits	Name	Description
1	RECOVER_FROM_HW_DEACTIVATION	Writing '1' to this bit will cause the UIM-io-ctrl to exit override state and reflect UIM signals from UART_DM to the UIM card. Setting both bits of this register to '1' will cause neither of the commands to be executed.
0	INITIATE_HW_DEACTIVATION	Initiates a HW deactivation sequence when writing '1' to this bit. This executes the sequence on the uim_data, uim_clk & uim_rst_n lines and alerts the PMIC to cut the power-supply to the card afterwards. Setting both bits of this register to '1' will cause neither of the commands to be executed.

0x078AF188 BLSP1_BLSP_UART0_UART_DM_UIM_IO_STATUS**Type:** R**Clock:** AHB_CLK**Reset State:** 0x00000002**BLSP1_BLSP_UART0_UART_DM_UIM_IO_STATUS**

Bits	Name	Description
2	UIM_IO_WRITE_IN_PROGRESS	Asserted after UIM-write (UIM_CFG/UIM_CMD register write) while write is being propagated into the UIM-IO-controller. An additional read/write to UIM_CFG/UIM_CMD register should be done while high. De-asserted when write finishes propagating.
1	UIM_DEACTIVATION_STATUS	Indicates whether a HW deactivation sequence was performed and sequence recovery hasn't been executed yet. This bit is set when the sequence is initiated either by a HW trigger (batt_alarm, card-removal) or a SW trigger ('initiate_shutdown_sequence' bit in CMD register). Cleared after executing a 'recover_from_hw_deactivation' command.
0	CARD_PRESENCE	Indication on card presence in slot. High when card present, low otherwise.

0x078AF18C BLSP1_BLSP_UART0_UART_DM_UIM_IRQ_ISR**Type:** R**Clock:** AHB_CLK**Reset State:** 0x00000000**BLSP1_BLSP_UART0_UART_DM_UIM_IRQ_ISR**

Bits	Name	Description
4	UIM_IO_WRITE_DONE	Asserted when UIM-write had finished. Cleared by writing '1' to matching field in UART_DM_UIM_IRQ_CLR.

BLSP1_BLSP_UART0_UART_DM_UIM_IRQ_ISR (cont.)

Bits	Name	Description
3	HW_SEQUENCE_FINISH	Asserted when UIM-controller's HW-deactivation sequence has been initiated and finished. Cleared by writing '1' to matching field in UART_DM_UIM_IRQ_CLR.
2	BATT_ALARM	Set to high when battery-alarm indication received. Cleared by writing '1' to matching field in UART_DM_UIM_IRQ_CLR.
1	UIM_CARD_INSERTION	Set to high when card insertion indication received. Cleared by writing '1' to matching field in UART_DM_UIM_IRQ_CLR.
0	UIM_CARD_REMOVAL	Set to high when card removal indication received. Cleared by writing '1' to matching field in UART_DM_UIM_IRQ_CLR.

0x078AF190 BLSP1_BLSP_UART0_UART_DM_UIM_IRQ_MISR**Type:** R**Clock:** AHB_CLK**Reset State:** 0x00000000**BLSP1_BLSP_UART0_UART_DM_UIM_IRQ_MISR**

Bits	Name	Description
4	UIM_IO_WRITE_DONE	See UART_DM_UIM_IRQ_ISR for field descriptions.
3	HW_SEQUENCE_FINISH	See UART_DM_UIM_IRQ_ISR for field descriptions.
2	BATT_ALARM	See UART_DM_UIM_IRQ_ISR for field descriptions.
1	UIM_CARD_INSERTION	See UART_DM_UIM_IRQ_ISR for field descriptions.
0	UIM_CARD_REMOVAL	See UART_DM_UIM_IRQ_ISR for field descriptions.

0x078AF194 BLSP1_BLSP_UART0_UART_DM_UIM_IRQ_CLR**Type:** W**Clock:** AHB_CLK**Reset State:** Undefined**BLSP1_BLSP_UART0_UART_DM_UIM_IRQ_CLR**

Bits	Name	Description
4	UIM_IO_WRITE_DONE	Clears matching bit in UART_DM_UIM_IRQ_ISR/MISR.
3	HW_SEQUENCE_FINISH	Clears matching bit in UART_DM_UIM_IRQ_ISR/MISR.
2	BATT_ALARM	Clears matching bit in UART_DM_UIM_IRQ_ISR/MISR.
1	UIM_CARD_INSERTION	Clears matching bit in UART_DM_UIM_IRQ_ISR/MISR.
0	UIM_CARD_REMOVAL	Clears matching bit in UART_DM_UIM_IRQ_ISR/MISR.

0x078AF198 BLSP1_BLSP_UART0_UART_DM_UIM_IRQ_IMR**Type:** RW**Clock:** AHB_CLK**Reset State:** 0x00000000**BLSP1_BLSP_UART0_UART_DM_UIM_IRQ_IMR**

Bits	Name	Description
4	UIM_IO_WRITE_DONE	When set, matching bit in UART_DM_UIM_IRQ_MISR will reflect the matching bit in UART_DM_UIM_IRQ_ISR. Otherwise, the matching bit in UART_DM_UIM_IRQ_MISR will be a constant '0'.
3	HW_SEQUENCE_FINISH	When set, matching bit in UART_DM_UIM_IRQ_MISR will reflect the matching bit in UART_DM_UIM_IRQ_ISR. Otherwise, the matching bit in UART_DM_UIM_IRQ_MISR will be a constant '0'.
2	BATT_ALARM	When set, matching bit in UART_DM_UIM_IRQ_MISR will reflect the matching bit in UART_DM_UIM_IRQ_ISR. Otherwise, the matching bit in UART_DM_UIM_IRQ_MISR will be a constant '0'.
1	UIM_CARD_INSERTION	When set, matching bit in UART_DM_UIM_IRQ_MISR will reflect the matching bit in UART_DM_UIM_IRQ_ISR. Otherwise, the matching bit in UART_DM_UIM_IRQ_MISR will be a constant '0'.
0	UIM_CARD_REMOVAL	When set, matching bit in UART_DM_UIM_IRQ_MISR will reflect the matching bit in UART_DM_UIM_IRQ_ISR. Otherwise, the matching bit in UART_DM_UIM_IRQ_MISR will be a constant '0'.

0x078AF19C BLSP1_BLSP_UART0_UART_DM_UIM_IRQ_IMR_SET**Type:** W**Clock:** AHB_CLK**Reset State:** Undefined**BLSP1_BLSP_UART0_UART_DM_UIM_IRQ_IMR_SET**

Bits	Name	Description
4	UIM_IO_WRITE_DONE	Write '1' to set the matching bit in UART_DM_UIM_IRQ_IMR.
3	HW_SEQUENCE_FINISH	Write '1' to set the matching bit in UART_DM_UIM_IRQ_IMR.
2	BATT_ALARM	Write '1' to set the matching bit in UART_DM_UIM_IRQ_IMR.
1	UIM_CARD_INSERTION	Write '1' to set the matching bit in UART_DM_UIM_IRQ_IMR.
0	UIM_CARD_REMOVAL	Write '1' to set the matching bit in UART_DM_UIM_IRQ_IMR.

0x078AF1A0 BLSP1_BLSP_UART0_UART_DM_UIM_IRQ_IMR_CLR**Type:** W**Clock:** AHB_CLK**Reset State:** Undefined**BLSP1_BLSP_UART0_UART_DM_UIM_IRQ_IMR_CLR**

Bits	Name	Description
4	UIM_IO_WRITE_DONE	Write '1' to clear the matching bit in UART_DM_UIM_IRQ_IMR.
3	HW_SEQUENCE_FINISH	Write '1' to clear the matching bit in UART_DM_UIM_IRQ_IMR.
2	BATT_ALARM	Write '1' to clear the matching bit in UART_DM_UIM_IRQ_IMR.
1	UIM_CARD_INSERTION	Write '1' to clear the matching bit in UART_DM_UIM_IRQ_IMR.
0	UIM_CARD_REMOVAL	Write '1' to clear the matching bit in UART_DM_UIM_IRQ_IMR.

20.3 BLSP1_BLSP_UART1_UART_DM**0x078B0000 BLSP1_BLSP_UART1_UART_DM_MR1****Type:** RW**Clock:** AHB_CLK**Reset State:** 0x00000000

The UART_DM_MR1 register is the UART mode register #1. It is used, along with UART_DM_MR2, to configure the operational mode of the UART.

BLSP1_BLSP_UART1_UART_DM_MR1

Bits	Name	Description
31:8	AUTO_RFR_LEVEL1	These bits are used, along with field AUTO_RFR_LEVEL0 to program the level in the receive FIFO at which the Ready-For-Receive signal (RFR_N) toggles, if programmed to do so (see RX_RDY_CTL field of this register). When RX-FIFO level is greater than the value in fields AUTO_RFR_LEVEL1 and AUTO_RFR_LEVEL0, signal will indicate Not-Ready (RFR_N high). Otherwise signal will indicate Ready RFR_N low. Configured value should be higher/equal than RFRW value, otherwise the watermark will never be reached. Value of this register is in words and can be programmed from 1 to 2^{RAM_ADDR_WIDTH}. - Only RAM_ADDR_WIDTH + 1:8 bits of this field are generated.

BLSP1_BLSP_UART1_UART_DM_MR1 (cont.)

Bits	Name	Description
7	RX_RDY_CTL	Setting (1) this bit enables the automatic ready-for-receiving (RFR_N) control for the receiver. RFR_N is turned off, or set (1), when a valid start bit is received and the RX channel FIFO is at the level programmed in bits [31:8,5:0] of this register. When the FIFO level falls back to the programmed level, the RFR_N signal is turned on, or clear (0). When this feature is off, the RFR_N signal can be used by normal signal port bit manipulation (see UART_DM_CR register).
6	CTS_CTL	When this bit is set(1), the transmitter checks the CTS_N input to determine whether to begin transmission of a new character. If CTS_N is low, the character is sent. Otherwise the transmitter continues marking until CTS_N goes low, then the next character is transmitted. A change on CTS_N during the transmission of a character has no effect on that character. When this bit is clear(0), the CTS_N input for the channel has no effect on the transmitter.
5:0	AUTO_RFR_LEVEL0	See the description of field AUTO_RFR_LEVEL1.

0x078B0004 BLSP1_BLSP_UART1_UART_DM_MR2**Type:** RW**Clock:** AHB_CLK**Reset State:** 0x00000000

See description of UART_DM_MR1 register.

BLSP1_BLSP_UART1_UART_DM_MR2

Bits	Name	Description
10	RFR_CTS_LOOPBACK	Enables internal loopback between RFR_N of RX channel and CTS_N of TX channel if set. For this feature to be enabled, data loopback must also be enabled (LOOPBACK bit of this register).
9	RX_ERROR_CHAR_OFF	When this bit is set, characters received with parity or framing errors get discarded and not moved to the RX channel. Otherwise they enter RX channel as any valid character.
8	RX_BREAK_ZERO_CHAR_OFF	When this bit is set, the zero character received at rx_break event get discarded and not moved to the RX channel. Otherwise it enters RX-channel as any valid character.
7	LOOPBACK	When set - data output of transmitter is fed back to the data input of the receiver, i.e. TX-pin is connected to RX-pin. Might be used with flow-control loopback (RFR_CTS_LOOPBACK bit above).

BLSP1_BLSP_UART1_UART_DM_MR2 (cont.)

Bits	Name	Description
6	ERROR_MODE	This bit controls the operation of the two status register bits for the RX channel (parity or framing error and received break, see UART_DM_SR register). - When clear (0), the UART operates in character mode and the status bits apply only to the last character received. - When set (1), the UART operates in block mode and both bits are sticky, i.e. they go high once an error/break-event is detected and go low only after reset error status command is issued (see UART_DM_CR register).
5:4	BITS_PER_CHAR	These bits determine how many data bits are transmitted or received per character, not including the start, stop, and parity bits. 0x0: ENUM_5_BITS (5 bits) 0x1: ENUM_6_BITS (6 bits) 0x2: ENUM_7_BITS (7 bits) 0x3: ENUM_8_BITS (8 bits)
3:2	STOP_BIT_LEN	This field programs the duration of the stop bit that is appended to each transmitted character. The stop bit duration can be 9/16, 1, 1+ 9/16, and 2 bit times. The receiver only samples the stop bit once, regardless of the programmed stop bit length. This configuration only applies to UART/IrDA interfaces, for SIM interface see SIM_STOP_BIT_LEN field of UART_DM_SIM_CFG register. 0x0: ENUM_0_563 (0.563 (9/16) bit times) 0x1: ENUM_1_000_BIT_TIME (1.000 bit time) 0x2: ENUM_1_563 (1.563 (1+9/16) bit times) 0x3: ENUM_2_000_BIT_TIMES (2.000 bit times)
1:0	PARITY_MODE	These bits determine which parity mode is used. The user can select between odd, even, space, or no parity. 0x0: NO_PARITY (no parity - last data-bit is followed by a stop-bit) 0x1: ODD_PARITY (odd parity - parity-bit ensures number of ones in data-bits+parity-bit is odd) 0x2: EVEN_PARITY (even parity - parity-bit ensures number of ones in data-bits+parity-bit is even) 0x3: SPACE_PARITY (space parity - parity is always '0')

0x078B0008 BLSP1_BLSP_UART1_UART_DM_CSR_SR_DEPRECATED**Type:** RW**Clock:** AHB_CLK**Reset State:** Undefined

This register is deprecated and remains only for backward-compatibility. Use UART_DM_CSR or UART_DM_SR.

BLSP1_BLSP_UART1_UART_DM_CSR_SR_DEPRECATED

Bits	Name	Description
31:0	UART_DM_CSR_SR_DEPRECATED	

0x078B0010 BLSP1_BLSP_UART1_UART_DM_CR_MISR_DEPRECATED**Type:** RW**Clock:** AHB_CLK**Reset State:** Undefined

This register is deprecated and remains only for backward-compatibility. Use UART_DM_CR or UART_DM_MISR.

BLSP1_BLSP_UART1_UART_DM_CR_MISR_DEPRECATED

Bits	Name	Description
31:0	UART_DM_CR_MISR_DEPRECATED	

0x078B0014 BLSP1_BLSP_UART1_UART_DM_IMR_ISR_DEPRECATED**Type:** RW**Clock:** AHB_CLK**Reset State:** Undefined

This register is deprecated and remains only for backward-compatibility. Use UART_DM_IMR or UART_DM_ISR.

BLSP1_BLSP_UART1_UART_DM_IMR_ISR_DEPRECATED

Bits	Name	Description
31:0	UART_DM_IMR_ISR_DEPRECATED	

0x078B0018 BLSP1_BLSP_UART1_UART_DM_IPR**Type:** RW**Clock:** AHB_CLK**Reset State:** 0xFFFFFFFF9F

The UART_DM_IPR register is the UART interrupt programming register.

BLSP1_BLSP_UART1_UART_DM_IPR

Bits	Name	Description
31:7	STALE_TIMEOUT_MSB	The stale character time-out duration field contains a number from 1 to $2^{30} - 1$. This number determines how many character times must elapse before a stale event is generated. This character time is defined as 10 times the bit rate. It is reset whenever a data bit is received. Never set this fields to 0 while RX channel is operational. Do not re-configure while an RX-transfer is active. Note the discontinuity in the bit assignments.
6	SAMPLE_DATA	Currently not in use. Setting (1) to this bit enables the sample data mode. In this mode start bit is sampled along with the data of the UART character. Works only in 'Packet mode' (See the CR register, CHANNEL_COMMAND bit for more information).
5	RESERVED	
4:0	STALE_TIMEOUT_LSB	This bitfield is the LSbits of the STALE_TIMEOUT bitfield.

0x078B001C BLSP1_BLSP_UART1_UART_DM_TFWR**Type:** RW**Clock:** AHB_CLK**Reset State:** 0x00000000

The UART_DM_TFWR register is the UART transmit FIFO watermark register.

BLSP1_BLSP_UART1_UART_DM_TFWR

Bits	Name	Description
31:0	TFW	These bits contain a number, between 0 and $2^{\text{RAM_ADDR_WIDTH}} - 1$, that determines the level of the transmit (TX) FIFO at which space_avail_req signal is asserted in BAM mode or TXLEV interrupt is asserted in non-BAM mode. The signal/interrupt is asserted when the FIFO level (number of TX words in the FIFO) is less than or equal to the value in TFWR. Notice that the value is in words. See UART_DM_ISR register for TXLEV interrupt description. - Only RAM_ADDR_WIDTH -1:0 bits are generated.

0x078B0020 BLSP1_BLSP_UART1_UART_DM_RFWR**Type:** RW**Clock:** AHB_CLK**Reset State:** Undefined

The UART_DM_RFWR register is the UART receive FIFO watermark register.

BLSP1_BLSP_UART1_UART_DM_RFWR

Bits	Name	Description
31:0	RFW	These bits contain a number, between 0 and $2^{\text{RAM_ADDR_WIDTH}} - 1$, that determines the level of the receive (RX) FIFO at which data_avail_req signal is asserted in BAM mode or RXLEV interrupt is asserted in non-BAM mode. The signal/interrupt is asserted when the FIFO level (number of words in the RX FIFO) is greater than the value in RFWR. Configured value should be lower/equal than AUTO_RFR_LEVEL value, otherwise the watermark will never be reached. Notice that the value is in words. See UART_DM_ISR register for TXLEV interrupt description - Only RAM_ADDR_WIDTH -1:0 bits are generated.

0x078B0024 BLSP1_BLSP_UART1_UART_DM_HCR**Type:** RW**Clock:** AHB_CLK**Reset State:** 0x00000000

Configures Hunt-Character value.

BLSP1_BLSP_UART1_UART_DM_HCR

Bits	Name	Description
7:0	DATA	The UART_DM_HCR register is the UART hunt character register, which contains the character for which the receiver looks to assert a hunt character received interrupt.

0x078B0034 BLSP1_BLSP_UART1_UART_DM_DMRX**Type:** RW**Clock:** AHB_CLK**Reset State:** 0x00000000

Initializes RX transfer and configures maximal length in bytes of the transfer.

BLSP1_BLSP_UART1_UART_DM_DMRX

Bits	Name	Description
24:0	RX_DM_CRCI_CHARS	A write to RX_DM_CRCI_CHARS initializes RX transfer in SW (FIFO) mode, with written value configuring the maximal length in bytes of the transfer. In BAM-mode, only the value held in this register impacts the data-flow and not the write-event. A write-value of 0 is not valid. Read of UART_DM_DMRX register is returns the number of characters that were received since the end of the last RX transfer. To read back the last value written to this register use UART_DM_DMRX_DBG register. Refer to UART_DM_BCR register description for additional constraints on the value to be written to this register.

0x078B0038 BLSP1_BLSP_UART1_UART_DM_IRDA_RX_TOTAL_SNAP_DEPRECATED**Type:** RW**Clock:** AHB_CLK**Reset State:** Undefined

This register is deprecated and remains only for backward-compatibility. Use UART_DM_IRDA or UART_DM_RX_TOTAL_SNAP.

BLSP1_BLSP_UART1_UART_DM_IRDA_RX_TOTAL_SNAP_DEPRECATED

Bits	Name	Description
31:0	UART_DM_IRDA_RX_TOTAL_SNAP_DEPRECATED	

0x078B003C BLSP1_BLSP_UART1_UART_DM_DMEN**Type:** RW**Clock:** AHB_CLK**Reset State:** 0x00000000

The UART_DM_DMEN register defines the DMA used by the core and the data packing mode for TX and RX channels.

BLSP1_BLSP_UART1_UART_DM_DMEN

Bits	Name	Description
5	RX_SC_ENABLE	- Set (1) for work in Single-Character mode for RX channel (every character received is zero-padded into a word). - Clearing this bit requires resetting the receiver.

BLSP1_BLSP_UART1_UART_DM_DMEN (cont.)

Bits	Name	Description
4	TX_SC_ENABLE	- Set (1) for work in Single-Character mode for TX channel (Only LSB byte of each word get transmitted). - Clearing this bit requires resetting the transmitter.
3	RX_BAM_ENABLE	- Set (1) this bit to enable RX BAM interface (BAM-mode). - Clear (0) this bit to disable RX BAM interface (SW/FIFO mode). Clearing this bit requires resetting the receiver (see UART_DM_CR register).
2	TX_BAM_ENABLE	- Set (1) this bit to enable TX BAM interface (BAM-mode). - Clear (0) this bit to disable TX BAM interface (SW/FIFO mode). Clearing this bit requires resetting the transmitter(see UART_DM_CR register).
1	RESERVED_1	
0	RESERVED_2	

0x078B0040 BLSP1_BLSP_UART1_UART_DM_NO_CHARS_FOR_TX**Type:** RW**Clock:** AHB_CLK**Reset State:** 0x00000000

Initialization of TX transfer in SW/FIFO mode.

BLSP1_BLSP_UART1_UART_DM_NO_CHARS_FOR_TX

Bits	Name	Description
23:0	TX_TOTAL_TRANS_LEN	The total number of characters for transmission. Before writing a new value, the TX FIFO must be empty as indicated by TX_READY interrupt (see UART_DM_ISR register for description) or after a TX-channel reset. It is used by the TX-channel in SW/FIFO mode to calculate how many characters to transmit in the last word. Any additional writes to the TX FIFO beyond TX_TOTAL_TRANS_LEN characters will be discarded. A TX_READY interrupt is triggered after TX_TOTAL_TRANS_LEN number of characters were moved from the TX FIFO to the unpacking buffer (not all may have been transmitted at that point though).

0x078B0044 BLSP1_BLSP_UART1_UART_DM_BADR**Type:** RW**Clock:** AHB_CLK**Reset State:** Undefined

Configures FIFO division between RX-FIFO and TX-FIFO.

BLSP1_BLSP_UART1_UART_DM_BADR

Bits	Name	Description
31:2	RX_BASE_ADDR	RX FIFO base address. Both FIFOs use the same RAM ($2^{\text{RAM_ADDR_WIDTH}}$, 32-bit entries). This register controls the division of the memory between the RX and TX FIFOs. The division must be a multiple of 4 entries for legacy reasons, therefore bits 1:0 of this register are not used. The size of the TX FIFO equals RX_BASE_ADDR. The size of the RX FIFO is $2^{\text{RAM_ADDR_WIDTH}} - \text{RX_BASE_ADDR}$. * The default is $\text{RX_BASE_ADDR} = 2^{(\text{RAM_ADDR_WIDTH} - 1)}$ * Only RAM_ADDR_WIDTH -1:2 bits are generated.

0x078B004C BLSP1_BLSP_UART1_UART_DM_TXFS**Type:** R**Clock:** AHB_CLK**Reset State:** Undefined

TX channel status register.

BLSP1_BLSP_UART1_UART_DM_TXFS

Bits	Name	Description
31:14	TX_FIFO_STATE_MSB	The msb of TX_FIFO_STATE bitfield. - Only RAM_ADDR_WIDTH +4 +3 -1:14 bits are generated.
13:10	TX_ASYNC_FIFO_STATE	TX asynchronous fifo status - number of characters in the TX asynchronous FIFO.
9:7	TX_BUFFER_STATE	TX unpacking buffer status - number of bytes in the unpacking buffer
6:0	TX_FIFO_STATE_LSB	TX FIFO status - number of words in the TX FIFO

0x078B0050 BLSP1_BLSP_UART1_UART_DM_RXFS**Type:** R**Clock:** AHB_CLK**Reset State:** Undefined

RX channel status register.

BLSP1_BLSP_UART1_UART_DM_RXFS

Bits	Name	Description
31:14	RX_FIFO_STATE_MSB	The msb of TX_FIFO_STATE bitfield. Only RAM_ADDR_WIDTH +4 +3 -1:14 bits are generated.
13:10	RX_ASYNC_FIFO_STATE	RX asynchronous fifo status - number of characters in the RX asynchronous FIFO.
9:7	RX_BUFFER_STATE	RX packing buffer status - number of bytes in the packing buffer
6:0	RX_FIFO_STATE_LSB	The number of entries in the RX FIFO that have at least one valid word. It is incremented each time a word is moved from the packing register to the RX FIFO. It is decremented each time a word is read from the RX FIFO. Note: The Uart does not keep track of non-valid characters in each word.

0x078B0060 BLSP1_BLSP_UART1_UART_DM_MISR_MODE**Type:** RW**Clock:** WR_CLK**Reset State:** Undefined**BLSP1_BLSP_UART1_UART_DM_MISR_MODE**

Bits	Name	Description
31:2	RESERVED	
1:0	MODE	0x0: DISABLED (Disabled) 0x1: ENABLED_TX_TEST (Enabled, TX test) 0x2: ENABLED_RX_TEST (Enabled, RX test)

0x078B0064 BLSP1_BLSP_UART1_UART_DM_MISR_RESET**Type:** W**Clock:** WR_CLK**Reset State:** Undefined**BLSP1_BLSP_UART1_UART_DM_MISR_RESET**

Bits	Name	Description
31:1	RESERVED	
0	RESET	

0x078B0068 BLSP1_BLSP_UART1_UART_DM_MISR_EXPORT**Type:** RW**Clock:** WR_CLK**Reset State:** Undefined**BLSP1_BLSP_UART1_UART_DM_MISR_EXPORT**

Bits	Name	Description
31:1	RESERVED	
0	EXPORT	When clear (0), testbus is driven by what is sent into the MISR. (i.e. what is the result of the muxing of all the input data streams with <BLOCK>_TEST_MODE) When set (1), testbus is driven by the current state of the MISR. (debug typically)

0x078B006C BLSP1_BLSP_UART1_UART_DM_MISR_VAL**Type:** R**Clock:** WR_CLK**Reset State:** Undefined**BLSP1_BLSP_UART1_UART_DM_MISR_VAL**

Bits	Name	Description
9:0	VAL	Current MISR state

0x078B0070 BLSP1_BLSP_UART1_UART_DM_TF_RF_DEPRECATED**Type:** RW**Clock:** AHB_CLK**Reset State:** Undefined

This register is deprecated and remains only for backward-compatibility. Use UART_DM_TF or UART_DM_RF.

BLSP1_BLSP_UART1_UART_DM_TF_RF_DEPRECATED

Bits	Name	Description
31:0	UART_DM_TF_RF_DEPRECATED	

0x078B0074 BLSP1_BLSP_UART1_UART_DM_TF_RF_2_DEPRECATED**Type:** RW**Clock:** AHB_CLK**Reset State:** Undefined

This register is deprecated and remains only for backward-compatibility. Use UART_DM_TF_2 or UART_DM_RF_2.

BLSP1_BLSP_UART1_UART_DM_TF_RF_2_DEPRECATED

Bits	Name	Description
31:0	UART_DM_TF_RF_2_DEPR ECATED	

0x078B0078 BLSP1_BLSP_UART1_UART_DM_TF_RF_3_DEPRECATED**Type:** RW**Clock:** AHB_CLK**Reset State:** Undefined

This register is deprecated and remains only for backward-compatibility. Use UART_DM_TF_3 or UART_DM_RF_3.

BLSP1_BLSP_UART1_UART_DM_TF_RF_3_DEPRECATED

Bits	Name	Description
31:0	UART_DM_TF_RF_3_DEPR ECATED	

0x078B007C BLSP1_BLSP_UART1_UART_DM_TF_RF_4_DEPRECATED**Type:** RW**Clock:** AHB_CLK**Reset State:** Undefined

This register is deprecated and remains only for backward-compatibility. Use UART_DM_TF_4 or UART_DM_RF_4.

BLSP1_BLSP_UART1_UART_DM_TF_RF_4_DEPRECATED

Bits	Name	Description
31:0	UART_DM_TF_RF_4_DEPR ECATED	

0x078B0080 BLSP1_BLSP_UART1_UART_DM_SIM_CFG**Type:** RW**Clock:** AHB_CLK**Reset State:** 0x00000000

The UART_DM_SIM_CFG register is used to configure the SIM interface for the UART.

* This register is generated only when SIM_GLUE_GEN generic equals 1.

BLSP1_BLSP_UART1_UART_DM_SIM_CFG

Bits	Name	Description
17	UIM_TX_MODE	When this bit is set (1), any re-transmission signal from the UIM card will be ignored (The transmission portion of the SIM interface operates in block mode T=1). When clear (0), re-transmission will be executed upon request (the transmission portion of the SIM interface operates in asynchronous transfer mode T=0). Note that the operation of this function is conditional upon SIM_SEL field being set (1) to designate the UIM_IF mode of operation.
16	UIM_RX_MODE	When this bit is set (1), error-signal indication will not be generated by the receiver (the receive portion of the SIM interface operates in block mode T=1). When clear (0), error signal will be generated when a character error is detected (the receive portion of the SIM interface operates in asynchronous transfer mode T=0). Note that the operation of this function is conditional upon SIM_SEL field being set (1) to designate the UIM_IF mode of operation.
15:8	SIM_STOP_BIT_LEN	In UIM_IF mode, this field programs the duration of the stop bit that is appended to each transmitted character. The stop bit duration can be 1 to 254 bit times. The receiver only samples the stop bit once, regardless of the programmed stop bit length. 0x1: ENUM_1_BIT_TIMES (1 bit times) 0x2: ENUM_2_BIT_TIMES (2 bit times) 11 111110: 254 bit times)
7	SIM_CLK_ON	When this bit is set(1), the UIM_CLK is turned on at either the TD8 or TD4 frequency, as indicated by bit6(UIM_CLK_TD8_SEL). When this bit is cleared, the UIM_CLK is stopped either LOW or HIGH, as indicated by bit 5(UIM_CLK_STOP_HIGH).
6	SIM_CLK_TD8_SEL	This bit selects the UIM_CLK frequency at which the UIM_CLK is turned on. 0x1: TD8 (UIM_CLK runs at the TD8 frequency (same frequency as SIM-SRC_CLK)) 0x0: TD4 (UIM_CLK runs at the TD4 frequency (1/2 the frequency of SIM-SRC_CLK))
5	SIM_CLK_STOP_HIGH	This bit indicates at what point - high or low - the UIM_CLK will stop. 0x1: HIGH (high) 0x0: LOW (low)

BLSP1_BLSP_UART1_UART_DM_SIM_CFG (cont.)

Bits	Name	Description
4	RESERVED	
3	MASK_RX	Setting (1) this bit masks off any 0 on the RX line, which prevents the receiver from seeing the START bit. Set (1) this bit if you do not want to receive the data you transmit.
2	SWAP_D	Setting (1) this bit causes the UIM_IF to swap the 8 data bits (making MSB into LSB, bit 6 into bit1, and so on) before it is read by the microprocessor.
1	INV_D	Set (1) this bit to cause the UIM_IF to invert the 8 data bit before it is read by the microprocessor.
0	SIM_SEL	When set (1), UART_DM will operate in SIM/UIM interface mode. When cleared (0), SIM/UIM interface is disabled.

0x078B00A0 BLSP1_BLSP_UART1_UART_DM_CSR**Type:** RW**Clock:** AHB_CLK**Reset State:** 0x00000000

The UART_DM_CSR register is the UART clock selection register. This register is used to determine the bit rate for the transmitter and receiver. The transmitter and receiver can be clocked at different rates.

Data-rate of receiver/transmitter is UART-CLK frequency divided by the integer matching the configured value below:

0xF = 16

0xE = 32

0xD = 48

0xC = 64

0xB = 96

0xA = 128

0x9 = 192

0x8 = 256

0x7 = 384

0x6 = 512

0x5 = 768

0x4 = 1536

0x3 = 3072

0x2 = 6144

0x1 = 12288

0x0 = 24576

BLSP1_BLSP_UART1_UART_DM_CSR

Bits	Name	Description
7:4	UART_RX_CLK_SEL	Use the values above to select the receive bit rate.
3:0	UART_TX_CLK_SEL	Use the values above to select the transmit bit rate.

0x078B00A4 BLSP1_BLSP_UART1_UART_DM_SR**Type:** R**Clock:** AHB_CLK**Reset State:** Undefined

The UART_DM_SR register is the UART status register. This register is used to obtain the current state of the UART subsystem. This register is updated asynchronously.

BLSP1_BLSP_UART1_UART_DM_SR

Bits	Name	Description
11:10	TRANS_END_TRIGGER	RX-transfer-end event type. Stores trigger for last transfer end: '00' for no trigger, '01' for DMRX-event, '10' for timeout stale, '11' for SW stale.
9	TRANS_ACTIVE	Transfer Active bit.
8	RX_BREAK_START_LAST	When break start is detected, this bit is set (1). When break end was detected, this bit is reset (0).
7	HUNT_CHAR	When set (1), this bit indicates that the programmed character in the UART_DM_HCR register was received. If programmed, this condition causes the RXHUNT bit to be set (1) in the UART_DM_ISR register. This bit is cleared by issuing a reset error status command.

BLSP1_BLSP_UART1_UART_DM_SR (cont.)

Bits	Name	Description
6	RX_BREAK	When set (1), this bit indicates that an all-zero character of the programmed length was received without a stop bit. If a break begins in the middle of a character, it is detected if the condition remains until the end of the next character time. Only one all zero character is written into the FIFO upon receiving the break. After the RX input returns high for at least half a bit time, then more characters are accepted by the receiver. Whenever the break condition starts or stops as defined above, the RXBREAK bit is set (1) in the UART_DM_ISR register. This bit is appended to the character in the receive FIFO.
5	PAR_FRAME_ERR	A parity error occurs if parity is programmed to be checked and an incoming character is received with incorrect parity. A framing error occurs when the RX input is low while the stop bit is being sampled. This bit is sampled in the middle of the first stop bit position, regardless of the programmed length of the stop bit. This error bit is appended to the character in the receive FIFO.
4	UART_OVERRUN	An overrun error occurs if one or more incoming characters are lost. This happens when the receive FIFO is full, the packing buffer is full and another character has been received in the shift register. The character in the receive shift register is lost. Upon overrun, this status bit remains high until cleared by a reset error status command.
3	TXEMT	This bit is set (1) when the transmitter under-runs. It is set after the transmission of the stop bit at the end of a character, if there are no characters waiting to be sent. It is reset when a character is written to the transmit FIFO.
2	TXRDY	This bit indicates that the transmit FIFO has space in it. It is reset when the FIFO becomes full and set when space becomes available again. Any characters that are written to the FIFO while this bit is low is lost.
1	RXFULL	This bit is set (1) when the receive FIFO becomes full. It is reset when the CPU reads a word.
0	RXRDY	This bit is set (1) when there is a word in the receive FIFO waiting to be read. It is reset when the last word currently stored in the FIFO is read.

0x078B00A8 BLSP1_BLSP_UART1_UART_DM_CR**Type:** RW**Clock:** AHB_CLK**Reset State:** 0x00000000

The UART_DM_CR register is the UART command register. This register is used to issue specific commands to the UART subsystem. This register is updated asynchronously.

CAUTION Do not reset the transmitter and disable it at the same time. Do not reset the receiver and disable it at the same time.

Channel-commands:

0x00 - Null Command - Does nothing.

0x01 - Reset Receiver - Resets the receiver as if a hardware reset were issued. The receiver is disabled and the FIFO, packing buffer and shift registers are flushed.

0x02 - Reset Transmitter - Resets the transmitter as if a hardware reset were issued. The transmitter signal goes high (marking) and the FIFO, unpacking register and shift register are flushed.

0x03 - Reset error status - Clears the overrun error and hunt char received status bits in both the character and block error modes. In the block error mode, it clears the error status and received break.

0x04 - Reset break change interrupt - Clears the break change interrupt status bit.

0x05 - Start break - Forces the transmitter signal low. The transmitter must be enabled. If the transmitter is busy, the break is started when all characters in the transmit FIFO and the transmit shift register have been completely sent.

0x06 - Stop break - If executed while channel is breaking, this command causes the transmitter signal to go high. The signal remains high for at least one bit time before sending out a new character.

0x07 - Reset CTS_N - Clears ISR bit 5.

0x08 - Reset stale interrupt - Clears the stale interrupt.

0x09 - Packet mode - Turns on the sample data mode, which causes the receiver to sample the receive data stream at 16 times the programmed baud rate. The data is sampled with the start of the start bit, or the first data bit, and continued until the marking state. To exit this state, write 1100 in the command field.

0x0A - test_parity_on - Internal use only.

0x0B - test_frame_on - Internal use only.

0x0C - Mode reset - Turns off the sample data mode.

0x0D - Set RFR - Asserts the ready for receiving signal (actual flow-control port is active-low (rfr_n) and will go low following this command).

0x0E - Reset RFR - Deasserts the ready for receiving signal (actual flow-control port is active-low (rfr_n) and will go high following this command).

0x0F - uart_reset_int - Internal use only.

0x10 - Reset TX_ERROR - Clears TX_ERROR

0x11 - Clear TX_DONE - Clears the TX_DONE interrupt (ISR bit 9)

0x12 - Reset break start interrupt - Clears the break start interrupt status bit.

0x13 - Reset break end interrupt - Clears the break end interrupt status bit.

- 0x14 - Reset par_frame_err interrupt - Clears the par_frame_err interrupt status bit.
- 0x15 - Start RX BAM IFC - Initialize producer sidebands towards BAM after reset.
- 0x16 - Start TX BAM IFC - Initialize consumer sidebands towards BAM after reset.
- 0x17 - WWT counter enable - Enable WWT counter & IRQ.
- 0x18 - WWT counter disable - Disable WWT counter & IRQ.
- 0x19 - Clear TX_WR_ERROR IRQ - Clears TX write error interrupt status bit.
- 0x1A - Clear RX_RD_ERROR IRQ - Clears RX read error interrupt status bit.
- 0x1B - Clear TX_COMP IRQ - Clears TX complete interrupt status bit.
- 0x1C - Clear WWT IRQ - Clears WWT timeout interrupt status bit.

General-commands:

- 0x0 - Null Command - Does nothing.
- 0x1 - CR Protection Enable - Enables CR HW protection. When two consecutive writes to the CR are detected, the second write is delayed until the command of the first write is finished. The delay is done by de-asserting the AHB ready and this ensures that the first command completes and the second one will be executed right afterward.
- 0x2 - CR Protection Disable - Disables CR HW protection. SW is responsible for managing delay between writes to the CR register.
- 0x3 - Reset TX-Ready interrupt - Clears the TX_READY interrupt.
- 0x4 - SW Force Stale - Causes a stale event (even if stale events are disabled).
- 0x5 - Enable Stale Event - Enables the stale event mechanism
- 0x6 - Disable Stale Event - Enables the stale event mechanism
- 0x7 - RESERVED

BLSP1_BLSP_UART1_UART_DM_CR

Bits	Name	Description
11	CHANNEL_COMMAND_MSB	This is the msb of the CHANNEL_COMMAND bitfield.
10:8	GENERAL_COMMAND	Writing the appropriate value to these bits executes the commands that are listed in Table .
7:4	CHANNEL_COMMAND_LSB	Writing the appropriate value to these bits along with bit 11,executes the channel-command as listed in the register decription.
3	UART_TX_DISABLE	This command terminates the operation of the transmitter after any character in the transmit shift register is sent.

BLSP1_BLSP_UART1_UART_DM_CR (cont.)

Bits	Name	Description
2	UART_TX_EN	This command enables the operation of the transmitter.
1	UART_RX_DISABLE	This command immediately terminates the operation of the channel receiver and any incoming characters are lost. None of the receiver status bits are affected by this command and characters that are already in the receive FIFO remain there.
0	UART_RX_EN	This command enables the channel receiver.

0x078B00AC BLSP1_BLSP_UART1_UART_DM_MISR**Type:** R**Clock:** AHB_CLK**Reset State:** 0x00000000**BLSP1_BLSP_UART1_UART_DM_MISR**

Bits	Name	Description
16:0	UART_MISR	The UART_DM_MISR register is the masked interrupt status register. A read from this register returns the 'AND' of the UART_DM_ISR and the UART_DM_IMR registers. This register is updated asynchronously. This bit field is $\text{misr} \leq (\text{isr}(16 \text{ DOWNT0 } 7) \text{ AND } \text{imr}(16 \text{ DOWNT0 } 7)) \text{ \& '0' \& } (\text{isr}(5 \text{ DOWNT0 } 0) \text{ AND } \text{imr}(5 \text{ DOWNT0 } 0)).$

0x078B00B0 BLSP1_BLSP_UART1_UART_DM_IMR**Type:** RW**Clock:** AHB_CLK**Reset State:** 0x00000000

The UART_DM_IMR register is the UART interrupt mask register. This register is used to enable the corresponding functions in the UART_DM_ISR register. Setting (1) a bit in the UART_DM_IMR register causes an interrupt to be generated, if the corresponding bit in the UART_DM_ISR register is set. Clearing (0) a bit in the UART_DM_IMR register causes the setting of the corresponding bit in the UART_DM_ISR register to have no effect on the interrupt signal. (See UART_DM_ISR register)

Bit 6 of the UART_DM_IMR register, CURRENT_CTS, is slightly different. If the current value of the CTS is one (1), no interrupt is generated. However, the user can use bit 6 in this register to mask the CTS value when reading the UART_DM_MISR register or as a general-purpose bit.

BLSP1_BLSP_UART1_UART_DM_IMR

Bits	Name	Description
16	WWT_IRQ	Masking bit of the matching field interrupt bit (See UART_DM_ISR for field descriptions).
15	TXCOMP_IRQ	Masking bit of the matching field interrupt bit (See UART_DM_ISR for field descriptions).
14	RX_RD_ERROR_IRQ	Masking bit of the matching field interrupt bit (See UART_DM_ISR for field descriptions).
13	TX_WR_ERROR_IRQ	Masking bit of the matching field interrupt bit (See UART_DM_ISR for field descriptions).
12	PAR_FRAME_ERR_IRQ	Masking bit of the matching field interrupt bit (See UART_DM_ISR for field descriptions).
11	RXBREAK_END	Masking bit of the matching field interrupt bit (See UART_DM_ISR for field descriptions).
10	RXBREAK_START	Masking bit of the matching field interrupt bit (See UART_DM_ISR for field descriptions).
9	TX_DONE	Masking bit of the matching field interrupt bit (See UART_DM_ISR for field descriptions).
8	TX_ERROR	Masking bit of the matching field interrupt bit (See UART_DM_ISR for field descriptions).
7	TX_READY	Masking bit of the matching field interrupt bit (See UART_DM_ISR for field descriptions).
6	CURRENT_CTS	Masking bit of the matching field interrupt bit (See UART_DM_ISR for field descriptions).
5	DELTA_CTS	Masking bit of the matching field interrupt bit (See UART_DM_ISR for field descriptions).
4	RXLEV	Masking bit of the matching field interrupt bit (See UART_DM_ISR for field descriptions).
3	RXSTALE	Masking bit of the matching field interrupt bit (See UART_DM_ISR for field descriptions).
2	RXBREAK_CHANGE	Masking bit of the matching field interrupt bit (See UART_DM_ISR for field descriptions).
1	RXHUNT	Masking bit of the matching field interrupt bit (See UART_DM_ISR for field descriptions).
0	TXLEV	Masking bit of the matching field interrupt bit (See UART_DM_ISR for field descriptions).

0x078B00B4 BLSP1_BLSP_UART1_UART_DM_ISR**Type:** R**Clock:** AHB_CLK**Reset State:** Undefined

The UART_DM_ISR register is the UART interrupt status register. This register provides the current status of the possible interrupt conditions. If one of these bits is set (1), and the corresponding bit in the UART_DM_IMR register is set (1), an interrupt is generated (with the exception of bit 6, CURRENT_CTS). If the corresponding bit in the UART_DM_IMR register is clear (0), setting one of these bits has no effect on the UART interrupt request signal. This register is updated asynchronously.

BLSP1_BLSP_UART1_UART_DM_ISR

Bits	Name	Description
16	WWT_IRQ	WWT timeout reached interrupt. Fired after WWT timer was enabled and no data was received during the configured time that follows. Cleared by issuing CHANNEL_COMMAND = CLEAR_WWT_IRQ (UART_DM_CR register).
15	TXCOMP_IRQ	TX-complete interrupt. Fired when TX channel finished transmitting all of TX-transfer's data and no new transfer was initiated. Cleared by issuing CHANNEL_COMMAND = CLEAR_TX_COMP_IRQ (UART_DM_CR register).
14	RX_RD_ERROR_IRQ	RX-FIFO empty read error. Fired when AHB master is trying to read from an empty RX-FIFO. Cleared by issuing CHANNEL_COMMAND = CLEAR_RX_RD_ERROR_IRQ (UART_DM_CR register).
13	TX_WR_ERROR_IRQ	TX-FIFO full write error. Fired when AHB master is trying to write into a full TX-FIFO. Cleared by issuing CHANNEL_COMMAND = CLEAR_TX_WR_ERROR_IRQ (UART_DM_CR register).
12	PAR_FRAME_ERR_IRQ	This bit is asserted in the same condition at which PAR_FRAME_ERR status bit is asserted (See UART_DM_SR register). Cleared by issuing CHANNEL_COMMAND = RESET_PAR_FRAME_ERR_IRQ (UART_DM_CR register).
11	RXBREAK_END	This bit is set (1) if the end of a break condition is detected. Cleared by issuing CHANNEL_COMMAND = RESET_BREAK_END_IRQ (UART_DM_CR register).
10	RXBREAK_START	This bit is set (1) if the start of a break condition is detected. Cleared by issuing CHANNEL_COMMAND = RESET_BREAK_START_IRQ (UART_DM_CR register).
9	TX_DONE	This bit is used only in UIM/SIM_IF mode. Set (1) this bit when the last byte in the transmitter has been sent. Cleared by issuing CHANNEL_COMMAND = CLEAR_TX_DONE_IRQ (UART_DM_CR register). - This bit is generated only when SIM_GLUE_GEN generic equals 1.

BLSP1_BLSP_UART1_UART_DM_ISR (cont.)

Bits	Name	Description
8	TX_ERROR	Only used in UIM_IF mode. Set (1) this bit to indicate that there has been parity error during transmission. Cleared by issuing CHANNEL_COMMAND = CLEAR_TX_ERROR_IRQ (UART_DM_CR register). - This bit is generated only when SIM_GLUE_GEN generic equals 1.
7	TX_READY	This bit, when set(1), indicates that: 1. TX FIFO is empty. 2. The value written in NO_CHARS_FOR_TX register equals the number of bytes written to the TX FIFO since the register was updated. Note: There may be characters in the unpack buffer or in the shift register. Cleared by issuing GENERAL_COMMAND = RESET_TX_READY_IRQ (UART_DM_CR register)
6	CURRENT_CTS	This bit indicates the current state of the CTS input, high ('1') meaning peer is not ready for data reception, low ('0') - peer is ready. The toggling of this bit does not generate an interrupt.
5	DELTA_CTS	This bit, when set (1), indicates that the CTS input has changed states. Cleared by issuing CHANNEL_COMMAND = RESET_CTS_N (UART_DM_CR register).
4	RXLEV	This bit is set when a word is loaded into the receive FIFO that brings the total number of words in the FIFO above the programmed watermark level in the FIFO watermark register (UART_DM_RFWR). This bit is cleared after enough words have been read to bring the level equal to or below the programmed watermark level.
3	RXSTALE	This bit indicates that an RX-transfer has finished (See UART_DM HPG for more information). Cleared by issuing CHANNEL_COMMAND = RESET_STALE_IRQ (UART_DM_CR register).
2	RXBREAK_CHANGE	This bit is set (1) if the start or end of a break condition is detected. Cleared by issuing CHANNEL_COMMAND = RESET_BREAK_CHANGE_IRQ (UART_DM_CR register). A detected break is defined as an all-zeros character with an invalid stop bit AND LOW PARITY. The end of a break is defined as a mark condition (1) at least 1/2 bit width. A detected break occupies one position in the Rx FIFO.
1	RXHUNT	This bit is set (1) when an incoming character matches the value programmed into the UART_DM_HCR register. Cleared by issuing CHANNEL_COMMAND = RESET_ERROR_STATUS (UART_DM_CR register).

BLSP1_BLSP_UART1_UART_DM_ISR (cont.)

Bits	Name	Description
0	TXLEV	This bit is set (1) when a word which is transferred from the transmit FIFO to the transmit shift register brings the total number of words in the FIFO below or equal to the programmed watermark level in the UART_DM_TFWR register. This bit is cleared (0) after enough words have been written to the FIFO to bring the level above the programmed watermark level. Since this bit does not have a Clear command, during interrupt handling this bit must be masked (using the matching bit in UART_DM_IMR) until writing additional data for transmission is required. After data is written, this interrupt-bit should be unmasked.

0x078B00B8 BLSP1_BLSP_UART1_UART_DM_IRDA**Type:** RW**Clock:** AHB_CLK**Reset State:** 0x00000000

The UART_DM_IRDA register enables the IRDA function, selects a loopback, and allows inversion of RX and TX data. This register also controls the IRDA transceiver that optionally interfaces between the UART and the DP_RX_DATA and DP_TX_DATA pins.

The UART proper is a standard RX/TX configuration that converts the serial pin input and output lines to 8-bit data for the microprocessor. The configuration contains receive and transmit blocks, a control block, a bit rate generator for RX and TX, and FIFO interfaces with the microprocessor data bus. The IRDA register provides the means of switching IRDA circuits into the serial I/O lines and the IRDA function is normally switched out. The IRDA modulator/demodulator may be enabled or disabled under microprocessor control. When enabled, the RX and TX data paths have individual inversion select bits, so external true or inverted drivers may be used in the signal path.

* The register is generated only when IRDA_IFC_GEN generic equals 1.

BLSP1_BLSP_UART1_UART_DM_IRDA

Bits	Name	Description
4	MEDIUM_RATE_EN	- Set (1) for 1/4 bit-time pulse length (Medium rate) - Clear (0) for 3/16 bit-time pulse length (Low rate)
3	IRDA_LOOPBACK	- This bit loops the IRDA modulated TX line back into the IRDA RX line. The normal UART loopback mode must be off in order to see this effect.
2	INVERT_IRDA_TX	This bit inverts the polarity of the IRDA modulated signal on the DP_TX_DATA pin. - Set (1) this bit for an inverted polarity. - Clear (0) this bit for a non-inverted polarity.

BLSP1_BLSP_UART1_UART_DM_IRDA (cont.)

Bits	Name	Description
1	INVERT_IRDA_RX	This bit inverts the polarity of the IRDA received signal on the DP_RX_DATA pin. - Set (1) this bit for inverted the polarity. - Clear (0) this bit for non-inverted polarity.
0	IRDA_EN	- Set (1) this bit to enable the IRDA transceiver. - Clear (0) this bit to bypass the IRDA transceiver and ignore the other bits of this register.

0x078B00BC BLSP1_BLSP_UART1_UART_DM_RX_TOTAL_SNAP**Type:** R**Clock:** AHB_CLK**Reset State:** 0x00000000**BLSP1_BLSP_UART1_UART_DM_RX_TOTAL_SNAP**

Bits	Name	Description
23:0	RX_TOTAL_BYTES	?RX_TOTAL_SNAP register holds the number of characters received since the end of last Rx transfer. It is updated at the end of an Rx transfer (see UART_DM HPG for more information). Rx transfer ends when one of the conditions is met: - The number of characters which were received since the end of the previous transfer equals the value which was written to DMRX at ?Transfer initialization?. - A stale event occurred (flush operation already performed if was needed)

0x078B00C0 BLSP1_BLSP_UART1_UART_DM_WWT_TIMEOUT**Type:** RW**Clock:** AHB_CLK**Reset State:** 0x00000000**BLSP1_BLSP_UART1_UART_DM_WWT_TIMEOUT**

Bits	Name	Description
24:0	WWT_TIMEOUT	timeout value for WWT mechanism.

0x078B00C4 BLSP1_BLSP_UART1_UART_DM_CLK_CTRL**Type:** RW**Clock:** AHB_CLK**Reset State:** 0x00000000**BLSP1_BLSP_UART1_UART_DM_CLK_CTRL**

Bits	Name	Description
23	UART_IRDA_CLK_CGC_OPEN	Forces UART IRDA-clock (uart_tx_clk) CGC to open when set.
22	UART_SIM_CLK_CGC_OPEN	Forces UART SIM-clock (uart_tx_clk) CGC to open when set.
21	UART_RX_CLK_CGC_OPEN	Forces UART RX-clock (uart_tx_clk) CGC to open when set.
20	UART_TX_CLK_CGC_OPEN	Forces UART TX-clock (uart_tx_clk) CGC to open when set.
19	RESERVED_1	
18	RESERVED_2	
17	RESERVED_3	
16	RESERVED_4	
15	RESERVED_5	
14	AHB_RX_BAM_CLK_CGC_OPEN	Forces AHB RX-BAM-clock (hrx_bam_clk) CGC to open when set.
13	AHB_TX_BAM_CLK_CGC_OPEN	Forces AHB TX-BAM-clock (htx_bam_clk) CGC to open when set.
12	RESERVED_6	
11	RESERVED_7	
10	AHB_RX_CLK_CGC_OPEN	Forces AHB RX-clock (hrx_clk) CGC to open when set.
9	AHB_TX_CLK_CGC_OPEN	Forces AHB TX-clock (htx_clk) CGC to open when set.
8	AHB_WR_CLK_CGC_OPEN	Forces AHB write-clock (hwr_clk) CGC to open when set.
7	RESERVED_8	
6	RESERVED_9	
5	RX_ENABLE_CGC_OPT	CGCs will gate AHB & UART clocks due to RX-DISABLE command/RX-RESET command only when set.
4	TX_ENABLE_CGC_OPT	CGCs will gate AHB & UART clocks due to TX-DISABLE command/TX-RESET command only when set.
3	RESERVED_10	
2	RESERVED_11	
1	RESERVED_12	

BLSP1_BLSP_UART1_UART_DM_CLK_CTRL (cont.)

Bits	Name	Description
0	AHB_CLK_CGC_CLOSE	Setting this bit to '1' will completely gate AHB-clock to the core. Only allowed register acces during the duration this bit is set is the access to reset this bit. All other register accesses are not allowed and might result in errors. When set to '0', AHB clock is not gated at the root of the core and can be gated by the other gating features.

0x078B00C8 BLSP1_BLSP_UART1_UART_DM_BCR**Type:** RW**Clock:** AHB_CLK**Reset State:** 0x00000000**BLSP1_BLSP_UART1_UART_DM_BCR**

Bits	Name	Description
5	RX_DMRX_1BYTE_RES_EN	Allows DMRX values other than a multiple of 16 if set.
4	RX_STALE_IRQ_DMRX_EQUAL	If set, Stale irq will be asserted when (DMRX==valid_char_count) and stale is not enabled.
3	RESERVED	
2	RX_DMRX_LOW_EN	DMRX lower than valid_char_cnt logic is enabled if bit is set
1	STALE_IRQ_EMPTY	Stale interrupt when RX-FIFO is empty will fire if bit is set.
0	TX_BREAK_DISABLE	TX-pin value when transmitter disabled while in break state is '0' if set, else '1'

0x078B00CC BLSP1_BLSP_UART1_UART_DM_RX_TRANS_CTRL**Type:** RW**Clock:** AHB_CLK**Reset State:** 0x00000000**BLSP1_BLSP_UART1_UART_DM_RX_TRANS_CTRL**

Bits	Name	Description
2	RX_DMRX_CYCLIC_EN	If set - Auto-Re-Activated RX-transfer will be re-activated with a DMRX value from the DMRX register, making a DMRX-transfer-end-event possible. Otherwise, Auto-Re-Activated RX-transfer cannot end with a DMRX event unless its value is written to the DMRX register throughout the transfer.
1	RX_TRANS_AUTO_RE_ACTIVATE	RX-transfer will be automatically re-activated after last data of previous transfer was read - if set. Otherwise, transfers are only activated by writing to DMRX register.

BLSP1_BLSP_UART1_UART_DM_RX_TRANS_CTRL (cont.)

Bits	Name	Description
0	RX_STALE_AUTO_RE_EN	RX-Stale automatic re-enable - on if set.

0x078B00D4 BLSP1_BLSP_UART1_UART_DM_FSM_STATUS**Type:** R**Clock:** AHB_CLK**Reset State:** 0x00000000**BLSP1_BLSP_UART1_UART_DM_FSM_STATUS**

Bits	Name	Description
29:28	TX_COMP_FSM	Value of TX-COMP FSM state: '00' for empty, '01' non-empty, '10' for empty-check, '11' for empty-check-fail.
27	RESERVED_1	
26:24	RX_PACK_FSM	Value of RX-packing FSM state: '000' for idle, '001' for fifo_write, '010' for wait_ack, '011' for flush_check, '100' for flushing, '101' for flush_done.
22:23	RESERVED_2	
21:20	RX_TRANS_FSM	Value of RX-transfer FSM state: '00' for inactive, '01' for active, '10' for active_dmr, '11' for active_flush.
19	RESERVED_3	
18:16	TX_TRANS_FSM	Value of TX-transfer FSM state: '000' for idle, '001' for bam_idle, '010' for active, '011' for inactive, '100' for last_trap, '101' for nwd.
15	RESERVED_4	
14:12	RX_PRO_TRANS_END_FSM	Value of RX_PRO_TRANS_END FSM state: '000' for disabled, '001' for idle, '010' for 2char_dphase, '011' for 1char_waiting, '100' for 1char dphase, '101' for mess_only_dphase.
11	RESERVED_5	
10:8	RX_PRO_ACTIVE_FSM	Value of RX_PRO_ACTIVE FSM state: '000' for disabled, '001' for isactive, '010' for wait_check_1, '011' for wait_check_1, '100' for pf_pending, '101' for wait_update_1, '110' for wait_update_2, '111' for inactive?
7	RESERVED_6	
6:4	TX_CON_TRANS_END_FSM	Value of TX_CON_TRANS_END FSM state: '000' for disabled, '001' for idle, '010' for 2char_dphase, '011' for 1char_waiting, '100' for 1char dphase, '101' for mess_only_dphase, '110' for normal_dphase.
3:1	RESERVED_7	
0	RX_TRANSFER_ACTIVE	1 is RX transfer is currently active, 0 if inactive.

0x078B00DC BLSP1_BLSP_UART1_UART_DM_GENERICS**Type:** R**Clock:** AHB_CLK**Reset State:** 0x00000000**BLSP1_BLSP_UART1_UART_DM_GENERICS**

Bits	Name	Description
7	GENERIC_BAM_IFC	Value of 'BAM_IFC_GEN' generic.
6	GENERIC_DM_IFC	Value of 'DM_IFC_GEN' generic.
5	GENERIC_IRDA_IFC	Value of 'IRDA_IFC_GEN' generic
4	GENERIC_SIM_GLUE	Value of 'SIM_GLUE_GEN' generic
3:0	GENERIC_RAM_ADDR_WIDTH	Value of 'RAM_ADDR_WIDTH' generic.

0x078B0100 BLSP1_BLSP_UART1_UART_DM_TF**Type:** W**Clock:** AHB_CLK**Reset State:** Undefined

In order to support bursts of 4, 0x70-0x7C address space is reserved for the UART_DM TX FIFO.

BLSP1_BLSP_UART1_UART_DM_TF

Bits	Name	Description
31:0	UART_TF	The UART_TF register is the UART transmit FIFO register. This register is used to access the channel transmit FIFO. If the FIFO is full at the time of writing, the characters are lost and an interrupt is generated (see UART_DM_ISR register).

0x078B0104 BLSP1_BLSP_UART1_UART_DM_TF_2**Type:** W**Clock:** AHB_CLK**Reset State:** Undefined**BLSP1_BLSP_UART1_UART_DM_TF_2**

Bits	Name	Description
31:0	UART_TF	See UART_DM_TF register.

0x078B0108 BLSP1_BLSP_UART1_UART_DM_TF_3**Type:** W**Clock:** AHB_CLK**Reset State:** Undefined**BLSP1_BLSP_UART1_UART_DM_TF_3**

Bits	Name	Description
31:0	UART_TF	See UART_DM_TF register.

0x078B010C BLSP1_BLSP_UART1_UART_DM_TF_4**Type:** W**Clock:** AHB_CLK**Reset State:** Undefined**BLSP1_BLSP_UART1_UART_DM_TF_4**

Bits	Name	Description
31:0	UART_TF	See UART_DM_TF register.

0x078B0110 BLSP1_BLSP_UART1_UART_DM_TF_5**Type:** W**Clock:** AHB_CLK**Reset State:** Undefined**BLSP1_BLSP_UART1_UART_DM_TF_5**

Bits	Name	Description
31:0	UART_TF	See UART_DM_TF register.

0x078B0114 BLSP1_BLSP_UART1_UART_DM_TF_6**Type:** W**Clock:** AHB_CLK**Reset State:** Undefined**BLSP1_BLSP_UART1_UART_DM_TF_6**

Bits	Name	Description
31:0	UART_TF	See UART_DM_TF register.

0x078B0118 BLSP1_BLSP_UART1_UART_DM_TF_7**Type:** W**Clock:** AHB_CLK**Reset State:** Undefined**BLSP1_BLSP_UART1_UART_DM_TF_7**

Bits	Name	Description
31:0	UART_TF	See UART_DM_TF register.

0x078B011C BLSP1_BLSP_UART1_UART_DM_TF_8**Type:** W**Clock:** AHB_CLK**Reset State:** Undefined**BLSP1_BLSP_UART1_UART_DM_TF_8**

Bits	Name	Description
31:0	UART_TF	See UART_DM_TF register.

0x078B0120 BLSP1_BLSP_UART1_UART_DM_TF_9**Type:** W**Clock:** AHB_CLK**Reset State:** Undefined**BLSP1_BLSP_UART1_UART_DM_TF_9**

Bits	Name	Description
31:0	UART_TF	See UART_DM_TF register.

0x078B0124 BLSP1_BLSP_UART1_UART_DM_TF_10**Type:** W**Clock:** AHB_CLK**Reset State:** Undefined**BLSP1_BLSP_UART1_UART_DM_TF_10**

Bits	Name	Description
31:0	UART_TF	See UART_DM_TF register.

0x078B0128 BLSP1_BLSP_UART1_UART_DM_TF_11**Type:** W**Clock:** AHB_CLK**Reset State:** Undefined**BLSP1_BLSP_UART1_UART_DM_TF_11**

Bits	Name	Description
31:0	UART_TF	See UART_DM_TF register.

0x078B012C BLSP1_BLSP_UART1_UART_DM_TF_12**Type:** W**Clock:** AHB_CLK**Reset State:** Undefined**BLSP1_BLSP_UART1_UART_DM_TF_12**

Bits	Name	Description
31:0	UART_TF	See UART_DM_TF register.

0x078B0130 BLSP1_BLSP_UART1_UART_DM_TF_13**Type:** W**Clock:** AHB_CLK**Reset State:** Undefined**BLSP1_BLSP_UART1_UART_DM_TF_13**

Bits	Name	Description
31:0	UART_TF	See UART_DM_TF register.

0x078B0134 BLSP1_BLSP_UART1_UART_DM_TF_14**Type:** W**Clock:** AHB_CLK**Reset State:** Undefined**BLSP1_BLSP_UART1_UART_DM_TF_14**

Bits	Name	Description
31:0	UART_TF	See UART_DM_TF register.

0x078B0138 BLSP1_BLSP_UART1_UART_DM_TF_15

Type: W
Clock: AHB_CLK
Reset State: Undefined

BLSP1_BLSP_UART1_UART_DM_TF_15

Bits	Name	Description
31:0	UART_TF	See UART_DM_TF register.

0x078B013C BLSP1_BLSP_UART1_UART_DM_TF_16

Type: W
Clock: AHB_CLK
Reset State: Undefined

BLSP1_BLSP_UART1_UART_DM_TF_16

Bits	Name	Description
31:0	UART_TF	See UART_DM_TF register.

0x078B0140 BLSP1_BLSP_UART1_UART_DM_RF

Type: R
Clock: AHB_CLK
Reset State: Undefined

The UART_DM_RF register is the UART receive FIFO register, which is used to access the channel receive FIFO. In order to support bursts of 16, 0x140-0x17C address space is reserved for the UART_DM RX FIFO.

BLSP1_BLSP_UART1_UART_DM_RF

Bits	Name	Description
31:0	UART_RF	This register returns the next word in the receive FIFO. After the read is completed, the FIFO read pointer is updated to make the next word available.

0x078B0144 BLSP1_BLSP_UART1_UART_DM_RF_2

Type: R
Clock: AHB_CLK
Reset State: Undefined

BLSP1_BLSP_UART1_UART_DM_RF_2

Bits	Name	Description
31:0	UART_RF	see UART_DM_RF register.

0x078B0148 BLSP1_BLSP_UART1_UART_DM_RF_3**Type:** R**Clock:** AHB_CLK**Reset State:** Undefined**BLSP1_BLSP_UART1_UART_DM_RF_3**

Bits	Name	Description
31:0	UART_RF	see UART_DM_RF register.

0x078B014C BLSP1_BLSP_UART1_UART_DM_RF_4**Type:** R**Clock:** AHB_CLK**Reset State:** Undefined**BLSP1_BLSP_UART1_UART_DM_RF_4**

Bits	Name	Description
31:0	UART_RF	see UART_DM_RF register.

0x078B0150 BLSP1_BLSP_UART1_UART_DM_RF_5**Type:** R**Clock:** AHB_CLK**Reset State:** Undefined**BLSP1_BLSP_UART1_UART_DM_RF_5**

Bits	Name	Description
31:0	UART_RF	see UART_DM_RF register.

0x078B0154 BLSP1_BLSP_UART1_UART_DM_RF_6

Type: R
Clock: AHB_CLK
Reset State: Undefined

BLSP1_BLSP_UART1_UART_DM_RF_6

Bits	Name	Description
31:0	UART_RF	see UART_DM_RF register.

0x078B0158 BLSP1_BLSP_UART1_UART_DM_RF_7

Type: R
Clock: AHB_CLK
Reset State: Undefined

BLSP1_BLSP_UART1_UART_DM_RF_7

Bits	Name	Description
31:0	UART_RF	see UART_DM_RF register.

0x078B015C BLSP1_BLSP_UART1_UART_DM_RF_8

Type: R
Clock: AHB_CLK
Reset State: Undefined

BLSP1_BLSP_UART1_UART_DM_RF_8

Bits	Name	Description
31:0	UART_RF	see UART_DM_RF register.

0x078B0160 BLSP1_BLSP_UART1_UART_DM_RF_9

Type: R
Clock: AHB_CLK
Reset State: Undefined

BLSP1_BLSP_UART1_UART_DM_RF_9

Bits	Name	Description
31:0	UART_RF	see UART_DM_RF register.

0x078B0164 BLSP1_BLSP_UART1_UART_DM_RF_10

Type: R
Clock: AHB_CLK
Reset State: Undefined

BLSP1_BLSP_UART1_UART_DM_RF_10

Bits	Name	Description
31:0	UART_RF	see UART_DM_RF register.

0x078B0168 BLSP1_BLSP_UART1_UART_DM_RF_11

Type: R
Clock: AHB_CLK
Reset State: Undefined

BLSP1_BLSP_UART1_UART_DM_RF_11

Bits	Name	Description
31:0	UART_RF	see UART_DM_RF register.

0x078B016C BLSP1_BLSP_UART1_UART_DM_RF_12

Type: R
Clock: AHB_CLK
Reset State: Undefined

BLSP1_BLSP_UART1_UART_DM_RF_12

Bits	Name	Description
31:0	UART_RF	see UART_DM_RF register.

0x078B0170 BLSP1_BLSP_UART1_UART_DM_RF_13

Type: R
Clock: AHB_CLK
Reset State: Undefined

BLSP1_BLSP_UART1_UART_DM_RF_13

Bits	Name	Description
31:0	UART_RF	see UART_DM_RF register.

0x078B0174 BLSP1_BLSP_UART1_UART_DM_RF_14

Type: R
Clock: AHB_CLK
Reset State: Undefined

BLSP1_BLSP_UART1_UART_DM_RF_14

Bits	Name	Description
31:0	UART_RF	see UART_DM_RF register.

0x078B0178 BLSP1_BLSP_UART1_UART_DM_RF_15

Type: R
Clock: AHB_CLK
Reset State: Undefined

BLSP1_BLSP_UART1_UART_DM_RF_15

Bits	Name	Description
31:0	UART_RF	see UART_DM_RF register.

0x078B017C BLSP1_BLSP_UART1_UART_DM_RF_16

Type: R
Clock: AHB_CLK
Reset State: Undefined

BLSP1_BLSP_UART1_UART_DM_RF_16

Bits	Name	Description
31:0	UART_RF	see UART_DM_RF register.

0x078B0180 BLSP1_BLSP_UART1_UART_DM_UIM_CFG

Type: RW
Clock: AHB_CLK
Reset State: 0x00001703

Configuration register for UIM-controller (linked with this UART_DM). Reset value relevant to UIM-controller HW-reset and not a UART_DM HW-reset. Writing to this register requires waiting until UIM-WRITE is done (indication by status-bit or interrupt) before attempting an additional write/read one of UIM_CFG/UIM_CMD registers.

BLSP1_BLSP_UART1_UART_DM_UIM_CFG

Bits	Name	Description
15	BATT_ALARM_QUICK_DROP_EN	When set, deactivation sequence resulted from battery-alarm will drop all UIM-lines immediately. Otherwise, lines will be dropped gradually.
14	TESTBUS_EN	When set, UIM-controller will output testbus signals on the testbus port. Otherwise, testbus port will be driven with zeros.
13	SW_RESET	When set, UIM-controller will be at reset state.
12	MODE18	Configures PAD's working reference voltage (set to '0' for 3V, '1' for 1.8V).
11	RESERVED_1	
10	PMIC_ALARM_EN	Enables messaging on the alarm-line to notify PMIC to drop card's voltage supply. Set to enable messaging, clear to disable.
9	BATT_ALARM_TRIGGER_EN	Enables triggering of deactivation sequence when battery-alarm detected. Set to enable triggering, clear to disable.
8	UIM_RMV_TRIGGER_EN	Enables triggering of deactivation sequence when card-removal detected. Set to enable triggering, clear to disable.
7	RESERVED_2	
6	UIM_CARD_EVENTS_ENABLE	When cleared, UIM-controller will not react to any card-event (insertion/removal). When set, UIM-controller reacts to these events and considered active. This bit should be set only after/during UIM_PRESENT_POLARITY bit is configured so UIM-controller will not interpret card-events incorrectly due to a wrong polarity configuration (the reset-value of the polarity might not match the actual one).
5	UIM_PRESENT_POLARITY	Determines value interpretation of uim_present signal. Setting this bit to '1' indicates that a uim-card present in the uim-slot will return the value '1' on uim_present line & '0' when a card is not present. Setting this bit to '0' indicates that a uim-card present in the uim-slot will return the value '0' on uim_present line & '1' when a card is not present.
4:0	EVENT_DEBOUNCE_TIME	Configures debounce-time of card-removal/insertion events (in sleep-clk cycles).

0x078B0184 BLSP1_BLSP_UART1_UART_DM_UIM_CMD**Type:** W**Clock:** AHB_CLK**Reset State:** Undefined

Command register for UIM-controller (linked with this UART_DM). Writing to this register requires waiting until UIM-WRITE is done (indication by status-bit or interrupt) before attempting an additional write/read one of UIM_CFG/UIM_CMD registers.

BLSP1_BLSP_UART1_UART_DM_UIM_CMD

Bits	Name	Description
1	RECOVER_FROM_HW_DEACTIVATION	Writing '1' to this bit will cause the UIM-io-ctrl to exit override state and reflect UIM signals from UART_DM to the UIM card. Setting both bits of this register to '1' will cause neither of the commands to be executed.
0	INITIATE_HW_DEACTIVATION	Initiates a HW deactivation sequence when writing '1' to this bit. This executes the sequence on the uim_data, uim_clk & uim_rst_n lines and alerts the PMIC to cut the power-supply to the card afterwards. Setting both bits of this register to '1' will cause neither of the commands to be executed.

0x078B0188 BLSP1_BLSP_UART1_UART_DM_UIM_IO_STATUS**Type:** R**Clock:** AHB_CLK**Reset State:** 0x00000002**BLSP1_BLSP_UART1_UART_DM_UIM_IO_STATUS**

Bits	Name	Description
2	UIM_IO_WRITE_IN_PROGRESS	Asserted after UIM-write (UIM_CFG/UIM_CMD register write) while write is being propagated into the UIM-IO-controller. An additional read/write to UIM_CFG/UIM_CMD register should be done while high. De-asserted when write finishes propagating.
1	UIM_DEACTIVATION_STATUS	Indicates whether a HW deactivation sequence was performed and sequence recovery hasn't been executed yet. This bit is set when the sequence is initiated either by a HW trigger (batt_alarm, card-removal) or a SW trigger ('initiate_shutdown_sequence' bit in CMD register). Cleared after executing a 'recover_from_hw_deactivation' command.
0	CARD_PRESENCE	Indication on card presence in slot. High when card present, low otherwise.

0x078B018C BLSP1_BLSP_UART1_UART_DM_UIM_IRQ_ISR**Type:** R**Clock:** AHB_CLK**Reset State:** 0x00000000**BLSP1_BLSP_UART1_UART_DM_UIM_IRQ_ISR**

Bits	Name	Description
4	UIM_IO_WRITE_DONE	Asserted when UIM-write had finished. Cleared by writing '1' to matching field in UART_DM_UIM_IRQ_CLR.

BLSP1_BLSP_UART1_UART_DM_UIM_IRQ_ISR (cont.)

Bits	Name	Description
3	HW_SEQUENCE_FINISH	Asserted when UIM-controller's HW-deactivation sequence has been initiated and finished. Cleared by writing '1' to matching field in UART_DM_UIM_IRQ_CLR.
2	BATT_ALARM	Set to high when battery-alarm indication received. Cleared by writing '1' to matching field in UART_DM_UIM_IRQ_CLR.
1	UIM_CARD_INSERTION	Set to high when card insertion indication received. Cleared by writing '1' to matching field in UART_DM_UIM_IRQ_CLR.
0	UIM_CARD_REMOVAL	Set to high when card removal indication received. Cleared by writing '1' to matching field in UART_DM_UIM_IRQ_CLR.

0x078B0190 BLSP1_BLSP_UART1_UART_DM_UIM_IRQ_MISR**Type:** R**Clock:** AHB_CLK**Reset State:** 0x00000000**BLSP1_BLSP_UART1_UART_DM_UIM_IRQ_MISR**

Bits	Name	Description
4	UIM_IO_WRITE_DONE	See UART_DM_UIM_IRQ_ISR for field descriptions.
3	HW_SEQUENCE_FINISH	See UART_DM_UIM_IRQ_ISR for field descriptions.
2	BATT_ALARM	See UART_DM_UIM_IRQ_ISR for field descriptions.
1	UIM_CARD_INSERTION	See UART_DM_UIM_IRQ_ISR for field descriptions.
0	UIM_CARD_REMOVAL	See UART_DM_UIM_IRQ_ISR for field descriptions.

0x078B0194 BLSP1_BLSP_UART1_UART_DM_UIM_IRQ_CLR**Type:** W**Clock:** AHB_CLK**Reset State:** Undefined**BLSP1_BLSP_UART1_UART_DM_UIM_IRQ_CLR**

Bits	Name	Description
4	UIM_IO_WRITE_DONE	Clears matching bit in UART_DM_UIM_IRQ_ISR/MISR.
3	HW_SEQUENCE_FINISH	Clears matching bit in UART_DM_UIM_IRQ_ISR/MISR.
2	BATT_ALARM	Clears matching bit in UART_DM_UIM_IRQ_ISR/MISR.
1	UIM_CARD_INSERTION	Clears matching bit in UART_DM_UIM_IRQ_ISR/MISR.
0	UIM_CARD_REMOVAL	Clears matching bit in UART_DM_UIM_IRQ_ISR/MISR.

0x078B0198 BLSP1_BLSP_UART1_UART_DM_UIM_IRQ_IMR**Type:** RW**Clock:** AHB_CLK**Reset State:** 0x00000000**BLSP1_BLSP_UART1_UART_DM_UIM_IRQ_IMR**

Bits	Name	Description
4	UIM_IO_WRITE_DONE	When set, matching bit in UART_DM_UIM_IRQ_MISR will reflect the matching bit in UART_DM_UIM_IRQ_ISR. Otherwise, the matching bit in UART_DM_UIM_IRQ_MISR will be a constant '0'.
3	HW_SEQUENCE_FINISH	When set, matching bit in UART_DM_UIM_IRQ_MISR will reflect the matching bit in UART_DM_UIM_IRQ_ISR. Otherwise, the matching bit in UART_DM_UIM_IRQ_MISR will be a constant '0'.
2	BATT_ALARM	When set, matching bit in UART_DM_UIM_IRQ_MISR will reflect the matching bit in UART_DM_UIM_IRQ_ISR. Otherwise, the matching bit in UART_DM_UIM_IRQ_MISR will be a constant '0'.
1	UIM_CARD_INSERTION	When set, matching bit in UART_DM_UIM_IRQ_MISR will reflect the matching bit in UART_DM_UIM_IRQ_ISR. Otherwise, the matching bit in UART_DM_UIM_IRQ_MISR will be a constant '0'.
0	UIM_CARD_REMOVAL	When set, matching bit in UART_DM_UIM_IRQ_MISR will reflect the matching bit in UART_DM_UIM_IRQ_ISR. Otherwise, the matching bit in UART_DM_UIM_IRQ_MISR will be a constant '0'.

0x078B019C BLSP1_BLSP_UART1_UART_DM_UIM_IRQ_IMR_SET**Type:** W**Clock:** AHB_CLK**Reset State:** Undefined**BLSP1_BLSP_UART1_UART_DM_UIM_IRQ_IMR_SET**

Bits	Name	Description
4	UIM_IO_WRITE_DONE	Write '1' to set the matching bit in UART_DM_UIM_IRQ_IMR.
3	HW_SEQUENCE_FINISH	Write '1' to set the matching bit in UART_DM_UIM_IRQ_IMR.
2	BATT_ALARM	Write '1' to set the matching bit in UART_DM_UIM_IRQ_IMR.
1	UIM_CARD_INSERTION	Write '1' to set the matching bit in UART_DM_UIM_IRQ_IMR.
0	UIM_CARD_REMOVAL	Write '1' to set the matching bit in UART_DM_UIM_IRQ_IMR.

0x078B01A0 BLSP1_BLSP_UART1_UART_DM_UIM_IRQ_IMR_CLR**Type:** W**Clock:** AHB_CLK**Reset State:** Undefined**BLSP1_BLSP_UART1_UART_DM_UIM_IRQ_IMR_CLR**

Bits	Name	Description
4	UIM_IO_WRITE_DONE	Write '1' to clear the matching bit in UART_DM_UIM_IRQ_IMR.
3	HW_SEQUENCE_FINISH	Write '1' to clear the matching bit in UART_DM_UIM_IRQ_IMR.
2	BATT_ALARM	Write '1' to clear the matching bit in UART_DM_UIM_IRQ_IMR.
1	UIM_CARD_INSERTION	Write '1' to clear the matching bit in UART_DM_UIM_IRQ_IMR.
0	UIM_CARD_REMOVAL	Write '1' to clear the matching bit in UART_DM_UIM_IRQ_IMR.

20.4 BLSP1_BLSP_QUP0_QUP**0x078B5000 BLSP1_BLSP_QUP0_QUP_CONFIG****Type:** RW**Clock:** crif_clk**Reset State:** 0x00000000

Unless otherwise stated, register bits written return the value when read.

a. The contents of this register should only be changed when in the RESET_STATE (see the QUP_STATE register).

b. The value written to the QUP output FIFO, is shifted left toward the MSB before passing it to the mini-core as follows:

o N equals 8 or less - shift 24

o N equals 16 to 9 - shift 16

o N equals 24 to 17 - shift 8

o N equals 32 to 25 - no shift

BLSP1_BLSP_QUP0_QUP_CONFIG

Bits	Name	Description
16	EN_EXT_OUT_FLAG	Enable Extended OUTPUT/INPUT_SERVICE_FLAG interrupt generation. When this bit is set, the behavior of the *_SERVICE_FLAG and MAX_OUTPUT_DONE_FLAG changes. Service flag and the done flag are not set till the last SPI or I2C transfer has completed. a) For SPI: 1. The shifting needs to complete 2. Clock stopped if CLK_ALWAYS_ON=0, 3. CS, if used, de-asserted b) For I2C, 1. The ACK needs to complete 2. STOP condition done generated (Attention need to be put in synchronizing the last transaction with STOP tag) 3. clock and data wires return to high condition When this bit is clear, the legacy behavior is applicable. Under legacy, the DONE_FLAG is set when: 1. MX_*_COUNT is reached 2. The output FIFO is empty 3. Last bit was sent on the link Under legacy, the flag interrupt is generated before the last transfer is complete on the external interface.
15	CORE_EXTRA_CLK_ON_EN	Enable additional transfer based qup_core_clk gating for power save. When set to 0, clock on is turned on unconditionally. When set to 1, dynamic clock on is turned on.
14	FIFO_CLK_GATE_EN	When set to 0, fifo clock is mostly on - legacy mode. When set to 1, fifo clock is turned off dynamically.
13	CORE_CLK_ON_EN	When set to 0, clock on is turned on unconditionally. When set to 1, dynamic clock on is turned on and driven high conditionally.
12	APP_CLK_ON_EN	When set to 0, clock on is turned on unconditionally. When set to 1, dynamic clock on is turned on and driven high conditionally.
11:8	MINI_CORE	value: 0000 null core value: 0001 SPI core value: 0010 I2C master controller value: 0011 reserved value: 0100 reserved value: 0101 reserved value: 0110 reserved value: 0111 reserved See Note 1.
7	NO_INPUT	qup_data_in is not used and the value is a 'don't care'. The QUP input FIFO is always empty. The input service interrupt and INPUT_SERVICE_FLAG bit are never set. See notes (a) and (b) above.

BLSP1_BLSP_QUP0_QUP_CONFIG (cont.)

Bits	Name	Description
6	NO_OUTPUT	qup_data_out is not used and is held at the value zero. The QUP output FIFO is always empty. The output service flag and OUTPUT_SERVICE_FLAG bit are never set. qup_data_out is still driven when SPI_CS#_N is asserted. The setting for NO_TRI_STATE still applies. See notes (a) and (b) above.
5	QUP_HREADY_CTRL	When set to 0, qup will not stall the AHB bus (legacy mode). When set to 1, qup may stall the AHB bus until register access is done.
4:0	N	Number of logical bits N in the mini-core that constitutes a single transfer. The value zero indicates N equals one. The value of all ones indicates N equals 32. The value in this register must be 3 (i.e., N is 4 or higher in order for the STATE field to be set to RUN_STATE). See note (a) above.

0x078B5004 BLSP1_BLSP_QUP0_QUP_STATE**Type:** RW**Clock:** crif_clk**Reset State:** 0x0000001C**BLSP1_BLSP_QUP0_QUP_STATE**

Bits	Name	Type	Description
6	I2C_FLUSH	RW	Flushing an i2c transfer requires using version 2 tags. FLUSH should be used in BAM mode only. Setting this bit to 1 will flush all tags and tag related data besides EOT until FLUSH STOP tag is encountered. Setting this bit to 0 has no impact. Reading this bit returns the flush operation status - 0 = ongoing, 1 = done.
5	WAIT_FOR_EOT	RW	Only applicable when moving to RUN_STATE using command descriptor in SPI mode with NO_OUTPUT. Setting this bit to 1 will stall the done_toggle_com indication until basi_trans_end_pro is asserted. This will prevent the next command descriptor to start prematurely.
4	I2C_MAST_GEN	R	Read only. Reflects the RTL generic setting for GEN_I2C_MASTER_MINI_CORE.
3	SPI_GEN	R	Read only. Reflects the RTL generic setting for GEN_SPI_MINI_CORE.
2	STATE_VALID	R	Read only. If and only if set to one, writes to the STATE field is allowed or reads from the STATE field is valid. Writes to this bit is ignored.

BLSP1_BLSP_QUP0_QUP_STATE (cont.)

Bits	Name	Type	Description
1:0	STATE	RW	When clear (00), the mini-core and related logic is held in RESET_STATE. When set to '01', the mini-core and related logic is released from reset and enters the RUN_STATE. When set to '11', the mini-core and related logic enters the PAUSE_STATE at the next appropriate point in time. Writing (10) to this field clears these two bits. For PAUSE_STATE to RESET_STATE transition, two writes of (10) are required for the transition to complete.

0x078B5008 BLSP1_BLSP_QUP0_QUP_IO_MODES**Type:** RW**Clock:** crif_clk**Reset State:** 0x000000A5

Unless otherwise stated, register bits written return the value when read.

Notes:

a. 'Packing' occurs as follows:

- N equals 8 or less - pack four values into each QUP input FIFO word.
- N equals 16 to 9 - pack two values into each QUP input FIFO word.
- N equals 32 to 17 - no packing. Each mini-core value is moved to an QUP input FIFO word.

b. 'Un-Packing' occurs as follows:

- N equals 8 or less - un-pack four values from each QUP output FIFO word.
- N equals 16 to 9 - un-pack two values from each QUP output FIFO word.
- N equals 32 to 17 - no packing. Each QUP output FIFO value is moved to the mini-core.

c. INPUT_MODE and OUTPUT_MODE should be both in BAM_Mode or both in non BAM_Mode.

BLSP1_BLSP_QUP0_QUP_IO_MODES

Bits	Name	Type	Description
16	OUTPUT_BIT_SHIFT_EN	RW	If set, enables the QUP output FIFO block to do bit shifting on the output data.
15	PACK_EN	RW	Indicates data values from the mini-core are to be packed before placement in the QUP input FIFO. See note (a) above.
14	UNPACK_EN	RW	Indicates data values taken from the QUP output FIFO should be unpacked before passing to the mini-core. See note (b) above.

BLSP1_BLSP_QUP0_QUP_IO_MODES (cont.)

Bits	Name	Type	Description
13:12	INPUT_MODE	RW	Value 00: FIFO_Mode Value 01: Block_Mode Value 10: Reserved Value 11: BAM_Mode (note if the RTL generic BLOCK_SIZE_INPUT = 0, then only FIFO_Mode is available)
11:10	OUTPUT_MODE	RW	Value 00: FIFO_Mode Value 01: Block_Mode Value 10: Reserved Value 11: BAM_Mode (note if the RTL generic BLOCK_SIZE_OUTPUT = 0, then only FIFO_Mode is available)
9:7	INPUT_FIFO_SIZE	R	Read only, actual value set by RTL generic. Value 000: 2X BLOCK_SIZE Value 001: 4X BLOCK_SIZE Value 010: 8X BLOCK_SIZE Value 011: 16X BLOCK_SIZE
6:5	INPUT_BLOCK_SIZE	R	Read only. Actual value set by RTL generic. Indicates the block size associated with Block_Mode for input. Value 00: 4 Bytes (FIFO_Mode only) Value 01: 16 Bytes Value 10: 32 Bytes Value 11: Reserved
4:2	OUTPUT_FIFO_SIZE	R	Read only. Actual value set by RTL GENERIC. Value 000: 2X BLOCK_SIZE Value 001: 4X BLOCK_SIZE Value 010: 8X BLOCK_SIZE Value 011: 16X BLOCK_SIZE
1:0	OUTPUT_BLOCK_SIZE	R	Read only. Actual value set by RTL GENERIC. Value 00: 04 Bytes (FIFO mode only) Value 01: 16 Bytes Value 10: 32 Bytes Value 11: Reserved

0x078B500C BLSP1_BLSP_QUP0_QUP_SW_RESET**Type:** W**Clock:** crif_clk**Reset State:** 0x00000000

A write to this register resets the entire QUP core and all mini-cores. The internal registers are brought back to their reset values. Reading of this register returns zero. The AHB clock domain reset will stay asserted until either the spi_clk or i2c_clk domains are reset. It is prohibited to write to any QUP register during this period and failing to do so will cause an ERROR response on the

AHB bus. In order to avoid this SW should poll the QUP_STATE[STATE_VALID] bit until it is asserted.

BLSP1_BLSP_QUP0_QUP_SW_RESET

Bits	Name	Description
31:0	QUP_SW_RESET	

0x078B5014 BLSP1_BLSP_QUP0_QUP_TRANSFER_CANCEL

Type: RW

Clock: crif_clk

Reset State: 0x00000000

BLSP1_BLSP_QUP0_QUP_TRANSFER_CANCEL

Bits	Name	Type	Description
15:8	TRANSFER_CANCEL_ID	RW	This field include the id of the transfer that is scheduled for cancellation. This field is valid when asserting TRANSFER_CANCEL bit. A value of 0xFF will cancel the current transfer regardless of its LOCAL_ID. Canceling an i2c transfer requires using version 2 tags.
7	TRANSFER_CANCEL	W	Setting this bit to 1 will cause the HW to go to pause_state when the LOCAL_ID = TRANSFER_CANCEL_ID. This would allow flushing the rest of the transfer in order to implement cancel API. Canceling an i2c transfer requires using version 2 tags.

0x078B5018 BLSP1_BLSP_QUP0_QUP_OPERATIONAL

Type: RW

Clock: crif_clk

Reset State: 0x000000C0

Refer to the chapter on 'QUP Software Considerations' regarding the correct interpretation of the various service interrupts generated by QUP and their relation to these register bits.

BLSP1_BLSP_QUP0_QUP_OPERATIONAL

Bits	Name	Description
15	NWD	Notify When Done (NWD). Read only status bit which when set indicates a NWD acknowledgement is outstanding. This bit is set by QUP hardware when NWD is signaled from BAM. This bit is cleared by QUP hardware when the NWD request is acknowledged by assertion of done_toggle. This bit is always unconditionally cleared in RESET_STATE.

BLSP1_BLSP_QUP0_QUP_OPERATIONAL (cont.)

Bits	Name	Description
14	DONE_TOGGLE	Read only status bit which provides the current state of the side-band done_toggle signal sent to BAM. At each NWD acknowledgement, this bit toggles.
13	IN_BLOCK_READ_REQ	Read only. When set by hardware, indicates QUP input FIFO has BLOCK_SIZE_INPUT amount of data ready for reading. Valid only in Block_Mode.
12	OUT_BLOCK_WRITE_REQ	Read only. When set by hardware, indicates QUP output FIFO needs BLOCK_SIZE_OUTPUT amount of data to be written. Valid only in Block_Mode.
11	MAX_INPUT_DONE_FLAG	Read only. When set by hardware, indicates QUP input FIFO has reached the programmed QUP_MX_INPUT_COUNT value. Valid in FIFO_Mode (only if MX_READ_COUNT is non-zero), Block_Mode and Bam_Mode (for debug only).
10	MAX_OUTPUT_DONE_FLAG	Read only. When set by hardware, indicates QUP output FIFO has reached the programmed QUP_MX_OUTPUT_COUNT value. Valid in Block_Mode and Bam_Mode (for debug only).
9	INPUT_SERVICE_FLAG	When set by hardware, indicates QUP input FIFO has an outstanding input service request. At the point in time this bit is set, the hardware also asserts qup_irq. Writing a 'zero' to this bit does nothing. Writing a 'one' to this bit clears it and acknowledges software has or will read the data. Valid in all modes , recommended to be masked in BAM mode
8	OUTPUT_SERVICE_FLAG	When set by hardware, indicates QUP output FIFO has an outstanding output service request. At the point in time this bit is set, the hardware also asserts qup_irq. Writing a 'zero' to this bit does nothing. Writing a 'one' to this bit clears it and acknowledges software has or will read the data. Valid in all modes , recommended to be masked in BAM mode
7	INPUT_FIFO_FULL	Read only. When set, indicates the input FIFO is full and can accept no more data from the QUP mini-core.
6	OUTPUT_FIFO_FULL	Read only. When set, indicates the output FIFO is full and can accept no more CRIF writes.
5	INPUT_FIFO_NOT_EMPTY	Read only. When set, indicates the input FIFO has at least one value in it to be read. When clear, indicates the input FIFO is empty.
4	OUTPUT_FIFO_NOT_EMPTY	Read only. When set, indicates the output FIFO has at least one value in it to be shifted out. When clear, indicates the output FIFO is empty.

0x078B501C BLSP1_BLSP_QUP0_QUP_ERROR_FLAGS**Type:** RW**Clock:** crif_clk**Reset State:** 0x00000000

All bits in this register are set by hardware and remain set until cleared by software. Writing a 'one' to a bit clears it while writing a 'zero' leaves the bit unchanged.

BLSP1_BLSP_QUP0_QUP_ERROR_FLAGS

Bits	Name	Description
5	OUTPUT_OVER_RUN_ERR	Indicates the output FIFO was full when a CRIF write was attempted to the FIFO. The write is discarded.
4	INPUT_UNDER_RUN_ERR	Indicates the input FIFO was empty when a CRIF read was attempted to the FIFO. The read value returns is indeterminate.
3	OUTPUT_UNDER_RUN_ERR	Indicates the output FIFO was empty when an output shift operation required a value.
2	INPUT_OVER_RUN_ERR	Indicates the input FIFO was full when an input shift operation provided a new value. When this happens, the new value is discarded.

0x078B5020 BLSP1_BLSP_QUP0_QUP_ERROR_FLAGS_EN

Type: RW

Clock: crif_clk

Reset State: 0x0000003C

Each bit in this register controls the setting of a corresponding error flag. If set, each bit enables generation of qup_irq and the setting of the corresponding error flag in the QUP_ERROR_FLAGS register for the specified error case. At reset, all error enable bits are set to '1'.

BLSP1_BLSP_QUP0_QUP_ERROR_FLAGS_EN

Bits	Name	Description
5	OUTPUT_OVER_RUN_ERR_EN	If set, enables output over run error generation.
4	INPUT_UNDER_RUN_ERR_EN	If set, enables input under run error generation.
3	OUTPUT_UNDER_RUN_ERR_EN	If set, enables output under run error generation.
2	INPUT_OVER_RUN_ERR_EN	If set, enables input over run error generation.

0x078B5028 BLSP1_BLSP_QUP0_QUP_OPERATIONAL_MASK

Type: RW

Clock: crif_clk

Reset State: 0x00000000

This register masks several QUP_OPERATIONAL flags from creating an interrupt .

BLSP1_BLSP_QUP0_QUP_OPERATIONAL_MASK

Bits	Name	Description
9	INPUT_SERVICE_MASK	If set, this flag in QUP_OPERATIONAL does not cause an interrupt but provides status only.
8	OUTPUT_SERVICE_MASK	If set, this flag in QUP_OPERATIONAL does not cause an interrupt but provides status only.

0x078B5100 BLSP1_BLSP_QUP0_QUP_MX_OUTPUT_COUNT

Type: RW

Clock: crif_clk

Reset State: 0x00000000

Unless otherwise stated, register bits written return the value when read. This register only applies to Block_Mode. The counter decrements for each output transfer when the STATE field (see QUP_STATE register) is moved from the RESET_STATE to the RUN_STATE. The PAUSE_STATE does not effect the count.

Notes:

a. Allows the total number of Mini Core transfers to be less than an exact multiple of OUTPUT_BLOCK_SIZE. Any additional outputs are discarded.

BLSP1_BLSP_QUP0_QUP_MX_OUTPUT_COUNT

Bits	Name	Type	Description
31	MX_CONFIG_DURING_RUN	W	Write only value. When reset to zero the MX_OUTPUT_COUNT value only applies in reset_state (legacy behavior). When set to one the MX_OUTPUT_COUNT value will also apply in run_state and allow writing the transaction data in a few portions when not knowing the transaction size in advance. Reconfiguration during run is only allowed after the last portion has ended (i.e. MAX_OUTPUT_DONE_FLAG was asserted)
15:0	MX_OUTPUT_COUNT	RW	Number of writes of size N to the mini-core per RUN_STATE. A value of zero indicates the output count function is not enabled for use. See note (a) above.

0x078B5104 BLSP1_BLSP_QUP0_QUP_MX_OUTPUT_CNT_CURRENT**Type:** R**Clock:** crif_clk**Reset State:** 0x00000000**BLSP1_BLSP_QUP0_QUP_MX_OUTPUT_CNT_CURRENT**

Bits	Name	Description
15:0	MX_OUTPUT_CNT_CURRENT	Current value of QUP_MX_OUTPUT_COUNT.

0x078B510C BLSP1_BLSP_QUP0_QUP_OUTPUT_FIFO_WORD_CNT**Type:** R**Clock:** crif_clk**Reset State:** 0x00000000

This register holds the number of words in the output FIFO at a given time. Note the fields in this register are dynamically updated hence when this register is read during a FIFO operation, the value of this register could have been altered by the time SW gets the read value. Recommended for debug purposes.

BLSP1_BLSP_QUP0_QUP_OUTPUT_FIFO_WORD_CNT

Bits	Name	Description
8:0	OUTPUT_FIFO_WORD_CNT	Number of words in the output FIFO.

**0x078B5110+ BLSP1_BLSP_QUP0_QUP_OUTPUT_FIFOc, c=[0..15]
0x4*c****Type:** W**Clock:** crif_clk**Reset State:** 0x00000000

Note that consecutive writes to this address keeps filling up the output FIFO. The max address to address the output FIFO is 0x14C.

BLSP1_BLSP_QUP0_QUP_OUTPUT_FIFOc

Bits	Name	Description
31:0	OUTPUT	Value to be shifted out.

0x078B5150 BLSP1_BLSP_QUP0_QUP_MX_WRITE_COUNT**Type:** RW**Clock:** crif_clk**Reset State:** 0x00000000

What the QUP_MX_OUTPUT_COUNT register means to Block_Mode and Bam_Mode, this register means the same to FIFO_mode. If this register is non-zero, then the qup_irq is asserted after shifting in the number of shifts specified by this register.

Unlike the QUP_MX_OUTPUT_COUNT register case, the SW should not program the QUP_MX_WRITE_COUNT value to be more than the 'actual depth' of the FIFO (BLOCK_SIZE_OUTPUT * FIFO_SIZE_OUTPUT).

BLSP1_BLSP_QUP0_QUP_MX_WRITE_COUNT

Bits	Name	Type	Description
31	MX_CONFIG_DURING_RUN	W	Write only value. When reset to zero the MX_WRITE_COUNT value only applies in reset_state (legacy behavior). When set to one the MX_WRITE_COUNT value will also apply in run_state and allow writing the transaction data in a few portions when not knowing the transaction size in advance. Reconfiguration during run is only allowed after the last portion has ended (i.e. MAX_OUTPUT_DONE_FLAG was asserted)
15:0	MX_WRITE_COUNT	RW	The number of 'writes' of size N. This is used only if the core is in FIFO_Mode.

0x078B5154 BLSP1_BLSP_QUP0_QUP_MX_WRITE_CNT_CURRENT**Type:** R**Clock:** crif_clk**Reset State:** 0x00000000**BLSP1_BLSP_QUP0_QUP_MX_WRITE_CNT_CURRENT**

Bits	Name	Description
15:0	MX_WRITE_CNT_CURRENT	Current value of QUP_MX_WRITE_COUNT counter.

0x078B5200 BLSP1_BLSP_QUP0_QUP_MX_INPUT_COUNT**Type:** RW**Clock:** crif_clk**Reset State:** 0x00000000

Unless otherwise stated, register bits written return the value when read. This register only applies to Block_Mode and Bam_Mode (for non-balanced SPI). The counter decrements for each input transfer when the STATE field (see QUP_STATE register) is moved from the RESET_STATE to the RUN_STATE. The PAUSE_STATE does not affect the count.

Notes:

a. Allows the number of shift register transfers to be less than an exact multiple of INPUT_BLOCK_SIZE. When count reached, remainder of INPUT_BLOCK_SIZE is filled with zeroes.

BLSP1_BLSP_QUP0_QUP_MX_INPUT_COUNT

Bits	Name	Type	Description
31	MX_CONFIG_DURING_RUN	W	Write only value. When reset to zero the MX_INPUT_COUNT value only applies in reset_state (legacy behavior). When set to one the MX_INPUT_COUNT value will also apply in run_state and allow writing the transaction data in a few portions when not knowing the transaction size in advance. Reconfiguration during run is only allowed after the last portion has ended (i.e. MAX_INPUT_DONE_FLAG was asserted)
15:0	MX_INPUT_COUNT	RW	Number of reads of size N from the mini-core per RUN_STATE. A value of zero indicates the input count function is not enabled for use. See note (a) above.

0x078B5204 BLSP1_BLSP_QUP0_QUP_MX_INPUT_CNT_CURRENT

Type: R

Clock: crif_clk

Reset State: 0x00000000

BLSP1_BLSP_QUP0_QUP_MX_INPUT_CNT_CURRENT

Bits	Name	Description
15:0	MX_INPUT_CNT_CURRENT	Current value of QUP_MX_INPUT_COUNT.

0x078B5208 BLSP1_BLSP_QUP0_QUP_MX_READ_COUNT

Type: RW

Clock: crif_clk

Reset State: 0x00000000

What the QUP_MX_INPUT_COUNT register means to Block_Mode and Bam_Mode, this register means the same to FIFO_mode. If this register is non-zero, then the qup_input_service_irq is asserted after shifting in the number of shifts specified by this register.

Unlike the QUP_MX_INPUT_COUNT register case, the SW should not program the QUP_MX_READ_COUNT value to be more than the 'actual depth' of the FIFO (BLOCK_SIZE_INPUT * FIFO_SIZE_INPUT). If the SPI mini-core and slave operation is enabled, then an InputOverRun error will result.

BLSP1_BLSP_QUP0_QUP_MX_READ_COUNT

Bits	Name	Type	Description
31	MX_CONFIG_DURING_RUN	W	Write only value. When reset to zero the MX_READ_COUNT value only applies in reset_state (legacy behavior). When set to one the MX_READ_COUNT value will also apply in run_state and allow writing the transaction data in a few portions when not knowing the transaction size in advance. Reconfiguration during run is only allowed after the last portion has ended (i.e. MAX_INPUT_DONE_FLAG was asserted)
15:0	MX_READ_COUNT	RW	The number of 'reads' of size N. This is used only if the core is in FIFO_Mode.

0x078B520C BLSP1_BLSP_QUP0_QUP_MX_READ_CNT_CURRENT

Type: R

Clock: crif_clk

Reset State: 0x00000000

BLSP1_BLSP_QUP0_QUP_MX_READ_CNT_CURRENT

Bits	Name	Description
15:0	MX_READ_CNT_CURRENT	Current value of QUP_MX_READ_COUNT counter.

0x078B5214 BLSP1_BLSP_QUP0_QUP_INPUT_FIFO_WORD_CNT

Type: R

Clock: crif_clk

Reset State: 0x00000000

This register holds the number of words in the input FIFO at a given time. Note the fields in this register are dynamically updated. Hence, when this register is read during a FIFO operation, the value of this register could have been altered by the time SW gets the read value. Recommended for debug purposes.

BLSP1_BLSP_QUP0_QUP_INPUT_FIFO_WORD_CNT

Bits	Name	Description
8:0	INPUT_FIFO_WORD_CNT	Number of words in the input FIFO.

0x078B5218+ BLSP1_BLSP_QUP0_QUP_INPUT_FIFOc, c=[0..15]
0x4*c

Type: R

Clock: crif_clk

Reset State: Undefined

Consecutive reads to this address reads the input FIFO contents on a FIFO basis. The max address to read the input FIFO is 0x254.

BLSP1_BLSP_QUP0_QUP_INPUT_FIFOc

Bits	Name	Description
31:0	INPUT	Value shifted in.

0x078B5300 BLSP1_BLSP_QUP0_SPI_CONFIG

Type: RW

Clock: crif_clk

Reset State: 0x00000000

Unless otherwise stated, register bits written return the value when read.

The contents of this register should only be changed when in the RESET_STATE. See the QUP_STATE register. Both NO_OUTPUT and NO_INPUT set to zero provides default full duplex operation. Setting both these bits to one is not a legal operational state.

BLSP1_BLSP_QUP0_SPI_CONFIG

Bits	Name	Type	Description
10	HS_MODE	RW	When set, SPI HS_MODE is enabled. When clear, SPI HS_MODE is disabled.
9	INPUT_FIRST	RW	When set, INPUT FIRST SPI protocol is used. When clear, OUTPUT FIRST SPI protocol is used.
8	LOOP_BACK	RW	Loopback is only valid in non-HS mode. Always clear for normal operation. If set, loop back on SPI_DATA_MO_SI under MASTER operation or SPI_DATA_MI_SO under SLAVE operation.
5	SLAVE_OPERATION	R	This register is writable only when the hardware signal spi_slave_en is asserted. When set, the SPI is configured for SLAVE operation. Zero indicates MASTER operation.

0x078B5304 BLSP1_BLSP_QUP0_SPI_IO_CONTROL**Type:** RW**Clock:** crif_clk**Reset State:** 0x00000000

The contents of this register should only be changed when in the RESET_STATE. See the QUP_STATE register.

Unless otherwise stated, register bits written return the value when read.

BLSP1_BLSP_QUP0_SPI_IO_CONTROL

Bits	Name	Description
11	FORCE_CS	When this bit is set, the chip select is asserted unconditionally with no relationship to QUP_STATE or transaction state.
10	CLK_IDLE_HIGH	When set to 1, spi_clk will be high when spi is idle (spi_cs is not asserted). When cleared to 0, spi_clk will be low when spi is idle (spi_cs is not asserted).
9	CLK_ALWAYS_ON	When MASTER operation is used, run SPI_CLK during IDLE.
8	MX_CS_MODE	If this bit is set, then for a given RUN state, the associated chip select will be asserted for the first N-bit transfer and will be kept asserted till the last programmed transfer. Applies only in MASTER mode. The number of transfers is determined by SPI_MX_OUTPUT_COUNT and/or SPI_MX_INPUT_COUNT registers.
7:4	CS_N_POLARITY	These four bits control the polarity of four SPI_CS#_N respectively. Setting any of this bit to '1', makes the associated SPI_CS#_N active HIGH. This field is a 'don't care' in SLAVE operation.
3:2	CS_SELECT	When MASTER operation is used, controls the assertion of SPI_CS_#N pins. Selects SPI_CS_N if 00, SPI_CS1_N if 01, SPI_CS2_N if 10, or SPI_CS3_N if 11. The deselected ChipSelects will be driven to inactive state dictated by the field CS_N_POLARITY. This field is a 'don't care' in SLAVE operation.
1	TRISTATE_CS	When set, drives Z on all SPI_CS#_N lines. When clear, enables normal functionality on these lines. This bit can be used to support deployment of the core as one of the masters in a multi-master configuration.
0	NO_TRI_STATE	When set, spi_data_out is not taken tri-state when SPI_CS#_N is de-asserted. This bit is normally set for MASTER operation but may be optionally left cleared.

0x078B5308 BLSP1_BLSP_QUP0_SPI_ERROR_FLAGS**Type:** RW**Clock:** crif_clk**Reset State:** 0x00000000

All bits in this register are set by hardware and remain set until cleared by software. Writing a 'one' to a bit clears it while writing a 'zero' leaves the bit unchanged.

BLSP1_BLSP_QUP0_SPI_ERROR_FLAGS

Bits	Name	Description
3	TRANSFER_CANCEL_DONE	This bit is set (1) when the SPI controller has completed canceling the transfer that was scheduled for cancellation (LOCAL_ID = TRANSFER_CANCEL_ID).
2	TRANSFER_CANCEL_ID_MATCH	This bit is set (1) when the SPI controller is ready for cancel operation: it has found the transfer that was scheduled for cancellation (LOCAL_ID = TRANSFER_CANCEL_ID) and the logic is done with previous activity.
1	SPI_SLV_CLK_OVER_RUN_ERR	Number of SPI_CLK ticks which occurred while SPI_CS#_N was asserted was greater than the programmed value of N. When this happens, only the first N bits are passed to the SPI input FIFO and the output shift value is held at zero.
0	SPI_SLV_CLK_UNDER_RUN_ERR	Number of SPI_CLK ticks which occurred while SPI_CS#_N was asserted was less than the programmed value of N. When this occurs, the bits which were received are passed to the SPI input FIFO and the CURRENT output bit is forced to zero.

0x078B530C BLSP1_BLSP_QUP0_SPI_ERROR_FLAGS_EN

Type: RW

Clock: crif_clk

Reset State: 0x0000000C

Each bit in this register controls the setting of a corresponding error flag. If set, each bit enables generation of spi_error_irq and the setting of the corresponding error flag in the SPI_ERROR_FLAGS register for the specified error case. At reset, all error enable bits are set to '1'.

BLSP1_BLSP_QUP0_SPI_ERROR_FLAGS_EN

Bits	Name	Description
3	TRANSFER_CANCEL_DONE_EN	If set, enables generating the transfer cancel done indication.
2	TRANSFER_CANCEL_ID_MATCH_EN	If set, enables generating the transfer cancel ID match indication.
1	SPI_SLV_CLK_OVER_RUN_ERR_EN	If set, enables clock over run error generation.
0	SPI_SLV_CLK_UNDER_RUN_ERR_EN	If set, enables clock under run error generation.

0x078B5310 BLSP1_BLSP_QUP0_SPI_DEASSERT_WAIT**Type:** RW**Clock:** crif_clk**Reset State:** 0x00000000

This register holds the de-assertion wait time of SPI_CS#_N between consecutive N-bit transfers in MASTER operation. The wait time is specified in number of cycles of cc_spi_master_clk.

BLSP1_BLSP_QUP0_SPI_DEASSERT_WAIT

Bits	Name	Description
5:0	DEASSERT_WAIT	Number cc_spi_master_clk ticks associated with the deasserted time of SPI_CS#_N. Only applies to MASTER operation. For SLAVE operation, this field is a 'don't care'. A value of zero indicates SPI_CS#_N remains de-asserted for exactly one clock tick. A value of one, indicates two ticks, etc. All ones indicates 64 ticks.

0x078B5314 BLSP1_BLSP_QUP0_SPI_MASTER_LOCAL_ID**Type:** RW**Clock:** crif_clk**Reset State:** 0x00000000**BLSP1_BLSP_QUP0_SPI_MASTER_LOCAL_ID**

Bits	Name	Description
15:8	EXTENDED_ID	SW may use this field to specify the current sub-transfer ID to be read when TRANSFER_CANCEL_ID_MATCH is asserted. This value has no impact on HW.
7:0	LOCAL_ID	SW specifies the current transfer ID. HW will use this value to match with TRANSFER_CANCEL_ID.

0x078B5318 BLSP1_BLSP_QUP0_SPI_MASTER_COMMAND**Type:** W**Clock:** crif_clk**Reset State:** 0x00000000**BLSP1_BLSP_QUP0_SPI_MASTER_COMMAND**

Bits	Name	Description
0	RESET_CANCEL_FSM	This command should be given only when the CANCEL_FSM_STATE is in CANCEL_PENDING_STATE or PAUSE_WAIT_STATE and when the core is in PAUSE. The main use case for this bit is to allow withdrawing a cancel command.

0x078B531C BLSP1_BLSP_QUP0_SPI_MASTER_STATUS**Type:** R**Clock:** crif_clk**Reset State:** 0x00000000**BLSP1_BLSP_QUP0_SPI_MASTER_STATUS**

Bits	Name	Description
2:0	CANCEL_FSM_STATE	This 3-bit field informs the microprocessor of the state of the SPI cancel logic. Internal use only

0x078B5400 BLSP1_BLSP_QUP0_I2C_MASTER_CLK_CTL**Type:** RW**Clock:** crif_clk**Reset State:** 0x00000000

The I2C_CLK_CTL register is a read/write register that controls clock divider values.

BLSP1_BLSP_QUP0_I2C_MASTER_CLK_CTL

Bits	Name	Description
28	SCL_EXT_FORCE_LOW	When set to 0, SCL generation maintains legacy behavior. When set to 1, SCL state machine will go to the FORCED_LOW_STATE if force_low is asserted near (within 5 i2c_clk cycles = 5 * 52ns = 260ns $\approx$ MAX(Tsu;DAT)) the clock edge.
27:26	SDA_NOISE_REJECTION	Allows adding extra sampling levels on SDA to reject short low pulses. This value specifies how many TCXO cycles of logic low on SDA would be considered as valid logic low. 0x0 - legacy mode, 0x01 - one cycle wide low pulse is rejected, 0x2 - two cycles wide low pulse is rejected, 0x3 - three cycles wide low pulse is rejected
25:24	SCL_NOISE_REJECTION	Allows adding extra sampling levels on SCL to reject short low pulses. This value specifies how many TCXO cycles of logic low on SCL would be considered as valid logic low. 0x0 - legacy mode, 0x01 - one cycle wide low pulse is rejected, 0x2 - two cycles wide low pulse is rejected, 0x3 - three cycles wide low pulse is rejected
23:16	HIGH_TIME_DIVIDER_VALUE	Allows setting SCL duty cycle to non 50%. If this value is zero then legacy mode is used. If this value is non-zero then it will be used as the SCL high time counter and FS_DIVIDER_VALUE will be used as the low time counter. Minimum value is 0x7.
10:8	HS_DIVIDER_VALUE	The value in this register represents the clock period multiplier in high-speed (HS) mode. Minimum value is 0x3. $I2C_HS_CLK = I2C_CLK / (3 * (HS_DIVIDER_VALUE + 1) + 2)$

BLSP1_BLSP_QUP0_I2C_MASTER_CLK_CTL (cont.)

Bits	Name	Description
7:0	FS_DIVIDER_VALUE	The value in this register represents the clock period multiplier in fast/standard (FS) mode. Minimum value is 0x7. When HIGH_TIME_DIVIDER_VALUE=0: $I2C_FS_CLK = I2C_CLK / (2 * (FS_DIVIDER_VALUE + 3))$ When HIGH_TIME_DIVIDER_VALUE!=0: $I2C_FS_CLK = I2C_CLK / (FS_DIVIDER_VALUE + HIGH_TIME_DIVIDER_VALUE + 6)$

0x078B5404 BLSP1_BLSP_QUP0_I2C_MASTER_STATUS**Type:** RW**Clock:** crif_clk**Reset State:** 0x0C000000

The I2C_STATUS is a status register. Writing any value clears the status bits.

BLSP1_BLSP_QUP0_I2C_MASTER_STATUS

Bits	Name	Type	Description
30:28	CANCEL_FSM_STATE	R	This 3-bit field informs the microprocessor of the state of the I2C cancel logic. Internal use only 0x0: IDLE_STATE (IDLE_STATE) 0x1: CANCEL_PENDING_STATE (CANCEL_PENDING_STATE) 0x2: PAUSE_STATE (PAUSE_STATE) 0x3: PAUSE_WAIT_STATE (PAUSE_WAIT_STATE) 0x4: SEND_STOP_STATE (SEND_STOP_STATE) 0x5: FLUSH_STATE (FLUSH_STATE) 0x6: SEND_IRQ_STATE (SEND_IRQ_STATE)
27	I2C_SCL	R	Logic state of I2C bus serial clock wire.
26	I2C_SDA	R	Logic state of I2C bus serial data wire.
25	INVALID_READ_SEQ	RW	Interrupt source. This bit is set (1) when a MI_REC tag does not follow a START tag for an I2C READ. Not applicable for Halcyon.
24	INVALID_READ_ADDR	RW	Interrupt source. In version 1 tags this bit is set (1) when the I2C controller is trying to receive data from a non-existent I2C slave (address). In version 2 tags this bit is set (1) when the I2C controller is trying to access a non-existent I2C slave (address).
23	INVALID_TAG	RW	Interrupt source. This bit is set (1) when the I2C controller is trying to process data from the output FIFO that is improperly tagged.

BLSP1_BLSP_QUP0_I2C_MASTER_STATUS (cont.)

Bits	Name	Type	Description
22:20	INPUT_FSM_STATE	R	This 3-bit field informs the microprocessor of the state of the I2C MASTER INPUT FSM block. Internal use only 0x0: V1_V2_RESET_STATE (RESET_STATE value. Applies for both version 1 and version 2 tags) 0x1: V1_V2_IDLE_STATE (IDLE_STATE value. Applies for both version 1 and version 2 tags) 0x2: V1_V2_READ_LAST_BYTE_STATE (READ_LAST_BYTE_STATE value. Applies for both version 1 and version 2 tags) 0x3: V1_V2_MI_REC_STATE (MI_REC_STATE value. Applies for both version 1 and version 2 tags) 0x4: V1_V2_DEC_STATE (DEC_STATE value. Applies for both version 1 and version 2 tags) 0x5: V1_V2_STORE_STATE (STORE_STATE value. Applies for both version 1 and version 2 tags) 0x6: V2_WR_TAG_STATE (WR_TAG_STATE value. Only applicable in version 2 tags) 0x7: V2_WAIT_TAG_STATE (WAIT_TAG_STATE value. Only applicable in version 2 tags)

BLSP1_BLSP_QUP0_I2C_MASTER_STATUS (cont.)

Bits	Name	Type	Description
19:16	OUTPUT_FSM_STATE	R	This 3-bit field informs the microprocessor of the state of the I2C MASTER OUTPUT FSM block. Internal use only 0x0: V1_V2_RESET_STATE (RESET_STATE value. Applies for both version 1 and version 2 tags) 0x1: V1_V2_IDLE_STATE (IDLE_STATE value. Applies for both version 1 and version 2 tags) 0x2: V1_V2_DECODE_STATE (DECODE_STATE value. Applies for both version 1 and version 2 tags) 0x3: V1_V2_SEND_STATE (SEND_STATE value. Applies for both version 1 and version 2 tags) 0x4: V1_V2_MI_REC_STATE (MI_REC_STATE value. Applies for both version 1 and version 2 tags) 0x5: V1_V2_NOP_STATE (NOP_STATE value. Applies for both version 1 and version 2 tags) 0x6: V1_V2_INVALID_STATE (INVALID_STATE value. Applies for both version 1 and version 2 tags) 0x7: V1_PEEK_STATE (PEEK_STATE value. Only applicable in version 1 tags) 0x8: V1_SEND_R_STATE (SEND_R_STATE value. Only applicable in version 1 tags) 0xB: V2_GET_BYTE_STATE (GET_BYTE_STATE value. Only applicable in version 2 tags) 0xC: V2_GET_WRITE_BYTE_STATE (GET_WRITE_BYTE_STATE value. Only applicable in version 2 tags) 0xD: V2_SPECIALITY_STATE (SPECIALITY_STATE value. Only applicable in version 2 tags) 0xE: V2_WAIT_FLUSH_STOP_STATE (WAIT_FLUSH_STOP_STATE value. Only applicable in version 2 tags) 0xF: V2_STOP_STATE (STOP_STATE value. Only applicable in version 2 tags)
15:13	CLK_STATE	R	This 3-bit field informs the microprocessor of the state of the I2C clk_control block. Internal use only 0x0: RESET_BUSIDLE_STATE (RESET_BUSIDLE_STATE) 0x1: NOT_MASTER_STATE (NOT_MASTER_STATE) 0x2: HIGH_STATE (HIGH_STATE) 0x3: LOW_STATE (LOW_STATE) 0x4: HIGH_WAIT_STATE (HIGH_WAIT_STATE) 0x5: FORCED_LOW_STATE (FORCED_LOW_STATE) 0x6: HS_ADDR_LOW_STATE (HS_ADDR_LOW_STATE) 0x7: DOUBLE_BUFFER_WAIT_STATE (DOUBLE_BUFFER_WAIT_STATE)

BLSP1_BLSP_QUP0_I2C_MASTER_STATUS (cont.)

Bits	Name	Type	Description
12:10	DATA_STATE	R	This 3-bit field informs the microprocessor of the state of the I2C data_control block. Internal use only 0x0: RESET_WAIT_STATE (RESET_WAIT_STATE) 0x1: TX_ADDR_STATE (TX_ADDR_STATE) 0x2: TX_DATA_STATE (TX_DATA_STATE) 0x3: TX_HS_ADDR_STATE (TX_HS_ADDR_STATE) 0x4: TX_10_BIT_ADDR_STATE (TX_10_BIT_ADDR_STATE) 0x5: TX_2ND_BYTE_STATE (TX_2ND_BYTE_STATE) 0x6: RX_DATA_STATE (RX_DATA_STATE) 0x7: RX_NACK_HOLD_STATE (RX_NACK_HOLD_STATE)
9	BUS_MASTER	R	This bit is set (1) when the I2C controller is the present bus master.
8	BUS_ACTIVE	R	This bit is set (1) when the bus is in use by this, or any other controller.
7:6	FAILED	RW	Interrupt source. This 2-bit field contains the failure information of the present I2C transfers. If transmitting, failed[1] contains the information regarding the byte that has been transmitted. Failed[0] contains the information of the byte awaiting transmission if there is a byte pipelined. If no byte is pipelined, ignore this bit. If receiving, failed[1] contains the information of the byte received and stored in the buffer, and failed[0] should be ignored. For example: Value 00: Byte n transmitted successfully, byte n+1 to begin transmission Value 01: Byte n transmitted successfully, byte n+1 errored: type of error indicated in other STATUS bits. (queued invalid write, for example) Value 10: Byte n errored: type of error indicated in other STATUS bits, byte n+1 to begin transmission (byte n+1 would have to be a valid address byte for this condition to occur) Value 11: Byte n errored: type of error indicated in other STATUS bits, byte n+1 discarded as well (if the byte was a data byte destined for a NACKed address, data is useless, therefore discarded)
5	INVALID_WRITE	RW	Interrupt source. This bit is set (1) when software writes data to the I2C_DATA register that should be flagged as an address but is not.
4	ARB_LOST	RW	Interrupt source. This bit is set (1) when the controller loses arbitration for the bus. If this bit gets set (1) during the transmission, all data has been lost, and the microprocessor must re-request its transmission.

BLSP1_BLSP_QUP0_I2C_MASTER_STATUS (cont.)

Bits	Name	Type	Description
3	PACKET_NACKED	RW	Interrupt source. This bit is set high when a NACK is received from a slave. If a high speed master code is sent and there is an ACK from a slave, then this bit is set (1) to indicate that the high speed mode can not be entered. If the high speed mode is accepted by the slave, then a NACK is performed and this bit is not set (1).
2	BUS_ERROR	RW	Interrupt source. This bit is set (1) when an unexpected START or STOP condition is detected. This returns the controller to its reset state.
1	TRANSFER_CANCEL_DONE	RW	Only applicable when using version 2 tags. Interrupt source. This bit is set (1) when the I2C controller has completed canceling the transfer that was scheduled for cancellation (LOCAL_ID = TRANSFER_CANCEL_ID).
0	TRANSFER_CANCEL_ID_MATCH	RW	Only applicable when using version 2 tags. Interrupt source. This bit is set (1) when the I2C controller is ready for cancel operation: it has found the transfer that was scheduled for cancellation (LOCAL_ID = TRANSFER_CANCEL_ID) and the logic is done with previous activity.

0x078B5408 BLSP1_BLSP_QUP0_I2C_MASTER_CONFIG**Type:** RW**Clock:** crif_clk**Reset State:** 0x00000000**BLSP1_BLSP_QUP0_I2C_MASTER_CONFIG**

Bits	Name	Description
0	EN_VERSION_TWO_TAG	When set, version 2 of the mini-core is enabled.

0x078B540C BLSP1_BLSP_QUP0_I2C_MASTER_BUS_CLEAR**Type:** RW**Clock:** crif_clk**Reset State:** 0x00000000

BLSP1_BLSP_QUP0_I2C_MASTER_BUS_CLEAR

Bits	Name	Description
0	CLEAR	This command should be given only when the i2c mincore is idle. When in doubt SW can use SW_RESET to reset the minicore. When set, an I2C 'Bus Clear' executed per the I2C standard. A bus clear consists of nine I2C clock ticks with data wire left not-driven. Has the effect of clearing any slave which has lost read sync. After the clear is complete, the hardware sets the CLEAR bit to zero. The bit can not be cleared by software. After a bus clear, both clock and data lines should be high. If not, an external slave has hung the bus by holding down one or both lines. Reset the slave to clear if possible. Any slave reset mechanism is beyond the scope of the I2C mini core.

0x078B5410 BLSP1_BLSP_QUP0_I2C_MASTER_LOCAL_ID**Type:** R**Clock:** crif_clk**Reset State:** 0x00000000**BLSP1_BLSP_QUP0_I2C_MASTER_LOCAL_ID**

Bits	Name	Description
7:0	LOCAL_ID	Read only value capturing the ID byte of the last NOP LOCAL tag. Only applicable when using V2 tags

0x078B5414 BLSP1_BLSP_QUP0_I2C_MASTER_COMMAND**Type:** RW**Clock:** crif_clk**Reset State:** 0x00000000**BLSP1_BLSP_QUP0_I2C_MASTER_COMMAND**

Bits	Name	Description
0	RESET_CANCEL_FSM	This command should be given only when the CANCEL_FSM_STATE is in CANCEL_PENDING_STATE or PAUSE_WAIT_STATE and when the core is in PAUSE. The main use case for this bit is to allow withdrawing a cancel command in case the transfer was flushed (QUP_STATE[I2C_FLUSH] was set) due to I2C NACK.

20.5 BLSP1_BLSP_QUP1_QUP

0x078B6000 BLSP1_BLSP_QUP1_QUP_CONFIG

Type: RW

Clock: crif_clk

Reset State: 0x00000000

Unless otherwise stated, register bits written return the value when read.

a. The contents of this register should only be changed when in the RESET_STATE (see the QUP_STATE register).

b. The value written to the QUP output FIFO, is shifted left toward the MSB before passing it to the mini-core as follows:

o N equals 8 or less - shift 24

o N equals 16 to 9 - shift 16

o N equals 24 to 17 - shift 8

o N equals 32 to 25 - no shift

BLSP1_BLSP_QUP1_QUP_CONFIG

Bits	Name	Description
16	EN_EXT_OUT_FLAG	Enable Extended OUTPUT/INPUT_SERVICE_FLAG interrupt generation. When this bit is set, the behavior of the *_SERVICE_FLAG and MAX_OUTPUT_DONE_FLAG changes. Service flag and the done flag are not set till the last SPI or I2C transfer has completed. a) For SPI: 1. The shifting needs to complete 2. Clock stopped if CLK_ALWAYS_ON=0, 3. CS, if used, de-asserted b) For I2C, 1. The ACK needs to complete 2. STOP condition done generated (Attention need to be put in synchronizing the last transaction with STOP tag) 3. clock and data wires return to high condition When this bit is clear, the legacy behavior is applicable. Under legacy, the DONE_FLAG is set when: 1. MX_*_COUNT is reached 2. The output FIFO is empty 3. Last bit was sent on the link Under legacy, the flag interrupt is generated before the last transfer is complete on the external interface.
15	CORE_EXTRA_CLK_ON_EN	Enable additional transfer based qup_core_clk gating for power save. When set to 0, clock on is turned on unconditionally. When set to 1, dynamic clock on is turned on.

BLSP1_BLSP_QUP1_QUP_CONFIG (cont.)

Bits	Name	Description
14	FIFO_CLK_GATE_EN	When set to 0, fifo clock is mostly on - legacy mode. When set to 1, fifo clock is turned off dynamically.
13	CORE_CLK_ON_EN	When set to 0, clock on is turned on unconditionally. When set to 1, dynamic clock on is turned on and driven high conditionally.
12	APP_CLK_ON_EN	When set to 0, clock on is turned on unconditionally. When set to 1, dynamic clock on is turned on and driven high conditionally.
11:8	MINI_CORE	value: 0000 null core value: 0001 SPI core value: 0010 I2C master controller value: 0011 reserved value: 0100 reserved value: 0101 reserved value: 0110 reserved value: 0111 reserved See Note 1.
7	NO_INPUT	qup_data_in is not used and the value is a 'don't care'. The QUP input FIFO is always empty. The input service interrupt and INPUT_SERVICE_FLAG bit are never set. See notes (a) and (b) above.
6	NO_OUTPUT	qup_data_out is not used and is held at the value zero. The QUP output FIFO is always empty. The output service flag and OUTPUT_SERVICE_FLAG bit are never set. qup_data_out is still driven when SPI_CS#_N is asserted. The setting for NO_TRI_STATE still applies. See notes (a) and (b) above.
5	QUP_HREADY_CTRL	When set to 0, qup wil not stall the AHB bus (legacy mode). When set to 1, qup may stall the AHB bus until register access is done.
4:0	N	Number of logical bits N in the mini-core that constitutes a single transfer. The value zero indicates N equals one. The value of all ones indicates N equals 32. The value in this register must be 3 (i.e., N is 4 or higher in order for the STATE field to be set to RUN_STATE). See note (a) above.

0x078B6004 BLSP1_BLSP_QUP1_QUP_STATE**Type:** RW**Clock:** crif_clk**Reset State:** 0x0000001C

BLSP1_BLSP_QUP1_QUP_STATE

Bits	Name	Type	Description
6	I2C_FLUSH	RW	Flushing an i2c transfer requires using version 2 tags. FLUSH should be used in BAM mode only. Setting this bit to 1 will flush all tags and tag related data besides EOT until FLUSH STOP tag is encountered. Setting this bit to 0 has no impact. Reading this bit returns the flush operation status - 0 = ongoing, 1 = done.
5	WAIT_FOR_EOT	RW	Only applicable when moving to RUN_STATE using command descriptor in SPI mode with NO_OUTPUT. Setting this bit to 1 will stall the done_toggle_com indication until basi_trans_end_pro is asserted. This will prevent the next command descriptor to start prematurely.
4	I2C_MAST_GEN	R	Read only. Reflects the RTL generic setting for GEN_I2C_MASTER_MINI_CORE.
3	SPI_GEN	R	Read only. Reflects the RTL generic setting for GEN_SPI_MINI_CORE.
2	STATE_VALID	R	Read only. If and only if set to one, writes to the STATE field is allowed or reads from the STATE field is valid. Writes to this bit is ignored.
1:0	STATE	RW	When clear (00), the mini-core and related logic is held in RESET_STATE. When set to '01', the mini-core and related logic is released from reset and enters the RUN_STATE. When set to '11', the mini-core and related logic enters the PAUSE_STATE at the next appropriate point in time. Writing (10) to this field clears these two bits. For PAUSE_STATE to RESET_STATE transition, two writes of (10) are required for the transition to complete.

0x078B6008 BLSP1_BLSP_QUP1_QUP_IO_MODES**Type:** RW**Clock:** crif_clk**Reset State:** 0x000000A5

Unless otherwise stated, register bits written return the value when read.

Notes:

a. 'Packing' occurs as follows:

- N equals 8 or less - pack four values into each QUP input FIFO word.
- N equals 16 to 9 - pack two values into each QUP input FIFO word.
- N equals 32 to 17 - no packing. Each mini-core value is moved to an QUP input FIFO word.

b. 'Un-Packing' occurs as follows:

- N equals 8 or less - un-pack four values from each QUP output FIFO word.
- N equals 16 to 9 - un-pack two values from each QUP output FIFO word.
- N equals 32 to 17 - no packing. Each QUP output FIFO value is moved to the mini-core.

c. INPUT_MODE and OUTPUT_MODE should be both in BAM_Mode or both in non BAM_Mode.

BLSP1_BLSP_QUP1_QUP_IO_MODES

Bits	Name	Type	Description
16	OUTPUT_BIT_SHIFT_EN	RW	If set, enables the QUP output FIFO block to do bit shifting on the output data.
15	PACK_EN	RW	Indicates data values from the mini-core are to be packed before placement in the QUP input FIFO. See note (a) above.
14	UNPACK_EN	RW	Indicates data values taken from the QUP output FIFO should be unpacked before passing to the mini-core. See note (b) above.
13:12	INPUT_MODE	RW	Value 00: FIFO_Mode Value 01: Block_Mode Value 10: Reserved Value 11: BAM_Mode (note if the RTL generic BLOCK_SIZE_INPUT = 0, then only FIFO_Mode is available)
11:10	OUTPUT_MODE	RW	Value 00: FIFO_Mode Value 01: Block_Mode Value 10: Reserved Value 11: BAM_Mode (note if the RTL generic BLOCK_SIZE_OUTPUT = 0, then only FIFO_Mode is available)
9:7	INPUT_FIFO_SIZE	R	Read only, actual value set by RTL generic. Value 000: 2X BLOCK_SIZE Value 001: 4X BLOCK_SIZE Value 010: 8X BLOCK_SIZE Value 011: 16X BLOCK_SIZE
6:5	INPUT_BLOCK_SIZE	R	Read only. Actual value set by RTL generic. Indicates the block size associated with Block_Mode for input. Value 00: 4 Bytes (FIFO_Mode only) Value 01: 16 Bytes Value 10: 32 Bytes Value 11: Reserved
4:2	OUTPUT_FIFO_SIZE	R	Read only. Actual value set by RTL GENERIC. Value 000: 2X BLOCK_SIZE Value 001: 4X BLOCK_SIZE Value 010: 8X BLOCK_SIZE Value 011: 16X BLOCK_SIZE

BLSP1_BLSP_QUP1_QUP_IO_MODES (cont.)

Bits	Name	Type	Description
1:0	OUTPUT_BLOCK_SIZE	R	Read only. Actual value set by RTL GENERIC. Value 00: 04 Bytes (FIFO mode only) Value 01: 16 Bytes Value 10: 32 Bytes Value 11: Reserved

0x078B600C BLSP1_BLSP_QUP1_QUP_SW_RESET**Type:** W**Clock:** crif_clk**Reset State:** 0x00000000

A write to this register resets the entire QUP core and all mini-cores. The internal registers are brought back to their reset values. Reading of this register returns zero. The AHB clock domain reset will stay asserted until either the spi_clk or i2c_clk domains are reset. It is prohibited to write to any QUP register during this period and failing to do so will cause an ERROR response on the AHB bus. In order to avoid this SW should poll the QUP_STATE[STATE_VALID] bit until it is asserted.

BLSP1_BLSP_QUP1_QUP_SW_RESET

Bits	Name	Description
31:0	QUP_SW_RESET	

0x078B6014 BLSP1_BLSP_QUP1_QUP_TRANSFER_CANCEL**Type:** RW**Clock:** crif_clk**Reset State:** 0x00000000**BLSP1_BLSP_QUP1_QUP_TRANSFER_CANCEL**

Bits	Name	Type	Description
15:8	TRANSFER_CANCEL_ID	RW	This field include the id of the transfer that is scheduled for cancellation. This field is valid when asserting TRANSFER_CANCEL bit. A value of 0xFF will cancel the current transfer regardless of its LOCAL_ID. Canceling an i2c transfer requires using version 2 tags.
7	TRANSFER_CANCEL	W	Setting this bit to 1 will cause the HW to go to pause_state when the LOCAL_ID = TRANSFER_CANCEL_ID. This would allow flushing the rest of the transfer in order to implement cancel API. Canceling an i2c transfer requires using version 2 tags.

0x078B6018 BLSP1_BLSP_QUP1_QUP_OPERATIONAL**Type:** RW**Clock:** crif_clk**Reset State:** 0x000000C0

Refer to the chapter on 'QUP Software Considerations' regarding the correct interpretation of the various service interrupts generated by QUP and their relation to these register bits.

BLSP1_BLSP_QUP1_QUP_OPERATIONAL

Bits	Name	Description
15	NWD	Notify When Done (NWD). Read only status bit which when set indicates a NWD acknowledgement is outstanding. This bit is set by QUP hardware when NWD is signaled from BAM. This bit is cleared by QUP hardware when the NWD request is acknowledged by assertion of done_toggle. This bit is always unconditionally cleared in RESET_STATE.
14	DONE_TOGGLE	Read only status bit which provides the current state of the side-band done_toggle signal sent to BAM. At each NWD acknowledgement, this bit toggles.
13	IN_BLOCK_READ_REQ	Read only. When set by hardware, indicates QUP input FIFO has BLOCK_SIZE_INPUT amount of data ready for reading. Valid only in Block_Mode.
12	OUT_BLOCK_WRITE_REQ	Read only. When set by hardware, indicates QUP output FIFO needs BLOCK_SIZE_OUTPUT amount of data to be written. Valid only in Block_Mode.
11	MAX_INPUT_DONE_FLAG	Read only. When set by hardware, indicates QUP input FIFO has reached the programmed QUP_MX_INPUT_COUNT value. Valid in FIFO_Mode (only if MX_READ_COUNT is non-zero), Block_Mode and Bam_Mode (for debug only).
10	MAX_OUTPUT_DONE_FLAG	Read only. When set by hardware, indicates QUP output FIFO has reached the programmed QUP_MX_OUTPUT_COUNT value. Valid in Block_Mode and Bam_Mode (for debug only).
9	INPUT_SERVICE_FLAG	When set by hardware, indicates QUP input FIFO has an outstanding input service request. At the point in time this bit is set, the hardware also asserts qup_irq. Writing a 'zero' to this bit does nothing. Writing a 'one' to this bit clears it and acknowledges software has or will read the data. Valid in all modes , recommended to be masked in BAM mode
8	OUTPUT_SERVICE_FLAG	When set by hardware, indicates QUP output FIFO has an outstanding output service request. At the point in time this bit is set, the hardware also asserts qup_irq. Writing a 'zero' to this bit does nothing. Writing a 'one' to this bit clears it and acknowledges software has or will read the data. Valid in all modes , recommended to be masked in BAM mode
7	INPUT_FIFO_FULL	Read only. When set, indicates the input FIFO is full and can accept no more data from the QUP mini-core.

BLSP1_BLSP_QUP1_QUP_OPERATIONAL (cont.)

Bits	Name	Description
6	OUTPUT_FIFO_FULL	Read only. When set, indicates the output FIFO is full and can accept no more CRIF writes.
5	INPUT_FIFO_NOT_EMPTY	Read only. When set, indicates the input FIFO has at least one value in it to be read. When clear, indicates the input FIFO is empty.
4	OUTPUT_FIFO_NOT_EMPTY	Read only. When set, indicates the output FIFO has at least one value in it to be shifted out. When clear, indicates the output FIFO is empty.

0x078B601C BLSP1_BLSP_QUP1_QUP_ERROR_FLAGS**Type:** RW**Clock:** crif_clk**Reset State:** 0x00000000

All bits in this register are set by hardware and remain set until cleared by software. Writing a 'one' to a bit clears it while writing a 'zero' leaves the bit unchanged.

BLSP1_BLSP_QUP1_QUP_ERROR_FLAGS

Bits	Name	Description
5	OUTPUT_OVER_RUN_ERR	Indicates the output FIFO was full when a CRIF write was attempted to the FIFO. The write is discarded.
4	INPUT_UNDER_RUN_ERR	Indicates the input FIFO was empty when a CRIF read was attempted to the FIFO. The read value returns is indeterminate.
3	OUTPUT_UNDER_RUN_ERR	Indicates the output FIFO was empty when an output shift operation required a value.
2	INPUT_OVER_RUN_ERR	Indicates the input FIFO was full when an input shift operation provided a new value. When this happens, the new value is discarded.

0x078B6020 BLSP1_BLSP_QUP1_QUP_ERROR_FLAGS_EN**Type:** RW**Clock:** crif_clk**Reset State:** 0x0000003C

Each bit in this register controls the setting of a corresponding error flag. If set, each bit enables generation of qup_irq and the setting of the corresponding error flag in the QUP_ERROR_FLAGS register for the specified error case. At reset, all error enable bits are set to '1'.

BLSP1_BLSP_QUP1_QUP_ERROR_FLAGS_EN

Bits	Name	Description
5	OUTPUT_OVER_RUN_ERR_EN	If set, enables output over run error generation.
4	INPUT_UNDER_RUN_ERR_EN	If set, enables input under run error generation.
3	OUTPUT_UNDER_RUN_ERR_EN	If set, enables output under run error generation.
2	INPUT_OVER_RUN_ERR_EN	If set, enables input over run error generation.

0x078B6028 BLSP1_BLSP_QUP1_QUP_OPERATIONAL_MASK**Type:** RW**Clock:** crif_clk**Reset State:** 0x00000000

This register masks several QUP_OPERATIONAL flags from creating an interrupt .

BLSP1_BLSP_QUP1_QUP_OPERATIONAL_MASK

Bits	Name	Description
9	INPUT_SERVICE_MASK	If set, this flag in QUP_OPERATIONAL does not cause an interrupt but provides status only.
8	OUTPUT_SERVICE_MASK	If set, this flag in QUP_OPERATIONAL does not cause an interrupt but provides status only.

0x078B6100 BLSP1_BLSP_QUP1_QUP_MX_OUTPUT_COUNT**Type:** RW**Clock:** crif_clk**Reset State:** 0x00000000

Unless otherwise stated, register bits written return the value when read. This register only applies to Block_Mode. The counter decrements for each output transfer when the STATE field (see QUP_STATE register) is moved from the RESET_STATE to the RUN_STATE. The PAUSE_STATE does not effect the count.

Notes:

- a. Allows the total number of Mini Core transfers to be less than an exact multiple of OUTPUT_BLOCK_SIZE. Any additional outputs are discarded.

BLSP1_BLSP_QUP1_QUP_MX_OUTPUT_COUNT

Bits	Name	Type	Description
31	MX_CONFIG_DURING_RUN	W	Write only value. When reset to zero the MX_OUTPUT_COUNT value only applies in reset_state (legacy behavior). When set to one the MX_OUTPUT_COUNT value will also apply in run_state and allow writing the transaction data in a few portions when not knowing the transaction size in advance. Reconfiguration during run is only allowed after the last portion has ended (i.e. MAX_OUTPUT_DONE_FLAG was asserted)
15:0	MX_OUTPUT_COUNT	RW	Number of writes of size N to the mini-core per RUN_STATE. A value of zero indicates the output count function is not enabled for use. See note (a) above.

0x078B6104 BLSP1_BLSP_QUP1_QUP_MX_OUTPUT_CNT_CURRENT**Type:** R**Clock:** crif_clk**Reset State:** 0x00000000**BLSP1_BLSP_QUP1_QUP_MX_OUTPUT_CNT_CURRENT**

Bits	Name	Description
15:0	MX_OUTPUT_CNT_CURRENT	Current value of QUP_MX_OUTPUT_COUNT.

0x078B610C BLSP1_BLSP_QUP1_QUP_OUTPUT_FIFO_WORD_CNT**Type:** R**Clock:** crif_clk**Reset State:** 0x00000000

This register holds the number of words in the output FIFO at a given time. Note the fields in this register are dynamically updated hence when this register is read during a FIFO operation, the value of this register could have been altered by the time SW gets the read value. Recommended for debug purposes.

BLSP1_BLSP_QUP1_QUP_OUTPUT_FIFO_WORD_CNT

Bits	Name	Description
8:0	OUTPUT_FIFO_WORD_COUNT	Number of words in the output FIFO.

**0x078B6110+ BLSP1_BLSP_QUP1_QUP_OUTPUT_FIFOc, c=[0..15]
0x4*c****Type:** W**Clock:** crif_clk**Reset State:** 0x00000000

Note that consecutive writes to this address keeps filling up the output FIFO. The max address to address the output FIFO is 0x14C.

BLSP1_BLSP_QUP1_QUP_OUTPUT_FIFOc

Bits	Name	Description
31:0	OUTPUT	Value to be shifted out.

0x078B6150 BLSP1_BLSP_QUP1_QUP_MX_WRITE_COUNT**Type:** RW**Clock:** crif_clk**Reset State:** 0x00000000

What the QUP_MX_OUTPUT_COUNT register means to Block_Mode and Bam_Mode, this register means the same to FIFO_mode. If this register is non-zero, then the qup_irq is asserted after shifting in the number of shifts specified by this register.

Unlike the QUP_MX_OUTPUT_COUNT register case, the SW should not program the QUP_MX_WRITE_COUNT value to be more than the 'actual depth' of the FIFO (BLOCK_SIZE_OUTPUT * FIFO_SIZE_OUTPUT).

BLSP1_BLSP_QUP1_QUP_MX_WRITE_COUNT

Bits	Name	Type	Description
31	MX_CONFIG_DURING_RUN	W	Write only value. When reset to zero the MX_WRITE_COUNT value only applies in reset_state (legacy behavior). When set to one the MX_WRITE_COUNT value will also apply in run_state and allow writing the transaction data in a few portions when not knowing the transaction size in advance. Reconfiguration during run is only allowed after the last portion has ended (i.e. MAX_OUTPUT_DONE_FLAG was asserted)
15:0	MX_WRITE_COUNT	RW	The number of 'writes' of size N. This is used only if the core is in FIFO_Mode.

0x078B6154 BLSP1_BLSP_QUP1_QUP_MX_WRITE_CNT_CURRENT**Type:** R**Clock:** crif_clk**Reset State:** 0x00000000**BLSP1_BLSP_QUP1_QUP_MX_WRITE_CNT_CURRENT**

Bits	Name	Description
15:0	MX_WRITE_CNT_CURRENT	Current value of QUP_MX_WRITE_COUNT counter.

0x078B6200 BLSP1_BLSP_QUP1_QUP_MX_INPUT_COUNT**Type:** RW**Clock:** crif_clk**Reset State:** 0x00000000

Unless otherwise stated, register bits written return the value when read. This register only applies to Block_Mode and Bam_Mode (for non-balanced SPI). The counter decrements for each input transfer when the STATE field (see QUP_STATE register) is moved from the RESET_STATE to the RUN_STATE. The PAUSE_STATE does not affect the count.

Notes:

a. Allows the number of shift register transfers to be less than an exact multiple of INPUT_BLOCK_SIZE. When count reached, remainder of INPUT_BLOCK_SIZE is filled with zeroes.

BLSP1_BLSP_QUP1_QUP_MX_INPUT_COUNT

Bits	Name	Type	Description
31	MX_CONFIG_DURING_RUN	W	Write only value. When reset to zero the MX_INPUT_COUNT value only applies in reset_state (legacy behavior). When set to one the MX_INPUT_COUNT value will also apply in run_state and allow writing the transaction data in a few portions when not knowing the transaction size in advance. Reconfiguration during run is only allowed after the last portion has ended (i.e. MAX_INPUT_DONE_FLAG was asserted)
15:0	MX_INPUT_COUNT	RW	Number of reads of size N from the mini-core per RUN_STATE. A value of zero indicates the input count function is not enabled for use. See note (a) above.

0x078B6204 BLSP1_BLSP_QUP1_QUP_MX_INPUT_CNT_CURRENT**Type:** R**Clock:** crif_clk**Reset State:** 0x00000000**BLSP1_BLSP_QUP1_QUP_MX_INPUT_CNT_CURRENT**

Bits	Name	Description
15:0	MX_INPUT_CNT_CURRENT	Current value of QUP_MX_INPUT_COUNT.

0x078B6208 BLSP1_BLSP_QUP1_QUP_MX_READ_COUNT**Type:** RW**Clock:** crif_clk**Reset State:** 0x00000000

What the QUP_MX_INPUT_COUNT register means to Block_Mode and Bam_Mode, this register means the same to FIFO_mode. If this register is non-zero, then the qup_input_service_irq is asserted after shifting in the number of shifts specified by this register.

Unlike the QUP_MX_INPUT_COUNT register case, the SW should not program the QUP_MX_READ_COUNT value to be more than the 'actual depth' of the FIFO (BLOCK_SIZE_INPUT * FIFO_SIZE_INPUT). If the SPI mini-core and slave operation is enabled, then an InputOverRun error will result.

BLSP1_BLSP_QUP1_QUP_MX_READ_COUNT

Bits	Name	Type	Description
31	MX_CONFIG_DURING_RUN	W	Write only value. When reset to zero the MX_READ_COUNT value only applies in reset_state (legacy behavior). When set to one the MX_READ_COUNT value will also apply in run_state and allow writing the transaction data in a few portions when not knowing the transaction size in advance. Reconfiguration during run is only allowed after the last portion has ended (i.e. MAX_INPUT_DONE_FLAG was asserted)
15:0	MX_READ_COUNT	RW	The number of 'reads' of size N. This is used only if the core is in FIFO_Mode.

0x078B620C BLSP1_BLSP_QUP1_QUP_MX_READ_CNT_CURRENT**Type:** R**Clock:** crif_clk**Reset State:** 0x00000000

BLSP1_BLSP_QUP1_QUP_MX_READ_CNT_CURRENT

Bits	Name	Description
15:0	MX_READ_CNT_CURRENT	Current value of QUP_MX_READ_COUNT counter.

0x078B6214 BLSP1_BLSP_QUP1_QUP_INPUT_FIFO_WORD_CNT**Type:** R**Clock:** crif_clk**Reset State:** 0x00000000

This register holds the number of words in the input FIFO at a given time. Note the fields in this register are dynamically updated. Hence, when this register is read during a FIFO operation, the value of this register could have been altered by the time SW gets the read value. Recommended for debug purposes.

BLSP1_BLSP_QUP1_QUP_INPUT_FIFO_WORD_CNT

Bits	Name	Description
8:0	INPUT_FIFO_WORD_CNT	Number of words in the input FIFO.

**0x078B6218+ BLSP1_BLSP_QUP1_QUP_INPUT_FIFOc, c=[0..15]
0x4*c****Type:** R**Clock:** crif_clk**Reset State:** Undefined

Consecutive reads to this address reads the input FIFO contents on a FIFO basis. The max address to read the input FIFO is 0x254.

BLSP1_BLSP_QUP1_QUP_INPUT_FIFOc

Bits	Name	Description
31:0	INPUT	Value shifted in.

0x078B6300 BLSP1_BLSP_QUP1_SPI_CONFIG**Type:** RW**Clock:** crif_clk**Reset State:** 0x00000000

Unless otherwise stated, register bits written return the value when read.

The contents of this register should only be changed when in the RESET_STATE. See the QUP_STATE register. Both NO_OUTPUT and NO_INPUT set to zero provides default full duplex operation. Setting both these bits to one is not a legal operational state.

BLSP1_BLSP_QUP1_SPI_CONFIG

Bits	Name	Type	Description
10	HS_MODE	RW	When set, SPI HS_MODE is enabled. When clear, SPI HS_MODE is disabled.
9	INPUT_FIRST	RW	When set, INPUT FIRST SPI protocol is used. When clear, OUTPUT FIRST SPI protocol is used.
8	LOOP_BACK	RW	Loopback is only valid in non-HS mode. Always clear for normal operation. If set, loop back on SPI_DATA_MO_SI under MASTER operation or SPI_DATA_MI_SO under SLAVE operation.
5	SLAVE_OPERATION	R	This register is writable only when the hardware signal spi_slave_en is asserted. When set, the SPI is configured for SLAVE operation. Zero indicates MASTER operation.

0x078B6304 BLSP1_BLSP_QUP1_SPI_IO_CONTROL

Type: RW

Clock: crif_clk

Reset State: 0x00000000

The contents of this register should only be changed when in the RESET_STATE.

Unless otherwise stated, register bits written return the value when read.

BLSP1_BLSP_QUP1_SPI_IO_CONTROL

Bits	Name	Description
11	FORCE_CS	When this bit is set, the chip select is asserted unconditionally with no relationship to QUP_STATE or transaction state.
10	CLK_IDLE_HIGH	When set to 1, spi_clk will be high when spi is idle (spi_cs is not asserted). When cleared to 0, spi_clk will be low when spi is idle (spi_cs is not asserted).
9	CLK_ALWAYS_ON	When MASTER operation is used, run SPI_CLK during IDLE.
8	MX_CS_MODE	If this bit is set, then for a given RUN state, the associated chip select will be asserted for the first N-bit transfer and will be kept asserted till the last programmed transfer. Applies only in MASTER mode. The number of transfers is determined by SPI_MX_OUTPUT_COUNT and/or SPI_MX_INPUT_COUNT registers.
7:4	CS_N_POLARITY	These four bits control the polarity of four SPI_CS#_N respectively. Setting any of this bit to '1', makes the associated SPI_CS#_N active HIGH. This field is a 'don't care' in SLAVE operation.

BLSP1_BLSP_QUP1_SPI_IO_CONTROL (cont.)

Bits	Name	Description
3:2	CS_SELECT	When MASTER operation is used, controls the assertion of SPI_CS_#N pins. Selects SPI_CS_N if 00, SPI_CS1_N if 01, SPI_CS2_N if 10, or SPI_CS3_N if 11. The deselected ChipSelects will be driven to inactive state dictated by the field CS_N_POLARITY. This field is a 'don't care' in SLAVE operation.
1	TRISTATE_CS	When set, drives Z on all SPI_CS#_N lines. When clear, enables normal functionality on these lines. This bit can be used to support deployment of the core as one of the masters in a multi-master configuration.
0	NO_TRI_STATE	When set, spi_data_out is not taken tri-state when SPI_CS#_N is de-asserted. This bit is normally set for MASTER operation but may be optionally left cleared.

0x078B6308 BLSP1_BLSP_QUP1_SPI_ERROR_FLAGS**Type:** RW**Clock:** crif_clk**Reset State:** 0x00000000

All bits in this register are set by hardware and remain set until cleared by software. Writing a 'one' to a bit clears it while writing a 'zero' leaves the bit unchanged.

BLSP1_BLSP_QUP1_SPI_ERROR_FLAGS

Bits	Name	Description
3	TRANSFER_CANCEL_DONE	This bit is set (1) when the SPI controller has completed canceling the transfer that was scheduled for cancellation (LOCAL_ID = TRANSFER_CANCEL_ID).
2	TRANSFER_CANCEL_ID_MATCH	This bit is set (1) when the SPI controller is ready for cancel operation: it has found the transfer that was scheduled for cancellation (LOCAL_ID = TRANSFER_CANCEL_ID) and the logic is done with previous activity.
1	SPI_SLV_CLK_OVER_RUN_ERR	Number of SPI_CLK ticks which occurred while SPI_CS#_N was asserted was greater than the programmed value of N. When this happens, only the first N bits are passed to the SPI input FIFO and the output shift value is held at zero.
0	SPI_SLV_CLK_UNDER_RUN_ERR	Number of SPI_CLK ticks which occurred while SPI_CS#_N was asserted was less than the programmed value of N. When this occurs, the bits which were received are passed to the SPI input FIFO and the CURRENT output bit is forced to zero.

0x078B630C BLSP1_BLSP_QUP1_SPI_ERROR_FLAGS_EN**Type:** RW**Clock:** crif_clk**Reset State:** 0x0000000C

Each bit in this register controls the setting of a corresponding error flag. If set, each bit enables generation of spi_error_irq and the setting of the corresponding error flag in the SPI_ERROR_FLAGS register for the specified error case. At reset, all error enable bits are set to '1'.

BLSP1_BLSP_QUP1_SPI_ERROR_FLAGS_EN

Bits	Name	Description
3	TRANSFER_CANCEL_DONE_EN	If set, enables generating the transfer cancel done indication.
2	TRANSFER_CANCEL_ID_MATCH_EN	If set, enables generating the transfer cancel ID match indication.
1	SPI_SLV_CLK_OVER_RUN_ERR_EN	If set, enables clock over run error generation.
0	SPI_SLV_CLK_UNDER_RUN_ERR_EN	If set, enables clock under run error generation.

0x078B6310 BLSP1_BLSP_QUP1_SPI_DEASSERT_WAIT**Type:** RW**Clock:** crif_clk**Reset State:** 0x00000000

This register holds the de-assertion wait time of SPI_CS#_N between consecutive N-bit transfers in MASTER operation. The wait time is specified in number of cycles of cc_spi_master_clk.

BLSP1_BLSP_QUP1_SPI_DEASSERT_WAIT

Bits	Name	Description
5:0	DEASSERT_WAIT	Number cc_spi_master_clk ticks associated with the deasserted time of SPI_CS#_N. Only applies to MASTER operation. For SLAVE operation, this field is a 'don't care'. A value of zero indicates SPI_CS#_N remains de-asserted for exactly one clock tick. A value of one, indicates two ticks, etc. All ones indicates 64 ticks.

0x078B6314 BLSP1_BLSP_QUP1_SPI_MASTER_LOCAL_ID**Type:** RW**Clock:** crif_clk**Reset State:** 0x00000000**BLSP1_BLSP_QUP1_SPI_MASTER_LOCAL_ID**

Bits	Name	Description
15:8	EXTENDED_ID	SW may use this field to specify the current sub-transfer ID to be read when TRANSFER_CANCEL_ID_MATCH is asserted. This value has no impact on HW.
7:0	LOCAL_ID	SW specifies the current transfer ID. HW will use this value to match with TRANSFER_CANCEL_ID.

0x078B6318 BLSP1_BLSP_QUP1_SPI_MASTER_COMMAND**Type:** W**Clock:** crif_clk**Reset State:** 0x00000000**BLSP1_BLSP_QUP1_SPI_MASTER_COMMAND**

Bits	Name	Description
0	RESET_CANCEL_FSM	This command should be given only when the CANCEL_FSM_STATE is in CANCEL_PENDING_STATE or PAUSE_WAIT_STATE and when the core is in PAUSE. The main use case for this bit is to allow withdrawing a cancel command.

0x078B631C BLSP1_BLSP_QUP1_SPI_MASTER_STATUS**Type:** R**Clock:** crif_clk**Reset State:** 0x00000000**BLSP1_BLSP_QUP1_SPI_MASTER_STATUS**

Bits	Name	Description
2:0	CANCEL_FSM_STATE	This 3-bit field informs the microprocessor of the state of the SPI cancel logic. Internal use only

0x078B6400 BLSP1_BLSP_QUP1_I2C_MASTER_CLK_CTL**Type:** RW**Clock:** crif_clk**Reset State:** 0x00000000

The I2C_CLK_CTL register is a read/write register that controls clock divider values.

BLSP1_BLSP_QUP1_I2C_MASTER_CLK_CTL

Bits	Name	Description
28	SCL_EXT_FORCE_LOW	When set to 0, SCL generation maintains legacy behavior. When set to 1, SCL state machine will go to the FORCED_LOW_STATE if force_low is asserted near (within 5 SCL cycles = $5 * 52\text{ns} = 260\text{ns} \approx \text{MAX}(\text{Tsu}, \text{DAT})$) the clock edge.
27:26	SDA_NOISE_REJECTION	Allows adding extra sampling levels on SDA to reject short low pulses. This value specifies how many TCXO cycles of logic low on SDA would be considered as valid logic low. 0x0 - legacy mode, 0x01 - one cycle wide low pulse is rejected, 0x2 - two cycles wide low pulse is rejected, 0x3 - three cycles wide low pulse is rejected
25:24	SCL_NOISE_REJECTION	Allows adding extra sampling levels on SCL to reject short low pulses. This value specifies how many TCXO cycles of logic low on SCL would be considered as valid logic low. 0x0 - legacy mode, 0x01 - one cycle wide low pulse is rejected, 0x2 - two cycles wide low pulse is rejected, 0x3 - three cycles wide low pulse is rejected
23:16	HIGH_TIME_DIVIDER_VALUE	Allows setting SCL duty cycle to non 50%. If this value is zero then legacy mode is used. If this value is non-zero then it will be used as the SCL high time counter and FS_DIVIDER_VALUE will be used as the low time counter. Minimum value is 0x7.
10:8	HS_DIVIDER_VALUE	The value in this register represents the clock period multiplier in high-speed (HS) mode. Minimum value is 0x3. $\text{I2C_HS_CLK} = \text{I2C_CLK} / (3 * (\text{HS_DIVIDER_VALUE} + 1) + 2)$
7:0	FS_DIVIDER_VALUE	The value in this register represents the clock period multiplier in fast/standard (FS) mode. Minimum value is 0x7. When HIGH_TIME_DIVIDER_VALUE=0: $\text{I2C_FS_CLK} = \text{I2C_CLK} / (2 * (\text{FS_DIVIDER_VALUE} + 3))$ When HIGH_TIME_DIVIDER_VALUE!=0: $\text{I2C_FS_CLK} = \text{I2C_CLK} / (\text{FS_DIVIDER_VALUE} + \text{HIGH_TIME_DIVIDER_VALUE} + 6)$

0x078B6404 BLSP1_BLSP_QUP1_I2C_MASTER_STATUS**Type:** RW**Clock:** crif_clk**Reset State:** 0x0C000000

The I2C_STATUS is a status register. Writing any value clears the status bits.

BLSP1_BLSP_QUP1_I2C_MASTER_STATUS

Bits	Name	Type	Description
30:28	CANCEL_FSM_STATE	R	This 3-bit field informs the microprocessor of the state of the I2C cancel logic. Internal use only 0x0: IDLE_STATE (IDLE_STATE) 0x1: CANCEL_PENDING_STATE (CANCEL_PENDING_STATE) 0x2: PAUSE_STATE (PAUSE_STATE) 0x3: PAUSE_WAIT_STATE (PAUSE_WAIT_STATE) 0x4: SEND_STOP_STATE (SEND_STOP_STATE) 0x5: FLUSH_STATE (FLUSH_STATE) 0x6: SEND_IRQ_STATE (SEND_IRQ_STATE)
27	I2C_SCL	R	Logic state of I2C bus serial clock wire.
26	I2C_SDA	R	Logic state of I2C bus serial data wire.
25	INVALID_READ_SEQ	RW	Interrupt source. This bit is set (1) when a MI_REC tag does not follow a START tag for an I2C READ. Not applicable for Halcyon.
24	INVALID_READ_ADDR	RW	Interrupt source. In version 1 tags this bit is set (1) when the I2C controller is trying to receive data from a non-existent I2C slave (address). In version 2 tags this bit is set (1) when the I2C controller is trying to access a non-existent I2C slave (address).
23	INVALID_TAG	RW	Interrupt source. This bit is set (1) when the I2C controller is trying to process data from the output FIFO that is improperly tagged.
22:20	INPUT_FSM_STATE	R	This 3-bit field informs the microprocessor of the state of the I2C MASTER INPUT FSM block. Internal use only 0x0: V1_V2_RESET_STATE (RESET_STATE value. Applies for both version 1 and version 2 tags) 0x1: V1_V2_IDLE_STATE (IDLE_STATE value. Applies for both version 1 and version 2 tags) 0x2: V1_V2_READ_LAST_BYTE_STATE (READ_LAST_BYTE_STATE value. Applies for both version 1 and version 2 tags) 0x3: V1_V2_MI_REC_STATE (MI_REC_STATE value. Applies for both version 1 and version 2 tags) 0x4: V1_V2_DEC_STATE (DEC_STATE value. Applies for both version 1 and version 2 tags) 0x5: V1_V2_STORE_STATE (STORE_STATE value. Applies for both version 1 and version 2 tags) 0x6: V2_WR_TAG_STATE (WR_TAG_STATE value. Only applicable in version 2 tags) 0x7: V2_WAIT_TAG_STATE (WAIT_TAG_STATE value. Only applicable in version 2 tags)

BLSP1_BLSP_QUP1_I2C_MASTER_STATUS (cont.)

Bits	Name	Type	Description
19:16	OUTPUT_FSM_STATE	R	This 3-bit field informs the microprocessor of the state of the I2C MASTER OUTPUT FSM block. Internal use only 0x0: V1_V2_RESET_STATE (RESET_STATE value. Applies for both version 1 and version 2 tags) 0x1: V1_V2_IDLE_STATE (IDLE_STATE value. Applies for both version 1 and version 2 tags) 0x2: V1_V2_DECODE_STATE (DECODE_STATE value. Applies for both version 1 and version 2 tags) 0x3: V1_V2_SEND_STATE (SEND_STATE value. Applies for both version 1 and version 2 tags) 0x4: V1_V2_MI_REC_STATE (MI_REC_STATE value. Applies for both version 1 and version 2 tags) 0x5: V1_V2_NOP_STATE (NOP_STATE value. Applies for both version 1 and version 2 tags) 0x6: V1_V2_INVALID_STATE (INVALID_STATE value. Applies for both version 1 and version 2 tags) 0x7: V1_PEEK_STATE (PEEK_STATE value. Only applicable in version 1 tags) 0x8: V1_SEND_R_STATE (SEND_R_STATE value. Only applicable in version 1 tags) 0xB: V2_GET_BYTE_STATE (GET_BYTE_STATE value. Only applicable in version 2 tags) 0xC: V2_GET_WRITE_BYTE_STATE (GET_WRITE_BYTE_STATE value. Only applicable in version 2 tags) 0xD: V2_SPECIALITY_STATE (SPECIALITY_STATE value. Only applicable in version 2 tags) 0xE: V2_WAIT_FLUSH_STOP_STATE (WAIT_FLUSH_STOP_STATE value. Only applicable in version 2 tags) 0xF: V2_STOP_STATE (STOP_STATE value. Only applicable in version 2 tags)
15:13	CLK_STATE	R	This 3-bit field informs the microprocessor of the state of the I2C clk_control block. Internal use only 0x0: RESET_BUSIDLE_STATE (RESET_BUSIDLE_STATE) 0x1: NOT_MASTER_STATE (NOT_MASTER_STATE) 0x2: HIGH_STATE (HIGH_STATE) 0x3: LOW_STATE (LOW_STATE) 0x4: HIGH_WAIT_STATE (HIGH_WAIT_STATE) 0x5: FORCED_LOW_STATE (FORCED_LOW_STATE) 0x6: HS_ADDR_LOW_STATE (HS_ADDR_LOW_STATE) 0x7: DOUBLE_BUFFER_WAIT_STATE (DOUBLE_BUFFER_WAIT_STATE)

BLSP1_BLSP_QUP1_I2C_MASTER_STATUS (cont.)

Bits	Name	Type	Description
12:10	DATA_STATE	R	This 3-bit field informs the microprocessor of the state of the I2C data_control block. Internal use only 0x0: RESET_WAIT_STATE (RESET_WAIT_STATE) 0x1: TX_ADDR_STATE (TX_ADDR_STATE) 0x2: TX_DATA_STATE (TX_DATA_STATE) 0x3: TX_HS_ADDR_STATE (TX_HS_ADDR_STATE) 0x4: TX_10_BIT_ADDR_STATE (TX_10_BIT_ADDR_STATE) 0x5: TX_2ND_BYTE_STATE (TX_2ND_BYTE_STATE) 0x6: RX_DATA_STATE (RX_DATA_STATE) 0x7: RX_NACK_HOLD_STATE (RX_NACK_HOLD_STATE)
9	BUS_MASTER	R	This bit is set (1) when the I2C controller is the present bus master.
8	BUS_ACTIVE	R	This bit is set (1) when the bus is in use by this, or any other controller.
7:6	FAILED	RW	Interrupt source. This 2-bit field contains the failure information of the present I2C transfers. If transmitting, failed[1] contains the information regarding the byte that has been transmitted. Failed[0] contains the information of the byte awaiting transmission if there is a byte pipelined. If no byte is pipelined, ignore this bit. If receiving, failed[1] contains the information of the byte received and stored in the buffer, and failed[0] should be ignored. For example: Value 00: Byte n transmitted successfully, byte n+1 to begin transmission Value 01: Byte n transmitted successfully, byte n+1 errored: type of error indicated in other STATUS bits. (queued invalid write, for example) Value 10: Byte n errored: type of error indicated in other STATUS bits, byte n+1 to begin transmission (byte n+1 would have to be a valid address byte for this condition to occur) Value 11: Byte n errored: type of error indicated in other STATUS bits, byte n+1 discarded as well (if the byte was a data byte destined for a NACKed address, data is useless, therefore discarded)
5	INVALID_WRITE	RW	Interrupt source. This bit is set (1) when software writes data to the I2C_DATA register that should be flagged as an address but is not.
4	ARB_LOST	RW	Interrupt source. This bit is set (1) when the controller loses arbitration for the bus. If this bit gets set (1) during the transmission, all data has been lost, and the microprocessor must re-request its transmission.

BLSP1_BLSP_QUP1_I2C_MASTER_STATUS (cont.)

Bits	Name	Type	Description
3	PACKET_NACKED	RW	Interrupt source. This bit is set high when a NACK is received from a slave. If a high speed master code is sent and there is an ACK from a slave, then this bit is set (1) to indicate that the high speed mode can not be entered. If the high speed mode is accepted by the slave, then a NACK is performed and this bit is not set (1).
2	BUS_ERROR	RW	Interrupt source. This bit is set (1) when an unexpected START or STOP condition is detected. This returns the controller to its reset state.
1	TRANSFER_CANCEL_DONE	RW	Only applicable when using version 2 tags. Interrupt source. This bit is set (1) when the I2C controller has completed canceling the transfer that was scheduled for cancellation (LOCAL_ID = TRANSFER_CANCEL_ID).
0	TRANSFER_CANCEL_ID_MATCH	RW	Only applicable when using version 2 tags. Interrupt source. This bit is set (1) when the I2C controller is ready for cancel operation: it has found the transfer that was scheduled for cancellation (LOCAL_ID = TRANSFER_CANCEL_ID) and the logic is done with previous activity.

0x078B6408 BLSP1_BLSP_QUP1_I2C_MASTER_CONFIG**Type:** RW**Clock:** crif_clk**Reset State:** 0x00000000**BLSP1_BLSP_QUP1_I2C_MASTER_CONFIG**

Bits	Name	Description
0	EN_VERSION_TWO_TAG	When set, version 2 of the mini-core is enabled.

0x078B640C BLSP1_BLSP_QUP1_I2C_MASTER_BUS_CLEAR**Type:** RW**Clock:** crif_clk**Reset State:** 0x00000000

BLSP1_BLSP_QUP1_I2C_MASTER_BUS_CLEAR

Bits	Name	Description
0	CLEAR	This command should be given only when the i2c mincore is idle. When in doubt SW can use SW_RESET to reset the minicore. When set, an I2C 'Bus Clear' executed per the I2C standard. A bus clear consists of nine I2C clock ticks with data wire left not-driven. Has the effect of clearing any slave which has lost read sync. After the clear is complete, the hardware sets the CLEAR bit to zero. The bit can not be cleared by software. After a bus clear, both clock and data lines should be high. If not, an external slave has hung the bus by holding down one or both lines. Reset the slave to clear if possible. Any slave reset mechanism is beyond the scope of the I2C mini core.

0x078B6410 BLSP1_BLSP_QUP1_I2C_MASTER_LOCAL_ID**Type:** R**Clock:** crif_clk**Reset State:** 0x00000000**BLSP1_BLSP_QUP1_I2C_MASTER_LOCAL_ID**

Bits	Name	Description
7:0	LOCAL_ID	Read only value capturing the ID byte of the last NOP LOCAL tag. Only applicable when using V2 tags

0x078B6414 BLSP1_BLSP_QUP1_I2C_MASTER_COMMAND**Type:** RW**Clock:** crif_clk**Reset State:** 0x00000000**BLSP1_BLSP_QUP1_I2C_MASTER_COMMAND**

Bits	Name	Description
0	RESET_CANCEL_FSM	This command should be given only when the CANCEL_FSM_STATE is in CANCEL_PENDING_STATE or PAUSE_WAIT_STATE and when the core is in PAUSE. The main use case for this bit is to allow withdrawing a cancel command in case the transfer was flushed (QUP_STATE[I2C_FLUSH] was set) due to I2C NACK.

20.6 BLSP1_BLSP_QUP2_QUP

0x078B7000 BLSP1_BLSP_QUP2_QUP_CONFIG

Type: RW

Clock: crif_clk

Reset State: 0x00000000

Unless otherwise stated, register bits written return the value when read.

a. The contents of this register should only be changed when in the RESET_STATE (see the QUP_STATE register).

b. The value written to the QUP output FIFO, is shifted left toward the MSB before passing it to the mini-core as follows:

o N equals 8 or less - shift 24

o N equals 16 to 9 - shift 16

o N equals 24 to 17 - shift 8

o N equals 32 to 25 - no shift

BLSP1_BLSP_QUP2_QUP_CONFIG

Bits	Name	Description
16	EN_EXT_OUT_FLAG	Enable Extended OUTPUT/INPUT_SERVICE_FLAG interrupt generation. When this bit is set, the behavior of the *_SERVICE_FLAG and MAX_OUTPUT_DONE_FLAG changes. Service flag and the done flag are not set till the last SPI or I2C transfer has completed. a) For SPI: 1. The shifting needs to complete 2. Clock stopped if CLK_ALWAYS_ON=0, 3. CS, if used, de-asserted b) For I2C, 1. The ACK needs to complete 2. STOP condition done generated (Attention need to be put in synchronizing the last transaction with STOP tag) 3. clock and data wires return to high condition When this bit is clear, the legacy behavior is applicable. Under legacy, the DONE_FLAG is set when: 1. MX_*_COUNT is reached 2. The output FIFO is empty 3. Last bit was sent on the link Under legacy, the flag interrupt is generated before the last transfer is complete on the external interface.
15	CORE_EXTRA_CLK_ON_EN	Enable additional transfer based qup_core_clk gating for power save. When set to 0, clock on is turned on unconditionally. When set to 1, dynamic clock on is turned on.

BLSP1_BLSP_QUP2_QUP_CONFIG (cont.)

Bits	Name	Description
14	FIFO_CLK_GATE_EN	When set to 0, fifo clock is mostly on - legacy mode. When set to 1, fifo clock is turned off dynamically.
13	CORE_CLK_ON_EN	When set to 0, clock on is turned on unconditionally. When set to 1, dynamic clock on is turned on and driven high conditionally.
12	APP_CLK_ON_EN	When set to 0, clock on is turned on unconditionally. When set to 1, dynamic clock on is turned on and driven high conditionally.
11:8	MINI_CORE	value: 0000 null core value: 0001 SPI core value: 0010 I2C master controller value: 0011 reserved value: 0100 reserved value: 0101 reserved value: 0110 reserved value: 0111 reserved See Note 1.
7	NO_INPUT	qup_data_in is not used and the value is a 'don't care'. The QUP input FIFO is always empty. The input service interrupt and INPUT_SERVICE_FLAG bit are never set. See notes (a) and (b) above.
6	NO_OUTPUT	qup_data_out is not used and is held at the value zero. The QUP output FIFO is always empty. The output service flag and OUTPUT_SERVICE_FLAG bit are never set. qup_data_out is still driven when SPI_CS#_N is asserted. The setting for NO_TRI_STATE still applies. See notes (a) and (b) above.
5	QUP_HREADY_CTRL	When set to 0, qup wil not stall the AHB bus (legacy mode). When set to 1, qup may stall the AHB bus until register access is done.
4:0	N	Number of logical bits N in the mini-core that constitutes a single transfer. The value zero indicates N equals one. The value of all ones indicates N equals 32. The value in this register must be 3 (i.e., N is 4 or higher in order for the STATE field to be set to RUN_STATE). See note (a) above.

0x078B7004 BLSP1_BLSP_QUP2_QUP_STATE**Type:** RW**Clock:** crif_clk**Reset State:** 0x0000001C

BLSP1_BLSP_QUP2_QUP_STATE

Bits	Name	Type	Description
6	I2C_FLUSH	RW	Flushing an i2c transfer requires using version 2 tags. FLUSH should be used in BAM mode only. Setting this bit to 1 will flush all tags and tag related data besides EOT until FLUSH STOP tag is encountered. Setting this bit to 0 has no impact. Reading this bit returns the flush operation status - 0 = ongoing, 1 = done.
5	WAIT_FOR_EOT	RW	Only applicable when moving to RUN_STATE using command descriptor in SPI mode with NO_OUTPUT. Setting this bit to 1 will stall the done_toggle_com indication until basi_trans_end_pro is asserted. This will prevent the next command descriptor to start prematurely.
4	I2C_MAST_GEN	R	Read only. Reflects the RTL generic setting for GEN_I2C_MASTER_MINI_CORE.
3	SPI_GEN	R	Read only. Reflects the RTL generic setting for GEN_SPI_MINI_CORE.
2	STATE_VALID	R	Read only. If and only if set to one, writes to the STATE field is allowed or reads from the STATE field is valid. Writes to this bit is ignored.
1:0	STATE	RW	When clear (00), the mini-core and related logic is held in RESET_STATE. When set to '01', the mini-core and related logic is released from reset and enters the RUN_STATE. When set to '11', the mini-core and related logic enters the PAUSE_STATE at the next appropriate point in time. Writing (10) to this field clears these two bits. For PAUSE_STATE to RESET_STATE transition, two writes of (10) are required for the transition to complete.

0x078B7008 BLSP1_BLSP_QUP2_QUP_IO_MODES**Type:** RW**Clock:** crif_clk**Reset State:** 0x000000A5

Unless otherwise stated, register bits written return the value when read.

Notes:

a. 'Packing' occurs as follows:

- N equals 8 or less - pack four values into each QUP input FIFO word.
- N equals 16 to 9 - pack two values into each QUP input FIFO word.
- N equals 32 to 17 - no packing. Each mini-core value is moved to an QUP input FIFO word.

b. 'Un-Packing' occurs as follows:

- N equals 8 or less - un-pack four values from each QUP output FIFO word.
- N equals 16 to 9 - un-pack two values from each QUP output FIFO word.
- N equals 32 to 17 - no packing. Each QUP output FIFO value is moved to the mini-core.

c. INPUT_MODE and OUTPUT_MODE should be both in BAM_Mode or both in non BAM_Mode.

BLSP1_BLSP_QUP2_QUP_IO_MODES

Bits	Name	Type	Description
16	OUTPUT_BIT_SHIFT_EN	RW	If set, enables the QUP output FIFO block to do bit shifting on the output data.
15	PACK_EN	RW	Indicates data values from the mini-core are to be packed before placement in the QUP input FIFO. See note (a) above.
14	UNPACK_EN	RW	Indicates data values taken from the QUP output FIFO should be unpacked before passing to the mini-core. See note (b) above.
13:12	INPUT_MODE	RW	Value 00: FIFO_Mode Value 01: Block_Mode Value 10: Reserved Value 11: BAM_Mode (note if the RTL generic BLOCK_SIZE_INPUT = 0, then only FIFO_Mode is available)
11:10	OUTPUT_MODE	RW	Value 00: FIFO_Mode Value 01: Block_Mode Value 10: Reserved Value 11: BAM_Mode (note if the RTL generic BLOCK_SIZE_OUTPUT = 0, then only FIFO_Mode is available)
9:7	INPUT_FIFO_SIZE	R	Read only, actual value set by RTL generic. Value 000: 2X BLOCK_SIZE Value 001: 4X BLOCK_SIZE Value 010: 8X BLOCK_SIZE Value 011: 16X BLOCK_SIZE
6:5	INPUT_BLOCK_SIZE	R	Read only. Actual value set by RTL generic. Indicates the block size associated with Block_Mode for input. Value 00: 4 Bytes (FIFO_Mode only) Value 01: 16 Bytes Value 10: 32 Bytes Value 11: Reserved
4:2	OUTPUT_FIFO_SIZE	R	Read only. Actual value set by RTL GENERIC. Value 000: 2X BLOCK_SIZE Value 001: 4X BLOCK_SIZE Value 010: 8X BLOCK_SIZE Value 011: 16X BLOCK_SIZE

BLSP1_BLSP_QUP2_QUP_IO_MODES (cont.)

Bits	Name	Type	Description
1:0	OUTPUT_BLOCK_SIZE	R	Read only. Actual value set by RTL GENERIC. Value 00: 04 Bytes (FIFO mode only) Value 01: 16 Bytes Value 10: 32 Bytes Value 11: Reserved

0x078B700C BLSP1_BLSP_QUP2_QUP_SW_RESET**Type:** W**Clock:** crif_clk**Reset State:** 0x00000000

A write to this register resets the entire QUP core and all mini-cores. The internal registers are brought back to their reset values. Reading of this register returns zero. The AHB clock domain reset will stay asserted until either the spi_clk or i2c_clk domains are reset. It is prohibited to write to any QUP register during this period and failing to do so will cause an ERROR response on the AHB bus. In order to avoid this SW should poll the QUP_STATE[STATE_VALID] bit until it is asserted.

BLSP1_BLSP_QUP2_QUP_SW_RESET

Bits	Name	Description
31:0	QUP_SW_RESET	

0x078B7014 BLSP1_BLSP_QUP2_QUP_TRANSFER_CANCEL**Type:** RW**Clock:** crif_clk**Reset State:** 0x00000000**BLSP1_BLSP_QUP2_QUP_TRANSFER_CANCEL**

Bits	Name	Type	Description
15:8	TRANSFER_CANCEL_ID	RW	This field include the id of the transfer that is scheduled for cancellation. This field is valid when asserting TRANSFER_CANCEL bit. A value of 0xFF will cancel the current transfer regardless of its LOCAL_ID. Canceling an i2c transfer requires using version 2 tags.
7	TRANSFER_CANCEL	W	Setting this bit to 1 will cause the HW to go to pause_state when the LOCAL_ID = TRANSFER_CANCEL_ID. This would allow flushing the rest of the transfer in order to implement cancel API. Canceling an i2c transfer requires using version 2 tags.

0x078B7018 BLSP1_BLSP_QUP2_QUP_OPERATIONAL**Type:** RW**Clock:** crif_clk**Reset State:** 0x000000C0

Refer to the chapter on 'QUP Software Considerations' regarding the correct interpretation of the various service interrupts generated by QUP and their relation to these register bits.

BLSP1_BLSP_QUP2_QUP_OPERATIONAL

Bits	Name	Description
15	NWD	Notify When Done (NWD). Read only status bit which when set indicates a NWD acknowledgement is outstanding. This bit is set by QUP hardware when NWD is signaled from BAM. This bit is cleared by QUP hardware when the NWD request is acknowledged by assertion of done_toggle. This bit is always unconditionally cleared in RESET_STATE.
14	DONE_TOGGLE	Read only status bit which provides the current state of the side-band done_toggle signal sent to BAM. At each NWD acknowledgement, this bit toggles.
13	IN_BLOCK_READ_REQ	Read only. When set by hardware, indicates QUP input FIFO has BLOCK_SIZE_INPUT amount of data ready for reading. Valid only in Block_Mode.
12	OUT_BLOCK_WRITE_REQ	Read only. When set by hardware, indicates QUP output FIFO needs BLOCK_SIZE_OUTPUT amount of data to be written. Valid only in Block_Mode.
11	MAX_INPUT_DONE_FLAG	Read only. When set by hardware, indicates QUP input FIFO has reached the programmed QUP_MX_INPUT_COUNT value. Valid in FIFO_Mode (only if MX_READ_COUNT is non-zero), Block_Mode and Bam_Mode (for debug only).
10	MAX_OUTPUT_DONE_FLAG	Read only. When set by hardware, indicates QUP output FIFO has reached the programmed QUP_MX_OUTPUT_COUNT value. Valid in Block_Mode and Bam_Mode (for debug only).
9	INPUT_SERVICE_FLAG	When set by hardware, indicates QUP input FIFO has an outstanding input service request. At the point in time this bit is set, the hardware also asserts qup_irq. Writing a 'zero' to this bit does nothing. Writing a 'one' to this bit clears it and acknowledges software has or will read the data. Valid in all modes , recommended to be masked in BAM mode
8	OUTPUT_SERVICE_FLAG	When set by hardware, indicates QUP output FIFO has an outstanding output service request. At the point in time this bit is set, the hardware also asserts qup_irq. Writing a 'zero' to this bit does nothing. Writing a 'one' to this bit clears it and acknowledges software has or will read the data. Valid in all modes , recommended to be masked in BAM mode
7	INPUT_FIFO_FULL	Read only. When set, indicates the input FIFO is full and can accept no more data from the QUP mini-core.

BLSP1_BLSP_QUP2_QUP_OPERATIONAL (cont.)

Bits	Name	Description
6	OUTPUT_FIFO_FULL	Read only. When set, indicates the output FIFO is full and can accept no more CRIF writes.
5	INPUT_FIFO_NOT_EMPTY	Read only. When set, indicates the input FIFO has at least one value in it to be read. When clear, indicates the input FIFO is empty.
4	OUTPUT_FIFO_NOT_EMPTY	Read only. When set, indicates the output FIFO has at least one value in it to be shifted out. When clear, indicates the output FIFO is empty.

0x078B701C BLSP1_BLSP_QUP2_QUP_ERROR_FLAGS**Type:** RW**Clock:** crif_clk**Reset State:** 0x00000000

All bits in this register are set by hardware and remain set until cleared by software. Writing a 'one' to a bit clears it while writing a 'zero' leaves the bit unchanged.

BLSP1_BLSP_QUP2_QUP_ERROR_FLAGS

Bits	Name	Description
5	OUTPUT_OVER_RUN_ERR	Indicates the output FIFO was full when a CRIF write was attempted to the FIFO. The write is discarded.
4	INPUT_UNDER_RUN_ERR	Indicates the input FIFO was empty when a CRIF read was attempted to the FIFO. The read value returns is indeterminate.
3	OUTPUT_UNDER_RUN_ERR	Indicates the output FIFO was empty when an output shift operation required a value.
2	INPUT_OVER_RUN_ERR	Indicates the input FIFO was full when an input shift operation provided a new value. When this happens, the new value is discarded.

0x078B7020 BLSP1_BLSP_QUP2_QUP_ERROR_FLAGS_EN**Type:** RW**Clock:** crif_clk**Reset State:** 0x0000003C

Each bit in this register controls the setting of a corresponding error flag. If set, each bit enables generation of qup_irq and the setting of the corresponding error flag in the QUP_ERROR_FLAGS register for the specified error case. At reset, all error enable bits are set to '1'.

BLSP1_BLSP_QUP2_QUP_ERROR_FLAGS_EN

Bits	Name	Description
5	OUTPUT_OVER_RUN_ERR_EN	If set, enables output over run error generation.
4	INPUT_UNDER_RUN_ERR_EN	If set, enables input under run error generation.
3	OUTPUT_UNDER_RUN_ERR_EN	If set, enables output under run error generation.
2	INPUT_OVER_RUN_ERR_EN	If set, enables input over run error generation.

0x078B7028 BLSP1_BLSP_QUP2_QUP_OPERATIONAL_MASK**Type:** RW**Clock:** crif_clk**Reset State:** 0x00000000

This register masks several QUP_OPERATIONAL flags from creating an interrupt .

BLSP1_BLSP_QUP2_QUP_OPERATIONAL_MASK

Bits	Name	Description
9	INPUT_SERVICE_MASK	If set, this flag in QUP_OPERATIONAL does not cause an interrupt but provides status only.
8	OUTPUT_SERVICE_MASK	If set, this flag in QUP_OPERATIONAL does not cause an interrupt but provides status only.

0x078B7100 BLSP1_BLSP_QUP2_QUP_MX_OUTPUT_COUNT**Type:** RW**Clock:** crif_clk**Reset State:** 0x00000000

Unless otherwise stated, register bits written return the value when read. This register only applies to Block_Mode. The counter decrements for each output transfer when the STATE field (see QUP_STATE register) is moved from the RESET_STATE to the RUN_STATE. The PAUSE_STATE does not effect the count.

Notes:

- a. Allows the total number of Mini Core transfers to be less than an exact multiple of OUTPUT_BLOCK_SIZE. Any additional outputs are discarded.

BLSP1_BLSP_QUP2_QUP_MX_OUTPUT_COUNT

Bits	Name	Type	Description
31	MX_CONFIG_DURING_RUN	W	Write only value. When reset to zero the MX_OUTPUT_COUNT value only applies in reset_state (legacy behavior). When set to one the MX_OUTPUT_COUNT value will also apply in run_state and allow writing the transaction data in a few portions when not knowing the transaction size in advance. Reconfiguration during run is only allowed after the last portion has ended (i.e. MAX_OUTPUT_DONE_FLAG was asserted)
15:0	MX_OUTPUT_COUNT	RW	Number of writes of size N to the mini-core per RUN_STATE. A value of zero indicates the output count function is not enabled for use. See note (a) above.

0x078B7104 BLSP1_BLSP_QUP2_QUP_MX_OUTPUT_CNT_CURRENT**Type:** R**Clock:** crif_clk**Reset State:** 0x00000000**BLSP1_BLSP_QUP2_QUP_MX_OUTPUT_CNT_CURRENT**

Bits	Name	Description
15:0	MX_OUTPUT_CNT_CURRENT	Current value of QUP_MX_OUTPUT_COUNT.

0x078B710C BLSP1_BLSP_QUP2_QUP_OUTPUT_FIFO_WORD_CNT**Type:** R**Clock:** crif_clk**Reset State:** 0x00000000

This register holds the number of words in the output FIFO at a given time. Note the fields in this register are dynamically updated hence when this register is read during a FIFO operation, the value of this register could have been altered by the time SW gets the read value. Recommended for debug purposes.

BLSP1_BLSP_QUP2_QUP_OUTPUT_FIFO_WORD_CNT

Bits	Name	Description
8:0	OUTPUT_FIFO_WORD_COUNT	Number of words in the output FIFO.

**0x078B7110+ BLSP1_BLSP_QUP2_QUP_OUTPUT_FIFOc, c=[0..15]
0x4*c****Type:** W**Clock:** crif_clk**Reset State:** 0x00000000

Note that consecutive writes to this address keeps filling up the output FIFO. The max address to address the output FIFO is 0x14C.

BLSP1_BLSP_QUP2_QUP_OUTPUT_FIFOc

Bits	Name	Description
31:0	OUTPUT	Value to be shifted out.

0x078B7150 BLSP1_BLSP_QUP2_QUP_MX_WRITE_COUNT**Type:** RW**Clock:** crif_clk**Reset State:** 0x00000000

What the QUP_MX_OUTPUT_COUNT register means to Block_Mode and Bam_Mode, this register means the same to FIFO_mode. If this register is non-zero, then the qup_irq is asserted after shifting in the number of shifts specified by this register.

Unlike the QUP_MX_OUTPUT_COUNT register case, the SW should not program the QUP_MX_WRITE_COUNT value to be more than the 'actual depth' of the FIFO (BLOCK_SIZE_OUTPUT * FIFO_SIZE_OUTPUT).

BLSP1_BLSP_QUP2_QUP_MX_WRITE_COUNT

Bits	Name	Type	Description
31	MX_CONFIG_DURING_RUN	W	Write only value. When reset to zero the MX_WRITE_COUNT value only applies in reset_state (legacy behavior). When set to one the MX_WRITE_COUNT value will also apply in run_state and allow writing the transaction data in a few portions when not knowing the transaction size in advance. Reconfiguration during run is only allowed after the last portion has ended (i.e. MAX_OUTPUT_DONE_FLAG was asserted)
15:0	MX_WRITE_COUNT	RW	The number of 'writes' of size N. This is used only if the core is in FIFO_Mode.

0x078B7154 BLSP1_BLSP_QUP2_QUP_MX_WRITE_CNT_CURRENT**Type:** R**Clock:** crif_clk**Reset State:** 0x00000000**BLSP1_BLSP_QUP2_QUP_MX_WRITE_CNT_CURRENT**

Bits	Name	Description
15:0	MX_WRITE_CNT_CURRENT	Current value of QUP_MX_WRITE_COUNT counter.

0x078B7200 BLSP1_BLSP_QUP2_QUP_MX_INPUT_COUNT**Type:** RW**Clock:** crif_clk**Reset State:** 0x00000000

Unless otherwise stated, register bits written return the value when read. This register only applies to Block_Mode and Bam_Mode (for non-balanced SPI). The counter decrements for each input transfer when the STATE field (see QUP_STATE register) is moved from the RESET_STATE to the RUN_STATE. The PAUSE_STATE do not affect the count.

Notes:

a. Allows the number of shift register transfers to be less than an exact multiple of INPUT_BLOCK_SIZE. When count reached, remainder of INPUT_BLOCK_SIZE is filled with zeroes.

BLSP1_BLSP_QUP2_QUP_MX_INPUT_COUNT

Bits	Name	Type	Description
31	MX_CONFIG_DURING_RUN	W	Write only value. When reset to zero the MX_INPUT_COUNT value only applies in reset_state (legacy behavior). When set to one the MX_INPUT_COUNT value will also apply in run_state and allow writing the transaction data in a few portions when not knowing the transaction size in advance. Reconfiguration during run is only allowed after the last portion has ended (i.e. MAX_INPUT_DONE_FLAG was asserted)
15:0	MX_INPUT_COUNT	RW	Number of reads of size N from the mini-core per RUN_STATE. A value of zero indicates the input count function is not enabled for use. See note (a) above.

0x078B7204 BLSP1_BLSP_QUP2_QUP_MX_INPUT_CNT_CURRENT**Type:** R**Clock:** crif_clk**Reset State:** 0x00000000**BLSP1_BLSP_QUP2_QUP_MX_INPUT_CNT_CURRENT**

Bits	Name	Description
15:0	MX_INPUT_CNT_CURRENT	Current value of QUP_MX_INPUT_COUNT.

0x078B7208 BLSP1_BLSP_QUP2_QUP_MX_READ_COUNT**Type:** RW**Clock:** crif_clk**Reset State:** 0x00000000

What the QUP_MX_INPUT_COUNT register means to Block_Mode and Bam_Mode, this register means the same to FIFO_mode. If this register is non-zero, then the qup_input_service_irq is asserted after shifting in the number of shifts specified by this register.

Unlike the QUP_MX_INPUT_COUNT register case, the SW should not program the QUP_MX_READ_COUNT value to be more than the 'actual depth' of the FIFO (BLOCK_SIZE_INPUT * FIFO_SIZE_INPUT). If the SPI mini-core and slave operation is enabled, then an InputOverRun error will result.

BLSP1_BLSP_QUP2_QUP_MX_READ_COUNT

Bits	Name	Type	Description
31	MX_CONFIG_DURING_RUN	W	Write only value. When reset to zero the MX_READ_COUNT value only applies in reset_state (legacy behavior). When set to one the MX_READ_COUNT value will also apply in run_state and allow writing the transaction data in a few portions when not knowing the transaction size in advance. Reconfiguration during run is only allowed after the last portion has ended (i.e. MAX_INPUT_DONE_FLAG was asserted)
15:0	MX_READ_COUNT	RW	The number of 'reads' of size N. This is used only if the core is in FIFO_Mode.

0x078B720C BLSP1_BLSP_QUP2_QUP_MX_READ_CNT_CURRENT**Type:** R**Clock:** crif_clk**Reset State:** 0x00000000

BLSP1_BLSP_QUP2_QUP_MX_READ_CNT_CURRENT

Bits	Name	Description
15:0	MX_READ_CNT_CURRENT	Current value of QUP_MX_READ_COUNT counter.

0x078B7214 BLSP1_BLSP_QUP2_QUP_INPUT_FIFO_WORD_CNT**Type:** R**Clock:** crif_clk**Reset State:** 0x00000000

This register holds the number of words in the input FIFO at a given time. Note the fields in this register are dynamically updated. Hence, when this register is read during a FIFO operation, the value of this register could have been altered by the time SW gets the read value. Recommended for debug purposes.

BLSP1_BLSP_QUP2_QUP_INPUT_FIFO_WORD_CNT

Bits	Name	Description
8:0	INPUT_FIFO_WORD_CNT	Number of words in the input FIFO.

**0x078B7218+ BLSP1_BLSP_QUP2_QUP_INPUT_FIFOc, c=[0..15]
0x4*c****Type:** R**Clock:** crif_clk**Reset State:** Undefined

Consecutive reads to this address reads the input FIFO contents on a FIFO basis. The max address to read the input FIFO is 0x254.

BLSP1_BLSP_QUP2_QUP_INPUT_FIFOc

Bits	Name	Description
31:0	INPUT	Value shifted in.

0x078B7300 BLSP1_BLSP_QUP2_SPI_CONFIG**Type:** RW**Clock:** crif_clk**Reset State:** 0x00000000

Unless otherwise stated, register bits written return the value when read.

The contents of this register should only be changed when in the RESET_STATE. See the QUP_STATE register. Both NO_OUTPUT and NO_INPUT set to zero provides default full duplex operation. Setting both these bits to one is not a legal operational state.

BLSP1_BLSP_QUP2_SPI_CONFIG

Bits	Name	Type	Description
10	HS_MODE	RW	When set, SPI HS_MODE is enabled. When clear, SPI HS_MODE is disabled.
9	INPUT_FIRST	RW	When set, INPUT FIRST SPI protocol is used. When clear, OUTPUT FIRST SPI protocol is used.
8	LOOP_BACK	RW	Loopback is only valid in non-HS mode. Always clear for normal operation. If set, loop back on SPI_DATA_MO_SI under MASTER operation or SPI_DATA_MI_SO under SLAVE operation.
5	SLAVE_OPERATION	R	This register is writable only when the hardware signal spi_slave_en is asserted. When set, the SPI is configured for SLAVE operation. Zero indicates MASTER operation.

0x078B7304 BLSP1_BLSP_QUP2_SPI_IO_CONTROL

Type: RW

Clock: crif_clk

Reset State: 0x00000000

The contents of this register should only be changed when in the RESET_STATE. See the QUP_STATE register.

Unless otherwise stated, register bits written return the value when read.

BLSP1_BLSP_QUP2_SPI_IO_CONTROL

Bits	Name	Description
11	FORCE_CS	When this bit is set, the chip select is asserted unconditionally with no relationship to QUP_STATE or transaction state.
10	CLK_IDLE_HIGH	When set to 1, spi_clk will be high when spi is idle (spi_cs is not asserted). When cleared to 0, spi_clk will be low when spi is idle (spi_cs is not asserted).
9	CLK_ALWAYS_ON	When MASTER operation is used, run SPI_CLK during IDLE.
8	MX_CS_MODE	If this bit is set, then for a given RUN state, the associated chip select will be asserted for the first N-bit transfer and will be kept asserted till the last programmed transfer. Applies only in MASTER mode. The number of transfers is determined by SPI_MX_OUTPUT_COUNT and/or SPI_MX_INPUT_COUNT registers.

BLSP1_BLSP_QUP2_SPI_IO_CONTROL (cont.)

Bits	Name	Description
7:4	CS_N_POLARITY	These four bits control the polarity of four SPI_CS#_N respectively. Setting any of this bit to '1', makes the associated SPI_CS#_N active HIGH. This field is a 'don't care' in SLAVE operation.
3:2	CS_SELECT	When MASTER operation is used, controls the assertion of SPI_CS_#N pins. Selects SPI_CS_N if 00, SPI_CS1_N if 01, SPI_CS2_N if 10, or SPI_CS3_N if 11. The deselected ChipSelects will be driven to inactive state dictated by the field CS_N_POLARITY. This field is a 'don't care' in SLAVE operation.
1	TRISTATE_CS	When set, drives Z on all SPI_CS#_N lines. When clear, enables normal functionality on these lines. This bit can be used to support deployment of the core as one of the masters in a multi-master configuration.
0	NO_TRI_STATE	When set, spi_data_out is not taken tri-state when SPI_CS#_N is de-asserted. This bit is normally set for MASTER operation but may be optionally left cleared.

0x078B7308 BLSP1_BLSP_QUP2_SPI_ERROR_FLAGS**Type:** RW**Clock:** crif_clk**Reset State:** 0x00000000

All bits in this register are set by hardware and remain set until cleared by software. Writing a 'one' to a bit clears it while writing a 'zero' leaves the bit unchanged.

BLSP1_BLSP_QUP2_SPI_ERROR_FLAGS

Bits	Name	Description
3	TRANSFER_CANCEL_DONE	This bit is set (1) when the SPI controller has completed canceling the transfer that was scheduled for cancellation (LOCAL_ID = TRANSFER_CANCEL_ID).
2	TRANSFER_CANCEL_ID_MATCH	This bit is set (1) when the SPI controller is ready for cancel operation: it has found the transfer that was scheduled for cancellation (LOCAL_ID = TRANSFER_CANCEL_ID) and the logic is done with previous activity.
1	SPI_SLV_CLK_OVER_RUN_ERR	Number of SPI_CLK ticks which occurred while SPI_CS#_N was asserted was greater than the programmed value of N. When this happens, only the first N bits are passed to the SPI input FIFO and the output shift value is held at zero.
0	SPI_SLV_CLK_UNDER_RUN_ERR	Number of SPI_CLK ticks which occurred while SPI_CS#_N was asserted was less than the programmed value of N. When this occurs, the bits which were received are passed to the SPI input FIFO and the CURRENT output bit is forced to zero.

0x078B730C BLSP1_BLSP_QUP2_SPI_ERROR_FLAGS_EN**Type:** RW**Clock:** crif_clk**Reset State:** 0x0000000C

Each bit in this register controls the setting of a corresponding error flag. If set, each bit enables generation of spi_error_irq and the setting of the corresponding error flag in the SPI_ERROR_FLAGS register for the specified error case. At reset, all error enable bits are set to '1'.

BLSP1_BLSP_QUP2_SPI_ERROR_FLAGS_EN

Bits	Name	Description
3	TRANSFER_CANCEL_DONE_EN	If set, enables generating the transfer cancel done indication.
2	TRANSFER_CANCEL_ID_MATCH_EN	If set, enables generating the transfer cancel ID match indication.
1	SPI_SLV_CLK_OVER_RUN_ERR_EN	If set, enables clock over run error generation.
0	SPI_SLV_CLK_UNDER_RUN_ERR_EN	If set, enables clock under run error generation.

0x078B7310 BLSP1_BLSP_QUP2_SPI_DEASSERT_WAIT**Type:** RW**Clock:** crif_clk**Reset State:** 0x00000000

This register holds the de-assertion wait time of SPI_CS#_N between consecutive N-bit transfers in MASTER operation. The wait time is specified in number of cycles of cc_spi_master_clk.

BLSP1_BLSP_QUP2_SPI_DEASSERT_WAIT

Bits	Name	Description
5:0	DEASSERT_WAIT	Number cc_spi_master_clk ticks associated with the deasserted time of SPI_CS#_N. Only applies to MASTER operation. For SLAVE operation, this field is a 'don't care'. A value of zero indicates SPI_CS#_N remains de-asserted for exactly one clock tick. A value of one, indicates two ticks, etc. All ones indicates 64 ticks.

0x078B7314 BLSP1_BLSP_QUP2_SPI_MASTER_LOCAL_ID**Type:** RW**Clock:** crif_clk**Reset State:** 0x00000000**BLSP1_BLSP_QUP2_SPI_MASTER_LOCAL_ID**

Bits	Name	Description
15:8	EXTENDED_ID	SW may use this field to specify the current sub-transfer ID to be read when TRANSFER_CANCEL_ID_MATCH is asserted. This value has no impact on HW.
7:0	LOCAL_ID	SW specifies the current transfer ID. HW will use this value to match with TRANSFER_CANCEL_ID.

0x078B7318 BLSP1_BLSP_QUP2_SPI_MASTER_COMMAND**Type:** W**Clock:** crif_clk**Reset State:** 0x00000000**BLSP1_BLSP_QUP2_SPI_MASTER_COMMAND**

Bits	Name	Description
0	RESET_CANCEL_FSM	This command should be given only when the CANCEL_FSM_STATE is in CANCEL_PENDING_STATE or PAUSE_WAIT_STATE and when the core is in PAUSE. The main use case for this bit is to allow withdrawing a cancel command.

0x078B731C BLSP1_BLSP_QUP2_SPI_MASTER_STATUS**Type:** R**Clock:** crif_clk**Reset State:** 0x00000000**BLSP1_BLSP_QUP2_SPI_MASTER_STATUS**

Bits	Name	Description
2:0	CANCEL_FSM_STATE	This 3-bit field informs the microprocessor of the state of the SPI cancel logic. Internal use only

0x078B7400 BLSP1_BLSP_QUP2_I2C_MASTER_CLK_CTL**Type:** RW**Clock:** crif_clk**Reset State:** 0x00000000

The I2C_CLK_CTL register is a read/write register that controls clock divider values.

BLSP1_BLSP_QUP2_I2C_MASTER_CLK_CTL

Bits	Name	Description
28	SCL_EXT_FORCE_LOW	When set to 0, SCL generation maintains legacy behavior. When set to 1, SCL state machine will go to the FORCED_LOW_STATE if force_low is asserted near (within 5 SCL clock cycles = $5 * 52\text{ns} = 260\text{ns} \approx \text{MAX}(\text{Tsu}, \text{DAT})$) the clock edge.
27:26	SDA_NOISE_REJECTION	Allows adding extra sampling levels on SDA to reject short low pulses. This value specifies how many TCXO cycles of logic low on SDA would be considered as valid logic low. 0x0 - legacy mode, 0x01 - one cycle wide low pulse is rejected, 0x2 - two cycles wide low pulse is rejected, 0x3 - three cycles wide low pulse is rejected
25:24	SCL_NOISE_REJECTION	Allows adding extra sampling levels on SCL to reject short low pulses. This value specifies how many TCXO cycles of logic low on SCL would be considered as valid logic low. 0x0 - legacy mode, 0x01 - one cycle wide low pulse is rejected, 0x2 - two cycles wide low pulse is rejected, 0x3 - three cycles wide low pulse is rejected
23:16	HIGH_TIME_DIVIDER_VALUE	Allows setting SCL duty cycle to non 50%. If this value is zero then legacy mode is used. If this value is non-zero then it will be used as the SCL high time counter and FS_DIVIDER_VALUE will be used as the low time counter. Minimum value is 0x7.
10:8	HS_DIVIDER_VALUE	The value in this register represents the clock period multiplier in high-speed (HS) mode. Minimum value is 0x3. $\text{I2C_HS_CLK} = \text{I2C_CLK} / (3 * (\text{HS_DIVIDER_VALUE} + 1) + 2)$
7:0	FS_DIVIDER_VALUE	The value in this register represents the clock period multiplier in fast/standard (FS) mode. Minimum value is 0x7. When HIGH_TIME_DIVIDER_VALUE=0: $\text{I2C_FS_CLK} = \text{I2C_CLK} / (2 * (\text{FS_DIVIDER_VALUE} + 3))$ When HIGH_TIME_DIVIDER_VALUE!=0: $\text{I2C_FS_CLK} = \text{I2C_CLK} / (\text{FS_DIVIDER_VALUE} + \text{HIGH_TIME_DIVIDER_VALUE} + 6)$

0x078B7404 BLSP1_BLSP_QUP2_I2C_MASTER_STATUS**Type:** RW**Clock:** crif_clk**Reset State:** 0x0C000000

The I2C_STATUS is a status register. Writing any value clears the status bits.

BLSP1_BLSP_QUP2_I2C_MASTER_STATUS

Bits	Name	Type	Description
30:28	CANCEL_FSM_STATE	R	This 3-bit field informs the microprocessor of the state of the I2C cancel logic. Internal use only 0x0: IDLE_STATE (IDLE_STATE) 0x1: CANCEL_PENDING_STATE (CANCEL_PENDING_STATE) 0x2: PAUSE_STATE (PAUSE_STATE) 0x3: PAUSE_WAIT_STATE (PAUSE_WAIT_STATE) 0x4: SEND_STOP_STATE (SEND_STOP_STATE) 0x5: FLUSH_STATE (FLUSH_STATE) 0x6: SEND_IRQ_STATE (SEND_IRQ_STATE)
27	I2C_SCL	R	Logic state of I2C bus serial clock wire.
26	I2C_SDA	R	Logic state of I2C bus serial data wire.
25	INVALID_READ_SEQ	RW	Interrupt source. This bit is set (1) when a MI_REC tag does not follow a START tag for an I2C READ. Not applicable for Halcyon.
24	INVALID_READ_ADDR	RW	Interrupt source. In version 1 tags this bit is set (1) when the I2C controller is trying to receive data from a non-existent I2C slave (address). In version 2 tags this bit is set (1) when the I2C controller is trying to access a non-existent I2C slave (address).
23	INVALID_TAG	RW	Interrupt source. This bit is set (1) when the I2C controller is trying to process data from the output FIFO that is improperly tagged.
22:20	INPUT_FSM_STATE	R	This 3-bit field informs the microprocessor of the state of the I2C MASTER INPUT FSM block. Internal use only 0x0: V1_V2_RESET_STATE (RESET_STATE value. Applies for both version 1 and version 2 tags) 0x1: V1_V2_IDLE_STATE (IDLE_STATE value. Applies for both version 1 and version 2 tags) 0x2: V1_V2_READ_LAST_BYTE_STATE (READ_LAST_BYTE_STATE value. Applies for both version 1 and version 2 tags) 0x3: V1_V2_MI_REC_STATE (MI_REC_STATE value. Applies for both version 1 and version 2 tags) 0x4: V1_V2_DEC_STATE (DEC_STATE value. Applies for both version 1 and version 2 tags) 0x5: V1_V2_STORE_STATE (STORE_STATE value. Applies for both version 1 and version 2 tags) 0x6: V2_WR_TAG_STATE (WR_TAG_STATE value. Only applicable in version 2 tags) 0x7: V2_WAIT_TAG_STATE (WAIT_TAG_STATE value. Only applicable in version 2 tags)

BLSP1_BLSP_QUP2_I2C_MASTER_STATUS (cont.)

Bits	Name	Type	Description
19:16	OUTPUT_FSM_STATE	R	This 3-bit field informs the microprocessor of the state of the I2C MASTER OUTPUT FSM block. Internal use only 0x0: V1_V2_RESET_STATE (RESET_STATE value. Applies for both version 1 and version 2 tags) 0x1: V1_V2_IDLE_STATE (IDLE_STATE value. Applies for both version 1 and version 2 tags) 0x2: V1_V2_DECODE_STATE (DECODE_STATE value. Applies for both version 1 and version 2 tags) 0x3: V1_V2_SEND_STATE (SEND_STATE value. Applies for both version 1 and version 2 tags) 0x4: V1_V2_MI_REC_STATE (MI_REC_STATE value. Applies for both version 1 and version 2 tags) 0x5: V1_V2_NOP_STATE (NOP_STATE value. Applies for both version 1 and version 2 tags) 0x6: V1_V2_INVALID_STATE (INVALID_STATE value. Applies for both version 1 and version 2 tags) 0x7: V1_PEEK_STATE (PEEK_STATE value. Only applicable in version 1 tags) 0x8: V1_SEND_R_STATE (SEND_R_STATE value. Only applicable in version 1 tags) 0xB: V2_GET_BYTE_STATE (GET_BYTE_STATE value. Only applicable in version 2 tags) 0xC: V2_GET_WRITE_BYTE_STATE (GET_WRITE_BYTE_STATE value. Only applicable in version 2 tags) 0xD: V2_SPECIALITY_STATE (SPECIALITY_STATE value. Only applicable in version 2 tags) 0xE: V2_WAIT_FLUSH_STOP_STATE (WAIT_FLUSH_STOP_STATE value. Only applicable in version 2 tags) 0xF: V2_STOP_STATE (STOP_STATE value. Only applicable in version 2 tags)
15:13	CLK_STATE	R	This 3-bit field informs the microprocessor of the state of the I2C clk_control block. Internal use only 0x0: RESET_BUSIDLE_STATE (RESET_BUSIDLE_STATE) 0x1: NOT_MASTER_STATE (NOT_MASTER_STATE) 0x2: HIGH_STATE (HIGH_STATE) 0x3: LOW_STATE (LOW_STATE) 0x4: HIGH_WAIT_STATE (HIGH_WAIT_STATE) 0x5: FORCED_LOW_STATE (FORCED_LOW_STATE) 0x6: HS_ADDR_LOW_STATE (HS_ADDR_LOW_STATE) 0x7: DOUBLE_BUFFER_WAIT_STATE (DOUBLE_BUFFER_WAIT_STATE)

BLSP1_BLSP_QUP2_I2C_MASTER_STATUS (cont.)

Bits	Name	Type	Description
12:10	DATA_STATE	R	This 3-bit field informs the microprocessor of the state of the I2C data_control block. Internal use only 0x0: RESET_WAIT_STATE (RESET_WAIT_STATE) 0x1: TX_ADDR_STATE (TX_ADDR_STATE) 0x2: TX_DATA_STATE (TX_DATA_STATE) 0x3: TX_HS_ADDR_STATE (TX_HS_ADDR_STATE) 0x4: TX_10_BIT_ADDR_STATE (TX_10_BIT_ADDR_STATE) 0x5: TX_2ND_BYTE_STATE (TX_2ND_BYTE_STATE) 0x6: RX_DATA_STATE (RX_DATA_STATE) 0x7: RX_NACK_HOLD_STATE (RX_NACK_HOLD_STATE)
9	BUS_MASTER	R	This bit is set (1) when the I2C controller is the present bus master.
8	BUS_ACTIVE	R	This bit is set (1) when the bus is in use by this, or any other controller.
7:6	FAILED	RW	Interrupt source. This 2-bit field contains the failure information of the present I2C transfers. If transmitting, failed[1] contains the information regarding the byte that has been transmitted. Failed[0] contains the information of the byte awaiting transmission if there is a byte pipelined. If no byte is pipelined, ignore this bit. If receiving, failed[1] contains the information of the byte received and stored in the buffer, and failed[0] should be ignored. For example: Value 00: Byte n transmitted successfully, byte n+1 to begin transmission Value 01: Byte n transmitted successfully, byte n+1 errored: type of error indicated in other STATUS bits. (queued invalid write, for example) Value 10: Byte n errored: type of error indicated in other STATUS bits, byte n+1 to begin transmission (byte n+1 would have to be a valid address byte for this condition to occur) Value 11: Byte n errored: type of error indicated in other STATUS bits, byte n+1 discarded as well (if the byte was a data byte destined for a NACKed address, data is useless, therefore discarded)
5	INVALID_WRITE	RW	Interrupt source. This bit is set (1) when software writes data to the I2C_DATA register that should be flagged as an address but is not.
4	ARB_LOST	RW	Interrupt source. This bit is set (1) when the controller loses arbitration for the bus. If this bit gets set (1) during the transmission, all data has been lost, and the microprocessor must re-request its transmission.

BLSP1_BLSP_QUP2_I2C_MASTER_STATUS (cont.)

Bits	Name	Type	Description
3	PACKET_NACKED	RW	Interrupt source. This bit is set high when a NACK is received from a slave. If a high speed master code is sent and there is an ACK from a slave, then this bit is set (1) to indicate that the high speed mode can not be entered. If the high speed mode is accepted by the slave, then a NACK is performed and this bit is not set (1).
2	BUS_ERROR	RW	Interrupt source. This bit is set (1) when an unexpected START or STOP condition is detected. This returns the controller to its reset state.
1	TRANSFER_CANCEL_DONE	RW	Only applicable when using version 2 tags. Interrupt source. This bit is set (1) when the I2C controller has completed canceling the transfer that was scheduled for cancellation (LOCAL_ID = TRANSFER_CANCEL_ID).
0	TRANSFER_CANCEL_ID_MATCH	RW	Only applicable when using version 2 tags. Interrupt source. This bit is set (1) when the I2C controller is ready for cancel operation: it has found the transfer that was scheduled for cancellation (LOCAL_ID = TRANSFER_CANCEL_ID) and the logic is done with previous activity.

0x078B7408 BLSP1_BLSP_QUP2_I2C_MASTER_CONFIG**Type:** RW**Clock:** crif_clk**Reset State:** 0x00000000**BLSP1_BLSP_QUP2_I2C_MASTER_CONFIG**

Bits	Name	Description
0	EN_VERSION_TWO_TAG	When set, version 2 of the mini-core is enabled.

0x078B740C BLSP1_BLSP_QUP2_I2C_MASTER_BUS_CLEAR**Type:** RW**Clock:** crif_clk**Reset State:** 0x00000000

BLSP1_BLSP_QUP2_I2C_MASTER_BUS_CLEAR

Bits	Name	Description
0	CLEAR	This command should be given only when the i2c mincore is idle. When in doubt SW can use SW_RESET to reset the minicore. When set, an I2C 'Bus Clear' executed per the I2C standard. A bus clear consists of nine I2C clock ticks with data wire left not-driven. Has the effect of clearing any slave which has lost read sync. After the clear is complete, the hardware sets the CLEAR bit to zero. The bit can not be cleared by software. After a bus clear, both clock and data lines should be high. If not, an external slave has hung the bus by holding down one or both lines. Reset the slave to clear if possible. Any slave reset mechanism is beyond the scope of the I2C mini core.

0x078B7410 BLSP1_BLSP_QUP2_I2C_MASTER_LOCAL_ID**Type:** R**Clock:** crif_clk**Reset State:** 0x00000000**BLSP1_BLSP_QUP2_I2C_MASTER_LOCAL_ID**

Bits	Name	Description
7:0	LOCAL_ID	Read only value capturing the ID byte of the last NOP LOCAL tag. Only applicable when using V2 tags

0x078B7414 BLSP1_BLSP_QUP2_I2C_MASTER_COMMAND**Type:** RW**Clock:** crif_clk**Reset State:** 0x00000000**BLSP1_BLSP_QUP2_I2C_MASTER_COMMAND**

Bits	Name	Description
0	RESET_CANCEL_FSM	This command should be given only when the CANCEL_FSM_STATE is in CANCEL_PENDING_STATE or PAUSE_WAIT_STATE and when the core is in PAUSE. The main use case for this bit is to allow withdrawing a cancel command in case the transfer was flushed (QUP_STATE[I2C_FLUSH] was set) due to I2C NACK.

20.7 BLSP1_BLSP_QUP3_QUP

0x078B8000 BLSP1_BLSP_QUP3_QUP_CONFIG

Type: RW

Clock: crif_clk

Reset State: 0x00000000

Unless otherwise stated, register bits written return the value when read.

- a. The contents of this register should only be changed when in the RESET_STATE (see the QUP_STATE register).
- b. The value written to the QUP output FIFO, is shifted left toward the MSB before passing it to the mini-core as follows:
 - o N equals 8 or less - shift 24
 - o N equals 16 to 9 - shift 16
 - o N equals 24 to 17 - shift 8
 - o N equals 32 to 25 - no shift

BLSP1_BLSP_QUP3_QUP_CONFIG

Bits	Name	Description
16	EN_EXT_OUT_FLAG	Enable Extended OUTPUT/INPUT_SERVICE_FLAG interrupt generation. When this bit is set, the behavior of the *_SERVICE_FLAG and MAX_OUTPUT_DONE_FLAG changes. Service flag and the done flag are not set till the last SPI or I2C transfer has completed. a) For SPI: 1. The shifting needs to complete 2. Clock stopped if CLK_ALWAYS_ON=0, 3. CS, if used, de-asserted b) For I2C, 1. The ACK needs to complete 2. STOP condition done generated (Attention need to be put in synchronizing the last transaction with STOP tag) 3. clock and data wires return to high condition When this bit is clear, the legacy behavior is applicable. Under legacy, the DONE_FLAG is set when: 1. MX_*_COUNT is reached 2. The output FIFO is empty 3. Last bit was sent on the link Under legacy, the flag interrupt is generated before the last transfer is complete on the external interface.
15	CORE_EXTRA_CLK_ON_EN	Enable additional transfer based qup_core_clk gating for power save. When set to 0, clock on is turned on unconditionally. When set to 1, dynamic clock on is turned on.

BLSP1_BLSP_QUP3_QUP_CONFIG (cont.)

Bits	Name	Description
14	FIFO_CLK_GATE_EN	When set to 0, fifo clock is mostly on - legacy mode. When set to 1, fifo clock is turned off dynamically.
13	CORE_CLK_ON_EN	When set to 0, clock on is turned on unconditionally. When set to 1, dynamic clock on is turned on and driven high conditionally.
12	APP_CLK_ON_EN	When set to 0, clock on is turned on unconditionally. When set to 1, dynamic clock on is turned on and driven high conditionally.
11:8	MINI_CORE	value: 0000 null core value: 0001 SPI core value: 0010 I2C master controller value: 0011 reserved value: 0100 reserved value: 0101 reserved value: 0110 reserved value: 0111 reserved See Note 1.
7	NO_INPUT	qup_data_in is not used and the value is a 'don't care'. The QUP input FIFO is always empty. The input service interrupt and INPUT_SERVICE_FLAG bit are never set. See notes (a) and (b) above.
6	NO_OUTPUT	qup_data_out is not used and is held at the value zero. The QUP output FIFO is always empty. The output service flag and OUTPUT_SERVICE_FLAG bit are never set. qup_data_out is still driven when SPI_CS#_N is asserted. The setting for NO_TRI_STATE still applies. See notes (a) and (b) above.
5	QUP_HREADY_CTRL	When set to 0, qup wil not stall the AHB bus (legacy mode). When set to 1, qup may stall the AHB bus until register access is done.
4:0	N	Number of logical bits N in the mini-core that constitutes a single transfer. The value zero indicates N equals one. The value of all ones indicates N equals 32. The value in this register must be 3 (i.e., N is 4 or higher in order for the STATE field to be set to RUN_STATE). See note (a) above.

0x078B8004 BLSP1_BLSP_QUP3_QUP_STATE**Type:** RW**Clock:** crif_clk**Reset State:** 0x0000001C

BLSP1_BLSP_QUP3_QUP_STATE

Bits	Name	Type	Description
6	I2C_FLUSH	RW	Flushing an i2c transfer requires using version 2 tags. FLUSH should be used in BAM mode only. Setting this bit to 1 will flush all tags and tag related data besides EOT until FLUSH STOP tag is encountered. Setting this bit to 0 has no impact. Reading this bit returns the flush operation status - 0 = ongoing, 1 = done.
5	WAIT_FOR_EOT	RW	Only applicable when moving to RUN_STATE using command descriptor in SPI mode with NO_OUTPUT. Setting this bit to 1 will stall the done_toggle_com indication until basi_trans_end_pro is asserted. This will prevent the next command descriptor to start prematurely.
4	I2C_MAST_GEN	R	Read only. Reflects the RTL generic setting for GEN_I2C_MASTER_MINI_CORE.
3	SPI_GEN	R	Read only. Reflects the RTL generic setting for GEN_SPI_MINI_CORE.
2	STATE_VALID	R	Read only. If and only if set to one, writes to the STATE field is allowed or reads from the STATE field is valid. Writes to this bit is ignored.
1:0	STATE	RW	When clear (00), the mini-core and related logic is held in RESET_STATE. When set to '01', the mini-core and related logic is released from reset and enters the RUN_STATE. When set to '11', the mini-core and related logic enters the PAUSE_STATE at the next appropriate point in time. Writing (10) to this field clears these two bits. For PAUSE_STATE to RESET_STATE transition, two writes of (10) are required for the transition to complete.

0x078B8008 BLSP1_BLSP_QUP3_QUP_IO_MODES**Type:** RW**Clock:** crif_clk**Reset State:** 0x000000A5

Unless otherwise stated, register bits written return the value when read.

Notes:

a. 'Packing' occurs as follows:

- N equals 8 or less - pack four values into each QUP input FIFO word.
- N equals 16 to 9 - pack two values into each QUP input FIFO word.
- N equals 32 to 17 - no packing. Each mini-core value is moved to an QUP input FIFO word.

b. 'Un-Packing' occurs as follows:

- N equals 8 or less - un-pack four values from each QUP output FIFO word.
- N equals 16 to 9 - un-pack two values from each QUP output FIFO word.
- N equals 32 to 17 - no packing. Each QUP output FIFO value is moved to the mini-core.

c. INPUT_MODE and OUTPUT_MODE should be both in BAM_Mode or both in non BAM_Mode.

BLSP1_BLSP_QUP3_QUP_IO_MODES

Bits	Name	Type	Description
16	OUTPUT_BIT_SHIFT_EN	RW	If set, enables the QUP output FIFO block to do bit shifting on the output data.
15	PACK_EN	RW	Indicates data values from the mini-core are to be packed before placement in the QUP input FIFO. See note (a) above.
14	UNPACK_EN	RW	Indicates data values taken from the QUP output FIFO should be unpacked before passing to the mini-core. See note (b) above.
13:12	INPUT_MODE	RW	Value 00: FIFO_Mode Value 01: Block_Mode Value 10: Reserved Value 11: BAM_Mode (note if the RTL generic BLOCK_SIZE_INPUT = 0, then only FIFO_Mode is available)
11:10	OUTPUT_MODE	RW	Value 00: FIFO_Mode Value 01: Block_Mode Value 10: Reserved Value 11: BAM_Mode (note if the RTL generic BLOCK_SIZE_OUTPUT = 0, then only FIFO_Mode is available)
9:7	INPUT_FIFO_SIZE	R	Read only, actual value set by RTL generic. Value 000: 2X BLOCK_SIZE Value 001: 4X BLOCK_SIZE Value 010: 8X BLOCK_SIZE Value 011: 16X BLOCK_SIZE
6:5	INPUT_BLOCK_SIZE	R	Read only. Actual value set by RTL generic. Indicates the block size associated with Block_Mode for input. Value 00: 4 Bytes (FIFO_Mode only) Value 01: 16 Bytes Value 10: 32 Bytes Value 11: Reserved
4:2	OUTPUT_FIFO_SIZE	R	Read only. Actual value set by RTL GENERIC. Value 000: 2X BLOCK_SIZE Value 001: 4X BLOCK_SIZE Value 010: 8X BLOCK_SIZE Value 011: 16X BLOCK_SIZE

BLSP1_BLSP_QUP3_QUP_IO_MODES (cont.)

Bits	Name	Type	Description
1:0	OUTPUT_BLOCK_SIZE	R	Read only. Actual value set by RTL GENERIC. Value 00: 04 Bytes (FIFO mode only) Value 01: 16 Bytes Value 10: 32 Bytes Value 11: Reserved

0x078B800C BLSP1_BLSP_QUP3_QUP_SW_RESET**Type:** W**Clock:** crif_clk**Reset State:** 0x00000000

A write to this register resets the entire QUP core and all mini-cores. The internal registers are brought back to their reset values. Reading of this register returns zero. The AHB clock domain reset will stay asserted until either the spi_clk or i2c_clk domains are reset. It is prohibited to write to any QUP register during this period and failing to do so will cause an ERROR response on the AHB bus. In order to avoid this SW should poll the QUP_STATE[STATE_VALID] bit until it is asserted.

BLSP1_BLSP_QUP3_QUP_SW_RESET

Bits	Name	Description
31:0	QUP_SW_RESET	

0x078B8014 BLSP1_BLSP_QUP3_QUP_TRANSFER_CANCEL**Type:** RW**Clock:** crif_clk**Reset State:** 0x00000000**BLSP1_BLSP_QUP3_QUP_TRANSFER_CANCEL**

Bits	Name	Type	Description
15:8	TRANSFER_CANCEL_ID	RW	This field include the id of the transfer that is scheduled for cancellation. This field is valid when asserting TRANSFER_CANCEL bit. A value of 0xFF will cancel the current transfer regardless of its LOCAL_ID. Canceling an i2c transfer requires using version 2 tags.
7	TRANSFER_CANCEL	W	Setting this bit to 1 will cause the HW to go to pause_state when the LOCAL_ID = TRANSFER_CANCEL_ID. This would allow flushing the rest of the transfer in order to implement cancel API. Canceling an i2c transfer requires using version 2 tags.

0x078B8018 BLSP1_BLSP_QUP3_QUP_OPERATIONAL**Type:** RW**Clock:** crif_clk**Reset State:** 0x000000C0

Refer to the chapter on 'QUP Software Considerations' regarding the correct interpretation of the various service interrupts generated by QUP and their relation to these register bits.

BLSP1_BLSP_QUP3_QUP_OPERATIONAL

Bits	Name	Description
15	NWD	Notify When Done (NWD). Read only status bit which when set indicates a NWD acknowledgement is outstanding. This bit is set by QUP hardware when NWD is signaled from BAM. This bit is cleared by QUP hardware when the NWD request is acknowledged by assertion of done_toggle. This bit is always unconditionally cleared in RESET_STATE.
14	DONE_TOGGLE	Read only status bit which provides the current state of the side-band done_toggle signal sent to BAM. At each NWD acknowledgement, this bit toggles.
13	IN_BLOCK_READ_REQ	Read only. When set by hardware, indicates QUP input FIFO has BLOCK_SIZE_INPUT amount of data ready for reading. Valid only in Block_Mode.
12	OUT_BLOCK_WRITE_REQ	Read only. When set by hardware, indicates QUP output FIFO needs BLOCK_SIZE_OUTPUT amount of data to be written. Valid only in Block_Mode.
11	MAX_INPUT_DONE_FLAG	Read only. When set by hardware, indicates QUP input FIFO has reached the programmed QUP_MX_INPUT_COUNT value. Valid in FIFO_Mode (only if MX_READ_COUNT is non-zero), Block_Mode and Bam_Mode (for debug only).
10	MAX_OUTPUT_DONE_FLAG	Read only. When set by hardware, indicates QUP output FIFO has reached the programmed QUP_MX_OUTPUT_COUNT value. Valid in Block_Mode and Bam_Mode (for debug only).
9	INPUT_SERVICE_FLAG	When set by hardware, indicates QUP input FIFO has an outstanding input service request. At the point in time this bit is set, the hardware also asserts qup_irq. Writing a 'zero' to this bit does nothing. Writing a 'one' to this bit clears it and acknowledges software has or will read the data. Valid in all modes , recommended to be masked in BAM mode
8	OUTPUT_SERVICE_FLAG	When set by hardware, indicates QUP output FIFO has an outstanding output service request. At the point in time this bit is set, the hardware also asserts qup_irq. Writing a 'zero' to this bit does nothing. Writing a 'one' to this bit clears it and acknowledges software has or will read the data. Valid in all modes , recommended to be masked in BAM mode
7	INPUT_FIFO_FULL	Read only. When set, indicates the input FIFO is full and can accept no more data from the QUP mini-core.

BLSP1_BLSP_QUP3_QUP_OPERATIONAL (cont.)

Bits	Name	Description
6	OUTPUT_FIFO_FULL	Read only. When set, indicates the output FIFO is full and can accept no more CRIF writes.
5	INPUT_FIFO_NOT_EMPTY	Read only. When set, indicates the input FIFO has at least one value in it to be read. When clear, indicates the input FIFO is empty.
4	OUTPUT_FIFO_NOT_EMPTY	Read only. When set, indicates the output FIFO has at least one value in it to be shifted out. When clear, indicates the output FIFO is empty.

0x078B801C BLSP1_BLSP_QUP3_QUP_ERROR_FLAGS**Type:** RW**Clock:** crif_clk**Reset State:** 0x00000000

All bits in this register are set by hardware and remain set until cleared by software. Writing a 'one' to a bit clears it while writing a 'zero' leaves the bit unchanged.

BLSP1_BLSP_QUP3_QUP_ERROR_FLAGS

Bits	Name	Description
5	OUTPUT_OVER_RUN_ERR	Indicates the output FIFO was full when a CRIF write was attempted to the FIFO. The write is discarded.
4	INPUT_UNDER_RUN_ERR	Indicates the input FIFO was empty when a CRIF read was attempted to the FIFO. The read value returns is indeterminate.
3	OUTPUT_UNDER_RUN_ERR	Indicates the output FIFO was empty when an output shift operation required a value.
2	INPUT_OVER_RUN_ERR	Indicates the input FIFO was full when an input shift operation provided a new value. When this happens, the new value is discarded.

0x078B8020 BLSP1_BLSP_QUP3_QUP_ERROR_FLAGS_EN**Type:** RW**Clock:** crif_clk**Reset State:** 0x0000003C

Each bit in this register controls the setting of a corresponding error flag. If set, each bit enables generation of qup_irq and the setting of the corresponding error flag in the QUP_ERROR_FLAGS register for the specified error case. At reset, all error enable bits are set to '1'.

BLSP1_BLSP_QUP3_QUP_ERROR_FLAGS_EN

Bits	Name	Description
5	OUTPUT_OVER_RUN_ERR_EN	If set, enables output over run error generation.
4	INPUT_UNDER_RUN_ERR_EN	If set, enables input under run error generation.
3	OUTPUT_UNDER_RUN_ERR_EN	If set, enables output under run error generation.
2	INPUT_OVER_RUN_ERR_EN	If set, enables input over run error generation.

0x078B8028 BLSP1_BLSP_QUP3_QUP_OPERATIONAL_MASK**Type:** RW**Clock:** crif_clk**Reset State:** 0x00000000

This register masks several QUP_OPERATIONAL flags from creating an interrupt .

BLSP1_BLSP_QUP3_QUP_OPERATIONAL_MASK

Bits	Name	Description
9	INPUT_SERVICE_MASK	If set, this flag in QUP_OPERATIONAL does not cause an interrupt but provides status only.
8	OUTPUT_SERVICE_MASK	If set, this flag in QUP_OPERATIONAL does not cause an interrupt but provides status only.

0x078B8100 BLSP1_BLSP_QUP3_QUP_MX_OUTPUT_COUNT**Type:** RW**Clock:** crif_clk**Reset State:** 0x00000000

Unless otherwise stated, register bits written return the value when read. This register only applies to Block_Mode. The counter decrements for each output transfer when the STATE field (see QUP_STATE register) is moved from the RESET_STATE to the RUN_STATE. The PAUSE_STATE does not effect the count.

Notes:

- a. Allows the total number of Mini Core transfers to be less than an exact multiple of OUTPUT_BLOCK_SIZE. Any additional outputs are discarded.

BLSP1_BLSP_QUP3_QUP_MX_OUTPUT_COUNT

Bits	Name	Type	Description
31	MX_CONFIG_DURING_RUN	W	Write only value. When reset to zero the MX_OUTPUT_COUNT value only applies in reset_state (legacy behavior). When set to one the MX_OUTPUT_COUNT value will also apply in run_state and allow writing the transaction data in a few portions when not knowing the transaction size in advance. Reconfiguration during run is only allowed after the last portion has ended (i.e. MAX_OUTPUT_DONE_FLAG was asserted)
15:0	MX_OUTPUT_COUNT	RW	Number of writes of size N to the mini-core per RUN_STATE. A value of zero indicates the output count function is not enabled for use. See note (a) above.

0x078B8104 BLSP1_BLSP_QUP3_QUP_MX_OUTPUT_CNT_CURRENT**Type:** R**Clock:** crif_clk**Reset State:** 0x00000000**BLSP1_BLSP_QUP3_QUP_MX_OUTPUT_CNT_CURRENT**

Bits	Name	Description
15:0	MX_OUTPUT_CNT_CURRENT	Current value of QUP_MX_OUTPUT_COUNT.

0x078B810C BLSP1_BLSP_QUP3_QUP_OUTPUT_FIFO_WORD_CNT**Type:** R**Clock:** crif_clk**Reset State:** 0x00000000

This register holds the number of words in the output FIFO at a given time. Note the fields in this register are dynamically updated hence when this register is read during a FIFO operation, the value of this register could have been altered by the time SW gets the read value. Recommended for debug purposes.

BLSP1_BLSP_QUP3_QUP_OUTPUT_FIFO_WORD_CNT

Bits	Name	Description
8:0	OUTPUT_FIFO_WORD_COUNT	Number of words in the output FIFO.

**0x078B8110+ BLSP1_BLSP_QUP3_QUP_OUTPUT_FIFOc, c=[0..15]
0x4*c****Type:** W**Clock:** crif_clk**Reset State:** 0x00000000

Note that consecutive writes to this address keeps filling up the output FIFO. The max address to address the output FIFO is 0x14C.

BLSP1_BLSP_QUP3_QUP_OUTPUT_FIFOc

Bits	Name	Description
31:0	OUTPUT	Value to be shifted out.

0x078B8150 BLSP1_BLSP_QUP3_QUP_MX_WRITE_COUNT**Type:** RW**Clock:** crif_clk**Reset State:** 0x00000000

What the QUP_MX_OUTPUT_COUNT register means to Block_Mode and Bam_Mode, this register means the same to FIFO_mode. If this register is non-zero, then the qup_irq is asserted after shifting in the number of shifts specified by this register.

Unlike the QUP_MX_OUTPUT_COUNT register case, the SW should not program the QUP_MX_WRITE_COUNT value to be more than the 'actual depth' of the FIFO (BLOCK_SIZE_OUTPUT * FIFO_SIZE_OUTPUT).

BLSP1_BLSP_QUP3_QUP_MX_WRITE_COUNT

Bits	Name	Type	Description
31	MX_CONFIG_DURING_RUN	W	Write only value. When reset to zero the MX_WRITE_COUNT value only applies in reset_state (legacy behavior). When set to one the MX_WRITE_COUNT value will also apply in run_state and allow writing the transaction data in a few portions when not knowing the transaction size in advance. Reconfiguration during run is only allowed after the last portion has ended (i.e. MAX_OUTPUT_DONE_FLAG was asserted)
15:0	MX_WRITE_COUNT	RW	The number of 'writes' of size N. This is used only if the core is in FIFO_Mode.

0x078B8154 BLSP1_BLSP_QUP3_QUP_MX_WRITE_CNT_CURRENT**Type:** R**Clock:** crif_clk**Reset State:** 0x00000000**BLSP1_BLSP_QUP3_QUP_MX_WRITE_CNT_CURRENT**

Bits	Name	Description
15:0	MX_WRITE_CNT_CURRENT	Current value of QUP_MX_WRITE_COUNT counter.

0x078B8200 BLSP1_BLSP_QUP3_QUP_MX_INPUT_COUNT**Type:** RW**Clock:** crif_clk**Reset State:** 0x00000000

Unless otherwise stated, register bits written return the value when read. This register only applies to Block_Mode and Bam_Mode (for non-balanced SPI). The counter decrements for each input transfer when the STATE field (see QUP_STATE register) is moved from the RESET_STATE to the RUN_STATE. The PAUSE_STATE does not affect the count.

Notes:

a. Allows the number of shift register transfers to be less than an exact multiple of INPUT_BLOCK_SIZE. When count reached, remainder of INPUT_BLOCK_SIZE is filled with zeroes.

BLSP1_BLSP_QUP3_QUP_MX_INPUT_COUNT

Bits	Name	Type	Description
31	MX_CONFIG_DURING_RUN	W	Write only value. When reset to zero the MX_INPUT_COUNT value only applies in reset_state (legacy behavior). When set to one the MX_INPUT_COUNT value will also apply in run_state and allow writing the transaction data in a few portions when not knowing the transaction size in advance. Reconfiguration during run is only allowed after the last portion has ended (i.e. MAX_INPUT_DONE_FLAG was asserted)
15:0	MX_INPUT_COUNT	RW	Number of reads of size N from the mini-core per RUN_STATE. A value of zero indicates the input count function is not enabled for use. See note (a) above.

0x078B8204 BLSP1_BLSP_QUP3_QUP_MX_INPUT_CNT_CURRENT**Type:** R**Clock:** crif_clk**Reset State:** 0x00000000**BLSP1_BLSP_QUP3_QUP_MX_INPUT_CNT_CURRENT**

Bits	Name	Description
15:0	MX_INPUT_CNT_CURRENT	Current value of QUP_MX_INPUT_COUNT.

0x078B8208 BLSP1_BLSP_QUP3_QUP_MX_READ_COUNT**Type:** RW**Clock:** crif_clk**Reset State:** 0x00000000

What the QUP_MX_INPUT_COUNT register means to Block_Mode and Bam_Mode, this register means the same to FIFO_mode. If this register is non-zero, then the qup_input_service_irq is asserted after shifting in the number of shifts specified by this register.

Unlike the QUP_MX_INPUT_COUNT register case, the SW should not program the QUP_MX_READ_COUNT value to be more than the 'actual depth' of the FIFO (BLOCK_SIZE_INPUT * FIFO_SIZE_INPUT). If the SPI mini-core and slave operation is enabled, then an InputOverRun error will result.

BLSP1_BLSP_QUP3_QUP_MX_READ_COUNT

Bits	Name	Type	Description
31	MX_CONFIG_DURING_RUN	W	Write only value. When reset to zero the MX_READ_COUNT value only applies in reset_state (legacy behavior). When set to one the MX_READ_COUNT value will also apply in run_state and allow writing the transaction data in a few portions when not knowing the transaction size in advance. Reconfiguration during run is only allowed after the last portion has ended (i.e. MAX_INPUT_DONE_FLAG was asserted)
15:0	MX_READ_COUNT	RW	The number of 'reads' of size N. This is used only if the core is in FIFO_Mode.

0x078B820C BLSP1_BLSP_QUP3_QUP_MX_READ_CNT_CURRENT**Type:** R**Clock:** crif_clk**Reset State:** 0x00000000

BLSP1_BLSP_QUP3_QUP_MX_READ_CNT_CURRENT

Bits	Name	Description
15:0	MX_READ_CNT_CURRENT	Current value of QUP_MX_READ_COUNT counter.

0x078B8214 BLSP1_BLSP_QUP3_QUP_INPUT_FIFO_WORD_CNT**Type:** R**Clock:** crif_clk**Reset State:** 0x00000000

This register holds the number of words in the input FIFO at a given time. Note the fields in this register are dynamically updated. Hence, when this register is read during a FIFO operation, the value of this register could have been altered by the time SW gets the read value. Recommended for debug purposes.

BLSP1_BLSP_QUP3_QUP_INPUT_FIFO_WORD_CNT

Bits	Name	Description
8:0	INPUT_FIFO_WORD_CNT	Number of words in the input FIFO.

**0x078B8218+ BLSP1_BLSP_QUP3_QUP_INPUT_FIFOc, c=[0..15]
0x4*c****Type:** R**Clock:** crif_clk**Reset State:** Undefined

Consecutive reads to this address reads the input FIFO contents on a FIFO basis. The max address to read the input FIFO is 0x254.

BLSP1_BLSP_QUP3_QUP_INPUT_FIFOc

Bits	Name	Description
31:0	INPUT	Value shifted in.

0x078B8300 BLSP1_BLSP_QUP3_SPI_CONFIG**Type:** RW**Clock:** crif_clk**Reset State:** 0x00000000

Unless otherwise stated, register bits written return the value when read.

The contents of this register should only be changed when in the RESET_STATE. See the QUP_STATE register. Both NO_OUTPUT and NO_INPUT set to zero provides default full duplex operation. Setting both these bits to one is not a legal operational state.

BLSP1_BLSP_QUP3_SPI_CONFIG

Bits	Name	Type	Description
10	HS_MODE	RW	When set, SPI HS_MODE is enabled. When clear, SPI HS_MODE is disabled.
9	INPUT_FIRST	RW	When set, INPUT FIRST SPI protocol is used. When clear, OUTPUT FIRST SPI protocol is used.
8	LOOP_BACK	RW	Loopback is only valid in non-HS mode. Always clear for normal operation. If set, loop back on SPI_DATA_MO_SI under MASTER operation or SPI_DATA_MI_SO under SLAVE operation.
5	SLAVE_OPERATION	R	This register is writable only when the hardware signal spi_slave_en is asserted. When set, the SPI is configured for SLAVE operation. Zero indicates MASTER operation.

0x078B8304 BLSP1_BLSP_QUP3_SPI_IO_CONTROL

Type: RW

Clock: crif_clk

Reset State: 0x00000000

The contents of this register should only be changed when in the RESET_STATE. See the QUP_STATE register.

Unless otherwise stated, register bits written return the value when read.

BLSP1_BLSP_QUP3_SPI_IO_CONTROL

Bits	Name	Description
11	FORCE_CS	When this bit is set, the chip select is asserted unconditionally with no relationship to QUP_STATE or transaction state.
10	CLK_IDLE_HIGH	When set to 1, spi_clk will be high when spi is idle (spi_cs is not asserted). When cleared to 0, spi_clk will be low when spi is idle (spi_cs is not asserted).
9	CLK_ALWAYS_ON	When MASTER operation is used, run SPI_CLK during IDLE.
8	MX_CS_MODE	If this bit is set, then for a given RUN state, the associated chip select will be asserted for the first N-bit transfer and will be kept asserted till the last programmed transfer. Applies only in MASTER mode. The number of transfers is determined by SPI_MX_OUTPUT_COUNT and/or SPI_MX_INPUT_COUNT registers.

BLSP1_BLSP_QUP3_SPI_IO_CONTROL (cont.)

Bits	Name	Description
7:4	CS_N_POLARITY	These four bits control the polarity of four SPI_CS#_N respectively. Setting any of this bit to '1', makes the associated SPI_CS#_N active HIGH. This field is a 'don't care' in SLAVE operation.
3:2	CS_SELECT	When MASTER operation is used, controls the assertion of SPI_CS_#N pins. Selects SPI_CS_N if 00, SPI_CS1_N if 01, SPI_CS2_N if 10, or SPI_CS3_N if 11. The deselected ChipSelects will be driven to inactive state dictated by the field CS_N_POLARITY. This field is a 'don't care' in SLAVE operation.
1	TRISTATE_CS	When set, drives Z on all SPI_CS#_N lines. When clear, enables normal functionality on these lines. This bit can be used to support deployment of the core as one of the masters in a multi-master configuration.
0	NO_TRI_STATE	When set, spi_data_out is not taken tri-state when SPI_CS#_N is de-asserted. This bit is normally set for MASTER operation but may be optionally left cleared.

0x078B8308 BLSP1_BLSP_QUP3_SPI_ERROR_FLAGS**Type:** RW**Clock:** crif_clk**Reset State:** 0x00000000

All bits in this register are set by hardware and remain set until cleared by software. Writing a 'one' to a bit clears it while writing a 'zero' leaves the bit unchanged.

BLSP1_BLSP_QUP3_SPI_ERROR_FLAGS

Bits	Name	Description
3	TRANSFER_CANCEL_DONE	This bit is set (1) when the SPI controller has completed canceling the transfer that was scheduled for cancellation (LOCAL_ID = TRANSFER_CANCEL_ID).
2	TRANSFER_CANCEL_ID_MATCH	This bit is set (1) when the SPI controller is ready for cancel operation: it has found the transfer that was scheduled for cancellation (LOCAL_ID = TRANSFER_CANCEL_ID) and the logic is done with previous activity.
1	SPI_SLV_CLK_OVER_RUN_ERR	Number of SPI_CLK ticks which occurred while SPI_CS#_N was asserted was greater than the programmed value of N. When this happens, only the first N bits are passed to the SPI input FIFO and the output shift value is held at zero.
0	SPI_SLV_CLK_UNDER_RUN_ERR	Number of SPI_CLK ticks which occurred while SPI_CS#_N was asserted was less than the programmed value of N. When this occurs, the bits which were received are passed to the SPI input FIFO and the CURRENT output bit is forced to zero.

0x078B830C BLSP1_BLSP_QUP3_SPI_ERROR_FLAGS_EN**Type:** RW**Clock:** crif_clk**Reset State:** 0x0000000C

Each bit in this register controls the setting of a corresponding error flag. If set, each bit enables generation of spi_error_irq and the setting of the corresponding error flag in the SPI_ERROR_FLAGS register for the specified error case. At reset, all error enable bits are set to '1'.

BLSP1_BLSP_QUP3_SPI_ERROR_FLAGS_EN

Bits	Name	Description
3	TRANSFER_CANCEL_DONE_EN	If set, enables generating the transfer cancel done indication.
2	TRANSFER_CANCEL_ID_MATCH_EN	If set, enables generating the transfer cancel ID match indication.
1	SPI_SLV_CLK_OVER_RUN_ERR_EN	If set, enables clock over run error generation.
0	SPI_SLV_CLK_UNDER_RUN_ERR_EN	If set, enables clock under run error generation.

0x078B8310 BLSP1_BLSP_QUP3_SPI_DEASSERT_WAIT**Type:** RW**Clock:** crif_clk**Reset State:** 0x00000000

This register holds the de-assertion wait time of SPI_CS#_N between consecutive N-bit transfers in MASTER operation. The wait time is specified in number of cycles of cc_spi_master_clk.

BLSP1_BLSP_QUP3_SPI_DEASSERT_WAIT

Bits	Name	Description
5:0	DEASSERT_WAIT	Number cc_spi_master_clk ticks associated with the deasserted time of SPI_CS#_N. Only applies to MASTER operation. For SLAVE operation, this field is a 'don't care'. A value of zero indicates SPI_CS#_N remains de-asserted for exactly one clock tick. A value of one, indicates two ticks, etc. All ones indicates 64 ticks.

0x078B8314 BLSP1_BLSP_QUP3_SPI_MASTER_LOCAL_ID**Type:** RW**Clock:** crif_clk**Reset State:** 0x00000000**BLSP1_BLSP_QUP3_SPI_MASTER_LOCAL_ID**

Bits	Name	Description
15:8	EXTENDED_ID	SW may use this field to specify the current sub-transfer ID to be read when TRANSFER_CANCEL_ID_MATCH is asserted. This value has no impact on HW.
7:0	LOCAL_ID	SW specifies the current transfer ID. HW will use this value to match with TRANSFER_CANCEL_ID.

0x078B8318 BLSP1_BLSP_QUP3_SPI_MASTER_COMMAND**Type:** W**Clock:** crif_clk**Reset State:** 0x00000000**BLSP1_BLSP_QUP3_SPI_MASTER_COMMAND**

Bits	Name	Description
0	RESET_CANCEL_FSM	This command should be given only when the CANCEL_FSM_STATE is in CANCEL_PENDING_STATE or PAUSE_WAIT_STATE and when the core is in PAUSE. The main use case for this bit is to allow withdrawing a cancel command.

0x078B831C BLSP1_BLSP_QUP3_SPI_MASTER_STATUS**Type:** R**Clock:** crif_clk**Reset State:** 0x00000000**BLSP1_BLSP_QUP3_SPI_MASTER_STATUS**

Bits	Name	Description
2:0	CANCEL_FSM_STATE	This 3-bit field informs the microprocessor of the state of the SPI cancel logic. Internal use only

0x078B8400 BLSP1_BLSP_QUP3_I2C_MASTER_CLK_CTL**Type:** RW**Clock:** crif_clk**Reset State:** 0x00000000

The I2C_CLK_CTL register is a read/write register that controls clock divider values.

BLSP1_BLSP_QUP3_I2C_MASTER_CLK_CTL

Bits	Name	Description
28	SCL_EXT_FORCE_LOW	When set to 0, SCL generation maintains legacy behavior. When set to 1, SCL state machine will go to the FORCED_LOW_STATE if force_low is asserted near (within 5 SCL clock cycles = $5 * 52\text{ns} = 260\text{ns} \approx \text{MAX}(\text{Tsu}, \text{DAT})$) the clock edge.
27:26	SDA_NOISE_REJECTION	Allows adding extra sampling levels on SDA to reject short low pulses. This value specifies how many TCXO cycles of logic low on SDA would be considered as valid logic low. 0x0 - legacy mode, 0x01 - one cycle wide low pulse is rejected, 0x2 - two cycles wide low pulse is rejected, 0x3 - three cycles wide low pulse is rejected
25:24	SCL_NOISE_REJECTION	Allows adding extra sampling levels on SCL to reject short low pulses. This value specifies how many TCXO cycles of logic low on SCL would be considered as valid logic low. 0x0 - legacy mode, 0x01 - one cycle wide low pulse is rejected, 0x2 - two cycles wide low pulse is rejected, 0x3 - three cycles wide low pulse is rejected
23:16	HIGH_TIME_DIVIDER_VALUE	Allows setting SCL duty cycle to non 50%. If this value is zero then legacy mode is used. If this value is non-zero then it will be used as the SCL high time counter and FS_DIVIDER_VALUE will be used as the low time counter. Minimum value is 0x7.
10:8	HS_DIVIDER_VALUE	The value in this register represents the clock period multiplier in high-speed (HS) mode. Minimum value is 0x3. $\text{I2C_HS_CLK} = \text{I2C_CLK} / (3 * (\text{HS_DIVIDER_VALUE} + 1) + 2)$
7:0	FS_DIVIDER_VALUE	The value in this register represents the clock period multiplier in fast/standard (FS) mode. Minimum value is 0x7. When HIGH_TIME_DIVIDER_VALUE=0: $\text{I2C_FS_CLK} = \text{I2C_CLK} / (2 * (\text{FS_DIVIDER_VALUE} + 3))$ When HIGH_TIME_DIVIDER_VALUE!=0: $\text{I2C_FS_CLK} = \text{I2C_CLK} / (\text{FS_DIVIDER_VALUE} + \text{HIGH_TIME_DIVIDER_VALUE} + 6)$

0x078B8404 BLSP1_BLSP_QUP3_I2C_MASTER_STATUS**Type:** RW**Clock:** crif_clk**Reset State:** 0x0C000000

The I2C_STATUS is a status register. Writing any value clears the status bits.

BLSP1_BLSP_QUP3_I2C_MASTER_STATUS

Bits	Name	Type	Description
30:28	CANCEL_FSM_STATE	R	This 3-bit field informs the microprocessor of the state of the I2C cancel logic. Internal use only 0x0: IDLE_STATE (IDLE_STATE) 0x1: CANCEL_PENDING_STATE (CANCEL_PENDING_STATE) 0x2: PAUSE_STATE (PAUSE_STATE) 0x3: PAUSE_WAIT_STATE (PAUSE_WAIT_STATE) 0x4: SEND_STOP_STATE (SEND_STOP_STATE) 0x5: FLUSH_STATE (FLUSH_STATE) 0x6: SEND_IRQ_STATE (SEND_IRQ_STATE)
27	I2C_SCL	R	Logic state of I2C bus serial clock wire.
26	I2C_SDA	R	Logic state of I2C bus serial data wire.
25	INVALID_READ_SEQ	RW	Interrupt source. This bit is set (1) when a MI_REC tag does not follow a START tag for an I2C READ. Not applicable for Halcyon.
24	INVALID_READ_ADDR	RW	Interrupt source. In version 1 tags this bit is set (1) when the I2C controller is trying to receive data from a non-existent I2C slave (address). In version 2 tags this bit is set (1) when the I2C controller is trying to access a non-existent I2C slave (address).
23	INVALID_TAG	RW	Interrupt source. This bit is set (1) when the I2C controller is trying to process data from the output FIFO that is improperly tagged.
22:20	INPUT_FSM_STATE	R	This 3-bit field informs the microprocessor of the state of the I2C MASTER INPUT FSM block. Internal use only 0x0: V1_V2_RESET_STATE (RESET_STATE value. Applies for both version 1 and version 2 tags) 0x1: V1_V2_IDLE_STATE (IDLE_STATE value. Applies for both version 1 and version 2 tags) 0x2: V1_V2_READ_LAST_BYTE_STATE (READ_LAST_BYTE_STATE value. Applies for both version 1 and version 2 tags) 0x3: V1_V2_MI_REC_STATE (MI_REC_STATE value. Applies for both version 1 and version 2 tags) 0x4: V1_V2_DEC_STATE (DEC_STATE value. Applies for both version 1 and version 2 tags) 0x5: V1_V2_STORE_STATE (STORE_STATE value. Applies for both version 1 and version 2 tags) 0x6: V2_WR_TAG_STATE (WR_TAG_STATE value. Only applicable in version 2 tags) 0x7: V2_WAIT_TAG_STATE (WAIT_TAG_STATE value. Only applicable in version 2 tags)

BLSP1_BLSP_QUP3_I2C_MASTER_STATUS (cont.)

Bits	Name	Type	Description
19:16	OUTPUT_FSM_STATE	R	This 3-bit field informs the microprocessor of the state of the I2C MASTER OUTPUT FSM block. Internal use only 0x0: V1_V2_RESET_STATE (RESET_STATE value. Applies for both version 1 and version 2 tags) 0x1: V1_V2_IDLE_STATE (IDLE_STATE value. Applies for both version 1 and version 2 tags) 0x2: V1_V2_DECODE_STATE (DECODE_STATE value. Applies for both version 1 and version 2 tags) 0x3: V1_V2_SEND_STATE (SEND_STATE value. Applies for both version 1 and version 2 tags) 0x4: V1_V2_MI_REC_STATE (MI_REC_STATE value. Applies for both version 1 and version 2 tags) 0x5: V1_V2_NOP_STATE (NOP_STATE value. Applies for both version 1 and version 2 tags) 0x6: V1_V2_INVALID_STATE (INVALID_STATE value. Applies for both version 1 and version 2 tags) 0x7: V1_PEEK_STATE (PEEK_STATE value. Only applicable in version 1 tags) 0x8: V1_SEND_R_STATE (SEND_R_STATE value. Only applicable in version 1 tags) 0xB: V2_GET_BYTE_STATE (GET_BYTE_STATE value. Only applicable in version 2 tags) 0xC: V2_GET_WRITE_BYTE_STATE (GET_WRITE_BYTE_STATE value. Only applicable in version 2 tags) 0xD: V2_SPECIALITY_STATE (SPECIALITY_STATE value. Only applicable in version 2 tags) 0xE: V2_WAIT_FLUSH_STOP_STATE (WAIT_FLUSH_STOP_STATE value. Only applicable in version 2 tags) 0xF: V2_STOP_STATE (STOP_STATE value. Only applicable in version 2 tags)
15:13	CLK_STATE	R	This 3-bit field informs the microprocessor of the state of the I2C clk_control block. Internal use only 0x0: RESET_BUSIDLE_STATE (RESET_BUSIDLE_STATE) 0x1: NOT_MASTER_STATE (NOT_MASTER_STATE) 0x2: HIGH_STATE (HIGH_STATE) 0x3: LOW_STATE (LOW_STATE) 0x4: HIGH_WAIT_STATE (HIGH_WAIT_STATE) 0x5: FORCED_LOW_STATE (FORCED_LOW_STATE) 0x6: HS_ADDR_LOW_STATE (HS_ADDR_LOW_STATE) 0x7: DOUBLE_BUFFER_WAIT_STATE (DOUBLE_BUFFER_WAIT_STATE)

BLSP1_BLSP_QUP3_I2C_MASTER_STATUS (cont.)

Bits	Name	Type	Description
12:10	DATA_STATE	R	This 3-bit field informs the microprocessor of the state of the I2C data_control block. Internal use only 0x0: RESET_WAIT_STATE (RESET_WAIT_STATE) 0x1: TX_ADDR_STATE (TX_ADDR_STATE) 0x2: TX_DATA_STATE (TX_DATA_STATE) 0x3: TX_HS_ADDR_STATE (TX_HS_ADDR_STATE) 0x4: TX_10_BIT_ADDR_STATE (TX_10_BIT_ADDR_STATE) 0x5: TX_2ND_BYTE_STATE (TX_2ND_BYTE_STATE) 0x6: RX_DATA_STATE (RX_DATA_STATE) 0x7: RX_NACK_HOLD_STATE (RX_NACK_HOLD_STATE)
9	BUS_MASTER	R	This bit is set (1) when the I2C controller is the present bus master.
8	BUS_ACTIVE	R	This bit is set (1) when the bus is in use by this, or any other controller.
7:6	FAILED	RW	Interrupt source. This 2-bit field contains the failure information of the present I2C transfers. If transmitting, failed[1] contains the information regarding the byte that has been transmitted. Failed[0] contains the information of the byte awaiting transmission if there is a byte pipelined. If no byte is pipelined, ignore this bit. If receiving, failed[1] contains the information of the byte received and stored in the buffer, and failed[0] should be ignored. For example: Value 00: Byte n transmitted successfully, byte n+1 to begin transmission Value 01: Byte n transmitted successfully, byte n+1 errored: type of error indicated in other STATUS bits. (queued invalid write, for example) Value 10: Byte n errored: type of error indicated in other STATUS bits, byte n+1 to begin transmission (byte n+1 would have to be a valid address byte for this condition to occur) Value 11: Byte n errored: type of error indicated in other STATUS bits, byte n+1 discarded as well (if the byte was a data byte destined for a NACKed address, data is useless, therefore discarded)
5	INVALID_WRITE	RW	Interrupt source. This bit is set (1) when software writes data to the I2C_DATA register that should be flagged as an address but is not.
4	ARB_LOST	RW	Interrupt source. This bit is set (1) when the controller loses arbitration for the bus. If this bit gets set (1) during the transmission, all data has been lost, and the microprocessor must re-request its transmission.

BLSP1_BLSP_QUP3_I2C_MASTER_STATUS (cont.)

Bits	Name	Type	Description
3	PACKET_NACKED	RW	Interrupt source. This bit is set high when a NACK is received from a slave. If a high speed master code is sent and there is an ACK from a slave, then this bit is set (1) to indicate that the high speed mode can not be entered. If the high speed mode is accepted by the slave, then a NACK is performed and this bit is not set (1).
2	BUS_ERROR	RW	Interrupt source. This bit is set (1) when an unexpected START or STOP condition is detected. This returns the controller to its reset state.
1	TRANSFER_CANCEL_DONE	RW	Only applicable when using version 2 tags. Interrupt source. This bit is set (1) when the I2C controller has completed canceling the transfer that was scheduled for cancellation (LOCAL_ID = TRANSFER_CANCEL_ID).
0	TRANSFER_CANCEL_ID_MATCH	RW	Only applicable when using version 2 tags. Interrupt source. This bit is set (1) when the I2C controller is ready for cancel operation: it has found the transfer that was scheduled for cancellation (LOCAL_ID = TRANSFER_CANCEL_ID) and the logic is done with previous activity.

0x078B8408 BLSP1_BLSP_QUP3_I2C_MASTER_CONFIG**Type:** RW**Clock:** crif_clk**Reset State:** 0x00000000**BLSP1_BLSP_QUP3_I2C_MASTER_CONFIG**

Bits	Name	Description
0	EN_VERSION_TWO_TAG	When set, version 2 of the mini-core is enabled.

0x078B840C BLSP1_BLSP_QUP3_I2C_MASTER_BUS_CLEAR**Type:** RW**Clock:** crif_clk**Reset State:** 0x00000000

BLSP1_BLSP_QUP3_I2C_MASTER_BUS_CLEAR

Bits	Name	Description
0	CLEAR	This command should be given only when the i2c mincore is idle. When in doubt SW can use SW_RESET to reset the minicore. When set, an I2C 'Bus Clear' executed per the I2C standard. A bus clear consists of nine I2C clock ticks with data wire left not-driven. Has the effect of clearing any slave which has lost read sync. After the clear is complete, the hardware sets the CLEAR bit to zero. The bit can not be cleared by software. After a bus clear, both clock and data lines should be high. If not, an external slave has hung the bus by holding down one or both lines. Reset the slave to clear if possible. Any slave reset mechanism is beyond the scope of the I2C mini core.

0x078B8410 BLSP1_BLSP_QUP3_I2C_MASTER_LOCAL_ID**Type:** R**Clock:** crif_clk**Reset State:** 0x00000000**BLSP1_BLSP_QUP3_I2C_MASTER_LOCAL_ID**

Bits	Name	Description
7:0	LOCAL_ID	Read only value capturing the ID byte of the last NOP LOCAL tag. Only applicable when using V2 tags

0x078B8414 BLSP1_BLSP_QUP3_I2C_MASTER_COMMAND**Type:** RW**Clock:** crif_clk**Reset State:** 0x00000000**BLSP1_BLSP_QUP3_I2C_MASTER_COMMAND**

Bits	Name	Description
0	RESET_CANCEL_FSM	This command should be given only when the CANCEL_FSM_STATE is in CANCEL_PENDING_STATE or PAUSE_WAIT_STATE and when the core is in PAUSE. The main use case for this bit is to allow withdrawing a cancel command in case the transfer was flushed (QUP_STATE[I2C_FLUSH] was set) due to I2C NACK.

20.8 BLSP1_BLSP_QUP4_QUP

0x078B9000 BLSP1_BLSP_QUP4_QUP_CONFIG

Type: RW

Clock: crif_clk

Reset State: 0x00000000

Unless otherwise stated, register bits written return the value when read.

- a. The contents of this register should only be changed when in the RESET_STATE (see the QUP_STATE register).
- b. The value written to the QUP output FIFO, is shifted left toward the MSB before passing it to the mini-core as follows:
 - o N equals 8 or less - shift 24
 - o N equals 16 to 9 - shift 16
 - o N equals 24 to 17 - shift 8
 - o N equals 32 to 25 - no shift

BLSP1_BLSP_QUP4_QUP_CONFIG

Bits	Name	Description
16	EN_EXT_OUT_FLAG	Enable Extended OUTPUT/INPUT_SERVICE_FLAG interrupt generation. When this bit is set, the behavior of the *_SERVICE_FLAG and MAX_OUTPUT_DONE_FLAG changes. Service flag and the done flag are not set till the last SPI or I2C transfer has completed. a) For SPI: 1. The shifting needs to complete 2. Clock stopped if CLK_ALWAYS_ON=0, 3. CS, if used, de-asserted b) For I2C, 1. The ACK needs to complete 2. STOP condition done generated (Attention need to be put in synchronizing the last transaction with STOP tag) 3. clock and data wires return to high condition When this bit is clear, the legacy behavior is applicable. Under legacy, the DONE_FLAG is set when: 1. MX_*_COUNT is reached 2. The output FIFO is empty 3. Last bit was sent on the link Under legacy, the flag interrupt is generated before the last transfer is complete on the external interface.
15	CORE_EXTRA_CLK_ON_EN	Enable additional transfer based qup_core_clk gating for power save. When set to 0, clock on is turned on unconditionally. When set to 1, dynamic clock on is turned on.

BLSP1_BLSP_QUP4_QUP_CONFIG (cont.)

Bits	Name	Description
14	FIFO_CLK_GATE_EN	When set to 0, fifo clock is mostly on - legacy mode. When set to 1, fifo clock is turned off dynamically.
13	CORE_CLK_ON_EN	When set to 0, clock on is turned on unconditionally. When set to 1, dynamic clock on is turned on and driven high conditionally.
12	APP_CLK_ON_EN	When set to 0, clock on is turned on unconditionally. When set to 1, dynamic clock on is turned on and driven high conditionally.
11:8	MINI_CORE	value: 0000 null core value: 0001 SPI core value: 0010 I2C master controller value: 0011 reserved value: 0100 reserved value: 0101 reserved value: 0110 reserved value: 0111 reserved See Note 1.
7	NO_INPUT	qup_data_in is not used and the value is a 'don't care'. The QUP input FIFO is always empty. The input service interrupt and INPUT_SERVICE_FLAG bit are never set. See notes (a) and (b) above.
6	NO_OUTPUT	qup_data_out is not used and is held at the value zero. The QUP output FIFO is always empty. The output service flag and OUTPUT_SERVICE_FLAG bit are never set. qup_data_out is still driven when SPI_CS#_N is asserted. The setting for NO_TRI_STATE still applies. See notes (a) and (b) above.
5	QUP_HREADY_CTRL	When set to 0, qup wil not stall the AHB bus (legacy mode). When set to 1, qup may stall the AHB bus until register access is done.
4:0	N	Number of logical bits N in the mini-core that constitutes a single transfer. The value zero indicates N equals one. The value of all ones indicates N equals 32. The value in this register must be 3 (i.e., N is 4 or higher in order for the STATE field to be set to RUN_STATE). See note (a) above.

0x078B9004 BLSP1_BLSP_QUP4_QUP_STATE**Type:** RW**Clock:** crif_clk**Reset State:** 0x0000001C

BLSP1_BLSP_QUP4_QUP_STATE

Bits	Name	Type	Description
6	I2C_FLUSH	RW	Flushing an i2c transfer requires using version 2 tags. FLUSH should be used in BAM mode only. Setting this bit to 1 will flush all tags and tag related data besides EOT until FLUSH STOP tag is encountered. Setting this bit to 0 has no impact. Reading this bit returns the flush operation status - 0 = ongoing, 1 = done.
5	WAIT_FOR_EOT	RW	Only applicable when moving to RUN_STATE using command descriptor in SPI mode with NO_OUTPUT. Setting this bit to 1 will stall the done_toggle_com indication until basi_trans_end_pro is asserted. This will prevent the next command descriptor to start prematurely.
4	I2C_MAST_GEN	R	Read only. Reflects the RTL generic setting for GEN_I2C_MASTER_MINI_CORE.
3	SPI_GEN	R	Read only. Reflects the RTL generic setting for GEN_SPI_MINI_CORE.
2	STATE_VALID	R	Read only. If and only if set to one, writes to the STATE field is allowed or reads from the STATE field is valid. Writes to this bit is ignored.
1:0	STATE	RW	When clear (00), the mini-core and related logic is held in RESET_STATE. When set to '01', the mini-core and related logic is released from reset and enters the RUN_STATE. When set to '11', the mini-core and related logic enters the PAUSE_STATE at the next appropriate point in time. Writing (10) to this field clears these two bits. For PAUSE_STATE to RESET_STATE transition, two writes of (10) are required for the transition to complete.

0x078B9008 BLSP1_BLSP_QUP4_QUP_IO_MODES**Type:** RW**Clock:** crif_clk**Reset State:** 0x000000A5

Unless otherwise stated, register bits written return the value when read.

Notes:

a. 'Packing' occurs as follows:

- N equals 8 or less - pack four values into each QUP input FIFO word.
- N equals 16 to 9 - pack two values into each QUP input FIFO word.
- N equals 32 to 17 - no packing. Each mini-core value is moved to an QUP input FIFO word.

b. 'Un-Packing' occurs as follows:

- N equals 8 or less - un-pack four values from each QUP output FIFO word.
- N equals 16 to 9 - un-pack two values from each QUP output FIFO word.
- N equals 32 to 17 - no packing. Each QUP output FIFO value is moved to the mini-core.

c. INPUT_MODE and OUTPUT_MODE should be both in BAM_Mode or both in non BAM_Mode.

BLSP1_BLSP_QUP4_QUP_IO_MODES

Bits	Name	Type	Description
16	OUTPUT_BIT_SHIFT_EN	RW	If set, enables the QUP output FIFO block to do bit shifting on the output data.
15	PACK_EN	RW	Indicates data values from the mini-core are to be packed before placement in the QUP input FIFO. See note (a) above.
14	UNPACK_EN	RW	Indicates data values taken from the QUP output FIFO should be unpacked before passing to the mini-core. See note (b) above.
13:12	INPUT_MODE	RW	Value 00: FIFO_Mode Value 01: Block_Mode Value 10: Reserved Value 11: BAM_Mode (note if the RTL generic BLOCK_SIZE_INPUT = 0, then only FIFO_Mode is available)
11:10	OUTPUT_MODE	RW	Value 00: FIFO_Mode Value 01: Block_Mode Value 10: Reserved Value 11: BAM_Mode (note if the RTL generic BLOCK_SIZE_OUTPUT = 0, then only FIFO_Mode is available)
9:7	INPUT_FIFO_SIZE	R	Read only, actual value set by RTL generic. Value 000: 2X BLOCK_SIZE Value 001: 4X BLOCK_SIZE Value 010: 8X BLOCK_SIZE Value 011: 16X BLOCK_SIZE
6:5	INPUT_BLOCK_SIZE	R	Read only. Actual value set by RTL generic. Indicates the block size associated with Block_Mode for input. Value 00: 4 Bytes (FIFO_Mode only) Value 01: 16 Bytes Value 10: 32 Bytes Value 11: Reserved
4:2	OUTPUT_FIFO_SIZE	R	Read only. Actual value set by RTL GENERIC. Value 000: 2X BLOCK_SIZE Value 001: 4X BLOCK_SIZE Value 010: 8X BLOCK_SIZE Value 011: 16X BLOCK_SIZE

BLSP1_BLSP_QUP4_QUP_IO_MODES (cont.)

Bits	Name	Type	Description
1:0	OUTPUT_BLOCK_SIZE	R	Read only. Actual value set by RTL GENERIC. Value 00: 04 Bytes (FIFO mode only) Value 01: 16 Bytes Value 10: 32 Bytes Value 11: Reserved

0x078B900C BLSP1_BLSP_QUP4_QUP_SW_RESET**Type:** W**Clock:** crif_clk**Reset State:** 0x00000000

A write to this register resets the entire QUP core and all mini-cores. The internal registers are brought back to their reset values. Reading of this register returns zero. The AHB clock domain reset will stay asserted until either the spi_clk or i2c_clk domains are reset. It is prohibited to write to any QUP register during this period and failing to do so will cause an ERROR response on the AHB bus. In order to avoid this SW should poll the QUP_STATE[STATE_VALID] bit until it is asserted.

BLSP1_BLSP_QUP4_QUP_SW_RESET

Bits	Name	Description
31:0	QUP_SW_RESET	

0x078B9014 BLSP1_BLSP_QUP4_QUP_TRANSFER_CANCEL**Type:** RW**Clock:** crif_clk**Reset State:** 0x00000000**BLSP1_BLSP_QUP4_QUP_TRANSFER_CANCEL**

Bits	Name	Type	Description
15:8	TRANSFER_CANCEL_ID	RW	This field include the id of the transfer that is scheduled for cancellation. This field is valid when asserting TRANSFER_CANCEL bit. A value of 0xFF will cancel the current transfer regardless of its LOCAL_ID. Canceling an i2c transfer requires using version 2 tags.
7	TRANSFER_CANCEL	W	Setting this bit to 1 will cause the HW to go to pause_state when the LOCAL_ID = TRANSFER_CANCEL_ID. This would allow flushing the rest of the transfer in order to implement cancel API. Canceling an i2c transfer requires using version 2 tags.

0x078B9018 BLSP1_BLSP_QUP4_QUP_OPERATIONAL**Type:** RW**Clock:** crif_clk**Reset State:** 0x000000C0

Refer to the chapter on 'QUP Software Considerations' regarding the correct interpretation of the various service interrupts generated by QUP and their relation to these register bits.

BLSP1_BLSP_QUP4_QUP_OPERATIONAL

Bits	Name	Description
15	NWD	Notify When Done (NWD). Read only status bit which when set indicates a NWD acknowledgement is outstanding. This bit is set by QUP hardware when NWD is signaled from BAM. This bit is cleared by QUP hardware when the NWD request is acknowledged by assertion of done_toggle. This bit is always unconditionally cleared in RESET_STATE.
14	DONE_TOGGLE	Read only status bit which provides the current state of the side-band done_toggle signal sent to BAM. At each NWD acknowledgement, this bit toggles.
13	IN_BLOCK_READ_REQ	Read only. When set by hardware, indicates QUP input FIFO has BLOCK_SIZE_INPUT amount of data ready for reading. Valid only in Block_Mode.
12	OUT_BLOCK_WRITE_REQ	Read only. When set by hardware, indicates QUP output FIFO needs BLOCK_SIZE_OUTPUT amount of data to be written. Valid only in Block_Mode.
11	MAX_INPUT_DONE_FLAG	Read only. When set by hardware, indicates QUP input FIFO has reached the programmed QUP_MX_INPUT_COUNT value. Valid in FIFO_Mode (only if MX_READ_COUNT is non-zero), Block_Mode and Bam_Mode (for debug only).
10	MAX_OUTPUT_DONE_FLAG	Read only. When set by hardware, indicates QUP output FIFO has reached the programmed QUP_MX_OUTPUT_COUNT value. Valid in Block_Mode and Bam_Mode (for debug only).
9	INPUT_SERVICE_FLAG	When set by hardware, indicates QUP input FIFO has an outstanding input service request. At the point in time this bit is set, the hardware also asserts qup_irq. Writing a 'zero' to this bit does nothing. Writing a 'one' to this bit clears it and acknowledges software has or will read the data. Valid in all modes , recommended to be masked in BAM mode
8	OUTPUT_SERVICE_FLAG	When set by hardware, indicates QUP output FIFO has an outstanding output service request. At the point in time this bit is set, the hardware also asserts qup_irq. Writing a 'zero' to this bit does nothing. Writing a 'one' to this bit clears it and acknowledges software has or will read the data. Valid in all modes , recommended to be masked in BAM mode
7	INPUT_FIFO_FULL	Read only. When set, indicates the input FIFO is full and can accept no more data from the QUP mini-core.

BLSP1_BLSP_QUP4_QUP_OPERATIONAL (cont.)

Bits	Name	Description
6	OUTPUT_FIFO_FULL	Read only. When set, indicates the output FIFO is full and can accept no more CRIF writes.
5	INPUT_FIFO_NOT_EMPTY	Read only. When set, indicates the input FIFO has at least one value in it to be read. When clear, indicates the input FIFO is empty.
4	OUTPUT_FIFO_NOT_EMPTY	Read only. When set, indicates the output FIFO has at least one value in it to be shifted out. When clear, indicates the output FIFO is empty.

0x078B901C BLSP1_BLSP_QUP4_QUP_ERROR_FLAGS**Type:** RW**Clock:** crif_clk**Reset State:** 0x00000000

All bits in this register are set by hardware and remain set until cleared by software. Writing a 'one' to a bit clears it while writing a 'zero' leaves the bit unchanged.

BLSP1_BLSP_QUP4_QUP_ERROR_FLAGS

Bits	Name	Description
5	OUTPUT_OVER_RUN_ERR	Indicates the output FIFO was full when a CRIF write was attempted to the FIFO. The write is discarded.
4	INPUT_UNDER_RUN_ERR	Indicates the input FIFO was empty when a CRIF read was attempted to the FIFO. The read value returns is indeterminate.
3	OUTPUT_UNDER_RUN_ERR	Indicates the output FIFO was empty when an output shift operation required a value.
2	INPUT_OVER_RUN_ERR	Indicates the input FIFO was full when an input shift operation provided a new value. When this happens, the new value is discarded.

0x078B9020 BLSP1_BLSP_QUP4_QUP_ERROR_FLAGS_EN**Type:** RW**Clock:** crif_clk**Reset State:** 0x0000003C

Each bit in this register controls the setting of a corresponding error flag. If set, each bit enables generation of qup_irq and the setting of the corresponding error flag in the QUP_ERROR_FLAGS register for the specified error case. At reset, all error enable bits are set to '1'.

BLSP1_BLSP_QUP4_QUP_ERROR_FLAGS_EN

Bits	Name	Description
5	OUTPUT_OVER_RUN_ERR_EN	If set, enables output over run error generation.
4	INPUT_UNDER_RUN_ERR_EN	If set, enables input under run error generation.
3	OUTPUT_UNDER_RUN_ERR_EN	If set, enables output under run error generation.
2	INPUT_OVER_RUN_ERR_EN	If set, enables input over run error generation.

0x078B9028 BLSP1_BLSP_QUP4_QUP_OPERATIONAL_MASK**Type:** RW**Clock:** crif_clk**Reset State:** 0x00000000

This register masks several QUP_OPERATIONAL flags from creating an interrupt .

BLSP1_BLSP_QUP4_QUP_OPERATIONAL_MASK

Bits	Name	Description
9	INPUT_SERVICE_MASK	If set, this flag in QUP_OPERATIONAL does not cause an interrupt but provides status only.
8	OUTPUT_SERVICE_MASK	If set, this flag in QUP_OPERATIONAL does not cause an interrupt but provides status only.

0x078B9100 BLSP1_BLSP_QUP4_QUP_MX_OUTPUT_COUNT**Type:** RW**Clock:** crif_clk**Reset State:** 0x00000000

Unless otherwise stated, register bits written return the value when read. This register only applies to Block_Mode. The counter decrements for each output transfer when the STATE field (see QUP_STATE register) is moved from the RESET_STATE to the RUN_STATE. The PAUSE_STATE does not effect the count.

Notes:

- a. Allows the total number of Mini Core transfers to be less than an exact multiple of OUTPUT_BLOCK_SIZE. Any additional outputs are discarded.

BLSP1_BLSP_QUP4_QUP_MX_OUTPUT_COUNT

Bits	Name	Type	Description
31	MX_CONFIG_DURING_RUN	W	Write only value. When reset to zero the MX_OUTPUT_COUNT value only applies in reset_state (legacy behavior). When set to one the MX_OUTPUT_COUNT value will also apply in run_state and allow writing the transaction data in a few portions when not knowing the transaction size in advance. Reconfiguration during run is only allowed after the last portion has ended (i.e. MAX_OUTPUT_DONE_FLAG was asserted)
15:0	MX_OUTPUT_COUNT	RW	Number of writes of size N to the mini-core per RUN_STATE. A value of zero indicates the output count function is not enabled for use. See note (a) above.

0x078B9104 BLSP1_BLSP_QUP4_QUP_MX_OUTPUT_CNT_CURRENT**Type:** R**Clock:** crif_clk**Reset State:** 0x00000000**BLSP1_BLSP_QUP4_QUP_MX_OUTPUT_CNT_CURRENT**

Bits	Name	Description
15:0	MX_OUTPUT_CNT_CURRENT	Current value of QUP_MX_OUTPUT_COUNT.

0x078B910C BLSP1_BLSP_QUP4_QUP_OUTPUT_FIFO_WORD_CNT**Type:** R**Clock:** crif_clk**Reset State:** 0x00000000

This register holds the number of words in the output FIFO at a given time. Note the fields in this register are dynamically updated hence when this register is read during a FIFO operation, the value of this register could have been altered by the time SW gets the read value. Recommended for debug purposes.

BLSP1_BLSP_QUP4_QUP_OUTPUT_FIFO_WORD_CNT

Bits	Name	Description
8:0	OUTPUT_FIFO_WORD_COUNT	Number of words in the output FIFO.

**0x078B9110+ BLSP1_BLSP_QUP4_QUP_OUTPUT_FIFOc, c=[0..15]
0x4*c****Type:** W**Clock:** crif_clk**Reset State:** 0x00000000

Note that consecutive writes to this address keeps filling up the output FIFO. The max address to address the output FIFO is 0x14C.

BLSP1_BLSP_QUP4_QUP_OUTPUT_FIFOc

Bits	Name	Description
31:0	OUTPUT	Value to be shifted out.

0x078B9150 BLSP1_BLSP_QUP4_QUP_MX_WRITE_COUNT**Type:** RW**Clock:** crif_clk**Reset State:** 0x00000000

What the QUP_MX_OUTPUT_COUNT register means to Block_Mode and Bam_Mode, this register means the same to FIFO_mode. If this register is non-zero, then the qup_irq is asserted after shifting in the number of shifts specified by this register.

Unlike the QUP_MX_OUTPUT_COUNT register case, the SW should not program the QUP_MX_WRITE_COUNT value to be more than the 'actual depth' of the FIFO (BLOCK_SIZE_OUTPUT * FIFO_SIZE_OUTPUT).

BLSP1_BLSP_QUP4_QUP_MX_WRITE_COUNT

Bits	Name	Type	Description
31	MX_CONFIG_DURING_RUN	W	Write only value. When reset to zero the MX_WRITE_COUNT value only applies in reset_state (legacy behavior). When set to one the MX_WRITE_COUNT value will also apply in run_state and allow writing the transaction data in a few portions when not knowing the transaction size in advance. Reconfiguration during run is only allowed after the last portion has ended (i.e. MAX_OUTPUT_DONE_FLAG was asserted)
15:0	MX_WRITE_COUNT	RW	The number of 'writes' of size N. This is used only if the core is in FIFO_Mode.

0x078B9154 BLSP1_BLSP_QUP4_QUP_MX_WRITE_CNT_CURRENT**Type:** R**Clock:** crif_clk**Reset State:** 0x00000000**BLSP1_BLSP_QUP4_QUP_MX_WRITE_CNT_CURRENT**

Bits	Name	Description
15:0	MX_WRITE_CNT_CURRENT	Current value of QUP_MX_WRITE_COUNT counter.

0x078B9200 BLSP1_BLSP_QUP4_QUP_MX_INPUT_COUNT**Type:** RW**Clock:** crif_clk**Reset State:** 0x00000000

Unless otherwise stated, register bits written return the value when read. This register only applies to Block_Mode and Bam_Mode (for non-balanced SPI). The counter decrements for each input transfer when the STATE field (see QUP_STATE register) is moved from the RESET_STATE to the RUN_STATE. The PAUSE_STATE do not affect the count.

Notes:

a. Allows the number of shift register transfers to be less than an exact multiple of INPUT_BLOCK_SIZE. When count reached, remainder of INPUT_BLOCK_SIZE is filled with zeroes.

BLSP1_BLSP_QUP4_QUP_MX_INPUT_COUNT

Bits	Name	Type	Description
31	MX_CONFIG_DURING_RUN	W	Write only value. When reset to zero the MX_INPUT_COUNT value only applies in reset_state (legacy behavior). When set to one the MX_INPUT_COUNT value will also apply in run_state and allow writing the transaction data in a few portions when not knowing the transaction size in advance. Reconfiguration during run is only allowed after the last portion has ended (i.e. MAX_INPUT_DONE_FLAG was asserted)
15:0	MX_INPUT_COUNT	RW	Number of reads of size N from the mini-core per RUN_STATE. A value of zero indicates the input count function is not enabled for use. See note (a) above.

0x078B9204 BLSP1_BLSP_QUP4_QUP_MX_INPUT_CNT_CURRENT**Type:** R**Clock:** crif_clk**Reset State:** 0x00000000**BLSP1_BLSP_QUP4_QUP_MX_INPUT_CNT_CURRENT**

Bits	Name	Description
15:0	MX_INPUT_CNT_CURRENT	Current value of QUP_MX_INPUT_COUNT.

0x078B9208 BLSP1_BLSP_QUP4_QUP_MX_READ_COUNT**Type:** RW**Clock:** crif_clk**Reset State:** 0x00000000

What the QUP_MX_INPUT_COUNT register means to Block_Mode and Bam_Mode, this register means the same to FIFO_mode. If this register is non-zero, then the qup_input_service_irq is asserted after shifting in the number of shifts specified by this register.

Unlike the QUP_MX_INPUT_COUNT register case, the SW should not program the QUP_MX_READ_COUNT value to be more than the 'actual depth' of the FIFO (BLOCK_SIZE_INPUT * FIFO_SIZE_INPUT). If the SPI mini-core and slave operation is enabled, then an InputOverRun error will result.

BLSP1_BLSP_QUP4_QUP_MX_READ_COUNT

Bits	Name	Type	Description
31	MX_CONFIG_DURING_RUN	W	Write only value. When reset to zero the MX_READ_COUNT value only applies in reset_state (legacy behavior). When set to one the MX_READ_COUNT value will also apply in run_state and allow writing the transaction data in a few portions when not knowing the transaction size in advance. Reconfiguration during run is only allowed after the last portion has ended (i.e. MAX_INPUT_DONE_FLAG was asserted)
15:0	MX_READ_COUNT	RW	The number of 'reads' of size N. This is used only if the core is in FIFO_Mode.

0x078B920C BLSP1_BLSP_QUP4_QUP_MX_READ_CNT_CURRENT**Type:** R**Clock:** crif_clk**Reset State:** 0x00000000

BLSP1_BLSP_QUP4_QUP_MX_READ_CNT_CURRENT

Bits	Name	Description
15:0	MX_READ_CNT_CURRENT	Current value of QUP_MX_READ_COUNT counter.

0x078B9214 BLSP1_BLSP_QUP4_QUP_INPUT_FIFO_WORD_CNT**Type:** R**Clock:** crif_clk**Reset State:** 0x00000000

This register holds the number of words in the input FIFO at a given time. Note the fields in this register are dynamically updated. Hence, when this register is read during a FIFO operation, the value of this register could have been altered by the time SW gets the read value. Recommended for debug purposes.

BLSP1_BLSP_QUP4_QUP_INPUT_FIFO_WORD_CNT

Bits	Name	Description
8:0	INPUT_FIFO_WORD_CNT	Number of words in the input FIFO.

**0x078B9218+ BLSP1_BLSP_QUP4_QUP_INPUT_FIFOc, c=[0..15]
0x4*c****Type:** R**Clock:** crif_clk**Reset State:** Undefined

Consecutive reads to this address reads the input FIFO contents on a FIFO basis. The max address to read the input FIFO is 0x254.

BLSP1_BLSP_QUP4_QUP_INPUT_FIFOc

Bits	Name	Description
31:0	INPUT	Value shifted in.

0x078B9300 BLSP1_BLSP_QUP4_SPI_CONFIG**Type:** RW**Clock:** crif_clk**Reset State:** 0x00000000

Unless otherwise stated, register bits written return the value when read.

The contents of this register should only be changed when in the RESET_STATE. See the QUP_STATE register. Both NO_OUTPUT and NO_INPUT set to zero provides default full duplex operation. Setting both these bits to one is not a legal operational state.

BLSP1_BLSP_QUP4_SPI_CONFIG

Bits	Name	Type	Description
10	HS_MODE	RW	When set, SPI HS_MODE is enabled. When clear, SPI HS_MODE is disabled.
9	INPUT_FIRST	RW	When set, INPUT FIRST SPI protocol is used. When clear, OUTPUT FIRST SPI protocol is used.
8	LOOP_BACK	RW	Loopback is only valid in non-HS mode. Always clear for normal operation. If set, loop back on SPI_DATA_MO_SI under MASTER operation or SPI_DATA_MI_SO under SLAVE operation.
5	SLAVE_OPERATION	R	This register is writable only when the hardware signal spi_slave_en is asserted. When set, the SPI is configured for SLAVE operation. Zero indicates MASTER operation.

0x078B9304 BLSP1_BLSP_QUP4_SPI_IO_CONTROL

Type: RW

Clock: crif_clk

Reset State: 0x00000000

The contents of this register should only be changed when in the RESET_STATE. See the QUP_STATE register.

Unless otherwise stated, register bits written return the value when read.

BLSP1_BLSP_QUP4_SPI_IO_CONTROL

Bits	Name	Description
11	FORCE_CS	When this bit is set, the chip select is asserted unconditionally with no relationship to QUP_STATE or transaction state.
10	CLK_IDLE_HIGH	When set to 1, spi_clk will be high when spi is idle (spi_cs is not asserted). When cleared to 0, spi_clk will be low when spi is idle (spi_cs is not asserted).
9	CLK_ALWAYS_ON	When MASTER operation is used, run SPI_CLK during IDLE.
8	MX_CS_MODE	If this bit is set, then for a given RUN state, the associated chip select will be asserted for the first N-bit transfer and will be kept asserted till the last programmed transfer. Applies only in MASTER mode. The number of transfers is determined by SPI_MX_OUTPUT_COUNT and/or SPI_MX_INPUT_COUNT registers.

BLSP1_BLSP_QUP4_SPI_IO_CONTROL (cont.)

Bits	Name	Description
7:4	CS_N_POLARITY	These four bits control the polarity of four SPI_CS#_N respectively. Setting any of this bit to '1', makes the associated SPI_CS#_N active HIGH. This field is a 'don't care' in SLAVE operation.
3:2	CS_SELECT	When MASTER operation is used, controls the assertion of SPI_CS_#N pins. Selects SPI_CS_N if 00, SPI_CS1_N if 01, SPI_CS2_N if 10, or SPI_CS3_N if 11. The deselected ChipSelects will be driven to inactive state dictated by the field CS_N_POLARITY. This field is a 'don't care' in SLAVE operation.
1	TRISTATE_CS	When set, drives Z on all SPI_CS#_N lines. When clear, enables normal functionality on these lines. This bit can be used to support deployment of the core as one of the masters in a multi-master configuration.
0	NO_TRI_STATE	When set, spi_data_out is not taken tri-state when SPI_CS#_N is de-asserted. This bit is normally set for MASTER operation but may be optionally left cleared.

0x078B9308 BLSP1_BLSP_QUP4_SPI_ERROR_FLAGS**Type:** RW**Clock:** crif_clk**Reset State:** 0x00000000

All bits in this register are set by hardware and remain set until cleared by software. Writing a 'one' to a bit clears it while writing a 'zero' leaves the bit unchanged.

BLSP1_BLSP_QUP4_SPI_ERROR_FLAGS

Bits	Name	Description
3	TRANSFER_CANCEL_DONE	This bit is set (1) when the SPI controller has completed canceling the transfer that was scheduled for cancellation (LOCAL_ID = TRANSFER_CANCEL_ID).
2	TRANSFER_CANCEL_ID_MATCH	This bit is set (1) when the SPI controller is ready for cancel operation: it has found the transfer that was scheduled for cancellation (LOCAL_ID = TRANSFER_CANCEL_ID) and the logic is done with previous activity.
1	SPI_SLV_CLK_OVER_RUN_ERR	Number of SPI_CLK ticks which occurred while SPI_CS#_N was asserted was greater than the programmed value of N. When this happens, only the first N bits are passed to the SPI input FIFO and the output shift value is held at zero.
0	SPI_SLV_CLK_UNDER_RUN_ERR	Number of SPI_CLK ticks which occurred while SPI_CS#_N was asserted was less than the programmed value of N. When this occurs, the bits which were received are passed to the SPI input FIFO and the CURRENT output bit is forced to zero.

0x078B930C BLSP1_BLSP_QUP4_SPI_ERROR_FLAGS_EN**Type:** RW**Clock:** crif_clk**Reset State:** 0x0000000C

Each bit in this register controls the setting of a corresponding error flag. If set, each bit enables generation of spi_error_irq and the setting of the corresponding error flag in the SPI_ERROR_FLAGS register for the specified error case. At reset, all error enable bits are set to '1'.

BLSP1_BLSP_QUP4_SPI_ERROR_FLAGS_EN

Bits	Name	Description
3	TRANSFER_CANCEL_DONE_EN	If set, enables generating the transfer cancel done indication.
2	TRANSFER_CANCEL_ID_MATCH_EN	If set, enables generating the transfer cancel ID match indication.
1	SPI_SLV_CLK_OVER_RUN_ERR_EN	If set, enables clock over run error generation.
0	SPI_SLV_CLK_UNDER_RUN_ERR_EN	If set, enables clock under run error generation.

0x078B9310 BLSP1_BLSP_QUP4_SPI_DEASSERT_WAIT**Type:** RW**Clock:** crif_clk**Reset State:** 0x00000000

This register holds the de-assertion wait time of SPI_CS#_N between consecutive N-bit transfers in MASTER operation. The wait time is specified in number of cycles of cc_spi_master_clk.

BLSP1_BLSP_QUP4_SPI_DEASSERT_WAIT

Bits	Name	Description
5:0	DEASSERT_WAIT	Number cc_spi_master_clk ticks associated with the deasserted time of SPI_CS#_N. Only applies to MASTER operation. For SLAVE operation, this field is a 'don't care'. A value of zero indicates SPI_CS#_N remains de-asserted for exactly one clock tick. A value of one, indicates two ticks, etc. All ones indicates 64 ticks.

0x078B9314 BLSP1_BLSP_QUP4_SPI_MASTER_LOCAL_ID**Type:** RW**Clock:** crif_clk**Reset State:** 0x00000000**BLSP1_BLSP_QUP4_SPI_MASTER_LOCAL_ID**

Bits	Name	Description
15:8	EXTENDED_ID	SW may use this field to specify the current sub-transfer ID to be read when TRANSFER_CANCEL_ID_MATCH is asserted. This value has no impact on HW.
7:0	LOCAL_ID	SW specifies the current transfer ID. HW will use this value to match with TRANSFER_CANCEL_ID.

0x078B9318 BLSP1_BLSP_QUP4_SPI_MASTER_COMMAND**Type:** W**Clock:** crif_clk**Reset State:** 0x00000000**BLSP1_BLSP_QUP4_SPI_MASTER_COMMAND**

Bits	Name	Description
0	RESET_CANCEL_FSM	This command should be given only when the CANCEL_FSM_STATE is in CANCEL_PENDING_STATE or PAUSE_WAIT_STATE and when the core is in PAUSE. The main use case for this bit is to allow withdrawing a cancel command.

0x078B931C BLSP1_BLSP_QUP4_SPI_MASTER_STATUS**Type:** R**Clock:** crif_clk**Reset State:** 0x00000000**BLSP1_BLSP_QUP4_SPI_MASTER_STATUS**

Bits	Name	Description
2:0	CANCEL_FSM_STATE	This 3-bit field informs the microprocessor of the state of the SPI cancel logic. Internal use only

0x078B9400 BLSP1_BLSP_QUP4_I2C_MASTER_CLK_CTL**Type:** RW**Clock:** crif_clk**Reset State:** 0x00000000

The I2C_CLK_CTL register is a read/write register that controls clock divider values.

BLSP1_BLSP_QUP4_I2C_MASTER_CLK_CTL

Bits	Name	Description
28	SCL_EXT_FORCE_LOW	When set to 0, SCL generation maintains legacy behavior. When set to 1, SCL state machine will go to the FORCED_LOW_STATE if force_low is asserted near (within 5 i2c_clk cycles = $5 * 52\text{ns} = 260\text{ns} \approx \text{MAX}(\text{Tsu}, \text{DAT})$) the clock edge.
27:26	SDA_NOISE_REJECTION	Allows adding extra sampling levels on SDA to reject short low pulses. This value specifies how many TCXO cycles of logic low on SDA would be considered as valid logic low. 0x0 - legacy mode, 0x01 - one cycle wide low pulse is rejected, 0x2 - two cycles wide low pulse is rejected, 0x3 - three cycles wide low pulse is rejected
25:24	SCL_NOISE_REJECTION	Allows adding extra sampling levels on SCL to reject short low pulses. This value specifies how many TCXO cycles of logic low on SCL would be considered as valid logic low. 0x0 - legacy mode, 0x01 - one cycle wide low pulse is rejected, 0x2 - two cycles wide low pulse is rejected, 0x3 - three cycles wide low pulse is rejected
23:16	HIGH_TIME_DIVIDER_VALUE	Allows setting SCL duty cycle to non 50%. If this value is zero then legacy mode is used. If this value is non-zero then it will be used as the SCL high time counter and FS_DIVIDER_VALUE will be used as the low time counter. Minimum value is 0x7.
10:8	HS_DIVIDER_VALUE	The value in this register represents the clock period multiplier in high-speed (HS) mode. Minimum value is 0x3. $\text{I2C_HS_CLK} = \text{I2C_CLK} / (3 * (\text{HS_DIVIDER_VALUE} + 1) + 2)$
7:0	FS_DIVIDER_VALUE	The value in this register represents the clock period multiplier in fast/standard (FS) mode. Minimum value is 0x7. When HIGH_TIME_DIVIDER_VALUE=0: $\text{I2C_FS_CLK} = \text{I2C_CLK} / (2 * (\text{FS_DIVIDER_VALUE} + 3))$ When HIGH_TIME_DIVIDER_VALUE!=0: $\text{I2C_FS_CLK} = \text{I2C_CLK} / (\text{FS_DIVIDER_VALUE} + \text{HIGH_TIME_DIVIDER_VALUE} + 6)$

0x078B9404 BLSP1_BLSP_QUP4_I2C_MASTER_STATUS**Type:** RW**Clock:** crif_clk**Reset State:** 0x0C000000

The I2C_STATUS is a status register. Writing any value clears the status bits.

BLSP1_BLSP_QUP4_I2C_MASTER_STATUS

Bits	Name	Type	Description
30:28	CANCEL_FSM_STATE	R	This 3-bit field informs the microprocessor of the state of the I2C cancel logic. Internal use only 0x0: IDLE_STATE (IDLE_STATE) 0x1: CANCEL_PENDING_STATE (CANCEL_PENDING_STATE) 0x2: PAUSE_STATE (PAUSE_STATE) 0x3: PAUSE_WAIT_STATE (PAUSE_WAIT_STATE) 0x4: SEND_STOP_STATE (SEND_STOP_STATE) 0x5: FLUSH_STATE (FLUSH_STATE) 0x6: SEND_IRQ_STATE (SEND_IRQ_STATE)
27	I2C_SCL	R	Logic state of I2C bus serial clock wire.
26	I2C_SDA	R	Logic state of I2C bus serial data wire.
25	INVALID_READ_SEQ	RW	Interrupt source. This bit is set (1) when a MI_REC tag does not follow a START tag for an I2C READ. Not applicable for Halcyon.
24	INVALID_READ_ADDR	RW	Interrupt source. In version 1 tags this bit is set (1) when the I2C controller is trying to receive data from a non-existent I2C slave (address). In version 2 tags this bit is set (1) when the I2C controller is trying to access a non-existent I2C slave (address).
23	INVALID_TAG	RW	Interrupt source. This bit is set (1) when the I2C controller is trying to process data from the output FIFO that is improperly tagged.
22:20	INPUT_FSM_STATE	R	This 3-bit field informs the microprocessor of the state of the I2C MASTER INPUT FSM block. Internal use only 0x0: V1_V2_RESET_STATE (RESET_STATE value. Applies for both version 1 and version 2 tags) 0x1: V1_V2_IDLE_STATE (IDLE_STATE value. Applies for both version 1 and version 2 tags) 0x2: V1_V2_READ_LAST_BYTE_STATE (READ_LAST_BYTE_STATE value. Applies for both version 1 and version 2 tags) 0x3: V1_V2_MI_REC_STATE (MI_REC_STATE value. Applies for both version 1 and version 2 tags) 0x4: V1_V2_DEC_STATE (DEC_STATE value. Applies for both version 1 and version 2 tags) 0x5: V1_V2_STORE_STATE (STORE_STATE value. Applies for both version 1 and version 2 tags) 0x6: V2_WR_TAG_STATE (WR_TAG_STATE value. Only applicable in version 2 tags) 0x7: V2_WAIT_TAG_STATE (WAIT_TAG_STATE value. Only applicable in version 2 tags)

BLSP1_BLSP_QUP4_I2C_MASTER_STATUS (cont.)

Bits	Name	Type	Description
19:16	OUTPUT_FSM_STATE	R	This 3-bit field informs the microprocessor of the state of the I2C MASTER OUTPUT FSM block. Internal use only 0x0: V1_V2_RESET_STATE (RESET_STATE value. Applies for both version 1 and version 2 tags) 0x1: V1_V2_IDLE_STATE (IDLE_STATE value. Applies for both version 1 and version 2 tags) 0x2: V1_V2_DECODE_STATE (DECODE_STATE value. Applies for both version 1 and version 2 tags) 0x3: V1_V2_SEND_STATE (SEND_STATE value. Applies for both version 1 and version 2 tags) 0x4: V1_V2_MI_REC_STATE (MI_REC_STATE value. Applies for both version 1 and version 2 tags) 0x5: V1_V2_NOP_STATE (NOP_STATE value. Applies for both version 1 and version 2 tags) 0x6: V1_V2_INVALID_STATE (INVALID_STATE value. Applies for both version 1 and version 2 tags) 0x7: V1_PEEK_STATE (PEEK_STATE value. Only applicable in version 1 tags) 0x8: V1_SEND_R_STATE (SEND_R_STATE value. Only applicable in version 1 tags) 0xB: V2_GET_BYTE_STATE (GET_BYTE_STATE value. Only applicable in version 2 tags) 0xC: V2_GET_WRITE_BYTE_STATE (GET_WRITE_BYTE_STATE value. Only applicable in version 2 tags) 0xD: V2_SPECIALITY_STATE (SPECIALITY_STATE value. Only applicable in version 2 tags) 0xE: V2_WAIT_FLUSH_STOP_STATE (WAIT_FLUSH_STOP_STATE value. Only applicable in version 2 tags) 0xF: V2_STOP_STATE (STOP_STATE value. Only applicable in version 2 tags)
15:13	CLK_STATE	R	This 3-bit field informs the microprocessor of the state of the I2C clk_control block. Internal use only 0x0: RESET_BUSIDLE_STATE (RESET_BUSIDLE_STATE) 0x1: NOT_MASTER_STATE (NOT_MASTER_STATE) 0x2: HIGH_STATE (HIGH_STATE) 0x3: LOW_STATE (LOW_STATE) 0x4: HIGH_WAIT_STATE (HIGH_WAIT_STATE) 0x5: FORCED_LOW_STATE (FORCED_LOW_STATE) 0x6: HS_ADDR_LOW_STATE (HS_ADDR_LOW_STATE) 0x7: DOUBLE_BUFFER_WAIT_STATE (DOUBLE_BUFFER_WAIT_STATE)

BLSP1_BLSP_QUP4_I2C_MASTER_STATUS (cont.)

Bits	Name	Type	Description
12:10	DATA_STATE	R	This 3-bit field informs the microprocessor of the state of the I2C data_control block. Internal use only 0x0: RESET_WAIT_STATE (RESET_WAIT_STATE) 0x1: TX_ADDR_STATE (TX_ADDR_STATE) 0x2: TX_DATA_STATE (TX_DATA_STATE) 0x3: TX_HS_ADDR_STATE (TX_HS_ADDR_STATE) 0x4: TX_10_BIT_ADDR_STATE (TX_10_BIT_ADDR_STATE) 0x5: TX_2ND_BYTE_STATE (TX_2ND_BYTE_STATE) 0x6: RX_DATA_STATE (RX_DATA_STATE) 0x7: RX_NACK_HOLD_STATE (RX_NACK_HOLD_STATE)
9	BUS_MASTER	R	This bit is set (1) when the I2C controller is the present bus master.
8	BUS_ACTIVE	R	This bit is set (1) when the bus is in use by this, or any other controller.
7:6	FAILED	RW	Interrupt source. This 2-bit field contains the failure information of the present I2C transfers. If transmitting, failed[1] contains the information regarding the byte that has been transmitted. Failed[0] contains the information of the byte awaiting transmission if there is a byte pipelined. If no byte is pipelined, ignore this bit. If receiving, failed[1] contains the information of the byte received and stored in the buffer, and failed[0] should be ignored. For example: Value 00: Byte n transmitted successfully, byte n+1 to begin transmission Value 01: Byte n transmitted successfully, byte n+1 errored: type of error indicated in other STATUS bits. (queued invalid write, for example) Value 10: Byte n errored: type of error indicated in other STATUS bits, byte n+1 to begin transmission (byte n+1 would have to be a valid address byte for this condition to occur) Value 11: Byte n errored: type of error indicated in other STATUS bits, byte n+1 discarded as well (if the byte was a data byte destined for a NACKed address, data is useless, therefore discarded)
5	INVALID_WRITE	RW	Interrupt source. This bit is set (1) when software writes data to the I2C_DATA register that should be flagged as an address but is not.
4	ARB_LOST	RW	Interrupt source. This bit is set (1) when the controller loses arbitration for the bus. If this bit gets set (1) during the transmission, all data has been lost, and the microprocessor must re-request its transmission.

BLSP1_BLSP_QUP4_I2C_MASTER_STATUS (cont.)

Bits	Name	Type	Description
3	PACKET_NACKED	RW	Interrupt source. This bit is set high when a NACK is received from a slave. If a high speed master code is sent and there is an ACK from a slave, then this bit is set (1) to indicate that the high speed mode can not be entered. If the high speed mode is accepted by the slave, then a NACK is performed and this bit is not set (1).
2	BUS_ERROR	RW	Interrupt source. This bit is set (1) when an unexpected START or STOP condition is detected. This returns the controller to its reset state.
1	TRANSFER_CANCEL_DONE	RW	Only applicable when using version 2 tags. Interrupt source. This bit is set (1) when the I2C controller has completed canceling the transfer that was scheduled for cancellation (LOCAL_ID = TRANSFER_CANCEL_ID).
0	TRANSFER_CANCEL_ID_MATCH	RW	Only applicable when using version 2 tags. Interrupt source. This bit is set (1) when the I2C controller is ready for cancel operation: it has found the transfer that was scheduled for cancellation (LOCAL_ID = TRANSFER_CANCEL_ID) and the logic is done with previous activity.

0x078B9408 BLSP1_BLSP_QUP4_I2C_MASTER_CONFIG**Type:** RW**Clock:** crif_clk**Reset State:** 0x00000000**BLSP1_BLSP_QUP4_I2C_MASTER_CONFIG**

Bits	Name	Description
0	EN_VERSION_TWO_TAG	When set, version 2 of the mini-core is enabled.

0x078B940C BLSP1_BLSP_QUP4_I2C_MASTER_BUS_CLEAR**Type:** RW**Clock:** crif_clk**Reset State:** 0x00000000

BLSP1_BLSP_QUP4_I2C_MASTER_BUS_CLEAR

Bits	Name	Description
0	CLEAR	This command should be given only when the i2c mincore is idle. When in doubt SW can use SW_RESET to reset the minicore. When set, an I2C 'Bus Clear' executed per the I2C standard. A bus clear consists of nine I2C clock ticks with data wire left not-driven. Has the effect of clearing any slave which has lost read sync. After the clear is complete, the hardware sets the CLEAR bit to zero. The bit can not be cleared by software. After a bus clear, both clock and data lines should be high. If not, an external slave has hung the bus by holding down one or both lines. Reset the slave to clear if possible. Any slave reset mechanism is beyond the scope of the I2C mini core.

0x078B9410 BLSP1_BLSP_QUP4_I2C_MASTER_LOCAL_ID**Type:** R**Clock:** crif_clk**Reset State:** 0x00000000**BLSP1_BLSP_QUP4_I2C_MASTER_LOCAL_ID**

Bits	Name	Description
7:0	LOCAL_ID	Read only value capturing the ID byte of the last NOP LOCAL tag. Only applicable when using V2 tags

0x078B9414 BLSP1_BLSP_QUP4_I2C_MASTER_COMMAND**Type:** RW**Clock:** crif_clk**Reset State:** 0x00000000**BLSP1_BLSP_QUP4_I2C_MASTER_COMMAND**

Bits	Name	Description
0	RESET_CANCEL_FSM	This command should be given only when the CANCEL_FSM_STATE is in CANCEL_PENDING_STATE or PAUSE_WAIT_STATE and when the core is in PAUSE. The main use case for this bit is to allow withdrawing a cancel command in case the transfer was flushed (QUP_STATE[I2C_FLUSH] was set) due to I2C NACK.

20.9 BLSP1_BLSP_QUP5_QUP

0x078BA000 BLSP1_BLSP_QUP5_QUP_CONFIG

Type: RW

Clock: crif_clk

Reset State: 0x00000000

Unless otherwise stated, register bits written return the value when read.

- a. The contents of this register should only be changed when in the RESET_STATE (see the QUP_STATE register).
- b. The value written to the QUP output FIFO, is shifted left toward the MSB before passing it to the mini-core as follows:
 - o N equals 8 or less - shift 24
 - o N equals 16 to 9 - shift 16
 - o N equals 24 to 17 - shift 8
 - o N equals 32 to 25 - no shift

BLSP1_BLSP_QUP5_QUP_CONFIG

Bits	Name	Description
16	EN_EXT_OUT_FLAG	Enable Extended OUTPUT/INPUT_SERVICE_FLAG interrupt generation. When this bit is set, the behavior of the *_SERVICE_FLAG and MAX_OUTPUT_DONE_FLAG changes. Service flag and the done flag are not set till the last SPI or I2C transfer has completed. a) For SPI: 1. The shifting needs to complete 2. Clock stopped if CLK_ALWAYS_ON=0, 3. CS, if used, de-asserted b) For I2C, 1. The ACK needs to complete 2. STOP condition done generated (Attention need to be put in synchronizing the last transaction with STOP tag) 3. clock and data wires return to high condition When this bit is clear, the legacy behavior is applicable. Under legacy, the DONE_FLAG is set when: 1. MX_*_COUNT is reached 2. The output FIFO is empty 3. Last bit was sent on the link Under legacy, the flag interrupt is generated before the last transfer is complete on the external interface.
15	CORE_EXTRA_CLK_ON_EN	Enable additional transfer based qup_core_clk gating for power save. When set to 0, clock on is turned on unconditionally. When set to 1, dynamic clock on is turned on.

BLSP1_BLSP_QUP5_QUP_CONFIG (cont.)

Bits	Name	Description
14	FIFO_CLK_GATE_EN	When set to 0, fifo clock is mostly on - legacy mode. When set to 1, fifo clock is turned off dynamically.
13	CORE_CLK_ON_EN	When set to 0, clock on is turned on unconditionally. When set to 1, dynamic clock on is turned on and driven high conditionally.
12	APP_CLK_ON_EN	When set to 0, clock on is turned on unconditionally. When set to 1, dynamic clock on is turned on and driven high conditionally.
11:8	MINI_CORE	value: 0000 null core value: 0001 SPI core value: 0010 I2C master controller value: 0011 reserved value: 0100 reserved value: 0101 reserved value: 0110 reserved value: 0111 reserved See Note 1.
7	NO_INPUT	qup_data_in is not used and the value is a 'don't care'. The QUP input FIFO is always empty. The input service interrupt and INPUT_SERVICE_FLAG bit are never set. See notes (a) and (b) above.
6	NO_OUTPUT	qup_data_out is not used and is held at the value zero. The QUP output FIFO is always empty. The output service flag and OUTPUT_SERVICE_FLAG bit are never set. qup_data_out is still driven when SPI_CS#_N is asserted. The setting for NO_TRI_STATE still applies. See notes (a) and (b) above.
5	QUP_HREADY_CTRL	When set to 0, qup wil not stall the AHB bus (legacy mode). When set to 1, qup may stall the AHB bus until register access is done.
4:0	N	Number of logical bits N in the mini-core that constitutes a single transfer. The value zero indicates N equals one. The value of all ones indicates N equals 32. The value in this register must be 3 (i.e., N is 4 or higher in order for the STATE field to be set to RUN_STATE). See note (a) above.

0x078BA004 BLSP1_BLSP_QUP5_QUP_STATE**Type:** RW**Clock:** crif_clk**Reset State:** 0x0000001C

BLSP1_BLSP_QUP5_QUP_STATE

Bits	Name	Type	Description
6	I2C_FLUSH	RW	Flushing an i2c transfer requires using version 2 tags. FLUSH should be used in BAM mode only. Setting this bit to 1 will flush all tags and tag related data besides EOT until FLUSH STOP tag is encountered. Setting this bit to 0 has no impact. Reading this bit returns the flush operation status - 0 = ongoing, 1 = done.
5	WAIT_FOR_EOT	RW	Only applicable when moving to RUN_STATE using command descriptor in SPI mode with NO_OUTPUT. Setting this bit to 1 will stall the done_toggle_com indication until basi_trans_end_pro is asserted. This will prevent the next command descriptor to start prematurely.
4	I2C_MAST_GEN	R	Read only. Reflects the RTL generic setting for GEN_I2C_MASTER_MINI_CORE.
3	SPI_GEN	R	Read only. Reflects the RTL generic setting for GEN_SPI_MINI_CORE.
2	STATE_VALID	R	Read only. If and only if set to one, writes to the STATE field is allowed or reads from the STATE field is valid. Writes to this bit is ignored.
1:0	STATE	RW	When clear (00), the mini-core and related logic is held in RESET_STATE. When set to '01', the mini-core and related logic is released from reset and enters the RUN_STATE. When set to '11', the mini-core and related logic enters the PAUSE_STATE at the next appropriate point in time. Writing (10) to this field clears these two bits. For PAUSE_STATE to RESET_STATE transition, two writes of (10) are required for the transition to complete.

0x078BA008 BLSP1_BLSP_QUP5_QUP_IO_MODES**Type:** RW**Clock:** crif_clk**Reset State:** 0x000000A5

Unless otherwise stated, register bits written return the value when read.

Notes:

a. 'Packing' occurs as follows:

- N equals 8 or less - pack four values into each QUP input FIFO word.
- N equals 16 to 9 - pack two values into each QUP input FIFO word.
- N equals 32 to 17 - no packing. Each mini-core value is moved to an QUP input FIFO word.

b. 'Un-Packing' occurs as follows:

- N equals 8 or less - un-pack four values from each QUP output FIFO word.
- N equals 16 to 9 - un-pack two values from each QUP output FIFO word.
- N equals 32 to 17 - no packing. Each QUP output FIFO value is moved to the mini-core.

c. INPUT_MODE and OUTPUT_MODE should be both in BAM_Mode or both in non BAM_Mode.

BLSP1_BLSP_QUP5_QUP_IO_MODES

Bits	Name	Type	Description
16	OUTPUT_BIT_SHIFT_EN	RW	If set, enables the QUP output FIFO block to do bit shifting on the output data.
15	PACK_EN	RW	Indicates data values from the mini-core are to be packed before placement in the QUP input FIFO. See note (a) above.
14	UNPACK_EN	RW	Indicates data values taken from the QUP output FIFO should be unpacked before passing to the mini-core. See note (b) above.
13:12	INPUT_MODE	RW	Value 00: FIFO_Mode Value 01: Block_Mode Value 10: Reserved Value 11: BAM_Mode (note if the RTL generic BLOCK_SIZE_INPUT = 0, then only FIFO_Mode is available)
11:10	OUTPUT_MODE	RW	Value 00: FIFO_Mode Value 01: Block_Mode Value 10: Reserved Value 11: BAM_Mode (note if the RTL generic BLOCK_SIZE_OUTPUT = 0, then only FIFO_Mode is available)
9:7	INPUT_FIFO_SIZE	R	Read only, actual value set by RTL generic. Value 000: 2X BLOCK_SIZE Value 001: 4X BLOCK_SIZE Value 010: 8X BLOCK_SIZE Value 011: 16X BLOCK_SIZE
6:5	INPUT_BLOCK_SIZE	R	Read only. Actual value set by RTL generic. Indicates the block size associated with Block_Mode for input. Value 00: 4 Bytes (FIFO_Mode only) Value 01: 16 Bytes Value 10: 32 Bytes Value 11: Reserved
4:2	OUTPUT_FIFO_SIZE	R	Read only. Actual value set by RTL GENERIC. Value 000: 2X BLOCK_SIZE Value 001: 4X BLOCK_SIZE Value 010: 8X BLOCK_SIZE Value 011: 16X BLOCK_SIZE

BLSP1_BLSP_QUP5_QUP_IO_MODES (cont.)

Bits	Name	Type	Description
1:0	OUTPUT_BLOCK_SIZE	R	Read only. Actual value set by RTL GENERIC. Value 00: 04 Bytes (FIFO mode only) Value 01: 16 Bytes Value 10: 32 Bytes Value 11: Reserved

0x078BA00C BLSP1_BLSP_QUP5_QUP_SW_RESET**Type:** W**Clock:** crif_clk**Reset State:** 0x00000000

A write to this register resets the entire QUP core and all mini-cores. The internal registers are brought back to their reset values. Reading of this register returns zero. The AHB clock domain reset will stay asserted until either the spi_clk or i2c_clk domains are reset. It is prohibited to write to any QUP register during this period and failing to do so will cause an ERROR response on the AHB bus. In order to avoid this SW should poll the QUP_STATE[STATE_VALID] bit until it is asserted.

BLSP1_BLSP_QUP5_QUP_SW_RESET

Bits	Name	Description
31:0	QUP_SW_RESET	

0x078BA014 BLSP1_BLSP_QUP5_QUP_TRANSFER_CANCEL**Type:** RW**Clock:** crif_clk**Reset State:** 0x00000000**BLSP1_BLSP_QUP5_QUP_TRANSFER_CANCEL**

Bits	Name	Type	Description
15:8	TRANSFER_CANCEL_ID	RW	This field include the id of the transfer that is scheduled for cancellation. This field is valid when asserting TRANSFER_CANCEL bit. A value of 0xFF will cancel the current transfer regardless of its LOCAL_ID. Canceling an i2c transfer requires using version 2 tags.
7	TRANSFER_CANCEL	W	Setting this bit to 1 will cause the HW to go to pause_state when the LOCAL_ID = TRANSFER_CANCEL_ID. This would allow flushing the rest of the transfer in order to implement cancel API. Canceling an i2c transfer requires using version 2 tags.

0x078BA018 BLSP1_BLSP_QUP5_QUP_OPERATIONAL**Type:** RW**Clock:** crif_clk**Reset State:** 0x000000C0

Refer to the chapter on 'QUP Software Considerations' regarding the correct interpretation of the various service interrupts generated by QUP and their relation to these register bits.

BLSP1_BLSP_QUP5_QUP_OPERATIONAL

Bits	Name	Description
15	NWD	Notify When Done (NWD). Read only status bit which when set indicates a NWD acknowledgement is outstanding. This bit is set by QUP hardware when NWD is signaled from BAM. This bit is cleared by QUP hardware when the NWD request is acknowledged by assertion of done_toggle. This bit is always unconditionally cleared in RESET_STATE.
14	DONE_TOGGLE	Read only status bit which provides the current state of the side-band done_toggle signal sent to BAM. At each NWD acknowledgement, this bit toggles.
13	IN_BLOCK_READ_REQ	Read only. When set by hardware, indicates QUP input FIFO has BLOCK_SIZE_INPUT amount of data ready for reading. Valid only in Block_Mode.
12	OUT_BLOCK_WRITE_REQ	Read only. When set by hardware, indicates QUP output FIFO needs BLOCK_SIZE_OUTPUT amount of data to be written. Valid only in Block_Mode.
11	MAX_INPUT_DONE_FLAG	Read only. When set by hardware, indicates QUP input FIFO has reached the programmed QUP_MX_INPUT_COUNT value. Valid in FIFO_Mode (only if MX_READ_COUNT is non-zero), Block_Mode and Bam_Mode (for debug only).
10	MAX_OUTPUT_DONE_FLAG	Read only. When set by hardware, indicates QUP output FIFO has reached the programmed QUP_MX_OUTPUT_COUNT value. Valid in Block_Mode and Bam_Mode (for debug only).
9	INPUT_SERVICE_FLAG	When set by hardware, indicates QUP input FIFO has an outstanding input service request. At the point in time this bit is set, the hardware also asserts qup_irq. Writing a 'zero' to this bit does nothing. Writing a 'one' to this bit clears it and acknowledges software has or will read the data. Valid in all modes , recommended to be masked in BAM mode
8	OUTPUT_SERVICE_FLAG	When set by hardware, indicates QUP output FIFO has an outstanding output service request. At the point in time this bit is set, the hardware also asserts qup_irq. Writing a 'zero' to this bit does nothing. Writing a 'one' to this bit clears it and acknowledges software has or will read the data. Valid in all modes , recommended to be masked in BAM mode
7	INPUT_FIFO_FULL	Read only. When set, indicates the input FIFO is full and can accept no more data from the QUP mini-core.

BLSP1_BLSP_QUP5_QUP_OPERATIONAL (cont.)

Bits	Name	Description
6	OUTPUT_FIFO_FULL	Read only. When set, indicates the output FIFO is full and can accept no more CRIF writes.
5	INPUT_FIFO_NOT_EMPTY	Read only. When set, indicates the input FIFO has at least one value in it to be read. When clear, indicates the input FIFO is empty.
4	OUTPUT_FIFO_NOT_EMPTY	Read only. When set, indicates the output FIFO has at least one value in it to be shifted out. When clear, indicates the output FIFO is empty.

0x078BA01C BLSP1_BLSP_QUP5_QUP_ERROR_FLAGS**Type:** RW**Clock:** crif_clk**Reset State:** 0x00000000

All bits in this register are set by hardware and remain set until cleared by software. Writing a 'one' to a bit clears it while writing a 'zero' leaves the bit unchanged.

BLSP1_BLSP_QUP5_QUP_ERROR_FLAGS

Bits	Name	Description
5	OUTPUT_OVER_RUN_ERR	Indicates the output FIFO was full when a CRIF write was attempted to the FIFO. The write is discarded.
4	INPUT_UNDER_RUN_ERR	Indicates the input FIFO was empty when a CRIF read was attempted to the FIFO. The read value returns is indeterminate.
3	OUTPUT_UNDER_RUN_ERR	Indicates the output FIFO was empty when an output shift operation required a value.
2	INPUT_OVER_RUN_ERR	Indicates the input FIFO was full when an input shift operation provided a new value. When this happens, the new value is discarded.

0x078BA020 BLSP1_BLSP_QUP5_QUP_ERROR_FLAGS_EN**Type:** RW**Clock:** crif_clk**Reset State:** 0x0000003C

Each bit in this register controls the setting of a corresponding error flag. If set, each bit enables generation of qup_irq and the setting of the corresponding error flag in the QUP_ERROR_FLAGS register for the specified error case. At reset, all error enable bits are set to '1'.

BLSP1_BLSP_QUP5_QUP_ERROR_FLAGS_EN

Bits	Name	Description
5	OUTPUT_OVER_RUN_ERR_EN	If set, enables output over run error generation.
4	INPUT_UNDER_RUN_ERR_EN	If set, enables input under run error generation.
3	OUTPUT_UNDER_RUN_ERR_EN	If set, enables output under run error generation.
2	INPUT_OVER_RUN_ERR_EN	If set, enables input over run error generation.

0x078BA028 BLSP1_BLSP_QUP5_QUP_OPERATIONAL_MASK**Type:** RW**Clock:** crif_clk**Reset State:** 0x00000000

This register masks several QUP_OPERATIONAL flags from creating an interrupt .

BLSP1_BLSP_QUP5_QUP_OPERATIONAL_MASK

Bits	Name	Description
9	INPUT_SERVICE_MASK	If set, this flag in QUP_OPERATIONAL does not cause an interrupt but provides status only.
8	OUTPUT_SERVICE_MASK	If set, this flag in QUP_OPERATIONAL does not cause an interrupt but provides status only.

0x078BA100 BLSP1_BLSP_QUP5_QUP_MX_OUTPUT_COUNT**Type:** RW**Clock:** crif_clk**Reset State:** 0x00000000

Unless otherwise stated, register bits written return the value when read. This register only applies to Block_Mode. The counter decrements for each output transfer when the STATE field (see QUP_STATE register) is moved from the RESET_STATE to the RUN_STATE. The PAUSE_STATE does not effect the count.

Notes:

- a. Allows the total number of Mini Core transfers to be less than an exact multiple of OUTPUT_BLOCK_SIZE. Any additional outputs are discarded.

BLSP1_BLSP_QUP5_QUP_MX_OUTPUT_COUNT

Bits	Name	Type	Description
31	MX_CONFIG_DURING_RUN	W	Write only value. When reset to zero the MX_OUTPUT_COUNT value only applies in reset_state (legacy behavior). When set to one the MX_OUTPUT_COUNT value will also apply in run_state and allow writing the transaction data in a few portions when not knowing the transaction size in advance. Reconfiguration during run is only allowed after the last portion has ended (i.e. MAX_OUTPUT_DONE_FLAG was asserted)
15:0	MX_OUTPUT_COUNT	RW	Number of writes of size N to the mini-core per RUN_STATE. A value of zero indicates the output count function is not enabled for use. See note (a) above.

0x078BA104 BLSP1_BLSP_QUP5_QUP_MX_OUTPUT_CNT_CURRENT**Type:** R**Clock:** crif_clk**Reset State:** 0x00000000**BLSP1_BLSP_QUP5_QUP_MX_OUTPUT_CNT_CURRENT**

Bits	Name	Description
15:0	MX_OUTPUT_CNT_CURRENT	Current value of QUP_MX_OUTPUT_COUNT.

0x078BA10C BLSP1_BLSP_QUP5_QUP_OUTPUT_FIFO_WORD_CNT**Type:** R**Clock:** crif_clk**Reset State:** 0x00000000

This register holds the number of words in the output FIFO at a given time. Note the fields in this register are dynamically updated hence when this register is read during a FIFO operation, the value of this register could have been altered by the time SW gets the read value. Recommended for debug purposes.

BLSP1_BLSP_QUP5_QUP_OUTPUT_FIFO_WORD_CNT

Bits	Name	Description
8:0	OUTPUT_FIFO_WORD_COUNT	Number of words in the output FIFO.

**0x078BA110+ BLSP1_BLSP_QUP5_QUP_OUTPUT_FIFOc, c=[0..15]
0x4*c****Type:** W**Clock:** crif_clk**Reset State:** 0x00000000

Note that consecutive writes to this address keeps filling up the output FIFO. The max address to address the output FIFO is 0x14C.

BLSP1_BLSP_QUP5_QUP_OUTPUT_FIFOc

Bits	Name	Description
31:0	OUTPUT	Value to be shifted out.

0x078BA150 BLSP1_BLSP_QUP5_QUP_MX_WRITE_COUNT**Type:** RW**Clock:** crif_clk**Reset State:** 0x00000000

What the QUP_MX_OUTPUT_COUNT register means to Block_Mode and Bam_Mode, this register means the same to FIFO_mode. If this register is non-zero, then the qup_irq is asserted after shifting in the number of shifts specified by this register.

Unlike the QUP_MX_OUTPUT_COUNT register case, the SW should not program the QUP_MX_WRITE_COUNT value to be more than the 'actual depth' of the FIFO (BLOCK_SIZE_OUTPUT * FIFO_SIZE_OUTPUT).

BLSP1_BLSP_QUP5_QUP_MX_WRITE_COUNT

Bits	Name	Type	Description
31	MX_CONFIG_DURING_RUN	W	Write only value. When reset to zero the MX_WRITE_COUNT value only applies in reset_state (legacy behavior). When set to one the MX_WRITE_COUNT value will also apply in run_state and allow writing the transaction data in a few portions when not knowing the transaction size in advance. Reconfiguration during run is only allowed after the last portion has ended (i.e. MAX_OUTPUT_DONE_FLAG was asserted)
15:0	MX_WRITE_COUNT	RW	The number of 'writes' of size N. This is used only if the core is in FIFO_Mode.

0x078BA154 BLSP1_BLSP_QUP5_QUP_MX_WRITE_CNT_CURRENT**Type:** R**Clock:** crif_clk**Reset State:** 0x00000000**BLSP1_BLSP_QUP5_QUP_MX_WRITE_CNT_CURRENT**

Bits	Name	Description
15:0	MX_WRITE_CNT_CURRENT	Current value of QUP_MX_WRITE_COUNT counter.

0x078BA200 BLSP1_BLSP_QUP5_QUP_MX_INPUT_COUNT**Type:** RW**Clock:** crif_clk**Reset State:** 0x00000000

Unless otherwise stated, register bits written return the value when read. This register only applies to Block_Mode and Bam_Mode (for non-balanced SPI). The counter decrements for each input transfer when the STATE field (see QUP_STATE register) is moved from the RESET_STATE to the RUN_STATE. The PAUSE_STATE do not affect the count.

Notes:

a. Allows the number of shift register transfers to be less than an exact multiple of INPUT_BLOCK_SIZE. When count reached, remainder of INPUT_BLOCK_SIZE is filled with zeroes.

BLSP1_BLSP_QUP5_QUP_MX_INPUT_COUNT

Bits	Name	Type	Description
31	MX_CONFIG_DURING_RUN	W	Write only value. When reset to zero the MX_INPUT_COUNT value only applies in reset_state (legacy behavior). When set to one the MX_INPUT_COUNT value will also apply in run_state and allow writing the transaction data in a few portions when not knowing the transaction size in advance. Reconfiguration during run is only allowed after the last portion has ended (i.e. MAX_INPUT_DONE_FLAG was asserted)
15:0	MX_INPUT_COUNT	RW	Number of reads of size N from the mini-core per RUN_STATE. A value of zero indicates the input count function is not enabled for use. See note (a) above.

0x078BA204 BLSP1_BLSP_QUP5_QUP_MX_INPUT_CNT_CURRENT**Type:** R**Clock:** crif_clk**Reset State:** 0x00000000**BLSP1_BLSP_QUP5_QUP_MX_INPUT_CNT_CURRENT**

Bits	Name	Description
15:0	MX_INPUT_CNT_CURRENT	Current value of QUP_MX_INPUT_COUNT.

0x078BA208 BLSP1_BLSP_QUP5_QUP_MX_READ_COUNT**Type:** RW**Clock:** crif_clk**Reset State:** 0x00000000

What the QUP_MX_INPUT_COUNT register means to Block_Mode and Bam_Mode, this register means the same to FIFO_mode. If this register is non-zero, then the qup_input_service_irq is asserted after shifting in the number of shifts specified by this register.

Unlike the QUP_MX_INPUT_COUNT register case, the SW should not program the QUP_MX_READ_COUNT value to be more than the 'actual depth' of the FIFO (BLOCK_SIZE_INPUT * FIFO_SIZE_INPUT). If the SPI mini-core and slave operation is enabled, then an InputOverRun error will result.

BLSP1_BLSP_QUP5_QUP_MX_READ_COUNT

Bits	Name	Type	Description
31	MX_CONFIG_DURING_RUN	W	Write only value. When reset to zero the MX_READ_COUNT value only applies in reset_state (legacy behavior). When set to one the MX_READ_COUNT value will also apply in run_state and allow writing the transaction data in a few portions when not knowing the transaction size in advance. Reconfiguration during run is only allowed after the last portion has ended (i.e. MAX_INPUT_DONE_FLAG was asserted)
15:0	MX_READ_COUNT	RW	The number of 'reads' of size N. This is used only if the core is in FIFO_Mode.

0x078BA20C BLSP1_BLSP_QUP5_QUP_MX_READ_CNT_CURRENT**Type:** R**Clock:** crif_clk**Reset State:** 0x00000000

BLSP1_BLSP_QUP5_QUP_MX_READ_CNT_CURRENT

Bits	Name	Description
15:0	MX_READ_CNT_CURRENT	Current value of QUP_MX_READ_COUNT counter.

0x078BA214 BLSP1_BLSP_QUP5_QUP_INPUT_FIFO_WORD_CNT**Type:** R**Clock:** crif_clk**Reset State:** 0x00000000

This register holds the number of words in the input FIFO at a given time. Note the fields in this register are dynamically updated. Hence, when this register is read during a FIFO operation, the value of this register could have been altered by the time SW gets the read value. Recommended for debug purposes.

BLSP1_BLSP_QUP5_QUP_INPUT_FIFO_WORD_CNT

Bits	Name	Description
8:0	INPUT_FIFO_WORD_CNT	Number of words in the input FIFO.

**0x078BA218+BLSP1_BLSP_QUP5_QUP_INPUT_FIFOc, c=[0..15]
0x4*c****Type:** R**Clock:** crif_clk**Reset State:** Undefined

Consecutive reads to this address reads the input FIFO contents on a FIFO basis. The max address to read the input FIFO is 0x254.

BLSP1_BLSP_QUP5_QUP_INPUT_FIFOc

Bits	Name	Description
31:0	INPUT	Value shifted in.

0x078BA300 BLSP1_BLSP_QUP5_SPI_CONFIG**Type:** RW**Clock:** crif_clk**Reset State:** 0x00000000

Unless otherwise stated, register bits written return the value when read.

The contents of this register should only be changed when in the RESET_STATE. See the QUP_STATE register. Both NO_OUTPUT and NO_INPUT set to zero provides default full duplex operation. Setting both these bits to one is not a legal operational state.

BLSP1_BLSP_QUP5_SPI_CONFIG

Bits	Name	Type	Description
10	HS_MODE	RW	When set, SPI HS_MODE is enabled. When clear, SPI HS_MODE is disabled.
9	INPUT_FIRST	RW	When set, INPUT FIRST SPI protocol is used. When clear, OUTPUT FIRST SPI protocol is used.
8	LOOP_BACK	RW	Loopback is only valid in non-HS mode. Always clear for normal operation. If set, loop back on SPI_DATA_MO_SI under MASTER operation or SPI_DATA_MI_SO under SLAVE operation.
5	SLAVE_OPERATION	R	This register is writable only when the hardware signal spi_slave_en is asserted. When set, the SPI is configured for SLAVE operation. Zero indicates MASTER operation.

0x078BA304 BLSP1_BLSP_QUP5_SPI_IO_CONTROL

Type: RW

Clock: crif_clk

Reset State: 0x00000000

The contents of this register should only be changed when in the RESET_STATE. See the QUP_STATE register.

Unless otherwise stated, register bits written return the value when read.

BLSP1_BLSP_QUP5_SPI_IO_CONTROL

Bits	Name	Description
11	FORCE_CS	When this bit is set, the chip select is asserted unconditionally with no relationship to QUP_STATE or transaction state.
10	CLK_IDLE_HIGH	When set to 1, spi_clk will be high when spi is idle (spi_cs is not asserted). When cleared to 0, spi_clk will be low when spi is idle (spi_cs is not asserted).
9	CLK_ALWAYS_ON	When MASTER operation is used, run SPI_CLK during IDLE.
8	MX_CS_MODE	If this bit is set, then for a given RUN state, the associated chip select will be asserted for the first N-bit transfer and will be kept asserted till the last programmed transfer. Applies only in MASTER mode. The number of transfers is determined by SPI_MX_OUTPUT_COUNT and/or SPI_MX_INPUT_COUNT registers.

BLSP1_BLSP_QUP5_SPI_IO_CONTROL (cont.)

Bits	Name	Description
7:4	CS_N_POLARITY	These four bits control the polarity of four SPI_CS#_N respectively. Setting any of this bit to '1', makes the associated SPI_CS#_N active HIGH. This field is a 'don't care' in SLAVE operation.
3:2	CS_SELECT	When MASTER operation is used, controls the assertion of SPI_CS_#N pins. Selects SPI_CS_N if 00, SPI_CS1_N if 01, SPI_CS2_N if 10, or SPI_CS3_N if 11. The deselected ChipSelects will be driven to inactive state dictated by the field CS_N_POLARITY. This field is a 'don't care' in SLAVE operation.
1	TRISTATE_CS	When set, drives Z on all SPI_CS#_N lines. When clear, enables normal functionality on these lines. This bit can be used to support deployment of the core as one of the masters in a multi-master configuration.
0	NO_TRI_STATE	When set, spi_data_out is not taken tri-state when SPI_CS#_N is de-asserted. This bit is normally set for MASTER operation but may be optionally left cleared.

0x078BA308 BLSP1_BLSP_QUP5_SPI_ERROR_FLAGS**Type:** RW**Clock:** crif_clk**Reset State:** 0x00000000

All bits in this register are set by hardware and remain set until cleared by software. Writing a 'one' to a bit clears it while writing a 'zero' leaves the bit unchanged.

BLSP1_BLSP_QUP5_SPI_ERROR_FLAGS

Bits	Name	Description
3	TRANSFER_CANCEL_DONE	This bit is set (1) when the SPI controller has completed canceling the transfer that was scheduled for cancellation (LOCAL_ID = TRANSFER_CANCEL_ID).
2	TRANSFER_CANCEL_ID_MATCH	This bit is set (1) when the SPI controller is ready for cancel operation: it has found the transfer that was scheduled for cancellation (LOCAL_ID = TRANSFER_CANCEL_ID) and the logic is done with previous activity.
1	SPI_SLV_CLK_OVER_RUN_ERR	Number of SPI_CLK ticks which occurred while SPI_CS#_N was asserted was greater than the programmed value of N. When this happens, only the first N bits are passed to the SPI input FIFO and the output shift value is held at zero.
0	SPI_SLV_CLK_UNDER_RUN_ERR	Number of SPI_CLK ticks which occurred while SPI_CS#_N was asserted was less than the programmed value of N. When this occurs, the bits which were received are passed to the SPI input FIFO and the CURRENT output bit is forced to zero.

0x078BA30C BLSP1_BLSP_QUP5_SPI_ERROR_FLAGS_EN**Type:** RW**Clock:** crif_clk**Reset State:** 0x0000000C

Each bit in this register controls the setting of a corresponding error flag. If set, each bit enables generation of spi_error_irq and the setting of the corresponding error flag in the SPI_ERROR_FLAGS register for the specified error case. At reset, all error enable bits are set to '1'.

BLSP1_BLSP_QUP5_SPI_ERROR_FLAGS_EN

Bits	Name	Description
3	TRANSFER_CANCEL_DONE_EN	If set, enables generating the transfer cancel done indication.
2	TRANSFER_CANCEL_ID_MATCH_EN	If set, enables generating the transfer cancel ID match indication.
1	SPI_SLV_CLK_OVER_RUN_ERR_EN	If set, enables clock over run error generation.
0	SPI_SLV_CLK_UNDER_RUN_ERR_EN	If set, enables clock under run error generation.

0x078BA310 BLSP1_BLSP_QUP5_SPI_DEASSERT_WAIT**Type:** RW**Clock:** crif_clk**Reset State:** 0x00000000

This register holds the de-assertion wait time of SPI_CS#_N between consecutive N-bit transfers in MASTER operation. The wait time is specified in number of cycles of cc_spi_master_clk.

BLSP1_BLSP_QUP5_SPI_DEASSERT_WAIT

Bits	Name	Description
5:0	DEASSERT_WAIT	Number cc_spi_master_clk ticks associated with the deasserted time of SPI_CS#_N. Only applies to MASTER operation. For SLAVE operation, this field is a 'don't care'. A value of zero indicates SPI_CS#_N remains de-asserted for exactly one clock tick. A value of one, indicates two ticks, etc. All ones indicates 64 ticks.

0x078BA314 BLSP1_BLSP_QUP5_SPI_MASTER_LOCAL_ID**Type:** RW**Clock:** crif_clk**Reset State:** 0x00000000**BLSP1_BLSP_QUP5_SPI_MASTER_LOCAL_ID**

Bits	Name	Description
15:8	EXTENDED_ID	SW may use this field to specify the current sub-transfer ID to be read when TRANSFER_CANCEL_ID_MATCH is asserted. This value has no impact on HW.
7:0	LOCAL_ID	SW specifies the current transfer ID. HW will use this value to match with TRANSFER_CANCEL_ID.

0x078BA318 BLSP1_BLSP_QUP5_SPI_MASTER_COMMAND**Type:** W**Clock:** crif_clk**Reset State:** 0x00000000**BLSP1_BLSP_QUP5_SPI_MASTER_COMMAND**

Bits	Name	Description
0	RESET_CANCEL_FSM	This command should be given only when the CANCEL_FSM_STATE is in CANCEL_PENDING_STATE or PAUSE_WAIT_STATE and when the core is in PAUSE. The main use case for this bit is to allow withdrawing a cancel command.

0x078BA31C BLSP1_BLSP_QUP5_SPI_MASTER_STATUS**Type:** R**Clock:** crif_clk**Reset State:** 0x00000000**BLSP1_BLSP_QUP5_SPI_MASTER_STATUS**

Bits	Name	Description
2:0	CANCEL_FSM_STATE	This 3-bit field informs the microprocessor of the state of the SPI cancel logic. Internal use only

0x078BA400 BLSP1_BLSP_QUP5_I2C_MASTER_CLK_CTL**Type:** RW**Clock:** crif_clk**Reset State:** 0x00000000

The I2C_CLK_CTL register is a read/write register that controls clock divider values.

BLSP1_BLSP_QUP5_I2C_MASTER_CLK_CTL

Bits	Name	Description
28	SCL_EXT_FORCE_LOW	When set to 0, SCL generation maintains legacy behavior. When set to 1, SCL state machine will go to the FORCED_LOW_STATE if force_low is asserted near (within 5 SCL clock cycles = $5 * 52\text{ns} = 260\text{ns} \approx \text{MAX}(\text{Tsu}, \text{DAT})$) the clock edge.
27:26	SDA_NOISE_REJECTION	Allows adding extra sampling levels on SDA to reject short low pulses. This value specifies how many TCXO cycles of logic low on SDA would be considered as valid logic low. 0x0 - legacy mode, 0x01 - one cycle wide low pulse is rejected, 0x2 - two cycles wide low pulse is rejected, 0x3 - three cycles wide low pulse is rejected
25:24	SCL_NOISE_REJECTION	Allows adding extra sampling levels on SCL to reject short low pulses. This value specifies how many TCXO cycles of logic low on SCL would be considered as valid logic low. 0x0 - legacy mode, 0x01 - one cycle wide low pulse is rejected, 0x2 - two cycles wide low pulse is rejected, 0x3 - three cycles wide low pulse is rejected
23:16	HIGH_TIME_DIVIDER_VALUE	Allows setting SCL duty cycle to non 50%. If this value is zero then legacy mode is used. If this value is non-zero then it will be used as the SCL high time counter and FS_DIVIDER_VALUE will be used as the low time counter. Minimum value is 0x7.
10:8	HS_DIVIDER_VALUE	The value in this register represents the clock period multiplier in high-speed (HS) mode. Minimum value is 0x3. $\text{I2C_HS_CLK} = \text{I2C_CLK} / (3 * (\text{HS_DIVIDER_VALUE} + 1) + 2)$
7:0	FS_DIVIDER_VALUE	The value in this register represents the clock period multiplier in fast/standard (FS) mode. Minimum value is 0x7. When HIGH_TIME_DIVIDER_VALUE=0: $\text{I2C_FS_CLK} = \text{I2C_CLK} / (2 * (\text{FS_DIVIDER_VALUE} + 3))$ When HIGH_TIME_DIVIDER_VALUE!=0: $\text{I2C_FS_CLK} = \text{I2C_CLK} / (\text{FS_DIVIDER_VALUE} + \text{HIGH_TIME_DIVIDER_VALUE} + 6)$

0x078BA404 BLSP1_BLSP_QUP5_I2C_MASTER_STATUS**Type:** RW**Clock:** crif_clk**Reset State:** 0x0C000000

The I2C_STATUS is a status register. Writing any value clears the status bits.

BLSP1_BLSP_QUP5_I2C_MASTER_STATUS

Bits	Name	Type	Description
30:28	CANCEL_FSM_STATE	R	This 3-bit field informs the microprocessor of the state of the I2C cancel logic. Internal use only 0x0: IDLE_STATE (IDLE_STATE) 0x1: CANCEL_PENDING_STATE (CANCEL_PENDING_STATE) 0x2: PAUSE_STATE (PAUSE_STATE) 0x3: PAUSE_WAIT_STATE (PAUSE_WAIT_STATE) 0x4: SEND_STOP_STATE (SEND_STOP_STATE) 0x5: FLUSH_STATE (FLUSH_STATE) 0x6: SEND_IRQ_STATE (SEND_IRQ_STATE)
27	I2C_SCL	R	Logic state of I2C bus serial clock wire.
26	I2C_SDA	R	Logic state of I2C bus serial data wire.
25	INVALID_READ_SEQ	RW	Interrupt source. This bit is set (1) when a MI_REC tag does not follow a START tag for an I2C READ. Not applicable for Halcyon.
24	INVALID_READ_ADDR	RW	Interrupt source. In version 1 tags this bit is set (1) when the I2C controller is trying to receive data from a non-existent I2C slave (address). In version 2 tags this bit is set (1) when the I2C controller is trying to access a non-existent I2C slave (address).
23	INVALID_TAG	RW	Interrupt source. This bit is set (1) when the I2C controller is trying to process data from the output FIFO that is improperly tagged.
22:20	INPUT_FSM_STATE	R	This 3-bit field informs the microprocessor of the state of the I2C MASTER INPUT FSM block. Internal use only 0x0: V1_V2_RESET_STATE (RESET_STATE value. Applies for both version 1 and version 2 tags) 0x1: V1_V2_IDLE_STATE (IDLE_STATE value. Applies for both version 1 and version 2 tags) 0x2: V1_V2_READ_LAST_BYTE_STATE (READ_LAST_BYTE_STATE value. Applies for both version 1 and version 2 tags) 0x3: V1_V2_MI_REC_STATE (MI_REC_STATE value. Applies for both version 1 and version 2 tags) 0x4: V1_V2_DEC_STATE (DEC_STATE value. Applies for both version 1 and version 2 tags) 0x5: V1_V2_STORE_STATE (STORE_STATE value. Applies for both version 1 and version 2 tags) 0x6: V2_WR_TAG_STATE (WR_TAG_STATE value. Only applicable in version 2 tags) 0x7: V2_WAIT_TAG_STATE (WAIT_TAG_STATE value. Only applicable in version 2 tags)

BLSP1_BLSP_QUP5_I2C_MASTER_STATUS (cont.)

Bits	Name	Type	Description
19:16	OUTPUT_FSM_STATE	R	This 3-bit field informs the microprocessor of the state of the I2C MASTER OUTPUT FSM block. Internal use only 0x0: V1_V2_RESET_STATE (RESET_STATE value. Applies for both version 1 and version 2 tags) 0x1: V1_V2_IDLE_STATE (IDLE_STATE value. Applies for both version 1 and version 2 tags) 0x2: V1_V2_DECODE_STATE (DECODE_STATE value. Applies for both version 1 and version 2 tags) 0x3: V1_V2_SEND_STATE (SEND_STATE value. Applies for both version 1 and version 2 tags) 0x4: V1_V2_MI_REC_STATE (MI_REC_STATE value. Applies for both version 1 and version 2 tags) 0x5: V1_V2_NOP_STATE (NOP_STATE value. Applies for both version 1 and version 2 tags) 0x6: V1_V2_INVALID_STATE (INVALID_STATE value. Applies for both version 1 and version 2 tags) 0x7: V1_PEEK_STATE (PEEK_STATE value. Only applicable in version 1 tags) 0x8: V1_SEND_R_STATE (SEND_R_STATE value. Only applicable in version 1 tags) 0xB: V2_GET_BYTE_STATE (GET_BYTE_STATE value. Only applicable in version 2 tags) 0xC: V2_GET_WRITE_BYTE_STATE (GET_WRITE_BYTE_STATE value. Only applicable in version 2 tags) 0xD: V2_SPECIALITY_STATE (SPECIALITY_STATE value. Only applicable in version 2 tags) 0xE: V2_WAIT_FLUSH_STOP_STATE (WAIT_FLUSH_STOP_STATE value. Only applicable in version 2 tags) 0xF: V2_STOP_STATE (STOP_STATE value. Only applicable in version 2 tags)
15:13	CLK_STATE	R	This 3-bit field informs the microprocessor of the state of the I2C clk_control block. Internal use only 0x0: RESET_BUSIDLE_STATE (RESET_BUSIDLE_STATE) 0x1: NOT_MASTER_STATE (NOT_MASTER_STATE) 0x2: HIGH_STATE (HIGH_STATE) 0x3: LOW_STATE (LOW_STATE) 0x4: HIGH_WAIT_STATE (HIGH_WAIT_STATE) 0x5: FORCED_LOW_STATE (FORCED_LOW_STATE) 0x6: HS_ADDR_LOW_STATE (HS_ADDR_LOW_STATE) 0x7: DOUBLE_BUFFER_WAIT_STATE (DOUBLE_BUFFER_WAIT_STATE)

BLSP1_BLSP_QUP5_I2C_MASTER_STATUS (cont.)

Bits	Name	Type	Description
12:10	DATA_STATE	R	This 3-bit field informs the microprocessor of the state of the I2C data_control block. Internal use only 0x0: RESET_WAIT_STATE (RESET_WAIT_STATE) 0x1: TX_ADDR_STATE (TX_ADDR_STATE) 0x2: TX_DATA_STATE (TX_DATA_STATE) 0x3: TX_HS_ADDR_STATE (TX_HS_ADDR_STATE) 0x4: TX_10_BIT_ADDR_STATE (TX_10_BIT_ADDR_STATE) 0x5: TX_2ND_BYTE_STATE (TX_2ND_BYTE_STATE) 0x6: RX_DATA_STATE (RX_DATA_STATE) 0x7: RX_NACK_HOLD_STATE (RX_NACK_HOLD_STATE)
9	BUS_MASTER	R	This bit is set (1) when the I2C controller is the present bus master.
8	BUS_ACTIVE	R	This bit is set (1) when the bus is in use by this, or any other controller.
7:6	FAILED	RW	Interrupt source. This 2-bit field contains the failure information of the present I2C transfers. If transmitting, failed[1] contains the information regarding the byte that has been transmitted. Failed[0] contains the information of the byte awaiting transmission if there is a byte pipelined. If no byte is pipelined, ignore this bit. If receiving, failed[1] contains the information of the byte received and stored in the buffer, and failed[0] should be ignored. For example: Value 00: Byte n transmitted successfully, byte n+1 to begin transmission Value 01: Byte n transmitted successfully, byte n+1 errored: type of error indicated in other STATUS bits. (queued invalid write, for example) Value 10: Byte n errored: type of error indicated in other STATUS bits, byte n+1 to begin transmission (byte n+1 would have to be a valid address byte for this condition to occur) Value 11: Byte n errored: type of error indicated in other STATUS bits, byte n+1 discarded as well (if the byte was a data byte destined for a NACKed address, data is useless, therefore discarded)
5	INVALID_WRITE	RW	Interrupt source. This bit is set (1) when software writes data to the I2C_DATA register that should be flagged as an address but is not.
4	ARB_LOST	RW	Interrupt source. This bit is set (1) when the controller loses arbitration for the bus. If this bit gets set (1) during the transmission, all data has been lost, and the microprocessor must re-request its transmission.

BLSP1_BLSP_QUP5_I2C_MASTER_STATUS (cont.)

Bits	Name	Type	Description
3	PACKET_NACKED	RW	Interrupt source. This bit is set high when a NACK is received from a slave. If a high speed master code is sent and there is an ACK from a slave, then this bit is set (1) to indicate that the high speed mode can not be entered. If the high speed mode is accepted by the slave, then a NACK is performed and this bit is not set (1).
2	BUS_ERROR	RW	Interrupt source. This bit is set (1) when an unexpected START or STOP condition is detected. This returns the controller to its reset state.
1	TRANSFER_CANCEL_DONE	RW	Only applicable when using version 2 tags. Interrupt source. This bit is set (1) when the I2C controller has completed canceling the transfer that was scheduled for cancellation (LOCAL_ID = TRANSFER_CANCEL_ID).
0	TRANSFER_CANCEL_ID_MATCH	RW	Only applicable when using version 2 tags. Interrupt source. This bit is set (1) when the I2C controller is ready for cancel operation: it has found the transfer that was scheduled for cancellation (LOCAL_ID = TRANSFER_CANCEL_ID) and the logic is done with previous activity.

0x078BA408 BLSP1_BLSP_QUP5_I2C_MASTER_CONFIG**Type:** RW**Clock:** crif_clk**Reset State:** 0x00000000**BLSP1_BLSP_QUP5_I2C_MASTER_CONFIG**

Bits	Name	Description
0	EN_VERSION_TWO_TAG	When set, version 2 of the mini-core is enabled.

0x078BA40C BLSP1_BLSP_QUP5_I2C_MASTER_BUS_CLEAR**Type:** RW**Clock:** crif_clk**Reset State:** 0x00000000

BLSP1_BLSP_QUP5_I2C_MASTER_BUS_CLEAR

Bits	Name	Description
0	CLEAR	This command should be given only when the i2c mincore is idle. When in doubt SW can use SW_RESET to reset the minicore. When set, an I2C 'Bus Clear' executed per the I2C standard. A bus clear consists of nine I2C clock ticks with data wire left not-driven. Has the effect of clearing any slave which has lost read sync. After the clear is complete, the hardware sets the CLEAR bit to zero. The bit can not be cleared by software. After a bus clear, both clock and data lines should be high. If not, an external slave has hung the bus by holding down one or both lines. Reset the slave to clear if possible. Any slave reset mechanism is beyond the scope of the I2C mini core.

0x078BA410 BLSP1_BLSP_QUP5_I2C_MASTER_LOCAL_ID**Type:** R**Clock:** crif_clk**Reset State:** 0x00000000**BLSP1_BLSP_QUP5_I2C_MASTER_LOCAL_ID**

Bits	Name	Description
7:0	LOCAL_ID	Read only value capturing the ID byte of the last NOP LOCAL tag. Only applicable when using V2 tags

0x078BA414 BLSP1_BLSP_QUP5_I2C_MASTER_COMMAND**Type:** RW**Clock:** crif_clk**Reset State:** 0x00000000**BLSP1_BLSP_QUP5_I2C_MASTER_COMMAND**

Bits	Name	Description
0	RESET_CANCEL_FSM	This command should be given only when the CANCEL_FSM_STATE is in CANCEL_PENDING_STATE or PAUSE_WAIT_STATE and when the core is in PAUSE. The main use case for this bit is to allow withdrawing a cancel command in case the transfer was flushed (QUP_STATE[I2C_FLUSH] was set) due to I2C NACK.

21 USB2

21.1 USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS

0x078D9000 USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_ID

Type: R

Clock: CC_USB_OTG_HS_CLK

Reset State: 0x0042FA05

The USB_OTG_HS_ID register is the identification register. It provides a simple way to determine if the USB-HS USB 2.0 core is provided in the system. The ID register identifies the USB-HS USB 2.0 core and its revision.

USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_ID

Bits	Name	Description
31:24	RESERVED_BITS31_24	These bits are reserved and should be set to zero.
23:16	REVISION	This field contains the revision number of the core - 0x42.
15:8	NID	This field contains the complement version of ID[7:0] - 0xFA.
7:0	ID	This field is the configuration number. This number is set to 0x05 and indicates that the peripheral is the USB-HS USB 2.0 core.

0x078D9004 USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_HWGENERAL

Type: R

Clock: CC_USB_OTG_HS_CLK

Reset State: 0x000005C2

The USB_OTG_HS_HWGENERAL register contains the general hardware parameters.

USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_HWGENERAL

Bits	Name	Description
31:11	RESERVED_BITS31_11	

USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_HWGENERAL (cont.)

Bits	Name	Description
10:9	SM	VUSB_HS_PHY_SERIAL = 2
8:6	PHYM	VUSB_HS_PHY_TYPE = 7
5:4	PHYW	VUSB_HS_PHY16_8 = 0
3	BWT	This bit is reserved for internal testing = 0
2:1	CLCK	VUSB_HS_CLOCK_CONFIGURATION = 1
0	RT	VUSB_HS_RESET_TYPE = 0

0x078D9008 USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_HWHOST**Type:** R**Clock:** CC_USB_OTG_HS_CLK**Reset State:** 0x10020001

The USB_OTG_HS_HWHOST register contains the host hardware parameters.

USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_HWHOST

Bits	Name	Description
31:24	TTPER	VUSB_HS_TT_PERIODIC_CONTEXTS
23:16	TTASY	VUSB_HS_TT_ASYNC_CONTEXTS
15:4	RESERVED_BITS15_4	Clear (0) these bits.
3:1	NPORT	VUSB_HS_NUM_PORT-1
0	HC	VUSB_HS_HOST

0x078D900C USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_HWDEVICE**Type:** R**Clock:** CC_USB_OTG_HS_CLK**Reset State:** 0x00000011

The USB_OTG_HS_HWDEVICE register contains the device hardware parameters.

USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_HWDEVICE

Bits	Name	Description
31:6	RESERVED_BITS31_6	
5:1	DEVEP	VUSB_HS_DEV_EP
0	DC	Device capable; [VUSB_HS_DEV/=0]

0x078D9010 USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_HWTXBUF**Type:** R**Clock:** CC_USB_OTG_HS_CLK**Reset State:** 0x80070A08

The USB_OTG_HS_HWTXBUF register contains the Tx buffer hardware parameters.

USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_HWTXBUF

Bits	Name	Description
31	TXLCR	This bit is fixed to 1'b1 so that the local context register's are included in the design. This means that the DMA context is implemented in FlipFlops.
30:24	RESERVED_BITS30_24	Clear (0) these bits.
23:16	TXCHANADD	VUSB_HS_TX_CHAN_ADD - Defines the number of address lines needed per Endpoint per the TX latency buffer. It's reset value is taken from a GENERIC value passed to the core.
15:8	TXADD	VUSB_HS_TX_ADD - Defines the number of address lines needed per the entire TX latency buffer. It's reset value is taken from a GENERIC value passed to the core.
7:0	TXBURST	VUSB_HS_TX_BURST - Defines the data burst length of the AHB master interface in Quad-words (4-byte increments) of the TX data. It's reset value is taken from a GENERIC value passed to the core. Note that the actual burst lengths will depend on the settings of USB_OTG_HS_AHB_MODE and USB_OTG_HS_AHB_BURST registers.

0x078D9014 USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_HWRXBUF**Type:** R**Clock:** CC_USB_OTG_HS_CLK**Reset State:** 0x00000808

The USB_OTG_HS_HWRXBUF register contains the Rx buffer hardware parameters.

USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_HWRXBUF

Bits	Name	Description
31:16	RESERVED_BITS31_16	Clear (0) these bits).
15:8	RX_ADD	VUSB_HS_RX_ADD - Defines the number of address lines needed per the entire RX latency buffer. It's reset value is taken from a GENERIC value passed to the core.

USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_HWRXBUF (cont.)

Bits	Name	Description
7:0	RX_BURST	VUSB_HS_RX_BURST - Defines the data burst length of the AHB master interface in Quad-words (4-byte increments) of the RX data. It's reset value is taken from a GENERIC value passed to the core. Note that the actual burst lengths will depend on the settings of USB_OTG_HS_AHB_MODE and USB_OTG_HS_AHB_BURST registers.

**0x078D9040+ USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_SCRATCH_RAMn, n=[0..15]
0x4*n****Type:** RW**Clock:** CC_USB_OTG_HS_CLK**Reset State:** 0x00000000

USB_OTG_HS_SCRATCH_RAMn registers are 16 32bit scratch registers. Required for passing USB software information between different images.

USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_SCRATCH_RAMn

Bits	Name	Description
31:0	SCRATCH_REGISTER	32 bit scratch register

0x078D9080 USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_GPTIMER0LD**Type:** RW**Clock:** CC_USB_OTG_HS_CLK**Reset State:** 0x00000000

The USB_OTG_HS_GPTIMER0LD register is the general purpose timer #0 load register. This register contains the timer duration or load value. See the GPTIMER0CTRL register for a description of the timer functions.

USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_GPTIMER0LD

Bits	Name	Description
31:24	RESERVED_BITS31_24	Clear (0) these bits.
23:0	GPTLD	General purpose timer load value. This field is the value to be loaded into the GPTCNT countdown timer on a reset action. This value in this register represents the time in microseconds minus 1 for the timer duration. Example: for a one millisecond timer, load 1000-1=999 or 0x0003E7. Note: Max value is 0xFFFFF or 16.777215 seconds

0x078D9084 USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_GPTIMER0CTRL**Type:** RW**Clock:** CC_USB_OTG_HS_CLK**Reset State:** 0x00000000

The USB_OTG_HS_GPTIMER0CTRL register is the general purpose timer #0 control register. This register contains the control for the timer and a data field can be queried to determine the running count value. This timer has a granularity of 1 ms and can be programmed to a little over 16 seconds. There are two modes supported by this timer: the first is a one-shot and the second is a looped count, which is described in the register table below. When the timer counter value transitions to zero, an interrupt can be generated though the use of the timer interrupts in the USBTS and USBINTR registers.

USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_GPTIMER0CTRL

Bits	Name	Description
31	GTPRUN	General purpose timer run Read/write value 0 = Timer stop value 1 = Timer run This bit enables the general-purpose timer to run. Setting or clearing this bit will not have an effect on the GPTCNT.
30	GPTRST	General purpose timer reset Write-only value 0 = No action value 1 = Load counter value Writing a one to this bit will reload the GPTCNT with the value in GPTLD.
29:25	RESERVED_BITS29_25	Clear (0) these bits.
24	GPTMODE	General purpose timer mode Read/write value 0 = One shot value 1 = Repeat This bit selects between a single timer countdown and a looped count down. In the one-shot mode, the timer will count down to zero, generate an interrupt, and stop until the counter is reset by the software. In the repeat mode, the timer will count down to zero, generate an interrupt, and automatically reload the counter to begin again.
23:0	GPTCNT	General purpose timer counter Read-only This field is the value of the running timer.

0x078D9088 USB_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_GPTIMER1LD**Type:** RW**Clock:** CC_USB_OTG_HS_CLK**Reset State:** 0x00000000

The USB_OTG_HS_GPTIMER1LD register is the general purpose timer #1 control register. This register contains the timer duration or load value. See the GPTIMER0LD register for a description of the timer functions.

USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_GPTIMER1LD

Bits	Name	Description
31:24	RESERVED_BITS31_24	Clear (0) these bits.
23:0	GPTLD	General purpose timer load value. This field is the value to be loaded into the GPTCNT countdown timer on a reset action. This value in this register represents the time in microseconds minus 1 for the timer duration. Example: for a one millisecond timer, load 1000-1=999 or 0x0003E7. Note: Max value is 0xFFFFF or 16.777215 seconds.

0x078D908C USB_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_GPTIMER1CTRL**Type:** RW**Clock:** CC_USB_OTG_HS_CLK**Reset State:** 0x00000000

The USB_OTG_HS_GPTIMER1CTRL register is the general purpose timer #1 control register. See the description of the USB_OTG_HS_GPTIMER0CTRL register for details about this register.

USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_GPTIMER1CTRL

Bits	Name	Description
31	GTPRUN	General purpose timer run Read/write value 0 = Timer stop value 1 = Timer run This bit enables the general-purpose timer to run. Setting or clearing this bit will not have an effect on the GPTCNT.
30	GPTRST	General purpose timer reset Write-only value 0 = No action value 1 = Load counter value Writing a one to this bit will reload the GPTCNT with the value in GPTLD.

USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_GPTIMER1CTRL (cont.)

Bits	Name	Description
29:25	RESERVED_BITS29_25	Clear (0) these bits.
24	GPTMODE	General purpose timer mode Read/write value 0 = One shot value 1 = Repeat This bit selects between a single timer countdown and a looped count down. In the one-shot mode, the timer will count down to zero, generate an interrupt, and stop until the counter is reset by the software. In the repeat mode, the timer will count down to zero, generate an interrupt, and automatically reload the counter to begin again.
23:0	GPTCNT	General purpose timer counter Read-only This field is the value of the running timer.

0x078D9090 USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_AHB_BURST**Type:** RW**Clock:** CC_USB_OTG_HS_CLK**Reset State:** 0x00000000

The USB_OTG_HS_AHB_BURST register determines the AHB master mode of the USB HS Core.

NOTE If the USB_OTG_HS_AHB_MODE is configured to be 0x0 (the AHB Transactor is used), this register must be 0x0 as well, otherwise the AHB Master bus behavior is undefined. Use values other than 0x0 only if setting USB_OTG_HS_AHB_MODE to 0x1.

USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_AHB_BURST

Bits	Name	Description
31:3	RESERVED_BITS31_3	Should be set to zero.

USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_AHB_BURST (cont.)

Bits	Name	Description
2:0	AHB_BURST	AMBA AHB BURST. This is a r/w field that selects the following options for the m_hburst signal of the AMBA master interface: - 3'b000: INCR burst of unspecified length - 3'b001: INCR4, non-multiple transfers of INCR4 will be decomposed into singles - 3'b010: INCR8, non-multiple transfers of INCR8, will be decomposed into INCR4 or singles - 3'b011: INCR16, non-multiple transfers of INCR16, will be decomposed into INCR8, INCR4 or singles - 3'b100: This value is reserved and should not be used - 3'b101: INCR4, non-multiple transfers of INCR4 will be decomposed into smaller unspecified length bursts - 3'b110: INCR8, non-multiple transfers of INCR8 will be decomposed into smaller unspecified length bursts - 3'b111: INCR16, non-multiple transfers of INCR16 will be decomposed into smaller unspecified length bursts In all cases where the unspecified length burst is allowed, singles access may also occur, this is mostly true when the transaction is not 32-bit aligned. Two consecutive single accesses should not happen. When a INCRx burst size is selected and the transfer is not multiple of the INCRx burst, the burst is decomposed in the different ways. With AHBBRST[2] = 1, the smaller bursts will be unspecified length. with AHBBRST[2] = 0, the smaller bursts will be smaller INCRx or singles. For example, lets say that it's required at a given time, to transfer 22 words of information, for the following values of AHBBRST the master sequence will be: - 3'b101: INCR4+ INCR4 +INCR4+ INCR4 +INCR4+ INCR4 unspec. length - 3'b110: INCR8+INCR8+INCR4+ INCR4 unspec. length - 3'b111: INCR16+INCR4+ INCR4 unspec. length - 3'b001: INCR4+ INCR4 +INCR4+ INCR4 +INCR4+SINGLE+SINGLE - 3'b010: INCR8+INCR8+INCR4+SINGLE+SINGLE - 3'b011: INCR16+INCR4+SINGLE+SINGLE This field after reset is set to a default value that can be configured in the file vusb_hs_cfg.vhd.
		The AHBBRST field is only used if the AMBA-AHB system interface has been selected. It has no effect for cores featuring BVCI interface. In the later case the read will return zeros. When this field is different from zero, the value of the fields TXBURST /RXBURST in register BURSTSIZE 160h, will be ignored by the controller. Internally the BURSTSIZE will be set to the value of the INCRx AMBA burst. Since this has a direct relation with the burst sizes you must be careful with AHB burst selected. Although the TXBURST / RXBURST are bypassed, this register can still be written / read with no effect, while the AHBBRST field is non-zero.

0x078D9094 USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_XTOR_STS**Type:** R**Clock:** CC_USB_OTG_HS_CLK**Reset State:** 0x00000000

The USB_OTG_HS_XTOR_STS register is currently a placeholder for future status bits from the AHB2AHB Transactor.

USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_XTOR_STS

Bits	Name	Description
31:2	RESERVED_BITS31_2	Not used currently.
1	GRANT_STOLEN	Reports whether the arbiter removed the hgrant signal prior to completing a transaction. This is currently supported in WRITES ONLY. This bit can be cleared by writing a '1' to the GRANT_STOLEN_CLEAR bit in the USB_OTG_HS_AHB_MODE register. To enable this bit again, write a '0' to the GRANT_STOLEN_CLEAR bit in the USB_OTG_HS_AHB_MODE register.
0	RESERVED_BIT0	Not used currently.

0x078D9098 USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_AHB_MODE**Type:** RW**Clock:** CC_USB_OTG_HS_CLK**Reset State:** 0x00000008**USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_AHB_MODE**

Bits	Name	Description
31	ASYNC_BRIDGES_BYPASS	Default is '0'. When '0' the asynchronous bridge on the master AHB interface is used. When '1', it is bypassed. The bridge on the slave AHB is always used.
30:5	RESERVED_BITS30_5	Not used currently.
4	INCR_OVERRIDE	Valid only if the Transactor is bypassed: When '1', all INCR bursts from the USB Core will be internally transformed into SINGLE transfers. When '0', if the USB Core issues an INCR burst, it will propagate to the external master AHB port.

USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_AHB_MODE (cont.)

Bits	Name	Description
3:2	HPROT_MODE	When '00' the HPROT signal out of the USB Wrapper is '0001', and all transactions are non-posted. When '01' the HPROT signal out of the USB Wrapper is '0101', and all transactions are posted. When '10' the HPROT signal out of the USB Wrapper alternates according to the context of the AHB bus access. Control structures are non-posted while data transfer is posted. When '11', reserved value, but currently maps to non-posted (same as '00'). Default value is '10'
1	GRANT_STOLEN_CLEAR	Clears the grant stolen field of the USB_OTG_HS_XTOR_STS register. To enable this bit again, write '0' after clearing the GRANT_STOLEN ('1').
0	XTOR_BYPASS	When this bit is set (1), the AHB Transactor is bypassed, and the USB HS Core's AHB Master interface is directly connected to the Qualcomm AHB system. In this case, the USB_OTG_HS_AHB_BURST register value will determine the bus characteristics. When this bit is reset (0), the AHB Transactor is used to connect the USB HS Core to the Qualcomm AHB system. Default value is '0'

0x078D909C USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_GEN_CONFIG**Type:** RW**Clock:** CC_USB_OTG_HS_CLK**Reset State:** 0x00000830

The USB_OTG_HS_GEN_CONFIG register is used to configure various features that have been added to the HS USB Core.

USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_GEN_CONFIG

Bits	Name	Description
31:16	RESERVED	Reserved
15	SYS_CLK_SW_EN	This bit is applicable only in SPS Device mode. When this bit is set then USB core always voting for USB_SYSTEM_CLK. Default value is 0 - USB core doesn't request USB_SYSTEM_CLK when in Low Power Mode.
14	TESTMUX_SEL_4	see TESTMUX_SEL_3_0 the first 4 bits of this register.
13	USB_BAM_DISABLE	This bit disables the bam logic inside the USB and makes him work in Legacy mode.
12	DMA_HPROT_CTRL	When this bit is set Link Controller always does non-posted dQH writes.

USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_GEN_CONFIG (cont.)

Bits	Name	Description
11	ISO_FIX_EN	This bit enables fix for Isochronous bug in CI core (CR--0000135251).
10	DSC_PE_RST_EN	This bit enables an automatic reset of Device PE State Machine on disconnection event when operating as device. This reset is a HW fix for CR-000940.
9	HOST_SIM_TIMERS_EN_S USP	When this bit is set (1), the timers used for the USB suspend process short for faster simulation and ATE time. When this bit is clear(0), the timers used for the USB suspend process are according to the USB specification.
8	HOST_SIM_TIMERS_EN_S TD	When this bit is set (1), the timers used for the USB reset on the ULPI are short for faster simulation and ATE time. When this bit is clear(0), the timers used for the USB reset on the ULPI are according to the USB specification.
7	PE_RX_BUF_PENDING_EN	This is only valid in Device Mode. Setting this bit will cause to store a Transaction Status Tag in the Pending register instead of RX Buffer if the RX Buffer is full. The Tag will move from Pending register to RX Buffer as soon as it becomes not full.
6	STREAM_RX_BYPASS_EN ABLE	This is only valid in Device Mode. If SDIS bit is set (bit #4 of USB_OTG_HS_USBMODE (0x1A8)), i.e. streaming mode is disabled, setting this bit will cause the RX traffic to override the SDIS bit, and to receive in streaming mode. TX will still be in non-streaming mode.
5	ULPI_SERIAL_EN	This bit must be set to enable operation of ULPI Serial FS/LS mode. Default state is '1' - ULPI Serial mode is supported.
4	PE_DP_TXFIFO_IDLE_FOR CE	This is only valid in Device Mode. Setting this bit to '1' forces the dp_tx_fifo_cmd_dev to be equal to PE_DP_TXFIFO_IDLE when Device PE state machine in REPORT_NAK state and the RX Buffer is full. This bit is used to enable fix of CR-001612. Reset value is '1'.

USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_GEN_CONFIG (cont.)

Bits	Name	Description
3:0	TESTMUX_SEL_3_0	With TESTMUX_SEL_4 select one of the following test buses: Value 00001 Key state machines 00010 dma_eng_0 dma_dev_sm_1 00011 dma_eng_1 dma_context 00100 dma_eng_2 dma_mem_arb 00101 prot_eng_0 00110 prot_eng_1 00111 prot_eng_2 01000 port_ctrl_0 01001 port_ctrl_1 01010 tx_buffer 01011 rx_buffer 01100 otg 01101 hsic_test_bus1 01110 hsic_test_bus2 10000 dma_eng_3 dma_dev_sm_2 10001 dma_eng_4 dma_traf others zeros

0x078D90A0 USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_GEN_CONFIG_2**Type:** RW**Clock:** CC_USB_OTG_HS_CLK**Reset State:** 0x003AFF60

Register for the chicken bits. The USB_OTG_HS_GEN_CONFIG_2 register is used to configure various features that have been added to the HS USB Core. By default, all chicken bits are off and the fix is applicable.

USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_GEN_CONFIG_2

Bits	Name	Type	Description
31:26	RESERVED_BITS31_26	R	Clear (0) these bits.
25	BVALID_SW_SEL	RW	BVALID_FROM_PHY = 0, BVALID_FROM_REG = 1, default value 0 - BVALID_FROM_PHY
24	BVALID_SW	RW	B-Device Session Valid Indicator, default value 0
23	AVALID_SW_SEL	RW	AVALID_FROM_PHY = 0, AVALID_FROM_REG = 1, default value 0 - AVALID_FROM_PHY
22	AVALID_SW	RW	A-Device Session Valid Indicator, default value 0
21	IDDIG_SW_SEL	RW	IDDIG_FROM_PHY = 0, IDDIG_FROM_REG = 1, default value 1 - IDDIG value is determined by register value

USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_GEN_CONFIG_2 (cont.)

Bits	Name	Type	Description
20	IDDIG_SW	RW	HOST = 0, DEVICE = 1, default value 1 - Device
19	ULPI_TX_PKT_EN_CLR_FI X	RW	fix to QCTDD00284866 - Race between Port Reset start and SOF transmission in HSIC Host, default value 1 - enable
18	TRANSACTOR_NONPOST_WRITE_DIS	RW	chicken bit for QCTDD00257652 - This feature make the usb state machines to wait till the transactor finishes to process the transfers of specific descriptors. When bit is 1 this feature is disabled, default value 0 - enable
17	FPR_CTRL	RW	chicken bit that enables SW to clear the FPR bit in PORTSC register in Host mode instead of fixed Host Resume duration of 21 ms. This bit should be set to 1 to enable SW control of FPR. When bit is 0 then Host Resume is fixed 21 ms. Default value is 0 for backward compatibility.
16	SW_RESET_ALT_EN	RW	
15	DPSE_DMSE_HV_INTR_EN	RW	chicken bit that enables hv dpse/dmse interrupts to use usb low power mode wakeup mechanism. Default value is 1 - enable use of hv dpse/dmse interrupts.
14	USB_HW_SW_EVENTS_TO G_EN	RW	chicken bit that enables toggling of usb_otg_hs_sw_events and usb_otg_hs_hw_events buses. Default value is 1 - toggling enabled.
13	SYS_CLK_HOST_DEV_GATE_EN	RW	Chicken bit for the internal host/device clock gating. when this bit is set when usb operates in host mode the device logic will be turned off and vice versa, when this bit is cleared it disables internal system clock gating. Default value is 1 - internal clock gating is enabled.
12	LINESTATE_DIFF_WAKEUP_EN	RW	chicken bit for CR-0000155486. when this bit is set the USB enables the HW fix for Race condition between the attempt to enter LPM and USB bus reset. Default value is 1.
11	ULPI_LPM_PEND_EN	RW	chicken bit for CR-0000153486. when this bit is set the USB enables the HW fix for Function Control and LPM race condition. Default value is 1.
10	RX_FULL_NAK_EN	RW	chicken bit for CR-0000152878. when this bit is set the USB will respond with NAK's for Host Tokens with very slow AHB and when Streaming mode is enabled. Default value is 1.
9	ENDLESS_TD_EN	RW	chicken bit for CR-0000152976. When this bit is set Performance enhancements for 'infinite' Producer pipe (out endpoint with eTD that points to itself) are enabled. Default value is 1.
8	SCRATCH_RAM_EN	RW	chicken bit for CR-0000149922. When this bit is set the use of scratch ram is enabled and the SW can read/write from addresses 0x040 - 0x07c. when this bit is clear SW can no longer access those registers. Default value is 1.

USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_GEN_CONFIG_2 (cont.)

Bits	Name	Type	Description
7	SESS_VLD_CTRL_EN	RW	When this bit is set then bit 25 of USBCMD register controls sess_vld signal inside the Link Controller. When this bit is clear then Link Controller receives sess_vld directly from PHY. Default value is 0.
6	CI_T_WTSUSRSTHS_EN	RW	When this bit is 0 then Device Port Control State Machine waits 2.5 us from USB Reset detection until starting driving Chirp K. When this bit is 1 then Device Port Control State Machine waits 1.5 ms from USB Reset detection until starting driving Chirp K. Default value is 1 - legacy behavior.
5	CI_T_UCH_EN	RW	When this bit is 0 then Device Port Control State Machine drives Chirp K for 1 ms. When this bit is 1 then Device Port Control State Machine drives Chirp K for 2ms. Default value is 1 - legacy behavior.
4	DP_RESET	RW	chicken bit for fix CI2687: When the OTG core is acting as a Host, and VBUS is turned off, and the attached Device attempts to perform a Session Request Protocol by using Data-line Pulsing, it will not be recognized by the Host. Also, when doing HNP and becoming a Host, a SE0 is forced in the line causing the OPT TD5.4 test to fail, without the software workaround.
3	ZLP_PRIME	RW	chicken bit for fix CI2655: When using ISO IN endpoints with MULT=3 and low bandwidth system bus access, the controller may enter into a wait loop situation without warning the software. Due to the low bandwidth the last packet from a mult3 sequence may not be fetched in time before the last token IN is received (for that frame/endpoint). This will cause the controller to reply with a zero length packet (ZLP), breaking the prime sequence.
2	NO_SOF_RX_FIFO_FULL	RW	chicken bit for fix CI2581: During normal operation, if the RX Fifo becomes full and the protocol engine needs to send a command to the DMA state machine, it will wait in that state until the RX Fifo becomes not full. As the protocol state machine also handles the SOF generation, the Soft will no longer be sent. If one SOF is missed, the Host controller will issue a false babble detection. If more than 3.125ms are elapsed without Soft the peripheral will recognize the idle bus as a USB reset.
1	WRONG_OPMODE_SUSP	RW	chicken bit for fix CI1274: When the Controller enters a Suspend state it asserts pomade with the wrong value, This causes no issue in actual usage.
0	RESUME_END_INTER	RW	chicken bit for fix CI1179: Working as host, when doing resume a port change interrupt was fired at the end of resume. According to the EHCI spec no interrupt should be fired.

0x078D9100 USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_CAPLENGTH**Type:** R**Clock:** CC_USB_OTG_HS_CLK**Reset State:** 0x01000040

The USB_OTG_HS_CAPLENGTH register is the capability register length. It is used to indicate which offset to add to the register base address at the beginning of the operational register.

USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_CAPLENGTH

Bits	Name	Description
31:16	HCIVERSION	BCD encoding of the EHCI revision number supported by this host controller. The most significant byte of this register represents a major revision and the least significant byte is the minor revision.
7:0	CAPLENGTH	Offset at beginning of operational register

0x078D9104 USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_HCSPARAMS**Type:** R**Clock:** CC_USB_OTG_HS_CLK**Reset State:** 0x00010011

The USB_OTG_HS_HCSPARAMS register contains the host control structural parameters. The port steering logic capabilities are described in this register.

USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_HCSPARAMS

Bits	Name	Description
31:28	RESERVED_BITS31_28	Clear (0) these bits.
27:24	N_TT	Number of transaction translators This field indicates the number of embedded transaction translators associated with the USB2.0 host controller. For a multi-port host, this field will always equal 0001. For all other implementations, N_TT = 0000. This in a non-EHCI field to support embedded TT.
23:20	N_PTT	Number of ports per transaction translator This field indicates the number of ports assigned to each transaction translator within the USB2.0 host controller. For a multi-port host this field will always equal N_PORTS. For all other implementations, N_PTT = 0000. This in a non-EHCI field to support embedded TT.
19:17	RESERVED_BITS19_17	Clear (0) these bits.

USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_HCSPARAMS (cont.)

Bits	Name	Description
16	PI	Port indicator This bit indicates whether the ports support port indicator control. When set (1), the port status and control registers include a read/writable field for controlling the state of the port indicator. This field will always be set (1).
15:12	N_CC	Number of companion controllers This field indicates the number of companion controllers associated with this USB2.0 host controller. A zero in this field indicates there are no internal companion controllers. Port-ownership hand-off is not supported. A value larger than zero in this field indicates that there are companion USB1.1 host controller(s). Port-ownership hand-off are supported. High-, full-, and low-speed devices are supported on the host controller root ports. In this implementation, this field will always be clear (0).
11:8	N_PCC	Number of ports per companion controller This field indicates the number of ports supported per internal companion controller. It is used to indicate the port routing configuration to the system software. For example, if N_PORTS has a value of 6 and N_CC has a value of 2, then N_PCC could have a value of 3. The convention is that the first N_PCC ports are assumed to be routed to companion controller 1, the next N_PCC ports to companion controller 2, and so on. In the previous example, the N_PCC could have been 4, where the first 4 are routed to companion controller 1 and the last two are routed to companion controller 2. The number in this field must be consistent with N_PORTS and N_CC. In this implementation, this field will always be clear (0).
7:5	RESERVED_BITS7_5	Clear (0) these bits.
4	PPC	Port power control This field indicates whether the host controller implementation includes port power control. Set (1) indicates that the ports have port power switches. Clear (0) indicates that the ports do not have port power switches. The value of this field affects the functionality of the port power field in each port status and control register. This bit will always be clear (0) for a device only implementation.
3:0	N_PORTS	Number of downstream ports This field specifies the number of physical downstream ports implemented on this host controller. The value of this field determines how many port registers are addressable in the operational register. Valid values are in the range of 1h to 0h. A zero in this field is undefined. The number of ports for a host implementation is configured from 1 to 8. This field will always be set (1) for device-only implementation.

0x078D9108 USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_HCCPARAMS**Type:** R**Clock:** CC_USB_OTG_HS_CLK**Reset State:** 0x00000006

The USB_OTG_HS_HCCPARAMS register contains the host control capability parameters. This register identifies multiple mode control (time-base bit functionality) addressing capability.

USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_HCCPARAMS

Bits	Name	Description
31:16	RESERVED_BITS31_16	Clear (0) these bits.
15:8	EECP	EHCI extended capabilities pointer Default = 0 This optional field indicates the existence of a capabilities list. A value of 00h indicates no extended capabilities are implemented. A non-zero value in this register indicates the offset in PCI configuration space of the first EHCI extended capability. The pointer value must be 40h or greater if implemented to maintain the consistency of the PCI header defined for this class of device. For this implementation, all of these bits are clear (0).
7:4	IST	Isochronous scheduling threshold Default = implementation dependent This field indicates, relative to the current position of the executing host controller, where the software can reliably update the isochronous schedule. When bit [7] is clear (0), the value of the least significant 3 bits indicates the number of microframes a host controller can hold a set of isochronous data structures (one or more) before flushing the state. When bit [7] is set (1), then host software assumes the host controller may cache an isochronous data structure for an entire frame. All of the bits in this field will always be clear (0).
3	RESERVED_BIT3	Clear (0) this bit.
2	ASP	Asynchronous schedule park capability Default = 1 If this bit is set (1), then the host controller supports the park feature for high-speed queue heads in the asynchronous schedule. The feature can be disabled, or enabled and set to a specific level by using the (ASPE) and (ASP[1:0]) fields in the USB_OTG_HS_USBCMD register. This field will always be set (1).

USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_HCCPARAMS (cont.)

Bits	Name	Description
1	PFL	Programmable frame list flag If this bit is clear (0), then the system software must use a frame list length of 1024 elements with this host controller. The FS[2:0] field in the USBCMD register is read-only and must be cleared (0). If this bit is set (1), then the system software can specify and use a smaller frame list, and configure the host controller using the FS[2:0] field in the USBCMD register. The frame list must always be aligned on a 4k-page boundary. This requirement ensures that the frame list is always physically contiguous. This bit in this field will always be set (1).
0	ADC	64-bit addressing capability This field will always be clear (0). No 64-bit addressing capability is supported.

0x078D9120 USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_DCIVERSION**Type:** R**Clock:** CC_USB_OTG_HS_CLK**Reset State:** 0x00000001

The USB_OTG_HS_DCIVERSION register contains the device interface version number. The device controller interface conforms to the two-byte BCD encoding of the interface version number contained in this register.

USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_DCIVERSION

Bits	Name	Description
15:0	DCIVERSION	Device interface version number

0x078D9124 USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_DCCPARAMS**Type:** R**Clock:** CC_USB_OTG_HS_CLK**Reset State:** 0x00000190

The USB_OTG_HS_DCCPARAMS register contains the device control capability parameters. These fields describe the overall host/device capability of the controller.

USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_DCCPARAMS

Bits	Name	Description
31:9	RESERVED_BITS31_9	Clear (0) these bits.

USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_DCCPARAMS (cont.)

Bits	Name	Description
8	HC	Host capable When this bit is set (1), this controller is capable of operating as an EHCI-compatible USB 2.0 host controller.
7	DC	Device capable When this bit is set (1), this controller is capable of operating as a USB 2.0 device.
6:5	RESERVED_BITS6_5	Clear (0) these bits.
4:0	DEN	Device endpoint number This field indicates the number of endpoints build into the device controller. If this controller is not device capable, then this field will be all zeroes. Valid values for this field are 0 through 16.

0x078D9140 USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_USBCMD**Type:** RW**Clock:** CC_USB_OTG_HS_CLK**Reset State:** 0x00080000

The USB_OTG_HS_USBCMD register is the USB command register. The serial bus host/device controller executes the command indicated in this register.

USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_USBCMD

Bits	Name	Description
31	RST_CTRL	Default value = 0. Set to 1 to block operational reset to xcvr (ser and ulpi) clock domains.
30	ULPI_STP_CTRL	Default value = 0. Set to 1 to block the ulpi_stp signal from going out to ULPI PHY
29	ASYNC_INTR_CTRL	Default value = 0. Set to 1 to allow the async interrupt out from the HS core.
28	SE0_GLITCH_FIX_CTRL	Default value = 0. Set to 1 to activate the SE0 glitch fix mechanism
27	FS_3_WIRE_2_WIRE_SELECT	Default value = 0. Set this bit for enabling the two wire interface on the fs_dat and fs_se0 pins
26	ULPI_SER3_NOT6_SEL	ULPI serial 3 bits select. Read/write Read: Current status of serial data bus wide Write: SW writes '1' to this bit to request 3 pins ULPI data wide, or '0' to request 6 bit data wide in FsLsSerial Mode.
25	SESS_VLD_CTRL	Default value = 0. Set this bit to enable Link Controller operation after switching interface from Serial to ULPI.
24	RESERVED_BIT24	Clear (0) these bit

USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_USBCMD (cont.)

Bits	Name	Description
23:16	ITC	Interrupt threshold control Read/write Default 08h The system software uses this field to set the maximum rate at which the host/device controller will issue interrupts. This field contains the maximum interrupt interval measured in micro-frames. Valid values are shown below. Value Maximum interrupt interval 00h Immediate (no threshold) 01h 1 micro-frame 02h 2 micro-frames 04h 4 micro-frames 08h 8 micro-frames 10h 16 micro-frames 20h 32 micro-frames 40h 64 micro-frames
15	FS2	This field is Read/Write only if Programmable Frame List Flag in the HCCPARAMS registers is set to one. This field specifies the size of the frame list that controls which bits in the Frame Index Register should be used for the Frame List Current index. Note that this field is made up from USBCMD bits 15, 3 and 2. Values meaning 000 1024 elements (4096 bytes) Default value 001 512 elements (2048 bytes) 010 256 elements (1024 bytes) 011 128 elements (512 bytes) 100 64 elements (256 bytes) 101 32 elements (128 bytes) 110 16 elements (64 bytes) 111 8 elements (32 bytes) Only the host controller uses this field.
14	ATDTW	Add dTD tripwire Read/write (Device mode only) This bit is used as a semaphore to ensure the to proper addition of a new dTD to an active (primed) endpoint's linked list. This bit is set (1) and cleared (0) by the software. This bit shall also be cleared (0) by the hardware when the state machine is a hazard region for which adding a dTD to a primed endpoint may go unrecognized. For more information on the use of this bit, see the Device Operational Model section of the USB-HS DEV reference manual.

USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_USBCMD (cont.)

Bits	Name	Description
13	SUTW	Setup tripwire (device mode only) Read/write This bit is used as a semaphore to ensure that the setup data payload of 8 bytes is extracted from a QH by the DCD without being corrupted. If the setup lockout mode is off (see USBMODE) then there exists a hazard when new setup data arrives while the DCD is copying the setup data payload from the QH for a previous setup packet. This bit is set (1) and cleared (0) by software, and will be cleared (0) by hardware when a hazard exists. For more information on the use of this bit, see the Device Operational Model section of the USB-HS DEV reference manual.
12	RESERVED_BITS12	Clear (0) this bit
11	ASPE	Asynchronous schedule park mode enable (OPTIONAL) Read/write If the asynchronous park capability (ASP) bit in the HCCPARAMS register is set (1), then this bit defaults to a 1h and is R/W. Otherwise, the bit must be cleared (0) and is RO. The software uses this bit to enable or disable the park mode. value 1 = Park mode is enabled. value 0 = Park mode is disabled. This field is set (1) in this implementation.
10	RESERVED_BIT10	Clear (0) this bit.
9:8	ASP	Asynchronous schedule park mode count (OPTIONAL) Read/write If the Asynchronous park capability (ASP) bit in the HCCPARAMS register is set (1), then this field defaults to 3h and is R/W. Otherwise it defaults to zero and is RO. This field contains a count of the number of successive transactions the host controller is allowed to execute from a high-speed queue head on the asynchronous schedule before continuing traversal of the Asynchronous schedule. See Section 4.10.3.2 for full operational details. Valid values are 1h to 3h. The software must not clear (0) this bit when the ASPE bit in this register is set (1), as this will result in undefined behavior. This field is set to 3h in this implementation.
7	LR	Light host/device controller reset (OPTIONAL) Read only Not implemented. This bit will always be clear (0).

USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_USBCMD (cont.)

Bits	Name	Description
6	IAA	Interrupt on async advance doorbell Read/write This bit is used as a doorbell by the software to tell the host controller to issue an interrupt the next time it advances the asynchronous schedule. The software must set (1) this bit to ring the doorbell. When the host controller has evicted all appropriate cached schedule states, it sets (1) the interrupt on the async advance status (AAI) bit in the USBSTS register. If the interrupt on async advance enable (AAE) bit in the USBINTR register is set (1), then the host controller will assert an interrupt at the next interrupt threshold. The host controller clears (0) this bit after it has set (1) the interrupt on async advance status (AAI) bit in the USBSTS register. The software should not set (1) this bit when the asynchronous schedule is inactive. Doing so will yield undefined results. This bit is only used in the host mode. Setting (1) this bit when the device mode is selected will produce undefined results.
5	ASE	Asynchronous Schedule Enable Read/write Default = 0b This bit controls whether the host controller skips processing the asynchronous schedule. value 0 = Do not process the asynchronous schedule value 1 = Use the ASYNCLISTADDR register to access the asynchronous schedule. Only the host controller uses this bit.
4	PSE	Periodic schedule enable Read/write Default 0b This bit controls whether the host controller skips processing the periodic schedule. value 0 = Do not process the periodic schedule value 1 = Use the PERIODICLISTBASE register to access the periodic schedule. Only the host controller uses this bit.

USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_USBCMD (cont.)

Bits	Name	Description
3:2	FS	This field is Read/Write only if Programmable Frame List Flag in the HCCPARAMS registers is set to one. This field specifies the size of the frame list that controls which bits in the Frame Index Register should be used for the Frame List Current index. Note that this field is made up from USBCMD bits 15, 3 and 2. Values meaning 000 1024 elements (4096 bytes) Default value 001 512 elements (2048 bytes) 010 256 elements (1024 bytes) 011 128 elements (512 bytes) 100 64 elements (256 bytes) 101 32 elements (128 bytes) 110 16 elements (64 bytes) 111 8 elements (32 bytes) Only the host controller uses this field.
1	RST	Controller reset (RESET) Read/write The software uses this bit to reset the controller. This bit is cleared (0) by the host/device controller when the reset process is complete. The software cannot terminate the reset process early by clearing (0) this bit. Host controller: When the software sets (1) this bit, the host controller resets its internal pipelines, timers, counters, state machines, and so on to their initial values. Any transaction currently in progress on the USB is immediately terminated. A USB reset is not driven on downstream ports. The software should not set (1) this bit when the HCHalted bit in the USBSTS register is clear (0). Attempting to reset an actively running host controller will result in undefined behavior. Device controller: When the software sets (1) this bit, the device controller resets its internal pipelines, timers, counters, state machines, and so on to their initial values. Setting this bit when the device is in the attached state is not recommended, since the effect on an attached host is undefined. In order to ensure that the device is not in an attached state before initiating a device controller reset, all primed endpoints should be flushed and the USBCMD run/stop bit should be cleared (0).

USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_USBCMD (cont.)

Bits	Name	Description
0	RS	Run/stop Read/Write Default 0b value 1 = Run value 0 = Stop Host controller: When this bit is set (1), the host controller proceeds with the execution of the schedule. The host controller continues execution as long as this bit remains set (1). When this bit is clear (0), the host controller completes the current transaction on the USB and then halts. The HC halted bit in the status register indicates when the host controller has finished the transaction and has entered the stopped state. The software should not set (1) this bit unless the host controller is in the halted state (that is, the HCHalted bit in the USBSTS register is set (1)). Device controller: Setting (1) this bit will cause the device controller to enable a pull-up on D+ and initiate an attach event. This control bit is not directly connected to the pull-up enable, as the pull-up will become disabled upon transitioning into high-speed mode. The software should use this bit to prevent an attach event before the device controller has been properly initialized. Clearing (0) this bit will cause a detach event.

0x078D9144 USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_USBSTS**Type:** RW**Clock:** CC_USB_OTG_HS_CLK**Reset State:** 0x00000000

The USB_OTG_HS_USBSTS register is the USB status register. This register indicates various states of the host/device controller and any pending interrupts. This register does not indicate status resulting from a transaction on the serial bus. The software clears certain bits in this register by setting (1) them.

USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_USBSTS

Bits	Name	Description
31	ULPI_INTR	Default value = 0. This bit is set when Interrupt during ULPI -Serial mode or ULPI Interrupt during LPM occurs. Writing a 1 to this bit will clear it.
30	PHY_SESS_VLD_CHG	This bit is set when PHY_SESS_VLD bit changes its value.
29	PHY_SESS_VLD	This bit presents the SESS_VLD status of PHY.

USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_USBSTS (cont.)

Bits	Name	Description
28	PHY_ALT_INT	This bit is asserted when a non-USB interrupt from PHY is detected. This interrupt is used for procedures like Battery Charging. This bit is set when bit 7 (alt_int) of RX CMD is high. Writing a 1 to this bit will clear it.
27	ASYNC_WAKEUP_INTR	This bit is asserted when async wakeup interrupt is detected. Writing a 1 to this bit will clear it.
26	RESERVED_BIT26	Clear (0) this bit.
25	TI1	General purpose timer interrupt 1 (GPTINT1) Read/write control This bit is set (1) when the counter in the GPTIMER1CTRL (non-EHCI) register transitions to zero. Setting (1) this bit will clear it.
24	TI0	General purpose timer interrupt 0 (GPTINT0) Read/write control This bit is set (1) when the counter in the GPTIMER0CTRL (non-EHCI) register transitions to zero. Setting (1) this bit will clear it.
23:20	RESERVED_BITS23_20	Clear (0) these bits.
19	UPI	USB host periodic interrupt (USBHSTPERINT) Read/write control This bit is set (1) by the host controller when the cause of an interrupt is a completion of a USB transaction, where the transfer descriptor (TD) has an interrupt on complete (IOC) bit set (1) and the TD was from the periodic schedule. This bit is also set (1) by the host controller when a short packet is detected AND the packet is on the periodic schedule. A short packet is when the actual number of bytes received was less than the expected number of bytes. This bit is not used by the device controller and will always be clear (0).
18	UAI	USB host asynchronous interrupt (USBHSTASYNCINT) Read/write control This bit is set (1) by the host controller when the cause of an interrupt is a completion of a USB transaction, where the transfer descriptor (TD) has an interrupt on complete (IOC) bit set AND the TD was from the asynchronous schedule. This bit is also set (1) by the host when a short packet is detected AND the packet is on the asynchronous schedule. A short packet is when the actual number of bytes received was less than the expected number of bytes. This bit is not used by the device controller and will always be clear (0).
17	RESERVED_BIT17	Clear (0) these bits.

USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_USBSTS (cont.)

Bits	Name	Description
16	NAKI	NAK interrupt bit Read only This bit is set (1) by the hardware when, for a particular endpoint, both the Tx/Rx endpoint NAK bit and the corresponding Tx/Rx endpoint NAK enable bit are set (1). This bit is automatically cleared (0) by the hardware when all of the enabled Tx/Rx endpoint NAK bits are cleared (0).
15	AS	Asynchronous schedule status Read only Default = 0 This bit reports the current real status of the asynchronous schedule. When cleared (0), the asynchronous schedule status is disabled. And, if set (1), the status is enabled. The host controller is not required to immediately disable or enable the asynchronous schedule when software transitions the asynchronous schedule enable bit in the USBCMD register. When this bit and the asynchronous schedule enable bit are the same value, the asynchronous schedule is either enabled (1) or disabled (0). This bit is only used by the host controller.
14	PS	Periodic schedule status Read only Default = 0 This bit reports the current real status of the periodic schedule. When cleared (0), the periodic schedule is disabled. And, if set (1), the status is enabled. The host controller is not required to immediately disable or enable the periodic schedule when software transitions the periodic schedule enable bit in the USBCMD register. When this bit and the periodic schedule enable bit are the same value, the periodic schedule is either enabled (1) or disabled (0). This bit is only used by the host controller.
13	RCL	Reclamation Read only Default = 0 This is a read-only status bit that is used to detect an empty asynchronous schedule. This bit is only used by the host controller.
12	HCH	HC halted Read only Default = 1 This bit is a clear (0) whenever the run/stop bit is set (1). The host controller sets (1) this bit after it has stopped executing, because of the run/stop bit being cleared (0), either by the software or by the host controller hardware (for example, an internal error). This bit is only used by the host controller.
11	RESERVED_BIT11	Clear (0) this bit.

USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_USBSTS (cont.)

Bits	Name	Description
10	ULPII	ULPI interrupt Read/write control Default = 0 When the ULPI viewport is present in the design, an event completion will set (1) this interrupt. This bit is used by both the host and device controllers. It is only present in designs where the configuration constant VUSB_HS_PHY_ULPI = 1.
9	RESERVED_BIT9	Clear (0) this bit.
8	SLI	DC suspend Read/write control Default = 0 When a device controller enters a suspend state from an active state, this bit will be set (1). The device controller clears (0) the bit upon exiting from a suspend state. This bit is only used by the device controller.
7	SRI	SOF received Read/write control Default = 0 When the device controller detects a start of (micro) frame, this bit will be set (1). When a SOF is extremely late, the device controller will automatically set (1) this bit to indicate that an SOF was expected. Therefore, this bit will be set (1) roughly every 1 ms in the device FS mode and every 125 ms in the HS mode, and will be synchronized to the actual SOF that is received. Since the device controller is initialized to FS before connect, this bit will be set (1) at an interval of 1 ms during the prelude to connect and chirp. In the host mode, this bit will be set (1) every 125 us and can be used by the host controller driver as a time base. The software writes a 1 to this bit to clear it. This is a non-EHCI status bit.
6	URI	USB reset received Read/write control Default = 0 When the device controller detects a USB reset and enters the default state, this bit will be set (1). The software can set (1) this bit to clear the USB reset received status bit. This bit is only used by the device controller.

USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_USBSTS (cont.)

Bits	Name	Description
5	AAI	Interrupt on async advance Read/write control Default = 0 The system software can force the host controller to issue an interrupt the next time the host controller advances the asynchronous schedule by setting (1) the interrupt on async advance doorbell bit in the USBCMD register. This status bit indicates the assertion of that interrupt source. This bit is only used by the host controller.
4	SEI	System error Read/write control This interrupt is triggered when there is an AHB error (HRESP = ERROR) on the AHB Master interface.
3	FRI	Frame list rollover Read/write control The host controller sets (1) this bit when the frame list index rolls over from its maximum value to zero. The exact value at which the rollover occurs depends on the frame list size. For example, if the frame list size (as programmed in the frame list size field of the USBCMD register) is 1024, the frame index register rolls over every time FRINDEX [1:3] toggles. Similarly, if the size is 512, the host controller sets (1) this bit to a one every time FHINDEX [12] toggles. This bit is only used by the host controller.
2	PCI	Port change detect Read/write control The host controller sets (1) this bit when a connect status occurs on any port, a port enable/disable change occurs on any port, or the force port resume bit is set (1) as the result of a J-K transition on the suspended port. The device controller sets (1) this bit when the port controller enters the full or high-speed operational state. When the port controller exits the full or high-speed operation states due to reset or suspend events, the notification mechanisms are the USB reset received bit and the DC suspend bits, respectively. This bit is not EHCI compatible.
1	UEI	USB error interrupt (USBERRINT) Read/write control When completion of a USB transaction results in an error condition, this bit is set (1) by the host/device controller. This bit is set (1) along with the USBINT bit, if the TD on which the error interrupt occurred also had its interrupt-on-complete (IOC) bit set (1). See Section (Reference Host Operation Model: Transfer/Transaction Based Interrupt ? i.e. 4.15.1 in EHCI [5]) for a complete list of host error interrupt conditions. See section Device Error Matrix in the USB-HS DEV reference manual. The device controller detects resume signaling only.

USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_USBSTS (cont.)

Bits	Name	Description
0	UI	USB interrupt (USBINT) Read/write control This bit is set (1) by the host/device controller when the cause of an interrupt is a completion of a USB transaction where the transfer descriptor (TD) has an interrupt-on-complete (IOC) bit set (1). This bit is also set (1) by the host/device controller when a short packet is detected. A short packet is when the actual number of bytes received was less than the expected number of bytes.

0x078D9148 USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_USBINTR**Type:** RW**Clock:** CC_USB_OTG_HS_CLK**Reset State:** 0x00000000

The USB_OTG_HS_USBINTR register is the USB interrupt enable register. The interrupts to the software are enabled with this register. An interrupt is generated when a bit is set (1) and the corresponding interrupt is active. The USB status register (USBSTS) still shows interrupt sources, even if they are disabled by the USBINTR register, which allows polling of interrupt events by the software.

USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_USBINTR

Bits	Name	Description
31	ULPI_INTR_EN	Default value =0. When this bit is a 1 and ULPI_INTR is a 1, the controller will issue an interrupt. The interrupt is acknowledged by software clearing the ULPI_INTR bit.
30	PHY_SESS_VLD_CHG_EN	Default value =0. When this bit is set then Link Controller will issue an interrupt when PHY_SESS_VLD changes its value.
29:26	RESERVED_BITS29_26	Clear (0) these bits.
25	TIE1	General purpose timer interrupt enable 1 When this bit is set (1), and the GPTINT1 bit in the USBSTS register is set (1), the controller will issue an interrupt. The interrupt is acknowledged by the software clearing (0) the GPTINT1 bit.
24	TIE0	General purpose timer interrupt enable 0 When this bit is set (1), and the GPTINT0 bit in the USBSTS register is set (1), the controller will issue an interrupt. The interrupt is acknowledged by the software clearing (0) the GPTINT0 bit.
23:20	RESERVED_BITS23_20	Clear (0) these bits.

USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_USBINTR (cont.)

Bits	Name	Description
19	UPIE	USB host periodic interrupt enable When this bit is set (1), and the USBHSTPERINT bit in the USBSTS register is set (1), the host controller will issue an interrupt at the next interrupt threshold. The interrupt is acknowledged by the software clearing (0) the USBHSTPERINT bit.
18	UAIE	USB host asynchronous interrupt enable When this bit is set (1), and the USBHSTASYNCINT bit in the USBSTS register is set (1), the host controller will issue an interrupt at the next interrupt threshold. The interrupt is acknowledged by the software clearing (0) the USBHSTASYNCINT bit.
17	RESERVED_BIT17	Clear (0) this bit.
16	NAKE	NAK interrupt enable This bit is set (1) by the software if it wants to enable the hardware interrupt for the NAK interrupt bit. If both this bit and the corresponding NAK interrupt bit are set (1), a hardware interrupt is generated.
15:11	RESERVED_BITS15_11	Clear (0) these bits.
10	ULPIE	ULPI enable When this bit is set (1), and the ULPI Interrupt bit in the USBSTS register transitions, the controller will issue an interrupt. The interrupt is acknowledged by the software setting (1) the ULPI interrupt bit. This bit is used by both the host and device controllers. It is only present in designs where configuration constant VUSB_HS_PHY_ULPI = 1.
9	RESERVED_BIT9	Clear (0) this bit.
8	SLE	Sleep enable When this bit is set (1) and the DC suspend bit in the USBSTS register transitions, the device controller will issue an interrupt. The interrupt is acknowledged by the software setting (1) the DC suspend bit. This bit is only used by the device controller.
7	SRE	SOF received enable When this bit is set (1) and the SOF received bit in the USBSTS register is set (1), the device controller will issue an interrupt. The interrupt is acknowledged by the software clearing (0) the SOF received bit.
6	URE	USB reset enable When this bit is set (1) and the USB reset received bit in the USBSTS register is set (1), the device controller will issue an interrupt. The interrupt is acknowledged by the software clearing (0) the USB reset received bit.

USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_USBINTR (cont.)

Bits	Name	Description
5	AAE	Interrupt on async advance enable When this bit is set (1) and the interrupt on async advance bit in the USBSTS register is set (1), the host controller will issue an interrupt at the next interrupt threshold. The interrupt is acknowledged by the software clearing (0) the interrupt on async advance bit. This bit is only used by the host controller.
4	SEE	System error enable When this bit is set (1) and the system error bit in the USBSTS register is set (1), the host/device controller will issue an interrupt. The interrupt is acknowledged by the software clearing (0) the system error bit.
3	FRE	Frame list rollover enable When this bit is set (1) and the frame list rollover bit in the USBSTS register is set (1), the host controller will issue an interrupt. The interrupt is acknowledged by the software clearing (0) the frame list rollover bit. This bit is only used by the host controller.
2	PCE	Port change detect enable When this bit is set (1) and the port change detect bit in the USBSTS register is set (1), the host/device controller will issue an interrupt. The interrupt is acknowledged by the software clearing (0) the port change detect bit.
1	UEE	USB error interrupt enable When this bit is set (1) and the USBERRINT bit in the USBSTS register is set (1), the host controller will issue an interrupt at the next interrupt threshold. The interrupt is acknowledged by the software clearing (0) the USBERRINT bit in the USBSTS register.
0	UE	USB interrupt enable When this bit is set (1) and the USBINT bit in the USBSTS register is set (1), the host/device controller will issue an interrupt at the next interrupt threshold. The interrupt is acknowledged by the software clearing (0) the USBINT bit.

0x078D914C USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_FRINDEX**Type:** RW**Clock:** CC_USB_OTG_HS_CLK**Reset State:** 0x00000000

The USB_OTG_HS_FRINDEX register is the USB frame index register. This register is used by the host controller to index the periodic frame list. The register updates every 125 microseconds (once each micro-frame). Bits [N:3] are used to select a particular entry in the periodic frame list during periodic schedule execution. The number of bits used for the index depends on the size of the frame list as set by the system software in the frame list size field in the USBCMD register.

This register must be written as a DWord. Byte writes produce undefined results. This register cannot be written unless the host controller is in the 'halted' state, as indicated by the HC halted bit. A write to this register while the run/stop bit is set (1) produces undefined results. Writes to this register also affect the SOF value.

In the device mode, this register is read-only and the device controller updates the FRINDEX [13:3] register from the frame number indicated by the SOF marker. Whenever a SOF is received by the USB bus, FRINDEX [13:3] will be checked against the SOF marker. If FRINDEX [13:3] is different from the SOF marker, FRINDEX [13:3] will be set to the SOF value and FRINDEX [2:0] will be cleared (0) (that is, SOF for a 1-ms frame). If FRINDEX [13:3] is equal to the SOF value, FRINDEX [2:0] will be incremented (that is, SOF for 125-ms micro-frame).

USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_FRINDEX

Bits	Name	Description
31:14	RESERVED_BITS31_14	Clear (0) these bits.
13:0	FRINDEX	Frame index The value in this register increments at the end of each time frame (for example, a micro-frame). Bits [N:3] are used for the frame list current index. This means that each location of the frame list is accessed 8 times (frames or micro-frames) before moving to the next index. The following data shows the values of N based on the value of the frame list size field in the USBCMD register when used in host mode. USBCMD [Frame list size] number Elements N 000b (1024) 12 001b (512) 11 010b (256) 10 011b (128) 9 100b (64) 8 101b (32) 7 110b (16) 6 111b (8) 5 In the device mode, the value is the current frame number of the last frame transmitted. It is not used as an index. In either mode, bits 2:0 indicate the current microframe.

0x078D9154 USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_PERIODICLISTBASE

Type: RW

Clock: CC_USB_OTG_HS_CLK

Reset State: 0x00000000

BANKED=host

The USB_OTG_HS_PERIODICLISTBASE register is the periodic list base address register. This 32-bit register contains the beginning address of the periodic frame list in the system memory. The HCD loads this register prior to starting the schedule execution by the host controller. The memory

structure referenced by this physical memory pointer is assumed to be 4-Kbyte aligned. The contents of this register are combined with the frame index register (FRINDEX) to enable the host controller to step through the periodic frame list in sequence.

NOTE This device is shared between the host controller and device controller operation. For host controller operation, this is the USB_OTG_HS_PERIODICLISTBASE register. For device controller operation, this is the USB_OTG_HS_DEVICEADDR register.

USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_PERIODICLISTBASE

Bits	Name	Description
31:12	PERBASE	Base address (low) These bits correspond to memory address signals [31:12], respectively. Only used by the host controller.
11:0	RESERVED_BITS11_0	This field must be written as zeros. During runtime, the values of these bits are undefined.

0x078D9154 USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_DEVICEADDR

Type: RW

Clock: CC_USB_OTG_HS_CLK

Reset State: 0x00000000

BANKED=device

The USB_OTG_HS_DEVICEADDR register is the USB device address register. The upper seven bits of this register represent the device address. After any controller reset or a USB reset, the device address is set to the default address (0). The default address will match all incoming addresses. The software shall reprogram the address after receiving a SET_ADDRESS descriptor.

The USBADRA is used to accelerate the SET_ADDRESS sequence by allowing the DCD to preset the USBADR register before the status phase of the SET_ADDRESS descriptor.

NOTE This device is shared between the host controller and device controller operations. For host controller operation, this is the USB_OTG_HS_PERIODICLISTBASE register. For device controller operation, this is the USB_OTG_HS_DEVICEADDR register.

USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_DEVICEADDR

Bits	Name	Description
31:25	USBADR	Device address These bits correspond to the USB device address.

USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_DEVICEADDR (cont.)

Bits	Name	Description
24	USBADRA	Device address advance Default = 0 When this bit is clear (0), any writes to USBADR are instantaneous. When this bit is set (1) at the same time or before USBADR is written, the write to the USBADR field is staged and held in a hidden register. After an IN occurs on endpoint 0 and is ACKed, USBADR will be loaded from the holding register. The hardware will automatically clear (0) this bit on the following conditions: 1. IN is ACKed to endpoint 0 (USBADR is updated from staging register). 2. OUT/SETUP occur to endpoint 0 (USBADR is not updated). 3. Device reset occurs (USBADR is reset to 0). Note: After the status phase of the SET_ADDRESS descriptor, the DCD has 2 ms to program the USBADR field. This mechanism will ensure this specification is met when the DCD can not write of the device address within 2 ms from the SET_ADDRESS status phase. If the DCD writes the USBADR with USBADRA = 1 after the SET_ADDRESS data phase (before the prime of the status phase), the USBADR will be programmed instantly at the correct time and meet the 2-ms USB requirement.
23:0	RESERVED_BITS23_0	These bits must be written as zeros. During runtime, the values of these bits are undefined.

0x078D9158 USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_ASYNCLISTADDR**Type:** RW**Clock:** CC_USB_OTG_HS_CLK**Reset State:** 0x00000000

BANKED=host

The USB_OTG_HS_ASYNCLISTADDR register is the next asynchronous list address register. This 32-bit register contains the address of the next asynchronous queue head to be executed by the host. Bits [4:0] of this register cannot be modified by the system software and will always return a zero when read.

NOTE The USB_OTG_HS_ASYNCLISTADDR register and the USB_OTG_HS_ENDPOINTLISTADDR register are shared between the host controller and device controller operations. For the host controller, this is the USB_OTG_HS_ASYNCLISTADDR register. For the device controller, this is the USB_OTG_HS_ENDPOINTLISTADDR register.

USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_ASYNCLISTADDR

Bits	Name	Description
31:5	ASYBASE	Link pointer low (LPL). These bits correspond to memory address signals [31:5], respectively. This field may only reference a queue head (QH). This field is only used by the host controller.
4:0	RESERVED_BITS4_0	The values of these bits has no effect on circuit operation.

0x078D9158 USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_ENDPOINTLISTADDR**Type:** RW**Clock:** CC_USB_OTG_HS_CLK**Reset State:** 0x00000000

BANKED=device

The USB_OTG_HS_ENDPOINTLISTADDR register is the endpointlist address register. In the device mode, this register contains the address of the top of the endpoint list in system memory. Bits [10:0] of this register cannot be modified by the system software and will always return a zero when read.

NOTE The USB_OTG_HS_ASYNCLISTADDR register and the USB_OTG_HS_ENDPOINTLISTADDR register are shared between the host controller and device controller operations. For the host controller, this is the USB_OTG_HS_ASYNCLISTADDR register. For the device controller, this is the USB_OTG_HS_ENDPOINTLISTADDR register.

USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_ENDPOINTLISTADDR

Bits	Name	Description
31:11	EPBASE	Endpoint list pointer (low) These bits correspond to memory address signals [31:11], respectively. This field will reference a list of up to 32 queue heads (QH). That is, one queue head per endpoint & direction.
10:0	RESERVED_BITS10_0	The values of these bits has no effect on circuit operation.

0x078D915C USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_TTCTRL**Type:** RW**Clock:** CC_USB_OTG_HS_CLK**Reset State:** 0x00000000

The USB_OTG_HS_TTCTRL register is the TT status and control register. This register contains parameters needed for internal TT operations.

NOTE This register is not used in the device controller operation.

USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_TTCTRL

Bits	Name	Description
31	RESERVED_BIT31	Not used.
30:24	TTHA	Not used.
23:0	RESERVED_BITS23_0	Not used.

0x078D9160 USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_BURSTSIZE**Type:** RW**Clock:** CC_USB_OTG_HS_CLK**Reset State:** 0x00001010

The USB_OTG_HS_BURSTSIZE register is the programmable burst size register. This register is used to control dynamically change the burst size used during data movement on the initiator (master) interface.

USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_BURSTSIZE

Bits	Name	Description
31:16	RESERVED_BITS31_16	The value of these bits has no effect on circuit operation.
15:8	TXPBURST	Programmable Tx burst length This register represents the maximum length of a the burst in 32-bit words while moving data from system memory to the USB bus.
7:0	RXPBURST	Programmable Rx burst length This register represents the maximum length of a the burst in 32-bit words while moving data from the USB bus to system memory.

0x078D9164 USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_TXFILLTUNING**Type:** RW**Clock:** CC_USB_OTG_HS_CLK**Reset State:** 0x00000000

The USB_OTG_HS_TXFILLTUNING register is the host transmit pre-buffer packet tuning register. The fields in this register control performance tuning associated with how the host controller posts data to the Tx latency FIFO before moving the data onto the USB bus. The specific areas of performance include the how much data to post into the FIFO and an estimate for how long that operation should take in the target system.

Definitions:

T0 = Standard packet overhead

T1 = Time to send data payload

Tff = Time to fetch packet into TX FIFO up to specified level.

Ts = Total packet flight time (send-only) packet

Ts = T0 + T1

Tp = Total packet time (fetch and send) packet

Tp = Tff + T0 + T1

Upon discovery of a transmit (OUT/SETUP) packet in the data structures, the host controller checks to ensure Tp remains before the end of the [micro] frame. If so it proceeds to pre-fill the TX FIFO. If at anytime during the pre-fill operation the time remaining the [micro] frame is < Ts, then the packet attempt ceases and the packet is tried at a later time. Although this is not an error condition and the host controller will eventually recover, a mark will be made the scheduler health counter to note the occurrence of a 'back-off' event. When a back-off event is detected, the partial packet fetched may need to be discarded from the latency buffer to make room for periodic traffic that will begin after the next SOF. Too many back-off events can waste bandwidth and power on the system bus, and thus should be minimized (not necessarily eliminated). Back-offs can be minimized with use of the TSCHEALTH (Tff), as described in the register table.

USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_TXFILLTUNING

Bits	Name	Description
31:22	RESERVED_BITS31_22	The value of these bits has no effect on circuit operation.
21:16	TXFIFOTHRES	FIFO burst threshold Default = 2 This register controls the number of data bursts that are posted to the TX latency FIFO in the host mode before the packet begins on to the bus. The minimum value is 2 and this value should be as low as possible to maximize USB performance. A higher value can be used in systems with unpredictable latency and/or insufficient bandwidth, where the FIFO may underrun because the data transferred from the latency FIFO to USB occurs before it can be replenished from system memory. This value is ignored if the stream disable bit in USBMODE register is set (1).
15:13	RESERVED_BITS15_13	The value of these bits has no effect on circuit operation.
12:8	TXSCHHEALTH	Scheduler health counter Read/write to clear Default = 0 This register increments when the host controller fails to fill the Tx latency FIFO to the level programmed by TXFIFOTHRES before running out of time to send the packet before the next start-of-frame. This health counter measures the number of times this occurs to provide feedback to selecting a proper TXSCHHEALTH. Writing to this register will clear the counter and this counter will be at a maximum at 31.

USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_TXFILLTUNING (cont.)

Bits	Name	Description
7:0	TXSCHOH	Scheduler overhead Default = 0 This register adds an additional fixed offset to the schedule time estimator described above as Tff. As an approximation, the value chosen for this register should limit the number of back-off events captured in the TXSCHHEALTH to less than 10 per second in a highly-utilized bus. Choosing a value that is too high for this register is not desired, as it can needlessly reduce USB utilization. The time unit represented in this register is 1.267 ms when a device is connected in the high-speed mode for OTG and SPH. The time unit represented in this register is 6.333 ms when a device is connected in the low/full speed mode for OTG and SPH. The time unit represented in this register is always 1.267 times the MPH product.

0x078D9170 USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_ULPI_VIEWPORT**Type:** RW**Clock:** CC_USB_OTG_HS_CLK**Reset State:** 0x08000000

The USB_OTG_HS_ULPI_VIEWPORT register provides indirect access to the ULPI PHY register set. Although the core performs access to the ULPI PHY register set, there may be extraordinary circumstances where software may need direct access.

CAUTION Writes to the ULPI through the viewport can substantially harm standard USB operations. Currently, no usage model has been defined where the software should need to execute writes directly to the ULPI - see the exception regarding optional features below.

Executing read operations though the ULPI viewport should have no harmful side effects to standard USB operations.

NOTE The ULPI viewport is only synthesized in the design if the constant VUSB_HS_PHY_ULPI is set (1). If the ULPI interface is not enabled, this register will always read zeros.

Two operations can be performed with the ULPI viewport: wake-up and read/write operations. The wake-up operation is used to put the ULPI interface into normal operation mode and re-enable the clock, if necessary. A wake-up operation is required before accessing the registers when the ULPI interface is operating in the low power mode, serial mode, or car kit mode.

The ULPI state can be determined by reading the sync state bit (ULPISS). If this bit is set (1), then the ULPI interface is running in the normal operation mode and can accept read/write operations. If the ULPISS is clear (0), then read/write operations will not be able to execute. Undefined behavior will result if ULPISS = 0, and a read or write operation is performed.

To execute a wake-up operation, write all 32-bits of the ULPI Viewport where ULPIPORT is constructed appropriately, and the ULPIWU bit is set (1) and the ULPIRUN bit is clear (0). Poll the ULPI viewport until ULPIWU is zero for the operation to complete.

To execute a read or write operation, write all 32-bits of the ULPI viewport where ULPIDATWR, ULPIADDR, ULPIPORT, and ULPIRW are constructed appropriately, and the ULPIRUN bit is set (1). Poll the ULPI viewport until ULPIRUN is zero for the operation to complete. Once ULPIRUN is zero, the ULPIDATRD will be valid if the operation was a read.

The polling method above could also be replaced and an interrupt driven using the ULPI interrupt defined in the USBTS and USBINTR registers. When a wake-up or read/write operation completes, the ULPI interrupt will be set (1).

Several optional features may need to be enabled or disabled by the system software as part of system configuration. These bits are contained in the interface and OTG control registers of the ULPI PHY register set. These registers also contain bits that are controlled by the link dynamically and, therefore, should only be modified by the system software using the set/clear access method. Direct writes to these registers could have harmful side effects to the standard USB operations. The following bits are optional bits:

* Bits 3 through 7 in the interface control register

* Bits 6 and 7 in the OTG control register.

USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_ULPI_VIEWPORT

Bits	Name	Description
31	ULPIWU	ULPI wake-up Setting (1) this bit begins the wake-up operation. The bit will automatically transition to 0 after the wake-up is complete. Once this bit is set (1), the driver can not clear it back to 0. Note: The driver must never execute a wake-up and a read/write operation at the same time.
30	ULPIRUN	ULPI read/write run Setting (1) this bit will begin the read/write operation. The bit will automatically transition to 0 after the read/write is complete. Once this bit is set (1), the driver can not clear it back to 0. Note: The driver must never execute a wakeup and a read/write operation at the same time.
29	ULPIRW	ULPI read/write control This bit selects between running a read or write operation. value 0 = Read value 1 = Write
28	ULPIFORCE	ULPI read/write force This bit enables forcing register access during RX packet value 0 = register access is not forced during RX packet (default) value 1 = register access is forced during RX packet reception

USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_ULPI_VIEWPORT (cont.)

Bits	Name	Description
27	ULPISS	ULPI sync state Read Only value 1 = Normal sync state value 0 = In another state (for example, car kit, serial, low power) This bit represents the state of the ULPI interface. Before reading this bit, the ULPIPORT field should be set accordingly if used with the multi-port host. Otherwise, this field should always remain 0.
26:24	ULPIPORT	ULPI port number For the wakeup or read/write operation to be executed, this value selects the port number the ULPI PHY is attached to in the multi-port host. The valid range is 0 to 7. This field should always be written as a 0 for the non-multi port products.
23:16	ULPIADDR	ULPI data address When a read or write operation is commanded, the address of the operation is written to this field.
15:8	ULPIDATRD	ULPI data read Read only After a read operation completes, the result is placed in this field.
7:0	ULPIDATWR	ULPI data write When a write operation is commanded, the data to be sent is written to this field.

0x078D9178 USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_ENDPTNAK**Type:** RW**Clock:** CC_USB_OTG_HS_CLK**Reset State:** 0x00000000

The USB_OTG_HS_ENDPTNAK register is the endpoint NAK register.

USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_ENDPTNAK

Bits	Name	Description
23:16	EPTN	Tx endpoint NAK Each tx endpoint has 1 bit in this field. The bit is set (1) when the device sends a NAK handshake on a received IN token for the corresponding endpoint. Bit 15 - Endpoint #15 Bit 1 - Endpoint #1 Bit 0 - Endpoint #0

USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_ENDPTNAK (cont.)

Bits	Name	Description
7:0	EPRN	Rx endpoint NAK Each rx endpoint has 1 bit in this field. The bit is set (1) when the device sends a NAK handshake on a received OUT or PING token for the corresponding endpoint. Bit 15 - Endpoint #15 Bit 1 - Endpoint #1 Bit 0 - Endpoint #0

0x078D917C USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_ENDPTNAKEN**Type:** RW**Clock:** CC_USB_OTG_HS_CLK**Reset State:** 0x00000000

The USB_OTG_HS_ENDPTNAKEN register is the endpoint NAK enable register.

USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_ENDPTNAKEN

Bits	Name	Description
23:16	EPTNE	Tx endpoint NAK enable Each bit is an enable bit for the corresponding Tx endpoint NAK bit. If this bit is set (1) and the corresponding Tx endpoint NAK bit is set (1), the NAK Interrupt bit is set (1). Bit 15 - Endpoint #15 Bit 1 - Endpoint #1 Bit 0 - Endpoint #0
7:0	EPRNE	Rx endpoint NAK enable Each bit is an enable bit for the corresponding Rx endpoint NAK bit. If this bit is set (1) and the corresponding Rx endpoint NAK bit is set (1), the NAK Interrupt bit is set (1). Bit 15 - Endpoint #15 Bit 1 - Endpoint #1 Bit 0 - Endpoint #0

0x078D9184 USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_PORTSC**Type:** RW**Clock:** CC_USB_OTG_HS_CLK**Reset State:** 0xCC000004

This is the USB_OTG_HS_PORTSC register.

USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_PORTSC

Bits	Name	Description
31:30	PTS	Parallel transceiver select Read/write This register bit pair is used in conjunction with the configuration constant VUSB_HS_PHY_TYPE to control which parallel transceiver interface is selected. If VUSB_HS_PHY_TYPE is set for 0, 1, 2, or 3, then this bit is read only. If VUSB_HS_PHY_TYPE is 4, 5, 6, or 7, then this bit is read/write. This field is reset to the following values: 00 if VUSB_HS_PHY_TYPE = 0,4 - UTMI/UTMI+ 01 if VUSB_HS_PHY_TYPE = 1,5 - Reserved 10 if VUSB_HS_PHY_TYPE = 2,6 - ULPI 11 if VUSB_HS_PHY_TYPE = 3,7 - Serial/1.1 PHY (FS only) This bit is not defined in the EHCI specification.
29	STS	Serial transceiver select Read/write This register bit is used in conjunction with the configuration constant VUSB_HS_PHY_SERIAL to control whether the parallel or serial transceiver interface is selected for FS and LS operation. The serial interface engine can be used in combination with the UTMI+ or ULPI physical interface to provide FS/LS signaling instead of the parallel interface. - If VUSB_HS_PHY_SERIAL is 0 or 1, then this bit is read only. - If VUSB_HS_PHY_SERIAL is 2 or 3, then this bit is read/write. This bit has no effect unless the parallel transceiver select is set to UTMI+ or ULPI. The Serial/1.1 physical interface will use the serial interface engine for FS/LS signaling regardless of this bit value. Note: This bit was reserved for future operation, and is now adding for dynamic use of the serial engine in accord with UMTI+ and ULPI characterization logic. This bit is not defined in the EHCI specification.
28	PTW	Parallel transceiver width Read/write This register bit is used in conjunction with the configuration constant VUSB_HS_PHY16_8 to control the data bus width of the UTMI transceiver interface. - If VUSB_HS_PHY16_8 is 0 or 1, then this bit is read only. - If VUSB_HS_PHY16_8 is 2 or 3, then this bit is read/write. This bit is reset to 1 if VUSB_HS_PHY16_8 selects a default UTMI interface width of 16-bits, else it is reset to 0. - Writing this bit to 0 selects the 8-bit [60MHz] UTMI interface. - Writing this bit to 1 selects the 16-bit [30MHz] UTMI interface. This bit has no effect if the serial interface is selected. This bit is not defined in the EHCI specification.

USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_PORTSC (cont.)

Bits	Name	Description
27:26	PSPD	Port speed Read only This register field indicates the speed at which the port is operating. For HS mode operation in the host controller and HS/FS operation in the device controller, the port routing steers data to the protocol engine. For FS and LS mode operation in the host controller, the port routing steers data to the protocol engine with the embedded transaction translator. value 00 = Full speed value 01 = Low speed value 10 = High speed This bit is not defined in the EHCI specification.
25	SPRT	Short Port Reset Time. shortens port reset time for simulation.
24	PFSC	Port force full speed connect Read/write Default = 0 Setting (1) this bit will force the port to only connect at full speed. It also disables the chirp sequence that allows the port to identify itself as high speed. This is useful for testing FS configurations with a HS host, hub or device. This bit is not defined in the EHCI specification.
23	PHCD	PHY low power suspend - clock disable (PLPSCD) Read/write Default = 0 value 1 = Disable the PHY clock. value 0 = Enable the PHY clock. Reading this bit will indicate the status of the PHY clock. Note: The PHY clock cannot be disabled if it is being used as the system clock. In the device mode, The PHY can be put into low power suspend - clock disable when the device is not running (USBCMD run/stop = 0) or the host has signaled suspend (PORTSC SUSPEND = 1). Low power suspend will be cleared automatically when the host has signaled resume. Before forcing a resume from the device, the device controller driver must clear this bit. In the host mode, the PHY can be put into low power suspend - clock disable when the downstream device has been put into the suspend mode or when no downstream device is connected. Low power suspend is completely under the control of the software. See 10 System Level Issues and Core Configuration for more discussion on clock disable and power-down issues. This bit is not defined in the EHCI specification.

USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_PORTSC (cont.)

Bits	Name	Description
22	WKOC	Wake on over-current enable (WKOC_E) Read/write Default = 0 Setting (1) this bit enables the port to be sensitive to over-current conditions as wake-up events. This bit is clear (0) if the port power (PP) bit (bit 12) in this register is clear (0). This bit is output from the controller as signal pwrctl_wake_ovrcurr_en (OTG/host core only) for use by an external power control circuit.
21	WKDS	Wake on disconnect enable (WKDSCNNT_E) Read/write Default = 0 Setting (1) this bit enables the port to be sensitive to device disconnects as wake-up events. This field is clear (0) if the port power (PP) bit (bit 12) in this register is clear (0) or in the device mode. This bit is output from the controller as signal pwrctl_wake_dscnt_en (OTG/host core only) for use by an external power control circuit.
20	WKN	Wake on connect enable (WKNNT_E) Read/write Default = 0 Setting (1) this bit enables the port to be sensitive to device connects as wake-up events. This field is clear (0) if the port power (PP) bit (bit 12) in this register is clear (0) or in the device mode. This bit is output from the controller as signal pwrctl_wake_dscnt_en (OTG/host core only) for use by an external power control circuit.

USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_PORTSC (cont.)

Bits	Name	Description
19:16	PTC	Port test control Read/write Default = 0000 Any other value than zero indicates that the port is operating in the test mode. Value Specific test 0000b TEST_MODE_DISABLE 0001b J_STATE 0010b K_STATE 0011b SE0 (host) / NAK (device) 0100b Packet 0101b FORCE_ENABLE_HS 0110b FORCE_ENABLE_FS 0111b FORCE_ENABLE_LS 1000b to 1111b Reserved The FORCE_ENABLE_FS and FORCE_ENABLE_LS tests are extensions to the test mode support, as specified in the EHCI specification. Writing the PTC field to any of the FORCE_ENABLE_{HS/FS/LS} values will force the port into the connected and enabled state at the selected speed. Writing the PTC field back to TEST_MODE_DISABLE will allow the port state machines to progress normally from that point. Note: Low speed operations are not supported as a peripheral device.
15:14	PIC	Port indicator control Read/write Default = 0 Writing to this field has no effect if the P_INDICATOR bit in the HCSPARAMS register is a zero. If the P_INDICATOR bit is set (1), then this field has the following meanings: Bit value Meaning value 00 Port indicators are off value 01 Amber value 10 Green value 11 Undefined This field is output from the controller as signals port_ind_ctl_1 and port_ind_ctl_0 for use by an external LED driving circuit.

USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_PORTSC (cont.)

Bits	Name	Description
13	PO	Port owner Read only Port owner hand-off is not implemented in this design, therefore this bit will always read back as clear (0). Default = 0 The EHCI definition is include here for reference: - This bit unconditionally goes clear (0) when the configured bit in the CONFIGFLAG register makes a 0-to-1 transition. - This bit unconditionally goes set (1) whenever the configured bit is clear (0). The system software uses this field to release ownership of the port to a selected host controller (in the event that the attached device is not a high-speed device). The software sets (1) this bit when the attached device is not a high-speed device. When this bit is set (1), it indicates that an internal companion controller owns and controls the port.
12	PP	Port power (PP) Read/write or read-only The function of this bit depends on the value of the port power switching (PPC) field in the HCSPARAMS register. - PPC = 0: PP is read-only. A device controller with no OTG capability does not have port power control switches. - PPC = 1: PP is read/write. The host/OTG controller requires port power control switches. This bit represents the current setting of the switch (0 = off, 1 = on). When power is not available on a port (and PP equals a 0), the port is non-functional and will not report attaches, detaches, and so on. When an over-current condition is detected on a powered port and PPC is set (1), the PP bit in each affected port may be transitioned by the host controller driver from a one to a zero, removing power from the port. This feature is implemented in the host/OTG controller (PPC = 1). In a device-only implementation, port power control is not necessary. So, PPC and PP = 0.

USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_PORTSC (cont.)

Bits	Name	Description
11:10	LS	Line status Read-only These bits reflect the current logical levels of the D+ (bit 11) and D- (bit 10) signal lines: Value Meaning value 00 SE0 value 10 J-state value 01 K-state value 11 Undefined In the host mode, the use of linestate by the host controller driver is not necessary (unlike EHCI), because the port controller state machine and the port routing manage the connection of LS and FS. In the device mode, the use of linestate by the device controller driver is not necessary.
9	HSP	High-speed port Read-only Default = 0 When the bit is set (1), the host/device connected to the port is in the high-speed mode. When this bit is clear (0), the host/device connected to the port is not in a high-speed mode. Note: HSP is redundant with PSPD(27:26), but will remain in the design for compatibility. This bit is not defined in the EHCI specification.
8	PR	Port reset This bit is clear (0) if the port power (PP) bit (bit 12) of this register is clear (0). Host mode: - Read/write - 1 = Port is in reset - 0 = Port is not in reset - Default = 0 When the software sets (1) this bit, the bus-reset sequence as defined in the USB Specification Revision 2.0 is started. This bit will automatically clear (go to 0) after the reset sequence is complete. Note: This behavior is different from EHCI, where the host controller driver is required to clear (0) this bit after the reset duration is timed in the driver. Device mode: This bit is a read-only status bit. Device reset from the USB bus is also indicated in the USBSTS register.

USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_PORTSC (cont.)

Bits	Name	Description
7	SUSP	Suspend Host mode: - Read/write - 1 = Port in suspend state - 0 = Port not in suspend state - Default = 0 The port enabled bit and suspend bit of this register define the port states: Bit value Port state value 0x Disable value 10 Enable value 11 Suspend When in the suspend state, downstream propagation of data is blocked on this port, except for port reset. The blocking occurs at the end of the current transaction if a transaction was in progress when this bit set (1). In the suspend state, the port is sensitive to resume detection. Note that the bit status does not change until the port is suspended and that there may be a delay in suspending a port if there is a transaction currently in progress on the USB. The host controller will unconditionally clear (0) this bit when the software clears (0) the force port resume bit. The host controller ignores a write of zero to this bit. If the host software sets (1) this bit when the port is not enabled (port enabled bit is a zero), the results are undefined. This field is clear (0) if port power (PP) is clear (0) in the host mode. Device mode: - Read only - 1 = Port in suspend state - 0 = Port not in suspend state - Default = 0 In the device mode, this bit is a read-only status bit.

USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_PORTSC (cont.)

Bits	Name	Description
6	FPR	Force port resume - Read/write - 1 = Resume detected/driven on port - 0 = No resume (K-state) detected/driven on port - Default = 0 Host mode: The software sets (1) this bit to drive resume signaling. The host controller sets (1) this bit if a J-to-K transition is detected while the port is in the suspend state. When this bit transitions to a one because a J-to-K transition is detected, the port change detect bit in the USBSTS register is also set (1). This bit will automatically clear (go to 0) after the resume sequence is complete. This behavior is different from EHCI, where the host controller driver is required to clear (0) this bit after the resume duration is timed in the driver. Note that, when the host controller owns the port, the resume sequence follows the defined sequence documented in the USB Specification Revision 2.0. The resume signaling (full-speed 'K') is driven on the port as long as this bit remains set (1). This bit will remain set (1) until the port has switched to the high-speed idle. Clearing (0) this bit has no affect, because the port controller will time the resume operation and clear (0) the bit when the port control state switches to HS or FS idle. This bit is clear if the port power (PP) is clear (0) in the host mode. This bit is not-EHCI compatible. Device mode: After the device has been in the suspend state for 5 ms or more, the software must set (1) this bit to drive resume signaling before clearing. The device controller will set (1) this bit if a J-to-K transition is detected while the port is in the suspend state. The bit will be cleared (0) when the device returns to normal operation. Also, when this bit transitions to a one because a J-to-K transition was detected, the port change detect bit in the USBSTS register is also set (1).
5	OCC	Over-current change Read/write control Default = 0 This bit gets is set (1) when there is a change to over-current active. The software clears this bit by setting (1) this bit position. For host/OTG implementations, the user can provide over-current detection to the vbus_pwr_fault input for this condition. For device-only implementations, this bit shall always be clear (0).

USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_PORTSC (cont.)

Bits	Name	Description
4	OCA	Over-current active Read-only Default = 0 value 1 = This port currently has an over-current condition. value 0 = This port does not have an over-current condition. This bit will automatically transition from set (1) to clear (0) when the over current condition is removed. For host/OTG implementations, the user can provide over-current detection to the vbus_pwr_fault input for this condition. For device-only implementations, this bit shall always be clear (0).
3	PEC	Port enable/disable change Read/write control value 1 = Port enabled/disabled status has changed value 0 = No change Default = 0 Host mode: For the root hub, this bit gets set (1) only when a port is disabled due to disconnect on the port or due to the appropriate conditions existing at the EOF2 point (See Chapter 11 of the USB Specification). The software clears this bit by writing a one to it. This field is clear (0) if the port power (PP) bit in the register is clear (0). Device mode: The device port is always enabled (this bit will be zero).
2	PE	Port enabled/disabled Read/write value 1 = Enable value 0 = Disable Default 0 Host mode: Ports can only be enabled by the host controller as a part of the reset and enable. The software cannot enable a port by setting (1) this field. Ports can be disabled by either a fault condition (disconnect event or other fault condition) or by the host software. Note that the bit status does not change until the port state actually changes. There may be a delay in disabling or enabling a port, due to other host controller and bus events. When the port is disabled, (0) downstream propagation of data is blocked except for reset. This field is clear (0) if the port power (PP) bit is cleared (0) in the host mode. Device mode: The device port is always enabled (this bit will be set [1])

USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_PORTSC (cont.)

Bits	Name	Description
1	CSC	Connect status change Read/write control value 1 = Change in current connect status value 0 = No change Default = 0 Host mode: Indicates that a change has occurred in the port's current connect status. The host/device controller sets (1) this bit for all changes to the port device connect status, even if the system software has not cleared an existing connect status change. For example, the insertion status changes twice before system software has cleared the changed condition, the hub hardware will be 'setting' an already-set bit (that is, the bit will remain set). The software clears this bit by writing a one to it. This field is zero if the port power (PP) bit in this register is zero in the host mode. Device mode: This bit is undefined in the device controller mode.
0	CCS	Current connect status Read-only Host mode: value 1 = Device is present on port value 0 = No device is present Default = 0 This value reflects the current state of the port, and may not correspond directly to the event that caused the connect status change bit (bit 1) to be set (1). This field is clear if the port power (PP) bit in this register is clear (0) in host mode. Device mode: value 1 = Attached value 0 = Not attached Default = 0 If this bit is set (1), this indicates that the device successfully attached, and is operating in either high speed or full speed, as indicated by the high speed port bit in this register. If this bit is clear (0), this indicates that the device did not attach successfully or was forcibly disconnected by the software clearing (0) the run bit in the USBCMD register. It does not state the device being disconnected or suspended.

0x078D91A4 USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_OTGSC**Type:** RW**Clock:** CC_USB_OTG_HS_CLK**Reset State:** 0x00000E20

USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_OTGSC

Bits	Name	Type	Description
31	RESERVED_BIT31	RW	Clear (0) this bit.
30	DPIE	RW	Data pulse interrupt enable
29	FIELD_1MSE	RW	1 millisecond timer interrupt enable - read/write
28	BSEIE	RW	B session end interrupt enable Read/write Setting (1) this bit enables the B session end interrupt.
27	BSVIE	RW	B session valid interrupt enable Read/write Setting (1) this bit enables the B session valid interrupt.
26	ASVIE	RW	A session valid interrupt enable Read/write Setting (1) this bit enables the A session valid interrupt.
25	AVVIE	RW	A Vbus valid interrupt enable Read/write Setting (1) this bit enables the A Vbus valid interrupt.
24	IDIE	RW	USB ID interrupt enable Read/write Setting (1) this bit enables the USB ID interrupt.
23	RESERVED_BIT23	RW	Clear (0) this bit.
22	DPIS	RW	Data pulse interrupt status Read/write to clear This bit is set (1) when data bus pulsing occurs on DP or DM. Data bus pulsing is only detected when USBMODE.CM = Host (11) and PORTSC(0). PortPower = Off (0). The software must write a one to clear this bit.
21	FIELD_1MSS	RW	1 millisecond timer interrupt status Read/write to clear This bit is set (1) once every millisecond. The software must write a one to clear this bit.
20	BSEIS	RW	B session end interrupt status Read/write to clear This bit is set (1) when the Vbus has fallen below the B session end threshold. The software must write a one to clear this bit
19	BSVIS	RW	B session valid interrupt status Read/write to clear This bit is set (1) when Vbus has either risen above or fallen below the B session valid threshold (0.8 VDC). The software must write a one to clear this bit.

USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_OTGSC (cont.)

Bits	Name	Type	Description
18	ASVIS	RW	A session valid interrupt status Read/write to clear This bit is set (1) when Vbus has either risen above or fallen below the A session valid threshold (0.8 VDC). The software must write a one to clear this bit.
17	AVVIS	RW	A Vbus valid interrupt status Read/write to clear This bit is set (1) when Vbus has either risen above or fallen below the Vbus valid threshold (4.4 VDC) on an A device. The software must write a one to clear this bit.
16	IDIS	RW	USB ID interrupt status Read/write This bit is set (1) when a change on the ID input has been detected. The software must write a one to clear this bit.
15	RESERVED_BIT15	RW	Clear (0) this bit.
14	DPS	RW	Data bus pulsing status Read-only If this bit is set (1), it indicates that the data bus pulsing is being detected on the port.
13	FIELD_1MST	RW	1 millisecond timer toggle Read-only This bit toggles once per millisecond.
12	BSE	R	B session end Read-only Indicates that the Vbus is below the B session end threshold.
11	BSV	RW	B session valid Read-only Indicates that the Vbus is above the B session valid threshold.
10	ASV	RW	A session valid Read-only Indicates that the Vbus is above the A session valid threshold.
9	AVV	RW	A Vbus valid Read-only Indicates that the Vbus is above the A Vbus valid threshold.

USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_OTGSC (cont.)

Bits	Name	Type	Description
8	ID	RW	USB ID Read-only value 0 = A device value 1 = B device
7	HABA	RW	Hardware assist B-disconnect to A-connect Read/write value 0 = Disabled value 1 = Enable automatic B-disconnect to A-connect sequence.
6	HADP	RW	Hardware assist data-pulse Write to set
5	IDPU	RW	ID pull-up Read/write This bit provides control over the ID pull-up register. value 0 = Off value 1 = On (default) When this bit is clear (0), the ID input will not be sampled.
4	DP	RW	Data pulsing Read/write Setting (1) this bit causes the pull-up on DP to be asserted for data pulsing during SRP.
3	OT	RW	OTG termination Read/write This bit must be set (1) when the OTG device is in the device mode. This controls the pull-down on DM.
2	HAAR	RW	Hardware assist auto-reset Read/write value 0 = Disabled value 1 = Enable automatic reset after connect on host port.
1	VC	RW	Vbus charge Read/write Setting (1) this bit causes the Vbus line to be charged. This is used for Vbus pulsing during SRP.
0	VD	RW	Vbus discharge Read/write Setting (1) this bit causes the Vbus to discharge through a resistor.

0x078D91A8 USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_USBMODE**Type:** RW**Clock:** CC_USB_OTG_HS_CLK**Reset State:** 0x00000000

The USB_OTG_HS_USBMODE register is the USB device mode register.

USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_USBMODE

Bits	Name	Description
31:6	RESERVED_BITS31_6	Clear (0) these bits.
5	VBPS	Vbus power select value 0 - Output is 0 value 1 - Output is 1 This bit is connected to the vbus_pwr_select output and can be used for any generic control, but is named to be used by logic that selects between an on-chip Vbus power source (charge pump) and an off-chip source in systems when both are available.
4	SDIS	Stream disable mode value 0 = Inactive (default) value 1 = Active Device mode: Setting (1) this bit disables double priming on both the Rx and Tx for low bandwidth systems. This mode ensures that, when the Rx and Tx buffers are sufficient to contain an entire packet, the standard double buffering scheme is disabled to prevent overruns/underruns in bandwidth-limited systems. Note: In the high speed mode, a NYET handshake will respond to all packets received when the stream disable is active. Host mode: Setting (1) this bit ensures that overruns/underruns of the latency FIFO are eliminated for low bandwidth systems where the Rx and Tx buffers are sufficient to contain the entire packet. Enabling the stream disable also has the effect of ensuring that the Tx latency is filled to capacity before the packet is launched onto the USB. Note: Time duration to pre-fill the FIFO becomes significant when the stream disable is active. See TXFILLTUNING and TXTTFILLTUNING [MPH only] to characterize the adjustments needed for the scheduler when using this feature. Note: The use of this feature substantially limits of the overall USB performance.
3	SLOM	Setup lockout mode. In the device mode, this bit controls the behavior of the setup lock mechanism. See Control Endpoint Operation Model. value 0 = Setup lockouts on (default) value 1 = Setup lockouts off (DCD requires use of a setup data buffer tripwire in USBCMD)

USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_USBMODE (cont.)

Bits	Name	Description
2	ES	Endian select Read/write This bit can change the byte ordering of the transfer buffers to match the host microprocessor bus architecture. The bit fields in the microprocessor interface and the DMA data structures (including the setup buffer within the device QH) are unaffected by the value of this bit, because they are based upon 32-bit words. value 0 = Little endian (default) - first byte referenced in the least significant byte of a 32-bit word. value 1 = Big endian - first byte referenced in most significant byte of a 32-bit word.
1:0	CM	Controller mode Read/write once The controller mode is defaulted to the proper mode for host-only and device-only implementations. For those designs that contain both host and device capability, the controller will default to an idle state and will need to be initialized to the desired operating mode after reset. For combination host/device controllers, this register can only be written once after reset. If it is necessary to switch modes, the software must reset the controller by writing to the RESET bit in the USBCMD register before reprogramming this register. value 00 = Idle [default for combination host/device] value 01 = Reserved value 10 = Device controller [default for device only controller] value 11 = Host controller [default for host only controller]

0x078D91AC USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_ENDPTSETUPSTAT**Type:** RW**Clock:** CC_USB_OTG_HS_CLK**Reset State:** 0x00000000

The USB_OTG_HS_ENDPTSETUPSTAT register is the endpoint set-up status register.

USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_ENDPTSETUPSTAT

Bits	Name	Description
31:16	RESERVED_BITS31_16	Clear (0) these bits.

USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_ENDPTSETUPSTAT (cont.)

Bits	Name	Description
15:0	ENDPTSETUPSTAT	Set-up endpoint status For every set-up transaction that is received, a corresponding bit in this register is set (1). The software must clear or acknowledge the setup transfer by setting (1) a respective bit after it has read the setup data from the queue head. The response to a set-up packet as in the order of operations and total response time is crucial to limit bus time outs while the set-up lock-out mechanism is engaged. See Managing Endpoints in the Device Operational Model. This register is only used in the device mode.

0x078D91B0 USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_ENDPTPRIME**Type:** RW**Clock:** CC_USB_OTG_HS_CLK**Reset State:** 0x00000000

The USB_OTG_HS_ENDPTPRIME register is the endpoint initialization register. This register is only used in the device mode.

USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_ENDPTPRIME

Bits	Name	Description
31:16	PETB	Prime endpoint transmit buffer For each endpoint, a corresponding bit is used to request that a buffer is prepared for a transmit operation in order to respond to a USB IN/INTERRUPT transaction. The software should set (1) the corresponding bit when posting a new transfer descriptor to an endpoint. The hardware will automatically use this bit to begin parsing for a new transfer descriptor from the queue head and prepare a transmit buffer. The hardware will clear (0) this bit when the associated endpoint(s) is (are) successfully primed. Note: These bits will be momentarily set (1) by the hardware during hardware re-priming operations when a dTD is retired, and the dQH is updated. PETB[15] - Endpoint #15 PETB[1] - Endpoint #1 PETB[0] - Endpoint #0

USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_ENDPTPRIME (cont.)

Bits	Name	Description
15:0	PERB	Prime endpoint receive buffer For each endpoint, a corresponding bit is used to request that a buffer is prepared for a receive operation for when a USB host initiates a USB OUT transaction. The software should set (1) the corresponding bit whenever posting a new transfer descriptor to an endpoint. The hardware will automatically use this bit to begin parsing for a new transfer descriptor from the queue head and prepare a receive buffer. The hardware will clear (0) this bit when the associated endpoint(s) is (are) successfully primed. Note: These bits will be momentarily set (1) by the hardware during hardware re-priming operations when a dTD is retired, and the dQH is updated. Bit 15 - Endpoint #15 Bit 1 - Endpoint #1 Bit 0 - Endpoint #0

0x078D91B4 USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_ENDPTFLUSH**Type:** RW**Clock:** CC_USB_OTG_HS_CLK**Reset State:** 0x00000000

The USB_OTG_HS_ENDPTFLUSH register is the endpoint de-initialize register. This register is only used in the device mode.

USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_ENDPTFLUSH

Bits	Name	Description
31:16	FETB	Flush endpoint transmit buffer Setting (1) a bit in this register will cause the associated endpoint to clear any primed buffers. If a packet is in progress for one of the associated endpoints, that transfer will continue until completion. The hardware will clear this register after the endpoint flush operation is successful. FETB[15] - Endpoint #15 FETB[1] - Endpoint #1 FETB[0] - Endpoint #0
15:0	FERB	Flush endpoint receive buffer Setting a bit (1) will cause the associated endpoint to clear any primed buffers. If a packet is in progress for one of the associated endpoints, that transfer will continue until completion. The hardware will clear this register after the endpoint flush operation is successful. Bit 15 - Endpoint #15 Bit 1 - Endpoint #1 Bit 0 - Endpoint #0

0x078D91B8 USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_ENDPTSTAT**Type:** R**Clock:** CC_USB_OTG_HS_CLK**Reset State:** 0x00000000

The USB_OTG_HS_ENDPTSTAT register is the endpoint status register. This register is only used in the device mode.

USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_ENDPTSTAT

Bits	Name	Description
31:16	ETBR	Endpoint transmit buffer ready One bit for each endpoint indicates the status of the respective endpoint buffer. This bit is set (1) by the hardware as a response to receiving a command from a corresponding bit in the ENDPTPRIME register. There will always be a delay between setting (1) a bit in the ENDPTPRIME register and the endpoint indicating that it is ready. This delay time varies based upon the current USB traffic and the number of bits set (1) in the ENDPTPRIME register. The buffer ready status is cleared by a USB reset, by the USB DMA system, or through the ENDPTFLUSH register. Note: These bits will be momentarily cleared by the hardware during hardware endpoint re-priming operations when a dTD is retired, and the dQH is updated. ETBR[15] - Endpoint #15 ETBR[1] - Endpoint #1 ETBR[0] - Endpoint #0
15:0	ERBR	Endpoint receive buffer ready One bit for each endpoint indicates the status of the respective endpoint buffer. This bit is set (1) by the hardware as a response to receiving a command from a corresponding bit in the ENDPTPRIME register. There will always be a delay between setting (1) a bit in the ENDPTPRIME register and the endpoint indicating that it is ready. This delay time varies based upon the current USB traffic and the number of bits set (1) in the ENDPTPRIME register. The buffer ready status is cleared by a USB reset, by the USB DMA system, or through the ENDPTFLUSH register. Note: These bits will be momentarily cleared by the hardware during hardware endpoint re-priming operations when a dTD is retired, and the dQH is updated. ERBR[15] - Endpoint #15 ERBR[1] - Endpoint #1 ERBR[0] - Endpoint #0

0x078D91BC USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_ENDPTCOMPLETE**Type:** RW**Clock:** CC_USB_OTG_HS_CLK**Reset State:** 0x00000000

The register is the endpoint complete register. This register is only used in the device mode.

USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_ENDPTCOMPLETE

Bits	Name	Description
31:16	ETCE	Endpoint transmit complete event Each bit indicates that a transmit event (IN/INTERRUPT) occurred and the software should read the corresponding endpoint queue to determine the endpoint status. If the corresponding IOC bit is set (1) in the transfer descriptor, then this bit will be set (1) simultaneously with the USBINT. Writing a one will clear the corresponding bit in this register. ETCE[15] - Endpoint #15 ETCE[1] - Endpoint #1 ETCE[0] - Endpoint #0
15:0	ERCE	Endpoint receive complete event Each bit indicates that a received event (OUT/SETUP) occurred and the software should read the corresponding endpoint queue to determine the transfer status. If the corresponding IOC bit is set (1) in the transfer descriptor, then this bit will be set (1) simultaneously with the USBINT. Writing a one will clear the corresponding bit in this register. ERCE[15] - Endpoint #15 ERCE[1] - Endpoint #1 ERCE[0] - Endpoint #0

0x078D91C0 USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_ENDPTCTRL0**Type:** RW**Clock:** CC_USB_OTG_HS_CLK**Reset State:** 0x00800080

The USB_OTG_HS_ENDPTCTRL0 register is the endpoint control 0 register. Every device will implement endpoint0 as a control endpoint.

USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_ENDPTCTRL0

Bits	Name	Description
31:24	RESERVED_BITS31_24	Clear (0) these bits.

USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_ENDPTCTRL0 (cont.)

Bits	Name	Description
23	TXE	Tx endpoint enable value 1 = Enabled Endpoint0 is always enabled. Read only
22:20	RESERVED_BITS22_20	Clear (0) these bits.
19:18	TXT	Tx endpoint type Read only value 00 = Control Endpoint0 is fixed as a control end point.
17	RESERVED_BIT17	Clear (0) this bit.
16	TXS	Tx endpoint stall Read/write value 0 = End point OK (default) value 1 = End point stalled The software can set (1) this bit to force the endpoint to return a STALL handshake to the host. It will continue returning STALL until the bit is cleared (0) by the software or it will automatically be cleared (0) upon receipt of a new SETUP request. After receiving a SETUP request, this bit will continue to be cleared (0) by the hardware until the associated ENDPTSETUPSTAT bit is cleared (0). Note: There is a slight delay (50 clocks maximum) between the ENDPTSETUPSTAT being cleared (0) and the hardware continuing to clear (0) this bit. In most systems, it is unlikely that the DCD software will observe this delay. However, should the DCD observe that the stall bit is not set (1) after writing a one to it, then continually write this stall bit until it is set (1) or until a new setup has been received by checking the associated ENDPTSETUPSTAT bit.
15:8	RESERVED_BITS15_8	Clear (0) these bits.
7	RXE	Rx endpoint enable value 1 = Enabled Endpoint0 is always enabled. Read only
6:4	RESERVED_BITS6_4	Clear (0) these bits.
3:2	RXT	Rx endpoint type Read only value 00 = Control Endpoint0 is fixed as a control end point.
1	RESERVED_BIT1	Clear (0) this bit.

USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_ENDPTCTRL0 (cont.)

Bits	Name	Description
0	RXS	Rx endpoint stall Read/write value 0 = End point OK (default) value 1 = End point stalled The software can set (1) this bit to force the endpoint to return a STALL handshake to the host. It will continue returning STALL until the bit is cleared (0) by the software or it will automatically be cleared (0) upon receipt of a new SETUP request. After receiving a SETUP request, this bit will continue to be cleared (0) by the hardware until the associated ENDPTSETUPSTAT bit is cleared (0). Note: There is a slight delay (50 clocks maximum) between the ENDPTSETUPSTAT being cleared (0) and the hardware continuing to clear (0) this bit. In most systems, it is unlikely that the DCD software will observe this delay. However, should the DCD observe that the stall bit is not set (1) after writing a one to it, then continually write this stall bit until it is set (1) or until a new setup has been received by checking the associated ENDPTSETUPSTAT bit.

**0x078D91C0+USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_ENDPTCTRLn, n=[1..15]
0x4*n****Type:** RW**Clock:** CC_USB_OTG_HS_CLK**Reset State:** 0x00000000

The USB_OTG_HS_ENDPTCTRLn register is the endpoint control n register. There is an ENDPTCTRLn register for each endpoint in a device.

CAUTION If one endpoint direction is enabled and the paired endpoint of the opposite direction is disabled, then the unused direction type must be changed from the default control-type to any other type (such as bulk-type). Leaving an unconfigured endpoint control will cause undefined behavior for the data PID tracking on the active endpoint/direction.

USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_ENDPTCTRLn

Bits	Name	Description
31:24	RESERVED_BITS31_24	Clear (0) these bits.
23	TXE	Tx endpoint enable value 0 = Disabled (default) value 1 = Enabled An endpoint should be enabled only after it has been configured.

USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_ENDPTCTRLn (cont.)

Bits	Name	Description
22	TXR	Tx data toggle reset (WS) value 1 = Reset PID sequence Whenever a configuration event is received for this endpoint, the software must set (1) this bit in order to synchronize the data PIDs between the host and device.
21	TXI	Tx data toggle inhibit value 0 = PID sequencing enabled (default) value 1 = PID sequencing disabled This bit is only used for testing and should always be cleared (0). Setting (1) this bit will cause this endpoint to ignore the data toggle sequence and always transmit DATA0 for a data packet.
20	RESERVED_BIT20	Clear (0) this bit.
19:18	TXT	Tx endpoint type value 00 = Control value 01 = Isochronous value 10 = Bulk value 11 = Interrupt
17	TXD	Tx endpoint data source value 0 = Dual port memory buffer/DMA engine (default) This bit should always be cleared (0).
16	TXS	Tx endpoint stall value 0 = End point OK value 1 = End point stalled This bit will be cleared (0) automatically upon receipt of a SETUP request if this endpoint is configured as a control endpoint, and this bit will continue to be cleared (0) by the hardware until the associated ENDPTSETUPSTAT bit is cleared (0). The software can set (1) this bit to force the endpoint to return a STALL handshake to the host. This control will continue to STALL until this bit is either cleared (0) by the software or automatically cleared (0) as described above for control endpoints. Note (control endpoint types only): There is a slight delay (50 clocks maximum) between the ENDPTSETUPSTAT being cleared (0) and the hardware continuing to clear (0) this bit. In most systems, it is unlikely that the DCD software will observe this delay. However, should the DCD observe that the stall bit is not set (1) after writing a one to it, then continually write this stall bit until it is set (1) or until a new setup has been received by checking the associated ENDPTSETUPSTAT bit.
15:8	RESERVED_BITS15_8	Clear (0) these bits.
7	RXE	Rx endpoint enable value 0 = Disabled (default) value 1 = Enabled An endpoint should be enabled only after it has been configured.

USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_ENDPTCTRLn (cont.)

Bits	Name	Description
6	RXR	Rx data toggle reset (WS) Write 1 = Reset PID sequence Whenever a configuration event is received for this endpoint, the software must set (1) this bit in order to synchronize the data PIDs between the host and the device.
5	RXI	Rx data toggle inhibit value 0 = Disabled (default) value 1 = Enabled This bit is only used for testing and should always be cleared (0). Setting (1) this bit will cause this endpoint to ignore the data toggle sequence and always accept a data packet regardless of their data PID.
4	RESERVED_BIT4	Clear (0) this bit.
3:2	RXT	Rx endpoint type value 00 = Control value 01 = Isochronous value 10 = Bulk value 11 = Interrupt
1	RXD	Rx endpoint data sink value 0 = Dual port memory buffer/DMA engine (default) This bit should always be cleared (0).
0	RXS	Rx endpoint stall value 0 = End point OK value 1 = End point stalled This bit will be cleared (0) automatically upon receipt of a SETUP request if this endpoint is configured as a control endpoint, and this bit will continue to be cleared (0) by the hardware until the associated ENDPTSETUPSTAT bit is cleared (0). The software can set (1) this bit to force the endpoint to return a STALL handshake to the host. This control will continue to STALL until this bit is either cleared (0) by the software or automatically cleared (0) as described above for control endpoints. Note (control endpoint types only): There is a slight delay (50 clocks maximum) between the ENDPTSETUPSTAT being cleared (0) and the hardware continuing to clear (0) this bit. In most systems, it is unlikely that the DCD software will observe this delay. However, should the DCD observe that the stall bit is not set (1) after writing a one to it, then continually write this stall bit until it is set (1) or until a new setup has been received by checking the associated ENDPTSETUPSTAT bit.

**0x078D91FC+USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_ENDPT_PIPE_IDn, n=[1..15]
0x4*n****Type:** RW**Clock:** CC_USB_OTG_HS_CLK**Reset State:** 0x001F001F

The USB_OTG_HS_ENDPT_PIPE_IDn register is the endpoint pipe number register. There is an USB_OTG_HS_ENDPT_PIPE_IDn register for each endpoint in a device.

Note: reset value of TX_PIPE_ID and RX_PIPE_ID is 0x1F, which means that Endpoint is not mapped to any pipe.

USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_ENDPT_PIPE_IDn

Bits	Name	Description
31:21	RESERVED_BITS31_21	Clear (0) these bits.
20:16	TX_PIPE_ID	This field indicate the pipe number that this tx end point (n) will use in pipe mode.
15:5	RESERVED_BITS15_5	Clear (0) these bits.
4:0	RX_PIPE_ID	This field indicate the pipe number that this rx end point (n) will use in pipe mode.

0x078D9240 USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_PHY_CTRL**Type:** RW**Clock:** CC_USB_OTG_HS_CLK**Reset State:** 0x388C3C3A

The USB_OTG_HS_PHY_CTRL register is used to configure various features in the Synopsys 28nm PHY.

USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_PHY_CTRL

Bits	Name	Type	Description
31	RESERVED_BITS31	R	Clear (0) th bits.
30	USB2_PHY_CLK_REF_DIV 2	RW	PHY should get 19.2Mhz on the CLKCORE/XO. When the ref_clk is 38.4Mhz the clock divider in the HS PHY wrapper is enabled to divide the frequency by 2. Default value is '0' - ref_clk is not divided.
29:26	USB2_PHY_PLLPTUNE	RW	connects to port PLLPTUNE of the SYPS HS PHY- PLL Proportional path tune. Default value - 4'b1110

USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_PHY_CTRL (cont.)

Bits	Name	Type	Description
25:24	USB2_PHY_PLLITNUE	RW	connects to port PLLITUNE of the SYPS HS PHY- PLL integral path tune. Default value - 2'b00
23	USB2_PHY_VREGBYPASS	RW	connects to port VREGBYPASS of the SYPS HS PHY- This signal controls the VReg18 bypass mode. 1: the VReg18 is bypassed. 1.8V power supply needs to be provided on VDDH0. 0: the VReg18 is not bypassed. A 3.3V supply needs to be provided on VDDH0 and the internal 1.8V supply is generated by the VREG18. Default value - 1'b1
22	USB2_PHY_PLLBTUNE	RW	connects to port PLLBTUNE of the SYPS HS PHY- PLL Bandwidth adjustment, this adjustment doubles the bandwidth of the PLL. 1: Double bandwidth mode enabled 0: Design Default If PLLBTUNE = '1' is set, then FSEL[2:0] must be set to 3'b010 and a 24MHZ reference clock must be used. Default value - 1'b0
21	USB2_PHY_CLAMP_MPM_DPSE_DMSE_EN_N	RW	Clamp enable for DPSEHV and DMSEHV interrupts level shifters from USB VDD180 domain to VDDPAD MPM in usb2 phy wrapper. When set (1), VLS is active and These HV interrupts are translated from 1.8V domain to MPM Vdd domain. When clear (0), VLS is clamped to inactive state. Default value is 0. SW should set this bit before Entering Retention only if there is no PHY power collapse.
20	USB2_PHY_DMSE_INTEN	RW	Enable HV DMSE pin interrupt from phy. This interrupt translated with level shifter for wakeup from retention when VDDCX is nominal and TCXO is running.
19	USB2_PHY_DMSEHV_CLAMP_EN	RW	Clamp enable for DMSEHV interrupt level shifter from VDDPAD domain to VDDCX domain. When set (1), VLS is active and DMSEHV interrupt is translated from 1.8V domain to Vddcx domain. When clear (0), VLS is clamped to zero. Default value is 1. SW should clear this bit before power collapse of 1.8V domain of PHY.
18	USB2_PHY_ULPI_CLK_EN	RW	Enables the ulpi_clk from the PHY. Default value is 1- clock is enabled
17	USB2_PHY_REFCLKOUT_EN	RW	Enables the refclock_out from the Primary PHY (the refclkock_out of the primary PHY can be used as a reference clock of the secondary PHY. Default Value is 0
16	RESERVED_BIT16	R	Clear (0) this bit

USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_PHY_CTRL (cont.)

Bits	Name	Type	Description
15	USB2_PHY_USE_CLKCORE	RW	Configure whether the XO or the CLKCORE of the HS PHY should be used as a reference clock. When external clock (XO) is used as a ref_clk, USE_CLKCORE should be 0. When CLKCORE is used as the ref_clk, USE_CLKCORE=1. Default is '0' - using XO as ref_clk and not the CLKCORE
14	USB2_PHY_DPSE_INTEN	RW	Enable HV DPSE pin interrupt from phy. This interrupt translated with level shifter for wakeup from retention when VDDCX is nominal and TCXO is running.
13	USB2_PHY_DPSEHV_CLAMP_EN	RW	Clamp enable for DPSEHV interrupt level shifter from VDDPAD domain to VDDCX domain. When set (1), VLS is active and DPSEHV interrupt is translated from 1.8V domain to Vddcx domain. When clear (0), VLS is clamped to zero. Default value is 1. SW should clear this bit before power collapse of 1.8V domain of PHY.
12	USB2_PHY_IDHV_CLAMP_EN	RW	Clamp enable for IDHV interrupt level shifter from VDDPAD domain to VDDCX domain. When set (1), VLS is active and IDHV interrupt is translated from 1.8V domain to Vddcx domain. When clear (0), VLS is clamped high. Default value is 1. SW should clear this bit before power collapse of 1.8V domain of PHY.
11	USB2_PHY_OTGSESSVLDHV_CLAMP_EN	RW	Clamp enable for OTGSESSVLDHV interrupt level shifter from VDDPAD domain to VDDCX domain. When set (1), VLS is active and OTGSESSVLDHV interrupt is translated from 1.8V domain to Vddcx domain. When clear (0), VLS is clamped to zero. Default value is 1. SW should clear this bit before power collapse of 1.8V domain of PHY.
10	PHY_MPM_HV_CLAMP_EN	RW	Clamp enable for IDHV and OTGSESSVLDHV interrupts level shifters from USB VDD180 domain to VDDPAD MPM in usb2 phy wrapper. When set (1), VLS is active and These HV interrupts are translated from 1.8V domain to MPM Vdd domain. When clear (0), VLS is clamped to inactive state. Default value is 1. SW should clear this bit before power collapse of 1.8V domain of PHY.
9	USB2_PHY_OTGSESSVLDHV_INTEN	RW	Enable HV session valid interrupt from phy. This interrupt translated with level shifter for wakeup from retention when VDDCX is nominal and TCXO is running.
8	USB2_PHY_IDHV_INTEN	RW	Enable HV id pin interrupt from phy. This interrupt translated with level shifter for wakeup from retention when VDDCX is nominal and TCXO is running.
7	USB2_PHY_ULPI_POR	RW	Reset for ULPI PHY Wrapper and ULPI clock domain logic in usb2_phy_wrapper.
6:4	USB2_PHY_FSEL	RW	Reference Clock Frequency Select 011 (Assumes 19.2MHz default)

USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_PHY_CTRL (cont.)

Bits	Name	Type	Description
3	HOST_PORTCTRL_FORCE_SUSEN	RW	Chicken bit for CR-0000153908 - when this bit set the core will enter to low power mode when portctrl host sm enter to suspend mode. Default value is 1.
2	USB2_PHY_SIDDQ	RW	IDDQ Test Enable.
1	USB2_PHY_RETEN	RW	Retention mode enable/disable
0	USB2_PHY_POR	RW	Power-On-Reset (Analog configuration needs to happen before POR is set to 0)

0x078D9244 USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_GENERIC1**Type:** R**Clock:** CC_USB_OTG_HS_CLK**Reset State:** 0x00000000

The USB_OTG_HS_GENERIC1 register. reset is 0xFFFFFFFF for generics because the value of these registers changes per core instance. SW can read these registers to identify the real HW parameters of the core instance.

USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_GENERIC1

Bits	Name	Description
31:16	USB_HS_TX_DEPTH	TX_DEPTH is the number of words in the TX buffer. This number is calculated by the number of bytes allocated per Endpoint divided by 4, times the number of endpoints. In HS USB this value is usually 512 bytes per EP, and for FS ONLY USB it is usually 64 Bytes per EP.
15:0	USB_HS_RX_DEPTH	RX_DEPTH is the number of words in the RX buffer. This number is usually 256, but can support powers of 2 up to 4096.

0x078D9248 USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_GENERIC2**Type:** R**Clock:** CC_USB_OTG_HS_CLK**Reset State:** 0x00000000

The USB_OTG_HS_GENERIC2 register. reset is 0xFFFFFFFF for generics because the value of these registers changes per core instance. SW can read these registers to identify the real HW parameters of the core instance.

USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_GENERIC2

Bits	Name	Description
31:20	RESERVED_BITS31_19	
19	USE_HBM	USE_HBM generic specifies if HBM(host bam manager) block is supported in RTL. The HBM block allow AHB transfers through the BAM in host mode. '0' - BAM isn't used in host mode. '1' - BAM can be used in host mode.
18	USE_EXT_HSIC	USE_EXT_HSIC generic specifies if the USB core can use both external HSIC and ULPI phy's. the HSIC phy will be accessed through ulpi_1 bus. '0' - Existing design (per USE_HSIC generic). '1' - Both ULPI and HSIC phy's are supported (USE_HSIC generic is redundant).
17	RESERVED	Reserved
16	USE_FS_SIE	USE_FS_SIE generic specifies if FS serial interface logic is supported in RTL. '0' - FS serial interface logic is not supported. '1' - FS serial interface logic is supported
15	USE_SPS_AHB2AHB	USE_SPS_AHB2AHB generic specifies if AHB2AHB bridge is connected between BAM and SPS Fabric AHB. only applicable when USE_SPS = 1. 1 - the core has AHB2AHB bridge between BAM and SPS Fabric AHB, 0 - the BAM connected directly to SPS Fabric AHB
14	LPM_SUPPORT	The LPM_SUPPORT generic specifies if USB support Link Power Management. 1 - the core support Link Power Management, 0 - the core doesn't support Link Power Management
13:9	USB_HS_DEV_EP	The number of USB endpoints. Valid values for the number of Endpoints are 4, 8, and 16
8:3	MAX_PIPES	The number of simultaneous parallel pipes supported by the BAM. Supported values are 2 to 30.
2	USE_SPS	The USE_SPS generic specifies if Bam is connected to USB core, 1 - Bam is connected to USB core, 0 - USB core doesn't include BAM instance
1	USE_HSIC	The USE_HSIC generic specifies if an HSIC is connected to USB core. 1 - the core is connected to HSIC phy, 0 - the core is connected to ULPI phy
0	UTMI_PHY_SW_IF_EN	The UTMI_PHY_SW_IF_EN generic specifies if UTMI phy Register Interface is enabled. 1 - UTMI phy Register Interface is enabled, 0 - UTMI phy Register Interface is disabled

0x078D9250 USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_L1_EP_CTRL**Type:** RW**Clock:** CC_USB_OTG_HS_CLK**Reset State:** 0x0000FFFF

The USB_OTG_HS_L1_EP_CTRL register enables/disables transition to L1 and exit from L1 state when specific TX endpoints are primed.

USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_L1_EP_CTRL

Bits	Name	Description
31:16	TX_EP_PRIME_L1_EXIT	Those bit's enables Remote Wakeup in L1 state when SW starts Priming The specific Endpoint.
15:0	TX_EP_PRIME_L1_EN	Control bit's that enables/disables transition to L1 when the specific TX Endpoint is active.

0x078D9254 USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_L1_CONFIG

Type: RW

Clock: CC_USB_OTG_HS_CLK

Reset State: 0x00000000

The USB_OTG_HS_L1_CONFIG register is used to configure various L1 features.

USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_L1_CONFIG

Bits	Name	Description
31:13	RESERVED_BITS31_13	Clear (0) these bits.
12	L1_SUSP_SLEEP_SEL	Applicable only with USB2.0 phy. control bit that selects how the phy would enter ULPI low power mode, 0 - by writing to suspendM bit in func_ctrl register, 1 - by writing to sleepM bit in func_ctrl register. default value 0 - suspendM
11	PLL_PWR_DWN_EN	Control bit that enables/disables power down of 480 MHz PLL in L1 state. default value 0 -disable
10	PHY_LPM_EN	Control bit that enable/disables entering ULPI Low Power Mode in L1 state. default value 0 -disable
9	GATE_AHB_CLK_EN	Control bit that enable/disables clock request signaling for usb_ahb_clk in L1 state, currently this clock is not gated so this bit has no effect.
8	GATE_FS_XCVR_CLK_EN	Control bit that enable/disables clock gating of usb_fs_xcvr_clk in L1 state, Applicable only with HSIC phy. default value 0 -disable
7	GATE_SYS_CLK_EN	Control bit that enable/disables clock gating of usb_system_clk in L1 state. default value 0 -disable
6	GATE_XCVR_CLK_EN	Control bit that enable/disables power-down of 480 MHz PLL in L1 state, Applicable only with HSIC phy. default value 0 -disable
5	L1_REMOTE_WAKEUP_EN	Control bit that enables/disables Remote Wakeup in L1 state. When this bit is low, then Link Controller never initiates Remote Wakeup in L1 state. When this bit is high, Link Controller can initiate Remote Wakeup.

USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_L1_CONFIG (cont.)

Bits	Name	Description
4	LPM_EN	Control bit that enables/disables LPM support. When this bit is zero a full backward compatibility is ensured - no LPM support and no response for LPM Extended Transaction
3:0	PLL_TURNOFF_MIN_HIRD	Specifying a minimum expected HIRD value from Host that enables HW mechanism for turning off the 480 MHz PLL. The default value is 50us. Each value adds up 10 us to the previous value. the maximum value is 200us.

0x078D9260 USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_LPM_ATTRIBUTES**Type:** R**Clock:** CC_USB_OTG_HS_CLK**Reset State:** 0x00000000

The USB_OTG_HS_LPM_ATTRIBUTES register store the bmAttribute Field of the LPM transaction.

USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_LPM_ATTRIBUTES

Bits	Name	Description
31:5	RESERVED_BITS31_5	Clear (0) these bits.
4	BREMOTEWAKE	A value of one (1B) in this field enables the addressed device to wake the host upon any meaningful application-specific event (e.g. an interrupt for a device with one or more interrupt endpoints). A value of zero (0B) disables the device from initiating remote wake.
3:0	HIRD	Host Initiated Resume Duration. The HIRD value is a 4-bit encoded value. The value of 0000B is interpreted as 50 us. Each value up adds 75 us to the previous value. Based on this rule, the maximum value resume drive time is at encoding value 1111B which represents 1.2ms

0x078D9264 USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_LPM_TIMERS**Type:** RW**Clock:** CC_USB_OTG_HS_CLK**Reset State:** 0x00000078

Register that control different times during LinkPM, NOT Applicable with HSIC phy.

USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_LPM_TIMERS

Bits	Name	Description
11:0	LINKPM_RESUME_TIME	How much time the device make sure that the host drives resume in multiplies of 33.33 ns.Default reset value is 12'b000001111000 which means if the device will count less than 4us of resume it will disregard the resume signal. NOT Applicable with HSIC phy.

0x078D9270 USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_HW_VER**Type:** R**Clock:** CC_USB_OTG_HS_CLK**Reset State:** 0x00000000

The USB_OTG_HS_HW_VER register contains HW versioning information according to the guidelines of the IP Catalog methodology.

USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_HW_VER

Bits	Name	Description
31:28	MAJOR	Indicates different interface version. Major version changes are not backward compatible.
27:16	MINOR	Indicates expanded functionality. Minor version adds functionality while being backward compatibility for existing features.
15:0	STEP	Indicates a change in the HW which is not intended to impact SW compatibility.

0x078D9278 USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_PHY_SEC_CTRL**Type:** RW**Clock:** CC_USB_OTG_HS_CLK**Reset State:** 0x000C3C32

The USB_OTG_HS_PHY_SEC_CTRL register is used to configure various features in the secondary Synopsys 28nm PHY.

USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_PHY_SEC_CTRL

Bits	Name	Type	Description
31:22	RESERVED_BITS31_22	R	Clear (0) these bits.

USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_PHY_SEC_CTRL (cont.)

Bits	Name	Type	Description
21	USB2_PHY_CLAMP_MPM_DPSE_DMSE_EN_N_2	RW	Clamp enable for DPSEHV and DMSEHV interrupts level shifters from USB VDD180 domain to VDDPAD MPM in usb2 phy wrapper. When set (1), VLS is active and These HV interrupts are translated from 1.8V domain to MPM Vdd domain. When clear (0), VLS is clamped to inactive state. Default value is 0. SW should set this bit before Entering Retention only if there is no PHY power collapse.
20	USB2_PHY_DMSE_INTEN_2	RW	Enable HV DMSE pin interrupt from phy. This interrupt translated with level shifter for wakeup from retention when VDDCX is nominal and TCXO is running.
19	USB2_PHY_DMSEHV_CLAMP_EN_2	RW	Clamp enable for DMSEHV interrupt level shifter from VDDPAD domain to VDDCX domain. When set (1), VLS is active and DMSEHV interrupt is translated from 1.8V domain to Vddcx domain. When clear (0), VLS is clamped to zero. Default value is 1. SW should clear this bit before power collapse of 1.8V domain of PHY.
18	USB2_PHY_ULPI_CLK_EN_2	RW	Enables the ulpi_clk from the PHY. Default value is 1- clock is enabled
17	RESERVED_BIT17	RW	Clear (0) this bit
16	USB2_PHY_HS2_DIG_CLAMP_N_2	RW	Clamping the Low Voltage interface between DVDD of the PHY and Vddcx (Reset value is 0 clamp)
15	USB2_PHY_USE_CLKCORE_E_2	RW	Configure whether the XO or the CLKCORE of the HS PHY should be used as a reference clock (Default Value is 0 which means Use the XO)
14	USB2_PHY_DPSE_INTEN_2	RW	Enable HV DPSE pin interrupt from phy. This interrupt translated with level shifter for wakeup from retention when VDDCX is nominal and TCXO is running.
13	USB2_PHY_DPSEHV_CLAMP_EN_2	RW	Clamp enable for DPSEHV interrupt level shifter from VDDPAD domain to VDDCX domain. When set (1), VLS is active and DPSEHV interrupt is translated from 1.8V domain to Vddcx domain. When clear (0), VLS is clamped to zero. Default value is 1. SW should clear this bit before power collapse of 1.8V domain of PHY.
12	USB2_PHY_IDHV_CLAMP_EN_2	RW	Clamp enable for IDHV interrupt level shifter from VDDPAD domain to VDDCX domain. When set (1), VLS is active and IDHV interrupt is translated from 1.8V domain to Vddcx domain. When clear (0), VLS is clamped high. Default value is 1. SW should clear this bit before power collapse of 1.8V domain of PHY.

USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_PHY_SEC_CTRL (cont.)

Bits	Name	Type	Description
11	USB2_PHY_OTGSESSVLDHV_CLAMP_EN_2	RW	Clamp enable for OTGSESSVLDHV interrupt level shifter from USB VDD180 domain to VDDCX domain. When set (1), VLS is active and OTGSESSVLDHV interrupt is translated from 1.8V domain to Vddcx domain. When clear (0), VLS is clamped to zero. Default value is 1. SW should clear this bit before power collapse of 1.8V domain of PHY.
10	PHY_MPM_HV_CLAMP_EN_2	RW	Clamp enable for IDHV and OTGSESSVLDHV interrupts level shifters from USB VDD180 domain to VDDPAD MPM in usb2 phy wrapper. When set (1), VLS is active and these HV interrupts are translated from 1.8V domain to MPM Vdd domain. When clear (0), VLS is clamped to inactive state. Default value is 1. SW should clear this bit before power collapse of 1.8V domain of PHY.
9	USB2_PHY_OTGSESSVLDHV_INTEN_2	RW	Enable HV session valid interrupt from phy. This interrupt translated with level shifter for wakeup from retention when VDDCX is nominal and TCXO is running.
8	USB2_PHY_IDHV_INTEN_2	RW	Enable HV id pin interrupt from phy. This interrupt translated with level shifter for wakeup from retention when VDDCX is nominal and TCXO is running.
7	USB2_PHY_ULPI_POR_2	RW	Reset for ULPI PHY Wrapper and ULPI clock domain logic in usb2_phy_wrapper.
6:4	USB2_PHY_FSEL_2	RW	Reference Clock Frequency Select 011 (Assumes 19.2MHz default)
3	RESERVED_BIT	R	Clear this bit
2	USB2_PHY_SIDDQ_2	RW	IDDQ Test Enable.
1	USB2_PHY_RETEN_2	RW	Retention mode enable/disable
0	USB2_PHY_POR_2	RW	Power-On-Reset (Analog configuration needs to happen before POR is set to 0)

0x078D9340 USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_BTO_TIME_HS**Type:** RW**Clock:** CC_USB_OTG_HS_CLK**Reset State:** 0x000B40A2

Register that controls the bus timeout value For HS Hostdevice

USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_BTO_TIME_HS

Bits	Name	Type	Description
31:22	RESERVED_BITS31_22	R	

USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_BTO_TIME_HS (cont.)

Bits	Name	Type	Description
21:11	BTO_TIME_HS_HST	RW	the bus timeout value for HS Host in multiplies of 16 ns. Default reset value is 11'b00010100010 which means 2.7 us (1296 HS bit times).
10:0	BTO_TIME_HS_DEV	RW	the bus timeout value for HS Device in multiplies of 16 ns. Default reset value is 11'b00101101000 which means 6 us (2880 HS bit times).

0x078D9344 USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_BTO_TIME_FS_LS**Type:** RW**Clock:** CC_USB_OTG_HS_CLK**Reset State:** 0x002B20CC

Register that controls the bus timeout value For FSS

USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_BTO_TIME_FS_LS

Bits	Name	Type	Description
31:22	RESERVED_BITS31_22	R	
21:11	BTO_TIME_LS	RW	the bus timeout value for LS Device in multiplies of 16 ns. Default reset value is 11'b10101100100 which means 22.7 us (35 LS bit times).
10:0	BTO_TIME_FS	RW	the bus timeout value for FS Host/Device in multiplies of 16 ns. Default reset value is 11'b00011001100 which means 3.33 us (41 FS bit times).

0x078D9348 USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_QSB_VALUES**Type:** RW**Clock:** CC_USB_OTG_HS_CLK**Reset State:** 0x00000028

configurable QSB values

USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_QSB_VALUES

Bits	Name	Description
6	MEMTYPE_VALUE_EN	Enabling the hprot values
5:4	MEMTYPE_FOR_RDATA	hprot[3:2] for read data. should be 2 by default so memtype is 2 (cachable)
3:2	MEMTYPE_FOR_WDATA	hprot[3:2] for write data during transaction or block. should be 2 by default so memtype is 2 (cachable)

USB2_HSIC_USB_OTG_HS_BASE_USB_OTG_HS_QSB_VALUES (cont.)

Bits	Name	Description
1:0	MEMTYPE_FOR_NON_BUF_ACCESS	hprot[3:2] for last write data. should be 0 by default so memtype is 0 (strongly ordered)

21.2 USB2_HSIC_USB_OTG_HS_BAM**21.2.1 USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM****0x078C4000 USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_CTRL****Type:** RW**Clock:** bam_clk**Reset State:** 0x00020000

BAM Control register allows global controls for the BAM.

USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_CTRL

Bits	Name	Description
20	BAM_MESS_ONLY_CANCEL_WB	When active (1) - BAM will DROP former descriptor write-backing and Interrupt firing when former descriptor has been closed (accumulated=acknowledged) and ack_on_success with EOT and size 0 arrived (messaging only). When legacy mode is used (0) - This said a case then the former descriptor has been closed (accumulated=acknowledged) and ack_on_success with EOT and size 0 arrived (messaging only). In this case the FIFO-pointer has been already advanced before the ack_on_success arrives and therefore we do write-back to former descriptor and fire interrupt. 1'b1 - Drop WB and Interrupt 1'b0 - legacy.
19	CACHE_MISS_ERR_RESP_EN	When set to '1', upon local ahb access results with cache miss, the bam_ndp will not stall the bus, and finish the access with error response. This bit is relevant for BAM_NDP only.
18:17	LOCAL_CLK_GATING	These Bits enables power saving by using a local clock gating cell. Bit 17: 1'b1 - CGC is on, so that the clock is controlled by the HW - this is the reset value. 1'b0 - CGC is off so that the clock is free running Bit 18: Reserved bit for future improvements.

USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_CTRL (cont.)

Bits	Name	Description
16	IBC_DISABLE	This Bit enables power saving by disabling the inactivity clock timer/counter. This is useful when: 1) The BAM is Enabled but idling for long periods. 2) The BAM is Enabled and running but the inactivity timers/counters are not required by the SW for operating the BAM. When the BAM is Disabled, it automatically shuts down those timers to save power. 1'b1 - Enable Power Saving Mode 1'b0 - Disable Power Saving Mode
15	BAM_CACHED_DESC_STORE	This Bit enables storage of a cached Descriptor into RAM when doing Context Switch and loading it back when returning to this pipe. This is relevant for System Modes Only. BAM to BAM Consumer mode doesn't have this option. Enabling this makes the Context Switch process slightly longer but reduces AHB Descriptor Fetch time later on. It also reduces the number of descriptor fetches done by the BAM if BAM is treating more than 2 pipes. 1'b1 - Enabled 1'b0 - Disabled Available in BAM only
14:13	BAM_DESC_CACHE_SEL	This Selector chooses which Descriptor Caching mode will be used. This is relevant for System Producer or Consumer modes only. 2'b00 - Cache when current descriptor has less than 64 bytes left. 2'b01 - Cache when current descriptor has less than 128 bytes left. 2'b10 - Immediate Auto cache - cache will happen whenever there are no descriptors cached. This might cause unnecessary descriptors reads when switching between more than 2 pipes. It is the most effective when each pipe processes more bytes than there are in an average descriptor (small descriptors, big block size), or when working with 2 or less pipes. 2'b11 - Cache when descriptor is needed only. This results in 1 descriptor present in cache at most. Available in BAM only
11:5	RESERVED	Reserved
4	BAM_EN_ACCUM	When Enabled, BAM will accumulate data written by the peripheral in Direct Mode into INCR8 Bursts. This is an optimization feature for Producer Direct Mode cases. 1'b1 - Enabled 1'b0 - Disabled Available in BAM only

USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_CTRL (cont.)

Bits	Name	Description
1	BAM_EN	After reset the BAM wakes up in Disabled Mode. While Disabled, BAM operates in Legacy mode (all the transfers from the peripheral go around the BAM as if it doesn't exist) Software Enables this bit to allow BAM operation. This bit doesn't affect the PIPE enable bit in BAM_P_CTRL register. The only possibility to Disable the BAM after it has been enabled is by Hardware or Software reset. 1'b1 - Enabled 1'b0 - Disabled
0	BAM_SW_RST	This will reset the BAM & all Pipes. Software may use this to reset the BAM. As long as the bit remains high the BAM remains in reset state. Software should clear this bit 1'b1 - Reset state 1'b0 - Normal state

0x078C4040 USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_TIMER**Type:** R**Clock:** bam_clk**Reset State:** 0x00000000

This register counts the idle time of the BAM (all the pipes are in idle state).

USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_TIMER

Bits	Name	Description
15:0	TIMER	Inactivity timer counter. 16 bits = 64k*0.1ms ~ 6 seconds

0x078C4044 USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_TIMER_CTRL**Type:** RW**Clock:** bam_clk**Reset State:** 0x00000000

This register sets various controls and the threshold of the inactivity for interrupt triggering after long idle time.

The register can be written and read only when INACTIVITY_TIMERS_SUPPORTED generic equals to 1.

USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_TIMER_CTRL

Bits	Name	Description
31	TIMER_RST	To begin using the inactivity timer the SW should write the threshold and write 1 to this bit. writing the value 0 will hold the timer reset until value 1 is written again. Then the timer will start down counting from the TIMER_THRESOLD value 1'b1 - Active 1'b0 - Reset
30	TIMER_RUN	Will be used along with TIMER_MODE Not implemented at this time. Set to zero(0)
29	TIMER_MODE	Not implemented at this time. Set to zero(0)
15:0	TIMER_TRSHLD	Inactivity timer threshold for interrupt generation.

0x078C4008 USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_DESC_CNT_TRSHLD**Type:** RW**Clock:** bam_clk**Reset State:** 0x00000001

This Register holds a threshold value for the counter summing the Size of the Descriptors Provided. This value is Global for all pipes but it is relevant for System Producer BAM mode and Both Consumer modes, where Descriptors summing takes place in order to provide the information to the peripheral via pipe bytes free (producer) or pipe bytes available interfaces.

USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_DESC_CNT_TRSHLD

Bits	Name	Description
15:0	CNT_TRSHLD	Threshold value. The maximum allowed value is 32kByte. The minimum allowed value is 1 (value 0 is not allowed). Available in BAM only

0x078C4014 USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_IRQ_STTS**Type:** R**Clock:** bam_clk**Reset State:** 0x00000000

Interrupt status register describing the reason for the ongoing interrupt. This is an informative only register and therefore read only.

Clearing the interrupt bits is done by writing to BAM_IRQ_CLR register.

USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_IRQ_STTS

Bits	Name	Description
4	BAM_TIMER_IRQ	Inactivity timer has reached its threshold value Interrupt
3	BAM_EMPTY_IRQ	The interrupt indicates the BAMs buffer is empty. This bit can become high only when BAM_DATA_ERASE or BAM_DATA_FLUSH is high in BAM_TEST_BUS_SEL register.
2	BAM_ERROR_IRQ	This Indicates a Fatal Error has happened in the BAM. Currently this might happen due to an UnSuccessful Pipe Reset operation.
1	BAM_HRESP_ERR_IRQ	This interrupt Indicates that an Erroneous HResponse has been received by the AHB Master. Additional Information about the error may be read at BAM_AHB_MASTER_ERR_* Registers. Clearing this interrupt also clears the BAM_AHB_MASTER_ERR_* Registers. Available in BAM only

0x078C4018 USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_IRQ_CLR**Type:** W**Clock:** bam_clk**Reset State:** 0x00000000

Writing to this register causes the interrupt to clear.

USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_IRQ_CLR

Bits	Name	Description
4	BAM_TIMER_CLR	Clear Inactivity timer interrupt by writing this bit 1'b1 - Clear 1'b0 - Unchanged
3	BAM_EMPTY_CLR	1'b1 - Clear 1'b0 - Unchanged Available in BAM-Lite only
2	BAM_ERROR_CLR	1'b1 - Clear 1'b0 - Unchanged
1	BAM_HRESP_ERR_CLR	1'b1 - Clear 1'b0 - Unchanged Available in BAM only

0x078C401C USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_IRQ_EN**Type:** RW**Clock:** bam_clk**Reset State:** 0x00000000

Enable bits for the Interrupts in the BAM Interrupt Status register.

USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_IRQ_EN

Bits	Name	Description
4	BAM_TIMER_EN	Enables Inactivity timer interrupt by writing this bit 1'b1 - Enable 1'b0 - Disable
3	BAM_EMPTY_EN	1'b1 - Enable 1'b0 - Disable Available in BAM-Lite only
2	BAM_ERROR_EN	1'b1 - Enable 1'b0 - Disable
1	BAM_HRESP_ERR_EN	1'b1 - Enable 1'b0 - Disable Available in BAM only

0x078C407C USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_CNFG_BITS**Type:** RW**Clock:** bam_clk**Reset State:** 0x00000000

This Register holds the BAM configuration bits for bug fixes. It is highly recommended to follow the directions for each bit and set it accordingly.

USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_CNFG_BITS

Bits	Name	Description
31	AOS_OVERFLOW_PRVNT	This bit enables overflow prevention when Sideband block accumulates ack_on_success updates from peripherals to values higher than 0x7fff. '1' - enable ack-on_success overflow prevention. If current ack_on_success update causes overflow accept_ack_on_success_toggle will not approve the update until SB accumulator is cleared. '0' - disable ack-on_success overflow prevention. Available in BAM-NDP only. Default value is '0' (Legacy). Recommended value is '1'.

USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_CNFG_BITS (cont.)

Bits	Name	Description
30	MULTIPLE_EVENTS_DESC_AVAIL_EN	This is an enable bit for supporting multiple events received by a Producer BAM2BAM and makes BAM-NDP to notify a correct desc_cache_avail bus. It fixed CR: QCTDD00209370.
29	MULTIPLE_EVENTS_SIZE_EN	This is an enable bit for supporting multiple events received by a Producer BAM2BAM and makes BAM-NDP delivering correct descriptor sizes on local_ahb bit. It fixed CR: QCTDD00209379.
28	BAM_ZLT_W_CD_SUPPORT	Recommended value: 1 Enables support of ZLT (i.e. a zero-sized descriptor) before a command-descriptor. Specifically it supports a case on a pipe without any data-descriptors but with ZLT descriptors and Command-descriptors only. This allows putting NWD bit in the ZLT descriptor (instead of in the CMD) which results in better performance than using the NWD in the command-descriptor. Available in BAM-Lite Only.
27	BAM_CD_ENABLE	Recommended value: 1 Enables the command descriptor feature.
26	BAM_AU_ACCUMED	Recommended value: 1 This bit fixes a bug in the Ack Update state machine, where an overflow happened while counting descriptors and reaching more than 64kB of calculated sizes. 1'b1 - Enable Fix 1'b0 - Disable Available in BAM/BAM-NDP only
25	BAM_PSM_P_HD_DATA	Recommended value: 1 This bit allows pipe state machine to ignore retransmission requests if a pipe has just been initialized and process those as a regular fetch request. (consumer modes only). When this bit disabled, BAM could fetch descriptors for a pipe which was reset and no descriptors were added yet, if a retransmission request followed after the reset.. 1'b1 - Enable Fix 1'b0 - Disable Available in BAM only
24	BAM_REG_P_EN	Recommended value: 1 This bit fixes the pipe configuration signals mux for the current active pipe in 2 pipes BAM. When disabled, internal state machines might get into enabled states while the pipe is disabled. This would typically happen after pipe reset. 1'b1 - Enable 1'b0 - Disable Available in BAM/BAM-NDP only

USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_CNFG_BITS (cont.)

Bits	Name	Description
23	BAM_WB_DSC_AVL_P_RST	Recommended value: 1 This bit allows the Writeback state machine to reset the vector indicating there're available descriptors when a pipe reset occurs. If disabled, BAM might fetch descriptors after resetting and reconfiguring a pipe, even though no Event (descriptors) was provided.. 1'b1 - Enable 1'b0 - Disable Available in BAM only
22	BAM_WB_RETR_SVPNT	Recommended value: 1 This bit fixes a bug where a pipe which was reset, still stored its retransmission save point, but into the illegal's pipe address space, thus hurting the last pipe of the BAM if the BAM had a total 4, 8 or 16 pipes. This is relevant for Producer to System modes only. (CR-0000151585) 1'b1 - Enabled 1'b0 - Disable Available in BAM/BAM-NDP only
21	BAM_WB_CSW_ACK_IDL	Recommended value: 1 This bit allows the Writeback state machine to switch into idle state prior to visiting disabled state. This is needed when context switching from mode X to another pipe of mode X is well. This is required to fix a bug in the 2 pipes BAM. 1'b1 - Enable 1'b0 - Disable Available in BAM/BAM-NDP only
20	BAM_WB_BLK_CSW	Recommended value: 1 When Enabled, this bit does not allow context switch to happen in the Writeback state machine until it has created a descriptor. This is relevant when the descriptor fifo is becoming full and there's no space to create a descriptor, while another pipe is context switching. This might result in the descriptor not to be created ever, if it was the last one for that pipe. Relevant for Producer BAM-to-BAM mode only. 1'b1 - Enable 1'b0 - Disable Available in BAM only
19	BAM_WB_P_RES	Recommended value: 1 This bit allows to reset all internal registers of the Writeback state machine when performing pipe reset. 1'b1 - 1'b0 - Disable Available in BAM only

USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_CNFG_BITS (cont.)

Bits	Name	Description
18	BAM_SI_P_RES	Recommended value: 1 This bit allows to reset all internal registers of the Sideband Inform state machine when performing pipe reset. 1'b1 - Enable 1'b0 - Disable Available in BAM only
17	BAM_AU_P_RES	Recommended value: 1 This bit allows to reset all internal registers of the Ack Update state machine when performing pipe reset. 1'b1 - Enable 1'b0 - Disable Available in BAM only
16	BAM_PSM_P_RES	Recommended value: 1 This bit allows to reset all internal registers of the Pipe state machine when performing pipe reset.. 1'b1 - Enable 1'b0 - Disable Available in BAM only
15	BAM_PSM_CSW_REQ	Recommended value: 1 This bit forces the context switch request from pipe state machine to RAM controller not to last longer than the slave requested. (2 Pipes BAM bug fix). 1'b1 - Enable 1'b0 - Disable Available in BAM only
14	BAM_SB_CLK_REQ	Recommended value: 1 This bit allows the clock request from the sideband block to propagate into the BAM's common clock request. 1'b1 - Propagate Sideband Clock Request 1'b0 - Disable Available in BAM only
13	BAM_IBC_DISABLE	Recommended value: 1 This bit helps to save power by allowing the BAM to keep the inactivity base counter in reset when BAM is disabled or when SW configures IBC_DISABLE bit high. Irrelevant field for BAM-NDP. 1'b1 - Enable Power Saving 1'b0 - Disable Power Saving

USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_CNFG_BITS (cont.)

Bits	Name	Description
12	BAM_NO_EXT_P_RST	Recommended value: 1 This bit allows the BAM / BAM Lite to ignore the externally connected blocks (eDML) when doing pipe reset. The BAM, once instructed to pipe reset, first thing lets the externally connected block know a reset is needed. Then it waits for the externally connected block to Acknowledge it is ready for the pipe reset (meaning it doesn't push any data for the reseted pipe) and then the BAM Lite completes the pipe reset operation internally. When disabled, the BAM doesn't require any Acknowledge from the external block to perform pipe reset. 1'b1 - Enable external block pipe reset 1'b0 - Disable - ignore external block pipe reset
11	BAM_FULL_PIPE	Recommended value: 0 This enables the BAM support for a BAM to BAM Producer which insists to write to a full pipe. When 0, BAM might issue data overflow if producers write to a full pipe. When 1 BAM will not allow this and lower HReady when peripheral tries to do so. Once space is freed in the pipe, Hready will rise and the flow will continue. 1'b1 - Enable 1'b0 - Disable Available in BAM only
3	RESERVED	Reserved
2	BAM_PIPE_CNFG	Recommended value: 1 Pipe SM upgrade for writing EOT bit to the previous descriptor. It's invoked only when EOB arrives in the end of a descriptor. It is highly recommended to set this bit high. Leaving it low might cause incorrect Pipe Bytes Free value reported to peripheral in rare cases. 1'b1 - Enable 1'b0 - Disable Available in BAM only
1:0	RESERVED	Reserved

0x078C5000 USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_REVISION**Type:** R**Clock:** bam_clk**Reset State:** Undefined

This Register holds a hard coded revision number of the BAM RTL. This corresponds to the BAM target number in the IMS. Also additional BAM settings are reflected here.

USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_REVISION

Bits	Name	Description
31:24	INACTIV_TMR_BASE	This indicates the width of the inactivity timers base counter.
23	CMD_DESC_EN	This indicates BAM has Command Descriptor feature enabled in HW.
22:21	DESC_CACHE_DEPTH	This field indicates the Per-Pipe-Descriptor-Cache. could be 1-4: 2'b00 - Descriptor Cache Depth of 1. 2'b01 - Descriptor Cache Depth of 2. 2'b10 - Descriptor Cache Depth of 3. 2'b11 - Descriptor Cache Depth of 4. Valid only for BAM_NDP.
20	NUM_INACTIV_TMRS	This bit is checked only if INACTIV_TMRS_EXST = 1. This bit indicates how many inactivity timers are implemented according to value of INACTIVITY_TIMERS_SUPPORTED generic: 1'b1 - One timer for global BAM activity (INACTIVITY_TIMERS_SUPPORTED = 1) 1'b0 - MAX_PIPES timers for each pipe activity (INACTIVITY_TIMERS_SUPPORTED = 2)
19	INACTIV_TMRS_EXST	This indicates BAM has inactivity timers. 1'b1 - Timers available (INACTIVITY_TIMERS_SUPPORTED = 1 or 2) 1'b0 - No timers available (INACTIVITY_TIMERS_SUPPORTED = 0)
18	HIGH_FREQUENCY_BAM	This indicates BAM has Double Sampling on the Ack On Success interface. 1'b1 - Ack On Success double sampled 1'b0 - No double sampling of Ack On Success
17	BAM_HAS_NO_BYPASS	This field indicates BAM has no bypass on the AHB Data bus, from the Peripheral AHB Master to the fabric. When Bypass is available, it is the default routing when BAM is disabled. 1'b1 - No Bypass 1'b0 - Bypass Exists
16	RESERVED	Reserved
15	RESERVED	Reserved
14	AXI_ACTIVE	This field indicates BAM_NDP uses internal AXI bridge on the master port.

USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_REVISION (cont.)

Bits	Name	Description
13:12	CE_BUFFER_SIZE	This field indicates the size (in Data words) of the buffer which stores the command elements. Each command element includes 4 words. 2'b00 - 4 Words (one command element). 2'b10 - 8 Words (2 command elements). 2'b11 - 16 Words (4 command elements). Valid Only if CMD_DESC_EN = 1'b1.
11:8	NUM_EES	This indicates the Number of Interrupt Lines (Execution Environments) supported by the BAM. When 1 EE is supported, only the BAM_IRQ_SRC*_EE0 registers exist. When 4 EEs are supported, all BAM_IRQ_SRC*_EE _n registers exist for n=[0..3].
7:0	RESERVED	Reserved

0x078C5008 USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_NUM_PIPES**Type:** R**Clock:** bam_clk**Reset State:** Undefined

This Register holds a the hard coded number of pipes in the BAM. Since each BAM may have a different number of pipes instantiated in HW, this register allows the SW to know this parameter. The reset state value doesn't have any meaning in this register since it is a constant.

USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_NUM_PIPES

Bits	Name	Description
31:24	RESERVED	Reserved
23:16	RESERVED	Reserved
15:14	BAM_DATA_ADDR_BUS_WIDTH	This field indicates BAM data bus address width: 2'b00 - 32bit address width 2'b01 - 36bit address width 2'b1x - Reserved for future widening
7:0	BAM_NUM_PIPES	This field contains the number of pipes available in this BAM

0x078C5024 USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_AHB_MASTER_ERR_CTRL**Type:** R**Clock:** bam_clk**Reset State:** 0x00000000

This Register holds the information about the data error in the AHB Buffer during bus Access.

The value stored is the value of the first error occurred.

Clear the BAM_HRESP_ERR_IRQ interrupt to clear the recorded error.

Available in BAM only

USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_AHB_MASTER_ERR_CTRL

Bits	Name	Description
22:18	BAM_ERR_HVMID	HVMID
17	BAM_ERR_DIRECT_MODE	DIRECT MODE
16:12	BAM_ERR_HCID	HCID
11:8	BAM_ERR_HPROT	HPROT
7:5	BAM_ERR_HBURST	HBURST
4:3	BAM_ERR_HSIZE	HSIZE
2	BAM_ERR_HWRITE	HWRITE
1:0	BAM_ERR_HTRANS	HTRANS

0x078C5028 USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_AHB_MASTER_ERR_ADDR**Type:** R**Clock:** bam_clk**Reset State:** 0x00000000

This Register holds the information about the data error in the AHB Buffer during bus Access.

The value stored is the value of the first error occurred.

Clear the BAM_HRESP_ERR_IRQ interrupt to clear the recorded error.

Available in BAM only

USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_AHB_MASTER_ERR_ADDR

Bits	Name	Description
31:0	BAM_ERR_ADDR	HADDR

0x078C502C USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_AHB_MASTER_ERR_DATA**Type:** R**Clock:** bam_clk**Reset State:** 0x00000000

This Register holds the information about the data error in the AHB Buffer during bus Access.

The value stored is the value of the first error occurred.

Clear the BAM_HRESP_ERR_IRQ interrupt to clear the recorded error.

Available in BAM only

USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_AHB_MASTER_ERR_DATA

Bits	Name	Description
31:0	BAM_ERR_DATA	HDATA

0x078C5100 USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_AHB_MASTER_ERR_ADDR_LSB**Type:** R**Clock:** bam_clk**Reset State:** 0x00000000

This Register holds the information about the data error in the AHB Buffer during bus Access.

The value stored is the value of the first error occurred.

Clear the BAM_HRESP_ERR_IRQ interrupt to clear the recorded error.

Available in BAM only

Relevant only on 36 bit address BAM.

USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_AHB_MASTER_ERR_ADDR_LSB

Bits	Name	Description
31:0	BAM_ERR_ADDR	32 LSB of HADDR

0x078C5104 USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_AHB_MASTER_ERR_ADDR_MSB**Type:** R**Clock:** bam_clk**Reset State:** 0x00000000

This Register holds the information about the data error in the AHB Buffer during bus Access.

The value stored is the value of the first error occurred.

Clear the BAM_HRESP_ERR_IRQ interrupt to clear the recorded error.

Available in BAM only

Relevant only on 36 bit address BAM.

USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_AHB_MASTER_ERR_ADDR_MSB

Bits	Name	Description
3:0	BAM_ERR_ADDR	4 MSB of Address (bits 35 to 32) of HADDR

0x078C6000 USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_TRUST_REG

Type: RW

Clock: bam_clk

Reset State: 0x00000000

This Register holds the BAM Trusted configurations.

This register cannot be reset by any SW reset. Only HW reset applies.

USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_TRUST_REG

Bits	Name	Description
13	LOCK_EE_CTRL	This bit controls if the EE setting defined in the TRUST registers will be taken into account for the pipe lock grouping decoding. 1'b0 - Only P_LOCK_GROUP is checked for pipe locking. 1'b1 - Both BAM_P_EE and P_LOCK_GROUP are checked for pipe locking.
12:8	RESERVED	Reserved
7	BAM_RST_BLOCK	When enabled, the BAM Global SW reset will not be available for usage. This is in order to deny Global SW reset requests by the Global security group. 1'b1 - Enable 1'b0 - Disable

USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_TRUST_REG (cont.)

Bits	Name	Description
2:0	BAM_EE	This Field Indicates the EE # (0-7) to which the BAM Interrupt belongs to. The pipe interrupts will fire on the EE # set at BAM_P_TRUST_REGn registers. The number of supported EEs is reflect in the NUM_EES field in the BAM_REVISION register. 3'b000 - EE0 3'b001 - EE1 3'b010 - EE2 3'b011 - EE3 3'b100 - EE4 3'b101 - EE5 3'b110 - EE6 3'b111 - EE7

**0x078C6020+ USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_P_TRUST_REGn, n=[0..30]
0x4*n**

Type: RW

Clock: bam_clk

Reset State: 0x00000000

This register holds the BAM Trusted configurations.

This register cannot be reset by any SW reset. Only HW reset applies.

USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_P_TRUST_REGn

Bits	Name	Description
12:8	RESERVED	Reserved
7:3	BAM_P_SUP_GROUP	Super-Group is the upper division of the pipes compared to the pipe-group division. When any pipe locks or unlocks the BAM, all the pipes which are in a different super-group will NOT be affected. Only the pipes within the same super-group of the locker are affected. Usually pipes which belongs to different peripherals which share the same BAM, would be in a different super-groups

USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_P_TRUST_REGn (cont.)

Bits	Name	Description
2:0	BAM_P_EE	This Field Indicates the EE # (0-7) to which the Pipe Interrupt belongs to. The BAM interrupt will fire to the EE # set at BAM_P_TRUST_REGn registers. The number of supported EEs is reflect in the NUM_EES field in the BAM_REVISION register. 3'b000 - EE0 3'b001 - EE1 3'b010 - EE2 3'b011 - EE3 3'b100 - EE4 3'b101 - EE5 3'b110 - EE6 3'b111 - EE7

**0x078C7000+ USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_IRQ_SRCS_EEn, n=[0..7]
4096*n****Type:** R**Clock:** bam_clk**Reset State:** 0x00000000

This Interrupt Status Register provides a single interrupt bit per each pipe in the BAM and a single bit for the BAM (not pipe related) interrupt. This register is read only and provides fast indication for the software to decide which pipe triggered the interrupt. After checking this register, SW reads the BAM_P_IRQ_STTSn register for the pipe interrupt reason and BAM_IRQ_STTS for the BAM interrupt reason.

This register has an alias - BAM_IRQ_SRCS register.

Registers with n=[1..7] exist only when the BAM supports multiple EEs (NUM_EES field in the BAM_REVISION register).

USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_IRQ_SRCS_EEn

Bits	Name	Description
31	BAM_IRQ	BAM Global interrupt
30:0	P_IRQ	An interrupt bit per each pipe in the BAM.

**0x078C7004+ USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_IRQ_SRCS_MSK_EEn, n=[0..7]
4096*n****Type:** RW**Clock:** bam_clk**Reset State:** 0x00000000

Interrupt Sources Mask register allows to mask the interrupts by their source, whether BAM or a specific pipe.

This register has an alias - BAM_IRQ_SRCS_MSK register.

Registers with n=[1..7] exist only when the BAM supports multiple EEs (NUM_EES field in the BAM_REVISION register).

USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_IRQ_SRCS_MSK_EEn

Bits	Name	Description
31	BAM_IRQ_MSK	BAM Global Interrupt Mask 0x1: ENABLE_BAM_INTERRUPT (Enable BAM interrupt) 0x0: DISABLE_BAM_INTERRUPT (Disable BAM interrupt)
30:0	P_IRQ_MSK	An Interrupt Mask bit per each pipe in the BAM. 0x1: ENABLE_PIPE_INTERRUPT (Enable Pipe interrupt) 0x0: DISABLE_PIPE_INTERRUPT (Disable Pipe interrupt)

0x078C7008+ USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_IRQ_SRCS_UNMASKED_EEn, 4096*n, n=[0..7]

Type: R

Clock: bam_clk

Reset State: 0x00000000

The register shows the interrupts sources like (BAM_IRQ_SRCS) although the interrupts from the BAM and the pipes are not masked on the lower level.

This register has an alias - BAM_IRQ_SRCS_UNMASKED register.

Registers with n=[1..7] exist only when the BAM supports multiple EEs (NUM_EES field in the BAM_REVISION register).

USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_IRQ_SRCS_UNMASKED_EEn

Bits	Name	Description
31	BAM_IRQ_UNMASKED	BAM Global Interrupt after Mask
30:0	P_IRQ_UNMASKED	An Interrupt bit per each pipe in the BAM, after Mask

0x078C700C+ USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_PIPE_ATTR_EEn, n=[0..7] 4096*n

Type: R

Clock: bam_clk

Reset State: 0x00000000

The register shows the pipes attributed to a specific EE, and an indication if BAM is enabled.

The reset value written above is true only if EE>0. For BAM_PIPE_ATTR_EE0 the default value is: Bits [MAX_PIPES-1:0] = 1 , Bits [31: MAX_PIPES] = 0 .

USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_PIPE_ATTR_EEn

Bits	Name	Description
31	BAM_ENABLED	1'b1 - BAM is Enabled. 1'b0 - BAM Disabled.
30:0	P_ATTR	If bit 'i' == 1'b1 - Pipe 'i' is attributed to this EE.

0x078C7010 USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_IRQ_SRCS

Type: R

Clock: bam_clk

Reset State: 0x00000000

This Interrupt Status Register provides a single interrupt bit per each pipe in the BAM and a single bit for the BAM (not pipe related) interrupt. This register is read only and provides fast indication for the software to decide which pipe triggered the interrupt. After checking out this register, SW reads the BAM_P_IRQ_STTSn register for the pipe interrupt reason and BAM_IRQ_STTS for the BAM interrupt reason.

This register points to the physical BAM_IRQ_SRCS_EE0 register.

USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_IRQ_SRCS

Bits	Name	Description
31	BAM_IRQ	BAM Global interrupt
30:0	P_IRQ	An interrupt bit per each pipe in the BAM.

0x078C7014 USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_IRQ_SRCS_MSK

Type: RW

Clock: bam_clk

Reset State: 0x00000000

Interrupt Sources Mask register allows to mask the interrupts by their source, whether BAM or a specific pipe.

This register points to the physical BAM_IRQ_SRCS_MSK_EE0 register.

USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_IRQ_SRCS_MSK

Bits	Name	Description
31	BAM_IRQ_MSK	BAM Global Interrupt Mask 0x1: ENABLE_BAM_INTERRUPT (Enable BAM interrupt) 0x0: DISABLE_BAM_INTERRUPT (Disable BAM interrupt)
30:0	P_IRQ_MSK	An Interrupt Mask bit per each pipe in the BAM. 0x1: ENABLE_PIPE_INTERRUPT (Enable Pipe interrupt) 0x0: DISABLE_PIPE_INTERRUPT (Disable Pipe interrupt)

0x078C7018 USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_IRQ_SRCS_UNMASKED**Type:** R**Clock:** bam_clk**Reset State:** 0x00000000

The register shows the interrupts sources like (BAM_IRQ_SRCS) although the interrupts from the BAM and the pipes are not masked on the lower level.

This register points to the physical BAM_IRQ_SRCS_UNMASKED_EE0 register.

USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_IRQ_SRCS_UNMASKED

Bits	Name	Description
31	BAM_IRQ_UNMASKED	BAM Global Interrupt after Mask
30:0	P_IRQ_UNMASKED	An Interrupt bit per each pipe in the BAM, after Mask

**0x078D7000+ USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_P_CTRLn, n=[0..30]
4096*n****Type:** RW**Clock:** bam_clk**Reset State:** 0x00000000

Pipe Control register provides various controls for the pipe.

USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_P_CTRLn

Bits	Name	Description
20:16	P_LOCK_GROUP	Pipe's lock group. Upon a lock request on this pipe, all pipes related to different pipe group and different EE (see PIPE_TRUST_REG) will be locked.

USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_P_CTRLn (cont.)

Bits	Name	Description
11	P_WRITE_NWD	BAM-Lite feature. Applicable to BAM2BAM producer pipes only. When this bit is set for producer B2B pipe, NWD bit (bit number 28 in the second word of descriptor) will be written with EOT into the generated descriptor for the consumer usage. This bit is not applicable in pipes belong to a Peripheral which uses messaging_only (e.g. USB2).
10:9	P_PREFETCH_LIMIT	Bam Lite feature Selects the number of bytes to be prefetched by the Bam Lite, once a peripheral requests to read data from a pipe. This feature is relevant for Consumer modes only. 2'b00 - 32 bytes at most (INCR8) 2'b01 - 16 bytes at most (INCR4) 2'b10 - 4 bytes at most (SINGLE) Available in BAM-Lite only
8:7	P_AUTO_EOB_SEL	Bam Lite feature. Selects the number of bytes to be processed before creating End Of Block indication internally by the BAM. See P_AUTO_EOB. Descriptors generated by this feature (bam to bam producer mode) will always be of the chosen size, unless an End Of Transfer, causes a smaller descriptor to be created. 2'b00 - 512 Bytes 2'b01 - 256 Bytes 2'b10 - 128 Bytes 2'b11 - 64 Bytes Available in BAM-Lite only
6	P_AUTO_EOB	BAM Lite feature. When Enabled, the BAM Lite will ignore any incoming End Of Block indications from the peripheral (if any). Instead, the BAM Lite will generate it's own End Of Block indications internally, every P_AUTO_EOB_SEL bytes. This feature is relevant for Producer Bam 2 Bam modes only. It is a must for peripherals which are not able to produce End Of Block indications. 1'b1 - Enable 1'b0 - Disable Available in BAM-Lite only
5	P_SYS_MODE	This bit sets the pipe to work in System Mode, where the Software is the second side of the pipe. 1'b1 - System mode. 1'b0 - BAM to BAM mode.

USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_P_CTRLn (cont.)

Bits	Name	Description
4	P_SYS_STRM	Streaming Enable. Relevant to System Producer BAM mode only. When enabled, BAM will not close descriptors with End Of Transfer notifications received from peripheral, but continue writing the data to the same descriptor. 1'b1 - Enable Streaming Mode 1'b0 - None-Streaming Mode. This field is NOT supported in BAM-NDP (i.e. no streaming mode in NDP).
3	P_DIRECTION	This bit denotes pipe direction. 1'b1 - The pipe can be written (producer mode) 1'b0 - The pipe can be read (consumer mode)
1	P_EN	1'b1 - Enable Pipe 1'b0 - Disable Pipe

**0x078D7004+ USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_P_RSTn, n=[0..30]
4096*n****Type:** W**Clock:** bam_clk**Reset State:** 0x00000000

Pipe Reset register provides Pipe Reset ability. Software needs to write 1 to this register to make the Pipe Reset. Writing 0 to this register will finish the Pipe Reset. Writing zero is needed prior to Peripheral reconfiguration for the same pipe.

Note: Software can invoke pipe reset only after it make sure that the pipe is 'quiet'.

('quiet' - No data transaction or command-descriptor is active at the time.

BAM_P_RSTn, n=[0..30]

USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_P_RSTn

Bits	Name	Description
0	P_SW_RST	This resets the pipe and it's registers, (Both Flip-Flops and RAM). Software can invoke pipe reset only after it make sure that the pipe is 'quiet'. 1'b1 - Reset 1'b0 - Do Nothing

**0x078D7008+ USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_P_HALTn, n=[0..30]
4096*n****Type:** RW**Clock:** bam_clk**Reset State:** 0x00000000

Pipe Halt register Enables/Disables the Halt Sequence. Refer to the Software Procedures chapter for the Halt Sequence Procedure.

This is a self-modifying register.

USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_P_HALTn

Bits	Name	Description
1	P_PROD_HALTED	This bit is used to inform the Consumer about the Producer being in Halt mode now. This is a part of the Halt procedure. See the Software Procedures chapter. 1'b1 - Sets this bit to 1 1'b0 - Does Nothing This bit will be cleared by the HW.
0	P_HALT	When Enabled, the Pipe will enter Halt Mode. 1'b1 - Enable Halt 1'b0 - Disable Halt For B2B Producer pipes, this bit is self modifying. It doesn't need the SW to clear it. See the Software Procedure chapter for the detailed flow.

0x078D7010+ USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_P_IRQ_STTSn, n=[0..30] 4096*n

Type: R

Clock: bam_clk

Reset State: 0x00000000

Pipe Interrupt status register describing the reason for the ongoing interrupt. This is an informative only register and therefore read only. It also has informative bits which are not interrupts.

Clearing the interrupt bits is done by writing to P_IRQ_CLR register.

USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_P_IRQ_STTSn

Bits	Name	Description
7	P_HRESP_ERR_IRQ	This interrupt Indicates that an Erroneous HResponse has been received by the AHB Master. Additional Information about the error may be read at BAM_AHB_MASTER_ERR_* Registers. Clearing this interrupt also clears the BAM_AHB_MASTER_ERR_* Registers. Available in BAM only
6	P_PIPE_RST_ERROR_IRQ	Unsuccessful Pipe Reset operation. Pipe reset timer expired.
5	P_TRNSFR_END_IRQ	For producer this interrupt happens whenever the last AHB burst of a transfer happens For Consumer when ack_on_success closes a transfer

USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_P_IRQ_STTSn (cont.)

Bits	Name	Description
4	P_ERR_IRQ	Pipe Error interrupt indicates that an error has happened. TBD: Error reason is not yet defined. Interrupt not in use.
3	P_OUT_OF_DESC_IRQ	Out of descriptors interrupt has several meanings depending on the operation mode: Producer - no free space in the descriptors fifo for adding a descriptor Consumer - no descriptors in the descriptors fifo for processing IO Vectors Producer - No descriptors to read (and therefore no space for data to write to the pipe) IO Vectors Consumer - No descriptors to read (and therefore no data to read from pipe)
2	P_WAKE_IRQ	Wake peripheral interrupt signals the driver to wake up the peripheral (USB for example) for transmission via the current pipe. This interrupt is issued whenever an Event comes to the specific pipe.
1	P_TIMER_IRQ	Inactivity timer has reached its threshold value Interrupt
0	P_PRCSD_DESC_IRQ	This Interrupt rises when BAM finishes processing an IO Vector which has INT bit selected

**0x078D7014+ USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_P_IRQ_CLRn, n=[0..30]
4096*n****Type:** W**Clock:** bam_clk**Reset State:** 0x00000000

Writing to this register causes the interrupt to clear.

BAM_P_IRQ_CLRn, n=[0..30]

USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_P_IRQ_CLRn

Bits	Name	Description
7	P_HRESP_ERR_CLR	1'b1 - Clear 1'b0 - Unchanged
6	P_PIPE_RST_ERROR_CLR	1'b1 - Clear 1'b0 - Unchanged
5	P_TRNSFR_END_CLR	1'b1 - Clear 1'b0 - Unchanged
4	P_ERR_CLR	1'b1 - Clear 1'b0 - Unchanged
3	P_OUT_OF_DESC_CLR	1'b1 - Clear 1'b0 - Unchanged

USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_P_IRQ_CLRn (cont.)

Bits	Name	Description
2	P_WAKE_CLR	Clear Wake peripheral interrupt by writing this bit 1'b1 - Clear 1'b0 - Unchanged
1	P_TIMER_CLR	Clear Inactivity timer interrupt by writing this bit 1'b1 - Clear 1'b0 - Unchanged
0	P_PRCSD_DESC_CLR	Clear Descriptor Processed interrupt by writing this bit 1'b1 - Clear 1'b0 - Unchanged

**0x078D7018+ USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_P_IRQ_ENn, n=[0..30]
4096*n****Type:** RW**Clock:** bam_clk**Reset State:** 0x00000000

Enable bits for the Interrupts in the Pipe Interrupt Status register.

USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_P_IRQ_ENn

Bits	Name	Description
7	P_HRESP_ERR_EN	1'b1 - Enable 1'b0 - Disable
6	P_PIPE_RST_ERROR_EN	1'b1 - Enable 1'b0 - Disable
5	P_TRNSFR_END_EN	1'b1 - Enable 1'b0 - Disable
4	P_ERR_EN	1'b1 - Enable 1'b0 - Disable
3	P_OUT_OF_DESC_EN	1'b1 - Enable 1'b0 - Disable
2	P_WAKE_EN	Enable Wake peripheral interrupt 1'b1 - Enable 1'b0 - Disable
1	P_TIMER_EN	Enable Inactivity timer Interrupt 1'b1 - Enable 1'b0 - Disable
0	P_PRCSD_DESC_EN	Enable Descriptor Processed Interrupt 1'b1 - Enable 1'b0 - Disable

**0x078D701C+USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_P_TIMERn, n=[0..30]
4096*n****Type:** R**Clock:** bam_clk**Reset State:** 0x00000000

This register counts the idle time of the pipe.

BAM_P_TIMERn, n=[0..30]

USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_P_TIMERn

Bits	Name	Description
15:0	P_TIMER	Inactivity timer counter. 16 bits = 64k*0.1ms ~ 6 seconds

**0x078D7020+ USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_P_TIMER_CTRLn, n=[0..30]
4096*n****Type:** RW**Clock:** bam_clk**Reset State:** 0x00000000

This register sets various controls and the threshold of the inactivity for interrupt triggering after long idle time.

The resolution of the pipe inactivity timers are in units set by a common counter. This common counter is controlled by a PARAMETER value of the BAM, and uses a separate clock, the inactivity_timers_clk. This clock can be slower than the bam_clk. The intent of the design is to use the sleep_clk, which is an always on 32kHz clock. This allows the bam_clk to be turned-off, while the base counter keeps counting. The width of this counter, as well of the rollover value are set using BAM PARAMETER. These values, taking the inactivity_timers_clk frequency define the pulse frequency that the base counter gives, and the threshold value can be calculated and set to interrupt after a specific time, regardless of the bam_clk frequency, and independent of clock power save features of the bam_clk. The timers continue counting even if the clock is turned off due to BAM dynamic hardware clock request voting.

The timer configuration is based on the inactivity_timers_clk period and the INACTIVITY_TIMER_WIDTH parameter constant. These parameters should be taken into account when setting this register. For example: for inactivity_timer_clk period of 1us and parameter is 3 and P_TIMER_THRSHLD is 10 will indicate the $2^3 * 1\mu s * 10$ which is 80us of inactivity in a pipe before sending an interrupt.

The general formula is: $2^{\text{INACTIVITY_TIMER_WIDTH}} * \text{clock_period} * \text{P_TIMER_TRSHLD}$.

USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_P_TIMER_CTRLn

Bits	Name	Description
31	P_TIMER_RST	To begin using the inactivity timer the SW should write the threshold and write 1 to this bit. writing the value 0 will hold the timer reset until value 1 is written again. Then the timer will start down counting from the P_TIMER_THRESHOLD value 1'b1 - Active 1'b0 - Reset
30	P_TIMER_RUN	Will be used along with P_TIMER_MODE Not implemented at this time. Set to zero(0)
29	P_TIMER_MODE	Not implemented at this time. Set to zero(0)
15:0	P_TIMER_TRSHLD	Inactivity timer threshold for interrupt generation.

**0x078D7024+ USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_P_PRDCR_SDBNDn, n=[0..30]
4096*n****Type:** R**Clock:** bam_clk**Reset State:** 0x00000000

This Register reflects the current status of the Producer Sideband interface as the peripheral sees it and some additional internal bits related to its management.

USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_P_PRDCR_SDBNDn

Bits	Name	Description
24	BAM_P_SB_UPDATED	sideband_inform_bytes_avail_updated_producer. relevant for system mode only. The sideband-Inform block which responsible for reading descriptor has updated the SB block (which responsible for publishing to peripheral) with all of its read descriptors.
20	BAM_P_TOGGLE	Pipe Bytes Free Toggle value. The toggle polarity of the publishment (each publishment changes its polarity).
19:16	BAM_P_CTRL	Pipe Bytes Free Ctrl value. The possible values are only: 0000 or 0001 (means desc-fifo-full).
15:0	BAM_P_BYTES_FREE	Pipe Bytes Free value. The accumulated bytes_free which are available for peripheral to use

**0x078D7028+ USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_P_CNMSMR_SDBNDn, n=[0..30]
4096*n****Type:** R**Clock:** bam_clk**Reset State:** 0x00000000

This Register reflects the current status of the Consumer Sideband interface as the peripheral sees it and some additional internal bits related to its management.

USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_P_CNMSMR_SDBNDn

Bits	Name	Description
24	BAM_P_SB_UPDATED	sideband_inform_bytes_avail_updated_consumer. The sideband-Inform block which responsible for reading descriptor has updated the SB block (which responsible for publishing to peripheral) with all of its read descriptors.
23	BAM_P_WAIT_4_ACK	waiting_for_ack_bytes_avail_r. BAM is waiting for Peripheral to acknowledge the former publish.
22	BAM_P_ACK_TOGGLE	ack_bytes_avail_toggle. The Peripheral's acknowledgement for the publish. This is a toggling signal (each ack will be inverted to its former).
21	BAM_P_ACK_TOGGLE_R	ack_bytes_avail_toggle_r. The Peripheral's sampled acknowledgement for the publish. This is a toggling signal (each ack will be inverted to its former).
20	BAM_P_TOGGLE	Pipe Bytes Avail Toggle value. The toggle polarity of the publishment (each publishment changes its polarity).
19:16	BAM_P_CTRL	Pipe Bytes Avail Ctrl value. The possible values of BAM_P_CTRL field are: 0100 - Means immediate command (could be only in bam-ndp). 0001 - Means End-Of-Transaction (EOT). 0011 - means End-Of-Transaction with Notify-When-Done (NWD). 0000 - Neither Immediate-command nor EOT nor NWD.
15:0	BAM_P_BYTES_AVAIL	Pipe Bytes AVAIL value. The size of the current publish.

**0x078D7800+ USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_P_SW_OFSTSn, n=[0..30]
4096*n****Type:** RW**Clock:** bam_clk**Reset State:** 0x00000000

This register denotes the pointer Offset of the first Not Acknowledged Descriptor. Meaning all Descriptors prior to this one have been processed by the BAM. This register is only used by the Software. After receiving an interrupt, software reads this register in order to know what descriptors has been processed.

When the BAM works as Producer, Software will usually read this value after Descriptor Processed Interrupt or after the End of Transfer Interrupt.

When the BAM works as Consumer, Software will usually read this value after Descriptor Processed Interrupt.

NOTE This is non relevant in BAM to BAM modes.

NOTE Although being Writable, Software should never write to this register.

NOTE When using 2 pipes BAM, REVISION 0x03, this registers is Read Only.

BAM_P_SW_OFSTSn, n=[0..30]

USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_P_SW_OFSTSn

Bits	Name	Description
31:16	SW_OFST_IN_DESC	Offset inside the Descriptor. This value is used by the Software only when a BAM is working as System Producer. Then the SW can read this value to know how many Bytes were written to the current descriptor if it was not closed down. This is useful for Streaming mode. This Field is NOT available in BAM-NDP
15:0	SW_DESC_OFST	Descriptor FIFO offset.

0x078D7804+ USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_P_AU_PSM_CNTXT_1_n, n=[0..30] 4096*n

Type: RW

Clock: bam_clk

Reset State: 0x00000000

Internal Context of Ack Update state machine.

NOTE When using 2 pipes BAM, REVISION 0x03, this registers is Read Only.

BAM_P_AU_PSM_CNTXT_1_n, n=[0..30]

USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_P_AU_PSM_CNTXT_1_n

Bits	Name	Description
31:16	AU_PSM_ACCUMED	PSM: The acknowledged (EOB/T) number of bytes since last Write Event. AU: The accumulation of the descriptor sizes for the present ack_on_success event. This is never bigger than the ACKED field. Once the ACCUMED values are bigger than Event Threshold, an Event will be issued, the ACKED value will decrease by the ACCUMED size and this value will be zeroed. This also serves as the Bytes Consumed value in the Event.
15:0	AU_ACKED	AU: The left over of the number of bytes that were received via the ack_on_success, that did not cause a descriptor closure. This value is down counting upon issuing Read Event. PSM:

**0x078D7808+ USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_P_PSM_CNTXT_2_n, n=[0..30]
4096*n****Type:** RW**Clock:** bam_clk**Reset State:** 0x00000000

Internal Context of Pipe & Writeback state machines.

NOTE When using 2 pipes BAM, REVISION 0x03, this registers is Read Only.

BAM_P_PSM_CNTXT_2_n, n=[0..30]

USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_P_PSM_CNTXT_2_n

Bits	Name	Description
31	PSM_DESC_VALID	The Descriptor Valid bit provided by the Writeback state machine to Pipe state machine.
30	PSM_DESC_IRQ	The Descriptor Interrupt bit provided by the Writeback state machine to pipe state machine.
29	PSM_DESC_IRQ_DONE	The Descriptor Interrupt needed selection bit.
28:25	PSM_GENERAL_BITS	General Context Switch Bits
24:22	PSM_CONS_STATE	State of the Pipe Consumer state machine.
21:19	PSM_PROD_SYS_STATE	State of the Pipe Producer System state machine.
18:16	PSM_PROD_B2B_STATE	State of the Pipe Producer BAM to BAM state machine.
15:0	PSM_DESC_SIZE	The Descriptor Size provided by the Writeback state machine to pipe state machine.

**0x078D780C+USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_P_PSM_CNTXT_3_n, n=[0..30]
4096*n****Type:** RW**Clock:** bam_clk**Reset State:** 0x00000000

Internal Context of Pipe & Writeback state machines.

NOTE When using 2 pipes BAM, REVISION 0x03, this registers is Read Only.

BAM_P_PSM_CNTXT_3_n, n=[0..30]

USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_P_PSM_CNTXT_3_n

Bits	Name	Description
31:0	PSM_DESC_ADDR	The Data Address provided in the current descriptor from the Writeback state machine.

**0x078D7810+ USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_P_PSM_CNTXT_4_n, n=[0..30]
4096*n****Type:** RW**Clock:** bam_clk**Reset State:** 0x00000000

Internal Context of Pipe & Writeback state machines.

NOTE When using 2 pipes BAM, REVISION 0x03, this registers is Read Only.

BAM_P_PSM_CNTXT_4_n, n=[0..30]

USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_P_PSM_CNTXT_4_n

Bits	Name	Description
31:16	PSM_DESC_OFST	The running Descriptor Fifo Offset of the Writeback state machine
15:0	PSM_SAVED_ACCUMED_SIZE	The Saved Accumulated Size from the Pipe state machine.

**0x078D7814+ USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_P_PSM_CNTXT_5_n, n=[0..30]
4096*n****Type:** RW**Clock:** bam_clk**Reset State:** 0x00000000

Internal Context of Pipe & Writeback state machines.

NOTE When using 2 pipes BAM, REVISION 0x03, this registers is Read Only.

BAM_P_PSM_CNTXT_5_n, n=[0..30]

USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_P_PSM_CNTXT_5_n

Bits	Name	Description
31:16	PSM_BLOCK_BYTE_CNT	The Byte count inside the current Block.
15:0	PSM_OFST_IN_DESC	The Offset inside a register. This is used by pipe state machine to compute offset inside a register after retransmission

**0x078D7818+ USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_P_EVNT_REGn, n=[0..30]
4096*n****Type:** RW**Clock:** bam_clk**Reset State:** 0x00000000

This Register is where Read/Write events are sent to. If current BAM works as Producer, it will receive Read events into this register from the Consumer. If the Consumer is another BAM, it will

write read events to this register. If the Consumer is the Software, it will prepare descriptors in the descriptor FIFO and update the DESC_FIFO_PEER_OFST value in this register.

If current BAM works as Consumer, opposite behavior is assumed.

BAM_P_EVNT_REGn, n=[0..30]

USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_P_EVNT_REGn

Bits	Name	Description
31:16	P_BYTES_CONSUMED	This field is used only in the B2B mode. It indicates the number of bits consumed by the Consumer. Consumer sends this value to the Producer as a part of a read event, for the Producer to know how many bytes were consumed. When working in System mode, this value should be set to zero. BAM to BAM Producer writes zero in this value.
15:0	P_DESC_FIFO_PEER_OFST	This field indicates the Descriptor Fifo Offset Pointer of the Peer BAM (The BAM or the Software which creates descriptors). For example when current BAM works as consumer for the current pipe, Producer BAM (or the software) updates this value to let the current pipe know about the new descriptors added in the Descriptor Fifo.

0x078D781C+USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_P_DESC_FIFO_ADDRn, n=[0..30] 4096*n

Type: RW

Clock: bam_clk

Reset State: 0x00000000

This register stores the Address of the Descriptor Fifo beginning.

NOTE This register is used by all modes.

BAM_P_DESC_FIFO_ADDRn, n=[0..30]

USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_P_DESC_FIFO_ADDRn

Bits	Name	Description
31:0	P_DESC_FIFO_ADDR	Address of the Descriptors Fifo. This address must be 8 bytes aligned.

0x078D7820+ USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_P_FIFO_SIZESn, n=[0..30] 4096*n

Type: RW

Clock: bam_clk

Reset State: 0x00000000

The least significant field is the Descriptor Fifo Size.

The most significant field is the Data Fifo Size.

NOTE This Descriptor Fifo Size is used by all modes. The Data Fifo Size is non relevant in System Modes.

BAM_P_FIFO_SIZES_n, n=[0..30]

USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_P_FIFO_SIZES_n

Bits	Name	Description
31:16	P_DATA_FIFO_SIZE	Size of the Data Address Space. ALWAYS write 16'b0 to this field of P_DATA_FIFO_SIZE unless you work in BAM2BAM producer mode (also known as peer mode). NOTE: Data fifo size should not be bigger than 32KB.
15:0	P_DESC_FIFO_SIZE	Size of the Descriptors Fifo. Must be 8 byte aligned. Each descriptor is 8 bytes. Size of the descriptor fifo must contain an integer number of Descriptors. Size of descriptor fifo must be 16 bytes and above.

0x078D7824+ USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_P_DATA_FIFO_ADDR_n, n=[0..30] 4096*n

Type: RW

Clock: bam_clk

Reset State: 0x00000000

This register stores the Beginning of the Data Address space.

This register is non relevant in System Modes.

BAM_P_DATA_FIFO_ADDR_n, n=[0..30]

USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_P_DATA_FIFO_ADDR_n

Bits	Name	Description
31:0	P_DATA_FIFO_ADDR	Data Address Space beginning.

0x078D7828+ USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_P_EVTN_GEN_TRSHLD_n, n=[0..30] 4096*n

Type: RW

Clock: bam_clk

Reset State: 0x00000000

This register configures the threshold value for Read/Write event generation by the BAM towards another BAM. When aBAM pushes/pops this number of bytes to/from the pipe an event to the

other BAM will be generated. A counter summing up the bytes pushed/popped is reset when event is issued, thus restarting count for the threshold.

Events are also issued after End Of Transfer received from the peripheral (Producer case), non regarding whether the pushed bytes count has reached the threshold or not.

This register is non relevant in System Modes.

BAM_P_EVNT_GEN_TRSHLDn, n=[0..30]

USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_P_EVNT_GEN_TRSHLDn

Bits	Name	Description
15:0	P_TRSHLD	Threshold value. The maximum allowed value is 32kByte.

0x078D782C+USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_P_EVNT_DEST_ADDRn, n=[0..30] 4096*n

Type: RW

Clock: bam_clk

Reset State: 0x00000000

This register stores the Event Destination Address which is the address of BAM_P_EVNT_REG in the BAM on the other side of the Pipe.

Producer BAM will send Write Events to this address.

Consumer BAM will send Read Events to this address.

This is relevant only for BAM to BAM modes. In System modes, BAM doesn't issue Write/Read events.

BAM_P_EVNT_DEST_ADDRn, n=[0..30]

USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_P_EVNT_DEST_ADDRn

Bits	Name	Description
31:0	P_EVNT_DEST_ADDR	Address of the Event Register in the peer BAM (second end of the pipe). Driver will configure this when initializing the pipe.

0x078D7830+ USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_P_DF_CNTXT_n, n=[0..30] 4096*n

Type: RW

Clock: bam_clk

Reset State: 0x00000000

This register is used on in BAM-NDP core.

BAM_P_DF_CNTXT_n, n=[0..30]

USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_P_DF_CNTXT_n

Bits	Name	Description
31:16	WB_ACCUMULATED	Relevant in B2B Producer mode only. This field stores the current running accumulation of the total amount of data that has been written to the data fifo. When it goes bigger then event_threshold a write event occurs.
15:0	DF_DESC_OFST	Holds the descriptor offset of the Descriptor-Fetcher block which is responsible for fetching descriptors before publishing them to the peripheral.

0x078D7834+ USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_P_RETR_CNTXT_n, n=[0..30] 4096*n

Type: RW

Clock: bam_clk

Reset State: 0x00000000

Internal Context stored for retransmission.

This is being stored by Ack Update state machine in consumer mode or by the Writeback state machine in producer mode.

This is being loaded by the Writeback state machine when retransmission happens.

NOTE When using 2 pipes BAM, REVISION 0x03, this registers is Read Only.

BAM_P_RETR_CNTXT_n, n=[0..30]

USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_P_RETR_CNTXT_n

Bits	Name	Description
31:16	RETR_DESC_OFST	Descriptor Fifo Offset for retransmission.
15:0	RETR_OFST_IN_DESC	The Offset inside the Descriptor for retransmission.

0x078D7838+ USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_P_SI_CNTXT_n, n=[0..30] 4096*n

Type: RW

Clock: bam_clk

Reset State: 0x00000000

Internal Context of Sideband Inform state machine.

NOTE When using 2 pipes BAM, REVISION 0x03, this registers is Read Only.

BAM_P_SI_CNTXT_n, n=[0..30]

USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_P_SI_CNTXT_n

Bits	Name	Description
15:0	SI_DESC_OFST	The Descriptor Fifo Offset pointer stored by the Sideband Inform state machine.

**0x078D7900+ USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_P_PSM_CNTXT_3_LSBn, n=[0..30]
4096*n****Type:** RW**Clock:** bam_clk**Reset State:** 0x00000000

Internal Context of Pipe & Writeback state machines.

Relevant only on 36 bit address BAM.

NOTE When using 2 pipes BAM, REVISION 0x03, this registers is Read Only.

BAM_P_PSM_CNTXT_3_n, n=[0..30]

USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_P_PSM_CNTXT_3_LSBn

Bits	Name	Description
31:0	PSM_DESC_ADDR	32 LSB of The Data Address provided in the current descriptor from the Writeback state machine.

**0x078D7904+ USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_P_PSM_CNTXT_3_MSBn, n=[0..30]
4096*n****Type:** RW**Clock:** bam_clk**Reset State:** 0x00000000

Internal Context of Pipe & Writeback state machines.

Relevant only on 36 bit address BAM.

NOTE When using 2 pipes BAM, REVISION 0x03, this registers is Read Only.

BAM_P_PSM_CNTXT_3_n, n=[0..30]

USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_P_PSM_CNTXT_3_MSBn

Bits	Name	Description
3:0	PSM_DESC_ADDR	4 MSB of address (bits 35 to 32) of the Data Address provided in the current descriptor from the Writeback state machine.

0x078D7910+ USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_P_DESC_FIFO_ADDR_LSBn,
4096*n n=[0..30]

Type: RW

Clock: bam_clk

Reset State: 0x00000000

This register stores the LSB of Address of the Descriptor Fifo beginning.

Relevant only on 36 bit address BAM.

NOTE This register is used by all modes.

BAM_P_DESC_FIFO_ADDRn, n=[0..30]

USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_P_DESC_FIFO_ADDR_LSBn

Bits	Name	Description
31:0	P_DESC_FIFO_ADDR	32 LSB of address of the Descriptors Fifo. This address must be 8 bytes aligned.

0x078D7914+ USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_P_DESC_FIFO_ADDR_MSBn,
4096*n n=[0..30]

Type: RW

Clock: bam_clk

Reset State: 0x00000000

This register stores the MSB of Address of the Descriptor Fifo beginning.

Relevant only on 36 bit address BAM.

NOTE This register is used by all modes.

BAM_P_DESC_FIFO_ADDRn, n=[0..30]

USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_P_DESC_FIFO_ADDR_MSBn

Bits	Name	Description
3:0	P_DESC_FIFO_ADDR	4 MSB of address (bits 35 to 32) of Address of the Descriptors Fifo. This address must be 8 bytes aligned.

0x078D7920+ USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_P_DATA_FIFO_ADDR_LSBn,
4096*n n=[0..30]

Type: RW

Clock: bam_clk

Reset State: 0x00000000

This register stores the Beginning of the Data Address space.

This register is non relevant in System Modes.

Relevant only on 36 bit address BAM.

BAM_P_DATA_FIFO_ADDRn, n=[0..30]

USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_P_DATA_FIFO_ADDR_LSBn

Bits	Name	Description
31:0	P_DATA_FIFO_ADDR	32 LSB of Data Address Space beginning. Relevant only on 36 bit address BAM.

0x078D7924+ USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_P_DATA_FIFO_ADDR_MSBn,
4096*n n=[0..30]

Type: RW

Clock: bam_clk

Reset State: 0x00000000

This register stores the Beginning of the Data Address space.

This register is non relevant in System Modes.

Relevant only on 36 bit address BAM.

BAM_P_DATA_FIFO_ADDRn, n=[0..30]

USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_P_DATA_FIFO_ADDR_MSBn

Bits	Name	Description
3:0	P_DATA_FIFO_ADDR	4 MSB of Address (bits 35 to 32) of Data Address Space beginning. Relevant only on 36 bit address BAM.

0x078D7930+ USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_P_EVNT_DEST_ADDR_LSBn,
4096*n n=[0..30]

Type: RW

Clock: bam_clk

Reset State: 0x00000000

This register stores the Event Destination Address which is the address of BAM_P_EVNT_REG in the BAM on the other side of the Pipe.

Producer BAM will send Write Events to this address.

Consumer BAM will send Read Events to this address.

This is relevant only for BAM to BAM modes. In System modes, BAM doesn't issue Write/Read events.

Relevant only on 36 bit address BAM.

BAM_P_EVNT_DEST_ADDRn, n=[0..30]

USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_P_EVNT_DEST_ADDR_LSBn

Bits	Name	Description
31:0	P_EVNT_DEST_ADDR	32 LSB of Address of the Event Register in the peer BAM (second end of the pipe). Driver will configure this when initializing the pipe. Relevant only on 36 bit address BAM.

0x078D7934+ USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_P_EVNT_DEST_ADDR_MSBn,
4096*n n=[0..30]

Type: RW

Clock: bam_clk

Reset State: 0x00000000

This register stores the Event Destination Address which is the address of BAM_P_EVNT_REG in the BAM on the other side of the Pipe.

Producer BAM will send Write Events to this address.

Consumer BAM will send Read Events to this address.

This is relevant only for BAM to BAM modes. In System modes, BAM doesn't issue Write/Read events.

Relevant only on 36 bit address BAM.

BAM_P_EVNT_DEST_ADDRn, n=[0..30]

USB2_HSIC_USB_OTG_HS_BAM_BAM_BAM_P_EVNT_DEST_ADDR_MSBn

Bits	Name	Description
3:0	P_EVNT_DEST_ADDR	4 MSB of Address (bits 35 to 32) of the Event Register in the peer BAM (second end of the pipe). Driver will configure this when initializing the pipe. Relevant only on 36 bit address BAM.

22 PDM

WEB_TCXO4 includes three PDMs and one MND_clock_devider, which are controlled using the following Regs. the are clocked with HCLK

0x00068000 PDM_WEB_TCXO4_TCXO_PDM0_TOP_CTL

Type: RW

Clock: HCLK

Reset State: 0x00000000

Top level control register of PDM0 used for OE, Output polarity ctl and clock enable of PDM0

PDM_WEB_TCXO4_TCXO_PDM0_TOP_CTL

Bits	Name	Description
2	TCXO4_CLK_DISABLE	Set (1) this bit to disable the TCXO4 clock, and Clear (0) the bit to enable the TXCO4 clock.PDM0
1	PDM0_POLARITY	This bit defines the polarity of PDM0. Set (1) this bit to invert the sense of the PDM0 output, meaning that the PDM value will indicate the number of 0's in every 2^16 cycles. Clearing (0) this bit will keep the PDM in it's native polarity, meaning that the PDM value will indicate the numbed of 1's in every 2^16 cycles.
0	PDM0_OE	

0x00068004 PDM_WEB_TCXO4_PDM0_CTL

Type: RW

Clock: HCLK

Reset State: 0x00008000

The PDM0_CTL register is the pulse density modulation register number 0. This register determines the density of pulses on the PDM0 pin.

PDM_WEB_TCXO4_PDM0_CTL

Bits	Name	Description
15:0	PDM0_DAT	the PDM0_DAT, after inverting the MSbit (1's complement), determines the number of 1's in 2^{16} TCXO4 cycles for PDM0. When the PDM0_POLARITY is set (1), the output sense is inverted

0x00069000 PDM_WEB_TCXO4_TCXO_PDM1_TOP_CTL**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

Top level control register of PDM1 used for OE, Output polarity ctl and clock enable of PDM1

PDM_WEB_TCXO4_TCXO_PDM1_TOP_CTL

Bits	Name	Description
2	TCXO4_CLK_DISABLE1	Set (1) this bit to disable the TCXO4 clock, and Clear (0) the bit to enable the TXCO4 clock. PDM1
1	PDM1_POLARITY	This bit defines the polarity of PDM1. Set (1) this bit to invert the sense of the PDM1 output, meaning that the PDM value will indicate the number of 0's in every 2^{16} cycles. Clearing (0) this bit will keep the PDM in it's native polarity, meaning that the PDM value will indicate the numbed of 1's in every 2^{16} cycles.
0	PDM1_OE	

0x00069004 PDM_WEB_TCXO4_PDM1_CTL**Type:** RW**Clock:** HCLK**Reset State:** 0x00000080

The PDM1_CTL register is the pulse density modulation register number 1. This register determines the density of pulses on the PDM1 pin.

PDM_WEB_TCXO4_PDM1_CTL

Bits	Name	Description
7:0	PDM1_DAT	the PDM1_DAT, after inverting the MSbit (1's complement), determines the number of 1's in 2^8 TCXO4 cycles for PDM1. When the PDM1_POLARITY is set (1), the output sense is inverted

0x0006A000 PDM_WEB_TCXO4_TCXO_PDM2_TOP_CTL**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

Top level control register of PDM2 used for OE, Output polarity ctl and clock enable of PDM2

PDM_WEB_TCXO4_TCXO_PDM2_TOP_CTL

Bits	Name	Description
2	PDM2_CLK_DISABLE	Set (1) this bit to disable the PDM2 clock, and Clear (0) the bit to enable the PDM2 clock.
1	PDM2_POLARITY	This bit defines the polarity of PDM2. Set (1) this bit to invert the sense of the PDM2 output, meaning that the PDM value will indicate the number of 0's in every 2 ¹⁶ cycles. Clearing (0) this bit will keep the PDM in it's native polarity, meaning that the PDM value will indicate the number of 1's in every 2 ¹⁶ cycles.
0	PDM2_OE	

0x0006A004 PDM_WEB_TCXO4_PDM2_CTL**Type:** RW**Clock:** HCLK**Reset State:** 0x00008000

The PDM2_CTL register is the pulse density modulation register number 2. This register determines the density of pulses on the PDM2 pin.

PDM_WEB_TCXO4_PDM2_CTL

Bits	Name	Description
15:0	PDM2_DAT	the PDM2_DAT, after inverting the MSbit (1's complement), determines the number of 1's in 2 ¹⁶ PDM2_clk cycles for PDM2. When the PDM2_POLARITY is set (1), the output sense is inverted

0x0006B000 PDM_WEB_TCXO4_GP_MN_CLK_MDIV**Type:** RW**Clock:** HCLK**Reset State:** 0x00000000

The GP_MN_CLK_MDIV register is the MDIV specification register for the GP M/N clock. For all GP_MN registers, this signal is generated by dividing the frequency of the TCXO signal by an M/N counter. M,N and D have the following specifications and constraints:

- * The maximum value of M is 511.
- * The maximum value of N is 8191.
- * $N > 2M$.
- * The value of M cannot exceed the value of N.
- * N and M cannot have any common divider else than 1.
- * D is allowed to be between M and N-M => $(M < D < N - M)$.

PDM_WEB_TCXO4_GP_MN_CLK_MDIV

Bits	Name	Description
8:0	GP_M_VALUE	This register defines the value of M for the M/N counter, which generates the GP_MN signal.

0x0006B004 PDM_WEB_TCXO4_GP_MN_CLK_NDIV

Type: RW
Clock: HCLK
Reset State: 0x00000000

The GP_MN_CLK_NDIV register is the NDIV specification register for the GP M/N clock.

PDM_WEB_TCXO4_GP_MN_CLK_NDIV

Bits	Name	Description
12:0	GP_N_VALUE	This register defines the value $-(N-M)$, using 1'Bits complement value of N minus M for the M/N counter, which generates the GP_MN signal.

0x0006B008 PDM_WEB_TCXO4_GP_MN_CLK_DUTY

Type: RW
Clock: HCLK
Reset State: 0x00000000

The GP_MN_CLK_DUTY register is the duty cycle specification register for the GP M/N clock.

PDM_WEB_TCXO4_GP_MN_CLK_DUTY

Bits	Name	Description
12:0	GP_D_VALUE	This register defines the value of D, which determines the duty cycle of the GP M/N signal. D determines the # on low(0) cycles in an N-cycles interval. The useful value of D must be in the range of M to N minus M.

23 PRNG

0x00022000 PRNG_PRNG_DATA_OUT

Type: R

Reset State: 0x00000000

PRNG_PRNG_DATA_OUT

Bits	Name	Description
31:0	PRNG_DATA_OUT	Pseudo Random Number output. If the FIFO is empty, reading this register will return 0.

0x00022004 PRNG_PRNG_STATUS

Type: R

Reset State: 0x00000000

PRNG_PRNG_STATUS

Bits	Name	Description
31:10	RESERVED	
9:8	CURRENT_OPERATION	0x0: IDLE (IDLE) 0x1: GENERATE (GENERATE) 0x2: INSTANTIATE (INSTANTIATE) 0x3: RESEED (RESEED)
7	GENERATE_REQUIRED	Indicates that a generate is, or soon will be occurring. This occurs when the generate counter != 0. This is only valid when the PRNG is enabled (PRNG_EN = 1)
6	RESEED_REQUIRED	Indicates that a reseed operation is, or soon will be occurring. This occurs when the reseed counter = reseed freq. This is only valid when the PRNG is enabled (PRNG_EN = 1)
5	INSTANTIATE_REQUIRED	Indicates that an instantiate operation or soon will be occurring. This occurs when the instantiate counter = 0. This is only valid when the PRNG is enabled (PRNG_EN = 1)

PRNG_PRNG_STATUS (cont.)

Bits	Name	Description
4	RING_OSC3_HEALTHY	Indicates that ring oscillator 3 is healthy. Only valid when RING_OSC3_CFG is set to one of the feedback settings (not valid when set to one of the FORCE inputs). When set to one of the FORCE inputs, this is reset to 0.
3	RING_OSC2_HEALTHY	Indicates that ring oscillator 2 is healthy. Only valid when RING_OSC2_CFG is set to one of the feedback settings (not valid when set to one of the FORCE inputs). When set to one of the FORCE inputs, this is reset to 0.
2	RING_OSC1_HEALTHY	Indicates that ring oscillator 1 is healthy. Only valid when RING_OSC1_CFG is set to one of the feedback settings (not valid when set to one of the FORCE inputs). When set to one of the FORCE inputs, this is reset to 0.
1	RING_OSC0_HEALTHY	Indicates that ring oscillator 0 is healthy. Only valid when RING_OSC0_CFG is set to one of the feedback settings (not valid when set to one of the FORCE inputs). When set to one of the FORCE inputs, this is reset to 0.
0	DATA_AVAIL	Indicates that data is available on the PRNG_DATA_OUT FIFO.

0x00022010 PRNG_PRNG_ENTROPY_CNTR**Type:** R**Reset State:** 0x00000000**PRNG_PRNG_ENTROPY_CNTR**

Bits	Name	Description
31:0	ENTROPY_CNTR	The current value of the entropy counter. Counts DOWN. Usually this is 0 except when actually doing an instantiate or reseed operation.

0x00022014 PRNG_PRNG_SAMPLE_CNTR**Type:** R**Reset State:** 0x00000000**PRNG_PRNG_SAMPLE_CNTR**

Bits	Name	Description
31:0	SAMPLE_CNTR	The number of clocks since the last sample of the LFSRs. Counts DOWN. Usually this is 0 except when actually doing an instantiate or reseed operation.

0x00022018 PRNG_PRNG_GEN_CNTR**Type:** R**Reset State:** 0x00000000**PRNG_PRNG_GEN_CNTR**

Bits	Name	Description
31:0	PRNG_GEN_CNTR	Current value of the generate counter

0x0002201C PRNG_PRNG_RESEED_CNTR**Type:** R**Reset State:** 0x00000000**PRNG_PRNG_RESEED_CNTR**

Bits	Name	Description
31:0	RESEED_CNTR	Current value of the reseed counter

0x00022020 PRNG_PRNG_INSTANTIATE_CNTR**Type:** R**Reset State:** 0x00000000**PRNG_PRNG_INSTANTIATE_CNTR**

Bits	Name	Description
31:0	INSTANTIATE_CNTR	The current number of reseed operations since the last reseed.

0x00022100 PRNG_PRNG_LFSR_CFG**Type:** RW**Reset State:** 0x00000000**PRNG_PRNG_LFSR_CFG**

Bits	Name	Description
31:17	RESERVED	
16	LFSR_OVRID_ON	Enables entropy collectors while FIFOFULL flag is asserted. When disabled, entropy collectors are disabled after FIFOFULL flag is on.
15	LFSR3_EN	Enables the output of LFSR3. When disabled, the LFSR is reset to 0 and disables the clock to the LFSR.

PRNG_PRNG_LFSR_CFG (cont.)

Bits	Name	Description
14:12	RING_OSC3_CFG	Configuration for Ring Oscillator 3 00X: FORCE0. This is used for testability 01X: FORCE1. This is used for testability. 0x4: FEEDBACK_POINT_0 (feedback point 0 (slowest)) 0x5: FEEDBACK_POINT_1 (feedback point 1) 0x6: FEEDBACK_POINT_2 (feedback point 2) 0x7: FEEDBACK_POINT_3 (feedback point 3 (fastest))
11	LFSR2_EN	Enables the output of LFSR2. When disabled, the LFSR is reset to 0 and disables the clock to the LFSR.
10:8	RING_OSC2_CFG	Configuration for Ring Oscillator 2 00X: FORCE0. This is used for testability 01X: FORCE1. This is used for testability. 0x4: FEEDBACK_POINT_0 (feedback point 0 (slowest)) 0x5: FEEDBACK_POINT_1 (feedback point 1) 0x6: FEEDBACK_POINT_2 (feedback point 2) 0x7: FEEDBACK_POINT_3 (feedback point 3 (fastest))
7	LFSR1_EN	Enables the output of LFSR1. When disabled, the LFSR is reset to 0 and disables the clock to the LFSR.
6:4	RING_OSC1_CFG	Configuration for Ring Oscillator 1 00X: FORCE0. This is used for testability 01X: FORCE1. This is used for testability. 0x4: FEEDBACK_POINT_0 (feedback point 0 (slowest)) 0x5: FEEDBACK_POINT_1 (feedback point 1) 0x6: FEEDBACK_POINT_2 (feedback point 2) 0x7: FEEDBACK_POINT_3 (feedback point 3 (fastest))
3	LFSR0_EN	Enables the output of LFSR0. When disabled, the LFSR is reset to 0 and disables the clock to the LFSR.
2:0	RING_OSC0_CFG	Configuration for Ring Oscillator 0 00X: FORCE0. This is used for testability 01X: FORCE1. This is used for testability. 0x4: FEEDBACK_POINT_0 (feedback point 0 (slowest)) 0x5: FEEDBACK_POINT_1 (feedback point 1) 0x6: FEEDBACK_POINT_2 (feedback point 2) 0x7: FEEDBACK_POINT_3 (feedback point 3 (fastest))

0x00022104 PRNG_PRNG_CONFIG**Type:** RW**Reset State:** 0x00000000

PRNG_PRNG_CONFIG

Bits	Name	Description
31:6	RESERVED	
5:2	TEST_OUT_SEL	Selects the appropriate test output to be put out on prng_rosc_test_out signal: 0XXX: Output 0 0x8: ROSC0_DIV_32 (rosc0_div_32) 0x9: ROSC1_DIV_32 (rosc1_div_32) 0xA: ROSC2_DIV_32 (rosc2_div_32) 0xB: ROSC3_DIV_32 (rosc3_div_32) 0xC: ROSC0_SAMP (rosc0_samp) 0xD: ROSC1_SAMP (rosc1_samp) 0xE: ROSC2_SAMP (rosc2_samp) 0xF: ROSC3_SAMP (rosc3_samp)
1	PRNG_EN	Enable signal for the PRNG indicating that all setup is complete. Ring oscillators and the LFSR will still operate as per PRNG_LFSR_CFG, however when this is 0, the PRNG when the PRNG is disabled it will return to the IDLE state once its current operation has been completed, with the current operation possibly requiring space to be created in the output fifo for data that must be generated in the completion of the current operation. This space can be created by reading from the PRNG_DATA_OUT register. 0x0: DISABLED (disabled) 0x1: ENABLED (enabled)
0	SW_RESET	Software initiated reset. Automatically cleared when the write operation is complete. Reset is internally held for 2 clock cycles after this register is written and so no operations/accesses should be attempted until after this recovery time.

0x00022108 PRNG_PRNG_PERSONALIZATION_STRING**Type:** RW**Reset State:** 0x00000000**PRNG_PRNG_PERSONALIZATION_STRING**

Bits	Name	Description
31:0	PERSONALIZATION_STRING	The personalization string. This may be all 0, but also may be obtained from other non-regular processes such as time between key presses, searcher noise (if present), or other system events.

0x00022110 PRNG_PRNG_NUM_ENTROPY**Type:** RW**Reset State:** 0x00000400**PRNG_PRNG_NUM_ENTROPY**

Bits	Name	Description
31:0	NUM_ENTROPY_BYTES	Indicates the number of entropy bytes to be used during instantiate and reseed processes. Note that the minimum value of this field should be 16 for this design, although no enforcement is done in HW. Larger values provide more randomness. NIST SP800-90 specifies the maximum value to be 235 BITS.

0x00022114 PRNG_PRNG_SAMPLE_FREQ**Type:** RW**Reset State:** 0x00000400**PRNG_PRNG_SAMPLE_FREQ**

Bits	Name	Description
31:0	SAMPLE_FREQ	The number of clocks between samples of the LFSR.

0x00022118 PRNG_PRNG_GEN_WORDS**Type:** RW**Reset State:** 0x00000008**PRNG_PRNG_GEN_WORDS**

Bits	Name	Description
31:0	REQUESTED_WORDS	The number of words per generate operation

0x0002211C PRNG_PRNG_RESEED_FREQ**Type:** RW**Reset State:** 0x00010000**PRNG_PRNG_RESEED_FREQ**

Bits	Name	Description
31:0	RESEED_FREQ	The number of generate operations before a reseed is initiated. The minimum number of generate operations before a reseed is initiated is 2.

0x00022120 PRNG_PRNG_INSTANTIATE_FREQ**Type:** RW**Reset State:** 0x00010000**PRNG_PRNG_INSTANTIATE_FREQ**

Bits	Name	Description
31:0	INSTANTIATE_FREQ	The number of reseed operations before an instantiate is initiated.

24 SMMU

24.1 SMMU_SS_LOCAL_SMMU_SS_LOCAL

0x01EF0000 SMMU_SS_LOCAL_SMMU_INTR_SEL_SEC

Type: RW

Clock: PCNOC_CLK

Reset State: 0x00000000

This register is used to steer one or multiple out of 32 context bank interrupts from App TCU to secure interrupt output

SMMU_SS_LOCAL_SMMU_INTR_SEL_SEC

Bits	Name	Description
31:0	INT_APP_CXT_EN	Setting nth bit will steer nth context bank interrupt to secure interrupt output

0x01EF0008 SMMU_SS_LOCAL_SMMU_MISC_TIE_OFF_SEC

Type: RW

Clock: PCNOC_CLK

Reset State: 0x00000000

This register is used to drive static values to APP and GFX TCUs

SMMU_SS_LOCAL_SMMU_MISC_TIE_OFF_SEC

Bits	Name	Description
3	APP_INTEG_SEC_OVRRIDE	Value of INTEG_SEC_OVRRIDE Port of APP TCU
2	GFX_INTEG_SEC_OVRRIDE	Value of INTEG_SEC_OVRRIDE Port of GFX TCU
1	APP_SPNIDEN	Value of SPNIDEN Port of APP TCU

SMMU_SS_LOCAL_SMMU_MISC_TIE_OFF_SEC (cont.)

Bits	Name	Description
0	GFX_SPNIDEN	Value of SPNIDEN Port of GFX TCU

0x01EF0010 SMMU_SS_LOCAL_SMMU_CATS_64_BIT_CTL_SEC**Type:** RW**Clock:** PCNOC_CLK**Reset State:** 0x00000000

This register is used to control CATS Mux for 64 bit TBUs

SMMU_SS_LOCAL_SMMU_CATS_64_BIT_CTL_SEC

Bits	Name	Description
18:9	CATS_SID	SID Value
8	CATS_SID_EN	If 1 enables CATS SID generation from Register, If 0 CATS SID generation will be generated from upper 3 bits of 512M CATS space
7:5	CATS_512M_REMAP	Upper Address Bits to remap 512M CATS space in 4G space
4:1	CATS_TBU_NUM	TBU ID for which CATS Mux is Enabled. Valid Values Are: 0 = PCNOC TBU CATS, 4 = QDSP TBU CATS, 5 = GPS TBU CATS, 6 = DIME TBU CATS, 7 = PRONTO TBU CATS
0	CATS_EN	Enable CATS Mux for One of the 64 bit TBUs

0x01EF0018 SMMU_SS_LOCAL_SMMU_CATS_128_BIT_CTL_SEC**Type:** RW**Clock:** PCNOC_CLK**Reset State:** 0x00000000

This register is used to control CATS Mux for 128 bit TBUs

SMMU_SS_LOCAL_SMMU_CATS_128_BIT_CTL_SEC

Bits	Name	Description
18:9	CATS_SID	SID Value
8	CATS_SID_EN	If 1 enables CATS SID generation from Register, If 0 CATS SID generation will be generated from upper 3 bits of 512M CATS space
7:5	CATS_512M_REMAP	Upper Address Bits to remap 512M CATS space in 4G space

SMMU_SS_LOCAL_SMMU_CATS_128_BIT_CTL_SEC (cont.)

Bits	Name	Description
4:1	CATS_TBU_NUM	TBU ID for which CATS Mux is Enabled Valid Values Are: 1 = VFE TBU CATS, 2 = VENUS TBU CATS, 3 = MDP TBU CATS, 8 = JPEG TBU CATS, 9 = GFX TBU CATS
0	CATS_EN	Enable CATS Mux for One of the 128 bit TBUs

0x01EF0020 SMMU_SS_LOCAL_SMMU_APP_TCU_PTW_VMID_SEC**Type:** RW**Clock:** PCNOC_CLK**Reset State:** 0x00000000

This register is used to provide VMID for APP TCU PTW Interface

SMMU_SS_LOCAL_SMMU_APP_TCU_PTW_VMID_SEC

Bits	Name	Description
4:0	VMID	VMID Value for APP TCU PTW AXI Interface

0x01EF0028 SMMU_SS_LOCAL_SMMU_GFX_TCU_PTW_VMID_SEC**Type:** RW**Clock:** PCNOC_CLK**Reset State:** 0x00000000

This register is used to provide VMID for GFX TCU PTW Interface

SMMU_SS_LOCAL_SMMU_GFX_TCU_PTW_VMID_SEC

Bits	Name	Description
4:0	VMID	VMID Value for GFX TCU PTW AXI Interface

0x01EF1000 SMMU_SS_LOCAL_SMMU_INTR_SEL_NSHYP**Type:** RW**Clock:** PCNOC_CLK**Reset State:** 0x00000000

This register is used to steer one or multiple out of 32 context bank interrupts from App TCU to non-secure hypervisor interrupt output

SMMU_SS_LOCAL_SMMU_INTR_SEL_NSHYP

Bits	Name	Description
31:0	INT_APP_CXT_EN	Setting nth bit will steer nth context bank interrupt to non-secure hypervisor interrupt output. If same nth bit is set in SMMU_INTR_SEL_SEC, then nth context bank interrupt won't be steered to non-secure hypervisor interrupt output

0x01EF1008 SMMU_SS_LOCAL_SMMU_GFX_AHB_4K_APERTURE_NSHYP**Type:** RW**Clock:** PCNOC_CLK**Reset State:** 0x00000000

This register is used to access GFX TCU registers from Private Oxili AHB port

SMMU_SS_LOCAL_SMMU_GFX_AHB_4K_APERTURE_NSHYP

Bits	Name	Description
31:0	GFX_AHB_4K_APERTURE	Aperture of 4KB to access GFX TCU registers from Oxili Private AHB port. Bits [11:0] are don't care. Bits [31:12] should be set as per the memory map

0x01EF1010 SMMU_SS_LOCAL_SMMU_MSA_BYPASS_ERR_ADDR_NSHYP**Type:** RW**Clock:** PCNOC_CLK**Reset State:** 0x00000000

This register is used to set Modem TBU output bus address in case of VA-PA mismatch

SMMU_SS_LOCAL_SMMU_MSA_BYPASS_ERR_ADDR_NSHYP

Bits	Name	Description
31:0	MSA_BYPASS_ERR_ADDR	Bus Address for VA-PA mismatch. This address should be set to a reserved location in SNOC so that error response can be generated from SNOC

0x01EF1018 SMMU_SS_LOCAL_SMMU_MSA_BYPASS_INTR_EN_NSHYP**Type:** RW**Clock:** PCNOC_CLK**Reset State:** 0x00000000

This register is used to enable Modem TBUs VA-PA mismatch interrupts

SMMU_SS_LOCAL_SMMU_MSA_BYPASS_INTR_EN_NSHYP

Bits	Name	Description
2	QDSP_MSA_VA_MISMATCH_H_IRQ_EN	QDSP VA-PA mismatch interrupt Enable
1	GPS_MSA_VA_MISMATCH_IRQ_EN	GPS VA-PA mismatch interrupt Enable
0	DIME_MSA_VA_MISMATCH_IRQ_EN	DIME VA-PA mismatch interrupt Enable

0x01EF2000 SMMU_SS_LOCAL_SMMU_INTR_SEL_NS**Type:** RW**Clock:** PCNOC_CLK**Reset State:** 0x00000000

This register is used to steer one or multiple out of 32 context bank interrupts from App TCU to non-secure interrupt output

SMMU_SS_LOCAL_SMMU_INTR_SEL_NS

Bits	Name	Description
31:0	INT_APP_CXT_EN	Setting nth bit will steer nth context bank interrupt to non-secure interrupt output. If same nth bit is set in SMMU_INTR_SEL_SEC OR SMMU_INTR_SEL_NSHYP, then nth context bank interrupt won't be steered to non-secure interrupt output.

00x01EF4000 SMMU_SS_LOCAL_SMMU_SID2VMID_SMR_JPEG_n, n=[0..1]+0x8*n**Type:** RW**Clock:** PCNOC_CLK**Reset State:** 0x00000000

This register is used to configure SID2VMID for JPEG TBU

SMMU_SS_LOCAL_SMMU_SID2VMID_SMR_JPEG_n

Bits	Name	Description
6	VALID	If 1: This register has valid values and will be used for matching. If 0: This register has invalid values and will not be used for matching
5:2	VMID	VMID value in case of match
1	SID_MASK	Setting nth bit will mask nth SID bit for comparison
0	SID_MATCH	Match value of SID

0x01EF4010 SMMU_SS_LOCAL_SMMU_SID2VMID_FAULT_VMID_JPEG**Type:** RW**Clock:** PCNOC_CLK**Reset State:** 0x00000000

This register is used to set VMID in case of no match

SMMU_SS_LOCAL_SMMU_SID2VMID_FAULT_VMID_JPEG

Bits	Name	Description
3:0	VMID	No Match VMID

0x01EF4018 SMMU_SS_LOCAL_SMMU_SID2VMID_FAULT_INTR_EN_JPEG**Type:** RW**Clock:** PCNOC_CLK**Reset State:** 0x00000000

This register is used to enable SID2VMID Fault Interrupt

SMMU_SS_LOCAL_SMMU_SID2VMID_FAULT_INTR_EN_JPEG

Bits	Name	Description
0	FAULT_INTR_EN	Setting this enables SID2VMID no match and multi match interrupt

0x01EF4020 SMMU_SS_LOCAL_SMMU_SID2VMID_FAULT_INTR_STTS_JPEG**Type:** R**Clock:** PCNOC_CLK**Reset State:** 0x00000000

This register is used for SID2VMID Fault Interrupt Status

SMMU_SS_LOCAL_SMMU_SID2VMID_FAULT_INTR_STTS_JPEG

Bits	Name	Description
2	FAULT_SID	SID which caused nomatch OR multimatch Fault
1	FAULT_NO_MATCH	No Match Interrupt
0	FAULT_MULTI_MATCH	Multi Match Interrupt

0x01EF4028 SMMU_SS_LOCAL_SMMU_SID2VMID_FAULT_INTR_CLR_JPEG**Type:** W**Clock:** PCNOC_CLK**Reset State:** 0x00000000

This register is used to clear SID2VMID Fault Interrupt

SMMU_SS_LOCAL_SMMU_SID2VMID_FAULT_INTR_CLR_JPEG

Bits	Name	Description
0	FAULT_INTR_CLR	Writing 1 will clear SID2VMID nomatch/multimatch Interrupt

**0x01EF5000+ SMMU_SS_LOCAL_SMMU_SID2VMID_SMR_VFE_n, n=[0..3]
0x8*n****Type:** RW**Clock:** PCNOC_CLK**Reset State:** 0x00000000

This register is used to configure SID2VMID for VFE TBU

SMMU_SS_LOCAL_SMMU_SID2VMID_SMR_VFE_n

Bits	Name	Description
8	VALID	If 1: This register has valid values and will be used for matching. If 0: This register has invalid values and will not be used for matching
7:4	VMID	VMID value in case of match
3:2	SID_MASK	Setting nth bit will mask nth SID bit for comparison
1:0	SID_MATCH	Match value of SID

0x01EF5020 SMMU_SS_LOCAL_SMMU_SID2VMID_FAULT_VMID_VFE**Type:** RW**Clock:** PCNOC_CLK**Reset State:** 0x00000000

This register is used to set VMID in case of no match

SMMU_SS_LOCAL_SMMU_SID2VMID_FAULT_VMID_VFE

Bits	Name	Description
3:0	VMID	No Match VMID

0x01EF5028 SMMU_SS_LOCAL_SMMU_SID2VMID_FAULT_INTR_EN_VFE**Type:** RW**Clock:** PCNOC_CLK**Reset State:** 0x00000000

This register is used to enable SID2VMID Fault Interrupt

SMMU_SS_LOCAL_SMMU_SID2VMID_FAULT_INTR_EN_VFE

Bits	Name	Description
0	FAULT_INTR_EN	Setting this enables SID2VMID no match and multi match interrupt

0x01EF5030 SMMU_SS_LOCAL_SMMU_SID2VMID_FAULT_INTR_STTS_VFE**Type:** R**Clock:** PCNOC_CLK**Reset State:** 0x00000000

This register is used for SID2VMID Fault Interrupt Status

SMMU_SS_LOCAL_SMMU_SID2VMID_FAULT_INTR_STTS_VFE

Bits	Name	Description
3:2	FAULT_SID	SID which caused nomatch OR multimatch Fault
1	FAULT_NO_MATCH	No Match Interrupt
0	FAULT_MULTI_MATCH	Multi Match Interrupt

0x01EF5038 SMMU_SS_LOCAL_SMMU_SID2VMID_FAULT_INTR_CLR_VFE**Type:** W**Clock:** PCNOC_CLK**Reset State:** 0x00000000

This register is used to clear SID2VMID Fault Interrupt

SMMU_SS_LOCAL_SMMU_SID2VMID_FAULT_INTR_CLR_VFE

Bits	Name	Description
0	FAULT_INTR_CLR	Writing 1 will clear SID2VMID nomatch/multimatch Interrupt

**0x01EF6000+ SMMU_SS_LOCAL_SMMU_SID2VMID_SMR_MDP_n, n=[0..1]
0x8*n****Type:** RW**Clock:** PCNOC_CLK**Reset State:** 0x00000000

This register is used to configure SID2VMID for MDP TBU

SMMU_SS_LOCAL_SMMU_SID2VMID_SMR_MDP_n

Bits	Name	Description
6	VALID	If 1: This register has valid values and will be used for matching. If 0: This register has invalid values and will not be used for matching
5:2	VMID	VMID value in case of match
1	SID_MASK	Setting nth bit will mask nth SID bit for comparison
0	SID_MATCH	Match value of SID

0x01EF6010 SMMU_SS_LOCAL_SMMU_SID2VMID_FAULT_VMID_MDP**Type:** RW**Clock:** PCNOC_CLK**Reset State:** 0x00000000

This register is used to set VMID in case of no match

SMMU_SS_LOCAL_SMMU_SID2VMID_FAULT_VMID_MDP

Bits	Name	Description
3:0	VMID	No Match VMID

0x01EF6018 SMMU_SS_LOCAL_SMMU_SID2VMID_FAULT_INTR_EN_MDP**Type:** RW**Clock:** PCNOC_CLK**Reset State:** 0x00000000

This register is used to enable SID2VMID Fault Interrupt

SMMU_SS_LOCAL_SMMU_SID2VMID_FAULT_INTR_EN_MDP

Bits	Name	Description
0	FAULT_INTR_EN	Setting this enables SID2VMID no match and multi match interrupt

0x01EF6020 SMMU_SS_LOCAL_SMMU_SID2VMID_FAULT_INTR_STTS_MDP

Type: R
Clock: PCNOC_CLK
Reset State: 0x00000000

This register is used for SID2VMID Fault Interrupt Status

SMMU_SS_LOCAL_SMMU_SID2VMID_FAULT_INTR_STTS_MDP

Bits	Name	Description
2	FAULT_SID	SID which caused nomatch OR multimatch Fault
1	FAULT_NO_MATCH	No Match Interrupt
0	FAULT_MULTI_MATCH	Multi Match Interrupt

0x01EF6028 SMMU_SS_LOCAL_SMMU_SID2VMID_FAULT_INTR_CLR_MDP

Type: W
Clock: PCNOC_CLK
Reset State: 0x00000000

This register is used to clear SID2VMID Fault Interrupt

SMMU_SS_LOCAL_SMMU_SID2VMID_FAULT_INTR_CLR_MDP

Bits	Name	Description
0	FAULT_INTR_CLR	Writing 1 will clear SID2VMID nomatch/multimatch Interrupt

**0x01EF7000+ SMMU_SS_LOCAL_SMMU_SID2VMID_SMR_VENUS_n, n=[0..16]
0x8*n**

Type: RW
Clock: PCNOC_CLK
Reset State: 0x00000000

This register is used to configure SID2VMID for VENUS TBU

SMMU_SS_LOCAL_SMMU_SID2VMID_SMR_VENUS_n

Bits	Name	Description
20	VALID	If 1: This register has valid values and will be used for matching. If 0: This register has invalid values and will not be used for matching
19:16	VMID	VMID value in case of match
15:8	SID_MASK	Setting nth bit will mask nth SID bit for comparison

SMMU_SS_LOCAL_SMMU_SID2VMID_SMR_VENUS_n (cont.)

Bits	Name	Description
7:0	SID_MATCH	Match value of SID

0x01EF7088 SMMU_SS_LOCAL_SMMU_SID2VMID_FAULT_VMID_VENUS**Type:** RW**Clock:** PCNOC_CLK**Reset State:** 0x00000000

This register is used to set VMID in case of no match

SMMU_SS_LOCAL_SMMU_SID2VMID_FAULT_VMID_VENUS

Bits	Name	Description
3:0	VMID	No Match VMID

0x01EF7090 SMMU_SS_LOCAL_SMMU_SID2VMID_FAULT_INTR_EN_VENUS**Type:** RW**Clock:** PCNOC_CLK**Reset State:** 0x00000000

This register is used to enable SID2VMID Fault Interrupt

SMMU_SS_LOCAL_SMMU_SID2VMID_FAULT_INTR_EN_VENUS

Bits	Name	Description
0	FAULT_INTR_EN	Setting this enables SID2VMID no match and multi match interrupt

0x01EF7098 SMMU_SS_LOCAL_SMMU_SID2VMID_FAULT_INTR_STTS_VENUS**Type:** R**Clock:** PCNOC_CLK**Reset State:** 0x00000000

This register is used for SID2VMID Fault Interrupt Status

SMMU_SS_LOCAL_SMMU_SID2VMID_FAULT_INTR_STTS_VENUS

Bits	Name	Description
9:2	FAULT_SID	SID which caused nomatch OR multimatch Fault

SMMU_SS_LOCAL_SMMU_SID2VMID_FAULT_INTR_STTS_VENUS (cont.)

Bits	Name	Description
1	FAULT_NO_MATCH	No Match Interrupt
0	FAULT_MULTI_MATCH	Multi Match Interrupt

0x01EF70A0 SMMU_SS_LOCAL_SMMU_SID2VMID_FAULT_INTR_CLR_VENUS**Type:** W**Clock:** PCNOC_CLK**Reset State:** 0x00000000

This register is used to clear SID2VMID Fault Interrupt

SMMU_SS_LOCAL_SMMU_SID2VMID_FAULT_INTR_CLR_VENUS

Bits	Name	Description
0	FAULT_INTR_CLR	Writing 1 will clear SID2VMID nomatch/multimatch Interrupt

**0x01EF8000+ SMMU_SS_LOCAL_SMMU_SID2VMID_SMR_PRONTO_n, n=[0..12]
0x8*n****Type:** RW**Clock:** PCNOC_CLK**Reset State:** 0x00000000

This register is used to configure SID2VMID for PRONTO TBU

SMMU_SS_LOCAL_SMMU_SID2VMID_SMR_PRONTO_n

Bits	Name	Description
14	VALID	If 1: This register has valid values and will be used for matching. If 0: This register has invalid values and will not be used for matching
13:10	VMID	VMID value in case of match
9:5	SID_MASK	Setting nth bit will mask nth SID bit for comparison
4:0	SID_MATCH	Match value of SID

0x01EF8068 SMMU_SS_LOCAL_SMMU_SID2VMID_FAULT_VMID_PRONTO**Type:** RW**Clock:** PCNOC_CLK**Reset State:** 0x00000000

This register is used to set VMID in case of no match

SMMU_SS_LOCAL_SMMU_SID2VMID_FAULT_VMID_PRONTO

Bits	Name	Description
3:0	VMID	No Match VMID

0x01EF8070 SMMU_SS_LOCAL_SMMU_SID2VMID_FAULT_INTR_EN_PRONTO**Type:** RW**Clock:** PCNOC_CLK**Reset State:** 0x00000000

This register is used to enable SID2VMID Fault Interrupt

SMMU_SS_LOCAL_SMMU_SID2VMID_FAULT_INTR_EN_PRONTO

Bits	Name	Description
0	FAULT_INTR_EN	Setting this enables SID2VMID no match and multi match interrupt

0x01EF8078 SMMU_SS_LOCAL_SMMU_SID2VMID_FAULT_INTR_STTS_PRONTO**Type:** R**Clock:** PCNOC_CLK**Reset State:** 0x00000000

This register is used for SID2VMID Fault Interrupt Status

SMMU_SS_LOCAL_SMMU_SID2VMID_FAULT_INTR_STTS_PRONTO

Bits	Name	Description
6:2	FAULT_SID	SID which caused nomatch OR multimatch Fault
1	FAULT_NO_MATCH	No Match Interrupt
0	FAULT_MULTI_MATCH	Multi Match Interrupt

0x01EF8080 SMMU_SS_LOCAL_SMMU_SID2VMID_FAULT_INTR_CLR_PRONTO**Type:** W**Clock:** PCNOC_CLK**Reset State:** 0x00000000

This register is used to clear SID2VMID Fault Interrupt

SMMU_SS_LOCAL_SMMU_SID2VMID_FAULT_INTR_CLR_PRONTO

Bits	Name	Description
0	FAULT_INTR_CLR	Writing 1 will clear SID2VMID nomatch/multimatch Interrupt

**0x01EF9000+ SMMU_SS_LOCAL_SMMU_SID2VMID_SMR_PERIPH_n, n=[0..46]
0x8*n****Type:** RW**Clock:** PCNOC_CLK**Reset State:** 0x00000000

This register is used to configure SID2VMID for PERIPH TBU

SMMU_SS_LOCAL_SMMU_SID2VMID_SMR_PERIPH_n

Bits	Name	Description
24	VALID	If 1: This register has valid values and will be used for matching. If 0: This register has invalid values and will not be used for matching
23:20	VMID	VMID value in case of match
19:10	SID_MASK	Setting nth bit will mask nth SID bit for comparison
9:0	SID_MATCH	Match value of SID

0x01EF9178 SMMU_SS_LOCAL_SMMU_SID2VMID_FAULT_VMID_PERIPH**Type:** RW**Clock:** PCNOC_CLK**Reset State:** 0x00000000

This register is used to set VMID in case of no match

SMMU_SS_LOCAL_SMMU_SID2VMID_FAULT_VMID_PERIPH

Bits	Name	Description
3:0	VMID	No Match VMID

0x01EF9180 SMMU_SS_LOCAL_SMMU_SID2VMID_FAULT_INTR_EN_PERIPH**Type:** RW**Clock:** PCNOC_CLK**Reset State:** 0x00000000

This register is used to enable SID2VMID Fault Interrupt

SMMU_SS_LOCAL_SMMU_SID2VMID_FAULT_INTR_EN_PERIPH

Bits	Name	Description
0	FAULT_INTR_EN	Setting this enables SID2VMID no match and multi match interrupt

0x01EF9188 SMMU_SS_LOCAL_SMMU_SID2VMID_FAULT_INTR_STTS_PERIPH**Type:** R**Clock:** PCNOC_CLK**Reset State:** 0x00000000

This register is used for SID2VMID Fault Interrupt Status

SMMU_SS_LOCAL_SMMU_SID2VMID_FAULT_INTR_STTS_PERIPH

Bits	Name	Description
11:2	FAULT_SID	SID which caused nomatch OR multimatch Fault
1	FAULT_NO_MATCH	No Match Interrupt
0	FAULT_MULTI_MATCH	Multi Match Interrupt

0x01EF9190 SMMU_SS_LOCAL_SMMU_SID2VMID_FAULT_INTR_CLR_PERIPH**Type:** W**Clock:** PCNOC_CLK**Reset State:** 0x00000000

This register is used to clear SID2VMID Fault Interrupt

SMMU_SS_LOCAL_SMMU_SID2VMID_FAULT_INTR_CLR_PERIPH

Bits	Name	Description
0	FAULT_INTR_CLR	Writing 1 will clear SID2VMID nomatch/multimatch Interrupt

**0x01EFA000+ SMMU_SS_LOCAL_SMMU_SID2VMID_SMR_DIME_n, n=[0..1]
0x8*n****Type:** RW**Clock:** PCNOC_CLK**Reset State:** 0x00000000

This register is used to configure SID2VMID for DIME TBU

SMMU_SS_LOCAL_SMMU_SID2VMID_SMR_DIME_n

Bits	Name	Description
6	VALID	If 1: This register has valid values and will be used for matching. If 0: This register has invalid values and will not be used for matching
5:2	VMID	VMID value in case of match
1	SID_MASK	Setting nth bit will mask nth SID bit for comparison
0	SID_MATCH	Match value of SID

0x01EFA010 SMMU_SS_LOCAL_SMMU_SID2VMID_FAULT_VMID_DIME**Type:** RW**Clock:** PCNOC_CLK**Reset State:** 0x00000000

This register is used to set VMID in case of no match

SMMU_SS_LOCAL_SMMU_SID2VMID_FAULT_VMID_DIME

Bits	Name	Description
3:0	VMID	No Match VMID

0x01EFA018 SMMU_SS_LOCAL_SMMU_SID2VMID_FAULT_INTR_EN_DIME**Type:** RW**Clock:** PCNOC_CLK**Reset State:** 0x00000000

This register is used to enable SID2VMID Fault Interrupt

SMMU_SS_LOCAL_SMMU_SID2VMID_FAULT_INTR_EN_DIME

Bits	Name	Description
0	FAULT_INTR_EN	Setting this enables SID2VMID no match and multi match interrupt

0x01EFA020 SMMU_SS_LOCAL_SMMU_SID2VMID_FAULT_INTR_STTS_DIME**Type:** R**Clock:** PCNOC_CLK**Reset State:** 0x00000000

This register is used for SID2VMID Fault Interrupt Status

SMMU_SS_LOCAL_SMMU_SID2VMID_FAULT_INTR_STTS_DIME

Bits	Name	Description
2	FAULT_SID	SID which caused nomatch OR multimatch Fault
1	FAULT_NO_MATCH	No Match Interrupt
0	FAULT_MULTI_MATCH	Multi Match Interrupt

0x01EFA028 SMMU_SS_LOCAL_SMMU_SID2VMID_FAULT_INTR_CLR_DIME**Type:** W**Clock:** PCNOC_CLK**Reset State:** 0x00000000

This register is used to clear SID2VMID Fault Interrupt

SMMU_SS_LOCAL_SMMU_SID2VMID_FAULT_INTR_CLR_DIME

Bits	Name	Description
0	FAULT_INTR_CLR	Writing 1 will clear SID2VMID nomatch/multimatch Interrupt

**0x01EFB000+SMMU_SS_LOCAL_SMMU_SID2VMID_SMR_GPS_n, n=[0..1]
0x8*n****Type:** RW**Clock:** PCNOC_CLK**Reset State:** 0x00000000

This register is used to configure SID2VMID for GPS TBU

SMMU_SS_LOCAL_SMMU_SID2VMID_SMR_GPS_n

Bits	Name	Description
6	VALID	If 1: This register has valid values and will be used for matching. If 0: This register has invalid values and will not be used for matching
5:2	VMID	VMID value in case of match
1	SID_MASK	Setting nth bit will mask nth SID bit for comparison
0	SID_MATCH	Match value of SID

0x01EFB010 SMMU_SS_LOCAL_SMMU_SID2VMID_FAULT_VMID_GPS**Type:** RW**Clock:** PCNOC_CLK**Reset State:** 0x00000000

This register is used to set VMID in case of no match

SMMU_SS_LOCAL_SMMU_SID2VMID_FAULT_VMID_GPS

Bits	Name	Description
3:0	VMID	No Match VMID

0x01EFB018 SMMU_SS_LOCAL_SMMU_SID2VMID_FAULT_INTR_EN_GPS**Type:** RW**Clock:** PCNOC_CLK**Reset State:** 0x00000000

This register is used to enable SID2VMID Fault Interrupt

SMMU_SS_LOCAL_SMMU_SID2VMID_FAULT_INTR_EN_GPS

Bits	Name	Description
0	FAULT_INTR_EN	Setting this enables SID2VMID no match and multi match interrupt

0x01EFB020 SMMU_SS_LOCAL_SMMU_SID2VMID_FAULT_INTR_STTS_GPS**Type:** R**Clock:** PCNOC_CLK**Reset State:** 0x00000000

This register is used for SID2VMID Fault Interrupt Status

SMMU_SS_LOCAL_SMMU_SID2VMID_FAULT_INTR_STTS_GPS

Bits	Name	Description
2	FAULT_SID	SID which caused nomatch OR multimatch Fault
1	FAULT_NO_MATCH	No Match Interrupt
0	FAULT_MULTI_MATCH	Multi Match Interrupt

0x01EFB028 SMMU_SS_LOCAL_SMMU_SID2VMID_FAULT_INTR_CLR_GPS**Type:** W**Clock:** PCNOC_CLK**Reset State:** 0x00000000

This register is used to clear SID2VMID Fault Interrupt

SMMU_SS_LOCAL_SMMU_SID2VMID_FAULT_INTR_CLR_GPS

Bits	Name	Description
0	FAULT_INTR_CLR	Writing 1 will clear SID2VMID nomatch/multimatch Interrupt

**0x01EFC000+ SMMU_SS_LOCAL_SMMU_SID2VMID_SMR_QDSP_n, n=[0..1]
0x8*n****Type:** RW**Clock:** PCNOC_CLK**Reset State:** 0x00000000

This register is used to configure SID2VMID for QDSP TBU

SMMU_SS_LOCAL_SMMU_SID2VMID_SMR_QDSP_n

Bits	Name	Description
7	VALID	If 1: This register has valid values and will be used for matching. If 0: This register has invalid values and will not be used for matching
6:2	VMID	VMID value in case of match
1	SID_MASK	Setting nth bit will mask nth SID bit for comparison
0	SID_MATCH	Match value of SID

0x01EFC010 SMMU_SS_LOCAL_SMMU_SID2VMID_FAULT_VMID_QDSP**Type:** RW**Clock:** PCNOC_CLK**Reset State:** 0x00000000

This register is used to set VMID in case of no match

SMMU_SS_LOCAL_SMMU_SID2VMID_FAULT_VMID_QDSP

Bits	Name	Description
4:0	VMID	No Match VMID

0x01EFC018 SMMU_SS_LOCAL_SMMU_SID2VMID_FAULT_INTR_EN_QDSP**Type:** RW**Clock:** PCNOC_CLK**Reset State:** 0x00000000

This register is used to enable SID2VMID Fault Interrupt

SMMU_SS_LOCAL_SMMU_SID2VMID_FAULT_INTR_EN_QDSP

Bits	Name	Description
0	FAULT_INTR_EN	Setting this enables SID2VMID no match and multi match interrupt

0x01EFC020 SMMU_SS_LOCAL_SMMU_SID2VMID_FAULT_INTR_STTS_QDSP**Type:** R**Clock:** PCNOC_CLK**Reset State:** 0x00000000

This register is used for SID2VMID Fault Interrupt Status

SMMU_SS_LOCAL_SMMU_SID2VMID_FAULT_INTR_STTS_QDSP

Bits	Name	Description
2	FAULT_SID	SID which caused nomatch OR multimatch Fault
1	FAULT_NO_MATCH	No Match Interrupt
0	FAULT_MULTI_MATCH	Multi Match Interrupt

0x01EFC028 SMMU_SS_LOCAL_SMMU_SID2VMID_FAULT_INTR_CLR_QDSP**Type:** W**Clock:** PCNOC_CLK**Reset State:** 0x00000000

This register is used to clear SID2VMID Fault Interrupt

SMMU_SS_LOCAL_SMMU_SID2VMID_FAULT_INTR_CLR_QDSP

Bits	Name	Description
0	FAULT_INTR_CLR	Writing 1 will clear SID2VMID nomatch/multimatch Interrupt

**0x01EFD000+SMMU_SS_LOCAL_SMMU_SID2VMID_SMR_GFX_n, n=[0..1]
0x8*n****Type:** RW**Clock:** PCNOC_CLK**Reset State:** 0x00000000

This register is used to configure SID2VMID for GFX TBU

SMMU_SS_LOCAL_SMMU_SID2VMID_SMR_GFX_n

Bits	Name	Description
7	VALID	If 1: This register has valid values and will be used for matching. If 0: This register has invalid values and will not be used for matching
6:2	VMID	VMID value in case of match
1	SID_MASK	Setting nth bit will mask nth SID bit for comparison
0	SID_MATCH	Match value of SID

0x01EFD010 SMMU_SS_LOCAL_SMMU_SID2VMID_FAULT_VMID_GFX**Type:** RW**Clock:** PCNOC_CLK**Reset State:** 0x00000000

This register is used to set VMID in case of no match

SMMU_SS_LOCAL_SMMU_SID2VMID_FAULT_VMID_GFX

Bits	Name	Description
4:0	VMID	No Match VMID

0x01EFD018 SMMU_SS_LOCAL_SMMU_SID2VMID_FAULT_INTR_EN_GFX**Type:** RW**Clock:** PCNOC_CLK**Reset State:** 0x00000000

This register is used to enable SID2VMID Fault Interrupt

SMMU_SS_LOCAL_SMMU_SID2VMID_FAULT_INTR_EN_GFX

Bits	Name	Description
0	FAULT_INTR_EN	Setting this enables SID2VMID no match and multi match interrupt

0x01EFD020 SMMU_SS_LOCAL_SMMU_SID2VMID_FAULT_INTR_STTS_GFX**Type:** R**Clock:** PCNOC_CLK**Reset State:** 0x00000000

This register is used for SID2VMID Fault Interrupt Status

SMMU_SS_LOCAL_SMMU_SID2VMID_FAULT_INTR_STTS_GFX

Bits	Name	Description
2	FAULT_SID	SID which caused nomatch OR multimatch Fault
1	FAULT_NO_MATCH	No Match Interrupt
0	FAULT_MULTI_MATCH	Multi Match Interrupt

0x01EFD028 SMMU_SS_LOCAL_SMMU_SID2VMID_FAULT_INTR_CLR_GFX**Type:** W**Clock:** PCNOC_CLK**Reset State:** 0x00000000

This register is used to clear SID2VMID Fault Interrupt

SMMU_SS_LOCAL_SMMU_SID2VMID_FAULT_INTR_CLR_GFX

Bits	Name	Description
0	FAULT_INTR_CLR	Writing 1 will clear SID2VMID nomatch/multimatch Interrupt

25 Qualcomm Debug Subsystem Wrapper

25.1 CXATBFUNNEL_32W2SP

The Funnel is used when there is more than one trace source.

25.1.1 QDSS_WRAPPER_ATB_FUN2X1_CXATBFUNNEL_32W2SP

The Funnel is used when there is more than one trace source.

0x00868000 QDSS_WRAPPER_ATB_FUN2X1_CTRL_REG

Type: RW

Reset State: 0x00000300

The Funnel Control Register enables the slave ports and defines the hold time of the slave ports. Hold time refers to the number of transactions that are output on the funnel master port from the same slave while that slave port atvalidsx is HIGH. Hold time does not refer to clock cycles in this context

QDSS_WRAPPER_ATB_FUN2X1_CTRL_REG

Bits	Name	Description
11:8	HT	The formatting scheme can easily become inefficient if fast switching occurs, so, where possible, this must be minimized. If a source has nothing to transmit, then another source is selected irrespective of the minimum number of transactions. The maximum value that can be entered is 0xE and this equates to 15 cycles. 0xF is reserved. 0x0: HOLD_TIME_1 (1 transactions hold time) 0x1: HOLD_TIME_2 (2 transactions hold time) 0x2: HOLD_TIME_3 (3 transactions hold time) 0x3: HOLD_TIME_4 (4 transactions hold time) 0x4: HOLD_TIME_5 (5 transactions hold time) 0x5: HOLD_TIME_6 (6 transactions hold time) 0x6: HOLD_TIME_7 (7 transactions hold time) 0x7: HOLD_TIME_8 (8 transactions hold time) 0x8: HOLD_TIME_9 (9 transactions hold time) 0x9: HOLD_TIME_10 (10 transactions hold time) 0xA: HOLD_TIME_11 (11 transactions hold time) 0xB: HOLD_TIME_12 (12 transactions hold time) 0xC: HOLD_TIME_13 (13 transactions hold time) 0xD: HOLD_TIME_14 (14 transactions hold time) 0xE: HOLD_TIME_15 (15 transactions hold time)
1	ENS1	Setting this bit enables this input, or slave, port. If the bit is not set then this has the effect of excluding the port from the priority selection scheme. The reset value is all clear, that is, all ports disabled. 0x1: ENABLE (Slave port enabled) 0x0: DISABLE (Slave port disabled)
0	ENS0	Setting this bit enables this input, or slave, port. If the bit is not set then this has the effect of excluding the port from the priority selection scheme. The reset value is all clear, that is, all ports disabled. 0x1: ENABLE (Slave port enabled) 0x0: DISABLE (Slave port disabled)

0x00868004 QDSS_WRAPPER_ATB_FUN2X1_PRIORITY_CTRL_REG**Type:** RW**Reset State:** 0x00000000

The Priority Control Register defines the order in which inputs are selected. Each 3-bit field represents a priority for each particular slave interface. Location 0 has the priority value for the first slave port. Location 1 is the priority value for the second slave port, Location 2 is the third, down to location 7, which has the priority value of the eighth slave port. Values represent the priority value for each port number. If you want to give highest priority to a particular slave port,

the corresponding port must be programmed with the lowest value. Typically this is likely to be a port that has more important data or that has a small FIFO and is therefore likely to overflow.

If you want to give lowest priority to a particular slave port, the corresponding slave port must be programmed with the highest value. Typically this is likely to be a device that has a large FIFO that is less likely to overflow or a source that has information that is of lower importance.

QDSS_WRAPPER_ATB_FUN2X1_PRIORITY_CTRL_REG

Bits	Name	Description
5:3	PRIPORT1	Priority value of the 2nd port. 0x0: PRORITY_LEVEL_0 (This value represents the highest priority.) 0x1: PRORITY_LEVEL_1 (2nd highest priority) 0x2: PRORITY_LEVEL_2 (3rd highest priority) 0x3: PRORITY_LEVEL_3 (4th highest priority) 0x4: PRORITY_LEVEL_4 (5th highest priority) 0x5: PRORITY_LEVEL_5 (6th highest priority) 0x6: PRORITY_LEVEL_6 (7th highest priority) 0x7: PRORITY_LEVEL_7 (This value represents the lowest priority.)
2:0	PRIPORT0	Priority value of the first slave port. The value written into this location is the value that you want to assign the first slave port. 0x0: PRORITY_LEVEL_0 (This value represents the highest priority.) 0x1: PRORITY_LEVEL_1 (2nd highest priority) 0x2: PRORITY_LEVEL_2 (3rd highest priority) 0x3: PRORITY_LEVEL_3 (4th highest priority) 0x4: PRORITY_LEVEL_4 (5th highest priority) 0x5: PRORITY_LEVEL_5 (6th highest priority) 0x6: PRORITY_LEVEL_6 (7th highest priority) 0x7: PRORITY_LEVEL_7 (This value represents the lowest priority.)

00x00868F00 QDSS_WRAPPER_ATB_FUN2X1_ITCTRL

Type: RW

Reset State: 0x00000000

This register is used to enable topology detection. For more information see the CoreSight Architecture Specification. This register enables the component to switch from a functional mode, the default behavior, to integration mode where the inputs and outputs of the component can be directly controlled for the purpose of integration testing and topology solving

NOTE When a device has been in integration mode, it might not function with the original behavior. After performing integration or topology detection, you must reset the system to ensure correct behavior of CoreSight and other connected system components that are affected by the integration or topology detection.

QDSS_WRAPPER_ATB_FUN2X1_ITCTRL

Bits	Name	Description
0	INTEGRATION_MODE	Allows the component to switch from functional mode to integration mode or back. If no integration functionality is implemented, this register must read as zero. 0x1: INTEGRATION_MODE (Enable integration mode) 0x0: FUNCTIONAL_MODE (Disable integration mode)

0x00868FA0 QDSS_WRAPPER_ATB_FUN2X1_CLAIMSET**Type:** RW**Reset State:** 0x0000000F

This is used in conjunction with Claim Tag Clear Register, CLAIMCLR. This register forms one half of the Claim Tag value. This location allows individual bits to be set, write, and returns the number of bits that can be set, read.

QDSS_WRAPPER_ATB_FUN2X1_CLAIMSET

Bits	Name	Type	Description
3:0	CLAIMSET_W	W	Each bit is considered separately: 0 = no effect, 1 = set this bit in the claim tag.
3:0	CLAIMSET_R	R	This claim tag bit is implemented 0xF: CLAIM_TAG_IMPLEMENTED_BITS (These bits are present within the claim tag field)

0x00868FA4 QDSS_WRAPPER_ATB_FUN2X1_CLAIMCLR**Type:** RW**Reset State:** 0x00000000

This register is used in conjunction with Claim Tag Set Register, CLAIMSET. This register forms one half of the Claim Tag value. This location enables individual bits to be cleared, write, and returns the current Claim Tag value, read.

QDSS_WRAPPER_ATB_FUN2X1_CLAIMCLR

Bits	Name	Type	Description
3:0	CLAIMCLR_W	W	Each bit is considered separately: 0 = no effect, 1 = clear this bit in the claim tag.
3:0	CLAIMCLR_R	R	The value present reflects the current setting of the Claim Tag.

0x00868FB0 QDSS_WRAPPER_ATB_FUN2X1_LOCKACCESS**Type:** W**Reset State:** Undefined

This is used to enable write access to device registers.

QDSS_WRAPPER_ATB_FUN2X1_LOCKACCESS

Bits	Name	Description
31:0	ACCESS_W	A write of 0xC5ACCE55 enables further write access to this device. An invalid write will have the affect of removing write access. 0xC5ACCE55: UNLOCK (Unlock the protection register to enable write access) 0xDEADBEEF: LOCK (invalid write value to enable the protection register and block write access)

0x00868FB4 QDSS_WRAPPER_ATB_FUN2X1_LOCKSTATUS**Type:** R**Reset State:** 0x00000003

This indicates the status of the Lock control mechanism. This lock prevents accidental writes by code under debug. This register must always be present although there might not be any lock-access control mechanism. The lock mechanism, where present and locked, must block write accesses to any control register, except the Lock Access Register. For most components this covers all registers except for the Lock Access Register 0xFB0

QDSS_WRAPPER_ATB_FUN2X1_LOCKSTATUS

Bits	Name	Description
2	LOCKTYPE	Indicates if the Lock Access Register (0xFB0) is implemented as 8-bit or 32-bit 0x0: ENUM_32_BIT (This component implements a 32-bit Lock Access Register)
1	LOCKGRANT	Returns the current status of the Lock 0x0: ACCESS_PERMITTED (Write access is allowed to this device) 0x1: DEVICE_LOCKED (Write access to the component is blocked. All writes to control registers are ignored. Reads are permitted)
0	LOCKEXIST	Indicates that a lock control mechanism exists for this device 0x0: LOCK_NOT_PRESENT (No lock control mechanism exists, writes to the Lock Access register (0xFB0) will be ignored) 0x1: LOCK_PRESENT (Lock control mechanism is present)

0x00868FB8 QDSS_WRAPPER_ATB_FUN2X1_AUTHSTATUS**Type:** R**Reset State:** 0x00000000

Reports the required **security** level and current status of those enables. Where functionality changes on a given **security** level then this change in status must be reported in this register

QDSS_WRAPPER_ATB_FUN2X1_AUTHSTATUS

Bits	Name	Description
7:6	SNID	Indicates the security level for secure non-invasive debug 0x0: ENUM_2_B00 (Functionality not implemented or controlled elsewhere)
5:4	SID	Indicates the security level for secure invasive debug 0x0: ENUM_2_B00 (Functionality not implemented or controlled elsewhere)
3:2	NSNID	Indicates the security level for non-secure non-invasive debug 0x0: ENUM_2_B00 (Functionality not implemented or controlled elsewhere)
1:0	NSID	Indicates the security level for non-secure invasive debug 0x0: ENUM_2_B00 (Functionality not implemented or controlled elsewhere)

0x00868FC8 QDSS_WRAPPER_ATB_FUN2X1_DEVID**Type:** R**Reset State:** 0x00000032

This indicates the capabilities of the CoreSight Funnel.

QDSS_WRAPPER_ATB_FUN2X1_DEVID

Bits	Name	Description
7:4	SCHEME	This value indicates the priority scheme implemented within the Funnel 0x3: STATIC_PRIORITY (Slave ports may be programmed to have higher or lower priority with respect to each other)
3:0	RESERVED	Reserved

0x00868FCC QDSS_WRAPPER_ATB_FUN2X1_DEVTYPE**Type:** R**Reset State:** 0x00000012

It provides a debugger with information about the component when the Part Number field is not recognized. The debugger can then report this information.

QDSS_WRAPPER_ATB_FUN2X1_DEVTYPE

Bits	Name	Description
7:4	SUB_TYPE	sub-classification within the major category 0x1: FUNNEL_ROUTER
3:0	MAJOR_TYPE	Major classification grouping for this debug/trace component 0x2: TRACE_LINK

0x00868FE0 QDSS_WRAPPER_ATB_FUN2X1_PERIPHID0

Type: R

Reset State: 0x00000008

Part of the set of Peripheral Identification registers. Contains part of the designer specific part number.

QDSS_WRAPPER_ATB_FUN2X1_PERIPHID0

Bits	Name	Description
7:0	PART_NUMBER_BITS7TO0	Bits [7:0] of the component's part number. This is selected by the designer of the component. 0x8: CORESIGHT_FUNNEL_PART_NUMBER_7_0 (Lowest 8 bits of the Part Number (0x100))

0x00868FE4 QDSS_WRAPPER_ATB_FUN2X1_PERIPHID1

Type: R

Reset State: 0x000000B9

Part of the set of Peripheral Identification registers. Contains part of the designer specific part number and part of the designer identity.

QDSS_WRAPPER_ATB_FUN2X1_PERIPHID1

Bits	Name	Description
7:4	JEP106_BITS3TO0	Bits 3:0 of the JEDEC identity code indicating the designer of the component (along with the continuation code) 0xB: ARM_JEP106_IDENTITY_CODE_7_0 (Lowest 4 bits of the JEP106 Identity Code)

QDSS_WRAPPER_ATB_FUN2X1_PERIPHID1 (cont.)

Bits	Name	Description
3:0	PART_NUMBER_BITS11TO8	Bits [11:8] of the component's part number. This is selected by the designer of the component. 0x9: CORESIGHT_FUNNEL_PART_NUMBER_11_8 (Upper 4 bits of the Part Number (0x100))

0x00868FE8 QDSS_WRAPPER_ATB_FUN2X1_PERIPHID2**Type:** R**Reset State:** 0x0000002B

Part of the set of Peripheral Identification registers. Contains part of the designer identity and the product revision.

QDSS_WRAPPER_ATB_FUN2X1_PERIPHID2

Bits	Name	Description
7:4	REVISION	The Revision field is an incremental value starting at 0x0 for the first design of this component. This only increases by 1 for both major and minor revisions and is simply used as a look-up to establish the exact major/minor revision. 0x2: R2P0 (This device is at r2p0)
3	JEDEC	Always set. Indicates that a JEDEC assigned value is used 0x1: JEDEC_IDENTITY (The designer ID is specified by JEDEC (http://www.jedec.org))
2:0	JEP106_BITS6TO4	Bits 6:4 of the JEDEC identity code indicating the designer of the component (along with the continuation code) 0x3: ARM_JEP106_IDENTITY_CODE_6_4 (Upper 3 bits of the JEP106 Identity Code)

0x00868FEC QDSS_WRAPPER_ATB_FUN2X1_PERIPHID3**Type:** R**Reset State:** 0x00000000

Part of the set of Peripheral Identification registers. Contains the RevAnd and Customer Modified fields.

QDSS_WRAPPER_ATB_FUN2X1_PERIPHID3

Bits	Name	Description
7:4	REVAND	This field indicates minor errata fixes specific to this design, for example metal fixes after implementation. In most cases this field is zero. It is recommended that component designers ensure this field can be changed by a metal fix if required, for example by driving it from registers that reset to zero. 0x0: NO_MODS (Indicates that there have been no metal fixes to this component)
3:0	CUSTOMER_MODIFIED	Where the component is reusable IP, this value indicates if the customer has modified the behavior of the component. In most cases this field is zero. 0x0: NO_MODS (Indicates that there have been no modifications made)

0x00868FD0 QDSS_WRAPPER_ATB_FUN2X1_PERIPHID4**Type:** R**Reset State:** 0x00000004

Part of the set of Peripheral Identification registers. Contains part of the designer identity and the memory footprint indicator.

QDSS_WRAPPER_ATB_FUN2X1_PERIPHID4

Bits	Name	Description
7:4	FOURKB_COUNT	This is a 4-bit value that indicates the total contiguous size of the memory window used by this component in powers of 2 from the standard 4KB. If a component only requires the standard 4KB then this should read as 0x0, 4KB only, for 8KB set to 0x1, 16KB == 0x2, 32KB == 0x3, and so on. 0x0: ENUM_4KB (Indicates that the device only occupies 4kB of memory)
3:0	JEP106_CONT	JEDEC continuation code indicating the designer of the component (along with the identity code) 0x4: ARM_JEP106_CONTINUATION_CODE (Indicates that ARM's JEDEC identity code is on the 5th bank)

0x00868FD4 QDSS_WRAPPER_ATB_FUN2X1_PERIPHID5**Type:** RW**Reset State:** Undefined

Reserved

QDSS_WRAPPER_ATB_FUN2X1_PERIPHID5

Bits	Name	Description
31:0	PERIPHID5	

0x00868FD8 QDSS_WRAPPER_ATB_FUN2X1_PERIPHID6**Type:** RW**Reset State:** Undefined

Reserved

QDSS_WRAPPER_ATB_FUN2X1_PERIPHID6

Bits	Name	Description
31:0	PERIPHID6	

0x00868FDC QDSS_WRAPPER_ATB_FUN2X1_PERIPHID7**Type:** RW**Reset State:** Undefined

Reserved

QDSS_WRAPPER_ATB_FUN2X1_PERIPHID7

Bits	Name	Description
31:0	PERIPHID7	

0x00868FF0 QDSS_WRAPPER_ATB_FUN2X1_COMPID0**Type:** R**Reset State:** 0x0000000D

A component identification register, that indicates that the identification registers are present.

QDSS_WRAPPER_ATB_FUN2X1_COMPID0

Bits	Name	Description
7:0	PREAMBLE	Contains bits [24:31] of the component identification 0xD: ENUM_OD (Identification value)

0x00868FF4 QDSS_WRAPPER_ATB_FUN2X1_COMPID1**Type:** R**Reset State:** 0x00000090

A component identification register, that indicates that the identification registers are present. This register also indicates the component class.

QDSS_WRAPPER_ATB_FUN2X1_COMPID1

Bits	Name	Description
7:4	CLASS	Class of the component. E.g. ROM table, CoreSight component etc. 0x9: CORESIGHT_COMPONENT (Indicates the component is a CoreSight component)
3:0	PREAMBLE	Contains bits [19:16] of the component identification 0x0: ENUM_0 (Identification value)

0x00868FF8 QDSS_WRAPPER_ATB_FUN2X1_COMPID2**Type:** R**Reset State:** 0x00000005

A component identification register, that indicates that the identification registers are present.

QDSS_WRAPPER_ATB_FUN2X1_COMPID2

Bits	Name	Description
7:0	PREAMBLE	Contains bits [15:8] of the component identification 0x5: ENUM_05 (Identification value)

0x00868FFC QDSS_WRAPPER_ATB_FUN2X1_COMPID3**Type:** R**Reset State:** 0x000000B1

A component identification register, that indicates that the identification registers are present.

QDSS_WRAPPER_ATB_FUN2X1_COMPID3

Bits	Name	Description
7:0	PREAMBLE	Contains bits [7:0] of the component identification 0xB1: B1 (Identification value)

25.2 CXATBFUNNEL_32W8SP

The Funnel is used when there is more than one trace source.

25.2.1 QDSS_WRAPPER_ATB_FUN8X1_CXATBFUNNEL_32W8SP

The Funnel is used when there is more than one trace source.

0x00869000 QDSS_WRAPPER_ATB_FUN8X1_CTRL_REG

Type: RW

Reset State: 0x00000300

The Funnel Control Register enables the slave ports and defines the hold time of the slave ports. Hold time refers to the number of transactions that are output on the funnel master port from the same slave while that slave port atvalidsx is HIGH. Hold time does not refer to clock cycles in this context

QDSS_WRAPPER_ATB_FUN8X1_CTRL_REG

Bits	Name	Description
11:8	HT	The formatting scheme can easily become inefficient if fast switching occurs, so, where possible, this must be minimized. If a source has nothing to transmit, then another source is selected irrespective of the minimum number of transactions. The maximum value that can be entered is 0xE and this equates to 15 cycles. 0xF is reserved. 0x0: HOLD_TIME_1 (1 transactions hold time) 0x1: HOLD_TIME_2 (2 transactions hold time) 0x2: HOLD_TIME_3 (3 transactions hold time) 0x3: HOLD_TIME_4 (4 transactions hold time) 0x4: HOLD_TIME_5 (5 transactions hold time) 0x5: HOLD_TIME_6 (6 transactions hold time) 0x6: HOLD_TIME_7 (7 transactions hold time) 0x7: HOLD_TIME_8 (8 transactions hold time) 0x8: HOLD_TIME_9 (9 transactions hold time) 0x9: HOLD_TIME_10 (10 transactions hold time) 0xA: HOLD_TIME_11 (11 transactions hold time) 0xB: HOLD_TIME_12 (12 transactions hold time) 0xC: HOLD_TIME_13 (13 transactions hold time) 0xD: HOLD_TIME_14 (14 transactions hold time) 0xE: HOLD_TIME_15 (15 transactions hold time)

QDSS_WRAPPER_ATB_FUN8X1_CTRL_REG (cont.)

Bits	Name	Description
7	ENS7	Setting this bit enables this input, or slave, port. If the bit is not set then this has the effect of excluding the port from the priority selection scheme. The reset value is all clear, that is, all ports disabled. 0x1: ENABLE (Slave port enabled) 0x0: DISABLE (Slave port disabled)
6	ENS6	Setting this bit enables this input, or slave, port. If the bit is not set then this has the effect of excluding the port from the priority selection scheme. The reset value is all clear, that is, all ports disabled. 0x1: ENABLE (Slave port enabled) 0x0: DISABLE (Slave port disabled)
5	ENS5	Setting this bit enables this input, or slave, port. If the bit is not set then this has the effect of excluding the port from the priority selection scheme. The reset value is all clear, that is, all ports disabled. 0x1: ENABLE (Slave port enabled) 0x0: DISABLE (Slave port disabled)
4	ENS4	Setting this bit enables this input, or slave, port. If the bit is not set then this has the effect of excluding the port from the priority selection scheme. The reset value is all clear, that is, all ports disabled. 0x1: ENABLE (Slave port enabled) 0x0: DISABLE (Slave port disabled)
3	ENS3	Setting this bit enables this input, or slave, port. If the bit is not set then this has the effect of excluding the port from the priority selection scheme. The reset value is all clear, that is, all ports disabled. 0x1: ENABLE (Slave port enabled) 0x0: DISABLE (Slave port disabled)
2	ENS2	Setting this bit enables this input, or slave, port. If the bit is not set then this has the effect of excluding the port from the priority selection scheme. The reset value is all clear, that is, all ports disabled. 0x1: ENABLE (Slave port enabled) 0x0: DISABLE (Slave port disabled)
1	ENS1	Setting this bit enables this input, or slave, port. If the bit is not set then this has the effect of excluding the port from the priority selection scheme. The reset value is all clear, that is, all ports disabled. 0x1: ENABLE (Slave port enabled) 0x0: DISABLE (Slave port disabled)

QDSS_WRAPPER_ATB_FUN8X1_CTRL_REG (cont.)

Bits	Name	Description
0	ENS0	Setting this bit enables this input, or slave, port. If the bit is not set then this has the effect of excluding the port from the priority selection scheme. The reset value is all clear, that is, all ports disabled. 0x1: ENABLE (Slave port enabled) 0x0: DISABLE (Slave port disabled)

0x00869004 QDSS_WRAPPER_ATB_FUN8X1_PRIORITY_CTRL_REG**Type:** RW**Reset State:** 0x00000000

The Priority Control Register defines the order in which inputs are selected. Each 3-bit field represents a priority for each particular slave interface. Location 0 has the priority value for the first slave port. Location 1 is the priority value for the second slave port, Location 2 is the third, down to location 7, which has the priority value of the eighth slave port. Values represent the priority value for each port number. If you want to give highest priority to a particular slave port, the corresponding port must be programmed with the lowest value. Typically this is likely to be a port that has more important data or that has a small FIFO and is therefore likely to overflow.

If you want to give lowest priority to a particular slave port, the corresponding slave port must be programmed with the highest value. Typically this is likely to be a device that has a large FIFO that is less likely to overflow or a source that has information that is of lower importance.

QDSS_WRAPPER_ATB_FUN8X1_PRIORITY_CTRL_REG

Bits	Name	Description
23:21	PRIPORT7	Priority value of the eighth port. The value written into this location is the value that you want to assign the eighth slave port. If a priority value has been entered for multiple different slave ports then the arbitration logic selects the lowest port number of them. 0x0: PRORITY_LEVEL_0 (This value represents the highest priority.) 0x1: PRORITY_LEVEL_1 (2nd highest priority) 0x2: PRORITY_LEVEL_2 (3rd highest priority) 0x3: PRORITY_LEVEL_3 (4th highest priority) 0x4: PRORITY_LEVEL_4 (5th highest priority) 0x5: PRORITY_LEVEL_5 (6th highest priority) 0x6: PRORITY_LEVEL_6 (7th highest priority) 0x7: PRORITY_LEVEL_7 (This value represents the lowest priority.)

QDSS_WRAPPER_ATB_FUN8X1_PRIORITY_CTRL_REG (cont.)

Bits	Name	Description
20:18	PRIPORT6	Priority value of the 7th port. 0x0: PRORITY_LEVEL_0 (This value represents the highest priority.) 0x1: PRORITY_LEVEL_1 (2nd highest priority) 0x2: PRORITY_LEVEL_2 (3rd highest priority) 0x3: PRORITY_LEVEL_3 (4th highest priority) 0x4: PRORITY_LEVEL_4 (5th highest priority) 0x5: PRORITY_LEVEL_5 (6th highest priority) 0x6: PRORITY_LEVEL_6 (7th highest priority) 0x7: PRORITY_LEVEL_7 (This value represents the lowest priority.)
17:15	PRIPORT5	Priority value of the 6th port. 0x0: PRORITY_LEVEL_0 (This value represents the highest priority.) 0x1: PRORITY_LEVEL_1 (2nd highest priority) 0x2: PRORITY_LEVEL_2 (3rd highest priority) 0x3: PRORITY_LEVEL_3 (4th highest priority) 0x4: PRORITY_LEVEL_4 (5th highest priority) 0x5: PRORITY_LEVEL_5 (6th highest priority) 0x6: PRORITY_LEVEL_6 (7th highest priority) 0x7: PRORITY_LEVEL_7 (This value represents the lowest priority.)
14:12	PRIPORT4	Priority value of the 5th port. 0x0: PRORITY_LEVEL_0 (This value represents the highest priority.) 0x1: PRORITY_LEVEL_1 (2nd highest priority) 0x2: PRORITY_LEVEL_2 (3rd highest priority) 0x3: PRORITY_LEVEL_3 (4th highest priority) 0x4: PRORITY_LEVEL_4 (5th highest priority) 0x5: PRORITY_LEVEL_5 (6th highest priority) 0x6: PRORITY_LEVEL_6 (7th highest priority) 0x7: PRORITY_LEVEL_7 (This value represents the lowest priority.)
11:9	PRIPORT3	Priority value of the 4th port. 0x0: PRORITY_LEVEL_0 (This value represents the highest priority.) 0x1: PRORITY_LEVEL_1 (2nd highest priority) 0x2: PRORITY_LEVEL_2 (3rd highest priority) 0x3: PRORITY_LEVEL_3 (4th highest priority) 0x4: PRORITY_LEVEL_4 (5th highest priority) 0x5: PRORITY_LEVEL_5 (6th highest priority) 0x6: PRORITY_LEVEL_6 (7th highest priority) 0x7: PRORITY_LEVEL_7 (This value represents the lowest priority.)

QDSS_WRAPPER_ATB_FUN8X1_PRIORITY_CTRL_REG (cont.)

Bits	Name	Description
8:6	PRIPORT2	Priority value of the 3rd port. 0x0: PRORITY_LEVEL_0 (This value represents the highest priority.) 0x1: PRORITY_LEVEL_1 (2nd highest priority) 0x2: PRORITY_LEVEL_2 (3rd highest priority) 0x3: PRORITY_LEVEL_3 (4th highest priority) 0x4: PRORITY_LEVEL_4 (5th highest priority) 0x5: PRORITY_LEVEL_5 (6th highest priority) 0x6: PRORITY_LEVEL_6 (7th highest priority) 0x7: PRORITY_LEVEL_7 (This value represents the lowest priority.)
5:3	PRIPORT1	Priority value of the 2nd port. 0x0: PRORITY_LEVEL_0 (This value represents the highest priority.) 0x1: PRORITY_LEVEL_1 (2nd highest priority) 0x2: PRORITY_LEVEL_2 (3rd highest priority) 0x3: PRORITY_LEVEL_3 (4th highest priority) 0x4: PRORITY_LEVEL_4 (5th highest priority) 0x5: PRORITY_LEVEL_5 (6th highest priority) 0x6: PRORITY_LEVEL_6 (7th highest priority) 0x7: PRORITY_LEVEL_7 (This value represents the lowest priority.)
2:0	PRIPORT0	Priority value of the first slave port. The value written into this location is the value that you want to assign the first slave port. 0x0: PRORITY_LEVEL_0 (This value represents the highest priority.) 0x1: PRORITY_LEVEL_1 (2nd highest priority) 0x2: PRORITY_LEVEL_2 (3rd highest priority) 0x3: PRORITY_LEVEL_3 (4th highest priority) 0x4: PRORITY_LEVEL_4 (5th highest priority) 0x5: PRORITY_LEVEL_5 (6th highest priority) 0x6: PRORITY_LEVEL_6 (7th highest priority) 0x7: PRORITY_LEVEL_7 (This value represents the lowest priority.)

0x00869F00 QDSS_WRAPPER_ATB_FUN8X1_ITCTRL**Type:** RW**Reset State:** 0x00000000

This register is used to enable topology detection. For more information see the CoreSight Architecture Specification. This register enables the component to switch from a functional mode, the default behavior, to integration mode where the inputs and outputs of the component can be directly controlled for the purpose of integration testing and topology solving

Note

When a device has been in integration mode, it might not function with the original behavior. After performing integration or topology detection, you must reset the system to ensure correct behavior of CoreSight and other connected system components that are affected by the integration or topology detection.

QDSS_WRAPPER_ATB_FUN8X1_ITCTRL

Bits	Name	Description
0	INTEGRATION_MODE	Allows the component to switch from functional mode to integration mode or back. If no integration functionality is implemented, this register must read as zero. 0x1: INTEGRATION_MODE (Enable integration mode) 0x0: FUNCTIONAL_MODE (Disable integration mode)

0x00869FA0 QDSS_WRAPPER_ATB_FUN8X1_CLAIMSET

Type: RW

Reset State: 0x0000000F

This is used in conjunction with Claim Tag Clear Register, CLAIMCLR. This register forms one half of the Claim Tag value. This location allows individual bits to be set, write, and returns the number of bits that can be set, read.

QDSS_WRAPPER_ATB_FUN8X1_CLAIMSET

Bits	Name	Type	Description
3:0	CLAIMSET_W	W	Each bit is considered separately: 0 = no effect, 1 = set this bit in the claim tag.
3:0	CLAIMSET_R	R	This claim tag bit is implemented 0xF: CLAIM_TAG_IMPLEMENTED_BITS (These bits are present within the claim tag field)

0x00869FA4 QDSS_WRAPPER_ATB_FUN8X1_CLAIMCLR

Type: RW

Reset State: 0x00000000

This register is used in conjunction with Claim Tag Set Register, CLAIMSET. This register forms one half of the Claim Tag value. This location enables individual bits to be cleared, write, and returns the current Claim Tag value, read.

QDSS_WRAPPER_ATB_FUN8X1_CLAIMCLR

Bits	Name	Type	Description
3:0	CLAIMCLR_W	W	Each bit is considered separately: 0 = no effect, 1 = clear this bit in the claim tag.
3:0	CLAIMCLR_R	R	The value present reflects the current setting of the Claim Tag.

0x00869FB0 QDSS_WRAPPER_ATB_FUN8X1_LOCKACCESS**Type:** W**Reset State:** Undefined

This is used to enable write access to device registers.

QDSS_WRAPPER_ATB_FUN8X1_LOCKACCESS

Bits	Name	Description
31:0	ACCESS_W	A write of 0xC5ACCE55 enables further write access to this device. An invalid write will have the affect of removing write access. 0xC5ACCE55: UNLOCK (Unlock the protection register to enable write access) 0xDEADBEEF: LOCK (invalid write value to enable the protection register and block write access)

0x00869FB4 QDSS_WRAPPER_ATB_FUN8X1_LOCKSTATUS**Type:** R**Reset State:** 0x00000003

This indicates the status of the Lock control mechanism. This lock prevents accidental writes by code under debug. This register must always be present although there might not be any lock-access control mechanism. The lock mechanism, where present and locked, must block write accesses to any control register, except the Lock Access Register. For most components this covers all registers except for the Lock Access Register 0xFB0

QDSS_WRAPPER_ATB_FUN8X1_LOCKSTATUS

Bits	Name	Description
2	LOCKTYPE	Indicates if the Lock Access Register (0xFB0) is implemented as 8-bit or 32-bit 0x0: ENUM_32_BIT (This component implements a 32-bit Lock Access Register)

QDSS_WRAPPER_ATB_FUN8X1_LOCKSTATUS (cont.)

Bits	Name	Description
1	LOCKGRANT	Returns the current status of the Lock 0x0: ACCESS_PERMITTED (Write access is allowed to this device) 0x1: DEVICE_LOCKED (Write access to the component is blocked. All writes to control registers are ignored. Reads are permitted)
0	LOCKEXIST	Indicates that a lock control mechanism exists for this device 0x0: LOCK_NOT_PRESENT (No lock control mechanism exists, writes to the Lock Access register (0xFB0) will be ignored) 0x1: LOCK_PRESENT (Lock control mechanism is present)

0x00869FB8 QDSS_WRAPPER_ATB_FUN8X1_AUTHSTATUS**Type:** R**Reset State:** 0x00000000

Reports the required security level and current status of those enables. Where functionality changes on a given security level then this change in status must be reported in this register

QDSS_WRAPPER_ATB_FUN8X1_AUTHSTATUS

Bits	Name	Description
7:6	SNID	Indicates the security level for secure non-invasive debug 0x0: ENUM_2_B00 (Functionality not implemented or controlled elsewhere)
5:4	SID	Indicates the security level for secure invasive debug 0x0: ENUM_2_B00 (Functionality not implemented or controlled elsewhere)
3:2	NSNID	Indicates the security level for non-secure non-invasive debug 0x0: ENUM_2_B00 (Functionality not implemented or controlled elsewhere)
1:0	NSID	Indicates the security level for non-secure invasive debug 0x0: ENUM_2_B00 (Functionality not implemented or controlled elsewhere)

0x00869FC8 QDSS_WRAPPER_ATB_FUN8X1_DEVID**Type:** R**Reset State:** 0x00000038

This indicates the capabilities of the CoreSight Funnel.

QDSS_WRAPPER_ATB_FUN8X1_DEVID

Bits	Name	Description
7:4	SCHEME	This value indicates the priority scheme implemented within the Funnel 0x3: STATIC_PRIORITY (Slave ports may be programmed to have higher or lower priority with respect to each other)
3:0	RESERVED	Reserved

0x00869FCC QDSS_WRAPPER_ATB_FUN8X1_DEVTYPE**Type:** R**Reset State:** 0x00000012

It provides a debugger with information about the component when the Part Number field is not recognized. The debugger can then report this information.

QDSS_WRAPPER_ATB_FUN8X1_DEVTYPE

Bits	Name	Description
7:4	SUB_TYPE	sub-classification within the major category 0x1: FUNNEL_ROUTER
3:0	MAJOR_TYPE	Major classification grouping for this debug/trace component 0x2: TRACE_LINK

0x00869FE0 QDSS_WRAPPER_ATB_FUN8X1_PERIPHID0**Type:** R**Reset State:** 0x00000008

Part of the set of Peripheral Identification registers. Contains part of the designer specific part number.

QDSS_WRAPPER_ATB_FUN8X1_PERIPHID0

Bits	Name	Description
7:0	PART_NUMBER_BITS7TO0	Bits [7:0] of the component's part number. This is selected by the designer of the component. 0x8: CORESIGHT_FUNNEL_PART_NUMBER_7_0 (Lowest 8 bits of the Part Number (0x100))

0x00869FE4 QDSS_WRAPPER_ATB_FUN8X1_PERIPHID1**Type:** R**Reset State:** 0x000000B9

Part of the set of Peripheral Identification registers. Contains part of the designer specific part number and part of the designer identity.

QDSS_WRAPPER_ATB_FUN8X1_PERIPHID1

Bits	Name	Description
7:4	JEP106_BITS3TO0	Bits 3:0 of the JEDEC identity code indicating the designer of the component (along with the continuation code) 0xB: ARM_JEP106_IDENTITY_CODE_7_0 (Lowest 4 bits of the JEP106 Identity Code)
3:0	PART_NUMBER_BITS11TO8	Bits [11:8] of the component's part number. This is selected by the designer of the component. 0x9: CORESIGHT_FUNNEL_PART_NUMBER_11_8 (Upper 4 bits of the Part Number (0x100))

0x00869FE8 QDSS_WRAPPER_ATB_FUN8X1_PERIPHID2**Type:** R**Reset State:** 0x0000002B

Part of the set of Peripheral Identification registers. Contains part of the designer identity and the product revision.

QDSS_WRAPPER_ATB_FUN8X1_PERIPHID2

Bits	Name	Description
7:4	REVISION	The Revision field is an incremental value starting at 0x0 for the first design of this component. This only increases by 1 for both major and minor revisions and is simply used as a look-up to establish the exact major/minor revision. 0x2: R2P0 (This device is at r2p0)
3	JEDEC	Always set. Indicates that a JEDEC assigned value is used 0x1: JEDEC_IDENTITY (The designer ID is specified by JEDEC (http://www.jedec.org))
2:0	JEP106_BITS6TO4	Bits 6:4 of the JEDEC identity code indicating the designer of the component (along with the continuation code) 0x3: ARM_JEP106_IDENTITY_CODE_6_4 (Upper 3 bits of the JEP106 Identity Code)

0x00869FEC QDSS_WRAPPER_ATB_FUN8X1_PERIPHID3**Type:** R**Reset State:** 0x00000000

Part of the set of Peripheral Identification registers. Contains the RevAnd and Customer Modified fields.

QDSS_WRAPPER_ATB_FUN8X1_PERIPHID3

Bits	Name	Description
7:4	REVAND	This field indicates minor errata fixes specific to this design, for example metal fixes after implementation. In most cases this field is zero. It is recommended that component designers ensure this field can be changed by a metal fix if required, for example by driving it from registers that reset to zero. 0x0: NO_MODS (Indicates that there have been no metal fixes to this component)
3:0	CUSTOMER_MODIFIED	Where the component is reusable IP, this value indicates if the customer has modified the behavior of the component. In most cases this field is zero. 0x0: NO_MODS (Indicates that there have been no modifications made)

0x00869FD0 QDSS_WRAPPER_ATB_FUN8X1_PERIPHID4**Type:** R**Reset State:** 0x00000004

Part of the set of Peripheral Identification registers. Contains part of the designer identity and the memory footprint indicator.

QDSS_WRAPPER_ATB_FUN8X1_PERIPHID4

Bits	Name	Description
7:4	FOURKB_COUNT	This is a 4-bit value that indicates the total contiguous size of the memory window used by this component in powers of 2 from the standard 4KB. If a component only requires the standard 4KB then this should read as 0x0, 4KB only, for 8KB set to 0x1, 16KB == 0x2, 32KB == 0x3, and so on. 0x0: ENUM_4KB (Indicates that the device only occupies 4kB of memory)
3:0	JEP106_CONT	JEDEC continuation code indicating the designer of the component (along with the identity code) 0x4: ARM_JEP106_CONTINUATION_CODE (Indicates that ARM's JEDEC identity code is on the 5th bank)

0x00869FD4 QDSS_WRAPPER_ATB_FUN8X1_PERIPHID5**Type:** RW**Reset State:** Undefined

Reserved

QDSS_WRAPPER_ATB_FUN8X1_PERIPHID5

Bits	Name	Description
31:0	PERIPHID5	

0x00869FD8 QDSS_WRAPPER_ATB_FUN8X1_PERIPHID6**Type:** RW**Reset State:** Undefined

Reserved

QDSS_WRAPPER_ATB_FUN8X1_PERIPHID6

Bits	Name	Description
31:0	PERIPHID6	

0x00869FDC QDSS_WRAPPER_ATB_FUN8X1_PERIPHID7**Type:** RW**Reset State:** Undefined

Reserved

QDSS_WRAPPER_ATB_FUN8X1_PERIPHID7

Bits	Name	Description
31:0	PERIPHID7	

0x00869FF0 QDSS_WRAPPER_ATB_FUN8X1_COMPID0**Type:** R**Reset State:** 0x0000000D

A component identification register, that indicates that the identification registers are present.

QDSS_WRAPPER_ATB_FUN8X1_COMPID0

Bits	Name	Description
7:0	PREAMBLE	Contains bits [24:31] of the component identification 0xD: ENUM_0D (Identification value)

0x00869FF4 QDSS_WRAPPER_ATB_FUN8X1_COMPID1**Type:** R**Reset State:** 0x00000090

A component identification register, that indicates that the identification registers are present. This register also indicates the component class.

QDSS_WRAPPER_ATB_FUN8X1_COMPID1

Bits	Name	Description
7:4	CLASS	Class of the component. E.g. ROM table, CoreSight component etc. 0x9: CORESIGHT_COMPONENT (Indicates the component is a CoreSight component)
3:0	PREAMBLE	Contains bits [19:16] of the component identification 0x0: ENUM_0 (Identification value)

0x00869FF8 QDSS_WRAPPER_ATB_FUN8X1_COMPID2**Type:** R**Reset State:** 0x00000005

A component identification register, that indicates that the identification registers are present.

QDSS_WRAPPER_ATB_FUN8X1_COMPID2

Bits	Name	Description
7:0	PREAMBLE	Contains bits [15:8] of the component identification 0x5: ENUM_05 (Identification value)

0x00869FFC QDSS_WRAPPER_ATB_FUN8X1_COMPID3**Type:** R**Reset State:** 0x000000B1

A component identification register, that indicates that the identification registers are present.

QDSS_WRAPPER_ATB_FUN8X1_COMPID3

Bits	Name	Description
7:0	PREAMBLE	Contains bits [7:0] of the component identification 0xB1: B1 (Identification value)

25.3 QDSS_WRAPPER_QDSS_WRAPPER

0x0086C000+ QDSS_WRAPPER_HWEVENT_EXPD_COLn_OEN_REG, n=[0..15]
0x4*n

Type: RW

Clock: PDBGCLK

Reset State: 0x00000000

HWEVENT MUX COLn Enable

QDSS_WRAPPER_HWEVENT_EXPD_COLn_OEN_REG

Bits	Name	Description
31:0	HWEVENT_COLUMN_ENB	Enables the Selected Column MUXn output,1= enables the COLn MUX output,0=disables the COL MUX output

0x0086C040+ QDSS_WRAPPER_HWEVENT_EXPD_COLn_MUX_SEL_LSB_REG, n=[0..15]
0x8*n

Type: RW

Clock: PDBGCLK

Reset State: 0x00000000

HWEVENT MUX COLn Select LSB

QDSS_WRAPPER_HWEVENT_EXPD_COLn_MUX_SEL_LSB_REG

Bits	Name	Description
31:30	EXPD_COL_MUX_EVENT15_SEL	Mux Select bits for event15 of the COLn Mux
29:28	EXPD_COL_MUX_EVENT14_SEL	Mux Select bits for event14 of the COLn Mux
27:26	EXPD_COL_MUX_EVENT13_SEL	Mux Select bits for event13 of the COLn Mux
25:24	EXPD_COL_MUX_EVENT12_SEL	Mux Select bits for event12 of the COLn Mux
23:22	EXPD_COL_MUX_EVENT11_SEL	Mux Select bits for event11 of the COLn Mux

QDSS_WRAPPER_HWEVENT_EXPD_COLn_MUX_SEL_LSB_REG (cont.)

Bits	Name	Description
21:20	EXPD_COL_MUX_EVENT10_SEL	Mux Select bits for event10 of the COLn Mux
19:18	EXPD_COL_MUX_EVENT9_SEL	Mux Select bits for event9 of the COLn Mux
17:16	EXPD_COL_MUX_EVENT8_SEL	Mux Select bits for event8 of the COLn Mux
15:14	EXPD_COL_MUX_EVENT7_SEL	Mux Select bits for event7 of the COLn Mux
13:12	EXPD_COL_MUX_EVENT6_SEL	Mux Select bits for event6 of the COLn Mux
11:10	EXPD_COL_MUX_EVENT5_SEL	Mux Select bits for event5 of the COLn Mux
9:8	EXPD_COL_MUX_EVENT4_SEL	Mux Select bits for event4 of the COLn Mux
7:6	EXPD_COL_MUX_EVENT3_SEL	Mux Select bits for event3 of the COLn Mux
5:4	EXPD_COL_MUX_EVENT2_SEL	Mux Select bits for event2 of the COLn Mux
3:2	EXPD_COL_MUX_EVENT1_SEL	Mux Select bits for event1 of the COLn Mux
1:0	EXPD_COL_MUX_EVENT0_SEL	Mux Select bits for event0 of the COLn Mux

**0x0086C044+ QDSS_WRAPPER_HWEVENT_EXPD_COLn_MUX_SEL_MSB_REG, n=[0..15]
0x8*n**

Type: RW

Clock: PDBGCLK

Reset State: 0x00000000

HWEVENT MUX COLn Select MSB

QDSS_WRAPPER_HWEVENT_EXPD_COLn_MUX_SEL_MSB_REG

Bits	Name	Description
31:30	EXPD_COL_MUX_EVENT31_SEL	Mux Select bits for event31 of the COLn Mux
29:28	EXPD_COL_MUX_EVENT30_SEL	Mux Select bits for event30 of the COLn Mux
27:26	EXPD_COL_MUX_EVENT29_SEL	Mux Select bits for event29 of the COLn Mux

QDSS_WRAPPER_HWEVENT_EXPd_COLn_MUX_SEL_MSB_REG (cont.)

Bits	Name	Description
25:24	EXPd_COL_MUX_EVENT28_SEL	Mux Select bits for event28 of the COLn Mux
23:22	EXPd_COL_MUX_EVENT27_SEL	Mux Select bits for event27 of the COLn Mux
21:20	EXPd_COL_MUX_EVENT26_SEL	Mux Select bits for event26 of the COLn Mux
19:18	EXPd_COL_MUX_EVENT25_SEL	Mux Select bits for event25 of the COLn Mux
17:16	EXPd_COL_MUX_EVENT24_SEL	Mux Select bits for event24 of the COLn Mux
15:14	EXPd_COL_MUX_EVENT23_SEL	Mux Select bits for event23 of the COLn Mux
13:12	EXPd_COL_MUX_EVENT22_SEL	Mux Select bits for event22 of the COLn Mux
11:10	EXPd_COL_MUX_EVENT21_SEL	Mux Select bits for event21 of the COLn Mux
9:8	EXPd_COL_MUX_EVENT20_SEL	Mux Select bits for event20 of the COLn Mux
7:6	EXPd_COL_MUX_EVENT19_SEL	Mux Select bits for event19 of the COLn Mux
5:4	EXPd_COL_MUX_EVENT18_SEL	Mux Select bits for event18 of the COLn Mux
3:2	EXPd_COL_MUX_EVENT17_SEL	Mux Select bits for event17 of the COLn Mux
1:0	EXPd_COL_MUX_EVENT16_SEL	Mux Select bits for event16 of the COLn Mux

0x0086C100 QDSS_WRAPPER_IRQ_MUX_REG**Type:** RW**Clock:** PDBGCLK**Reset State:** 0x00000000

IRQ HWEVENT MUX Config register

QDSS_WRAPPER_IRQ_MUX_REG

Bits	Name	Description
31	IRQ_SPI_MSB_MUX_OEN	SPI MSB IRQ HWEVENT Enable, 1= enables the SPI_MSB_MUX output
30:15	SPARE0	Reserved

QDSS_WRAPPER_IRQ_MUX_REG (cont.)

Bits	Name	Description
14:7	IRQ_SPI_MSB_MUX_SEL	IRQ MUX selection, selects one out of (255-24) SPIs, this MUX output is mapped on HWE: a53_intc_spi_msb_irq_hwevent
6:0	IRQ_PPI_SPI_MUX_SEL	IRQ MUX selection, selects one out of 60 PPIs +24 SPIs from A53SS. This MUX output is mapped on A53 hwevents: a53_intc_ppi_muxout_hwevent. Select Line Encoding (1-15) core0 ppi (16-30) core1 ppi (31-45) core2 ppi (46-60) core3 ppi (64-87) spi[23:0] others reserved

0x0086C104 QDSS_WRAPPER_HWE_SPDM_MUX_REG**Type:** RW**Clock:** PDBGCLK**Reset State:** 0x00000000

SPDM HWEVENT MUX Config register

QDSS_WRAPPER_HWE_SPDM_MUX_REG

Bits	Name	Description
31	HWE_SPDM_MUX_OEN	SPDM HWEVENT Enable, 1= enables the SPDM HWE MUX output, 0=disables the SPDM HWE MUX output
30:9	SPARE1	Reserved
8:0	HWE_SPDM_MUX_SEL	SPDM HWEVENT MUX selection, selects one out of 512 HWEVENTS which is routed to SPDMs spdm_cy_clk_a[15] realtime monitor

0x0086CF00 QDSS_WRAPPER_CS_HWE_EXPD_ITCTL**Type:** R**Clock:** PDBGCLK**Reset State:** 0x00000000

QDSS WrapperIntegration Control (Coresight Programmer's Model) Register

QDSS_WRAPPER_CS_HWE_EXPD_ITCTL

Bits	Name	Description
31:0	RFU	Reads as zero since no integration functionality is implemented.

0x0086CFA0 QDSS_WRAPPER_CS_HWE_EXPD_CLAIMSET**Type:** R**Clock:** PDBGCLK**Reset State:** 0x00000000

QDSS WrapperClaim Tag Set (Coresight Programmer's Model) Register

QDSS_WRAPPER_CS_HWE_EXPD_CLAIMSET

Bits	Name	Description
31:0	RFU	Reads as zero since no integration functionality is implemented.

0x0086CFA4 QDSS_WRAPPER_CS_HWE_EXPD_CLAIMCLR**Type:** R**Clock:** PDBGCLK**Reset State:** 0x00000000

QDSS WrapperClaim Tag Clear (Coresight Programmer's Model) Register

QDSS_WRAPPER_CS_HWE_EXPD_CLAIMCLR

Bits	Name	Description
31:0	RFU	Reads as zero since no integration functionality is implemented.

0x0086CFB0 QDSS_WRAPPER_CS_HWE_EXPD_LOCKACCESS**Type:** W**Clock:** PDBGCLK**Reset State:** Undefined

QDSS WrapperLock Access (Coresight Programmer's Model) Register

QDSS_WRAPPER_CS_HWE_EXPD_LOCKACCESS

Bits	Name	Description
31:0	LOCKACCESS	Write access code. A write of 0xC5ACCE55 enables further writes to the QDSS Coresight CSR Slave from software.

0x0086CFB4 QDSS_WRAPPER_CS_HWE_EXPD_LOCKSTATUS**Type:** R**Clock:** PDBGCLK**Reset State:** 0x00000003

QDSS WrapperLock Status (Coresight Programmer's Model) Register

QDSS_WRAPPER_CS_HWE_EXPD_LOCKSTATUS

Bits	Name	Description
31:3	RFU	Reserved for Future Use.
2	FIELD_8_BIT_LOCK	Reads as zero since this component does not implement an 8 bit lock access register.
1	ACCESS	Access permission. This bit reflects the inversion of the current state of the UNLOCK bit within this component. 0x1: HIGH (Write access blocked. Reads permitted.) 0x0: LOW (Access permitted.)
0	LOCK_CONTROL	Reads as one since lock exists for this component.

0x0086CFB8 QDSS_WRAPPER_CS_HWE_EXPD_AUTHSTATUS**Type:** R**Clock:** PDBGCLK**Reset State:** 0x00000000

QDSS WrapperAuthentication Status (Coresight Programmer's Model) Register

QDSS_WRAPPER_CS_HWE_EXPD_AUTHSTATUS

Bits	Name	Description
31:0	RFU	Reads as zero since no integration functionality is implemented.

0x0086CFC8 QDSS_WRAPPER_CS_HWE_EXPD_DEVICEID

Type: R
Clock: PDBGCLK
Reset State: 0x00000000

QDSS WrapperDevice Identification (Coresight Programmer's Model) Register

QDSS_WRAPPER_CS_HWE_EXPD_DEVICEID

Bits	Name	Description
31:8	RFU	Reserved for Future Use.
7:0	DEVID	This register is implementation defined for each Part Number and Designer. Unused for this component so reads as zeros.

0x0086CFCC QDSS_WRAPPER_CS_HWE_EXPD_DEVICETYPE

Type: R
Clock: PDBGCLK
Reset State: 0x00000004

QDSS WrapperDevice Type (Coresight Programmer's Model) Register

QDSS_WRAPPER_CS_HWE_EXPD_DEVICETYPE

Bits	Name	Description
31:8	RFU	Reserved for Future Use.
7:4	SUBTYPE	Reads as zeros meaning Other sub type.
3:0	MAJTYPE	Reads as 0x4 meaning Debug Control major type.

0x0086CFD0 QDSS_WRAPPER_CS_HWE_EXPD_PERIPHID4

Type: R
Clock: PDBGCLK
Reset State: 0x0000000X

QDSS WrapperPeripheral ID4 (Coresight Programmer's Model) Register

QDSS_WRAPPER_CS_HWE_EXPD_PERIPHID4

Bits	Name	Description
31:8	RFU	Reserved for Future Use.
7:4	FIELD_4KB_COUNT	Number of 4kB blocks. Reads as zero or 4kB block size.

QDSS_WRAPPER_CS_HWE_EXPD_PERIPHID4 (cont.)

Bits	Name	Description
3:0	JEP106_CONTINUATION	JEP106 continuation code. Reads as TARGETID[11:8] from MSM JTAG ID fuses [11:8]. This field should be set to bits [10:7] of Qualcomm's JEDEC assigned JEP106 code.

0x0086CFD4 QDSS_WRAPPER_CS_HWE_EXPD_PERIPHID5**Type:** R**Clock:** PDBGCLK**Reset State:** 0x00000000

QDSS WrapperPeripheral ID5 (Coresight Programmer's Model) Register

QDSS_WRAPPER_CS_HWE_EXPD_PERIPHID5

Bits	Name	Description
31:0	RFU	Reads as zero per Coresight Architecture.

0x0086CFD8 QDSS_WRAPPER_CS_HWE_EXPD_PERIPHID6**Type:** R**Clock:** PDBGCLK**Reset State:** 0x00000000

QDSS WrapperPeripheral ID6 (Coresight Programmer's Model) Register

QDSS_WRAPPER_CS_HWE_EXPD_PERIPHID6

Bits	Name	Description
31:0	RFU	Reads as zero per Coresight Architecture.

0x0086CFDC QDSS_WRAPPER_CS_HWE_EXPD_PERIPHID7**Type:** R**Clock:** PDBGCLK**Reset State:** 0x00000000

QDSS WrapperPeripheral ID7 (Coresight Programmer's Model) Register

QDSS_WRAPPER_CS_HWE_EXPD_PERIPHID7

Bits	Name	Description
31:0	RFU	Reads as zero per Coresight Architecture.

0x0086CFE0 QDSS_WRAPPER_CS_HWE_EXPD_PERIPHID0**Type:** R**Clock:** PDBGCLK**Reset State:** 0x00000040

QDSS WrapperPeripheral ID0 (Coresight Programmer's Model) Register

QDSS_WRAPPER_CS_HWE_EXPD_PERIPHID0

Bits	Name	Description
31:8	RFU	Reserved for Future Use.
7:0	PARTNUM	PartNumber[7:0]. [7:6] is Subsystem/core family ID (e.g. denote QDSS family or generation). [5:4] is Subsystem/core configuration options (e.g. denote cache options, etc.). [3:2] is Subsystem/core fuse options. [1:0] is Subsystem/core future use field. Reads as 0x40.

0x0086CFE4 QDSS_WRAPPER_CS_HWE_EXPD_PERIPHID1**Type:** R**Clock:** PDBGCLK**Reset State:** 0x000000X4

QDSS WrapperPeripheral ID1 (Coresight Programmer's Model) Register

QDSS_WRAPPER_CS_HWE_EXPD_PERIPHID1

Bits	Name	Description
31:8	RFU	Reserved for Future Use.
7:4	JEP106_IDENTITY_3_0	JEP106 Identity Code[3:0]. Reads as TARGETID[4:1] from MSM JTAG ID fuses [4:1] which will always be 0x0 for Qualcomm.
3:0	PARTNUM	PartNumber[11:8]. [11:8] is 0x4 meaning Qualcomm designed Coresight component in QDSS. Reads as 0x4.

0x0086CFE8 QDSS_WRAPPER_CS_HWE_EXPD_PERIPHID2**Type:** R**Clock:** PDBGCLK**Reset State:** 0b000000000000000000000000000011XXX

QDSS WrapperPeripheral ID2 (Coresight Programmer's Model) Register

QDSS_WRAPPER_CS_HWE_EXPD_PERIPHID2

Bits	Name	Description
31:8	RFU	Reserved for Future Use.
7:4	MAJREV	Major revision number of this peripheral. Reads as 0x0 for this version of QDSS.
3	JEDEC	Reads as one which indicates that a JEDEC assigned value is used.
2:0	JEP106_IDENTITY_6_4	JEP106 Identity Code[6:4]. Reads as TARGETID[7:5] from MSM JTAG ID fuses [7:5] which will always be 3'b111 for Qualcomm.

0x0086CFEC QDSS_WRAPPER_CS_HWE_EXPD_PERIPHID3**Type:** R**Clock:** PDBGCLK**Reset State:** 0x00000000

QDSS WrapperPeripheral ID3 (Coresight Programmer's Model) Register

QDSS_WRAPPER_CS_HWE_EXPD_PERIPHID3

Bits	Name	Description
31:8	RFU	Reserved for Future Use.
7:4	REV_AND	This field indicates minor errata fixes specific to this design. Reads as zeros for this version of QDSS.
3:0	CUSTOMER_MODIFIED	Implementation defined. Always reads as zeros for QDSS.

0x0086CFF0 QDSS_WRAPPER_CS_HWE_EXPD_COMPID0**Type:** R**Clock:** PDBGCLK**Reset State:** 0x0000000D

QDSS WrapperComponent ID0 (Coresight Programmer's Model) Register

QDSS_WRAPPER_CS_HWE_EXPD_COMPID0

Bits	Name	Description
31:8	RFU	Reserved for Future Use.
7:0	PREAMBLE_7_0	Coresight mandates this read as 0x0D.

0x0086CFF4 QDSS_WRAPPER_CS_HWE_EXPD_COMPID1**Type:** R**Clock:** PDBGCLK**Reset State:** 0x00000090

QDSS WrapperComponent ID1 (Coresight Programmer's Model) Register

QDSS_WRAPPER_CS_HWE_EXPD_COMPID1

Bits	Name	Description
31:8	RFU	Reserved for Future Use.
7:4	PREAMBLE_15_12	Coresight mandates this read as 0x9 or Coresight class component.
3:0	PREAMBLE_11_8	Coresight mandates this read as 0x0.

0x0086CFF8 QDSS_WRAPPER_CS_HWE_EXPD_COMPID2**Type:** R**Clock:** PDBGCLK**Reset State:** 0x00000005

QDSS WrapperComponent ID2 (Coresight Programmer's Model) Register

QDSS_WRAPPER_CS_HWE_EXPD_COMPID2

Bits	Name	Description
31:8	RFU	Reserved for Future Use.
7:0	PREAMBLE_23_16	Coresight mandates this read as 0x05.

0x0086CFFC QDSS_WRAPPER_CS_HWE_EXPD_COMPID3**Type:** R**Clock:** PDBGCLK**Reset State:** 0x000000B1

QDSS WrapperComponent ID3 (Coresight Programmer's Model) Register

QDSS_WRAPPER_CS_HWE_EXPD_COMPID3

Bits	Name	Description
31:8	RFU	Reserved for Future Use.
7:0	PREAMBLE_31_24	Coresight mandates this read as 0xB1.

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[MDSS_MDP_VP_0_DSPP_1_GC_ARLUT_GREEN_OFFSET_N, 3649](#)
[MDSS_MDP_VP_0_DSPP_1_GC_ARLUT_GREEN_SLOPE_N, 3649](#)
[MDSS_MDP_VP_0_DSPP_1_GC_ARLUT_GREEN_STARTX_N, 3648](#)
[MDSS_MDP_VP_0_DSPP_1_GC_ARLUT_RED_OFFSET_N, 3648](#)
[MDSS_MDP_VP_0_DSPP_1_GC_ARLUT_RED_SLOPE_N, 3648](#)
[MDSS_MDP_VP_0_DSPP_1_GC_ARLUT_RED_STARTX_N, 3647](#)
[MDSS_MDP_VP_0_DSPP_1_HIST_COLLECT_CTL, 3635](#)
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[MDSS_MDP_VP_0_DSPP_1_HIST_SEL, 3637](#)
[MDSS_MDP_VP_0_DSPP_1_HIST_SUM, 3637](#)
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[MDSS_MDP_VP_0_DSPP_1_PA_LUTV, 3638](#)
[MDSS_MDP_VP_0_DSPP_1_PA_SAT_ADJUST, 3639](#)
[MDSS_MDP_VP_0_DSPP_1_PA_VAL_ADJUST, 3640](#)
[MDSS_MDP_VP_0_DSPP_1_PCC_PLANE*c*_B_COEFF, *c*=\[0..2\], 3630](#)
[MDSS_MDP_VP_0_DSPP_1_PCC_PLANE*c*_BB_COEFF, *c*=\[0..2\], 3632](#)
[MDSS_MDP_VP_0_DSPP_1_PCC_PLANE*c*_BG_COEFF, *c*=\[0..2\], 3632](#)
[MDSS_MDP_VP_0_DSPP_1_PCC_PLANE*c*_CONSTANT_COEFF, *c*=\[0..2\], 3629](#)
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MDSS_MDP_VP_0_DSPP_1_PCC_PLANEc_GG_COEFF,
 c=[0..2], 3631
 MDSS_MDP_VP_0_DSPP_1_PCC_PLANEc_R_COEFF,
 c=[0..2], 3629
 MDSS_MDP_VP_0_DSPP_1_PCC_PLANEc_RB_COEFF,
 c=[0..2], 3631
 MDSS_MDP_VP_0_DSPP_1_PCC_PLANEc_RG_COEFF,
 c=[0..2], 3631
 MDSS_MDP_VP_0_DSPP_1_PCC_PLANEc_RGB_COEFF_0,
 c=[0..2], 3632
 MDSS_MDP_VP_0_DSPP_1_PCC_PLANEc_RGB_COEFF_1,
 c=[0..2], 3633
 MDSS_MDP_VP_0_DSPP_1_PCC_PLANEc_RR_COEFF,
 c=[0..2], 3630
 MDSS_MDP_VP_0_DSPP_1_SIX_ZONE_ADJ_CURVE_P0,
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 MDSS_MDP_VP_0_DSPP_1_SIX_ZONE_ADJ_CURVE_P1,
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 MDSS_MDP_VP_0_DSPP_1_SIX_ZONE_THRESHOLDS, 3641
 MDSS_MDP_VP_0_DSPP_1_SKIN_ADJUST_PWL0, 3642
 MDSS_MDP_VP_0_DSPP_1_SKIN_ADJUST_PWL1, 3642
 MDSS_MDP_VP_0_DSPP_1_SKIN_REGION_HUE, 3642
 MDSS_MDP_VP_0_DSPP_1_SKIN_REGION_SAT, 3643
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 MDSS_MDP_VP_0_DSPP_1_SKY_ADJUST_PWL0, 3644
 MDSS_MDP_VP_0_DSPP_1_SKY_ADJUST_PWL1, 3644
 MDSS_MDP_VP_0_DSPP_1_SKY_REGION_HUE, 3644
 MDSS_MDP_VP_0_DSPP_1_SKY_REGION_SAT, 3645
 MDSS_MDP_VP_0_DSPP_1_SKY_REGION_VAL, 3645
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 MDSS_MDP_VP_0_DSPP_2_DITHER_MATRIX_ROW2, 3658
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 MDSS_MDP_VP_0_DSPP_2_FOLIAGE_ADJUST_PWL0, 3669
 MDSS_MDP_VP_0_DSPP_2_FOLIAGE_ADJUST_PWL1, 3669
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 MDSS_MDP_VP_0_DSPP_2_FOLIAGE_REGION_SAT, 3670
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 MDSS_MDP_VP_0_DSPP_2_GC_ARLUT_BLUE_SLOPE_N,
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 MDSS_MDP_VP_0_DSPP_2_GC_ARLUT_BLUE_STARTX_N,
 3673
 MDSS_MDP_VP_0_DSPP_2_GC_ARLUT_GREEN_OFFSET_N,
 3673
 MDSS_MDP_VP_0_DSPP_2_GC_ARLUT_GREEN_SLOPE_N,
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 MDSS_MDP_VP_0_DSPP_2_GC_ARLUT_GREEN_STARTX_N,
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 MDSS_MDP_VP_0_DSPP_2_GC_ARLUT_RED_OFFSET_N,
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 MDSS_MDP_VP_0_DSPP_2_GC_ARLUT_RED_SLOPE_N,
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 MDSS_MDP_VP_0_DSPP_2_GC_ARLUT_RED_STARTX_N,
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 MDSS_MDP_VP_0_DSPP_2_HIST_SEL, 3660
 MDSS_MDP_VP_0_DSPP_2_HIST_SUM, 3661
 MDSS_MDP_VP_0_DSPP_2_HIST_V_DATA_n_BIN_i,
 i=[0..255], 3674
 MDSS_MDP_VP_0_DSPP_2_HIST_V_DATA_n, 3661
 MDSS_MDP_VP_0_DSPP_2_OP_MODE, 3651
 MDSS_MDP_VP_0_DSPP_2_PA_CONT_ADJUST, 3663
 MDSS_MDP_VP_0_DSPP_2_PA_HUE_ADJUST, 3662
 MDSS_MDP_VP_0_DSPP_2_PA_LUTV_IDX_i, i=[0..127], 3674
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 MDSS_MDP_VP_0_DSPP_2_PA_SAT_ADJUST, 3663
 MDSS_MDP_VP_0_DSPP_2_PA_VAL_ADJUST, 3663
 MDSS_MDP_VP_0_DSPP_2_PCC_PLANEc_B_COEFF,
 c=[0..2], 3654
 MDSS_MDP_VP_0_DSPP_2_PCC_PLANEc_BB_COEFF,
 c=[0..2], 3656
 MDSS_MDP_VP_0_DSPP_2_PCC_PLANEc_BG_COEFF,
 c=[0..2], 3655
 MDSS_MDP_VP_0_DSPP_2_PCC_PLANEc_CONSTANT_COEFF,
 c=[0..2], 3653
 MDSS_MDP_VP_0_DSPP_2_PCC_PLANEc_G_COEFF,
 c=[0..2], 3653
 MDSS_MDP_VP_0_DSPP_2_PCC_PLANEc_GG_COEFF,
 c=[0..2], 3655
 MDSS_MDP_VP_0_DSPP_2_PCC_PLANEc_R_COEFF,
 c=[0..2], 3653
 MDSS_MDP_VP_0_DSPP_2_PCC_PLANEc_RB_COEFF,
 c=[0..2], 3655
 MDSS_MDP_VP_0_DSPP_2_PCC_PLANEc_RG_COEFF,
 c=[0..2], 3654
 MDSS_MDP_VP_0_DSPP_2_PCC_PLANEc_RGB_COEFF_0,
 c=[0..2], 3656
 MDSS_MDP_VP_0_DSPP_2_PCC_PLANEc_RGB_COEFF_1,
 c=[0..2], 3656
 MDSS_MDP_VP_0_DSPP_2_PCC_PLANEc_RR_COEFF,
 c=[0..2], 3654
 MDSS_MDP_VP_0_DSPP_2_SIX_ZONE_ADJ_CURVE_P0,
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 MDSS_MDP_VP_0_DSPP_2_SIX_ZONE_ADJ_CURVE_P1,
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 MDSS_MDP_VP_0_DSPP_2_SIX_ZONE_THRESHOLDS, 3665
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G_TRANSP_HIGH0, 3443
MDSS_MDP_VP_0_LAYER_0_BLEND0_F-
G_TRANSP_HIGH1, 3443
MDSS_MDP_VP_0_LAYER_0_BLEND0_F-
G_TRANSP_LOW0, 3442
MDSS_MDP_VP_0_LAYER_0_BLEND0_F-
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MDSS_MDP_VP_0_LAYER_0_BLEND1_B-
G_TRANSP_HIGH0, 3448
MDSS_MDP_VP_0_LAYER_0_BLEND1_B-
G_TRANSP_HIGH1, 3449
MDSS_MDP_VP_0_LAYER_0_BLEND1_B-
G_TRANSP_LOW0, 3448
MDSS_MDP_VP_0_LAYER_0_BLEND1_B-
G_TRANSP_LOW1, 3448
MDSS_MDP_VP_0_LAYER_0_BLEND1_FG_ALPHA, 3446
MDSS_MDP_VP_0_LAYER_0_BLEND1_F-
G_TRANSP_HIGH0, 3447
MDSS_MDP_VP_0_LAYER_0_BLEND1_F-
G_TRANSP_HIGH1, 3447
MDSS_MDP_VP_0_LAYER_0_BLEND1_F-
G_TRANSP_LOW0, 3446
MDSS_MDP_VP_0_LAYER_0_BLEND1_F-
G_TRANSP_LOW1, 3447
MDSS_MDP_VP_0_LAYER_0_BLEND1_OP, 3445
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MDSS_MDP_VP_0_LAYER_0_BLEND2_B-
G_TRANSP_HIGH0, 3453
MDSS_MDP_VP_0_LAYER_0_BLEND2_B-
G_TRANSP_HIGH1, 3453
MDSS_MDP_VP_0_LAYER_0_BLEND2_B-
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MDSS_MDP_VP_0_LAYER_0_BLEND2_B-
G_TRANSP_LOW1, 3452
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MDSS_MDP_VP_0_LAYER_0_BLEND2_F-
G_TRANSP_HIGH0, 3451
MDSS_MDP_VP_0_LAYER_0_BLEND2_F-
G_TRANSP_HIGH1, 3452
MDSS_MDP_VP_0_LAYER_0_BLEND2_F-
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MDSS_MDP_VP_0_LAYER_0_BLEND2_F-
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G_TRANSP_HIGH0, 3457
MDSS_MDP_VP_0_LAYER_0_BLEND3_B-
G_TRANSP_HIGH1, 3457
MDSS_MDP_VP_0_LAYER_0_BLEND3_B-
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MDSS_MDP_VP_0_LAYER_0_BLEND3_FG_ALPHA, 3454
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G_TRANSP_HIGH0, 3456
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G_TRANSP_HIGH1, 3456
MDSS_MDP_VP_0_LAYER_0_BLEND3_F-
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MDSS_MDP_VP_0_LAYER_0_BLEND3_F-
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SOR_BLEND_TRANSP_HIGH1, 3462
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SOR_BLEND_TRANSP_LOW0, 3461
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MDSS_MDP_VP_0_LAYER_0_CURSOR_IMG_SIZE, 3458
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FO_WATERMARK_0, 3468
MDSS_MDP_VP_0_LAYER_0_CURSOR_REQPRIORITY_FI-
FO_WATERMARK_1, 3469
MDSS_MDP_VP_0_LAYER_0_CURSOR_REQPRIORITY_FI-
FO_WATERMARK_2, 3470
MDSS_MDP_VP_0_LAYER_0_CURSOR_SIZE, 3458
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PACK_RD_PLANE0, 3467
MDSS_MDP_VP_0_LAYER_0_CURSOR_START_XY, 3460
MDSS_MDP_VP_0_LAYER_0_CURSOR_STRIDE, 3458
MDSS_MDP_VP_0_LAYER_0_CURSOR_XY, 3459
MDSS_MDP_VP_0_LAYER_0_GC_ARLUT_BLUE_OFF-
SET_N, 3465
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MDSS_MDP_VP_0_LAYER_0_GC_ARLUT_BLUE_STARTX-
_N, 3465
MDSS_MDP_VP_0_LAYER_0_GC_ARLUT_GREEN_OFF-
SET_N, 3464
MDSS_MDP_VP_0_LAYER_0_GC_AR-
LUT_GREEN_SLOPE_N, 3464
MDSS_MDP_VP_0_LAYER_0_GC_AR-
LUT_GREEN_STARTX_N, 3464

MDSS_MDP_VP_0_LAYER_0_GC_ARLUT_RED_OFFSET_N, 3463
 MDSS_MDP_VP_0_LAYER_0_GC_ARLUT_RED_SLOPE_N, 3463
 MDSS_MDP_VP_0_LAYER_0_GC_ARLUT_RED_STARTX_N, 3463
 MDSS_MDP_VP_0_LAYER_0_GC_MISR_CTRL, 3470
 MDSS_MDP_VP_0_LAYER_0_GC_MISR_SIGNATURE, 3471
 MDSS_MDP_VP_0_LAYER_0_OP_MODE, 3438
 MDSS_MDP_VP_0_LAYER_0_OUT_SIZE, 3439
 MDSS_MDP_VP_0_LAYER_1_BLEND0_BG_ALPHA, 3475
 MDSS_MDP_VP_0_LAYER_1_BLEND0_B-G_TRANSP_HIGH0, 3477
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 MDSS_MDP_VP_0_LAYER_1_BLEND0_B-G_TRANSP_LOW0, 3476
 MDSS_MDP_VP_0_LAYER_1_BLEND0_B-G_TRANSP_LOW1, 3477
 MDSS_MDP_VP_0_LAYER_1_BLEND0_FG_ALPHA, 3474
 MDSS_MDP_VP_0_LAYER_1_BLEND0_F-G_TRANSP_HIGH0, 3476
 MDSS_MDP_VP_0_LAYER_1_BLEND0_F-G_TRANSP_HIGH1, 3476
 MDSS_MDP_VP_0_LAYER_1_BLEND0_F-G_TRANSP_LOW0, 3475
 MDSS_MDP_VP_0_LAYER_1_BLEND0_F-G_TRANSP_LOW1, 3475
 MDSS_MDP_VP_0_LAYER_1_BLEND0_OP, 3473
 MDSS_MDP_VP_0_LAYER_1_BLEND1_BG_ALPHA, 3479
 MDSS_MDP_VP_0_LAYER_1_BLEND1_B-G_TRANSP_HIGH0, 3481
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 MDSS_MDP_VP_0_LAYER_1_BLEND1_B-G_TRANSP_LOW0, 3481
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 MDSS_MDP_VP_0_LAYER_1_BLEND1_FG_ALPHA, 3479
 MDSS_MDP_VP_0_LAYER_1_BLEND1_F-G_TRANSP_HIGH0, 3480
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 MDSS_MDP_VP_0_LAYER_1_BLEND1_F-G_TRANSP_LOW0, 3479
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 MDSS_MDP_VP_0_LAYER_1_BLEND1_OP, 3478
 MDSS_MDP_VP_0_LAYER_1_BLEND2_BG_ALPHA, 3484
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 MDSS_MDP_VP_0_LAYER_1_BLEND2_B-G_TRANSP_HIGH1, 3486
 MDSS_MDP_VP_0_LAYER_1_BLEND2_B-G_TRANSP_LOW0, 3485
 MDSS_MDP_VP_0_LAYER_1_BLEND2_B-G_TRANSP_LOW1, 3486
 MDSS_MDP_VP_0_LAYER_1_BLEND2_FG_ALPHA, 3483
 MDSS_MDP_VP_0_LAYER_1_BLEND2_F-G_TRANSP_HIGH0, 3485
 MDSS_MDP_VP_0_LAYER_1_BLEND2_F-G_TRANSP_HIGH1, 3485
 MDSS_MDP_VP_0_LAYER_1_BLEND2_F-G_TRANSP_LOW0, 3484
 MDSS_MDP_VP_0_LAYER_1_BLEND2_F-G_TRANSP_LOW1, 3484
 MDSS_MDP_VP_0_LAYER_1_BLEND2_OP, 3482
 MDSS_MDP_VP_0_LAYER_1_BLEND3_BG_ALPHA, 3488
 MDSS_MDP_VP_0_LAYER_1_BLEND3_B-G_TRANSP_HIGH0, 3490
 MDSS_MDP_VP_0_LAYER_1_BLEND3_B-G_TRANSP_HIGH1, 3491
 MDSS_MDP_VP_0_LAYER_1_BLEND3_B-G_TRANSP_LOW0, 3490
 MDSS_MDP_VP_0_LAYER_1_BLEND3_B-G_TRANSP_LOW1, 3490
 MDSS_MDP_VP_0_LAYER_1_BLEND3_FG_ALPHA, 3488
 MDSS_MDP_VP_0_LAYER_1_BLEND3_F-G_TRANSP_HIGH0, 3489
 MDSS_MDP_VP_0_LAYER_1_BLEND3_F-G_TRANSP_HIGH1, 3489
 MDSS_MDP_VP_0_LAYER_1_BLEND3_F-G_TRANSP_LOW0, 3488
 MDSS_MDP_VP_0_LAYER_1_BLEND3_F-G_TRANSP_LOW1, 3489
 MDSS_MDP_VP_0_LAYER_1_BLEND3_OP, 3487
 MDSS_MDP_VP_0_LAYER_1_BORDER_COLOR_0, 3473
 MDSS_MDP_VP_0_LAYER_1_BORDER_COLOR_1, 3473
 MDSS_MDP_VP_0_LAYER_1_CURSOR_BASE_ADDR_SW_STATUS, 3499
 MDSS_MDP_VP_0_LAYER_1_CURSOR_BASE_ADDR, 3493
 MDSS_MDP_VP_0_LAYER_1_CURSOR_BLEND_CONFIG, 3494
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 MDSS_MDP_VP_0_LAYER_1_CURSOR_BLEND_TRANSP_LOW0, 3494
 MDSS_MDP_VP_0_LAYER_1_CURSOR_BLEND_TRANSP_LOW1, 3495
 MDSS_MDP_VP_0_LAYER_1_CURSOR_FETCH_CONFIG, 3499
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 MDSS_MDP_VP_0_LAYER_1_CURSOR_REQPRIORITY_FIFO_WATERMARK_0, 3501
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A Exhibit 1

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2. **COMPLIANCE WITH LAWS; APPLICABLE LAW.** You agree to comply with all applicable local, international and national laws and regulations and with U.S. Export Administration Regulations, as they apply to the subject matter of this Agreement. This Agreement is governed by the laws of the State of California, excluding California's choice of law rules.

3. **CONTRACTING PARTIES.** If the Materials are downloaded on any computer owned by a corporation or other legal entity, then this Agreement is formed by and between QTI and such entity. The individual accepting the terms of this Agreement represents and warrants to QTI that they have the authority to bind such entity to the terms and conditions of this Agreement.

4. **MISCELLANEOUS PROVISIONS.** This Agreement, together with all exhibits attached hereto, which are incorporated herein by this reference, constitutes the entire agreement between QTI and You and supersedes all prior negotiations, representations and agreements between the parties with respect to the subject matter hereof. No addition or modification of this Agreement shall be effective unless made in writing and signed by the respective representatives of QTI and You. The restrictions, limitations, exclusions and conditions set forth in this Agreement shall apply even if QTI or any of its affiliates becomes aware of or fails to act in a manner to address any violation or failure to comply therewith. You hereby acknowledge and agree that the restrictions, limitations, conditions and exclusions imposed in this Agreement on the rights granted in this Agreement are not a derogation of the benefits of such rights. You further acknowledges that, in the absence of such restrictions, limitations, conditions and exclusions, QTI would not have entered into this Agreement with You. Each party shall be responsible for and shall bear its own expenses in connection with this Agreement. If any of the provisions of this Agreement are determined to be invalid, illegal, or otherwise unenforceable, the remaining provisions shall remain in full force and effect. This Agreement is entered into solely in the English language, and if for any reason any other language version is prepared by any party, it shall be solely for convenience and the English version shall govern and control all aspects. If You are located in the province of Quebec, Canada, the following applies: The Parties hereby confirm they have requested this Agreement and all related documents be prepared in English.